

Research continuation and complete source proofs

Split-Zero Cohomology and Arithmetic Weight Control

13 September 2026

This reader integrates the Split-Zero carrier, finite arithmetic operator comparisons, the historical packet ghost, chain retractions, arithmetic moments, Gaussian resolvents, full kernel layers, symmetric tensor frontiers, measured support branches, the original theta norm-volume identities, quartet derivative costs, gamma-reference determinants, and the complete two-Stieltjes-space and spectral-sum comparisons, the full relative connection, strict normal energy, the conormal derivative with its original residue maps, nested relative-frame transitions and density, the full limiting derivative graph, cyclic arithmetic metric comparisons, every retained conormal depth, exterior trace amplification, the exact equality and coupling maps, positive exterior formulas for arithmetic interpolation volumes, the original source and relation Toda flows with all four loss terms, exact endpoint comparisons, certified theta-seed input with explicit infinite tails, gamma convolution and its full arithmetic coefficient maps, the certified seed coefficients, finite source-to-quotient metric transfer, and the complete complex amplitude in the sum fibres. This edition also proves the original arithmetic monic-norm estimate, the balanced endpoint bound, exact products and their sharp scalar optimization, full confluent transfer, finite Hermitian window transport, the arithmetic central-volume upper comparison, and reflected norm/conormal kernel transport. Every original source note and the complete independent calculations supporting these statements accompany the proof chapters. The displayed upper and lower volume estimates retain their explicitly calculated gap.

The continuation also constructs the compact-support image and its dual, logarithmic specialization, exact invariant-subspace coupling, polynomial-exponential cohomology, its complete period determinant, and the translation maps carrying the original arithmetic operator and metric through this family. It also proves the finite-free SGA trace, derived collision modules and constituent period curvature, and joins the full Hochschild critical-comparison kernel to that curvature and to the original-metric Laplacian control. It further gives the complete periodized source, density and tail estimates, finite-circle curvature and quotient compensation, the circle observation diamond, and the residue-constituent source and curvature maps. It further includes the complete Holonomy source and the exact selected descent, quotient, generator and relation-completion proofs, together with their full independent mathematical reviews. It retains the original arithmetic functions, measures, scalar actions, support fibres, and complete jets. Part I gives the local continuation with proofs. Part II preserves the complete selected source notes, their exact revisions, and their chronology.

Contents

I	Integrated mathematical continuation	37
1	Exact Split-Zero carriers and the local spectral bridge	37
1.1	Scalar support and independent channel amplitudes	37
1.2	Tesserine sectors and a supported critical fixed locus	39
1.3	Exact sector, holonomy and positive-moment maps	40
1.4	The theta-packet form and its scalar comparison	42
1.5	Projective zeta at all zeros and at its pole	42
1.6	Matrix packets and their full relation module	43
2	The historical packet ghost and its full arithmetic jet quotient	44
3	Historical Hurwitz reconstruction and the complete Laurent jet bridge	48
3.1	Source receipts and exact locators	48
3.2	Every compact Hurwitz input is determined by one simple-zero germ	49
3.3	The complete zero cluster removes the simplicity restriction	52
3.4	The historical period map preserves every finite spectral jet	53
3.5	Exact conjugate-reciprocal involution on the jets and conormal	55
3.6	Period collisions: the exact unlabelled image and its missing fibres	56
3.7	Two prime periods recover the complete finite spectral algebra	57
3.8	Split-Zero support is preserved by the exact maps	59
3.9	All-order Hurwitz motion transported to the period and tesserine coordinates	60
3.10	Mathematical scope	61
4	Historical support branches, ideal completion, and measured observations	61
4.1	Exact ordinary and continuous Split-Zero branches	62
4.2	The exact frame comparison retaining all ordinary branches	64
4.3	A fully explicit nontrivial frame with algebraic branches and no frame points	66
4.4	Atomic observation and the exact support data it removes	67
4.5	The exact operator-algebraic branch map	68
5	Reconstruction with changes of support index	70
5.1	The internal quotient and the actual comparison kernel	71
5.2	The earlier reconstruction applied to every original source level	72
6	The original theta complex and its arithmetic packet	73
6.1	Source spaces and fixed arithmetic factors	74
6.2	The exact operator range, including its inverse	75
6.3	The original extension supplies the boundary coefficient	76
6.4	Actual representative corrections and rank-two control	77
7	Certified moments from the original arithmetic theta source	78
7.1	Source polynomials and the exact reflection map	78
7.2	An absolutely convergent double series for every entry	79
7.3	A complete finite truncation certificate	80
7.4	Exact adjoint, parity, and finite-boundary checks	81
7.5	Reproducible certified finite arithmetic data	82
8	Exact source-metric optimization and aggregate arithmetic control	83

8.1	The fixed arithmetic maps and their boundary	83
8.2	A source-density calculation with its topology retained	84
8.3	Every positive packet metric has an exact finite source lift	84
8.4	Sharp control with all nilpotent jets present	86
8.5	The exact tensor cost of unrestricted metrics	88
8.6	A stronger bound on the actual orthogonal sequence	88
8.7	Original supported maps at every independent source and coefficient face . .	89
9	The actual theta minimum as a confluent arithmetic kernel	90
9.1	Original functions, maps, coordinates, and measures	90
9.2	Exact constrained minimum over the admitted source	91
9.3	Monic arithmetic polynomials and their exact recurrence	93
9.4	Exact relative weight control from the same kernel	94
9.5	Exact change when one more theta direction is admitted	95
9.6	The original measure supplies a full tail recurrence	95
9.7	Metric dependence and the precise mathematical gain	97
9.8	Propagation through the original finite source tower	98
10	The full jet kernel, finite resolvents, and the actual analytic tail	98
10.1	A finite compression in the original Mellin coordinate	99
10.2	Exact quadrature and the full confluent kernel	100
10.3	The infinite multiplication operator and its genuine tail	101
11	Quantitative convergence of the arithmetic Gaussian resolvents	103
11.1	The Gaussian rule inherits the original exponential moments	104
11.2	Exact moments give a quantitative analytic-strip estimate	104
11.3	Every confluent derivative and the original finite algebra	105
12	The exact terminal step and its retained boundary energy	108
12.1	A step written entirely in the original polynomial packet	108
12.2	The complete boundary-energy increment	109
12.3	The exact rational terminal step and its polynomial continuation	110
12.4	The extra jets required to recover the metric from the tail	111
12.5	Actual quartet packets close the energy in two volume ratios	112
12.6	What the recurrence proves about boundary energy	113
12.7	An exact positive-moment counterexample to consecutive energy monotonicity	113
13	The exact metric departure and the cost of complete jets	119
13.1	A quantitative constraint on every retained nilpotent chain	120
14	Coherent packets, joint tensor layers, and their exact spectral cost	121
14.1	The arithmetic unit and the coherent one-variable maps	121
14.2	The original reflection and the exact diagonal coefficient	122
14.3	The joint total-degree source and its relation layer	122
14.4	Nested packets on the joint source: the additional projection	124
14.5	Positive layer rank and the complete tensor spectral departure	124
15	Density of the original joint boundaries and determinant dissipation	126
15.1	Polynomial density with the original weighted measures	126
15.2	Strong limits of the actual source representatives	127

15.3	The accumulated determinant cost of the same boundary sequence	129
16	The actual kernel layer, its derivative image, and its quotient	130
16.1	The original arithmetic source and its constrained minimum	130
16.2	The complete next-layer frame and its theta primitives	131
16.3	The exact incidence map and the derivative-image rank	132
16.4	The whole quotient of the next layer by the derivative image	133
16.5	The derivative between consecutive relation quotients	134
16.6	The exact original-theta norm and volume bridge	134
16.7	The complete scalar dictionary for the boundary energy	136
17	Original orthogonal shells and the exact projective-ratio algebra	137
17.1	Two sections of the same polynomial filtration	137
17.2	The full correspondence between the two image spaces	139
17.3	The algebra generated by the original coordinate ratio	140
17.4	A complete finite witness for unequal shell ranks	142
18	The original frontier energy, determinant growth, and the entire border fibre	144
18.1	Original moment coordinates and the relation graph	144
18.2	Exact relative loss and the source derivative energy	146
18.3	The entire small border operator, including its zero extension	147
18.4	A quantified link between the frontier bound and determinant cost	148
19	The signed symmetric summand and its complete spectral cost	149
19.1	The chain projector and its completed and finite targets	149
19.2	Orbit coordinates, their Gram, and all local nilpotents	151
19.3	Complete symmetric spectral sums and their exact departure	152
19.4	The cost in the original symmetric relation layer	154
20	Arithmetic quartet parity and the exact positive frontier operator	155
20.1	The actual two frontier matrices	155
20.2	The original linear reflection separates the two frontiers	156
20.3	The complete positive operator and the sharper arithmetic bound	157
20.4	The same positive frontier on the actual symmetric summand	159
21	Quartet parity in the original derivative energy and volume loss	162
21.1	Original maps, their spaces, and the reflection	162
21.2	The single original boundary block	163
21.3	Complete determinant constraints, including zero cases	164
21.4	Exact energy-volume equality for the original one-variable layer	165
22	The exact two-Stieltjes-space representation of the theta packet	166
22.1	The quadratic algebra and both reflection branches	166
22.2	The original measures and a Hilbert space isomorphism	168
22.3	The two monic families and their original norms	169
22.4	Both full jet kernels and the exact weight operator	170
22.5	The original tesserine coordinates and support lift	171
22.6	The complete rank-one half-line boundary and its derivative cost	172
23	The exact gamma reference and the arithmetic Gram determinants	173
23.1	The original measure, phase, and packet cancellation	173

23.2	Gamma norms from a generating integral	174
23.3	Finite arithmetic determinants in the retained metrics	175
23.4	Full confluent kernel and the original coordinates	176
23.5	The finite map to Romik's gamma parameter	177
23.6	Exact scope of the exponential-weight theorem	178
23.7	Primary source locators	178
24	The symmetric spectral sum and its unequal Stieltjes pair	179
24.1	Original polynomial coordinates and the reflection ideals	179
24.2	The entire defect is the original paired packet	180
24.3	The original unit and all coordinate factorials	182
24.4	The actual matrix weight and its positive half-line pair	183
24.5	Full-jet kernels for the actual constrained minimum	184
24.6	Original primitives and supported maps	185
25	The sum connection on both half-line branches and its exact domain	186
25.1	The actual amplitude, its mass, and the original relative fibre	186
25.2	The complete matrix connection and its actual graph	187
25.3	A branch-preserving quadratic map for the entire matrix weight	188
25.4	The meeting point of the two branches is part of the domain	190
25.5	The even specialization and the unchanged scalar contraction	190
25.6	The actual arithmetic normal contribution is strictly positive	191
25.7	Exact reconstruction from the actual finite normal image	193
26	The original conormal derivative and its finite section commutator	195
26.1	The first thickening and the dual-number map	195
26.2	The original monic section and the exact finite correction	197
26.3	The full arithmetic unit and the original finite representatives	199
26.4	The one-variable residue functional and the original theta relation	200
26.5	Supported algebra and coefficient maps	201
26.6	The arithmetic unit and complete tensor partial residues	202
27	Exact transitions between nested relative frames	204
27.1	Original sources and coefficient inclusions	204
27.2	The original weights and their orthogonal transition	205
27.3	Full source matrices, graph maps and arithmetic observation	206
27.4	Every variable-frame term	208
27.5	Nested chains and both quadratic branches	209
27.6	Exact zero and endpoint scopes	210
27.7	One independent exact original-coordinate calibration	210
28	Density of the actual relative frames and the limiting derivative graph	212
28.1	An exponential bound with the original arithmetic factors	212
28.2	Every centred fibre and its exponential moment	214
28.3	A complete weighted polynomial density proof	214
28.4	The multiplier, literal frames, and their strong limit	215
28.5	A fixed original source and the complete derivative domain	216
28.6	Invariant frames, sparse frames, and all original zero cases	219

28.7	Frame covariance and the exact half-line transport	220
29	The actual normal-image limit, its graph, and its arithmetic observation	221
29.1	Exact contractions of the finite normal images	222
29.2	The algebraic limit and its two exact metric presentations	223
29.3	Original arithmetic jets and the size of reconstruction	223
29.4	The split lift retains the algebraic source of the zero limit	225
30	The cyclic sum at every conormal depth	226
30.1	Original coordinates and the exact collided annihilator	226
30.2	Unscaled invariant bases and a full-unit retraction	228
30.3	The derivative tower, scalar first jets, and the unit obstruction	229
30.4	The finite cyclic derivative and its exact Weyl correction	231
30.5	All first jets and the symmetric conormal kernel	231
30.6	Cyclic relation response and the full product Jacobian	233
30.7	Trace pairings and all four complementary operator blocks	234
30.8	Exact two-factor symmetric presentation at every depth	236
30.9	Split lifts, source provenance and quantitative scope	237
31	The cyclic sum source inside the full connection	238
31.1	Complete polynomial moments and the original finite metric	238
31.2	A full real-exponential family of the same information identity	239
31.3	All cyclic normal energy is retained under frame enlargement	240
31.4	The canonical metric difference has its actual conormal lift	241
32	Exterior trace equality, the spectral projector, and quantitative coupling	242
32.1	1. Original objects and exact domains	242
32.2	2. Exact trace maximum and equality theorem	243
32.3	3. Sharp quantitative trace slack	244
32.4	4. Exact morphism from the projector difference to the coupling	245
32.5	5. Sharp original two-dimensional family	246
32.6	6. Critical generalized blocks at the trace maximum	246
32.7	The original quartet, its critical blocks, and a canonical equality example	248
32.8	7. Integration scope	249
33	The cyclic control as an exact change of arithmetic interpolation volume	249
33.1	The exact volume map and the reflected quartet bound	250
33.2	The positive exterior minor sum and its actual source vector	251
33.3	The exact first admitted degree	253
34	The original source and relation Toda flows with their exact control maps	253
34.1	Original arithmetic objects and the strip of real tilts	254
34.2	The source relation and its Laplace observation	256
34.3	Both monic determinants and their exact determinant-line map	257
34.4	Toda derivatives and the full curvature operators	257
34.5	The rank-one update, trace phase and original rank-two radius	259
34.6	Exact maps to the CV omitted determinant and positive minors	260
34.7	Both exact losses, their equality cases and the telescope	261
34.8	The first admitted degree and the empty quotient	262

34.9	The complete join with the original spectral coupling	264
34.10	Crosswalk and exact continuing arithmetic quantity	265
35	The original theta input and certified infinite-series tails	265
35.1	The original theta function and its Mellin transform	266
35.2	Mellin division, the dagger sign, and the tilted observation	267
35.3	The complete double series and an explicit omitted tail	268
35.4	The second moment and the first tensor coefficient	270
35.5	Certified numerical enclosures and their executable scope	271
36	An exact comparison of equality, volume contraction, and its discarded square	272
37	The original gamma convolution and the exact arithmetic coefficient maps	274
37.1	Source, attribution, and original coordinates	274
37.2	The gamma reference with its actual mass	276
37.3	The sum projection and exact coefficient powers	277
37.4	The supplied analytic generating join, with its branches	278
37.5	The complete density expansion for the original bounded references	280
37.6	Original-coordinate finite Grams and every tilt derivative	281
37.7	The original cyclic quotient and the exact cochain correction	282
37.8	Determinant indices and the original Toda substitution	283
38	Certified theta moments and the original gamma arithmetic coefficients	284
38.1	The theta input and its transform, with the original factors	285
38.2	Every derivative polynomial and the exact higher-moment map	286
38.3	The incomplete-gamma square and a proved omitted tail	287
38.4	The coefficient morphism at the actual gamma reference	288
38.5	Certified infinite quantities and exact rational error transport	290
38.6	Exact tensor coefficient interface through degree six	292
39	The first three original source costs at every tensor degree	293
40	Finite arithmetic Gram transfer with the original minimum quotient metric	295
40.1	The original source, relation and coordinate maps	296
40.2	Full Gamma mass and the finite coefficient evaluation	297
40.3	Finite coefficient errors and polynomial-test bounds	299
40.4	Exact least lifts and the boundary-resolvent remainder	300
40.5	A two-degree map for the actual consecutive ratio	301
40.6	Certified finite envelopes and quotient-dimensional bounds	302
40.7	Exact calibration and a bounded relative-volume counterexample	303
40.8	Verification scope	306
41	The complete complex amplitude in gamma sum descent	306
41.1	Original objects and the question answered	306
41.2	Unitary transport of the actual conditional projection	307
41.3	Strictness for an actual quartet, with its exact constant	308
41.4	Full complex coefficient transform and exact variance norm	308
41.5	The exact induced change of the original finite quotient metric	309
41.6	Support, the original unit, and the domain of arithmetic observation	310
41.7	Exact polynomial calibration with the original gamma mass	312

41.8	Provenance and consequence	313
42	Exact endpoint windows: Gamma coefficients, paired Schur volumes, and intrinsic sensitivity	313
42.1	The original source and the complete finite input map	313
42.2	Canonical lifts and one arbitrary-window block	315
42.3	Endpoint roots and exact paired-volume cancellations	316
42.4	Exact projection between the canonical representative spaces	317
42.5	Window sensitivity and exact cancellation of mass changes	319
42.6	Rational interval certificates for the original input	322
42.7	Boundary cases and the retained cochain map	323
42.8	Verification scope and exact auxiliary calibration	325
43	Complete arithmetic endpoint derivation and exact source audit	327
43.1	Arithmetic endpoint delivery: complete source intake and mathematical audit	328
44	The original leading-coefficient line and its arithmetic relation image	335
45	Exact window products, sharp endpoint optimization, and original relation costs	338
45.1	Original source and a proof of its exact radius identity	338
45.2	Window product, endpoint extremum, and phases	340
45.3	The sharp scalar bound with only the combined budget	341
45.4	Schur windows and the reduced source–relation determinant	342
45.5	Full confluent transfer and its two-determinant window	343
45.6	An arithmetic lower cost on one relation block	345
45.7	Mass-seven polynomial calibration and zero cases	348
46	The complete balanced four-volume threshold calculation	349
46.1	The two endpoint calculations give a four-volume threshold of four	349
47	The complete balanced-window arithmetic proof	357
47.1	Balanced arithmetic norm windows and the strengthened endpoint threshold	357
48	PR 24: complete implemented corrected proof	359
48.1	Confluent transfer for the original source–relation volume	359
49	An arithmetic lower-envelope comparison for central quotient volumes: full jets, finite determinant bounds, and an exact quadratic benchmark	365
49.1	Original source and every jet coordinate	365
49.2	The lower envelope as an explicit positive comparison form	367
49.3	An upper determinant depending only on the two Laplace values	368
49.4	The interval kernel and the actual comparison map	369
49.5	The explicit scalar bound and its finite gap	369
49.6	A full-jet quadratic benchmark and its exact relation family	372
49.7	Source representatives and support	374
49.8	Verification and provenance	374
50	Reflected relation norms and the exact common refinement of conormal algebras	375
50.1	The original reflection and the norm polynomial	375
50.2	Full jets at the union of the original and reflected roots	376
50.3	A common refinement with the original first conormal algebra	376
50.4	Square-zero structure, reflected sectors, and supported zero	377
50.5	Reflected kernel transport and every power of the norm kernel	378

	50.6	Provenance and verification scope	379
51		Exact optimization of the arithmetic full-jet determinant weights	380
	51.1	The original source and the signed full-jet minors	380
	51.2	Global maximizers, including every zero coefficient	381
	51.3	A complete monotone scalar description of the interior optimum	383
	51.4	Evaluable enclosures and the exact maximum gain	385
	51.5	The optimized original arithmetic bound, including boundary weights	386
	51.6	Joint even moments and exact optimization of their diagonal family	388
	51.7	Verification boundary and continuity	390
52		Original relation moments and their positive-form comparison	391
	52.1	The original Hilbert objects and their maps	391
	52.2	The source correction, all cross-pairings, and exterior area	392
	52.3	Sharp determinant bound, exact remainder, and all boundary cases	392
	52.4	The comparator join in the original metric	394
	52.5	Literal Gaussian calibration and source corrections	395
	52.6	Rank information with every zero direction retained	395
	52.7	The entire comparison excess is the source contraction defect	396
	52.8	Attribution and exact source scope	397
53		The first canonical window, its full phase product, and both arithmetic block budgets	397
	53.1	Original source and the actual first-degree step	398
	53.2	The complete first-window product and both finite budgets	399
	53.3	Actual comparable norm windows with complete constants	400
	53.4	The full relation cost and its retained zero directions	401
	53.5	Provenance and verification scope	402
54		Exact arithmetic restriction certificates and projection products	402
	54.1	The original source, quotient, and canonical lifts	403
	54.2	Return, action, and the entire original relation correction	403
	54.3	The exact source projection and principal-angle morphism	404
	54.4	The arithmetic remainder bound at every finite degree	405
	54.5	Finite comparison matrices and the complete trace error	406
	54.6	Constructing rational brackets for the actual relation	408
	54.7	The four-volume comparison with its literal finite constants	408
	54.8	Central rank, both endpoint returns, and all inverse powers	409
	54.9	Exact auxiliary calculation and retained proof dependencies	412
55		Deligne machinery inside Split-Zero cohomology	412
	55.1	Deligne's image comparison inside the Split-Zero source complexes	412
56		Exact determinant of the polynomial-exponential period comparison	425
	56.1	Exact determinant of the polynomial-exponential period comparison	425
57		Translation of the exponential comparison: full coefficient, period, metric, and finite-field maps	429
	57.1	The original fibre and the auxiliary translation	429
	57.2	Periods and the retained exponential factor	430
	57.3	Translation of the full singular connection and boundary character	431

57.4	The original positive forms and their control operators	431
57.5	The finite-field comparison with its scalar character	432
57.6	Specialization and split support	433
58	Primary-source correction to the Deligne comparison	433
58.1	Primary-source correction to the Deligne comparison	433
59	SGA trace, exact constituent curvature, and the derived collision fibre	435
59.1	SGA trace, exact constituent curvature, and the derived collision fibre	435
60	The SGA trace-period source: exact coefficient and source corrections	442
60.1	The SGA trace-period source: exact coefficient and source corrections	442
61	Hochschild trace, Laplacian, and specialization: source integration	444
61.1	Hochschild trace, Laplacian, and specialization: source integration	444
62	The period curvature of the retained critical-comparison kernel	446
62.1	The period curvature of the retained critical-comparison kernel	446
63	The original Laplacian control along the exact period deformation	456
63.1	The original Laplacian control along the exact period deformation	456
64	Periodized source: exact scope and source definitions	460
64.1	The periodized-source proof and its completed comparison	460
65	Complete original periodized source control note	460
65.1	Result and scope	461
65.2	1. Source intake and the Deligne operation	461
65.3	2. Retain the split coefficients and arithmetic source	462
65.4	3. An exact diagonal logarithmic coordinate map	462
65.5	4. Periodization and the full zeroth-mode map	463
65.6	5. The original Gram is recovered with an exponential error	463
65.7	6. Exact recovery from all nonzero modes	465
65.8	7. Transfer through the same full-jet quotient	465
65.9	8. The four-volume coefficient is preserved up to a calculated error	466
65.10	9. The periodic Laplacian and its exact arithmetic boundary	466
65.11	10. Retain the critical comparison kernel and its cone	467
65.12	11. The exact fixed-circle completion and its lost packet directions	468
65.13	12. A finite Fourier cutoff with an explicit source-side error	469
65.14	13. What has and has not been added	470
65.15	References and evidence	470
66	Complete periodized arithmetic source and finite-cutoff proofs	470
66.1	Periodized source control: complete comparison calculations	470
67	Complete periodized source density, tails and completion	479
67.1	Density and exact completion of the original periodized arithmetic source	479
68	Finite-circle curvature and complete quotient compensation	487
68.1	Finite sampling, residue curvature and the original control form	487
69	Complete circle and critical-observation diamond	493
69.1	The exact diamond of the circle and tensor-critical observations	493
70	Residue constituent source and exact derivative maps	500
70.1	Current reading of the delivered residue-constituent note	500

71	Complete original residue constituent curvature note	501
71.1	Scope and source identifications	501
71.2	The deformation is the residue rank-one map	502
71.3	Exact constituent-period curvature	503
71.4	Transfer to the original theta metric	505
71.5	The period realization of the arithmetic Laplacian	506
71.6	Translation, cohomological support, and exact scope	507
71.7	Verification and provenance	508
71.8	References and supplied-source locators	508
72	Complete residue constituent curvature extension	509
72.1	Residue endomorphisms, metric variation, and the forced period Laplacian . .	509
73	Holonomy descent: exact public source reading	517
73.1	Reading the holonomy delivery with its exact domains and verification	517
74	The complete original holonomy-descent control note	518
74.1	Result and scope	518
74.2	1. Fixed inputs, maps, and the marked split base	518
74.3	2. All circle holonomies recover the full source	519
74.4	3. Every finite cyclic cover is an exact character decomposition	520
74.5	4. Common arithmetic representatives and the exact descent cost	521
74.6	5. Sharp quadratic control of the descent cost	522
74.7	6. Endpoints and a genuinely finite phase construction	524
74.8	7. The original action and relation primitive descend together	526
74.9	8. Quantitative transport of the new residue-constituent coefficient	526
74.10	9. What premature completion would forget	527
74.11	10. Exact calibrations and the remaining arithmetic estimate	527
74.12	Source basis and attribution	528
75	The complete typed holonomy-descent addendum	529
75.1	Holonomy descent: the inverse domain, positive root masses, and sharp source realization	529
76	The complete phase quotient and polynomial density	533
76.1	Independent completed-phase quotient review	533
77	Continuous holonomy cochains and the original relation metric	538
77.1	Original coefficient spaces and actual phase norms	539
77.2	The continuous cochain splitting and its full contraction	540
77.3	The minimum norm of the common image	541
77.4	The positive relation gap and its mean-zero decomposition	542
77.5	Actual sampled Hilbert spaces, including zero weights	542
77.6	Density and the complete fibrewise Hilbert quotient	543
77.7	The complete primary kernel before phase integration	544
77.8	The measurable completed field and its vanishing map	545
77.9	The source-to-completion cochain map	545
77.10	The receiving supported zero and all marked fibres	546
77.11	Exact outcome and scope	546

78	Holonomy descent and the original arithmetic generator	546
78.1	Holonomy descent cost and the original arithmetic generator	546
79	Exact phase observation and its full degree cost	552
79.1	Exact fixed-phase observation and its degree cost	552
80	The complete continuous relation closure	554
81	Complete independent proof of the original continuous relation completion	558
81.1	Independent review: original continuous relation completion	558
82	Arithmetic determinant transport on the original tau complex	563
82.1	Original objects and the fixed comparison	563
82.2	Canonical sections, determinant lines and full variations	564
82.3	The signed four-endpoint term on one retained source	566
82.4	The full spectral and actual-overlap bound for the correction	567
82.5	Source dependencies and precise status	571
83	Exact identification of the retained AT extraction hash	571
84	Exact relation-window spectral transport on the original tau complex	574
84.1	The original source and its four windows	574
84.2	Full spectra and an explicit source isometry	575
84.3	The spectral bound with every eigenvalue retained	576
85	The one-dimensional quotient: exact angle and nonlinear transport	580
85.1	Complete proof dependencies	581
86	Complete arithmetic cohomology over the tau-base	581
87	Complete Gamma-reference lower bound and arithmetic determinant comparison . . .	589
88	Complete Deligne comparison and its actual tau-base cohomology	600
89	The original tau-base weight line in the complete exponential reflection flow	611
90	Complete independent tau-base arithmetic determinant transport proof	615
91	Actual tau-base scaling and its complete exponential cohomology flow	621
92	Complete relation-window spectral proof and literal-flag strictness	628
93	The complete original arithmetic Taylor-unit formal gauge	635
94	Complete independent review of the actual Taylor-unit formal gauge	640
95	Complete arithmetic determinant variations and their original determinant lines . . .	642
96	All-degree angle transport on the original arithmetic four-volume	647
96.1	Fixed arithmetic source, maps and forms	647
96.2	Relation energy, Gram ratios and the full principal angles	648
96.3	The actual four-volume has at most $2q - 1$ nonzero angles	649
96.4	The moment identity forces strict positivity	649
96.5	A nonlinear endpoint estimate in every degree	650
96.6	Complete source and mathematical scope	651
97	Complete four-window spectral sharpness on the original polynomial flags	651
98	Complete independent four-window sharpness proof and fixed-metric optimum	657
99	Complete analytic pole classes, connection, periods and endpoint transport	663
100	Complete independent review of analytic pole-residue transport	673
101	Complete all-degree principal-angle proof and literal tensor theta primitive	676
102	Complete independent principal-angle volume proof and integrated metric comparison	683

103	Complete independent acceptance proof of the literal tensor theta primitive	687
104	Which completion retains the arithmetic jets?	688
104.1	Logarithmic coordinates and the original measure	689
104.2	Exponential weights and exact jet norms	690
104.3	The energy term carried by the weighted topology	691
104.4	The tesserine coordinate in the actual weight form	691
II	Complete counterfactual, mixed-support and boundary calculations	692
105	The object, its actual elements, and its assembly	692
106	The absolute base and the two distinct zero fibres	694
107	Exactly what survives and how every offcritical zero enters	695
108	The norm observations are values on this same object	696
109	One anchored offcritical assembly of the original tau complex	696
109.1	The source-indexed complex and its canonical directed limit	696
109.2	The exact Mellin identification, including every actual zero	697
109.3	The anchored sheaf diagram and its retained diagonal and kernel	698
109.4	Full finite observations and all tensor jets of the same diagram	700
109.5	All proper source and original leg-support fibres, and the exact actions	702
109.6	The exact attachment to the single arithmetic tensor algebra	703
110	The single arithmetic carrier and its divisor-tensor gluing	705
110.1	A definition that does not presuppose a list of zeros	705
110.2	The exact divisor arrows and their arithmetic units	706
110.3	One differential graded arithmetic object	707
110.4	The finite original source and metric diagrams	708
110.5	The cyclic sources and every conormal depth in the same diagram	709
110.6	Reflection and the retained split zero	712
111	Full mixed carriers, their faces, prime branches, and depth maps	713
112	All independent coefficient faces attached to the original theta source	723
112.1	The original source and its full finite arithmetic sections	723
112.2	The independent coefficient face category and its exact quotient	725
112.3	Independent slot-local leg presentations	726
112.4	Arithmetic dilation on every face and its full source boundary	727
112.5	Packet enlargement with every coefficient face retained	728
112.6	The original negacyclic masks and coefficient Frobenius	728
112.7	The independent outer source label, including empty coefficient support	730
112.8	Exact enlargement from offcritical support to all nontrivial packets	732
112.9	Literal comparison with the earlier finite source and its cocycle	734
112.10	The shifted coefficient quotient on the original theta complexes	735
113	The empty-packet endpoint of the exact source diagram	736
114	The exact original cochain attachment on every independent coefficient face	736
114.1	Original spaces, independent faces, and their exact cochain maps	737
114.2	The complete original section and its Euler boundary	738
114.3	Full-unit divisor transport on every face	739

114.4	The exact dilation homotopy on the original source	740
114.5	A strict cochain map that can actually be balanced	741
114.6	Every proper source retains its residual class	742
114.7	Naturality to the independent mixed observation vertices	743
114.8	The full coherent and tensor comparison on all faces	744
114.9	Finite-stage continuity and the attachment category	745
115	The nilpotent filtration on every independent arithmetic face	746
115.1	Original packets and the filtration	746
115.2	Independent masks and the arithmetic action	747
115.3	Tensor products with every coefficient and sign retained	748
115.4	The exact Tate-scaling comparison	749
115.5	The actual length-two quotient retains its induced degrees	751
116	Strict filtered control of the original length-two jet extension	753
116.1	Uniqueness of the centered nilpotent filtration	753
116.2	Translation to a specified constituent degree	753
116.3	The original extension and its arithmetic action	754
116.4	Induced subobject and quotient filtrations	754
116.5	Added Frobenius and the exact shifted intertwiners	755
116.6	The nilpotent link and its Tate-typed isomorphism	756
116.7	The explicit relative filtration on this extension	756
116.8	The original extension class is retained	757
116.9	Transport from the full original packet to its actual quotient	758
117	The same arithmetic discrepancy in the mixed filtration and the source boundary . . .	761
118	Fixed objects and original source maps	763
119	The full lattice and its marked fibre	764
120	A genuine finite-field mixed extension	765
121	Exact division and the full boundary connection	766
122	The full primary jets at the boundary	767
123	Exact ordered period determinant	768
124	Exact comparison with the original source metric	769
125	Compatibility with independently retained mixed faces	770
126	Completed single-primary boundary calculation and the continuing mixed-support calculation	770
126.1	The exact primary specialization, connection and period matrix	770
126.2	The computed boundary and its exact marked-source maps	771
126.3	The exact finite-field boundary on the same primary phase	773
126.4	Continuation through the full mixed-support family	774
127	Joint trace control on the original source and relation spaces	775
127.1	All four endpoints as operators on one source	776
127.2	Ranks, multiplicities and the retained overlap	776
127.3	A bound that uses cancellation before estimating	777
128	Combining the existing spectral and overlap controls on the original source	780
128.1	The literal graph augmentation on this finite coefficient source	785

128.2	The complete original relation rows in the graph endpoint control	787
129	The actual single-primary marked family and its complete boundary monodromy . . .	790
129.1	The actual packet and the exact Pascal map	790
129.2	The two boundary coefficients in the unchanged basis	791
129.3	The original ordered contours and their exact translation	792
129.4	Complete monodromy, its finite cover, and the boundary invariants	793
129.5	The same full marked fibre retains the arithmetic nilpotent	794
129.6	The complete Taylor unit and the original theta source	794
129.7	The finite-cover lattice and the original source metric	796
129.8	Support labels and the scope of the computed boundary	797
130	The finite-field sheaf of the single-primary phase	797
130.1	Specified coefficient specialization and all constants	797
130.2	The actual power-map splitting, including its zero stalk	798
130.3	Compact-support cohomology and the negative Gauss line	799
130.4	The exact decomposition on the original parameter	800
130.5	Inertia at zero, including a zero constant specialization	800
130.6	Purity proved by the exact Gauss calculation	801
131	Ordered tensor primary packets: invariant characters and the actual arithmetic action	802
131.1	The original ordered family and its period matrix	802
131.2	Full monodromy and the exact character projectors	804
131.3	The arithmetic action moves the invariant character	805
131.4	The complete compressed dilation, including every return power	806
131.5	The full Taylor unit and its exact original-source projector	807
131.6	The original terminal class on its exact invariant subsequence	808
131.7	The exact rank of arithmetic departure from the invariant plane	810
132	Ordered tensors of the actual single-primary finite-field sheaf	811
132.1	The original coefficient map and the common parameter	811
132.2	Every ordered character tuple and its exact constant	812
132.3	Original inertia, its tame projector, and multiplicities	812
132.4	Swan conductor and the finite-cover logarithm	814
132.5	Frobenius on the invariant space and exact descent	814
132.6	Exact weight and the two cancellation cases	816
132.7	Exact counts of the arithmetic orbit blocks	816
132.8	The terminal character, its orbit, and the original scalar	817
132.9	The shared integral deck-character model and its two base changes	818
132.10	The multiplicity-one case and its exact scalar comparison	820
133	The retained primary cyclic lattice at its specified coefficient prime	821
133.1	The actual coefficient localization and the unchanged coordinates	821
133.2	The primitive lattice and the exact factorial inclusion	822
133.3	Exact integer depths, quotient coordinates, and reduction	824
133.4	The exact original chain-support map for these depths	826
133.5	The retained boundary-character labels and the full-stage obstruction	828
134	Original cyclic coefficients and the tensor boundary maps	830

134.1	The cyclic exponent with its retained coefficient map	830
134.2	The same cyclic map carries the tensor boundary projector	833
134.3	Finite-field boundary and the actual coefficient-specialization restriction . . .	835
135	The actual amplified class and the exact phase exclusion error	838
135.1	The full amplified primary factor and its nonzero observation	838
135.2	The exact phase generator and its boundary on that class	839
135.3	The full-jet inverse of the boundary comparison	840
135.4	The scaling of the proved global phase controls	840
135.5	What the computed quotient repairs do to the amplified generator	841
135.6	The exact cancellation inside the original derivative energy	842
135.7	A finite moment upper expression for the same generator error	843
135.8	Original spaces and the actual source at both degrees	844
135.9	The exact relation map between the two sections	845
135.10	Exact generator difference, with its literal primitive	845
135.11	Exact energy identities before any inequality	846
135.12	Actual source derivative moments and a uniform comparison	847
135.13	Generator-difference bounds in the original quotient metric	848
135.14	An explicit length choice makes both comparison errors small	849
135.15	The complete quartet boundary cannot satisfy the same estimate	849
135.16	At the minimal cutoff the gluing error is identically zero	850
135.17	Split-base typing, provenance, and independent check	853
135.18	The actual match to the amplification calculation	853
136	The actual arithmetic amplification and its proved error scales	853
136.1	The invariant that the tensor construction amplifies	854
136.2	The exact original window map into that control	855
136.3	Uniform norm errors and their precise amplified rates	855
136.4	The available upper functions on the identical window	857
136.5	What the fixed-packet degree limit preserves	858
137	Deligne's final amplification: the exact contradiction	859
137.1	The proved bound being reused	859
137.2	Constructing the object to which the earlier exclusion applies	860
137.3	Apply the earlier bound to the newly constructed product	860
137.4	The finite contradiction, without relying on informal limiting language	861
137.5	Why repeating a bound is not sufficient by itself	861
137.6	The logical pattern recorded for the actual Split-Zero object	862
138	The matching passage in Weil II	862
139	The actual final bootstrap in Weil II	863
139.1	Conventions, original weight, and the universal assertion	863
139.2	The two earlier inputs: a strict bound and local integral shifts	864
139.3	The exact open-extension map used by the induction	865
139.4	The product and pencil, with all three reduction hypotheses	865
139.5	Why the middle coefficient sheaf has pure constituents	866
139.6	Vanishing cycles detect the nonconstant weights	867

139.7	Leray transfers the surviving error only once	868
139.8	The nonzero class survives the two exact injections	869
139.9	Removing the reduction hypotheses preserves the offending eigenvalue	869
139.10	A finite contradiction, followed by the exact dual lower bound	870
139.11	The mixed-sheaf theorem retains and controls the boundary	871
139.12	The exact contradiction mechanism available for comparison	872
140	What has been reconstructed, and what has been decided	872
140.1	Complete source coverage and its limits	873
140.2	Source order and the surviving mathematical content	873
141	The original absolute base and the arithmetic scalar square	874
142	The actual cohomological route of a hypothetical zero	875
143	What the purity construction does, and the precise remaining calculation	876
144	1. Objects, provenance, and cochain conventions	877
145	2. The original pushforward and the complete restriction fibre	878
146	3. Naturality and the right-adjoint comparison	879
147	4. The dual of the restriction fibre and its nonsplit extension	880
148	5. Calculation on the original theta source and every mask	881
149	6. Finite jets, their full kernel, and the retained topology	882
150	7. Original scaling, generator, and dual actions	885
151	8. Full finite residue arrow and its chain degree	886
152	9. Tensor compatibility, including the dual Koszul sign	888
153	10. Exact consequences for continuation	890
154	Finite dilation of the original arithmetic packet over the tau base	891
154.1	0. Scope and exact source correspondence	891
154.2	1. The original spaces, maps, and topology	891
154.3	2. The finite arithmetic packet and its unchanged Taylor unit	893
154.4	3. The finite dilation cocycle as an actual convergent source integral	894
154.5	4. All nilpotents, the Mellin multiplier, and an incomplete-gamma evaluation	896
154.6	5. Actual two-leg homotopies and the support-face comparison	897
154.7	6. The exact bridge to the already constructed global section	899
154.8	7. Tensor powers with every cohomological sign and jet coefficient	901
154.9	8. The complete dx boundary pairing, its derivative, and tensor mass	903
154.10	9. Source pins and status of the calculation	906
155	The original balanced Mellin kernel: full polynomial comparison and meromorphic remainder	906
155.1	Original objects and full-plane Euler inverses	906
155.2	Every polynomial kernel and cokernel is computed on the actual theta sequence	908
155.3	The balanced kernel has invertible polynomial action	908
155.4	Canonical finite spectral part and an explicit nonzero kernel class	910
156	The exact derived comparison at every finite spectral thickening	911
157	An actual proper source label with a finite spectral kernel	913
158	The actual arithmetic packet through mixed-support synchronization	915
158.1	The original mixed-support object, with its unit	916

158.2	A proved split injection and its exact retraction	916
158.3	The exact coefficient Frobenius comparison	917
158.4	The original arithmetic action, full jets, and modulus	918
158.5	The trace obstruction also survives the pure coefficient factor	919
159	The multiplicity boundary floor	919
159.1	The original source, packet algebra, and arithmetic section	920
159.2	Theta relation spaces and the exact rank-two boundary	921
159.3	Relative control spectrum and its invariant trace	923
159.4	The invariant-subspace projection argument	925
159.5	Exact consequences for a putative off-line zero	927
160	Arithmetic symmetric frontier cost and determinant growth	928
160.1	The retained arithmetic source and its canonical finite metric	929
160.2	The actual next relation and its recurrence bound	931
160.3	Exact symmetric restriction with every orbit factor	933
160.4	The aggregate spectral-position restriction	934
160.5	Coordinate invariance and determinant growth	935
160.6	The exact signed eigenline witness	936
161	The original conormal map, the arithmetic Jacobian, and complete symmetric trace fibres	937
161.1	Source pin and the unchanged arithmetic objects	937
161.2	The exact conormal domain and its arithmetic observation	939
161.3	The entire conormal image of an actual zero packet	940
161.4	The full mixed derivative and its arithmetic trace	941
161.5	Restriction to the actual signed symmetric cochain image	942
161.6	Intrinsic symmetric trace: the exact comparison of its multiplicities	942
161.7	Consequence for the actual RH counterfactual, with every multiplicity	944
161.8	The relative fibres at the expected sum weight still retain negative directions .	945
162	The off-line counterfactual inside the original holonomy construction	946
162.1	The retained arithmetic objects and every phase	946
162.2	A uniform phasewise lower bound for the original control	947
162.3	The exact energy forced into the retained theta boundary	948
162.4	The full lattice test for the original Laplacian boundary	949
162.5	What these computations decide	950
163	Actual bounds for the retained four-window costs	950
163.1	Every term of the inherited lower bound	950
163.2	A proved upper bound for these same four original volumes	951
163.3	The exact compatible interval for the signed arithmetic correction	953
163.4	The holonomy energy is tied to the same endpoint contractions	957
163.5	The sharper finite comparison retains its exact error	965
164	The same arithmetic source in every bound	965
165	An explicit original relation that the source isometry does not preserve	966
166	Status at the preceding proof increment	967
167	The original coherent quotient and the derived classes it does not remove	967
167.1	The original sheaf and its canonical local summands	968

167.2	Every ordinary coherent observation factors through the actual zero scheme	969
167.3	A complete local derived calculation	970
167.4	The residue morphism into the original pointed-base dual	972
167.5	Proper labels and the precise consequence for purity comparison	973
168	The canonical derived dual and the retained original tau complex	974
168.1	The global splitting is an actual sheaf map	974
168.2	A canonical local dual without a choice of meromorphic basis	975
168.3	The natural local derived bidual retains every kernel class	976
168.4	The exact global target of the local Ext calculation	977
168.5	The original tau complex, its dilation, and its retained degree zero	977
169	Exact arithmetic quotient transport and its unavoidable window defect	979
169.1	The retained source, forms, and arithmetic maps	979
169.2	An invertible repair preserving every arithmetic coefficient	980
169.3	The exact quotient contraction and its original metric	981
169.4	A positive original-moment obstruction and a sharp repair minimum	981
169.5	Square-zero conjugation error and the signed arithmetic bound	982
169.6	The literal finite relation complex and its computed homotopy	984
170	The closest unitary preserving the actual arithmetic observation	984
170.1	Original objects and forms	985
170.2	Invertible blocks and strict leakage	986
170.3	Closest repairs	986
170.4	AW's specified phases and a determinant bound for the repair cost	987
170.5	Full coefficient action and the exact defect	988
170.6	No invertible relation-preserving exact intertwiner	990
171	Actual boundary tilts and uniform convolution tails	990
171.1	1. Objects and the exact retained expression	990
171.2	2. A centered partial-sum bound from the original floor formula	990
171.3	3. Explicit two-sided Gamma bounds	991
171.4	4. A global explicit actual boundary-tilt bound	992
171.5	5. Boundary moments and finite derivative orders	993
171.6	6. A uniform convolution envelope with the literal tilted mass	993
172	Internal derivation of the exact Binet identity	994
172.1	1. Defining integrals and the finite Euler formula	994
172.2	2. Nonvanishing and the holomorphic Gamma logarithm	995
172.3	3. Beta product, duplication, and the factor $\sqrt{\pi}$	996
172.4	4. The bounded integral remainder	997
172.5	5. Equality of second derivatives	997
172.6	6. The recurrence fixes the linear constant	998
172.7	7. Duplication fixes the additive constant	998
173	Uniform original source comparison and the two growing norm windows	999
173.1	A uniform convolution bound at the actual exponential boundary	999
173.2	Pointwise comparison of the original convolution	1000
173.3	Full finite polynomial forms, with the original coordinate maps	1000

173.4	Uniform arithmetic monic norms	1001
173.5	The two original windows with full finite errors	1002
173.6	The resulting actual four-window lower estimate	1002
174	Fixed-packet asymptotics of the complete original Gamma jet determinant	1003
174.1	The two singularities with the original coordinate and constants	1004
174.2	The complete off-critical blocks	1005
174.3	Critical full jets and all mixed blocks	1006
174.4	The entire determinant and the original quotient volume	1007
174.5	The original tensor-Gamma reference with its full mass	1008
175	A fixed denominator in the Gamma weight	1009
175.1	Explicit comparison of the weights	1009
175.2	The exact morphism on the original jet constraints	1010
175.3	Passage to the asymptotic coefficient	1011
175.4	Application to the actual arithmetic weight	1012
176	The original arithmetic classes under Hilbert completion	1013
176.1	Vanishing of the entire quotient Gram	1013
176.2	Exact arithmetic observations and theta primitives	1014
176.3	The exact algebraic quotient and its Hausdorff reduction	1015
176.4	The coefficient for the complete counterfactual quartet	1016
177	The actual arithmetic form passes to every original quotient	1017
178	A finite computable comparison for the four arithmetic volumes	1019
179	The strengthened restriction and its exact reach	1020
180	The original convolution reference has a compulsory spectral spread	1021
181	The certified theta moments as observations of the total original object	1022
181.1	From the original theta boundary to the finite moment matrices	1023
181.2	The reciprocal action on every full original packet	1023
181.3	The full half-trace and the exact global moment limit	1025
181.4	The certified finite restriction on this attached observation	1026
181.5	The precise exclusion for an offcritical separator	1027
182	Original theta Hankel witness: independent proof and full packet transport	1029
182.1	WBR1. Source and scalar factors	1029
182.2	WBR2. Positive density and explicit growth	1030
182.3	WBR3. Reciprocal zeros and coefficient map	1031
182.4	WBR4. Finite criterion and full infinite-tail proof	1032
182.5	WBR5. Actual packet and its complete representative	1033
182.6	WBR6. Reciprocal action, inverse jet, and literal theta boundary	1033
182.7	WBR7. Full finite algebra, half-trace, and scalar radical	1034
182.8	WBR8. Exact critical-line comparison	1035
182.9	WBR9. Computation and integration record	1036
183	Complete product proof for the original theta entire function	1036
183.1	A sublinear growth bound and its zeros	1036
183.2	The product and a lower bound on selected circles	1037
183.3	The zero-free factor is constant	1038

183.4	Logarithmic coefficients with their full multiplicities	1038
184	The original integral and its two omitted regions	1039
185	Exact propagation to the original Hankel matrices	1040
186	Quantitative rational form of the counterfactual separator	1041
187	The completed finite certificate	1043
188	Scope and fixed inputs	1044
189	The original absolute base and its actual fibres	1045
190	The theta complex over the marked point	1046
190.1	Actual packets and the finite observation kernel	1047
191	The complete dilation and relation action	1047
192	The weight-matched external family over the same base	1048
192.1	The cyclic inclusion and its complete family defect	1049
193	Deligne's step, with the actual parameter and action maps	1049
194	The original boundary pairing, before its supported quotient	1051
195	One common source for the signed arithmetic/Gamma correction	1051
196	An exact signed four-endpoint calculation	1052
196.1	The equal-rank cancellation is calculated before estimation	1053
197	The retained quadratic term and a finite upper certificate	1054
198	All inputs are the actual one-factor arithmetic moments	1054
198.1	A closed first-pair calculation with the original relation cancellation	1055
199	Holonomy recovery and the exact original metric correction	1056
200	What is newly established and what remains	1056
III	Complete source, dual and transcript calculations	1058
201	Verified typed maps in the original τ -base construction	1058
201.1	Source pins and exact scope	1058
201.2	1. The structural map does not perform Boolean localization	1058
201.3	2. The global pushforward and its canonical derived restriction	1059
201.4	3. The actual arithmetic cohomology and its three support fibres	1060
201.5	4. The right adjoint is computed algebraically on that same chart	1060
201.6	5. Exact categorical and all-support kernels	1062
201.7	6. Dual support arrows: the exact reversal and its limit	1062
201.8	7. A completed exact jet map that makes the eigenline wording precise	1063
201.9	Audit uses	1064
201.10	8. The original arithmetic section completes the source eigenline lift	1064
202	Scope and source pins	1068
203	1. Original objects, signs, and both index sets	1068
204	2. The complete complexes and every support arrow	1069
205	3. Cohomology, with an explicit quotient and inverse	1069
206	4. What the reversed bottom actually contains	1070
207	5. Strict natural splitting fails, but the derived splitting has a concrete roof	1070
208	6. The original action and the actual defect of this roof	1071
209	7. Continuity and the finite residue inclusion remain attached	1073

210	Appendix A: complete retraction on the original theta source	1073
210.1	A.1. Original spaces and the fixed auxiliary functions	1073
210.2	A.2. Theta continuity and recovery on both exterior regions	1074
210.3	A.3. Fixed contraction and full Schwartz estimates	1075
210.4	A.4. The original moment correction and exact splitting	1076
210.5	1. Original arithmetic objects and analytic identities	1077
210.6	2. Full actual jet packets, reflection, and residue pairing	1078
210.7	3. The exact additional grading and its inverse	1079
210.8	4. Exact obstruction to a literal simultaneous identification	1080
210.9	5. Residue Jacobian, adjoint, and trace contraction	1081
210.10	6. A proved continuous section on the original theta target	1082
210.11	7. An exact action on the original four-point theta sheaf	1083
210.12	8. Compatibility with the original right-adjoint dual map	1084
210.13	9. Exact scope of the completed comparison	1085
210.14	Provenance	1085
211	Completed Mellin-realization calculations on the original theta source	1085
211.1	Retained objects and source equations	1085
211.2	Exact Fourier calculation for the retained theta source	1086
211.3	The raw two-leg syzygy at every source order	1088
211.4	Operator balancing removes the raw tensor relation, and retains honest joint cycles	1089
211.5	An explicit free submodule in the raw realization kernel	1090
211.6	Source pins and transcript locators	1092
212	Exact weighted theta-source constants	1092
212.1	Original objects and domains	1092
212.2	Weighted lines and exact finite moments	1093
212.3	Derivative and Fourier-cutoff Grams, including the cross terms	1094
212.4	A finite choice of comparison parameters and its exact scope	1095
213	Circle resolvent with all arithmetic jets	1096
213.1	Original operator and its domain	1096
213.2	The resolvent comparison and the complete nonresonant packet	1097
213.3	The resonant block with its full kernel and nilpotents	1097
213.4	The original Laplacian, including its two boundary terms	1098
214	Exact fluid-cutoff transport into the retained theta complex	1099
214.1	1. Original spaces and the cutoff map	1099
214.2	2. The explicit map to the original uncentered theta target	1099
214.3	3. Constant parabolic scaling, with the same viscosity	1100
214.4	4. Full actual-zero jets and the right-adjoint dual	1101
214.5	5. The unchanged source Weil scalar is invariant under this transport	1102
214.6	6. Time-dependent concentration and its exact extra force	1102
214.7	Proven scope	1103
215	Finished calculations on the retained physical and arithmetic formulas	1103
215.1	1. The modular scattering coefficient retains every zero multiplicity	1103

215.2	2. Channel redistribution is exact and need not be positive	1105
215.3	3. Equal marginal histories leave a phase family even after cross-energy measurement	1105
215.4	4. The arithmetic occupation map, modular clock, and trace-class boundary	1106
215.5	5. The force pullback has a real terminal-control issue	1107
216	Exact source-viscosity and cutoff-clock morphism	1108
217	Complete mathematical continuation ledger	1110
218	Overview	1111
219	Mathematical continuation audit of the original tau-base programme	1111
220	Acquisition and complete source coverage	1113
221	Shared transcript acquisition and coverage	1113
222	Early context: U0001–U0015	1114
223	Complete early-context audit: U0001–U0015	1114
223.1	Findings that matter for the current tau continuation	1114
223.2	Completed calculation supplied for integration	1114
223.3	Passage-by-passage ledgers	1115
223.4	Complete coverage table	1115
223.5	Acquisition and audit limits	1122
224	Opening passage ledger: U0001–U0007	1123
224.1	Opening-context coverage and mathematical continuation audit	1123
225	Central passage ledger: U0008–U0012	1134
225.1	Early-context local audit: U0008–U0012	1134
226	Physical passage ledger: U0013–U0015	1142
226.1	Retained physical bridge and modular-scattering precursor audit	1142
227	Formation of the original base: U0016–U0028	1148
228	Formation of the original tau-base program: passage audit U0016–U0028	1148
228.1	Coverage and source pins	1148
228.2	Status of every user turn	1148
228.3	Passage-by-passage findings	1150
228.4	Exact continuation route already supplied by the later source	1162
228.5	Prioritized redo instructions for integration	1163
228.6	Completed tractable calculation delivered to root	1163
229	Typed route: U0029–U0040	1164
230	Passage-by-passage typed-route audit: U0029–U0040	1164
230.1	Coverage and source provenance	1164
230.2	Controlling continuation	1164
230.3	Every user-response episode	1165
230.4	Exact passage ledger	1166
230.5	Completed work delivered with this audit	1180
230.6	Prioritized continuation for integration	1180
230.7	Retained-turn index	1180
231	Actual tau cohomology: U0041–U0054	1186
232	Passage audit: actual tau cohomology and continuation, U0041–U0054	1186

232.1	AT41: U0041 $\rightarrow$ A1458	1186
232.2	AT42: U0042 $\rightarrow$ A1474	1187
232.3	AT43: U0043 $\rightarrow$ A1523	1187
232.4	AT44: U0044 $\rightarrow$ A1523	1188
232.5	AT45: U0045 $\rightarrow$ A1523	1189
232.6	AT46: U0046 $\rightarrow$ progress responses	1189
232.7	AT47: U0047 $\rightarrow$ A1644	1190
232.8	AT48: U0048 $\rightarrow$ A1694	1191
233	U0049–U0054: complete later-control audit	1192
234	Later control continuation audit: U0049–U0054	1192
234.1	Scope, coverage and source pins	1192
234.2	Episode ledger	1192
234.3	Two completed source-comparison calculations for the audit	1198
234.4	Integration judgment and priority	1200
235	Late analytic control: U0055–U0067	1201
236	Late mathematical continuation audit: U0055–U0067	1201
236.1	Reading and evidence scope	1201
236.2	Original controlling objects	1201
236.3	Episode coverage	1201
236.4	L01. Cyclic sum source: genuine comparison, unfinished growing-degree estimate	1203
236.5	L02. Exterior amplification is completed; it does not constitute the upper estimate	1204
236.6	L03. Toda determinants and phase are calculated; seed numerics do not certify packet asymptotics	1205
236.7	L04. Gamma descent preserves the source but its one-sided bound cannot control consecutive ratios	1205
236.8	L05. The missing arithmetic norm-window estimate is actually proved	1206
236.9	L06. Finite restriction certificates do not control their growing source gaps	1207
236.10	L07. Two-trace determinant majorant completes a finite problem, not the arithmetic growth problem	1207
236.11	L08. Auxiliary compactification has exact boundary and curvature; degree is insufficient	1208
236.12	L09. Exponential family: valid special-fibre map with an explicitly nontransferred purity bound	1209
236.13	L10. Logarithmic constituent cohomology and negative coupling are completed	1210
236.14	L11. SGA trace and period determinant are completed; the common factor cancels	1210
236.15	L12. Finite circle recovery is proved; promised resolvent and growing constants were not calculated	1211
236.16	L13. Holonomy reconstruction explicitly preserves A_τ versus Boolean restriction	1212
236.17	L14. The requested second alternative is legitimately answered at the auxiliary-inference scope	1213
236.18	Newly finished calculations	1214

236.19	Prioritized continuation addressed to the original session	1214
237	Supplementary continuation to the original web session	1215
238	Continue the actual tau-base construction after the transcript audit	1215
238.1	The exact source to retain	1215
238.2	Calculations completed in this delivery	1215
238.3	First continuation: calculate the signed arithmetic correction on the original source	1216
238.4	Second continuation: carry the actual relation boundary through the geometric comparison	1217
238.5	Third continuation: finish the precise global balanced-kernel calculation . . .	1218
238.6	Required result	1218
239	Continuation postscript: analytic pole maps completed	1219
240	Postscript to the sealed supplementary continuation	1219
IV	Complete original-source analytic transport and moment certification	1220
241	Which analytic pole classes belong to the original theta source?	1221
241.1	1. Original source, fixed parameters, and the exact norm	1221
241.2	2. Complete Hilbert admission, including zeros on the observation line	1221
241.3	3. Its entire cohomology and exact inclusion into the analytic family	1222
241.4	4. Strong-Schwartz admission is an actual full-zero packet construction	1222
241.5	5. Polynomial coordinates, source maps and every retained boundary defect .	1223
241.6	6. The actual theta Gram and finite source cost	1225
241.7	7. The connection in the same arithmetic numerator frame	1225
241.8	8. Exact recovery of the old canonical quotient metric	1226
241.9	Sources and verification scope	1228
242	Counterfactual RH search on the original split-supported theta input	1235
242.1	Scope	1235
242.2	1. Original input and support	1235
242.3	2. Coefficients with the original mass retained	1236
242.4	3. A finite certificate contradicting RH	1236
242.5	4. Work through the off-line counterfactual: construct its negative direction .	1237
242.6	5. Connect the witness to the original finite packet, including its jets	1239
242.7	6. The performed search uses the original theta seed, not zero ordinates . . .	1239
242.8	7. Finite verification and the next decisive object	1240
242.9	Source and execution record	1241
243	Certified original-theta Hankel calculation	1242
243.1	1. Original integral and coefficients	1242
243.2	2. Explicit positive omitted regions	1243
243.3	3. Validated finite integration and its exact export	1243
243.4	4. Interval coefficient and LDL proof	1244
243.5	5. Recorded values and finite conclusion	1245
243.6	Reproduction	1245
244	Independent theta-tail proof and certificate code review	1246

244.1	1. Exact objects and retained integral	1246
244.2	2. Positivity and the exact majorant	1246
244.3	3. Omitted summands on the retained interval	1246
244.4	4. The complete integration tail	1247
244.5	5. The exact remainder and interval addition	1248
244.6	6. Entire finite integrand and its literal code expression	1248
244.7	7. Complete tail-code and exact-export review	1249
244.8	8. Original logarithmic coefficient recurrence	1249
244.9	9. Unpivoted LDL proof and the meaning of positivity	1250
244.10	10. Reviewed source and observed certificate	1251
244.11	11. Acceptance and scope of the completed proof	1251
245	Independent verification of the positive theta tails	1252
245.1	1. Exact integrand and the disjoint omitted regions	1252
245.2	2. Omitted n -sum on $0 \leq y \leq L$	1252
245.3	3. Omitted y -interval, including every $n \geq 1$	1253
245.4	4. Denominator signs at the actual parameters	1253
245.5	5. Callback analyticity	1254
246	Independent review of the theta error bounds and finite rational witness	1255
246.1	1. Retained integral and complete positive tail bounds	1255
246.2	2. Recurrence, interval propagation, and finite matrix signs	1256
246.3	3. Reciprocal-zero map and the full polynomial form	1257
246.4	4. Separator, explicit cutoff, and all edge cases	1257
246.5	5. Rational coefficient accuracy and interval detection	1258
246.6	6. Relation to the recorded finite certificate	1259
247	Root acceptance of the original theta certificate and complete proof closure	1260
247.1	Exact accepted calculation	1260
247.2	Independent integer arithmetic and actual replay	1260
247.3	Written-proof corrections and closure	1261
248	Independent final review: the full genus-zero proof and the positive theta tails	1262
248.1	1. Exact sources and the original analytic input	1262
248.2	2. Counting all zeros, with their multiplicities	1264
248.3	3. Construction of the literal product and the selected-circle bound	1264
248.4	4. The exact argument forcing the zero-free quotient to be constant	1265
248.5	5. The logarithmic coefficient map retains every multiplicity	1266
248.6	6. Verification of the positive tails and the exact omitted regions	1266
248.7	7. Analytic callback and source-to-proof comparison	1267
248.8	8. Acceptance and scope	1268
249	SplitZero: maps, support fibres, ideals, and exact corrections	1269
249.1	Conventions	1269
249.2	1. Exact universal morphisms	1269
249.3	2. Intrinsic support and fibre modules	1269
249.4	3. Complete ideal classification	1270
249.5	4. Correcting the monoid-semiring presentation	1270

249.6	5. Scope within the Zeta programme	1271
249.7	Source records	1271
V	Exact original-source metric and recursive propagation proofs	1272
250	Source enclosures and convergent signed projector control	1272
250.1	The unchanged polynomial and arithmetic source	1272
250.2	Both scaled secants, with the original boundary terms	1273
250.3	The canonical projector is exactly the original difference	1274
250.4	Restriction losses and the full cross trace	1275
250.5	A specified signed approximation and its exact residual	1276
250.6	The original primitive and the residual at a proper support	1277
251	Recursive propagation of the original signed metric bounds	1279
251.1	The original source and all four quotient volumes	1280
251.2	The simultaneous bounds on that integrand	1280
251.3	Carrying the refined bound through both endpoint coordinates	1282
251.4	Carrying the constructed signed residual into the endpoint	1283
VI	The original phase graph, mixed residual, relation rows and signed arithmetic return	1285
252	Original full-quartet critical values and every ordered tensor sum	1286
252.1	The exact polynomial, primitive and local leading coefficients	1286
252.2	A positive sum of squares proves that the rectangle does not collapse	1287
252.3	An exact real-part bound, retaining the original constant	1288
252.4	Every ordered tensor critical point, collision and full local rank	1288
253	The calculated phase operator on the original full jets and theta source	1291
253.1	The complete jet map and its inverse	1291
253.2	The phase operator is invertible on every original tensor block	1291
253.3	The original full Taylor unit and the literal theta boundary	1292
253.4	The original norm retains its complete comparison factor	1293
253.5	Exact phase cross-pairing on that same original source	1294
253.6	The full quartet connection and its exact lattice obstruction	1295
254	The full mixed phase boundary on the original arithmetic source	1297
254.1	One finite matrix for the whole boundary, including mixed pairings	1297
254.2	The arithmetic and phase boundary on the identical terminal class	1299
254.3	The exact attachment to the four original polynomial windows	1300
254.4	Original mixed support and the existing analytic period comparison	1303
255	Original objects and the fixed comparison	1305
256	Canonical sections, determinant lines and full variations	1306
257	The signed four-endpoint term on one retained source	1307
258	An exact finite-source bound for the correction	1308
258.1	The same control in the original nonlinear endpoint coordinate	1310
259	Source dependencies and precise status	1312

260	The exact gamma reference and the arithmetic Gram determinants	1313
260.1	The original measure, phase, and packet cancellation	1313
260.2	Gamma norms from a generating integral	1313
260.3	Finite arithmetic determinants in the retained metrics	1315
260.4	Full confluent kernel and the original coordinates	1316
260.5	The finite map to Romik's gamma parameter	1317
260.6	Exact scope of the exponential-weight theorem	1317
260.7	Primary source locators	1318
261	1. Result and exact source boundary	1319
262	2. The base, source, and both kinds of quotient	1319
263	3. A quantitative local arithmetic mass lemma	1321
263.1	3.1 Uniform polynomial growth on a fixed strip	1321
263.2	3.2 Propagate a nonzero value to the critical-line interval	1321
264	4. A lower envelope for the actual convolution, after three factors	1322
264.1	4.1 Local mass of the unchanged g/h	1322
264.2	4.2 The positive compact convolution factor	1322
265	5. Upper moments and exact lower monic norms	1323
266	6. The norm-window estimate is proved	1324
267	7. Compose the proved bound with the formalizer's endpoint theorem	1324
268	8. The remaining four volumes retain their finite jet maps	1325
269	9. The volume window is also two explicit retained-boundary costs	1325
270	10. Deligne reading and what has actually been transferred	1326
271	11. Proof and computation status	1327
272	References and pinned inputs	1327
273	Actual boundary tilts and uniform convolution tails	1328
273.1	1. Objects and the exact retained expression	1328
273.2	2. A centered partial-sum bound from the original floor formula	1328
273.3	3. Explicit two-sided Gamma bounds	1329
273.4	4. A global explicit actual boundary-tilt bound	1330
273.5	5. Boundary moments and finite derivative orders	1330
273.6	6. A uniform convolution envelope with the literal tilted mass	1331
274	Internal derivation of the exact Binet identity	1332
274.1	1. Defining integrals and the finite Euler formula	1332
274.2	2. Nonvanishing and the holomorphic Gamma logarithm	1333
274.3	3. Beta product, duplication, and the factor $\sqrt{\pi}$	1334
274.4	4. The bounded integral remainder	1335
274.5	5. Equality of second derivatives	1335
274.6	6. The recurrence fixes the linear constant	1336
274.7	7. Duplication fixes the additive constant	1336
275	Uniform original source comparison and the two growing norm windows	1337
275.1	A uniform convolution bound at the actual exponential boundary	1337
275.2	Pointwise comparison of the original convolution	1338
275.3	Full finite polynomial forms, with the original coordinate maps	1338

275.4	Uniform arithmetic monic norms	1339
275.5	The two original windows with full finite errors	1340
275.6	The resulting actual four-window lower estimate	1340
276	An arithmetic lower-envelope comparison for central quotient volumes: full jets, finite determinant bounds, and an exact quadratic benchmark	1342
276.1	Original source and every jet coordinate	1342
276.2	The lower envelope as an explicit positive comparison form	1343
276.3	An upper determinant depending only on the two Laplace values	1345
276.4	The interval kernel and the actual comparison map	1345
276.5	The explicit scalar bound and its finite gap	1346
276.6	A full-jet quadratic benchmark and its exact relation family	1349
276.7	Source representatives and support	1350
276.8	Verification and provenance	1351
277	The first canonical window, its full phase product, and both arithmetic block budgets	1352
277.1	Original source and the actual first-degree step	1352
277.2	The complete first-window product and both finite budgets	1353
277.3	Actual comparable norm windows with complete constants	1354
277.4	The full relation cost and its retained zero directions	1355
277.5	Provenance and verification scope	1356
278	Result and scope	1357
279	1. Fixed inputs, maps, and the marked split base	1357
280	2. All circle holonomies recover the full source	1358
281	3. Every finite cyclic cover is an exact character decomposition	1359
281.1	The finite locally free coefficient cover	1360
282	4. Common arithmetic representatives and the exact descent cost	1360
282.1	The cochain comparison and its full contractible complement	1361
283	5. Sharp quadratic control of the descent cost	1362
283.1	The arithmetic hypothesis is supplied by the existing correlation bound	1363
284	6. Endpoints and a genuinely finite phase construction	1363
284.1	Exponentially convergent quadrature for the quotient itself	1364
284.2	Finite frequency truncation with the phase retained	1365
285	7. The original action and relation primitive descend together	1365
286	8. Quantitative transport of the new residue-constituent coefficient	1366
287	9. What premature completion would forget	1367
288	10. Exact calibrations and the remaining arithmetic estimate	1367
289	Source basis and attribution	1368
290	Density and exact completion of the original periodized arithmetic source	1369
290.1	1. The unchanged theta amplitude and its canceled zeros	1369
290.2	2. A proved exponential tail, with the original constants	1370
290.3	3. Pointwise convolution, strict positivity, and the full scalar-circle map	1371
290.4	4. Polynomial density with an explicit analytic strip and uniqueness	1372
290.5	5. The exact closed relation space, quotient metric, and kernel	1373
290.6	6. Full repeated-root blocks and the canonical metric limit	1375

291	The original theta complex and its arithmetic packet	1378
291.1	Source spaces and fixed arithmetic factors	1378
291.2	The exact operator range, including its inverse	1379
291.3	The original extension supplies the boundary coefficient	1380
291.4	Actual representative corrections and rank-two control	1381
292	An exact low-height exclusion from the original theta boundary	1383
292.1	The original source and its exact endpoint mass	1383
292.2	A rational upper bound on the unscaled second moment	1384
292.3	Uniform exclusion in the closed low-height rectangle	1385
293	The certified theta moments as observations of the total original object	1386
293.1	From the original theta boundary to the finite moment matrices	1386
293.2	The reciprocal action on every full original packet	1387
293.3	The full half-trace and the exact global moment limit	1388
293.4	The certified finite restriction on this attached observation	1389
293.5	The precise exclusion for an offcritical separator	1390
294	The original full packet at its formal boundary	1393
294.1	The fixed polynomial and the two complete complexes	1393
294.2	Complete remainder, inverse, and chain homotopies	1394
294.3	The exact connection and the constant determinant	1396
294.4	The original arithmetic source and its Jacobian	1397
294.5	The precise attachment to the mixed support carriers	1398
294.6	All ordered tensor factors and formal character blocks	1399
294.7	Horizontal formal projectors and the unchanged source form	1400
295	The actual sectorial comparison of the original full packet	1402
295.1	A specified algebraic Borel germ for every original column	1402
295.2	Continuation along the original critical-value rays	1403
295.3	The exact sectorial gauge, not only its asymptotic series	1404
295.4	Evaluation as actual lifted contours, with all gamma factors	1405
295.5	The exact integer comparison to the original global rays	1405
295.6	Actual transitions and their exact critical-value support	1406
295.7	The exact original source and ordered tensors	1407
295.8	The signed path of every actual transition entry	1408
295.9	The mixed split-support comparison and its complete fibres	1408
295.10	Every original source metric factor under the transition	1410
296	Recursive propagation of the original signed metric bounds	1411
296.1	The original source and all four quotient volumes	1411
296.2	The simultaneous bounds on that integrand	1412
296.3	Carrying the refined bound through both endpoint coordinates	1413
296.4	Carrying the constructed signed residual into the endpoint	1414
297	Source enclosures and convergent signed projector control	1416
297.1	The unchanged polynomial and arithmetic source	1416
297.2	Both scaled secants, with the original boundary terms	1416
297.3	The canonical projector is exactly the original difference	1418

297.4	Restriction losses and the full cross trace	1419
297.5	A specified signed approximation and its exact residual	1420
297.6	The original primitive and the residual at a proper support	1421
298	Exact relation-window spectral transport on the original tau complex	1424
298.1	The original source and its four windows	1424
298.2	Full spectra and an explicit source isometry	1425
298.3	The spectral bound with every eigenvalue retained	1426
299	The one-dimensional quotient: exact angle and nonlinear transport	1429
299.1	Complete proof dependencies	1430
300	Joint trace control on the original source and relation spaces	1431
300.1	All four endpoints as operators on one source	1431
300.2	Ranks, multiplicities and the retained overlap	1432
300.3	A bound that uses cancellation before estimating	1432
301	Exact comparison of inherited EW transport with AGT and AA	1436
301.1	1. Exact source pins and reading scope	1436
301.2	2. Original source, parameters, metrics and maps	1437
301.3	3. Existing angle proof and the exact changed pairing	1439
301.4	4. Full general-window consequence of the crossed pairing	1440
301.5	5. Strict positivity: prior theorem and alternate later proof	1442
301.6	6. The theta primitive is the same original cochain	1443
301.7	7. Attribution text and cumulative integration scope	1444
302	The off-line counterfactual inside the original holonomy construction	1445
302.1	The retained arithmetic objects and every phase	1445
302.2	A uniform phasewise lower bound for the original control	1446
302.3	The exact energy forced into the retained theta boundary	1447
302.4	The full lattice test for the original Laplacian boundary	1448
302.5	What these computations decide	1448
303	Exact endpoint windows: Gamma coefficients, paired Schur volumes, and intrinsic sensitivity	1449
303.1	The original source and the complete finite input map	1449
303.2	Canonical lifts and one arbitrary-window block	1451
303.3	Endpoint roots and exact paired-volume cancellations	1452
303.4	Exact projection between the canonical representative spaces	1453
303.5	Window sensitivity and exact cancellation of mass changes	1454
303.6	Rational interval certificates for the original input	1457
303.7	Boundary cases and the retained cochain map	1459
303.8	Verification scope and exact auxiliary calibration	1460
304	Complete Gamma-reference lower bound and arithmetic determinant comparison	1464
305	The original source and relation Toda flows with their exact control maps	1476
305.1	Original arithmetic objects and the strip of real tilts	1476
305.2	The source relation and its Laplace observation	1478
305.3	Both monic determinants and their exact determinant-line map	1479
305.4	Toda derivatives and the full curvature operators	1480

305.5	The rank-one update, trace phase and original rank-two radius	1481
305.6	Exact maps to the CV omitted determinant and positive minors	1482
305.7	Both exact losses, their equality cases and the telescope	1483
305.8	The first admitted degree and the empty quotient	1484
305.9	The complete join with the original spectral coupling	1486
305.10	Crosswalk and exact continuing arithmetic quantity	1487
306	Combining the existing spectral and overlap controls on the original source	1488
306.1	The literal graph augmentation on this finite coefficient source	1492
306.2	The complete original relation rows in the graph endpoint control	1494
307	Actual bounds for the retained four-window costs	1498
307.1	Every term of the inherited lower bound	1498
307.2	A proved upper bound for these same four original volumes	1499
307.3	The exact compatible interval for the signed arithmetic correction	1500
307.4	The holonomy energy is tied to the same endpoint contractions	1504
307.5	The sharper finite comparison retains its exact error	1512
308	Continuous holonomy cochains and the original relation metric	1514
308.1	Original coefficient spaces and actual phase norms	1514
308.2	The continuous cochain splitting and its full contraction	1515
308.3	The minimum norm of the common image	1516
308.4	The positive relation gap and its mean-zero decomposition	1517
308.5	Actual sampled Hilbert spaces, including zero weights	1517
308.6	Density and the complete fibrewise Hilbert quotient	1518
308.7	The complete primary kernel before phase integration	1519
308.8	The measurable completed field and its vanishing map	1520
308.9	The source-to-completion cochain map	1520
308.10	The receiving supported zero and all marked fibres	1521
308.11	Exact outcome and scope	1521
309	The literal phase graph as a weighted source and its exact Schur correction	1522
309.1	The graph factorization and its finite-rank correction	1523
309.2	The two exact least-norm sections	1524
309.3	The positive residual and its exact source meaning	1525
309.4	Literal transitions between polynomial windows	1526
309.5	All four signed endpoints, with the finite correction evaluated	1526
309.6	The exact scalar source, its fibre residual and all cross convolutions	1528
310	The scalar density of the original phase graph source	1530
310.1	Exact fibre and three-convolution formulas	1530
310.2	Uniform actual tail constants and the phase insertion	1531
310.3	All-vector bounds on the original growing windows	1533
310.4	The exact signed four-endpoint consequence	1534
311	The strict mixed-fibre component of the original phase observation	1537
311.1	The exact kernel for every mixed tensor	1538
312	The exact phase-graph map to the original averaged quotient and its completion	1540
312.1	The positive cost of adding the literal defect	1540

312.2	Composition with the accepted all-holonomy quotient	1541
312.3	Exponential moments and the complete graph quotient	1543
312.4	The degree limit and the exact obstruction to a uniform comparison	1545
313	The sealed strict mixed residual inside the completed graph quotient	1547
314	Exact generator control on the finite graph and its strict mixed quotient	1550
315	The original relation coefficients, their complete tail, and mixed source subtraction	1552
315.1	An exact analysis map for the unchanged relation source	1552
315.2	A strict mixed subtraction in the endpoint upper bound	1554
315.3	Return to the original arithmetic endpoint	1555
316	The full correlations of the original relation rows	1556
316.1	A finite source map for all the retained correlations	1556
316.2	Every omitted correlation and an explicit finite remainder	1557
316.3	Both negative corrections in the same original inverse	1558
317	The finite generator converges to the realized completed generator	1560
318	The current arithmetic route and its next calculation	1561
318.1	A single chain of exact comparison maps	1561
318.2	Topology, duality and the two sides of a weight estimate	1562
318.3	Provenance and mathematical status	1563
318.4	The original theta norm and the positive arithmetic frontier	1565
318.5	The symmetric summand, the actual relation layer, and support	1566
318.6	The original derivative cost and both positive-half-line kernels	1567
318.7	The gamma determinant and the retained spectral-sum fibre	1568
318.8	The sum derivative, its strict normal energy and the full conormal algebra	1569
318.9	The full relative limit and the original arithmetic observation	1570
318.10	Every relation depth and its full cyclic complement	1571
318.11	Exterior amplification and the arithmetic volume to estimate	1571
318.12	The original Toda volumes and the complete four-term loss	1572
318.13	Certified theta input and a calculated comparison of the two gaps	1573
318.14	Gamma convolution, certified higher input, and the complete quotient comparison	1574
318.15	Arithmetic endpoint estimates and the full reflected relation map	1576
VII	Complete source proof notes and chronology	1614
S1	PR 8: Finite operator comparison	1615
S1.1	Exact finite operator quotients of the tau-base theta cohomology	1615
S2	PR 9: Chain descent	1627
S2.1	Chain-level arithmetic descent over the tau base	1627
S3	PR 10: Global retraction	1637
S3.1	A global retraction of the original theta complex over the tau base	1637
S4	PR 11: Homotopy control	1647
S4.1	Homotopy choices and source-preserving arithmetic control	1647
S5	PR 13: Orthogonal boundary control	1653
S5.1	Orthogonal theta boundaries and rank-two arithmetic control	1653

S6	PR 17: Spectral-sum descent and the complete relative arithmetic fibre	1659
S6.1	Spectral-sum descent with the relative arithmetic fibre retained	1659
S7	Derived Split-Zero mathematics	1665
S7.1	SplitZero: reconstruction, internal homology, and balanced finite jets	1665
S8	PR 14: Coherent arithmetic interpolation and joint tensor boundary control	1671
S8.1	Coherent arithmetic interpolation and joint tensor boundary control	1671
S9	Full-jet interpolation kernels and the actual next relation layer	1682
S9.1	Full-jet interpolation kernels and the actual next relation layer	1682
S10	Arithmetic frontier estimates and symmetric tensor control	1689
S10.1	Arithmetic frontier estimates and symmetric tensor control	1689
S11	Complete spectral-sum descent: relative arithmetic fibres, dual resolution and all nilpo- tents	1699
S11.1	Spectral-sum descent with the relative arithmetic fibre retained	1699
S12	PR 20: Arithmetic sum density, the full relative connection and conormal derivatives .	1711
S12.1	Quantitative sum-fibre connection and the conormal arithmetic trace	1711
S13	Cyclic arithmetic sum control and the complete contracted relation tower	1722
S13.1	Sum-generated arithmetic cohomology, scalar control, and exact conormal depth	1722
S14	Exterior trace amplification with the original arithmetic Gram and full quartet multi- plicities	1736
S14.1	Result and scope	1736
S14.2	1. GitHub corrections integrated at a pinned revision	1737
S14.3	2. The original scalar and arithmetic source	1738
S14.4	3. The existing canonical metric and its rank-two operator	1739
S14.5	4. Exterior control: the allowance does not multiply by exterior degree	1740
S14.6	5. The exterior maps live on the original cochain source	1741
S14.7	6. A determinant line measures the sum of the positive defects	1743
S14.8	7. Spectral and metric projections are linked explicitly	1744
S14.9	8. Exact sum counting for a nonreal quartet	1745
S14.10	9. The quantitative target after this construction	1746
S14.11	10. Deligne architecture and the boundary retained in this step	1747
S14.12	11. Verification and reproducibility	1748
S14.13	References and provenance	1748
S15	Source and relation volume dynamics for the original arithmetic control	1749
S15.1	Results and scope	1749
S15.2	1. Integration of the new formalization report	1750
S15.3	2. Original base, coefficient square, and arithmetic source	1750
S15.4	3. One arithmetic function generates both determinant flows	1752
S15.5	4. The quotient determinant is the source/boundary volume ratio	1753
S15.6	5. The two classical Toda identities have their actual arithmetic input	1754
S15.7	6. The phase in the rank-two control is a scalar volume derivative	1755
S15.8	7. The exact scalar formula and a two-step upper bound	1756
S15.9	8. What the exterior lower bound forces on these arithmetic volumes	1757
S15.10	9. The Laplace input can be evaluated from the original theta source	1758
S15.11	10. Exact finite checks, an essential phase example, and current endpoint . . .	1759

S15.12	References and source boundary	1760
S16	Supplied conormal derivative, symmetric restriction and uncompiled tower draft . . .	1761
S17	PR 21: conormal tower and cyclic depth formalization with its complete mathematical note	1771
S17.1	Original conormal and cyclic relation towers	1771
S18	The coordinating Zeta contribution: determinant losses and spectral coupling	1776
S18.1	Exact join: determinant losses and spectral coupling	1776
S19	Gamma convolution descent and the original arithmetic coefficient correction	1778
S19.1	1. Scope and the inspected collaboration state	1778
S19.2	2. The original arithmetic source and its split maps	1778
S19.3	3. Carry the repository's gamma reference through every tensor power	1779
S19.4	4. The sum projection and its complete polynomial complement	1780
S19.5	5. Insert the complete arithmetic multiplier, not a substituted gamma measure	1781
S19.6	6. Exact arithmetic coefficients at every tensor degree	1783
S19.7	7. Feed the result into the two original Toda determinants	1785
S19.8	8. The current upper-control expression, with all reference costs exposed	1786
S19.9	9. A proved all-degree tail estimate for the actual arithmetic moments	1787
S19.10	10. What has and has not been concluded	1788
S19.11	References and exact source roles	1788
S20	Arithmetic endpoint bounds: complete original norm proof	1789
S20.1	1. Result and exact source boundary	1789
S20.2	2. The base, source, and both kinds of quotient	1790
S20.3	3. A quantitative local arithmetic mass lemma	1791
S20.4	4. A lower envelope for the actual convolution, after three factors	1792
S20.5	5. Upper moments and exact lower monic norms	1793
S20.6	6. The norm-window estimate is proved	1793
S20.7	7. Compose the proved bound with the formalizer's endpoint theorem	1794
S20.8	8. The remaining four volumes retain their finite jet maps	1794
S20.9	9. The volume window is also two explicit retained-boundary costs	1795
S20.10	10. Deligne reading and what has actually been transferred	1795
S20.11	11. Proof and computation status	1796
S20.12	References and pinned inputs	1796
S21	Complete supplied endpoint window product continuation	1797
S22	Complete supplied arithmetic norm continuation	1805
S23	PR 23: complete original endpoint and multiplicity criterion	1812
S23.1	Sharper endpoint-only criterion and the scope of exterior iteration	1812
S24	The complete original endpoint and Gamma join	1815
S24.1	Endpoint degree selection in the actual gamma-corrected coordinates	1815
S25	PR 24: complete original confluent transfer proof	1817
S25.1	Confluent transfer for the original source-relation volume	1817
S26	Complete scalar volume comparison and arithmetic maps	1823
S26.1	An explicit scalar comparison for the original arithmetic quotient volumes	1823
S27	Complete all-degree division and affine raw-jet estimates	1829

S27.1	Exact finite remainder bound and affine raw-jet transport	1829
S28	The complete corrected endpoint and multiplicity criterion	1834
S28.1	Sharper endpoint-only criterion and the scope of exterior iteration	1834
S29	Complete original endpoint window product proof	1837
S29.1	Exact consecutive-window control and endpoint determinants	1837
S30	Complete original endpoint restriction control	1841
S30.1	1. Result, inherited input, and scope	1841
S30.2	2. Preserve the original base and source	1842
S30.3	3. Construct the canonical degree transport and its adjoint	1843
S30.4	4. Record the arithmetic action and the actual boundary correction	1844
S30.5	5. Exact volume as a restriction spectrum	1845
S30.6	6. An exterior expansion retains every full-jet basis weight	1845
S30.7	7. A certified finite logarithmic upper bound	1846
S30.8	8. The off-line alternative forces a precise restriction phenomenon	1848
S30.9	9. Exact calibration retaining nilpotence and mass	1848
S30.10	10. Integration with the Deligne-oriented programme	1849
S30.11	References and provenance	1849
S31	Complete original two-trace relation control	1851
S32	Complete original polynomial-exponential cohomology comparison	1860
S33	The original complete comparable-window norm proof	1871
S33.1	Arithmetic norm review and comparable-degree windows	1871
S34	The original complete consecutive-window proof	1873
S34.1	Consecutive losses and the arithmetic endpoint threshold	1873
S35	Complete supplied specialization and curvature continuation	1876
S36	Complete supplied logarithmic constituent and coupling calculation	1885
S37	Complete exact restriction fixture and its source maps	1894
S37.1	Exact lower-source gap and two-pair restriction certificates	1894
S38	SGA trace, the retained diagonal, and the arithmetic period determinant	1903
S38.1	1. The completed calculation and its role	1903
S38.2	2. Source intake and mathematical categories	1903
S38.3	3. The arithmetic object that is retained	1904
S38.4	4. SGA 5's relative duality, calculated on this polynomial algebra	1905
S38.5	5. The whole derived diagonal is retained	1906
S38.6	6. The preceding exponential family and its exact coefficient connections	1907
S38.7	7. Complete evaluation of the determinant of periods	1908
S38.8	8. Recover the complete cyclic spectrum from that determinant family	1909
S38.9	9. Two explicit calibrations	1910
S38.10	10. Transport the original metric and control through the same periods	1910
S38.11	11. The source-specific remaining term stays explicit	1911
S38.12	12. Relation to Deligne's finite-field family	1912
S38.13	13. Verification and reproducibility	1912
S38.14	References and source status	1913
S39	The original theta complex inside the Hochschild trace realization	1914

S39.1 The original theta complex inside the Hochschild trace realization 1914

S40 A quantitative Laplacian bound from the original control form 1921

S40.1 A quantitative Laplacian bound from the original control form 1921

S41 Specialization, invariant hulls, and the original curvature cost 1923

S41.1 Specialization, invariant hulls, and the original curvature cost 1923

Part I

Integrated mathematical continuation

1 Exact Split-Zero carriers and the local spectral bridge

Throughout this section, semiring homomorphisms preserve additive zero and multiplicative unit. Let R be a nonzero commutative ring and $\mathbb{B} = (\{0, 1\}, \vee, \wedge)$. Define

$$G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset G(R).$$

Addition and multiplication of supported elements are those of R , including when their result is zero. The additional element satisfies

$$\tau + r = r + \tau = r, \quad \tau + \tau = \tau, \quad \tau r = r\tau = \tau, \quad \tau^2 = \tau.$$

Thus the semiring zero is τ , and $e \neq \tau$.

1.1 Scalar support and independent channel amplitudes

Proposition 1.1 (Pair carrier and scalar reflection). *There is a semiring isomorphism*

$$\eta_R : G(R) \longrightarrow \{(0, 0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}, \quad \tau \mapsto (0, 0), \quad r \mapsto (r, 1).$$

The coordinate maps

$$\begin{aligned} p_R : G(R) &\rightarrow R, & p_R(\tau) &= 0, & p_R(r) &= r, \\ \chi_R : G(R) &\rightarrow \mathbb{B}, & \chi_R(\tau) &= 0, & \chi_R(r) &= 1 \end{aligned}$$

are semiring homomorphisms. The map p_R is universal among semiring homomorphisms from $G(R)$ to a ring.

Proof. In the product semiring, two supported pairs have sum $(r + s, 1)$ and product $(rs, 1)$. These identities retain support when the amplitude vanishes. The pair $(0, 0)$ is an additive identity and an absorbing multiplicative zero. This proves closure and all operation identities, and the displayed map is bijective. Its coordinate projections give p_R, χ_R . If $f : G(R) \rightarrow A$ has ring target, then $e + e = e$ gives $f(e) + f(e) = f(e)$, hence $f(e) = 0_A$. Its restriction to the supported copy of R is therefore a unital ring homomorphism $h : R \rightarrow A$. The equality $f = hp_R$ holds on R and on τ , and determines h . Conversely every such h gives hp_R . $\square$

The larger carrier $R \times \mathbb{B}$ contains $(1, 0)$; its exact relation to $G(R)$ is the inclusion above. The pair presentation requires $R \times \{1\}$.

For a nontrivial bounded distributive lattice L , define

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), & (r, \lambda)(s, \mu) &= (rs, \lambda \wedge \mu). \end{aligned}$$

Write $z_\lambda = (0, \lambda)$, $\tau_L = z_{0_L}$, and $e_L = z_{1_L}$.

Proposition 1.2 (Lattice reflection with its original coefficient ring). *The displayed carrier is a commutative semiring. The projections*

$$p_L : G_L(R) \rightarrow R, \quad \chi_L : G_L(R) \rightarrow L$$

are respectively the universal ring reflection and the support homomorphism. In particular the amplitude codomain of $G_{\mathbb{B}^2}(\mathbb{C})$ is $\mathbb{C}$.

Proof. A summand with nonzero amplitude has top support, so its sum has top support. A factor with lower support has zero amplitude, so a product of lower support has zero amplitude. These facts prove closure. All laws restrict from the product semiring; distributivity uses that of L . Both projections preserve operations, zero and unit. If $f : G_L(R) \rightarrow A$ has ring target, $e_L + z_\lambda = e_L$ gives $f(z_\lambda) = 0$ by cancellation in A . Restriction to the supported copy of R is a ring homomorphism h and $f = hp_L$. Conversely each h gives this factorization. Surjectivity of p_L gives uniqueness. $\square$

Let $j : \mathbb{C} \rightarrow G(\mathbb{C})$ include the supported copy. It preserves addition, multiplication and unit; it takes ordinary zero to e , whereas the semiring zero is τ . Put

$$M = G(\mathbb{C})^2, \quad p^{(2)} = (p_{\mathbb{C}}, p_{\mathbb{C}}) : M \rightarrow \mathbb{C}^2, \quad \chi^{(2)} = (\chi_{\mathbb{C}}, \chi_{\mathbb{C}}) : M \rightarrow \mathbb{B}^2.$$

For $\epsilon \in \mathbb{B}$ let $u_0 = \tau, u_1 = e$.

Proposition 1.3 (Exact scalar-lattice and two-amplitude embeddings). *There are injective semiring homomorphisms*

$$b : G_{\mathbb{B}^2}(\mathbb{C}) \hookrightarrow M, \quad B : G_{\mathbb{B}^2}(\mathbb{C}^2) \hookrightarrow M$$

defined by

$$\begin{aligned} b(c, (1, 1)) &= (j(c), j(c)), \\ b(0, (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}), \\ B((a, b), (1, 1)) &= (j(a), j(b)), \\ B((0, 0), (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}). \end{aligned}$$

Their amplitude and support identities are

$$p^{(2)}b = \Delta_{\mathbb{C}}p_{\mathbb{B}^2}, \quad \chi^{(2)}b = \chi_{\mathbb{B}^2}, \quad p^{(2)}B = p_{\mathbb{B}^2}, \quad \chi^{(2)}B = \chi_{\mathbb{B}^2},$$

where $\Delta_{\mathbb{C}}(c) = (c, c)$.

Proof. The pair presentation identifies M with

$$\{((a, b), (\lambda_1, \lambda_2)) \in \mathbb{C}^2 \times \mathbb{B}^2 : \lambda_1 = 0 \Rightarrow a = 0, \quad \lambda_2 = 0 \Rightarrow b = 0\},$$

with componentwise operations. The map b restricts the product semiring map $(c, \lambda) \mapsto ((c, c), \lambda)$. Lower support in its source forces $c = 0$, so it lands in the displayed subset. It retains the amplitude and both support bits and is injective. The map B restricts the identity coordinate map on $\mathbb{C}^2 \times \mathbb{B}^2$; lower support in its source forces both amplitudes to vanish, so it too lands in M and is injective. The prescriptions agree at the top-support zero. Both maps preserve zero and unit. Projection proves all four identities. $\square$

The element $(j(1), \tau) \in M$ is outside both images. There is no function $r : M \rightarrow G_{\mathbb{B}^2}(\mathbb{C}^2)$ satisfying both $p_{\mathbb{B}^2}r = p^{(2)}$ and $\chi_{\mathbb{B}^2}r = \chi^{(2)}$: it would send that element to $((1, 0), (1, 0))$, outside its target carrier. This identifies the precise obstruction to a reverse map preserving both projections.

The synchronized carrier has the exact map

$$G(\mathbb{C}^2) \cong G(\mathbb{C}) \times_{\mathbb{B}} G(\mathbb{C}) \hookrightarrow M.$$

It takes τ to (τ, τ) and a supported (a, b) to $(j(a), j(b))$. Equality of the two support characters leaves exactly those cases, proving bijectivity onto the fibre product. Operations agree componentwise. Its image has masks $(0, 0), (1, 1)$, while the other masks in M carry independent mixed amplitudes.

The map $p^{(2)}$ is the universal ring reflection of M . Indeed, (e, e) is additively idempotent, so every ring-target map kills it. Adding (e, e) to $x \in M$ gives the fully supported element of amplitude $p^{(2)}x$. The map is consequently determined by a ring homomorphism from the supported copy of $\mathbb{C}^2$, proving the universal property. A semiring isomorphism $G_{\mathbb{B}^2}(\mathbb{C}) \cong M$ would induce a ring isomorphism $\mathbb{C} \cong \mathbb{C}^2$. This is impossible because $(1, 0)$ is a nontrivial idempotent of $\mathbb{C}^2$, and $\mathbb{C}$ has only 0, 1. The positive relation between the original carriers is b , with diagonal amplitude projection.

1.2 Tesseract sectors and a supported critical fixed locus

Reserve τ for split absence and use κ for conjugation. Let

$$\sigma(s) = 1 - s, \quad \kappa(s) = \bar{s}, \quad K_4 = \{1, \sigma, \kappa, \sigma\kappa\}.$$

The real tesseract algebra and its complexification are

$$A_{\mathbb{R}} = \mathbb{R}[K_4] = \mathbb{R}[\sigma, \kappa]/(\sigma^2 - 1, \kappa^2 - 1), \quad A_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} A_{\mathbb{R}}.$$

The polynomial algebra in this presentation is commutative.

Proposition 1.4 (Character algebra and quartet coordinates). *For $\epsilon, \delta \in \{1, -1\}$ let*

$$E_{\epsilon\delta} = \frac{1}{4}(1 + \epsilon\sigma)(1 + \delta\kappa).$$

Character evaluation gives an algebra isomorphism

$$\mathcal{F} : A_{\mathbb{C}} \rightarrow \mathbb{C}^4, \quad x \mapsto (\text{ev}_{\epsilon,\delta}(x))_{\epsilon,\delta},$$

with inverse $(d_{\epsilon\delta}) \mapsto \sum d_{\epsilon\delta} E_{\epsilon\delta}$. For the exact quartet element

$$Q(s) = s \cdot 1 + (1 - s)\sigma + \bar{s}\kappa + (1 - \bar{s})\sigma\kappa, \quad s = \beta + i\gamma,$$

the four character values are

$$c_{++} = 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1), \quad c_{--} = 4i\gamma.$$

Projection to the last two specified characters is a unital algebra map $\pi : A_{\mathbb{C}} \rightarrow \mathbb{C}^2$. The composite πQ is a real affine bijection

$$\mathbb{C} \rightarrow \mathbb{R} \times i\mathbb{R}, \quad s \mapsto (a, b) = (c_{-+}(s), c_{--}(s)), \quad s = \frac{1}{2} + \frac{1}{4}(a + b).$$

Proof. Evaluation of $E_{\epsilon\delta}$ at (ϵ', δ') is $\frac{1}{4}(1 + \epsilon'\sigma)(1 + \delta'\kappa)$. It is one when both signs agree and zero otherwise. This proves the two inverse formulas and all orthogonal idempotent identities, retaining the factors of four. Character evaluation preserves the algebra operations. Directly,

$$\begin{aligned} c_{++} &= s + (1 - s) + \bar{s} + (1 - \bar{s}) = 2, \\ c_{+-} &= s + (1 - s) - \bar{s} - (1 - \bar{s}) = 0, \\ c_{-+} &= s - (1 - s) + \bar{s} - (1 - \bar{s}) = 2(s + \bar{s}) - 2 = 4\beta - 2, \\ c_{--} &= s - (1 - s) - \bar{s} + (1 - \bar{s}) = 2(s - \bar{s}) = 4i\gamma. \end{aligned}$$

The proposed inverse recovers $\beta + i\gamma$. Conversely every real a and purely imaginary b is recovered by substituting that inverse into the two character expressions. $\square$

The quartet image has real affine dimension two. Its real linear span has dimension three, spanned by E_{++}, E_{-+}, iE_{--} , retaining the constant character coordinate. The real tesseract algebra has dimension four. The maps $Q, \mathcal{F}, \pi$ and the displayed affine inverse specify the exact relationship of these objects.

Proposition 1.5 (Supported sector lift and fixed locus). *Define*

$$\Psi : \mathbb{C} \rightarrow M, \quad \Psi(s) = (j(c_{-+}(s)), j(c_{--}(s))).$$

It has $\chi^{(2)}\Psi(s) = (1, 1)$ for all s . The $G(\mathbb{C})$ -linear automorphisms

$$T_{\sigma}(x, y) = ((-1)x, (-1)y), \quad T_{\kappa}(x, y) = (x, (-1)y)$$

of the free semimodule M give

$$\Psi(\sigma s) = T_{\sigma}\Psi(s), \quad \Psi(\kappa s) = T_{\kappa}\Psi(s).$$

Here $-1 = j(-1)$ is a supported unit. Their product satisfies

$$\text{Fix}(T_{\sigma}T_{\kappa}) = \{\tau, e\} \times G(\mathbb{C}), \quad \Psi^{-1} \text{Fix}(T_{\sigma}T_{\kappa}) = \{s : \Re s = \frac{1}{2}\}.$$

Proof. The sector formulas give $(a, b)(1 - s) = (-a, -b)$ and $(a, b)(\bar{s}) = (a, -b)$. Multiplication by $j(-1)$ negates supported amplitudes and fixes τ . It preserves addition, commutes with scalar multiplication and squares to identity. This proves the automorphism and equivariance statements. Its fixed points are τ and supported $j(c)$ with $c = -c$. Since $2 \neq 0$ in $\mathbb{C}$, the latter is precisely e . The product action is $((-1)x, y)$, proving its fixed locus. On the image of Ψ the first coordinate is supported and is fixed exactly when $4\beta - 2 = 0$. $\square$

The scalar-lattice channel atoms have images $b(z_{10}) = (e, \tau)$ and $b(z_{01}) = (\tau, e)$. The separate support swap has fixed masks $(0, 0), (1, 1)$. Since $\chi^{(2)}\Psi$ is constantly $(1, 1)$, it cannot detect the critical line by that swap. The support action induced by $T_\sigma T_\kappa$ is identity; its amplitude action is $(-a, b)$. The preceding proposition supplies the exact fixed-locus test with the first-channel supported zero retained.

1.3 Exact sector, holonomy and positive-moment maps

Fix a prime p , $\ell_p = \log p > 0$, and $u \in S^1$. For an arithmetic packet, take $u = \psi(\text{Frob}_p)$ with $L/\mathbb{Q}$ finite abelian, p unramified, and $\psi : \text{Gal}(L/\mathbb{Q}) \rightarrow S^1$ a character. All powers use $p^w = \exp(w\ell_p)$ with the positive real logarithm.

Proposition 1.6 (Sector-to-holonomy map with constants retained). *The analytic function*

$$H_{p,u} : \mathbb{C}^2 \rightarrow \mathbb{C}^\times, \quad H_{p,u}(a, b) = u \exp\left(\frac{\ell_p}{4}(a + b)\right)$$

gives a function $H_{p,u}p^{(2)}$ on M . On the quartet image,

$$H_{p,u}(c_{-+}(s), c_{--}(s)) = up^{s-1/2} =: z_{p,u}(s),$$

$$\frac{\log |z_{p,u}(s)|}{\ell_p} = \frac{c_{-+}(s)}{4} = \Re s - \frac{1}{2}, \quad |z_{p,u}(s)|^2 - 1 = \exp\left(\frac{\ell_p}{2}c_{-+}(s)\right) - 1.$$

Proof. The exact inverse for the sector coordinates gives $s - \frac{1}{2} = (a + b)/4$. Substitution proves the exponential identity. On that image a is real and b is purely imaginary. Since $|u| = 1$, taking absolute values yields $\exp(\ell_p a/4)$. Taking its real logarithm and, separately, its square proves the two remaining identities. $\square$

The radial exponent is $c_{-+}/4$; the exponent $c_{-+}/2$ belongs to the squared modulus. The map $H_{p,u}$ is used with its analytic-function type, without any assertion that it preserves semiring addition.

Proposition 1.7 (Rank-one packet). *For $z \in \mathbb{C}$ define*

$$v_2(z) = \begin{pmatrix} 1 \\ z \end{pmatrix}, \quad T_2(z) = v_2(z)v_2(z)^* = \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix}, \quad P_z(w) = w - \bar{z}.$$

Then $T_2(z)$ is Hermitian positive semidefinite of rank one and

$$\ker T_2(z) = \mathbb{C}(-\bar{z}, 1)^\top.$$

It is Toeplitz exactly when $|z| = 1$, equivalently when the unique root of P_z lies on S^1 . For $z = z_{p,u}(s)$ this is equivalent to $\Re s = \frac{1}{2}$.

Proof. For $x \in \mathbb{C}^2$, $x^*T_2(z)x = |v_2(z)^*x|^2 \geq 0$. The vector $v_2(z)$ is nonzero and hence its outer product has rank one. The adjoint kernel equation is $x_0 + \bar{z}x_1 = 0$, giving the displayed space. A two-by-two matrix is Toeplitz exactly when its diagonal entries agree, here $1 = |z|^2$. The root $\bar{z}$ has modulus $|z|$. The final equivalence follows from the preceding proposition and $\ell_p > 0$. $\square$

Theorem 1.8 (Positive finite packets and their exact diagonal defect). *Let $r \geq 1$, $\alpha_j > 0$, and $z_j \in \mathbb{C}$. For $n \geq 1$ put*

$$v_n(z) = (1, z, \dots, z^{n-1})^\top, \quad M_n = \sum_{j=1}^r \alpha_j v_n(z_j) v_n(z_j)^*.$$

The matrix M_n is Hermitian positive semidefinite. For $n \geq 3$, it is Toeplitz exactly when every $|z_j| = 1$. With matrix indices starting at zero,

$$\mathcal{D}(M_n) := (M_n)_{22} - 2(M_n)_{11} + (M_n)_{00} = \sum_{j=1}^r \alpha_j (|z_j|^2 - 1)^2.$$

For these packets and $n \geq 3$, vanishing of $\mathcal{D}$ already characterizes Toeplitzness. If the z_j are pairwise distinct and $r \leq n$, the rank is r . If $r < n$, coefficient vectors identify the kernel with

$$\left(\prod_{j=1}^r (w - \bar{z}_j) \right) \mathbb{C}[w]_{\leq n-1-r} \subset \mathbb{C}[w]_{\leq n-1}.$$

In particular $r = n - 1$ gives corank one and exactly the conjugated roots $\bar{z}_1, \dots, \bar{z}_{n-1}$.

Proof. The identity

$$x^* M_n x = \sum_j \alpha_j |v_n(z_j)^* x|^2$$

proves positivity and shows that its kernel is the simultaneous kernel of all $v_n(z_j)^*$. The first three diagonal entries are $\sum \alpha_j$, $\sum \alpha_j |z_j|^2$, and $\sum \alpha_j |z_j|^4$. Their displayed linear combination gives the defect without rescaling any weight. Toeplitzness equates these entries, forcing zero defect. Every weight is positive and every square nonnegative, hence all $|z_j| = 1$. Conversely that condition gives $(M_n)_{k\ell} = \sum_j \alpha_j z_j^{k-\ell}$, a Toeplitz matrix. The same argument proves the zero-defect characterization.

Let V have columns $v_n(z_j)$. Its first r rows have Vandermonde determinant $\prod_{i < j} (z_j - z_i) \neq 0$, so V has rank r . The established kernel identity is $\ker M_n = \ker V^*$, proving the rank. For $P_x(w) = \sum_{k=0}^{n-1} x_k w^k$, the kernel equations say $P_x(\bar{z}_j) = 0$. Distinct linear factors divide P_x exactly when these equations hold: division by one factor preserves the other root equations since that factor is nonzero at those roots. This proves the product and degree bound. At $r = n - 1$, the remaining factor is a scalar; for a nonzero kernel vector it is nonzero, so there are no further roots. $\square$

The size bound is sharp for general positive packets. At $n = 2$, the data $(\alpha_1, \alpha_2) = (4, 1)$ and $(z_1, z_2) = (1/2, 2)$ give

$$M_2 = 4 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This is positive definite and Toeplitz, while neither atom is on S^1 . At size three the diagonal is $(5, 5, 65/4)$ and the defect is $45/4$. One atom retains the two-by-two criterion. Zero-weight atoms are invisible; repeated atoms combine their weights before applying the distinct-column rank count.

The exact sector defect is therefore

$$\mathcal{D}(M_n) = \sum_{j=1}^r \alpha_j \left[\exp \left(\frac{\log p_j}{2} c_{-+}(s_j) \right) - 1 \right]^2.$$

The function chain

$$M \xrightarrow{p^{(2)}} \mathbb{C}^2 \xrightarrow{H_{p,u}} \mathbb{C}^\times \xrightarrow{z \mapsto v_n(z) v_n(z)^*} \text{Herm}_{\geq 0}(n)$$

has explicit maps at every step. Positive finite summation supplies the preceding theorem.

1.4 The theta-packet form and its scalar comparison

Proposition 1.9 (Eigenpair form quotient and sector channel). *For every complex d -by- d matrix A , Hermitian matrix G , and nonzero eigenvector v with $Av = \rho v$, let $W = A^*G + GA - G$. Then*

$$v^*Wv = (2\Re\rho - 1)v^*Gv.$$

*On the domain of these data where $v^*Gv \neq 0$, the quotient*

$$\omega(A, G, v, \rho) = \frac{v^*Wv}{v^*Gv} = 2\Re\rho - 1 = \frac{c_{-+}(\rho)}{2}$$

is real. Its associated holonomy satisfies

$$|z_{p,u}(\rho)|^2 = \exp(\ell_p\omega), \quad z_{p,u}(\rho) = u \exp\left(\frac{\ell_p}{4}(c_{-+}(\rho) + c_{--}(\rho))\right).$$

Proof. Since $(Av)^* = \bar{\rho}v^*$,

$$v^*(A^*G + GA - G)v = (Av)^*Gv + v^*G(Av) - v^*Gv = (\bar{\rho} + \rho - 1)v^*Gv.$$

This proves the unquotiented identity even when the right factor is zero. Division on the stated domain gives ω , and $c_{-+}(\rho) = 4\Re\rho - 2$ gives the factor $1/2$. The holonomy identities follow from the exact sector exponential above. $\square$

This is the typed map from an eigenpair of the form packet to its Split-Zero sector and local moment matrix. The moment matrix is positive for every parameter, so that positivity cannot supply a further zero-defect conclusion. For $\rho = 1, u = 1$,

$$z = \sqrt{p}, \quad T_2(z) = \begin{pmatrix} 1 & \sqrt{p} \\ \sqrt{p} & p \end{pmatrix} \geq 0,$$

and its diagonal entries differ. On the form side $A = (1), G = (1), v = (1)$ give $W = (1)$ and $\omega = 1$; the exact maps take these data to the same off-critical packet. Any arithmetic identity forcing the actual family of W to vanish on its eigenvectors must therefore be established from that family's construction. These maps retain that requirement rather than replacing it by Gram positivity.

1.5 Projective zeta at all zeros and at its pole

A pair in $G(\mathbb{C})^2$ is unimodular exactly when at least one amplitude is nonzero. Such a coordinate is a supported unit. If both amplitudes are zero, every linear combination has zero amplitude and cannot be the unit. Let $\mathbb{P}_{\text{um}}^1(G(\mathbb{C}))$ denote unimodular pairs modulo common unit multiplication.

Proposition 1.10 (Projective amplitude section and reciprocal). *The map*

$$p_* : \mathbb{P}_{\text{um}}^1(G(\mathbb{C})) \rightarrow \mathbb{P}^1(\mathbb{C}), \quad [x : y] \mapsto [p(x) : p(y)]$$

has an injective section

$$\iota : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad [a : b] \mapsto [j(a) : j(b)],$$

whose image is the full-support component. With

$$\mathcal{R} : G(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad x \mapsto [1 : x], \quad r : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad c \mapsto [1 : c],$$

one has $p_\mathcal{R} = rp_{\mathbb{C}}$ and $\mathcal{R}j = \iota r$.*

Proof. Unimodularity ensures the projected pair is nonzero. A common supported unit projects to a nonzero complex scalar, so p_* is well defined. A nonzero ordinary pair becomes a unimodular supported pair; changing its representative lifts to common supported unit multiplication. Thus ι is well defined, and $p_*\iota$ is identity. Every full-support pair is the supported lift of its amplitude pair. The pair $(1, x)$ is always unimodular, and projection of its coordinates proves both remaining identities. $\square$

In particular $\mathcal{R}(e) = [1 : e] = \infty_e$ and $\mathcal{R}(\tau) = [1 : \tau] = \infty_\tau$. Both project to $[1 : 0]$, while their Boolean projective supports are $[1 : 1]$ and $[1 : 0]$. Similarly $[e : 1]$ and $[\tau : 1]$ project to the ordinary zero while retaining different supports.

Proposition 1.11 (Two exact charts for the supported zeta map). *For the meromorphic zeta function with its simple pole of residue one at 1, the full-support lift is*

$$Z_{\text{sp}} : \mathbb{C} \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad Z_{\text{sp}}(s) = \begin{cases} [j(\zeta(s)) : 1], & s \neq 1, \\ [1 : e], & s = 1. \end{cases}$$

It equals $\iota\hat{\zeta}$. At every zeta zero ρ its value is $[e : 1]$. Near the pole it has the chart

$$Z_{\text{sp}}(s) = [1 : j(h(s))], \quad h(s) = 1/\zeta(s), \quad h(1) = 0, \quad h'(1) = 1.$$

The reciprocal is a finite-amplitude coordinate away from the zeta zeros, including its extension at 1. At a zero of order m , it has a pole of order m .

Proof. Put $t = s - 1$ and write $\zeta(1+t) = t^{-1} + g(t)$ with g holomorphic near zero. Then $h(1+t) = t/(1+tg(t))$. Its denominator has value one at zero, so this is holomorphic there with value zero and derivative one. On the overlap where $\zeta \neq 0$, common multiplication by $j(\zeta(s))$ gives $[1 : j(h(s))] = [j(\zeta(s)) : 1]$. This proves the transition and pole value. The first chart directly gives $[e : 1]$ at each zero. At a zero of order m , write $\zeta(\rho+t) = t^m u(t)$ with $u(0) \neq 0$. Its reciprocal is $t^{-m}u(t)^{-1}$, proving its pole order; the first chart retains the original order m . $\square$

The swap $J_*[x : y] = [y : x]$ is induced by an invertible permutation matrix and commutes with ι, p_* . Hence J_*Z_{sp} is the global reciprocal projective map, sending zeta zeros to ∞_e and the pole to $[e : 1]$. Analytic statements use the complex structure transported to the full-support component by ι ; no topology on the extra support components is needed.

1.6 Matrix packets and their full relation module

Theorem 1.12 (Block packet with explicit hypotheses). *Let V be a complex Hilbert space of finite dimension $d > 0$ with fixed positive-definite Hermitian inner product. Let $g \in \text{GL}(V)$ and $m \geq 1$. Use that metric and its orthogonal direct sums for all adjoints. Define*

$$W_m(g) : V \rightarrow V^{m+1}, \quad v \mapsto (v, gv, \dots, g^m v), \quad \mathbb{T}_m(g) = W_m(g)W_m(g)^*,$$

$$D_m(g) : V^m \rightarrow V^{m+1}, \quad \begin{cases} (D_m(g)y)_0 = -g^*y_1, \\ (D_m(g)y)_k = y_k - g^*y_{k+1} \quad (1 \leq k \leq m-1), \\ (D_m(g)y)_m = y_m. \end{cases}$$

The matrix $\mathbb{T}_m(g)$ is positive semidefinite and

$$\ker \mathbb{T}_m(g) = \text{im } D_m(g), \quad \text{rank } \mathbb{T}_m(g) = d, \quad \text{corank } \mathbb{T}_m(g) = md.$$

It is block Toeplitz exactly when g is unitary for the fixed metric.

Proof. The Gram factorization gives positivity and $\ker \mathbb{T}_m = \ker W_m^*$. The first component of $W_m v$ is v , so W_m is injective and the kernel dimension is $(m+1)d - d = md$. The adjoint is

$$W_m^*(v_0, \dots, v_m) = \sum_{k=0}^m (g^*)^k v_k.$$

Substitution of $D_m(y)$ gives

$$-g^* y_1 + \sum_{k=1}^{m-1} ((g^*)^k y_k - (g^*)^{k+1} y_{k+1}) + (g^*)^m y_m = 0.$$

Thus its image lies in the kernel. If $D_m(y) = 0$, the last component gives $y_m = 0$, and backward substitution gives each preceding $y_k = 0$. Its image dimension is md , proving equality with the kernel. The block of indices j, k is $g^j (g^*)^k$. For unitary g this is g^{j-k} , hence block Toeplitz. Conversely block Toeplitzness equates blocks $(0, 0), (1, 1)$, giving $I = gg^*$. Invertibility yields $g^* = g^{-1}$ and unitarity. $\square$

At $m = 0$ the only block is I , so the bound $m \geq 1$ is essential. At $m = 1$, the full kernel is $\{(-g^* y, y) : y \in V\}$; its vector polynomials are $(wI - g^*)y$. Taking determinant gives the exact scalar map

$$wI - g^* \mapsto \det(wI - g^*) = \overline{\det(\bar{w}I - g)}.$$

The equality follows by conjugating the determinant expansion and transposing. It retains conjugate eigenvalues and multiplicities. For the standard metric and $J_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ with $t > 0$, the polynomial is $(w - 1)^2$ but

$$J_t J_t^* = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix} \neq I.$$

Its block packet consequently fails Toeplitzness, although the determinant roots lie on the unit circle. The kernel remains $\{(-J_t^* y, y) : y \in \mathbb{C}^2\}$, retaining the matrix information that determinant forgets.

2 The historical packet ghost and its full arithmetic jet quotient

The source of the packet construction in this section is the response dated 24 June 2026, 11:13:04 AM in the SSD export *Globalization Semiring Analysis*, whose exact path, hash and response-line locators are recorded in `logbook/historical_split_transcripts_20260912.md`. The source proves the permutation-packet calculation. We give that calculation completely, then construct its comparison with the full local algebras of the arithmetic packet. In particular, we retain the nilpotent jet operators and distinguish the source's complex-linear permutation from arithmetic conjugation by explicit maps.

Let Z be a finite set of complex numbers stable under $c(\rho) = \bar{\rho}$ and $\sigma(\rho) = 1 - \rho$. Assign a positive integer m_ρ constant on each orbit. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A[f] = [sf].$$

We use the original coordinate s , without replacing its multiplicative operator by its semisimple part. Put

$$L_Z = \bigoplus_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad e_{\rho,k} = t_\rho^k \quad (0 \leq k < m_\rho)$$

in the indicated direct summand. The coefficient inner product makes all $e_{\rho,k}$ orthonormal; it is an explicitly chosen reference inner product, not the theta-source metric.

Proposition 2.1 (Original-coordinate identification with all jets). *The map*

$$T : E_h \longrightarrow L_Z, \quad T[f]_\rho = \sum_{k=0}^{m_\rho-1} \frac{f^{(k)}(\rho)}{k!} t_\rho^k$$

is a complex-algebra isomorphism. In these coordinates

$$TAT^{-1} = A_{ss} + N, \quad A_{ss}e_{\rho,k} = \rho e_{\rho,k}, \quad Ne_{\rho,k} = \begin{cases} e_{\rho,k+1} & k+1 < m_\rho, \\ 0 & k+1 = m_\rho. \end{cases}$$

Proof. Taylor multiplication modulo $t_\rho^{m_\rho}$ proves multiplicativity. Its kernel consists exactly of polynomials divisible by every $(s - \rho)^{m_\rho}$ and hence by their product h , since the factors are pairwise coprime. For completeness an inverse is explicit. Set $h_\rho = h/(s - \rho)^{m_\rho}$, and let a_ρ be the Taylor polynomial of $1/h_\rho(s)$ at ρ through degree $m_\rho - 1$. Then $E_\rho = [a_\rho h_\rho]$ has residue 1 at the ρ factor and 0 at every other factor. Thus

$$T^{-1}e_{\rho,k} = E_\rho[(s - \rho)^k]$$

gives the inverse. Multiplication by $s = \rho + t_\rho$ gives the displayed operator, including its last nilpotent column. $\square$

In every subsequent operator formula on L_Z , the symbol A denotes the transported operator $TAT^{-1} : L_Z \rightarrow L_Z$ of Proposition 2.1. The polynomial-coordinate quotient at the end is equipped with the original multiplication-by- s operator.

Define complex-linear permutations C_0, S_0 and a complex-linear parity operator H on this complete basis by

$$C_0e_{\rho,k} = e_{\bar{\rho},k}, \quad S_0e_{\rho,k} = e_{1-\rho,k}, \quad He_{\rho,k} = (-1)^k e_{\rho,k}.$$

Also let K be the conjugate-linear map $K(\sum a_{\rho,k}e_{\rho,k}) = \sum \bar{a}_{\rho,k}e_{\rho,k}$. All four maps commute, square to the identity, and preserve products in L_Z , with K conjugate-linear on scalars.

Proposition 2.2 (Typed arithmetic involutions). *Under T , ordinary conjugation of polynomial coefficients is the conjugate-linear map $\mathcal{C} = C_0K$, and pullback by $s \mapsto 1 - s$ is the complex-linear map $\mathcal{S} = S_0H$. In particular*

$$C_0 = \mathcal{C}K, \quad S_0 = \mathcal{S}H, \quad \mathcal{C}A = A\mathcal{C}, \quad \mathcal{S}AS = I - A.$$

Here the equality involving $\mathcal{C}$ is an equality of conjugate-linear maps. Moreover $C_0N = NC_0$, $S_0N = NS_0$, and $HN = -NH$.

Proof. If $f^\#(s) = \overline{f(\bar{s})}$, then $(f^\#)^{(k)}(\rho)/k! = \overline{f^{(k)}(\bar{\rho})/k!}$, giving $\mathcal{C} = C_0K$. The k th derivative of $f(1 - s)$ at ρ is $(-1)^k f^{(k)}(1 - \rho)$, giving $\mathcal{S} = S_0H$. The ideal (h) is preserved by both operations: conjugation fixes the monic polynomial h , and $h(1 - s) = (-1)^{\deg h} h(s)$. Conjugating coefficients fixes the indeterminate s , whereas substituting $1 - s$ replaces multiplication by s by multiplication by $1 - s$. This proves the two identities for A . The assertions about N follow on each basis vector, including the vector of largest jet degree, which both sides send to zero. $\square$

Theorem 2.3 (Historical ghost with complete multiplicities). *Set*

$$R = C_0S_0, \quad \Gamma = C_0 - S_0, \quad P_- = \frac{I - R}{2}, \quad P_+ = \frac{I + R}{2}.$$

These satisfy

$$\Gamma^* = \Gamma, \quad \Gamma^2 = 4P_-, \quad \Gamma e_{\rho,k} = 0 \iff \Re \rho = \frac{1}{2}.$$

On a nonreal off-axis quartet of common multiplicity m , Γ has eigenvalues -2 and 2 each with multiplicity m , and 0 with multiplicity $2m$. Furthermore

$$\text{rank } P_- = 2m, \quad \text{Tr}(\Gamma^*\Gamma) = 8m.$$

On a critical-line orbit $\Gamma = P_- = 0$ at every jet degree.

Proof. The permutations C_0, S_0 are commuting selfadjoint involutions in the declared reference inner product. Consequently $(C_0 - S_0)^2 = 2I - 2C_0S_0$. The equality $C_0e_{\rho,k} = S_0e_{\rho,k}$ says exactly that the two labelled basis vectors have the same root, namely $\bar{\rho} = 1 - \rho$, which is equivalent to $2\Re\rho = 1$.

On an off-axis nonreal quartet order the roots as $(\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho})$. At every fixed jet degree the four joint (C_0, S_0) eigenspaces have characters $(1, 1), (1, -1), (-1, 1), (-1, -1)$. Explicit joint eigenvectors are

$$e_{\rho,k} + \epsilon e_{\bar{\rho},k} + \delta e_{1-\rho,k} + \epsilon \delta e_{1-\bar{\rho},k}, \quad \epsilon, \delta \in \{1, -1\}.$$

They form a basis: their pairwise inner products are zero and each has squared norm 4. The eigenvalue of Γ is $\epsilon - \delta$. Thus its squared eigenvalues are 0, 4, 4, 0, giving the rank and trace with every one of the m jet degrees included. On a critical orbit the two permutations agree on every basis vector. $\square$

For real off-axis orbits the same proof gives rank m and squared trace $4m$, since the orbit has two elements. Actual nontrivial zeta zeros in $0 < \Re s < 1$ are nonreal: for real $0 < s < 1$ the alternating series $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is strictly positive by grouping successive pairs, and $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$. For completeness the paired terms satisfy

$$(2n-1)^{-s} - (2n)^{-s} = s \int_{2n-1}^{2n} x^{-s-1} dx.$$

They converge uniformly on each compact subset of $\Re s > 0$, because on such a subset $|s| \leq M$, $\Re s \geq \delta > 0$, and the absolute value is at most $M(2n-1)^{-1-\delta}$. Thus η is analytic there. The product identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ follows first in $\Re s > 1$ by splitting even terms and then by analytic continuation. Division is used here only on the real interval $0 < s < 1$, where the denominator is strictly negative. Hence the quartet formula applies to off-axis arithmetic packets.

Proposition 2.4 (The exact group-algebra quotient). *For $W = \langle c, \sigma \mid c^2 = \sigma^2 = 1, c\sigma = \sigma c \rangle$ the homomorphism*

$$q : \mathbb{C}[W] \longrightarrow \mathbb{C}[u]/(u^2 - 1), \quad c \longmapsto u, \quad \sigma \longmapsto u$$

is surjective, with

$$I_q := \ker q = (c - \sigma)\mathbb{C}[W] = \mathbb{C}(c - \sigma) \oplus \mathbb{C}(1 - c\sigma).$$

Under the representation $c \mapsto C_0, \sigma \mapsto S_0, I_q L_Z = P_- L_Z$. Thus $L_Z/P_- L_Z$ is the largest quotient of the W -representation on which $C_0 = S_0$.

Proof. In the basis $(1, c, \sigma, c\sigma)$, q sends $a + bc + d\sigma + ec\sigma$ to $(a + e) + (b + d)u$. Its kernel is therefore spanned by $c - \sigma$ and $1 - c\sigma$. Multiplication by c sends $c - \sigma$ to $1 - c\sigma$, and multiplication by any generator stays in their span. This proves the ideal formula. Since $P_- = \Gamma^2/4$ and $\Gamma = \Gamma P_-$, their images agree and equal $I_q L_Z$. A quotient kills I_q precisely when it kills this image; this proves its universal property without any invariance assumption about the arithmetic generator. $\square$

The generator acts on the complete local rings. The next calculation constructs exactly the extra kernel needed to descend that action.

Theorem 2.5 (Arithmetic saturation of the historical ghost quotient). *Put*

$$Be_{\rho,k} = (\Re\rho - \tfrac{1}{2})e_{\rho,k}, \quad Z_{\text{crit}} = \{\rho \in Z : \Re\rho = \tfrac{1}{2}\}, \quad L_{\text{off}} = \bigoplus_{\rho \notin Z_{\text{crit}}} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}).$$

The smallest A -invariant subspace containing $I_q L_Z$ is exactly

$$I_q L_Z + A(I_q L_Z) = L_{\text{off}}.$$

On each off-axis orbit this sum is direct. The largest quotient on which both the original generator A descends and $C_0 = S_0$ is therefore the exact algebra and W -module quotient

$$L_Z \longrightarrow L_{Z_{\text{crit}}}, \quad (f_\rho)_\rho \longmapsto (f_\rho)_{\rho \in Z_{\text{crit}}}.$$

Its kernel is L_{off} , including every local multiplicity. In polynomial coordinates this is the quotient

$$\mathbb{C}[s]/(h) \longrightarrow \mathbb{C}[s]/(h_{\text{crit}}), \quad [f]_h \longmapsto [f]_{h_{\text{crit}}}, \quad h_{\text{crit}} = \prod_{\rho \in Z_{\text{crit}}} (s - \rho)^{m_\rho},$$

whose kernel is $(h_{\text{crit}})/(h)$; the target is zero when $h_{\text{crit}} = 1$.

Proof. The permutation R sends ρ to $1 - \bar{\rho}$, preserving its imaginary part and reversing $\Re \rho - 1/2$. It preserves the jet index and commutes with N . Accordingly $RB = -BR$ and

$$A = \frac{1}{2}I + B + iY + N, \quad Ye_{\rho,k} = \Im \rho e_{\rho,k}, \quad RY = YR, \quad RN = NR.$$

This is an exact identity for A , with N retained. If $v \in P_-L_Z$, then $(\frac{1}{2}I + iY + N)v \in P_-L_Z$ whereas $Bv \in P_+L_Z$. Therefore

$$P_+Av = Bv \quad (v \in P_-L_Z).$$

On each off-axis orbit all diagonal entries of B are nonzero, so $B : P_-L_Z \rightarrow P_+L_Z$ is an isomorphism on that orbit, its inverse being the restriction of the original diagonal operator B^{-1} . For $w = w_- + w_+$ on that orbit, take $v = B^{-1}w_+$; then $P_+Av = w_+$ and $w - Av \in P_-L_Z$. Thus every w lies in $P_-L_Z + A(P_-L_Z)$. If $Av \in P_-L_Z$ with $v \in P_-L_Z$, then $Bv = P_+Av = 0$ and $v = 0$, proving directness.

On a critical orbit $P_- = 0$. Both A and all permutations preserve the direct sum of off-axis factors, so the subspace just calculated is invariant and is precisely the smallest A -invariant subspace containing P_-L_Z . If a linear map F intertwines A with an operator on its target and satisfies $FC_0 = FS_0$, its kernel contains I_qL_Z and is A -invariant. It consequently contains L_{off} . Conversely the displayed coordinate projection intertwines A , preserves products and has $C_0 = S_0$ on its target. This proves the universal quotient assertion, also for targets carrying all the stated structures. The polynomial quotient and its kernel follow from $h_{\text{crit}} \mid h$ and Proposition 2.1. $\square$

Proposition 2.6 (Exact displacement and nilpotent contribution). *In the declared reference coefficient metric, let $W_{\text{ref}} = A^* + A - I$. Then*

$$W_{\text{ref}} = 2B + N + N^*, \quad \frac{W_{\text{ref}} - RW_{\text{ref}}R}{2} = 2B, \quad \frac{W_{\text{ref}} + RW_{\text{ref}}R}{2} = N + N^*.$$

For the original tessarine coordinate $c_{-+}(\rho) = 4(\Re \rho - 1/2)$, the first displayed component equals multiplication by $c_{-+}/2$.

Proof. The adjoint of A_{ss} is diagonal with entries $\bar{\rho}$, so $A_{\text{ss}}^* + A_{\text{ss}} - I = 2B$. Because R is a self-adjoint permutation commuting with N , it also commutes with N^* . Conjugation by R reverses B . Adding and subtracting the two resulting identities proves both formulas, including the entire nilpotent term. The tessarine identification uses $c_{-+} = 2(2\Re \rho - 1) = 4B$ with its original factor 4. $\square$

The precise map from this reference calculation to a theta-source metric retains the full metric matrix. Let G_E be that positive Hermitian matrix in the original polynomial basis of E_h ; then in the declared coefficient basis of L_Z it is

$$G_L = (T^{-1})^* G_E T^{-1}.$$

Writing the original polynomial-coordinate generator as A_E and the transported one as $A_L = TA_ET^{-1}$ for this formula, one obtains

$$\begin{aligned} W_E &= A_E^* G_E + G_E A_E - G_E, \\ W_L &= A_L^* G_L + G_L A_L - G_L = (T^{-1})^* W_E T^{-1}, \\ W_L &= G_L W_{\text{ref}} + [A_L^*, G_L], \quad [A_L^*, G_L] = A_L^* G_L - G_L A_L^*. \end{aligned}$$

The first two identities follow by inserting $A_L = TA_ET^{-1}$ and cancelling adjacent factors $T^{-1}T$ and $T^*(T^{-1})^*$. The last identity follows by expanding $G_L(A_L^* + A_L - I) + (A_L^* G_L - G_L A_L^*)$. Thus both nilpotent and metric-commutator contributions are present in the exact arithmetic weight; no zero-location estimate is supplied by discarding either term.

Finally form the split-zero semiring $G(\text{End}_{\mathbb{C}} L_Z)$, using the associative endomorphism ring. The original ghost has supported lift $\Gamma^\bullet$; when the comparison cancels it is $e = 0^\bullet$, whereas the absent comparison is τ . Conjugation by T extends to a semiring isomorphism

$$G(\text{End}_{\mathbb{C}} E_h) \longrightarrow G(\text{End}_{\mathbb{C}} L_Z), \quad X^\bullet \longmapsto (TXT^{-1})^\bullet, \quad \tau \longmapsto \tau.$$

It preserves addition, composition, the unit, e , and τ directly. Thus the complete jet comparison preserves supported cancellation. Theorem 2.5 states exactly which arithmetic factors are removed if that cancellation is imposed as a generator-compatible quotient. It does not assert that those factors are absent from the original arithmetic cohomology.

The quotient itself induces an endomorphism map on an exact specified domain. Define

$$\mathcal{D} = \{X \in \text{End}_{\mathbb{C}} L_Z : X(L_{\text{off}}) \subseteq L_{\text{off}}\}.$$

It is a unital subalgebra, and the induced quotient action is a surjective unital homomorphism

$$d : \mathcal{D} \longrightarrow \text{End}_{\mathbb{C}} L_{Z_{\text{crit}}}, \quad \ker d = \text{Hom}_{\mathbb{C}}(L_Z, L_{\text{off}}).$$

Indeed invariance of L_{off} is exactly what makes the quotient action well defined; the kernel condition says exactly that every value of X lies in L_{off} . Any endomorphism of the critical direct summand lifts by making it zero on the off-axis direct summand, proving surjectivity. Thus

$$G(d) : G(\mathcal{D}) \longrightarrow G(\text{End}_{\mathbb{C}} L_{Z_{\text{crit}}})$$

is the explicit split-zero descent map. Both A and Γ belong to $\mathcal{D}$; it sends $A^\bullet$ to the supported critical generator, $\Gamma^\bullet$ to the supported zero e , and τ to τ . This statement includes the zero target algebra when the critical summand is absent.

3 Historical Hurwitz reconstruction and the complete Laurent jet bridge

This section reconstructs a compact probability measure from the complete local Hurwitz zero cluster at any nontrivial zeta zero. It also proves the complete finite-jet relation with the marked Laurent period, including the involution coefficients and the exact collision quotient. Two distinct prime periods recover the entire finite spectral algebra with every multiplicity retained.

3.1 Source receipts and exact locators

1. The original *Split-Zero CUE Hurwitz phase space* source has SHA256 95CCA3275F2012E32F9E506EC0211273D5E7DA1FD72AD90F7167AA37094E94CC. Its exact excerpts accompany this edition in `sources/historical/hurwitz_laurent/`: `compact_hurwitz_and_residues.tex`

reproduces source lines 722–856; `zero_motion_and_bernstein.tex` reproduces lines 1110–1257; and `all_moments_claim.tex` reproduces lines 1907–1912. The statements below retain the original coordinate s , displacement t , support coordinate y , support bound L , and moments $m_n(\mu)$.

2. The *Mathematical Functor Construction* transcript has SHA256 8F94D3A8172F8585837A4E4F6060B5F8DAC38CF627C9689335F13792B7710D85. Its complete mathematical proof block, source lines 14142–15894, is retained as `sources/historical/hurwitz_laurent/marked_period_laurent_conormal_proofs.md`. Theorems 3, 7, 11 and 13 give the marked Laurent augmentation and period maps; Theorems 15–18 give the packet and Laurent involution; Theorems 20–25 give the marked conormal chart. Within that source, lines 15143–15227 prove $J_u^* = J_{-u}$, lines 15397–15502 give the $d \log |a_\infty|$ chart, and lines 15504–15631 give the augmentation-conormal diagram.
3. The corresponding standalone *Terzo-CC-CVS-JT deformation intertwiner* witness has SHA256 BAF7DB6506710CA9AA5B7705C7D5687B4E66536C08A9FE77549ABBC041BC9044. Its Laurent deformation appears at source lines 490–542 and its conormal calculation at lines 638–760. The complete proofs needed here are reproduced below and in the accompanying transcript excerpt.

The historical source gives the forward zero-motion recursion and the marked conormal map. The proofs below give the inverse moment recursion, the complete multiple-zero version, the full nilpotent Laurent isomorphism, and its collision quotient. Source line numbers and hashes identify the particular source editions; these comparisons make no claim of global mathematical priority.

3.2 Every compact Hurwitz input is determined by one simple-zero germ

Keep the original family

$$Z_{\mu,t}(s) = \int_{[0,L]} \zeta(s, 1+ty) d\mu(y), \quad m_n(\mu) = \int_{[0,L]} y^n d\mu(y), \quad m_0(\mu) = 1.$$

Here μ is a Borel probability measure, L is any finite support bound, and t is the original displacement; no rescaling of either variable is made. For $|t|L < 1$, put

$$b_n(s) = \frac{(-1)^n}{n!} (s)_n \zeta(s+n), \quad Z_{\mu,t}(s) = \sum_{n \geq 0} m_n(\mu) b_n(s) t^n. \quad (\text{H1})$$

The expansion is the source's compact Taylor formula. To verify its analytic scope directly, for $\text{Re } s > 1$ expand each $(k+ty)^{-s}$ in its binomial Taylor series. On compact s -sets and $|t|L \leq r < 1$, the ratios $|ty|/k \leq r$ give normally convergent double sums. Integration is termwise valid. The identity extends meromorphically in s ; near a nontrivial zero ρ all displayed coefficients are holomorphic. Joint holomorphicity and normal convergence in the strip can be checked directly: for a compact set K in $0 < \text{Re } s < 1$, the product

$$\left| \frac{(s)_n}{n!} \right| = \prod_{j=1}^n \left| 1 + \frac{s-1}{j} \right| \leq \prod_{j=1}^n \left(1 + \frac{C_K}{j} \right) \leq e^{C_K} (n+1)^{C_K}$$

is polynomially bounded uniformly in $s \in K$. The factor $\zeta(s+n)$ is uniformly bounded for $n \geq 1$ by the absolutely convergent Dirichlet series at $1 + \min_{s \in K} \text{Re } s$. Since $m_n \leq L^n$, the terms of (H1) are bounded by a constant times $(n+1)^{C_K} (L|t|)^n$. Thus the series is normally convergent when $L|t| < 1$, including on a surrounding compact annulus for any contour used below. Cauchy estimates give the same local derivative control. For $n \geq 1$, $\text{Re}(\rho+n) > 1$ and none of $\rho, \rho+1, \dots, \rho+n-1$ is zero, so

$$b_n(\rho) \neq 0. \quad (\text{H2})$$

The zeta factor in (H2) is nonzero because on $\operatorname{Re} z > 1$ the logarithmic Euler-product series

$$\sum_{p \text{ prime}} \sum_{k \geq 1} \frac{p^{-kz}}{k}$$

converges absolutely, and its exponential is $\zeta(z)$ there. This conclusion uses the actual zero's strip $0 < \operatorname{Re} \rho < 1$; it does not assume RH.

Fix a simple nontrivial zero ρ , so $\zeta(\rho) = 0$ and $\zeta'(\rho) \neq 0$. The analytic implicit function theorem produces the unique local holomorphic germ

$$\rho_\mu(t) = \rho + w_\mu(t), \quad w_\mu(t) = \sum_{j \geq 1} c_j(\mu) t^j, \quad Z_{\mu,t}(\rho_\mu(t)) = 0. \quad (\text{H3})$$

This is the existing source theorem. Its simplicity hypothesis is retained explicitly, rather than silently extending a scalar branch through a multiple zero.

The inverse recurrence. For each integer $N \geq 1$, the input moment is recovered by the exact finite formula

$$m_N(\mu) = -\frac{1}{b_N(\rho)} \sum_{n=0}^{N-1} m_n(\mu) [t^{N-n}] b_n \left(\rho + \sum_{j=1}^N c_j(\mu) t^j \right). \quad (\text{H4})$$

Thus $(c_1, \dots, c_N)$ determines $(m_1, \dots, m_N)$ and conversely. In particular, two compact probability inputs have the same zero-trajectory jet through order N exactly when their first N moments agree.

Proof. Substitute (H3) in (H1). Its coefficient of t^N is

$$0 = \sum_{n=0}^N m_n(\mu) [t^{N-n}] b_n(\rho + w_\mu(t)).$$

The $n = N$ summand is exactly $m_N(\mu) b_N(\rho)$. It contains no positive-order coefficient of w . Every other summand uses only $m_0, \dots, m_{N-1}$ and $c_1, \dots, c_N$. Division is legitimate by (H2), proving (H4). In the same coefficient equation the only occurrence of c_N is $\zeta'(\rho) c_N$, from the $n = 0$ summand, because every other Taylor term has either a positive power of t already or at least two positive-order factors of w . This proves the forward triangular recurrence and both directions of the finite-jet assertion by induction. $\square$

The first two inverse formulas, with every sign retained, are

$$m_1 = \frac{\zeta'(\rho) c_1}{\rho \zeta(\rho + 1)}, \quad (\text{H5})$$

$$m_2 = -\frac{2}{\rho(\rho + 1)\zeta(\rho + 2)} \left[\zeta'(\rho) c_2 + \frac{\zeta''(\rho)}{2} c_1^2 - m_1(\zeta(\rho + 1) + \rho \zeta'(\rho + 1)) c_1 \right]. \quad (\text{H6})$$

These follow by $b_1(s) = -s\zeta(s + 1)$ and $b_2(s) = s(s + 1)\zeta(s + 2)/2$, not by changing the Hurwitz convention.

Exact measure reconstruction on the original interval. Assume $L > 0$. For every integer $M \geq 1$ and $0 \leq j \leq M$, set

$$\lambda_{M,j} = \binom{M}{j} \sum_{k=0}^{M-j} (-1)^k \binom{M-j}{k} \frac{m_{j+k}(\mu)}{L^{j+k}}, \quad \mu_M = \sum_{j=0}^M \lambda_{M,j} \delta_{Lj/M}. \quad (\text{H7})$$

Then every $\lambda_{M,j} \geq 0$, their sum is one, and μ_M converges weakly to μ . Thus (H4) and (H7) give an explicit reconstruction of the entire input from the zero germ, with the original support coordinate and bound present in every formula.

Proof. Expanding $(1 - y/L)^{M-j}$ proves

$$\lambda_{M,j} = \int_{[0,L]} \binom{M}{j} (y/L)^j (1 - y/L)^{M-j} d\mu(y).$$

This proves positivity, and the binomial theorem proves the sum is one. At fixed original y , let J have binomial law with M trials and success probability y/L . Its original-coordinate variable LJ/M has mean y and variance

$$\mathbb{E}[(LJ/M - y)^2] = \frac{y(L - y)}{M} \leq \frac{L^2}{4M}. \quad (\text{H8})$$

For continuous f on $[0, L]$ and any $\delta > 0$, split the expectation according to $|LJ/M - y| \leq \delta$. Uniform continuity and Chebyshev's inequality give

$$|\mathbb{E}f(LJ/M) - f(y)| \leq \omega_f(\delta) + 2\|f\|_\infty \frac{L^2}{4M\delta^2},$$

uniformly in y . First take $M \rightarrow \infty$, then $\delta \rightarrow 0$, and integrate against μ . This is weak convergence. For Lipschitz f , Cauchy-Schwarz gives the explicit bound $\text{Lip}(f)L/(2\sqrt{M})$. If $L = 0$, the input is exactly δ_0 , all positive moments vanish, and the original zero branch is constant; this exceptional fibre is retained separately. $\square$

Injectivity without a prespecified common support bound. If two compactly supported probability measures have the same zero germ at the chosen simple ρ , (H4) gives equal moments. Choose any common finite support bound for the two measures. Their measures (H7) are identical at every M , and their weak limits are therefore identical. Hence the map

$$\left\{ \begin{array}{c} \text{compact probability measures} \\ \text{on } [0, \infty) \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{holomorphic zero germs} \\ \text{at the fixed simple } \rho \end{array} \right\},$$

$$\mu \longmapsto \rho_\mu$$

is injective onto its actual image. This statement asserts no surjectivity onto arbitrary holomorphic germs.

The exact support endpoint is also recovered:

$$\sup \text{supp } \mu = \lim_{n \rightarrow \infty} m_n(\mu)^{1/n}. \quad (\text{H9})$$

Indeed, if the endpoint is L_μ , then $m_n \leq L_\mu^n$. For any $0 < a < L_\mu$, positive mass lies in $(a, L_\mu]$, so $m_n \geq a^n \mu((a, L_\mu])$. Taking n th roots and then $a \uparrow L_\mu$ proves the claim. The zero endpoint gives δ_0 and both sides zero.

The completed residue comparison has its own exact factorization. Define

$$\Lambda_{\mu,t}(s) = \pi^{-s/2} \Gamma(s/2) Z_{\mu,t}(s).$$

Then, for the same displacement domain $|t|L < 1$,

$$\text{Res}_{s=1} \Lambda_{\mu,t} = 1, \quad \text{Res}_{s=0} \Lambda_{\mu,t} = -1 - 2tm_1(\mu), \quad (\text{HRes1})$$

and consequently

$$\mathfrak{B}(\mu, t) := \text{Res}_{s=0} \Lambda_{\mu,t} + \text{Res}_{s=1} \Lambda_{\mu,t} = -2tm_1(\mu). \quad (\text{HRes2})$$

Proof. For $\operatorname{Re} \mathfrak{a} > 0$ and $\operatorname{Re} s > 1$, termwise integration of the absolutely convergent geometric series gives

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \int_0^\infty x^{s-1} \frac{e^{-\mathfrak{a}x}}{1 - e^{-x}} dx.$$

At zero the integrand's second factor is $x^{-1} + (1/2 - \mathfrak{a}) + O(x)$, uniformly on compact subsets of the indicated parameter half-plane. Subtract these two terms on $(0, 1)$. The remaining integrals define a function $H(s, \mathfrak{a})$ holomorphic for $\operatorname{Re} s > -1$, and

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \frac{1}{s-1} + \frac{1/2 - \mathfrak{a}}{s} + H(s, \mathfrak{a}).$$

The tail integral is holomorphic there because of exponential decay. Since $\Gamma(1) = 1$ and $\Gamma(s) = s^{-1} + O(1)$, the formula proves $\operatorname{Res}_{s=1} \zeta(s, \mathfrak{a}) = 1$ and $\zeta(0, \mathfrak{a}) = 1/2 - \mathfrak{a}$. For $\mathfrak{a} = 1 + ty$, the parameters stay in a compact subset of that half-plane as y ranges over the original support. Uniformity therefore permits integration of both coefficients against the probability measure. It gives residue one at $s = 1$ and $Z_{\mu,t}(0) = -1/2 - tm_1(\mu)$. Finally $\pi^{-1/2}\Gamma(1/2) = 1$, $\Gamma(s/2) = 2/s + O(1)$, and $\pi^{-s/2} = 1 + O(s)$ give (HRes1) and their sum (HRes2). $\square$

Thus the residue map factors through the explicitly recovered first moment:

$$\rho_\mu \mapsto \frac{\zeta'(\rho)c_1(\mu)}{\rho\zeta(\rho+1)} = m_1(\mu) \mapsto -2tm_1(\mu).$$

The full germ retains every higher moment through (H4), with measure reconstruction given by (H7).

3.3 The complete zero cluster removes the simplicity restriction

The preceding source theorem used a simple scalar branch. The full cluster gives an inverse for **every** nontrivial zero, with its original multiplicity retained. Fix a nontrivial zero ρ of exact order $r \geq 1$. Choose a closed disc about ρ entirely inside $0 < \operatorname{Re} s < 1$, whose boundary Γ contains no zero and whose interior contains no zero of ζ other than ρ . Thus the entire disc also excludes the pole at 1. Shrink the complex t -disc so $Z_{\mu,t}$ is nonzero on Γ and has exactly r zeros inside, with multiplicity. The normal-convergence estimate following (H1) supplies joint holomorphicity and uniform convergence to ζ on Γ ; Rouché's theorem then proves the exact zero count.

For $1 \leq j \leq r$ define the holomorphic power-sum germs

$$q_j(t) = \frac{1}{2\pi i} \oint_\Gamma (s - \rho)^j \frac{\partial_s Z_{\mu,t}(s)}{Z_{\mu,t}(s)} ds.$$

They are exactly the sums of the j th powers of the displacements of the r zeros, counted with multiplicity, by the residue theorem. Set $e_0(t) = 1$ and recursively

$$je_j(t) = \sum_{i=1}^j (-1)^{i-1} e_{j-i}(t) q_i(t).$$

Newton's identities show that the polynomial

$$P_\mu(s, t) = \sum_{j=0}^r (-1)^j e_j(t) (s - \rho)^{r-j} = (s - \rho)^r + \sum_{N \geq 1} P_N(s) t^N \quad (\text{H10})$$

has precisely the full zero cluster as its divisor, and every $P_N(s)$ has degree at most $r - 1$ in the original displacement $s - \rho$. The original point ρ and monic leading coefficient one are both retained. The polynomial is independent of any enumeration or Puiseux branch choices.

There is a jointly holomorphic unit $U_\mu(s, t)$ on a smaller disc such that

$$Z_{\mu,t}(s) = U_\mu(s, t)P_\mu(s, t), \quad U_\mu(s, t) = \sum_{N \geq 0} U_N(s)t^N, \quad U_0(s) = \frac{\zeta(s)}{(s - \rho)^r}. \quad (\text{H11})$$

For completeness, joint holomorphicity does not need to be assumed from the root parametrization. Choose an inner circle still enclosing all zeros for small t ; on that circle the quotient Z/P has nonzero denominator. Its Cauchy integral against $1/(\zeta - s)$ defines a jointly holomorphic function of (s, t) inside, since the integrand is jointly holomorphic on the fixed contour. For each fixed t , the zeros of Z and P agree with their exact multiplicities, so the quotient has removable singularities at the roots and the Cauchy integral equals Z/P . The quotient has no zero, because the two divisors already coincide. This proves (H11). The contour variable ζ here denotes a complex integration variable, not the Riemann zeta function.

Now the input moments have the exact recursive recovery

$$m_N(\mu) = \frac{1}{b_N(\rho)} \sum_{k=1}^N U_{N-k}(\rho)P_k(\rho), \quad (\text{H12})$$

followed by the exact analytic division

$$U_N(s) = \frac{m_N(\mu)b_N(s) - \sum_{k=1}^N U_{N-k}(s)P_k(s)}{(s - \rho)^r}. \quad (\text{H13})$$

In (H13) the numerator vanishes to order at least r ; its holomorphic extension at ρ has value equal to its r th derivative there divided by $r!$. Thus the formula retains every derivative needed by a multiple zero, rather than dividing a nonvanishing germ by a zero.

Proof. Comparing the coefficient of t^N in (H1) and (H11) gives

$$m_N(\mu)b_N(s) = U_N(s)(s - \rho)^r + \sum_{k=1}^N U_{N-k}(s)P_k(s).$$

Evaluation at $s = \rho$ and the nonvanishing (H2) prove (H12). Rearranging the same analytic equality proves (H13), including the divisibility and the value at the centre. Starting from the completely known U_0 , the pair (H12)–(H13) is triangular and recovers every moment from the polynomial-cluster germ.

Two measures with the same polynomial-cluster jet through order N therefore have the same moments through N , by induction. Conversely, equal moments through N give $Z_{\mu,t} - Z_{\nu,t} = O(t^{N+1})$ as holomorphic functions on the fixed circle. Since both denominators remain nonzero there, their logarithmic derivatives, contour power sums, Newton coefficients and hence polynomials (H10) agree through order N . This proves the converse without choosing scalar zero branches. Finally (H7) reconstructs the measure, so the cluster germ at any nontrivial zero determines the entire compact probability input. $\square$

For $r = 1$, $P_N(s) = -c_N$, so (H12)–(H13) reduce to the inverse branch recurrence. For $r > 1$, the contour construction keeps all r zeros and their multiplicities even when individual zeros have fractional-power trajectories. This is a fully proved extension of the historical source's simple-zero statement.

3.4 The historical period map preserves every finite spectral jet

The transcript's marked Frobenius character is

$$z_{p,\omega}(s) = \omega p^{s-1/2}, \quad p > 1, \quad \omega \in S^1, \quad a = \log p > 0. \quad (\text{J1})$$

For the original prime-circle construction, p is a rational prime and $\omega = \psi(\text{Frob}_p)$; the proof below only uses the displayed positive a and unit-modulus ω . It retains the mark and phase.

Fix $\rho \in \mathbb{C}$ and an integer multiplicity $m \geq 1$, and write

$$z = \omega p^{\rho-1/2}, \quad E_{\rho,m} = \mathbb{C}[X]/(X - \rho)^m, \quad L_{z,m} = \mathbb{C}[U, U^{-1}]/(U - z)^m.$$

The original spectral coordinate remains X , and the original Laurent period generator remains U .

Exact jet isomorphism. There is a complex-algebra isomorphism

$$\Phi_{\rho,m} : L_{z,m} \xrightarrow{\sim} E_{\rho,m}, \quad [U] \mapsto z \sum_{k=0}^{m-1} \frac{a^k (X - \rho)^k}{k!}. \quad (\text{J2})$$

Its inverse is

$$[X] \mapsto \rho + \frac{1}{a} \sum_{k=1}^{m-1} \frac{(-1)^{k+1}}{k} \left(\frac{U - z}{z} \right)^k. \quad (\text{J3})$$

For every $0 \leq k < m$, (J2) induces on the k th associated conormal grade the exact multiplier

$$[(U - z)^k] \mapsto (za)^k [(X - \rho)^k]. \quad (\text{J4})$$

Proof. The target of (J2) is a unit: it is z times a polynomial with constant term one and nilpotent remaining part. Its difference from z lies in $(X - \rho)$, so its m th power vanishes. The Laurent universal property and the quotient therefore define (J2). In $\mathbb{C}[t]/t^m$, truncated $\exp(t)$ and $\log(1 + t)$ are inverse operations. This identity can be proved without convergence: formal differentiation shows $(\log(\exp t))' = 1$ and its constant term is zero; likewise $\exp(\log(1 + t))/(1 + t)$ has derivative zero and constant term one. Reducing the resulting formal identities modulo t^m gives (J3) and both composites. Finally

$$\Phi(U - z) = za(X - \rho) + (X - \rho)^2 \left(z \sum_{j=2}^{m-1} \frac{a^j (X - \rho)^{j-2}}{j!} \right),$$

so its k th power modulo $(X - \rho)^{k+1}$ is exactly $(za)^k (X - \rho)^k$. For $m = 1$, the sums in (J3) are empty and both rings are the specified scalar fibres, as required. $\square$

For a finite primary spectral block $A = \rho I + N$ with $N^m = 0$, this means that its actual period operator is exactly

$$\omega \exp(a(A - \tfrac{1}{2}I)) = z \sum_{k=0}^{m-1} \frac{a^k N^k}{k!}. \quad (\text{J5})$$

There is no deletion of nilpotent multiplicity in this map. Indeed, $\exp(aN) - I = N(aI + NB(N))$; its second factor is an invertible polynomial in N , so its powers have exactly the same kernels as the corresponding powers of N . This proves preservation of all Jordan block lengths locally.

3.5 Exact conjugate-reciprocal involution on the jets and conormal

Define the original spectral involution

$$\rho^\dagger = 1 - \bar{\rho}, \quad z^\dagger = \omega p^{\rho^\dagger - 1/2} = 1/\bar{z}. \quad (\text{J6})$$

On polynomial classes define the conjugate-linear algebra isomorphism

$$\mathcal{S}_E : E_{\rho,m} \longrightarrow E_{\rho^\dagger,m}, \quad [f(X)] \longmapsto [\bar{f}(1 - X)]. \quad (\text{J7})$$

On Laurent classes define

$$\mathcal{S}_L : L_{z,m} \longrightarrow L_{z^\dagger,m}, \quad \left[\sum c_n U^n \right] \longmapsto \left[\sum \bar{c}_n U^{-n} \right]. \quad (\text{J8})$$

Both have square the identity after returning to the original labelled block, and the full finite-jet diagram commutes:

$$\boxed{\mathcal{S}_E \Phi_{\rho,m} = \Phi_{\rho^\dagger,m} \mathcal{S}_L.} \quad (\text{J9})$$

Proof. The polynomial generator transforms by

$$X - \rho \longmapsto 1 - X - \bar{\rho} = -(X - \rho^\dagger).$$

The Laurent generator transforms by the exact identity

$$U - z \longmapsto U^{-1} - \bar{z} = -\bar{z} U^{-1} (U - z^\dagger). \quad (\text{J10})$$

The factor $-\bar{z}U^{-1}$ is a unit, so the respective m th-power ideals map to one another and both maps are defined. Applying each twice fixes constants and the original generator. To check (J9), it suffices to check the Laurent generator and constants. Its left image is

$$\bar{z} \sum_{k=0}^{m-1} \frac{(-a)^k (X - \rho^\dagger)^k}{k!},$$

which is the inverse of $z^\dagger \exp(a(X - \rho^\dagger))$ modulo $(X - \rho^\dagger)^m$ and hence the right image. $\square$

For clarity, the entire nonlinear mixing of Laurent Taylor coefficients under (J8) is explicit. With $t' = U - z^\dagger$, for $1 \leq k < m$ one has

$$\boxed{\mathcal{S}_L((U - z)^k) = \sum_{\ell=k}^{m-1} (-1)^\ell \binom{\ell-1}{\ell-k} \bar{z}^{k+\ell} (t')^\ell.} \quad (\text{J11})$$

Proof. By (J10), the expression is

$$(-\bar{z})^k U^{-k} (t')^k = (-1)^k \bar{z}^{2k} (t')^k (1 + \bar{z}t')^{-k}.$$

The formal binomial expansion $(1 + v)^{-k} = \sum_{j \geq 0} (-1)^j \binom{k+j-1}{j} v^j$, truncated when the total degree reaches m , gives (J11). The constant basis vector maps to one separately. $\square$

In particular, the conormal maps are

$$[X - \rho] \longmapsto -[X - \rho^\dagger], \quad [U - z] \longmapsto -\bar{z}^2 [U - z^\dagger], \quad (\text{J12})$$

and their compatibility with (J4) is an exact equality of coefficients:

$$-\bar{z}^2 (z^\dagger a) = -\bar{z}a = \bar{z}\bar{a}(-1).$$

The higher grade k has factors $(-1)^k$ and $(-1)^k \bar{z}^{2k}$ respectively, with the corresponding $(za)^k$ comparison. This supplies the all-order anti-involution/conormal morphism requested by the historical construction; it does not erase the higher terms in (J11).

On the real analytic radial chart $s = \beta + i\gamma$, the same exact map gives

$$\nu = \log |z| = (\beta - \tfrac{1}{2}) \log p, \quad d\nu = (\log p) d\beta, \quad c_{-+} = 4(\beta - \tfrac{1}{2}), \quad \nu = \frac{\log p}{4} c_{-+}. \quad (\text{J13})$$

The source's marked transverse chart $j_S(\nu) = [((e^\nu, 1, \dots, 1), 0)]$ pulls its displayed form $\alpha = d \log |a_\infty|$ back to $d\nu$, since $\log(e^\nu) = \nu$. Thus the composite character→radial chart has pullback $\alpha = (\log p) d\beta$, with the original prime and tessarine factor retained. The local identity requires no conjecture about zeros. A global identification with any separately defined Weil quadratic form is not asserted by this local chart calculation; the exact algebraic and chart maps proved here are the morphisms actually available from this source block.

3.6 Period collisions: the exact unlabelled image and its missing fibres

Let Z be a finite set of distinct spectral points ρ , with original multiplicities $m_\rho \geq 1$, and

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[X]/h(X) \cong \prod_{\rho \in Z} E_{\rho, m_\rho}.$$

The last map is the CRT map $f \mapsto (f \bmod (X - \rho)^{m_\rho})$; pairwise coprimality gives its inverse by the Bezout identity. No multiplicity is discarded.

Set $z_\rho = \omega p^{\rho-1/2}$. The equality of two period values is exactly

$$z_\rho = z_\sigma \iff \rho - \sigma \in \frac{2\pi i}{\log p} \mathbb{Z}. \quad (\text{J14})$$

The proof is the exact kernel $2\pi i \mathbb{Z}$ of the complex exponential. In particular, a fixed prime's character cannot automatically be treated as globally injective on a zero packet. No noncollision claim for actual zeta zeros is inserted here.

For each distinct period value z , put $M_z = \max_{\rho: z_\rho=z} m_\rho$ and

$$P(U) = \prod_z (U - z)^{M_z}, \quad L_P = \mathbb{C}[U, U^{-1}]/P(U).$$

There is an injective algebra map

$$\Phi_Z : L_P \hookrightarrow E_h \quad (\text{J15})$$

whose ρ th component is Laurent reduction modulo $(U - z_\rho)^{m_\rho}$ followed by (J2). Its image consists exactly of the tuples which, after (J3) on each labelled primary block, have the same Laurent Taylor coefficients at a common period value through every order visible in both blocks. More explicitly, if $a_\rho \in E_{\rho, m_\rho}$, write

$$\Phi_{\rho, m_\rho}^{-1}(a_\rho) = \sum_{k=0}^{m_\rho-1} b_{\rho, k} (U - z_\rho)^k.$$

Then (a_ρ) lies in the image exactly when

$$b_{\rho,k} = b_{\sigma,k} \quad \text{whenever } z_\rho = z_\sigma \text{ and } 0 \leq k < \min(m_\rho, m_\sigma). \quad (\text{J16})$$

The exact dimension and cokernel dimension as complex vector spaces are

$$\dim \operatorname{im} \Phi_Z = \sum_z M_z, \quad \dim \operatorname{coker} \Phi_Z = \sum_\rho m_\rho - \sum_z M_z. \quad (\text{J17})$$

Proof. Use CRT also on P . For a fixed common value z , a Laurent polynomial vanishes on every corresponding spectral block exactly when it is divisible by $(U - z)^{m_\rho}$ for every such ρ , which is equivalent to divisibility by $(U - z)^{M_z}$. Therefore the kernel of the original Laurent evaluation map is exactly (P) , giving (J15). Reduction of one degree- $< M_z$ polynomial to each smaller truncation gives (J16). Conversely choose any one block of maximal multiplicity M_z ; its Laurent coefficients define the common polynomial, and (J16) forces all other components to be its truncations. These constructions prove the image equality, while the dimensions follow from the explicit polynomial bases. $\square$

Keeping the labels restores a full exact isomorphism:

$$\prod_{\rho \in Z} L_{z_\rho, m_\rho} \xrightarrow{\Pi \Phi_{\rho, m_\rho}} E_h. \quad (\text{J18})$$

Thus the historical period map and the current finite arithmetic jet object have a proved relation at both the labelled and unlabelled levels. The collision compatibility space is exactly the image (J16). An explicit quotient records every lost independent coefficient: choose one maximal block ρ_z over each z , with $m_{\rho_z} = M_z$, and define the complex-linear map

$$\Delta_Z : E_h \longrightarrow \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho}, \quad (a_\rho) \longmapsto (b_{\rho,k} - b_{\rho_z,k})_{\substack{z, \rho \neq \rho_z \\ 0 \leq k < m_\rho}}. \quad (\text{J19})$$

It is surjective: set all maximal-block coefficients to zero and prescribe each remaining block's coefficients to be the desired output. Its kernel is exactly (J16). Consequently

$$0 \longrightarrow L_P \xrightarrow{\Phi_Z} E_h \xrightarrow{\Delta_Z} \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho} \longrightarrow 0 \quad (\text{J20})$$

is an exact sequence of complex vector spaces. The first map is an algebra homomorphism; the last map is explicitly linear and is not incorrectly promoted to a ring homomorphism. Its target has the cokernel dimension in (J17).

If Z is stable under $\rho \mapsto 1 - \bar{\rho}$ with the same multiplicities, the maps (J7) and (J8), together with permutation of the labelled factors, give conjugate-linear involutions on both sides of (J18). Formula (J9) proves that (J18) intertwines them. The unlabelled construction (J15) is also stable: $z \mapsto 1/\bar{z}$ permutes the common period values and preserves M_z ; (J10) carries the defining ideal of one value to that of the paired value.

3.7 Two prime periods recover the complete finite spectral algebra

Retain a finite nonempty set Z of distinct spectral points with the original orders $m_\rho \geq 1$, and $E_h = \mathbb{C}[X]/h(X)$, where $h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}$. Choose distinct rational primes p, q and fixed marks $\omega_p, \omega_q \in S^1$. Put

$$a = \log p, \quad b = \log q, \quad u_\rho = \omega_p e^{a(\rho-1/2)}, \quad v_\rho = \omega_q e^{b(\rho-1/2)}.$$

Define the units $\mathfrak{A}_p, \mathfrak{A}_q \in E_h$ by their complete original spectral CRT components:

$$\begin{aligned} (\mathfrak{A}_p)_\rho &= u_\rho \sum_{k=0}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \\ (\mathfrak{A}_q)_\rho &= v_\rho \sum_{k=0}^{m_\rho-1} \frac{b^k (X - \rho)^k}{k!}. \end{aligned} \quad (\text{P1})$$

The period-algebra map therefore has the exact type

$$\mathcal{P}_{p,q} : \mathbb{C}[U^{\pm 1}, V^{\pm 1}] \longrightarrow E_h, \quad U \longmapsto \mathfrak{A}_p, \quad V \longmapsto \mathfrak{A}_q. \quad (\text{P2})$$

This map uses the actual periods of the original generator; the notation $\mathcal{P}_{p,q}$ leaves the theta map Θ unchanged.

Theorem 3.1 (Complete two-prime Laurent presentation). *The map (P2) is surjective, with every local order retained. More precisely, in $\mathcal{L} = \mathbb{C}[U^{\pm 1}, V^{\pm 1}]$, put*

$$\begin{aligned} B_\rho(U) &= v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j, \\ I_\rho &= ((U - u_\rho)^{m_\rho}, V - B_\rho(U)). \end{aligned} \quad (\text{P3})$$

Here $\binom{b/a}{0} = 1$, and for $j \geq 1$ the coefficient is $(b/a)(b/a - 1) \cdots (b/a - j + 1)/j!$. The ideals I_ρ are pairwise comaximal, and

$$\begin{aligned} \ker \mathcal{P}_{p,q} &= \bigcap_{\rho \in Z} I_\rho = \prod_{\rho \in Z} I_\rho, \\ \mathcal{L} / \bigcap_{\rho \in Z} I_\rho &\xrightarrow{\sim} E_h. \end{aligned} \quad (\text{P4})$$

The map on each primary quotient is the original finite exponential map (J2), with the second generator determined by (P3).

Proof. First the two marked periods separate distinct complex points. If their values at ρ, σ agree, there exist integers k, l with

$$a(\rho - \sigma) = 2\pi i k, \quad b(\rho - \sigma) = 2\pi i l.$$

Eliminating $\rho - \sigma$ gives $k \log q = l \log p$, hence $q^k = p^l$. Prime valuations in the positive rationals give $k = l = 0$, including negative integer exponents. Thus $\rho = \sigma$. Each fixed phase cancels only within its own prime character; neither mark was changed.

For a fixed ρ , set $y = (U - u_\rho)/u_\rho$. The quotient $\mathcal{L}/I_\rho$ is exactly $\mathbb{C}[y]/y^{m_\rho}$: its generators satisfy

$$U = u_\rho(1 + y), \quad V = v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} y^j.$$

Both expressions are units, since their constant terms are nonzero and their remaining parts are nilpotent. The Laurent inverses therefore impose no extra relation. These expressions give a map from the Laurent quotient to $\mathbb{C}[y]/y^{m_\rho}$, whose inverse sends y to $(U - u_\rho)/u_\rho$; their composites fix the displayed generators.

The map to E_{ρ, m_ρ} is

$$y \longmapsto \sum_{k=1}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \quad X \longmapsto \rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} y^j \quad (\text{P5})$$

for it and its inverse, respectively. The finite logarithm and exponential identities proved in (J2)–(J3) verify both composites. The identity $(1+y)^{b/a} = \exp((b/a) \log(1+y))$, reduced modulo y^{m_ρ} , verifies the prescribed V -image. For completeness its binomial coefficients are fixed by the formal differential equation $(1+y)F' = (b/a)F$, $F(0) = 1$: $(j+1)[y^{j+1}]F = (b/a-j)[y^j]F$. This recurrence gives exactly the coefficients in (P3). Thus I_ρ is precisely the kernel of the ρ -component of (P2), including order one.

For two different labels, either $u_\rho \neq u_\sigma$, or their v -values differ by the separation already proved. In the first case the powers of the two distinct linear U -factors have a polynomial Bezout identity, proving $I_\rho + I_\sigma = \mathcal{L}$. In the second case, $(V - v_\rho)^{m_\rho} \in I_\rho$: modulo I_ρ , $V - v_\rho = B_\rho(U) - v_\rho$ is a multiple of $U - u_\rho$. The corresponding powers of the two distinct linear V -factors again give a Bezout identity. Hence all primary kernels are pairwise comaximal.

The Chinese-remainder map now gives

$$\mathcal{L} / \bigcap_{\rho} I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} \mathcal{L} / I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} E_{\rho, m_{\rho}} \xleftarrow{\sim} E_h. \quad (\text{P6})$$

For clarity, the first map is onto because Bezout identities give classes ϵ_ρ equal to one in factor ρ and zero in every other factor; summing ϵ_ρ times prescribed representatives recovers any tuple. Its kernel is the intersection by definition. Pairwise comaximality also gives intersection equal to product, first for two ideals by multiplying a Bezout identity by an element of the intersection, and then by induction. These arguments prove every assertion in (P4).

The original coordinate itself has the explicit inverse image

$$[X] \mapsto \sum_{\rho \in Z} \epsilon_\rho \left(\rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j \right). \quad (\text{P7})$$

Each term is exactly (P5) in its own primary factor; the other factors vanish through ϵ_ρ . Thus the inverse retains the original generator and all of its nilpotent jets. $\square$

This presentation is an exact statement for each finite packet. The single-prime collision quotient (J20) and the two-prime map (P6) specify their respective information. The factorwise split lifts below retain their individual support labels.

3.8 Split-Zero support is preserved by the exact maps

For any ring R , keep the original $G(R) = \{\tau\} \sqcup R^\bullet$, global zero τ , and supported zero $e_R = 0_R^\bullet$. Every ring map $f : R \rightarrow S$ gives

$$G(f)(\tau) = \tau, \quad G(f)(r^\bullet) = f(r)^\bullet. \quad (\text{S1})$$

This preserves addition and multiplication: if one input is τ , use its defining identity/absorber laws, and if both are supported, use the ring homomorphism identities with the supported tag retained. In particular $G(f)(e_R) = e_S \neq \tau$, even if f has a nontrivial kernel.

Apply (S1) separately to each map (J2), and then take the product of these lifted semiring isomorphisms. The resulting map is

$$\prod_{\rho \in Z} G(L_{z_\rho, m_\rho}) \xrightarrow{\sim} \prod_{\rho \in Z} G(E_{\rho, m_\rho}). \quad (\text{S2})$$

Every individual supported-zero coordinate is retained. The support map on both sides is the full vector of presence indicators in $\{0, 1\}^Z$; under (S2) it is unchanged.

For a finite nonempty index set $\mathcal{I}$, the comparison with applying G once to a product ring is the map

$$\begin{aligned} \iota_R : G \left(\prod_{i \in \mathcal{I}} R_i \right) &\longrightarrow \prod_{i \in \mathcal{I}} G(R_i), \\ \tau &\longmapsto (\tau)_i, \quad (r_i)_i^\bullet \longmapsto (r_i^\bullet)_i. \end{aligned} \quad (\text{S4})$$

It is an injective unital semiring homomorphism. For two supported inputs, the operations agree coordinate by coordinate by the ring laws. When a τ occurs, its additive-identity and multiplicative-absorber laws also agree coordinate by coordinate. The two sorts of image are disjoint because $\mathcal{I}$ is nonempty; equality of supported images recovers every r_i . Its support image is exactly the empty and full support vectors. For ring maps $f_i : R_i \rightarrow S_i$, direct evaluation proves

$$\left(\prod_i G(f_i) \right) \iota_R = \iota_S G \left(\prod_i f_i \right). \quad (\text{S5})$$

This is the exact natural comparison with the factorwise map (S2). If $\mathcal{I}$ is empty, the same comparison has the two-element semiring $G(0)$ as source and the one-element empty product as target, so it is not injective; its unique map identifies e and τ . The product map (S2) itself remains the identity between empty products in that case.

For the unlabelled map, a source factor $G(L_{z,M_z})$ maps to all spectral factors over z , using (J2) after the quotient $L_{z,M_z} \twoheadrightarrow L_{z,m_\rho}$. This is injective because one target block has multiplicity M_z . Its support image consists exactly of the vectors which are constant on every collision class $q^{-1}(z)$, where $q : Z \rightarrow \{z_\rho\}$ is $\rho \mapsto z_\rho$. A source τ gives τ in every such factor, while a supported element gives a supported element in every factor, including factors in which its amplitude becomes zero. The support embedding is exactly

$$q^{-1} : \mathcal{P}(\{z_\rho\}) \hookrightarrow \mathcal{P}(Z), \quad (\text{S3})$$

which preserves empty set, unions and intersections. The direct-image map $q_* : \mathcal{P}(Z) \rightarrow \mathcal{P}(\{z_\rho\})$ preserves unions and has $q_* q^{-1} = \text{id}$, but need not preserve intersections. For two distinct labels ρ, σ in the same collision class, $\{\rho\} \cap \{\sigma\} = \emptyset$ while $q_* \{\rho\} \cap q_* \{\sigma\} = \{z\}$. Thus (S3), rather than an unsupported lattice quotient isomorphism, is the exact support relation; the amplitude compatibility is (J16).

For a same-centre truncation $m' \leq m$, the quotient

$$L_{z,m} \twoheadrightarrow L_{z,m'}, \quad E_{\rho,m} \twoheadrightarrow E_{\rho,m'}$$

commutes with (J2) because the finite power series simply reduce to the corresponding lower nilpotence order. Its lifted map sends the supported nonzero class $(U - z)^{m'}$ to e when $m' < m$, and never to τ . This supplies a concrete example of the original distinction between a vanished jet amplitude and absent support, inside the current arithmetic finite-jet packet.

3.9 All-order Hurwitz motion transported to the period and tesserine coordinates

For the actual zero branch (H3), the exact period germ is

$$z_\mu(t) = \omega p^{\rho_\mu(t)-1/2} = z_\rho \exp \left((\log p) \sum_{n \geq 1} c_n(\mu) t^n \right). \quad (\text{C1})$$

Its N th coefficient is

$$[t^N]z_\mu(t) = z_\rho \sum_{k=1}^N \frac{(\log p)^k}{k!} \sum_{\substack{n_1+\dots+n_k=N \\ n_j \geq 1}} \prod_{j=1}^k c_{n_j}(\mu) \quad (N \geq 1). \quad (\text{C2})$$

The empty coefficient is z_ρ . Conversely, using the unique logarithm germ vanishing at $z_\mu/z_\rho = 1$,

$$c_N(\mu) = \frac{1}{\log p} [t^N] \log(z_\mu(t)/z_\rho). \quad (\text{C3})$$

Equations (C2)–(C3) are inverse because the same formal exponential/logarithm calculation used in (J2)–(J3) applies at every finite order. Combining (C3) with (H4) proves that the marked local period germ also determines the entire compact Hurwitz input.

For real t in the germ's domain,

$$\begin{aligned} \log |z_\mu(t)| &= (\log p) \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right), \\ c_{-+}(\rho_\mu(t)) &= 4 \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right). \end{aligned} \quad (\text{C4})$$

All coefficients, including tangential ones, are retained in (C1)–(C3). The real projection in (C4) is explicitly the radial/tessierine observable; its loss of imaginary coefficients is a specified map. For $t \neq 0$, the Hurwitz family generally lacks the original functional equation, so $1 - \rho_\mu(t)$ is not asserted to be another zero of that family. The tessierine coordinate formula is the proved coordinate map evaluated at the actual branch, with this spectral distinction preserved.

3.10 Mathematical scope

The maps (J2)–(J20) identify the full finite arithmetic algebra with its labelled Laurent jets and calculate the exact quotient of unlabelled single-prime observations. Equations (H12)–(H13) recover all moments from the actual zero cluster, and (H7) reconstructs the entire measure. These statements retain the original multiplicities, support coordinates and marked periods.

The constructions give exact algebraic and analytic comparison maps. They do not supply the uniform estimate on the arithmetic moment matrices required by the weight-control calculation elsewhere in this reader.

The accompanying source records retain the historical locators and independent proof reviews. The proofs in this section specify their own hypotheses and maps in full.

4 Historical support branches, ideal completion, and measured observations

This chapter gives the complete branch, ideal-completion, atomic-kernel, and operator-algebra proofs extracted and continued from the historical Split-Zero support program. Its mathematical report and independent audit are, respectively, `logbook/historical_next_responses_20260912.md` and `logbook/frame_branch_independent_audit_20260912.md`. The equation identifiers B, I, M, A and O are local to this chapter.

Source locators and comparison. Exact complete response bodies are staged at `sources/historical/next_responses/preboolean_prime_branches.md` (source lines 23137–23956; response

heading 23104; exported timestamp 7/6/2026, 12:23:35 PM), sources/historical/next_response/preboolean_arithmetic_branch_maps.md (source lines 25162–25715; heading 25058; timestamp 7/6/2026, 7:58:41 PM), and sources/historical/next_responses/pointfree_frame_arithmetic_claim.md (source lines 25754–26221; heading 25721; timestamp 7/6/2026, 10:01:27 PM). Their shared original source has SHA-256 8d1780a24df3bb31db981b5e437558de31e6d94bf21bb54103a0a336d6a73091. The respective excerpt SHA-256 values are f5f11dd6b26a3a9d6a4c77cf2f7b86898759a19fa740c993d5749c8d47c02618, e84873955876b2be5f7738f746b4e133fc479cfe726bcb85d477f9aec7d6d10e, and 57c6662732b723e451a495f0224507f61e3a969956178018af7633ae06797ac0. For each excerpt, its first line is the first indicated original source line, so every original locator has the exact excerpt locator obtained by subtracting that first source line and adding one.

The core corpus comparison is split_zero_projective_monads_surcomplex/split_support_geometry_arithmetic_curve_v11.tex, SHA-256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d. Its lines 1772–2054 already prove the ordinary prime-filter branches and their arithmetic maps; lines 2001–2018 explicitly impose arbitrary-join preservation for locale points. The support and measured realizations were also compared at lines 5006–5096 and 5346–5378. Accordingly the ordinary branch classification is a surviving historical construction, while the ideal quotient, explicit descent witnesses, atomic kernel and full L^∞ correspondence below supply the additional proved maps.

The final historical response, source lines 25982–26020 (excerpt lines 229–267), equates ordinary amplitude-fixed semiring branches with frame points without imposing arbitrary-join preservation. The exact correction is the classification (B1)–(B5), the frame quotient (I2), and the point-descent criterion (I3)–(I10). The measure-algebra construction then exhibits the kernel pairs that ordinary branches can distinguish. No external KMS classification or Morita equivalence is used as a theorem of this chapter.

4.1 Exact ordinary and continuous Split-Zero branches

The results below are stated in classical set theory. The ultrafilter construction explicitly uses Zorn’s lemma. No claim about existence or nonexistence of points in a constructive foundation is inferred from that use of choice; such an internal comparison would also have to specify its truth-value object and its available joins.

Let R be a nonzero commutative ring and let L be a nontrivial bounded distributive lattice. Retain the source carrier and all its exceptional coordinates:

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subseteq R \times L,$$

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu),$$

$$\tau_L = (0, 0_L), \quad e_L = (0, 1_L), \quad 1 = (1_R, 1_L).$$

Write $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $e = 0_R^\bullet$ distinct from global zero τ . The indicated subset is the carrier throughout.

Here $\mathbb{B} = \{0, 1\}$ has join and meet as its semiring operations. The supported copy in $G(R)$ has $r^\bullet + s^\bullet = (r + s)^\bullet$ and $r^\bullet s^\bullet = (rs)^\bullet$; its global zero satisfies $\tau + x = x + \tau = x$ and $\tau x = x\tau = \tau$. The map $G_{\mathbb{B}}(R) \rightarrow G(R)$ defined by $(0, 0) \mapsto \tau$, $(r, 1) \mapsto r^\bullet$ is a bijection and preserves these operations in each support case, hence is a unital semiring isomorphism. To verify the general carrier, the displayed operations are the restrictions of the product semiring $R \times L$. This subset is closed: a sum of nonzero amplitude has top support because at least one input has nonzero amplitude; a product of nonzero amplitude has two nonzero input amplitudes and hence top support. Every zero-amplitude

output belongs to the zero fibre. The zero and unit belong to the subset. Thus associativity, commutativity and distributivity restrict from the product, including all supported cancellations. All homomorphisms below preserve zero and unit. Write $z_\lambda = (0, \lambda)$.

Theorem B1 — all ordinary amplitude-fixed branches. There is a natural bijection

$$\mathrm{Hom}_{\mathrm{DLat}_{0,1}}(L, \mathbb{B}) \xrightarrow{\sim} \{\beta : G_L(R) \rightarrow G(R) \text{ unital semiring map} : \beta(r, 1_L) = r^\bullet\}. \quad (\text{B1})$$

It sends ϵ to

$$\beta_\epsilon(r, 1_L) = r^\bullet, \quad \beta_\epsilon(0, \lambda) = \begin{cases} \tau & \epsilon(\lambda) = 0, \\ e & \epsilon(\lambda) = 1. \end{cases} \quad (\text{B2})$$

The map is surjective as a semiring map; its exact kernel congruence is equality of amplitudes together with equality of selected support values:

$$(r, \lambda) \sim_\epsilon (s, \mu) \iff r = s \text{ and } \epsilon(\lambda) = \epsilon(\mu). \quad (\text{B3})$$

Proof. The source formulas agree at $(0, 1_L)$ since $\epsilon(1_L) = 1$. They preserve zero and unit. On the zero fibre, preservation of addition and multiplication is precisely preservation of join and meet. If one argument has nonzero amplitude, its lattice coordinate is 1_L ; adding a zero-fibre argument leaves it unchanged and multiplying by that argument gives $(0, \lambda)$, matching respectively addition by τ or e , and multiplication by τ or e in $G(R)$. With both arguments at full support, even if their sum or product cancels to amplitude zero, both outputs are supported and ring identities prove preservation. This verifies all exceptional products and cancellations.

Conversely, $(0, \lambda)$ is additively idempotent. The only additive idempotents of $G(R)$ are τ and e : a supported $r^\bullet + r^\bullet = r^\bullet$ implies $2r = r$ in the ring, hence $r = 0$. Therefore every amplitude-fixed β sends the zero fibre into $\{\tau, e\}$. Identifying this two-element subsemiring with $\mathbb{B}$ gives a map ϵ preserving bottom, top, meet and join. Formula (B2) reconstructs β ; both constructions are inverse. Every amplitude and τ occurs in the image, and inspecting (B2) gives exactly (B3). $\square$

Now let L be a frame: all joins exist and finite meets distribute over arbitrary joins. Its zero fibre has specified infinitary joins

$$\bigvee_{i \in I}^{\mathrm{fib}} (0, \lambda_i) = (0, \bigvee_{i \in I} \lambda_i).$$

Theorem B2 — exactly the continuous branches. Under (B1), the subset of branches preserving all these zero-fibre joins is exactly

$$\mathrm{Hom}_{\mathrm{Frm}}(L, \mathbb{B}). \quad (\text{B4})$$

Indeed the corresponding equality for (B2) says exactly $\epsilon(\bigvee_i \lambda_i) = \bigvee_i \epsilon(\lambda_i)$, and finite meets are already preserved. This is the precise additional structure required by the transcript's final frame formulation. Its ordinary algebraic maps retain a larger, explicitly identified category.

The classification extends to every map between original lattice-split carriers. For commutative unital rings R, A and nontrivial bounded distributive lattices L, M ,

$$\mathrm{Hom}_{\mathrm{Semi}}(G_L(R), G_M(A)) \cong \mathrm{Hom}_{\mathrm{Ring}}(R, A) \times \mathrm{Hom}_{\mathrm{DLat}_{0,1}}(L, M). \quad (\text{B5})$$

The pair (h, f) gives $(r, \lambda) \mapsto (h(r), f(\lambda))$. Conversely an arbitrary semiring map sends each additively idempotent z_λ to amplitude zero, say $z_{f(\lambda)}$; this preserves finite joins, meets and bottom. The identity $e_L = (1_R, 1_L) + (-1_R, 1_L)$ makes its image have top support, because one summand is the target unit; additive idempotence makes its amplitude zero. Thus $e_L \mapsto e_M$ and f preserves

top. Now $(r, 1_L) + e_L = (r, 1_L)$ forces its image to have top support. Write that image $(h(r), 1_M)$. The supported operations give a unital ring map h , and the formulas recover every source element, consistently at e_L . Coordinate composition proves naturality and both inverse constructions. In particular the entire two-factor datum is recoverable from the semiring map; scalar amplitude and support are not freely interchanged.

4.2 The exact frame comparison retaining all ordinary branches

For the underlying bounded distributive lattice of a frame L , let $\text{Idl}(L)$ be the set of lattice ideals. An ideal is a nonempty lower set closed under finite joins. Its bottom is $\{0_L\}$, its top is L , its finite meet is intersection, and the join of a family of ideals is

$$\bigvee_{\alpha} I_{\alpha} = \{x \in L : x \leq y_1 \vee \cdots \vee y_n \text{ for some finite } n \geq 0, y_j \in \bigcup_{\alpha} I_{\alpha}\}. \quad (\text{I1})$$

The empty finite join is 0_L ; this retains the bottom ideal in every formula.

Theorem I1 — frame quotient and principal-ideal section. The maps

$$q : \text{Idl}(L) \rightarrow L, \quad I \mapsto \bigvee I, \quad \eta : L \rightarrow \text{Idl}(L), \quad \lambda \mapsto \downarrow \lambda \quad (\text{I2})$$

have the following exact properties:

1. $\text{Idl}(L)$ is a frame and q is a surjective frame homomorphism.
2. η is an injective bounded-lattice homomorphism, $q\eta = \text{id}_L$, and $q(I) \leq \lambda \iff I \subseteq \eta(\lambda)$.
3. The kernel congruence of q is exactly $I \sim J \iff \bigvee I = \bigvee J$. No claim that η preserves arbitrary joins is made.

Proof. Formula (I1) is an ideal and is the least ideal containing every I_{α} ; intersections give infima. To prove frame distributivity, let $x \in I \cap \bigvee_{\alpha} J_{\alpha}$. Choose $x \leq y_1 \vee \cdots \vee y_n$ with $y_j \in J_{\alpha_j}$. In the distributive lattice,

$$x = x \wedge (y_1 \vee \cdots \vee y_n) = (x \wedge y_1) \vee \cdots \vee (x \wedge y_n).$$

Each $x \wedge y_j$ belongs to $I \cap J_{\alpha_j}$, proving $I \cap \bigvee_{\alpha} J_{\alpha} \subseteq \bigvee_{\alpha} (I \cap J_{\alpha})$; the reverse inclusion follows from containment. Thus $\text{Idl}(L)$ is a frame.

The supremum of the right side of (I1) equals $\bigvee_{\alpha} \bigvee I_{\alpha}$, proving arbitrary-join preservation by q , including the empty family. Since L is a frame,

$$(\bigvee I) \wedge (\bigvee J) = \bigvee_{i \in I, j \in J} (i \wedge j) = \bigvee (I \cap J).$$

The last equality holds because each $i \wedge j$ is in the intersection, and each $x \in I \cap J$ is the term $x \wedge x$. Hence q preserves finite meets and the top. Principal ideals satisfy $\downarrow \lambda \cap \downarrow \mu = \downarrow (\lambda \wedge \mu)$ and $\downarrow \lambda \vee \downarrow \mu = \downarrow (\lambda \vee \mu)$; they also preserve bottom/top. Their suprema are the original elements, proving $q\eta = \text{id}_L$, injectivity and surjectivity. The adjunction equivalence and the kernel description follow immediately from the definition of supremum. $\square$

Theorem I2 — all algebraic branches become frame points before the quotient. There are mutually inverse bijections

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \cong \text{Hom}_{\text{Frm}}(\text{Idl}(L), \mathbb{B}), \quad (\text{I3})$$

given by

$$\widehat{\epsilon}(I) = \bigvee_{\lambda \in I} \epsilon(\lambda) \quad \text{and} \quad \epsilon = \widehat{\epsilon}\eta. \quad (I4)$$

A frame point $\widehat{\epsilon}$ in (I3) factors through q if and only if its original ϵ is a frame point of L . Thus the exact inclusion of point sets is

$$q^* : \text{Pt}(L) \hookrightarrow \text{Pt}(\text{Idl}(L)), \quad \epsilon \mapsto \epsilon q, \quad (I5)$$

and its image is the points constant on the kernel congruence in Theorem I1.

Proof. In (I4), the value is 1 precisely when the ideal meets the prime filter $\epsilon^{-1}(1)$. For a join of ideals, a value 1 is witnessed by an element below a finite join of source elements; finite primeness then gives an element of value 1 in one of those source ideals. The reverse implication follows by containment, proving arbitrary-join preservation. For intersections, if I and J each have a value 1 element i, j , their meet $i \wedge j \in I \cap J$ has value 1; the reverse implication is immediate. Top and bottom follow from $\epsilon(1) = 1, \epsilon(0) = 0$. Therefore $\widehat{\epsilon}$ is a frame homomorphism.

Every ideal is the join in $\text{Idl}(L)$ of its principal ideals, so an arbitrary frame point is completely determined by its restriction along η , giving both inverse identities. If ϵ preserves arbitrary joins, $\widehat{\epsilon}(I) = \epsilon(\bigvee I) = (\epsilon q)(I)$. Conversely if $\widehat{\epsilon} = \chi q$ for a frame map χ , then $\epsilon = \widehat{\epsilon}\eta = \chi q\eta = \chi$, so ϵ is a frame map. A set map constant on the fibres of surjective q factors uniquely; if that map is a frame map, the induced factor preserves arbitrary joins and finite meets by lifting families through q . This proves the final image claim. $\square$

Theorem I3 — original Split-Zero comparison and exact congruence. Every bounded-lattice map $f : L \rightarrow M$ induces

$$G_f(R) : G_L(R) \rightarrow G_M(R), \quad (r, \lambda) \mapsto (r, f(\lambda)). \quad (I6)$$

For the nontrivial frames in Theorem I1 this gives a split pair of unital semiring maps

$$G_L(R) \xrightarrow{G_\eta(R)} G_{\text{Idl}(L)}(R) \xrightarrow{G_q(R)} G_L(R), \quad G_q G_\eta = \text{id}. \quad (I7)$$

They preserve τ and supported e as distinct elements. The exact kernel of $G_q(R)$ consists of equal amplitudes and equal suprema of zero-fibre ideals:

$$(r, I) \sim (s, J) \iff r = s \text{ and } \bigvee I = \bigvee J. \quad (I8)$$

Proof. A nonzero amplitude in $G_L(R)$ has full support and maps to full support because $f(1_L) = 1_M$; hence (I6) lands in the exact target carrier. Finite join and meet preservation establish the semiring identities coordinate by coordinate, including zero-amplitude cancellation. Formula (I7) follows from $q\eta = \text{id}_L$, and (I8) is equality of the two coordinates of the image. The bottom and top of the nontrivial source and target lattices differ; their images under these bounded maps are again bottom and top, proving the exact zero statements. $\square$

Thus an ordinary branch which fails zero-fibre join continuity is a fully explicit frame point **upstairs**, and the arithmetic map fails to descend through a proved frame quotient **downstairs**. The original support object and all its algebraic branches remain related by the maps (I2)–(I8).

The two available lifts have an exact comparison, which prevents a split algebra map from being mistaken for a split frame map. Given an ordinary $\epsilon : L \rightarrow \mathbb{B}$, both

$$\epsilon q : \text{Idl}(L) \rightarrow \mathbb{B}, \quad \widehat{\epsilon} : \text{Idl}(L) \rightarrow \mathbb{B} \quad (I9)$$

are bounded-lattice maps and agree on every principal ideal. The first descends through q by definition; the second preserves arbitrary joins by Theorem I2. They coincide on all ideals exactly when

ϵ preserves arbitrary joins. The forward implication follows because $\epsilon q = \widehat{\epsilon}$ and surjectivity of the frame map q force its factor ϵ to preserve joins; the converse is (I4). This also yields two amplitude-fixed Split-Zero branch maps out of $G_{\text{Idl}(L)}(R)$, with exactly the same criterion for equality.

There is a further exact localic structure. The map $j = \eta q$ is a nucleus on $\text{Idl}(L)$: it is monotone, $I \subseteq j(I)$ because each $i \in I$ lies below $\bigvee I$, it satisfies $j^2 = j$ because $q\eta = \text{id}_L$, and it preserves finite meets because both q and η do. Its fixed ideals are exactly the principal ideals. Their inherited meet and nucleus-applied join

$$\bigvee_{\alpha}^{\text{fix}} \downarrow \lambda_{\alpha} = j \left(\bigvee_{\alpha}^{\text{Idl}(L)} \downarrow \lambda_{\alpha} \right) = \downarrow \left(\bigvee_{\alpha}^L \lambda_{\alpha} \right) \quad (\text{I10})$$

identify the fixed-point frame exactly with the original L . This proves the quotient/sub-locale correspondence at the level of operations, while retaining the precise failure of η to preserve the ambient ideal-frame joins.

4.3 A fully explicit nontrivial frame with algebraic branches and no frame points

Let L be the Lebesgue probability measure algebra of $[0, 1]$: measurable subsets modulo null symmetric difference. Write $[A]$ for the class of A , keep the original measure $\mu(A)$, and use union, intersection and complement for finite Boolean operations. These operations are well defined modulo null sets, since changing their inputs changes any output only within the union of those null differences. The exact order is $[A] \leq [C]$ if and only if $\mu(A \setminus C) = 0$.

Lemma M1 — completeness and atomlessness. This L is a complete Boolean algebra, hence a frame, and has no atoms.

Proof. For an arbitrary family $([A_i])_{i \in I}$, let

$$b = \sup \{ \mu(A_{i_1} \cup \dots \cup A_{i_n}) : n \geq 0 \}.$$

Choose finite unions F_n with measures tending to b , and put $B = \bigcup_n F_n$. Replacing F_n by the union of the first n such finite unions makes them increasing without changing their defining kind, so $\mu(B) = b$. For each original A_i , the finite-union supremum bounds $\mu(F_n \cup A_i) \leq b$, and continuity from below gives $\mu(B \cup A_i) \leq b = \mu(B)$. Thus $A_i \setminus B$ is null. If $[C]$ is another upper bound, every one of the countably many constituent sets in B is contained in C modulo a null set; hence $B \setminus C$ is null. This proves that $[B]$ is the supremum. Infima follow by complements.

In a complete Boolean algebra finite meets distribute over arbitrary joins: if $v = \bigvee_i (a \wedge b_i)$, then every $b_i \leq \neg a \vee v$; taking joins and intersecting with a gives $a \wedge \bigvee_i b_i \leq v$. The reverse inequality is immediate. Thus L is a frame.

For a nonzero $[A]$, the function $F(t) = \mu(A \cap [0, t])$ is continuous because $|F(t) - F(u)| \leq |t - u|$. It takes the values 0 and $\mu(A) > 0$ at the endpoints. There is therefore t with $F(t) = \mu(A)/2$. The class of $A \cap [0, t]$ is strictly between 0 and $[A]$, so no nonzero class is an atom. $\square$

Theorem M2 — exact frame points of a complete Boolean algebra. If B is any complete Boolean algebra, its frame points correspond bijectively to its atoms:

$$\text{Hom}_{\text{Frm}}(B, \mathbb{B}) \cong \text{At}(B), \quad a \mapsto \epsilon_a(b) = \mathbf{1}_{\{a \leq b\}}. \quad (\text{M1})$$

Proof. An atom below $\bigvee_i b_i$ lies below some b_i : otherwise every $a \wedge b_i = 0$, while frame distributivity gives $a = a \wedge \bigvee_i b_i = \bigvee_i (a \wedge b_i) = 0$, a contradiction. Thus ϵ_a preserves arbitrary joins. It preserves finite meets, top and bottom directly.

Conversely, a frame map $\epsilon : B \rightarrow \mathbb{B}$ into the two-element Boolean algebra preserves complements, since complements are characterized by their finite meet 0 and join 1. By De Morgan it therefore also

preserves arbitrary meets. Put $a = \bigwedge \{b : \epsilon(b) = 1\}$. Its image is 1, so $a \neq 0$. If $0 < c \leq a$ and $\epsilon(c) = 0$, then $\epsilon(\neg c) = 1$, forcing $a \leq \neg c$ and hence $c = 0$, a contradiction. Consequently $\epsilon(c) = 1$, which by definition of a implies $a \leq c$. Thus a is an atom. Every element of the filter contains a ; conversely if $a \leq b$, then $\epsilon(b) = 1$. This proves the inverse formulas and their uniqueness. $\square$

In particular the measure algebra of Lemma M1 has **no frame points**.

It nevertheless has ordinary lattice characters. Here is an explicit witnessed failure of the required continuity. Let $D_n = [0, 2^{-n}]$ as original measurable subsets of $[0, 1]$; their classes are nonzero and any finite intersection is the smallest selected D_n . The filter generated by these classes is proper. Order proper filters containing it by inclusion; the union of every chain is proper, so Zorn's lemma gives a maximal proper filter F . In a Boolean algebra, maximal proper filters contain exactly one of $b, \neg b$: if neither lay in F , adjoining b would produce 0, so some $f \in F$ satisfies $f \wedge b = 0$ and hence $f \leq \neg b$, which already puts $\neg b \in F$. Its characteristic function is therefore a bounded-lattice homomorphism $\epsilon_F : L \rightarrow \mathbb{B}$.

Every $\epsilon_F([D_n]) = 1$, while every $\epsilon_F([D_n^c]) = 0$. But

$$\bigvee_{n \geq 1} [D_n^c] = [[0, 1] \setminus \{0\}] = 1_L, \quad \epsilon_F(1_L) = 1 \neq \bigvee_{n \geq 1} 0 = 0. \quad (\text{M2})$$

Thus (B2) gives an actual surjective semiring branch $\beta_{\epsilon_F} : G_L(R) \rightarrow G(R)$ preserving amplitudes, even though the frame has no point. Formula (M2) gives its exact failed zero-fibre join equation. The point $\widehat{\epsilon}_F$ of $\text{Idl}(L)$ does not descend through q . More explicitly, set

$$J = \bigvee_{n \geq 1}^{\text{Idl}(L)} \downarrow [D_n^c].$$

Then $q(J) = 1_L = q(L)$, but $\widehat{\epsilon}_F(J) = 0$ and $\widehat{\epsilon}_F(L) = 1$. This is an explicit pair in the kernel congruence of q distinguished by the upstairs point; it proves the obstruction to descent at an exact source/target pair.

In fact **every** ordinary branch on this measure algebra fails already countable-join preservation. For its ultrafilter F , start at $a_0 = 1$. Split a representative of a_n into two equal-measure pieces by the continuous cumulative-measure construction in Lemma M1. Exactly one of their classes belongs to F ; call it a_{n+1} . The resulting decreasing sequence satisfies $a_n \in F$ and $\mu(a_n) = 2^{-n}$. Its infimum is 0 by continuity of measure. Consequently $b_n = \neg a_n$ increases to 1 and each $\epsilon_F(b_n) = 0$, again contradicting countable-join preservation. The ideal $I = \bigcup_n \downarrow b_n$ then satisfies $q(I) = q(L) = 1$, $\widehat{\epsilon}_F(I) = 0$, and $\widehat{\epsilon}_F(L) = 1$. This gives a kernel-congruence witness for each individual algebraic branch, not merely for the displayed dyadic choice.

4.4 Atomic observation and the exact support data it removes

For a complete Boolean algebra B , put

$$b_{\text{at}} = \bigvee_{a \in \text{At}(B)} a, \quad b_{\text{na}} = \neg b_{\text{at}}.$$

The frame-point evaluation map is

$$\text{ev} : B \rightarrow \mathcal{P}(\text{At}(B)), \quad x \mapsto \{a : a \leq x\}. \quad (\text{A1})$$

Theorem A1 — exact atomic quotient. This is a surjective complete Boolean-algebra homomorphism. Its kernel congruence is

$$\text{ev}(x) = \text{ev}(y) \iff x \wedge b_{\text{at}} = y \wedge b_{\text{at}}, \quad (\text{A2})$$

and its inverse image of the empty set is the full principal ideal $\downarrow b_{\text{na}}$. The complete decomposition retaining that ideal is

$$B \xrightarrow{\sim} (\downarrow b_{\text{at}}) \times (\downarrow b_{\text{na}}), \quad x \mapsto (x \wedge b_{\text{at}}, x \wedge b_{\text{na}}), \quad (\text{A3})$$

with inverse $(u, v) \mapsto u \vee v$ and respective local units $b_{\text{at}}, b_{\text{na}}$.

Proof. Each coordinate of (A1) is the frame point in (M1), so it preserves arbitrary joins and finite meets. Complements are preserved because an atom lies below exactly one of $x, \neg x$. For any subset S of atoms, the element $\bigvee_{a \in S} a$ maps exactly to S , by pairwise disjointness of distinct atoms and the arbitrary-join argument in (M1). Thus evaluation is surjective. Distributivity gives $x \wedge b_{\text{at}} = \bigvee_{a \in \text{At}(B)} (x \wedge a) = \bigvee_{a \leq x} a$, which proves (A2) and the empty-fibre claim. The two complementary components in (A3) reconstruct x by distributivity, and disjointness of $b_{\text{at}}, b_{\text{na}}$ proves both inverse identities and preservation of complete Boolean operations. $\square$

The complement on either local Boolean factor $\downarrow b$ in (A3) is explicitly $u \mapsto b \wedge \neg u$; its local top is b , not the global top of B unless $b = 1$.

This supplies an exact relation to the source's adelic measure-algebra support. In a finite measure-algebra realization, frame-point evaluation detects precisely its measure atoms; (A3) retains the entire nonatomic component rather than treating lack of points as lack of support. For a discrete countable measure with positive measure at each singleton, the singleton classes are the atoms and $b_{\text{at}} = 1$; evaluation is an isomorphism to the power set of the original places. For the Lebesgue example, $b_{\text{at}} = 0, b_{\text{na}} = 1$, and every frame-point evaluation is absent while the entire nontrivial Boolean carrier persists.

When applying $G_{\text{ev}}(R)$, retain the exceptional case: if there are no atoms, $\mathcal{P}(\text{At}(B))$ is the one-element lattice, so the exact carrier $G_{\mathcal{P}(\emptyset)}(R)$ is isomorphic to R , and this quotient identifies τ_B and e_B with ordinary zero. It is the explicit ring-reflection map on this observation, not evidence that $\tau_B = e_B$ in the original carrier. The split comparison (I7) has nontrivial lattices on both sides and preserves them as distinct; it remains available when the point-observation quotient collapses them.

Concretely, $\Psi : G_{\{*\}}(R) \rightarrow R, (r, *) \mapsto r$, is bijective and preserves both operations because the unique support coordinate is fixed. In the atomless case its composition is the exact identity

$$\Psi \circ G_{\text{ev}}(R) = p_B : G_B(R) \rightarrow R, \quad p_B(r, \lambda) = r. \quad (\text{A4})$$

4.5 The exact operator-algebraic branch map

The core manuscript's L^∞ realization permits an additional precise bridge to state spaces. Work with the same finite probability measure space, complete measure algebra B , and the complex algebra $M = L^\infty(\Omega, \mu; \mathbb{C})$ equipped with essential-supremum norm and complex conjugation. The original measure and every atomic mass are retained.

Theorem O1 — all Boolean branches extend exactly to multiplicative states. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(B, \mathbb{B}) \cong \{\chi : M \rightarrow \mathbb{C} : \chi \text{ is a unital complex-linear } * \text{-homomorphism}\}. \quad (\text{O1})$$

The correspondence is specified on every projection by

$$\chi \epsilon(\mathbf{1}_A) = \epsilon([A]), \quad (\text{O2})$$

and on arbitrary $f \in M$ by the uniform extension of this finite formula:

$$\chi_\epsilon \left(\sum_{j=1}^n c_j \mathbf{1}_{A_j} \right) = \sum_{j=1}^n c_j \epsilon([A_j]), \quad (O3)$$

where the A_j form the specified measurable partition of Ω modulo null sets. Every such χ is positive and has norm 1.

Proof. Exactly one member of a finite partition has ϵ -value 1: finite-join preservation gives at least one, and pairwise disjointness with meet preservation gives at most one. Thus (O3) selects one of the original values c_j . Refining two partitions by all intersections shows the formula is independent of the simple-function presentation. The unique selected intersection gives additivity and multiplicativity for two simple functions, while conjugating their values gives star preservation. The selected value has modulus at most the essential supremum of the simple function, so the map is contractive.

Every bounded measurable complex function has simple functions tending to it in essential-supremum norm: partition a square containing its essential range into finitely many squares of diameter at most 2^{-n} and use one original value in each nonempty square. The resulting evaluations are Cauchy by contractivity. Their limit defines a unique continuous extension independent of the approximating sequence. Addition, multiplication and conjugation commute with these uniform limits, proving the required homomorphism identities. Positivity follows from $f = g^*g$ with $g = \sqrt{f}$ for every nonnegative measurable f ; hence $\chi(f) = |\chi(g)|^2 \geq 0$. Contractivity and $\chi(1) = 1$ prove norm 1.

Conversely a unital star homomorphism sends projections to real idempotents 0 or 1. Since $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B$ and $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A \cap B}$, its projection values form the required Boolean lattice map. It is contractive because $\|f\|_\infty^2 \mathbf{1} - f^*f$ is the square of a bounded measurable function, so positivity gives $|\chi(f)|^2 \leq \|f\|_\infty^2$. It agrees with (O3) on simple functions, and continuity forces equality on all of M . This proves both inverse constructions. $\square$

Theorem O2 — normal branches, exact atomic masses, and failure of normality. For the correspondence (O1), the following are equivalent:

1. χ preserves suprema of bounded increasing nets of nonnegative elements of M (normality).
2. ϵ is a frame point of the complete measure algebra.
3. There is a measure atom a such that

$$\chi(f) = \frac{1}{\mu(a)} \int_a f d\mu \quad (f \in M). \quad (O4)$$

In particular the atomless Lebesgue example has multiplicative states for all of its ordinary branches, and none is normal; its exact countable failure is exhibited by (M2).

All bounded increasing nets used here have a supremum in the specified M . An explicit proof for nonnegative nets retains the finite measure: put $b = \sup_i \int_\Omega f_i d\mu$. Choose indices with integrals tending to b , and use directedness to replace them by an increasing dominating sequence. Choose representatives outside one common null set for the countably many sequence comparisons, so this selected sequence is pointwise increasing there. Its pointwise supremum g has integral b . For every fixed original index i , each finite maximum with a sequence member lies below a net member, so $\int_\Omega \max(g, f_i) d\mu \leq b = \int_\Omega g d\mu$ by monotone convergence. Thus $f_i \leq g$ almost everywhere. Every upper bound of the original net bounds the selected sequence and hence g , proving that g is the required least upper bound.

Proof. The supremum in M of the finite-union projections from an arbitrary family $([A_i])$ is the projection $\mathbf{1}_{\bigvee_i [A_i]}$: Lemma M1's countable essential union is an upper bound, and its increasing finite selected unions show it is the least upper bound also among arbitrary measurable nonnegative functions. Normality applied to this increasing net therefore gives preservation of its full projection

join. Together with (O2), this is precisely frame continuity of ϵ , proving $1 \Rightarrow 2$. By Theorem M2, a frame point corresponds to a unique atom a .

Every bounded measurable f is essentially constant on a . To see this, use the finite measurable partitions of its essential range from the proof of Theorem O1. Each partition has exactly one cell whose intersection with a is nonzero, and by atomhood that intersection equals a modulo null sets. Its diameter tends to 0, so the simple approximants restrict to constants c_n on a , with $|c_n - c_m|$ bounded by their uniform difference. Their limit c is the almost-everywhere value of f on a . Formula (O3) selects exactly these constants and gives $\chi(f) = c$; integration over the original atom gives precisely (O4), with the factor $1/\mu(a)$ retained. Thus $2 \Rightarrow 3$.

Finally let a bounded increasing net $f_i \geq 0$ have supremum f in M , and let c_i be its respective constants on a . These constants increase; put $c = \sup_i c_i$. Every f_i is bounded above by the function equal to c on a and to f off a ; hence the least-upper-bound property of f forces its constant on a to be at most c . Conversely each $f_i \leq f$ forces $c_i \leq f|_a$, and taking their supremum gives the reverse inequality. Thus $\chi(f) = \sup_i \chi(f_i)$. This proves $3 \Rightarrow 1$, including every original coefficient and atomic mass. $\square$

Equations (B1),(O1),(O2) form the promised typed connection between algebraic Split-Zero branches and operator-algebraic states. They identify exactly which branches can be normal observations of the measured support.

5 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R -modules V_l , with linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f) , where $\alpha : L \rightarrow L'$ preserves bottom and joins with R -linear fibre maps

$$f_l : V_l \rightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l. \quad (\text{D1})$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)} f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 5.1 (Full support-index reconstruction). *The diagram category just defined is equivalent to the category of $G(R)$ -semimodules and their $G(R)$ -linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx). \quad (\text{D2})$$

Its global zero is $(\perp, 0)$. On morphisms it is $\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$.

The inverse sends a semimodule M to

$$\begin{aligned} L_M &= eM, \quad l \vee k = l + k, \quad \perp = 0_M, \\ V_l &= \{m \in M : em = l\}, \quad \rho_{lk}(m) = m + k \quad (l + k = k). \end{aligned} \quad (\text{D3})$$

The natural comparison maps are

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \mapsto m, \quad m \mapsto (em, m), \quad (\text{D4})$$

and the label-and-fibre isomorphism

$$L \longrightarrow e\mathcal{T}(L, V), \quad l \mapsto (l, 0), \quad x \in V_l \mapsto (l, x). \quad (\text{D5})$$

Proof. For (D2), threefold addition transports both parenthesizations to $l \vee k \vee j$. Linearity and composition of the ρ 's make both amplitudes $\rho_l x + \rho_k y + \rho_j z$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l . Their addition law at a vector in fibre l is the ordinary R -module addition in V_l ; in particular $0_R^\bullet$ gives $(l, 0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha, f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2 = e$, $e + e = e$, and $1 + e = 1$ in $G(R)$. Thus every $l \in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m \in V_l$,

$$m + l = (1 + e)m = m, \quad m + (-1_R)^\bullet m = em = l.$$

The inherited addition in V_l , with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r \in R$, $e(r^\bullet m) = em = l$, since $er^\bullet = e$. Its scalar action satisfies every R -module law inherited from the semimodule, and $0_R m = em = l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M .

If $l + k = k$, then $e(m + k) = l + k = k$, so ρ_{lk} has the correct codomain. Its zero law is $l + k = k$; its additivity uses $k + k = k$. Also $r^\bullet k = k$, proving R -linearity. Identity follows from $m + l = m$, and composition from $k + j = j$. A split-linear map $F : M \rightarrow N$ takes eM to eN , with $\alpha(l) = F(l)$. It preserves joins and bottom, restricts to R -linear maps on each fibre, and $F(m + k) = F(m) + F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

because $e(m + n) = l \vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ . It is a split-linear isomorphism. In the reconstructed total, $e(l, x) = (l, 0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l, x) + (k, 0) = (k, \rho_{lk}x)$, recovering the given ρ . For a morphism (α, f) , (D5) takes $(l, 0)$ to $(\alpha(l), 0)$; for F , (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. $\square$

5.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \rightarrow \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \quad (\text{D6})$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N , let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \rightarrow D$ of R -module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \text{im } d_C^{\text{prev}}$ inside Z_C , and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \quad [z] \mapsto [z] \quad (\text{D7})$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \rightarrow \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C ; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \rightarrow N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \quad (\text{D8})$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear map satisfying the respective condition has its unique factorization through the displayed subsemimodule. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

5.2 The earlier reconstruction applied to every original source level

We now evaluate (D2), (D6), and (D7) on the original theta source, including the finite polynomial source levels used in (TC2) and (TB.24)–(TB.26). Let $\mathcal{W}$ contain every complex linear subspace W of the original V , ordered by inclusion; no closure or invariance under D is imposed for this cochain construction. Fix $n \geq 1$ and $I_n = \{0, \dots, n-1\}$. Its nonbottom labels are

$$\lambda = (W, S; A), \quad W \in \mathcal{W}, \quad \emptyset \neq S \subseteq \{+, -\}, \quad A \subseteq I_n. \quad (\text{D9})$$

Adjoin one distinct bottom $\perp$, with zero fibre. The join of two nonbottom labels is $(W + W', S \cup S'; A \cup A')$. The bottom is strictly below even $(0, S; \emptyset)$. In degrees zero and one put

$$\begin{aligned} C_{W,S;A}^{X,0} &= \left(\bigoplus_{+ \in S} W \oplus \bigoplus_{- \in S} \widehat{W} \right)^A, & C_{W,S;A}^{X,1} &= \mathcal{B}_X^A, \\ d_{W,S;A}(\phi_i, \psi_i)_i &= (\Theta \phi_i - \Theta \widehat{\psi}_i)_i. \end{aligned} \quad (\text{D10})$$

Here $X = U$ or Ω as in (MCF41); missing legs use typed zeros. The transition includes old subspaces and coordinates and inserts typed zeros in newly present legs or coefficients. Fourier linearity gives $\widehat{W} \subseteq \widehat{W + W'}$; hence all transitions have their stated domains. Substitution into the differential proves the cochain identity, and inclusions compose literally. Thus (D2) applies in each degree to this actual diagram. At every nonbottom label with $A = \emptyset$ the amplitude fibre is zero, but its element is $(W, S; \emptyset, 0)$, while the global zero is $(\perp, 0)$. Applying (D2) proves they remain distinct and gives their exact joins. No coefficient-mask colimit is taken.

For each W , put $Q_{W,X} = \mathcal{B}_X / \Theta W$. The full cohomology and the comparison to the full source are

$$H^0(C_{W,S;A}^X) = \begin{cases} W^A, & S = \{+, -\}, \\ 0, & |S| = 1, \end{cases} \quad H^1(C_{W,S;A}^X) = Q_{W,X}^A, \quad (D11)$$

$$0 \longrightarrow (V/W)^A \xrightarrow{\eta_W^A} Q_{W,X}^A \xrightarrow{\pi_W^A} (T_X Q)^A \longrightarrow 0, \quad \eta_W[v] = [\Theta v]_W.$$

Indeed the source differential has image $(\Theta W)^A$. A single-leg differential is injective. At a joint slot the inverse coordinates $u = \phi - \hat{\psi}$, $v = (\phi + \hat{\psi})/2$, $\phi = v + u/2$, $\psi = \hat{v} - \hat{u}/2$ identify its kernel with $v \in W$. For the last sequence, injectivity of Θ makes η_W injective; a class killed by π_W is represented by Θv for some $v \in V$, and every element of $T_X Q$ has a representative in $\mathcal{B}_X$. This proves exactness, coordinatewise also for empty A . The coequalizer (D6) uses precisely $(\Theta W)^A$, and (D7) identifies the kernel of the full-source comparison with $(V/W)^A$ through this same η_W^A . This is the all-label fibre kernel of (D8); with unchanged support label, the global-zero categorical kernel contains only its bottom fibre. Both maps are retained.

For $W \subseteq W'$ the source transition therefore induces

$$0 \longrightarrow (W'/W)^A \xrightarrow{[v] \mapsto [\Theta v]_W} Q_{W,X}^A \longrightarrow Q_{W',X}^A \longrightarrow 0. \quad (D12)$$

The proof uses the same kernel computation with W' in place of V . Thus source refinement has its actual nonzero comparison kernel before any full-source quotient is applied.

The critical-strip enlargement also has its original typed map:

$$0 \longrightarrow C_{W,S;A}^U \longrightarrow C_{W,S;A}^\Omega \longrightarrow (T_L Q)^A[-1] \longrightarrow 0, \quad L = \{\Re s = 1/2\}. \quad (D13)$$

The first map is identity on source legs and inclusion on the central term. Its degree-zero quotient is zero. Its degree-one quotient is $\mathcal{B}_\Omega^A / \mathcal{B}_U^A$, since the same ΘW lies in both. The quotient map q identifies this space with $(T_\Omega Q / T_U Q)^A$. For any annihilator $p = p_U p_L$ with its factors rooted in the disjoint sets U, L , choose $a p_U + b p_L = 1$. The maps $b(D)p_L(D)$ and $a(D)p_U(D)$ split each killed class into its U and L parts. Their intersection is zero by the same Bezout identity; uniqueness makes the splitting independent of the annihilator. Thus the quotient is exactly $(T_L Q)^A$, with every multiplicity retained. All three complexes keep the same outer and coefficient labels.

The original arithmetic operator has the following additional type on this larger source diagram. If $DW \subseteq W'$, then

$$D_{W,W'} : C_{W,S;A}^X \longrightarrow C_{W',S;A}^X, \quad (\phi_i, \psi_i)_i \longmapsto (D\phi_i, (1-D)\psi_i)_i, \quad (F_i)_i \longmapsto (DF_i)_i. \quad (D14)$$

Fourier gives $(1-D)\widehat{W} = \widehat{DW} \subseteq \widehat{W'}$, and $\Theta D = D\Theta$ proves the cochain identity. Moreover $D\mathcal{B}_X \subseteq \mathcal{B}_X$ because D commutes with each polynomial annihilator of a quotient class. Thus every displayed map has its original domain. For D -invariant W , take $W' = W$ to recover precisely (MCF34). For the actual finite source levels $W_m = \text{span}\{\phi_*, D\phi_*, \dots, D^m\phi_*\}$, one instead has $DW_m \subseteq W_{m+1}$ and must use (D14) with that codomain. Linear independence of these vectors follows from $H_{P(D)\phi_*} = P$, so $\dim(W_{m+1}/W_m) = 1$ exactly. Dilation likewise gives a cochain map from W to $\mathcal{R}_a W$, using $\mathcal{R}_a$ on the plus source and central term and $a\mathcal{R}_{1/a}$ on the minus source. The Fourier identity proves its stated minus codomain $\widehat{\mathcal{R}_a W}$.

6 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R , while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m , including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

6.1 Source spaces and fixed arithmetic factors

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x.$$

Use the original Fourier, theta and Mellin transforms

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{A1})$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 6.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \rightarrow \mathcal{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \text{Re } s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{A2})$$

For the exact original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \quad H_{P(D)\phi_*} = P, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP. \quad (\text{A3})$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\widehat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) \phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k) |\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\text{Re } s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) =$

$O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2}\{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. $\square$

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Let $Q = \mathcal{B}/\Theta V$ and $q : \mathcal{B} \rightarrow Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \quad (\text{A4})$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q . This follows from injectivity of theta and $\text{im } d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-} , with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

6.2 The exact operator range, including its inverse

Proposition 6.2 (Source and target Euler inverses). *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Then

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)S_\rho = 1, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{A5})$$

The operator $D - \rho$ is injective. For $0 < \text{Re } \rho < 1$, the same formula, extended evenly, gives

$$(D - \rho)V = \ker(H(\cdot)(\rho)). \quad (\text{A6})$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \text{Re } \rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N - \sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D - \rho)S_\rho \phi = \phi$ on $\mathbb{R}$ gives $(1 - \rho) \int S_\rho \phi = \int \phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V . $\square$

For any polynomial $h(t) = \prod_i (t - \rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H), \quad E_h = \mathbb{C}[t]/(h). \quad (\text{A7})$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho y} / k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \quad (\text{A8})$$

Theorem 6.3 (Finite arithmetic kernel, quotient and packet section). *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{A9})$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu , and set

$$R_{\text{ref}} u = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M} F, \quad \sigma = q R_{\text{ref}}.$$

Then

$$\mathcal{M} R_{\text{ref}} u = (g/h) R_Z(u), \quad J_Z R_{\text{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \quad (\text{A10})$$

The projector is $\sigma \bar{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h \Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\text{ref}} = j_h v \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ . Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. $\square$

6.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E , define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$D R_{\text{ref}} - R_{\text{ref}} A = f_0 \ell, \quad f_0 = \Theta\phi_*. \quad (\text{A11})$$

Indeed $t R_Z(u) - R_Z(tu)$ is divisible by the monic h , and its degree is at most d ; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathcal{B}_Z = \Theta V + R_{\text{ref}} E$ fits into the actual extension

$$0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J_Z} E \rightarrow 0. \quad (\text{A12})$$

It splits complex-linearly through R_{ref} ; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_* \ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathcal{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\simeq} [V \rightarrow \mathcal{B}_Z] \longrightarrow [V \rightarrow \mathcal{B}]. \quad (\text{A13})$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

6.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below uses these specified metrics. Put $\phi_j = D^j \phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j \phi_j$. For $F_m = \Theta \Phi_m : \mathbb{C}^{m+1} \rightarrow \mathcal{H}$, define

$$\begin{aligned} H_m &= F_m^* F_m, & C_m &= F_m^* R_{\text{ref}}, & B_m &= -H_m^{-1} C_m, \\ \Pi_m &= F_m H_m^{-1} F_m^*, & R_m &= R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{aligned} \quad (\text{A14})$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \rightarrow \mathbb{C}^{m+1}$,

$$\begin{aligned} (R_{\text{ref}} + F_m B)^* (R_{\text{ref}} + F_m B) &= G_m + (B - B_m)^* H_m (B - B_m), \\ G_m &= R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{aligned} \quad (\text{A15})$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0 \ell + S_m B_m - I_m B_m A, \quad D R_m - R_m A = F_{m+1} K_m. \quad (\text{A16})$$

Integration by parts in dx , with the source boundary term $[x \bar{F} H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \quad (\text{A17})$$

In particular rank $W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \bar{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy-Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \quad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \quad (\text{A18})$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{A19})$$

The next sections compute the actual arithmetic moment problem behind G_m , sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

7 Certified moments from the original arithmetic theta source

The moment matrices in the arithmetic kernel formulas can be calculated directly from the specified theta source. This section gives an exact incomplete-gamma series, proves an explicit bound for the entire omitted double tail, and applies directed interval arithmetic to the retained terms. The construction retains the original factor $g = 2\xi$, the measure dx , the Euler operator $D = -x\partial_x$, and every polynomial coefficient. No information about the location of a zeta zero enters this calculation.

7.1 Source polynomials and the exact reflection map

Keep

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_j = D^j \Theta \phi_*, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx). \quad (\text{AM.1})$$

Introduce polynomials with their original integer coefficients by

$$P_0(z) = 4z^2 - 6z, \quad P_{j+1}(z) = 2z(P_j(z) - P'_j(z)), \quad Q_j(z) = \sum_{k=0}^j (-1)^k \binom{j}{k} P_k(z). \quad (\text{AM.2})$$

Write p_{ja} and q_{ja} for the coefficients of z^a . Both polynomials have degree $d_j = j + 2$, with leading coefficients 2^{j+2} and $(-1)^j 2^{j+2}$, respectively, and both have zero constant coefficient. The formula defining Q_j is an explicit change of polynomial coordinates; neither family is discarded.

Define the linear involution on the actual Hilbert observation by

$$\mathcal{T} : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}_{>0}, dx), \quad (\mathcal{T}F)(x) = x^{-1}F(1/x). \quad (\text{AM.3})$$

It is unitary and self-adjoint, satisfies $\mathcal{T}^2 = 1$, and, on the strong Schwartz space $\mathcal{B}$,

$$D\mathcal{T} = \mathcal{T}(1 - D), \quad \mathcal{T}f_0 = f_0, \quad \mathcal{T}f_j = (1 - D)^j f_0. \quad (\text{AM.4})$$

Consequently, the complete source series and its reflected series are

$$f_j(x) = 2 \sum_{n \geq 1} P_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \tilde{f}_j(x) := (1 - D)^j f_0(x) = 2 \sum_{n \geq 1} Q_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad (\text{AM.5})$$

and $f_j(1/x) = x \tilde{f}_j(x)$.

Proof. For $z = \pi x^2$, applying $-x\partial_x$ to $P(z)e^{-z}$ gives $2z(P - P')e^{-z}$. This proves the recurrence and (AM.5). The leading coefficients and zero constant coefficients follow by induction. Under the Fourier convention $\widehat{\phi}(\xi) = \int \phi(x) e^{-2\pi i x \xi} dx$, differentiating the Fourier transform of the Gaussian gives

$$\widehat{x^2 e^{-\pi x^2}}(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}}(\xi) = \left(\xi^4 - \frac{3\xi^2}{\pi} + \frac{3}{4\pi^2} \right) e^{-\pi \xi^2}.$$

Multiplication by the exact coefficients $4\pi^2, -6\pi$ therefore gives $\widehat{\phi}_* = \phi_*$. Its value at zero is zero, and its integral is $\widehat{\phi}_*(0) = 0$. The theta-Fourier identity already proved for the source V then gives $\mathcal{T}f_0 = f_0$.

Substitution $u = 1/x$ gives $\int |x^{-1}F(1/x)|^2 dx = \int |F(u)|^2 du$, so $\mathcal{T}$ is unitary. It is an involution and hence self-adjoint. Direct differentiation proves $D\mathcal{T} = \mathcal{T}(1 - D)$, including the constant 1 arising from the original factor x^{-1} . Induction proves the other identities in (AM.4). Local uniform convergence of each differentiated Gaussian series follows on $x \geq \delta > 0$ by a polynomial times Gaussian majorant. The series gives decay of all Euler derivatives at infinity, and its exact reflected expression gives the corresponding decay at zero. Thus these operations apply to the original functions in $\mathcal{B}$. $\square$

7.2 An absolutely convergent double series for every entry

For integers $i, j \geq 0$, retain the raw moment entry

$$H_{ij} = \langle D^i f_0, D^j f_0 \rangle_{L^2(dx)} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j |2\xi(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{AM.6})$$

The equality is the original Mellin Plancherel identity. In particular H_{ij} is real because the source functions are real. For $L \geq 0$ and $\lambda > 0$, set

$$I_L(\lambda) = \int_1^\infty x^{2L} e^{-\lambda x^2} dx = \frac{\Gamma(L + 1/2, \lambda)}{2\lambda^{L+1/2}}, \quad \Gamma(a, \lambda) = \int_\lambda^\infty u^{a-1} e^{-u} du. \quad (\text{AM.7})$$

Theorem 7.1 (Exact original arithmetic moment series). *With $\lambda_{nm} = \pi(n^2 + m^2)$, the following double series converges absolutely, including after the finite coefficient sums are replaced by their absolute values:*

$$H_{ij} = 2 \sum_{n,m \geq 1} \sum_{a=1}^{d_i} \sum_{b=1}^{d_j} (p_{ia} p_{jb} + q_{ia} q_{jb}) \pi^{a+b} n^{2a} m^{2b} \cdot \frac{\Gamma(a + b + 1/2, \lambda_{nm})}{\lambda_{nm}^{a+b+1/2}}. \quad (\text{AM.8})$$

Equivalently, it is the Gram integral on the positive half-domain

$$H_{ij} = \int_1^\infty (f_i(x) f_j(x) + \tilde{f}_i(x) \tilde{f}_j(x)) dx. \quad (\text{AM.9})$$

Thus the precise map to the positive Gram presentation is

$$\mathbb{C}[s] \longrightarrow L^2([1, \infty), dx)^{\oplus 2}, \quad P \longmapsto (P(D)f_0, P(1-D)f_0)|_{[1, \infty)}. \quad (\text{AM.10})$$

It preserves the original polynomial inner product (AM.6) and is injective. The signs in (AM.8) are retained; the positivity is the Gram positivity of this specified map.

Proof. Split the integral in (AM.6) at $x = 1$. On $(0, 1)$, set $x = 1/u$. The two factors u supplied by (AM.5) cancel the Jacobian u^{-2} exactly. This gives (AM.9). For absolute convergence, write

$$C_j = \sum_{a=1}^{d_j} |p_{ja}| \pi^a, \quad E_j = \sum_{a=1}^{d_j} |q_{ja}| \pi^a. \quad (\text{AM.11})$$

For $n, x \geq 1$, the polynomial bounds are

$$|P_j(\pi n^2 x^2)| \leq C_j n^{2d_j} x^{2d_j}, \quad |Q_j(\pi n^2 x^2)| \leq E_j n^{2d_j} x^{2d_j}.$$

Since $n^2 + m^2 \geq 2$ and $x^2 \geq 1$,

$$(n^2 + m^2)x^2 \geq \frac{1}{2}(n^2 + m^2) + x^2.$$

The absolute integrand series is therefore bounded by

$$4(C_i C_j + E_i E_j) x^{2(d_i + d_j)} e^{-\pi x^2} \left(\sum_{n \geq 1} n^{2d_i} e^{-\pi n^2/2} \right) \left(\sum_{m \geq 1} m^{2d_j} e^{-\pi m^2/2} \right),$$

which is integrable on $[1, \infty)$. This proves all required sum-integral interchanges. Substitution $u = \lambda x^2$ gives (AM.7). The two theta factors contribute 4, and (AM.7) contributes $1/2$, leaving exactly the factor 2 in (AM.8).

For complex polynomial coefficients, conjugate the coefficients of the first factor in the Gram integral. Equation (AM.9) then proves the inner-product assertion in (AM.10). If a nonzero polynomial P were in its kernel, its first component would vanish on $[1, \infty)$. Put $U(z) = \sum_j P_j(z)c_j$ when $P(s) = \sum_j c_j s^j$. The distinct degrees and nonzero leading coefficients of the P_j imply $U \neq 0$. Multiplying the vanishing theta series by $e^{\pi x^2}/2$ gives

$$U(\pi x^2) = - \sum_{n \geq 2} U(\pi n^2 x^2) e^{-\pi(n^2-1)x^2}.$$

The right side decays exponentially times a fixed polynomial as $x \rightarrow \infty$, by the same Gaussian bound. A nonzero polynomial on the left cannot have that decay. This contradiction proves injectivity. $\square$

7.3 A complete finite truncation certificate

Let $H_{ij}^{[N]}$ be (AM.8) with both theta indices restricted to $1 \leq n, m \leq N$, keeping all coefficients p_{ia}, q_{ia} . For the calculation of this entry choose an integer $N \geq \max(d_i, d_j)$. Define, for $k = i, j$,

$$\begin{aligned} r_{k,N} &= \left(\frac{N+2}{N+1} \right)^{2d_k} e^{-\pi(2N+3)}, & T_{k,N} &= \frac{(N+1)^{2d_k}}{1 - r_{k,N}}, \\ A_{k,N} &= \sum_{n=1}^N n^{2d_k} e^{-\pi(n^2-1)} + T_{k,N} e^{-\pi((N+1)^2-1)}. \end{aligned} \quad (\text{AM.12})$$

All these numbers are specified positive real constants. The choice of N ensures $r_{k,N} < 1$, since

$$\log r_{k,N} < \frac{2d_k}{N+1} - \pi(2N+3) < 2 - 5\pi < 0.$$

Theorem 7.2 (Rigorous complete double-tail bound). *For every $i, j \geq 0$ and $N \geq \max(i, j) + 2$,*

$$\boxed{|H_{ij} - H_{ij}^{[N]}| \leq \mathcal{E}_{ij,N} := 4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) \cdot I_{d_i+d_j}(\pi((N+1)^2 + 1))}. \quad (\text{AM.13})$$

In particular the bound tends to zero as $N \rightarrow \infty$, for each fixed pair i, j , and provides an explicit finite certificate for every raw arithmetic moment.

Proof. For $x \geq 1$, the ratio of successive terms of $n^{2d_k} e^{-\pi n^2 x^2}$ is at most

$$\left(\frac{n+1}{n} \right)^{2d_k} e^{-\pi(2n+1)}.$$

The logarithm of this expression has derivative $-2d_k/(n(n+1)) - 2\pi < 0$ for positive real n . For $n \geq N+1$ every ratio is consequently at most $r_{k,N}$. Summing the geometric majorant yields

$$\sum_{n > N} n^{2d_k} e^{-\pi n^2 x^2} \leq T_{k,N} e^{-\pi(N+1)^2 x^2}, \quad \sum_{n \geq 1} n^{2d_k} e^{-\pi n^2 x^2} \leq A_{k,N} e^{-\pi x^2}. \quad (\text{AM.14})$$

For the second inequality, divide by $e^{-\pi x^2}$, bound each retained factor $e^{-\pi(n^2-1)x^2}$ at $x = 1$, and apply the first inequality to the tail.

Write $F_{k,N}$ for the finite P_k -series and $\tilde{F}_{k,N}$ for the finite Q_k -series. Equations (AM.11) and (AM.14) give

$$\begin{aligned} |f_k(x)|, |F_{k,N}(x)| &\leq 2C_k x^{2d_k} A_{k,N} e^{-\pi x^2}, \\ |f_k(x) - F_{k,N}(x)| &\leq 2C_k x^{2d_k} T_{k,N} e^{-\pi(N+1)^2 x^2}. \end{aligned}$$

The reflected inequalities have exactly E_k in place of C_k . In each product of (AM.9), use the exact decomposition $ab - a_N b_N = (a - a_N)b + a_N(b - b_N)$. The resulting bound on the absolute difference of integrands is

$$4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) x^{2(d_i + d_j)} e^{-\pi((N+1)^2 + 1)x^2}.$$

Its integral is exactly the right side of (AM.13). Finally, the $A_{k,N}$ remain bounded for fixed k , the $T_{k,N}$ grow at most polynomially, and the integral decays exponentially in $(N + 1)^2$. For completeness, with $L = d_i + d_j$ fixed and $\lambda \geq 2L$, the inequality $\log x \leq (x^2 - 1)/2$ for $x \geq 1$ gives

$$I_L(\lambda) \leq e^{-\lambda} \int_1^\infty e^{-(\lambda-L)(x^2-1)} dx \leq \frac{e^{-\lambda}}{2(\lambda-L)}.$$

Here the last step uses $x^2 - 1 \geq 2(x - 1)$. This proves the asserted limit with an additional elementary upper bound when desired. $\square$

The finite matrix $(H_{ij}^{[N]})_{0 \leq i, j \leq m}$ is itself positive definite for every $N \geq 1$. Indeed it is the Gram matrix of $(F_{j,N}, \tilde{F}_{j,N})$ on the same half-domain; the injectivity proof in (AM.10) applies verbatim to the finite sum. This fact does not assign an ordering to its difference from the full matrix. The relation to the full matrix is the entrywise enclosure (AM.13), whose signs and error bounds are explicit.

7.4 Exact adjoint, parity, and finite-boundary checks

On $\mathcal{B}$ in $L^2(dx)$, integration by parts gives $\langle DF, H \rangle = \langle F, (1 - D)H \rangle$. Therefore the complete moments satisfy

$$H_{i+1,j} + H_{i,j+1} = H_{ij}, \quad H_{ij} = \sum_{k=0}^i (-1)^k \binom{i}{k} H_{0,j+k}, \quad H_{01} = \frac{1}{2} H_{00}. \quad (\text{AM.15})$$

All boundary terms vanish because the original functions have rapid decay at both ends. More explicitly,

$$(Df_i)f_j + f_i(Df_j) - f_i f_j = -\partial_x(x f_i f_j),$$

whose integral on $(0, \infty)$ is zero. Moving each of the i copies of D in the first argument to the second gives the middle identity; symmetry gives the last one.

The finite sums retain an exact boundary correction at $x = 1$:

$$H_{i+1,j}^{[N]} + H_{i,j+1}^{[N]} - H_{ij}^{[N]} = F_{i,N}(1)F_{j,N}(1) - \tilde{F}_{i,N}(1)\tilde{F}_{j,N}(1). \quad (\text{AM.16})$$

To prove this, use $DF_{j,N} = F_{j+1,N}$ for the first component and $D\tilde{F}_{j,N} = \tilde{F}_{j,N} - \tilde{F}_{j+1,N}$ for the second. Applying the preceding integration-by-parts calculation on $[1, \infty)$ gives the first displayed endpoint product for the first component and its negative for the second. The endpoints at infinity vanish term by term. In particular the exact finite identity $H_{01}^{[N]} = H_{00}^{[N]}/2$ follows even before the omitted tail is enclosed, since $F_{0,N} = \tilde{F}_{0,N}$.

One may also retain the exact triangular coordinate map

$$B_{ja} = \begin{cases} \binom{j}{a} (-1/2)^{j-a}, & a \leq j, \\ 0, & a > j, \end{cases} \quad C = BHB^T. \quad (\text{AM.17})$$

This is the Gram matrix of $u_j = (D - 1/2)^j f_0$, with inverse coordinate matrix $(B^{-1})_{ja} = \binom{j}{a} (1/2)^{j-a}$. No original moment is lost. Since $\mathcal{T}u_j = (-1)^j u_j$, unitarity proves $C_{ij} = 0$ for odd $i + j$. The operator $D - 1/2$ is skew-symmetric on this displayed strong Schwartz domain, so for $i + j = 2q$,

$$C_{ij} = (-1)^{i+q} \|u_q\|^2, \quad \|u_q\|^2 = \int_{\mathbb{R}} t^{2q} |2\xi(1/2 + it)|^2 \frac{dt}{2\pi} > 0. \quad (\text{AM.18})$$

For the first equality move all i powers to the second argument, and compare with moving the q powers in $\langle u_q, u_q \rangle$. Mellin Plancherel gives the second equality. Strict positivity follows from the nonzero original function and the polynomial injectivity already proved.

7.5 Reproducible certified finite arithmetic data

The companion program `scripts/check_arithmetic_moments.py` implements the exact integer recurrence (AM.2), evaluates every retained term of (AM.8) by Arb directed ball arithmetic, and enlarges its radius by an upward-rounded value of (AM.13). Thus the final balls enclose the infinite-source entries, including rounding error and both omitted theta tails. The default run uses $0 \leq i, j \leq 5, N = 7$, and 110 decimal working digits; its machine-readable output is `checks/arithmetic_moments.json`.

Let $\Delta_k = \det(H_{ij})_{0 \leq i, j < k}$ and $\Delta_0 = 1$. The squared norm h_k of the monic degree- k polynomial for the original measure μ is

$$h_k = \frac{\Delta_{k+1}}{\Delta_k}. \quad (\text{AM.19})$$

Indeed the coefficients below the leading term solve the Gram system making that monic polynomial orthogonal to degrees below k . Substitution of that unique solution into its squared norm gives the Schur complement of the leading k -by- k Gram block. The block determinant identity gives (AM.19). The interval calculation certifies strict positivity of all six leading principal determinants recorded in the companion certificate and all six corresponding h_k ; those are the actual arithmetic norms used by the orthogonal recurrence, through the finite degrees stated.

Here are explicit certified intervals from that run. Each row means the entire interval $[c - r, c + r]$, with the printed decimal centre c and radius r interpreted exactly. Rounding the stored Arb balls to the displayed decimals was performed outwards.

raw moment	c	r
H_{00}	1.279007247846485140480	$4.67 \cdot 10^{-22}$
H_{01}	0.6395036239232425702398	$3.33 \cdot 10^{-23}$
H_{02}	-12.73579749060893715027	$2.81 \cdot 10^{-21}$
H_{11}	13.37530111453217972051	$2.57 \cdot 10^{-21}$
H_{12}	6.687650557266089860256	$2.85 \cdot 10^{-22}$
H_{22}	389.8820543216399270934	$1.65 \cdot 10^{-20}$

Symmetry supplies the other entries of this original three-by-three moment block. The six original monic squared norms are enclosed by

k	c for h_k	r
0	1.279007247846485140480	$4.67 \cdot 10^{-22}$
1	13.05554930257055843539	$2.69 \cdot 10^{-21}$
2	250.0089768190499145175	$4.88 \cdot 10^{-20}$
3	6694.666685525627506773	$4.76 \cdot 10^{-19}$
4	224298.1953314520556463	$9.95 \cdot 10^{-18}$
5	9383436.471794727053469	$2.54 \cdot 10^{-16}$

The analytic double-tail bound is less than $2.961369 \cdot 10^{-63}$ for every entry of the six-by-six matrix, and is less than $1.150030 \cdot 10^{-83}$ for H_{00} . These bounds account for omitted source terms; the displayed intervals additionally include finite-term arithmetic and decimal conversion errors.

The checker also evaluates (AM.15) and the independent exact finite-boundary identity (AM.16), and detects removal of either the reflected half-domain, the theta factor 2, or the adjoint constant 1. Its checks execute identically under ordinary Python and Python's optimization flag. Finite interval certificates do not assert an infinite-degree weight estimate; (AM.8)–(AM.14) give the exact arithmetic input and its complete error control for calculating further degrees.

8 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

8.1 The fixed arithmetic maps and their boundary

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx). \end{aligned}$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta \phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ , and retain

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \dots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$R_{\text{ref}} u = S_{h_Z}^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad JF = j_h \mathcal{M}F, \quad \sigma_Z = q R_{\text{ref}},$$

$$J R_{\text{ref}} = 1_E, \quad J\Theta = 0, \quad D R_{\text{ref}} - R_{\text{ref}} A = f_0 \ell_Z, \quad \ell_Z(u) = [t^{d-1}] R_Z(u). \quad (\text{TC1})$$

Here $S_\rho^{\mathcal{B}} F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S_{h_Z}^{\mathcal{B}}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^L c_j \phi_j, \quad F_L = \Theta \Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathcal{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \quad C : E \longrightarrow \mathbb{C}^{L+1}. \quad (\text{TC2})$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \rightarrow \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1} . Directly,

$$DR - RA = \Theta K_R, \quad K_R = \phi_* \ell_Z + \Phi_{L+1}(S_L C - I_L C A) : E \longrightarrow V. \quad (\text{TC3})$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^*R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

because $[x\overline{F(x)}H(x)]_0^\infty = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^*G_R + G_RA - G_R = -(R^*\Theta K_R + (\Theta K_R)^*R). \quad (\text{TC4})$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

8.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \quad (\mathcal{U}F)(y) = e^{y/2}F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\text{Im } z| < \pi/4$. On every closed narrower strip, $\text{Re}(e^{2z}) \geq c_a e^{2\text{Re } z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\text{Re } z \rightarrow +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1}f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2}f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\text{Im } z = a$ sign t has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \quad (\text{TC5})$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \bar{v} d\mu$ has a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2 + it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g . It follows that

$$\overline{\bigcup_{L \geq 0} F_L \mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{TC6})$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

8.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \quad \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m. \quad (\text{TC7})$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \quad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \quad (\text{TC8})$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \rightarrow 1$ strongly. Since E is finite dimensional, $R_m \rightarrow 0$ in $\text{Hom}(E, \mathcal{H})$, and $G_m \rightarrow 0$ in matrix norm.

Exact fixed-level image. At every specified level m , the entire image of the source-Gram map is

$$\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \rightarrow \mathbb{C}^{m+1}\} = \{G_m + \Delta : \Delta \succeq 0, \text{rank } \Delta \leq m+1\}. \quad (\text{TC8a})$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \quad (\text{TC8b})$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m+1$ has a factorization $\Delta = T^* T$ with $T : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ , retain its nonnegative eigenvalues, and insert their square roots into $m+1$ rows. Taking $C = B_m + H_m^{-1/2} T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E , there exist a finite L and a coefficient map $C : E \rightarrow \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \quad JR = 1, \quad qR = \sigma_Z, \quad R^* R = G. \quad (\text{TC9})$$

The construction is as follows. Choose $m \geq d-1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1} . Put

$$\Delta = G - G_m > 0, \quad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \quad (\text{TC10})$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E . The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \quad R_m^* F_m = 0,$$

so its Gram is

$$R^* R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ . This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\text{Ext}_{\mathbb{C}[t]}^1(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E . The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \quad [p(t)] \longmapsto [p(D)\phi_*]. \quad (\text{TC10a})$$

It is well-defined and $\mathbb{C}[t]$ -linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \quad [\psi] \longmapsto j_h \left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*} \right). \quad (\text{TC10b})$$

The denominator is

$$\mathcal{M}\phi_*(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\operatorname{Re} s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z ; this proves well-definedness of κ_h and $\kappa_h\iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z . For the first factor $t - \rho$, its source inverse on $x > 0$ is

$$S_\rho\psi(x) = x^{-\rho} \int_x^\infty \psi(y)y^{\rho-1} dy = - \int_0^1 \psi(tx)t^{\rho-1} dt.$$

The second expression and $0 < \operatorname{Re} \rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\operatorname{Re} \rho - 1}$. The first expression gives Schwartz decay at infinity. Integration of $(D - \rho)S_\rho\psi = \psi$ on the full real line gives $(1 - \rho) \int S_\rho\psi = \int \psi = 0$, so $S_\rho\psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s - \rho)$, so the remaining zero jets permit iteration through every factor of h_Z . Consequently $\ker(j_h\mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$, where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h\kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D -equivariant section: an intertwiner $T : E \rightarrow \mathcal{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathcal{B}$. To see this injectivity, $(D - \rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathcal{B}$ force $c = 0$, and one iterates over the factors of h_Z . The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

8.4 Sharp control with all nilpotent jets present

For a positive definite G , set

$$W_A(G) = A^*G + GA - G, \quad \epsilon_A(G) = \sup_{u \neq 0} \frac{|u^*W_A(G)u|}{u^*Gu}, \quad \eta_Z = \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|. \quad (\text{TC11})$$

For every eigenvector $Au = \rho u$,

$$u^*W_A(G)u = (2 \operatorname{Re} \rho - 1)u^*Gu. \quad (\text{TC12})$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G . We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t}e^{tA^*}G_0e^{tA}, \\ G_\epsilon &= \int_{-\infty}^\infty e^{-\epsilon|t|}F(t) dt, \\ W_A(G_\epsilon) &= \epsilon \int_{-\infty}^\infty \operatorname{sign}(t)e^{-\epsilon|t|}F(t) dt. \end{aligned} \quad (\text{TC13})$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A - \rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho. \quad (\text{TC14})$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\bar{\rho} + \sigma - 1$ has absolute value at most η_Z . This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^*F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \quad (\text{TC15})$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \quad (\text{TC16})$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at} dt = k! a^{-k-1}$ show that

$$G_\epsilon = \sum_{\rho, \sigma \in Z} \sum_{i=0}^{m_\rho-1} \sum_{j=0}^{m_\sigma-1} \frac{I_{i+j}(\bar{\rho} + \sigma - 1, \epsilon)}{i!j!} \cdot (N_\rho^i P_\rho)^* G_0 (N_\sigma^j P_\sigma). \quad (\text{TC17})$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \rightarrow \mathbb{C}^{L+1}}} \epsilon_A((R_{\text{ref}} + F_L C)^*(R_{\text{ref}} + F_L C)) = \eta_Z. \quad (\text{TC18})$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA} u\|_G \leq e^{(1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA} u\|_G \leq e^{(-1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t , hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z , and use (TC13) on that summand. Its control is strictly smaller than η_Z . The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

8.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^n I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \quad W_{A_n, n}(G) = A_n^* G + G A_n - nG. \quad (\text{TC19})$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \dots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \text{Spec } A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \quad (\text{TC20})$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \dots, \rho)$ for a zero attaining η_Z . Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\inf_{G > 0} \sup_{v \neq 0} \frac{|v^* W_{A_n, n}(G) v|}{v^* G v} = n\eta_Z. \quad (\text{TC21})$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

8.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \bar{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \rightarrow \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \rightarrow E.$$

The source boundary of R_m lies in the span of $f_0, \dots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \operatorname{rank} W_m \leq 2. \quad (\text{TC22})$$

Reflection stability gives

$$\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0. \quad (\text{TC23})$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m -self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \max(2 \operatorname{Re} \rho - 1, 0) = \sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}|. \quad (\text{TC24})$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A -invariant flag, keeping all generalized eigenvectors. Choose a G_m -orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \rightarrow E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ , repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal

entries $\delta_j = 2 \operatorname{Re} \rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m . Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \leq P \leq I$ and $H \leq H_+$,

$$\sum_{\delta_j > 0} \delta_j = \operatorname{Tr}(PH) \leq \operatorname{Tr}(PH_+) \leq \operatorname{Tr}(H_+) = \epsilon_m. \quad (\text{TC25})$$

Each trace inequality follows by inserting the positive square root of P or H_+ , so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+ , the same argument gives $\sum_j \max(\delta_j, 0) \leq r_+ \epsilon_A(G)$. In the present source construction $r_+ \leq 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\operatorname{Im} c_m)^2. \quad (\text{TC26})$$

Indeed the two nonzero eigenvalues are those of $-\begin{pmatrix} c_m & v_m \\ u_m & c_m \end{pmatrix}$, and (TC23) gives $\operatorname{Re} c_m = 0$. If $a_{m+1} = (1 - \Pi_m)f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \quad u_m = \eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right). \quad (\text{TC27})$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right) v_m - (\operatorname{Im} c_m)^2 \geq \left(\sum_{\rho \in Z} m_\rho \left| \operatorname{Re} \rho - \frac{1}{2} \right| \right)^2. \quad (\text{TC28})$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

8.7 Original supported maps at every independent source and coefficient face

The passage from a finite source correction C to its representative, Gram, and boundary retains the earlier exact full-source formula

$$C \mapsto R_{\text{ref}} + \Theta \Phi_L C \mapsto (qR, JR, R^* R, K_R) = (\sigma_Z, 1_E, G_R, K_R). \quad (\text{TC29})$$

Here is its complete map before the full-source quotient. Set $T = \Phi_L C : E \rightarrow V$, so $R = s_h + \Theta T$ by (MCF42). At an arbitrary original source subspace $W \subseteq V$, define $\sigma_{R,W}(u) = [Ru]_W \in Q_{W,\Omega}$. Then

$$\sigma_{R,W} - \sigma_{h,W} = \eta_W([T(\cdot)]_W), \quad \pi_W \sigma_{R,W} = \sigma_Z. \quad (\text{TC29a})$$

These follow by subtracting the actual functions and using (D11). Consequently a permitted full-source correction is invisible in Q but retains precisely its class in V/W at a smaller source. The map $\sigma_{R,W}$ is injective, since its composite with π_W is the injective σ_Z .

If $DW \subseteq W'$, the generator comparison has its exact type

$$\overline{D}_{W,W'} \sigma_{R,W} - \sigma_{R,W'} A = \eta_{W'}([K_R(\cdot)]_{W'}), \quad K_R = \phi_* \ell_Z + DT - TA. \quad (\text{TC29b})$$

Indeed (TC3) is the source-function equality $DR - RA = \Theta K_R$; taking its class modulo $\Theta W'$ proves the formula. The left action exists by (D14), so no invariant source subspace is presumed here. Its kernel class vanishes exactly when $K_R(u) \in W'$ for the vector in question, by the injectivity of $\eta_{W'}$.

For every $a > 0$ the full corrected dilation primitive is

$$\begin{aligned} c_{R,a} &= \Psi_{a,h} + \mathcal{R}_a T - T A_{a,h} : E \rightarrow V, \\ \mathcal{R}_a R - R A_{a,h} &= \Theta c_{R,a}, \\ \overline{\mathcal{R}}_a^{W \rightarrow \mathcal{R}_a W} \sigma_{R,W} - \sigma_{R,\mathcal{R}_a W} A_{a,h} &= \eta_{\mathcal{R}_a W}([c_{R,a}] \mathcal{R}_a W). \end{aligned} \quad (\text{TC29c})$$

The first two equalities follow by substituting $R = s_h + \Theta T$ into (MCF24) and using $\mathcal{R}_a \Theta = \Theta \mathcal{R}_a$. The last follows by quotienting by $\Theta \mathcal{R}_a W$. Expanding the first expression and applying (MCF25) gives

$$c_{R,ab} = \mathcal{R}_a c_{R,b} + c_{R,a} A_{b,h}. \quad (\text{TC29d})$$

In the expansion, the terms $\mathcal{R}_a T A_{b,h}$ cancel with their negative, leaving $\mathcal{R}_{ab} T - T A_{ab,h}$ and the original primitive $\Psi_{ab,h}$. This proves the law with its complete source action and ordered arithmetic product.

At a label $(W, S; \mathfrak{a})$, with $\mathfrak{a} \subseteq I_n$, apply all these maps independently to each coordinate $u_i \in E$ for $i \in \mathfrak{a}$. The proper-source residual is then in $(V/W')^{\mathfrak{a}}$ or $(V/\mathcal{R}_a W)^{\mathfrak{a}}$, respectively. When a primitive belongs to the declared source, its two single-leg lifts are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$; their difference is the entire diagonal $(K_R u, \widehat{K_R u})$. If it does not belong, its class in (D11) is retained instead. The labels $W, S, \mathfrak{a}$ are unchanged by evaluating a zero amplitude, and an empty $\mathfrak{a}$ retains the outer label; external absence is still $(\perp, 0)$. Zero insertion commutes with the linear formulas, as does the coefficient Frobenius using the same sign in source and target. Every corrected metric and boundary formula (TC4)–(TC28) therefore uses the same original functions as before, while (TC29a)–(TC29d) record their full mixed-source comparison.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

9 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

9.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x \partial_x.$$

Let $\mathcal{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b |D^k F(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta \phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j \phi_*$ and $f_j = D^j f_0 = \Theta \phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \quad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2}F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity} . Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathcal{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g , including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \dots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \quad \varepsilon = (j_h v)^{-1} \in E^\times, \quad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h , including the factor 2 in $g = 2\xi$. It commutes with A . For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu . Write $R_{\text{ref}} : E \rightarrow \mathcal{B}$ for the source note's fixed finite section R_0 . This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m . The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}} = I_E, \quad J_Z \Theta = 0. \quad (\text{TB.1})$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad d\nu_Z(t) = |v(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{TB.2})$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

9.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\text{ref}} u + \sum_{j=0}^m f_j (Bu)_j, \quad B : E \longrightarrow \mathbb{C}^{m+1}. \quad (\text{TB.3})$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N , with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \quad C_N P = j_h P. \quad (\text{TB.4})$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j d\nu_Z(t), \quad 0 \leq i, j \leq N. \quad (\text{TB.5})$$

It is positive definite by (TB.2). Define

$$K_N = C_N(H_N^\nu)^{-1}C_N^*. \quad (\text{TB.6})$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 9.1 (The actual source minimum). *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1}C_N^*K_N^{-1}U_\varepsilon u, \quad \mathcal{M}R_m u = v p_{N,u}, \quad (\text{TB.7})$$

$$\boxed{G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon}. \quad (\text{TB.8})$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \quad (\text{TB.9})$$

has degree at most m . Its coefficients give exactly the map $B_m : E \rightarrow \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M}R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^m (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \quad (\text{TB.10})$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h , and the degree bound makes its quotient a polynomial of degree at most m . Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p + z)^* H_N^\nu (p + z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \quad (\text{TB.11})$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $qR_m = \sigma_Z$ and $J_Z R_m = I_E$. $\square$

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

9.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z , with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E , no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 9.2 (The retained vertical-line recurrence). *There are $b_n \in \mathbb{C}$, with $b_n + \bar{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \quad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \geq 1). \quad (\text{TB.12})$$

Proof. On the original line $s = 1/2 + it$, one has $\bar{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \leq n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1 - s)q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \bar{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. $\square$

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \quad (\text{TB.13})$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^N \frac{v_n v_n^*}{\kappa_n}. \quad (\text{TB.14})$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z, w) = \sum_{n=0}^N \frac{q_n(z) \overline{q_n(w)}}{\kappa_n}. \quad (\text{TB.15})$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0 \leq k < m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 9.3 (Confluent kernel boundary formula). *For every $N \geq d - 1$,*

$$\boxed{AK_N + K_N A^* - K_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N}}. \quad (\text{TB.16})$$

Before taking any jets the corresponding polynomial identity is

$$(z + \bar{w} - 1)\mathcal{K}_N(z, w) = \frac{q_{N+1}(z) \overline{q_N(w)} + q_N(z) \overline{q_{N+1}(w)}}{\kappa_N}. \quad (\text{TB.17})$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n , because $j_h(sq_n) = A(j_hq_n)$. Substitute these relations into

$$AK_N + K_N A^* - K_N = \sum_{n=0}^N \frac{(Av_n)v_n^* + v_n(Av_n)^* - v_n v_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n + \bar{b}_n = 1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncanceled term is the $n = N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z, w . This is a polynomial identity; no division by $z + \bar{w} - 1$ is performed at its zero set, and differentiating it proves the full confluent version. $\square$

9.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m = A^* G_m + G_m A - G_m = -\{R_m^*(DR_m - R_m A) + (DR_m - R_m A)^* R_m\}. \quad (\text{TB.18})$$

The boundary term $[x\bar{F}H]_0^\infty$ vanishes for $F, H \in \mathcal{B}$. Put

$$L_N = AK_N + K_N A^* - K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m = U_\varepsilon^* K_N^{-1} L_N K_N^{-1} U_\varepsilon.} \quad (\text{TB.19})$$

For clarity, multiplication gives

$$K_N^{-1} L_N K_N^{-1} = K_N^{-1} A + A^* K_N^{-1} - K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N = v_{N+1}^* K_N^{-1} v_{N+1}, \quad \beta_N = v_N^* K_N^{-1} v_N, \quad \gamma_N = v_N^* K_N^{-1} v_{N+1}. \quad (\text{TB.20})$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\boxed{\epsilon_m = \frac{|\operatorname{Re} \gamma_N| + \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.} \quad (\text{TB.21})$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1} U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y / (y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N} \begin{pmatrix} \gamma_N & \beta_N \\ \alpha_N & \bar{\gamma}_N \end{pmatrix}.$$

Its trace and determinant give

$$\lambda_\pm = \frac{\operatorname{Re} \gamma_N \pm \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.$$

Cauchy-Schwarz in K_N^{-1} gives $|\gamma_N|^2 \leq \alpha_N \beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re} \gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \bar{\rho}$, with the multiplicities retained, $2 \operatorname{Re} \operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1} L_N) = \frac{2 \operatorname{Re} \gamma_N}{\kappa_N}. \quad (\text{TB.22})$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{TB.23})$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \geq \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

9.5 Exact change when one more theta direction is admitted

The step $m \mapsto m + 1$ means $N \mapsto N + 1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \geq 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{w w^*}{k}. \quad (\text{TB.24})$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1} w w^* K_N^{-1}}{k + \theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1} w w^* K_N^{-1} U_\varepsilon}{k + \theta}, \quad (\text{TB.25})$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k + \theta} > 0. \quad (\text{TB.26})$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \neq 0$, the determinant formula follows by extending $K_N^{-1} w$ to a basis, in which the rank-one map $K_N^{-1} w w^* / k$ has one possible nonzero eigenvalue θ/k . Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h , as in (TB.9). This gives a source polynomial P_u of degree at most $m + 1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1} , where $W_{m+1} = \operatorname{span}(\phi_*, D\phi_*, \dots, D^{m+1}\phi_*)$. The same source transition on the precise packet domain is $Q_{W_m, \Omega} \rightarrow Q_{W_{m+1}, \Omega}$, where $Q_{W, \Omega} = \mathcal{B}_\Omega / \Theta W$. Its kernel is $\Theta W_{m+1} / \Theta W_m$, identified through $[v] \mapsto [\Theta v]_{W_m}$ with the original one-dimensional space W_{m+1} / W_m . The unchanged target inclusion $\mathcal{B}_\Omega \hookrightarrow \mathcal{B}$ embeds this sequence in the original full-target quotient sequence. At an independent coefficient mask $\mathfrak{a}$, the kernel is exactly $(W_{m+1} / W_m)^\mathfrak{a}$, with its outer source and leg labels retained by (D12). The scalar e gives the zero at that same label; τ gives external absence. These spaces satisfy $DW_m \subseteq W_{m+1}$, so their generator is the actual adjacent-stage map (D14), not an assumed endomorphism of W_m . The full surviving primitive and the resulting arithmetic identity are proved in (TB.37)–(TB.39). None of these typed source maps changes the Grams in (TB.24)–(TB.26).

9.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make

this measure invariant under $s \mapsto 1 - s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1 - s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$sQ_n = Q_{n+1} + \frac{1}{2}Q_n - \frac{h_n}{h_{n-1}}Q_{n-1} \quad (n \geq 1). \quad (\text{TB.27})$$

For $n = 0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \quad d_n u = \frac{1}{h_n} \int_{\mathbb{R}} \overline{Q_n(1/2 + it)} r(u)(1/2 + it) d\mu(t). \quad (\text{TB.28})$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g \sum_{n>m} Q_n d_n u, \quad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n.} \quad (\text{TB.29})$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n , and use the adjoint relation $s^* = 1 - s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n} d_{n+1} = d_{n-1} + d_n(\frac{1}{2}I - A) - \delta_{n0}\ell_Z.} \quad (\text{TB.30})$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1} d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1}(d_{m+1}^* d_m + d_m^* d_{m+1}).} \quad (\text{TB.31})$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 9.4 (The full source defect and its observation of the packet). *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{DR_m - R_m A = -\frac{h_{m+1}}{h_m} Q_m(D) f_0 d_{m+1} - Q_{m+1}(D) f_0 d_m.} \quad (\text{TB.35})$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \quad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \quad (\text{TB.36})$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^m Q_j d_j$ and compute

$$sr_m - r_m A = \ell_Z - \sum_{j=0}^m (sQ_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j . For $0 \leq j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z . The coefficient at Q_m is $-h_{m+1}d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed primitive belongs to V , since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D)f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \geq 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k . Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \geq 0$. In particular every term of the positive series in (TB.29) vanishes on u , so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. $\square$

9.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1 - \bar{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \rightarrow E_{Z^\dagger}$ defined in Section 318. The original arithmetic residue matrix S , satisfying $S^* = -S$ and $A^* S + S A = S$, is still related to its dual metric by

$$G_m^D = S^* G_m^{-1} S, \quad W_m^D = -S^* G_m^{-1} W_m G_m^{-1} S. \quad (\text{TB.32})$$

The original arithmetic Weil form, with $J_g = M_{j_h, g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \quad (\text{TB.33})$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \rightarrow 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \longrightarrow +\infty, \quad \prod_{m=m_0}^M \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \longrightarrow 0. \quad (\text{TB.34})$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \rightarrow 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z -moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, <https://dlmf.nist.gov/18.2>. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

9.8 Propagation through the original finite source tower

The finite spaces W_m used after (TB.26) have the precise source-domain maps (D14), because $DW_m \subseteq W_{m+1}$. Let $T_m = \Phi_m B_m : E \rightarrow W_m$, so $R_m = s_h + \Theta T_m$. Retain every outer leg label and every independent coefficient mask $\mathfrak{a} \subseteq I_n$. Then

$$\begin{aligned} 0 \rightarrow (W_{m+1}/W_m)^\mathfrak{a} &\rightarrow Q_{W_m, \Omega}^\mathfrak{a} \rightarrow Q_{W_{m+1}, \Omega}^\mathfrak{a} \rightarrow 0, \\ \sigma_{m, W_m}^\mathfrak{a} &= \sigma_{h, W_m}^\mathfrak{a}, \quad \sigma_{m+1, W_{m+1}}^\mathfrak{a} = \sigma_{m, W_{m+1}}^\mathfrak{a}, \end{aligned} \quad (\text{TB.37})$$

where $\sigma_{m, W}u = [R_m u]_W$ and $A^\mathfrak{a}$ denotes the coordinatewise direct sum of the original packet operator A . The exact sequence is (D12); its kernel map is $[v_i] \mapsto [\Theta v_i]_{W_m}$, not an erasure of a coefficient face. The first section equality holds because $T_m(E) \subseteq W_m$. The second holds because the actual difference $R_{m+1} - R_m$ belongs to ΘW_{m+1} , as proved by the numerator difference following (TB.26).

The entire primitive in (TB.35), with f_0 replaced by ϕ_* , belongs to W_{m+1} . Hence (TC29b) and (TB.37) give the adjacent-level arithmetic identity

$$\overline{D}_{W_m, W_{m+1}} \sigma_{m, W_m}^\mathfrak{a} = \sigma_{m+1, W_{m+1}}^\mathfrak{a} A^\mathfrak{a}. \quad (\text{TB.38})$$

This is an equality between the declared quotient spaces, obtained from the actual nonzero source boundary in (TB.35). It does not discard the lower source term or identify W_m with a D -invariant source. Before passing to W_{m+1} , that boundary retains its class $[K_{R_m}]_{W_m}$ in $(V/W_m)^\mathfrak{a}$. In particular, modulo W_m its exact remaining term is

$$[K_{R_m} u]_{W_m} = -[D^{m+1} \phi_*]_{W_m} d_m u, \quad (\text{TB.39})$$

since Q_m has degree m and Q_{m+1} is monic. Linear independence of the source iterates shows that this displayed generator of W_{m+1}/W_m is nonzero. Thus the retained source transition measures exactly the row d_m , while the full primitive in W_{m+1} retains also d_{m+1} as in (TB.35). The injective two-row observation (TB.36) therefore remains compatible with, and more informative than, this one transition.

All maps here also commute with the original inclusion from $\mathcal{B}_U$ to $\mathcal{B}_\Omega$ when the packet lies in U . For a packet containing critical-line centres the codomain is the specified $\mathcal{B}_\Omega$, whose quotient over $\mathcal{B}_U$ is the full $(T_L Q)^\mathfrak{a}$ in (D13). The Gram formulas (TB.7)–(TB.8), rank-one updates (TB.25)–(TB.26), and full source formula (TB.35) are unchanged actual functions at these domains. This propagates the later mixed-face and critical-line precision through the original arithmetic source tower.

10 The full jet kernel, finite resolvents, and the actual analytic tail

Retain the arithmetic measure, coordinates and maps of Section 9:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = |g(s)/h(s)|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

In particular, every selected zero has its complete multiplicity in h , and c is the coordinate column of the unit in the retained monomial basis of E . Write q_k for the monic orthogonal polynomials of ν_Z , $\kappa_k = \|q_k\|_{\nu_Z}^2$, and $a_k = \kappa_k / \kappa_{k-1} > 0$ for $k \geq 1$. The exact recurrence is

$$s q_k = q_{k+1} + b_k q_k - a_k q_{k-1}, \quad b_k + \bar{b}_k = 1, \quad (\text{KR.1})$$

where the last term is omitted at $k = 0$.

10.1 A finite compression in the original Mellin coordinate

Fix $n \geq d$, so $N = n - 1$ is the degree bound in the preceding kernel construction. On the coefficient space with basis $q_0, \dots, q_{n-1}$, put

$$H_n = \text{diag}(\kappa_0, \dots, \kappa_{n-1}), \quad (J_n)_{k,k} = b_k, \quad (J_n)_{k+1,k} = 1, \quad (J_n)_{k,k+1} = -a_{k+1}. \quad (\text{KR.2})$$

All unspecified entries are zero, and indices start at zero. Multiplication of the displayed matrices proves

$$J_n^* H_n + H_n J_n = H_n. \quad (\text{KR.3})$$

The isometry from this coefficient space to its polynomial image is $u \mapsto \sum_k q_k u_k$. For a standard coefficient Hermitian form, the corresponding isometry is

$$\mathcal{I}_n u = \sum_{k=0}^{n-1} \frac{q_k}{\sqrt{\kappa_k}} u_k, \quad B_n = H_n^{1/2} J_n H_n^{-1/2}. \quad (\text{KR.4})$$

This explicitly stated auxiliary coordinate map leaves A , the packet coordinates, q_k , and κ_k unchanged. The matrix B_n has lower off-diagonal entry $\sqrt{a_{k+1}}$, upper off-diagonal entry $-\sqrt{a_{k+1}}$, and satisfies $B_n^* + B_n = I$.

Lemma 10.1 (Nodes and endpoint resolvents). *The characteristic polynomial of J_n , and hence of B_n , is $q_n(z)$. Its roots $\lambda_{n,1}, \dots, \lambda_{n,n}$ are distinct and lie on $\text{Re } z = 1/2$. There are positive numbers $\omega_{n,j}$ and $\sigma_{n,j}$ such that*

$$\begin{aligned} m_n(z) &= \kappa_0 e_0^* (zI - B_n)^{-1} e_0 = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \omega_{n,j} &= \kappa_0, \\ F_n(z) &= e_{n-1}^* (zI - B_n)^{-1} e_{n-1} = \frac{q_{n-1}(z)}{q_n(z)} = \sum_{j=1}^n \frac{\sigma_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \sigma_{n,j} &= 1. \end{aligned} \quad (\text{KR.5})$$

Here m_n and F_n are respectively the initial and terminal observations of the same finite compression.

Proof. Expanding the tridiagonal determinant along its last row gives $D_{k+1}(z) = (z - b_k)D_k(z) + a_k D_{k-1}(z)$, with $D_0 = 1$, $D_1 = z - b_0$. This is (KR.1), proving $D_n = q_n$. The matrix $(B_n - \frac{1}{2}I)/i$ is Hermitian. If an eigenvector has first coordinate zero, its first row equation forces its second coordinate to be zero; recurrence through the rows forces every coordinate to vanish. Thus each eigenspace has dimension at most one. Hermitian diagonalization gives a complete eigenbasis, hence all eigenvalues are simple. The same argument beginning at the last row shows every eigenvector has nonzero last coordinate. If P_j are its rank-one orthogonal eigenprojections, set $\omega_{n,j} = \kappa_0 e_0^* P_j e_0 > 0$ and $\sigma_{n,j} = e_{n-1}^* P_j e_{n-1} > 0$. Their sums and the two spectral expansions follow from $\sum_j P_j = I$. The last diagonal cofactor of $zI - B_n$ is $q_{n-1}(z)$, proving the ratio in (KR.5). $\square$

Define $\delta(z) = |\text{Re } z - 1/2|$. For $\delta(z) > 0$, (KR.3)–(KR.4) give $\|(zI - B_n)^{-1}\| \leq \delta(z)^{-1}$: for a vector u , the real part of $u^*(zI - B_n)u$ is $(\text{Re } z - 1/2)\|u\|^2$, and Cauchy–Schwarz then gives the lower singular-value bound. In particular $|F_n(z)| \leq \delta(z)^{-1}$, and

$$\text{Re } F_n(z) = (\text{Re } z - \tfrac{1}{2}) \sum_j \frac{\sigma_{n,j}}{|z - \lambda_{n,j}|^2}. \quad (\text{KR.6})$$

10.2 Exact quadrature and the full confluent kernel

Theorem 10.2 (A polynomial formula valid at every packet centre). *Put*

$$L_{n,j}(z) = \frac{q_n(z)}{(z - \lambda_{n,j})q'_n(\lambda_{n,j})}. \quad (\text{KR.7})$$

This denotes the polynomial obtained by cancelling the displayed linear factor. For every polynomial P of degree at most $2n - 1$,

$$\int P(s) d\nu_Z = \sum_j \omega_{n,j} P(\lambda_{n,j}). \quad (\text{KR.8})$$

Consequently the complete arithmetic kernel is

$$K_{n-1} = \sum_{j=1}^n \frac{L_{n,j}(A) c c^* L_{n,j}(A)^*}{\omega_{n,j}}. \quad (\text{KR.9})$$

Formula (KR.9) includes all Taylor derivatives at all selected zeros, even when a selected critical zero equals a quadrature node.

Proof. For degree below n , the vector $P(J_n)e_0$ records exactly the coefficients of P in the monic orthogonal basis: in the successive multiplications no term of degree n occurs. Its initial coefficient is $\kappa_0^{-1} \int P d\nu_Z$. Thus the spectral expansion in (KR.5) proves (KR.8) in that degree range. For a general P in the asserted range, polynomial division gives $P = q_n U + R$, where $\deg U \leq n - 1$ and $\deg R < n$. On the line, the conjugate of $U(s)$ is again a polynomial in s , namely $\overline{U(1 - \bar{s})}$. Orthogonality therefore gives $\int q_n U d\nu_Z = 0$. The same multiple vanishes at all the nodes, proving (KR.8).

The polynomials $L_{n,j}$ form the interpolation basis of $\mathcal{P}_{n-1}$. Apply (KR.8) to $\overline{L_{n,j}(s)} L_{n,k}(s)$, interpreted as a polynomial on the line; its degree is at most $2n - 2$. Their Gram matrix is $\text{diag}(\omega_{n,j})$. In this basis the residue map has column j $L_{n,j} = L_{n,j}(A)c$. The definition $K = CH^{-1}C^*$ now proves (KR.9). This is a basis change in the polynomial source only. Polynomial evaluation at A retains its nilpotent powers, so the formula gives the complete jets without any distinct-centre or simple-zero assumption. $\square$

To identify the quadrature endpoints exactly, one has

$$\sigma_{n,j} = \frac{\omega_{n,j} |q_{n-1}(\lambda_{n,j})|^2}{\kappa_{n-1}} = \frac{\kappa_{n-1}}{\omega_{n,j} |q'_n(\lambda_{n,j})|^2}. \quad (\text{KR.10})$$

Indeed an eigenvector of B_n at λ has coordinates proportional to $\overline{q_k(\lambda)}/\sqrt{\kappa_k}$: substitution into the recurrence proves this assertion, including the endpoint row. The squared ratio of its last and first coordinates proves the first equality. The preceding polynomial kernel formula gives $\mathcal{K}_{n-1}(\lambda_{n,j}, \lambda_{n,j}) = 1/\omega_{n,j}$. In the original Christoffel–Darboux polynomial identity, set $w = \lambda_{n,j}$, divide its common polynomial factor $z - \lambda_{n,j}$, then evaluate at that point. This gives

$$\frac{1}{\omega_{n,j}} = \frac{q'_n(\lambda_{n,j}) \overline{q_{n-1}(\lambda_{n,j})}}{\kappa_{n-1}}.$$

Its squared modulus gives the second equality in (KR.10).

Theorem 10.3 (The exact resolvent metric for the packet). *Suppose, for this formula, that no selected centre equals one of the specified n quadrature nodes. Define maps on the original space E by*

$$Q_n = q_n(A), \quad X_{n,j} = (A - \lambda_{n,j}I)^{-1}c, \quad S_n = \sum_j \sigma_{n,j} X_{n,j} X_{n,j}^*, \quad r_n = F_n(A)c. \quad (\text{KR.11})$$

Then Q_n is invertible, $S_n > 0$, and

$$\boxed{K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*} \quad (\text{KR.12})$$

In particular, the original relative-control quantity at $m = n - 1 - d$ is exactly

$$\epsilon_m = \sup_{y \neq 0} \frac{|y^*(cr_n^* + r_n c^*)y|}{y^* S_n y}. \quad (\text{KR.13})$$

The identity is valid whenever that index belongs to the original source sequence; $m = -1$ is its already specified fixed-section extension. Every off-line packet obeys the nonintersection condition for every n . At an intersecting critical centre, (KR.9) gives the exact polynomial continuation, with no deletion of that centre or its jets.

Proof. The finite algebra E has an inverse for a polynomial precisely when the polynomial is nonzero at every centre: the Euclidean algorithm gives an inverse modulo h , or, equivalently, the finite Taylor inverse at each centre gives the inverse under the complete Chinese-remainder map. This proves the asserted invertibilities. Substitution of (KR.7) and (KR.10) into (KR.9) proves the first identity in (KR.12). It proves $S_n > 0$, since the residue map is onto at $n \geq d$, hence $K_{n-1} > 0$. The identity $AX_{n,j} = \lambda_{n,j} X_{n,j} + c$, together with $\lambda_{n,j} + \bar{\lambda}_{n,j} = 1$, proves

$$AS_n + S_n A^* - S_n = c \left(\sum_j \sigma_{n,j} X_{n,j} \right)^* + \left(\sum_j \sigma_{n,j} X_{n,j} \right) c^*.$$

The partial fractions in (KR.5), evaluated in the finite algebra, identify the sum with $F_n(A)c$. Because Q_n commutes with A , both kernel and its weight form are congruent, with the same factor κ_{n-1}^{-1} , to S_n and its displayed weight form. The invertible change $y = Q_n^* x$ proves equality of their Rayleigh suprema. The previous exact map from K_{n-1} to G_m then proves (KR.13). All of these maps are stated in the original quotient coordinates; the Taylor expansion of each rational factor includes every power through $m_\rho - 1$. $\square$

The scalar form of (KR.13), with every inverse taken on the same retained packet coordinates, is

$$\alpha_n = r_n^* S_n^{-1} r_n, \quad \beta_n = c^* S_n^{-1} c, \quad \gamma_n = c^* S_n^{-1} r_n,$$

$$\boxed{\epsilon_m = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}} \quad (\text{KR.13a})$$

Indeed the nonzero eigenvalues of $S_n^{-1/2}(cr_n^* + r_n c^*)S_n^{-1/2}$ coincide, with multiplicities, with the nonzero eigenvalues of $\begin{pmatrix} \gamma_n & \beta_n \\ \alpha_n & \bar{\gamma}_n \end{pmatrix}$: factor the former through its two displayed vectors and interchange the two rectangular factors. Its characteristic polynomial gives $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Cauchy-Schwarz gives $|\gamma_n|^2 \leq \alpha_n \beta_n$, so their greatest absolute value is (KR.13a), including zero or one-dimensional rank. The full congruence (KR.12) specifies exactly how the original q_n , κ_{n-1} and local unit enter the metric whose relative form is evaluated here.

10.3 The infinite multiplication operator and its genuine tail

Let $\mathcal{L} = L^2(\nu_Z)$, and let M be multiplication by s , with domain $\{f : sf \in \mathcal{L}\}$. It is a closed operator and $M^* = I - M$. These statements follow directly from the multiplication formula; its resolvent off the line is multiplication by $(z - s)^{-1}$, of norm at most $\delta(z)^{-1}$. The auxiliary vectors $\psi_k = q_k / \sqrt{\kappa_k}$ form a complete orthonormal basis. Completeness follows from the positive exponential moment of the actual arithmetic measure by the Fourier-annihilator proof already given for the theta measures.

Let P_n project onto $\psi_0, \dots, \psi_{n-1}$, and put $Q_n^{\text{tail}} = I - P_n$; this projection is distinct from the polynomial map $Q_n = q_n(A)$ in (KR.11). Define

$$M_n^{\text{tail}} = Q_n^{\text{tail}} M Q_n^{\text{tail}} \quad \text{on } Q_n^{\text{tail}} \text{Dom } M, \quad \zeta_n(z) = \langle \psi_n, (z - M_n^{\text{tail}})^{-1} \psi_n \rangle. \quad (\text{KR.14})$$

This tail resolvent exists at every off-line z , with norm at most $\delta(z)^{-1}$. Here is the domain justification. Every polynomial belongs to $\text{Dom } M$, so P_n preserves that domain and $Q_n^{\text{tail}} \text{Dom } M = \text{Dom } M \cap Q_n^{\text{tail}} \mathcal{L}$. The operator $(M - \frac{1}{2}I)/i$ is self-adjoint multiplication by t . Its off-diagonal blocks for $P_n \oplus Q_n^{\text{tail}}$ extend to a bounded, finite-rank self-adjoint operator, by (KR.1). Subtracting this operator preserves self-adjointness: for a self-adjoint T and bounded self-adjoint C , the definition of the adjoint gives $(T - C)^* = T^* - C$ on exactly $\text{Dom } T^*$. The resulting diagonal operator reduces the two projections. Its tail block is therefore self-adjoint on the displayed tail domain. Multiplication by i and addition of $1/2$ gives the claimed tail resolvent.

Theorem 10.4 (Exact finite-section, terminal, and tail comparison). *For $\delta(z) > 0$, let*

$$m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t).$$

Then

$$H_n^{\text{obs}}(z) = \frac{\zeta_n(z)}{1 + a_n F_n(z) \zeta_n(z)}, \quad (\text{KR.15})$$

$$m(z) - m_n(z) = \frac{(-1)^n \kappa_n \zeta_n(z)}{q_n(z)^2 [1 + a_n F_n(z) \zeta_n(z)]} = \frac{(-1)^n}{q_n(z)^2} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t). \quad (\text{KR.16})$$

Every denominator in these formulas is nonzero off the original Mellin line. The terminal finite observations additionally obey the exact Schur recurrence

$$F_1(z) = \frac{1}{z - b_0}, \quad F_{n+1}(z) = \frac{1}{z - b_n + a_n F_n(z)}. \quad (\text{KR.17})$$

Proof. Under $\mathcal{I}_n \oplus I$, the block operator M is

$$\begin{pmatrix} B_n & -\sqrt{a_n} e_{n-1} \langle \psi_n, \cdot \rangle \\ \sqrt{a_n} \psi_n e_{n-1}^* & M_n^{\text{tail}} \end{pmatrix}. \quad (\text{KR.18})$$

The signs follow directly from (KR.1); both coupling maps are bounded on the displayed summands. Eliminate the tail component in $z - M$. The finite Schur complement is $zI - B_n + a_n \zeta_n(z) e_{n-1} e_{n-1}^*$. For $\text{Re } z > 1/2$, the real part of its quadratic form is at least $\delta(z) \|u\|^2$, because $\text{Re } \zeta_n(z) > 0$; on the other side all signs reverse. The sign of this scalar real part follows either from the self-adjoint tail resolvent or from the resolvent identity $\text{Re} \langle v, (z - M_n^{\text{tail}})^{-1} v \rangle = (\text{Re } z - 1/2) \| (z - M_n^{\text{tail}})^{-1} v \|^2$. Thus the Schur complement is invertible and $1 + a_n \zeta_n F_n \neq 0$. Direct multiplication verifies the rank-one inverse formula

$$R_n - \frac{a_n \zeta_n}{1 + a_n \zeta_n F_n} R_n e_{n-1} e_{n-1}^* R_n, \quad R_n = (zI - B_n)^{-1}. \quad (\text{KR.19})$$

The analogous bottom Schur calculation, paired against ψ_n , gives (KR.15), since $\langle \psi_n, (z - M)^{-1} \psi_n \rangle = H_n^{\text{obs}}(z)$.

The two endpoint cofactors of $zI - B_n$ give

$$e_0^* R_n e_{n-1} = \frac{(-1)^{n-1} \sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}, \quad e_{n-1}^* R_n e_0 = \frac{\sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}. \quad (\text{KR.20})$$

These cofactors are the products of the successive upper and lower entries, respectively; the formula also covers $n = 1$. Pair (KR.19) against $\sqrt{\kappa_0} e_0$, the coefficient vector of the unchanged function 1. Substitute (KR.20) and $a_n \kappa_{n-1} = \kappa_n$. This proves the first equality in (KR.16); (KR.15) proves the second. Finally (KR.17) is (KR.1) divided by q_{n+1} , equivalently the last-coordinate finite Schur complement. $\square$

There is also a direct polynomial check on every factor in (KR.16). The polynomial

$$p_n(z) = \int \frac{q_n(z) - q_n(s)}{z - s} d\nu_Z(t)$$

has degree below n . At a quadrature node its ratio residue is $p_n(\lambda_{n,j})/q'_n(\lambda_{n,j}) = \int L_{n,j} d\nu_Z = \omega_{n,j}$, so $m_n = p_n/q_n$. Orthogonality to the polynomial $(q_n(z) - q_n(s))/(z - s)$ gives

$$m - m_n = \frac{1}{q_n(z)^2} \int \frac{q_n(s)^2}{z - s} d\nu_Z.$$

The polynomial $(-1)^n \overline{q_n(1 - \bar{s})}$ is monic and has the same orthogonality conditions as q_n ; uniqueness makes it equal to q_n . On the line this says $q_n(s)^2 = (-1)^n |q_n(s)|^2$, giving (KR.16) again. In particular the explicit error inequality is

$$|m(z) - m_n(z)| \leq \frac{\kappa_n}{\delta(z) |q_n(z)|^2}. \quad (\text{KR.21})$$

The initial and terminal observations have now been related by the actual block maps (KR.18)–(KR.20), the tail (KR.14), and the exact remainder (KR.16). The relative form uses the terminal observation through (KR.11)–(KR.13). An estimate for the initial observation can be propagated through these equalities while retaining the factors q_n , κ_n , a_n , and the tail denominator. None of those arithmetic quantities is discarded in passing from one observation to the other.

11 Quantitative convergence of the arithmetic Gaussian resolvents

Retain the actual measure and every coordinate of Section 10:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

The polynomial h contains the complete multiplicities of its selected actual zeros. The quotient g/h is therefore entire, including at the selected critical zeros. Use the initial Gaussian observation of (KR.5) and (KR.8),

$$m_n(z) = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \quad m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad \mathcal{E}_n(z) = m(z) - m_n(z), \quad \delta(z) = |\operatorname{Re} z - \tfrac{1}{2}|. \quad (\text{GC.1})$$

The Gaussian weights are positive and the distinct nodes lie on the original Mellin line, as proved in Section 10. Its exact quadrature (KR.8) and exact error (KR.16) are the inputs below. No asymptotic for q_n or κ_n is assumed.

For each fixed $0 < a < \pi/2$, the number

$$E_a = \max \left\{ \int e^{at} d\nu_Z(t), \int e^{-at} d\nu_Z(t) \right\} \quad (\text{GC.2})$$

is finite. Here is the connection to the proved arithmetic decay. The actual theta calculation in Section 104 gives $|g(1/2 + it)| \leq C_b e^{-b|t|}$ for every $0 < b < \pi/4$. Choose $a/2 < b < \pi/4$. Outside a sufficiently large compact interval, the monic polynomial h of degree d satisfies $|h(1/2 + it)| \geq c_h (1 + |t|)^d$. Thus the density multiplied by $e^{a|t|}$ is bounded there by a constant times $(1 + |t|)^{-2d} e^{-(2b-a)|t|}$. On the compact interval the entire quotient is bounded. Both parts are integrable, proving (GC.2) with the original factor $dt/(2\pi)$ retained.

11.1 The Gaussian rule inherits the original exponential moments

Lemma 11.1 (Exponential moment domination). *For every real v with $|v| \leq a$, write $\lambda_{n,j} = 1/2 + it_{n,j}$. Then*

$$\sum_j \omega_{n,j} e^{vt_{n,j}} \leq \int e^{vt} d\nu_Z(t) \leq E_a. \quad (\text{GC.3})$$

Proof. Let H be the real Hermite interpolation polynomial of degree at most $2n-1$ whose values and first derivatives at the distinct real nodes $t_{n,j}$ equal those of $f(t) = e^{vt}$. Its existence and uniqueness follow from the invertibility of the interpolation map: a polynomial in its kernel has a double zero at each of the n nodes, hence is zero in degree below $2n$; domain and target both have dimension $2n$. For a real t outside the nodes the interpolation remainder is

$$f(t) - H(t) = \frac{f^{(2n)}(\xi)}{(2n)!} \prod_j (t - t_{n,j})^2 \geq 0 \quad (\text{GC.4})$$

for some real ξ between t and the nodes. To prove the remainder, subtract $C \prod_j (x - t_{n,j})^2$ from $f(x) - H(x)$, choosing C to produce an additional zero at $x = t$. The resulting function has the n double zeros and the additional zero. Repeated Rolle's theorem gives a zero of its $2n$ -th derivative; solving at that zero gives (GC.4). The last sign follows from $f^{(2n)}(x) = v^{2n} e^{vx} \geq 0$. At a node the remainder is zero.

Integrate the pointwise inequality. The exponential and every polynomial are integrable by (GC.2). Since H is a polynomial in $t = (s - 1/2)/i$ of degree at most $2n - 1$, (KR.8) yields

$$\int H(t) d\nu_Z(t) = \sum_j \omega_{n,j} H(t_{n,j}) = \sum_j \omega_{n,j} e^{vt_{n,j}}.$$

This proves the first inequality in (GC.3). Pointwise convexity of e^{vt} in $v \in [-a, a]$, followed by integration, proves the second. No bound on the largest quadrature node has been inserted. $\square$

11.2 Exact moments give a quantitative analytic-strip estimate

Define

$$\Phi(u) = \int e^{iut} d\nu_Z(t), \quad \Phi_n(u) = \sum_j \omega_{n,j} e^{iut_{n,j}}, \quad \Delta_n(u) = \Phi(u) - \Phi_n(u). \quad (\text{GC.5})$$

These functions are holomorphic on $|\operatorname{Im} u| < a$. Indeed, on each smaller closed strip, every differentiated integrand is dominated by a constant times $e^{at} + e^{-at}$, because $|t|^k e^{b|t|} \leq C_{k,b,a}(e^{at} + e^{-at})$ when $b < a$. Dominated convergence proves all the derivatives of Φ ; the finite sum is entire. Equation (GC.3), applied at $v = -\operatorname{Im} u$, gives

$$|\Delta_n(u)| \leq 2E_a \quad (|\operatorname{Im} u| < a), \quad \Delta_n^{(k)}(0) = 0 \quad (0 \leq k < 2n). \quad (\text{GC.6})$$

The derivative identities are exactly the moments in (KR.8).

Lemma 11.2 (The complete strip bound). *For $|\operatorname{Im} u| < a$,*

$$|\Delta_n(u)| \leq 2E_a \left| \tanh\left(\frac{\pi u}{4a}\right) \right|^{2n}. \quad (\text{GC.7})$$

Proof. The explicit holomorphic coordinate maps

$$w = \tanh\left(\frac{\pi u}{4a}\right), \quad u = \frac{2a}{\pi} \operatorname{Log} \frac{1+w}{1-w}$$

are inverse bijections between the strip and the unit disk. For $|w| < 1$, the fraction lies in the right half-plane, so its principal logarithm has imaginary part in $(-\pi/2, \pi/2)$, which proves the asserted strip range. Compose $\Delta_n/(2E_a)$ with this inverse. By (GC.6), the resulting disk function is bounded by one and has a zero of order at least $2n$ at zero. Divide by w^{2n} , filling in its removable value at zero. On each circle $|w| = r < 1$, its modulus is at most r^{-2n} ; the maximum principle gives this bound inside the circle. Letting r increase to one proves the required bound by $|w|^{2n}$, hence (GC.7). $\square$

Theorem 11.3 (Direct off-line convergence with an explicit rate). *For every $0 < a < \pi/2$, every $n \geq 1$, and every z off the original Mellin line,*

$$|m(z) - m_n(z)| \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\delta(z)}{\pi}\right) (4n)^{-2a\delta(z)/\pi}. \quad (\text{GC.8})$$

On a compact set K disjoint from that line the same bound holds uniformly after replacing $\delta(z)$ everywhere by $\delta_K = \min_{z \in K} \delta(z) > 0$.

Proof. Put $w = z - 1/2$. On $\text{Re } w > 0$, the elementary Laplace identity and Fubini's theorem give

$$\mathcal{E}_n(z) = \int_0^\infty e^{-wu} \Delta_n(u) du. \quad (\text{GC.9})$$

For example the absolute integral for the original measure before subtraction is at most $\nu_Z(\mathbb{R})/\text{Re } w$, so the interchange is justified. On $\text{Re } w < 0$, the exact orientation is

$$\mathcal{E}_n(z) = - \int_0^\infty e^{wu} \Delta_n(-u) du. \quad (\text{GC.10})$$

These follow respectively from integrating $e^{-(w-it)u}$ and $-e^{(w-it)u}$. Set $b = \pi/(4a)$. Equation (GC.7) gives

$$|\mathcal{E}_n(z)| \leq 2E_a I_n(\delta(z)), \quad I_n(\delta) = \int_0^\infty e^{-\delta u} \tanh(bu)^{2n} du. \quad (\text{GC.11})$$

For $0 < x < 1$, direct integration gives

$$\log \frac{1-x}{1+x} = -2 \int_0^x \frac{dv}{1-v^2} \leq -2x.$$

Taking $x = e^{-2bu}$ yields $\tanh(bu)^{2n} \leq e^{-4ne^{-2bu}}$. With $\beta = \delta/(2b) = 2a\delta/\pi$, the same variable change gives

$$\begin{aligned} I_n(\delta) &\leq \frac{1}{2b} \int_0^1 x^{\beta-1} e^{-4nx} dx \\ &\leq \frac{1}{2b} (4n)^{-\beta} \Gamma(\beta), \quad \Gamma(\beta) = \int_0^\infty y^{\beta-1} e^{-y} dy. \end{aligned} \quad (\text{GC.12})$$

Substitution in (GC.11) proves (GC.8). On the compact set one uses $e^{-\delta(z)u} \leq e^{-\delta_K u}$ in (GC.11) and then (GC.12) at δ_K ; no monotonicity of the gamma factor is assumed. This proves locally uniform convergence on both components of the off-line domain. $\square$

11.3 Every confluent derivative and the original finite algebra

Write $F^{[k]} = F^{(k)}/k!$. The exact error (KR.16), with its vertical-line sign, gives for all $k \geq 0$

$$\mathcal{E}_n^{[k]}(z) = (-1)^n \sum_{j=0}^k (q_n^{-2})^{[k-j]}(z) (-1)^j \int \frac{|q_n(s)|^2}{(z-s)^{j+1}} d\nu_Z(t). \quad (\text{GC.13})$$

Every derivative under this integral is justified on a compact off-line neighborhood by its positive distance from the line and the finite measure $|q_n|^2 \nu_Z$. The product rule for Taylor coefficients has no binomial factors; those are precisely cancelled by the convention $F^{[k]} = F^{(k)}/k!$. For completeness its polynomial factor is explicitly

$$(q_n^{-2})^{[r]}(z) = \frac{(-1)^r}{q_n(z)^2} \sum_{r_1+\dots+r_n=r} \prod_{j=1}^n \frac{r_j+1}{(z-\lambda_{n,j})^{r_j}}. \quad (\text{GC.14})$$

This follows by multiplying the Taylor series of the n original factors $(z - \lambda_{n,j})^{-2}$, retaining every cross term.

Differentiating (GC.9)–(GC.10) gives the independent estimate

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{2E_a}{k!} \int_0^\infty u^k e^{-\delta(z)u} \tanh(bu)^{2n} du. \quad (\text{GC.15})$$

The integrable majorant $u^k e^{-\delta_K u}$ on a smaller off-line compact neighborhood justifies this differentiation. If $0 < \sigma < \delta$, the exponential series gives $u^k e^{-(\delta-\sigma)u} \leq k!/(\delta-\sigma)^k$. Equations (GC.11)–(GC.12) prove

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{4aE_a}{\pi(\delta-\sigma)^k} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi}, \quad 0 < \sigma < \delta \leq \delta(z). \quad (\text{GC.16})$$

Taking $\delta = \delta_K$ makes this a simultaneous compact-uniform bound on every fixed finite list of derivatives.

Here is the complete algebra transport, including the part of the original packet outside this resolvent domain. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{d_\rho}, \quad H_\rho(s) = \frac{h(s)}{(s - \rho)^{d_\rho}}, \quad U_\rho(s) = \sum_{k=0}^{d_\rho-1} (H_\rho^{-1})^{[k]}(\rho) (s - \rho)^k,$$

and define the actual polynomial remainder

$$e_\rho = j_h(H_\rho U_\rho) \in E. \quad (\text{GC.17})$$

Its remainder is one modulo $(s - \rho)^{d_\rho}$ and zero modulo every other factor of h . The distinct factors are coprime, and a polynomial divisible by each is divisible by their product: successive use of the Bezout identity proves this last statement. It follows directly that

$$e_\rho e_\eta = 0 \ (\rho \neq \eta), \quad e_\rho^2 = e_\rho, \quad \sum_{\rho \in Z} e_\rho = 1. \quad (\text{GC.18})$$

Restriction of a polynomial to its truncated Taylor coefficients therefore gives the exact algebra isomorphism

$$E \longrightarrow \bigoplus_{\rho \in Z} \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{d_\rho}), \quad j_h P \longmapsto \left(\sum_{k < d_\rho} P^{[k]}(\rho) \epsilon_\rho^k \right)_\rho. \quad (\text{GC.19})$$

Its inverse sends the coordinate ϵ_ρ^k to $e_\rho(s - \rho)^k$; composition is the identity by the just proved congruences. Multiplication is preserved because the Taylor coefficient of a product is the exact convolution of the two coefficient lists, truncated only by the displayed ideal.

Let $Z_{\text{off}} = \{\rho \in Z : \text{Re } \rho \neq 1/2\}$ and $Z_{\text{line}} = Z \setminus Z_{\text{off}}$. Define

$$e_{\text{off}} = \sum_{\rho \in Z_{\text{off}}} e_\rho, \quad e_{\text{line}} = 1 - e_{\text{off}}, \quad E_{\text{off}} = e_{\text{off}} E, \quad E_{\text{line}} = e_{\text{line}} E. \quad (\text{GC.20})$$

Then $E = E_{\text{off}} \oplus E_{\text{line}}$ as algebras with their respective idempotent units. The projection $u \mapsto e_{\text{off}}u$ has exactly kernel E_{line} and commutes with the original multiplication operator A . Under (GC.19), its restriction at every retained centre is $A_\rho = \rho I + N_\rho$, where multiplication by e_ρ gives $N_\rho^{d_\rho} = 0$ and retains all lower powers. This constructs the precise map from the full packet to the off-line resolvent domain, together with its full kernel and the operator on that kernel. An empty off-line set gives the zero summand, with the full original packet in the kernel.

For a function f holomorphic near Z_{off} , define

$$f(A_{\text{off}}) = \sum_{\rho \in Z_{\text{off}}} \sum_{k=0}^{d_\rho-1} f^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \quad (\text{GC.21})$$

Here $e_\rho(A)$ means multiplication by the explicitly constructed remainder (GC.17). Equation (GC.19) proves that (GC.21) is a unital algebra homomorphism on the off-line summand. No diagonalizability of A has entered this construction.

In particular, finite geometric expansion in each nilpotent block proves

$$\begin{aligned} m(A_{\text{off}}) &= \int (A_{\text{off}} - sI)^{-1} d\nu_Z(t), \\ m_n(A_{\text{off}}) &= \sum_j \omega_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1}, \\ \mathcal{E}_n(A_{\text{off}}) &= \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \mathcal{E}_n^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \end{aligned} \quad (\text{GC.22})$$

For the first equality the expansion is

$$(A - sI)^{-1} e_\rho(A) = \sum_{k < d_\rho} \frac{(-1)^k (A - \rho I)^k e_\rho(A)}{(\rho - s)^{k+1}}.$$

Each coefficient is integrable since $\delta(\rho) > 0$, and its integral is exactly $m^{[k]}(\rho)$. This proves the integral identity, rather than assigning a formal meaning to it.

For a concrete bound keep any fixed positive Hermitian form G_* on the original coefficient space and restrict it to E_{off} . When that summand is nonzero, put

$$\delta_* = \min_{\rho \in Z_{\text{off}}} \delta(\rho) > 0, \quad C_{\rho,k} = \left\| (A - \rho I)^k e_\rho(A)|_{E_{\text{off}}} \right\|_{G_*}.$$

The triangle inequality and (GC.16) give, for every $0 < \sigma < \delta_*$,

$$\|m(A_{\text{off}}) - m_n(A_{\text{off}})\|_{G_*} \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi} \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \frac{C_{\rho,k}}{(\delta_* - \sigma)^k}. \quad (\text{GC.23})$$

Thus the full initial resolvent converges in the original finite coefficient space, with every nilpotent contribution controlled. Applying the displayed operators to the retained vector $e_{\text{off}}c$ gives the corresponding vector statement with the additional factor $\|e_{\text{off}}c\|_{G_*}$.

The polynomial kernel (KR.9) continues to act on all of E , including E_{line} and collisions with quadrature nodes. The off-line estimates above concern the initial observation m_n . Its exact map to the terminal observation F_n is the block and tail comparison (KR.14)–(KR.20), with the factors q_n, κ_n, a_n and the tail denominator still present. The estimate (GC.23) may be inserted into those identities; it does not remove their arithmetic factors or the critical-line summand of (GC.20).

12 The exact terminal step and its retained boundary energy

All measures, packet coordinates and maps in this section are the original arithmetic ones. Thus $s = 1/2 + it$, $d\nu_Z = |g(s)/h(s)|^2 dt/(2\pi)$, $E = \mathbb{C}[s]/(h)$, $A = M_s$, and $c = j_h 1$. Every selected zero occurs with its complete multiplicity in h . Retain the monic q_n , their actual norms κ_n , and

$$a_n = \frac{\kappa_n}{\kappa_{n-1}} > 0, \quad q_{n+1}(s) = (s - b_n)q_n(s) + a_n q_{n-1}(s), \quad b_n + \bar{b}_n = 1. \quad (\text{KT.1})$$

No recurrence coefficient is replaced by a limiting value.

12.1 A step written entirely in the original polynomial packet

For $n \geq d = \deg h$, put

$$v_j = q_j(A)c, \quad K = K_{n-1} = \sum_{j=0}^{n-1} \frac{v_j v_j^*}{\kappa_j} > 0, \quad M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}. \quad (\text{KT.2})$$

Use the positive form $\langle x, y \rangle_n = x^* M_n y$, conjugate-linear in its first argument, and define

$$u_n = v_n, \quad w_n = v_{n-1}, \quad \alpha_n = \langle w_n, w_n \rangle_n, \quad \beta_n = \langle u_n, u_n \rangle_n, \quad \gamma_n = \langle u_n, w_n \rangle_n. \quad (\text{KT.3})$$

The arithmetic relative allowance at source level $m = n - d - 1$ will be denoted by $\epsilon^{(n)}$. The fixed-section extension at $m = -1$ has already been specified; the original corrected sequence has $n \geq d + 1$. The exact kernel boundary identity gives

$$\epsilon^{(n)} = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}, \quad \operatorname{Re} \gamma_n = \operatorname{Re} \operatorname{tr} A - \frac{d}{2} =: g_Z. \quad (\text{KT.4})$$

Indeed the kernel weight form is $(u_n w_n^* + w_n u_n^*)/\kappa_{n-1}$. Its possible nonzero generalized eigenvalues are obtained by reversing the two rectangular factors, giving $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Taking the trace after multiplication by K^{-1} gives the second identity. Cauchy-Schwarz proves the nonnegativity of the radicand and justifies the maximum absolute value in (KT.4).

Theorem 12.1 (Universal scalar and metric step). *Write $a = a_n$, and retain the exact next vector*

$$t_n = (A - b_n I)u_n + a_n w_n = v_{n+1}. \quad (\text{KT.5})$$

Then

$$M_{n+1} = \frac{1}{a} \left(M_n - \frac{M_n u_n u_n^* M_n}{a + \beta_n} \right), \quad (\text{KT.6})$$

$$\begin{aligned} \alpha_{n+1} &= \frac{\beta_n}{a + \beta_n}, \\ \gamma_{n+1} &= \frac{\langle t_n, u_n \rangle_n}{a + \beta_n}, \\ \beta_{n+1} &= \frac{1}{a} \left(\langle t_n, t_n \rangle_n - \frac{|\langle u_n, t_n \rangle_n|^2}{a + \beta_n} \right). \end{aligned} \quad (\text{KT.7})$$

Every denominator is strictly positive. These identities hold also when any selected critical centre equals a quadrature node, and when $q_n(A)$ is singular.

The exact determinant changes of the kernel and original theta metric are

$$\frac{\det K_n}{\det K_{n-1}} = 1 + \frac{\beta_n}{a_n}, \quad \frac{\det G_{m+1}}{\det G_m} = \frac{a_n}{a_n + \beta_n} = 1 - \alpha_{n+1}. \quad (\text{KT.8})$$

Proof. The unchanged kernel update is $K_n = K_{n-1} + u_n u_n^* / \kappa_n$. Its inverse is verified by multiplication:

$$K_n^{-1} = K^{-1} - \frac{K^{-1} u_n u_n^* K^{-1}}{\kappa_n + u_n^* K^{-1} u_n}.$$

Insert $\kappa_n = a \kappa_{n-1}$ and the definition of M_n ; this gives (KT.6). At the next step the ordered vectors in (KT.3) are $u_{n+1} = t_n$, $w_{n+1} = u_n$. Pair each ordered pair with (KT.6). The three resulting expressions are precisely (KT.7), including their conjugations. The denominator is positive because $a > 0$, $\beta_n \geq 0$. The determinant of the rank-one update is $1 + u_n^* K^{-1} u_n / \kappa_n = 1 + \beta_n / a$; this follows directly by expanding the determinant multilinearly in its columns. The original metric is $G_m = U_\varepsilon^* K_{n-1}^{-1} U_\varepsilon$, with the same fixed invertible local unit at adjacent levels. Its determinant ratio is therefore the inverse kernel ratio. Equation (KT.7) gives the last equality in (KT.8). All calculations use polynomial residues at A ; no inverse of $q_n(A)$ has been used. $\square$

12.2 The complete boundary-energy increment

Suppose first that $\beta_n > 0$. The exact orthogonal projection for $\langle \cdot, \cdot \rangle_n$ is

$$\begin{aligned} P_n x &= x - u_n \frac{\langle u_n, x \rangle_n}{\beta_n}, & p_n &= P_n w_n, & z_n &= P_n A u_n, \\ \Delta_n &= \alpha_n \beta_n - |\gamma_n|^2 = \beta_n \|p_n\|_n^2, & \Omega_n &= \beta_n \|z_n\|_n^2. \end{aligned} \quad (\text{KT.9})$$

The projection records a specified map inside the original packet; $A u_n$, its removed coefficient, and its remaining vector z_n are all retained in (KT.9).

Theorem 12.2 (Exact transverse energy law). *At every such step,*

$$\Delta_{n+1} = \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}. \quad (\text{KT.10})$$

Consequently

$$\Delta_{n+1} - \Delta_n = \frac{\beta_n}{a_n(a_n + \beta_n)} (\|z_n\|_n^2 + 2a_n \operatorname{Re} \langle z_n, p_n \rangle_n - a_n \beta_n \|p_n\|_n^2). \quad (\text{KT.11})$$

The exact energy law gives the following two-sided bounds by the sharp triangle inequalities:

$$\frac{|a_n \sqrt{\Delta_n} - \sqrt{\Omega_n}|}{\sqrt{a_n(a_n + \beta_n)}} \leq \sqrt{\Delta_{n+1}} \leq \frac{a_n \sqrt{\Delta_n} + \sqrt{\Omega_n}}{\sqrt{a_n(a_n + \beta_n)}}. \quad (\text{KT.12})$$

The actual relative allowance is recovered exactly by

$$\epsilon^{(n)} = |g_Z| + \sqrt{g_Z^2 + \Delta_n}. \quad (\text{KT.13})$$

For a reflection-stable packet with its multiplicities, $g_Z = 0$, so the energy law calculates the square of the actual allowance without a trace term.

Proof. Let $T = \langle t_n, t_n \rangle_n$ and $v = \langle u_n, t_n \rangle_n$. Substitution of (KT.7) gives

$$\alpha_{n+1} \beta_{n+1} - |\gamma_{n+1}|^2 = \frac{\beta_n T - |v|^2}{a_n(a_n + \beta_n)} = \frac{\beta_n \|P_n t_n\|_n^2}{a_n(a_n + \beta_n)}.$$

Since $P_n u_n = 0$, the exact recurrence (KT.5) gives $P_n t_n = P_n A u_n + a_n P_n w_n = z_n + a_n p_n$. This proves (KT.10). Expand the squared norm and subtract $\Delta_n = \beta_n \|p_n\|_n^2$ to prove (KT.11). The triangle inequality and its reverse, applied to these two specific vectors, prove (KT.12). Finally (KT.4) gives (KT.13). Its real trace term is unchanged with n ; equivalently the same trace computation can be performed at each adjacent positive kernel. Reflection permutes all roots and multiplicities, so $2 \operatorname{Re} \operatorname{tr} A = d$ in that case. $\square$

If $\beta_n = 0$, positivity of M_n forces $u_n = v_n = 0$. Then $\gamma_n = 0$, the current kernel weight is zero, $K_n = K_{n-1}$, and (KT.5)–(KT.7) give the entire exceptional step:

$$t_n = a_n w_n, \quad \alpha_{n+1} = \gamma_{n+1} = 0, \quad \beta_{n+1} = a_n \alpha_n, \quad \Delta_n = \Delta_{n+1} = 0, \quad \epsilon^{(n)} = \epsilon^{(n+1)} = 0. \quad (\text{KT.14})$$

Moreover $v_n = 0$ is exactly the divisibility $h \mid q_n$. Since every root of q_n is simple and lies on the Mellin line, this particular event forces the selected packet to consist of simple line roots. It does not remove the resulting packet or the next vectors in (KT.14).

12.3 The exact rational terminal step and its polynomial continuation

Where $q_n(A)$ and $q_{n+1}(A)$ are invertible, use the actual terminal maps and Gram from Section 10:

$$T_n = F_n(A), \quad r_n = T_n c, \quad Q_n = q_n(A), \quad K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*.$$

The map Q_n is the exact isometry

$$Q_n : (E, S_n^{-1}) \longrightarrow (E, M_n), \quad Q_n c = u_n, \quad Q_n r_n = w_n. \quad (\text{KT.15})$$

Indeed $Q_n^* M_n Q_n = S_n^{-1}$ follows by inverting the displayed Gram congruence, and the vector identity uses $F_n = q_{n-1}/q_n$. This proves the precise relation between the rational and polynomial forms of (KT.3)–(KT.13).

The terminal recurrence gives

$$T_{n+1} = (A - b_n I + a_n T_n)^{-1}, \quad \boxed{S_{n+1} = T_{n+1} (c c^* + a_n S_n) T_{n+1}^*}. \quad (\text{KT.16})$$

For a direct full-jet proof, the scalar kernel of a terminal observation is

$$H_{F_n}(z, w) = \frac{F_n(z) + \overline{F_n(w)}}{z + \overline{w} - 1}.$$

The quotient denotes its removable polynomial/rational continuation where numerator and denominator both vanish. It is identically the positive resolvent Gram $\sum_j \sigma_{n,j} / [(z - \lambda_{n,j})(\overline{w} - \overline{\lambda}_{n,j})]$. Substitution of $F_{n+1} = (z - b_n + a_n F_n)^{-1}$ gives

$$H_{F_{n+1}}(z, w) = F_{n+1}(z) \overline{F_{n+1}(w)} [1 + a_n H_{F_n}(z, w)]. \quad (\text{KT.17})$$

This identity holds first away from its removable sets and hence through them by continuation. Taking the complete residue on both variables proves (KT.16). It also follows from the original kernel update and $Q_{n+1}^{-1} Q_n = T_{n+1}$. Thus a quadrature collision is handled by the polynomial formulas (KT.2)–(KT.14), which retain all coordinates; (KT.15) states exactly the isometry on the open set where the rational presentation is defined.

In this presentation the determinant still retains the entire polynomial factor:

$$\frac{\det S_{n+1}}{\det S_n} = \frac{a_n^{d-1} (a_n + \beta_n)}{|\det(A - b_n I + a_n T_n)|^2}, \quad \det(A - b_n I + a_n T_n) = \prod_{\rho \in Z} \left(\frac{q_{n+1}(\rho)}{q_n(\rho)} \right)^{m_\rho}. \quad (\text{KT.18})$$

The determinant of $a_n S_n + c c^*$ gives the first formula. The second follows by evaluating the polynomial ratio on each full triangular Taylor block $\rho I + N_\rho$. The determinant has the stated multiplicity even when the block is not semisimple.

12.4 The extra jets required to recover the metric from the tail

There is an exact map from the scalar tail identity to the matrix energy, with a specified number of Taylor coefficients. For (KT.19)–(KT.22), take centres at which q_n is nonzero, so F_n is holomorphic there. Every off-line centre meets this condition. At a critical quadrature intersection the universal polynomial formulas (KT.2)–(KT.14) retain the complete packet. The terminal function satisfies $\overline{F_n(1 - \bar{z})} = -F_n(z)$, by its positive spectral sum on the line. Thus

$$H_{F_n}(z, w) = \frac{F_n(z) - F_n(1 - \bar{w})}{z - (1 - \bar{w})}. \quad (\text{KT.19})$$

For centres ρ, σ , and Taylor orders $0 \leq k < m_\rho, 0 \leq l < m_\sigma$, let $S_n^{\text{jet}}(\rho, k; \sigma, l)$ denote the coefficient of $(z - \rho)^k (\bar{w} - \bar{\sigma})^l$ in this kernel. In particular, at a reflection pair,

$$\boxed{S_n^{\text{jet}}(\rho, k; \sigma, l) = (-1)^l F_n^{[k+l+1]}(\rho) \quad \text{if } \rho = 1 - \bar{\sigma}.} \quad (\text{KT.20})$$

Here $F^{[j]} = F^{(j)}/j!$. To prove the formula, put $z = \rho + x, 1 - \bar{w} = \rho - y$ in (KT.19), and expand

$$\frac{x^r - (-y)^r}{x + y} = \sum_{j=0}^{r-1} x^{r-1-j} (-y)^j.$$

The coefficient of $x^k y^l$ is supplied exactly by $r = k + l + 1, j = l$, proving (KT.20) without dividing by zero.

For $D_{\rho\sigma} = \rho + \bar{\sigma} - 1 \neq 0$, the corresponding complete coefficient is

$$\begin{aligned} S_n^{\text{jet}}(\rho, k; \sigma, l) &= \sum_{i=0}^k F_n^{[i]}(\rho) \frac{(-1)^{k-i+l} \binom{k-i+l}{k-i}}{D_{\rho\sigma}^{k-i+l+1}} \\ &\quad + \sum_{j=0}^l \overline{F_n^{[j]}(\sigma)} \frac{(-1)^{k+l-j} \binom{k+l-j}{k}}{D_{\rho\sigma}^{k+l-j+1}}. \end{aligned} \quad (\text{KT.21})$$

This is the full product of the numerator Taylor series and the geometric expansion of $(D_{\rho\sigma} + x + y)^{-1}$. Both constant terms in the numerator are retained. If $V_Z : E \rightarrow \bigoplus_\rho \mathbb{C}^{m_\rho}$ is the fixed complete Taylor-coordinate map, then

$$S_n^{\text{jet}} = V_Z S_n V_Z^*, \quad S_n = V_Z^{-1} S_n^{\text{jet}} (V_Z^{-1})^*. \quad (\text{KT.22})$$

This follows by applying V_Z to each resolvent column in the positive Gram representation of S_n .

To specify the domain of the analytic extraction, let $h_{\text{off}} = \prod_{\text{Re } \rho \neq 1/2} (s - \rho)^{m_\rho}$. The full original algebra projects onto $E_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}})$; the kernel is the retained critical-line CRT summand, with its original action of A . Above the off-line summand use the doubled algebra $\widehat{E}_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}}^2)$. Its exact reduction sequence of $\mathbb{C}[s]$ -modules is

$$0 \longrightarrow E_{\text{off}} \xrightarrow{[p] \mapsto [h_{\text{off}} p]} \widehat{E}_{\text{off}} \xrightarrow{[p] \mapsto [p]} E_{\text{off}} \longrightarrow 0. \quad (\text{KT.23})$$

The first map is well-defined and injective because $h_{\text{off}}^2 \mid h_{\text{off}} p$ holds exactly when $h_{\text{off}} \mid p$; its image is precisely the kernel of the reduction. All maps commute with multiplication by s . The reduction is also a unital algebra map. The image of the injection is the square-zero ideal $(h_{\text{off}})/(h_{\text{off}}^2)$: the product of the images of $[p]$ and $[q]$ is $[h_{\text{off}}^2 pq] = 0$. This records its complete inherited multiplication as well as its module isomorphism with E_{off} . The nilpotent lengths in the middle term are exactly $2m_\rho$, and reduction retains the lower m_ρ coefficients. At reflected actual zeros, the full multiplicities are equal, by the functional equation and conjugation of g . Thus (KT.20) uses at most order $2m_\rho - 1$,

exactly a coefficient of this doubled algebra. Formula (KT.21) uses only the stated lower coefficients. Equations (KT.20)–(KT.23) prove the precise passage from terminal Taylor data to the full metric.

Write $\mathcal{E}_n = m - m_n$ for the initial observation error, and retain the genuine tail ζ_n from (KR.14). For every off-line point its real part has the sign of $\operatorname{Re} z - 1/2$, so $\zeta_n \neq 0$. The exact identity (KR.16) also gives

$$\mathcal{E}_n(z) = \frac{(-1)^n \kappa_n}{q_n(z)^2} H_n^{\text{obs}}(z), \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z.$$

The real part of H_n^{obs} has that same strict sign: its defining measure is positive of total mass one. Hence $\mathcal{E}_n(z) \neq 0$ off the line as well. Solving the actual tail equation, before taking any residues, therefore gives the exact extraction

$$F_n(z) = \frac{1}{a_n} \left(\frac{(-1)^n \kappa_n}{q_n(z)^2 \mathcal{E}_n(z)} - \frac{1}{\zeta_n(z)} \right). \quad (\text{KT.24})$$

It holds in the complete doubled algebra at the off-line centres. Every reciprocal there is explicit: if $f(\rho + x) = \sum_{j=0}^{2m_\rho-1} f_j x^j$ modulo x^{2m_ρ} , with $f_0 \neq 0$, the inverse coefficients are

$$b_0 = f_0^{-1}, \quad b_k = -f_0^{-1} \sum_{j=1}^k f_j b_{k-j} \quad (1 \leq k < 2m_\rho). \quad (\text{KT.25})$$

Multiplication of the two finite series proves all coefficients. The already proved bounds (GC.13)–(GC.16) apply to every one of the error coefficients required here. Equations (KT.24)–(KT.25), then (KT.20)–(KT.22), give a complete finite extraction of the terminal metric on E_{off} from the initial error and genuine tail data. In the retained monomial quotient coordinates, set $A_{\text{off}} = M_s$, $\pi_{\text{off}}([p]_h) = [p]_{h_{\text{off}}}$, and $c_{\text{off}} = [1] = \pi_{\text{off}} c$. Explicitly, retain the same ν_Z , q_n , nodes and terminal weights, and put

$$S_n^{\text{off}} = \sum_j \sigma_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1} c_{\text{off}} c_{\text{off}}^* (A_{\text{off}}^* - \bar{\lambda}_{n,j} I)^{-1}. \quad (\text{KT.26})$$

For the original quotient map $\pi_{\text{off}} : E \rightarrow E_{\text{off}}$, the resolvent intertwining identity gives $S_n^{\text{off}} = \pi_{\text{off}} S_n \pi_{\text{off}}^*$ whenever the full rational Gram is defined. The right side of (KT.26) remains defined at every off-line centre even when the full packet has a critical quadrature collision. No replacement of the measure by $|g/h_{\text{off}}|^2 dt / (2\pi)$ occurs in this projection. The full polynomial quadrature formula, applied to these projected residue columns, proves even at such a collision that

$$\pi_{\text{off}} K_{n-1} \pi_{\text{off}}^* = \kappa_{n-1}^{-1} q_n (A_{\text{off}}) S_n^{\text{off}} q_n^* (A_{\text{off}})^*. \quad (\text{KT.27})$$

These extraction formulas retain the inverse error, the inverse tail, q_n , κ_n , and a_n , including their cancellations. The initial error bound alone is not used to replace any of those terms. The polynomial metric step (KT.6) continues to include the critical-line summand, where the off-line resolvent domain ends.

12.5 Actual quartet packets close the energy in two volume ratios

Assume here that the selected actual packet is closed under both $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \bar{\rho}$, with the complete multiplicities. These are exact invariances of the specified divisor; no value of $\operatorname{Re} \rho$ has been imposed.

Theorem 12.3 (Two consecutive arithmetic volume contractions). *Let G_{-1} be the fixed-section metric already identified with $U_\varepsilon^* K_{d-1}^{-1} U_\varepsilon$, and define the actual positive determinant ratios*

$$r_m = \frac{\det G_m}{\det G_{m-1}} \quad (m \geq 0).$$

Then $0 < r_m \leq 1$, and for every $m \geq 0$, with $n = d + m + 1$,

$$\epsilon_m^2 = \frac{\kappa_{d+m+1}}{\kappa_{d+m}} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{KT.28})$$

This is the original relative allowance and the original theta Gram sequence, including their local unit and all multiplicities.

Proof. Conjugation invariance of the divisor makes every coefficient of the monic h real. The entire function g obeys $g(\bar{s}) = \overline{g(s)}$, so the unchanged density $|g(1/2 + it)/h(1/2 + it)|^2$ is even in t . For a monic orthogonal polynomial q , form $q^c(s) = \overline{q(\bar{s})}$. Substitution $t \mapsto -t$ and conjugation of each orthogonality equation show that q^c is monic with the same degree and the same lower-degree orthogonality. Positivity of the polynomial Gram proves uniqueness, so $q^c = q$. Thus every q_n has real coefficients in the original variable s . The coefficient of q_n in sq_n is therefore real as well; (KT.1) proves its exact value $b_n = 1/2$. This is a consequence of the given packet symmetries, with every other recurrence coefficient retained.

The original companion matrix A , vector c , residue columns v_j , and Gram matrices K_{n-1}, M_n are real. Therefore the scalar γ_n in (KT.3) is real. The second specified symmetry gives $g_Z = 0$ in (KT.4), so $\gamma_n = 0$ exactly. Hence $\epsilon_m^2 = \alpha_n \beta_n$. At the preceding and following steps (KT.7)–(KT.8) give

$$\alpha_n = 1 - r_m, \quad \alpha_{n+1} = 1 - r_{m+1}, \quad \beta_n = \frac{a_n \alpha_{n+1}}{1 - \alpha_{n+1}}.$$

All these denominators are positive: $G_m > 0$ and $a_n/(a_n + \beta_n) > 0$. Substitution proves (KT.28), including the exceptional zero-column steps. The inequality $0 < r_m \leq 1$ follows directly from the same determinant formula. At $m = 0$ the preceding metric is precisely the stated G_{-1} , so no earlier undefined source level is used. $\square$

12.6 What the recurrence proves about boundary energy

The unchanged arithmetic source has supplied the explicit energy increment (KT.11). Its three terms are the squared transverse A -image, its real pairing with the retained terminal vector, and the specified negative term $-a_n \beta_n \|p_n\|_n^2$. Thus the source recurrence produces an exactly computable boundary energy, including its sign-determining cross term. No sign for their sum has been assumed. Every term is a polynomial residue, an actual moment Gram inverse, or a retained recurrence coefficient. The rational route to those same quantities is (KT.15)–(KT.25). The original theta representative and its boundary are recovered from the preceding kernel-minimum formula and its explicit division by h . Consequently all updates are maps between the original source representatives, with the same quotient class and supported zero. For each support label the displayed linear maps lift by $(\lambda, x) \mapsto (\lambda, Tx)$; external absence remains τ , and a zero vector in that fibre remains its supported zero throughout.

12.7 An exact positive-moment counterexample to consecutive energy monotonicity

This section concerns the moment-to-packet construction for the explicitly specified positive measure below. It proves that positivity, the vertical orthogonal-polynomial recurrence, reflection stability of the packet, and the terminal rank-two identity together allow an increase of the relative weight energy at consecutive degrees. The measure is specified by its nine atoms; no assertion identifies it with the arithmetic measure $d\nu_Z(t) = |g(\frac{1}{2} + it)/h(\frac{1}{2} + it)|^2 dt/(2\pi)$. The exact correspondence with that construction will be the same polynomial residue map, multiplication operator, kernel sum and Hermitian defect, with the measure retained as part of the data throughout.

12.7.1 The measure and every moment needed below

Keep the original coordinate $s = \frac{1}{2} + it$. Define a probability measure ν on this line by the following table. In a row containing two signs, the displayed weight is assigned to each of the two atoms.

t	weight at each atom	
0	115/288	
$\pm 1/4$	29/120	
$\pm 1/2$	13/240	
$\pm 3/4$	11/2520	
± 1	1/6720	(TEC.1)

All nine weights are strictly positive, and

$$\frac{115}{288} + \frac{29}{60} + \frac{13}{120} + \frac{11}{1260} + \frac{1}{3360} = 1. \quad (\text{TEC.2})$$

Write $\mu_k = \int t^k d\nu$ and $v = 1/16$. The paired atoms give $\mu_k = 0$ for every odd integer $k \geq 1$, including $k = 1, 3, 5, 7, 9$. For positive integers r , the even-moment formula, with all original atom weights retained, is

$$\mu_{2r} = 4^{-2r} \left(\frac{29}{60} 1^{2r} + \frac{13}{120} 2^{2r} + \frac{11}{1260} 3^{2r} + \frac{1}{3360} 4^{2r} \right). \quad (\text{TEC.3})$$

Here the factor 4^{-2r} comes from the explicitly displayed atom locations in (TEC.1). The four contributions and their sums are

r	$\frac{29}{60} 1^{2r}$	$\frac{13}{120} 2^{2r}$	$\frac{11}{1260} 3^{2r}$	$\frac{1}{3360} 4^{2r}$	sum	
1	29/60	13/30	11/140	1/210	1	
2	29/60	26/15	99/140	8/105	3	
3	29/60	104/15	891/140	128/105	15	
4	29/60	416/15	8019/140	2048/105	105	
5	29/60	1664/15	72171/140	32768/105	939	(TEC.4)

Consequently

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8, \mu_{10}) = (1, v, 3v^2, 15v^3, 105v^4, 939v^5). \quad (\text{TEC.5})$$

Equations (TEC.2)–(TEC.5), together with cancellation of each odd moment between the two atoms, verify all moments through degree ten.

12.7.2 The actual monic orthogonal polynomials of this measure

Use the inner product

$$\langle P, Q \rangle_\nu = \int \overline{P(\frac{1}{2} + it)} Q(\frac{1}{2} + it) d\nu(t), \quad x = s - \frac{1}{2}. \quad (\text{TEC.6})$$

The following table retains the monic polynomials in the original s -coordinate and gives their actual squared norms.

k	$q_k(s)$	$\kappa_k = \langle q_k, q_k \rangle_\nu$	
0	1	1	
1	x	v	
2	$x^2 + v$	$2v^2$	
3	$x^3 + 3vx$	$6v^3$	
4	$x^4 + 6vx^2 + 3v^2$	$24v^4$	
5	$x^5 + 10vx^3 + 15v^2x$	$114v^5$	(TEC.7)

Proof. For an explicit verification, define real polynomials

$$\begin{aligned} P_0(t) &= 1, & P_1(t) &= t, & P_2(t) &= t^2 - v, \\ P_3(t) &= t^3 - 3vt, & P_4(t) &= t^4 - 6vt^2 + 3v^2, \\ P_5(t) &= t^5 - 10vt^3 + 15v^2t. \end{aligned} \quad (\text{TEC.8})$$

Direct substitution gives $q_k(\frac{1}{2} + it) = i^k P_k(t)$. Thus

$$\langle q_j, q_k \rangle_\nu = (-i)^j i^k \int P_j(t) P_k(t) d\nu(t). \quad (\text{TEC.9})$$

If j, k have different parity, the integrand on the right is odd and its integral is zero. The remaining six off-diagonal integrals, using (TEC.5), are precisely

$$\begin{aligned} \int P_0 P_2 d\nu &= v(1 - 1) = 0, \\ \int P_0 P_4 d\nu &= v^2(3 - 6 + 3) = 0, \\ \int P_2 P_4 d\nu &= v^3(15 - 21 + 9 - 3) = 0, \\ \int P_1 P_3 d\nu &= v^2(3 - 3) = 0, \\ \int P_1 P_5 d\nu &= v^3(15 - 30 + 15) = 0, \\ \int P_3 P_5 d\nu &= v^4(105 - 195 + 135 - 45) = 0. \end{aligned} \quad (\text{TEC.10})$$

For the diagonal integrals, direct polynomial multiplication gives

$$\begin{aligned} \int P_0^2 d\nu &= 1, & \int P_1^2 d\nu &= v, \\ \int P_2^2 d\nu &= v^2(3 - 2 + 1) = 2v^2, \\ \int P_3^2 d\nu &= v^3(15 - 18 + 9) = 6v^3, \\ \int P_4^2 d\nu &= v^4(105 - 180 + 126 - 36 + 9) = 24v^4, \\ \int P_5^2 d\nu &= v^5(939 - 2100 + 1950 - 900 + 225) = 114v^5. \end{aligned} \quad (\text{TEC.11})$$

In (TEC.9), the diagonal phase is $(-i)^k i^k = 1$. These equations therefore prove every entry of the Gram matrix asserted in (TEC.7). Each polynomial in that table is monic. A nonzero polynomial of degree at most five has at most five distinct roots, so it cannot vanish at all nine atoms in (TEC.1). Positivity of the weights makes the inner product positive definite on polynomials of degree at most five. If two monic polynomials of the same degree were both orthogonal to every lower degree polynomial, their difference would have lower degree and zero squared norm. Positive definiteness forces that difference to vanish. Thus (TEC.7) consists of the actual monic orthogonal polynomials and norms of the stated measure, including the fifth norm. $\square$

Polynomial multiplication in (TEC.7) also proves

$$sq_k = q_{k+1} + \frac{1}{2}q_k - kvq_{k-1} \quad (0 \leq k \leq 4), \quad (\text{TEC.12})$$

where the last term is omitted for $k = 0$. For $1 \leq k \leq 4$, the displayed coefficient is exactly $a_k = \kappa_k / \kappa_{k-1} = kv = k/16 > 0$. Thus (TEC.12) has the required vertical sign and $b_k + \overline{b_k} = 1$, with $b_k = 1/2$. The fifth norm in (TEC.7) is $\kappa_5 = 114/16^5 = 57/524288$; its value was derived from the actual tenth moment $939/16^5$.

12.7.3 The retained packet and its exact evaluation isomorphism

Set

$$h(s) = (s - \frac{3}{4})(s - \frac{1}{4}) = s^2 - s + \frac{3}{16}, \quad E = \mathbb{C}[s]/(h), \quad j_h : \mathbb{C}[s] \longrightarrow E. \quad (\text{TEC.13})$$

Both packet centres lie in $0 < \text{Re } s < 1$, and the reflection $\rho \mapsto 1 - \bar{\rho}$ exchanges them. They are distinct, and each has its complete multiplicity one. In the retained monomial basis $(1, s)$ of E , the unit and multiplication operator are

$$c = j_h 1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = M_s = \begin{pmatrix} 0 & -3/16 \\ 1 & 1 \end{pmatrix}. \quad (\text{TEC.14})$$

Indeed $s^2 = s - 3/16$ in this quotient, which proves both columns of the multiplication matrix.

The complete evaluation map is

$$T : E \longrightarrow \mathbb{C} \oplus \mathbb{C}, \quad [p] \longmapsto (p(3/4), p(1/4)),$$

$$T = \begin{pmatrix} 1 & 3/4 \\ 1 & 1/4 \end{pmatrix}, \quad T^{-1} = \begin{pmatrix} -1/2 & 3/2 \\ 2 & -2 \end{pmatrix}. \quad (\text{TEC.15})$$

Both evaluations vanish on the ideal (h) , so the map is well defined. Evaluations preserve addition, multiplication and the unit. For any pair (u_+, u_-) , the polynomial

$$2u_+(s - \frac{1}{4}) + 2u_-(\frac{3}{4} - s) \quad (\text{TEC.16})$$

evaluates to that pair. It has the coefficients given by T^{-1} , and multiplication of the two matrices in (TEC.15) gives the identity. This proves the typed algebra isomorphism, including its inverse. Evaluation of sp also proves

$$TA = \tilde{A}T, \quad \tilde{A} = \text{diag}(3/4, 1/4), \quad Tc = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (\text{TEC.17})$$

In the original quotient E one has $x^2 = v$. Reducing the explicit polynomials of (TEC.7), with no deletion of their supported residue classes, gives

k	$v_k := j_h q_k$ in the monomial basis $(1, s)$	
0	$(1, 0)^T$	
1	$(-1/2, 1)^T$	
2	$(1/8, 0)^T$	
3	$(-1/8, 1/4)^T$	
4	$(5/128, 0)^T$	
5	$(-13/256, 13/128)^T$	(TEC.18)

For example $q_3 = 4vx = x/4$, $q_4 = 10v^2 = 5/128$, and $q_5 = 26v^2x = (13/128)x$ in E . These calculations establish every entry of (TEC.18) directly from the original ideal relation.

12.7.4 The original kernels, terminal defects and exact energies

For $1 \leq N \leq 4$, define

$$K_N = \sum_{k=0}^N \frac{v_k v_k^*}{\kappa_k},$$

$$L_N = AK_N + K_N A^* - K_N,$$

$$\epsilon_N = \left\| K_N^{-1/2} L_N K_N^{-1/2} \right\|. \quad (\text{TEC.19})$$

All adjoints in (TEC.19) are conjugate transposes in the explicitly retained monomial coordinates. The columns v_0, v_1 are linearly independent. If a nonzero column u had $u^* K_N u = 0$, the positive sum in (TEC.19) would force both $v_0^* u = 0$ and $v_1^* u = 0$, a contradiction. Thus each K_N is positive definite, so its positive inverse square root and the displayed Hermitian operator norm are well defined.

Using (TEC.7) and (TEC.18), the successive kernels are

$$\begin{aligned} K_1 &= \begin{pmatrix} 5 & -8 \\ -8 & 16 \end{pmatrix}, & K_2 &= \begin{pmatrix} 7 & -8 \\ -8 & 16 \end{pmatrix}, \\ K_3 &= \begin{pmatrix} 53/3 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}, & K_4 &= \begin{pmatrix} 131/6 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}. \end{aligned} \quad (\text{TEC.20})$$

In particular the last step is the exact positive rank-one update

$$K_4 - K_3 = \frac{v_4 v_4^*}{\kappa_4} = \frac{25}{6} c c^*. \quad (\text{TEC.21})$$

Multiplication by the matrix in (TEC.14) gives

$$L_3 = \begin{pmatrix} -20/3 & 20/3 \\ 20/3 & 0 \end{pmatrix}, \quad L_4 = \begin{pmatrix} -65/6 & 65/6 \\ 65/6 & 0 \end{pmatrix}. \quad (\text{TEC.22})$$

These defects satisfy the original terminal identity, with its exact coefficients and signs:

$$L_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N} \quad (N = 3, 4). \quad (\text{TEC.23})$$

For $N = 3$, the numerator calculated from (TEC.18) is

$$\begin{pmatrix} -5/512 & 5/512 \\ 5/512 & 0 \end{pmatrix}, \quad \kappa_3 = 3/2048.$$

For $N = 4$, it is

$$\begin{pmatrix} -65/16384 & 65/16384 \\ 65/16384 & 0 \end{pmatrix}, \quad \kappa_4 = 3/8192.$$

Division gives exactly the two matrices in (TEC.22), proving (TEC.23) without an appeal to a numerical rank calculation.

Theorem 12.4 (Strict increase at two consecutive permitted levels). *For the positive measure (TEC.1) and the complete reflection-stable packet (TEC.13), the energies in (TEC.19) satisfy*

$$\epsilon_3^2 = \frac{25}{99}, \quad \epsilon_4^2 = \frac{4225}{15136}, \quad \epsilon_4^2 - \epsilon_3^2 = \frac{39875}{1498464} > 0. \quad (\text{TEC.24})$$

In the indexing $N = d + m$, with $d = \dim_{\mathbb{C}} E = 2$, these are the consecutive levels $m = 1$ and $m = 2$, both with $m \geq 0$.

Proof. Transport the two forms by the proved isomorphism (TEC.15):

$$\begin{aligned} \tilde{K}_3 &= T K_3 T^* = \begin{pmatrix} 20/3 & -2/3 \\ -2/3 & 20/3 \end{pmatrix}, \\ \tilde{K}_4 &= T K_4 T^* = \begin{pmatrix} 65/6 & 7/2 \\ 7/2 & 65/6 \end{pmatrix}. \end{aligned} \quad (\text{TEC.25})$$

By (TEC.17), the transported defect is

$$\tilde{L}_N = T L_N T^* = \tilde{A} \tilde{K}_N + \tilde{K}_N \tilde{A}^* - \tilde{K}_N. \quad (\text{TEC.26})$$

For either matrix $\tilde{K} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ in (TEC.25), one has

$$\tilde{L} = \text{diag}(a/2, -a/2), \quad \det(\tilde{L} - \lambda\tilde{K}) = \lambda^2(a^2 - b^2) - a^2/4. \quad (\text{TEC.27})$$

To check that this calculation gives the norm defined in the original coordinates, set $H = K^{-1/2}LK^{-1/2}$. Then

$$\det(H - \lambda I) = \det(K)^{-1} \det(L - \lambda K), \quad \det(T(L - \lambda K)T^*) = |\det T|^2 \det(L - \lambda K). \quad (\text{TEC.28})$$

The two determinant factors are nonzero and independent of λ . Consequently the eigenvalues of the Hermitian matrix H equal the roots of (TEC.27), with their multiplicities. The roots are the opposite real numbers $\pm a/(2\sqrt{a^2 - b^2})$. A Hermitian matrix has operator norm equal to the greatest absolute value of its eigenvalues, so

$$\epsilon^2 = \frac{a^2}{4(a^2 - b^2)}. \quad (\text{TEC.29})$$

For $N = 3$, substitution of $a = 20/3, b = -2/3$ gives $\epsilon_3^2 = (400/9)/(4 \cdot 396/9) = 25/99$. For $N = 4$, substitution of $a = 65/6, b = 7/2$ gives $\epsilon_4^2 = (4225/36)/(4 \cdot 3784/36) = 4225/15136$. Their difference, retaining the two denominators, is

$$\frac{4225 \cdot 99 - 25 \cdot 15136}{15136 \cdot 99} = \frac{418275 - 378400}{1498464} = \frac{39875}{1498464} > 0. \quad (\text{TEC.30})$$

Both energies are nonnegative, and their squared values have strictly positive difference, so $\epsilon_4 > \epsilon_3$. The original coordinate characteristic polynomials provide the same calculation:

$$\begin{aligned} \det(L_3 - \lambda K_3) &= 176\lambda^2 - 400/9, \\ \det(L_4 - \lambda K_4) &= (3784/9)\lambda^2 - 4225/36. \end{aligned} \quad (\text{TEC.31})$$

Finally $N = 3 = 2 + 1$ and $N = 4 = 2 + 2$, proving the stated degree range without using the lower degree $N = 1$. $\square$

12.7.5 The exact scope of the obstruction

For this measure the construction is the composite of the polynomial source with the original residue quotient

$$\mathcal{P}_N \hookrightarrow \mathbb{C}[s] \xrightarrow{j_h} E, \quad E \xrightarrow{T} \mathbb{C} \oplus \mathbb{C}, \quad (\text{TEC.32})$$

and multiplication is intertwined exactly by (TEC.17). Its source inner product is (TEC.6); its monic source Gram matrix is $\text{diag}(\kappa_0, \dots, \kappa_N)$; its kernel is the full sum (TEC.19). Thus it has precisely the vertical recurrence, positive kernel update, complete packet quotient and terminal defect maps appearing in the arithmetic moment construction. Equations (TEC.24) and (TEC.30) prove that these data alone do not imply nonincreasing consecutive energy. An argument for that property for ν_Z must therefore establish and use additional identities or estimates of its stated arithmetic density. The present example establishes the indicated obstruction to a universal recurrence-based argument; it asserts no consecutive increase for ν_Z .

The companion exact checker verifies 88 rational and complex-rational identities in both ordinary Python and optimized Python. It checks the atom weights and relevant moments, source Gram matrix, packet isomorphism, residue columns, kernel updates, terminal identities and generalized characteristic polynomials. Those finite checks supplement the complete computations above.

13 The exact metric departure and the cost of complete jets

Retain the original packet $E = \mathbb{C}[s]/h$, its polynomial coordinates and $A = M_s$. Write its eigenvalues with their algebraic multiplicities as $\rho_1, \dots, \rho_d$, and put $\delta_j = \operatorname{Re} \rho_j - 1/2$. No eigenvalue or repeated factor is discarded. For a positive Hermitian matrix G in these coordinates define

$$\begin{aligned} W_G &= A^*G + GA - G, \quad S_G = G^{-1/2}W_GG^{-1/2}, \quad \epsilon_G = \|S_G\|_{\text{op}}, \\ \mathfrak{d}_G(A) &= \operatorname{Tr}(G^{-1}A^*GA) - \sum_{j=1}^d |\rho_j|^2. \end{aligned} \quad (\text{MD1})$$

The positive square root is used only for the explicit isometry $x \mapsto G^{1/2}x$ from the specified metric space (E, G) to the Euclidean coordinate space. The original A, G , their entries and their maps remain those in (MD1).

Theorem 13.1 (Exact departure identity). *The number $\mathfrak{d}_G(A)$ is real and nonnegative, and*

$$\operatorname{Tr} S_G = 2 \sum_j \delta_j, \quad \operatorname{Tr}(S_G^2) = 4 \sum_j \delta_j^2 + 2\mathfrak{d}_G(A). \quad (\text{MD2})$$

*It vanishes precisely when A is normal for the specified metric, that is, when $A(G^{-1}A^*G) = (G^{-1}A^*G)A$. For the original orthogonal representatives R_m , their matrices G_m and rank-two forms W_m , one has exactly*

$$\epsilon_m = \left| \sum_j \delta_j \right| + \sqrt{2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j \right)^2}. \quad (\text{MD3})$$

For a reflection-stable packet this becomes

$$\boxed{\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A)}. \quad (\text{MD4})$$

Proof. Let $C = G^{1/2}AG^{-1/2}$. Then $S_G = C + C^* - I$ and $\operatorname{Tr}(G^{-1}A^*GA) = \operatorname{Tr}(C^*C)$. For completeness, choose a unit eigenvector of C , extend it to an orthonormal basis, and repeat on the lower-right compression. Induction gives a unitary matrix U with upper-triangular $T = U^*CU$ whose diagonal lists the same eigenvalues with their algebraic multiplicities. Unitary invariance of the trace gives

$$\mathfrak{d}_G(A) = \sum_{i < j} |T_{ij}|^2 \geq 0. \quad (\text{MD5})$$

If it is zero, T is diagonal and C is normal. Conversely, an upper-triangular normal matrix is diagonal: its first row norm equals its first column norm, eliminating the strict first-row entries, and induction eliminates every remaining such entry. Thus normality of C is equivalent to the stated equality for A , through the specified isometry.

Expanding $(C + C^* - I)^2$ under the trace gives

$$\operatorname{Tr}(S_G^2) = 2 \operatorname{Tr}(C^*C) + 2 \operatorname{Re} \operatorname{Tr}(C^2) - 4 \operatorname{Re} \operatorname{Tr} C + d.$$

The traces of C and C^2 are respectively $\sum_j \rho_j$ and $\sum_j \rho_j^2$, even in the presence of nilpotents. Since $2|\rho|^2 + 2 \operatorname{Re} \rho^2 - 4 \operatorname{Re} \rho + 1 = 4(\operatorname{Re} \rho - 1/2)^2$, this proves (MD2). The trace formula follows directly from $S_G = C + C^* - I$.

For G_m , Section 6 proved $\text{rank } S_{G_m} \leq 2$. List its two possibly nonzero real eigenvalues a, b , appending zero if necessary (also when $d = 1$). Equations (MD2) give

$$a + b = 2 \sum_j \delta_j, \quad (a - b)^2/4 = 2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - (\sum_j \delta_j)^2.$$

Thus the radicand is nonnegative, and $\max(|a|, |b|) = |(a + b)/2| + |a - b|/2$, proving (MD3). Reflection-stability pairs δ with $-\delta$, with the same multiplicities, proving (MD4). $\square$

The actual arithmetic kernel of Section 9 also gives this quantity without changing the local unit. Indeed $G_m = U_\varepsilon^* K_N^{-1} U_\varepsilon$, $N = d + m$, and $U_\varepsilon A = A U_\varepsilon$. Cyclicity of trace gives

$$\mathfrak{d}_{G_m}(A) = \text{Tr}(K_N A^* K_N^{-1} A) - \sum_j |\rho_j|^2. \quad (\text{MD6})$$

To verify the cancellation, substitute $G_m^{-1} = U_\varepsilon^{-1} K_N (U_\varepsilon^*)^{-1}$ in (MD1), commute A with U_ε and A^* with U_ε^* , and cancel the outer similar factors inside the trace. This calculation removes no source representative: their exact recovery remains (TB.7)–(TB.9).

13.1 A quantitative constraint on every retained nilpotent chain

For a selected root ρ of full order r , let

$$I_\rho : \mathbb{C}[z]/z^r \longrightarrow E$$

be the original Chinese-remainder inclusion: the ρ -component is the given Taylor polynomial in $z = s - \rho$, and every other component is zero. Its Gram matrix is $G_\rho = I_\rho^* G I_\rho$, in the basis $1, z, \dots, z^{r-1}$. Put $N_\rho z^k = z^{k+1}$, with $z^r = 0$. The exact intertwining relation is $A I_\rho = I_\rho(\rho I + N_\rho)$.

Theorem 13.2 (Full-jet metric inequalities). *For every positive original metric G , write $\delta = \text{Re } \rho - 1/2$. Then $\epsilon_G \geq 2|\delta|$, and for all $0 \leq k < k + \ell < r$,*

$$\frac{(G_\rho)_{k+\ell, k+\ell}}{(G_\rho)_{k, k}} \leq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^\ell. \quad (\text{MD7})$$

In particular, for $r \geq 2$,

$$\kappa_2(G_\rho) \geq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^{-(r-1)}. \quad (\text{MD8})$$

The bracket is strictly positive when $r \geq 2$. Here κ_2 is the ratio of the largest and smallest eigenvalues in the specified Taylor coordinates.

Proof. The definition of ϵ_G is the two-sided form inequality $-\epsilon_G G \leq W_G \leq \epsilon_G G$. On an eigenvector it gives $\epsilon_G \geq 2|\delta|$. Pull this same inequality back by I_ρ . Let $L = G_\rho^{1/2} N_\rho G_\rho^{-1/2}$ and $T = 2\delta I + L + L^*$. The pulled-back inequality gives $\|T\| \leq \epsilon_G$. Nilpotence retains $\text{Tr } L = \text{Tr}(L^2) = 0$, whence

$$\text{Tr } T = 2r\delta, \quad \|L\|_F^2 = \frac{1}{2} \text{Tr}(L + L^*)^2 = \frac{1}{2} \text{Tr } T^2 - 2r\delta^2 \leq \frac{r}{2} (\epsilon_G^2 - 4\delta^2). \quad (\text{MD9})$$

The first equality for the Frobenius norm follows by expanding the square and using the two zero traces. The final inequality uses the r real eigenvalues of T , each bounded in modulus by ϵ_G . For the original metric norm, $\|N_\rho\|_{G_\rho} = \|L\| \leq \|L\|_F$. Apply its ℓ -th power to the specified basis vector z^k . Since $N_\rho^\ell z^k = z^{k+\ell}$, the resulting squared norm inequality is exactly (MD7).

When $r \geq 2$, $N_\rho \neq 0$, so (MD9) makes the bracket strictly positive. Put $k = 0, \ell = r - 1$. The Rayleigh inequalities give $\kappa_2(G_\rho) \geq (G_\rho)_{0,0}/(G_\rho)_{r-1,r-1}$; combine with (MD7) to obtain (MD8). $\square$

All entries in (MD7) for the original sequence are explicitly $I_\rho^* U_\varepsilon^* K_N^{-1} U_\varepsilon I_\rho$. Thus this theorem supplies constraints on the actual moment calculation, with its original Taylor factors. At a critical root of order $r \geq 2$, a sequence with relative control approaching zero necessarily has the stated diverging Taylor-metric condition number. The surviving jets have not been erased; the inequalities calculate their relative metric scales. This is compatible with the complete graph and weighted-jet maps in Section 104, which retain those same coordinates.

14 Coherent packets, joint tensor layers, and their exact spectral cost

This section integrates the complete arithmetic interpolation construction of the 12 September coherent-interpolation delivery, whose source revision is 3bff1b2bd87dfa2592f65c5ccc231f6e e9541dfc, and proves its comparison with the preceding kernel and departure calculations. All zeros in a packet retain their complete multiplicities. A reflected packet is required only in statements explicitly using its reflection. The original spaces $V, \mathcal{B}$, map Θ , generator $D = -x\partial_x$, and Hilbert observation $L^2(dx)$ remain fixed.

14.1 The arithmetic unit and the coherent one-variable maps

For $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, write

$$d = \deg h, \quad E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1}, \quad g = 2\xi. \quad (\text{CT.1})$$

Complete multiplicities make v_h invertible. Let $F_h \in \mathcal{B}$ be the original inverse satisfying $h(D)F_h = f_0$ and $\mathcal{M}F_h = g/h$, and put $\mathcal{T}_h P = P(D)F_h$. The integral construction of F_h , its estimates at both ends, and injectivity of the Mellin transform are proved in Section 6. They give

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (\text{CT.2})$$

Let $C_N : \mathcal{P}_N \rightarrow E_h$ be the full remainder map in the original monomial coordinates, let $M_{h,N}$ be the Gram matrix of $\mathcal{P}_N$ for $d\nu_h = |g(1/2 + it)/h(1/2 + it)|^2 dt/(2\pi)$, and use U_a for multiplication by a on E_h . The two kernel matrices occurring in the constructions have the exact types

$$K_N = C_N M_{h,N}^{-1} C_N^*, \quad J_{h,N} = U_{v_h} C_N, \quad \mathcal{C}_{h,N} = U_{v_h} K_N U_{v_h}^*. \quad (\text{CT.3})$$

Consequently their actual representative and metric coincide:

$$\begin{aligned} M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1} &= M_{h,N}^{-1} C_N^* K_N^{-1} U_{\varepsilon_h}, \\ G_{h,n} &= \mathcal{C}_{h,n+d}^{-1} = U_{\varepsilon_h}^* K_{n+d}^{-1} U_{\varepsilon_h}. \end{aligned} \quad (\text{CT.4})$$

Indeed invert the congruence in (CT.3) and cancel the adjacent inverse factors. The first line is exactly the polynomial numerator in (TB.7)–(TB.9). Its difference from $\text{rem}_h(\varepsilon_h u)$ is divisible by h and has quotient of degree at most n . Applying (CT.2) gives the same original theta correction. Thus the arithmetic unit remains in every coefficient map and every source representative.

For completeness, the minimum assertion follows directly without an existence assumption. For any positive Gram matrix M and surjective J , set $G = (JM^{-1}J^*)^{-1}$ and $L = M^{-1}J^*G$. Then $JL = I$, $L^*ML = G$, and for $Jz = 0$, $z^*ML = z^*J^*G = 0$. Every feasible vector is $Lu + z$, so

$$(Lu + z)^* M (Lu + z) = u^* G u + z^* M z. \quad (\text{CT.5})$$

Positivity makes this the unique minimum. In (CT.3), positivity holds because a nonzero polynomial cannot vanish almost everywhere on the original integration line; surjectivity follows from complete polynomial remainders. The kernel is exactly $h\mathcal{P}_n$ when $N = n + d$. Hence $R_{h,n} = (I - P_{L_n})s_h$, where $s_h = \mathcal{T}_h \text{rem}_h(\varepsilon_h \cdot)$ and $L_n = \Theta \text{span}(\phi_*, \dots, D^n \phi_*)$.

If $h \mid H$ are full packets, put $m = H/h$. The factors h, m are coprime. The original CRT injection $\iota : E_h \rightarrow E_H$ inserts zero jets at the additional centres, and satisfies $A_H \iota = \iota A_h$. Multiplication $P \mapsto mP$ is an isometry of the declared polynomial images, since $(g/H)mP = (g/h)P$; it satisfies

$$\mathcal{T}_H(mP) = \mathcal{T}_h P, \quad J_H \mathcal{T}_H(mP) = \iota J_h \mathcal{T}_h P. \quad (\text{CT.6})$$

At the added centres the second identity follows from the full zeros of g/h ; at the old centres it is the identity of entire functions. Both numerators for $s_H \iota = \mathcal{T}_H(m \operatorname{rem}_h(\varepsilon_h \cdot))$ have degree below $\deg H$ and the same complete jets, proving equality. Since L_n is independent of the packet,

$$R_{H,n} \iota = R_{h,n}, \quad \iota^* G_{H,n} \iota = G_{h,n}, \quad \iota^* W_{H,n} \iota = W_{h,n}. \quad (\text{CT.7})$$

These are congruences, with no interchange of inversion and compression. Their split maps at a retained support λ are $(\lambda, u) \mapsto (\lambda, \iota u)$; all added zero jets are represented zeros at that support.

14.2 The original reflection and the exact diagonal coefficient

The original arithmetic density $|g(1/2 + it)|^2$ is even in t , since the real Taylor coefficients of g give $g(1/2 - it) = \overline{g(1/2 + it)}$. If Q_j is the monic orthogonal polynomial for that density in the variable s , the polynomial $(-1)^j Q_j(1 - s)$ is monic and orthogonal to all lower degrees, by the change $t \mapsto -t$. Uniqueness therefore gives

$$Q_j(1 - s) = (-1)^j Q_j(s), \quad \int t |Q_j(1/2 + it)|^2 |g(1/2 + it)|^2 \frac{dt}{2\pi} = 0. \quad (\text{CT.8})$$

The diagonal recurrence coefficient is thus exactly $\langle Q_j, sQ_j \rangle / \|Q_j\|^2 = 1/2$. For lower degrees the identity $D^* + D = I$ eliminates every coefficient except the adjacent one and gives its sign and actual norm ratio. More explicitly, sQ_j is orthogonal to Q_i for $i \leq j - 2$, because its inner product is the inner product of Q_j with $(1 - s)Q_i$. For $i = j - 1$ the leading term gives coefficient $-\|Q_j\|^2 / \|Q_{j-1}\|^2$. This proves, with the original vectors and norms,

$$Du_j = u_{j+1} + \frac{1}{2}u_j - \frac{\mathfrak{h}_j}{\mathfrak{h}_{j-1}}u_{j-1}, \quad u_j = Q_j(D)f_0, \quad \mathfrak{h}_j = \|u_j\|^2. \quad (\text{CT.9})$$

The final term is omitted at $j = 0$. The even density in (CT.8) is the additional reason the imaginary part of the diagonal coefficient vanishes; the adjoint identity by itself fixes only its real part.

For a packet stable under $\rho \mapsto 1 - \bar{\rho}$, including the same complete orders, the original maps are

$$\mathfrak{j}_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}, \quad \mathfrak{j}_{\mathcal{B}} F(x) = x^{-1} \overline{F(1/\bar{x})}.$$

The second is antiunitary for dx , and $D\mathfrak{j}_{\mathcal{B}} = \mathfrak{j}_{\mathcal{B}}(I - D)$. The identities $h^\dagger = (-1)^d h$ and $g^\dagger = g$ imply $(g/h)^\dagger = (-1)^d g/h$; they give $\mathfrak{j}_{\mathcal{B}} s_h = s_h \mathfrak{j}_h$ by the same complete jets and degree bounds as above. Equation (CT.9), or invariance of the polynomial space under $s \mapsto 1 - s$, shows $\mathfrak{j}_{\mathcal{B}} L_n = L_n$. The unique minimum (CT.5) then commutes with these maps. If $\mathfrak{j}_h u = C_h \bar{u}$,

$$C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h(I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}. \quad (\text{CT.10})$$

Substitution into a quadratic form proves that an upper bound and its reflected lower bound use this same $G_{h,n}$.

14.3 The joint total-degree source and its relation layer

Fix $k \geq 1$, $M \geq k(d - 1)$, and set

$$E^{(k)} = E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{P}_M^{(k)} = \mathbb{C}[s_1, \dots, s_k]_{\leq M}, \quad \mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}.$$

Here $\leq M$ means total degree. The full jet map is reduction by each $h(s_j)$, followed by multiplication by $v_h^{\otimes k}$, and its kernel is

$$\mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)). \quad (\text{CT.11})$$

Successive monic division in the declared order $s_1, \dots, s_k$ gives $P = \sum_j h(s_j)P_j$ in this kernel, with $\deg P_j \leq M - d$. Division lowers total degree at every replacement, so this bound holds at every division step. In the original complex $[V \rightarrow \mathcal{B}]$, placed in degrees 0, 1, the primitive for the j -th term uses $(-1)^{j-1}\phi_*$ in the j -th factor and F_h in the others, with the polynomial $P_j(D_1, \dots, D_k)$ applied to the whole tensor. The preceding $j - 1$ factors have degree one, so the tensor differential supplies the further factor $(-1)^{j-1}$. Its boundary is exactly $+\mathcal{T}_h^{(k)}(h(s_j)P_j)$.

Write $\mathcal{B}_M = \mathcal{T}_h^{(k)}\mathcal{I}_{h,k,M}$ inside $\mathcal{H}_k = L^2(d^k x)$. The arithmetic moment matrix in the retained monomials is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s^{\alpha_j}} s^{\beta_j} d\nu_h, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (\text{CT.12})$$

Product integration proves the first formula. A polynomial vanishing almost everywhere for this product measure is zero, by successive one-variable uniqueness, proving positivity. The complete remainder basis has total degree at most $k(d - 1)$, proving surjectivity. Thus (CT.5) proves

$$R_M = (I - P_{\mathcal{B}_M})s_h^{\otimes k}, \quad J^{(k)}R_M = I, \quad q^{(k)}R_M = \sigma_h^{\otimes k}. \quad (\text{CT.13})$$

The kernel dimension is $\binom{M+k}{k} - d^k$. Since $\mathcal{T}_h^{(k)}$ is injective by its entire Mellin multiplier, the orthogonal relation layer $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ has exactly

$$\dim \mathcal{E}_M = \binom{M+k}{k-1}. \quad (\text{CT.14})$$

Define all of the following maps on the original $E^{(k)}$:

$$B_M = D^{(k)}R_M - R_MA^{(k)}, \quad Y_M = P_{\mathcal{E}_M}R_M, \quad C_M = (I - P_{\mathcal{B}_M})B_M. \quad (\text{CT.15})$$

The complete jet of B_M is zero and its numerator has degree at most $M + 1$, so $B_M(E^{(k)}) \subseteq \mathcal{B}_{M+1}$. Both Y_M, C_M therefore have target $\mathcal{E}_M$. Integration by parts on these rapidly decreasing functions gives $(D^{(k)})^* + D^{(k)} = kI$. Substitution into $R_M^*(D^{(k)} + (D^{(k)})^*)R_M = kG_M$, followed by orthogonality to $\mathcal{B}_M$, proves

$$\begin{aligned} W_M &= (A^{(k)})^*G_M + G_MA^{(k)} - kG_M = -(Y_M^*C_M + C_M^*Y_M), \\ \Delta_M &= G_M - G_{M+1} = Y_M^*Y_M \succeq 0. \end{aligned} \quad (\text{CT.16})$$

For the second identity use the orthogonal decomposition of $\mathcal{B}_{M+1}$ and $R_{M+1} = R_M - Y_M$. The actual quotient map $[z] \mapsto (I - P_{\mathcal{B}_M})z$ identifies $\mathcal{B}_{M+1}/\mathcal{B}_M$ with $\mathcal{E}_M$, with inverse the inclusion followed by quotient. A transported boundary is zero at its next support and remains represented there; this map sends no supported input to external τ .

For a reflection-stable packet, put $C_{h,k} = C_h^{\otimes k}$. The original product reflection acts on a numerator by $(-1)^{kd}\overline{P(1 - \overline{s_1}, \dots, 1 - \overline{s_k})}$. This retains total degree and preserves the ideal in (CT.11), with its displayed factor. Its action on $\mathcal{H}_k$ is antiunitary. The unique minimum (CT.5) therefore commutes with reflection and gives

$$C_{h,k}^*G_MC_{h,k} = \overline{G_M}, \quad A^{(k)}C_{h,k} = C_{h,k}(kI - \overline{A^{(k)}}), \quad C_{h,k}^*W_MC_{h,k} = -\overline{W_M}. \quad (\text{CT.16a})$$

If $W_M u = \lambda G_M u$, conjugation followed by $(C_{h,k}^*)^{-1}$ in the last identity gives $W_M C_{h,k} \overline{u} = -\lambda G_M C_{h,k} \overline{u}$. All generalized eigenvalues are real because $G_M > 0$. This invertible antilinear

map pairs their full multiplicities. The source-layer maps themselves retain the reflection types. If $j_E = j_h^{\otimes k}$ and j_E is the restriction of the original function reflection to $\mathcal{E}_M$, then

$$j_E Y_M = Y_M j_E, \quad j_E C_M = -C_M j_E. \quad (\text{CT.16b})$$

Both boundary projections commute with reflection, proving the first identity. For the second use $D^{(k)} j_H = j_H(kI - D^{(k)})$, $A^{(k)} j_E = j_E(kI - A^{(k)})$, and $j_H R_M = R_M j_E$ in (CT.15). Expanding its two terms gives $j_H B_M = -B_M j_E$; the projection then proves (CT.16b).

14.4 Nested packets on the joint source: the additional projection

Let $h \mid H$, $q = \deg(H/h)$, $N = M + kq$, and $\iota_k = \iota_{hH}^{\otimes k}$. Multiplication of a numerator by $\prod_{j=1}^k (H/h)(s_j)$ gives a degree-preserving-with-shift map $\mathcal{P}_M^{(k)} \rightarrow \mathcal{P}_N^{(k)}$, with

$$\mathcal{T}_H^{(k)}(\prod_j (H/h)(s_j)P) = \mathcal{T}_h^{(k)}P, \quad J_H^{(k)} \mathcal{T}_h^{(k)}P = \iota_k J_h^{(k)} \mathcal{T}_h^{(k)}P. \quad (\text{CT.17})$$

The second equality follows on pure monomials from (CT.6) and then by linearity. In particular it maps the smaller relation space into the larger one in the same $\mathcal{H}_k$. Let

$$\mathcal{F} = \mathcal{B}_{H,k,N} \cap \mathcal{B}_{h,k,M}^\perp, \quad Z = P_{\mathcal{F}} R_{h,k,M}.$$

Since $s_H^{\otimes k} \iota_k = s_h^{\otimes k}$, the exact comparison is

$$\begin{aligned} R_{H,k,N} \iota_k &= R_{h,k,M} - Z, \\ \iota_k^* G_{H,k,N} \iota_k &= G_{h,k,M} - Z^* Z, \\ \iota_k^* W_{H,k,N} \iota_k &= W_{h,k,M} - ((A_h^{(k)})^* Z^* Z + Z^* Z A_h^{(k)} - k Z^* Z). \end{aligned} \quad (\text{CT.18})$$

Indeed nested orthogonal projections commute, and their difference is $P_{\mathcal{F}}$; apply them to the common section to prove the first line. Pythagoras proves the second; intertwining of $A^{(k)}$ and direct expansion prove the third. Also

$$\dim \mathcal{F} = \binom{N+k}{k} - (\deg H)^k - \binom{M+k}{k} + d^k. \quad (\text{CT.19})$$

This is the difference of the exact kernel dimensions. For $k = 1$, $M = n + d$, both boundary spaces are the original L_n , so $\mathcal{F} = 0$ and (CT.18) recovers (CT.7). For joint tensors (CT.18) retains every additional allowed boundary and calculates its effect on the control form.

14.5 Positive layer rank and the complete tensor spectral departure

List $\rho_1, \dots, \rho_d$ with all algebraic multiplicities, put $\delta_i = \text{Re } \rho_i - 1/2$, $\sigma_1 = \sum_i \delta_i$, $\sigma_2 = \sum_i \delta_i^2$, and define

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2}, \quad \epsilon_M = \|S_M\|_{\text{op}}, \quad r_M = \text{rank } \Delta_M, \\ \mathfrak{d}_M^{(k)} &= \text{Tr}(G_M^{-1} (A^{(k)})^* G_M A^{(k)}) - \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \rho_{i_j} \right|^2, \\ \mathcal{D}_k &= \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \delta_{i_j} \right|. \end{aligned} \quad (\text{CT.20})$$

All sums are ordered and retain repeated entries. The isometry $x \mapsto G_M^{1/2} x$ in this calculation leaves the original jet coordinates, source functions, metric and layer maps unchanged. For a map into $\mathcal{E}_M$, the Frobenius symbol below denotes its Hilbert–Schmidt norm for the layer’s inherited $L^2(d^k x)$ inner product.

Theorem 14.1 (Exact tensor departure and relative layer cost). *One has $\mathfrak{d}_M^{(k)} \geq 0$, and exactly*

$$\begin{aligned} \text{Tr } S_M &= 2kd^{k-1}\sigma_1, \\ \text{Tr}(S_M^2) &= 4\{kd^{k-1}\sigma_2 + k(k-1)d^{k-2}\sigma_1^2\} + 2\mathfrak{d}_M^{(k)}. \end{aligned} \quad (\text{CT.21})$$

The positive and negative inertias of W_M each obey

$$n_+(W_M), n_-(W_M) \leq \min(\text{rank } Y_M, \text{rank } C_M), \quad \text{rank } Y_M = r_M \leq \binom{M+k}{k-1}. \quad (\text{CT.22})$$

The complete ordered spectral cost satisfies

$$\boxed{\mathcal{D}_k \leq \frac{1}{2}\|S_M\|_1 \leq \|Y_M G_M^{-1/2}\|_F \|C_M G_M^{-1/2}\|_F}, \quad \mathcal{D}_k \leq r_M \epsilon_M. \quad (\text{CT.23})$$

In particular, the actual source matrices satisfy

$$\mathcal{D}_k^2 \leq \text{Tr}(G_M^{-1} \Delta_M) \text{Tr}(G_M^{-1} C_M^* C_M). \quad (\text{CT.24})$$

If the packet is reflection-stable, put $p_M = n_+(W_M) = n_-(W_M)$. Then

$$p_M \leq \min\{r_M, \text{rank } C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k \leq p_M \epsilon_M. \quad (\text{CT.24a})$$

In this case (CT.21) becomes $\text{Tr}(S_M^2) = 4kd^{k-1}\sigma_2 + 2\mathfrak{d}_M^{(k)}$. When $p_M > 0$, it implies

$$\epsilon_M^2 \geq \frac{2kd^{k-1}\sigma_2 + \mathfrak{d}_M^{(k)}}{p_M}. \quad (\text{CT.25})$$

Proof. Triangularize A_h over $\mathbb{C}$. The tensor product of this basis makes $A^{(k)}$ triangular, with diagonal entries $\sum_j \rho_{i_j}$. Thus these are all its eigenvalues with algebraic multiplicity; its nilpotent entries have not been discarded. Now apply the unitary triangularization argument of (MD5) to $T = G_M^{1/2} A^{(k)} G_M^{-1/2}$. The sum of squared absolute values of its strictly upper triangular entries is exactly $\mathfrak{d}_M^{(k)}$, proving nonnegativity. Since $S_M = T + T^* - kI$, expansion of its squared trace gives

$$\text{Tr}(S_M^2) = 4 \sum_{i_1, \dots, i_k} \left(\sum_j \delta_{i_j} \right)^2 + 2\mathfrak{d}_M^{(k)}.$$

The k diagonal terms in this ordered sum contribute $kd^{k-1}\sigma_2$; the $k(k-1)$ ordered cross terms contribute $k(k-1)d^{k-2}\sigma_1^2$. The linear trace is computed in the same way. This proves (CT.21), including $k = 1$.

Write $\hat{Y} = Y_M G_M^{-1/2}$ and $\hat{C} = C_M G_M^{-1/2}$. Then $S_M = -(\hat{Y}^* \hat{C} + \hat{C}^* \hat{Y})$. Its form is zero on $\ker \hat{Y}$. A positive definite subspace of dimension greater than $\text{rank } \hat{Y}$ intersects this kernel nontrivially, a contradiction; the same argument holds for a negative definite subspace and with $\hat{C}$ in place of $\hat{Y}$. Sylvester inertia under the displayed invertible congruence proves the first part of (CT.22). The identity $\Delta_M = Y_M^* Y_M$ gives equality of their kernels, hence their ranks, proving the rest.

In the unitary triangular basis of T , the diagonal of S_M consists of $2 \sum_j \delta_{i_j}$. For any Hermitian matrix S , its spectral decomposition gives $|S_{ii}| \leq \sum_a |\lambda_a| |U_{ia}|^2$. Sum over i to obtain $\sum_i |S_{ii}| \leq \sum_a |\lambda_a| = \|S\|_1$. This proves the first inequality in (CT.23). For finite matrices B, C with a common row space, $\|B^* C\|_1 \leq \|B\|_F \|C\|_F$: choose a singular-value decomposition of $B^* C$, use its left and right singular vectors to express its trace norm as $|\text{Tr}(UB^* C)|$ for a unitary U , and apply Cauchy-Schwarz to the entries of BU^* and C . The triangle inequality now proves the second inequality of (CT.23). Finally (CT.22) bounds the number of nonzero eigenvalues of S_M by $2r_M$, so $\|S_M\|_1 \leq 2r_M \epsilon_M$. Squaring the middle bound and writing its two Frobenius norms as traces proves (CT.24).

For a reflected packet $\sigma_1 = 0$, with every multiplicity paired. The exact reflection (CT.16a) pairs the nonzero eigenvalues of S_M , giving equal positive and negative counts p_M . Equation (CT.22) proves the bounds on p_M , and there are exactly $2p_M$ nonzero eigenvalues of S_M , each of square at most ϵ_M^2 . Their trace norm is at most $2p_M\epsilon_M$, proving (CT.24a); their squared trace proves (CT.25). If $p_M = 0$, the pairing implies $W_M = 0$ directly, and (CT.21) states that all the nonnegative terms vanish; no division by p_M is made. $\square$

Every quantity in (CT.24) is attached to the same arithmetic moment matrix and its next original relation layer. In particular, $0 \preceq \Delta_M \prec G_M$, because $G_{M+1} > 0$, so all eigenvalues of $G_M^{-1/2} \Delta_M G_M^{-1/2}$ lie in $[0, 1)$, and its trace is strictly smaller than r_M when $r_M > 0$. Its actual value is retained in (CT.24). Equations (CT.21)–(CT.25) calculate necessary relative layer costs and the full nilpotent departure for the constructed tensor family. They assert no unproved upper bound on these arithmetic quantities.

There is an exact determinant refinement of the same source loss. For $r_M > 0$, put

$$\pi_M = \frac{\det G_{M+1}}{\det G_M}, \quad \tau_M = \text{Tr}(G_M^{-1} \Delta_M), \quad \chi_M = \text{Tr}(G_M^{-1} C_M^* C_M).$$

Then

$$1 - \pi_M \leq \tau_M \leq r_M(1 - \pi_M^{1/r_M}) < r_M, \quad \boxed{\mathcal{D}_k^2 \leq \tau_M \chi_M \leq r_M(1 - \pi_M^{1/r_M}) \chi_M.} \quad (\text{CT.26})$$

To prove every factor, let $t_1, \dots, t_{r_M} \in (0, 1)$ be the positive eigenvalues of $G_M^{-1/2} \Delta_M G_M^{-1/2}$. The other eigenvalues are zero, so $\tau_M = \sum_i t_i$ and $\pi_M = \prod_i (1 - t_i)$. Induction on the number of factors proves $\prod_i (1 - t_i) \geq 1 - \sum_i t_i$. For positive numbers with a fixed sum, their product is at most the product with all entries equal: a maximum exists on the closed nonnegative simplex, lies in its interior when the sum is positive, and any unequal pair can be replaced by its average to increase the product, since $((a+b)/2)^2 - ab = (a-b)^2/4$. Apply this fact to $1 - t_i$ to obtain $\pi_M \leq (1 - \tau_M/r_M)^{r_M}$, proving both bounds. The last two inequalities in (CT.26) now follow from (CT.24). For $r_M = 1$, $\tau_M = 1 - \pi_M$ exactly; for $r_M = 0$, $\Delta_M = 0$, $\pi_M = 1$, and $W_M = 0$, so this case requires no exponent $1/r_M$.

15 Density of the original joint boundaries and determinant dissipation

Retain the complete packet h , its degree $d \geq 1$, the arithmetic unit v_h , and every original map in Section 14. In particular

$$s_j = \frac{1}{2} + it_j, \quad v_h = g/h, \quad g = 2\xi, \quad d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad E^{(k)} = E_h^{\otimes k}, \quad \mathcal{H}_k = L^2(d^k x). \quad (\text{JD.1})$$

The one-variable cyclic-density argument in the coherent-interpolation source has a full joint-source extension. We prove it here, including the exact relation between the disappearing Hilbert representatives and their unchanged finite jets.

15.1 Polynomial density with the original weighted measures

Lemma 15.1. *Let μ be a finite positive Borel measure on $\mathbb{R}^k$ such that $\int e^{a \sum_j |t_j|} d\mu < \infty$ for some $a > 0$. The complex polynomials in $t_1, \dots, t_k$ are dense in $L^2(\mu)$.*

Proof. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial. Define the finite complex measure $d\eta = \bar{f} d\mu$. For $0 < b < a/2$, Cauchy–Schwarz gives

$$\int e^{b \sum_j |t_j|} d|\eta| \leq \|f\|_{L^2(\mu)} \left(\int e^{2b \sum_j |t_j|} d\mu \right)^{1/2} < \infty. \quad (\text{JD.2})$$

Thus $\Phi(z) = \int e^{i \sum_j z_j t_j} d\eta(t)$ is holomorphic in the connected tube $|\operatorname{Im} z_j| < b$. On every smaller closed tube, each differentiated integrand is dominated by the integrable function in (JD.2), multiplied by a constant; the strict exponent margin absorbs each monomial. This justifies every derivative and the convergent Taylor series on small polydiscs. All derivatives at zero vanish by polynomial orthogonality. The Taylor series therefore vanishes near zero, and successive overlapping polydiscs along any path in the tube show $\Phi = 0$ throughout it. In particular the Fourier transform of η vanishes on $\mathbb{R}^k$.

For clarity, uniqueness here follows from an explicit approximate identity. Put

$$\gamma_\epsilon(x) = (4\pi\epsilon)^{-k/2} e^{-|x|^2/(4\epsilon)} = (2\pi)^{-k} \int e^{-\epsilon|u|^2} e^{iu \cdot x} du. \quad (\text{JD.3})$$

The second equality is the product Gaussian integral, obtained in one variable by differentiating its Fourier integral in x , integrating by parts in u , and evaluating its value at zero. The Gaussian integrability and finiteness of $|\eta|$ justify Fubini. Consequently $\eta * \gamma_\epsilon = 0$. For any continuous compactly supported φ , integration and Fubini give $\int (\varphi * \gamma_\epsilon) d\eta = 0$. These convolutions converge uniformly to φ : split the integral at a fixed small radius, use uniform continuity within it, and use the Gaussian tail and boundedness outside it. Hence $\int \varphi d\eta = 0$. Uniqueness of finite Borel measures from their continuous compactly supported integrals gives $\eta = 0$, and therefore $f = 0$ almost everywhere. The orthogonal complement of the polynomial closure is zero, proving the lemma. $\square$

The actual measure ν_h has every exponential moment of order less than $\pi/2$, with its full factor $dt/(2\pi)$, as proved in (GC.2). Multiplying it by any fixed polynomial in t preserves the same open range of exponential moments: choose a strictly larger exponent still below $\pi/2$ and absorb the polynomial into the difference. Product integration therefore verifies the lemma for both measures

$$\nu_h^{\otimes k}, \quad d\mu_{h,k}(t) = |h(s_1)|^2 \prod_{j=1}^k d\nu_h(t_j). \quad (\text{JD.4})$$

No factor h has been discarded. Its finitely many zeros on the first integration line have measure zero. The multiplication map

$$U_h : L^2(\mu_{h,k}) \longrightarrow L^2(\nu_h^{\otimes k}), \quad U_h P = h(s_1)P \quad (\text{JD.5})$$

is a unitary surjection: its norm identity follows directly from (JD.4), and its inverse is division by $h(s_1)$ away from that null set, with an arbitrary value on the null set. Polynomials in $s_j = 1/2 + it_j$ are precisely polynomials in t_j , since these coordinate changes are invertible. Thus (JD.5) and the lemma prove density of $h(s_1)\mathbb{C}[s_1, \dots, s_k]$ in $L^2(\nu_h^{\otimes k})$. This dense space is contained in the original joint ideal $(h(s_1), \dots, h(s_k))$.

15.2 Strong limits of the actual source representatives

The Mellin Plancherel map, with its original sign and measure, is

$$(\mathcal{M}_k F)(t) = \int_{(0,\infty)^k} F(x) \prod_{j=1}^k x_j^{-1/2+it_j} d^k x, \quad \mathcal{M}_k : \mathcal{H}_k \longrightarrow L^2 \left(\frac{d^k t}{(2\pi)^k} \right). \quad (\text{JD.6})$$

Indeed the substitutions $x_j = e^{u_j}$ send F to $e^{\sum_j u_j/2} F(e^{u_1}, \dots, e^{u_k})$ isometrically in $L^2(d^k u)$, and (JD.6) becomes its Fourier transform with kernel $e^{it \cdot u}$. The inverse has kernel $e^{-it \cdot u}$ and factor $(2\pi)^{-k}$. Multiplication by $\prod_j v_h(s_j)$ is a unitary surjection from $L^2(\nu_h^{\otimes k})$ to the target in (JD.6). For surjectivity divide by this product almost everywhere: the entire nonzero function v_h has only isolated zeros on the line, and the product zero set is null. Composing with the inverse of (JD.6) gives the unitary extension

$$\mathcal{U}_{h,k} : L^2(\nu_h^{\otimes k}) \longrightarrow \mathcal{H}_k, \quad \mathcal{U}_{h,k} P = \mathcal{T}_h^{(k)} P \quad \text{for every polynomial } P. \quad (\text{JD.7})$$

The polynomial identity follows from the actual Mellin transform of F_h and D_j , so this is an extension of the original source map, not a change of representative.

Theorem 15.2 (Joint boundary density and exact finite-jet graph closure). *For each $k \geq 1$, the original total-degree boundary spaces $\mathcal{B}_M$ satisfy*

$$\overline{\bigcup_{M \geq k(d-1)} \mathcal{B}_M} = \mathcal{H}_k, \quad P_{\mathcal{B}_M} \longrightarrow I \quad \text{strongly.} \quad (\text{JD.8})$$

For the actual coherent minima of (CT.13), in the original fixed finite monomial tensor coordinates,

$$\|R_M\|_{E^{(k)} \rightarrow \mathcal{H}_k} \longrightarrow 0, \quad G_M \downarrow 0, \quad \det G_M \longrightarrow 0. \quad (\text{JD.9})$$

Here $G_M > 0$ at every finite M , and the middle limit is both decreasing in the positive-semidefinite order and convergent in matrix operator norm. Every finite jet is still retained: $J^{(k)} R_M = I$ for every M .

Define J_{pol} on the dense original source $\mathcal{D}_{h,k} = \mathcal{T}_h^{(k)} \mathbb{C}[s_1, \dots, s_k]$ by its complete arithmetic jets, including multiplication by $v_h^{\otimes k}$. Its graph has exactly the closure

$$\overline{\text{Graph}(J_{\text{pol}})} = \mathcal{H}_k \oplus E^{(k)}. \quad (\text{JD.10})$$

As a closed linear relation this has the split exact sequence

$$0 \longrightarrow E^{(k)} \xrightarrow{u \mapsto (0,u)} \overline{\text{Graph}(J_{\text{pol}})} \xrightarrow{(F,u) \mapsto F} \mathcal{H}_k \longrightarrow 0, \quad F \longmapsto (F, 0). \quad (\text{JD.11})$$

Proof. Every polynomial in $h(s_1) \mathbb{C}[s_1, \dots, s_k]$ belongs to one finite total-degree ideal $\mathcal{I}_{h,k,M}$. Its density proved in (JD.5), followed by the unitary map (JD.7), proves the first assertion in (JD.8). Given $F \in \mathcal{H}_k$ and $\epsilon > 0$, choose $b \in \mathcal{B}_N$ with $\|F - b\| < \epsilon$. For every $M \geq N$, the distance from F to $\mathcal{B}_M$ is at most that norm, so $\|(I - P_{\mathcal{B}_M})F\| < \epsilon$. This proves strong convergence.

Apply it to each of the d^k images of the fixed monomial basis under $s_h^{\otimes k}$. The identity $R_M = (I - P_{\mathcal{B}_M})s_h^{\otimes k}$ makes each column tend to zero. For a vector with coordinate norm one, Cauchy-Schwarz bounds its image by the square root of the sum of the squared column norms, proving the first limit in (JD.9). Since $G_M = R_M^* R_M$ and $G_M - G_{M+1} = Y_M^* Y_M$, the norm limit and monotonicity follow. The determinant limit follows by continuity of the determinant in these same finite coordinates. Positivity and $J^{(k)} R_M = I$ are the proved finite interpolation identities; none of these finite algebraic equations is lost in taking the Hilbert norm limit.

The kernel of J_{pol} is exactly the union of the $\mathcal{B}_M$, so its graph closure contains $\mathcal{H}_k \oplus \{0\}$ by (JD.8). For any $u \in E^{(k)}$, the vectors $(R_M u, u)$ belong to its graph and converge to $(0, u)$ by (JD.9). The closure is a linear subspace, and therefore contains every (F, u) ; the reverse inclusion is its declared ambient space. This proves (JD.10), and the maps, exactness and section in (JD.11) follow immediately by evaluating their two coordinates. In particular the closure is a relation with vertical fibre $E^{(k)}$, and it cannot be the graph of a single-valued operator because $d^k > 0$. $\square$

There are corresponding exact limits for the original weight form and both interpolation kernels. Write $U = U_{v_h^{\otimes k}}$, let K_M be the remainder-only polynomial kernel on the joint source, and let $\mathcal{C}_M = G_M^{-1} = UK_M U^*$ be the complete arithmetic interpolation matrix. In the same fixed coordinates,

$$\begin{aligned} \|W_M\| &\leq (2\|A^{(k)}\| + k)\|G_M\| \longrightarrow 0, \\ \lambda_{\min}(\mathcal{C}_M) &= \|G_M\|^{-1} \longrightarrow \infty, \\ \lambda_{\min}(K_M) &\geq \frac{1}{\|U\|^2\|G_M\|} \longrightarrow \infty. \end{aligned} \tag{JD.9a}$$

The first line follows by the triangle inequality applied to the definition of W_M . Positivity of G_M proves the second. For the third, use $K_M = U^{-1}\mathcal{C}_M U^{-*}$ and $\|U^{-*}z\| \geq \|z\|/\|U\|$. Thus these coercive limits include every retained local jet, and the arithmetic unit remains in the comparison. The relative weight is related to its absolute form by the exact congruence

$$W_M = G_M^{1/2} S_M G_M^{1/2}, \quad \lambda_{\min}(G_M)\epsilon_M \leq \|W_M\| \leq \|G_M\|\epsilon_M. \tag{JD.9b}$$

The upper bound is submultiplicativity; applying it instead to $S_M = G_M^{-1/2} W_M G_M^{-1/2}$ gives the lower bound. Equations (JD.9a)–(JD.9b) retain the metric factors needed to pass from the proved absolute limits to the relative quantity.

The complementary quotient of the same closed relation is also explicit:

$$\frac{\overline{\text{Graph}(J_{\text{pol}})}}{\mathcal{H}_k \oplus \{0\}} \xrightarrow{\sim} E^{(k)}, \quad [(F, u)] \mapsto u, \quad u \mapsto [(0, u)]. \tag{JD.11a}$$

The kernel, well-definedness and both inverse identities follow by subtracting the two displayed pairs. With the product Hilbert norm the quotient norm is exactly $\|u\|$, since $\inf_{H \in \mathcal{H}_k} (\|F - H\|^2 + \|u\|^2)^{1/2} = \|u\|$, attained at $H = F$. Thus every completed-boundary class retains its original finite jet through an explicit isometric quotient.

This limit has an exact Split-Zero morphism. For a complex vector space X , use its one-support lift $\hat{X} = \{\tau_X\} \sqcup X^\bullet$, with τ_X the additive identity, vector addition within $X^\bullet$, and the original $G(\mathbb{C})$ -scalar action. In particular $0_X^\bullet \neq \tau_X$. Any linear map $T : X \rightarrow Y$ lifts to $\hat{T}(\tau_X) = \tau_Y$ and $\hat{T}(u^\bullet) = (Tu)^\bullet$; direct substitution checks addition and every scalar, including the supported scalar zero and external scalar zero. Give the supported fibre its norm topology and τ_X its isolated topology. Equation (JD.9) then says exactly

$$\hat{R}_M \longrightarrow \hat{0}, \quad \hat{0}(\tau_E) = \tau_{\mathcal{H}}, \quad \hat{0}(u^\bullet) = 0_{\mathcal{H}}^\bullet \quad (u \in E^{(k)}). \tag{JD.12}$$

Thus the limiting morphism retains the support of every input. Equation (JD.11) identifies all finite jets in the vertical fibre of the closed relation; it supplies the full typed relation between these supported Hilbert limits and the unchanged algebraic jets.

15.3 The accumulated determinant cost of the same boundary sequence

Continue to use the original ordered spectral sum and layer energy

$$\mathcal{D}_k = \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k (\text{Re } \rho_{i_j} - 1/2) \right|, \quad \chi_M = \text{Tr}(G_M^{-1} C_M^* C_M), \quad \tau_M = \text{Tr}(G_M^{-1} (G_M - G_{M+1})). \tag{JD.13}$$

The list $\rho_1, \dots, \rho_d$ includes every algebraic multiplicity. Define $\pi_M = \det G_{M+1} / \det G_M$. For integers $k(d-1) \leq a < b$, the following exact finite inequality holds whenever $\mathcal{D}_k > 0$:

$$\boxed{\mathcal{D}_k^2 \sum_{M=a}^{b-1} \frac{1}{\chi_M} \leq \sum_{M=a}^{b-1} \tau_M \leq \log \frac{\det G_a}{\det G_b}.} \tag{JD.14}$$

All denominators in this statement are proved positive. Indeed (CT.24) gives $\mathcal{D}_k^2 \leq \tau_M \chi_M$, with both factors nonnegative; strict positivity of the left side forces strict positivity of each factor. Let $t_{M,j} \in [0, 1)$ be the full eigenvalue list of $G_M^{-1/2}(G_M - G_{M+1})G_M^{-1/2}$, including zeros. The finite original matrices give exactly

$$\tau_M = \sum_j t_{M,j}, \quad \pi_M = \prod_j (1 - t_{M,j}), \quad -\log \pi_M - \tau_M = \sum_j \int_0^{t_{M,j}} \frac{u}{1-u} du \geq 0. \quad (\text{JD.15})$$

The first two identities are congruence and determinant identities; the last follows by integrating the derivative of $-\log(1-t) - t$. Sum (CT.24) divided by χ_M , then sum (JD.15). The product of the determinant ratios telescopes without a change of coordinates, proving (JD.14).

If $\mathcal{D}_k = 0$, every repeated tuple $(i, \dots, i)$ has zero summand, hence $\text{Re } \rho_i = 1/2$ for every retained zero. Conversely these equalities give $\mathcal{D}_k = 0$. The second inequality in (JD.14) and the exact remainder (JD.15) hold also in this case; no quotient $0/0$ is introduced. Equation (JD.9) proves that the right-hand side tends to infinity as $b \rightarrow \infty$ for each fixed a . Equations (JD.13)–(JD.15) retain the actual rate through χ_M , the complete relative Gram losses and their explicit logarithmic remainder. They impose no unproved upper estimate on that rate.

The same sequence supplies two further exact energy identities:

$$G_a = \sum_{M=a}^{\infty} Y_M^* Y_M, \quad \sum_{M=a}^{\infty} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M=a}^{\infty} \tau_M = \infty. \quad (\text{JD.16})$$

The first series converges in matrix operator norm, by telescoping $G_M - G_{M+1} = Y_M^* Y_M$ and the proved limit $G_M \rightarrow 0$. Taking traces of its finite partial sums proves the second identity; all summands are nonnegative. To prove the third, suppose instead that $\sum_M \tau_M$ were finite. Then beyond some index all $\tau_M \leq 1/2$, so every $0 \leq t_{M,j} \leq 1/2$. Integrating $(1-u)^{-1} \leq 2$ over $[0, t_{M,j}]$ gives $-\log(1-t_{M,j}) \leq 2t_{M,j}$. Hence the tail sum of $-\log \pi_M$ would be bounded by $2 \sum_M \tau_M < \infty$. The finite initial terms are finite because $G_M > 0$. Telescoping would therefore bound $\log(\det G_a / \det G_b)$ uniformly in b , contradicting $\det G_b \rightarrow 0$. This proves the divergence, including packets entirely on the critical line and every retained nilpotent chain. The equality $\tau_M = \|Y_M G_M^{-1/2}\|_{\text{HS}}^2$ identifies the relative energy in this divergence with the same layer map appearing in the convergent absolute energy series.

16 The actual kernel layer, its derivative image, and its quotient

This section integrates the complete kernel-layer delivery of 12 September 2026. Its archive hash and member hashes are recorded in `sources/web_kernel_layer_delivery/LOCAL_STAGING_PROVENANCE.json`. The source note's formulas (9)–(28) identify the interpolation kernel with an explicit basis of the next original theta relation layer. We first prove those maps in the retained coordinates and then calculate the exact derivative image and the entire quotient of that layer. The new quotient calculation retains the part of the layer not reached by the summed Euler derivative.

16.1 The original arithmetic source and its constrained minimum

Keep $g = 2\xi$, a nonempty full packet $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, and $d = \deg h$. Fix $k \geq 1$, and set

$$E = E_h^{\otimes k}, \quad E_h = \mathbb{C}[s]/(h), \quad A = \sum_{j=1}^k A_{h,j}, \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times. \quad (\text{KL.1})$$

All tensor coordinates use the original monomial remainders. The function F_h , with Mellin transform g/h , is the original test function satisfying $h(D)F_h = f_0$. The exact source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{M}_k \mathcal{T}(P) = P(s_1, \dots, s_k) \prod_{j=1}^k v_h(s_j), \quad s_j = \frac{1}{2} + it_j. \quad (\text{KL.2})$$

The Mellin multiplier proves injectivity on polynomials. Its Hilbert norm is the polynomial norm for the unchanged product measure

$$d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad \nu_h^{\otimes k}. \quad (\text{KL.3})$$

Let $\mathcal{P}_N$ be the polynomials of total degree at most N , and let $\mathcal{J}_N : \mathcal{P}_N \rightarrow E$ be full reduction by $h(s_1), \dots, h(s_k)$, followed by multiplication by the original unit $v_h^{\otimes k}$. For $N \geq k(d-1)$ this is onto, since the monomial remainder basis has at most that total degree. Fix an integer $N \geq k(d-1)$ for the remainder of this section; all subsequent levels $N+1, N+2$ satisfy the same threshold. Its kernel and its original theta image are

$$\mathcal{I}_N = \mathcal{P}_N \cap (h(s_1), \dots, h(s_k)), \quad \mathcal{L}_N = \mathcal{T}\mathcal{I}_N. \quad (\text{KL.4})$$

Let $p_j(s)$ be the actual monic orthogonal polynomial of ν_h , and retain $\kappa_j = \|p_j\|_{\nu_h}^2 > 0$. All moments exist by the proved arithmetic decay. The Gram is strictly positive because the density is positive except at a discrete set; this proves existence and uniqueness by finite orthogonal projection. For multi-indices set

$$p_\alpha = \prod_j p_{\alpha_j}, \quad \kappa_\alpha = \prod_j \kappa_{\alpha_j}, \quad a_\alpha = \bigotimes_j (v_h[p_{\alpha_j}]_h). \quad (\text{KL.5})$$

The monic triangular change preserves the full space of degree at most N , and product integration makes its Gram diagonal. It follows, by direct constrained minimization, that the original canonical representative and its metric are

$$\mathcal{K}_N = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha} > 0, \quad G_N = \mathcal{K}_N^{-1}, \quad R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (\text{KL.6})$$

Indeed the full jet of this vector is $\mathcal{K}_N G_N u = u$. If a coefficient vector has zero full jet, its inner product with the displayed coefficient vector is zero. Every other representative of that same jet is its sum with such a vector; its squared norm is therefore $u^* G_N u$ plus the squared norm of the added relation. This proves the unique minimum, its metric, and $R_N \perp \mathcal{L}_N$. These are the original maps $J^{(k)} R_N = I$ and $q^{(k)} R_N = \sigma$. The notation $\mathcal{K}_N$ here includes the local unit. To specify the comparison in the full tensor dimension, put

$$r_\alpha = \bigotimes_j [p_{\alpha_j}]_h, \quad K_N^{\text{rem}} = \sum_{|\alpha| \leq N} \frac{r_\alpha r_\alpha^*}{\kappa_\alpha}.$$

Then $\mathcal{K}_N = U_{v_h^{\otimes k}} K_N^{\text{rem}} U_{v_h^{\otimes k}}^*$, as an equality of operators on $E_h^{\otimes k}$. When $k = 1$, K_N^{rem} is exactly the earlier remainder-only kernel K_N .

16.2 The complete next-layer frame and its theta primitives

Write $\Sigma_n = \{\alpha \in \mathbb{N}^k : |\alpha| = n\}$ and $\mathbf{H}_n = \mathbb{C}^{\Sigma_n}$. The coordinate identification sends $x \in \mathbf{H}_n$ to the homogeneous polynomial $\sum_\alpha x_\alpha z^\alpha$; these z_j are coefficient variables and are not substituted for the original Mellin variables s_j . For $\beta \in \Sigma_{N+1}$, define the actual functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \mathcal{E}_N x = \sum_\beta x_\beta e_\beta, \quad \mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp. \quad (\text{KL.7})$$

The full jet of e_β is zero, hence it is in $\mathcal{L}_{N+1}$. The new-degree polynomial is orthogonal to all old polynomials and $R_N a_\beta \perp \mathcal{L}_N$, so $e_\beta \in \mathcal{E}_N$. The leading total-degree terms of the e_β are the distinct monomials s^β , proving independence by the injectivity of $\mathcal{T}$. For any element of $\mathcal{E}_N$, subtract its full degree- $N+1$ combination of the e_β . The remainder has zero jet and degree at most N , hence is in $\mathcal{L}_N$. It is also orthogonal to that space, so it is zero. This proves the exact isomorphism

$$\mathcal{E}_N : \mathcal{H}_{N+1} \xrightarrow{\sim} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1}/\mathcal{L}_N. \quad (\text{KL.8})$$

The inverse of the second arrow is $[f] \mapsto (I - P_{\mathcal{L}_N})f$.

Every vector in (KL.7) has a primitive in the original tensor complex. Its numerator polynomial is

$$p_\beta - \sum_{|\alpha| \leq N} p_\alpha \frac{a_\alpha^* G_N a_\beta}{\kappa_\alpha}. \quad (\text{KL.9})$$

Successive monic division in the ordered variables $s_1, \dots, s_k$ expresses this zero-jet polynomial as $\sum_i h(s_i) Q_{i,\beta}$, where $\deg Q_{i,\beta} \leq N+1-d$. At each replacement the leading power of the divided variable decreases and total degree does not increase, proving the bound. After the last division, the remainder has degree below d in every variable; its full jet is zero, so uniqueness of the complete remainder forces it to vanish. In the i -th tensor factor replace F_h by $(-1)^{i-1} \phi_*$, and apply $Q_{i,\beta}(D_1, \dots, D_k)$ to the whole tensor. The preceding $i-1$ degree-one factors give the differential sign $(-1)^{i-1}$. Together with $h(D)F_h = \Theta \phi_*$, the boundary of this primitive is exactly $+\mathcal{T}(h(s_i)Q_{i,\beta})$. Summing proves the asserted primitive with all ordered signs retained.

Put

$$\mathbf{U}_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad \mathbf{D}_{N+1} = \text{diag}(\kappa_\beta), \quad \mathbf{H}_N^{\text{lay}} = \mathcal{E}_N^* \mathcal{E}_N.$$

Expanding (KL.7) and using old/new polynomial orthogonality proves

$$\mathbf{H}_N^{\text{lay}} = \mathbf{D}_{N+1} + \mathbf{U}_N^* G_N \mathbf{U}_N > 0, \quad \mathcal{E}_N^* R_N = -\mathbf{U}_N^* G_N. \quad (\text{KL.10})$$

Define the projection loss $Y_N = P_{\mathcal{E}_N} R_N$. In particular its exact formula is $Y_N = -\mathcal{E}_N (\mathbf{H}_N^{\text{lay}})^{-1} \mathbf{U}_N^* G_N$. Its rank is exactly $\text{rank } \mathbf{U}_N$, because every other displayed factor is injective or invertible on its declared space.

16.3 The exact incidence map and the derivative-image rank

Define the actual old top-degree columns and their original norms by

$$\mathbf{A}_N = (a_\alpha)_{\alpha \in \Sigma_N} : \mathcal{H}_N \rightarrow E, \quad \mathbf{D}_N = \text{diag}(\kappa_\alpha)_{\Sigma_N}. \quad (\text{KL.11})$$

Define the coordinate incidence map by

$$\mathbf{T}_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1}, \quad (\mathbf{T}_n x)_\beta = \sum_{i: \beta_i > 0} x_{\beta - \mathbf{e}_i}. \quad (\text{KL.12})$$

On the specified homogeneous coefficient polynomials this is multiplication by $\ell(z) = z_1 + \dots + z_k$. A product of two nonzero polynomials over $\mathbb{C}$ is nonzero: choose a monomial order compatible with products and multiply the two leading terms. Since $\ell \neq 0$, $\mathbf{T}_n$ is injective.

The full unprojected derivative defect and its projected layer map are

$$B_N = D^{(k)} R_N - R_N A, \quad C_N = (I - P_{\mathcal{L}_N}) B_N : E \rightarrow \mathcal{E}_N.$$

They have the exact coordinate factorization

$$\boxed{C_N = \mathcal{E}_N \mathbf{T}_N \mathbf{D}_N^{-1} \mathbf{A}_N^* G_N.} \quad (\text{KL.13})$$

To prove it, multiply the finite numerator in (KL.6) by $s_1 + \cdots + s_k$. Monicity makes its coefficient at the new top polynomial p_β equal to $\sum_{i:\beta_i>0} a_{\beta-\mathbf{e}_i}^* G_N u / \kappa_{\beta-\mathbf{e}_i}$, precisely the row on the right side of (KL.13). The numerator of $R_N A u$ has degree at most N . Subtract the right side of (KL.13) from $B_N u$: its numerator has degree at most N and its full jet is zero, so it is an original relation in $\mathcal{L}_N$. Orthogonal projection proves (KL.13), including the original quotient statement $[B_N u] = [C_N u]$ in $\mathcal{B}^{(k)} / \mathcal{L}_N$.

Theorem 16.1 (Exact derivative rank and its full kernel). *With every original jet column retained,*

$$\boxed{\text{rank } C_N = \text{rank } A_N, \quad \ker C_N = \mathcal{K}_N \ker A_N^*} \quad (\text{KL.14})$$

Define the original weight form $W_N = A^* G_N + G_N A - k G_N$. It obeys the refined bound

$$\text{rank } W_N \leq 2 \min\{\text{rank } A_N, \text{rank } U_N\}. \quad (\text{KL.15})$$

Proof. Both $\mathcal{E}_N$ and $\mathbf{T}_N$ are injective; D_N and G_N are invertible. Thus (KL.13) has rank $\text{rank } A_N^*$, which equals the rank of its conjugate transpose. Its kernel is $G_N^{-1} \ker A_N^* = \mathcal{K}_N \ker A_N^*$, with the displayed invertible coordinate map retained. The original Green identity and orthogonality give $W_N = -(Y_N^* C_N + C_N^* Y_N)$. Each summand has rank at most the lesser of the two map ranks; the rank of a sum is at most the sum of its ranks. Use (KL.10) and (KL.14) to obtain (KL.15). No nonzero homogeneous-leading-term assertion has been substituted for an actual column a_α . $\square$

16.4 The whole quotient of the next layer by the derivative image

For $k \geq 2$, let H'_n be the homogeneous polynomials of degree n in $z_1, \dots, z_{k-1}$. There is a split exact sequence of vector spaces

$$0 \longrightarrow H_n \xrightarrow{\mathbf{T}_n} H_{n+1} \xrightarrow{\rho_n} H'_{n+1} \longrightarrow 0, \quad \rho_n f = f(z_1, \dots, z_{k-1}, -\sum_{i < k} z_i). \quad (\text{KL.16})$$

For a complete proof and inverse, divide f by the monic linear polynomial $\ell = z_k + \sum_{i < k} z_i$ in the last variable. Every replacement retains total degree, giving uniquely

$$f = \ell q + \iota r, \quad q \in H_n, \quad r = \rho_n f \in H'_{n+1}, \quad (\text{KL.17})$$

where ι includes a polynomial independent of z_k . For uniqueness set $z_k = -\sum_{i < k} z_i$ first to obtain $r = 0$, then use injectivity of multiplication by ℓ . This also proves surjectivity, exactness and the specified section ι . The ordered last coordinate and the signs in the substitution remain part of the map.

Theorem 16.2 (Exact unused-layer quotient). *Set $Z_N = \text{im}(D_N^{-1} A_N^*) \subseteq H_N$. Then the actual quotient is*

$$\boxed{\mathcal{E}_N / \text{im } C_N \simeq (H_N / Z_N) \oplus H'_{N+1}} \quad (\text{KL.18})$$

For an original layer representative $e = \mathcal{E}_N f$, divide $f = \ell q + \iota r$ by (KL.17); the map sends $[e]$ to $([q], r)$. Its inverse sends $([q], r)$ to $[\mathcal{E}_N(\ell q + \iota r)]$.

Proof. The invertibility of G_N in (KL.13) gives $\text{im } C_N = \mathcal{E}_N \mathbf{T}_N Z_N$. Changing e by such an element changes exactly q by an element of Z_N , while leaving r unchanged. Thus the first map is well-defined. Changing the representative of $[q]$ has the same effect on the inverse, so the inverse is well-defined. The unique division (KL.17) proves both inverse identities. For $k = 1$, $\mathbf{T}_N$ is an isomorphism between the one-dimensional shell spaces and the same statement holds with $H'_{N+1} = 0$. $\square$

In particular $\dim H'_{N+1} = \binom{N+k-1}{k-2}$ for $k \geq 2$, by counting monomials; the first summand has dimension $\binom{N+k-1}{k-1} - \text{rank } A_N$. Their sum is exactly the original layer dimension minus $\text{rank } C_N$. The quotient therefore retains both the full-jet rank loss and the explicitly complementary relation directions. For every coordinate vector, (KL.9) supplies its original theta primitive. The quotient is of actual test-function relations, with those primitives, rather than only a quotient of dimensions.

16.5 The derivative between consecutive relation quotients

The original Euler sum has $D^{(k)}\mathcal{L}_N \subseteq \mathcal{L}_{N+1}$, because it multiplies the numerator polynomial by $s_1 + \cdots + s_k$ and preserves the ideal in (KL.4). Hence it induces the well-typed map

$$\partial_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [D^{(k)}f]. \quad (\text{KL.19})$$

Under the exact layer coordinates of (KL.8), its matrix is

$$\partial_N = \mathcal{E}_{N+1} \mathsf{T}_{N+1} \mathcal{E}_N^{-1}, \quad (\text{KL.20})$$

where each $\mathcal{E}$ is followed or preceded by its quotient isomorphism as in (KL.8). Write $e_\beta^{(N)}$ for (KL.7) at level N , and $e_{\beta+\mathbf{e}_i}^{(N+1)}$ for the next layer. The actual recurrence $sp_j = p_{j+1} + b_j p_j - (\kappa_j/\kappa_{j-1})p_{j-1}$ gives the complete retained difference

$$\begin{aligned} D^{(k)}e_\beta^{(N)} - \sum_i e_{\beta+\mathbf{e}_i}^{(N+1)} = & \mathcal{T} \left[\left(\sum_i b_{\beta_i} \right) p_\beta - \sum_{i:\beta_i > 0} \frac{\kappa_{\beta_i}}{\kappa_{\beta_i-1}} p_{\beta-\mathbf{e}_i} \right] \\ & - D^{(k)}R_N a_\beta + R_{N+1} \sum_i a_{\beta+\mathbf{e}_i}. \end{aligned}$$

The final term in the one-variable recurrence is omitted at $j = 0$. Every term on the right has numerator degree at most $N + 1$. Each term on the left has zero full jet, so their difference is in $\mathcal{L}_{N+1}$. This proves (KL.20) with all lower polynomial coefficients retained. Applying the explicit ordered division in (KL.9) to this displayed numerator supplies its original theta primitive.

It follows that ∂_N is injective and its cokernel is exactly H'_{N+2} , with maps (KL.16)–(KL.17) at index $N + 1$. Thus the original derivative connects consecutive relation quotient types by a proved injection; it has not been made an endomorphism of a fixed quotient. The ordinary relation-admission map has the same declared source and target:

$$\text{adm}_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [f] = 0.$$

Equation (KL.19) instead sends the representative to its next derivative class. These are explicitly different typed maps of the same original representative, related by (KL.20) and the inclusion of their actual numerator relations.

All maps here have their original supported lifts. At a fixed support label λ , a linear map T acts by $(\lambda, x) \mapsto (\lambda, Tx)$. The relation-admission map therefore sends a killed class to the receiving fibre's supported zero. The external zero τ remains external absence. Each displayed kernel and quotient records its full supported vector space and each derivative retains its consecutive source and target; no passage to the Hilbert closure is used.

16.6 The exact original-theta norm and volume bridge

Specialize to $k = 1$. We now distinguish the degree-indexed G_N of (KL.6) from the correction-indexed original metric by writing

$$\mathbf{G}_m = G_{d+m} \quad (m \geq -1), \quad \mathbf{R}_m = R_{d+m}, \quad r_m = \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{m-1}} \quad (m \geq 0). \quad (\text{KL.21})$$

At $m = -1$, $\mathcal{P}_{d-1}$ contains no nonzero multiple of h . Thus the original representative is exactly the fixed section

$$s_h u = \mathbf{R}_{-1} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u), \quad \varepsilon_h = v_h^{-1}, \quad \mathbf{G}_{-1} = s_h^* s_h.$$

This fixes the initial determinant in (KL.21), without introducing a negative-degree theta polynomial. Put $L_{-1} = 0$ and

$$f_j = D^j f_0, \quad L_j = \text{span}(f_0, \dots, f_j), \quad u_j = (I - P_{L_{j-1}})f_j, \quad \mathfrak{h}_j = \|u_j\|^2 > 0 \quad (j \geq 0). \quad (\text{KL.22})$$

These $u_j = Q_j(D)f_0$ are the monic orthogonal theta vectors for the original measure $|g(1/2 + it)|^2 dt/(2\pi)$. Their definition does not depend on h . The monic p_j and their norms κ_j continue to use the original packet-dependent measure $|g/h|^2 dt/(2\pi)$. Multiplication by the unchanged monic polynomial h is the exact isometry connecting their source polynomial spaces:

$$P \mapsto hP, \quad \|hP\|_{\nu_h}^2 = \|P\|_{|g|^2 dt/(2\pi)}^2, \quad \mathcal{T}_h(hP) = P(D)f_0. \quad (\text{KL.23})$$

This follows by cancellation of the pointwise factor $|h|^2$ in the integral; the identities extend across the discrete zeros by their entire expressions. It is an isometry onto the specified subspace of multiples of h , with its degree shifted by the full degree $d = \sum_\rho m_\rho$.

Theorem 16.3 (Original norm as an exact relative Gram volume). *For every $j \geq 0$, including the initial theta vector,*

$$\boxed{\kappa_{d+j} = \mathfrak{h}_j r_j}. \quad (\text{KL.24})$$

Consequently, for $m \geq 0$,

$$\boxed{\frac{\kappa_{d+m+1}}{\kappa_{d+m}} = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} \frac{r_{m+1}}{r_m}, \quad \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{-1}} = \prod_{j=0}^m \frac{\kappa_{d+j}}{\mathfrak{h}_j}}. \quad (\text{KL.25})$$

All ratios have positive denominators and all packet multiplicities are retained.

Proof. At the degree level $N = d + j - 1$, define $e_{d+j} = \mathcal{T}_h p_{d+j} - \mathbf{R}_{j-1} a_{d+j}$. Its numerator has zero full jet, is monic of degree $d + j$, and is divisible by the original monic h . By ordinary monic division its quotient is monic of degree j . Under (KL.23) this is f_j plus a vector in L_{j-1} . By (KL.7) it is orthogonal to that space. The defining orthogonal projection in (KL.22) therefore proves

$$e_{d+j} = u_j, \quad \mathfrak{h}_j = \kappa_{d+j} + a_{d+j}^* \mathbf{G}_{j-1} a_{d+j}. \quad (\text{KL.26})$$

When $j = 0$, the quotient of the numerator by h is the constant 1, so the same argument explicitly gives $e_d = f_0 = u_0$; no previous theta layer is required.

For the full unit cancellation put $v_l = [p_l]_h = p_l(A_h)[1]_h$, and $K_{l-1} = \sum_{i=0}^{l-1} v_i v_i^* / \kappa_i$, using the original monomial remainder coordinates. At $l = d + j$,

$$a_l = U_{v_h} v_l, \quad \mathbf{G}_{j-1} = U_{\varepsilon_h}^* K_{l-1}^{-1} U_{\varepsilon_h}, \quad a_l^* \mathbf{G}_{j-1} a_l = v_l^* K_{l-1}^{-1} v_l. \quad (\text{KL.27})$$

Both cancellations use $U_{\varepsilon_h} U_{v_h} = I$; hence they preserve every derivative of the original local unit at every full jet. The degree update $K_l = K_{l-1} + v_l v_l^* / \kappa_l$ has determinant ratio $1 + v_l^* K_{l-1}^{-1} v_l / \kappa_l$. To verify this scalar identity directly, expand $\det(I + xy^*)$ multilinearly in its columns: all terms with two updated columns vanish since those columns are multiples of x , leaving $1 + y^* x$. The fixed-unit congruences in (KL.27) then give

$$r_j = \frac{\kappa_l}{\kappa_l + v_l^* K_{l-1}^{-1} v_l} = \frac{\kappa_{d+j}}{\mathfrak{h}_j}.$$

This also proves $0 < r_j \leq 1$, including the case $v_l = 0$. Divide two consecutive identities and multiply the first $m + 1$ identities to obtain (KL.25). $\square$

For two full packets h, H , their original norm sequences are therefore connected at the same theta degree j by the exact scalar equality

$$\frac{\kappa_{\deg h+j}^{(h)}}{r_j^{(h)}} = \mathfrak{h}_j = \frac{\kappa_{\deg H+j}^{(H)}}{r_j^{(H)}}. \quad (\text{KL.28})$$

When $h \mid H$, the corresponding source isometry is $P \mapsto (H/h)P$, since $(g/H)(H/h)P = (g/h)P$. At the old centres this equality preserves every jet, and at each added centre g/h has the complete zero order, so it realizes exactly the CRT insertion of zero jets. The reference numerator has degree below $\deg H$ and is its unique full remainder. Both corrected representatives then subtract the same projection onto the packet-independent L_j , proving the coherent equality $\mathbf{R}_{H,j} \iota_{hH} = \mathbf{R}_{h,j}$, including $j = -1$. Thus (KL.28) connects precisely these original coherent source maps and their fixed-section metrics.

16.7 The complete scalar dictionary for the boundary energy

For $m \geq 0$, set $n = d + m + 1$ and retain the terminal notation

$$M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}, \quad \alpha_n = v_{n-1}^* M_n v_{n-1}, \quad \beta_n = v_n^* M_n v_n, \quad \gamma_n = v_n^* M_n v_{n-1}.$$

Define the original theta projection rows by

$$d_j = \mathfrak{h}_j^{-1} u_j^* s_h : E_h \rightarrow \mathbb{C}, \quad \mathbf{R}_m = s_h - \sum_{j=0}^m u_j d_j, \quad c_m = d_m \mathbf{G}_m^{-1} d_{m+1}^*. \quad (\text{KL.29})$$

Here an expression $u_j^* F$ denotes the Hilbert inner product $\langle u_j, F \rangle$, conjugate-linear in its first argument. The row d_j therefore retains the original unnormalized vector and the exact factor $\mathfrak{h}_j^{-1}$.

The complete row and scalar comparison is

$$\boxed{\begin{aligned} d_m &= -\frac{v_{n-1}^* K_{n-1}^{-1} U_{\varepsilon_h}}{\kappa_{n-1}}, \\ d_{m+1} &= -\frac{v_n^* K_{n-1}^{-1} U_{\varepsilon_h}}{\mathfrak{h}_{m+1}}, \\ d_m \mathbf{G}_m^{-1} d_m^* &= \frac{\alpha_n}{\kappa_{n-1}}, \\ d_{m+1} \mathbf{G}_m^{-1} d_{m+1}^* &= \frac{\kappa_{n-1} \beta_n}{\mathfrak{h}_{m+1}^2}, \\ \mathfrak{h}_{m+1} c_m &= \overline{\gamma_n}. \end{aligned}} \quad (\text{KL.30})$$

To prove the first row, the polynomial numerator of s_h has degree below d . In (KL.29), each u_j has numerator hQ_j , monic of degree $d+j$. Thus the coefficient of degree $d+m = n-1$ in the numerator of $\mathbf{R}_m$ is $-d_m$. The finite expansion (KL.6) gives the same coefficient as $a_{n-1}^* \mathbf{G}_m / \kappa_{n-1}$. Cancelling the same units as in (KL.27) proves the first row, also at $m = 0$. For the second row use $u_{m+1} = e_n$, so (KL.10) gives $u_{m+1}^* \mathbf{R}_m = -a_n^* \mathbf{G}_m$. The orthogonality of u_{m+1} to L_m also gives $u_{m+1}^* \mathbf{R}_m = u_{m+1}^* s_h = \mathfrak{h}_{m+1} d_{m+1}$. This proves its formula. Finally $\mathbf{G}_m^{-1} = U_{v_h} K_{n-1} U_{v_h}^*$; insert both rows, cancel both unit pairs and the adjacent kernel inverse. The three remaining products are exactly those in (KL.30), including the conjugation in the last line.

Theorem 16.4 (Exact original-norm formula for the reflected packet). *Suppose the original full packet is stable under $\rho \mapsto 1 - \bar{\rho}$. For its actual relative allowance ϵ_m ,*

$$\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2. \quad (\text{KL.31})$$

If the packet is also stable under ordinary conjugation, then $c_m = 0$, so its complete formula becomes

$$\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{KL.32})$$

Proof. The terminal identity (KT.4), with the original full multiplicities, gives $\text{Re } \gamma_n = \text{Re tr } A_h - d/2 = 0$: reflected centres contribute complementary real parts with equal orders. Its two possible nonzero generalized eigenvalues then yield $\epsilon_m^2 = \alpha_n \beta_n - |\gamma_n|^2$. The preceding and succeeding rank-one kernel updates, also proved directly in (KL.27), give

$$\alpha_n = 1 - r_m, \quad \beta_n = \frac{\kappa_n}{\kappa_{n-1}} \frac{1 - r_{m+1}}{r_{m+1}}.$$

Insert (KL.25) into the product $\alpha_n \beta_n$ and the last line of (KL.30) into $|\gamma_n|^2$. Every positive ratio is retained, and this is (KL.31).

For the final assertion, conjugation stability makes h real in the original monomial coordinates. The real Taylor coefficients of g give $(g/h)(1/2 - it) = \overline{(g/h)(1/2 + it)}$, so the actual ν_h is even in t . Its monomial Gram matrix is real: the integrand for entry $s^i s^j$ at $-t$ is the complex conjugate of its integrand at t . Therefore the unique solution of the real Gram linear equations for the monic p_l has real coefficients. Polynomial division by the real monic h makes each v_l real, and the finite sum for K_{n-1} is real. Its inverse and M_n are real as well. Consequently γ_n is real. The already proved reflection trace identity makes its real part zero, hence $\gamma_n = 0$. Equation (KL.30), with $\mathfrak{h}_{m+1} > 0$, proves $c_m = 0$. No division by a jet column, an off-line displacement, or a boundary energy has been used; all zero-column cases remain included. $\square$

The source PR15 denotes the relative determinant at correction m by $\delta_m = \det \mathbf{G}_{m+1} / \det \mathbf{G}_m$. Its exact index map is therefore $\delta_m = r_{m+1}$, with $\delta_{-1} = r_0$. Its cross-term row product is the c_m of (KL.29). Equations (KL.24)–(KL.31) prove the full map from its original-theta scalar formula to the terminal-kernel formula, including both monic norm families, the unit congruences, the conjugation, the fixed initial section, and the complete orders. These equalities calculate the same actual boundary energy at every finite stage. They do not supply an estimate tending to zero; that requires further information about these now explicitly related arithmetic quantities.

The packet-independent norms in (KL.24)–(KL.32) also have the complete positive-half-line formulas (SP.18)–(SP.21) and the gamma-reference determinant formulas (TG.16)–(TG.19). Those sections prove the original affine and quadratic coordinate maps, their inverses, both measures and every phase. Consequently these formulas can be substituted directly into the identities here; the actual unit, packet determinant ratios, full zero orders and source degree indices remain unchanged.

17 Original orthogonal shells and the exact projective-ratio algebra

The source's degree shell consists of its original orthogonal product polynomials. This section proves its full comparison with the homogeneous monomial shell, including every lower-degree coefficient. It also calculates the algebra generated by the ratio of the two original tensor coordinates. The arithmetic local unit and all multiplicities remain in the resulting linear maps.

17.1 Two sections of the same polynomial filtration

Retain a complete nonempty arithmetic packet

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(g/h), \quad g = 2\xi. \quad (\text{KPC.1})$$

The complete orders make v_h a unit. For $k \geq 1$, write $E^{(k)} = E_h^{\otimes k}$ in its original monomial tensor coordinates and use the original full-jet map

$$J(P) = U_{v_h^{\otimes k}}[P]_{(h(s_1), \dots, h(s_k))}, \quad J : \mathbb{C}[s_1, \dots, s_k] \longrightarrow E^{(k)}. \quad (\text{KPC.2})$$

Here U_a is multiplication by the stated element a . The integration variables remain $s_j = 1/2 + it_j$, with

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{KPC.3})$$

Let $p_j(s) = \sum_{r=0}^j p_{j,r} s^r$, with $p_{j,j} = 1$, be the original monic orthogonal polynomials for this measure. Retain $\kappa_j = \int |p_j(s)|^2 d\nu_h > 0$. For a multi-index α , put

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad a_\alpha = J(p_\alpha). \quad (\text{KPC.4})$$

All moment integrals in these definitions use the original density.

For $N \geq 0$, let $\mathcal{P}_{\leq N}$ be the polynomials of total degree at most N , let $\mathcal{P}_{\leq -1} = 0$, and let $\mathcal{H}_N$ be the homogeneous polynomials of degree N . The highest-degree map $\ell_N : \mathcal{P}_{\leq N} \rightarrow \mathcal{H}_N$ has kernel $\mathcal{P}_{\leq N-1}$. It therefore identifies $\mathcal{H}_N$ with the associated-graded space $\mathcal{P}_{\leq N}/\mathcal{P}_{\leq N-1}$. There are two specified sections:

$$\begin{aligned} i_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha s^\alpha, \\ O_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha p_\alpha. \end{aligned} \quad (\text{KPC.5})$$

Theorem 17.1 (The exact lower-degree correction). *Both maps in (KPC.5) are linear sections of ℓ_N . Their difference $L_N = O_N - i_N$ has target $\mathcal{P}_{\leq N-1}$ and is exactly*

$$L_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) = \sum_{|\gamma| < N} \left(\sum_{\substack{|\alpha|=N \\ \gamma_j \leq \alpha_j \ (1 \leq j \leq k)}} c_\alpha \prod_{j=1}^k p_{\alpha_j, \gamma_j} \right) s^\gamma. \quad (\text{KPC.6})$$

The image of O_N is the orthogonal complement of $\mathcal{P}_{\leq N-1}$ in $\mathcal{P}_{\leq N}$ for the original product measure, and its basis in (KPC.5) has Gram $\text{diag}(\kappa_\alpha)_{|\alpha|=N}$. Define the two full residue maps on the same $\mathcal{H}_N$ by

$$A_N^{\text{hom}} = J i_N, \quad A_N^{\text{orth}} = J O_N.$$

Then, with $J_{<N} = J|_{\mathcal{P}_{\leq N-1}}$,

$$A_N^{\text{orth}} = A_N^{\text{hom}} + J_{<N} L_N. \quad (\text{KPC.7})$$

In particular the original orthogonal-shell columns of A_N^{orth} are exactly a_α .

Proof. Expand each factor in p_α . Its coefficient of s^γ is $\prod_j p_{\alpha_j, \gamma_j}$ when $\gamma_j \leq \alpha_j$ for all j , and is zero otherwise. If $|\alpha| = |\gamma| = N$, these inequalities force $\gamma = \alpha$, whose coefficient is one. This proves both section identities and the complete expression (KPC.6). For a monomial s^γ of degree less than $N = |\alpha|$, at least one $\gamma_j < \alpha_j$. In the product integral for $\langle s^\gamma, p_\alpha \rangle$, that factor is zero by one-variable orthogonality. All product integrals are justified by their finite moments and Cauchy-Schwarz. Distinct product polynomials are mutually orthogonal, and their norms multiply to κ_α . Their leading monomials make them independent. Their number is $\dim \mathcal{P}_{\leq N} - \dim \mathcal{P}_{\leq N-1}$, so they exhaust the indicated orthogonal complement. Applying the unchanged linear map J to $O_N = i_N + L_N$ proves (KPC.7). This application includes the fixed multiplication by $v_h^{\otimes k}$. For $N = 0$ the sum in (KPC.6) is empty and $L_0 = 0$, so the same proof includes the initial shell. $\square$

The associated-graded target is explicit as well. Put $F_N E^{(k)} = J(\mathcal{P}_{\leq N})$ and $F_{-1} E^{(k)} = 0$. Projection onto $F_N E^{(k)} / F_{N-1} E^{(k)}$, composed with either map in (KPC.7), gives the same surjection

$$\overline{A}_N : \mathcal{H}_N \twoheadrightarrow F_N E^{(k)} / F_{N-1} E^{(k)}. \quad (\text{KPC.8})$$

Indeed their difference lands in $F_{N-1} E^{(k)}$, and $\mathcal{P}_{\leq N} = i_N \mathcal{H}_N + \mathcal{P}_{\leq N-1}$ proves surjectivity. If $Z_N = \ker \overline{A}_N$ and $K_N^a = \ker A_N^a$ for $a \in \{\text{hom}, \text{orth}\}$, then $K_N^a \subseteq Z_N$ and the exact sequence is

$$0 \longrightarrow Z_N / K_N^a \longrightarrow \mathcal{H}_N / K_N^a \longrightarrow \mathcal{H}_N / Z_N \longrightarrow 0. \quad (\text{KPC.9})$$

The middle space is identified with $\text{im } A_N^a$ by $[c] \mapsto A_N^a c$, the last space with the target of (KPC.8), and the first space with $\text{im } A_N^a \cap F_{N-1} E^{(k)}$. These identifications follow directly from their kernels and are bijective: for the first image, membership in F_{N-1} is exactly the condition $c \in Z_N$. Thus equality of the graded maps retains, rather than removes, the two potentially different lower-filtration image spaces. In particular, once $N-1 \geq k(d-1)$, complete monomial remainders show $F_{N-1} E^{(k)} = E^{(k)}$, so (KPC.8) has zero target; the full maps (KPC.7) still have their stated images and kernels.

17.2 The full correspondence between the two image spaces

Theorem 17.2 (Common quotient and exact graph correspondence). *Write $I_h = \text{im } A_N^{\text{hom}}$, $I_o = \text{im } A_N^{\text{orth}}$, $K_h = \ker A_N^{\text{hom}}$, and $K_o = \ker A_N^{\text{orth}}$. Define*

$$Q_N = \mathcal{H}_N / (K_h + K_o), \quad \Gamma_N = \{(A_N^{\text{hom}} c, A_N^{\text{orth}} c) : c \in \mathcal{H}_N\}. \quad (\text{KPC.10})$$

The maps $q_h : I_h \rightarrow Q_N$ and $q_o : I_o \rightarrow Q_N$, given respectively by $A_N^{\text{hom}} c \mapsto [c]$ and $A_N^{\text{orth}} c \mapsto [c]$, are well-defined surjections, with exact sequences

$$\begin{aligned} 0 \longrightarrow A_N^{\text{hom}}(K_o) \longrightarrow I_h \xrightarrow{q_h} Q_N \longrightarrow 0, \\ 0 \longrightarrow A_N^{\text{orth}}(K_h) \longrightarrow I_o \xrightarrow{q_o} Q_N \longrightarrow 0. \end{aligned} \quad (\text{KPC.11})$$

One has the exact vector-space identifications

$$\boxed{\Gamma_N = I_h \times_{Q_N} I_o \cong \mathcal{H}_N / (K_h \cap K_o).} \quad (\text{KPC.12})$$

The projection $\Gamma_N \rightarrow I_h$ has kernel $\{0\} \oplus A_N^{\text{orth}}(K_h)$; the projection onto I_o has kernel $A_N^{\text{hom}}(K_o) \oplus \{0\}$.

Proof. Changing a representative of an element of I_h changes c by K_h , hence leaves its class modulo $K_h + K_o$ unchanged. The same argument applies to I_o . The formulas give surjectivity. If $q_h(A_N^{\text{hom}} c) = 0$, write $c = c_h + c_o$ with $c_h \in K_h, c_o \in K_o$. Then $A_N^{\text{hom}} c = A_N^{\text{hom}} c_o$. Conversely every such vector maps to zero. This proves the first kernel formula in (KPC.11); interchanging the two maps proves the second.

The paired map from $\mathcal{H}_N$ onto Γ_N has kernel exactly $K_h \cap K_o$, proving the last identification in (KPC.12). Every paired image belongs to the fibre product. Conversely, suppose $(A_N^{\text{hom}} c, A_N^{\text{orth}} b)$ belongs to the fibre product. Then $c - b = c_h + c_o$ for some $c_h \in K_h, c_o \in K_o$. Put $e = c - c_h = b + c_o$. It has the two desired images, so the pair belongs to Γ_N . This proves equality with the full fibre product, not merely an inclusion. Taking the kernel of each coordinate projection gives the asserted remaining formulas. $\square$

The quotient in (KPC.10) also has its precise universal property. If linear maps $f_h : I_h \rightarrow Z$ and $f_o : I_o \rightarrow Z$ satisfy $f_h A_N^{\text{hom}} = f_o A_N^{\text{orth}}$, their common composite on $\mathcal{H}_N$ vanishes on both K_h and K_o , hence on their sum. It therefore factors uniquely through Q_N , and that factor gives both original maps after the surjections in (KPC.11). Likewise, a linear map $T : I_h \rightarrow I_o$ satisfying $T A_N^{\text{hom}} = A_N^{\text{orth}}$ exists exactly when $K_h \subseteq K_o$. Necessity follows by evaluating on K_h ; under that inclusion the formula $T(A_N^{\text{hom}} c) = A_N^{\text{orth}} c$ is well-defined and proves sufficiency. It is an isomorphism exactly

when $K_h = K_o$. Thus (KPC.11)–(KPC.12) give the full typed correspondence even when no such isomorphism exists.

There is also an explicit polynomial deformation of the sections. For a complex parameter z , define, without division by z ,

$$p_j^{(z)}(s) = \sum_{r=0}^j p_{j,r} z^{j-r} s^r, \quad O_{N,z}(c) = \sum_{|\alpha|=N} c_\alpha \prod_j p_{\alpha_j}^{(z)}(s_j). \quad (\text{KPC.13})$$

For $|\gamma| < N$, its coefficient of s^γ is exactly the coefficient in (KPC.6) multiplied by $z^{N-|\gamma|}$; at degree N it is c_γ . Hence $O_{N,0} = i_N$, $O_{N,1} = O_N$, and $\ell_N O_{N,z} = I$ for every z . The maps $JO_{N,z}$ are matrices polynomial in the parameter, and (KPC.8) is unchanged throughout. A rank condition $\text{rank } JO_{N,z} \leq r$ is exactly the vanishing of every $(r+1)$ -minor. No constancy of their rank follows from this deformation. In particular (KPC.13) retains every original coefficient at $z = 1$; it does not replace the original measure or its orthogonal minimum.

17.3 The algebra generated by the original coordinate ratio

Now take $k = 2$ and retain $h(0) \neq 0$. Every original nontrivial zero in the arithmetic packet has $0 < \text{Re } \rho < 1$, so this domain includes that packet. Work in the original algebra

$$E^{(2)} = \mathbb{C}[s_1, s_2]/(h(s_1), h(s_2)), \quad \zeta = s_1 s_2^{-1}. \quad (\text{KPC.14})$$

The inverse is defined because s and $h(s)$ are coprime; the Euclidean identity for these polynomials supplies an inverse modulo h . All ordered pairs of centres and their multiplicities are retained.

Theorem 17.3 (Complete ratio algebra and homogeneous-shell rank). *For every distinct ratio $r = \rho/\sigma$, put*

$$M_r = \max_{\rho/\sigma=r} (m_\rho + m_\sigma - 1), \quad b = \sum_r M_r, \quad \mu_\zeta(Z) = \prod_r (Z - r)^{M_r}. \quad (\text{KPC.15})$$

Then evaluation at ζ has exactly kernel (μ_ζ) , and gives the unital algebra embedding

$$\mathbb{C}[Z]/(\mu_\zeta) \hookrightarrow E^{(2)}, \quad [P] \mapsto P(\zeta). \quad (\text{KPC.16})$$

Its image is the subalgebra $\mathbb{C}[\zeta]$. For every $N \geq 0$, the full arithmetic homogeneous-shell map is

$$\sum_{j=0}^N c_j s_1^j s_2^{N-j} \mapsto v_h^{\otimes 2} s_2^N \sum_{j=0}^N c_j \zeta^j. \quad (\text{KPC.17})$$

Its kernel consists exactly of the coefficient polynomials $\sum_{j=0}^N c_j Z^j$ divisible by μ_ζ , and

$$\boxed{\text{rank } A_N^{\text{hom}} = \min\{N+1, b\}.} \quad (\text{KPC.18})$$

The arithmetic unit and s_2^N in (KPC.17) are retained invertible multiplication maps. Formula (KPC.18) does not replace the orthogonal-shell map (KPC.7).

Proof. The factors $(s - \rho)^{m_\rho}$ of h are pairwise coprime. For each factor, a Euclidean identity with the product of the others gives a polynomial equal to one modulo that factor and zero modulo all others. Their residue classes are mutually orthogonal idempotents and sum to one: each of these assertions holds modulo every factor, hence modulo their product. They give the full Chinese-remainder isomorphism, with inverse obtained by multiplying each component representative by its idempotent and adding. Tensoring these explicit maps gives

$$E^{(2)} \cong \prod_{(\rho, \sigma)} \mathbb{C}[x, y]/(x^{m_\rho}, y^{m_\sigma}), \quad x = s_1 - \rho, \quad y = s_2 - \sigma. \quad (\text{KPC.19})$$

In each component the basis is the full set $x^a y^c$, $0 \leq a < m_\rho$, $0 \leq c < m_\sigma$. The inverse of $\sigma + y$ is explicitly $\sum_{j=0}^{m_\sigma-1} (-1)^j y^j / \sigma^{j+1}$, as multiplication and cancellation of consecutive powers verify.

Write $m = m_\rho$, $n = m_\sigma$. In this component the ratio retains the exact identity

$$\zeta - \frac{\rho}{\sigma} = \frac{\sigma x - \rho y}{\sigma(\sigma + y)}. \quad (\text{KPC.20})$$

Every monomial of total degree $m + n - 1$ vanishes, since one exponent must reach m or n . Thus the $(m + n - 1)$ -st power of (KPC.20) is zero. Its preceding power has the exact value

$$\left(\zeta - \frac{\rho}{\sigma}\right)^{m+n-2} = \binom{m+n-2}{m-1} \frac{\sigma^{m-1}(-\rho)^{n-1}}{\sigma^{2(m+n-2)}} x^{m-1} y^{n-1} \neq 0. \quad (\text{KPC.21})$$

Indeed only the displayed monomial survives in the numerator power; in the inverse-denominator power only its constant coefficient can multiply that monomial without making it zero. The displayed coefficient is nonzero over $\mathbb{C}$ because $\rho, \sigma \neq 0$. For $m = n = 1$ the preceding power is the zeroth power one, and the first power is zero, so this case is included. The local nilpotent index is therefore exactly $m + n - 1$.

For any nonzero polynomial $P(Z)$, factor at the local ratio r as $P(Z) = (Z - r)^e B(Z)$ with $B(r) \neq 0$. The element $B(\zeta)$ is invertible: after division by its nonzero constant $B(r)$, it is one plus a nilpotent polynomial in $\zeta - r$, whose finite geometric series is an inverse. Hence $P(\zeta) = 0$ in that component exactly when $e \geq m + n - 1$. Across all ordered components, the condition at each coincident ratio is its maximum exponent, and conditions at distinct ratios are coprime. This proves that the global evaluation kernel is precisely (μ_ζ) , including the zero polynomial. It proves (KPC.16) and its stated image.

Each homogeneous monomial satisfies $s_1^j s_2^{N-j} = s_2^N \zeta^j$ inside the original algebra. Applying the original arithmetic unit proves (KPC.17). Both multiplication factors there are invertible, so its kernel is exactly the asserted restricted polynomial kernel. For $N < b$, no nonzero polynomial of degree at most N is divisible by the monic μ_ζ . For $N \geq b$, all such multiples are $\mu_\zeta \mathbb{C}[Z]_{\leq N-b}$, which has dimension $N - b + 1$, since multiplication by the nonzero polynomial is injective. Subtracting this dimension from $N + 1$ gives (KPC.18) in both cases. $\square$

In the retained monomial tensor coordinates, set $S_1 = A_h \otimes I$, $S_2 = I \otimes A_h$, $c = [1] \otimes [1]$, and $Z_{\text{mat}} = S_1 S_2^{-1}$. Then (KPC.17) is exactly

$$(c_0, \dots, c_N) \mapsto U_{v_h^{\otimes 2}} S_2^N \sum_{j=0}^N c_j Z_{\text{mat}}^j c. \quad (\text{KPC.22})$$

This retains each complete Taylor block under (KPC.19) rather than replacing either coordinate by its eigenvalue. Multiplication by $v_h^{\otimes 2} s_2^N$ is a linear isomorphism from $\mathbb{C}[\zeta]$ to its displayed image, with inverse multiplication by $s_2^{-N} (v_h^{\otimes 2})^{-1}$ on that image. The image after this multiplication is not asserted to be a unital subalgebra.

The original summed generator also has an exact map between adjacent shells. Write $A^{(2)} = M_{s_1+s_2}$ on the retained algebra and $U = v_h^{\otimes 2}$. Then

$$A^{(2)}(U s_2^N f(\zeta)) = U s_2^{N+1} (1 + \zeta) f(\zeta). \quad (\text{KPC.22a})$$

This follows from $s_1 + s_2 = s_2(1 + \zeta)$ and commutativity of the original multiplication maps. Multiplication by $1 + Z$ sends degree at most N to degree at most $N + 1$, and sends a multiple of μ_ζ to another such multiple. Therefore (KPC.22a) is well-defined on the complete source quotients in (KPC.17). It is an adjacent-shell action; it does not assert that $\mathbb{C}[\zeta]$ alone is invariant under $A^{(2)}$. For example, if the packet has two distinct centres ρ, σ , the diagonal pairs (ρ, ρ) and (σ, σ) both have ratio one, while the constant terms of their summed generators are 2ρ and 2σ . A polynomial in the ratio has the same constant term at both pairs, so it cannot equal that summed coordinate. Formula (KPC.22a) retains the missing second-coordinate multiplication explicitly.

Theorem 17.4 (Exact algebra-retraction criterion). *In the local algebra $R_{\rho,\sigma} = \mathbb{C}[x, y]/(x^m, y^n)$, with $\rho, \sigma \neq 0$, the ratio-subalgebra inclusion $\mathbb{C}[\zeta] \hookrightarrow R_{\rho,\sigma}$ admits a unital complex-algebra retraction exactly when $\min(m, n) = 1$. In the full original tensor algebra $E_h^{\otimes 2}$, the inclusion (KPC.16) admits a unital complex-algebra retraction exactly when h is squarefree.*

Proof. First suppose $m = 1$, so $x = 0$ and $\zeta = \rho/(\sigma + y)$. This element is a unit, and $y = \rho\zeta^{-1} - \sigma$. Its inverse belongs to the ratio subalgebra: $\zeta = r + w$, with $r = \rho/\sigma \neq 0$ and $w^n = 0$, has inverse $\sum_{j=0}^{n-1} (-1)^j w^j / r^{j+1}$. Thus the ratio subalgebra contains y and is the whole local algebra. If $n = 1$, then $y = 0$ and $x = \sigma\zeta - \rho$, giving the same conclusion. In both cases the inclusion is an isomorphism and its inverse is the asserted retraction.

Now suppose $m, n \geq 2$, and put $L = m + n - 1$. The exact local index proved above identifies the ratio subalgebra with $B = \mathbb{C}[t]/(t^L)$, by $t \mapsto \zeta - \rho/\sigma$. In any putative retraction $f : R_{\rho,\sigma} \rightarrow B$, the images of x and y have zero constant term because their powers vanish. If the coefficient of t in $f(x)$ were a nonzero number a , the coefficient of t^m in $f(x)^m$ would be $a^m \neq 0$. Since $m < L$, that term survives in B , contradicting $x^m = 0$. The same argument using $n < L$ shows that $f(y)$ also has zero linear coefficient. The exact expression (KPC.20) then makes $f(\zeta - \rho/\sigma)$ have zero linear coefficient: its numerator lies in (t^2) and its denominator is a unit. A retraction would send that element to t . This contradiction proves the local converse.

If h is squarefree, (KPC.19) identifies the full tensor algebra with $\mathbb{C}^{\mathbb{Z} \times \mathbb{Z}}$. The ratio subalgebra is the algebra of functions constant on every fibre of $(\rho, \sigma) \mapsto \rho/\sigma$. Indeed (KPC.15) has exponent one at each ratio, and ordinary polynomial interpolation gives every assignment of values on the distinct ratios. Choose one ordered pair in each such fibre. Restriction of a function to those chosen pairs, then assigning that value to the entire corresponding fibre, is a unital algebra map back to this subalgebra. It is the identity on the subalgebra, proving existence of a retraction.

Conversely, let $M = \max_{\rho} m_{\rho} \geq 2$. The ratio-one component in (KPC.15) is

$$B_1 = \mathbb{C}[t]/(t^{2M-1}), \quad \zeta \mapsto 1 + t. \quad (\text{KPC.22b})$$

This is because $\rho/\sigma = 1$ for nonzero centres exactly when $\rho = \sigma$, and the maximum of $2m_{\rho} - 1$ is $2M - 1$. Compose a proposed global retraction with projection onto B_1 . A unital map from the product (KPC.19) into B_1 selects exactly one local factor. To justify that statement, an idempotent of B_1 has constant term zero or one. If it is zero, the idempotent lies in the nilpotent ideal (t) , and an idempotent which is nilpotent is zero, since all its positive powers equal itself. If the constant term is one, apply this argument to one minus the idempotent. Hence the only idempotents are zero and one. The mutually orthogonal source-factor idempotents sum to one, so exactly one maps to one and all others map to zero, proving the selection statement.

Suppose the selected factor is indexed by (ρ, σ) . Images of its nilpotent coordinates have zero constant term, so the constant term of its ratio image is ρ/σ . Equation (KPC.22b) forces this ratio to be one; thus $\rho = \sigma$, with common multiplicity $m = m_{\rho} \leq M < 2M - 1$. The relation $x^m = 0$ again forces the coefficient of t in the image of x to vanish, since a nonzero coefficient would produce a nonzero t^m term in the target. The same holds for y . Their ratio minus one therefore has zero linear coefficient by (KPC.20), contradicting its required image t in (KPC.22b). This proves the global converse while retaining all source-factor idempotents and the original ratio embedding. $\square$

17.4 A complete finite witness for unequal shell ranks

The following positive polynomial model tests the proposed passage from homogeneous rank to the original orthogonal rank. It is not a claim that $1/2$ is an arithmetic zero. Use the specified measure

$$\nu_{\text{ex}} = \delta_{1/2-i} + \delta_{1/2} + \delta_{1/2+i} \quad \text{on the variable } s, \quad h_{\text{ex}}(s) = s - \frac{1}{2}, \quad v_{\text{ex}} = 1. \quad (\text{KPC.23})$$

All three masses are exactly one. Through degree two the monic orthogonal polynomials and norms are

$$\begin{array}{c|c|c} j & p_j(s) & \kappa_j \\ \hline 0 & 1 & 3 \\ 1 & s - \frac{1}{2} & 2 \\ 2 & (s - \frac{1}{2})^2 + \frac{2}{3} & \frac{2}{3}. \end{array} \quad (\text{KPC.24})$$

To verify the complete Gram, their three value columns are

$$(1, 1, 1)^T, \quad (-i, 0, i)^T, \quad (-1/3, 2/3, -1/3)^T.$$

Their pairwise Hermitian inner products are zero, and their squared norms are respectively 3, $1 + 0 + 1 = 2$, and $1/9 + 4/9 + 1/9 = 2/3$. Thus the moment form through degree two is positive definite, as required for the source and next layer used below.

Take $k = 2$, $N = 1$. Here $d = 1$ and the threshold $N \geq k(d - 1) = 0$ holds. The original tensor residue algebra is $E_{\text{ex}}^{(2)} = \mathbb{C}$, with $s_1 = s_2 = 1/2$, and its full jet map is evaluation at that ordered pair. For $c = c_{10}s_1 + c_{01}s_2$, the two section and image maps are exactly

$$\begin{aligned} i_1(c) &= c_{10}s_1 + c_{01}s_2, \\ O_1(c) &= c_{10}(s_1 - \frac{1}{2}) + c_{01}(s_2 - \frac{1}{2}), \\ L_1(c) &= -\frac{1}{2}(c_{10} + c_{01}), \\ A_1^{\text{hom}}(c) &= \frac{1}{2}(c_{10} + c_{01}), \quad A_1^{\text{orth}}(c) = 0. \end{aligned} \quad (\text{KPC.25})$$

Thus the lower coefficient in (KPC.7) cancels the entire homogeneous image. The ranks are one and zero. The kernels are $K_h = \{(c_{10}, c_{01}) : c_{10} + c_{01} = 0\}$ and $K_o = \mathbb{C}^2$, so the common quotient in (KPC.10) is zero and the full correspondence is $\mathbb{C} \oplus \{0\}$. The ratio is $\zeta = 1$, its minimal polynomial is $Z - 1$, and (KPC.18) gives the correct homogeneous rank $\min(2, 1) = 1$. Equation (KPC.25) proves explicitly why it gives no equality with the orthogonal rank.

The actual finite minimum and next layer in this model can also be retained. In the source with product measure $\nu_{\text{ex}}^{\otimes 2}$, the degree-at-most-one orthogonal basis is p_0p_0, p_1p_0, p_0p_1 , with norms 9, 6, 6 and jets 1, 0, 0. Hence the full interpolation kernel is $K_1 = 1/9$, the metric is $G_1 = 9$, and its minimum is the constant polynomial $R_1u = u$. Multiplication by the original sum $s_1 + s_2$ acts on the packet by $A = 1$. Consequently

$$B_1u = (s_1 + s_2 - 1)u \in \mathcal{L}_1, \quad C_1 = (I - P_{\mathcal{L}_1})B_1 = 0, \quad W_1 = A^*G_1 + G_1A - 2G_1 = 0. \quad (\text{KPC.26})$$

Indeed $\mathcal{L}_1$ is spanned by $s_1 - 1/2, s_2 - 1/2$, the complete degree-one relation kernel. Thus the derivative rank zero is verified directly without presupposing the shell-rank formula.

At the next shell, ordered as $(2, 0), (1, 1), (0, 2)$, the original jets and norm diagonal are

$$U = \begin{pmatrix} 2/3 & 0 & 2/3 \end{pmatrix}, \quad D_{\text{next}} = \text{diag}(2, 4, 2).$$

The actual layer vectors are $e_{20} = (s_1 - 1/2)^2$, $e_{11} = (s_1 - 1/2)(s_2 - 1/2)$, and $e_{02} = (s_2 - 1/2)^2$. Each has zero jet and is orthogonal to $\mathcal{L}_1$, either by its product moments or by the layer construction. Their Gram is exactly

$$H = D_{\text{next}} + U^*G_1U = \begin{pmatrix} 6 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 6 \end{pmatrix}. \quad (\text{KPC.27})$$

For example the diagonal squared norm of e_{20} is $(1 + 0 + 1)(1 + 1 + 1) = 6$, and its pairing with e_{02} is $(-1 + 0 - 1)^2 = 4$; the odd cross terms vanish. The eigenvalues of the outer two-coordinate block are 10, 2, and the middle eigenvalue is 4, proving positivity. The next full kernel and metric therefore are

$$K_2 = \frac{1}{9} + \frac{(2/3)^2}{2} + \frac{(2/3)^2}{2} = \frac{5}{9}, \quad G_2 = \frac{9}{5}. \quad (\text{KPC.28})$$

The representative itself is the explicit polynomial

$$R_2 u = \left[1 + \frac{3}{5} \left((s_1 - \frac{1}{2})^2 + (s_2 - \frac{1}{2})^2 \right) \right] u. \quad (\text{KPC.28a})$$

For a direct verification, it has jet u . Its values for u equal to one are $-1/5$ at the four product nodes where both coordinates differ from $1/2$, $2/5$ at the four nodes where exactly one differs, and one at the central node. Its squared norm is therefore $4/25 + 16/25 + 1 = 9/5$. It is orthogonal to the five relation basis vectors $s_1 - 1/2, s_2 - 1/2, e_{20}, e_{11}, e_{02}$. The first two and e_{11} have an odd coordinate, so their pairings vanish by the explicit sign symmetry of the nine-node measure. For e_{20} , only the two noncentral first-coordinate values contribute, and for each of them the sum of the three values of $R_2(1)$ is $-1/5 + 2/5 - 1/5 = 0$. Its pairing is thus zero; the same computation with the coordinates interchanged treats e_{02} . These five independent zero-jet polynomials span the kernel in the six-dimensional degree-at-most-two source, whose jet map is onto the one-dimensional target. Every other feasible polynomial differs by a vector in this kernel. Pythagoras and positive definiteness therefore prove the unique-minimum assertion directly, including its actual norm. Thus the nonzero original relation layer and its positive metric loss are retained alongside the zero derivative layer. This complete finite calculation rejects only the proposed equality between homogeneous and orthogonal-shell ranks. The original arithmetic shell continues to be given by (KPC.4)–(KPC.7), and the exact graph and quotient maps relating its two presentations are (KPC.9)–(KPC.12).

Every linear map $T : V \rightarrow W$ in these comparisons has a support-preserving Split-Zero lift on the following specified carriers at a fixed support λ :

$$\widehat{V}_\lambda = \{\tau_V\} \sqcup (\{\lambda\} \times V), \quad \widehat{W}_\lambda = \{\tau_W\} \sqcup (\{\lambda\} \times W).$$

Addition has external τ as identity and uses vector addition inside the supported fibre. The scalar action is $r^\bullet(\lambda, v) = (\lambda, rv)$, while the external scalar zero sends every input to external absence and every scalar sends external absence to external absence. The lift maps τ_V to τ_W and (λ, v) to (λ, Tv) . Its addition identity follows from $T(v + w) = Tv + Tw$; its supported-scalar identity follows from $T(rv) = rTv$, including $r = 0$. The external cases follow from the definitions. Thus every displayed kernel vector maps to the represented zero at λ , and the graph, quotient and lower-degree maps retain this support even when the two ranks differ.

18 The original frontier energy, determinant growth, and the entire border fibre

The symmetric-frontier delivery of 12 September computes coordinates on the actual next relation layer. We retain these coordinates and prove their exact relation to the accumulated source energy of Section 15. The calculation also identifies the entire zero fibre of the delivery's smaller border operator, including its nilpotent extension.

18.1 Original moment coordinates and the relation graph

Keep a nonempty full-order packet h , its degree $d, g = 2\xi, E = \mathbb{C}[s]/h, v = j_h(g/h)$, and $A = M_s$. The positive measure remains

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}.$$

Let p_n be the monic orthogonal polynomial obtained from $1, s, \dots, s^n$ without dividing by its norm, and let $\omega_n = \|p_n\|_{\nu_h}^2 > 0$. The original generator has the exact recurrence

$$sp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad a_n = \omega_n / \omega_{n-1} \ (n \geq 1), \quad a_0 = 0, \quad b_n + \bar{b}_n = 1. \quad (\text{FE.1})$$

Here $p_{-1} = 0$. To verify the coefficients, multiplication by s has adjoint multiplication by $1 - s$ on the original line. Its pairing against degrees below $n - 1$ vanishes, the leading coefficient is one, and the pairing against p_{n-1} is $-\omega_n$. Its diagonal is $\int s|p_n|^2 d\nu_h/\omega_n$, which proves the last identity and retains its possibly nonzero imaginary part.

Fix $k \geq 1$ and $M \geq k(d - 1)$, and use $A_k = \sum_j A_j$ on the unchanged $E_k = E^{\otimes k}$. For $\alpha \in \mathbb{N}^k$, put

$$p_\alpha = \prod_j p_{\alpha_j}(s_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = v^{\otimes k}[p_\alpha] \in E_k.$$

All remainder coordinates, including every nilpotent jet and the entire local arithmetic unit, are retained in z_α . The original source map $\mathcal{T}^{(k)}$ is the map in (CT.11)–(CT.13). Its orthogonal polynomial images have norms ω_α . The constrained minimum gives exactly

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}, \quad R_M u = \sum_{|\alpha| \leq M} \mathcal{T}^{(k)} p_\alpha \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (\text{FE.2})$$

This is (CT.5) in the explicitly declared orthogonal source basis; no moment mass or polynomial norm has been changed.

Order the indices β of total degree $M + 1$, and put

$$r = \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\omega_\beta), \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \quad T_+ = [\mathcal{T}^{(k)} p_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k.$$

The two source maps have the same column ordering. Since the image of T_+ is orthogonal to the entire degree- M image,

$$b = T_+ - R_M F : \mathbb{C}^r \longrightarrow \mathcal{E}_M, \quad H = b^* b = \Omega + F^* G_M F, \quad b^* R_M = -F^* G_M. \quad (\text{FE.3})$$

Here $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ is the original relation layer. The complete jets of b vanish; both of its terms are orthogonal to $\mathcal{B}_M$. Taking the highest orthogonal degree of bu recovers $T_+ u$, proving injectivity. For any boundary of degree $M + 1$, subtract the image under b of its highest coefficient. The difference lies in $\mathcal{B}_M$. This gives the inverse on the orthogonal layer and proves surjectivity. It also gives an original primitive: divide the numerator of each column of b in the fixed order by $h(s_1), \dots, h(s_k)$. The primitive in factor j has coefficient $(-1)^{j-1}$; its tensor differential contributes the same sign, giving the original positive boundary.

Define the second frontier matrix by

$$E_+ = [w_\beta], \quad w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}. \quad (\text{FE.4})$$

The notation E_+ denotes a matrix $\mathbb{C}^r \rightarrow E_k$. From (FE.2), the highest coefficient of $D^{(k)} R_M - R_M A_k$ is $\Omega^{-1} E_+^* G_M$. Projection onto the graph (FE.3) therefore proves, with the original types and signs,

$$Y_M = -b H^{-1} F^* G_M, \quad C_M = b \Omega^{-1} E_+^* G_M, \quad \Delta_M = Y_M^* Y_M = G_M F H^{-1} F^* G_M. \quad (\text{FE.5})$$

The notation C_M is the original derivative-layer map in (CT.15), not the inverse interpolation matrix $\mathcal{C}_M$.

Applying (FE.1) to (FE.2) cancels every internal neighbouring pair, because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonals give exactly $k C_M$. The remaining pairs have top degree $M + 1$, proving

$$\mathcal{Z}_M = A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{FE.6})$$

Alternatively (FE.5) inserted into $W_M = -(Y_M^* C_M + C_M^* Y_M)$ gives exactly (FE.6). Thus both the polynomial cancellation and its actual boundary primitive yield the same original form.

18.2 Exact relative loss and the source derivative energy

For the next formulas only, use the displayed finite isometries from the specified metric coordinate spaces and define

$$U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}, \quad P = U U^*, \quad K_U = U^* U. \quad (\text{FE.7})$$

The original F, E_+, Ω, G_M are still given by (FE.1)–(FE.5), and can be recovered by the inverse coordinate maps in (FE.7). In particular no polynomial is assigned norm one and no arithmetic coefficient is suppressed.

Theorem 18.1 (Frontier energy and every relative Gram eigenvalue). *In these explicitly related coordinates,*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = U V^* + V U^*, \\ T_M &= G_M^{-1/2} \Delta_M G_M^{-1/2} = U (I + K_U)^{-1} U^* = P (I + P)^{-1}, \\ G_M^{-1/2} G_{M+1} G_M^{-1/2} &= (I + P)^{-1}. \end{aligned} \quad (\text{FE.8})$$

If $\lambda_1, \dots, \lambda_q > 0$ are all positive eigenvalues of P , including multiplicity, then

$$\begin{aligned} q &= \text{rank } F = \text{rank } Y_M = \text{rank } \Delta_M, \\ \tau_M &= \text{Tr}(G_M^{-1} \Delta_M) = \sum_{i=1}^q \frac{\lambda_i}{1 + \lambda_i}, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M} = \prod_{i=1}^q (1 + \lambda_i)^{-1}, \\ \chi_M &= \text{Tr}(G_M^{-1} C_M^* C_M) = \|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{FE.9})$$

The empty product equals one if $q = 0$. In addition $\text{rank } C_M = \text{rank } E_+$.

Proof. Equation (FE.6) proves the first line of (FE.8). The identity $H = \Omega^{1/2}(I + U^* U) \Omega^{1/2}$ substituted into (FE.5) proves the second line's first expression. The equality $(I + U U^*) U = U (I + U^* U)$, multiplied by the indicated invertible factors, identifies it with $P (I + P)^{-1}$. Subtract it from I to get the third line. The positive matrices $I + K_U$ and $I + P$ are invertible directly from their quadratic forms.

The graph map b is injective, and H, Ω, G_M are invertible, so (FE.5) proves both asserted ranks. Equivalently, the formula for T_M identifies its positive eigenvalues with $\lambda_i / (1 + \lambda_i)$, retaining zeros on $\ker P$. The trace and determinant formulas now follow from the finite spectral decomposition in these same coordinates. Finally

$$C_M G_M^{-1/2} = b \Omega^{-1/2} V^*, \quad (b \Omega^{-1/2})^* (b \Omega^{-1/2}) = I + U^* U.$$

Its squared Hilbert–Schmidt norm is $\text{Tr}(V(I + U^* U)V^*)$, which expands to the last line of (FE.9). This is the energy of the original source-layer map for its inherited $L^2(d^k x)$ inner product. $\square$

The exact remainder in the accumulated determinant calculation is therefore also a frontier formula:

$$-\log \pi_M - \tau_M = \sum_{i=1}^q \left(\log(1 + \lambda_i) - \frac{\lambda_i}{1 + \lambda_i} \right) = \sum_{i=1}^q \int_0^{\lambda_i} \frac{x}{(1+x)^2} dx. \quad (\text{FE.10})$$

Differentiate each displayed scalar expression and evaluate at zero to prove the equality, including every eigenvalue and its sign. Thus (JD.14) uses the exact original derivative cost $\|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2$, and (JD.16) is precisely the pair of identities

$$\sum_{M \geq a} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M \geq a} \sum_i \frac{\lambda_{M,i}}{1 + \lambda_{M,i}} = \infty. \quad (\text{FE.11})$$

Both use the already proved actual boundary density; the second retains its full inverse-interpolation cost.

18.3 The entire small border operator, including its zero extension

Let $n = d^k$, and define

$$B = [U, V] : \mathbb{C}^{2r} \rightarrow \mathbb{C}^n, \quad K = B^*B, \quad J_0 = \begin{pmatrix} 0 & I_r \\ I_r & 0 \end{pmatrix}, \quad A_0 = J_0 K, \quad H_0 = K^{1/2} J_0 K^{1/2}. \quad (\text{FE.12})$$

The symbol A_0 here is a finite border operator, not the arithmetic generator A or A_k . This distinction is accompanied by the exact maps below. The source's unscaled border operator is

$$J[F, E_+]^* G_M[F, E_+], \quad J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

With $D_\Omega = \text{diag}(\Omega^{1/2}, \Omega^{1/2})$, it is exactly $D_\Omega^{-1} A_0 D_\Omega$: substitution of (FE.7) gives

$$[F, E_+]^* G_M[F, E_+] = D_\Omega K D_\Omega.$$

This states the full coordinate similarity, including zero.

Put $N = \ker K = \ker B$ and $\mathcal{R} = \text{ran } K$. They are orthogonal complements since K is Hermitian, and $K^{1/2}$ restricts to an invertible positive map on $\mathcal{R}$. There is an exact sequence of spaces with their specified operators:

$$0 \longrightarrow (N, 0) \longrightarrow (\mathbb{C}^{2r}, A_0) \xrightarrow{K^{1/2}} (\mathcal{R}, H_0|_{\mathcal{R}}) \longrightarrow 0. \quad (\text{FE.13})$$

Indeed $K^{1/2} A_0 = H_0 K^{1/2}$, its kernel is N , and its image is $\mathcal{R}$. On N , $A_0 = 0$. The map

$$Q|_{\mathcal{R}} = B(K^{1/2}|_{\mathcal{R}})^{-1}, \quad Q|_N = 0 \quad (\text{FE.14})$$

is a partial isometry onto $\text{ran } B$: for $x, y \in \mathcal{R}$, substitution of $B^*B = K$ gives $\langle Qx, Qy \rangle = \langle x, y \rangle$. It satisfies $B = QK^{1/2}$ and

$$S_M = B J_0 B^* = Q H_0 Q^*, \quad S_M|_{(\text{ran } B)^\perp} = 0. \quad (\text{FE.15})$$

Thus the restriction of H_0 to $\mathcal{R}$ and that of S_M to $\text{ran } B$ are unitarily intertwined by an explicitly proved map; both remaining orthogonal complements and their zero actions are specified.

Theorem 18.2 (Exact zero fibre of the border comparison). *Let $Z_0 = \ker(H_0|_{\mathcal{R}})$ and $q_0 = \dim Z_0$. The complete generalized zero eigenspace of A_0 is*

$$\mathcal{N}_0 = (K^{1/2})^{-1}(Z_0), \quad A_0^2|_{\mathcal{N}_0} = 0. \quad (\text{FE.16})$$

Define $L = (K^{1/2}|_{\mathcal{R}})^{-1} : \mathcal{R} \rightarrow \mathcal{R}$. For its restriction to $z \in Z_0$, set

$$Lz = (K^{1/2}|_{\mathcal{R}})^{-1}z, \quad N_0 z = J_0 K^{1/2} z \in N. \quad (\text{FE.17})$$

The map $N_0 : Z_0 \rightarrow N$ is injective, and the exact vector space decomposition $\mathcal{N}_0 = N \oplus LZ_0$ gives

$$A_0(n + Lz) = N_0 z. \quad (\text{FE.18})$$

Consequently its zero Jordan data consist of exactly q_0 blocks of size two and $\dim N - q_0$ blocks of size one. Its nonzero Jordan blocks have size one and coincide, with full multiplicities, with the nonzero eigenvalues of S_M .

Proof. If $A_0^m x = 0$, (FE.13) gives $H_0^m K^{1/2} x = 0$. Hermitian spectral decomposition of $H_0|_{\mathcal{R}}$ identifies its generalized zero space with Z_0 , proving that x belongs to the preimage in (FE.16). Conversely, if $K^{1/2} x \in Z_0$, then $K^{1/2} A_0 x = 0$, so $A_0 x \in N$ and $A_0^2 x = 0$. This proves (FE.16) exactly.

For $z \in Z_0$, one has $K^{1/2}J_0K^{1/2}z = H_0z = 0$, proving $N_0z \in N$. Since J_0 is invertible and $K^{1/2}$ is injective on $\mathcal{R}$, N_0 is injective. Equation $K^{1/2}Lz = z$ gives the direct decomposition, and $A_0Lz = J_0KLz = J_0K^{1/2}z$ proves (FE.18). Choose a basis of Z_0 , use its injective image as part of a basis of N , and extend it in N . Each pair Lz, N_0z is a size-two chain, and the remaining basis vectors of N are size-one chains. This also proves $q_0 \leq \dim N$ without a rank assumption.

For a nonzero eigenvalue λ , take an eigenvector $z \in \mathcal{R}$ of H_0 . If $x = Lz$, then $n = A_0x - \lambda x \in N$, so $x + n/\lambda$ is an eigenvector of A_0 lifting z . The lift is unique because A_0 vanishes on N and $\lambda \neq 0$. This supplies every eigenvector of the Hermitian quotient and leaves no nontrivial Jordan chain at a nonzero eigenvalue: applying (FE.13) to such a chain would give one for the Hermitian quotient, and a residual chain contained in N is impossible at nonzero λ . Finally (FE.15) identifies those eigenvalues with the actual relative form. All claims include their algebraic multiplicities. $\square$

In particular,

$$\dim \ker S_M = n - \text{rank } B + q_0, \quad \dim \ker A_0 = 2r - \text{rank } B, \quad \text{mult}_0(A_0) = 2r - \text{rank } B + q_0. \quad (\text{FE.19})$$

The first equality is (FE.15), the second uses the invertibility of J_0 , and the third counts (FE.18). Thus replacing the source's border calculation by a Hermitian one has an exact kernel, quotient, complement and zero extension, all explicitly retained here. At any fixed retained support λ , each displayed linear map has its Split-Zero lift $(\lambda, x) \mapsto (\lambda, Tx)$, with external $\tau \mapsto \tau$. A size-two chain in (FE.18) is therefore the exact supported chain

$$(\lambda, Lz) \mapsto (\lambda, N_0z) \mapsto (\lambda, 0). \quad (\text{FE.19a})$$

The last value is the represented zero in that fibre. For $z \neq 0$, injectivity of N_0 proves that the intermediate value has nonzero amplitude; neither map sends it to external absence.

18.4 A quantified link between the frontier bound and determinant cost

Let $\lambda_M = \|UU^*\|_{\text{op}}$, and retain the original positive recurrence quantity

$$\Gamma_M = \max_{|\alpha|=M} \sum_{j=1}^k \{a_{\alpha_j+1} + (k-1)a_{\alpha_j}\} > 0. \quad (\text{FE.20})$$

The strict inequality follows from $a_{\alpha_j+1} > 0$. For clarity, its upper estimate uses an exact matrix comparison. Let $Z = [z_\alpha]_{|\alpha|=M}$, $\Omega_M = \text{diag}(\omega_\alpha)$, and let $L_{\alpha\beta} = a_{\beta_j}$ for $\alpha = \beta - e_j$, summed over those j . Then $E_+ = ZL$, and $\Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is the displayed sum in (FE.20): $\omega_\alpha a_{\alpha_j+1} / \omega_{\alpha+e_j} = 1$ for each outgoing edge, and the other predecessors contribute $\sum_{l \neq j} a_{\alpha_l}$. For a matrix with nonnegative entries, an eigenvector component of largest absolute value bounds the eigenvalue's absolute value by its largest row sum. The matrix is similar to the positive Hermitian matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$, so

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_M Z \Omega_M^{-1} Z^* \preceq \Gamma_M \mathcal{C}_M.$$

It follows that $VV^* \preceq \Gamma_M I$. For any vector x , Cauchy–Schwarz applied to (FE.8) gives $|x^* S_M x| \leq 2\|U^* x\| \|V^* x\|$. Thus

$$\epsilon_M \leq 2\sqrt{\Gamma_M \lambda_M}. \quad (\text{FE.21})$$

Every actual eigenline survives in the full tensor family. If $\delta_* = \max_{\rho \in Z} |\text{Re } \rho - 1/2|$, its repeated tensor gives $\epsilon_M \geq 2k\delta_*$. Combining this with (FE.9) and (FE.21) proves, at every finite degree,

$$\lambda_M \geq \frac{k^2 \delta_*^2}{\Gamma_M}, \quad \frac{\det G_M}{\det G_{M+1}} \geq 1 + \frac{k^2 \delta_*^2}{\Gamma_M}. \quad (\text{FE.22})$$

Indeed the determinant inverse is $\prod_i (1 + \lambda_i)$, which is at least $1 + \lambda_M$. For integers $k(d-1) \leq a < b$, multiplication, followed by the increasing logarithm, gives the explicit source constraint

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{k^2 \delta_*^2}{\Gamma_M} \right). \quad (\text{FE.23})$$

When $\delta_* = 0$, this remains a valid inequality with zero right side and no division by a zero spectral quantity. It is an exact necessary growth relation for the constructed arithmetic matrices, not an assumed upper estimate on that growth.

The original derivative energy supplies a second exact growth bound without the recurrence maximum. By (FE.8), $\|Y_M G_M^{-1/2}\|_{\text{op}}^2 = \lambda_M / (1 + \lambda_M)$. By (FE.9), $\|C_M G_M^{-1/2}\|_{\text{op}}^2 \leq \chi_M$. The original cross-pairing formula, the operator triangle inequality, and submultiplicativity therefore give

$$\epsilon_M^2 \leq 4 \frac{\lambda_M}{1 + \lambda_M} \chi_M. \quad (\text{FE.24})$$

If $\delta_* > 0$, combine it with the retained eigenline $\epsilon_M \geq 2k\delta_*$. The finite number $\lambda_M / (1 + \lambda_M)$ is strictly below one, so

$$\chi_M > k^2 \delta_*^2, \quad \lambda_M \geq \frac{k^2 \delta_*^2}{\chi_M - k^2 \delta_*^2}, \quad \log \frac{\det G_M}{\det G_{M+1}} \geq -\log \left(1 - \frac{k^2 \delta_*^2}{\chi_M} \right). \quad (\text{FE.25})$$

Every denominator and logarithm in (FE.25) has just been proved valid. The middle inequality follows by rearranging (FE.24); the last uses $\det G_M / \det G_{M+1} \geq 1 + \lambda_M$. Summing it over any $k(d-1) \leq a < b$ gives the corresponding finite accumulated determinant inequality. For $\delta_* = 0$, (FE.24) remains valid, and (FE.25) is left un-divided; no 0/0 value is assigned to an original zero-energy layer.

For the actual conjugation-and-reflection packets, the further exact original-layer calculation in Section 21 proves $\epsilon_M^2 \leq [\lambda_M / (1 + \lambda_M)] \zeta_M \leq [\lambda_M / (1 + \lambda_M)] \chi_M$. Here ζ_M is the operator-norm square of the same derivative map whose trace cost is χ_M . Its proof uses the two actual reflection parities in that layer. For $k = 1$, it proves the exact equality $\epsilon_M^2 = (1 - \pi_M) \chi_M$, retaining both zero frontier cases. Thus (FE.24)–(FE.25) retain their general hypotheses, while (QD.8)–(QD.15) give the stronger arithmetic conclusions and all their finite accumulated determinant bounds.

19 The signed symmetric summand and its complete spectral cost

This section proves the cochain, coordinate, and spectral comparisons for the symmetric summand in the arithmetic frontier delivery. The original packet is $E = \mathbb{C}[t]/h$, of dimension $d \geq 1$, and its operator is $A = M_t$. The tensor degree is $k \geq 1$. All orders in $h = \prod_\rho (t - \rho)^{m_\rho}$ remain present. The original joint representative, metric, and layer maps of (CT.12)–(CT.16) are denoted by R_M, G_M, Y_M, C_M , with $M \geq k(d-1)$. The source's symmetric construction is used through the maps proved below; its orbit sums keep their literal factorials.

19.1 The chain projector and its completed and finite targets

The completion in this subsection is specifically the completed projective tensor product $\widehat{\otimes}_\pi$ of the strong-Schwartz locally convex spaces. The global-retraction source, equations (16), (19), and (24), constructs continuous maps

$$\Lambda : \mathcal{B} \longrightarrow V, \quad \Lambda \Theta = 1, \quad s : Q \longrightarrow \mathcal{B}, \quad s[F] = (1 - \Theta \Lambda) F, \quad qs = 1. \quad (\text{SS.1})$$

Its specified cutoffs and moment projection occur in $\Lambda = P_V (1 - A_\alpha B_\beta)^{-1} Y_{\alpha, \beta}$. The convergence and continuity of this formula are proved in that source; no Hilbert-space replacement of these topologies

is used here. Put $C = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $P = sq$ as a cochain map with degree-zero part zero, and $H^1 = \Lambda$. Then in the two degrees respectively, $\Lambda\Theta = 1$ and $\Theta\Lambda = 1 - sq$ prove

$$dH + Hd = 1 - P. \quad (\text{SS.2})$$

For homogeneous inputs define the chain permutation by

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i||x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (\text{SS.3})$$

Here the degrees in the exponent belong to the original inputs. An adjacent interchange contributes $(-1)^{ab}$ for degrees a, b . Interchanging twice contributes $(-1)^{2ab} = 1$; disjoint interchanges commute; and the two three-interchange braid words both contribute $(-1)^{ab+ac+bc}$. Thus (SS.3) is a representation of S_k . On two adjacent factors, $d(x \otimes y) = dx \otimes y + (-1)^a x \otimes dy$. After interchange the coefficients of $dy \otimes x$ and $y \otimes dx$ are respectively $(-1)^{ab}$ and $(-1)^{ab+b}$, on either side of the cochain equation. This proves $dT_\pi = T_\pi d$, first for generators and then for every permutation. Each map is continuous and therefore extends to $C^{\widehat{\otimes}_\pi k}$.

The signed average and its complementary map are

$$S_k = \frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi, \quad 1 - S_k. \quad (\text{SS.4})$$

For each γ , exactly $k!$ pairs (π, σ) satisfy $\pi\sigma = \gamma$, and their signed coefficient is $\text{sgn}(\gamma)/(k!)^2$. Hence $S_k^2 = S_k$. In top degree every factor has degree one, so $T_\pi = \text{sgn}(\pi)P_\pi$ and S_k is the ordinary symmetric projection there. The unsigned chain average has top part $k!^{-1} \sum_{\pi} \text{sgn}(\pi)P_\pi$; this exact equation specifies its alternating target.

To verify the completed cohomology directly, extend the finite sum

$$H_k = \sum_{j=1}^k P^{\widehat{\otimes}_\pi(j-1)} \widehat{\otimes}_\pi H \widehat{\otimes}_\pi 1^{\widehat{\otimes}_\pi(k-j)}. \quad (\text{SS.5})$$

The degree -1 factor contributes the Koszul sign $(-1)^{\sum_{i < j} |x_i|}$. Expanding $dH_k + H_k d$, the terms in which the differential and homotopy occupy different slots cancel in pairs. The term in slot j is $P^{\widehat{\otimes}(j-1)} \otimes (1 - P) \otimes 1^{\widehat{\otimes}(k-j)}$ by (SS.2). Summing telescopes to $1 - P^{\widehat{\otimes} k}$. Continuity extends this identity from the dense algebraic tensor product. Replace H_k by the explicit equivariant homotopy

$$\overline{H}_k = \frac{1}{k!} \sum_{\pi} T_\pi H_k T_\pi^{-1}. \quad (\text{SS.6})$$

The operator $P^{\widehat{\otimes} k}$ commutes with every permutation, so (SS.6) has the same homotopy identity and commutes with S_k . The maps $q^{\widehat{\otimes}_\pi k}$ and $s^{\widehat{\otimes}_\pi k}$, restricted to the images of (SS.4), now give continuous inverse cohomology maps. Thus

$$H^r(\text{im } S_k) = \begin{cases} (Q^{\widehat{\otimes}_\pi k})^{S_k}, & r = k, \\ 0, & r \neq k. \end{cases} \quad (\text{SS.7})$$

The complementary cohomology is the image of $1 - \frac{1}{k!} \sum_{\pi} P_\pi$ on the same completed Q -tensor. The full cochain isomorphism and its inverse are $x \mapsto (S_k x, (1 - S_k)x)$ and $(u, v) \mapsto u + v$. Equations (SS.5)–(SS.7) prove the completion statement using the supplied continuous contraction; they require no exactness assertion for arbitrary completed tensor products.

Here is the finite-rank cochain map that supports the finite traces. The full packet jet $J_h : \mathcal{B} \rightarrow E$ annihilates ΘV , and descends to a continuous $j_h : Q \rightarrow E$. The original packet inclusion $\sigma_h : E \rightarrow Q$ satisfies $j_h \sigma_h = 1$, as $J_h R_M = 1$ and $q R_M = \sigma_h$ in one variable. Set $\iota_h = s \sigma_h : E \rightarrow \mathcal{B}$ and define

$$\Pi_h^0 = 0, \quad \Pi_h^1 = \iota_h J_h, \quad J_h \iota_h = 1. \quad (\text{SS.8})$$

This is a continuous finite-rank cochain projector: $\Pi_h^1 \Theta = 0$ and $(\Pi_h^1)^2 = \Pi_h^1$. It commutes with permutations after taking identical tensor powers. Consequently $\Pi_h^{\otimes k} S_k$ and $\Pi_h^{\otimes k} (1 - S_k)$ are the finite cochain projectors with ranks $\binom{k+d-1}{d-1}$ and $d^k - \binom{k+d-1}{d-1}$. The projector S_k on the full infinite complex itself has not been assigned either finite rank. The finite inclusion and retraction on cohomology are precisely $\sigma_h^{\otimes k}$ and $j_h^{\otimes k}$, restricted to the respective permutation summands. Their composite on the packet is the identity.

For $U(a) = e^{(\log a)A}$, $a > 0$, the arithmetic action obeys $J_h U_{\mathcal{B}}(a) = U(a) J_h$. Thus $J_h U_{\mathcal{B}}(a) \iota_h = U(a)$. If $L : F \rightarrow X$, $R : X \rightarrow F$, and F is finite dimensional, the trace of the finite-rank map LR is $\text{Tr}_F(RL)$: expanding LRx in a basis of F expresses both as the sum of the same coefficient functionals evaluated on the same image vectors. Apply this identity to (SS.8) and its tensor powers. Expanding a permutation trace in a tensor basis contracts one matrix index around each cycle, proving

$$\text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|}). \quad (\text{SS.9})$$

This proof uses all entries of $U(a)$ and applies to its nonsemisimple action. The cochain supertrace carries $(-1)^k$. The complementary finite supertrace is exactly $(-1)^k$ times $(\text{Tr } U(a))^k$ minus the trace in (SS.9). No infinite trace is assigned by these equations.

At a fixed permutation-invariant support fibre, every displayed linear map lifts by $(\lambda, x) \mapsto (\lambda, Tx)$. For arbitrary masks the support of (SS.4) is the specified orbit join $c(\lambda) = \bigvee_{\pi} \pi \lambda$, and the amplitude maps include the original transports into that fibre. The two projection maps then decompose the transported vector in the fibre over $c(\lambda)$; their sum is that transport. Their sum has not been declared the identity on a different original fibre. The upstream kernel is exactly the kernel of its specified transport $V_{\lambda} \rightarrow V_{c(\lambda)}$. At invariant masks the transport is the identity and the decomposition above applies. Every vanishing projected amplitude remains the supported zero in its target fibre; external absence is sent to external absence.

19.2 Orbit coordinates, their Gram, and all local nilpotents

Let $e_1, \dots, e_d$ denote the original power basis with indices shifted by one. For $n \in \mathbb{N}^d$, $|n| = k$, let b_n be the sum of the distinct basis tensors having occupancy n . Define

$$N = \binom{k+d-1}{d-1}, \quad I : \mathbb{C}^N \longrightarrow E^{\otimes k}, \quad I e_n = b_n, \quad D_o = I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, \quad L = D_o^{-1} I^*. \quad (\text{SS.10})$$

Distinct tensor basis vectors are orthonormal for this coordinate calculation; hence D_o records exactly their orbit sizes. It is separate from the arithmetic Gram. One has $LI = 1$ and $IL = P_{\text{sym}} = k!^{-1} \sum P_{\pi}$, since either map replaces every coefficient on an orbit by its average. Here $\text{Sym}^k E$ denotes this invariant image. Its comparison with the symmetric tensor quotient is the quotient map restricted to that image: it sends b_n to $(k! / \prod_i n_i!) e_1^{n_1} \cdots e_d^{n_d}$. The inverse sends this quotient monomial to $(\prod_i n_i! / k!) b_n$. These formulas prove the isomorphism and retain the exact diagonal factors between the two presentations.

Permutation invariance of the arithmetic constraints and measure makes the unique constrained minimum satisfy $R_M P_{\pi} = P_{\pi} R_M$; applying the inverse permutation to a feasible minimizer gives a feasible vector of the same norm, and uniqueness identifies it. Thus G_M and G_M^{-1} commute with P_{sym} . Retain the matrices

$$G_M^s = I^* G_M I, \quad W_M^s = I^* W_M I, \quad A^s = L A^{(k)} I, \quad A^{(k)} I = I A^s, \quad (G_M^s)^{-1} = L G_M^{-1} L^*. \quad (\text{SS.11})$$

The last equality includes both D_o^{-1} factors in L . Indeed its product with $I^* G_M I$ is

$$L G_M^{-1} (L^* I^*) G_M I = L G_M^{-1} P_{\text{sym}} G_M I = LI = 1.$$

This proves the inverse formula without exchanging arbitrary compression and inversion.

In these unscaled coordinates, if $Ae_j = \sum_i A_{ij}e_i$, the full generator is

$$A^{(k)}b_n = \left(\sum_i n_i A_{ii} \right) b_n + \sum_{\substack{i \neq j \\ n_j > 0}} (n_i + 1) A_{ij} b_{n+e_i-e_j}. \quad (\text{SS.12})$$

For a specified target tensor of occupancy $n + e_i - e_j$, each of its $n_i + 1$ entries equal to e_i can have come from an e_j in a source tensor, and these are exactly the contributing preimages. The diagonal term counts its n_i unchanged positions. This proves every coefficient in (SS.12). Using the original quotient's complete CRT isomorphism, set $E_\rho = \mathbb{C}[x_\rho]/x_\rho^{m_\rho}$, $t = \rho + x_\rho$. The explicit CRT map is reduction modulo each $(t - \rho)^{m_\rho}$; its inverse is the unique degree $< d$ polynomial with the prescribed full jets. Applying its tensor power and restriction through I, L is the exact change from the original orbit coordinates to the following direct sum:

$$\text{Sym}^k E \simeq \bigoplus_{\substack{\ell_\rho \geq 0 \\ \sum_\rho \ell_\rho = k}} \bigotimes_\rho \text{Sym}^{\ell_\rho} E_\rho, \quad w_\ell = \prod_\rho \binom{\ell_\rho + m_\rho - 1}{m_\rho - 1}. \quad (\text{SS.13})$$

The isomorphism groups the unscaled tensor orbit sums by the number of factors in each CRT summand; these grouped basis vectors exhaust the full basis, which proves the direct sum and its dimensions.

On the summand ℓ , the generator is $\lambda_\ell + N_\ell$, where $\lambda_\ell = \sum_\rho \ell_\rho \rho$ and N_ℓ is the sum of the multiplications by the x_ρ in their original slots. Put $h_\ell = \sum_\rho \ell_\rho (m_\rho - 1)$. Every application raises the total local exponent by one, so $N_\ell^{h_\ell+1} = 0$. On the tensor of local constants,

$$N_\ell^{h_\ell} \left(\bigotimes_\rho 1^{\otimes \ell_\rho} \right) = \frac{h_\ell!}{\prod_\rho ((m_\rho - 1)!)^{\ell_\rho}} \bigotimes_\rho (x_\rho^{m_\rho-1})^{\otimes \ell_\rho} \neq 0. \quad (\text{SS.14})$$

The coefficient counts the ordered choices of increments in each slot, each of which must be incremented exactly $m_\rho - 1$ times. Therefore the nilpotent index is exactly $h_\ell + 1$. Even when different ℓ have the same λ_ℓ , their direct summands in (SS.13) are retained separately. Traces may kill individual nilpotent diagonal contributions, while the complete operators (SS.12)–(SS.14) retain them.

19.3 Complete symmetric spectral sums and their exact departure

List $\rho_1, \dots, \rho_d$ with every algebraic multiplicity and put

$$\delta_i = \text{Re } \rho_i - \frac{1}{2}, \quad \sigma_1 = \sum_i \delta_i, \quad \sigma_2 = \sum_i \delta_i^2, \quad a_k = \binom{k+d-1}{d}, \quad b_k = \binom{k+d-1}{d+1}. \quad (\text{SS.15})$$

Binomial coefficients with lower index exceeding the upper index are zero. Thus these formulas include $k = 1$ and $d = 1$. Triangularize A . The induced ordered orbit basis is also triangular under (SS.12), with diagonal $\lambda_n = \sum_i n_i \rho_i$, once for each $n \in \mathbb{N}^d$, $|n| = k$. For example, order the monomials compatibly with the triangular basis's index weight; each off-diagonal substitution in (SS.12) decreases that weight. Consequently

$$\det(z - A^s) = \prod_{|n|=k} (z - \lambda_n), \quad \sum_{k \geq 0} q^k \text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \prod_{i=1}^d (1 - qa^{\rho_i})^{-1}. \quad (\text{SS.16})$$

The second identity is an identity of formal power series: expand each geometric series and select its occupancy. On grouping repeated roots it is $\prod_\rho (1 - qa^\rho)^{-m_\rho}$, and the coefficient for grouped

occupancy ℓ is precisely w_ℓ from (SS.13). This gives the exact multiplicity dictionary between the cycle trace (SS.9) and the full local jets.

Define the symmetric absolute spectral cost and metric departure by

$$\begin{aligned}\mathcal{D}_k^s &= \sum_{|n|=k} \left| \sum_i n_i \delta_i \right| = \sum_{|\ell|=k} w_\ell \left| \sum_\rho \ell_\rho (\operatorname{Re} \rho - \tfrac{1}{2}) \right|, \\ \mathfrak{d}_M^s &= \operatorname{Tr}((G_M^s)^{-1} (A^s)^* G_M^s A^s) - \sum_{|n|=k} |\lambda_n|^2, \\ S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2}, \quad \epsilon_M^s = \|S_M^s\|_{\text{op}}.\end{aligned}\tag{SS.17}$$

The square root is used only as the displayed isometry for computing the original Hermitian pencil; it changes none of the orbit vectors, source functions, arithmetic constants, or entries of G_M^s .

Theorem 19.1 (Exact occupancy moments and symmetric departure). *The full unweighted occupancies satisfy*

$$\begin{aligned}\sum_{|n|=k} \sum_i n_i \delta_i &= a_k \sigma_1, \\ \sum_{|n|=k} \left(\sum_i n_i \delta_i \right)^2 &= (a_k + b_k) \sigma_2 + b_k \sigma_1^2.\end{aligned}\tag{SS.18}$$

In the original arithmetic orbit Gram,

$$\begin{aligned}\mathfrak{d}_M^s &\geq 0, \quad \operatorname{Tr} S_M^s = 2a_k \sigma_1, \\ \operatorname{Tr}((S_M^s)^2) &= 4\{(a_k + b_k) \sigma_2 + b_k \sigma_1^2\} + 2\mathfrak{d}_M^s.\end{aligned}\tag{SS.19}$$

In particular, if $\bar{\delta} = \sigma_1/d$, the exact mean-square spread of the occupancy offsets is

$$\sum_{|n|=k} \left(\sum_i n_i \delta_i - k\bar{\delta} \right)^2 = \frac{Nk(k+d)}{d(d+1)} \left(\sigma_2 - \frac{\sigma_1^2}{d} \right).\tag{SS.20}$$

Proof. The product $\prod_i (1 - q_i)^{-1}$ enumerates every occupancy. Apply $q_i \partial_{q_i}$ and then set all $q_i = q$. Its coefficient of q^k gives $\sum n_i = [q^k] q(1 - q)^{-d-1} = a_k$. For two distinct indices, applying the two derivatives gives $\sum n_i n_j = [q^k] q^2 (1 - q)^{-d-2} = b_k$. Applying $q_i^2 \partial_{q_i}^2$ gives $\sum n_i (n_i - 1) = 2b_k$. These coefficient identities follow by multiplying geometric series, or by induction from $(1 - q)^{-r} = \sum_{m \geq 0} \binom{m+r-1}{r-1} q^m$. Therefore $\sum n_i^2 = a_k + 2b_k$. Expanding the square, the cross terms have coefficient b_k ; collecting the diagonal terms gives (SS.18). The identities $a_k = Nk/d$ and $a_k + b_k = Nk(k+d)/(d(d+1))$ follow directly from the factorials, including the stated edge cases. Expanding the centered square proves (SS.20).

Let $T = (G_M^s)^{1/2} A^s (G_M^s)^{-1/2}$. A unitary upper triangularization exists by induction: choose a unit eigenvector as the first basis vector, complete it to an orthonormal basis, and repeat on the remaining diagonal block. If its strictly upper triangular part is N_T , its Hilbert–Schmidt squared norm is exactly $\mathfrak{d}_M^s$; the diagonal entries are the complete λ_n . This proves nonnegativity while retaining every nonzero off-diagonal entry. Since $S_M^s = T + T^* - kI$, its diagonal is $2 \sum_i n_i \delta_i$; each strictly upper triangular entry and its adjoint contribute twice their squared absolute value to the squared trace. Thus $\operatorname{Tr}((S_M^s)^2) = 4 \sum_n (n \cdot \delta)^2 + 2\mathfrak{d}_M^s$. Equation (SS.18) proves (SS.19). $\square$

For comparison with (CT.20)–(CT.21), the ordered tensor has an occupancy n exactly $k! / \prod_i n_i!$ times. The permutation complement therefore has that multiplicity minus one. Its characteristic

polynomial, trace, and absolute spectral cost are obtained with these nonnegative integer multiplicities. In particular

$$\mathcal{D}_k = \mathcal{D}_k^s + \sum_{|n|=k} \left(\frac{k!}{\prod_i n_i!} - 1 \right) |n \cdot \delta|. \quad (\text{SS.21})$$

This equality retains every contribution omitted by the symmetric inclusion. Moreover the permutation projection is orthogonal also for G_M , because it commutes with that Gram. Its image and kernel are invariant under $A^{(k)}$. Thus the full metric operator is the orthogonal direct sum of its symmetric and complementary restrictions; the squared Hilbert–Schmidt norms and the eigenvalue-square sums add separately. The full departure in (CT.20) is exactly $\mathfrak{d}_M^s$ plus the corresponding complementary departure, each nonnegative. This proves the full operator comparison as well as the scalar identity (SS.21).

19.4 The cost in the original symmetric relation layer

Restrict the original source maps without changing their Hilbert targets:

$$Y_M^s = Y_M I, \quad C_M^s = C_M I, \quad \Delta_M^s = G_M^s - G_{M+1}^s = (Y_M^s)^* Y_M^s, \quad W_M^s = -((Y_M^s)^* C_M^s + (C_M^s)^* Y_M^s). \quad (\text{SS.22})$$

Their images lie in the invariant original relation layer, whose dimension is $r_{\text{lay}} = p_{\leq k}(M+1)$. Indeed an unscaled symmetric polynomial basis in exact degree $M+1$ is indexed by its partitions into at most k parts. The invariant full-jet map is onto at every $M \geq k(d-1)$: average a complete tensor remainder to lift each invariant jet. The difference of the two kernel dimensions is therefore the number of the new degree $M+1$ basis vectors. The injective polynomial realization and orthogonal layer quotient transport this dimension to the actual source layer, proving the assertion with its original relation ideal. Put

$$r_M^s = \text{rank } Y_M^s, \quad \tau_M^s = \text{Tr}((G_M^s)^{-1} \Delta_M^s), \quad \chi_M^s = \text{Tr}((G_M^s)^{-1} (C_M^s)^* C_M^s). \quad (\text{SS.23})$$

Then the precise relative source costs are

$$\begin{aligned} n_{\pm}(W_M^s) &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}\}, \\ \mathcal{D}_k^s &\leq \frac{1}{2} \|S_M^s\|_1 \leq \sqrt{\tau_M^s \chi_M^s}, \quad \mathcal{D}_k^s \leq r_M^s \epsilon_M^s. \end{aligned} \quad (\text{SS.24})$$

To prove the inertia bound, the Hermitian form in (SS.22) vanishes on $\ker Y_M^s$. Any strictly positive or strictly negative subspace of dimension greater than $\text{rank } Y_M^s$ intersects that kernel nontrivially, a contradiction. Apply the same argument to C_M^s . In the triangular basis used in (SS.19), the absolute diagonal sum of S_M^s is $2\mathcal{D}_k^s$. Its spectral expansion gives $|S_{ii}| \leq \sum_j |\lambda_j| |U_{ij}|^2$; summing over i bounds this sum by the trace norm. Finally, for any two finite-domain Hilbert-space maps B, C , their singular-vector decomposition and entrywise Cauchy–Schwarz prove $\|B^* C\|_1 \leq \|B\|_{\text{HS}} \|C\|_{\text{HS}}$. Apply it to $B = Y_M^s (G_M^s)^{-1/2}$ and $C = C_M^s (G_M^s)^{-1/2}$ in (SS.22). The squared norms are τ_M^s, χ_M^s . The inertia bound gives at most $2r_M^s$ nonzero eigenvalues, proving the last assertion.

If the packet is stable, with full orders, under $\rho \mapsto 1 - \bar{\rho}$, its original antiunitary reflection commutes with permutations. Restricting (CT.16a) to the invariant space pairs every nonzero eigenvalue of the Hermitian pencil with its negative. Thus $\sigma_1 = 0$, and with $p_M^s = n_+(W_M^s) = n_-(W_M^s)$,

$$\begin{aligned} p_M^s &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}, \lfloor N/2 \rfloor\}, \\ \mathcal{D}_k^s &\leq p_M^s \epsilon_M^s, \\ p_M^s (\epsilon_M^s)^2 &\geq \frac{2Nk(k+d)}{d(d+1)} \sigma_2 + \mathfrak{d}_M^s. \end{aligned} \quad (\text{SS.25})$$

There are $2p_M^s$ nonzero eigenvalues, each of absolute value at most ϵ_M^s . Bounding their absolute and squared sums and using (SS.19) proves both inequalities. They involve no division by p_M^s ; when it vanishes, (SS.19) forces every displayed nonnegative term to vanish.

Let $\delta_* = \max_i |\delta_i|$. If it is positive, $|n \cdot \delta| \leq k\delta_*$ yields the additional explicit lower cost

$$\mathcal{D}_k^s \geq \frac{(a_k + b_k)\sigma_2 + b_k\sigma_1^2}{k\delta_*}. \quad (\text{SS.26})$$

Indeed square each occupancy offset, bound it by $k\delta_*|n \cdot \delta|$, and sum using (SS.18). For a reflected packet this becomes $\mathcal{D}_k^s \geq N(k + d)\sigma_2/(d(d + 1)\delta_*)$. Each repeated eigenvector $v^{\otimes k}$ is nonzero and invariant, and its exact original Rayleigh quotient is $2k\delta_i$. Hence $\epsilon_M^s \geq 2k\delta_*$. Equations (SS.25)–(SS.26) additionally measure the cumulative cost of every occupancy, with all multiplicities and original layer ranks. If any $m_\rho > 1$, (SS.14) gives a nonzero nilpotent on the all- ρ summand. An operator with that Jordan block cannot be unitarily diagonal, so the sum of squares defining $\mathfrak{d}_M^s$ is strictly positive for every positive G_M^s . This retains a further finite-metric contribution even when all $\delta_i = 0$.

Finally the density proved in (JD.8) passes to the invariant source by the bounded permutation projection. The canonical representatives are the restrictions $R_M I$; hence $G_M^s \downarrow 0$ and $\det G_M^s \rightarrow 0$, while the full jets remain the identity at every finite M . With $\pi_M^s = \det G_{M+1}^s / \det G_M^s$, the positive eigenvalues t_j of $(G_M^s)^{-1/2} \Delta_M^s (G_M^s)^{-1/2}$ lie in $(0, 1)$, their number is r_M^s , and $\tau_M^s = \sum t_j$, $\pi_M^s = \prod (1 - t_j)$. The arithmetic mean–geometric mean inequality therefore gives

$$(\mathcal{D}_k^s)^2 \leq \tau_M^s \chi_M^s \leq r_M^s (1 - (\pi_M^s)^{1/r_M^s}) \chi_M^s \quad (r_M^s > 0). \quad (\text{SS.27})$$

For $r_M^s = 0$, (SS.22) gives $\Delta_M^s = W_M^s = 0$, without using an exponent $1/r_M^s$. Since $-\log(1 - t) - t = \int_0^t u/(1 - u) du$, the same finite source sequence also obeys, for integers $k(d - 1) \leq a < b$ and $\mathcal{D}_k^s > 0$,

$$(\mathcal{D}_k^s)^2 \sum_{M=a}^{b-1} (\chi_M^s)^{-1} \leq \sum_{M=a}^{b-1} \tau_M^s \leq \log \frac{\det G_a^s}{\det G_b^s}. \quad (\text{SS.28})$$

Positivity of every denominator follows from (SS.24). Telescoping proves the last inequality. Moreover $\sum_{M=a}^\infty \tau_M^s = \infty$: otherwise the tail has $t_j \leq \tau_M^s \leq 1/2$, and $-\log(1 - t_j) \leq 2t_j$ would keep the telescoped logarithm bounded, contrary to $\det G_M^s \rightarrow 0$. These identities transfer the entire determinant and relative-energy calculation to the source's smaller invariant family through its exact inclusion; they supply no assumed arithmetic upper estimate.

20 Arithmetic quartet parity and the exact positive frontier operator

Let h be a nonempty packet of actual zeros with their full orders, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Their composition also preserves the packet. Retain $d = \deg h$, $g = 2\xi$, $v_h = g/h$, $v = j_h v_h$, and the actual measure $d\nu_h = |v_h(1/2 + it)|^2 dt/(2\pi)$. Throughout this section $k \geq 1$, $M \geq k(d - 1)$, and

$$E_k = (\mathbb{C}[s]/h)^{\otimes k}, \quad A_k = \sum_{j=1}^k A_{h,j}, \quad \mathcal{C}_M = G_M^{-1}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{AP.1})$$

The metric G_M is the actual total-degree minimum of (CT.12)–(CT.13). All coordinates below remain its original monomial tensor coordinates.

20.1 The actual two frontier matrices

Let q_j be the original monic orthogonal polynomials for ν_h , and $\kappa_j = \int |q_j|^2 d\nu_h$. Thus κ_0 retains the mass of the original measure. For a multi-index α , put

$$q_\alpha = \prod_{j=1}^k q_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad z_\alpha = U_{v^{\otimes k}}[q_\alpha] \in E_k. \quad (\text{AP.2})$$

The source's notation p_j, ω_j denotes these same q_j, κ_j : their monic leading terms and identical lower-degree orthogonality determine the polynomial uniquely, and then determine its norm. No polynomial or measure is divided by its norm. The triangular polynomial change of basis and product integration in (CT.12) give exactly

$$\mathcal{C}_M = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\kappa_\alpha}. \quad (\text{AP.3})$$

Indeed this basis has Gram diagonal κ_α , and its full interpolation matrix has columns z_α ; substitution in $JM^{-1}J^*$ proves (AP.3), retaining $v^{\otimes k}$ in every column.

Write $a_j = \kappa_j / \kappa_{j-1}$ for $j \geq 1$, $a_0 = 0$, and retain the ordered frontier $|\beta| = M + 1$. Its size is $r = \binom{M+k}{k-1}$. Define maps with the same declared source $\mathbb{C}^r$:

$$\begin{aligned} \Omega &= \text{diag}(\kappa_\beta)_{|\beta|=M+1}, & F &= [z_\beta]_{|\beta|=M+1}, \\ w_\beta &= \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, & E_+ &= [w_\beta]_{|\beta|=M+1}. \end{aligned} \quad (\text{AP.4})$$

The original recurrence $sq_j = q_{j+1} + b_j q_j - a_j q_{j-1}$, with $b_j + \bar{b}_j = 1$, applied to (AP.3) gives

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{AP.5})$$

To check the cancellation completely, an adjacent pair $\alpha, \alpha + e_j$ both of degree at most M has coefficients κ_α^{-1} and $-a_{\alpha_j+1} \kappa_{\alpha+e_j}^{-1}$, whose sum is zero. Its transposed conjugate cancels in the same way. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, which cancels the last term. The remaining frontier pair has $\beta = \alpha + e_j$ and coefficient $\kappa_\alpha^{-1} = a_{\beta_j} / \kappa_\beta$; summing these pairs is exactly (AP.4)–(AP.5). The same sum gives the full inverse-metric update

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (\text{AP.6})$$

20.2 The original linear reflection separates the two frontiers

Conjugation closure makes h real, and reflection closure then gives $h(1-s) = (-1)^d h(s)$. The original functional equation gives

$$v_h(1-s) = (-1)^d v_h(s), \quad q_j(1-s) = (-1)^j q_j(s). \quad (\text{AP.7})$$

For the second identity, the actual density is even in t . The polynomial $(-1)^j q_j(1-s)$ is monic, and substitution $t \mapsto -t$ verifies its orthogonality to every lower degree. The difference from q_j has degree less than j and is orthogonal to itself, so positivity proves it is zero. This proves (AP.7) with the original leading coefficient.

Let $\mathfrak{c} : E_k \rightarrow E_k$ be the complex-linear map substituting $s_j \mapsto 1 - s_j$ in every factor. It is well-defined because the ideal $(h(s_1), \dots, h(s_k))$ is preserved, and $\mathfrak{c}^2 = I$. Let C denote its matrix in the retained original basis. The original function map

$$(\mathfrak{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}) \quad (\text{AP.8})$$

is a linear unitary involution for $d^k x$, by the change of variables $y_j = x_j^{-1}$. Its Mellin transform replaces s_j by $1 - s_j$. Consequently, on an original numerator,

$$\mathfrak{t}_k \mathcal{T}_h^{(k)} P = (-1)^{kd} \mathcal{T}_h^{(k)} P(1 - s_1, \dots, 1 - s_k). \quad (\text{AP.9})$$

Equations (AP.7) and (AP.2) show that its full jet is $\mathfrak{c}\mathcal{J}P$, with exactly the displayed unit sign. Total degree and the original relation ideal are preserved. Applying $\mathfrak{t}_k$ to the unique minimum for a prescribed jet therefore gives the unique minimum for its $\mathfrak{c}$ -image. Uniqueness follows from the orthogonal decomposition (CT.5). Hence

$$\mathfrak{t}_k R_M = R_M C, \quad C^* G_M C = G_M, \quad C z_\alpha = (-1)^{kd+|\alpha|} z_\alpha. \quad (\text{AP.10})$$

In particular, with $\eta = (-1)^{kd+M+1}$,

$$CF = \eta F, \quad CE_+ = -\eta E_+. \quad (\text{AP.11})$$

Every term defining w_β has degree M , which proves the second identity without suppressing any coefficient.

Declare the following maps between the original positive spaces and their coordinate Hilbert spaces:

$$P = G_M^{1/2} C G_M^{-1/2}, \quad U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}. \quad (\text{AP.12})$$

Here $G_M^{1/2}$ and $\Omega^{1/2}$ are the explicit isometries from the declared original metric spaces to Euclidean coordinate spaces, with their displayed inverses. Every expression in (AP.12) retains both original matrices. From (AP.10), P is unitary and $P^2 = I$; hence $P^* = P$. Equations (AP.11) imply $PU = \eta U$, $PV = -\eta V$, and thus

$$U^* V = (PU)^*(PV) = -U^* V = 0. \quad (\text{AP.13})$$

20.3 The complete positive operator and the sharper arithmetic bound

Put $A = U^* U$, $B = V^* V$, retaining their source type $\mathbb{C}^r$. Equivalently,

$$A = \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, \quad B = \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}. \quad (\text{AP.14})$$

These positive-semidefinite matrices are built from the two original frontiers; A is distinct from the arithmetic generator A_k .

Theorem 20.1 (Exact positive frontier operator). *The actual relative weight satisfies*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = UV^* + VU^*, \\ \boxed{\epsilon_{k,M} &= \|UV^*\|, \quad \epsilon_{k,M}^2 = \lambda_{\max}(A^{1/2} B A^{1/2}),} \quad p_M = \text{rank}(A^{1/2} B^{1/2}), \\ \text{Tr}(S_M^2) &= 2 \text{Tr}(AB). \end{aligned} \quad (\text{AP.15})$$

Its positive and negative inertias both equal p_M . Define the original incidence bound and volume ratio by

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad \lambda_{k,M} = \|U\|^2, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M}. \quad (\text{AP.16})$$

Then

$$\boxed{\epsilon_{k,M}^2 \leq \Gamma_{k,M} \lambda_{k,M} \leq \Gamma_{k,M} (\pi_M^{-1} - 1).} \quad (\text{AP.17})$$

All formulas include the zero-rank cases and every full arithmetic jet. No growth estimate for $\Gamma_{k,M}$ or π_M is assumed.

Proof. Equation (AP.5) proves the first equality in (AP.15). Decompose the retained coordinate Hilbert space into the two orthogonal eigenspaces of the explicit Hermitian involution P . Relative to the order $(\eta, -\eta)$, the operator has blocks

$$S_M = \begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}, \quad T = UV^*|_{\ker(P+\eta I)}. \quad (\text{AP.18})$$

Indeed U^* vanishes on the opposite eigenspace and V^* vanishes on the first, by (AP.11). Squaring gives diagonal blocks TT^*, T^*T . For each nonzero singular value σ of T , choose unit singular vectors x, y with $Ty = \sigma x$, $T^*x = \sigma y$; then (x, y) and $(x, -y)$ have eigenvalues $\sigma, -\sigma$. The remaining eigenspaces are $\ker T^* \oplus \ker T$, with eigenvalue zero. These vectors form the full orthogonal decomposition by the spectral theorem for T^*T , including all its zero vectors. It follows that $\epsilon = \|T\| = \|UV^*\|$ and both inertias equal $\text{rank}(UV^*)$.

Now $(UV^*)(UV^*)^* = UBU^*$. Its nonzero eigenvalues and rank are those of $B^{1/2}AB^{1/2}$: the maps $UB^{1/2}$ and its adjoint transfer each nonzero eigenvector, with inverse divided by that eigenvalue. The latter matrix is $T_0^*T_0$, with $T_0 = A^{1/2}B^{1/2}$, while $T_0T_0^* = A^{1/2}BA^{1/2}$. Their full positive spectra agree by the same maps, and their ranks equal $\text{rank } T_0$. This proves the norm and rank formulas, including rank zero. Finally (AP.13) kills $(UV^*)^2$ and $(VU^*)^2$. Taking the trace of the remaining two products gives the last line of (AP.15).

For the incidence estimate, let $Z = [z_\alpha]_{|\alpha|=M}$ and $\Omega_M = \text{diag}(\kappa_\alpha)$. Define $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, and zero otherwise; summation applies if more than one specified incidence contributes. Then $E_+ = ZL$. The matrix $N = \Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is exactly

$$\sum_{j=1}^k \left(a_{\alpha_j+1} + \sum_{l \neq j} a_{\alpha_l} \right) = \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (\text{AP.19})$$

To verify each factor, its j -th contributing frontier is $\beta = \alpha + e_j$, and $\kappa_\alpha a_{\beta_j} / \kappa_\beta = 1$. The predecessors of this β supply $\sum_l a_{\beta_l}$, which is precisely the displayed summand. Every eigenvalue of N has modulus at most the largest row sum: in an eigenvector choose a coordinate of largest absolute value and apply that row equation. The matrix N is similar, by $\Omega_M^{1/2}$, to the Hermitian positive matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$. Its eigenvalues therefore lie in $[0, \Gamma_{k,M}]$. Undoing the congruence and using (AP.3) proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z \Omega_M^{-1} Z^* \preceq \Gamma_{k,M} \mathcal{C}_M, \quad VV^* \preceq \Gamma_{k,M} I. \quad (\text{AP.20})$$

Consequently $\epsilon^2 = \|UV^*\|^2 \leq \|U\|^2 \|V\|^2 \leq \Gamma_{k,M} \lambda_{k,M}$. Finally (AP.6) gives

$$\pi_M^{-1} = \det(I + UU^*) = \prod_j (1 + \lambda_j(UU^*)) \geq 1 + \lambda_{k,M}. \quad (\text{AP.21})$$

All eigenvalues here are nonnegative, proving the second bound in (AP.17). If $U = 0$ or $V = 0$, (AP.15) gives zero weight directly; all determinant denominators remain positive. $\square$

The arithmetic departure has an exact expression in these same two frontier Grams. List the original ρ_i with every algebraic multiplicity and put $\delta_i = \text{Re } \rho_i - 1/2$. Comparing (AP.15) with the fully proved tensor trace (CT.21) gives

$$\mathfrak{d}_M^{(k)} = \text{Tr}(AB) - 2 \sum_{i_1, \dots, i_k} (\delta_{i_1} + \dots + \delta_{i_k})^2. \quad (\text{AP.22})$$

This retains the complete nonsemisimple generator in the left-hand definition of (CT.20). For $k = 1$, a four-distinct-point tessarine orbit of common full order μ has the further exact chain, with $M = d + m$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_{1,M}^2 = \lambda_{\max}(A^{1/2}BA^{1/2}) \leq \Gamma_{1,M}(\pi_M^{-1} - 1). \quad (\text{AP.23})$$

The first inequality, original character, multiplicity and critical supported-zero case are proved in (R9); (AP.15)–(AP.17) prove the rest. Thus the original character observation is related to an explicit positive frontier operator through the original metric, full interpolation unit and declared isometries.

20.4 The same positive frontier on the actual symmetric summand

The cochain projector used here is exactly $S_k = (k!)^{-1} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, with the original Koszul permutations T_π . Its top-degree action is the ordinary permutation projection, because T_π there equals $\text{sgn}(\pi) P_\pi$. Equations (SS.1)–(SS.9) prove its continuous cochain maps, completed projective-tensor cohomology, and finite packet projector, including the complementary summand. We now use its finite invariant image with the original total-degree representatives R_M .

Write $e_1, \dots, e_d$ for the original power basis of E , with indices shifted by one, and retain each occupancy $n \in \mathbb{N}^d$, $|n| = k$. Let b_n be the literal sum of the distinct basis tensors of occupancy n . Define the maps and dimensions by

$$\begin{aligned} N_s &= \binom{k+d-1}{d-1}, & I : \mathbb{C}^{N_s} &\longrightarrow E_k, & Ie_n &= b_n, \\ D_o &= I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, & L &= D_o^{-1} I^*, & P_s &= IL = \frac{1}{k!} \sum_{\pi} P_\pi, & LI &= I_{N_s}. \end{aligned} \quad (\text{AP.24})$$

Indeed the distinct original tensor basis vectors are orthonormal for this coordinate calculation; hence their disjoint orbit sums have precisely the displayed diagonal Gram D_o . The map L averages coefficients on each orbit, proving all the compositions in (AP.24). In particular $P_s^* = P_s$ and $P_s^2 = P_s$. The factorials in D_o are retained separately from the arithmetic Gram.

Permutation invariance of the original product measure, relation ideal, total degree and full interpolation unit gives $R_M P_\pi = P_\pi R_M$, by applying the unitary inverse permutation to each constrained minimum. Thus G_M and G_M^{-1} commute with P_s . Put

$$\begin{aligned} R_M^s &= R_M I, & G_M^s &= I^* G_M I, & W_M^s &= I^* W_M I, \\ A_k^s &= L A_k I, & A_k I &= I A_k^s, & W_M^s &= (A_k^s)^* G_M^s + G_M^s A_k^s - k G_M^s, \\ I^* G_M &= G_M^s L, & G_M I &= L^* G_M^s, & \mathcal{K}_M^s &:= (G_M^s)^{-1} = L G_M^{-1} L^*. \end{aligned} \quad (\text{AP.25})$$

These are the same representatives and matrices as (SS.10)–(SS.11). For completeness, $I^* G_M = I^* P_s G_M = I^* G_M P_s = I^* G_M IL$, proving the first rectangular identity and then its adjoint. The generator commutes with every factor permutation, so it preserves the image of I , proving its intertwining identity and the displayed weight formula. Finally

$$(L G_M^{-1} L^*)(I^* G_M I) = L G_M^{-1} P_s G_M I = LI = I_{N_s},$$

which proves the inverse formula with both factors D_o^{-1} present. Positivity follows since I is injective and $G_M \succ 0$. The notation $\mathcal{K}_M^s$ refers only to this inverse kernel; it will not denote a reflection.

Keep the complete, possibly redundant frontier of (AP.4), with its unchanged domain $\mathbb{C}^r$ and norm matrix Ω . Its symmetric target maps are

$$\begin{aligned} F_s &= L F : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, & E_s &= L E_+ : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, \\ \mathcal{Z}_M^s &= F_s \Omega^{-1} E_s^* + E_s \Omega^{-1} F_s^*, & W_M^s &= G_M^s \mathcal{Z}_M^s G_M^s, \\ \mathcal{K}_{M+1}^s &= \mathcal{K}_M^s + F_s \Omega^{-1} F_s^*. \end{aligned} \quad (\text{AP.26})$$

The rectangular identities (AP.25), applied to $I^* G_M \mathcal{Z}_M G_M I$, prove the weight formula. Applying L and L^* to (AP.6) proves the inverse update. It uses the same orbit maps at both source degrees. Thus the update concerns the actual next symmetric minimum, not a newly selected positive form.

The original linear reflection C commutes with every factor permutation, since it substitutes the same $1 - s_j$ in each factor. Its invariant-coordinate matrix is

$$\begin{aligned} C_s &= L C I, & C I &= I C_s, & C_s L &= L C, & C_s^2 &= I_{N_s}, \\ C_s^* G_M^s C_s &= G_M^s, & C_s F_s &= \eta F_s, & C_s E_s &= -\eta E_s, & \eta &= (-1)^{kd+M+1}. \end{aligned} \quad (\text{AP.27})$$

For example $CI = P_s CI = ILCI$, and $C_s L = LCP_s = LP_s C = LC$; these prove the first intertwiners and then the involution. The metric equation follows by substituting $IC_s = CI$ into $I^* G_M I$ and using (AP.10). Applying L to (AP.11) proves both parities with the full arithmetic unit and its original sign.

Define the explicit isometries and the two retained frontier Grams by

$$\begin{aligned} P_M^s &= (G_M^s)^{1/2} C_s (G_M^s)^{-1/2}, \\ U_s &= (G_M^s)^{1/2} F_s \Omega^{-1/2}, \quad V_s = (G_M^s)^{1/2} E_s \Omega^{-1/2}, \\ A_s &= U_s^* U_s, \quad B_s = V_s^* V_s. \end{aligned} \tag{AP.28}$$

The superscript on P_M^s distinguishes this transported reflection from the fixed permutation projection P_s in (AP.24). Equation (AP.27) proves that P_M^s is a Hermitian involution and that U_s, V_s have opposite eigenvalues $\eta, -\eta$. Their ranges are consequently orthogonal, so $U_s^* V_s = 0$.

Theorem 20.2 (Exact symmetric arithmetic frontier). *For the actual signed symmetric cohomology summand, the original Hermitian pencil has*

$$\begin{aligned} S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2} = U_s V_s^* + V_s U_s^*, \\ \boxed{\epsilon_M^s &= \|U_s V_s^*\|, \quad (\epsilon_M^s)^2 = \lambda_{\max}(A_s^{1/2} B_s A_s^{1/2}),} \\ p_M^s &= \text{rank}(A_s^{1/2} B_s^{1/2}), \quad n_+(W_M^s) = n_-(W_M^s) = p_M^s, \\ \text{Tr}((S_M^s)^2) &= 2 \text{Tr}(A_s B_s). \end{aligned} \tag{AP.29}$$

With the original $\Gamma_{k,M}$ of (AP.16), put

$$\lambda_M^s = \|U_s\|^2, \quad \pi_M^s = \frac{\det G_{M+1}^s}{\det G_M^s}.$$

Then the same source sequence satisfies

$$\boxed{(\epsilon_M^s)^2 \leq \Gamma_{k,M} \lambda_M^s \leq \Gamma_{k,M} ((\pi_M^s)^{-1} - 1), \quad \lambda_M^s \leq \lambda_{k,M}.} \tag{AP.30}$$

All formulas retain every occupancy and nilpotent direction in (SS.12)–(SS.14), as well as the ungrouped frontier multiplicities.

Proof. Equation (AP.26) gives the first line of (AP.29). Decomposing $\mathbb{C}^{N_s}$ into the two orthogonal eigenspaces of P_M^s gives exactly the block operator (AP.18), now with $T_s = U_s V_s^*$ between those two spaces. The singular-vector maps and their inverses proved after (AP.18) apply to these declared matrices without any rank assumption. They give opposite eigenvalues for each positive singular value, $\text{rank}(A_s^{1/2} B_s^{1/2})$, and all remaining eigenvalues zero. The same two explicit composites $U_s B_s^{1/2}$ and its adjoint identify the positive eigenvalues with those of $A_s^{1/2} B_s A_s^{1/2}$. Finally $U_s^* V_s = 0$ kills the two square cross factors; the traces of the other two products both equal $\text{Tr}(A_s B_s)$. This proves every assertion in (AP.29).

Applying L and L^* to the original inequality (AP.20), and using (AP.25), proves

$$E_s \Omega^{-1} E_s^* \preceq \Gamma_{k,M} \mathcal{K}_M^s, \quad V_s V_s^* \preceq \Gamma_{k,M} I_{N_s}. \tag{AP.31}$$

Thus $\|U_s V_s^*\|^2 \leq \|U_s\|^2 \|V_s\|^2 \leq \Gamma_{k,M} \lambda_M^s$. Taking determinants of the exact inverse update in (AP.26) gives

$$(\pi_M^s)^{-1} = \det(I_{N_s} + U_s U_s^*) = \det(I_r + A_s) \geq 1 + \lambda_M^s, \quad 0 < \pi_M^s \leq 1. \tag{AP.32}$$

The determinant equality follows either by the two rectangular composites and their nonzero spectra, or by multiplying the two block factors for $\det(I + XY) = \det(I + YX)$. The inequality multiplies the nonnegative factors $1 + \lambda_j(A_s)$, exactly as in (AP.21). This proves the first two bounds in (AP.30).

For the final bound, the literal orbit maps give

$$\begin{aligned} L^* G_M^s L &= P_s G_M P_s \preceq G_M, \\ A_s &= \Omega^{-1/2} F^* P_s G_M P_s F \Omega^{-1/2} \preceq \Omega^{-1/2} F^* G_M F \Omega^{-1/2} = A. \end{aligned} \quad (\text{AP.33})$$

Indeed G_M commutes with the orthogonal projection P_s , so their omitted cross terms vanish and $G_M - P_s G_M P_s = (I - P_s) G_M (I - P_s) \succeq 0$. Taking the largest Rayleigh quotient of the two positive matrices in (AP.33) proves $\lambda_M^s \leq \lambda_{k,M}$. No comparison of arbitrary compressed inverse matrices is used. $\square$

There is also an exact comparison of the two relative operators. The map

$$Q_M = G_M^{1/2} I (G_M^s)^{-1/2} : \mathbb{C}^{N_s} \longrightarrow \mathbb{C}^{d^k}$$

has $Q_M^* Q_M = I_{N_s}$, and the inverse formula in (AP.25) gives $Q_M Q_M^* = P_s$. Direct multiplication therefore gives

$$S_M^s = Q_M^* S_M Q_M, \quad U_s = Q_M^* U, \quad V_s = Q_M^* V, \quad \epsilon_M^s \leq \epsilon_{k,M}. \quad (\text{AP.34})$$

Here $G_M^{1/2}$ commutes with P_s by its spectral decomposition. The full S_M also commutes with P_s , since both A_k and G_M do. Consequently (AP.34) is its invariant orthogonal restriction; the complementary operator remains on $\ker P_s$. The norm inequality follows directly by applying S_M to the unit vector $Q_M u$ and then the contraction Q_M^* .

If desired, the source frontier can be grouped with every factorial retained. For an orbit $\mathcal{O}$ of degree- $M + 1$ multi-indices, let $m_j(\beta) = |\{l : \beta_l = j\}|$ and

$$n_{\mathcal{O}} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!}, \quad \hat{f}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lz_{\beta}, \quad \hat{e}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lw_{\beta}, \quad \hat{\omega}_{\mathcal{O}} = n_{\mathcal{O}} \kappa_{\beta}. \quad (\text{AP.35})$$

The product norm κ_{β} is constant on an orbit. Equivariance of the full unit, polynomials and incidence sums implies that Lz_{β} and Lw_{β} are individually constant on the same orbit. Hence, for example,

$$\sum_{\beta \in \mathcal{O}} \frac{(Lz_{\beta})(Lw_{\beta})^*}{\kappa_{\beta}} = \frac{\hat{f}_{\mathcal{O}} \hat{e}_{\mathcal{O}}^*}{\hat{\omega}_{\mathcal{O}}}. \quad (\text{AP.36})$$

The same calculation with both columns of the first or second type proves the corresponding kernel and incidence identities. Equations (AP.35)–(AP.36) are the exact map to the source's orbit-factorial presentation. The ungrouped matrices in (AP.26)–(AP.34) continue to retain all their columns.

All zero-rank cases follow without inversion of A_s or B_s . If their square-root product vanishes, then $p_M^s = 0$, $W_M^s = 0$, and $\epsilon_M^s = 0$, even if the kernel update is nonzero. If $\pi_M^s = 1$, (AP.32) forces $U_s = 0$, with the same conclusion. The dimension N_s is always at least one; if it is one, a Hermitian involution has only one nonzero eigenspace, so one of U_s, V_s vanishes and the weight is zero. This argument also covers a zero-dimensional reflection eigenspace in a larger target. It neither removes the retained quotient nor asserts existence of a particular actual packet. For $k = 1$, every orbit cardinality is one and $I = L = I_d$; all the displayed formulas recover (AP.15)–(AP.21) exactly.

Finally the symmetric source is still $R_M I$, and its two actual boundary maps are $Y_M I$ and $C_M I$ in (SS.22). The original relation-graph map, followed by the bounded permutation projection on its Hilbert target, realizes the redundant frontier used here. Grouping it by (AP.35) gives the invariant relation layer of dimension $p_{\leq k}(M + 1)$, as proved in (SS.22)–(SS.24). Its primitive is the original fixed-order division primitive followed by S_k ; the cochain identity in (SS.3)–(SS.4) preserves its boundary and all signs. At invariant support fibres the same maps lift with that support; vanishing amplitudes remain the corresponding supported zero. For other masks the original orbit-joined transport and its kernel remain as stated after (SS.9). Thus (AP.30) is a proved bound for the same

finite arithmetic cohomology summand, with no assumed asymptotic estimate for either $\Gamma_{k,M}$ or π_M^s .

Section 21 carries these same two parities through the original graph frame into the actual derivative layer. The equality $-\widehat{Y}^*\widehat{C} = UV^*$ proved there supplies the exact original derivative cost for the positive operator here. Sections 22 and 24 calculate its one-factor and symmetric-square counterparts through explicit quadratic algebra maps, with all local nilpotents and both reflection components.

21 Quartet parity in the original derivative energy and volume loss

This continuation retains the original maps of (FE.3)–(FE.9) and the actual quartet reflection of (AP.7)–(AP.13). Their combination proves an exact block identity for the two maps into the original theta relation layer. It yields a stronger derivative energy estimate than (FE.24), including an operator-norm version and the resulting finite accumulated determinant bounds.

21.1 Original maps, their spaces, and the reflection

Fix a nonempty packet of actual zeros, with every selected zero's full multiplicity, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Keep

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad g = 2\xi, \quad v_h = g/h, \quad v = j_h v_h, \quad E_k = (\mathbb{C}[s]/h)^{\otimes k}.$$

Here $k \geq 1$, $M \geq k(d-1)$, and the original generator is $A_k = \sum_{j=1}^k A_{h,j}$. Let $G = G_M > 0$ be the actual degree- M minimum metric in the original monomial remainder basis, and let $R = R_M : E_k \rightarrow \mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$ be its original representative, so $R^*R = G$.

For the unchanged measure $d\nu_h(t) = |v_h(1/2 + it)|^2 dt / (2\pi)$, let q_j be the monic orthogonal polynomials and $\kappa_j = \|q_j\|_{\nu_h}^2$. Write $q_\beta = \prod_j q_{\beta_j}$, $\kappa_\beta = \prod_j \kappa_{\beta_j}$, and $z_\beta = U_{v \otimes k}[q_\beta]$. With the original ordering of $|\beta| = M+1$, retain

$$\begin{aligned} r &= \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\kappa_\beta) > 0, \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \\ E_+ &= [w_\beta] : \mathbb{C}^r \rightarrow E_k, \quad w_\beta = \sum_{j:\beta_j>0} \frac{\kappa_{\beta_j}}{\kappa_{\beta_j-1}} z_{\beta-e_j}, \\ T_+ &= [\mathcal{T}_h^{(k)} q_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k, \quad b = T_+ - RF : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M, \\ H &= b^*b = \Omega + F^*GF > 0. \end{aligned} \tag{QD.1}$$

The space $\mathcal{E}_M$ is the actual next theta relation layer with its inherited L^2 norm. The column Gram of T_+ is Ω ; its image is perpendicular to the old polynomial space containing $R(E_k)$. Thus expanding its Gram proves H in (QD.1). The leading degree map gives the inverse of b : each layer vector's highest polynomial coefficient determines it, and subtracting that graph image leaves a vector both in the old relation space and perpendicular to it. In particular b is a bijection onto the original layer, with no arbitrary Hilbert replacement.

The two original maps into that same layer and their weight are

$$\begin{aligned} Y &= -bH^{-1}F^*G : E_k \rightarrow \mathcal{E}_M, \quad C_D = b\Omega^{-1}E_+^*G : E_k \rightarrow \mathcal{E}_M, \\ \Delta &= Y^*Y = G - G_{M+1}, \quad W = A_k^*G + GA_k - kG = -(Y^*C_D + C_D^*Y). \end{aligned} \tag{QD.2}$$

The notation C_D is explicitly the map C_M of (FE.5), and does not denote the inverse interpolation matrix. Its formula records the top polynomial coefficient of the original Euler defect; the graph

projection and the original theta primitive in (FE.3)–(FE.5) supply its actual source realization. The weight identity also follows directly by inserting the two displayed maps and cancelling the middle $b^*b = H$.

Let $C : E_k \rightarrow E_k$ replace each residue variable s_j by $1 - s_j$, and let

$$(\mathbf{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}).$$

The original packet symmetries and functional equation give $h(1 - s) = (-1)^d h(s)$, $v_h(1 - s) = (-1)^d v_h(s)$, and $q_j(1 - s) = (-1)^j q_j(s)$. For the last identity, the reflected polynomial with its sign restored is monic and has the same lower-degree orthogonality for the even density; uniqueness of the monic orthogonal polynomial proves equality. The variable change $x_j \mapsto x_j^{-1}$ proves that $\mathbf{t}_k$ is a unitary involution for $d^k x$, with Mellin action $s_j \mapsto 1 - s_j$. Thus it preserves total degree, the relation ideal and the unique constrained minimum. Writing $\eta = (-1)^{kd+M+1}$, one obtains

$$\mathbf{t}_k R = RC, \quad C^2 = I, \quad C^*GC = G, \quad CF = \eta F, \quad CE_+ = -\eta E_+, \quad \mathbf{t}_k b = \eta b. \quad (\text{QD.3})$$

The last equality uses both $\mathbf{t}_k T_+ = \eta T_+$ and $CF = \eta F$. It states the exact target reflection on the whole original relation layer, since b is onto.

For explicit coordinate isometries put

$$\begin{aligned} P &= G^{1/2}CG^{-1/2}, \quad P^* = P, \quad P^2 = I, \quad U = G^{1/2}F\Omega^{-1/2}, \quad V = G^{1/2}E_+\Omega^{-1/2}, \\ \hat{Y} &= YG^{-1/2}, \quad \hat{C} = C_D G^{-1/2} : \mathbb{C}^{d^k} \rightarrow \mathcal{E}_M, \quad J_b = bH^{-1/2} : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M. \end{aligned} \quad (\text{QD.4})$$

The matrix $G^{1/2}$ is the isometry from (E_k, G) to its Euclidean coordinate space, with inverse $G^{-1/2}$. Similarly $\Omega^{1/2}$ records the original polynomial norms, and J_b is a unitary map with inverse $H^{-1/2}b^*$. No commutation of H with Ω is assumed. In these exact coordinates,

$$\hat{Y} = -J_b H^{-1/2} F^* G^{1/2}, \quad \hat{C} = J_b H^{1/2} \Omega^{-1} E_+^* G^{1/2}. \quad (\text{QD.5})$$

21.2 The single original boundary block

Theorem 21.1 (Exact quartet derivative block and its energy bound). *The two original domain parities and their entire block relation are*

$$YC = \eta Y, \quad C_D C = -\eta C_D, \quad \hat{Y}P = \eta \hat{Y}, \quad \hat{C}P = -\eta \hat{C}, \quad -\hat{Y}^* \hat{C} = UV^*. \quad (\text{QD.6})$$

Let

$$\begin{aligned} S &= G^{-1/2}WG^{-1/2}, \quad \epsilon = \|S\|_{\text{op}}, \quad \lambda = \|UU^*\|_{\text{op}}, \quad \pi = \frac{\det G_{M+1}}{\det G}, \\ \zeta &= \|\hat{C}\|_{\text{op}}^2, \quad \chi = \|\hat{C}\|_{\text{HS}}^2. \end{aligned} \quad (\text{QD.7})$$

Then all these actual finite quantities satisfy

$$\boxed{\epsilon = \|\hat{Y}^* \hat{C}\|_{\text{op}} = \|UV^*\|_{\text{op}}, \quad \epsilon^2 \leq \frac{\lambda}{1 + \lambda} \zeta \leq \frac{\lambda}{1 + \lambda} \chi.} \quad (\text{QD.8})$$

The operator-norm cost and the trace cost are exactly

$$\begin{aligned} \hat{C}^* \hat{C} &= V(I + U^*U)V^*, \quad \zeta = \lambda_{\max}(V(I + U^*U)V^*), \\ \chi &= \text{Tr}(V(I + U^*U)V^*) = \|V\|_{\text{HS}}^2 + \|VU^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{QD.9})$$

In particular the operator-norm replacement of χ in (QD.8) is valid without a rank assumption.

Proof. From $C^*GC = G$ and $C^2 = I$ one gets $GC = C^*G$. Thus $F^*GC = (CF)^*G = \eta F^*G$, and similarly $E_+^*GC = -\eta E_+^*G$. Substitution in (QD.2) proves the first two parities, and the coordinate change proves the next two. Equations (QD.2) and $b^*b = H$ give, in the order shown,

$$\hat{Y}^*\hat{C} = -G^{1/2}FH^{-1}H\Omega^{-1}E_+^*G^{1/2} = -UV^*.$$

Decompose $\mathbb{C}^{d^k} = \ker(P - \eta I) \oplus \ker(P + \eta I)$, orthogonally. The first map in (QD.6) annihilates the second summand and the second map annihilates the first. Consequently $X = \hat{Y}^*\hat{C}$ maps the second summand into the first and vanishes on the first. The actual weight is

$$S = -(X + X^*) = -\begin{pmatrix} 0 & X_{+-} \\ X_{+-}^* & 0 \end{pmatrix}.$$

Squaring gives diagonal blocks $X_{+-}X_{+-}^*$ and $X_{+-}^*X_{+-}$. Their largest eigenvalues both equal $\|X_{+-}\|^2$ when this map is nonzero, by its singular value decomposition; if it is zero both blocks vanish. Hence $\|S\| = \|X\|$, including empty parity summands and all zero ranks.

The retained Gram identity in (QD.1) is $H = \Omega^{1/2}(I + U^*U)\Omega^{1/2}$. Substitution gives

$$\hat{Y}^*\hat{Y} = U(I + U^*U)^{-1}U^* = UU^*(I + UU^*)^{-1}.$$

The second equality follows by multiplying $(I + UU^*)U = U(I + U^*U)$ by its positive invertible factors. Thus $\|\hat{Y}\|^2 = \lambda/(1 + \lambda)$, also when $\lambda = 0$. Submultiplicativity in the exact identity $\epsilon = \|\hat{Y}^*\hat{C}\|$ proves the first inequality of (QD.8). On the other hand $\hat{C} = b\Omega^{-1/2}V^*$, so its Gram is exactly $V\Omega^{-1/2}H\Omega^{-1/2}V^* = V(I + U^*U)V^*$. This proves (QD.9). A positive finite matrix has largest eigenvalue at most its trace, proving $\zeta \leq \chi$ and the second inequality. $\square$

This is an actual-source strengthening of (FE.24): its general factor 4 is replaced by 1 under the proved original quartet parity. The retained derivative map still lands in the original $\mathcal{E}_M$; only the displayed exact isometries were used to calculate its norm.

21.3 Complete determinant constraints, including zero cases

Set $\delta_* = \max_{\rho \in Z} |\operatorname{Re} \rho - 1/2|$ and $q_* = 4k^2\delta_*^2$. The name q_* denotes this nonnegative scalar, not an orthogonal polynomial. All original packet multiplicities remain in E_k, A_k, G_M .

Theorem 21.2 (Quartet derivative cost and accumulated volume). *At every admitted degree, including $\delta_* = 0$,*

$$q_* \leq \epsilon^2 \leq \frac{\lambda}{1 + \lambda} \zeta \leq (1 - \pi) \zeta \leq (1 - \pi) \chi. \quad (\text{QD.10})$$

If $\delta_ > 0$, the exact source quantities necessarily obey*

$$\begin{aligned} \zeta &> q_*, \quad \chi \geq \zeta > q_*, \quad \lambda \geq \frac{q_*}{\zeta - q_*} \geq \frac{q_*}{\chi - q_*}, \\ \log \frac{\det G_M}{\det G_{M+1}} &\geq -\log \left(1 - \frac{q_*}{\zeta_M} \right) \geq -\log \left(1 - \frac{q_*}{\chi_M} \right). \end{aligned} \quad (\text{QD.11})$$

For $\delta_ > 0$ and any integers $k(d - 1) \leq a < b$, this gives*

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} -\log \left(1 - \frac{4k^2\delta_*^2}{\zeta_M} \right) \geq \sum_{M=a}^{b-1} -\log \left(1 - \frac{4k^2\delta_*^2}{\chi_M} \right). \quad (\text{QD.12})$$

Every denominator in (QD.11)–(QD.12) is positive, and every logarithm has a strictly positive argument, by the preceding inequalities.

Proof. For every selected centre ρ , the original finite algebra has a nonzero eigenvector of A_h with eigenvalue ρ : in its local factor use $(s - \rho)^{m_\rho - 1}$ and put zero in the other full CRT factors. Its k -fold tensor is nonzero and has eigenvalue $k\rho$ for A_k . The quotient of $W = A_k^*G + GA_k - kG$ on that original vector is exactly $2k \operatorname{Re} \rho - k$. The operator norm of its Hermitian relative matrix is at least the absolute value of this Rayleigh quotient. Taking the largest displacement and squaring proves $q_* \leq \epsilon^2$, with no semisimplicity assumption.

The inverse update is $G_{M+1}^{-1} = G_M^{-1} + F\Omega^{-1}F^*$. Conjugating by the original metric isometry gives

$$\pi = \det(I + UU^*)^{-1} = \prod_i (1 + \lambda_i)^{-1},$$

where all positive eigenvalues of UU^* occur with their multiplicities; the empty product is one. Since the product is at least $1 + \lambda$, $\pi \leq (1 + \lambda)^{-1}$, and therefore $\lambda/(1 + \lambda) \leq 1 - \pi$. Insert (QD.8) and $\zeta \leq \chi$ to prove (QD.10).

When $q_* > 0$, (QD.10) implies $\lambda > 0$ and $\zeta > 0$. Because $\lambda/(1 + \lambda) < 1$, it then implies $\zeta > q_*$. Rearranging its first cost inequality gives $\lambda(\zeta - q_*) \geq q_*$, proving the lower bound for λ . The function $x \mapsto q_*/(x - q_*)$ decreases for $x > q_*$, proving its comparison with the bound using χ . Finally $\pi^{-1} \geq 1 + \lambda \geq \zeta/(\zeta - q_*)$. Apply the increasing logarithm, and use $\zeta \leq \chi$, to obtain (QD.11). Adding the finite inequalities telescopes their original determinants and proves (QD.12). $\square$

At $\delta_* = 0$, (QD.10) remains the complete undivided statement. No value is assigned to $0/0$ when a derivative cost is zero. If $F = 0$, then $U = Y = 0$, $\lambda = 0$, $\pi = 1$ and $W = 0$; the derivative map C_D and its cost can still be nonzero. Conversely $\zeta = 0$ means $C_D = 0$, and the injectivity of $b\Omega^{-1/2}$ gives $V = E_+ = 0$; then $W = 0$ while the metric can still contract through F . These are retained source cases, and neither is excluded by division.

The same proof also gives the stagewise estimate with q_* replaced by the actual ϵ_M^2 whenever $\epsilon_M > 0$, since then $\zeta_M > \epsilon_M^2$. Thus a positive boundary cost remains detected even when every selected centre has real part $1/2$; no nilpotent contribution has been removed by using a zero spectral displacement.

For comparison with (FE.22)–(FE.23), retain the actual positive recurrence quantity

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad a_j = \kappa_j/\kappa_{j-1}, \quad a_0 = 0.$$

The proved arithmetic incidence bound (AP.20) is $VV^* \preceq \Gamma_{k,M}I$, and therefore $\epsilon_M^2 = \|UV^*\|^2 \leq \Gamma_{k,M}\lambda_M$. It yields, for all $\delta_* \geq 0$, the finite consequence

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{4k^2\delta_*^2}{\Gamma_{k,M}} \right). \quad (\text{QD.13})$$

Here $\Gamma_{k,M} > 0$ because every $a_{\alpha_j+1} > 0$. This is the explicit accumulated determinant consequence of the already proved quartet operator estimate (AP.17), whereas (QD.8)–(QD.12) additionally use the original derivative energy and its stronger operator-norm cost. Neither chain asserts an upper asymptotic estimate for those arithmetic quantities.

21.4 Exact energy–volume equality for the original one-variable layer

Theorem 21.3 (One-variable arithmetic equality). *For $k = 1$, at every admitted original degree M ,*

$$\boxed{\zeta_M = \chi_M, \quad \epsilon_M^2 = \frac{\lambda_M}{1 + \lambda_M} \chi_M = (1 - \pi_M) \chi_M.} \quad (\text{QD.14})$$

These equalities include $F = 0$, $E_+ = 0$, and $\delta_* = 0$. Whenever $\chi_M > 0$, they give the exact original determinant identity

$$\log \frac{\det G_M}{\det G_{M+1}} = -\log \left(1 - \frac{\epsilon_M^2}{\chi_M} \right), \quad 0 \leq \epsilon_M^2 < \chi_M. \quad (\text{QD.15})$$

Proof. For $k = 1$, (QD.1) has $r = 1$ at every M . The actual graph isometry $J_b : \mathbb{C} \rightarrow \mathcal{E}_M$ therefore proves that this original relation layer is one-dimensional. Let $e = J_b 1$, so $\|e\| = 1$. There are unique row maps $y, c : \mathbb{C}^d \rightarrow \mathbb{C}$ with $\widehat{Y} = ey$ and $\widehat{C} = ec$. Their actual cross-pairing is $\widehat{Y}^* \widehat{C} = y^* c$. Its norm is $\|y\| \|c\|$: the upper bound is submultiplicativity, and if both rows are nonzero, evaluating on the unit vector $c^*/\|c\|$ attains it. If either row is zero the claimed equality is zero on both sides. The positive matrix $c^* c$ has its only possible positive eigenvalue $\|c\|^2$; thus its largest eigenvalue and trace agree, including $c = 0$. Equations (QD.8)–(QD.9) now prove $\epsilon_M^2 = \|\widehat{Y}\|^2 \chi_M = \lambda_M \chi_M / (1 + \lambda_M)$ and $\zeta_M = \chi_M$.

The matrix U has one column, so UU^* has rank at most one. Its determinant product consequently gives $\pi_M = (1 + \lambda_M)^{-1}$, including $U = 0$. This proves the last equality in (QD.14). If $\chi_M > 0$, division gives $1 - \epsilon_M^2/\chi_M = \pi_M > 0$, and applying the logarithm proves (QD.15). If $\chi_M = 0$, one retains (QD.14); it sets $\epsilon_M = 0$ and imposes no additional value on π_M . In particular the possible contraction from F remains present in the zero derivative-cost case. $\square$

The equality (QD.14) is proved for the actual one-variable arithmetic source and its original one-dimensional relation layer. Thus the coefficient one in (QD.8) is an exact source identity throughout this family, at all nonzero or zero frontier ranks; no auxiliary model is required to test it.

22 The exact two-Stieltjes-space representation of the theta packet

We retain the actual function $g(s) = 2\xi(s)$, a nonempty finite packet of its nontrivial zeros with their complete orders, and the original monic packet polynomial h . The packet is closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$, hence also under $\rho \mapsto 1 - \rho$. No zero in this packet is real: the argument with the alternating eta series in Section 2 applies to the whole real interval $0 < s < 1$. In particular $g(1/2) \neq 0$. Throughout this section all polynomial degrees, powers of $1/2$, and local nilpotent orders are retained.

22.1 The quadratic algebra and both reflection branches

Put

$$z = s - \frac{1}{2}, \quad x = -z^2, \quad d = \deg h = 2D. \quad (\text{SP.1})$$

The even degree follows by partitioning the packet into reflection pairs $\{\rho, 1 - \rho\}$; each pair has the same order at its two members and neither member is $1/2$. For each such pair choose one member, write $\alpha_\rho = \rho - 1/2$, and put

$$r_\rho = -\alpha_\rho^2, \quad H(x) = \prod_{\{\rho, 1-\rho\}} (x - r_\rho)^{m_\rho}. \quad (\text{SP.2})$$

The product is independent of the chosen member. Distinct reflection pairs have distinct r_ρ , because equal squares imply equal or opposite α_ρ . Thus H is monic of degree D , $H(0) \neq 0$, and every original order occurs unchanged in H . Multiplication of the two factors in each pair proves

$$h(s) = (-1)^D H\left(-\left(s - \frac{1}{2}\right)^2\right). \quad (\text{SP.3})$$

Conjugation permutes the factors of H , so H has real coefficients. Define $A_H = \mathbb{C}[x]/(H)$, let X denote multiplication by x in its basis $1, x, \dots, x^{D-1}$, and let c_H be the column of its unit.

Theorem 22.1 (A full algebra isomorphism with its original coordinates). *There is a unital complex algebra isomorphism*

$$\begin{aligned} \mathcal{T} : \mathbb{C}[s]/(h) &\longrightarrow \mathbb{C}[x, z]/(H(x), z^2 + x) = A_H \oplus zA_H, \\ [s] &\longmapsto \frac{1}{2} + z. \end{aligned} \quad (\text{SP.4})$$

The inverse sends x to $-(s - 1/2)^2$ and z to $s - 1/2$. The multiplication in the displayed direct sum is

$$(a, b)(c, e) = (ac - xbe, ae + bc). \quad (\text{SP.5})$$

Here the direct sum describes the underlying module, with the specified multiplication. In the ordered target basis $1, x, \dots, x^{D-1}, z, zx, \dots, zx^{D-1}$, the matrix T of $\mathcal{T}$ has column n , $0 \leq n < 2D$, equal to the coefficients of

$$\begin{aligned} a_n(x) &= \sum_{2j \leq n} \binom{n}{2j} 2^{-(n-2j)} (-x)^j, \\ b_n(x) &= \sum_{2j+1 \leq n} \binom{n}{2j+1} 2^{-(n-2j-1)} (-x)^j. \end{aligned} \quad (\text{SP.6})$$

Its determinant and the original multiplication matrix are exactly

$$\det T = (-1)^{D(D-1)/2}, \quad TAT^{-1} = \frac{1}{2}I_{2D} + \begin{pmatrix} 0 & -X \\ I_D & 0 \end{pmatrix}, \quad A = M_s. \quad (\text{SP.7})$$

Proof. The substitutions in (SP.4) and its inverse send the two defining relations into zero by (SP.3) and $z^2 = -x$. They compose to the identity on the generators and therefore on both algebras. Reduction first by the monic polynomial $z^2 + x$ and then by the monic polynomial $H(x)$ gives a unique expression $a + zb$ with $\deg a, \deg b < D$. Multiplication and the relation $z^2 = -x$ give (SP.5). Expanding $(1/2 + z)^n$ and separating even and odd powers proves (SP.6). In the interlaced basis $1, z, x, zx, \dots, x^{D-1}, zx^{D-1}$, this change of basis is triangular with diagonal entry $(-1)^{\lfloor n/2 \rfloor}$. Their product is $(-1)^{D(D-1)} = 1$. Moving all the odd basis members after the even members has $D(D-1)/2$ inversions, proving the stated determinant. The columns of the inverse are explicitly the original polynomials $-(s - 1/2)^2$ and $(s - 1/2)(-s - 1/2)^2$. Finally multiplication of $a + zb$ by $1/2 + z$ gives $a/2 - xb + z(a + b/2)$, proving (SP.7). $\square$

Here is the complete local meaning of the two branches, including all derivatives. In the factor belonging to $r = -\alpha^2$, of order m , write $\eta = x - r$ and

$$B_r = \mathbb{C}[\eta]/(\eta^m), \quad Z_r = \alpha \sum_{j=0}^{m-1} \binom{1/2}{j} \left(-\frac{\eta}{\alpha^2}\right)^j. \quad (\text{SP.8})$$

The formal binomial identity gives $Z_r^2 = \alpha^2 - \eta = -x$ in B_r . Its constant term is $\alpha \neq 0$, so Z_r is invertible. There are inverse algebra maps

$$\begin{aligned} B_r[z]/(z^2 + x) &\longrightarrow B_r \times B_r, & a + zb &\longmapsto (a + Z_rb, a - Z_rb), \\ (u, v) &\longmapsto \frac{u+v}{2} + z \frac{u-v}{2Z_r}. \end{aligned} \quad (\text{SP.9})$$

The two idempotents are $(1 \pm z/Z_r)/2$. Each component has length m , and corresponds respectively to the original zero $s = 1/2 + \alpha$ or $s = 1/2 - \alpha$. To check the identification on the full local algebras, in the first original factor put $w = s - (1/2 + \alpha)$. The map is $\eta = -2\alpha w - w^2$, with inverse $w = Z_r(\eta) - \alpha$. Its linear coefficient -2α is nonzero, and the two compositions are identities modulo the respective m -th powers. For the second factor replace α by $-\alpha$. This proves the full local isomorphisms without replacing a local algebra by evaluation at its centre. Taking their products gives (SP.4).

The entire functional equation $g(1-s) = g(s)$ also has an exact form in this variable. There is an entire function

$$\Phi(x) = \sum_{n=0}^{\infty} \frac{(-1)^n g^{(2n)}(1/2)}{(2n)!} x^n, \quad g(s) = \Phi\left(-\left(s - \frac{1}{2}\right)^2\right). \quad (\text{SP.10})$$

Indeed the Taylor series of the entire even function $g(1/2 + z)$ contains only even powers, and converges for every z ; its even-power series therefore has infinite radius in $x = -z^2$. By the local isomorphisms above, Φ vanishes to exactly order m_ρ at each r_ρ . Consequently

$$V(x) = (-1)^D \frac{\Phi(x)}{H(x)} \quad \text{is entire,} \quad v_h(s) := \frac{g(s)}{h(s)} = V(x), \quad v = j_h v_h, \quad \mathcal{T}(v) = (V \bmod H, 0). \quad (\text{SP.11})$$

The element $V \bmod H$ is a unit in A_H , because the removed orders were complete. We write U_V for its multiplication matrix, with all its Taylor coefficients modulo H retained.

22.2 The original measures and a Hilbert space isomorphism

The two original measures are

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad d\nu_h(t) = |v_h(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}. \quad (\text{SP.12})$$

Both densities are even. They have all polynomial moments, by the actual theta decay already established for the kernel construction. They are positive except at a discrete set, hence give a positive norm to every nonzero polynomial. Their pushforwards under $t \mapsto t^2$ are, on $x > 0$, exactly

$$\begin{aligned} d\lambda_0(x) &= \frac{|\Phi(x)|^2}{2\pi\sqrt{x}} dx, \\ d\lambda_h(x) &= \frac{|V(x)|^2}{2\pi\sqrt{x}} dx, \quad d\lambda_0(x) = |H(x)|^2 d\lambda_h(x). \end{aligned} \quad (\text{SP.13})$$

For example the two half-lines contribute $2 dt = dx/\sqrt{x}$; this proves the constants in (SP.13). The formulas use the entire quotients at removable zeros.

Theorem 22.2 (Both Hilbert components with an explicit inverse). *For either pair $(\nu, \lambda) = (\nu_h, \lambda_h)$ or (μ, λ_0) , the map*

$$\begin{aligned} \mathcal{U} : L^2(\lambda) \oplus L^2(x\lambda) &\longrightarrow L^2(\nu), \\ (a, b) &\longmapsto [t \mapsto a(t^2) + itb(t^2)] \end{aligned} \quad (\text{SP.14})$$

is unitary. Its inverse, defined for almost every $x > 0$, is

$$a(x) = \frac{f(\sqrt{x}) + f(-\sqrt{x})}{2}, \quad b(x) = \frac{f(\sqrt{x}) - f(-\sqrt{x})}{2i\sqrt{x}}. \quad (\text{SP.15})$$

Multiplication by the original coordinate $s = 1/2 + it$ is carried to $(a, b) \mapsto (a/2 - xb, a + b/2)$. Its exact domain is

$$a \in L^2(\lambda) \cap L^2(x\lambda), \quad b \in L^2(x\lambda) \cap L^2(x^2\lambda). \quad (\text{SP.16})$$

Reflection $f(t) \mapsto f(-t)$ becomes $(a, b) \mapsto (a, -b)$.

Proof. The two parity components are orthogonal. More explicitly their sum of squared absolute values at t and $-t$ is $2|a(t^2)|^2 + 2t^2|b(t^2)|^2$; integrating proves $\|\mathcal{U}(a, b)\|_\nu^2 = \|a\|_\lambda^2 + \|b\|_{x\lambda}^2$. The same identity applied to the even and odd parts of an arbitrary $f \in L^2(\nu)$ proves that (SP.15) belongs to the stated direct sum and is its inverse. There is no atom at $t = 0$, so no extra value is required there. Multiplication gives $(1/2 + it)(a + itb) = a/2 - xb + it(a + b/2)$. On the original space its domain is $f \in L^2(\nu)$ with $tf \in L^2(\nu)$, since $|1/2 + it|^2 = 1/4 + t^2$. Applying the same parity norm identity to tf gives exactly (SP.16), including both conditions on each component. The reflection formula follows by substitution. $\square$

Multiplication by the original h is also preserved: from $L^2(\mu)$ to $L^2(\nu_h)$ it is an isometry, and in (SP.14) it is $(a, b) \mapsto ((-1)^D H a, (-1)^D H b)$. It is onto as an L^2 map, since division by h off its finitely many real-line zeros has precisely the inverse norm identity; those points have measure zero. Its action on polynomial degree, however, remains the actual multiplication by the monic polynomial of degree $2D$. Thus no finite polynomial subspace is identified with one of a different degree without this explicit multiplication map.

22.3 The two monic families and their original norms

Let P_n be the monic orthogonal polynomial of degree n for λ_h , and let R_n be the monic polynomial for $x\lambda_h$. Set

$$p_n = \int |P_n|^2 d\lambda_h, \quad u_n = \int x |R_n|^2 d\lambda_h. \quad (\text{SP.17})$$

All these numbers are strictly positive; the zeroth ones retain the two masses. Orthogonal projection of x^n onto lower powers in the positive moment matrix proves existence and uniqueness. Since that matrix is real, all coefficients are real.

Theorem 22.3 (Monicity, the Christoffel transform, and exact norm ratios). *The original vertical-line monic polynomials are*

$$q_{2n}(s) = (-1)^n P_n(x), \quad q_{2n+1}(s) = (-1)^n z R_n(x), \quad \kappa_{2n} = p_n, \quad \kappa_{2n+1} = u_n. \quad (\text{SP.18})$$

For every $n \geq 0$, $P_n(0) \neq 0$, and

$$c_n = \frac{P_{n+1}(0)}{P_n(0)} < 0, \quad R_n(x) = \frac{P_{n+1}(x) - c_n P_n(x)}{x}, \quad u_n = -c_n p_n. \quad (\text{SP.19})$$

Consequently the original recurrence coefficients obey

$$a_{2n+1} = -c_n \quad (n \geq 0), \quad a_{2n} = \frac{p_n}{-c_{n-1} p_{n-1}} \quad (n \geq 1). \quad (\text{SP.20})$$

The same statements with $\lambda_0, P_n^0, R_n^0, p_n^0, u_n^0$ give the original theta norms $\mathfrak{h}_j$ exactly:

$$\mathfrak{h}_{2n} = p_n^0, \quad \mathfrak{h}_{2n+1} = u_n^0 = -c_n^0 p_n^0, \quad \frac{\mathfrak{h}_{2n+1}}{\mathfrak{h}_{2n}} = -c_n^0, \quad \frac{\mathfrak{h}_{2n+2}}{\mathfrak{h}_{2n+1}} = \frac{p_{n+1}^0}{-c_n^0 p_n^0}. \quad (\text{SP.21})$$

Proof. Since $x = -z^2$, the leading coefficient of each polynomial in (SP.18) in the original variable s is one. Lower-degree even polynomials correspond to lower-degree polynomials in x , and the odd ones correspond to z times lower-degree polynomials in x . Their mixed inner products vanish by parity. Equation (SP.14) therefore proves all the required orthogonality and norm identities, and uniqueness identifies the original q_j .

If $n > 0$ and $P_n(0) = 0$, write $P_n = xQ$, with Q real, monic and of degree $n - 1$. Orthogonality to Q would give $0 = \int x |Q|^2 d\lambda_h > 0$, a contradiction. For $n = 0$, $P_0(0) = 1$. The numerator in (SP.19) vanishes at zero; its quotient by x is monic of degree n . For each polynomial L of degree below n , its inner product with L for $x\lambda_h$ is $\int (P_{n+1} - c_n P_n) \bar{L} d\lambda_h = 0$. This proves the asserted polynomial formula. Since $R_n - P_n$ has lower degree, $\int R_n P_n d\lambda_h = p_n$, while $\int R_n P_{n+1} d\lambda_h = 0$. Multiplying its numerator by R_n and integrating proves $u_n = -c_n p_n$, and positivity proves $c_n < 0$. Taking successive quotients of (SP.18) gives (SP.20). The proof uses only positivity and the actual measure identity, so its application to λ_0 proves (SP.21). $\square$

22.4 Both full jet kernels and the exact weight operator

Fix an original source degree $N \geq 2D - 1$, and put $n_e = \lfloor N/2 \rfloor$, $n_o = \lfloor (N - 1)/2 \rfloor$. Use the actual residue map $p \mapsto v_j h p$ from degree at most N , with its norm from ν_h , onto the original $E_h = \mathbb{C}[s]/h$. Write G_N^{src} for its minimum metric in the original monomial coordinates. Thus at $N = d + m$ it is precisely the earlier G_m , including the fixed-section endpoint $m = -1$. Define the full columns

$$f_n = U_V P_n(X) c_H, \quad e_n = U_V R_n(X) c_H, \quad K_e = \sum_{n=0}^{n_e} \frac{f_n f_n^*}{p_n}, \quad K_o = \sum_{n=0}^{n_o} \frac{e_n e_n^*}{u_n}. \quad (\text{SP.22})$$

No column is replaced by scalar evaluation: $P_n(X)$, $R_n(X)$, and U_V act on the whole length- D algebra. Since $n_e, n_o \geq D - 1$, the first D monic polynomials in each family give a basis modulo H . The unit U_V is invertible, so $K_e, K_o > 0$. Set $G_e = K_e^{-1}$, $G_o = K_o^{-1}$.

Theorem 22.4 (The complete metric and a positive half-line comparison). *In the coordinates (SP.4), the actual minimum metric is*

$$T^{-*} G_N^{\text{src}} T^{-1} = \begin{pmatrix} G_e & 0 \\ 0 & G_o \end{pmatrix}, \quad \det G_N^{\text{src}} = \det G_e \det G_o. \quad (\text{SP.23})$$

For $W_N = A^* G_N^{\text{src}} + G_N^{\text{src}} A - G_N^{\text{src}}$, its full transported form and original relative allowance are

$$T^{-*} W_N T^{-1} = \begin{pmatrix} 0 & G_o - G_e X \\ G_o - X^* G_e & 0 \end{pmatrix}, \quad \epsilon_{N-d} = \left\| G_e^{-1/2} (G_o - G_e X) G_o^{-1/2} \right\|_{\text{op}}. \quad (\text{SP.24})$$

All zero eigenvalues and both signs of the relative form are retained: they are the positive and negative singular values of the displayed D -by- D matrix, with the remaining zero eigenvalues of total multiplicity twice its nullity.

Proof. For completeness, if a surjective linear map from a coefficient space with positive Gram matrix M to a target has matrix J , its minimum metric is $(JM^{-1}J^*)^{-1}$. Indeed the vector $M^{-1}J^*(JM^{-1}J^*)^{-1}y$ has image y ; it is orthogonal in M to $\ker J$, since its inner product with w in that kernel is $y^*(JM^{-1}J^*)^{-1}Jw = 0$. All other preimages differ by such a w , so Pythagoras proves the formula. The polynomial basis (SP.18) has diagonal source Gram matrix κ_j . Under T , its residue columns are $((-1)^n f_n, 0)$ or $(0, (-1)^n e_n)$. Thus the inverse minimum metric is exactly $\text{diag}(K_e, K_o)$, proving the first identity. The second follows from $|\det T| = 1$, proved in (SP.7).

Substitute (SP.7) and (SP.23) into the definition of W_N . The diagonal $1/2$ terms cancel its subtracted metric; multiplying the remaining two block matrices gives the first formula in (SP.24). Taking the congruence by $\text{diag}(G_e^{-1/2}, G_o^{-1/2})$ gives a Hermitian matrix $\begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$. If $Bv = \sigma u$ and $B^*u = \sigma v$, then (u, v) and $(u, -v)$ have eigenvalues σ and $-\sigma$. Singular value decomposition supplies all these vectors; the two kernels supply the remaining vectors. Its operator norm is consequently $\|B\|$. An invertible coordinate change preserves the quotient $|y^*W_N y|/(y^*G_N y)$; this proves the original relative allowance, including its zeros. $\square$

The positive comparison in (SP.24) can also be written entirely inside one copy of A_H . Define

$$R = G_e^{-1/2} G_o G_e^{-1/2} > 0, \quad L = G_e^{1/2} X G_e^{-1/2}. \quad (\text{SP.25})$$

Then

$$\epsilon_{N-d}^2 = \left\| (R - L) R^{-1} (R - L)^* \right\|_{\text{op}}. \quad (\text{SP.26})$$

To verify this without changing a metric implicitly, observe that the matrix B in (SP.24) is $(R - L) G_e^{1/2} G_o^{-1/2}$. The product of the last factor with its adjoint is R^{-1} , so $BB^* = (R - L) R^{-1} (R - L)^*$.

This proves (SP.26). In particular $\epsilon_{N-d} = 0$ holds precisely when $G_o = G_e X$. In that case X is similar to the positive Hermitian matrix R , hence is diagonalizable with positive real spectrum. The minimal polynomial of multiplication by x on A_H is H : a polynomial killing this operator kills the unit, and is therefore divisible by H . Diagonalizability then implies that H is squarefree. Thus a repeated original zero forces a strictly positive finite allowance for every N in this theorem. The complete operator in (SP.26) describes this effect of the retained nilpotents; no bound on its limiting size has been assumed.

Both finite updates are explicit. When N is even, passing to $N + 1$ leaves K_e fixed and adds $e_{n_o+1}e_{n_o+1}^*/u_{n_o+1}$ to K_o ; when N is odd it leaves K_o fixed and adds $f_{n_e+1}f_{n_e+1}^*/p_{n_e+1}$ to K_e . In either case, writing K for the changing block, v for the full added column and k for its positive norm, the determinant ratio is

$$\frac{\det G_{N+1}^{\text{src}}}{\det G_N^{\text{src}}} = \left(1 + \frac{v^* K^{-1} v}{k}\right)^{-1}. \quad (\text{SP.27})$$

Indeed $K + vv^*/k = K^{1/2}(I + ww^*)K^{1/2}$ with $w = K^{-1/2}v/\sqrt{k}$; the middle matrix is identity on $w^\perp$ and has eigenvalue $1 + \|w\|^2$ on its span. This argument also proves (SP.27) when $v = 0$, with ratio one.

22.5 The original tessierine coordinates and support lift

For an original zero $\rho = \beta + i\gamma$, let $\delta = \beta - 1/2$. The coordinate introduced here is exactly

$$r_\rho = \gamma^2 - \delta^2 - 2i\delta\gamma. \quad (\text{SP.28})$$

Retain the group-algebra element and its original characters

$$\begin{aligned} Q(\rho) &= \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa \quad \text{in } \mathbb{C}[K_4], \\ c_{++} &= 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1) = 4\delta, \quad c_{--} = 4i\gamma. \end{aligned} \quad (\text{SP.29})$$

Direct multiplication, with all constants unchanged, gives

$$r_\rho - \bar{r}_\rho = -4i\delta\gamma = -\frac{c_{-+}c_{--}}{4}. \quad (\text{SP.30})$$

Every actual γ here is nonzero. It follows that r_ρ is real if and only if $\beta = 1/2$; in that event $r_\rho = \gamma^2 > 0$. Thus the selected actual packet lies on the critical line exactly when the roots of its complete H are positive real numbers. Their multiplicities are still exactly the original ones. On an eigenvector of A at ρ , the relative Rayleigh quotient of W_N is $2\beta - 1$, because $u^* A^* G u + u^* G A u - u^* G u = (2\beta - 1)u^* G u$. Hence

$$|c_{-+}|/2 = 2|\delta| \leq \epsilon_{N-d}. \quad (\text{SP.31})$$

This is the one-factor spectral bound; the tensor and full-orbit multiplicity bounds proved earlier remain in their original scope.

Finally each algebra isomorphism (SP.4) and (SP.9) induces its exact split-zero isomorphism: send τ to τ and a supported amplitude a to the supported image of a . The semiring laws follow directly from the algebra homomorphism laws and absorption by τ ; the inverse is the same lift of the displayed inverse. In particular the target of the lift of (SP.9) is $G(B_r \times B_r)$, whose common external support is retained. The map to $G(B_r) \times G(B_r)$ sends τ to (τ, τ) and a supported pair (u, v) to its two supported coordinates. It is an injective semiring homomorphism, since both original amplitudes and the external point are recovered; its image consists exactly of the two coordinates both absent or both supported. The two mixed-support fibres remain explicit in the product target. Character evaluation $G(\mathbb{C}[K_4]) \rightarrow G(\mathbb{C})$ likewise sends the supported $Q(\rho)$ to supported c_{-+} . At a critical zero this image is the supported zero e , and the common external τ remains its separate absent fibre throughout these exact maps.

22.6 The complete rank-one half-line boundary and its derivative cost

The two positive kernels in (SP.22) have a further exact compatibility identity. Their finite discrepancy retains the full local algebra, even though its matrix rank is at most one. Use the kernels at the specified original degree N , and keep every column f_n, e_n defined in (SP.22). For the two possible degree parities one has

$$K_e - XK_o = \begin{cases} f_n e_n^* / p_n, & N = 2n, \\ -f_{n+1} e_n^* / u_n, & N = 2n + 1. \end{cases} \quad (\text{SP.32})$$

Here the matrices K_e, K_o in the first row have upper indices $n, n-1$, and those in the second row have upper indices n, n . In particular these are different finite kernels at the two parities.

Proof. The full residue recurrence and its conjugate give the one-factor inverse-metric boundary identity

$$AC_N + C_N A^* - C_N = \frac{z_{N+1}(a_{N+1} z_N)^* + (a_{N+1} z_N) z_{N+1}^*}{\kappa_{N+1}}, \quad C_N = (G_N^{\text{src}})^{-1}, \quad z_j = U_v[q_j]_h.$$

Indeed substitute $C_N = \sum_{j=0}^N z_j z_j^* / \kappa_j$ and $Az_j = z_{j+1} + b_j z_j - a_j z_{j-1}$. The diagonal cancels because $b_j + \bar{b}_j = 1$; each interior adjacent term cancels because $a_{j+1} / \kappa_{j+1} = 1 / \kappa_j$. Only the displayed last pair remains. Transform this identity by T on the left and T^* on the right. By (SP.7) and (SP.23), its upper right block is $K_e - XK_o$.

If $N = 2n$, the two transformed columns are $Tz_N = ((-1)^n f_n, 0)$ and $Tz_{N+1} = (0, (-1)^n e_n)$. Their signs cancel, and $a_{2n+1} / \kappa_{2n+1} = 1 / p_n$, giving the first row. If $N = 2n + 1$, they are $Tz_N = (0, (-1)^n e_n)$ and $Tz_{N+1} = ((-1)^{n+1} f_{n+1}, 0)$. Their product has the minus sign, and $a_{2n+2} / \kappa_{2n+2} = 1 / u_n$, giving the second row. No column has been assumed nonzero. This proves (SP.32) on the whole length- D algebra at both degree parities. $\square$

For the following scalar contractions use the same current metrics G_e, G_o . Equation (SP.32), multiplied on the left by G_e and on the right by G_o , gives the complete outer-product formula for $G_o - G_e X$. Its exact norm is

$$\epsilon_{N-d}^2 = \begin{cases} (f_n^* G_e f_n)(e_n^* G_o e_n) / p_n^2, & N = 2n, \\ (f_{n+1}^* G_e f_{n+1})(e_n^* G_o e_n) / u_n^2, & N = 2n + 1. \end{cases} \quad (\text{SP.33})$$

To prove this, the matrix in (SP.24) becomes the outer product of $G_e^{1/2} f$ and $G_o^{1/2} e$, divided by the displayed positive norm, with the displayed parity sign. For vectors a, b , the norm of ab^* is $\|a\| \|b\|$: Cauchy-Schwarz gives the upper bound, and the vector $b / \|b\|$ attains it when $b \neq 0$. If either vector is zero the equality remains true. This proves (SP.33). Thus the full relative weight has at most the two nonzero eigenvalues $\epsilon_{N-d}, -\epsilon_{N-d}$, and all its other eigenvalues are zero. The full packet and its nilpotents still enter each of the four quadratic contractions in (SP.33).

Let χ_N denote the trace cost of the original one-factor derivative map at source degree N , as in (QD.7), and put $\pi_N = \det G_{N+1}^{\text{src}} / \det G_N^{\text{src}}$. Define

$$\begin{array}{c|c|c} & \lambda_N & \chi_N \\ \hline N = 2n & \frac{e_n^* G_o e_n}{p_n} & \frac{u_n + e_n^* G_o e_n}{p_n^2} (f_n^* G_e f_n) \\ N = 2n + 1 & \frac{f_{n+1}^* G_e f_{n+1}}{p_{n+1}} & \frac{p_{n+1} + f_{n+1}^* G_e f_{n+1}}{u_n^2} (e_n^* G_o e_n). \end{array} \quad (\text{SP.34})$$

These formulas compute the actual cost, rather than choosing one. In the original one-column graph frame the next-layer Gram is $H_{\text{lay}} = \kappa_{N+1} + z_{N+1}^* G_N^{\text{src}} z_{N+1}$, and the derivative row has coefficient $a_{N+1} z_N^* G_N^{\text{src}} / \kappa_{N+1}$. Taking its squared Hilbert–Schmidt norm against the original metric gives $\chi_N = H_{\text{lay}}(z_N^* G_N^{\text{src}} z_N) / \kappa_N^2$. Inserting the two transformed columns just used proves (SP.34), including every original norm and sign. The determinant update (SP.27) now gives, without any division by a zero column,

$$\pi_N = (1 + \lambda_N)^{-1}, \quad \epsilon_{N-d}^2 = (1 - \pi_N) \chi_N. \quad (\text{SP.35})$$

Substitution from the two rows of (SP.34) verifies the second identity against (SP.33). If the new column vanishes then $\lambda_N = 0, \pi_N = 1$ while the derivative cost may persist; if the predecessor column vanishes then $\chi_N = 0$ while the new column may still contract the metric. These are the two original zero-frontier cases, carried through the full half-line kernel formulas.

23 The exact gamma reference and the arithmetic Gram determinants

The norm sequence in (KL.24) is attached to the actual measure

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi. \quad (\text{TG.1})$$

We construct an explicit gamma reference for this measure, prove its polynomial norms, and retain the full arithmetic multiplier in finite Gram determinants. This provides another exact formula for the original boundary allowance; no asymptotic estimate for an arithmetic determinant is assumed.

23.1 The original measure, phase, and packet cancellation

The defining identity for the completed function gives, on the original line $s = 1/2 + it$,

$$\begin{aligned} g(\tfrac{1}{2} + it) &= \Gamma(\tfrac{1}{4} + it/2) a(t), \\ a(t) &= -\pi^{-1/4} (t^2 + \tfrac{1}{4}) e^{-it \log \pi/2} \zeta(\tfrac{1}{2} + it). \end{aligned} \quad (\text{TG.2})$$

Indeed $s(s-1) = -(t^2 + 1/4)$ and $\pi^{-s/2} = \pi^{-1/4} e^{-it \log \pi/2}$. Both the minus sign and this phase remain in the multiplier.

For $\lambda > 0$, define

$$d\sigma_\lambda(t) = |\Gamma(\lambda + it/2)|^2 \frac{dt}{2\pi}. \quad (\text{TG.3})$$

In particular

$$d\mu = |a|^2 d\sigma_{1/4}, \quad d\nu_h = |a_h|^2 d\sigma_{1/4}, \quad a_h(t) = \frac{g(1/2 + it)/h(1/2 + it)}{\Gamma(1/4 + it/2)}. \quad (\text{TG.4})$$

The last expression defines the values at every zero of the original full packet h ; away from those points it equals $a(t)/h(1/2 + it)$. The function g/h is entire because the complete order of each selected zero is removed. Gamma has no zero and has no pole on the real t axis. Thus (TG.4) retains the actual cancellation at each critical packet centre, without introducing a pole into the measure.

Multiplication gives surjective Hilbert isometries

$$\begin{aligned} U_a : L^2(\mu) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto aF, \\ U_{a_h} : L^2(\nu_h) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto a_h F. \end{aligned} \quad (\text{TG.5})$$

The integral identity proves the isometry. The zero set of each multiplier on the real axis is discrete, since it comes from a nonzero analytic function. Given $H \in L^2(\sigma_{1/4})$, define $F = H/a$, respectively

H/a_h , off this null set and give F arbitrary finite values on it. Then $\int |F|^2 d\mu = \int |H|^2 d\sigma_{1/4}$, respectively the packet identity, proving surjectivity and the stated inverse. These are maps between the indicated Hilbert spaces. Their images of polynomial subspaces are exactly $a\mathbb{C}[t]$ and $a_h\mathbb{C}[t]$, with the displayed multipliers retained. Moreover the packet isometry from (KL.23) satisfies the exact diagram

$$U_{a_h}(h(1/2 + it)P(t)) = U_a P(t). \quad (\text{TG.6})$$

23.2 Gamma norms from a generating integral

We give the integral derivation to fix all constants independently of different conventions for Meixner–Pollaczek polynomials. The beta integral, with $r = e^x$, gives

$$\int_{\mathbb{R}} \frac{e^{\lambda x}}{(1 + e^x)^{2\lambda}} e^{iux} dx = \frac{\Gamma(\lambda + iu)\Gamma(\lambda - iu)}{\Gamma(2\lambda)}. \quad (\text{TG.7})$$

For completeness the beta integral follows by applying the change of variables $v = r_1 + r_2$, $y = r_1/(r_1 + r_2)$ to the product of the two absolutely convergent gamma integrals; the Jacobian is v . The conditions $\text{Re}(\lambda \pm iu) > 0$ hold for real u . The function on the left before Fourier transformation and its second derivative are integrable: at each end they are bounded by a constant times $e^{-\lambda|x|}$. Two integrations by parts therefore make its Fourier transform integrable. Fourier inversion applies. One direct justification is to multiply the transform by $e^{-\varepsilon u^2}$, apply Fubini, and use the Gaussian approximate identity; dominated convergence removes that factor because the transform is integrable. Using its evenness and then $t = 2u$, we obtain

$$\int_{\mathbb{R}} e^{iyt} d\sigma_{\lambda}(t) = \frac{2\Gamma(2\lambda)}{(2 \cosh y)^{2\lambda}} = 2^{1-2\lambda} \Gamma(2\lambda) (\cosh y)^{-2\lambda}. \quad (\text{TG.8})$$

The factor 2 in the numerator is the Jacobian of the original change $t = 2u$.

The following integral proves the required exponential integrability.

The function $(2 \cosh(x/2))^{-2\lambda}$ extends holomorphically to $|\text{Im } x| < \pi$, using the branch positive on the real axis. On each closed narrower strip it is bounded by $Ce^{-\lambda|\text{Re } x|}$. In (TG.7), for $u > 0$, shift the contour to $\mathbb{R} + ic$, where $0 < c < \pi$. The vertical sides of the truncated rectangle tend to zero by that bound. For $u < 0$, shift down instead. It follows that

$$|\Gamma(\lambda + iu)|^2 \leq C_{\lambda,c} e^{-c|u|}.$$

Consequently $\int e^{v|t|} d\sigma_{\lambda}(t) < \infty$ for every $0 < v < \pi/2$, by choosing $2v < c < \pi$. This proves holomorphic continuation of both sides of (TG.8) to $|\text{Im } y| < \pi/2$, where they agree by the identity theorem and use the branch inherited from the real axis.

For z near zero use the logarithms that vanish at $z = 0$ and set

$$F_{\lambda}(z, t) = (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} = \sum_{n \geq 0} P_n^{(\lambda)}(t/2; \pi/2) z^n. \quad (\text{TG.9})$$

For real z near zero this is $(1 + z^2)^{-\lambda} \exp(t \arctan z)$. Apply (TG.8) at $y = -i(\arctan z + \arctan w)$, where z, w are real and sufficiently small. The identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

then proves

$$\int_{\mathbb{R}} F_{\lambda}(z, t) F_{\lambda}(w, t) d\sigma_{\lambda}(t) = 2^{1-2\lambda} \Gamma(2\lambda) (1 - zw)^{-2\lambda}. \quad (\text{TG.10})$$

All derivatives in z, w at zero can be passed through the integral: on a smaller closed bidisc their integrands are bounded by a polynomial in $|t|$ times $e^{c|t|}$, with $c < \pi/2$. Thus comparison of the Taylor coefficients in (TG.10) is justified, not formal.

Define the monic polynomial and its unchanged squared norm by

$$b_n^{(\lambda)}(t) = n!P_n^{(\lambda)}(t/2; \pi/2), \quad \gamma_n^{(\lambda)} = 2^{1-2\lambda}n!\Gamma(n+2\lambda). \quad (\text{TG.11})$$

The coefficient of t^n in (TG.9) is $1/n!$, since $\arctan z = z + O(z^3)$. Consequently $b_n^{(\lambda)}$ is monic. Equation (TG.10) gives the full orthogonality identity

$$\int_{\mathbb{R}} b_n^{(\lambda)}(t)b_j^{(\lambda)}(t)d\sigma_\lambda(t) = \gamma_n^{(\lambda)}\delta_{nj}. \quad (\text{TG.12})$$

All these polynomials have real coefficients, as is also clear from the real form of the generating function. Differentiating (TG.9) gives $(1+z^2)\partial_z F_\lambda = (t-2\lambda z)F_\lambda$. Its coefficient of z^n proves

$$\begin{aligned} b_0^{(\lambda)} &= 1, & b_1^{(\lambda)} &= t, \\ b_{n+1}^{(\lambda)} &= tb_n^{(\lambda)} - n(n+2\lambda-1)b_{n-1}^{(\lambda)}, & n &\geq 1, \\ \gamma_n^{(\lambda)}/\gamma_{n-1}^{(\lambda)} &= n(n+2\lambda-1), & n &\geq 1. \end{aligned} \quad (\text{TG.13})$$

For the actual reference $\lambda = 1/4$, abbreviate $b_n = b_n^{(1/4)}$ and $\gamma_n = \gamma_n^{(1/4)}$. Thus

$$\boxed{\gamma_n = \sqrt{2}n!\Gamma(n+\frac{1}{2}), \quad \gamma_{n+1}/\gamma_n = (n+1)(n+\frac{1}{2}).} \quad (\text{TG.14})$$

These formulas agree, with the displayed Jacobian and monic factors, with NIST DLMF §18.19, equations (18.19.8)–(18.19.9), and §18.22, equation (18.22.8).

23.3 Finite arithmetic determinants in the retained metrics

For $N \geq 0$ let $N = \text{diag}(\gamma_0, \dots, \gamma_N)$, and keep both matrices

$$\begin{aligned} \mathbf{B}_N &= \left(\int_{\mathbb{R}} b_i(t)b_j(t)|a(t)|^2 d\sigma_{1/4}(t) \right)_{0 \leq i,j \leq N}, \\ \mathbf{J}_N &= {}_N^{-1/2} \mathbf{B}_N {}_N^{-1/2}, \quad D_N = \det \mathbf{J}_N, \quad D_{-1} = 1. \end{aligned} \quad (\text{TG.15})$$

The diagonal factors are explicit invertible coordinate maps; $\mathbf{B}_N = {}_N^{1/2} \mathbf{J}_N {}_N^{1/2}$ recovers every original inner product. The integrals converge with no unproved growth estimate. Indeed, integration of the step function $\lfloor x \rfloor$ gives, initially for $\text{Re } s > 1$, and then by analytic continuation for $\text{Re } s > 0, s \neq 1$,

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx.$$

The initial equality follows by integrating each constant step on $[n, n+1)$, or by summation by parts of the absolutely convergent Dirichlet series. The last integral converges absolutely and locally uniformly in the asserted half-plane since $0 \leq \{x\} < 1$. On $s = 1/2 + it$, it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4}$, because $|s/(s-1)| = 1$. Equation (TG.2) therefore bounds $|a(t)|$ by a polynomial of degree three. The exponential integrability just proved gives all its polynomial moments. The packet multiplier has the same property on the tail because h is monic, and is bounded on each compact interval by its cancelled analytic expression in (TG.4). Each matrix is positive definite: the integral of the squared modulus of a nonzero polynomial is positive, because a vanishes only at isolated real points.

Theorem 23.1 (The actual theta norms as arithmetic determinants). *For all $n \geq 0$, with the initial value included,*

$$\mathfrak{h}_n = \gamma_n \frac{D_n}{D_{n-1}}, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} = (n+1)(n+\frac{1}{2}) \frac{D_{n+1}D_{n-1}}{D_n^2}. \quad (\text{TG.16})$$

Proof. Let $Q_n(t)$ be the monic polynomial orthogonal to degrees below n in $L^2(\mu)$. Its existence and uniqueness follow from the positive definite Gram matrix. Expressing $Q_0, \dots, Q_N$ in $b_0, \dots, b_N$ gives an upper triangular matrix with diagonal entries one. The determinant of the Gram matrix is therefore unchanged, and orthogonality gives $\det \mathbf{B}_N = \prod_{j=0}^N \|Q_j\|_\mu^2$.

The algebra isomorphism $\Psi : \mathbb{C}[s] \rightarrow \mathbb{C}[t]$, $(\Psi p)(t) = p(1/2 + it)$, preserves the original integral inner product. On the affine space of degree- n monic polynomials the map $p \mapsto i^{-n} \Psi p$ is its monic version and preserves the norm because $|i^{-n}| = 1$. It takes all lower-degree polynomial subspaces onto themselves. Thus the original monic theta residual has squared norm $\mathfrak{h}_n = \|Q_n\|_\mu^2$. The degree-dependent phase is used on this affine space; it is not asserted to define a linear map on the entire polynomial algebra.

Finally $D_N = \det \mathbf{B}_N / \prod_{j=0}^N \gamma_j$. Divide this equality at $N = n$ by the equality at $N = n-1$, taking an empty product to be one. This proves the first formula. Dividing consecutive formulas and using (TG.14) proves the second. $\square$

Define $\mathbf{B}_N^{(h)}, \mathbf{J}_N^{(h)}, D_N^{(h)}$ by (TG.15) with the entire-cancelled multiplier a_h of (TG.4). The same proof, with no restriction on whether the packet centres are critical, gives

$$\kappa_n^{(h)} = \gamma_n \frac{D_n^{(h)}}{D_{n-1}^{(h)}}. \quad (\text{TG.17})$$

Combining (TG.17) and (TG.16) with (KL.24) yields the exact packet-to-reference determinant map, for every $m \geq 0$,

$$r_m^{(h)} = \frac{(d+m)! \Gamma(d+m+1/2)}{m! \Gamma(m+1/2)} \frac{D_{d+m}^{(h)}}{D_{d+m-1}^{(h)}} \frac{D_{m-1}}{D_m}. \quad (\text{TG.18})$$

Every determinant is strictly positive and the full degree $d = \sum_\rho m_\rho$ is retained.

For a packet stable under both ordinary conjugation and $\rho \mapsto 1 - \bar{\rho}$, and for every $m \geq 0$, (KL.32) becomes

$$\epsilon_m^2 = (m+1)(m+\frac{1}{2}) \frac{D_{m+1}D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{TG.19})$$

This is the exact same finite-stage allowance. Its factor $(m+1)(m+1/2)$ is explicit; the arithmetic determinant quotient remains in the formula.

23.4 Full confluent kernel and the original coordinates

Set $\hat{b}_j(s) = i^j b_j((s-1/2)/i)$. These are monic polynomials in the original variable s . With $\hat{\mathbf{r}}_N = \text{diag}(1, i, \dots, i^N)$, their actual packet Gram matrix is

$$\mathbf{B}_{N,s}^{(h)} = \hat{\mathbf{r}}_N^* \mathbf{B}_N^{(h)} \hat{\mathbf{r}}_N. \quad (\text{TG.20})$$

Indeed $\hat{b}_j(1/2 + it) = i^j b_j(t)$; taking the conjugate in the first argument gives exactly the displayed congruence, including every phase.

Let $\mathbf{C}_N$ have columns $[\hat{b}_j]_h \in E_h = \mathbb{C}[s]/(h)$, in the original monomial remainder coordinates. For $N \geq d-1$,

$$K_N = \mathbf{C}_N (\mathbf{B}_{N,s}^{(h)})^{-1} \mathbf{C}_N^*, \quad \mathbf{G}_{N-d} = U_{\varepsilon_h}^* K_N^{-1} U_{\varepsilon_h}. \quad (\text{TG.21})$$

Here $\mathbf{G}_{-1}$ is the fixed initial-section metric, and $\varepsilon_h = (j_h(g/h))^{-1}$ is the same full local unit as before. To prove the first identity, choose the actual monic orthogonal polynomial basis through degree N . If T is its invertible coefficient matrix in the $\hat{b}_j$ basis, then

$$T^* \mathbf{B}_{N,s}^{(h)} T = \text{diag}(\kappa_0, \dots, \kappa_N).$$

Inverting gives $(\mathbf{B}_{N,s}^{(h)})^{-1} = T \text{diag}(\kappa_j^{-1}) T^*$. The columns of $\mathbf{C}_N T$ are the full remainders $[p_j]_h$. Substitution proves exactly $K_N = \sum_{j=0}^N [p_j]_h [p_j]_h^* / \kappa_j$, the original interpolation kernel. The first d monic polynomials span all remainders, so $\mathbf{C}_N$ is onto and K_N is positive definite. The second identity is the fixed-unit minimum-lift formula already proved in (KL.27), with the exact index $N = d + m$. No single remainder column is required to be nonzero or invertible.

For a centre ρ , put $t_\rho = (\rho - 1/2)/i$ and write $f^{[\ell]} = f^{(\ell)}/\ell!$. The full Taylor coordinates of these columns are explicitly

$$\hat{b}_j^{[\ell]}(\rho) = i^{j-\ell} b_j^{[\ell]}(t_\rho), \quad 0 \leq \ell < m_\rho. \quad (\text{TG.22})$$

This follows by differentiating the affine argument ℓ times. CRT converts these full Taylor coordinates to the monomial remainder coordinates, and multiplication by U_{v_h} retains all derivatives of g/h . Thus (TG.21) computes the entire confluent kernel, including nilpotent orders and critical centres.

23.5 The finite map to Romik's gamma parameter

Romik's expansion uses $P_n^{(3/4)}(t/2; \pi/2)$, with the unchanged variable $t/2$; see his §3 and Appendix A.2. The exact finite change to the parameter in (TG.3) is

$$\begin{aligned} b_n^{(1/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{1/2}{k} b_{n-2k}^{(3/4)}(t), \\ b_n^{(3/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{-1/2}{k} b_{n-2k}^{(1/4)}(t). \end{aligned} \quad (\text{TG.23})$$

To prove these formulas, divide the two generating functions: $F_{1/4}(z, t) = (1 + z^2)^{1/2} F_{3/4}(z, t)$. The binomial series at zero and coefficient comparison prove the first formula. Multiplication by $(1 + z^2)^{-1/2}$ proves the inverse. Each finite coefficient matrix is triangular with diagonal one, preserves parity, and is invertible. Applied to the same measure, it transports Gram matrices by its full congruence and transports every Taylor column by the same coefficient matrix. Consequently the kernel expression (TG.21) is invariant under this change of polynomial basis.

The two reference measures themselves have the exact Hilbert-space map

$$L^2(\sigma_{1/4}) \longrightarrow L^2(\sigma_{3/4}), \quad F(t) \longmapsto \frac{\Gamma(1/4 + it/2)}{\Gamma(3/4 + it/2)} F(t). \quad (\text{TG.24})$$

The gamma quotient has neither zeros nor poles for real t . Cancellation of its squared modulus proves the isometry and its inverse; this isometry has a multiplier, whereas (TG.23) is the finite polynomial coordinate map. Their respective domains and metrics have both been given explicitly.

Romik's coefficient asymptotics concern the expansion of Ξ in a fixed gamma-orthogonal basis. Inoue's §2 similarly treats expansion coefficients and convergence. In the present construction the arithmetic norms are the determinants (TG.16). Equations (TG.20)–(TG.24) are the exact maps connecting those polynomial presentations to these determinants; no assertion that their coefficient asymptotics already estimate D_n is used.

23.6 Exact scope of the exponential-weight theorem

The original theorem of Lubinsky–Mhaskar–Saff does include tail order one. In their 1988 paper, Definition 2.1 requires $W = e^{-Q}$, with Q even and continuous, Q' existing on $(0, \infty)$, $xQ'(x)$ bounded at zero, and, eventually,

$$Q' > 0, \quad x^2|Q'''|/Q' \leq C, \quad 1 + xQ''/Q' \rightarrow \alpha > 0. \quad (\text{TG.25})$$

Theorem 2.3 admits a finite generalized Jacobi factor, a lower-degree polynomial exponential, and a bounded nonnegative factor $\Psi(t) \rightarrow 1$. Its Theorem 2.4 replaces the polynomial exponential by V , but requires $\log V(t)/Q(t) \rightarrow 0$, as well as bounded reciprocal approximation on the expanding interval by polynomials of degree $o(n)$. The precise statements are on printed pages 67–70; their proof is on pages 80–81. These hypotheses concern the full weight, not its gamma factor alone.

We prove a concrete obstruction to inserting the actual arithmetic multiplier into those displayed hypotheses. Hardy’s theorem supplies real numbers $T_j \rightarrow \infty$ with $g(1/2 + iT_j) = 0$. Removing a fixed full polynomial packet leaves an infinite subsequence of these zeros. At each remaining T_j , the multiplier a_h of (TG.4) has the local form

$$a_h(t) = (t - T_j)^{\ell_j} v_j(t), \quad \ell_j \geq 1, \quad v_j(T_j) \neq 0, \quad (\text{TG.26})$$

where v_j is analytic in a neighbourhood of T_j . This follows from the Taylor series at an isolated zero, with its entire order ℓ_j retained.

Let $B(t) > 0$ be any continuous comparison amplitude near all sufficiently large T_j , and suppose it has no zeros there. For every j and every $\eta > 0$, continuity and (TG.26) give a nonempty interval $I_{j,\eta}$ about T_j on which

$$|a_h(t)|/B(t) < \eta. \quad (\text{TG.27})$$

Thus a decomposition with this ratio as the bounded multiplier Ψ cannot satisfy $\Psi \rightarrow 1$. The obstruction holds on intervals of positive measure, so changing the value only at each zero does not repair it. A finite generalized Jacobi factor has only finitely many real zeros or singularities; choosing the subsequence beyond them leaves (TG.27) unchanged.

For the logarithmic hypothesis choose any eventually positive continuous Q . At a remaining T_j , shrink the neighbourhood so that $Q(t) \leq Q(T_j) + 1$, and choose $\eta = \exp(-j(Q(T_j) + 1))$ in (TG.27). On the resulting positive-measure interval,

$$\frac{\log(|a_h(t)|/B(t))}{Q(t)} < -j \quad (t \neq T_j). \quad (\text{TG.28})$$

Hence this ratio cannot be V with $\log V/Q \rightarrow 0$. An additional factor $\Psi \rightarrow 1$ is bounded away from zero and infinity on the tail and does not remove the contradiction. This disproves the indicated application of Theorems 2.3–2.4, including almost-everywhere modifications of the weight. It makes no claim that other asymptotic methods are unavailable. The exact positive Gram matrices (TG.15)–(TG.21) retain the arithmetic information at all those zeros and are available for such further analysis.

23.7 Primary source locators

The content-reading route and source hashes are recorded in `work/theta_norm_literature_route_20260912.json` and `work/theta_norm_literature_sources/download_receipt.json`. The principal public locators are:

- D. S. Lubinsky, H. N. Mhaskar and E. B. Saff, *A proof of Freud’s conjecture for exponential weights*, Constructive Approximation 4 (1988), 65–83, Definition 2.1, Theorems 2.3–2.5 and §4: <https://math.vanderbilt.edu/saffeb/texts/81.pdf>.

- Dan Romik, *Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases*, §3.1–3.3 and Appendix A.2: <https://www.math.ucdavis.edu/~romik/data/uploads/papers/riemannxi-final.pdf>.
- Hiroto Inoue, *Expansion of Dirichlet L-function on the critical line in Meixner–Pollaczek polynomials*, arXiv:1412.1220, §2.1–2.2: <https://arxiv.org/abs/1412.1220>.
- NIST DLMF, equations (18.19.8)–(18.19.9), (18.22.8): <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22>.
- G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, Comptes rendus 158 (1914), 1012–1014, §1–3. The exact page images and transcriptions are linked in the machine-readable source route.

24 The symmetric spectral sum and its unequal Stieltjes pair

The complete PR 17 note, *Spectral-sum descent with the relative arithmetic fibre retained*, supplies the invariant relation ideal and the original matrix weight used below. We prove their further reflection decomposition, its full paired-packet defect, and the minimum-metric and weight formulas. The coordinate x in this section belongs to the spectral sum. We write x_h for the one-factor coordinate of Section 22; their exact relation on the defect will be proved.

24.1 Original polynomial coordinates and the reflection ideals

Keep $g = 2\xi$, the same nonempty quartet-closed packet h with every full order, $d = 2D$, and

$$h(s) = (-1)^D H(-(s - \tfrac{1}{2})^2), \quad A_H = \mathbb{C}[x_h]/(H), \quad v_h = g/h, \quad v = j_h v_h. \quad (\text{SSP.1})$$

In particular $D \geq 1$ and $H(0) \neq 0$, as proved in (SP.1)–(SP.3). Write

$$\begin{aligned} S &= s_1 + s_2, & r &= s_1 - s_2, & \Delta &= r^2, & Z &= S - 1, & x &= -Z^2, \\ s_1 &= \tfrac{1}{2} + (Z + r)/2, & s_2 &= \tfrac{1}{2} + (Z - r)/2. \end{aligned} \quad (\text{SSP.2})$$

The swap fixes S and negates r . Thus substitution gives a filtered algebra isomorphism $\mathbb{C}[S, \Delta] \rightarrow \mathbb{C}[s_1, s_2]^{S_2}$, with $\deg S = 1$, $\deg \Delta = 2$: expansion in r keeps precisely its even powers, and the inverse coordinate map in (SSP.2) keeps the factors $1/2$.

Write $h(1/2 + y) = \sum_{j=0}^D b_j y^{2j}$, with $b_D = 1$. Define the following entire finite polynomials:

$$\begin{aligned} A_0(x, \Delta) &= \sum_{j=0}^D b_j 4^{-j} \sum_{\ell=0}^j \binom{2j}{2\ell} (-x)^{j-\ell} \Delta^\ell, \\ A_1(x, \Delta) &= \sum_{j=1}^D b_j 4^{-j} \sum_{\ell=0}^{j-1} \binom{2j}{2\ell+1} (-x)^{j-\ell-1} \Delta^\ell, \quad Q_\Delta = \Delta A_1. \end{aligned} \quad (\text{SSP.3})$$

Binomial expansion, with $x = -Z^2$, proves exactly

$$\begin{aligned} h(s_1) &= A_0 + rZ A_1, & h(s_2) &= A_0 - rZ A_1, \\ A_0(0, \Delta) &= (-1)^D H(-\Delta/4), & A_1(0, \Delta) &= \frac{(-1)^{D+1}}{2} H'(-\Delta/4). \end{aligned} \quad (\text{SSP.4})$$

For the last identity one can also differentiate the difference $h(1/2 + (Z + r)/2) - h(1/2 + (Z - r)/2)$ at $Z = 0$, and divide by $2r$; both sides are polynomials, so the identity includes $r = 0$. The polynomial $A_0(0, \Delta)$ has degree D and leading coefficient 4^{-D} . No localization at r , Δ , Z , or x has been made.

Theorem 24.1 (The original symmetric algebra with both parity modules). *Let $\mathcal{E} = (E_h \otimes E_h)^{S_2}$, $E_h = \mathbb{C}[s]/(h)$. There is an exact algebra isomorphism*

$$\mathcal{E} \cong \mathbb{C}[x, Z, \Delta]/(Z^2 + x, A_0, ZQ_\Delta). \quad (\text{SSP.5})$$

Put

$$B_e = \mathbb{C}[x, \Delta]/(A_0, xQ_\Delta), \quad B_o = \mathbb{C}[x, \Delta]/(A_0, Q_\Delta). \quad (\text{SSP.6})$$

The underlying module of (SSP.5) is $B_e \oplus ZB_o$. The two maps and the product are

$$\begin{aligned} \beta : B_e &\rightarrow B_o, & [a] &\mapsto [a], & \alpha : B_o &\rightarrow B_e, & [b] &\mapsto [xb], \\ (a, b)(c, e) &= (ac - \alpha(be), \beta(a)e + \beta(c)b), \\ \alpha\beta &= xI_{B_e}, & \beta\alpha &= xI_{B_o}. \end{aligned} \quad (\text{SSP.7})$$

The map β is a unital algebra homomorphism; α is a homomorphism of $\mathbb{C}[x, \Delta]$ -modules. Its full multiplicative relation is $\alpha(b)\alpha(c) = x\alpha(bc)$, and $\alpha(1) = x$. Reflection in both original variables acts as $(a, b) \mapsto (a, -b)$. The dimensions are

$$\dim B_e = D(D + 1), \quad \dim B_o = D^2. \quad (\text{SSP.8})$$

Proof. In the polynomial algebra $\mathbb{C}[S, r]$, the original ideal $(h(s_1), h(s_2))$ is (H_0, rH_1) , where $H_0 = A_0(-Z^2, \Delta)$ and $H_1 = ZA_1(-Z^2, \Delta)$. The factors $1/2$ in the sum and difference of the original generators give both ideal containments. An invariant polynomial in this ideal can be written $H_0a + rH_1b$. Average this expression under $r \mapsto -r$. The result is $H_0a_e + \Delta H_1b_o$, where a_e and b_o are polynomials in S, Δ ; the odd part of b is rb_o . Conversely both invariant generators lie in the original ideal. Every invariant class in the quotient has an invariant lift by this same averaging. Therefore

$$\mathcal{E} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1). \quad (\text{SSP.9})$$

Replace S by $1 + Z$ and adjoin $x = -Z^2$; this proves (SSP.5) with inverse given by the original substitutions.

Before imposing A_0, ZQ_Δ , division by $Z^2 + x$ gives the free module $\mathbb{C}[x, \Delta] \oplus Z\mathbb{C}[x, \Delta]$. Multiplying the two remaining generators by arbitrary elements gives

$$A_0(a + Zb) + ZQ_\Delta(c + Ze) = A_0a - xQ_\Delta e + Z(A_0b + Q_\Delta c). \quad (\text{SSP.10})$$

Its even and odd coefficient ideals are precisely those in (SSP.6). This proves the module decomposition, including both directions of its defining relations. Multiplication by Z is $(a, b) \mapsto (-\alpha b, \beta a)$, so direct multiplication proves (SSP.7) and the two compositions. All operations are well-defined by the displayed ideals.

Reflection sends Z, r to $-Z, -r$, and fixes x, Δ . The one-factor decomposition (SP.4) has reflection spaces of dimensions D, D . On the invariant tensor square the positive space is the sum of their two symmetric squares, with dimension $2D(D + 1)/2$. The negative space consists of their mixed symmetrized tensors, with dimension D^2 . They are exactly the even and odd coefficient modules in (SSP.10), proving (SSP.8), including $D = 1$. $\square$

24.2 The entire defect is the original paired packet

Theorem 24.2 (Two exact sequences with every original local order). *The map α is injective and β is surjective. Regard A_H as a $\mathbb{C}[x, \Delta]$ -module through $x \mapsto 0, \Delta \mapsto -4x_h$. The two exact sequences of modules are*

$$\begin{aligned} 0 &\longrightarrow B_o \xrightarrow{\alpha} B_e \xrightarrow{\rho_0} A_H \longrightarrow 0, & \rho_0[a] &= [a(0, -4x_h)], \\ 0 &\longrightarrow A_H \xrightarrow{\iota_0} B_e \xrightarrow{\beta} B_o \longrightarrow 0, & \iota_0[q] &= [Q_\Delta q(-\Delta/4)]. \end{aligned} \quad (\text{SSP.11})$$

In particular both defects retain the original polynomial H and every order in it. On the full algebra $\mathcal{E}$, multiplication by the original sum is

$$A_S = I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}. \quad (\text{SSP.12})$$

Its full kernel and cokernel at $S = 1$ are

$$\ker(A_S - I) = \{(\iota_0 q, 0) : q \in A_H\}, \quad \text{coker}(A_S - I) \cong A_H, \quad [(a, b)] \mapsto \rho_0 a. \quad (\text{SSP.13})$$

Proof. Let $\mathcal{R} = \mathbb{C}[x, \Delta]$. Since $A_0(0, \Delta) \neq 0$, the prime polynomial x does not divide A_0 . If $xb = A_0c + xQ_\Delta e$, then $x(b - Q_\Delta e) = A_0c$ forces $c = xc_1$, whence $b = A_0c_1 + Q_\Delta e$. This proves injectivity of α without dividing by x in either quotient. Surjectivity of β follows from its quotient definition. The image of α is xB_e , because every polynomial representative of an even class is also an odd coefficient representative. Therefore

$$\text{coker } \alpha \cong \mathcal{R}/(A_0, x) \cong \mathbb{C}[\Delta]/((-1)^D H(-\Delta/4)) \cong A_H. \quad (\text{SSP.14})$$

The last isomorphism sends Δ to $-4x_h$; its inverse sends x_h to $-\Delta/4$. This proves the first sequence, with all constants.

We justify the cancellation required for the second sequence. The quotient $\mathcal{R}/(A_0, Q_\Delta) = B_o$ has finite dimension D^2 , already proved in (SSP.8). If a nonconstant irreducible polynomial divided both A_0, Q_Δ , then it would divide every polynomial annihilator of x and every polynomial annihilator of Δ in this quotient. Such nonzero univariate annihilators exist by linear dependence of the powers in the finite vector space. Over $\mathbb{C}$, a common irreducible divisor of a nonzero polynomial in x alone and one in Δ alone is impossible: their irreducible factors are respectively $x - a$ and $\Delta - b$. Thus $\gcd(A_0, Q_\Delta) = 1$ in $\mathcal{R}$.

The kernel of β consists exactly of the classes $[Q_\Delta p]$ in B_e . Multiplication by Q_Δ gives a surjection

$$\mathcal{R}/(A_0, x) \longrightarrow \ker \beta, \quad [p] \longmapsto [Q_\Delta p]. \quad (\text{SSP.15})$$

It is well-defined since $A_0 Q_\Delta = x Q_\Delta = 0$ in B_e . If $Q_\Delta p = A_0 c + x Q_\Delta e$, the coprimality just proved forces $A_0 \mid p - xe$. Hence $[p] = 0$ in $\mathcal{R}/(A_0, x)$, proving injectivity. Compose (SSP.15) with the inverse last map of (SSP.14) to obtain exactly ι_0 in (SSP.11). Finally multiplication by Z is the off-diagonal part of (SSP.12). Its kernel has odd component zero by injectivity of α , and even component $\ker \beta$. Its cokernel has zero odd quotient by surjectivity of β , and even quotient $\text{coker } \alpha$. This proves (SSP.13). $\square$

The full multiplication on the second defect is also explicit. Put

$$\mathfrak{c}_H = 2(-1)^D x_h H'(x_h) \pmod{H}.$$

Then, for every $q, t \in A_H$,

$$\iota_0(q)\iota_0(t) = \iota_0(\mathfrak{c}_H qt). \quad (\text{SSP.15a})$$

Indeed $xQ_\Delta = 0$ gives $Q_\Delta^2 = Q_\Delta Q_\Delta(0, \Delta)$ in B_e . Equation (SSP.4), followed by $\Delta = -4x_h$, identifies $Q_\Delta(0, -4x_h)$ with $\mathfrak{c}_H$. Substitution in the two products proves (SSP.15a). Thus the module isomorphism ι_0 transports the inherited ideal multiplication to $q \star t = \mathfrak{c}_H qt$, retaining the derivative and its nilpotent values.

The generator in (SSP.11) is the full class $Q_\Delta = \Delta A_1$ modulo (A_0, xQ_Δ) . Reducing its coefficient at $x = 0$ before forming this class would change the map. In particular (SSP.4) can vanish on repeated local fibres through H' ; the proved injective map ι_0 still retains those fibres. Equation (SSP.15) specifies the complete mixed nilpotent class that carries them.

24.3 The original unit and all coordinate factorials

The original full unit on $\mathcal{E}$ is

$$v^{(2)} = [v_h(s_1)v_h(s_2)] \in \mathcal{E}^\times. \quad (\text{SSP.16})$$

This bracket denotes its complete double Taylor class modulo $(h(s_1), h(s_2))$, followed by the invariant identification; the values of all admitted derivatives are retained. Since $d = 2D$, $v_h(1-s) = v_h(s)$. Thus this unit has the form $(u_e, 0)$ under (SSP.5), and its two multiplication maps are

$$U_e = M_{u_e} : B_e \rightarrow B_e, \quad U_o = M_{\beta(u_e)} : B_o \rightarrow B_o. \quad (\text{SSP.17})$$

Both are invertible, by multiplication with the components of the original inverse unit. Their exact intertwiners and defect actions are

$$\begin{aligned} U_e \alpha &= \alpha U_o, & \beta U_e &= U_o \beta, & \rho_0 U_e &= M_{V(x_h)^2} \rho_0, \\ U_e \iota_0 &= \iota_0 M_{V(x_h)^2}, & V(x_h) &= (-1)^D \Phi(x_h)/H(x_h). \end{aligned} \quad (\text{SSP.18})$$

Indeed the first two equations are the module multiplication laws in (SSP.7). The algebra homomorphism $\mathcal{E} \rightarrow A_H$ given by $(a, b) \mapsto \rho_0 a$ is the original specialization $s_2 = 1 - s_1$, followed by its even quotient: $Z = 0$, $\Delta = 4(s_1 - 1/2)^2 = -4x_h$. It sends (SSP.16) to $v_h(s_1)v_h(1-s_1) = V(x_h)^2$, proving the third equation with all jets. For the last equation, $xQ_\Delta = 0$ implies

$$u_e Q_\Delta q(-\Delta/4) = Q_\Delta u_e(0, \Delta) q(-\Delta/4). \quad (\text{SSP.19})$$

The same specialization proves (SSP.18). In particular the factor $g = 2\xi$ occurs in each original unit factor and is present in the squared defect unit.

Here are explicit finite coordinate maps to the unscaled symmetric basis used earlier. Let $T : E_h \rightarrow A_H \oplus z_h A_H$ be the exact matrix (SP.6), with $z_h = s - 1/2$. Use the bases $a_i = x_h^i$ and $b_i = z_h x_h^i$, $0 \leq i < D$, in that target. The new symmetric basis is ordered in these three groups:

$$\begin{aligned} a_i \odot a_j \ (i \leq j), & \quad b_i \odot b_j \ (i \leq j), & \quad a_i \odot b_j \ (0 \leq i, j < D), \\ u \odot v = u \otimes v + v \otimes u \ (u \neq v), & \quad u \odot u = u \otimes u. \end{aligned} \quad (\text{SSP.20})$$

The first two groups are the basis of B_e , and the last, after removal of its displayed Z , is the basis of B_o . These identifications follow by the inverse substitutions in (SSP.2) and (SP.4): swap invariance leaves even powers of r , and reflection leaves even powers of Z in the first two groups and odd powers in the last. Replace $Z^{2j} r^{2a}$ by $(-x)^j \Delta^a$, and $Z^{2j+1} r^{2a}$ by $Z(-x)^j \Delta^a$. This finite coefficient prescription gives each basis polynomial explicitly, without choosing values at roots.

Let I_{old} and I_{new} insert the literal orbit sums in, respectively, the original monomial tensor basis and the basis (SSP.20) before passage to its invariant space. Put

$$\begin{aligned} D_{\text{old}} &= I_{\text{old}}^* I_{\text{old}}, & L_{\text{old}} &= D_{\text{old}}^{-1} I_{\text{old}}^*, \\ D_{\text{new}} &= I_{\text{new}}^* I_{\text{new}}, & L_{\text{new}} &= D_{\text{new}}^{-1} I_{\text{new}}^*, \\ T_s &= L_{\text{new}}(T \otimes T) I_{\text{old}}, & T_s^{-1} &= L_{\text{old}}(T^{-1} \otimes T^{-1}) I_{\text{new}}. \end{aligned} \quad (\text{SSP.21})$$

Each diagonal entry of D_{old} , D_{new} is one for a repeated basis vector and two for two distinct vectors. These are the original orbit cardinalities. The products IL are both the permutation average $(I + \text{swap})/2$; it commutes with $T \otimes T$. Hence the two displayed matrices compose to identity. This proves all factors in the coordinate maps. Moreover

$$|\det T_s| = |\det T|^{d+1} = 1. \quad (\text{SSP.22})$$

To verify the exponent, triangularize T over $\mathbb{C}$; on symmetric tensors its diagonal entries are the products of its diagonal entries indexed by $i \leq j$. Each original diagonal entry occurs $d + 1$ times. The similarity inducing that triangularization and the final permutation into (SSP.20) do not change the absolute determinant. Equation (SP.7) then proves (SSP.22). No tensor basis vector has been divided by its length.

24.4 The actual matrix weight and its positive half-line pair

Retain the original density

$$w_h(t) = |v_h(1/2 + it)|^2 / (2\pi). \quad (\text{SSP.23})$$

It is even and has the exponential and polynomial integrability proved for the theta source. For a fixed source degree M , put $L_M = \lfloor M/2 \rfloor$ and keep every relative direction $0 \leq a \leq L_M$. The PR 17 matrix weight is exactly

$$W_{ac}(u) = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv. \quad (\text{SSP.24})$$

For each u , this finite matrix is real Hermitian and strictly positive. Indeed its quadratic form on a nonzero vector c is one half of the integral of $|\sum_a c_a (-v^2)^a|^2$ against the two positive densities. Their zeros are discrete, while that polynomial is nonzero, proving strict positivity. The actual decay gives finite integrals. Evenness of w_h gives $W(-u) = W(u)$, with every matrix entry retained.

For a symmetric original numerator

$$P(S, \Delta) = \sum_{a=0}^{L_M} \Delta^a f_a(S), \quad \deg f_a \leq M - 2a,$$

the original coordinate change $u = t_1 + t_2$, $v = t_1 - t_2$ has $dt_1 dt_2 = du dv / 2$, and $S = 1 + iu$, $\Delta = -v^2$. Expansion and Fubini, justified by the actual moment bounds, give

$$\|\mathcal{T}_h^{(2)} P\|^2 = \int_{\mathbb{R}} f(1 + iu)^* W(u) f(1 + iu) du. \quad (\text{SSP.25})$$

Here $\mathcal{T}_h^{(2)} P = P(D_1 + D_2, (D_1 - D_2)^2) F_h^{\otimes 2}$; its Mellin multiplier is $P(S, \Delta) v_h(s_1) v_h(s_2)$. Thus (SSP.25) includes both factors $1/(2\pi)$, the real Jacobian $1/2$, and every relative-coordinate sign.

Define the matrix-valued measure on $x > 0$ by

$$d\Lambda(x) = W(\sqrt{x}) \frac{dx}{\sqrt{x}}. \quad (\text{SSP.26})$$

For a matrix measure $d\Lambda = W_\Lambda dx$, its squared L^2 norm is $\int f^* W_\Lambda f dx$; every component has the declared dimension $L_M + 1$. The full unitary map and inverse are

$$\begin{aligned} \mathcal{U}_\Sigma : L^2(\Lambda) \oplus L^2(x\Lambda) &\longrightarrow L^2(\mathbb{R}, W(u) du), \\ (A, B) &\longmapsto f(1 + iu) = A(u^2) + iuB(u^2), \\ A(x) &= \frac{f(1 + i\sqrt{x}) + f(1 - i\sqrt{x})}{2}, \\ B(x) &= \frac{f(1 + i\sqrt{x}) - f(1 - i\sqrt{x})}{2i\sqrt{x}}. \end{aligned} \quad (\text{SSP.27})$$

To prove the norm identity, add the two quadratic forms at u and $-u$. The cross terms cancel and give $2A^*WA + 2u^2B^*WB$. The substitution $x = u^2$ proves the direct-sum norm with exactly (SSP.26). The same calculation on the parity parts of an arbitrary f proves that the displayed inverse belongs to both asserted spaces. There is no atom at zero, so the almost-everywhere inverse requires no value there. Multiplication by $S = 1 + iu$ is

$$(A, B) \longmapsto (A - xB, A + B), \quad \begin{cases} A \in L^2(\Lambda) \cap L^2(x\Lambda), \\ B \in L^2(x\Lambda) \cap L^2(x^2\Lambda). \end{cases} \quad (\text{SSP.28})$$

Its exact domain follows by adding $\|uf\|^2$ to $\|f\|^2$, since $|1 + iu|^2 = 1 + u^2$; the same parity calculation gives the two additional weighted norms.

The finite source image in (SSP.27) is precisely

$$\deg A_a \leq \left\lfloor \frac{M-2a}{2} \right\rfloor, \quad \deg B_a \leq \left\lfloor \frac{M-2a-1}{2} \right\rfloor \quad (0 \leq a \leq L_M). \quad (\text{SSP.29})$$

A negative degree bound means the zero polynomial. In particular the last odd relative direction can have a zero allowed polynomial space while its ambient Hilbert component is still retained. The exact coefficient maps are obtained by expanding each original $S^j = (1+Z)^j$:

$$(1+Z)^j = \sum_{2\ell \leq j} \binom{j}{2\ell} (-x)^\ell + Z \sum_{2\ell+1 \leq j} \binom{j}{2\ell+1} (-x)^\ell. \quad (\text{SSP.30})$$

The inverse replaces x by $-(S-1)^2$, Z by $S-1$ in $\sum_a \Delta^a(A_a(x) + ZB_a(x))$. This proves both (SSP.29) and preservation of the original weighted total-degree filtration.

24.5 Full-jet kernels for the actual constrained minimum

Fix $M \geq 2(d-1) = 4D-2$. Let $\mathcal{I}_e = \{(a, j) \in \mathbb{N}^2 : 2a+2j \leq M\}$, and $\mathcal{I}_o = \{(a, j) \in \mathbb{N}^2 : 2a+2j+1 \leq M\}$. The positive source Gram matrices and full residue columns are

$$\begin{aligned} (M_e)_{(a,j),(c,\ell)} &= \int_0^\infty x^{j+\ell} d\Lambda_{ac}(x), & F_{a,j} &= u_e[\Delta^a x^j]_{B_e}, \\ (M_o)_{(a,j),(c,\ell)} &= \int_0^\infty x^{j+\ell+1} d\Lambda_{ac}(x), & E_{a,j} &= \beta(u_e)[\Delta^a x^j]_{B_o}. \end{aligned} \quad (\text{SSP.31})$$

Use the exact bases (SSP.20) for these columns and let J_e, J_o be the corresponding column matrices. Each Gram is strictly positive: a nonzero coefficient vector gives a nonzero polynomial vector in (SSP.29), whose squared norm is positive for the strictly positive matrix density. All its moments are finite by (SSP.24)–(SSP.26).

The two jet maps are onto at the stated degree. Indeed the original symmetric orbit basis $s_1^i s_2^j + s_1^j s_2^i$ for $i < j < d$, and the diagonal basis $s_1^i s_2^i$, has degree at most $2(d-1)$. Its full residue map is onto $\mathcal{E}$, and multiplication by the actual unit is invertible. Reflection preserves the degree filtration and this unit, so parity projection proves surjectivity onto each of B_e, B_o separately. Equivalently the concrete basis (SSP.20) has maximum degrees $4D-4, 4D-2$, and $4D-3$ in its three groups. This verifies the endpoint $M = 4D-2$, including $D=1, M=2$.

Define the actual kernels and their inverse metrics by

$$K_e = J_e M_e^{-1} J_e^*, \quad K_o = J_o M_o^{-1} J_o^*, \quad G_e = K_e^{-1} > 0, \quad G_o = K_o^{-1} > 0. \quad (\text{SSP.32})$$

All columns here are full polynomial remainders with the full unit. In particular their matrix sizes are $D(D+1)$ and D^2 , respectively.

Theorem 24.3 (The same original minimum and its unequal relative blocks). *Let R_M^s be the original symmetric canonical representative in the unscaled monomial orbit coordinates, $G_M^s = (R_M^s)^* R_M^s$, and A_M^s its original sum generator. Then*

$$\begin{aligned} T_s^{-*} G_M^s T_s^{-1} &= \text{diag}(G_e, G_o), & \det G_M^s &= \det G_e \det G_o, \\ T_s A_M^s T_s^{-1} &= I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}. \end{aligned} \quad (\text{SSP.33})$$

The original weight and its exact allowance satisfy

$$\begin{aligned} W_M^s &= (A_M^s)^* G_M^s + G_M^s A_M^s - 2G_M^s, \\ T_s^{-*} W_M^s T_s^{-1} &= \begin{pmatrix} 0 & \beta^* G_o - G_e \alpha \\ G_o \beta - \alpha^* G_e & 0 \end{pmatrix}, \end{aligned} \quad (\text{SSP.34})$$

$$\epsilon_M^s = \left\| G_e^{-1/2} (\beta^* G_o - G_e \alpha) G_o^{-1/2} \right\|_{\text{op}}.$$

If the displayed rectangular block has rank p , the relative weight has p positive and p negative eigenvalues, given by its positive singular values and their negatives. Its zero multiplicity is

$$D(D+1) + D^2 - 2p = D + 2(D^2 - p). \quad (\text{SSP.35})$$

Thus the dimension difference, all additional zero directions, and all nonzero multiplicities are retained.

Proof. The actual product source has the permutation-invariant Hilbert norm and full invariant jet map specified before (SSP.25). For a symmetric target, apply swap to its unique full minimum. The image and norm are unchanged, so uniqueness makes that minimum symmetric. Therefore minimizing over precisely the invariant degree- M polynomials gives the original R_M^s , not a new family of positive forms.

The exact coefficient transformation (SSP.30) has source Gram $\text{diag}(M_e, M_o)$; mixed terms vanish by the proved parity isometry. Its full jet map in coordinates (SSP.21) is $\text{diag}(J_e, J_o)$. For a positive source Gram M and surjective map J , the vector

$$M^{-1}J^*(JM^{-1}J^*)^{-1}y$$

has image y and is M -orthogonal to $\ker J$. Every other preimage differs by a kernel vector, and its squared norm is the squared norm of this vector plus the positive squared norm of that difference. Thus its minimum metric is exactly $(JM^{-1}J^*)^{-1}$. Apply this calculation to the two declared blocks to prove the metric congruence in (SSP.33). The coordinate representative is $R_M^s T_s^{-1}$, and its two source coefficient blocks are explicitly $M_e^{-1}J_e^*G_e$ and $M_o^{-1}J_o^*G_o$. Hence both its source function and full jets agree with the original minimum. Equation (SSP.22) proves the determinant equality. Equation (SSP.12) proves the generator similarity.

Multiplying the two generator blocks with the metric in (SSP.33) cancels the two identity terms against $2G_M^s$. This gives exactly the middle line of (SSP.34). The relative congruence by $\text{diag}(G_e^{-1/2}, G_o^{-1/2})$ has the form $\begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$ with the rectangular block displayed there. For each positive singular value σ and corresponding unit singular vectors u, v , its vectors (u, v) and $(u, -v)$ have eigenvalues $\sigma, -\sigma$. The remaining kernel is $\ker B^* \oplus \ker B$, with dimensions $D(D+1) - p$ and $D^2 - p$. This proves its norm, all signs, and (SSP.35). The coordinate map

$$\text{diag}(G_e^{1/2}, G_o^{1/2})T_s(G_M^s)^{-1/2}$$

is unitary by (SSP.33) and intertwines the two relative Hermitian matrices. Thus these are the complete relative eigenvalues of the original arithmetic weight. No inverse of α, β, B , or either frontier rank has been used. $\square$

In particular zero allowance is exactly the finite equation $\beta^*G_o = G_e\alpha$, with both rectangular maps retained. It does not follow from positivity of the two source Gram matrices. The map from those positive matrices to this equation is the fully proved difference block (SSP.34). Every original zero ρ , of any order m_ρ , has an eigenvector represented in its local factor by $(s - \rho)^{m_\rho - 1}$. Its square tensor is symmetric, nonzero, and has sum eigenvalue 2ρ . Its original Rayleigh quotient is therefore

$$\frac{u^*W_M^s u}{u^*G_M^s u} = 4 \operatorname{Re} \rho - 2, \quad 4 \left| \operatorname{Re} \rho - \frac{1}{2} \right| \leq \epsilon_M^s. \quad (\text{SSP.36})$$

This retains the exact tensor factor and every original local direction in the full metric; it makes no assumption about the limiting size of that metric difference.

24.6 Original primitives and supported maps

For any invariant relation written $H_0A + \Delta H_1B$, the original identity is

$$H_0A + \Delta H_1B = h(s_1) \frac{A + rB}{2} + h(s_2) \frac{A - rB}{2}. \quad (\text{SSP.37})$$

The two original cochain primitives apply the first coefficient to $\phi_* \otimes F_h$ and the negative of the second coefficient to $F_h \otimes \phi_*$. The second tensor differential contributes its own minus sign, so their total boundary is precisely (SSP.37). For an even relation $A_0a + xQ_\Delta b$, take $A = a$, $B = -Zb$; for an odd relation $Z(A_0a + Q_\Delta b)$, take $A = Za$, $B = b$. Substitution of $Z^2 = -x$ proves both assertions with their full signs. Thus the two coefficient ideals in (SSP.6) have the original theta primitives, including the factor xQ_Δ in the even ideal. At a finite admitted degree the polynomial identity is the same original relation. For completeness, divide a relation P of total degree at most M first by the monic $h(s_1)$, then divide its remainder by the monic $h(s_2)$. The final remainder is zero because P lies in the original ideal. This gives $P = h(s_1)C_1 + h(s_2)C_2$, with $\deg C_1, \deg C_2 \leq M - d$. Average under swap and put $B_1 = (C_1 + \text{swap}(C_2))/2$, $B_2 = \text{swap}(B_1)$. Then $A = B_1 + B_2$ is even in r , and $B = (B_1 - B_2)/r$ is a polynomial even in r . Their degrees satisfy $\deg A \leq M - d$ and $\deg B \leq M - d - 1$, and their coefficients in (SSP.37) are exactly B_1, B_2 . This proves the original finite-degree primitive bounds, including a zero coefficient when its degree bound is negative.

Every algebra isomorphism in (SSP.5), (SSP.9), and (SSP.14) has the exact G -lift: it fixes external τ and sends a supported amplitude to its supported algebra image. The algebra-homomorphism laws and the absorbing rule for τ prove the semiring laws; the inverse is the lift of the displayed inverse. The algebra maps β, ρ_0 have the same functorial lift. The module maps α, ι_0 and all Hilbert, jet, kernel and representative maps above have their fixed-label linear lifts $(\lambda, v) \mapsto (\lambda, Tv)$, fixing external absence. Their exact module actions are the ones proved in (SSP.7), (SSP.11), and (SSP.18). A class killed in either exact sequence goes to the zero in its receiving supported fibre; the represented kernel and cokernel A_H retain every local order and the unit V^2 . The direct sum in (SSP.6) is equipped with the multiplication (SSP.7); its parity components have not been assigned independent external supports by this map.

25 The sum connection on both half-line branches and its exact domain

This section integrates the complete sum-connection delivery, whose source note is retained in the appendices at its declared PR 20 revision 2144c358c2b41aa235b15cf0fa472289aec8673f. We prove its mass-dependent derivative identity in the original source and then carry the full relative connection into the quadratic coordinate. Both branches, all relative coefficients, and the condition where the two branches meet are retained. Only the stated dagger stability is needed until evenness is explicitly imposed.

25.1 The actual amplitude, its mass, and the original relative fibre

Keep $g = 2\xi$, $v_h = g/h$, the entire quotient at the full selected zero orders, and the original $F_h \in \mathcal{B}$ with Mellin transform v_h . Let the packet be stable under $\rho \mapsto 1 - \bar{\rho}$. The empty polynomial $h = 1$ is also allowed for this analytic calculation; no nonzero finite jet quotient is inferred in that case. Set

$$a_h^{\text{SC}}(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h^{\text{SC}}(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt = \|F_h\|_{L^2(dx)}^2. \quad (\text{FC.1})$$

The superscript distinguishes the delivered spectral amplitude from the gamma-reference multiplier in (TG.4). Their exact relation is $a_h^{\text{SC}}(t) = \Gamma(1/4 + it/2) a_h^{\text{TG}}(t) / \sqrt{2\pi}$. Taking its squared modulus gives precisely the two original measure formulas; neither mass nor phase is replaced.

For $k \geq 2$ use

$$u = \sum_i t_i, \quad y_i = t_i - u/k \ (i < k), \quad y_k = - \sum_{i < k} y_i, \quad t_i = u/k + y_i, \quad \Psi(u, \mathbf{y}) = \prod_i a_h^{\text{SC}}(t_i). \quad (\text{FC.2})$$

The Jacobian has absolute value one. One way to check its constant is to use coordinates $(u, t_1, \dots, t_{k-1})$, whose inverse sets $t_k = u - \sum_{i < k} t_i$ and has determinant of absolute value one. The replacement $t_i = y_i + u/k$, $i < k$, is triangular with diagonal one. For $k = 2$, $v = t_1 - t_2 = 2y_1$, so the earlier measure $du dv/2$ is unchanged.

The test-space estimates imply that $e^{r/2} F_h(e^r)$ and all its derivatives and polynomial multiples are Schwartz on $\mathbb{R}$. Indeed arbitrary powers of e^r bound both tails, and the logarithmic derivatives are the stated D -derivatives of F_h . Fourier differentiation and integration by parts therefore show that a_h^{SC} and all its derivatives are Schwartz. These facts justify the product integrals and differentiated integrals below.

Dagger stability gives $\overline{a_h^{\text{SC}}} = (-1)^d a_h^{\text{SC}}$. Thus this amplitude and its derivative have the same constant real or imaginary phase. Off the discrete zero set, $w'_h = 2\overline{a_h^{\text{SC}}}(a_h^{\text{SC}})'$, with the product real, and consequently

$$I_h := \int \frac{|w'_h|^2}{w_h} = 4 \int |(a_h^{\text{SC}})'|^2 = 4 \int_0^\infty (\log x)^2 |F_h(x)|^2 dx, \quad \int \overline{a_h^{\text{SC}}}(a_h^{\text{SC}})' = 0. \quad (\text{FC.3})$$

Values assigned to the quotient on its discrete zero set do not change these integrals. The second equality is the original Mellin Plancherel identity; the last integral is $\frac{1}{2} \int w'_h = 0$ by Schwartz decay. In particular every number in (FC.3) is finite.

Put $m(u) = \|\Psi(u, \cdot)\|^2 = w_h^{*k}(u) > 0$. Positivity holds at each u : the factors have only discrete zeros, whose inverse images in a relative fibre are a countable union of null hyperplanes. The integral of m is exactly μ_h^k . Let $\Pi_{0,u}$ be orthogonal projection onto the line spanned by $\Psi(u, \cdot)$, and let $n(u) = (1 - \Pi_{0,u})\partial_u \Psi$. Constant phase gives $\langle \Psi, \partial_u \Psi \rangle = m'/2$, so

$$n = \partial_u \Psi - \frac{m'}{2m} \Psi, \quad \boxed{\int \frac{|m'|^2}{m} + 4 \int \|n\|^2 = \frac{\mu_h^{k-1}}{k} I_h}. \quad (\text{FC.4})$$

To prove the equality, differentiate the product in (FC.2). Its derivative is

$$k^{-1} \sum_i (a_h^{\text{SC}})'(t_i) \prod_{j \neq i} a_h^{\text{SC}}(t_j).$$

Before the common factor k^{-2} , each of the k diagonal terms in its squared norm has integral $\mu_h^{k-1} \|(a_h^{\text{SC}})'\|^2$, and every distinct-index term contains $|\int \overline{a_h^{\text{SC}}}(a_h^{\text{SC}})'|^2 = 0$. Thus its integral is $\mu_h^{k-1} I_h/(4k)$. The orthogonal line and complementary components have squared norms $|m'|^2/(4m)$ and $\|n\|^2$; integration proves (FC.4).

25.2 The complete matrix connection and its actual graph

Choose a fixed finite independent homogeneous relative polynomial family θ_α , of degrees ℓ_α , in the $k-1$ independent variables $z_i = s_i - S/k$. At source degree M , retain exactly

$$P(\mathbf{s}) = \sum_\alpha \theta_\alpha(\mathbf{z}) f_\alpha(S), \quad \deg f_\alpha \leq M - \ell_\alpha, \quad S = k/2 + iu, \quad c_P(u) = (f_\alpha(k/2 + iu))_\alpha. \quad (\text{FC.5})$$

The family can be the whole admitted relative monomial family or the specified invariant family. Its cardinality r and all degree bounds remain part of the source; it is not held fixed while M changes. In $\mathcal{H} = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$, define

$$\begin{aligned} j_u c &= \Psi(u, \mathbf{y}) \sum_\alpha \theta_\alpha(i\mathbf{y}) c_\alpha, & W &= j_u^* j_u > 0, & B &= j_u^* j'_u, & T &= (j'_u)^* j'_u, \\ \Pi &= j_u W^{-1} j_u^*, & \Gamma &= W^{-1} B, & N &= (1 - \Pi) j'_u, & \mathcal{N} &= N^* N = T - B^* W^{-1} B. \end{aligned} \quad (\text{FC.6})$$

Positivity follows from independence of the relative polynomials and the almost-everywhere non-vanishing just used. Schwartz estimates on compact u -intervals prove smoothness of these finite matrices and of j_u in operator norm. Direct differentiation gives

$$\begin{aligned} \partial_u(j_u c) &= j_u(c' + \Gamma c) + Nc, & j_u^* N &= 0, \\ \Gamma^* W + W \Gamma &= W', & \mathcal{N} &\succeq 0, \\ \|\partial_u(j_u c)\|^2 &= (c' + \Gamma c)^* W (c' + \Gamma c) + c^* \mathcal{N} c. \end{aligned} \quad (\text{FC.7})$$

For example $j_u \Gamma = \Pi j'_u$; adding the complementary term proves the first identity. Differentiating $W = j_u^* j_u$ proves metric compatibility, and the projection identity proves the normal Gram and positivity. In this original fixed frame the common phase makes $B = B^*$, hence $B = W'/2$. After a variable change of frame $j^C = jC(u)$, one instead keeps the full formulas

$$\begin{aligned} W^C &= C^* W C, & B^C &= C^* B C + C^* W C', \\ \Gamma^C &= C^{-1} \Gamma C + C^{-1} C', & N^C &= N C, & \mathcal{N}^C &= C^* \mathcal{N} C. \end{aligned} \quad (\text{FC.8})$$

They follow by the product rule and the unchanged projection image. For instance $C = e^{iu} I$ gives $B^C = B + iW$, so the fixed-frame identity $B = W'/2$ is not silently imposed on a changed frame.

The actual derivative graph is the image of $P \mapsto (j_u(c'_P + \Gamma c_P), N c_P)$. Addition is an isometric injection on that graph by (FC.7); its image is precisely $\partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P$. Here $\mathcal{U}_k$ is the original Mellin-Fourier isometry followed by (FC.2). On the test space it obeys

$$\mathcal{U}_k D^{(k)} = (k/2 + iu) \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -I. \quad (\text{FC.9})$$

The first two formulas follow by integration by parts and differentiation of the Mellin integral. Applying $-x_i \partial_{x_i}$ to $\log x_i$ gives the last formula, including $1/k$. Therefore addition of both graph components, followed by $-i \mathcal{U}_k^{-1}$ and the original full jet, recovers $J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P$. The bare derivative lands in the image of graph addition; to land in the pair graph itself one uses the actual map $v \mapsto (\Pi_u \partial_u v, (1 - \Pi_u) \partial_u v)$.

25.3 A branch-preserving quadratic map for the entire matrix weight

Put $x = u^2 > 0$, $v = \sqrt{x}$, and retain the ordered branches $u = v$ and $u = -v$. These letters denote the spectral sum coordinate, not the original positive Mellin variables in (FC.9). For any coefficient function define

$$c(v) = a(x) + ivb(x), \quad c(-v) = a(x) - ivb(x), \quad \mathbf{M}(x) = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}. \quad (\text{FC.10})$$

Its inverse is $a = (c(v) + c(-v))/2$, $b = (c(v) - c(-v))/(2iv)$. No relation between $W(v)$ and $W(-v)$ is assumed. Write $W_{\pm} = W(\pm v)$. The full positive paired density is

$$\begin{aligned} \mathbf{H}(x) &= \mathbf{M}^* \frac{\text{diag}(W_+, W_-)}{2v} \mathbf{M} \\ &= \frac{1}{2\sqrt{x}} \begin{pmatrix} W_+ + W_- & i\sqrt{x}(W_+ - W_-) \\ -i\sqrt{x}(W_+ - W_-) & x(W_+ + W_-) \end{pmatrix} > 0. \end{aligned} \quad (\text{FC.11})$$

The inverse and the change of variable on each half-line prove the unitary equality

$$\int_{\mathbb{R}} c(u)^* W(u) c(u) du = \int_0^\infty \begin{pmatrix} a \\ b \end{pmatrix}^* \mathbf{H}(x) \begin{pmatrix} a \\ b \end{pmatrix} dx. \quad (\text{FC.12})$$

Here $\begin{pmatrix} a \\ b \end{pmatrix}^*$ means the adjoint of the two-block column. Positivity follows from the invertibility of M at every $x > 0$; surjectivity on these L^2 spaces follows from its pointwise inverse and the same norm identity. Both branch weights and their off-diagonal difference remain in (FC.11).

For the precise polynomial image in (FC.5), write $f_\alpha(S) = \sum_n f_{\alpha,n} S^n$. Then

$$\begin{aligned} a_\alpha(x) &= \sum_n f_{\alpha,n} \sum_{2j \leq n} \binom{n}{2j} (k/2)^{n-2j} (-x)^j, \\ b_\alpha(x) &= \sum_n f_{\alpha,n} \sum_{2j+1 \leq n} \binom{n}{2j+1} (k/2)^{n-2j-1} (-x)^j. \end{aligned} \quad (\text{FC.13})$$

Their degree bounds are respectively $\lfloor (M - \ell_\alpha)/2 \rfloor$ and $\lfloor (M - \ell_\alpha - 1)/2 \rfloor$, with a negative bound meaning a zero component. The inverse is exactly $f_\alpha(S) = a_\alpha(-(S - k/2)^2) + (S - k/2)b_\alpha(-(S - k/2)^2)$. Expanding the original powers proves both formulas and their inverse, including every $k/2$ coefficient. Thus (FC.12) retains the full filtered source.

Theorem 25.1 (The complete paired connection and normal energy). *Define*

$$\begin{aligned} \Gamma_b(x) &= \text{diag} \left(\frac{\Gamma(v)}{2v}, -\frac{\Gamma(-v)}{2v} \right), \\ A(x) &= M^{-1} \Gamma_b M + M^{-1} M' = M^{-1} \Gamma_b M + \text{diag}(0, (2x)^{-1} I), \\ Q(x) &= M^* \frac{\text{diag}(\mathcal{N}(v), \mathcal{N}(-v))}{2v} M \succeq 0, \quad \nabla_x = \partial_x + A. \end{aligned} \quad (\text{FC.14})$$

For $q = (a, b)^T$, the exact differentiated norm is

$$\int_{\mathbb{R}} \|\partial_u(j_u c(u))\|^2 du = 4 \int_0^\infty x (\nabla_x q)^* H \nabla_x q dx + \int_0^\infty q^* Q q dx. \quad (\text{FC.15})$$

The transformed coefficient derivative and its metric identities are

$$\begin{aligned} J(x) \nabla_x, \quad J &= \begin{pmatrix} 0 & 2ixI \\ -2iI & 0 \end{pmatrix}, \quad J^2 = 4xI, \quad J^* H J = 4xH, \\ A^* H + H A &= H' + \frac{1}{2x} H. \end{aligned} \quad (\text{FC.16})$$

Consequently $A - (4x)^{-1} I$ is the connection matrix compatible with the full density H . The explicit Jacobian term in (FC.16) remains in the transported original derivative.

Proof. Let $F = (c(v), c(-v))^T = Mq$ and $K = \text{diag}(2vI, -2vI)$. The two original covariant derivative values are exactly $K(F' + \Gamma_b F) = KM \nabla_x q$. The minus sign on the second branch is from $d(-v)/dx = -1/(2v)$. Multiplying the explicit inverse of M gives $M^{-1}KM = J$ and $M^{-1}M' = \text{diag}(0, 1/(2x)I)$. The block-scalar matrix K commutes with the two-block weight and its squared modulus is $4xI$. This proves the first energy term and the first line of (FC.16). Applying (FC.12) to the original normal Gram proves the second energy term; (FC.7) then proves (FC.15).

For metric compatibility set $D = \text{diag}(W_+, W_-)/(2v)$. The original equality $\Gamma^* W + W \Gamma = W'$, with the two opposite derivatives of v , gives $\Gamma_b^* D + D \Gamma_b = D' + D/(2x)$. The additional term is the derivative of the factor $1/(2v)$. Substitution of $H = M^* D M$ and the two terms defining A proves the last line of (FC.16). Subtracting $1/(4x)I$ from the matrix and its adjoint removes exactly that extra term, proving the last assertion. $\square$

Multiplication by the original spectral sum has the exact paired matrix

$$S = k/2 I + \begin{pmatrix} 0 & -xI \\ I & 0 \end{pmatrix}. \quad (\text{FC.17})$$

This follows by multiplying $a + iub$ by $k/2 + iu$. On smooth compactly supported sections in $x > 0$, the commutator is $[J\nabla_x, S] = iI$. To prove it with all variable-frame terms present, conjugate to the two branch coordinates used in the proof: multiplication is $\text{diag}(k/2 + iv, k/2 - iv)$, the connection part commutes with this block-scalar matrix, and multiplication of its derivative by K gives iI on both blocks. Conjugating back proves the claim. Through (FC.9) this is the same original pair $D^{(k)}, \mathcal{L}_k$, with its original sign.

25.4 The meeting point of the two branches is part of the domain

For the L^2 identity (FC.12) no value at zero is required. For the derivative domain it is essential. Define the original domain to consist of coefficient functions c that are locally absolutely continuous on $\mathbb{R}$, have finite norm in (FC.12), and have finite derivative energy in (FC.7). Equivalently these are the coefficient functions for which $j_u c(u)$ has an L^2 weak derivative: on each compact interval the smooth left inverse $W^{-1}j_u^*$ and strict positivity of W prove both directions by the product rule. Its exact image under (FC.10) consists of locally absolutely continuous q on $(0, \infty)$ with the two finite integrals in (FC.12), (FC.15), and the joining condition

$$\lim_{x \downarrow 0} (a(x) + i\sqrt{x} b(x)) = \lim_{x \downarrow 0} (a(x) - i\sqrt{x} b(x)). \quad (\text{FC.18})$$

These finite one-sided limits exist under the stated integrability conditions. Indeed W is bounded above and below by positive constants near zero and Γ is bounded there. The two finite integrals imply that each branch coefficient and its ordinary u -derivative are square-integrable near zero. The fundamental theorem for locally absolutely continuous functions and Cauchy–Schwarz then give their finite endpoint limits. Their union has a weak derivative in L^2 precisely when those limits agree; integration by parts against a compactly supported test function shows that a mismatch contributes their nonzero jump times a delta distribution. This proves (FC.18) and both directions of the domain correspondence. For the original polynomial image (FC.13), the condition holds automatically. In general, finite energy on the two open branches alone does not impose it: a function constant with different values on the two sides near zero has zero ordinary derivative there, while its joined distributional derivative has that nonzero jump. The matching condition preserves the original domain without discarding either branch.

25.5 The even specialization and the unchanged scalar contraction

If the original packet also has conjugation closure, w_h is even. For the symmetric square with relative frame $1, \Delta, \Delta^2, \dots$, the actual j_u is even in u , since its two amplitude factors are interchanged by $u \mapsto -u$ and amplitude evenness. Thus W and $\mathcal{N}$ are even and Γ is odd. Set $L(x) = W(\sqrt{x})/\sqrt{x}$, $L_N(x) = \mathcal{N}(\sqrt{x})/\sqrt{x}$, and $\Gamma_x = \Gamma(\sqrt{x})/(2\sqrt{x})$. Equations (FC.14)–(FC.16) reduce, by direct block substitution, to

$$\begin{aligned} H &= \text{diag}(L, xL), & Q &= \text{diag}(L_N, xL_N), & A &= \text{diag}(\Gamma_x, \Gamma_x + (2x)^{-1}I), \\ \int \|\partial_u(j_u c)\|^2 du &= \int_0^\infty [4x(a' + \Gamma_x a)^* L(a' + \Gamma_x a) + (b + 2x(b' + \Gamma_x b))^* L(b + 2x(b' + \Gamma_x b)) \\ &\quad + a^* L_N a + x b^* L_N b] dx. \end{aligned} \quad (\text{FC.19})$$

These are exactly the two measures and the complete relative family of the preceding symmetric-square section. Its unequal finite quotient components are obtained from this same filtered source, using its already proved full jet and actual minimum.

For any $k \geq 2$ under the same quartet assumption, the scalar sum density m is even. Define the actual half-line density $\rho_k(x) = m(\sqrt{x})/\sqrt{x}$. It retains total mass $\int_0^\infty \rho_k = \mu_h^k$. The source

contraction (FC.4) has the precise radial form

$$4 \int_0^\infty x \frac{|\rho'_k(x) + \rho_k(x)/(2x)|^2}{\rho_k(x)} dx + 4 \int_0^\infty \frac{\|n(\sqrt{x})\|^2}{\sqrt{x}} dx = \frac{\mu_h^{k-1}}{k} I_h. \quad (\text{FC.20})$$

To verify it, $m(u) = u\rho_k(u^2)$ for $u > 0$, so $m'(u) = \rho_k(x) + 2x\rho'_k(x)$. Changing variables in $2 \int_0^\infty m'^2/m du$ gives its first term. Reflection of all relative variables is a unitary map of each fibre and carries $n(u)$ to $-n(-u)$, so its norm is even; changing variables gives the second term. This proves (FC.20) with the stated drift and the entire relative contribution. It is the transported source estimate, with its mass and domain. The complete matrix energy remains (FC.15), on the actual graph and with its normal Gram, rather than being assigned the scalar constant-column bound.

All linear maps on the displayed filtered sources and graph images have the original fixed-support lift: they preserve the support label, send a supported zero to the corresponding supported zero, and send external absence to external absence. Their additive and scalar identities follow from the proved linear maps. In particular the arithmetic observation is taken after graph addition and the specified inverse Mellin isometry; no arithmetic jet is assigned to an arbitrary independent measurable normal section.

25.6 The actual arithmetic normal contribution is strictly positive

We now use the actual zero set of the amplitude in (FC.1). Hardy's theorem that infinitely many zeros of the Riemann zeta function lie on the critical line, with the original source and reading record retained in Section 23, implies that

$$\mathcal{Z}_h = \{T \in \mathbb{R} : a_h^{\text{SC}}(T) = 0\} \text{ is infinite and discrete.} \quad (\text{FC.21})$$

Indeed a finite polynomial h removes only its selected finite set of complete zero orders. Every remaining critical-line zero is still a zero of v_h , and the nonzero entire function v_h has a discrete zero set. This also applies to $h = 1$. At any $T \in \mathcal{Z}_h$, keep its entire order $\ell_T \geq 1$ and its original leading coefficient

$$a_h^{\text{SC}}(T + \delta) = c_T \delta^{\ell_T} + O(\delta^{\ell_T+1}), \quad (a_h^{\text{SC}})'(T + \delta) = \ell_T c_T \delta^{\ell_T-1} + O(\delta^{\ell_T}), \quad c_T = \frac{(a_h^{\text{SC}})^{(\ell_T)}(T)}{\ell_T!} \neq 0. \quad (\text{FC.22})$$

These are expansions of the original amplitude, including its fixed phase and its factor $1/\sqrt{2\pi}$.

Theorem 25.2 (Strict scalar residual and every finite normal direction). *For the actual amplitude and every integer $k \geq 2$,*

$$\int_{\mathbb{R}} \|n(u)\|^2 du > 0, \quad \int_{\mathbb{R}} \frac{|m'(u)|^2}{m(u)} du < \frac{\mu_h^{k-1}}{k} I_h. \quad (\text{FC.23})$$

For $k \geq 3$, $\|n(u)\| > 0$ for every real u . For any nonempty finite independent relative polynomial frame of (FC.5), its full normal Gram satisfies

$$\begin{aligned} \mathcal{N}(u) &> 0 \quad \text{for every } u \in \mathbb{R}, & k &\geq 3, \\ \mathcal{N}(u) &> 0 \quad \text{for every } u \notin \mathcal{Z}_h + \mathcal{Z}_h, & k &= 2. \end{aligned} \quad (\text{FC.24})$$

In particular $\mathcal{N}(u) > 0$ almost everywhere for every $k \geq 2$, since $\mathcal{Z}_h + \mathcal{Z}_h$ is countable. No lower bound uniform in u , the relative degree, or the packet is asserted by these strict inequalities.

Proof. First fix $T \in \mathcal{Z}_h$ and choose $b_2, \dots, b_k$ outside $\mathcal{Z}_h$. Put $u_0 = T + \sum_{i=2}^k b_i$ and $A_b = \prod_{i=2}^k a_h^{\text{SC}}(b_i) \neq 0$. Within that fixed original sum fibre consider the transverse path

$$t_1 = T + \delta, \quad t_i = b_i \quad (2 \leq i < k), \quad t_k = b_k - \delta.$$

It changes the relative variables while retaining $u = u_0$. The derivative ∂_u is still the original derivative at fixed relative variables in (FC.2), and its product formula gives

$$\begin{aligned}\Psi(u_0, \mathbf{y}(\delta)) &= c_T A_b \delta^{\ell_T} + O(\delta^{\ell_T+1}), \\ \partial_u \Psi(u_0, \mathbf{y}(\delta)) &= \frac{\ell_T}{k} c_T A_b \delta^{\ell_T-1} + O(\delta^{\ell_T}).\end{aligned}\tag{FC.25}$$

Only the derivative of the first amplitude has order $\ell_T - 1$; every other product term retains the full factor $a_h^{\text{SC}}(T + \delta)$ and has order at least ℓ_T . Since $m(u_0) > 0$ and m is smooth, subtracting $m'(u_0)\Psi/(2m(u_0))$ does not change that nonzero leading coefficient. Thus $n(u_0, \cdot)$ is nonzero at nearby points of this path and, by continuity, on a nonempty open subset of the entire relative fibre. Its squared fibre norm is strictly positive. Differentiated Schwartz estimates show that Ψ and $\partial_u \Psi$ are continuous as functions of u with values in $\mathcal{H}$. Smoothness and positivity of m therefore make n continuous in that Hilbert norm. Its squared norm stays positive on some interval around u_0 , proving positive integrated residual. Equation (FC.4) proves (FC.23).

For $k \geq 3$, the same construction works at every prescribed u_0 . Choose $b_3, \dots, b_{k-1}$ outside the zero set if that list is nonempty. Choose b_2 outside both $\mathcal{Z}_h$ and the translate-reflection of that set which would make $b_k = u_0 - T - \sum_{i=2}^{k-1} b_i$ a zero. These two discrete sets cannot exhaust the real line. This proves the pointwise scalar statement.

To prove the full matrix assertion, fix u and suppose $N_u c = 0$. The exact projection formula gives $j'_u c = j_u d$, where $d = \Gamma_u c$ is a finite coefficient vector. Write $\vartheta_c(\mathbf{y}) = \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}$, and similarly ϑ_d . Equality in $\mathcal{H}$ says

$$(\partial_u \Psi) \vartheta_c = \Psi \vartheta_d.$$

Both sides have smooth representatives in the relative variables, so their almost-everywhere equality is equality everywhere: a nonzero continuous difference at a point would remain nonzero on a set of positive measure.

Suppose $k \geq 3$. For each $T \in \mathcal{Z}_h$, restrict to the affine hyperplane $t_1 = u/k + y_1 = T$, of dimension $k - 2 \geq 1$. Each of the other original coordinates is a nonconstant affine function on this hyperplane. The points where one of their amplitude factors vanishes are consequently a countable union of proper affine hyperplanes of measure zero. Their complement is dense and is open near each of its points. At any such point p , apply (FC.25). In the asserted smooth identity the left side has leading coefficient $\ell_T c_T A_b \vartheta_c(p)/k$ in degree $\ell_T - 1$, whereas the right side has order at least ℓ_T . Hence $\vartheta_c(p) = 0$. Continuity proves its vanishing on the entire real hyperplane. Division in y_1 then shows that the polynomial ϑ_c has factor $y_1 - (T - u/k)$: the remainder polynomial in the other variables vanishes on all their real values, and therefore is identically zero. There are infinitely many distinct such T , while the degree in y_1 is finite. Thus $\vartheta_c = 0$, and independence of the original relative polynomials gives $c = 0$. This proves injectivity of N_u and therefore $\mathcal{N}(u) > 0$ on the whole finite coefficient space.

For $k = 2$, take $u \notin \mathcal{Z}_h + \mathcal{Z}_h$. At every $T \in \mathcal{Z}_h$, the other amplitude $a_h^{\text{SC}}(u - T)$ is nonzero. The same expansion forces the univariate polynomial ϑ_c to vanish at $y_1 = T - u/2$. These are infinitely many distinct points, so again $\vartheta_c = 0$ and $c = 0$. The discrete subset $\mathcal{Z}_h$ of $\mathbb{R}$ is countable, hence so is its set of pairwise sums. This proves every assertion in (FC.24). $\square$

The consequences hold for the actual full derivative graph and both quadratic branches. For every nonzero admitted source polynomial P ,

$$0 < \int_{\mathbb{R}} c_P(u)^* \mathcal{N}(u) c_P(u) du < \infty,\tag{FC.26}$$

$$\mathcal{Q}(x) > 0 \quad \text{for all } x > 0 \text{ when } k \geq 3, \quad \mathcal{Q}(x) > 0 \quad \text{for almost every } x > 0 \text{ when } k = 2.$$

Indeed a nonzero vector polynomial c_P is nonzero outside a finite set on the real line, and (FC.24) supplies strict positivity almost everywhere. The integral is finite because $\|N_u c_P\| \leq \|j'_u c_P\|$, and

the latter is a polynomial multiple of differentiated Schwartz columns. The assertion about Q follows from its invertible congruence (FC.14). For $k = 2$, the exceptional positive x for which either $\sqrt{x}$ or $-\sqrt{x}$ belongs to $\mathcal{Z}_h + \mathcal{Z}_h$ form a countable set. Under every declared smooth invertible frame, $N_u^C c_P^C(u) = N_u c_P(u)$ and $\mathcal{N}^C = C^* \mathcal{N} C$. Thus the full normal energy and strictness are preserved by the exact coefficient isomorphism $c_P^C = C^{-1} c_P$, including its original filtered image.

The excluded two-factor fibre can actually vanish. For the quartet packet, the original amplitude is even, so

$$\partial_u \Psi(0, y) = \frac{1}{2} ((a_h^{\text{SC}})'(y) a_h^{\text{SC}}(-y) + a_h^{\text{SC}}(y) (a_h^{\text{SC}})'(-y)) = 0. \quad (\text{FC.27})$$

Since the relative frame is independent of u , this gives $j'_0 = 0$, $N_0 = 0$, and $\mathcal{N}(0) = 0$. Also $m'(0) = 0$, hence $n(0) = 0$ exactly. The positive total residual and the almost-everywhere full matrix assertion therefore retain this original zero fibre. They prove that every nonzero actual finite polynomial direction has a positive normal contribution while leaving its complete size in (FC.15) and (FC.26).

25.7 Exact reconstruction from the actual finite normal image

Fix $k \geq 2$, the same actual packet, and one of the finite relative families in (FC.5). Retain its original degree bound M and the literal polynomial basis

$$\begin{aligned} \mathcal{I}_M &= \{(\alpha, n) : 0 \leq n \leq M - \ell_\alpha\}, \quad q_M = |\mathcal{I}_M|, \quad P_{\alpha, n}(\mathbf{s}) = \theta_\alpha(\mathbf{z}) S^n, \\ P_p &= \sum_{(\alpha, n) \in \mathcal{I}_M} p_{\alpha, n} P_{\alpha, n}, \quad c_{P_p}(u) = C_M(u) p, \quad (C_M(u))_{\beta, (\alpha, n)} = \delta_{\alpha\beta} (k/2 + iu)^n. \end{aligned} \quad (\text{FC.28})$$

Thus C_M is a specified rectangular polynomial matrix, not a change of relative frame. Each of its columns retains the original power of S . Its derivative is $(C'_M)_{\beta, (\alpha, n)} = \delta_{\alpha\beta} i n (k/2 + iu)^{n-1}$ for $n \geq 1$, and zero for $n = 0$. Independence of the relative family over $\mathbb{C}$, followed by comparison of the powers of S , proves uniqueness of these coefficients. The source is the exact space $\mathcal{P}_M^\theta = \text{span}\{P_{\alpha, n}\}$; it is the full original admitted source when that family is the complete admitted relative basis, and the declared summand otherwise. If $\mathcal{I}_M$ is empty, all the finite source and image spaces below are zero and their maps are the unique maps of zero spaces. We write the nonempty case explicitly.

Let $\mathcal{K} = L^2(\mathbb{R}_u; \mathcal{H})$, with the original measures $du d\mathbf{y}$. Define the normal and tangential maps by their actual amplitudes,

$$\begin{aligned} \mathbf{N}_M &: \mathbb{C}^{q_M} \longrightarrow \mathcal{K}, & (\mathbf{N}_M p)(u) &= N_u C_M(u) p, \\ \mathbf{T}_M &: \mathbb{C}^{q_M} \longrightarrow \mathcal{K}, & (\mathbf{T}_M p)(u) &= j_u (C'_M(u) + \Gamma_u C_M(u)) p, \\ H_M^N &= \mathbf{N}_M^* \mathbf{N}_M = \int_{\mathbb{R}} C_M(u)^* \mathcal{N}(u) C_M(u) du > 0, & \mathcal{R}_M^N &= \text{ran } \mathbf{N}_M. \end{aligned} \quad (\text{FC.29})$$

The coefficient adjoints in this subsection use the displayed literal basis and its ordinary Euclidean coordinate inner product. The computed metric pulled back from the normal image is H_M^N . The original source-amplitude metric and their exact comparison operator remain

$$\begin{aligned} H_M^0 &= \int_{\mathbb{R}} C_M(u)^* W(u) C_M(u) du > 0, & K_M^{N:0} &= (H_M^0)^{-1} H_M^N, \\ (K_M^{N:0})^* H_M^0 &= H_M^0 K_M^{N:0} = H_M^N, & p^* H_M^N q &= p^* H_M^0 K_M^{N:0} q. \end{aligned} \quad (\text{FC.29a})$$

Indeed $W > 0$ and a nonzero vector polynomial $C_M p$ give $H_M^0 > 0$, with finite entries by the original moment bounds. Direct multiplication proves the remaining identities, so $K_M^{N:0}$ is self-adjoint and strictly positive in the H_M^0 inner product. This is the exact operator between the two retained forms; no identity of their norms is assumed. All columns of $\mathbf{N}_M$ and $\mathbf{T}_M$ belong to $\mathcal{K}$. For the first map, projection contraction gives $\|N_u C_M p\| \leq \|j'_u C_M p\|$, whose square is integrable by the polynomially weighted Schwartz estimates. For the second, it is the projection by Π_u of $\partial_u(j_u C_M p)$, and the same

estimates apply. Equation (FC.26) proves strict positivity of H_M^N for every nonzero coefficient vector. In particular the normal map is injective on the whole original finite source, including nonzero polynomials whose original arithmetic class is a supported zero.

Theorem 25.3 (The actual normal image recovers the source and both graph parts). *The inverse on $\mathcal{R}_M^N$, and its precise ambient formula, are*

$$\begin{aligned} L_M^N &= (H_M^N)^{-1} \mathbf{N}_M^* : \mathcal{K} \rightarrow \mathbb{C}^{q_M}, & \mathbf{N}_M^* r &= \int_{\mathbb{R}} C_M(u)^* N_u^* r(u) du, \\ L_M^N \mathbf{N}_M &= I_{q_M}, & \Pi_M^N &= \mathbf{N}_M L_M^N \text{ is the orthogonal projection onto } \mathcal{R}_M^N. \end{aligned} \quad (\text{FC.30})$$

The full original derivative graph is $\mathcal{G}_M = \text{ran}(\mathbf{T}_M, \mathbf{N}_M)$. It is recovered from its normal image by the exact linear bijection

$$\begin{aligned} \text{Rec}_M : \mathcal{R}_M^N &\longrightarrow \mathcal{G}_M, & r &\longmapsto (\mathbf{T}_M L_M^N r, r), \\ \text{Rec}_M^{-1}(t, r) &= r, & p &= L_M^N r, & P &= P_{L_M^N r}. \end{aligned} \quad (\text{FC.31})$$

Its complete metric identity retains both components:

$$\begin{aligned} H_M^\partial &= \mathbf{T}_M^* \mathbf{T}_M + H_M^N, & \|(\mathbf{T}_M p, \mathbf{N}_M p)\|^2 &= p^* H_M^\partial p, \\ \|\text{Rec}_M r\|^2 &= \|r\|^2 + \|\mathbf{T}_M L_M^N r\|^2 = (L_M^N r)^* H_M^\partial (L_M^N r). \end{aligned} \quad (\text{FC.32})$$

In particular reconstruction is not asserted to preserve the normal norm alone; (FC.32) specifies its full metric.

Proof. Every entry of the integral in (FC.30) converges by Cauchy–Schwarz, since it pairs r with one of the finitely many L^2 columns of $\mathbf{N}_M$. Pairing against an arbitrary coefficient vector proves that this integral is its Hilbert adjoint. Therefore $L_M^N \mathbf{N}_M = (H_M^N)^{-1} H_M^N = I$. Multiplication shows that Π_M^N is self-adjoint and idempotent, that its image is $\mathcal{R}_M^N$, and that it fixes every vector of that image. This proves the projection and inverse claims. The image is finite dimensional and hence closed; all displayed maps are bounded. If $r = \mathbf{N}_M p$, (FC.30) makes its reconstructed pair exactly $(\mathbf{T}_M p, \mathbf{N}_M p)$. Conversely every pair in $\mathcal{G}_M$ has this form. Injectivity of $\mathbf{N}_M$ gives both uniqueness of p and the inverse in (FC.31). The two graph components are pointwise orthogonal by $j_u^* N_u = 0$; their squared norms give (FC.32). Their addition consequently is the original derivative amplitude and has the same full graph norm, as already proved in (FC.7). $\square$

The full arithmetic derivative observation can now be specified on this actual normal image. Put

$$\begin{aligned} I &= (h(s_1), \dots, h(s_k)), & E &= \mathbb{C}[\mathbf{s}]/I = E_h^{\otimes k}, & U(\mathbf{s}) &= \prod_i v_h(s_i), \\ v^{(k)} &= j_I U \in E^\times, & \partial_S^{\text{rel}} &= \frac{1}{k} \sum_i \partial_{s_i}, & \beta_h &= (v^{(k)})^{-1} j_I (\partial_S^{\text{rel}} U). \end{aligned} \quad (\text{FC.33})$$

Here j_I means the complete polynomial Hermite class in the original product remainder coordinates. It retains every order in each factor. The entire function U is a unit at the finite packet; its inverse and every derivative used in β_h are taken in those full local algebras. Thus this formula uses no global division across a zero of v_h . If $h = 1$, the target E is the zero algebra and all arithmetic observations below are its unique zero map; the preceding analytic reconstruction continues unchanged.

For a nonempty packet define the finite matrix $J_M^{\log} : \mathbb{C}^{q_M} \rightarrow E$ by its original source columns. Mellin differentiation and (FC.9) give their exact formulas

$$\begin{aligned} (J_M^{\log})_{\alpha, n} &= J^{(k)} \mathcal{L}_k T_h^{(k)} P_{\alpha, n} = j_I (\partial_S^{\text{rel}} (U P_{\alpha, n})) \\ &= n v^{(k)} [\theta_\alpha S^{n-1}]_I + \beta_h v^{(k)} [\theta_\alpha S^n]_I. \end{aligned} \quad (\text{FC.34})$$

For $n = 0$ the first term is defined to be zero, and no negative power is used. To verify the last equality, the original coordinate map gives $\partial_S^{\text{rel}} S = 1$ and $\partial_S^{\text{rel}} z_i = 0$. Hence the relative polynomial does not acquire a derivative term, while differentiating S^n gives precisely its original factor n . The other product-rule term is the full β_h just defined. Equivalently $\beta_h = k^{-1} \sum_i j_I(v'_h/v_h)(s_i)$, with the quotient taken only in its actual unit neighbourhoods. Both factors $g = 2\xi$ and every Taylor coefficient remain in these columns.

The exact observation and the complete commuting source path are

$$\begin{aligned} \text{Obs}_M^{\log} : \mathcal{R}_M^N &\rightarrow E, & \text{Obs}_M^{\log}(r) &= \mathbf{J}_M^{\log}(H_M^N)^{-1} \mathbf{N}_M^* r, \\ \text{Obs}_M^{\log}(\mathbf{N}_M p) &= J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P_p, \\ \mathcal{R}_M^N &\xrightarrow{\text{Rec}_M} \mathcal{G}_M \xrightarrow{\text{Add}} i\mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta, \\ i\mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta &\xrightarrow{-i\mathcal{U}_k^{-1}} \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta \xrightarrow{J^{(k)}} E. \end{aligned} \tag{FC.35}$$

Substitute $r = \mathbf{N}_M p$ in (FC.30)–(FC.31). Graph addition gives $\partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P_p = i\mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} P_p$, and the last two arrows give exactly the columns in (FC.34). This proves equality of the two descriptions of the observation. All functions in the penultimate space are actual finite tensor test functions, since logarithmic multiplication preserves $\mathcal{B}$. Consequently this is an arithmetic observation on the specified finite normal image with a proved source inverse. The ambient formula for L_M^N also defines an orthogonal projection, but no arithmetic-jet interpretation of an arbitrary unrelated measurable normal section is required by (FC.35).

The reconstruction is unchanged under a declared smooth invertible relative frame. In that frame its coefficient matrix is $C_M^C = C(u)^{-1} C_M$, and the exact identities in (FC.8) give

$$\begin{aligned} N^C C_M^C &= N C_M, & j^C((C_M^C)' + \Gamma^C C_M^C) &= j(C_M' + \Gamma C_M), \\ (H_M^N)^C &= H_M^N, & (\mathbf{N}_M)^C &= \mathbf{N}_M, & (\mathbf{T}_M)^C &= \mathbf{T}_M. \end{aligned} \tag{FC.36}$$

The derivative of C^{-1} cancels exactly against the $C^{-1}C'$ connection term. Thus the original normal-image inverse, the full reconstructed graph and the arithmetic observation are the same maps on the same source after its explicit coefficient transport. Their fixed-support linear lifts retain the original label and all supported zeros. In particular a nonzero polynomial relation can have a nonzero normal image whose reconstructed arithmetic derivative is the full conormal response (FC.34); the source relation is retained before either quotient observation.

26 The original conormal derivative and its finite section commutator

The sum-connection source constructs differentiation on the first thickening of the original packet ideal. We prove its algebra map to dual numbers, then compute the exact finite endomorphism obtained from the original monic remainder section and from its full arithmetic unit. The conormal row and the section correction have different domains; their explicit factorization, complete ranks, and residue comparison are retained throughout.

26.1 The first thickening and the dual-number map

Let $k \geq 1$, and retain a nonempty packet of actual zeros with all complete orders,

$$h(s) = \prod_{\rho \in \mathcal{Z}} (s - \rho)^{m_\rho}, \quad d = \deg h \geq 1, \quad r_0 = |\mathcal{Z}|, \quad g = 2\xi, \quad v_h = g/h. \tag{CW.1}$$

Write $h_i = h(s_i)$, and put

$$\begin{aligned} P &= \mathbb{C}[s_1, \dots, s_k], \quad I = (h_1, \dots, h_k), \quad E = P/I, \quad E^{[2]} = P/I^2, \quad N = I/I^2, \\ \pi : E^{[2]} &\rightarrow E, \quad \partial_\Sigma = \frac{1}{k} \sum_{i=1}^k \partial_{s_i}, \quad \delta : E^{[2]} \rightarrow E, \quad [p] \mapsto [\partial_\Sigma p]. \end{aligned} \quad (\text{CW.2})$$

The derivative maps I^2 into I by the product rule, so δ has exactly the stated type. For each i , the class of h_i maps to zero under π , while $\delta[h_i] = h'_i/k \neq 0$ in the retained remainder basis. Thus factoring δ through π would fail on this explicit class. The full algebra map (CW.6) below retains both its quotient value and derivative. Monic division in each original variable gives a unique expansion

$$p = \sum_{\mathbf{a} \in \mathbb{N}^k} h_1^{a_1} \cdots h_k^{a_k} p_{\mathbf{a}}, \quad \deg_{s_i} p_{\mathbf{a}} < d. \quad (\text{CW.3})$$

Only finitely many terms occur. Existence follows by repeated one-variable division, which commutes for distinct variables. For uniqueness, the leading monomials $\prod_i s_i^{da_i+b_i}$, $0 \leq b_i < d$, of the corresponding products are all distinct; comparison in a monomial order, followed by subtraction, proves independence. This also proves that the h_i are algebraically independent and that P is free over $\mathbb{C}[h_1, \dots, h_k]$ with the original remainder basis.

Consequently the original conormal coefficient map is the E -module isomorphism

$$\iota_N : E^k \xrightarrow{\sim} N, \quad (a_i) \mapsto \left[\sum_i h_i \tilde{a}_i \right]_{I^2}, \quad \dim E = d^k, \quad \dim E^{[2]} = (k+1)d^k. \quad (\text{CW.4})$$

Changing a lift $\tilde{a}_i$ adds an element of I^2 . Modulo I^2 , (CW.3) has exactly its constant coefficient and its k linear h_i -coefficients, proving every assertion in (CW.4). In particular $N^2 = 0$ in $E^{[2]}$. The derivative obeys

$$\delta(ab) = \pi(a)\delta(b) + \pi(b)\delta(a), \quad \delta\iota_N(a_i) = \frac{1}{k} \sum_i h'_i a_i. \quad (\text{CW.5})$$

These identities follow before taking the indicated quotients; the terms still containing h_i vanish only in E .

Theorem 26.1 (A full algebra bridge for the coefficient derivative). *Let $B = E[\epsilon]/(\epsilon^2)$, where ϵ is a formal infinitesimal, and let $\mathfrak{j} = (h'_1, \dots, h'_k)$ be the indicated ideal of E . There is a unital algebra map*

$$\Phi : E^{[2]} \rightarrow B, \quad a \mapsto \pi(a) + \epsilon\delta(a). \quad (\text{CW.6})$$

The derivative induces $\bar{\partial}_\Sigma : E \rightarrow E/\mathfrak{j}$, and its image and kernel are exactly

$$\begin{aligned} \text{im } \Phi &= \{q + \epsilon t : [t]_{\mathfrak{j}} = \bar{\partial}_\Sigma q\}, \\ \ker \Phi &= \iota_N \ker \left(E^k \xrightarrow{(a_i) \mapsto k^{-1} \sum_i h'_i a_i} E \right). \end{aligned} \quad (\text{CW.7})$$

There is the exact sequence of complex vector spaces

$$0 \longrightarrow \ker(\delta|_N) \longrightarrow E^{[2]} \xrightarrow{\Phi} B \xrightarrow{\Psi} E/\mathfrak{j} \longrightarrow 0, \quad \Psi(q + \epsilon t) = [t]_{\mathfrak{j}} - \bar{\partial}_\Sigma q. \quad (\text{CW.8})$$

The map Φ is an algebra homomorphism; Ψ is given its displayed complex-linear type. The complete ranks and dimensions are

$$\begin{aligned} \text{rank}(\delta|_N) &= d^k - (d - r_0)^k, \\ \dim \ker \Phi &= (k-1)d^k + (d - r_0)^k, \\ \dim \text{im } \Phi &= 2d^k - (d - r_0)^k. \end{aligned} \quad (\text{CW.9})$$

Proof. Equation (CW.5) gives

$$\Phi(ab) = \pi(a)\pi(b) + \epsilon\{\pi(a)\delta(b) + \pi(b)\delta(a)\} = \Phi(a)\Phi(b),$$

and $\delta(1) = 0$, proving the unital algebra claim. Changing a representative of $q \in E$ by $\sum_i h_i p_i$ changes its derivative modulo I by $k^{-1} \sum_i h'_i [p_i] \in \mathfrak{j}$. Thus $\bar{\partial}_\Sigma$ is well-defined, with its product rule along $E \rightarrow E/\mathfrak{j}$. Every $\Phi(a)$ consequently satisfies the condition in (CW.7).

Conversely let that condition hold. Take the unique original monic remainder $\sigma(q) \in P$, of degree below d in every variable. Then

$$t - \delta[\sigma(q)]_{I^2} = \frac{1}{k} \sum_i h'_i a_i$$

for some $a_i \in E$, by the definition of $\mathfrak{j}$. The element $[\sigma(q)]_{I^2} + \iota_N(a_i)$ has image $q + \epsilon t$. This proves the image equality. Its kernel requires $\pi(a) = 0$, hence $a \in N$, and then exactly the vanishing of the row in (CW.5). This proves the kernel statement. The map Ψ is onto, since $\Psi(\epsilon t) = [t]$, and its kernel is the already computed image of Φ . This proves (CW.8) with its exact map types.

For the ranks, retain an ordered local tensor block at centres ρ_i , of full lengths m_i . In its coordinates

$$L = \mathbb{C}[z_1, \dots, z_k] / (z_1^{m_1}, \dots, z_k^{m_k}), \quad z_i = s_i - \rho_i,$$

write $h(s_i) = z_i^{m_i} c_i(z_i)$, $c_i(0) \neq 0$. Modulo the original local ideal,

$$h'_i = m_i c_i(0) z_i^{m_i-1}, \quad h'_i a_i = m_i c_i(0) z_i^{m_i-1} a_i|_{z_i=0}. \quad (\text{CW.10})$$

Thus the original input constant coefficient in the i -th variable is sent into that variable's top nilpotent direction; all remaining variables and the factor $m_i c_i(0)$ are retained. The row image is the ideal spanned by monomials having at least one exponent $m_i - 1$. Its dimension is $\prod_i m_i - \prod_i (m_i - 1)$. The excluded monomials remain in the complementary vector space and in $E/\mathfrak{j}$. Sum over every ordered tuple of centres. The two sums are

$$\sum_{\rho_1, \dots, \rho_k} \prod_i m_{\rho_i} = d^k, \quad \sum_{\rho_1, \dots, \rho_k} \prod_i (m_{\rho_i} - 1) = (d - r_0)^k. \quad (\text{CW.11})$$

This proves the first line of (CW.9). Subtract it from $\dim N = kd^k$ for the kernel, then subtract that kernel from $\dim E^{[2]}$ for the image, proving the other two. These counts include every simple and repeated local factor. $\square$

When the packet has the original reflection $a^\dagger(\mathbf{s}) = \overline{a(1 - \mathbf{s})}$, its ideals are stable because $h^\dagger = (-1)^d h$. The derivative satisfies $\delta(a^\dagger) = -(\delta a)^\dagger$. Set $(q + \epsilon t)^\dagger = q^\dagger - \epsilon t^\dagger$. Then $\Phi(a^\dagger) = \Phi(a)^\dagger$, by (CW.6). This retains the sign of the infinitesimal and the original conjugate-linear reflection.

26.2 The original monic section and the exact finite correction

Let $\sigma : E \rightarrow P$ denote the multivariate monic remainder section of (CW.3), and set $j = [\sigma]_{I^2} : E \rightarrow E^{[2]}$. It is complex-linear, with $\pi j = I_E$. Let $S = \sum_i s_i$, and write Z and $\widehat{Z}$ for multiplication by $S - k/2$ on E and $E^{[2]}$, respectively. The original product rule gives the rectangular operator identity

$$Z\delta - \delta\widehat{Z} = -\pi, \quad D_0 = \delta j : E \rightarrow E. \quad (\text{CW.12})$$

The original finite arithmetic generator is $A_k = Z + (k/2)I_E$.

For each i , let $\ell_i : E \rightarrow E_{\widehat{i}} := \bigotimes_{j \neq i} E_h$ extract the coefficient of s_i^{d-1} in $\sigma(u)$. Let $c_i : E_{\widehat{i}} \rightarrow E$ insert the constant polynomial one in the i -th variable. If $k = 1$, the target of ℓ_1 is $\mathbb{C}$, and c_1 is its unit column. These maps give the exact section defect

$$\begin{aligned} \mathbf{t}_Z &:= \widehat{Z}j - jZ = \iota_N(c_1 \ell_1, \dots, c_k \ell_k) : E \rightarrow N, \\ [Z, D_0] &= -I_E + \mathbf{J}, \quad \mathbf{J} = \delta \mathbf{t}_Z = \frac{1}{k} \sum_i M_{h'_i} c_i \ell_i. \end{aligned} \quad (\text{CW.13})$$

Indeed multiplying the original remainder by s_i creates only its top s_i^d term outside the admitted coordinate range. Monic division subtracts exactly $h_i \sigma_i(\ell_i u)$. Adding these terms for $S - k/2$ proves the first formula. Compose (CW.12) with j and use $\widehat{Z}j = jZ + \mathbf{t}_Z$; this proves the commutator with its minus sign. Applying (CW.5) proves its last formula. In particular this is a specified endomorphism of the original finite vector space, obtained by a declared section of π .

Theorem 26.2 (All projectors, eigenvalues and ranks of the section correction). *The original coordinate endomorphisms*

$$P_i = \frac{1}{d} M_{h'_i} c_i \ell_i, \quad P_i^2 = P_i, \quad P_i P_j = P_j P_i, \quad \text{rank } P_i = d^{k-1} \quad (\text{CW.14})$$

are commuting idempotents. For $T \subset \{1, \dots, k\}$, put

$$P_T = \prod_{i \in T} P_i \prod_{i \notin T} (I - P_i). \quad (\text{CW.15})$$

Their pairwise products are zero when their labels differ, and $\sum_T P_T = I_E$. On their ranges the full correction has

$$J = \frac{d}{k} \sum_i P_i, \quad J|_{\text{ran } P_T} = \frac{d|T|}{k} I. \quad (\text{CW.16})$$

For $d > 1$, its eigenvalue dj/k has multiplicity $\binom{k}{j} (d-1)^{k-j}$, for every $0 \leq j \leq k$. For all $d \geq 1$,

$$\text{Tr } J = d^k, \quad \text{rank } J = d^k - (d-1)^k, \quad \ker J = \bigcap_i \ker \ell_i. \quad (\text{CW.17})$$

For $d = 1$, every $P_i = I$, $J = I$, and only the $j = k$ eigenspace is nonzero, of dimension one.

Proof. The polynomial h'_i has degree $d-1$ and leading coefficient d . Therefore

$$\ell_i M_{h'_i} c_i = d I_{E_i}. \quad (\text{CW.18})$$

Substitution proves $P_i^2 = P_i$. It also proves that $M_{h'_i} c_i$ is injective, and hence that the rank is d^{k-1} . Operators in distinct variables are tensor products of their respective one-variable maps, so they commute. The one-variable space is the direct sum of $\mathbb{C}h'$ and the polynomials of degree at most $d-2$: the former has nonzero top coefficient d , and every polynomial splits by subtracting that top coefficient times h'/d . These are the image of P_i and the kernel of ℓ_i , of dimensions one and $d-1$.

Taking tensor products of these direct decompositions proves all the identities for P_T and gives $\dim \text{ran } P_T = (d-1)^{k-|T|}$ when $d > 1$. On this range exactly the P_i for $i \in T$ act as identity, proving (CW.16). Summing over $|T| = j$ gives the asserted multiplicity. Its zero space is the all-complement range, of dimension $(d-1)^k$, proving the rank and kernel in (CW.17). Finally

$$\text{Tr } J = \frac{d}{k} \sum_i \text{rank } P_i = \frac{d}{k} k d^{k-1} = d^k. \quad (\text{CW.19})$$

For $d = 1$, the one-variable complement is zero and its image is the entire one-dimensional space. Only the all-image tensor survives. This proves the stated endpoint directly, without assigning a positive multiplicity to a zero-dimensional space. $\square$

These are algebraic idempotents in the retained coordinates. For any actual positive minimum metric G_M , their exact metric adjoints and coordinate transports are

$$P_i^{\dagger G_M} = G_M^{-1} P_i^* G_M, \quad \tilde{P}_i = G_M^{1/2} P_i G_M^{-1/2}, \quad \tilde{J} = G_M^{1/2} J G_M^{-1/2}. \quad (\text{CW.20})$$

The transports retain the products, ranks and algebraic spectrum. For each i , the individual $\tilde{P}_i$ is Hermitian precisely when $P_i^* G_M = G_M P_i$, whereas $\tilde{J}$ is Hermitian precisely when $J^* G_M = G_M J$. None of these metric equalities is assumed. The same isometry carries the original weight to

$$G_M^{-1/2} W_M G_M^{-1/2} = \tilde{Z}^* + \tilde{Z}, \quad \tilde{Z} = G_M^{1/2} Z G_M^{-1/2}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{CW.21})$$

It carries (CW.13) to the identical commutator with all three operators transported. Thus the algebraic correction and the original Hermitian weight have an explicit common coordinate map, with every adjoint still attached to its actual metric.

The full conormal row rank in (CW.9) and the section rank in (CW.17) are connected by exactly

$$E \xrightarrow{(c_i \ell_i)_i} E^k \xrightarrow{\iota_N} N \xrightarrow{\delta|_N} E. \quad (\text{CW.22})$$

Its composite is J . The first map retains the specific highest-coordinate coefficients of the monic section; the full conormal row accepts arbitrary k input coefficients. This proves the relationship between the two ranks without replacing one domain by the other.

26.3 The full arithmetic unit and the original finite representatives

Keep the complete Taylor classes

$$\hat{U} = \left[\prod_i v_h(s_i) \right]_{I^2} \in (E^{[2]})^\times, \quad U = \pi(\hat{U}) \in E^\times, \quad \beta_h = U^{-1} \delta \hat{U} \in E. \quad (\text{CW.23})$$

Finite multivariate Hermite interpolation supplies these classes from the entire original functions near every ordered packet point. Their values are nonzero at those points because the removed orders were complete. One can also prove invertibility in the thickening directly: lift U^{-1} to b ; then $\hat{U}b = 1 + n$, $n \in N$, and $b(1 - n)$ is the inverse since $N^2 = 0$. Thus this thickening retains the original unit and its first relation derivatives.

Theorem 26.3 (The unit-conjugated finite Weyl identity). *Define the actual unit section and its derivative by*

$$j_U = \hat{U} j U^{-1} : E \rightarrow E^{[2]}, \quad \pi j_U = I_E, \quad D_U = \delta j_U = M_{\beta_h} + M_U D_0 M_U^{-1}. \quad (\text{CW.24})$$

Then, with all multiplication operators in the original basis,

$$\begin{aligned} \mathbf{t}_U &:= \hat{Z} j_U - j_U Z = M_{\hat{U}} \mathbf{t}_Z M_U^{-1} : E \rightarrow N, \\ [Z, D_U] &= -I_E + J_U, \quad J_U = \delta \mathbf{t}_U = M_U J M_U^{-1}. \end{aligned} \quad (\text{CW.25})$$

The complete spectrum, multiplicities, ranks and trace in (CW.16)–(CW.19) therefore hold for J_U as well.

Proof. The section equation follows from $\pi j = I$ and $\pi(\hat{U}) = U$. Apply (CW.5) to $\hat{U} j(U^{-1}u)$. Its two terms are $(\delta \hat{U})U^{-1}u$ and $U D_0(U^{-1}u)$, which prove (CW.24). The multiplication maps $\hat{Z}$, $M_{\hat{U}}$ commute, as do Z , M_U^{-1} . This proves the formula for $\mathbf{t}_U$. Its image lies in N , so the product-rule term $(\delta \hat{U})\pi(\mathbf{t}_Z M_U^{-1}u)$ vanishes exactly. The other term is $U J M_U^{-1}u$, proving the last equality in (CW.25). Finally M_{β_h} commutes with multiplication by Z . Conjugating (CW.13) by M_U proves the commutator. Similarity gives the entire stated spectrum and its zero space, without making an orthogonality claim. $\square$

These sections have an original analytic realization. Let

$$\mathcal{T}_h^{(k)} p = p(D_1, \dots, D_k) F_h^{\otimes k}, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad R_{\text{ref}}^{(k)} u = \mathcal{T}_h^{(k)} \sigma(U^{-1}u). \quad (\text{CW.26})$$

The original Mellin transform gives $\mathcal{M}_k \mathcal{T}_h^{(k)} p = (\prod_i v_h(s_i))p$. Its thickened Taylor class is $\widehat{U}[p]_{I^2}$, so the thickened class of $R_{\text{ref}}^{(k)} u$ is exactly $j_U u$. Multiplication by $\mathcal{L}_k$ differentiates this Mellin transform by ∂_Σ . Hence

$$J^{(k)} \mathcal{L}_k R_{\text{ref}}^{(k)} = \mathbf{D}_U. \quad (\text{CW.27})$$

This is the finite endomorphism of the specified original section, with its actual logarithmic source observable.

For the actual degree- M minimum, $M \geq k(d-1)$, retain its original coefficient Gram $\mathcal{M}_M > 0$ and full jet matrix $\mathcal{J}_M$, including multiplication by U . Its numerator and source function are exactly

$$p_{M,u} = \mathcal{M}_M^{-1} \mathcal{J}_M^* G_M u, \quad G_M = (\mathcal{J}_M \mathcal{M}_M^{-1} \mathcal{J}_M^*)^{-1}, \quad R_M u = \mathcal{T}_h^{(k)} p_{M,u}. \quad (\text{CW.28})$$

Indeed its jet is u , and its coefficient vector is $\mathcal{M}_M$ -orthogonal to $\ker \mathcal{J}_M$. Every other preimage differs by such a kernel vector; Pythagoras proves this same original constrained minimum. Define the concrete thickened section and admitted relation by

$$j_M u = \widehat{U}[p_{M,u}]_{I^2}, \quad a_M = j_M - j_U : E \rightarrow N, \quad \mathbf{D}_M = \delta j_M = \mathbf{D}_U + \delta a_M. \quad (\text{CW.29})$$

The image statement follows from $\pi j_M = \pi j_U = I_E$. Its numerator is the original allowed relation $p_{M,u} - \sigma(U^{-1}u) \in I$, not an independently chosen vector in the conormal layer. The corresponding exact identities are

$$\begin{aligned} J^{(k)} \mathcal{L}_k R_M &= \mathbf{D}_M, & [Z, \mathbf{D}_M] &= -I_E + \mathbf{J}_M^{\text{sec}}, \\ \mathbf{J}_M^{\text{sec}} &:= \delta(\widehat{Z} j_M - j_M Z) = \mathbf{J}_U + [Z, \delta a_M], & \text{Tr } \mathbf{J}_M^{\text{sec}} &= d^k. \end{aligned} \quad (\text{CW.30})$$

The first follows by the same original Mellin differentiation. For the second, compose (CW.12) with j_M . For the third, subtract (CW.25) and use $\delta \widehat{Z} a_M = Z a_M^\delta$, where $a_M^\delta = \delta a_M$, since $\pi a_M = 0$. This gives exactly the displayed commutator difference. Its trace is zero by the cyclic identity for finite matrices; (CW.19) gives the final trace. Only the specified fixed-section correction is assigned the ranks (CW.17). Every change to the actual minimum is retained through the computed map a_M and its commutator.

26.4 The one-variable residue functional and the original theta relation

Take $k = 1$, write $\ell : E_h \rightarrow \mathbb{C}$ for the coefficient of s^{d-1} , and let $U = j_h v_h$. The product derivative $g' = h'v_h + hv_h'$ gives

$$\mathbf{J}_U = M_{j_h g'} c_1 \ell M_U^{-1}, \quad \mathbf{J}_U u = (j_h g') \ell(U^{-1}u), \quad \text{rank } \mathbf{J}_U = 1, \quad \text{Tr } \mathbf{J}_U = d. \quad (\text{CW.31})$$

The rank follows either from (CW.25) or from the nonzero column $j_h g' = U h'$ and the nonzero functional. The trace of this rank-one map is $\ell(U^{-1} j_h g') = \ell(h') = d$.

For the original reflection-stable packet, put $f^\dagger(s) = \overline{f(1-\bar{s})}$. Its residue pairing is the well-defined form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{\widetilde{f}^\dagger(s) \widetilde{u}(s)}{g(s)} ds = \ell(U^{-1} f^\dagger u). \quad (\text{CW.32})$$

Changing either polynomial representative by h changes the integrand by a function holomorphic at all selected centres: $h^\dagger = (-1)^d h$, and $g = hv_h$, with v_h nonvanishing there. To prove the last equality, take $p = \text{rem}_h(U^{-1} f^\dagger u)$, of degree below d . The residues of the displayed quotient equal those of p/h . In the partial-fraction expansion of p/h , the coefficient of $1/s$ at infinity is the sum of its coefficients at the simple-pole terms; terms of pole order at least two have no $1/s$ coefficient. Since h is monic of degree d , that coefficient is exactly the original coefficient $\ell(p)$. This proves (CW.32),

keeping every higher local pole before the residue is taken. Consequently the finite correction is precisely

$$J_U u = (j_h g') \mathcal{R}_Z(1, u), \quad \text{Tr } J_U = \mathcal{R}_Z(1, j_h g') = \sum_{\rho \in Z} m_\rho = d. \quad (\text{CW.33})$$

The source map producing the same Jacobian is the original logarithmic theta observation:

$$J_h(\log x \Theta \phi) = j_h(g') j_h(H_\phi), \quad \mathcal{M} \Theta \phi = g H_\phi. \quad (\text{CW.34})$$

Differentiate the Mellin product before taking jets. Its $g H'_\phi$ term vanishes modulo the original h ; its other term is exactly (CW.34). For $\phi = P(D)\phi_*$, one has $H_\phi = P$, so the input ranges through all original finite coefficients. Thus this map's finite image is the full ideal $(j_h g')$, of dimension r_0 by (CW.10), rather than a selected one-dimensional input line. The exact connection to (CW.31) is its precomposition with

$$E_h \xrightarrow{\ell M_U^{-1}} \mathbb{C} \xrightarrow{c_1} E_h. \quad (\text{CW.35})$$

The rank-one correction and the rank- r_0 multiplication map therefore use the same Jacobian with their distinct input maps explicitly specified. Applying (CW.32) to (CW.34) gives, with the full conjugation,

$$\mathcal{R}_Z(f, j_h(g') j_h P) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho). \quad (\text{CW.36})$$

Indeed the residue of g'/g at a zero of order m_ρ is m_ρ , by differentiating its local product.

There is also an exact global source map

$$\nu : V \rightarrow Q = \mathcal{B}/\Theta V, \quad \nu(\phi) = q(\log x \Theta \phi). \quad (\text{CW.37})$$

Logarithmic multiplication preserves $\mathcal{B}$, by its defining two-sided polynomial bounds: $|\log x| \leq x + x^{-1}$ and $D^j((\log x)F) = (\log x)D^j F - jD^{j-1}F$ for $j \geq 1$ give every required bound from the original ones. Thus this map is defined. For the original $D = -x\partial_x$, $[D, \log x] = -I$ and $D\Theta = \Theta D$. Therefore

$$D_Q \nu - \nu D_V = q[D, \log x]\Theta = -q\Theta = 0. \quad (\text{CW.38})$$

For $a > 0$, let $\Lambda_a F(x) = F(ax)$. The identities $\Theta \Lambda_a = \Lambda_a \Theta$ and $(\log x)\Lambda_a - \Lambda_a(\log x) = -\log(a)\Lambda_a$ similarly give $\nu \Lambda_a = \Lambda_a \nu$. This proves scaling equivariance on the exact source V and target Q . Since $J_h \Theta = 0$, there is the induced full jet map $\bar{J}_h : Q \rightarrow E_h$ with $\bar{J}_h q = J_h$; its value on ν is exactly (CW.34).

26.5 Supported algebra and coefficient maps

Write $\text{pr}_0, \text{pr}_\epsilon : B \rightarrow E$ for the constant and infinitesimal coefficient maps. The first is a unital algebra homomorphism and the second is complex-linear, with the exact product rule

$$\text{pr}_\epsilon(bc) = \text{pr}_0(b) \text{pr}_\epsilon(c) + \text{pr}_0(c) \text{pr}_\epsilon(b). \quad (\text{CW.39})$$

The strongest typed lift of (CW.6) is the semiring map

$$G(E^{[2]}) \xrightarrow{G(\Phi)} G(B), \quad \pi^\tau = G(\text{pr}_0)G(\Phi), \quad \delta^\tau = (\text{pr}_\epsilon)^\tau G(\Phi). \quad (\text{CW.40})$$

Here $G(\Phi)$ and $G(\text{pr}_0)$ fix external τ and send supported amplitudes to their supported algebra images. The map $(\text{pr}_\epsilon)^\tau$ is the stated split-linear coefficient map; (CW.39) specifies its precise relationship to multiplication. Its value on the supported unit of B is the supported zero e_E .

Every vector in the algebraic kernel in (CW.7) maps under $G(\Phi)$ to supported zero, while external absence maps to external absence. A supported conormal element n maps to $(\epsilon \delta n)^\bullet$, retaining its

coefficient derivative; constant projection gives e_E . The sections, coefficient maps, exact sequences, and original representatives above lift at each fixed support by $(\lambda, u) \mapsto (\lambda, Tu)$. Thus the first thickening, its full kernel and quotient, and the source-selected endomorphism retain their distinct receiving fibres and the common external τ .

If $h = 1$, then $I = P$, $E = E^{[2]} = N = 0$, and all finite maps in (CW.2) have zero source and target. The source function is still the original theta seed $F_h = \Theta\phi_*$, whose map to this arithmetic quotient is zero. This states that case explicitly; the projector formulas dividing by d above have the declared domain $d \geq 1$.

26.6 The arithmetic unit and complete tensor partial residues

Return to the nonempty original packet of (CW.1), with $d \geq 1$ and $k \geq 1$. The unit itself has the following exact image under the algebra bridge already proved above.

Corollary 26.4 (The full unit and its first relation derivative). *For the original $\widehat{U}, U, \beta_h$ of (CW.23),*

$$\begin{aligned}\Phi(\widehat{U}) &= U + \epsilon\delta\widehat{U} = U(1 + \epsilon\beta_h), \\ \Phi(\widehat{U}^{-1}) &= \Phi(\widehat{U})^{-1} = U^{-1}(1 - \epsilon\beta_h).\end{aligned}\tag{CW.41}$$

Proof. The first identity is (CW.6) evaluated on the full thickened unit. The definition $\beta_h = U^{-1}\delta\widehat{U}$ gives its displayed factorization. Since $\epsilon^2 = 0$, $(1 + \epsilon\beta_h)(1 - \epsilon\beta_h) = 1$. The map Φ is unital and multiplicative, so applying it to $\widehat{U}\widehat{U}^{-1} = 1$ proves the second identity with its minus sign. In particular $\Phi(\widehat{U}a) = \Phi(\widehat{U})\Phi(a)$ transports the original arithmetic product through this same algebra map. $\square$

Retain all the original complete orders, $g = 2\xi$ and $v_h = g/h$, and write

$$\begin{aligned}E_h &= \mathbb{C}[s]/(h), & E &= E_h^{\otimes k}, & E_{\widehat{i}} &= \bigotimes_{j \neq i} E_h, \\ v_i &= j_h(v_h) \text{ in the } i\text{-th factor}, & \gamma_i &= j_h(g') \text{ in the } i\text{-th factor}, & U &= \prod_i v_i.\end{aligned}$$

Every v_i is the original complete invertible jet. Let $\ell_h : E_h \rightarrow \mathbb{C}$ extract the coefficient of s^{d-1} from the unique one-variable monic remainder. Let $c_i : E_{\widehat{i}} \rightarrow E$ insert the constant polynomial one in the i -th factor, and let $\ell_i : E \rightarrow E_{\widehat{i}}$ apply ℓ_h in that factor and identity in every other factor. Thus ℓ_i extracts the original coefficient of s_i^{d-1} , retaining every other-variable coordinate. For $k = 1$, $E_{\widehat{i}} = \mathbb{C}$, and c_i is the unit column. Let M_a denote multiplication by the displayed element in its declared algebra.

Define the coefficient-valued partial residue map

$$\mathcal{R}_i : E \longrightarrow E_{\widehat{i}}, \quad \mathcal{R}_i(u) = \sum_{\rho \in \mathbb{Z}} \text{Res}_{s_i=\rho} \frac{\widetilde{u}(s_1, \dots, s_k)}{g(s_i)} ds_i.\tag{CW.42}$$

The coefficients in the other original variables are taken modulo their original ideals after the indicated residue. Each residue uses the usual positively oriented complex s_i -coordinate; the residue notation carries no additional $2\pi i$ factor. Equivalently, each residue equals $1/(2\pi i)$ times the corresponding positively oriented contour integral.

This map is well-defined on E . A change by $h(s_i)p$ in its own variable changes the integrand by $p/v_h(s_i)$, which is holomorphic at every selected centre because the removed zero orders were complete. A change by $h(s_j)p$, $j \neq i$, produces that same factor $h(s_j)$ in the remaining coefficient algebra, hence zero in $E_{\widehat{i}}$. Linearity then covers the full original ideal. The finite sum of coefficient residues is complex-linear and $E_{\widehat{i}}$ -linear.

The exact coefficient formula is

$$\boxed{\mathcal{R}_i = \ell_i M_{v_i}^{-1}}.\tag{CW.43}$$

To prove it, let p_i be the degree-below- d remainder in s_i of $v_i^{-1}u$, retaining the full coefficient algebra $E_{\widehat{\mathcal{C}}}$. The residue in (CW.42) equals the corresponding residue of $p_i/h(s_i)$: the difference in the numerator vanishes to the complete selected orders, so division by h is locally holomorphic. Every coefficient of p_i/h has an ordinary partial-fraction expansion. The coefficient of s_i^{-1} at infinity is the sum of the finite simple-pole coefficients; higher-pole terms contribute no s_i^{-1} coefficient. Since h is monic and $\deg_{s_i} p_i < d$, that coefficient is precisely $\ell_i(p_i)$. This proof applies coefficientwise in the finite algebra $E_{\widehat{\mathcal{C}}}$ and gives (CW.43) with every other-variable nilpotent coefficient retained.

Define the original partial correction endomorphisms

$$C_i = M_{\gamma_i} c_i \mathcal{R}_i : E \longrightarrow E. \quad (\text{CW.44})$$

They satisfy

$$\begin{array}{l} C_i^2 = dC_i, \quad C_i C_j = C_j C_i, \\ \text{rank } C_i = d^{k-1}, \quad \text{Tr } C_i = d^k. \end{array} \quad (\text{CW.45})$$

Indeed $g' = h'v_h + hv'_h$ gives $\gamma_i = v_i h'_i$ in the original i -th quotient, so

$$\mathcal{R}_i M_{\gamma_i} c_i = \ell_i M_{v_i}^{-1} M_{\gamma_i} c_i = \ell_i M_{h'_i} c_i = dI_{E_{\widehat{\mathcal{C}}}}. \quad (\text{CW.46})$$

The last equality uses the literal degree $d - 1$ and leading coefficient d of the original derivative h' . This proves the first formula in (CW.45). It also proves that $M_{\gamma_i} c_i$ is injective and $\mathcal{R}_i$ is onto. The factorization (CW.44) therefore has rank $\dim E_{\widehat{\mathcal{C}}} = d^{k-1}$. Alternatively a specified right inverse for $\mathcal{R}_i$ is $a \mapsto v_i s_i^{d-1} c_i(a)$, using (CW.43). The two rectangular factors in (CW.44) have traces $\text{Tr}_E(AB) = \text{Tr}_{E_{\widehat{\mathcal{C}}}}(BA)$; apply (CW.46) to obtain $d d^{k-1} = d^k$. Operators belonging to distinct variables are tensor products of their respective one-variable maps with identity on the other factors. Their compositions therefore commute, proving the remaining identity.

These operators give precisely the correction of the original full-unit section:

$$J_U = \frac{1}{k} \sum_{i=1}^k C_i. \quad (\text{CW.47})$$

For proof, retain the complete thickened unit $\widehat{U}$, its residue $U = \prod_i v_i$, and its full first derivative $\beta_h = U^{-1} \delta \widehat{U}$. The original source-selected operator is still

$$\begin{aligned} D_U &= M_{\beta_h} + M_U D_0 M_U^{-1}, & [Z, D_U] &= -I_E + J_U, \\ J_U &= M_U \left(\frac{1}{k} \sum_i M_{h'_i} c_i \ell_i \right) M_U^{-1}. \end{aligned} \quad (\text{CW.48})$$

Write $U = v_i U_{\widehat{\mathcal{C}}}$, with $U_{\widehat{\mathcal{C}}} = \prod_{j \neq i} v_j$. The maps c_i, ℓ_i are $E_{\widehat{\mathcal{C}}}$ -linear, so the operator $M_{U_{\widehat{\mathcal{C}}}}$ commutes through their indicated source and target actions, and its inverse then cancels with exactly that same full other-variable unit. Thus each summand in (CW.48) is

$$M_U M_{h'_i} c_i \ell_i M_U^{-1} = M_{v_i h'_i} c_i \ell_i M_{v_i}^{-1} = M_{\gamma_i} c_i \mathcal{R}_i = C_i. \quad (\text{CW.49})$$

This proves (CW.47). Every v_i retains its original value, and M_{β_h} remains in D_U . It commutes with the original multiplication operator $Z = M_{\sum_i s_i - k/2}$, which is why it contributes zero to that specified commutator.

For completeness, retaining all k residue variables gives

$$\begin{aligned} \mathcal{R}_{\text{all}} : E &\longrightarrow \mathbb{C}, \\ \mathcal{R}_{\text{all}}(u) &= \sum_{\rho_1, \dots, \rho_k \in \mathcal{Z}} \text{Res}_{s_k = \rho_k} \cdots \text{Res}_{s_1 = \rho_1} \frac{\widetilde{u}(\mathbf{s})}{\prod_i g(s_i)} ds_1 \cdots ds_k. \end{aligned} \quad (\text{CW.50})$$

Here the formula means the ordered iterated coefficient residues, with each original variable positively oriented. Each variable operates on a different tensor factor, so swapping their evaluation order gives the identical scalar coefficient and introduces no sign. Formula (CW.43) in each factor proves

$$\mathcal{R}_{\text{all}} = \ell_h^{\otimes k} M_U^{-1}, \quad \boxed{C_1 \cdots C_k(u) = \left(\prod_i \gamma_i \right) \mathcal{R}_{\text{all}}(u).} \quad (\text{CW.51})$$

The map $\ell_h^{\otimes k} : E_h^{\otimes k} \rightarrow \mathbb{C}$ is the tensor product of the original one-variable highest-coefficient functionals. Tensoring the one-variable factorization (CW.44) proves the second equality directly, so it keeps all repeated-root and mixed nilpotent data before the final scalar functional is evaluated. The column and functional are both nonzero: their contraction is

$$\begin{aligned} \mathcal{R}_{\text{all}} \left(\prod_i \gamma_i \right) &= \prod_{i=1}^k \left(\sum_{\rho \in \mathcal{Z}} \text{Res}_{s_i=\rho} \frac{g'(s_i)}{g(s_i)} ds_i \right) \\ &= \prod_{i=1}^k \left(\sum_{\rho} m_{\rho} \right) = d^k. \end{aligned} \quad (\text{CW.52})$$

This is the rank-one full-product correction, with its exact nonzero eigenvalue and trace d^k . Its square is d^k times itself by (CW.51)–(CW.52). The residue of g'/g is m_{ρ} because $g(s) = (s - \rho)^{m_{\rho}} u_{\rho}(s)$ with $u_{\rho}(\rho) \neq 0$; hence $g'/g = m_{\rho}/(s - \rho) + u'_{\rho}/u_{\rho}$. This gives the complete multiplicity proof used in (CW.52).

At $d = 1$, (CW.46) gives $C_i = I_E$ on the one-dimensional E ; consequently $J_U = I_E$, and (CW.51) is the same identity. At $k = 1$, every formula reduces exactly to (CW.31)–(CW.33), including the original residue functional $\mathcal{R}_{\mathcal{Z}}(1, u)$, rank one and trace d . For $h = 1$, the original arithmetic quotient is zero, as already stated above; the positive-degree residue coefficient formulas have the declared domain $d \geq 1$. All repeated fibres, complete arithmetic units, and their original source and target maps have been retained.

27 Exact transitions between nested relative frames

This continuation starts with the original amplitude, relative coordinates, finite source powers, graph and arithmetic observation of FC.1–36. It proves the maps produced when the relative frame is enlarged. Every finite weight, original source coefficient and normal component is retained. The published source is used unchanged.

27.1 Original sources and coefficient inclusions

Fix the original packet polynomial $h, g = 2\xi$, the full entire quotient $v_h = g/h$, and the integer $k \geq 2$. Keep

$$a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad u = \sum_{i=1}^k t_i, \quad t_i = u/k + y_i, \quad y_k = -\sum_{i < k} y_i, \quad \Psi(u, \mathbf{y}) = \prod_{i=1}^k a(u/k + y_i). \quad (\text{NT.1})$$

The ambient fibre is $\mathcal{H} = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$, and the full ambient space is $\mathcal{K} = L^2(\mathbb{R}, du; \mathcal{H})$. Thus the original Jacobian $du d\mathbf{y}$ and the amplitude mass are unchanged. The original relative variables are $z_i = s_i - S/k$, $S = \sum_i s_i$, so $z_i = iy_i$ on this fibre.

Let the finite independent homogeneous relative families at levels $M \leq L$ have column counts r_M, r_L and degrees $\ell_{\alpha}^M, \ell_{\beta}^L$. Their inclusion is the literal constant matrix $E = E_{LM} : \mathbb{C}^{r_M} \rightarrow \mathbb{C}^{r_L}$

satisfying

$$\theta_\alpha^M(\mathbf{z}) = \sum_\beta \theta_\beta^L(\mathbf{z}) E_{\beta\alpha}, \quad j_M(u) = j_L(u)E, \quad j_a(u)c = \Psi(u, \mathbf{y}) \sum_\alpha \theta_\alpha^a(i\mathbf{y})c_\alpha. \quad (\text{NT.2})$$

Comparison of homogeneous components gives $E_{\beta\alpha} = 0$ unless $\ell_\beta^L = \ell_\alpha^M$. Independence of the smaller family makes E injective. For an actual nested monomial family, E is its original zero-extension matrix; no scaling of columns is introduced.

For $a = M, L$ retain the complete admitted source index set and its literal coefficient matrix,

$$\begin{aligned} \mathcal{I}_a &= \{(\alpha, n) : 0 \leq n \leq a - \ell_\alpha^a\}, & q_a &= |\mathcal{I}_a|, \\ P_p^a &= \sum_{(\alpha, n) \in \mathcal{I}_a} p_{\alpha, n} \theta_\alpha^a(\mathbf{z}) S^n, & (C_a(u))_{\beta, (\alpha, n)} &= \delta_{\alpha\beta} (k/2 + iu)^n. \end{aligned} \quad (\text{NT.3})$$

Define the source inclusion $F = F_{LM} : \mathbb{C}^{q_M} \rightarrow \mathbb{C}^{q_L}$ by $F_{(\beta, m), (\alpha, n)} = E_{\beta\alpha} \delta_{mn}$. Every nonzero entry is admitted since $L \geq M$ and the relative degrees agree. Direct expansion gives the exact identities

$$P_{Fp}^L = P_p^M, \quad C_L F = E C_M, \quad C'_L F = E C'_M. \quad (\text{NT.4})$$

Uniqueness of the relative-polynomial expansion and of each power of S proves that F is injective. Every factor $(k/2 + iu)^n$ and its derivative $in(k/2 + iu)^{n-1}$ is retained.

27.2 The original weights and their orthogonal transition

At every real u define, with the ambient Hilbert adjoint,

$$\begin{aligned} W_a &= j_a^* j_a > 0, & B_a &= j_a^* j'_a, & \Gamma_a &= W_a^{-1} B_a, & \Pi_a &= j_a W_a^{-1} j_a^*, \\ N_a &= (I - \Pi_a) j'_a, & \mathcal{N}_a &= N_a^* N_a, & a &= M, L. \end{aligned} \quad (\text{NT.5})$$

These are the original FC matrices. Their positivity and smoothness follow from independence of the relative family and the original Schwartz amplitude: a nonzero relative polynomial cannot vanish almost everywhere where Ψ is nonzero. The inverse matrices therefore exist at every real u , including $u = 0$.

The weighted coefficient retraction and coefficient projection are

$$R = W_M^{-1} E^* W_L : \mathbb{C}^{r_L} \rightarrow \mathbb{C}^{r_M}, \quad Q = ER : \mathbb{C}^{r_L} \rightarrow \mathbb{C}^{r_L}. \quad (\text{NT.6})$$

The symbol Q here is a coefficient projection; Π_a remains the ambient orthogonal projection. From (NT.2),

$$\begin{aligned} W_M &= E^* W_L E, & B_M &= E^* B_L E, & RE &= I, & Q^2 &= Q, & Q^* W_L &= W_L Q, \\ \Gamma_M &= R \Gamma_L E, & \Pi_L \Pi_M &= \Pi_M \Pi_L = \Pi_M. \end{aligned} \quad (\text{NT.7})$$

The first identities follow by multiplication of $j_M = j_L E$ and $j'_M = j'_L E$. For the projection identities, Π_a is self-adjoint, idempotent and has image $V_a = \text{ran } j_a$; $V_M \subseteq V_L$ implies that Π_L fixes $\text{ran } \Pi_M$, and taking adjoints proves the other order.

Theorem 27.1 (The complete connection transition). *Define the coefficient transition and its ambient image by*

$$K = K_{LM} = \Gamma_L E - E \Gamma_M = (I - Q) \Gamma_L E, \quad A = A_{LM} = j_L K. \quad (\text{NT.8})$$

Then $K : \mathbb{C}^{r_M} \rightarrow \mathbb{C}^{r_L}$ has image in the W_L -orthogonal complement of $\text{ran } E$, and

$$\begin{aligned} E^* W_L K &= 0, & A &= (\Pi_L - \Pi_M) j'_M, & N_M &= A + N_L E, \\ \text{ran } A &\subseteq V_L \cap V_M^\perp, & \text{ran } (N_L E) &\subseteq V_L^\perp, & A^* N_L E &= 0. \end{aligned} \quad (\text{NT.9})$$

Consequently the exact positive transition form is

$$\boxed{\mathcal{N}_M = E^* \mathcal{N}_L E + \Delta_{LM}}, \quad \begin{aligned} \Delta_{LM} &= A^* A = K^* W_L K \succeq 0 \\ &= E^* \Gamma_L^* W_L \Gamma_L E - \Gamma_M^* W_M \Gamma_M. \end{aligned} \quad (\text{NT.10})$$

For each $c \in \mathbb{C}^{r_M}$ the following equivalences and rank bound are exact:

$$c^* \Delta_{LM} c = 0 \iff Kc = 0 \iff Ac = 0, \quad \text{rank } \Delta_{LM} = \text{rank } K \leq \min(r_M, r_L - r_M). \quad (\text{NT.11})$$

Proof. Equation (NT.7) gives $E^* W_L \Gamma_L E = W_M \Gamma_M$; subtracting $E^* W_L E \Gamma_M$ proves $E^* W_L K = 0$. The original identity $j_a \Gamma_a = \Pi_a j'_a$ gives $j_L K = \Pi_L j'_L E - \Pi_M j'_M = (\Pi_L - \Pi_M) j'_M$. Splitting $I - \Pi_M = (\Pi_L - \Pi_M) + (I - \Pi_L)$ and using $j'_M = j'_L E$ proves $N_M = A + N_L E$. The commuting projection identities show that $\Pi_L - \Pi_M$ is the orthogonal projection onto $V_L \cap V_M^\perp$. This proves the two range statements and their orthogonality. Taking the Gram of this orthogonal sum proves the first line of (NT.10). Expanding $K = \Gamma_L E - E \Gamma_M$, using $E^* W_L \Gamma_L E = W_M \Gamma_M$ and its adjoint, gives the second line. Positivity of W_L implies that the quadratic form of $K^* W_L K$ vanishes exactly on $\ker K$. Injectivity of j_L gives $\ker A = \ker K$. The image of K lies in a space of dimension $r_L - r_M$; rank-nullity now proves (NT.11). $\square$

For a locally absolutely continuous original coefficient $c(u)$, write $\nabla_a = \partial_u + \Gamma_a$. The full derivative has the three mutually orthogonal components

$$\begin{aligned} \nabla_L(Ec) &= E \nabla_M c + Kc, \\ \partial_u(j_M c) &= j_M \nabla_M c + Ac + N_L Ec, \\ \|\partial_u(j_M c)\|^2 &= \|\nabla_M c\|_{W_M}^2 + c^* \Delta_{LM} c + c^* E^* \mathcal{N}_L Ec, \\ \|\nabla_L(Ec)\|_{W_L}^2 &= \|\nabla_M c\|_{W_M}^2 + c^* \Delta_{LM} c. \end{aligned} \quad (\text{NT.12})$$

The first equality is the product rule with constant E and (NT.8). The derivative formula follows from (NT.9) and the original connection decomposition. The image of its first term lies in V_M , the second in $V_L \cap V_M^\perp$, and the third in $V_L^\perp$. Their orthogonality proves every energy identity without discarding any coefficient derivative.

27.3 Full source matrices, graph maps and arithmetic observation

Define the actual maps on the original coefficient bases by

$$\begin{aligned} \mathbf{N}_a p &= N_a C_a p, & \mathbf{T}_a p &= j_a(C'_a + \Gamma_a C_a) p, & \mathbf{A}_{LM} p &= A_{LM} C_M p, \\ H_a^0 &= \int C_a^* W_a C_a du, & H_a^N &= \mathbf{N}_a^* \mathbf{N}_a, & H_a^\partial &= \mathbf{T}_a^* \mathbf{T}_a + H_a^N, \\ H_{LM}^{\text{tr}} &= \mathbf{A}_{LM}^* \mathbf{A}_{LM} = \int C_M^* \Delta_{LM} C_M du. \end{aligned} \quad (\text{NT.13})$$

Here the displayed equalities of functions take place in $\mathcal{K}$. All integrals are finite: each term is an orthogonal projection of an original polynomially weighted derivative amplitude, and the FC Schwartz estimates apply before any projection. Equations (NT.4), (NT.9), and (NT.12) give

$$\begin{aligned} \mathbf{N}_M &= \mathbf{A}_{LM} + \mathbf{N}_L F, & \mathbf{T}_L F &= \mathbf{T}_M + \mathbf{A}_{LM}, \\ \mathbf{T}_M^* \mathbf{A}_{LM} &= 0, & \mathbf{A}_{LM}^* \mathbf{N}_L F &= 0, & \mathbf{T}_M^* \mathbf{N}_L F &= 0, \\ H_M^0 &= F^* H_L^0 F, & H_M^N &= F^* H_L^N F + H_{LM}^{\text{tr}}, & F^* \mathbf{T}_L^* \mathbf{T}_L F &= \mathbf{T}_M^* \mathbf{T}_M + H_{LM}^{\text{tr}}, & H_M^\partial &= F^* H_L^\partial F. \end{aligned} \quad (\text{NT.14})$$

These identities follow first pointwise, then by integration. In particular, the positive amount transferred into the larger tangential image is precisely the amount removed from the normal form for the same original polynomial.

Let $\widehat{\Pi}_a$ denote the bounded orthogonal projection on $\mathcal{K}$ obtained by applying $\Pi_a(u)$ pointwise. Measurability follows from smoothness, and contraction follows by integrating the pointwise contraction. Define

$$\begin{aligned}\mathcal{U}_{LM}(t, n) &= (t + \widehat{\Pi}_L n, (I - \widehat{\Pi}_L)n), \quad t \in \text{ran } \widehat{\Pi}_M, \quad n \in \ker \widehat{\Pi}_M, \\ \mathcal{U}_{LM}^{-1}(t_L, n_L) &= (\widehat{\Pi}_M t_L, (I - \widehat{\Pi}_M)t_L + n_L).\end{aligned}\tag{NT.15}$$

The domain and codomain are respectively the orthogonal decompositions of $\mathcal{K}$ at levels M and L . The inverse formula is verified by the nested projection identities. Orthogonality of t , $\widehat{\Pi}_L n$ and $(I - \widehat{\Pi}_L)n$ shows that $\mathcal{U}_{LM}$ is unitary between these sum spaces. On the actual finite derivative graph it is exactly

$$\mathcal{U}_{LM}(\mathbf{T}_M p, \mathbf{N}_M p) = (\mathbf{T}_L F p, \mathbf{N}_L F p).\tag{NT.16}$$

Graph addition on either side is the same original function $\partial_u(j_M C_M p) = \partial_u(j_L C_L F p)$, so (NT.16) also supplies the exact map between the two presentations.

For completeness the finite normal maps are injective on nonempty actual polynomial sources. The FC.21–27 proof uses the infinite discrete set $\mathcal{Z}_h$ of unremoved critical-line zeros, retaining each original order ℓ_T and coefficient $c_T = a^{(\ell_T)}(T)/\ell_T!$. If $N_a(u)c = 0$, then $\Psi'\vartheta_c = \Psi\vartheta_d$ with finite relative polynomials and $d = \Gamma_a c$. On a fibre through $t_1 = T$ where all other amplitude factors are nonzero, the leading terms in the original transverse coordinate δ are

$$\Psi = c_T A_b \delta^{\ell_T} + O(\delta^{\ell_T+1}), \quad \Psi' = \frac{\ell_T}{k} c_T A_b \delta^{\ell_T-1} + O(\delta^{\ell_T}).\tag{NT.17}$$

Thus ϑ_c vanishes on that hyperplane. When $k \geq 3$ the other nonzero factors can be chosen in a dense subset of each hyperplane, so polynomial continuity gives vanishing on every such hyperplane; infinitely many distinct T force $\vartheta_c = 0$. For $k = 2$ and $u \notin \mathcal{Z}_h + \mathcal{Z}_h$, every remaining second factor $a(u - T)$ is nonzero; infinitely many distinct univariate roots give the same conclusion. Since $\mathcal{Z}_h + \mathcal{Z}_h$ is countable, $\mathcal{N}_a > 0$ almost everywhere for every finite nonempty frame. A nonzero vector polynomial C_{ap} is nonzero outside a finite set: one nonzero component polynomial suffices to bound its common real zero set. Integrating proves $H_a^N > 0$. This argument also includes nonzero source relations whose arithmetic class is zero.

Set $\mathcal{R}_M^N = \text{ran } \mathbf{N}_M$ and $\mathcal{R}_{L|M}^N = \text{ran}(\mathbf{N}_L F)$. The concrete normal transition and its inverse on these specified images are

$$\begin{aligned}\mathcal{Q}_{LM} : \mathcal{R}_M^N &\longrightarrow \mathcal{R}_{L|M}^N, & \mathcal{Q}_{LM} r &= (I - \widehat{\Pi}_L)r, & \mathcal{Q}_{LM} \mathbf{N}_M p &= \mathbf{N}_L F p, \\ H_{L|M}^N &= F^* H_L^N F > 0, & \mathcal{Q}_{LM}^{-1} r_L &= \mathbf{N}_M (H_{L|M}^N)^{-1} F^* \mathbf{N}_L^* r_L, \\ \|r\|^2 - \|\mathcal{Q}_{LM} r\|^2 &= \|\mathbf{A}_{LM} p\|^2 \quad (r = \mathbf{N}_M p).\end{aligned}\tag{NT.18}$$

Positivity of $H_{L|M}^N$ follows from injectivity of F and of $\mathbf{N}_L$. Substitution of $r_L = \mathbf{N}_L F p$ proves the inverse formula in both orders. Thus $\mathcal{Q}_{LM}$ is a contraction and a linear isomorphism between these finite images, with its complete norm loss specified. No inverse bound independent of the frame level is inserted.

The bounded ambient extension obtained from the exact smaller normal image inverse is

$$\begin{aligned}V_{LM} &= \mathbf{N}_L F (H_M^N)^{-1} \mathbf{N}_M^* = (I - \widehat{\Pi}_L) \Pi_M^N : \mathcal{K} \rightarrow \mathcal{K}, \\ \Pi_M^N &= \mathbf{N}_M (H_M^N)^{-1} \mathbf{N}_M^*, & \|V_{LM}\| &\leq 1, & V_{LM}|_{\mathcal{R}_M^N} &= \mathcal{Q}_{LM}.\end{aligned}\tag{NT.18a}$$

The equality follows by multiplying the first line of (NT.18) by $(H_M^N)^{-1} \mathbf{N}_M^*$. The two factors on the right are orthogonal projections and hence contractions. The smaller normal image is fixed by Π_M^N , proving the restriction statement. For $L = M$ this ambient extension is Π_M^N and restricts to the identity precisely on its declared image.

Retain the original arithmetic target, full quotient unit and derivative,

$$\begin{aligned}\mathfrak{A}_h &= \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)), \\ U &= \prod_i v_h(s_i), \quad v = [U], \quad \delta = \frac{1}{k} \sum_i \partial_{s_i}, \quad \beta = v^{-1}[\delta U], \\ (j_a^{\log})_{\alpha, n} &= nv[\theta_\alpha^a S^{n-1}] + \beta v[\theta_\alpha^a S^n].\end{aligned}\tag{NT.19}$$

At $n = 0$ the first term is zero; no negative power is taken. The full Hermite class, multiplicities, $g = 2\xi$ and unit derivatives are retained. The identity $\delta S = 1, \delta z_i = 0$ proves these columns by the product rule applied before taking the quotient. Equation (NT.4) then proves $j_L^{\log} F = j_M^{\log}$ entry by entry. On the actual normal image put $\text{Obs}_a^{\log} r = j_a^{\log}(H_a^N)^{-1} \mathbf{N}_a^* r$. The exact arithmetic transition is

$$\boxed{\text{Obs}_L^{\log} \mathcal{Q}_{LM} = \text{Obs}_M^{\log} \quad \text{on } \mathcal{R}_M^N.}\tag{NT.20}$$

Indeed both sides on $r = \mathbf{N}_M p$ are $j_M^{\log} p$ by (NT.18)–(NT.19). Their original source path is graph addition, followed by $-i\mathcal{U}_k^{-1}$ and the complete original arithmetic jet; (NT.16) preserves that path exactly. For $h = 1$ the arithmetic target is the zero algebra and the observation is its unique zero map; every analytic identity above continues unchanged.

To retain any declared target metric as well, choose its fixed original quotient basis and let $G_M^{\text{tar}}, G_L^{\text{tar}} > 0$ be the two declared Hermitian matrices in that same basis. The coordinate identity $\tau_{LM} : \mathfrak{A}_h \rightarrow \mathfrak{A}_h$ has the exact comparison operator

$$\begin{aligned}D_{LM}^{\text{tar}} &= (G_M^{\text{tar}})^{-1} G_L^{\text{tar}}, \quad (D_{LM}^{\text{tar}})^* G_M^{\text{tar}} = G_M^{\text{tar}} D_{LM}^{\text{tar}} = G_L^{\text{tar}}, \\ \|\text{Obs}_L^{\log} \mathcal{Q}_{LM} \mathbf{N}_M p\|_{G_L^{\text{tar}}}^2 &= p^* (j_M^{\log})^* G_L^{\text{tar}} j_M^{\log} p, \\ \|\text{Obs}_L^{\log} \mathcal{Q}_{LM} \mathbf{N}_M p\|_{G_L^{\text{tar}}}^2 - \|\text{Obs}_M^{\log} \mathbf{N}_M p\|_{G_M^{\text{tar}}}^2 &= p^* (j_M^{\log})^* (G_L^{\text{tar}} - G_M^{\text{tar}}) j_M^{\log} p.\end{aligned}\tag{NT.20a}$$

Direct multiplication proves the comparison identities, and (NT.20) proves the two observed-form identities. This specifies both target forms through the actual coordinate map without assigning a sign to their difference. In the zero target all these are the unique zero forms and maps, with no positive-dimensional inverse needed.

27.4 Every variable-frame term

Let $G_M(u), G_L(u)$ be smooth invertible matrices of their declared sizes. The changed frame and coefficients are $j_a^G = j_a G_a, c_a^G = G_a^{-1} c_a$. The same inclusion now has the generally variable matrix

$$E^G = G_L^{-1} E G_M, \quad R^G = G_M^{-1} R G_L, \quad \Gamma_a^G = G_a^{-1} \Gamma_a G_a + G_a^{-1} G'_a.\tag{NT.21}$$

For any such variable inclusion the exact formulas are

$$\begin{aligned}\Gamma_M^G &= R^G (\Gamma_L^G E^G + (E^G)'), \quad K^G = (E^G)' + \Gamma_L^G E^G - E^G \Gamma_M^G, \\ K^G &= G_L^{-1} K G_M, \quad A^G = A G_M, \quad N_a^G = N_a G_a, \quad \Delta_{LM}^G = G_M^* \Delta_{LM} G_M.\end{aligned}\tag{NT.22}$$

To prove the first equality, differentiate $j_M^G = j_L^G E^G$ before taking its mixed Gram: $B_M^G = (E^G)^* B_L^G E^G + (E^G)^* W_L^G (E^G)'$. Multiplication by $(W_M^G)^{-1}$ gives it. In the second line, differentiate $G_L^{-1} E G_M$ and substitute both transformed connections. The terms containing G'_L and G'_M cancel separately, leaving $G_L^{-1} K G_M$. The original projection images are unchanged; applying $I - \Pi_a$ to $(j_a G_a)'$ therefore gives $N_a G_a$. This proves the remaining statements and every variable-inclusion version of (NT.9)–(NT.12). In particular the term $(E^G)'$ cannot be deleted from the transition.

For the same original polynomial source, $C_a^G = G_a^{-1} C_a$ and $C_L^G F = E^G C_M^G$. Its tangential and normal amplitudes obey

$$\mathbf{N}_a^G = \mathbf{N}_a, \quad \mathbf{T}_a^G = \mathbf{T}_a, \quad \mathbf{A}_{LM}^G = \mathbf{A}_{LM}.\tag{NT.23}$$

The first and third follow from (NT.22); in the second, the derivative of G_a^{-1} cancels the connection term $G_a^{-1}G'_a$. Thus the literal source matrices in (NT.14), the normal-image transition, its inverse and the arithmetic observation are all the same maps after their explicit coefficient transport. The transformed coefficient functions are exactly this image of the original filtered source; their degree bounds are not replaced by a new polynomial frame convention.

27.5 Nested chains and both quadratic branches

For a third level $T \geq L$ retain $E_{TM} = E_{TL}E_{LM}$ and $F_{TM} = F_{TL}F_{LM}$. Expansion of (NT.8) proves

$$\begin{aligned} K_{TM} &= K_{TL}E_{LM} + E_{TL}K_{LM}, & A_{TM} &= A_{TL}E_{LM} + A_{LM}, \\ \Delta_{TM} &= \Delta_{LM} + E_{LM}^* \Delta_{TL} E_{LM}, & H_{TM}^{\text{tr}} &= H_{LM}^{\text{tr}} + F_{LM}^* H_{TL}^{\text{tr}} F_{LM}. \end{aligned} \quad (\text{NT.24})$$

The two ambient summands of A_{TM} lie in $V_T \cap V_L^\perp$ and $V_L \cap V_M^\perp$; they are orthogonal. Taking their Gram proves the third identity, and integration with (NT.4) proves the fourth. Induction gives the same mutually orthogonal sum and positive telescoping formula along every finite nested chain. The actual graph maps and normal maps compose:

$$\mathcal{U}_{TL}\mathcal{U}_{LM} = \mathcal{U}_{TM}, \quad \mathcal{Q}_{TL}\mathcal{Q}_{LM} = \mathcal{Q}_{TM} \quad \text{on the corresponding original source image.} \quad (\text{NT.25})$$

For the normal maps this follows directly from $(I - \hat{\Pi}_T)(I - \hat{\Pi}_L) = I - \hat{\Pi}_T$. For the graph maps, both compositions are the two orthogonal projections of the same graph-added vector. These proofs retain the intermediate positive transition forms. The ambient normal extensions also compose on all of $\mathcal{K}$: $V_{TL}V_{LM} = V_{TM}$. Indeed their matrix product contains $(H_L^N)^{-1}\mathbf{N}_L^*\mathbf{N}_L = I_{q_L}$, leaving $\mathbf{N}_T F_{TL} F_{LM} (H_M^N)^{-1}\mathbf{N}_M^*$.

Now put $x = u^2 > 0$, $v = \sqrt{x}$ and, at each level, keep

$$M_a = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}, \quad H_a = M_a^* \frac{\text{diag}(W_a(v), W_a(-v))}{2v} M_a, \quad Q_a^n = M_a^* \frac{\text{diag}(\mathcal{N}_a(v), \mathcal{N}_a(-v))}{2v} M_a. \quad (\text{NT.26})$$

Let $\mathcal{A}_a = M_a^{-1} \text{diag}(\Gamma_a(v)/(2v), -\Gamma_a(-v)/(2v)) M_a + M_a^{-1} M'_a$ be the complete paired connection. Its derivative includes the original $M_a^{-1} M'_a = \text{diag}(0, I/(2x))$. Write $\tilde{E} = \text{diag}(E, E)$ and

$$\begin{aligned} \mathcal{K}_{LM} &= \mathcal{A}_L \tilde{E} - \tilde{E} \mathcal{A}_M = M_L^{-1} \text{diag}\left(\frac{K(v)}{2v}, -\frac{K(-v)}{2v}\right) M_M, \\ H_M &= \tilde{E}^* H_L \tilde{E}, \quad \tilde{E}^* H_L \mathcal{K}_{LM} = 0, \\ \boxed{Q_M^n &= \tilde{E}^* Q_L^n \tilde{E} + 4x \mathcal{K}_{LM}^* H_L \mathcal{K}_{LM}.} \end{aligned} \quad (\text{NT.27})$$

The identity $M_L \tilde{E} = \text{diag}(E, E) M_M$ proves the first two transport assertions by substitution. Applying (NT.10) on both branches gives the normal identity: multiplication by $4x = 4v^2$ cancels precisely the squares of $1/(2v)$ in $\mathcal{K}_{LM}$, leaving $M_M^* \text{diag}(\Delta_{LM}(v), \Delta_{LM}(-v)) M_M/(2v)$. Thus both original signs and both Jacobian factors are retained. For $\boldsymbol{\eta}_M = (a_M, b_M)^t$ the paired inclusion is $\boldsymbol{\eta}_L = \tilde{E} \boldsymbol{\eta}_M$ and

$$(\partial_x + \mathcal{A}_L) \tilde{E} \boldsymbol{\eta}_M = \tilde{E} (\partial_x + \mathcal{A}_M) \boldsymbol{\eta}_M + \mathcal{K}_{LM} \boldsymbol{\eta}_M. \quad (\text{NT.28})$$

Weighted orthogonality in (NT.27) shows that its tangential energy gain is exactly the normal energy loss in that equation. It follows pointwise, before integration, that the complete FC.15 energy is the same on the two included presentations.

In variable frames the paired inclusion is instead $\mathcal{E}(x) = M_L^{-1} \text{diag}(E^G(v), E^G(-v)) M_M$. The corresponding transition is exactly $\mathcal{E}' + \mathcal{A}_L^G \mathcal{E} - \mathcal{E} \mathcal{A}_M^G$; applying (NT.22) on both branches and the product rule proves this formula, including its derivative. The transported original density retains $(\mathcal{A}_a)^* H_a + H_a \mathcal{A}_a = H'_a + H_a/(2x)$ at both levels, since this identity is obtained by differentiating the same two original branch densities. No evenness assumption is used in (NT.26)–(NT.28).

27.6 Exact zero and endpoint scopes

The identities above hold at every real u . If a coefficient vector or admitted polynomial is zero, each displayed linear map sends it to its corresponding zero. If the smaller frame is empty, its coefficient and source spaces are zero and each inclusion, transition and inverse-on-zero-image is the unique such map. For equal frames with an invertible constant E , the weighted projection Q is the identity, so $K = A = \Delta_{LM} = 0$; the normal map is transported exactly by E . For unequal frames, (NT.11) is the precise equality criterion; strict positivity of the actual finite $\mathcal{N}_L$ does not force strict positivity of Δ_{LM} . The maps in (NT.9) and their Gram identity give the exact relation in every such case.

For a quartet packet and $k = 2$, the original amplitude is even. At $u = 0$ its product derivative is $\frac{1}{2}(a'(y)a(-y) + a(y)a'(-y)) = 0$. Every fixed relative frame therefore has $j'_a(0) = \Gamma_a(0) = N_a(0) = 0$ and $K_{LM}(0) = A_{LM}(0) = \Delta_{LM}(0) = 0$. Under a smooth variable frame, $\Gamma_a^G(0)$ can have the retained term $G_a(0)^{-1}G'_a(0)$; (NT.22) still gives $K^G(0) = N_a^G(0) = 0$. The integrated finite-source normal inverse in (NT.18) remains valid because its positivity is an almost-everywhere and integrated assertion, proved in (NT.17).

Let $\mathcal{D}_a$ denote the coefficient functions locally absolutely continuous on $\mathbb{R}$ for which $\int c^* W_a c$ and $\int \|\partial_u(j_a c)\|^2$ are finite. For the constant inclusion, the exact domain correspondence is

$$E\mathcal{D}_M = \mathcal{D}_L \cap \{c_L : c_L(u) \in \text{ran } E \text{ for almost every } u\}. \quad (\text{NT.29})$$

The forward inclusion follows from $j_L E c = j_M c$ and constant E . Conversely, use any constant left inverse of E , for example $(E^* E)^{-1} E^*$, to obtain a locally absolutely continuous c_M from c_L . The closed fixed subspace condition and continuity give $c_L = E c_M$ everywhere. The same amplitude and derivative identities then give both required finite integrals. This proves both inclusions without identifying distinct coefficient metrics.

On the quadratic branches the original domain has the exact trace condition

$$\lim_{x \downarrow 0} (a_M(x) + i\sqrt{x} b_M(x)) = \lim_{x \downarrow 0} (a_M(x) - i\sqrt{x} b_M(x)). \quad (\text{NT.30})$$

The analogous condition after $\tilde{E}$ is equivalent because E is injective. The finite norm and energy on each open branch give finite traces: near zero, W_M has positive upper and lower bounds and Γ_M is bounded, so the coefficient and its ordinary derivative are both locally square-integrable. The fundamental theorem and Cauchy-Schwarz give the one-sided limits. Integration by parts then shows that unequal limits give their jump times δ_0 , whereas equality joins them into the original weak-derivative domain. This proves the trace assertion. For smooth invertible gauges through zero the matching condition is transported by their common value at zero. For gauges specified only on the punctured branches one must retain the original condition with $G_M(v)c_M^G(v)$ and $G_M(-v)c_M^G(-v)$; it is exactly (NT.30) applied to the original coefficient. No new endpoint value is assigned by the half-line formulas alone.

Finally every displayed linear source, graph and observation map has the original fixed-support lift: on a fixed support label it acts by the proved linear map, preserves a supported zero as a supported zero, and sends external absence to external absence. Its additivity and scalar laws follow directly from linearity. In particular a nonzero polynomial relation is carried by $P_{Fp}^L = P_p^M$ before any full Hermite quotient, and its arithmetic derivative is the unchanged value (NT.19)–(NT.20).

27.7 One independent exact original-coordinate calibration

This finite calibration has ambient space $\mathbb{C}^5$ with its standard Hermitian form. It checks the proved transition using literal polynomial matrices, without identifying this finite model with the arithmetic

amplitude (NT.1). Keep

$$p(u) = 1 + u, \quad q(u) = 1 + u + u^2, \quad j_L(u) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & 2p & -p \\ 2q & -q & q \end{pmatrix}, \quad E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

$$j_M(u) = j_L(u)E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & p \\ 3q & 0 \end{pmatrix}. \quad (\text{NT.31})$$

The first three rows make j_L injective, and E is injective. For $a = (1, 2, -1)^t$ and $b = (2, -1, 1)^t$, direct multiplication gives

$$W_L = I_3 + p^2 aa^t + q^2 bb^t > 0, \quad W_M = \begin{pmatrix} 2 + 9q^2 & 1 \\ 1 & 2 + p^2 \end{pmatrix} > 0,$$

$$B_L = pp'aa^t + qq'bb^t = W'_L/2, \quad B_M = \text{diag}(9qq', pp') = W'_M/2,$$

$$W_M(1) - W_M(-1) = \begin{pmatrix} 72 & 0 \\ 0 & 4 \end{pmatrix}, \quad [W_M(0), W_M(1)] = \begin{pmatrix} 0 & -69 \\ 69 & 0 \end{pmatrix}, \quad (\text{NT.32})$$

$$W_L(1) - W_L(-1) = 4aa^t + 8bb^t, \quad [W_L(0), W_L(1)] = \begin{pmatrix} 0 & 25 & -15 \\ -25 & 0 & -5 \\ 15 & 5 & 0 \end{pmatrix}.$$

Positivity follows from the squared norm of the injective original maps. Every displayed difference is nonzero, so both original weights are non-even and have noncommuting values.

At $u = 1$, put $J = j_L(1)$ and $D = j'_M(1)$. Their matrices and the actual smaller connection are

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 4 & -2 \\ 6 & -3 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 9 & 0 \end{pmatrix}, \quad (\text{NT.33})$$

$$W_L(1) = \begin{pmatrix} 41 & -10 & 14 \\ -10 & 26 & -17 \\ 14 & -17 & 14 \end{pmatrix}, \quad W_M(1) = \begin{pmatrix} 83 & 1 \\ 1 & 6 \end{pmatrix}, \quad \Gamma_M(1) = \frac{1}{497} \begin{pmatrix} 486 & -2 \\ -81 & 166 \end{pmatrix}.$$

An independent construction of the transition projection starts with

$$v = (171, -277, -603, 440, 48)^t.$$

Multiplication verifies $v = J(171, -277, -603)^t$, $(JE)^*v = 0$, and $v^*v = 665483$. The larger and smaller column spaces have dimensions three and two. Their orthogonal difference is therefore exactly the line spanned by v , so

$$\Pi_L(1) - \Pi_M(1) = \frac{vv^*}{665483}, \quad A(1) = \frac{8}{665483}v \begin{pmatrix} 54 & 55 \end{pmatrix},$$

$$N_L(1)E = \frac{1}{1339} \begin{pmatrix} -1458 & -146 \\ 459 & -202 \\ -567 & 92 \\ 54 & 55 \\ 225 & 6 \end{pmatrix}. \quad (\text{NT.34})$$

For the last matrix define $T = 1339^{-1} \begin{pmatrix} 1458 & 146 \\ -459 & 202 \\ 567 & -92 \end{pmatrix}$. Direct multiplication gives $W_L(1)T = J^*D = \begin{pmatrix} 54 & 2 \\ -27 & 4 \\ 27 & -2 \end{pmatrix}$. Thus $D - JT$ is exactly the displayed matrix $N_L(1)E$. It is orthogonal to every column of J and hence to v . This verifies both normal components independently of the proposed connection difference. Their full Grams are

$$A(1)^*A(1) = \frac{1}{665483} \begin{pmatrix} 186624 & 190080 \\ 190080 & 193600 \end{pmatrix}, \quad E^*\mathcal{N}_L(1)E = \frac{1}{1339} \begin{pmatrix} 2025 & 54 \\ 54 & 55 \end{pmatrix}. \quad (\text{NT.35})$$

Take the original coefficient curve $c(u) = (1 + iu, u^2 - i)^t$. Its directly differentiated vector at one is

$$w = Dc(1) + JEc'(1) = (i, 2, 2 + i, 5 - i, 9 + 18i)^t, \quad w^*w = 441. \quad (\text{NT.36})$$

The three components in (NT.12), with their original values, are

$$t = \frac{1}{497} \begin{pmatrix} 484 + 985i \\ 1079 - 247i \\ 1563 + 738i \\ 2158 - 494i \\ 4356 + 8865i \end{pmatrix}, \quad N_L(1)Ec(1) = \frac{1}{1339} \begin{pmatrix} -1604 - 1312i \\ 257 + 661i \\ -475 - 659i \\ 109 - i \\ 231 + 219i \end{pmatrix}, \quad (\text{NT.37})$$

$$A(1)c(1) = \frac{8(109 - i)}{665483}v,$$

$$w = t + A(1)c(1) + N_L(1)Ec(1), \quad \boxed{441 = \frac{217065}{497} + \frac{58496}{51191} + \frac{320}{103}}.$$

Multiplication of the explicit vectors proves their sum and their squared norms; their orthogonality was proved from the original column spaces above. All three energies are strictly positive. The number 441 is this pointwise original energy. Multiplying c by any smooth compactly supported scalar function equal to one near $u = 1$ retains the same pointwise value and derivative and makes the corresponding integrated energy finite.

The companion exact script `nested_relative_frame_fixture_20260912.py` retains these original matrices and verifies 24 rational-function identities and nonzero witnesses, with identical successful records in ordinary and optimized Python. Its full matrix receipts and independent sample review are pinned in `nested_relative_frame_fixture_20260912_manifest.json`. The full general proof is (NT.1)–(NT.30); the single concrete family above supplies an independent nonzero calculation of the transition and the surviving normal component.

28 Density of the actual relative frames and the limiting derivative graph

This working continuation proves the nested-frame limit for the original arithmetic amplitude. The input is the sum-connection construction (FC.1)–(FC.36), retained at the source hash recorded in the accompanying receipt. All statements below use the original measure, full packet, relative coordinates, polynomial source, and derivative. The proof uses an exponential moment that is established directly from the arithmetic amplitude. No recurrence or Christoffel asymptotic is assumed.

28.1 An exponential bound with the original arithmetic factors

Let

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad h(s) = \prod_{\rho \in \mathcal{Z}} (s - \rho)^{m_\rho},$$

$$d = \deg h, \quad v_h(s) = g(s)/h(s), \quad a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w(t) = |a(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w(t) dt. \quad (\text{ND.1})$$

Here the set of selected zeros is finite and each selected multiplicity is the full multiplicity in g . The quotient is its entire extension at those zeros. The empty product $h = 1$ is allowed. The packet is stable under $\rho \mapsto 1 - \bar{\rho}$, as in FC.1. For a monic polynomial this says $h(1 - \bar{s}) = (-1)^d \bar{h}(s)$; the corresponding identity for g has factor one. Hence on the original line

$$\overline{a(t)} = (-1)^d a(t), \quad \overline{a^{(j)}(t)} = (-1)^d a^{(j)}(t). \quad (\text{ND.2})$$

There is no change of phase or division by the mass μ_h .

Theorem 28.1 (Actual amplitude and all derivative tails). *For every integer $j \geq 0$ and every real b with $0 < b < \pi/4$, there is a finite constant $C_{h,j,b}$ such that*

$$|a^{(j)}(t)| \leq C_{h,j,b} e^{-b|t|} \quad (t \in \mathbb{R}). \quad (\text{ND.3})$$

The amplitude is a nonzero entire function of t . In particular $0 < \mu_h < \infty$, all polynomially weighted derivative integrals used below exist, and its real zero set is discrete.

Proof. We give the tail argument, including the gamma factor. For $\lambda > 0$, $\tau \in \mathbb{R}$, and $0 < \phi < \pi/2$, rotate Euler's gamma integral in the sector with angle $\theta = \text{sgn}(\tau)\phi$, taking $\theta = 0$ when $\tau = 0$. The principal power is holomorphic on the open sector. On the circle of radius R the exponential has modulus at most $e^{-R \cos \phi}$; its product with the power and arc length tends to zero as $R \rightarrow \infty$. The small circular contribution is bounded by a constant times ε^λ and tends to zero. Cauchy's theorem therefore gives

$$\begin{aligned} \Gamma(\lambda + i\tau) &= e^{i\theta(\lambda+i\tau)} \int_0^\infty r^{\lambda+i\tau-1} e^{-e^{i\theta}r} dr, \\ |\Gamma(\lambda + i\tau)| &\leq e^{-\phi|\tau|} \Gamma(\lambda) (\cos \phi)^{-\lambda}. \end{aligned} \quad (\text{ND.4})$$

At $\tau = 0$ the second bound follows directly from Euler's integral. Its constant is bounded when λ ranges over a compact subinterval of $(0, \infty)$.

For completeness, summing the counting function $[x]$ on $\Re s > 1$ gives $\zeta(s) = s \int_1^\infty [x] x^{-s-1} dx$. Splitting $[x] = x - \{x\}$ yields

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (\text{ND.5})$$

The last integral is absolutely and locally uniformly convergent on $\Re s > 0$; its differentiated integrals converge on compact subsets because $\int_1^\infty x^{-1-\sigma} (\log x)^n dx$ is finite for every $\sigma > 0$. Thus (ND.5) continues the same zeta function to that half-plane away from $s = 1$. For $s = \sigma + it$, $1/4 \leq \sigma \leq 3/4$, we have $|s/(s-1)| \leq 5$, $|s|/\sigma \leq 1 + 4|t|$, and consequently $|\zeta(s)| \leq 6 + 4|t|$.

Write $z = t + iv$ with $|v| \leq 1/4$, so that $s = 1/2 + iz = (1/2 - v) + it$. In (ND.4) take $\lambda = (1/2 - v)/2 \in [1/8, 3/8]$ and $\tau = t/2$. The modulus of $\pi^{-s/2}$ is exactly $\pi^{-(1/2-v)/2}$. The two polynomial factors in g , followed by (ND.5), therefore give the bound

$$|g(1/2 + iz)| \leq C_\phi (1 + |t|)^3 e^{-\phi|t|/2} \quad (|\Im z| \leq 1/4). \quad (\text{ND.6})$$

For $|t| \geq \max(1, 2 \max_{\rho \in \mathcal{Z}} |\Im \rho|)$, each factor $|s - \rho|$ is at least $|t|/2$. Thus $|h(s)| \geq 2^{-d} |t|^d$; for the empty product the lower bound is one and the maximum over roots is unnecessary. Division by this original polynomial, without removing its remaining unit, bounds v_h on the unbounded part of the strip. The complementary closed rectangle is compact and the entire quotient is bounded there, including every cancelled root. Choose $2b < \phi < \pi/2$. The polynomial factor times $e^{-(\phi/2-b)|t|}$ is bounded. It follows that $|a(z)| \leq C_{h,b} e^{-b|\Re z|}$ on this strip, with the original factor $1/\sqrt{2\pi}$ included in $C_{h,b}$. The Cauchy estimate on the circle of radius $1/8$ about a real t now gives

$$|a^{(j)}(t)| \leq j! 8^j C_{h,b} e^{b/8} e^{-b|t|}.$$

This proves (ND.3). Entirety follows from cancellation at the specified finite roots. The function cannot vanish identically: the formula for g is nonzero on the real interval $s > 1$, and division by a nonzero polynomial preserves nonidentity. The identity theorem makes the real zeros isolated. Positivity and finiteness of the mass follow from nonidentity and (ND.3). $\square$

28.2 Every centred fibre and its exponential moment

Fix $k \geq 2$, put $r = k - 1$, and retain the real coordinates

$$u = \sum_{i=1}^k t_i, \quad y_i = t_i - u/k \quad (1 \leq i \leq r), \quad y_k = -\sum_{i=1}^r y_i, \quad t_i = u/k + y_i,$$

$$S = k/2 + iu, \quad z_i = s_i - S/k = iy_i, \quad \Psi_u(y) = \prod_{i=1}^k a(u/k + y_i), \quad d\mu_u(y) = |\Psi_u(y)|^2 dy. \quad (\text{ND.7})$$

The independent relative variables are $y_1, \dots, y_r$. The determinant of the map with ordered variables $(u, y_1, \dots, y_r) \mapsto (t_1, \dots, t_k)$ is $(-1)^r$. Indeed moving the first column to the last contributes $(-1)^r$, and the resulting block determinant is $\det \begin{pmatrix} I_r & \mathbf{1}/k \\ -\mathbf{1}^T & 1/k \end{pmatrix} = 1/k + r/k = 1$. In particular the original product measure is exactly $du dy$. For $k = 2$, $v = t_1 - t_2 = 2y_1$ gives $du dv/2$. Neither the centre $k/2$ nor this Jacobian is changed.

For every fixed u , each of the k affine functions $y \mapsto u/k + y_i$ is nonconstant. The inverse image of a real zero of a is a proper affine hyperplane. There are at most countably many such hyperplanes, each of Lebesgue measure zero. Consequently $\Psi_u \neq 0$ almost everywhere for every u , including any fibre on which a particular normal derivative vanishes. The orders of the zeros remain the orders of the original product; the measure assertion does not modify the product on a hyperplane.

For every $0 < \eta < \pi/2$, choose b so that $\eta < 2b < \pi/2$. Since $\sum_{i=1}^k |t_i| \geq \|y\|_1 - r|u|/k$, (ND.3) proves the explicit bound

$$\begin{aligned} \int_{\mathbb{R}^r} e^{\eta\|y\|_1} d\mu_u(y) &\leq C_{h,0,b}^{2k} e^{2br|u|/k} \int_{\mathbb{R}^r} e^{-(2b-\eta)\|y\|_1} dy \\ &= C_{h,0,b}^{2k} e^{2br|u|/k} \left(\frac{2}{2b-\eta} \right)^r < \infty. \end{aligned} \quad (\text{ND.8})$$

This bound is uniform on compact u -intervals. It proves both finite moments and $0 < m_k(u) := \mu_u(\mathbb{R}^r) < \infty$ for every u . The coordinate change and Tonelli's theorem retain the full mass identity

$$\int_{\mathbb{R}} m_k(u) du = \mu_h^k. \quad (\text{ND.9})$$

At fixed relative coordinate the exact derivative is

$$\Psi'_u(y) = \frac{1}{k} \sum_{i=1}^k a'(u/k + y_i) \prod_{j \neq i} a(u/k + y_j). \quad (\text{ND.10})$$

Every derivative of this product, multiplied by any polynomial in (u, y) , is square integrable in $du dy$. To verify this claim directly, every derivative is a finite sum of products of derivatives of the original $a(t_i)$, with the powers of $1/k$ from (ND.7) retained. Each product is bounded by a constant times $e^{-b \sum |t_i|}$ by (ND.3). A polynomial in (u, y) becomes a polynomial in $t_1, \dots, t_k$ under the displayed invertible linear map; each of its monomials is integrable against $e^{-2b \sum |t_i|}$. The same bound on a compact u -interval proves continuity, and all orders of differentiation, of the relevant functions with values in $L^2(dy)$, by dominated convergence.

28.3 A complete weighted polynomial density proof

Theorem 28.2 (Exponential moments imply the required density). *Let μ be a finite positive Borel measure on $\mathbb{R}^r$ such that $\int e^{\eta\|y\|_1} d\mu(y) < \infty$ for some $\eta > 0$. Complex polynomials in all r coordinates are dense in $L^2(\mu)$.*

Proof. All polynomials belong to this space by the exponential moment. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial, using inner products conjugate linear in the first entry, and define the finite complex measure $d\nu = \bar{f} d\mu$. For $0 \leq 2\tau < \eta$, Cauchy-Schwarz gives

$$\int e^{\tau\|y\|_1} d|\nu|(y) \leq \|f\|_{L^2(\mu)} \left(\int e^{2\tau\|y\|_1} d\mu(y) \right)^{1/2} < \infty. \quad (\text{ND.11})$$

Every monomial moment of ν is zero. Fix $\xi \in \mathbb{R}^r \setminus \{0\}$. The function

$$F_\xi(z) = \int e^{iz\xi \cdot y} d\nu(y)$$

is holomorphic on the connected strip $|\Im z| \|\xi\|_\infty < \eta/2$. To see the differentiability, restrict to a compact substrip and choose $\tau < \eta/2$ strictly larger than its bound on $|\Im z| \|\xi\|_\infty$. Every polynomial factor from differentiation is bounded by the remaining exponential slack in (ND.11), which justifies differentiation under the integral. In a disc about zero, the exponential series is absolutely integrable by the same bound. Its n -th coefficient is $i^n \int (\xi \cdot y)^n d\nu(y)/n! = 0$, since this is a polynomial moment. Thus F_ξ vanishes near zero and throughout the strip by the holomorphic identity theorem. In particular $\hat{\nu}(\xi) = F_\xi(1) = 0$. The zero moment also gives $\hat{\nu}(0) = 0$.

Here is the Fourier uniqueness argument needed for this finite complex measure. For $\varepsilon > 0$ put

$$G_\varepsilon(x) = (4\pi\varepsilon)^{-r/2} e^{-|x|^2/(4\varepsilon)} = (2\pi)^{-r} \int_{\mathbb{R}^r} e^{-\varepsilon|\xi|^2} e^{i\xi \cdot x} d\xi. \quad (\text{ND.12})$$

The equality follows in one variable by differentiating the Gaussian Fourier integral and integrating by parts: its derivative is $-x/(2\varepsilon)$ times itself and its value at zero is $\sqrt{\pi/\varepsilon}$. The latter Gaussian integral follows by squaring it and using polar coordinates. Taking products gives (ND.12). Since ν is finite and the frequency Gaussian is integrable, Fubini's theorem gives, with the same positive Fourier sign,

$$(G_\varepsilon * \nu)(x) = (2\pi)^{-r} \int_{\mathbb{R}^r} e^{-\varepsilon|\xi|^2} e^{i\xi \cdot x} \hat{\nu}(-\xi) d\xi = 0.$$

For every bounded uniformly continuous function b , another absolutely convergent Fubini integral yields $\int (b * G_\varepsilon) d\nu = 0$. Moreover $b * G_\varepsilon \rightarrow b$ uniformly: in its difference from b , the integration over $|x| < \delta$ is bounded by the uniform modulus of continuity at δ , while the complementary integral is at most $2\|b\|_\infty \int_{|x| \geq \delta} G_\varepsilon$, which tends to zero after the substitution $x = \sqrt{\varepsilon}z$. Thus $\int b d\nu = 0$ for all such b .

For each nonempty closed set $F \subset \mathbb{R}^r$, the bounded Lipschitz functions $b_n(x) = \max(0, 1 - n \text{dist}(x, F))$ converge pointwise to 1_F . Dominated convergence against $|\nu|$ proves $\nu(F) = 0$; this also holds for the empty closed set. In particular $\nu(\mathbb{R}^r) = 0$. The Borel sets with zero ν -measure form a Dynkin system: complements have measure zero by this total mass identity, and countable disjoint unions have measure zero by countable additivity. Closed sets form a pi-system generating the Borel sigma-algebra. The pi-lambda theorem therefore gives $\nu(A) = 0$ for every Borel set A . Applying this to the positive and negative level sets of $\Re f$ and $\Im f$ shows $f = 0$ almost everywhere. Finally, the orthogonal complement of the closed polynomial span is zero, so the Hilbert orthogonal decomposition proves that span is all of $L^2(\mu)$. $\square$

28.4 The multiplier, literal frames, and their strong limit

Let $\mathcal{H} = L^2(\mathbb{R}^r, dy)$. For each fixed u , the exact multiplication map is

$$\begin{aligned} \mathcal{M}_u : L^2(\mu_u) &\longrightarrow \mathcal{H}, & q &\longmapsto \Psi_u q, \\ \mathcal{M}_u^{-1} F(y) &= \begin{cases} F(y)/\Psi_u(y), & \Psi_u(y) \neq 0, \\ 0, & \Psi_u(y) = 0. \end{cases} \end{aligned} \quad (\text{ND.13})$$

It is a unitary map: its squared norm is exactly $\int |q|^2 |\Psi_u|^2 dy$, and the inverse has weighted squared norm $\int |F|^2 dy$ because the zero set is null. These are maps between the displayed weighted and unweighted spaces. No claim of bounded multiplication by $1/\Psi_u$ on unweighted $\mathcal{H}$ is made or used. Assigning zero on null hyperplanes only selects a representative of a Hilbert class; every analytic amplitude and all of its pointwise zero orders remain those of (ND.7).

For $m \geq 0$ use the full monomial family

$$\begin{aligned} \mathcal{A}_m &= \{\alpha \in \mathbb{N}^r : |\alpha| \leq m\}, \quad \theta_\alpha(z) = z^\alpha, \quad \theta_\alpha(iy) = i^{|\alpha|} y^\alpha, \\ j_m(u) : \mathbb{C}^{\mathcal{A}_m} &\rightarrow \mathcal{H}, \quad j_m(u)c = \Psi_u \sum_{\alpha \in \mathcal{A}_m} i^{|\alpha|} y^\alpha c_\alpha, \end{aligned} \quad (\text{ND.14})$$

$$V_m(u) = \text{ran } j_m(u), \quad W_m(u) = j_m(u)^* j_m(u), \quad \Pi_m(u) = j_m(u) W_m(u)^{-1} j_m(u)^*.$$

All moments defining these matrices are finite by (ND.8). If $c \neq 0$, its displayed polynomial is nonzero. By continuity it is nonzero in a nonempty open set, which has positive μ_u -measure because the density is positive almost everywhere. Thus $W_m(u) > 0$ for every u . Multiplication proves that Π_m is self-adjoint, idempotent, and has range V_m , so it is precisely the orthogonal projection onto that range. The inverse here is the positive source Gram W_m^{-1} ; no inverse of a possibly zero normal Gram is used.

For $n \geq m$, let E_{nm} be literal zero-padding of the coefficient array. The complete inclusion and Gram maps are

$$\begin{aligned} j_n E_{nm} &= j_m, \quad W_m = E_{nm}^* W_n E_{nm}, \quad E_{\ell n} E_{nm} = E_{\ell m}, \\ V_m &\subset V_n, \quad \Pi_n \Pi_m = \Pi_m \Pi_n = \Pi_m. \end{aligned} \quad (\text{ND.15})$$

The first line follows term by term in the original polynomial sum. The second line follows from the nested ranges and orthogonal complements. In particular $\Pi_n - \Pi_m$ is the orthogonal projection onto $V_n \cap V_m^\perp$.

Theorem 28.3 (Full relative frames on every actual fibre). *For every $u \in \mathbb{R}$ and every $F \in \mathcal{H}$,*

$$\lim_{m \rightarrow \infty} \|F - \Pi_m(u)F\|_{\mathcal{H}} = 0. \quad (\text{ND.16})$$

For every finite m , however, $\|I - \Pi_m(u)\|_{\mathcal{H} \rightarrow \mathcal{H}} = 1$.

Proof. The density theorem applies to μ_u by (ND.8), separately for each real u . The phases $i^{|\alpha|}$ are nonzero constants and hence the literal monomials in (ND.14) have precisely the full polynomial span. Applying the onto isometry (ND.13) proves $\bigcup_m V_m(u) = \mathcal{H}$. Given F and $\varepsilon > 0$, choose $v \in V_M(u)$ with $\|F - v\| < \varepsilon$. For $m \geq M$, orthogonal projection minimizes distance, so $\|F - \Pi_m F\| \leq \|F - v\| < \varepsilon$. This proves (ND.16). The Hilbert space $L^2(\mathbb{R}^r)$ is infinite dimensional: indicators of infinitely many disjoint finite positive-measure balls are linearly independent. Each V_m has finite dimension. Its orthogonal complement therefore contains a unit vector, on which $I - \Pi_m$ has norm one. Orthogonal projection contraction gives the opposite inequality, proving the asserted operator norm exactly. $\square$

28.5 A fixed original source and the complete derivative domain

Fix one original polynomial $P(s_1, \dots, s_k)$ of total degree D . In the exact linear coordinates $S = \sum s_i$, $z_i = s_i - S/k$, $\sum z_i = 0$, it has the unique expansion

$$P = \sum_{|\alpha| \leq D} z^\alpha f_\alpha(S), \quad \deg f_\alpha \leq D - |\alpha|. \quad (\text{ND.17})$$

Uniqueness is coefficient comparison in the independent variables $(S, z_1, \dots, z_r)$; substitution by an invertible linear map preserves total degree. For $m \geq D$, retain an admitted source degree $M_m \geq$

$\max(D, m)$ and the full relative family $\mathcal{A}_m$. The fixed source has coefficient column

$$c_{P,m}(u)_\alpha = \begin{cases} f_\alpha(k/2 + iu), & |\alpha| \leq D, \\ 0, & D < |\alpha| \leq m. \end{cases} \quad c_{P,n} = E_{nm} c_{P,m}. \quad (\text{ND.18})$$

Every original constraint $\deg f_\alpha \leq M_m - |\alpha|$ is respected; neither the fixed polynomial nor its coefficient degree is changed as the frame grows. One may in particular take $M_m = m$ for all $m \geq D$, which is the full degree- m source of FC.28.

Put

$$\begin{aligned} p_u(y) &= P(k/2 + iu, iy_1, \dots, iy_r), \quad A_P(u, y) = \Psi_u(y) p_u(y) = j_m(u) c_{P,m}(u), \\ \partial_S^{\text{rel}} &= \frac{1}{k} \sum_{i=1}^k \partial_{s_i}, \quad F_P(u, y) = \Psi'_u(y) p_u(y), \\ \partial_u A_P &= F_P + i \Psi_u(\partial_S^{\text{rel}} P)_u. \end{aligned} \quad (\text{ND.19})$$

The notation $P(S, z)$ in these formulas denotes the exact substitution $s_i = S/k + z_i$; $z_k = -\sum_{i < k} z_i$. At fixed relative coordinates $dS/du = i$, so the displayed derivative includes exactly the factor i/k on the sum of the original coordinate derivatives. By (ND.3) and (ND.10), $A_P, F_P, \partial_u A_P$ belong to $\mathcal{K} = L^2(\mathbb{R}_u; \mathcal{H})$. In fact all their polynomially weighted derivatives belong to $\mathcal{K}$. In particular A_P lies in the actual derivative domain

$$\mathcal{D}_u = \{A \in \mathcal{K} : \partial_u A \in \mathcal{K} \text{ in the distributional sense}\}. \quad (\text{ND.20})$$

The smooth derivative in (ND.19) equals the distributional derivative, as follows by integration by parts against compactly supported test functions and the preceding integrability bounds.

We also retain its original Mellin realization and sign. If $\mathcal{M}f(s) = \int_0^\infty f(x) x^{s-1} dx$, the original test vector F_h has $\mathcal{M}F_h = v_h$, and the polynomial source is $\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}$, where $D_i = -x_i \partial_{x_i}$. The normalized critical-line transform has factor $(2\pi)^{-k/2}$, followed by (ND.7). Integration by parts gives $\mathcal{M}(D_i f) = s_i \mathcal{M}f$, so its image is exactly A_P , with the original product of a 's. Writing $x_i = e^{q_i}$, the Mellin integral becomes the Fourier transform with kernel $e^{it_i q_i}$ of $e^{(q_1 + \dots + q_k)/2} f(e^{q_1}, \dots, e^{q_k})$. The change of variable has squared norm exactly $\int |f|^2 d\mathbf{x}$; Fourier Plancherel and the Jacobian one in (ND.7) give the isometry $\mathcal{U}_k$ in FC.9. On this original test domain,

$$\begin{aligned} \mathcal{U}_k \mathcal{T}_h^{(k)} P &= A_P, \quad \mathcal{U}_k D^{(k)} = S \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \\ D^{(k)} &= \sum_i D_i, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -I. \end{aligned} \quad (\text{ND.21})$$

The second derivative identity follows by differentiating the Fourier integral with $\partial t_i / \partial u = 1/k$; the commutator follows from $-x_i \partial_{x_i} \log x_i = -1$, summed with the displayed coefficient $1/k$. All these operations are on the original test vectors and their logarithmic derivatives. Their spectral images have the established L^2 derivatives, so no new domain or arbitrary assignment of an arithmetic jet is introduced.

In the degree- m frame define the original matrices and maps

$$\begin{aligned} B_m &= j_m^* j'_m, \quad \Gamma_m = W_m^{-1} B_m, \quad N_m = (I - \Pi_m) j'_m, \quad \mathcal{N}_m = N_m^* N_m, \\ t_m(P) &= j_m(c'_{P,m} + \Gamma_m c_{P,m}), \quad r_m(P) = N_m c_{P,m}. \end{aligned} \quad (\text{ND.22})$$

Every term is the FC.6 term in the literal enlarged frame. Since $j_m \Gamma_m = \Pi_m j'_m$, the product rule gives

$$\begin{aligned} t_m(P) &= \Pi_m \partial_u A_P, \quad r_m(P) = (I - \Pi_m) \partial_u A_P = (I - \Pi_m) F_P, \\ \partial_u A_P &= t_m(P) + r_m(P), \quad \langle t_m(P, u), r_m(P, u) \rangle_{\mathcal{H}} = 0, \\ \|\partial_u A_P\|_{\mathcal{H}}^2 &= \|t_m(P)\|_{\mathcal{H}}^2 + \|r_m(P)\|_{\mathcal{H}}^2. \end{aligned} \quad (\text{ND.23})$$

For the second equality, the coefficient derivative in (ND.19) belongs to $V_m(u)$, so its orthogonal complement is zero. This equality identifies a single fixed function F_P , independent of frame degree, which controls every residual.

Theorem 28.4 (Vanishing of the fixed-source actual normal residual). *For every fixed original polynomial P and every real u ,*

$$\begin{aligned} \|r_m(P, u)\|_{\mathcal{H}} &\longrightarrow 0, & \|t_m(P, u) - \partial_u A_P(u)\|_{\mathcal{H}} &\longrightarrow 0, \\ \|r_m(P)\|_{\mathcal{H}} &\longrightarrow 0, & \|t_m(P) - \partial_u A_P\|_{\mathcal{H}} &\longrightarrow 0. \end{aligned} \quad (\text{ND.24})$$

The fibre convergence of both norms is uniform on compact u -intervals. The derivative graph converges, in the original $\mathcal{H} \oplus \mathcal{H}$ norm, to $(\partial_u A_P, 0)$, while its sum is exactly $\partial_u A_P$ at every degree.

Proof. For each u , $F_P(u) \in \mathcal{H}$, and (ND.16) applied to this vector gives the first limit through (ND.23). Furthermore,

$$0 \leq \|r_m(P, u)\|^2 \leq \|F_P(u)\|^2, \quad \int_{\mathbb{R}} \|F_P(u)\|^2 du < \infty. \quad (\text{ND.25})$$

All functions here are measurable; in fact j_m, j'_m, W_m are smooth in u by the compact-interval bounds after (ND.10), and W_m^{-1} is smooth since its determinant is everywhere positive. The dominated convergence theorem proves the integrated residual limit. Since $t_m - \partial_u A_P = -r_m$, it proves the two tangential limits and the asserted graph convergence as well.

For $n \geq m$, (ND.15) gives the exact orthogonal decomposition

$$\begin{aligned} r_m(P, u) &= r_n(P, u) + (\Pi_n - \Pi_m)F_P(u), \\ \|r_m(P, u)\|^2 - \|r_n(P, u)\|^2 &= \|(\Pi_n - \Pi_m)F_P(u)\|^2 \geq 0. \end{aligned} \quad (\text{ND.26})$$

The two terms are orthogonal because the first lies in $V_n^\perp$ and the second in V_n . Each squared residual is continuous by the same smoothness argument. On a compact set K the open sets $\{u : \|r_m(P, u)\|^2 < \varepsilon\}$ are increasing and cover K . A finite subcover, followed by taking its largest index, makes the squared residual less than ε everywhere on K . This proves compact uniform convergence without a global uniform-in-source assertion. $\square$

For a fixed finite source core $\text{span}\{P_1, \dots, P_q\}$ let $R_m c = \sum_j c_j r_m(P_j)$. Cauchy-Schwarz gives

$$\|R_m\|^2 \leq \sum_{j=1}^q \|r_m(P_j)\|_{\mathcal{H}}^2 \rightarrow 0, \quad \|R_m^* R_m\| \rightarrow 0, \quad R_m^* R_m - R_n^* R_n \succeq 0 \ (n \geq m). \quad (\text{ND.27})$$

The last inequality is (ND.26) integrated for each linear combination of the fixed sources. These are exactly the normal Gram matrices restricted by literal inclusion of that source core. The core and its coefficient norm are fixed in this statement. The full degree-varying source dimension, its minimum arithmetic norm, and the size of its normal-image inverse are not replaced by this finite-core limit.

The exact translation to the original source-amplitude metric is also available on this same fixed core. For an independent core put $(G_0)_{ij} = \langle A_{P_i}, A_{P_j} \rangle_{\mathcal{H}}$. It is independent of m and strictly positive: a nonzero linear combination of the original polynomials is nonzero on an open subset of the real critical-line coordinates, and the original product amplitude is nonzero almost everywhere there. Its squared integral is therefore positive. With $H_m = R_m^* R_m$, define

$$K_m = G_0^{-1} H_m, \quad K_m^* G_0 = G_0 K_m = H_m, \quad \|R_m\|_{G_0 \rightarrow \mathcal{H}}^2 \leq \lambda_{\min}(G_0)^{-1} \sum_j \|r_m(P_j)\|_{\mathcal{H}}^2 \longrightarrow 0.$$

The first equalities follow by matrix multiplication, and the bound follows from $c^* G_0 c \geq \lambda_{\min}(G_0) \|c\|^2$ and the same column Cauchy-Schwarz inequality. Multiplication by the positive

square root $G_0^{1/2}$ is the isometry from this original coefficient metric to the Euclidean metric. It carries K_m to the positive matrix $G_0^{-1/2} H_m G_0^{-1/2}$; its largest eigenvalue is exactly $\|R_m\|_{G_0 \rightarrow \mathcal{H}}^2$. Thus K_m also tends to zero in its original metric operator norm. These explicit congruence and comparison maps preserve both the source-amplitude metric and the normal metric.

The arithmetic maps themselves are retained by the same inclusions. For a nonempty packet set $I = (h(s_1), \dots, h(s_k))$, let j_I denote the full finite jet map, and put $U(\mathbf{s}) = \prod_i v_h(s_i)$. At every selected root $v_h(\rho) \neq 0$, since its full zero order has been divided out; hence $v = j_I U$ is a unit in the full tensor jet algebra. The original source and logarithmic derivative observations are

$$J(\mathcal{T}_h^{(k)} P) = v[P], \quad J\mathcal{L}_k \mathcal{T}_h^{(k)} P = j_I(\partial_S^{\text{rel}}(UP)). \quad (\text{ND.28})$$

Indeed the Mellin transform of $\mathcal{T}_h^{(k)} P$ is UP ; insertion of $k^{-1} \sum \log x_i$ differentiates it by ∂_S^{rel} . At a selected tuple the algebra of germs modulo the full powers is finite; a germ with nonzero constant term is invertible by its finite geometric expansion in the nilpotent maximal ideal. This proves the asserted unit property including all mixed jet orders. Since (ND.18) is the identity on P , both quantities in (ND.28) are identical at every degree. No passage of this jet map through a Hilbert-space limit is required for, or asserted by, the density theorem. If $h = 1$, the quotient by $I = (1)$ is the zero algebra and both finite arithmetic targets are zero.

28.6 Invariant frames, sparse frames, and all original zero cases

Let G be any fixed subgroup of the factor permutation group S_k . Its action on $\sum y_i = 0$, expressed in the first r coordinates, is real linear with determinant of absolute value one. To check the determinant, the action on the full t variables is a permutation matrix with determinant ± 1 ; conjugating by (ND.7) leaves u fixed and acts linearly on y , so the remaining block has determinant ± 1 . The product Ψ_u is unchanged by this action. Therefore

$$(\mathbf{U}_g F)(y) = F(g^{-1}y), \quad \mathbf{R}_G = \frac{1}{|G|} \sum_{g \in G} \mathbf{U}_g \quad (\text{ND.29})$$

defines unitary group actions on both $\mathcal{H}$ and $L^2(\mu_u)$; they are intertwined by $\mathcal{M}_u$. The adjoint of $\mathbf{U}_g$ is $\mathbf{U}_{g^{-1}}$, so $\mathbf{R}_G^* = \mathbf{R}_G$. Counting the $|G|$ products $gh = \ell$ for each ℓ gives $\mathbf{R}_G^2 = \mathbf{R}_G$. Its image is the invariant subspace: its image is fixed under every group element, and its value on a fixed vector is that vector. Thus it is exactly the invariant orthogonal projection.

Let $V_m^G(u)$ be Ψ_u times all G -invariant relative polynomials of degree at most m , using any full independent basis of that space with its original coefficients. The degree is preserved by the linear action and by its finite average. Applying $\mathbf{R}_G$ to polynomial approximants in the density theorem proves density in the invariant weighted subspace. Intertwining by (ND.13) therefore gives

$$\overline{\bigcup_m V_m^G(u)} = \mathcal{H}^G, \quad \Pi_m^G(u) \xrightarrow{\text{strongly}} \mathbf{R}_G. \quad (\text{ND.30})$$

To verify the second formula for an arbitrary F , decompose it as $\mathbf{R}_G F + (I - \mathbf{R}_G)F$. The second part is orthogonal to every V_m^G , and the first is approximated by their dense union by the same best-approximation argument as (ND.16). If the original polynomial P is G -invariant, then $A_P, F_P, \partial_u A_P$ are invariant, since the group fixes u and Ψ_u . Its expansion in a full homogeneous invariant basis retains the bound $\deg f_\alpha \leq D - \ell_\alpha$: apply coefficient comparison in powers of S and the homogeneous relative grading to (ND.17). Consequently (ND.23)–(ND.27) hold for these actual invariant frames, with Π_m^G in place of Π_m . In particular their fixed-source residual also tends to zero pointwise, on compact u -intervals, and in $\mathcal{H}$. This includes full symmetric orbit frames without changing the unscaled orbit coefficients.

For an arbitrary nested family of finite polynomial spaces E_m , let $V_m^E(u) = \mathcal{M}_u E_m$ and $V_\infty^E(u) = \overline{\bigcup_m V_m^E(u)}$. Its exact strong projection limit is

$$\Pi_m^E(u) \xrightarrow{\text{strongly}} Q_u^E := \text{proj}_{V_\infty^E(u)}. \quad (\text{ND.31})$$

Here is a proof requiring no density assumption. For $n \geq m$, orthogonality of the nested projections gives $\|\Pi_n^E F - \Pi_m^E F\|^2 = \|\Pi_n^E F\|^2 - \|\Pi_m^E F\|^2$. The squared norms increase and are bounded by $\|F\|^2$. Thus the projections applied to F are Cauchy. Their limit belongs to V_∞^E , and its difference from F is orthogonal to each V_m^E , hence to their closure. This proves (ND.31) and identifies the exact projection map. A fixed source admitted from some level onward is a sum of fixed members of that relative polynomial space times polynomials in S . Its coefficient derivative replaces each such polynomial by i times its S -derivative and therefore belongs to the same relative space. The residual limit is consequently precisely $(I - Q_u^E)\partial_u A_P = (I - Q_u^E)F_P$. This convergence is also in $\mathcal{K}$, by pointwise strong convergence and the integrable bound $4\|F_P(u)\|^2$ for the squared difference from its limiting residual. The limit field is measurable as a pointwise norm limit of measurable projection fields.

The projection in (ND.31) need not be the identity. A concrete actual example uses $E_m = \mathbb{C}$ at every level. Set $b_u = m_k(u)^{-1} \int y_1 d\mu_u(y)$, which is finite by (ND.8). The nonzero vector $F = \Psi_u(y)(y_1 - b_u)$ has zero inner product with Ψ_u , so every $\Pi_m^E F = 0$. It is nonzero because μ_u has positive density almost everywhere and the affine hyperplane $y_1 = b_u$ is null. This gives the exact projection and an explicit failed approximation, without asserting that the sparse space is unrelated to the full one.

All zero and empty cases have concrete maps. The zero source has $A_P = F_P = t_m = r_m = 0$ at every degree. A nonzero source whose arithmetic class $[P]$ or $v[P]$ vanishes is still the same nonzero polynomial in (ND.17), with the same amplitude and convergence; (ND.28) retains its full arithmetic class. The packet $h = 1$ gives the actual $2\xi/\sqrt{2\pi}$ amplitude and satisfies every analytic estimate above. No actual fibre has zero mass for $k \geq 2$, by the hyperplane argument following (ND.7). One may append an initial empty frame $V_{-1} = \{0\}$ with $\Pi_{-1} = 0$; the full nonempty sequence starts with the constant frame at degree zero. Empty source arrays have the unique zero maps.

The original exceptional two-factor fibre is retained. If the packet makes a even, as for the quartet in FC.27, then a' is odd and (ND.10) gives

$$\Psi'_0(y) = \frac{1}{2}(a'(y)a(-y) + a(y)a'(-y)) = 0, \quad j'_m(0) = 0, \quad N_m(0) = 0, \quad r_m(P, 0) = 0. \quad (\text{ND.32})$$

The derivative of the coefficient polynomial may be nonzero; (ND.23) keeps that term in the tangential component. The density proof and the limiting derivative graph require no positive inverse for $\mathcal{N}_m(0)$. Their only fibre inverse is W_m^{-1} , which remains strictly positive there. No conclusion about one-factor relative frames is hidden in the assumption $k \geq 2$: for $k = 1$ the relative coordinate space has one point, the multiplication image is one-dimensional when $a(u) \neq 0$ and is zero when $a(u) = 0$. At such a zero it is not onto the scalar ambient Hilbert space, and the pointwise density-to-identity assertion (ND.16) was explicitly proved only for $k \geq 2$.

28.7 Frame covariance and the exact half-line transport

Let $C_m(u)$ be any smooth invertible change of relative frame. Use $\tilde{j}_m = j_m C_m$ and $\tilde{c}_{P,m} = C_m^{-1} c_{P,m}$. The product and projection remain the same, while the complete connection and normal formulas are

$$\begin{aligned} \tilde{W}_m &= C_m^* W_m C_m, & \tilde{B}_m &= C_m^* B_m C_m + C_m^* W_m C'_m, \\ \tilde{\Gamma}_m &= C_m^{-1} \Gamma_m C_m + C_m^{-1} C'_m, & \tilde{N}_m &= N_m C_m, & \tilde{N}_m \tilde{c}_{P,m} &= r_m(P), \\ \tilde{j}_m (\tilde{c}'_{P,m} + \tilde{\Gamma}_m \tilde{c}_{P,m}) &= t_m(P), & \tilde{E}_{nm} &= C_n^{-1} E_{nm} C_m. \end{aligned} \quad (\text{ND.33})$$

The first two lines follow by differentiating $j_m C_m$ and using the unchanged projection image; its complementary projection annihilates $j_m C'_m$. Differentiating $C_m^{-1} c_{P,m}$ cancels the $C_m^{-1} C'_m$ term and

proves the tangential identity. Finally $\tilde{j}_n \tilde{E}_{nm} = \tilde{j}_m$, so these are the exact typed inclusion maps for the same nested images. Arbitrary growth of C_m is not used to assign an independent coefficient domain: the admitted columns are precisely the pullbacks of (ND.18), and the actual amplitudes and graph norms remain unchanged.

For clarity the residual limit can also be transported into the original two-branch half-line representation without an evenness assumption. For $R \in \mathcal{K}$, let $x = u^2 > 0$, $v = \sqrt{x}$, and set

$$\begin{aligned} R_+(x) &= R(v), \quad R_-(x) = R(-v), \quad e_R(x) = \frac{1}{2}(R_+(x) + R_-(x)), \quad o_R(x) = \frac{R_+(x) - R_-(x)}{2i\sqrt{x}}, \\ R_+ &= e_R + i\sqrt{x} o_R, \quad R_- = e_R - i\sqrt{x} o_R, \\ \|R\|_{\mathcal{K}}^2 &= \int_0^\infty \frac{\|R_+\|^2 + \|R_-\|^2}{2\sqrt{x}} dx = \int_0^\infty \left(\frac{\|e_R\|^2}{\sqrt{x}} + \sqrt{x} \|o_R\|^2 \right) dx. \end{aligned} \tag{ND.34}$$

The first equality is the positive Jacobian on each branch; the second is the parallelogram identity after the displayed exact inverse. These formulas define an onto isometry between the full-line space and the displayed two weighted half-line spaces. Thus (ND.24) carries the actual residual to zero in the sum of both original weighted norms, and carries the full derivative graph to the same transported derivative. Pointwise fibre convergence holds at every $x > 0$ on both branches. At the meeting point $u = 0$, the original smooth amplitudes have their original trace; no new independent half-line boundary values are introduced. The proof does not assert convergence of $\partial_u r_m$, which would require control of a different derivative. It proves exactly the fixed-source convergence of the normal and tangential parts of the original first derivative, with their sum and arithmetic source fixed at every degree.

29 The actual normal-image limit, its graph, and its arithmetic observation

Keep the finite dagger-stable packet, its entire quotient $v_h = g/h$, $g = 2\xi$, and an integer $k \geq 2$. The empty packet $h = 1$ is included. We retain the original amplitude and coordinates

$$a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad t_i = u/k + y_i, \quad y_k = -\sum_{i < k} y_i, \quad \Psi(u, \mathbf{y}) = \prod_{i=1}^k a(t_i), \quad S = k/2 + iu. \tag{NN.1}$$

The original measures are $du d\mathbf{y}$, with the unit Jacobian proved in (FC.2). No mass, phase, residue unit or source coefficient is replaced. Write $\mathcal{H} = L^2(d\mathbf{y})$ and $\mathcal{K} = L^2(du; \mathcal{H})$.

For each integer $m \geq 0$, let $\mathcal{P}_m$ be the complete original polynomial source of total degree at most m . Its relative frame consists of every monomial $\theta_\alpha(\mathbf{z})$ of degree $\ell_\alpha \leq m$ in the $k-1$ independent variables $z_i = s_i - S/k$, $i < k$, with $z_k = -\sum_{i < k} z_i$ retained as the dependent coordinate. Its literal source basis consists of $\theta_\alpha S^n$, $0 \leq n \leq m - \ell_\alpha$, as in (FC.28). Keep its coefficient map $C_m(u)$, frame $j_m(u)$, orthogonal projection $\Pi_m(u)$, connection $\Gamma_m(u)$, and normal map $N_m(u)$ of (FC.6). Thus the actual amplitude, normal and tangential source maps are

$$\begin{aligned} A_m p &= j_m C_m p = \Psi P_p, \\ N_m p &= N_m C_m p = (I - \Pi_m) \partial_u A_m p, \\ T_m p &= j_m (C'_m + \Gamma_m C_m) p = \Pi_m \partial_u A_m p. \end{aligned} \tag{NN.2}$$

All three maps have values in the same original space $\mathcal{K}$. The last two equalities include the derivative of every original power of S ; its factor is inS^{n-1} , not nS^{n-1} .

Let $Q_{\ell m}$ be the literal coefficient inclusion for $\mathcal{P}_m \subseteq \mathcal{P}_\ell$, $\ell \geq m$, and $E_{\ell m}$ its relative-frame inclusion. The coefficient identities and amplitude identity are

$$C_\ell Q_{\ell m} = E_{\ell m} C_m, \quad j_\ell E_{\ell m} = j_m, \quad A_\ell Q_{\ell m} = A_m. \tag{NN.3}$$

They follow by comparing each literal column $\theta_\alpha S^n$. In particular these are inclusions of the original sources, and not an identification of coefficient spaces of different dimensions.

The weighted-density theorem for the amplitude (NN.1) proves $\Pi_m(u) \rightarrow I$ strongly for every u , and proves for every fixed original polynomial P that

$$\begin{aligned} r_m(P) &:= (I - \Pi_m)\partial_u(\Psi P) \longrightarrow 0 \quad \text{in } \mathcal{K}, \\ t_m(P) &:= \Pi_m\partial_u(\Psi P) \longrightarrow \partial_u(\Psi P) \quad \text{in } \mathcal{K}. \end{aligned} \quad (\text{NN.4})$$

Here and below $m \geq \deg P$. The full density proof, including its original exponential estimate, the division across the selected full zero orders, all fibre zero sets and the invariant-source variant, is given in the preceding relative-density section. Equivalently, the first statement of (NN.4) is obtained by applying those actual fibre projections to the one fixed function $\partial_u(\Psi P) \in \mathcal{K}$, then applying dominated convergence to the bound $\|(I - \Pi_m)\partial_u(\Psi P)\| \leq \|\partial_u(\Psi P)\|$. Thus no unproved estimate about a sequence of changing source polynomials is used here.

29.1 Exact contractions of the finite normal images

Put $\mathcal{R}_m^N = \text{ran } N_m$, and retain $H_m^N = N_m^* N_m > 0$ and $L_m^N = (H_m^N)^{-1} N_m^*$ of (FC.29)–(FC.30). Their positivity and the exact source inverse hold for the actual arithmetic amplitude, including $h = 1$. All these source spaces contain the constant monomial and are nonzero. Define the bounded orthogonal projection on $\mathcal{K}$

$$(\mathbb{P}_m f)(u) = \Pi_m(u) f(u). \quad (\text{NN.5})$$

The finite-frame entries are smooth, so this is a measurable contraction; self-adjointness and idempotence hold after integrating their pointwise identities. The relative inclusions give $\mathbb{P}_\ell \mathbb{P}_m = \mathbb{P}_m = \mathbb{P}_m \mathbb{P}_\ell$ for $\ell \geq m$.

The exact transition from one normal image to the next is

$$\begin{aligned} V_{\ell m} : \mathcal{R}_m^N &\longrightarrow \mathcal{R}_\ell^N, \\ V_{\ell m} r &= (I - \mathbb{P}_\ell) r = N_\ell Q_{\ell m} L_m^N r. \end{aligned} \quad (\text{NN.6})$$

To prove both its codomain and its second expression, write uniquely $r = N_m p = (I - \mathbb{P}_m)\partial_u A_m p$. The nested projection identity and (NN.3) give

$$(I - \mathbb{P}_\ell) r = (I - \mathbb{P}_\ell)\partial_u A_m p = N_\ell Q_{\ell m} p.$$

Injectivity follows from injectivity of N_ℓ and of the literal inclusion. Moreover

$$\|V_{\ell m}\| \leq 1, \quad V_{mm} = I, \quad V_{nm} = V_{n\ell} V_{\ell m} \quad (n \geq \ell \geq m). \quad (\text{NN.7})$$

Projection contraction proves the first assertion; the source formula and $L_\ell^N N_\ell = I$ prove the others. The source and transition square commutes exactly:

$$L_\ell^N V_{\ell m} = Q_{\ell m} L_m^N. \quad (\text{NN.8})$$

In particular these changing normal images have a specified algebraic relation even though their Hilbert subspaces are not asserted to be nested by the ambient identity map.

Their complete energy difference on the fixed source is

$$\begin{aligned} r_m(P) &= (\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P) + r_\ell(P), \\ \|r_m(P)\|^2 - \|r_\ell(P)\|^2 &= \|(\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P)\|^2, \\ \|t_\ell(P)\|^2 - \|t_m(P)\|^2 &= \|(\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P)\|^2. \end{aligned} \quad (\text{NN.9})$$

Indeed $\mathbb{P}_\ell - \mathbb{P}_m$, $\mathbb{P}_m$, and $I - \mathbb{P}_\ell$ are mutually orthogonal projections. Apply their orthogonal decomposition to the unchanged derivative amplitude. The full nested-frame connection identities in the preceding section identify this middle component in the original coefficients and also under each stated variable frame change.

29.2 The algebraic limit and its two exact metric presentations

Consider the algebraic direct limit of the injective transitions $V_{\ell m}$. A class represented by $r \in \mathcal{R}_m^N$ is sent to the original polynomial with coefficients $L_m^N r$. Equation (NN.8) proves that the map is well-defined. It is onto $\mathcal{P}_\infty = \mathbb{C}[s_1, \dots, s_k]$, since every polynomial lies in a finite source. If its polynomial is zero then $r = N_m L_m^N r = 0$, so it is injective. We have proved the exact complex-linear isomorphism

$$\operatorname{colim}_{m, V_{\ell m}} \mathcal{R}_m^N \xrightarrow{\sim} \mathcal{P}_\infty. \quad (\text{NN.10})$$

Because (NN.7) is contractive, the limit seminorm is well-defined by

$$\|[r]\|_{N, \infty} = \lim_{\ell \rightarrow \infty} \|V_{\ell m} r\| = \inf_{\ell \geq m} \|V_{\ell m} r\|. \quad (\text{NN.11})$$

Representing the same class at a later index leaves the limit unchanged; representing two classes at a common index proves homogeneity and the triangle inequality. For the fixed polynomial given by (NN.10), (NN.4) proves

$$\|[r]\|_{N, \infty} = 0 \quad \text{for every class.} \quad (\text{NN.12})$$

Thus its Hausdorff seminorm quotient, and its completion, are the zero vector space. This conclusion retains the nonzero algebraic space in (NN.10), together with the exact zero quotient map.

For comparison retain the complete derivative graph $\mathcal{G}_m = \operatorname{ran}(\mathbf{T}_m, \mathbf{N}_m)$ with its original sum norm. Its transitions are

$$W_{\ell m}(\mathbf{T}_m p, \mathbf{N}_m p) = (\mathbf{T}_\ell Q_{\ell m} p, \mathbf{N}_\ell Q_{\ell m} p). \quad (\text{NN.13})$$

Uniqueness of the source proves that these maps are well-defined, injective and compose. Orthogonality and (NN.3) show that both sides have squared norm $\|\partial_u A_m p\|^2$, so every transition is isometric. Addition of the two components commutes with transitions and gives the exact isometry

$$\operatorname{colim}_{m, W_{\ell m}} \mathcal{G}_m \xrightarrow{\text{Add}} \{\partial_u(\Psi P) : P \in \mathcal{P}_\infty\} \subset \mathcal{H}. \quad (\text{NN.14})$$

Its extension identifies the Hilbert completion with the closure of this specified derivative image in the original $\mathcal{H}$. The derivative image is injective in its polynomial source. Indeed $\Psi P \in L^2(du; \mathcal{H})$ and a zero weak derivative in u make every scalar pairing with an $\mathcal{H}$ vector a constant distribution in u . Such a constant belongs to $L^2(du)$ only when zero. Hence $\Psi P = 0$. The nonzero entire one-factor amplitude has a discrete real zero set, so Ψ is nonzero almost everywhere in the original coordinates; a polynomial vanishing almost everywhere there is the zero polynomial.

The algebraic reconstruction maps of (FC.31) give the commuting identity

$$W_{\ell m} \operatorname{Rec}_m = \operatorname{Rec}_\ell V_{\ell m}. \quad (\text{NN.15})$$

It follows by substituting $r = N_m p$. Thus the two algebraic limits are explicitly isomorphic through the original source, while (NN.12) and (NN.14) compute their two distinct retained metrics and their exact completion maps.

29.3 Original arithmetic jets and the size of reconstruction

For a nonempty packet retain the original finite algebra $E = \mathbb{C}[\mathbf{s}]/(h(s_1), \dots, h(s_k))$, its complete Hermite coordinates, and $U = \prod_i v_h(s_i)$. Its jet is a unit. The fixed polynomial observation is the actual derivative jet

$$\mathcal{J}^{\log} P = j_I \left(\frac{1}{k} \sum_i \partial_{s_i}(UP) \right) \in E. \quad (\text{NN.16})$$

It includes the entire product derivative of U at every retained nilpotent order. Formula (FC.34) gives its exact finite matrices $J_m^{\log}$, including the original Mellin factors and the $-i\mathcal{U}_k^{-1}$ arrow from the derivative amplitude. Their coefficient inclusions give $J_\ell^{\log} Q_{\ell m} = J_m^{\log}$, whence

$$\text{Obs}_\ell^{\log} V_{\ell m} = \text{Obs}_m^{\log}, \quad \text{Obs}_m^{\log} = J_m^{\log} L_m^N. \quad (\text{NN.17})$$

These equalities follow from (NN.8), so they define on (NN.10) exactly the original algebraic map $\mathcal{J}^{\log}$.

This map is nonzero, with an explicit original source witness. Choose any retained tuple $\rho = (\rho_1, \dots, \rho_k)$, set $\sigma = \sum_i \rho_i$, and take $P_\rho = S - \sigma$. The complete jet evaluation at the constant coordinate of that tuple is

$$(\mathcal{J}^{\log} P_\rho)(\rho) = U(\rho) \neq 0. \quad (\text{NN.18})$$

The product-rule term multiplying P_ρ vanishes at that tuple; the other term is $U \partial_S^{\text{rel}} S = U$, with its full $1/k$ retained. The nonvanishing follows from taking every selected zero to its full order in h .

There is also a permutation-invariant witness that is itself an original theta relation. Take $P_{\text{rel}} = \sum_i h(s_i) \in I$. Its original full jet is zero, while its retained derivative is

$$\mathcal{J}^{\log} P_{\text{rel}} = \frac{1}{k} U_k \sum_i [h'(s_i)]_I \neq 0. \quad (\text{NN.18a})$$

The term differentiating U still has the factor P_{rel} and is zero in E . If $d = 1$, the sum of derivatives is exactly k . If $d \geq 2$, its degree- $d - 1$ standard remainder terms are the distinct monomials ds_i^{d-1} ; their nonzero coefficients cannot cancel. Multiplication by the full unit U_k proves the nonvanishing in both cases, including all confluent orders. Each $h(s_i)$ term has the original theta primitive with the two matching Koszul signs, so this witness retains the actual relation before its supported-zero quotient. It is available in the full invariant source as well as in the full tensor source.

Fix a genuine original target metric as follows. Choose a degree $m_* \geq k(d - 1)$. The original jet map $J_* p = j_I(U P_p)$ on $\mathcal{P}_{m_*}$ is onto: full monic remainders have total degree at most $k(d - 1)$, and multiplication by $j_I U$ is invertible. With its original amplitude metric $H_{m_*}^0 = A_{m_*}^* A_{m_*}$, the retained minimum-source metric is

$$G_* = (J_*(H_{m_*}^0)^{-1} J_*^*)^{-1} > 0. \quad (\text{NN.19})$$

Indeed for a target vector q , the source $(H_{m_*}^0)^{-1} J_*^* G_* q$ maps to q ; it is orthogonal in the original source metric to $\ker J_*$. Its squared norm is $q^* G_* q$, and adding any kernel vector adds its nonnegative squared norm. This proves the stated actual minimum and its constants.

The exact sizes of the finite inverse, graph reconstruction and arithmetic observation are

$$\begin{aligned} \|L_m^N\|_{N \rightarrow 0}^2 &= \lambda_{\max}((H_m^N)^{-1/2} H_m^0 (H_m^N)^{-1/2}), \\ \|\text{Rec}_m\|^2 &= \lambda_{\max}((H_m^N)^{-1/2} H_m^\partial (H_m^N)^{-1/2}), \\ \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2 &= \lambda_{\max}((H_m^N)^{-1/2} (J_m^{\log})^* G_* J_m^{\log} (H_m^N)^{-1/2}). \end{aligned} \quad (\text{NN.20})$$

Here H_m^0 , H_m^N , and H_m^∂ are the three original forms in (FC.29)–(FC.32). For a nonzero coefficient vector p , the denominator of each norm ratio is $p^* H_m^N p$; the respective numerators are $p^* H_m^0 p$, $p^* H_m^\partial p$, and $p^* (J_m^{\log})^* G_* J_m^{\log} p$. The change of variable $b = (H_m^N)^{1/2} p$, with its exact inverse, proves the displayed eigenvalue formulas without altering any of these original forms.

All three norms tend to infinity as $m \rightarrow \infty$. For the first two use any fixed nonzero polynomial: its original amplitude and its derivative amplitude both have fixed positive norm, whereas (NN.4) makes its normal norm tend to zero and (FC.26) keeps that norm strictly positive at every finite degree. For the third use the single fixed polynomial P_ρ in (NN.18); its G_* target norm is a fixed positive number. Each operator norm is bounded below by this specified ratio for every sufficiently large m , proving divergence. Equivalently, (NN.17) is a nonzero algebraic observation on the limit, and cannot descend continuously through its zero Hausdorff quotient.

Every changing actual minimum metric is retained as well. Define G_m by the same minimum-source construction at degree $m \geq m_*$. Its original-source inclusion makes $G_m \preceq G_*$: the earlier minimum lift is an allowed later lift with exactly the same amplitude norm. Set $B_m = G_*^{-1/2} G_m G_*^{-1/2} > 0$. The precise comparison is

$$\begin{aligned} q^* G_m q &= (G_*^{1/2} q)^* B_m (G_*^{1/2} q), \quad 0 < B_m \preceq I, \\ \|\text{Obs}_m^{\log}\|_{N \rightarrow G_m}^2 &= \lambda_{\max}((H_m^N)^{-1/2} (J_m^{\log})^* G_m J_m^{\log} (H_m^N)^{-1/2}), \\ \lambda_{\min}(B_m) \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2 &\leq \|\text{Obs}_m^{\log}\|_{N \rightarrow G_m}^2 \leq \lambda_{\max}(B_m) \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2. \end{aligned} \quad (\text{NN.21})$$

The first equality is direct multiplication. Its minimum and maximum eigenvalue inequalities applied to $q = J_m^{\log} p$, followed by the same Rayleigh supremum, prove every remaining assertion. Thus the divergence proved for the fixed actual metric is transported to the changing actual metric by an explicit positive operator; its displayed eigenvalues have not been assigned an unproved asymptotic.

29.4 The split lift retains the algebraic source of the zero limit

For a complex vector space V , use the split semimodule $G(V) = \{\tau\} \sqcup V$ over $G(\mathbb{C})$. The external τ is the additive identity, addition of supported vectors is their original vector addition, supported scalar multiplication is the original one, and multiplication involving external τ is external τ . Its supported zero is $e_V = 0_V \neq \tau$. For a linear map $f : V \rightarrow W$, the map $G(f)$ sends τ to τ and a supported vector v to the supported vector $f(v)$. Checking separately external and supported inputs proves additivity and compatibility with every scalar. Composition is the original composition on supported inputs and fixes τ .

Consequently (NN.10), (NN.15), and (NN.17) give exact split-semimodule maps. In particular the seminorm quotient in (NN.12) has the split lift

$$G(\mathcal{P}_\infty) \longrightarrow G(0) = \{\tau, e_0\}, \quad \tau \longmapsto \tau, \quad P \longmapsto e_0 \quad (P \in \mathcal{P}_\infty). \quad (\text{NN.22})$$

Its two fibres are exactly $\{\tau\}$ and the entire supported polynomial source. The algebraic arithmetic map is simultaneously $G(\mathcal{J}^{\log}) : G(\mathcal{P}_\infty) \rightarrow G(E)$. For a nonempty packet it cannot factor through (NN.22): evaluating a proposed factorization on the supported zero polynomial forces the image of e_0 to be e_E , whereas (NN.18) supplies a polynomial whose supported image has nonzero amplitude. This is an exact factorization obstruction between the explicitly constructed maps, with the original source, supported zeros and external absence retained.

For $h = 1$, the algebraic target E is zero, so its arithmetic observation is the zero map and its split lift does factor through (NN.22). The amplitude, density theorem, injective finite normal maps, inverse/graph divergence and the complete graph limit remain unchanged. The arithmetic norm formulas involving a nonzero G_* target are then replaced by their unique zero-target maps. For a full invariant source under a fixed finite permutation group, all constructions restrict through the exact Reynolds maps of the density section; the derivative and source are invariant, $P_\rho = S - \sigma$ is invariant, and the arithmetic target is the corresponding invariant subspace with the restriction of its original target form. Thus the same proofs retain that invariant source and its original metric rather than adding noninvariant vectors. To verify the asserted minimum metric on this subspace, permutations act unitarily on the original tensor amplitude space because they preserve its product measure and its identical one-factor amplitudes. The full jet map is equivariant. Averaging an arbitrary lift of an invariant target with the literal factor $1/|\mathfrak{G}|$ gives an invariant lift of the same target, so the invariant jet map is onto. Uniqueness of the full minimum shows that every permutation fixes that minimum lift. It is consequently the minimum in the invariant source as well, with exactly the restriction of G_m or G_* .

30 The cyclic sum at every conormal depth

This section continues the complete cyclic-sum source, especially its Sections 1–3 and 7–8, and the separately supplied conormal-tower proof. It retains the original packet, arithmetic unit, tensor variables and reflection. The new constructions are the retraction and invariant dimensions at every depth, the exact scalar first-jet image, its full conormal comparison, and the finite cyclic Weyl correction with its complementary map.

30.1 Original coordinates and the exact collided annihilator

Let $k \geq 1$, and let the nonempty original monic packet be

$$h(s) = \prod_{\nu=1}^a (s - \rho_\nu)^{m_\nu}, \quad d = \sum_{\nu=1}^a m_\nu \geq 1. \quad (\text{CC.1})$$

The ρ_ν are distinct, and every m_ν is the full order retained by the packet. In the arithmetic application $g = 2\xi$, $v_h = g/h$, and the chosen packet consists of zeros of g with those full orders. Thus $v_h(\rho_\nu) \neq 0$. For the original arithmetic reflection and pairing, retain the delivered packet hypothesis: the map $\rho \mapsto 1 - \bar{\rho}$ permutes the packet and preserves each full multiplicity. This is the hypothesis used by the involution in (CC.42)–(CC.48). The polynomial-algebra statements below apply to the displayed original roots without any assumption that they lie on the critical line. Put

$$\begin{aligned} P &= \mathbb{C}[s_1, \dots, s_k], \quad h_i = h(s_i), \quad I = (h_1, \dots, h_k), \quad S = \sum_{i=1}^k s_i, \\ B_r &= P/I^r, \quad A_r = B_r^{\mathfrak{S}_k} \quad (r \geq 1), \\ U_r &= \left[\prod_{i=1}^k v_h(s_i) \right]_{I^r} \in A_r^\times. \end{aligned} \quad (\text{CC.2})$$

An entire function in (CC.2) means its full finite Taylor class on each local factor. These classes are compatible under every quotient map. Their invertibility follows from their nonzero constant terms: in a finite local algebra a class $u_0 + n$, with $u_0 \neq 0$ and $n^N = 0$, has inverse $u_0^{-1} \sum_{j=0}^{N-1} (-n/u_0)^j$. This fixes U_r without replacing any of its Taylor coefficients.

For an ordered tuple $\mathbf{t} = (\nu_1, \dots, \nu_k)$, write

$$\begin{aligned} \lambda_{\mathbf{t}} &= \sum_i \rho_{\nu_i}, & D_{\mathbf{t}} &= \sum_i (m_{\nu_i} - 1), & M_{\mathbf{t}} &= \max_i m_{\nu_i}, \\ L_{\mathbf{t},r} &= 1 + D_{\mathbf{t}} + (r-1)M_{\mathbf{t}}, & \ell_{\lambda,r} &= \max_{\lambda_{\mathbf{t}}=\lambda} L_{\mathbf{t},r}, \\ \Lambda_k &= \{\lambda_{\mathbf{t}}\}, & \chi_r(T) &= \prod_{\lambda \in \Lambda_k} (T - \lambda)^{\ell_{\lambda,r}}, & q_r &= \deg \chi_r. \end{aligned} \quad (\text{CC.3})$$

The letter T is the polynomial indeterminate in χ_r ; its evaluation in the original algebra is $T = S$. Equal sums are grouped before taking the maximum.

Theorem 30.1 (Full local depth and the exact scalar contraction). *There are algebra isomorphisms in the original shifted coordinates*

$$B_r \longrightarrow \prod_{\mathbf{t} \in \{1, \dots, a\}^k} \mathbb{C}[z_1, \dots, z_k] / (z_1^{m_{\nu_1}}, \dots, z_k^{m_{\nu_k}})^r, \quad s_i \longmapsto \rho_{\nu_i} + z_i. \quad (\text{CC.4})$$

In the $\mathbf{t}$ -factor an exact basis consists of the monomials

$$z^\alpha, \quad \sum_i \left\lfloor \frac{\alpha_i}{m_{\nu_i}} \right\rfloor \leq r-1. \quad (\text{CC.5})$$

The nilpotency order of $S - \lambda_{\mathbf{t}} = \sum_i z_i$ on that factor is exactly $L_{\mathbf{t},r}$. Consequently

$$I^r \cap \mathbb{C}[S] = (\chi_r(S)), \quad C_r := \mathbb{C}[T]/(\chi_r) \xrightarrow{\alpha_r} A_r, \quad [F] \mapsto [F(S)] \quad (\text{CC.6})$$

is an injective unital algebra map. The original arithmetic coefficient map on the cyclic source is $\eta_r = M_{U_r, \alpha_r} : C_r \rightarrow A_r$. It is injective and intertwines multiplication by T and S .

Proof. For $r = 1$, separate-variable Chinese remainder division gives $I = \bigcap_{\mathbf{t}} J_{\mathbf{t}}$, where $J_{\mathbf{t}} = ((s_i - \rho_{\nu_i})^{m_{\nu_i}})_i$. Two different tuples give comaximal ideals: in a coordinate where their roots differ the two one-variable powers are coprime, and their Bezout identity already gives 1 in the sum of the two ideals. For finitely many pairwise comaximal ideals, intersection equals product. Therefore $I^r = (\prod J_{\mathbf{t}})^r = \prod J_{\mathbf{t}}^r = \bigcap J_{\mathbf{t}}^r$. The same Bezout identity, or its power expansion, proves that the powers remain comaximal. This proves (CC.4).

The local ideal is monomial. A monomial belongs to its r -th power precisely when some non-negative integers b_i , with $\sum b_i = r$, satisfy $m_{\nu_i} b_i \leq \alpha_i$ for every i . Such b_i exist precisely when the sum of the displayed floors is at least r . This proves (CC.5), including independence. Write $\alpha_i = m_{\nu_i} b_i + c_i$, $0 \leq c_i < m_{\nu_i}$. The greatest surviving total degree is $D_{\mathbf{t}} + (r-1)M_{\mathbf{t}}$: put every allowed block b_i in one coordinate attaining $M_{\mathbf{t}}$, and put $c_i = m_{\nu_i} - 1$ in every coordinate. All monomials of larger degree vanish. In $(\sum z_i)^{D_{\mathbf{t}}+(r-1)M_{\mathbf{t}}}$ that chosen monomial has the nonzero coefficient $(D_{\mathbf{t}} + (r-1)M_{\mathbf{t}})! / \prod \alpha_i!$. Distinct monomials cannot cancel. The asserted exact order follows. The annihilator in $\mathbb{C}[T]$ of evaluation on this factor is therefore $((T - \lambda_{\mathbf{t}})^{L_{\mathbf{t},r}})$. Intersecting these ideals takes the maximum at each repeated λ , proving (CC.6). The class of $F(S)$ is invariant, and multiplication by the unit U_r preserves injectivity and the stated intertwining. $\square$

Proposition 30.2 (Degree, scalar powers and all endpoints). *For the nonempty packet,*

$$\begin{aligned} q_r &\geq k(d-1) + (r-1)d + 1, \\ \chi_r \mid \chi_1^r, \quad \chi_r = \chi_1^r &\iff (r=1) \text{ or } (k=1) \text{ or } (m_{\nu} = 1 \ \forall \nu). \end{aligned} \quad (\text{CC.7})$$

These are assertions about the complete collided polynomials.

Proof. Repeated monic division in the unchanged variables gives the unique finite expansion

$$F = \sum_{\mathbf{b} \in \mathbb{N}^k, 0 \leq c_i < d} a_{\mathbf{b}, \mathbf{c}} \prod_i h_i^{b_i} s_i^{c_i}. \quad (\text{CC.8})$$

The leading monomials have the pairwise distinct exponent vectors $d\mathbf{b} + \mathbf{c}$, so a finite dependence is impossible. Equivalently P is free over $\mathbb{C}[h_1, \dots, h_k]$ with the displayed remainder basis. Membership in I^r is exactly the vanishing of the coefficients with $\sum b_i < r$. It follows that the highest homogeneous part of every element of I^r belongs to $(s_1^d, \dots, s_k^d)^r$. Indeed each highest-degree basis term has that property, and its lower terms have smaller total degree.

Put $D = k(d-1) + (r-1)d$. The monomial with exponents $(rd-1, d-1, \dots, d-1)$ survives the corresponding monomial ideal and has degree D . For each $0 \leq j \leq D$, choose a divisor of this monomial of degree j . Its floor sum is still at most $r-1$, and its coefficient in S^j is a nonzero multinomial coefficient. Thus the leading homogeneous part S^j of a monic polynomial of degree j in S cannot belong to that monomial ideal. Such a polynomial cannot lie in I^r . Apply this to $\chi_r(S)$ to obtain the lower bound.

For every tuple, $M_{\mathbf{t}} \leq 1 + D_{\mathbf{t}} = L_{\mathbf{t},1}$, so $L_{\mathbf{t},r} \leq rL_{\mathbf{t},1}$. Taking maxima proves divisibility. Each of the three stated equality cases follows directly from (CC.3). Conversely suppose $r, k > 1$, and put $M = \max_{\nu} m_{\nu} > 1$. Choose a root ρ of order M and its pure k -tuple. At $\lambda = k\rho$, every tuple has $D_{\mathbf{t}} \leq k(M-1)$ and $M_{\mathbf{t}} \leq M$; the pure tuple attains both bounds. Therefore

$$r\ell_{\lambda,1} - \ell_{\lambda,r} = (r-1)(k-1)(M-1) > 0. \quad (\text{CC.9})$$

This proves strict divisibility even when other tuples collide there. $\square$

For one root of order m , (CC.3) reads $\chi_r(T) = (T - k\rho)^{1+k(m-1)+(r-1)m}$. For $k = 1$, $\chi_r = h^r$ and α_r is an isomorphism. For $d = 1$, $\chi_r = (T - k\rho)^r$, but the invariant algebra can have a complement to the cyclic image: for $k \geq 2$, $r = 3$, the two symmetric degree-two polynomials $\sum z_i^2$ and $\sum_{i < j} z_i z_j$ are independent. The empty packet $h = 1$ is treated separately: $I = P$, $B_r = A_r = C_r = 0$, $\chi_r = 1$, and every displayed finite linear map is the zero map. No formula involving a chosen root or q_r^{-1} is applied in that case. The arithmetic source itself need not be zero merely because its finite packet projection is zero.

The attaining tuple can change with depth. As a declared algebraic example, take roots $1/2 + it$, $t = 0, 1, 2, 3$, of orders $6, 4, 4, 1$, and $k = 2$. At $\lambda = 1 + 3i$ the only unordered pairs are $(0, 3)$ and $(1, 2)$; they give respectively $6r$ and $4r + 3$. Thus $\ell_{\lambda,r} = \max(6r, 4r + 3)$. This example asserts no location or multiplicity of an arithmetic zero.

30.2 Unscaled invariant bases and a full-unit retraction

For an occupation vector $\mathbf{n} = (n_1, \dots, n_a)$ with $\sum n_\nu = k$, put $\lambda_{\mathbf{n}} = \sum n_\nu \rho_\nu$. In the factor for a fixed ordered tuple of that occupation, its stabilizer is $\prod_\nu \mathfrak{S}_{n_\nu}$. For each orbit of the surviving monomials (CC.5), take the sum of the distinct monomials in that orbit, with coefficient one. These are the original unscaled invariant basis elements. Their exact number is

$$a_{\mathbf{n},r} = \sum_{p=0}^{r-1} [t^p x_1^{n_1} \cdots x_a^{n_a}] \prod_{\nu=1}^a \prod_{j=0}^{r-1} (1 - x_\nu t^j)^{-m_\nu}, \quad a_{\lambda,r} = \sum_{\lambda_{\mathbf{n}}=\lambda} a_{\mathbf{n},r}. \quad (\text{CC.10})$$

Coefficient extraction here is finite formal algebra. Each variable in a group of order m_ν has a pair of labels (j, c) , $j \geq 0$, $0 \leq c < m_\nu$, representing the original exponent $m_\nu j + c$. An orbit is a multiset of these labels, and the surviving condition is that their j 's sum to at most $r - 1$. The geometric series for each label proves (CC.10). An invariant in the full product (CC.4) is determined by its entry at one ordered tuple per occupation; the other entries are its variable-permuted translates. The stabilizer condition is precisely the one just imposed. This proves both the claimed basis and

$$\begin{aligned} \dim B_r &= d^k \binom{r+k-1}{k}, \\ \dim A_r &= \sum_{\lambda} a_{\lambda,r} = \sum_{p=0}^{r-1} [t^p x^k] \prod_{j=0}^{r-1} (1 - xt^j)^{-d}, \\ a_{\mathbf{n},1} &= \prod_{\nu} \binom{n_\nu + m_\nu - 1}{m_\nu - 1}. \end{aligned} \quad (\text{CC.11})$$

For the first equality use (CC.8): there are $\binom{r+k-1}{k}$ vectors $\mathbf{b}$ with $\sum b_i < r$, and d^k possible remainders. For the second alternative expression, permutation acts on that same basis by permuting its k pairs (b_i, c_i) ; its unscaled orbit sums give the stated series.

Theorem 30.3 (Constructive retraction at every depth). *There is an explicit $\mathbb{C}[T]$ -linear retraction $\Pi_r : A_r \rightarrow C_r$ of η_r , with the complete arithmetic unit retained. Put $K_r = \ker \Pi_r$ and $Q_r = 1 - \eta_r \Pi_r$. Then*

$$\begin{aligned} C_r \oplus K_r &\rightarrow A_r, & (c, v) &\mapsto \eta_r(c) + v, \\ A_r &\rightarrow C_r \oplus K_r, & w &\mapsto (\Pi_r w, Q_r w) \end{aligned} \quad (\text{CC.12})$$

are inverse $\mathbb{C}[T]$ -module isomorphisms. In particular $A_r/\eta_r C_r \rightarrow K_r$, $[w] \mapsto Q_r w$ is an isomorphism, inverse to inclusion followed by quotient.

Proof. For each λ , use the exact CRT idempotent $e_\lambda(T)$ in C_r , and let $A_{\lambda,r} = e_\lambda(S)A_r$. Write $N = S - \lambda$ on this summand. Choose an ordered tuple attaining $\ell = \ell_{\lambda,r}$. Choose a coordinate of maximal multiplicity in it, put all $r - 1$ allowed blocks in that coordinate, and put every remainder at its

maximum $m_{\nu_i} - 1$. Call the resulting exponent α^* . Thus $|\alpha^*| = \ell - 1$. Let $\theta : A_{\lambda,r} \rightarrow \mathbb{C}$ extract the coefficient of z^{α^*} in that specific local factor, in the basis (CC.5). Define, with $X = T - \lambda$,

$$\begin{aligned}\Pi_\lambda^0(v) &= \sum_{j=0}^{\ell-1} \theta(N^{\ell-1-j}v) X^j \pmod{X^\ell}, \\ w_\lambda &= e_\lambda(S)U_r, \quad p_\lambda(X) = \Pi_\lambda^0(w_\lambda), \\ \Pi_\lambda(v) &= p_\lambda(X)^{-1} \Pi_\lambda^0(v).\end{aligned}\tag{CC.13}$$

The equality $N^\ell = 0$ gives $\Pi_\lambda^0(Nv) = X\Pi_\lambda^0(v)$, coefficient by coefficient: the constant term on the left is zero and the other coefficients shift by one. Also

$$p_\lambda(0) = \left(\prod_i v_h(\rho_{\nu_i}) \right) \frac{(\ell-1)!}{\prod_i \alpha_i^*!} \neq 0.\tag{CC.14}$$

To verify (CC.14), the selected term in $N^{\ell-1}$ has that multinomial coefficient; multiplying by a positive-degree term of the unit would exceed the greatest surviving total degree. The idempotent $e_\lambda(S)$ is exactly one on this local factor. Set $t = p_\lambda/p_\lambda(0) - 1$. Its exact inverse is $p_\lambda(0)^{-1} \sum_{j=0}^{\ell-1} (-t)^j$. This retains every coefficient of p_λ . The intertwining now gives $\Pi_\lambda(e_\lambda(S)U_r F(S)) = F(\lambda + X) \pmod{X^\ell}$. Take the product of these maps and use scalar CRT to define Π_r . It is the asserted retraction. Its kernel is S -stable. The identities $Q_r \eta_r = 0$, $\Pi_r Q_r = 0$, and $Q_r|_{K_r} = 1$ prove every map and inverse in (CC.12). $\square$

For any polynomial or entire f , multiplication by $f(S)$ has on $A_{\lambda,r}$ the form $f(\lambda)1$ plus a polynomial without constant term in the nilpotent N . Powers of a nilpotent map have trace zero, as is seen in a basis adapted to its kernel filtration. The same statement holds on the two invariant summands in (CC.12). Thus

$$\begin{aligned}\mathrm{Tr}_{A_r} f(S) &= \sum_\lambda a_{\lambda,r} f(\lambda), \\ \mathrm{Tr}_{C_r} f(T) &= \sum_\lambda \ell_{\lambda,r} f(\lambda), \\ \mathrm{Tr}_{K_r} f(S) &= \sum_\lambda (a_{\lambda,r} - \ell_{\lambda,r}) f(\lambda).\end{aligned}\tag{CC.15}$$

This also proves $a_{\lambda,r} \geq \ell_{\lambda,r}$. All nilpotent coordinates remain in (CC.4), (CC.10) and (CC.13); the trace formula follows from those actual operators.

30.3 The derivative tower, scalar first jets, and the unit obstruction

Denote the quotient maps by $\pi_r : B_{r+1} \rightarrow B_r$ and $\rho_r : C_{r+1} \rightarrow C_r$. The original derivative is

$$\delta_r : B_{r+1} \longrightarrow B_r, \quad [F] \longmapsto \left[\frac{1}{k} \sum_i \partial_{s_i} F \right], \quad \partial_r : C_{r+1} \longrightarrow C_r, \quad [F] \longmapsto [F'].\tag{CC.16}$$

Each is a complex linear map. Differentiating a product of $r+1$ elements of I leaves at least r such factors, so $\partial_S I^{r+1} \subset I^r$. Formula (CC.3) gives $\ell_{\lambda,r+1} \geq \ell_{\lambda,r} + 1$. Thus the derivative of a multiple of χ_{r+1} is a multiple of χ_r , proving the second descent as well. On representatives,

$$\begin{aligned}\delta_r \alpha_{r+1} &= \alpha_r \partial_r, & \pi_r \alpha_{r+1} &= \alpha_r \rho_r, \\ \delta_r(xy) &= \pi_r(x) \delta_r(y) + \pi_r(y) \delta_r(x), & \pi_r \delta_{r+1} &= \delta_r \pi_{r+1}.\end{aligned}\tag{CC.17}$$

The last equality is a map $B_{r+2} \rightarrow B_r$. The symmetric subspaces are preserved because the derivative averages all k coordinates with the literal factor $1/k$. Every δ_r is onto: the invertible coordinates

$S, z_1, \dots, z_{k-1}$, where $z_i = s_i - S/k$, have $\partial_S S = 1$ and $\partial_S z_i = 0$. Integrating powers of S gives a polynomial primitive of any polynomial representative. Averaging such a primitive proves surjectivity on the invariant subspaces too.

The same monic expansion (CC.8) identifies

$$I^j/I^{j+1} \cong (P/I)[\zeta_1, \dots, \zeta_k]_j, \quad \zeta_i \mapsto [h_i], \quad \text{gr } \delta = \frac{1}{k} \sum_i [h'_i] \partial_{\zeta_i}. \quad (\text{CC.18})$$

Indeed differentiating $h^{\mathbf{b}} r_{\mathbf{b}}$ in degree $j = |\mathbf{b}|$ gives $b_i h^{\mathbf{b}-e_i} h'_i r_{\mathbf{b}}/k$ in degree $j-1$; differentiating $r_{\mathbf{b}}$ stays in degree j . Both the integer b_i and the original polynomial h'_i survive.

Introduce a formal square-zero parameter ϵ , with its ordinary coefficient algebra meaning. The two maps

$$\begin{aligned} \Phi_r : B_{r+1} &\longrightarrow B_r[\epsilon]/(\epsilon^2), & b &\longmapsto \pi_r b + \epsilon \delta_r b, \\ \phi_r : C_{r+1} &\longrightarrow C_r[\epsilon]/(\epsilon^2), & F &\longmapsto \rho_r F + \epsilon \partial_r F \end{aligned} \quad (\text{CC.19})$$

are unital algebra homomorphisms by (CC.17) and its one-variable product rule. Extending α_r coefficientwise to α_r^ϵ gives the exact square $\Phi_r \alpha_{r+1} = \alpha_r^\epsilon \phi_r$. The coefficient projections of these algebra maps are precisely the quotient and derivative maps in (CC.16).

Put $R(T) = \prod_{\lambda \in \Lambda_k} (T - \lambda)$ and $\psi_r = \chi_r R$. Then

$$\begin{aligned} \ker \phi_r &= (\psi_r)/(\chi_{r+1}), & \dim \text{im } \phi_r &= q_r + |\Lambda_k|, \\ \ker \rho_r &= (\chi_r)/(\chi_{r+1}) \cong \mathbb{C}[T]/(\chi_{r+1}/\chi_r), & \partial_r(\ker \rho_r) &= \chi'_r C_r. \end{aligned} \quad (\text{CC.20})$$

For the first assertion, F and F' vanish modulo $(T - \lambda)^{\ell_{\lambda,r}}$ exactly when F has order at least $\ell_{\lambda,r} + 1$ there. The order implication in (CC.16) gives $\psi_r \mid \chi_{r+1}$, so the asserted kernel is well defined and has dimension $q_{r+1} - q_r - |\Lambda_k|$. This proves the image dimension and the injective factorization $\mathbb{C}[T]/(\psi_r) \hookrightarrow C_r[\epsilon]/(\epsilon^2)$. Multiplication by χ_r gives the displayed isomorphism onto $\ker \rho_r$; its kernel is exactly (χ_{r+1}/χ_r) . Differentiating $\chi_r F$ modulo χ_r gives $\chi'_r F$. At each λ , multiplication by χ'_r maps the input constant to a nonzero multiple of $(T - \lambda)^{\ell_{\lambda,r}-1}$. Its image has one dimension per λ , proving all image and kernel claims even at repeated collided roots.

Write

$$\beta_r = U_r^{-1} \delta_r U_{r+1} \in A_r, \quad \Phi_r(U_{r+1}) = U_r(1 + \epsilon \beta_r), \quad \Phi_r(U_{r+1}^{-1}) = U_r^{-1}(1 - \epsilon \beta_r). \quad (\text{CC.21})$$

The first formula is the definition of the full logarithmic derivative class, the second follows directly from (CC.19), and the third follows by multiplication and $\epsilon^2 = 0$. In particular,

$$\delta_r \eta_{r+1} = \eta_r \partial_r + M_{\delta_r U_{r+1}} \alpha_r \rho_r. \quad (\text{CC.22})$$

This formula retains the full derivative of the arithmetic unit.

There is a complete decomposition of its last term using (CC.12):

$$\begin{aligned} b_r &= \Pi_r(\delta_r U_{r+1}) \in C_r, & n_r &= Q_r(\delta_r U_{r+1}) \in K_r, \\ \mathcal{N}_r : C_r &\longrightarrow K_r, & \mathcal{N}_r(F) &= \alpha_r(F) n_r, \\ \delta_r \eta_{r+1} &= \eta_r(\partial_r + M_{b_r} \rho_r) + \mathcal{N}_r \rho_r. \end{aligned} \quad (\text{CC.23})$$

Here K_r is a $\mathbb{C}[T]$ -submodule, so multiplication by $\alpha_r(F)$ preserves it. Both Π_r and Q_r intertwine this multiplication; applying them to the last term of (CC.22) proves (CC.23). To state the obstruction independently of the chosen retraction, let $\mathbf{q}_r : A_r \rightarrow A_r/\eta_r C_r$ be the module quotient and put

$$\Omega_r = \mathbf{q}_r M_{\delta_r U_{r+1}} \alpha_r : C_r \longrightarrow A_r/\eta_r C_r. \quad \mathbf{q}_r \delta_r \eta_{r+1} = \Omega_r \rho_r, \quad \Omega_r = 0 \iff \beta_r \in \alpha_r(C_r). \quad (\text{CC.24})$$

To prove the equivalence, evaluate Ω_r at 1. Its vanishing says $\delta_r U_{r+1} = U_r \alpha_r(\gamma_r)$ for a unique $\gamma_r \in C_r$, precisely the condition on β_r . Conversely that equality makes every value of Ω_r zero by closure of the subalgebra $\alpha_r(C_r)$. Under (CC.12), Ω_r corresponds exactly to $\mathcal{N}_r$. When it vanishes, $b_r = \gamma_r$, and the exact pulled-back derivative is $\partial_r + M_{\gamma_r} \rho_r$. The formulas construct both the obstruction and its strongest scalar derivative correspondence without deleting the complementary directions.

30.4 The finite cyclic derivative and its exact Weyl correction

The quotient derivative has two different depths in its domain and codomain. To compute its finite endomorphism on the prescribed scalar coordinates, let $s_r : C_r \rightarrow C_{r+1}$ send a class to the same polynomial of degree less than q_r given by monic division by χ_r . This is a complex-linear section of ρ_r . Put

$$\begin{aligned} L_r(F) &= [T^{q_r-1}]F_0(T), & d_r &= \partial_r s_r : C_r \rightarrow C_r, \\ D_r^{\text{red}} &= d_r + M_{b_r}, & A_r^C &= M_T : C_r \rightarrow C_r, \end{aligned} \quad (\text{CC.25})$$

where F_0 is that unchanged monic remainder. The scalar $L_r(F)$ retains its original coefficient; it is not a functional defined by dividing by a norm. The complete differentiated arithmetic map on this section is

$$\delta_r \eta_{r+1} s_r = \eta_r D_r^{\text{red}} + \mathcal{N}_r : C_r \longrightarrow A_r. \quad (\text{CC.26})$$

Thus the reduced derivative and its complementary component have explicit maps into the original arithmetic algebra.

Theorem 30.4 (Finite sum-coordinate Weyl relation). *For the original monic remainder coordinates,*

$$\begin{aligned} [A_r^C - k/2, D_r^{\text{red}}] &= -I_{C_r} + J_r, & J_r(F) &= [\chi_r'] L_r(F), \\ \text{rank } J_r &= 1, & J_r^2 &= q_r J_r, & \text{Tr } J_r &= q_r. \end{aligned} \quad (\text{CC.27})$$

The full arithmetic correction is the image of the original retained relation:

$$\begin{aligned} \rho_r([\chi_r]_{\chi_{r+1}} L_r(F)) &= 0, \\ \delta_r(U_{r+1} \alpha_{r+1}([\chi_r] L_r(F))) &= \eta_r([\chi_r'] L_r(F)). \end{aligned} \quad (\text{CC.28})$$

No complementary derivative is suppressed in (CC.26).

Proof. Since TF_0 has degree at most q_r , its exact monic remainder is $TF_0 - \chi_r L_r(F)$. Differentiating it and reducing gives

$$d_r A_r^C F = [F_0 + TF_0' - \chi_r' L_r(F)], \quad A_r^C d_r F = [TF_0'].$$

Their difference proves (CC.27), because multiplication by b_r commutes with multiplication by T . The vector $[\chi_r']$ has degree $q_r - 1$ and leading coefficient $q_r \neq 0$. Thus $L_r([\chi_r']) = q_r$; the operator is rank one, its square is q_r times itself, and its trace is that same scalar, either in the remainder basis or by the rank-one matrix formula. For (CC.28), the quotient kills χ_r , and the product rule leaves $U_r \alpha_r(\chi_r') L_r(F)$; the term with $\delta_r U_{r+1}$ still contains $\alpha_r(\chi_r) = 0$. The original relation itself remains in B_{r+1} before this application. Finally $\mathcal{N}_r$ is $\mathbb{C}[T]$ -linear by (CC.23). Consequently the full relation obtained by applying S to (CC.26) and subtracting (CC.26) applied to TF is exactly $-\eta_r + \eta_r J_r$; this proves the intertwining including the complementary map. $\square$

At $r = 1, k = 1, C_1 = \mathbb{C}[s]/(h)$, L_1 is the coefficient of s^{d-1} , and (CC.27) gives $[M_{s-1/2}, d_1 + M_{b_1}] = -I + [h'] L_1$. In the original weighted coordinates $u = v_h F$, its conjugated correction is $u \mapsto v_h h' L_1(v_h^{-1} u)$. Multiplication by v_h and by v_h^{-1} are the explicit inverse coordinate maps; the expression is the one-variable full-unit correction in the preceding finite Weyl calculation. Formula (CC.28) gives its all-depth extension using the actual relation layer rather than differentiating a quotient zero.

30.5 All first jets and the symmetric conormal kernel

Set $E = B_1, E^{[2]} = B_2, A_h = \mathbb{C}[s]/(h)$, and write $c_i = [h'(s_i)] \in E$. Let a continue to denote the number of distinct roots and put $b = d - a = \deg \gcd(h, h')$. The conormal coordinates are the actual

isomorphism

$$\begin{aligned} \iota : E^k &\longrightarrow I/I^2, & (v_i) &\longmapsto \left[\sum_i h_i \tilde{v}_i \right], \\ \partial_N : E^k &\longrightarrow E, & (v_i) &\longmapsto \frac{1}{k} \sum_i c_i v_i. \end{aligned} \quad (\text{CC.29})$$

Here $\tilde{v}_i$ is its original separate-variable monic remainder. Changing a lift changes the displayed sum by I^2 . Equation (CC.8) proves injectivity, surjectivity and the inverse coefficient extraction. Applying the product rule gives exactly $\delta_1 \iota = \partial_N$. The full conormal differential and the specified contraction are the explicit maps

$$E^k \longrightarrow \bigoplus_i E \, ds_i \longrightarrow E, \quad (v_i) \longmapsto \sum_i c_i v_i \, ds_i \longmapsto \frac{1}{k} \sum_i c_i v_i. \quad (\text{CC.30})$$

Thus the intermediate cotangent module and every coefficient of the sum direction are retained.

Theorem 30.5 (Exact ranks and complete first-jet observations). *Put $T_h = \mathbb{C}[s]/(h, h')$. Then*

$$\begin{aligned} \text{im } \partial_N &= (c_1, \dots, c_k), & \text{coker } \partial_N &\cong T_h^{\otimes k}, \\ \text{rank}_{\mathbb{C}} \partial_N &= d^k - b^k, & \dim \ker \partial_N &= (k-1)d^k + b^k. \end{aligned} \quad (\text{CC.31})$$

Equip $E \oplus E^k$ with multiplication $(x, v)(y, w) = (xy, xw + yv)$, unit $(1, 0)$. The maps

$$\begin{aligned} J_{\text{all}} : B_2 &\longrightarrow E \oplus E^k, & [F] &\longmapsto ([F], ([\partial_i F])_i), \\ J_S : B_2 &\longrightarrow E[\epsilon]/(\epsilon^2), & [F] &\longmapsto [F] + \epsilon \delta_1 [F], \\ \mathcal{L} : E \oplus E^k &\longrightarrow E[\epsilon]/(\epsilon^2), & (x, v) &\longmapsto x + \frac{\epsilon}{k} \sum_i v_i \end{aligned} \quad (\text{CC.32})$$

are unital algebra homomorphisms and satisfy $J_S = \mathcal{L} J_{\text{all}}$. Their kernels and ranks are

$$\begin{aligned} \ker J_S &= \iota(\ker \partial_N), & \dim \text{im } J_S &= 2d^k - b^k, \\ \ker J_{\text{all}} &= \iota \left(\bigoplus_i \text{ann}_E(c_i) \right), & \dim \ker J_{\text{all}} &= kb d^{k-1}, \\ \dim \text{im } J_{\text{all}} &= (k+1)d^k - kb d^{k-1}. \end{aligned} \quad (\text{CC.33})$$

For squarefree h , J_{all} is an isomorphism onto the displayed square-zero algebra.

Proof. The row in (CC.29) has precisely the ideal of its entries as image. Quotienting by these entries in the separate tensor factors gives $T_h^{\otimes k}$, with the map induced by the original polynomial quotients. The Euclidean algorithm gives $(h, h') = (\gcd(h, h'))$, so $\dim T_h = b$. This proves the cokernel and rank in (CC.31), and rank-nullity on the kd^k -dimensional source proves its kernel dimension.

Each coordinate derivative maps I^2 into I ; hence the maps in (CC.32) are well defined. The product rule for each derivative gives exactly the specified square-zero multiplication. The factor $1/k$ gives $J_S = \mathcal{L} J_{\text{all}}$. An element in either jet kernel has zero ordinary quotient, so it is a conormal class. Applying (CC.29) gives its sum derivative, while applying each coordinate derivative gives $(c_i v_i)_i$. This proves the asserted kernels. Multiplication by h' on A_h has cokernel T_h , hence has rank $d - b$ and kernel dimension b . Tensoring the kernel and image with the other $k - 1$ vector-space factors shows that $\dim \text{ann}_E(c_i) = b d^{k-1}$. The dimension of B_2 is $(k+1)d^k$ by (CC.11), which proves both image dimensions. If $b = 0$, the full map has zero kernel and equal source and target dimensions; its inverse is multiplicative because it is the inverse of a bijective algebra homomorphism. $\square$

The images also have explicit coordinates. Let $j : E \rightarrow B_2$ choose the original remainder with $\deg_{s_i} < d$. Set $d_0(x) = \delta_1 j(x)$ and $D_0(x) = ([\partial_i j(x)])_i$. Directly from (CC.29),

$$\begin{aligned} \text{im } J_S &= \{x + \epsilon(d_0(x) + v) : x \in E, v \in (c_1, \dots, c_k)\}, \\ \text{im } J_{\text{all}} &= \{(x, D_0(x) + (c_i v_i)_i) : x \in E, (v_i) \in E^k\}. \end{aligned} \quad (\text{CC.34})$$

Every lift differs from $j(x)$ by exactly one conormal vector, which proves both inclusions of each image equality. In particular $\ker J_S = J_{\text{all}}^{-1}(\ker \mathcal{L})$, where $\ker \mathcal{L} = \{(0, v) : \sum_i v_i = 0\}$. This identifies precisely the relative directions removed by the sum contraction after all first derivatives have been retained.

The same calculation restricts to the permutation invariants. On E^k the simultaneous action is $(\sigma v)_i = \sigma(v_{\sigma^{-1}i})$; all the maps above intertwine this action. Evaluation at the first component gives the explicit vector-space isomorphism

$$(E^k)^{\mathfrak{S}_k} \longrightarrow A_h \otimes (A_h^{\otimes(k-1)})^{\mathfrak{S}_{k-1}}. \quad (\text{CC.35})$$

For its inverse, start with a tensor fixed by the stabilizer of index one. For each i , choose a permutation carrying one to i and use its translate as component i . Two choices differ by the stabilizer, so the component is independent of that choice. The resulting tuple is invariant, proving the inverse in full. When $k = 1$, the tensor with zero factors is $\mathbb{C}$ and both maps are the identity on A_h .

For a vector space of dimension n , the invariant tensors in its k -fold tensor power have a basis of unscaled sums over words with prescribed occupation numbers in a fixed basis. There are $\binom{n+k-1}{k}$ such occupations for $n \geq 1$, and zero for $n = 0, k \geq 1$. This proves the dimensions used here directly. Taking invariants in the exact cokernel sequence of (CC.31) remains exact: average a lift by $k!^{-1} \sum_{\sigma} \sigma$ to lift any invariant image. Consequently

$$\begin{aligned} \text{coker } \partial_N^{\text{sym}} &\cong (T_h^{\otimes k})^{\mathfrak{S}_k}, \\ \dim(E^k)^{\mathfrak{S}_k} &= d \binom{d+k-2}{k-1}, \\ \text{rank } \partial_N^{\text{sym}} &= \binom{d+k-1}{k} - \binom{b+k-1}{k}, \\ \dim \ker \partial_N^{\text{sym}} &= d \binom{d+k-2}{k-1} - \binom{d+k-1}{k} + \binom{b+k-1}{k}. \end{aligned} \quad (\text{CC.36})$$

The final binomial is defined to be zero when $b = 0$. For the declared polynomial fixture $h = s^2, k = 2$, the row is $(v_1, v_2) \mapsto s_1 v_1 + s_2 v_2$; its image is the span of $s_1, s_2, s_1 s_2$, of dimension three, and its kernel has dimension five. On invariants the domain has dimension four, target dimension three, image span $s_1 + s_2, s_1 s_2$, and kernel dimension two. These are exact quotient computations in the stated algebraic fixture, without an assertion about arithmetic roots.

In the original cochain convention a permutation of k factors of cochain degree one contributes $\text{sgn}(\sigma)$. Thus the top-degree action of $k!^{-1} \sum_{\sigma} \text{sgn}(\sigma) T_{\sigma}$ is $k!^{-1} \sum_{\sigma} \sigma$ on these tensor coefficients: the two signs multiply to one. This is the explicit identification with the prescribed symmetric cochain summand. Its top-degree supertrace multiplies each ordinary trace by the retained factor $(-1)^k$.

30.6 Cyclic relation response and the full product Jacobian

Fix the monic-division identity $\chi_1(S) = \sum_i h_i B_i(\mathbf{s})$ in the original polynomial ring; it exists by (CC.6) and (CC.8). Define

$$\begin{aligned} \mathcal{B} : C_1 &\longrightarrow E^k, & F &\longmapsto ([B_i]_I \alpha_1(F))_i, \\ \partial_N \mathcal{B}(F) &= \alpha_1([\chi'_1]F), & \delta_1[U \chi_1(S) F(S)]_{I^2} &= U_1 \alpha_1([\chi'_1]F). \end{aligned} \quad (\text{CC.37})$$

The first map is well defined and $\mathbb{C}[T]$ -linear because each coefficient multiplies the well-defined class $\alpha_1(F)$. Differentiate the displayed division identity with $k^{-1} \sum_i \partial_i$. Terms still containing h_i vanish modulo I ; the derivative of S is exactly one. This proves the first identity. Differentiating the full-unit relation proves the second, since terms with an undifferentiated χ_1 vanish only at the final reduction. These maps give the exact conormal-to-cyclic Jacobian comparison, including $\ker(\partial_N \mathcal{B}) = \ker M_{\chi_1'}$, since α_1 is injective. Its rank is $|\Lambda_k|$ by (CC.20), and its kernel has dimension $q_1 - |\Lambda_k|$.

The full product Jacobian has a different explicit factorization through the same original conormal row. Define

$$\begin{aligned} J_{\text{prod}} &= \prod_i c_i \in E, \\ \mathcal{P} : E &\longrightarrow E^k, & v &\longmapsto \left(v \prod_{j \neq i} c_j \right)_i, \\ \partial_N \mathcal{P} &= M_{J_{\text{prod}}} : E \rightarrow E. \end{aligned} \tag{CC.38}$$

Indeed each of the k summands in the row is $J_{\text{prod}} v / k$. This proves the factorization with the original factor $1/k$. It is permutation equivariant. Its exact kernel is $\mathcal{P}^{-1}(\ker \partial_N) = \text{ann}_E(J_{\text{prod}})$. Also $\ker \mathcal{P} = \bigcap_i \text{ann}_E(\prod_{j \neq i} c_j)$; these equalities follow component by component and retain the intermediate directions rather than declaring the two observations interchangeable.

In a local factor $\mathbb{C}[z]/(z^m)$, the literal factorization $h(\rho + z) = z^m u_\rho(z)$ gives $h'(\rho + z) = m u_\rho(0) z^{m-1}$ modulo z^m . Thus in an ordered product factor, multiplication by J_{prod} sends a constant to $\prod_i m_i u_{\rho_i}(0) \prod_i z_i^{m_i-1}$, with nonzero coefficient, and annihilates the maximal ideal. There is one image dimension for each of the a^k ordered root tuples. Therefore

$$\text{rank } M_{J_{\text{prod}}} = a^k, \quad \ker M_{J_{\text{prod}}} = \sqrt{0_E}, \quad \dim \sqrt{0_E} = d^k - a^k. \tag{CC.39}$$

To check the kernel assertion globally, a class is nilpotent precisely when its constant coefficient vanishes at each local factor: the local maximal ideals are nilpotent and the product is finite. This is exactly the computed annihilator. Under permutation invariants, one image dimension remains for each occupation of the distinct roots, giving rank $\binom{a+k-1}{k}$. The invariant kernel is $(\sqrt{0_E})^{\mathfrak{S}_k}$, either by restriction or by averaging a lift of each invariant image. The factorization (CC.38) continues to hold on these invariant spaces.

The arithmetic unit remains on every observed conormal vector:

$$\delta_1(U_2 \iota(v)) = U_1 \partial_N(v), \quad U_1 \partial_N \mathcal{P} = M_{U_1 J_{\text{prod}}}. \tag{CC.40}$$

The first formula follows from (CC.17), since $\pi_1 \iota(v) = 0$. Multiplication by U_1^{-1} is its explicit inverse change on the target, so the kernels and ranks remain those proved above while all actual Taylor coefficients stay in the map.

30.7 Trace pairings and all four complementary operator blocks

For a monic one-variable polynomial p , all finite residue loops below have positive orientation. The functional $\lambda_p(F) = \sum_{\zeta} \text{Res}_{T=\zeta} F(T) p(T)^{-1} dT$ is well defined on $\mathbb{C}[T]/(p)$: adding pQ adds an entire polynomial differential at every loop. At a root of order m , write $p = (T - \zeta)^m c(T)$, where $c(\zeta) \neq 0$. Then $p'/p = m/(T - \zeta) + c'/c$. Hence

$$\lambda_p(p'F) = \sum_{\zeta} m_{\zeta} F(\zeta) = \text{Tr}_{\mathbb{C}[T]/(p)} M_F. \tag{CC.41}$$

For the last equality, in each basis $1, T - \zeta, \dots, (T - \zeta)^{m-1}$, the multiplication matrix has diagonal $F(\zeta)$ repeated m times; positive nilpotent powers have zero diagonal. This proves the trace identity with all original multiplicities.

The packet reflection gives the conjugate-linear algebra involution

$$F^\dagger(\mathbf{s}) = \overline{F(1 - \overline{s}_1, \dots, 1 - \overline{s}_k)}$$

on each B_r , because it permutes the root factors of h with their full orders. On the scalar algebra its restriction is $F^{\dagger k}(T) = \overline{F(k - \overline{T})}$. The sum set and all the exponents in (CC.3) are permuted by $\lambda \mapsto k - \overline{\lambda}$; hence $\chi_r^{\dagger k} = (-1)^{q_r} \chi_r$. This proves well-definedness on each quotient and $\alpha_r(F)^\dagger = \alpha_r(F^{\dagger k})$. Consequently the cyclic pairing

$$R_{\chi_r}(F, G) = \lambda_{\chi_r}(F^{\dagger k} G) \quad \text{satisfies} \quad R_{\chi_r}(F, \chi'_r G) = \sum_{\lambda} \ell_{\lambda, r} \overline{F(k - \overline{\lambda})} G(\lambda). \quad (\text{CC.42})$$

This is (CC.41) applied to the same full quotient, with its specified involution.

For the original one-factor arithmetic pairing, put $v_h = [v_h]_h$, $\varepsilon_h = v_h^{-1}$, and $\mathcal{R}_Z(f, u) = \lambda_h(\varepsilon_h f^\dagger u)$. The identity $g' = h'v_h + hv'_h$ gives $j_h(g') = h'v_h$. Thus the complete unit calculation is

$$\begin{aligned} \mathcal{R}_Z(f, j_h(g')P) &= \lambda_h(\varepsilon_h f^\dagger v_h h'P) \\ &= \lambda_h(f^\dagger h'P) = \text{Tr}_{A_h} M_{f^\dagger P}. \end{aligned} \quad (\text{CC.43})$$

The product functional $\lambda_h^{\otimes k}$ consequently satisfies $\lambda_h^{\otimes k}(J_{\text{prod}} F) = \text{Tr}_E M_F$. To verify it without a tensor restriction on F , expand F in the original separate-variable monomial basis and apply (CC.41) factor by factor; both sides are linear in every coefficient. The product $U_1^{-1} U_1$ gives the same exact unit cancellation for its original weighted arithmetic version. This describes the full product trace and (CC.38) supplies its map through the sum-direction conormal observation.

Let $\chi_{A, r}(T) = \prod_{\lambda} (T - \lambda)^{a_{\lambda, r}}$. Equation (CC.15) gives the genuine quotient trace of every entire $f(S)$ on $T_r = A_r / \eta_r C_r$:

$$\text{Tr}_{T_r} f(S) = \sum_{\lambda} \text{Res}_{T=\lambda} f(T) \left(\frac{\chi'_{A, r}}{\chi_{A, r}} - \frac{\chi'_r}{\chi_r} \right) dT. \quad (\text{CC.44})$$

Its proof is the local logarithmic derivative calculation (CC.41) with exponents $a_{\lambda, r} - \ell_{\lambda, r}$. It retains the module T_r ; its formula is a trace of its induced operator, not a replacement of its nilpotent coordinates.

For an arbitrary linear endomorphism $M : A_r \rightarrow A_r$, use the fixed full-unit splitting (CC.12). Write $\iota_K : K_r \rightarrow A_r$ for inclusion and regard Q_r as a map with codomain K_r . Its full transformed matrix is

$$\begin{pmatrix} \Pi_r M \eta_r & \Pi_r M \iota_K \\ Q_r M \eta_r & Q_r M \iota_K \end{pmatrix} : C_r \oplus K_r \longrightarrow C_r \oplus K_r. \quad (\text{CC.45})$$

Applying the two inverse maps of (CC.12) on either side of M proves every block. In these original coordinates,

$$\text{Tr}_{A_r} M = \text{Tr}_{C_r}(\Pi_r M \eta_r) + \text{Tr}_{K_r}(Q_r M \iota_K). \quad (\text{CC.46})$$

The cross maps in (CC.45) remain part of the transformed operator. For completeness, invariance of trace under these inverse changes follows from $\text{Tr}(XY) = \text{Tr}(YX)$, proved by expanding $\sum_{i,j} X_{ij} Y_{ji}$. The quotient T_r has the induced map $[v] \mapsto [Mv]$ exactly when $M\eta_r C_r \subseteq \eta_r C_r$, equivalently $Q_r M \eta_r = 0$. Necessity follows by applying a well-defined quotient map to $[\eta_r c] = 0$; sufficiency follows by comparing any two representatives. In that case (CC.12) identifies its map with $Q_r M \iota_K$. This proves the exact quotient criterion and the always-defined complementary compression when it fails.

For a multiplier $M = M_w$ with $w \in A_r$, this criterion is

$$M_w \eta_r C_r \subseteq \eta_r C_r \iff w \in \alpha_r(C_r). \quad (\text{CC.47})$$

Indeed its value on $\eta_r(1) = U_r$ gives necessity after multiplication by U_r^{-1} . Conversely subalgebra closure proves sufficiency. If $w = f(S)$, both cross maps in (CC.45) are zero by $\mathbb{C}[T]$ -linearity, proving (CC.44) directly. The full weighted vector pairing is instead the explicitly typed trace

$$\mathrm{Tr}_{A_r} M_{(\eta_r F)^\dagger \eta_r G} = \mathrm{Tr}_{A_r} M_{U_r^\dagger U_r \alpha_r (F^\dagger_k G)}. \quad (\text{CC.48})$$

Multiplicativity of the involution proves this identity before any trace calculation. Insert that unchanged multiplier into all four blocks of (CC.45) to compute its full decomposition. In particular $U_r^\dagger U_r$ has an induced quotient multiplier precisely when it satisfies (CC.47); no membership is assumed from invariance under permutations. If an original positive Hermitian form is represented by H on A_r , the same coordinates give its full pullback

$$\begin{pmatrix} \eta_r^* H \eta_r & \eta_r^* H \iota_K \\ \iota_K^* H \eta_r & \iota_K^* H \iota_K \end{pmatrix}. \quad (\text{CC.49})$$

This follows by evaluating the form on $\eta_r c + \iota_K v$ and $\eta_r c' + \iota_K v'$; all four terms appear. The algebraic retraction entails no orthogonality assertion for that original metric.

30.8 Exact two-factor symmetric presentation at every depth

For $k = 2$, retain $S = s_1 + s_2$, $R = s_1 - s_2$ and $\Delta = R^2$, with inverse $s_1 = (S + R)/2$, $s_2 = (S - R)/2$. Write the original polynomial as $h(s) = \sum_{j=0}^d c_j s^j$, $c_d = 1$, and define

$$\begin{aligned} H_0(S, \Delta) &= \sum_{j=0}^d c_j 2^{-j} \sum_{2a \leq j} \binom{j}{2a} S^{j-2a} \Delta^a, \\ H_1(S, \Delta) &= \sum_{j=1}^d c_j 2^{-j} \sum_{2a+1 \leq j} \binom{j}{2a+1} S^{j-2a-1} \Delta^a. \end{aligned} \quad (\text{CC.50})$$

Binomial expansion gives exactly $h(s_1) = H_0 + R H_1$, $h(s_2) = H_0 - R H_1$, including $R = 0$. Therefore the linear change of generators with coefficients $1/2$ gives $I = (H_0, R H_1)$ in $\mathbb{C}[S, R]$. For every $r \geq 1$ the complete invariant relation ideal is

$$\begin{aligned} J_r &= \left(H_0^{r-j} \Delta^{\lceil j/2 \rceil} H_1^j : 0 \leq j \leq r \right) \subset \mathbb{C}[S, \Delta], \\ A_r &\cong \mathbb{C}[S, \Delta]/J_r, \quad C_r \hookrightarrow \mathbb{C}[S, \Delta]/J_r, \quad [F(T)] \mapsto [F(S)]. \end{aligned} \quad (\text{CC.51})$$

To prove the ideal identity, express any element of I^r as a sum of the generators $H_0^{r-j} R^j H_1^j$ times arbitrary polynomials in S, R . Average under $R \mapsto -R$. For even j , the coefficient's even part is a polynomial in S, Δ , giving $\Delta^{j/2}$. For odd j , its odd part is R times such a polynomial, giving $\Delta^{(j+1)/2}$. This proves containment in J_r . Each displayed generator is conversely an invariant member of I^r , proving equality. Every invariant quotient class has an invariant lift by the same average with coefficient $1/2$. Thus the induced algebra map is onto and has exactly this kernel. The injection in (CC.51) is precisely (CC.6) in those coordinates; its kernel before taking C_r is (χ_r) .

At depth one this recovers the literal presentation $\mathbb{C}[S, \Delta]/(H_0, \Delta H_1)$, with the relative factor Δ intact. Under the earlier quartet assumptions, its H_0, H_1 are exactly those of (SSP.9): both equal the even and odd coefficients of the same unchanged polynomial $h((S + R)/2)$. The further substitution $Z = S - 1$, $x = -Z^2$ is its original (SSP.2) map, so the cyclic injection sends $F(T)$ to $F(1 + Z)$ in the complete SSP algebra. Its weighted module injection sends it to $[v_h((S + R)/2) v_h((S - R)/2)] F(S)$; that product is invariant, and its finite Taylor class is exactly U_r . The inverse SSP coordinate formulas, and (CC.12) for the cyclic summand, retain the full relative and parity modules at every step.

The two parity modules themselves have exact formulas at every depth. Under those same quartet assumptions, put $\mathcal{R} = \mathbb{C}[x, \Delta]$ and use the unchanged polynomials A_0, A_1 of (SSP.3), so that $H_0 =$

$A_0(-Z^2, \Delta)$, $H_1 = ZA_1(-Z^2, \Delta)$. For all indices within $0 \leq j \leq r$, define

$$\begin{aligned} E_{2a} &= A_0^{r-2a} \Delta^a (-x)^a A_1^{2a}, \\ O_{2a+1} &= A_0^{r-2a-1} \Delta^{a+1} (-x)^a A_1^{2a+1}, \\ \mathfrak{a}_{e,r} &= (E_{2a}, xO_{2a+1}) \subset \mathcal{R}, & \mathfrak{a}_{o,r} &= (E_{2a}, O_{2a+1}) \subset \mathcal{R}, \\ B_{e,r} &= \mathcal{R}/\mathfrak{a}_{e,r}, & B_{o,r} &= \mathcal{R}/\mathfrak{a}_{o,r}. \end{aligned} \tag{CC.51a}$$

Then $A_r = B_{e,r} \oplus ZB_{o,r}$ as a module in these specified coordinates. Its exact maps and product are

$$\begin{aligned} \beta_r^{\text{par}} : B_{e,r} &\rightarrow B_{o,r}, & [a] &\mapsto [a], \\ \gamma_r^{\text{par}} : B_{o,r} &\rightarrow B_{e,r}, & [b] &\mapsto [xb], \\ (a, b)(c, e) &= (ac - \gamma_r^{\text{par}}(be), \beta_r^{\text{par}}(a)e + \beta_r^{\text{par}}(c)b), \\ \gamma_r^{\text{par}} \beta_r^{\text{par}} &= xI_{B_{e,r}}, & \beta_r^{\text{par}} \gamma_r^{\text{par}} &= xI_{B_{o,r}}. \end{aligned} \tag{CC.51b}$$

The superscript distinguishes these parity maps from the full logarithmic unit class β_r defined in (CC.21). For a direct proof, substitute $Z^2 = -x$ into each generator of (CC.51). Even $j = 2a$ gives E_{2a} , and odd $j = 2a + 1$ gives ZO_{2a+1} . Before imposing them the module is freely $\mathcal{R} \oplus Z\mathcal{R}$, by monic division in Z . Multiplication by $c + Zd$ shows that their even coefficient ideal is exactly (E_{2a}, xO_{2a+1}) , and their odd coefficient ideal is exactly (E_{2a}, O_{2a+1}) . This proves both directions of the claimed decomposition. The containments between those ideals and their multiples by x prove well-definedness of $\beta_r^{\text{par}}, \gamma_r^{\text{par}}$; multiplication in $\mathcal{R}[Z]/(Z^2 + x)$ proves every sign, product and composition in (CC.51b). At $r = 1$ these are exactly (SSP.6)–(SSP.7), because $E_0 = A_0$ and $O_1 = \Delta A_1 = Q_\Delta$. The original unit is still the full class of the product in (CC.2), carried through these inverse coefficient maps.

30.9 Split lifts, source provenance and quantitative scope

For a complex vector space V , retain $G(V) = \{\tau\} \sqcup \{v^\bullet : v \in V\}$, with $e_V = 0_V^\bullet$, τ the additive identity, and $v^\bullet + w^\bullet = (v + w)^\bullet$. The scalar action is $c^\bullet v^\bullet = (cv)^\bullet$, and an absent scalar or vector produces τ . For a complex-linear map $L : V \rightarrow W$, its exact lift sends $\tau \mapsto \tau$, $v^\bullet \mapsto (Lv)^\bullet$. It preserves addition and scalar action: on two supported inputs this is the linearity of L , and cases with τ follow from the given identity and absorbing rules. For an algebra homomorphism, the same formula also preserves supported products and units. These formulas apply to every map above with its proved type; a derivative uses the linear lift, while the maps (CC.19) and (CC.32) use the multiplicative lift.

In particular, for a conormal vector $n = \iota(v)$,

$$G(\pi_1)(n^\bullet) = e_E, \quad G(J_S)(n^\bullet) = (\epsilon \partial_N v)^\bullet, \quad G(\delta_1)(n^\bullet) = (\partial_N v)^\bullet. \tag{CC.52}$$

The common source is the retained thickening. Every arrow fixes external absence, and every supported kernel vector maps to the receiving supported zero. Equations (CC.37)–(CC.40) give the actual original-unit morphisms connecting cyclic, full conormal and product-Jacobian observations before this quotient.

The exact source map accompanying this section records hashes and line locators for the complete delivered cyclic note, the complete conormal pasted source, and the earlier proof fragments. The cyclic note’s depth-one retraction and Sections 7–8 are proved here at every original depth, with complete invariant dimensions, the scalar-power equality classification, the higher-degree lower bound, and the full-unit complement in (CC.24). Equations (CC.45)–(CC.48) provide the precise operator interpretation for the general weighted trace. Equations (CC.29)–(CC.36) prove the first-jet and invariant rank calculations supplied in the pasted source. The uncompiled status of its embedded Lean text describes that historical delivered draft. The separately recorded PR 21 successor has its own checked CI provenance; it does not make the new written proofs of this chapter kernel-checked. No new Lean execution, zero-location calculation, or claimed arithmetic asymptotic is part

of these proofs. The formulas give the exact finite algebra, retained relation derivatives and trace maps used by the adjoining cyclic metric calculation. The quantitative interpolation metric remains the original one, with its degree cost from (CC.7) and all its cross terms from (CC.49).

31 The cyclic sum source inside the full connection

Keep the original nonempty dagger-stable packet h , its full orders, $g = 2\xi$, $v_h = g/h$, and $k \geq 2$. Use exactly (NN.1), $S = k/2 + iu$, and $m_k(u) = w_h^{*k}(u) > 0$, with $w_h = |a_h|^2$ and $a_h(t) = v_h(1/2 + it)/\sqrt{2\pi}$. The complete cyclic-sum source note is retained in the source appendices. Its annihilator is

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{\ell_{\lambda}}, \quad \ell_{\lambda} = \max_{\sum_i \rho_i = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) \right), \quad C_{\chi} = \mathbb{C}[S]/(\chi). \quad (\text{CM.1})$$

Keep the full algebra $E = E_h^{\otimes k}$, the original unit $U_k = v_h^{\otimes k}$, the injective algebra map $\alpha : C_{\chi} \rightarrow E^{S_k}$, and the separate module injection $\eta = M_{U_k} \alpha$. Their exact construction, retraction, complement, nilpotent orders and trace comparison are proved in the cyclic conormal section and the complete delivered source. In particular the original analytic source is the injective map

$$\begin{aligned} \mathcal{V}_k(P) &= P(D_1 + \cdots + D_k) F_h^{\otimes k}, \\ \mathcal{U}_k \mathcal{V}_k(P) &= \Psi_{c_P}, \quad c_P(u) = P(k/2 + iu), \\ J^{(k)} \mathcal{V}_k(P) &= \eta[P]_{\chi}. \end{aligned} \quad (\text{CM.2})$$

Here $\mathcal{U}_k$ is the original Mellin–Fourier isometry and the change to the original sum and relative coordinates. Its norm is $\int |c_P|^2 m_k du$, with its original mass μ_h^k .

31.1 Complete polynomial moments and the original finite metric

Put $\mu_j = \int t^j |a_h|^2 dt$, $\nu_j = \int t^j |a'_h|^2 dt$, and $\theta_j = \int t^j \overline{a_h} a'_h dt$. The fixed arithmetic phase makes $\overline{a_h} a'_h = w'_h/2$; integration by parts with the original decay gives

$$\theta_0 = 0, \quad \theta_j = -\frac{j}{2} \mu_{j-1} \quad (j \geq 1). \quad (\text{CM.3})$$

Define the formal series $\mathcal{M}(z) = \sum \mu_j z^j / j!$ and $\mathcal{N}(z) = \sum \nu_j z^j / j!$. The entire finite moment arrays are

$$\begin{aligned} M_{j,k} &= \int u^j m_k(u) du = j! [z^j] \mathcal{M}(z)^k, \\ B_{j,k} &= \int u^j \|\partial_u \Psi(u, \cdot)\|^2 du \\ &= j! [z^j] \left(\frac{1}{k} \mathcal{N}(z) \mathcal{M}(z)^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}(z)^k \right). \end{aligned} \quad (\text{CM.4})$$

For completeness, the multinomial expansion of $(t_1 + \cdots + t_k)^j$ and Fubini give the first identity. The derivative is $k^{-1} \sum_i a'_h(t_i) \prod_{r \neq i} a_h(t_r)$. There are k diagonal terms, each carrying k^{-2} , and $k(k-1)$ distinct-index terms with that same factor. Their formal series are respectively $\mathcal{N} \mathcal{M}^{k-1} / k$ and $(k-1) \Theta(z)^2 \mathcal{M}^{k-2} / k$, where $\Theta(z) = -z \mathcal{M}(z) / 2$ by (CM.3). This proves the second identity with every cross term and the original k^{-2} retained.

In the literal source coordinates $1, S, \dots, S^N$, write $c_j(u) = (k/2 + iu)^j$. Its original amplitude

and derivative Gram matrices are, entry by entry,

$$\begin{aligned} H_{ij}^0 &= \int \bar{c}_i c_j m_k du, \\ H_{ij}^\partial &= \int \bar{c}_i c_j \|\Psi'\|^2 du + \int \bar{c}_i c'_j m_k du \\ &\quad + \frac{1}{2} \int \bar{c}_i c'_j m'_k du + \frac{1}{2} \int \bar{c}'_i c_j m'_k du, \quad c'_j = ij(k/2 + iu)^{j-1} \quad (j \geq 1), \quad c'_0 = 0. \end{aligned} \quad (\text{CM.5})$$

Each integrand coefficient is obtained by multiplying the displayed original polynomials. Integrating its u^r coefficient uses $M_{r,k}$, $B_{r,k}$, or $\int u^r m'_k = -rM_{r-1,k}$, with zero for $r = 0$. This is an explicit finite matrix construction from the original one-factor arrays; it keeps the two separate cross terms. Both forms are positive definite. The amplitude map is injective by $\Psi \neq 0$ almost everywhere. A zero derivative would give an $L^2(du; \mathcal{H})$ function constant in u , hence zero, and the same argument then gives the zero polynomial.

Let p_j be the original monic orthogonal polynomials for m_k , with full norms $\omega_j = \int |p_j(k/2 + iu)|^2 m_k(u) du$, and $b_j = [p_j]_\chi$ in the fixed remainder coordinates. For $N \geq \deg \chi - 1$, put

$$K_N = \sum_{j=0}^N b_j b_j^* / \omega_j, \quad G_N = K_N^{-1}, \quad T_N u = \sum_{j=0}^N \text{coeff}(p_j) \frac{b_j^* G_N u}{\omega_j}. \quad (\text{CM.6})$$

Thus T_N is the actual coefficient matrix of the canonical cyclic representative, and its two complete forms are

$$\mathcal{R}_N = \mathcal{V}_k T_N, \quad \mathcal{R}_N^* \mathcal{R}_N = T_N^* H^0 T_N = G_N, \quad (\partial_u \mathcal{U}_k \mathcal{R}_N)^* (\partial_u \mathcal{U}_k \mathcal{R}_N) = T_N^* H^\partial T_N. \quad (\text{CM.7})$$

The first equality on forms follows by multiplying (CM.6) in the orthogonal polynomial coordinates; the second is the definition of the original derivative Gram (CM.5). The latter has not been replaced by the constant-column Fisher estimate.

31.2 A full real-exponential family of the same information identity

The original-amplitude density theorem proves $|a_h^{(j)}(t)| \leq C_{h,j,b} e^{-b|t|}$ for every $b < \pi/4$. It follows, by choosing $2b > |\theta|$, that for each real $|\theta| < \pi/2$ the following are finite actual integrals:

$$M(\theta) = \int e^{\theta t} |a_h(t)|^2 dt > 0, \quad N(\theta) = \int e^{\theta t} |a'_h(t)|^2 dt. \quad (\text{CM.8})$$

The same estimates with a slightly larger real exponential on compact subsets prove differentiation under the integral for every polynomial power. Hence M, N are holomorphic for $|\Re z| < \pi/2$, and their Taylor coefficients at zero are the original arrays in (CM.4). Using Fubini with this actual exponential, rather than only formal coefficients, gives

$$\begin{aligned} \int e^{\theta u} m_k(u) du &= M(\theta)^k, \\ B_k(\theta) &:= \int e^{\theta u} \|\Psi'(u)\|^2 du = \frac{1}{k} N(\theta) M(\theta)^{k-1} + \frac{k-1}{4k} \theta^2 M(\theta)^k. \end{aligned} \quad (\text{CM.9})$$

In this calculation $\int e^{\theta t} \bar{a}_h a'_h = -\theta M(\theta)/2$. The diagonal and distinct-index product terms are therefore exactly those used in (CM.4).

The exact Hilbert-space multiplier is

$$L^2(e^{\theta t} dt) \xrightarrow{f \mapsto e^{\theta t/2} f} L^2(dt), \quad f \mapsto e^{\theta t/2} f, \quad \text{inverse } q \mapsto e^{-\theta t/2} q. \quad (\text{CM.10})$$

It is unitary by direct integration. For the original amplitude set $a_\theta = e^{\theta t/2} a_h$. Its mass and derivative energy are

$$\|a_\theta\|^2 = M(\theta), \quad \|a'_\theta\|^2 = N(\theta) - \frac{\theta^2}{4} M(\theta) > 0. \quad (\text{CM.11})$$

Expansion of a'_θ , followed by the cross integral above, proves the equality. If its derivative norm were zero, the nonzero L^2 function a_θ would be constant, which is impossible; this proves strict positivity. The product amplitude is exactly $\Psi_\theta = e^{\theta u/2} \Psi$, its density is $m_{k,\theta} = e^{\theta u} m_k$, and its relative line projection is the same Π_0 , since the multiplier is a nonzero scalar on each fibre. For $n_k = (I - \Pi_0) \Psi'$, its normal component is consequently $n_{k,\theta} = e^{\theta u/2} n_k$. Applying the complete product-rule proof of (FC.4) to this specific amplitude, including its mass, proves

$$\int e^{\theta u} \frac{(m'_k + \theta m_k)^2}{m_k} du + 4 \int e^{\theta u} \|n_k(u)\|^2 du = \frac{4}{k} M(\theta)^{k-1} \left(N(\theta) - \frac{\theta^2}{4} M(\theta) \right). \quad (\text{CM.12})$$

All square quotients are taken away from the already stated fibre zero sets; $m_k > 0$ itself everywhere. The multiplier keeps the same zeros of the original one-factor amplitude. Its normal component is nonzero on the same fibres as that in (FC.21)–(FC.27), so the second integral is strictly positive. No mass has been assigned the value one. The arithmetic maps (CM.1)–(CM.2) still use the original U_k ; (CM.10) specifies the exact weighted Hilbert comparison used solely to evaluate the displayed original exponential integrals.

31.3 All cyclic normal energy is retained under frame enlargement

The cyclic source uses the single relative coefficient 1 at every scalar degree N . Its line projection is Π_0 . The complete relative frame of total degree m has projection Π_m . For any fixed cyclic polynomial P , $m \geq \deg P$, the actual derivative amplitude has the three mutually orthogonal components

$$\begin{aligned} d(P) &= \partial_u(\Psi c_P) = t_0(P) + b_m(P) + r_m(P), \\ t_0(P) &= \Pi_0 d(P) = \Psi \left(i c_{P'} + \frac{m'_k}{2m_k} c_P \right), \\ b_m(P) &= (\Pi_m - \Pi_0) d(P), \quad r_m(P) = (I - \Pi_m) d(P), \\ n_k c_P &= b_m(P) + r_m(P). \end{aligned} \quad (\text{CM.13})$$

Here $c_{P'}(u) = P'(k/2 + iu)$. The derivative of the cyclic coefficient is $i c_{P'}$; multiplication by c_P commutes with the fibre projections. Hence projecting its derivative gives precisely the first two formulas, and subtracting proves the last one. The nesting $\Pi_0 \leq \Pi_m$ proves orthogonality of all three components. With the original global norms this gives

$$\|n_k c_P\|^2 = \|b_m(P)\|^2 + \|r_m(P)\|^2, \quad \|d(P)\|^2 = \|t_0(P)\|^2 + \|b_m(P)\|^2 + \|r_m(P)\|^2. \quad (\text{CM.14})$$

The proved density limit (NN.4) now yields the exact convergence

$$r_m(P) \rightarrow 0, \quad b_m(P) \rightarrow n_k c_P \quad \text{in } \mathcal{K}. \quad (\text{CM.15})$$

Thus the complete fixed cyclic normal vector is the limit of the added relative tangential vectors, with its entire positive norm retained by (CM.14). This is the explicit map between the constant relative frame of the new cyclic construction and the enlarging full relative frames. It makes no substitution of one frame for another. For a fixed finite collection of original cyclic polynomials, the corresponding restricted Gram matrices converge as well: each entry is a pairing of finitely many convergent $\mathcal{K}$ vectors. No statement for a sequence of changing canonical polynomials is inferred from this fixed-source convergence.

31.4 The canonical metric difference has its actual conormal lift

For $N \geq \deg \chi - 1$, let R_N^{full} be the original full-source minimum on the whole arithmetic quotient E , and let $\mathcal{R}_N$ be (CM.7). This full minimum exists at this degree: $\deg \chi - 1 \geq k(d - 1)$. Here is a direct proof with the original monic polynomial retained. For every $0 \leq q \leq k(d - 1)$, choose $0 \leq \alpha_i < d$ with $\sum_i \alpha_i = q$. The monomial $\mathbf{s}^\alpha$ in S^q has coefficient $q! / \prod_i \alpha_i! \neq 0$. It is already a standard remainder monomial. Every term with some exponent at least d , when reduced by the original monic $h(s_i)$, strictly lowers total degree. Thus the remainder of S^q has total degree exactly q , with that same nonzero coefficient. Remainders of lower powers have strictly lower degree. A nonzero polynomial in S of degree at most $k(d - 1)$ therefore cannot annihilate S in the full algebra. Since χ is its exact annihilator, the asserted degree inequality follows. Every monic multivariate remainder now belongs to the original full source at degree N , so the full jet map is onto. Set

$$\Delta_N = \mathcal{R}_N - R_N^{\text{full}}\eta, \quad E_N = \Delta_N^* \Delta_N \succeq 0. \quad (\text{CM.16})$$

Both representatives have the same full original jets. Their difference is therefore an actual polynomial-source boundary. The full minimum is orthogonal to that boundary, proving the complete Gram and control identities

$$\begin{aligned} G_N &= \eta^* G_N^{\text{full}} \eta + E_N, \\ W_N - \eta^* W_N^{\text{full}} \eta &= A_\chi^* E_N + E_N A_\chi - k E_N, \quad A_\chi = M_S. \end{aligned} \quad (\text{CM.17})$$

The second equality follows by substituting the first and using $A_k \eta = \eta A_\chi$. For an actual eigenvector $A_\chi x = \lambda x$, its exact contribution is

$$x^*(W_N - \eta^* W_N^{\text{full}} \eta)x = (2\Re \lambda - k) \|\Delta_N x\|^2. \quad (\text{CM.18})$$

This is the full effect of the positive Gram difference on that original eigenline, including either sign of $2\Re \lambda - k$.

Write $\Delta_N x = \mathcal{T}_h^{(k)} Q_x$ in the original polynomial coordinates. Since its full jet is zero and U_k is a unit, $Q_x \in I = (h(s_1), \dots, h(s_k))$. Fixed-order monic division supplies $Q_x = \sum_i h(s_i) Q_{i,x}$, linearly in x . Denote the fixed conormal map of this ideal by $\partial_I : I/I^2 \rightarrow E$, $[Q] \mapsto [k^{-1} \sum_i \partial_{s_i} Q]_I$. It is independent of the source cutoff N . The retained conormal map and its derivative observation are

$$\begin{aligned} \kappa_N : C_\chi &\longrightarrow I/I^2, \quad x \longmapsto [Q_x]_{I^2}, \\ \partial_I \kappa_N(x) &= \frac{1}{k} \sum_i h'(s_i) [Q_{i,x}]_I, \\ J^{(k)} \mathcal{L}_k \Delta_N x &= U_k \partial_I \kappa_N(x), \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i. \end{aligned} \quad (\text{CM.19})$$

The first expression is independent of the displayed division because it is the actual class of the uniquely specified polynomial Q_x . The derivative of I^2 lies in I ; its product rule gives the second expression and proves independence of every conormal coordinate choice. For the last equality differentiate the full $U Q_x$: the term $(\partial_S^{\text{rel}} U) Q_x$ lies in I , whereas the other term is the stated U_k times the complete conormal row. Mellin differentiation supplies exactly $\mathcal{L}_k$, as in (FC.34). Thus the actual canonical-minimum difference has an explicit map into the original conormal layer and then to its full arithmetic derivative, with every selected order and unit retained.

The original quotient sends the supported class of Q_x to its supported zero. The conormal lift in (CM.19) is retained before that quotient, so its possibly nonzero differentiated observation is obtained from the stated original relation, not from differentiating the scalar supported zero. For $Q_x \in I^2$ the conormal class and this derivative vanish; for general $Q_x \in I$ their full kernels are related exactly by $\ker \kappa_N = \{x : Q_x \in I^2\}$ and $\ker(J^{(k)} \mathcal{L}_k \Delta_N) = \kappa_N^{-1}(\ker \partial_I)$, since U_k is invertible. The original conormal kernel is computed in the conormal section. The external τ maps to τ along every split lift.

32 Exterior trace equality, the spectral projector, and quantitative coupling

Date: 2026-09-12. This is an independent finite-dimensional proof for the exterior-amplification integration. All norms and adjoints below use the original Hermitian form. No arithmetic asymptotic estimate, Lean execution, or external-source verification is asserted.

32.1 1. Original objects and exact domains

Let $\mathcal{C}$ be a finite-dimensional complex vector space with the positive-definite Hermitian form

$$\langle x, y \rangle_G = x^* G y$$

(conjugate-linear in the first variable). Let $k \in \mathbb{R}$, let $A \in \text{End}_{\mathbb{C}}(\mathcal{C})$, and set

$$A^\sharp = G^{-1} A^* G, \quad H = A^\sharp + A - k I_{\mathcal{C}}.$$

Suppose the actual operator H has rank at most two and trace zero. Its eigenvalues, with zero eigenvalues retained, are $+\epsilon, -\epsilon, 0, \dots, 0$ for a uniquely determined $\epsilon \geq 0$.

Let $V = \mathcal{C}_{>}$ be the direct sum of the **full generalized eigenspaces** of A for the eigenvalues satisfying $\text{Re } \lambda > k/2$. Let $W = V^{\perp_G}$, and let $P : \mathcal{C} \rightarrow \mathcal{C}$ be the G -orthogonal projection with image V and kernel W . Let Q be the spectral projector with image V and kernel the direct sum of all the other full generalized eigenspaces of A .

For specificity, this last projector is an exact polynomial in the original operator. Factor the minimal polynomial as $m_A = m_+ m_-$, where m_+ contains exactly the selected eigenvalues with their original minimal-polynomial exponents and m_- contains all the remaining eigenvalues with their original exponents. The two factors are coprime. Choose polynomials a, b with

$$a(z)m_+(z) + b(z)m_-(z) = 1.$$

Then

$$Q = b(A)m_-(A).$$

Indeed, the polynomial $b m_-$ is congruent to one modulo m_+ and to zero modulo m_- ; applying it to each full generalized eigenspace gives the asserted identity and zero maps. This also proves $Q^2 = Q$, $AQ = QA$, and independence from the choice of the Bezout polynomials.

All block maps below use the original orthogonal direct sum $\mathcal{C} = V \oplus W$, without a change in either restricted Hermitian form. Since $A(V) \subseteq V$, define

$$B = A|_V : V \rightarrow V, \quad D = (1 - P)A|_W : W \rightarrow W, \quad T = PA|_W : W \rightarrow V.$$

Then the exact block identities are

$$A = \begin{pmatrix} B & T \\ 0 & D \end{pmatrix}, \quad A^\sharp = \begin{pmatrix} B^\sharp & 0 \\ T^\sharp & D^\sharp \end{pmatrix}, \quad H = \begin{pmatrix} B^\sharp + B - k I_V & T \\ T^\sharp & D^\sharp + D - k I_W \end{pmatrix}. \quad (\text{E1})$$

In particular, as maps $W \rightarrow V$,

$$PH|_W = PA|_W = T. \quad (\text{E2})$$

This equality includes the proof of the vanishing term: $PA^\sharp|_W = ((1 - P)AP)^\sharp|_W = 0$, because $(1 - P)AP = 0$.

Write

$$L = \text{Tr}_{\mathcal{C}}(PH) = \text{Tr}_V(B^\sharp + B - k I_V) = \sum_{\text{Re } \lambda > k/2} m_A^{\text{alg}}(\lambda)(2 \text{Re } \lambda - k), \quad (\text{E3})$$

where $m_A^{\text{alg}}(\lambda)$ is the original algebraic multiplicity of λ in A . The final equality follows by taking the trace of B , whose characteristic polynomial consists exactly of the selected full generalized blocks, and by $\text{Tr } B^\sharp = \overline{\text{Tr } B}$. Thus $L > 0$ exactly when $V \neq 0$.

If $\epsilon = 0$, then $H = 0$. For every eigenvector $x \neq 0$ of A ,

$$0 = \langle x, Hx \rangle_G = (2 \operatorname{Re} \lambda - k) \langle x, x \rangle_G,$$

so all eigenvalues have real part $k/2$, and $V = 0$. Hence the case $L > 0$ automatically has $\epsilon > 0$. Also $V \neq C$ in that case, since $P = I$ would give $L = \text{Tr } H = 0$.

32.2 2. Exact trace maximum and equality theorem

Theorem E4. With these definitions and $L > 0$,

$$0 < L \leq \epsilon,$$

and the following five statements are equivalent:

1. $L = \epsilon$.
2. $P = Q$.
3. V reduces A : both V and $V^\perp_G$ are A -invariant.
4. $PA = AP$.
5. $T = 0$.

Moreover, each statement is equivalent to the positive H -eigenline being contained in V and the negative H -eigenline being contained in W .

Proof. Choose vectors u_+, u_- on those two eigenlines with

$$\langle u_+, u_+ \rangle_G = \langle u_-, u_- \rangle_G = 1, \quad \langle u_+, u_- \rangle_G = 0.$$

This records the eigenline projectors using the existing form and does not replace that form. The full operator, including its zero eigenspace, is

$$Hx = \epsilon u_+ \langle u_+, x \rangle_G - \epsilon u_- \langle u_-, x \rangle_G. \quad (\text{E4a})$$

Put $a = \|Pu_+\|_G^2$, $b = \|Pu_-\|_G^2$. Then

$$L = \epsilon(a - b), \quad 0 \leq a, b \leq 1. \quad (\text{E4b})$$

This proves $L \leq \epsilon$. Equality holds precisely when $a = 1$ and $b = 0$. Orthogonality of P identifies these equalities with $u_+ \in V$ and $u_- \in W$. In that case (E4a) gives $PH(1 - P) = 0$, and (E2) gives $T = 0$.

By (E1), $T = 0$, reduction of A , and $PA = AP$ are equivalent. If $P = Q$, the identity $AQ = QA$ already proved implies $PA = AP$. Conversely, if $T = 0$, the decomposition in (E1) is a block diagonal decomposition of the original A . The spectrum of B , with multiplicities, is the selected spectrum, while the characteristic polynomial of D contains exactly the remaining factors: the full characteristic polynomial is the product of the two block characteristic polynomials. Applying the defining projector polynomial to these blocks gives $Q = I_V \oplus 0_W = P$.

Finally suppose $T = 0$. Equation (E1) shows that P commutes with H . The two nonzero eigenspaces of H are one-dimensional. The restriction of an idempotent commuting with H to either such eigenspace is therefore either identity or zero. It follows that $a, b \in \{0, 1\}$. Since $\epsilon(a - b) = L > 0$, necessarily $a = 1, b = 0$, and hence $L = \epsilon$. This proves every stated implication. $\square$

The hypothesis $L > 0$ matters for the converse from reduction to $L = \epsilon$. For example, an empty selected subspace has $P = Q = 0$ and reduces A , while $L = 0$ may hold with $\epsilon > 0$. The theorem concerns exactly the nonempty right-half-plane subspace that appears in (E3).

32.3 3. Sharp quantitative trace slack

For a map $F : W \rightarrow V$, set $\|F\|_{\text{HS},G}^2 = \text{Tr}_W(F^\sharp F)$. This is the Hilbert-Schmidt norm with the original restricted metrics on both its domain and codomain.

Keep u_+, u_- as above and set

$$c = \langle Pu_+, Pu_- \rangle_G, \quad M = \begin{pmatrix} a & c \\ \bar{c} & b \end{pmatrix}. \quad (\text{E5})$$

The matrix M is the compression of the original orthogonal projection P to $E = \text{span}(u_+, u_-)$, expressed in that specified eigenbasis. For $z = (z_+, z_-)$,

$$z^* M z = \|P(z_+ u_+ + z_- u_-)\|_G^2, \quad z^* (I_2 - M) z = \|(1 - P)(z_+ u_+ + z_- u_-)\|_G^2. \quad (\text{E6})$$

Both matrices are positive semidefinite. In particular,

$$ab - |c|^2 \geq 0, \quad (1 - a)(1 - b) - |c|^2 \geq 0. \quad (\text{E7})$$

Theorem E8. The exact identity is

$$\epsilon^2 - L^2 - \|T\|_{\text{HS},G}^2 = \epsilon^2 (\det M + \det(I_2 - M)). \quad (\text{E8})$$

Consequently,

$$\epsilon^2 - L^2 \geq \|T\|_{\text{HS},G}^2, \quad \epsilon - L \geq \frac{\|T\|_{\text{HS},G}^2}{\epsilon + L}. \quad (\text{E9})$$

Proof. By (E2) and selfadjointness of H ,

$$\begin{aligned} \|T\|_{\text{HS},G}^2 &= \text{Tr}_V(PH(1 - P)HP|_V) \\ &= \text{Tr}_C(PH^2P) - \text{Tr}_C((PHP)^2). \end{aligned} \quad (\text{E10})$$

Formula (E4a) gives $H^2x = \epsilon^2 u_+ \langle u_+, x \rangle_G + \epsilon^2 u_- \langle u_-, x \rangle_G$. Thus the first trace on the right is $\epsilon^2(a + b)$. The compressed operator is the difference of the two rank-one maps generated by Pu_+ and Pu_- , with coefficient ϵ . Squaring these maps and taking their traces gives

$$\text{Tr}_C((PHP)^2) = \epsilon^2(a^2 + b^2 - 2|c|^2).$$

For clarity, the cross trace used here is $\text{Tr}((v \langle v, \cdot \rangle_G)(w \langle w, \cdot \rangle_G)) = |\langle v, w \rangle_G|^2$, directly by composition of the two rank-one maps. Therefore

$$\|T\|_{\text{HS},G}^2 = \epsilon^2(a + b - a^2 - b^2 + 2|c|^2). \quad (\text{E11})$$

Subtract this expression and $L^2 = \epsilon^2(a - b)^2$ from ϵ^2 . The result is

$$\epsilon^2(1 - a - b + 2ab - 2|c|^2) = \epsilon^2((ab - |c|^2) + ((1 - a)(1 - b) - |c|^2)),$$

which is exactly (E8). Inequalities (E7) prove its nonnegativity, and division by the positive number $\epsilon + L$ proves the second inequality in (E9). $\square$

Equality in the first inequality of (E9) holds precisely when M has eigenvalues zero and one. Indeed, its eigenvalues μ_1, μ_2 lie in $[0, 1]$, and the two nonnegative determinants both vanish precisely when $\mu_1 \mu_2 = 0$ and $(1 - \mu_1)(1 - \mu_2) = 0$. This is equivalent to the two eigenvalues being zero and one. It also means that E reduces P and that $P|_E$ has rank one: a compression eigenvector with eigenvalue zero satisfies $\|Px\|^2 = 0$, while a compression eigenvector with eigenvalue one satisfies $\|(1 - P)x\|^2 = 0$, by (E6). The reverse implication follows by restricting a rank-one orthogonal projection on E .

The same inequality can be expressed by exact commutators on $\mathcal{C}$:

$$[P, A] = \begin{pmatrix} 0 & T \\ 0 & 0 \end{pmatrix}, \quad [P, H] = \begin{pmatrix} 0 & T \\ -T^\sharp & 0 \end{pmatrix},$$

so

$$\epsilon^2 - L^2 \geq \|[P, A]\|_{\text{HS},G}^2 = \frac{1}{2} \|[P, H]\|_{\text{HS},G}^2. \quad (\text{E12})$$

32.4 4. Exact morphism from the projector difference to the coupling

Because Q has image V and restricts to identity on V , there is a unique linear map $X : W \rightarrow V$ such that

$$Q = \begin{pmatrix} I_V & X \\ 0 & 0 \end{pmatrix}, \quad Q - P = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}. \quad (\text{E13})$$

The equation $AQ = QA$, with (E1), yields the **exact Sylvester morphism**

$$\mathcal{S}_{B,D} : \text{Hom}_{\mathbb{C}}(W, V) \longrightarrow \text{Hom}_{\mathbb{C}}(W, V), \quad \mathcal{S}_{B,D}(X) = BX - XD = T. \quad (\text{E14})$$

It is an isomorphism. Here is an explicit proof and inverse retaining the original metrics. Set $\beta = \min_{\lambda \in \sigma(B)} \text{Re } \lambda$ and $\delta = \max_{\mu \in \sigma(D)} \text{Re } \mu$. By the full spectral selection, $\beta > k/2 \geq \delta$. The original operator exponentials have polynomial-times-exponential bounds because each nilpotent part in each generalized eigenspace has finite order. More explicitly, with the spectral projectors $\Pi_{\lambda}^B, \Pi_{\mu}^D$ and their nilpotency orders $\nu_{\lambda}^B, \nu_{\mu}^D$, the exact formulas are

$$\begin{aligned} e^{-tB} &= \sum_{\lambda \in \sigma(B)} e^{-t\lambda} \sum_{j=0}^{\nu_{\lambda}^B-1} \frac{(-t)^j}{j!} (B - \lambda I)^j \Pi_{\lambda}^B, \\ e^{tD} &= \sum_{\mu \in \sigma(D)} e^{t\mu} \sum_{\ell=0}^{\nu_{\mu}^D-1} \frac{t^{\ell}}{\ell!} (D - \mu I)^{\ell} \Pi_{\mu}^D. \end{aligned} \quad (\text{E15})$$

These formulas follow by expanding the exponential of scalar plus nilpotent on every generalized eigenspace; every sum in the nilpotent part terminates at the displayed order. Consequently the following integral converges in the original operator norm, for every $Y : W \rightarrow V$:

$$\mathcal{S}_{B,D}^{-1}(Y) = \int_0^{\infty} e^{-tB} Y e^{tD} dt. \quad (\text{E16})$$

Differentiating the integrand and integrating its derivative from zero to infinity proves

$$B \left(\int_0^{\infty} e^{-tB} Y e^{tD} dt \right) - \left(\int_0^{\infty} e^{-tB} Y e^{tD} dt \right) D = Y.$$

There is no upper-end boundary term, by $\beta - \delta > 0$ and the finite polynomial factors in (E15). If $BZ - ZD = 0$, differentiation shows that $e^{-tB} Z e^{tD}$ is constantly Z , while the same decay shows its limit is zero. Thus $Z = 0$, proving uniqueness and the asserted isomorphism.

Equip $\text{Hom}(W, V)$ with its original Hilbert-Schmidt norm, and define its actual positive smallest singular value

$$s_{B,D} = \min_{\|Y\|_{\text{HS},G}=1} \|BY - YD\|_{\text{HS},G} > 0. \quad (\text{E17})$$

The minimum exists because the unit sphere is compact, and is positive because (E14) is injective. The identity (E13) gives $\|X\|_{\text{HS},G} = \|Q - P\|_{\text{HS},G}$. Combining (E9), (E14), and (E17) proves

$$\epsilon^2 - L^2 \geq \|T\|_{\text{HS},G}^2 \geq s_{B,D}^2 \|Q - P\|_{\text{HS},G}^2. \quad (\text{E18})$$

Thus the difference between the algebraic spectral projector and the orthogonal projection is related to the trace defect by a proved, invertible, typed map.

An explicit bound on the inverse is also available with all original operator norms retained. Write

$$\kappa_{B,D} = \int_0^{\infty} \|e^{-tB}\|_{\text{op},G} \|e^{tD}\|_{\text{op},G} dt < \infty.$$

The ideal inequality $\|RYS\|_{\text{HS}} \leq \|R\|_{\text{op}}\|Y\|_{\text{HS}}\|S\|_{\text{op}}$ follows by diagonalizing the positive operators $R^\sharp R$ and $SS^\sharp$ in their respective existing Hermitian spaces and bounding their largest eigenvalues. Applying it under (E16) yields

$$\|Q - P\|_{\text{HS},G} \leq \kappa_{B,D}\|T\|_{\text{HS},G} \leq \kappa_{B,D}\sqrt{\epsilon^2 - L^2}. \quad (\text{E19})$$

If desired, (E15) gives the fully specified finite upper bound

$$\kappa_{B,D} \leq \sum_{\lambda,\mu} \sum_{j=0}^{\nu_\lambda^B-1} \sum_{\ell=0}^{\nu_\mu^D-1} \frac{(j+\ell)! \|(B-\lambda I)^j \Pi_\lambda^B\|_{\text{op},G} \|(D-\mu I)^\ell \Pi_\mu^D\|_{\text{op},G}}{j! \ell! (\text{Re } \lambda - \text{Re } \mu)^{j+\ell+1}}. \quad (\text{E20})$$

The scalar integral used is $\int_0^\infty t^n e^{-at} dt = n!/a^{n+1}$ for $a > 0$, proved by the substitution $v = at$ and n integrations by parts beginning with $\int_0^\infty e^{-v} dv = 1$. Every denominator in (E20) is positive. No bound on nonnormal operator norms has been replaced by a bound on eigenvalue real parts.

32.5 5. Sharp original two-dimensional family

Fix $k \in \mathbb{R}$, $\ell > 0$, and $t \in \mathbb{C}$. On $\mathbb{C}^2$ with its specified form $G = I_2$, define

$$A = \begin{pmatrix} k/2 + \ell/2 & t \\ 0 & k/2 - \ell/2 \end{pmatrix}, \quad H = A^* + A - kI_2 = \begin{pmatrix} \ell & t \\ \bar{t} & -\ell \end{pmatrix}. \quad (\text{E21})$$

Here $H^2 = (\ell^2 + |t|^2)I_2$, so $\epsilon = \sqrt{\ell^2 + |t|^2}$. The selected full generalized spectral subspace is exactly $V = \mathbb{C}(1, 0)$, since the two eigenvalues $k/2 \pm \ell/2$ are distinct. The original orthogonal projection is $P = \text{diag}(1, 0)$. Hence

$$L = \ell, \quad T = t : \mathbb{C} \rightarrow \mathbb{C}, \quad Q = \begin{pmatrix} 1 & t/\ell \\ 0 & 0 \end{pmatrix}, \quad s_{B,D} = \ell. \quad (\text{E22})$$

The formula for Q is also obtained directly from the polynomial $(A - (k/2 - \ell/2)I_2)/\ell$; thus its stated kernel is the full other eigenspace, with no orthogonality assumption. Direct calculation gives

$$\epsilon^2 - L^2 = |t|^2 = \|T\|_{\text{HS},G}^2 = s_{B,D}^2 \|Q - P\|_{\text{HS},G}^2. \quad (\text{E23})$$

Both quantitative constants in (E18) are attained by this family, including every $t \neq 0$. The equality theorem gives $L = \epsilon$ precisely when $t = 0$, exactly when $P = Q$ and the selected subspace reduces A .

32.6 6. Critical generalized blocks at the trace maximum

The equality theorem also constrains every full critical generalized block, including its original nilpotent structure.

Theorem E24. Suppose $L > 0$ and $L = \epsilon$. For every eigenvalue λ of the original operator A with $\text{Re } \lambda = k/2$, let

$$Z_\lambda = \ker(A - \lambda I)^{\nu_\lambda}$$

be its full generalized eigenspace, where ν_λ is its original exponent in the minimal polynomial. Then

$$HZ_\lambda = 0, \quad A|_{Z_\lambda} = \lambda I_{Z_\lambda}, \quad A^\sharp|_{Z_\lambda} = (k - \lambda)I_{Z_\lambda} = \bar{\lambda}I_{Z_\lambda}. \quad (\text{E24})$$

In particular, Z_λ reduces A , its spectral projector equals its G -orthogonal projection, and the original eigenvalue λ is semisimple. Consequently, the presence of even one critical-line Jordan chain of length greater than one implies the strict inequality

$$\epsilon > L \quad (\text{E25})$$

whenever $L > 0$.

Proof. At equality, Theorem E4 gives $P = Q$, $u_+ \in V$, and $u_- \in W$. Hence the restriction of the original Hermitian defect to the original orthogonal complement is

$$H|_W : w \longmapsto -\epsilon u_- \langle u_-, w \rangle_G. \quad (\text{E26})$$

It is negative semidefinite and has rank one. Every critical generalized eigenspace is in $\ker Q = W$, by the defining polynomial for Q . This conclusion uses the proved equality $P = Q$; it makes no claim that the original unselected spectral subspace was orthogonal before that equality was proved.

Fix such a λ , and let R_λ be the original G -orthogonal projection onto Z_λ . The subspace Z_λ is A -invariant. With $A_\lambda = A|_{Z_\lambda}$, its compressed Hermitian form is therefore

$$R_\lambda H|_{Z_\lambda} = A_\lambda^\sharp + A_\lambda - kI_{Z_\lambda}. \quad (\text{E27})$$

The adjoint on the right uses the original form restricted to Z_λ ; (E27) follows from the defining adjoint identity for two vectors in that subspace. The trace is

$$\text{Tr}_{Z_\lambda}(R_\lambda H|_{Z_\lambda}) = 2 \text{Re Tr } A_\lambda - k \dim Z_\lambda = (2 \text{Re } \lambda - k) \dim Z_\lambda = 0. \quad (\text{E28})$$

Here $A_\lambda = \lambda I + N_\lambda$ with N_λ nilpotent, so $\text{Tr } A_\lambda = \lambda \dim Z_\lambda$. To justify the nilpotent trace directly, choose a basis compatible with the filtration $\ker N_\lambda^j$; the map N_λ lowers that filtration and its matrix has zero diagonal.

Choose any G -orthonormal basis $z_1, \dots, z_r$ of Z_λ , solely to evaluate the trace in its existing form. Equations (E26) and (E28) give

$$0 = \sum_{j=1}^r \langle z_j, H z_j \rangle_G = -\epsilon \sum_{j=1}^r |\langle u_-, z_j \rangle_G|^2. \quad (\text{E29})$$

Since $\epsilon > 0$, every displayed square is zero. Formula (E26) now gives $H z_j = 0$ for every basis vector, and hence $H Z_\lambda = 0$.

The defining equation $H = A^\sharp + A - kI$ implies, for every $z \in Z_\lambda$,

$$A^\sharp z = kz - Az \in Z_\lambda. \quad (\text{E30})$$

Thus Z_λ is invariant under both A and $A^\sharp$. Its orthogonal complement is A -invariant: if $w \perp_G Z_\lambda$ and $z \in Z_\lambda$, then $\langle z, Aw \rangle_G = \langle A^\sharp z, w \rangle_G = 0$. It is also $A^\sharp$ -invariant by the corresponding identity with A and $A^\sharp$ exchanged. This proves the asserted reduction and shows that the restricted adjoint in (E27) is the actual restriction of $A^\sharp$.

On Z_λ put $N = A_\lambda - \lambda I$. It is nilpotent by the definition of this full generalized eigenspace, and (E30) gives

$$N^\sharp = kI - A_\lambda - \bar{\lambda}I = \lambda I - A_\lambda = -N, \quad (\text{E31})$$

because $\lambda + \bar{\lambda} = k$. A nilpotent skew-adjoint map on a positive-definite Hermitian space is zero, as the following calculation proves. If $N \neq 0$, take its nilpotency order $r \geq 2$, choose v with $N^{r-1}v \neq 0$, and set $y = N^{r-1}v$, $z = N^{r-2}v$. Then $Nz = y \neq 0$ and $Ny = 0$, whereas

$$\|y\|_G^2 = \langle Nz, y \rangle_G = \langle z, N^\sharp y \rangle_G = -\langle z, Ny \rangle_G = 0, \quad (\text{E32})$$

contradicting positive definiteness. Consequently $N = 0$, proving (E24).

Because Z_λ reduces A , the original orthogonal direct sum $\mathcal{C} = Z_\lambda \oplus Z_\lambda^{\perp_G}$ gives a block diagonal matrix for A . The complement contains exactly the other generalized eigenspaces: its characteristic polynomial is the quotient of the full characteristic polynomial by that of the full λ -generalized block, so it has no eigenvalue λ . The polynomial spectral projector for λ is therefore identity on Z_λ and zero on its orthogonal complement. It equals R_λ , as claimed. Finally, (E4) already proves $L \leq \epsilon$; (E24)

rules out equality whenever a critical Jordan chain of length greater than one is present. This gives (E25). $\square$

Distinct critical eigenspaces are also mutually orthogonal in the original form. For $x \in Z_\lambda, y \in Z_\mu$, (E24) gives

$$\bar{\lambda}\langle x, y \rangle_G = \langle Ax, y \rangle_G = \langle x, A^\sharp y \rangle_G = \bar{\mu}\langle x, y \rangle_G.$$

If $\lambda \neq \mu$, the inner product vanishes. Therefore the full critical subspace $Z = \bigoplus_{\operatorname{Re} \lambda = k/2} Z_\lambda$ is an original orthogonal sum, reduces A , and has the exact diagonal action $\sum_\lambda \lambda R_\lambda$, with $H|_Z = 0$. The shifted operator $A|_Z - (k/2)I_Z$ is skew-adjoint. This describes the full critical block and its projectors, retaining all its original eigenvalues and multiplicities.

32.7 The original quartet, its critical blocks, and a canonical equality example

Apply the preceding maps to the actual cyclic module and canonical Gram G_N of the complete exterior source note. For the exact packet $h(s) = \prod_{\varepsilon, \eta = \pm 1} (s - (\frac{1}{2} + \varepsilon\delta + i\eta\gamma))^m$, $\delta, \gamma > 0$, its distinct sum roots and full lengths are

$$\lambda_{a,b} = k/2 + \delta(2a - k) + i\gamma(2b - k), \quad 0 \leq a, b \leq k, \quad \ell_k = 1 + k(m - 1), \quad q = \ell_k(k + 1)^2. \quad (\text{E33})$$

Indeed the four occupation counts $(t, a - t, b - t, k - a - b + t)$ exist for every integer $\max(0, a + b - k) \leq t \leq \min(a, b)$; conversely each tuple has these two margins. Equality of sums forces both margins to agree because δ, γ are nonzero real numbers. The top surviving local monomial has degree $k(m - 1)$ and coefficient $(k(m - 1))! / ((m - 1)!)^k$, proving the stated full nilpotent length. Summing positive real defects over these actual cyclic blocks gives

$$L = 2\delta\ell_k(k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}^{\text{cyc}}, \quad N \geq q - 1. \quad (\text{E34})$$

For $k = 2n$ the sum of $2a - k$ over $a > k/2$ is $n(n + 1)$; for $k = 2n + 1$ it is $(n + 1)^2$, proving the floor and its constant. If k is even and $m > 1$, each $a = k/2$ block is critical and has length $\ell_k > 1$. The full cyclic local algebra is a single Jordan block in powers of $S - \lambda_{a,b}$, so (E24)–(E25) prove the strict inequality $\epsilon_{h,k,N}^{\text{cyc}} > L$ at every admitted N . For a larger packet containing this quartet, the same sums survive as CRT factors and their maximal nilpotent lengths are at least ℓ_k . Thus (E34) remains a lower bound; its admission threshold is the larger packet's actual $N \geq \deg \chi_{h,k} - 1$.

Equality can occur for a specified positive canonical moment measure. Here is a complete calibration, with no claim about zeta zeros. Take $k = 1, \mu, \sigma > 0$ and

$$d\nu(u) = \frac{\mu}{\sqrt{2\pi\sigma}} e^{-u^2/(2\sigma^2)} du, \quad S = \frac{1}{2} + iu, \quad \chi = (S - \frac{1}{2})^2 - \sigma^2, \quad N = 1. \quad (\text{E35})$$

The Gaussian integral and one integration by parts give mass μ , mean zero and second moment $\mu\sigma^2$; odd moments vanish by the original reflection $u \mapsto -u$. Consequently the original monic orthogonal polynomials through degree two are $p_0 = 1, p_1 = S - \frac{1}{2}, p_2 = (S - \frac{1}{2})^2 + \sigma^2$, with $\omega_0 = \mu, \omega_1 = \mu\sigma^2$. In the fixed remainder coordinates $(1, S)$ the column map $B = [b_0, b_1] = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}$ is invertible, and the actual canonical Gram is $G = B^{-*} \operatorname{diag}(\mu, \mu\sigma^2) B^{-1}$. Reduction gives $b_2 = 2\sigma^2 b_0$, whence

$$a = \mu\sigma^2, \quad d = 4\mu\sigma^4, \quad c = 0, \quad \epsilon = 2\sigma = L. \quad (\text{E36})$$

The formula follows directly from $\epsilon^2 = (ad - |c|^2)/\omega_1^2$ and the two simple roots $1/2 \pm \sigma$. All original masses and coordinates are retained. For $0 < \sigma < 1/2$ both roots lie in the open critical strip, but they are declared calibration roots. Thus positive measure and canonical interpolation alone cannot supply universal strictness; (E14)–(E18) identify the exact projector and coupling maps governing equality for each original arithmetic measure.

The full exterior cochain idempotent also has its complementary cochain idempotent. For an idempotent e commuting with d , the maps $x \mapsto (ex, (1 - e)x)$ and $(u, v) \mapsto u + v$ are inverse cochain maps, since $e(1 - e) = 0$ and both summands are d -stable. They induce the same inverse maps on cycles modulo boundaries. Ordinary traces here are taken on the retained finite arithmetic submodules: the signed tensor permutation preserves the finite packet, so restriction of e and $1 - e$ gives finite invariant summands and inclusion followed by projection is the stated idempotent. In a basis of these two summands, a commuting finite operator is block diagonal, proving additivity of its ordinary trace; its top cochain sign is the original $(-1)^{kp}$. The splitting of the entire possibly infinite-dimensional theta cohomology itself supplies these maps but no ordinary finite trace.

32.8 7. Integration scope

Equations (E3), (E4), (E8), (E18), and (E24) may be applied directly whenever the exterior-amplification construction supplies its actual finite Hermitian form, its original multiplication operator, and its proved rank-two trace-zero Hermitian defect. All constants $G, k, \epsilon, L, s_{B,D}$ remain attached to that specific construction and degree. The proof establishes an equality classification, an exact projector morphism, a sharp quantitative slack identity, and strictness in the presence of a retained critical Jordan chain. It supplies no uniform estimate for $s_{B,D}$, $\kappa_{B,D}$, or ϵ as arithmetic parameters change; such an estimate would require a further calculation on those original varying objects.

Independent algebra check: the determinant-slack calculation (E8)–(E11), its equality classification, and the sharp family (E21)–(E23) were separately recomputed by the bounded `determinant_slack_check` agent. That agent then independently checked all steps of (E24)–(E32), including the passage from zero compressed trace to ambient annihilation and the use of full primary spaces in identifying the projectors. Both derivations agreed with the formulas above. The projector conclusion applies to each full primary space, including every Jordan block at its eigenvalue; it asserts nothing about arbitrary coordinate projectors onto individually selected copies of a repeated eigenspace. These are second mathematical derivations, not Lean or numerical certificates.

33 The cyclic control as an exact change of arithmetic interpolation volume

Retain a nonempty reflection-stable packet with every original order, the tensor degree $k \geq 1$, the cyclic algebra $C = \mathbb{C}[S]/(\chi)$ of dimension $q \geq 1$, and its fixed monic remainder coordinates. Let $p_j(S)$ be the original monic orthogonal polynomials on $S = k/2 + iu$ for the measure $m_{h,k}(u) du$, and put

$$b_j = [p_j]_\chi, \quad \omega_j = \int_{\mathbb{R}} |p_j(k/2 + iu)|^2 m_{h,k}(u) du > 0, \quad K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad G_N = K_N^{-1}. \quad (\text{CV.1})$$

Here $N \geq q - 1$; the original $\omega_0 = \mu_h^k$ remains. All determinants below are taken in these same coordinates. Set $K_{-1} = 0$ and

$$D_j = \det K_j, \quad \tilde{K}_N = K_{N-1} + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad (\text{CV.2})$$

$$a = b_N^* G_N b_N, \quad d = b_{N+1}^* G_N b_{N+1}, \quad c = b_N^* G_N b_{N+1}.$$

The matrix $\tilde{K}_N$ records the actual polynomial subspace spanned by $p_0, \dots, p_{N-1}, p_{N+1}$, with its inherited norm: the degree- N orthogonal direction is omitted and the degree- $(N + 1)$ direction is retained. It can be singular. No inverse for it is used.

Proposition 33.1 (Exact adjacent-volume identities). *The following equalities hold, including $N = q - 1$:*

$$\begin{aligned} 1 - \frac{D_{N-1}}{D_N} &= \frac{a}{\omega_N}, \\ \frac{D_{N+1}}{D_N} - 1 &= \frac{d}{\omega_{N+1}}, \\ \frac{\tilde{D}_N}{D_N} &= \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d}{\omega_{N+1}}\right) + \frac{|c|^2}{\omega_N \omega_{N+1}}. \end{aligned} \quad (\text{CV.3})$$

Consequently the original rank-two control allowance satisfies

$$\boxed{(\epsilon_{h,k,N}^{\text{cyc}})^2 = \frac{\omega_{N+1}}{\omega_N} \frac{D_{N+1} + D_{N-1} - D_N - \tilde{D}_N}{D_N}}. \quad (\text{CV.4})$$

Equivalently, retaining the entire complex cross term,

$$(\epsilon_{h,k,N}^{\text{cyc}})^2 = \frac{\omega_{N+1}}{\omega_N} \left[\left(1 - \frac{D_{N-1}}{D_N}\right) \left(\frac{D_{N+1}}{D_N} - 1\right) - \frac{|c|^2}{\omega_N \omega_{N+1}} \right]. \quad (\text{CV.5})$$

Proof. The first q monic polynomials give a triangular basis modulo χ , so K_N is positive definite in the stated range. For an invertible K and columns x, y , direct block elimination of $\begin{pmatrix} K & x \\ -y^* & 1 \end{pmatrix}$ in its two possible orders gives $\det(K + xy^*) = \det K(1 + y^* K^{-1}x)$. Applying this with $K = K_N$ and the displayed signed increments gives the first two equations, even if the matrix after subtraction is singular. For completeness the same block elimination with two columns proves $\det(K + UV^*) = \det K \det(I_2 + V^* K^{-1}U)$. Take

$$U = \left(-\frac{b_N}{\sqrt{\omega_N}}, \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}\right), \quad V = \left(\frac{b_N}{\sqrt{\omega_N}}, \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}\right).$$

The positive square roots only specify this auxiliary factorization; the original columns, coordinates and norms in (CV.1) are unchanged. The resulting two-by-two determinant is

$$\det \begin{pmatrix} 1 - a/\omega_N & c/\sqrt{\omega_N \omega_{N+1}} \\ -\bar{c}/\sqrt{\omega_N \omega_{N+1}} & 1 + d/\omega_{N+1} \end{pmatrix},$$

which proves the third equation with its plus sign. Subtract that equation from the sum of the first two determinant ratios as ordered in (CV.4). The resulting value is $(ad - |c|^2)/(\omega_N \omega_{N+1})$. The original control formula is $\epsilon^2 = (ad - |c|^2)/\omega_N^2$, proving (CV.4)–(CV.5). In particular its numerator is nonnegative by the Gram determinant of the two original vectors b_N, b_{N+1} in G_N . For $N = q - 1$, K_{N-1} has rank $q - 1$, so $D_{N-1} = 0$ and $a = \omega_N$; every equation still holds. For $q = 1$ the two-vector Gram determinant is zero, consistently with the trace-zero one-dimensional control. There is no positive-dimensional determinant or Gram inverse to apply when $h = 1$; the coefficient quotient is then zero and its unique maps have the split lift $\{\tau, e\}$. $\square$

33.1 The exact volume map and the reflected quartet bound

Let $\mathcal{P}_N$ be the span of $p_0, \dots, p_N$ with its original integral norm, and let $T_N : \mathcal{P}_N \rightarrow C$ be reduction modulo χ . In the displayed polynomial basis its matrix is $[b_0 \cdots b_N]$, and its source Gram is $\text{diag}(\omega_0, \dots, \omega_N)$. The adjoint is therefore the explicit map

$$T_N^* x = \sum_{j=0}^N p_j \frac{b_j^* x}{\omega_j}, \quad T_N T_N^* = K_N, \quad (\text{CV.6})$$

where the star uses the fixed Euclidean pairing on the written target coordinates, solely to specify this typed adjoint. The canonical right inverse is $T_N^* G_N$, whose original source Gram is G_N . Taking its q th exterior power maps the fixed unscaled coordinate wedge to a vector of squared norm $\det G_N = D_N^{-1}$ in $\bigwedge^q \mathcal{P}_N$. The exact exterior source map is

$$\bigwedge^q \mathcal{V}_{h,k} : \bigwedge^q \mathcal{P}_N \longrightarrow \bigwedge^q (\mathcal{B}^{\widehat{\otimes} k}).$$

It preserves the determinant norm D_N^{-1} , because the original source map preserves every pairwise integral inner product. To pass to the original tensor cochain source, use instead $\mathcal{V}_{h,k}^{\otimes q} \text{Alt}_q$, where $\text{Alt}_q(v_1 \wedge \cdots \wedge v_q) = \sum_{\pi \in S_q} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(q)}$. The exterior quotient satisfies $\text{wed Alt}_q = q! I$: every permuted summand contributes its permutation sign twice. Expanding the tensor inner product gives $q!$ copies of its determinant, so the literal tensor-source squared norm is $q! D_N^{-1}$. Its target is $\mathcal{B}^{\widehat{\otimes} kq}$ in top cochain degree kq . This proves the asserted interpolation-volume interpretation without changing its target metric or source mass. Replacing T_N by its restriction to the omitted-direction subspace gives exactly $\widetilde{K}_N$; the construction needs no assertion that this restricted map is onto.

Now take the packet polynomial to be exactly

$$h(s) = \prod_{\varepsilon, \eta \in \{+1, -1\}} \left(s - \left(\frac{1}{2} + \varepsilon \delta + i \eta \gamma \right) \right)^m$$

for a complete actual nonreal quartet $\frac{1}{2} \pm \delta \pm i\gamma$, $\delta, \gamma > 0$, of common order m , and set $\ell_k = 1 + k(m-1)$. Its cyclic annihilator has $q = \ell_k(k+1)^2$. The complete exterior trace calculation gives $L_k \leq \epsilon$, where

$$L_k = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{CV.7})$$

Substitution in (CV.4), with all quantities nonnegative, proves the exact necessary arithmetic-volume inequality

$$\boxed{D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N \geq \frac{\omega_N}{\omega_{N+1}} D_N L_k^2, \quad N \geq \ell_k(k+1)^2 - 1.} \quad (\text{CV.8})$$

The cross term has been transferred to the actual omitted-direction determinant by (CV.3); it has not been discarded. Thus (CV.4) supplies an exact expression for the still unestimated quantity

$$\frac{\epsilon^2}{k^6} = \frac{\omega_{N+1}}{k^6 \omega_N} \frac{D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N}{D_N}. \quad (\text{CV.9})$$

Every term comes from the specified arithmetic convolution measure and the complete quotient. No decay rate for (CV.9) is asserted here. The formula identifies which signed determinant combination, and which orthogonal-norm ratio at growing degree, the next analytic calculation has to estimate together. For a larger reflection-stable packet containing this quartet, the quartet sums remain present and their full nilpotent lengths are at least ℓ_k . Indeed the same ordered quartet factors embed as CRT summands of the larger tensor algebra, and the largest nilpotent length at a fixed sum is a maximum over all such summands. Their positive trace contribution is therefore at least (CV.7). Inequality (CV.8) remains valid, with every K_j, D_j, ω_j formed from that larger packet and the admission threshold replaced by its actual $N \geq \deg \chi_{h,k} - 1$. No equality for that larger degree is inferred from the quartet subpacket.

33.2 The positive exterior minor sum and its actual source vector

Write $\Delta_N = D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N$. For an increasing finite index set J , let B_J be the matrix whose columns are the original b_j , in that increasing order, and put $\Omega_J = \prod_{j \in J} \omega_j$, with the empty

product equal to one. For every $q \geq 2$ and $N \geq q - 1$,

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i}. \quad (\text{CV.10})$$

For $q = 2$ the index set I is empty and the sum has its single displayed two-column term. For $q = 1$ set this sum to zero; then $\Delta_N = 0$ exactly. Thus the signed combination in (CV.4) is an explicit sum of positive exterior contributions, retaining every original column and squared polynomial norm.

Proof. First take any finite set L of column indices and form $K_L = \sum_{j \in L} b_j b_j^* / \omega_j$. Multilinearity of its q columns expands its determinant as

$$\det K_L = \sum_{(j_1, \dots, j_q) \in L^q} \det[b_{j_1}, \dots, b_{j_q}] \prod_{r=1}^q \overline{(b_{j_r})_r} / \omega_{j_r}.$$

If two indices coincide, the determinant in that summand is zero. For a fixed increasing set $J = \{i_1 < \dots < i_q\}$, the remaining terms are indexed by its permutations. Their sum is

$$\frac{\det B_J}{\Omega_J} \sum_{\sigma \in \mathfrak{S}_q} \text{sgn}(\sigma) \prod_{r=1}^q \overline{(b_{i_{\sigma(r)}})_r} = \frac{|\det B_J|^2}{\Omega_J}.$$

This proves directly, including singular K_L , $\det K_L = \sum_{J \subseteq L, |J|=q} |\det B_J|^2 / \Omega_J$. Apply it to the four index sets

$$\{0, \dots, N+1\}, \quad \{0, \dots, N-1\}, \quad \{0, \dots, N\}, \quad \{0, \dots, N-1, N+1\}.$$

For each q -element subset of the first set, the coefficients in the signed combination defining Δ_N are respectively $1 + 1 - 1 - 1 = 0$ if it contains neither N nor $N+1$, $1 - 1 = 0$ if it contains exactly one of them, and 1 if it contains both. The surviving subsets have exactly the form $I \cup \{N, N+1\}$ in (CV.10). If $q = 1$, no subset contains both indices and the result is zero. No inverse for a smaller or omitted-direction Gram matrix has occurred, so this proof also applies at $N = q - 1$. $\square$

There is a precise source map for the positive sum. Let $e_C = [1] \wedge [S] \wedge \dots \wedge [S^{q-1}]$ be the fixed coordinate wedge in $\wedge^q C$, of squared norm one for the coordinate Euclidean pairing used in (CV.6). For $q \geq 2$ define

$$\mathcal{E}_N = \left(\bigwedge_{i=1}^{q-2} \mathcal{P}_{N-1} \right) \wedge p_N \wedge p_{N+1} \subseteq \bigwedge^q \mathcal{P}_{N+1}, \quad \Lambda_N = \left(\bigwedge^q T_{N+1} \right) |_{\mathcal{E}_N}.$$

The increasing vectors $e_I = (\bigwedge_{i \in I} p_i) \wedge p_N \wedge p_{N+1}$ form an orthogonal basis of $\mathcal{E}_N$ with their original squared norms $\Omega_{I \cup \{N, N+1\}}$. They satisfy $\Lambda_N e_I = \det B_{I \cup \{N, N+1\}} e_C$. Consequently the adjoint in these explicitly specified source and target pairings gives the original, unscaled vector

$$\zeta_N := \Lambda_N^* e_C = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\Omega_{I \cup \{N, N+1\}}} e_I, \quad (\text{CV.10a})$$

$$\|\zeta_N\|^2 = \Delta_N, \quad \Lambda_N \zeta_N = \Delta_N e_C, \\ \|\mathcal{V}_{h,k}^{\otimes q} \text{Alt}_q \zeta_N\|^2 = q! \Delta_N.$$

The first two equalities follow by inserting the displayed orthogonal basis and its full norms. The last equality follows from the pairwise isometry of $\mathcal{V}_{h,k}$ and the unscaled alternating-map calculation following (CV.6). Its target is the original tensor source $\mathcal{B}^{\widehat{\otimes} kq}$ in degree kq . For $q = 1$ take $\mathcal{E}_N = \{0\}$, $\Lambda_N = 0$, and $\zeta_N = 0$, so that all three zero identities remain exact. Thus (CV.10) comes with an explicitly typed original arithmetic source vector and the literal tensor factorial. Combining it with (CV.4) gives $\epsilon^2 = (\omega_{N+1} / \omega_N) \|\zeta_N\|^2 / D_N$; no estimate as k or N grows is inferred from this equality.

33.3 The exact first admitted degree

At $N = q - 1$ expand the original monic annihilator in the original orthogonal basis:

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\int \overline{p_j(k/2 + iu)} \chi(k/2 + iu) m_{h,k}(u) du}{\omega_j}. \quad (\text{CV.11})$$

This is a finite polynomial identity: both leading coefficients are one, and orthogonality gives every displayed coefficient, without a change of measure. The reduction map T_{q-1} is a linear isomorphism, since the q original monic remainders are a basis. Its inverse is therefore its canonical minimum section. Consequently $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $0 \leq i, j < q$. Reduction of (CV.11) gives $b_q = -\sum_{j<q} \gamma_j b_j$ and hence

$$a = \omega_{q-1}, \quad c = -\gamma_{q-1} \omega_{q-1}, \quad d = \sum_{j<q} |\gamma_j|^2 \omega_j, \quad (\text{CV.12})$$

$$\boxed{\epsilon_{h,k,q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\|\chi\|^2 - \omega_q - |\gamma_{q-1}|^2 \omega_{q-1}}{\omega_{q-1}}}.$$

The first expression follows by substituting the preceding three exact Gram entries into (CV.4); the second follows by Pythagoras applied to (CV.11). Reflection makes c purely imaginary, so the same is true of γ_{q-1} ; it has not been set to zero. Let $\mathcal{P}_{q-2}$ be the original integral orthogonal projection onto $\mathcal{P}_{q-2}$, with $\mathcal{P}_{-1} = 0$ for $q = 1$. Then the numerator in (CV.12) is exactly $\|\mathcal{V}_{h,k} \mathcal{P}_{q-2} \chi\|^2$ by the original source isometry. Meanwhile the full $\mathcal{V}_{h,k} \chi$ has zero cyclic and full jets, since $\chi(S) \in I$, and the original fixed-order polynomial division gives its theta primitive as in (CM.19). These are different specified source maps: projecting that boundary is an integral orthogonal operation and has the cyclic reduction $T_{q-2}(\mathcal{P}_{q-2} \chi) = \sum_{j<q-1} \gamma_j b_j$, where T_{q-2} is the restriction of T_{q-1} to $\mathcal{P}_{q-2}$; its actual full arithmetic jet is $J^{(k)} \mathcal{V}_{h,k} \mathcal{P}_{q-2} \chi = \eta(\sum_{j<q-1} \gamma_j b_j)$. For $q = 1$ the first map is the unique map from $\mathcal{P}_{-1} = 0$. This exact formula retains its possibly nonzero observation. For $q = 1$ the sum is empty and both sides are zero. The degree- q annihilator and its projection have thus given a complete original-source formula at the first admitted degree, with no large-degree estimate added.

34 The original source and relation Toda flows with their exact control maps

This is a standalone continuation of the identities CV.1–CV.12. It retains the original polynomial quotient, arithmetic measure, monic coordinates, arithmetic unit, source relations and complex phase. The delivered `Tau_Toda_Volume_Control/NOTE.tex` was read in full from its original archive; its SHA256 is

027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375.

The calculations below prove the exact maps underlying the determinant quotient, its first and second tilt derivatives, the omitted-direction determinant, the positive minor sum and the finite upper bound. No asymptotic arithmetic estimate is asserted.

34.1 Original arithmetic objects and the strip of real tilts

The original spaces and transforms are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx. \end{aligned}$$

The series and its Euler derivatives converge uniformly at large x with every power of decay. For small x , Poisson summation with Fourier kernel $e^{-2\pi ixy}$ gives $\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x)$; both zero terms vanish because $\phi(0) = \hat{\phi}(0) = 0$. The Fourier transform preserves the stated two zero constraints and evenness. This formula proves every small- x seminorm after Euler differentiation, hence $\Theta : V \rightarrow \mathcal{B}$. It also commutes with D on these spaces by termwise differentiation. Mellin transforms of elements of $\mathcal{B}$ are entire, and integration by parts gives $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$; both endpoint terms vanish with their original coefficients.

Let h be a monic, nonempty packet of actual nontrivial zeros of

$$g(s) = 2\zeta(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with their complete orders. Assume the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Set $d = \deg h$ and retain

$$v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = [v_h]_h \in E_h^\times, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}. \quad (\text{TVB.1})$$

Here the jet of an entire function f means its unique polynomial Hermite remainder of degree less than d : in the local factor $\mathbb{C}[t]/(t^{m_\rho})$, it is $\sum_{j < m_\rho} f^{(j)}(\rho) t^j / j!$, with $t = s - \rho$. The coprime factors of h give the isomorphism to their product by monic division and Bezout's identity, which supplies the unique global remainder. The constant $v_h(\rho)$ is nonzero at each selected zero because its full order was divided out. If the local jet is $\sum a_j t^j$, its inverse coefficients are $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$ for $n < m_\rho$. Multiplication verifies this inverse in every local factor, and the product isomorphism gives the inverse of v_h .

The original F_h has a construction within $\mathcal{B}$. For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy.$$

The equality is the vanishing Mellin value. For any desired power bound at infinity choose $A > \Re \rho$ with $|F(y)| \leq C_A y^{-A}$ for $y \geq 1$; the first integral is bounded by $C_A x^{-A}/(A - \Re \rho)$ there. At zero choose B with $B + \Re \rho > 0$ and $|F(y)| \leq C_B y^B$ for $0 < y \leq 1$; the second integral is bounded by $C_B x^B/(B + \Re \rho)$. The exact identity $(D - \rho)S_\rho F = F$ now gives all Euler derivative bounds inductively, so $S_\rho F \in \mathcal{B}$. Mellin integration by parts proves $\mathcal{M}(S_\rho F) = \mathcal{M}F/(s - \rho)$, with its removable value at $s = \rho$ supplied by the derivative of the numerator. For the original seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V$, direct substitution $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \Re s > -2.$$

For $\Re s > 1$, absolute convergence permits the two-sided theta sum to be integrated termwise, yielding $\mathcal{M}(\Theta\phi_0) = 2\zeta(s)\mathcal{M}\phi_0(s) = g(s)$; both sides continue to the same entire function. Apply S_ρ successively to $\Theta\phi_0$ over the full list of roots of h , including each repeated occurrence. At every stage the next vanishing Mellin value holds by its remaining order. This constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$

and $h(D)F_h = \Theta\phi_0$; the latter also follows by Mellin injectivity, which is Fourier injectivity after $x = e^y$. Fix $k \geq 1$, set $c = k/2$, $S = s_1 + \cdots + s_k$, and put $I = (h(s_1), \dots, h(s_k)) \subset \mathbb{C}[s_1, \dots, s_k]$. Let χ be the original monic generator of $\{P \in \mathbb{C}[S] : P(S) \in I\}$. Its degree is $q \geq 1$ and

$$C = \mathbb{C}[S]/(\chi), \quad A = M_S : C \longrightarrow C. \quad (\text{TVB.2})$$

The coordinate basis of C is always $1, S, \dots, S^{q-1}$. The polynomial exists because multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$ has an annihilating polynomial. Division by its least-degree monic annihilator proves that its entire annihilating ideal is (χ) . In particular the injection

$$\eta : C \longrightarrow (E_h^{\otimes k})^{\mathfrak{S}_k}, \quad [P] \longmapsto v_h^{\otimes k}[P(S)]_I \quad (\text{TVB.3})$$

has exactly zero kernel: multiplication by the displayed unit is invertible and the preceding annihilator ideal describes the other factor's kernel. The original tensor source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad D = -x\partial_x, \quad \mathcal{M}F_h = v_h.$$

On the original Mellin line its scalar observation has squared norm

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad \int m_k = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{TVB.4})$$

Indeed the tensor Mellin–Plancherel density is $\prod_i w_h(t_i) d^k t$. In coordinates $u = \sum_i t_i$, $y = (t_1, \dots, t_{k-1})$, the inverse is $(y, u) \mapsto (y_1, \dots, y_{k-1}, u - \sum y_i)$. With (y, u) in this order its determinant is 1; with (u, y) its determinant is $(-1)^{k-1}$. Integrating the original density over y gives (TVB.4), with no scalar divisor.

For every compact substrip $|\Re\theta| \leq r < \pi/2$, all integrals

$$M_h(\theta) = \int e^{\theta t} w_h(t) dt, \quad Z(\theta) = M_h(\theta)^k = \int e^{\theta u} m_k(u) du \quad (\text{TVB.5})$$

and all their polynomially weighted derivatives converge locally uniformly. Here is an elementary sufficient bound. Summation by parts for the alternating Dirichlet series, whose partial sums have modulus at most one, gives $|\eta(s)| \leq |s|/\Re s$ when $\Re s > 0$. On $\Re s = 1/2$, the denominator in $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ has modulus at least $\sqrt{2} - 1$. For $0 < \varphi < \pi/2$, rotate the defining gamma integral to the ray with argument $\text{sgn}(t)\varphi$. The large arc vanishes by $e^{-R\cos\varphi}$ and the small arc by the positive real part $1/4$ of its exponent. Its absolute value is bounded by

$$|\Gamma(1/4 + it/2)| \leq \Gamma(1/4)(\cos\varphi)^{-1/4} e^{-\varphi|t|/2}.$$

For sufficiently large $|t|$, the monic polynomial h satisfies $|h(1/2 + it)| \geq C_h(1 + |t|)^d$ with $C_h > 0$. The two factors $s, s - 1$, the preceding zeta bound and the gamma bound consequently give

$$w_h(t) \leq C_{h,\varphi}(1 + |t|)^{6-2d} e^{-\varphi|t|}$$

outside a compact interval. On that interval v_h is entire, since the selected orders are complete. Choose $r < \varphi < \pi/2$ and absorb every fixed polynomial in the remaining exponential decay. This proves the asserted domination and differentiation under the integral. Fubini gives the exact product M_h^k in (TVB.5). The nonzero entire function v_h has a discrete real-line zero set; therefore $w_h > 0$ almost everywhere. Its convolutions are positive almost everywhere (in fact at every u for $k \geq 2$), since their fibre integrands are positive off a finite union of measure-zero inverse images of this discrete set. Thus the norm of every nonzero polynomial is positive.

In what follows θ is real, $|\theta| < \pi/2$, and the source pairing, conjugate-linear in its first argument, is exactly

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (\text{TVB.6})$$

The map $f(u) \mapsto e^{\theta u/2} f(u)$ is an onto isometry from $L^2(e^{\theta u} m_k du)$ to $L^2(m_k du)$, with inverse $f \mapsto e^{-\theta u/2} f$. It commutes with multiplication by $c + iu$. This specifies the exact Hilbert comparison for the tilt. The polynomial algebra, its reduction and η are unchanged. When the target of $\mathcal{V}$ is observed at this tilt, its declared tensor norm is

$$\|F\|_{k,\theta}^2 = \int_{\mathbb{R}^k} e^{\theta \sum_i t_i} |\mathcal{M}_{\text{unit}} F(t_1, \dots, t_k)|^2 d^k t,$$

where $\mathcal{M}_{\text{unit}}$ is the original Mellin–Plancherel map with a factor $(2\pi)^{-1/2}$ in each coordinate. The map $f(t) \mapsto e^{\theta \sum_i t_i/2} f(t)$ is an onto isometry from this weighted Mellin Hilbert space to the original one, with the opposite exponential as inverse. The restriction to $\mathcal{V}P$ is exactly (TVB.6), by the same sum-fibre integral as (TVB.4). Thus every tilted exterior norm below has a specified target norm; at $\theta = 0$ it is the original tensor-source norm.

34.2 The source relation and its Laplace observation

Write $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$. For $N \geq q - 1$, put $n = N + 1$, $m = N - q + 1$. Literal monic division gives the exact sequence

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{T_N} C \longrightarrow 0, \quad M_\chi Q = \chi Q, \quad T_N P = [P]_\chi. \quad (\text{TVB.7})$$

Its kernel is precisely the indicated image, and its first q monomial columns prove surjectivity. Denote that kernel by $\mathcal{D}_N = \chi \mathcal{P}_{m-1}$.

This sequence has an explicit map to the original theta boundaries. To see every sign, choose the quotients Q_i from successive monic division, in the order $s_1, \dots, s_k$, so that

$$\chi(S) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{TVB.8})$$

The remainder is zero by the definition of χ and uniqueness of the tensor remainder basis. Put $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. It is even, vanishes at zero and has integral zero. Direct gamma integration of its two terms gives $\mathcal{M}(\Theta \phi_0) = g$, hence $h(D)F_h = \Theta \phi_0$ by Mellin injectivity on the original rapidly decreasing source. For every polynomial P , an actual primitive of $\mathcal{V}(\chi P)$ in the tensor complex $(V \xrightarrow{\Theta} \mathcal{B})^{\otimes k}$ is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{TVB.9})$$

The preceding $i - 1$ factors have cochain degree one. The tensor differential therefore contributes $(-1)^{i-1}$ at the i th factor, cancelling the written sign and yielding (TVB.8). Euler differentiation preserves the source conditions and commutes with Θ , so all terms have the indicated domains. Conversely the full jet of $\mathcal{V}P$ is $\eta T_N P$; injectivity in (TVB.3) shows that the admitted polynomial source has no further zero-jet elements. In particular every boundary vector in (TVB.7) retains its positive source norm before the quotient sends it to represented zero.

For any polynomial P , define its Laplace observation by

$$\mathcal{L}(P)(\theta) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Differentiation under (TVB.5) proves the typed intertwining $\mathcal{L}(SP) = (c + i\partial_\theta) \mathcal{L}(P)$. Coefficient-wise conjugation gives the exact Gram observation

$$Z_\chi = \bar{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0. \quad (\text{TVB.10})$$

Thus the two signs and both polynomial factors arise from $\langle M_\chi 1, M_\chi 1 \rangle_\theta$. This is a map on source functions, with no replacement of the higher contracted ideal by a power of χ .

34.3 Both monic determinants and their exact determinant-line map

Set

$$\mathfrak{D}_j = \det[Z^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{B}_j = \det[Z_\chi^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{D}_0 = \mathfrak{B}_0 = 1. \quad (\text{TVB.11})$$

All nonempty determinants are positive by polynomial positivity. The coefficient map from $1, S, \dots, S^{j-1}$ to $1, u, \dots, u^{j-1}$ is explicitly

$$S^a = (c + iu)^a = \sum_{b=0}^a \binom{a}{b} c^{a-b} i^b u^b. \quad (\text{TVB.12})$$

Its determinant is $i^{j(j-1)/2}$ and its Gram determinant multiplier is exactly one. Therefore $\mathfrak{D}_j$ is also the original S -monomial Gram determinant. The same calculation applies to the relation-weight pairing $\langle P, Q \rangle_{\chi, \theta} = \langle \chi P, \chi Q \rangle_\theta$ and $\mathfrak{B}_j$. This proves the coordinate correspondence without changing the original monic polynomial.

Let $p_j(S; \theta)$ be the monic source orthogonal polynomials, and $r_j(S; \theta)$ the monic polynomials for this relation-weight pairing. Retain their original squared norms

$$\omega_j = \|p_j\|_\theta^2 = \mathfrak{D}_{j+1}/\mathfrak{D}_j, \quad \nu_j = \|\chi r_j\|_\theta^2 = \mathfrak{B}_{j+1}/\mathfrak{B}_j. \quad (\text{TVB.13})$$

The equations follow by triangular monic Gram elimination, whose diagonal change is one. In particular $\omega_0 = M_h(\theta)^k$. The full mass has not been divided out.

Give C its written coordinate Euclidean pairing only when specifying adjoints. Let M_N be the original monomial source Gram, and J_N its reduction matrix. The minimum lift is

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = M_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N. \quad (\text{TVB.14})$$

Here the last expression specifies polynomial coefficients. It satisfies $T_N R_N = I$, $\text{im } R_N \perp \mathcal{D}_N$ and $R_N^* R_N = G_N$ in the original source pairing. Indeed the first equality is matrix multiplication, and pairing with any vector in $\ker J_N$ gives zero. Every other lift is $R_N x + B$ for $B \in \mathcal{D}_N$, so its squared norm is $x^* G_N x + \|B\|^2$. This proves the minimum assertion and uniqueness.

Use the source basis $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$. Its change from the n source monomials is triangular with diagonal one. Replacing its first q columns by $R_N[1], \dots, R_N[S^{q-1}]$ adds boundary columns and again has determinant one. The resulting Gram is the direct sum of G_N and the relation monomial Gram. Consequently

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_m}, \quad D_N^{\text{CV}} := \det K_N = V_N^{-1}. \quad (\text{TVB.15})$$

More precisely the map $\det C \otimes \det \mathcal{P}_{m-1} \rightarrow \det \mathcal{P}_N$ sends the written coordinate wedges to the wedge of these quotient lifts followed by M_χ of the relation monomials; its coordinate determinant is 1. Putting the kernel wedge first instead contributes the explicit sign $(-1)^{qm}$. Its norm identity is $\det G_N \mathfrak{B}_m = \mathfrak{D}_n$. Composing the wedge with $\mathcal{V}^{\otimes n} \text{Alt}_n$ multiplies its squared norm by exactly $n!$: expanding the unscaled alternating tensor sum gives $n!$ copies of the Gram determinant. The original tensor target is $\mathcal{B}^{\widehat{\otimes} kn}$ in cochain degree kn .

34.4 Toda derivatives and the full curvature operators

For either positive real u -weight in (TVB.11), write $Q_j(u)$ for its real monic orthogonal polynomials and h_j for their norms. Orthogonality and leading coefficients prove

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1} \ (j \geq 1), \quad a_0 = 0.$$

For $i \leq j-2$, differentiating $\langle Q_i, Q_j \rangle = 0$ shows that $\partial_\theta Q_j$ is orthogonal to Q_i , because uQ_i and $\partial_\theta Q_i$ have degree at most $j-1$. For $i = j-1$ the same identity gives its coefficient $-a_j$. Thus

$\partial_\theta Q_j = -a_j Q_{j-1}$ and $h'_j = b_j h_j$. To compute b'_j , differentiate $h_j b_j = \langle Q_j, u Q_j \rangle$. The two polynomial derivative terms contribute $-2a_j h_j$. The derivative of the measure contributes $\|u Q_j\|^2 = h_j(a_{j+1} + b_j^2 + a_j)$. After subtracting $h'_j b_j$ this gives $b'_j = a_{j+1} - a_j$. The monic Gram determinant is $\prod_{j < n} h_j$, hence

$$(\log \mathfrak{D}_n)'' = \frac{\omega_n}{\omega_{n-1}} \quad (n \geq 1), \quad (\log \mathfrak{B}_m)'' = \frac{\nu_m}{\nu_{m-1}} \quad (m \geq 1). \quad (\text{TVB.16})$$

Both 0th logarithms have second derivative zero. Multiplication by the square of the determinant proves each Hankel-Toda equation with coefficient one. Set

$$\alpha_{N+1} = \omega_{N+1}/\omega_N, \quad \beta_m = \nu_m/\nu_{m-1} \quad (m \geq 1), \quad \beta_0 = 0, \quad \ell_N = \log V_N.$$

Then $\ell''_N = \alpha_{N+1} - \beta_m$.

There is a finer exact source-map identity. Let $X = (S - c)/i$, which is multiplication by the real variable u in (TVB.6). Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ to $\mathcal{D}_N$ in that pairing. Differentiating $T_N R_N = I$ gives $R'_N x \in \mathcal{D}_N$. Differentiating orthogonality against each fixed boundary polynomial gives

$$R'_N = -P_{\mathcal{D}_N} X R_N =: -I_N, \quad O_N := (1 - P_N) X R_N. \quad (\text{TVB.17})$$

Every derivative is an actual boundary with the primitive (TVB.9). Differentiating its norm, including the derivative of the measure, gives

$$G'_N = R_N^* X R_N, \quad G''_N = R_N^* X^2 R_N - 2R_N^* X P_{\mathcal{D}_N} X R_N.$$

The projection to $\text{im } R_N$ is $R_N G_N^{-1} R_N^*$ and, together with $P_{\mathcal{D}_N}$, sums to P_N . Subtracting the square of the first derivative therefore proves

$$G''_N - G'_N G_N^{-1} G'_N = O_N^* O_N - I_N^* I_N. \quad (\text{TVB.18})$$

This is an equality of maps $C \rightarrow C^{\text{anti}*}$, where $C^{\text{anti}*}$ consists of conjugate-linear functionals on C . For example $O_N^* O_N$ sends x to the functional $y \mapsto \langle O_N y, O_N x \rangle_\theta$; this assignment is complex-linear in x . The fixed coordinate Euclidean pairing identifies that antidual with the written matrix coordinates.

Write $b_j = [p_j]_\chi$, and $a = b_N^* G_N b_N$, $d_+ = b_{N+1}^* G_N b_{N+1}$. The canonical next boundary has the particularly exact factorization

$$\mathfrak{b}_N := p_{N+1} - R_N b_{N+1} = \chi r_m, \quad \nu_m = \omega_{N+1} + d_+. \quad (\text{TVB.19})$$

To prove it, χr_m is monic of degree $N + 1$ and orthogonal to $\mathcal{D}_N$. Subtract p_{N+1} ; the difference lies in $\mathcal{P}_N \cap \mathcal{D}_N^\perp$ and has reduction $-b_{N+1}$. Uniqueness of R_N proves the first equality; the two terms in its definition are orthogonal, giving the second. The coefficient of p_N in $R_N x$ is $b_N^* G_N x / \omega_N$. The leading coefficient of X as an S -polynomial is $1/i = -i$. Consequently the full maps are

$$O_N x = -i p_{N+1} \frac{b_N^* G_N x}{\omega_N}, \quad I_N x = -i \chi r_{m-1} \frac{b_{N+1}^* G_N x}{\nu_{m-1}} \quad (m \geq 1), \quad I_N = 0 \quad (m = 0). \quad (\text{TVB.20})$$

For the second equation, $X \chi r_j$ is an old boundary if $j \leq m - 2$. For $j = m - 1$ its coefficient of χr_m is $-i$. Moving X across the pairing conjugates this coefficient to i , and $\langle \chi r_m, R_N x \rangle = -b_{N+1}^* G_N x$ by (TVB.19). Division by the original norm ν_{m-1} gives exactly the second sign in (TVB.20).

Both maps have rank at most one. For $N \geq q$, put $\delta_N = V_N/V_{N-1}$. The rank-one calculation in the next section gives $a/\omega_N = 1 - \delta_N$ and $\delta_N = \omega_N/\nu_{m-1}$. Taking traces in (TVB.20) gives the separate, positive source costs

$$\begin{aligned} \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1} (1 - \delta_N), \\ \text{Tr}(G_N^{-1} I_N^* I_N) &= \beta_m - \alpha_{N+1} \delta_N = \frac{\nu_m - \omega_{N+1}}{\nu_{m-1}}. \end{aligned} \quad (\text{TVB.21})$$

Their difference is $\alpha_{N+1} - \beta_m$ as required. Thus the individual positive map costs each retain the common quantity $\alpha_{N+1}\delta_N$; assigning them separately the two Toda coefficients would lose this quantity. At $N = q - 1$, the outer cost is α_q and the inner cost is zero. If a single notation for the cost identities is wanted at this endpoint, define only the scalar convention $\delta_{q-1}^{\text{end}} := \det K_{q-2} / \det K_{q-1} = 0$; the included column has leverage $a/\omega_{q-1} = 1$. Together with $\beta_0 = 0$ the cost identities are $\alpha_q(1 - \delta_{q-1}^{\text{end}}) = \alpha_q$ and $\beta_0 - \alpha_q\delta_{q-1}^{\text{end}} = 0$. This convention is not a quotient involving V_{q-2} and does not introduce an inverse for K_{q-2} or a norm ν_{-1} . All statements concern the original polynomial source and its original boundary subspace.

34.5 The rank-one update, trace phase and original rank-two radius

The source-orthogonal basis gives

$$K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad K_{N+1} = K_N + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}. \quad (\text{TVB.22})$$

The first q monic remainder columns are a triangular basis; K_N is positive definite precisely in the admitted range. Block elimination of $\begin{pmatrix} K & U \\ -V^* & I \end{pmatrix}$ in either order proves $\det(K + UV^*) = \det K \det(I + V^* K^{-1} U)$. Applying it to the positive update and the subtraction of b_N from K_N yields, for $N \geq q$,

$$\frac{a}{\omega_N} = 1 - \delta_N, \quad \frac{d_+}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1, \quad \delta_N = \frac{\omega_N}{\nu_{m-1}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_m}. \quad (\text{TVB.23})$$

The last two identities also follow directly from (TVB.15). Each δ is positive and at most one, since increasing the source space weakly decreases every minimum squared norm. The exact inverse update is

$$G_{N+1} = G_N - \frac{G_N b_{N+1} b_{N+1}^* G_N}{\omega_{N+1} + d_+}. \quad (\text{TVB.24})$$

Multiplying this expression by K_{N+1} proves it directly; the denominator is $\nu_m > 0$ by (TVB.19).

Reflection of the tensor algebra sends S to $k - \bar{S}$ and preserves its annihilator ideal. Monicity gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic. The exact algebra isomorphism

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P(S)] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q(u)] = [Q((S - c)/i)] \quad (\text{TVB.25})$$

transports the original ideal by its leading unit i^q . Its coefficient matrix E is (TVB.12) in dimension q . Thus A transports to $cI + iT$, where $T = M_u$ in the real quotient. Its Gram transports to $E^{-*} G_N E^{-1}$, its kernel to $E K_N E^*$, and $\det G_N$ is unchanged because $|\det E| = 1$. Write

$$\sigma = \text{Tr } T = \frac{\text{Tr } A - cq}{i} \in \mathbb{R}, \quad \psi_N = \sigma - \ell'_N.$$

Reality of $\hat{\chi}$, the weight and its monic orthogonal polynomials makes all the u -coordinate source and quotient Grams real. Let $Q_j(u) = i^{-j} p_j(c + iu)$ and $d_j = [Q_j]$. Then $E b_j = i^j d_j$ and both source norms are ω_j .

The leading coefficient of the u -polynomial minimum lift of x is $d_N^* G_N x / \omega_N$, where in this calculation G_N denotes the transported real-coordinate Gram. The difference $X R_N - R_N T$ has zero reduction, and subtracting $(Q_{N+1} - R_N d_{N+1}) d_N^* G_N / \omega_N$ cancels its highest coefficient. Its remainder is an old boundary. Pair with R_N ; old boundaries vanish and $R_N^* (Q_{N+1} - R_N d_{N+1}) = -G_N d_{N+1}$. Taking the trace after multiplication by G_N^{-1} proves

$$\ell'_N = \sigma - \frac{d_N^* G_N d_{N+1}}{\omega_N}, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i \psi_N. \quad (\text{TVB.26})$$

The second equation is back in the fixed original S coordinates: the phase is $\bar{i}^N i^{N+1} = i$. Both signs are retained.

For completeness, the u recurrence proved above transports to $Sp_j = p_{j+1} + (c + ib_j^{\text{rec}})p_j - \alpha_j p_{j-1}$ in the original source coordinates. Inserting this in (TVB.22) telescopes all interior terms and gives

$$AK_N + K_N A^* - kK_N = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}.$$

For $W_N = A^*G_N + G_N A - kG_N$, multiplication on both sides by G_N gives the corresponding identity for W_N . Its Hermitian relative operator is $H_N = G_N^{-1/2}W_N G_N^{-1/2}$. It has rank at most two and trace $2\Re \text{Tr } A - kq = 0$. The sum of the squares of its eigenvalues is $2(ad_+ - |z_N|^2)/\omega_N^2$, where $z_N = b_N^* G_N b_{N+1} = i\omega_N \psi_N$. This last assertion follows by expanding the square of $(xy^* + yx^*)/\omega_N$ for $x = G_N^{1/2}b_N, y = G_N^{1/2}b_{N+1}$; its trace is $[2ad_+ + 2\Re(z_N^2)]/\omega_N^2$. Its nonzero eigenvalues are therefore $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2. \quad (\text{TVB.27})$$

The first expression is valid already at $N = q - 1$.

34.6 Exact maps to the CV omitted determinant and positive minors

Retain $D_j^{\text{CV}} = \det K_j$, put $K_{-1} = 0$, and form

$$\tilde{K}_N = K_{N-1} + \frac{b_{N+1}b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad \Delta_N = D_{N+1}^{\text{CV}} + D_{N-1}^{\text{CV}} - D_N^{\text{CV}} - \tilde{D}_N.$$

This is the actual quotient kernel of the source with p_N omitted and p_{N+1} included. No inverse of it is used. The two-column determinant formula preceding (TVB.23), with one negative and one positive update, proves

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} = \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d_+}{\omega_{N+1}}\right) + \frac{|z_N|^2}{\omega_N \omega_{N+1}}, \quad \epsilon_N^2 = \alpha_{N+1} \frac{\Delta_N}{D_N^{\text{CV}}}. \quad (\text{TVB.28})$$

In particular, for $N \geq q$, the complete phase map is the exact nonnegative difference

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} - \frac{\delta_N}{\delta_{N+1}} = \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.29})$$

The sign of ψ_N is retained by the first derivative in (TVB.26); the omitted determinant observes its square.

For any finite column set L , determinant multilinearity gives

$$\det \left(\sum_{j \in L} \frac{b_j b_j^*}{\omega_j} \right) = \sum_{\substack{J \subseteq L \\ |J|=q}} \frac{|\det B_J|^2}{\prod_{j \in J} \omega_j}, \quad B_J = [b_j]_{j \in J \text{ increasing}}.$$

Terms with repeated indices vanish. The $q!$ remaining permutations of a fixed increasing index set yield its determinant times its conjugate, which proves the displayed identity including singular kernels. Apply it to the four column sets defining Δ_N . A set containing neither of the two last indices has coefficient $1 + 1 - 1 - 1 = 0$; one containing exactly one has coefficient $1 - 1 = 0$; one containing both has coefficient one. Hence

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \quad (\text{TVB.30})$$

For $q = 1$ it is zero. This proof applies at $N = q - 1$. For $q \geq 2$ define the actual source subspace $\mathcal{E}_N = (\bigwedge^{q-2} \mathcal{P}_{N-1}) \wedge p_N \wedge p_{N+1}$ and $\Lambda_N = (\bigwedge^q T_{N+1})|_{\mathcal{E}_N}$. For the fixed coordinate wedge $e_C = 1 \wedge S \wedge \cdots \wedge S^{q-1}$, whose coordinate Euclidean squared norm is one, the adjoint is

$$\zeta_N := \Lambda_N^* e_C = \sum_{|I|=q-2} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \left(\bigwedge_{i \in I} p_i \right) \wedge p_N \wedge p_{N+1}. \quad (\text{TVB.31})$$

The displayed source wedges are orthogonal with the written full product norms. Therefore $\|\zeta_N\|_\theta^2 = \Delta_N$ and $\Lambda_N \zeta_N = \Delta_N e_C$. At $\theta = 0$ its literal original tensor-source image has squared norm $q! \Delta_N$ under $\mathcal{V}^{\otimes q} \text{Alt}_q$. At other real tilts use the onto Hilbert comparison after (TVB.6) on each scalar source factor; the same determinant norm and factorial hold in that specified tilted observation. For $q = 1$ set $\mathcal{E}_N = 0$, $\Lambda_N = 0$, $\zeta_N = 0$.

34.7 Both exact losses, their equality cases and the telescope

For $N \geq q$ define

$$v_N = -\log \delta_N, \quad s_N = (v_N + v_{N+1})/2, \quad t_N = (v_{N+1} - v_N)/2, \quad \mathcal{R}_N = e^{2s_N} = \frac{V_{N-1}}{V_{N+1}} \geq 1.$$

Expanding $(1 - e^{-v_N})(e^{v_{N+1}} - 1)$ gives $2e^{t_N} \cosh s_N - e^{2t_N} - 1$. Completing this scalar square, with both terms retained, yields

$$\boxed{\epsilon_N^2 = \alpha_{N+1} \sinh^2 s_N - \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 - \psi_N^2.} \quad (\text{TVB.32})$$

Thus the exact upper bound and its equality conditions are

$$\epsilon_N \leq \sqrt{\alpha_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}, \quad \text{equality} \iff \psi_N = 0 \text{ and } \delta_N(2 - \delta_{N+1}) = 1. \quad (\text{TVB.33})$$

Indeed $\alpha_{N+1} > 0$ and both lost squares in (TVB.32) are nonnegative. Multiplication of $e^{t_N} = \cosh s_N$ by $2\sqrt{\delta_N \delta_{N+1}}$ gives precisely the displayed equation. This includes the case $\mathcal{R}_N = 1$, when both contractions equal one and $\epsilon_N = \psi_N = 0$ by (TVB.27). The corresponding equality with the full positive minor source is

$$\frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N} - \frac{\|\zeta_N\|_\theta^2}{D_N^{\text{CV}}} = (e^{t_N} - \cosh s_N)^2 + \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.34})$$

It follows from (TVB.28), (TVB.31) and (TVB.32), so its minor equality has the same two necessary and sufficient conditions. In particular equality is a fully specified source-volume equation, not a consequence of positivity alone.

There is a direct formulation in the actual volumes, without logarithmic variables. Put $U_N = \sqrt{\alpha_{N+1}}(\mathcal{R}_N - 1)/(2\sqrt{\mathcal{R}_N})$. Since

$$e^{t_N} = \frac{V_N}{\sqrt{V_{N-1} V_{N+1}}}, \quad \cosh s_N = \frac{V_{N-1} + V_{N+1}}{2\sqrt{V_{N-1} V_{N+1}}},$$

equation (TVB.32) gives the exact two losses

$$U_N^2 - \epsilon_N^2 = \alpha_{N+1} \frac{(2V_N - V_{N-1} - V_{N+1})^2}{4V_{N-1} V_{N+1}} + \frac{|z_N|^2}{\omega_N^2}. \quad (\text{TVB.34a})$$

Thus the volume equality is the arithmetic midpoint $2V_N = V_{N-1} + V_{N+1}$ together with $z_N = 0$. Equal consecutive logarithmic contractions alone do not give this midpoint: their extra equation $V_N^2 = V_{N-1} V_{N+1}$ implies it only if all three positive volumes are equal, by the square $(\sqrt{V_{N-1}} -$

$\sqrt{V_{N+1}})^2$. In the original inverse-volume coordinates $D_- = D_{N-1}^{\text{cv}}$, $D_0 = D_N^{\text{cv}}$, $D_+ = D_{N+1}^{\text{cv}}$ and $E_N = \tilde{D}_N$, the same identities become

$$\begin{aligned} E_N D_0 - D_- D_+ &= \frac{D_0^2 |z_N|^2}{\omega_N \omega_{N+1}}, \\ \frac{D_0(D_+ - D_-)^2}{4D_- D_+} - \Delta_N &= \frac{(2D_- D_+ - D_0(D_- + D_+))^2}{4D_0 D_- D_+} + E_N - \frac{D_- D_+}{D_0}. \end{aligned} \quad (\text{TVB.34b})$$

Substitute $D_j = V_j^{-1}$ in (TVB.34a), or multiply out its two squares, to verify every denominator. Both terms on the right are nonnegative by the first equation. Equality is equivalently $D_0 = 2D_- D_+ / (D_- + D_+)$ and $E_N = (D_- + D_+)/2$; it then gives $\Delta_N = E_N - D_0$. In particular phase zero at these degrees records a positive omitted-direction determinant $E_N = D_- D_+ / D_0$, with its exact map specified in (TVB.28), rather than a zero determinant.

The algebraic exterior lower bound applies at each real tilt as well as at zero. Here is a direct proof. In the Hermitian matrix H_N above let P be the orthogonal projection to the image, under $G_N^{1/2}$, of the sum of all full generalized eigenspaces of A with $\Re \lambda > c$. Invariance of that subspace makes the transformed A upper triangular in its orthogonal direct sum with the complement, without asserting that the spectral projection itself is orthogonal. Taking the trace of its first diagonal block gives

$$\text{Tr}(PH_N) = L := \sum_{\Re \lambda > c} \ell_\lambda (2\Re \lambda - k).$$

Since H_N has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros, $\text{Tr}(PH_N) \leq \epsilon_N$: its positive eigenline contributes at most ϵ_N and its negative eigenline contributes a nonpositive number. All algebraic multiplicities ℓ_λ are retained. Thus $L \leq \epsilon_N$ for the same tilted minimum metric, without choosing another quotient metric.

Combining this with (TVB.33), using monotonicity of $\sinh$ on $[0, \infty)$ and its inverse, proves

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.35})$$

For $N = N_0 + 2j$, $0 \leq j < r$, with $N_0 \geq q$, every denominator in one logarithm is the numerator in the next. Summing the exact logs and then their lower bounds proves

$$\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N_0+2j+1}}} \right), \quad r \geq 1. \quad (\text{TVB.36})$$

Every term is at the same fixed real tilt, packet and tensor degree. No relation between different k or a limiting value of either determinant is used. Under $\mathcal{V}$ the minimum representatives at all degrees retain jets η and their original quotient classes; each difference of two such lifts is the explicit boundary in (TVB.7)–(TVB.9).

34.8 The first admitted degree and the empty quotient

At $N = q - 1$, the first q monic remainder columns are a basis, so $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Expand the original polynomial, with its coefficients retained,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle_\theta}{\omega_j}. \quad (\text{TVB.37})$$

Reduction gives $b_q = -\sum_{j<q} \gamma_j b_j$. Hence

$$a = \omega_{q-1}, \quad z_{q-1} = -\gamma_{q-1} \omega_{q-1}, \quad d_+ = \sum_{j<q} |\gamma_j|^2 \omega_j = \nu_0 - \omega_q, \quad \gamma_{q-1} = -i\psi_{q-1}.$$

The last equality follows from (TVB.26), which does not require K_{q-2}^{-1} . Substitution gives the exact CV.12-Toda endpoint

$$\epsilon_{q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - \psi_{q-1}^2. \quad (\text{TVB.38})$$

There is no V_{q-2} here. The omitted determinant instead records $\tilde{D}_{q-1}/D_{q-1}^{\text{CV}} = \psi_{q-1}^2/\alpha_q$. The polynomial projection $\mathbf{P}_{q-2}\chi = \sum_{j<q-1} \gamma_j p_j$ has the actual jet $\eta \sum_{j<q-1} \gamma_j b_j$, whereas χ itself has zero jet and primitive (TVB.9). Its squared source norm in (TVB.6) is the numerator in (TVB.38). This is the exact map from the whole relation to its lower orthogonal projection, including the possibly nonzero resulting observation. For $q = 1$, the sum is empty, the lower source is $\mathcal{P}_{-1} = 0$, and $\epsilon_N = 0$ at every admitted degree because its one-dimensional relative control is trace zero.

The endpoint projection and the minor vector have a stronger exact typed correspondence. Assume $q \geq 2$, let $D = D_{q-1}^{\text{CV}} = \prod_{j<q} \omega_j^{-1}$, and set $I_j = \{0, \dots, q-2\} \setminus \{j\}$ for $0 \leq j \leq q-2$. The determinant of the first q monic remainder columns is 1. Substituting $b_q = -\sum_{l<q} \gamma_l b_l$ in the minor with columns $I_j \cup \{q-1, q\}$ leaves only $l = j$. Moving b_j from its last position to position j takes $q-1-j$ transpositions; with the preceding minus sign this proves

$$\det B_{I_j \cup \{q-1, q\}} = (-1)^{q-j} \gamma_j. \quad (\text{TVB.38a})$$

Write $e_j = (\bigwedge_{i \in I_j} p_i) \wedge p_{q-1} \wedge p_q$. The explicit conjugate-linear isomorphism

$$\mathcal{H} : \mathcal{P}_{q-2} \longrightarrow \mathcal{E}_{q-1}, \quad \mathcal{H} \left(\sum_{j<q-1} a_j p_j \right) = \sum_{j<q-1} (-1)^{q-j} \overline{a_j} \frac{\omega_j D}{\omega_q} e_j \quad (\text{TVB.38b})$$

has inverse $\sum t_j e_j \mapsto \sum (-1)^{q-j} \omega_q \overline{t_j} p_j / (\omega_j D)$. The source and target basis vectors are orthogonal and $\|e_j\|_\theta^2 = \omega_q / (D \omega_j)$. Consequently

$$\|\mathcal{H}P\|_\theta^2 = \frac{D}{\omega_q} \|P\|_\theta^2, \quad \zeta_{q-1} = \mathcal{H}(\mathbf{P}_{q-2}\chi), \quad \Delta_{q-1} = \frac{D}{\omega_q} \|\mathbf{P}_{q-2}\chi\|_\theta^2. \quad (\text{TVB.38c})$$

This follows term by term from (TVB.31) and (TVB.38a). The actual alternating tensor image has squared norm $q! D \|\mathbf{P}_{q-2}\chi\|_\theta^2 / \omega_q$ in the specified tilted tensor observation, and in the original source norm at $\theta = 0$. The top coefficient is retained separately: $\tilde{D}_{q-1} = D |\gamma_{q-1}|^2 \omega_{q-1} / \omega_q$. For $q = 1$ the map $\mathcal{H}$ is the unique map between zero spaces, with the same zero identities; the top-coefficient formula still records the entire omitted determinant.

For completeness the finite upper bound is strict when $q = 1$ and $N \geq 1$. Consecutive source polynomials have no common root: their three-term recurrence with its nonzero coefficient $\alpha_j > 0$ would propagate such a root backwards to $p_0 = 1$. For any nonzero quotient $q \geq 1$, both b_N, b_{N+1} therefore cannot vanish, since a root of χ would then be common to both polynomials. When $N \geq q$, K_{N-1} is positive definite and $K_{N+1} - K_{N-1}$ is a nonzero positive semidefinite matrix. Conjugating by $K_{N-1}^{-1/2}$ and multiplying its eigenvalues proves $D_{N+1}^{\text{CV}} > D_{N-1}^{\text{CV}}$, hence $\mathcal{R}_N > 1$ and $U_N > 0$. At $q = 1$ this is strictly larger than the already proved $\epsilon_N = 0$.

Finally let $h = 1$, so $\chi = 1$, $q = 0$, and $C = 0$. The analytic function g , w_1 and all source norms still exist. For every $N \geq 0$ the source sequence is multiplication by one from $\mathcal{P}_N$ onto itself followed by the zero quotient. Then $m = N+1$, $Z_\chi = Z$, $\mathfrak{B}_m = \mathfrak{D}_m$, and the determinant of the zero-dimensional Gram is the empty determinant $V_N = 1$. Its logarithm and derivatives are zero. The minimum lift, its curvature maps, and the relative control are the unique zero maps, with $\epsilon_N = L = 0$. The two Toda coefficients in their curvature difference are equal, so that difference is zero. No positive-dimensional inverse is introduced. The support lift retains $G(0) = \{\tau, e\}$: every represented polynomial, including each nonzero source relation, maps to e , while τ maps to τ .

34.9 The complete join with the original spectral coupling

This section is the exact join supplied in PARENT_EXACT_JOIN.md, SHA256

71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370.

Its two inputs are the delivered Toda equation (28), proved here as (TVB.32), and the complete original-metric exterior identity E8. The latter source is retained in the cyclic companion at commit 161089942cce70a09fbc441d08ba4d36c1fd810c. We reproduce the finite identity and its typed coupling map here to make the join complete; no novelty or limiting estimate is claimed.

All quantities in this section are at the original observation $\theta = 0$, with $N \geq q$. Write $A^\sharp = G_N^{-1} A^* G_N$ and $\mathcal{H}_N = A^\sharp + A - kI$, acting on the original (C, G_N) . Let C_+ be the full generalized spectral subspace with $\Re \lambda > c$, let $C_- = C_+^{\perp_{G_N}}$, and let $P : C \rightarrow C$ be their orthogonal projection onto C_+ . Suppose first $C_+ \neq 0$. Its trace L from (TVB.35) is positive, so $\epsilon_N > 0$. Choose G_N -unit eigenvectors u_+, u_- of $\mathcal{H}_N$ at $\epsilon_N, -\epsilon_N$; selfadjointness makes them orthogonal. They specify only the coefficient injection

$$E : \mathbb{C}^2 \longrightarrow C, \quad E(z_+, z_-) = z_+ u_+ + z_- u_-, \quad E^\sharp E = I_2, \quad M = E^\sharp P E. \quad (\text{TVB.39})$$

Both M and $I_2 - M$ are positive semidefinite, since their quadratic forms are respectively the squared original norms of $P E z$ and $(1 - P) E z$. In the original orthogonal decomposition define

$$B = A|_{C_+}, \quad D^{\text{op}} = (1 - P)A|_{C_-}, \quad C = PA|_{C_-} : C_- \longrightarrow C_+.$$

Invariance of C_+ gives $A = \begin{pmatrix} B & C \\ 0 & D^{\text{op}} \end{pmatrix}$. Its adjoint has the conjugate-transpose blocks in the original restricted metrics, so $P\mathcal{H}_N|_{C_-} = C$ exactly.

Put $a_+ = \|Pu_+\|^2$, $a_- = \|Pu_-\|^2$, and $z = \langle Pu_+, Pu_- \rangle_{G_N}$. The compression M has these two diagonal entries and off-diagonal entries $z, \bar{z}$. The original rank-two spectral expansion gives $L = \epsilon_N(a_+ - a_-)$ and

$$\begin{aligned} \|C\|_{\text{HS}, G_N}^2 &= \text{Tr}(P\mathcal{H}_N^2 P) - \text{Tr}((P\mathcal{H}_N P)^2) \\ &= \epsilon_N^2(a_+ + a_- - a_+^2 - a_-^2 + 2|z|^2). \end{aligned}$$

The cross trace in the second equation is $|\langle v, w \rangle|^2$ for the two rank-one maps $v\langle v, \cdot \rangle$ and $w\langle w, \cdot \rangle$, as follows by composing them. Subtract the displayed norm and $\epsilon_N^2(a_+ - a_-)^2$ from ϵ_N^2 ; direct expansion proves

$$\epsilon_N^2 - L^2 = \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}. \quad (\text{TVB.40})$$

Both determinants are nonnegative by the quadratic forms after (TVB.39). Adding (TVB.40) to (TVB.32) cancels precisely the intermediate ϵ_N^2 and gives all four nonnegative losses

$$\boxed{\alpha_{N+1} \sinh^2 s_N - L^2 = \alpha_{N+1}(e^{t_N} - \cosh s_N)^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}.} \quad (\text{TVB.41})$$

Every adjoint, projection, norm and coefficient is for the same original minimum metric as the Toda volumes.

The spectral projector has an exact map to the coupling. Factor the minimal polynomial of A as $m_+ m_-$ over the selected and remaining full generalized blocks, with their original exponents. Bezout polynomials a, b with $am_+ + bm_- = 1$ give $Q = b(A)m_-(A)$: it is identity on each selected block and zero on each other block. Thus $Q^2 = Q$, $AQ = QA$, and in the preceding orthogonal decomposition $Q = \begin{pmatrix} I & Y \\ 0 & 0 \end{pmatrix}$ for a unique map $Y : C_- \rightarrow C_+$. Block multiplication proves

$$\mathcal{S} : \text{Hom}(C_-, C_+) \longrightarrow \text{Hom}(C_-, C_+), \quad \mathcal{S}(Y) = BY - YD^{\text{op}} = C, \quad Q - P = \begin{pmatrix} 0 & Y \\ 0 & 0 \end{pmatrix}. \quad (\text{TVB.42})$$

This morphism is invertible with its full original nilpotent parts:

$$\mathcal{S}^{-1}(Z) = \int_0^\infty e^{-tB} Z e^{tD^{\text{op}}} dt. \quad (\text{TVB.43})$$

Indeed every selected eigenvalue has real part greater than c and every remaining one has real part at most c . On a full block at λ the first exponential is exactly $e^{-t\lambda} \sum_{j < \ell_\lambda} (-t)^j (B - \lambda I)^j / j!$, with its corresponding projector, and the second has the same formula with positive t on its remaining blocks. Summing these finite expressions gives a polynomial times a strictly decaying exponential for their product in (TVB.43). This proves convergence in the original operator norm. Differentiating the integrand gives $-B e^{-tB} Z e^{tD^{\text{op}}} + e^{-tB} Z e^{tD^{\text{op}}} D^{\text{op}}$. Its integral has lower endpoint Z and zero upper endpoint, proving the inverse identity. If $BY - Y D^{\text{op}} = 0$, the same product with $Z = Y$ is constant and decays to zero, hence $Y = 0$. This proves uniqueness as well. The space C_- is nonzero: otherwise $C_+ = C$ and $L = \text{Tr } \mathcal{H}_N = 0$, contradicting $L > 0$. Together with $C_+ \neq 0$ this makes the unit sphere in $\text{Hom}(C_-, C_+)$ nonempty. The actual smallest singular value $s_{B,D} > 0$ of $\mathcal{S}$ in the original Hilbert-Schmidt pairing exists by compactness of its unit sphere and injectivity. Consequently $\|C\|_{\text{HS}, G_N} \geq s_{B,D} \|Q - P\|_{\text{HS}, G_N}$. No bound on $s_{B,D}$ independent of the arithmetic inputs is used.

Retaining the phase and coupling in (TVB.41) gives the finite strengthening

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{\sqrt{L^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2}}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.44})$$

It follows by subtracting only the nonnegative imbalance and compression-determinant terms and applying the same inverse sinh as in (TVB.35). Summation over the disjoint indices $N = N_0 + 2j$ has the same exact endpoints as (TVB.36), with this degree-dependent square root in each summand. If $C_+ = 0$, use (TVB.32) directly; no positive-eigenline compression is asserted when $\epsilon_N = 0$. At $N = q - 1$, the scalar term instead uses the separate endpoint expression (TVB.38). These identities supply no asymptotic estimate and change no arithmetic class.

34.10 Crosswalk and exact continuing arithmetic quantity

CV.1–CV.3 are (TVB.22)–(TVB.24), with the additional proved identification $D_N^{\text{CV}} = \mathfrak{B}_m / \mathfrak{D}_{N+1}$. CV.4–CV.5 become (TVB.27)–(TVB.29), where the whole complex cross term is $i\omega_N(\sigma - \ell'_N)$. CV.6 is the minimum lift (TVB.14) and its determinant-line map after (TVB.15). CV.10–CV.10a are (TVB.30)–(TVB.31), now with the exact two-loss identity (TVB.34). CV.11–CV.12 are precisely (TVB.37)–(TVB.38). The new operator refinement is (TVB.20)–(TVB.21): it computes both individual rank-one curvature costs, including their shared retained quantity $\alpha_{N+1} \delta_N$.

For an actual quartet polynomial the exterior multiplicities and their lower trace L remain those of that exact polynomial; for a larger packet they remain a lower contribution to its actual full trace. All admission thresholds here use its actual $q = \deg \chi$. The finite formulas prove an upper estimate by the exact source and relation determinants and prove the telescope (TVB.36). They do not estimate the growth of α_{N+1} or the joint determinant ratio as k and N grow. That continuing arithmetic question is the original expression (TVB.27), equivalently the positive-minor source norm in (TVB.28)–(TVB.31), with both losses (TVB.32) available without changing any of its inputs.

35 The original theta input and certified infinite-series tails

This continuation supplies the omitted-tail and ball-arithmetic certificate for the analytic input of Section 9 of *Source-boundary volume dynamics for the arithmetic control*. Every theta coefficient, Fourier factor, conjugation, and complex branch used below is specified. The original seed has total

mass $M_1(0)$; the measure is never divided by that mass. The result computes analytic input of the source determinant flow. Its exact connection with a finite zero packet is given below by Mellin division and the multiplier $h(1/2 + it)$.

35.1 The original theta function and its Mellin transform

Put $D = -z\partial_z$, let $\mathcal{S}_{\pi/4} = \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \pi/4\}$, and use the restriction of the principal logarithm to this sector. Define

$$\vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{TI.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n=1}^{\infty} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{TI.2})$$

On a compact subset of the sector, each differentiated summand is bounded by a polynomial in n times e^{-an^2} with $a > 0$. Consequently both series define holomorphic functions there, and all the displayed derivatives are termwise valid.

Lemma 35.1 (The exact theta inversion). *On $\mathcal{S}_{\pi/4}$,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{TI.3})$$

For each $0 \leq \alpha < \pi/4$, every integer $j \geq 0$, and every real $A > 0$, $D^j f_0(z) = O(|z|^{-A})$ as $|z| \rightarrow \infty$ and $D^j f_0(z) = O(|z|^A)$ as $|z| \rightarrow 0$, uniformly for $|\arg z| \leq \alpha$.

Proof. For $x > 0$, the Fourier transform with kernel $e^{-2\pi i \xi t}$ of $e^{-\pi x^2 t^2}$ is $x^{-1}e^{-\pi \xi^2/x^2}$. For completeness, the value at $\xi = 0$ follows from $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$, whose square is evaluated in polar coordinates. Differentiation in ξ and integration by parts give the differential equation $I'(\xi) = -2\pi \xi I(\xi)/x^2$; the stated transform is its solution with the stated initial value. Periodize the Gaussian over integer translates. The resulting smooth periodic function has the absolutely convergent Fourier series with coefficients $x^{-1}e^{-\pi n^2/x^2}$. Evaluation at zero gives $\vartheta(x) = x^{-1}\vartheta(1/x)$. Both sides are holomorphic in the connected sector, so the identity theorem gives the first equation in (TI.3).

For the linear involution $(Rf)(z) = z^{-1}f(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this identity to $\vartheta = R\vartheta$ proves the second equation. The real coefficients in (TI.2) give the third. On a closed subsector, $\Re(z^2) \geq |z|^2 \cos(2\alpha) > 0$; the series and each derivative therefore have Gaussian decay at infinity, with a fixed polynomial factor. The second equation in (TI.3), differentiated using $DR = R(1 - D)$, transports that bound to zero and proves all the endpoint assertions. $\square$

For functions with these endpoint bounds set

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{TI.4})$$

The Mellin integral is entire. When $\Re s > 1$, the sum of the absolute integrals of the terms in (TI.2) is finite, because it is bounded by a constant depending on s times $\sum n^{-\Re s}$. Thus

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2}\zeta(s) (4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)) \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) =: g(s). \end{aligned} \quad (\text{TI.5})$$

The first line contains the factor 2 of (TI.2) and the factor $1/2$ from the substitution $u = \pi n^2 x^2$. The second follows from $\Gamma(z+1) = z\Gamma(z)$, which follows by integration by parts. Equation (TI.5) gives the entire continuation of its right side. In particular $g = 2\zeta$ with the original convention in the source note. Substituting $x \mapsto 1/x$ in (TI.4), using (TI.3), proves

$$g(1-s) = g(s), \quad \overline{g(\bar{s})} = g(s). \quad (\text{TI.6})$$

35.2 Mellin division, the dagger sign, and the tilted observation

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a monic finite product of actual zeros of g , with the selected orders at most their actual orders and with the selected multiset invariant under $\rho \mapsto 1 - \bar{\rho}$. The complete-order packets used in the source note satisfy these conditions. Write $d = \deg h$ and

$$v_h = g/h, \quad h^{\dagger}(s) = \overline{h(1 - \bar{s})}, \quad v_h^{\dagger}(s) = \overline{v_h(1 - \bar{s})}. \quad (\text{TI.7})$$

Removable singularities are filled in. The leading coefficient of $h^{\dagger}$ is $(-1)^d$, and its zero multiset is that of h ; therefore

$$h^{\dagger} = (-1)^d h, \quad v_h^{\dagger} = (-1)^d v_h. \quad (\text{TI.8})$$

Proposition 35.2 (Construction without an endpoint assumption). *For every $\sigma \in \mathbb{R}$ the integral*

$$F_h(z) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} v_h(s) z^{-s} ds \quad (\text{TI.9})$$

converges on the sector, is independent of σ , defines a holomorphic function there, and has the rapid endpoint bounds in the first lemma on each closed subsector. It satisfies

$$\mathcal{M}F_h = v_h, \quad h(D)F_h = f_0, \quad F_h(1/\bar{z}) = (-1)^d \bar{z} \overline{F_h(z)}. \quad (\text{TI.10})$$

Proof. Rotate the Mellin integral of f_0 to the ray $e^{i\beta}\mathbb{R}_{>0}$, where $|\beta| < \pi/4$. The endpoint estimates make both closing arcs vanish, and give

$$g(s) = e^{i\beta s} \int_0^{\infty} f_0(e^{i\beta} x) x^{s-1} dx.$$

For σ in any compact interval, the absolute integral on the right is uniformly bounded. Choosing the sign of β to be that of t proves $|g(\sigma + it)| \leq C e^{-|\beta||t|}$. For sufficiently large $|t|$, the explicit finite product h is bounded below by $c(1 + |t|)^d$ on the same strip. On the remaining compact rectangle, v_h is bounded because its singularities are removable. These facts give the same exponential estimate for v_h , with a different constant. It is valid for every $|\beta| < \pi/4$ and every compact interval of σ .

In (TI.9), choose $|\beta| > |\arg z|$. Absolute convergence and local uniform convergence of every differentiated integral follow. Horizontal contour segments tend to zero by the same estimate, proving independence of σ . Multiplication by s^j computes $D^j F_h$; moving σ arbitrarily far to the right or left gives respectively the asserted infinity or zero bounds.

The change $x = e^y$ turns (TI.9) on the line $\Re s = \sigma$ into Fourier inversion for $e^{\sigma y} F_h(e^y)$. The exponentially decreasing function $v_h(\sigma + it)$ and its derivatives have the required integrability, and inversion gives $\mathcal{M}F_h = v_h$. The Fourier inversion and Plancherel conventions here can be verified by inserting $e^{-\varepsilon t^2}$ in the inverse integral: Fubini gives convolution with the Gaussian approximate identity of integral one; letting $\varepsilon \downarrow 0$ gives inversion, and applying the same calculation to the inner product gives the factor $1/(2\pi)$ in Plancherel. Thus no change of Fourier normalization is implicit. Multiplication by $h(s)$ in (TI.9) is $h(D)$ on the source. Fourier inversion of (TI.5) proves $h(D)F_h = f_0$.

For the conjugate-linear involution $(\mathcal{R}F)(x) = x^{-1} \overline{F(1/\bar{x})}$, substitution gives $\mathcal{M}(\mathcal{R}F)(s) = \overline{\mathcal{M}F(1 - \bar{s})}$. Equations (TI.8) and Mellin uniqueness imply $\mathcal{R}F_h = (-1)^d F_h$ on the positive axis. Both sides extend holomorphically to the sector as $z^{-1} \overline{F_h(1/\bar{z})}$ and $(-1)^d F_h(z)$. Conjugating the resulting identity gives the last equation in (TI.10). $\square$

Here are the exact Hilbert-space maps. The change of variables

$$U : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}, dy), \quad (UF)(y) = e^{y/2} F(e^y) \quad (\text{TI.11})$$

is unitary, with inverse $u \mapsto [x \mapsto x^{-1/2} u(\log x)]$. On the rapid sector functions constructed above define, for real $|\theta| < \pi/2$,

$$(E_{\theta}F)(y) = e^{(y+i\theta/2)/2} F(e^{y+i\theta/2}). \quad (\text{TI.12})$$

This map has codomain $L^2(\mathbb{R}, dy)$; it is the analytic translation of UF on this specified domain. Its norm is not asserted to equal the untilded norm. Shifting the y contour by $i\theta/2$ in (TI.4), with the closing vertical edges vanishing by the proved endpoint bounds, gives

$$\mathcal{F}_+(E_\theta F_h)(t) = e^{\theta t/2} v_h(1/2 + it). \quad (\text{TI.13})$$

Indeed $y = z - i\theta/2$ contributes $e^{ity} = e^{\theta t/2} e^{itz}$. In the source expression the factor $e^{i\theta/4}$ remains present.

Define the original, unscaled density and its mass by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad \mu_h = M_h(0). \quad (\text{TI.14})$$

The exponential estimate above proves that the integral defining M_h and each of its derivatives converge locally uniformly on $|\Re\theta| < \pi/2$: for a compact substrip choose $2|\beta| > \sup |\Re\theta|$. By (TI.11)–(TI.13) and Plancherel,

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2} x)|^2 dx = 2 \int_1^\infty |F_h(e^{i\theta/2} x)|^2 dx \quad (\theta \text{ real}). \quad (\text{TI.15})$$

For the second equality the last identity in (TI.10) gives

$$F_h(e^{i\theta/2}/x) = (-1)^d e^{-i\theta/2} x \overline{F_h(e^{i\theta/2} x)}.$$

After taking absolute squares, the factor x^2 cancels the Jacobian of $x \mapsto 1/x$. The dagger sign is retained before that operation. Dagger stability alone gives this endpoint identity. If the selected multiset is additionally stable under ordinary conjugation, then $v_h(\bar{s}) = \overline{v_h(s)}$, so $w_h(-t) = w_h(t)$ and $M_h(-\theta) = M_h(\theta)$. This proves the precise extra symmetry needed for evenness.

The typed relation between seed and packet input is also explicit:

$$F_h \xrightarrow{h(D)} f_0 \quad v_h(1/2 + it) \xrightarrow{h(1/2 + it)} g(1/2 + it), \quad w_1(t) = |h(1/2 + it)|^2 w_h(t). \quad (\text{TI.16})$$

The arrow in the first display means the linear differential map on the rapid sector-function space; (TI.10) and (TI.13) prove its commutation with the transform. The scalar equality of densities holds even at selected line zeros by continuity; it does not require division of a measured zero value by zero.

35.3 The complete double series and an explicit omitted tail

For $h = 1$ we have $F_1 = f_0$. Set $c_1 = -6$, $c_2 = 4$ and, for real θ first,

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}), \quad \alpha_{rs} = r + s + \frac{1}{2}. \quad (\text{TI.17})$$

For $\Re C > 0$ define

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}. \quad (\text{TI.18})$$

The power uses the holomorphic logarithm of the right half-plane, real on its positive axis. To check (TI.18), first substitute $u = Cx^2$ for positive real C . Both sides extend holomorphically to the right half-plane: the integral by dominated convergence and the right side by the same continued incomplete gamma integral with its ray deformable inside that half-plane. The identity theorem then gives (TI.18) there with the stated branch.

Theorem 35.3 (The retained factor and the locally uniform tail). *For every θ with $|\Re\theta| < \pi/2$,*

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(\alpha_{rs}, C_{mn}(\theta))}{C_{mn}(\theta)^{\alpha_{rs}}}. \quad (\text{TI.19})$$

Let $M_{1,J}$ be exactly this sum with $1 \leq m, n \leq J$, for an integer $J \geq 0$. Fix $0 \leq \theta_0 < \pi/2$ and $B \geq 0$, and put

$$d_0 = \pi e^{-B} \cos \theta_0, \quad L = J + 1, \quad \rho_a = e^{-d_0(2a+1)}, \quad (\text{TI.20})$$

$$H_p(a, d_0) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{matrix} j \\ b \end{matrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{TI.21})$$

$$A_a(d_0, B) = 6\pi e^B H_2(a, d_0) + 4\pi^2 e^{2B} H_4(a, d_0). \quad (\text{TI.22})$$

Here $\left\{ \begin{matrix} j \\ b \end{matrix} \right\}$ is determined by $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$, $\left\{ \begin{matrix} j \\ 0 \end{matrix} \right\} = 0$ for $j > 0$, $\left\{ \begin{matrix} 0 \\ b \end{matrix} \right\} = 0$ for $b > 0$, and $\left\{ \begin{matrix} j+1 \\ b \end{matrix} \right\} = b \left\{ \begin{matrix} j \\ b \end{matrix} \right\} + \left\{ \begin{matrix} j \\ b-1 \end{matrix} \right\}$, with out-of-range indices zero. For every $|\Re\theta| \leq \theta_0$ and $|\Im\theta| \leq B$,

$$|M_1(\theta) - M_{1,J}(\theta)| \leq \mathcal{E}_J(\theta_0, B) := 16A_1(d_0, B)A_L(d_0, B)I_8(d_0(1+L^2)). \quad (\text{TI.23})$$

Every quantity on the right is explicit and positive. It tends to zero as $J \rightarrow \infty$. In particular (TI.23) supplies the requested locally uniform real-tilt bound by taking $B = 0$.

Proof. For real θ , substitute (TI.2) into the last expression in (TI.15). A product contributes $2 \cdot 2 \cdot 2 = 8$ before integration: the endpoint factor 2 and the two original theta coefficients 2. Equation (TI.18) contributes $1/2$. The remaining factor is precisely 4, and the powers of the two dilations give $e^{i\theta r} e^{-i\theta s}$. This establishes the coefficients and phases in (TI.19), subject to the absolute convergence now proved.

For integers $a \geq 1$, $p \geq 0$, and $x \geq 1$,

$$\begin{aligned} \sum_{n \geq a} n^p e^{-d_0 n^2 x^2} &\leq e^{-d_0 a^2 x^2} \sum_{j \geq 0} (a+j)^p e^{-d_0(2a+1)j} \\ &= e^{-d_0 a^2 x^2} H_p(a, d_0). \end{aligned}$$

The inequality uses $(a+j)^2 - a^2 = 2aj + j^2 \geq (2a+1)j$ and $x^2 \geq 1$. For the equality expand $(a+j)^p$ by the binomial theorem and use $j^r = \sum_b \left\{ \begin{matrix} r \\ b \end{matrix} \right\} (j)_b$. That last identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$. Thus (TI.21) is a finite positive rational function, not an unevaluated infinite sum.

For complex θ in the rectangle, let $f^\pm(x) = f_0(e^{\pm i\theta/2} x)$ and let $f_J^\pm$ contain only $1 \leq n \leq J$ in the original series. Since

$$\Re(\pi n^2 e^{\pm i\theta}) = \pi n^2 e^{\mp \Im\theta} \cos(\Re\theta) \geq d_0 n^2,$$

and $|e^{\pm i\Re\theta}| \leq e^{rB}$, the positive bound just proved implies

$$\begin{aligned} |f^\pm(x)|, |f_J^\pm(x)| &\leq 2A_1(d_0, B) x^4 e^{-d_0 x^2}, \\ |f^\pm(x) - f_J^\pm(x)| &\leq 2A_L(d_0, B) x^4 e^{-d_0 L^2 x^2}. \end{aligned}$$

Here $x^2 \leq x^4$ retains a bound for the original -6 term; no coefficient of the exact series is changed. These inequalities prove absolute integrability of the full expanded double series, locally uniformly on the rectangle. They also prove

$$2 \int_1^\infty |f^+(x)f^-(x) - f_J^+(x)f_J^-(x)| dx \leq 16A_1(d_0, B)A_L(d_0, B) \int_1^\infty x^8 e^{-d_0(1+L^2)x^2} dx,$$

because the product difference is $(f^+ - f_J^+)f^- + f_J^+(f^- - f_J^-)$. The right side is (TI.23). For fixed $d_0 > 0$, formula (TI.21) bounds $H_p(L, d_0)$ by a fixed constant times $(1 + L)^p$: its denominators are bounded away from zero uniformly for $L \geq 1$. Moreover $I_8(b) \leq e^{-b} \int_1^\infty x^8 e^{-d_0(x^2-1)} dx$ for $b \geq d_0$. Hence (TI.23) tends to zero exponentially in L^2 .

For complex θ , $C_{mn}(\theta)$ stays in the right half-plane by the displayed inequality. The integral $2 \int_1^\infty f^+(x)f^-(x) dx$ is holomorphic on the whole strip. It equals M_1 on the real interval by (TI.15) and conjugation of f_0 . Holomorphic uniqueness proves equality on the strip. The same locally uniform bounds justify (TI.19) there and finish the proof. $\square$

The tail controls differentiated moments as well. If the closed disc $\{z : |z - \theta| \leq r\}$ lies in the rectangle of the theorem, Cauchy's integral formula gives, for every integer $j \geq 0$,

$$|M_1^{(j)}(\theta) - M_{1,J}^{(j)}(\theta)| \leq \frac{j!}{r^j} \mathcal{E}_J(\theta_0, B). \quad (\text{TI.24})$$

The bound includes arbitrary finite derivative orders used by a Hankel determinant. It does not assert a useful estimate when the derivative order or determinant dimension itself grows without bound.

35.4 The second moment and the first tensor coefficient

Let $H = (D - 1/2)f_0$. With $y = \pi n^2 x^2$ and $P(y) = 8y^2 - 12y$, literal differentiation gives

$$\begin{aligned} (D - 1/2)(P(y)e^{-y}) &= (-2yP'(y) + 2yP(y) - P(y)/2) e^{-y} \\ &= (16y^3 - 60y^2 + 30y)e^{-y}. \end{aligned} \quad (\text{TI.25})$$

Thus

$$H(x) = \sum_{n \geq 1} \sum_{r=1}^3 d_r (\pi n^2)^r x^{2r} e^{-\pi n^2 x^2}, \quad (d_1, d_2, d_3) = (30, -60, 16). \quad (\text{TI.26})$$

The unitary map (TI.11) satisfies $U(D - 1/2)F = -(UF)'$ by differentiation. Integration by parts in the Fourier transform therefore gives $\mathcal{F}_+(UH)(t) = it g(1/2 + it)$. It follows, with the original $1/(2\pi)$ factor, that

$$M_1''(0) = \frac{1}{2\pi} \int_{\mathbb{R}} t^2 |g(1/2 + it)|^2 dt = \int_0^\infty |H(x)|^2 dx = 2 \int_1^\infty H(x)^2 dx. \quad (\text{TI.27})$$

For the last equality, $DR = R(1 - D)$ and $Rf_0 = f_0$ give $RH = -H$, that is $H(1/x) = -xH(x)$. Its square gives precisely the same Jacobian cancellation as before, with the new minus sign retained before squaring.

Let H_J restrict (TI.26) to $n \leq J$ and put $T_J = 2 \int_1^\infty H_J(x)^2 dx$. Then

$$T_J = \sum_{m,n=1}^J \sum_{r,s=1}^3 d_r d_s (\pi m^2)^r (\pi n^2)^s \frac{\Gamma(r+s+1/2, \pi(m^2+n^2))}{(\pi(m^2+n^2))^{r+s+1/2}}. \quad (\text{TI.28})$$

There is no coefficient 4 in this last expression: (TI.26) already contains its fully differentiated coefficients, and the endpoint factor 2 cancels the $1/2$ in (TI.18). Set

$$B_a = 30\pi H_2(a, \pi) + 60\pi^2 H_4(a, \pi) + 16\pi^3 H_6(a, \pi). \quad (\text{TI.29})$$

The proof of (TI.23), now with degree six in x , gives

$$|M_1''(0) - T_J| \leq \mathcal{T}_J := 4B_1 B_{J+1} I_{12}(\pi(1 + (J+1)^2)). \quad (\text{TI.30})$$

In detail, $|H|, |H_J| \leq B_1 x^6 e^{-\pi x^2}$ and $|H - H_J| \leq B_{J+1} x^6 e^{-\pi(J+1)^2 x^2}$; apply $|H^2 - H_J^2| \leq |H - H_J|(|H| + |H_J|)$ and the factor 2 of (TI.27).

For tensor degree $k \geq 1$, let $m_k = w_1^{*k}$ with its total mass μ_1^k , where $\mu_1 = M_1(0)$. Absolute integrability and Fubini give $Z_k(\theta) = \int e^{\theta u} m_k(u) du = M_1(\theta)^k$. The function w_1 is even by (TI.6), so $Z'_k(0) = M'_1(0) = 0$. The degree-zero monic source polynomial is 1, of squared norm $h_0 = \mu_1^k$, and its monic degree-one orthogonal polynomial in the original sum variable u is u , of squared norm $h_1 = Z''_k(0) = k\mu_1^{k-1}M''_1(0)$. Therefore the first recurrence coefficient is exactly

$$a_1^{(k)} = \frac{h_1}{h_0} = k \frac{M''_1(0)}{M_1(0)}. \quad (\text{TI.31})$$

The quotient of norms in (TI.31) is the coefficient in the monic three-term recurrence. It leaves both norms and the measure mass specified, and performs no replacement of m_k by a probability law.

35.5 Certified numerical enclosures and their executable scope

The accompanying script `toda_theta_input_tail_check_20260912.py` evaluates (TI.19), (TI.28), (TI.23), and (TI.30) at $J = 8$ with `python-flint 0.9.0`, using Arb real balls and Acb rectangular complex balls at 256-bit precision and one arithmetic thread. All integer and rational input, including $\theta = 1/10$, is supplied exactly; π , exponentials, powers, and incomplete gamma functions are enclosed by those ball operations. The incomplete gamma call has argument C and parameter α , that is `C.gamma_upper(alpha)`. Its branch is the one in (TI.18). The documented ball and endpoint semantics used here are in [the Arb interface](#) and [the Acb interface](#).

The script adds a symmetric error ball whose radius is the upward rounded upper endpoint of the proved tail bound. Thus the result encloses the infinite sum, rather than only its finite arithmetic. The finite real-tilt sum is real by the proved conjugation identity; the script additionally checks that its imaginary ball contains zero. The positive real mass enclosure is then used in interval division for (TI.31). The following are rational decimal intervals:

$$\begin{aligned} M_1(0) &\in [1.2790072478464851404795335922671932744916, \\ &\quad 1.2790072478464851404795335922671932744919], \\ M_1(1/10) &\in [1.3459071784582497969682205923576298051604, \\ &\quad 1.3459071784582497969682205923576298051607], \\ M''_1(0) &\in [13.0555493025705584353926846581231936824343, \\ &\quad 13.0555493025705584353926846581231936824346], \\ a_1^{(k)}/k &\in [10.2075647534857216888657612522375551749964, \\ &\quad 10.2075647534857216888657612522375551749967]. \end{aligned}$$

Every width is $3 \cdot 10^{-40}$. The computed positive tail bounds satisfy, respectively,

$$\mathcal{E}_8(0,0) < 7 \cdot 10^{-107}, \quad \mathcal{E}_8(1/10,0) < 3 \cdot 10^{-106}, \quad \mathcal{T}_8 < 4 \cdot 10^{-103}. \quad (\text{TI.32})$$

The receipt preserves both the displayed decimal intervals and exact rational dyadic lower and upper endpoints of each resulting ball. Its decimal endpoints are generated by outward rounding at 40 decimal places, with one additional unit on each side, and are subsequently checked by rigorous ball comparisons. Normal and optimized execution produce identical mathematical records. The requested negative-control mode inserts the false inequality $a_1^{(k)}/k < 1$ and must raise an error; it uses an explicit checked exception rather than a Python assertion, so optimization cannot remove that check.

The numerical certificate uses the standard correctness contract of Arb/Acb ball arithmetic together with the written infinite-tail proof. It is not inferred from agreement of two numerical precisions. Its output concerns the three specified infinite analytic-seed quantities and their recurrence

ratio. For $h = 1$ the cyclic arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the actual quotient map sends each polynomial to zero, while the positive seed source norm and (TI.31) remain. For a selected packet h , the precise passage from the present input to w_h is (TI.16), obtained from the globally defined Mellin division g/h . No packet determinant or bound as k tends to infinity is certified by these four scalar intervals.

36 An exact comparison of equality, volume contraction, and its discarded square

This calculation extends the positive Gaussian calibration of E35–E36 through two further source degrees. Every measure, phase, coefficient and quotient basis is specified. It is a calibration of the same constructions; the roots below are declared coefficients, with no assertion that they are zeros of the arithmetic function.

Fix $\mu > 0, \sigma > 0, k = 1$ and $c = 1/2$. Put

$$d\nu(u) = \frac{\mu}{\sqrt{2\pi\sigma}} e^{-u^2/(2\sigma^2)} du, \quad x = S - c, \quad \chi(S) = x^2 - \sigma^2, \quad C = \mathbb{C}[S]/(\chi). \quad (\text{TC1})$$

The original source pairing is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) d\nu(u)$. We use both the original quotient basis $(1, S)$ and the displayed basis $(1, x)$. The exact coordinate matrix, sending columns in the latter basis to columns in the former, is

$$L_{\text{coord}} = \begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix}, \quad A_S = \begin{pmatrix} 0 & \sigma^2 - c^2 \\ 1 & 2c \end{pmatrix}, \quad A_x = \begin{pmatrix} c & \sigma^2 \\ 1 & c \end{pmatrix}, \quad A_S L_{\text{coord}} = L_{\text{coord}} A_x. \quad (\text{TC2})$$

Here A_x represents multiplication by S in the basis $(1, x)$; multiplication by x is $A_x - cI$. Thus $G_{N,x} = L_{\text{coord}}^* G_{N,S} L_{\text{coord}}$ and $K_{N,S} = L_{\text{coord}} K_{N,x} L_{\text{coord}}^*$. In particular $\det L_{\text{coord}} = 1$ preserves the specified determinant, while the matrices and source polynomials retain their full coordinate transformations.

Integration by parts, with the Gaussian boundary term zero, gives $\int u^{2j} d\nu = \mu(2j-1)!!\sigma^{2j}$ and all odd moments zero: indeed the zeroth integral is μ , and differentiating $u^{2j-1}e^{-u^2/(2\sigma^2)}$ gives the recurrence $m_{2j} = (2j-1)\sigma^2 m_{2j-2}$. The first four monic source polynomials and their full squared norms are consequently

j	$p_j(S)$	ω_j
0	1	μ
1	x	$\mu\sigma^2$
2	$x^2 + \sigma^2$	$2\mu\sigma^4$
3	$x^3 + 3\sigma^2 x$	$6\mu\sigma^6$

(TC3)

For example $p_2(c + iu) = -(u^2 - \sigma^2)$ and $p_3(c + iu) = -i(u^3 - 3\sigma^2 u)$. Substitution of the moments proves every pairwise orthogonality and every displayed norm; the phases -1 and $-i$ are retained before taking the pairings.

In the basis $(1, x)$ the quotient columns $b_j = [p_j]_\chi$ are

$$b_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 2\sigma^2 \\ 0 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 0 \\ 4\sigma^2 \end{pmatrix}. \quad (\text{TC4})$$

For completeness, the minimum-source metric is obtained by writing a source polynomial as $\sum_{j=0}^N z_j p_j$. Its norm is $\sum \omega_j |z_j|^2$ and its quotient is $\sum b_j z_j$. The constrained minimizer is $z_j = \omega_j^{-1} b_j^* G_{N,x} v$, where $K_{N,x} = \sum b_j b_j^* / \omega_j$ and $G_{N,x} = K_{N,x}^{-1}$. Indeed these coefficients have quotient v , and their inner product with every quotient-kernel coefficient vector is zero. Pythagoras then

proves uniqueness and minimum norm $v^*G_{N,x}v$. Set $V_N = \det G_{N,x} = \det G_{N,S}$. Directly from (TC3)–(TC4),

$$\begin{aligned} G_{1,x} &= \text{diag}(\mu, \mu\sigma^2), \\ G_{2,x} &= \text{diag}(\mu/3, \mu\sigma^2), & (V_1, V_2, V_3) &= \mu^2\sigma^2(1, 1/3, 1/11). \\ G_{3,x} &= \text{diag}(\mu/3, 3\mu\sigma^2/11), \end{aligned} \quad (\text{TC5})$$

This is also exactly the source/relation determinant formula. On the original vertical line $\chi(c + iu) = -(u^2 + \sigma^2)$, so its zeroth relation norm and its next monic relation norm are

$$\nu_0 = \int (u^2 + \sigma^2)^2 d\nu = 6\mu\sigma^4, \quad \nu_1 = \int u^2(u^2 + \sigma^2)^2 d\nu = 22\mu\sigma^6. \quad (\text{TC6})$$

The two relation columns $\chi, \chi x$ are orthogonal by parity. For $n = 2, 3, 4$, the source Gram determinant is $\mathfrak{D}_n = \prod_{j < n} \omega_j$; the corresponding relation determinants are $\mathfrak{B}_0 = 1, \mathfrak{B}_1 = \nu_0, \mathfrak{B}_2 = \nu_0\nu_1$. These identities follow by the triangular monic changes of basis. They give $\mathfrak{D}_{N+1}/\mathfrak{B}_{N-1} = V_N$ for each $N = 1, 2, 3$, including the literal empty determinant at $N = 1$.

Let $H_N = A_x + G_{N,x}^{-1}A_x^*G_{N,x} - 2cI$. Multiplication of the displayed matrices yields

$$H_1 = \begin{pmatrix} 0 & 2\sigma^2 \\ 2 & 0 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0 & 4\sigma^2 \\ 4/3 & 0 \end{pmatrix}, \quad \epsilon_1^2 = 4\sigma^2, \quad \epsilon_2^2 = 16\sigma^2/3. \quad (\text{TC7})$$

Each H_N is self-adjoint in its stated positive metric; its squared matrix is the corresponding scalar multiple of I . Thus these are the squares of its operator norm, or equivalently $\text{Tr}(H_N^2)/2$. The spectral aggregate for A_x is $L_{\text{spec}} = 2\sigma$, since its eigenvalues are $c + \sigma, c - \sigma$. Hence equality $\epsilon_1 = L_{\text{spec}}$ holds at the first admitted degree, whereas $\epsilon_2 > L_{\text{spec}}$ holds at the next. No freedom to choose another metric was used in either calculation.

To compute the auxiliary derivative at the original observation, replace $d\nu$ by $e^{\theta u}d\nu$ for real θ and set $\ell_N(\theta) = \log \det G_N(\theta)$ in either specified quotient basis. The precise real-coordinate moment Grams are $H_{rs}(\theta) = \int u^{r+s}e^{\theta u}d\nu$ and $H_{rs}^\chi(\theta) = \int u^{r+s}(u^2 + \sigma^2)^2e^{\theta u}d\nu$. Their matrices at $-\theta$ are conjugated by $\text{diag}(1, -1, 1, \dots)$ from those at θ . The exact map to original monomials is $S^j = \sum_{r=0}^j \binom{j}{r} c^{j-r} i^r u^r$. On dimension n its triangular determinant is $i^{n(n-1)/2}$, of modulus one. It therefore identifies these determinants with the original source and relation determinants, including the relation multiplier χ . Both determinants are even functions of θ . Their positive determinant ratio is therefore even, giving $\ell'_N(0) = 0$. Also $\sigma_T := \text{Tr}((A_x - cI)/i) = 0$. This proves the vanishing of the phase subtraction at zero tilt; it does not impose evenness at a nonzero tilt.

Finally set $\delta_N = V_N/V_{N-1}$ at the two degrees where both metrics exist, and set $\mathcal{R}_2 = V_1/V_3 = (\delta_2\delta_3)^{-1}$. Equations (TC3) and (TC5) give

$$\delta_2 = 1/3, \quad \delta_3 = 3/11, \quad a_3 = \omega_3/\omega_2 = 3\sigma^2, \quad \mathcal{R}_2 = 11. \quad (\text{TC8})$$

The complete Toda expression and its finite upper bound are respectively

$$a_3(1 - \delta_2)(\delta_3^{-1} - 1) = 16\sigma^2/3, \quad a_3 \frac{(\mathcal{R}_2 - 1)^2}{4\mathcal{R}_2} = 75\sigma^2/11. \quad (\text{TC9})$$

To identify the exact lost term, retain $s = (\log 3 + \log(11/3))/2 = \log(11)/2$ and $t = (\log(11/3) - \log 3)/2$. Then $e^t = \sqrt{11}/3$ and $\cosh s = 6/\sqrt{11}$, whence

$$a_3(e^t - \cosh s)^2 = 49\sigma^2/33 = 75\sigma^2/11 - 16\sigma^2/3. \quad (\text{TC10})$$

Thus the entire gap between the exact arithmetic-construction radius in this specified calibration and its Toda upper expression is precisely the retained imbalance square. The phase square is zero

for the proved parity reason. Both metric degrees, the quotient matrix, the source and relation volumes, and the original mass μ occur explicitly in this comparison.

There is also an explicit comparison with the spectral and orthogonal projectors in the exterior equality proof. In the $(1, x)$ coordinates let $v_+ = (\sigma, 1)^\top$ and $v_- = (-\sigma, 1)^\top$. They are eigenvectors of A_x for $c + \sigma$ and $c - \sigma$. Directly,

$$v_+^* G_{1,x} v_- = 0, \quad v_+^* G_{2,x} v_- = 2\mu\sigma^2/3. \quad (\text{TC11})$$

At $N = 2$ the orthogonal projector to $\mathbb{C}v_+$ and the algebraic spectral projector along $\mathbb{C}v_-$ are respectively

$$P = \frac{v_+ v_+^* G_{2,x}}{v_+^* G_{2,x} v_+} = \begin{pmatrix} 1/4 & 3\sigma/4 \\ 1/(4\sigma) & 3/4 \end{pmatrix}, \quad Q = \frac{A_x - (c - \sigma)I}{2\sigma} = \begin{pmatrix} 1/2 & \sigma/2 \\ 1/(2\sigma) & 1/2 \end{pmatrix}. \quad (\text{TC12})$$

These formulas prove $P^2 = P$, $P^{\dagger G_{2,x}} = P$, $Q^2 = Q$, and $\text{im } P = \text{im } Q = \mathbb{C}v_+$. Their exact kernels are $\ker P = \mathbb{C}(-3\sigma, 1)^\top = v_+^{\perp G_{2,x}}$ and $\ker Q = \mathbb{C}v_-$: direct multiplication proves the two zero values, while $Pv_+ = Qv_+ = v_+$ proves rank one. In particular $Pv_- = v_+/2$. They also retain the change to original coordinates: both matrices are sent to $L_{\text{coord}}(\cdot)L_{\text{coord}}^{-1}$ and paired with $G_{2,S} = L_{\text{coord}}^{-*} G_{2,x} L_{\text{coord}}^{-1}$. For $X = Q - P$, the exact Hilbert–Schmidt pairing gives

$$\|X\|_{\text{HS}, G_{2,x}}^2 = \text{Tr}(G_{2,x}^{-1} X^* G_{2,x} X) = 1/3, \quad \epsilon_2^2 - L_{\text{spec}}^2 = L_{\text{spec}}^2 \|X\|_{\text{HS}, G_{2,x}}^2 = 4\sigma^2/3. \quad (\text{TC13})$$

For example the two diagonal entries of $G_{2,x}^{-1} X^* G_{2,x} X$ are $1/4$ and $1/12$, which proves the trace value without changing its pairing. The last equality follows also from (TC7). Thus the two explicit gaps have different exact source maps: (TC13) measures the spectral/orthogonal projector discrepancy, while (TC10) measures the square discarded by the Toda upper bound. Both maps and both positive quantities are calculated in the original specified metric.

37 The original gamma convolution and the exact arithmetic coefficient maps

37.1 Source, attribution, and original coordinates

This chapter proves the coefficient and cochain maps needed to integrate *Tau Gamma Convolution Descent*, whose original NOTE.tex has SHA256

7cf984d87de405b815250b44454c4ee27c8ab46574da245b377ebc6e49327fa2.

The analytic generating identities (GC.24)–(GC.26) were supplied by the *Zeta Function Literature & Local Corpus Foundation* task on 13 September 2026. The proofs here verify and integrate that supplied join; they do not claim its discovery. The original gamma generating integral and monic family are the existing TG construction, with the classical Meixner–Pollaczek parameters retained.

All scalar source coordinates remain

$$s = \frac{1}{2} + it, \quad S = s_1 + \cdots + s_k = \frac{k}{2} + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GC.1})$$

Here $k \geq 1$ is an integer. We use inner products conjugate-linear in the first argument. For a polynomial, an overline over its name means coefficientwise conjugation. In particular $\overline{P(c + iu)} = \overline{P}(c - iu)$ when c, u are real.

37.1.1 The analytic source and its actual theta boundary

Let $D = -x\partial_x$. Let V be the complex vector space of even Schwartz functions ϕ on $\mathbb{R}$ with $\phi(0) = 0$ and $\int_{\mathbb{R}} \phi(x) dx = 0$. Let $\mathcal{B}$ be the smooth functions on $\mathbb{R}_{>0}$ such that for every $a > 0$ and every integer $j \geq 0$,

$$D^j F(x) = O(x^a) \quad (x \downarrow 0), \quad D^j F(x) = O(x^{-a}) \quad (x \rightarrow \infty).$$

The original theta map and cochain grading are

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1. \quad (\text{GC.2})$$

The sum and its differentiated sums converge at infinity by the Schwartz bounds. Poisson summation gives its corresponding bounds at zero: the zero Fourier coefficient vanishes by $\int \phi = 0$, and the original zero term vanishes by $\phi(0) = 0$. Differentiation preserves both conditions, because $D\phi(0) = 0$ and integration by parts gives $\int D\phi = \int \phi = 0$. Thus $D\Theta = \Theta D$ on these spaces. The rapid bounds make the Mellin integral

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx$$

entire for $F \in \mathcal{B}$ and give $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$, including the vanished endpoint terms.

For clarity the seed and all its constants can be constructed directly. Put

$$\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad f_0 = D(D-1)\vartheta = \Theta\phi_0. \quad (\text{GC.3})$$

Gaussian integration gives $\int 4\pi^2 x^4 e^{-\pi x^2} dx = 3$ and $\int 6\pi x^2 e^{-\pi x^2} dx = 3$ on $\mathbb{R}$, and $\phi_0(0) = 0$, so $\phi_0 \in V$. The Gaussian Fourier transform in the convention $\widehat{f}(y) = \int f(x)e^{-2\pi ixy} dx$ is $e^{-\pi y^2}$: differentiation under the integral and integration by parts give $\widehat{f}' = -2\pi y\widehat{f}$, and the integral at zero is one. Periodizing this Gaussian gives an absolutely convergent Fourier series with those computed coefficients. Evaluating it at zero proves $\vartheta(1/x) = x\vartheta(x)$ for $x > 0$. Holomorphic continuation gives the same identity for $|\arg z| < \pi/4$, where both theta series converge normally on compact subsets.

For $RF(x) = x^{-1}F(1/x)$, direct differentiation gives $DR = R(1-D)$. Hence $D(D-1)$ commutes with R and

$$f_0(1/z) = zf_0(z). \quad (\text{GC.4})$$

On every closed narrower sector, each derivative of f_0 at infinity is bounded by a polynomial times $e^{-c|z|^2}$ for some $c > 0$. Equation (GC.4), differentiated finitely many times, proves rapid decay at zero too. In particular $f_0 \in \mathcal{B}$.

For $\Re s > 1$, absolute convergence permits termwise Mellin integration of $2 \sum_{n \geq 1} \phi_0(nx)$. The change $y = \pi n^2 x^2$ gives

$$\begin{aligned} \mathcal{M}f_0(s) &= \pi^{-s/2} \zeta(s) \{4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)\} \\ &= s(s-1)\pi^{-s/2} \Gamma(s/2) \zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GC.5})$$

The first line retains the theta factor two and the integration Jacobian $1/2$. The gamma recurrence proves the second line. The Mellin integral gives its entire continuation.

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a finite monic packet of actual nontrivial zeros of g , each selected with its complete order. The empty packet is $h = 1$. Write $d = \deg h$ and $v_h = g/h$. At each selected centre, cancelling its exact local factor constructs the analytic value of v_h ; hence v_h is entire. It is nonzero at each selected centre and is not identically zero.

We record the bounds that justify the integrals below. For any closed real interval of σ , any $0 < \beta < \pi/4$, and any $j \geq 0$,

$$|v_h(\sigma + it)| \leq C_{j,\beta,h}(1 + |t|)^{-j} e^{-\beta|t|}. \quad (\text{GC.6})$$

To prove this first for g , rotate its Mellin contour to $z = e^{i\beta}x$ for $t \geq 0$ and to $z = e^{-i\beta}x$ for $t < 0$. The sector estimates after (GC.4) make the two closing arcs vanish, uniformly for the stated σ interval. The rotation contributes $e^{\mp\beta t}$ and a unit-modulus factor depending on σ . Repeated integration by parts along that ray gives every displayed power of $(1 + |t|)^{-1}$; all ray endpoint terms vanish by the same estimates. For $|t|$ larger than twice the moduli of all packet centres relative to that σ interval, monicity gives a lower bound for $|h(\sigma + it)|$ by a positive constant times $|t|^d$. On the remaining compact rectangle the cancelled entire quotient is bounded. This proves (GC.6) for v_h .

For any real σ define

$$F_h(x) = \frac{1}{2\pi} \int_{\mathbb{R}} v_h(\sigma + it) x^{-\sigma - it} dt. \quad (\text{GC.7})$$

Horizontal rectangle estimates from (GC.6) prove independence of σ . Differentiating in x multiplies the integrand by a power of $\sigma + it$ and remains absolutely convergent. Choosing σ arbitrarily large positive at infinity and arbitrarily large negative at zero proves $F_h \in \mathcal{B}$. Fourier inversion in $\log x$, with the displayed $1/(2\pi)$ constant, gives $\mathcal{M}F_h = v_h$. Consequently Mellin injectivity gives

$$h(D)F_h = f_0 = \Theta\phi_0. \quad (\text{GC.8})$$

Define the actual density, mass, and Laplace integral by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt. \quad (\text{GC.9})$$

Equation (GC.6) proves holomorphy of M_h on $|\Re\theta| < \pi/2$, with $M_h^{(j)}(\theta) = \int t^j e^{\theta t} w_h(t) dt$ there. Indeed any compact substrip is dominated using a β for which 2β is larger than its real width. The mass is positive because v_h is a nonzero analytic function; its real-line zeros are discrete. Every original spectral moment and its 2π factor are retained.

37.2 The gamma reference with its actual mass

Fix $\lambda > 0$ and write $\alpha = 2\lambda$. Define

$$d\sigma_\lambda(t) = r_\lambda(t) dt, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda). \quad (\text{GC.10})$$

The product of two gamma integrals, changed by $(a, b) = (ry, (1 - r)y)$ with Jacobian y , proves the beta integral. Its substitution $r/(1 - r) = e^x$ gives

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{\Gamma(\lambda + iv)\Gamma(\lambda - iv)}{\Gamma(2\lambda)}.$$

The integrand before Fourier transformation and its second derivative are integrable. Two integrations by parts make its transform integrable, so Fourier inversion is valid. Alternatively, multiplying that transform by $e^{-\varepsilon v^2}$ gives an absolutely integrable Gaussian approximation and proves the same inversion by dominated convergence. The substitution $t = 2v$, with Jacobian two, therefore proves

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GC.11})$$

The beta integrand extends holomorphically to $|\Im x| < \pi$ using the branch positive on $\mathbb{R}$. On every closed narrower strip it is bounded by a constant times $e^{-\lambda|\Re x|}$. Shifting the contour up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_{\lambda,b} e^{-b|v|}$ for every $b < \pi$. Thus σ_λ has exponential moments of every order less than $\pi/2$ in the original t variable.

Fourier uniqueness for finite measures and multiplication of (GC.11) give the literal sum density

$$r_{\lambda,k} := r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (\text{GC.12})$$

Its mass is c_λ^k . Analytic continuation justified by the proved exponential moments gives

$$Z_{\lambda,k}^\Gamma(\theta) = \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-k\alpha}, \quad |\Re \theta| < \pi/2. \quad (\text{GC.13})$$

The power uses the analytic logarithm of $\cos \theta$ equal to zero at zero; this cosine has no zeros on the stated strip.

For $|z| < 1$, let

$$\begin{aligned} \mathcal{G}_\lambda(z, t) &= (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} \\ &= \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n. \end{aligned} \quad (\text{GC.14})$$

Both logarithms are the principal logarithms and vanish at zero, since $1 \pm iz$ have positive real part. For real z, w near zero, the Laplace transform and the elementary cosine addition identity give

$$\int \mathcal{G}_\lambda(z, t) \mathcal{G}_\lambda(w, t) d\sigma_\lambda(t) = c_\lambda (1 - zw)^{-\alpha}.$$

On a sufficiently small closed bidisc all derivatives of the integrands are bounded by a polynomial times $e^{a|t|}$ with $a < \pi/2$. The exponential moment proof permits their integration. Coefficient comparison then proves, with the monic leading coefficient obtained from $\arctan z = z + O(z^3)$,

$$\begin{aligned} \int b_n^{(\lambda)} b_j^{(\lambda)} d\sigma_\lambda &= c_\lambda n! (\alpha)_n \delta_{nj}, \\ b_0^{(\lambda)} &= 1, \quad b_1^{(\lambda)} = t, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \end{aligned} \quad (\text{GC.15})$$

The recurrence follows directly by differentiating (GC.14): $(1 + z^2) \partial_z \mathcal{G}_\lambda = (t - \alpha z) \mathcal{G}_\lambda$. The polynomials have real coefficients.

They are complete in $L^2(r_{\lambda,k} du)$ at parameter $k\lambda$. In fact, an orthogonal vector f defines the finite complex measure $f(u) r_{\lambda,k}(u) du$. Cauchy-Schwarz and exponential integrability give its Fourier transform on a nonempty complex strip. All derivatives at zero vanish by polynomial orthogonality, so analytic uniqueness and Fourier uniqueness imply $f = 0$ almost everywhere. This argument uses the full measure mass.

37.3 The sum projection and exact coefficient powers

On the actual spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du),$$

let $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$. Fubini and (GC.12) prove that it is an isometry. The affine fibre chart has Jacobian one, so its adjoint for $k \geq 2$ is

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GC.16})$$

The denominator is positive everywhere. Fibre Cauchy-Schwarz and Fubini give $\|C_\Sigma F\| \leq \|F\|$, so the formula defines the indicated adjoint on all of $\mathcal{H}_k$. When $k = 1$, both maps are the identity. In every case $C_\Sigma U_\Sigma = 1$ and $P_\Sigma = U_\Sigma C_\Sigma$ is the orthogonal projection. The maps

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n, \quad n \in \ker C_\Sigma \quad (\text{GC.17})$$

are inverse Hilbert isometries. This retains the entire relative fibre.

Multiplying the k generating functions gives a finite identity at each coefficient:

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1+\dots+n_k=n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \quad (\text{GC.18})$$

Pairing it with a product polynomial and using (GC.15) leaves only its total degree $n = \sum_i n_i$. The pairing at that degree is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$ and the sum norm is $c_\lambda^k n! (k\alpha)_n$. Completeness therefore proves

$$C_\Sigma \prod_i b_{n_i}^{(\lambda)} = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}. \quad (\text{GC.19})$$

The squared norm of the retained complementary polynomial is exactly

$$c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right], \quad (\text{GC.20})$$

by the orthogonal decomposition (GC.17).

For the original arithmetic density define, for every $\lambda > 0$,

$$\begin{aligned} A_{h,\lambda}(t) &= \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda} = |A_{h,\lambda}|^2, \quad w_h = B_{h,\lambda} r_\lambda, \\ c_{h,n} &= \frac{\int b_n^{(\lambda)}(t) w_h(t) dt}{c_\lambda n! (\alpha)_n}, \quad \mathcal{A}_h(z) = \sum_{n \geq 0} (\alpha)_n c_{h,n} z^n, \\ d_{h,k,n} &= [z^n] \mathcal{A}_h(z)^k. \end{aligned} \quad (\text{GC.21})$$

The notation $\mathcal{A}_h$ retains the fixed parameter λ . The complex amplitude $A_{h,\lambda}$ and the generating function $\mathcal{A}_h$ are different specified maps. The integrals in (GC.21) converge by (GC.6), before any convergence claim about the power series. Each $d_{h,k,n}$ is the finite expression

$$d_{h,k,n} = \sum_{n_1+\dots+n_k=n} \prod_i (\alpha)_{n_i} c_{h,n_i}. \quad (\text{GC.22})$$

Let $m_{h,k} = w_h^{*k}$. Absolute moments and Fubini give, by (GC.18),

$$\int_{\mathbb{R}} b_n^{(k\lambda)}(u) m_{h,k}(u) du = c_\lambda^k n! d_{h,k,n}. \quad (\text{GC.23})$$

This finite identity holds for every $\lambda > 0$ and every n ; it does not require an infinite product to be interchanged with an integral. In particular $c_{h,0} = \mu_h/c_\lambda$ and the total mass is μ_h^k .

37.4 The supplied analytic generating join, with its branches

Theorem 37.1 (The exact analytic generating function). *For the original arithmetic measure and every $\lambda > 0$, the series in (GC.21) converges holomorphically on $|z| < 1$ and is exactly*

$$\mathcal{A}_h(z) = \frac{1}{c_\lambda} (1 + z^2)^{-\lambda} M_h(\arctan z). \quad (\text{GC.24})$$

Here $\arctan z = (\log(1 + iz) - \log(1 - iz))/(2i)$, and every logarithm is the branch inherited from zero. For real $|\theta| < \pi/4$,

$$M_h(\theta) = c_\lambda \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta). \quad (\text{GC.25})$$

With the original tensor mass and reference mass,

$$\frac{Z_{h,k}(\theta)}{Z_{\lambda,k}^\Gamma(\theta)} = \mathcal{A}_h(\tan \theta)^k, \quad Z_{h,k}(\theta) = \int e^{\theta u} m_{h,k}(u) du = M_h(\theta)^k. \quad (\text{GC.26})$$

Proof. For $|z| < 1$, both $1 \pm iz$ have positive real part. Their principal logs are analytic, and their sum is the analytic logarithm of $1 + z^2$ vanishing at zero. Direct substitution in (GC.14) gives

$$\mathcal{G}_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GC.27})$$

There is no omitted phase in this identity: the $it/2$ coefficient multiplies $\log(1 - iz) - \log(1 + iz) = -2i \arctan z$.

The ratio $w = (1 + iz)/(1 - iz)$ has

$$\Re w = \frac{1 - |z|^2}{|1 - iz|^2} > 0, \quad \tan(2\Re \arctan z) = \frac{2\Re z}{1 - |z|^2}.$$

The argument of w lies in $(-\pi/2, \pi/2)$ and equals $2\Re \arctan z$. Thus for every $0 \leq r < 1$,

$$|z| \leq r \implies |\Re \arctan z| \leq \arctan r < \pi/4, \quad |\mathcal{G}_\lambda(z, t)| \leq (1 - r^2)^{-\lambda} e^{\arctan(r)|t|}. \quad (\text{GC.28})$$

For the angle inequality use monotonicity of arctangent and $2|\Re z|/(1 - |z|^2) \leq 2r/(1 - r^2)$, together with $2\arctan r \in [0, \pi/2)$. For the modulus inequality use $|1 + z^2| \geq 1 - r^2$ and real $\lambda > 0$.

Equation (GC.6) makes the right side of (GC.28) integrable against $w_h(t) dt$, uniformly on every compact subdisc. Dominated integration therefore defines a holomorphic function $c_\lambda^{-1} \int \mathcal{G}_\lambda(z, t) w_h(t) dt$ on the whole unit disc. One explicit justification is Morera's theorem: domination permits exchanging the integral with each triangle contour, and the inner holomorphic contour integral is zero. For its derivatives on $|z| \leq r$, choose $r < R < 1$ and apply Cauchy's derivative formula on circles of radius $R - r$; the kernel derivative $|\partial_z^n \mathcal{G}_\lambda(z, t)|$ is bounded by $n!(R - r)^{-n}(1 - R^2)^{-\lambda} e^{\arctan(R)|t|}$. This integrable bound justifies every derivative interchange.

At zero, (GC.14) now gives

$$\mathcal{A}_h^{(n)}(0) = \frac{1}{c_\lambda} \int b_n^{(\lambda)}(t) w_h(t) dt = n!(\alpha)_n c_{h,n}. \quad (\text{GC.29})$$

Thus the Taylor series of this holomorphic integral is exactly the original coefficient series. Substitution of (GC.27) in the integral proves (GC.24). All its constants occur before the integration.

For real $|\theta| < \pi/4$, $|\tan \theta| < 1$, $\arctan(\tan \theta) = \theta$ on the specified branch, and $1 + \tan^2 \theta = \sec^2 \theta > 0$. Solving (GC.24) for $M_h(\theta)$ gives (GC.25). Finally the absolute integral of $\prod_i e^{\theta t_i} w_h(t_i)$ is finite throughout $|\Re \theta| < \pi/2$. Fubini gives $Z_{h,k} = M_h^k$ with no mass division. Divide the k th power of (GC.25) by (GC.13) to obtain (GC.26). At zero this ratio is explicitly $(\mu_h/c_\lambda)^k$. $\square$

The inverse also has a precise complex domain. If $|\Re \theta| < \pi/4$, then

$$|\tan \theta|^2 = \frac{\cosh(2\Im \theta) - \cos(2\Re \theta)}{\cosh(2\Im \theta) + \cos(2\Re \theta)} < 1.$$

The denominator is positive and the branch satisfies $\arctan(\tan \theta) = \theta$, first on a neighbourhood of zero and then on this connected strip by analytic uniqueness. Hence (GC.25)–(GC.26) extend to that strip with the analytic branch of the cosine power in (GC.13). No logarithm of $\mathcal{A}_h$ is asserted: it can have complex zeros, whereas all real-tilt integrals $M_h(\theta)$ are strictly positive.

For each finite derivative there is also an exact moment formula. Write $b_n^{(\lambda)}(t) = \sum_{j=0}^n b_{n,j} t^j$. Then

$$n!(\alpha)_n c_{h,n} = \frac{1}{c_\lambda} \sum_{j=0}^n b_{n,j} M_h^{(j)}(0), \quad d_{h,k,n} = \frac{1}{n!} \left. \frac{d^n}{dz^n} \mathcal{A}_h(z)^k \right|_{z=0}. \quad (\text{GC.30})$$

The first identity follows by the finite polynomial expansion and (GC.9); the second is Taylor's formula, whose Leibniz sum is exactly (GC.22). The triangular coefficients $b_{n,n} = 1$ give an invertible finite map between moments through degree n and coefficients through degree n , with the full diagonal factors $c_\lambda n!(\alpha)_n$ retained.

37.5 The complete density expansion for the original bounded references

The finite identities and analytic join above hold for every $\lambda > 0$. For the original full-quartet reference choose $\lambda = 1/4$. More generally choose the actual shifted reference

$$\lambda = \frac{1}{4} + L, \quad L \text{ an integer}, \quad L \geq \max(0, 3 - d). \quad (\text{GC.31})$$

These choices admit the following proved bounded multiplier and hence the complete L^2 density expansion.

Indeed at $s = 1/2 + it$ the defining g identity gives

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (\text{GC.32})$$

This expression is continued through each selected zero by the already defined entire quotient. Integrating the step function $[x]$, initially for $\Re s > 1$, gives $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$. The last integral is holomorphic for $\Re s > 0$, so the identity continues there away from 1. On the original line it gives $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For a nonempty packet put $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$; for $h = 1$ put $T_h = 1$. For $|t| \geq T_h$, every original packet factor has modulus at least $|t|/2$. Gamma recurrence then gives

$$\begin{aligned} A_{h,1/4+L}(t) &= \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}, \\ B_{h,1/4+L}(t) &\leq \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} |t|^{6-2d-2L}. \end{aligned} \quad (\text{GC.33})$$

For the constant 25, the two squared numerator estimates at $|t| \geq 1$ are $(t^2 + 1/4)^2 \leq 25t^4/16$ and $|\zeta(s)|^2 \leq 16t^2$. Each shift factor has modulus at least $|t|/2$. The exponent in (GC.33) is nonpositive for (GC.31). Thus a completely specified finite bound is

$$C_{h,L} = \max \left\{ \sup_{|t| \leq T_h} B_{h,1/4+L}(t), \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} T_h^{6-2d-2L} \right\} > 0. \quad (\text{GC.34})$$

The compact term is finite by continuity of the cancelled multiplier. This is a mathematical bound with a specified supremum, not a claim that a numerical packet supremum has already been enclosed.

For these references $B \in L^2(d\sigma_\lambda)$, because the reference has finite mass and $0 \leq B \leq C_{h,L}$. Fibre integration of $w_h = Br_\lambda$ proves

$$B_{h,\lambda;k} = C_\Sigma(B^{\otimes k}), \quad m_{h,k} = r_{\lambda,k} B_{h,\lambda;k}, \quad 0 \leq B_{h,\lambda;k} \leq C_{h,L}^k. \quad (\text{GC.35})$$

For $k \geq 2$ it is positive at every sum coordinate: the zeros of B form a countable discrete set, and their forbidden fibre coordinates are a countable union of affine hyperplanes of measure zero. Every remaining reference factor is positive. For $k = 1$ the original isolated zeros remain.

Combining (GC.23), completeness, and the sum norm in (GC.15) gives the full, convergent $L^2(r_{\lambda,k} du)$ identity

$$B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (\text{GC.36})$$

In particular the norm of the retained relative density vector is

$$\|B^{\otimes k} - U_\Sigma B_{h,\lambda;k}\|^2 = \left(\int B^2 d\sigma_\lambda \right)^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (\text{GC.37})$$

This is Pythagoras for (GC.17), so every norm and mass factor is retained. A truncation of (GC.36) is not asserted pointwise positive.

37.6 Original-coordinate finite Grams and every tilt derivative

Let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ for $N \geq 0$, with original ordered basis $1, S, \dots, S^N$, and set $c = k/2$. The arithmetic source Gram and its reference are

$$\begin{aligned} (\mathbf{M}_{h,k,N}(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} m_{h,k}(u) du, \\ (\mathbf{M}_{\lambda,k,N}^\Gamma(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} r_{\lambda,k}(u) du, \quad 0 \leq i, j \leq N. \end{aligned} \quad (\text{GC.38})$$

They are holomorphic entrywise for $|\Re \theta| < \pi/2$ and positive definite at every real tilt in that strip: a nonzero polynomial has only finitely many real-line roots and each measure is positive almost everywhere. Let $\mathbf{H}_{h,k,N}$ denote the corresponding real- u monomial Hankel matrix. The exact matrix of the unchanged coordinate substitution is

$$(\mathbf{L}_{N+1})_{aj} = \begin{cases} \binom{j}{a} c^{j-a} i^a, & 0 \leq a \leq j \leq N, \\ 0, & a > j, \end{cases} \quad \det \mathbf{L}_{N+1} = i^{N(N+1)/2}. \quad (\text{GC.39})$$

Expanding each original S monomial proves

$$\mathbf{M}_{h,k,N} = \mathbf{L}_{N+1}^* \mathbf{H}_{h,k,N} \mathbf{L}_{N+1} \quad (\text{GC.40})$$

and the identical congruence for the reference. Since the displayed determinant has modulus one, it also proves exact equality of the two coordinate determinants. The congruence, rather than a relabeling of a matrix, preserves every complex phase in its entries.

For an integer $r \geq 0$, expand the specified polynomial uniquely as

$$u^r (c - iu)^i (c + iu)^j = \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} b_n^{(k,\lambda)}(u). \quad (\text{GC.41})$$

The monic triangular basis makes these coefficients finite, unique, and computable by polynomial subtraction in the written order. Differentiation of (GC.38) and (GC.23) now give the exact formula

$$\partial_\theta^r (\mathbf{M}_{h,k,N})_{ij} \big|_{\theta=0} = c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.42})$$

The derivative interchange is justified by the exponential bounds after (GC.9), or directly by Fubini on the k -fold product. Since $d_{h,k,n}$ uses exactly the finite coefficient power (GC.22), the entire r th derivative matrix is determined by $c_{h,0}, \dots, c_{h,2N+r}$. At zero tilt $r = 0$, the list ends at $2N$. There is no truncation error in this finite statement.

Equation (GC.42) supplies each derivative at zero from its stated finite coefficient list. Evaluation at a nonzero tilt is related to those derivatives by the following exact analytic Taylor identity, valid for $|\theta| < \pi/2$:

$$(\mathbf{M}_{h,k,N}(\theta))_{ij} = \sum_{r \geq 0} \frac{\theta^r}{r!} c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.43})$$

Absolute convergence follows by bounding the absolute exponential series by $e^{|\theta||u|}$ times the original polynomial modulus. For larger imaginary components within the full holomorphic strip, analytic continuation of these entries is the defined original Laplace integral (GC.38). This specifies the relation between all finite coefficient jets and the full tilted metric.

37.7 The original cyclic quotient and the exact cochain correction

For a nonempty packet let $E_h = \mathbb{C}[s]/(h)$ and $\mathcal{E}_h = E_h^{\otimes k}$. Let a be the element $s_1 + \cdots + s_k$ of this finite algebra. The kernel of $\mathbb{C}[S] \rightarrow \mathcal{E}_h, P \mapsto P(a)$, is generated by its unique monic polynomial χ . It is nonzero since the powers of a lie in a finite-dimensional vector space. Put $q = \deg \chi \geq 1$, $C = \mathbb{C}[S]/(\chi)$, and $A = M_S : C \rightarrow C$. The induced map $C \rightarrow \mathcal{E}_h$ is injective by the definition of the kernel. For each selected zero, the complete Taylor value of g/h is nonzero at its centre; its residue $v_h = j_h(g/h) \in E_h$ is consequently a unit in every local factor and in E_h . Thus the original arithmetic injection

$$\eta_{h,k} : C \hookrightarrow \mathcal{E}_h, \quad [P] \mapsto v_h^{\otimes k} P(a) \quad (\text{GC.44})$$

retains the entire local unit and every selected multiplicity.

The original analytic polynomial map is

$$\mathcal{V}_{h,k} P = P(D_1 + \cdots + D_k) F_h^{\otimes k}. \quad (\text{GC.45})$$

Its tensor Mellin transform is $P(s_1 + \cdots + s_k) \prod_i v_h(s_i)$. Taking complete selected jets gives precisely (GC.44) after the quotient π_χ . No modulus operation has been applied to that arithmetic jet map.

Successive monic divisions, in the order $s_1, \dots, s_k$, provide specific polynomials $Q_i(s_1, \dots, s_k)$ with

$$\chi(s_1 + \cdots + s_k) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{GC.46})$$

To prove this, divide in s_1 , retain its remainder, divide that remainder in s_2 , and continue. The final remainder has degree less than d in every coordinate. Its residue in $\mathcal{E}_h$ is zero by the defining property of χ , and the tensor remainder monomials form a basis, so every coefficient is zero. This proves the displayed identity and fixes the division order.

The operators D_i commute on the tensor complex. For a polynomial P , define the actual cochain of degree $k - 1$ by

$$\begin{aligned} \mathcal{P}_{\chi,h,k}(P) = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \cdots + D_k) \\ \cdot (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \end{aligned} \quad (\text{GC.47})$$

Every summand belongs to the indicated tensor source, since D preserves V and $\mathcal{B}$. The first $i - 1$ factors have cochain degree one. The tensor differential on the i th factor contributes $(-1)^{i-1}$, cancelling the written sign. The other factors have zero differential. Equations (GC.8) and (GC.46) therefore give

$$d_{\text{tot}} \mathcal{P}_{\chi,h,k}(P) = \mathcal{V}_{h,k}(\chi P). \quad (\text{GC.48})$$

This identifies the cochain primitive, its sign, and its boundary in their different cochain degrees.

For the full quotient at finite source degree take $N \geq q - 1$ and $m = N - q + 1$. With $\mathcal{P}_{-1} = 0$, monic division gives the exact sequence in original coordinates

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad J_N P = [P]_\chi. \quad (\text{GC.49})$$

The kernel assertion is divisibility, and the first q monomial columns prove surjectivity. Let B_χ be the $(N+1) \times m$ matrix of M_χ in the original ordered bases. Its entries are $[S^i](\chi(S)S^j)$; it is independent of the metric and tilt. The relation Gram and every zero-tilt derivative are therefore

$$\mathbf{B}_{h,k,N}(\theta) = B_\chi^* \mathbf{M}_{h,k,N}(\theta) B_\chi, \quad \mathbf{B}_{h,k,N}^{(r)}(0) = B_\chi^* \mathbf{M}_{h,k,N}^{(r)}(0) B_\chi. \quad (\text{GC.50})$$

Equivalently its polynomial integrand is $u^r |\chi(c+iu)|^2 (c-iu)^i (c+iu)^j$, where $0 \leq i, j < m$. Its degree is at most $r + 2q + 2(m-1) = r + 2N$, so the relation derivative uses the same list through $2N + r$

as the source derivative. Both complex polynomial factors, their conjugates, and all χ coefficients remain in this formula.

For a fixed real admitted tilt write $M_h = \mathbf{M}_{h,k,N}(\theta)$ and $M_\Gamma = \mathbf{M}_{\lambda,k,N}^\Gamma(\theta)$. Their canonical quotient Grams and minimum lifts are

$$G_h = (J_N M_h^{-1} J_N^*)^{-1}, \quad R_h = M_h^{-1} J_N^* G_h, \quad G_\Gamma = (J_N M_\Gamma^{-1} J_N^*)^{-1}, \quad R_\Gamma = M_\Gamma^{-1} J_N^* G_\Gamma. \quad (\text{GC.51})$$

Surjectivity of J and positivity of the source Grams prove every inverse exists. These formulas give $JR_h = JR_\Gamma = I_C$ and make each lift orthogonal to $\ker J$ in its own metric, proving its minimum property by the orthogonal Pythagorean identity. For $m > 0$ put

$$K = (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_\Gamma : C \longrightarrow \mathbb{C}^m. \quad (\text{GC.52})$$

The relation Gram is positive definite because B_χ is injective. The section $R_\Gamma - B_\chi K$ still has quotient identity and is orthogonal to the relation space in M_h . The preceding uniqueness gives the exact section map

$$\boxed{R_h = R_\Gamma - B_\chi K.} \quad (\text{GC.53})$$

At $N = q - 1$, $m = 0$, take K to be the zero map to the zero vector space and omit the inverse in (GC.52); (GC.53) remains the identity of the unique degree-less-than- q representatives.

For $u_0 \in C$, let $\kappa_{u_0}(S)$ be the polynomial whose coefficient vector is Ku_0 . Combining (GC.48) and (GC.53) proves the exact cochain correction

$$\boxed{\mathcal{V}_{h,k}(R_h u_0 - R_\Gamma u_0) = -\mathcal{V}_{h,k}(\chi \kappa_{u_0}) = d_{\text{tot}}(-\mathcal{P}_{\chi,h,k}(\kappa_{u_0})).} \quad (\text{GC.54})$$

Thus applying $\mathcal{V}$ gives the boundary in degree k ; the explicit negative cochain on the right is its primitive in degree $k - 1$. Passing to the original arithmetic quotient sends this boundary to zero. Explicitly, for the original set of labels $\mathcal{L}$ and vector spaces E_ℓ , write the supplied carrier as $\{\tau\} \sqcup \bigsqcup_{\ell \in \mathcal{L}} (\{\ell\} \times E_\ell)$. A family of linear maps $L_\ell : E_\ell \rightarrow F_\ell$ has the exact support lift $\tau \mapsto \tau$ and $(\ell, v) \mapsto (\ell, L_\ell v)$, with $e_\ell = (\ell, 0)$ in each target fibre. Substitution proves that these lifts preserve identities and composition. Applied to the original quotient, the relation boundary in (GC.54) therefore has image e_ℓ , while the external τ has image τ . The source vector and its primitive remain available before that quotient.

37.8 Determinant indices and the original Toda substitution

For real admitted θ , write $\mathfrak{D}_n$ for the original degree-less-than- n source Gram determinant and $\mathfrak{B}_m$ for the relation Gram determinant on m multiplication columns. Empty determinants are one. The analogous reference determinants carry the superscript Γ . For $N \geq q - 1$, set

$$V_N = \det G_h, \quad V_N^\Gamma = \det G_\Gamma, \quad X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_m = \mathfrak{B}_m / \mathfrak{B}_m^\Gamma.$$

The quotient-first coefficient matrix with columns $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$ has determinant one: its columns have successively distinct degrees and all their leading coefficients are one. Replacing its first q columns by the corresponding minimum sections only subtracts relation columns, so it still has determinant one. Orthogonality then gives the exact block determinant formula

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}} = V_N^\Gamma T_N, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}, \quad N \geq q - 1. \quad (\text{GC.55})$$

In particular the first full quotient has denominator $\mathfrak{B}_0 = 1$ and requires no relation inverse.

For completeness the reference source determinant is

$$\mathfrak{D}_n^\Gamma(\theta) = c_\chi^{kn} \prod_{j=0}^{n-1} j!(k\alpha)_j (\cos \theta)^{-n(k\alpha+n-1)}, \quad a_n^\Gamma(\theta) = n(n + k\alpha - 1) \sec^2 \theta. \quad (\text{GC.56})$$

Here a_n is the ratio of consecutive monic squared norms in the real- u coordinate; the norm-preserving phase of (GC.39) gives the same ratios in the original S coordinate. A direct proof of the Hankel identity used to obtain (GC.56) is as follows. Let p_j be the monic orthogonal polynomials for an $e^{\theta u}$ -tilted positive measure, let h_j be their squared norms, and write $up_j = p_{j+1} + b_j p_j + a_j p_{j-1}$, with $p_{-1} = 0$ and $a_0 = 0$. For $j \geq 1$, pairing with p_{j-1} gives $a_j = h_j/h_{j-1}$, and pairing with p_j gives $h'_j = b_j h_j$ for every $j \geq 0$. Differentiating orthogonality for $j \geq 1$ shows $\partial_\theta p_j = -a_j p_{j-1}$: its pairings with degrees at most $j-2$ vanish, and the remaining pairing is $-h_j$. For $j \geq 1$, write $p_j = u^j + c_j u^{j-1} + \dots$, and set $c_0 = 0$ for $p_0 = 1$. The derivative statement for p_0 is $\partial_\theta p_0 = 0$. Thus $c'_j = -a_j$ and $b_j = c_j - c_{j+1}$ hold for every $j \geq 0$, and consequently $b'_j = a_{j+1} - a_j$. Since $\mathfrak{D}_n = \prod_{j < n} h_j$,

$$(\log \mathfrak{D}_n)'' = \sum_{j < n} b'_j = a_n = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1} / \mathfrak{D}_n^2, \quad n \geq 1.$$

For the reference, the determinant values at $n=0$ and $n=1$ are 1 and (GC.13). The right side of the proposed formula (GC.56) has second logarithmic derivative $n(k\alpha + n - 1) \sec^2 \theta$. Its consecutive ratio has exactly the same value, with constants $n!(k\alpha)_n / ((n-1)!(k\alpha)_{n-1})$ and cosine exponent -2 . Positive induction in the displayed determinant identity proves (GC.56) for every n .

The reference relation determinant is the determinant of the moment matrix with entries

$$\partial_\theta^{i+j} \{ \bar{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \}, \quad 0 \leq i, j < m. \quad (\text{GC.57})$$

Differentiation under the integral gives the unchanged weight $|\chi(c + iu)|^2 r_{\lambda,k}(u)$ in this u -coordinate matrix. The relation version of (GC.39)–(GC.40) has determinant of modulus one, proving that (GC.57) is the original S -coordinate relation determinant as stated. It contains the actual χ with all coefficients.

At $\theta = 0$ put

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

The consecutive full-quotient formulas have domain $N \geq q$ and give, by literal substitution in (GC.55)–(GC.56),

$$a_{N+1} = (N+1)(N+k\alpha)Q_N, \quad \frac{V_{N-1}}{V_{N+1}} = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}. \quad (\text{GC.58})$$

The domain $N \geq q$ is necessary because V_{N-1} occurs. The separate original endpoint at $N = q - 1$ is not replaced by a q -dimensional metric at degree $q - 2$. At every fixed admissible real tilt the phase derivative remains the exact equality

$$\partial_\theta \log V_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N, \quad N \geq q - 1. \quad (\text{GC.59})$$

All these derivatives are of positive real-tilt determinants. Formula (GC.42) gives their underlying source and relation entry derivatives with the precise finite coefficient indices; their quotient and determinant operations use those original matrices.

For $h = 1$ the finite arithmetic quotient is zero and $\chi = 1$. Its relation and source spaces coincide at every degree N , so $\mathfrak{B}_{N+1} = \mathfrak{D}_{N+1}$, $V_N = V_N^\Gamma = T_N = 1$. The quotient lift has zero-dimensional domain and the cochain correction (GC.54) is the zero map on that domain. The nonzero analytic density, its mass μ_1^k , its coefficient series and (GC.24)–(GC.43) remain as constructed. The support lift is $G(0) = \{\tau, e\}$; it does not produce a nonzero finite spectral quotient from the analytic seed.

38 Certified theta moments and the original gamma arithmetic coefficients

This continuation proves and evaluates the fourth and sixth moments of the original theta seed, then transports the enclosures through the exact coefficient map for the reference parameter $\lambda = 13/4$

in *An exact gamma pushforward for the arithmetic Toda control*. The seed retains $g = 2\xi$, the density $|g(1/2 + it)|^2/(2\pi)$, the Fourier coordinate t , and its actual total mass. The reference retains the argument $\lambda + it/2$ and its different actual mass. We give the full analytic formulas, an explicit infinite-square tail for every finite derivative order, and the exact rational enclosures used by the executable certificate.

38.1 The theta input and its transform, with the original factors

On $\mathcal{S} = \{z \neq 0 : |\arg z| < \pi/4\}$ use the principal logarithm, and put

$$D = -z\partial_z, \quad L = D - \frac{1}{2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{GS.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{GS.2})$$

The series and each differentiated series converge locally uniformly: on a compact subset of $\mathcal{S}$, every summand is bounded by a polynomial in n times e^{-an^2} for some $a > 0$.

Lemma 38.1 (Inversion and both endpoints). *On the specified sector,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{GS.3})$$

Every D -derivative of f_0 decays faster than every power both at zero and at infinity on a closed proper subsector.

Proof. For $x > 0$, Fourier transformation with kernel $e^{-2\pi i \eta u}$ maps $e^{-\pi x^2 u^2}$ to $x^{-1}e^{-\pi \eta^2/x^2}$. The value at $\eta = 0$ follows from the Gaussian integral, whose square in polar coordinates is one. Differentiation in η and integration by parts give $I'(\eta) = -2\pi \eta I(\eta)/x^2$, which proves the stated formula with that initial value. Periodize this Gaussian over integer translates. Its Fourier coefficients are the displayed rapidly decreasing transform values; its absolutely convergent Fourier series evaluated at zero proves $\vartheta(x) = x^{-1}\vartheta(1/x)$. Holomorphic uniqueness extends the identity to $\mathcal{S}$.

For $(RF)(z) = z^{-1}F(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this to $R\vartheta = \vartheta$ proves the second equality in (GS.3). The real series coefficients prove the third. On $|\arg z| \leq \beta < \pi/4$ one has $\Re z^2 \geq |z|^2 \cos(2\beta) > 0$; the series and each derivative therefore have Gaussian decay at infinity with a fixed polynomial factor. Inversion and $DR = R(1 - D)$ transport all these bounds to zero, with every derivative retained. $\square$

Define, on these rapid sector functions,

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (UF)(y) = e^{y/2}F(e^y), \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{GS.4})$$

Here $U : L^2(\mathbb{R}_{>0}, dx) \rightarrow L^2(\mathbb{R}, dy)$ is a surjective linear isometry: $x = e^y$ proves the norm identity, and its inverse is $u \mapsto [x \mapsto x^{-1/2}u(\log x)]$. The endpoint bounds prove that the Mellin integral is entire. For $\Re s > 1$, absolute integration of (GS.2) is justified by $\sum_{n \geq 1} n^{-\Re s} < \infty$. Substitution $v = \pi n^2 x^2$ then gives

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2}\zeta(s)\{4\Gamma(s/2+2) - 6\Gamma(s/2+1)\} \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GS.5})$$

The factor 2 in (GS.2) and the substitution factor $1/2$ both occur before the first equality. Integration by parts proves the gamma recurrence used in the second equality. The entire Mellin integral continues the right side. Substitution $x \mapsto 1/x$ and real conjugation in (GS.4) prove

$$g(1-s) = g(s), \quad g(\bar{s}) = \overline{g(s)}. \quad (\text{GS.6})$$

In particular the following original density is even:

$$w(t) = \frac{|g(1/2 + it)|^2}{2\pi}, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt. \quad (\text{GS.7})$$

All moments exist. To see the exponential bound directly, rotate the Mellin contour to the ray $e^{i\beta}\mathbb{R}_{>0}$ with $|\beta| < \pi/4$. The endpoint bounds make the closing arcs vanish, and the rotated formula has the factor $e^{i\beta s}$. On the line $\Re s = 1/2$, choosing the sign of β to match $\Im s$ yields $|g(1/2 + it)| \leq C_\beta e^{-|\beta||t|}$. Thus $\int e^{a|t|} w(t) dt < \infty$ for every $0 < a < \pi/2$. Evenness gives $\mu_{2r+1} = 0$.

38.2 Every derivative polynomial and the exact higher-moment map

Let $H_r = L^r f_0$ for integers $r \geq 0$. Direct differentiation of (GS.4) gives $ULF = -(UF)'$ on the rapid sector functions. Integration by parts in $\mathcal{F}_+$, with the endpoint bounds already proved, therefore gives the commuting transform identity

$$\mathcal{F}_+(UH_r)(t) = (it)^r g(1/2 + it). \quad (\text{GS.8})$$

Every sign and power i^r belongs to this map.

Theorem 38.2 (Exact higher-moment formula). *Define polynomials $P_r \in \mathbb{Q}[y]$ by*

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y), \quad (\text{GS.9})$$

$$P_r(y) = \sum_{a=1}^{r+2} p_{r,a} y^a, \quad p_{r+1,a} = 2p_{r,a-1} - (2a + \tfrac{1}{2})p_{r,a}, \quad (\text{GS.10})$$

where coefficients outside their indicated range are zero. Then

$$H_r(x) = \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \mu_{2r} = \int_0^\infty |H_r(x)|^2 dx = 2 \int_1^\infty H_r(x)^2 dx. \quad (\text{GS.11})$$

The first four complete polynomials are

$$\begin{aligned} P_0 &= 8y^2 - 12y, \\ P_1 &= 16y^3 - 60y^2 + 30y, \\ P_2 &= 32y^4 - 224y^3 + 330y^2 - 75y, \\ P_3 &= 64y^5 - 720y^4 + 2116y^3 - 1635y^2 + \tfrac{375}{2}y. \end{aligned} \quad (\text{GS.12})$$

Proof. With $y = \pi n^2 x^2$, the operator D is $-2y\partial_y$. Consequently $L(P(y)e^{-y}) = \{-2yP'(y) + (2y - 1/2)P(y)\}e^{-y}$. This proves (GS.9), the series identity in (GS.11), and (GS.10) by collecting each specified coefficient. It also proves inductively that the degree is $r+2$, the leading coefficient is 2^{r+3} , and the constant coefficient is zero. Applying the recurrence three times gives the complete rational lists (GS.12).

Fourier Plancherel in the convention (GS.4) has factor $1/(2\pi)$: $\int |u|^2 dy = (2\pi)^{-1} \int |\mathcal{F}_+ u|^2 dt$. For the Schwartz functions here this follows by inserting $e^{-\varepsilon t^2}$ into Fourier inversion, applying Fubini, and evaluating the Gaussian kernel of integral one; letting $\varepsilon \downarrow 0$ gives inversion and the inner-product identity. Applying it to (GS.8) and the unitary U proves the first integral equality in (GS.11).

The relation $DR = R(1 - D)$ gives $RL = -LR$, while $Rf_0 = f_0$. Thus $RH_r = (-1)^r H_r$, or $H_r(1/x) = (-1)^r x H_r(x)$ for $x > 0$. In the integral over $(0, 1)$ substitute $x = 1/v$. The factor v^2 from the squared identity cancels the Jacobian v^{-2} and yields exactly the integral over $(1, \infty)$. The sign $(-1)^r$ is retained before squaring. All values $H_r(x)$ are real because their series has real coefficients, which finishes (GS.11). $\square$

The numerical driver in the delivered Gamma note writes $q_0 = 4y^2 - 6y$ and uses the same linear recurrence for q_r . Its exact coefficient map to this presentation is $q_r \mapsto P_r = 2q_r$, with inverse $P_r \mapsto P_r/2$. Thus $H_r = 2 \sum_n q_r(\pi n^2 x^2) e^{-\pi n^2 x^2}$. At the integral level (GS.11) consequently gives a factor 8 in front of the sum using $q_r q_r I_{2a+2b}$. At the gamma-function level below it gives a factor 4 in front of $q_r q_r \Gamma/C^\alpha$. The coefficient map accounts for both presentations without changing the theta input or its mass.

38.3 The incomplete-gamma square and a proved omitted tail

For real $C > 0$ and integers $p \geq 0$ put

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}, \quad \Gamma(a, C) = \int_C^\infty v^{a-1} e^{-v} dv. \quad (\text{GS.13})$$

Substitution $v = Cx^2$ proves the equality. Only positive real arguments and their positive real powers occur here. The companion recurrence, including its endpoint term, is

$$I_{p+2}(C) = \frac{p+1}{2C} I_p(C) + \frac{e^{-C}}{2C}. \quad (\text{GS.14})$$

It follows by integration by parts in (GS.13), whose upper endpoint vanishes exponentially.

Let $H_{r,J}$ restrict the original series (GS.11) to $1 \leq n \leq J$, where $J \geq 0$ is an integer. Put $T_{r,J} = 2 \int_1^\infty H_{r,J}^2 dx$. The entire finite expression is

$$T_{r,J} = \sum_{m,n=1}^J \sum_{a,b=1}^{r+2} p_{r,a} p_{r,b} (\pi m^2)^a (\pi n^2)^b \frac{\Gamma(a+b+1/2, \pi(m^2+n^2))}{[\pi(m^2+n^2)]^{a+b+1/2}}. \quad (\text{GS.15})$$

Here the endpoint factor 2 in (GS.11) cancels the factor 1/2 in (GS.13); the coefficients $p_{r,a}$ already contain the complete original theta factor. Empty sums at $J = 0$ have value zero.

To specify the tail by finite positive expressions, define Stirling numbers of the second kind by $\left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} = 1$, zero when one index is zero and the other positive, zero outside $0 \leq b \leq j$, and

$$\left\{ \begin{smallmatrix} j+1 \\ b \end{smallmatrix} \right\} = b \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} j \\ b-1 \end{smallmatrix} \right\}.$$

For $a \geq 1, d > 0$, integers $p \geq 0$, set

$$\rho_a = e^{-d(2a+1)}, \quad H_p(a, d) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{GS.16})$$

$$A_r(a) = \sum_{b=1}^{r+2} |p_{r,b}| \pi^b H_{2b}(a, \pi). \quad (\text{GS.17})$$

Theorem 38.3 (Every finite derivative order has an explicit square tail). *For every integer $r \geq 0$ and $J \geq 0$, with $K = J + 1$,*

$$|\mu_{2r} - T_{r,J}| \leq E_{r,J} := 4A_r(1)A_r(K)I_{4r+8}(\pi(1+K^2)). \quad (\text{GS.18})$$

The right side is a finite positive expression and tends to zero as $J \rightarrow \infty$. In particular the infinite version of (GS.15) is absolutely convergent and equals μ_{2r} .

Proof. For $x \geq 1$, integers $a \geq 1$ and $p \geq 0$, write $n = a + j$. The inequality $(a + j)^2 - a^2 = 2aj + j^2 \geq (2a + 1)j$ proves

$$\sum_{n \geq a} n^p e^{-\pi n^2 x^2} \leq e^{-\pi a^2 x^2} \sum_{j \geq 0} (a + j)^p \rho_a^j = e^{-\pi a^2 x^2} H_p(a, \pi). \quad (\text{GS.19})$$

For the equality, expand $(a + j)^p$ and then use $j^v = \sum_b \left\{ \begin{smallmatrix} v \\ b \end{smallmatrix} \right\} (j)_b$, where $(j)_b = j(j-1) \cdots (j-b+1)$. This identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed Stirling recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$, which proves the last equality in (GS.19).

Since $x^{2b} \leq x^{2r+4}$ for every original exponent $b \leq r+2$, (GS.19) gives the three explicit inequalities

$$\begin{aligned} |H_r(x)|, |H_{r,J}(x)| &\leq A_r(1) x^{2r+4} e^{-\pi x^2}, \\ |H_r(x) - H_{r,J}(x)| &\leq A_r(K) x^{2r+4} e^{-\pi K^2 x^2}. \end{aligned}$$

The same estimate for the sum of absolute values of all summands proves absolute integrability of the expanded double square. In particular termwise integration in the infinite formula is legitimate. Now use $|H_r^2 - H_{r,J}^2| \leq |H_r - H_{r,J}|(|H_r| + |H_{r,J}|)$ and the endpoint factor 2 in (GS.11). The resulting constant is exactly 4, the power is $4r + 8$, and integration gives (GS.18).

For fixed r , the denominators in (GS.16) are bounded away from zero uniformly for integers $a \geq 1$; thus $H_p(a, \pi)$ is at most a constant times $(1 + a)^p$, and $A_r(a)$ has at most polynomial growth. For $C \geq \pi$,

$$I_p(C) \leq e^{-C} \int_1^\infty x^p e^{-\pi(x^2-1)} dx.$$

The last integral is finite. Hence the factor with $C = \pi(1 + K^2)$ dominates that polynomial growth and forces $E_{r,J} \rightarrow 0$, proving the remaining assertions. $\square$

In the two new cases the coefficients in (GS.17) are completely specified by

$$\begin{aligned} A_2(a) &= 75\pi H_2 + 330\pi^2 H_4 + 224\pi^3 H_6 + 32\pi^4 H_8, \\ A_3(a) &= \frac{375}{2}\pi H_2 + 1635\pi^2 H_4 + 2116\pi^3 H_6 + 720\pi^4 H_8 + 64\pi^5 H_{10}, \end{aligned} \quad (\text{GS.20})$$

where each H_p on the right means $H_p(a, \pi)$. These absolute coefficients enter the error estimate only. Every original negative coefficient stays in the exact square (GS.15).

38.4 The coefficient morphism at the actual gamma reference

For $\lambda > 0$ put

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad \alpha = 2\lambda. \quad (\text{GS.21})$$

We prove the constants needed for the coefficient map. The beta integral follows from the product of two gamma integrals by the substitution $v = x_1 + x_2$, $u = x_1/(x_1 + x_2)$ with Jacobian v . After $u/(1-u) = e^x$ it reads

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{|\Gamma(\lambda + iv)|^2}{\Gamma(2\lambda)}. \quad (\text{GS.22})$$

The integrand before transformation and its first two derivatives decay exponentially at both ends. Two integrations by parts therefore give an integrable Fourier transform, so Fourier inversion applies with the same Gaussian justification as above. Using its evenness and substituting $t = 2v$ gives

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GS.23})$$

The factor 2 from $t = 2v$ is present in c_λ . The function $(2 \cosh(x/2))^{-2\lambda}$ is holomorphic on $|\Im x| < \pi$ with the branch positive on the real line and decays exponentially on each closed narrower strip. Shifting the contour in (GS.22) up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_c e^{-c|v|}$ for each $0 < c < \pi$. Thus the density in (GS.21) has exponential moments for every exponent of absolute value less than $\pi/2$. Analytic continuation of (GS.23) yields

$$\int e^{\theta t} r_\lambda(t) dt = c_\lambda (\cos \theta)^{-\alpha} \quad (|\theta| < \pi/2). \quad (\text{GS.24})$$

Define the monic polynomials by the local generating function

$$F_\lambda(z, t) = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n, \quad (\text{GS.25})$$

$$b_0 = 1, \quad b_1 = t, \quad b_{n+1} = tb_n - n(n + \alpha - 1)b_{n-1}. \quad (\text{GS.26})$$

The logarithms in (GS.25) vanish at zero. For real z close to zero, $F_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}$. Equation (GS.24) and $\cos(\arctan z + \arctan v) = (1 - zv)/\sqrt{(1 + z^2)(1 + v^2)}$ give

$$\int F_\lambda(z, t) F_\lambda(v, t) r_\lambda(t) dt = c_\lambda (1 - zv)^{-\alpha}. \quad (\text{GS.27})$$

Exponential integrability permits every derivative at $(0, 0)$ under the integral. Coefficient comparison proves

$$\int b_n^{(\lambda)}(t) b_j^{(\lambda)}(t) r_\lambda(t) dt = c_\lambda n! (\alpha)_n \delta_{nj}. \quad (\text{GS.28})$$

The highest coefficient of (GS.25) is $t^n/n!$, so these polynomials are monic. Finally differentiating (GS.25) gives $(1 + z^2) \partial_z F_\lambda = (t - \alpha z) F_\lambda$; coefficient comparison proves (GS.26), including all its constants.

The actual amplitude linking the two measures is

$$A_{1,\lambda}(t) = \frac{g(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{1,\lambda}(t) = |A_{1,\lambda}(t)|^2, \quad w(t) = B_{1,\lambda}(t) r_\lambda(t). \quad (\text{GS.29})$$

Gamma has no zeros and no poles at the displayed arguments. Here is a direct proof on their half-plane. For $\Re z > 0$, repeated integration by parts gives $B(z, n + 1) = n!/[z(z + 1) \cdots (z + n)]$. Substitution $u = v/n$ in the beta integral and $(1 - v/n)^n \leq e^{-v}$ prove $n^z B(z, n + 1) \rightarrow \Gamma(z)$ by dominated convergence. The reciprocal finite products converge to

$$ze^{\gamma z} \prod_{j \geq 1} (1 + z/j) e^{-z/j}, \quad \gamma = \lim_{n \rightarrow \infty} \left(\sum_{j=1}^n j^{-1} - \log n \right).$$

Integral comparison proves existence of the real limit. The logarithm of the tail product converges absolutely because $\log(1 + z/j) - z/j = O(j^{-2})$, and every factor is nonzero on $\Re z > 0$. The product is therefore finite and nonzero; by the previous limit it is $1/\Gamma(z)$. The original gamma integral is holomorphic on this half-plane, which also excludes poles there. The phase-preserving source map is multiplication by $A_{1,\lambda}$ from $L^2(w dt)$ to $L^2(r_\lambda dt)$; the exact norm identity follows by (GS.29). It is onto: the zero set of g on the line is discrete because g is entire and $g(2) > 0$ by (GS.5). Division by the amplitude off that null set gives the inverse and the same norm equality.

For completeness the full source phase at the chosen parameter is

$$A_{1,13/4}(t) = \frac{-\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it)}{(1/4 + it/2)(5/4 + it/2)(9/4 + it/2)}. \quad (\text{GS.30})$$

It follows from (GS.5), $s(s-1) = -(t^2 + 1/4)$, and three gamma recurrences. The elementary identity $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ first follows by summation by parts for $\Re s > 1$ and extends to $\Re s > 0$, $s \neq 1$, because its integral converges absolutely and locally uniformly there. On the line in (GS.30) it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 4\sqrt{t^2 + 1/4}$. The squared modulus of the denominator in (GS.30) is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, so

$$0 \leq B_{1,13/4}(t) \leq \frac{1024}{\sqrt{\pi}} \frac{(t^2 + 1/4)^2}{(t^2 + 25/4)(t^2 + 81/4)} \leq \frac{1024}{\sqrt{\pi}}. \quad (\text{GS.31})$$

The last denominator exceeds the numerator by $26t^2 + 253/2 > 0$ after expansion. Thus $B_{1,13/4}$ belongs to $L^2(r_{13/4} dt)$, and its gamma coefficients have their literal Hilbert-space type.

Write $c = c_{13/4}$ and $c_j = c_{1,j}$. Equations (GS.28)–(GS.29) give the linear coefficient map on the moment functional:

$$c_j = \frac{\int b_j^{(13/4)}(t) w(t) dt}{c j! (13/2)_j}, \quad c = 2^{-11/2} \Gamma(13/2) = \frac{10395\sqrt{\pi}}{2048\sqrt{2}}. \quad (\text{GS.32})$$

The last identity follows by six gamma recurrences from $\Gamma(1/2) = \sqrt{\pi}$. At this exact parameter the complete polynomials needed are

$$\begin{aligned} b_0 &= 1, & b_1 &= t, & b_2 &= t^2 - \frac{13}{2}, \\ b_3 &= t^3 - \frac{43}{2}t, & b_4 &= t^4 - 47t^2 + \frac{663}{4}, \\ b_5 &= t^5 - 85t^3 + \frac{3931}{4}t, \\ b_6 &= t^6 - \frac{275}{2}t^4 + \frac{13801}{4}t^2 - \frac{69615}{8}. \end{aligned} \quad (\text{GS.33})$$

Substituting successively into (GS.26) proves every line. Evenness gives $c_1 = c_3 = c_5 = 0$, and the exact even coefficient map becomes

$$\begin{aligned} c_0 &= \mu_0/c, \\ c_2 &= (2\mu_2 - 13\mu_0)/(195c), \\ c_4 &= (4\mu_4 - 188\mu_2 + 663\mu_0)/(377910c), \\ c_6 &= (8\mu_6 - 1100\mu_4 + 27602\mu_2 - 69615\mu_0)/(2737957950c). \end{aligned} \quad (\text{GS.34})$$

Here $2!(13/2)_2 = 195/2$, $4!(13/2)_4 = 188955/2$, and $6!(13/2)_6 = 1368978975/4$. These factors show directly how every denominator in (GS.34) arises. The invertible triangular moment map has the complete inverse

$$\begin{aligned} \mu_0 &= cc_0, \\ \mu_2 &= (195cc_2 + 13\mu_0)/2, \\ \mu_4 &= (377910cc_4 + 188\mu_2 - 663\mu_0)/4, \\ \mu_6 &= (2737957950cc_6 + 1100\mu_4 - 27602\mu_2 + 69615\mu_0)/8. \end{aligned} \quad (\text{GS.35})$$

Substitution in (GS.34) proves that the displayed maps are mutual inverses. In particular the mass in (GS.7) remains μ_0 , while the reference mass in (GS.21) remains c .

38.5 Certified infinite quantities and exact rational error transport

The companion checker evaluates (GS.15) for $r = 0, 1, 2, 3$ and $J = 8$ in Arb real balls at 256-bit precision, with one arithmetic thread and python-flint 0.9.0. All polynomial input is exact rational arithmetic; π , exponentials, powers and gamma functions are enclosed by Arb. The upper-incomplete-gamma call is `C.gamma_upper(alpha)`, with argument C and order α , precisely as in (GS.13). Each finite

ball is enlarged by a symmetric ball with radius the upward-rounded upper endpoint of (GS.18). It therefore encloses the infinite quantity. The computed tails satisfy the rational inequalities

$$E_{0,8} < 7 \cdot 10^{-107}, \quad E_{1,8} < 4 \cdot 10^{-103}, \quad E_{2,8} < 2 \cdot 10^{-99}, \quad E_{3,8} < 14 \cdot 10^{-96}. \quad (\text{GS.36})$$

Each inequality is tested as a strict rigorous ball comparison.

The resulting exact rational decimal intervals are:

$$\begin{aligned} \mu_0 &\in [1.2790072478464851404795335922671932744916, \\ &\quad 1.2790072478464851404795335922671932744919], \\ \mu_2 &\in [13.0555493025705584353926846581231936824343, \\ &\quad 13.0555493025705584353926846581231936824346], \\ \mu_4 &\in [383.2743417173642425544072374791308324300288, \\ &\quad 383.2743417173642425544072374791308324300291], \\ \mu_6 &\in [17946.5273016052076235667646919834204772848931, \\ &\quad 17946.5273016052076235667646919834204772848934], \\ c &\in [6.3614260045872192926298535487331092297380, \\ &\quad 6.3614260045872192926298535487331092297383], \\ c_0 &\in [0.2010566886927858658099161632202187798614, \\ &\quad 0.2010566886927858658099161632202187798617], \\ c_2 &\in [0.0076454429994864011881338514807939967436, \\ &\quad 0.0076454429994864011881338514807939967439], \\ c_4 &\in [-0.0000305169208037620437467248395269085390, \\ &\quad -0.0000305169208037620437467248395269085387], \\ c_6 &\in [-0.0000003851554876709816618220743079695913, \\ &\quad -0.0000003851554876709816618220743079695910]. \end{aligned}$$

Each interval has width $3 \cdot 10^{-40}$. The coefficient balls are computed directly from the infinite-moment balls in (GS.34), before their conversion to the displayed decimal intervals. Interval dependency can enlarge enclosures; it cannot invalidate inclusion. The signs $c_4 < 0$ and $c_6 < 0$ follow from their negative upper endpoints and are preserved.

Here is the exact outward conversion used by the checker. If the resulting Arb ball is Q , let ℓ and u be its outward dyadic endpoints and set $d = 40$,

$$L_Q = 10^{-d}(\lfloor 10^d \ell \rfloor - 1), \quad U_Q = 10^{-d}(\lceil 10^d u \rceil + 1). \quad (\text{GS.37})$$

The checker then verifies $Q > L_Q$ and $Q < U_Q$ by rigorous ball comparison. The receipt stores L_Q, U_Q both as decimal strings and exact rational strings, and stores the dyadic endpoints as exact rationals. Thus every display can be checked with integer arithmetic after ball generation. The same receipt records the tail's exact upward dyadic rational.

The executable finite checks additionally verify all exact rational polynomial lists in (GS.12) and (GS.33) and the four exact rational denominator factors following (GS.34). They compare the ball evaluations of the polynomial and integer numerator forms in (GS.34) for overlap, and similarly compare the endpoint-containing recurrence (GS.14) at each argument pair used. These checks detect convention mistakes; the analytic certificate is the ball enclosure enlarged by the proved infinite tail, independently of any truncation agreement.

Four fault modes inject specific false mathematical assertions: discarding the inversion factor two preserves μ_4 ; assigning reference mass one preserves c_0 ; c_4 is nonnegative; and the cutoff-zero sum with zero error encloses the full positive seed mass. The already certified disjoint intervals reject each assertion. These checks use explicit conditionals and a nonzero exit status; Python optimization

cannot remove them. Normal and optimized runs produce the same mathematical results and each fault produces one recorded rejection.

The source note's two floating truncations remain its original numerical evidence. The intervals above independently certify their displayed values through the original theta series and (GS.18). The correctness contract for the elementary ball operations and outward endpoints is documented in the [python-flint Arb interface](#). No proof of that arithmetic library's implementation is asserted.

38.6 Exact tensor coefficient interface through degree six

The map from the certified one-factor data to every finite tensor degree can also be stated without changing the mass or sum variable. Let $k \geq 1$, $u = t_1 + \cdots + t_k$, and $m_k = w^{*k}$. Fubini gives $\int m_k = \mu_0^k$, and exponential integrability proves all the moments below are finite. Multiplying the generating functions (GS.25) gives the finite polynomial identity

$$b_n^{(k\lambda)}(u) = \sum_{n_1 + \cdots + n_k = n} \frac{n!}{n_1! \cdots n_k!} \prod_{j=1}^k b_{n_j}^{(\lambda)}(t_j). \quad (\text{GS.38})$$

Its proof is literal multiplication: the exponents of $1 \mp iz$ add to $-k\lambda \pm iu/2$, and the coefficient of degree n has only finitely many terms.

Use $\lambda = 13/4$, $\alpha = 13/2$, $a_j = (\alpha)_j c_j$, and the formal coefficient series $\mathcal{A}(z) = \sum_{j \geq 0} a_j z^j$. For any coefficient appearing below only finitely many c_j are needed. Integration of (GS.38), using (GS.32) separately in each factor, proves

$$\int b_n^{(k\lambda)}(u) m_k(u) du = c^k n! d_{k,n}, \quad d_{k,n} = [z^n] \mathcal{A}(z)^k. \quad (\text{GS.39})$$

No interchange of an infinite formal series with an integral is required. Evenness gives the complete first four nonzero expressions

$$\begin{aligned} d_{k,0} &= c_0^k, \\ d_{k,2} &= k c_0^{k-1} a_2, \\ d_{k,4} &= k c_0^{k-1} a_4 + \binom{k}{2} c_0^{k-2} a_2^2, \\ d_{k,6} &= k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3. \end{aligned} \quad (\text{GS.40})$$

For degree six the partitions into even positive degrees are 6, 4 + 2 and 2 + 2 + 2; their distinct-factor choices have respectively k , $k(k-1)$ and $\binom{k}{3}$ possibilities. This proves all three coefficients, including the mixed term and its sign. The analogous enumeration proves degree two and four. Since $c_0 > 0$, even the displayed negative powers at $k = 1, 2$ are defined; their accompanying zero combinatorial factors give the correct absent terms.

Equations (GS.34)–(GS.35), (GS.39)–(GS.40) are the exact map from the certified original moments to these all- k low-degree arithmetic gamma moments, retaining c^k and the coordinate u . For example their degree-zero identity is $c^k d_{k,0} = \mu_0^k$. They give interval evaluations at every specified finite integer k by rational and ball arithmetic with the already enclosed inputs. They assert no uniform estimate when polynomial degree grows. For this actual seed $h = 1$, the finite arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the quotient map sends every polynomial to zero because every polynomial lies in the ideal (1). The analytic source measure and all the coefficients above remain nonzero before that quotient. No nonzero packet quotient is inferred from these seed calculations.

39 The first three original source costs at every tensor degree

Retain the exact theta seed, its Fourier coordinate and its measure

$$w(t) dt = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt.$$

Equations (GS.1)–(GS.18) prove that w is even, has positive mass μ_0 , and has finite moments of every order. The certificate of (GS.36)–(GS.37) encloses the original $\mu_0, \mu_2, \mu_4, \mu_6$. This section transports those existing certified integrals by exact rational arithmetic. It performs no further quadrature.

For each integer $k \geq 1$ put $u = t_1 + \cdots + t_k$, $m_k = w^{*k}$ and $M_j(k) = \int u^j m_k(u) du$. Fubini and evenness give the full first four even sum moments

$$\begin{aligned} M_0(k) &= \mu_0^k, \\ M_2(k) &= k\mu_2\mu_0^{k-1}, \\ M_4(k) &= k\mu_4\mu_0^{k-1} + 3k(k-1)\mu_2^2\mu_0^{k-2}, \\ M_6(k) &= k\mu_6\mu_0^{k-1} + 15k(k-1)\mu_4\mu_2\mu_0^{k-2} + 15k(k-1)(k-2)\mu_2^3\mu_0^{k-3}. \end{aligned} \quad (\text{GS.41})$$

Here every missing term integrates to zero because at least one factor has an odd moment. For M_4 , the surviving exponent partitions are 4 and $2 + 2$. The second has $\binom{k}{2} 4!/(2!2!) = 3k(k-1)$ terms. For M_6 the partitions are 6, $4 + 2$ and $2 + 2 + 2$; their numbers of terms are respectively k , $k(k-1)6!/(4!2!) = 15k(k-1)$ and $\binom{k}{3} 6!/(2!2!2!) = 15k(k-1)(k-2)$. The second-moment expansion has precisely k surviving terms. These finite counts prove (GS.41) with its full mass factors. Because $\mu_0 > 0$, the displayed powers are defined even when a zero combinatorial coefficient accompanies a negative exponent.

All nonzero polynomial norms in $L^2(m_k du)$ are strictly positive. Indeed the nonnegative density m_k has positive mass μ_0^k , so its positivity set has positive Lebesgue measure. A nonzero polynomial has finitely many real roots, as repeated division by its distinct linear factors cannot exceed its degree. Its squared modulus times m_k is therefore positive on a set of positive measure, and its integral is positive. This proves every positivity and division below for the actual source.

Put $\nu = \mu_2/\mu_0$. With the norms taken in the unchanged measure $m_k du$, the first four monic orthogonal polynomials and all their squared norms are

$$\begin{aligned} p_0(u) &= 1, & h_0(k) &= M_0(k), \\ p_1(u) &= u, & h_1(k) &= M_2(k), \\ p_2(u) &= u^2 - k\nu, & h_2(k) &= M_4(k) - \frac{M_2(k)^2}{M_0(k)}, \\ p_3(u) &= u^3 - \frac{M_4(k)}{M_2(k)}u, & h_3(k) &= M_6(k) - \frac{M_4(k)^2}{M_2(k)}. \end{aligned} \quad (\text{GS.42})$$

To verify orthogonality of p_2 , its pairing with 1 is $M_2 - k\nu M_0 = 0$, and its pairing with u is zero by evenness. For p_3 , parity gives zero against 1 and p_2 ; its pairing with u is $M_4 - (M_4/M_2)M_2 = 0$. Expanding each squared polynomial gives the norms in (GS.42). Uniqueness follows because the difference of two monic orthogonal polynomials of the same degree lies in the lower-degree space, is orthogonal to itself, and hence is zero by the preceding positivity argument.

Define the positive original monic source costs $a_j^{(k)} = h_j(k)/h_{j-1}(k)$. These are exactly the lower coefficients in the real-coordinate three-term recurrence: in up_j , its pairing with p_{j-1} equals the pairing of p_j with up_{j-1} , which is h_j because $up_{j-1} - p_j$ has lower degree. Pairings with lower degrees vanish by the same argument. The degree- j coefficient vanishes by evenness. Therefore $up_j = p_{j+1} + a_j^{(k)} p_{j-1}$. For $j = 3$, the next monic polynomial p_4 exists by finite Gram orthogonalization, whose Gram is positive definite by the proved polynomial norm property; its full formula is not

needed to determine the already specified coefficient h_3/h_2 . Substituting (GS.41) into (GS.42) gives

$$\begin{aligned} a_1^{(k)} &= k\nu, \\ a_2^{(k)} &= \frac{\mu_4}{\mu_2} + (2k-3)\nu, \\ a_3^{(k)} &= \frac{M_6(k) - M_4(k)^2/M_2(k)}{M_4(k) - M_2(k)^2/M_0(k)}. \end{aligned} \quad (\text{GS.43})$$

For the second equality one obtains $M_4/M_2 = \mu_4/\mu_2 + 3(k-1)\nu$ and $M_2/M_0 = k\nu$, whose exact difference is the displayed cost. For an explicit expanded version of the third equality, write $r_4 = \mu_4/\mu_0$ and $r_6 = \mu_6/\mu_0$, solely as the indicated scalar quotients of the original moments. Then

$$a_3^{(k)} = \frac{r_6 - r_4^2/\nu + 9(k-1)r_4\nu + 3(k-1)(2k-7)\nu^3}{r_4 + (2k-3)\nu^2}. \quad (\text{GS.44})$$

To check every term, divide the numerator of (GS.43) by $k\mu_0^k$ using (GS.41). The square term contributes $r_4^2/\nu + 6(k-1)r_4\nu + 9(k-1)^2\nu^3$; subtracting it from $r_6 + 15(k-1)r_4\nu + 15(k-1)(k-2)\nu^3$ gives the numerator in (GS.44). The denominator of (GS.43), divided by the same $k\mu_0^k$, is $r_4 + (2k-3)\nu^2$. Thus both original norms and all mass factors are recovered from these scalar abbreviations.

The original spectral coordinate is $S = k/2 + iu$. Let $c_k = k/2$ and define the algebra isomorphism

$$\Psi_k : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad (\Psi_k P)(u) = P(c_k + iu), \quad \Psi_k^{-1}p(S) = p((S - c_k)/i). \quad (\text{GS.45})$$

Direct substitution proves the two inverse identities. Give $\mathbb{C}[S]$ the source norm $\int |P(c_k + iu)|^2 m_k(u) du$; then Ψ_k is a linear isometry onto the polynomial source in the real coordinate. On the degree- j monic affine space, its phase-preserving monic inverse is $p \mapsto P_j = i^j \Psi_k^{-1}p$. Since $|i^j| = 1$, this map preserves the original norm and has inverse $P \mapsto i^{-j} \Psi_k P$ on that affine space. In particular the complete first four spectral monic polynomials are

$$\begin{aligned} P_0 &= 1, \quad P_1 = S - c_k, \quad P_2 = (S - c_k)^2 + k\nu, \\ P_3 &= (S - c_k)^3 + \left\{ \frac{\mu_4}{\mu_2} + 3(k-1)\nu \right\} (S - c_k). \end{aligned} \quad (\text{GS.46})$$

Their squared norms are exactly h_0, h_1, h_2, h_3 from (GS.42). Multiplying the real recurrence by i^{j+1} proves the spectral recurrence with its actual lower sign,

$$SP_j = P_{j+1} + c_k P_j - a_j^{(k)} P_{j-1}. \quad (\text{GS.47})$$

Thus the positive norm ratio $a_j^{(k)}$ and the negative lower coefficient in the S presentation have their exact phase map.

To connect these costs to the existing Gamma determinant corrections, let D_n be the determinant of the source Gram on $1, u, \dots, u^{n-1}$, with $D_0 = 1$, and let D_n^Γ use the Gamma sum measure $r_{13/4}^{*k} du$. Its mass is $c_{13/4}^k$, and its monic norms are $c_{13/4}^k j!(13k/2)_j$, by (GS.23), Fourier uniqueness and (GS.28). Specifically Fourier transformation of the convolution gives $c_{13/4}^k \cosh^{-13k/2} y$; division by the mass constant in (GS.23) identifies its density as a positive scalar times $r_{13k/4}$, proving the asserted norms without any change to u . Monic orthogonalization is triangular with diagonal one, so $D_n = \prod_{j < n} h_j$ and the analogous Gamma product holds. Define $X_n = D_n/D_n^\Gamma$ and

$$Q_N = \frac{X_{N+2}X_N}{X_{N+1}^2}, \quad a_{N+1}^{(k)} = (N+1)(N+13k/2)Q_N. \quad (\text{GS.48})$$

Dividing the two consecutive determinant products proves the second identity. In particular, with $\eta = \mu_4/\mu_2$,

$$\begin{aligned} Q_0 &= \frac{2\nu}{13}, \\ Q_1(k) &= \frac{\eta + (2k-3)\nu}{13k+2} = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \eta - \frac{43}{13}\nu, \\ Q_2(k) &= \frac{2a_3^{(k)}}{39k+12}. \end{aligned} \tag{GS.49}$$

Every equality is obtained by substituting (GS.43) into (GS.48). No volume of a nonzero packet quotient enters these source costs.

The pure rational checker reads the exact dyadic moment endpoints from the pinned GS.36 certificate and propagates them through (GS.43)–(GS.49) using `fractions.Fraction` exclusively. For intervals $[a, b]$ and $[c, d]$, it uses endpoint addition, $[a - d, b - c]$ for subtraction, the minimum and maximum of the four endpoint products for multiplication, and multiplication by $[1/d, 1/c]$ for division after checking $c > 0$. These rules follow from linearity in each variable and the decreasing map $x \mapsto 1/x$ on the positive axis. Thus they preserve inclusion without any floating-point operation. Applying (GS.37) with exact rational floor and ceiling yields, among the saved intervals,

$$\begin{aligned} \nu &\in [10.2075647534857216888657612522375551749964, \\ &\quad 10.2075647534857216888657612522375551749967], \\ Q_0 &\in [1.5703945774593417982870401926519315653839, \\ &\quad 1.5703945774593417982870401926519315653842], \\ a_2^{(1)} &\in [19.1496329281085602472067784658376980738791, \\ &\quad 19.1496329281085602472067784658376980738794], \\ \rho &\in [-4.4062857337815667270988244239412754068822, \\ &\quad -4.4062857337815667270988244239412754068819]. \end{aligned}$$

All four widths are $3 \cdot 10^{-40}$. The receipt also encloses μ_4/μ_2 , $\mu_4\mu_0 - \mu_2^2$, $a_2^{(2)}$, $Q_1(1)$, $a_3^{(1)}$ and $Q_2(1)$. In particular $\rho < 0$ and (GS.49) give the exact fixed-degree all- k comparison

$$0 < Q_1(1) \leq Q_1(k) < Q_0, \quad Q_1(k+1) - Q_1(k) = \frac{-13\rho}{(13k+2)(13k+15)} > 0. \tag{GS.50}$$

The positive lower bound follows either from $h_2/h_1 > 0$ or its certified $k = 1$ enclosure; the difference follows by literal subtraction in (GS.49). The computation concerns original source polynomial degrees at most three, at every specified tensor degree. The analytic seed still has quotient $\mathbb{C}[S]/(1) = 0$, its quotient determinant is the empty determinant one, and every nonzero source norm above remains before that zero quotient.

40 Finite arithmetic Gram transfer with the original minimum quotient metric

This continuation proves a finite coefficient-to-metric map, its exact boundary-resolvent remainder, and a two-degree determinant formula for the consecutive volume ratio in the arithmetic Toda control. The original Gamma mass, tensor sum, arithmetic amplitude, quotient coordinates and relation polynomial remain in every formula. A bounded positive-weight calibration at the end proves that the relative Gamma correction can increase with degree even while the actual minimum volume decreases. That calibration is explicitly an auxiliary measure, with no assertion that its polynomial is an annihilator of actual zeta zeros.

The sources read for this calculation are the complete `RESEARCH_NOTE.md` and `NOTE.tex` in the delivered Tau Gamma Convolution Descent, and the original arithmetic definitions and proofs TVB.1–TVB.38 in `toda_cv_exact_bridge_20260912.tex`. The finite transfer below is at the original observation $\theta = 0$. Its proof does not replace a nonzero exponential tilt by a polynomial test. All interval claims here are finite rational inequalities; no uniform estimate as packet, tensor degree or polynomial degree grow is asserted.

40.1 The original source, relation and coordinate maps

Let $h(s)$ be a monic nonempty packet of actual zeros of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s),$$

including the complete order at each selected zero. Put $v_h = g/h$, $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$, and $\mu_h = \int_{\mathbb{R}} w_h(t) dt$. The quotient is entire: local Taylor division removes exactly the stated vanishing order. Its value at each selected zero is nonzero. The full Taylor remainder $[v_h]_h$ is therefore a unit of $E_h = \mathbb{C}[s]/(h)$. Indeed in $\mathbb{C}[t]/(t^m)$ the inverse of $\sum_{j < m} a_j t^j$, $a_0 \neq 0$, has coefficients $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$; the Chinese remainder map from coprime factors is the exact algebra isomorphism obtained by Bezout's identity. This retains every jet and multiplicity.

The original tensor coordinate is

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GMT.1})$$

The source observation is the conjugate-linear-first pairing

$$\langle P, Q \rangle_h = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k. \quad (\text{GMT.2})$$

The change $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Thus Fubini gives (GMT.2) from the literal tensor Mellin observation without a divisor or an average coordinate. The complex Mellin amplitude remains

$$\prod_{i=1}^k \frac{v_h(1/2 + it_i)}{\sqrt{2\pi}} P(c + i \sum_i t_i).$$

Let $I = (h(s_1), \dots, h(s_k))$ and let χ be the monic generator of $I \cap \mathbb{C}[S]$, of degree q . Existence follows from an annihilating polynomial of multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$. Division by its least-degree monic member proves that every member of the intersection is a multiple of χ . For $N \geq q-1$ define $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and the exact maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0, \quad M_\chi P = \chi P, \quad \mathcal{C} = \mathbb{C}[S]/(\chi), \quad J_N P = [P]_\chi. \quad (\text{GMT.3})$$

Here $\mathcal{P}_{-1} = 0$. All coordinate vectors of $\mathcal{C}$ are in the written basis $1, S, \dots, S^{q-1}$. Its map to the original tensor jet is

$$\eta[P] = [v_h]_h^{\otimes k} [P(S)]_I.$$

Its kernel is zero by the preceding intersection ideal and invertible unit. No Gamma coefficient calculation changes this map.

Write B_N for the coefficient matrix of the columns $\chi, \chi S, \dots, \chi S^{N-q}$ in $\mathcal{P}_N$. The matrix $[E_N, B_N]$, where E_N includes the first q monomials, is triangular with diagonal one. Thus its determinant is exactly one in this order; putting the relation block first contributes $(-1)^{q(N-q+1)}$.

The original theta map is $\Theta\phi(x) = \sum_{n \neq 0} \phi(nx)$, with $D = -x\partial_x$, and F_h satisfies $\mathcal{M}F_h = v_h$, $h(D)F_h = \Theta\phi_0$, where $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. The exact primitive attached to a column χP is

retained as follows. Successive monic division in $s_1, \dots, s_k$ gives $\chi(S) = \sum_i h(s_i) Q_i(\mathbf{s})$. Its tensor primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) P(\sum_j D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GMT.4})$$

The differential at the i th position has sign $(-1)^{i-1}$, so differentiation gives precisely the original source vector $\mathcal{V}_{h,k}(\chi P)$. This proves the relation-to-boundary map, including its sign. The construction of F_h uses the actual zero-division operator $S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ at a vanishing Mellin value. The two integrals give respectively every large- and small- x power bound. Differentiation gives $(D - \rho) S_\rho F = F$, hence all Euler derivative bounds and the Mellin quotient by $s - \rho$. Starting from $\Theta \phi_0$ and successively dividing the complete root list constructs the displayed F_h . The seed identity follows by substituting $y = \pi x^2$ in its Mellin integral, which gives $\mathcal{M}\phi_0(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2)$ for $\Re s > -2$, followed by the absolutely convergent theta sum for $\Re s > 1$ and analytic continuation.

On each original support label ℓ , a boundary still maps to $(\ell, 0) = e_\ell$; external absence τ still maps to τ . The linear polynomial maps (GMT.3)–(GMT.4) precede the quadratic map $R \mapsto R^* H R$ from source representatives to Hermitian forms. For the empty packet $h = 1$, the analytic source remains, but $\chi = 1, q = 0, \mathcal{C} = 0, G_N$ is the zero-dimensional form and its determinant is one. Every quotient formula below then has its empty-block interpretation.

40.2 Full Gamma mass and the finite coefficient evaluation

For $\lambda > 0$ set

$$\alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{GMT.5})$$

The beta integral with its coordinate $x = 2y$ gives

$$|\Gamma(\lambda + it/2)|^2 = 2\Gamma(2\lambda) \int_{\mathbb{R}} e^{ity} (2 \cosh y)^{-2\lambda} dy.$$

Fourier inversion therefore gives $\int e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}$. Both functions are integrable: the right integrand and all its derivatives decay exponentially, so integration by parts bounds its Fourier transform by every inverse power. Exponential integrability of r_λ on $|a| < \pi/2$ follows by rotating the gamma integral by any φ with $|a| < \varphi < \pi/2$; the squared modulus has a bound $\Gamma(\lambda)^2 (\cos \varphi)^{-2\lambda} e^{-\varphi|t|}$. Analytic continuation now proves

$$\int e^{au} r_{\lambda,k}(u) du = C_k (\cos a)^{-\beta}, \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{C_k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GMT.6})$$

The last identity is Fourier uniqueness with the exact product mass.

Let $b_j^{(\lambda)}$ denote the monic polynomials with

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GMT.7})$$

These agree with $j! P_j^{(\lambda)}(t/2; \pi/2)$ in the classical generating function at [DLMF 18.23.7](#). The next proof fixes their scale without relying on that identification. Differentiating (GMT.7) gives

$$b_{j+1}^{(\lambda)} = t b_j^{(\lambda)} - j(j + \alpha - 1) b_{j-1}^{(\lambda)}.$$

For small real z, w , integration of the product of generating functions by (GMT.6) gives $c_\lambda(1-zw)^{-\alpha}$, since $\cos(\arctan z + \arctan w) = (1-zw)/\sqrt{(1+z^2)(1+w^2)}$. Local exponential domination justifies every coefficient derivative. Thus

$$\int b_i^{(\lambda)} b_j^{(\lambda)} r_\lambda = \delta_{ij} c_\lambda j! (\alpha)_j, \quad \int b_i^{(k\lambda)} b_j^{(k\lambda)} r_{\lambda,k} = \delta_{ij} C_k j! (\beta)_j. \quad (\text{GMT.8})$$

Put $B(t) = |v_h(1/2 + it)/\Gamma(\lambda + it/2)|^2$, so that $w_h = Br_\lambda$ with its full complex amplitude still present in the Mellin source. One may use the bounded reference constructed in the Gamma descent: $\lambda = 1/4$ for degree at least three, or $\lambda = 1/4 + L$, $L \geq \max(0, 3 - \deg h)$ otherwise. The relevant boundedness follows directly from the compact entire quotient, $|h(1/2 + it)| \geq (|t|/2)^{\deg h}$ beyond twice the root radius, $|\zeta(1/2 + it)| \leq 2(1 + |t|)$, and the nonvanishing factors $\prod_{j < L} (j + 1/4 + it/2)$ in the gamma shift. In particular $B \in L^2(r_\lambda dt)$. All statements of this section also hold for any specified nonnegative nonzero $B \in L^2(r_\lambda dt)$. For that domain write $w_B = Br_\lambda$ and $m_{B,k} = w_B^{*k}$; at the arithmetic input these equal the original $w_h, m_{h,k}$. Cauchy-Schwarz with (GMT.6) gives all polynomial moments. The calibration below uses this latter domain before it is explicitly capped to a bounded function.

The sum adjoint is the actual conditional integral

$$C_\Sigma F(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GMT.9})$$

Its adjoint inclusion $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$ is isometric by Fubini; hence $C_\Sigma U_\Sigma = 1$ and $F \mapsto (C_\Sigma F, (1 - U_\Sigma C_\Sigma)F)$ has inverse $(f, n) \mapsto U_\Sigma f + n$ on $L^2(r_{\lambda,k}) \oplus \ker C_\Sigma$. At $k = 1$ both maps are the identity. The normal component and every support label remain in this paired map. The specified sum density is $m_{B,k} = r_{\lambda,k} B_k$, $B_k = C_\Sigma B^{\otimes k}$; in particular it is $m_{h,k}$ at the arithmetic input.

Define the real coefficients $c_{B,j}$ (the original $c_{h,j}$ for the arithmetic multiplier $B = B_{h,\lambda}$) and formal coefficient power

$$c_{B,j} = \frac{\int b_j^{(\lambda)} w_B}{c_\lambda j! (\alpha)_j}, \quad A(z) = \sum_{j \geq 0} (\alpha)_j c_{B,j} z^j, \quad d_n = [z^n] A(z)^k. \quad (\text{GMT.10})$$

Multiplying the finite generating coefficients in (GMT.7) gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing with $B^{\otimes k}$ gives the exact finite identity

$$\int b_n^{(k\lambda)}(u) m_{B,k}(u) du = C_k n! d_n. \quad (\text{GMT.11})$$

Every term is integrable by Cauchy-Schwarz; no infinite formal-series product was interchanged with an integral.

For a specified finite list of polynomials $p_0, \dots, p_r$ in $\mathcal{P}_N$, expand the actual tests

$$\overline{p_a(c + iu)} p_b(c + iu) = \sum_{n=0}^{2N} L_{ab,n} b_n^{(k\lambda)}(u).$$

This expansion is a finite algorithm: subtract the top coefficient times the monic polynomial of the same degree, then continue in descending degree through zero. The resulting remainder is exactly zero, by the decreasing degree. Equations (GMT.10)–(GMT.11) prove

$$\boxed{H(p)_{ab} = C_k \sum_{n=0}^{2N} n! L_{ab,n} d_n.} \quad (\text{GMT.12})$$

The coefficient-to-form map is homogeneous of degree k in $c_{B,0}, \dots, c_{B,2N}$. Its full coordinate type is $\mathbb{R}^{2N+1} \rightarrow \text{Herm}(\mathcal{P}_N)$ on the actual moment vectors, followed by congruence $H \mapsto P^*HP$ for a polynomial coefficient matrix P . The nonnegative density has positive mass. Its positivity set has positive Lebesgue measure, and a nonzero polynomial has only finitely many real roots. Thus the integral of its squared modulus is strictly positive, proving positivity of every finite source Gram and of every injective relation Gram. Choosing $p_a = S^a$ gives H_N ; choosing the columns of B_N gives exactly $B_N^* H_N B_N$. Choosing actual representatives gives their actual Gram. Every relation column has degree at most N , including its complete χ factor.

One may instead evaluate in the monic Gamma columns $p_j^\Gamma(S) = i^j b_j^{(k\lambda)}((S - c)/i)$. Their coefficient matrix U_N is triangular with diagonal one, and their reference Gram is

$$U_N^* H_N^\Gamma U_N = \text{diag}\{C_k j! (\beta)_j : 0 \leq j \leq N\}. \quad (\text{GMT.13})$$

This is a stated invertible coordinate map, with all squared norms retained. In particular it divides neither the measure nor the polynomials by their mass or norms. The original reduction matrix becomes $J_N U_N$; the original source form is recovered by $U_N^{-*}(\cdot) U_N^{-1}$.

40.3 Finite coefficient errors and polynomial-test bounds

Finite errors propagate before any inverse is taken. For real central coefficients $\widehat{c}_{B,j}$ and radii $e_j \geq 0$ enclosing the actual $c_{B,j}$ for $0 \leq j \leq 2N$, define the finite polynomials

$$\widehat{A} = \sum_{j=0}^{2N} (\alpha)_j \widehat{c}_{B,j} z^j, \quad A_+ = \sum_{j=0}^{2N} (\alpha)_j |\widehat{c}_{B,j}| z^j, \quad E_+ = \sum_{j=0}^{2N} (\alpha)_j e_j z^j,$$

and set

$$\widehat{d}_n = [z^n] \widehat{A}^k, \quad e_n^{(k)} = [z^n] \{(A_+ + E_+)^k - A_+^k\}. \quad (\text{GMT.14})$$

The multinomial expansion of each difference of products groups terms by the nonempty set of factors containing a coefficient error. Taking absolute values term by term proves $|d_n - \widehat{d}_n| \leq e_n^{(k)}$ for $n \leq 2N$. Thus the Hermitian central matrix $\widehat{H}$ from (GMT.12) satisfies

$$|(H - \widehat{H})_{ab}| \leq \varepsilon_{ab} := C_k \sum_{n=0}^{2N} n! |L_{ab,n}| e_n^{(k)}. \quad (\text{GMT.15})$$

The masses in this formula are exact constants. An interval evaluation of those constants is propagated by the same product rule; it cannot be replaced by setting $C_k = 1$. Exact polynomial coefficients of χ remain fixed in $L_{ab,n}$; any numerical enclosure of such coefficients must likewise be propagated before reporting a matrix certificate.

There is also a strictly polynomial-test refinement of the L^2 tail bound in the supplied descent. Write $B_L = \sum_{j \leq L} c_{B,j} b_j^{(\lambda)}$ and $E_k = B_k - C_\Sigma(B_L^{\otimes k})$. Orthogonality proves $\|B_L\| \leq \|B\|$. Expanding the tensor difference into k terms and using contraction of C_Σ proves $\|E_k\| \leq k \|B\|^{k-1} \|B - B_L\|$ with the original measures. Equation (GMT.11) applied degree by degree shows that E_k is orthogonal to all polynomials of degree at most L . Therefore

$$\left| \int \overline{p_a(c + iu)} p_b(c + iu) E_k(u) r_{\lambda,k}(u) du \right| \leq \|E_k\| \left(C_k \sum_{n > L} n! (\beta)_n |L_{ab,n}|^2 \right)^{1/2}. \quad (\text{GMT.16})$$

This follows by discarding only the orthogonal test components whose pairings are proved zero, and then applying Cauchy–Schwarz and (GMT.8). It is exactly zero at $L \geq 2N$. In particular a signed truncated density has the same finite positive Gram at that degree; pointwise positivity of the truncation is not required by this map.

An intrinsic finite polynomial-test bound is also available. The reference evaluation kernel is

$$\mathcal{K}_N^\Gamma(u) = \sum_{j=0}^N \frac{b_j^{(k\lambda)}(u)^2}{C_k j! (\beta)_j}.$$

Expansion in the original monic basis and Cauchy–Schwarz give $|P(c + iu)|^2 \leq \|P\|_\Gamma^2 \mathcal{K}_N^\Gamma(u)$. It follows that the form induced by a real error E_k obeys

$$|\langle P, P \rangle_{E_k}| \leq \|P\|_\Gamma^2 \|E_k\|_{L^2(r_{\lambda,k})} \|\mathcal{K}_N^\Gamma\|_{L^2(r_{\lambda,k})}. \quad (\text{GMT.17})$$

The last squared norm is an exact Gamma integral of a degree- $4N$ polynomial, evaluable by (GMT.8) or (GMT.12) with the reference weight. This is a finite evaluation-functional estimate, retaining every Christoffel-kernel denominator and the full mass.

40.4 Exact least lifts and the boundary-resolvent remainder

For a positive definite source matrix H in the fixed source coordinates, define

$$K(H) = J_N H^{-1} J_N^*, \quad G(H) = K(H)^{-1}, \quad R(H) = H^{-1} J_N^* G(H). \quad (\text{GMT.18})$$

Surjectivity of J_N makes $K(H)$ positive definite. Multiplication gives $J_N R(H) = 1$ and $B_N^* H R(H) = 0$. Every other lift is $R(H)x + B_N y$, whose squared norm is $x^* G(H)x + y^* B_N^* H B_N y$. This proves minimum, uniqueness and $R(H)^* H R(H) = G(H)$.

Let H_0 and H_1 be two actual positive finite source forms on the same polynomial space. In particular they may be the Gamma and arithmetic forms already constructed. Put

$$H_t = H_0 + tE, \quad E = H_1 - H_0, \quad 0 \leq t \leq 1, \quad A_0 = B_N^* H_0 B_N, \quad F = B_N^* E B_N, \quad C = B_N^* E R_0,$$

where $R_0 = R(H_0)$ and $G_0 = G(H_0)$. The whole interval of forms is positive because it is the convex combination of the two actual positive forms. Thus every nonempty $A_t = A_0 + tF$ is positive. The complete, signed resolvent formulas are

$$\boxed{R_t = R_0 - tB_N A_t^{-1} C, \quad G_t = G_0 + tR_0^* E R_0 - t^2 C^* A_t^{-1} C.} \quad (\text{GMT.19})$$

To prove the first equation, its reduction is the identity and $B_N^* H_t R_0 = tC$, while $B_N^* H_t B_N = A_t$. Consequently its boundary pairing is zero, so (GMT.18) identifies it. Expanding its full H_t norm gives $R_0^* H_t R_0 - 2t^2 C^* A_t^{-1} C + t^2 C^* A_t^{-1} A_t A_t^{-1} C$. The two negative terms and one positive term yield exactly (GMT.19). In particular the unscaled matrix $C^* A_t^{-1} C$ is positive semidefinite and vanishes precisely when $C = 0$. Pairing $A_t^{-1} > 0$ with every column of C proves this criterion. The full quadratic loss $t^2 C^* A_t^{-1} C$ vanishes precisely when $t = 0$ or $C = 0$; for an empty boundary block it is the zero form throughout.

The representative difference is a literal boundary with coefficient vector $-tA_t^{-1} Cx$; inserting that polynomial into (GMT.4) gives its full original theta primitive. The same jets ηx persist. No statement of metric nonidentity leaves its comparison map unspecified.

For completeness the exact derivative identities are

$$R'_t = -B_N A_t^{-1} B_N^* E R_t, \quad G'_t = R_t^* E R_t, \quad G''_t = -2(B_N^* E R_t)^* A_t^{-1} (B_N^* E R_t). \quad (\text{GMT.20})$$

Differentiate $J_N R_t = 1$ and $B_N^* H_t R_t = 0$ to obtain the first formula. The derivative R'_t is a boundary, hence its two pairings with R_t in the derivative of $R_t^* H_t R_t$ vanish. This proves the second formula. Differentiating it and substituting the first proves the third. This is a proved concavity statement along the actual finite metric segment, with the entire boundary coupling retained.

The determinant-line calculation has no hidden source-size exponent. The change $[E_N, B_N] \mapsto [R(H), B_N]$ adds boundary columns to the first q columns and has determinant one. Its Gram is block diagonal. Therefore

$$V_N(H) = \det G(H) = \frac{\det H}{\det(B_N^* H B_N)}. \quad (\text{GMT.21})$$

For the original arithmetic and Gamma forms this is exactly $V_N = V_N^\Gamma T_N$, with the same $T_N = X_{N+1}/Y_{N-q+1}$ as the supplied descent. Formula (GMT.19) gives its complete matrix-level map and its nonlinear remainder.

40.5 A two-degree map for the actual consecutive ratio

Fix $N \geq q$. In original source monomials split degree $N + 1$ into degrees $0, \dots, N - 1$ followed by S^N, S^{N+1} , and write

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_-^{-1} C_D, \quad J_+ = [J_-, J_D], \quad F_D = J_D - J_- H_-^{-1} C_D. \quad (\text{GMT.22})$$

Here $H_- = H_{N-1}$, $H_+ = H_{N+1}$, W is 2×2 , and $F_D : \mathbb{C}^2 \rightarrow \mathcal{C}$. Its columns are the quotient classes of the two specified polynomials after subtracting their actual orthogonal projections onto $\mathcal{P}_{N-1}$. The block map

$$(x, y) \mapsto (x - H_-^{-1} C_D y, y)$$

has determinant one and carries the source Gram to $\text{diag}(H_-, W)$. Positivity proves $W > 0$ and, by applying the reduction map to this exact orthogonal direct sum,

$$K_+ = K_- + F_D W^{-1} F_D^*, \quad Z_D = F_D^* G_- F_D \succeq 0, \quad G_- = K_-^{-1}. \quad (\text{GMT.23})$$

Block elimination of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in its two orders proves $\det(I + AB) = \det(I + BA)$. Applying this identity to (GMT.23) gives the exact scalar maps

$$\boxed{\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W} = 1 + \text{Tr}(W^{-1} Z_D) + \frac{\det Z_D}{\det W}.} \quad (\text{GMT.24})$$

The last equality is the complete characteristic polynomial of a 2×2 matrix. All denominators are positive. This gives the actual monotonicity because both eigenvalues of $W^{-1/2} Z_D W^{-1/2}$ are non-negative.

The same block also gives the adjacent source recurrence coefficient. In the order in (GMT.22), the monic residual of S^N has squared norm W_{11} , and eliminating it from the residual of S^{N+1} gives squared norm $\det W / W_{11}$. Therefore

$$\boxed{a_{N+1} = \frac{\omega_{N+1}}{\omega_N} = \frac{\det W}{W_{11}^2}.} \quad (\text{GMT.25})$$

The first residual is monic because the subtracted projection has smaller degree; so is the second after its additional elimination. The norms have not been rescaled. The coefficient map $S^j = \sum_{l \leq j} \binom{j}{l} c^{j-l} i^l u^l$ has determinant $i^{j(j-1)/2}$ in dimension j and hence Gram determinant multiplier one, which proves that these are also the source Hankel-Toda norms.

Apply (GMT.22)–(GMT.25) separately to the Gamma reference and the arithmetic source, retaining their same J_+ and χ . Then

$$\boxed{\frac{T_{N-1}}{T_{N+1}} = \frac{\det(W + Z_D) / \det W}{\det(W^\Gamma + Z_D^\Gamma) / \det W^\Gamma}.} \quad (\text{GMT.26})$$

Every entry is a finite polynomial in the one-factor coefficients through degree $2N + 2$, followed by explicitly stated inverses of positive matrices. In the supplied Toda upper estimate, substitute the two positive scalars $a_{N+1} = \det W / W_{11}^2$ and $\mathcal{R}_N = \det(W + Z_D) / \det W$. Its full right side becomes

$$\sqrt{\frac{\det W}{W_{11}^2} \frac{\det(W + Z_D) / \det W - 1}{2\sqrt{\det(W + Z_D) / \det W}}}. \quad (\text{GMT.27})$$

This is an exact substitution into that existing inequality; its discarded phase and volume-imbalance squares remain the ones proved in TVB.32–TVB.34a. Formula (GMT.27) alone does not remove them or assert equality. The two-degree determinant formula itself also works at $q = 0$, where F_D and Z_D are zero and the ratio is one.

40.6 Certified finite envelopes and quotient-dimensional bounds

Suppose a finite coefficient calculation supplies the entry bounds (GMT.15), or the proved polynomial-test bounds (GMT.16). Let $v_i > 0$ be any specified positive rational weights and set

$$\Delta = \text{diag}(\delta_i), \quad \delta_i = \sum_j \varepsilon_{ij} \frac{v_j}{v_i}, \quad H^{\text{lo}} = \hat{H} - \Delta, \quad H^{\text{hi}} = \hat{H} + \Delta. \quad (\text{GMT.28})$$

The radii may be replaced by symmetric rational upper radii. For each pair i, j , the inequality $2|x_i||x_j| \leq (v_j/v_i)|x_i|^2 + (v_i/v_j)|x_j|^2$ gives $|x^*(H - \hat{H})x| \leq x^*\Delta x$ after summation. This proves $H^{\text{lo}} \preceq H \preceq H^{\text{hi}}$ without using a pointwise lower bound on B .

The positive-pivot certificate for H^{lo} is entirely finite. Eliminate successively its first diagonal pivot, recording

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \mapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \mapsto (x - a^{-1}b^*y, y).$$

The displayed congruence is invertible. Positive recorded rational pivots prove positivity by induction; nonpositive pivots reject that particular enclosure. Actual matrix positivity is already proved by the source integral. For any sequence of valid entry intervals tending to their actual entries, the lower envelopes tend to that positive matrix. Its least unit-sphere quadratic value is positive by compactness, and convergence in the finite sum of entry errors eventually makes every lower envelope positive. Hence the certificate process succeeds at every fixed finite degree when its actual input intervals are refined; no success or interval refinement is asserted for absent arithmetic data.

For a positive certified lower envelope, calculate the actual finite quotient forms $G^{\text{lo}}, G^{\text{hi}}$ by (GMT.18), or by the equivalent Schur formula

$$G(H) = E_N^* H E_N - E_N^* H B_N (B_N^* H B_N)^{-1} B_N^* H E_N. \quad (\text{GMT.29})$$

The formula follows by substituting $E_N - B_N (B_N^* H B_N)^{-1} B_N^* H E_N$ for the unique least lift. The same radii certify the representative coefficients themselves. Write $\hat{R} = R(\hat{H})$, $A^{\text{lo}} = B_N^* H^{\text{lo}} B_N$ and define, using any rational upper bounds for the displayed complex moduli,

$$\begin{aligned} s_{ai} &= \sum_{r,l} |(B_N)_{ra}| \varepsilon_{rl} |\hat{R}_{li}|, \\ k_{ab} &= \frac{1}{2} ((A^{\text{lo}})_{aa}^{-1} + (A^{\text{lo}})_{bb}^{-1}), \\ e_{ri}^R &= \sum_{a,b} |(B_N)_{ra}| k_{ab} s_{bi}. \end{aligned}$$

Then the full original-coordinate error and its quadratic loss obey

$$\boxed{|(R(H) - \hat{R})_{ri}| \leq e_{ri}^R, \quad \hat{R}^* H \hat{R} - G(H) = (R(H) - \hat{R})^* H (R(H) - \hat{R})}. \quad (\text{GMT.29a})$$

To prove the entry bound set $A = B_N^* H B_N$ and $S_E = B_N^* (H - \widehat{H}) \widehat{R}$. Formula (GMT.19) gives $R(H) - \widehat{R} = -B_N A^{-1} S_E$. The triangle inequality gives $|(S_E)_{ai}| \leq s_{ai}$. Inversion reverses positive order: conjugating $A \succeq A^{\text{lo}}$ by $(A^{\text{lo}})^{-1/2}$, inverting its eigenvalues, and conjugating back gives $A^{-1} \preceq (A^{\text{lo}})^{-1}$. Positivity of each two-coordinate Gram minor of A^{-1} gives $|A_{ab}^{-1}| \leq \sqrt{A_{aa}^{-1} A_{bb}^{-1}} \leq k_{ab}$, where the last step is the arithmetic-geometric mean inequality and the proved diagonal bounds. Matrix multiplication now proves the coefficientwise bound in (GMT.29a). For its second identity, $\widehat{R} - R(H)$ is an actual boundary, orthogonal to $R(H)$ in H . Expanding $\widehat{R}^* H \widehat{R}$ proves the equality.

This loss has an explicit q -dimensional rational upper form. Choose positive rational weights v_i in the original quotient coordinates and set

$$\tau_{ij} = \sum_{a,b} s_{ai} |(A^{\text{lo}})^{-1}|_{ab} s_{bj}, \quad \Delta_Q = \text{diag} \left\{ \sum_j \tau_{ij} \frac{v_j}{v_i} \right\}.$$

The same entrywise quadratic inequality as (GMT.28) proves

$$0 \preceq \widehat{R}^* H \widehat{R} - G(H) \preceq \Delta_Q. \quad (\text{GMT.29b})$$

Indeed the middle matrix is $S_E^* A^{-1} S_E \preceq S_E^* (A^{\text{lo}})^{-1} S_E$. Its value on x is at most $\sum_{i,j} \tau_{ij} |x_i| |x_j| \leq x^* \Delta_Q x$. Every source row, relation coefficient and quotient coordinate is retained in these sums. At an empty boundary block both error forms are zero. Applying $\mathcal{V}_{h,k}$ to each representative difference gives the theta primitive (GMT.4), so the certified coefficient error also has its original source and quotient interpretation.

Minimizing the two proven source inequalities over the same affine space of representatives gives

$$G^{\text{lo}} \preceq G(H) \preceq G^{\text{hi}}, \quad \det G^{\text{lo}} \leq V_N(H) \leq \det G^{\text{hi}}. \quad (\text{GMT.30})$$

For the determinant assertion, conjugate by the positive square root of the lower form and multiply the eigenvalues, all of which are at least one; apply the same argument for the upper form.

Construct the degree $N - 1$ envelope by taking the corresponding principal submatrix of the same degree $N + 1$ envelope. Then

$$\frac{\det G_-^{\text{lo}}}{\det G_+^{\text{hi}}} \leq \frac{V_{N-1}}{V_{N+1}} \leq \frac{\det G_-^{\text{hi}}}{\det G_+^{\text{lo}}}. \quad (\text{GMT.31})$$

Division by the exact positive Gamma ratio from (GMT.26) gives the certified interval for T_{N-1}/T_{N+1} . These are determinants of size q , with no independent numerator/denominator error estimate on the larger source and relation determinants.

For example a relative certificate $(1 - \eta)\widehat{H} \preceq H \preceq (1 + \eta)\widehat{H}$, with a recorded rational $0 \leq \eta < 1$ and $\widehat{H} > 0$, gives exactly

$$(1 - \eta)^q \det G(\widehat{H}) \leq V_N(H) \leq (1 + \eta)^q \det G(\widehat{H}),$$

and a common such certificate for the two nested degrees gives ratio multipliers $((1 - \eta)/(1 + \eta))^q$ and its reciprocal. This follows because $R(aH) = R(H)$ and $G(aH) = aG(H)$ for $a > 0$, as direct substitution in (GMT.18) proves. The mass is scaled only in this explicitly declared comparison, never assigned a new value in the original arithmetic source.

40.7 Exact calibration and a bounded relative-volume counterexample

Take the explicitly auxiliary one-factor weight

$$\lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = \frac{1}{2}, \quad C_2 = \frac{1}{4}, \quad B(t) = 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = (S - 1)^2 - 2. \quad (\text{GMT.32})$$

The positive polynomial B lies in $L^2(r_1)$ by (GMT.6). It is not the function $|g/h\Gamma|^2$ of a claimed zeta packet. Since $b_2^{(1)} = t^2 - 2$, its coefficients are $c_{B,0} = 3/2$, $c_{B,2} = 1/4$ and all others zero. Therefore

$$A(z) = \frac{3}{2} + \frac{3}{2}z^2, \quad (d_0, d_2, d_4) = \left(\frac{9}{4}, \frac{9}{2}, \frac{9}{4}\right), \quad B_2(u) = \frac{54}{35} + \frac{39}{280}u^2 + \frac{3}{1120}u^4. \quad (\text{GMT.33})$$

To verify the last equality, use $b_2^{(2)} = u^2 - 4$, $b_4^{(2)} = u^4 - 32u^2 + 72$ in $\sum_n d_n b_n^{(2)}/(4)_n$ and expand all three terms. The one-factor mass is $3/4$ and the sum mass is $9/16$. Repeated multiplication by t in the recurrence (GMT.8) gives the one-factor reference even moments through degree eight as $(1/2, 1, 8, 136, 3968)$. Inserting $B = 1 + t^2/4$ and expanding $(t_1 + t_2)^n$ with the full binomial coefficients gives sum moments

$$(\mu_0, \mu_1, \dots, \mu_8) = (9/16, 0, 9/2, 0, 117, 0, 5472, 0, 385272). \quad (\text{GMT.34})$$

This is also (GMT.11)–(GMT.12); the two calculations independently fix its mass and tensor cross term.

The original degree-three source Gram and reference Gram are

$$H_3 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix}, \quad H_3^\Gamma = \frac{1}{4} \begin{pmatrix} 1 & 1 & -3 & -11 \\ 1 & 5 & 5 & -55 \\ -3 & 5 & 65 & 65 \\ -11 & -55 & 65 & 1685 \end{pmatrix}. \quad (\text{GMT.35})$$

Every entry follows by expanding $(1 - iu)^a(1 + iu)^b$ against (GMT.34), or against the reference moments. The complete original relation and reduction matrices are

$$B_3 = \begin{pmatrix} -1 & 0 \\ -2 & -1 \\ 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{pmatrix}. \quad (\text{GMT.36})$$

These follow from $S^2 = 2S + 1$ and $S^3 = 5S + 2$ modulo χ ; multiplication verifies $J_3 B_3 = 0$ and $\det[E_3, B_3] = 1$. Direct substitution into (GMT.18) gives

$$G_1 = \frac{1}{16} \begin{pmatrix} 9 & 9 \\ 9 & 81 \end{pmatrix}, \quad G_3 = \begin{pmatrix} 81/244 & 81/244 \\ 81/244 & 175041/80764 \end{pmatrix},$$

$$R_3 = \begin{pmatrix} 117/122 & -801/40382 \\ -5/61 & 18963/20191 \\ 5/122 & -907/40382 \\ 0 & 7/331 \end{pmatrix}, \quad G_3^\Gamma = \begin{pmatrix} 5/38 & 5/38 \\ 5/38 & 2255/4142 \end{pmatrix}. \quad (\text{GMT.37})$$

These values can be checked without trusting an inverse algorithm: multiply to obtain $J_3 R_3 = I$, $B_3^* H_3 R_3 = 0$ and $R_3^* H_3 R_3 = G_3$, then use the uniqueness proved in (GMT.18).

For the degree-one to degree-three split, the full two-degree data are

$$W = \begin{pmatrix} 81 & 243 \\ 243 & 3159 \end{pmatrix}, \quad F_D = \begin{pmatrix} 10 & 2 \\ 0 & 28 \end{pmatrix}, \quad Z_D = \frac{1}{4} \begin{pmatrix} 225 & 675 \\ 675 & 16137 \end{pmatrix}. \quad (\text{GMT.38})$$

Equation (GMT.22), evaluated with (GMT.35)–(GMT.36), proves these entries by ordinary rational multiplication. Consequently

$$\frac{V_1}{V_3} = \frac{20191}{4860}, \quad a_3 = 30, \quad \frac{V_1^\Gamma}{V_3^\Gamma} = \frac{2071}{450}, \quad \boxed{\frac{T_1}{T_3} = \frac{100955}{111834} < 1}. \quad (\text{GMT.39})$$

Here $V_1^\Gamma = 1/4$, $V_3^\Gamma = 225/4142$, obtained by taking the two displayed 2×2 determinants. Thus no unrecorded cancellation of reference masses is used.

The phenomenon has a completely specified bounded positive version. Define $B^{[200]}(t) = \min(1 + t^2/4, 10001)$ and $w^{[200]} = B^{[200]}r_1$. This changes the auxiliary weight only; it leaves $S = 1 + iu$, χ , the reference Gamma mass and the unscaled convolution operation fixed. For $0 \leq j \leq 6$,

$$\int |t|^j (w - w^{[200]})(t) dt \leq \frac{1}{4} \int_{|t| > 200} |t|^{j+2} r_1(t) dt < \frac{8!}{2^{92}} =: \epsilon_*. \quad (\text{GMT.40})$$

Here is a proof with no floating evaluation. For every integer $r \geq 0$, $x^r \leq r! 2^r e^{x/2}$; on $x > 200$ multiply by $e^{(x-200)/2} \geq 1$. Equation (GMT.6) and evenness give

$$\int_{|t| > 200} |t|^r r_1(t) dt \leq 2c_1 r! 2^r e^{-100} (\cos 1)^{-2} \leq 4r! 2^r e^{-100}.$$

The estimate $\cos 1 \geq 1/2$ follows from $1 - \cos 1 = \int_0^1 \sin x dx \leq \int_0^1 x dx = 1/2$, using $|\sin x| \leq x$ on this interval. Since $e > 2$, $e^{-100} < 2^{-100}$. Use $r = j + 2 \leq 8$ and the factor $1/4$ to obtain (GMT.40).

For $0 \leq j \leq 6$, the absolute j th one-factor moment of w is at most $A_* = \mu_0(w) + \mu_6(w) = 3/4 + 1128 = 4515/4$, because $|t|^j \leq 1 + |t|^6$. The exact tensor difference is $(w - w^{[200]}) \otimes w + w^{[200]} \otimes (w - w^{[200]})$. Expanding $|t_1 + t_2|^n \leq \sum_j \binom{n}{j} |t_1|^j |t_2|^{n-j}$ proves a sum-moment difference at degree $n \leq 6$ of at most $2^{n+1} A_* \epsilon_* \leq 128 A_* \epsilon_*$. The coefficient absolute sum in $(1 - iu)^a (1 + iu)^b$ is at most $2^{a+b} \leq 64$ for $a, b \leq 3$. Hence every source entry changes by at most $8192 A_* \epsilon_* < 1/40000$. The last inequality is an integer comparison after substitution of A_*, ϵ_* . Equation (GMT.28) with four weights equal to one now proves

$$H_3 - 10^{-4}I \prec H_3^{[200]} \prec H_3 + 10^{-4}I. \quad (\text{GMT.41})$$

Strictness follows because the error in each row sum is strictly less than 10^{-4} . The four leading principal minors of the lower matrix are

$$\frac{703}{1250}, \quad \frac{253068751}{100000000}, \quad \frac{102481960035937}{500000000000}, \quad \frac{4980299350501627807501}{10000000000000000}.$$

They are positive; their successive ratios are exactly the positive pivots in the elimination proving (GMT.28). Strict source inequalities give strict quotient inequalities: evaluate the lower form on the actual minimum lift, and the actual form on the upper minimum lift. Each lift of a nonzero quotient vector is nonzero, so the positive source difference has a strictly positive quadratic value. Computing (GMT.29) for these two rational endpoint matrices, and the degree-one principal submatrices, gives the strict ratio enclosure

$$\frac{20694424798404596925036}{4984219548001747192501} < \frac{V_1^{[200]}}{V_3^{[200]}} < \frac{20703435010591353075036}{4980299350501627807501} < \frac{2071}{450}. \quad (\text{GMT.42})$$

The first two strict comparisons follow from that strict minimization and (GMT.31); the final comparison follows by positive-denominator cross multiplication. The rational envelope computation is reproduced by the exact checker accompanying this proof. Division by the fixed positive Gamma ratio therefore proves $T_1^{[200]}/T_3^{[200]} < 1$ for a function bounded between 1 and 10001. This is an explicit obstruction to deriving relative volume monotonicity from positivity and boundedness of a multiplier alone. Its exact relation to the arithmetic problem is the same coefficient, relation, least-lift and determinant maps (GMT.12), (GMT.18), (GMT.24) and (GMT.26). It does not assert that the actual arithmetic multiplier has this behavior.

40.8 Verification scope

The accompanying exact checker constructs moments in two ways: Jacobi recurrence and the literal two-factor binomial expansion, then compares them with the Gamma coefficient formula. It verifies the original S -coordinate Gram, every displayed lift and relation identity, the full boundary-resolvent loss, five points on its actual positive matrix segment, both two-degree formulas and the rational envelope calculation. Its deliberate formula changes remove the mass, omit the tensor cross term, reverse the boundary-loss sign, remove the conjugation, or remove one power in the Schur denominator. Each change is rejected by the corresponding exact check in ordinary and optimized Python. The written proofs establish the all-degree identities; these finite checks provide reproducible regression evidence. There is no Lean execution or analytic interval claim for a nonempty zeta packet in this verification.

41 The complete complex amplitude in gamma sum descent

41.1 Original objects and the question answered

The delivered Gamma Convolution Descent note constructs the conditional expectation of the squared arithmetic amplitude. We construct the exact comparison for its full complex amplitude. The two resulting source Grams add to the original arithmetic Gram. We prove a strict positive contribution of the retained relative component for every full quartet, and give its exact map into the original minimum quotient metric. All measures below retain their masses. The variable is the original sum $u = t_1 + \cdots + t_k$, and the arithmetic variable remains $S = k/2 + iu$.

Fix $\lambda > 0$ and an integer $k \geq 1$. Retain

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GP.1})$$

Thus $\int r_{\lambda,k} = c_\lambda^k$. The beta-integral proof of this identity, with its original Jacobian $t = 2v$, is retained in the cumulative reader, TG.7–TG.8, and the delivered note, (5)–(9). In particular

$$\int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}, \quad |\theta| < \pi/2. \quad (\text{GP.2})$$

Let h be the original finite polynomial packet of complete orders of actual zeros of $g = 2\xi$, and include $h = 1$ when considering the analytic seed. Define the entire quotient $v_h = g/h$ and keep

$$A(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B(t) = |A(t)|^2, \quad w(t) = B(t)r_\lambda(t), \quad \mu_h = \int w(t) dt. \quad (\text{GP.3})$$

Use one of the references proved in the delivered note for which $B \leq C_h < \infty$. In particular a full quartet of degree $d \geq 4$ uses its original $\lambda = 1/4$. The seed uses $\lambda = 13/4$ and $C_1 = 1024/\sqrt{\pi}$. The statements involving decay explicitly use the quartet reference. The function A is nonzero and analytic on the real line: v_h is a nonzero entire function and reciprocal gamma is entire and nonzero on this line. Its real zeros form a discrete set. No value at a removed packet zero is assigned using an uncanceled fraction.

Put $d\nu(\mathbf{t}) = \prod_i r_\lambda(t_i) d\mathbf{t}$ and $d\nu_\Sigma(u) = r_{\lambda,k}(u) du$. In $\mathcal{H} = L^2(\nu)$ and $\mathcal{H}_\Sigma = L^2(\nu_\Sigma)$ write

$$Uf(\mathbf{t}) = f(\sum_i t_i), \quad C = U^*, \quad P = UC, \quad a(\mathbf{t}) = \prod_i A(t_i), \quad (\text{GP.4})$$

$$b_k = C(|a|^2), \quad a_k = C(a), \quad d_k = b_k - |a_k|^2.$$

For $k = 1$, $C = U = I$, $a_1 = A$, $b_1 = B$, $d_1 = 0$. For $k \geq 2$ the conditional integral CF is precisely

$$(CF)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GP.5})$$

These are probability measures on individual fibres, derived by dividing by their stated fibre mass; neither original measure is replaced by a probability measure. Fubini proves $CU = I$ and the isometry of U . Jensen's inequality in (GP.5) proves $d_k \geq 0$ and $0 \leq b_k \leq C_h^k$. For $k \geq 2$, $b_k(u) > 0$ at every real u : the discrete zeros of each factor exclude only countably many hyperplanes in this fibre, which have zero Lebesgue measure.

41.2 Unitary transport of the actual conditional projection

Theorem 41.1 (Complete amplitude comparison). *Use the actual arithmetic spaces*

$$\mathcal{H}^h = L^2(\mathbb{R}^k, |a|^2 d\nu), \quad \mathcal{H}_\Sigma^h = L^2(\mathbb{R}, b_k d\nu_\Sigma).$$

The maps

$$T : \mathcal{H}^h \longrightarrow \mathcal{H}, \quad F \longmapsto aF, \quad T_\Sigma : \mathcal{H}_\Sigma^h \longrightarrow \mathcal{H}_\Sigma, \quad f \longmapsto \sqrt{b_k}f \quad (\text{GP.6})$$

are surjective isometries. The positive square root in T_Σ is a specified comparison map; the original amplitude in T remains complex. The actual arithmetic sum inclusion $U_h f = f(\sum t_i)$ and its adjoint C_h^{sum} satisfy

$$C_h^{\text{sum}} F = \frac{C(|a|^2 F)}{b_k}, \quad J = TU_h T_\Sigma^{-1} = M_a U M_{b_k^{-1/2}}, \quad J^* = M_{b_k^{-1/2}} C M_{\bar{a}}, \quad J^* J = I. \quad (\text{GP.7})$$

All quotients here are defined almost everywhere. In particular their operator extensions are bounded even when the individual inverse multiplier is unbounded. The transported arithmetic projection is

$$Q = TP_h T^{-1} = JJ^*, \quad QF = aU \left(\frac{C(\bar{a}F)}{b_k} \right), \quad P_h = U_h C_h^{\text{sum}}. \quad (\text{GP.8})$$

Thus P and Q are related by the displayed exact transport from the original arithmetic spaces.

Proof. The squared norms in (GP.6) agree by direct substitution. The zero set of a is a union of null hyperplanes; b_k is positive almost everywhere, including $k = 1$. Division off those null sets constructs the two inverses with their exact squared norms. Applying (GP.5) to the arithmetic fibre density gives C_h^{sum} . Pairing $M_a U M_{b_k^{-1/2}} f$ with F proves the stated adjoint. The identity $C(|a|^2 U f) = b_k f$ then gives $J^* J = I$. Conjugating P_h gives (GP.8). These norm identities first hold for bounded functions supported on finite-measure sets on which the denominators are bounded below; truncation and the isometries extend them to the indicated complete Hilbert spaces. This also proves that no unbounded multiplier composition is being asserted on a larger domain. $\square$

Theorem 41.2 (Exact fibre angle and retained defect). *Define $\gamma_k = a_k / \sqrt{b_k}$ almost everywhere. Then*

$$U^* J = M_{\gamma_k}, \quad J = U M_{\gamma_k} + N, \quad U^* N = 0, \quad N^* N = M_{1-|\gamma_k|^2} = M_{d_k/b_k}. \quad (\text{GP.9})$$

For every real u at which $b_k > 0$, the two fibre projections onto 1 and $a/\sqrt{b_k}$ obey

$$\text{Tr}_u(P_u - Q_u)^2 = 2(1 - |\gamma_k(u)|^2), \quad \|[P_u, Q_u]\|_{\text{HS},u}^2 = 2|\gamma_k(u)|^2(1 - |\gamma_k(u)|^2). \quad (\text{GP.10})$$

These are finite-rank traces in the specified fibre Hilbert space, not a claim that the global decomposable operators are trace class. The complete unscaled variance also has the formula

$$d_k(u) = \frac{1}{2} \iint |a(\mathbf{t}) - a(\mathbf{s})|^2 d\nu_u(\mathbf{t}) d\nu_u(\mathbf{s}), \quad (\text{GP.11})$$

where both ν_u are the exact conditional measures in (GP.5).

Proof. Applying C to Jf gives $\gamma_k f$. Orthogonal decomposition with P gives (GP.9), and taking squared norms uses $J^*J = I$. On a fibre let $e_0 = 1$ and $v = a/\sqrt{b_k}$, so both have norm one and $\langle e_0, v \rangle = \gamma_k$, using the inner product antilinear in its first argument. If $|\gamma_k| < 1$, put $e_1 = (v - \gamma_k e_0)/\sqrt{1 - |\gamma_k|^2}$. In the displayed orthonormal comparison frame, without altering either original vector, the two projections are

$$P_u = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q_u = \begin{pmatrix} |\gamma_k|^2 & \gamma_k \sqrt{1 - |\gamma_k|^2} \\ \overline{\gamma_k} \sqrt{1 - |\gamma_k|^2} & 1 - |\gamma_k|^2 \end{pmatrix}.$$

Matrix multiplication proves (GP.10); both vanish when $|\gamma_k| = 1$. Expanding the square in (GP.11), and using the unit total mass of each conditional measure, gives $C(|a|^2) - |Ca|^2$. No phase is discarded in this calculation. $\square$

41.3 Strictness for an actual quartet, with its exact constant

For a full quartet retain $\lambda = 1/4$, degree $d \geq 4$ and $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. The delivered proof gives

$$|A(t)|^2 \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d} \quad (|t| \geq T_h). \quad (\text{GP.12})$$

Here is its direct calculation. In the unchanged amplitude,

$$A(t) = -\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it) / h(1/2 + it).$$

The continuation formula $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ gives $|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For $|t| \geq T_h$, every retained root factor has modulus at least $|t|/2$, so $|h| \geq (|t|/2)^d$. Also $(t^2 + 1/4)^2 \leq 25t^4/16$ and $4(1 + |t|)^2 \leq 16t^2$ for $|t| \geq 1$. Multiplication proves (GP.12). Consequently $A(t) \rightarrow 0$ at both real ends.

Theorem 41.3 (Strict relative contribution on every sum fibre). *For every such actual quartet, every $k \geq 2$, and every real u ,*

$$d_k(u) > 0, \quad |\gamma_k(u)| < 1. \quad (\text{GP.13})$$

Thus the relative component is nonzero at every fibre. It is given exactly by (GP.9) and (GP.11); strictness does not assign a uniform positive lower bound to it.

Proof. Fix u . If $d_k(u) = 0$, (GP.11), or equality in the variance identity, says that the continuous function a is constant almost everywhere on the fibre. Its conditional density is strictly positive everywhere, so continuity makes it constant everywhere. For $k = 2$, this would make $A(t)A(u - t)$ constant in t . Its limit is zero by (GP.12) and boundedness of A , whereas some value is nonzero: avoid the two discrete sets of zeros of $A(t)$ and $A(u - t)$. For $k > 2$, choose fixed $t_2, \dots, t_{k-1}$ at nonzeros of A and vary $t_1 = t$, $t_k = u - t - \sum_{2 \leq i < k} t_i$. The identical argument makes the product both zero and nonzero, a contradiction. The assertion for γ_k follows by dividing by the strictly positive b_k already proved. This proof uses only the explicit quartet tail, not a conjecture about positions of unselected zeros. $\square$

41.4 Full complex coefficient transform and exact variance norm

Put $\alpha = 2\lambda$ and keep the original monic polynomials

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}, \quad \|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n. \quad (\text{GP.14})$$

The specified gamma measure has exponential moments. Its polynomials are complete in L^2 : for a vector orthogonal to them, multiplying it by the measure gives a finite complex measure by Cauchy-Schwarz; the same inequality with exponential moments makes its Fourier transform holomorphic

on a strip about the real axis. All derivatives at zero vanish. Analytic uniqueness and Fourier uniqueness make the measure, hence the vector, zero. The same proof applies to the sum measure with mass c_λ^k .

Define complex coefficients, retaining conjugation conventions,

$$a_j = \frac{\int b_j^{(\lambda)}(t)A(t)r_\lambda(t)dt}{c_\lambda j!(\alpha)_j}, \quad \mathfrak{A}(z) = \sum_{j \geq 0} (\alpha)_j a_j z^j, \quad e_{k,n} = [z^n] \mathfrak{A}(z)^k. \quad (\text{GP.15})$$

The square-integrable vector A has these orthogonal coefficients. The series in (GP.15) may be used coefficientwise; no interchange of an infinite product with an integral is required below.

Theorem 41.4 (Complex convolution and its exact lost norm). *With all original masses,*

$$a_k(u) = \sum_{n \geq 0} \frac{e_{k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u) \quad \text{in } L^2(r_{\lambda,k} du),$$

$$\int d_k r_{\lambda,k} = \mu_h^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |e_{k,n}|^2. \quad (\text{GP.16})$$

For every polynomial F in the original S coordinate and every real $|\theta| < \pi/2$, the exact weighted version is

$$\|a U(e^{\theta u/2} F(k/2 + iu))\|^2 = \int e^{\theta u} |F(k/2 + iu)|^2 |a_k(u)|^2 r_{\lambda,k}(u) du$$

$$+ \int e^{\theta u} |F(k/2 + iu)|^2 d_k(u) r_{\lambda,k}(u) du. \quad (\text{GP.17})$$

Proof. Multiplication of the finite-degree coefficients of the generating functions gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing this finite identity with a gives $c_\lambda^k n! e_{k,n}$. Division by the original sum norm $c_\lambda^k n! (k\alpha)_n$, followed by completeness, gives the first identity. Pythagoras for $P = UC$ gives the second since $\|a\|^2 = \mu_h^k$. Finally conditional expectation commutes with multiplication by a function of the sum. Apply the same orthogonal decomposition to $aU(e^{\theta u/2} F)$ to obtain (GP.17). Boundedness by C_h^k , (GP.2), and polynomial growth prove all integrals finite, including any fixed number of theta derivatives. $\square$

41.5 The exact induced change of the original finite quotient metric

Fix the actual annihilator $\chi = \chi_{h,k}$, degree $q > 0$, and $N \geq q - 1$. Use original polynomial spaces $E = \mathcal{P}_N(S)$, $E_\partial = \mathcal{P}_{N-q}(S)$, where a negative degree means the zero space, and $C_\chi = \mathbb{C}[S]/(\chi)$. Let $B_\chi : E_\partial \rightarrow E$ be multiplication by χ and $\pi : E \rightarrow C_\chi$ the quotient. Use the original monomial bases, or any explicitly transported fixed bases, in every matrix below. For a coefficient column x denote its polynomial by F_x . Define three source matrices by

$$x^* M_h x = \int e^{\theta u} |F_x(k/2 + iu)|^2 b_k r_{\lambda,k} du,$$

$$x^* M_{\text{coh}} x = \int e^{\theta u} |F_x(k/2 + iu)|^2 |a_k|^2 r_{\lambda,k} du,$$

$$x^* M_{\text{rel}} x = \int e^{\theta u} |F_x(k/2 + iu)|^2 d_k r_{\lambda,k} du. \quad (\text{GP.18})$$

These matrices are elements of $\text{Herm}(E)$, equivalently linear maps $E \rightarrow E^\vee$ with $E^\vee$ the conjugate dual under $x \mapsto [y \mapsto y^* M x]$. The associated linear observation maps are $F \mapsto aU(e^{\theta u/2} F)$, $F \mapsto$

$U(e^{\theta u/2} a_k F)$, and $F \mapsto (I - P)aU(e^{\theta u/2} F)$. The Gram operation itself has type $\text{Hom}_{\mathbb{C}}(E, \mathcal{H}) \rightarrow \text{Herm}(E)$, $R \mapsto R^* R$. Thus

$$M_h = M_{\text{coh}} + M_{\text{rel}}. \quad (\text{GP.19})$$

The original arithmetic matrix M_h is unchanged.

Lemma 41.5. M_{coh} is positive definite for every packet/reference admitted above, every $k \geq 1$, every finite N , and real $|\theta| < \pi/2$. For the quartets of (GP.12) and $k \geq 2$, M_{rel} is also positive definite.

Proof. Put $f = Ar_{\lambda}$. It is a nonzero L^1 function, since A is bounded and nonzero almost everywhere. The numerator of a_k is f^{*k} . If this convolution were zero almost everywhere, its Fourier transform $(\hat{f})^k$ would vanish everywhere, whence $\hat{f} = 0$ and $f = 0$ by Fourier uniqueness, a contradiction. Consequently $|a_k|^2 r_{\lambda,k}$ is positive on a set of positive Lebesgue measure. A nonzero polynomial on the indicated affine line has only finitely many real zeros, so its squared norm is strictly positive. This proves the first assertion. The second follows from $d_k(u) > 0$ for every u , proved in (GP.13). $\square$

Let $R_h, R_{\text{coh}} : C_{\chi} \rightarrow E$ be the least-norm sections for M_h, M_{coh} , and write $G_h = R_h^* M_h R_h$, $G_{\text{coh}} = R_{\text{coh}}^* M_{\text{coh}} R_{\text{coh}}$. They exist uniquely: for any section R_0 , subtract $B_{\chi}(B_{\chi}^* M B_{\chi})^{-1} B_{\chi}^* M R_0$; the boundary Gram is invertible because B_{χ} is injective and M positive definite. An empty boundary contributes the zero map.

Theorem 41.6 (Exact two positive terms in the quotient change). *In these original coordinates put*

$$K = (B_{\chi}^* M_h B_{\chi})^{-1} B_{\chi}^* M_{\text{rel}} R_{\text{coh}}. \quad (\text{GP.20})$$

It is a map $C_{\chi} \rightarrow E_{\partial}$, or the unique zero map when the boundary is empty. Then

$$R_h = R_{\text{coh}} - B_{\chi} K, \quad \boxed{G_h = G_{\text{coh}} + K^* B_{\chi}^* M_{\text{coh}} B_{\chi} K + R_h^* M_{\text{rel}} R_h.} \quad (\text{GP.21})$$

For an actual full quartet, $k \geq 2$, and $q > 0$, this gives $G_h - G_{\text{coh}} > 0$ for every admitted N and theta. More precisely, with the actual positive matrix

$$E_q = G_{\text{coh}}^{-1/2} (K^* B_{\chi}^* M_{\text{coh}} B_{\chi} K + R_h^* M_{\text{rel}} R_h) G_{\text{coh}}^{-1/2}, \quad (\text{GP.22})$$

the original determinants satisfy

$$\det G_h = \det G_{\text{coh}} \det(I + E_q). \quad (\text{GP.23})$$

These comparison conjugations do not replace either quotient metric.

Proof. $B_{\chi}^* M_{\text{coh}} R_{\text{coh}} = 0$, so the formula for the M_h section using R_{coh} as initial section gives (GP.20) and the first identity. Expand $R_h^* M_{\text{coh}} R_h$ about R_{coh} . The two cross terms vanish by the stated orthogonality. Add $R_h^* M_{\text{rel}} R_h$ using (GP.19); this proves (GP.21) with both terms retained. R_h is injective because $\pi R_h = I$. Thus strict positivity of M_{rel} makes its displayed pullback positive definite on the nonzero quotient. Conjugation by the actual positive square root of G_{coh} proves (GP.23). For $N = q - 1$ the first positive term is zero and the second remains as stated. For $q = 0$ the original quotient is zero: both section maps are zero, and the empty determinant is one. The nonzero analytic source does not supply an extra quotient dimension. $\square$

41.6 Support, the original unit, and the domain of arithmetic observation

The original cyclic source is $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$. In its Mellin realization its full image is

$$\prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} a(\mathbf{t}) P(k/2 + i \sum_i t_i). \quad (\text{GP.24})$$

Multiplication by the displayed gamma factors is an isometry $\mathcal{H} \rightarrow L^2(d\mathbf{t})$. Equations (GP.9) and (GP.17) split the vector before this multiplication and reconstruct it by addition. On the image graph

$$\mathcal{G}_N = \{(U(a_k F), (I - P)aUF) : F \in \mathcal{P}_N(S)\}$$

addition $\text{add}_N : \mathcal{G}_N \rightarrow aU(\mathcal{P}_N)$ is injective, since the squared norm of the reconstructed vector is the strictly positive polynomial norm M_h . It is onto the displayed image by construction. Denote the multiplication in (GP.24) by $M_\Gamma : \mathcal{H} \rightarrow L^2(d\mathbf{t})$ and the original unitary Mellin map by $\mathcal{M}_{\text{unit}}$. The graph observation is exactly $J^{(k)}\mathcal{M}_{\text{unit}}^{-1}M_\Gamma \text{add}_N$, and likewise with $q^{(k)}$. These are well-defined linear maps because the reconstructed image is precisely the original admitted Mellin source. Their values on the graph of P are exactly

$$J^{(k)}\mathcal{V}P = \eta_{h,k}\pi P, \quad q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}\pi P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{GP.25})$$

This specifies the domain of the observation; no arithmetic jet is assigned to an arbitrary independent measurable relative vector. The graph in this display uses theta zero. At a nonzero admitted theta, multiply both graph components by the same $e^{\theta u/2}$; its inverse multiplies by $e^{-\theta u/2}$ before the original inverse Mellin map. Both maps are isometries between the declared tilted and untilted spaces.

Here is the full cochain map for the relation. In the original tensor algebra

$$\mathbb{C}[s_1, \dots, s_k]/(h(s_1), \dots, h(s_k)),$$

the cyclic annihilator satisfies $\chi(s_1 + \dots + s_k) = 0$. Successive monic division in the fixed order $s_1, \dots, s_k$ gives unique quotients at each division step and polynomials Q_i with

$$\chi(s_1 + \dots + s_k) = \sum_{i=1}^k h(s_i)Q_i(s_1, \dots, s_k).$$

The final remainder is zero because the monomials of degree less than $\deg h$ in each variable are a basis of that tensor algebra. In $[V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ put $D = -x\partial_x$ and $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. It is even, $\phi_0(0) = 0$ and $\int_{\mathbb{R}} \phi_0 = 0$: the two even Gaussian integrals are $4\pi^2 \cdot 3/(4\pi^2)$ and $6\pi \cdot 1/(2\pi)$, both equal to three. For $\Re s > 1$, termwise Mellin integration of $\Theta\phi_0(x) = \sum_{n \neq 0} \phi_0(nx)$ yields $2\zeta(s)\mathcal{M}\phi_0(s)$. Direct integration with $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \pi^{-s/2}(2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

Thus $\mathcal{M}(\Theta\phi_0) = g$ there and by continuation on the original Mellin domain. The original inverse Mellin definition of F_h then gives $h(D)F_h = \Theta\phi_0$. The precise primitive is

$$\mathcal{P}_\chi F = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) F(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GP.26})$$

Euler differentiation preserves the even Schwartz source and its two conditions: its value at zero vanishes and its integral is the original integral by integration by parts. It also preserves rapid decay at both multiplicative ends and commutes with Θ . Every term therefore lies in the asserted cochain domain. The preceding $i-1$ degree-one factors give the tensor differential sign $(-1)^{i-1}$, cancelling the written sign. Hence, with no sign omitted,

$$d\mathcal{P}_\chi F = \mathcal{V}(\chi F), \quad d[-\mathcal{P}_\chi K] = \mathcal{V}(R_h - R_{\text{coh}}). \quad (\text{GP.27})$$

This proves the cochain correction map on C_χ in (GP.21), including its zero value when the boundary is empty.

Every linear map on this graph and every original source or quotient has its original fixed-support lift: $(\ell, x) \mapsto (\ell, Lx)$, and external absence maps to external absence. The maps preserve addition and scalar action because the underlying maps are linear. For an actual relation the quotient value is $(\ell, 0)$, its supported zero, with the source and the two graph components still reconstructible. The norm has type $\mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ and is positively homogeneous; its square is quadratic. Trace has type $\text{End}_{\mathbb{C}}(E) \rightarrow \mathbb{C}$ and is linear; its restriction to Hermitian endomorphisms is real-linear. The Gram map and squared modulus are quadratic on the underlying real vector spaces, and trace composed with a Gram map is quadratic. The exact linear observation maps producing the source Grams and the quotient values are (GP.18), (GP.21), and (GP.24)–(GP.25).

41.7 Exact polynomial calibration with the original gamma mass

The following auxiliary calibration tests the complete complex-phase identities by rational arithmetic. It is a declared polynomial amplitude, with all fixed moments finite in the original gamma measure. It does not invoke the bounded packet estimate (GP.12). Take $\lambda = 1$, $k = 2$, $c_1 = 1/2$, $c_1^2 = 1/4$, and $A(t) = 1 + it/2$. The original sum coordinate is $S = 1 + iu$. Direct conditional polynomial calculation gives

$$\begin{aligned} a_2(u) &= \frac{6}{5} + \frac{iu}{2} - \frac{u^2}{20}, \\ b_2(u) &= \frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120}, \\ d_2(u) &= \frac{18}{175} + \frac{13u^2}{1400} + \frac{u^4}{5600} > 0. \end{aligned} \tag{GP.28}$$

Here are enough exact polynomial identities to prove the calculation. For the one-factor parameter $\alpha = 2$, $b_2^{(1)}(t) = t^2 - 2$. For the sum parameter 4, the second and fourth monic polynomials are $u^2 - 4$ and $u^4 - 32u^2 + 72$. The conditional addition formula yields

$$\begin{aligned} C(t_1 t_2) &= (u^2 - 4)/5, \\ C(t_1^2) &= C(t_2^2) = 3(u^2 - 4)/10 + 2, \\ C(t_1^2 t_2^2) &= \frac{3}{70}(u^4 - 32u^2 + 72) + \frac{6}{5}(u^2 - 4) + 4. \end{aligned}$$

Indeed the respective coefficients are $2^2/(4)_2$, $(2)_2/(4)_2$, and $(2)_2^2/(4)_4$. Expanding $(1 + it_1/2)(1 + it_2/2)$ and $(1 + t_1^2/4)(1 + t_2^2/4)$ using these identities gives the first two lines of (GP.28). Subtracting the squared modulus gives its last line. This is the complete complex subtraction, including its imaginary linear coefficient.

For the auxiliary annihilator $\chi(S) = (S - 1)^2 - 2$, use the original S monomials at $N = 1$, where the boundary is empty. The sum gamma moments of orders zero, two, four, six are respectively $1/4, 1, 14, 376$, obtained by differentiating $\frac{1}{4} \cos^{-4} \theta$. Substitution in (GP.18) gives

$$\begin{aligned} G_h &= \begin{pmatrix} 9/16 & 9/16 \\ 9/16 & 81/16 \end{pmatrix}, \quad G_{\text{coh}} = \begin{pmatrix} 21/40 & 21/40 \\ 21/40 & 189/40 \end{pmatrix}, \\ G_h - G_{\text{coh}} &= \begin{pmatrix} 3/80 & 3/80 \\ 3/80 & 27/80 \end{pmatrix}, \\ \det G_h &= 81/32, \quad \det G_{\text{coh}} = 441/200, \quad \det(G_h - G_{\text{coh}}) = 9/800. \end{aligned} \tag{GP.29}$$

For example $\int b_2 r_{1,2} = 9/16$ and $\int u^2 b_2 r_{1,2} = 9/2$, while the corresponding coherent moments are $21/40$ and $21/5$. Odd moments vanish, so the original off-diagonal S entries equal the masses. These calculations prove every entry above. They exhibit the strictly positive contribution in the unchanged quotient coordinates.

The accompanying checker additionally derives moments from the Laplace function, solves the independent product-moment equations, and checks all moments through degree eight and the section/quotient identities through $N = 4$. It uses a separate two-atom conditional fibre of original masses two and three to test the complex adjoint and both projection traces. Its mathematical controls change a conjugate transpose, omit an original mass factor, or omit the relative quotient term. Those finite checks supplement the proofs above and do not estimate an actual growing-degree zeta quotient.

41.8 Provenance and consequence

The source constants, packet, conditional density, and coefficient addition identity come from the complete delivered Gamma Convolution Descent note, equations (2), (5)–(19), (24)–(27), and (32). The original unit and theta relation primitive come from the retained cyclic source construction and TVB in the cumulative reader. The classical generating function is [NIST DLMF 18.23.7](#), with $b_n^{(\lambda)}(t) = n!P_n^{(\lambda)}(t/2; \pi/2)$. The exact specialization and norms are proved in TG.7–TG.14. The bounded amplitude is used here to prove the full phase transport, fibre projection formulas, strict quartet variance, complex coefficient transform, and two-term positive quotient identity. These equations provide additional exact data for the same Toda quotient. They do not give a growing-degree estimate for its consecutive volume ratio. That ratio remains the actual determinant expression of the retained metric, now with the explicit contribution (GP.21)–(GP.23).

42 Exact endpoint windows: Gamma coefficients, paired Schur volumes, and intrinsic sensitivity

This continuation evaluates the original endpoint quantity of the complete written criterion at source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f of PR 23. The source criterion and the supplied *Endpoint degree selection in the actual gamma-corrected coordinates* were read completely, together with GC.1–59 and GMT.1–42, including GMT.29a–b. The present work proves the arbitrary-window Schur map, its exact determinant cancellations, a finite rational interval certificate, and a new sensitivity estimate through the actual angles between canonical least-lift spaces. The finite input list grows through degree $2(n+r)$. Every statement below is finite. No estimate for that growing list or its arithmetic conditioning is presumed.

42.1 The original source and the complete finite input map

Let $h(s)$ be the given monic packet of actual zeros, with their complete orders, of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s).$$

Retain its entire quotient $v_h = g/h$ and original positive density

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (\text{EW.1})$$

The complete analytic construction in GC.2–9 proves that w_h has every polynomial moment and exponential moments of every order less than $\pi/2$. Its mechanism is the theta seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, the equality $\mathcal{M}\Theta\phi_0 = g$, contour rotation through every angle $|\beta| < \pi/4$, and exact local division by h . In particular $|v_h(\sigma + it)|$ is bounded, uniformly for σ in each fixed compact real interval, by $C_{j,\beta,h}(1 + |t|)^{-j}e^{-\beta|t|}$ for every $j \geq 0$ and $0 < \beta < \pi/4$. These proofs and their original constants are retained in the accompanying complete GC source. The zero set of $v_h(1/2 + it)$ is discrete, so $w_h > 0$ almost everywhere. Its convolution has positive mass μ_h^k .

Throughout this paper

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k, \quad (EW.2)$$

$$\langle P, Q \rangle_H = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du.$$

The pairing is conjugate-linear in its first argument. The sum chart $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Fubini therefore proves both the convolution in (EW.1) and the full mass μ_h^k , with no average coordinate or mass division. Every nonzero polynomial has finitely many real-line zeros, so its squared norm in (EW.2) is strictly positive.

Fix the original monic relation $\chi \in \mathbb{C}[S]$, of degree $q \geq 1$. For the packet source it generates the exact intersection $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ under $S = \sum s_i$. Define $\mathcal{C} = \mathbb{C}[S]/(\chi)$ in the ordered coordinates $1, S, \dots, S^{q-1}$. The quotient map is the actual remainder map. Its injection into the tensor arithmetic jets is

$$\eta : \mathcal{C} \longrightarrow (\mathbb{C}[s]/(h))^{\otimes k}, \quad [P] \longmapsto [v_h]_h^{\otimes k} P(s_1 + \cdots + s_k). \quad (EW.3)$$

Complete local cancellation makes the constant term of v_h nonzero at every selected centre. Each local truncated power series is invertible by recursively solving its product with an inverse to equal one. The Chinese remainder isomorphism then proves that $[v_h]_h$ is a unit. The unfactored polynomial map $\mathbb{C}[S] \rightarrow (\mathbb{C}[s]/(h))^{\otimes k}$ in (EW.3) consequently has kernel equal to the stated intersection ideal (χ) . Factoring by that ideal gives the displayed η with kernel zero, proving injectivity without discarding a jet.

For $N \geq q - 1$, put $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{P}_{-1} = 0$. Let H_N be its source matrix in the original monomials. Let $J_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}^q$ be monic remainder and $B_N : \mathbb{C}^{N-q+1} \rightarrow \mathbb{C}^{N+1}$ be multiplication by χ . Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0. \quad (EW.4)$$

In particular B_N is injective, J_N is surjective, and $\ker J_N = \text{im } B_N$.

For later coefficient evaluation keep all Gamma constants:

$$\lambda > 0, \quad \alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (EW.5)$$

Let the real monic polynomials $b_j^{(\lambda)}$ be specified by

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 + z^2)^{-\lambda} e^{t \arctan z}, \quad |z| < 1, \quad b_{j+1} = t b_j - j(j + \alpha - 1) b_{j-1}, \quad (EW.6)$$

where the logarithms defining the generating function are the branches from zero, $b_0 = 1$, and $b_1 = t$. Differentiation of the displayed generating function proves the recurrence and monicity. The beta integral and Fourier inversion give the literal convolution mass and Laplace transform

$$r_{\lambda,k} = r_{\lambda}^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \int e^{\theta u} r_{\lambda,k}(u) du = C_k (\cos \theta)^{-\beta}, \quad |\Re \theta| < \pi/2. \quad (EW.7)$$

Their complete beta-integral proof with the changes $t = 2v$ and $u = \sum t_i$ is GC.10–15. In particular the reference monic norms at $\theta = 0$ are $C_k j! (\beta)_j$. They also follow directly by integrating the product of two functions (EW.6), which gives $C_k (1 - zw)^{-\beta}$, and comparing its finite derivatives.

Define the given one-factor coefficients and their finite powers by

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_\lambda j! (\alpha)_j}, \quad d_l = \sum_{j_1 + \cdots + j_k = l} \prod_{a=1}^k (\alpha)_{j_a} c_{h,j_a}. \quad (EW.8)$$

Multiplying the k generating functions (EW.6) and comparing the coefficient of z^l gives the finite polynomial identity

$$b_l^{(k\lambda)}(\sum t_i) = \sum_{j_1+\dots+j_k=l} \frac{l!}{\prod j_i!} \prod_i b_{j_i}^{(\lambda)}(t_i).$$

Integrating every term against the full product measure proves

$$\int b_l^{(k\lambda)}(u) m_{h,k}(u) du = C_k l! d_l. \quad (\text{EW.9})$$

All sums here are finite and all their terms are integrable. For any specified polynomials $p_a \in \mathcal{P}_M$, perform monic polynomial subtraction in the actual u variable to find the unique coefficients $L_{ab,l}$ in

$$\overline{p_a(c+iu)} p_b(c+iu) = \sum_{l=0}^{2M} L_{ab,l} b_l^{(k\lambda)}(u).$$

The complete coefficient-to-Gram map is therefore

$$H(p)_{ab} = C_k \sum_{l=0}^{2M} l! L_{ab,l} d_l. \quad (\text{EW.10})$$

It is a degree- k real polynomial map from the actual coefficient vector to Hermitian forms. The precise tests contain $c = k/2$, the two phases $\pm i$, and, when a relation column is used, both $\chi(c+iu)$ and $\overline{\chi}(c-iu)$. In particular (EW.10) constructs H_M from $c_{h,0}, \dots, c_{h,2M}$ and the stated k, λ . At $l = 0$, $c_{h,0} = \mu_h/c_\lambda$, so the same map gives $H_{00} = \mu_h^k$.

42.2 Canonical lifts and one arbitrary-window block

For each admitted degree define the exact matrices and maps

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : \mathbb{C}^q \longrightarrow \mathbb{C}^{N+1}, \quad V_N = \det G_N. \quad (\text{EW.11})$$

Positivity of H_N and surjectivity of J_N prove $K_N > 0$. Multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. For every $x \in \mathbb{C}^q$, any lift is $R_N x + B_N y$, and its norm is exactly $x^* G_N x + y^* B_N^* H_N B_N y$. This proves the unique minimum property and $R_N^* H_N R_N = G_N$.

Let E_N include the first q monomials. The square coefficient matrix $[E_N, B_N]$ has determinant one: its columns are ordered by successive degree and each leading coefficient is one. Subtracting boundary columns to replace E_N by R_N retains determinant one. Its Gram is block diagonal, so

$$V_N = \frac{\det H_N}{\det(B_N^* H_N B_N)}. \quad (\text{EW.12})$$

Empty determinants are one. This fixes the relation determinant and its index $N - q + 1$ in the original quotient coordinates.

Take integers

$$n \geq q, \quad r \geq 1, \quad m = n + r, \quad d = r + 1. \quad (\text{EW.13})$$

In $\mathcal{P}_m$ split the ordered monomials into degrees $0, \dots, n-1$ and the d appended columns $S^n, \dots, S^m$. Write

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad J_m = [J_{n-1}, J_D], \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EW.14})$$

The map

$$\Psi : \mathbb{C}^n \oplus \mathbb{C}^d \longrightarrow \mathbb{C}^{m+1}, \quad (x, y) \longmapsto (x - H_{n-1}^{-1} C_D y, y) \quad (\text{EW.15})$$

is invertible with determinant one. Direct multiplication gives $\Psi^* H_m \Psi = \text{diag}(H_{n-1}, W)$ and $J_m \Psi = [J_{n-1}, F]$. Thus $W > 0$; F maps the appended orthogonal residual polynomials to their original quotient classes. No quotient is independently chosen.

Let W_j be the first j rows and columns of W , F_j its first j quotient columns, and $Z_j = F_j^* G_{n-1} F_j$ for $1 \leq j \leq d$. The same map (EW.15), restricted to these columns, gives the exact positive-kernel update

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*. \quad (\text{EW.16})$$

Eliminating the two diagonal blocks of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in the two orders proves $\det(I + AB) = \det(I + BA)$. Use this identity in (EW.16) to obtain

$$\mathcal{R}_j := \frac{V_{n-1}}{V_{n+j-1}} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad \mathcal{R}_0 = 1. \quad (\text{EW.17})$$

All the determinants on the right are of size j , whereas its equivalent kernel calculation has size q . The matrices Z_j are positive semidefinite. Diagonalizing $W_j^{-1/2} Z_j W_j^{-1/2}$ proves $\mathcal{R}_j \geq 1$. Nested minimization also proves $G_{N+1} \preceq G_N$, so $\mathcal{R}_j$ is nondecreasing in j .

Let $p_N(S)$ be the unique monic polynomial of degree N orthogonal to $\mathcal{P}_{N-1}$, and let ω_N be its original squared norm. Existence and uniqueness follow by solving the positive lower-degree projection problem; its leading coefficient remains one. The first diagonal entry $w = W_1$ is ω_n . Eliminating the first r residual columns from the final one gives $\omega_m = \det W_d / \det W_r$. Consequently

$$U := \frac{\omega_m}{\omega_n} = \frac{\det W_d}{\det W_r w}, \quad \mathcal{R} := \frac{V_{n-1} V_n}{V_{m-1} V_m} = \frac{\mathcal{R}_r \mathcal{R}_d}{\mathcal{R}_1} \geq 1. \quad (\text{EW.18})$$

The last inequality also follows by multiplying $V_{n-1}/V_{m-1} \geq 1$ and $V_n/V_m \geq 1$. Both numerator and denominator norms carry their literal masses.

For the Gamma source (EW.7), the same J_m and χ give matrices W^Γ , F^Γ , Z_j^Γ by the identical maps. For their original source determinants put $\mathfrak{D}_j = \det H_{j-1}$ for $j \geq 1$ and $\mathfrak{D}_0 = 1$. For the relation determinant on exactly j columns put $\mathfrak{B}_j = \det(B_{q+j-1}^* H_{q+j-1} B_{q+j-1})$ for $j \geq 1$ and $\mathfrak{B}_0 = 1$. The reference determinants use the same columns and carry the superscript Γ . Writing the supplied corrections as $X_j = \mathfrak{D}_j / \mathfrak{D}_j^\Gamma$, $Y_j = \mathfrak{B}_j / \mathfrak{B}_j^\Gamma$ and $T_N = X_{N+1} / Y_{N-q+1}$, the relation (EW.12) proves

$$U = (n+1)_r (n+\beta)_r \frac{X_{m+1} X_n}{X_m X_{n+1}}, \quad (\text{EW.19})$$

$$\mathcal{R} = \frac{V_{n-1}^\Gamma V_n^\Gamma}{V_{m-1}^\Gamma V_m^\Gamma} \frac{T_{n-1} T_n}{T_{m-1} T_m}.$$

Indeed $\omega_j = C_k j! (\beta)_j X_{j+1} / X_j$ follows by taking successive source determinants in the original monic basis. Taking the ratio gives the first line with its exact rising factorials. The new symbol $\mathcal{R}_j$ in (EW.17) has no effect on the original positive arithmetic correction T_N .

42.3 Endpoint roots and exact paired-volume cancellations

Put $a = 1/(2r)$. The original endpoint expression is

$$\mathcal{E}_{n,r} = U^a \sinh(a \log \mathcal{R}) = \frac{(U\mathcal{R})^a - (U/\mathcal{R})^a}{2}. \quad (\text{EW.20})$$

All roots are the unique positive real roots. The equality follows from the definition of the real hyperbolic sine and positivity. It evaluates the existing quantity, preserving its two original norm endpoints and all four original volume endpoints.

There is a further exact cancellation inside its two root arguments. Set

$$p = \det W_r, \quad s = \det W_d, \quad e = w + Z_1, \quad D = \det(W_r + Z_r) \det(W_d + Z_d).$$

The scalar Z_1 is real nonnegative, and all four quantities p, s, e, D are strictly positive. Substituting (EW.17)–(EW.18) without changing a coordinate gives

$$U = \frac{s}{pw}, \quad \mathcal{R} = \frac{Dw}{pse}, \quad \boxed{U\mathcal{R} = \frac{D}{p^2e}, \quad \frac{U}{\mathcal{R}} = \frac{s^2e}{w^2D}}. \quad (\text{EW.21})$$

Thus the common highest source determinant cancels from the first argument; the common prefix source determinant cancels from the second. The first and last original monic norms have not been set equal or removed from the underlying source. Formula (EW.21) is the exact rational map explaining these cancellations.

It is sometimes useful to retain the small difference in (EW.20) as one positive numerator. With $P = (U\mathcal{R})^a$ and $Q = (U/\mathcal{R})^a$, the finite factorization $P^{2r} - Q^{2r} = (P - Q) \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j$ proves

$$\boxed{\mathcal{E}_{n,r} = \frac{U(\mathcal{R}^2 - 1)}{2^{2r-1} \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j}}. \quad (\text{EW.22})$$

The denominator remains positive at $\mathcal{R} = 1$, and the numerator is then exactly zero. This representation preserves the sign of the endpoint quantity and avoids subtracting two enclosures before their common positive factor has been identified.

Every entry in (EW.14) is determined by (EW.10) through degree $2m = 2(n + r)$. Each J_N and B_N is computed by monic division or multiplication by the same χ . Equations (EW.11)–(EW.22) therefore prove the complete finite map

$$\boxed{(c_{h,0}, \dots, c_{h,2(n+r)}; k, \lambda, \chi) \mapsto (H_m, J_m, W, F, G_{n-1}) \mapsto \mathcal{E}_{n,r}}. \quad (\text{EW.23})$$

It consists of stated polynomial operations, inverses of matrices whose positivity has been proved, determinants, and positive roots.

For reference the endpoint selection follows from the original degree-wise estimate by a finite argument. Write $x_j = \frac{1}{2} \log(V_{n+j-1}/V_{n+j+1}) \geq 0$ and $A_j = \sqrt{\omega_{n+j+1}/\omega_{n+j}}$. The local estimate is $0 \leq \epsilon_{n+j} \leq A_j \sinh x_j$. If an x_j is zero, the minimum allowance is zero. Otherwise $(\log \sinh x)'' = -1/\sinh^2 x < 0$ proves finite Jensen, hence $\prod_j \sinh x_j \leq \sinh(\sum_j x_j/r)^r$. The actual products give $\prod_j A_j = \sqrt{U}$ and $2 \sum_j x_j = \log \mathcal{R}$. Multiplying the local estimates and taking the positive r th root therefore proves

$$\min_{0 \leq j < r} \epsilon_{n+j} \leq \mathcal{E}_{n,r}. \quad (\text{EW.24})$$

The local arithmetic trace inequality and its retained loss terms are proved in the complete source TVB.32–34a; (EW.24) is their exact finite selection, not an added local estimate or an asymptotic claim.

42.4 Exact projection between the canonical representative spaces

The following result supplies more information than the separate determinant values. Use the common ambient space $(\mathcal{P}_m, \langle \cdot, \cdot \rangle_H)$, with its original source inner product. Let $\iota_j : \mathbb{C}^{j+1} \rightarrow \mathbb{C}^{m+1}$ be monomial inclusion. The least-lift space and its exact isometry are

$$L_j = \iota_j R_j \mathbb{C}^q \subset \mathcal{P}_m, \quad \mathcal{J}_j : (\mathbb{C}^q, G_j) \longrightarrow L_j, \quad x \longmapsto \iota_j R_j x. \quad (\text{EW.25})$$

Here $\mathcal{J}_j^* H_m \mathcal{J}_j = G_j$ proves the word isometry; its inverse on L_j is the original remainder map. The original arithmetic correction T_N retains its meaning in (EW.19).

Theorem 42.1 (Canonical projection and all its moving directions). *For $q - 1 \leq a < b \leq m$, let $\Pi_b : \mathcal{P}_m \rightarrow \mathcal{P}_m$ be the ambient orthogonal projection with range L_b , including that range by its fixed polynomial inclusion. It obeys*

$$\boxed{\Pi_b \iota_a R_a x = \iota_b R_b x \quad (x \in \mathbb{C}^q)}. \quad (\text{EW.26})$$

Its restriction $L_a \rightarrow L_b$ is invertible. Its squared singular values, with precisely the source inner products in (EW.25), are the eigenvalues σ_i^2 of

$$G_a^{-1/2} G_b G_a^{-1/2}, \quad 0 < \sigma_i \leq 1. \quad (\text{EW.27})$$

At most $s_{a,b} = \min(q, b - a)$ of these values are less than one, and

$$\begin{aligned} \prod_i \sigma_i^2 &= V_b / V_a, \\ \|\Pi_a - \Pi_b\|_1 &= 2 \sum_i \sqrt{1 - \sigma_i^2}, \\ \|\Pi_a - \Pi_b\|_1 &\leq \min\{2\sqrt{s_{a,b} \log(V_a/V_b)}, 2s_{a,b}\}. \end{aligned} \quad (\text{EW.28})$$

The trace norm is for operators in the stated ambient inner product.

Proof. Both representatives in (EW.26) have remainder x . Their difference belongs to $\chi\mathcal{P}_{b-q}$ and is therefore orthogonal to L_b by (EW.11). The second representative belongs to L_b ; this proves the orthogonal projection identity. The same identity and injectivity of each lift show that the restriction is invertible. In the quotient coordinates it is the identity map from metric G_a to metric G_b . The explicit maps from Euclidean coordinates are $u \mapsto \iota_a R_a G_a^{-1/2} u$ and $v \mapsto \iota_b R_b G_b^{-1/2} v$. Their matrices give the squared singular values (EW.27), and no original form or mass has been changed. A projection is a contraction, so these values lie in $(0, 1]$. Taking its determinant proves the first line of (EW.28).

The boundary space at degree b extends the one at degree a by exactly $b - a$ independent monic columns. Vectors in L_a are already orthogonal to the older boundary. Imposing orthogonality to these extra $b - a$ columns has rank at most $b - a$. Its kernel is $L_a \cap L_b$: any vector in that kernel is in $\mathcal{P}_b$ and orthogonal to its full boundary, hence is a minimum representative. The intersection therefore has dimension at least $q - (b - a)$ when this number is positive. A unit vector in L_a is unchanged in norm by Π_b precisely when it belongs to L_b , by orthogonal Pythagoras. Thus at most $s_{a,b}$ singular values are less than one.

For a complete trace-norm calculation choose orthonormal eigenvectors $x_i \in L_a$ for the squared restriction and put $y_i = \Pi_b x_i / \sigma_i \in L_b$. Orthogonality gives $\langle x_i, y_j \rangle = \sigma_j \delta_{ij}$ and the y_i form an orthonormal basis of L_b . If $\sigma_i = 1$, then $y_i = x_i$. If $\sigma_i < 1$, set $z_i = (y_i - \sigma_i x_i) / \sqrt{1 - \sigma_i^2}$. Directly taking all these inner products proves that the planes $\text{span}(x_i, z_i)$ are mutually orthogonal and orthogonal to the common directions. In such a plane the matrix of $\Pi_a - \Pi_b$ is exactly

$$\begin{pmatrix} 1 - \sigma_i^2 & -\sigma_i \sqrt{1 - \sigma_i^2} \\ -\sigma_i \sqrt{1 - \sigma_i^2} & -(1 - \sigma_i^2) \end{pmatrix}.$$

Its eigenvalues are $\pm \sqrt{1 - \sigma_i^2}$. On the common directions and the orthogonal complement of $L_a + L_b$ the difference is zero. This proves the exact trace-norm sum. Finally $1 - x \leq -\log x$ for $0 < x \leq 1$, because the derivative of $x - 1 - \log x$ is $1 - 1/x$ and its value at one is zero. Cauchy-Schwarz on the at most $s_{a,b}$ nonzero terms gives the first bound in (EW.28); bounding each square root by one gives the second bound independently. $\square$

42.5 Window sensitivity and exact cancellation of mass changes

The next calculation holds for every continuously differentiable family of positive source matrices $H_m(t)$, using all their principal submatrices and the fixed χ, J_j . Thus it applies both to the actual coefficient forms and to a segment between their certified finite enclosures. Write $E = \dot{H}_m$ and define the real spectral spread

$$\mathfrak{o}_H(E) = \lambda_{\max}(H_m^{-1/2} E H_m^{-1/2}) - \lambda_{\min}(H_m^{-1/2} E H_m^{-1/2}) \geq 0. \quad (\text{EW.29})$$

The two congruences in this expression specify its exact coordinate map; the source H_m remains in every evaluation.

Embed the coefficient vector of p_j in $\mathcal{P}_m$ and set

$$P_j = \frac{\iota_j p_j p_j^* \iota_j^*}{\omega_j}, \quad Q_j = \iota_j R_j G_j^{-1} R_j^* \iota_j^*. \quad (\text{EW.30})$$

These are contravariant matrices; their products $P_j H_m$ and $Q_j H_m$ are respectively the ambient orthogonal projections onto the monic orthogonal line and L_j . To verify this, multiply each by itself and use $p_j^* H_j p_j = \omega_j$ or $R_j^* H_j R_j = G_j$. The resulting operators are idempotent and self-adjoint in H_m and have the stated ranges. The matrices $H_m^{1/2} P_j H_m^{1/2}$ and $H_m^{1/2} Q_j H_m^{1/2}$ are their exact Euclidean isometric conjugates.

Differentiating the norm of the monic polynomial gives $\dot{\omega}_j = p_j^* \dot{H}_j p_j$: its derivative has degree less than j , and both cross terms vanish by its original orthogonality. Differentiating $J_j R_j = I$ makes $\dot{R}_j$ a boundary, so the same expansion gives $\dot{G}_j = R_j^* \dot{H}_j R_j$. The determinant derivative follows by expanding the determinant multilinearly in its columns, or by differentiating $\det(G_j + t \dot{G}_j) = \det G_j \det(I + t G_j^{-1} \dot{G}_j)$. Thus

$$\frac{d}{dt} \log \omega_j = \text{Tr}(P_j E), \quad \frac{d}{dt} \log V_j = \text{Tr}(Q_j E). \quad (\text{EW.31})$$

All these formulas use the original coordinates before the final cyclic trace.

Put $\mathcal{L} = \log \mathcal{R} \geq 0$ and $s = \min(q, r)$. The endpoint derivatives are exactly

$$\begin{aligned} (\log U)' &= \text{Tr}((P_m - P_n)E), \\ \mathcal{L}' &= \text{Tr}((Q_{n-1} + Q_n - Q_{m-1} - Q_m)E), \\ \mathcal{E}'_{n,r} &= \frac{1}{2r} \left\{ \mathcal{E}_{n,r} (\log U)' + U^{1/(2r)} \cosh(\mathcal{L}/(2r)) \mathcal{L}' \right\}. \end{aligned} \quad (\text{EW.32})$$

These equalities also hold at $\mathcal{R} = 1$. They identify the exact paired sensitivity rather than separate errors of large source and relation determinants.

Theorem 42.2 (Intrinsic finite sensitivity). *Throughout every such positive continuously differentiable family,*

$$|(\log U)'| \leq \mathfrak{o}_H(E), \quad |\mathcal{L}'| \leq \mathfrak{o}_H(E) \min\{2s, \sqrt{2s\mathcal{L}}\}. \quad (\text{EW.33})$$

For any two positive source matrices $H^{(0)}, H^{(1)}$ in the same original coordinates, let

$$\kappa = \frac{\lambda_{\max}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})}{\lambda_{\min}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})} \geq 1. \quad (\text{EW.34})$$

Their original window values satisfy

$$|\log(U^{(1)}/U^{(0)})| \leq \log \kappa, \quad |\sqrt{\mathcal{L}^{(1)}} - \sqrt{\mathcal{L}^{(0)}}| \leq \sqrt{s/2} \log \kappa. \quad (\text{EW.35})$$

In particular these sensitivity bounds contain no independent source-size determinant exponent. They vanish for any scalar change $H^{(1)} = tH^{(0)}$, $t > 0$, exactly as the original endpoint does.

Proof. Let $\Lambda = H_m^{-1/2} E H_m^{-1/2}$. Subtracting $\frac{1}{2}(\lambda_{\max}\Lambda + \lambda_{\min}\Lambda)I$ leaves operator norm $\mathfrak{o}_H(E)/2$. This scalar subtraction does not change either trace in (EW.32): the two monic-line projections each have trace one and the four least-lift projections each have trace q , with the displayed signs summing to zero.

The two monic orthogonal lines at degrees $n < m$ are orthogonal in the common ambient source. Their projection difference has eigenvalues 1, -1 and zeros, hence trace norm two. For any self-adjoint A and B , diagonalizing A gives $|\mathrm{Tr}(AB)| \leq \sum_i |\lambda_i(A)| \|B\|$. This proves the first bound in (EW.33).

For the second trace split its four projections as $(\Pi_{n-1} - \Pi_{m-1}) + (\Pi_n - \Pi_m)$. Both index gaps are exactly r . Put $\ell_1 = \log(V_{n-1}/V_{m-1})$ and $\ell_2 = \log(V_n/V_m)$, so $\ell_1 + \ell_2 = \mathcal{L}$. Equation (EW.28), the triangle inequality for the trace norm, and $\sqrt{\ell_1} + \sqrt{\ell_2} \leq \sqrt{2\mathcal{L}}$ give

$$\|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 2\sqrt{2s\mathcal{L}}, \quad \|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 4s.$$

Multiplying either bound by $\mathfrak{o}_H(E)/2$ proves the second bound of (EW.33).

For a continuously differentiable path and $\varepsilon > 0$, the exact bound $|\mathcal{L}'| \leq \mathfrak{o}_H(E)\sqrt{2s\mathcal{L}}$ implies

$$\left| \frac{d}{dt} \sqrt{\mathcal{L} + \varepsilon} \right| \leq \sqrt{s/2} \mathfrak{o}_H(E).$$

Integrate on the compact path and let ε decrease to zero. This proves the square-root Lipschitz assertion along a path, including points at which $\mathcal{L} = 0$; no division by zero is used.

Use the straight positive segment $H(t) = (1-t)H^{(0)} + tH^{(1)}$, $0 \leq t \leq 1$. Let $\ell_i > 0$ be the eigenvalues of $(H^{(0)})^{-1/2}H^{(1)}(H^{(0)})^{-1/2}$. Congruence by $(H^{(0)})^{1/2}$ and then its displayed orthonormal eigenbasis gives the generalized eigenvalues of $(E, H(t))$ exactly as

$$\frac{\ell_i - 1}{1 + t(\ell_i - 1)}.$$

These are the eigenvalues of $H(t)^{-1/2}EH(t)^{-1/2}$: both follow from the same equation $Ex = \lambda H(t)x$. The displayed function is increasing in ℓ_i , with derivative $[1 + t(\ell_i - 1)]^{-2}$. Hence

$$\int_0^1 \mathfrak{o}_{H(t)}(E) dt = \log \ell_{\max} - \log \ell_{\min} = \log \kappa.$$

Integrating the already proved differential inequalities proves (EW.35). Finally direct substitution in (EW.11) gives $R_N(tH) = R_N(H)$, $G_N(tH) = tG_N(H)$, and $\omega_j(tH) = t\omega_j(H)$. Thus U and $\mathcal{R}$ are exactly unchanged by that explicitly stated common scaling. The source mass remains $t\mu_h^k$ in the scaled comparison; it was never reassigned the value one. $\square$

The independent projection review supplied a stronger exact version from the same angles. For $s \geq 1$, it is

$$|\mathcal{L}'| \leq 2s \mathfrak{o}_H(E) \sqrt{1 - e^{-\mathcal{L}/(2s)}}. \quad (\text{EW.35a})$$

To prove it, list $x_i = -\log \sigma_i^2$ for each of the two projection pairs used above, omitting their zero terms and then appending zeros to make exactly $2s$ entries. Their sum is $\mathcal{L}$. The exact trace-norm formula and the scalar cancellation prove $|\mathcal{L}'| \leq \mathfrak{o}_H(E) \sum_i f(x_i)$, where $f(x) = \sqrt{1 - e^{-x}}$. For $x > 0$,

$$f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} < 0.$$

Continuity at zero extends concavity there: apply the concavity inequality to positive arguments increased by ε and then let $\varepsilon \downarrow 0$. Finite Jensen on the full $2s$ -term list proves (EW.35a). Since $1 - e^{-x} \leq x$ and $1 - e^{-x} \leq 1$, it implies both bounds on $|\mathcal{L}'|$ in (EW.33).

Define the explicit increasing map on the actual loss variable by

$$\Phi_s : [0, \infty) \longrightarrow [0, \infty), \quad \Phi_s(L) = 2 \operatorname{arcosh}(e^{L/(4s)}), \quad \Phi_s^{-1}(v) = 4s \log \cosh(v/2).$$

Here $\operatorname{arcosh} x = \log(x + \sqrt{x^2 - 1})$ for $x \geq 1$, so both maps and their inverses have specified real branches. For $L > 0$, $\Phi'_s(L) = [2s\sqrt{1 - e^{-L/(2s)}}]^{-1}$. Apply (EW.35a) to $\Phi_s(\mathcal{L} + \varepsilon)$. Its derivative in absolute value is at most $\mathfrak{o}_H(E) \sqrt{(1 - e^{-\mathcal{L}/(2s)})/(1 - e^{-(\mathcal{L}+\varepsilon)/(2s)})} \leq \mathfrak{o}_H(E)$. Integrating and letting $\varepsilon \downarrow 0$, then using the exact straight-segment length already proved, gives

$$|\Phi_s(\mathcal{L}^{(1)}) - \Phi_s(\mathcal{L}^{(0)})| \leq \log \kappa. \quad (\text{EW.35b})$$

This conclusion includes zero losses by continuity. For $s = 0$, the only loss is zero and define the one-point map $\Phi_0(0) = 0$; the projection and endpoint comparisons are their zero identities.

For $s \geq 1$, the stronger enclosure itself has an exact algebraic form. Given any $\rho \geq \kappa$ and the original $\mathcal{R}^{(0)} \geq 1$, put

$$z_0 = (\mathcal{R}^{(0)})^{1/(4s)} + \sqrt{(\mathcal{R}^{(0)})^{1/(2s)} - 1}, \quad z_- = \max\{1, z_0/\sqrt{\rho}\}, \quad z_+ = \sqrt{\rho} z_0, \quad (\text{EW.35c})$$

$$\left(\frac{z_- + z_-^{-1}}{2} \right)^{4s} \leq \mathcal{R}^{(1)} \leq \left(\frac{z_+ + z_+^{-1}}{2} \right)^{4s}.$$

Indeed $z_0 = \exp(\Phi_s(\mathcal{L}^{(0)})/2)$, (EW.35b) encloses its counterpart between z_- and z_+ , and $z \mapsto (z + z^{-1})/2$ is increasing for $z \geq 1$. The identity $\mathcal{R} = \cosh(\Phi_s(\log \mathcal{R})/2)^{4s}$ proves (EW.35c). At the same time $U^{(0)}/\rho \leq U^{(1)} \leq \rho U^{(0)}$ by (EW.35). Thus monotonicity in (EW.20) supplies an endpoint enclosure using only nonnegative real roots, rational operations and integer powers when the central $U, \mathcal{R}$ and a bound ρ are rational. For example any proved rational Loewner comparison $aH^{(0)} \preceq H^{(1)} \preceq bH^{(0)}$, $0 < a \leq b$, gives $\rho = b/a$: the extreme Rayleigh quotients lie between a and b . This is the exact map from a positive-matrix certificate to the sharper window certificate, with the original mass and relation unchanged.

The square-root assertion is a Lipschitz assertion, and does not assert ordinary differentiability at every zero of $\mathcal{L}$. For an exact auxiliary illustration take $q = 1$, $\chi = S$, $n = 1$, $r = 1$, and the positive original-coordinate matrix

$$H_2(t) = \begin{pmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ t & 0 & 1 \end{pmatrix}, \quad |t| < 1.$$

Its leading pivots are $1, 1, 1 - t^2$, its remainder is evaluation at $S = 0$, and (EW.11) gives $V_0 = V_1 = 1$, $V_2 = 1 - t^2$. Thus $\mathcal{L} = -\log(1 - t^2)$ and $\sqrt{\mathcal{L}}/|t| \rightarrow 1$ as $t \rightarrow 0$. Its square root has a corner. This positive-form example belongs exactly to the finite family domain of (EW.33)–(EW.35); it is not claimed to be an arithmetic moment source. The regularized proof above includes it.

For completeness, (EW.35) has a directly usable endpoint enclosure. Writing $d_H = \log \kappa$ and $b = \sqrt{s/2} d_H$, it proves

$$e^{-d_H} U^{(0)} \leq U^{(1)} \leq e^{d_H} U^{(0)}, \quad (\text{EW.36})$$

$$(\max\{0, \sqrt{\mathcal{L}^{(0)}} - b\})^2 \leq \mathcal{L}^{(1)} \leq (\sqrt{\mathcal{L}^{(0)}} + b)^2.$$

The function $U^{1/(2r)} \sinh(\mathcal{L}/(2r))$ is increasing in each positive U and nonnegative $\mathcal{L}$, as direct differentiation shows. Substitution of the two endpoints in (EW.36) therefore bounds the actual endpoint expression. This is a finite stability theorem, including its exact $s = \min(q, r)$ dependence.

42.6 Rational interval certificates for the original input

Here is an explicit finite algorithm which uses enclosures of the original coefficients; it does not replace them by an arbitrary moment matrix. For clarity first suppose the constants in (EW.5) and the polynomial tests in (EW.10) are exact. Given real intervals $|c_{h,j} - \hat{c}_j| \leq e_j$ for $0 \leq j \leq 2m$, set

$$\begin{aligned}\hat{A}(z) &= \sum_{j=0}^{2m} (\alpha)_j \hat{c}_j z^j, \\ A_+(z) &= \sum_{j=0}^{2m} (\alpha)_j |\hat{c}_j| z^j, \quad E_+(z) = \sum_{j=0}^{2m} (\alpha)_j e_j z^j, \\ \hat{d}_l &= [z^l] \hat{A}(z)^k, \quad e_l^{(k)} = [z^l] \{(A_+(z) + E_+(z))^k - A_+(z)^k\}.\end{aligned}\tag{EW.37}$$

The multinomial expansion groups each difference of products by the nonempty set of factors carrying an error. Taking absolute values term by term proves $|d_l - \hat{d}_l| \leq e_l^{(k)}$. Thus the Hermitian central matrix and entry radii

$$\hat{H}_{ab} = C_k \sum_l l! L_{ab,l} \hat{d}_l, \quad \varepsilon_{ab} = C_k \sum_l l! |L_{ab,l}| e_l^{(k)}\tag{EW.38}$$

enclose the actual source. Rational upper bounds for the moduli give rational radii. An uncertain real constant, including C_k , is propagated by interval multiplication before the final matrix is reported. For $|x - \hat{x}| \leq e_x$ and $|y - \hat{y}| \leq e_y$, the exact elementary rule is $|xy - \hat{x}\hat{y}| \leq |\hat{x}|e_y + |\hat{y}|e_x + e_x e_y$. Repeated use handles every finite product in (EW.8)–(EW.10).

The original relation need not have rational coefficients. To retain it exactly while producing rational certificates, use its quotient-first coordinate map at each endpoint degree N :

$$A_N = [E_N, B_N] : \mathbb{C}^q \oplus \mathbb{C}^{N-q+1} \longrightarrow \mathbb{C}^{N+1}, \quad \det A_N = 1, \quad \tilde{H}_N = A_N^* H_N A_N.\tag{EW.39}$$

The first component of $A_N^{-1}P$ is exactly $J_N P$ by monic division. Supply rational enclosures of the real and imaginary parts of the actual χ coefficients when they are not exact rational numbers. Ordinary complex interval addition, multiplication and conjugation in (EW.39) give rational entry enclosures of the actual $\tilde{H}_N$. For a complex interval a modulus radius is bounded rationally, for example by the sum of its two coordinate radii. Conjugation changes the sign of the imaginary coordinate; it is applied to the actual first factor. This procedure never changes χ , its degree, its monicity or the original quotient coordinates. Intervals for λ and its positive constants, if required, are propagated in the same finite operations and in the specified recurrence (EW.6).

Here is a rational matrix certificate valid for either H_m or any of these $\tilde{H}_N$. Choose a Hermitian rational centre $\hat{M}$ and symmetric rational radii ε_{ab} enclosing its actual error. Choose any positive rational v_a and form

$$\Delta = \text{diag} \left(\sum_b \varepsilon_{ab} \frac{v_b}{v_a} \right), \quad M^- = \hat{M} - \Delta, \quad M^+ = \hat{M} + \Delta.\tag{EW.40}$$

The pairwise inequality $2|x_a||x_b| \leq (v_b/v_a)|x_a|^2 + (v_a/v_b)|x_b|^2$ proves $M^- \preceq M \preceq M^+$. Rational complex arithmetic now eliminates each leading pivot by the congruence

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \longmapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \longmapsto (x - a^{-1}b^*y, y).\tag{EW.41}$$

Recording strictly positive rational pivots proves $M^- > 0$ by induction and hence also $M^+ > 0$. A nonpositive pivot rejects that particular proposed certificate. Every actual matrix is already positive by its source integral and, for (EW.39), its invertible congruence. At a fixed finite degree, convergent valid entry intervals eventually give positive lower matrices: the least unit-sphere value of the

actual quadratic form is positive by compactness, and the finite sum of entry errors tends to zero. This proves the finite certificate's convergence without assuming or claiming that a growing-degree arithmetic input has already been enclosed.

For the two norm endpoints take the corresponding principal submatrices of the certified H_m^-, H_m^+ . Define ω_j^-, ω_j^+ by their exact monic Schur complements. The squared norm of any fixed monic polynomial is between its two enclosing forms. Minimizing over the same affine monic set proves

$$0 < \omega_j^- \leq \omega_j \leq \omega_j^+, \quad j = n, m. \quad (\text{EW.42})$$

For each of $N = n - 1, n, m - 1, m$, use (EW.39)–(EW.41) and split its first q and last $N - q + 1$ coordinates. The Schur complements of the latter block in $\tilde{H}_N^-, \tilde{H}_N^+$ give rational positive forms G_N^-, G_N^+ . Their affine fibre has fixed first coordinate x , by the exact map (EW.39). Minimization therefore proves

$$G_N^- \preceq G_N \preceq G_N^+, \quad 0 < V_N^- := \det G_N^- \leq V_N \leq \det G_N^+ =: V_N^+. \quad (\text{EW.43})$$

For the determinant inequality, conjugate by the square root of the smaller positive form and multiply the resulting eigenvalues, each at least one. At $N = q - 1$ the boundary block is empty and the Schur complement is the whole q -dimensional form.

All numbers in the following enclosure are rational and positive:

$$\begin{aligned} U_- &= \omega_m^- / \omega_n^+, & U_+ &= \omega_m^+ / \omega_n^-, \\ \mathcal{R}_- &= \max \left\{ 1, \frac{V_{n-1}^- V_n^-}{V_{m-1}^+ V_m^+} \right\}, & \mathcal{R}_+ &= \frac{V_{n-1}^+ V_n^+}{V_{m-1}^- V_m^-}. \end{aligned} \quad (\text{EW.44})$$

They satisfy $U_- \leq U \leq U_+$ and $1 \leq \mathcal{R}_- \leq \mathcal{R} \leq \mathcal{R}_+$. The use of the lower bound one follows from the proved nested minimization, rather than an interval rounding choice.

Positive roots require no floating logarithm certificate. For any positive rational x and integer $2r$, find rational numbers $l, u > 0$ such that $l^{2r} \leq x \leq u^{2r}$. Doubling upper or lower powers of two supplies an initial interval, and ordinary rational bisection makes it arbitrarily narrow. Each decision is a positive-denominator integer comparison. Strict monotonicity of $t \mapsto t^{2r}$ for $t > 0$ proves the enclosure and convergence. Apply this algorithm to

$$U_- \mathcal{R}_-, \quad U_- / \mathcal{R}_-, \quad U_+ \mathcal{R}_+, \quad U_+ / \mathcal{R}_+.$$

Write their certified root intervals as $[l_1, u_1], \dots, [l_4, u_4]$. The function in (EW.20) is increasing in $U > 0$ and $\mathcal{R} \geq 1$: its U derivative is its nonnegative value times $1/(2rU)$, and its $\mathcal{R}$ derivative is $U^{1/(2r)} \cosh(\log \mathcal{R} / (2r)) / (2r\mathcal{R}) > 0$. Therefore the complete rational endpoint certificate is

$$\boxed{\max\{0, (l_1 - u_2)/2\} \leq \mathcal{E}_{n,r} \leq (u_3 - l_4)/2.} \quad (\text{EW.45})$$

Alternatively (EW.22) gives a positive numerator and denominator certificate from the same root brackets. Every matrix pivot, integer power comparison, relation coefficient and source mass used in this algorithm is inspectable. It makes no claim for coefficient intervals which have not been provided.

42.7 Boundary cases and the retained cochain map

At $r = 1$, the block has two columns, $W_r = w$, and $\mathcal{R} = \mathcal{R}_2$ because the two copies of $\mathcal{R}_1$ cancel. Equations (EW.18)–(EW.22) reduce exactly to the two-degree formula GMT.24–27. At $n = q$, the base source degree is $q - 1$ and its boundary dimension is zero; its quotient metric is simply the source Gram. All formulas above still use positive matrices of their specified sizes. No degree $q - 2$ full-quotient form is inserted.

The actual positive polynomial source has an additional strictness property for $q \geq 1$: every two-degree quotient-volume ratio is strictly greater than one. Here is the complete algebraic reason. Let $\pi_j(u)$ be the real monic orthogonal family of the original positive real- u measure. Multiplication by u is self-adjoint. Pairing $u\pi_j$ with lower orthogonal polynomials and using their degrees gives

$$u\pi_j = \pi_{j+1} + b_j\pi_j + a_j\pi_{j-1}, \quad b_j \in \mathbb{R}, \quad a_j = \omega_j/\omega_{j-1} > 0 \quad (j \geq 1).$$

For $j = 0$ the last term is absent. The exact monic phase map $p_j(S) = i^j \pi_j((S - c)/i)$ has the same squared norm, because $|i^j| = 1$, and gives

$$Sp_j = p_{j+1} + (c + ib_j)p_j - a_j p_{j-1}.$$

If a nonconstant χ divided p_j and p_{j+1} , this recurrence and $a_j \neq 0$ would make it divide p_{j-1} . Induction would make it divide $p_0 = 1$, a contradiction. For the two appended columns at degrees $N, N+1$ over $\mathcal{P}_{N-1}$, the residuals are p_N and $p_{N+1} + zp_N$, where the original coefficient is exactly $z = \langle p_N, S^{N+1} \rangle_H / \omega_N$. Indeed their difference from those polynomials has smaller degree, and both residuals are orthogonal to $\mathcal{P}_{N-1}$. If their full quotient residual map F_2 were zero, both p_N and p_{N+1} would have zero remainder, contradicting the divisibility argument. Thus $Z_2 = F_2^* G_{N-1} F_2$ is a nonzero positive semidefinite matrix, and (EW.17) gives $V_{N-1}/V_{N+1} > 1$ for every $N \geq q$. The finite product

$$\mathcal{R} = \prod_{j=0}^{r-1} \frac{V_{n+j-1}}{V_{n+j+1}} > 1 \quad (q \geq 1, n \geq q, r \geq 1) \quad (\text{EW.46a})$$

is its exact overlapping telescope. This proves strict positivity of the actual $\mathcal{E}_{n,r}$ and of $\mathcal{L}$ at each admitted nonempty arithmetic window; it supplies no uniform lower bound as degrees or packets grow. The more general positive-form sensitivity theorem includes the zero-loss boundary separately.

The empty packet is a separate declared case: $h = 1$, $\chi = 1$, $q = 0$, and $\mathcal{C} = 0$. The analytic density and mass remain those of (EW.1). Put $V_N = 1$ even at the empty degree $N = -1$; all quotient matrices, projections and F_j have their empty dimensions. For $n \geq 0$, $r \geq 1$ the source norms exist, $\mathcal{R} = 1$, and $\mathcal{E}_{n,r} = 0$. If $n = 0$, use the empty base source H_{-1} , with determinant one and no inverse operation; the block $W = H_r$ then has its original monomial coordinates. Its first entry is the full mass. The projection and volume sensitivity statements become their zero-map identities, while the source-norm sensitivity remains the proved rank-two calculation.

Finally the projection comparison (EW.26) has an explicit original theta representative. For $a < b$ and $x \in \mathcal{C}$, monic division gives a unique polynomial $\kappa_x \in \mathcal{P}_{b-q}$ with

$$\iota_a R_a x - \iota_b R_b x = \chi \kappa_x. \quad (\text{EW.46})$$

Its coefficients are obtained by the same exact division, so the map $x \mapsto \kappa_x$ is linear. Let $\mathcal{V}_{h,k} P = P(D_1 + \cdots + D_k) F_h^{\otimes k}$, where $D = -x\partial_x$, $MF_h = v_h$, and $h(D)F_h = \Theta\phi_0$. Successive division in $s_1, \dots, s_k$ supplies specific Q_i with $\chi(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$. The final tensor remainder vanishes because its monomials form a basis of $(\mathbb{C}[s]/(h))^{\otimes k}$ and χ annihilates the original sum. Thus the exact cochain of degree $k-1$

$$\mathcal{P}_\chi(\kappa_x) = \sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) \kappa_x(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}) \quad (\text{EW.47})$$

satisfies

$$d_{\text{tot}} \mathcal{P}_\chi(\kappa_x) = \mathcal{V}_{h,k} (\iota_a R_a x - \iota_b R_b x). \quad (\text{EW.48})$$

Indeed the tensor differential at position i contributes $(-1)^{i-1}$ because its preceding factors have cochain degree one; this cancels the written sign, and $h(D_i)F_h = \Theta\phi_0$ then proves the equality term

by term. The arithmetic jet of both representatives is the same ηx of (EW.3). In the original support carrier $\{\tau\} \sqcup \bigsqcup_{\ell}(\{\ell\} \times E_{\ell})$, a fibrewise linear map sends (ℓ, v) to $(\ell, L_{\ell}v)$ and τ to τ . Thus the displayed boundary maps to $e_{\ell} = (\ell, 0)$ while external absence remains τ . The endpoint's representative changes, projection angles, and volume changes have therefore all been related to the original typed maps and their full theta primitives.

42.8 Verification scope and exact auxiliary calibration

The companion checker uses the explicitly auxiliary weight

$$\begin{aligned} \lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = 1/2, \quad C_2 = 1/4, \\ B(t) = 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = S^2 - 2S - 1. \end{aligned} \quad (\text{EW.49})$$

This is a specified positive $L^2(r_1)$ multiplier and the same finite coefficient-to-Gram domain as (EW.8)–(EW.10). It is not asserted to be an actual zeta packet. Its exact connection to the arithmetic problem is the complete sequence of maps (EW.10), (EW.11), (EW.14)–(EW.23), and (EW.39)–(EW.45), with its auxiliary inputs written explicitly.

For this fixture $b_2^{(1)} = t^2 - 2$, so $c_{B,0} = 3/2$, $c_{B,2} = 1/4$, and every other coefficient is zero. Consequently $\sum_j (2)_j c_{B,j} z^j = 3/2 + 3z^2/2$ and its square has coefficients $d_0 = 9/4$, $d_2 = 9/2$, $d_4 = 9/4$. Its one-factor mass is $3/4$, the literal sum mass is $9/16$, and the reference sum mass is $1/4$. The checker independently constructs Gamma moments from the monic recurrence, inserts $1 + t^2/4$ in each factor, and expands every power of $t_1 + t_2$ with its full binomial coefficient. It also constructs the coefficient-side moments from (EW.9); agreement compares two exact finite calculations, including the tensor cross terms.

In this auxiliary example gamma recurrence gives a further exact all-degree check. Since $|\Gamma(2 + it/2)|^2 = (1 + t^2/4)|\Gamma(1 + it/2)|^2$, one has $Br_1 = r_2$. The displayed masses give $c_2 = 3/4$, $c_4 = 315/8$, and hence

$$m = r_2 * r_2 = r_4/70, \quad r_1 * r_1 = r_2/3, \quad \omega_j = \frac{9}{16} j!(8)_j, \quad U = (n+1)_r(n+8)_r. \quad (\text{EW.50})$$

The first two equalities follow from the literal convolution law (EW.7), with coefficients $(3/4)^2/(315/8) = 1/70$ and $(1/2)^2/(3/4) = 1/3$. The monic original-coordinate map is $p_j(S) = i^j b_j^{(4)}((S-1)/i)$, whose squared norm is exactly the third quantity by (EW.7). Taking the stated endpoint ratio proves the final formula.

The exact selected windows computed in the accompanying rational receipt are

(n, r)	U	$\mathcal{R}$
(2, 1)	30	20191/4860
(2, 2)	1320	3073295611/216513000
(3, 2)	2640	295913127219469/36104825790000

(EW.51)

Here is a full independent finite derivation of the entries in that table, using the original quotient kernel rather than a Schur block. Let $x = S - 1$ solely in the stated quotient-coordinate map

$$A : \mathbb{C}_{(1,x)}^2 \longrightarrow \mathbb{C}_{(1,S)}^2, \quad (a, b) \longmapsto (a - b, b), \quad A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \det A = 1.$$

It takes the identical relation to $x^2 - 2$, so its residue equation is exactly $x^2 = 2$. It preserves the determinant of the quotient form by the congruence $G_N^x = A^* G_N A$. The monic source family in (EW.50) obeys $p_{j+1} = xp_j + j(j+7)p_{j-1}$, by applying the exact phase map to (EW.6). Orthogonal expansion of the original source inverse in (EW.11) proves the finite quotient kernel formula

$$(G_N^x)^{-1} = \sum_{j=0}^N \frac{[p_j]_{(1,x)} [p_j]_{(1,x)}^*}{\omega_j}. \quad (\text{EW.52})$$

Indeed the matrix of the monic orthogonal columns in the original source is invertible triangular with diagonal one, takes H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, and sends the reduction columns to their written remainders. Substituting that inverse congruence into $J_N H_N^{-1} J_N^*$ gives (EW.52) term by term. The recurrence and $x^2 = 2$ give the complete required list

j	$[p_j]_{(1,x)}$	ω_j
0	(1, 0)	9/16
1	(0, 1)	9/2
2	(10, 0)	81
3	(0, 28)	2430
4	(356, 0)	106920
5	(0, 1588)	6415200

(EW.53)

For example the successive remainders are obtained from $x^2 + 8 = 10, 10x + 18x = 28x, 28x^2 + 300 = 356$, and $356x + 44 \cdot 28x = 1588x$. The positive kernels from (EW.52) are diagonal in this explicitly transported quotient basis, with

N	$((G_N^x)^{-1})_{00}$	$((G_N^x)^{-1})_{11}$	V_N
1	16/9	2/9	81/32
2	244/81	2/9	729/488
3	244/81	662/1215	98415/161528
4	56102/13365	662/1215	16238475/37139524
5	56102/13365	376069/400950	2679348375/10549111519

(EW.54)

Each diagonal entry is the indicated finite sum of rational terms in (EW.52), and V_N is the reciprocal of their product because $\det A = 1$. Substituting (EW.53)–(EW.54) in (EW.18) gives all three rows of (EW.51) by positive-denominator multiplication.

There is also an executed coefficient-box certificate, before any matrix error is chosen. Put $\varepsilon = 10^{-12}$ and retain $n = 2, r = 1, m = 3$, the same reference mass and the same χ . Allow the complete specified one-factor input box

$$|c_{B,0} - 3/2| \leq \varepsilon, \quad |c_{B,2} - 1/4| \leq \varepsilon, \quad c_{B,j} = 0 \quad (j = 1, 3, 4, 5, 6). \quad (\text{EW.55})$$

Every point of this box is realized by the specified positive multiplier $B_{a,b}(t) = a + bt^2$, where $a = c_{B,0} - 2c_{B,2}$ and $b = c_{B,2}$. They obey $a \geq 1 - 3\varepsilon > 0$ and $b \geq 1/4 - \varepsilon > 0$. Its coefficient formula follows by $a + bt^2 = (a + 2b)b_0^{(1)} + bb_2^{(1)}$. The interior input $a = 1 + \varepsilon, b = 1/4 - \varepsilon/4$ has one-factor mass $3/4 + \varepsilon/4$ and sum mass $(3/4 + \varepsilon/4)^2$. Thus neither the box nor its test input replaces the zeroth coefficient by a unit mass.

For (EW.55), the complete coefficient-power error formula (EW.37) is exactly

$$\begin{aligned} e_0^{(2)} &= 3\varepsilon + \varepsilon^2, & e_2^{(2)} &= 21\varepsilon + 12\varepsilon^2, & e_4^{(2)} &= 18\varepsilon + 36\varepsilon^2, \\ e_1^{(2)} &= e_3^{(2)} = e_5^{(2)} = e_6^{(2)} = 0. \end{aligned} \quad (\text{EW.56})$$

Indeed the central polynomial is $3/2 + (3/2)z^2$ and its error polynomial is $\varepsilon + 6\varepsilon z^2$; expanding the full square difference gives the three displayed coefficients. The upper corner of the coefficient box attains all three positive differences, so its quadratic error terms cannot be omitted.

Expanding all sixteen original tests $(1 - iu)^a(1 + iu)^b, 0 \leq a, b \leq 3$, in the monic Gamma basis of (EW.6), then using (EW.38)–(EW.40) with four weights equal to one, gives

$$\hat{H}_3 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix},$$

$$\begin{aligned}\Delta_{00} &= 54\varepsilon + 28\varepsilon^2, & \Delta_{11} &= (1029\varepsilon + 873\varepsilon^2)/2, \\ \Delta_{22} &= (2109\varepsilon + 1773\varepsilon^2)/2, & \Delta_{33} &= 29754\varepsilon + 32878\varepsilon^2.\end{aligned}\tag{EW.57}$$

These diagonal sums retain the absolute value of every test coefficient. The real coefficients occur at even degrees and the imaginary coefficients at odd degrees; the latter pair exactly to zero against every member of the specified even coefficient box. The complete complex expansions, including those zero pairings, are recorded in the certificate.

Here is an elementary exact positive-definiteness certificate for the whole box. Rational multiplication verifies

$$\hat{H}_3^{-1} = \frac{1}{2430} \begin{pmatrix} 8019 & -1863 & 351 & -27 \\ -1863 & 1501 & -147 & 29 \\ 351 & -147 & 39 & -3 \\ -27 & 29 & -3 & 1 \end{pmatrix}, \quad \text{Tr } \hat{H}_3^{-1} = 956/243 < 4.$$

The central form is positive by its actual measure, or by the four original monic norms in (EW.53). Its inverse is positive, so each of its eigenvalues is at most its trace. Consequently $\hat{H}_3 \succ I_4/4$. Substitution of $\varepsilon = 10^{-12}$ into the four full polynomials (EW.57) gives $0 < \Delta_{jj} < 10^{-6}$ by integer comparison. Hence

$$\hat{H}_3 - \Delta \succ (1/4 - 10^{-6})I_4 \succ 0, \quad \hat{H}_3 - \Delta \preceq H_3 \preceq \hat{H}_3 + \Delta \tag{EW.58}$$

for every actual multiplier in (EW.55). This proves the positive matrix enclosure before its inverses are evaluated. The accompanying receipt also records each exact rational elimination pivot separately.

Use the original J_N and χ at $N = 1, 2, 3$ in (EW.42)–(EW.45). The resulting rational interval for every endpoint value in this whole coefficient box is

$$\frac{10247872526420120792621067}{2^{81}} \leq \mathcal{E}_{2,1} \leq \frac{10247872551311149445602506}{2^{81}}. \tag{EW.59}$$

The four endpoint arguments in (EW.45) are computed from the written rational matrices $\hat{H}_3 \pm \Delta$ by their monic and quotient Schur formulas; the receipt gives every one of these exact rational values and its two squared-root comparisons. Thus (EW.59) is certified by integer powers and the minimum principle, with no rounding assertion about a transcendental logarithm. The interior positive multiplier and all four box corners are independently reconstructed by the literal one-factor moment recurrence and tensor binomial expansion, and pass the same original-coordinate entry bounds.

All original- S Gram entries are computed from $(1 - iu)^a(1 + iu)^b$. The checker's window certificates include the exact reduction matrix, source norms, quotient metrics, the $r + 1$ Schur block and all its prefixes, both cancellations in (EW.21), positive rational matrix envelopes, and the integer power comparisons for (EW.45). The written proof establishes the all-degree statements; each executed finite check is reported separately in the accompanying receipt. The base fixture executes 416 distinct mathematical checks across fifteen windows; its normal and optimized runs pass and six changed-formula runs in each mode are rejected after executing the complete suite. The separate supplement executes 329 mathematical checks and two dependency-hash checks; its four changed-formula runs in each mode are also rejected after the complete suite, including the omitted quadratic coefficient error at an attained box corner. The independent remainder-kernel calibration executes the six scalar comparisons in (EW.51). These counts describe their separate scopes; repeated modes and changed-formula runs do not add new mathematical checks. Numerical examples do not stand in for the arithmetic input or a growing-window bound.

43 Complete arithmetic endpoint derivation and exact source audit

Source role. Full nine-section derivation, constants, endpoint domains and metric maps. Current provenance correction: fresh normal and optimized JSON outputs have identical bytes, but the delivered arithmetic

file has 989 LF bytes and the fresh file has 1,016 bytes with 27 CRLF endings. Replacing each CRLF by LF gives the delivered bytes exactly; parsed JSON values agree. Thus the later reproduced claim of byte identity with the delivered file is superseded. The full byte-to-text map and separately retained hashes are in `end-point_restriction_json_equality_correction_20260913.md` and its JSON companion.

Exact revision. `d0afdde7ebb41d1415ffca8ba61a6099a30bbaf4593576388ddb85895b6241b9`

Complete source. `sources/local_endpoint_continuations/arithmetic_endpoint_intake_20260913.md`

Source SHA-256. `d0afdde7ebb41d1415ffca8ba61a6099a30bbaf4593576388ddb85895b6241b9`

43.1 Arithmetic endpoint delivery: complete source intake and mathematical audit

This audit covers the complete 13 September 2026 arithmetic endpoint delivery. It proves the displayed dyadic norm estimate for the original arithmetic measure, supplies the intermediate constants and typed maps, and records exactly what the finite replay establishes. The source's equation numbers are retained as references. This file does not modify the source archive, the 478-page Gamma paper, a release, or a publication branch.

43.1.1 1. Exact source and execution record

The archive `Tau_Arithmetic_Endpoint_Bounds_2026-09-13.zip` has 781925 bytes and SHA-256 `c32ca4bebc86c6b7c7ffb58533a49b75e2140b00b31cdb6602cea6d8537ad9f6`. All 32 archive members were extracted with path traversal, absolute path, and conflicting-existing-byte checks into `sources/web_arithmetic_endpoint_delivery/Tau_Arithmetic_Endpoint_Bounds`. Every extracted byte agrees with its archive member. The original archive's 31-entry manifest checks every member other than that manifest itself.

The complete mathematical Markdown (28116 bytes, SHA `8fa97624a55c2f1baad54e465e4637c584329f24f6c198bb5d168a443811a280`) and complete 840-line TeX (31714 bytes, SHA `f776e7934f026342f48732a74f14d57470b8aae1cef76ebdee5f831bd49dedf8`) were read. The complete second user paste, README, handoff, programme state, source review, checks description, Python checker, reader builder, manifest, all JSON receipts and all text logs were read. The add-only patch was completely parsed; all 15 added files reconstruct their corresponding delivered files byte for byte, including the explicitly recorded missing final newline of `checks/source-reading.json`. No patch was applied to any checkout.

The supplied HTML contains all 225 mathematical nodes from the Markdown in the same order and with exactly the same TeX annotation strings. Its 226th MathML element is the duplicate g/h in the table of contents. All 48 numbered displays are present. The supplied desktop, mobile and theorem PNGs were each actually viewed; the title and theorem regions shown are readable. Those three PNGs and browser-layout measurements remain historical evidence; this intake does not claim a fresh browser rendering or inspection of portions absent from those screenshots.

The checker `check_endpoint_bounds.py` has SHA `dcf57f151fd34865907bfa80f8fc063e5d46dbf69201eb ee7e0a9b19b26fd2b7`. Complete inspection found only finite SymPy/unittest calculations and stdout output. Four sequential fresh processes were run with one thread in the BLAS/OpenMP environment, with bytecode writing disabled and an isolated working directory:

- Normal Python: all 18 named test methods passed.
- `python -O`: the same 18 methods passed; the full JSON stdout is byte-identical to the normal run and to the delivered successful payload.
- Normal `--fail-control`: exit 1 with the declared explicit exception.
- Optimized `--fail-control`: the same declared exit 1.

The failure option executes before the test suite is loaded. These are two guard-failure controls with **zero test methods executed and zero mathematical formula mutants**. They do not test mutation sensitivity of the analytic argument. All 37 entries in the inherited Gamma manifest were independently checked against the retained original Gamma files. The inherited 17-method executions are historical in this intake; their earlier root replay remains a separate record.

The fresh receipts are `work/arithmetic_endpoint_intake_replay_20260913/REPLAY.json` and `HTML_MATH_AUDIT.json`. The replay helper is `work/replay_arithmetic_endpoint_intake_20260913.py`. Its first byte-parser attempt stopped before all executions on the standard no-final-newline marker; the parser was repaired to retain that marker's exact semantics, after which all byte comparisons and all four executions completed. No delivered code or proof was repaired to obtain passing tests.

43.1.2 2. Original measure and the two exact quotient maps

Let

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s), \quad h(s) = \prod_{\rho \in Z} (s-\rho)^{m_\rho}, \quad d = \deg h.$$

The packet is fixed and finite, and each selected multiplicity is the actual full order of the zero of the entire nonzero function g . Hence the unique holomorphic continuation v_h defined by $g = hv_h$ is entire and nonzero. Define the original density and mass

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k.$$

No symmetry of the selected packet is required for the analytic norm theorem. Put $c = k/2$. The source norm of $P \in \mathbb{C}[S]$ is

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du.$$

It is positive for every nonzero polynomial: the density is positive almost everywhere, and a nonzero polynomial has only finitely many real-line zeros. All moments exist by the exponential estimate proved below. Finite-dimensional projection therefore gives a unique monic minimizer p_n , with positive squared norm $\omega_{h,k,n}$, for each $n \geq 0$.

Writing $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$ and $\mathcal{P}_j = 0$ for $j < 0$, the leading coefficient map gives

$$0 \longrightarrow \mathcal{P}_{n-1} \longrightarrow \mathcal{P}_n \xrightarrow{\text{lc}_n} \mathbb{C} \longrightarrow 0, \quad n \geq 0.$$

For monic χ of degree $q \geq 1$, Euclidean division proves, for every $n \geq q-1$,

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times \chi} \mathcal{P}_n \xrightarrow{\pi_\chi} \mathbb{C}[S]/(\chi) \longrightarrow 0.$$

The first arrow is injective because $\mathbb{C}[S]$ is a domain; its image consists exactly of the degree- $\leq n$ multiples of χ . The remainder has degree $< q \leq n+1$, proving surjectivity of the last arrow at the stated endpoint. For $n \geq q$, multiplication by χ induces the exact isomorphism

$$\overline{\times \chi} : \mathcal{P}_{n-q}/\mathcal{P}_{n-q-1} \xrightarrow{\sim} \mathcal{P}_n/\mathcal{P}_{n-1}, \quad \text{lc}_n(\chi Q) = \text{lc}_{n-q} Q.$$

Required typing correction to the sentence after source (13): this graded isomorphism requires $n \geq q$. At $n = q-1$ its proposed source is zero while the target leading line has dimension one. The exact sequence at $n = q-1$ remains valid, with zero relation source and $\mathcal{P}_{q-1} \cong \mathbb{C}[S]/(\chi)$. Thus there is an exact endpoint description, not an identification of unequal spaces.

In each original admitted support fibre λ , the lifted map is $(\lambda, P) \mapsto (\lambda, [P]_\chi)$, and it sends τ to τ . Its kernel inside that fibre is precisely the polynomial relation space above, and a relation maps to $(\lambda, 0)$. The leading coefficient square commutes because multiplication by the monic polynomial preserves the literal coefficient. The original arithmetic map remains

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h).$$

These are the delivered original theta-source identities. The present norm proof uses their explicitly specified norm integral and does not replace the Taylor unit or the target of either map.

43.1.3 3. Full local mass argument and all compact cases

For $|T| \geq 10$ let $f_T(z) = \zeta(1/2 + iT + iz)$. The Joukowski image of $|w| = 8$ is the ellipse with semiaxes $a = 65/16$, $b_0 = 63/16$. On and inside it, the image coordinate s satisfies

$$-55/16 \leq \Re s \leq 71/16, \quad |\Im s - T| \leq 65/16, \quad |\Im s| \geq 95/16 > 0.$$

The pole at $s = 1$ is consequently outside an open neighbourhood of the filled ellipse. For $\sigma = \Re s \geq 1/4$, summing $\lfloor x \rfloor$ on $\sigma > 1$, then using locally uniform absolute convergence on $\sigma > 0$, gives

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx, \quad \left| \int_1^\infty \{x\} x^{-s-1} dx \right| \leq \frac{1}{\sigma} \leq 4.$$

The denominator $|s-1|$ stays bounded away from zero on this set and $|s| = O(1+|T|)$, so this part has a uniform $O(1+|T|)$ bound. For $\sigma \leq 1/4$, the functional equation and vertical-strip Gamma bound give

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s),$$

$$|\sin(\pi s/2) \Gamma(1-s)| \leq C(1+|\Im s|)^{1/2-\sigma}.$$

Here $\Re(1-s) \in [3/4, 71/16]$, so the preceding integral estimate bounds $\zeta(1-s)$ by $C'(1+|T|)$. The power is at most $3/2 + 55/16 = 79/16 < 5$. All factors $2^\sigma \pi^{\sigma-1}$ have fixed compact real range. Consequently one finite $C_0 \geq 1$ gives

$$|f_T(z)| \leq M_T = C_0(1+|T|)^6$$

on the full ellipse. The open neighbourhood noted above validates Cauchy's formula on each closed unit disc centred in $[-1, 1]$: the discs lie in $|\Re z| \leq 2$, $|\Im z| \leq 1$, and $4/a^2 + 1/b_0^2 < 1$ (both semiaxes exceed three). Hence $\sup_{[-1,1]} |f'_T| \leq M_T$.

Let $r_* = (3 + \sqrt{13})/2$, so $r_* - r_*^{-1} = 3$, $r_* < 4$, and $z_* = (-ir_* + (-ir_*)^{-1})/2 = -3i/2$. The absolutely convergent Möbius series at real part two gives

$$|\zeta(2+iT)| \geq c_* := \zeta(2)^{-1} > 0.$$

The holomorphic function $F_T(w) = f_T((w+w^{-1})/2)$ on $1 < |w| < 8$ has inner-boundary bound $m_T = \max_{[-1,1]} |f_T| > 0$ and outer-boundary bound M_T . Subtract from $\log |F_T|$ the harmonic function linear in $\log |w|$ with those boundary values. The subharmonic maximum principle, including any zeros of F_T , gives

$$c_* \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}.$$

Therefore $m_T \geq c_*^{1/\vartheta} M_T^{-(1/\vartheta-1)} \geq c_*^3 M_T^{-2}$, since $0 < c_* < 1$, $M_T \geq 1$, $1/\vartheta < 3$. A maximizing point of the interval has a direction into the interval with at least one unit of available length. Since $m_T \leq M_T$, the length $\ell = m_T/(2M_T) \leq 1/2$ fits in that direction. The derivative bound implies $|f_T| \geq m_T/2$ on it. Thus

$$I(T) := \int_{T-1}^{T+1} |\zeta(1/2+it)|^2 dt \geq \ell(m_T/2)^2 = \frac{m_T^3}{8M_T} \geq \frac{c_*^9}{8C_0^7} (1+|T|)^{-42}.$$

On $[-10, 10]$, write $I(T) = \int_{-1}^1 |\zeta(1/2+i(T+x))|^2 dx$. Continuity and uniform boundedness on a compact set give continuity in T . If any such integral were zero, its continuous nonnegative integrand would vanish throughout a whole interval, and the identity theorem on the pole-free neighbourhood of the critical line would force ζ identically zero, contradicting $\zeta(2) > 0$. Hence $I_{\min} = \min_{[-10,10]} I(T) > 0$. The literal choice

$$c_\zeta = \min\{I_{\min}, c_*^9/(8C_0^7)\} > 0$$

proves source (23) for every real T , including the transition endpoints.

43.1.4 4. Carry every original factor to the convolution lower bound

Stirling's formula on the vertical line and the fact that Gamma has neither zeros nor poles at $1/4 + it/2$ show that

$$c_\Gamma := \inf_{t \in \mathbb{R}} |\Gamma(1/4 + it/2)|^2 (1+|t|)^{1/2} e^{\pi|t|/2} > 0.$$

Indeed the expression has a positive finite limit as $|t| \rightarrow \infty$; on the intervening compact set it is continuous and strictly positive. This supplies the stated global Gamma lower bound, including $t = 0$.

Let $H_h = \prod_\rho (1 + |\rho - 1/2|)^{m_\rho}$. The triangle inequality gives $|h(1/2 + it)| \leq H_h(1 + |t|)^d$. Applying this to the entire identity $g = hv_h$ gives $|v_h|^2 \geq |g|^2/[H_h^2(1 + |t|)^{2d}]$ even at a selected centre: the denominator used is strictly positive and is never the zero value of h . On the line, $|s(s-1)|^2 = (t^2 + 1/4)^2$ and $|\pi^{-s/2}|^2 = \pi^{-1/2}$. Moreover

$$\frac{(t^2 + 1/4)^2}{(1 + |t|)^{1/2}} \geq \frac{1}{16\sqrt{2}}.$$

For $|t| \leq 1$ this follows by numerator $\geq 1/16$ and denominator $\leq \sqrt{2}$; for $|t| \geq 1$ it follows from $|t|^4/\sqrt{2|t|} \geq 1/\sqrt{2}$. Thus a literal permissible source constant is

$$c_h^{(0)} = \frac{c_\Gamma}{32\sqrt{2}\pi^{3/2}H_h^2}, \quad \alpha_0 = \pi/2,$$

and

$$w_h(t) \geq c_h^{(0)} e^{-\alpha_0|t|} (1 + |t|)^{-2d} |\zeta(1/2 + it)|^2.$$

When $|t - T| \leq 1$, $|t| \leq |T| + 1$ and $1 + |t| \leq 2(1 + |T|)$. Set

$$B_h = 42 + 2d, \quad c_h^{(1)} = c_h^{(0)} e^{-\alpha_0} 2^{-2d} c_\zeta > 0.$$

Integrating proves the exact lower local mass

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\alpha_0|T|} (1 + |T|)^{-B_h}.$$

For completeness, the analytic upper properties used next also follow from the actual density. For sufficiently large $|t|$, each factor $|1/2 + it - \rho| \geq |t|/2$. The upper Gamma estimate, the real-part-1/2 integral estimate for zeta above, and this lower bound for h give an exponential $e^{-\alpha_0|t|}$ times a fixed polynomial upper bound for w_h . On compact sets its entire definition is continuous and bounded. It is therefore bounded and integrable and has finite Laplace values for $|b| < \alpha_0$. Its zero set on the real line is discrete, because v_h is nonzero entire, so $w_h > 0$ almost everywhere.

For each real u , the product $w_h(t)w_h(u - t)$ is positive almost everywhere, integrable, and has a positive integral. In addition

$$\|(w_h * w_h)(\cdot + a) - (w_h * w_h)\|_\infty \leq \|w_h\|_\infty \|w_h(\cdot + a) - w_h\|_1 \rightarrow 0.$$

Hence $m_{h,2}$ is continuous and strictly positive. The quantities

$$b_h = \min_{|v| \leq 1} m_{h,2}(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0, \quad c_h = b_h c_h^{(1)} > 0$$

are actual compact observations of the original measure. Restricting the positive convolution integral gives

$$m_{h,3}(u) = \int w_h(t) m_{h,2}(u - t) dt \geq b_h \int_{u-1}^{u+1} w_h(t) dt \geq c_h e^{-\alpha_0|u|} (1 + |u|)^{-B_h}.$$

For $k > 3$, the exact convolution integral is over $v_4, \dots, v_k \in \mathbb{R}$, with first factor $m_{h,3}(u - \sum v_a)$. Tonelli applies. Restrict it to $[-1, 1]^{k-3}$, where $|\sum v_a| \leq k - 3$, and integrate each remaining factor. This proves, including $k = 3$ with an empty product,

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(|u|+k-3)} (1 + |u| + k - 3)^{-B_h}.$$

Every factor in the omitted complement remains in the original nonnegative integral. There is no conclusion at $k = 1, 2$ by substituting a negative number of factors. The density still has mass μ_h^k .

43.1.5 5. Exact monic norm and literal uniform constant

Fix $0 < b < \alpha_0$, and define $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$. Tonelli and the definition of convolution give

$$\int e^{\pm bu} m_{h,k}(u) du = M_h(\pm b)^k, \quad \int e^{b|u|} m_{h,k}(u) du \leq M_h(b)^k + M_h(-b)^k.$$

The nonnegative exponential series implies $|u|^{2j} \leq (2j)! b^{-2j} e^{b|u|}$. The literal monic trial polynomial $(S - c)^j$ becomes $i^j u^j$ on the original integration line. Consequently

$$\omega_{h,k,j} \leq (2j)! b^{-2j} [M_h(b)^k + M_h(-b)^k], \quad j \geq 0.$$

The Legendre constant can be derived without dropping its leading coefficient. Let

$$P_j(x) = \frac{1}{2^j j!} \frac{d^j}{dx^j} (x^2 - 1)^j, \quad \ell_j = \frac{(2j)!}{2^j (j!)^2}.$$

Integration by parts j times proves its orthogonality against every polynomial of degree below j , because the boundary terms vanish to the required orders. The same integration gives

$$\int_{-1}^1 P_j(x)^2 dx = \frac{\ell_j}{2^j} \int_{-1}^1 (1 - x^2)^j dx = \frac{\ell_j}{2^j} \frac{2^{2j+1} (j!)^2}{(2j+1)!} = \frac{2}{2j+1}.$$

The beta integral in the middle equality follows by $x = 2y - 1$ and integration by parts for $\int_0^1 y^j (1 - y)^j dy = (j!)^2 / (2j+1)!$. Thus the monic polynomial $L^j \ell_j^{-1} P_j(u/L)$ has squared norm

$$\mathfrak{l}_j(L) = \frac{2L^{2j+1}}{2j+1} \left(\frac{2^j (j!)^2}{(2j)!} \right)^2.$$

Any other polynomial with the same leading coefficient differs by a lower-degree polynomial. Orthogonality gives an exact sum of squared norms, proving minimality and uniqueness. A monic S -polynomial maps to a u -polynomial with leading coefficient i^j . The explicit inverse image of the interval minimizer is

$$\frac{i^j L^j}{\ell_j} P_j \left(\frac{S - c}{iL} \right),$$

whose S^j coefficient is $i^j i^{-j} = 1$. This proves the claimed lower bound at the original coordinate type. Multiplication by the inside and outside characteristic functions is the exact isometry

$$L^2(m_{h,k} du) \longrightarrow L^2([-L, L], m_{h,k} du) \oplus L^2(\mathbb{R} \setminus [-L, L], m_{h,k} du),$$

with inverse addition after extension by zero. Applying the lower envelope to its first component gives

$$\omega_{h,k,j} \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(L+k-3)} (1 + L + k - 3)^{-B_h} \mathfrak{l}_j(L), \quad k \geq 3, j \geq 0, L > 0.$$

Taking $j = n$, $L = n$ here and $j = 2n$ above gives exactly source (35). To verify the constant in source (36), put

$$M_* = \max\{1, M_h(b), M_h(-b)\}, \quad C_c = \max(1, c_h^{-1}), \quad C_\vartheta = \max(1, \vartheta_h^{-1}).$$

The central binomial estimate $\binom{2n}{n} \leq 4^n$ gives $\mathfrak{l}_n(n) \geq 2n^{2n+1} / ((2n+1)4^n)$. Combining the factor 2 from the two Laplace values with this prefactor, before taking roots, yields

$$\left(\frac{2(4n)!}{\mathfrak{l}_n(n)} \right)^{1/(2n)} \leq 32n(2 + 1/n)^{1/(2n)} \leq 64n.$$

The first inequality uses $(4n)! \leq (4n)^{4n}$. For the second, $2 + 1/n \leq 4^n$ for every integer $n \geq 1$. For $n \geq k \geq 3$, the remaining factors satisfy

$$M_*^{k/(2n)} \leq \sqrt{M_*}, \quad c_h^{-1/(2n)} \leq \sqrt{C_c}, \quad \vartheta_h^{-(k-3)/(2n)} \leq \sqrt{C_\vartheta},$$

$$e^{\alpha_0(n+k-3)/(2n)} \leq e^{\alpha_0}, \quad (1+n+k-3)^{B_h/(2n)} \leq (2n)^{B_h/(2n)} \leq 2^{B_h/2} e^{B_h/(2e)}.$$

The final inequality follows by differentiating $\log x/x$, whose maximum on $x > 0$ is $1/e$. The factor b^{-4n} becomes exactly b^{-2} . Multiplying proves the literal sufficient constant

$$C_h = 64b^{-2} e^{\alpha_0} 2^{B_h/2} e^{B_h/(2e)} \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}$$

and the theorem

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n, \quad n \geq k \geq 3.$$

The packet and the chosen b determine a finite constant; neither k nor n enters it. The compact minima are proved positive, and they have not been numerically enclosed. The same argument gives source (37) for integers $k \geq 3, n \geq 0, r \geq 1$ and $L > 0$, with numerator degree $n+r$, exponent $1/(2r)$, and all factors exactly as printed. An endpoint selection additionally requires $n \geq q$.

43.1.6 6. Exact composition with the delivered endpoint theorem

The complete immutable PR23 note was read at commit c720f40530eed2f5969dbabe94dfd3fddc0f507f, from work/endpoint_criterion_pr23_c720f405_20260913.md. Only its endpoint theorem is used here. Its original local source estimate is

$$0 \leq \epsilon_N \leq A_N \sinh x_N, \quad A_N = \sqrt{\omega_{N+1}/\omega_N} > 0, \quad x_N = \frac{1}{2} \log(V_{N-1}/V_{N+1}) \geq 0.$$

For a window $N = n, \dots, n+r-1, n \geq q, r \geq 1$, if one $x_N = 0$, the corresponding nonnegative allowance is zero. Otherwise $(\log \sinh x)'' = -1/\sinh^2 x < 0$, so finite Jensen gives

$$\prod \sinh x_N \leq \sinh \left(\frac{1}{r} \sum x_N \right)^r.$$

The minimum allowance raised to r is bounded by the product of the allowances. Both exact products telescope:

$$\prod A_N = \sqrt{\omega_{n+r}/\omega_n}, \quad 2 \sum x_N = \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}.$$

Taking nonnegative roots proves source (38). The finite minimum is attained inside the existing family. Choosing $r = n \geq \max(q, k), k \geq 3$, and using the just-proved norm theorem yields source (39) with its original C_h .

For the packet consisting exactly of a nonreal off-line quartet with common actual multiplicity m , put $\ell = 1 + k(m-1)$. The inherited aggregate exterior result supplies

$$q_k = \ell(k+1)^2, \quad L_{h,k} = 2\delta\ell(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}$$

at every admitted degree. The elementary finite sum behind the count is $\sum_{a > k/2} (2a-k) = \lfloor (k+1)^2/4 \rfloor$. If k is even, $4 \lfloor (k+1)^2/4 \rfloor = k(k+2) \geq k(k+1)$; if k is odd, it equals $(k+1)^2 \geq k(k+1)$. Hence $q_k \geq k$ and $L_{h,k}/q_k \geq \delta k/2$. Applying (39) at $n = q_k$ gives

$$\mathcal{B}_{h,k} := \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}} \geq 2q_k \operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right).$$

For $a > 0$, $\operatorname{arsinh}(ak) = \log k + \log(a + \sqrt{a^2 + k^{-2}})$; its second summand tends to $\log(2a)$. Dividing proves the source's threshold $\liminf \mathcal{B}_{h,k}/(q_k \log k) \geq 2$. This implication uses the existing exterior lower theorem; it does not assert the existence of such a quartet. Later exact-radius improvements belong to the parallel product lane and are not asserted to have been proved by the present delivered source.

43.1.7 7. Complete block boundary isomorphism and determinant transport

Here $q \geq 1$, $i < j$, and $i \geq q - 1$. In the polynomial source let $\mathcal{D}_a = \ker(\pi_\chi|_{\mathcal{P}_a})$. In the actual Hilbert source these are their images under the injective source map. Let $R_i : E \rightarrow \mathcal{P}_i$ be the unique minimum norm section of π_χ , and $G_i = R_i^* R_i$, with all adjoints taken in the stated original source metric and the fixed remainder coordinates. Let

$$Tz = \sum_{a=i+1}^j z_a p_a, \quad F = \pi_\chi T = [b_{i+1}, \dots, b_j], \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j).$$

The original orthogonality gives $T^*T = \Omega$, $T^*R_i = 0$, and $T^*\mathcal{D}_i = 0$. Since $R_i \perp \mathcal{D}_i$, the map $B = T - R_i F$ takes values in $\mathcal{D}_j \cap \mathcal{D}_i^\perp$. It has zero remainder because $\pi_\chi R_i = I$. Its coefficients along the basis $p_{i+1}, \dots, p_j$ are exactly z , so it is injective. For any d in that intersection, choose z to be these upper coefficients of d . Then $d - Bz \in \mathcal{P}_i$, has zero remainder, and is perpendicular to $\mathcal{D}_i$. Thus $d - Bz = 0$. This proves surjectivity and its explicit inverse.

The exact Gram and pairing identities are

$$B^*B = H := \Omega + F^*G_iF > 0, \quad B^*R_i = -F^*G_i.$$

The section

$$\tilde{R} = R_i + BH^{-1}F^*G_i$$

has remainder identity, is perpendicular to $\mathcal{D}_i$, and satisfies $B^*\tilde{R} = -F^*G_i + HH^{-1}F^*G_i = 0$. Since $\mathcal{D}_j = \mathcal{D}_i \oplus \text{im } B$ is the proved orthogonal decomposition, $\tilde{R} = R_j$. Expanding its Gram gives

$$G_j = G_i - G_iFH^{-1}F^*G_i.$$

An alternative direct kernel derivation retains the same maps: in the orthogonal monic basis, $K_i = G_i^{-1} = \sum_{a=0}^i b_a b_a^* / \omega_a$, hence

$$K_j = K_i + F\Omega^{-1}F^*, \quad \frac{V_i}{V_j} = \det(I + \Omega^{-1}F^*G_iF).$$

The determinant identity follows from $V_i/V_j = \det K_j / \det K_i$ and $\det(I + AB) = \det(I + BA)$. To specify its positive spectral type, $\Omega^{-1}F^*G_iF$ is conjugate by $\Omega^{1/2}$ to the Hermitian positive semidefinite matrix $\Omega^{-1/2}F^*G_iF\Omega^{-1/2}$. Its eigenvalues are precisely the generalized eigenvalues of the pair (F^*G_iF, Ω) . Therefore the determinant equals $\prod_a (1 + \lambda_a) > 0$. Applying it to $(i, j) = (q - 1, 2q - 1)$ and $(q, 2q)$, then taking real logarithms, proves source (47). Integrating $(1 + x)^{-1} \leq 1$ from 0 to λ_a gives $\log(1 + \lambda_a) \leq \lambda_a$, proving (48) with its two separately typed matrices.

In original theta source coordinates, each column of B is the source image of a uniquely divisible polynomial χQ . Thus its original cochain primitive is the inherited fixed-order division primitive for that Q , and linear combination gives a primitive for every block column. The exact transport is

$$\mathbb{C}^{j-i} \xrightarrow{B} \mathcal{D}_j / \mathcal{D}_i \hookrightarrow \mathcal{B}^{\hat{\otimes} k} / \mathcal{D}_i \longrightarrow \mathcal{B}^{\hat{\otimes} k} / \mathcal{D}_j.$$

The final composite vanishes because the representative lies in $\mathcal{D}_j$. Its marked lift remains the receiving fibre's supported zero. The direct-sum proof and the displayed Gram keep all source norms and pairings before this projection.

43.1.8 8. The confluent formula's exact domain and full-order bridge

The complete immutable PR24 note was fetched from commit dfcbb5cbf7fec8c9301fe242c13e741d762013e into work/pr24_confluent_transfer_dfcbb5_20260913.md (14007 bytes, SHA d1046b75762e4248ded8f0b901c1ac12dc01065e4e684ca2d166b6a535425763) and read in full. Its source determinant identity, source (41), gives source (42) simply by substitution of $N = q - 1, q, 2q - 1, 2q$, for which $a = N - q + 1$ is exactly $0, 1, q, q + 1$. Each numerator and denominator retains its product of q original monic norms. The common raw Vandermonde cancels in this quotient because it occurs twice on each side and is nonzero.

There is a domain precision needed when applying that raw-jet presentation to arbitrary finite packets. Define the exact ring map $\Psi P = P(c + iu)$, and put $\psi = i^{-q}\Psi\chi$, which is monic. The norm multiplier is always

$$\Pi(u) = \overline{\chi}(c - iu)\chi(c + iu) = \overline{\psi}(u)\psi(u),$$

of degree $2q$, real and nonnegative on the real line. The formula $\Pi = \psi^2$ with every node length doubled requires ψ real, equivalently invariance of the original root multiset under $S \mapsto k - \overline{S}$. The exact quartet satisfies this. For an arbitrary packet, take the union of the roots of ψ and their conjugates, adding their orders whenever nodes coincide. These are the actual full orders of Π . The raw jet map then has kernel exactly (Π) by polynomial divisibility and has dimension $2q$. Its Vandermonde is

$$\det V = \left(\prod_a \prod_{d=0}^{r_a-1} d! \right) \prod_{a < b} (z_b - z_a)^{r_a r_b}$$

in the fixed displayed node and derivative order. The coefficient d in multiplication-by- u on raw jets remains $(Xv)_{a,d} = z_a v_{a,d} + dv_{a,d-1}$.

The transfer proof itself applies to this general Π : if a linear combination of $Q_a, \dots, Q_{a+2q-1}$ has all these raw jets zero, it equals ΠH with $\deg H < a$. Orthogonality gives $0 = \int \Pi |H|^2 m_{h,k}$, which forces $H = 0$. Thus the actual jet matrix F_a is invertible. Dividing the unique raw-jet cancellation of Q_{a+2q} by Π gives the monic relation-weight polynomial. Its norm is $-d_{a,0}\omega_a > 0$, and shifting the $2q$ source columns gives $\det F_{a+1} / \det F_a = -d_{a,0}$, with sign $(-1)^{2q-1} = -1$. Multiplication yields the relation/source determinant ratio; the original Schur determinant then gives (41). Therefore the analytic theorem remains valid for all fixed finite packets, and the literal doubled-node description is applied only on its proven symmetry domain. The parallel endpoint-product proof is expanding this transfer in its complete original coordinate dictionary.

43.1.9 9. Source-review claims and research integration status

The delivered source review records selected readings from a six-part S20 archive, its 216580466-byte concatenation and hash, and the discrepancy between an old TeX <=2 and a current page record <2 in Deligne 3.2.10, together with an explicit squared-eigenvalue correction. That external archive and its selected passages are **not included** among these 32 members. This intake has verified the complete delivered report and its manifest bytes; it does not independently reassert reassembly or inspection of files that it has not opened. The arithmetic estimate proved here uses the stated classical zeta/Gamma/Legendre facts and no finite-field purity theorem.

The resulting deliverable establishes a full written norm theorem at the original measure and a finite exact block-volume comparison. Its only newly identified literal source correction is the endpoint domain of the graded multiplication isomorphism; the raw-confluent symmetry domain is made explicit when applying the cited upstream presentation beyond quartets. Neither correction changes the dyadic norm proof or its constant. The theorem composes with the existing finite endpoint criterion exactly as shown. The growing-degree upper estimate for the quotient volumes is not a theorem of this delivery. The independently developing exact-radius and balanced-window improvements are separate additions; they should receive their own proof bodies and provenance before the next cumulative publication.

Classical primary references for the exact functions used above are [DLMF 25.4, Reflection Formulas](#), [DLMF 5.11, Asymptotic Expansions](#), and [DLMF 18.5, Explicit Representations](#). The independent analytic review supplies its actual source-access record and a second full derivation.

44 The original leading-coefficient line and its arithmetic relation image

This proof supplies the exact endpoint domain and metric of the multiplication map in the paragraph following equation (13) of the delivered *Arithmetic Endpoint Bounds*. The arithmetic monic-norm theorem retains all of its original constants and its domain $n \geq k \geq 3$.

Fix a finite packet h , a positive integer k , and the original source map

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad D = -x\partial_x, \quad \mathcal{M}F_h = (2\xi)/h.$$

Its original Hermitian form, conjugate-linear in the first argument, is

$$\langle P, Q \rangle_{h,k} = \int_{\mathbb{R}} \overline{P(k/2 + iu)} Q(k/2 + iu) m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}, \quad w_h(t) = \frac{|((2\xi)/h)(1/2 + it)|^2}{2\pi}. \quad (\text{AL.1})$$

The actual mass is $\int m_{h,k} = \mu_h^k > 0$. The exponential upper bound proved in the delivered arithmetic note makes every polynomial norm finite. The density is positive almost everywhere; a nonzero polynomial has finitely many zeros on the integration line. Thus (AL.1) is positive definite on every finite polynomial space.

Let $\mathcal{P}_a = \mathbb{C}[S]_{\leq a}$ for $a \geq 0$ and $\mathcal{P}_a = 0$ for $a < 0$. For $a \geq 0$ put $\mathcal{L}_a = \mathcal{P}_a/\mathcal{P}_{a-1}$ with its original quotient norm. The leading-coefficient map is denoted lc_a . Let p_a be the unique monic minimizer in (AL.1), and put $\omega_a = \|p_a\|_{h,k}^2 > 0$. Its existence follows by projecting S^a onto the finite-dimensional subspace $\mathcal{P}_{a-1}$, and its uniqueness follows from positivity. The defining orthogonality is

$$p_a \perp \mathcal{P}_{a-1}, \quad \|z[p_a]\|_{\mathcal{L}_a}^2 = |z|^2 \omega_a, \quad z \in \mathbb{C}. \quad (\text{AL.2})$$

Here $[p_a] = [S^a]$; this equality is in the degree quotient and does not identify their polynomial representatives in the original source.

Fix the original monic arithmetic relation polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$ and the original remainder algebra $E = \mathbb{C}[S]/(\chi)$. Write π_χ for polynomial reduction.

Proposition 44.1 (Exact multiplication and leading-coefficient maps). *For $n \geq q - 1$ the following sequence is exact:*

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{AL.3})$$

For $n \geq q$ there is a commuting diagram of exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{P}_{n-q-1} & \longrightarrow & \mathcal{P}_{n-q} & \xrightarrow{\text{lc}_{n-q}} & \mathbb{C} \longrightarrow 0 \\ & & \downarrow \times\chi & & \downarrow \times\chi & & \downarrow \text{id} \\ 0 & \longrightarrow & \mathcal{P}_{n-1} & \longrightarrow & \mathcal{P}_n & \xrightarrow{\text{lc}_n} & \mathbb{C} \longrightarrow 0. \end{array} \quad (\text{AL.4})$$

It induces the coefficient-preserving isomorphism

$$\overline{m}_\chi : \mathcal{L}_{n-q} \xrightarrow{\sim} \mathcal{L}_n, \quad [Q] \longmapsto [\chi Q]. \quad (\text{AL.5})$$

At $n = q - 1$ the relation source in (AL.3) is zero and $\pi_\chi : \mathcal{P}_{q-1} \rightarrow E$ is an isomorphism. The degree map from that relation source to $\mathcal{L}_{q-1}$ is the zero map from 0 to a one-dimensional space.

Proof. Multiplication by a nonzero polynomial in the integral domain $\mathbb{C}[S]$ is injective. Its degree formula is $\deg(\chi Q) = q + \deg Q$ for $Q \neq 0$, so its image in $\mathcal{P}_n$ consists exactly of the multiples of χ in that space. Euclidean division by the monic polynomial χ gives a unique remainder of degree below q . That remainder belongs to $\mathcal{P}_n$ precisely throughout the admitted range $n \geq q - 1$. This proves (AL.3), including its endpoint.

For $n \geq q$, both displayed leading-coefficient rows are exact by their definitions, and the degree formula shows that the two multiplication maps have the indicated targets. Since χ is monic,

$$\text{lc}_n(\chi Q) = \text{lc}_{n-q}(Q), \quad Q \in \mathcal{P}_{n-q}. \quad (\text{AL.6})$$

This proves commutativity of the right square; the inclusion square commutes because both composites are literal multiplication by χ . Lower-degree inputs map to $\mathcal{P}_{n-1}$, so (AL.5) is well-defined.

The two leading-coefficient identifications with $\mathbb{C}$ take it to the identity in (AL.4), proving that it is an isomorphism and specifying its inverse: $z[S^n] \mapsto z[S^{n-q}]$.

At $n = q - 1$, $\mathcal{P}_{-1} = \mathcal{P}_{-2} = 0$. The exact endpoint diagram instead is

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow 0 & & \\ 0 & \longrightarrow & \mathcal{P}_{q-2} & \longrightarrow & \mathcal{P}_{q-1} & \xrightarrow{\text{lc}_{q-1}} & \mathbb{C} & \longrightarrow & 0. \end{array} \quad (\text{AL.7})$$

Every square commutes, and the degree target has dimension one even when $q = 1$, for which it is the constant line $\mathcal{P}_0/0$. This proves the claimed endpoint map and the exact domain restriction $n \geq q$ for (AL.5). $\square$

Proposition 44.2 (Exact source metric and arithmetic image). *For $n \geq q$, the norm, adjoint, and original polynomial representative of (AL.5) are given by*

$$\|\overline{m}_\chi\|^2 = \frac{\omega_n}{\omega_{n-q}}, \quad \overline{m}_\chi^*[p_n] = \frac{\omega_n}{\omega_{n-q}}[p_{n-q}], \quad \overline{m}_\chi[p_{n-q}] = [p_n]. \quad (\text{AL.8})$$

With the actual source correction

$$r_n = \chi p_{n-q} - p_n \in \mathcal{P}_{n-1}, \quad (\text{AL.9})$$

one has the exact identities

$$\|\mathcal{V}_{h,k}(\chi p_{n-q})\|^2 = \omega_n + \|\mathcal{V}_{h,k}r_n\|^2, \quad \pi_\chi(r_n) = -\pi_\chi(p_n), \quad \pi_\chi(\chi p_{n-q}) = 0. \quad (\text{AL.10})$$

The unquotiented leading coefficient of every relation is transported by (AL.6), and its arithmetic image is the original supported zero.

Proof. Both χp_{n-q} and p_n are monic of degree n . Their difference therefore has degree at most $n - 1$, proving (AL.9), and their classes in $\mathcal{L}_n$ agree. Formula (AL.2) gives

$$\frac{\|\overline{m}_\chi(z[p_{n-q}])\|^2}{\|z[p_{n-q}]\|^2} = \frac{|z|^2 \omega_n}{|z|^2 \omega_{n-q}} = \frac{\omega_n}{\omega_{n-q}} \quad (z \neq 0).$$

This proves the operator norm in (AL.8). The adjoint identity follows by testing the defining equation against an arbitrary scalar multiple of $[p_{n-q}]$ and using the positive real quotient weights in (AL.2). Orthogonality $p_n \perp r_n$ gives the first identity of (AL.10) with the literal original source norms. Reducing (AL.9) modulo χ gives its other two identities.

To state the supported image with its complete type, for each original admitted label λ and the fixed admitted degree n , define the map on $\{\tau\} \sqcup \{(\lambda, P) : P \in \mathcal{P}_n\}$ with target $\{\tau\} \sqcup \{(\lambda, x) : x \in E\}$ by

$$\tilde{\pi}_\chi(\lambda, P) = (\lambda, \pi_\chi P), \quad \tilde{\pi}_\chi(\tau) = \tau.$$

It sends a relation $(\lambda, \chi Q)$ to $(\lambda, 0_E)$, while external absence is fixed. The kernel inside this finite source fibre is exactly $\chi \mathcal{P}_{n-q}$, the image of multiplication in (AL.3). The same formula on the full fibre $\mathbb{C}[S]$ has kernel $\chi \mathbb{C}[S]$; its restriction along $\mathcal{P}_n \hookrightarrow \mathbb{C}[S]$ is the finite map just defined, because $(\chi \mathbb{C}[S]) \cap \mathcal{P}_n = \chi \mathcal{P}_{n-q}$ by the degree formula. The original arithmetic map satisfies

$$J^{(k)} \mathcal{V}_{h,k} = \eta_{h,k} \pi_\chi, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_k) 1, \quad v_h = j_h((2\xi)/h). \quad (\text{AL.11})$$

Composing (AL.3) with this displayed original map sends each relation to zero at that receiving fibre; every Taylor-unit coefficient stays in $\eta_{h,k}$. Before that projection, (AL.10) records its full source norm and the exact correction to the minimum-norm leading representative. $\square$

For the empty arithmetic packet $\chi = 1, q = 0$, the algebra E is zero. Multiplication is the identity $\mathcal{P}_n \rightarrow \mathcal{P}_n$ and the induced line map is the identity for every $n \geq 0$, with norm one; (AL.8)–(AL.10) reduce to $\omega_n/\omega_n = 1$ and $r_n = 0$. Its analytic source and actual mass μ_h^k remain in (AL.1). This case uses no positive-rank quotient inverse.

The coefficient-preserving isomorphism, its original metric in (AL.8), the representative correction in (AL.10), and the endpoint map (AL.7) give the complete relation between the degree-line norm and its arithmetic relation image. All identities hold at the original coordinate $S = k/2 + iu$; no change to the norm estimate or the source generator is required.

45 Exact window products, sharp endpoint optimization, and original relation costs

This continuation proves the product refinement in the delivered four-endpoint calculation, then solves the associated scalar problem with only the combined logarithmic budget retained. It gives explicit maps from these quantities to the original source, relation and Schur determinants. The new arithmetic consequence uses the proved norm estimate of the accompanying *Arithmetic endpoint bounds* source. Its analytic proof remains an explicit dependency; the finite polynomial calculations below do not certify that analytic theorem or an upper estimate for the growing arithmetic relation volumes.

45.1 Original source and a proof of its exact radius identity

Fix the original monic packet h of actual selected zeros, with all their orders, of $g = 2\xi$, and put

$$v_h = g/h, \quad w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad m_{h,k} = w_h^{*k}, \quad c = k/2, \quad \int m_{h,k} = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{EP.1})$$

The complete source construction proves positivity almost everywhere and all polynomial moments, also after multiplication by $e^{\theta u}$ for real $|\theta| < \pi/2$. Retain that literal density and the pairing

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_{h,k}(u) du. \quad (\text{EP.2})$$

It is conjugate-linear in its first entry. Every nonzero polynomial has positive squared norm since its zero set on the line is finite. Let χ be the original degree- q monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$, where $S = \sum s_i$. In this section $q \geq 1$ and the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Reflection preserves this ideal and gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$ by monicity. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic.

Write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and $C = \mathbb{C}[S]/(\chi)$ in the coordinates $1, S, \dots, S^{q-1}$. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0 \quad (N \geq q-1). \quad (\text{EP.3})$$

Let H_N be the Gram in the stated monomial coordinates. The exact least-lift matrices, with their full types, are

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N, \quad V_N = \det G_N. \quad (\text{EP.4})$$

Surjectivity of J_N proves $K_N > 0$. If B_N is the multiplication matrix for χ , direct multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Every lift of x is $R_N x + B_N y$ and has squared norm $x^* G_N x + y^* B_N^* H_N B_N y$. This proves both the minimum property and $R_N^* H_N R_N = G_N$. Inclusions of the polynomial spaces therefore give $G_{N+1} \preceq G_N$ and $V_{N+1} \leq V_N$.

Let p_j be the monic orthogonal family in S , with squared norms $\omega_j > 0$, and let $b_j = [p_j]_X$. The monic triangular coefficient matrix conjugates H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, so

$$K_N = \sum_{j=0}^N b_j b_j^* / \omega_j. \quad (\text{EP.5})$$

For $N \geq q$ define $\delta_N = V_N / V_{N-1} \in (0, 1]$. The determinant identity $\det(I + XY) = \det(I + YX)$ follows by eliminating the two diagonal blocks of $\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ in the two orders. Applied to (EP.5), it gives

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - \delta_N, \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1. \quad (\text{EP.6})$$

Both follow respectively by subtracting the last term of K_N and adding the next term; all the earlier kernels inverted here are positive.

The exact algebra map

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q] = [Q((S - c)/i)] \quad (\text{EP.7})$$

has coefficient matrix E triangular with diagonal $1, i, \dots, i^{q-1}$. Thus $|\det E| = 1$. The quotient Gram becomes $E^{-*} G_N E^{-1}$ and its determinant stays V_N . Put $A = M_S : C \rightarrow C$ and $T = M_u$ in the real quotient, so $\Phi A \Phi^{-1} = cI + iT$ and $\sigma = \text{Tr } T = (\text{Tr } A - cq)/i \in \mathbb{R}$. The real monic polynomials $Q_j = i^{-j} p_j(c + iu)$ have norms ω_j . Pairing uQ_j with Q_l proves

$$uQ_j = Q_{j+1} + a_j^{(0)} Q_j + (\omega_j / \omega_{j-1}) Q_{j-1}, \quad a_j^{(0)} \in \mathbb{R}; \quad Sp_j = p_{j+1} + (c + ia_j^{(0)}) p_j - (\omega_j / \omega_{j-1}) p_{j-1}. \quad (\text{EP.8})$$

The last term is absent for $j = 0$. Indeed pairing with degrees below $j - 1$ vanishes by self-adjoint multiplication by u , the leading coefficient of Q_{j+1} is one, and pairing with Q_{j-1} gives the displayed norm ratio.

For completeness the phase has its original derivative interpretation. In the real coordinates let $d_j = [Q_j]$, and keep the transported matrices denoted temporarily by R, G . Differentiating the minimum norm shows $G' = R^* u R$: the two terms containing R' vanish because $J_N R' = 0$ and R is orthogonal to $\ker J_N$. The leading coefficient of Rx is $d_N^* Gx / \omega_N$. The polynomial

$$uRx - RTx - (Q_{N+1} - Rd_{N+1}) d_N^* Gx / \omega_N$$

is in $\ker J_N$: its leading degree cancels and its reduction is zero. Pair it with R and use $R^* Q_{N+1} = 0$. Taking the trace gives

$$\partial_\theta \log V_N = \sigma - d_N^* G d_{N+1} / \omega_N, \quad \phi_N := \sigma - \partial_\theta \log V_N, \quad b_N^* G_N b_{N+1} = i \omega_N \phi_N. \quad (\text{EP.9})$$

The factor i is $\overline{i^N} i^{N+1}$; no phase has been removed. Differentiation is justified by the exponential moment domination of the original source.

Substitute (EP.8) into (EP.5). Terms with adjacent indices cancel because ω_j / ω_{j-1} is the exact recurrence coefficient; diagonal terms cancel against $2cK_N$. Thus

$$AK_N + K_N A^* - 2cK_N = (b_{N+1} b_N^* + b_N b_{N+1}^*) / \omega_N. \quad (\text{EP.10})$$

The relative Hermitian defect $\mathcal{H}_N = G_N^{-1/2} (A^* G_N + G_N A - 2cG_N) G_N^{-1/2}$ has rank at most two and trace zero. For $x = G_N^{1/2} b_N$, $y = G_N^{1/2} b_{N+1}$ its matrix is $(xy^* + yx^*) / \omega_N$. Expanding its square gives trace $2(\|x\|^2 \|y\|^2 - |x^* y|^2) / \omega_N^2$ because $x^* y = i \omega_N \phi_N$. Its nonzero eigenvalues, if present, are $\epsilon_N, -\epsilon_N$ with $\epsilon_N > 0$; at rank zero define $\epsilon_N = 0$. Consequently (EP.6) proves the full identity

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1). \quad (\text{EP.11})$$

For $q = 1$ the trace-zero defect is the zero map; the identity retains its possibly nonzero phase. We never form V_{q-2} .

45.2 Window product, endpoint extremum, and phases

Fix $n \geq q$, $r \geq 1$, $m = n + r$, and put

$$d_j = \delta_{n+j} \ (0 \leq j \leq r), \quad U = \omega_m / \omega_n, \quad \mathcal{A} = U^{1/(2r)}, \quad E_* = \min_{0 \leq j < r} \epsilon_{n+j}. \quad (\text{EP.12})$$

Multiplying every term of (EP.11), including both endpoint terms, gives the exact overlapping product

$$\mathcal{K} := \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = U \frac{V_n}{V_m} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (\text{EP.13})$$

Indeed the norm ratios telescope to U and $\prod_{j=1}^r d_j = V_m / V_n$. Each interior numerator occurs twice. As $E_*^{2r} \leq \prod \epsilon_{n+j}^2 \leq \mathcal{K}$ and $0 \leq 1 - d_j < 1$, this proves

$$E_* \leq (UV_n / V_m)^{1/(2r)}. \quad (\text{EP.14})$$

For $r \geq 2$ write $s = r - 1$ and retain all four endpoint ratios

$$\alpha = d_0 = V_n / V_{n-1}, \quad \beta = d_r = V_m / V_{m-1}, \quad t = \prod_{j=1}^s d_j = V_{m-1} / V_n. \quad (\text{EP.15})$$

For $0 < t < 1$, set $v_j = -\log d_j$. The function $f(v) = \log(1 - e^{-v})$ has $f''(v) = -e^{-v} / (1 - e^{-v})^2 < 0$ on $v > 0$. The tangent-line inequality $f(v) \leq f(\bar{v}) + f'(\bar{v})(v - \bar{v})$ follows by integrating the decreasing derivative on either side of $\bar{v}$. Summing at $\bar{v} = (-\log t) / s$ and exponentiating proves

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (\text{EP.16})$$

If one $d_j = 1$, the left side is zero; if $t = 1$, all are one and both sides are zero. For $t < 1$ with all $d_j < 1$, strict concavity shows equality exactly when all $d_j = t^{1/s}$. Insert this result in (EP.13) to obtain

$$E_* \leq \mathcal{U} := \mathcal{A} \left(\frac{(1 - \alpha)(1 - \beta)}{\beta t} \right)^{1/(2r)} (1 - t^{1/(r-1)})^{(r-1)/r}. \quad (\text{EP.17})$$

At $r = 1$ the separate, zero-safe formula is $E_* \leq \mathcal{A} \sqrt{(1 - \alpha)(\beta^{-1} - 1)}$. The four endpoint slots then overlap and no $t^{1/0}$ is defined.

This bound is attained in the relaxed scalar data class: choose the equal interior contractions and, when $\alpha, \beta, t < 1$, put $f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0$ and

$$\omega_{n+j+1} / \omega_{n+j} = \mathcal{U}^2 / f_j, \quad \epsilon_{n+j} = \mathcal{U}, \quad \phi_{n+j} = 0. \quad (\text{EP.18})$$

Their product is U by (EP.17), so both norm endpoints are respected and every radius identity holds. If an endpoint contraction is one, or $t = 1$, every compatible sequence has a zero radius at some step, hence $E_* = 0$ and the zero bound is attained. At $r = 1$ choose the sole norm ratio U . These scalar data are related to the arithmetic objects by the evaluation map (EP.4)–(EP.12); the proof of scalar attainment does not assert that this evaluation map is surjective.

For strictly interior contractions define the exact loss

$$\mathcal{J} = s \log(1 - t^{1/s}) - \sum_{j=1}^s \log(1 - d_j) \geq 0. \quad \mathcal{K} = \mathcal{U}^{2r} e^{-2\mathcal{J}}. \quad (\text{EP.19})$$

It also has the proved quantitative lower bound

$$\mathcal{J} \geq \frac{t}{2(1-t)^2} \sum_{j=1}^s \left(v_j + \frac{\log t}{s} \right)^2. \quad (\text{EP.20})$$

To prove this, all v_j and their mean lie in $(0, -\log t]$, where $-f''(v) \geq t/(1-t)^2$. Integrating the derivative twice gives the tangent inequality with the displayed quadratic remainder. The linear terms sum to zero. This proves (EP.20), including $s = 1$, where both sides vanish. If an interior $d_j = 1$ the exact product (EP.13) applies with value zero; no logarithm of zero is evaluated.

Define the actual phase polynomial

$$P(y) = \prod_{j=0}^{r-1} (y + \phi_{n+j}^2), \quad y \geq 0. \quad P(E_*^2) \leq \mathcal{K}. \quad (\text{EP.21})$$

It is continuous, tends to infinity, and is strictly increasing on $[0, \infty)$: for $y_2 > y_1 \geq 0$ every factor increases, and if a factor vanishes at y_1 , the product at y_2 is strictly positive. The actual product guarantees $\mathcal{K} \geq P(0)$. There is therefore exactly one $y_* \geq 0$ with $P(y_*) = \mathcal{K}$, and $E_* \leq \sqrt{y_*}$. If $\mathcal{K} = P(0)$, then $y_* = 0$. If an externally supplied number were smaller than $P(0)$, it would be incompatible with the displayed radius data; it would not define a nonnegative threshold. In particular at $\mathcal{K} = 0$ at least one phase vanishes and $y_* = 0$. Positive rational enclosures for the phases' squares and $\mathcal{K}$ lead to polynomial evaluation and rational bisection; positive powers and products suffice for each comparison. Using $\mathcal{U}^{2r}$ in place of $\mathcal{K}$ gives the valid endpoint-only version, also with its phase retained.

45.3 The sharp scalar bound with only the combined budget

Put

$$\mathcal{B} = \log \frac{V_{n-1}V_n}{V_{m-1}V_m} = -\log \left(d_0 d_r \prod_{j=1}^{r-1} d_j^2 \right) \geq 0. \quad (\text{EP.22})$$

The product in (EP.13), divided by U , is the function

$$F(d) = \frac{(1-d_0)(1-d_r) \prod_{j=1}^{r-1} (1-d_j)^2}{d_r \prod_{j=1}^{r-1} d_j}. \quad (\text{EP.23})$$

For $\mathcal{B} > 0$ its unique maximum over these scalar contractions with this budget is attained at

$$d_0 = \frac{1}{1+2x}, \quad d_r = 1-2x, \quad d_j = \frac{1-x}{1+x} \quad (1 \leq j < r), \quad (\text{EP.24})$$

where $x \in (0, 1/2)$ is the unique solution of

$$\mathcal{B} = 2 \operatorname{artanh}(2x) + 4(r-1) \operatorname{artanh} x. \quad (\text{EP.25})$$

Here $\operatorname{artanh} x = \frac{1}{2} \log((1+x)/(1-x))$. The exact maximum and consequent bound are

$$F_{\max} = \frac{4^r x^{2r}}{(1-4x^2)(1-x^2)^{r-1}}, \quad \boxed{E_* \leq \mathcal{U} \leq \mathcal{A} \frac{2x}{[(1-4x^2)(1-x^2)^{r-1}]^{1/(2r)}}}. \quad (\text{EP.26})$$

The middle expression for $r = 1$ uses the separate expression after (EP.17).

Here is a full extremization proof. Write $v_j = -\log d_j$. The feasible simplex is $v_0 + v_r + 2 \sum_{j=1}^{r-1} v_j = \mathcal{B}$, $v_j \geq 0$, a compact set. The product vanishes at every boundary point and is positive at interior points, so a maximum is interior. Its logarithm is

$$f(v_0) + f(v_r) + 2 \sum_{j=1}^{r-1} f(v_j) + v_r + \sum_{j=1}^{r-1} v_j. \quad (\text{EP.27})$$

Its Hessian is strictly negative diagonal, so it is strictly concave. The Lagrange multiplier equations are

$$\frac{1}{e^{v_0} - 1} = \lambda, \quad 1 + \frac{1}{e^{v_r} - 1} = \lambda, \quad 1 + \frac{2}{e^{v_j} - 1} = 2\lambda \quad (1 \leq j < r). \quad (\text{EP.28})$$

They imply $\lambda > 1$ and yield (EP.24) with $x = 1/(2\lambda)$. The right side of (EP.25) is strictly increasing from zero to infinity because its derivative is $4/(1 - 4x^2) + 4(r - 1)/(1 - x^2) > 0$. Thus it has exactly one solution and determines the unique maximum. Substitution into (EP.23) gives (EP.26). The value $\mathcal{B} = 0$ forces all $d_j = 1$, and all bounds are zero; it is represented by the continuous endpoint $x = 0$. The norm-ratio construction (EP.18), now using these contractions and $\mathcal{A}F_{\max}^{1/(2r)}$, proves sharpness in the same scalar class for this reduced data.

We prove the comparison with the earlier hyperbolic-sine bound directly. For $a, b \in (0, 1]$ and $z = -\frac{1}{2} \log(ab)$,

$$\sinh^2 z - (1 - a)(b^{-1} - 1) = \frac{(1 - 2a + ab)^2}{4ab} \geq 0. \quad (\text{EP.29})$$

Expansion after multiplying by $4ab$ proves the equality term by term. Also $(\log \sinh z)'' = -1/\sinh^2 z < 0$ for $z > 0$, so the same tangent-line proof used for (EP.16) shows $\prod \sinh z_j \leq \sinh((\sum z_j)/r)^r$. If any $z_j = 0$, the product is zero and the inequality is immediate. Taking $a = d_j, b = d_{j+1}$ gives $\sum z_j = \mathcal{B}/2$, and hence

$$\mathcal{A}F_{\max}^{1/(2r)} \leq \mathcal{A} \sinh(\mathcal{B}/(2r)). \quad (\text{EP.30})$$

For $r > 1, \mathcal{B} > 0$ the inequality is strict. At a maximizer every $z_j > 0$. Equality in the concavity step requires all $z_j = z > 0$, so $d_j d_{j+1} = e^{-2z} =: p < 1$. Equality in (EP.29) requires $d_j = (1 + p)/2$ for every $0 \leq j < r$. For $r > 1$ this implies $d_0 = d_1$, whence $p = ((1 + p)/2)^2$ and $(1 - p)^2 = 0$, a contradiction. For $r = 1$, (EP.25)–(EP.26) give exactly $\sinh(\mathcal{B}/2)$. For geometric contractions $d_j = e^{-v}$ the fixed-endpoint bound is $2\mathcal{A} \sinh(v/2)$ and the earlier bound is $\mathcal{A} \sinh v$; their ratio is exactly $\cosh(v/2)$.

The optimizer can be certified without a transcendental equation. The real endpoint datum $R = e^{\mathcal{B}} \geq 1$ gives

$$(1 + 2x)(1 + x)^{2(r-1)} = R(1 - 2x)(1 - x)^{2(r-1)}. \quad (\text{EP.31})$$

The ratio of the two positive sides before subtraction is strictly increasing on $[0, 1/2)$ by (EP.25), so rational bisection isolates its unique root using integer products when R is rational. For a general actual endpoint, certified rational bounds $1 \leq R_- \leq R \leq R_+$ give enclosing roots $x(R_-) \leq x(R) \leq x(R_+)$ by the same monotonicity. Positive-power bisection then encloses (EP.26). This specifies the computational map exactly.

45.4 Schur windows and the reduced source-relation determinant

Split the source monomials of $\mathcal{P}_m$ at degree $n - 1$ and set

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EP.32})$$

The triangular map $(x, y) \mapsto (x - H_{n-1}^{-1} C_D y, y)$, from $\mathbb{C}^n \oplus \mathbb{C}^{r+1}$ onto $\mathbb{C}^{m+1}$, has determinant one and takes H_m to $\text{diag}(H_{n-1}, W)$ and the remainder map to $[J_{n-1}, F]$. Consequently $W > 0$. For its first j columns put $W_j = W[1 : j, 1 : j]$, $F_j = F[:, 1 : j]$, $Z_j = F_j^* G_{n-1} F_j$. Block inversion and the determinant identity from (EP.6) prove

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*, \quad R_j := V_{n-1}/V_{n+j-1} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad R_0 = 1. \quad (\text{EP.33})$$

This R_j is a scalar and is distinct from the typed least-lift R_N by its explicit definition; the scalar is obtained from that least-lift Gram by determinant. Write

$$\begin{aligned} w &= (W_1)_{11}, \quad p = \det W_r, \quad s_+ = \det W_{r+1}, \quad e = w + (Z_1)_{11}, \\ H_+ &= \det(W_{r+1} + Z_{r+1}), \quad H_- = \det(W_r + Z_r). \end{aligned} \quad (\text{EP.34})$$

All six quantities are positive. Successive monic Schur complements give $w = \omega_n$ and $\omega_m = s_+/p$. Thus the exact bridge from the original EW matrices to the new endpoint quantities is

$$\begin{aligned} U &= s_+/(pw), \quad \alpha = w/e, \quad \beta = H_-s_+/(pH_+), \quad t = ep/(wH_-), \\ U \frac{V_n}{V_m} &= \frac{H_+}{pe}, \quad \mathcal{B} = \log \frac{H_-H_+w}{ps_+e}. \end{aligned} \quad (\text{EP.35})$$

For example $\alpha = 1/R_1$, $\beta = R_r/R_{r+1}$ and $t = R_1/R_r$, and substitution gives each displayed equality. Every individual contraction can also be recovered as $d_j = R_j/R_{j+1}$ for $0 \leq j \leq r$, so the exact product, the loss $\mathcal{J}$ and every comparison use the same block. For $r = 1$, $p = w$, $H_- = e$, $t = 1$, and the separate single-step formula applies. At $n = q$ the base relation space is empty and $G_{q-1} = H_{q-1}$; no inverse of an empty deficient kernel occurs.

To prove the determinant cancellation independently, define

$$\mathfrak{D}_j = \det H_{j-1}, \quad \mathfrak{D}_0 = 1, \quad \mathfrak{B}_l = \det(B_{q+l-1}^* H_{q+l-1} B_{q+l-1}), \quad \mathfrak{B}_0 = 1. \quad (\text{EP.36})$$

The coefficient matrix with first columns the q quotient monomials and last columns $\chi, S\chi, \dots, S^{N-q}\chi$ is triangular with diagonal one. Subtracting boundary columns to replace the first ones by R_N keeps its determinant one and makes the Gram block diagonal. Hence

$$\omega_N = \mathfrak{D}_{N+1}/\mathfrak{D}_N, \quad V_N = \mathfrak{D}_{N+1}/\mathfrak{B}_{N-q+1}, \quad \Lambda_N := \omega_N/V_N = \mathfrak{B}_{N-q+1}/\mathfrak{D}_N. \quad (\text{EP.37})$$

The norm identity uses the analogous monic orthogonal triangular basis. Equations (EP.14), (EP.35), and (EP.37) now prove

$$E_* \leq \left(\frac{\Lambda_m}{\Lambda_n} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{m-q+1} \mathfrak{D}_n}{\mathfrak{D}_m \mathfrak{B}_{n-q+1}} \right)^{1/(2r)} = \left(\frac{H_+}{pe} \right)^{1/(2r)}. \quad (\text{EP.38})$$

For $N \geq q$, multiplication by χ in (EP.3) also sends the original leading-coefficient line at degree $N - q$ isomorphically to the leading-coefficient line of its relation at degree N . Its leading coefficient is one; the source norm of that relation is exactly the form defining $\mathfrak{B}$. The quotient J_N sends the same relation to zero. These give the maps relating the determinant and quotient observations.

For comparison only, multiplying the original source measure by a positive scalar a sends $H_N \mapsto aH_N$, $G_N \mapsto aG_N$, $\omega_N \mapsto a\omega_N$, $V_N \mapsto a^q V_N$, and $\Lambda_N \mapsto a^{1-q} \Lambda_N$. This follows directly from (EP.4) and (EP.37). All window ratios displayed above are unchanged; the actual measure throughout remains (EP.1).

45.5 Full confluent transfer and its two-determinant window

We now construct the transfer determinants in (EP.38) from their full raw derivative maps. Retain $\psi = \hat{\chi} = i^{-q}\chi(c + iu)$ and $\Pi = \psi^2$ under the reflection hypothesis of (EP.7). If χ has distinct roots λ_a of complete lengths ℓ_a , let $\zeta_a = (\lambda_a - c)/i$ and $r_a = 2\ell_a$. The width is $d = \sum_a r_a = 2q$. The map

$$\mathcal{J}_2 f = (f^{(e)}(\zeta_a))_{a, 0 \leq e < r_a} : \mathbb{C}[u] \longrightarrow \bigoplus_a \mathbb{C}^{r_a} \quad (\text{EP.38a})$$

uses the fixed order of nodes and then increasing raw derivative orders. Its kernel is (II) by Taylor divisibility and the coprimality of distinct linear factors. In these coordinates a product uses

$\sum_{b=0}^e \binom{e}{b} f^{(b)} g^{(e-b)}$, and multiplication by u has the block formula $(X_{\Pi}v)_{a,e} = \zeta_a v_{a,e} + e v_{a,e-1}$. The original coordinate map has the exact raw-jet expression

$$(\partial_u^e \Phi P)(\zeta_a) = i^e P^{(e)}(\lambda_a), \quad \mathbb{C}[S]/(\chi^2) \xrightarrow{\sim} \mathbb{C}[u]/(\Pi), \quad [P] \mapsto [P(c + iu)], \quad (\text{EP.38b})$$

since $\Phi(\chi^2) = i^{2q}\Pi$. Every $e!$ and i^e is retained.

Let $v_j = \mathcal{J}_2 Q_j$ for the same source polynomials used in (EP.8), and form

$$\mathbf{F}_a = [v_a, \dots, v_{a+d-1}], \quad \mathbf{V} = [\mathcal{J}_2 1, \dots, \mathcal{J}_2 u^{d-1}]. \quad (\text{EP.38c})$$

Their full raw Vandermonde determinant is

$$\det \mathbf{V} = \left(\prod_a \prod_{e=0}^{r_a-1} e! \right) \prod_{a < b} (\zeta_b - \zeta_a)^{r_a r_b} \neq 0. \quad (\text{EP.38d})$$

To prove the formula without changing node orders, replace the monomial columns successively by the monic polynomials $\prod_{b < a} (u - \zeta_b)^{r_b} (u - \zeta_a)^e$. This is a triangular column change of determinant one, ordered by total degree. Its jet matrix is block lower triangular. At its (a, e) diagonal entry, the derivative is $e! \prod_{b < a} (\zeta_a - \zeta_b)^{r_b}$. Multiplying these entries proves (EP.38d), including repeated orders. The matrix $\mathbf{F}_0$ equals $\mathbf{V}$ times the actual monic coefficient matrix; the determinants, rather than all entries, agree.

Every $\mathbf{F}_a$ is invertible. A vector in its kernel gives a polynomial $Q \in \text{span}(Q_a, \dots, Q_{a+d-1})$ with $Q = \Pi H$ and $\deg H < a$. Source orthogonality gives $0 = \langle H, Q \rangle = \int \Pi |H|^2 d\mu_\theta$. The polynomial Π is positive off its finitely many real roots, and the original source density is positive almost everywhere, so $H = 0$. For $a = 0$ divisibility and degree already force $Q = 0$. Solving $\mathbf{F}_a z_a = v_{a+d}$ yields a monic polynomial

$$\hat{Q}_a = \frac{Q_{a+d} - \sum_{j=0}^{d-1} (z_a)_j Q_{a+j}}{\Pi}. \quad (\text{EP.38e})$$

It has degree a and is orthogonal for the weight $\Pi d\mu_\theta$: pair its numerator with any polynomial of smaller degree. The unique monic orthogonal polynomial for this real positive weight is real, so this quotient is real. Pairing its numerator with itself after division, or with $\hat{Q}_a$, leaves only the Q_a term and gives

$$\nu_a := \int \Pi |\hat{Q}_a|^2 d\mu_\theta = -(z_a)_0 \omega_a > 0. \quad (\text{EP.38f})$$

All source monic polynomials have been left at their literal norms. The original monic relation polynomial is $\chi(S) i^a \hat{Q}_a((S - c)/i)$, whose value on the source line is $i^{a+q} \psi(u) \hat{Q}_a(u)$ and whose norm is exactly ν_a . This is the explicit source map to the relation norm in (EP.36).

The column shift gives $\mathbf{F}_{a+1} = \mathbf{F}_a[e_1, \dots, e_{d-1}, z_a]$. Its determinant multiplier is $(-1)^{d-1} (z_a)_0 = -(z_a)_0$ because $d = 2q$. Multiplying (EP.38f) from index zero proves

$$\boxed{\frac{\det \mathbf{F}_a}{\det \mathbf{V}} = \prod_{j=0}^{a-1} \frac{\nu_j}{\omega_j} = \frac{\mathfrak{B}_a}{\mathfrak{D}_a}}. \quad (\text{EP.38g})$$

The source and relation determinant identities follow from their monic orthogonal triangular matrices; the S - u coefficient matrices have determinant of modulus one, including the common i^q on each relation column. Equation (EP.38g) therefore holds for the original determinants, with their complex node factors intact.

For $a = N - q + 1$, substitution into (EP.37) gives

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det \mathbf{V}}{\det \mathbf{F}_a}, \quad \Lambda_N = \frac{\det \mathbf{F}_a}{\det \mathbf{V} \prod_{j=a}^{N-1} \omega_j}. \quad (\text{EP.38h})$$

In particular with $a = n - q + 1$ the two-volume norm criterion is

$$E_* \leq \left[\frac{\det F_{a+r}}{\det F_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}. \quad (\text{EP.38i})$$

For $q = 1$ the product has zero terms and value one. The expression inside the root is positive by its exact equality to Λ_m/Λ_n ; no absolute value conceals a complex phase.

The complete transfer of the jet multiplication map is also explicit. Put $K_a = [e_1, \dots, e_{d-1}, z_a]$ and $T_a = F_a^{-1} X_{\Pi} F_a$. Then $T_{a+1} = K_a^{-1} T_a K_a$, and the source recurrence gives $z_a = (T_a - a_{a+d-1}^{(0)} I) e_{d-1} - (\omega_{a+d-1}/\omega_{a+d-2}) e_{d-2}$. Here $d \geq 2$, so both indices exist. Differentiating monic orthogonality gives $\partial_{\theta} Q_j = -(\omega_j/\omega_{j-1}) Q_{j-1}$ for $j \geq 1$ and $\partial_{\theta} Q_0 = 0$: its pairings below $j - 1$ vanish and its Q_{j-1} pairing is $-\omega_j$. The norm derivative is $\partial_{\theta} \log \omega_j = a_j^{(0)}$ by the same minimum argument as (EP.9). Internal shifted columns have zero trace, so

$$\partial_{\theta} \log V_N = \sum_{j=a}^N a_j^{(0)} + \frac{\omega_a}{\omega_{a-1}} e_0^T F_a^{-1} v_{a-1}. \quad (\text{EP.38j})$$

At $a = 0$ omit the last term entirely. This follows by writing $F'_a = F_a L_a - (\omega_a/\omega_{a-1}) v_{a-1} e_0^T$, where $(L_a)_{j-1,j} = -\omega_{a+j}/\omega_{a+j-1}$ and all other entries vanish. Taking the trace of $F_a^{-1} F'_a$ in (EP.38h) proves (EP.38j). It supplies the phase in (EP.9) through the same complete jet map.

For the Gamma reference use the same Π , nodes, row order, and raw derivatives, and denote the reference by a superscript Γ . With $X_a = \mathfrak{D}_a/\mathfrak{D}_a^{\Gamma}$ and $Y_a = \mathfrak{B}_a/\mathfrak{B}_a^{\Gamma}$, division of (EP.38g) gives

$$\frac{\det F_a}{\det F_a^{\Gamma}} = \frac{Y_a}{X_a}, \quad \frac{V_N}{V_N^{\Gamma}} = \frac{X_{N+1}}{Y_{N-q+1}}. \quad (\text{EP.38k})$$

The invariant endpoint values still depend only on original moments through degree $2m$ by (EP.32)–(EP.38). In contrast the individual entries of F_a , with $a = N - q + 1$, involve Q_{N+q} , which is obtained by a projection using moments through $2N + 2q - 1$. The determinant identity proves their exact cancellation in the invariant answer. It does not identify every raw entry with a function of the shorter input list.

For a non-reflection-stable polynomial, define $\chi^{\dagger}(S) = (-1)^q \bar{\chi}(k - S)$, where bars conjugate coefficients. Then $\Phi\chi = i^q \psi$ and $\Phi\chi^{\dagger} = i^q \bar{\psi}$, so the exact extension of the norm-jet map has domain $\mathbb{C}[S]/(\chi\chi^{\dagger}) \xrightarrow{\sim} \mathbb{C}[u]/(\bar{\psi}\psi)$ under the same Φ . It uses $\Pi = \bar{\psi}\psi$ instead of ψ^2 . Its nodes are the union of the roots of ψ and their conjugates, and the orders add at every overlap; their sum is again $2q$. The positivity, division, raw-jet, and determinant proofs above apply to this actual monic Π , with the just-proved changed quotient domain. The original relation map still multiplies by χ , whose squared norm weight is $|\psi|^2$. This supplies the exact general norm map. The original radius and phase argument (EP.7)–(EP.11) is stated under its reflection hypothesis and is not extended by this norm observation alone.

45.6 An arithmetic lower cost on one relation block

The complete analytic theorem in the delivered arithmetic endpoint source proves, for the actual density (EP.1) at $\theta = 0$,

$$(\omega_{h,k,2n}/\omega_{h,k,n})^{1/(2n)} \leq C_h n \quad (n \geq k \geq 3). \quad (\text{EP.39})$$

Its retained literal sufficient constant, from equation (36) there, is

$$C_h = 64b^{-2} e^{\pi/2} 2^{B_h/2} e^{B_h/(2e)} \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}, \quad 0 < b < \pi/2, \quad (\text{EP.40})$$

where $B_h = 42 + 2 \deg h$, $M_* = \max(1, M_h(b), M_h(-b))$, $M_h(b) = \int e^{bt} w_h(t) dt$, $\vartheta_h = \int_{-1}^1 w_h(t) dt$, and $c_h = b_h c_h^{(1)} > 0$ with $b_h = \min_{|u| \leq 1} (w_h * w_h)(u)$ and the original local-mass constant $c_h^{(1)}$ of that source. No norm estimate is assumed in this paper in place of that proof. Its complete source is retained in the dependency record, with the independent analytic audit.

For a packet consisting exactly of an off-line quartet $1/2 \pm \delta \pm i\gamma$, $\delta, \gamma > 0$, with common full multiplicity m_0 , the original exterior calculation gives

$$\begin{aligned} q_k &= [1 + k(m_0 - 1)](k + 1)^2, \\ L_{h,k} &= 2\delta[1 + k(m_0 - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor, \\ L_{h,k}/q_k &\geq \delta k/2, \quad L_{h,k} \leq \epsilon_{h,k,N}. \end{aligned} \tag{EP.41}$$

Here is the complete finite spectral calculation behind (EP.41). In each tensor local algebra write $s_j = \rho_j + z_j$ with $z_j^{m_0} = 0$. The sum nilpotent $\sum z_j$ has exact order $1 + k(m_0 - 1)$: its top nonzero power has coefficient $[k(m_0 - 1)!]/[(m_0 - 1)!]^k$ at $\prod z_j^{m_0-1}$, and every higher monomial has an exponent at least m_0 . The distinct sum centres are $c + \delta(2a - k) + i\gamma(2b - k)$, $0 \leq a, b \leq k$. Every such pair occurs: the number of simultaneous plus signs can be any integer between $\max(0, a + b - k)$ and $\min(a, b)$, a nonempty interval. Distinct pairs give distinct centres since $\delta, \gamma > 0$. The cyclic annihilator is the least common multiple of these local annihilators, giving the displayed q_k with the full common order.

For any admitted G_N , conjugate $A - cI$ to the Euclidean metric by $G_N^{1/2}$ and put it in unitary upper triangular form, ordering the eigenvalues with positive real part first. Such a form follows inductively by choosing an eigenvector and applying induction to the quotient orthogonal block. Its Hermitian sum has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros by (EP.10)–(EP.11). The trace of its compression onto the positive-real diagonal slots is twice the sum of those positive real parts. This trace is at most ϵ_N : in an eigenbasis of the Hermitian sum it equals $\epsilon_N \langle P e_+, e_+ \rangle - \epsilon_N \langle P e_-, e_- \rangle \leq \epsilon_N$ for the orthogonal projection P onto those slots. The sum of the positive integers $2a - k$, $0 \leq a \leq k$, is $\lfloor (k + 1)^2/4 \rfloor$, by separately summing the even and odd arithmetic progressions. Multiplying by 2δ , the $k + 1$ imaginary centres, and the full local length proves exactly $L_{h,k}$ and its bound by ϵ_N . Finally $\lfloor (k + 1)^2/4 \rfloor \geq k(k + 1)/4$, checked in the same two parities, proves $L_{h,k}/q_k \geq \delta k/2$. Thus the last inequality in (EP.41) holds at every admitted degree. Retaining the exact $L_{h,k}$ makes the following deduction explicit. Set $n = r = q_k$ in (EP.13), and abbreviate $q = q_k$, $L = L_{h,k}$. Every $d_j < 1$ in this window because $L > 0$ and (EP.11). Taking logarithms in (EP.13) and using (EP.39) proves

$$\begin{aligned} \log \frac{V_q}{V_{2q}} &\geq 2q \log \frac{L}{C_h q} - \log(1 - d_0) - \log(1 - d_q) - 2 \sum_{j=1}^{q-1} \log(1 - d_j) \\ &\quad + \sum_{j=0}^{q-1} \log(1 + \phi_{q+j}^2/L^2). \end{aligned} \tag{EP.42}$$

Indeed $\prod(\epsilon^2 + \phi^2) \geq L^{2q} \prod(1 + \phi^2/L^2)$, whereas $\log U \leq 2q \log(C_h q)$. All added terms on the right of (EP.42) after its first one are nonnegative, with the contraction penalty strictly positive. For $q > 1$, (EP.16) gives the additional endpoint-only lower penalty $-\log(1 - \alpha) - \log(1 - \beta) - 2(q - 1) \log(1 - t^{1/(q-1)})$. In particular the actual two-volume threshold is

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k}/V_{2q_k})}{q_k \log k} \geq 2. \tag{EP.43}$$

This follows by replacing L/q_k by $\delta k/2$ in (EP.42) and dividing by $q_k \log k$. The constants are independent of k . The first norm window also gives the exact determinant lower bound $\Lambda_{2q}/\Lambda_q \geq L^{2q}$ by (EP.38).

Here is the exact receiving relation layer. For $i \geq q - 1$, $j > i$, let $\mathbf{T} = [p_{i+1}, \dots, p_j]$ be source columns, $\mathbf{F} = [b_{i+1}, \dots, b_j]$, and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. Then

$$\mathbf{b} = \mathbf{T} - R_i \mathbf{F} : \mathbb{C}^{j-i} \rightarrow \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathcal{D}_N = \chi \mathcal{P}_{N-q}, \quad \mathbf{b}^* \mathbf{b} = \Omega + \mathbf{F}^* G_i \mathbf{F}. \quad (\text{EP.44})$$

Reduction vanishes termwise. Each column is orthogonal to $\mathcal{D}_i$ because $p_{i+a} \perp \mathcal{P}_i$ and $R_i \perp \mathcal{D}_i$. Their highest degrees are distinct and monic. Successive subtraction of these highest coefficients spans the dimension- $j - i$ new relation layer and proves that $\mathbf{b}$ is an isomorphism onto that layer. Orthogonality of the two summands gives its exact Gram. Kernel addition as in (EP.5) yields

$$\frac{V_i}{V_j} = \det(I + \Omega^{-1/2} \mathbf{F}^* G_i \mathbf{F} \Omega^{-1/2}) = \prod_{a=1}^{s_0} (1 + \lambda_a), \quad s_0 = \text{rank } \mathbf{F} \leq \min(q, j - i). \quad (\text{EP.45})$$

The positive eigenvalues λ_a are those of the written positive semidefinite matrix; zero eigenvalues contribute one. At $(i, j) = (q, 2q)$, (EP.42) thus gives

$$\sum_a \log(1 + \lambda_a) \geq 2q \log \frac{L}{C_h q}, \quad \lambda_{\max} \geq \left(\frac{L}{C_h q} \right)^2 - 1 \geq \frac{\delta^2 k^2}{4C_h^2} - 1. \quad (\text{EP.46})$$

For the second assertion, pad the eigenvalue list to length q with zeros, define $\lambda_{\max}$ as the maximum of this full list, and bound the determinant by $(1 + \lambda_{\max})^q$. If its right side is negative, the assertion still follows from $\lambda_{\max} \geq 0$. For sufficiently large k the determinant lower bound exceeds one, so the layer has positive rank.

The representative change is determined, not discarded: projection onto the new orthogonal relation layer in (EP.44) gives

$$R_j = R_i + \mathbf{b}(\Omega + \mathbf{F}^* G_i \mathbf{F})^{-1} \mathbf{F}^* G_i. \quad (\text{EP.47})$$

Indeed $\mathbf{b}^* R_i = -\mathbf{F}^* G_i$, so the right side is a lift orthogonal to both old and new relations, hence the unique minimum lift. Its arithmetic jet remains $\eta x = [v_h]_h^{\otimes k} x(S)$, whose injectivity follows because the full local unit has a recursive power-series inverse and (χ) is exactly the sum-polynomial kernel. For an actual relation $\chi \kappa$, repeated monic division gives polynomials Q_a with $\chi(\sum s_a) = \sum_a h(s_a) Q_a(\mathbf{s})$. The original cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa \left(\sum D_b (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \right) \quad (\text{EP.48})$$

has differential $\mathcal{V}_{h,k}(\chi \kappa)$, since $h(D)F_h = \Theta \phi_0$ and the tensor differential contributes the same sign $(-1)^{a-1}$, with all other differentials zero. Thus its quotient image is the supported zero $(\ell, 0)$ at the same label ℓ . The fibre map sends (ℓ, x) to $(\ell, R_N x)$ and the external absence τ to τ . This identifies the retained source cost in (EP.44)–(EP.46) with the relation sent to that supported zero through the actual maps.

The four-volume budget is linked exactly to the two-volume cost by

$$\mathcal{B} = 2 \log(V_n/V_m) + \log(\beta/\alpha). \quad (\text{EP.49})$$

The boundary term remains signed. The two-volume threshold alone does not bound it from below. A different, balanced window connects the original four endpoints without that signed comparison: taking $n = q$, $r = q - 1$ yields the central cost $\log(V_q/V_{2q-1})$, and monotonicity proves

$$\log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} \geq 2 \log \frac{V_q}{V_{2q-1}}. \quad (\text{EP.50})$$

The complete proof in the accompanying *Four-volume threshold* continuation, read in full, establishes

$$(\omega_{h,k,n+r}/\omega_{h,k,n})^{1/(2r)} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n, \quad (\text{EP.50a})$$

with the literal sufficient constant

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) M_* \max(1, \vartheta_h^{-1}) \max(1, b^{-3}) 6^{B_h/3}. \quad (\text{EP.50b})$$

It derives this from the original lower convolution envelope and the exact monic Legendre norm, retaining every factor before its $2r$ -th root. Its full source proof is included as a dependency. Applying (EP.13) now at $n = q, r = q - 1$ gives the following complete version with its retained penalties:

$$\begin{aligned} C_{\text{mid}} := \log \frac{V_q}{V_{2q-1}} &\geq 2(q-1) \log \frac{L}{C_h^{\text{bal}} q} - \log(1 - d_0) - \log(1 - d_{q-1}) \\ &\quad - 2 \sum_{j=1}^{q-2} \log(1 - d_j) + \sum_{j=0}^{q-2} \log(1 + \phi_{q+j}^2/L^2), \quad d_j = \delta_{q+j}. \end{aligned} \quad (\text{EP.50c})$$

All $d_j < 1$ follow again from $L > 0$ in (EP.11). The original four-volume budget has the exact positive decomposition

$$\log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = 2C_{\text{mid}} + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}}. \quad (\text{EP.50d})$$

Both extra terms are nonnegative. Dropping only nonnegative terms from (EP.50c)–(EP.50d) and using $L/q \geq \delta k/2$ proves

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 4. \quad (\text{EP.50e})$$

Indeed the numerator is at least $4(q_k - 1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q_k - 1)/q_k \rightarrow 1$. For the actual central relation block $(i, j) = (q, 2q - 1)$, its full matrix in (EP.45) has size $q - 1$, so the same determinant argument gives

$$\lambda_{\text{max}}^{\text{mid}} \geq \left(L / (C_h^{\text{bal}} q) \right)^2 - 1 \geq \delta^2 k^2 / (4(C_h^{\text{bal}})^2) - 1. \quad (\text{EP.50f})$$

The norm bound, the new product penalties, the central relation eigenvalues, and the original four-volume budget are therefore connected by these explicit maps. No upper estimate for those growing arithmetic relation matrices is proved by this lower-cost calculation.

45.7 Mass-seven polynomial calibration and zero cases

The independent finite fixture is the measure $7e^{-u^2/2} du / \sqrt{2\pi}$ and quotient $\mathbb{C}[u]/(u^2 + 1)$. Integration by parts gives monic Hermite polynomials $Q_0 = 1, Q_1 = u, Q_{j+1} = uQ_j - jQ_{j-1}$ and norms $\omega_j = 7j!$: the derivative identity $Q'_j = jQ_{j-1}$ follows by induction in the recurrence, and integration by parts proves orthogonality and $\|Q_j\|^2 = j\|Q_{j-1}\|^2$ starting from mass seven. In the quotient basis $(1, u)$ use

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A = \frac{1}{2}I + iT. \quad (\text{EP.51})$$

Its original- S relation is $\chi(S) = (S - 1/2)^2 - 1$. The exact map from the $(1, S)$ coefficients to $(1, u)$ is $E = \begin{pmatrix} 1 & 1/2 \\ 0 & i \end{pmatrix}$; $G_N^S = E^* G_N^u E$ and $|\det E| = 1$. Thus all fixture volumes are the original- S volumes as well. Kernel summation (EP.5), in the transported coordinates, gives

$$\begin{aligned} G_2^u &= \text{diag}(7/3, 7), \quad G_3^u = \text{diag}(7/3, 21/11), \quad G_4^u = \text{diag}(42/43, 21/11), \\ (V_1, V_2, V_3, V_4, V_5) &= (49, 49/3, 49/11, 882/473, 980/1333). \end{aligned} \quad (\text{EP.52})$$

For a diagonal $G = \text{diag}(g_0, g_1)$, $A^*G + GA - G = i(GT - T^T G)$ has relative squared radius $(g_0 + g_1)^2 / (g_0 g_1)$. The resulting exact values are

$$(\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = (16/3, 400/99, 4225/946), \quad U_{2,3} = 60, \quad (\alpha, \beta, t) = (1/3, 110/279, 54/473). \quad (\text{EP.53})$$

Reflection $u \mapsto -u$ conjugates the source and quotient matrices at tilt θ to those at $-\theta$ by diagonal sign matrices. Their determinants are even, and $\text{Tr } T = 0$, so $\phi_N = 0$ at zero tilt for this fixture. This proves that the direct defect and the radius identities test the same allowance.

All finite values in (EP.52)–(EP.53) are derived in the checker from literal Gaussian moments $7(2j - 1)!!$, monomial Gram matrices, monic polynomial division, and (EP.4), independently of assigned quotient matrices. Further windows check (EP.13), (EP.35), and (EP.38), while exact scalar extremizers check the new formula and phase boundary cases. These are finite auxiliary calculations. They do not assert that the Gaussian fixture is the arithmetic convolution measure; (EP.1)–(EP.7), (EP.32)–(EP.38) give the explicit common maps and the distinct specified source inputs.

For the mass-seven window, the exact inputs to the new reduced-budget root calculation are $U = 60$ and $e^B = 630509/1080$. The rational procedure (EP.31), followed by the positive sixth-root comparisons in (EP.26), gives the explicit certificate

$$\frac{719736229856370310835738215}{2^{88}} \leq \mathcal{AF}_{\max}^{1/6} \leq \frac{719736229856370310835738217}{2^{88}}. \quad (\text{EP.54})$$

Its value is approximately 2.325593185504684, with that decimal used only for orientation. The check record contains the two rational brackets for x , their exact comparisons in (EP.31), and the two sixth-power comparisons determining (EP.54).

The primary checker executes 415 exact auxiliary mathematical checks normally and under python -0. Its six actual formula changes alter the literal mass, the repeated interior contraction, the terminal ratio orientation, the Schur cancellation, the retained phase, and a raw derivative factorial; each changed run executes the full suite and is rejected in both modes. An independent scalar checker executes 2,453 exact checks in its unmodified runs, and its five changed-formula runs are rejected in both modes. Repeated modes and changed-formula runs do not add mathematical checks to those scopes. Their full source, complete records, independent mathematical reviews, and the eight exact proof dependencies accompany this article.

For the empty quotient, declare $q = 0$, $\chi = 1$, $C = 0$ and $V_N = 1$ including $N = -1$. Then $\epsilon_N = \phi_N = 0$, all quotient maps and relation-kernel update columns have their empty dimensions, and the exact product is zero. The formulas requiring $q \geq 1$ are used only in their stated range; the cochain boundary map and original source mass still exist. For $q = 1$ the transfer norm bands, when used, are empty products equal to one. At an endpoint contraction equal to one all zero-radius and phase cases are handled by (EP.11), (EP.13), and (EP.21) before logarithms.

46 The complete balanced four-volume threshold calculation

Source role. Complete owner calculation with the original arithmetic norm and volume, updated at its endpoint proof by the full AW/SP signed metric control.

Exact revision. 315a039b8cfbf8d54716f13cd9f67e2e8b1a146431eaf2c6045be040c8af8c2a

Complete source. `sources/owner_endpoint_terminal/published/workbenches/tau-arithmetic-endpoint-bounds/FOUR_VOLUME_THRESHOLD.md`

Source SHA-256. 315a039b8cfbf8d54716f13cd9f67e2e8b1a146431eaf2c6045be040c8af8c2a

46.1 The two endpoint calculations give a four-volume threshold of four

46.1.1 Result and motivation

The arithmetic endpoint note bounds the monic source norms. The subsequent window-product calculation retains the repeated occurrence of every interior volume contraction. Composing them on the window one degree shorter than the original window improves the original four-volume lower threshold from 2 to 4:

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 4.$$

This statement concerns the original arithmetic family belonging to a fixed packet consisting exactly of a full off-line quartet, with every multiplicity retained. It is a necessary volume-growth consequence for such a packet, not an assertion that an off-line zero exists. No upper bound contradicting this growth is established here.

This note is a written continuation of the two supplied calculations. It is not included in the frozen Toda-Gamma edition DOI [10.5281/zenodo.22730756](https://doi.org/10.5281/zenodo.22730756).

46.1.2 1. Original source, constants, and coordinates

Keep $g = 2\xi$, the fixed monic packet polynomial h , and

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad M_h(b) = \int e^{bt} w_h(t) dt.$$

The mass remains μ_h^k . The monic degree- j source norm is $\omega_{h,k,j}$ for the observation $S = k/2 + iu$. In this coordinate map the leading coefficient acquires the phase i^j , of squared modulus one. Neither the source measure nor the generator is replaced.

The arithmetic endpoint proof gives constants $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, and $B_h = 42 + 2 \deg h$, such that

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\pi(|u|+k-3)/2} (1 + |u| + k - 3)^{-B_h}, \quad k \geq 3. \quad (1)$$

Here (1) is the proved lower envelope of the actual threefold-and-higher convolution, not a newly postulated comparison measure. Its proof is in the [supplied complete arithmetic note](#).

Choose one $0 < b < \pi/2$ and define bounding constants

$$X_h = \max\{1, M_h(b), M_h(-b)\}, \quad K_h = \max\{1, \vartheta_h^{-1}\}.$$

Taking maxima here only bounds constants; it performs no rescaling of the source, its quotient, or either Gram determinant.

46.1.3 2. The balanced-window norm estimate, with an explicit constant

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq n$, one has

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq C_h^{\text{bal}} n, \quad (2)$$

where the following literal finite constant suffices:

$$C_h^{\text{bal}} = 256e^\pi \max\{1, 3/c_h\} X_h K_h \max\{1, b^{-3}\} 6^{B_h/3}. \quad (3)$$

To prove this, the trial polynomial $(S - k/2)^j$ and exponential moments give the unchanged upper estimate

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k).$$

The exact monic Legendre norm on $[-n, n]$ is

$$\mathfrak{l}_n(n) = \frac{2n^{2n+1}}{2n+1} \left(\frac{2^n}{\binom{2n}{n}} \right)^2 \geq \frac{2n^{2n+1}}{(2n+1)4^n}.$$

This follows from the Legendre leading coefficient and squared norm in [DLMF Table 18.3.1](#). Combining (1) with this least-norm calculation gives the exact finite bound

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b)^k + M_h(-b)^k) e^{\pi(n+k-3)/2} (1+n+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2r)}. \quad (4)$$

For completeness, use $(2(n+r))! \leq (4n)^{2(n+r)}$, $M_h(b)^k + M_h(-b)^k \leq 2X_h^k$, $\vartheta_h^{-(k-3)} \leq K_h^k$, and $1+n+k-3 \leq 2n$. After taking the root, the factorial and Legendre factors are bounded by

$$4n 8^{n/r} \left(\frac{2n+1}{c_h n} \right)^{1/(2r)}.$$

The remaining factors are

$$b^{-(n+r)/r} (X_h K_h)^{k/(2r)} e^{\pi(n+k-3)/(4r)} (2n)^{B_h/(2r)}.$$

Now $n/r \leq 2$, $(n+r)/r \leq 3$, $k/(2r) \leq 1$, and $(n+k-3)/(4r) \leq 1$. Also $2n+1 \leq 3n$, and $\log(2n)/n$ decreases for $n \geq 3$, so $(2n)^{B_h/(2r)} \leq 6^{B_h/3}$. These inequalities prove (2)–(3). The diagonal $r = n$ recovers the scale of the supplied theorem. The needed new case is $r = n-1$, which lies in the same balanced range.

46.1.4 3. The exact product on the same canonical metrics

Let $V_N = \det G_N$, $\delta_N = V_N/V_{N-1} \in (0, 1]$, and retain the calculated real phase $\phi_N = \sigma - \partial_\theta \log V_N$. The original radius identity is

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N)(\delta_{N+1}^{-1} - 1), \quad N \geq q.$$

Multiply at $N = n, \dots, n+r-1$, without discarding intermediate factors:

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - \delta_n)(1 - \delta_{n+r}) \prod_{j=1}^{r-1} (1 - \delta_{n+j})^2. \quad (5)$$

The original phase remains on the left. Each factor on the right after the ratio belongs to $[0, 1]$. The exterior theorem bounds the aggregate spectral defect $L_{h,k}$ by every ϵ_N . Thus (5) proves

$$L_{h,k}^{2r} \leq \frac{\omega_{h,k,n+r} V_n}{\omega_{h,k,n} V_{n+r}}. \quad (6)$$

This deduction needs neither a new metric nor an assumption that the phase vanishes. A unit contraction makes a factor in (5) zero and directly forces $L_{h,k} = 0$; the inequality (6) still holds. The first admitted degree $q-1$ is not used in any radius identity here.

46.1.5 4. Central window and the factor four

For the exact quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$ and common order m , keep

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

In particular $L_{h,k}/q_k \geq \delta k/2$. Put $q = q_k$ and choose $n = q, r = q-1$. All hypotheses of the proved balanced-window estimate are satisfied: $q \geq k \geq 3$, $q-1 \geq q/2$, and every radius index is at least q . Equations (2) and (6) yield

$$\log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q} \geq 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (7)$$

Write $C_k = \log(V_q/V_{2q-1})$. The original four-volume budget is exactly

$$\begin{aligned} \mathcal{B}_{h,k} &= \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} \\ &= 2C_k + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}} \geq 2C_k. \end{aligned} \quad (8)$$

Both last terms are nonnegative by the original rank-one kernel update. Substitution of (7) proves

$$\mathcal{B}_{h,k} \geq 4(q_k - 1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (9)$$

46.1.6 The signed metric control at these same four endpoints

The AW calculation supplies the full-spectrum estimate and its condition-number consequence. The SP calculation supplies the additional actual-overlap refinement. Their exact comparison ACM.1–13 proves that they act on the same source. We carry both calculations into the budget (8), rather than replacing its four signed terms by four separate errors. Here the integration coordinate remains the original u in $S = k/2 + iu$; the coordinate called y in ACM is exactly this u . The new variable $x \in [0, 1]$ is only the metric interpolation parameter.

Put $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{H} = \mathcal{P}_{2q}$ in its original ordered monomials, $v(u) = (1, S, \dots, S^{2q})$, and retain the full arithmetic density $m_1(u) = w_h^{*k}(u)$. The original Gamma comparison has the literal density and mass

$$\begin{aligned} r_a(u) &= \frac{|\Gamma(a + iu/2)|^2}{2\pi}, & c_a &= 2^{1-2a}\Gamma(2a), \\ m_0(u) &= \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}(u), & c_{1/4} &= \sqrt{2\pi}, & \int_{\mathbb{R}} m_0(u) du &= (\sqrt{2\pi})^k. \end{aligned} \quad (\text{FV.M1})$$

The complete Gamma convolution proof retained in this reader gives $m_0 = r_{1/4}^{*k}$. Thus (FV.M1) uses its original mass; the arithmetic mass is still μ_h^k . Define

$$m_x = (1 - x)m_0 + xm_1, \quad M_x = \int_{\mathbb{R}} v(u)^* v(u) m_x(u) du, \quad C_x = M_x^{-1}(M_1 - M_0). \quad (\text{FV.M2})$$

Both endpoint densities are positive almost everywhere and have all moments, so every M_x is positive: a nonzero polynomial has only finitely many zeros on the integration line. No matrix or density is divided by its mass.

Let $I_N : \mathbb{C}[S]_{\leq N} \hookrightarrow \mathcal{H}$ be coefficient inclusion and $B_N : \mathbb{C}[S]_{\leq N-q} \rightarrow \mathbb{C}[S]_{\leq N}$ be multiplication by the original monic χ . Negative-degree spaces are zero. Put $H_N(x) = I_N^* M_x I_N$, and use the original remainder J_N to set $G_N(x) = (J_N H_N(x)^{-1} J_N^*)^{-1}$ and $V_N(x) = \det G_N(x)$. The maps to the common source and their orthogonal projections are exactly

$$\begin{aligned} \tilde{B}_N &= I_N B_N, & P_N &= I_N H_N(x)^{-1} I_N^* M_x, \\ Q_N &= \tilde{B}_N (\tilde{B}_N^* M_x \tilde{B}_N)^{-1} \tilde{B}_N^* M_x, & Q_{q-1} &= 0. \end{aligned} \quad (\text{FV.M3})$$

The formula for Q_{q-1} uses the zero relation domain, with empty determinant one. For each other N , injectivity of multiplication by monic χ proves invertibility of its relation Gram. Direct multiplication proves $P_N^2 = P_N$, $Q_N^2 = Q_N$ and self-adjointness in M_x , with ranges $\mathcal{P}_N$ and $\chi\mathcal{P}_{N-q}$. These formulas are also the exact typed identities $\tilde{B}_N = B_N^{\text{SP}} = I_N B_N^{\text{AW}}$.

The coefficient matrix $[1, S, \dots, S^{q-1}, \chi, S\chi, \dots, S^{N-q}\chi]$ is triangular with diagonal one. Its Gram has determinant $\det H_N(x)$. Eliminating the relation block leaves its Schur complement $G_N(x)$: indeed for a fixed remainder z , minimizing over the relation coefficients solves the relation normal equations and gives precisely $z^* G_N(x) z$. Consequently

$$\det G_N(x) = \frac{\det H_N(x)}{\det(B_N^* H_N(x) B_N)}, \quad \frac{d}{dx} \log V_N(x) = \text{Tr}((P_N - Q_N)C_x). \quad (\text{FV.M4})$$

For the derivative, multilinearity of determinant gives $d \log \det H = \text{Tr}(H^{-1} dH)$. Insert $dH_N = I_N^*(M_1 - M_0)I_N$ in the numerator and $B_N^* dH_N B_N$ in the denominator. Cyclicity of the finite trace and $M_x C_x = M_1 - M_0$ give the two terms in (FV.M4).

Define the original signed function and its two projection sums by

$$\begin{aligned} F(x) &= \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q), & A_x &= U_x - W_x. \end{aligned} \quad (\text{FV.M5})$$

Keeping the signs of all four terms in (FV.M4) gives

$$F'(x) = \text{Tr}(A_x C_x) = \int_{\mathbb{R}} (m_1(u) - m_0(u)) v(u) A_x M_x^{-1} v(u)^* du, \quad (\text{FV.M6})$$

$$\mathcal{B}_{h,k} = F(1) = \mathcal{B}_{h,k}^{\Gamma} + \Delta_{h,k}, \quad \mathcal{B}_{h,k}^{\Gamma} = F(0), \quad \Delta_{h,k} = \int_0^1 F'(x) dx.$$

The scalar density uses the full inverse M_x^{-1} , not an ordinary Euclidean projection kernel. The integral identity follows by substituting $M_1 - M_0 = \int v^* v (m_1 - m_0)$ in the finite trace; its entries are integrable moments. The density is real because $A_x M_x^{-1}$ is Hermitian. It is signed, and its m_x integral equals $\text{Tr } A_x = 0$, as proved next.

The relation subspaces for $q, 2q-1, 2q$ are nested, of dimensions $1, q, q+1$. Hence U_x has eigenvalue two on $\chi \mathcal{P}_{q-1} \ominus \chi \mathcal{P}_0$ (dimension $q-1$), eigenvalue one on $\chi \mathcal{P}_0$ and $\chi \mathcal{P}_q \ominus \chi \mathcal{P}_{q-1}$ (one dimension each), and eigenvalue zero on its q -dimensional complement. The degree flag decomposes W_x identically: its two summands have overlap $\mathcal{P}_{2q-1} \ominus \mathcal{P}_q$ of dimension $q-1$, and two one-dimensional end pieces. Thus both U_x, W_x have rank $q+1$, trace $2q$ and trace of the square $4q-2$. These statements include $q=1$, when the overlap dimension $q-1$ is zero.

Let $c_1(x) \leq \dots \leq c_{2q+1}(x)$ be the eigenvalues of C_x in M_x , let $d_x = c_{2q+1}(x) - c_1(x)$, and put $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. Retain both exact pointwise expressions

$$S_{\text{AW}}(x) = c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)),$$

$$S_{\text{SP}}(x) = \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}, \quad (\text{FV.M7})$$

$$\mathfrak{I}_{h,k} = \int_0^1 \min \{ S_{\text{AW}}(x), S_{\text{SP}}(x) \} dx.$$

Here is the full finite proof of their applicability at this earlier endpoint. For a rank- r orthogonal projection E , its diagonal in an orthonormal eigenbasis of C_x lies in $[0, 1]$ and sums to r . Moving this diagonal mass to the lowest or highest eigenvalues gives $\sum_{j=1}^r c_j \leq \text{Tr}(EC_x) \leq \sum_{j=2q+2-r}^{2q+1} c_j$. Each of U_x, W_x is its range projection, of rank $q+1$, plus its eigenvalue-two projection, of rank $q-1$. Applying the two inequalities to these two summands and subtracting the resulting extrema gives $|F'(x)| \leq S_{\text{AW}}(x)$, with the multiplicities displayed in (FV.M7).

Dimension gives $s_x \geq 1$. Let E_x be the M_x -orthogonal projection onto the actual intersection of the two ranges. Each of U_x, W_x dominates its own range projection, which dominates E_x ; therefore $U_x - E_x$ and $W_x - E_x$ are positive and each has trace $2q - s_x$. The triangle inequality for their trace norms gives $\|A_x\|_1 \leq 4q - 2s_x$. The full spectra just proved also give

$$\text{Tr } A_x = 0, \quad \text{Tr } A_x^2 = 8q - 4 - 2 \text{Tr}(U_x W_x), \quad \|A_x\|_1^2 \leq (2q+1) \text{Tr } A_x^2. \quad (\text{FV.M8})$$

The last inequality is Cauchy-Schwarz on all $2q+1$ real eigenvalues. Subtract $(c_1 + c_{2q+1})I/2$ from C_x in the trace; the first identity leaves F' unchanged. In an eigenbasis of A_x , each diagonal entry of the centered C_x has absolute value at most $d_x/2$. Thus $|F'(x)| \leq d_x \|A_x\|_1/2 \leq S_{\text{SP}}(x)$. Both estimates hold at every x on the same original metric. Their pointwise minimum is integrable: all Grams and inverse Grams are smooth, all ordered eigenvalues are continuous, and s_x is Borel since its level sets are determined by vanishing and nonvanishing matrix minors. Consequently (FV.M6) gives the finite two-sided enclosure

$$\boxed{\mathcal{B}_{h,k}^{\Gamma} - \mathfrak{I}_{h,k} \leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\Gamma} + \mathfrak{I}_{h,k}.} \quad (\text{FV.M9})$$

For the positive original relative endomorphism $D = M_0^{-1} M_1$ on $(\mathcal{H}, M_0)$, let $b_1 \leq \dots \leq b_{2q+1}$ be all its eigenvalues. Direct factorization $M_x = M_0((1-x)I + xD)$ gives $C_x = ((1-x)I + xD)^{-1}(D - I)$ and hence

$$c_j(x) = \frac{b_j - 1}{1 - x + x b_j}, \quad \int_0^1 c_j(x) dx = \log b_j,$$

$$\mathfrak{I}_{h,k} \leq \min \left\{ \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \int_0^1 S_{\text{SP}}(x) dx \right\} \leq (2q-1) \log \frac{b_{2q+1}}{b_1}. \quad (\text{FV.M10})$$

The scalar fraction is increasing in b with derivative $(1-x+xb)^{-2}$, proving the stated eigenvalue order throughout the path. Integrating its logarithmic derivative proves the integral. For the last bound use $s_x \geq 1$ in (FV.M7) and integrate d_x . The common consequence is the earlier AW bound; the pointwise minimum retains both its spectral refinement and the additional SP overlap.

The recursive metric transport proof RMT1–22 also applies here on the actual moment source and supplies a further propagated refinement. Its coordinate map is $y \mapsto u = y$, leaving $S = k/2 + iu$ unchanged. Equation (FV.M1) equals its reference measure (RMT2)–(RMT3), including $c_{1/4}^k = (\sqrt{2\pi})^k$; the arithmetic measure is identically $w_h^{*k} du$ in both. Therefore all entries of its M_x equal (FV.M2). Its included relation matrix is $I_N B_N = \tilde{B}_N$ in (FV.M3), and its quotient is this same J_N . Substitution proves equality of both Gram inverses, every P_N, Q_N , and the four volumes individually. In particular its $\mathcal{B}(x)$ equals the original $F(x)$ in (FV.M5), with no reversal of an endpoint or alteration of a sign. Its quantities E_q, O_q, d_x are precisely S_{AW}, S_{SP}, d_x here.

The measure hypothesis of that result holds literally, rather than being assigned to an arbitrary coefficient form. Its required positivity $F(x) > 0$ can also be checked in these coordinates: if $F(x) = 0$, the two crossed nested ratios $(q-1, 2q)$ and $(q, 2q-1)$ both have determinant ratio one. Their least-norm sections, included in the common $\mathcal{H}$ by I_i, I_j , then agree, because $G_i - G_j = (I_i R_i - I_j R_j)^* M_x (I_i R_i - I_j R_j)$ is positive and determinant one forces equality of the two forms. In the first crossed pair the literal remainder section contains 1, which would be orthogonal to $\chi \mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \sum_j \bar{a}_j (k-S)^j$ for $\chi(S) = \sum_j a_j S^j$ equals $\overline{\chi(S)}$ on the same line. Orthogonality to $\chi \chi^{\#k}$ would force $\int |\chi(S)|^2 m_x(u) du = 0$, contrary to the positive degree-2q moment Gram. This retains the full original relation polynomial.

The further two entries in the current recursive control are calculated on these same operators. Retain

$$\begin{aligned} T_q(x) &= \frac{d_x}{2} \|U_x - W_x\|_1, \\ \Sigma_x &= \text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{2q+1}, \\ S_{HS}(x) &= \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}. \end{aligned} \tag{FV.M15a}$$

The midpoint-centering argument proving (FV.M8) already proves $|F'(x)| \leq T_q(x) \leq S_{SP}(x)$ before either trace-norm majorant is taken. Both inequalities are retained. For the second new entry, C_x is self-adjoint in M_x because $M_x C_x = M_1 - M_0$ is Hermitian. Its centered operator $C_x - \text{Tr}(C_x)I/(2q+1)$ has squared Hilbert–Schmidt norm exactly $\Sigma_x \geq 0$, by expanding its square and trace. Since $\text{Tr}(U_x - W_x) = 0$, replacing C_x by this centered operator leaves the trace pairing $F'(x)$ unchanged. Cauchy–Schwarz on the finite matrix entries in an M_x -orthonormal basis therefore gives $|F'(x)| \leq S_{HS}(x)$, with the other squared norm exactly the trace in (FV.M8). This basis is used only to prove the Hilbert-space inequality: both endpoint forms and every source-coordinate operator remain the original ones. The expressions in (FV.M15a) equal RMT10a–RMT10b after the coefficient identifications already proved above.

Put $a = 2q - 1$. The full-angle estimate of RMT11, whose two complete crossed angle lists and original zero angle are retained in that proof, therefore gives the simultaneous quantities

$$\begin{aligned} A_q(x) &= a\sqrt{1 - e^{-F(x)/a}} d_x, \\ \mathfrak{J}_{h,k} &= \int_0^1 \min\{S_{AW}(x), T_q(x), S_{SP}(x), S_{HS}(x), A_q(x)\} dx \leq \mathfrak{J}_{h,k}, \\ \mathfrak{J}_{h,k}^{\text{nl}} &= \int_0^1 \min\left\{d_x, \frac{S_{AW}(x)}{a\sqrt{1 - e^{-F(x)/a}}}, \frac{T_q(x)}{a\sqrt{1 - e^{-F(x)/a}}}, \right. \\ &\quad \left. \frac{S_{SP}(x)}{a\sqrt{1 - e^{-F(x)/a}}}, \frac{S_{HS}(x)}{a\sqrt{1 - e^{-F(x)/a}}}\right\} dx \leq \log \frac{b_{2q+1}}{b_1}. \end{aligned} \tag{FV.M15}$$

The exact source identities above identify all five retained estimates individually with RMT9–RMT13, including the exact trace-norm and centered Hilbert–Schmidt entries. Each bounds the same $|F'(x)|$, so their pointwise minimum does also. In particular it is at most the earlier three-entry minimum and the two-entry $\mathfrak{J}_{h,k}$; none of their full proofs is discarded. Strict positivity and compactness of the fixed path keep the displayed denominators nonzero. RMT13 proves measurability of the actual range intersection, and (FV.M10) computes the last integral of d_x . Integrating gives $|\Delta_{h,k}| \leq \mathfrak{J}_{h,k}$ directly.

For completeness the scalar coordinate used to propagate this refinement is $\mathcal{H}_a(B) = 2 \text{arcosh}(e^{B/(2a)})$. It has derivative $1/(a\sqrt{1 - e^{-B/a}})$ and inverse $2a \log \cosh(z/2)$, by direct differentiation and substitution on the positive real branches. Apply this derivative to the five bounds in (FV.M15) and integrate, exactly as in the

same-source RMT17, to obtain

$$|\mathcal{H}_a(\mathcal{B}_{h,k}) - \mathcal{H}_a(\mathcal{B}_{h,k}^\Gamma)| \leq \mathfrak{I}_{h,k}^{\text{nl}}. \quad (\text{FV.M16})$$

The constructed signed interval (ISM31)–(ISM37), also propagated in (RMT19)–(RMT22), is on these identical original forms. Its frequency y is this u , its endomorphism X_x is exactly $C_x = M_x^{-1}(M_1 - M_0)$, and its $\mathcal{A}_x$ is exactly $U_x - W_x = A_x$ by (FV.M3)–(FV.M6). Its included relation matrix is $I_N B_N$ and its minimum section is $I_N R_N(x)$: both have the same remainder, are orthogonal to the same relation range, and uniqueness of the positive minimum proves equality. The bare symbol $C : E \rightarrow \mathcal{H}$ in ISM denotes the literal remainder lift; it is not the endomorphism C_x here. Thus its scalar correction $\mathcal{C}_{\text{ISM}} = \mathcal{B}(M_1) - \mathcal{B}(M_0)$ equals $\Delta_{h,k}$, with the original Gamma-to-arithmetic orientation.

Fix an integer $p_N \geq 0$ solely as a finite truncation order and set $L_{\text{ISM}} = L_{\text{RMT}} = p_N$. This symbol changes none of the original polynomial cutoffs N , spectral quantity $L_{h,k}$, or parameters k, u, x . Retain the original positive eigenvalues $b_1, \dots, b_n$ of $D = M_0^{-1} M_1$, where $n = 2q + 1$, and define the exact finite operators

$$\begin{aligned} B_x^\circ &= (1 - x)I + xD, & \alpha_x &= \frac{2(1 - x) + x(b_1 + b_n)}{2}, \\ H_x^\circ &= I - B_x^\circ/\alpha_x, & C_x^{[p_N]} &= \alpha_x^{-1} \sum_{r=0}^{p_N} (H_x^\circ)^r (D - I), \\ R_x^{[p_N]} &= (H_x^\circ)^{p_N+1} C_x, & E_x^{[p_N]} &= R_x^{[p_N]} - \frac{\text{Tr } R_x^{[p_N]}}{n} I. \end{aligned} \quad (\text{FV.M16a})$$

Multiplying the finite geometric sum by $I - H_x^\circ$ gives $C_x - C_x^{[p_N]} = R_x^{[p_N]}$. Every factor is a real function of D and self-adjoint in the original M_x metric. Trace zero of A_x therefore gives the exact signed decomposition $F'(x) = \text{Tr}(C_x^{[p_N]} A_x) + \text{Tr}(E_x^{[p_N]} A_x)$. Define the actual center and residual, preserving their signs, by

$$\begin{aligned} j_{p_N} &= \int_0^1 \text{Tr}(C_x^{[p_N]} A_x) dx, \\ e_{p_N} &= \int_0^1 \sqrt{\text{Tr}((E_x^{[p_N]})^2) \text{Tr}(A_x^2)} dx, \\ \vartheta &= \frac{b_n - b_1}{b_n + b_1}, & K_D &= \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \end{aligned} \quad (\text{FV.M16b})$$

They are exactly J_{p_N}, ϵ_{p_N} of ISM36 and j_{p_N}, e_{p_N} of RMT20. Cauchy–Schwarz applied to the last trace in the exact decomposition and then integration proves

$$j_{p_N} - e_{p_N} \leq \Delta_{h,k} \leq j_{p_N} + e_{p_N}, \quad 0 \leq e_{p_N} \leq \vartheta^{p_N+1} \sqrt{K_D(8q-6)} \longrightarrow 0. \quad (\text{FV.M16c})$$

For the stated bound, the eigenvalues of H_x° have absolute value at most $\vartheta < 1$, as the midpoint of the extreme eigenvalues $1 - x + xb_1$ and $1 - x + xb_n$ is precisely α_x . The original eigenvalues of C_x are $(b_j - 1)/(1 - x + xb_j)$. Their absolute values are at most $|b_j - 1|/\min\{1, b_j\}$. Centering the residual subtracts the nonnegative square of its trace divided by n from its squared Hilbert–Schmidt norm. This proves $\text{Tr}((E_x^{[p_N]})^2) \leq \vartheta^{2p_N+2} K_D$. Both positive U_x, W_x dominate their range projections, whose intersection has dimension at least one. Positivity of the trace product gives $\text{Tr}(U_x W_x) \geq 1$, so (FV.M8) gives $\text{Tr}(A_x^2) \leq 8q - 6$, proving (FV.M16c). All of these are continuous finite matrix integrands. Their exact integrals are the objects used here; a numerical integration must retain its own integration error. Convergence is for this fixed pair of original source forms. It asserts no estimate uniform in k . The complete original-source and proper-support proof for this construction is (ISM38)–(ISM43) in Section 250. To compare its primitive with (FV.M12), instantiate only that primitive calculation with the ordered pair (M_1, M_x) . Its section difference is then exactly $\rho_{\text{ISM},1} - \rho_{\text{ISM},0} = I_N(R_N(x) - R_N(1))$, so its boundary has precisely the orientation in (FV.M12). At $x = 0$ this is the negative of the section difference for the original ordered pair (M_0, M_1) ; linearity multiplies its primitive and proper-source residual by -1 . This typed primitive comparison leaves the original Gamma-to-arithmetic orientation of $\Delta_{h,k}$ and the signed estimates above unchanged.

Write $H_0 = \mathcal{H}_a(\mathcal{B}_{h,k}^\Gamma)$ and retain the complete simultaneous endpoints

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{lo}} &= \max\left\{0, \mathcal{B}_{h,k}^\Gamma - \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{\max\{0, H_0 - \mathfrak{J}_{h,k}^{\text{nl}}\}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} - e_{p_N} \right\}, \\ \mathcal{B}_{h,k}^{\text{hi}} &= \min\left\{ \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{H_0 + \mathfrak{J}_{h,k}^{\text{nl}}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} + e_{p_N} \right\}. \end{aligned} \quad (\text{FV.M17})$$

The inverse is increasing and extends continuously by zero at zero. Applying it to (FV.M16) and intersecting the resulting interval with $|\Delta_{h,k}| \leq \mathfrak{J}_{h,k}$ and the oriented interval (FV.M16c) proves, for every fixed truncation order $p_N \geq 0$,

$$\mathcal{B}_{h,k}^{\text{lo}} \leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\text{hi}} \leq \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k} \leq \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k}. \quad (\text{FV.M18})$$

These formulas use the complete proof in Section 251, with the actual measure and coefficient correspondences just proved. The preceding generic signed matrix calculation (FV.M3)–(FV.M10) remains valid on its own stated domain; the full-angle refinement here is specialized to the actual original arithmetic/Gamma moment path.

Finally let the two unchanged arithmetic endpoint losses be $E_0 = \log(V_{q-1}/V_q)$ and $E_1 = \log(V_{2q-1}/V_{2q})$. Equation (8), with (FV.M6), gives the literal central identity and bounds

$$\begin{aligned} 2C_k &= \mathcal{B}_{h,k}^\Gamma + \Delta_{h,k} - E_0 - E_1, \\ \max\left\{0, \frac{\mathcal{B}_{h,k}^{\text{lo}} - E_0 - E_1}{2}\right\} &\leq C_k \leq \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}, \\ 4(q-1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q} + E_0 + E_1 &\leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\text{hi}}. \end{aligned} \quad (\text{FV.M11})$$

The first line is substitution in (8); the second uses (FV.M18) and $C_k \geq 0$; the third uses the first inequality in (7) without discarding either endpoint loss. All expressions retain the original full packet h , its exact χ , and the constants in (1)–(4). This gives an actual finite control for the old four-volume expression. It supplies no uniform estimate of $\mathfrak{J}_{h,k}$ as the packet order grows, so the original lower-growth conclusion (9) remains intact.

The comparison is induced by identity polynomial maps on every $\mathcal{P}_N$ and the identity on $\mathbb{C}[S]/(\chi)$. For the original source realization $\mathcal{V}P = P(\sum D_i)F_h^{\otimes k}$ and complete unit map $\eta[P] = [g/h]_h^{\otimes k} P(\sum s_i)$, these identities commute exactly with ηJ_N ; the values and every raw derivative of each polynomial are unchanged. The least lift at x is $R_N(x) = H_N(x)^{-1} J_N^* G_N(x)$, so $J_N(R_N(x) - R_N(1)) = 0$. Monic division gives a unique coefficient map $\kappa_N(x)$ with $R_N(x) - R_N(1) = B_N \kappa_N(x)$. Retain the original fixed-order tensor division $\chi(\sum s_i) = \sum_i h(s_i) Q_i(\mathbf{s})$. On a quotient vector z , the complete original theta primitive of this difference is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D})(\kappa_N(x)z) (\sum_j D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{FV.M12})$$

The tensor differential contributes the second sign $(-1)^{i-1}$ and $h(D_i)F_h = \Theta\phi_0$, so its boundary is exactly $\mathcal{V}(R_N(x) - R_N(1))z$. Thus the Gram comparison is attached to an explicit original source map and primitive; it does not identify the interpolated norm with the original arithmetic norm. For each unchanged outer label and independent coefficient face, retain the actual declared subcomplex $C_\lambda^\bullet \subseteq C_{\text{full}}^\bullet$ and its degree- $(k-1)$ source C_λ^{k-1} . Write Ξ for the primitive in (FV.M12). Whenever $d\Xi$ belongs to C_λ^k , its exact residual is $[d\Xi] \in C_\lambda^k / d(C_\lambda^{k-1})$. The full top-degree comparison has the exact kernel

$$\ker[H^k(C_\lambda^\bullet) \longrightarrow H^k(C_{\text{full}}^\bullet)] = \frac{d(C_{\text{full}}^{k-1}) \cap C_\lambda^k}{d(C_\lambda^{k-1})}. \quad (\text{FV.M13})$$

Here both complexes have top degree k , so all degree- k elements are cycles. A class $[z]$ maps to zero precisely when $z = d\Xi$ for some $\Xi \in C_{\text{full}}^{k-1}$; it is already zero in the declared source precisely when $z = d\Xi_\lambda$ for a declared primitive. This proves (FV.M13), including its injective inclusion into $H^k(C_\lambda^\bullet)$. Equivalently the

primitive map has domain $\{\Xi \in C_{\text{full}}^{k-1} : d\Xi \in C_{\lambda}^k\}$ and kernel $C_{\lambda}^{k-1} + \ker d$, so quotienting by this exact kernel gives the same residual space. This retains possible cancellations among tensor slots. For one leg with unchanged top source $\mathcal{B}$ and injective $\Theta : V \rightarrow \mathcal{B}$, it becomes the exact sequence

$$0 \longrightarrow V/W \xrightarrow{[v] \mapsto [\Theta v]} \mathcal{B}/\Theta W \longrightarrow \mathcal{B}/\Theta V \longrightarrow 0. \quad (\text{FV.M14})$$

The first map is injective because $\Theta v \in \Theta W$ forces $v \in W$; its image is the kernel of the quotient map, which is surjective by definition. Thus a relation with primitive in the declared source maps to its supported zero. A general full-source primitive retains the exact residual (FV.M13) on every coefficient face, rather than declaring its proper-source class zero. Identity polynomial maps retain the outer labels; external τ still maps to τ and an empty coefficient face retains its outer label.

Since h is fixed and $q_k \rightarrow \infty$, division by $q_k \log k$ gives the claimed $\liminf$ of at least 4. An upper limit strictly below 4 would already conflict with (9); no such upper estimate has been obtained here. The two-volume conclusion at $n = r = q$ is also retained: $\liminf \log(V_q/V_{2q})/(q \log k) \geq 2$.

46.1.7 5. Source maps and verification scope

All representatives remain those of the supplied arithmetic construction: $J^{(k)}R_N = \eta_{h,k}$ and $q^{(k)}R_N = \sigma_h^{\otimes k} \eta_{h,k}$. Restricting the degree window does not alter their unit, jets, fibre labels, or cohomology classes. The supported-zero relation and the external absence τ retain their original separate maps. In particular, no repeated exterior iteration or erased multiplicity cost occurs in (5)–(9).

The analytic envelope and original norm theorem are from the supplied Arithmetic Endpoint Bounds note. Equation (5) is the subsequent supplied window-product calculation. Equations (2)–(4) and their composition (7)–(9) are the present continuation. The two independent review lanes record finite tests separately from written analytic proof; no new Lean certificate or interval enclosure of C_h^{bal} is asserted.

47 The complete balanced-window arithmetic proof

Source role. Complete owner balanced-window proof and its exact parameter domain.

Exact revision. a3bb84329d384071b4ac4829731e26dbe2586b719548a660bae6eefb34f5e00c

Complete source. `sources/owner_endpoint_terminal/published/workbenches/tau-arithmetic-endpoint-bounds/BALANCED_WINDOW_PROOF.md`

Source SHA-256. a3bb84329d384071b4ac4829731e26dbe2586b719548a660bae6eefb34f5e00c

47.1 Balanced arithmetic norm windows and the strengthened endpoint threshold

This is a written analytic extension of the supplied arithmetic endpoint note, not a Lean certificate or a numerical enclosure. The fixed packet, original source, Taylor unit, measure and total mass are unchanged. It is intended for the next integration cut, not the already frozen E publication.

47.1.1 Original measure and constants

Fix a monic finite divisor h of $g=2x$, removing its selected zeros with their full orders, and put $d=\deg h$. On the original line $S=k/2+iu$, use

$$w_h(t) = |(g/h)(1/2+it)|^{2/(2\pi i)}, \quad m_h(h,k) = w_h^{k-1}, \\ \omega_{h,k}(j) = \inf_{P \text{ monic, degree } j} \int_{\mathbb{R}} |P(k/2+iu)|^2 m_h(h,k)(u) du.$$

The mass is μ_h^{k-1} , not one. The original note proves, for $a=\pi/2$ and $B=42+2d$,

$$m_h(h,k)(u) \geq c_h \vartheta_h^{k-3} \exp[-a(|u|+k-3)] (1+|u|+k-3)^{-B}, \quad k \geq 3,$$

where $c_h = b_h c_h^{k-1} > 0$, $b_h = \min_{|v| \leq 1} (w_h^{k-1}(v)) > 0$, and $\vartheta_h = \int_{-1,1} w_h^{k-1} > 0$. Fix $0 < b < a$ and retain both actual finite exponential moments $M_+ = \int \exp(bt) w_h^{k-1}(t) dt$ and $M_- = \int \exp(-bt) w_h^{k-1}(t) dt$. Define $X = \max(1, M_+, M_-)$ and $K = \max(1, \vartheta_h^{k-1})$. None of these definitions normalizes the source.

47.1.2 Exact preceding estimate

The monic Lebesgue minimum, allowing any leading coefficient of modulus one, is

$$\text{ell}_n(L) = [2 L^{(2n+1)/(2n+1)}] [2^n (n!)^{2/(2n)!}]^2.$$

This includes the original leading phase i^n under $S=k/2+iu$. For integers $n \geq 0$, $r \geq 1$, $k \geq 3$ and any $L > 0$, the trial-monomial exponential upper bound and interval Legendre lower bound give

$$\begin{aligned} A_{\text{h,k,n,r}} &:= (\omega_{\text{h,k,n+r}}/\omega_{\text{h,k,n}})^{1/(2r)} \\ &\leq [(2(n+r))! b^{-(2(n+r))} (M_+^{+k} M_-^{-k}) \\ &\quad \exp[a(L+k-3)] (1+L+k-3)^B \\ &\quad / (c_h \text{vartheta}_h^{(k-3)} \text{ell}_n(L))]^{1/(2r)}. \end{aligned} \quad (\text{BW1})$$

For the balanced-window theorem take integers $n \geq k \geq 3$ and $n/2 \leq r \leq n$, and set $L=n$. Then

$$A_{\text{h,k,n,r}} \leq \text{Cbal}_h n, \quad (\text{BW2})$$

$$\begin{aligned} \text{Cbal}_h &= 256 \exp(\pi i) \max(1, 3/c_h) \times K \\ &\quad \max(1, b^{(-3)}) 6^{(B/3)}. \end{aligned} \quad (\text{BW3})$$

Proof of the uniform constant The central-binomial bound gives $\text{ell}_n(n) \geq 2 n^{(2n+1)/[(2n+1)4^n]}$. Also $(2(n+r))! \leq (4n)^{(2(n+r))}$ and $M_+^{+k} M_-^{-k} \leq 2X^k$. Combining these before taking roots cancels the factor two. The factorial, Legendre and c_h factors after the $2r$ -th root are at most

$$\begin{aligned} n * 4 * 8^{(n/r)} * [(2n+1)/(c_h n)]^{1/(2r)} \\ \leq 256 n \max(1, 3/c_h). \end{aligned}$$

Here $n/r \leq 2$, $(2n+1)/n \leq 3$, and $0 < 1/(2r) \leq 1$. The remaining factors obey

$$\begin{aligned} X^{(k/(2r))} &\leq X, \\ \text{vartheta}_h^{(-(k-3)/(2r))} &\leq K, \\ b^{(-(n+r)/r)} &\leq \max(1, b^{(-3)}), \\ \exp[a(n+k-3)/(2r)] &\leq \exp(2a) = \exp(\pi i). \end{aligned}$$

Finally $1+n+k-3 \leq 2n$ and $2r \geq n$. The function $\log(2x)/x$ decreases for $x \geq 3$, since its derivative is $[1 - \log(2x)]/x^2 < 0$. Hence

$$(1+n+k-3)^{(B/(2r))} \leq (2n)^{(B/n)} \leq 6^{(B/3)}.$$

Multiplication proves BW2-BW3. This proof works for the original arithmetic density without an evenness hypothesis. The constants depend on fixed h and b only, never n, k, r .

47.1.3 Composition with the exact consecutive-window product

This paragraph assumes the separately reviewed exact-window theorem for the original canonical quotient family, including its nonnegative contractions and zero-contraction cases. Write $V_N = \det G_N > 0$ and $\Lambda_N = \omega_N/V_N$. Its simple product corollary is

$$\begin{aligned} \min_{(n \leq N \leq n+r)} \epsilon_N \\ \leq (\Lambda_{n+r}/\Lambda_n)^{1/(2r)} \\ = A_{\text{h,k,n,r}} (V_n/V_{n+r})^{1/(2r)}. \end{aligned} \quad (\text{BW4})$$

Do not infer BW4 from BW1 alone. It uses the original rank-two Toda identities and the retained source/quotient maps. If every allowance on the window has the exterior lower bound $L_{\text{h,k}}$, BW2-BW4 imply

$$\log(V_n/V_{n+r}) \geq 2r \log[L_{\text{h,k}}/(\text{Cbal}_h n)]. \quad (\text{BW5})$$

This holds even if the right side is negative; canonical volume monotonicity independently gives a nonnegative left side.

For the packet consisting exactly of the hypothetical off-line quartet $1/2 \pm \delta \pm i \gamma$, with fixed $\delta, \gamma > 0$ and common multiplicity $m \geq 1$, retain

$$q = q_k = [1+k(m-1)](k+1)^2,$$

$$L_-(h, k) = 2\delta [1+k(m-1)](k+1)\text{floor}((k+1)^2/4),$$

$$q \geq k \geq 3, \quad L_-(h, k)/q \geq \delta k/2.$$

Take $n=q$ and $r=q-1$. Since $q \geq 3$, $q/2 \leq q-1 \leq q$; the degrees $q, \dots, 2q-2$ are admitted and the norm endpoint is $2q-1$, not $2q$. Define the central two-volume cost

$$C_k = \log(V_q/V_{(2q-1)}).$$

Then

$$C_k \geq 2(q-1) \log[\delta k/(2 C_{bal_h})]. \quad (\text{BW6})$$

Canonical monotonicity gives $V_{(q-1)} \geq V_q$ and $V_{(2q-1)} \geq V_{(2q)}$. Consequently each outer cost dominates this same central cost:

$$\log(V_{(q-1)}/V_{(2q-1)}) \geq C_k,$$

$$\log(V_q/V_{(2q)}) \geq C_k.$$

For the original four-volume budget $B_k = \log[V_{(q-1)}V_q/(V_{(2q-1)}V_{(2q)})]$,

$$B_k \geq 2 C_k \geq 4(q-1) \log[\delta k/(2 C_{bal_h})],$$

$$\liminf_{(k \rightarrow \infty)} B_k/(q_k \log k) \geq 4. \quad (\text{BW7})$$

Indeed $q_k \geq (k+1)^2$ tends to infinity, so $(q_k-1)/q_k$ tends to one; fixed positive $\delta/(2C_{bal_h})$ contributes only $O_h(q_k)$. The diagonal choice $r=q$ also yields $\log(V_q/V_{(2q)}) \geq 2q \log[\delta k/(2C_{bal_h})]$, but the overlapping shorter window proves the stronger four-volume threshold.

The supplied note's older threshold 2 remains correct under its weaker sinh endpoint estimate. BW7 is a strengthened synthesis using BW4, not a correction to that valid implication. No upper estimate on B_k or C_k follows. A separately proved $\limsup B_k/(q_k \log k) < 4$ would contradict this quartet; no such upper estimate and no RH proof is asserted here.

47.1.4 Reference and evidence boundary

The analytic ingredients are the zeta reflection formula (DLMF 25.4.2), the gamma vertical-strip estimate, uniform for bounded real part (DLMF 5.11.9), and the Legendre orthogonality/leading coefficient (DLMF 18.3). The annulus propagation, convolution restriction, BW1-BW3 and BW5-BW7 are written derivations. Finite regression checks support algebraic fixtures only, not the analytic quantifiers or arithmetic zero hypotheses.

48 PR 24: complete implemented corrected proof

Source role. Complete corrected proof. The sealed proposal bytes equal the terminal adopted corrected note; both provenance identities are retained with their exact adoption receipts.

Exact revision. 47eb717484d92c96a062a0990a4567a0929e33e1038742e538b610fcc157b9d6

Complete source. sources/owner_endpoint_review/review_pr24/proposed/RESEARCH_NOTE.md

Source SHA-256. 47eb717484d92c96a062a0990a4567a0929e33e1038742e538b610fcc157b9d6

48.1 Confluent transfer for the original source–relation volume

48.1.1 Scope and current-main integration

This add-only analytic continuation starts from main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f, after the merge of PR #21 and its workflow-trigger correction. The PR #22 mathematical note was read at 8112 10d24b80813a08972ca23f919db015383137; its strict verification was in progress at that reading. This contribution neither modifies that branch nor treats pending checks as a certificate.

The source is the delivered Toda-volume continuation, following the cyclic and exterior-trace notes. Its full LaTeX was read directly. The new result is a fixed-width, full-jet transfer for the SAME canonical

source/relation quotient metric. It applies the classical Christoffel transformation; it is not a claim to a new general orthogonal-polynomial theorem or to RH.

Keep $G(\mathbb{Z}) \rightarrow \mathbb{Z}$, with its infinite target, and its coefficient square to $G(\mathbb{C}) \rightarrow \mathbb{C}$. The structural base remains b_τ . Every supported coefficient map below retains $e=0^\bullet$ and external τ separately. The original theta complex is $C_+ = [V \rightarrow B]$, with $D = -x d/dx$, $Q = B/\Theta V$ and $g=2$. No scalar, source function or measure is assigned a new normalization.

For a nonempty finite packet h of actual zeros with full orders, retain F_h with Mellin transform g/h . For the doubled-original-root transfer below, require the same dagger stability as the original source: its cyclic annihilator satisfies the polynomial identity $\text{conj}(\chi)(k-S) = (-1)^q \chi(S)$. The complete reflection-stable quartet packets used in the final target satisfy this hypothesis. It does not follow from being an arbitrary finite subpacket of zeros. At tensor degree $k \geq 1$, keep the actual cyclic sum module

```
C=C[S]/(chi),  q=deg chi >= 1,  c=k/2,  A=M_S,
eta[P]=upsilon_h^(tensor k) P(sum_i M_(s_i))1,
upsilon_h=j_h(g/h),
Vcal P=P(D_1+...+D_k)(F_h^(tensor k)).
```

The original equations are

```
J^(k) Vcal=eta pi_chi,
q^(k) Vcal=sigma_h^(tensor k) eta pi_chi.
```

Its norm is the actual integral against $m_k = w_h^{\{k\}}$, where

$$w_h(t) = |(g/h)(1/2 + i t)|^{2/(2\pi)}, \quad \text{integral } m_k = \mu_h^k.$$

The auxiliary measure is $dm_\theta(u) = \exp(\theta u) m_k(u) du$, $|\theta| < \pi/2$. Its mass is $M_h(\theta)^k$. θ in this expression is the real tilt parameter, not the theta-summation map. The exponential-moment bound from the source makes all finite derivatives below valid. At tilt zero the metric is the original arithmetic metric.

For $h=1$ the finite arithmetic quotient is zero. Its determinant is the empty determinant 1; the analytic seed and its mass remain, and $G(0)$ still has e and τ . The positive-dimensional inverses below require $q \geq 1$.

48.1.2 1. Exact coordinate and raw-jet maps

Use the ring isomorphism

$$\text{Psi}: C[S] \rightarrow C[u], \quad P \mapsto P(c+iu), \quad u \mapsto (S-c)/i.$$

Thus $d_u \text{Psi}(P) = i \text{Psi}(P')$, and multiplication by S remains $c+iu$. Define the original norm polynomial

$$\text{Pi}(u) = \text{conj}(\chi)(c-iu) \chi(c+iu).$$

Its leading coefficient is literally $(-i)^q i^q = 1$. It is real and nonnegative on $\mathbb{R}$. Reflection gives $\chi^\dagger = (-1)^q \chi$, hence

```
psi(u)=i^(-q) chi(c+iu) is real and monic,
Pi=psi^2,  Psi(chi)=i^q psi.
```

The factor i^q remains in the source multiplication map. For distinct roots λ_a of χ , of complete lengths ℓ_a , put

$$\zeta_a = (\lambda_a - c)/i, \quad r_a = 2\ell_a, \quad r = \sum r_a = 2q.$$

Explicitly, the coordinate isomorphism Psi induces $C[S]/\chi^2 \rightarrow C[u]/\text{Pi}$ because $\text{Psi}(\chi^2) = i^{2q} \text{Pi}$, and the raw jet conversion is $(d_u^d \text{Psi}(P))(\zeta_a) = i^d P^{(d)}(\lambda_a)$. This is the actual map used below, including every derivative factor. Without the displayed dagger identity, Pi need not equal psi^2 and its roots need not be the original ζ_a with doubled orders; that unsupported general-packet version is not asserted here.

Define raw, NOT factorial-divided, derivative coordinates

$$J_2 f = (f^d)(\zeta_a)_{(a, 0 \leq d \leq r_a)}.$$

Their kernel is (Π) . Their product uses the full binomial rule. Multiplication by u has matrix

$$(X_{\Pi} v)_{(a,d)} = \zeta_a v_{(a,d)} + d v_{(a,d-1)}, \quad v_{(a,-1)} = 0.$$

All nilpotent orders and the coefficient d are retained.

Let Q_j be the real monic orthogonal source polynomials for $d\mu_\theta$, obtained by subtracting lower-degree projections, and write

$$\begin{aligned} \omega_j &= \int Q_j^2 d\mu_\theta > 0, \\ a_j &= \omega_j / \omega_{j-1}, \quad a_0 = 0, \\ b_j &= \int u Q_j^2 d\mu_\theta / \omega_j. \end{aligned}$$

The source recurrence is

$$u Q_j = Q_{(j+1)} + b_j Q_j + a_j Q_{(j-1)}, \quad Q_{(-1)} = 0.$$

No Q_j is divided by its norm. Set

$$\begin{aligned} v_j &= J_2 Q_j, \\ F_n &= [v_n, \dots, v_{(n+r-1)}], \\ V &= [J_2 1, \dots, J_2 u^{(r-1)}]. \end{aligned}$$

Both are r by r . The raw confluent determinant is

$$\det V = \left(\prod_a \prod_{d=0}^{r_a-1} d! \right) \prod_{a < b} (\zeta_b - \zeta_a)^{(r_a - r_b)}.$$

The fixed row order is part of this formula. $F_0 = V$ times the actual unit-triangular polynomial coefficient matrix; only their determinants are equal.

48.1.3 2. Invertibility and the original relation polynomial

Invertibility proof. Suppose F lies in $\text{span}(Q_n, \dots, Q_{(n+r-1)})$ and $J_2 F = 0$. Then $F = \Pi H$ with $\deg H < n$. Orthogonality gives

$$0 = \langle H, F \rangle = \int \Pi |H|^2 d\mu_\theta.$$

The integrand is nonnegative and its weight is positive almost everywhere except finitely many points. Thus $H=0$, then $F=0$. For $n=0$ the degree argument suffices. Hence every F_n is invertible. This does NOT estimate its inverse uniformly.

Solve

$$d_n = F_n^{(-1)} v_{(n+r)}, \quad \text{vartheta}_n = -(d_n)_0.$$

The polynomial

$$\hat{Q}_n = (Q_{(n+r)} - \sum_{j=0}^{r-1} (d_n)_j Q_{(n+j)}) / \Pi$$

exists by exact raw-jet divisibility and is monic of degree n . Pairing its numerator with every polynomial of degree below n proves that $\hat{Q}_n$ is precisely the monic orthogonal polynomial for the ORIGINAL relation weight $\Pi d\mu_\theta$. Its norm is

$$\begin{aligned} \nu_n &= \int \hat{Q}_n^2 \Pi d\mu_\theta \\ &= -(d_n)_0 \omega_n = \omega_n \text{vartheta}_n > 0. \end{aligned} \quad (1)$$

Indeed only the Q_n column pairs with the degree- n quotient; its coefficient is $-(d_n)_0$. This proves $\text{vartheta}_n > 0$ with arbitrary nonreal and repeated nodes.

Let $K_n = [e_1, \dots, e_{(r-1)}, d_n]$. Then

$$\begin{aligned} F_{n+1} &= F_n K_n, \\ \det K_n &= (-1)^{(r-1)} (d_n)_\theta = \text{vartheta}_n. \end{aligned} \quad (2)$$

The sign uses the displayed even $r=2q$. On the same jet algebra,

$$\begin{aligned} T_n &= F_n^{(-1)} X_{\text{Pi}} F_n, \\ T_{n+1} &= K_n^{(-1)} T_n K_n, \\ d_n &= (T_n - b_{n+r-1}) I_{e_{r-1}} - a_{n+r-1} e_{r-2}. \end{aligned} \quad (3)$$

Thus the full operator and its jets have an explicit discrete transport, rather than only a determinant observation.

48.1.4 3. Fixed-width determinant and quotient-volume formulas

Let D_n be the n by n source Hankel determinant, and B_n the corresponding determinant for Pi dm_θ . $D_0 = B_0 = 1$. Orthogonality gives

$$D_n = \text{product}_{(j < n)} \omega_j, \quad B_n = \text{product}_{(j < n)} \nu_j.$$

Multiplying (1) and (2) proves the complete confluent Christoffel identity

$$B_n/D_n = \text{product}_{(j < n)} \text{vartheta}_j = \det F_n / \det V. \quad (4)$$

This is a proof from the source division and norm, not a distinct-root approximation. Raw derivative factorials remain in $\det V$.

For $N \geq q-1$ set $n = N-q+1$. The source sequence is still

$$\emptyset \rightarrow P_{(N-q)} \xrightarrow{-\chi} P_N \rightarrow C[S]/\chi \rightarrow \emptyset.$$

Its canonical least-norm quotient Gram G_N satisfies the existing Schur identity $\det G_N = D_{(N+1)}/B_n$. Consequently

$$V_N := \det G_N = (\text{product}_{(j=n)}^{(n+q-1)} \omega_j) \det V / \det F_n. \quad (5)$$

The constant coefficient map $S = c + i u$ has determinant of modulus one on each remainder basis, so these are the original quotient determinants, with that coordinate map retained.

For $n \geq 1$,

$$\begin{aligned} \delta_N &= V_N/V_{(N-1)} = \omega_N/\nu_{(n-1)} \\ &= \omega_N/(\omega_{(n-1)} \text{vartheta}_{(n-1)}), \end{aligned} \quad (6)$$

$$\begin{aligned} R_N &= V_{(N-1)}/V_{(N+1)} \\ &= \omega_{(n-1)} \omega_n \text{vartheta}_{(n-1)} \text{vartheta}_n \\ &\quad / (\omega_N \omega_{(N+1)}). \end{aligned} \quad (7)$$

The transfer width $2q$ does not grow with N . The source recurrence and high-degree polynomial evaluation are still required; no computational-complexity or conditioning claim is inferred from the width alone.

48.1.5 4. One additional solve retains the full phase

Differentiating actual source orthogonality under $\exp(\theta u)$ gives

$$d_\theta Q_j = -a_j Q_{(j-1)}, \quad d_\theta \log \omega_j = b_j.$$

For the first identity, every coefficient below $j-1$ vanishes by differentiated orthogonality; pairing with $Q_{(j-1)}$ gives $-\omega_j$. Let L_n have entries $(L_n)(j-1, j) = -a_{(n+j)}$ for $1 \leq j \leq r$, all other entries zero. Then

$$F_n' = F_n L_n - a_n v_{(n-1)} e_\theta^T. \quad (8)$$

At $n=0$ the last term is zero. Trace of L_n is zero. Hence

$$\begin{aligned}
(\log \det F_n)' &= -a_n e_{\theta}^T F_n^{(-1)} v_{(n-1)}, \\
(\log V_N)' &= \sum_{j=n}^{n+q-1} b_j \\
&\quad + a_n e_{\theta}^T F_n^{(-1)} v_{(n-1)}. \tag{9}
\end{aligned}$$

Retain $T=(A-cl)/i$ and $\sigma=\text{Tr } T$, which is real. The previous source cross term is exactly

$$(b_N^* G_N b_{(N+1)})/\omega_N = i(\sigma - (\log V_N)').$$

Here b_N in that formula denotes the remainder vector of the S-monic source polynomial, not the scalar recurrence b_j . The factor i and the distinction of types remain. The phase is $\phi_N = \sigma - (\log V_N)'$. At zero tilt for a full quartet, evenness gives $\phi_N=0$; it is not set to zero at a general tilt.

48.1.6 5. The positive gaps are norms of actual source vectors

Put

$$Z_j = \psi \hat{Q}_j - Q_{(j+q)} \text{ in } P_{(j+q-1)}.$$

The leading terms cancel. Source orthogonality therefore proves

$$\Delta_j := \|Z_j\|^2 = \nu_j - \omega_{(j+q)} \geq 0. \tag{10}$$

The first term $\psi \hat{Q}_j$ is, through the explicit unit $i^{(-q)}$, an original chi relation. Its quotient is its supported zero. The difference Z_j instead has class $-[Q_{(j+q)}]$ in the original quotient. Its norm is not assigned to an absent element.

The SAME canonical rank-two control is $W_N = A^* G_N + G_N A - k G_N$. Combining its previously proved radius identity with (6) gives, for $n \geq 1$,

$$\epsilon_N^2 = \Delta_{(n-1)} \Delta_n / (\nu_{(n-1)} \omega_N) - \phi_N^2. \tag{11}$$

Thus every factor in the upper bound

$$\epsilon_N \leq \sqrt{\Delta_{(n-1)} \Delta_n / (\nu_{(n-1)} \omega_N)}$$

comes from a specified positive source norm. Both gaps are computed by the transfer:

$$\begin{aligned}
\Delta_{(n-1)} &= \omega_{(n-1)} \vartheta_{(n-1)} - \omega_N, \\
\Delta_n &= \omega_n \vartheta_n - \omega_{(N+1)}.
\end{aligned}$$

The previous two-step bound is now

$$\begin{aligned}
\epsilon_N &\leq \sqrt{\omega_{(N+1)} / \omega_N} \\
&\quad (R_{N-1}) / (2 \sqrt{R_N}), \tag{12}
\end{aligned}$$

with R_N from (7). At the first degree $N=q-1$, use the separate expression

$$\epsilon_{(q-1)}^2 = (\nu_{\theta} - \omega_q) / \omega_{(q-1)} - \phi_{(q-1)}^2.$$

No inverse at the previous rank-deficient degree is introduced.

48.1.7 6. Keep the actual conormal quotient, not a repeated first ideal

The helper algebra for the norm computation is $C[u]/\Pi$, isomorphic through $S=c+iu$ to $C[S]/\chi_1^2$. The actual depth-two object is $C[S]/\chi_2$, defined using the literal pullback of the original I^2 . The merged conormal work gives $\chi_2 \mid \chi_1^2$. The actual maps are

$$C[S]/\chi_1^2 \twoheadrightarrow C[S]/\chi_2 \twoheadrightarrow C[S]/\chi_1. \tag{13}$$

Their kernels are respectively $(\chi_2)/(\chi_1^2)$ and $(\chi_1)/(\chi_2)$. In raw jets they truncate at the indicated full orders and intertwine multiplication by S . Differentiation into depth one also commutes, since $\chi_1 \mid \chi_2'$. The coordinate relation $d_u \Psi = i \Psi d_S$ is retained.

For $h = (s - \rho)^2$ and tensor degree two, $X = S - 2\rho$ gives

$$C[X]/X^6 \twoheadrightarrow C[X]/X^5 \twoheadrightarrow C[X]/X^3.$$

The first kernel is the class of X^5 . The retained depth-two class X^3 has derivative $3X^2$ at depth one. These maps use one original scalar support. Their split lifts send quotient-killed classes to e at the receiving label and fix external τ .

The full multiplier $U = \text{product}_i(g/h)(s_i)$ retains the checked equation $\delta(U)P = U \delta(P) + (\delta U)P$. The second term is not discarded or presumed cyclic. The $2q$ -jet norm calculation is not counted as construction of new actual spectral multiplicities.

48.1.8 7. Residual-controlled numerical interface

For an actual F and a specified candidate inverse Y , suppose a justified bound gives $\|I - YF\|_\infty \leq \eta < 1$. For candidate $\hat{d}$ and residual $r = v - F \hat{d}$, the exact Neumann argument proves

$$\|\hat{d} - d\|_\infty \leq \|Y\|_\infty \|r\|_\infty / (1 - \eta). \quad (14)$$

This bounds the first coordinates used in (1) and (9). Arithmetic application requires certified input enclosures for the moments, source polynomials and zeros. Floating-point output alone is not a certificate.

Positive enclosures of ω and ϑ bound (7). Since $(R-1)^2/(4R)$ is increasing for $R \geq 1$, (12) yields a valid upper enclosure of ϵ^2 . The full phase can improve it through (11). Finite positivity/invertibility is not substituted for a uniform estimate of these residuals or inverse norms.

48.1.9 8. Exact calibration and analytic target

For a declared Gaussian measure of literal mass seven and variance one, $\psi = u^2 + 1$, $q = 2, n = 2, N = 3$, the calculation gives

```
(omega_0, ..., omega_4) = (7, 7, 14, 42, 168),
vartheta_1 = 22, vartheta_2 = 86/3,
Qhat_1 = u, Qhat_2 = u^2 - 11/3,
Delta_1 = 112, Delta_2 = 700/3,
G_3 = diag(7/3, 21/11), epsilon_3^2 = 400/99.
```

An independent quotient-Gram calculation with $A = (1/2)I + i M_u$ gives the same result. The nonreal u -roots are algebraic fixtures, not zeros of ξ ; the nonzero allowance explicitly tests that the method does not manufacture purity. Repeated nodes, complex quartets, asymmetric atomic tilt derivatives and mass changes are covered by the expanded checks.

For the actual quartet retain

$$q_k = [1 + k(m-1)](k+1)^2, \\ L(h, k) = 2 \delta [1 + k(m-1)](k+1) \lfloor (k+1)^2/4 \rfloor.$$

The source exterior/filtration result gives $L(h, k) \leq \epsilon(h, k, N)$, and $L(h, k)/(k q_k) \geq \delta/2$. Thus $\epsilon = o(k q_k)$, with fixed h and admitted $N \geq q_k$, is sufficient for the intended contradiction. The new representation attaches that exact target to TWO consecutive source-to-relation transfer coefficients and ONE phase solve. It has NOT proved that asymptotic bound.

No new Lean execution or arithmetic interval certificate is claimed. All new mathematical claims have the written proofs above at the retained source inputs. The original arithmetic function, unit, action, θ relations, residue pairing and trace remain unchanged.

48.1.10 References and checks

Classical background: C. Krattenthaler (2021), ‘A determinant identity for moments of orthogonal polynomials that implies Uvarov’s formula for the orthogonal polynomials of rationally related densities’, arXiv:2103.03969; NIST DLMF Section 18.2. The proofs above supply the particular confluent and arithmetic interfaces rather than attributing them wholesale to those references.

The expanded conversation delivery includes the complete LaTeX/HTML reader and twenty exact finite regression methods; both normal and optimized runs passed with matching records, and both deliberate-failure controls failed. The preceding Toda archive’s 35 manifest entries verified and its 24-method checker passed fresh normal and optimized runs. These are stated verification scopes, not new Lean certificates. The public core checker in this directory independently tests the principal raw-jet, transfer, norm, determinant, volume and phase interfaces.

49 An arithmetic lower-envelope comparison for central quotient volumes: full jets, finite determinant bounds, and an exact quadratic benchmark

This calculation constructs upper bounds for the original central volume $\log(V_q/V_{2q-1})$. Its input is the proved lower envelope and Laplace estimate of the complete *Arithmetic Endpoint Bounds* source, equations (29)–(32). The comparison acts on the original polynomial and quotient maps, including all multiplicities and phases. The original matrix bound retains an explicit positive matrix: its arithmetic moments stop at degree $2q$, while its comparison moments continue to $4q - 2$. A second bound uses only the supplied envelope constants and two Laplace values. The signed four-endpoint comparison is now propagated into (AU.31a)–(AU.31b); it retains the endpoint losses and uses actual moments through degree $4q$. The explicit scalar estimate has quadratic, rather than logarithmic, growth per quotient direction. An exact auxiliary calculation determines that scale and supplies a monic family relating its full jets to the original quartet jets. No bound contradicting the quartet lower estimate is asserted.

49.1 Original source and every jet coordinate

Fix a complete quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$, and common complete zero order $m \geq 1$ of $g = 2\xi$. Keep its monic polynomial h , its entire quotient $v_h = g/h$, and

$$w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad k \geq 3. \quad (\text{AU.1})$$

Write $c = k/2$ and keep the original observation $S = c + iu$. The source form is $\int \overline{P(c + iu)} Q(c + iu) m_k(u) du$, conjugate-linear in its first argument. Its constant norm is μ_h^k . The coefficient map and its inverse are

$$\Psi : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad P \longmapsto P(c + iu), \quad f \longmapsto f((S - c)/i). \quad (\text{AU.2})$$

These are inverse ring homomorphisms by substitution. Multiplication by S is carried to $c + iM_u$, and $\partial_u^d \Psi(P) = i^d \Psi(P^{(d)})$. On the ordered degree- N coefficient spaces the matrix is triangular with diagonal $1, i, \dots, i^N$, so its determinant has modulus one. The original quotient volumes therefore have exactly the same values in the transported remainder basis; no mass or norm is divided out.

Let $\ell = 1 + k(m - 1)$ and use the ordered distinct grid

$$\lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \zeta_{ab} = (\lambda_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{AU.3})$$

The original cyclic relation and its transported monic polynomial are

$$q = \ell(k + 1)^2, \quad \chi(S) = \prod_{a,b} (S - \lambda_{ab})^\ell, \quad \psi(u) = i^{-q} \chi(c + iu) = \prod_{a,b} (u - \zeta_{ab})^\ell. \quad (\text{AU.4})$$

The grid is invariant under complex conjugation in the u chart, so ψ is real monic. Equivalently the original coefficient identity is $\bar{\chi}(k - S) = (-1)^q \chi(S)$. In particular the original norm multiplier is $\bar{\chi}(c - iu)\chi(c + iu) = \psi(u)^2$, with the source multiplication still carrying the unit i^q .

For clarity the preceding cyclic degree also follows from the original tensor algebra. In a tensor of k length- m local factors, the sum of the k nilpotent generators has power $k(m - 1)$ nonzero: its coefficient on the product of their top powers is $(k(m - 1))!/((m - 1)!)^k$, nonzero in $\mathbb{C}$. Its next power is zero by total degree. Every grid sum in (AU.3) occurs, since the two signs in each quartet factor can be chosen independently. Direct products of local factors give the same maximal length ℓ at every grid sum. Distinct pairs have distinct real and imaginary parts. The minimal polynomial of the original sum is therefore exactly (AU.4), with no lost multiplicity.

Put $\mathcal{E} = \mathbb{C}[u]/(\psi)$, with ordered remainder coefficients $1, u, \dots, u^{q-1}$. For $N \geq q - 1$ let J_N be remainder and B_N multiplication by ψ in the ordered monomials. Monic division gives the exact sequence

$$0 \longrightarrow \mathbb{C}[u]_{\leq N-q} \xrightarrow{B_N} \mathbb{C}[u]_{\leq N} \xrightarrow{J_N} \mathcal{E} \longrightarrow 0. \quad (\text{AU.5})$$

The source on the left is zero if $N = q - 1$. Precisely, the coefficient isomorphisms are $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}[u]_{\leq N}$, $a \mapsto \sum_{j=0}^N a_j u^j$, and $\kappa : \mathbb{C}^q \rightarrow \mathcal{E}$, $x \mapsto [\sum_{j < q} x_j u^j]$. The polynomial quotient map is $\kappa J_N \iota_N^{-1}$ and the polynomial multiplication map is $\iota_N B_N \iota_{N-q}^{-1}$. When a matrix formula is evaluated on a polynomial below, these displayed coefficient isomorphisms are the specified identifications. The original S chart uses the same coefficient isomorphisms with u^j replaced by S^j . The complete raw derivative map is

$$\mathbf{E}_N f = (f^{(d)}(\zeta_{ab}))_{a,b,0 \leq d < \ell}, \quad (\mathbf{E}_N)_{(a,b,d),j} = \begin{cases} j! \zeta_{ab}^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d. \end{cases} \quad (\text{AU.6})$$

Taylor expansion at each distinct centre shows that its kernel is $(\psi) \cap \mathbb{C}[u]_{\leq N}$. Let $\mathbf{Z} = \mathbf{E}_{q-1}$. Then $\mathbf{E}_N = \mathbf{Z} J_N$ and $\mathbf{Z}$ is invertible. Indeed a degree below q polynomial with all those zero raw jets is divisible by degree- q ψ and is zero. For the fixed row order,

$$\det \mathbf{Z} = \left(\prod_{a,b} \prod_{d=0}^{\ell-1} d! \right) \prod_{(a,b) < (a',b')} (\zeta_{a'b'} - \zeta_{ab})^{\ell^2}. \quad (\text{AU.7})$$

One proof first regards all roots as distinct variables in the ordinary Vandermonde determinant, groups each cluster, and replaces its rows by successive divided differences. Each row operation divides by exactly the corresponding within-cluster difference. Sending the nodes in a cluster to its centre replaces its divided-difference row of order d by $f^{(d)}/d!$, by the finite Taylor expansion of each monomial. Multiplying those rows by $d!$ proves (AU.7). This is an identity of polynomials after removal of the indicated factors; no source norm or zero multiplicity is approximated. In the original S coordinates the same raw jet of order d has its literal factor i^d from (AU.2).

For any positive source matrix H_N , the least quotient form is

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \quad R_N = H_N^{-1} J_N^* G_N. \quad (\text{AU.8})$$

Surjectivity proves $K_N > 0$. Multiplication proves $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Thus every lift $R_N x + B_N y$ has norm $x^* G_N x + y^* B_N^* H_N B_N y$, proving the minimum property. In raw jets the kernel is $\mathbf{K}_N = \mathbf{E}_N H_N^{-1} \mathbf{E}_N^*$, and $\det \mathbf{K}_N = |\det \mathbf{Z}|^2 / V_N$. Hence

$$C_k := \log(V_q / V_{2q-1}) = \log \det \mathbf{K}_{2q-1} - \log \det \mathbf{K}_q. \quad (\text{AU.9})$$

All factorials in (AU.7) cancel only in this specified ratio.

49.2 The lower envelope as an explicit positive comparison form

The supplied arithmetic proof establishes $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, $B = 42 + 2 \deg h = 42 + 8m$, and

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad \alpha = \pi/2. \quad (\text{AU.10})$$

Set $a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}$ and $d_k = k - 2 > 0$. Define the comparison measure, retaining its literal mass, by

$$d\nu_k(u) = a_k e^{-\alpha|u|} (d_k + |u|)^{-B} du. \quad (\text{AU.11})$$

The identity map $L^2(m_k du) \rightarrow L^2(\nu_k)$ is well defined and a contraction, since (AU.10) gives the squared-norm inequality for every measurable function. It restricts to the identity on each polynomial space and commutes with multiplication by $c + iu$, remainder (AU.5), and every raw jet (AU.6). The comparison is therefore a map on the same quotient, not a declaration that the measures coincide.

Every moment of ν_k is finite, and the following integral evaluates it with positive integrands:

$$\nu_{k,j} := \int_{\mathbb{R}} u^j d\nu_k(u) = \begin{cases} \frac{2a_k j!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-d_k t}}{(\alpha + t)^{j+1}} dt, & j \text{ even}, \\ 0, & j \text{ odd}. \end{cases} \quad (\text{AU.12})$$

To prove it, repeated integration by parts gives $\int_0^\infty t^{B-1} e^{-yt} dt = (B-1)! y^{-B}$ for $y > 0$. Insert $y = d_k + u$ on the positive half-line. Tonelli applies because each integrand is nonnegative there. Integrating $u^j e^{-(\alpha+t)u}$ gives $j!(\alpha + t)^{-j-1}$ by the same integration by parts. Evenness supplies the factor two or the vanishing odd moment. The bound $\nu_{k,j} \leq 2a_k d_k^{-B} j! \alpha^{-j-1}$ for even j also proves finiteness directly. In particular $\nu_{k,0} > 0$ is retained, and generally differs from μ_h^k .

Let $H_N^\nu = (\nu_{k,i+j})_{0 \leq i,j \leq N}$. Its quadratic form is the ν_k integral of the polynomial's squared modulus. The weight is strictly positive everywhere, so a nonzero polynomial has positive norm. Thus $0 < H_N^\nu \preceq H_N$. Minimizing on the identical affine fibres in (AU.5) proves

$$0 < G_N^\nu \preceq G_N, \quad K_N \preceq K_N^\nu, \quad K_N^\nu = E_N (H_N^\nu)^{-1} E_N^*. \quad (\text{AU.13})$$

The inverse inequality follows also by conjugating H_N by $(H_N^\nu)^{-1/2}$ and inverting its positive eigenvalues.

Theorem 49.1 (Actual low-degree data control the higher quotient volume). *For the original quartet source and $k \geq 3$, the finite upper bound is*

$$0 \leq C_k \leq \log \det K_{2q-1}^\nu - \log \det K_q. \quad (\text{AU.14})$$

Its actual arithmetic inputs are the moments through degree $2q$ and the stated envelope constants. The comparator moments through $4q - 2$ are the explicit positive integrals (AU.12).

Proof. The first inequality follows by minimizing the same quotient norm in successively larger polynomial spaces. For the second use (AU.13) at $N = 2q - 1$ and (AU.9). Determinant is increasing on positive Hermitian forms: conjugate by the square root of the smaller one and multiply eigenvalues at least one. H_q uses precisely the original moments $0, \dots, 2q$, and all the matrices in the second term of (AU.14) have already specified finite algebraic maps from them. No higher actual moment enters. Equivalently, if $\mathfrak{D}_j^\nu = \det H_{j-1}^\nu$ and $\mathfrak{B}_j^\nu$ is the Gram determinant of the original ψ times $1, u, \dots, u^{j-1}$, the quotient-first coefficient matrix $[1, u, \dots, u^{q-1}, \psi, u\psi, \dots]$ is triangular with diagonal one. Orthogonal elimination of its relation block gives

$$V_{2q-1}^\nu = \mathfrak{D}_{2q}^\nu / \mathfrak{B}_q^\nu, \quad C_k \leq \log V_q + \log \mathfrak{B}_q^\nu - \log \mathfrak{D}_{2q}^\nu. \quad (\text{AU.15})$$

This proves the same upper bound in the exact source/relation determinants. Both the source mass and relation multiplication factors remain in their respective determinants. $\square$

The existing Gamma coefficient map also makes the cutoff in (AU.14) explicit. With $\lambda > 0$, $\alpha_\lambda = 2\lambda$, $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda)$ and $C_\lambda = c_\lambda^k$, define the original monic Gamma polynomials by

$$\sum_{j \geq 0} b_j^{(\lambda)}(u) z^j / j! = (1 + z^2)^{-\lambda} e^{u \arctan z}, \quad c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha_\lambda)_j}.$$

Multiplying the k generating functions and taking finite derivatives at zero gives

$$d_l = \sum_{j_1 + \dots + j_k = l} \prod_a (\alpha_\lambda)_{j_a} c_{h,j_a}, \quad \int b_l^{(k\lambda)} m_k = C_\lambda l! d_l. \quad (\text{AU.16})$$

All finite terms are integrable by the supplied exponential moments. Expand $u^{i+j} = \sum_{l \leq i+j} L_{ij,l} b_l^{(k\lambda)}(u)$ by monic subtraction. Then $(H_q)_{ij} = C_\lambda \sum_l l! L_{ij,l} d_l$. Thus the actual coefficient list $c_{h,0}, \dots, c_{h,2q}$ suffices for (AU.14). At zero degree it gives $C_\lambda c_{h,0}^k = \mu_h^k$, retaining the source mass.

49.3 An upper determinant depending only on the two Laplace values

Fix $0 < b < \pi/2$ and set $M_k = M_h(b)^k + M_h(-b)^k$, with the original $M_h(\pm b) = \int e^{\pm bu} w_h(u) du$. Positivity and convolution give $\int e^{b|u|} m_k(u) du \leq M_k$. The exponential series therefore gives $\int |u|^{2j} m_k(u) du \leq M_k (2j)! b^{-2j}$. For arbitrary $\tau_0, \dots, \tau_q > 0$ define

$$D_q(\tau) = M_k \left(\sum_{j=0}^q \tau_j \right) \text{diag} \left(\frac{(2j)!}{b^{2j} \tau_j} \right)_{j=0}^q. \quad (\text{AU.17})$$

For $f = \sum a_j u^j$, Cauchy-Schwarz applied to $\sqrt{\tau_j}$ and $a_j u^j / \sqrt{\tau_j}$ proves

$$|f(u)|^2 \leq \left(\sum \tau_j \right) \sum |a_j|^2 |u|^{2j} / \tau_j.$$

Integration proves $H_q \preceq D_q(\tau)$. Inverting and applying the same full jet map proves

$$\mathbf{K}_q \succeq \frac{1}{M_k \sum \tau_j} \mathbf{A}_q(\tau), \quad \mathbf{A}_q(\tau) = \mathbf{E}_q \text{diag} \left(\frac{\tau_j b^{2j}}{(2j)!} \right) \mathbf{E}_q^* > 0. \quad (\text{AU.18})$$

Consequently an upper bound without any unevaluated actual polynomial moment is

$$C_k \leq q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det \mathbf{K}_{2q-1}^\nu - \log \det \mathbf{A}_q(\tau). \quad (\text{AU.19})$$

Every positive choice of τ gives a proved bound, and the infimum over such choices remains a bound. Multiplying all τ_j by the same positive number cancels between the two written determinant factors; it changes neither measure. No optimizer or numerical enclosure is assumed in (AU.19).

The lower determinant has the exact positive expansion

$$\det \mathbf{A}_q(\tau) = \sum_{\substack{I \subset \{0, \dots, q\} \\ |I|=q}} |\det(\mathbf{E}_q)_{[:,I]}|^2 \prod_{j \in I} \frac{\tau_j b^{2j}}{(2j)!}. \quad (\text{AU.20})$$

Expand each determinant multilinearly in columns in the product $\mathbf{E}_q D \mathbf{E}_q^*$: terms with repeated column indices vanish by alternation, and grouping permutations of the same subset gives (AU.20). Its subset $I = \{0, \dots, q-1\}$ is strictly positive by (AU.7). This expression retains full raw-jet minors and factorials without estimating an inverse by its individual entries.

49.4 The interval kernel and the actual comparison map

For $T > 0$, set

$$a_k(T) = a_k e^{-\alpha T} (d_k + T)^{-B}. \quad (\text{AU.21})$$

Restriction to $[-T, T]$ followed by the identity on polynomials maps the actual source to $L^2([-T, T], a_k(T) du)$ contractively. The same is true from $L^2(\nu_k)$, by the pointwise lower bound on that interval. Let $H_N^I(T)$ be the Lebesgue interval Gram. Then $a_k(T) H_N^I(T) \preceq H_N^\nu \preceq H_N$. Let P_j denote the Legendre polynomial specified by Rodrigues' formula, $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Integration by parts j times proves its orthogonality to lower degrees; direct evaluation of the resulting beta integral gives its squared norm $2/(2j + 1)$. Its highest coefficient is $(2j)!/(2^j (j!)^2)$. The endpoint boundary terms vanish because $(x^2 - 1)^j$ has zeros of order j at both endpoints. Applying these constants to $P_j(u/T)$ gives the full interval jet kernel

$$(\mathbb{K}_N^I(T))_{(a,b,d),(a',b',e)} = \sum_{j=0}^N \frac{2j+1}{2T^{1+d+e}} P_j^{(d)}(\zeta_{ab}/T) \overline{P_j^{(e)}(\zeta_{a'b'}/T)}. \quad (\text{AU.22})$$

This follows by writing the inverse Gram in the orthogonal Legendre basis and applying (AU.6); each derivative contributes its stated T^{-d} . In particular

$$\mathbb{K}_{2q-1} \preceq \mathbb{K}_{2q-1}^\nu \preceq a_k(T)^{-1} \mathbb{K}_{2q-1}^I(T). \quad (\text{AU.23})$$

Equations (AU.14) and (AU.19) remain valid if their $\log \det \mathbb{K}_{2q-1}^\nu$ is replaced by $\log \det \mathbb{K}_{2q-1}^I(T) - q \log a_k(T)$. The global envelope version is at least as strong as each such interval version by the proved positive-matrix inequality (AU.23).

The restriction map has the exact orthogonal decomposition $f \mapsto (f1_{[-T,T]}, f1_{\mathbb{R} \setminus [-T,T]})$ in the original weighted L^2 source; its two squared norms add to the original norm. The interval bound drops a nonnegative term in that equality. It does not assign the outside component to external absence.

49.5 The explicit scalar bound and its finite gap

The following explicit scalar estimate is the independent full calculation VC.1–43, composed here with the preceding original quotient maps. Put $R_0 = \sqrt{\delta^2 + \gamma^2}$ and define

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = kR_0/T \leq 1/2, \quad N = 2q - 1, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \quad A_N = \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \\ B_q(r) &= \left(\frac{1+r}{1-r} \right)^q, \quad \mathcal{C}_{k,q} = \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{AU.24})$$

All values are positive. In particular $q \geq k$ and $D \geq 2R_0$ give the stated bound on r . These choices bound constants; they do not change the original density.

Theorem 49.2 (A full finite Hermitian upper bound). *In the transported ordered quotient coordinates, and by the explicit congruence below also in the original S coordinates,*

$$\boxed{K_q \preceq K_{2q-1} \preceq \mathcal{C}_{k,q} K_q, \quad 0 \leq C_k \leq q \log \mathcal{C}_{k,q}.} \quad (\text{AU.25})$$

Proof. Let $\mathbb{T}_j$ be the coefficient matrix of the original map $\Psi(P) = P(c + iu)$ in degree j . Write $H_j^S, J_j^S, K_j^S, G_j^S$ for matrices in the original S monomials. Then

$$H_j^S = \mathbb{T}_j^* H_j \mathbb{T}_j, \quad J_j \mathbb{T}_j = \mathbb{T}_{q-1} J_j^S, \quad K_j = \mathbb{T}_{q-1} K_j^S \mathbb{T}_{q-1}^*.$$

The first identity is the source substitution, the second is monic division, and substitution in (AU.8) gives the third. Each $\det T_j$ has modulus one, as proved in (AU.2). All source norms on S polynomials in this proof use H_j^S . The exact auxiliary coordinate map used in this proof is $\Psi_T P(x) = P(c + iTx)$, with inverse $f((S - c)/(iT))$. Its target measure is $Tm_k(Tx)dx$ and retains mass μ_h^k by substitution. The target monic relation is $\chi_T(x) = (iT)^{-q}\chi(c + iTx)$, so $\Psi_T(\chi Q) = (iT)^q\chi_T\Psi_T Q$. Monic division therefore proves $\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_{\chi} P$. At every original centre λ , the full order- d raw jet is $(\Psi_T P)^{(d)}((\lambda - c)/(iT)) = (iT)^d P^{(d)}(\lambda)$. The coefficient matrix has diagonal $(iT)^j$ at degree j ; its quotient determinant is $(iT)^{q(q-1)/2}$. Consequently the target quotient form is the original form transported by inverse congruence, and its determinant is $T^{-q(q-1)}V_j$ at every source degree j . This factor cancels in the central ratio, with all phases and masses still present in the map.

For a polynomial $f(x) = \sum f_j x^j$ put $\|f\|_{\text{coeff},1} = \sum |f_j|$. The Rodrigues formula above, expanded by the product rule, gives

$$P_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a.$$

The coefficient norm of a product is at most the product of the two coefficient norms by convolution and the triangle inequality. Since $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$, this proves $\|P_j\|_{\text{coeff},1} \leq \sum_a \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j$. The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$. In the orthogonal expansion $f = \sum_{j=0}^N f_j^L P_j$, Cauchy-Schwarz and the squared norm $2/(2j+1)$ give $|f_j^L| \leq \sqrt{(2j+1)/2} \|f\|_{L^2([-1,1])}$. Summing 4^j therefore proves

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}, \quad \deg f \leq N. \quad (\text{AU.26})$$

List all roots $z_1, \dots, z_q$ of χ_T , with repetitions; each has modulus at most r . Define e_a, h_j by

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j.$$

The second identity is formal multiplication of geometric series. It gives $|e_a| \leq \binom{q}{a} r^a$ and $|h_j| \leq \binom{q+j-1}{j} r^j$: the latter counts exponent lists of total degree j by their $q-1$ separating bars. For $0 \leq s \leq q-1$, multiply those inverse series through order s . It proves that the quotient of x^{q+s} by χ_T is exactly $Q_s = \sum_{j=0}^s h_j x^{s-j}$ and its remainder is $x^{q+s} - \chi_T Q_s$, of degree below q . Removing the leading term of $\chi_T Q_s$ removes precisely one from its coefficient norm. All its remaining coefficients are the negative remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \\ &\leq ((1+r)/(1-r))^q = B_q(r). \end{aligned}$$

At degrees below q the remainder is the original monomial and has coefficient norm one, also bounded by $B_q(r)$. Linearity proves

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{AU.27})$$

Every root was retained in this proof. Its smaller literal finite constant $\max\{1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1\}$ can also replace $B_q(r)$.

For v of degree at most $q-1$, each monomial is bounded by $\max\{1, |x|^{q-1}\}$. Using $\max(1, y) \leq 1+y$ for $y \geq 0$ and the proved Laplace moment bound gives

$$\int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2. \quad (\text{AU.28})$$

On the other hand, restriction to $[-T, T]$ and (AU.21) prove $\|P\|_{H_N^S}^2 \geq Ta_k(T)\|\Psi_T P\|_{L^2([-1,1])}^2$. Take $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (AU.26)–(AU.28) and the exact division map now give

$$\|\text{rem}_{\chi} P\|_{H_{q-1}^S}^2 \leq C_{k,q} \|P\|_{H_N^S}^2. \quad (\text{AU.29})$$

The map estimated here is $s\pi_{\chi,N} : \mathcal{P}_N^S \rightarrow \mathcal{P}_{q-1}^S$, where $\pi_{\chi,N}$ is the polynomial quotient map and $s : \mathbb{C}[S]/(\chi) \rightarrow \mathcal{P}_{q-1}^S$ selects the unique remainder representative. Their exact identities are $\pi_{\chi,q-1}s = I$ and $\pi_{\chi,q-1}s\pi_{\chi,N} = \pi_{\chi,N}$. The composite $s\pi_{\chi,N}$ equals the identity on polynomials of degree below q .

Apply (AU.29) to the least lift $R_N^S x$ from the original matrices. Its remainder lifts the same x already at degree $q-1$, hence $x^* G_q^S x \leq x^* G_{q-1}^S x \leq C_{k,q} x^* G_N^S x$. Conversely degree inclusion gives $G_N^S \preceq G_q^S$. Conjugating by $(G_q^S)^{-1/2}$, bounding its positive eigenvalues and then inverting proves (AU.25), including the determinant exponent q , first for K_j^S and then by the displayed congruence for K_j . It also proves $C_{k,q} \geq 1$. $\square$

Retain the fixed constants

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_{\text{env}} = \max\{1, 2/(3c_h D)\}.$$

Because $bT \geq 2q$, the factorial ratio in (AU.24) is at most one. Also $M_k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that $U_k(T) \leq 3X^k$. The displayed A_N satisfies $A_{2q-1}^2 \leq (2q/9)256^q$. On $[0, 1/2]$ the derivative of $\log((1+r)/(1-r))$ is $2/(1-r^2) \leq 8/3$, so

$$\log B_q(r)^2 \leq \frac{16}{3}qr = \frac{16R_0}{3D}k. \quad (\text{AU.30})$$

Use also $\vartheta_h^{-(k-3)} \leq K^k$, $1 + Dq + k - 3 \leq 1 + (D+1)q$, and $e^{-3\alpha} \leq 1$. Substitution in (AU.24)–(AU.25) proves the finite scalar bound

$$\begin{aligned} 0 \leq C_k \leq \mathcal{U}_k := & (\alpha D + \log 256)q^2 \\ & + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ & + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}. \end{aligned} \quad (\text{AU.31})$$

All of its constants are independent of k and q for fixed h, b . The exact finite bounds (AU.14), (AU.19), and $q \log C_{k,q}$ can be retained together by taking their minimum; each independently bounds the same C_k with the proved maps.

The signed endpoint calculation now improves this earlier central estimate on the same original source. Retain its definitions $\mathcal{B}_{h,k}^\Gamma$, $\mathcal{J}_{h,k}$, the recursively refined upper endpoint $\mathcal{B}_{h,k}^{\text{hi}}$, and the arithmetic endpoint terms $E_0 = \log(V_{q-1}/V_q)$, $E_1 = \log(V_{2q-1}/V_{2q})$ from (FV.M1)–(FV.M18), Section 46.1.6. The exact coefficient map Ψ in (AU.2) carries its S source to this u source. Its quotient coefficient map τ_{q-1} sends $[P(S)]$ to $[P(c + iu)]$; because $\Psi(\chi Q) = i^q \psi \Psi(Q)$, it has the exact remainder identity $J_N^u \tau_N = \tau_{q-1} J_N^S$. Thus $K_N^u = \tau_{q-1} K_N^S \tau_{q-1}^*$ and $V_N^u = |\det \tau_{q-1}|^{-2} V_N^S = V_N^S$. The identity is valid at both densities of the interpolation; their literal masses, the factor i^q in relation multiplication and every raw-derivative factor i^d are retained. Define

$$\mathcal{U}_k^{\text{signed}} = \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}. \quad (\text{AU.31a})$$

Equations (AU.9) and (FV.M11) now prove $0 \leq C_k \leq \mathcal{U}_k^{\text{signed}}$ by substitution, with no endpoint term discarded. Together with the existing proved bounds this gives

$$\begin{aligned} 0 \leq C_k \leq \widehat{\mathcal{U}}_k := \min \Big\{ & \log \det K_{2q-1}^\nu - \log \det K_q, \quad q \log C_{k,q}, \mathcal{U}_k, \quad \mathcal{U}_k^{\text{signed}}, \\ & q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det K_{2q-1}^\nu - \log \det A_q(\tau) \Big\}, \quad \tau_j > 0. \end{aligned} \quad (\text{AU.31b})$$

Each entry is an upper bound for the same C_k , respectively by (AU.14), (AU.25), (AU.31), (FV.M11), and (AU.19). Their minimum is therefore an upper bound; (AU.23) still gives each interval-kernel variant with its full T powers. The new signed entry includes the actual-moment full-angle refinement and its nonlinear transport in (FV.M15)–(FV.M18). It uses the actual original source moments through degree $4q$, the relative eigenvalues and the actual projection overlap in (FV.M7). The lower-degree arithmetic cutoff of (AU.14) is unchanged for its own entry; it is not asserted for the new signed entry.

The complete central lower estimate in the balanced-window/factor-four source is $C_k \geq \mathcal{L}_k = 2(q-1)(\log k + \log(\delta/(2C_h^{\text{bal}})))$. The current explicitly computed gap is the literal function

$$\begin{aligned} \mathcal{U}_k - \mathcal{L}_k &= (\alpha D + \log 256)q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &\quad + Bq \log(1 + (D+1)q) + q \log C_{\text{env}} \\ &\quad - 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \end{aligned} \quad (\text{AU.32})$$

The propagated upper bound gives the additional actual finite gap $0 \leq \widehat{\mathcal{U}}_k - \mathcal{L}_k \leq \mathcal{U}_k - \mathcal{L}_k$: the first inequality follows from $\mathcal{L}_k \leq C_k \leq \widehat{\mathcal{U}}_k$, and the second from the minimum in (AU.31b). The function in (AU.32) still concerns the two displayed scalar upper and lower estimates. It is not an asserted exact value of the unknown C_k . Since $q = \ell(k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Dividing every term of (AU.31)–(AU.32) by q^2 proves

$$\lim_{k \rightarrow \infty} \frac{\mathcal{U}_k - \mathcal{L}_k}{q_k^2} = \alpha D + \log 256 > 0, \quad \limsup_{k \rightarrow \infty} C_k/q_k^2 \leq \alpha D + \log 256. \quad (\text{AU.33})$$

Dividing the upper estimate by $q \log k$ gives the exact leading factor $(\alpha D + \log 256)q/\log k$. It grows as $(\alpha D + \log 256)k^2/\log k$ for $m = 1$ and as $(\alpha D + \log 256)(m-1)k^3/\log k$ for $m > 1$. The division step itself has only the linear-in- k logarithmic cost (AU.30). The two quadratic contributions in this scalar proof come from the full coefficient estimate (AU.26) and the least density over the interval in (AU.21). The positive matrices in (AU.14), (AU.19), and (AU.22) retain the directional information discarded in those scalar estimates, with their exact source and quotient maps already proved. No upper asymptotic for those sharper determinants is inserted.

49.6 A full-jet quadratic benchmark and its exact relation family

Here the auxiliary source is explicitly Lebesgue measure on $[-1, 1]$ and the auxiliary relation is u^q . Its mass is two. These data are inserted into the same sequence (AU.5), kernel (AU.8), and central ratio (AU.9). No assertion that they are an arithmetic density is made. Write D_d for its d by d monomial determinant, $D_0 = 1$, and $B_j(q)$ for the j by j determinant of the relation measure $u^{2q} du$ on that interval. The quotient-first triangular determinant calculation proving (AU.15) gives exactly

$$\frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} = \frac{D_{q+1}B_q(q)}{B_1(q)D_{2q}}, \quad B_1(q) = \frac{2}{2q+1}. \quad (\text{AU.40})$$

The orthogonal Legendre calculation gives

$$D_d = \prod_{j=0}^{d-1} \frac{2}{2j+1} \left(\frac{2^j}{\binom{2j}{j}} \right)^2. \quad (\text{AU.41})$$

For $q = 2s$, parity splits the relation Gram into two s by s blocks, and

$$B_{2s}(2s) = C_s(2s+1/2)C_s(2s+3/2), \quad C_s(x) = \frac{\prod_{j=0}^{s-1} (j!)^2}{\prod_{a,b=0}^{s-1} (x+a+b)}. \quad (\text{AU.42})$$

Indeed the even block has entries $1/(2s+a+b+1/2)$ and the odd block entries $1/(2s+a+b+3/2)$. The Cauchy determinant formula used here has numerator $\prod_{a<b}(b-a)^2 = \prod_j (j!)^2$. It follows by multiplying the determinant by all its denominators: the numerator is alternating separately in its two lists of variables, so contains their two Vandermonde factors. Its degree is their total degree, and comparing one residue inductively gives constant one. Substitution gives (AU.42) with no absolute-value or sign ambiguity.

Theorem 49.3 (An explicit quadratic scale in the full-origin jet fibre). *For every even integer $q \geq 2$,*

$$\left| \log \frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} - \kappa_* q^2 \right| \leq 12q \log(8q), \quad \kappa_* := \frac{9}{2} \log 3 - 6 \log 2 = \frac{3}{2} \log(27/16) > 0. \quad (\text{AU.43})$$

Proof. The elementary binomial bounds $4^j/(2j+1) \leq \binom{2j}{j} \leq 4^j$ hold because the central coefficient is the largest of the $2j+1$ coefficients summing to 4^j . Therefore (AU.41) gives

$$\log D_d = -d(d-1) \log 2 + E_d, \quad |E_d| \leq d \log(4d) \quad (d \geq 1). \quad (\text{AU.44})$$

At $s \geq 1$, expand the numerator of $C_s(2s)$ as $2 \sum_{i=1}^{s-1} (s-i) \log i$. The function $f(x) = (1-x) \log x$ is increasing on $(0, 1]$, has integral $-3/4$, and is nonpositive. Its right Riemann sum differs from its integral by a number between zero and $(\log s + 1)/s$: the difference is bounded above by $-\int_0^{1/s} f(x) dx \leq (\log s + 1)/s$, using $-f(x) \leq -\log x$. It follows that this numerator is $s^2 \log s - 3s^2/2$ plus an error of absolute value at most $s \log s + 2s$.

The denominator's logarithm is $s^2 \log s + \sum_{a,b=0}^{s-1} \log(2 + a/s + b/s)$. Both partial derivatives of the latter function lie in $[0, 1/2]$. Its left-square Riemann sum differs from its integral by at most $1/s$, hence the sum by at most s . Two integrations give

$$\int_0^1 \int_0^1 \log(2+x+y) dx dy = 18 \log 2 - 9 \log 3 - 3/2.$$

For example use the second antiderivative $F(t) = t^2 \log t/2 - 3t^2/4$ and calculate $F(4) - 2F(3) + F(2)$. Thus

$$|\log C_s(2s) - 9s^2 \log(3/4)| \leq s \log s + 3s.$$

For $c > 0$, the change to $C_s(2s+c)$ has logarithm $-\sum_{a,b} \log(1+c/(2s+a+b))$, between $-cs/2$ and zero. Using $c = 1/2, 3/2$ in (AU.42) proves

$$\left| \log B_q(q) - \frac{9}{2} q^2 \log(3/4) \right| \leq q \log(q/2) + \frac{7}{2} q. \quad (\text{AU.45})$$

Insert (AU.44)–(AU.45) into (AU.40). The quadratic coefficient is $3 \log 2 + (9/2) \log(3/4) = \kappa_*$. The remaining absolute error is at most

$$3q \log 2 + (q+1) \log(4(q+1)) + 2q \log(8q) \\ + q \log(q/2) + \frac{7}{2} q + \log((2q+1)/2).$$

For $q \geq 2$, use $q+1 \leq 3q/2$, $4(q+1) \leq 8q$, $\log(q/2) \leq \log(8q)$ and $\log((2q+1)/2) \leq \log(8q)$. The two constant multiples of q are bounded together by $3q \log(8q)$, since $3 \log(16) > 3 \log 2 + 7/2$. Their sum is then at most $8q \log(8q)$, hence certainly the displayed $12q \log(8q)$. This proves (AU.43) for every indicated finite q , not only along a limit. $\square$

There is an exact algebraic relation of this benchmark's quotient to every original grid quotient. Define

$$\psi_t(u) = \prod_{a,b} (u - t\zeta_{ab})^\ell, \quad \mathcal{A} = \mathbb{C}[t, u]/(\psi_t(u)). \quad (\text{AU.46})$$

Monic division over $\mathbb{C}[t]$ proves that $1, u, \dots, u^{q-1}$ form a free $\mathbb{C}[t]$ basis: division gives spanning and a nonzero multiple of a monic degree- q polynomial cannot have smaller degree. Every fibre has dimension q . The fibre at $t = 1$ is the original transported quotient; at $t = 0$ it is $\mathbb{C}[u]/u^q$. For $t \neq 0$,

$$\mathcal{A}_t \xrightarrow{\sim} \mathcal{A}_1, \quad [f(u)] \mapsto [f(tu)], \quad [g(u)] \mapsto [g(u/t)] \quad (\text{AU.47})$$

are inverse maps, since $\psi_t(tu) = t^q \psi_1(u)$. Their full raw jet equation is $(d/du)^d f(tu)|_{u=\zeta_{ab}} = t^d f^{(d)}(t\zeta_{ab})$, so the coefficient d in multiplication's raw-jet Jordan blocks and each order ℓ remain for $t \neq 0$. At $t = 0$ monic division gives the complete merged order q , rather than an assertion that separate evaluation rows stay invertible at coincident points.

For any fixed positive polynomial source H_N , remainder matrices $J_N(t)$ are polynomial in t by the monic subtraction algorithm. They contain I_q in their first q columns. Thus $K_N(t) = J_N(t)H_N^{-1}J_N(t)^*$ is positive for every real t , including zero, and the exact central function obeys

$$C(t) = \log \det K_{2q-1}(t) - \log \det K_q(t), \quad C'(t) = \text{Tr}(K_{2q-1}^{-1}K'_{2q-1} - K_q^{-1}K'_q), \quad (\text{AU.48})$$

where $K'_N = J'_N H_N^{-1} J_N^* + J_N H_N^{-1} (J'_N)^*$. The determinant derivative follows by its multilinear expansion. This gives both the free family and the exact finite source-dependent map along it. The estimate (AU.43) belongs to its stated Lebesgue special fibre. Equations (AU.46)–(AU.48) do not transfer that estimate to $t = 1$ without the displayed change in the full kernel.

49.7 Source representatives and support

The source polynomial maps throughout are the original $\mathcal{V}_{h,k}P = P(D_1 + \dots + D_k)F_h^{\otimes k}$, $D = -x\partial_x$, $\mathcal{M}F_h = g/h$. The arithmetic inclusion is the original $\eta[P] = [g/h]_h^{\otimes k} P(\sum s_i)$, after the explicit chart (AU.2). Each local Taylor constant of g/h at a selected complete zero is nonzero. Recursively solving multiplication of truncated Taylor series therefore proves this factor is a unit. It does not alter the original cyclic kernel (AU.4). The comparison to ν_k changes the norm used in an estimate, while its identity polynomial map leaves η , every raw jet, and the original relations unchanged.

For two polynomial representatives with the same quotient, monic division gives their difference χQ . In the tensor variables the original annihilator identity supplies $\chi(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$, with Q_i obtained by successive fixed-order monic divisions. The final remainder is zero because the monomial tensor basis is independent and the left side annihilates the sum. If $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$ and $h(D)F_h = \Theta\phi_0$, the actual primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) Q(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{AU.49})$$

The tensor differential at position i contributes the second $(-1)^{i-1}$, cancelling the displayed one and giving exactly $\mathcal{V}_{h,k}(\chi Q)$. Thus every quotient comparison retains the same cohomology class with an explicit original primitive. On the supported carrier a fibre map sends (λ, v) to (λ, Lv) and sends external τ to τ . The killed relation has image $(\lambda, 0) = e_\lambda$. No comparison identifies that supported zero with external absence.

49.8 Verification and provenance

The complete arithmetic source (including the proof of its envelope), the exact endpoint-window source, the full PR24 note and scope correction/review, and the owner's complete balanced-window/factor-four proof were read for this calculation. Its new work is (AU.11)–(AU.23), the scalar estimate (AU.24)–(AU.33), and the full calculation (AU.40)–(AU.48). The constants $c_h, \vartheta_h, M_h(\pm b)$ retain their supplied definitions; no numerical enclosure of actual quartet data is claimed. The accompanying finite rational checker verifies the displayed maps and determinant formulas on declared auxiliary positive sources. It does not certify the analytic source by sampled values and makes no Lean claim.

50 Reflected relation norms and the exact common refinement of conormal algebras

Fix an integer $k \geq 1$, retain the original coordinate S and the real number $c = k/2$, and let $\chi \in \mathbb{C}[S]$ be monic of degree $q \geq 1$. The intended application uses the actual cyclic annihilator with every root order retained. All the algebra below holds for every such χ . It completes the comparison raised by the full-jet transfer construction: the squared norm of an original relation, its first conormal algebra, and their common refinement have explicit maps even for a packet which is not closed under reflection. The reflected factors and their phases are displayed throughout.

50.1 The original reflection and the norm polynomial

Let a bar over a polynomial mean conjugation of its coefficients. Define the conjugate-linear involution and its reflected monic polynomial by

$$f^*(S) = \bar{f}(k - S), \quad \chi^\dagger(S) = (-1)^q \chi^*(S), \quad N_\chi(S) = \chi(S) \chi^\dagger(S). \quad (\text{RF.1})$$

The factor $(-1)^q$ remains part of this definition. In particular $\chi^* = (-1)^q \chi^\dagger$ is the original involution image. The change of polynomial coordinates and the full leading phase are

$$\Psi : \mathbb{C}[S] \xrightarrow{\sim} \mathbb{C}[u], \quad \Psi(f)(u) = f(c + iu), \quad \Psi^{-1}(F)(S) = F((S - c)/i), \quad \psi = i^{-q} \Psi(\chi). \quad (\text{RF.2})$$

Proposition 50.1. *The polynomial $\chi^\dagger$ is monic of degree q , $(\chi^\dagger)^\dagger = \chi$, and $N_\chi^* = N_\chi$. The original-coordinate identities are*

$$\Psi(f^*) = \overline{\Psi(f)}, \quad \Psi(\chi) = i^q \psi, \quad \Psi(\chi^\dagger) = i^q \bar{\psi}, \quad \Psi(N_\chi) = i^{2q} \psi \bar{\psi}. \quad (\text{RF.3})$$

Consequently the monic polynomial $\Pi = \psi \bar{\psi}$ has real coefficients and satisfies $\Pi(u) = |\psi(u)|^2 \geq 0$ for real u . The following three conditions are equivalent:

$$\chi^\dagger = \chi, \quad \bar{\chi}(k - S) = (-1)^q \chi(S), \quad \bar{\psi} = \psi. \quad (\text{RF.4})$$

Under these equivalent conditions, $N_\chi = \chi^2$ and $\Pi = \psi^2$.

Proof. The leading coefficient of $\bar{\chi}(k - S)$ is $(-1)^q$, which proves monicity with its stated phase. Since k is real, $(f^*)^* = f$ and $(fg)^* = f^*g^*$. Applying these equations twice, including both real factors $(-1)^q$, gives $(\chi^\dagger)^\dagger = \chi$ and $N_\chi^* = (-1)^{2q} \chi^\dagger \chi = N_\chi$. Direct substitution gives $\Psi(f^*)(u) = \bar{f}(c - iu) = \overline{\Psi(f)(u)}$. Conjugating $\Psi(\chi) = i^q \psi$ therefore gives $\Psi(\chi^*) = (-i)^q \bar{\psi}$. Multiplication by $(-1)^q$ yields the third equation of (RF.3); multiplication by $\Psi(\chi)$ yields the fourth. The two leading coefficients in Π equal one, and conjugating its product reverses its factors, proving that its coefficients are real. Evaluation on the real axis proves its value there. Finally (RF.3) and bijectivity of Ψ prove the equivalence (RF.4), including the factor i^q . $\square$

The exact norm map can also be written without suppressing its phase. For the original positive density $m_{h,k}(u)$ and any polynomial Q , put $\tilde{Q} = \Psi(Q)$. Polynomial integrability is supplied by the original arithmetic source construction. Pointwise (RF.3) gives

$$\Psi(\chi Q) = i^q \psi \tilde{Q}, \quad \int_{\mathbb{R}} |\chi(c + iu) Q(c + iu)|^2 m_{h,k}(u) du = \int_{\mathbb{R}} \Pi(u) |\tilde{Q}(u)|^2 m_{h,k}(u) du. \quad (\text{RF.5})$$

Thus multiplication by χ on the original polynomial source is carried by Ψ to i^q times multiplication by ψ . Both integrals use the original density and its literal mass. The equality of integrands is $|i^q|^2 |\psi \tilde{Q}|^2 = \Pi |\tilde{Q}|^2$, and proves the complete norm comparison.

50.2 Full jets at the union of the original and reflected roots

Let a_λ be the order of λ as a root of χ , with value zero at other points. The order of λ in $\chi^\dagger$ is

$$b_\lambda = a_{k-\bar{\lambda}}, \quad r_\lambda = a_\lambda + b_\lambda. \quad (\text{RF.6})$$

To prove this, write $\chi(S) = \prod_\alpha (S - \alpha)^{a_\alpha}$ and substitute it in (RF.1). Each reflected factor is $k - S - \bar{\alpha} = -(S - (k - \bar{\alpha}))$; their q minus signs cancel the displayed $(-1)^q$. Orders at coincident roots add when the two polynomials are multiplied. In particular $\sum_\lambda r_\lambda = 2q$.

Theorem 50.2. *There are explicit isomorphisms*

$$\mathcal{A}_N := \mathbb{C}[S]/(N_\chi) \xrightarrow{\Psi_N} \mathbb{C}[u]/(\Pi), \quad [P] \mapsto [P(c + iu)], \quad (\text{RF.7})$$

and, for any fixed ordering of the distinct roots in (RF.6),

$$\mathcal{J}_N : \mathcal{A}_N \xrightarrow{\sim} \bigoplus_{\lambda: r_\lambda > 0} \mathbb{C}^{r_\lambda}, \quad [P] \mapsto (P^{(d)}(\lambda))_{\lambda, 0 \leq d < r_\lambda}. \quad (\text{RF.8})$$

The product in each raw-derivative block is $(v \cdot w)_d = \sum_{j=0}^d \binom{d}{j} v_j w_{d-j}$. If $\zeta_\lambda = (\lambda - c)/i$, the same raw jets in the u coordinate are exactly

$$(\Psi P)^{(d)}(\zeta_\lambda) = i^d P^{(d)}(\lambda). \quad (\text{RF.9})$$

Multiplication by S in (RF.8) is the block map $(M_S v)_d = \lambda v_d + d v_{d-1}$, with $v_{-1} = 0$.

Proof. Equation (RF.3) maps the ideal (N_χ) onto (Π) because i^{2q} is an invertible constant. Applying Ψ^{-1} proves both well-definedness and inverse for (RF.7). A polynomial has its first r raw derivatives zero at λ exactly when $(S - \lambda)^r$ divides it: its Taylor expansion is $\sum_d P^{(d)}(\lambda)(S - \lambda)^d/d!$. For distinct roots the corresponding factors are relatively prime. Repeated polynomial division, or Bezout's identity applied successively to those factors, shows that simultaneous divisibility is divisibility by their product N_χ . This proves the exact kernel in (RF.8). Representatives of degree less than $2q$ are unique by monic division, so the source and target of the induced injection both have dimension $2q$; it is therefore onto. Leibniz's rule gives the stated block product, including every binomial coefficient. The chain rule proves (RF.9), and differentiating SP gives $(SP)^{(d)}(\lambda) = \lambda P^{(d)}(\lambda) + d P^{(d-1)}(\lambda)$. These calculations also show that all maps retain the entire nilpotent root order. $\square$

There is a canonical original arithmetic quotient map $\pi_N : \mathcal{A}_N \rightarrow \mathcal{A}_\chi := \mathbb{C}[S]/(\chi)$. Its exact kernel and its coefficient-module parametrization are

$$0 \longrightarrow \mathbb{C}[S]/(\chi^\dagger) \xrightarrow{\iota_N} \mathcal{A}_N \xrightarrow{\pi_N} \mathcal{A}_\chi \longrightarrow 0, \quad \iota_N[f] = [\chi f]. \quad (\text{RF.10})$$

Indeed $N_\chi \mid \chi f$ is equivalent to $\chi^\dagger \mid f$ by cancellation in $\mathbb{C}[S]$. The image consists precisely of the multiples of χ modulo N_χ , and reduction modulo χ is surjective. This proves exactness, injectivity and all dimensions. The map ι_N is a $\mathbb{C}[S]$ -module map; the quotient maps are algebra homomorphisms.

50.3 A common refinement with the original first conormal algebra

Retain the original first conormal algebra $\mathcal{A}_2 = \mathbb{C}[S]/(\chi^2)$ and its exact sequence

$$0 \longrightarrow \mathbb{C}[S]/(\chi) \xrightarrow{[f] \mapsto [\chi f]} \mathcal{A}_2 \longrightarrow \mathcal{A}_\chi \longrightarrow 0. \quad (\text{RF.11})$$

Let g be the monic greatest common divisor of $\chi, \chi^\dagger$, and put

$$d = \chi g, \quad \ell = \chi^2 \chi^\dagger / g. \quad (\text{RF.12})$$

These polynomials retain all original roots: at a point λ their orders are respectively $a_\lambda + \min(a_\lambda, b_\lambda)$ and $a_\lambda + \max(a_\lambda, b_\lambda)$.

Theorem 50.3. *The following sequence is exact as a sequence of $\mathbb{C}[S]$ -modules:*

$$0 \longrightarrow \mathbb{C}[S]/(\ell) \xrightarrow{j} \mathcal{A}_2 \oplus \mathcal{A}_N \xrightarrow{t} \mathbb{C}[S]/(d) \longrightarrow 0, \quad j[P] = ([P], [P]), \quad t([P], [Q]) = [P - Q]. \quad (\text{RF.13})$$

The first map is an algebra isomorphism onto the fibre product

$$\boxed{\mathbb{C}[S]/(\ell) \simeq \mathcal{A}_2 \times_{\mathbb{C}[S]/(\chi g)} \mathcal{A}_N.} \quad (\text{RF.14})$$

The projections of this fibre product followed by reduction modulo χ agree. All the maps in (RF.10)–(RF.14) commute with the original multiplication operator M_S .

Proof. In the polynomial ring, the intersection of two principal ideals has at every root the maximum of their orders; their sum has the minimum. The intersection assertion follows immediately from divisibility at every root. For the sum, divide the two polynomials by their monic greatest common divisor. The remaining factors are coprime, and the Euclidean algorithm supplies a polynomial linear combination equal to one. Multiplying back proves that their ideal sum is the gcd ideal. Applying this to χ^2 and $\chi\chi^\dagger$ gives the polynomials ℓ and d in (RF.12). Thus j is well-defined and its kernel is zero. The difference map t is well-defined since d divides both defining polynomials; it is onto because $t([P], 0) = [P]$. If $P - Q$ belongs to their ideal sum, write $P - Q = \chi^2 A + N_\chi B$. Then $T = P - \chi^2 A = Q + N_\chi B$ maps to $([P], [Q])$ under j . This proves $\ker t = \text{im } j$ without choosing different roots or changing any order. The image consists of exactly those pairs with equal reduction modulo d , and its multiplication is componentwise. Therefore the injection is the algebra isomorphism (RF.14). Since χ divides d , the final reductions agree. Multiplication by S commutes with taking every polynomial residue, with $P \mapsto \chi P$, and with the difference of representatives; this proves the final assertion directly. $\square$

On the two kernel $\mathbb{C}[S]$ -modules of the quotient maps to $\mathcal{A}_\chi$, the comparison map has an especially explicit description. Reduction to $\mathbb{C}[S]/(\chi g)$ sends $[\chi f]$ from either algebra to the same residue class. Its image is $(\chi)/(\chi g)$, canonically isomorphic to $\mathbb{C}[S]/(g)$ by $[f] \mapsto [\chi f]$. Thus the exact diagram of kernel maps is the pair of quotient maps

$$\mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[S]/(g) \longleftarrow \mathbb{C}[S]/(\chi^\dagger), \quad (\text{RF.15})$$

with their literal multiplication-by- χ parametrizations in (RF.11) and (RF.10). This is the relation between the first conormal kernel and the kernel recorded by the relation norm.

50.4 Square-zero structure, reflected sectors, and supported zero

The kernel in (RF.11) has square zero, because products of its representatives contain χ^2 . The kernel in (RF.10) has square zero exactly when $\chi^\dagger = \chi$. To prove necessity, its element $[\chi]$ would have square zero, requiring $\chi\chi^\dagger \mid \chi^2$, and hence $\chi^\dagger \mid \chi$. Equal degree and monicity force equality. Sufficiency follows from the same divisibility. The exact common refinement for both cases remains (RF.13)–(RF.15).

For an explicit complementary case suppose $g = 1$. Choose the polynomials u, v supplied by the Euclidean algorithm with $u\chi + v\chi^\dagger = 1$. Then the algebra map

$$\mathcal{A}_N \xrightarrow{\sim} \mathcal{A}_\chi \times \mathcal{A}_{\chi^\dagger}, \quad [P] \longmapsto ([P]_\chi, [P]_{\chi^\dagger}), \quad e_{\text{ref}} = [u\chi]_{N_\chi} \quad (\text{RF.16})$$

is an isomorphism. Its kernel before factoring is the intersection $(\chi) \cap (\chi^\dagger) = (N_\chi)$, and for arbitrary residues represented by a, b , the polynomial $av\chi^\dagger + bu\chi$ has those two residues. This proves surjectivity and the inverse formula. The element e_{ref} has coordinates $(0, 1)$, proving $e_{\text{ref}}^2 = e_{\text{ref}}$. Its ideal is the kernel of π_N and is isomorphic as a unital algebra with its local unit e_{ref} to $\mathcal{A}_{\chi^\dagger}$. In these

coordinates M_S acts by the original residue of S in each factor. Equations (RF.14)–(RF.16) therefore calculate the reflected component and its exact map to the original arithmetic quotient.

To state the Split-Zero observation explicitly, attach any original support label λ_0 to each of these vector spaces. A linear map $L : E \rightarrow F$ has the support-preserving lift

$$\tilde{L}(\tau) = \tau, \quad \tilde{L}(\lambda_0, x) = (\lambda_0, Lx). \quad (\text{RF.17})$$

Composition satisfies $\widetilde{\tilde{L}_2 \tilde{L}_1} = \tilde{L}_2 \tilde{L}_1$ by evaluation on τ and on each labelled vector. Addition and scalar action within a fibre are preserved because L is linear. Every vector of the kernels in (RF.10), (RF.11), or (RF.13) maps to the receiving supported zero $(\lambda_0, 0)$; the input and all its pre-quotient coordinates remain defined. In particular, the nonzero algebra idempotent e_{ref} in (RF.16) has zero amplitude in $\mathcal{A}_\chi$, and its labelled image is $(\lambda_0, 0)$. These two values and the external point τ have been related by the explicit quotient, rather than identified in the source.

The first conormal algebra used here is exactly $\mathbb{C}[S]/(\chi^2)$. If a tensor construction supplies a different annihilator, its divisibility maps require that actual polynomial. No replacement of a tensor annihilator is made by (RF.12): (RF.13) is the complete comparison for the two polynomials explicitly stated here.

50.5 Reflected kernel transport and every power of the norm kernel

Write $I_2 = \ker(\mathcal{A}_2 \rightarrow \mathcal{A}_\chi)$ and $I_N = \ker \pi_N$. The coefficient parametrizations in (RF.10) and (RF.11) prove

$$\text{Ann}_{\mathbb{C}[S]} I_2 = (\chi), \quad \text{Ann}_{\mathbb{C}[S]} I_N = (\chi^\dagger). \quad (\text{RF.18})$$

Indeed a polynomial p annihilates the cyclic generator $[\chi]$ of the second kernel precisely when $\chi\chi^\dagger \mid p\chi$, equivalently $\chi^\dagger \mid p$; the first assertion follows with χ^2 in place of $\chi\chi^\dagger$. Thus the original $\mathbb{C}[S]$ -action on I_N descends to $\mathcal{A}_\chi$ exactly when $\chi^\dagger \mid \chi$, or equivalently $\chi^\dagger = \chi$. The original action always descends to $\mathcal{A}_{\chi^\dagger}$.

There is nevertheless a direct reflected identification of both kernel spaces, retaining the exact scalar action:

$$F : I_2 \xrightarrow{\sim} I_N, \quad F([\chi f]_{\chi^2}) = [\chi f^*]_{N_\chi}, \quad G([\chi h]_{N_\chi}) = [\chi h^*]_{\chi^2}. \quad (\text{RF.19})$$

If $\chi \mid f_1 - f_2$, then $\chi^\dagger \mid (f_1 - f_2)^*$ by (RF.1), proving that F is well-defined. If $\chi^\dagger \mid h_1 - h_2$, then $\chi \mid (h_1 - h_2)^*$, proving the assertion for G . The identity $\star^2 = 1$ proves $GF = 1$ and $FG = 1$. Both maps preserve addition and conjugate complex scalars. For every $p \in \mathbb{C}[S]$ their complete intertwining law is

$$F(p(S)x) = p^*(S)F(x), \quad FM_S = (kI - M_S)F. \quad (\text{RF.20})$$

This follows by applying $\star$ to the product pf . The labelled lift sends τ to τ and (λ_0, x) to (λ_0, Fx) ; its scalar law is $F(\alpha x) = \bar{\alpha}F(x)$. It is therefore a conjugate-linear fibre map with the original support retained.

The exact product on the coefficient module of I_N is

$$[f] \diamond [h] = [\chi fh]_{\chi^\dagger}, \quad \iota_N([f] \diamond [h]) = \iota_N[f] \iota_N[h]. \quad (\text{RF.21})$$

Changing f or h by a multiple of $\chi^\dagger$ changes the right-hand representative by a multiple of that polynomial. The second equality is the identity $\chi(\chi fh) = \chi^2 fh$ modulo N_χ . This proves the full product transport; no algebra-isomorphism assertion is imposed on the semilinear map F .

Theorem 50.4. *For every integer $t \geq 1$, retaining all root orders from (RF.6),*

$$\dim_{\mathbb{C}} I_N^t = \sum_{\lambda: r_\lambda > 0} \max\{b_\lambda - (t-1)a_\lambda, 0\}. \quad (\text{RF.22})$$

The kernel is nilpotent exactly when every reflected root occurs among the original roots. In this case its exact nilpotence index is

$$\max_{\lambda: b_\lambda > 0} \left\lceil \frac{a_\lambda + b_\lambda}{a_\lambda} \right\rceil. \quad (\text{RF.23})$$

In general its persistent part is the algebra

$$I_N^\infty := \bigcap_{t \geq 1} I_N^t \simeq \prod_{\lambda: a_\lambda = 0, b_\lambda > 0} \mathbb{C}[S]/((S - \lambda)^{b_\lambda}). \quad (\text{RF.24})$$

The isomorphism is the restriction of the original CRT residue map. An empty product here denotes the zero algebra. Its unit when nonzero is the idempotent with value one in those reflected-only blocks and zero in all other blocks. Projection to $\mathcal{A}_\chi$ sends it to zero, and its labelled image is supported zero.

Proof. In the block at λ put $z = S - \lambda$ and write $\chi = z^{a_\lambda} U_\lambda$ with $U_\lambda(\lambda) \neq 0$. The original polynomial U_λ remains part of the parametrization. It is invertible in $\mathbb{C}[S]/(z^{a_\lambda + b_\lambda})$: its constant coefficient is nonzero, and the finite geometric expansion in its nilpotent nonconstant part gives an inverse. Therefore the ideal generated by χ^t in this block is exactly (z^{ta_λ}) . Its basis is $z^{ta_\lambda}, \dots, z^{a_\lambda + b_\lambda - 1}$ when the lower index is smaller than the upper length, and is empty otherwise. This proves (RF.22), including $a_\lambda = 0$.

If $a_\lambda = 0 < b_\lambda$, that ideal is the whole nonzero block for every t , so nilpotence is impossible. Otherwise each block with $b_\lambda > 0$ vanishes precisely when $ta_\lambda \geq a_\lambda + b_\lambda$. Taking the largest of these exact integer thresholds proves (RF.23). The maximum is over a nonempty set because $\deg \chi^\dagger = q \geq 1$. There are finitely many blocks. The blocks with $a_\lambda > 0$ all vanish after a common finite threshold, whereas the others persist unchanged. The CRT isomorphism and its inverse from Bezout consequently give (RF.24) and the stated idempotent. Reduction modulo χ omits precisely the blocks where $a_\lambda = 0$, proving its amplitude image and the assertion about its support-preserving lift. $\square$

For a complete overlapping example keep $k = 2, c = 1$ and put $A = S - (2 + i), B = S - i$, so $A = B - 2$. For $\chi = A^2 B$ the retained polynomials are

$$\chi^\dagger = AB^2, \quad g = AB, \quad N_\chi = A^3 B^3, \quad d = A^3 B^2, \quad \ell = A^4 B^3. \quad (\text{RF.25})$$

In particular $\chi^2 - N_\chi = A^3 B^2(A - B) = -2A^3 B^2$ has degree five, so the square of $[\chi]$ in the degree-six norm algebra is nonzero. Its cube is zero because $N_\chi \mid A^6 B^3$. Formula (RF.22) gives dimensions 3, 1, 0 for the first three powers, with every original root order retained. If representatives P, Q satisfy $P - Q = dR$, their common refinement is explicitly

$$T = P + \frac{1}{2} R \chi^2 = Q + \frac{1}{2} R N_\chi, \quad (\text{RF.26})$$

since $N_\chi - \chi^2 = 2d$. Its residues are P modulo χ^2 and Q modulo N_χ . Thus the full mixed fibre-product map is available even though the norm kernel and the square-zero conormal kernel have different products.

50.6 Provenance and verification scope

This derivation follows the explicit dagger-stability correction to the full-jet transfer note at revision `dfcbb55cbf7fec8c9301fe242c13e741d762013e`. The endpoint product continuation retains the full transfer calculation for $\Pi = \psi\bar{\psi}$. The present note proves the original-coordinate helper algebra, both kernel modules, and their exact gcd/lcm refinement, including their raw jets and original action. These are polynomial algebra proofs with all domains and maps displayed. The source norm identity (RF.5) uses the unchanged arithmetic density; it does not estimate the growth of any quotient volume or establish a zero-location claim.

51 Exact optimization of the arithmetic full-jet determinant weights

This is a separate continuation of the sealed arithmetic volume upper estimate AU.1–33 and AU.40–49. It solves the weight optimization in AU.19 completely, including boundary optima, zero polynomial coefficients, the sole nonunique case, and an evaluable one-dimensional characterization. The signed raw-jet minors, source coordinate phases, factorials and arithmetic masses remain explicit. The improvement over equal weights is proved to be less than one natural-logarithm unit for every degree within that specified family. The continuation (OW.30)–(OW.40) proves a stronger source estimate by keeping the even moments together. It then solves the entire resulting diagonal family using the actual moments through degree $2q$. Its exact comparison with AU.19 removes a term of order $q \log q$; the earlier less-than-one theorem retains its precise, narrower domain.

51.1 The original source and the signed full-jet minors

Keep $g = 2\xi$, the original complete monic zero packet h , and

$$w_h(t) = |(g/h)(1/2 + it)|^2 / (2\pi), \quad m_k = w_h^{*k}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad S = c + iu, \quad c = k/2. \quad (\text{OW.1})$$

The source polynomial norm is $\int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du$ and its mass is μ_h^k . For the original degree- q monic relation χ put

$$\psi(u) = i^{-q} \chi(c + iu) = \sum_{j=0}^q \psi_j u^j, \quad \psi_q = 1, \quad \psi_j = \frac{i^{j-q}}{j!} \chi^{(j)}(c). \quad (\text{OW.2})$$

The last identity is the polynomial Taylor expansion at the original centre c . The exact ring map $P(S) \mapsto P(c + iu)$ has inverse $f(u) \mapsto f((S - c)/i)$ and intertwines multiplication by S with $c + iM_u$. Its raw derivative of order d at $\zeta = (\lambda - c)/i$ is $i^d P^{(d)}(\lambda)$. No coefficient phase or derivative factorial has been reassigned.

List the distinct roots ζ_a of ψ , in the original order, with their complete lengths ℓ_a , so $\sum_a \ell_a = q \geq 1$. For $N \geq q - 1$ the full raw-jet matrix is

$$(E_N)_{(a,d),j} = \begin{cases} j! \zeta_a^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d, \end{cases} \quad 0 \leq d < \ell_a, \quad 0 \leq j \leq N. \quad (\text{OW.3})$$

Taylor expansion proves that its kernel consists of polynomials divisible by ψ . Thus $Z := E_{q-1}$ is invertible by degree, and $E_q(\psi_0, \dots, \psi_q)^T = 0$. Write $\Delta_j = \det E_q[:, \{0, \dots, q\} \setminus \{j\}]$, with the retained increasing column order. The precise identity is

$$\Delta_j = (-1)^{q-j} \psi_j \det Z \quad (0 \leq j \leq q). \quad (\text{OW.4})$$

To prove the sign, append any row of E_q as a final row and expand the resulting zero determinant along that row. It shows that $((-1)^j \Delta_j)_{j=0}^q$ lies in the kernel of E_q . That kernel has dimension one because its first q columns form Z . It is therefore a scalar multiple of the displayed monic coefficient vector. Its last coordinate is $(-1)^q \Delta_q = (-1)^q \det Z$, which proves (OW.4). In particular a zero coefficient gives exactly a zero minor, rather than a deleted column or shortened jet space.

The raw confluent determinant, in the same row order, is

$$\det Z = \left(\prod_a \prod_{d=0}^{\ell_a-1} d! \right) \prod_{a < a'} (\zeta_{a'} - \zeta_a)^{\ell_a \ell_{a'}}. \quad (\text{OW.5})$$

For completeness take the ordinary Vandermonde determinant with temporarily distinct nodes grouped into the prescribed clusters. Successive divided-difference row operations remove its

within-cluster Vandermonde factors. The finite Taylor expansion of each monomial shows that its order- d divided-difference row tends to the raw derivative divided by $d!$. Multiplication of these rows by $d!$ gives (OW.5), with the cross-cluster factors still in their original order. Equivalently this proves the identity after cancellation of polynomial factors, so no analytic source approximation is involved.

Fix $0 < b < \pi/2$ and retain the two literal Laplace values

$$M_k = M_h(b)^k + M_h(-b)^k, \quad M_h(b) = \int_{\mathbb{R}} e^{bt} w_h(t) dt, \quad \beta_j = \frac{b^{2j}}{(2j)!} > 0. \quad (\text{OW.6})$$

For positive auxiliary weights τ_j the matrix used in AU.19 is $A_q(\tau) = E_q \text{diag}(\tau_j \beta_j) E_q^*$. Cauchy–Binet and (OW.4) prove

$$\det A_q(\tau) = |\det Z|^2 \sum_{j=0}^q a_j \prod_{l \neq j} \tau_l, \quad (\text{OW.7})$$

$$a_j = |\psi_j|^2 \prod_{l \neq j} \beta_l = \left(\prod_{l=0}^q \beta_l \right) \eta_j, \quad \eta_j = \frac{|\psi_j|^2}{\beta_j} = \frac{(2j)!}{(j!)^2 b^{2j}} |\chi^{(j)}(c)|^2.$$

In particular $a_q > 0$ and $\eta_q = (2q)! b^{-2q} > 0$. To verify Cauchy–Binet directly, expand the determinant multilinearly in the columns of E_q and then its adjoint. Repeated column choices vanish by alternation; collecting each q -element set gives its minor times its conjugate and the product of its diagonal weights. This yields exactly the first line of (OW.7). The signed identity is proved before taking the squared modulus.

For arbitrary positive τ , let $s = \sum_j \tau_j$ and $t_j = \tau_j/s$. The map $(\tau_j) \mapsto (s, (t_j))$ is a bijection from the positive cone to $(0, \infty)$ times the open simplex, with inverse $\tau_j = s t_j$. The determinant in (OW.7) acquires exactly the factor s^q . Thus the $q \log s$ in AU.19 cancels exactly. This changes auxiliary optimization coordinates and does not change m_k , its mass, or either source or quotient norm. Multiplication of all a_j by the common positive product in (OW.7) multiplies the objective by that product, with the maximizing weights unchanged. It suffices to solve the following finite optimization with all its original coefficients retained.

51.2 Global maximizers, including every zero coefficient

For $a_0, \dots, a_q \geq 0$ with $a_q > 0$, define

$$\mathcal{S}_q = \{\tau \in \mathbb{R}^{q+1} : \tau_j \geq 0, \sum_j \tau_j = 1\}, \quad D_a(\tau) = \sum_{j=0}^q a_j \prod_{l \neq j} \tau_l, \quad d(a) = \max_{\tau \in \mathcal{S}_q} D_a(\tau). \quad (\text{OW.8})$$

The objective is a polynomial and the simplex is compact. Its value at equal weights is positive, so a positive maximum exists. Put $I = \{j : a_j > 0\}$, $p = |I|$, $A = \max_j a_j$, and $S_a = \sum_j a_j$.

Theorem 51.1 (Complete boundary and uniqueness classification). *For $q \geq 2$ the maximizing weights are precisely the following.*

1. If $A \geq S_a - A$, choose its index j . Unless exactly two coefficients are positive and equal, the maximizer is unique: $\tau_j = 0$, $\tau_i = 1/q$ for every $i \neq j$, and $d(a) = A/q^q$.
2. If exactly two coefficients $a_j = a_l = A$ are positive, every maximizer, and only a maximizer, satisfies $\tau_j + \tau_l = 1/q$ and $\tau_i = 1/q$ at the other $q - 1$ indices. Its value is again A/q^q .
3. If $A < S_a - A$, then $p \geq 3$ and the maximizer is unique and has every coordinate strictly positive. At a zero coefficient it satisfies $\tau_j = 1/q$.

For $q = 1$, $D_a = a_0 \tau_1 + a_1 \tau_0$ is linear: it is maximized at the endpoint selecting its larger coefficient, or everywhere if the two coefficients are equal. Its value is $\max(a_0, a_1)$.

Proof. An elementary determinant representation proves the needed concavity. Set $v_i = \sqrt{a_i/a_q}$ for $0 \leq i < q$ and

$$M(\tau) = \text{diag}(\tau_0, \dots, \tau_{q-1}) + \tau_q vv^*. \quad D_a(\tau) = a_q \det M(\tau). \quad (\text{OW.9})$$

For positive first q weights, the rank-one determinant expansion gives $\det M = \prod_{i < q} \tau_i (1 + \tau_q \sum_{i < q} |v_i|^2 / \tau_i)$. This is (OW.9); since both sides are polynomials it holds at boundary weights too. Every $M(\tau)$ is positive semidefinite on the simplex, and $D_a > 0$ precisely when it is positive definite. For a real affine line $M(t) = M_0 + tE$ through positive definite matrices, differentiating the inverse identity gives $(M^{-1})' = -M^{-1}EM^{-1}$. The determinant's multilinear derivative therefore proves

$$(\log \det M)' = \text{Tr}(M^{-1}E), \quad (\log \det M)'' = -\text{Tr}((M^{-1/2}EM^{-1/2})^2) \leq 0. \quad (\text{OW.10})$$

The final matrix is Hermitian, so its squared eigenvalues are nonnegative, and equality holds exactly when $E = 0$.

If $p \geq 3$, at least two coordinates of v are nonzero. The linear map from τ to $M(\tau)$ is injective: an off-diagonal entry between two such coordinates forces a zero increment of τ_q whenever the matrix increment is zero; then its diagonal forces every other increment to be zero. Thus $\log D_a$ is strictly concave on its positive domain, which is convex as the inverse image of the positive definite cone. There can be at most one maximizer with positive objective.

If a simplex point has at least two zero coordinates, every monomial in D_a contains a zero, so its objective is zero. A positive boundary maximum must therefore have exactly one zero coordinate, say $\tau_j = 0$, and then $D_a = a_j \prod_{i \neq j} \tau_i$. It requires $a_j > 0$. For q nonnegative numbers of sum one, concavity of $\log x$ (whose second derivative is $-x^{-2}$) gives product at most q^{-q} , with equality exactly when they all equal $1/q$; a zero factor has product zero. Hence every positive boundary maximum is the candidate just described. At this point

$$\partial_i D_a = a_j / q^{q-1} \quad (i \neq j), \quad \partial_j D_a = (S_a - a_j) / q^{q-1}. \quad (\text{OW.11})$$

If $S_a - a_j > a_j$, the feasible path $\tau_j = \varepsilon$, $\tau_i = (1-\varepsilon)/q$ for $i \neq j$ has positive derivative $(S_a - 2a_j)/q^{q-1}$ at zero. That boundary point cannot maximize. Conversely, if $S_a - a_j \leq a_j$, the gradient of $\log D_a$ at the candidate has all active entries equal and the inactive entry no larger. Its scalar product with any feasible displacement is nonpositive. Concavity along the segment to any point with positive objective gives $\log D_a(\tau) \leq \log D_a(\tau^*) + \nabla \log D_a(\tau^*) \cdot (\tau - \tau^*) \leq \log D_a(\tau^*)$. This derivative is valid at the boundary because $M(\tau^*) > 0$ and remains positive in a neighborhood. Points of zero objective cannot exceed it. This proves the boundary criterion and, for $p \geq 3$, uniqueness from strict concavity.

If $p = 1$, $D_a = A \prod_{i \neq j} \tau_i$ and the same product inequality gives the stated unique maximizer. If $p = 2$, with positive coefficients at j, l , factor the polynomial exactly as

$$D_a(\tau) = \left(\prod_{i \notin \{j, l\}} \tau_i \right) (a_j \tau_l + a_l \tau_j). \quad (\text{OW.12})$$

At fixed $\tau_j + \tau_l$, unequal coefficients allocate that sum entirely to the larger coefficient's complementary variable. The remaining q factors of total sum one have the unique equal product maximum. If the two coefficients coincide, their sum is one factor instead, and equality in that same product inequality gives exactly the listed segment of maximizers.

When $A < S_a - A$, no boundary candidate can maximize, so compactness gives an interior maximum and strict concavity makes it unique. It remains to verify the stated zero coordinate value; that follows from the complete interior stationarity calculation below. The case $q = 1$ is immediate from its displayed linear polynomial. $\square$

No artificial coefficient phase is required to apply (OW.9) to the original raw-jet matrix. In fact $E_q = Z[I_q, -\psi_{<q}]$ by monic remainder, so

$$A_q(\tau) = ZB_0^{1/2}(\text{diag}(\tau_0, \dots, \tau_{q-1}) + \tau_q z z^*)B_0^{1/2}Z^*, \quad B_0 = \text{diag}(\beta_0, \dots, \beta_{q-1}), \quad z_i = -\psi_i \sqrt{\beta_q/\beta_i}. \quad (\text{OW.13})$$

Its $|z_i|^2$ equals η_i/η_q . Define the diagonal unitary $U_{ii} = -\psi_i/|\psi_i|$ when $\psi_i \neq 0$ and $U_{ii} = 1$ otherwise. Then $z = Uv$, with $v_i = \sqrt{\eta_i/\eta_q}$, and its central matrix is $UM(\tau)U^*$. This is the exact phase-preserving congruence connecting the auxiliary determinant proof to the original columns; all the phase entries remain in U .

51.3 A complete monotone scalar description of the interior optimum

Assume $q \geq 2$ and $A < S_a - A$, so the preceding theorem gives its unique interior maximizer. For positive weights put $P = \prod_j \tau_j$ and $S = \sum_j a_j/\tau_j$; then $D_a = PS$. Differentiating every factor gives $\partial_i D_a = P(S/\tau_i - a_i/\tau_i^2)$. The Lagrange multiplier for $\sum \tau_i = 1$ is qD_a because $\sum_i \tau_i \partial_i D_a = qD_a$, each monomial having degree q . Consequently its complete equations are

$$a_i = S\tau_i(1 - q\tau_i). \quad (\text{OW.14})$$

If $a_i = 0$, positivity gives $\tau_i = 1/q$. At a positive coefficient let $y_i = 1 - q\tau_i$. The equation shows $0 < y_i < 1$, and counting the zero coordinates gives

$$\sum_{i \in I} y_i = 1, \quad a_i = c_* y_i(1 - y_i), \quad c_* = S/q > 0, \quad \tau_i = (1 - y_i)/q. \quad (\text{OW.15})$$

Here c_* is the optimizer's scalar, distinct from the source centre $c = k/2$ in (OW.1). Conversely any positive solution of (OW.15), together with $\tau_i = 1/q$ at zero coefficients, has sum one and satisfies (OW.14): indeed $\sum_i a_i/\tau_i = qc_* \sum_{i \in I} y_i = qc_*$. Concavity then proves that this stationary point is the global maximizer, and uniqueness has already been established.

For $0 \leq t \leq 1/A$ and $i \in I$ define the continuous functions

$$f_i(t) = \frac{1 - \sqrt{1 - a_i t}}{2} = \frac{a_i t}{2(1 + \sqrt{1 - a_i t})}, \quad F(t) = \sum_{i \in I} f_i(t), \quad t = 4/c_*. \quad (\text{OW.16})$$

All square roots are the nonnegative real roots. The second formula is its exact rationalization, valid also at $t = 0$. Each f_i is strictly increasing on this interval, has value zero at zero, and takes values at most $1/2$.

Theorem 51.2 (The unique scalar root and the complete maximizing value). *Under $A < S_a - A$ exactly one of the following descriptions applies.*

1. If $F(1/A) \geq 1$, there is a unique $t_* > 0$ in $(0, 1/A]$ with $F(t_*) = 1$. Set $y_i = f_i(t_*)$ at every positive coefficient.
2. If $F(1/A) < 1$, the largest coefficient has a unique index j . There is a unique $t_* \in (0, 1/A)$ solving

$$H(t) := \sum_{i \in I \setminus \{j\}} \frac{a_i}{A} \frac{1 + \sqrt{1 - At}}{1 + \sqrt{1 - a_i t}} = 1. \quad (\text{OW.17})$$

The function H is strictly decreasing, with $H(0) = (S_a - A)/A > 1$ and $H(1/A) = 2F(1/A) - 1 < 1$. Set $y_j = 1 - f_j(t_*)$ and $y_i = f_i(t_*)$ for $i \neq j$.

In either case the weights are those in (OW.15), with zeros handled as stated, and the exact value is

$$d(a) = \frac{4}{q^q t_*} \prod_{i \in I} (1 - y_i). \quad (\text{OW.18})$$

Proof. In (OW.15) at most one y_i can exceed $1/2$ because their sum is one. If none does, its quadratic equations have precisely the smaller roots $f_i(t)$, and their sum is $F(t) = 1$. Continuity, strict increase and $F(0) = 0$ prove existence and uniqueness exactly in the first listed case.

If $y_j > 1/2$, each other y_i is strictly smaller than $1 - y_j < 1/2$, because $p \geq 3$ gives at least two other positive terms. The function $x(1 - x)$ is strictly increasing on $(0, 1/2)$. It follows that $a_i < c_* y_j(1 - y_j) = a_j$, so the large root must occur at the unique largest coefficient. Its sum equation is $\sum_{i \neq j} f_i(t) = f_j(t)$. At $t > 0$ dividing by the positive f_j and using the second formula in (OW.16) gives exactly (OW.17).

If $F(1/A) < 1$, the maximum is unique already from the endpoint values: two largest coefficients would contribute $1/2 + 1/2$ and the other positive terms would make $F(1/A) > 1$. For $a_i < A$ and $0 < t < 1/A$, direct logarithmic differentiation gives

$$\frac{d}{dt} \log \frac{1 + \sqrt{1 - At}}{1 + \sqrt{1 - a_i t}} = \frac{(1 - a_i t)^{-1/2} - (1 - At)^{-1/2}}{2t} < 0. \quad (\text{OW.19})$$

Thus every term of H decreases strictly. Its value at zero is given by continuity, and $f_j(1/A) = 1/2$ gives the second endpoint value. The intermediate value theorem proves its unique root. The constructed y has positive entries summing to one and solves the quadratic equations, so (OW.15) proves it is the optimum. Conversely if $F(1/A) \geq 1$, an index with a large root would have to be the unique largest one; its decreasing H would have $H(1/A) = 2F(1/A) - 1 \geq 1$ and hence no root inside $(0, 1/A)$. At equality, the largest root is exactly $1/2$ and coincides with the first construction. This proves completeness of the cases.

Finally $\prod_j \tau_j = q^{-(q+1)} \prod_{i \in I} (1 - y_i)$ and $S = qc_* = 4q/t_*$. Multiplication gives (OW.18), with the full original dimension exponent q retained. $\square$

The zero-coefficient assertion can also be proved globally without interior differentiation. If $p \geq 2$ and $z = q + 1 - p > 0$, write $s = \sum_{i \in I} \tau_i$ and $\tau_i = s\sigma_i$ on I . The polynomial factors as $(\prod_{i \notin I} \tau_i) s^{p-1} D_{a|I}(\sigma)$. For fixed s , the first product is at most $((1 - s)/z)^z$. The derivative of $\log((1 - s)^z s^{p-1})$ vanishes only at $s = (p - 1)/q$, and its second derivative is strictly negative. The endpoints have zero product. Hence every maximum has $\tau_i = 1/q$ at zero coefficients, and

$$d(a) = \frac{(p - 1)^{p-1}}{q^q} d_{p-1}(a|I), \quad (\text{OW.20})$$

where the last maximum uses the explicitly displayed positive-index simplex of dimension $p - 1$. If $z = 0$, the same formula has factor one. For $p = 1$, the first theorem gives the value directly, with the positive coordinate zero and the other q weights $1/q$. Thus the reduction has an explicit map and determinant power, without changing the dimension of the original arithmetic quotient.

For three positive coefficients a, b, d satisfying the strict triangle inequalities, the solution even has a rational expression. Put $\Delta = 2ab + 2ad + 2bd - a^2 - b^2 - d^2 > 0$. Then

$$\tau_a = \frac{2a(b + d - a)}{q\Delta}, \quad \tau_b = \frac{2b(a + d - b)}{q\Delta}, \quad \tau_d = \frac{2d(a + b - d)}{q\Delta}, \quad (\text{OW.21})$$

$$\tau_i = 1/q \quad (a_i = 0), \quad \boxed{d(a) = \frac{4abd}{q^q \Delta}}.$$

Indeed $\Delta = 4ab - (a + b - d)^2 > 0$ because $0 < a + b - d < 2 \min(a, b) \leq 2\sqrt{ab}$. On the positive-index simplex the proposed coordinates $\sigma_a = a(b + d - a)/\Delta$ and its cyclic counterparts are positive and sum to one. For its quadratic polynomial $a\sigma_b\sigma_d + b\sigma_a\sigma_d + d\sigma_a\sigma_b$, direct substitution makes every gradient coordinate $2abd/\Delta$. Euler's degree-two identity then gives its value abd/Δ ; concavity proves maximality. Equation (OW.20) supplies exactly the powers and the weights in (OW.21).

51.4 Evaluable enclosures and the exact maximum gain

For exact rational nonnegative input coefficients, the branch comparison at $1/A$ uses only finitely many nonnegative square roots. Equality is permitted as the first branch's endpoint. In the strict cases, (OW.16) or (OW.17) has a continuous strictly monotone scalar function on a known compact interval with opposite endpoint signs. Rational bisection therefore gives its unique root. Square roots themselves admit rational brackets by squaring positive rationals and bisecting. If an exact trial point is a root, it already solves the problem. Otherwise convergent radical brackets eventually resolve the nonzero sign at that trial point. A positive lower bracket for t_* is obtained after finitely many bisections because $t_* > 0$. Substitution in (OW.18) then yields convergent value brackets; every denominator is bounded away from zero on those root brackets. Boundary maximizers require only the exact coefficient comparison $A \geq S_a - A$ and their explicit product formula.

There is also a uniform certificate that avoids deciding a branch from uncertain coefficients. For two nonnegative coefficient lists, the product inequality used in (OW.11) gives, at every simplex point,

$$|D_a(\tau) - D_{\hat{a}}(\tau)| \leq q^{-q} \sum_j |a_j - \hat{a}_j|.$$

Taking a maximizing point for each list in turn proves

$$\boxed{|d(a) - d(\hat{a})| \leq q^{-q} \sum_j |a_j - \hat{a}_j|}. \quad (\text{OW.22})$$

The maximum is also increasing in each coefficient because each monomial is nonnegative. Thus nonnegative rational boxes $a^- \leq a \leq a^+$ give $d(a^-) \leq d(a) \leq d(a^+)$, and their value width is at most $q^{-q} \sum (a_j^+ - a_j^-)$. These statements remain valid if a corner list is all zero, where its maximum is exactly zero, or its reference last coefficient vanishes: a positive reference coefficient can be selected by the explicit simultaneous permutation of coefficient and simplex indices, whose inverse permutation restores every original index. No unresolved arithmetic equality must be assumed to construct the enclosure. For the original list $a_q > 0$ its lower value eventually stays positive, so logarithms can also be enclosed by their ordinary monotone integral $\log x = \int_1^x dt/t$.

A rational candidate has a direct exact value certificate, so no radical equality needs to be decided merely to enclose the answer. At any simplex point τ with $D := D_a(\tau) > 0$, calculate the full polynomial gradient $g_i = \partial_i D_a(\tau)$ and set

$$h = \max_i g_i - qD \geq 0.$$

The sign follows from $\sum_i \tau_i g_i = qD$. Concavity and the simplex constraint give, for every point of positive objective,

$$\log D_a(\sigma) \leq \log D + \frac{\sum_i g_i(\sigma_i - \tau_i)}{D} \leq \log D + h/D.$$

Therefore

$$\boxed{D \leq d(a) \leq De^{h/D}, \quad h < D \implies d(a) \leq \frac{D^2}{D - h}}. \quad (\text{OW.22a})$$

The last inequality follows from $e^x = \sum_{n \geq 0} x^n/n! \leq \sum_{n \geq 0} x^n = (1 - x)^{-1}$ for $0 \leq x < 1$. The polynomial gradient is defined also at a boundary point; the argument is valid there because its matrix (OW.9) is positive definite. Thus an exactly stationary rational candidate, including a boundary maximum with the correct inactive gradient, has $h = 0$ and certifies its exact value.

Here is a terminating way to combine this certificate with scalar evaluation, including an unresolved equality $F(1/A) = 1$. Use rational brackets for every square root. If the endpoint interval separates from one, select the proved branch and bisect its scalar root. If an endpoint interval still contains one, also form the candidate from its smaller-root values at $1/A$. At any scalar trial

whose comparison interval contains zero, form its candidate as well, instead of making an unsupported sign decision. Rational midpoints of the root brackets give nonnegative y approximations; replace the corresponding nonnegative auxiliary weights by their ratios to their positive sum, making an exact rational simplex point. One may keep zero-coefficient weights exactly $1/q$ and apply the positive-index rescaling (OW.20) only to the remaining coordinates. These are the explicit positive-cone coordinate maps already proved, and no source measure changes. Test (OW.22a) at each such candidate.

As square-root intervals narrow, a strictly nonzero branch or bisection sign is eventually determined. If a tested equality is exact, its candidate tends instead to the already proved optimum. In a fixed strict branch, successive root brackets tend to its unique root. In all cases the resulting candidates tend to the maximizer or to a member of the stated maximizing segment. There $D > 0$ and $h = 0$ by (OW.11) or (OW.14), and D, h are continuous functions of the candidate. Thus eventually $h < D$ and the rational certificate width $D^2/(D - h) - D$ is smaller than any prescribed positive rational tolerance. Every returned interval is certified solely by the rational polynomial gradient and (OW.22a), independently of the numerical branch-selection history.

Theorem 51.3 (The finite gain is uniformly less than one). *Let $\tau_j^{\text{eq}} = 1/(q + 1)$ and $S_a = \sum a_j > 0$. For every $q \geq 1$,*

$$D_a(\tau^{\text{eq}}) = \frac{S_a}{(q + 1)^q}, \quad \frac{S_a}{(q + 1)^q} \leq d(a) \leq \frac{S_a}{q^q}, \quad (\text{OW.23})$$

$$0 \leq \mathfrak{g}(a) := \log \frac{d(a)}{D_a(\tau^{\text{eq}})} \leq q \log(1 + 1/q) < 1.$$

The lower equality holds precisely when all coefficients are equal. The upper non-strict equality holds precisely when exactly one coefficient is positive.

Proof. The equal-weight value is obtained by substituting each weight. Every product of q selected nonnegative weights has sum at most one and is therefore at most q^{-q} . Summing with coefficients a_j gives the upper bound. If just one coefficient is positive, its boundary maximizer attains that bound. If two or more are positive, equality would require every corresponding product to attain q^{-q} . For the term omitting j , that requires $\tau_j = 0$ and $\tau_i = 1/q$ at every other index. Two distinct positive terms cannot meet those two incompatible conditions. Thus the upper equality has exactly the stated case.

If equal weights maximize, they are an interior point. Differentiating the polynomial there gives $\partial_i D_a = (q + 1)^{-(q-1)}(S_a - a_i)$, which must be constant in i under the sum constraint. Thus every a_i is equal. Conversely equal coefficients give equal gradient coordinates at equal weights, and the determinant concavity proves maximality for $q \geq 2$; for $q = 1$ it is the constant linear polynomial. Finally $\log(1 + x) < x$ for $x > 0$, since $\log(1 + x) = \int_0^x (1 + t)^{-1} dt < \int_0^x dt = x$. Apply this with $x = 1/q$ to prove the strict upper bound one. $\square$

One additional fully explicit lower bracket for the gain is

$$\mathfrak{g}(a) \geq \max\{0, q \log(1 + 1/q) + \log(A/S_a)\}. \quad (\text{OW.24})$$

It follows by testing the best boundary point. In the boundary regime its second expression is the exact gain. When $p \geq 2$, testing equal weights on the positive-index simplex in (OW.20) also gives $\mathfrak{g}(a) \geq q \log(1 + 1/q) + (p - 1) \log((p - 1)/p)$, with every zero retained through the factor in (OW.20).

51.5 The optimized original arithmetic bound, including boundary weights

To specify the analytic comparator used in the original bound, retain the proved constants $c_h > 0$, $\vartheta_h > 0$, $B = 42 + 2 \deg h$, $\alpha = \pi/2$, and $k \geq 3$. Its measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k - 2 + |u|)^{-B} du \leq m_k(u) du. \quad (\text{OW.25})$$

Let H_N^ν and H_N be the two monomial source Grams and $K_N^{\nu, \text{raw}} = E_N(H_N^\nu)^{-1}E_N^*$, with the original E_N and ψ . Their positive definiteness follows from their positive almost-everywhere densities and the finite zero set of every nonzero polynomial. Contraction of the identity polynomial map gives $H_N^\nu \preceq H_N$, so $K_N^{\text{raw}} \preceq K_N^{\nu, \text{raw}}$. If J_N is monic remainder, the original quotient metric is $G_N = (J_N H_N^{-1} J_N^*)^{-1}$, $V_N = \det G_N$. Every least lift is $H_N^{-1} J_N^* G_N x$; its difference from any other lift is orthogonal to it, as multiplication by J_N verifies. Thus these are the original quotient norms. Since $E_N = Z J_N$, $\det K_N^{\text{raw}} = |\det Z|^2 / V_N$, exactly as in the signed jet map above. The comparison measure defines V_N^ν by the same formula, with its own literal mass.

The exponential-series inequality and convolution give $\int |u|^{2j} m_k(u) du \leq M_k / \beta_j$. Weighted Cauchy-Schwarz, with positive weights of sum one, then gives $H_q \preceq M_k \text{diag}(1/(\tau_j \beta_j))$. Inversion and the original raw-jet map prove $K_q^{\text{raw}} \succeq M_k^{-1} A_q(\tau)$. Together with the upper comparison at degree $2q - 1$ this proves the original AU.19 bound on the open simplex:

$$C_k := \log(V_q / V_{2q-1}) \leq q \log M_k + \log \det K_{2q-1}^{\nu, \text{raw}} - \log \det A_q(\tau). \quad (\text{OW.26})$$

At a boundary maximizer in the first theorem its omitted coefficient is positive, so (OW.7) shows $\det A_q(\tau^*) > 0$. Use the explicit positive regularization $\tau^\varepsilon = (1 - \varepsilon)\tau^* + \varepsilon\tau^{\text{eq}}$ for $0 < \varepsilon < 1$. The raw matrices converge entry by entry to the positive definite $A_q(\tau^*)$. Taking the limit of their quadratic-form inequality preserves $K_q^{\text{raw}} \succeq M_k^{-1} A_q(\tau^*)$. Continuity of the positive determinant therefore extends (OW.26) to the boundary optimum. No expression $1/\tau_j$ is evaluated at zero; the boundary statement is proved on the dual raw kernel.

Let $B_\beta = \prod_{j=0}^q \beta_j$. Substitution of the exact maximum (OW.7), followed by the already proved raw-jet determinant conversion, gives the sharpened original bound

$$C_k \leq \mathcal{U}_{\text{opt}} := q \log M_k - \log B_\beta - \log d(\eta) - \log V_{2q-1}^\nu. \quad (\text{OW.27})$$

Here $d(\eta)$ is completely evaluated by the first theorem and (OW.16)–(OW.18), including every coefficient zero and the actual factorials in (OW.7). The factor $|\det Z|^2$ cancels between the two displayed raw kernels only after its signed minors and phase map have been proved. The original source mass stays in M_k , and the comparison mass stays in V_{2q-1}^ν . If desired the latter determinant is computed in its exact source/relation form $V_{2q-1}^\nu = \det H_{2q-1}^\nu / \det(B_{2q-1}^* H_{2q-1}^\nu B_{2q-1})$, where B_{2q-1} is multiplication by the same monic ψ . The quotient-first monomial-plus-relation coefficient matrix has determinant one; eliminating its relation block proves this formula.

The equal-weight upper expression from the identical estimate is

$$\mathcal{U}_{\text{eq}} = q \log M_k - \log B_\beta - \log \left(\frac{\sum_j \eta_j}{(q+1)^q} \right) - \log V_{2q-1}^\nu.$$

Thus its exact improvement is

$$\mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} = \mathfrak{g}(\eta), \quad 0 \leq \mathfrak{g}(\eta) \leq q \log(1 + 1/q) < 1. \quad (\text{OW.28})$$

This is an absolute bound, not a bound per coordinate or per degree. For the original complete quartet, $q = [1 + k(m-1)](k+1)^2$ tends to infinity. Dividing (OW.28) by $q \log k$ or by q^2 proves that the difference tends to zero on either scale. This conclusion requires no bound at all on the unknown comparison determinant's own asymptotic size.

For that quartet the gain is strictly positive and strictly smaller than the stated bound. Indeed its sum roots in the u chart are $(2b-k)\gamma - i(2a-k)\delta$, $0 \leq a, b \leq k$, each with the full length $1 + k(m-1)$. They are stable under negation, so comparing leading coefficients gives $\psi(-u) = (-1)^q \psi(u)$. Hence coefficients of the opposite parity vanish, while $\psi_q = 1$ is nonzero. The coefficients η_j cannot all be equal. Also at least one root is nonzero because $\delta, \gamma > 0$, so ψ is not the monomial u^q and at least two coefficients are nonzero. Applying both equality classifications in (OW.23) proves

$$0 < \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} < q \log(1 + 1/q) < 1 \quad \text{for every admitted complete quartet.} \quad (\text{OW.29})$$

The nonmonomial assertion follows directly from the factorizations: the roots of u^q are all zero, whereas the displayed root list contains a nonzero root. Its exact replacement here is the signed co-factor map (OW.4), whose nonzero extra coefficient yields the additional positive determinant term. No relation or jet is removed.

51.6 Joint even moments and exact optimization of their diagonal family

Keep the source, chart, polynomial and every raw-jet row from (OW.1)–(OW.7). For $0 \leq j \leq q$ write

$$\mu_{2j} := \int_{\mathbb{R}} u^{2j} m_k(u) du > 0, \quad \mu_0 = \mu_h^k, \quad S_q(b) := \sum_{j=0}^q \beta_j \mu_{2j}. \quad (\text{OW.30})$$

The inequalities are strict because the source density is positive almost everywhere and $u^{2j} > 0$ off zero. Finiteness follows from (OW.6) and its exponential-series majorant. These are moments in the original u chart; the zeroth moment remains the full k -fold source mass.

For a coefficient vector $p = (p_0, \dots, p_q)^T$ the ordinary complex Cauchy–Schwarz inequality gives the pointwise statement

$$\left| \sum_{j=0}^q p_j u^j \right|^2 \leq \left(\sum_{j=0}^q \frac{|p_j|^2}{\beta_j} \right) \left(\sum_{j=0}^q \beta_j u^{2j} \right) \leq \left(\sum_{j=0}^q \frac{|p_j|^2}{\beta_j} \right) \cosh(bu).$$

The first inequality uses the two vectors $p_j / \sqrt{\beta_j}$ and $\sqrt{\beta_j} u^j$, with complex conjugation in the inner product; the second uses the full nonnegative even exponential series. Its finite tail is strictly positive for every $u \neq 0$. Integration, including both original Laplace values, therefore proves

$$\begin{aligned} H_q &\preceq S_q(b) \operatorname{diag}(\beta_j^{-1}) \prec \frac{M_k}{2} \operatorname{diag}(\beta_j^{-1}), \\ 0 < S_q(b) &< \int_{\mathbb{R}} \cosh(bu) m_k(u) du = \frac{M_h(b)^k + M_h(-b)^k}{2} = \frac{M_k}{2}. \end{aligned} \quad (\text{OW.31})$$

The second matrix inequality is strict positive definiteness because it is a strictly positive scalar difference times a positive diagonal matrix. The convolution identity here follows by integrating $e^{b(t_1 + \dots + t_k)}$ against $\prod_i w_h(t_i) dt_i$; its nonnegative integrand permits iterated integration. The identical calculation applies at $-b$.

Inverting (OW.31) and applying the unchanged full raw-jet matrix gives

$$K_q^{\text{raw}} \succeq S_q(b)^{-1} E_q \operatorname{diag}(\beta_j) E_q^*.$$

The determinant identity (OW.7) at the literal weights all equal to one is $|\det Z|^2 B_\beta \sum_j \eta_j$. The original numerator comparison $K_{2q-1}^{\text{raw}} \preceq K_{2q-1}^{\nu, \text{raw}}$ is unchanged. Hence

$$\begin{aligned} C_k &\leq \mathcal{U}_{\text{fin}} := q \log S_q(b) - \log B_\beta - \log \left(\sum_j \eta_j \right) - \log V_{2q-1}^\nu, \\ \mathcal{U}_{\text{fin}} &< \mathcal{U}_{\text{cosh}} := q \log(M_k/2) - \log B_\beta - \log \left(\sum_j \eta_j \right) - \log V_{2q-1}^\nu. \end{aligned} \quad (\text{OW.32})$$

Thus the pointwise finite sum supplies its own proved source bound; it does not use the older scalar estimate $M_k \sum \tau_j$ from AU.19. Direct subtraction, with no moment or factorial discarded, gives

$$\begin{aligned} \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{cosh}} &= q \log(2(q+1)), \\ \mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{fin}} &= q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} > 0. \end{aligned} \quad (\text{OW.33})$$

The positivity follows from $\mathfrak{g}(\eta) < 1$ and $q \log(2(q+1)) \geq \log 4 > 1$ for $q \geq 1$. For the last elementary inequality, $\log 4 = 2 \int_1^2 dt/t > 2(2-1)/2 = 1$.

The same argument determines a larger family completely. To avoid identifying these new diagonal weights with the cofactor coefficients a_j in (OW.7), denote them by $r_j > 0$ and put

$$s(r) := \sum_{j=0}^q r_j \mu_{2j}, \quad H_q \preceq s(r) \operatorname{diag}(r_j^{-1}), \quad K_q^{\text{raw}} \succeq s(r)^{-1} E_q \operatorname{diag}(r_j) E_q^*. \quad (\text{OW.34})$$

Indeed pointwise Cauchy-Schwarz with $p_j/\sqrt{r_j}$ and $\sqrt{r_j} u^j$, followed by integration, gives the middle inequality for every coefficient vector p . Inversion reverses the positive definite matrix order: if $X \preceq Y$, then $Y^{-1/2} X Y^{-1/2} \preceq I$ and its positive eigenvalues are at most one, so $X^{-1} \succeq Y^{-1}$. Congruence by E_q proves the final inequality. This explicitly specifies the source-to-raw-jet map.

The positive-cone map and its inverse are

$$(r_j) \mapsto \left(s(r), t_j = \frac{r_j \mu_{2j}}{s(r)} \right), \quad (s, t) \mapsto r_j = \frac{s t_j}{\mu_{2j}}. \quad (\text{OW.35})$$

They are mutually inverse bijections between $\mathbb{R}_{>0}^{q+1}$ and $(0, \infty) \times \mathcal{S}_q^\circ$. Positivity of every moment proves each division is defined; summing the displayed t_j proves the simplex constraint, and substituting either composite restores every original index. This map acts only on auxiliary diagonal weights. It does not divide the source measure by its mass or alter a polynomial, jet, source coordinate, or quotient coordinate.

Put $\kappa_j = |\psi_j|^2 \mu_{2j}$, so $\kappa_q = \mu_{2q} > 0$. Applying (OW.4) and Cauchy-Binet before any phase cancellation gives

$$\begin{aligned} \frac{\det(E_q \operatorname{diag}(r_j) E_q^*)}{s(r)^q} &= |\det Z|^2 \sum_{j=0}^q |\psi_j|^2 \prod_{l \neq j} \frac{r_l}{s(r)} \\ &= |\det Z|^2 \left(\prod_{j=0}^q \mu_{2j}^{-1} \right) D_\kappa(t). \end{aligned} \quad (\text{OW.36})$$

The product has exactly $q+1$ moments. In each degree- q monomial its missing moment is supplied by the factor μ_{2j} in κ_j ; thus it includes, without a changed exponent, $\mu_0 = \mu_h^k$. Every original coefficient phase still occurs in (OW.4) and (OW.13), and each zero ψ_j gives its precise zero coefficient κ_j .

Combining (OW.34)–(OW.36) with the numerator comparison and the exact raw-jet determinant conversion gives the complete optimized bound

$$\boxed{C_k \leq \mathcal{U}_{\text{mom}} := \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log V_{2q-1}^\nu.} \quad (\text{OW.37})$$

The earlier global optimizer theorems apply to the literal list $\kappa_0, \dots, \kappa_q$, so (OW.8)–(OW.22a) calculate $d(\kappa)$ with all its boundary cases, scalar root, moment dependence and raw polynomial multiplicities retained. They also prove that (OW.37) is the infimum over the entire positive diagonal family (OW.34). For completeness at a boundary optimum t^* the limiting raw matrix is $E_q \operatorname{diag}(t_j^*/\mu_{2j}) E_q^*$, whose determinant is positive by (OW.36). Regularize t^* by $(1-\varepsilon)t^* + \varepsilon t^{\text{eq}}$ and use $r_j = t_j/\mu_{2j}$, so $s(r) = 1$. For every $\varepsilon > 0$ (OW.34) holds, and the raw matrices converge entrywise to that positive definite limit. Their quadratic-form inequalities and determinants pass to the limit exactly as in (OW.26). No inverse of a zero boundary weight is evaluated.

The finite even-series weights $r_j = \beta_j$ correspond under (OW.35) to the explicit interior point

$$t_j^\beta = \frac{\beta_j \mu_{2j}}{S_q(b)}, \quad D_\kappa(t^\beta) = \left(\prod_{j=0}^q \mu_{2j} \right) \frac{B_\beta \sum_j \eta_j}{S_q(b)^q}. \quad (\text{OW.38})$$

This equality follows term by term from the defining polynomial; it also follows by setting $r = \beta$ in (OW.36). Since its value is positive, all the following logarithms are defined. Their exact comparison is

$$\begin{aligned}\mathcal{U}_{\text{fin}} - \mathcal{U}_{\text{mom}} &= \log \frac{d(\kappa)}{D_{\kappa}(t^{\beta})} \geq 0, \\ \mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{mom}} &= q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} + \log \frac{d(\kappa)}{D_{\kappa}(t^{\beta})}.\end{aligned}\tag{OW.39}$$

The last logarithm is exactly computable from the proved optimizer; no less-than-one claim is made for it relative to the generally nonuniform point t^{β} . If the moment simplex instead uses its own equal point t^{eq} , its gain is precisely $\mathfrak{g}(\kappa) < 1$ by (OW.23). These different reference points are related by the explicit map (OW.35), rather than silently identified.

The exact removed term $q \log(2(q+1))$ has order $q \log q$. For the original $q = [1+k(m-1)](k+1)^2$ and fixed multiplicity m , its ratio to $q \log k$ tends to 2 if $m = 1$ and to 3 if $m > 1$. Indeed $q = (k+1)^2$ in the first case; in the second $q/k^3 = (m-1+1/k)(1+1/k)^2 \rightarrow m-1 > 0$. Taking logarithms proves both limits, while $\log(2(q+1))/q \rightarrow 0$ proves its ratio to q^2 tends to zero. The literal comparison (OW.39) retains all additional finite moment gains. It does not determine their growth or the comparison determinant's growth. In particular it does not subtract a term from AU's separately derived interval scalar bound without a proved comparison to that bound. It gives a stronger, fully specified source inequality and its original quotient-volume consequence using the actual finite moments.

There is also an exact finite comparison to the full original low-degree kernel, retaining all mixed moments. Let $\psi = (\psi_0, \dots, \psi_q)^T$ and let the coefficient matrix T have columns $1, u, \dots, u^{q-1}, \psi$ in the monomial basis of degree at most q . Its upper block is the identity and its last diagonal entry is $\psi_q = 1$, so $\det T = 1$. The last vector spans the relation kernel of J_q . In the Gram $T^* H_q T$, eliminate that last block by its positive scalar $\psi^* H_q \psi$. The resulting Schur complement is the Gram of the least lifts of the first q quotient coordinates: subtracting the orthogonal projection onto the relation is exactly the minimization defining G_q . Taking determinants therefore proves

$$\begin{aligned}V_q &= \frac{\det H_q}{\psi^* H_q \psi}, \quad \mathcal{U}_{\text{low}} := \log V_q - \log V_{2q-1}^{\nu}, \\ 0 \leq \mathcal{U}_{\text{mom}} - \mathcal{U}_{\text{low}} &= \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log \det H_q + \log(\psi^* H_q \psi).\end{aligned}\tag{OW.40}$$

The sign follows directly from the limiting raw-kernel inequality used in (OW.37): its determinant gives $|\det Z|^2/V_q \geq |\det Z|^2(\prod_j \mu_{2j}^{-1})d(\kappa)$. Its cancellation is legitimate because Z is invertible by (OW.3). The numerator comparison also gives $C_k \leq \mathcal{U}_{\text{low}}$. Thus (OW.40) records the exact nonnegative finite loss of the fully optimized diagonal source estimate relative to the full original H_q kernel. All entries of H_q are actual moments through degree $2q$ and every determinant is on its declared original space; the formula imposes no estimate on the loss's asymptotic growth.

51.7 Verification boundary and continuity

The sealed AU article is an input and remains unmodified. All new mathematics is written in (OW.1)–(OW.40): a signed full-jet identity, global maximum classification, monotone scalar evaluation, complete uncertain-input enclosures, the exact gain within AU.19, and the stronger joint-moment source family with its complete optimum. The auxiliary weight simplex and all its coordinate maps are specified separately from the original arithmetic source; its supported relations, Taylor unit, and quotient action are unchanged because every matrix uses the same E_N, J_N, ψ and (OW.2). A quotient-killed class remains the supported zero at its original label, and the induced fibre map fixes the external absence τ ; the optimization variables τ_j in this paper are real auxiliary weights with their own declared domain (OW.8).

Finite fixtures accompany the written proof and test its exact maps and optimum formulas on declared coefficient lists. They do not infer an arithmetic asymptotic from numerical optimization, assert new zero data, or replace the complete global proofs above.

52 Original relation moments and their positive-form comparison

This supplement gives the complete finite proof behind the incoming relation-moment certificate, records its two domain corrections, and constructs the exact source factorization for an upper comparison kernel. Every quotient coordinate, original source norm, zero eigenvalue, and source label is retained. The comparison kernel is additional data on the existing coefficient space; it does not replace the arithmetic norm. The arithmetic construction of that kernel is supplied by the cumulative paper's arithmetic upper-comparison proof.

52.1 The original Hilbert objects and their maps

Fix a monic polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$ and retain $E = \mathbb{C}[S]/(\chi)$ with ordered remainder basis $1, S, \dots, S^{q-1}$. The original multiplication endomorphism is $A = M_S$. Fix the original monic orthogonal polynomial columns p_n , their literal positive squared norms ω_n , and $b_n = [p_n]_\chi$. The degree- N source $\mathcal{P}_N = (\mathbb{C}^{N+1}, O_N)$ has $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and quotient map $B_N = [b_0, \dots, b_N]$. For $N \geq q - 1$, its first q columns form an invertible triangular matrix, so B_N is surjective and

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad V_N = \det G_N > 0. \quad (\text{RM.1})$$

Here K_N is a positive form on the conjugate dual coefficient space and G_N is the original quotient inner product, conjugate-linear in its first argument. Direct multiplication proves $B_N R_N = 1$ and $R_N^* O_N R_N = G_N$. If $u \in \ker B_N$, then $u^* O_N R_N x = u^* B_N^* G_N x = 0$. Thus R_N is the actual least-norm lift.

Fix $q - 1 \leq i \leq j$, set $r = j - i$, and write

$$F = [b_{i+1}, \dots, b_j], \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j), \quad D = K_j - K_i = F \Omega^{-1} F^*. \quad (\text{RM.2})$$

For $r = 0$ these maps have their literal zero-dimensional domains. The identity $I : E_i = (E, G_i) \rightarrow E_j = (E, G_j)$ has adjoint $T = K_i G_j : E_j \rightarrow E_i$, since $G_i T = G_j$. Consequently

$$Q = T I \in \text{End}(E_i), \quad Z = Q^{-1} - 1 = D G_i = F F^\dagger, \quad F^\dagger = \Omega^{-1} F^* G_i. \quad (\text{RM.3})$$

In particular $G_i Z = G_i D G_i \succeq 0$, so Z is positive self-adjoint in the original G_i . With its full q eigenvalues $\lambda_a \geq 0$,

$$g_a = (1 + \lambda_a)^{-1}, \quad \frac{V_i}{V_j} = \det(1 + Z) = \prod_{a=1}^q (1 + \lambda_a). \quad (\text{RM.4})$$

All zero λ_a contribute the factor one and remain in the dimension. If $Z = 0$, then $V_i = V_j > 0$, the ratio equals one, and its logarithm equals zero. This corrects the incoming note's phrase "the volume is zero."

For the real scalar k in the arithmetic action identity, put $H_N = K_N A^* G_N + A - k1$. Regard H_j on E_i via the specified coordinate identity and denote that endomorphism by $\hat{H}_j$. Calculation in the original coordinates gives

$$\begin{aligned} H_i Q - Q \hat{H}_j &= (K_i A^* G_i + A - k1) K_i G_j - K_i G_j (K_j A^* G_j + A - k1) \\ &= A K_i G_j - K_i G_j A = [A, Q]. \end{aligned}$$

Multiplying $[A, Q^{-1}] = -Q^{-1}[A, Q]Q^{-1}$ out proves

$$[A, Z] = \hat{H}_j(1 + Z) - (1 + Z)H_i. \quad (\text{RM.5})$$

Thus the cost and action are related even when their eigenspaces differ.

52.2 The source correction, all cross-pairings, and exterior area

The map $\Phi = R_i F : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{P}_i$ has image Gram $\Phi^* O_i \Phi = F^* G_i F$. Trace cyclicity and Hermitian symmetry give

$$t_1 = \text{Tr } Z = \sum_{n=i+1}^j \frac{b_n^* G_i b_n}{\omega_n}, \quad (\text{RM.6})$$

$$t_2 = \text{Tr } Z^2 = \sum_{n,m=i+1}^j \frac{|b_n^* G_i b_m|^2}{\omega_n \omega_m}, \quad (\text{RM.7})$$

$$\mathcal{A}_2 = \frac{t_1^2 - t_2}{2} = \sum_{i < n < m \leq j} \frac{(b_n^* G_i b_n)(b_m^* G_i b_m) - |b_n^* G_i b_m|^2}{\omega_n \omega_m}. \quad (\text{RM.8})$$

The diagonal terms cancel in $t_1^2 - t_2$; each unordered pair occurs twice, proving the last identity with its literal factor $1/2$. In unscaled exterior bases endowed with determinant Gram matrices,

$$\|\Phi\|_{\text{HS}}^2 = t_1, \quad \left\| \bigwedge^2 \Phi \right\|_{\text{HS}}^2 = \mathcal{A}_2. \quad (\text{RM.9})$$

Indeed the source vector $e_n \wedge e_m$ has squared norm $\omega_n \omega_m$ and its image has exactly the numerator in (RM.8). The alternating tensor map $u \wedge v \mapsto u \otimes v - v \otimes u$ has image Gram twice the exterior Gram. Applying this map in both source and target yields the commutative exterior/tensor square and cancels that same factor in the Hilbert–Schmidt ratio; neither Gram is identified with the other without the factor.

Let $L : \mathcal{P}_i \rightarrow \mathcal{P}_j$ pad by zero and let $N : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{P}_j$ be the actual new-column inclusion. Their images are orthogonal and $B_j N = F$, $B_j L = B_i$. Hence the retained relation map

$$\mathfrak{b} = N - L R_i F, \quad B_j \mathfrak{b} = 0, \quad \mathfrak{b}^* O_j \mathfrak{b} = \Omega + F^* G_i F. \quad (\text{RM.10})$$

This proves the relation identity and its full norm. Under the original source realization $\mathcal{V}$, the equality $J^{(k)} \mathcal{V} = \eta \pi_\chi$ makes both components have the same arithmetic jets. The original internal quotient sends $(\ell_j, \mathfrak{b}v)$ to $(\ell_j, 0)$, preserving the receiving label. Its lift sends external absence τ to τ . The coefficient amplitude map $p_R : G(R) \rightarrow R$ sends both $0_R^\bullet$ and τ to 0_R , and the exact fibre $p_R^{-1}(0_R) = \{0_R^\bullet, \tau\}$ records precisely the information lost by this subsequent observation. Likewise a dependent admitted wedge remains an admitted source vector whose observed squared area is $0^\bullet$.

52.3 Sharp determinant bound, exact remainder, and all boundary cases

For $q \geq 2$, positivity gives $t_1^2/q \leq t_2 \leq t_1^2$ by Cauchy–Schwarz and the nonnegative pair products. Define

$$d = \sqrt{\frac{qt_2 - t_1^2}{q-1}}, \quad a = \frac{t_1 + (q-1)d}{q}, \quad b = \frac{t_1 - d}{q}. \quad (\text{RM.11})$$

Thus $0 \leq d \leq t_1$, $0 \leq b \leq a$, and direct expansion proves $a + (q-1)b = t_1$ and $a^2 + (q-1)b^2 = t_2$. For any eigenvalue x , the other $q-1$ eigenvalues give $t_2 \geq x^2 + (t_1 - x)^2/(q-1)$. Solving that quadratic yields $x \leq a$. Consequently $Q \succeq (1+a)^{-1}1$ in G_i .

Here is a direct algebraic proof of the entire interpolation remainder. For $s \geq 1$, let

$$H_s(x) = \frac{1}{s+b} - \frac{x-b}{(s+b)^2} + \frac{(x-b)^2}{(s+a)(s+b)^2}. \quad (\text{RM.12})$$

Its value and derivative agree with $1/(s+x)$ at b , and its value agrees at a . Multiplication by the common denominator proves, identically,

$$\frac{1}{s+x} - H_s(x) = \frac{(a-x)(x-b)^2}{(s+x)(s+b)^2(s+a)}. \quad (\text{RM.13})$$

Integrate $1/s - 1/(s+x)$ over $[1, \infty)$ to obtain $\log(1+x)$. Integration of $1/s - H_s(x)$ gives its quadratic interpolant

$$P(x) = \log(1+b) + \frac{x-b}{1+b} + c(x-b)^2, \\ c = \frac{\log(1+a) - \log(1+b) - (a-b)/(1+b)}{(a-b)^2} < 0 \quad (a > b). \quad (\text{RM.14})$$

The coefficient integrals converge: the quadratic coefficient is $-\int_1^\infty ((s+a)(s+b)^2)^{-1} ds$, and the other two are the displayed logarithm and reciprocal. Therefore, for $0 \leq x \leq a$,

$$P(x) - \log(1+x) = (a-x)(x-b)^2 \int_1^\infty \frac{ds}{(s+x)(s+b)^2(s+a)} \geq 0. \quad (\text{RM.15})$$

The trace of any quadratic depends only on q, t_1, t_2 . Those values equal the moments of $a, b, \dots, b$, so summing (RM.15) proves

$$\log(V_i/V_j) \leq U_2 := \log(1+a) + (q-1)\log(1+b). \quad (\text{RM.16})$$

The remainder in (RM.15) is strictly positive away from $x = a, b$. If m eigenvalues equal a , the first trace identity gives $ma + (q-m)b = a + (q-1)b$, hence $(m-1)(a-b) = 0$. Equality for $a > b$ therefore occurs exactly once at a and $q-1$ times at b . That spectrum realizes the prescribed moments, proving sharpness from these data. For $a = b$, zero variance forces $Z = b1$ and equality already holds. For $q = 1$, the ratio is exactly $1 + t_1$. For $q = 0$, its empty determinant is one. The case $i = j$ gives $Z = 0$ at every admitted dimension.

With $t_3 = \text{Tr } Z^3$, set

$$C_3 = -t_3 + (a+2b)t_2 - (2ab+b^2)t_1 + qab^2 = \text{Tr}((a1-Z)(Z-b1)^2) \geq 0. \quad (\text{RM.17})$$

Finite spectral summation of (RM.15) proves the exact loss

$$U_2 - \log \det(1+Z) = \int_1^\infty \frac{\text{Tr}((a1-Z)(Z-b1)^2(s1+Z)^{-1})}{(s+b)^2(s+a)} ds. \quad (\text{RM.18})$$

Every factor is a function of the same G_i -self-adjoint endomorphism. Since $(s+x)(s+b)^2(s+a) \leq (s+a)^4$, the loss is at least $C_3 \int_1^\infty (s+a)^{-4} ds = C_3/(3(1+a)^3)$. Also Jensen's scalar inequality, whose strictness follows from $\frac{d^2}{dx^2} \log(1+x) = -(1+x)^{-2} < 0$, gives $U_2 \leq q \log(1+t_1/q)$, strictly unless $a = b$.

For $q \geq 2$ and $t_1 > 0$, $\mathcal{A}_2 = (q-1)b(t_1 - qb/2)$ and $0 \leq b \leq t_1/q$ imply

$$\frac{\mathcal{A}_2}{(q-1)t_1} \leq b \leq \frac{2\mathcal{A}_2}{(q-1)t_1}, \quad \frac{V_i}{V_j} \leq (1+t_1) \left(1 + \frac{2\mathcal{A}_2}{(q-1)t_1}\right)^{q-1}. \quad (\text{RM.19})$$

When $\mathcal{A}_2 = 0 < t_1$, all eigenvalue pair products vanish, so exactly one eigenvalue is positive and the ratio is $1 + t_1$.

52.4 The comparator join in the original metric

Let U_j be the actual explicit upper kernel supplied by the arithmetic comparison construction, expressed in the same quotient coordinates. That construction proves $U_j \succeq K_j \succeq K_i$ as dual forms. Define from the available lower-degree arithmetic data and this comparator

$$D_+ = U_j - K_i, \quad Z_+ = D_+ G_i, \quad t_1^+ = \text{Tr } Z_+, \quad t_2^+ = \text{Tr } Z_+^2. \quad (\text{RM.20})$$

Then $G_i(Z_+ - Z) = G_i(U_j - K_j)G_i \succeq 0$. Thus both operators are positive self-adjoint in G_i and $Z_+ \succeq Z$ there. For completeness, ordered eigenvalues increase under this order: for an orthonormal eigenbasis of either operator, the top- m span gives the lower inequality below; every m -dimensional subspace intersects the span of eigenvectors numbered $m, \dots, q$, which gives the upper inequality. This proves the max-min characterization

$$\lambda_m(X) = \max_{\dim M=m} \min_{v \in M \setminus 0} \frac{\langle v, Xv \rangle_{G_i}}{\langle v, v \rangle_{G_i}}.$$

Applying the pointwise form inequality inside this characterization yields $\lambda_m(Z_+) \geq \lambda_m(Z)$ for each of the original q indices. In particular, for $q \geq 2$, with a_+, b_+ obtained by (RM.11) from t_1^+, t_2^+ ,

$$\frac{V_i}{V_j} \leq \det(1 + Z_+) = \det(U_j G_i) \leq (1 + a_+)(1 + b_+)^{q-1}. \quad (\text{RM.21})$$

No monotonicity of the sharp bound as a function of two separately varied trace numbers was used; the first inequality follows from the actual form order, and the second follows from (RM.16) for Z_+ . For $q \geq 2$, $Q \succeq (1 + a_+)^{-1}1$ is a derived finite gap. For $q = 1$, the comparator determinant is $1 + t_1^+$ and $Q \succeq (1 + t_1^+)^{-1}1$; for $q = 0$, the empty determinant is one. When $i \leq q$, the actual input to K_i, G_i uses moments only through $2i \leq 2q$; U_j uses the explicit comparison measure's moments through $2j \leq 4q$ in the two endpoint blocks. Formula (RM.21) supplies a certificate without introducing the unknown actual high-degree moments into its inputs.

There is an exact source map behind this comparison. Interpret D_+ as a positive Hermitian form on the conjugate dual coefficient space $E^\vee$. Let

$$\mathcal{C}_+ = E^\vee / \ker D_+, \quad \langle [\alpha], [\beta] \rangle_+ = \alpha^* D_+ \beta, \quad J_+[\alpha] = D_+ \alpha \in E_i. \quad (\text{RM.22})$$

These expressions are independent of representatives. Positivity implies the only zero-norm classes are the removed radical: diagonalize the positive Hermitian form in the stated coordinate space to prove $\alpha^* D_+ \alpha = 0 \iff D_+ \alpha = 0$. This diagonalization proves a fact about that form and does not alter G_i . The adjoint is $J_+^\dagger v = [G_i v]$, hence $J_+ J_+^\dagger = Z_+$ and

$$\Phi_+ = R_i J_+ : \mathcal{C}_+ \rightarrow \mathcal{P}_i, \quad \|\Phi_+\|_{\text{HS}}^2 = t_1^+, \quad \left\| \bigwedge^2 \Phi_+ \right\|_{\text{HS}}^2 = ((t_1^+)^2 - t_2^+)/2. \quad (\text{RM.23})$$

To prove the second identity in any Gram basis, diagonalize the positive endomorphism $\Phi_+^\dagger \Phi_+$ on its existing source Hilbert space. Its exterior eigenvalues are all unordered products. Its nonzero eigenvalues agree, with multiplicity, with those of $\Phi_+ \Phi_+^\dagger$ by the mutually inverse maps Φ_+ and $\lambda^{-1} \Phi_+^\dagger$ on each nonzero eigenspace. Finally R_i is an isometry from E_i into $\mathcal{P}_i$, which proves (RM.23), retaining the zero eigenspaces.

The original correction factors through this source by an exact contraction. For $\alpha \in \ker D_+$, (RM.2) and $D \preceq D_+$ imply $\alpha^* F \Omega^{-1} F^* \alpha = 0$, so $F^* \alpha = 0$. Thus $Fv \in \text{im } D_+$ for every v . There is a unique class $Cv = [u]$ solving $D_+ u = Fv$. This defines a linear map

$$C : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{C}_+, \quad J_+ C = F, \quad \Phi_+ C = \Phi. \quad (\text{RM.24})$$

Let $s = u^* D_+ u \geq 0$. The equality $D_+ u = Fv$ and Cauchy-Schwarz in Ω give $s = |u^* Fv| \leq (u^* F \Omega^{-1} F^* u)^{1/2} (v^* \Omega v)^{1/2} \leq s^{1/2} (v^* \Omega v)^{1/2}$. Dividing when $s > 0$, with the case $s = 0$ immediate, proves $\|Cv\|_+^2 \leq v^* \Omega v$. The full kernel is $\ker C = \ker F$ by (RM.24) and uniqueness of the class.

Exterior functoriality gives $\bigwedge^2 \Phi = (\bigwedge^2 \Phi_+)(\bigwedge^2 C)$, preserving every admitted pair. This is the exact morphism from the old source correction to the new positive-form source. The original relation $N - L\Phi$ from (RM.10) is retained, with its original supported-zero image.

The cost-action defect of the comparator also has an explicit form. Put $\Delta = U_j - K_j$. Then $Z_+ = Z + \Delta G_i$, and direct expansion using $\hat{H}_j = A + K_j A^* G_j - k1$ gives

$$\begin{aligned} [A, Z_+] &= \hat{H}_j(1 + Z_+) - (1 + Z_+)H_i + \mathcal{E}, \\ \mathcal{E} &= [A, \Delta G_i] - \hat{H}_j \Delta G_i + \Delta G_i H_i \\ &= \Delta A^* G_i - K_j A^* G_j \Delta G_i. \end{aligned} \quad (\text{RM.25})$$

The second line defines the exact defect; substituting the displayed control matrices and cancelling equal terms proves the last expression. It is not discarded or asserted small.

52.5 Literal Gaussian calibration and source corrections

For the declared polynomial calibration, retain $S = 1 + ix$, literal Gaussian mass seven, variance one, $p_{n+1} = (S - 1)p_n + np_{n-1}$ and $\omega_n = 7n!$. For $q = 3$, write $y = S - 1$ only as the explicit polynomial $S - 1$, without changing the remainder basis. The relevant columns, expressed in the fixed basis $1, S, S^2$, are

$$\begin{aligned} b_0 &= (1, 0, 0)^t, \quad b_1 = (-1, 1, 0)^t, \quad b_2 = (2, -2, 1)^t, \quad b_3 = (-3, 3, 0)^t, \\ b_4 &= (9, -12, 6)^t, \quad b_5 = (-15, 15, 0)^t, \quad b_6 = (60, -90, 45)^t. \end{aligned}$$

These follow by the literal recurrence and division by $(S - 1)^3$. Substitution into (RM.1)–(RM.3) gives

$$Z_{3,6} = \frac{1}{8} \begin{pmatrix} -29 & -35 & 84 \\ 42 & 48 & -126 \\ -21 & -21 & 69 \end{pmatrix}. \quad (\text{RM.26})$$

The direct recurrence gives $G_3 = \begin{pmatrix} 7 & 7 & 0 \\ 7 & 49/5 & 28/5 \\ 0 & 28/5 & 126/5 \end{pmatrix}$, so (RM.26) is its literal coefficient matrix in the retained metric. Its invariant conclusions are

$$t_1 = 11, \quad t_2 = 3365/32, \quad t_3 = 273947/256, \quad \mathcal{A}_2 = 507/64, \quad \det(1 + Z) = 5145/256. \quad (\text{RM.27})$$

The characteristic polynomial factors with roots $3/4$, $(41 - 7\sqrt{34})/8$, $(41 + 7\sqrt{34})/8$; all are positive, distinct, and retain $q = 3$. The two-trace extremal roots are $a = 11/3 + 7\sqrt{127}/12$, $b = 11/3 - 7\sqrt{127}/24$. Since the actual three roots are distinct and the equality spectrum in (RM.16) has two equal roots, its inequality is strict. Since $a > b$, its improvement over $3 \log(1 + 11/3)$ is also strict. Rational logarithm enclosures in the companion replay give the numerical values in the incoming note without relying on decimals for either strictness argument.

In dimension two, the same literal recurrence gives $(t_1, t_2) = (2, 5/2)$ for $(i, j) = (1, 3)$ and $(7/4, 37/16)$ for $(2, 4)$. The respective (a, b) are $(3/2, 1/2)$ and $(3/2, 1/4)$, so (RM.16) is exact and yields ratios $15/4$ and $25/8$.

The incoming window expressions contain $1/(2r)$ and a minimum over $i \leq N < j$. Their domain is $i < j$. The original and comparator maps retain $i = j$ as well; scalar formulas keep their explicitly stated dimension and positive-trace domains, with zero cost handled separately. No minimum over an empty window is defined, and no inverse below the first admissible degree $q - 1$ is introduced. The complete first-degree and consecutive-window proofs remain in the cumulative reader.

52.6 Rank information with every zero direction retained

Let $W = \text{im } F \subset E_i$ and $\rho = \dim W$. Since $\langle v, Zv \rangle_{G_i} = \|F^\dagger v\|_\Omega^2$, one has

$$\ker Z = \ker F^\dagger = W^{\perp_{G_i}}, \quad E_i = W \oplus^{\perp_{G_i}} W^{\perp_{G_i}}. \quad (\text{RM.28})$$

To check invariance and positivity on W , $Z = FF^\dagger$ has image in W , and the only vector of W in its kernel is zero by (RM.28). The restriction Z_W is therefore positive definite on its inherited metric, and the literal inclusion/projection maps give

$$Z = Z_W \oplus 0_{q-\rho}, \quad Q = (1_W + Z_W)^{-1} \oplus 1_{q-\rho}, \quad \det(1_E + Z) = \det(1_W + Z_W) 1^{q-\rho}. \quad (\text{RM.29})$$

The source kernel is also retained. Put $\mathcal{U} = (\ker F)^\perp$ in the original source. Its orthogonal projection $P_{\mathcal{U}}$ satisfies $F = F|_{\mathcal{U}}P_{\mathcal{U}}$. The map $F|_{\mathcal{U}} : \mathcal{U} \rightarrow W$ is a bijection: injectivity is the orthogonal intersection with $\ker F$, and surjectivity follows by decomposing any source preimage. Thus it identifies the source Hilbert quotient $(\mathbb{C}^r, \Omega)/\ker F$ with W by a specified map; the norms on these objects are related by that map's retained positive Gram, rather than being asserted equal. Its composition with R_i is precisely the old least-lift correction on the quotient source. For $v \in \ker F$, the original relation (RM.10) is Nv , so the kernel directions remain actual newly admitted theta relations, with their norm $v^*\Omega v$ and their receiving supported-zero labels.

Any proved integer upper bound $\rho \leq R \leq q$ gives a sharper finite determinant certificate. The ρ positive eigenvalues and $R - \rho$ expressly retained zeros have the same t_1, t_2 . For $R \geq 2$ set

$$d_R = \sqrt{\frac{Rt_2 - t_1^2}{R-1}}, \quad a_R = \frac{t_1 + (R-1)d_R}{R}, \quad b_R = \frac{t_1 - d_R}{R}. \quad (\text{RM.30})$$

Applying the already proved scalar argument to that R -entry list gives

$$\det(1_E + Z) \leq (1 + a_R)(1 + b_R)^{R-1} 1^{q-R}. \quad (\text{RM.31})$$

This operation does not identify or remove the $q - R$ original directions. The full decomposition (RM.29) records their action by Q and Z . For $R = 1$, the exact ratio is $1 + t_1$; for $R = 0$, it is one. One always has $\rho \leq \min(q, j - i)$ from the literal dimensions of F . Additional parity or relation identities may give a better proved bound; their full proofs must supply that information. The present argument makes no such unproved rank assertion for an arithmetic packet.

For fixed feasible t_1, t_2 , this bound is at least as sharp as the q -dimensional bound. Indeed the R -entry extremizer $a_R, b_R, \dots, b_R$ with $q - R$ additional zeros is feasible for the q -dimensional moment problem. The optimality already proved for that problem yields

$$\log(1 + a_R) + (R - 1) \log(1 + b_R) \leq \log(1 + a) + (q - 1) \log(1 + b).$$

The known rank bound belongs to the actual cost Z . To apply (RM.30)–(RM.31) to Z_+ , its own rank must be proved; the inequality $Z_+ \succeq Z$ does not imply $\text{rank } Z_+ \leq \text{rank } Z$. Its exact source in (RM.22) gives $\text{rank } Z_+ = \dim \mathcal{C}_+ = \text{rank } D_+$, which is the correct replacement.

52.7 The entire comparison excess is the source contraction defect

The contraction in (RM.24) has a concrete adjoint:

$$C^\dagger[\alpha] = \Omega^{-1} F^* \alpha. \quad (\text{RM.32})$$

It is well-defined because $\ker D_+ \subset \ker F^*$ was proved above. For $Cv = [u]$, its defining equation gives $\langle Cv, [\alpha] \rangle_+ = u^* D_+ \alpha = v^* F^* \alpha = \langle v, \Omega^{-1} F^* \alpha \rangle_\Omega$, proving the adjoint identity. Define $\mathcal{L} = 1_{\mathcal{C}_+} - CC^\dagger$. Then

$$\langle [\alpha], \mathcal{L}[\beta] \rangle_+ = \alpha^* (D_+ - F\Omega^{-1}F^*)\beta = \alpha^* (U_j - K_j)\beta, \quad (\text{RM.33})$$

$$\mathcal{L} \succeq 0, \quad J_+ \mathcal{L} J_+^\dagger = (U_j - K_j)G_i = Z_+ - Z. \quad (\text{RM.34})$$

The first equality follows by (RM.32) and the definition of the adjoint; the second follows by (RM.2) and (RM.20). The right-hand form is nonnegative by the actual upper-kernel comparison. Multiplying out the left side of (RM.34) gives $J_+ J_+^\dagger - J_+ CC^\dagger J_+^\dagger = Z_+ - FF^\dagger$, proving its last equality.

Thus the complete excess of the comparator over the arithmetic cost is retained as a specific source contraction defect.

For clarity, the arithmetic upper kernel itself is formed on the original polynomial coordinates. Write $X_j = [[S^0]_\chi, \dots, [S^j]_\chi]$ and let H_j^m be the full monomial Gram for the actual source measure $m_{h,k}(u)du$, with $S = k/2 + iu$. The arithmetic upper-comparison proof constructs the explicit positive measure

$$d\nu_{h,k}(u) = a_k e^{-\alpha|u|} (k - 2 + |u|)^{-B} du, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h = 42 + 8m,$$

$$a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}, \quad k \geq 3,$$

where $m \geq 1$ is the full common multiplicity of the four zeros in the quartet and $\deg h = 4m$. Its literal constants are $c_h > 0$ and $\vartheta_h = \int_{-1}^1 w_h(t)dt > 0$. That complete proof establishes $d\nu_{h,k} \leq m_{h,k}(u)du$. For any polynomial coefficient vector v , $v^* H_j^m v = \int |\sum v_n (k/2 + iu)^n|^2 d\nu_{h,k} \leq v^* H_j^m v$. To prove the inverse inequality without a metric change, the exact completion of the square gives, for either positive form H ,

$$\ell^* H^{-1} \ell = \max_v \{2 \operatorname{Re}(\ell^* v) - v^* H v\},$$

the maximum attained at $v = H^{-1} \ell$. The form inequality therefore implies $(H_j^m)^{-1} \preceq (H_j^\nu)^{-1}$. Consequently

$$U_j = X_j (H_j^\nu)^{-1} X_j^*, \quad K_j = X_j (H_j^m)^{-1} X_j^* \preceq U_j. \quad (\text{RM.35})$$

The expression for K_j agrees exactly with (RM.1): the invertible triangular monic-polynomial coefficient matrix transports H_j^m to O_j and X_j to B_j , so the middle matrices cancel in this composite. All factors $(k/2 + iu)^n$ remain in these Grams. The measure inequality and its constants are supplied by the complete arithmetic upper-comparison proof retained with this supplement; no bound on the actual omitted high-degree moments is assumed.

52.8 Attribution and exact source scope

The general trace-constrained determinant bound is classical. The author-hosted copy of Grone, Johnson, Marques de Sá and Wolkowicz, *Improving Hadamard's inequality*, Linear and Multilinear Algebra 16 (1984), 305–322, identifies the dimension/trace/trace-square optimization; see <https://www.math.uwaterloo.ca/~hwolkowi/henry/reports/PAPER17.pdf> and <https://doi.org/10.1080/03081088408817634>. The complete argument (RM.11)–(RM.19) above is independently supplied here. The added comparison consists of the exact maps (RM.20)–(RM.25).

The unchanged incoming archive has SHA256

ec9324d794e839fac855eccea6a0c16e5e
736022441ac993a464df1f96b01d39

Its full note, markdown, handoff, checker, patch, source metadata and all 37 original file members remain preserved. The note's actual high-degree moment requirements remain stated; its unproved asymptotic example is recorded as motivation in the source, without supplying any missing arithmetic estimate. No Lean, actual zeta-packet quadrature, uniform trace-growth bound, or RH conclusion is asserted by this finite supplement.

53 The first canonical window, its full phase product, and both arithmetic block budgets

This chapter joins the original first-degree source calculation (CV.11)–(CV.12) and (TVB.37)–(TVB.38) to the consecutive-window proof of PR 25 at the exact head 66fac5e7885a40cd4873

e74b754907320f05b067. The complete original and proposed corrected sources accompany this chapter. Their correction retains all zero generalized eigenvalues. The source radius, original norm envelope and finite propagation are proved below with their original domains. The full first-window phase product has a single terminal contraction and two copies of each interior contraction; its first step contains no preceding quotient inverse.

53.1 Original source and the actual first-degree step

Retain $g = 2\xi$, the complete fixed packet h , integer tensor degree $k \geq 1$, and the original density

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad S = c + iu, \quad c = k/2. \quad (\text{CJ.1})$$

Its source inner product is the integral against $m_{h,k}(u)du$; the original monic orthogonal polynomials p_n have squared norms $\omega_n > 0$, including $\omega_0 = \mu_h^k$. Use the original dagger-stable monic cyclic polynomial χ of degree $q \geq 1$, the full quotient $E = \mathbb{C}[S]/(\chi)$ and the original remainder basis. Its columns, kernels and metrics are

$$\begin{aligned} b_n &= [p_n]_\chi, \quad B_N = [b_0, \dots, b_N], \quad O_N = \text{diag}(\omega_0, \dots, \omega_N), \\ K_N &= B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N \quad (N \geq q-1). \end{aligned} \quad (\text{CJ.2})$$

Surjectivity of B_N and positivity of O_N prove $K_N > 0$. The section $R_N = O_N^{-1} B_N^* G_N$ satisfies $B_N R_N = I$ and $R_N^* O_N R_N = G_N$; its orthogonality to $\ker B_N$ proves that it is the unique minimum lift. Source inclusion proves $G_{N+1} \preceq G_N$ and $0 < V_{N+1} \leq V_N$.

Write the original expansion, retaining every coefficient,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle}{\omega_j}, \quad \nu_0 = \|\chi\|^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j. \quad (\text{CJ.3})$$

The first q remainder columns form an invertible triangular matrix, so $b_a^* G_{q-1} b_b = \delta_{ab} \omega_b$ for $a, b < q$. Reduction of (CJ.3) gives $b_q = -\sum_{j < q} \gamma_j b_j$ and hence $b_q^* G_{q-1} b_q = \nu_0 - \omega_q$. The exact rank-one determinant formula therefore gives

$$\delta_q := \frac{V_q}{V_{q-1}} = \left(1 + \frac{b_q^* G_{q-1} b_q}{\omega_q} \right)^{-1} = \frac{\omega_q}{\nu_0}. \quad (\text{CJ.4})$$

The rank-one determinant identity follows by taking a basis whose first vector is the update vector u when it is nonzero. In that basis uv^* has entries only in its first row; $I + uv^*$ is therefore upper triangular with diagonal entries $1 + v^*u, 1, \dots, 1$. Its determinant is $1 + v^*u$. The zero update gives the same identity directly.

Let ϕ_N denote the retained real phase of the original theta deformation, with the sign convention of (TVB.26): $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. Its reality follows in the literal chart $S = c + iu$ as follows. Write $Q_n(u) = i^{-n} p_n(c + iu)$ and $\hat{\chi}(u) = i^{-q} \chi(c + iu)$. Dagger stability gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$, so $\hat{\chi}$ is real monic. The source measure is real and positive; the monic orthogonality equations therefore give real Q_n . If E is the exact coefficient map $P(S) \mapsto P(c + iu)$, then $E b_n = i^n [Q_n]_{\hat{\chi}}$, while the transported kernel and metric are $E K_N E^*$ and $E^{-*} G_N E^{-1}$. The latter has real entries because the phases in each kernel summand occur together with their conjugates. Thus the original cross pairing is i times a real number, with its original factor ω_N retained. This proves the stated reality and specifies the exact chart map rather than changing the metric. At the first degree that identity reads $\gamma_{q-1} = -i\phi_{q-1}$. The original two-column Hermitian control formula (CV.4), before specialization, is

$$\epsilon_N^2 = \frac{a_N d_N - |z_N|^2}{\omega_N^2}, \quad a_N = b_N^* G_N b_N, \quad z_N = b_N^* G_N b_{N+1}, \quad d_N = b_{N+1}^* G_N b_{N+1}. \quad (\text{CJ.5})$$

Here z_N is purely imaginary in the stated dagger-stable source. The original boundary-control identity on (E, G_N) is

$$H_N = G_N^{-1}(A^*G_N + G_NA - kG_N) = \frac{b_{N+1}b_N^*G_N + b_Nb_{N+1}^*G_N}{\omega_N}, \quad A = M_S.$$

For completeness this identity follows directly from the original monic recurrence. Self-adjointness of multiplication by u in the integral pairing and orthogonality give $uQ_n = Q_{n+1} + \beta_n Q_n + \alpha_n Q_{n-1}$, where $\beta_n \in \mathbb{R}$ and $\alpha_n = \omega_n/\omega_{n-1}$ for $n \geq 1$; the lower term is zero at $n = 0$. Indeed all coefficients below degree $n - 1$ vanish by pairing with uQ_j of degree at most $n - 1$, and pairing with Q_{n-1} gives the displayed norm ratio. In the original S coordinates this is $Sp_n = p_{n+1} + (c + i\beta_n)p_n - \alpha_n p_{n-1}$. Reduce it modulo χ and insert it into $AK_N + K_NA^* - kK_N$. The diagonal coefficients are $c + i\beta_n + c - i\beta_n - k = 0$. For each internal adjacent pair the coefficient is $\omega_n^{-1} - \alpha_{n+1}\omega_{n+1}^{-1} = 0$. The remaining two terms are $(b_{N+1}b_N^* + b_Nb_{N+1}^*)/\omega_N$. Multiplication on the right by the original G_N proves the written boundary-control formula with its signs. It has rank at most two, is self-adjoint in G_N , and has trace $(z_N + \bar{z}_N)/\omega_N = 0$. Squaring its two summands and taking the trace gives $[z_N^2 + \bar{z}_N^2 + 2a_N d_N]/\omega_N^2 = 2(a_N d_N - |z_N|^2)/\omega_N^2$. For $q \geq 2$ its full spectrum is $\epsilon_N, -\epsilon_N$ and $q - 2$ zeros, with $\epsilon_N \geq 0$; these first two entries may themselves be zero. For $q = 1$ trace zero makes the entire control zero. This proves (CJ.5), including the cross-pairing. At $N = q - 1$ the three entries are respectively $\omega_{q-1}, -\gamma_{q-1}\omega_{q-1}, \nu_0 - \omega_q$. Substitution proves

$$\epsilon_{q-1}^2 + \phi_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}. \quad (\text{CJ.6})$$

For $q = 1$, the lower projection in (CV.12) is the zero source and $\epsilon_{q-1} = 0$; the retained phase identity and (CJ.6) still hold. No G_{q-2} occurs in this calculation.

Define $\Lambda_N = \omega_N/V_N$. Combining (CJ.4)–(CJ.6) gives the exact first step

$$(\epsilon_{q-1}^2 + \phi_{q-1}^2)\Lambda_{q-1} = \Lambda_q(1 - \delta_q). \quad (\text{CJ.7})$$

Indeed its left side is $(\nu_0 - \omega_q)/V_{q-1}$, and $\nu_0/V_{q-1} = \omega_q/V_q$ by (CJ.4). This proves the identity with its single contraction and full original phase.

53.2 The complete first-window product and both finite budgets

At each regular degree $N \geq q$, the proved original radius identity (EP.11), equivalently (TVB.27), is

$$(\epsilon_N^2 + \phi_N^2)\Lambda_N = \Lambda_{N+1}(1 - \delta_N)(1 - \delta_{N+1}), \quad \delta_N = V_N/V_{N-1}. \quad (\text{CJ.8})$$

Every δ_N lies in $(0, 1]$. Multiply (CJ.7) by the consecutive identities (CJ.8) at $N = q, \dots, j - 1$. All Λ_N are positive, so canceling their identical intermediate factors is valid. It yields, for every $j \geq q$,

$$\prod_{N=q-1}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_{q-1}}{\omega_{q-1} V_j} (1 - \delta_j) \prod_{N=q}^{j-1} (1 - \delta_N)^2. \quad (\text{CJ.9})$$

For $j = q$ the interior product is empty and equal to one, so this is precisely the first step. For $j > q$, each $\delta_N, q \leq N < j$, occurs once from the step ending at N and once from the step beginning there. The terminal δ_j occurs once. This proves every exponent in (CJ.9). For a regular initial degree $i \geq q$, the identical multiplication instead gives

$$\prod_{N=i}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_i}{\omega_i V_j} (1 - \delta_i)(1 - \delta_j) \prod_{N=i+1}^{j-1} (1 - \delta_N)^2, \quad j > i. \quad (\text{CJ.10})$$

These two expressions explicitly relate the first and regular windows to the same source, phase and quotient volumes.

For the original off-line quartet let its common multiplicity be $m \geq 1$, its real displacement be $\delta > 0$ and its positive imaginary coordinate be $\gamma > 0$. The complete tensor calculation gives, with no omitted repeated root,

$$q = [1 + k(m-1)](k+1)^2, \quad L = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2. \quad (\text{CJ.11})$$

The original Hermitian compression proof gives $0 < L \leq \epsilon_N$ at every admitted degree: on the full sum of generalized eigenspaces with $\Re \lambda > k/2$, its trace is L . An orthogonal projection onto that subspace in the original metric has trace pairing at most ϵ_N with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N$ and zeros. This is the same source metric as (CJ.2). Thus each contraction occurring in (CJ.9)–(CJ.10) is strictly between zero and one. A value one for any δ_N would make a positive radius product zero.

For either original length- q block $(i, j) = (q-1, 2q-1)$ or $(q, 2q)$, taking logarithms of its exact product now gives

$$\log(V_i/V_j) = \sum_{N=i}^{j-1} \log(\epsilon_N^2 + \phi_N^2) - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j}, \quad (\text{CJ.12})$$

where the entire contraction penalty is

$$\begin{aligned} \mathcal{P}_{q-1, 2q-1} &= -\log(1 - \delta_{2q-1}) - 2 \sum_{N=q}^{2q-2} \log(1 - \delta_N), \\ \mathcal{P}_{q, 2q} &= -\log(1 - \delta_q) - \log(1 - \delta_{2q}) - 2 \sum_{N=q+1}^{2q-1} \log(1 - \delta_N). \end{aligned} \quad (\text{CJ.13})$$

Both sums retain their literal first and last degrees and are nonnegative. Since $\epsilon_N \geq L$, the exact finite bound is

$$\log(V_i/V_j) \geq 2q \log L - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (\text{CJ.14})$$

Every phase term remains; none is assigned the value zero.

53.3 Actual comparable norm windows with complete constants

The original arithmetic lower-envelope proof gives, for $k \geq 3$,

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h. \quad (\text{CJ.15})$$

Its complete analytic proof is retained in the arithmetic source appendix and AU.1–AU.12. Fix $0 < b < \pi/2$, and keep $M_h(\pm b) = \int e^{\pm bt} \omega_h(t) dt$. Set the explicit positive constants

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ M &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{M}/b. \end{aligned} \quad (\text{CJ.16})$$

For every integer $n \geq k \geq 3$ the literal monic norms obey

$$(\ell_h n)^{2n} \leq \omega_n \leq (U_h n)^{2n}. \quad (\text{CJ.17})$$

Here is the full envelope calculation. The exact minimum squared norm of a monic polynomial on $[-n, n]$ is

$$\frac{2n^{2n+1}}{2n+1} \left[\frac{2^n (n!)^2}{(2n)!} \right]^2 \geq \frac{2}{3} n^{2n} 4^{-n}.$$

The last inequality uses $(2n)!/(n!)^2 \leq 4^n$ and $2n/(2n+1) \geq 2/3$. The original chart's factor i^n has modulus one; its monic polynomial space is exactly this extremal space through the inverse substitution. On $[-n, n]$, (CJ.15) is at least $c_h \vartheta_h^{k-3} e^{-\alpha(k-3+n)} (k-2+n)^{-B}$. Use, in their stated directions,

$$\vartheta_h^{k-3} \geq t_h^n, \quad e^{-\alpha(k-3+n)} \geq e^{-2\alpha n}, \quad (k-2+n)^{-B} \geq (2n)^{-B} \geq 2^{-nB}, \quad 2c_h/3 \geq c_0^n.$$

The elementary $2n \leq 2^n$ holds at $n = 1$ and follows by induction from $2(n+1) \leq 4n$ for $n \geq 1$. Multiplication proves the lower bound (CJ.17). For the upper bound use the monic trial polynomial $(S - c)^n$, the nonnegative exponential series, and convolution to obtain $\omega_n \leq (2n)! b^{-2n} (M_h(b)^k + M_h(-b)^k)$. This is at most $(2n)^{2n} b^{-2n} 2M^n$, which is at most $(4\sqrt{M}n/b)^{2n}$ since $2 \leq 4^n$. This proves both inequalities with the original mass retained.

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq 2n$, divide the two instances of (CJ.17) and take the positive $2r$ -th root:

$$\left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)} \leq n U_h^{1+n/r} \ell_h^{-n/r} (1+r/n)^{1+n/r} \leq C_h^{\text{win}} n, \quad (CJ.18)$$

$$C_h^{\text{win}} = 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2.$$

Indeed $1 + n/r \leq 3$, $n/r \leq 2$, and $1 + r/n \leq 3$. For each block above, $r = q$ and $n = q$ or $q - 1$. Equation (CJ.11) gives $q - 1 \geq k$ for $k \geq 3$, and $q/2 \leq q \leq 2(q - 1)$ for these q . Both norm factors are therefore at most $C_h^{\text{win}} q$. Combining (CJ.11), (CJ.14) and (CJ.18), with $D_h = \delta/(2C_h^{\text{win}}) > 0$, proves for each block

$$\log(V_i/V_j) \geq 2q \log(D_h k) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (CJ.19)$$

Adding the two bounds and dividing by $q \log k$ proves the four-volume lower limit at least four. This includes the complete finite penalties before taking that consequence.

53.4 The full relation cost and its retained zero directions

For any admitted $i < j$, set $F = [b_{i+1}, \dots, b_j]$ and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. The exact update is $K_j - K_i = F\Omega^{-1}F^*$, so

$$V_i/V_j = \det(I + F\Omega^{-1}F^*G_i) = \det(I + \Omega^{-1}F^*G_iF). \quad (CJ.20)$$

The rectangular determinant equality follows by eliminating either off-diagonal block of the original block matrix

$$\begin{pmatrix} I & F \\ -\Omega^{-1}F^*G_i & I \end{pmatrix}.$$

On the new-column source $\mathbb{C}^{j-i}$ with metric Ω , the second cost is positive self-adjoint. Its eigenvalues are nonnegative, including their zero multiplicities. Write $L_{i,j} : \mathbb{C}^{i+1} \rightarrow \mathbb{C}^{j+1}$ for padding in the same monic orthogonal basis, and $T_{\text{new}} : \mathbb{C}^{j-i} \rightarrow \mathbb{C}^{j+1}$ for the coordinate inclusion into positions $i+1, \dots, j$. The original relation map from $(\mathbb{C}^{j-i}, \Omega)$ to $(\mathcal{P}_j, O_j)$ is $\mathfrak{b} = T_{\text{new}} - L_{i,j}R_iF$. The new columns and the lower source are orthogonal, and $R_i^*O_iR_i = G_i$, so

$$B_j\mathfrak{b} = 0, \quad \mathfrak{b}^*O_j\mathfrak{b} = \Omega + F^*G_iF. \quad (CJ.21)$$

Set $\mathcal{D}_N = \ker B_N = \chi\mathcal{P}_{N-q}$, with the zero space when $N = q - 1$. The new-coordinate component of $\mathfrak{b}$ is the identity; it is therefore injective. Its image lies in $\mathcal{D}_j \cap (LD_i)^\perp$, whose dimension is $j - i$, by

the dimensions of the two nested relation spaces. Thus it is an isomorphism onto that entire relation layer. Reduction sends its labelled vectors to their receiving supported zero; the lifted reduction fixes external absence τ .

At zero tilt the full quartet gives an even original density. Its u -polynomial $\psi = i^{-q}\chi(c + iu)$ satisfies $\psi(-u) = (-1)^q\psi(u)$ because its complete root multiset is stable under negation. Uniqueness of the monic orthogonal polynomial gives $Q_n(-u) = (-1)^nQ_n(u)$: substituting $-u$ preserves all original integral pairings and the adjusted leading coefficient. Monic division modulo ψ preserves parity. When k is even, q in (CJ.11) is odd. In the first block the q new polynomials $Q_q, \dots, Q_{2q-1}$ contain $(q+1)/2$ odd polynomials, while the odd quotient subspace has dimension $(q-1)/2$. Their reduction columns are dependent. The original $S = c + iu$ basis map and the nonzero column phases i^n preserve rank. Thus at least one of the q cost eigenvalues in that block is zero. It remains in the full determinant as a factor one; (CJ.19)–(CJ.21) remain valid with every original dimension and nilpotent direction retained.

For either length- q block let $\lambda_a \geq 0$ be all its cost eigenvalues. Equations (CJ.19)–(CJ.20) imply

$$\prod_{a=1}^q (1 + \lambda_a) \geq (D_h k)^{2q}, \quad \sum_{a=1}^q (1 + \lambda_a)^s \geq q(D_h k)^{2s} \quad (s > 0). \quad (\text{CJ.22})$$

The second assertion is the arithmetic-geometric mean inequality for the q positive numbers $(1 + \lambda_a)^s$; it also obeys the trivial lower bound q . Here is its exact restriction map. Give the same coefficient space E its two original metrics and let $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$ be the identity coefficient map. Its Hilbert adjoint is $T_{i,j} : (E, G_j) \rightarrow (E, G_i)$ with matrix $G_i^{-1}G_j$, because $x^*G_jy = x^*G_i(G_i^{-1}G_j)y$ for every x, y . Thus $Q = T_{i,j}I_{i,j}$ acts on (E, G_i) and satisfies

$$Q^{-1} = G_j^{-1}G_i = K_jG_i = I + F\Omega^{-1}F^*G_i.$$

It is positive self-adjoint in that original metric; its quadratic form is $x^*G_iK_jG_ix > 0$ for $x \neq 0$. Consequently its eigenvalues are exactly the $1 + \lambda_a$ in (CJ.22), with all zero cost directions giving eigenvalue one. Hence (CJ.22) is also an inverse-restriction trace bound in its specified original metric. No individual positive cost lower bound or strict-positive rank claim is inferred.

53.5 Provenance and verification scope

The complete cumulative sources CV and TVB prove the radius and phase map used here, and the complete arithmetic source and AU prove the actual density envelope (CJ.15). PR 25 supplies the comparable-window constants and first-degree propagation. The full first-window phase product (CJ.9), its finite penalties (CJ.13)–(CJ.19), and their connection to the retained relation costs are written out here. The owner review at the cited head requires a nonnegative-eigenvalue correction and workflow-trigger corrections; this chapter preserves that status and the complete proposed parity proof. It performs no local Lean execution and makes no new formal-verification claim. Its finite algebra and analytic envelope calculations belong to this written proof.

54 Exact arithmetic restriction certificates and projection products

This article joins the original arithmetic lower-envelope comparison to the canonical restriction maps. It proves a positive arithmetic return gap at every finite degree, carries entry errors through the actual quotient, and identifies all loss traces with observations of two original source projections. The balanced product gives a complementary lower estimate on every inverse spectral power in both endpoint windows. The complete arithmetic envelope and balanced monic-norm proofs are retained as explicit source dependencies; their constants appear below. Every finite matrix and restriction consequence used here is proved.

54.1 The original source, quotient, and canonical lifts

Retain $g = 2\xi$, the monic packet h of selected zeros with their complete orders, $v_h = g/h$, and

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad c = k/2. \quad (\text{ERJ.1})$$

The original pairing, conjugate-linear in its first entry, is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) m_k(u) du$. Its constant norm is μ_h^k . The arithmetic construction supplies positivity almost everywhere and all polynomial moments. A nonzero polynomial has finitely many zeros on this line, so its norm is positive. Let $S = s_1 + \dots + s_k$, let the monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ be χ , and put $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, $A = M_S$. The ordered quotient coordinates are $1, S, \dots, S^{q-1}$. For $N \geq q - 1$ write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$, with negative degree spaces zero. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{ERJ.2})$$

In monomial coordinates let H_N be the source Gram. In the original monic orthogonal basis $p_0, \dots, p_N$, put $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and $B_N = [b_0, \dots, b_N]$, $b_n = [p_n]_\chi$. All p_n stay monic; $\omega_0 = \mu_h^k$. The monic triangular coefficient change has determinant one and gives the equal constructions

$$\begin{aligned} K_N &= J_N H_N^{-1} J_N^* = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \\ R_N &= O_N^{-1} B_N^* G_N : (E, G_N) \longrightarrow (\mathcal{P}_N, O_N), \quad K_N = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n}. \end{aligned} \quad (\text{ERJ.3})$$

Here and below a formula with O_N uses monic source coordinates; its monomial version conjugates by that explicit coefficient change. Surjectivity of B_N proves $K_N > 0$. Multiplication gives $B_N R_N = I$ and $R_N^* O_N R_N = G_N$. For $z \in \ker B_N$, $z^* O_N R_N = z^* B_N^* G_N = 0$. Thus every lift $R_N x + z$ has squared norm $x^* G_N x + z^* O_N z$. This proves the unique minimum property, $\text{im } R_N = \mathcal{D}_N^\perp$, and, for $i \leq j$, $G_j \preceq G_i$ and $K_i \preceq K_j$.

The actual analytic representative is $\mathcal{V}P = P(D_1 + \dots + D_k) F_h^{\otimes k}$, where $D = -x\partial_x$, $\mathcal{M}F_h = v_h$, and $h(D)F_h = \Theta\phi_0$. Its norm is the pairing in (ERJ.1). Its full-jet map is $J^{(k)}\mathcal{V} = \eta J_N$, where $\eta[P] = [v_h]_h^{\otimes k} P(A_k)1$. Each local factor of $[v_h]_h$ has nonzero constant coefficient, so its inverse is the finite geometric expansion in its nilpotent part. The kernel of $P \mapsto P(A_k)1$ is exactly (χ) by its definition. These two facts prove that η is injective, retaining every local derivative and arithmetic unit.

54.2 Return, action, and the entire original relation correction

Let $L_{i,j} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$ pad monic coefficients, and let $P_{i,j}$ be the orthogonal projection retaining their first $i + 1$ entries. Then $B_j L_{i,j} = B_i$ and $L_{i,j}^* O_j L_{i,j} = O_i$. The original identity transport $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$, $x \mapsto x$, is a contraction and intertwines A . Its metric adjoint is

$$T_{i,j} = K_i G_j : (E, G_j) \longrightarrow (E, G_i), \quad L_{i,j} R_i T_{i,j} = P_{i,j} R_j, \quad T_{i,j} T_{j,l} = T_{i,l}. \quad (\text{ERJ.4})$$

Indeed $x^* G_j y = x^* G_i T_{i,j} y$. The first $i+1$ coordinates of R_j are $O_i^{-1} B_i^* G_j$, exactly those of $R_i T_{i,j}$. The composition identity is $K_i G_j K_j G_l = K_i G_l$. For endomorphism types put $U_{i,j} = T_{i,j} I_{i,j}$ on (E, G_i) and $\hat{U}_{i,j} = I_{i,j} T_{i,j}$ on (E, G_j) . Their coordinate matrices are $K_i G_j$, and the original invertible identity transport satisfies $I_{i,j} U_{i,j} = \hat{U}_{i,j} I_{i,j}$. An endomorphism power or eigenvalue of T below means these specified similar endomorphisms, with all metrics retained. As an underlying endomorphism of E , $T = T_{i,j}$ is self-adjoint and positive in both displayed metrics: $G_i T = G_j$ and $G_j T = G_j K_i G_j$. It obeys $0 < T \preceq I$ in each metric. Its eigenvalues g_a lie in $(0, 1]$. The squared singular values of $I_{i,j}$,

of its adjoint in the reverse Hilbert types, and of the restricted canonical lift in (ERJ.4) are these same g_a . Consequently

$$H_{i,j}^{\text{loss}} = I - T_{i,j}, \quad \mathcal{L}_{i,j} := \log(V_i/V_j) = -\log \det T_{i,j} = -\sum_{a=1}^q \log g_a. \quad (\text{ERJ.5})$$

Write $W_N = A^*G_N + G_NA - kG_N$, $H_N = K_NW_N$, and $V_N = AK_N + K_NA^* - kK_N$. Direct expansion proves

$$AT - TA = H_iT - TH_j = V_iG_j - K_iW_j. \quad (\text{ERJ.6})$$

The two $K_iA^*G_j$ terms and the two kK_iG_j terms cancel. If $\epsilon_N = \|H_N\|_{G_N}$, contraction of the adjoint gives $\|AT - TA\|_{(E,G_j) \rightarrow (E,G_i)} \leq \epsilon_i + \epsilon_j$. Thus the original action accompanies the return through its exact commutator, including on any chosen return eigenspace. The representative change is the entire map

$$\begin{aligned} \Delta_{i,j} &= L_{i,j}R_i - R_j = L_{i,j}R_iH_{i,j}^{\text{loss}} - (I - P_{i,j})R_j, \\ B_j\Delta_{i,j} &= 0, \quad \text{im } \Delta_{i,j} \subseteq \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \Delta_{i,j}^*O_j\Delta_{i,j} = G_i - G_j. \end{aligned} \quad (\text{ERJ.7})$$

Both terms in the difference have remainder $H_{i,j}^{\text{loss}}$. Orthogonality to $\mathcal{D}_i$ follows for both complete lifts. Since R_j is orthogonal to $\mathcal{D}_j$, the equality $L_{i,j}R_i = R_j + \Delta_{i,j}$ is an orthogonal sum and proves the Gram.

For the explicit theta primitive keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, with F_h of degree one and ϕ_0 of degree zero. The original compatible chain action D commutes with this differential. Successive monic division gives $\chi(\sum s_a) = \sum_a h(s_a)Q_a(\mathbf{s})$ in the tensor ideal. For any relation $\chi\kappa$ the cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa(\sum_b D_b) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \quad (\text{ERJ.8})$$

has differential $\mathcal{V}(\chi\kappa)$: applying the tensor differential gives a second factor $(-1)^{a-1}$ and $h(D_a)F_h = \Theta\phi_0$; all other terms have zero differential. Apply this fixed linear division primitive to every column of (ERJ.7). On an active split fibre the maps are $x^\bullet \mapsto x^\bullet$ and $x^\bullet \mapsto (Tx)^\bullet$, with $\tau \mapsto \tau$. They are $G(\mathbb{C})$ -linear by the original fibre operations and linearity of the displayed maps. The relation image in the internal quotient is the supported zero $(\ell_j, 0)$; the external τ is retained. The reverse maps compose by (ERJ.4), while their action comparison is (ERJ.6). No relation, support label, or primitive is deleted.

54.3 The exact source projection and principal-angle morphism

Use the common original Hilbert space $\mathcal{P}_j$ and its isometries $\ell_i = L_{i,j}R_i : (E, G_i) \rightarrow \mathcal{P}_j$ and $\ell_j = R_j : (E, G_j) \rightarrow \mathcal{P}_j$. Let Π_i, Π_j be the orthogonal endomorphism projections onto these two images. Because $R_j \perp \mathcal{D}_j \supseteq \mathcal{D}_i$, its lower-degree projection lies in $\ell_i E$. Also $\ell_i x - \ell_j x \in \mathcal{D}_j$. Hence

$$\Pi_i \ell_j = P_{i,j} \ell_j = \ell_i T, \quad \Pi_j \ell_i = \ell_j, \quad (\Pi_i \Pi_j)|_{\ell_i E} = \ell_i T \ell_i^{-1}, \quad (\Pi_j \Pi_i)|_{\ell_j E} = \ell_j T \ell_j^{-1}. \quad (\text{ERJ.9})$$

These are similarities on the indicated full images. The range of $\Pi_i \Pi_j$ is $\ell_i E$ because its restriction there is invertible. Its kernel is $\ker \Pi_j$: the reverse inclusion follows from invertibility of $\Pi_i : \ell_j E \rightarrow \ell_i E$. This gives its direct sum into that image and kernel, with exactly q nonzero eigenvalues g_a and $j+1-q$ zero eigenvalues. Reversing the pair proves the corresponding assertion for $\Pi_j \Pi_i$.

Choose a G_i -orthonormal eigenbasis x_a of T . Put $u_a = \ell_i x_a$ and $v_a = \ell_j x_a / \sqrt{g_a}$. Since $G_j = G_i T$, both families are orthonormal and $\langle u_a, v_b \rangle = \delta_{ab} \sqrt{g_a}$ by (ERJ.9). If $g_a = 1$ then $u_a = v_a$. Otherwise $w_a = (v_a - \sqrt{g_a} u_a) / \sqrt{1 - g_a}$ makes mutually orthonormal pairs (u_a, w_a) . On each such plane,

$$\Pi_i = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Pi_j = \begin{pmatrix} g_a & \sqrt{g_a(1-g_a)} \\ \sqrt{g_a(1-g_a)} & 1-g_a \end{pmatrix}. \quad (\text{ERJ.10})$$

Their difference has trace zero and square $(1 - g_a)I_2$. Both vanish on the common orthogonal complement, and the shared directions with $g_a = 1$ contribute zero. Therefore, for every real $s > 0$ and in particular each integer $m \geq 1$,

$$\mathrm{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2s} = 2 \sum_a (1 - g_a)^s, \quad \mathrm{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2m} = 2 \mathrm{Tr}_E ((H_{i,j}^{\mathrm{loss}})^m). \quad (\mathrm{ERJ}.11)$$

Functional powers here use the proved positive spectral decomposition. The restricted map $\mathcal{R} = \Pi_i|_{\ell_j E}$ is the bijection $\ell_i T_{i,j} \ell_j^{-1} : \ell_j E \rightarrow \ell_i E$. For every $s > 0$ the exact inverse singular-value-power observation is

$$\sum_a \sigma_a(\mathcal{R}^{-1})^{2s} = \sum_a \sigma_a(T_{i,j}^{-1})^{2s} = \sum_a \sigma_a(I_{i,j}^{-1})^{2s} = \mathrm{Tr} U_{i,j}^{-s} = \sum_a g_a^{-s}. \quad (\mathrm{ERJ}.11a)$$

This follows from $I_{i,j}^\dagger I_{i,j} = U_{i,j}$ and the equal singular values of a map and its adjoint; inversion reciprocates the singular values. These are Schatten observations of the original typed maps, including when $0 < 2s < 1$, where no norm triangle inequality is asserted. Only the bijection between the canonical images is inverted; the full projection products retain their ambient zeros. For a return eigenvector x of eigenvalue g the original lower portion satisfies

$$\frac{\|P_{i,j} R_j x\|_{\mathcal{O}_j}^2}{\|R_j x\|_{\mathcal{O}_j}^2} = g, \quad \frac{\|(I - P_{i,j}) R_j x\|_{\mathcal{O}_j}^2}{\|R_j x\|_{\mathcal{O}_j}^2} = 1 - g. \quad (\mathrm{ERJ}.12)$$

Indeed its first squared norm is $x^* G_j K_i G_j x = g x^* G_j x$; orthogonal decomposition proves the second. Its old quotient coordinate is $g x$. Applying the invertible T^{-1} recovers x ; the complete representative difference is exactly (ERJ.7).

54.4 The arithmetic remainder bound at every finite degree

For the actual quartet $1/2 \pm \delta \pm i\gamma$ of common order $m \geq 1$, $\delta, \gamma > 0$, retain

$$\ell = 1 + k(m - 1), \quad q = \ell(k + 1)^2, \quad \chi(S) = \prod_{a,b=0}^k (S - c - (2a - k)\delta - i(2b - k)\gamma)^\ell. \quad (\mathrm{ERJ}.13)$$

The top nonzero power of the sum of k local length- m nilpotents is $k(m - 1)$, with coefficient $(k(m - 1))! / ((m - 1)!)^k$ on the product of their top powers; its next power vanishes by total degree. Each displayed grid sum occurs and the distinct pairs have distinct real and imaginary parts. This proves the complete order and degree in (ERJ.13) by taking the least common multiple of local annihilators.

The complete arithmetic-envelope source proves, for $k \geq 3$,

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k - 2 + |u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h, \quad \vartheta_h = \int_{-1}^1 w_h > 0. \quad (\mathrm{ERJ}.14)$$

Here $c_h > 0$ is its original local-mass convolution constant. Fix $0 < b < \pi/2$, $M_h(\pm b) = \int e^{\pm b t} w_h(t) dt$ and $M_k = M_h(b)^k + M_h(-b)^k$. Positivity and convolution give $\int e^{b|u|} m_k \leq M_k$, and hence $\int |u|^{2d} m_k \leq M_k (2d)! b^{-2d}$. Set $R_0 = \sqrt{\delta^2 + \gamma^2}$, $D = \max(1, 2/b, 2R_0)$, $T = Dq$, $r_0 = kR_0/T \leq 1/2$, and retain

$$\begin{aligned} a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(k-3+T)} (k - 2 + T)^{-B}, \\ U_k(T) &= \mu_h^k + M_k \frac{(2q - 2)!}{(bT)^{2q-2}}, \quad A_N = \sqrt{(2N + 1)/2} \frac{4^{N+1} - 1}{3}, \\ B_q(r_0) &= \left(\frac{1 + r_0}{1 - r_0} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r_0)^2}{T a_k(T)}. \end{aligned} \quad (\mathrm{ERJ}.15)$$

The coordinate map $\Psi_T P(x) = P(c + iTx)$ has inverse $f \mapsto f((S - c)/(iT))$ and target measure $Tm_k(Tx)dx$, whose mass remains μ_h^k . Put $\chi_T = (iT)^{-q}\chi(c + iTx)$. Then $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$, so remainder commutes with this map. Every original raw derivative of order d is multiplied by $(iT)^d$. The quotient coefficient determinant is $(iT)^{q(q-1)/2}$, and its transported form determinant is $T^{-q(q-1)}V_N$; this identical factor cancels in a volume ratio.

List all q roots z_a of χ_T with repetition. Their moduli are at most r_0 . Write $E(t) = \prod_a (1 - z_a t)$ and $E(t)^{-1} = \sum_{l \geq 0} h_l t^l$. For every integer $s \geq 0$, multiplication of these formal inverse series through order s proves that $Q_s(x) = \sum_{l=0}^s h_l x^{s-l}$ is the quotient of x^{q+s} by χ_T , and that $x^{q+s} - \chi_T Q_s$ has degree below q . Convolution gives $\|\chi_T\|_1 \leq (1 + r_0)^q$ and $|h_l| \leq \binom{q+l-1}{l} r_0^l$, where $\|\cdot\|_1$ is the coefficient absolute sum. The sole coefficient of $\chi_T Q_s$ at degree at least q is its leading coefficient one. Its remaining coefficients are the negative remainder. Therefore

$$\|\text{rem}_{\chi_T} x^{q+s}\|_1 \leq (1 + r_0)^q \sum_{l=0}^s \binom{q+l-1}{l} r_0^l - 1 \leq B_q(r_0). \quad (\text{ERJ.16})$$

The last bound sums the q geometric series. At degrees below q the remainder has norm one. Linearity proves $\|\text{rem}_{\chi_T} f\|_1 \leq B_q(r_0)\|f\|_1$ for every polynomial f , without a degree restriction on s .

Rodrigues' formula for Legendre polynomials gives orthogonality, squared norm $2/(2n+1)$, and coefficient norm at most $\binom{2n}{n} \leq 4^n$. For the norm, integrate by parts n times; the endpoint terms vanish because $(x^2 - 1)^n$ has order n there, and the remaining beta integral gives $2/(2n+1)$. For the coefficient bound use $P_n(x) = 2^{-n} \sum_{a=0}^n \binom{n}{a}^2 (x-1)^{n-a} (x+1)^a$, then sum the coefficient absolute values. Cauchy-Schwarz on each orthogonal coefficient and summation of 4^n yield $\|f\|_1 \leq A_N \|f\|_{L^2[-1,1]}$ for $\deg f \leq N$. For $\deg v \leq q-1$, each monomial at u/T has modulus at most $\max(1, |u/T|^{q-1})$, so $\int |v(u/T)|^2 m_k \leq U_k(T) \|v\|_1^2$. Finally (ERJ.14) restricted to $[-T, T]$ gives $\|P\|^2 \geq T a_k(T) \|\Psi_T P\|_{L^2[-1,1]}^2$. Composing these actual maps proves, for every $N \geq q-1$,

$$\|\text{rem}_\chi P\|^2 \leq C_N \|P\|^2, \quad G_{q-1} \preceq C_N G_N, \quad C_j^{-1} K_j \preceq K_i \preceq K_j \quad (q-1 \leq i \leq j). \quad (\text{ERJ.17})$$

Apply the first inequality to the least lift $R_N x$ to obtain the second. Then use $G_j \preceq G_i \preceq G_{q-1}$ and inversion of positive forms. Taking $i = j$ proves $C_j \geq 1$. Thus the two original endpoint pairs $(q-1, 2q-1)$ and $(q, 2q)$ have the explicit positive gaps C_{2q-1}^{-1} and C_{2q}^{-1} .

54.5 Finite comparison matrices and the complete trace error

All following comparisons use the original quotient coordinates. For positive source brackets $H_N^- \preceq H_N \preceq H_N^+$, inversion and congruence give

$$L_N = J_N(H_N^+)^{-1} J_N^* \preceq K_N \preceq U_N = J_N(H_N^-)^{-1} J_N^*. \quad (\text{ERJ.18})$$

Their positivity follows from surjectivity of J_N . One actual upper comparator uses the density $\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}$ from (ERJ.14). Its even moment of degree d is

$$\int u^d \nu_k(u) du = \frac{2c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} d!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-(k-2)t}}{(\alpha+t)^{d+1}} dt; \quad \int u^d \nu_k(u) du = 0 \quad (d \text{ odd}). \quad (\text{ERJ.19})$$

Insert $(k-2+u)^{-B} = (B-1)!^{-1} \int_0^\infty t^{B-1} e^{-(k-2+u)t} dt$ on the positive half-line; Tonelli and $\int_0^\infty u^d e^{-yu} du = d! y^{-d-1}$ prove the formula. Thus $H_N^\nu \preceq H_N$ is an explicit strictly positive source. It supplies U_j using comparison moments through $2j$, at most $4q$ at the second endpoint; actual low-degree source data for i stop at $2i \leq 2q$. These cutoffs keep the actual and comparison integrals attached to their stated maps.

Given $0 < L_i \preceq K_i \preceq K_j \preceq U_j$, set $\tau = \text{Tr}(L_i^{-1}U_j)$. The Hermitian matrix $L_i^{-1/2}U_jL_i^{-1/2}$ has all q eigenvalues at least one. Each is at most $\tau - q + 1$ because the remaining $q - 1$ contribute at least $q - 1$. It follows that

$$U_j \preceq (\tau - q + 1)L_i, \quad g := \frac{1}{\tau - q + 1} > 0, \quad K_i \succeq gK_j. \quad (\text{ERJ.20})$$

This coefficient is optimal from the trace and the lower bound one: $\text{diag}(\tau - q + 1, 1, \dots, 1)$ attains it. An upper rational $\tau_+ \geq \tau$ gives the certified rational gap $(\tau_+ - q + 1)^{-1}$.

Now take rational Hermitian brackets for both K_i, K_j as in (ERJ.18), and define the exact rational quantities

$$H_* = I - L_iU_j^{-1}, \quad t_m^* = \text{Tr } H_*^m, \quad d = \text{Tr}(U_iL_j^{-1}) - \text{Tr}(L_iU_j^{-1}), \quad r = 1 - g. \quad (\text{ERJ.21})$$

Here g is certified for the reference pair (L_i, U_j) itself. For increasing generalized eigenvalues the exact minimum-maximum formula is

$$\lambda_a(F, G) = \min_{\dim W=a} \max_{0 \neq x \in W} \frac{x^*Fx}{x^*Gx}. \quad (\text{ERJ.22})$$

For $G = I$, the span of the first a eigenvectors attains λ_a ; every a -dimensional W intersects the span of eigenvectors $a, \dots, q$, whose dimension is $q - a + 1$, and hence its maximum is at least λ_a . The invertible substitution $y = G^{1/2}x$ proves the general case with subspace dimensions preserved. The pointwise Rayleigh quotients for (L_i, U_j) , (K_i, K_j) , and (U_i, L_j) are ordered. Their extrema are therefore ordered. If $\alpha_a = \lambda_a(L_i, U_j)$ and $\beta_a = \lambda_a(K_i, K_j)$, this proves

$$g \leq \alpha_a \leq \beta_a \leq 1, \quad 0 \leq \sum_a (\beta_a - \alpha_a) \leq d. \quad (\text{ERJ.23})$$

The sum identity uses similarity of FG^{-1} to its positive Hermitian representative. The kernels F, G are forms on the dual quotient coefficient space: $G^{-1}F$ there and FG^{-1} on the quotient are similar through the invertible map G . These β_a are the actual g_a of the return. For $r > 0$, integration of the derivative of x^m on $[0, r]$ gives the complete moment enclosure

$$a_m := \max(0, t_m^* - mr^{m-1}d) \leq s_m := \text{Tr}((H_{i,j}^{\text{loss}})^m) \leq t_m^* \leq qr^m. \quad (\text{ERJ.24})$$

Indeed apply $0 \leq u^m - v^m \leq mr^{m-1}(u - v)$ to $u = 1 - \alpha_a, v = 1 - \beta_a$ and sum (ERJ.23). If $r = 0$, all $\alpha_a = \beta_a = 1$ and both loss operators vanish; the exact certificate is zero.

Integrating the geometric series on $[0, x]$ proves $-\log(1 - x) = \sum_{m \geq 1} x^m/m$ for $0 \leq x < 1$. All terms are nonnegative. Its tail is bounded by $x^{p+1}/((p+1)(1-x))$. Thus for every integer $p \geq 1$,

$$\sum_{m=1}^p \frac{a_m}{m} \leq \mathcal{L}_{i,j} \leq \sum_{m=1}^p \frac{t_m^*}{m} + \frac{t_{p+1}^*}{(p+1)g}. \quad (\text{ERJ.25})$$

The width of the two stated endpoints is at most

$$\frac{d(1 - r^p)}{g} + \frac{qr^{p+1}}{(p+1)g}. \quad (\text{ERJ.26})$$

The first term sums $r^{m-1}d$ using (ERJ.24); the second uses $t_{p+1}^* \leq qr^{p+1}$. This accounts for the finite matrix error and series tail separately with their original dimension.

The sharper coefficient

$$c_p(r) = \sum_{l \geq 0} \frac{r^l}{p+1+l} = \frac{-\log g - \sum_{m=1}^p r^m/m}{r^{p+1}} \leq \frac{1}{(p+1)g} \quad (\text{ERJ.27})$$

can replace the last coefficient in (ERJ.25): the tail at $x \leq r$ is at most $c_p(r)x^{p+1}$. To certify its logarithm, write exactly $x = 2^e y$ for rational $x > 0$, integer e , and $1 \leq y < 2$. For $t = (y - 1)/(y + 1)$, integration of $2/(1 - t^2)$ gives

$$\log y = 2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} + E_M, \quad 0 \leq E_M \leq \frac{2t^{2M+1}}{(2M+1)(1-t^2)}. \quad (\text{ERJ.28})$$

At $t = 1/3$ the same formula encloses $\log 2$. Multiply that interval by e , reversing endpoints when $e < 0$, and add the $\log y$ interval. If $\log g \in [\ell_-, \ell_+]$ has width ε , replacing $-\log g$ by $-\ell_-$ in (ERJ.27) adds at most $\varepsilon t_{p+1}^*/r^{p+1} \leq q\varepsilon$ to the upper endpoint. The $r = 0$ case was handled before this division.

54.6 Constructing rational brackets for the actual relation

Let an actual Hermitian source matrix M have a rational Hermitian center M_0 and symmetric rational radii $\varepsilon_{ab} \geq 0$ with $|(M - M_0)_{ab}| \leq \varepsilon_{ab}$. For fixed positive rational w_a set $D_{aa} = \sum_b \varepsilon_{ab} w_b / w_a$. Squaring $\sqrt{t}x - y/\sqrt{t}$ proves $2xy \leq tx^2 + t^{-1}y^2$. Therefore

$$\begin{aligned} |z^*(M - M_0)z| &\leq \sum_{a,b} \varepsilon_{ab} |z_a| |z_b| \\ &\leq \frac{1}{2} \sum_{a,b} \varepsilon_{ab} \left(\frac{w_b}{w_a} |z_a|^2 + \frac{w_a}{w_b} |z_b|^2 \right) = z^* D z. \end{aligned} \quad (\text{ERJ.29})$$

Hence $M_0 - D \preceq M \preceq M_0 + D$. Exact Hermitian elimination tests positivity: a first pivot $d_0 > 0$ gives $d_0|x + d_0^{-1}b^*y|^2 + y^*(C - bb^*/d_0)y$. Induction proves positivity precisely when all successive pivots and their Schur complements are positive.

For the actual, possibly nonrational coefficients of χ , use the exact ordered coefficient matrix

$$\mathcal{A}_N = [1, S, \dots, S^{q-1}, \chi, S\chi, \dots, S^{N-q}\chi]. \quad (\text{ERJ.30})$$

It is triangular with diagonal one. At $N = q - 1$ its relation block is empty. Monic division gives $J_N \mathcal{A}_N = [I_q, 0] =: J_0$. Consequently, in the original remainder coordinates,

$$M_N = \mathcal{A}_N^* H_N \mathcal{A}_N, \quad K_N = J_0 M_N^{-1} J_0^*. \quad (\text{ERJ.31})$$

Intervals for the original source entries and original coefficients of χ enter the literal finite products for M_N ; interval addition, multiplication, and conjugation contain those products. The sum of the real and imaginary radii bounds a complex error. Use conjugate centers and symmetrically enlarged radii, then (ERJ.29) and exact positive pivots. The two inverse compressions of the resulting source brackets yield rational L_N, U_N by (ERJ.18). No center polynomial replaces the actual quotient map in (ERJ.31).

For a fixed actual pair and fixed w_a , refining source and coefficient intervals makes the radii and D tend to zero. Positivity of the actual M_N ensures eventual positive lower pivots. The cofactor formula proves continuity of inversion where the determinant is nonzero; thus $L_N, U_N \rightarrow K_N, d \rightarrow 0$, and the reference gap in (ERJ.20) tends to a positive number. For any requested positive width, first make d/g sufficiently small and then choose p so the geometric tail in (ERJ.26) is smaller than the remaining width. The rational logarithm error can be decreased independently. This proves convergence of the finite algebraic certificate for each fixed actual pair. The quantities controlling that convergence remain explicit when the degree is allowed to grow.

54.7 The four-volume comparison with its literal finite constants

The arithmetic pair losses add to $\mathcal{B}_{h,k} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$. Equations (ERJ.15)–(ERJ.17) prove

$$\mathcal{B}_{h,k} \leq q \log C_{2q-1} + q \log C_{2q}, \quad \frac{C_{2q}}{C_{2q-1}} = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2 \leq \frac{147}{5} < 32. \quad (\text{ERJ.32})$$

Indeed the first ratio is at most $5/3$, and the ratio inside the square is $4 + 3/(4^{2q} - 1) \leq 21/5$.

The signed metric control now bounds these same two return losses before the scalar envelope is taken. Use $\mathcal{B}_{h,k}^\Gamma$ and $\mathfrak{I}_{h,k}$ and the recursively refined endpoints $\mathcal{B}_{h,k}^{\text{lo}}, \mathcal{B}_{h,k}^{\text{hi}}$, as defined and proved in (FV.M1)–(FV.M18), Section 46.1.6. Its common monomial source $H_N(1)$ is exactly the original H_N in (ERJ.3): both have entries $\int (c + iu)^a (c + iu)^b m_k(u) du$. Its quotient J_N is monic remainder by this same χ . If T_N is the monic triangular coefficient matrix of $p_0, \dots, p_N$, then $O_N = T_N^* H_N T_N$, $B_N^{\text{ERJ}} = J_N T_N$, and direct substitution gives $B_N^{\text{ERJ}} O_N^{-1} (B_N^{\text{ERJ}})^* = J_N H_N^{-1} J_N^*$. Hence no change of quotient coordinates occurs in the comparison. In particular the literal refinement of (ERJ.32) is

$$\mathcal{B}_{h,k}^{\text{lo}} \leq \mathcal{B}_{h,k} \leq \min\{q \log C_{2q-1} + q \log C_{2q}, \mathcal{B}_{h,k}^{\text{hi}}\}. \quad (\text{ERJ.32a})$$

The lower bound combines nonnegativity of the two original return losses with (FV.M18); the upper bound combines that equation with (ERJ.32). Each rational trace certificate in (ERJ.25) is retained as well: the two upper endpoints can be summed and included as a further entry in the minimum. The source and relation errors in (ERJ.26)–(ERJ.31) continue to be computed for those original matrices.

This also controls the earlier inverse-power observations, not merely their determinant. Let $g_{a,j}$, $a = 0, 1$, $1 \leq j \leq q$, be all eigenvalues of the two original returns, and let

$$U_{h,k}^{\text{joint}} := \min\{q \log C_{2q-1} + q \log C_{2q}, \mathcal{B}_{h,k}^{\text{hi}}\}.$$

They lie in $(0, 1]$, and (ERJ.5) gives the exact identity $\sum_{a,j} -\log g_{a,j} = \mathcal{B}_{h,k}$. For every real $s > 0$, put $z_{a,j} = g_{a,j}^{-s} - 1 \geq 0$. The expansion of $\prod(1 + z_{a,j})$ contains $1 + \sum z_{a,j}$ and only further nonnegative terms. Consequently

$$\begin{aligned} \text{Tr } U_{q-1,2q-1}^{-s} + \text{Tr } U_{q,2q}^{-s} &\leq 2q - 1 + e^{s\mathcal{B}_{h,k}} \leq 2q - 1 + e^{sU_{h,k}^{\text{joint}}}, \\ g_{a,j} &\geq e^{-\mathcal{B}_{h,k}} \geq e^{-U_{h,k}^{\text{joint}}}. \end{aligned} \quad (\text{ERJ.32b})$$

The individual lower bound follows because each nonnegative $-\log g_{a,j}$ is at most their full sum. All powers are of the typed return endomorphisms in (ERJ.4), so (ERJ.11a) identifies them with the inverse singular-value observations of the original restricted lifts. Their action defect and theta primitive remain (ERJ.6)–(ERJ.8). This finite bound is compatible with the central lower estimates below; it does not assign an order-uniform bound to $U_{h,k}^{\text{joint}}$.

Set $X_0 = \max(1, M_h(b), M_h(-b))$, $K_0 = \max(1, \vartheta_h^{-1})$ and $C_{\text{env}} = \max(1, 2/(3c_h D))$. Since $bT \geq 2q$, the factorial ratio in U_k is at most one; also $M_k = 2 \int \cosh(bu) m_k \geq 2\mu_h^k$. Thus $U_k(T) \leq 3X_0^k$. The stated coefficient constant obeys $A_{2q-1}^2 \leq (2q/9)256^q$. On $0 \leq r_0 \leq 1/2$, the derivative of $\log((1 + r_0)/(1 - r_0))$ is at most $8/3$, so $\log B_q(r_0)^2 \leq 16R_0 k/(3D)$. Substitution into C_{2q-1} , using $\vartheta_h^{-(k-3)} \leq K_0^k$ and $k - 2 + Dq \leq 1 + (D + 1)q$, proves

$$\begin{aligned} q \log C_{2q-1} &\leq \mathcal{U}_k, \quad \mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32, \\ \mathcal{U}_k &= (\alpha D + \log 256)q^2 + \left(\log(X_0 K_0) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D + 1)q) + q \log C_{\text{env}}. \end{aligned} \quad (\text{ERJ.33})$$

All constants are independent of k for the fixed packet and b . As $q = [1 + k(m - 1)](k + 1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. The displayed scalar majorant divided by q^2 tends to $2(\alpha D + \log 256)$; divided by $q \log k$ it grows. The finite matrix and trace bounds (ERJ.18)–(ERJ.28) observe the same original four volumes and retain the directional data before this scalar estimate.

54.8 Central rank, both endpoint returns, and all inverse powers

Keep the original phase ϕ_N and relative Hermitian radius ϵ_N . For a reflection-stable packet the exact monic recurrence is $S p_n = p_{n+1} + (c + i a_n) p_n - (\omega_n / \omega_{n-1}) p_{n-1}$, $a_n \in \mathbb{R}$. It follows by transporting

to u with $p_n(c + iu) = i^n Q_n(u)$ and pairing self-adjoint multiplication by u with every lower monic orthogonal polynomial. Substitution into (ERJ.3) cancels the internal adjacent terms and gives

$$V_N = (b_{N+1}b_N^* + b_Nb_{N+1}^*)/\omega_N, \quad b_N^*G_Nb_{N+1} = i\omega_N\phi_N. \quad (\text{ERJ.34})$$

The pairing in the second formula is pure imaginary: in the real u coordinates both the quotient Gram and the remainder columns are real, and the source factors contribute $\overline{i^N}i^{N+1} = i$. This defines the retained real phase; it equals the original phase in the monic norm derivative formula. The relative Hermitian matrix has rank at most two and trace zero. Its square has trace $2[(b_N^*G_Nb_N)(b_{N+1}^*G_Nb_{N+1}) - |b_N^*G_Nb_{N+1}|^2]/\omega_N^2$, giving eigenvalues $\epsilon_N, -\epsilon_N$ and the remaining zeros. The rank-one determinant update in (ERJ.3) gives, for $N \geq q$, $b_N^*G_Nb_N/\omega_N = 1 - \delta_N$ and $b_{N+1}^*G_Nb_{N+1}/\omega_{N+1} = \delta_{N+1}^{-1} - 1$, where $\delta_N = V_N/V_{N-1}$. Therefore

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N}(1 - \delta_N)(\delta_{N+1}^{-1} - 1). \quad (\text{ERJ.35})$$

The determinant update follows by eliminating $\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ in its two block orders to prove $\det(I + XY) = \det(I + YX)$. Multiplying (ERJ.35) for $N = q, \dots, 2q - 2$ proves

$$\prod_{N=q}^{2q-2} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_{2q-1}V_q}{\omega_qV_{2q-1}}(1 - d_0)(1 - d_{q-1}) \prod_{a=1}^{q-2} (1 - d_a)^2, \quad d_a = \delta_{q+a}. \quad (\text{ERJ.36})$$

Each internal contraction occurs twice and both endpoint contractions occur once.

For (ERJ.13) the complete positive-real spectral compression gives

$$L = 2\delta\ell(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2, \quad 0 < L \leq \epsilon_N. \quad (\text{ERJ.37})$$

Here twice the sum of positive real parts of the eigenvalues of $A - cI$ equals L : sum the positive integers $2a - k$, giving $\lfloor (k+1)^2/4 \rfloor$, then include all $k+1$ imaginary centres and their full length ℓ . A unitary upper triangular form in the actual G_N metric, with these eigenvalues first, shows that this is the trace of the Hermitian defect compressed to their invariant subspace. A projection paired with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N, 0, \dots, 0$ has trace at most ϵ_N , proving the stated bound. The integer inequality $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$ follows in each parity.

The complete balanced monic-norm source proves

$$(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{bal}} n \quad (n \geq k \geq 3, n/2 \leq r \leq n), \quad (\text{ERJ.38})$$

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) X_0 K_0 \max(1, b^{-3}) 6^{B/3}.$$

This is its literal sufficient constant from the original envelope and monic Legendre minimum, with its full analytic proof included in the dependency record. Apply it at $n = q, r = q - 1$ and put $X = L/(C_h^{\text{bal}}q)$. Every d_a in (ERJ.36) is strictly below one, since otherwise its right side is zero while every left factor is at least $L^2 > 0$. Define the entire nonnegative penalty

$$P_{\text{mid}} = -\log(1 - d_0) - \log(1 - d_{q-1}) - 2 \sum_{a=1}^{q-2} \log(1 - d_a) + \sum_{a=0}^{q-2} \log(1 + \phi_{q+a}^2/L^2). \quad (\text{ERJ.39})$$

Taking logarithms in (ERJ.36) proves

$$C_{\text{mid}} := \log(V_q/V_{2q-1}) \geq 2(q-1) \log X + P_{\text{mid}}. \quad (\text{ERJ.40})$$

Indeed every factor $\epsilon_N^2 + \phi_N^2$ is at least $L^2(1 + \phi_N^2/L^2)$, and (ERJ.38) bounds the norm ratio.

The central update is the full matrix

$$K_{2q-1} - K_q = \sum_{a=q+1}^{2q-1} b_a b_a^* / \omega_a, \quad \text{rank}(K_{2q-1} - K_q) \leq q - 1. \quad (\text{ERJ.41})$$

Its positive Hermitian representative shows that the central return has at least one exact unit eigenvalue. List one such value separately and the other $q - 1$ as $\gamma_1, \dots, \gamma_{q-1}$, keeping any additional unit values. Their product is $e^{-C_{\text{mid}}}$. For every real $s > 0$, convexity of e^{sx} gives

$$\begin{aligned} \text{Tr } T_{q,2q-1}^{-s} &= 1 + \sum_{a=1}^{q-1} \gamma_a^{-s} \\ &\geq 1 + (q-1)e^{sC_{\text{mid}}/(q-1)} \\ &\geq \max \left\{ q, 1 + (q-1)X^{2s}e^{sP_{\text{mid}}/(q-1)} \right\}. \end{aligned} \quad (\text{ERJ.42})$$

For completeness $e^y \geq 1 + y$ follows by differentiating $e^y - 1 - y$, whose minimum is zero at $y = 0$. Apply it to $y_a = -s \log \gamma_a - \bar{y}$, where $\bar{y} = -(s/(q-1)) \sum_a \log \gamma_a$: $\sum_a e^{-s \log \gamma_a} = e^{\bar{y}} \sum_a e^{y_a} \geq (q-1)e^{\bar{y}}$. The q bound also follows termwise from $\gamma_a \leq 1$. The actual rank $\rho = \text{rank}(K_{2q-1} - K_q)$ preserves exactly $q - \rho$ unit eigenvalues, because $I - T = (K_{2q-1} - K_q)G_{2q-1}$ has rank ρ . Write $A_+ = \max\{0, 2(q-1) \log X + P_{\text{mid}}\}$. If $\rho > 0$, applying the same exponential argument to just the ρ eigenvalues below one proves the stronger exact-rank consequence

$$\text{Tr } T_{q,2q-1}^{-s} \geq q - \rho + \rho e^{sC_{\text{mid}}/\rho} \geq q - \rho + \rho e^{sA_+/\rho}. \quad (\text{ERJ.42a})$$

For fixed $a \geq 0$, $r(e^{a/r} - 1)$ decreases as $r > 0$ increases: its derivative is $e^{a/r}(1 - a/r) - 1 \leq 0$, since the derivative of $e^t(1 - t)$ is $-te^t \leq 0$ for $t \geq 0$. Thus $\rho \leq q - 1$ proves that (ERJ.42a) improves (ERJ.42). At $\rho = 0$ the central return is the identity, its cost is zero, and the trace is exactly q . There is no division by ρ in that case. By (ERJ.22), the ordered eigenvalues of each pair (K_{q-1}, K_{2q-1}) and (K_q, K_{2q}) are at most those of (K_q, K_{2q-1}) : their Rayleigh quotients decrease respectively in the numerator and by increase of the denominator. Since x^{-s} is decreasing, *each* original endpoint return obeys the final bound in (ERJ.42), and the stronger rank-refined bound (ERJ.42a), by the same ordered comparison. Its minimum eigenvalue also obeys

$$g_{\min}^{(a)} \leq \min \left\{ 1, X^{-2}e^{-P_{\text{mid}}/(q-1)} \right\} \leq \min \left\{ 1, \frac{4(C_h^{\text{bal}})^2}{\delta^2 k^2} \right\}, \quad a = 0, 1. \quad (\text{ERJ.43})$$

The first inequality follows from $\gamma_{\min}^{q-1} \leq \prod \gamma_a$ and (ERJ.40), followed by the same ordered comparison. Equations (ERJ.9)–(ERJ.12) and (ERJ.7)–(ERJ.8) give its exact source, action, and theta interpretation. The identity

$$\mathcal{B}_{h,k} = 2C_{\text{mid}} + \log(V_{q-1}/V_q) + \log(V_{2q-1}/V_{2q}) \quad (\text{ERJ.44})$$

has nonnegative endpoint terms. Combining (ERJ.37)–(ERJ.40) proves

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4,$$

because its lower bound is $4(q-1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q-1)/q \rightarrow 1$. The consecutive-first-window continuation separately proves the length- q products at the two actual endpoints, including the single terminal factor at initial degree $q - 1$, and gives its stronger per-window inverse-power estimate. Its original constant and phase formula stay in that complete companion proof; (ERJ.42) retains the exact central unit eigenvalue and full central penalty.

54.9 Exact auxiliary calculation and retained proof dependencies

The accompanying complete fixture proof uses $S = 1 + iu$, $\chi = S^2 - 2S - 1$ and $r_\lambda(u) = |\Gamma(\lambda + iu/2)|^2/(2\pi)$. Its source is $m = r_4/70$ of mass $9/16$; its smaller source is $\nu = (54/35)(r_2/3)$ of mass $27/70$. The full identity

$$m(u) = \left(\frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120} \right) \frac{r_2(u)}{3} \quad (\text{ERJ.45})$$

follows from two applications of $\Gamma(z+1) = z\Gamma(z)$, and proves their source contraction. The complete fixture appendix derives all moments, original S Grams, remainder matrices, extrema, and rational logarithm endpoints. Its remainder norm squared is $763/108$, attained by the original polynomial $-245/36 + (791/108)S - (7/9)S^2 + (7/27)S^3$ in both degree three and degree four. The resulting source gap is $108/763$, while the two actual minimum return eigenvalues equal $135/331$. The exact four-volume ratio is $3073295611/216513000$. These quantities are proved in the full fixture text retained with this article, rather than inferred from a decimal computation.

The existing fixture executed 303 distinct mathematical checks in each unchanged normal and optimized run. Seven literal formula changes in each mode were rejected after the complete calculation, giving 16 actual jobs in total. The preserved source and full execution JSON have hashes recorded in this article's manifest. This completion did not rerun that unchanged suite. Its exact source/Gamma fixture is connected to the arithmetic construction by (ERJ.2)–(ERJ.4), (ERJ.17)–(ERJ.20); the density (ERJ.45) and its masses remain its declared auxiliary inputs.

For the empty packet $h = 1$, $E = 0$, the quotient determinants are empty determinants one and every quotient trace and four-volume loss is zero. The original analytic source and mass still exist. The positive inverses and $q - 1$ spectral averages above are used only in their stated domains; the actual quartet has $q \geq (k + 1)^2$ and $k \geq 3$. At $i = j$, return is the identity, its relation correction and projection difference vanish, and the exact loss is zero.

The complete source dependencies are the original *Canonical restriction and the arithmetic endpoint volume; An arithmetic lower-envelope comparison for central quotient volumes* (AU.1–49); *Exact window products, sharp endpoint optimization, and original relation costs* (EP.1–54), with its complete balanced-window proof dependency; the endpoint-window proof (EW.1–59); the complete 303-check fixture proof; and the *Consecutive first-window join* (CJ.1–22). The accompanying manifest pins their actual files and the new independent mathematical review. The global cumulative reader carries these full proofs and the new article in their stated order.

55 Deligne machinery inside Split-Zero cohomology

55.1 Deligne's image comparison inside the Split-Zero source complexes

This sidebar integrates the delivered specialization-curvature, exponential-comparison, and logarithmic-constituent-control notes. Their logarithmic cohomology and constituent-coupling calculations are reproduced with complete proofs and explicitly joined to the existing Split-Zero complexes; they are attributed to those web contributions. The sidebar supplies the maps connecting the compact-support comparison, its dual, the Split-Zero internal quotient, and the compactified curve of arithmetic representatives. It also calculates the logarithmic de Rham complex of that curve. In particular, it identifies which cohomology groups the exceptional directions occupy and the divisor twist in their support-compatible dual.

The source reference is P. Deligne, *La conjecture de Weil II*, Publications Mathématiques de l'IHÉS **52** (1980), 137–252, https://www.numdam.org/item/PMIHES_1980__52__137_0/. The specific weight comparison is in (3.3.4)–(3.3.6), printed page 206; the preceding geometric argument is in (3.2.8)–(3.2.13). The source-language S20 records for printed pages 201–207 were read in full, including the strict inequality in (3.2.10) and the squared eigenvalue in (3.2.13). The complete

S20 typed transcription is retained separately from this integration. The symbol Q_0 in (DS7)–(DS8) denotes a finite-field cardinality; later Q_0 denotes the explicitly defined quotient Gram, as in the supplied logarithmic note; q denotes the dimension of the actual arithmetic packet.

55.1.1 The comparison image, its entire cone, and its dual

Let $c : C^\bullet \rightarrow D^\bullet$ be a cochain map of complexes of vector spaces over a field K . All differentials raise degree by one. Write $Z_C^m = \ker d_C^m$, $B_C^m = \text{im } d_C^{m-1}$, and similarly for D . Define

$$K_c^m = \{x \in Z_C^m : c(x) \in B_D^m\}, \quad I_c^m = \text{im}(H^m(c) : H^m(C) \rightarrow H^m(D)). \quad (\text{DS1})$$

There are exact sequences with the displayed maps:

$$\begin{aligned} 0 \longrightarrow K_c^m / B_C^m &\xrightarrow{[x] \mapsto [x]} H^m(C) \xrightarrow{[x] \mapsto [cx]} I_c^m \longrightarrow 0, \\ 0 \longrightarrow I_c^m &\longrightarrow H^m(D) \longrightarrow H^m(D) / I_c^m \longrightarrow 0. \end{aligned} \quad (\text{DS2})$$

Indeed, $B_C^m \subset K_c^m$ because c commutes with the differentials. The kernel of the second arrow in the first sequence is exactly the classes represented by K_c^m , and every element of I_c^m has the indicated preimage. The second sequence is the defining inclusion and quotient. Thus both the lost source classes and the extra target classes are present.

The single complex retaining both sides is

$$\text{Cone}(c)^m = D^m \oplus C^{m+1}, \quad d(y, x) = (d_D y + cx, -d_C x). \quad (\text{DS3})$$

Expansion gives $d^2(y, x) = (d_D^2 y + d_D cx - cd_C x, d_C^2 x) = 0$. The maps $y \mapsto (y, 0)$ and $(y, x) \mapsto x$ give the exact sequence of complexes $0 \rightarrow D \rightarrow \text{Cone}(c) \rightarrow C[1] \rightarrow 0$, where $C[1]^m = C^{m+1}$ has differential $-d_C$. To make the relevant cohomology segment explicit, a cone cocycle obeys $d_C x = 0$ and $cx = -d_D y$, so its projection represents a class in K_c^{m+1} / B_C^{m+1} . Conversely, if $cx = d_D v$, then $(-v, x)$ is a cone cocycle. A projected class is zero precisely when subtracting a cone boundary leaves a representative $(y, 0)$ with $d_D y = 0$. Finally $(y, 0)$ is a cone boundary precisely when $y = d_D u + cv$ with $d_C v = 0$. This proves the exact segment

$$H^m(C) \xrightarrow{H^m(c)} H^m(D) \longrightarrow H^m(\text{Cone}(c)) \longrightarrow H^{m+1}(C) \xrightarrow{H^{m+1}(c)} H^{m+1}(D). \quad (\text{DS4})$$

Every chain action commuting with c acts on (DS1)–(DS4), since it preserves both cycles and boundaries; on the cone its action is the direct sum of the two specified actions.

For a finite-dimensional linear map $f : V \rightarrow W$, its ordinary linear transpose $f^\vee : W^\vee \rightarrow V^\vee$ gives a perfect pairing

$$\text{im } f \times \text{im } f^\vee \longrightarrow K, \quad (f(v), f^\vee \lambda) \longmapsto \lambda(f(v)). \quad (\text{DS5})$$

If v is changed by $\ker f$, the value is unchanged. If λ is changed by $\ker f^\vee$, its restriction to $\text{im } f$ is unchanged. A nonzero $f(v)$ is detected by some functional on W , and a nonzero $f^\vee \lambda$ is detected by some v , proving nondegeneracy on both sides. In particular

$$(\text{im } f)^\vee \simeq \text{im } f^\vee, \quad \ker f^\vee = (\text{im } f)^\perp, \quad \text{im } f^\vee = (\ker f)^\perp. \quad (\text{DS6})$$

For invertible actions T_V, T_W with $fT_V = T_W f$, the dual actions are $(T_V^{-1})^\vee, (T_W^{-1})^\vee$; their induced actions on the two images make the evaluation in (DS5) invariant. For an additive action A and a specified scalar b , the action $bI - A^\vee$ on the dual instead satisfies $\langle Av, \lambda \rangle + \langle v, (bI - A^\vee)\lambda \rangle = b\langle v, \lambda \rangle$. These are two explicitly defined action conventions, with different types of weight parameter.

Deligne's comparison takes

$$C = R\Gamma_c(X, \mathcal{F}), \quad D = R\Gamma(X, \mathcal{F}), \quad c = c_{X, \mathcal{F}}, \quad I_c^m = \text{im } c_m. \quad (\text{DS7})$$

Here $X = X_0 \otimes_{\mathbb{F}_{Q_0}} \bar{\mathbb{F}}_{Q_0}$, X_0 is smooth of pure dimension N and finite type, and $\mathcal{F}_0$ is a smooth $\mathbb{Q}_\ell$ -sheaf, $\ell \neq \text{char } \mathbb{F}_{Q_0}$, pure of weight n . In the source theorem the upper weight bound on H_c^m is $n + m$. The dual of $\mathcal{F}$ has weight $-n$, and Poincaré duality gives the exact degree and twist

$$H^m(X, \mathcal{F})^\vee \simeq H_c^{2N-m}(X, \mathcal{F}^\vee)(N). \quad (\text{DS8})$$

The right side has upper weight $-n + (2N - m) - 2N = -n - m$ by the already established compact-support theorem. Dualizing gives the lower weight bound $n + m$ on H^m . Since c_m is Frobenius equivariant, choose a basis first adapted to its kernel in the source and then to its image in the target. The corresponding action matrices are block triangular. Every eigenvalue on I_c^m therefore occurs on both H_c^m and H^m , with its algebraic conjugates. The two bounds force its weight to equal $n + m$. This proves the image step (3.3.6) from the two stated source bounds and proves exactly why (DS7) retains the cone (DS3).

The tensor comparison has an equally explicit algebraic map. For finite-dimensional $f_a : V_a \rightarrow W_a$ ($1 \leq a \leq r$),

$$\text{im} \left(\bigotimes_{a=1}^r f_a \right) = \bigotimes_{a=1}^r \text{im } f_a \subseteq \bigotimes_{a=1}^r W_a. \quad (\text{DS9})$$

The tensor map sends a pure tensor to the pure tensor of its images, so its image is contained in the right side. Conversely these image pure tensors span the right side and each has a preimage; inclusivity into the target follows by extending a basis of each image to a basis of W_a . Thus tensor powers preserve the same comparison image. For complexes, the tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{\deg x} x \otimes dy$; swapping the factors has the sign $(-1)^{\deg x \deg y}$. In characteristic zero a specified symmetry projection is $\frac{1}{r!} \sum_{\pi \in \mathfrak{S}_r} \pi$ with those graded signs. Its idempotence follows because each group element occurs exactly $r!$ times in the double sum. No unspecific tensor eigenvalue is substituted for the eigenvalue on this actual image.

55.1.2 The internal Split-Zero lift of the same maps

For a complex vector space V , set $S(V) = \{\tau\} \sqcup V^\bullet$. Supported addition and scalar multiplication are the original operations: $v^\bullet + w^\bullet = (v + w)^\bullet$ and $a^\bullet v^\bullet = (av)^\bullet$. External τ is the additive identity and is absorbing for scalar multiplication. The supported zero is $e_V = 0_V^\bullet$. Every linear map $f : V \rightarrow W$ has the canonical lift

$$S(f)(\tau) = \tau, \quad S(f)(v^\bullet) = f(v)^\bullet. \quad (\text{DS10})$$

Checking the two supported inputs and inputs containing τ proves additivity, the scalar law and composition. For $B \subseteq V$, $S(V) \rightarrow S(V/B)$ sends $b^\bullet$ to $e_{V/B}$ for every $b \in B$, and sends τ to τ . Two supported vectors have the same image exactly when their difference lies in B . This is the coequalizer of inclusion $S(B) \rightarrow S(V)$ and the map $b^\bullet \mapsto e_V, \tau \mapsto \tau$: any map equalizing these has equal values on $v^\bullet$ and $(v + b)^\bullet$, by additivity, and hence has a unique factorization through the displayed quotient.

Applying (DS10) to all maps in (DS1)–(DS4) lifts their source kernels, images, quotients and actions. A differential squared is the supported-zero map on represented vectors and still sends τ to τ . The ordinary kernel $\ker H^m(c)$ is carried by the represented fibre of $S(H^m(c))^{-1}(e)$; the preimage of external absence is only $\{\tau\}$. These two fibres are disjoint. Their exact common carrier is $S(\ker H^m(c)) = \{\tau\} \sqcup (\ker H^m(c))^\bullet$, with the inclusion of each of the two fibres. If $p_V : S(V) \rightarrow V$ sends $v^\bullet$ to v and τ to 0, this carrier is precisely $p_V^{-1}(\ker H^m(c))$. Thus the amplitude projection proves the two inclusions while retaining the two distinct receiving zeros. With a join semilattice of supports, carry out the construction in each fibre and use the original transition maps; (DS1)–(DS4) commute with those transitions because their definitions use the commuting cochain maps. No support label is discarded.

The coefficient comparison remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad p_R(a^\bullet) = a, \quad p_R(\tau) = 0. \quad (\text{DS11})$$

It commutes on every supported integer and on τ . In particular the linear/coherent calculation below is the amplitude image of a specified Split-Zero diagram; its supported exceptional kernel remains available before applying p .

55.1.3 The original source relation produces the curve

Fix the actual cyclic object $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, and the original action $A = M_S$. The monic source spaces $\mathcal{P}_N$, their positive Gram matrices O_N , and their full remainder maps $B_N : \mathcal{P}_N \rightarrow E$ give

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N. \quad (\text{DS12})$$

Take $i \leq j$ with both remainder maps onto, and retain the source inclusion L_{ij} with $B_j L_{ij} = B_i$ and $L_{ij}^* O_j L_{ij} = O_i$. Put

$$\Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j, \quad R(z) = R_j + z \Delta. \quad (\text{DS13})$$

The equalities follow directly: $B_N R_N = I$, $R_j^* O_j \Delta = G_j B_j \Delta = 0$, and expansion of $(L_{ij} R_i)^* O_j (L_{ij} R_i)$ gives $G_j + \Delta^* O_j \Delta$. For the kernel order, the actual source projection $P = L_{ij} O_i^{-1} L_{ij}^* O_j$ obeys $P^2 = P$ and $P^* O_j = O_j P$ by the retained source isometry. Therefore $K_j - K_i = B_j (I - P) O_j^{-1} (I - P)^* B_j^* \succeq 0$; expanding the right side gives $B_j O_j^{-1} B_j^* - B_i O_i^{-1} B_i^*$. Therefore

$$B_j R(z) = I, \quad R(z)^* O_j R(z) = G_j + |z|^2 C. \quad (\text{DS14})$$

The source is the original theta source with its arithmetic coefficient unit and every full jet; the new variable z parametrizes only the comparison and does not replace S .

Let $E_0 = \ker \Delta$, $E_1 = E_0^{\perp G_j}$, $r = \dim E_1$, and let P_0, P_1 be the specified projections. On $X = \mathbb{P}_{\mathbb{C}}^1$ with coordinates $[u : v]$, finite coordinate $z = v/u$ and infinity coordinate $w = u/v$, set $D = \{\infty\}$ and

$$\mathcal{E} = (E_0 \otimes \mathcal{O}_X) \oplus (E_1 \otimes \mathcal{O}_X(-D)). \quad (\text{DS15})$$

The source morphism is

$$\Psi(x_0 + x_1 \otimes (u, v)) = R_j x_0 + u R_j x_1 + v \Delta x_1. \quad (\text{DS16})$$

It is fibrewise injective. At $u \neq 0$, apply B_j and use $E_0 \cap E_1 = 0$; at $u = 0$, apply B_j to force $x_0 = 0$ and then use injectivity of $\Delta|_{E_1}$. Thus the original source Gram defines a smooth positive metric on this bundle, including infinity. The observation $\alpha = B_j \Psi$ is identity in the finite frame and $P_0 + w P_1$ in the infinity frame. Consequently

$$0 \rightarrow \mathcal{E} \xrightarrow{\alpha} E \otimes \mathcal{O}_X \rightarrow i_* E_1 \rightarrow 0, \quad (\text{DS17})$$

where $i : D \hookrightarrow X$. In fact its image lattice is $\mathcal{M} = E_0 \mathcal{O}_X + E \mathcal{O}_X(-D)$; at infinity it is $E_0 \mathcal{O} + w E \mathcal{O}$.

Tensoring the two locally free terms of (DS17) with the residue field at infinity gives $[E \xrightarrow{P_0} E]$ in degrees $-1, 0$. Its cohomology is E_1 in both degrees. Intrinsically the negative term is $E_1 \otimes \mathcal{O}(-D)|_D$. This calculation is the derived fibre of the skyscraper cokernel, and explicitly retains the Tor term that ordinary right-exact specialization would omit. Dualizing (DS17) gives

$$0 \rightarrow E^* \mathcal{O}_X \rightarrow \mathcal{E}^\vee \rightarrow i_* (E_1^* \otimes \mathcal{O}(D)|_D) \rightarrow 0. \quad (\text{DS18})$$

To check its terminal term, resolve $\mathcal{O}_D$ by $[\mathcal{O}(-D) \rightarrow \mathcal{O}]$ and apply $\mathcal{H}om(-, \mathcal{O})$. The resulting cokernel is $\mathcal{O}(D)|_D$. The two boundary factors pair by evaluation and $\mathcal{O}(-D)|_D \otimes \mathcal{O}(D)|_D \rightarrow \mathbb{C}_D$.

55.1.4 A calculated compact-support/ordinary de Rham comparison

Transport the trivial connection through α on $X \setminus D$. All connection complexes here are complexes of sheaves of $\mathbb{C}$ -vector spaces: the differentials satisfy the Leibniz rule and are not asserted to be $\mathcal{O}_X$ -linear. In the infinity frame it is

$$\nabla = d + P_1 \frac{dw}{w}, \quad \mathcal{K} = [\mathcal{E} \xrightarrow{\nabla} \mathcal{E} \otimes \Omega_X^1(\log D)] \quad \text{in degrees } 0, 1. \quad (\text{DS19})$$

Indeed $\alpha^{-1}d\alpha = P_1 dw/w$. Its residue is P_1 , its horizontal monodromy is $\exp(-2\pi i P_1) = I$, and the map α is a morphism to $\mathcal{K}_{\text{ord}} = [E\mathcal{O}_X \xrightarrow{d} E\Omega_X^1(\log D)]$.

Here is the complete local calculation of the extension at infinity. On analytic power series the E_0 differential, with dw/w as the one-form frame, is $\sum a_n w^n \mapsto \sum n a_n w^n$. Its kernel and cokernel are the constant terms, both E_0 . The E_1 differential is

$$\sum_{n \geq 0} b_n w^n \mapsto \sum_{n \geq 0} (n+1) b_n w^n. \quad (\text{DS20})$$

It is invertible: its inverse divides the n th coefficient by $n+1$, which preserves every disk of convergence. Thus the E_1 part has zero derived stalk at infinity. On a punctured disk, integrating a Laurent series term by term divides its w^n coefficient by n for $n \neq 0$; the only one-form class remaining is dw/w with integral $2\pi i$ around a positively oriented w -circle. These statements, together with the usual local primitive on a disk, identify the E_0 part with $Rj_* \underline{E}_0$ and the E_1 part with $j_! \underline{E}_1$, where $j : X \setminus D \hookrightarrow X$. The identification for $j_!$ is the extension-by-zero map of flat sections; its stalk cohomology is an isomorphism on the open disk and is zero on both sides at infinity by (DS20). The identification for Rj_* is the restriction map to the punctured disk and the two Laurent calculations. This constructs the comparison on the present curve rather than assigning compact support by name.

Its entire global cohomology is

$$\mathbb{H}^0(X, \mathcal{K}) = E_0, \quad \mathbb{H}^1(X, \mathcal{K}) = 0, \quad \mathbb{H}^2(X, \mathcal{K}) = E_1 \otimes \mathbb{T}, \quad \mathbb{T} = H^1(X, \Omega_X^1). \quad (\text{DS21})$$

For a direct verification, $\Omega_X^1(\log D) = \mathcal{O}_X(-1)$. The two terms of $\mathcal{K}$ are therefore $E_0 \mathcal{O} \oplus E_1 \mathcal{O}(-1)$ and $E_0 \mathcal{O}(-1) \oplus E_1 \mathcal{O}(-2)$. Using the two affine charts, polynomial division of Laurent polynomials proves

$$\begin{aligned} H^0(\mathcal{O}) &= \mathbb{C}, & H^1(\mathcal{O}) &= 0, & H^0(\mathcal{O}(-1)) &= H^1(\mathcal{O}(-1)) = 0, \\ H^0(\mathcal{O}(-2)) &= 0, & H^1(\mathcal{O}(-2)) &= \mathbb{C}. \end{aligned} \quad (\text{DS22})$$

More explicitly, after identifying $\mathcal{O}(-2)$ with Ω_X^1 , a Laurent one-form on the overlap has every term except dw/w removable by a one-form on one of the charts. The coefficient of dw/w is not removable and is its stated one-dimensional quotient. Retain $\mathbb{T}$ as this quotient line, with its original generator $[dz/z] = -[dw/w]$. The double complex for these two terms has only the entries E_0 in total degree zero and $E_1 \otimes \mathbb{T}$ in total degree two. There is no differential to or from them, proving (DS21) without an uncomputed higher differential.

Taking the quotient of the two complexes in (DS19) gives

$$\mathcal{Q}_D = [i_* E_1 \xrightarrow{0} i_* E_1 dw/w] \quad \text{in degrees } 0, 1. \quad (\text{DS23})$$

The zero differential follows by reducing $w \partial_w$ modulo w . The termwise short exact sequence $0 \rightarrow \mathcal{K} \rightarrow \mathcal{K}_{\text{ord}} \rightarrow \mathcal{Q}_D \rightarrow 0$ and the fact $\mathbb{H}^0(\mathcal{K}_{\text{ord}}) = E$, $\mathbb{H}^{>0}(\mathcal{K}_{\text{ord}}) = 0$ give

$$0 \rightarrow E_0 \rightarrow E \rightarrow E_1 \rightarrow 0, \quad E_1 dw/w \xrightarrow{\delta} E_1 \otimes \mathbb{T} \quad \text{an isomorphism.} \quad (\text{DS24})$$

Fix the Čech convention $s_V - s_U$ for the overlap, with U the finite chart and V the infinity chart, and the total-complex differential $\delta + (-1)^p \nabla$ on Čech degree p . Lifting a boundary one-form as

$(0, v dw/w)$ makes its connecting class $v[dw/w]$; thus $\delta(v dw/w) = v \otimes [dw/w] = -v \otimes [dz/z]$. In the original finite coordinate $dw/w = -dz/z$. Both that sign and the chosen positive orientation of the w -circle are retained. Under (DS21), the ordinary comparison image is exactly E_0 in degree zero and zero in degrees one and two. The full E_1 occurs in its boundary comparison and in degree two.

55.1.5 The divisor twist in the support-compatible dual

The bundle dual in (DS18) and the dual de Rham comparison are joined by the exact additional map of line bundles $\mathcal{O}(-D) \hookrightarrow \mathcal{O}$. Define

$$\mathcal{E}^{\text{cd}} = \mathcal{E}^\vee \otimes \mathcal{O}(-D), \quad \mathcal{K}^{\text{cd}} = [\mathcal{E}^{\text{cd}} \xrightarrow{\nabla^{\text{cd}}} \mathcal{E}^{\text{cd}} \otimes \Omega_X^1(\log D)]. \quad (\text{DS25})$$

The dual connection has residue $-P_1^\vee$. Multiplication of its local frame by w , the local generator of $\mathcal{O}(-D)$, adds I to the residue. Hence the residue of (DS25) is $P_0^\vee$, and

$$\mathcal{E}^{\text{cd}} = E_0^* \mathcal{O}(-D) \oplus E_1^* \mathcal{O}, \quad \frac{m}{\mathbb{H}^m(\mathcal{K}^{\text{cd}})} \left| \begin{array}{ccc} 0 & 1 & 2 \\ E_1^* & 0 & E_0^* \otimes \mathbb{T} \end{array} \right. \quad (\text{DS26})$$

The proof is exactly the power-series calculation (DS20) and the four line-bundle groups (DS22), now with the two summands exchanged.

Evaluation is a morphism $\mathcal{E} \otimes \mathcal{E}^{\text{cd}} \rightarrow \mathcal{O}(-D)$. The connections obey the Leibniz identity for evaluation, since the dual connection cancels the original connection term. Wedge product therefore gives a cochain map

$$\mathcal{K} \otimes_{\mathbb{C}} \mathcal{K}^{\text{cd}} \longrightarrow [\mathcal{O}(-D) \xrightarrow{d} \Omega_X^1] =: \mathcal{K}_c(\mathbb{C}). \quad (\text{DS27})$$

On a degree-zero tensor it is evaluation; on degree-one tensors it is evaluation times the retained one-form; on two one-forms it is zero because $\Omega_X^2 = 0$. These prescriptions and the tensor sign in (DS9) verify the cochain identity in every degree. The degreewise evaluation and wedge maps factor through their $\mathcal{O}_X$ -balanced tensor products; the displayed complex tensor product is over $\mathbb{C}$, so its differential is defined. Define the trace on $\mathbb{H}^2(\mathcal{K}_c(\mathbb{C}))$ by

$$\text{Tr}_D([dz/z]) = 1, \quad \text{Tr}_D([dw/w]) = -1 = -\frac{1}{2\pi i} \int_{|w|=\epsilon}^{\text{counterclockwise}} \frac{dw}{w}. \quad (\text{DS28})$$

Holomorphic one-forms on either chart have zero overlap residue, so this is well-defined on (DS22). The induced pairings are

$$E_0 \times E_0^* \rightarrow \mathbb{C}, \quad E_1 \times E_1^* \rightarrow \mathbb{C}, \quad (\text{DS29})$$

ordinary evaluation after applying the retained trace on $\mathbb{T}$, in complementary degrees 0, 2 and 2, 0. They are perfect, since a nonzero vector is detected by a linear functional. Thus (DS25)–(DS29) prove the support-compatible duality, including the dimension-one degree change and its actual boundary line factor. The finite-field analogue of the trace target is $\overline{\mathbb{Q}}_\ell(-1)$; the characteristic-zero calculation here has the explicit residue normalization (DS28).

55.1.6 Retaining the arithmetic action through the exceptional fibre

Transporting the original A through α gives

$$\hat{A}_\infty = (P_0 + w^{-1}P_1)A(P_0 + wP_1) = \begin{pmatrix} A_{00} & wA_{01} \\ w^{-1}A_{10} & A_{11} \end{pmatrix}. \quad (\text{DS30})$$

It extends regularly on $\mathcal{E}$ exactly when $AE_0 \subseteq E_0$, since its sole possible negative power is $w^{-1}P_1AP_0$. The explicit repair on the same ambient arithmetic target is

$$F_A = \sum_{a=0}^{q-1} A^a E_0, \quad \mathcal{M}_A = F_A \mathcal{O}_X + E \mathcal{O}_X(-D). \quad (\text{DS31})$$

Cayley–Hamilton puts $A^q E_0$ in the displayed sum, proving invariance; every invariant subspace containing E_0 contains this sum. Similarly every coherent A -stable submodule of $E\mathcal{O}$ containing $\mathcal{M}$ contains $\sum_{a=0}^{q-1} A^a \mathcal{M}$, which is exactly $\mathcal{M}_A$ in (DS31). There are exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{M} \rightarrow \mathcal{M}_A \rightarrow i_*(F_A/E_0) \rightarrow 0, \\ 0 \rightarrow \mathcal{M}_A \rightarrow E\mathcal{O}_X \rightarrow i_*(E/F_A) \rightarrow 0. \end{aligned} \quad (\text{DS32})$$

The second is A equivariant on every term. The connection d on $E\mathcal{O}$ preserves $\mathcal{M}_A$ logarithmically, because $d(wv) = w dv + v dw$ lies in $\mathcal{M}_A \otimes \Omega^1(\log D)$. It commutes with the constant original A . Define its two-term complex $\mathcal{K}_A$ with the same degree convention as (DS19). Calculation with any vector-space complement of F_A proves, intrinsically,

$$\mathbb{H}^0(\mathcal{K}_A) = F_A, \quad \mathbb{H}^1(\mathcal{K}_A) = 0, \quad \mathbb{H}^2(\mathcal{K}_A) = (E/F_A) \otimes \mathbb{T}, \quad (\text{DS33})$$

with actions $A|_{F_A}$ and $\bar{A}$, respectively. The identifications are independent of that auxiliary complement: the degree-zero map is inclusion into E and the degree-two map is the connecting quotient map in (DS32), as in (DS24).

Its dual lattice has a coordinate-free formula. Put $F_A^\perp = \{\lambda \in E^* : \lambda(F_A) = 0\}$. Then

$$\mathcal{M}_A^\vee(-D) = F_A^\perp \mathcal{O}_X + E^* \mathcal{O}_X(-D). \quad (\text{DS34})$$

Indeed a meromorphic functional belongs to the left side precisely when its value on every element of $F_A \mathcal{O} + wE\mathcal{O}$ lies in $w\mathcal{O}$. This first forces it to be holomorphic, by testing wE , and then forces its constant term to annihilate F_A . This is exactly the right side. Fix the additive dual action as $-A^\vee$, so evaluation has the zero action on its one-dimensional target. The displayed lattice is invariant under this action because F_A is invariant. Equations (DS27)–(DS29) therefore become the perfect equivariant pairings $F_A \times F_A^*$ and $(E/F_A) \times F_A^\perp$. For the original real integer k , replacing the dual action by $kI - A^\vee$ puts the scalar k action on the evaluation target, as checked in (DS6). The passage to the metric antidual is the specified conjugate-linear map $\mathfrak{c} : E^\vee \rightarrow E^\vee$, $\mathfrak{c}(\lambda)(v) = \overline{\lambda(v)}$. In the fixed polynomial basis it conjugates the functional's coefficient column, and hence $\mathfrak{c}(kI - A^T) = (kI - A^*)\mathfrak{c}$. This proves the exact coefficient map between the linear dual and the original metric antidual convention. For every polynomial p , an adapted basis to F_A proves

$$\text{Tr } p(A) = \text{Tr } p(A|_{F_A}) + \text{Tr } p(\bar{A}). \quad (\text{DS35})$$

This retains the complete arithmetic trace across the calculated cohomological degrees. There is no need to presume that the metric kernel E_0 was an arithmetic constituent.

The source cost of (DS31) remains visible. If $s_A = \dim F_A - \dim E_0$, choose a basis first for E_0 , then for F_A/E_0 , and then for E/F_A . The local inclusion $\mathcal{M} \rightarrow \mathcal{M}_A$ has diagonal blocks I, wI_{s_A}, I and determinant w^{s_A} . If the metric on $\mathcal{M}_A$ is transported on the punctured chart from the original source metric on $\mathcal{M}$, the determinant identity is

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w). \quad (\text{DS36})$$

It follows by taking determinants of $G_{\mathcal{M}} = J^* G_A J$. Thus the action-stable complex (DS33) retains the exceptional contribution through an explicitly measured metric pole.

55.1.7 The same curvature, its source map, and the weight interface

For $\mathcal{L} = (\det \mathcal{E})^\vee$ with its original source metric, put

$$Z = G_j^{-1}(G_i - G_j) = (K_j - K_i)G_i. \quad (\text{DS37})$$

The second equality is $K_j G_i - I = (K_j - K_i) G_i$. The first expression proves self-adjointness and positivity in G_j ; the second proves the same in G_i , because $G_i Z = G_i (K_j - K_i) G_i \succeq 0$ and $K_j - K_i \succeq 0$. No zero eigenvalue is removed. Differentiating $\log \det(G_j + |z|^2 C)$ yields

$$c_1(\mathcal{L}) = \frac{i}{2\pi} \kappa(|z|^2) dz \wedge d\bar{z}, \quad \kappa(t) = \text{Tr}(Z(I + tZ)^{-2}). \quad (\text{DS38})$$

To verify every coefficient, set $f(t) = \log \det(I + tZ)$. Then $f'(t) = \text{Tr}(Z(I + tZ)^{-1})$ and $f'(t) + t f''(t) = \kappa(t)$. Since $\partial \bar{\partial} f(|z|^2) = (f'(|z|^2) + |z|^2 f''(|z|^2)) dz \wedge d\bar{z}$, this proves (DS38) with the stated sign. For a positive eigenvalue λ , $\int_0^\infty \lambda(1 + t\lambda)^{-2} dt = 1$; for a zero eigenvalue the integrand is zero. Since $i dz \wedge d\bar{z} = 2dx \wedge dy$, polar integration gives the degree and the logarithmic moment

$$\int_X c_1(\mathcal{L}) = r, \quad \log \frac{\det G_i}{\det G_j} = \int_0^1 \log(1/t) \kappa(t) dt. \quad (\text{DS39})$$

For the second equality, write $M(t) = t f'(t)$, so $M' = \kappa$ and $M(0) = 0$. Nonnegative Fubini gives $\int_0^1 \log(1/t) \kappa(t) dt = \int_0^1 t^{-1} M(t) dt = f(1) - f(0)$.

The full source observation is also retained. With

$$P_z = R(z) G(z)^{-1} R(z)^* O_j, \quad \mathcal{N}_z = (I - P_z) \Delta, \quad G(z) = G_j + |z|^2 C, \quad (\text{DS40})$$

orthogonal projection gives

$$\mathcal{N}_z^* O_j \mathcal{N}_z = C - |z|^2 C G(z)^{-1} C, \quad \text{Tr}(G(z)^{-1} \mathcal{N}_z^* O_j \mathcal{N}_z) = \kappa(|z|^2). \quad (\text{DS41})$$

Indeed $R(z)^* O_j \Delta = \bar{z} C$; substitute into $\Delta^* O_j (I - P_z) \Delta$. Factoring $G(z) = G_j (I + |z|^2 Z)$ in the trace proves the second equality. Applying the actual remainder map gives

$$B_j P_z \Delta = \bar{z} G(z)^{-1} C, \quad B_j \mathcal{N}_z = -\bar{z} G(z)^{-1} C. \quad (\text{DS42})$$

Their sum is the original theta relation. In the split lift its quotient value is the supported zero e , with both prequotient pieces still given by (DS42).

The curvature jets are the retained relation moments:

$$\kappa^{(n)}(0) = (-1)^n (n+1)! \text{Tr}(Z^{n+1}), \quad t_1 = \kappa(0), \quad t_2 = -\frac{1}{2} \kappa'(0), \quad \mathcal{A}_2 = \frac{1}{2} (t_1^2 - t_2). \quad (\text{DS43})$$

These follow by differentiating each rational eigenvalue term in (DS38); finiteness of the matrix makes the differentiation exact. The dimension-one divisor degree in (DS39), the arithmetic trace in (DS35), and the source metric pole in (DS36) are now related by specified maps on the same compactified curve.

For completeness, exponentiation supplies the precise multiplicative observer of the original additive action, for every real t :

$$F_t = \exp(tA) : E \rightarrow E, \quad F_t|_{F_A} = \exp(tA|_{F_A}), \quad \bar{F}_t = \exp(t\bar{A}). \quad (\text{DS44})$$

On the full generalized eigenspace of A at ρ , write $A = \rho I + N$, $N^m = 0$ with its actual order. Then $F_t = e^{t\rho} \sum_{a=0}^{m-1} t^a N^a / a!$; in particular it retains the nilpotent terms and has eigenvalue modulus $e^{t \text{Re } \rho}$. Choosing $t = \log Q_0$ gives the exact number $Q_0^{\text{Re } \rho}$. This is the proved operator map connecting the additive spectral parameter and a multiplicative weight observer. It does not provide a finite-field descent or an étale Frobenius structure. Those structures in (DS7) have the specific types $X_0/\mathbb{F}_{Q_0}$, a smooth $\mathbb{Q}_\ell$ -sheaf and its geometric Frobenius action; the constructed characteristic-zero object has instead the fully calculated complexes (DS19), (DS25), (DS33) and the commuting action (DS44). Its complete comparison to those source operations is (DS1)–(DS9), (DS24), (DS27), and (DS34), with their actual kernels.

Finally the original positive metric supplies its precise additive weight measurement on every invariant inclusion $I_F : F \hookrightarrow E$:

$$G_F = I_F^* G(z) I_F, \quad W_F = A_F^* G_F + G_F A_F - k G_F, \quad \text{Tr}(G_F^{-1} W_F) = 2 \text{Re Tr } A_F - k \dim F. \quad (\text{DS45})$$

The identity follows by cyclicity of the finite trace, with $A I_F = I_F A_F$. Thus k remains the original tensor parameter, and all constituents of E are retained through (DS32)–(DS35). Equations (DS38)–(DS43) make the current endpoint estimate an estimate on a specified source-induced curvature measure. Its total degree alone does not bound its logarithmic moment: for the explicitly declared matrix family $G_j = I_r$, $C = L I_r$, $L > 0$, these quantities are respectively r and $r \log(1 + L)$, and the mass on $0 \leq t \leq 1/L$ is exactly $r/2$ by direct integration. The original arithmetic source, its relation map (DS13), its full dual comparison (DS25)–(DS34), and the actual moment observers (DS43) therefore remain the concrete data for the continuing curvature calculation.

55.1.8 Every divisor depth and the unchanged conormal directions

For a specified A -invariant subspace $F \subseteq E$ and an integer $m \geq 1$, retain

$$\mathcal{M}_{F,m} = F \mathcal{O}_X + E \mathcal{O}_X(-mD), \quad \mathcal{K}_{F,m} = [\mathcal{M}_{F,m} \xrightarrow{d} \mathcal{M}_{F,m} \otimes \Omega_X^1(\log D)]. \quad (\text{DS46})$$

The local coefficient differential on $w^m(E/F)$ is $w \partial_w + m$. The inclusion $\mathcal{K}_{F,m} \hookrightarrow \mathcal{K}_{F,1}$ has quotient supported at D , with coefficient basis $w, w^2, \dots, w^{m-1}$ in each of degrees zero and one and differential

$$w^a v \mapsto a w^a v dw/w \quad (1 \leq a < m). \quad (\text{DS47})$$

Define $h(w^a v dw/w) = a^{-1} w^a v$ on degree one and $h = 0$ on degree zero. Then $dh + hd = I$ on each displayed basis element, so this entire quotient is contractible. Thus the inclusion is a quasi-isomorphism, equivariant for the original A , and

$$\mathbb{H}^0(\mathcal{K}_{F,m}) = F, \quad \mathbb{H}^1(\mathcal{K}_{F,m}) = 0, \quad \mathbb{H}^2(\mathcal{K}_{F,m}) = (E/F) \otimes \mathbb{T}. \quad (\text{DS48})$$

This preserves every higher coefficient until its explicit differential (DS47) contracts it. In original arithmetic degree k the shifted complex is $\mathcal{K}_{F,m}[-k]$, so the two surviving degrees are $k, k+2$ and

$$\text{Str}(p(A) \mid R\Gamma \mathcal{K}_{F,m}[-k]) = (-1)^k \text{Tr}_E p(A). \quad (\text{DS49})$$

The equality follows from (DS35); multiplication by $p(A)$ acts trivially on the factor $\mathbb{T}$.

The exact higher-depth dual is

$$\mathcal{M}_{F,m}^\vee(-D) = F^{\text{ann}} \mathcal{O}_X((m-1)D) + E^* \mathcal{O}_X(-D) \subset E^* \otimes \mathcal{O}_X(*D). \quad (\text{DS50})$$

At infinity, testing a meromorphic functional on $w^m E$ permits poles of order at most $m-1$, and testing on F forces every coefficient of powers $w^{1-m}, \dots, w^0$ to annihilate F . Its remaining coefficients are $w E^* \mathcal{O}$. This proves (DS50), including its ambient meromorphic sheaf. In a coefficient complement its residues are 1 on F^* and $1-m$ on $(E/F)^*$. The natural inclusion of the depth-one dual into (DS50) has quotient basis $w^{1-m}, \dots, w^{-1}$ in both degrees, with differential $w^a v \mapsto a w^a v dw/w$ and inverse a^{-1} , so it is contractible. Therefore its dual cohomology is F^{ann} in degree zero and $F^* \otimes \mathbb{T}$ in degree two. Evaluation into $\mathcal{O}(-D)$ and (DS27) give the same perfect complementary-degree pairings. All negative powers, residues and the divisor twist have been retained.

The independent conormal depth has the explicit map

$$\frac{\mathbb{C}[w] \otimes \mathcal{P}}{(w^m) + I^a(\mathbb{C}[w] \otimes \mathcal{P})} \xrightarrow{\sim} (\mathbb{C}[w]/(w^m)) \otimes (\mathcal{P}/I^a), \quad [f(w)P] \mapsto [f] \otimes [P], \quad (\text{DS51})$$

where $\mathcal{P} = \mathbb{C}[s_1, \dots, s_k]$ and $I = (h(s_1), \dots, h(s_k))$. Reduction of the w -degree writes the left side uniquely as $\sum_{b=0}^{m-1} w^b [P_b]_{I^a}$. This supplies the inverse and proves injectivity as well as surjectivity.

Restriction in either depth commutes with the other. Differentiation in w has the well-defined type from depth $m + 1$ to depth m ; differentiation in s_j has type from I^{a+1} quotients to I^a quotients, because $\partial_{s_j} I^{a+1} \subset I^a$. On pure tensors the two derivatives commute, hence on their full spans. Lifting these maps by (DS10) retains the same support label and the receiving supported zero in both depth directions.

Pullback of (DS38) under the actual map $z \mapsto z^m$ has radial density

$$\kappa_m(t) = m^2 t^{m-1} \kappa(t^m), \quad t = |z|^2. \quad (\text{DS52})$$

Indeed differentiate $f(t^m)$ twice in the combination $f'_m + t f''_m$ used in (DS38). Substitution $s = t^m$ proves $\int_0^\infty \kappa_m(t) dt = mr$ and $\int_0^1 \log(1/t) \kappa_m(t) dt = \int_0^1 \log(1/s) \kappa(s) ds$. Thus increasing divisor depth multiplies degree by m while leaving the original endpoint logarithmic moment fixed.

55.1.9 The invariant quotient gives an exact coupling subtraction

The following quantitative calculation is the delivered logarithmic constituent-control result, now proved in the same source notation. Take $I : F \hookrightarrow E$ and $\pi : E \twoheadrightarrow E/F$ with $AI = IA_F$ and $\pi A = A_Q \pi$, and put $G(t) = G_0 + tC$, $G_0 = G_j$, $0 \leq t \leq 1$. The cases $F = 0$, E use the empty determinant 1, empty trace 0, and the unique maps on the zero-dimensional space; all formulas below still hold. For any initial coefficient lift $\overline{H}$, define

$$H_0 = \overline{H} - I(I^* G_0 I)^{-1} I^* G_0 \overline{H}, \quad \pi H_0 = I_{E/F}, \quad I^* G_0 H_0 = 0. \quad (\text{DS53})$$

These last two identities follow from $\pi I = 0$ and matrix multiplication. Changing $\overline{H}$ by IU cancels the change in (DS53); thus the orthogonal lift itself is independent of the initial lift. In the original bases define

$$\begin{aligned} G_{F,0} &= I^* G_0 I, \quad C_F = I^* C I, \quad D_F = I^* C H_0, \\ Q_0 &= H_0^* G_0 H_0, \quad C_Q = H_0^* C H_0, \quad G_F(t) = G_{F,0} + t C_F, \\ H_t &= H_0 - t I G_F(t)^{-1} D_F. \end{aligned} \quad (\text{DS54})$$

Here Q_0 is the original quotient Gram, not the finite-field cardinality used in (DS7). Every inverse is defined: I and H_0 are injective on their respective coefficient spaces, and $G(t) > 0$. In coordinates $[I, H_0]$ the original metric is exactly

$$\begin{pmatrix} G_F(t) & t D_F \\ t D_F^* & Q_0 + t C_Q \end{pmatrix}. \quad (\text{DS55})$$

The replacement $H_0 \mapsto H_t$ is the upper triangular coefficient change $\begin{pmatrix} I & -t G_F(t)^{-1} D_F \\ 0 & I \end{pmatrix}$ of determinant 1. It gives

$$I^* G(t) H_t = 0, \quad \pi H_t = I_{E/F}, \quad Q(t) := H_t^* G(t) H_t = Q_0 + t C_Q - t^2 D_F^* G_F(t)^{-1} D_F > 0. \quad (\text{DS56})$$

Expansion proves the formula, and injectivity of H_t proves positivity. It is also the quotient norm: any other lift of v is $H_t v + Ix$, whose squared norm is $v^* Q(t) v + x^* G_F(t) x$ by orthogonality. The unique minimum is therefore attained at $H_t v$.

Define the actual coupling map

$$\beta_t = G_F(t)^{-1} I^* C H_t : E/F \rightarrow F. \quad (\text{DS57})$$

Differentiate $\pi H_t = I$ to write $H'_t = I U_t$; differentiating $I^* G(t) H_t = 0$ then gives $I^* C H_t + G_F(t) U_t = 0$. Therefore $H'_t = -I \beta_t$. In the derivative of $Q(t) = H_t^* G(t) H_t$ the two terms containing H'_t vanish by orthogonality. Differentiating the remaining term gives

$$Q'(t) = H_t^* C H_t, \quad Q''(t) = -2 \beta_t^* G_F(t) \beta_t. \quad (\text{DS58})$$

The factor two follows because $I^*CH_t = G_F(t)\beta_t$ and its adjoint supply the two equal terms. The exact integral remainder for $Q(1)$ is $Q(1) = Q(0) + Q'(0) + \int_0^1 (1-t)Q''(t)dt$; insert (DS56) at the two endpoints to get

$$\Pi_F := D_F^* G_F(1)^{-1} D_F = 2 \int_0^1 (1-t) \beta_t^* G_F(t) \beta_t dt. \quad (\text{DS59})$$

This identifies the finite subtraction with an integrated squared map on the original quotient source.

The action defect is the unique map $a_t : E/F \rightarrow F$ with

$$AH_t - H_t A_Q = I a_t, \quad a'_t = \beta_t A_Q - A_F \beta_t. \quad (\text{DS60})$$

The first left side has zero image under π , so it lies in $\text{im } I$, which proves existence and uniqueness. Differentiating and inserting $H'_t = -I\beta_t$ proves the second formula, using injectivity of I . Thus the nilpotent extension data are retained in a_t even when the two constituent actions have equal eigenvalues.

For clarity the entire Sylvester operator has a finite exact description. On a pair of primary spaces let $A_F = \lambda I + N_F$, $A_Q = \mu I + N_Q$, with $N_F^a = 0$ and $N_Q^b = 0$. Put $\mathcal{N}(X) = XN_Q - N_F X$. The two multiplications commute, so the binomial expansion makes $\mathcal{N}^{a+b-1} = 0$. If $\mu \neq \lambda$, the inverse of $X \mapsto XA_Q - A_F X$ is

$$Y \mapsto \sum_{r=0}^{a+b-2} \frac{(-1)^r}{(\mu - \lambda)^{r+1}} \mathcal{N}^r(Y). \quad (\text{DS61})$$

Multiplying the finite geometric series by $(\mu - \lambda)I + \mathcal{N}$ leaves identity and a zero terminal power. At $\mu = \lambda$, the operator is exactly $\mathcal{N}$; its kernel is the space $\{X : XN_Q = N_F X\}$ and its image is the span of the explicit matrices $E_{uv}N_Q - N_F E_{uv}$. Matrix-unit expansion proves that this span is its whole image. This keeps the kernel and every coefficient of the image without a nonexistent inverse.

The same original control $W(t) = A^*G(t) + G(t)A - kG(t)$ obeys

$$\begin{aligned} I^*W(t)I &= A_F^*G_F(t) + G_F(t)A_F - kG_F(t), \\ H_t^*W(t)H_t &= A_Q^*Q(t) + Q(t)A_Q - kQ(t). \end{aligned} \quad (\text{DS62})$$

The first identity uses invariance of F . For the second insert (DS60); both terms containing Ia_t vanish by (DS56). On the original theta source write $r(z) = \mathcal{V}R(z)$ and retain its actual primitive $D^{(k)}r(z) - r(z)A = dK(z)$, with $K(z) = (1-z)K_j + zK_i$ in the inherited primitive notation. Then exact multiplication gives

$$D^{(k)}r(z)H_t - r(z)H_t A_Q = dK(z)H_t + r(z)Ia_t, \quad t = |z|^2. \quad (\text{DS63})$$

Here $D^{(k)}$ differentiates the original source variables, not the comparison parameter z or t . The first term is killed by the original theta quotient. The second has its retained image in the arithmetic constituent F and is killed by the further quotient $E \rightarrow E/F$. Formula (DS63) proves the precise map between these two successive supported-zero operations.

Put $Q_+ = Q_0 + C_Q$, $\Xi_F = Q_+^{-1}\Pi_F$. Then Ξ_F is self-adjoint and nonnegative for Q_+ , and $Q(1) = Q_+ - \Pi_F > 0$ proves that all its eigenvalues belong to $[0, 1)$. Taking determinants in (DS55)–(DS56), with the fixed coefficient determinant $|\det[I, H_0]|^2$ cancelled only between the same two original endpoints, gives

$$\begin{aligned} \log \frac{V_i}{V_j} &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} + \log \det(I - \Xi_F) \\ &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}. \end{aligned} \quad (\text{DS64})$$

For each eigenvalue $0 \leq x < 1$, integrating the geometric series $\sum_{a \geq 0} x^a = (1 - x)^{-1}$ from zero to x proves the logarithm series. There are finitely many eigenvalues, so their sum yields the second equality and every term is nonnegative. Its first p terms therefore give the stated upper estimate, and these estimates decrease to the exact original endpoint volume.

There is also an explicit remainder from the same matrices. If $\rho = \lambda_{\max}(\Xi_F) < 1$, then for every $p \geq 0$,

$$0 \leq \sum_{r=p+1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r} \leq \frac{\text{Tr}(\Xi_F^{p+1})}{(p+1)(1-\rho)}. \quad (\text{DS65})$$

Indeed $\text{Tr}(\Xi_F^{p+1+a}) \leq \rho^a \text{Tr}(\Xi_F^{p+1})$ on each nonnegative eigenvalue, and $p+1+a \geq p+1$; summing the resulting geometric series proves the inequality. The value ρ is the largest generalized eigenvalue of (Π_F, Q_+) of this same source pair, with no independently assumed uniform gap. On the zero-dimensional quotient $E/F = 0$, define $\rho = 0$; every trace and both sides of (DS65) are then zero, so the same formula includes that endpoint.

The two positive terms in (DS64) are the logarithmic determinants of the compressed positive operators $Z_F = G_{F,0}^{-1} C_F$ and $Z_Q = Q_0^{-1} C_Q$. Consequently the already proved two-trace majorant, applied to each of these specified matrices in its own full dimension, can be combined with every finite subtraction in (DS64). This integration retains the extension map (DS60), the quotient source (DS63), and the original total curvature (DS39) together; the cohomological constituent decomposition does not remove its measurable coupling.

55.1.10 The degree-one exponential family joins the same coefficient object

The delivered exponential-comparison note constructs the additional degree-one family from the unchanged polynomial $\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a$. Its immediate join to the present cohomology calculation is as follows. Use new independent parameters u, t and retain every coefficient of

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1}, \quad \Phi_t(S) = \Phi(S) - tS, \quad R = \mathbb{C}[u, t]. \quad (\text{DS66})$$

The two-term coefficient complex is

$$\mathcal{C} = [R[S] \xrightarrow{\mathcal{D}} R[S]dS], \quad \mathcal{D}P = (uP' + (\chi - t)P)dS. \quad (\text{DS67})$$

For a nonzero polynomial P of degree a and leading coefficient $b \in R$, $\mathcal{D}P/dS$ has degree $a + q$ and leading coefficient b . Successively subtracting $\mathcal{D}(bS^a)/dS$ cancels each highest term in a polynomial of degree at least q and terminates. Uniqueness follows from the same degree calculation. Hence $R[S] = \text{im}(\mathcal{D}/dS) \oplus R[S]_{<q}$ as coefficient modules, $H^0(\mathcal{C}) = 0$, and $\mathcal{H} = H^1(\mathcal{C})$ is free with the original basis $[1], \dots, [S^{q-1}]$. At $u = t = 0$ the differential is multiplication by the literal χ , so $\mathcal{H}/(u, t)\mathcal{H} = E$ with every nilpotent jet.

The commutators are $[\mathcal{D}/dS, \partial_t] = I$ and $[\mathcal{D}/dS, -S/u] = -I$ after inverting u . They prove that $\nabla_t = \partial_t - S/u$ descends through the differential. Define $\mathcal{R} : E \rightarrow E$ by sending a remainder to its S^{q-1} coefficient times $[1]$. Reducing S^q with the relation $[\chi - t] = 0$ shows in the displayed basis that

$$\nabla_t = \partial_t - \frac{A + t\mathcal{R}}{u}, \quad (-u\nabla_t) \bmod (u, t) = A. \quad (\text{DS68})$$

The regular operator $-u\partial_t + A + t\mathcal{R}$ preserves $(u, t)\mathcal{H}$: on uv its derivative contribution is $-u^2\partial_t v$, and on tv it is $-uv - ut\partial_t v$. This verifies the asserted specialization as an actual induced operator, rather than specializing a singular connection symbol.

The same family now also has the full u -connection. Use the literal division (XD.u1) for this χ , with matrices $C_\chi(t)$, B_χ and $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$. The connections on the two terms of (DS67) are

$$\partial_u - \Phi_t/u^2 + 1/u \quad \text{and} \quad \partial_u - \Phi_t/u^2, \quad (\text{DS70})$$

in degrees zero and one. Their chain identity is the commutator $[u\partial_S + \chi - t, \partial_u - \Phi_t/u^2] = -(u\partial_S + \chi - t)/u$, proved before quotienting in (XD.u4). On an ordered product of k such polynomial factors, the same calculation in cochain degree j gives

$$\nabla_u^{(j)} = \partial_u - \frac{\sum_i \Phi_{t_i}(S_i)}{u^2} + \frac{k-j}{u}. \quad (\text{DS71})$$

Each summand of the exterior differential has commutator $-L_i/u$; the adjacent-degree scalar difference is $+L_i/u$, so all terms cancel with their original exterior signs. This includes the non-top degrees rather than imposing the top formula on them.

There is an exact source lift for the fixed cyclic observer already used in (DS69). On parameter-dependent vectors of its original admitted theta-source domain define

$$\nabla_{u,N}^{\text{src}} = \partial_u + r_N \Omega_u j_E, \quad \nabla_{t,N}^{\text{src}} = \partial_t + r_N \Omega_t j_E, \quad \Omega_t = -(A + t\mathcal{R})/u. \quad (\text{DS72})$$

The domain is the original one on which j_E , $D^{(k)}$, r_N and the boundary primitive are defined, with holomorphic parameter coefficients on a branch of $u \neq 0$. The maps j_E, r_N are constant in these parameters. Multiplying by $j_E r_N = I$ gives

$$j_E \nabla_{a,N}^{\text{src}} = (\partial_a + \Omega_a) j_E, \quad \nabla_{a,N}^{\text{src}} r_N = r_N (\partial_a + \Omega_a), \quad a = u, t. \quad (\text{DS73})$$

The source splitting is the exact identity $x = r_N j_E x + (x - r_N j_E x)$, with its second term in $\ker j_E$. On that kernel (DS72) is ordinary parameter differentiation. Since $r_N j_E r_N j_E = r_N j_E$, the full source curvature equals $r_N (\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t]) j_E = 0$ by (XD.u6). The original source differential still gives, for each $a = u, t$,

$$[D^{(k)}, \nabla_{a,N}^{\text{src}}] = r_N [A, \Omega_a] j_E + dK_N \Omega_a j_E. \quad (\text{DS74})$$

To prove this, insert $D^{(k)} r_N = r_N A + dK_N$ and $j_E D^{(k)} = A j_E$ into the commutator; $D^{(k)}$ commutes with parameter differentiation. Every term has the original source target, and $K_N \Omega_a j_E$ is the displayed full primitive. Thus the parameter-flat source observation and the original Euler action are related by their exact commutator, retaining its coefficient part and theta boundary. No source norm or full Taylor unit in r_N has been changed.

The logarithmic comparison retains the original exact coefficient sequence $0 \rightarrow F \rightarrow E \rightarrow E/F \rightarrow 0$, with the outer terms occurring in (DS48). The common full E is obtained through its actual outer comparison complexes: $\mathbb{H}^0(\mathcal{K}_{E,1}) = E$ and $\mathbb{H}^2(\mathcal{K}_{0,1}) = E \otimes \mathbb{T}$. The maps to and from the mixed complex are $0 \rightarrow F \rightarrow E$ and $E \otimes \mathbb{T} \rightarrow (E/F) \otimes \mathbb{T} \rightarrow 0$. At the exponential special fibre the cochain map $\mathbb{C}[S] \xrightarrow{\chi} \mathbb{C}[S] dS \rightarrow E[-1]$ is zero in degree zero and $P dS \mapsto [P]_\chi$ in degree one. Its kernel complex is the isomorphism $\mathbb{C}[S] \xrightarrow{\chi} \chi \mathbb{C}[S] dS$ and is contractible by division by χ on its image. This proves the special-fibre quasi-isomorphism. The resulting map to the full coefficient object of the logarithmic outer complexes is the identity on $[1], \dots, [S^{q-1}]$, intertwining A . Composing with $E \rightarrow E\mathcal{O}_X$, $v \mapsto v \otimes 1$, gives the map into the original constant target of (DS32). It retains the coefficient extension and its map a_t in (DS60). The original theta comparison $j_E r_N = I$, $j_E D^{(k)} = A j_E$ makes this identity a source comparison as well:

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t\mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t\mathcal{R}) = dK_N. \quad (\text{DS69})$$

Both equalities follow by multiplying and using $j_E r_N = I$; the added terms cancel exactly in the second equality. Thus the source primitive is unchanged and the full kernel $\ker j_E$ remains. The exact period equation (XD.u5), chain terms (DS70)–(DS71), and source connection and commutator (DS72)–(DS74) now develop this same family in both parameters. The finite-field boundary extension and the full single-primary calculation are included with their exact specialization hypotheses. Equations (DS66)–(DS74) give the coefficient, cohomological and source-level join used here.

56 Exact determinant of the polynomial-exponential period comparison

56.1 Exact determinant of the polynomial-exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix $u \in \mathbb{C}^*$ and one argument of u . All the following comparisons retain this u , the original coordinate S , and the ordered remainder basis $1, S, \dots, S^{q-1}$. Set

$$D_{u,t}P = uP' + (\chi - t)P, \quad H_{u,t} = \mathbb{C}[S]/D_{u,t}\mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let $\text{red}_{u,t}$ be its unique representative of degree less than q . Let $A(t)$ be the multiplication matrix of S on $\mathbb{C}[S]/(\chi - t)$ in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case $q = 1$.

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{jb}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

56.1.1 Uniform estimates on the fixed unbounded contours

For a compact set $K \subset \mathbb{C}^{q+1}$ of the parameters $(c_0, \dots, c_{q-1}, t)$, put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies $S^d/(du) = -r^d/(d|u|)$. For $r \geq 1$ every other exponent has degree at most $d-1$, so

$$\text{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in S : a derivative in c_a contributes $S^{a+1}/((a+1)u)$ and a derivative in t contributes $-S/u$. For any fixed derivative order its absolute value on the tail is bounded by $Cr^L e^{-\alpha r^d}$, with C, L uniform on K . The required majorant is integrable, since substitution $x = \alpha r^d$ gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and t ; every parameter derivative may be taken under the

integral. The same bound gives $e^{\Phi_t(S)/u}P(S) \rightarrow 0$ at the infinite endpoint of each ray, for every polynomial P . Consequently

$$\int_{\Gamma_j} e^{\Phi_t(S)/u} D_{u,t} P dS = u \int_{\Gamma_j} d(e^{\Phi_t(S)/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is $-e^{\Phi_t(0)/u}P(0) = -P(0)$; the coefficients $-1, +1$ of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

56.1.2 An exact trace comparison for twisted reduction

For $1 \leq m \leq q$ and $0 \leq b < q$, ordinary monic division gives unique polynomials $Q_{m,b}, R_{m,b}$ satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If $m + b < q$, take $Q_{m,b} = 0$. Otherwise $\deg Q_{m,b} = m + b - q \leq q - 1$, and the differential itself gives

$$\text{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below q . In every nonzero case,

$$\deg Q'_{m,b} \leq m + b - q - 1 \leq b - 1. \quad (\text{XD10})$$

Thus the coefficient of S^b in the derivative correction vanishes. Let $T_m(u, c, t)$ have column b equal to the coordinates of the left side of (XD9). The columns $R_{m,b}$ are exactly those of $A(t)^m$. Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\text{Tr } T_m(u, c, t) = \text{Tr } A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set $B_a = T_{a+1}/((a+1)u)$ for $0 \leq a < q$. Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \text{Tr } B_a = \frac{\text{Tr } A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in c, t for this fixed nonzero u .

56.1.3 The full singular parameter connection on this same cyclic family

The original determinant (XD19) is retained. Its complete parameter connection is now computed on the same coefficient quotient, before taking traces. For every original column $0 \leq b < q$, perform the unique monic division

$$\Phi_t(S)S^b = (\chi(S) - t)Q_b^\Phi(S, t) + R_b^\Phi(S, t), \quad \deg R_b^\Phi < q, \quad \deg Q_b^\Phi \leq b + 1. \quad (\text{XD.u1})$$

Write $C_\chi(t)$ for the matrix of the remainders and B_χ for the matrix of the derivatives $(Q_b^\Phi)'$, in the unchanged basis. The differential gives the exact, already reduced identity

$$\text{red}_{u,t}(\Phi_t S^b) = R_b^\Phi - u(Q_b^\Phi)', \quad C_\chi(t) = \Phi_t(A(t)). \quad (\text{XD.u2})$$

The second equality uses ordinary multiplication in $\mathbb{C}[S]/(\chi - t)$; the first retains its differential correction. Comparing division coefficients above degree q shows that only the constant coefficient of Q_b^Φ can depend on t : both $-tS^{b+1}$ and $-tQ_b^\Phi$ have degree at most q . Differentiation removes that constant. Its leading term is $S^{b+1}/(q+1)$. Therefore

$$(B_\chi)_{bb} = \frac{b+1}{q+1}, \quad (B_\chi)_{rb} = 0 \ (r > b), \quad \partial_t B_\chi = 0, \quad \text{Tr } B_\chi = q/2. \quad (\text{XD.u3})$$

On degrees zero and one of the original two-term complex the connections are respectively

$$\nabla_u^0 = \partial_u - \Phi_t/u^2 + 1/u, \quad \nabla_u^1 = \partial_u - \Phi_t/u^2. \quad (\text{XD.u4})$$

Indeed, writing $L = u\partial_S + \chi - t$, direct commutators give $[L, \partial_u] = -\partial_S$ and $[L, -\Phi_t/u^2] = -(\chi - t)/u$. Hence $\nabla_u^1 L = L \nabla_u^0$. Reduction by (XD.u2) proves the full top connection and period differential:

$$\Omega_u = -C_\chi(t)/u^2 + B_\chi/u, \quad \nabla_u = \partial_u + \Omega_u, \quad \partial_u \Pi = \Pi \Omega_u, \quad \Omega_t = -A(t)/u, \quad \partial_t \Pi = \Pi \Omega_t. \quad (\text{XD.u5})$$

All these statements hold locally on every retained branch of $u \neq 0$. One may deform the labelled contours within the same decay sectors; the tail estimate (XD5), locally uniform in $u \neq 0$, removes the connecting arcs. Differentiation in u inserts $-\Phi_t/u^2$, so (XD.u2) proves the asserted period equation without changing a ray orientation. The scalar commutators $[\partial_u - \Phi_t/u^2, \partial_t - S/u] = 0$, with the same degree-zero correction, prove flatness. Equivalently the retained matrices obey

$$\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] = 0. \quad (\text{XD.u6})$$

Taking the exterior power of (XD.u5) gives $\partial_u \det \Pi = (-\mathcal{F}/u^2 + q/(2u)) \det \Pi$, where $\mathcal{F}$ is precisely (XD13). If $\zeta = e^{2\pi i/(q+1)}$, $\beta_b = (b+1)/(q+1)$ and $W_{jb} = \zeta^{j(b+1)} - 1$, its retained constant is

$$K_q = (q+1)^{-q/2} e^{\pi i q/2} \det W \prod_{b=0}^{q-1} \Gamma(\beta_b), \quad \det \Pi = K_q u^{q/2} e^{\mathcal{F}/u}. \quad (\text{XD.u7})$$

This is exactly (XD16)–(XD19): each original column factor is $[(q+1)u]^{\beta_b} e^{\pi i \beta_b} \Gamma(\beta_b)/(q+1)$ on the specified branch, and $\sum_b \beta_b = q/2$. It adds the full singular connection to the earlier determinant proof; the determinant itself is the same proved result.

The packet polynomial h in the marked-product calculation and the cyclic polynomial $\chi_{h,k}$ have distinct specified coefficient maps. Formula (XD.u1) applies literally to the present χ . For $k=1$, $\chi_{h,1} = h$ via $S \mapsto s$, and every displayed matrix is the matrix C_h, B_h under that exact variable isomorphism. For general k the product lattice is instead connected to E at the marked fibre by the original injection $\eta_k[P] = v_h^{\otimes k} P(\sum_i s_i)1$. Its full source identity is $q_{\mathcal{B}}^{\otimes k} \mathcal{V}_{h,k} P = \sigma_h^{\otimes k} \eta_k[P]$. In the unweighted polynomial product family its cyclic relation has the explicit parameter image $[\chi(S)P(S)] = \sum_i [t_i Q_i - u \partial_i Q_i]$ whenever $\chi(S)P(S) = \sum_i h(s_i)Q_i$ in the original polynomial ring. Thus the two connections have a common specified marked-fibre map; no equality of the two families or their ranks is inserted.

56.1.4 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix A and integer $n \geq 1$, the product rule and cyclic invariance of trace give

$$d \text{Tr } A^n = \sum_{r=0}^{n-1} \text{Tr}(A^r (dA) A^{n-1-r}) = n \text{Tr}(A^{n-1} dA). \quad (\text{XD14})$$

Because $A(t)$ acts on the actual quotient by $\chi - t$, it satisfies $\chi(A(t)) - tI_q = 0$. Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt + \text{Tr}((\chi(A(t)) - tI_q) dA(t)) = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt. \quad (\text{XD15})$$

At $c = t = 0$ the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u) e^{il\theta_0} (\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d} \Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting $x = r^d/(d|u|)$ on each oriented ray in (XD3). In particular all $a_l(u)$ are positive real numbers. For $V_{jl} = \zeta^{jl} - 1$ one may include $j = 0$ in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here $1 \leq l, m < d$, so the two individual nonconstant sums vanish, and the first sum is $d\delta_{lm}$. The all-ones rank-one matrix has eigenvalue q on the all-ones vector and zero on its $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[\frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase: $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{il\theta_0}$.

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join $(0, 0)$ to (c, t) by (sc, st) , $0 \leq s \leq 1$. Equations (XD12) give $\Pi' = \Pi B(s)$, with continuous $B(s)$, and (XD15) gives $\text{Tr } B(s) = \mathcal{F}'(sc, st)/u$. For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = \det \Pi \text{Tr } B. \quad (\text{XD20})$$

Consequently $\det \Pi \exp(-\mathcal{F}/u)$ has derivative zero. At the starting point A is nilpotent and $\mathcal{F}(0, 0) = 0$, so its value is $\det \Pi^{\text{mon}}(u)$. This proves (XD19). Equation (XD18) proves its nonvanishing for every c, t , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \text{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by Π^{-1} followed by the specified degree-below- q section.

56.1.5 Exact determinant transport to the arithmetic representative

Retain the original positive Gram G_N on the cyclic coefficient space E , its determinant $V_N = \det G_N$, and the original maps $r_N : E \longrightarrow \mathcal{B}^{\otimes k}$ and j_E on the admitted test-function domain with $j_E r_N = I_E$. Identify E with $H_{u,t}$ by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \longrightarrow \mathbb{C}^q, \quad x \longmapsto \Pi j_E x, \quad \mathbb{C}^q \longrightarrow r_N(E), \quad y \longmapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively $r_N j_E$ on $r_N(E)$ and $\Pi(j_E r_N) \Pi^{-1}$ on $\mathbb{C}^q$. On the full admitted domain the first observation has precisely $\ker j_E$ as its kernel; all original theta boundaries killed by j_E remain in that kernel.

With the explicitly fixed coordinate metric I_q on the target, put

$$H_{\Pi} = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_{\Pi}. \quad (\text{XD23})$$

These operators satisfy $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_{\Pi} > 0$. Taking determinants in (XD19) gives the exact identities

$$\det H_{\Pi} = C_q(u) \exp\left(2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\log \det B_N^{\text{per}} = \log C_q(u) + 2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N. \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same u, c, t for two admitted degrees $i \leq j$, subtraction proves

$$\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j). \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on V_i/V_j is inferred from this determinant identity.

57 Translation of the exponential comparison: full coefficient, period, metric, and finite-field maps

57.1 The original fibre and the auxiliary translation

The September 13 exponential comparison starts with the actual arithmetic polynomial and sum operator

$$\chi(S) = S^q + \sum_{b=0}^{q-1} c_b S^b, \quad q > 0, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S,$$

and the specified primitive

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{b=0}^{q-1} \frac{c_b S^{b+1}}{b+1}.$$

Its cohomology is the cokernel of $D_{u,t} = u\partial_S + \chi(S) - t$, with the omitted differential symbol understood as the coefficient identification with one-forms. All formulas below retain this original object at $a = 0$. The parameter $a \in \mathbb{C}$ defines an auxiliary comparison family; a translated polynomial is not asserted to be a new zeta zero packet.

Set

$$\chi_a(S) = \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \quad T_a P(X) = P(X + a). \quad (\text{DT.1})$$

Thus $\Phi_a(0) = 0$, and no additive constant in the potential is lost. Use the same substitution on coefficients of one-forms, with $dS = dX$.

Proposition 57.1 (The coefficient and connection maps). *The map T_a is an isomorphism of the two-term polynomial complexes for χ_a and χ , over $\mathbb{C}[u, t]$. It induces an isomorphism of their free rank- q cohomology modules. On the special fibre $(u, t) = (0, 0)$,*

$$T_a : E_a = \mathbb{C}[S]/(\chi_a) \xrightarrow{\sim} E, \quad T_a A_a T_a^{-1} = A + aI. \quad (\text{DT.2})$$

Every primary multiplicity and nilpotent Jordan block is retained. For $u \neq 0$, the descended parameter connections satisfy

$$T_a \nabla_t^a T_a^{-1} = \nabla_t - \frac{a}{u} I, \quad \nabla_t = \partial_t - \frac{A(t)}{u}, \quad (\text{DT.3})$$

where $A(t)$ is the original companion matrix for $\chi - t$.

Proof. Substitution gives, for every polynomial P ,

$$T_a(uP' + (\chi_a - t)P) = u\partial_X(T_aP) + (\chi(X) - t)T_aP.$$

The inverse is substitution by $-a$. A polynomial of degree n with leading coefficient $b \neq 0$ is sent by either differential to a polynomial of degree $n + q$ with the same leading coefficient. Successive subtraction of such images therefore gives a unique representative of degree less than q , over the coefficient ring. This proves the stated freeness and the base-change identification. On the special fibre, $T_a(SP) = (X + a)T_aP$, which proves DT.2. In particular, $(A_a - (\lambda + a)I)^m$ is conjugate to $(A - \lambda I)^m$ for every m , proving the multiplicity statement without discarding the nilpotent part. On arbitrary polynomial representatives, the connection is $\partial_t - S/u$. Since T_a is independent of t , its conjugate is $\partial_t - (X + a)/u$. The commutator with the differential vanishes, so this equality descends and yields DT.3. $\square$

In the fixed bases $1, S, \dots, S^{q-1}$ and $1, X, \dots, X^{q-1}$, the matrix of T_a is

$$(C_a)_{rb} = \begin{cases} \binom{b}{r} a^{b-r}, & 0 \leq r \leq b < q, \\ 0, & r > b, \end{cases} \quad C_a^{-1} = C_{-a}, \quad \det C_a = 1. \quad (\text{DT.4})$$

The inverse and composition rules follow by composing the literal substitutions. In particular, $C_a A_a(t) C_a^{-1} = A(t) + aI$.

57.2 Periods and the retained exponential factor

Fix $u \neq 0$, put $d = q + 1$, and use the original rays

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \Gamma_j = -\ell_0 + \ell_j, \quad 1 \leq j \leq q,$$

oriented as in the exponential comparison. Let $\Pi(u, t)$ denote its period matrix, whose (j, b) entry is $\int_{\Gamma_j} e^{(\Phi(X) - tX)/u} X^b dX$. Let Π_a use the identical labelled ray prescription for Φ_a in the S -plane.

Proposition 57.2 (Exact period translation). *Define*

$$h_a(t) = -\Phi(-a) - ta, \quad f_a(u, t) = \exp(h_a(t)/u). \quad (\text{DT.5})$$

Then

$$\Pi_a(u, t) = f_a(u, t) \Pi(u, t) C_a. \quad (\text{DT.6})$$

The map $f_a T_a$ intertwines the two parameter connections in DT.3.

Proof. The substitution $S = X + a$ gives the exact identity

$$\Phi_a(S) - tS = \Phi(X) - tX + h_a(t).$$

It changes the contour to $\Gamma_j - a$. This translated contour may be moved to Γ_j without changing the integral. Indeed, truncate the two rays at radius R , join their translated endpoints by segments of length $|a|$, and apply Cauchy's theorem to the resulting closed path. The entire integrand has no finite singularity. On each joining segment its leading exponent is $-R^d/(d|u|) + O(R^{d-1})$, uniformly along that segment; a polynomial factor grows at most polynomially. Its integral tends to zero. The two portions at the finite starting point have opposite orientation and cancel. Absolute convergence on the original and translated rays follows from the same estimate. Finally, $(X + a)^b$ is the column expansion DT.4. This proves DT.6 including the scalar and all phases.

Since $\partial_t f_a = -(a/u)f_a$, for every section v one has

$$\nabla_t(f_a T_a v) = f_a \left(\nabla_t T_a v - \frac{a}{u} T_a v \right) = f_a T_a \nabla_t^a v.$$

The scalar f_a is nonzero on $u \neq 0$; no extension of this exponential gauge to $u = 0$ is used. DT.2 is the separately proved coefficient specialization at that fibre. $\square$

57.3 Translation of the full singular connection and boundary character

The same retained scalar $h_a(t) = -\Phi(-a) - ta$ also controls the u direction. Write ∇_u^a for the degree-one connection of the translated phase and ∇_u for that of the original phase, both including their exact matrices (XD.u1)–(XD.u5). Literal substitution in the full polynomial complex gives

$$T_a \nabla_u^a T_a^{-1} = \nabla_u - \frac{h_a(t)}{u^2} I, \quad \nabla_u(f_a T_a v) = f_a T_a \nabla_u^a v. \quad (\text{DT.14})$$

Indeed $T_a \Phi_t^a T_a^{-1} = \Phi_t + h_a(t)$ and T_a is independent of u . Differentiating the unchanged exponential gives $\partial_u f_a = -h_a(t) f_a / u^2$, proving the second identity term by term. In degree zero each side additionally contains the same $1/u$ term. Together with DT.3 and DT.6 this proves a flat two-parameter intertwiner on $u \neq 0$, with its singular scalar retained. In matrix form, if Ω_u^a, Ω_u are the two original remainder matrices,

$$C_a \Omega_u^a C_a^{-1} = \Omega_u - h_a(t) I / u^2. \quad (\text{DT.15})$$

It follows equally by differentiating DT.6 in u , using invertibility of the original period matrix. Its determinant has the scalar f_a^q , in agreement with the complete constant and phase in (XD.u7).

Before fixing $u = u_0$ in the finite-field calculation, the same polynomial substitution is a morphism over the actual parameter scheme. Use the specified coefficient map $R_0[a] \rightarrow \mathbb{F}_Q$ of characteristic $p > d = q + 1$, the same $\ell \neq p$, and the nontrivial additive character ψ fixed in the finite-field comparison below. Write $\mathcal{U} = \mathbb{G}_{m, \mathbb{F}_Q, u} \times \mathbb{A}_{\mathbb{F}_Q, t}^1$, and let $\pi_X : \mathbb{A}_X^1 \times \mathcal{U} \rightarrow \mathcal{U}$ and $\pi_S : \mathbb{A}_S^1 \times \mathcal{U} \rightarrow \mathcal{U}$ be the literal projections. Define the full derived complexes

$$\mathcal{F} = R\pi_{X!} \mathcal{L}_\psi((\Phi(X) - tX)/u), \quad \mathcal{F}_a = R\pi_{S!} \mathcal{L}_\psi((\Phi_a(S) - tS)/u).$$

The base-defined isomorphism $\sigma_a(X, u, t) = (X + a, u, t)$ satisfies $\pi_S \sigma_a = \pi_X$. Its exact derived compact-support direct-image identity, in every cohomology degree, is

$$\mathcal{F}_a \simeq \mathcal{F} \otimes \mathcal{L}_\psi(h_a(t)/u). \quad (\text{DT.16})$$

The Artin–Schreier torsor addition proving DT.12 works over this coefficient ring without any change, and the character line is pulled back from the parameter scheme, so it factors out of the compact-support complex. Restricting to a specified (u_0, t_0) is exactly DT.12–DT.13. At fixed t_0 , its possible singular boundary character is therefore the full $\mathcal{L}_\psi(h_a(t_0)/u)$; it cannot be replaced by its value at a nonzero u_0 . This is the precise family map used when passing from the translated single-primary phase to its local boundary representation.

57.4 The original positive forms and their control operators

The period matrices used below are invertible on the stated domain $u \neq 0$. Here is a complete verification needed for the determinant ratios. At the monomial potential, the substitution $x = r^d / (d|u|)$ in the absolutely convergent ray integral gives

$$\Pi_{j,b}^{\text{mon}} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} (\zeta^{j(b+1)} - 1), \quad \zeta = e^{2\pi i/d},$$

where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx > 0$ for $s > 0$. The column factors are all nonzero. A dependence among the remaining columns would give a polynomial $\sum_{l=1}^q b_l (Z^l - 1)$ of degree at most q , vanishing at all $q + 1$ roots of unity, including 1. This polynomial is zero, so every $b_l = 0$.

Join the monomial coefficients to the given (c, t) by a straight path with real parameter $s \in [0, 1]$, keeping u fixed. On this compact coefficient path the leading negative ray term dominates all lower terms uniformly: for sufficiently large r the absolute value of any fixed coefficient derivative of the integrand is at most $C r^L \exp[-r^d / (2d|u|)]$. This is integrable by the same gamma substitution; hence

differentiation under every contour integral is valid. A derivative in c_b multiplies the integrand by $S^{b+1}/((b+1)u)$, and one in t multiplies it by $-S/u$. The unique reduction proved above, together with the vanishing of $\int_{\Gamma_j} e^{(\Phi-tS)/u} D_{u,t} P dS$, thus gives $\Pi'(s) = \Pi(s)B(s)$ with a continuous matrix $B(s)$. That integral vanishes because it is u times an exact derivative; both infinite endpoint values are zero and the two finite endpoint values cancel with the specified contour signs. The determinant identity

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi)\Pi B) = (\det \Pi) \text{Tr} B$$

holds also at singular matrices. Therefore $\det \Pi(s) = \det \Pi(0) \exp(\int_0^s \text{Tr} B(v) dv) \neq 0$. This proves invertibility without removing repeated-root parameters. Equation DT.6 proves it for the translated matrix as well.

Let G_N be any one of the original canonical arithmetic Gram matrices, and retain its index N , cochain degree k , and control form $W_N = A^* G_N + G_N A - k G_N$. The exact pullback along DT.2 is

$$G_N^{[a]} = C_a^* G_N C_a. \quad (\text{DT.7})$$

This is a transported specified form. No claim identifies it with a newly computed canonical theta form for a translated polynomial.

Proposition 57.3 (Control and determinant transport). *For the transported form and the original value of k ,*

$$W_N^{[a]} = A_a^* G_N^{[a]} + G_N^{[a]} A_a - k G_N^{[a]} = C_a^* (W_N + (a + \bar{a}) G_N) C_a, \quad (\text{DT.8})$$

$$\text{Tr}((G_N^{[a]})^{-1} W_N^{[a]}) = \text{Tr}(G_N^{-1} W_N) + 2q \text{Re } a. \quad (\text{DT.9})$$

Write $H_\Pi = \Pi^* \Pi$ and $B_N^{\text{per}} = G_N^{-1} H_\Pi$. Then

$$B_N^{\text{per},[a]} := (G_N^{[a]})^{-1} \Pi_a^* \Pi_a = |f_a|^2 C_a^{-1} B_N^{\text{per}} C_a. \quad (\text{DT.10})$$

Consequently, for the same (u, t) and the same original indices,

$$\frac{\det B_j^{\text{per},[a]}}{\det B_i^{\text{per},[a]}} = \frac{\det B_j^{\text{per}}}{\det B_i^{\text{per}}} = \frac{\det G_i}{\det G_j}. \quad (\text{DT.11})$$

Proof. Insert $C_a A_a C_a^{-1} = A + aI$ into the first definition of DT.8. Taking the inverse Gram and using invariance of trace under similarity gives DT.9. Equation DT.6 gives $\Pi_a^* \Pi_a = |f_a|^2 C_a^* H_\Pi C_a$, proving DT.10. Its determinant contains the factor $|f_a|^{2q}$; this same factor occurs at both indices and cancels in their ratio. The remaining period Gram also cancels, giving DT.11. Thus the original endpoint determinant cost is retained by this exact comparison. $\square$

57.5 The finite-field comparison with its scalar character

Let $R_0 \subset \mathbb{C}$ be the finitely generated coefficient ring of the original exponential construction, and adjoin a . For a specified map $R_0[a] \rightarrow \mathbb{F}_Q$ of characteristic $p > d$, and specified $u_0 \in \mathbb{F}_Q^\times$, $t_0 \in \mathbb{F}_Q$, reduce every displayed coefficient through that map. Put

$$f(X) = \frac{\Phi(X) - t_0 X}{u_0}, \quad \gamma_a = \frac{-\Phi(-a) - t_0 a}{u_0} \in \mathbb{F}_Q.$$

Use the Artin–Schreier rank-one sheaf convention in which the geometric-Frobenius trace at $x \in \mathbb{F}_{Q^n}$ is $\psi(\text{Tr}_{\mathbb{F}_{Q^n}/\mathbb{F}_p} f(x))$, for a fixed nontrivial additive character ψ valued in $\overline{\mathbb{Q}_\ell}$, $\ell \neq p$.

Proposition 57.4 (Translation and the constant character line). *For the isomorphism $\sigma_a : X \mapsto S = X + a$,*

$$\sigma_a^* \mathcal{L}_\psi((\Phi_a - t_0 S)/u_0) \simeq \mathcal{L}_\psi(f) \otimes \mathcal{L}_\psi(\gamma_a). \quad (\text{DT.12})$$

Hence compactly supported cohomology in every degree is identified with that of $\mathcal{L}_\psi(f)$, tensored with the displayed constant character line. Each geometric-Frobenius eigenvalue is multiplied by

$$\zeta_a = \psi(\text{Tr}_{\mathbb{F}_Q/\mathbb{F}_p} \gamma_a). \quad (\text{DT.13})$$

This is a root of unity, so it preserves each complex absolute value and hence the purity weights of this finite-field realization.

Proof. The potential identity in the period proof holds after the specified coefficient reduction. The tensor identity of Artin–Schreier character sheaves can be obtained directly from their covers: if $y_1^p - y_1 = f$ and $y_2^p - y_2 = \gamma_a$, then $(y_1 + y_2)^p - (y_1 + y_2) = f + \gamma_a$. The addition map identifies the contracted product of the two $\mathbb{F}_p$ -torsors with the third torsor. On the character lines, the addition action becomes the product of their characters, which gives DT.12. Pullback by the base-defined isomorphism σ_a preserves compact supports and commutes with Frobenius. Tensoring with a line pulled back from the base tensors the compact-support complex with that line: this can be checked on a cochain resolution before taking cohomology. Our stated trace convention makes its Frobenius scalar DT.13. The additive group of $\mathbb{F}_p$ has exponent p , so $\zeta_a^p = 1$, proving the absolute-value assertion. $\square$

These formulas compose with the exponential-family application of Deligne, *La conjecture de Weil II*, §§3.7.2–3.7.4, to the degree- d polynomial with nonzero leading coefficient. The source for that application, including the boundary chart and its hypotheses, is retained in the exponential comparison note; the present proof establishes the additional translation maps themselves.

57.6 Specialization and split support

The polynomial chain isomorphism T_a is defined before either complex or finite-field realization. Its coefficient map, the specialization $u = t = 0$, and the inclusion of the original fibre $a = 0$ induce maps under the same split-zero functor G . For a ring map $\varphi : B \rightarrow B'$, this functor acts by

$$G(\varphi)(\tau) = \tau, \quad G(\varphi)(b^\bullet) = \varphi(b)^\bullet.$$

Therefore a supported coefficient specialized to zero maps to $e = 0^\bullet$, and an absent coefficient maps to τ . The square with the amplitude maps $p_B, p_{B'}$ commutes, since both composites take τ to zero and $b^\bullet$ to $\varphi(b)$. On each already specified receiving module fibre, the coefficient chain relation specializes to the relation $\chi P = 0$ without changing its label. The functorial map thus retains both the supported kernel and external absence.

The translation formulas specify exactly how the two realizations respond to the same coefficient operation: DT.2 shifts the arithmetic sum operator, DT.9 changes its metric trace by $2q \text{Re } a$, DT.10 retains the analytic exponential scalar, and DT.13 supplies the finite-field character scalar. These are complete comparison maps. They do not identify a finite-field Frobenius with A , and they leave the original theta metric estimate as the same determinant quantity in DT.11. The original arithmetic construction remains the specified fibre $a = 0$ throughout.

58 Primary-source correction to the Deligne comparison

58.1 Primary-source correction to the Deligne comparison

This dated correction supersedes the earlier reported internal numbering mismatch in the final citation of Weil II, Section 3.7.4. The original delivered note, S20 page records, historical typed witnesses,

and sealed intake remain preserved. The correction is based on the Zeta literature task's complete retained primary-source review of 13 September 2026.

The final citation on printed page 216 is **(I.8.11), with Roman I**. It refers to **Weil I, Section 8.11, printed page 306**, rather than to Weil II, Corollary 1.8.11. Weil I, Section 8.10, reduces to one variable and the monomial $Q = x^d$; Section 8.11 handles that case, and Sections 8.12–8.13 continue the Euler-characteristic calculation. The citation is resolved. This does not assert that Section 8.11 alone states the entire general purity theorem used in Weil II.

The printed page 216 itself uses S'_0 in the connectedness sentence. The preceding parameter sheaves are defined on the good open S_0 . That open is connected: it is a nonempty open subset of an irreducible affine parameter space, hence is irreducible. The printed prime remains part of the literal source typography while the application retains the explicitly defined good open.

The high-resolution printed display has the three terms

$$\begin{aligned} R^i f_! \mathcal{F}_0(\Psi Q_S) &= R^i \bar{f}_* j_! \mathcal{F}_0(\Psi Q_S) \\ &= R^i \bar{f}_* j_! \mathcal{F}_0(\Psi Q_S). \end{aligned}$$

The two final terms both have $\bar{f}_*$. The identity $\bar{f}_! = \bar{f}_*$ for the proper compactification explains a possible intermediate step, but replacing one of the printed stars by an exclamation mark would reconstruct that step. It would not transcribe a visible distinction. The repeated display and the prime are already present in the printed source.

The same primary review confirms the strict weight inequality < 2 in Lemma 3.2.10 on printed page 203. Section 3.2.13 explicitly uses $w_q(\alpha^2)$ and then divides the resulting bound by two. Both the strict inequality and the tensor square of the same eigenvalue remain essential to those deductions; losses in older typed witnesses must not enter the current proof.

On printed page 178, Corollary 1.8.12 requires a lisse ι -mixed sheaf on a connected scheme of finite type over the finite field, pure at one point, and concludes pointwise purity. Normality is not a hypothesis of that corollary; mixedness is. Constant rank alone does not supply its hypotheses.

The verified exponential theorem (3.7.2.3), printed pages 215–216, applies to the actual specialized potential $(\Phi - tS)/u$. Its degree is $d = q + 1$, its leading coefficient is the unit $1/(du)$ on $u \neq 0$, and the specified characteristic $p > d$ makes d prime to p . The leading projective zero locus in $\mathbb{P}^0$ is empty and smooth. Thus compact-support cohomology is concentrated in degree one, has dimension q , and is pure of weight one. Repeated critical roots do not obstruct these hypotheses. Lemma 3.7.3 supplies the lisse family through the actual compactification and its local-acyclicity calculation, including infinity. These are applications of the stated historical theorem. The exact source and period comparisons in the adjoining proofs retain their own operators and original theta metrics.

The retained primary review records visual reading of complete Weil II printed pages 178, 202, 203, 206, 215 and 216, including a higher-resolution check of page 216. Its bounded collaborator checked complete Weil I printed pages 305–306 and the cited case. This correction records their actual inspection scope; it does not claim a new audit of every foundational theorem in either paper. The evidence includes the complete written review, source URLs and exact source hashes, with no reproduced scan or rendered page.

Primary sources: [Deligne, *La conjecture de Weil II*, printed pages 178, 203, 215–216](#); [Deligne, *La conjecture de Weil I*, printed pages 305–306](#).

59 SGA trace, exact constituent curvature, and the derived collision fibre

59.1 SGA trace, exact constituent curvature, and the derived collision fibre

This calculation extends the constituent period map in the delivered *SGA Trace Period Control*, equations (54a)–(54c). Its period determinant is the same determinant already proved in (XD1)–(XD26), with every original coefficient, ray and phase retained. We calculate the full normal derivative, its kernel and its norm. The result applies to the actual invariant constituents, including repeated primary factors.

59.1.1 The retained period map and arithmetic source

Keep the original monic polynomial and its increasing-power frame:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S, \quad q \geq 1, \quad e_0 = [1], \quad \ell([P]) = [S^{q-1}] \operatorname{rem}_\chi P. \quad (\text{SC1})$$

Thus $\ell : E \rightarrow \mathbb{C}$ is a specified linear functional and $\mathcal{R} = e_0 \ell : E \rightarrow E$ has rank one. Fix $u \in \mathbb{C}^*$, one argument of u , and the actual potential and contours

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad \theta_j = \frac{\arg u + \pi + 2\pi j}{q+1}, \quad \Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}. \quad (\text{SC2})$$

Here ℓ_j^{ray} is the outward ray $S = r e^{i\theta_j}$, $r \geq 0$; it is distinct from the coefficient functional ℓ . The period matrix is

$$\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{SC3})$$

The full convergence and determinant proof is (XD4)–(XD21). To state its precise retained inputs here, on a compact coefficient set K its tail is bounded by $r^L \exp(-r^{q+1}/(2(q+1)|u|))$ after $r \geq \max(1, 2(q+1)M_K)$, where $M_K = \sum_a \sup_K |c_a|/(a+1) + \sup_K |t|$. Differentiation is therefore valid in every coefficient direction. Integration by parts cancels the two origin terms, and reduction of S^q uses $[\chi - t] = 0$ with no derivative correction. It follows that

$$\Pi'(t) = -\frac{1}{u} \Pi(t)(A + t\mathcal{R}). \quad (\text{SC4})$$

The proven determinant is

$$\det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\operatorname{Tr} \Phi_t(A + t\mathcal{R})}{u}\right), \quad |\det \Pi_{\text{mon}}(u)|^2 = (2\pi|u|)^q > 0. \quad (\text{SC5})$$

The complex initial matrix is explicitly

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{SC6})$$

The product form and the full complex phase in (SC5)–(SC6) are reused from (XD18)–(XD19). The evaluated squared constant is the additional gamma-product evaluation in the delivered SGA note. Here is every factor in that evaluation. Set $d = q+1$. The matrix $V_{jl} = e^{2\pi i j l/d} - 1$ has $V^* V = d(I_q + J_q)$ by the root-of-unity sum in (XD17), so $|\det V|^2 = d^{q+1}$. Since $2 \sum_{l=1}^{d-1} l/d = q$, its column factors give

$$|\det \Pi_{\text{mon}}|^2 = d^{q+1} \frac{(d|u|)^q}{d^{2q}} \left(\prod_{l=1}^{d-1} \Gamma(l/d) \right)^2. \quad (\text{SC6a})$$

The gamma multiplication formula, DLMF (5.5.6), is

$$\Gamma(dz) = (2\pi)^{(1-d)/2} d^{dz-1/2} \prod_{a=0}^{d-1} \Gamma(z + a/d). \quad (\text{SC6b})$$

At $z = 1/d$, its left side and the final product factor are $\Gamma(1) = 1$, so the remaining product equals $(2\pi)^{q/2} d^{-1/2}$, which is DLMF (5.5.7). In (SC6a) the complete power of d is $(q+1) + q - 2q - 1 = 0$. This proves $|\det \Pi_{\text{mon}}|^2 = (2\pi|u|)^q$ without a frame change or an omitted phase. Thus every $\Pi(t)$ is invertible, including at collisions. The full reference is <https://dlmf.nist.gov/5.5.E7>.

For every original admitted source degree N , retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{V} R_N, \quad j_E r_N = I_E. \quad (\text{SC7})$$

The source identities and primitive have their original types $j_E D^{(k)} = A j_E$ and $D^{(k)} r_N - r_N A = dK_N^{\text{prim}}$. The exact parameter action on the admitted source domain is

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t \mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t \mathcal{R}) = dK_N^{\text{prim}}. \quad (\text{SC8})$$

Indeed insertion of $j_E r_N = I_E$ proves each equality by multiplication; the added terms cancel in the last difference. In particular the normal map calculated below has an explicit lift to this source, with the complete original primitive and kernel of j_E retained.

59.1.2 Every invariant constituent has a nonzero top coefficient

Let $I : F \hookrightarrow E$ be an actual injective linear map whose image is A -invariant, with $AI = I A_F$, and put $p = \dim F$. For the moment suppose $0 < p < q$. The image $I(F)$ is an ideal of E : invariance under A implies invariance under every polynomial in A , and every element of E is such a polynomial applied to e_0 . Its inverse image in $\mathbb{C}[S]$ is an ideal containing (χ) . It is generated by a unique monic divisor g of χ ; writing $\chi = gh$ gives $\deg h = p$ and $\deg g = q - p$. The exact map is

$$\mathbb{C}[S]/(h) \longrightarrow I(F), \quad [a]_h \longmapsto [ga]_\chi. \quad (\text{SC9})$$

It is onto by the definition of (g) , and is injective because $gh \mid ga$ in the polynomial domain implies $h \mid a$. Consequently $[g], [Sg], \dots, [S^{p-1}g]$ is an ordered basis of $I(F)$. Its last member is monic of degree $q - 1$, while every earlier member has smaller degree. Hence the restricted functional $L = \ell I : F \rightarrow \mathbb{C}$ is onto. Also $e_0 \notin I(F)$, since an ideal containing the unit is the whole algebra. We have proved

$$\begin{aligned} \mathcal{R}I &= e_0 L, \quad \text{rank}(\mathcal{R}I) = 1, \quad \ker(\mathcal{R}I) = \ker L, \quad \dim \ker L = p - 1, \\ \mathbb{C}e_0 \cap I(F) &= \{0\} \quad \text{inside } E. \end{aligned} \quad (\text{SC10})$$

These arguments never replace a repeated factor by its square-free part. In the basis (SC9), L is exactly the last coordinate; in the original chosen basis of F it remains the literal row ℓI .

59.1.3 The exact normal derivative at every parameter

Give the target $\mathbb{C}^q$ the period-coordinate Hermitian metric I_q specified in the delivered note. Define

$$\begin{aligned} Y(t) &= \Pi(t)I, \quad H(t) = Y(t)^* Y(t), \quad P(t) = Y(t)H(t)^{-1}Y(t)^*, \\ N(t) &= I_q - P(t), \quad z(t) = N(t)\Pi(t)e_0. \end{aligned} \quad (\text{SC11})$$

Every $Y(t)$ is injective, so $H(t) > 0$. Multiplication shows $P(t)^2 = P(t) = P(t)^*$ and $P(t)Y(t) = Y(t)$; thus $N(t)$ is the orthogonal projection onto the complement of the displayed period subspace. The vector $z(t)$ is nonzero for every t : otherwise $\Pi(t)e_0$ lies in $\Pi(t)I(F)$, and invertibility of $\Pi(t)$ would give $e_0 \in I(F)$, contradicting (SC10).

Using invariance under the original A , equation (SC4) yields

$$Y'(t) = -\frac{1}{u}Y(t)A_F - \frac{t}{u}\Pi(t)e_0L, \quad \boxed{N(t)Y'(t) = -\frac{t}{u}z(t)L.} \quad (\text{SC12})$$

This holds for every complex t , with the same fixed constituent F of the original A . We have not assumed that F is invariant under $A + t\mathcal{R}$. Formula (SC12) computes exactly its failure of invariance after passing through the original periods. For $t \neq 0$ the normal derivative has image $\mathbb{C}z(t)$, kernel $\ker L$ and rank one. At $t = 0$ it vanishes. Differentiating (SC4) once more gives

$$Y''(0) = \frac{1}{u^2}\Pi(0)A^2I - \frac{1}{u}\Pi(0)\mathcal{R}I, \quad N(0)Y''(0) = -\frac{1}{u}z(0)L. \quad (\text{SC13})$$

The first term has image inside $Y(0)F$ because $A^2I = IA_F^2$. The normal acceleration therefore has rank exactly one, image $\mathbb{C}z(0)$ and kernel $\ker L$, including all repeated-root cases.

The actual original source lift of its parameter defect is $r_N e_0 L : F \rightarrow r_N(E)$. The map $x \mapsto N(t)\Pi(t)j_E x$ sends it to $z(t)L$. Thus its source, coefficient and period forms have the commuting factorization

$$F \xrightarrow{L} \mathbb{C} \xrightarrow{a \mapsto ar_N e_0} r_N(E) \xrightarrow{j_E} E \xrightarrow{N(t)\Pi(t)} \mathbb{C}^q. \quad (\text{SC14})$$

At the original Split-Zero support, every linear arrow lifts by $x^\bullet \mapsto f(x)^\bullet$ and $\tau \mapsto \tau$. In particular $\ker L$ maps to the supported zero; external absence retains its distinct image τ . No new zero-adjunction is used.

59.1.4 Curvature is an exact rank-one expression

Write $\partial_t = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{t}} = \frac{1}{2}(\partial_x + i\partial_y)$ for $t = x + iy$. Since Y is holomorphic, $\partial_t H = Y^*Y'$, $\partial_{\bar{t}} H = Y'^*Y$ and $\partial_{\bar{t}}\partial_t H = Y'^*Y'$. Differentiating $H^{-1}H = I_p$ gives $\partial_{\bar{t}}H^{-1} = -H^{-1}Y'^*YH^{-1}$. The determinant derivative and cyclic invariance of trace therefore give

$$\begin{aligned} \partial_{\bar{t}}\partial_t \log \det H &= \text{Tr}(H^{-1}Y'^*Y' - H^{-1}Y'^*YH^{-1}Y'^*Y') \\ &= \text{Tr}(H^{-1}Y'^*NY'). \end{aligned} \quad (\text{SC15})$$

There is no logarithm-branch choice: $\det H > 0$. Insert the full normal map (SC12), use $N^* = N = N^2$, and retain the rank-one factors to obtain

$$\boxed{\partial_{\bar{t}}\partial_t \log \det H(t) = |t|^2 c_F(t), \quad c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L H(t)^{-1} L^* > 0.} \quad (\text{SC16})$$

Strict positivity follows because $z(t) \neq 0$, $L \neq 0$ and $H(t)^{-1} > 0$. Thus this curvature vanishes exactly at $t = 0$ and vanishes there to precisely quadratic order; it is strictly positive at every $t \neq 0$. For the dual determinant line with metric $(\det H)^{-1}$ the convention $c_1 = -(i/(2\pi))\partial\bar{\partial} \log h$ gives

$$c_1 = \frac{i}{2\pi} |t|^2 c_F(t) dt \wedge d\bar{t}. \quad (\text{SC17})$$

In these coordinates $i dt \wedge d\bar{t} = 2 dx \wedge dy$; this records the orientation and factor explicitly.

For completeness the exact formula supplies a convergent Taylor statement. Entries of Π have convergent complex power series; entries of H , P , N and c_F are real analytic because $\det H$ is everywhere positive. On any closed disk $|t| \leq \varepsilon$, continuity of the first real derivatives gives a finite bound $M_\varepsilon = \sup \|\nabla c_F\|$. Integrating the derivative along the real segment $s \mapsto st$ proves $|c_F(t) - c_F(0)| \leq M_\varepsilon |t|$. Consequently

$$|\partial_{\bar{t}}\partial_t \log \det H(t) - |t|^2 c_F(0)| \leq M_\varepsilon |t|^3. \quad (\text{SC18})$$

This estimate follows from the actual analytic functions in (SC11), and (SC16) supplies the stronger identity at all parameters.

One may check the projector's first derivatives directly: $\partial_t P = NY'H^{-1}Y^*$ and $\partial_{\bar{t}} P = YH^{-1}Y'^*N$. They vanish at zero by (SC12), explaining the retained stationary tangent space at that parameter without deleting its acceleration.

59.1.5 Both original theta and period metrics remain explicit

On F retain the positive original theta Gram $G_{N,F} = I^*G_N I$ and the positive comparison operator

$$B_{N,F}(t) = G_{N,F}^{-1}H(t), \quad \mathcal{O}_{N,F}(t) = \log \frac{\det H(t)}{\det G_{N,F}}. \quad (\text{SC19})$$

Here $G_{N,F}$ is fixed as t varies. Thus (SC15)–(SC18) hold unchanged with $\mathcal{O}_{N,F}$ in place of $\log \det H$. For real t at zero and $u > 0$, (SC12) gives $H'(0) = -(A_F^*H(0) + H(0)A_F)/u$, hence

$$-u\mathcal{O}'_{N,F}(0) - kp = 2 \operatorname{Re} \operatorname{Tr} A_F - kp. \quad (\text{SC20})$$

The original constant k and its generalized eigenspace multiplicities are retained. On the full positive-real-defect constituent this is the existing aggregate defect $L_{h,k}$.

The squared Hilbert–Schmidt norm of the normal acceleration in the original domain metric, and its one nonzero squared singular value, are both

$$a_{N,F} = \frac{\|z(0)\|_{I_q}^2}{|u|^2} L G_{N,F}^{-1} L^*. \quad (\text{SC21})$$

Indeed the adjoint of $zL : (F, G_{N,F}) \rightarrow (\mathbb{C}^q, I_q)$ is $G_{N,F}^{-1}L^*z^*$, so the product and its trace give (SC21). The unique nonzero eigenvalue equals its trace because the product has rank one. The curvature coefficient is instead the exact observer

$$c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L B_{N,F}(t)^{-1} G_{N,F}^{-1} L^*. \quad (\text{SC22})$$

This follows from $H = G_{N,F}B_{N,F}$; it is the required map between the two metric measurements. It does not replace either metric by an unweighted coordinate norm.

Finally transport the original metric on the whole period target by $\tilde{G}_N = \Pi^{-*}G_N\Pi^{-1}$. Its projection onto YF is

$$P_N^\theta = Y G_{N,F}^{-1} Y^* \tilde{G}_N = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{SC23})$$

Direct multiplication proves its idempotence and its $\tilde{G}_N$ -self-adjointness. With $z_N^\theta = (I_q - P_N^\theta)\Pi e_0$, its squared target norm is the literal theta Schur complement

$$\|z_N^\theta\|_{\tilde{G}_N}^2 = e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{SC24})$$

The right side is positive because it is the norm of the nonzero G_N -orthogonal projection of e_0 away from $I(F)$. For this moving target metric $Y^* \tilde{G}_N Y = G_{N,F}$ is constant. Its differentiated metric terms are explicitly

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N (A + t\mathcal{R}) \Pi^{-1} / u, \quad \partial_{\bar{t}} \tilde{G}_N = \Pi^{-*} (A + t\mathcal{R})^* G_N \Pi^{-1} / \bar{u}. \quad (\text{SC25})$$

They cancel $Y^* \tilde{G}_N Y'$ and its conjugate derivative in the product rule. Thus the constant pulled-back theta Gram and the nonzero curvature (SC16) are connected by the exact metric congruence (SC23)–(SC25), with every term retained.

For $F = 0$, use the empty Gram determinant 1, empty trace 0 and the zero normal map. For $F = E$, $P = I_q$ and $N = 0$ identically, so (SC15) is zero: the full determinant logarithm in (SC5) is the real part of a holomorphic function plus its stated constant. The strictly positive coefficient in (SC16) belongs precisely to $0 < p < q$. This curvature and the original theta comparison are fully calculated observables; no uniform arithmetic upper bound is asserted by these finite-dimensional identities.

59.1.6 The full two-parameter curvature in the same original frames

Allow the original $u \neq 0$ to vary on a retained contour branch, keeping the same χ , F , I , G_N , and target metric. Use $C_\chi(t)$ and B_χ from the literal division (XD.u1). For $a = u, t$ put

$$\Omega_u = -C_\chi(t)/u^2 + B_\chi/u, \quad \Omega_t = -(A + t\mathcal{R})/u, \quad Y_a = \Pi\Omega_a I, \quad Z_a = N\Pi\Omega_a I. \quad (\text{SC.u1})$$

The convergence and full reduction in (XD.u1)–(XD.u5) prove these are precisely $\partial_a Y$ and its original normal projection. For every $a, b \in \{u, t\}$, the complete Hermitian matrix of curvature coefficients is

$$\partial_{\bar{a}}\partial_b \log \det H = \text{Tr}(H^{-1}Z_a^*Z_b). \quad (\text{SC.u2})$$

For the proof, holomorphy gives $\partial_b H = Y^*Y_b$, $\partial_{\bar{a}} H = Y_a^*Y$, and $\partial_{\bar{a}}\partial_b H = Y_a^*Y_b$. The derivative of H^{-1} is $-H^{-1}Y_a^*YH^{-1}$. Substitution in the logarithmic determinant derivative gives $\text{Tr}(H^{-1}Y_a^*(I - YH^{-1}Y^*)Y_b)$. Since $N^2 = N = N^*$, this is (SC.u2). In particular every linear combination has value $\text{Tr}(H^{-1}(\sum_b v_b Z_b)^*(\sum_b v_b Z_b)) \geq 0$, using the original positive H . This proves the mixed coefficients as well as both diagonal ones, with the original one-form ordering $du \wedge d\bar{u}$, $dt \wedge d\bar{u}$, $du \wedge d\bar{t}$, $dt \wedge d\bar{t}$.

At $t = 0$, $C_\chi(0) = \Phi(A)$ is a polynomial in the original action, so $C_\chi(0)I = I\Phi(A_F)$. Consequently the order-two term has zero normal projection there, while the complete first-order term remains:

$$Z_u(u, 0) = u^{-1}N\Pi B_\chi I, \quad Z_t(u, 0) = 0, \quad \partial_{\bar{u}}\partial_u \log \det H(u, 0) = |u|^{-2} \text{Tr}(H^{-1}I^*B_\chi^*\Pi N\Pi B_\chi I). \quad (\text{SC.u3})$$

Thus the earlier stationary t tangent retains the calculated u normal map. No invariance of F under B_χ is assumed. For $t \neq 0$, the complete expression in (SC.u1) retains both $-C_\chi(t)/u^2$ and B_χ/u before taking its squared norm.

The positive original source Gram is constant in both parameters. The transported metric therefore has the exact derivatives

$$\partial_a \tilde{G}_N = -\Pi^{-*}G_N\Omega_a\Pi^{-1}, \quad \partial_a \tilde{G}_N = -\Pi^{-*}\Omega_a^*G_N\Pi^{-1}, \quad a = u, t. \quad (\text{SC.u4})$$

They follow from $\partial_a \Pi^{-1} = -\Omega_a\Pi^{-1}$; holomorphy gives $\partial_a \Pi^{-*} = 0$. In the product rule for $Y^*\tilde{G}_N Y$ these terms cancel the derivatives of Y and Y^* , giving exactly the fixed $I^*G_N I$. Formula (SC25) is its $a = t$ case. The full original source comparison is (DS72)–(DS74); in particular its theta boundary $dK_N\Omega_a j_E$ remains in the Euler commutator. All curvature coefficients (SC.u2) also apply to $\log \det(G_{N,F}^{-1}H)$, since the determinant of $G_{N,F}$ is constant in both parameters.

For every neighboring inclusion J in (SC31), the identity $h_J = v_{N,J}b_{N,J}$ in (SC35) now holds on this same two-parameter branch. Its u and mixed logarithmic derivatives are the corresponding formula (SC.u2) with $I = J$, $H = J^*\Pi^*\Pi J$, and its own actual normal projection. For a zero-column inclusion its determinant is 1 and every derivative is zero; for the full inclusion the normal projection is zero. This propagates the singular connection through all three retained minor maps rather than replacing a rectangular minor by the full determinant.

59.1.7 The full primary matrix now evaluates every original ideal volume

The completed single-primary proof specializes this calculation to the original $h(s) = (s-\rho)^m$, $t = 0$, with $n = m + 1$, $c = -(-\rho)^n/n$ and $y = s - \rho$. Its exact Pascal matrix is $P_{ab} = \binom{b}{a}\rho^{b-a}$ for $a \leq b$, zero otherwise; $e_s = e_y P$ and P^{-1} has $-\rho$ in place of ρ . Retain its full ordered period matrix

$$\begin{aligned} \Pi_s &= e^{c/u} W D(u) P, \quad W_{jb} = e^{2\pi i j(b+1)/n} - 1, \\ D(u)_{bb} &= d_b u^{\beta_b}, \quad \beta_b = (b+1)/n, \quad d_b = n^{\beta_b-1} e^{\pi i \beta_b} \Gamma(\beta_b). \end{aligned} \quad (\text{SC.u5})$$

The proof SP.10–15 of this formula is included in full: it uses the original contour translation, the decaying endpoint connectors and the gamma radial integral, so both c and all phases in (SC.u5) have their original values.

For an original invariant constituent of dimension p , put $r = m - p$ and let J_r have the literal polynomial columns $y^r, y^{r+1}, \dots, y^{m-1}$ in the original s basis. Every ideal of the primary algebra is $(y^r)/(y^m)$ by division in $\mathbb{C}[y]$, so its specified inclusion has the unique form $I = J_r C_F$ for an invertible constant p -by- p matrix C_F . Let W_r select the columns $r, \dots, m - 1$ from W , and D_r the same diagonal entries from D . Since PJ_r is the literal inclusion of those translated coordinates,

$$Y = e^{c/u} W_r D_r C_F, \quad H = e^{2\operatorname{Re}(c/u)} C_F^* D_r^* (W_r^* W_r) D_r C_F. \quad (\text{SC.u6})$$

The exact root-of-unity sum gives $W_r^* W_r = n(I_p + \mathbf{1}_p \mathbf{1}_p^*)$: including the zero row adds zero, and expansion of the three nonconstant character sums leaves $n\delta_{ab} + n$. This matrix has eigenvalue $n(p + 1)$ on $\mathbf{1}_p$ and eigenvalue n on its perpendicular complement, so its determinant is $n^p(p + 1)$. Taking determinants in the unchanged matrices therefore gives

$$\det H(u, 0) = |\det C_F|^2 n^p (p + 1) e^{2p \operatorname{Re}(c/u)} \prod_{b=r}^{m-1} |d_b|^2 |u|^{2\beta_b}. \quad (\text{SC.u7})$$

No frame determinant, contour phase or Taylor coefficient in the original source Gram $G_{N,F} = I^* G_N I$ is removed. The determinant of the actual comparison is exactly the right side divided by $\det G_{N,F}$; its complete matrix is $G_{N,F}^{-1}$ times (SC.u6). For $p = 0$ all products and determinants are empty and equal one, and for $p = m$ this is the full determinant formula.

In particular $Y(F) = \operatorname{im} W_r$ is constant as u varies on $u \neq 0$: D_r and C_F are invertible. Thus its normal u derivative and its u curvature vanish on this exact primary slice. Equivalently, $PB_h P^{-1} = \operatorname{diag}(\beta_b)$ preserves $PJ_r(F)$ in (SC.u3). This evaluates that previously retained B_h/u normal term on the actual primary ideals. The t -normal map and its full acceleration are still (SC12)–(SC13); the tensor boundary requires the ordered character products calculated in TP/TPF rather than propagating the one-factor invariant-space dimension.

59.1.8 The exact derived base change at a collision

The trace and the positive-degree diagonal groups in the delivered SGA note have a further explicit specialization map. Retain the whole one-parameter algebra over $T = \mathbb{C}[t]$:

$$\mathcal{E} = T[S]/(\chi(S) - t) \simeq \mathbb{C}[S], \quad t \mapsto \chi(S), \quad M = \mathcal{E}/\chi' \mathcal{E}. \quad (\text{SC26})$$

The isomorphism sends $[P(t, S)]$ to $P(\chi(S), S)$; its inverse sends S to its original class. Monic division proves that $\mathcal{E}$ is free of rank q over T . Since χ' is a nonzero polynomial over $\mathbb{C}$, multiplication by χ' is injective on $\mathcal{E} \simeq \mathbb{C}[S]$. Consequently

$$0 \longrightarrow \mathcal{E} \xrightarrow{\chi'} \mathcal{E} \longrightarrow M \longrightarrow 0 \quad (\text{SC27})$$

is a free T -resolution of M . The diagonal resolution gives $HH_0(\mathcal{E}/T) = \mathcal{E}$, $HH_{2a+1}(\mathcal{E}/T) = M$ and $HH_{2a+2}(\mathcal{E}/T) = 0$ for $a \geq 0$. For clarity, that resolution is free over $\mathcal{B} = \mathcal{E} \otimes_T \mathcal{E}$ with alternating maps $X - Y$ and $\mathcal{C}_H = (H(X) - H(Y))/(X - Y)$, followed by the multiplication augmentation. Over $(T[Y]/H(Y))[X]$ both factors are monic and therefore non-zero-divisors; their product is $H(X)$ modulo $H(Y)$. Cancellation proves each alternating kernel equals the preceding image. Tensoring by the diagonal sets $X = Y$ and gives the maps $0, \chi', 0, \chi', \dots$, proving the stated groups.

Fix $t_0 \in \mathbb{C}$, put $\kappa = T/(t - t_0)$ and $E_{t_0} = \mathbb{C}[S]/(\chi - t_0)$. Tensoring (SC27) gives the complete kernel-and-cokernel sequence

$$\begin{aligned} 0 \longrightarrow \operatorname{Tor}_1^T(M, \kappa) \longrightarrow E_{t_0} \xrightarrow{\chi'} E_{t_0} \\ \longrightarrow M \otimes_T \kappa \longrightarrow 0. \end{aligned} \quad (\text{SC28})$$

It computes the actual specialized diagonal groups:

$$\begin{aligned} HH_{2a+1}(E_{t_0}/\mathbb{C}) &= M \otimes_T \kappa = E_{t_0}/(\chi'), \\ HH_{2a+2}(E_{t_0}/\mathbb{C}) &= \mathrm{Tor}_1^T(M, \kappa) = \mathrm{ann}_{E_{t_0}}(\chi'). \end{aligned} \quad a \geq 0. \quad (\text{SC29})$$

Every arrow is induced by the same polynomial multiplication, so these are equivariant maps for the original S -action.

There is also a representative-level description of the Tor map. Resolve κ by $[T \xrightarrow{t-t_0} T]$ to identify $\mathrm{Tor}_1^T(M, \kappa)$ with the kernel of $t - t_0$ on M . If $[v]_{\chi'}$ belongs to this kernel, write

$$(\chi - t_0)v = \chi'w. \quad [v]_{\chi'} \longmapsto [w]_{\chi-t_0}. \quad (\text{SC30})$$

Replacing v by $v + \chi'a$ replaces w by $w + (\chi - t_0)a$, so the map is well-defined. Its image is annihilated by χ' . If its image is zero, $w = (\chi - t_0)a$ and cancellation of the nonzero polynomial $\chi - t_0$ gives $v = \chi'a$, proving injectivity. Conversely an annihilated class $[w]_{\chi-t_0}$ satisfies $\chi'w = (\chi - t_0)v$ for a polynomial v , giving its preimage. These operations commute with multiplication by S .

At a root λ of multiplicity m in $\chi - t_0$, its local algebra is $\mathbb{C}[x]/(x^m)$, $x = S - \lambda$, and $\chi' = mx^{m-1}a(x)$ with the original invertible local factor a (this equality is in the local algebra: before reduction, the derivative of $x^m a(x)$ also contains the term $x^m a'(x)$). The odd group has local module $\mathbb{C}[x]/(x^{m-1})$ and the even group has local module $(x)/(x^m)$; both have dimension $m - 1$, with their full induced $S = \lambda + x$ action. The map $[b]_{x^{m-1}} \mapsto [xb]_{x^m}$ is an explicit module isomorphism between these local modules, using the stated local coordinate. For $m = 1$ both groups vanish. Thus the zero even group of the family and the nonzero even group of a collision fibre are related by the exact Tor term (SC28), not by omitting specialization data. The coefficient maps in the residue construction commute with base change; this calculation retains the additional derived term required when their kernels are specialized. Lifting every arrow at its original Split-Zero support keeps its receiving supported zero and external absence distinct as in (SC14).

59.1.9 The exact neighboring-minor identity and its theta factors

The ideal basis in (SC9) supplies three actual polynomial inclusions. Write their increasing-power coefficient matrices as

$$J_g = [g, Sg, \dots, S^{p-1}g], \quad J_- = [g, Sg, \dots, S^{p-2}g], \quad J_+ = [J_g, e_0]. \quad (\text{SC31})$$

For $p = 1$, J_- has zero columns. All three are injective on their stated coefficient spaces; the first two assertions follow from their distinct monic leading degrees, and the last from $e_0 \notin I(F)$. Let $C = (I^*I)^{-1}I^*J_g$. Since the columns of J_g lie in I , this satisfies $IC = J_g$; it is invertible because both matrices are bases of that same subspace. The exact frame equations are

$$H_g(t) := (\Pi J_g)^* \Pi J_g = C^* H(t) C, \quad LC = (0, \dots, 0, 1), \quad h_g(t) = |\det C|^2 \det H(t), \quad (\text{SC32})$$

where $h_J(t) = \det((\Pi J)^* \Pi J)$ for each of the three inclusions. The scalar $|\det C|^2$ has been retained; it is independent of t , so it has zero logarithmic derivatives.

The Gram of ΠJ_+ has block form

$$\begin{pmatrix} H_g & (\Pi J_g)^* \Pi e_0 \\ (\Pi e_0)^* \Pi J_g & \|\Pi e_0\|^2 \end{pmatrix}. \quad (\text{SC33})$$

Eliminate its off-diagonal blocks using triangular matrices of determinant one. Its lower-right Schur complement is the squared norm of the orthogonal projection of Πe_0 away from $\mathrm{im} \Pi J_g$, which is $\|z(t)\|^2$ by (SC11). Thus $h_+ = h_g \|z\|^2$. The last diagonal entry of H_g^{-1} , computed by the adjugate

formula, is its upper-left principal minor divided by h_g , namely h_-/h_g . By (SC32) that entry is exactly $LH^{-1}L^*$. Formula (SC16) becomes

$$\partial_{\bar{t}}\partial_t \log h_g(t) = \frac{|t|^2}{|u|^2} \frac{h_+(t)h_-(t)}{h_g(t)^2}. \quad (\text{SC34})$$

This identity retains all three original polynomial maps. It is the precise neighboring-minor equation joining the period calculation to the source-volume and Toda calculations; it does not identify differently ordered or differently weighted families.

For each same inclusion $J = J_-, J_g, J_+$, define its original theta volume and its positive comparison determinant by

$$v_{N,J} = \det(J^*G_NJ), \quad b_{N,J}(t) = \det((J^*G_NJ)^{-1}J^*\Pi(t)^*\Pi(t)J). \quad (\text{SC35})$$

Injectivity and positivity show that both numbers are positive. For the empty inclusion both are 1. Taking the determinant of the literal product gives $h_J = v_{N,J}b_{N,J}$, and therefore

$$c_F(t) = \frac{1}{|u|^2} \frac{v_{N,J_+}v_{N,J_-}}{v_{N,J_g}^2} \frac{b_{N,J_+}(t)b_{N,J_-}(t)}{b_{N,J_g}(t)^2}. \quad (\text{SC36})$$

Every original theta mass and coefficient inclusion remains in the v factors; every period comparison remains in the b factors. This proves the metric dictionary required to use (SC34) within that original source programme.

For $p = q - 1$, the divisor g is monic linear. The square matrix $J_+ = [g, Sg, \dots, S^{q-2}g, e_0]$ has determinant $(-1)^{q-1}$: expand along its last column, and the remaining minor on rows $1, \dots, q - 1$ is upper triangular with every diagonal entry 1. Thus, with that sign retained,

$$\det(\Pi J_+) = (-1)^{q-1} \det \Pi, \quad h_+(t) = (2\pi|u|)^q \exp\left(2 \operatorname{Re} \frac{\operatorname{Tr} \Phi_t(A + t\mathcal{R})}{u}\right). \quad (\text{SC37})$$

The top minor in (SC34) is therefore the explicitly evaluated full determinant in this codimension-one case. The remaining two minors and their exact theta factors are still those in (SC31)–(SC36).

60 The SGA trace-period source: exact coefficient and source corrections

60.1 The SGA trace-period source: exact coefficient and source corrections

The complete adjoining source is *SGA trace, the retained diagonal, and the arithmetic period determinant*, with its original equations 1–62, including 21a, 28a and 54a–54c. Its delivered Markdown and generated LaTeX remain byte-exact source witnesses. Six explicitly recorded replacements produce the corrected full LaTeX. The changes retain every proof paragraph and all original polynomial coefficients, operators, source metrics, ray phases and orientations. This introduction states the coefficient types and the source qualifications needed when reading the unaltered witness.

60.1.1 The relative trace and its coefficient specialization

For a commutative ring R and the original monic polynomial χ of degree $q > 0$, set

$$T = R[t], \quad H(S) = \chi(S) - t, \quad E_t = T[S]/(H). \quad (\text{SGAC1})$$

The ordered basis $1, S, \dots, S^{q-1}$ makes E_t a free T -module of rank q . Division by the monic H proves existence of the displayed representative; a nonzero multiple of H cannot have degree below q , proving uniqueness. Thus multiplication by P is a finite-free T -endomorphism, whose trace has the exact type

$$\mathrm{Tr}_{E_t/R[t]}(M_P) \in R[t]. \quad (\text{SGAC2})$$

In the delivered equation (1), the base R must therefore be read as $R[t]$, already printed correctly in equation (21). Finiteness over $R[t]$ alone does not assert finiteness over R .

Retain the source's functional $\lambda_t([P]) = [S^{q-1}] \text{rem}_H P$ and its polynomial divided difference $\mathcal{C}_H(X, Y) = (H(X) - H(Y))/(X - Y)$ in $E_t \otimes_T E_t$. The displayed fraction is polynomial division. Its X^{q-1} coefficient is one and $(X - Y)\mathcal{C}_H = 0$. Since $P(X) - P(Y)$ is divisible by $X - Y$, these two identities give

$$(\lambda_t \otimes 1)((P \otimes 1)\mathcal{C}_H) = P, \quad \mu(\mathcal{C}_H) = \chi'(S). \quad (\text{SGAC3})$$

Consequently $\mathcal{C}_H$ is the actual coevaluation tensor for the residue pairing $(x, y) \mapsto \lambda_t(xy)$. Its contraction with multiplication by P is

$$T \xrightarrow{1 \mapsto \mathcal{C}_H} E_t \otimes_T E_t \xrightarrow{M_P \otimes 1} E_t \otimes_T E_t \xrightarrow{\mu} E_t \xrightarrow{\lambda_t} T, \quad \mathrm{Tr}_{E_t/T} M_P = \lambda_t(\chi'P). \quad (\text{SGAC4})$$

No derivative or discriminant is inverted by this composite.

For the parameter-universal polynomial $\chi_{\mathbf{c}}(S) = S^q + \sum_{a=0}^{q-1} c_a S^a$, the trace bases in delivered equations (2) and (35) are $\mathbb{C}[\mathbf{c}, t]$. In particular the complete formula is

$$\mathcal{F}(\mathbf{c}, t) = \mathrm{Tr}_{E_t/\mathbb{C}[\mathbf{c}, t]}(M_{\Phi_t}), \quad \Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS. \quad (\text{SGAC5})$$

For any coefficient evaluation $f : T \rightarrow T'$, monic division gives the explicit basis-preserving isomorphism $E_t \otimes_T T' \rightarrow T'[S]/(f(H))$ by $[P] \otimes a \mapsto a[f(P)]$. It has the inverse that sends each of the same q basis elements to its tensor basis element. Every multiplication-matrix entry evaluates under f ; summing its diagonal therefore proves $f(\mathrm{Tr}_{E_t/T} M_P) = \mathrm{Tr}_{E_t \otimes_T T'/T'} M_{f(P)}$. This is the exact family-to-fibre trace map. At specified complex coefficient and parameter values it gives the finite-dimensional $\mathbb{C}$ -linear trace, retaining the marked original polynomial.

60.1.2 The differential coefficient ring and the metric paragraph

In both terms of delivered equation (28), the universal ring is $\mathbb{C}[\mathbf{c}, u, t, S]$. The differential is the same coefficient-linear map

$$D(P) = (uP' + (\chi_{\mathbf{c}} - t)P)dS, \quad [\mathbb{C}[\mathbf{c}, u, t, S] \xrightarrow{D} \mathbb{C}[\mathbf{c}, u, t, S]dS]. \quad (\text{SGAC6})$$

At the fixed original coefficient value this specializes to the source's $\mathbb{C}[u, t, S]$ complex. Differentiation in S and coefficient evaluation commute, so this is an actual map of the two-term complexes. The leading monic term gives the finite reduction: for a term of degree at least q , subtract the corresponding coefficient multiple of $D(S^m)$; its derivative contribution has degree $m - 1$, while its leading polynomial term has degree $m + q$. Each subtraction lowers the unreduced leading degree and uses no division by a parameter. The resulting remainder is unique because the leading term of $D(P)$ for nonzero P has degree $\deg P + q$. Every subtraction commutes with coefficient evaluation. The range of D retains its source type as a linear relation; for example $D(S) = (u + S(\chi_{\mathbf{c}} - t))dS$ gives the original relation $[S(\chi_{\mathbf{c}} - t)] = -u[1]$.

The generated paragraph following equation (50) suffered Markdown emphasis damage. Its exact mathematical sentence is:

Conversely $\Pi^* \Pi$ is the Gram for the separately specified coordinate metric I_q , and its comparison operator is $G_N^{-1} \Pi^* \Pi$.

Here $u \neq 0$ and $\Pi = \Pi(\mathbf{c}, u, t)$ is the original period matrix. Its determinant is nonzero by the complete evaluation in equations (38)–(41) of the adjoining note. With the original positive Gram G_N retained, the metric transport is unchanged:

$$\tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}, \quad \Pi^* \tilde{G}_N \Pi = G_N, \quad G_N (G_N^{-1} \Pi^* \Pi) = \Pi^* \Pi. \quad (\text{SGAC7})$$

The two equalities are direct multiplication, with $\Pi^{-*} = (\Pi^{-1})^*$. Thus they specify both the original-metric isometry and the comparison with the separately stated coordinate metric, with all Gram factors retained.

60.1.3 The independently inspected primary source

The current independent primary witness is the original French SGA 5 scan, printed pages 120–130 and 136–137. The expected English master was not located in the bounded search. The delivered note’s statements about its author’s supplied-master reading remain historical source statements; this integration does not count the missing English bytes as another complete reading.

Within SGA 5’s noetherian setup, formula (6.8.4) assumes that the ambient smooth map is proper. Formula (6.8.5) uses the finite-flat composite, regular immersion and smooth ambient map; ambient properness is not an additional hypothesis of that formula. Its printed qualification concerning the proof/reference is retained in the complete primary review. The monic calculation SGAC1–SGAC4 supplies the displayed trace identity over arbitrary commutative R directly.

Printed page 128 has three $N_{Y/S}$ tokens where the preceding definition and its Hom isomorphism give the conormal $N_{Y/X} = I/I^2$. This is an annotated source typography correction. In the present monic model $I = (H)$ and the exact conormal differential is

$$I/I^2 \longrightarrow E_t dS, \quad [HP] \longmapsto [P] \chi'(S) dS. \quad (\text{SGAC8})$$

It is well-defined because changing HP by H^2Q changes its derivative by a term divisible by H after reduction. Equivalently $d(HP) = P\chi'dS + H dP$ reduces to the displayed image. The generator $[H]$ and the derivative χ' retain their actual roles in the trace composite SGAC4. Literal reference scans remain reference-only; this correction does not silently alter them.

The six exact source replacements, their occurrence counts and intermediate hashes are recorded in `cumulative_sga_embedding_corrections_public_20260913.json`. The dictionary also binds the unchanged delivered TeX and its complete corrected descendant. It records no changed mathematical test result and makes no additional arithmetic growth claim.

61 Hochschild trace, Laplacian, and specialization: source integration

61.1 Hochschild trace, Laplacian, and specialization: source integration

The three complete notes adjoining this introduction are the *Hochschild comparison*, equations H1–H42, the *Laplacian parabola*, equations P1–P8, and the *Specialization research note*, equations S1–S28. Their full definitions, proofs, source references, exceptions, and verification qualifications are retained. They were reviewed at commit `a4a87844c257485ae6d669f5b3eb6d96b9e8b8d8` and merged with the identical tree at `bf0ef873dfbfbcb3bd0ec36f3442e403403eb8`. The supplied earlier draft and checkpoint descriptions remain historical source statements. The accompanying owner receipts record the accepted reviewed-head checks and exact merged-source comparison.

The comparison uses Connes and Consani’s *Hochschild homology, trace map and ζ -cycles*, printed pages 83–101. The owner’s complete source review read that chapter and visually inspected printed pages 89–92, 94–95 and 100. Its exact reading extent and the primary published-offprint hash are

retained in the evidence. No additional full-chapter inspection or local formal execution is claimed by this integration.

61.1.1 The precise trace image and the original source

The factorization in H6 is read with its stated restricted target:

$$V \xrightarrow{j} j(V) \xrightarrow{\text{Tr}} \mathcal{B}, \quad \text{Tr } j|_V = \Theta, \quad \mathcal{E} = \frac{1}{2}\mathcal{U}\Theta, \quad (\mathcal{U}F)(u) = u^{1/2}F(u).$$

The target $j(V)$ records the qualified trace image; it does not assert an unrestricted strong-Schwartz-valued trace on every Hochschild class. The isometry is precisely $\mathcal{U} : L^2(du) \rightarrow L^2(du/u)$, whose norm equality is $\int |u^{1/2}F(u)|^2 du/u = \int |F(u)|^2 du$.

The Gaussian $g_0 = e^{-\pi u^2}$ belongs to the larger even Schwartz space, and its image under $D^2 - D$ is the original $\phi_* = (4\pi^2 u^4 - 6\pi u^2)e^{-\pi u^2} \in V$. The full cochain comparison is H12, with both ordered legs in H13. Its Fourier identity is exactly $\mathcal{F}D = (1 - D)\mathcal{F}$; physical parity instead commutes with D . The original factor $\mathcal{M}\Theta\phi_* = 2\xi$ remains.

Before applying the cited earlier continuous retraction, H15 retains the map and kernel

$$\mathcal{B}/\Theta V \longrightarrow \mathcal{B}/\overline{\Theta V}, \quad \ker = \overline{\Theta V}/\Theta V.$$

The retraction theorem is an earlier analytic input, separately identified in the full note. A finite matrix calculation does not prove its closed-image conclusion.

61.1.2 The finite kernel and the retained zero mode

The critical-line comparison H19 retains the plus-sign Mellin convention, its action equations H20, the entire off-critical generalized kernel H22, and the critical recovery map

$$\Xi_\rho[f] = \sum_{j < m} \frac{i^{-j}}{j!} f^{(j)}(\gamma) \epsilon^j, \quad \rho = \frac{1}{2} + i\gamma.$$

The reciprocal multiplier proof applies to all nilpotent powers; the surviving derivatives recover every critical jet. Killed represented coefficients have value $(\lambda, 0)$ in the receiving support fibre. The map does not send them to structural absence.

The raw zero coefficient $\xi(1/2)L^{-1/2}$ is nonzero and does not extend smoothly to $L = 0$. H34–H37 retain the extra smooth scalar coordinate first, with reconstruction $L^{-1/2}a(L)$ in the normalized Fourier basis and circle-function value $a(L)/L$. The covering factor $\sqrt{n}$ gives the coordinate action $a(L) \mapsto a(nL)$.

The subsequent original-source Gaussian average is H38. Its large- T convergence is in the stated local smooth-flat section topology, with the full bound

$$p_{N,m,[0,M]}(\Pi_{\neq 0} \widehat{\Sigma \mathcal{E} \phi_T}) \leq C_{N,m,M}(1+T)^m e^{-2\pi^2 T/M^2}, \quad T \geq 1.$$

The source does not assert uniform Schwartz seminorms for ϕ_T as $T \rightarrow \infty$. The limiting pair $(0, \xi(1/2))$, local smooth multiplication, and the closed-module argument in H41 prove that the retained zero line becomes a supported boundary in the resulting quotient.

The primary-source limit correction on printed page 100 must remain explicit:

$$\zeta(1/2 \pm 2\pi i/L) \longrightarrow \zeta(1/2) \neq 0 \quad \text{as } L \longrightarrow \infty.$$

The printed $L \rightarrow 0$ at this step is erroneous. The owner checked the actual rendered published page; the correction is present in the complete detailed comparison note and is preserved here.

61.1.3 The same arithmetic metric and its exceptional fibre

H29–H31 and P1–P8 retain the original positive Gram G , action A , and boundary form $W_k = A^*G + GA - kG$. The Laplacian $A^2 - kA$ is related to the source metrics through the exact adjoint and energy identities, including all nilpotent jet terms. The two-sided original-metric control yields the full numerical-range parabola. Rank two alone does not make its relative error small. The source’s proposed growing-packet limit $\varepsilon_k/k \rightarrow 0$ is explicitly unproved; it is not added to the established estimates.

The specialization note retains the source-coordinate metric O_j , $C = \delta R^* O_j \delta R = G_i - G_j$, and the exact cost/return transformation. Its all-order curvature coefficients, telescoping identity, logarithmic integral and finite certificate are all included, through S28. Their successful fixed-block stopping degree is not asserted to be uniform over changing packets.

The local comparison $\alpha(w) = P_0 + wP_1$, the action pole $w^{-1}P_1AP_0$, and the derived-fibre line $E_1 \otimes \mathcal{O}(-1)|_\infty$ remain in the full source. For the original cyclic quotient, the exact hull and quotient are $F_A = (d)/(\chi)$ and $E/F_A \cong \mathbb{C}[S]/(d)$, with $d = \gcd(\chi, p_1, \dots, p_m)$ and $s_A = r - \deg d$. Transporting the original metric retains $\det G_A = |w|^{-2s_A} \det G_{\mathcal{M}}$ and the negative boundary atom $-s_A \delta_\infty$ in its curvature current.

The accompanying evidence records the owner’s written review, actual finite executions and accepted exact-head CI. The analytic maps, heat limit, coherent-sheaf consequences and numerical-range proof remain identified as written mathematics; they are not silently included in the selected Lean declarations.

Primary reference: [Connes and Consani, *Hochschild homology, trace map and \$\zeta\$ -cycles*, *Proceedings of Symposia in Pure Mathematics* 105 \(2023\), pages 83–101.](#)

62 The period curvature of the retained critical-comparison kernel

62.1 The period curvature of the retained critical-comparison kernel

This section joins the original theta representatives, the critical-line comparison of the Hochschild note (H16)–(H25), and the period curvature (SC1)–(SC37). The single-packet identification is proved first, including its full arithmetic unit. We then calculate the factorwise critical comparison on every cyclic tensor packet. All tensor targets below use the completed projective tensor product of the programme’s locally convex spaces; its finite surviving image is proved injective by a continuous retraction, without an assertion of tensor exactness for arbitrary spaces.

The proof-bearing comparison source is *The original theta complex inside the Hochschild trace realization*, (H1)–(H25), at the retained PR27 revision a4a87844c257485ae6d669f5b3eb6d96b9e8b8d8; its original analytic constructions are retained in the cumulative source. The arithmetic section and unit maps are (A7)–(A13), (CT.1)–(CT.5), and (TVB.1)–(TVB.3). The calculations here prove the additional maps between those specified objects.

62.1.1 The single-packet polynomial and its complete arithmetic unit

Retain the original nonempty packet, with every selected zero’s complete order,

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad g(s) = 2\xi(s), \quad v_h(s) = g(s)/h(s), \quad d = \deg h. \quad (\text{PC1})$$

The packet is the same one used by the original source, including its reflection symmetry when that symmetry is used in its metric. Put $E_h = \mathbb{C}[s]/(h)$, and let j_h mean the full Hermite remainder. Its

local coordinate at ρ is $x = s - \rho$, with basis $1, x, \dots, x^{m_\rho-1}$. Thus

$$v_{h,\rho} = \sum_{a=0}^{m_\rho-1} \frac{v_h^{(a)}(\rho)}{a!} x^a, \quad v_h = j_h(v_h), \quad \varepsilon_h = v_h^{-1}. \quad (\text{PC2})$$

This inverse exists with all its coefficients retained. Indeed write $h(s) = (s - \rho)^{m_\rho} h_\rho(s)$ and $g(s) = (s - \rho)^{m_\rho} g_\rho(s)$. Both remaining factors are nonzero at ρ , so the constant coefficient $a_0 = g_\rho(\rho)/h_\rho(\rho)$ is nonzero. If the coefficients of $v_{h,\rho}$ are a_a , its inverse coefficients satisfy

$$b_0 = a_0^{-1}, \quad b_n = -a_0^{-1} \sum_{a=1}^n a_a b_{n-a} \quad (1 \leq n < m_\rho). \quad (\text{PC3})$$

Multiplication makes the constant coefficient one and each other coefficient through degree $m_\rho - 1$ zero. The coprime polynomial factors of h give the Chinese-remainder isomorphism; assembling these inverses proves (PC2). The notation $j_h(h/g)$ for ε_h uses its holomorphic germ at every selected zero; no global entire reciprocal is asserted.

For $k = 1$, the sum variable is literally $S = s_1$. The isomorphism $\mathbb{C}[S] \rightarrow \mathbb{C}[s_1]$, $S \mapsto s_1$, pulls $(h(s_1))$ back to $(h(S))$. The monic annihilator defining the cyclic packet is consequently

$$\begin{aligned} \chi_{h,1}(S) &= h(S), \quad E = \mathbb{C}[S]/(h(S)), \quad A = M_S, \quad q = d, \\ \iota_1 : E &\xrightarrow{\sim} E_h, \quad [P(S)] \mapsto [P(s)], \quad \eta_h = M_{v_h} \iota_1. \end{aligned} \quad (\text{PC4})$$

The equality includes every exponent m_ρ . The inverse of η_h is $\iota_1^{-1} M_{\varepsilon_h}$. It intertwines the two original multiplication actions because multiplication by the unit commutes with s . Its actual multiplicative rule is $\eta_h(x)\eta_h(y) = v_h\eta_h(xy)$ and $\eta_h(1) = v_h$. Thus its source and target polynomial algebras and its module-map type remain explicit.

Use the original spaces and maps

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0\}, \quad Q = \mathcal{B}/\Theta V, \quad \mathbf{q} : \mathcal{B} \rightarrow Q, \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}F_h = v_h, \quad h(D)F_h = \Theta\phi_*. \end{aligned} \quad (\text{PC5})$$

The source construction of F_h is the iterated original Euler inverse $S_\rho F = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$ on the vanishing Mellin fibre, as proved in (A5)–(A7) and (TVB.1). Let $\mathcal{T}_h P = P(D)F_h$, $\pi_h P = [P]_h$, and $J_h = j_h \mathcal{M} : \mathcal{B} \rightarrow E_h$, with the retained $\sigma_h : E_h \rightarrow Q$. The exact source relations are

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h = \eta_h \pi_h, \quad \mathbf{q} \mathcal{T}_h = \sigma_h \eta_h \pi_h. \quad (\text{PC6})$$

Here and below π_h uses the variable identification (PC4). The first identity follows from $h(D)F_h = \Theta\phi_*$ and commutation with D . Integration by parts gives $\mathcal{M}\mathcal{T}_h P = v_h P$, proving the second identity. For the third, the retained section in (A10) is $\sigma_h = \mathbf{q} R_{\text{ref}}$, with $R_{\text{ref}} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u)$. Write $P = P_0 + hT$, $\deg P_0 < d$. Then $R_{\text{ref}} \eta_h[P] = \mathcal{T}_h P_0$, and the difference from $\mathcal{T}_h P$ is the displayed theta boundary $\Theta(T(D)\phi_*)$. This proves the precise equality with that section, rather than identifying two sections merely from equal jets. Since $J_h \Theta = 0$, its descended map is $\bar{J}_h : Q \rightarrow E_h$, defined by $\bar{J}_h \mathbf{q} = J_h$; the section identity is $\bar{J}_h \sigma_h = I_{E_h}$. Also $D\mathcal{T}_h P = \mathcal{T}_h(SP)$ and (PC6) imply $D_Q \sigma_h \eta_h = \sigma_h \eta_h A$. Surjectivity of η_h therefore proves $D_Q \sigma_h = \sigma_h M_s$ with its exact domains.

For each admitted original polynomial degree N , retain its actual remainder matrix B_N , source Gram O_N , and least representative

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{T}_h R_N. \quad (\text{PC7})$$

The source Gram is for the original density $|v_h(1/2+iy)|^2 dy/(2\pi)$ and the original monic coordinates. Since $B_N R_N = I$, (PC6) gives

$$j_E := \eta_h^{-1} J_h : \mathcal{B} \rightarrow E, \quad j_E r_N = I, \quad J_h r_N = \eta_h, \quad q r_N = \sigma_h \eta_h, \quad j_E D = A j_E. \quad (\text{PC8})$$

The last identity uses $J_h D = M_s J_h$ and the intertwining in (PC4). In particular the coefficient observer in (SC7) is j_E . In full Mellin-jet coordinates the same Gram is $\eta_h^{-*} G_N \eta_h^{-1}$; multiplying back by η_h^* and η_h recovers exactly G_N . The unit is present in both the source map and the coordinate congruence.

62.1.2 The weighted critical comparison and every recovered jet

Reserve $y \in \mathbb{R}$ for the critical-line coordinate, distinct from the complex period parameter t . The retained continuous map is

$$Q_{\text{crit}} = \mathcal{S}(\mathbb{R}) / \overline{\zeta(1/2+iy) \mathcal{S}(\mathbb{R})}, \quad \Gamma[F] = [\mathcal{M}F(1/2+iy)], \quad M = M_{1/2+iy}, \quad \Gamma D_Q = M\Gamma. \quad (\text{PC9})$$

It is induced by the plus-sign Mellin restriction: the theta image restricts to $2\zeta(1/2+iy)\mathcal{M}\phi(1/2+iy)$. The continuity and strong-Schwartz domains are those proved in (H16)–(H20); continuity also makes it factor through the closed theta quotient. Let $Z_c = \{\rho \in Z : \Re \rho = 1/2\}$ and $Z_o = Z \setminus Z_c$. Retain the complete factors

$$h_c = \prod_{\rho \in Z_c} (S - \rho)^{m_\rho}, \quad h_o = \prod_{\rho \in Z_o} (S - \rho)^{m_\rho}, \quad h = h_c h_o. \quad (\text{PC10})$$

An empty product is 1. Chinese remainders identify $E = E_c \oplus E_o$, where E_c is the ideal $(h_o)/(h)$ and E_o is the ideal $(h_c)/(h)$, with projections π_c, π_o . The notation also retains the isomorphisms to the products of their local primary algebras, including their nilpotent coordinates.

Define the actual weighted comparison

$$\tilde{\sigma}_h = \sigma_h \eta_h : E \rightarrow Q, \quad \mathcal{C}_h = \Gamma \tilde{\sigma}_h : E \rightarrow Q_{\text{crit}}. \quad (\text{PC11})$$

Then

$$\boxed{\ker \mathcal{C}_h = E_o, \quad \mathcal{C}_h A = M \mathcal{C}_h.} \quad (\text{PC12})$$

To prove the kernel, first take an off-critical root ρ . The reciprocal $(1/2+iy-\rho)^{-1}$ and all its derivatives have polynomial bounds on the real axis. Multiplication by it preserves Schwartz space, the zeta ideal and its closure, and is inverse to $M-\rho$ on Q_{crit} . On a primary block $(A-\rho)^{m_\rho} = 0$, intertwining therefore forces the whole block to zero. Multiplication by v_h preserves each CRT block and does not change this conclusion.

For a critical root $\rho = 1/2 + i\gamma$ retain the evaluation

$$\Xi_\rho[f] = \sum_{j=0}^{m_\rho-1} \frac{i^{-j}}{j!} f^{(j)}(\gamma) x^j \in \mathbb{C}[x]/(x^{m_\rho}), \quad x = S - \rho. \quad (\text{PC13})$$

Every derivative used here vanishes on the zeta ideal and, by continuity, on its closure. The chain rule supplies i^j when differentiating $\mathcal{M}F(1/2+iy)$, so (PC6) and $\bar{J}_h \sigma_h = I$ give the full formula

$$\Xi_\rho \mathcal{C}_h[P] = v_{h,\rho}[P]_\rho = \sum_{j < m_\rho} \sum_{a=0}^j \frac{v_h^{(a)}(\rho)}{a!} \frac{P^{(j-a)}(\rho)}{(j-a)!} x^j. \quad (\text{PC14})$$

In particular the critical observer recovers the actual unit on $e_0 = [1]$, rather than the constant one. Assemble the Ξ_ρ by the retained CRT map and then ι_1^{-1} into $\Xi_c : Q_{\text{crit}} \rightarrow E_c$. Put $v_c = \pi_c \iota_1^{-1}(v_h)$

and $\varepsilon_c = \pi_{c\iota_1^{-1}}(\varepsilon_h)$; these are inverse units in the ideal algebra E_c , whose identity is $e_c = \pi_c(e_0)$. Multiplication by this inverse defines the continuous map

$$\mathcal{R}_c := M_{\varepsilon_c} \Xi_c, \quad \mathcal{R}_c \mathcal{C}_h = \pi_c. \quad (\text{PC15})$$

This is a left inverse on the entire critical block, proving (PC12). For an empty critical packet both maps have zero-dimensional target. When $E_c \neq 0$, (PC14) proves $\mathcal{C}_h(e_0) \neq 0$. The finite image $W_h = \mathcal{C}_h(E_c)$ is preserved by M , and $\mathcal{C}_h|_{E_c}$ and $\mathcal{R}_c|_{W_h}$ are inverse maps.

62.1.3 The comparison kernel is a holomorphic period subbundle

Use exactly the polynomial $\chi = h$ in (SC1)–(SC6), its potential

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad h(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad u \in \mathbb{C}^*. \quad (\text{PC16})$$

Fix the original argument of u and rays $\theta_j = (\arg u + \pi + 2\pi j)/(q+1)$, with $\Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}$ and $\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS$. The original monomial factors are

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{PC17})$$

They have nonzero determinant: a column dependence of their last matrix gives a polynomial of degree at most q vanishing at all $q+1$ roots of unity. Its coefficients must be zero. Along any compact coefficient path the negative leading ray term dominates the lower-degree terms by an integrable bound $C r^L \exp(-r^{q+1}/(2(q+1)|u|))$. Differentiation under the integral is therefore valid. The original polynomial reduction $uP' + (h-t)P$ has leading degree $\deg P + q$ and the same leading coefficient, so it has a unique remainder below degree q . Its integral vanishes by the exact derivative and cancellation of the two origin values. Coefficient variation thus gives $\Pi' = \Pi B$ along a path, and the adjugate determinant identity gives $\det \Pi(s) = \det \Pi(0) \exp(\int \text{Tr } B)$. This proves invertibility at every t , including all collisions.

In the unchanged coefficient basis set

$$\ell([P]) = [S^{q-1}] \text{rem}_h P, \quad \mathcal{R} = e_0 \ell, \quad A_t = A + t\mathcal{R}, \quad \Pi'(t) = -\Pi(t)A_t/u. \quad (\text{PC18})$$

The last equality follows by differentiating in t : for the final column the sole reduction is $[h-t] = 0$, obtained from the differential of the constant polynomial one. Define maps on the period target, with range still in the original critical quotient,

$$\mathcal{C}_{h,t} = \mathcal{C}_h \Pi(t)^{-1}, \quad A_t^{\text{per}} = \Pi(t)A_t \Pi(t)^{-1}, \quad \mathcal{K}_{h,t} = \ker \mathcal{C}_{h,t} = \Pi(t)E_o. \quad (\text{PC19})$$

The equality of kernels follows from invertibility and (PC12). The inverse matrix is holomorphic, so this is a holomorphic kernel bundle of constant rank, with the indicated global frame from E_o . Its quotient is the fixed finite image W_h ; more explicitly $\mathcal{C}_{h,t} \Pi(t)|_{E_c} = \mathcal{C}_h|_{E_c}$ and $\Pi(t)|_{E_c} \mathcal{R}_c$ is a right inverse on W_h .

The exact action defect is

$$\boxed{\mathcal{C}_{h,t} A_t^{\text{per}} - M \mathcal{C}_{h,t} = t \mathcal{C}_h(e_0) \ell \Pi(t)^{-1}.} \quad (\text{PC20})$$

Indeed multiplying the first term gives $\mathcal{C}_h(A + t\mathcal{R})\Pi^{-1}$, and the part containing A cancels $M \mathcal{C}_h \Pi^{-1}$ by (PC12). This proves the formula before restricting its domain.

Put $F = E_o$, $p = \dim F$. If $0 < p < q$, it is the ideal $(h_c)/(h)$, with ordered basis $[h_c], [Sh_c], \dots, [S^{p-1}h_c]$. The last member is monic of degree $q - 1$, so $\ell|F$ is onto with kernel dimension $p - 1$. An ideal containing e_0 would be all of E , hence $e_0 \notin F$. It follows from (PC14) and (PC20) that, for $t \neq 0$, the restriction of this action defect to $\mathcal{K}_{h,t}$ has

$$\text{im} = \mathbb{C}\mathcal{C}_h(e_0), \quad \ker = \Pi(t)(F \cap \ker \ell), \quad \text{rank} = 1. \quad (\text{PC21})$$

At $t = 0$ it is zero. If $F = 0$ its restricted domain is zero. If $F = E$, the comparison $\mathcal{C}_h$ is zero and every displayed defect is zero. These statements cover all actual sector possibilities without asserting that an off-critical zero exists.

62.1.4 The full curvature and both original metric observations

Take any specified coefficient inclusion $I : F \hookrightarrow E$; put $L = \ell I$, and for $0 < p < q$ define

$$Y = \Pi I, \quad H = Y^*Y, \quad P = YH^{-1}Y^*, \quad N = I_q - P, \quad z = N\Pi e_0. \quad (\text{PC22})$$

This is exactly the kernel bundle (PC19), measured in the fixed period-coordinate Gram I_q . The Gram is positive because Y is injective. Multiplication proves that P is its orthogonal projection. Since $e_0 \notin F$ and Π is invertible, $z(t) \neq 0$ for every t . Original invariance $AI = IA_F$ gives

$$Y' = -YA_F/u - (t/u)\Pi e_0 L, \quad NY' = -(t/u)zL, \quad N(0)Y''(0) = -z(0)L/u. \quad (\text{PC23})$$

The final equation follows by differentiating (PC18); the term $\Pi(0)A^2I/u^2$ lies inside the kernel frame and is removed by $N(0)$. Thus the first normal derivative vanishes at zero and the normal acceleration has rank one, with its original u factor.

For $t = x + iy_t$ use $\partial_t = (\partial_x - i\partial_{y_t})/2$ and its conjugate. Holomorphicity gives $\partial_t H = Y^*Y'$, $\partial_{\bar{t}} H = Y'^*Y$, and $\partial_{\bar{t}} \partial_t H = Y'^*Y'$. Differentiate $H^{-1}H = I$ and the determinant to obtain

$$\begin{aligned} \partial_{\bar{t}} \partial_t \log \det H &= \text{Tr}(H^{-1}Y'^*Y' - H^{-1}Y'^*YH^{-1}Y^*Y') \\ &= \text{Tr}(H^{-1}Y'^*NY') = \frac{|t|^2}{|u|^2} \|z\|^2 LH^{-1}L^*. \end{aligned} \quad (\text{PC24})$$

Every factor in the last coefficient is strictly positive except the explicit $|t|^2$: $z \neq 0$, $L \neq 0$ and $H^{-1} > 0$. The coefficient is real analytic, so the curvature vanishes at zero to exactly quadratic order. In the determinant-dual metric $(\det H)^{-1}$ its first Chern form is $\frac{i}{2\pi} \frac{|t|^2}{|u|^2} \|z\|^2 LH^{-1}L^* dt \wedge d\bar{t}$, with $i dt \wedge d\bar{t} = 2dx \wedge dy_t$. This proves that the SC curvature is the curvature of this actual critical-comparison kernel in the specified fixed period metric.

The original theta Gram remains separately and explicitly present. Let $G_{N,F} = I^*G_N I$, and set

$$B_{N,F} = G_{N,F}^{-1}H, \quad \tilde{G}_N = \Pi^{-*}G_N\Pi^{-1}, \quad P_N^\theta = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{PC25})$$

Thus $H = G_{N,F}B_{N,F}$ and $Y^*\tilde{G}_N Y = G_{N,F}$. Both identities follow by direct multiplication. The squared normal norm of Πe_0 for the moving target metric is exactly

$$e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{PC26})$$

It is positive because it is the squared norm of the nonzero G_N -orthogonal component of e_0 outside F . From (PC18), $\partial_t \Pi^{-1} = A_t \Pi^{-1}/u$, so

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N A_t \Pi^{-1}/u, \quad \partial_{\bar{t}} \tilde{G}_N = \Pi^{-*} A_t^* G_N \Pi^{-1}/\bar{u}. \quad (\text{PC27})$$

In the product rule for $Y^*\tilde{G}_N Y$ these terms cancel the derivatives of Y and Y^* exactly. Hence its Gram and determinant are constant in the same holomorphic frame. Formula (PC24) and this cancellation are related by the full displayed metric congruence, including all original theta masses and unit factors. For $F = 0$ the determinant is 1; for $F = E$, $N = 0$. Both endpoint curvatures are therefore zero.

62.1.5 The singular connection on the retained critical kernel

The fixed- u formulas above now extend on each original branch of $u \neq 0$ using (XD.u1)–(XD.u5), with $\chi = h$ at $k = 1$. Their coefficient identification is exactly (PC4), and the full v_h in (PC6)–(PC8) remains unchanged. Put $\Omega_u = -C_h(t)/u^2 + B_h/u$ in these original coordinates. The observer and its full kernel obey

$$C_{h,u,t} = C_h \Pi(u, t)^{-1}, \quad \ker C_{h,u,t} = \Pi(u, t) E_o, \quad \partial_u C_{h,u,t} = C_h (C_h(t)/u^2 - B_h/u) \Pi^{-1}. \quad (\text{PC.u1})$$

Invertibility proves the first two identities; differentiating $\Pi^{-1} \Pi = I$ with (XD.u5) proves the third. The displayed critical target, all local inverse units in (PC15), and the right inverse $\Pi|_{E_c} \mathcal{R}_c$ are unchanged.

For $Y = \Pi I$, $H = Y^* Y$ and $N = I - Y H^{-1} Y^*$ from (PC22), the normal u map is

$$N \partial_u Y = N \Pi (-C_h(t)/u^2 + B_h/u) I. \quad (\text{PC.u2})$$

The full curvature and both metric derivatives are exactly (SC.u2)–(SC.u4), now on this particular kernel inclusion. To check the marked boundary parameter $t = 0$ directly, $C_h(0) = \Phi_h(A)$ preserves the ideal E_o , so the normal image of its order-two term is zero; its retained first-order term is $u^{-1} N \Pi B_h I$. Thus

$$\partial_{\bar{u}} \partial_u \log \det H(u, 0) = |u|^{-2} \text{Tr}(H^{-1} I^* B_h^* \Pi^* N \Pi B_h I), \quad \partial_{\bar{u}} \partial_t \log \det H(u, 0) = 0. \quad (\text{PC.u3})$$

The second equality follows because the normal t map is zero at $t = 0$, as already proved in (PC23). Empty and full kernels have zero normal maps and use the same empty determinant convention. At other t the complete two terms of (PC.u2), their cross pairing and the mixed u, t coefficient remain.

For each cyclic tensor packet (PC29), apply this calculation to the literal polynomial $\chi_{h,k}$ with its degree q_k , not to a product of one-factor period matrices. Replace h, E_o, I, C_h by $\chi_{h,k}, F_k, I_k, C_k$ in (PC.u1)–(PC.u3). Their proofs use precisely the already proved ideal and intertwining identities (PC34)–(PC38), so this gives the full u derivative of the same critical observation and the two-parameter curvature of every existing kernel minor. The marked-product period map instead has domain $E_h^{\otimes k}$ and restricts to E_k through the exact weighted injection η_k in (PC29). Its retained rectangular period form is $\eta_k^* \Pi_{\text{prod}}^* \Pi_{\text{prod}} \eta_k$. These typed maps and the parameter cyclic relation in (XD.u7)'s following paragraph carry the two families together; their different coefficient domains are never silently identified.

62.1.6 The exact completed tensor comparison and its collision jets

We now retain an arbitrary original integer $k \geq 1$. Set

$$B_k = E_h^{\otimes k} = \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)), \quad S = s_1 + \dots + s_k, \quad U_k = v_h^{\otimes k}, \quad (\text{PC28})$$

and retain the original cyclic algebra and its two distinct maps. The variable isomorphism between its tensor coefficient carriers is $\iota_k = \iota_1^{\otimes k} : E^{\otimes k} \xrightarrow{\sim} B_k$. Thus

$$E_k = \mathbb{C}[S] / (\chi_{h,k}), \quad A_k = M_S, \quad \alpha_k[P] = [P(S)] \in B_k^{\mathfrak{S}_k}, \quad \eta_k = M_{U_k} \alpha_k. \quad (\text{PC29})$$

The map α_k is unital; η_k is the original unit-weighted module injection. Define the fully typed comparison $\bar{\alpha}_k = \iota_k^{-1} \alpha_k : E_k \rightarrow E^{\otimes k}$. Then $\eta_k = \eta_h^{\otimes k} \bar{\alpha}_k$, since $\eta_h = M_{v_h} \iota_1$. Their source polynomial is exactly

$$\chi_{h,k}(S) = \prod_{\lambda} (S - \lambda)^{m_{\lambda}}, \quad m_{\lambda} = \max_{\substack{\rho_1 + \dots + \rho_k = \lambda \\ \rho_i \in \mathbb{Z}}} \left(1 + \sum_{i=1}^k (m_{\rho_i} - 1) \right). \quad (\text{PC30})$$

All tuples are ordered in the full tensor algebra; equal sums are grouped by their literal equality. To verify the exponent, the local factor at a tuple has coordinates $x_i = s_i - \rho_i$ and quotient by $(x_i^{m_{\rho_i}})$.

The nilpotent sum $x_1 + \cdots + x_k$ has highest nonzero power $D = \sum_i (m_{\rho_i} - 1)$: that power contains the monomial $\prod_i x_i^{m_{\rho_i} - 1}$ with the nonzero coefficient $D! / \prod_i (m_{\rho_i} - 1)!$. Higher powers vanish. Taking the maximum over equal sums proves (PC30) and the injectivity of α_k . The invariant unit vector already has that same annihilator, so passage to symmetric invariants does not alter it.

The coefficient map of the same collision-jet calculation. The nonzero factorial in the preceding proof also determines the entire coefficient-specialization map. Fix an ordered tuple $\rho = (\rho_1, \dots, \rho_k)$ occurring in (PC30), and retain $m_i = m_{\rho_i}$, $\lambda = \sum_i \rho_i$, $D = \sum_i (m_i - 1)$, and the original coordinates $x_i = s_i - \rho_i$. On the actual retained coefficient ring $A \subset \mathbb{C}$ carrying these coordinates and the full Taylor unit and inverse, the local map of (PC29) is

$$\begin{aligned} \mathcal{E}_A &= A[x_1, \dots, x_k] / (x_i^{m_i})_{i=1}^k, & J &= \sum_i x_i, & S &= \lambda + J, \\ \mathcal{C}_A &= A[S] / ((S - \lambda)^{D+1}), & \eta_\rho[P] &= U_\rho P(\lambda + J), & U_\rho &= \prod_i j_{\rho_i}(v_h). \end{aligned} \quad (\text{PC30a})$$

Every original coefficient of $j_{\rho_i}(v_h)$, including its inverse, is retained. This is a module map for the displayed original sum coordinate, since multiplication by S on the source is multiplication by $\lambda + J$ on the target. Its underlying unweighted ring map has value one at one, and its weighted map has the original value U_ρ at one.

For the specified coefficient map $\alpha : A \rightarrow k_0$ where it is defined, with finite target of characteristic $p > \max_i m_i$, put $\mathcal{R} = A_{\ker \alpha}$. The image of α is a finite field, so its kernel is maximal, and the map extends by $a/b \mapsto \alpha(a)/\alpha(b)$. The characteristic-zero domain $\mathcal{R}$ has every integer prime to p as a unit and p as a nonzero nonunit. Thus all the following maps are over this specified coefficient localization, with no extension to a larger CRT stage being presumed.

For every $0 \leq r \leq D$ retain the complete vector

$$V_r = \sum_{\substack{0 \leq b_i \leq m_i \\ \sum_i b_i = r}} \frac{x^{\mathbf{b}}}{\prod_i b_i!}, \quad W_r = U_\rho V_r, \quad J^r = r! V_r, \quad JW_r = (r+1)W_{r+1}. \quad (\text{PC30b})$$

Here $W_{D+1} = 0$. Each denominator is a unit in $\mathcal{R}$. To verify the first multiplication identity, expand $(\sum_i x_i)^r$: the surviving monomial $x^{\mathbf{b}}$ has exactly $r! / \prod_i b_i!$ ordered assignments. For the second, its degree- $r+1$ coefficient is $\sum_{i: b_i > 0} b_i / \prod_i b_i! = (r+1) / \prod_i b_i!$. Multiplication by the original unit proves the formula for W_r . A tuple of every degree $r \leq D$ exists by successively filling the capacities $m_i - 1$; choose one pivot $\mathbf{b}^*(r)$ for each such degree. The functionals $\lambda_r(x) = \mathbf{b}^*(r)! [x^{\mathbf{b}^*(r)}]x$ satisfy $\lambda_r(V_t) = \delta_{rt}$. Consequently $\pi(x) = \sum_r \lambda_r(x) V_r$ is a projection onto $\Lambda = \bigoplus_{r=0}^D \mathcal{R} V_r$. Subtracting this projection leaves exactly the span F of the nonpivot monomials. Conjugating by the entire unit therefore proves

$$\begin{aligned} \mathcal{E}_{\mathcal{R}} &= \Lambda_U \oplus U_\rho F, & \Lambda_U &= \bigoplus_{r=0}^D \mathcal{R} W_r, \\ L_U &:= \eta_\rho(\mathcal{C}_{\mathcal{R}}) = \bigoplus_{r=0}^D r! \mathcal{R} W_r, & \eta_\rho((S - \lambda)^r) &= r! W_r, \\ \Lambda_U / L_U &\simeq \bigoplus_{r=0}^D \mathcal{R} / (p^{d_r}), & d_r &= \sum_{a \geq 1} \lfloor r/p^a \rfloor, & [\sum_r c_r W_r] &\longmapsto (c_r \bmod p^{d_r})_r. \end{aligned} \quad (\text{PC30c})$$

Counting the factors of p in each integer up to r gives $r! = p^{d_r} \varepsilon_r$ with $\varepsilon_r \in \mathcal{R}^\times$, proving the quotient formula. Its kernel and surjectivity follow in the displayed basis. In particular the cokernel into the full local tensor also retains the free summand $U_\rho F$. The containing lattice is the exact saturation:

$(L_U \otimes \text{Frac } \mathcal{R}) \cap \mathcal{E}_{\mathcal{R}} = \Lambda_U$. Indeed every factorial becomes invertible in the fraction field, and the free complement in the displayed splitting has zero component for an element in that rational span. No original vector has been divided by its factorial inside $\mathcal{E}_{\mathcal{R}}$.

Apply the actual map to $B = k_0$, or the quotient map to $B = \mathcal{R}/p\mathcal{R}$, and mark base changes by bars. All pivot maps and the full inverse unit base change, so the $\overline{W}_r$ remain independent. The exact result is

$$\begin{aligned} \overline{\eta}_{\rho}((S - \overline{\lambda})^r) &= \overline{r!} \overline{W}_r, & \ker \overline{\eta}_{\rho} &= ((S - \overline{\lambda})^p), \\ \text{im } \overline{\eta}_{\rho} &= \bigoplus_{r=0}^{\min(D, p-1)} B \overline{W}_r, & \text{ind}(M_{S-\overline{\lambda}} | \overline{\mathcal{C}}_{\mathcal{R}}) &= D+1, & \text{ind}(M_{\overline{J}} | \text{im } \overline{\eta}_{\rho}) &= \min(p, D+1). \end{aligned} \quad (\text{PC30d})$$

The kernel ideal is zero if $D < p$. The factorial is a unit below p and zero from p onwards, proving the image and kernel. In characteristic p , $\overline{J}^p = \sum_i \overline{x}_i^p = 0$, whereas $\overline{J}^{\min(p-1, D)} \overline{U}_{\rho} = \min(p-1, D)! \overline{W}_{\min(p-1, D)} \neq 0$. These identities prove the two different indices with their exact module targets. They are joined by the explicit free resolution and its connecting map

$$\begin{aligned} 0 \longrightarrow \mathcal{C}_{\mathcal{R}} \xrightarrow{\eta_{\rho}} \Lambda_U \longrightarrow \Lambda_U / L_U \longrightarrow 0, \\ \text{Tor}_1^{\mathcal{R}}(\Lambda_U / L_U, B) = \bigoplus_{r=p}^D B \epsilon_r, & \quad \partial \epsilon_r = (S - \overline{\lambda})^r \in \ker \overline{\eta}_{\rho}. \end{aligned} \quad (\text{PC30e})$$

In the retained bases its differential is $\text{diag}(0!, \dots, D!)$. Its base change has exactly the displayed kernel, proving this Tor group and its connecting map, with differential sign positive.

There is also an exact formula for collisions after a coefficient map actually defined on the full one-factor CRT data in (PC28)–(PC30), with $p > m_{\rho}$ for every $\rho \in Z$. Retain those idempotents and their original inverses, so their base changes still give the product of the ordered local factors. Let $\mathcal{Z}_{k,p} = \{\alpha(\sum_i \rho_i) : \rho \in Z^k\}$ be the set of attained residual sums. Define

$$\psi_{h,k}(X) = \prod_{\mu \in \mathcal{Z}_{k,p}} (X - \mu)^{n_{\mu}}, \quad n_{\mu} = \max_{\substack{\rho \in Z^k \\ \alpha(\sum_i \rho_i) = \mu}} \min \left(p, 1 + \sum_i (m_{\rho_i} - 1) \right). \quad (\text{PC30f})$$

Every maximum in this display is over a nonempty set. When Z is empty, $k \geq 1$ makes the tuple set empty, so the product is $\psi_{h,k} = 1$; both the original cyclic source $k_0[X]/(1)$ and its tensor image are zero. The global inequality on the orders is vacuous in this empty case. The multiplication matrix on each factor is $\mu + \overline{J}$, whose least annihilating power is (PC30d). More explicitly, write a nonzero polynomial as $P(X) = (X - \mu)^e Q(X)$ with $Q(\mu) \neq 0$. The matrix $Q(\mu + \overline{J})$ is the nonzero scalar $Q(\mu)$ times $I + H$, with H a polynomial in $\overline{J}$ having zero constant term and hence nilpotent. Its inverse is the finite geometric series $Q(\mu)^{-1} \sum_{j \geq 0} (-H)^j$, stopped after the nilpotence index. Thus $P(\mu + \overline{J}) = 0$ exactly when $\overline{J}^e = 0$. This proves that the least monic annihilating polynomial is the indicated power of $X - \mu$. On a product, a polynomial annihilates the operator exactly when it annihilates every factor; its monic annihilator is their least common multiple. Taking the maximum exponent at each equal residual sum proves (PC30f). The original reduced polynomial $\overline{\chi}_{h,k}$ still annihilates all factors, so $\psi_{h,k} \mid \overline{\chi}_{h,k}$. The original identity vector generates the polynomial image, and the full reduced unit is invertible; hence the precise specialized map of (PC29) has

$$\ker \overline{\eta}_k = (\psi_{h,k})/(\overline{\chi}_{h,k}), \quad \text{im } \overline{\eta}_k \simeq k_0[X]/(\psi_{h,k}), \quad [P] \longmapsto \overline{U}_k P(\overline{S}). \quad (\text{PC30g})$$

The type of this isomorphism is a module isomorphism; the unweighted map has the corresponding algebra isomorphism. Its kernel is the annihilator ideal just proved, and its image is the definition of that polynomial image, proving both statements. This retains new residual collisions instead of reducing the characteristic-zero maximum without its map. It makes no claim that a local coefficient

map extends to all the sum-CRT coefficients: the actual separating idempotent at degree $k \geq p$ in the original quartet corner gives p^{-1} by its divided difference (BTF.9), and forbids such an extension. The characteristic-zero formula (PC30), its critical tuple quotient below, and the completed complex comparison retain their original domains throughout.

Let $\chi_{c,k}$ be the same polynomial formed using only tuples from Z_c , with exponents r_λ given by the same maximum; an absent critical tuple at λ has $r_\lambda = 0$. When $Z_c = \emptyset$, define $\chi_{c,k} = 1$. Put $q_k = \deg \chi_{h,k}$ and $r_k = \deg \chi_{c,k}$. The tuple inclusion proves $r_\lambda \leq m_\lambda$ and hence $\chi_{c,k} \mid \chi_{h,k}$. The same nilpotence proof gives the exact injection

$$\alpha_{c,k} : \mathbb{C}[S]/(\chi_{c,k}) \hookrightarrow E_c^{\otimes k}, \quad [P] \mapsto P(A_c^{[k]})e_c^{\otimes k}, \quad \pi_c^{\otimes k} \bar{\alpha}_k = \alpha_{c,k} \pi_{\chi_{c,k}}. \quad (\text{PC31})$$

Here $A_c = A|E_c$ and $A_c^{[k]}$ is its sum on the ordered tensor factors. The identity vector is the displayed $e_c^{\otimes k}$, so the codomain is precisely the original CRT ideal tensor. For an empty critical factor both the source and target of $\alpha_{c,k}$ are zero. This identity keeps the full source polynomial $\chi_{h,k}$ and the actual quotient map to its critical tuple polynomial.

The target is the completed projective tensor space $Q_{\text{crit}}^{\hat{\otimes} \pi^k}$. There is no loss of the finite critical image on completion. In fact define $i_c = \mathcal{C}_h|E_c$ and use the continuous map $\mathcal{R}_c$ in (PC15). Their tensor maps extend continuously to the projective completions, and on pure tensors their composition is identity. Pure tensors span the finite-dimensional $E_c^{\otimes k}$, which is already complete, so

$$\mathcal{R}_c^{\hat{\otimes} k} i_c^{\otimes k} = I_{E_c^{\otimes k}}. \quad (\text{PC32})$$

This proves injectivity and a continuous retraction onto the finite image inside the completed target. Its associated continuous idempotent has that image as its kernel-complement, so the image is closed in the Hausdorff completed space. This argument uses the displayed maps and does not assume preservation of arbitrary exact sequences by completion.

The actual tensor comparison, with its original weighted inclusion, is

$$\mathcal{C}_k = \Gamma^{\hat{\otimes} k} \sigma_h^{\hat{\otimes} k} \eta_k = \mathcal{C}_h^{\hat{\otimes} k} \bar{\alpha}_k : E_k \rightarrow Q_{\text{crit}}^{\hat{\otimes} \pi^k}. \quad (\text{PC33})$$

Here $\sigma_h^{\hat{\otimes} k} \eta_k$ is the existing arithmetic cohomology inclusion. Applying the raw critical evaluations gives $\Xi_c^{\hat{\otimes} k} \mathcal{C}_k = (M_{v_c})^{\otimes k} \alpha_{c,k} \pi_{\chi_{c,k}}$. Applying their inverses (PC32) instead gives $\alpha_{c,k} \pi_{\chi_{c,k}}$. Every derivative coordinate has its product phase $i^{-\sum j_i} / \prod j_i!$ and the full tensor of the local units (PC14). The injectivity in (PC31)–(PC32) therefore proves the exact kernel and image dimension

$$F_k := \ker \mathcal{C}_k = (\chi_{c,k})/(\chi_{h,k}), \quad \dim F_k = q_k - r_k, \quad \dim \text{im } \mathcal{C}_k = r_k. \quad (\text{PC34})$$

To display all collision data, at an ambient sum λ write $x = S - \lambda$ and $\chi_{c,k}(S) = x^{r_\lambda} a_\lambda(x)$ with $a_\lambda(0) \neq 0$. The kernel in that original primary factor is the image of the explicit map

$$\mathbb{C}[x]/(x^{m_\lambda - r_\lambda}) \longrightarrow \mathbb{C}[x]/(x^{m_\lambda}), \quad [b] \longmapsto [x^{r_\lambda} a_\lambda(x)b]. \quad (\text{PC35})$$

It is injective: multiply by the recursively computed inverse of a_λ and cancel the displayed power of x . It is onto the local ideal by its definition. Thus all $m_\lambda - r_\lambda$ kernel jets remain, with action $S = \lambda + x$ and the local unit a_λ retained. If $\Re \lambda \neq k/2$, no all-critical tuple has that sum, so $r_\lambda = 0$ and the entire block belongs to the kernel. At a sum with real part $k/2$, the same exact formula still applies: an absent critical tuple kills the whole block, and a coinciding critical tuple retains precisely the quotient of length r_λ while the additional jets lie in (PC35). This is the complete relationship between factorwise critical selection and selection by the real part of the sum.

The original sum action also descends with its exact type. Multiplication by $1/2 + iy$ is continuous on the one-factor quotient and preserves its relation space. On the completed tensor target define the continuous operator

$$M^{[k]} = \sum_{a=1}^k I^{\hat{\otimes}(a-1)} \hat{\otimes} M \hat{\otimes} I^{\hat{\otimes}(k-a)}. \quad (\text{PC36})$$

On pure tensors this is the sum of the original k multipliers, and continuity extends it to the completed target. Since each η_h commutes with the original one-factor action, (PC12) gives $\mathcal{C}_k A_k = M^{[k]} \mathcal{C}_k$ on the cyclic image. At source level, the finite multinomial expansion of $P(D_1 + \cdots + D_k) F_h^{\otimes k}$ and (PC6) give $q^{\widehat{\otimes} k} \mathcal{V}_k P = \sigma_h^{\widehat{\otimes} k} \eta_k[P]$. Thus both the completed comparison and its sum action are induced by the original weighted source, with no replacement of U_k .

62.1.7 The same kernel curvature in every original cyclic tensor packet

Form the original period matrix $\Pi_k(t)$ of (SC1)–(SC6) with the literal polynomial $\chi_{h,k}$, its full increasing-power basis, the same specified $u \neq 0$, and its own degree q_k in the ray formula. Write $e_{0,k} = [1]$, $\ell_k[P] = [S^{q_k-1}] \text{rem}_{\chi_{h,k}} P$ and $A_{k,t} = A_k + t e_{0,k} \ell_k$. The preceding convergence, remainder and determinant argument applies to this polynomial with all its coefficients; hence $\Pi'_k = -\Pi_k A_{k,t}/u$ and it is invertible at every t . Define

$$\mathcal{C}_{k,t} = \mathcal{C}_k \Pi_k^{-1}, \quad A_{k,t}^{\text{per}} = \Pi_k A_{k,t} \Pi_k^{-1}, \quad \mathcal{K}_{k,t} = \Pi_k F_k. \quad (\text{PC37})$$

Equations (PC34) and (PC36) prove

$$\ker \mathcal{C}_{k,t} = \mathcal{K}_{k,t}, \quad \mathcal{C}_{k,t} A_{k,t}^{\text{per}} - M^{[k]} \mathcal{C}_{k,t} = t \mathcal{C}_k(e_{0,k}) \ell_k \Pi_k^{-1}. \quad (\text{PC38})$$

If $0 < r_k < q_k$, then $\mathcal{C}_k(e_{0,k}) \neq 0$ by (PC31)–(PC33). Put $p_k = q_k - r_k$. The kernel ideal has basis $[\chi_{c,k}], \dots, [S^{p_k-1} \chi_{c,k}]$, with the last member monic of degree $q_k - 1$. Consequently $\ell_k|F_k$ is onto, its kernel has dimension $p_k - 1$, and $e_{0,k} \notin F_k$. For every $t \neq 0$, the restriction of (PC38) to $\mathcal{K}_{k,t}$ has rank one, image $\mathbb{C} \mathcal{C}_k(e_{0,k})$, and full kernel $\Pi_k(F_k \cap \ker \ell_k)$. At zero it vanishes.

With any specified inclusion $I_k : F_k \hookrightarrow E_k$, put $Y_k = \Pi_k I_k$, $H_k = Y_k^* Y_k$, $N_k = I - Y_k H_k^{-1} Y_k^*$, $z_k = N_k \Pi_k e_{0,k}$ and $L_k = \ell_k I_k$. The calculations (PC23)–(PC24) now give

$$\partial_{\bar{t}} \partial_t \log \det H_k = \frac{|t|^2}{|u|^2} \|z_k\|^2 L_k H_k^{-1} L_k^*. \quad (\text{PC39})$$

Its coefficient is strictly positive for $0 < r_k < q_k$, and the normal acceleration at zero is $-z_k(0) L_k/u$ with rank one. This calculation includes the collision-kernel jets of (PC35). The complete neighboring-minor form is obtained with

$$J_g = [\chi_{c,k}, S\chi_{c,k}, \dots, S^{p_k-1} \chi_{c,k}], \quad J_- = [\chi_{c,k}, \dots, S^{p_k-2} \chi_{c,k}], \\ J_+ = [J_g, e_{0,k}], \quad h_J = \det((\Pi_k J)^* \Pi_k J). \quad (\text{PC40})$$

For $p_k = 1$, J_- is empty and $h_- = 1$. The Schur complement of the last column gives $h_+ = h_g \|z_k\|^2$; the adjugate formula for the last inverse-Gram entry gives $L_k H_k^{-1} L_k^* = h_-/h_g$ after the exact constant frame congruence of (SC32). Therefore

$$\partial_{\bar{t}} \partial_t \log h_g = \frac{|t|^2}{|u|^2} \frac{h_+ h_-}{h_g^2}. \quad (\text{PC41})$$

For the original tensor-source Gram $G_{N,k}$, each of these same inclusions has factors $v_{N,J} = \det(J^* G_{N,k} J)$ and $b_{N,J} = \det((J^* G_{N,k} J)^{-1} J^* \Pi_k^* \Pi_k J)$. The literal product identity $h_J = v_{N,J} b_{N,J}$ retains every theta mass, polynomial map and period factor in (PC41). The moving metric $\Pi_k^{-*} G_{N,k} \Pi_k^{-1}$ has constant kernel Gram $I_k^* G_{N,k} I_k$, with the full derivative cancellation (PC27), replacing A_t by $A_{k,t}$. These are the two exact metric observations on the same critical-comparison kernel.

If $Z_c = \emptyset$, then $r_k = 0$, $F_k = E_k$ and $\mathcal{C}_k = 0$; the full kernel bundle has zero curvature. If all packet zeros are critical, $\chi_{c,k} = \chi_{h,k}$, $F_k = 0$, and the empty-kernel curvature is zero. If both sectors occur, an all-critical tuple proves $r_k > 0$, while a repeated off-critical root has sum $k\rho$ of real part

different from $k/2$, proving $r_k < q_k$. Thus the strict case in (PC39) has an exact sector criterion at every original k .

Finally each coefficient arrow in this section has its original Split-Zero lift: $v^\bullet \mapsto (Tv)^\bullet$ and $\tau \mapsto \tau$, with the retained support label. The complete kernel (PC34), its smaller action-defect kernel in (PC38), and its cohomological source image remain separately represented. A vector killed by one of these comparisons lands at that receiving fibre's supported zero. The amplitude projection maps both that zero and external absence to ordinary zero, and the displayed split maps commute with that projection. This preserves the original support and the full arithmetic unit through the critical quotient, the completed tensor target, and the period curvature calculation.

63 The original Laplacian control along the exact period deformation

63.1 The original Laplacian control along the exact period deformation

We calculate the control parameter for the actual companion deformation, and then transport its complete numerical range through the periods. All metrics and coefficient vectors below are the original ones.

Fix the original cyclic algebra, admitted degree and positive Gram

$$\begin{aligned} E = \mathbb{C}[S]/(\chi), \quad q = \deg \chi \geq 1, \quad A = M_S, \quad G = G_N > 0, \quad k \in \mathbb{R}, \\ e_0 = [1], \quad \ell[P] = [S^{q-1}] \operatorname{rem}_\chi P. \end{aligned} \quad (\text{LC1})$$

Here χ is the original monic polynomial, $N \in \mathbb{Z}$ satisfies $N \geq q - 1$, and the deformation parameter is $t \in \mathbb{C}$. Put $R = e_0 \ell$, $A_t = A + tR$, and retain the actual control form and relative, Laplacian and shifted endomorphisms

$$W_t = A_t^* G + G A_t - kG, \quad X_t = G^{-1} W_t, \quad L_t = A_t^2 - kA_t, \quad B_t = A_t - \frac{k}{2} I. \quad (\text{LC2})$$

The control form has value $v^* W_t w = \langle v, X_t w \rangle_G$; thus its exact form-to-operator map is $X_t = G^{-1} W_t$. Here X_t is self-adjoint for $\langle v, w \rangle_G = v^* G w$. Its finite operator norm is defined by the supremum over the original nonzero vectors. Write $\epsilon_N = \|X_0\|_G$. This is an actual finite quantity, so $-\epsilon_N G \preceq W_0 \preceq \epsilon_N G$ follows from the spectral theorem for the displayed positive metric.

For the original reflection-stable source, its complete rank-two calculation (TVB.22)–(TVB.30) evaluates this same quantity. In the unchanged source-orthogonal columns b_j and norms ω_j , put

$$a = b_N^* G b_N, \quad d_+ = b_{N+1}^* G b_{N+1}, \quad z_N = b_N^* G b_{N+1}, \quad D_N = \det K_N.$$

For every integer $j \geq -1$ retain $K_j = \sum_{r=0}^j b_r b_r^* / \omega_r$, with the empty sum $K_{-1} = 0$ on E , and put $D_j = \det K_j$. The omitted-column kernel and determinant are $\tilde{K}_N = K_{N-1} + b_{N+1} b_{N+1}^* / \omega_{N+1}$ and $\tilde{D}_N = \det \tilde{K}_N$. At the first admitted degree $N = q - 1$, the first $q - 1$ monic remainder columns are independent, so K_{q-2} has rank $q - 1$ and $D_{q-2} = 0$. No inverse of this boundary kernel is used. The retained identities give

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1} \frac{\Delta_N}{D_N}, \quad \alpha_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \quad \Delta_N = D_{N+1} + D_{N-1} - D_N - \tilde{D}_N. \quad (\text{LC3})$$

The omitted-column determinant $\tilde{D}_N$, every boundary case and its positive source-minor sum are defined and proved in that full chapter. To check the radius here directly, the original control is $W_0 = G(b_{N+1} b_N^* + b_N b_{N+1}^*) G / \omega_N$. Its relative self-adjoint operator has rank at most two and trace zero; z_N is purely imaginary by the retained reflection phase. Expansion of its squared trace gives $2(ad_+ - |z_N|^2) / \omega_N^2$, so its two possible nonzero eigenvalues are exactly $\epsilon_N, -\epsilon_N$. The two-column determinant identity proves the second equality in (LC3). No estimate or limiting assumption is needed to evaluate it.

63.1.1 The exact additional radius of the companion term

The adjoint of R for the original metric is $R^{\dagger G} = G^{-1}\ell^*e_0^*G$. Thus

$$X_t - X_0 = tR + \bar{t}R^{\dagger G}. \quad (\text{LC4})$$

For $q > 1$, $\ell e_0 = 0$, hence $R^2 = 0$. Put

$$f = G^{-1}\ell^*, \quad a_0 = e_0^*G e_0 > 0, \quad b_0 = \ell G^{-1}\ell^* > 0, \quad \sigma_N = \sqrt{a_0 b_0}. \quad (\text{LC5})$$

The columns e_0, f are G -orthogonal because $e_0^*G f = \bar{\ell}e_0 = 0$. They are independent and their Gram is exactly $\text{diag}(a_0, b_0)$. In these original unscaled columns the matrix of (LC4) is

$$\begin{pmatrix} 0 & tb_0 \\ \bar{t}a_0 & 0 \end{pmatrix}. \quad (\text{LC6})$$

Indeed $Rf = b_0 e_0$, $Re_0 = 0$, $R^{\dagger G}e_0 = a_0 f$ and $R^{\dagger G}f = 0$. On their G -orthogonal complement both terms vanish. The characteristic polynomial on the displayed two-dimensional space is $x^2 - |t|^2 a_0 b_0$. Therefore the exact additional operator norm is $|t|\sigma_N$; all remaining eigenvalues are zero. This retains the two original vector lengths, with no choice of unit vectors.

For $q = 1$, instead $R = I_E$, so (LC4) equals $2 \operatorname{Re}(t)I_E$ and its exact norm is $2|\operatorname{Re} t|$. Consequently the entirely specified quantity

$$E_N(t) = \epsilon_N + \begin{cases} |t|\sigma_N, & q > 1, \\ 2|\operatorname{Re} t|, & q = 1 \end{cases} \quad \text{satisfies} \quad \|X_t\|_G \leq E_N(t), \quad -E_N(t)G \preceq W_t \preceq E_N(t)G. \quad (\text{LC7})$$

The inequality is the triangle inequality for the same operator norm. It is not an assertion that equality always holds. The two summands in (LC7) are the actual norms just computed, and the original ϵ_N remains available through (LC3). The full Laplacian change is also exact:

$$L_t - L_0 = t(AR + RA - kR) + t^2 R^2. \quad (\text{LC8})$$

Thus the quadratic term vanishes for $q > 1$, while for $q = 1$ it is the retained scalar $t^2 I_E$.

63.1.2 An explicit numerical-range enclosure with no missing input

Expansion of (LC2), without a commutation assumption, proves

$$L_t^*G - GL_t = A_t^*W_t - W_tA_t, \quad GL_t = W_tB_t - B_t^*GB_t - \frac{k^2}{4}G. \quad (\text{LC9})$$

For the second equality insert $W_t = B_t^*G + GB_t$, so the right side is $GB_t^2 - k^2G/4 = GL_t$. The first equality follows by expansion; the two middle terms $A_t^*GA_t$ cancel.

For each unchanged nonzero $v \in E$, define the homogeneous observations

$$z = \frac{v^*GL_tv}{v^*Gv}, \quad r = \frac{\|B_tv\|_G}{\|v\|_G}, \quad a + ib = \frac{\langle v, X_tB_tv \rangle_G}{\|v\|_G^2}. \quad (\text{LC10})$$

Cauchy-Schwarz and (LC7) give $a^2 + b^2 \leq E_N(t)^2 r^2$. The second identity in (LC9) gives $\operatorname{Re} z = a - r^2 - k^2/4$ and $\operatorname{Im} z = b$. Write $e = E_N(t)$. First, $a \leq er$ implies $\operatorname{Re} z \leq (e^2 - k^2)/4 - (r - e/2)^2$. Second, direct multiplication gives

$$e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - (\operatorname{Im} z)^2 = (a - e^2/2)^2 + e^2 r^2 - a^2 - b^2 \geq 0.$$

We have proved the actual enclosure

$$\boxed{\operatorname{Re} z \leq \frac{E_N(t)^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq E_N(t)^2 \left(\frac{E_N(t)^2 - k^2}{4} - \operatorname{Re} z \right)}. \quad (\text{LC11})$$

It includes every eigenvalue by taking its nonzero eigenvector; no diagonalizability is needed. It includes the case $E_N(t) = 0$. Every quantity in (LC11) is supplied by the original finite metric, the monic coefficient functional and the actual deformation parameter.

63.1.3 The complete period transport

For fixed $u \neq 0$, the full determinant proof makes the original period matrix $\Pi(u, t)$ invertible at every parameter. Put

$$\tilde{G} = \Pi^{-*} G \Pi^{-1}, \quad \tilde{A}_t = \Pi A_t \Pi^{-1}, \quad \tilde{L}_t = \Pi L_t \Pi^{-1}, \quad \tilde{W}_t = \Pi^{-*} W_t \Pi^{-1}. \quad (\text{LC12})$$

Direct multiplication gives $\tilde{W}_t = \tilde{A}_t^* \tilde{G} + \tilde{G} \tilde{A}_t - k \tilde{G}$ and $\tilde{L}_t = \tilde{A}_t^2 - k \tilde{A}_t$. For $x = \Pi v$, its norm and the complete numerical-range quotient are exactly those in (LC10). Its control quotient and adjoint-defect identity are likewise transported by the literal congruences in (LC12). Thus (LC11) applies unchanged in that period realization.

The transported fixed action is $\Pi A \Pi^{-1}$; the displayed deformed action differs from it by $t \Pi R \Pi^{-1}$, whose control contribution was computed in (LC4)–(LC7). The arithmetic source realizes precisely this change through $D_t^{(k)} = D^{(k)} + \text{tr}_N R j_E$, with $j_E r_N = I$ and the original boundary primitive retained, as proved in (SC7)–(SC8). These equalities supply the source, coefficient and period maps for the same Laplacian control. To retain its complete primitive, let $\mathcal{C}_{\text{prim}}$ be the original primitive term of the tensor complex and write $D_0 : \mathcal{C}_{\text{prim}} \rightarrow \mathcal{C}_{\text{prim}}$ for its original Euler action of cochain degree zero. Thus $K_N^{\text{prim}} : E \rightarrow \mathcal{C}_{\text{prim}}$ and $D^{(k)} d = d D_0$ on this term. The original boundary observation $j_E d = 0$ makes the added finite-rank action vanish there, so $D_t^{(k)} d = d D_0$ as well. Put $K = K_N^{\text{prim}}$. Using $D_t^{(k)} r_N = r_N A_t + dK$, expand both terms of the square:

$$\begin{aligned} (D_t^{(k)})^2 r_N &= D_t^{(k)} (r_N A_t) + D_t^{(k)} dK \\ &= (r_N A_t + dK) A_t + d D_0 K = r_N A_t^2 + d(K A_t + D_0 K). \end{aligned}$$

Subtract $k D_t^{(k)} r_N = k r_N A_t + k dK$ with its original sign. This proves the complete quadratic source comparison

$$\boxed{((D_t^{(k)})^2 - k D_t^{(k)}) r_N - r_N L_t = d(D_0 K_N^{\text{prim}} + K_N^{\text{prim}} A_t - k K_N^{\text{prim}}).} \quad (\text{LC12a})$$

The term $K_N^{\text{prim}} A_t = K_N^{\text{prim}} A + t K_N^{\text{prim}} R$ retains the full deformation contribution. All terms inside the displayed primitive have the same target $\mathcal{C}_{\text{prim}}$; the original differential, source action and boundary class remain explicit. No smallness uniform in packet size is claimed for the fully calculated ϵ_N or σ_N .

The same complete transport now varies in $u \neq 0$ as well as t . With $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$ from (XD.u1)–(XD.u5), the precise derivative equations are

$$\begin{aligned} \partial_u \tilde{G} &= -\Pi^{-*} G \Omega_u \Pi^{-1}, & \partial_u \tilde{G} &= -\Pi^{-*} \Omega_u^* G \Pi^{-1}, \\ \partial_u \tilde{L}_t &= \Pi[\Omega_u, L_t] \Pi^{-1}, & \partial_u \tilde{A}_t &= \Pi[\Omega_u, A_t] \Pi^{-1}. \end{aligned} \quad (\text{LC.u1})$$

Here G, A_t, L_t are independent of u . Each formula follows by differentiating its literal definition (LC12), inserting $\Pi_u = \Pi \Omega_u$ and $(\Pi^{-1})_u = -\Omega_u \Pi^{-1}$, and retaining both product terms. Thus the full numerical-range identity (LC10)–(LC12) holds at every such u with exactly the same original bound $E_N(t)$, while its moving metric has the displayed derivatives. This statement makes no bound on the unweighted target metric. For the original source lift the exact u commutator is $r_N[A, \Omega_u] j_E + d K_N^{\text{prim}} \Omega_u j_E$ by (DS74); the primitive is in the original cochain degree.

The actual comparison numbers used in (LC14)–(LC16) are now the eigenvalues $m(u, t), M(u, t)$ of $G^{-1} \Pi(u, t)^* \Pi(u, t)$. Their original proof by two quadratic-form inequalities applies pointwise without alteration. The full u and mixed curvatures of this same constituent are (SC.u1)–(SC.u4), with its actual inclusion I and Gram $I^* G I$. In particular the exact order-two and order-one normal terms must both be used at $t \neq 0$; the original t -curvature comparison (LC15) remains its own calculated matrix entry.

63.1.4 The exact finite comparison with constituent curvature

Let $I : F \hookrightarrow E$ be an actual injective coefficient map with $AI = IA_F$ and $0 < p := \dim F < q$. In particular $q > 1$, so the radius σ_N in (LC5) applies. Retain the original orthogonal projection and its two scalar observations:

$$\begin{aligned} G_F &= I^*GI, & P_F &= IG_F^{-1}I^*G, & L &= \ell I, & f &= G^{-1}\ell^*, \\ a_\perp &= e_0^*G(1 - P_F)e_0, & b_F &= LG_F^{-1}L^* = \ell IG_F^{-1}I^*\ell^*. \end{aligned} \quad (\text{LC13})$$

The injection and positive G give $G_F > 0$. Multiplication proves $P_F^2 = P_F$ and $P_F^*G = GP_F$, and its image is exactly $I(F)$. Consequently $a_\perp = \|(1 - P_F)e_0\|_G^2$ and $b_F = \|P_F f\|_G^2$, the second identity following from $I^*Gf = I^*\ell^*$, with the same column f as in (LC5).

Both numbers are strictly positive on this actual constituent. Indeed A -invariance implies invariance under every polynomial in A , so $I(F)$ is an ideal of E . Its inverse image in $\mathbb{C}[S]$ is (g) for a unique monic divisor g of χ . Writing $\chi = gh$, the map $[a]_h \mapsto [ga]_\chi$ is an isomorphism onto $I(F)$: it is onto by the ideal description, and $gh \mid ga$ implies $h \mid a$ by cancellation in the polynomial domain. Thus $\deg h = p$ and the complete original ideal has basis $[g], [Sg], \dots, [S^{p-1}g]$. The last vector is monic of degree $q - 1$, proving $L \neq 0$; positivity of G_F^{-1} gives $b_F > 0$. An ideal containing $e_0 = [1]$ is all of E . Properness therefore gives $e_0 \notin I(F)$ and $a_\perp > 0$. Every repeated factor remains in this ideal calculation.

At the same $u \neq 0$ and parameter t , let $m(t)$ and $M(t)$ be the actual smallest and largest eigenvalues of $B(t) = G^{-1}\Pi(t)^*\Pi(t)$. This operator is G -self-adjoint and positive because $v^*GB(t)v = \|\Pi(t)v\|_{I_q}^2 > 0$ for each nonzero v . Its finite spectrum therefore gives $0 < m(t) \leq M(t) < \infty$ and the quadratic-form inequalities $m(t)G \preceq \Pi(t)^*\Pi(t) \preceq M(t)G$. These are evaluated finite comparison quantities, with no bound uniform in the arithmetic inputs imposed on them.

Put $Y = \Pi I$, $H_F = Y^*Y$, and $P^{\text{per}} = YH_F^{-1}Y^*$, the orthogonal projection in the specified period-coordinate metric I_q . Its residual is $z = (1 - P^{\text{per}})\Pi e_0$. The exact comparison is

$$\boxed{m(t)a_\perp \leq \|z(t)\|_{I_q}^2 \leq M(t)a_\perp, \quad \frac{b_F}{M(t)} \leq LH_F(t)^{-1}L^* \leq \frac{b_F}{m(t)}.} \quad (\text{LC14})$$

To prove the first pair, orthogonal projection gives $\|z\|_{I_q}^2 = \min_{x \in F} \|\Pi(e_0 - Ix)\|_{I_q}^2$. The same minimization in the original metric has value $a_\perp$, attained at $x = G_F^{-1}I^*Ge_0$. The lower quadratic-form bound holds for every x , so its minimum is at least $m(t)a_\perp$. Evaluating the upper bound at the displayed original minimizer gives the upper inequality. Compression by I gives $m(t)G_F \preceq H_F \preceq M(t)G_F$. Inversion of positive forms, followed by the unchanged row L , proves the second pair. Explicitly, for any positive matrix T and column w , completing the square proves $w^*T^{-1}w = \max_x (2 \operatorname{Re}(w^*x) - x^*Tx)$. Applying this formula to the three positive matrices verifies the inverse order and both scalar factors without a commutation assumption.

The coefficient of the actual constituent curvature in (SC16) is $c_F(t) = \|z(t)\|_{I_q}^2 LH_F(t)^{-1}L^*/|u|^2$. For completeness, the period equation and $AI = IA_F$ give $Y' = -YA_F/u - t\Pi e_0 L/u$, hence $(1 - P^{\text{per}})Y' = -tzL/u$. Holomorphy of Y gives $\partial_t H_F = Y^*Y'$ and $\partial_{\bar{t}} H_F = Y'^*Y$; differentiating its inverse in the determinant derivative proves $\partial_{\bar{t}} \partial_t \log \det H_F = \operatorname{Tr}(H_F^{-1}Y'^*(1 - P^{\text{per}})Y')$. Substitution yields the full equality and the bounds

$$\begin{aligned} \partial_{\bar{t}} \partial_t \log \det H_F(t) &= |t|^2 c_F(t), \\ \boxed{\frac{m(t)}{M(t)} \frac{a_\perp b_F}{|u|^2} \leq c_F(t) \leq \frac{M(t)}{m(t)} \frac{a_\perp b_F}{|u|^2}.} \end{aligned} \quad (\text{LC15})$$

Every factor in (LC14) is positive, so multiplying its appropriate two lower or two upper bounds proves (LC15). In particular the coefficient is positive even at $t = 0$, while the actual curvature at that parameter is zero through its retained factor $|t|^2$.

The comparison to the Laplacian radius has an exact two-term loss. Orthogonality of P_F gives $a_0 = \|P_F e_0\|_G^2 + a_\perp$ and $b_0 = b_F + \|(1 - P_F)f\|_G^2$. Multiplying with every term retained therefore proves

$$\sigma_N^2 - a_\perp b_F = \|P_F e_0\|_G^2 b_0 + a_\perp \|(1 - P_F)G^{-1}\ell^*\|_G^2 \geq 0. \quad (\text{LC16})$$

Combining (LC15)–(LC16) and the exact additional norm (LC4)–(LC6) gives, on this same original constituent,

$$c_F(t) \leq \frac{M(t)}{m(t)} \frac{\sigma_N^2}{|u|^2}, \quad \partial_{\bar{t}} \partial_t \log \det H_F(t) \leq \frac{M(t)}{m(t)} \frac{\|X_t - X_0\|_G^2}{|u|^2}.$$

Thus the companion control, the original constituent projections, and its period curvature have an explicit finite comparison with the complete loss and both metric comparison eigenvalues retained. The proof assigns no uniform estimate to $M(t)/m(t)$ or σ_N .

64 Periodized source: exact scope and source definitions

64.1 The periodized-source proof and its completed comparison

The complete source note below contains P1–P64 together with P26a and P26b. Its source functions, coefficient frames, Fourier zero coordinate, remainder map and theta relations are retained in full. The adjoining PSA and PSD proof bodies give the detailed source and completion calculations used in its integration.

The standalone density notation is fixed by

$$g = 2\xi, \quad v_h = g/h, \quad w_h(t) = \frac{1}{2\pi} |v_h(1/2 + it)|^2.$$

Here g/h is the entire canceled quotient. PSD1–PSD14 prove its full canceled jets and exponential tails in these original constants. PSD18–PSD20 give the exact isometry and inverse between the original Hilbert-valued circle source and its scalar weighted sequence presentation. PSD21–PSD41 prove the density, complete ideal kernel, generator maps and fixed-period metric limit. For $k = 1$, every selected sampled root has positive weight by full-order cancellation; arbitrary other integer atoms are not assigned positive weights. For $k \geq 2$, the proved convolution positivity applies at every integer atom.

The finite-source observation keeps the full original remainder and its jets. The subsequent fixed-circle completion has the explicitly computed kernel $(p_L)/(\chi)$. The DC comparison supplies the further map from the critical jets to these reduced sample values. The source proofs retain the domains of each arrow and their different completion topologies.

PSA23–PSA25 and FC5–FC7, with the endpoint paragraph following FC13, are the same asymmetric source comparison. PSA27–PSA29 and FC19–FC20 are the same finite Fourier boundary calculation. PSD41 supplies the one-fold cancellation fact also used in DC. These are cross-references to common calculations, not additional theorem layers.

The newly received 16-method finite suite passed normally and with Python optimization; seven substantive source mutations were rejected in each mode. Those exact finite checks do not replace the analytic proofs. The supplied numerical calibrations remain numerical checks; no interval certificate or arithmetic asymptotic upper bound is asserted by the circle construction.

65 Complete original periodized source control note

Source role. complete original readable source ONCE, all66labels; convert with explicit original-label namespace The complete original source is reproduced once. The preceding public wrapper records the exact current reading. Original display tags retain their literal tokens with local namespace Periodized.

Exact revision. f903ae8473efdfcde0f45a83878837289d2b0604c1a34903276d7c840fe972b9

Complete source. sources/web_periodized_source_delivery/Tau_Periodized_Source_Control/RESEARCH_NOTE.md

Source SHA-256. f903ae8473efdfcde0f45a83878837289d2b0604c1a34903276d7c840fe972b9

Source metadata: author (MetaInlines)

Owner-directed SplitZero research continuation

Source metadata: date (MetaInlines)

2026-09-13

Source metadata: title (MetaInlines)

Periodized theta-source recovery and arithmetic endpoint control

65.1 Result and scope

This continuation joins the newly supplied Connes–Consani comparison to the original source-constrained arithmetic metrics. It proves a quantitative finite-period realization of those metrics, carries the original relation quotient through that realization, and calculates the unchanged boundary term between the circle Laplacian and the arithmetic Laplacian.

The principal estimate is on an explicitly specified finite source, not on a freely chosen quotient metric. Write M for its original Gram and $M(L)$ for the Gram of its full periodization, including the zeroth Fourier coefficient. For the two original exponential-weight Grams $M_{a,+}$, $M_{a,-}$,

$$-\frac{M_{a,+} + M_{a,-}}{e^{aL} - 1} \preceq M(L) - M \preceq \frac{M_{a,+} + M_{a,-}}{e^{aL} - 1}, \quad a, L > 0. \quad (\text{Periodized P1})$$

All matrices in (P1) are integrals of the actual source functions. For any fixed finite degree, this supplies a finite period at which the original full-jet quotient can be reconstructed with certified relative error. The four endpoint volumes acquire an error at most

$$2q \log \frac{1 + \eta}{1 - \eta}, \quad 0 < \eta < 1. \quad (\text{Periodized P2})$$

For fixed η , this is $O(q)$, below the existing $q \log k$ scale. The transfer estimate does not itself bound the periodized endpoint volumes as k grows. It is a quantitative, source-faithful realization of that remaining question.

A second theorem gives an exact weighted recovery from all nonzero circle modes. It retains a compensating Riemann-zeta factor because the Fourier lattice is summed literally, not counted once.

The present results have written proofs. The included exact algebraic regressions and numerical calibrations have narrower, separately recorded scopes. No RH conclusion, new Lean certificate, or novelty claim for Parseval, periodization, or positive-form comparison is made.

65.2 1. Source intake and the Deligne operation

The new pasted note reports the original identities

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{E}\phi = \frac{1}{2}\mathcal{U}\Theta\phi, \quad (\mathcal{U}F)(x) = x^{1/2}F(x), \quad (\text{Periodized P3})$$

and the critical comparison

$$\Gamma : Q = \mathcal{B}/\Theta V \longrightarrow \mathcal{S}(\mathbb{R})/\overline{\zeta(1/2 + it)\mathcal{S}(\mathbb{R})}. \quad (\text{Periodized P4})$$

Its finite-packet kernel is the full off-critical generalized block. We retain that theorem and its original jet retractions; we do not infer injectivity from the target’s spectral location. We use its augmented zeroth-mode object without asserting a new independent audit of the printed Proposition 5.2.

The supplied Deligne_FR.tex is a multi-work corpus. Its selected D022 passage, *Poids dans la cohomologie des variétés algébriques*, states strict compatibility of geometric weight filtrations with the actual maps. It also

retains the Thom–Gysin twist. The inline-work list in this uploaded file omits D032 (*Weil II*). For *Weil II* we separately reassembled the previously supplied S20 archive and read its current French page-local records, notably pp. 178, 203 and 206. The input archive and exact selected records are identified in SOURCE_REVIEW.md.

The operation used here is the one emphasized by *Weil II* 3.3.4–3.3.6: establish the estimates on a specified comparison image, retaining its kernel, duality and action. No geometric weight filtration is assigned to the analytical circle spaces below. Instead we construct the missing norm comparison and propagate it to the same original quotient; this avoids replacing a full packet by the image of (P4).

65.3 2. Retain the split coefficients and arithmetic source

The scalar construction and its arithmetic observation remain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (\text{Periodized P5})$$

The target of $p_{\mathbb{Z}}$ stays infinite. A coefficient-linear map f is lifted on a retained support fibre by

$$(\ell, v) \mapsto (\ell, fv), \quad \tau \mapsto \tau. \quad (\text{Periodized P6})$$

If $fv = 0$, its image is the receiving fibre zero $(\ell, 0)$. No new scalar zero is adjoined when a source degree, relation depth or tensor degree changes.

Keep the original spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int \phi = 0\}, \quad D = -x\partial_x, \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad g(s) = 2\xi(s). \quad (\text{Periodized P7})$$

On the finite sources used here, all functions are finite linear combinations of tensors of strong-Schwartz functions and their logarithmic derivatives. In particular every integral below, with any fixed real exponential weight in a logarithmic variable, is finite.

For a fixed nonempty packet polynomial h containing the full orders of its selected zeros, retain

$$\mathcal{M}F_h = g/h, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad h(D)F_h = \Theta\phi_*. \quad (\text{Periodized P8})$$

At tensor degree $k \geq 1$, the cyclic source and full arithmetic observation are

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad E = \mathbb{C}[S]/(\chi_{h,k}), \quad A = M_S, \quad (\text{Periodized P9})$$

$$J^{(k)}\mathcal{V}_{h,k} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h). \quad (\text{Periodized P10})$$

These are inputs from the existing programme. The present proofs of norm transfer require no assumption about the location of the roots of h or χ . The entire quotient g/h , its unit and the original source remain fixed.

65.4 3. An exact diagonal logarithmic coordinate map

For $k = 1$, put $\mathcal{K} = \mathbb{C}$. For $k > 1$, put $\mathcal{K} = L^2(\mathbb{R}^{k-1}, d\mathbf{z})$. Set

$$x_i = e^{r+z_i} \ (i < k), \quad x_k = e^r. \quad (\text{Periodized P11})$$

Thus $r = \log x_k, z_i = \log x_i - \log x_k$. The map from $(r, \mathbf{z})$ to $(\log x_1, \dots, \log x_k)$ has determinant of absolute value one. The exact Hilbert isometry is

$$(\mathcal{U}_k F)(r, \mathbf{z}) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r). \quad (\text{Periodized P12})$$

Its inverse substitutes (P11) and multiplies by $e^{-(kr+\sum z_i)/2}$. Direct change of variables proves

$$\|\mathcal{U}_k F\|_{L^2(dr;\mathcal{K})}^2 = \|F\|_{L^2(d^k x)}^2, \quad \mathcal{U}_k D^{(k)} F = (-\partial_r + k/2)\mathcal{U}_k F, \quad D^{(k)} = \sum_i D_i. \quad (\text{Periodized P13})$$

The fixed Fourier transform is

$$\widehat{\psi}(u) = \int_{\mathbb{R}} \psi(r) e^{iur} dr, \quad \|\psi\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} \|\widehat{\psi}(u)\|_{\mathcal{K}}^2 du. \quad (\text{Periodized P14})$$

Then $-\partial_r + k/2$ is carried to multiplication by $k/2 + iu$. The tensor-sum coordinate is still S , not its average.

For each finite source, ψ and its derivatives are $\mathcal{K}$ -valued Schwartz functions. For example weighted Cauchy-Schwarz and the original separate strong-Schwartz bounds show $\int e^{2a|r|} \|\partial_r^j \psi(r)\|_{\mathcal{K}}^2 dr < \infty$ for every fixed $a > 0, j$. One-dimensional Sobolev estimates then give the corresponding pointwise decay. These facts justify the Hilbert-valued Fourier and periodization manipulations below.

65.5 4. Periodization and the full zeroth-mode map

For $L > 0$, define

$$\mathcal{P}_L \psi(r) = \sum_{m \in \mathbb{Z}} \psi(r + mL), \quad 0 \leq r < L. \quad (\text{Periodized P15})$$

The series and its derivatives converge normally on the circle for the stated source. In the orthonormal Fourier basis $L^{-1/2} e^{-2\pi i n r / L}$, unfolding the integral gives

$$\widehat{\mathcal{P}_L \psi}(n) = L^{-1/2} \widehat{\psi}(2\pi n / L). \quad (\text{Periodized P16})$$

The zeroth coefficient is $L^{-1/2} \widehat{\psi}(0)$; the corresponding constant function has value $L^{-1} \widehat{\psi}(0)$.

The source-preserving augmented map is

$$\mathcal{A}_L \psi = \left((L^{-1/2} \widehat{\psi}(2\pi n / L))_{n \neq 0}, \widehat{\psi}(0) \right). \quad (\text{Periodized P17})$$

Its target metric is the direct sum of the ordinary sequence metric and $L^{-1} \langle \cdot, \cdot \rangle_{\mathcal{K}}$ on the last component. Reconstruction inserts $L^{-1/2} \widehat{\psi}(0)$ at index zero. Thus (P17), with that target metric, has exactly the same Gram as the full periodization. The coefficient object at $L = 0$ may retain the last coordinate, as in the supplied repair; no finite limit of its raw metric at $L = 0$ is asserted.

For $k = 1$, $F = \Theta \phi$, (P3) gives

$$\Sigma_{eL} \mathcal{E} \phi = \frac{1}{2} \mathcal{P}_L \mathcal{U}_1 \Theta \phi, \quad \widehat{\Sigma_{eL} \mathcal{E} \phi}(0) = \frac{\mathcal{M} \Theta \phi(1/2)}{2\sqrt{L}}. \quad (\text{Periodized P18})$$

For the unchanged ϕ_* , the right side is $\xi(1/2)/\sqrt{L}$, exactly the new note's retained mode. Working with $\mathcal{E}$ rather than the original theta map would multiply its Gram by $1/4$; this factor is not assigned the value one.

65.6 5. The original Gram is recovered with an exponential error

Fix any specified finite source map $T : \mathbb{C}^d \rightarrow \mathcal{B}^{\otimes k}$, and write $\Psi = \mathcal{U}_k T$. All matrices use its unchanged coefficient basis. Define

$$M = \Psi^* \Psi, \quad M(L) = (\mathcal{P}_L \Psi)^* \mathcal{P}_L \Psi, \quad Z_0 c = \widehat{\Psi c}(0), \quad (\text{Periodized P19})$$

$$M_{a,\pm} = (e^{\pm ar} \Psi)^* (e^{\pm ar} \Psi). \quad (\text{Periodized P20})$$

In original coordinates these are exactly

$$c^* M_{a,\pm} c = \int_{(0,\infty)^k} x_k^{\pm 2a} |Tc(\mathbf{x})|^2 d^k x. \quad (\text{Periodized P21})$$

Both weights remain; they are auxiliary observations of the same functions.

Define the matrix-valued correlation by

$$c^* \mathcal{C}(s) d = \int_{\mathbb{R}} \langle \Psi c(r), \Psi d(r+s) \rangle_{\mathcal{K}} dr. \quad (\text{Periodized P22})$$

Unfolding one period, with absolutely convergent products, proves

$$M(L) = \sum_{m \in \mathbb{Z}} \mathcal{C}(mL), \quad \mathcal{C}(0) = M. \quad (\text{Periodized P23})$$

For $s > 0$, weighted Cauchy–Schwarz gives, for the same coefficient vector c ,

$$|c^* \mathcal{C}(s) c| \leq e^{-as} \sqrt{(c^* M_{a,-} c)(c^* M_{a,+} c)}. \quad (\text{Periodized P24})$$

Indeed the product of $e^{-ar} \Psi c(r)$ and $e^{a(r+s)} \Psi c(r+s)$ is e^{as} times the original integrand. For negative s , exchange the two weights. Summing (P24) over both nonzero lattice directions gives

$$|c^*(M(L) - M)c| \leq \frac{2}{e^{aL} - 1} \sqrt{(c^* M_{a,-} c)(c^* M_{a,+} c)} \leq \frac{c^*(M_{a,-} + M_{a,+})c}{e^{aL} - 1}. \quad (\text{Periodized P25})$$

The polarization is not used to replace the cross-pairings by diagonal entries: the last quadratic inequality for every c is precisely the matrix statement (P1).

Parseval also gives the exact sampled formula

$$M(L)_{ij} = \frac{1}{L} \sum_{n \in \mathbb{Z}} \left\langle \widehat{\Psi e_i}(2\pi n/L), \widehat{\Psi e_j}(2\pi n/L) \right\rangle_{\mathcal{K}}. \quad (\text{Periodized P26})$$

For the actual cyclic source with polynomial columns $P_a(S)$, this Hilbert-valued formula becomes a scalar sampled arithmetic moment matrix. Partial Fourier Plancherel in the retained relative variables gives

$$\|\widehat{\mathcal{U}_k F_h^{\otimes k}}(u)\|_{\mathcal{K}}^2 = 2\pi m_{h,k}(u), \quad m_{h,k} = w_h^{*k}. \quad (\text{Periodized P26a})$$

Indeed the full Fourier amplitude is $\prod_i v_h(1/2 + it_i)$; integrating its squared modulus over the $k-1$ relative Fourier variables contributes $(2\pi)^{-(k-1)}$, whereas the convolution uses $(2\pi)^{-k}$. Their ratio is precisely 2π .

Consequently

$$M_N(L)_{ab} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(2\pi n/L) \overline{P_a(k/2 + 2\pi in/L)} P_b(k/2 + 2\pi in/L). \quad (\text{Periodized P26b})$$

For $k \geq 2$, this matrix is positive definite for every finite degree and every $L > 0$. The original w_h is nonnegative and positive off a discrete set. Its convolution $w_h * w_h$ is strictly positive everywhere, and further convolutions preserve strict positivity. A nonzero polynomial cannot vanish at all the infinitely many distinct points $k/2 + 2\pi in/L$. Every sampled weight is positive, proving the assertion. The large- L estimate is still needed for quantitative comparison with the original Gram, even though invertibility in these tensor degrees holds at all $L > 0$.

The nonzero-mode observation is related by

$$M^\times(L) = M(L) - \frac{1}{L} Z_0^* Z_0. \quad (\text{Periodized P27})$$

This is a rank-at-most- d correction; for $k = 1$ its rank is at most one. Subtracting it is not the same operation as forgetting an absent source. The map (P17) retains its entire coefficient.

65.7 6. Exact recovery from all nonzero modes

For $\delta > 0$, define

$$M_\delta = \frac{1}{2\pi} \int_{\mathbb{R}} |u|^\delta \widehat{\Psi}(u)^* \widehat{\Psi}(u) du. \quad (\text{Periodized P28})$$

Then

$$M_\delta = \frac{(2\pi)^\delta}{\zeta(1+\delta)} \int_0^\infty L^{-1-\delta} M^\times(L) dL. \quad (\text{Periodized P29})$$

Proof: evaluate the quadratic form on any c , use Tonelli on its nonnegative Fourier sum, and put $u = 2\pi n/L$ for $n > 0$. The resulting integral is

$$\int_0^\infty L^{-2-\delta} \|\widehat{\Psi}c(2\pi n/L)\|^2 dL = (2\pi n)^{-1-\delta} \int_0^\infty u^\delta \|\widehat{\Psi}c(u)\|^2 du. \quad (\text{Periodized P30})$$

The negative indices give the integral on the negative half-line. Summing $n^{-1-\delta}$ gives the displayed $\zeta(1+\delta)$; multiplication by $(2\pi)^\delta$ leaves the original $1/(2\pi)$. Equality of all quadratic forms proves (P29). The integral is taken on source Grams before applying the nonlinear map $M \mapsto (J_N M^{-1} J_N^*)^{-1}$. No identity obtained by averaging already-formed quotient Grams is claimed.

Dominated convergence gives $M_\delta \rightarrow M$ as $\delta \downarrow 0$. The zero mode stays in (P17); its isolated frequency does not contribute to the continuous Lebesgue integral. Inserting the raw zero-mode summand into (P29) instead would make its small- L integral diverge. This is why the explicit separation and reconstruction are necessary.

There is a finite-source rate. For any fixed $\delta_0 > 0$, set

$$\mathcal{L}_{\delta_0} = \frac{1}{2\pi} \int |\log |u|| (1 + |u|^{\delta_0}) \widehat{\Psi}(u)^* \widehat{\Psi}(u) du. \quad (\text{Periodized P31})$$

The logarithm is integrable at the origin and the tails are controlled by Schwartz decay. For $0 < \delta \leq \delta_0$, the mean value formula for $|u|^\delta$ proves

$$-\delta \mathcal{L}_{\delta_0} \preceq M_\delta - M \preceq \delta \mathcal{L}_{\delta_0}. \quad (\text{Periodized P32})$$

The zeta factor in (P29) counts nonzero Fourier lattice points on a positive real half-plane. It is not substituted for the source function $g = 2\xi$, and it supplies no assertion about the zero set of g .

65.8 7. Transfer through the same full-jet quotient

Now specialize T to the actual polynomial source at degree N , with its specified basis. Let $J_N : \mathcal{P}_N \rightarrow E$ be the unchanged full remainder map π_χ , followed by the retained inclusion η whenever the larger arithmetic target is used. Work in remainder coordinates on E ; η is not discarded or inverted as a rectangular determinant.

For $N \geq q - 1$, $M_N \succ 0$. Set

$$\kappa_{a,N} = \max_{c \neq 0} \frac{c^*(M_{a,+,N} + M_{a,-,N})c}{c^* M_N c}. \quad (\text{Periodized P33})$$

It is finite. Given $0 < \eta_0 < 1$, the explicit choice

$$L \geq \frac{1}{a} \log \left(1 + \frac{\kappa_{a,N}}{\eta_0} \right) \quad (\text{Periodized P34})$$

implies

$$(1 - \eta_0)M_N \preceq M_N(L) \preceq (1 + \eta_0)M_N. \quad (\text{Periodized P35})$$

In particular periodization is injective on this finite source. All hypotheses needed to take the inverse in the next equation have now been verified.

For either source metric, use the same constrained minimum:

$$K_N(L) = J_N M_N(L)^{-1} J_N^*, \quad G_N(L) = K_N(L)^{-1}, \quad C_N(L) = M_N(L)^{-1} J_N^* G_N(L). \quad (\text{Periodized P36})$$

Here $J_N C_N(L) = I_E$. At the original observation write G_N, C_N without L . Applying (P35) to every representative with a fixed remainder and then taking the infimum proves

$$\boxed{(1 - \eta_0)G_N \preceq G_N(L) \preceq (1 + \eta_0)G_N.} \quad (\text{Periodized P37})$$

This proof compares the actual same constrained representative sets; it does not infer a form estimate from an arbitrary inverse compression.

The representative correction is explicitly

$$C_N(L) - C_N \in \ker J_N = \chi \mathcal{P}_{N-q}, \quad C_N(L) - C_N = M_\chi \operatorname{div}_\chi(C_N(L) - C_N). \quad (\text{Periodized P38})$$

Its original theta primitive is also retained. Since $\chi(S)Q(S)$ belongs to the original ideal $(h(s_1), \dots, h(s_k))$, fixed-order monic division gives $\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s})$. Replace F_h in slot i by ϕ_* , apply $Q_i(D_1, \dots, D_k)$, and give that primitive coefficient $(-1)^{i-1}$. The tensor differential contributes a second $(-1)^{i-1}$, giving the indicated positive relation. After reconstruction its quotient value is the receiving supported zero. The relation itself, its coefficient, and its support label remain available upstream.

A common period works on all degrees up to an admitted D . Use $\kappa_{a,D}$ in (P34): all smaller source forms are restrictions through the actual polynomial inclusions, so their Rayleigh quotients are no larger. Taking $D = 2q + 2$ also covers the one- and two-degree raises used by the scaling and Laplacian maps.

65.9 8. The four-volume coefficient is preserved up to a calculated error

Write $V_N = \det G_N$, $V_N(L) = \det G_N(L)$. Then (P37) proves

$$q \log(1 - \eta_0) \leq \log V_N(L) - \log V_N \leq q \log(1 + \eta_0). \quad (\text{Periodized P39})$$

For a common period valid at all four endpoints, set

$$\mathcal{B}_{h,k}(L) = \log \frac{V_{q-1}(L)V_q(L)}{V_{2q-1}(L)V_{2q}(L)}. \quad (\text{Periodized P40})$$

The coefficients of the two numerator errors and the two denominator errors give exactly

$$\boxed{|\mathcal{B}_{h,k}(L) - \mathcal{B}_{h,k}| \leq 2q \log \frac{1 + \eta_0}{1 - \eta_0}.} \quad (\text{Periodized P41})$$

For each tensor degree choose the displayed $L(k)$ from the degree- $2q_k + 2$ source. A fixed $\eta_0 \in (0, 1)$ then gives the equality of asymptotic lower/upper thresholds whenever either is read in the extended real sense:

$$\frac{\mathcal{B}_{h,k}(L(k)) - \mathcal{B}_{h,k}}{q_k \log k} \longrightarrow 0. \quad (\text{Periodized P42})$$

Thus the inherited necessary threshold four for an exact off-line quartet is transferred to these specified finite-period source quotients. No bound on the growth of $\kappa_{a,2q_k+2}$ is asserted; it is the explicit cost entering the required circle size. The remaining upper estimate concerns the constructed $\mathcal{B}_{h,k}(L(k))$, not the unconstrained eigenvalue spacing of a free circle operator.

65.10 9. The periodic Laplacian and its exact arithmetic boundary

The circle operator on $L^2(\mathbb{R}/L\mathbb{Z}; \mathcal{K})$ is

$$D_L^{(k)} = -\partial_r + k/2, \quad \mathcal{L}_{L,k} = (D_L^{(k)})^2 - kD_L^{(k)} = \partial_r^2 - k^2/4. \quad (\text{Periodized P43})$$

On periodic Sobolev domains, $-i\partial_r$ is self-adjoint; equivalently its Fourier diagonal is the real sequence $-2\pi n/L$. The displayed Laplacian is self-adjoint and has eigenvalues

$$-\left(\frac{2\pi n}{L}\right)^2 - \frac{k^2}{4}, \quad n \in \mathbb{Z}. \quad (\text{Periodized P44})$$

The full zeroth mode is retained at $-k^2/4$.

Let $r_N(L) = \mathcal{V}_{h,k} C_N(L)$ be the coefficient representative and let

$$R_{N,L} = \mathcal{P}_L \mathcal{U}_k r_N(L), \quad B_{N,L} = \mathcal{P}_L \mathcal{U}_k (D_L^{(k)} r_N(L) - r_N(L) A). \quad (\text{Periodized P45})$$

The second map is the periodization of the original theta boundary, with its primitive supplied in (P38) and the original derivative comparison. Both maps have finite domain E ; their adjoints require no bounded extension of the global theta operator.

Direct intertwining gives

$$D_L^{(k)} R_{N,L} - R_{N,L} A = B_{N,L}, \quad R_{N,L}^* R_{N,L} = G_N(L), \quad (\text{Periodized P46})$$

$$A^* G_N(L) + G_N(L) A - k G_N(L) = -(R_{N,L}^* B_{N,L} + B_{N,L}^* R_{N,L}). \quad (\text{Periodized P47})$$

The second-order equation is

$$\boxed{\mathcal{L}_{L,k} R_{N,L} - R_{N,L} (A^2 - kA) = (D_L^{(k)} - k) B_{N,L} + B_{N,L} A.} \quad (\text{Periodized P48})$$

Proof: apply $D_L^{(k)}$ to (P46), replace $D_L^{(k)} R_{N,L}$ by $R_{N,L} A + B_{N,L}$, and subtract k times (P46). Every term remains in its stated periodic smooth domain.

Equation (P48) is the actual relation between the negative circle spectrum and the arithmetic Laplacian. It contains two boundary terms. Their quotient values vanish, but their representative norms and cross-pairings are not set to zero.

For example, on an arithmetic eigenvector $Av = \rho v$, the Fourier diagonal of (P43) proves

$$\left\| ((D_L^{(k)} - k) B_{N,L} + B_{N,L} A) v \right\| \geq |\operatorname{Im}(\rho(\rho - k))| \|R_{N,L} v\|. \quad (\text{Periodized P49})$$

This is a lower test for the retained boundary, not an upper estimate. It follows from $\|(T - z)y\| \geq |\operatorname{Im} z| \|y\|$ for the displayed self-adjoint Fourier diagonal $T = \mathcal{L}_{L,k}$.

The coefficient arithmetic action has type $\mathcal{J}_{N+1,L} D_L^{(k)} R_{N,L} : E \rightarrow E$ through the recovered original source coordinates. The finite periodic source itself is mapped to degree $N + 1$ by $D_L^{(k)}$. It is not asserted to be an invariant spectral subspace of the circle operator.

65.11 10. Retain the critical comparison kernel and its cone

The supplied comparison theorem (P4) gives, on a packet with its full multiplicities,

$$\ker(\Gamma\sigma_Z) = E_{Z,\text{off}}. \quad (\text{Periodized P50})$$

Its critical jet maps contain the explicit coefficients $i^{-j}/j!$. On the exact finite image the resulting sequence is

$$0 \rightarrow E_{Z,\text{off}} \rightarrow E_Z \xrightarrow{\Gamma\sigma_Z} \operatorname{im}(\Gamma\sigma_Z) \rightarrow 0. \quad (\text{Periodized P51})$$

The two-term cone $[E_Z \rightarrow \operatorname{im}(\Gamma\sigma_Z)]$, in degrees $-1, 0$, retains $E_{Z,\text{off}}$ in degree -1 . The arithmetic generator restricts to it. A represented off-critical class maps to a supported zero under this particular comparison, not to external absence.

For the finite-period construction, the original source map $\Phi_{N,L} = \mathcal{P}_L \mathcal{U}_k \mathcal{V}_{h,k}|_{\mathcal{P}_N}$ is injective when (P34) holds. Hence it has the actual inverse onto its image and the observation

$$\mathcal{J}_{N,L} : \text{im } \Phi_{N,L} \rightarrow E, \quad \mathcal{J}_{N,L} = J_N \Phi_{N,L}^{-1}. \quad (\text{Periodized P52})$$

The quotient of this image by $\Phi_{N,L}(\ker J_N)$ is exactly E , by (P52). This supplies the source-faithful norm computation. Mapping further to the critical-only quotient for $k = 1$ composes with (P4) and again has the kernel (P50); it cannot be omitted from the diagram.

The new theorem therefore does not infer RH by declaring every circle mode real. It retains the complete original packet before applying that further spectral quotient, and gives the exact boundary and metric costs of the comparison.

65.12 11. The exact fixed-circle completion and its lost packet directions

The preceding source-faithful quotient has a further completion map which can be calculated exactly. Fix $k \geq 2$, $L > 0$, and set

$$S_n = k/2 + 2\pi i n/L, \quad \nu_n = \frac{2\pi}{L} m_{h,k}(2\pi n/L) > 0. \quad (\text{Periodized P53})$$

The closure of the cyclic polynomial source is $\mathcal{H}_{\text{cyc},L} = \ell^2(\mathbb{Z}, \nu)$, through $P \mapsto (P(S_n))_n$. To prove density, use the retained exponential tail of w_h : for every $0 < b < \pi/2$, $e^{b|t|} w_h(t) \in L^1 \cap L^\infty$. The triangle inequality inside convolution gives

$$e^{b|u|} m_{h,k}(u) \leq (e^{b|\cdot|} w_h)^{*k}(u) \leq \|e^{b|\cdot|} w_h\|_\infty \|e^{b|\cdot|} w_h\|_1^{k-1}. \quad (\text{Periodized P54})$$

Thus the discrete measure and its polynomially weighted variants have a finite exponential moment. If an element is orthogonal to every polynomial, its associated finite complex measure has Fourier transform analytic in a strip, with every derivative zero at zero. The identity theorem and uniqueness of Fourier transforms force that measure to be zero. This proves density, and applies unchanged after multiplying the measure by $|\chi|^2$ away from its zero atoms.

Define the finite index set and squarefree polynomial

$$\mathcal{Z}_L = \{n \in \mathbb{Z} : \chi(S_n) = 0\}, \quad p_L(S) = \prod_{n \in \mathcal{Z}_L} (S - S_n). \quad (\text{Periodized P55})$$

The empty product is 1. Since $p_L \mid \chi$, the actual completion comparison is

$$E = \mathbb{C}[S]/\chi \xrightarrow{\theta_L} \mathcal{H}_{\text{cyc},L}/\overline{\chi \mathbb{C}[S]} \xrightarrow{\sim} \bigoplus_{n \in \mathcal{Z}_L} \mathbb{C}, \quad [P] \mapsto (P(S_n))_{n \in \mathcal{Z}_L}. \quad (\text{Periodized P56})$$

Its target has the metric $\sum_{n \in \mathcal{Z}_L} \nu_n |v_n|^2$, and

$$\ker \theta_L = (p_L)/(\chi). \quad (\text{Periodized P57})$$

Proof of the Hilbert quotient: every χ^P vanishes at $\mathcal{Z}_L$, so its closure is contained in that vanishing subspace. Conversely let f vanish at those atoms and put $g_n = f_n/\chi(S_n)$ elsewhere. Then g is square-integrable for $|\chi|^2 \nu$. Polynomial density for that weighted measure approximates g , and multiplication by χ approximates f in the original norm. Thus the closure is exactly the stated vanishing subspace. The algebraic kernel is divisibility by the squarefree p_L .

This comparison retains only the reduced values at those roots actually sampled by this circle. Its kernel contains all unsampled primary blocks and every nonconstant jet in a sampled multiple-root block. It is related to the supplied Schwartz critical-line comparison (P50), not identified with it: that comparison has a continuous range of critical frequencies and continuous derivative-evaluation retractions, whereas (P56) is a discrete Hilbert completion with the explicit kernel (P57).

There is also an exact endpoint limit for the same canonical periodic metrics. Let E use its fixed remainder basis and let $J_{\mathcal{Z}_L}$ be evaluation in that basis. Then

$$G_N(L) \downarrow J_{\mathcal{Z}_L}^* \text{diag}(\nu_n)_{n \in \mathcal{Z}_L} J_{\mathcal{Z}_L} \text{ as } N \rightarrow \infty. \quad (\text{Periodized P58})$$

For a given remainder class, the infimum over all polynomial representatives is the squared distance to the closure of $\chi\mathbb{C}[S]$, which is exactly its sampled value norm by (P56). The admitted relation spaces increase with degree, so these infima decrease to that distance. Polarization gives entrywise convergence, and finite dimensionality gives convergence of the matrices. Their limiting radical is the kernel in (P57).

The original e -lift of θ_L sends a killed represented class to $(\ell, 0)$. The class remains in $(p_L)/(\chi)$ before this quotient. The same single scalar construction supports these several kernels.

This determines the order-of-limits issue rather than ignoring it. Increasing the circle size according to (P34) before comparing a prescribed finite degree gives the original metric within (P37). Keeping L fixed and completing in polynomial degree gives (P58), which generally loses arithmetic directions. A proof cannot replace the former family by the latter completion while keeping the full packet unchanged.

For $k = 1$, the same argument applies on the positive support atoms $\{n : w_h(2\pi n/L) > 0\}$; the clean all-integer statement above is explicitly restricted to $k \geq 2$. The seed with $h = 1$ remains an analytic calibration, not a nonempty quotient.

65.13 12. A finite Fourier cutoff with an explicit source-side error

The fixed-circle Gram still has an infinite Fourier sum. That sum also admits a certified finite truncation without choosing a different source. For an integer $p \geq 1$, define the actual derivative Gram

$$H_{p,N} = \int_{\mathbb{R}} (1+r^2)(\partial_r^p \Psi_N(r))^* \partial_r^p \Psi_N(r) dr. \quad (\text{Periodized P59})$$

The source comparison is $\partial_r^p \mathcal{U}_k = (-1)^p \mathcal{U}_k(D^{(k)} - k/2)^p$. Consequently (P59) is the original integral with factor $1 + (\log x_k)^2$ and the displayed polynomial of the original generator; the sign $(-1)^p$ appears in the map and has squared modulus one in its Gram.

Integration by parts and weighted Cauchy–Schwarz give

$$\|\widehat{\Psi_N c}(u)\|_{\mathcal{K}}^2 \leq \pi |u|^{-2p} c^* H_{p,N} c \quad (u \neq 0). \quad (\text{Periodized P60})$$

Here $\int_{\mathbb{R}} (1+r^2)^{-1} dr = \pi$; no Fourier factor is absorbed into $H_{p,N}$.

For $J \geq 1$, let $M_N(L, J)$ be (P26) summed only over $|n| \leq J$, including $n = 0$. Every omitted matrix summand is positive, and

$$0 \preceq M_N(L) - M_N(L, J) \preceq \frac{1}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1} H_{p,N}. \quad (\text{Periodized P61})$$

Indeed the prefactor before the remaining series is $(2\pi/L)(L/(2\pi))^{2p}$; the estimate $\sum_{n>J} n^{-2p} \leq J^{1-2p}/(2p-1)$ gives exactly the displayed coefficient.

On the common degree- D source set

$$\lambda_{p,D} = \max_{c \neq 0} \frac{c^* H_{p,D} c}{c^* M_{DC} c}, \quad \eta_{\text{per}} = \frac{\kappa_{a,D}}{e^{aL} - 1}, \quad \eta_{\text{tail}} = \frac{\lambda_{p,D}}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1}. \quad (\text{Periodized P62})$$

Both constants come from the unchanged finite source. When their sum $\eta = \eta_{\text{per}} + \eta_{\text{tail}} < 1$, the established inequalities prove, simultaneously for all $N \leq D$,

$$(1 - \eta)M_N \preceq M_N(L, J) \preceq (1 + \eta)M_N. \quad (\text{Periodized P63})$$

Both L and J can be selected explicitly. For positive budgets $\eta_P + \eta_T < 1$, take

$$L \geq \frac{1}{a} \log(1 + \kappa_{a,D}/\eta_P), \quad J \geq \max \left\{ 1, D, \frac{L}{2\pi} \left(\frac{\lambda_{p,D}}{(2p-1)\eta_T} \right)^{1/(2p-1)} \right\}, \quad J \in \mathbb{Z}. \quad (\text{Periodized P64})$$

The same constrained-minimum and determinant arguments now apply to a genuinely finite sum of the original spectral-amplitude Grams. Its four-volume comparison still has the error in (P41), with the total η . Rounding of the sampled arithmetic values is a further input: its matrix enclosure must be added to (P63)

before treating a numerical calculation as a certificate. No such high-degree arithmetic enclosure is claimed in this delivery.

65.14 13. What has and has not been added

The contribution consists of: the diagonal multi-variable coordinate map; the full augmented periodic metric; the exponentially controlled original-source Gram recovery; the exact all-period nonzero-mode identity with its lattice factor; the source-preserving minimum-norm quotient transfer; its $O(q)$ four-volume error; and the exact periodic-to-arithmetic Laplacian boundary.

The predecessor period determinant, residue trace, Deligne exponential family, scalar Laplacian and critical-kernel constructions are retained as inputs, not recounted as newly proved results. The new finite-period estimates act on the same G_N that those constructions must compare.

The quantitative upper problem remains the growth of the periodized quotient volumes, equivalently the original endpoint volumes by (P41). A bound contradicting the inherited threshold four is not proved here. The supplied negative circle operator alone does not give such a bound because (P48) retains the original nonzero representative-level boundary.

65.15 References and evidence

- User-supplied Pasted `text(3).txt`, and PR27 HOCHSCHILD_COMPARISON.md, inspected at head a4a87844c257485ae6d669f5b3eb6d96b9e8b8d8. The precise finite-kernel, zeroth-mode and Laplacian statements retain their reported analytic versus Lean scopes.
- Connes and Consani, *Hochschild homology, trace map and zeta-cycles*, arXiv:2207.10419. Its abstract explicitly distinguishes its all-zero Laplacian realization and critical-only sheaf realization. No new erratum claim is made by this continuation.
- Deligne, *Poids dans la cohomologie des variétés algébriques*, selected D022 text in the newly supplied French corpus, especially the strictness statement and the displayed Thom–Gysin twist.
- Deligne, *La conjecture de Weil II*, the previously supplied S20 current French page-local records, pp. 178, 203 and 206. The archive’s exact reconstructed SHA-256 and reading scope are recorded separately.
- The periodization, Parseval and quadratic-form comparison ingredients are classical. All source-specific identities and estimates used above have proofs in the displayed coordinates.

The generated package contains no copies of the supplied literary corpus and no fonts. Its checks report exactly what was executed. It contains no proof-assistant claim for the analytical statements.

66 Complete periodized arithmetic source and finite-cutoff proofs

66.1 Periodized source control: complete comparison calculations

This companion retains the complete supplied periodized-source note, including P1–P64 and P26a–P26b, as its proof source. It supplies the source regularity, coordinate and image calculations used in reading those formulas, makes the finite Fourier projection part of the exact operator diagram, and retains the sharper one-sided error furnished by the omitted positive Gram. The adjoining density companion supplies the explicit definitions and full exponential-tail and completion proof. The original source, remainder basis, arithmetic unit, and relations are retained throughout.

The asymmetric cutoff calculation below is also the calculation in FC5–FC7 and the four-endpoint paragraph following FC13 of the adjoining periodized-curvature control bridge. PSA23–PSA25 record its source and determinant steps, with the later AW/SP spectral and actual-overlap control propagated through the same original and sampled coefficient maps, and PSA27–PSA29 record the same finite Fourier boundary diagram as FC19–FC20. These are audit derivations of the same mathematics, not additional results created by repeating the calculation.

Fix the original packet polynomial h , its full canceled zero orders, and $g = 2\xi$. Write $v_h = g/h$,

with removable values understood, and

$$w_h(t) = \frac{1}{2\pi} |v_h(1/2 + it)|^2, \quad m_{h,k} = w_h^{*k}. \quad (\text{PSA1})$$

These definitions must be inserted before the first use of v_h, w_h in the supplied standalone note. They specify the original density; no rescaling of g, h , or the tensor-sum coordinate is performed. The full proof that these functions have the required exponential tails, including canceled-root values, is in the adjoining density companion.

66.1.1 The original coordinates and the complete source domain

Let $k \geq 1$ and let F be a finite linear combination of tensors of strong-Schwartz functions on $(0, \infty)$ and their logarithmic derivatives. Use $\mathcal{K} = \mathbb{C}$ for $k = 1$ and $\mathcal{K} = L^2(\mathbb{R}^{k-1}, d\mathbf{z})$ otherwise. In the ordered coordinates $(r, z_1, \dots, z_{k-1})$, put

$$y_i = r + z_i \ (i < k), \quad y_k = r, \quad x_i = e^{y_i}. \quad (\text{PSA2})$$

The derivative matrix from $(r, \mathbf{z})$ to $(y_1, \dots, y_k)$ has first column all ones, last row $(1, 0, \dots, 0)$ and the identity in the remaining $k-1$ columns. Expansion along its last row gives determinant $(-1)^{k-1}$. Thus the positive density is $d^k x = e^{kr + \sum z_i} dr d\mathbf{z}$, with the orientation sign recorded separately. The map and its inverse are

$$\begin{aligned} (\mathcal{U}_k F)(r, \mathbf{z}) &= e^{(kr + \sum z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \\ (\mathcal{U}_k^{-1} \psi)(\mathbf{x}) &= (x_1 \cdots x_k)^{-1/2} \psi(\log x_k, \log x_1 - \log x_k, \dots, \log x_{k-1} - \log x_k). \end{aligned} \quad (\text{PSA3})$$

Substitution into the integral proves equality of the two squared norms and proves that these maps are mutually inverse Hilbert isometries. With $D_i = -x_i \partial_{x_i}$ and $D^{(k)} = \sum_i D_i$, simultaneous change of all y_i is exactly change of r with every z_i fixed. Differentiating the displayed factor, without altering the coordinate, gives

$$\mathcal{U}_k D^{(k)} = (-\partial_r + k/2) \mathcal{U}_k, \quad \partial_r^j \mathcal{U}_k = (-1)^j \mathcal{U}_k (D^{(k)} - k/2)^j. \quad (\text{PSA4})$$

For every $a > 0$ and integer $j \geq 0$, strong-Schwartz estimates on each original tensor factor imply

$$I_{a,j} := \int_{\mathbb{R}} e^{2a|r|} \|\partial_r^j \mathcal{U}_k F(r)\|_{\mathcal{K}}^2 dr < \infty. \quad (\text{PSA5})$$

Indeed after (PSA3)–(PSA4) this is the original $d^k x$ integral of $|(D^{(k)} - k/2)^j F|^2$ multiplied by $\max\{x_k^{2a}, x_k^{-2a}\}$. The maximum is bounded by the sum of those two weights, and every term is a finite integral of a finite sum of products of original strong-Schwartz functions. This proves (PSA5) on the stated source without asserting it for arbitrary Hilbert inputs.

For completeness it also yields normal convergence of periodization. For any Hilbert-valued H^1 function f on $[r, r+1]$, select a point where its squared norm is no larger than its integral, use the fundamental theorem of calculus, and apply Cauchy–Schwarz. The result is $\|f(r)\|^2 \leq 2 \int_r^{r+1} (\|f(t)\|^2 + \|f'(t)\|^2) dt$. Consequently, for $\psi = \mathcal{U}_k F$,

$$\|\partial_r^j \psi(r)\|_{\mathcal{K}}^2 \leq 2e^{2a} e^{-2a|r|} (I_{a,j} + I_{a,j+1}). \quad (\text{PSA6})$$

The bound is summable over $r + mL$, uniformly for r in a fixed period and for every derivative. Thus the periodization is smooth; the ensuing Fourier transforms, products, and unfolded correlations are absolutely convergent on each finite source.

66.1.2 The full circle image and its Fourier constants

Use the plus-sign transform $\widehat{\psi}(u) = \int \psi(r) e^{iur} dr$ and the orthonormal circle basis $e_n(r) = L^{-1/2} e^{-2\pi i n r / L}$. Unfolding its coefficient gives

$$\langle e_n, \mathcal{P}_L \psi \rangle = L^{-1/2} \int_{\mathbb{R}} \psi(r) e^{2\pi i n r / L} dr, \quad \mathcal{P}_L \psi = \sum_m \psi(r + mL). \quad (\text{PSA7})$$

Here the scalar basis is paired with the vector-valued function using the conjugate-linear first-slot convention. In particular the zero coefficient is $L^{-1/2} \widehat{\psi}(0)$, whereas the constant circle function is $L^{-1} \widehat{\psi}(0)$.

The augmented coefficient map sends ψ to the nonzero coefficients and the unscaled vector $\widehat{\psi}(0)$. Its last summand has metric $L^{-1} \langle \cdot, \cdot \rangle_{\mathcal{K}}$. The inverse coordinate map on this image inserts $L^{-1/2} \widehat{\psi}(0)$ at $n = 0$. Thus, for a finite column map $\Psi = \mathcal{U}_k T$ and $Z_0 c = \widehat{\Psi} c(0)$,

$$M(L) = M^\times(L) + L^{-1} Z_0^* Z_0. \quad (\text{PSA8})$$

For $k = 1$, $T = \Theta \phi$, the Connes–Consani map remains $\mathcal{E} \phi = \frac{1}{2} \mathcal{U}_1 \Theta \phi$. It therefore has zero coefficient $\mathcal{M} \Theta \phi(1/2) / (2\sqrt{L})$. For the original seed this is $\xi(1/2) / \sqrt{L}$. Its entire Gram is one quarter of the original theta-source Gram. The coefficient and metric formulas record that factor explicitly.

The scalar convolution formula retains the relative Hilbert fibre. For $\psi = \mathcal{U}_k F_h^{\otimes k}$, take the further plus-sign Fourier transform in $\mathbf{z}$. Its phase becomes

$$ur + \sum_{i < k} t_i z_i = \sum_{i < k} t_i y_i + \left(u - \sum_{i < k} t_i \right) y_k. \quad (\text{PSA9})$$

Fubini therefore gives the full amplitude $\prod_{i < k} v_h(1/2 + it_i) v_h(1/2 + i(u - \sum_{i < k} t_i))$. Plancherel in exactly the $k - 1$ relative variables gives

$$\begin{aligned} \|\widehat{\psi}(u)\|_{\mathcal{K}}^2 &= (2\pi)^{-(k-1)} \int_{\mathbb{R}^{k-1}} \prod_{i < k} |v_h(1/2 + it_i)|^2 |v_h(1/2 + i(u - \sum_{i < k} t_i))|^2 d\mathbf{t} \\ &= 2\pi m_{h,k}(u). \end{aligned} \quad (\text{PSA10})$$

By (PSA4), a polynomial column contributes its factor $P(k/2 + iu)$. Parseval and (PSA7) now prove the entire sampled matrix

$$M_N(L)_{ab} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(2\pi n/L) \overline{P_a(k/2 + 2\pi i n/L)} P_b(k/2 + 2\pi i n/L). \quad (\text{PSA11})$$

The density companion proves positive weights for $k \geq 2$ and hence strict positivity of this finite polynomial Gram for every $L > 0$. For $k = 1$, quantitative source comparison below supplies positivity once the stated period has been chosen; no assertion about positivity at every integer atom is substituted for that comparison.

66.1.3 The original-source correlation inequality

Define $M = \Psi^* \Psi$ and $M_{a,\pm} = (e^{\pm ar} \Psi)^* (e^{\pm ar} \Psi)$ in the unchanged coefficient basis. In original coordinates these are exactly the integrals weighted by $x_k^{\pm 2a}$. With $c^* \mathcal{C}(s) d = \int \langle \Psi c(r), \Psi d(r+s) \rangle dr$, unfolding pairs of translates gives

$$M(L) = \sum_{m \in \mathbb{Z}} \mathcal{C}(mL), \quad \mathcal{C}(0) = M, \quad \mathcal{C}(-s) = \mathcal{C}(s)^*. \quad (\text{PSA12})$$

For $s > 0$, insert e^{-ar} on the first factor and $e^{a(r+s)}$ on the second. Their product is e^{as} times the original integrand. Cauchy–Schwarz and the substitution $t = r + s$ give

$$|c^* \mathcal{C}(s)c| \leq e^{-as} (c^* M_{a,-} c)^{1/2} (c^* M_{a,+} c)^{1/2}. \quad (\text{PSA13})$$

The roles of the weights reverse for negative s . The two geometric series sum to $2/(e^{aL} - 1)$. Applying $2\sqrt{xy} \leq x + y$ to these actual two quadratic forms yields

$$-\frac{M_{a,+} + M_{a,-}}{e^{aL} - 1} \preceq M(L) - M \preceq \frac{M_{a,+} + M_{a,-}}{e^{aL} - 1}. \quad (\text{PSA14})$$

This proves an inequality on every coefficient vector, including complex cross-pairings, and hence the claimed matrix order.

66.1.4 All-period recovery before the quotient minimum

For $\delta > 0$, quadratic-form Tonelli and $u = 2\pi n/L$ for $n > 0$ give

$$\int_0^\infty L^{-2-\delta} \|\widehat{\Psi} c(2\pi n/L)\|^2 dL = (2\pi n)^{-1-\delta} \int_0^\infty u^\delta \|\widehat{\Psi} c(u)\|^2 du. \quad (\text{PSA15})$$

Negative n supplies the negative half-line. The sum of $n^{-1-\delta}$ over the positive integers is exactly $\zeta(1 + \delta)$, with no double counting of either half-line. Equality of quadratic forms proves

$$M_\delta := \frac{1}{2\pi} \int |u|^\delta \widehat{\Psi}(u)^* \widehat{\Psi}(u) du = \frac{(2\pi)^\delta}{\zeta(1 + \delta)} \int_0^\infty L^{-1-\delta} M^\times(L) dL. \quad (\text{PSA16})$$

For $0 < \delta \leq \delta_0$, the mean-value formula implies $||u|^\delta - 1| \leq \delta |\log |u|| (1 + |u|^{\delta_0})$. The right side is integrable against every finite-source Fourier Gram: the logarithm is locally integrable and the tails are Schwartz. Thus the exact matrix with this logarithmic integrand, denoted $\mathcal{L}_{\delta_0}$ in P31, obeys $-\delta \mathcal{L}_{\delta_0} \preceq M_\delta - M \preceq \delta \mathcal{L}_{\delta_0}$, proving convergence and its rate. If the nonzero $Z_0^* Z_0/L$ term were inserted into (PSA16), its small-period integral would be $\int_0^\infty L^{-2-\delta} dL$ times that matrix and would diverge. Formula (PSA8) is the exact reconstruction of the omitted coefficient. Formula (PSA16) averages source Grams; it makes no interchange of this integral with inverse compression.

66.1.5 The same finite quotient and its actual relation coefficient

Let $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi > 0$, and $\mathcal{P}_N = \{P : \deg P \leq N\}$, with its original basis. For $N \geq q - 1$, the monic remainder map $J_N : \mathcal{P}_N \rightarrow E$ is onto. The actual arithmetic observation in the larger tensor target is still ηJ_N , with $\eta[P] = v_h^{\otimes k} P(A_k)1$ and $v_h = j_h(g/h)$. We take determinants only in the stated remainder basis of E .

For any positive source Gram M_N , set

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad C_N = M_N^{-1} J_N^* G_N. \quad (\text{PSA17})$$

Surjectivity makes K_N positive definite. Direct multiplication gives $J_N C_N = I$. If $J_N P = v$, then $P = C_N v + z$ with $J_N z = 0$. The cross-term vanishes because $C_N^* M_N z = G_N J_N z = 0$. Therefore

$$P^* M_N P = v^* G_N v + z^* M_N z, \quad v^* G_N v = \min_{J_N P = v} P^* M_N P. \quad (\text{PSA18})$$

This proves both the actual minimum and its unique representative. Applying any source lower and upper bounds to this same affine set and then taking its minimum transfers those same factors to G_N .

In particular, (PSA14) and the actual Rayleigh quotient $\kappa_{a,N} = \max_{c \neq 0} c^* (M_{a,+} + M_{a,-,N}) c / (c^* M_N c)$ prove P34–P37. The constraint set and coefficient coordinates are unchanged;

the minimizing representatives may change and their difference is retained below. For the second representative $C_N(L)$ obtained from $M_N(L)$,

$$J_N(C_N(L) - C_N) = 0, \quad C_N(L) - C_N = M_\chi \operatorname{div}_\chi(C_N(L) - C_N). \quad (\text{PSA19})$$

Monic division here is applied column by column and yields the actual degree- $\leq N - q$ coefficient, with the zero coefficient if $N = q - 1$. The tensor relation is also explicit. In the original polynomial ring, fixed-order monic division in the variables $s_1, \dots, s_k$ expresses $\chi(\sum s_i)Q(\sum s_i)$ as $\sum_i h(s_i)Q_i$. Its remainder is zero because χ is the annihilator of the original tensor-sum cyclic action. Replace the i th F_h by ϕ_* , apply $Q_i(D_1, \dots, D_k)$, and attach $(-1)^{i-1}$. The tensor differential supplies the other $(-1)^{i-1}$; their product is $+1$, and $h(D_i)F_h = \Theta\phi_*$ supplies the original relation. Its arithmetic image is the receiving supported zero, with its source coefficient retained, and not external absence.

66.1.6 The finite frequency cutoff and its one-sided allowance

For $p \geq 1$, define the unchanged-source derivative Gram

$$H_{p,N} = \int (1 + r^2)(\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr. \quad (\text{PSA20})$$

By (PSA4), the original-coordinate source map is $(-1)^p(D^{(k)} - k/2)^p$ and its weight is $1 + (\log x_k)^2$. Integration by parts has no boundary term by (PSA6), and gives $\widehat{\partial_r^p \psi}(u) = (-iu)^p \widehat{\psi}(u)$. Weighted Cauchy-Schwarz with $\int (1 + r^2)^{-1} dr = \pi$ proves $\|\widehat{\Psi_N c}(u)\|^2 \leq \pi |u|^{-2p} c^* H_{p,N} c$ for $u \neq 0$.

Let $\Pi_{L,J}$ be the orthogonal projection onto precisely the Fourier modes $|n| \leq J$, including zero, and form $M_N(L, J) = (\Pi_{L,J} \mathcal{P}_L \Psi_N)^* (\Pi_{L,J} \mathcal{P}_L \Psi_N)$. The omitted Gram is positive. Summing both signs of n and using $\sum_{n>J} n^{-2p} \leq \int_J^\infty t^{-2p} dt$ proves

$$0 \preceq M_N(L) - M_N(L, J) \preceq \frac{1}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1} H_{p,N}. \quad (\text{PSA21})$$

The exact coefficient before the series was $(2\pi/L)(L/(2\pi))^{2p}$; (PSA21) retains the two-sided lattice factor and the original Plancherel convention.

Use the actual polynomial inclusion from degree N into degree D . It pulls back each of $M_D, M_{a,\pm,D}, H_{p,D}$ to the corresponding degree- N form. Consequently their Rayleigh bounds at D control all $N \leq D$. With the exact P62 quantities, write

$$\eta_{\text{per}} = \frac{\kappa_{a,D}}{e^{aL} - 1}, \quad \eta_{\text{tail}} = \frac{\lambda_{p,D}}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1}, \quad \lambda_{p,D} = \max_{c \neq 0} \frac{c^* H_{p,D} c}{c^* M_D c}. \quad (\text{PSA22})$$

Choose L and integer J by P64 for specified positive budgets whose sum is below one. These choices prove that $\eta_{\text{per}} + \eta_{\text{tail}} < 1$ after choosing the budgets strictly below one. Combining the two actual inequalities gives the sharper, asymmetric result

$$(1 - \eta_{\text{per}} - \eta_{\text{tail}})M_N \preceq M_N(L, J) \preceq (1 + \eta_{\text{per}})M_N. \quad (\text{PSA23})$$

The upper bound retains the positivity of the omitted Gram. It implies the supplied symmetric total-error bound, and also proves strict positivity of the finite sampled matrix. For $k \geq 2, J \geq D$ in P64 separately supplies more distinct positive sample atoms than the degree of a nonzero polynomial.

Define $G_N(L, J), C_N(L, J)$ by (PSA17) using this finite Gram. By (PSA18), the factors in (PSA23) pass unchanged to $G_N(L, J)$. The same proof of (PSA19) supplies its original theta correction. Taking determinants on the fixed q -dimensional remainder space gives

$$q \log(1 - \eta_{\text{per}} - \eta_{\text{tail}}) \leq \log \frac{\det G_N(L, J)}{\det G_N} \leq q \log(1 + \eta_{\text{per}}). \quad (\text{PSA24})$$

For $D = 2q + 2$, all four endpoint degrees and both generator raises remain among these actual restrictions. The common space for the four determinant terms themselves is $\mathcal{H} = \mathcal{P}_{2q}$. Let $I_N : \mathcal{P}_N \hookrightarrow \mathcal{H}$ and $I_{2q,D} : \mathcal{H} \hookrightarrow \mathcal{P}_D$ be the literal coefficient inclusions. Restriction gives

$$M^{(0)} = I_{2q,D}^* M_D I_{2q,D} = M_{2q}, \quad M^{(1)} = I_{2q,D}^* M_D(L, J) I_{2q,D} = M_{2q}(L, J). \quad (\text{PSA24a})$$

Set $\alpha = 1 - \eta_{\text{per}} - \eta_{\text{tail}} > 0$ and $\beta = 1 + \eta_{\text{per}}$. These two forms satisfy $\alpha M^{(0)} \preceq M^{(1)} \preceq \beta M^{(0)}$ by (PSA23). In this paragraph let $x \in [0, 1]$ be a metric path parameter, distinct from the physical x_i , and retain

$$M(x) = (1 - x)M^{(0)} + xM^{(1)}, \quad D_{L,J} = (M^{(0)})^{-1}M^{(1)}, \quad C(x) = M(x)^{-1}(M^{(1)} - M^{(0)}). \quad (\text{PSA24b})$$

These are the original and actual sampled forms, rather than a Gamma comparison. The finite algebra of the AW and SP calculations applies through the identity of this coefficient space, with their endpoint forms replaced by exactly $M^{(0)}, M^{(1)}$. We prove that algebra for this source here.

For $N \in \{q - 1, q, 2q - 1, 2q\}$ put $B_N = \times \chi : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ and define

$$\begin{aligned} P_N(x) &= I_N(I_N^* M(x) I_N)^{-1} I_N^* M(x), \\ Q_N(x) &= I_N B_N (B_N^* I_N^* M(x) I_N B_N)^{-1} B_N^* I_N^* M(x), \\ U(x) &= Q_{2q-1} + Q_{2q} - Q_q, \quad Q_{q-1} = 0, \\ W(x) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{PSA24c})$$

Every inverse is on the displayed actual domain. For $N = q - 1$ the relation domain is zero and its determinant is one. Multiplication verifies that these are the orthogonal projections onto the specified polynomial and relation subspaces in $M(x)$. The degree-less-than- q remainder section s_N and the monic relation columns give a frame $[s_N, B_N]$ of determinant one: its high-degree block is triangular with diagonal one. Block elimination in its Gram therefore gives

$$V_N(x) = \frac{\det(I_N^* M(x) I_N)}{\det(B_N^* I_N^* M(x) I_N B_N)}, \quad \frac{d}{dx} \log V_N(x) = \text{Tr}((P_N - Q_N)C(x)). \quad (\text{PSA24d})$$

This determinant is the same quotient determinant as (PSA17)–(PSA18), since its Schur complement is the minimum Gram over the same remainder fibres. The derivative follows by the finite determinant derivative and cyclicity of trace. In the relation-first orientation, the unchanged frame has determinant $(-1)^{q(N-q+1)}$ and its squared modulus is one, so no orientation factor has been removed. Consequently the four original signs give $\mathcal{B}'(x) = \text{Tr}((U - W)C(x))$.

The flags of relation subspaces have dimensions 1, q , $q + 1$, and the polynomial flag has dimensions q , $q + 1$, $2q$, $2q + 1$. Nested projections act as identity or zero on successive orthogonal pieces. Hence each of U, W has eigenvalues 0 of multiplicity q , 1 of multiplicity 2, and 2 of multiplicity $q - 1$. Thus each has rank $q + 1$, trace $2q$ and trace of the square $\text{Tr } U^2 = \text{Tr } W^2 = 4q - 2$. The eigenvalue-two multiplicity is zero at $q = 1$. Let $s(x) = \dim(\text{ran } U \cap \text{ran } W) \geq 1$, and write $c_1(x) \leq \dots \leq c_n(x)$ for the eigenvalues of $C(x)$, $n = 2q + 1$, and $d(x) = c_n(x) - c_1(x)$. Define

$$\begin{aligned} S_{\text{AW}}^{L,J}(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ S_{\text{SP}}^{L,J}(x) &= \frac{d(x)}{2} \min \left\{ 4q - 2s(x), \sqrt{(2q+1)(8q-4-2\text{Tr}(UW))} \right\}, \\ \mathcal{J}_{L,J} &= \int_0^1 \min \{ S_{\text{AW}}^{L,J}(x), S_{\text{SP}}^{L,J}(x) \} dx. \end{aligned} \quad (\text{PSA24e})$$

For an orthogonal projection P of rank r , its diagonal entries in a $C(x)$ -eigenbasis lie in $[0, 1]$ and sum to r . Moving a fixed diagonal mass from a smaller eigenvalue to a larger one increases its trace

pairing; hence $\sum_{j=1}^r c_j \leq \text{Tr}(PC) \leq \sum_{j=n-r+1}^n c_j$. Each of U, W is the sum of a rank- $q + 1$ range projection and a rank- $q - 1$ eigenvalue-two projection. Applying these bounds and subtracting their full extrema yields $|\text{Tr}((U - W)C)| \leq S_{\text{AW}}^{L,J}$.

Let E_L be the projection onto the actual intersection of the two ranges. The positive operators $U - E_L, W - E_L$ both have trace $2q - s(x)$, because each of U, W dominates its range projection. Thus $\|U - W\|_1 \leq 4q - 2s(x)$. Also $\text{Tr}(U - W) = 0$ and $\text{Tr}((U - W)^2) = 8q - 4 - 2\text{Tr}(UW)$. Cauchy-Schwarz on its n real eigenvalues gives the other trace-norm estimate in (PSA24e). Subtracting $(c_1 + c_n)I/2$ from C leaves the trace pairing unchanged and bounds its absolute value by $d(x)\|U - W\|_1/2$, as is seen in an orthonormal eigenbasis of $U - W$. This proves the SP bound on the same signed derivative, and therefore its pointwise minimum with the AW bound. Smooth positive forms give bounded projector matrices and continuous eigenvalues; rank strata are defined by matrix minors, so $s(x)$ is Borel. The minimum is integrable, and integration with its original signs gives the first bound below.

The exact trace norm and the centered Hilbert-Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(x) &= \frac{d(x)}{2} \|U - W\|_1, \\ \Sigma(x) &= \text{Tr}(C(x)^2) - \frac{(\text{Tr } C(x))^2}{2q + 1}, \\ S_{\text{HS}}(x) &= \sqrt{\Sigma(x) \text{Tr}((U - W)^2)}. \end{aligned} \tag{PSA24g}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(x))I/(2q+1)$ instead. The square of its Hilbert-Schmidt norm in the unchanged source metric is exactly $\Sigma(x)$. Hilbert-Schmidt Cauchy-Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(x)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a-10b and ISM25-26 through the already proved original coefficient and projection maps.

The actual source in (PSA24a) has additional moment structure. The first measure is $m_{h,k}(y) dy$, and the second is the unchanged positive atomic measure $\sum_{|n| \leq J} (2\pi/L) m_{h,k}(2\pi n/L) \delta_{2\pi n/L}$, including zero. Their degree- $2q$ Grams are exactly $M^{(0)}, M^{(1)}$ by (PSA10)-(PSA11) and (PSA21). Positivity was proved in (PSA23). Hence the complete RMT calculation applies with this original/sampled pair, with no change to any atom, mass or polynomial coefficient. For $a_q = 2q - 1$ put

$$\begin{aligned} A_{L,J}(x) &= a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}} d(x), \\ \mathcal{J}_{L,J}^{(5)} &= \int_0^1 \min\{S_{\text{AW}}^{L,J}(x), S_{\text{tr}}(x), S_{\text{SP}}^{L,J}(x), S_{\text{HS}}(x), A_{L,J}(x)\} dx, \\ \mathcal{I}_{L,J} &= \int_0^1 \min \left\{ d(x), \frac{S_{\text{AW}}^{L,J}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}}, \frac{S_{\text{SP}}^{L,J}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}}, \frac{S_{\text{HS}}(x)}{a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}} \right\} dx. \end{aligned} \tag{PSA24f}$$

RMT11 keeps both original q -angle lists of the crossed quotient pairs and proves $|\mathcal{B}'| \leq A_{L,J}$ on these measures. Its strict positivity argument applies also to this atomic endpoint: equality $\mathcal{B} = 0$ would place 1 orthogonal to $\chi \mathcal{P}_q$, and pairing with $\chi^{\#k}(S) = \sum_j \bar{\chi}_j(k - S)^j$ would give $0 = \int |\chi|^2 d\mu_x$, contradicting the positive Gram of the nonzero χ . Thus $\mathcal{B} > 0$ on the whole compact path. Taking the minimum of this angle derivative bound, the two earlier fully proved bounds, and both bounds

in (PSA24g) yields $|\mathcal{B}(1) - \mathcal{B}(0)| \leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{J}_{L,J}$. The generic two-form proof remains valid; this further inequality uses precisely the actual moment measures.

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues of $D_{L,J}$. The form comparison gives $\alpha \leq b_j \leq \beta$. Factoring $M(x) = M^{(0)}((1-x)I + xD_{L,J})$ proves $c_j(x) = (b_j - 1)/(1 - x + xb_j)$; the order follows from derivative $(1 - x + xb_j)^{-2} > 0$ with respect to b_j . Its integral is $\log b_j$. Therefore the full strengthened finite-circle error, with every original source constant, is

$$\begin{aligned} |\mathcal{B}_{h,k}(L, J) - \mathcal{B}_{h,k}| &\leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{J}_{L,J} \\ &\leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \\ &\leq (2q-1) \log \frac{1 + \eta_{\text{per}}}{1 - \eta_{\text{per}} - \eta_{\text{tail}}}. \end{aligned} \quad (\text{PSA25})$$

The first quantity is also at most $\int_0^1 S_{\text{SP}}^{L,J}(x) dx$. Each logarithmic ratio in the middle expression is at most $\log(\beta/\alpha)$ and its coefficient total is $2q-1$. This proves the last inequality, including the empty sum at $q=1$. It is at most $(2q-1) \log((1+\eta)/(1-\eta))$ for $\eta = \eta_{\text{per}} + \eta_{\text{tail}}$. The older bound with coefficient $2q$ follows from this stronger estimate; the asymmetric omitted-tail allowance is unchanged. Fixed positive budgets still give an $O(q_k)$ error, whose ratio to $q_k \log k$ tends to zero. No bound on the growth of the original $\kappa_{a,D}$ or $\lambda_{p,D}$ is inserted. The sampled values remain the exact analytic values with every zero mode, Fourier factor and tensor source coefficient retained.

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Use $\mathbf{M}_z = M(x)|_{x=z}$, $\mathbf{C}_z = C(x)|_{x=z}$, $\mathbf{A}_z = (U(x) - W(x))|_{x=z}$ and $\mathbf{D} = D_{L,J}$. These are the actual continuous/sampled moment forms in (PSA24a), and the RMT/ISM function $\mathcal{B}$ is precisely the original sampled-path four-volume function. The truncation ℓ_N differs from both the circle length L and the original derivative order p . Write $\mathbf{D} = (\mathbf{M}_0)^{-1} \mathbf{M}_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let $\mathbf{A}_z$ be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, \quad H_z^\circ = I - \frac{(1-z)I + z\mathbf{D}}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (\mathbf{D} - I), \quad R_z^{[\ell_N]} = (H_z^\circ)^{\ell_N+1} \mathbf{C}_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \quad (\text{PSA25ba})$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $\mathbf{C}_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator $\mathbf{D}$, hence self-adjoint in the original $\mathbf{M}_z = \mathbf{M}_0((1-z)I + z\mathbf{D})$ metric. Since $\text{Tr } \mathbf{A}_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} \mathbf{A}_z)$ plus $\text{Tr}(E_z^{[\ell_N]} \mathbf{A}_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} \mathbf{A}_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(\mathbf{A}_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \quad (\text{PSA25bb})$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_{\mathbf{D}} = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_{\mathbf{D}}(8q-6)} \longrightarrow 0. \quad (\text{PSA25bc})$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

This also improves the actual sampled endpoint itself in its nonlinear coordinate. Define $H_0 = 2 \operatorname{arcosh}(e^{\mathcal{B}_{h,k}/(2a_q)})$. RMT16’s derivative and inverse, on this identified moment path, give $|\mathcal{H}_{a_q}(\mathcal{B}_{h,k}(L, J)) - H_0| \leq \mathcal{I}_{L,J} \leq \log(b_n/b_1) \leq \log(\beta/\alpha)$. Combining the increasing inverse with the just-proved additive bound yields

$$\begin{aligned} \mathcal{B}_{h,k}(L, J) &\geq \max \left\{ 0, \mathcal{B}_{h,k} - \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{L,J}\}}{2}, \mathcal{B}_{h,k} + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}_{h,k}(L, J) &\leq \min \left\{ \mathcal{B}_{h,k} + \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{L,J}}{2}, \mathcal{B}_{h,k} + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned} \quad (\text{PSA25a})$$

All denominators in (PSA24f) are justified by the strict positivity above, with no uniform lower bound in tensor order presumed.

66.1.7 The finite observation map and both Laplacian boundary terms

Put $\Phi_{N,L,J} = \Pi_{L,J} \mathcal{P}_L \mathcal{U}_k \mathcal{V}_{h,k} |_{\mathcal{P}_N}$. Strict positivity in (PSA23) makes it injective. The exact observation has the stated image as domain:

$$\mathcal{J}_{N,L,J} : \operatorname{im} \Phi_{N,L,J} \longrightarrow E, \quad \mathcal{J}_{N,L,J} = J_N \Phi_{N,L,J}^{-1}. \quad (\text{PSA26})$$

On that image, the inverse can equivalently be written $(\Phi_{N,L,J}^* \Phi_{N,L,J})^{-1} \Phi_{N,L,J}^*$ in the unchanged coefficient coordinates. Its kernel is $\Phi_{N,L,J}(\ker J_N)$, so its quotient is exactly E with the metric (PSA17), retaining all jets. This construction uses finite dimensionality and the proved lower source bound; it asserts no bounded inverse on a larger completed source.

On the periodic H^1 domain let $D_L = -\partial_r + k/2$. The adjoint is $D_L^* = \partial_r + k/2$, so $D_L^* + D_L = k$. On periodic H^2 , $\mathcal{L}_{L,k} = D_L^2 - kD_L = \partial_r^2 - k^2/4$ is the self-adjoint Fourier diagonal with values $-(2\pi n/L)^2 - k^2/4$. The projection $\Pi_{L,J}$ commutes with both derivatives on these domains.

Let $r = \mathcal{V}_{h,k} C_N(L, J)$ and use the original arithmetic action $A = M_S$ on E . The finite representative and its actual boundary are

$$R = \Pi_{L,J} \mathcal{P}_L \mathcal{U}_k r, \quad B = \Pi_{L,J} \mathcal{P}_L \mathcal{U}_k (D^{(k)} r - rA). \quad (\text{PSA27})$$

The unperiodized difference has zero remainder at degree $N + 1$ and has the monic theta primitive already proved above. Commutation and (PSA4) prove $D_L R - RA = B$ and $R^* R = G_N(L, J)$. Substituting $RA = D_L R - B$ and using $D_L^* + D_L = k$ gives

$$A^* G_N(L, J) + G_N(L, J) A - k G_N(L, J) = -(R^* B + B^* R). \quad (\text{PSA28})$$

Applying D_L once more to the first identity and subtracting k times it gives, with every term retained,

$$\mathcal{L}_{L,k} R - R(A^2 - kA) = (D_L - k)B + BA. \quad (\text{PSA29})$$

This proves the finite-cutoff extension of P46–P48. Setting $\Pi_{L,J} = I$ gives the original full-period identities.

For $Av = \rho v$, (PSA29) is $(\mathcal{L}_{L,k} - \rho(\rho - k))Rv = ((D_L - k)B + BA)v$. On the original Fourier diagonal, the squared norm on the left is the sum of $|-(2\pi n/L)^2 - k^2/4 - \rho(\rho - k)|^2 \|(Rv)_n\|^2$. Every coefficient is at least $|\operatorname{Im}(\rho(\rho - k))|^2$, proving P49 also after the finite projection. The conclusion is

a lower bound on this actual boundary, not an upper bound on the arithmetic endpoint volume. The exact observation (PSA26), its one- and two-degree extensions, and (PSA27)–(PSA29) supply the map from the circle calculation to the arithmetic action. For those inverse observation extensions, take the common controlled source degree $D \geq N + 2$; the endpoint choice $D = 2q + 2$ already has this property for every endpoint $N \leq 2q$. The boundary identity itself uses the displayed smooth columns and needs injectivity only at its representative degree N .

66.1.8 The subsequent completion and the scope of verification

The fixed-circle limit keeps L fixed and completes the increasing polynomial source. The adjoining density proof supplies its typed isometry to $\ell^2(\mathbb{Z}, \nu)$, polynomial density for the original measure and its $|\chi|^2$ weight, and the exact quotient map $[P] \mapsto (P(S_n))_{\chi(S_n)=0}$. Its kernel is $(p_L)/(\chi)$, with p_L the squarefree product of the sampled roots, and the canonical Grams decrease to the sampled-value Gram. That proof retains all multiplicities in χ before taking the calculated kernel. The finite observation (PSA26) instead retains the entire remainder space at the specified N, L, J . These are related by the explicit quotient and completion arrows; an unsampled class or a nonconstant sampled jet maps to a supported zero under the latter arrow.

The complete supplied 16-method finite suite passed both normally and under Python optimization, with identical result records. Seven additional in-memory mutations each failed at one intended mathematical assertion in both modes, with no execution errors. These finite checks cover stated algebraic interfaces. The original zero-mode command-line negative is an unconditional failure-path control; the additional mutation removes the actual zero-mode Gram from its original complex-matrix calculation. The original raw checker and proof sources remain unchanged. The analytic proofs above and the adjoining density proof remain written mathematics; no Lean or interval certificate is inferred from those checks.

67 Complete periodized source density, tails and completion

67.1 Density and exact completion of the original periodized arithmetic source

67.1.1 1. The unchanged theta amplitude and its canceled zeros

Use the original Mellin transform

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx, \quad D = -x\partial_x, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{PSD1})$$

The source is the supplied strong-Schwartz function satisfying

$$v_h(s) := \mathcal{M}F_h(s) = \frac{g(s)}{h(s)}, \quad h(D)F_h = \Theta\phi_*, \quad \phi_*(x) = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}. \quad (\text{PSD2})$$

Here the quotient is its entire extension, retaining the given polynomial h . Every chosen root occurs in h with its full multiplicity in g ; no location assumption is imposed on those roots. Writing a possible leading coefficient explicitly, let

$$h(s) = a_h \prod_{\rho \in Z_h} (s - \rho)^{m_\rho}, \quad a_h \neq 0, \quad d = \sum_{\rho} m_\rho. \quad (\text{PSD3})$$

For the original monic packet, $a_h = 1$; retaining it in these formulas shows where that convention enters.

The theta factor is exact. For $\operatorname{Re}(s) > 1$, absolute convergence gives

$$\begin{aligned}\mathcal{M}\phi_*(s) &= \pi^{-s/2}(2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) \\ &= \frac{s(s-1)}{2}\pi^{-s/2}\Gamma(s/2), \\ \mathcal{M}(\Theta\phi_*)(s) &= 2\zeta(s)\mathcal{M}\phi_*(s) = g(s).\end{aligned}\tag{PSD4}$$

The factor two in Θ and the factor one-half in the Mellin transform of ϕ_* remain visible. Logarithmic integration by parts gives $\mathcal{M}(DF) = sMF$ for the given strong-Schwartz source; the boundary terms vanish at both ends. Thus (PSD2) has the same Mellin transform as its original theta right-hand side. The conditions $\phi_*(0) = 0$ and $\int_{\mathbb{R}} \phi_* = 2\mathcal{M}\phi_*(1) = 0$ also retain the two source moments.

At a selected zero ρ of order m , put

$$g(s) = (s - \rho)^m a(s), \quad h(s) = (s - \rho)^m b(s), \quad a(\rho)b(\rho) \neq 0.$$

The canceled value is therefore

$$v_h(\rho) = \frac{a(\rho)}{b(\rho)} = \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)} \neq 0.\tag{PSD5}$$

This value applies when a selected critical zero lies exactly on a sampled Fourier frequency. It must not be evaluated as an uncanceled $0/0$ or assigned zero. The entire jet is fixed as well: with

$$a_r = \frac{g^{(m+r)}(\rho)}{(m+r)!}, \quad b_r = \frac{h^{(m+r)}(\rho)}{(m+r)!}, \quad v_h(\rho + z) = \sum_{j \geq 0} c_j z^j,$$

one has

$$c_0 = a_0/b_0, \quad c_j = \frac{a_j - \sum_{r=1}^j b_r c_{j-r}}{b_0} \quad (j \geq 1).\tag{PSD6}$$

The arithmetic density used throughout the periodized note is exactly

$$w_h(t) = \frac{1}{2\pi} |v_h(1/2 + it)|^2, \quad m_{h,k} = \underbrace{w_h * \cdots * w_h}_{k \text{ factors}}.\tag{PSD7}$$

Because v_h is entire and not identically zero, its zeros on the vertical line form a discrete subset of the real parameter t . Consequently w_h is continuous, nonnegative, and strictly positive outside that discrete subset. Formula (PSD5) determines the values at every canceled critical root. No zero-location theorem is used.

67.1.2 2. A proved exponential tail, with the original constants

An elementary contour estimate supplies the needed gamma decay without substituting a numerical approximation. If $a > 0$, y is real, and $0 < \theta < \pi/2$, rotating the gamma-integral ray by $\operatorname{sign}(y)\theta$ gives

$$|\Gamma(a + iy)| \leq \Gamma(a)(\cos \theta)^{-a} e^{-\theta|y|}.\tag{PSD8}$$

For completeness, take the branch of z^{a+iy-1} in the sector between the positive ray and that rotated ray. Integrate $z^{a+iy-1}e^{-z}$ around its boundary between radii ϵ and R . For each fixed y , the small arc is bounded by a constant times ϵ^a and tends to zero. On the large arc $\operatorname{Re}(z) \geq R \cos(\theta) > 0$, so exponential decay dominates the power of R . The two ray integrals agree in the limit. Parametrizing the rotated ray by $z = e^{i \operatorname{sign}(y)\theta} r$ contributes the exact factor $e^{i \operatorname{sign}(y)\theta(a+iy)}$, whose modulus is

$e^{-\theta|y|}$. Taking the modulus inside the remaining integral gives $\int_0^\infty r^{a-1} e^{-r \cos \theta} dr = \Gamma(a)(\cos \theta)^{-a}$. At $y = 0$ the same bound follows from either ray.

On the original critical line, an elementary bound for zeta follows from the alternating Dirichlet series. For $\operatorname{Re}(s) > 0$, its partial summation formula is

$$\eta(s) = s \int_1^\infty A(x) x^{-s-1} dx, \quad A(x) = \sum_{1 \leq n \leq x} (-1)^{n-1} \in \{0, 1\}.$$

The integral converges locally uniformly there and agrees with the alternating series; the finite partial-summation boundary has modulus at most $N^{-\operatorname{Re}(s)}$ and tends to zero. For $s = 1/2 + it$,

$$|\eta(s)| \leq 2|s|, \quad |1 - 2^{1-s}| \geq \sqrt{2} - 1, \quad |\zeta(s)| \leq \frac{2|s|}{\sqrt{2} - 1}. \quad (\text{PSD9})$$

The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ first follows by absolutely convergent rearrangement for $\operatorname{Re}(s) > 1$, then by analytic continuation. Its denominator on the line has the displayed strictly positive lower bound.

Apply (PSD8) with $a = 1/4$, $y = t/2$, and retain $|\pi^{-s/2}|^2 = \pi^{-1/2}$. For every real t with $h(1/2 + it) \neq 0$, (PSD1), (PSD7), and (PSD9) give

$$w_h(t) \leq \frac{2\Gamma(1/4)^2}{\pi^{3/2}(\sqrt{2} - 1)^2 \sqrt{\cos \theta}} \frac{(t^2 + 1/4)^3}{|h(1/2 + it)|^2} e^{-\theta|t|}. \quad (\text{PSD10})$$

The power $(t^2 + 1/4)^3$ comes from $|s(s-1)|^2 |s|^2$; no asymptotic equality is asserted. Let

$$R = \max(\{1\} \cup \{2|\rho - 1/2| : \rho \in Z_h\}).$$

This definition also gives $R = 1$ for the retained analytic seed $h = 1$, whose selected-root set is empty; all product and tail formulas then use $d = 0$ and $a_h = 1$.

For $|t| \geq R$, each factor satisfies

$$|1/2 + it - \rho| \geq |t| - |\rho - 1/2| \geq |t|/2, \quad |h(1/2 + it)| \geq |a_h|(|t|/2)^d.$$

Thus, with the exact constant in front of the fraction in (PSD10) denoted by C_θ ,

$$w_h(t) \leq C_\theta \frac{4^d}{|a_h|^2} \frac{(t^2 + 1/4)^3}{|t|^{2d}} e^{-\theta|t|} \quad (|t| \geq R). \quad (\text{PSD11})$$

On the compact interval $[-R, R]$, the entire canceled quotient in (PSD2) is continuous, so $B_R = \max_{|t| \leq R} w_h(t) < \infty$. For any $0 < b < \pi/2$, choose $b < \theta < \pi/2$. The compact bound together with (PSD11) proves

$$A_b := \|e^{b|\cdot|} w_h\|_\infty < \infty, \quad B_b := \|e^{b|\cdot|} w_h\|_1 < \infty. \quad (\text{PSD12})$$

For example, source-dependent upper bounds are the maximum of $e^{bR} B_R$ and the supremum of the right side of (PSD11) multiplied by $e^{b|t|}$ on the two tails, and

$$B_b \leq 2R e^{bR} B_R + 2C_\theta \frac{4^d}{|a_h|^2} \int_R^\infty \frac{(T^2 + 1/4)^3}{T^{2d}} e^{-(\theta-b)T} dT. \quad (\text{PSD13})$$

The remaining integral is finite because $\theta - b > 0$. These are analytical constants of the original source; this audit has not computed their values or interval enclosures.

67.1.3 3. Pointwise convolution, strict positivity, and the full scalar-circle map

Convolution is taken with ordinary Lebesgue measure, with the factors $1/(2\pi)$ already contained in every w_h . The triangle inequality for $u = t_1 + \dots + t_k$ gives pointwise

$$e^{b|u|} m_{h,k}(u) \leq (e^{b|\cdot|} w_h)^{*k}(u) \leq A_b B_b^{k-1}. \quad (\text{PSD14})$$

For two factors this follows directly by bounding one factor by A_b and integrating the other. Induction gives the displayed expression for every k . The integrals are absolutely convergent for every u . The same bounds and continuity under translation give a continuous representative of the convolution. For instance, w_h is bounded, continuous, and tends to zero at infinity by (PSD11), hence uniformly continuous; dominated convergence against the integrable other factor proves continuity at every u .

For $k = 2$ and fixed u , the sets where $w_h(t) = 0$ or $w_h(u - t) = 0$ are discrete. There is a t_0 outside their union. The continuous nonnegative integrand is strictly positive on a neighborhood of t_0 , and therefore

$$(w_h * w_h)(u) > 0 \quad \text{for every real } u. \quad (\text{PSD15})$$

For the induction step, a continuous everywhere-positive convolution is integrated against w_h , which is positive on some open interval. The resulting convolution is strictly positive at every u . Consequently all (P53) weights are strictly positive for $k \geq 2$, including $n = 0$.

Retain the original diagonal coordinate isometry $\mathcal{U}_k$, the space $\mathcal{K}$, and the plus-sign Fourier transform from (P11)–(P14). Put

$$\psi_0 = \mathcal{U}_k F_h^{\otimes k}, \quad u_n = 2\pi n/L, \quad S_n = k/2 + iu_n, \quad b_n = \widehat{\psi}_0(u_n) \in \mathcal{K}.$$

In the simultaneous Fourier variables for $(r, z_1, \dots, z_{k-1})$, the amplitude is exactly

$$\prod_{j < k} v_h(1/2 + it_j) v_h \left(1/2 + i \left(u - \sum_{j < k} t_j \right) \right).$$

For each fixed u , partial Plancherel in the retained z variables therefore gives

$$\|\widehat{\psi}_0(u)\|_{\mathcal{K}}^2 = (2\pi)^{-(k-1)} \int_{\mathbb{R}^{k-1}} \prod_{j < k} |v_h(1/2 + it_j)|^2 \left| v_h \left(1/2 + i \left(u - \sum_{j < k} t_j \right) \right) \right|^2 d\mathbf{t} = 2\pi m_{h,k}(u). \quad (\text{PSD16})$$

The equality holds pointwise, not just almost everywhere: the original transformed source is Schwartz in the Hilbert-valued variable, and both sides are continuous; alternatively the preceding absolutely convergent integral computes the partial transform at each u . Thus no choice of an almost-everywhere convolution version can alter a lattice sample.

Let $H_{\text{orig}, \text{cyc}, L}$ denote the closure in the original $L^2(\mathbb{R}/L\mathbb{Z}; \mathcal{K})$ of the periodized polynomial source. Define the weighted scalar space

$$H_\nu = \ell^2(\mathbb{Z}, \nu), \quad \nu_n = \frac{2\pi}{L} m_{h,k}(u_n) = \frac{\|b_n\|_{\mathcal{K}}^2}{L}. \quad (\text{PSD17})$$

There is an exact isometry into the original Hilbert-valued circle space:

$$I_L : H_\nu \longrightarrow L^2(\mathbb{R}/L\mathbb{Z}; \mathcal{K}), \quad \widehat{I_L a}(n) = L^{-1/2} a_n b_n. \quad (\text{PSD18})$$

The hats on the right are coefficients in the unchanged orthonormal basis $L^{-1/2} \exp(-2\pi i n r / L)$. Parseval gives $\|I_L a\|^2 = \sum_n \nu_n |a_n|^2$. For a polynomial P , (P13) and (P16) give

$$I_L((P(S_n))_n) = \mathcal{P}_L \mathcal{U}_k \mathcal{V}_{h,k} P. \quad (\text{PSD19})$$

Polynomial density, proved next, identifies the range of (PSD18) with $H_{\text{orig}, \text{cyc}, L}$. Its inverse on that range is

$$a_n = \sqrt{L} \frac{\langle b_n, \hat{f}(n) \rangle_{\mathcal{K}}}{\|b_n\|_{\mathcal{K}}^2}, \quad (\text{PSD20})$$

with the inner product conjugate-linear in its first variable. Every denominator is positive for $k \geq 2$. These maps retain the amplitude b_n , the actual source measure, and every Fourier constant.

67.1.4 4. Polynomial density with an explicit analytic strip and uniqueness

Fix $k \geq 2$ and $L > 0$. By (PSD14), for each $0 < b < \pi/2$,

$$\nu_n \leq \frac{2\pi}{L} A_b B_b^{k-1} e^{-b|u_n|}. \quad (\text{PSD21})$$

For every fixed polynomial Q and $0 < c < b$,

$$\sum_{n \in \mathbb{Z}} e^{c|u_n|} |Q(S_n)|^2 \nu_n < \infty. \quad (\text{PSD22})$$

Indeed $|Q(k/2 + iu)|^2$ is bounded by a finite polynomial in $|u|$, and the remaining factor is $\exp(-(b-c)|u_n|)$ on a lattice with spacing $2\pi/L$. All polynomial moments exist.

Here is the density proof for any positive submeasure μ_n on a subset I of this lattice satisfying a finite exponential moment $\sum_{n \in I} e^{c|u_n|} \mu_n < \infty$ for some $c > 0$. Suppose $f \in \ell^2(I, \mu)$ is orthogonal to all polynomial sequences $P(S_n)$. For $0 < a < c/2$, Cauchy-Schwarz gives

$$\sum_{n \in I} |f_n| \mu_n e^{a|u_n|} \leq \left(\sum_{n \in I} |f_n|^2 \mu_n \right)^{1/2} \left(\sum_{n \in I} e^{2a|u_n|} \mu_n \right)^{1/2} < \infty. \quad (\text{PSD23})$$

Consequently

$$F(z) = \sum_{n \in I} \overline{f_n} \mu_n e^{iz u_n} \quad (\text{PSD24})$$

is holomorphic on $|\text{Im}(z)| < c/2$. On every smaller closed strip, the series and all its derivative series converge normally: choose a further positive margin inside $c/2$, and absorb each fixed power of $|u_n|$ into the exponential margin in (PSD23). The half-width is justified by the square in Cauchy-Schwarz; no full-width extension is needed.

For every $j \geq 0$, the exact polynomial $P_j(S) = ((S - k/2)/i)^j$ evaluates to u_n^j , so orthogonality gives

$$F^{(j)}(0) = i^j \sum_{n \in I} \overline{f_n} \mu_n u_n^j = 0. \quad (\text{PSD25})$$

The Taylor expansion and the identity theorem on the connected strip imply $F = 0$ there. Uniqueness can be proved directly with the actual lattice, rather than invoked abstractly: extend the coefficients by zero outside I . Since (PSD23) gives absolute summability, termwise integration is valid and

$$\frac{1}{L} \int_0^L F(x) e^{-i u_m x} dx = \sum_{n \in I} \overline{f_n} \mu_n \frac{1}{L} \int_0^L e^{2\pi i(n-m)x/L} dx = \begin{cases} \overline{f_m} \mu_m, & m \in I, \\ 0, & m \notin I. \end{cases} \quad (\text{PSD26})$$

The left side is zero. Positivity of each μ_m on I implies $f_m = 0$. The orthogonal complement of the polynomials is zero, proving their density.

Apply this first to $\mu = \nu$ and $I = \mathbb{Z}$. It proves the assertion used in (PSD19). Apply it again to

$$I = \{n : \chi(S_n) \neq 0\}, \quad \mu_n = |\chi(S_n)|^2 \nu_n. \quad (\text{PSD27})$$

The exponential moment follows from (PSD22) with $Q = \chi$. Zero atoms are removed rather than divided by. This establishes exactly the weighted density needed for (P56).

67.1.5 5. The exact closed relation space, quotient metric, and kernel

Let χ be the original nonconstant packet-sum polynomial of degree q , and set

$$\mathcal{Z}_L = \{n \in \mathbb{Z} : \chi(S_n) = 0\}, \quad p_L(S) = \prod_{n \in \mathcal{Z}_L} (S - S_n). \quad (\text{PSD28})$$

The set is finite because χ is a nonzero polynomial and the S_n are distinct. The empty product is one. These are exact sampled roots; no tolerance rule is introduced.

Every χP vanishes on $\mathcal{Z}_L$. Coordinate evaluation is continuous on H_ν , since $|f_n| \leq \nu_n^{-1/2} \|f\|_\nu$. Therefore the closure of $\chi \mathbb{C}[S]$ is contained in the closed subspace

$$V_L = \{f \in H_\nu : f_n = 0 \ (n \in \mathcal{Z}_L)\}.$$

Conversely take $f \in V_L$ and define $g_n = f_n / \chi(S_n)$ on its complement I . Then

$$\sum_{n \in I} |g_n|^2 |\chi(S_n)|^2 \nu_n = \sum_{n \in I} |f_n|^2 \nu_n < \infty.$$

By (PSD27), polynomials P_j approximate g in that weighted space. Multiplication by χ gives

$$\|\chi P_j - f\|_\nu^2 = \sum_{n \in I} |P_j(S_n) - g_n|^2 |\chi(S_n)|^2 \nu_n \longrightarrow 0. \quad (\text{PSD29})$$

Thus $\overline{\chi \mathbb{C}[S]} = V_L$. The orthogonal complement is the sequence space supported on $\mathcal{Z}_L$, and the Hilbert quotient is isometric to that subspace by restriction. Its metric is exactly $\sum_{n \in \mathcal{Z}_L} \nu_n |v_n|^2$.

The resulting algebraic comparison is the surjection

$$\theta_L : \mathbb{C}[S]/(\chi) \longrightarrow H_\nu / \overline{\chi \mathbb{C}[S]} \simeq \bigoplus_{n \in \mathcal{Z}_L} \mathbb{C}, \quad [P] \longmapsto (P(S_n))_{n \in \mathcal{Z}_L}. \quad (\text{PSD30})$$

It is well-defined because $\chi(S_n) = 0$. Surjectivity follows from the explicit Lagrange polynomials

$$\ell_n(S) = \prod_{m \in \mathcal{Z}_L, m \neq n} \frac{S - S_m}{S_n - S_m}, \quad \ell_n(S_m) = \delta_{nm}. \quad (\text{PSD31})$$

For an empty set the target is the zero vector space, and the map is the unique surjection onto it. Since the sampled roots are distinct, a polynomial vanishes on them if and only if it is divisible by p_L . Moreover p_L divides χ . Hence

$$\ker \theta_L = (p_L)/(\chi), \quad \text{rank } \theta_L = |\mathcal{Z}_L|, \quad \dim \ker \theta_L = q - |\mathcal{Z}_L|. \quad (\text{PSD32})$$

These formulas include the empty set, when $(p_L)/(\chi) = \mathbb{C}[S]/(\chi)$.

The generator intertwines exactly. With $A[P] = [SP]$ on the original algebra and $A_L(v_n) = (S_n v_n)$ on the finite target,

$$\theta_L A = A_L \theta_L. \quad (\text{PSD33})$$

The kernel is therefore an invariant submodule; it has not been removed before forming the comparison.

There is a precise cyclic presentation of this invariant kernel. Put $t_L = \chi/p_L$ and define

$$\mu_{p_L} : \mathbb{C}[S]/(t_L) \longrightarrow E, \quad [Q]_{(t_L)} \longmapsto [p_L Q]_{(\chi)}. \quad (\text{PSD33a})$$

This is well-defined because $p_L t_L = \chi$. Its image is $(p_L)/(\chi)$. If $p_L Q$ is divisible by $\chi = p_L t_L$, polynomial cancellation gives divisibility of Q by t_L , so the map is injective. No coprimality between p_L and t_L is used. Consequently the entire comparison is the exact sequence

$$0 \longrightarrow \mathbb{C}[S]/(\chi/p_L) \xrightarrow{\mu_{p_L}} \mathbb{C}[S]/(\chi) \xrightarrow{\theta_L} \mathbb{C}^{\mathcal{Z}_L} \longrightarrow 0. \quad (\text{PSD33b})$$

Multiplication commutes in the polynomial ring, hence $A\mu_{p_L} = \mu_{p_L} A_{t_L}$, with A_{t_L} the original multiplication by S on the source algebra. Together with (PSD33), this proves equivariance of both arrows.

The Fourier-side generator has the same typed action. Define

$$T_L a = (S_n a_n)_n, \quad \mathcal{D}(T_L) = \left\{ a \in H_\nu : \sum_n \nu_n |S_n a_n|^2 < \infty \right\}. \quad (\text{PSD33c})$$

It is closed: convergence of a and $T_L a$ in H_ν implies convergence of each coordinate, giving the displayed pointwise relation in the limit and its square summability. The coordinate projection $Q_{\mathcal{Z}_L}$ commutes with T_L on this domain and has range in it, because its support is finite. On polynomials, $T_L(P(S_n)) = ((SP)(S_n))$; under (PSD18) it is exactly the circle operator $-\partial_r + k/2$. It induces A_L on the quotient. Therefore the quadratic generator also intertwines exactly:

$$\theta_L(A^2 - kA) = (A_L^2 - kA_L)\theta_L, \quad S_n(S_n - k) = -k^2/4 - (2\pi n/L)^2. \quad (\text{PSD33d})$$

67.1.6 6. Full repeated-root blocks and the canonical metric limit

Factor the actual polynomial as $\chi(S) = a_\chi \prod_\lambda (S - \lambda)^{m_\lambda}$. The exact Chinese-remainder map is

$$\mathbb{C}[S]/(\chi) \xrightarrow{\sim} \bigoplus_\lambda \mathbb{C}[\epsilon_\lambda]/(\epsilon_\lambda^{m_\lambda}), \quad [P] \longmapsto \left(\sum_{j=0}^{m_\lambda-1} \frac{P^{(j)}(\lambda)}{j!} \epsilon_\lambda^j \right)_\lambda. \quad (\text{PSD34})$$

Under this map, A acts on each summand by $\lambda + \epsilon_\lambda$. At an unsampled root the whole summand lies in $\ker(\theta_L)$. At a sampled root $\lambda = S_n$, the comparison takes the constant coefficient of this full Taylor class, and its kernel is exactly

$$\epsilon_\lambda \mathbb{C}[\epsilon_\lambda]/(\epsilon_\lambda^{m_\lambda}). \quad (\text{PSD35})$$

In particular every nonconstant generalized-jet direction is retained in the explicit kernel. The induced action on the remaining constant coefficient is multiplication by λ ; (PSD33) follows in these

coordinates as well. This proves the full multiplicity statement rather than inferring it from a list of sampled eigenvalues.

The inverse in (PSD34) is explicit. Set $H_\lambda(S) = \chi(S)/(S - \lambda)^{m_\lambda}$. Its value at λ is nonzero. Let $V_\lambda(\epsilon)$ be the Taylor polynomial of $1/H_\lambda(\lambda + \epsilon)$ through degree $m_\lambda - 1$, and put $e_\lambda(S) = H_\lambda(S)V_\lambda(S - \lambda)$. Modulo the λ primary factor this polynomial equals one; modulo every other primary factor it equals zero. The inverse sends a tuple $F_\lambda(\epsilon)$ to $[\sum_\lambda e_\lambda(S)F_\lambda(S - \lambda)]_{(\chi)}$. Both composites are the identity: their differences are divisible by every pairwise coprime primary factor, hence by their product, which differs from χ only by its original nonzero leading coefficient.

At a sampled root, write $p_L(\lambda + \epsilon) = \epsilon u_\lambda(\epsilon)$, where $u_\lambda(0) = \prod_{\substack{\mu \text{ sampled} \\ \mu \neq \lambda}} (\lambda - \mu) \neq 0$. Multiplication by p_L identifies the t_L block $\mathbb{C}[\epsilon]/(\epsilon^{m_\lambda - 1})$ with (PSD35) by multiplication by ϵu_λ . At an unsampled root, $p_L(\lambda + \epsilon)$ is a unit and identifies the entire t_L block with the original full block. Thus the kernel generator has block length $m_\lambda - 1$ at each sampled root and length m_λ at each unsampled root, with the original eigenvalues retained.

Now retain the note's fixed remainder-coordinate basis on $E = \mathbb{C}[S]/(\chi)$, the degree- N polynomial source $\mathcal{P}_N$, and the canonical constrained minimum $G_N(L)$. For every $N \geq q - 1$, its source Gram is positive definite: the summands have positive weights and a nonzero polynomial has only finitely many zeros, whereas this lattice is infinite. The finite-dimensional minimum therefore exists uniquely and is precisely

$$e^* G_N(L) e = \min\{\|P\|_\nu^2 : P \in \mathcal{P}_N, [P] = e\}. \quad (\text{PSD36})$$

Choose the unique degree-less-than- q remainder R_e of e . The permitted representatives are $R_e + \chi \mathcal{P}_{N-q}$, with the relation space understood to be zero when $N = q - 1$. These relation spaces increase with N , so the forms $G_N(L)$ decrease in Loewner order. Their union is $\chi \mathbb{C}[S]$; their closure is the space computed in (PSD29). For a closed-subspace distance, approximation in that closure proves

$$\begin{aligned} \lim_{N \rightarrow \infty} e^* G_N(L) e &= \inf_{Q \in \mathbb{C}[S]} \|R_e + \chi Q\|_\nu^2 \\ &= \text{dist}(R_e, V_L)^2 = \sum_{n \in \mathcal{Z}_L} \nu_n |R_e(S_n)|^2. \end{aligned} \quad (\text{PSD37})$$

The equality with the infimum is exact: every polynomial Q appears at a finite degree, so the decreasing finite-degree infima converge to the infimum over the union; the norm is continuous, so replacing the union by its closure does not change that infimum.

Let $J_{\mathcal{Z}_L}$ be evaluation at the sampled roots in the same remainder basis. Equation (PSD37) gives the matrix limit

$$G_N(L) \downarrow G_\infty(L) := J_{\mathcal{Z}_L}^* \text{diag}(\nu_n)_{n \in \mathcal{Z}_L} J_{\mathcal{Z}_L}. \quad (\text{PSD38})$$

Polarization of (PSD37) gives every matrix entry; finite dimension then gives convergence in matrix norm. Since all sampled weights are positive,

$$\text{rad } G_\infty(L) = \ker J_{\mathcal{Z}_L} = (p_L)/(\chi). \quad (\text{PSD39})$$

Thus a nonzero repeated-jet class in (PSD35) has positive norm for every finite N , while its minimum norm tends to zero as N tends to infinity at this fixed L . More explicitly, choose such a class e and its minimizing representatives $P_{N,e}$. They retain the complete prescribed remainder, hence every original Taylor coefficient, and (PSD37) proves $\|P_{N,e}\|_\nu \rightarrow 0$. Any Taylor functional that is nonzero on e is consequently unbounded for this completed polynomial norm. This supplies the actual relation between retained finite-degree generalized jets and their fixed-circle completion.

There is a strict finite-degree strengthening. For $e \neq 0$, let $P_{N,e}$ be its nonzero minimizing polynomial in (PSD36). It has the prescribed values $R_e(S_n)$ on $\mathcal{Z}_L$. Hence

$$e^*(G_N(L) - G_\infty(L))e = \sum_{n \notin \mathcal{Z}_L} \nu_n |P_{N,e}(S_n)|^2 > 0. \quad (\text{PSD40})$$

The complement of the finite set $\mathcal{Z}_L$ is infinite, all its weights are positive, and a nonzero polynomial cannot vanish at every node there. Thus $G_N(L) - G_\infty(L)$ is positive definite for every finite $N \geq q - 1$, although it converges to zero. The proof gives no positive lower bound uniform in N . The limit has rank $|\mathcal{Z}_L|$; in particular it is positive definite exactly when all roots of χ are simple and sampled. When $\mathcal{Z}_L$ is empty, the exact sequence has kernel equal to all of E and $G_N(L)$ tends to zero.

For $k = 1$, the density and closed-relation quotient proofs apply to the positive support $\mathcal{I}_{h,L} = \{n : w_h(2\pi n/L) > 0\}$. The Hilbert space, all evaluations, and the definition of $\mathcal{Z}_L$ must be restricted to that support. On this support, the formula (PSD20) also gives the inverse of the source-to-sequence isometry. The all-integer positivity, the positive-definite matrix assertions for every finite degree at arbitrary fixed L , and the strict difference in (PSD40) have been proved here for $k \geq 2$. For $k = 1$, the source's separately proved large-period comparison (P34)–(P37) supplies the finite-degree injectivity used there; no assertion that every fixed circle has all integer atoms with positive weight is inserted into that result.

For the actual one-fold packet, $\chi_{h,1} = h$. If $h(S_n) = 0$ with $S_n = 1/2 + 2\pi in/L$, the full-order cancellation formula (PSD5) gives

$$w_h(2\pi n/L) = \frac{1}{2\pi} \left| \frac{g^{(m)}(S_n)}{h^{(m)}(S_n)} \right|^2 > 0, \quad m = \text{ord}_{S_n} h = \text{ord}_{S_n} g. \quad (\text{PSD41})$$

Consequently the sampled-root set $\mathcal{Z}_L = \{n : h(1/2 + 2\pi in/L) = 0\}$ is already contained in $\mathcal{I}_{h,L}$. Restricting the Hilbert space to its positive support therefore retains every sampled root of this original one-fold packet. Its reduced evaluation quotient is $\mathbb{C}^{\mathcal{Z}_L}$, with the original weights, and its algebraic kernel is $(p_L)/(h)$ by the same proof as (PSD28)–(PSD32). Each factor of p_L is a critical-line factor of h , occurring once, so p_L divides the full-multiplicity critical factor $\chi_{c,1}$. This is exactly the cancellation fact used by the accompanying critical-jet comparison (DC1)–(DC30). It asserts positivity at the selected sampled roots; it does not assert positivity at every other integer atom or invertibility of every high-degree Gram at arbitrary fixed L .

68 Finite-circle curvature and complete quotient compensation

68.1 Finite sampling, residue curvature and the original control form

We retain the polynomial source, remainder map, coefficient basis and arithmetic inner product from the periodized-source construction. The following calculation propagates its proved finite-source errors to the residue insertion and the original Laplacian. The quotient curvature calculation also incorporates the exact Schur-complement argument supplied by the coordinating Zeta task. Fix the original tensor degree $k \in \mathbb{Z}_{\geq 1}$ throughout the source construction.

Let $E = \mathbb{C}[S]/(\chi)$, with the original monic χ of degree $q \geq 1$, $A = M_S$, $e_0 = [1]$, $\ell[P] = [S^{q-1}] \text{rem}_\chi P$, and $R = e_0 \ell$. For $N \geq q - 1$, let $\mathcal{P}_N$ be the original polynomial coefficient space, $J_N : \mathcal{P}_N \rightarrow E$ the full remainder map, and $M_N > 0$ its original source Gram. Keep

$$G = G_N = (J_N M_N^{-1} J_N^*)^{-1}. \quad (\text{FC1})$$

All adjoints denoted by $*$ are conjugate transposes in these specified coefficient frames. For a positive form T , write $B^{\dagger T} = T^{-1} B^* T$ and $\|B\|_T$ for its induced operator norm.

68.1.1 The actual finite sampling error

Fix the common source degree $D \geq N$, $a > 0$, integer $p \geq 1$, circle length $L > 0$ and integer cutoff $J \geq 1$. In the original diagonal log coordinates $r = \log x_k$, $z_i = \log x_i - \log x_k$, retain

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_i z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad \mathcal{K} = L^2(\mathbb{R}^{k-1}, dz). \quad (\text{FC2})$$

The Jacobian is $d^k x = e^{kr + \sum_i z_i} dr dz$, proving directly that this map preserves the original norm. Its derivative satisfies $\mathcal{U}_k D^{(k)} = (-\partial_r + k/2)\mathcal{U}_k$. Let Ψ_D be the image of the degree- D source columns. Define

$$\begin{aligned} M_{a,\pm,D} &= (e^{\pm ar} \Psi_D)^* (e^{\pm ar} \Psi_D), \quad H_{p,D} = \int_{\mathbb{R}} (1+r^2) (\partial_r^p \Psi_D)^* (\partial_r^p \Psi_D) dr, \\ \kappa_{a,D} &= \max_{c \neq 0} \frac{c^* (M_{a,+D} + M_{a,-D}) c}{c^* M_{Dc}}, \quad \lambda_{p,D} = \max_{c \neq 0} \frac{c^* H_{p,D} c}{c^* M_{Dc}}, \\ \eta_P &= \frac{\kappa_{a,D}}{e^{aL} - 1}, \quad \eta_T = \frac{\lambda_{p,D}}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1}, \quad \delta = \eta_P + \eta_T. \end{aligned} \quad (\text{FC3})$$

These are finite observations of the original source; the weighted forms retain $x_k^{\pm 2a}$ and $1 + (\log x_k)^2$ in the original coordinates. For positive budgets $\varepsilon_P, \varepsilon_T$ with $\varepsilon_P + \varepsilon_T < 1$, choose the literal values of (P64):

$$\begin{aligned} L &\geq a^{-1} \log(1 + \kappa_{a,D}/\varepsilon_P), \\ J &\geq \max \left\{ 1, D, \frac{L}{2\pi} \left(\frac{\lambda_{p,D}}{(2p-1)\varepsilon_T} \right)^{1/(2p-1)} \right\}, \quad J \in \mathbb{Z}. \end{aligned} \quad (\text{FC3a})$$

Substitution into (FC3) proves $\eta_P \leq \varepsilon_P$ and $\eta_T \leq \varepsilon_T$, including $\lambda_{p,D} = 0$. Thus $0 \leq \delta < 1$ has been obtained from actual finite quantities. No estimate uniform in h, k, D is inserted.

Use the plus Fourier transform $\hat{\psi}(u) = \int \psi(r) e^{iur} dr$. Let $M_N(L, J)$ be the full source Gram

$$M_N(L, J)_{ab} = \frac{1}{L} \sum_{|n| \leq J} \left\langle \widehat{\Psi_N e_a}(2\pi n/L), \widehat{\Psi_N e_b}(2\pi n/L) \right\rangle_{\mathcal{K}}. \quad (\text{FC4})$$

The $n = 0$ term is included with its original factor L^{-1} . For clarity, the estimates producing (FC3) can be checked without an inverse operation. Unfolding periodization gives $M_N(L) - M_N = \sum_{m \neq 0} \mathcal{C}_N(mL)$. Weighted Cauchy-Schwarz gives $|c^* \mathcal{C}_N(s) c| \leq e^{-a|s|} \sqrt{c^* M_{a,-,N} c c^* M_{a,+,N} c}$. Summing both lattice directions bounds its absolute value by $c^* (M_{a,-,N} + M_{a,+,N}) c / (e^{aL} - 1)$. Also integration by parts and $\int (1+r^2)^{-1} dr = \pi$ give $\|\widehat{\Psi_N c}(u)\|^2 \leq \pi |u|^{-2p} c^* H_{p,N} c$. Summing the two tails $|n| > J$, with $\sum_{n > J} n^{-2p} \leq J^{1-2p}/(2p-1)$, yields exactly the coefficient of $\lambda_{p,D}$ in (FC3). Restriction from the degree- D source preserves all these quadratic-form bounds. Therefore

$$(1 - \eta_P - \eta_T) M_N \preceq M_N(L, J) \preceq (1 + \eta_P) M_N. \quad (\text{FC5})$$

The asymmetry records that the omitted Fourier tail is positive. Put $\alpha = 1 - \eta_P - \eta_T > 0$, $\beta = 1 + \eta_P$, and retain

$$H = G_N(L, J) = (J_N M_N(L, J)^{-1} J_N^*)^{-1}. \quad (\text{FC6})$$

Completing the square proves that $v^* G v$ is the minimum of $c^* M_N c$ over $J_N c = v$, and the same statement holds for H . Taking these minima in (FC5) proves

$$\alpha G \preceq H \preceq \beta G, \quad (1 - \delta) G \preceq H \preceq (1 + \delta) G. \quad (\text{FC7})$$

The coefficient minimizers are $C_N = M_N^{-1} J_N^* G$ and $C_N(L, J) = M_N(L, J)^{-1} J_N^* H$. Their difference belongs to $\chi \mathcal{P}_{N-q}$. Fixed monic division writes its polynomial as $\chi Q = \sum_i h(s_i) Q_i(\mathbf{s})$. The original tensor primitive replaces F_h by ϕ_* in slot i , applies $Q_i(D_1, \dots, D_k)$, and has coefficient $(-1)^{i-1}$. The tensor differential contributes the same sign. Thus this change of representative has an explicit primitive with every original summand retained.

68.1.2 A dimension-free relative estimate for the residue coefficient

Let $I : F \hookrightarrow E$ be an actual nonzero proper A -invariant constituent and set $L_F = \ell I$. For every positive form T define

$$\begin{aligned} T_F &= I^* T I, & P_{T,F} &= I T_F^{-1} I^* T, \\ a_F(T) &= \min_{x \in F} (e_0 - Ix)^* T (e_0 - Ix) = e_0^* T (1 - P_{T,F}) e_0, \\ b_F(T) &= L_F T_F^{-1} L_F^*, & c_F(T) &= a_F(T) b_F(T), & \sigma(T)^2 &= (e_0^* T e_0) (\ell T^{-1} \ell^*). \end{aligned} \quad (\text{FC8})$$

Here $q > 1$. To verify strict positivity, $I(F)$ is an ideal of E because it is invariant under every polynomial in A . Its polynomial preimage is (g) with $\chi = gh$, and the original ideal basis is $[g], \dots, [S^{p_F-1}g]$, where $p_F = \deg h = \dim F$. Its last member is monic of degree $q - 1$, so $L_F \neq 0$; properness gives $e_0 \notin I(F)$. These statements prove $a_F(T), b_F(T) > 0$ without discarding repeated factors.

The same constrained-minimum argument as in (FC7) gives $\alpha a_F(G) \leq a_F(H) \leq \beta a_F(G)$. Compression and inverse order give $\beta^{-1} b_F(G) \leq b_F(H) \leq \alpha^{-1} b_F(G)$. Inverse order follows, for example, from the exact identity $w^* T^{-1} w = \max_x (2 \operatorname{Re} w^* x - x^* T x)$. Multiplication proves the full coefficient estimate

$$\left| \frac{\alpha}{\beta} \leq \frac{c_F(H)}{c_F(G)} \leq \frac{\beta}{\alpha}, \quad \left| \log \frac{c_F(H)}{c_F(G)} \right| \leq \log \frac{\beta}{\alpha}. \quad (\text{FC9})$$

Applying the same two form bounds to e_0 and the inverse form to ℓ^* also proves

$$\frac{\alpha}{\beta} \sigma(G)^2 \leq \sigma(H)^2 \leq \frac{\beta}{\alpha} \sigma(G)^2. \quad (\text{FC10})$$

Every number is evaluated in its stated original or sampled metric. Neither (FC9) nor (FC10) has a dimension factor.

For the actual period matrix $\Pi(u, t)$, $u \neq 0$, put $T_{\text{per}}(t) = \Pi(t)^* \Pi(t)$ and let $m_H(t), M_H(t)$ be the least and greatest eigenvalues of $H^{-1} T_{\text{per}}(t)$. Their positivity follows from the nonzero full period determinant. Applying the same calculation, now between H and T_{per} , gives

$$\frac{m_H(t)}{M_H(t)} c_F(H) \leq c_F(T_{\text{per}}(t)) \leq \frac{M_H(t)}{m_H(t)} c_F(H). \quad (\text{FC11})$$

The period equation $\Pi' = -\Pi(A + tR)/u$ implies, with $Y = \Pi I$ and $P = Y(Y^* Y)^{-1} Y^*$, $(1 - P)Y' = -t(1 - P)\Pi e_0 L_F / u$. Differentiating $\log \det(Y^* Y)$ gives $\partial_t \log \det(Y^* Y) = \operatorname{Tr}((Y^* Y)^{-1} Y'^* (1 - P)Y')$. Consequently the precise connection to (FC9) is

$$\partial_t \log \det(Y^* Y) = \frac{|t|^2}{|u|^2} c_F(T_{\text{per}}(t)). \quad (\text{FC12})$$

Equations (FC9)–(FC12) retain the actual comparison spectrum; they do not substitute the sampled coefficient for the period coefficient.

There is also a useful exact endpoint bound. For two admitted degrees $i \leq j$, the original representative spaces increase, so $G_i \succeq G_j > 0$. Let θ and ζ_+ be the least and greatest eigenvalues of the actual relative operator $G_i^{-1} G_j$. Then $\theta G_i \preceq G_j \preceq \zeta_+ G_i$ with $0 < \theta \leq \zeta_+ \leq 1$. Applying the proved minimum and inverse comparisons with these two factors gives the first inequality below. All eigenvalues lie in $(0, 1]$, so their product is at most their least member. This proves the remaining inequalities and retains the sharper endpoint coefficient from (RCX21):

$$\left| \log \frac{c_F(G_j)}{c_F(G_i)} \right| \leq \log \frac{\zeta_+}{\theta} \leq -\log \theta \leq \log \frac{\det G_i}{\det G_j}. \quad (\text{FC13})$$

The last inequality retains the correct direction. It does not infer an upper volume estimate from a scalar curvature observation. For the four original endpoints, retain the same degree $D = 2q+2$ used

to control their two generator raises. Its literal coefficient inclusion $I_{2q,D} : \mathcal{P}_{2q} \hookrightarrow \mathcal{P}_D$ restricts the two original forms to $M^{(0)} = M_{2q}$ and $M^{(1)} = M_{2q}(L, J)$. The complete signed determinant proof (PSA24a)–(PSA25) applies through these exact restrictions and the unchanged remainder maps. In particular it is an original/sampled comparison, with no replacement by the Gamma density. Write $b_1 \leq \dots \leq b_{2q+1}$ for the eigenvalues of $D_{L,J} = (M^{(0)})^{-1}M^{(1)}$. They lie in $[\alpha, \beta]$ by (FC5). The common-source projection formulas in (PSA24c) use $I_N B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_{2q}$, where B_N itself has codomain $\mathcal{P}_N$; substituting $M_N = I_N^* M(x) I_N$ proves the equality of their relation Grams. The actual trace identity (PSA24d), full flag spectra, and retained cross-pairings prove

$$\begin{aligned} |\mathcal{B}(L, J) - \mathcal{B}| &\leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{J}_{L,J} \\ &\leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log(\beta/\alpha). \end{aligned} \quad (\text{FC13a})$$

Here $\mathcal{J}_{L,J}$ is the generic two-bound integral in (PSA24e), and $\mathcal{J}_{L,J}^{(5)}$ is the actual-moment five-bound integral in (PSA24f). Its original measure and its positive atomic measure are explicitly identified there, so the RMT angle theorem applies through those same coefficient maps. The full nonlinear and signed interval (PSA25a) consequently propagates at these same four endpoints. Explicitly, if $\mathcal{B}_0 = \mathcal{B}$ is the original endpoint, define $a_q = 2q-1$, $H_0 = 2 \operatorname{arccosh}(e^{\mathcal{B}_0/(2a_q)})$ and retain the five-entry radius $\mathcal{I}_{L,J}$ of (PSA24f). The constructed j_{ℓ_N}, e_{ℓ_N} in (PSA25ba)–(PSA25bc) use precisely $M_{2q}, M_{2q}(L, J)$, their original projectors and the same source inclusions; the finite-series index ℓ_N does not change the circle length. Thus the actual endpoint also obeys

$$\begin{aligned} \mathcal{B}(L, J) &\geq \max \left\{ 0, \mathcal{B}_0 - \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{L,J}\}}{2}, \mathcal{B}_0 + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}(L, J) &\leq \min \left\{ \mathcal{B}_0 + \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{L,J}}{2}, \mathcal{B}_0 + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned} \quad (\text{FC13b})$$

Each bound is the proved same-source specialization of RMT22; taking the intersection preserves its exact endpoint. The formula and complete proof there retain all four signs; (FC13a) substitutes the same forms and bounds each relative ratio by β/α with total coefficient $1 + 2(q-1) = 2q-1$. The residue coefficient inequality (FC13) stays dimension-free: its operator is on the quotient E , while (FC13a) concerns the full source $\mathcal{P}_{2q}$. The exact map between them is the minimum section $C_N = M_N^{-1} J_N^* G_N$ from (FC7), satisfying $J_N C_N = I_E$ and $C_N^* M_N C_N = G_N$. It therefore retains the quotient norm and its original relation-valued sampling difference; no equality of these two different trace functionals is assumed.

68.1.3 Transporting the control form through the sampling error

The control also depends on the unchanged operator, so we retain its complete error term. For $t \in \mathbb{C}$, retaining that tensor degree k , set

$$\begin{aligned} A_t &= A + tR, \quad T_t = A_t - \frac{k}{2}I, \quad W_G(t) = A_t^* G + G A_t - kG, \quad X_G(t) = G^{-1} W_G(t), \\ W_H(t) &= A_t^* H + H A_t - kH, \quad X_H(t) = H^{-1} W_H(t), \quad B = G^{-1}(H - G). \end{aligned} \quad (\text{FC14})$$

The matrix identities in this subsection and in (FC24)–(FC29) also hold for any real scalar k when considered on the displayed finite coefficient space; the source realization retains its given integer tensor degree. Then $B^{\dagger G} = B$, $\|B\|_G \leq \delta$, and $H = G(I + B)$. No coordinate vector or inner product has been rescaled. Direct multiplication proves the exact identity

$$(I + B)X_H(t) = X_G(t) + T_t^{\dagger G} B + B T_t. \quad (\text{FC15})$$

Let $\epsilon_G(t) = \|X_G(t)\|_G$, $\epsilon_H(t) = \|X_H(t)\|_H$, and $\tau_G(t) = \|T_t\|_G$. The two X operators are self-adjoint in their respective metrics, so their norms are the suprema of their absolute quadratic forms divided

by the respective squared norms. For every original v ,

$$|v^*(W_H(t) - W_G(t))v| = |\langle T_t v, Bv \rangle_G + \langle Bv, T_t v \rangle_G| \leq 2\delta\tau_G(t) v^* G v. \quad (\text{FC16})$$

Using (FC7) in both directions proves

$$\epsilon_H(t) \leq \frac{\epsilon_G(t) + 2\delta\tau_G(t)}{\alpha}, \quad \epsilon_G(t) \leq \beta\epsilon_H(t) + 2\delta\tau_G(t). \quad (\text{FC17})$$

Thus the control transfer retains the actual generator size as well as the relative metric error. Both quantities are finite observations of the original matrices. The exact LC additional radii $|t|\sigma(G)$ for $q > 1$, and $2|\operatorname{Re} t|$ for $q = 1$, remain available through (FC10).

For completeness, define the evaluated upper bound $e = \beta\epsilon_H(t) + 2\delta\tau_G(t)$ and $L_t = A_t^2 - kA_t$. Expansion gives $GL_t = W_G(t)T_t - T_t^*GT_t - k^2G/4$. For each nonzero original vector v , put $z = v^*GL_tv/(v^*Gv)$, $r = \|T_tv\|_G/\|v\|_G$ and $a + ib = \langle v, X_G(t)T_tv \rangle_G/(v^*Gv)$. Equation (FC17) gives $a^2 + b^2 \leq e^2r^2$, while the expansion gives $\operatorname{Re} z = a - r^2 - k^2/4$ and $\operatorname{Im} z = b$. The identities $er - r^2 = e^2/4 - (r - e/2)^2$ and

$$e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - b^2 = (a - e^2/2)^2 + e^2r^2 - a^2 - b^2$$

prove the actual original-metric enclosure

$$\operatorname{Re} z \leq \frac{e^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right). \quad (\text{FC18})$$

No division by e occurs, so the zero-radius case is included.

68.1.4 The finite Fourier boundary and the quotient compensation

Let Q_J be the orthogonal Fourier projection to $|n| \leq J$, with the full $\mathcal{K}$ fibre at each retained mode. It commutes with $D_L = -\partial_r + k/2$. Use the actual least-norm representative for $M_N(L, J)$ and write

$$\begin{aligned} r^{L,J} &= \mathcal{V}_{h,k} C_N(L, J), \quad \mathcal{R} = Q_J \mathcal{P}_L \mathcal{U}_k r^{L,J}, \\ \mathcal{B} &= Q_J \mathcal{P}_L \mathcal{U}_k (D^{(k)} r^{L,J} - r^{L,J} A). \end{aligned} \quad (\text{FC19})$$

The explicit polynomial-division primitive following (FC7) applies to this relation as well. Direct intertwining gives $D_L \mathcal{R} - \mathcal{R} A = \mathcal{B}$ and $\mathcal{R}^* \mathcal{R} = H$. Since $D_L^* + D_L = k$ on these smooth finite-mode columns, multiplication and then one further application of D_L prove

$$\begin{aligned} W_H(0) &= -(\mathcal{R}^* \mathcal{B} + \mathcal{B}^* \mathcal{R}), \\ (D_L^2 - kD_L) \mathcal{R} - \mathcal{R} (A^2 - kA) &= (D_L - k) \mathcal{B} + \mathcal{B} A. \end{aligned} \quad (\text{FC20})$$

In particular (FC17) transports an actual boundary control computed from (FC20) to the original Gram. Finite mode count does not assign interval enclosures to its arithmetic coefficients; such enclosures have to be proved for any numerical application.

Finally retain the exact coefficient quotient $E/I(F)$ and a constant lift matrix J_F of its fixed basis. Then $C = [I, J_F]$ is invertible. For $T_{\text{per}} = \Pi^* \Pi$ the least-period-norm lift and its Gram are

$$\begin{aligned} Q(t) &= J_F - I(I^* T_{\text{per}} I)^{-1} I^* T_{\text{per}} J_F, \\ K(t) &= J_F^* T_{\text{per}} J_F - J_F^* T_{\text{per}} I(I^* T_{\text{per}} I)^{-1} I^* T_{\text{per}} J_F. \end{aligned} \quad (\text{FC21})$$

Completing the square in $\|\Pi(J_F v + Ix)\|^2$ proves uniqueness, positivity, $\pi Q = \pi J_F$ and $I^* T_{\text{per}} Q = 0$. The change $J_F \mapsto J_F + IB_0$ cancels exactly in Q and K ; a quotient basis change $J_F \mapsto J_F D_0$ gives $K \mapsto D_0^* K D_0$. Block elimination in $C^* T_{\text{per}} C$ gives

$$\det(I^* T_{\text{per}} I) \det K = |\det C|^2 |\det \Pi|^2. \quad (\text{FC22})$$

The complete period determinant is a nonzero constant times a holomorphic exponential in t at fixed u . Its retained squared constant is $(2\pi|u|)^q$; hence its log modulus squared has zero mixed curvature. Differentiating (FC22) and using (FC12) proves

$$\partial_{\bar{t}}\partial_t \log \det K(t) = -\frac{|t|^2}{|u|^2} \mathfrak{c}_F(T_{\text{per}}(t)). \quad (\text{FC23})$$

Both actual curvatures vanish at $t = 0$, and their nonzero leading coefficients have opposite signs. More precisely applying $\partial_t\partial_{\bar{t}}$ to (FC12) at zero gives $\partial_t^2\partial_{\bar{t}}^2 \log \det(I^*T_{\text{per}}I)|_0 = \mathfrak{c}_F(T_{\text{per}}(0))/|u|^2$; the same derivative of $\log \det K$ is its negative. The respective $t^2\bar{t}^2$ Taylor coefficients are these numbers divided by $2!2!$.

Apply the same Schur formula to the unchanged G and sampled H to obtain K_G, K_H . The fibre minima in the same quotient give $\alpha K_G \preceq K_H \preceq \beta K_G$. The literal comparison endomorphism $K_G^{-1}K(t)$ satisfies $K_G(K_G^{-1}K(t)) = K(t)$ and has log determinant curvature (FC23). This supplies the metric morphism on the original quotient, together with its finite sampling error and its exact curvature compensation. No coefficient class, generalized eigenspace or support label is removed by these identities.

68.1.5 The exact commutator allowance for the same two metrics

The coordinating Zeta task supplied the following sharper transfer, which we prove in full for the actual G, H, A_t already defined. It retains the specific metric discrepancy that interacts with the generator. All adjoints in this subsection are for G . Put $\mathcal{C} = G^{-1}H = I + B$, using the same B as in (FC14). This is a positive G -self-adjoint endomorphism with spectrum in $[\alpha, \beta]$. This symbol denotes the metric endomorphism; the constant quotient frame $C = [I, J_F]$ in (FC22) remains unchanged. Subtract $\mathcal{C}X_G(t)$ from (FC15), in the displayed order:

$$X_H(t) - X_G(t) = \mathcal{C}^{-1}[T_t^{\dagger G}, B]. \quad (\text{FC24})$$

Indeed $X_G = T_t^{\dagger G} + T_t$, so the numerator of this difference is $T_t^{\dagger G}B - BT_t^{\dagger G}$.

Let $\mathcal{S} = \mathcal{C}^{1/2}$ be the unique positive G -self-adjoint square root. The finite spectral theorem constructs it by replacing each positive eigenvalue of the actual $\mathcal{C}$ by its positive square root. Uniqueness follows because a positive square root commutes with its square and, on each eigenspace of that square, positivity forces its eigenvalues to be that positive root. Thus no additional metric is selected. The exact map $\mathcal{S} : (E, H) \rightarrow (E, G)$ is an onto isometry, since $\mathcal{S}^*G\mathcal{S} = G\mathcal{S}^2 = G\mathcal{C} = H$. Its inverse is the actual $\mathcal{S}^{-1}$. In this comparison set

$$\begin{aligned} Y_H &= \mathcal{S}X_H\mathcal{S}^{-1} = \mathcal{S}^{-1}T_t^{\dagger G}\mathcal{S} + \mathcal{S}T_t\mathcal{S}^{-1}, \\ Z &= [\mathcal{S}, T_t]\mathcal{S}^{-1}, \quad Y_H - X_G = Z + Z^{\dagger G}. \end{aligned} \quad (\text{FC25})$$

The adjoint of Z is $\mathcal{S}^{-1}[T_t^{\dagger G}, \mathcal{S}]$; inserting it proves the last identity with both ordered terms present. The isometry gives $\|Y_H\|_G = \epsilon_H(t)$, so the reverse triangle inequality gives

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq 2\|[\mathcal{S}, T_t]\mathcal{S}^{-1}\|_G. \quad (\text{FC26})$$

The commutator $Y = [\mathcal{S}, T_t]$ satisfies the Sylvester equation $SY + Y\mathcal{S} = [\mathcal{C}, T_t] = [B, T_t]$. Its exact solution is

$$[\mathcal{S}, T_t] = \int_0^\infty e^{-r\mathcal{S}}[B, T_t]e^{-r\mathcal{S}} dr. \quad (\text{FC27})$$

To prove convergence, the original G -operator norm satisfies $\|e^{-r\mathcal{S}}\|_G \leq e^{-r\sqrt{\alpha}}$. Differentiating the integrand and integrating its two terms gives $SY + Y\mathcal{S} = [B, T_t]$, with value $[B, T_t]$ at zero

and limit zero at infinity. For uniqueness, in the spectral decomposition of $\mathcal{S}$, each matrix block of a homogeneous solution is multiplied by the positive sum of its two eigenvalues; all such blocks must vanish. Therefore the integral equals the stated commutator, rather than a separate solution chosen for a bound. Its norm is at most $\|[B, T_t]\|_G/(2\sqrt{\alpha})$, while $\|S^{-1}\|_G \leq \alpha^{-1/2}$. Substitution proves

$$\boxed{|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \alpha^{-1} \|[B, T_t]\|_G,} \quad (\text{FC28})$$

$$[B, T_t] = [B, A] + t[B, R], \quad [B, R]v = (Be_0)\ell(v) - e_0\ell(Bv).$$

The scalar $k/2$ commutes exactly, and both residue terms remain. In particular the fully evaluated number

$$e_{\text{comm}}(t) = \epsilon_H(t) + \|Z + Z^{\dagger G}\|_G \leq \epsilon_H(t) + \alpha^{-1} \|[B, A] + t[B, R]\|_G \quad (\text{FC29})$$

is another valid original-metric control radius in (FC18). Taking the lesser of this number and the radius there remains valid because both have just been proved to bound $\epsilon_G(t)$. If $[B, T_t] = 0$, its G -adjoint gives $[T_t^{\dagger G}, B] = 0$, and uniqueness in (FC27) gives $[S, T_t] = 0$. Equations (FC24)–(FC26) then prove $X_H = X_G$, $Y_H = X_G$ and $\epsilon_H = \epsilon_G$ exactly.

These identities are covariant under every invertible coordinate map $U : E' \rightarrow E$. Indeed $G' = U^*GU$, $H' = U^*HU$, $T'_t = U^{-1}T_tU$ and $B' = U^{-1}BU$ imply $S' = U^{-1}SU$ by positive-square-root uniqueness in G' . Every commutator and integrand above transforms by this same similarity, and every norm is preserved by the stated coordinate map. Thus (FC28) is a comparison of the two original source observations with their full generator; its arithmetic commutator still requires evaluation for any numerical or uniform endpoint application.

69 Complete circle and critical-observation diamond

69.1 The exact diamond of the circle and tensor-critical observations

This calculation uses the original packet h , the original cyclic polynomial $\chi = \chi_{h,k}$, and both actual observation maps. Its proof dependencies are the complete *Periodized theta-source recovery and arithmetic endpoint control*, (P1)–(P64), in `sources/web_periodized_source_delivery/Tau_Periodized_Source_Control/NOTE.tex`, and the complete period/critical-kernel bridge (PC1)–(PC41), in `work/period_critical_kernel_bridge_20260913.tex`. The former is the delivered source of SHA256

acf63a56d5f50cfaa23f57ee52a66df261ec241ce13fd81ccac39a3bbbddd3ff;

the latter's final typeset source has SHA256

7a7733e0d6910398b6de716604a82ef5287a55002e1874327490a70ccc3dd414.

Both complete proofs are retained beside this continuation. The analytic theta section and critical quotient retain the full Hochschild comparison (H1)–(H42) used there. All tensor completions below are the same completed projective tensor products as in (PC32).

69.1.1 The original source, two actual targets, and their inverses

Retain the original nonempty packet h , $k \geq 2$, and $L > 0$ for the positive-weight all-integer circle realization. The final paragraph of this subsection gives the exact $k = 1$ support modification. Write Z for the original selected zero set, with its complete multiplicities m_ρ . Put

$$R = \mathbb{C}[S], \quad E = R/(\chi), \quad q = \deg \chi, \quad A = M_S, \quad e_0 = [1]_\chi,$$

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{m_\lambda}, \quad m_\lambda = \max_{\substack{\rho_1 + \dots + \rho_k = \lambda \\ \rho_i \in Z}} \left(1 + \sum_{i=1}^k (m_{\rho_i} - 1) \right). \quad (\text{DC1})$$

Every tuple is ordered. Equal sums are combined only by their literal equality in $\mathbb{C}$. In the tuple-local algebra $\mathbb{C}[x_1, \dots, x_k]/(x_i^{m_{\rho_i}})$, the highest nonzero power of $x_1 + \dots + x_k$ has degree $D = \sum_i (m_{\rho_i} - 1)$ and coefficient $D! / \prod_i (m_{\rho_i} - 1)!$ on $\prod_i x_i^{m_{\rho_i} - 1}$. All higher powers vanish. Taking the maximum over equal sums proves (DC1), including every exponent. The nonempty original packet gives $q \geq 1$.

Use only tuples in $Z_c = \{\rho \in Z : \operatorname{Re} \rho = 1/2\}$ to define $a = \chi_{c,k}$. At an ambient root λ let r_λ be the same maximum over those all-critical tuples, and set $r_\lambda = 0$ if there is none. Thus

$$a(S) = \prod_{\lambda} (S - \lambda)^{r_\lambda}, \quad 0 \leq r_\lambda \leq m_\lambda, \quad a \mid \chi. \quad (\text{DC2})$$

If Z_c is empty, $a = 1$. This definition retains the full nilpotent length of the all-critical cyclic image.

The actual tensor-critical map is

$$C = \Gamma^{\widehat{\otimes} k} \sigma_h^{\widehat{\otimes} k} \eta_k : E \longrightarrow Q_{\text{crit}}^{\widehat{\otimes} k}, \quad \eta_k = M_{v_h^{\otimes k}} \alpha_k, \quad v_h = j_h((2\xi)/h). \quad (\text{DC3})$$

Its full map type includes $\iota_1 : \mathbb{C}[S]/h(S) \rightarrow E_h = \mathbb{C}[s]/h(s)$, $[P(S)] \mapsto [P(s)]$, and $\bar{\alpha}_k = (\iota_1^{-1})^{\otimes k} \alpha_k$. Consequently $\eta_k = \eta_h^{\otimes k} \bar{\alpha}_k$ with $\eta_h = M_{v_h} \iota_1$. Write $q_{\mathcal{B}} : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V$ for the original quotient map. The original source comparison is still $q_{\mathcal{B}}^{\widehat{\otimes} k} \mathcal{V}_{h,k} = \sigma_h^{\widehat{\otimes} k} \eta_k \pi_\chi$. No arithmetic unit is replaced by 1.

By (PC31)–(PC35), $\ker C = (a)/(\chi)$. Set $W_a = \operatorname{im} C$. The induced map and its actual inverse are

$$\begin{aligned} j_a : R/(a) &\xrightarrow{\sim} W_a, & j_a[P]_a &= C[P]_\chi, \\ j_a^{-1} &= \alpha_{c,k}^{-1} \mathcal{R}_c^{\widehat{\otimes} k} \Big|_{W_a}, & C &= j_a \pi_a. \end{aligned} \quad (\text{DC4})$$

Here $\pi_a : E \rightarrow R/(a)$ is polynomial reduction, and $\alpha_{c,k} : R/(a) \hookrightarrow E_c^{\otimes k}$ is the injection $[P] \mapsto P(A_c^{[k]}) e_c^{\otimes k}$, with e_c the identity of the original critical CRT ideal. The inverse in (DC4) has domain the image of this injection, because (PC33) proves $\mathcal{R}_c^{\widehat{\otimes} k} C = \alpha_{c,k} \pi_a$. In each one-factor critical coordinate,

$$\Xi_\rho[f] = \sum_{j=0}^{m_\rho-1} \frac{i^{-j}}{j!} f^{(j)}(\operatorname{Im} \rho) x^j, \quad \mathcal{R}_c = M_{\varepsilon_c} \Xi_c, \quad \varepsilon_c v_c = e_c. \quad (\text{DC5})$$

Thus (DC4) retains the full tensor product of the factors $i^{-j}/j!$ and the recursively inverted original local units. It is an isomorphism of finite-dimensional R -modules. The action on W_a is the restriction M_a of the original sum multiplier $M^{[k]}$, and $M_a j_a = j_a M_S$. No algebra multiplication on the ambient critical quotient is needed.

For the circle retain the original frequencies, scalar measure, and vector amplitude:

$$\begin{aligned} S_n &= k/2 + 2\pi i n/L, & \nu_n &= (2\pi/L) m_{h,k} (2\pi n/L) > 0, \\ b = p_L &= \prod_{n \in \mathcal{Z}_L} (S - S_n), & \mathcal{Z}_L &= \{n \in \mathbb{Z} : \chi(S_n) = 0\}, \\ \beta_n &= \widehat{\mathcal{U}_k F_h^{\otimes k}}(2\pi n/L), & \|\beta_n\|_{\mathcal{K}}^2 &= 2\pi m_{h,k} (2\pi n/L). \end{aligned} \quad (\text{DC6})$$

The frequencies are distinct, so b is squarefree and divides χ ; an empty product is 1. In the actual circle Fourier basis $L^{-1/2} e^{-2\pi i n r/L}$, the polynomial source has coefficient $L^{-1/2} \beta_n P(S_n)$. The scalar-to-vector map $c \mapsto (L^{-1/2} \beta_n c_n)_n$ has squared norm $\sum_n \nu_n |c_n|^2$, proving the isometry with the weighted cyclic space without rescaling any β_n or ν_n .

For completeness, the Hilbert quotient in (P56) is exactly

$$W_b = \ell^2(\mathbb{Z}, \nu) / \overline{\{(\chi(S_n) P(S_n))_n : P \in R\}} \xrightarrow{\sim} \bigoplus_{n \in \mathcal{Z}_L} \mathbb{C}, \quad \|v\|_{W_b}^2 = \sum_{n \in \mathcal{Z}_L} \nu_n |v_n|^2. \quad (\text{DC7})$$

Indeed the relation closure vanishes on $\mathcal{Z}_L$. For the reverse inclusion divide a vector vanishing there by $\chi(S_n)$ off that finite set; the result belongs to $\ell^2(|\chi|^2\nu)$. Polynomial density in this space follows from the original exponential moment (P54): an orthogonal vector defines a finite complex measure with analytic Fourier transform in a strip, whose derivatives at zero all vanish. The power series, the identity theorem, and Fourier uniqueness force the vector to vanish. Multiplying the approximating polynomials by χ proves the reverse inclusion. This also proves the stated isometric quotient. All finite vectors in (DC7) are attained by polynomial interpolation. The actual observation, its injection, and inverse are therefore

$$\begin{aligned} \theta = \theta_L : E &\rightarrow W_b, \quad \theta[P]_\chi = (P(S_n))_{n \in \mathcal{Z}_L}, \quad \ker \theta = (b)/(\chi), \\ j_b : R/(b) &\xrightarrow{\sim} W_b, \quad j_b[P]_b = (P(S_n))_{n \in \mathcal{Z}_L}, \quad \theta = j_b \pi_b, \\ j_b^{-1}(v) &= \left[\sum_{n \in \mathcal{Z}_L} v_n \prod_{\substack{m \in \mathcal{Z}_L \\ m \neq n}} \frac{S - S_m}{S_n - S_m} \right]_b. \end{aligned} \quad (\text{DC8})$$

Every denominator is nonzero. The action M_b on W_b is $(v_n) \mapsto (S_n v_n)$, so $M_b j_b = j_b M_S$ and $\theta A = M_b \theta$. The class $j_b[1]_b$ has squared norm $\sum_{n \in \mathcal{Z}_L} \nu_n$, which is positive exactly when $\mathcal{Z}_L$ is nonempty and is zero otherwise. This actual number has not been assigned the value one. For $k = 1$ the identical proof uses only the actual support $I_L = \{n : w_h(2\pi n/L) > 0\}$ and $\mathcal{Z}_L = \{n \in I_L : \chi(S_n) = 0\}$ in (DC6)–(DC8). All formulas below use that support-specific b and remain exact. For the actual full-order packet, this restriction omits no sampled root of h . At such a root ρ write $g(s) = (s - \rho)^{m_\rho} g_\rho(s)$ and $h(s) = (s - \rho)^{m_\rho} h_\rho(s)$, where $g_\rho(\rho) h_\rho(\rho) \neq 0$ by the original complete-order hypothesis. Hence $v_h(\rho) = g_\rho(\rho)/h_\rho(\rho) \neq 0$. For $\rho = 1/2 + 2\pi i n/L$ the actual weight is $\nu_n = |v_h(\rho)|^2/L > 0$, by $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$. Thus for $k = 1$ the polynomial b is also exactly the product over all sampled roots of $\chi = h$, even if some other lattice atoms have zero weight. No positivity of those other atoms is used.

69.1.2 The common quotient and the actual joint image

For any monic divisor $s \mid \chi$, denote the reduction by $\pi_s : E \rightarrow E_s = R/(s)$ and its kernel by $K_s = (s)/(\chi)$. In particular $E_1 = 0$, $K_1 = E$, $E_\chi = E$, and $K_\chi = 0$. Put

$$d = \gcd(a, b), \quad l = \text{lcm}(a, b) = ab/d, \quad d \mid a, b \mid l \mid \chi. \quad (\text{DC9})$$

All gcds and lcms are monic. Define the maps into the common quotient E_d by their full types:

$$f_a : W_a \rightarrow E_d, \quad f_a = \pi_d^a j_a^{-1}, \quad f_b : W_b \rightarrow E_d, \quad f_b = \pi_d^b j_b^{-1}, \quad f_a C = f_b \theta = \pi_d. \quad (\text{DC10})$$

Here $\pi_d^a : E_a \rightarrow E_d$ and $\pi_d^b : E_b \rightarrow E_d$ are reduction. Both maps are onto, commute with S , and are continuous on these finite-dimensional spaces. They give the actual fibre product

$$W_\vee = \{(x, y) \in W_a \oplus W_b : f_a x = f_b y\}, \quad J = (C, \theta) : E \rightarrow W_\vee. \quad (\text{DC11})$$

Thus the compatibility condition is expressed through the exact weighted inverse (DC4) and original-circle inverse (DC8).

There is an explicit isomorphism

$$\begin{aligned} j_l : E_l &\xrightarrow{\sim} W_\vee, \quad j_l[P]_l = (j_a[P]_a, j_b[P]_b), \quad J = j_l \pi_l, \\ \ker J &= K_l, \quad \dim W_\vee = \deg l = \deg a + \deg b - \deg d. \end{aligned} \quad (\text{DC12})$$

Well-definedness follows from $a, b \mid l$. If both components vanish, a and b divide P ; their prime-power factorizations show that l divides P . This proves injectivity. To prove surjectivity with an

explicit inverse, write $a = da'$, $b = db'$ and compute polynomials u, v with $ua' + vb' = 1$. Given compatible representatives X modulo a and Y modulo b , $Y - X = dT$ for some $T \in R$. Then

$$P = X + auT, \quad P - X = auT, \quad P - Y = -bvT. \quad (\text{DC13})$$

It has the required two residues. A different choice of representatives, Bezout coefficients, or T changes P by a multiple of both a and b , hence of l . This proves the well-defined inverse and the entire isomorphism. The Euclidean algorithm terminates because its successive nonzero remainders have strictly decreasing degrees, and its back substitution gives u, v . The same proof covers $a = 1$, $b = 1$, or $d = 1$.

The complete exact quotient sequence is

$$0 \longrightarrow E_l \xrightarrow{[P] \mapsto ([P]_a, [P]_b)} E_a \oplus E_b \xrightarrow{(x, y) \mapsto \pi_d^a x - \pi_d^b y} E_d \longrightarrow 0. \quad (\text{DC14})$$

The difference map is onto because its first component is onto; its kernel and first-map injectivity were just proved. Transport by j_a, j_b, j_l proves the identical sequence for the actual images. In parallel, the kernel diamond has

$$\begin{aligned} K_a \cap K_b &= K_l, & K_a + K_b &= K_d, \\ 0 \longrightarrow K_l &\xrightarrow{z \mapsto (z, -z)} K_a \oplus K_b \xrightarrow{(x, y) \mapsto x + y} K_d \longrightarrow 0. \end{aligned} \quad (\text{DC15})$$

The intersection follows from the same divisibility argument. Bezout applied to a, b gives $d = ua + vb$, which proves the equality of the sum with K_d . The displayed signs then prove exactness. Every map in (DC14)–(DC15) commutes with multiplication by S .

In particular $E_d = E/(K_a + K_b)$ is the largest common quotient of the original two observations in the following exact sense. For any vector space T and linear maps $g_a : W_a \rightarrow T$, $g_b : W_b \rightarrow T$ with $g_a C = g_b \theta$, their common composite vanishes on $K_a + K_b$. It therefore has a unique factor $g_d : E_d \rightarrow T$, defined by $g_d[P]_d = g_a C[P]_\chi$. It is well defined by that vanishing and satisfies $g_a = g_d f_a$, $g_b = g_d f_b$ by surjectivity. If the given maps intertwine specified actions, so does g_d , by applying the surjective map π_d . This is the common-quotient assertion with its domain and universality fully specified.

69.1.3 Every local length and both exact factorization criteria

At a root λ of χ , use $x = S - \lambda$ and retain

$$\begin{aligned} \chi &= x^{m_\lambda} c_\lambda(x), & a &= x^{r_\lambda} a_\lambda(x), & b &= x^{s_\lambda} b_\lambda(x), \\ c_\lambda(0) a_\lambda(0) b_\lambda(0) &\neq 0, & s_\lambda &\in \{0, 1\}, \\ d &= x^{\min(r_\lambda, s_\lambda)} d_\lambda(x), & l &= x^{\max(r_\lambda, s_\lambda)} l_\lambda(x), & d_\lambda(0) l_\lambda(0) &\neq 0. \end{aligned} \quad (\text{DC16})$$

These are the literal polynomial cofactors at λ , with no chosen constant rescaling. The source local ring is $R_\lambda = \mathbb{C}[x]/(x^{m_\lambda})$. For $s \in \{a, b, d, l\}$, write its exponent as $e_\lambda(s)$ and its cofactor as $s_\lambda^{\text{unit}}(x)$. The exact kernel inclusion is

$$\mathbb{C}[x]/(x^{m_\lambda - e_\lambda(s)}) \longrightarrow R_\lambda, \quad [B] \longmapsto [x^{e_\lambda(s)} s_\lambda^{\text{unit}}(x) B]. \quad (\text{DC17})$$

It is injective and its image is precisely the local component of K_s . Indeed the unit is invertible by recursive coefficient division; multiplying by that inverse and cancelling the stated power of x proves injectivity. Its image is the principal ideal by definition. The quotient is the reduction $R_\lambda \rightarrow \mathbb{C}[x]/(x^{e_\lambda(s)})$. All these maps intertwine the actual action $\lambda + x$. Thus the four quotient lengths are respectively $r_\lambda, s_\lambda, \min(r_\lambda, s_\lambda)$, and $\max(r_\lambda, s_\lambda)$, and the four kernel lengths are m_λ minus those lengths. No primary component or nilpotent jet is discarded from the source.

There is a map of the actual images $T_{ab} : W_a \rightarrow W_b$ with $\theta = T_{ab}C$ if and only if $b \mid a$. In that case it is unique and its complete formula is

$$T_{ab} = j_b \pi_b^a j_a^{-1}, \quad M_b T_{ab} = T_{ab} M_a. \quad (\text{DC18})$$

Necessity follows by applying the assumed factorization to K_a : it gives $K_a \subset K_b$. Since the class of a is in K_a , this means $a = bP + \chi Q$ for some P, Q , and $b \mid \chi$ then gives $b \mid a$. Sufficiency is the displayed reduction map; uniqueness follows because C is onto W_a . Intertwining follows from the corresponding property of every factor. The reverse factorization exists if and only if $a \mid b$, and is exactly

$$T_{ba} = j_a \pi_a^b j_b^{-1}, \quad C = T_{ba} \theta. \quad (\text{DC19})$$

The same proof establishes it even if an image is zero. Consequently the two observations have mutually inverse factorizations over E if and only if $a = b$. Since b is squarefree, (DC19) requires every critical cyclic exponent to be at most one and every such root to be actually sampled. Formula (DC18) instead requires each sampled root to occur in a ; it places no upper bound on its critical multiplicity.

Their extra observations are calculated by the original maps:

$$\begin{aligned} \theta(K_a) &\simeq K_a/K_l \simeq [a]_b E_b, & \dim \theta(K_a) &= \deg b - \deg d, \\ C(K_b) &\simeq K_b/K_l \simeq [b]_a E_a, & \dim C(K_b) &= \deg a - \deg d. \end{aligned} \quad (\text{DC20})$$

Here $[a]_b E_b$ is the ideal generated by the residue of a in $R/(b)$, and $[b]_a E_a$ is the ideal generated by the residue of b in $R/(a)$. Explicitly the first isomorphism sends the class of aP modulo K_l to $j_b[aP]_b$. Its kernel is exactly $K_a \cap K_b$, and all values in that ideal are attained. The ideal in $R/(b)$ is also the ideal generated by d , by Bezout and reduction modulo b , so its dimension is $\deg b - \deg d$. Interchanging a, b proves the second line. Alternatively (DC17) calculates the local lengths as $s_\lambda - \min(r_\lambda, s_\lambda)$ and $r_\lambda - \min(r_\lambda, s_\lambda)$.

This gives a concrete criterion for a circle value of a class killed by C . It occurs exactly at a sampled λ with $s_\lambda = 1$ and $r_\lambda = 0$. Let $e_\lambda \in E$ be the CRT idempotent equal to 1 in the λ -primary ring and 0 in every other primary ring. Polynomial Bezout produces this idempotent in the original E . Then

$$C e_\lambda = 0, \quad \theta e_\lambda = (\delta_{\lambda, S_n})_{n \in \mathcal{Z}_L}, \quad \|\theta e_\lambda\|^2 = \nu_n \quad (S_n = \lambda). \quad (\text{DC21})$$

Such a λ has real part $k/2$ because it is sampled. Its absence from a means precisely that no all-critical tuple has this sum; the ambient tuples realizing it have at least one off-critical member. The statement is a criterion and explicit map for actual roots of the retained packet; it asserts no existence of an off-critical zero. When $r_\lambda \geq 1$ and $s_\lambda = 1$, the circle sees the constant critical jet and all $r_\lambda - 1$ higher critical jets lie in $C(K_b)$. If mixed tuples raise m_λ above r_λ , the additional $m_\lambda - r_\lambda$ source jets remain in K_a and K_l . For an unsampled root every surviving critical jet lies in $C(K_b)$. These are all local possibilities, directly from (DC16)–(DC20).

69.1.4 The same period transport and all four action defects

Keep the period matrix $\Pi(t) = \Pi_k(t)$ of the original polynomial χ from (PC37), with its original ordered rays, $u \in \mathbb{C}^*$, and increasing-power basis. In particular its entries are the integrals of $S^j dS$ against $\exp((S^{q+1}/(q+1) + \sum_{\ell=0}^{q-1} c_\ell S^{\ell+1}/(\ell+1) - tS)/u)$ on those rays. It is invertible for every t , including repeated-root parameters, by the complete fixed-ray and determinant proof (PC16)–(PC18). Retain

$$\ell[P]_\chi = [S^{q-1}] \text{rem}_\chi P, \quad R_0 = e_0 \ell, \quad A_t = A + t R_0, \quad \Pi' = -\Pi A_t / u, \quad A_t^{\text{per}} = \Pi A_t \Pi^{-1}. \quad (\text{DC22})$$

Here R_0 is an endomorphism and is distinct from the polynomial ring R used above. The following indexed families use four formal roles $\mathcal{I} = \{\text{critical, circle, common, joint}\}$ with associated polynomials $(s_i)_{i \in \mathcal{I}} = (a, b, d, l)$. Subscripts a, b, d, l on the actual targets and observations below are short-hand for these four roles. In K_s , $\deg s$, and polynomial reduction, s means the associated polynomial. Thus even if two or all four polynomials coincide, their four actual target realizations and maps remain separately specified. Set $W_d = E_d$, $j_d = I$, $T_d = \pi_d$, $T_a = C$, $T_b = \theta$, $T_l = J$, and $W_l = W_\vee$. Let M_s be multiplication by S on W_s through j_s . For $s = l$ it is explicitly $(M_a, M_b)|_{W_\vee}$; this restriction is well defined by (DC10). For all four indices, $T_s = j_s \pi_s$, $T_s A = M_s T_s$, and $\ker T_s = K_s$.

Define the four actual period observations on the same target coordinate space $\mathbb{C}^q$ by

$$T_{s,t} = T_s \Pi^{-1}, \quad \mathcal{K}_{s,t} = \ker T_{s,t} = \Pi K_s, \quad s \in \{a, b, d, l\}. \quad (\text{DC23})$$

They are onto their fixed images, with holomorphic kernels and constant ranks $\deg s$. If j_s^{-1} is followed by any polynomial representative of degree less than $\deg s$, its composition with Π is an explicit holomorphic right inverse of $T_{s,t}$. The diagram remains exactly the same:

$$f_a T_{a,t} = f_b T_{b,t} = T_{d,t}, \quad T_{l,t} = (T_{a,t}, T_{b,t}), \quad \mathcal{K}_{a,t} \cap \mathcal{K}_{b,t} = \mathcal{K}_{l,t}, \quad \mathcal{K}_{a,t} + \mathcal{K}_{b,t} = \mathcal{K}_{d,t}. \quad (\text{DC24})$$

Every equality follows by composing its previously proved source equality with the invertible map Π^{-1} . For the transported original action $B_t = \Pi A \Pi^{-1}$ this is an exact diagram of R -modules: $T_{s,t} B_t = M_s T_{s,t}$. Indeed multiplication cancels the two middle factors $\Pi^{-1} \Pi$ and gives the original identity $T_s A = M_s T_s$. The deformed action A_t^{per} instead has the following fully calculated defects.

Before restricting any domain, the complete action defect is

$$\boxed{T_{s,t} A_t^{\text{per}} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}}. \quad (\text{DC25})$$

Indeed the first term is $T_s(A + t e_0 \ell) \Pi^{-1}$; the A part is $M_s T_s \Pi^{-1}$ by the original intertwining. This proves the formula without asserting invariance of a deformed kernel. The weights in its four target vectors are explicit:

$$T_a(e_0) = C(e_0), \quad T_b(e_0) = (1)_{n \in \mathcal{Z}_L}, \quad T_d(e_0) = [1]_d, \quad T_l(e_0) = (C(e_0), (1)_{n \in \mathcal{Z}_L}). \quad (\text{DC26})$$

The first vector has the raw jet recovery $(M_{v_c})^{\otimes k} \alpha_{c,k} [1]_a$ before applying the inverse units; the second has its original norm from (DC8). If $\deg s > 0$, $T_s(e_0) \neq 0$ because j_s is injective. If $\deg s = 0$, its target and every displayed defect are zero.

For any monic $v \mid \chi$ with $\deg v < q$, the ideal K_v has the exact basis $[v], [Sv], \dots, [S^{q-\deg v-1}v]$. Monicity gives different leading degrees, proving independence; division by χ/v proves spanning. Its last member has degree $q-1$ and leading coefficient 1, so $\ell|_{K_v}$ is onto. Consequently, for $t \neq 0$, $\deg s > 0$, and $\deg v < q$, restriction of the s -defect to $\mathcal{K}_{v,t}$ has

$$\begin{aligned} \text{im } (T_{s,t} A_t^{\text{per}} - M_s T_{s,t})|_{\mathcal{K}_{v,t}} &= \mathbb{C} T_s(e_0), \\ \ker (T_{s,t} A_t^{\text{per}} - M_s T_{s,t})|_{\mathcal{K}_{v,t}} &= \Pi(K_v \cap \ker \ell), \quad \text{rank} = 1. \end{aligned} \quad (\text{DC27})$$

For $v = \chi$ the restricted domain is zero; for $t = 0$ or $s = 1$ the whole restricted defect is zero. In particular the self-kernel defects have rank one precisely when $t \neq 0$ and $0 < \deg s < q$. Formula (DC27) also calculates the critical defect on the circle kernel and the circle defect on the critical kernel, with their full domains. The observations themselves on those crossed kernels retain (DC20): $T_{b,t}(\mathcal{K}_{a,t}) = \theta(K_a)$ and $T_{a,t}(\mathcal{K}_{b,t}) = C(K_b)$.

The factorizations (DC18)–(DC19), whenever their proved divisibility criterion holds, remain unchanged after period transport. Conversely a factorization at even one parameter t can be composed with $\Pi(t)$ and gives the original factorization, so its same divisibility criterion is necessary. The maps f_a, f_b carry the first two defects to the common defect, and the joint defect has the two original defects as components, because $f_a C(e_0) = f_b \theta(e_0) = [1]_d$. Thus both the observations and their entire failure of deformed-action intertwining commute with the exact diamond.

69.1.5 The exact extra circle summand

Squarefreeness of the actual circle polynomial gives a more explicit joint-image formula. Put $b_{\text{new}} = b/d$. Its roots are exactly the sampled roots absent from a , so $\gcd(a, b_{\text{new}}) = 1$ and $l = ab_{\text{new}}$. Denote by W_{new} the actual circle coordinate subspace on these roots, with its inherited metric $\sum_{S_n: b_{\text{new}}(S_n)=0} \nu_n |v_n|^2$ and diagonal action S_n . There is an exact isomorphism

$$\begin{aligned} W_V &\xrightarrow{\sim} W_a \oplus W_{\text{new}}, & (x, y) &\longmapsto (x, y_{\text{new}}), \\ (x, z) &\longmapsto (x, y), & y_n &= \begin{cases} (j_a^{-1}x)(S_n), & d(S_n) = 0, \\ z_n, & b_{\text{new}}(S_n) = 0. \end{cases} \end{aligned} \quad (\text{DC28})$$

Evaluation in the first case is independent of a polynomial representative because $a(S_n) = 0$. It gives exactly the shared reduction in (DC10); the two cases exhaust the sampled roots. Thus the displayed maps are mutually inverse. They intertwine the original actions, since evaluation of SP gives $S_n P(S_n)$. This is the actual-image transport of the CRT isomorphism $R/(l) \simeq R/(a) \oplus R/(b_{\text{new}})$. Its first factor retains the original unit-weighted critical map and all of its nilpotent jets; its second factor retains the actual new circle amplitudes and weights. It asserts an isomorphism of finite-dimensional R -modules. If one assigns the actual circle norm to the second component of $W_V \subset W_a \oplus W_b$, that norm under (DC28) is exactly $\sum_{d(S_n)=0} \nu_n |(j_a^{-1}x)(S_n)|^2 + \sum_{b_{\text{new}}(S_n)=0} \nu_n |z_n|^2$. The shared first sum remains present, so the module isomorphism has not silently been asserted to preserve a chosen direct-sum metric on $W_a \oplus W_{\text{new}}$.

The new circle coordinates have an explicit original polynomial source lift. For a prescribed vector $z = (z_n)_{b_{\text{new}}(S_n)=0}$, set

$$P_z(S) = a(S) \sum_{b_{\text{new}}(S_n)=0} \frac{z_n}{a(S_n)} \prod_{\substack{b_{\text{new}}(S_m)=0 \\ m \neq n}} \frac{S - S_m}{S_n - S_m}. \quad (\text{DC29})$$

Each $a(S_n)$ and each $S_n - S_m$ in a denominator is nonzero, by the proved coprimality and distinct sampled frequencies. The factor a puts $[P_z]_\chi$ in K_a . At a new sample the interpolation product gives exactly z_n ; at a shared sample the factor a gives zero. Consequently $C[P_z]_\chi = 0$ and $\theta[P_z]_\chi = (0, z)$, with squared circle norm $\sum \nu_n |z_n|^2$ on precisely the new samples. At source level the lift is the unchanged $\mathcal{V}_{h,k} P_z$, whose cohomology class is $\sigma_h^{\otimes k} \eta_k [P_z]_\chi$. Thus the circle summand in (DC28) has an actual unit-weighted arithmetic source, including every coefficient $a(S_n)^{-1}$. When there are no new samples, the sum and its lift are zero. For $k = 1$, every sampled root of $\chi = h$ is itself critical, so $b \mid a$ and this extra circle summand is always zero.

The differential equation of all four transported observations retains the same forcing as (DC25):

$$u \partial_t T_{s,t} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}, \quad s \in \{a, b, d, l\}. \quad (\text{DC30})$$

Differentiating $\Pi^{-1} \Pi = I$ and using (DC22) gives $(\Pi^{-1})' = A_t \Pi^{-1}/u$. Multiplication by $u T_s$ and the original relation $T_s A = M_s T_s$ prove (DC30). The fixed maps f_a, f_b carry these differential equations to the common one by (DC10); the joint equation is their paired equation. The original factors u, t , and the full four vectors in (DC26) remain in both formulas.

The entire diamond now also has its exact singular u equation. Use the division (XD.u1) for the same cyclic χ , and write $C_\chi(t), B_\chi$ in its unchanged increasing-power basis. On every retained contour branch let $T_{s,u,t} = T_s \Pi(u, t)^{-1}$. For all four formal roles,

$$\partial_u T_{s,u,t} = T_s (C_\chi(t)/u^2 - B_\chi/u) \Pi(u, t)^{-1}, \quad s \in \{a, b, d, l\}. \quad (\text{DC31})$$

This follows from $\partial_u \Pi^{-1} = -\Omega_u \Pi^{-1}$ and the original constant maps T_s , with $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$. The same composition gives kernels $\Pi(u, t) K_s$ and right inverses from (DC23), so all inter-sections, sums and exact quotients in (DC24) remain valid in both parameters. Multiplying (DC31)

by f_a, f_b gives the common equation because $f_a T_a = f_b T_b = T_d$; the joint equation has the same two components. The factorization maps (DC18)–(DC19) are parameter-independent and commute with this derivative for the same reason. This proves the full differential diamond.

The actual circle weights of (DC6)–(DC8) are unchanged. Let $D_\nu = \text{diag}(\nu_n)_{n \in \mathbb{Z}_L}$ in the stated circle coordinate basis, including an empty matrix for an empty sample set. Its exact pulled-back positive semidefinite form on the period coordinates is

$$\begin{aligned} H_b^{\text{obs}} &= \Pi^{-*} T_b^* D_\nu T_b \Pi^{-1}, \quad \ker H_b^{\text{obs}} = \Pi K_b, \\ \partial_u H_b^{\text{obs}} &= -\Pi^{-*} T_b^* D_\nu T_b \Omega_u \Pi^{-1}, \\ \partial_u H_b^{\text{obs}} &= -\Pi^{-*} \Omega_u^* T_b^* D_\nu T_b \Pi^{-1}. \end{aligned} \tag{DC32}$$

The kernel identity follows since all retained ν_n are strictly positive, so zero norm is precisely $T_b \Pi^{-1} x = 0$. The derivatives follow by the same inverse-matrix product rule, keeping the original weights in every term. Under (DC28), the shared and new circle sums therefore still use exactly those same ν_n ; the map itself is constant in u, t . The actual polynomial lift (DC29), all inverse-unit factors and the receiving support labels remain unchanged.

All coefficient arrows above have their original supported Split-Zero lifts: $v^\bullet \mapsto (Tv)^\bullet$ and $\tau \mapsto \tau$, with the specified support retained. The kernel inclusions (DC15), local maps (DC17), and period kernels (DC23) identify exactly the represented classes sent to the receiving supported zero. Projection to ordinary coefficients commutes with every displayed arrow. The common quotient and joint image therefore preserve the same support data while recording these different, explicitly calculated coefficient kernels.

70 Residue constituent source and exact derivative maps

70.1 Current reading of the delivered residue-constituent note

The complete delivered note, equations (RC1)–(RC53), is retained unchanged. The following statements accompany that historical source in the current integrated edition.

First, the translation source-byte comparison has already been resolved. The dated translation seal proves that the reviewed source becomes the final source by the single geometry change from a 25-millimetre margin to a 23-millimetre margin. Reversing that exact byte-string substitution reproduces the reviewed SHA-256 value. The later README records the completed delivery. The historical final paragraph of the delivered note therefore records an intake question that the existing seal answers.

Second, the precise derivative convention in (RC28) is the real axis $t = x \in \mathbb{R}$, with the displayed real $u > 0$. For $Y = \Pi I$ and $H_F = Y^* Y$, the period equation gives

$$Y'(0) = -Y(0) A_F / u, \quad \left. \frac{d}{dx} H_F(x) \right|_0 = -(A_F^* H_F(0) + H_F(0) A_F) / u.$$

Taking its logarithmic determinant gives $-2 \operatorname{Re} \operatorname{Tr} A_F / u$, precisely (RC28). The corresponding Wirtinger derivative is $-\operatorname{Tr} A_F / u$. The original fixed denominator $\det G_{N,F}$ contributes zero to both derivatives.

Third, (RC27) has the correct mixed fourth derivative and factorial. Its exact relation to real-axis differentiation is (RCX26)–(RCX32):

$$\psi_{t\bar{t}\bar{t}\bar{t}}(0) = \mathfrak{c}_F(H(0)) / |u|^2, \quad \left. \frac{d^4}{dx^4} \psi(x) \right|_0 = 2 \operatorname{Re} \phi^{(4)}(0) + 6 \mathfrak{c}_F(H(0)) / |u|^2.$$

Here $\phi(t) = \log \det(I^* \Pi(0)^* \Pi(t) I)$, and its full fourth derivative is evaluated by the displayed non-commutative period recursion in the companion. The coefficient of $t^2 \bar{t}^2$ is the mixed derivative divided by $2!2! = 4$.

Fourth, the source equation (RC41) is evaluated on the admitted strong-Schwartz tensor domain stable under the original $D^{(k)}$ and its square. The exact lift $\widehat{R} = r_N R j_E$ satisfies $j_E \widehat{R} = R j_E$ and $\widehat{R} d = 0$. The complete primitive relating the quadratic actions is

$$((D^{(k)} + t\widehat{R})^2 - k(D^{(k)} + t\widehat{R}))r_N - r_N(A_t^2 - kA_t) = d(D_0K + KA_t - kK),$$

where $D^{(k)}r_N - r_NA = dK$ and $D^{(k)}d = dD_0$. Every term has the original source and primitive target.

The shared general rank and curvature proofs were already integrated as (SC9)–(SC25); the source comparison in (RC32)–(RC33) is also (LC14)–(LC15). The new companion adds the universal endomorphism inverse, the exact metric differential and sharpened endpoint estimate, the complete higher period jets, and the forced Laplacian map. The delivered twelve-method suite has been executed in normal and optimized modes with six substantive negative-control failures. Its translation method checks the full q -dimensional trace shift; the constituent p -dimensional formula is proved by the full frame transport in (RCX36)–(RCX39).

71 Complete original residue constituent curvature note

Source role. complete original readable source ONCE, RC1–53; retain all original labels in a dedicated original-note namespace The complete original source is reproduced once. The preceding public wrapper records the exact current reading. Original display tags retain their literal tokens with local namespace Residue.

Exact revision. 9840f5e979dc2d8c38fe52f804c85fc7664a4eb33676db39706aaebcd9c0ae3b

Complete source. sources/web_residue_constituent_delivery/Tau_Residue_Constituent_Curvature/NOTE.md

Source SHA-256. 9840f5e979dc2d8c38fe52f804c85fc7664a4eb33676db39706aaebcd9c0ae3b

Source metadata: author (MetaInlines)

Research continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

Exact rank-one transport, a source-sensitive curvature coefficient, and the retained period correction

Source metadata: title (MetaInlines)

Residue-driven constituent curvature and the original arithmetic Laplacian

71.1 Scope and source identifications

This note continues the supplied *Deligne Split Cohomology* sidebar (DS1–69), *Exponential Period Determinant* (XD1–26), *Exponential Translation Comparison* (DT1–13), and the pasted SGA trace/period continuation. It uses their explicit coefficient and period maps. The additional results proved here concern the rank-one deformation on every nonzero proper invariant constituent, its exact period curvature, the change of that coefficient between the original source metrics, and the second-order realization of the arithmetic Laplacian. These are written proofs, supplemented by the included exact finite checker. They are not new Lean certificates or a proof of a uniform arithmetic endpoint bound.

Keep the original nonempty cyclic object, including its full multiplicities,

$$E = \mathbb{C}[S]/(\chi), \quad \chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad A = M_S. \quad (\text{Residue RC1})$$

The coordinate basis is $1, S, \dots, S^{q-1}$. When $E = E_{h,k}$ is the arithmetic packet, the source retains $g = 2\xi$, $\mathcal{M}F_h = g/h$, the full arithmetic Taylor unit in η , and the original source/relation norm. In particular, retain the specified maps on the admitted strong-Schwartz tensor domain $\mathcal{X}$:

$$r_N : E \longrightarrow \mathcal{X}, \quad j_E : \mathcal{X} \longrightarrow E, \quad j_E r_N = I_E, \quad j_E D^{(k)} = A j_E, \quad j_E d = 0, \quad G_N = r_N^* r_N \succ 0. \quad (\text{Residue RC2})$$

Here d is the original cochain differential and $D^{(k)}$ the original sum of scaling generators. Equation (RC2) is the supplied source comparison, not an assertion that the full-jet observation extends boundedly to the unweighted Hilbert completion (see *Exponential Intake*, §§4–5).

There are three distinct parameters. S is the arithmetic spectral coordinate; z is the earlier representative-curve parameter; t below is the exponential deformation parameter. The metric curve of DS13–14 is $G(z) = G_j + |z|^2(G_i - G_j)$. None of its radial derivatives is substituted for an exponential t -derivative.

For fixed $u \in \mathbb{C}^\times$, the supplied polynomial complex is

$$\mathcal{C}_{u,t} = [\mathbb{C}[S] \xrightarrow{\mathcal{D}_{u,t}} \mathbb{C}[S]dS], \quad \mathcal{D}_{u,t}P = (uP' + (\chi - t)P)dS. \quad (\text{Residue RC3})$$

Its degree-one cohomology is a coefficient-vector-space quotient with the specified degree-below- q frame. Multiplication by S does not descend as an algebra operation on this general fibre: the supplied identities $[\chi - t] = 0$ and $[S(\chi - t)] = -u[1]$ rule out that interpretation. The parameter connection does descend, and in the retained frame it is

$$\nabla_t = \partial_t - \frac{A(t)}{u}, \quad A(t) = A + t\mathcal{R}, \quad \mathcal{R} = e_0 e_{q-1}^T. \quad (\text{Residue RC4})$$

At $u = t = 0$ the complex specializes to the original quotient by χ . The regular operator $-u\nabla_t$ induces A on that special fibre; this is the type-correct specialization in DS68.

The fixed oriented contours of XD2 give a period matrix $\Pi(t) = \Pi(u, \mathbf{c}, t)$ satisfying

$$\Pi'(t) = -\frac{1}{u}\Pi(t)A(t), \quad \det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\mathcal{F}(\mathbf{c}, t)}{u}\right), \quad \mathcal{F} = \text{Tr } \Phi_t(A(t)). \quad (\text{Residue RC5})$$

XD4–7 supply the uniform contour estimates that justify holomorphy and differentiation. XD16–20 supply the nonzero monomial initial determinant and an adjugate proof valid even at a hypothetical singular matrix. Thus $\Pi(t)$ is invertible for every complex t , without deleting repeated-root fibres. The gamma multiplication formula simplifies XD18 to

$$\det(\Pi(t)^* \Pi(t)) = (2\pi|u|)^q \exp\left(2 \operatorname{Re} \frac{\mathcal{F}(\mathbf{c}, t)}{u}\right). \quad (\text{Residue RC6})$$

This last simplification is already in the supplied SGA continuation. It is not a new result of the present note. Bloch and Esnault (1999, §5) study period determinants of this polynomial-exponential type; the literal normalizations used here are the ones proved in XD1–26.

71.2 The deformation is the residue rank-one map

Define the original residue functional and its bilinear pairing by

$$\ell([P]) = [S^{q-1}] \operatorname{rem}_\chi P, \quad \beta(x, y) = \ell(xy). \quad (\text{Residue RC7})$$

In the power basis, $\beta(S^i, S^j)$ is zero for $i + j < q - 1$ and one for $i + j = q - 1$. Reversing its columns therefore produces a triangular matrix with diagonal one. Consequently

$$\det \beta = (-1)^{q(q-1)/2}, \quad \beta^b : E \xrightarrow{\sim} E^\vee. \quad (\text{Residue RC8})$$

This proof inverts no discriminant. It is the perfect residue pairing in the supplied finite-flat SGA calculation. Its contraction with multiplication by χ' gives the separate, potentially degenerate trace pairing:

$$\operatorname{Tr}_E M_P = \ell(\chi' P). \quad (\text{Residue RC9})$$

For completeness, the divided difference $\mathcal{C}_\chi(X, Y) = (\chi(X) - \chi(Y))/(X - Y)$ is the coevaluation tensor for (RC8): $(\ell \otimes 1)((P \otimes 1)\mathcal{C}_\chi) = P$ and its image under multiplication is χ' . The finite-dimensional evaluation-coevaluation trace then proves (RC9). Neither (RC8) nor (RC9) identifies the bilinear residue form with the positive Hermitian source form G_N .

In the fixed remainder frame the perturbation in (RC4) is exactly

$$\boxed{\mathcal{R}(v) = \ell(v) 1_E, \quad \mathcal{R} = 1_E \otimes \ell.} \quad (\text{Residue RC10})$$

It follows that the pair $(A, \mathcal{R})$ generates every coefficient endomorphism:

$$\boxed{\mathbb{C}\langle A, \mathcal{R} \rangle = \text{End}_{\mathbb{C}}(E).} \quad (\text{Residue RC11})$$

Proof. The vectors $A^i 1_E, 0 \leq i < q$, are the power basis. By (RC8), the functionals $\ell \circ A^j, 0 \leq j < q$, are a basis of $E^\vee$. Therefore the q^2 operators

$$A^i \mathcal{R} A^j = (A^i 1_E) \otimes (\ell \circ A^j) \quad (\text{Residue RC12})$$

are a vector-space basis of $\text{End}_{\mathbb{C}}(E)$. Every one belongs to the algebra generated by A and $\mathcal{R}$.

Now fix an actual nonzero proper invariant constituent, with its full maps,

$$0 \longrightarrow F \xrightarrow{\iota} E \xrightarrow{\pi} E/F \longrightarrow 0, \quad A\iota = \iota A_F, \quad 0 < p := \dim F < q. \quad (\text{Residue RC13})$$

The deformation has a specified transverse map

$$\boxed{\pi \mathcal{R} \iota = (\pi 1_E) \otimes \ell_F : F \longrightarrow E/F, \quad \ell_F = \ell \circ \iota.} \quad (\text{Residue RC14})$$

Both tensor factors are nonzero. If 1_E belonged to F , its entire A -orbit would belong to F , contradicting $F \neq E$. If $\ell_F = 0$, then for $v \in F$, invariance would give $\ell(A^j v) = 0$ for every j . Nondegeneracy of (RC8) would force every $v \in F$ to be zero. Thus

$$\boxed{\text{rank}(\pi \mathcal{R} \iota) = 1, \quad \ker(\pi \mathcal{R} \iota) = \ker \ell_F, \quad \text{im}(\pi \mathcal{R} \iota) = \mathbb{C} \pi 1_E.} \quad (\text{Residue RC15})$$

In particular the kernel has dimension $p - 1$. This is the strengthening of the supplied rank-at-most-one statement. It holds with all nilpotent jets retained. Equivalently, an invariant subspace is an ideal $(a)/(\chi)$ for a monic divisor $a \mid \chi$; the polynomial $S^{q-1-\deg a} a$ witnesses that ℓ_F is nonzero whenever $0 < \deg a < q$.

No nonzero proper *constant remainder-coordinate* subspace is invariant under $A(t)$ for every t . This assertion must not be called differential irreducibility: moving horizontal subbundles do exist. For example, $\Pi(t)^{-1} \Pi(0)F$ is horizontal. It is a different marked family from the fixed original constituent in (RC13).

71.3 Exact constituent-period curvature

Keep F and its inclusion fixed as in (RC13), and put

$$P_F(t) = \Pi(t)\iota, \quad H(t) = \Pi(t)^* \Pi(t), \quad H_F(t) = P_F(t)^* P_F(t) = \iota^* H(t) \iota. \quad (\text{Residue RC16})$$

The target metric in this section is exactly the fixed coordinate metric I_q of the period construction. Since $\Pi(t)$ is invertible, $P_F(t)$ has rank p everywhere. Its orthogonal projector is

$$P_F(t) = P_F(t) H_F(t)^{-1} P_F(t)^*. \quad (\text{Residue RC17})$$

Use (RC4) and $A\iota = \iota A_F$ before projecting:

$$P'_F(t) = -\frac{1}{u} P_F(t) A_F - \frac{t}{u} \Pi(t) \mathcal{R} \iota. \quad (\text{Residue RC18})$$

The first term is tangent to the same period subspace. Thus its complete normal derivative is

$$\mathcal{N}_F(t) := (I - \mathbf{P}_F(t))P'_F(t) = -\frac{t}{u}(I - \mathbf{P}_F(t))\Pi(t)1_E \ell_F. \quad (\text{Residue RC19})$$

The column $(I - \mathbf{P}_F(t))\Pi(t)1_E$ never vanishes: vanishing would imply $1_E \in F$. Equations (RC15) and (RC19) prove that this normal map is zero at $t = 0$ and has rank exactly one at every $t \neq 0$. Its kernel for $t \neq 0$ is $\ker \ell_F$. At the marked fibre,

$$(I - \mathbf{P}_F(0))P'_F(0) = 0, \quad (I - \mathbf{P}_F(0))P''_F(0) = -\frac{1}{u}(I - \mathbf{P}_F(0))\Pi(0)1_E \ell_F \neq 0. \quad (\text{Residue RC20})$$

The second identity follows by differentiating (RC5): $\Pi'' = \Pi A(t)^2/u^2 - \Pi \mathcal{R}/u$. At zero, $A^2 \iota = \iota A_F^2$, so the A^2 term vanishes under the stated normal projection. The surviving map has rank one.

71.3.1 The exact coefficient of curvature

For any positive Hermitian matrix M on E , define

$$q_{M,F}([1_E]) = \min_{f \in F} \|1_E - \iota f\|_M^2, \quad M_F = \iota^* M \iota, \quad (\text{Residue RC21})$$

and

$$\mathbf{c}_F(M) = q_{M,F}([1_E]) \ell_F M_F^{-1} \ell_F^*. \quad (\text{Residue RC22})$$

This scalar is strictly positive for (RC13). It is independent of the basis chosen in F , but it retains the marked quotient vector $[1_E]$ and the marked residue functional ℓ_F . The two factors are the actual quotient norm and the actual dual norm, not arbitrary positive constants.

Let $\psi_F(t) = \log \det H_F(t)$, where the determinant is of the $p \times p$ positive Gram. No rectangular determinant is used. Ordinary matrix differentiation, using holomorphy of P_F , gives

$$\begin{aligned} \partial_t \partial_{\bar{t}} \psi_F &= \text{Tr} [H_F^{-1} P_F'^* P_F' - H_F^{-1} P_F'^* P_F H_F^{-1} P_F^* P_F'] \\ &= \text{Tr} [H_F^{-1} \mathcal{N}_F^* \mathcal{N}_F]. \end{aligned} \quad (\text{Residue RC23})$$

The projector in (RC17) is self-adjoint and idempotent, which justifies the last equality. Further,

$$q_{H(t),F}([1_E]) = \|(I - \mathbf{P}_F(t))\Pi(t)1_E\|^2. \quad (\text{Residue RC24})$$

Substitute (RC19) and take the trace of its rank-one product. This yields an exact identity for all complex t :

$$\partial_t \partial_{\bar{t}} \log \det H_F(t) = \frac{|t|^2}{|u|^2} \mathbf{c}_F(H(t)). \quad (\text{Residue RC25})$$

Thus the dual determinant line of the period-image subbundle, with its specified induced metric, has Chern form

$$c_1((\det \mathcal{F}_{\text{per}})^\vee) = \frac{i}{2\pi} \frac{|t|^2}{|u|^2} \mathbf{c}_F(H(t)) dt \wedge d\bar{t}. \quad (\text{Residue RC26})$$

It is strictly positive away from $t = 0$ and has quadratic vanishing there. Equivalently,

$$\partial_t^2 \partial_{\bar{t}}^2 \log \det H_F(t)|_{t=0} = \frac{\mathbf{c}_F(H(0))}{|u|^2} > 0. \quad (\text{Residue RC27})$$

The coefficient of $t^2 \bar{t}^2$ in $\log \det H_F$ is therefore $\mathbf{c}_F(H(0))/(4|u|^2)$; the factor four comes from $2!2!$. This mixed derivative is not the second derivative along the real axis. For $F = 0$ or $F = E$, the relevant curvature is identically zero; (RC13) is essential for strict positivity.

71.3.2 The first derivative still records the aggregate defect

At $t = 0$ and real $u > 0$, (RC18) gives

$$-u \frac{d}{dt} \log \frac{\det H_F(t)}{\det G_{N,F}} \Big|_{t=0} - kp = 2 \operatorname{Re} \operatorname{Tr} A_F - kp, \quad G_{N,F} = \iota^* G_N \iota. \quad (\text{Residue RC28})$$

This is the supplied constituent-minor observer, not a new estimate. Taking $F = E_{>}$, the complete positive-real-defect generalized eigenspace, gives the existing aggregate spectral defect. Equations (RC25)–(RC27) add the exact transverse curvature carried by that same marked constituent. A derivative of the full determinant alone sees only the balanced total trace on a reflection-stable packet.

71.4 Transfer to the original theta metric

71.4.1 A source-space expression for the coefficient

Using (RC2), the same coefficient for the actual arithmetic metric is

$$\mathfrak{c}_F(G_N) = \left(\min_{f \in F} \|r_N(1_E - \iota f)\|^2 \right) \left(\sup_{0 \neq f \in F} \frac{|\ell_F(f)|^2}{\|r_N \iota f\|^2} \right). \quad (\text{Residue RC29})$$

The supremum equals the finite dual norm in (RC22). Every source norm, the arithmetic unit implicit in r_N , and every allowed theta correction remain in this expression.

The isometry from the original source metric into period coordinates is not $E \xrightarrow{\Pi} (\mathbb{C}^q, I_q)$. It is

$$(E, G_N) \xrightarrow{\Pi(t)} (\mathbb{C}^q, \tilde{G}_N(t)), \quad \tilde{G}_N(t) = \Pi(t)^{-*} G_N \Pi(t)^{-1}. \quad (\text{Residue RC30})$$

Consequently the corresponding quotient-unit/residue product is invariant when the subspace, unit, functional, and metric are all transported: $F \mapsto \Pi F$, $1_E \mapsto \Pi 1_E$, and $\ell \mapsto \ell \Pi^{-1}$. In particular it equals $\mathfrak{c}_F(G_N)$, independently of t . This does not identify the Chern form for a varying target metric with the fixed-target-metric form (RC26).

71.4.2 A finite condition-number bridge

Let $a_N(t) > 0$ and $b_N(t) > 0$ be the least and greatest eigenvalues of

$$B_N^{\text{per}}(t) = G_N^{-1} H(t). \quad (\text{Residue RC31})$$

This operator is positive and self-adjoint in G_N , so $a_N G_N \preceq H(t) \preceq b_N G_N$. Minimizing a quadratic form over a fixed coset preserves this order, whereas inversion of the restricted positive form reverses it. Therefore

$$\frac{a_N}{b_N} \mathfrak{c}_F(G_N) \leq \mathfrak{c}_F(H(t)) \leq \frac{b_N}{a_N} \mathfrak{c}_F(G_N). \quad (\text{Residue RC32})$$

Writing $\mathcal{K}_N = b_N/a_N$, the exact curvature has the two-sided source-sensitive enclosure

$$\frac{|t|^2}{|u|^2 \mathcal{K}_N(t)} \mathfrak{c}_F(G_N) \leq \partial_t \partial_{\bar{t}} \log \det H_F(t) \leq \frac{|t|^2 \mathcal{K}_N(t)}{|u|^2} \mathfrak{c}_F(G_N). \quad (\text{Residue RC33})$$

The calculated determinant of B_N^{per} determines its eigenvalue product, not $\mathcal{K}_N$. Equation (RC33) keeps the precise additional metric quantity required to compare the two curvature observations.

71.4.3 The existing endpoint certificate controls variation of this coefficient

For the original nested source degrees $i \leq j$, let $G_i \succeq G_j \succ 0$, $V_n = \det G_n$, and

$$\theta = \lambda_{\min}(G_i^{-1} G_j) \in (0, 1]. \quad (\text{Residue RC34})$$

Then $\theta G_i \preceq G_j \preceq G_i$. The quotient norms satisfy $\theta q_{G_i, F} \leq q_{G_j, F} \leq q_{G_i, F}$, and the squared restricted dual norms satisfy $\|\ell_F\|_{G_i^{-1}}^2 \leq \|\ell_F\|_{G_j^{-1}}^2 \leq \theta^{-1} \|\ell_F\|_{G_i^{-1}}^2$, with each inverse taken on the restricted form. Multiplying gives

$$\theta \mathbf{c}_F(G_i) \leq \mathbf{c}_F(G_j) \leq \theta^{-1} \mathbf{c}_F(G_i). \quad (\text{Residue RC35})$$

Every eigenvalue of $G_i^{-1} G_j$ lies in $(0, 1]$, and one equals θ . Their product is at most θ . Hence

$$\left| \log \frac{\mathbf{c}_F(G_j)}{\mathbf{c}_F(G_i)} \right| \leq -\log \theta \leq \log \frac{V_i}{V_j}. \quad (\text{Residue RC36})$$

This is a new use of the existing endpoint quantity. Any certified upper bound for the original $\log(V_i/V_j)$ immediately bounds the change of the specified residue/quotient coupling, without a separately assumed uniform spectral gap. It does not reverse into an upper bound for $\log(V_i/V_j)$ from the coupling alone.

71.5 The period realization of the arithmetic Laplacian

The quadratic polynomial of the original arithmetic generator is

$$\mathbf{L}_k = A^2 - kA. \quad (\text{Residue RC37})$$

For $k = 1$ this is the polynomial corresponding to Connes–Consani’s $H(1+H)$ under $H = -D$. A nonconstant parameter connection introduces an exact additional term at second order. From (RC5),

$$\Pi''(t) = \frac{1}{u^2} \Pi(t) A(t)^2 - \frac{1}{u} \Pi(t) \mathcal{R}. \quad (\text{Residue RC38})$$

Thus, on every fibre,

$$u^2 \Pi''(t) + ku \Pi'(t) + u \Pi(t) \mathcal{R} = \Pi(t) (A(t)^2 - kA(t)). \quad (\text{Residue RC39})$$

At zero this is the exact period comparison for (RC37). Omitting $u \Pi \mathcal{R}$ would give a false equation. The correction is the same residue map that supplies (RC15) and (RC25).

The original source lift of this correction is

$$\mathcal{R}_N^{\text{src}} = r_N \mathcal{R} j_E : \mathcal{X} \rightarrow \mathcal{X}, \quad j_E \mathcal{R}_N^{\text{src}} = \mathcal{R} j_E, \quad \mathcal{R}_N^{\text{src}} d = 0. \quad (\text{Residue RC40})$$

It is defined on the specified source domain. For a fixed $x \in \mathcal{X}$ let $Y_x(t) = \Pi(t) j_E x$. Applying (RC39) at zero and using (RC2) gives

$$(u^2 \partial_t^2 + ku \partial_t) Y_x|_0 + u \Pi(0) j_E \mathcal{R}_N^{\text{src}} x = \Pi(0) j_E ((D^{(k)})^2 - kD^{(k)}) x. \quad (\text{Residue RC41})$$

The original primitive for $D^{(k)} r_N - r_N A = dK_N$ is still present; the perturbed source operator $D^{(k)} + t \mathcal{R}_N^{\text{src}}$ has the same primitive, exactly as in DS69.

For the original Gram, retain

$$W_N = A^* G_N + G_N A - kG_N. \quad (\text{Residue RC42})$$

Direct expansion proves

$$\mathbf{L}_k^* G_N - G_N \mathbf{L}_k = A^* W_N - W_N A, \quad G_N \mathbf{L}_k = W_N A - A^* G_N A. \quad (\text{Residue RC43})$$

These are the preceding Laplacian-control identities, included here to fix the map to the present period calculation. Under (RC30), put $\tilde{A} = \Pi A \Pi^{-1}$, $\tilde{\mathbf{L}}_k = \Pi \mathbf{L}_k \Pi^{-1}$, and $\tilde{W}_N = \Pi^{-*} W_N \Pi^{-1}$. Substitution gives

$$\tilde{A}^* \tilde{G}_N + \tilde{G}_N \tilde{A} - k \tilde{G}_N = \tilde{W}_N, \quad (\text{Residue RC44})$$

and the two identities (RC43) in precisely the transported metric. Thus every original numerical-range/control estimate is transported by an actual isometry, not by exchanging G_N for $H(t)$.

The full primary action is retained:

$$\mathbb{L}_k|_{E_\rho} = \rho(\rho - k)I + (2\rho - k)N_\rho + N_\rho^2. \quad (\text{Residue RC45})$$

The finite residue algebra and the adelic Hochschild source are not being declared isomorphic. The latter reaches these finite coefficients through the original theta complex, its quotient, and j_E . In particular a theta boundary is killed by j_E ; the differentiated original relation producing the Jacobian trace is taken before that quotient. The finite algebra's higher Hochschild modules from the supplied periodic resolution, $HH_{2a+1} = E/(\chi')$ and $HH_{2a+2} = \text{ann}_E(\chi')$, remain separate from the operator $\mathcal{R}$ and from the adelic Hochschild groups.

71.6 Translation, cohomological support, and exact scope

71.6.1 Translation preserves curvature but changes the marked trace

The supplied DT1–13 comparison is

$$\chi_a(S) = \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \quad \Pi_a(t) = f_a(t)\Pi(t)C_a, \quad f_a(t) = e^{(-\Phi(-a)-ta)/u}, \quad (\text{Residue RC46})$$

where C_a is the Pascal substitution matrix, $\det C_a = 1$, and $C_a A_a C_a^{-1} = A + aI$. Transport the same constituent by $\iota_a = C_a^{-1}\iota$. Then

$$P_{F,a}(t) = f_a(t)P_F(t), \quad \det H_{F,a}(t) = |f_a(t)|^{2p} \det H_F(t). \quad (\text{Residue RC47})$$

Since f_a is nowhere zero and holomorphic,

$$\boxed{\partial_t \partial_{\bar{t}} \log \det H_{F,a}(t) = \partial_t \partial_{\bar{t}} \log \det H_F(t).} \quad (\text{Residue RC48})$$

But, with the original k fixed,

$$2 \operatorname{Re} \operatorname{Tr}(A_F + aI) - kp = (2 \operatorname{Re} \operatorname{Tr} A_F - kp) + 2p \operatorname{Re} a. \quad (\text{Residue RC49})$$

Thus curvature, unlike the first-derivative observer, does not retain this scalar normalization by itself. The exact scalar f_a and the marked arithmetic fibre $a = 0$ remain necessary parts of the comparison. This is not an assertion that translated polynomials are actual zeta packets. In the specified finite-field realization DT12–13 instead tensors with the constant Artin–Schreier character line, whose geometric-Frobenius scalar is a root of unity. The finite-field weights are preserved by that operation; it does not identify that Frobenius with A .

71.6.2 Cohomological degrees and supported-zero maps

For an invariant F , the supplied logarithmic lattice $\mathcal{M}_F = F\mathcal{O} + E\mathcal{O}(-D)$ gives

$$\mathbb{H}^0(\mathcal{K}_F) = F, \quad \mathbb{H}^1(\mathcal{K}_F) = 0, \quad \mathbb{H}^2(\mathcal{K}_F) = (E/F) \otimes \mathbb{T}. \quad (\text{Residue RC50})$$

Its support-compatible dual is $\mathcal{M}_F^\vee(-D) = F^{\text{ann}}\mathcal{O} + E^\vee\mathcal{O}(-D)$. The evaluation target includes that divisor twist. The full coefficient extension $0 \rightarrow F \rightarrow E \rightarrow E/F \rightarrow 0$ is retained through the outer comparison complexes; it is not reconstructed from the two middle cohomology groups alone. These are the corrected distinctions in DS25–34 and DS66–69.

The new transverse map (RC14) is therefore a specified map $\mathbb{H}^0(\mathcal{K}_F) \rightarrow E/F$, with the latter connected to degree two by the retained line $\mathbb{T}$. It is not silently a degree-zero chain endomorphism of $\mathcal{K}_F$. In particular, $\mathcal{R}$ need not preserve its lattice; (RC15) quantifies the failure.

Lift every coefficient map by the original support reconstruction, retaining the structural square

$$\begin{array}{ccc} G(\mathbb{Z}) & \longrightarrow & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \longrightarrow & \mathbb{C}. \end{array} \quad (\text{Residue RC51})$$

For each support label λ , the new map is

$$(\lambda, v) \longmapsto (\lambda, (\pi 1_E) \ell_F(v)), \quad \tau \longmapsto \tau. \quad (\text{Residue RC52})$$

If $v \in \ker \ell_F$, its receiving value is the supported zero at λ , not external absence. For the single active fibre,

$$S(f)^{-1}(\tau) = \{\tau\}, \quad S(f)^{-1}(e) = (\ker f)^\bullet, \quad p^{-1}(\ker f) = \{\tau\} \sqcup (\ker f)^\bullet. \quad (\text{Residue RC53})$$

This uses the corrected split-kernel formula in the supplied review. The original infinite arithmetic quotient, support transitions, and all source boundaries remain attached.

71.7 Verification and provenance

The included checker contains twelve independent finite test methods. Its fixtures are monic polynomials with rational coefficients, including repeated roots, and exact complex-rational matrices. It checks the residue pairing and trace, the q^2 endomorphism basis, the exact transverse rank and kernel, the curvature factorization, a bivariate fourth-order log-Gram jet with its factorial, the metric enclosures, the forced Laplacian equation, and the constituent translation comparison. The normal and optimized runs have their actual command and output records in checks/. Three deliberately false formulas are tested separately: zero transverse rank, omitted Laplacian forcing, and omitted $|t|^2$ in the curvature. These must fail in both modes.

The finite tests do not prove improper-integral convergence, the global analytic period theorem, Deligne's theorem, or uniform growth on arithmetic packets. Those parts have their written source proofs and stated hypotheses. No predecessor checker or Lean build is claimed as a new execution here. No remote repository or publication is modified by this package.

The dated *Deligne Primary Source Correction* supersedes the historical unresolved-citation claim in *Exponential Intake*: the reference is Roman **(I.8.11)** to Weil I, printed page 306. The strict weight inequality, the tensor square of the same eigenvalue, and the lisse/mixed hypotheses are retained. This note relies on that supplied correction for those printed-page identifications; it does not claim a fresh full audit of Weil II.

The intake receipt records the hashes actually measured from the uploaded files. All nine files listed in the manifest's copies array match their output hashes. The currently supplied README is newer than the README hash recorded in that manifest. The independent review also pins translation TeX hash 39163624..., whereas the delivered translation source and the manifest agree on bd1831cd.... The review's source-byte identity must therefore be reconciled before its acceptance is attributed to those exact delivered translation bytes; this discrepancy does not itself show that a displayed mathematical formula is false.

71.8 References and supplied-source locators

Bloch, S., & Esnault, H. (1999). *Gauß–Manin determinant connections and periods for irregular connections* [Preprint]. arXiv, math/9912095. The inspected §5 supplies the polynomial-exponential period context, not the new assertions (RC11), (RC15), (RC25), or (RC36).

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and ζ -cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83–101). American Mathematical Society. DOI: 10.1090/pspum/105/01896.

Deligne, P. (1980). La conjecture de Weil: II. *Publications Mathématiques de l'IHÉS*, 52, 137–252. DOI: 10.1007/BF02684780. The present note does not infer characteristic-zero spectral purity from finite-field purity.

National Institute of Standards and Technology. (n.d.). §5.5: Functional relations. In *Digital Library of Mathematical Functions*. Equation 5.5.6, Gauss's gamma multiplication formula, gives the constant in (RC6).

Deligne Split Cohomology. (2026, September 13). Supplied research sidebar, DS1–69. Source: deligne_split_sidebar_20260913.tex; relevant maps DS12–14, DS25–34, DS53–69.

Exponential Period Determinant. (2026, September 13). Supplied research note, XD1–26. Source: deligne_exponential_determinant_extension_20260913.tex.

Exponential Translation Comparison. (2026, September 13). Supplied research note, DT1–13. Source: Exponential_Translation_Comparison.tex.

Pasted markdown (20260913-034115). (2026, September 13). Supplied SGA trace/period continuation. Sections 2–3 give the residue coevaluation and periodic resolution; §§4–5 evaluate and differentiate the determinant; §§6–7 identify the constituent minor and transport the original metric.

Independent Deligne Review, Exponential Intake, and Deligne Primary Source Correction. (2026, September 13). Supplied review and correction records. Their historical execution and source-reading claims are not relabeled as new executions in this note.

72 Complete residue constituent curvature extension

72.1 Residue endomorphisms, metric variation, and the forced period Laplacian

The delivered residue–constituent note proves (RC1)–(RC53). Its transverse-rank and period-curvature calculation is the same exact calculation as (SC9)–(SC25), and its comparison with the original positive form is already (LC13)–(LC15). We retain these shared results and prove the additional endomorphism, variation, differential, and translation maps below. The complete preceding determinant and constituent proofs are included in the standalone edition. No original arithmetic source norm, coefficient, multiplicity, or contour factor is changed.

72.1.1 A universal inverse to the residue endomorphism basis

Let B be an arbitrary commutative ring with identity. Fix the monic polynomial and its original power-basis quotient

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a \in B[S], \quad q \geq 1, \quad E_B = B[S]/(\chi), \quad e_i = [S^i] \quad (0 \leq i < q). \quad (\text{RCX1})$$

Polynomial division by the monic polynomial proves that these e_i are a basis: subtract leading multiples until the degree is less than q ; a nonzero multiple of a monic degree- q polynomial cannot have degree less than q , even when B has zero divisors. Let $A = M_S$, $e = e_0$, and let $\ell : E_B \rightarrow B$ be the coefficient of S^{q-1} in this remainder. Set

$$R = e\ell, \quad \mathcal{B}_{ij} = \ell(e_i e_j), \quad 0 \leq i, j < q. \quad (\text{RCX2})$$

The matrix $\mathcal{B}$ is symmetric. Its entries with $i + j < q - 1$ are zero, and its antidiagonal entries are one. Reversing its columns therefore gives a triangular matrix with diagonal entries one. The column-reversal permutation has $q(q-1)/2$ inversions, so

$$\det \mathcal{B} = (-1)^{q(q-1)/2}, \quad \mathcal{B}^{-1} = (-1)^{q(q-1)/2} \text{adj}(\mathcal{B}). \quad (\text{RCX3})$$

In particular, this inverse uses no discriminant, root separation, or division by a coefficient. The pairing $\beta(x, y) = \ell(xy)$ induces an isomorphism $E_B \rightarrow \text{Hom}_B(E_B, B)$ with matrix $\mathcal{B}$.

For a matrix T in the original power basis, put $K(T) = T\mathcal{B}^{-1}$. The complete reconstruction is

$$\boxed{T = \sum_{i,j=0}^{q-1} K(T)_{ij} A^i R A^j, \quad K(T) = T\mathcal{B}^{-1}.} \quad (\text{RCX4})$$

Indeed $A^i e = e_i$, and the row ℓA^j has entries $\ell(e_j e_b) = \mathcal{B}_{jb}$. Thus the matrix of the displayed sum is $K(T)\mathcal{B} = T$. Conversely, a zero sum has $K\mathcal{B} = 0$, so $K = 0$. This proves, with explicit inverse, the isomorphism of free B -modules

$$B^{q \times q} \longrightarrow \text{End}_B(E_B), \quad (K_{ij}) \longmapsto \sum_{i,j} K_{ij} A^i R A^j. \quad (\text{RCX5})$$

Its image is contained in the unital B -algebra generated by A, R ; hence that algebra is all of $\text{End}_B(E_B)$. This includes $q = 1$, where $R = I$. For a ring map $B \rightarrow B'$, monic division, $\mathcal{B}$, its displayed inverse, and (RCX4) all map to their corresponding formulas over B' . Thus this is a universal coefficient and base-change identity, including every nonreduced fibre.

The residue dual rows give literal matrix units:

$$d^j = \sum_{b=0}^{q-1} (\mathcal{B}^{-1})_{jb} \ell A^b, \quad U_{ij} = e_i d^j = \sum_b (\mathcal{B}^{-1})_{jb} A^i R A^b. \quad (\text{RCX6})$$

The equality $d^j(e_k) = \delta_{jk}$ follows by multiplication of $\mathcal{B}^{-1}\mathcal{B}$. Consequently $U_{ij}U_{kl} = \delta_{jk}U_{il}$ and $\sum_i U_{ii} = I$. These equations specify the multiplication law, not merely a dimension count.

For the original complex arithmetic source, retain exactly

$$j_E r_N = I, \quad j_E D^{(k)} = A j_E, \quad j_E d = 0, \quad G_N = r_N^* r_N > 0. \quad (\text{RCX7})$$

The maps have domains $r_N : E \rightarrow \mathcal{X}$ and $j_E : \mathcal{X} \rightarrow E$ on the admitted strong-Schwartz source $\mathcal{X}$; d is its original boundary map. Every $T \in \text{End}_{\mathbb{C}}(E)$ has the canonical finite-rank source lift

$$\widehat{T} = r_N T j_E, \quad j_E \widehat{T} = T j_E, \quad \widehat{T} r_N = r_N T, \quad \widehat{T} d = 0, \quad \widehat{T} \widehat{U} = \widehat{T} \widehat{U}. \quad (\text{RCX8})$$

Each identity follows by inserting $j_E r_N = I$ or $j_E d = 0$. In particular $\widehat{T} = r_N j_E$, an idempotent on the full source with image $r_N E$. Thus the lift is a unital representation in the corner with identity $r_N j_E$, and is injective because $j_E \widehat{T} r_N = T$. No assertion that j_E is bounded on an unweighted completion is required or made. Taking $T = U_{ij}$ in (RCX8) gives every matrix unit at the original source, with the same boundary kernel. Taking $T = R$ gives the residue perturbation $r_N R j_E$ already used in (SC7)–(SC8).

72.1.2 The original quotient-unit and residue metric product

Now $B = \mathbb{C}$, and $I : F \hookrightarrow E$ is a fixed nonzero proper A -invariant inclusion of dimension p . Keep the original unit e and row $L = \ell I$. Both $e \notin I(F)$ and $L \neq 0$ follow directly from (RCX3): if $e \in I(F)$, all $A^i e$ belong to $I(F)$; if $L = 0$, invariance gives $\ell A^j I = 0$ for all j , and the invertible residue matrix forces $I = 0$.

For every positive Hermitian form M on the original E , define

$$\begin{aligned} M_F &= I^* M I, & P_M &= I M_F^{-1} I^* M, & r_M &= (1 - P_M) e, & w_M &= I M_F^{-1} L^*, \\ q_M &= r_M^* M r_M, & d_M &= L M_F^{-1} L^*, & \mathbf{c}_F(M) &= q_M d_M. \end{aligned} \quad (\text{RCX9})$$

The equations $P_M^2 = P_M$, $P_M^* M = M P_M$, and $\text{im } P_M = I(F)$ follow by multiplication. Thus $I^* M r_M = 0$. Completing the square gives

$$q_M = \min_{v \in F} (e - I v)^* M (e - I v), \quad d_M = w_M^* M w_M = \sup_{v \neq 0} \frac{|L v|^2}{v^* M_F v}. \quad (\text{RCX10})$$

For the supremum, write $x = M_F^{-1} L^*$; then $L v = x^* M_F v$, and Cauchy-Schwarz in M_F proves the upper bound $x^* M_F x = d_M$, attained at $v = x \neq 0$. The minimum is attained at $v = M_F^{-1} I^* M e$, with residual $r_M \neq 0$. Consequently $q_M, d_M > 0$, and $r_M^* M w_M = 0$. At $M = G_N$, (RCX10) has exactly the original source expression

$$\mathbf{c}_F(G_N) = \left(\min_{v \in F} \|r_N(e - I v)\|^2 \right) \left(\sup_{v \neq 0} \frac{|L v|^2}{\|r_N I v\|^2} \right). \quad (\text{RCX11})$$

In particular every original arithmetic Taylor unit and source mass encoded in r_N remains in the two displayed factors.

For any invertible coefficient map $C : E' \rightarrow E$, retain the transported data $I' = C^{-1}I$, $e' = C^{-1}e$, $\ell' = \ell C$, and $M' = C^*MC$. Direct substitution gives

$$M'_F = M_F, \quad P_{M'} = C^{-1}P_MC, \quad r_{M'} = C^{-1}r_M, \quad w_{M'} = C^{-1}w_M, \quad \mathfrak{c}_{F'}(M') = \mathfrak{c}_F(M). \quad (\text{RCX12})$$

For a positive real scalar s , the formulas separately give $q_{sM} = sq_M$, $d_{sM} = s^{-1}d_M$, and $\mathfrak{c}_F(sM) = \mathfrak{c}_F(M)$. These are proved homogeneity identities with both factors retained; no form is replaced by a rescaled form in the arithmetic construction.

72.1.3 An exact rank-two differential for metric variation

Let $M(s) > 0$ be a C^1 path on a closed real interval; a dot denotes its actual derivative. All inverses in (RCX9) are C^1 . Differentiating $q = r^*Mr$, the two terms with $\dot{r}$ vanish: $\dot{r} \in I(F)$, whereas r is M -orthogonal to $I(F)$. Differentiating M_F^{-1} gives $(M_F^{-1})^\cdot = -M_F^{-1}\dot{M}_FM_F^{-1}$. Hence

$$\dot{q} = r^*\dot{M}r, \quad \dot{d} = -w^*\dot{M}w, \quad \boxed{\frac{d}{ds} \log \mathfrak{c}_F(M(s)) = \frac{r^*\dot{M}r}{q} - \frac{w^*\dot{M}w}{d}}. \quad (\text{RCX13})$$

Define endomorphisms on the original E , with its original form,

$$Q_r = \frac{rr^*M}{q}, \quad Q_w = \frac{ww^*M}{d}, \quad Z_M = Q_r - Q_w, \quad X = M^{-1}\dot{M}. \quad (\text{RCX14})$$

Each Q is its indicated rank-one M -orthogonal projection. Their products in both orders vanish because $r^*Mw = 0$. It follows that Z_M has eigenvalues 1, -1 , and zero on the remaining $q - 2$ dimensions. The complete differential is

$$\boxed{\frac{d}{ds} \log \mathfrak{c}_F(M(s)) = \text{Tr}(Z_M X)}. \quad (\text{RCX15})$$

Indeed the trace of a column times a row is their row-column contraction, giving exactly (RCX13). This also proves that scalar metric directions cancel: $X = \alpha I$ gives $\text{Tr } Z_M = 0$, while the original scalar α and both rank-one terms remain specified.

The operator X is self-adjoint in M , since $MX = \dot{M}$ is Hermitian. Let $\lambda_-(s)$ and $\lambda_+(s)$ be its least and greatest real eigenvalues. Each quotient in (RCX13) is a Rayleigh quotient of this same X , so

$$\left| \frac{d}{ds} \log \mathfrak{c}_F(M(s)) \right| \leq \lambda_+(s) - \lambda_-(s). \quad (\text{RCX16})$$

The positive equality holds exactly when r lies in the greatest eigenspace and w in the least; negative equality reverses them. To verify the equality statement, diagonalize the M -self-adjoint operator: a Rayleigh quotient is a weighted mean of its eigenvalues, with nonnegative weights and positive total weight, and attains an extreme precisely on the corresponding eigenspace. Integrating the actual continuous differential yields

$$\log \frac{\mathfrak{c}_F(M(1))}{\mathfrak{c}_F(M(0))} = \int_0^1 \text{Tr}(Z_{M(s)} M(s)^{-1} \dot{M}(s)) ds. \quad (\text{RCX17})$$

This gives an exact observation on the metric path as well as the bound obtained by integrating (RCX16).

For the nested original forms $G_i \succeq G_j > 0$, put

$$T = G_i^{-1}G_j, \quad 0 < \theta = \lambda_{\min}(T) \leq \eta = \lambda_{\max}(T) \leq 1, \quad M(s) = G_i + s(G_j - G_i). \quad (\text{RCX18})$$

No diagonalization of an arithmetic source is performed here: the spectral theorem is used to evaluate the indicated relative operator. Algebraic multiplication gives

$$M(s) = G_i(I + s(T - I)), \quad M(s)^{-1}\dot{M}(s) = (I + s(T - I))^{-1}(T - I). \quad (\text{RCX19})$$

Thus its eigenvalues are $(\lambda - 1)/(1 + s(\lambda - 1))$. The derivative of this expression in λ is $1/(1 + s(\lambda - 1))^2 > 0$. The spread therefore integrates exactly to

$$\int_0^1 (\lambda_+(s) - \lambda_-(s)) ds = \log \eta - \log \theta. \quad (\text{RCX20})$$

Combining (RCX16)–(RCX20), and keeping $V_n = \det G_n$, proves the sharpened endpoint estimate

$$\left| \log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} \right| \leq \log \frac{\eta}{\theta} \leq -\log \theta \leq \log \frac{V_i}{V_j}. \quad (\text{RCX21})$$

For the last inequality, all q eigenvalues of T lie in $(0, 1]$ and one is θ , so $\det T \leq \theta$. Taking negative logarithms and $\det T = V_j/V_i$ proves it. This improves (RC36) by retaining the greatest eigenvalue η as well as the least. It assigns no unproved uniform bound to these actual finite quantities.

72.1.4 All original volume factors in the coupling

The ideal classification gives $I(F) = (g)/(\chi)$, with g monic, $\chi = gh$, and $\deg h = p$. Indeed A -invariance is invariance under every polynomial in A , so $I(F)$ is an ideal; its preimage in $\mathbb{C}[S]$ is generated by a monic divisor g of χ . The map $[a]_h \mapsto [ga]_\chi$ is bijective onto this ideal: surjectivity is its definition and injectivity follows from $gh \mid ga \Rightarrow h \mid a$. Keep its columns and frame map

$$J_g = [g, Sg, \dots, S^{p-1}g] = IC, \quad J_- = [g, \dots, S^{p-2}g], \quad J_+ = [J_g, e], \quad C = (I^*I)^{-1}I^*J_g. \quad (\text{RCX22})$$

Here C is invertible, $LC = e_{p-1}^T$, and the empty determinant for $p = 1$ is one. The last equality follows because $S^{p-1}g$ is monic of degree $q - 1$, while the preceding columns have smaller degree. All displayed columns are independent, including J_+ because $e \notin I(F)$. For $v_J(M) = \det(J^*MJ)$, the Schur complement for J_+ and the last diagonal inverse cofactor for $J_g^*MJ_g$ give

$$q_M = \frac{v_{J_+}(M)}{v_{J_g}(M)}, \quad d_M = \frac{v_{J_-}(M)}{v_{J_g}(M)}, \quad \mathfrak{c}_F(M) = \frac{v_{J_+}(M)v_{J_-}(M)}{v_{J_g}(M)^2}. \quad (\text{RCX23})$$

For the first identity, the Schur complement is $e^*Me - e^*MJ_g(J_g^*MJ_g)^{-1}J_g^*Me = q_M$. For the second, use $LC = e_{p-1}^T$ and $J_g^*MJ_g = C^*M_F C$; the inverse last cofactor is precisely the determinant of the first $p - 1$ Gram columns divided by the full determinant. These proofs retain the frame determinant $\det(J_g^*MJ_g) = |\det C|^2 \det M_F$.

Write $v_{n,J} = v_J(G_n)$. The exact endpoint observation is

$$\log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} = \log \frac{v_{j,J_+}}{v_{i,J_+}} + \log \frac{v_{j,J_-}}{v_{i,J_-}} - 2 \log \frac{v_{j,J_g}}{v_{i,J_g}}. \quad (\text{RCX24})$$

This supplies the original Toda-volume expression for the rank-two differential (RCX15). It follows by taking logarithms of the three positive factors in (RCX23); none is discarded. In particular a scalar form change by s contributes $(p + 1) \log s + (p - 1) \log s - 2p \log s = 0$ to (RCX24), exactly matching (RCX12), with every dimension retained.

72.1.5 The mixed fourth jet and the real-axis derivatives

Fix the original $u \in \mathbb{C}^\times$, the full period matrix $\Pi(t)$, and its equation $\Pi' = -\Pi(A + tR)/u$. The full determinant and contour convergence are proved in (XD1)–(XD26) and (SC1)–(SC6b), included before this section. Put $Y = \Pi I$, $H = \Pi^* \Pi$, $H_F = Y^* Y$, and $\psi(t) = \log \det H_F(t)$. With $P = Y H_F^{-1} Y^*$ and $z = (1 - P)\Pi e$, the exact equations are

$$(1 - P)Y' = -\frac{t}{u}zL, \quad \partial_t \partial_{\bar{t}} \psi = \frac{|t|^2}{|u|^2} \mathbf{c}_F(H(t)). \quad (\text{RCX25})$$

For completeness, $Y' = -Y A_F/u - t\Pi e L/u$ gives the first. Holomorphy gives $\partial_t H_F = Y^* Y'$; differentiate the inverse in $\partial_t \log \det H_F = \text{Tr}(H_F^{-1} Y^* Y')$. The result is $\text{Tr}(H_F^{-1} Y'^* (1 - P)Y')$. Substitution of the first equation, (RCX10), and the period quotient-norm minimum gives the second.

All entries are real analytic and $H_F > 0$, so ψ has a convergent expansion in $t, \bar{t}$ near zero. This can also be seen by treating the two variables independently in $\det(I^* \Pi(\bar{s})^* \Pi(t) I)$; its value at $(0, 0)$ is positive, so it has a holomorphic logarithm on a sufficiently small polydisc. Equation (RCX25) then gives

$$\begin{aligned} \psi_{t\bar{t}}(0) = \psi_{t\bar{t}\bar{t}}(0) = \psi_{t\bar{t}\bar{t}\bar{t}}(0) = 0, \quad \psi_{t\bar{t}t\bar{t}}(0) = \psi_{t\bar{t}\bar{t}t\bar{t}}(0) = 0, \\ \boxed{\psi_{t\bar{t}\bar{t}\bar{t}}(0) = \mathbf{c}_F(H(0))/|u|^2 > 0, \quad [t^2 \bar{t}^2] \psi = \mathbf{c}_F(H(0))/(4|u|^2).} \end{aligned} \quad (\text{RCX26})$$

The first row is obtained by applying the indicated derivatives to $t\bar{t} \mathbf{c}_F(H(t))/|u|^2$ and evaluating at zero. The factorial in the second row is $2!2! = 4$.

Here is the exact map to real-axis derivatives, including the terms a mixed derivative does not determine. Let

$$\phi(t) = \log \det(I^* \Pi(0)^* \Pi(t) I), \quad \phi(0) = \log \det H_F(0) \quad (\text{RCX27})$$

on its local nonzero determinant neighbourhood. Its derivatives of positive order are the pure t -derivatives of ψ . For $x \in \mathbb{R}$, $d/dx = \partial_t + \partial_{\bar{t}}$. Reality of ψ , the binomial theorem for commuting derivatives, and (RCX26) prove

$$\begin{aligned} \left. \frac{d^2}{dx^2} \psi(x) \right|_0 &= 2 \operatorname{Re} \phi''(0), \\ \left. \frac{d^4}{dx^4} \psi(x) \right|_0 &= 2 \operatorname{Re} \phi^{(4)}(0) + 6 \frac{\mathbf{c}_F(H(0))}{|u|^2}. \end{aligned} \quad (\text{RCX28})$$

Thus the mixed observation and the real derivatives are related by these exact equations, with the pure holomorphic jets retained. In particular the first equation and $\Pi''(0) = \Pi(0)A^2/u^2 - \Pi(0)R/u$ give

$$\phi'(0) = -\operatorname{Tr} A_F/u, \quad \phi''(0) = -\frac{1}{u} L H_F(0)^{-1} I^* H(0) e. \quad (\text{RCX29})$$

To check the second identity, the second logarithmic derivative is the trace of the second matrix derivative minus the square of the first logarithmic matrix derivative. The A_F^2/u^2 terms cancel exactly, leaving the displayed rank-one trace.

72.1.6 Every period derivative and the forced Laplacian

All higher pure jets in (RCX28) are given by an explicit finite noncommutative recursion in the original A, R, u . Set

$$Q_0(t) = I, \quad Q_{n+1}(t) = Q'_n(t) - \frac{1}{u}(A + tR)Q_n(t). \quad (\text{RCX30})$$

Induction using the product rule and the period equation proves $\Pi^{(n)}(t) = \Pi(t)Q_n(t)$ for every $n \geq 0$. Every Q_n is a polynomial in t with coefficients in the algebra over $\mathbb{C}[u, u^{-1}]$ generated by

A, R . In particular, with $A_t = A + tR$,

$$\begin{aligned} Q_1 &= -A_t/u, & Q_2 &= A_t^2/u^2 - R/u, \\ Q_3 &= -A_t^3/u^3 + (RA_t + 2A_tR)/u^2, \\ Q_4 &= A_t^4/u^4 - (RA_t^2 + 2A_tRA_t + 3A_t^2R)/u^3 + 3R^2/u^2. \end{aligned} \quad (\text{RCX31})$$

The order of RA_t and A_tR is retained: differentiating A_t^2 gives $RA_t + A_tR$, and the product $-A_tQ_2/u$ contributes one additional A_tR/u^2 . Differentiating Q_3 gives $-(RA_t^2 + A_tRA_t + A_t^2R)/u^3 + 3R^2/u^2$; the additional $-A_tQ_3/u$ contributes $A_t^4/u^4 - (A_tRA_t + 2A_t^2R)/u^3$. Adding them proves the displayed Q_4 , including the $3R^2$ term. For $q > 1$, $\ell e = 0$ and $R^2 = 0$; for $q = 1$, $R = I$ and that term remains $3I/u^2$.

To evaluate the remaining fourth jet entirely from the displayed data, put

$$D_n = H_F(0)^{-1} I^* H(0) Q_n(0) I \quad (1 \leq n \leq 4).$$

Then

$$\phi^{(4)}(0) = \text{Tr}(D_4 - 4D_1D_3 - 3D_2^2 + 12D_1^2D_2 - 6D_1^4). \quad (\text{RCX32})$$

Indeed the local matrix in (RCX27), after left multiplication by $H_F(0)^{-1}$, is $I + \sum_{n=1}^4 D_n t^n/n! + O(t^5)$. The convergent trace-log expansion $\text{Tr} \log(I + Z) = \sum_{m \geq 1} (-1)^{m+1} \text{Tr}(Z^m)/m$ follows by differentiating both sides near zero and using their common value zero at $Z = 0$. Collecting the t^4 coefficient gives respectively D_4 , $-4D_1D_3 - 3D_2^2$, $12D_1^2D_2$, and $-6D_1^4$ after multiplication by $4!$. Cyclic trace, rather than a commutation assumption, combines the three cubic terms. This proves (RCX32) and completely evaluates the real fourth derivative in (RCX28).

For the original tensor degree $k \geq 1$, let $L_t = A_t^2 - kA_t$. The following polynomial matrix identity also holds for every real k , as a separate algebraic parameter. Equation (RCX31) proves the exact differential realization

$$\boxed{u^2 \Pi'' + ku \Pi' + u \Pi R = \Pi L_t.} \quad (\text{RCX33})$$

The added term is the same $R = e\ell$ whose complete algebra and normal action were computed above. On the admitted source domain stable under $D^{(k)}$ and its square, let $\hat{R} = r_N R j_E$ and $Y_x(t) = \Pi(t) j_E x$ for a fixed source vector x . Applying (RCX33) at zero and twice using $j_E D^{(k)} = A j_E$ gives

$$(u^2 \partial_t^2 + ku \partial_t) Y_x|_0 + u \Pi(0) j_E \hat{R} x = \Pi(0) j_E ((D^{(k)})^2 - k D^{(k)}) x. \quad (\text{RCX34})$$

Every term has target $\mathbb{C}^q$ with its specified period coordinates; the source vector remains in the same admitted domain.

The primitive also has a complete formula. Let $\mathcal{C}_{\text{prim}}$ be the original primitive source, $D_0 : \mathcal{C}_{\text{prim}} \rightarrow \mathcal{C}_{\text{prim}}$ its Euler action, and retain

$$D^{(k)} d = d D_0, \quad D^{(k)} r_N - r_N A = dK, \quad K : E \rightarrow \mathcal{C}_{\text{prim}}.$$

Put $D_t = D^{(k)} + t\hat{R}$. Equations (RCX8) give $D_t d = d D_0$ and $D_t r_N = r_N A_t + dK$. Therefore direct expansion gives

$$\boxed{(D_t^2 - k D_t) r_N - r_N L_t = d(D_0 K + K A_t - k K).} \quad (\text{RCX35})$$

Indeed $D_t(r_N A_t) = (r_N A_t + dK) A_t$ and $D_t dK = d D_0 K$; subtract $k(r_N A_t + dK)$. This reproduces the complete primitive (LC12a) and attaches it to the new forced period equation. In particular the term tKR in $K A_t$ is present with its original sign.

72.1.7 Translation of the marked curvature and first derivative

For a fixed complex a , keep the actual substitution data

$$\begin{aligned} \chi_a(S) &= \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \\ (C_a)_{rb} &= \binom{b}{r} a^{b-r} \quad (r \leq b), \quad f_a(t) = \exp((- \Phi(-a) - ta)/u). \end{aligned} \quad (\text{RCX36})$$

The zero entries for $r > b$ are retained. The literal substitution $P(S) \mapsto P(X + a)$ gives $C_a^{-1} = C_{-a}$, $\det C_a = 1$, $C_a e = e$, and $\ell C_a = \ell$, because translation preserves the highest coefficient in degree below q . It follows in particular that $C_a R C_a^{-1} = R$. Also $C_a A_a(t) C_a^{-1} = A_t + aI$.

The original oriented exponential contour integral gives

$$\Pi_a(t) = f_a(t) \Pi(t) C_a. \quad (\text{RCX37})$$

To retain its analytic justification, substitute $S = X + a$. The exponent becomes $(\Phi(X) - tX - \Phi(-a) - ta)/u$, and the polynomial factor becomes $(X + a)^b$. The substituted contour is the original ray contour translated by $-a$. Truncate at radius R_0 and connect the corresponding end-points by segments of length $|a|$. Along each segment the real part of the leading exponent is $-R_0^{q+1}/((q+1)|u|) + O(R_0^q)$, uniformly on the segment. Its integral tends to zero even with the polynomial factor. Cauchy's theorem applies because the integrand is entire; the finite connectors cancel with the opposite ray orientations. This proves equality with the original contour and yields (RCX37) with every scalar and column coefficient present.

Transport the same constituent by $I_a = C_a^{-1}I$. The exact Gram and coupling relations are

$$\begin{aligned} Y_a &= f_a Y, \quad H_a = |f_a|^2 C_a^* H C_a, \quad H_{F,a} = |f_a|^2 H_F, \\ \mathfrak{c}_{F_a}(H_a) &= \mathfrak{c}_F(H), \quad \log \det H_{F,a} = \log \det H_F + 2p \operatorname{Re} \frac{-\Phi(-a) - ta}{u}. \end{aligned} \quad (\text{RCX38})$$

The coupling equality follows from the complete pullback (RCX12) and its proved scalar law. In the determinant identity the added function is the real part of an affine holomorphic function of t at fixed u , so its $\partial_t \partial_{\bar{t}}$ derivative and every pure or mixed $t, \bar{t}$ derivative of total order at least two vanish. All mixed curvatures, (RCX26), and the real derivatives in (RCX28) are therefore unchanged under this specified translation.

For real $u > 0$, the first real-axis derivative instead gives

$$-u \frac{d}{dx} \log \frac{\det H_{F,a}(x)}{\det(I_a^* C_a^* G_N C_a I_a)} \Big|_0 - kp = 2 \operatorname{Re} \operatorname{Tr} A_F - kp + 2p \operatorname{Re} a. \quad (\text{RCX39})$$

The denominator is exactly $\det(I^* G_N I)$, independent of x , and the numerator derivative follows from (RCX29) and (RCX38). This specifies the complete relation between translation, curvature, and the original trace defect. The original arithmetic object is the marked fibre $a = 0$; the comparison creates no claim that an auxiliary translated polynomial is an arithmetic zero packet.

72.1.8 Every u, t jet and the retained translation derivatives

The recursion (RCX30) now has its full two-parameter version on each original branch of $u \neq 0$. Use the exact matrices $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$ and $\Omega_t = -A_t/u$ from (XD.u1)–(XD.u6). Starting at $Q_{0,0} = I$, define

$$Q_{r+1,s} = \partial_u Q_{r,s} + \Omega_u Q_{r,s}, \quad Q_{r,s+1} = \partial_t Q_{r,s} + \Omega_t Q_{r,s}. \quad (\text{RCX.u1})$$

These prescriptions are compatible. The commutator of their two left-acting differential operators is multiplication by $\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] = 0$. Starting at I they therefore give the same matrix for every ordering with r u derivatives and s t derivatives. The product rule and $\partial_a \Pi = \Pi \Omega_a$ prove

by induction the exact identity $\partial_u^r \partial_t^s \Pi = \Pi Q_{r,s}$. All coefficient matrices remain in the algebra on the same remainder frame. In particular, retaining the order of every product,

$$\begin{aligned} Q_{1,0} &= -C_\chi/u^2 + B_\chi/u, \\ Q_{2,0} &= C_\chi^2/u^4 + (2C_\chi - C_\chi B_\chi - B_\chi C_\chi)/u^3 + (B_\chi^2 - B_\chi)/u^2, \\ Q_{1,1} &= C_\chi A_t/u^3 + (A_t - B_\chi A_t)/u^2. \end{aligned} \quad (\text{RCX.u2})$$

Indeed $\partial_u C_\chi = \partial_u B_\chi = 0$ and $\partial_u(-A_t/u) = A_t/u^2$. Expanding $\partial_u \Omega_u + \Omega_u^2$ and $\partial_u \Omega_t + \Omega_u \Omega_t$ gives exactly these three formulas. Setting $r = 0$ gives every earlier matrix (RCX30)–(RCX32), including its $3R^2/u^2$ term.

For $H = \Pi^* \Pi$ the exact derivative is $\partial_u H = H \Omega_u$ and $\partial_{\bar{u}} H = \Omega_u^* H$. Thus along an arbitrary specified differentiable curve $(u(s), t(s))$ the metric derivative in (RCX13)–(RCX17) is the actual Hermitian matrix

$$\dot{H} = \Omega_{\text{path}}^* H + H \Omega_{\text{path}}, \quad \Omega_{\text{path}} = \dot{u} \Omega_u + \dot{t} \Omega_t. \quad (\text{RCX.u3})$$

Substitution into (RCX13) supplies both original Rayleigh quotients and their exact signed difference, including the full singular u terms. The constituent Hermitian Hessian is (SC.u2) on the same I, H_F , so the pure t fourth-jet identity (RCX26)–(RCX28) is retained as that exact slice. Differentiating the already proved forced equation (RCX33) also gives its complete u descendant:

$$2u\Pi_{tt} + u^2\Pi_{utt} + k\Pi_t + ku\Pi_{ut} + \Pi R + u\Pi_u R = \Pi \Omega_u L_t. \quad (\text{RCX.u4})$$

The coefficient factors and every derivative term follow from the product rule; L_t and R are independent of u . Composing with the fixed original j_E gives the exact same identity on its admitted theta-source domain. Its original Euler-action connection defect is precisely (DS74), so no boundary primitive is discarded by this extension.

For the translated phase retain $h_a(t) = -\Phi(-a) - ta$. The exact identity (RCX38) holds in both parameters:

$$\psi_a(u, t) - \psi(u, t) = 2p \operatorname{Re}(h_a(t)/u). \quad (\text{RCX.u5})$$

This is pluriharmonic on $u \neq 0$, so all mixed holomorphic–antiholomorphic derivatives of this difference vanish. Its pure u derivatives are explicitly retained:

$$\partial_u(\psi_a - \psi) = -ph_a(t)/u^2, \quad \partial_u^2(\psi_a - \psi) = 2ph_a(t)/u^3, \quad \partial_u \partial_t(\psi_a - \psi) = pa/u^2. \quad (\text{RCX.u6})$$

They follow by differentiating $p(h_a/u + \overline{h_a/u})$ in Wirtinger coordinates, with $\partial_t h_a = -a$. Consequently the earlier vanishing of derivatives of order at least two concerns the fixed- u t derivatives only. The full translated singular connection is the proved map (DT.14)–(DT.16), and these additional pure derivatives are carried into (RCX.u1)–(RCX.u4) through that exact map.

72.1.9 Source chronology and exact comparison scope

The delivered note’s historical statement about translation source hashes is resolved by the pre-existing sealed translation manifest. The reviewed and final source SHA-256 values, respectively, are

```
39163624c54704c203a69481ff4d7df4262afa108ac1e0989171d5ba8d700c42
bd1831cda196602d34afc1fa5039d9d80041711fba72bc8b45c782f67c25bade
```

The exact one-field change is the geometry margin from 25 to 23 millimetres; reversing this one byte-string substitution reproduces the reviewed hash. The original dated seal records that verification, its four-page visual review, and the later README delivery. The raw delivered note remains unchanged.

The new finite suite checks actual examples of the residue basis, the transverse map, the mixed jet with factorial, the metric inequalities, the forced Laplacian, and translation. These examples supplement the full proofs above. The universal endomorphism identity, exact differential (RCX15), endpoint bound (RCX21), and source primitive (RCX35) supply explicit relations to the original coefficient, metric, and cochain objects. They do not provide an upper estimate uniform over the remaining arithmetic endpoint family; the original determinant quantity remains explicit in (RCX21) and (RCX24).

The concurrent finite-sampling calculation (FC1)–(FC23) acts on these same original forms. Its exact quotient Gram is $G_N = (J_N M_N^{-1} J_N^*)^{-1}$, and its finite-circle Gram is $H_N = (J_N M_N(L, J)^{-1} J_N^*)^{-1}$. The supplied source-error calculation gives the actual constants $\alpha = 1 - \eta_P - \eta_T > 0$, $\beta = 1 + \eta_P$, with $\alpha G_N \preceq H_N \preceq \beta G_N$. Here is the complete finite-dimensional map to our observation: minimizing both forms on $e + I(F)$ gives $\alpha q_{G_N} \leq q_{H_N} \leq \beta q_{G_N}$; restricting to $I(F)$ and reversing inverse order gives $\beta^{-1} d_{G_N} \leq d_{H_N} \leq \alpha^{-1} d_{G_N}$. Thus

$$\frac{\alpha}{\beta} \mathbf{c}_F(G_N) \leq \mathbf{c}_F(H_N) \leq \frac{\beta}{\alpha} \mathbf{c}_F(G_N). \quad (\text{RCX40})$$

This is exactly (FC9) in our retained unit/residue notation. Equations (RCX13)–(RCX17) further evaluate the full differential along the straight path between these particular forms; no commutation between the form and the generator is inserted. The original full-volume observer and all constituent minors remain those of (RCX21)–(RCX24).

73 Holonomy descent: exact public source reading

73.1 Reading the holonomy delivery with its exact domains and verification

The complete original H1–H66 derivation, including H49a–H49g, is retained with its original editable sources. Its source and quotient are $\Psi_N = \mathcal{U}_k \mathcal{V}|_{\mathcal{P}_N}$ and $J_N : \mathcal{P}_N \rightarrow \mathbb{C}[S]/(\chi_{h,k})$, with the unchanged $g = 2\xi$, unit, tensor degree, and original theta relations.

The phrase following (H49c) concerning the inverse domain is read as the selected closed strip $|\operatorname{Im} z| \leq s$ with $d_s < 1$, and a neighborhood of it. The correlation series itself extends to the larger strip $|\operatorname{Im} z| < aL$. HDI1–10 proves the exact inverse, its compression maps in the original metrics, the contour domain, and finite parameter choices. No raw note has been edited to conceal this clarification.

At fixed phase, the further completed quotient in (H64) uses the positive support of its actual sampled measure. For a selected one-fold root ρ of full multiplicity d_ρ , its weight is exactly

$$\nu_\rho = \frac{1}{L} \left| \frac{g^{(d_\rho)}(\rho)}{h^{(d_\rho)}(\rho)} \right|^2 > 0.$$

HDI11–21 retains the leading coefficient, every original Fourier amplitude, the kernel $(p_\theta)/(\chi)$, and the measurable direct-integral map with kernel all of E . The continuous-cochain comparison supplies the accompanying full source complex. Their zero coefficient outputs lift to represented zeros on the original labels, with external absence fixed. HDI22–24 gives a compactly supported source attaining the exact quadratic loss with literal mass 7; its explicit source map specifies the calibration's scope.

Fresh verification comprises the 22 newly supplied methods in ordinary and optimized Python, plus three separate substantive mutations in each mode. All positive methods pass, and all six mutations reach their intended formula or map rejection. The original negative launcher has an unconditional failure fallback, and its support branch compares fixed tuples. The complete raw code and historical logs are preserved; the independent controls and full execution receipts show the actual

corrected verification. Historical predecessor checks remain historical and are not counted as new runs.

The full original note, HDI, the continuous-cochain proof, and the unchanged-generator companion contain the mathematics. The original arithmetic volume upper estimate remains unproved by this delivery; no finite regression is represented as that estimate or as a proof of RH.

74 The complete original holonomy-descent control note

Source role. Complete original H1–H66 and H49a–H49g, reproduced once. The full public wrapper and HDI4–8 supply the exact selected-strip correction; no original formula or wording is replaced. This complete source appears once with every original tag token retained.

Exact revision. 900e7c16369a58093b16a03dbc449d41a100ee45dbe8515e30bc026e660ba393

Complete source. `sources/holonomy_complete_closure/intake/sources/holonomy_original/NOTE.md`

Source SHA-256. 900e7c16369a58093b16a03dbc449d41a100ee45dbe8515e30bc026e660ba393

Source metadata: author (MetaInlines)

Owner-directed SplitZero research continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: title (MetaInlines)

All-holonomy descent of the original theta quotient

74.1 Result and scope

The original arithmetic source is recovered exactly by the complete family of twisted circle periodizations. The new calculation retains what happens when minimization over the original theta relations is performed in that family. There is a positive, explicitly calculated difference between the original quotient metric and the average of the fibre quotient metrics. It is the norm of a family of original relations, not an unspecified error.

If every phase-source Gram lies between $(1 - \eta)M$ and $(1 + \eta)M$, where M is their exact mean and $0 \leq \eta < 1$, the calculated comparison is

$$(1 - \eta^2)G \preceq \overline{G} \preceq G, \quad G - \overline{G} = \int (C - C_\theta)^* M_\theta (C - C_\theta) \frac{d\theta}{2\pi}. \quad (\text{Holonomy H1})$$

Here C and C_θ are minimum-norm sections of the **same** original polynomial quotient. Every difference $C - C_\theta$ has its actual theta primitive. The factor η^2 is sharp for the stated matrix hypotheses. This is an application of positive-matrix variational methods, not a priority claim for operator concavity.

The family is joined to finite cyclic covers, the new residue-constituent coefficient, and the original endpoint-volume criterion. All these constructions remain over the marked split base. No finite-field weight theorem is applied to a Hilbert circle quotient. No uniform arithmetic upper bound contradicting the inherited quartet threshold is proved in this note.

74.2 1. Fixed inputs, maps, and the marked split base

The owner's scalar object, its support observation, and its arithmetic observation are retained:

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r, \quad b_R(\tau) = 0_{\mathbb{B}}, \quad b_R(r^\bullet) = 1_{\mathbb{B}}. \quad (\text{Holonomy H2})$$

The map (p_R, b_R) is injective. The arithmetic coefficient square is

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (\text{Holonomy H3})$$

The target of $p_{\mathbb{Z}}$ is infinite. The pointed-monoid base has structure map $\{\tau_0, 1\} \rightarrow U_{\times} G(R)$, taking τ_0 to τ and 1 to $1^{\bullet}$. Direct image to its point retains the coefficient object; it is not replaced by its Boolean localization.

On a fixed original support label ℓ , a coefficient-linear map f lifts by

$$(\ell, v) \mapsto (\ell, fv), \quad \tau \mapsto \tau. \quad (\text{Holonomy H4})$$

For a relation submodule B_{ℓ} , the quotient is $(\ell, v) \mapsto (\ell, [v])$. Its equality criterion is $v - v' \in B_{\ell}$. Thus $f(v) = 0$ gives the receiving fibre zero, not external absence. Every map constructed below has this lift. Products of phase fibres use the original product support diagram; on a common label the scalar Fourier matrices give linear equivalences. For unequal labels, sums first use the stated joins and transports. No injectivity of arbitrary support transports is assumed.

Retain the original theta data

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int \phi = 0\}, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi. \quad (\text{Holonomy H5})$$

The prior full-jet analytic construction supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$, for a selected finite packet h with every selected order complete. The present finite arithmetic module is

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k} > 0, \quad A = M_S, \quad S = s_1 + \cdots + s_k. \quad (\text{Holonomy H6})$$

Use its same source map and full inclusion:

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_{\chi}, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_{\chi}, \quad (\text{Holonomy H7})$$

$$\eta[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h). \quad (\text{Holonomy H8})$$

The unit in (H8), every nilpotent jet, and $g = 2\xi$ stay in these equations. For $N \geq q - 1$, let $P_N = \mathbb{C}[S]_{\leq N}$, $J_N = \pi_{\chi}|_{P_N}$, and $B_N = \ker J_N = \chi P_{N-q}$, where $P_{-1} = 0$. Fix their coefficient bases. A matrix denoted B below is the injective coefficient matrix for multiplication by this literal χ ; it is not a replacement of a higher contracted relation ideal by a power of χ .

The new source corrections are incorporated as follows. The supplied primary-source correction resolves the reference at Weil II 3.7.4 as Roman I.8.11 in Weil I. The strict bound in 3.2.10, the square in 3.2.13, and the lisse/mixed hypotheses in 1.8.12 remain. The supplied translation note keeps its scalar $\exp((- \Phi(-a) - ta)/u)$; the marked arithmetic fibre remains $a = 0$. The supplied residue note's exact rank-one and curvature identities are inherited inputs in Section 8, not discoveries of this continuation.

74.3 2. All circle holonomies recover the full source

Retain the predecessor's exact logarithmic coordinates

$$r = \log x_k, \quad z_i = \log x_i - \log x_k \ (i < k), \quad (\mathcal{U}_k F)(r, z) = e^{(kr + \sum z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r). \quad (\text{Holonomy H9})$$

Their logarithmic Jacobian is one. With $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$, this is an isometry into $L^2(\mathbb{R}, dr; \mathcal{K})$, and

$$\mathcal{U}_k D^{(k)} = (-\partial_r + k/2)\mathcal{U}_k. \quad (\text{Holonomy H10})$$

No relative variable is integrated out of the vector-valued source. Integrals in this note are defined inside each specified coefficient fibre, then lifted by (H4); no unconstructed integration operation on arbitrary mixed-support carriers is assumed. For $L > 0$ and $0 \leq \theta < 2\pi$, define

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad 0 \leq r < L. \quad (\text{Holonomy H11})$$

The original finite sources have sufficient weighted Schwartz bounds to differentiate and sum this expression on compact sets. It extends to the Hilbert-space transform by density. Its boundary and Fourier equations are

$$\mathcal{Z}_{L,\theta}\psi(r+L) = e^{-i\theta}\mathcal{Z}_{L,\theta}\psi(r), \quad \widehat{\mathcal{Z}_{L,\theta}\psi}(n) = L^{-1/2}\widehat{\psi}((2\pi n + \theta)/L), \quad (\text{Holonomy H12})$$

where $\widehat{\psi}(u) = \int \psi(r)e^{iur}dr$, and the target basis is $L^{-1/2}e^{-i(2\pi n + \theta)r/L}$. To check the factor, substitute $y = r + aL$ in the coefficient integral. The two phase factors give $e^{ia\theta}e^{-ia(2\pi n + \theta)} = 1$.

Exact isometry and inverse. The full transform is

$$\mathcal{Z}_L : L^2(\mathbb{R}; \mathcal{K}) \xrightarrow{\sim} \int_{[0, 2\pi)}^{\oplus} L^2([0, L]; \mathcal{K}) \frac{d\theta}{2\pi}. \quad (\text{Holonomy H13})$$

For $0 \leq r < L$ and $a \in \mathbb{Z}$, its inverse is

$$\psi(r + aL) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ia\theta} (\mathcal{Z}_L\psi)(r, \theta) d\theta. \quad (\text{Holonomy H14})$$

For each r , these are precisely the Fourier-series map and inverse on the sequence $(\psi(r + aL))_a$. Parseval followed by integration in r proves (H13). The measure $d\theta/(2\pi)$ is the actual Fourier inversion measure; no arithmetic mass has been assigned the value one.

For the fixed source-column map $\Psi_N = \mathcal{U}_k \mathcal{V}|_{P_N}$ put

$$M_N(\theta) = (\mathcal{Z}_{L,\theta}\Psi_N)^*(\mathcal{Z}_{L,\theta}\Psi_N), \quad M_N = \Psi_N^* \Psi_N. \quad (\text{Holonomy H15})$$

Then, exactly,

$$\int_0^{2\pi} M_N(\theta) \frac{d\theta}{2\pi} = M_N. \quad (\text{Holonomy H16})$$

The original norm is $\int |P(k/2 + iu)|^2 m_{h,k}(u) du$, where $m_{h,k} = w_h^{*k}$ and $w_h = |(g/h)(1/2 + iu)|^2/(2\pi)$. Partial Plancherel therefore gives

$$M_N(\theta)_{ab} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}((2\pi n + \theta)/L) \overline{P_a(k/2 + i(2\pi n + \theta)/L)} P_b(k/2 + i(2\pi n + \theta)/L). \quad (\text{Holonomy H17})$$

For $k \geq 2$ all these scalar weights are positive. Infinitely many distinct sample points then imply positive definiteness on every finite polynomial source. The uniform quantitative statements below instead use the displayed relative source bound, which by itself implies positive definiteness, including other admitted sources.

At $\theta = 0, n = 0$, retain the term $L^{-1}\widehat{\Psi}_N(0)^*\widehat{\Psi}_N(0)$. It is not discarded in (H15)–(H17). The phase variable is a character of the translation lattice. It is not the coefficient translation a of the exponential note: χ, A, η , and g are fixed throughout (H11)–(H17).

74.4 3. Every finite cyclic cover is an exact character decomposition

Let $m \geq 1$ and $\theta_j = 2\pi j/m$. For a function f on the circle of length mL , define

$$Q_j f(r) = \frac{1}{m} \sum_{a=0}^{m-1} e^{ia\theta_j} f(r + aL). \quad (\text{Holonomy H18})$$

Finite character sums prove $Q_j Q_l = \delta_{jl} Q_j$, $\sum_j Q_j = I$, and $Q_j f(r+L) = e^{-i\theta_j} Q_j f(r)$. The actual unitary map and inverse are

$$f \mapsto (\sqrt{m} Q_j f|_{[0, L]})_{j=0}^{m-1}, \quad f(r + aL) = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} e^{-ia\theta_j} g_j(r). \quad (\text{Holonomy H19})$$

The norm follows either by the finite Fourier identity or orthogonality of the Q_j on the long circle. Applied to the same original source,

$$Q_j \mathcal{Z}_{mL,0} \psi = \frac{1}{m} \mathcal{Z}_{L,\theta_j} \psi, \quad M_N(mL, 0) = \frac{1}{m} \sum_{j=0}^{m-1} M_N(L, \theta_j). \quad (\text{Holonomy H20})$$

The second equation is exact at every finite degree; the individual masses are not reset. For a general phase θ , the compatible refinement is

$$M_N(L, \theta) = \frac{1}{n} \sum_{a=0}^{n-1} M_N(L/n, (\theta + 2\pi a)/n). \quad (\text{Holonomy H21})$$

It follows equally by partitioning the sampled frequency lattice into congruence classes. Repeated refinements compose because these are literal partitions of the same lattice.

74.4.1 The finite locally free coefficient cover

Here is the algebraic cover that gives the finite characters, rather than an identification with an arithmetic Frobenius. Let

$$A_{\text{cov}} = \mathbb{C}[z, z^{-1}], \quad B_{\text{cov}} = \mathbb{C}[w, w^{-1}], \quad z \mapsto w^m. \quad (\text{Holonomy H22})$$

The pushforward coefficient module is free with its full basis $1, w, \dots, w^{m-1}$. Division of an integer exponent by m proves this also for negative powers. For $\zeta = e^{2\pi i/m}$, the deck action is $\gamma_a(w) = \zeta^{-a}w$. The projector $m^{-1} \sum_a \zeta^{ja} \gamma_a$ has image $A_{\text{cov}} w^j$; multiplication of those summands retains the extra factor z when indices cross m .

The derivative $m w^{m-1}$ is invertible. This proves this particular cover is finite étale, hence finite locally free. EGA I, 1.6.3 and 9.2, and EGA II, 6.1 describe the affine direct-image and finite-morphism setting; here the module and projectors are written explicitly. The base-change map is

$$B_{\text{cov}} \otimes_{A_{\text{cov}}} A' \xrightarrow{\sim} A'[w, w^{-1}]/(w^m - z'), \quad b \otimes a' \mapsto a'b(w). \quad (\text{Holonomy H23})$$

The circle restriction is $w = e^{-2\pi i r/(mL)}$, $z = e^{-2\pi i r/L}$ and yields (H18). Applying G to the ring map in (H22) commutes with (H3), and fixes τ and e . Character projectors act linearly on the coefficient fibres. A coefficient projected to zero is a represented fibre zero. These idempotent endomorphisms are not relabelled as the supported-zero scalar e .

This is a use of the finite-cover and same-image architecture, not a proof of arithmetic weight bounds from finite étaleness. No six-functor theorem for the split semiring category is asserted.

74.5 4. Common arithmetic representatives and the exact descent cost

Suppress N for this section. Let $J : P_N \rightarrow E$ be the original surjection, let $B : \mathbb{C}^r \hookrightarrow P_N$ be the coefficient matrix of $\chi_{P_{N-q}}$, and retain $M = \int M_\theta$. For each phase and the common source, define

$$G = (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \quad G_\theta = (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta. \quad (\text{Holonomy H24})$$

These formulas hold at $N = q - 1$ too: the relation column space then has dimension zero and the corresponding correction is zero. The empty-packet coefficient module has empty Gram determinant one and is handled separately; it does not erase the nonzero analytic seed.

The defining equations are

$$JC = JC_\theta = I_E, \quad B^*MC = 0, \quad B^*M_\theta C_\theta = 0. \quad (\text{Holonomy H25})$$

Consequently there is a unique coefficient map $X_\theta : E \rightarrow \mathbb{C}^r$ with

$$C - C_\theta = BX_\theta, \quad X_\theta = (B^*M_\theta B)^{-1}B^*M_\theta C. \quad (\text{Holonomy H26})$$

The last formula follows by applying B^*M_θ to the first. Thus every discrepancy is in the original relation submodule, and monic division followed by the existing tensor theta primitive realizes it in the original cochain complex.

Set

$$\overline{G} = \int G_\theta \frac{d\theta}{2\pi}, \quad \mathcal{D} = \int X_\theta^*(B^*M_\theta B)X_\theta \frac{d\theta}{2\pi}. \quad (\text{Holonomy H27})$$

The fundamental identity is

$$\boxed{G = \overline{G} + \mathcal{D}.} \quad (\text{Holonomy H28})$$

Proof. Write $C = C_\theta + BX_\theta$ and expand $C^*M_\theta C$. Both cross terms vanish by (H25), leaving $G_\theta + X_\theta^*B^*M_\theta BX_\theta$. Integrating gives $C^*MC = G$. Every integrand is positive semidefinite. Uniform source bounds justify the finite-dimensional integrals and their inverses; for continuous positive definite matrices on the compact phase circle these bounds also exist without a specified value.

Equation (H28) is the exact difference between first taking the common-source quotient and first minimizing separately in each phase. It replaces a proposed interchange of those operations with a calculated relation norm.

74.5.1 The cochain comparison and its full contractible complement

For m phases put

$$\mathcal{P}_{N,m}^{\text{eq}} = \{(P_j) \in P_N^m : JP_j \text{ is independent of } j\}. \quad (\text{Holonomy H29})$$

There are actual complexes in degrees 0, 1,

$$[B_N \hookrightarrow P_N] \xrightarrow{\Delta_m} [B_N^m \hookrightarrow \mathcal{P}_{N,m}^{\text{eq}}], \quad (\text{Holonomy H30})$$

where Δ_m repeats every coefficient. The reverse cochain map is $\text{av}_m(P_j) = m^{-1} \sum_j P_j$ in each degree. It satisfies $\text{av}_m \Delta_m = I$. The complement is

$$\mathcal{B}_{N,m}^0 = \{(b_j) \in B_N^m : \sum_j b_j = 0\}, \quad [\mathcal{B}_{N,m}^0 \xrightarrow{I} \mathcal{B}_{N,m}^0]. \quad (\text{Holonomy H31})$$

The inverse coefficient coordinates are $P_j \mapsto (\overline{P}, (P_j - \overline{P})_j)$ and $(P, (b_j)) \mapsto (P + b_j)_j$. Since all jets agree, each difference lies in B_N . They prove the asserted direct-sum decomposition of complexes, and the identity is the explicit contraction on the complement. Both complexes have the same degree-one E , with original arithmetic inclusion η . In original arithmetic degree k the complement has degrees $k-1, k$, so an endomorphism preserving this decomposition has its two complement traces with opposite cohomological signs.

The maps are strict for the stated polynomial filtration: repeating a polynomial does not lower its degree, and averaging a degree- N tuple remains degree at most N . More explicitly, $\Delta_m(P_N) = \text{im } \Delta_m \cap P_N^m$, and averaging the equal-jet tuples at degree N is onto P_N . The multiplication map $M_S : P_N \rightarrow P_{N+1}$ commutes with both operations; its quotient is the original A . At a fixed cutoff it is a degree-raising map, not a presumed endomorphism of P_N .

The complement's cohomology is zero, but its source norm is present in (H27)–(H28). Its contraction is not asserted isometric. Split reconstruction retains its represented vectors, the corresponding supported-zero maps, and external absence. Thus the exactness on the common image does not supply a missing norm estimate by itself.

74.6 5. Sharp quadratic control of the descent cost

Theorem. Let $M_\theta \succ 0$ be the specified source matrices with mean M , and suppose

$$(1 - \eta)M \preceq M_\theta \preceq (1 + \eta)M, \quad 0 \leq \eta < 1. \quad (\text{Holonomy H32})$$

Then for the original J and (H24),

$$\boxed{(1 - \eta^2)G \preceq \overline{G} \preceq G, \quad 0 \preceq \mathcal{D} \preceq \eta^2 G.} \quad (\text{Holonomy H33})$$

Proof of the upper inequality. It is (H28).

Proof of the lower inequality. For $x \in [1 - \eta, 1 + \eta]$, the complete scalar identity is

$$\frac{2 - x}{1 - \eta^2} - \frac{1}{x} = \frac{\eta^2 - (x - 1)^2}{x(1 - \eta^2)} \geq 0. \quad (\text{Holonomy H34})$$

The endomorphism $X_\theta = M^{-1}M_\theta$ is positive self-adjoint in the retained M -metric and has that spectral interval. Applying (H34) to this finite operator and composing with the displayed metric-dual map M^{-1} proves

$$M_\theta^{-1} \preceq \frac{2M^{-1} - M^{-1}M_\theta M^{-1}}{1 - \eta^2}. \quad (\text{Holonomy H35})$$

Averaging retains $\int M_\theta = M$ and gives

$$\overline{K} := \int JM_\theta^{-1}J^* \frac{d\theta}{2\pi} \preceq (1 - \eta^2)^{-1}K, \quad K = JM^{-1}J^* = G^{-1}. \quad (\text{Holonomy H36})$$

For each phase the block matrix

$$\begin{pmatrix} K_\theta & I_E \\ I_E & G_\theta \end{pmatrix} \succeq 0 \quad (\text{Holonomy H37})$$

is positive: its quadratic form is $(x + G_\theta y)^* K_\theta (x + G_\theta y)$. Averaging and taking the Schur complement gives $\overline{G} \succeq \overline{K}^{-1}$. Inverting (H36) proves the lower inequality of (H33). This proof does not change the original metric, discard the mean condition, or choose a new section to fit a desired form.

More generally, $aM \preceq M_\theta \preceq bM$, $0 < a \leq 1 \leq b$, gives

$$\overline{G} \succeq \frac{ab}{a+b-1}G, \quad G - \overline{G} \preceq \frac{(1-a)(b-1)}{a+b-1}G. \quad (\text{Holonomy H38})$$

The inverse chord is $x^{-1} \leq (a+b-x)/(ab)$ on $[a, b]$; the same proof applies.

These are variational positive-operator calculations related to the classical shorted-operator construction. The contribution here is the exact supported source, common jet map, family of theta primitives, and phase recovery to which they are applied.

74.6.1 The arithmetic hypothesis is supplied by the existing correlation bound

For $a > 0$, retain the two actual weighted Grams

$$M_{a,\pm} = (e^{\pm ar}\Psi_N)^*(e^{\pm ar}\Psi_N), \quad \kappa_{a,N} = \sup_{v \neq 0} \frac{v^*(M_{a,+} + M_{a,-})v}{v^*Mv}. \quad (\text{Holonomy H39})$$

Their original coordinate factors are $x_k^{\pm 2a}$. With $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r+s)dr$, unfolding gives

$$M_N(\theta) = \sum_{b \in \mathbb{Z}} e^{ib\theta} \mathcal{C}(bL). \quad (\text{Holonomy H40})$$

Weighted Cauchy–Schwarz bounds each quadratic correlation by $e^{-a|s|} \sqrt{(v^*M_{a,+}v)(v^*M_{a,-}v)}$. Summing both signs proves, uniformly in phase,

$$-\delta_L M \preceq M_N(\theta) - M \preceq \delta_L M, \quad \delta_L = \frac{\kappa_{a,N}}{e^{aL} - 1}. \quad (\text{Holonomy H41})$$

Thus any specified L with $\delta_L < 1$ supplies (H32). The passage from source error to the averaged quotient error is now quadratic: $\mathcal{D} \preceq \delta_L^2 G$. Growth of $\kappa_{a,N}$ with the arithmetic packet is not bounded by (H41).

74.7 6. Endpoints and a genuinely finite phase construction

Apply the uniform phase bound to one finite source containing all four endpoint spaces, for example the original degree $2q + 2$ source. Restriction gives the same η on every smaller source. Write $V_N = \det G_N$ and $\bar{V}_N = \det \bar{G}_N$.

Equation (H33) proves

$$q \log(1 - \eta^2) \leq \log \bar{V}_N - \log V_N \leq 0. \quad (\text{Holonomy H42})$$

For $\mathcal{B} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$ and its averaged-metric analogue $\bar{\mathcal{B}}$,

$$|\bar{\mathcal{B}} - \mathcal{B}| \leq 2q \log \frac{1}{1 - \eta^2}. \quad (\text{Holonomy H43})$$

The complete equality remains available:

$$V_N = \bar{V}_N \det(I + \bar{G}_N^{-1} \mathcal{D}_N). \quad (\text{Holonomy H44})$$

The endomorphism $\bar{G}_N^{-1} \mathcal{D}_N$ is positive self-adjoint for $\bar{G}_N$. Both its source map and its determinant contribution are specified. In particular, averaging the quotient metrics without the correction in (H44) is not an asserted exact reconstruction.

At m equally spaced phases define

$$M^{(m)} = \frac{1}{m} \sum_j M(L, \theta_j) = M(mL, 0), \quad G^{(m)} = (J(M^{(m)})^{-1} J^*)^{-1}, \quad \bar{G}^{(m)} = \frac{1}{m} \sum_j G(L, \theta_j). \quad (\text{Holonomy H45})$$

Every formula (H24)–(H28) holds exactly with this finite mean and its actual canonical section. To compare this finite construction with the original M , put

$$\delta_m = \frac{\kappa_{a,N}}{e^{amL} - 1}, \quad \nu_m = \frac{\delta_L + \delta_m}{1 - \delta_m}. \quad (\text{Holonomy H46})$$

For $\delta_m < 1$ and $\nu_m < 1$, triangle inequalities relative to M and $M^{(m)}$ give

$$(1 - \nu_m)M^{(m)} \preceq M(L, \theta_j) \preceq (1 + \nu_m)M^{(m)}. \quad (\text{Holonomy H47})$$

The theorem and the original quotient transport then prove

$$(1 - \nu_m^2)(1 - \delta_m)G \preceq \bar{G}^{(m)} \preceq (1 + \delta_m)G. \quad (\text{Holonomy H48})$$

Consequently the finite-phase four-volume error is at most

$$2q \log \frac{1 + \delta_m}{(1 - \delta_m)(1 - \nu_m^2)}. \quad (\text{Holonomy H49})$$

For fixed sufficiently large L , δ_m tends to zero exponentially with the number of phases. The remaining comparison cost tends to the proved quadratic bound, rather than to an unproved zero.

74.7.1 Exponentially convergent quadrature for the quotient itself

The exact finite-cover identity controls the source mean. There is also a direct, quantitative convergence estimate for the mean of the quotient metrics, with the original matrix inversions included.

Let $\kappa_{\pm}$ be the individual largest generalized eigenvalues of $(M_{a,\pm}, M)$. Weighted Cauchy–Schwarz on two distinct vectors proves

$$\|M^{-1} \mathcal{C}(nL)\|_M \leq \sqrt{\kappa_+ \kappa_-} e^{-a|n|L}. \quad (\text{Holonomy H49a})$$

Here $\|\cdot\|_M$ is the operator norm in the original source metric. Choose $s > 0$ with $s < aL$ and retain

$$d_s = \frac{2\sqrt{\kappa_+\kappa_-}}{e^{aL-s} - 1} < 1. \quad (\text{Holonomy H49b})$$

The correlation series defines a holomorphic matrix $\widetilde{M}(z)$ on $|\Im z| < aL$, agreeing with M_θ on the real axis, and $\|M^{-1}(\widetilde{M}(z) - M)\|_M \leq d_s$ on $|\Im z| \leq s$. This is a holomorphic extension of the real-axis Gram coefficients. It is not defined by conjugating the complex parameter in a new Gram formula.

Retain the common section C and original relation columns B from (H24), and define its holomorphic Schur complement

$$\widetilde{G}(z) = C^* \widetilde{M}(z) C - C^* \widetilde{M}(z) B (B^* \widetilde{M}(z) B)^{-1} B^* \widetilde{M}(z) C. \quad (\text{Holonomy H49c})$$

The inverse exists throughout that strip: its difference from $B^* M B$ has relative operator norm at most $d_s < 1$, so the Neumann series converges. On the real axis (H49c) is exactly G_θ . Orthogonality $B^* M C = 0$ gives diagonal bound $1 + d_s$, cross bounds d_s , and inverse bound $(1 - d_s)^{-1}$ in the stated source/quotient metrics. Consequently

$$\|G^{-1} \widetilde{G}(z)\|_G \leq 1 + d_s + \frac{d_s^2}{1 - d_s} = \frac{1}{1 - d_s}. \quad (\text{Holonomy H49d})$$

Shift the contour in its periodic Fourier coefficient to $\Im z = -s$ for positive frequency and $+s$ for negative frequency. The two vertical sides cancel by periodicity. This proves the coefficient bound $(1 - d_s)^{-1} e^{-s|n|}$. Root-of-unity averaging keeps only multiples of m , so summing those remaining coefficients gives

$$\left\| G^{-1} (\overline{G}^{(m)} - \overline{G}) \right\|_G \leq \varepsilon_{m,s} := \frac{2}{(1 - d_s)(e^{sm} - 1)}. \quad (\text{Holonomy H49e})$$

Both real matrices are Hermitian. Combining this norm inequality with the proved quadratic comparison gives the improved finite statement

$$(1 - \eta^2 - \varepsilon_{m,s})G \preceq \overline{G}^{(m)} \preceq (1 + \varepsilon_{m,s})G. \quad (\text{Holonomy H49f})$$

It is useful whenever its displayed lower factor is positive. Unlike a bare claim of convergence of phase sampling, (H49b)–(H49f) specify the source-weight cost, the analytic strip, the matrix inverse domain and the complete geometric-series tail. The finite four-volume error is then at most

$$2q \log \frac{1 + \varepsilon_{m,s}}{1 - \eta^2 - \varepsilon_{m,s}}. \quad (\text{Holonomy H49g})$$

The same maximal finite source supplies constants for all four endpoint restrictions. No off-axis value of ξ has been replaced by a phase character; this is a quadrature estimate for the same canonical quotient construction.

74.7.2 Finite frequency truncation with the phase retained

Write $0 \leq \theta < 2\pi$, $p \geq 1$, and

$$H_{p,N} = \int (1 + r^2) (\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr. \quad (\text{Holonomy H50})$$

Integration by parts and Cauchy-Schwarz prove $\|\widehat{\Psi_N v}(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For an integer $J \geq 2$, retain the frequencies $|n| \leq J$. Since $|n + \theta/(2\pi)| \geq |n| - 1$ on the omitted indices,

$$0 \preceq M(L, \theta) - M(L, \theta; J) \preceq \frac{1}{2p-1} \left(\frac{L}{2\pi(J-1)} \right)^{2p-1} H_{p,N}. \quad (\text{Holonomy H51})$$

The two tails and the factor π/L combine to give exactly this coefficient. A source-relative upper bound $H_{p,N} \preceq h_{p,N} M$ therefore supplies a uniform omitted error ϵ_J . If $\epsilon_J < 1 - \delta_L$, put $\xi_J = \epsilon_J/(1 - \delta_L)$. Then

$$(1 - \xi_J)G(L, \theta) \preceq G(L, \theta; J) \preceq G(L, \theta). \quad (\text{Holonomy H52})$$

Combining either (H48) or (H49f) with (H52) multiplies that lower factor by $1 - \xi_J$ and leaves its upper factor unchanged. The four-volume error is respectively (H49) or (H49g) with the extra term $2q \log(1/(1 - \xi_J))$. Rounded arithmetic samples require an additional supplied enclosure. No unevaluated sample or high-degree matrix has been certified by writing these bounds.

74.8 7. The original action and relation primitive descend together

Multiplication by S maps P_N to P_{N+1} . If $C_{N,\theta}$ is the actual phase section, monic division defines the unique $Q_{N,\theta}$ by

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}. \quad (\text{Holonomy H53})$$

Applying J_{N+1} proves divisibility; this is a finite polynomial identity. For the actual theta map the right-hand side has the inherited original tensor primitive. Denote its image by $\mathcal{B}_{N,\theta}$. The half-density and twisted periodization commute with $D^{(k)}$ as in (H10), so

$$D_{L,\theta}R_{N,\theta} - R_{N,\theta}A = \mathcal{B}_{N,\theta}, \quad D_{L,\theta} = -\partial_r + k/2. \quad (\text{Holonomy H54})$$

On the quasi-periodic domain both arguments have boundary multiplier $e^{-i\theta}$, whose modulus is one. Integration by parts therefore has no residual endpoint term. It gives

$$A^*G_{N,\theta} + G_{N,\theta}A - kG_{N,\theta} = -(R_{N,\theta}^*\mathcal{B}_{N,\theta} + \mathcal{B}_{N,\theta}^*R_{N,\theta}). \quad (\text{Holonomy H55})$$

The direct integral and finite character maps carry these identities with all phases and source primitives present. They do not make the theta boundary orthogonal to the arithmetic representative. The circle spectrum alone does not settle (H55).

The period differential from the uploaded residue note is compatible with this same A : $\Pi' = -\Pi(A + t\mathcal{R})/u$, $\mathcal{R} = 1_E \otimes \ell$, and

$$u^2\Pi'' + ku\Pi' + u\Pi\mathcal{R} = \Pi((A + t\mathcal{R})^2 - k(A + t\mathcal{R})). \quad (\text{Holonomy H56})$$

The extra $u\Pi\mathcal{R}$ is retained. The new phase parameter does not replace t or introduce a translated cyclic polynomial.

74.9 8. Quantitative transport of the new residue-constituent coefficient

The uploaded note proves the following on the marked cyclic fibre. Put

$$\ell([P]) = [S^{q-1}] \text{rem}_\chi P, \quad \beta(x, y) = \ell(xy), \quad \mathcal{R} = 1_E \otimes \ell. \quad (\text{Holonomy H57})$$

The residue form β is perfect, with determinant $(-1)^{q(q-1)/2}$. The maps $A^i\mathcal{R}A^j = (A^i1) \otimes (\ell A^j)$ span all endomorphisms. Thus for a fixed nonzero proper A -invariant inclusion $\iota : F \hookrightarrow E$, both $\pi 1_E$ and $\ell_F = \ell\iota$ are nonzero, and

$$\pi\mathcal{R}\iota = (\pi 1_E) \otimes \ell_F : F \rightarrow E/F \quad (\text{Holonomy H58})$$

has rank exactly one. These are inherited results, including repeated-root primary algebras.

For a positive form M on this same E , retain the coefficient

$$\mathfrak{c}_F(M) = \left(\min_{f \in F} \|1_E - \iota f\|_M^2 \right) \ell_F(\iota^* M \iota)^{-1} \ell_F^*. \quad (\text{Holonomy H59})$$

Apply (H33), without averaging this nonlinear coefficient. Minimizing the first factor and inverting the restriction in the second factor prove

$$(1 - \eta^2) \mathfrak{c}_F(G_N) \leq \mathfrak{c}_F(\overline{G}_N) \leq (1 - \eta^2)^{-1} \mathfrak{c}_F(G_N). \quad (\text{Holonomy H60})$$

In logarithms the absolute error is at most $-\log(1 - \eta^2)$, with no factor q . For the finite construction, if $aG \preceq M_0 \preceq bG$ are the explicit bounds from (H48), (H52), the factors are a/b and b/a .

For $H(t) = \Pi(t)^* \Pi(t)$ the inherited period curvature is

$$\partial_t \partial_{\bar{t}} \log \det(\iota^* H(t) \iota) = \frac{|t|^2}{|u|^2} \mathfrak{c}_F(H(t)). \quad (\text{Holonomy H61})$$

Its original first real derivative, for real $u > 0$, is

$$-u \frac{d}{dt} \log \frac{\det(\iota^* H(t) \iota)}{\det(\iota^* G_N \iota)} \Big|_{t=0} - k \dim F = 2\Re \operatorname{Tr} A_F - k \dim F. \quad (\text{Holonomy H62})$$

Equations (H60)–(H62) connect the new computable source average with the inherited curvature coefficient and the same marked trace observable. They do not convert control of a single coefficient into an upper bound for every endpoint volume. The original period/source condition number is still needed to compare $H(t)$ and G_N .

The translation note's exact map is retained:

$$\Pi_a = f_a \Pi C_a, \quad f_a = \exp((- \Phi(-a) - ta)/u), \quad C_a A_a C_a^{-1} = A + aI. \quad (\text{Holonomy H63})$$

Its nonzero holomorphic factor leaves the curvature in (H61) unchanged but changes the additive trace in (H62) by $2 \dim F \Re a$. No such translation is performed here; the marked value is $a = 0$. On a finite-field specialization the separate constant character line is the one in the supplied note, not an identification with the additive shift.

74.10 9. What premature completion would forget

For a fixed phase, the predecessor's discrete-completion argument gives the map

$$E \longrightarrow \ell^2(\nu_{L,\theta}) / \overline{\chi \mathbb{C}[S]} \simeq \bigoplus_{\chi(k/2 + i(2\pi n + \theta)/L) = 0} \mathbb{C}, \quad (\text{Holonomy H64})$$

with kernel $(p_{L,\theta})/(\chi)$, where $p_{L,\theta}$ is the squarefree product over the sampled roots. This follows from polynomial density for the exponentially weighted discrete measure and its $|\chi|^2$ multiple; it is a further Hilbert quotient, not (H24).

Only finitely many phases modulo 2π can sample a root of the fixed polynomial. Thus the direct integral of the **completed quotient targets** in (H64) is zero almost everywhere, even when the original E is nonzero. Its map from E has kernel all of E . In split coordinates that map sends represented inputs to the zero of their receiving support; it still fixes external absence. At every finite degree in (H24), $J C_\theta = I_E$ and the averaged Gram is positive definite.

The typed order of operations is therefore part of the result: the exact source isometry (H13), the original finite quotient (H24), and the later completion (H64) are accompanied by their actual maps and kernels. None is silently exchanged with another. The contractible complement in (H31) and the positive norm (H28) likewise coexist.

74.11 10. Exact calibrations and the remaining arithmetic estimate

The constant in (H33) is sharp. Take a coefficient source with the literal mass 7,

$$M_\pm = 7 \begin{pmatrix} 1 & \pm \eta \\ \pm \eta & 1 \end{pmatrix}, \quad M = \frac{1}{2}(M_+ + M_-) = 7I_2, \quad J = (1 \ 0), \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (\text{Holonomy H65})$$

These are also moment matrices of the two measures on the points $-1, 1$ with masses $7(1 \mp \eta)/2$ and $7(1 \pm \eta)/2$. Direct minimization gives

$$C = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C_+ = \begin{pmatrix} 1 \\ -\eta \end{pmatrix}, \quad C_- = \begin{pmatrix} 1 \\ \eta \end{pmatrix}, \quad G = 7, \quad G_{\pm} = 7(1 - \eta^2), \quad \mathcal{D} = 7\eta^2. \quad (\text{Holonomy H66})$$

Every difference is in the original kernel of J . This is a finite positive-source calibration, not a zeta packet. It proves that the quadratic loss cannot be reduced under only (H32) and the mean condition.

The new exact checker includes repeated-root polynomial quotient fixtures, complex Gram cross terms, finite character projectors, the cochain complement and contraction, source-mean equality, and the precise residue-coefficient comparison. It uses exact rational or Gaussian-rational arithmetic. Its finite checks do not certify the infinite theta integrals, the period determinant argument, or a general Lean theorem.

The arithmetic endpoint issue remains the inherited threshold-four problem for the fixed hypothetical quartet: the original source would require $\mathcal{B}_{h,k} \geq 4q_k \log(D_h k)$. Equation (H43) and its finite version transport that threshold with an explicit $O(q_k)$ allowance when the supplied error factors are fixed. They do not prove a contradicting upper bound for $\mathcal{B}_{h,k}$.

The progress here is a stronger, source-preserving descent: all holonomies recover the source exactly; finite covers assemble those holonomies coherently; the failure of independently minimized local representatives to glue is an explicit theta-relation norm; its effect is quadratically bounded; and the same bound reaches the new residue-curvature coefficient. The original complex, infinite arithmetic quotient, support labels, arithmetic action, and full jets remain throughout.

74.12 Source basis and attribution

The owner supplies the SplitZero, theta, canonical full-jet, periodization, and residue-constituent programme. This note supplies the displayed all-holonomy comparison, exact common-versus-local minimum correction, its sharp quadratic enclosure, and the explicit joins to that programme. No global novelty claim is made for Fourier-series decomposition, finite-cover character projectors, positive-matrix concavity, or Schur complements.

The new uploaded primary-source correction is accepted at its stated review scope. This session read its TeX, not the underlying scan anew. The selected uploaded EGA readings were I, 1.6.3 and 9.2.1–9.2.2, and II, 6.1.1–6.1.5; neither the entire EGA corpus nor the complete Deligne corpus has been audited. The explanatory Deligne sidebar’s comparison, cone, dual, and split-lift sections were read. The standalone Deligne_Split_Cohomology(2).tex is a wrapper; its sibling sidebar was read as the intended substantive input without silently editing its historical include path.

Primary external attribution:

- Deligne, P. (1980). *La conjecture de Weil II*. Publications mathématiques de l’IHÉS 52, 137–252. The supplied correction retains the precise roles of 1.8.12, 3.2.10, 3.2.13, 3.3.4–3.3.6 and 3.7.2–3.7.4.
- Grothendieck, A., and Dieudonné, J. *Éléments de géométrie algébrique*, the selected uploaded editable sections identified above. For the finite locally free definition, see also Stacks Project, Tag 02K9; for affine morphisms and their coefficient algebra, Tag 01S5.
- Zak, J. (1967). Finite translations in solid-state physics. *Physical Review Letters* 19, 1385–1387. DOI 10.1103/PhysRevLett.19.1385. The publisher abstract was inspected for historical attribution; (H11)–(H21) have proofs in this note rather than a claim to have audited the full paper.
- Anderson, W. N., Jr., and Trapp, G. E. (1975). Shorted operators II. *SIAM Journal on Applied Mathematics* 28, 60–71. DOI 10.1137/0128007. The publisher abstract and bibliographic page were inspected; the exact finite inequality used here is proved in (H34)–(H38).

Source files, byte hashes, exact selected ranges, execution receipts, and current-repository inspection are recorded separately. Literature source files and the owner’s unrelated corpus are not republished in this delivery.

75 The complete typed holonomy-descent addendum

75.1 Holonomy descent: the inverse domain, positive root masses, and sharp source realization

This addendum retains the original $\chi = \chi_{h,k}$, its quotient $E = \mathbb{C}[S]/(\chi)$, multiplication $A[P] = [SP]$, the original source g/h with $g = 2\xi$, and every measure in (H9)–(H17). It makes the inverse domain in (H49c) precise, proves the positive-support and shifted-density assertions used in (H64), and realizes the sharp quadratic loss with actual compactly supported source columns. The independent proof review identified the analytic-circle calibration. The full source-density proof PSD1–41 is an accompanying dependency.

75.1.1 The correlation strip and the domain of the quotient inverse

Fix the original source map $\Psi : \mathcal{P}_N \rightarrow L^2(\mathbb{R}; \mathcal{K})$, its Gram $M = \Psi^* \Psi \succ 0$, quotient $J : \mathcal{P}_N \twoheadrightarrow E$, and literal relation injection $B : \mathbb{C}^r \hookrightarrow \mathcal{P}_N$. Define

$$G = (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \quad M_B = B^*MB. \quad (\text{HDI1})$$

Then $JC = I$, $C^*MC = G$, $B^*MC = 0$. For $r = 0$, the relation metric and its inverse are the unique automorphism of $\{0\}$; every correction factoring through B is the zero map on its stated domain and codomain. For $r > 0$, $M_B \succ 0$. Retain $a > 0$, $M_{a,\pm} = (e^{\pm ar}\Psi)^*(e^{\pm ar}\Psi)$, and

$$\kappa_{\pm} = \sup_{v \neq 0} \frac{v^* M_{a,\pm} v}{v^* M v}, \quad K_a = \sqrt{\kappa_+ \kappa_-}, \quad \kappa = \sup_{v \neq 0} \frac{v^* (M_{a,+} + M_{a,-}) v}{v^* M v}. \quad (\text{HDI2})$$

These retain the factors $x_k^{\pm 2a}$ in the original coordinates. For $\mathcal{C}(v) = \int \Psi(r)^* \Psi(r+v) dr$, weighted Cauchy–Schwarz on two vectors gives

$$|\langle x, \mathcal{C}(v)y \rangle| \leq K_a e^{-a|v|} \|x\|_M \|y\|_M, \quad \|M^{-1}\mathcal{C}(nL)\|_M \leq K_a e^{-a|n|L}. \quad (\text{HDI3})$$

For $v > 0$, write the integrand as e^{-av} times the pairing of $e^{-ar}\Psi(r)x$ with $e^{a(r+v)}\Psi(r+v)y$. Their squared integrals are $x^* M_{a,-} x$ and $y^* M_{a,+} y$. For $v < 0$, use the opposite signs. The dual norm in the retained M -metric then proves the operator inequality. Thus

$$\widetilde{M}(z) = \sum_{n \in \mathbb{Z}} e^{inz} \mathcal{C}(nL), \quad d_s = \frac{2K_a}{e^{aL-s} - 1}, \quad \|M^{-1}(\widetilde{M}(z) - M)\|_M \leq d_s \quad (|\operatorname{Im} z| \leq s < aL). \quad (\text{HDI4})$$

The series and its derivative series converge normally on every closed substrip of $|\operatorname{Im} z| < aL$: powers of $|n|$ are absorbed by the remaining geometric decay. The zero coefficient is M ; summing both nonzero signs proves the displayed bound. This constructs a holomorphic, periodic extension of the real Gram coefficients, without conjugating the complex parameter in a Gram.

Choose $0 < s < aL$ with $d_s < 1$. Put $T = M^{-1}(\widetilde{M} - M)$. The maps and their adjoints are

$$\begin{aligned} C : (E, G) &\rightarrow (\mathcal{P}_N, M), & C^\dagger &= G^{-1}C^*M, & C^\dagger C &= I, \\ B : (\mathbb{C}^r, M_B) &\rightarrow (\mathcal{P}_N, M), & B^\dagger &= M_B^{-1}B^*M, & B^\dagger B &= I. \end{aligned} \quad (\text{HDI5})$$

They are isometries onto orthogonal subspaces by (HDI1). Consequently the four compressions of T by B, C have norm at most d_s , with these domain and target metrics. In particular,

$$B^* \widetilde{M} B = M_B(I + B^\dagger T B), \quad (I + B^\dagger T B)^{-1} = \sum_{j \geq 0} (-B^\dagger T B)^j \quad (\text{HDI6})$$

exists and has norm at most $(1 - d_s)^{-1}$ on the selected closed strip. Continuity of d_s , and its strict inequality, give $s' > s$ with $s' < aL$, $d_{s'} < 1$. Hence the inverse is holomorphic on a neighborhood

of the closed strip used by the contour calculation. The phrase *throughout that strip* after (H49c) means this selected strip, not the whole larger correlation strip $|\operatorname{Im} z| < aL$.

The corresponding Schur complement is

$$\tilde{G} = C^* \tilde{M} C - C^* \tilde{M} B (B^* \tilde{M} B)^{-1} B^* \tilde{M} C, \quad \|G^{-1} \tilde{G}\|_G \leq 1 + d_s + \frac{d_s^2}{1 - d_s} = \frac{1}{1 - d_s}. \quad (\text{HDI7})$$

The bound follows from (HDI5)–(HDI6), including the two cross compressions and their actual metrics. On the real axis, subtract $B(B^* M_\theta B)^{-1} B^* M_\theta C$ from C . This remains a section of J and is M_θ -orthogonal to $\ker J$. Adding any relation adds precisely its nonnegative squared norm, so this is the unique minimizing section C_θ , and its Gram is the displayed Schur complement.

For Fourier coefficients $\hat{G}_n = (2\pi)^{-1} \int_0^{2\pi} e^{-in\theta} \tilde{G}(\theta) d\theta$, shift to imaginary part $-s$ if $n > 0$, and $+s$ if $n < 0$. The just-proved neighborhood of holomorphy permits the contour, and periodicity cancels the two vertical sides. The exponential has modulus $e^{-s|n|}$. Therefore

$$\|G^{-1} \hat{G}_n\|_G \leq \frac{e^{-s|n|}}{1 - d_s}, \quad \left\| G^{-1} \left(\frac{1}{m} \sum_{j=0}^{m-1} G_{2\pi j/m} - \overline{G} \right) \right\|_G \leq \frac{2}{(1 - d_s)(e^{sm} - 1)}. \quad (\text{HDI8})$$

Absolute convergence permits averaging term by term. The finite character sum keeps precisely the indices divisible by m ; summing their two nonzero tails proves the second bound.

Explicit finite choices exist. Choose $s_0 > 0$, $0 < d_0 < 1$, $0 < \eta_0 < 1$, $0 < \varepsilon_0 < 1 - \eta_0^2$, and set

$$L = \frac{1}{a} \max \{s_0 + \log(1 + 2K_a/d_0), \log(1 + \kappa/\eta_0)\}, \quad (\text{HDI9})$$

$$m = \max \left\{ 1, \left\lceil \frac{1}{s_0} \log \left(1 + \frac{2}{(1 - d_0)\varepsilon_0} \right) \right\rceil \right\}.$$

The positive weighted source forms give $K_a > 0$, so $aL > s_0$. Monotonicity gives $d_{s_0} \leq d_0$, $\delta_L = \kappa/(e^{aL} - 1) \leq \eta_0$, and error at most ε_0 in (HDI8). With the delivered quadratic theorem,

$$(1 - \eta_0^2 - \varepsilon_0)G \preceq \frac{1}{m} \sum_{j=0}^{m-1} G_{2\pi j/m} \preceq (1 + \varepsilon_0)G. \quad (\text{HDI10})$$

Its lower factor is positive by the actual selection. These are finite source-dependent formulas; they do not bound growth of $M_{a,\pm}$ with the arithmetic packet. The generator companion HG gives the combined phase and frequency budget for its operators.

75.1.2 Selected-root weights and density on the shifted lattice

Fix $L > 0$, $k \geq 1$, and $0 \leq \theta < 2\pi$, and retain

$$u_n = \frac{2\pi n + \theta}{L}, \quad S_n = k/2 + iu_n, \quad \nu_n = \frac{2\pi}{L} m_{h,k}(u_n), \quad I_\theta = \{n : \nu_n > 0\}. \quad (\text{HDI11})$$

The source-density proof PSD1–22 supplies, for each $0 < b < \pi/2$, the actual finite constants and pointwise bound

$$A_b = \|e^{b|\cdot|} w_h\|_\infty, \quad B_b = \|e^{b|\cdot|} w_h\|_1, \quad m_{h,k}(u) \leq A_b B_b^{k-1} e^{-b|u|}. \quad (\text{HDI12})$$

The full PSD source accompanies this addendum; its Gamma factor, leading coefficient, and cancellation calculation are retained. The entire nonzero function g/h has a discrete zero set on the vertical line, so w_h is continuous and positive away from that discrete set. For a fixed u , choose t outside the zero sets of both $w_h(t)$ and $w_h(u - t)$. An interval of strictly positive convolution integrand follows

by continuity, so $m_{h,2}(u) > 0$. Induction integrates the preceding everywhere positive convolution against w_h on an interval of positivity. Thus $I_\theta = \mathbb{Z}$ for $k \geq 2$.

For $k = 1$, use I_θ . At a selected root ρ , let d_ρ be its full multiplicity in both g and h . Then

$$\begin{aligned} g(s) &= (s - \rho)^{d_\rho} a_\rho(s), \quad h(s) = (s - \rho)^{d_\rho} b_\rho(s), \quad a_\rho(\rho) b_\rho(\rho) \neq 0, \\ (g/h)(\rho) &= \frac{a_\rho(\rho)}{b_\rho(\rho)} = \frac{g^{(d_\rho)}(\rho)}{h^{(d_\rho)}(\rho)} \neq 0. \end{aligned} \quad (\text{HDI13})$$

The leading coefficient of h stays in b_ρ and its derivative. In the original monic packet convention $\chi_{h,1} = h$. If a nonzero leading coefficient a_h is displayed, the monic minimal polynomial is $a_h^{-1}h$ and its ideal is (h) ; the analytic source remains exactly g/h . Consequently every sampled root $\rho = S_n$ of $\chi_{h,1}$ has the exact positive weight

$$\nu_n = \frac{1}{L} \left| \frac{g^{(d_\rho)}(\rho)}{h^{(d_\rho)}(\rho)} \right|^2 > 0. \quad (\text{HDI14})$$

This is the full-order cancellation fact that justifies every root index printed in (H64), even for $k = 1$.

Here is the density proof with the phase retained. Let $\mu_n > 0$ on $I \subseteq I_\theta$ have $\sum_I e^{c|u_n|} \mu_n < \infty$ for some $c > 0$, and suppose $v \in \ell^2(I, \mu)$ is orthogonal to all polynomial sequences. For $0 < d < c/2$, Cauchy-Schwarz proves

$$\sum_I |v_n| \mu_n e^{d|u_n|} \leq \|v\|_\mu \left(\sum_I e^{2d|u_n|} \mu_n \right)^{1/2} < \infty. \quad (\text{HDI15})$$

Therefore $F(z) = \sum_I \bar{v}_n \mu_n e^{izu_n}$ is holomorphic for $|\operatorname{Im} z| < c/2$, with all derivative series normally convergent on smaller strips. The powers are absorbed by the remaining exponential margin. The original-coordinate polynomial $((S - k/2)/i)^j$ evaluates to u_n^j , so every derivative at zero vanishes. The identity theorem gives $F = 0$. Extend the coefficients by zero off I . Absolute summability and $u_n - u_m = 2\pi(n - m)/L$ give

$$\frac{1}{L} \int_0^L F(x) e^{-iu_m x} dx = \sum_I \bar{v}_n \mu_n \frac{1}{L} \int_0^L e^{2\pi i(n-m)x/L} dx = \begin{cases} \bar{v}_m \mu_m, & m \in I, \\ 0, & m \notin I. \end{cases} \quad (\text{HDI16})$$

The left side is zero, so $v = 0$. This proves density. Apply it to $\mu = \nu$ on I_θ , and then to $\mu_n = |\chi(S_n)|^2 \nu_n$ where $\chi(S_n) \neq 0$. For both, (HDI12) gives the required exponential moment because a fixed polynomial is absorbed by the difference of two positive exponential exponents.

Put $Z_\theta = \{n \in I_\theta : \chi(S_n) = 0\}$ and $p_\theta = \prod_{n \in Z_\theta} (S - S_n)$. Coordinate evaluation is continuous in $\mathcal{H}_\theta = \ell^2(I_\theta, \nu)$, with bound $\|f_n\| \leq \nu_n^{-1/2} \|f\|_\nu$. Thus the closure of $\chi \mathbb{C}[S]$ vanishes on Z_θ . Conversely, if f vanishes there, division by $\chi(S_n)$ on the complement puts f/χ in the second weighted space, with exactly the same squared norm. Polynomial approximation there, followed by multiplication by χ , approximates f in $\mathcal{H}_\theta$. Consequently

$$\begin{aligned} \overline{\chi \mathbb{C}[S]} &= \{f \in \mathcal{H}_\theta : f_n = 0 \ (n \in Z_\theta)\}, \\ \mathcal{T}_\theta &:= \mathcal{H}_\theta / \overline{\chi \mathbb{C}[S]} \cong \left(\mathbb{C}^{Z_\theta}, \sum_{n \in Z_\theta} \nu_n |a_n|^2 \right), \\ e_\theta : E &\twoheadrightarrow \mathcal{T}_\theta, \quad [P] \mapsto (P(S_n))_{n \in Z_\theta}, \quad \ker e_\theta = (p_\theta)/(\chi). \end{aligned} \quad (\text{HDI17})$$

The isometry is restriction, the orthogonal representative of the Hilbert quotient. The evaluation map is well-defined since χ vanishes at these nodes. The explicit Lagrange polynomial $\prod_{m \in Z_\theta, m \neq n} (S - S_m)/(S_n - S_m)$ proves surjectivity. Distinctness of the nodes proves the divisibility

description of the kernel. The empty product is one and its target is zero. Polynomial cancellation also gives the exact sequence, equivariant for multiplication by S ,

$$0 \longrightarrow \mathbb{C}[S]/(\chi/p_\theta) \xrightarrow{[Q] \mapsto [p_\theta Q]} E \xrightarrow{e_\theta} \mathcal{T}_\theta \longrightarrow 0. \quad (\text{HDI18})$$

No coprimality of $p_\theta, \chi/p_\theta$ is required. At a root λ of multiplicity d_λ , the original primary block is $\mathbb{C}[\epsilon]/(\epsilon^{d_\lambda})$. At a sampled root this map is its constant coefficient, with kernel $\epsilon\mathbb{C}[\epsilon]/(\epsilon^{d_\lambda})$; at an unsampled root the full block is in the kernel. This follows by substituting $S = \lambda + \epsilon$ into evaluation, using the full explicit Chinese-remainder isomorphism in PSD34–35.

For precision about the original source, let $b_n(\theta) = \widehat{\mathcal{U}_k F_h^{\otimes k}}(u_n)$. The source-to-sequence dictionary is

$$\widehat{I_\theta} a(n) = L^{-1/2} a_n b_n(\theta), \quad \|b_n(\theta)\|_{\mathcal{K}}^2 = 2\pi m_{h,k}(u_n), \quad a_n = \sqrt{L} \frac{\langle b_n(\theta), \widehat{f}(n) \rangle}{\|b_n(\theta)\|_{\mathcal{K}}^2} \quad (n \in I_\theta). \quad (\text{HDI19})$$

The Fourier basis is $L^{-1/2} e^{-i(2\pi n + \theta)r/L}$. Partial Plancherel on the retained relative variables gives the middle equality, as proved pointwise in PSD16. Parseval proves the isometry and its inverse on its range. Polynomial density identifies that range with the closed polynomial source. On polynomials the map is exactly $I_\theta(P(S_n)) = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V} P$ by the plus-sign Fourier convention. All amplitudes and the factor $L^{-1/2}$ remain in the comparison.

The measurable field can be specified without an unspecified choice of fibre coordinates. Embed $\mathcal{T}_\theta$ in $\ell^2(\mathbb{Z})$ by $a_n \mapsto \sqrt{\nu_n(\theta)} a_n$ on Z_θ , zero elsewhere. Its range projection is

$$Q_\theta = \text{diag}(1_{\{\nu_n(\theta) > 0, \chi(S_n(\theta)) = 0\}})_{n \in \mathbb{Z}}. \quad (\text{HDI20})$$

Each entry is Borel, since ν_n and $\chi(S_n)$ are continuous. The sections $Q_\theta e_n$ form a countable measurable generating family; approximating any fixed vector by finitely supported vectors proves strong measurability of $Q_\theta v$. A nonzero entry requires a root ρ with $\text{Re } \rho = k/2$ and $\theta \equiv L \text{Im } \rho \pmod{2\pi}$. There are only finitely many such phases. Thus the decomposable projection $f(\theta) \mapsto Q_\theta f(\theta)$ on $L^2(d\theta/(2\pi); \ell^2(\mathbb{Z}))$ is zero, because its squared norm is an integral supported on a finite set. It follows exactly that

$$\int^\oplus \mathcal{T}_\theta \frac{d\theta}{2\pi} = 0, \quad E \longrightarrow \int^\oplus \mathcal{T}_\theta \frac{d\theta}{2\pi}, \quad [P] \mapsto [e_\theta[P]]_{\text{a.e.}}, \quad \ker = E. \quad (\text{HDI21})$$

The section in this map is Borel by its explicit coordinates $1_{Z_\theta}(n) \sqrt{\nu_n(\theta)} P(S_n(\theta))$, and has zero norm. Its pointwise exceptional maps remain (HDI17)–(HDI18) with their positive weights; passage to the almost-everywhere class is the exact arrow removing the measure-zero section. The accompanying continuous-cochain comparison further records this operation on its source complex.

On every original support label ℓ , each coefficient map above lifts as $(\ell, v) \mapsto (\ell, f v)$, with $\tau \mapsto \tau$. In particular (HDI21) takes a represented input to the zero in its receiving fibre, with represented support still present, while external absence remains fixed. The computed Hilbert kernel does not erase that support.

75.1.3 A compactly supported source realizes the full sharp phase loss

Fix $L > 0$, $0 \leq \eta < 1$, and the literal mass 7. Choose two nonzero smooth functions f_0, g_0 with disjoint compact supports in $(0, L)$. Write $\alpha = \int |f_0|^2 > 0$, $\beta = \int |g_0|^2 > 0$, and set

$$f = f_0/\sqrt{\alpha}, \quad g_1 = g_0/\sqrt{\beta}, \quad \psi_0(r) = \sqrt{7} f(r), \quad \psi_1(r) = \sqrt{7} \left(\eta f(r-L) + \sqrt{1-\eta^2} g_1(r) \right). \quad (\text{HDI22})$$

Functions are extended by zero off their supports. The factors α, β explicitly describe the isometric coordinates of this constructed source; its Gram retains the mass 7. Disjoint supports give $\langle f, g_1 \rangle = 0$

and $\langle f, f(\cdot - L) \rangle = 0$, while $\|f\| = \|g_1\| = \|f(\cdot - L)\| = 1$. Direct integration therefore gives $M = 7I_2$ for $\Psi(v_0, v_1) = v_0\psi_0 + v_1\psi_1$. All exponential weighted Grams and derivative moments are finite, because these are smooth compactly supported columns.

On $0 \leq r < L$, only the $a = 0$ terms of f, g_1 and the $a = 1$ term of $f(\cdot - L)$ contribute to the actual holonomy transform $\sum_a e^{ia\theta}\psi(r + aL)$. Hence

$$\mathcal{Z}_{L,\theta}\psi_0 = \sqrt{7}f, \quad \mathcal{Z}_{L,\theta}\psi_1 = \sqrt{7}(\eta e^{i\theta}f + \sqrt{1-\eta^2}g_1), \quad M_\theta = 7 \begin{pmatrix} 1 & \eta e^{i\theta} \\ \eta e^{-i\theta} & 1 \end{pmatrix}. \quad (\text{HDI23})$$

Its mean is exactly $7I_2$. The off-diagonal relative matrix squares to $\eta^2 I_2$, so its relative eigenvalues are $1 - \eta, 1 + \eta$. This is an entire trigonometric matrix family, obtained from the displayed source with the original sign convention for the phase.

Take $J = \begin{pmatrix} 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \end{pmatrix}^T$. Solving $JC_\theta = 1$ and $B^*M_\theta C_\theta = 0$, and integrating the original relation discrepancy, gives

$$C = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C_\theta = \begin{pmatrix} 1 \\ -\eta e^{-i\theta} \end{pmatrix}, \quad X_\theta = \eta e^{-i\theta}, \quad (\text{HDI24})$$

$$G = 7, \quad G_\theta = 7(1 - \eta^2), \quad X_\theta^* B^* M_\theta B X_\theta = 7\eta^2, \quad \mathcal{D} = 7\eta^2.$$

Each formula follows by matrix multiplication, and $C - C_\theta = BX_\theta$ is the actual relation identity. Thus equality in (H33) holds on the full phase circle in this class of smooth compactly supported Zak sources. The explicit arrow $\Psi : \mathbb{C}^2 \rightarrow L^2(\mathbb{R})$ and its quotient J specify this calibration's domain. They do not identify these columns with a particular arithmetic packet $F_h^{\otimes k}$. The arithmetic application continues to use its original $\mathcal{U}_k\mathcal{V}$, J_N, χ and the theta primitives of (H7), (H26), and (H53)–(H55).

76 The complete phase quotient and polynomial density

76.1 Independent completed-phase quotient review

Date: 2026-09-13. Scope: the polynomial-density, completed-relation, kernel, and direct-integral assertions in H64. No source file was edited; no scans, program tests, Lean runs, or commutator calculations were performed.

76.1.1 Reviewed text and conclusion

The review uses the following complete supplied sources. The original relative source names below are retained for provenance; their delivered locations and full source hashes are recorded in the companion source manifest.

The relevant fixed text is:

- `web_holonomy_descent_delivery/Tau_Holonomy_Descent_Control/NOTE.md`, lines 67–90: the complete selected orders, original cyclic polynomial, and Taylor unit; lines 166–177: H17 and positivity for tensor degree at least two; lines 703–715: H64, the asserted kernel, and the almost-everywhere direct-integral statement.
- `web_holonomy_descent_delivery/Tau_Holonomy_Descent_Control/SOURCE_REVIEW.md`: the source-reading scope was read. This review adds no primary-source scan claim.
- `web_periodized_source_delivery/Tau_Periodized_Source_Control/RESEARCH_NOTE.md`, lines 88–113: the full-order packet and original cyclic source; lines 564–584: P53–P54 and the exponential-tail density argument; lines 586–612: P55–P57, the target metric, and the closure calculation; line 633: the positive-support qualification for tensor degree one.

- web_cyclic_sum_delivery/Tau_Cyclic_Sum_Control/RESEARCH_NOTE.md, lines 31–34: the full-order packet and unit $j_h(g/h)$; lines 54–70: the cyclic minimal polynomial and inclusion.
- web_arithmetic_endpoint_delivery/Tau_Arithmetic_Endpoint_Bounds/RESEARCH_NOTE.md, lines 95–105: the full-order packet and the entire quotient defining the actual density.

Conclusion. H64 and its kernel are correct for the stated original full-order arithmetic source at every integer tensor degree $k \geq 1$. For a general nonnegative sampled density, its right side must include only positive-mass sampled roots. At $k = 1$, this qualification makes no change to H64 for the actual source: the ideals $(\chi_{h,1}) = (h)$ agree, and every sampled root has positive mass by full-order cancellation with the original leading coefficient a_h retained. The direct-integral target is zero, with source kernel all of E , in both the general positive-support formulation and its arithmetic specialization. The proof below supplies the explicit metric and the density argument that are abbreviated in H64.

76.1.2 1. Coordinates, mass, and exponential moments

Fix the original integer $k \geq 1$, period $L > 0$, phase $0 \leq \theta < 2\pi$, and the original nonempty full-order packet h . Throughout,

$$c = k/2, \quad u_{n,\theta} = \frac{2\pi n + \theta}{L}, \quad S_{n,\theta} = c + iu_{n,\theta}, \quad n \in \mathbb{Z}, \quad (\text{CQ1})$$

$$v_h = g/h, \quad g = 2\xi, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \nu_{n,\theta} = \frac{2\pi}{L} m_{h,k}(u_{n,\theta}). \quad (\text{CQ2})$$

No mass is divided out. Put

$$I_\theta = \{n \in \mathbb{Z} : \nu_{n,\theta} > 0\}, \quad \mathcal{H}_\theta = \ell^2(I_\theta, \nu_\theta). \quad (\text{CQ3})$$

The notation $\ell^2(\nu)$ on all integers means this same Hilbert space after quotienting the zero semi-norm coordinates. A zero-mass coordinate is not an evaluation functional on that Hilbert space.

The predecessor's actual-source input, explicitly recorded immediately before P54, is that for each $0 < b < \pi/2$,

$$f_b(t) = e^{b|t|} w_h(t) \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R}). \quad (\text{CQ4})$$

This is an inherited estimate for the specified entire quotient, not an extra assumption about a new density. Its consequence for sampled measures is proved here. In the convolution integral, $|t_1 + \cdots + t_k| \leq |t_1| + \cdots + |t_k|$, so

$$e^{b|u|} m_{h,k}(u) \leq f_b^{*k}(u) \leq C_{b,h,k}, \quad C_{b,h,k} := \|f_b\|_\infty \|f_b\|_1^{k-1}. \quad (\text{CQ5})$$

The second inequality follows by integrating one bounded factor and the remaining $k - 1$ factors in L^1 . It includes $k = 1$. For any $0 \leq a < b$, the literal sum satisfies

$$\begin{aligned} \sum_{n \in I_\theta} \nu_{n,\theta} e^{a|u_{n,\theta}|} &\leq \frac{2\pi C_{b,h,k}}{L} \sum_{n \in \mathbb{Z}} e^{-(b-a)|2\pi n + \theta|/L} \\ &= \frac{2\pi C_{b,h,k}}{L} \frac{e^{-(b-a)\theta/L} + e^{-(b-a)(2\pi-\theta)/L}}{1 - e^{-2\pi(b-a)/L}} < \infty. \end{aligned} \quad (\text{CQ6})$$

The two series use $n \geq 0$ and $n \leq -1$, respectively. Formula CQ6 also includes $\theta = 0$ without deleting its zeroth term.

Let $\chi(S) = \sum_{j=0}^q \chi_j S^j$ be the literal monic original polynomial, of degree $q > 0$. For real u ,

$$|\chi(c + iu)| \leq C_\chi (1 + |u|)^q, \quad C_\chi = \sum_{j=0}^q |\chi_j| \max(1, c)^j. \quad (\text{CQ7})$$

Choose $0 < a_1 < a_2 < b$. The continuous function $(1 + t)^{2q} \exp(-(a_2 - a_1)t)$ is bounded on $t \geq 0$, since it tends to zero at infinity. Multiplying CQ7 by the exponential and using CQ6 proves

$$\sum_{n \in I_\theta} |\chi(S_{n,\theta})|^2 \nu_{n,\theta} e^{a_1 |u_{n,\theta}|} < \infty. \quad (\text{CQ8})$$

Thus the original discrete measure and its $|\chi|^2$ multiple each have a finite positive exponential moment. All their polynomial moments exist.

76.1.3 2. A complete polynomial-density proof on the shifted lattice

The following argument applies to either measure. Retain the shifted lattice CQ1, a subset I of its indices, and positive masses μ_n such that

$$\sum_{n \in I} \mu_n e^{a |u_{n,\theta}|} < \infty \quad \text{for some } a > 0. \quad (\text{CQ9})$$

Suppose f in $\ell^2(I, \mu)$ is orthogonal to all polynomial evaluation vectors. Since $u = (S - c)/i$, the evaluations of polynomials in S and in the real coordinate u span the same subspace. Conjugating the orthogonality equations if necessary gives

$$\sum_{n \in I} f_n \mu_n u_{n,\theta}^j = 0 \quad (j = 0, 1, 2, \dots). \quad (\text{CQ10})$$

Define

$$F(z) = \sum_{n \in I} f_n \mu_n e^{z u_{n,\theta}}, \quad |\Re z| < a/2. \quad (\text{CQ11})$$

For $0 \leq t < a/2$, Cauchy–Schwarz gives

$$\sum_{n \in I} |f_n| \mu_n e^{t |u_{n,\theta}|} \leq \|f\|_{\ell^2(\mu)} \left(\sum_{n \in I} \mu_n e^{2t |u_{n,\theta}|} \right)^{1/2} < \infty. \quad (\text{CQ12})$$

This proves locally uniform absolute convergence on the open strip in CQ11. For every derivative order j and every smaller closed strip, choose t' strictly between its width and $a/2$. The bound

$$\sup_{x \geq 0} x^j \exp(-(t' - t)x) < \infty$$

and CQ12 at t' prove locally uniform absolute convergence of the derivative series. Hence F is holomorphic and CQ10 says that all its derivatives at zero vanish. Its convergent Taylor series is zero on a disk about zero. Holomorphic uniqueness on the connected strip therefore gives $F(z) = 0$ on that entire strip.

There is no need to appeal to an unproved general Fourier-measure uniqueness statement here. CQ12 at $t = 0$ allows termwise integration, and for each $m \in I$,

$$\begin{aligned} 0 &= \frac{1}{L} \int_0^L F(it) e^{-it u_{m,\theta}} dt \\ &= \sum_{n \in I} f_n \mu_n \frac{1}{L} \int_0^L e^{2\pi i(n-m)t/L} dt = f_m \mu_m. \end{aligned} \quad (\text{CQ13})$$

Because $\mu_m > 0$, every coordinate f_m vanishes. The orthogonal complement of the polynomial span is zero, so that span is dense. CQ6 proves density in $\mathcal{H}_\theta$; CQ8 proves density in the Hilbert space for the positive atoms of $|\chi|^2\nu_\theta$.

76.1.4 3. The closed relation space, exact map, metric, and kernel

Define

$$Z_\theta = \{n \in I_\theta : \chi(S_{n,\theta}) = 0\}, \quad p_\theta(S) = \prod_{n \in Z_\theta} (S - S_{n,\theta}), \quad (\text{CQ14})$$

where an empty product is 1. The sample points are distinct, so Z_θ has at most q members and p_θ is a squarefree divisor of χ .

Let $\mathcal{R}_\theta$ be the closure in $\mathcal{H}_\theta$ of the evaluation vectors of $\chi\mathbb{C}[S]$. Every such vector vanishes at all indices in Z_θ . Each of those coordinates is continuous, since

$$|f_n| \leq \nu_{n,\theta}^{-1/2} \|f\|_{\mathcal{H}_\theta} \quad (n \in Z_\theta). \quad (\text{CQ15})$$

It follows that $\mathcal{R}_\theta$ is contained in the subspace of vectors vanishing on Z_θ . For the reverse inclusion, let f vanish there and set

$$g_n = \frac{f_n}{\chi(S_{n,\theta})} \quad (n \in I_\theta \setminus Z_\theta). \quad (\text{CQ16})$$

This vector is square-integrable for the weighted masses

$$\mu_n = |\chi(S_{n,\theta})|^2 \nu_{n,\theta},$$

because its squared norm is exactly $\sum_{I_\theta \setminus Z_\theta} |f_n|^2 \nu_{n,\theta}$. By Section 2 there are polynomials P_j tending to g in that weighted norm. Multiplication by the literal χ gives

$$\|\chi P_j - f\|_{\mathcal{H}_\theta}^2 = \sum_{n \in I_\theta \setminus Z_\theta} |P_j(S_{n,\theta}) - g_n|^2 |\chi(S_{n,\theta})|^2 \nu_{n,\theta} \longrightarrow 0. \quad (\text{CQ17})$$

At indices in Z_θ , both terms on the left vanish. Consequently

$$\mathcal{R}_\theta = \{f \in \mathcal{H}_\theta : f_n = 0 \text{ for all } n \in Z_\theta\}. \quad (\text{CQ18})$$

Restriction to Z_θ therefore induces the exact isometric isomorphism

$$\mathcal{H}_\theta / \mathcal{R}_\theta \xrightarrow{\sim} \left(\mathbb{C}^{Z_\theta}, \|(v_n)\|^2 = \sum_{n \in Z_\theta} \nu_{n,\theta} |v_n|^2 \right). \quad (\text{CQ19})$$

The quotient norm is exactly the displayed norm: the coordinate projection onto Z_θ is orthogonal, and the complementary projection has range $\mathcal{R}_\theta$.

Since adding χQ to a polynomial changes its evaluation vector by an element of $\mathcal{R}_\theta$, there is a well-defined linear comparison

$$T_\theta : E = \mathbb{C}[S]/(\chi) \longrightarrow \mathcal{H}_\theta / \mathcal{R}_\theta, \quad [P] \longmapsto (P(S_{n,\theta}))_{n \in Z_\theta}. \quad (\text{CQ20})$$

It is surjective. When Z_θ is nonempty, the literal Lagrange polynomials

$$L_n(S) = \prod_{m \in Z_\theta \setminus \{n\}} \frac{S - S_{m,\theta}}{S_{n,\theta} - S_{m,\theta}} \quad (\text{CQ21})$$

map to its coordinate basis; when it is empty, the target is zero. A polynomial vanishes at every distinct point indexed by Z_θ if and only if it is divisible by their squarefree product. Hence the exact kernel is

$$\ker T_\theta = (p_\theta)/(\chi) \subseteq E. \quad (\text{CQ22})$$

This retains the multiplicities in χ . In the primary decomposition of E , each unsampled primary block lies in the kernel; at a sampled root of multiplicity m , the kernel is the maximal ideal $(S - \rho)/(S - \rho)^m$, while its scalar value survives. CQ20 is a linear map to a Hilbert quotient; no multiplication of arbitrary square-summable sequences is required.

76.1.5 4. Why the original arithmetic source needs no extra positivity hypothesis

For $k \geq 2$, the actual entire nonzero function v_h has isolated zeros. Thus $w_h(t) > 0$ off a discrete subset of the real axis. For fixed u , both $w_h(t)$ and $w_h(u - t)$ are positive for almost every t ; their product is integrable by boundedness and integrability from CQ4. Its integral is strictly positive. Thus $m_{h,2}(u) > 0$ for every real u . Inductively, convolving the everywhere-positive $m_{h,k-1}$ with the almost-everywhere-positive nonzero w_h gives $m_{h,k}(u) > 0$. CQ5 makes the integrals finite. Therefore $I_\theta = \mathbb{Z}$ for $k \geq 2$ and CQ14 is exactly the index set displayed in H64.

For $k = 1$, the original cyclic algebra is $\mathbb{C}[S]/(h)$, with generator multiplication by S . Its annihilator ideal is exactly (h) : $h(M_S) = 0$, and any polynomial annihilating M_S also annihilates 1, so is divisible by the unchanged h . Consequently $(\chi_{h,1}) = (h)$. If $\chi_{h,1}$ denotes the monic minimal polynomial and a_h is the original leading coefficient of h , then $\chi_{h,1} = h/a_h$; this identifies the ideal generator and does not replace the original source $v_h = g/h$. Equality $\chi_{h,1} = h$ is asserted only when the source separately fixes $a_h = 1$.

Write the unchanged packet as

$$h(S) = a_h \prod_{\rho \in Z_h} (S - \rho)^{m_\rho}, \quad a_h \neq 0, \quad m_\rho = \text{ord}_\rho g. \quad (\text{CQ23})$$

For a selected root ρ , full-order removal gives the exact nonzero value

$$v_h(\rho) = \frac{g^{(m_\rho)}(\rho)}{h^{(m_\rho)}(\rho)} = \frac{g^{(m_\rho)}(\rho)}{a_h m_\rho! \prod_{\sigma \in Z_h \setminus \{\rho\}} (\rho - \sigma)^{m_\sigma}} \neq 0. \quad (\text{CQ24})$$

This follows by factoring $(S - \rho)^{m_\rho}$ from both g and the unchanged h ; the numerator derivative, original leading coefficient a_h , and every denominator factor are nonzero by their definitions. If a sample satisfies $S_{n,\theta} = \rho$, then $\Re \rho = 1/2$ and

$$\nu_{n,\theta} = \frac{2\pi}{L} w_h(\Im \rho) = \frac{1}{L} |v_h(\rho)|^2 = \frac{|g^{(m_\rho)}(\rho)|^2}{L |a_h|^2 (m_\rho!)^2 \prod_{\sigma \in Z_h \setminus \{\rho\}} |\rho - \sigma|^{2m_\sigma}} > 0. \quad (\text{CQ25})$$

Thus every root appearing in H64 has positive mass also at $k = 1$. Other zero-mass lattice coordinates are discarded by the Hilbert-space definition CQ3, but none removes a summand from H64. The full-order hypothesis at NOTE.md line 67 and the preserved unit at lines 85–90 already supply the needed fact. A useful exposition repair is to define I_θ and then state CQ25; it does not change H64's conclusion for this source.

For comparison, outside this arithmetic domain the omission of positive support can change the theorem. With $\nu_0 = 0$, $\nu_n = e^{-|n|}$ for $n \neq 0$, and $\chi(S) = S - S_{0,\theta}$, Section 3 gives zero completed quotient, whereas a sum over all sampled polynomial roots would give one copy of $\mathbb{C}$. This example establishes why the general formula must use CQ14; it is not a counterexample to the full-order arithmetic result.

76.1.6 5. The direct-integral target and its exact zero map

Retain the fixed polynomial χ and $c = k/2$. Let

$$\mathcal{F} = \{\theta \in [0, 2\pi) : \theta \equiv L\mathfrak{S}\rho \pmod{2\pi} \text{ for some root } \rho \text{ of } \chi \text{ with } \Re\rho = c\}. \quad (\text{CQ26})$$

Each such root specifies exactly one phase in the chosen half-open interval, and then exactly one integer n with $\rho = S_{n,\theta}$. Therefore $\mathcal{F}$ is finite, with at most q elements. For θ outside $\mathcal{F}$, the index set Z_θ is empty, so CQ19 is the zero Hilbert space.

A concrete measurable realization prevents ambiguity about the field of targets. Map CQ19 into the fixed Hilbert space $\ell^2(\mathbb{Z})$ by

$$(v_n)_{n \in Z_\theta} \mapsto (\mathbf{1}_{n \in Z_\theta} \sqrt{\nu_{n,\theta}} v_n)_{n \in \mathbb{Z}}. \quad (\text{CQ27})$$

This is an isometry onto the range of the diagonal orthogonal projection with diagonal entries $\mathbf{1}_{\nu_{n,\theta} > 0} \mathbf{1}_{\chi(S_{n,\theta})=0}$. Those entries are measurable: the first condition is measurable for the original continuous density, and the second is the zero set of a continuous function of θ . Hence these ranges define a measurable Hilbert field. Their direct integral has norm

$$\int_0^{2\pi} \sum_{n \in Z_\theta} \nu_{n,\theta} |v_n(\theta)|^2 \frac{d\theta}{2\pi}. \quad (\text{CQ28})$$

Every section is zero outside the finite set $\mathcal{F}$; its norm in CQ28 is zero. The direct integral is therefore the zero Hilbert space. For each $[P] \in E$, the evaluation sections of CQ20 are measurable through CQ27 and supported in $\mathcal{F}$. They consequently define the zero class in the direct integral. The induced map and its exact kernel are

$$E \longrightarrow \int_{[0,2\pi)}^\oplus (\mathcal{H}_\theta / \mathcal{R}_\theta) \frac{d\theta}{2\pi} = 0, \quad \ker = E. \quad (\text{CQ29})$$

Individual exceptional fibres can remain nonzero and carry the positive masses in CQ19; a finite set has zero measure for the actual phase measure $d\theta/(2\pi)$. CQ29 proves precisely the completed-target claim at NOTE.md line 715. It does not change the source isometry H13 or the finite quotient H24.

On a retained split support label ℓ , the actual induced map is $(\ell, v) \mapsto (\ell, 0)$, while external absence maps to external absence. This is the lift of CQ29 by the stated H4; its coefficient kernel is all of E , exactly as CQ29 records.

76.1.7 Review disposition

Accept H64, its displayed squarefree kernel, and the zero direct-integral conclusion for the original full-order packet. The recommended addition is explanatory: define the positive support and retained metric by CQ3, CQ14, CQ19; explicitly record the degree-one identity and positive sampled-root value CQ25. The unrestricted all-atom version is not justified for arbitrary other nonnegative densities. No source edit or mathematical test result is claimed by this review.

77 Continuous holonomy cochains and the original relation metric

This continuation proves the continuous-phase version of H29–H31 and the exact maps to the completed targets in H64 of the delivered *All-holonomy descent of the original theta quotient*. The delivered NOTE.tex was read completely, all 1152 lines; its SHA-256 is

444f82a8fa420e6f471ef448442085f3a93acb137069e1505d155faab051b881.

The archive SHA-256 is

f0506a27cd4c9012f43fa70c3d1f8f996bbb6db01b0eea2944b34dcf290ca8f0.

All source coordinates, masses, phases, relation columns, and full primary multiplicities below are retained. No finite test or Lean execution is used as a proof. The sampled-density argument uses the full analytic proof *Periodized source density review*, PSD8–PSD14, whose TeX SHA-256 is

51102189bcc7a45610464143a5c0bee21c08059c317024e6e856bc7de993fe74.

The completed-fibre calculation develops H64 and is also proved in the companion completed-quotient and holonomy-intake derivations. Its repetition here supplies the maps needed for this continuous common-image complex, including the mean-zero contraction and the precise weighted relation gap.

77.1 Original coefficient spaces and actual phase norms

Fix the original monic polynomial χ , its full degree $q \geq 1$, and an integer $N \geq q - 1$. Put

$$\begin{aligned} E &= \mathbb{C}[S]/(\chi), & P_N &= \mathbb{C}[S]_{\leq N}, & J &= J_N : P_N \longrightarrow E, \\ B_N &= \ker J = \chi P_{N-q}, & A &= M_S : E \longrightarrow E, \end{aligned} \quad (\text{HCD1})$$

where $P_{-1} = 0$. Use the original increasing-power coefficient bases. Let $B : \mathbb{C}^r \rightarrow P_N$, $r = N - q + 1$, be multiplication by the literal χ , so that $\text{im } B = B_N$. At $r = 0$ it is the unique map from the zero coefficient space.

Let $\Theta = [0, 2\pi)$ carry its actual Fourier phase measure $d\mu(\theta) = d\theta/(2\pi)$. Retain the measurable Hermitian phase source matrices M_θ and their mean

$$M = \int_{\Theta} M_\theta d\mu, \quad 0 < aI_{P_N} \preceq M_\theta \preceq bI_{P_N} < \infty \quad \text{almost everywhere,} \quad a, b > 0. \quad (\text{HCD2})$$

The original source isometry H13–H17 identifies M with the original source Gram, including its full constant mass. For the actual source, the bounds in (HCD2) are obtained from H39–H41, also proved in HG3, as follows. Let $\Psi_N : P_N \rightarrow L^2(\mathbb{R}; \mathcal{K})$ be the original finite source columns after the half-density map, with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$. Fix $\beta > 0$ for the retained weighted source Grams, whose finite entries are supplied by the original source regularity:

$$M_{\beta, \pm} = (e^{\pm\beta r} \Psi_N)^* (e^{\pm\beta r} \Psi_N), \quad \kappa_{\beta, N} = \max_{v \neq 0} \frac{v^* (M_{\beta, +} + M_{\beta, -}) v}{v^* M v}. \quad (\text{HCD2a})$$

These are the original weights $x_k^{\pm 2\beta}$; all original source measures remain present. The maximum is finite and positive because $M > 0$ and the two weighted Grams are finite positive forms. For $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r+s) dr$ and $s \geq 0$, write the integrand at v as the inner product of $e^{-\beta r} \Psi_N(r)v$ and $e^{\beta(r+s)} \Psi_N(r+s)v$, multiplied by $e^{-\beta s}$. Cauchy–Schwarz, followed by translation in the second integral, gives

$$|v^* \mathcal{C}(s) v| \leq e^{-\beta s} \sqrt{(v^* M_{\beta, -} v)(v^* M_{\beta, +} v)}.$$

For $-s$ the same bound follows from $\mathcal{C}(-s) = \mathcal{C}(s)^*$. Unfolding the literal Zak sum gives $M_\theta = \sum_{j \in \mathbb{Z}} e^{ij\theta} \mathcal{C}(jL)$. The correlation bound and polarization give absolute convergence of each matrix entry, uniformly in phase. Pairing j and $-j$, using $2\sqrt{uv} \leq u + v$, and summing the geometric series prove

$$-\delta_L M \preceq M_\theta - M \preceq \delta_L M, \quad \delta_L = \frac{\kappa_{\beta, N}}{e^{\beta L} - 1}. \quad (\text{HCD2b})$$

For any chosen $0 < \eta < 1$, the explicit finite circumference

$$L \geq \frac{1}{\beta} \log(1 + \kappa_{\beta,N}/\eta) \implies \delta_L \leq \eta < 1 \quad (\text{HCD2c})$$

therefore supplies (HCD2), with $a = (1 - \delta_L)\lambda_{\min}(M)$ and $b = (1 + \delta_L)\lambda_{\max}(M)$ in the original coefficient frame. The phase mean is still exactly M : every nonzero Fourier mode integrates to zero against $d\theta/(2\pi)$. The concrete choice $\eta = 1/2$ gives $L = \beta^{-1} \log(1 + 2\kappa_{\beta,N})$ and $\delta_L = 1/2$. These are actual finite-source bounds at the stated cutoff; no bound uniform in N or the arithmetic packet is asserted. The argument below also applies to any fixed L for which (HCD2) is already established.

Define the Hilbert space

$$\mathcal{P} = L^2(\Theta; P_N, M_\theta d\mu), \quad \|P\|_{\mathcal{P}}^2 = \int_{\Theta} P(\theta)^* M_\theta P(\theta) d\mu.$$

By (HCD2), its elements are exactly coefficientwise L^2 functions, with equivalent norms. Equality means equality almost everywhere. The coefficientwise Bochner integral $\text{av } P = \int P d\mu$ exists, since coefficientwise L^2 implies L^1 on this finite phase measure. Every fixed coefficient map commutes with this integral, entry by entry.

The closed relation space and common-image space are

$$\mathcal{B} = L^2(\Theta; B_N), \quad \mathcal{P}^{\text{eq}} = \{P \in \mathcal{P} : JP(\theta) = x \text{ a.e. for a fixed } x \in E\}. \quad (\text{HCD3})$$

They carry the restricted source norms. They are closed: J is continuous on coefficientwise L^2 , $B_N = \ker J$ is closed, and constant E -valued functions form a closed subspace. For the last assertion, averaging followed by repetition is a bounded idempotent with precisely that image.

77.2 The continuous cochain splitting and its full contraction

Use complexes in degrees 0, 1 with actual inclusion differentials:

$$\mathcal{C}_N = [B_N \hookrightarrow P_N], \quad \mathcal{C}_N = [\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}]. \quad (\text{HCD4})$$

Repetition $\Delta v(\theta) = v$ and averaging av are cochain maps. In degree one, $J \text{av } P = \int JP d\mu = x$; in degree zero the mean belongs to B_N . The equality $\text{av } \Delta = I$ uses the literal phase measure. Repetition is an isometry:

$$\|\Delta v\|_{\mathcal{P}}^2 = v^* \left(\int M_\theta d\mu \right) v = v^* M v.$$

Averaging is bounded with the explicit estimate

$$\|\text{av } P\|_M^2 \leq b \|\text{av } P\|_{\text{coeff}}^2 \leq b \int \|P(\theta)\|_{\text{coeff}}^2 d\mu \leq (b/a) \|P\|_{\mathcal{P}}^2. \quad (\text{HCD5})$$

This uses scalar Cauchy–Schwarz for the coefficient vector and does not replace the arithmetic mass by one.

Put $\mathcal{B}^0 = \{b \in \mathcal{B} : \text{av } b = 0\}$. The inverse continuous coefficient maps of complexes are

$$\begin{aligned} \mathcal{C}_N &\simeq \mathcal{C}_N \oplus [\mathcal{B}^0 \xrightarrow{I} \mathcal{B}^0], \\ P &\longmapsto (\text{av } P, P - \Delta \text{av } P), \quad (v, b^0) \longmapsto \Delta v + b^0. \end{aligned} \quad (\text{HCD6})$$

In degree one, $J(P - \Delta \text{av } P) = x - x = 0$ and its mean is zero. In degree zero every value already belongs to B_N . Both composites restore the original vector. Inclusion carries these components to inclusion and identity, respectively, proving the isomorphism of complexes.

The degree-minus-one map

$$h : \mathcal{P}^{\text{eq}} \longrightarrow \mathcal{B}, \quad hP = P - \Delta \text{av } P \quad (\text{HCD7})$$

satisfies $dh + hd = I - \Delta \text{av}$. In degree one this is $dhP = P - \Delta \text{av } P$; in degree zero it is $hdb = b - \Delta \text{av } b$. On the complementary complex it is the identity from degree one to degree zero. Boundedness follows from (HCD5) and the isometry of repetition. It is not asserted orthogonal or isometric.

The common quotient map is

$$j : \mathcal{P}^{\text{eq}} \longrightarrow E, \quad P \longmapsto JP(\theta) \text{ a.e.} \quad (\text{HCD8})$$

It is onto by repeating a right inverse of J , and its kernel is exactly $\mathcal{B}$. Inclusion is injective. Therefore $H^0(\mathcal{C}_N) = 0$ and $H^1(\mathcal{C}_N) = E$, with the isomorphism induced by the original full J . Repetition and averaging induce the identity on this E .

Multiplication by S maps P_N to P_{N+1} and B_N to B_{N+1} . Applied pointwise, it sends common image x to common image Ax and commutes with repetition, mean and h . These are maps between the stated cutoffs. On degree-one cohomology the action is the full companion A , including its nilpotent primary parts. No trace of the infinite-dimensional contractible complement is assumed.

77.3 The minimum norm of the common image

Retain the original and phasewise minimum sections

$$\begin{aligned} G &= (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \\ G_\theta &= (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta, \\ \overline{G} &= \int G_\theta d\mu. \end{aligned} \quad (\text{HCD9})$$

Surjectivity and positivity make every inverse defined. Continuity of matrix inversion proves measurability. For any fixed coefficient right inverse C_{ref} of J ,

$$\frac{a}{\|J\|_{\text{coeff}}^2} I_E \preceq G_\theta \preceq b \|C_{\text{ref}}\|_{\text{coeff}}^2 I_E. \quad (\text{HCD10})$$

Indeed any lift p of x has $\|x\| \leq \|J\| \|p\|$, proving the lower bound, and the particular lift $C_{\text{ref}}x$ proves the upper bound. The formula for C_θ then also gives boundedness. All following integrals and sections are consequently defined.

Direct multiplication gives $JC_\theta = I$, and for $b_0 \in B_N$,

$$b_0^* M_\theta C_\theta = b_0^* J^* G_\theta = 0. \quad (\text{HCD11})$$

The section $\mathcal{S}x(\theta) = C_\theta x$ has common image x . Any other P with that image has $P - \mathcal{S}x \in \mathcal{B}$. Phasewise orthogonality gives

$$\|P\|_{\mathcal{P}}^2 = x^* \overline{G} x + \|P - \mathcal{S}x\|_{\mathcal{P}}^2. \quad (\text{HCD12})$$

Thus the quotient metric of the continuous complex is exactly $\overline{G} > 0$ on its full E . For the constant-source complex the identical argument gives G . Repetition is isometric on sources, and its induced identity $(E, G) \rightarrow (E, \overline{G})$ has the additional continuous relation directions available in the target.

77.4 The positive relation gap and its mean-zero decomposition

For $r > 0$, put $X_\theta = (B^*M_\theta B)^{-1}B^*M_\theta C$. Both sections have right-inverse image I_E , so their difference lies in B_N and equals BX_θ ; multiplication by B^*M_θ proves the formula and uniqueness. At $r = 0$, both sections equal the unique inverse of J and X_θ is the zero map into the zero space. Expanding $C = C_\theta + BX_\theta$ and using (HCD11) proves

$$\boxed{G - \overline{G} = \mathcal{D} = \int (C - C_\theta)^* M_\theta (C - C_\theta) d\mu = \int X_\theta^* (B^* M_\theta B) X_\theta d\mu \succeq 0.} \quad (\text{HCD13})$$

The integral of $C^* M_\theta C$ is $C^* M C = G$. Every term is the norm of a field in the original relation image. Multiplication by the literal χ , followed by the inherited finite tensor theta primitive, realizes that field in the original cochain complex.

The relation to the particular mean-zero contraction is also exact. Define

$$\overline{C} = \int C_\theta d\mu, \quad a_C = C - \overline{C} : E \rightarrow B_N, \quad h_\theta = C_\theta - \overline{C} : E \rightarrow B_N. \quad (\text{HCD14})$$

Then $\int h_\theta d\mu = 0$, and h_θ is precisely the field h_θ . The original discrepancy is $C - C_\theta = a_C - h_\theta$ and need not have mean zero. Since a_C has relation-valued columns, phase and common-source orthogonality give

$$\begin{aligned} \int a_C^* M_\theta h_\theta d\mu &= -a_C^* M \overline{C} \\ &= -a_C^* M (C - a_C) = a_C^* M a_C. \end{aligned} \quad (\text{HCD15})$$

The adjoint cross term is the same Hermitian matrix. Expanding all terms of (HCD13) therefore proves

$$\boxed{\mathcal{D} = \int h_\theta^* M_\theta h_\theta d\mu - a_C^* M a_C.} \quad (\text{HCD16})$$

This retains the positive relation norm and the complete weighted cross terms. A mean-zero coefficient field is not presumed orthogonal to constant relations for the varying metric. At $N = q - 1$, $B_N = 0$, $C_\theta = C = J^{-1}$, $a_C = h_\theta = 0$ and $\overline{G} = G$.

77.5 Actual sampled Hilbert spaces, including zero weights

Fix the original circumference $L > 0$, tensor degree $k \geq 1$, and $c = k/2$. Retain

$$\begin{aligned} u_n(\theta) &= (2\pi n + \theta)/L, & s_n(\theta) &= c + iu_n(\theta), \\ w_n(\theta) &= \frac{2\pi}{L} m_{h,k}(u_n(\theta)) \geq 0, & n &\in \mathbb{Z}. \end{aligned} \quad (\text{HCD17})$$

The Fourier coefficient is literally $2\pi/L$, with $m_{h,k} = w_h^{*k}$, $w_h = |(g/h)(1/2 + iu)|^2/(2\pi)$ and $g = 2\xi$. Sampling gives

$$\tau_\theta P = (P(s_n(\theta)))_n, \quad \|\tau_\theta P\|_\theta^2 = \sum_n w_n(\theta) |P(s_n(\theta))|^2 = P^* M_\theta P. \quad (\text{HCD18})$$

This is the actual phase source Gram H17.

Define $\mathcal{H}_\theta$ to be weighted square-summable sequences modulo sequences supported where $w_n(\theta) = 0$. Thus zero weights, possible when $k = 1$, have an explicit null subspace. For $k \geq 2$,

positivity of the original convolution makes every sample weight positive. At $k = 1$, retain the exact ideal equality $(\chi_{h,1}) = (h)$. If $a_h \neq 0$ is the original leading coefficient of h , then $\chi_{h,1} = h/a_h$ is its monic generator, and $\chi_{h,1} = h$ in the original monic packet convention. Indeed a polynomial annihilates multiplication by S on $\mathbb{C}[S]/(h)$ if and only if its action on $[1]$ is zero, which is equivalent to belonging to (h) . The analytic source remains the original g/h , with the factor a_h retained in that quotient and its norm. Now take a complete selected root $\rho = 1/2 + iu_n(\theta)$ whose common order in g and h is m . The two Taylor expansions have leading coefficients $g^{(m)}(\rho)/m!$ and $h^{(m)}(\rho)/m!$, so the canceled entire quotient has the exact nonzero value

$$(g/h)(\rho) = \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)}, \quad w_n(\theta) = \frac{1}{L} \left| \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)} \right|^2 > 0. \quad (\text{HCD18a})$$

This retains both cancellation factorials and the full Fourier mass $(2\pi/L)(1/(2\pi))$. It is the selected-root calculation PSD5 and PSD41. Other $k = 1$ coordinates where g/h vanishes still give the explicit null subspace above. For a concrete measurable realization use the isometry $v \mapsto (\sqrt{w_n(\theta)}v_n)_n$ into ordinary $\ell^2(\mathbb{Z})$, with zero coordinates at zero weights. Its inverse on its range is $v_n = y_n/\sqrt{w_n(\theta)}$ at positive weights; other coordinates give the null class. This specifies the norm map without changing a mass. The range projections are measurable diagonal matrices; their images of the countable standard basis give a measurable field.

The original gamma-contour and alternating-zeta calculation PSD8–PSD13 proves, for every $0 < \gamma < \pi/2$, the finite constants

$$A_\gamma = \|e^{\gamma|\cdot|}w_h\|_\infty, \quad B_\gamma = \|e^{\gamma|\cdot|}w_h\|_1.$$

That calculation retains the leading coefficient of h , its full degree, and the continuous canceled values on the compact interval containing its selected roots. With $f_\gamma(v) = e^{\gamma|v|}w_h(v)$, the ordinary triangle inequality on the convolution variables and integration of the remaining factors give, for every real u ,

$$e^{\gamma|u|}m_{h,k}(u) \leq f_\gamma^{*k}(u) \leq A_\gamma B_\gamma^{k-1}. \quad (\text{HCD19a})$$

For the second inequality bound one factor by A_γ and integrate each of the other $k - 1$ factors with its literal Lebesgue measure. For $k = 1$ the expression is A_γ , with $B_\gamma^0 = 1$. For any $0 < 2\epsilon < \gamma$,

$$\sup_{0 \leq \theta < 2\pi} \sum_n w_n(\theta) e^{2\epsilon|u_n(\theta)|} < \infty. \quad (\text{HCD19})$$

Indeed $|u_n(\theta)| \geq (2\pi/L)(|n| - 1)$, so with $d = \gamma - 2\epsilon > 0$ the sum is at most

$$\frac{2\pi}{L} A_\gamma B_\gamma^{k-1} e^{2\pi d/L} \left(1 + \frac{2e^{-2\pi d/L}}{1 - e^{-2\pi d/L}} \right).$$

For every fixed polynomial Q , the constant $\sup_{u \in \mathbb{R}} |Q(c + iu)|^2 e^{-d|u|/2}$ is finite. Inserting it into the same geometric estimate, with $d/2$ in place of d , proves uniform exponential summability of $w_n(\theta)|Q(s_n(\theta))|^2$. This includes $Q = \chi$ and every fixed polynomial relation used below. All original constants in the weights are unchanged. The full source proof of PSD8–PSD14 is a dependency of this continuation; the remaining discrete density argument is given explicitly here with its shifted Fourier phase.

77.6 Density and the complete fibrewise Hilbert quotient

Here is a direct density proof with the zero weights retained. If $f \in \mathcal{H}_\theta$ is orthogonal to all polynomials, put $a_n = w_n(\theta)\overline{f_n}$. Cauchy–Schwarz and (HCD19) give $\sum |a_n|e^{\epsilon|u_n|} < \infty$. Hence $F(z) = \sum_n a_n e^{zu_n}$ is holomorphic on $|\Re z| < \epsilon$, with termwise derivatives on smaller closed strips.

Every derivative at zero is zero, since polynomials in S and in $(S - c)/i$ span the same space. The Taylor series and the identity theorem give $F = 0$ on the connected strip. On the imaginary axis,

$$F(it) = e^{it\theta/L} \sum_n a_n e^{2\pi i n t/L}.$$

The series is absolutely uniformly convergent. Multiplication by $e^{-it\theta/L} e^{-2\pi i n t/L}$ and integration over $0 \leq t \leq L$ give $a_m = 0$. At positive weights this implies $f_m = 0$; at zero weights the class is already null. Thus polynomials are dense. The identical argument proves polynomial density for the weights $w_n |\chi(s_n)|^2$.

Multiplication by the original polynomial is the isometry

$$U_\chi : \ell^2(w_n |\chi(s_n)|^2) \longrightarrow \mathcal{H}_\theta, \quad f \longmapsto (\chi(s_n) f_n)_n. \quad (\text{HCD20})$$

Its image is precisely the sequences vanishing at the positive-weight sampled roots of χ . To prove the converse inclusion, take such a sequence v and set $f_n = v_n / \chi(s_n)$ when $\chi(s_n) \neq 0$, and zero elsewhere. Its squared domain norm equals $\|v\|_\theta^2$. Thus the image is closed. Polynomial density in the domain now proves

$$\mathcal{R}_\theta := \overline{\chi \mathbb{C}[S]}^{\mathcal{H}_\theta} = \{v : v_n = 0 \text{ if } w_n(\theta) > 0, \chi(s_n(\theta)) = 0\}. \quad (\text{HCD21})$$

The receiving Hilbert quotient is exactly

$$\begin{aligned} \mathcal{Q}_\theta &= \mathcal{H}_\theta / \mathcal{R}_\theta \simeq \bigoplus_{\substack{n: \chi(s_n(\theta))=0 \\ w_n(\theta) > 0}} \mathbb{C}, \\ \|v\|_{\mathcal{Q}_\theta}^2 &= \sum_{\text{displayed } n} w_n(\theta) |v_n|^2. \end{aligned} \quad (\text{HCD22})$$

The quotient map is orthogonal coordinate restriction. No weight equal to zero is inverted.

77.7 The complete primary kernel before phase integration

Let Λ_θ^+ be the distinct sampled roots of χ with positive weight, and set $p_\theta(S) = \prod_{\lambda \in \Lambda_\theta^+} (S - \lambda)$, using $p_\theta = 1$ for an empty product. The finite map is

$$\rho_\theta : E \longrightarrow \mathcal{Q}_\theta, \quad [P]_\chi \longmapsto (P(\lambda))_{\lambda \in \Lambda_\theta^+}. \quad (\text{HCD23})$$

It is well defined, and is onto by Lagrange interpolation at these distinct points, using degree less than $|\Lambda_\theta^+| \leq q$. Vanishing at all those points is equivalent to divisibility by p_θ . Since $p_\theta \mid \chi$,

$$\ker \rho_\theta = (p_\theta) / (\chi) \subset E. \quad (\text{HCD24})$$

This specifies the zero-weight-corrected kernel in H64: a root at a zero-mass coordinate contributes no receiving Hilbert line.

Write the full original factorization as $\chi = \prod_{\lambda \in \Lambda} (S - \lambda)^{m_\lambda}$. The complete Taylor map is

$$\begin{aligned} E &\longrightarrow \prod_{\lambda \in \Lambda} \mathbb{C}[\varepsilon_\lambda] / (\varepsilon_\lambda^{m_\lambda}), \\ [P] &\longmapsto \left(\sum_{j=0}^{m_\lambda-1} \frac{P^{(j)}(\lambda)}{j!} \varepsilon_\lambda^j \right)_\lambda. \end{aligned} \quad (\text{HCD25})$$

For an explicit inverse, let $q_\lambda = \chi / (S - \lambda)^{m_\lambda}$ and let u_λ be the Taylor polynomial of $1/q_\lambda$ at λ through degree $m_\lambda - 1$. Then $e_\lambda = [u_\lambda q_\lambda]_\chi$ is one in the λ factor and zero in every other factor, by divisibility

by each original prime power. It follows that $\sum e_\lambda = 1$ and the factors are orthogonal idempotents. The inverse is $\sum_\lambda e_\lambda \sum_j a_{\lambda,j} (S - \lambda)^j$.

In these full primary coordinates, ρ_θ takes the constant coefficient of each active sampled factor. Its kernel in that factor is (ε_λ) , retaining every order $1, \dots, m_\lambda - 1$. For an unsampled factor or a zero-weight sampled factor, the kernel is the entire primary algebra, including its constant term. The original action is multiplication by $\lambda + \varepsilon_\lambda$; the receiving action is the simple scalar λ . Evaluation intertwines these actions with the displayed nilpotent kernel.

77.8 The measurable completed field and its vanishing map

Every sampled root has real part c . For a fixed root λ with $\Re \lambda = c$, its only possible phase is

$$\theta_\lambda = L\Im \lambda \pmod{2\pi}, \quad 0 \leq \theta_\lambda < 2\pi, \quad n_\lambda = (L\Im \lambda - \theta_\lambda)/(2\pi) \in \mathbb{Z}.$$

Thus the set Θ_χ of possible phases is finite. For $\theta \notin \Theta_\chi$, $\mathcal{Q}_\theta = 0$. At an exceptional phase the positive-weight condition in (HCD22) is still retained.

In the measurable realization after (HCD18), the projection onto $\mathcal{Q}_\theta$ has diagonal entries

$$\mathbf{1}_{\{w_n(\theta) > 0\}} \mathbf{1}_{\{\chi(s_n(\theta)) = 0\}}, \quad n \in \mathbb{Z}.$$

These are measurable and give an explicit field of projections. The direct integral satisfies

$$\mathcal{Q} = \int_{\Theta}^{\oplus} \mathcal{Q}_\theta d\mu = 0, \quad (\text{HCD26})$$

because every measurable section has squared norm supported on the finite measure-zero set Θ_χ . This is an almost-everywhere assertion; each exceptional nonzero receiving line is still exactly the weighted line in (HCD22).

For $x \in E$, the section $\theta \mapsto \rho_\theta x$ is measurable: it is given by polynomial evaluations and the displayed coordinate masks. It defines

$$\rho : E \longrightarrow \mathcal{Q}, \quad x \longmapsto [\theta \mapsto \rho_\theta x]_{\text{a.e.}}, \quad \ker \rho = E. \quad (\text{HCD27})$$

Every pointwise exceptional value has zero direct-integral class. This does not assert that those pointwise maps vanish, or that the original nonreduced coefficient algebra was zero.

77.9 The source-to-completion cochain map

The direct-integral source is the weighted space with squared norm $\int_{\Theta} \sum_n w_n(\theta) |v_n(\theta)|^2 d\mu$. Let $\mathcal{H}$ denote this space, and let $\mathcal{R}$ be its closed subspace of sections belonging to $\mathcal{R}_\theta$ almost everywhere. The coordinate projection above has kernel $\mathcal{R}$.

The measurable evaluation map is

$$\mathbb{T} : \mathcal{P} \longrightarrow \mathcal{H}, \quad P \longmapsto (P(\theta, s_n(\theta)))_{\theta, n}.$$

The integrated identity (HCD18) proves it is an isometry. Evaluation of a relation-valued section is in $\mathcal{R}$, because the original section is a multiple of the literal χ . Hence there is a cochain map

$$[\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}] \longrightarrow [\mathcal{R} \hookrightarrow \mathcal{H}], \quad (\text{HCD28})$$

using evaluation in both degrees. On a common image x , the receiving quotient value is $\rho_\theta x$, since representatives differ by an original relation. The induced degree-one map is exactly (HCD27).

The global relation subspace is also the closure of measurable finite polynomial relation sections. To prove this without a measurable choice of approximating polynomials, suppose $v \in \mathcal{H}$ is orthogonal to every section $\mathbf{1}_U(\theta) \chi(S) S^j$, for measurable U and $j \geq 0$. Its fibre inner product with

χS^j is integrable by Cauchy–Schwarz, and its integral over every U is zero, so that inner product is zero almost everywhere. The original exponential sample bounds make every fixed polynomial section square-integrable. Intersecting the countably many resulting full-measure sets, (HCD21) gives $v(\theta) \in \mathcal{Q}_\theta$ there. Conversely every such section is orthogonal to the relations. The orthogonal-complement characterization proves that their closed span is exactly $\mathcal{R}$.

Since $\mathcal{Q} = 0$, $\mathcal{R} = \mathcal{H}$ as Hilbert spaces, and the completed complex is the identity complex. Its degree-one cohomology is zero. The finite continuous common-image complex has positive Gram $\overline{G}$ on its full E by (HCD8)–(HCD12). The intervening map and its whole kernel are precisely (HCD23)–(HCD28).

77.10 The receiving supported zero and all marked fibres

Integrate in the stated coefficient fibre, then apply the original marked support construction. At a fixed original label ℓ , a linear map f has the lift

$$(\ell, x) \mapsto (\ell, f(x)), \quad \tau \mapsto \tau. \quad (\text{HCD29})$$

For (HCD27), every represented input maps to the receiving supported zero $(\ell, 0)$, and external absence maps to external absence τ . Their inverse images are respectively the full represented fibre and the singleton external absence. At a fixed phase, the represented kernel is the entire ideal (HCD24), with its full primary decomposition (HCD25). No integral on arbitrary mixed-support totals is presumed.

If the empty-packet case $q = 0$ is separately admitted, $E = 0$, $B_N = P_N$, and the finite and continuous coefficient complexes are identity complexes. The quotient Gram determinant is the empty determinant one. The analytic source and its literal norm remain present before the quotient. Its marked receiving object still has τ and the represented zero, with the same maps (HCD29).

77.11 Exact outcome and scope

The continuous common-image complex has the explicit bounded cochain contraction (HCD7) on its complement and retains the whole nonreduced quotient in degree one. Its minimum metric is $\overline{G}$, and its difference from the constant-source metric is the positive original relation norm (HCD13). The mean-zero decomposition retains the weighted cross term (HCD15), giving (HCD16).

The subsequent completion has receiving weighted value spaces (HCD22), primary kernels (HCD24)–(HCD25), measurable field (HCD26), and actual cochain map (HCD28). Its receiving coefficient direct integral is zero because only finitely many phases can sample a fixed polynomial root. Every exceptional phase, nilpotent component and zero-weight coordinate remains in the stated maps and kernels. No arithmetic volume asymptotic or RH conclusion is inferred.

78 Holonomy descent and the original arithmetic generator

78.1 Holonomy descent cost and the original arithmetic generator

This calculation joins the complete holonomy source construction to the original metric commutator and residue calculation. We retain the tensor degree $k \in \mathbb{Z}_{\geq 1}$, the monic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, all its primary multiplicities, the coefficient frame of $E = \mathbb{C}[S]/(\chi)$, $A = M_S$, $e_0 = [1]$, and $\ell[P] = [S^{q-1}] \text{rem}_\chi P$. Put $R = e_0 \ell$. The arithmetic inclusion $\eta[P] = v_h^{\otimes k} P(A_k)1$ and the source seed with Mellin transform g/h , $g = 2\xi$, remain fixed. The scalar error below is written δ , so it does not rename that inclusion. The phase parameter θ , period parameter t , and coefficient translation parameter remain separate original coordinates.

78.1.1 The actual relation norm as a metric endomorphism

Fix $N \geq q - 1$ and let $J : \mathcal{P}_N \rightarrow E$ be the original polynomial remainder map. Denote the literal multiplication matrix of χ by $B_{\text{rel}} : \mathbb{C}^{N+1-q} \hookrightarrow \mathcal{P}_N$. For $N = q - 1$ its domain is zero. Let Ψ_N be the original source columns after the half-density map

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad \mathcal{K} = L^2(\mathbb{R}^{k-1}, dz). \quad (\text{HG1})$$

The logarithmic Jacobian is $d^k x = e^{kr + \sum z_i} dr dz$. Thus the displayed map preserves the original source norm, and $\mathcal{U}_k D^{(k)} = (-\partial_r + k/2)\mathcal{U}_k$. For $L > 0$, define

$$\begin{aligned} (\mathcal{Z}_{L,\theta}\psi)(r) &= \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad 0 \leq r < L, \quad d\mu(\theta) = \frac{d\theta}{2\pi}, \\ M &= \Psi_N^* \Psi_N, \quad M_\theta = (\mathcal{Z}_{L,\theta} \Psi_N)^* (\mathcal{Z}_{L,\theta} \Psi_N). \end{aligned} \quad (\text{HG2})$$

For a square-integrable source, Fourier series in the translation index give $\int \|\mathcal{Z}_{L,\theta}\psi\|^2 d\mu = \|\psi\|^2$. Polarization gives $\int M_\theta d\mu = M$. The inverse is $\psi(r + aL) = \int e^{-ia\theta} \mathcal{Z}_{L,\theta}\psi(r) d\mu$, first as a Fourier coefficient for almost every r , and hence in the original Hilbert space. The actual finite columns have the weighted regularity established in the source construction, so their sums and derivatives converge in the stated norms.

Keep the two weighted source Grams and their actual comparison value

$$M_{a,\pm} = (e^{\pm ar} \Psi_N)^* (e^{\pm ar} \Psi_N), \quad \kappa = \max_{v \neq 0} \frac{v^* (M_{a,+} + M_{a,-}) v}{v^* M v}, \quad \delta = \frac{\kappa}{e^{aL} - 1}, \quad a > 0. \quad (\text{HG3})$$

Their original coordinate weights are $x_k^{\pm 2a}$. For $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r + s) dr$, weighted Cauchy-Schwarz gives, for $s \geq 0$, $|v^* \mathcal{C}(s) v| \leq e^{-as} \sqrt{(v^* M_{a,-} v)(v^* M_{a,+} v)}$; the bound at $-s$ follows by adjunction. Unfolding the periodization gives $M_\theta = \sum_{b \in \mathbb{Z}} e^{ib\theta} \mathcal{C}(bL)$. Sum the two tails and use $2\sqrt{uv} \leq u + v$ to obtain $-\delta M \preceq M_\theta - M \preceq \delta M$. Choose L as explicitly specified below so that $\delta < 1$. This proves positivity of every phase Gram in the admitted original source, including the cases where individual sampled weights vanish.

Retain its canonical sections and Grams:

$$\begin{aligned} G &= (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \\ G_\theta &= (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta, \quad \overline{G} = \int G_\theta d\mu. \end{aligned} \quad (\text{HG4})$$

Completing the square on $J^{-1}(v)$ proves these are exactly the minimum-norm representatives. In particular $JC = JC_\theta = I$ and $B_{\text{rel}}^* MC = B_{\text{rel}}^* M_\theta C_\theta = 0$. There is a unique X_θ with

$$\begin{aligned} C - C_\theta &= B_{\text{rel}} X_\theta, \quad X_\theta = (B_{\text{rel}}^* M_\theta B_{\text{rel}})^{-1} B_{\text{rel}}^* M_\theta C, \\ \mathcal{D} &= \int (C - C_\theta)^* M_\theta (C - C_\theta) d\mu, \quad G = \overline{G} + \mathcal{D}. \end{aligned} \quad (\text{HG5})$$

The equality follows by expanding $C = C_\theta + B_{\text{rel}} X_\theta$: the two cross terms vanish by the displayed orthogonality, and $\int C^* M_\theta C d\mu = C^* MC = G$. Empty relation columns give $C = C_\theta = J^{-1}$ and $\mathcal{D} = 0$ directly. Every $B_{\text{rel}} X_\theta v$ is multiplication by the original χ ; it has the original monic-division theta primitive in the retained tensor complex. In particular this positive norm is evaluated on actual relations before passing to their cohomology class.

Here is a complete proof of the quadratic bound used below. For $1 - \delta \leq x \leq 1 + \delta$,

$$\frac{2-x}{1-\delta^2} - \frac{1}{x} = \frac{\delta^2 - (x-1)^2}{x(1-\delta^2)} \geq 0. \quad (\text{HG6})$$

Apply the finite spectral theorem to the positive M -self-adjoint endomorphism $M^{-1}M_\theta$ and then the original dual map M^{-1} . This gives $M_\theta^{-1} \preceq (2M^{-1} - M^{-1}M_\theta M^{-1})/(1 - \delta^2)$. Consequently $\overline{K} = \int JM_\theta^{-1}J^*d\mu \preceq G^{-1}/(1 - \delta^2)$. Each block matrix with rows (G_θ^{-1}, I) and (I, G_θ) is positive: its quadratic form at (x, y) is $(x + G_\theta y)^*G_\theta^{-1}(x + G_\theta y)$. Averaging and taking the Schur complement proves $\overline{G} \succeq \overline{K}^{-1} \succeq (1 - \delta^2)G$. Together with (HG5) this proves

$$0 \preceq \mathcal{D} \preceq \delta^2 G, \quad \mathbf{D} = G^{-1}\mathcal{D} = \mathbf{D}^\dagger G, \quad 0 \preceq_G \mathbf{D} \preceq_G \delta^2 I, \quad G^{-1}\overline{G} = I - \mathbf{D}. \quad (\text{HG7})$$

The original inner product convention is $\langle v, w \rangle_G = v^* G w$. No metric or vector mass is replaced in this construction.

78.1.2 The full commutator receives the quadratic relation cost

Put $T_t = A + tR - kI/2$ for the original complex parameter t and real integer k . For any actual positive Gram H on E , let

$$X_H = T_t^{\dagger H} + T_t, \quad \epsilon_H(t) = \|X_H\|_H, \quad B = G^{-1}(H - G). \quad (\text{HG8})$$

The endomorphism B is G -self-adjoint. Suppose its actual spectral lower bound is $I + B \succeq_G \alpha I$ with $\alpha > 0$; the concrete holonomy and finite values of α are proved below. Let $\mathbf{S} = (I + B)^{1/2}$ be the positive square root in the G metric. It is the exact isometry $(E, H) \rightarrow (E, G)$ because $\mathbf{S}^* G \mathbf{S} = G(I + B) = H$. Direct multiplication gives

$$\begin{aligned} \mathbf{S} X_H \mathbf{S}^{-1} - X_G &= Z + Z^{\dagger G}, \quad Z = [\mathbf{S}, T_t] \mathbf{S}^{-1}, \\ [\mathbf{S}, T_t] &= \int_0^\infty e^{-r\mathbf{S}} [B, T_t] e^{-r\mathbf{S}} dr. \end{aligned} \quad (\text{HG9})$$

For the first equality expand the left side as $\mathbf{S}^{-1} T_t^{\dagger G} \mathbf{S} + \mathbf{S} T_t \mathbf{S}^{-1} - T_t^{\dagger G} - T_t$. Its two ordered terms are exactly $Z^{\dagger G}$ and Z . For the second equality, both sides solve the Sylvester equation $\mathbf{S} Y + Y \mathbf{S} = [B, T_t]$. The integral converges since $\|e^{-r\mathbf{S}}\|_G \leq e^{-r\sqrt{\alpha}}$. Differentiate the integrand to verify the equation, with its boundary value $[B, T_t]$ at zero and zero at infinity. In the spectral decomposition of $\mathbf{S}$, a homogeneous solution has each block multiplied by a strictly positive sum of two eigenvalues, so is zero. This proves the claimed integral equality, including repeated eigenvalues. Its norm is at most $\|[B, T_t]\|_G/(2\sqrt{\alpha})$. The isometry, reverse triangle inequality, and $\|\mathbf{S}^{-1}\|_G \leq \alpha^{-1/2}$ therefore prove

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \alpha^{-1} \|[B, A] + t[B, R]\|_G. \quad (\text{HG10})$$

This is the full actual-metric transfer supplied by the coordinating Zeta task, here proved for the present maps and parameters.

For the exact phase mean set $H = \overline{G}$. Equations (HG5)–(HG7) give $B = -\mathbf{D}$ and $\alpha = 1 - \delta^2$. Hence

$$\begin{aligned} |\epsilon_{\overline{G}}(t) - \epsilon_G(t)| &\leq \|Z + Z^{\dagger G}\|_G \leq \frac{\|[D, A] + t[D, R]\|_G}{1 - \delta^2}, \\ \frac{\|[D, A] + t[D, R]\|_G}{1 - \delta^2} &\leq \frac{\delta^2}{1 - \delta^2} \|A + tR - kI/2\|_G. \end{aligned} \quad (\text{HG11})$$

Here is the full spectral argument for the last inequality. For any G -self-adjoint B with distinct eigenvalues $b_1 < \dots < b_r$, retain its original spectral projections and put $P_j = \mathbf{1}_{(b_j, \infty)}(B)$. The exact decomposition is $B = b_1 I + \sum_{j=1}^{r-1} (b_{j+1} - b_j) P_j$. In the G -orthogonal decomposition into the image and kernel of P_j , the commutator $[P_j, T]$ has block rows $(0, T_{12})$ and $(-T_{21}, 0)$. Its norm is $\max(\|T_{12}\|, \|T_{21}\|) \leq \|T\|_G$, as follows by applying the two off-diagonal blocks to the orthogonal components of an arbitrary vector. Consequently

$$\|[B, T]\|_G \leq (b_r - b_1) \|T\|_G. \quad (\text{HG11a})$$

When $r = 1$, the sum is empty and the commutator is zero. This proof retains the complete eigenvalue data and every spectral subspace. Applying it to (HG7) gives the last inequality in (HG11); the exact commutator in its preceding line remains available. Thus the original source-relative phase error δ reaches the generator through a proved quadratic allowance. If $[D, T_t] = 0$, the integral in (HG9) is zero and the two control radii agree exactly. At $N = q - 1$ this equality holds for every t because $\mathcal{D} = 0$.

78.1.3 Exact residue contribution from two original Gram determinants

For any G -self-adjoint B in (HG8), retain $x = e_0$ and $y = G^{-1}\ell^*$. Then $\ell(v) = \langle y, v \rangle_G$ and $R = xy^{\dagger G}$. Define the maps $U, V : \mathbb{C}^2 \rightarrow E$ and the six actual scalar moments

$$\begin{aligned} U &= [Bx, x], & V &= [y, -By], & [B, R] &= UV^{\dagger G}, \\ a_0 &= \|Bx\|_G^2, & c_0 &= \|x\|_G^2, & d_0 &= \|y\|_G^2, & f_0 &= \|By\|_G^2, \\ p_0 &= \langle x, Bx \rangle_G \in \mathbb{R}, & r_0 &= \langle y, By \rangle_G \in \mathbb{R}. \end{aligned} \quad (\text{HG12})$$

Self-adjointness proves the two reality statements. The signed factorization follows on a vector v from $Bx\langle y, v \rangle_G - x\langle By, v \rangle_G = Bx\ell(v) - x\ell(Bv)$. Its two coefficient Grams are

$$\begin{aligned} P &= U^{\dagger G}U = \begin{pmatrix} a_0 & p_0 \\ p_0 & c_0 \end{pmatrix}, & Q &= V^{\dagger G}V = \begin{pmatrix} d_0 & -r_0 \\ -r_0 & f_0 \end{pmatrix}, \\ T_{\text{res}} &= \text{tr}(PQ) = a_0d_0 + c_0f_0 - 2p_0r_0, & \Delta_{\text{res}} &= (a_0c_0 - p_0^2)(d_0f_0 - r_0^2). \end{aligned} \quad (\text{HG13})$$

Both negative cross entries and their complete contribution remain. Let $K = UV^{\dagger G}$. The nonzero eigenvalues of $K^{\dagger G}K = V^{\dagger G}P^{\dagger G}Q^{\dagger G}V$, with multiplicities, are those of PQ . To see this without a rank assumption, compute the determinant of the block matrix with rows $(I, -zA_0)$ and (B_0, I) by either diagonal block elimination. This proves $\det(I + zA_0B_0) = \det(I + zB_0A_0)$ for rectangular maps. Apply it to $A_0 = V, B_0 = PV^{\dagger G}$. Applying the same identity in dimension two proves that PQ has the same characteristic polynomial as the positive semidefinite matrix $P^{1/2}QP^{1/2}$, including when P is singular. Its two eigenvalues are therefore real and nonnegative. Their sum and product are (HG13). It follows that

$$\|[B, R]\|_G^2 = \frac{T_{\text{res}} + \sqrt{T_{\text{res}}^2 - 4\Delta_{\text{res}}}}{2}. \quad (\text{HG14})$$

The square root is the nonnegative real one. The discriminant is nonnegative by the positive spectrum just proved, and both factors of Δ_{res} are nonnegative Gram determinants.

There is no division by a vector mass or determinant. If $T_{\text{res}} = 0$, positivity gives $K = 0$ and $\Delta_{\text{res}} = 0$. If $\Delta_{\text{res}} = 0 < T_{\text{res}}$, the rank is one and its norm is $\sqrt{T_{\text{res}}}$. If $\Delta_{\text{res}} > 0$, the rank is two; a zero discriminant retains both equal singular-value squares $T_{\text{res}}/2$. For $q = 1$, write $B = bI$ with b real. Then $a_0 = b^2c_0, p_0 = bc_0, f_0 = b^2d_0, r_0 = bd_0$, which give $T_{\text{res}} = \Delta_{\text{res}} = 0$ with the original masses unchanged. Repeated roots of χ do not enter any inversion in this argument.

For the holonomy application use $B = -D$ in (HG12)–(HG14). One may bound the full allowance by

$$\frac{\|[D, A] + t[D, R]\|_G}{1 - \delta^2} \leq \frac{\|[D, A]\|_G + |t|\sqrt{(T_{\text{res}} + \sqrt{T_{\text{res}}^2 - 4\Delta_{\text{res}}})/2}}{1 - \delta^2}. \quad (\text{HG15})$$

The left side retains cancellation between the arithmetic and residue commutators. The right side is its triangle-inequality bound, while (HG14) itself is an exact norm. The complete rank-two calculation and degenerate cases incorporate the coordinating Zeta task's residue proof.

78.1.4 Finite phase and frequency values with explicit choices

The preceding quantities also have a finite approximation on the same polynomial source. Let $\kappa_{\pm}$ be the largest generalized eigenvalues of $(M_{a,\pm}, M)$ and put $b_a = \sqrt{\kappa_+ \kappa_-}$. Weighted Cauchy–Schwarz on two distinct vectors proves $\|M^{-1}\mathcal{C}(nL)\|_M \leq b_a e^{-a|n|L}$. For $0 < s < aL$ define

$$d_s = \frac{2b_a}{e^{aL-s} - 1}, \quad \varepsilon = \frac{2}{(1 - d_s)(e^{sm} - 1)}, \quad H_m = \frac{1}{m} \sum_{j=0}^{m-1} G_{2\pi j/m}. \quad (\text{HG16})$$

When $d_s < 1$, the correlation series is holomorphic for $|\operatorname{Im} z| < aL$ and its source-relative difference from M has norm at most d_s on the closed smaller strip. Its holomorphic quotient is the literal Schur complement

$$\tilde{G}(z) = C^* \tilde{M}(z) C - C^* \tilde{M}(z) B_{\text{rel}} (B_{\text{rel}}^* \tilde{M}(z) B_{\text{rel}})^{-1} B_{\text{rel}}^* \tilde{M}(z) C. \quad (\text{HG17})$$

Here the star acts on the fixed coefficient maps; it does not conjugate z . Orthogonality for M gives diagonal bound $1 + d_s$, cross bounds d_s , and inverse bound $(1 - d_s)^{-1}$ in the original quotient and relation metrics, by the Neumann series. Therefore $\|G^{-1}\tilde{G}(z)\|_G \leq 1 + d_s + d_s^2/(1 - d_s) = (1 - d_s)^{-1}$. At real $z = \theta$ this is exactly G_θ . Shift the contour defining its n th Fourier coefficient to $\operatorname{Im} z = -s \operatorname{sign}(n)$. Periodicity cancels the vertical sides and bounds the coefficient by $(1 - d_s)^{-1} e^{-s|n|}$. Finite phase averaging keeps the multiples of m ; summing the nonzero such coefficients proves $\|G^{-1}(H_m - \overline{G})\|_G \leq \varepsilon$.

For integer $p \geq 1$ keep the actual derivative Gram $H_{p,N} = \int (1 + r^2)(\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr$ and its comparison value $h_p = \max_{v \neq 0} v^* H_{p,N} v / (v^* M v)$. The quasi-periodic Fourier basis is $L^{-1/2} e^{-i(2\pi n + \theta)r/L}$, whose coefficient is $L^{-1/2} \hat{\Psi}_N((2\pi n + \theta)/L)$ for the convention $\hat{\psi}(u) = \int \psi(r) e^{iur} dr$. Integration by parts and weighted Cauchy–Schwarz give $\|\widehat{\Psi}_N v(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For $J \geq 2$, sum the omitted $|n| > J$ with $|n + \theta/(2\pi)| \geq |n| - 1$. Comparing the sum from J with the integral from $J - 1$ proves the uniform relative source tail bound

$$\epsilon_J = \frac{h_p}{2p-1} \left(\frac{L}{2\pi(J-1)} \right)^{2p-1}, \quad 0 \preceq M_\theta - M_{\theta,J} \preceq \epsilon_J M, \quad \xi_J = \frac{\epsilon_J}{1 - \delta}. \quad (\text{HG18})$$

The full $\theta = 0, n = 0$ coefficient remains among the retained terms. When $\xi_J < 1$, minimizing on the same remainder fibres gives $(1 - \xi_J)G_\theta \preceq G_{\theta,J} \preceq G_\theta$. Consequently the actual finite mean and its constants satisfy

$$H_{m,J} = \frac{1}{m} \sum_j G_{2\pi j/m,J}, \quad \alpha = (1 - \xi_J)(1 - \delta^2 - \varepsilon), \quad \beta = 1 + \varepsilon, \quad (\text{HG19})$$

$$\alpha G \preceq H_{m,J} \preceq \beta G.$$

These constants can be made positive by explicit choices from the same original source. Choose budgets $0 < \delta_0 < 1, 0 < \varepsilon_0 < 1 - \delta_0^2$, and $0 < \xi_0 < 1$, and set

$$L = a^{-1} \log(1 + \kappa/\delta_0 + 4b_a), \quad s = aL - \log(1 + 4b_a),$$

$$m = \max\{1, \lceil s^{-1} \log(1 + 4/\varepsilon_0) \rceil\}, \quad (\text{HG20})$$

$$J = 2 + \left\lceil \frac{L}{2\pi} \left(\frac{h_p}{(2p-1)\xi_0(1-\delta)} \right)^{1/(2p-1)} \right\rceil.$$

Both weighted Grams are positive on this finite source, so $b_a > 0$. Thus $0 < s < aL, d_s = 1/2, \delta < \delta_0, \varepsilon \leq \varepsilon_0$, and $\xi_J \leq \xi_0$. Substitution proves $\alpha > 0$. These choices use actual weighted source integrals; they are explicit mathematical values without an assertion that numerical interval enclosures for those values have already been computed.

The relation cost and the finite tail remain separate exact summands:

$$B_{m,J} = G^{-1}(H_{m,J} - G) = -D + E_{m,J}, \quad E_{m,J} = G^{-1}(H_{m,J} - \overline{G}). \quad (\text{HG21})$$

Indeed $H_{m,J} \preceq H_m \preceq \overline{G} + \varepsilon G$, whereas $H_{m,J} \succeq (1 - \xi_J)(\overline{G} - \varepsilon G)$. Since $\overline{G} \preceq G$, these imply $-\rho G \preceq H_{m,J} - \overline{G} \preceq \varepsilon G$, where $\rho = \xi_J + (1 - \xi_J)\varepsilon \geq \varepsilon$ because $\varepsilon < 1$. Hence $\|E_{m,J}\|_G \leq \rho$. Use (HG10) with the proved α in (HG19) to obtain

$$\begin{aligned} |\epsilon_{H_{m,J}}(t) - \epsilon_G(t)| &\leq \alpha^{-1} \| -[D, A] - t[D, R] + [E_{m,J}, A] + t[E_{m,J}, R] \|_G, \\ \| -[D, T_t] + [E_{m,J}, T_t] \|_G &\leq \| [D, T_t] \|_G + (\rho + \varepsilon) \| T_t \|_G. \end{aligned} \quad (\text{HG22})$$

The second line uses (HG11a) with the proved spectral interval $[-\rho, \varepsilon]$ of $E_{m,J}$. The primary full commutator retains cancellation between both sources of discrepancy. Independently (HG19) gives the spectral interval $[\alpha - 1, \beta - 1]$ for $B_{m,J}$. Equation (HG11a) therefore yields the further valid allowance $(\beta - \alpha) \| T_t \|_G / \alpha$. The lesser of independently proved allowances is valid. Formula (HG14) applies without change to the actual $B_{m,J}$, including its finite tail, and computes its entire rank-at-most-two residue commutator.

78.1.5 The same finite observation retains the full action boundary

For each retained phase let $C_{\theta,J}$ be the canonical section for $M_{\theta,J}$. Monic division gives $SC_{\theta,J} - C_{\theta,J}A = \chi Q_{\theta,J}$ in $\mathcal{P}_{N+1}$, with all original coefficients. Apply the source, half-density, twisted periodization and projection to $|n| \leq J$ to these two columns. Denote the resulting representative and boundary by $\mathcal{R}_j, \mathcal{B}_j$. Then

$$\begin{aligned} D_{L,\theta_j} \mathcal{R}_j - \mathcal{R}_j A &= \mathcal{B}_j, & \mathcal{R}_j^* \mathcal{R}_j &= G_{\theta_j,J}, \\ A^* H_{m,J} + H_{m,J} A - k H_{m,J} &= -\frac{1}{m} \sum_j (\mathcal{R}_j^* \mathcal{B}_j + \mathcal{B}_j^* \mathcal{R}_j). \end{aligned} \quad (\text{HG23})$$

Here $D_{L,\theta_j} = -\partial_r + k/2$ on the quasi-periodic domain. The projection commutes with this operator. Integration by parts has zero boundary difference because both columns have boundary multiplier $e^{-i\theta_j}$ of modulus one. Thus $D^* + D = k$, which proves the second line by inserting the first and expanding both ordered terms. For complex t the complete control form adds $\bar{t} R^* H_{m,J} + t H_{m,J} R$ to its second line. This gives the actual finite phase/frequency observation entering $\epsilon_{H_{m,J}}(t)$ in (HG22), with its theta relation intact.

For a nonzero proper A -invariant inclusion $I : F \hookrightarrow E$, the residue coefficient is $\mathfrak{c}_F(T) = \min_x \|e_0 - Ix\|_T^2 \ell I(I^* T I)^{-1} I^* \ell^*$. Both factors are strictly positive: if $e_0 \in F$, cyclicity gives $F = E$; if $\ell|_F = 0$, invariance would force $\ell A^j|_F = 0$ for every j , and perfection of the residue pairing would give $F = 0$. Minimizing the first factor and inverting the second on the same F shows that $\alpha G \preceq T \preceq \beta G$ gives

$$\frac{\alpha}{\beta} \mathfrak{c}_F(G) \leq \mathfrak{c}_F(T) \leq \frac{\beta}{\alpha} \mathfrak{c}_F(G). \quad (\text{HG24})$$

For $T = \overline{G}$ use $\alpha = 1 - \delta^2$, $\beta = 1$; for $T = H_{m,J}$ use (HG19). The constant quotient, original unit, and original functional in this coefficient are unchanged. The exact period metric $\Pi(t)^* \Pi(t)$ retains its separately proved comparison to these source Grams and its curvature $|t|^2 \mathfrak{c}_F(\Pi^* \Pi) / |u|^2$.

The transfer also reaches the original Laplacian, including its numerical range. For $H = \overline{G}$ or $H = H_{m,J}$, use its actual S, Z from (HG9) and set $e = \epsilon_H(t) + \|Z + Z^\dagger\|_G$. Equation (HG10) proves $e \geq \epsilon_G(t)$. Larger proved allowances from (HG11) or (HG22) may replace the second summand. Keep $\mathcal{L}_t = (A + tR)^2 - k(A + tR) = T_t^2 - k^2 I/4$. Then the identity $X_G = T_t^{\dagger G} + T_t$ gives $G\mathcal{L}_t = GX_G T_t - T_t^* G T_t - k^2 G/4$. For an arbitrary nonzero original vector v , put $z = v^* G \mathcal{L}_t v / (v^* G v)$, $r = \|T_t v\|_G / \|v\|_G$, and $a_1 + ib_1 = \langle v, X_G T_t v \rangle_G / (v^* G v)$. Cauchy-Schwarz gives $a_1^2 + b_1^2 \leq e^2 r^2$, while the exact expansion gives $\text{Re } z = a_1 - r^2 - k^2/4$ and $\text{Im } z = b_1$. Completing the square in $er - r^2$ and

then using the exact identity

$$\begin{aligned} e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - b_1^2 &= (a_1 - e^2/2)^2 + e^2 r^2 - a_1^2 - b_1^2 \geq 0, \\ \operatorname{Re} z &\leq \frac{e^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq e^2 \left(\frac{e^2 - k^2}{4} - \operatorname{Re} z \right) \end{aligned} \quad (\text{HG25})$$

prove the stated enclosure for every vector in the original metric. It includes every eigenvalue by inserting its original eigenvector, with generalized eigenspaces left in the source. No division by e occurs, so $e = 0$ is included. This carries the actual quadratic holonomy allowance and finite relation boundary into the same Laplacian polynomial; it does not infer a bound on $\epsilon_H(t)$ from its finite presentation alone.

Finally every invertible original coordinate change $v = Wv'$ sends G, H to W^*GW, W^*HW , and sends B, D, T_t to their $W^{-1}(\cdot)W$ matrices. It sends x, y to $W^{-1}x, W^{-1}y$ and ℓ to ℓW . The positive square root in (HG9) transforms by the same similarity by uniqueness. Therefore the six moments, Gram determinants, all norms, and each ordered identity above are unchanged as values of the original objects. On a retained split support label λ , each coefficient map f acts by $(\lambda, v) \mapsto (\lambda, fv)$ and fixes external τ . A vanishing relation or commutator is the receiving supported zero. This applies also to the isometries and inverse maps above, so the support observation and arithmetic observation commute with every displayed comparison.

79 Exact phase observation and its full degree cost

79.1 Exact fixed-phase observation and its degree cost

Fix the original packet h , integer tensor degree $k \geq 1$, and $\chi = \chi_{h,k}$; retain $g = 2\xi$. This section varies only the polynomial cutoff N . The original source density and its measure are

$$w_h(u) = \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \quad m(u) = w_h^{*k}(u), \quad d\nu(u) = m(u) du, \quad S(u) = k/2 + iu. \quad (\text{HD1})$$

The full gamma/eta estimates in the periodized-source proof give $\int e^{b|u|} w_h(u) du < \infty$ for $0 < b < \pi/2$. For this same b the inequality $e^{b|u_1 + \dots + u_k|} \leq \prod_i e^{b|u_i|}$ and Tonelli's theorem give $\int e^{b|u|} m(u) du \leq (\int e^{b|u|} w_h(u) du)^k < \infty$. The measure has no atoms, and its density is positive almost everywhere: g/h has isolated real-line zeros after the removable selected roots are filled by their full derivatives. For $k \geq 2$ convolution is positive at every real argument, as proved by integrating the positive almost-everywhere factors. In particular every finite original polynomial Gram is positive.

Let M_N be that Gram in the exact coefficient frame $1, S, \dots, S^N$. For a fixed real u_0 put $S_0 = k/2 + iu_0$ and $a_N = (1, S_0, \dots, S_0^N)$. Retain the actual values

$$\Lambda_N(u_0) = a_N M_N^{-1} a_N^*, \quad c_N = M_N^{-1} a_N^* / \Lambda_N(u_0), \quad P_N^{u_0}(S) = \sum_{j=0}^N (c_N)_j S^j. \quad (\text{HD2})$$

The superscript labels this particular polynomial, rather than changing the original source space $\mathcal{P}_N$. The Cauchy-Schwarz inequality in the M_N metric gives

$$\begin{aligned} \Lambda_N(u_0) &= \sup_{P \in \mathcal{P}_N \setminus \{0\}} \frac{|P(S_0)|^2}{\int |P(S(u))|^2 d\nu(u)}, \\ P_N^{u_0}(S_0) &= 1, \quad \int |P_N^{u_0}(S(u))|^2 d\nu(u) = \Lambda_N(u_0)^{-1}. \end{aligned} \quad (\text{HD3})$$

Equality in the supremum follows by inserting $M_N^{-1}a_N^*$. This calculation retains every original mass and phase of M_N .

We prove $\Lambda_N(u_0) \uparrow \infty$, including at a root of χ . Monotonicity follows from the nested polynomial spaces. Suppose for contradiction their suprema had finite bound K . Then evaluation at S_0 would be a bounded linear functional on the polynomial subspace of $L^2(\nu)$, and would extend to its closure. The Hilbert space representation theorem supplies f in that closure with $P(S_0) = \int \bar{f}(u)P(S(u))d\nu(u)$ for every polynomial P . Since $S(u) = k/2 + iu$ is an invertible affine coordinate over $\mathbb{C}$, this includes all polynomials in u with their full coefficients. Define the finite complex measure $d\sigma = \bar{f}d\nu - \delta_{u_0}$. It has all polynomial moments zero. Cauchy–Schwarz gives

$$\int e^{b|u|/2}|f(u)|d\nu(u) \leq \|f\|_{L^2(\nu)} \left(\int e^{b|u|}d\nu(u) \right)^{1/2} < \infty. \quad (\text{HD4})$$

Thus $F(z) = \int e^{izu}d\sigma(u)$ is holomorphic in $|\text{Im } z| < b/2$. On every strictly smaller strip, dominated differentiation follows by absorbing each power of $|u|$ into the remaining exponential margin. All derivatives of F at zero vanish. Its power series and the identity theorem imply $F = 0$ on that connected strip, and in particular on the real axis.

For completeness, this forces the measure itself to vanish. For $t > 0$ use the Gaussian $\gamma_t(x) = (4\pi t)^{-1/2}e^{-x^2/(4t)}$. Its integrable Fourier representation is $\gamma_t(x) = (2\pi)^{-1} \int e^{-t\zeta^2}e^{-i\zeta x}d\zeta$, obtained by the Gaussian integral. Fubini then gives $(\sigma * \gamma_t)(x) = 0$ since $F(\zeta) = 0$. For each compactly supported continuous φ , $\varphi * \gamma_t \rightarrow \varphi$ uniformly: split the convolution into $|y| < \varepsilon$, controlled by uniform continuity, and its Gaussian tail, whose mass tends to zero. Integration against the finite measure σ therefore gives $\int \varphi d\sigma = 0$. The uniqueness of finite regular measures on continuous test functions gives $\sigma = 0$. But $\bar{f}\nu$ has zero mass at the singleton $\{u_0\}$ and δ_{u_0} has mass one there, a contradiction. This proves

$$\Lambda_N(u_0) \longrightarrow \infty, \quad \mathcal{U}_k \mathcal{V} P_N^{u_0} \longrightarrow 0 \text{ in the original source norm}, \quad P_N^{u_0}(S_0) = 1. \quad (\text{HD5})$$

In particular evaluation on the polynomial domain is not closable as an operator from its original Hilbert closure to $\mathbb{C}$: its graph has the limiting pair $(0, 1)$. This conclusion concerns that specific evaluation map; all finite polynomial maps (HD2) remain exact.

Now fix $L > 0$, $0 \leq \theta < 2\pi$, and an integer n_0 with $u_0 = (2\pi n_0 + \theta)/L$ and $m(u_0) > 0$. Its original sampled mass is $\omega_0 = (2\pi/L)m(u_0) > 0$. The exact phase source matrix is

$$(M_{N,\theta})_{ij} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m((2\pi n + \theta)/L) \overline{S((2\pi n + \theta)/L)^i} S((2\pi n + \theta)/L)^j. \quad (\text{HD6})$$

Its n_0 summand gives $M_{N,\theta} \succeq \omega_0 a_N^* a_N$, so

$$\lambda_{\max}(M_N^{-1}M_{N,\theta}) \geq \omega_0 \Lambda_N(u_0) \longrightarrow \infty. \quad (\text{HD7})$$

For every actual weighted source exponent $a > 0$, let $\kappa_{a,N} = \lambda_{\max}(M_N^{-1}(M_{a,+,N} + M_{a,-,N}))$. The proved correlation comparison $M_{N,\theta} \preceq (1 + \kappa_{a,N}/(e^{aL} - 1))M_N$ does not require that its error be less than one; it follows from summing the same convergent tails. Consequently

$$\boxed{\kappa_{a,N} \geq (e^{aL} - 1)(\omega_0 \Lambda_N(u_0) - 1), \quad \kappa_{a,N} \longrightarrow \infty.} \quad (\text{HD8})$$

The finite original matrix expression (HD2) is an explicit lower certificate for this degree cost. To ensure the correlation error $\kappa_{a,N}/(e^{aL_N} - 1) \leq \delta_0 < 1$ by that estimate, the literal choice must satisfy $L_N \geq a^{-1} \log(1 + \kappa_{a,N}/\delta_0)$. This derives a constraint on these actual correlation constants at fixed h, k ; it does not assert a growth rate with the distinct tensor parameter k , or exclude another arithmetic estimate.

The source isometry over all holonomies is the exact map that relates this divergent single-phase observation to the full source. Equations (HG2) and (HD3) give

$$\int \|\mathcal{Z}_{L,\vartheta} \mathcal{U}_k \mathcal{V} P_N^{u_0}\|^2 d\mu(\vartheta) = \Lambda_N(u_0)^{-1} \longrightarrow 0, \quad \|\mathcal{Z}_{L,\vartheta} \mathcal{U}_k \mathcal{V} P_N^{u_0}\|^2 \geq \omega_0. \quad (\text{HD9})$$

The finite columns are regular and define the displayed phase values; the full direct integral is an equivalence class modulo null phase sets. Thus these are simultaneous statements about the very same columns, with the original positive sampled mass retained. In particular point evaluation in the phase parameter cannot be extended as a bounded operator from this full source norm through the direct integral.

There is an exact polynomial-quotient criterion as well. Evaluation at S_0 factors through the original $J : \mathbb{C}[S] \rightarrow E$ if and only if $\chi(S_0) = 0$: necessity follows by evaluating χ , and sufficiency follows from divisibility of every difference in $\ker J = (\chi)$. In that case the induced map $E \rightarrow \mathbb{C}$, $[P] \mapsto P(S_0)$, has kernel $(S - S_0)/(\chi)$, where the notation means the image of the ideal $(S - S_0)$ in E . If S_0 has multiplicity r in χ , write $\chi = (S - S_0)^r q_0$ with $q_0(S_0) \neq 0$. The map on its local factor $\mathbb{C}[x]/(x^r)$ is the constant coefficient, with kernel $(x)/(x^r)$; all other primary factors map to zero. The full nilpotents and every cofactor remain in this map's domain and kernel. If $\chi(S_0) \neq 0$, evaluation is still the exact source map in (HD2)–(HD9); its value on χ explains precisely why it does not factor through E .

For $k \geq 2$, every n_0 has positive mass. For $k = 1$, a sampled selected root of the complete packet has positive mass because, if $h = (S - S_0)^r h_0$ and g has its complete order r there, then $(g/h)(S_0) = g^{(r)}(S_0)/(r!h_0(S_0)) \neq 0$. Other zero-mass samples are absent from the receiving discrete Hilbert space and make no claim in (HD7). For the divergence of $\kappa_{a,N}$ alone, one may choose any real u_0 where $m(u_0) > 0$ and then its unique phase modulo 2π for the fixed L ; such a point exists by (HD1). This gives (HD8) without asserting that every one-fold phase has a positive selected root.

Every displayed source, evaluation, quotient and inverse coefficient map lifts on an original split label λ by $(\lambda, v) \mapsto (\lambda, fv)$ and fixes τ . For each retained label the exact inverse images under this lift are

$$\tilde{f}^{-1}(\{(\lambda, 0)\}) = \{(\lambda, v) : v \in \ker f\}, \quad \tilde{f}^{-1}(\{\tau\}) = \{\tau\}. \quad (\text{HD10})$$

Thus the nonzero coefficient vectors in the source kernel remain represented inputs and map to the receiving represented zero. The external absence has its displayed singleton inverse image. No identification of a positive sampled mass, a represented zero, or external absence is used in the degree-cost calculation.

80 The complete continuous relation closure

This continues the fixed-packet calculation HD1–9 without changing its measure, polynomial coordinate, quotient, multiplicities or metric. It supplies the continuous-source counterpart of the already proved sampled quotient P53–58 and H64. Every limit below fixes the original packet h and tensor degree k ; no tensor-degree rate or Riemann-hypothesis conclusion is asserted.

80.0.1 Inputs and exact maps

Let $g = 2\xi$, $k \geq 1$, $c = k/2$, and retain

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m(u) = w_h^{*k}(u), \quad d\nu(u) = m(u) du, \quad S(u) = c + iu. \quad (\text{CR1})$$

HD1–4 prove for this actual measure that, for any fixed $0 < b < \pi/2$, $\int e^{b|u|} d\nu(u) < \infty$. The nonzero density is positive almost everywhere, and ν has no atoms. Its total mass remains μ_h^k . Fix the

original nonzero monic cyclic polynomial $\chi = \chi_{h,k}$, with degree $q \geq 1$ and every primary multiplicity retained. Define

$$H = L^2(\mathbb{R}, d\nu), \quad H_\chi = L^2(\mathbb{R}, |\chi(S(u))|^2 d\nu), \quad E = \mathbb{C}[S]/(\chi), \quad J : \mathbb{C}[S] \longrightarrow E. \quad (\text{CR2})$$

For $N \geq q - 1$, let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{B}_N = \chi \mathcal{P}_{N-q}$, with $\mathcal{P}_{-1} = 0$. The source maps, their kernels and their arithmetic class maps are exactly

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{CR3})$$

No coefficient of χ or of the polynomial source has been rescaled. The original source norm is the norm of H through the specified Mellin observation. In particular all conclusions transport back through that original isometry, rather than selecting another Hilbert norm.

80.0.2 Polynomial density, with its measure proof

We first prove the exact density fact used twice below. Let η be any of the two finite measures ν or $|\chi(S)|^2 \nu$. Both have a positive exponential moment. For the second, polynomial growth gives $|\chi(c + iu)|^2 \leq C(1 + |u|)^{2q}$; choosing $0 < a < b$ absorbs this factor into $e^{(b-a)|u|}$, so $\int e^{a|u|} d\eta < \infty$. This bound is a consequence for the fixed original measure, not a new source hypothesis. All polynomial moments consequently exist.

If f in $L^2(\eta)$ is orthogonal to all polynomials in S , it is orthogonal to all polynomials in u because $u = (S - c)/i$ is the exact inverse coordinate. Set $d\sigma = \bar{f} d\eta$. Cauchy-Schwarz gives

$$\int e^{a|u|/2} d|\sigma|(u) \leq \|f\|_{L^2(\eta)} \left(\int e^{a|u|} d\eta(u) \right)^{1/2} < \infty. \quad (\text{CR4})$$

Thus $F(z) = \int e^{izu} d\sigma(u)$ is holomorphic for $|\text{Im } z| < a/2$. On each smaller closed strip, powers of $|u|$ are absorbed by the remaining exponential margin, justifying every derivative under the integral. All derivatives at zero vanish by the orthogonality equations. Its Taylor series and the identity theorem give $F = 0$ on the connected strip, including the real axis.

Here is the required uniqueness step explicitly. For $t > 0$ put $\gamma_t(x) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$. Its Fourier integral is $\gamma_t(x) = (2\pi)^{-1} \int e^{-t\zeta^2} e^{-i\zeta x} d\zeta$. Fubini, justified by the finite variation of σ and the integrable Gaussian factor, gives $\sigma * \gamma_t = 0$. For every continuous compactly supported ϕ , $\phi * \gamma_t \rightarrow \phi$ uniformly: uniform continuity controls $|y| < \epsilon$, and the Gaussian tail controls its complement. Integrating against σ gives $\int \phi d\sigma = 0$. Uniqueness of finite regular measures on continuous compactly supported test functions gives $\sigma = 0$. Hence $f = 0$ η -almost everywhere. The orthogonal complement of the polynomial span is zero, proving

$$\overline{\mathbb{C}[S]}^H = H, \quad \overline{\mathbb{C}[S]}^{H_\chi} = H_\chi. \quad (\text{CR5})$$

80.0.3 The original relation multiplication is an onto isometry

The map and its inverse on equivalence classes are

$$U_\chi : H_\chi \longrightarrow H, \quad f \longmapsto \chi(S)f, \quad U_\chi^{-1}v(u) = \frac{v(u)}{\chi(S(u))} \quad (\chi(S(u)) \neq 0). \quad (\text{CR6})$$

At the finitely many real u where $\chi(S(u)) = 0$, choose the inverse value zero. These points have ν -measure zero, so the choice does not affect either Hilbert equivalence class. Direct integration proves

$$\|U_\chi f\|_H^2 = \int |f|^2 |\chi(S)|^2 d\nu = \|f\|_{H_\chi}^2, \quad \|U_\chi^{-1} v\|_{H_\chi}^2 = \|v\|_H^2. \quad (\text{CR7})$$

The two composites are identities almost everywhere. Polynomial density in H_χ and this actual multiplier therefore give

$$\boxed{\overline{\chi\mathbb{C}[S]}^H = H, \quad H/\overline{\chi\mathbb{C}[S]} = 0.} \quad (\text{CR8})$$

The coordinate-level comparison from the original algebraic quotient is $[P]_\chi \mapsto P + \overline{\chi\mathbb{C}[S]} = 0$, with kernel all of E . This identifies the map, not the two quotients. More precisely, complete both terms of $[\chi\mathbb{C}[S] \hookrightarrow \mathbb{C}[S]]$ in their inherited H norms. Both completions identify with H by CR5/CR8, and the extended inclusion is I_H . The explicit degree-minus-one contraction of $[H \xrightarrow{I_H} H]$ is I_H . Before completion, the cohomology is the full algebraic E ; its induced map to this completed cohomology is zero with the stated kernel.

80.0.4 Limit of the same finite canonical metric

Use the fixed increasing-power remainder section $s : E \rightarrow \mathcal{P}_{q-1}$, the inverse of J on that finite subspace. Let Π_N be the H -orthogonal projection onto the finite original relation subspace $\mathcal{B}_N$. The exact least-norm lift and quotient Gram are

$$R_N = (I - \Pi_N)s, \quad J_N R_N = I_E, \quad G_N = R_N^* R_N. \quad (\text{CR9})$$

To verify minimality directly, every lift of x is $s(x) + b$ with b in $\mathcal{B}_N$. The decomposition into $\mathcal{B}_N$ and its orthogonal complement shows that the unique minimizing lift is $R_N x$. The adjoints in the displayed Gram use the fixed original coefficient frame on E and the actual source H norm. If M_N is the original source Gram, the equivalent coordinate formulas are

$$G_N = (J_N M_N^{-1} J_N^*)^{-1}, \quad R_N = M_N^{-1} J_N^* G_N. \quad (\text{CR10})$$

Their equality with CR9 follows from $J_N R_N = I$ and source orthogonality to $\ker J_N$; the minimizer is unique. Every finite G_N is positive definite.

The nested $\mathcal{B}_N$ have dense union by CR8. For any v in H and $\epsilon > 0$, choose b in some $\mathcal{B}_n$ with $\|v - b\| < \epsilon$. For $N \geq n$, $\|(I - \Pi_N)v\| \leq \|v - b\| < \epsilon$. Thus Π_N converges strongly to I_H . Applying this to each of the q fixed columns of s proves

$$\boxed{G_N \downarrow 0, \quad V_N = \det G_N \longrightarrow 0 \quad (N \longrightarrow \infty; h, k \text{ fixed}).} \quad (\text{CR11})$$

The decrease is Loewner decrease, because enlarging the admitted relation space cannot increase a minimum. Convergence is in every fixed finite coordinate matrix norm: each column norm of R_N tends to zero, and $|(G_N)_{ij}| \leq \|R_N e_i\| \|R_N e_j\|$. Determinant convergence then follows from its polynomial expression. This proves no growth rate.

For each nonzero x in E , the literal sequence $R_N x$ tends to zero in H but has $J(R_N x) = x$ for every N . Therefore the densely defined map $J : \mathbb{C}[S] \subset H \rightarrow E$, with any fixed positive finite norm on its original coefficient target, is not closable. Every $(0, x)$ lies in the closure of its graph. This is stronger than one evaluation witness, and records exactly why the algebraic class map has no continuous extension to this completed Hilbert quotient. It does not make J zero on its original polynomial domain.

In fact the graph closure is all of H times E . Given (v, y) , choose polynomials p_j tending to v by CR5, and put $x_j = y - Jp_j$. For each fixed j , CR9–11 allow a cutoff N_j with $\|R_{N_j} x_j\|_H < 1/j$. Then

the polynomial $p_j + R_{N,j}x_j$ tends to v and its exact J value is y . This proves the full graph assertion without a bound uniform in j or an extension of J to H .

Under any fixed coefficient change $x = Ux'$, the matrices and source lifts transform by $G'_N = U^*G_NU$, $R'_N = R_NU$, and the identity on classes transforms by $J'_N = U^{-1}J_N$. Hence CR9–11 retain precisely the same morphisms in another coordinate frame; no N -dependent frame or metric normalization is used.

80.0.5 The phase quotient and averaging comparison

Fix the original circumference L . No assertion that every $k = 1$ phase source Gram is invertible is needed. Write T_θ for the original polynomial evaluation into its positive-weight sampled Hilbert space H_θ . The precise finite receiving space and comparison map are

$$Q_{N,\theta} = T_\theta(\mathcal{P}_N)/T_\theta(\mathcal{B}_N), \quad \rho_{N,\theta} : E \longrightarrow Q_{N,\theta}, \quad [P]_\chi \longmapsto [T_\theta P], \quad \ker \rho_{N,\theta} = J_N(\ker(T_\theta|_{\mathcal{P}_N})). \quad (\text{CR12a})$$

Both images are finite-dimensional closed subspaces of H_θ and carry its actual norm. The map is onto. It is well defined because any two degree- N lifts differ by $\mathcal{B}_N$. If $T_\theta P$ lies in $T_\theta \mathcal{B}_N$, subtract the corresponding b in $\mathcal{B}_N$ to obtain an element of $\ker T_\theta$ with the same J_N class; this proves the displayed kernel in both directions. Its pulled-back squared norm is the uniquely specified positive semidefinite coefficient form

$$x^* G_{N,\theta} x = \inf_{b \in \mathcal{B}_N} \|T_\theta(sx + b)\|_{H_\theta}^2. \quad (\text{CR12b})$$

This is the same canonical quotient form: when the phase source Gram is positive, $\rho_{N,\theta}$ is an isomorphism and CR12b gives exactly the positive Gram $(J_N M_{N,\theta}^{-1} J_N^*)^{-1}$. When it is singular, CR12a states the actual additional kernel and CR12b uses no inverse. No zero-weight coordinate is inverted and no new metric is selected.

The form is measurable in θ : for each x its infimum may be taken over a countable dense set of relation coefficients, since its finite quadratic norm is continuous in those coefficients. Each such value is measurable from the original sampled measure. Polarization gives the matrix entries. Using the particular lift $R_N x$ in CR12b and the original all-phase source identity proves

$$\int x^* G_{N,\theta} x \frac{d\theta}{2\pi} \leq \int \|T_\theta R_N x\|_{H_\theta}^2 \frac{d\theta}{2\pi} = \|R_N x\|_H^2 = x^* G_N x. \quad (\text{CR12c})$$

In particular these integrals are finite. Define the actual mean form $\bar{G}_N = \int G_{N,\theta} d\theta / (2\pi)$ and its difference $D_N = G_N - \bar{G}_N$. Then, for all original $k \geq 1$,

$$G_N = \bar{G}_N + D_N, \quad D_N \succeq 0, \quad 0 \preceq \bar{G}_N \preceq G_N. \quad (\text{CR12})$$

On an admitted positive phase family, D_N agrees with the complete relation-gap formula HCD13, including its phasewise least-norm sections; CR12a–c extend the comparison through the specified kernels without asserting those inverse formulas at singular phases.

Consequently both $\bar{G}_N \rightarrow 0$ and $D_N \rightarrow 0$ in fixed coefficient norm, without interchanging any infinite-dimensional inverse and integral. The relative matrix $G_N^{-1} D_N$ is not assigned a zero limit by this assertion: only the preceding absolute inequalities have been proved. The actual relative transfer remains the previously calculated metric/generator commutator, not this unscaled matrix limit.

The independent sampled density argument P53–58/CQ14–22 gives, at each fixed phase, the exact limit

$$G_{N,\theta} \downarrow K_\theta = J_{Z_\theta}^* \text{diag}(\nu_{n,\theta}) J_{Z_\theta}, \quad Z_\theta = \{n : \nu_{n,\theta} > 0, \chi(S_{n,\theta}) = 0\}, \quad (\text{CR13})$$

where $S_{n,\theta} = c + i(2\pi n + \theta)/L$ and $\nu_{n,\theta} = (2\pi/L)m((2\pi n + \theta)/L)$. Its exact kernel is $(p_\theta)/(\chi)$, with p_θ the squarefree product of the displayed sampled roots. Both full primary multiplicities and all masses remain in the domain/map. The target takes only the stated scalar values.

If Z_θ is nonempty, choose an original class x whose evaluation at one of those roots is one (the class of 1 already suffices). Its canonical single-phase/source relative cost satisfies the explicit inequality

$$\frac{x^* G_{N,\theta} x}{x^* G_N x} \geq \frac{\nu_{n,\theta}}{x^* G_N x} \rightarrow \infty. \quad (\text{CR14})$$

This strengthens HD's full-source evaluation bound to this precise quotient comparison. It needs an actual positive-mass sampled root; none is invented. For $k = 1$ full-order cancellation supplies positivity at any selected sampled root; for $k \geq 2$ the original convolution is positive everywhere. If Z_θ is empty, CR13 has zero limit but CR14 makes no relative-rate assertion.

For the fixed χ , the exceptional phases are finite. The completed phase quotient field is therefore zero almost everywhere for its actual phase measure, and its direct integral is zero. The exact source isometry over all phases, the finite quotient identity $J_N R_N = I$, CR11, CR12 and the exceptional positive quotient limit CR13 are all compatible through their displayed maps. No arithmetic class is declared absent before the specified completion. A retained split label is transported as $(\lambda, v) \mapsto (\lambda, fv)$; the zero comparison takes it to $(\lambda, 0)$, fixes τ , and does not identify that receiving zero with external absence.

80.0.6 Provenance and scope

Read in full before deriving this note: the owner HD1–9 source, SHA256 56c05d9be82fa356f82d71324f317f86325e761531767b57d4a085e91b7760fa; holonomy_completed_quotient_density_review_20260913.md; and holonomy_continuous_descent_review_20260913.tex. P53–58 and surrounding original-source definitions were read directly in their delivered research note. CR4–11 are proved here, not inferred from a numerical check. CR12–14 explicitly compose those results with the named existing phase maps.

This is a written argument, pending independent review. No new Lean, numerical arithmetic enclosure, publication, tensor- k estimate or claimed solution of the remaining arithmetic upper-bound problem accompanies it.

81 Complete independent proof of the original continuous relation completion

81.1 Independent review: original continuous relation completion

Date: 2026-09-13. Scope: CR1–14 in the parent's period_laplacian/CONTINUOUS_RELATION_COMPLETION.md, including the subsequently written CR12a–c pullback construction. This is a written mathematical review, not a numerical or Lean certificate. Only this new independent review directory is written. No source, owner, public, or remote artifact is edited. No new model examples are needed.

81.1.1 Disposition and the actual positivity issue

Accept CR1–11 and CR13–14 with the exact domains and positive-support maps specified there. Accept the revised unconditional CR12a–c and CR12. The original HCD2 assumption is a uniformly positive finite phase family; it is not, by itself, a proof of that assumption for every one-fold phase. Full-order cancellation proves positivity at a sampled selected root, not positivity of every sampled weight or invertibility of every finite one-fold source Gram. The pullback construction resolves that distinction without adding a hypothesis or choosing a new source norm.

For the actual source one can also prove, without uniform phase positivity, that every finite averaged pullback form is positive definite, and that these average forms decrease to zero. The full graph closure of the original class map is $H \times E$. Proofs and qualifications follow.

81.1.2 1. Continuous measure, density, and domains

Retain the original full-order packet h , tensor degree $k \geq 1$, $g = 2\xi$, $c = k/2$, $S(u) = c + iu$, and

$$w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi), m = w_h^{*k}, d\nu = m(u) du.$$

HD1 and P54 provide the positive exponential moments for this actual measure; P54 also provides the pointwise exponential bound needed for sampling. No mass normalization occurs: $\nu(\mathbb{R}) = \mu_h^k$. The quotient g/h is entire and nonzero after full-order removal. Its real-line zeros are isolated, so $w_h > 0$ almost everywhere. For $k \geq 2$, integrating the positive-almost-everywhere product in convolution proves $m(u) > 0$ for every real u . The inherited boundedness/integrability makes those convolution integrals finite. In every case ν is atomless and positive on every set of positive Lebesgue measure.

For either $\eta = \nu$ or $\eta = |\chi(S)|^2\nu$, a smaller positive exponential exponent absorbs the fixed polynomial factor. Every polynomial is in $L^2(\eta)$. With inner product $\int \bar{f} g d\eta$, orthogonality of f to all S -polynomials gives zero moments of $\sigma = \bar{f} \eta$ in u , since $u = (S - c)/i$ is invertible. Cauchy–Schwarz supplies the exponentially weighted finite variation needed in CR4. Hence $F(z) = \int \exp(izu) d\sigma(u)$ is holomorphic in the stated strip; dominated differentiation on each smaller strip is valid for every order. All derivatives at zero vanish. The identity theorem gives $F = 0$ on the strip.

The Gaussian uniqueness argument in CR4–5 has the correct Fourier signs: $\gamma_t(x - u)$ contributes $\exp(-i\xi x) \exp(i\xi u)$, so its integral against σ is zero. Finite variation permits Fubini; compactly supported continuous test functions are uniformly approximated by their Gaussian convolutions. Thus $\sigma = 0$, hence $f = 0$ almost everywhere. This proves both polynomial-density assertions without a moment-problem uniqueness hypothesis being silently added.

Put $H = L^2(\nu)$ and $H_\chi = L^2(|\chi(S)|^2\nu)$. The map

$$U_\chi : H_\chi \longrightarrow H, f \longmapsto \chi(S)f$$

is a linear onto isometry. The inverse $v/\chi(S)$ belongs to H_χ because its weighted squared norm is exactly $\|v\|_H^2$. The finitely many real zeros of $\chi(S)$ have zero ν -mass; inverse values there are irrelevant to both equivalence classes. This is not an assertion that unweighted multiplication by χ is bounded $H \longrightarrow H$. Its weighted domain is essential and is exactly the inherited norm of its relation image, so it does not replace the source metric.

Density in H_χ and this onto isometry prove $\text{closure}_H(\chi\mathbb{C}[S]) = H$. Completing the two terms of the inclusion complex $[\chi\mathbb{C}[S] \longrightarrow \mathbb{C}[S]]$ in their inherited H norms therefore gives $[H \xrightarrow{I} H]$, whose degree-minus-one contraction is I . The original algebraic quotient $E = \mathbb{C}[S]/(\chi)$ maps to its zero completed quotient with kernel all of E . In particular completion does not identify the original algebraic quotient with zero before the stated map.

81.1.3 2. The same finite canonical forms and the full graph

For $N \geq q - 1$, multiplication by the literal nonzero χ is injective on P_{N-q} and has image $B_N = \ker J_N$. At $N = q - 1$ this is the zero relation space. Polynomial evaluation in H is injective because a nonzero polynomial has only finitely many real-line zeros and the density is positive almost everywhere. Thus every finite original M_N is positive definite.

Let $s : E \rightarrow P_{q-1}$ be the fixed remainder section and let Π_N be orthogonal projection in the actual H norm onto B_N . Then

$$R_N = (I - \Pi_N)s, J_N R_N = I_E, G_N = R_N^* R_N.$$

The adjoint in this Gram uses the fixed coefficient inner product on E and the actual H inner product. Every lift is $sx + b$, $b \in B_N$; orthogonal decomposition gives unique minimality. In coefficient coordinates the unique minimizer is exactly

$$G_N = (J_N M_N^{-1} J_N^*)^{-1}, R_N = M_N^{-1} J_N^* G_N.$$

All inverses here are legitimate: $M_N > 0$ and J_N is onto. No phase positivity enters these continuous-source formulas.

The B_N increase and have dense union. Therefore $\Pi_N \rightarrow I_H$ strongly. Applying this to the finitely many fixed columns of s gives $R_N \rightarrow 0$ in the operator norm from any fixed finite norm on E to H , and hence $G_N \rightarrow 0$ in every fixed coordinate matrix norm. The minimum decreases when relations are enlarged, proving $G_N \downarrow 0$ in Loewner order. Each finite G_N remains positive definite, and $\det G_N \rightarrow 0$ since $q \geq 1$. No cutoff or tensor-degree rate follows.

For every $x \in E$, $(R_N x, J R_N x) \rightarrow (0, x)$. Thus the closure of graph J contains $\{0\} \times E$. It also contains (p, Jp) for every polynomial p , and is a closed linear subspace. Adding $(0, x - Jp)$ shows it contains (p, x) for every polynomial p and every x . Polynomial density now proves the exact stronger assertion

$$\text{closure}(\text{graph}(J : \mathbb{C}[S] \subset H \rightarrow E)) = H \times E.$$

This is full nonclosability, not just an unbounded-evaluation example. It is compatible with $J_N R_N = I_E$ at every finite cutoff. Under every fixed invertible coefficient change $x = Wx'$, the exact maps become $G'_N = W^* G_N W$, $R'_N = R_N W$, $J'_N = W^{-1} J_N$; all claims persist.

81.1.4 3. Unconditional finite phase pullback, including singular phases

Fix the original $L > 0$, put $u_n(\theta) = (2\pi n + \theta)/L$, and retain $\nu_n(\theta) = (2\pi/L)m(u_n(\theta))$. Let

$$I_\theta = \{n : \nu_n(\theta) > 0\}, H_\theta = \ell^2(I_\theta, \nu_\theta), T_\theta P = (P(c + iu_n(\theta)))_{n \in I_\theta}.$$

The pointwise exponential estimate P54/CQ5 and its lattice sum CQ6 prove every finite polynomial evaluation is in this original Hilbert space, including when $k = 1$. Zero-weight coordinates are null classes, never inverted evaluation functionals. Define

$$Q_{N,\theta} = T_\theta(P_N)/T_\theta(B_N)$$

with the quotient norm inherited from H_θ . Both images are finite-dimensional and closed, whether or not T_θ is injective. The map

$$\rho_{N,\theta} : E \rightarrow Q_{N,\theta}, [P] \mapsto [T_\theta P]$$

is well defined and onto. Its exact kernel is

$$\ker \rho_{N,\theta} = J_N(\ker(T_\theta|_{P_N})).$$

Indeed $T_\theta P = T_\theta b$, for some $b \in B_N$, exactly when $P - b$ is a sampling-null polynomial with the same class. Conversely such a null representative maps to zero. The unique pullback form is

$$x^* G_{N,\theta} x = \min_{b \in B_N} \|T_\theta(sx + b)\|^2.$$

The minimum exists because it is distance to a closed finite-dimensional relation image. Its minimizing observed vector is unique, although a minimizing coefficient polynomial need not be. Its radical is the exact displayed kernel. If $M_{N,\theta} > 0$, this is precisely the usual positive canonical Gram

$(J_N M_{N,\theta}^{-1} J_N^*)^{-1}$. At a singular phase that inverse expression is not asserted.

For each fixed x , the minimum equals the infimum over a countable dense set of complex rational relation coefficients. The norm at each such coefficient is measurable in θ , so the infimum is measurable; polarization makes all matrix entries measurable. Testing the particular original lift $R_N x$ and unfolding the literal phase measure gives

$$\begin{aligned} & \int x^* G_{N,\theta} x \frac{d\theta}{2\pi} \\ & \leq \int \|T_\theta R_N x\|^2 \frac{d\theta}{2\pi} \\ & = \int_{\mathbb{R}} |R_N x(c + iu)|^2 m(u) du = x^* G_N x. \end{aligned}$$

The normalization is exact: $(2\pi/L) d\theta/(2\pi) = du$ on each lattice interval. The integrals are finite. Off-diagonal entries are integrable by the positive-form inequality $|G_{ij}| \leq \sqrt{G_{ii}G_{jj}}$ and Cauchy-Schwarz in the actual phase measure. Thus for every original $k \geq 1$ and every $N \geq q - 1$, without universal phase invertibility,

$$\begin{aligned} 0 & \leq \overline{G}_N := \int G_{N,\theta} \frac{d\theta}{2\pi} \leq G_N, \\ D_N & := G_N - \overline{G}_N \geq 0. \end{aligned}$$

81.1.5 Exact relation-gap version without coefficient inverse assumptions

Let $\Pi_{N,\theta}$ be orthogonal projection in H_θ onto $T_\theta(B_N)$. Since $T_\theta R_N x$ and $T_\theta s x$ differ by that relation image, the minimizing observed lift is $(I - \Pi_{N,\theta})T_\theta R_N x$. Orthogonal decomposition gives

$$x^* D_N x = \int \|\Pi_{N,\theta} T_\theta R_N x\|^2 \frac{d\theta}{2\pi}.$$

This is an exact positive original-relation norm at every phase; it specializes to HCD13 on the admitted positive family. For an explicit measurable coefficient formula only, let B contain the literal relation columns, $F_\theta = B^* M_{N,\theta} B$, and $d_\theta = B^* M_{N,\theta} R_N$. Then the projected relation is $T_\theta B F_\theta^+ d_\theta$, where $+$ is the finite Moore–Penrose pseudoinverse. Here d_θ lies in the range of F_θ : a vector in $\ker F_\theta$ gives the zero observed relation and hence pairs to zero with $T_\theta R_N$. Thus this expression is precisely the projection, not a modified quotient. Pseudoinverse is Borel measurable, for example as the limit of $(F_\theta^2 + \epsilon I)^{-1} F_\theta$. Its observed squared norm is bounded by that of $T_\theta R_N$, so the integral exists even if the chosen coefficient representative is not uniformly bounded. At $B_N = 0$ the projection is zero and $D_N = 0$.

The mean-zero coefficient contraction and coefficientwise averaging of arbitrary fields in HCD5–16 still use the HCD2 uniform-equivalence hypothesis. The unconditional argument above does not silently extend those stronger coefficient-field assertions beyond their stated domain.

81.1.6 What is unconditional about one-fold positivity

For $k = 1$, the zeros of the actual nonzero entire g/h on the sampled vertical line form a discrete, hence countable, set. Their phases modulo 2π form a countable set. Outside it every lattice weight is positive; a nonzero polynomial cannot vanish at all the infinitely many distinct sample points. Therefore $M_{N,\theta} > 0$ for almost every phase, simultaneously for all finite N . For $k \geq 2$ the same statement holds at every phase by everywhere-positive convolution.

Consequently, for every nonzero fixed x , $x^* G_{N,\theta} x > 0$ almost everywhere and its integral is positive. This proves the additional actual-source statement

$$\overline{G}_N > 0 \text{ at every finite cutoff, and } \overline{G}_N \downarrow 0.$$

Monotonicity follows phasewise by enlarging relations; the limit follows from $\overline{G}_N \leq G_N$. Also $0 \leq D_N \leq G_N$ implies $D_N \rightarrow 0$. No monotonicity of D_N , no zero limit for $G_N^{-1} D_N$, and no uniform positive lower bound in θ is inferred. In particular positivity almost everywhere does not supply the HCD2 uniform lower constant.

81.1.7 4. Fixed-phase limits, full primary kernels, and surviving masses

The exponential sampled moment proof applies on I_θ , and also after multiplying the measure by $|\chi|^2$. Thus the sampled polynomial vectors are dense. The image of multiplication by χ is exactly the closed subspace vanishing at

$$Z_\theta = \{n \in I_\theta : \chi(c + iu_n(\theta)) = 0\}.$$

Division by χ on the complementary positive atoms proves the reverse inclusion isometrically in the weighted domain. Restriction to Z_θ is the orthogonal quotient map, with squared norm $\sum_{n \in Z_\theta} \nu_n(\theta) |v_n|^2$. No weight or phase is removed.

Increasing finite relation images have union equal to the evaluated polynomial relation space. Their distances therefore decrease to distance to its closure, including singular finite phase Grams. For every class x , this proves

$$x^* G_{N,\theta} x \downarrow \sum_{n \in Z_\theta} \nu_n(\theta) |x(S_n(\theta))|^2.$$

Polarization gives the matrix limit CR13 in the same fixed coefficient frame. Let p_θ be the square-free product of the displayed roots. The exact limiting radical is $(p_\theta)/(\chi)$. In a sampled primary block $\mathbb{C}[\epsilon]/(\epsilon^r)$, only the constant coefficient is received and the full ideal $(\epsilon)/(\epsilon^r)$ stays in the kernel. Every unsampled primary block stays entirely in the kernel. This uses the full original χ ; it does not replace its multiplicities by a reduced polynomial upstream.

At $k = 1$, the original cyclic minimal polynomial has ideal $(\chi) = (h)$. If a_h is the original leading coefficient, $\chi = h/a_h$ identifies the monic ideal generator, without replacing the density $|g/h|^2/(2\pi)$. For an actual selected root ρ of complete order r , write $h = (S - \rho)^r h_0$, with the unchanged $h_0(\rho) \neq 0$. Then

$$(g/h)(\rho) = g^{(r)}(\rho)/(r! h_0(\rho)) \neq 0,$$

$$\nu_n(\theta) = |g^{(r)}(\rho)|^2/(L(r!)^2 |h_0(\rho)|^2) > 0$$

whenever $S_n(\theta) = \rho$. Thus no sampled root of the actual one-fold χ loses its receiving line to a zero weight. This does not say that every phase samples such a root or that every other atom has positive weight. It is the exact narrower positivity needed for CR14.

If a displayed sampled root exists, take the original unit class $x = 1$ (or any class evaluating to one there). Since $G_{N,\theta} \geq K_\theta$, its numerator is at least the retained positive mass. Its continuous denominator is positive at every finite cutoff and tends to zero by CR11. This proves the quotient ratio in CR14 tends to infinity. If Z_θ is empty there is no such lower bound and no relative rate is asserted. No sampled root is invented.

Only finitely many phases can sample a root of the fixed χ : each root on $\text{Re } S = c$ specifies its unique phase $L \text{Im } \rho \bmod 2\pi$. The completed quotient field is therefore zero almost everywhere. Its measurable realization is the range in ordinary $\ell^2(\mathbb{Z})$ of the diagonal projection with entries $\mathbf{1}_{\nu_n(\theta) > 0} \mathbf{1}_{\chi(S_n(\theta))=0}$. Its direct integral is the zero Hilbert space, and the induced map from the whole original E has kernel E . Individual exceptional weighted fibres are not erased pointwise. The retained marked map is $(\text{label}, v) \mapsto (\text{label}, 0)$, with $\tau \mapsto \tau$; represented zero is not external absence.

81.1.8 5. Read scope and provenance pins

Read completely: the original and revised root note; the current HD source; the completed-phase density review; the continuous-descent review; and the complete delivered periodized-source research note. Source SHA-256 pins:

- source-workspace/work/holonomy_degree_cost_20260913.tex: 9A1BA6CD7E8BEE67C6D7868C74F2CF4F9D8C11FA255D9D278C7A2420CF0DACA6. This current version includes HD10. The parent's earlier 56c05d... pin concerns its earlier HD1–9 snapshot, not this current file.

- source-workspace/work/holonomy_completed_quotient_density_review_20260913.md: 20fC066345E4AD5860C0EEB02BB7E446A69CA0826CC8D8F2196B517BA91EA753.
- source-workspace/work/holonomy_continuous_descent_review_20260913.tex: E283A93B513CB99398CE5F43BBF230D1494BFB55D59243916B1E017852B376C6.
- source-workspace/output/split_zero_rh_tandem_2026-09-12/sources/web_periodized_source_delivery/Tau_Periodized_Source_Control/RESEARCH_NOTE.md: F903AE8473EFDCE0F45A83878837289D2B0604C1A34903276D7C840FE972B9.

Key source proof locations are HD1–5 and HD6–10; HCD2 for the uniform phase assumption, HCD9–16 for its positive-family gap formulas, HCD17–29 for sampled support, completion and marked maps; CQ4–8 for actual exponential sampled tails, CQ14–22 for the quotient and primary kernel, CQ23–25 for exact one-fold root masses, CQ26–29 for the direct integral; and P8–10, P26a–b, P53–58 in the delivered note.

This review makes no evaluated arithmetic enclosure, uniform source or tensor-degree estimate, public-publication, or proof-assistant claim.

82 Arithmetic determinant transport on the original tau complex

This calculation retains the arithmetic coefficient complex over the absolute pointed base $\mathbf{F}_{1,\tau} = \{\tau, 1\}$. It computes the signed arithmetic/reference determinant term on the original finite theta source. The new formulas are a complete first and second variation, an exact four-window projection trace, and a quantitative bound in terms of the actual common source Gram. No arithmetic source multiplier is assumed bounded above or bounded away from zero.

82.1 Original objects and the fixed comparison

The retained tau-base construction has the four-point chart $+, - < \eta < \sigma$ and the coefficient sheaf $\mathcal{T}$ with $(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0)$. Its two nonzero restrictions are Θ and $J\Theta = \Theta\mathcal{F}$. The actual projective resolution gives

$$A_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}], \quad H^1 A_\tau(\mathcal{T}) = Q = \mathcal{B}/\Theta V. \quad (\text{AT1})$$

These are the fixed source objects of the original Tau Base note, equations (18)–(25), supplied in the source closure. The tensor resolution in its equations (46)–(48) retains the original differential signs and identifies its top cohomology with $Q^{\otimes k}$. The operation here uses that pushforward, with its arithmetic classes. Its separately computed restriction to σ remains an available comparison; it is not substituted for (AT1).

Fix an original actual packet h of full zero orders of $g = 2\xi$, and a positive integer k . Retain the full tensor-sum operator, its original cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $E = \mathbb{C}[S]/(\chi)$ in the increasing-power remainder frame. The original construction supplies

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}P = \eta_{h,k}J_N P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad (\text{AT2})$$

where $v_h = j_h(g/h)$ is the full invertible Taylor unit, J_N is literal remainder modulo χ , and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the original injective full-jet observation, $E_h = \mathbb{C}[s]/(h)$. The original section $\sigma_h : E_h \hookrightarrow Q$ gives the separate arithmetic cohomology injection and equality

$$q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}J_N P, \quad \sigma_h^{\otimes k}\eta_{h,k} : E \hookrightarrow Q^{\otimes k}. \quad (\text{AT2a})$$

The supplied arithmetic-input and cyclic-sum proofs construct these maps; all operations below keep them fixed. In particular the sum coordinate is $S = k/2 + iu$, with no division by k .

For $N \geq q - 1$, set $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and let $B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ be multiplication by the literal monic χ . Write $s_N : E \rightarrow \mathcal{P}_N$ for the degree-less-than- q remainder section. Polynomial division proves

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad J_N s_N = I_E. \quad (\text{AT3})$$

The arithmetic norm has density $m_1(u) = m_{h,k}(u) = w_h^{*k}(u)$, $w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi)$. The retained Gamma reference has density

$$m_0(u) = r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a). \quad (\text{AT4})$$

Its mass is $(2\pi)^{k/2}$; the arithmetic mass is the original μ_h^k . The source proofs included with this calculation establish finite moments of every degree and positivity almost everywhere for both densities. The calculation below only integrates polynomials against these two actual measures and their sum. It needs no estimate on the pointwise ratio m_1/m_0 .

For $0 \leq t \leq 1$ define, in the unchanged coefficient frame,

$$\begin{aligned} m_t &= (1-t)m_0 + tm_1, \quad \Delta m = m_1 - m_0, \quad v_N(u) = (1, S(u), \dots, S(u)^N), \\ M_N(t) &= \int v_N(u)^* v_N(u) m_t(u) du, \quad \Delta M_N = M_N(1) - M_N(0), \\ H_N(t) &= B_N^* M_N(t) B_N. \end{aligned} \quad (\text{AT5})$$

Every $M_N(t)$ is positive definite: a nonzero polynomial has finitely many real zeros, and m_t is positive almost everywhere. B_N is injective, so $H_N(t)$ is positive definite on its actual relation domain. When $N = q - 1$, that domain is zero; its determinant is one, its unique endomorphism is its inverse, and all compositions through that domain are zero. The real parameter t interpolates norms on (AT3); it changes neither the absolute base nor a cochain map.

The reference density in (AT4) is exactly the density denoted $r_{1/4}^{*k}$ in SP: with the retained convolution constant $c_a = 2^{1-2a}\Gamma(2a)$ one has

$$c_{1/4} = \sqrt{2\pi}, \quad r_{1/4}^{*k}(u) = \frac{c_{1/4}^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi} = m_0(u). \quad (\text{AT4a})$$

The original Gamma convolution identity therefore identifies all moments and the mass $(2\pi)^{k/2}$, without a mass adjustment. Writing $x = t$ and $y = u$, the precise coefficient arrow to the common-source notation of SP is $B_N^{\text{SP}} = I_N B_N^{\text{AT}}$, from $\mathcal{P}_{N-q}$ to $H = \mathcal{P}_{2q}$. Consequently

$$(B_N^{\text{SP}})^* M_* B_N^{\text{SP}} = B_N^* M_N B_N, \quad Q_N^{\text{SP}} = I_N B_N H_N^{-1} B_N^* I_N^* M_* = Q_N. \quad (\text{AT4b})$$

These identities follow from $M_N = I_N^* M_* I_N$. They include the zero relation domain, where all compositions are zero.

82.2 Canonical sections, determinant lines and full variations

Theorem 82.1. *On the fixed exact sequence (AT3), its least-norm section and quotient Gram are*

$$\begin{aligned} R_N(t) &= s_N - B_N H_N(t)^{-1} B_N^* M_N(t) s_N, \\ J_N R_N(t) &= I_E, \quad B_N^* M_N(t) R_N(t) = 0, \\ G_N(t) &= R_N(t)^* M_N(t) R_N(t) = (J_N M_N(t)^{-1} J_N^*)^{-1}, \\ V_N(t) &:= \det G_N(t) = \frac{\det M_N(t)}{\det H_N(t)}. \end{aligned} \quad (\text{AT6})$$

In the original polynomial frame the exact first and second variations are

$$\begin{aligned}\dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, & \dot{G}_N &= R_N^* \Delta M_N R_N, \\ \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N \preceq 0.\end{aligned}\tag{AT7}$$

The exact logarithmic variation, with both trace terms retained, is

$$\begin{aligned}\frac{d}{dt} \log V_N &= \text{Tr}(M_N^{-1} \Delta M_N) - \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N) \\ &= \text{Tr}(G_N^{-1} R_N^* \Delta M_N R_N).\end{aligned}\tag{AT8}$$

Its second derivative is

$$\begin{aligned}\frac{d^2}{dt^2} \log V_N &= -2 \text{Tr}(G_N^{-1} R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N) \\ &\quad - \text{Tr}(G_N^{-1} \dot{G}_N G_N^{-1} \dot{G}_N) \leq 0.\end{aligned}\tag{AT9}$$

Proof. Every lift of $x \in E$ is $s_N x + B_N y$. Expanding its squared norm and completing its finite quadratic expression gives the unique minimizer $y = -H_N^{-1} B_N^* M_N s_N$. This proves the first two lines of (AT6). For the inverse expression, every polynomial vector p decomposes as $B_N y + R_N J_N p$. Orthogonality therefore gives $R_N^* M_N p = G_N J_N p$ for every p . Taking adjoints yields $M_N R_N = J_N^* G_N$. Substituting this into $J_N R_N = I_E$ gives $(J_N M_N^{-1} J_N^*) G_N = I_E$, which proves the inverse expression.

The determinant calculation preserves orientation. Put $r = N - q + 1$ and $L_N = [B_N \ s_N]$. In the ordered monomial bases, $[s_N \ B_N]$ has determinant one, since χ is monic and the lower diagonal block is triangular with diagonal one. Thus $\det L_N = (-1)^{qr}$. The upper-left block of $L_N^* M_N L_N$ is H_N , and its Schur complement is G_N . Consequently $\det H_N \det G_N = |(-1)^{qr}|^2 \det M_N = \det M_N$. Equivalently the map

$$\begin{aligned}\det \mathcal{P}_{N-q} \otimes \det E &\longrightarrow \det \mathcal{P}_N, \\ (y_1 \wedge \cdots \wedge y_r) \otimes (x_1 \wedge \cdots \wedge x_q) &\longmapsto B_N y_1 \wedge \cdots \wedge B_N y_r \wedge s_N x_1 \wedge \cdots \wedge s_N x_q\end{aligned}\tag{AT10}$$

is the fixed determinant-line isomorphism. Replacing any lift by a relation does not change its wedge. Orthogonal lifts give the product metric just computed. This proves the determinant identity including the empty relation space, without changing any scalar mass.

Differentiate $J_N R_N = I$; then $\dot{R}_N$ lies in $\text{im } B_N$. Differentiating $B_N^* M_N R_N = 0$ determines its unique coefficient and proves the first identity of (AT7). On differentiating $R_N^* M_N R_N$, the two terms involving $\dot{R}_N$ vanish by that orthogonality. A second differentiation gives the two equal negative terms in (AT7), since ΔM_N is Hermitian. This proves the claimed second derivative and its sign. Applying the finite determinant derivative to (AT6) gives the first equality of (AT8); applying it to G_N gives the second. One more derivative gives (AT9). The first trace there is the squared Hilbert–Schmidt norm of $H_N^{-1/2} B_N^* \Delta M_N R_N G_N^{-1/2}$; the second is the squared Hilbert–Schmidt norm of $G_N^{-1/2} \dot{G}_N G_N^{-1/2}$. These identities establish their signs and retain the original forms. All inverse matrices are smooth on a neighborhood of the compact interval $[0, 1]$, by positivity and continuity. No infinite-dimensional inverse has been used. $\square$

The derivative $\dot{R}_N$ in (AT7) is a literal polynomial relation. Applying the original $\mathcal{V}$ in (AT2) therefore retains its original theta primitive and sends its arithmetic class to zero; $J^{(k)} \mathcal{V} R_N(t) = \eta_{h,k}$ for every t . At each nonempty original support label A , the cochain identities lift by $(A, P) \mapsto (A, fP)$, while $\tau \mapsto \tau$. Thus a relation-valued derivative has a supported zero class at label A . Its source vector and its full nonnegative energy in (AT7)–(AT9) remain present. These formulas compute their contribution rather than deleting it after taking cohomology.

82.3 The signed four-endpoint term on one retained source

Let $N_* = 2q$. Every space in (AT3) for the four endpoints sits in $\mathcal{P}_{N_*}$ by its actual coefficient inclusion. Give this common space the form $M_*(t) = M_{N_*}(t)$. Write $P_N(t)$ for its orthogonal projection onto $\mathcal{P}_N$, and $Q_N(t)$ for its orthogonal projection onto the relation subspace $\mathcal{R}_N = \chi\mathcal{P}_{N-q}$. The latter is zero at $N = q - 1$. For the coefficient inclusion $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$, the formulas, with their domains retained, are

$$P_N = I_N M_N^{-1} I_N^* M_*, \quad Q_N = I_N B_N H_N^{-1} B_N^* I_N^* M_*, \quad C(t) = M_*(t)^{-1} \Delta M_*. \quad (\text{AT11})$$

Both P_N and Q_N are $M_*(t)$ -self-adjoint idempotents with their displayed images, verified by substitution. Since $\mathcal{R}_N \subset \mathcal{P}_N$, $P_N Q_N = Q_N P_N = Q_N$. Their difference is the orthogonal projection onto the finite canonical quotient section and has trace q . Cyclicity of finite matrix trace in (AT11) proves the exact common-source version of (AT8):

$$\frac{d}{dt} \log V_N(t) = \text{Tr}((P_N(t) - Q_N(t))C(t)). \quad (\text{AT12})$$

Put $i_0 = q - 1$, $i_1 = q$, $j_0 = 2q - 1$, and $j_1 = 2q$. Define the two positive operators

$$\begin{aligned} U(t) &= (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \\ W(t) &= (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \end{aligned} \quad (\text{AT13})$$

Each difference in this display is an orthogonal projection, because the two spaces in its pair are nested in the same Hilbert space. Each has rank q . The sums need not be projections and are not asserted to be orthogonal sums; they are positive operators with $\text{Tr } U = \text{Tr } W = 2q$.

Write

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{i_0}(t)V_{i_1}(t)}{V_{j_0}(t)V_{j_1}(t)}, \quad T_N = V_N(1)/V_N(0), \\ \Delta \mathcal{B} &= \mathcal{B}(1) - \mathcal{B}(0) = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}}. \end{aligned} \quad (\text{AT14})$$

Subtracting the four exact formulas (AT12), with their original signs, and integrating gives

$$\Delta \mathcal{B} = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \quad (\text{AT15})$$

This is the actual arithmetic correction in the audit's formula (54). It compares the two added relation windows with the two added polynomial windows. Concavity of each individual $\log V_N(t)$ in (AT9) does not assign a sign to their signed combination in (AT15); that combination has now been computed.

There is also an exact pointwise integral, using no division by the arithmetic density. Define

$$\begin{aligned} k_N(t, u) &= v_N(u)M_N(t)^{-1}v_N(u)^*, \\ r_N(t, u) &= |\chi(S(u))|^2 v_{N-q}(u)H_N(t)^{-1}v_{N-q}(u)^*, \\ d_N(t, u) &= k_N(t, u) - r_N(t, u), \quad r_{q-1}(t, u) = 0, \\ d_N(t, u) &= v_N(u)R_N(t)G_N(t)^{-1}R_N(t)^*v_N(u)^* \geq 0. \end{aligned} \quad (\text{AT16})$$

The last equality follows by writing the orthogonal splitting of M_N^{-1} as $B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*$; multiplying by M_N and evaluating on $\text{im } B_N \oplus \text{im } R_N$ proves that inverse identity. In particular $\int d_N(t, u)m_t(u) du = q$. Consequently (AT8) also reads

$$\frac{d}{dt} \log V_N(t) = \int d_N(t, u) \Delta m(u) du, \quad \Delta \mathcal{B} = \int_0^1 \int (d_{i_0} + d_{i_1} - d_{j_0} - d_{j_1})(t, u) \Delta m(u) du dt. \quad (\text{AT17})$$

All coefficients of these fixed-degree polynomials in u and $\bar{S}(u)$ are bounded for $t \in [0, 1]$. Since $|\Delta m| \leq m_0 + m_1$ and these measures have the required finite moments, absolute integrability proves the exchange of integrals. This remains valid at every zero of m_1 .

82.4 The full spectral and actual-overlap bound for the correction

Let $n = 2q + 1$ and let $b_1 \leq \dots \leq b_n$ be all eigenvalues of $D_* = M_*(0)^{-1}M_*(1)$ in its original $M_*(0)$ inner product. This operator is positive because $v^*M_*(0)D_*v = v^*M_*(1)v > 0$ for $v \neq 0$. Retain $\kappa_* = b_n/b_1$ and put

$$\begin{aligned} c_j(t) &= \frac{b_j - 1}{1 - t + tb_j}, & d(t) &= c_n(t) - c_1(t), \\ \mathcal{L}_q(D_*) &= \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \\ S_{AW}(t) &= c_{q+2}(t) - c_q(t) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(t) - c_j(t)), \\ s(t) &= \dim(\text{ran } U(t) \cap \text{ran } W(t)), \\ S_{SP}(t) &= \frac{d(t)}{2} \min \left\{ 4q - 2s(t), \sqrt{(2q+1)(8q-4-2\text{Tr}(U(t)W(t)))} \right\}, \\ \mathcal{J}_{h,k} &= \int_0^1 \min\{S_{AW}(t), S_{SP}(t)\} dt. \end{aligned}$$

(AT18a)

An empty sum for $q = 1$ is zero. Every quantity uses the unchanged common source $H = \mathcal{P}_{2q}$ and the actual relation maps in (AT11). The complete spectra and $\mathcal{L}_q$ bound were proved in AW7 and AW13–AW16; the actual overlap terms were proved in SP8–SP10. Their full proofs are propagated here to the original endpoint estimate, so neither calculation is replaced by a reference.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(t) &= \frac{d(t)}{2} \|U(t) - W(t)\|_1, \\ \Sigma(t) &= \text{Tr}(C(t)^2) - \frac{(\text{Tr } C(t))^2}{2q+1}, \\ S_{\text{HS}}(t) &= \sqrt{\Sigma(t) \text{Tr}((U(t) - W(t))^2)}. \end{aligned}$$

(AT18c)

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(t))I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(t)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(t)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

The original-moment angle calculation supplies a further simultaneous bound at this same claim

site. Put $a_q = 2q - 1$ and

$$\begin{aligned} A_{h,k}(t) &= a_q \sqrt{1 - e^{-B(t)/a_q}} d(t), \\ \mathcal{J}_{h,k}^{(5)} &= \int_0^1 \min\{S_{AW}(t), S_{Tr}(t), S_{SP}(t), S_{HS}(t), A_{h,k}(t)\} dt, \\ \mathcal{I}_{h,k} &= \int_0^1 \min \left\{ d(t), \frac{S_{AW}(t)}{a_q \sqrt{1 - e^{-B(t)/a_q}}}, \frac{S_{SP}(t)}{a_q \sqrt{1 - e^{-B(t)/a_q}}}, \right. \\ &\quad \left. \frac{S_{Tr}(t)}{a_q \sqrt{1 - e^{-B(t)/a_q}}}, \frac{S_{HS}(t)}{a_q \sqrt{1 - e^{-B(t)/a_q}}} \right\} dt. \end{aligned} \quad (\text{AT18b})$$

These specialize RMT1–RMT18 through $a_{\text{RMT}} = a_q$, $x = t$, $C_x = C(t)$ and $(U_x, W_x) = (U(t), W(t))$. The density and relation arrows were proved in (AT4a)–(AT4b), so every source moment and original cohomology map agrees. This application uses the actual moment measures (AT4)–(AT5). It does not impose moment structure on an arbitrary positive coefficient Gram.

Theorem 82.2. *The exact correction on the original finite source satisfies*

$$|\Delta \mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k} \leq \mathcal{L}_q(D_*) \leq (2q - 1) \log \kappa_*. \quad (\text{AT18})$$

Moreover $\mathcal{J}_{h,k} \leq \int_0^1 S_{SP}(t) dt$. The original Gamma and arithmetic endpoint quantities consequently obey

$$\max\{\mathcal{B}_{h,k}^\Gamma - \mathcal{J}_{h,k}^{(5)}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} - e_{\ell_N}\} \leq \mathcal{B}_{h,k}^{\text{ar}} \leq \min\{\mathcal{B}_{h,k}^\Gamma + \mathcal{J}_{h,k}^{(5)}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} + e_{\ell_N}\}. \quad (\text{AT19})$$

Here the constructed j_{ℓ_N}, e_{ℓ_N} are defined and proved in (AT19ba)–(AT19bc) below.

Proof. The relation flag $\chi \mathcal{P}_0 \subset \chi \mathcal{P}_{q-1} \subset \chi \mathcal{P}_q \subset H$ has dimensions $1, q, q+1, 2q+1$. On its successive orthogonal pieces the operator $U = Q_{2q-1} + Q_{2q} - Q_q$ has values $1, 2, 1, 0$ with multiplicities $1, q-1, 1, q$. On the orthogonal pieces of $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset H$, W has values $0, 1, 2, 1$ with multiplicities $q, 1, q-1, 1$. Nested orthogonal projections act as identity or zero on these pieces, which proves the assertions including $q = 1$. In particular

$$\begin{aligned} \text{rank } U &= \text{rank } W = q + 1, \quad \text{Tr } U = \text{Tr } W = 2q, \quad \text{Tr } U^2 = \text{Tr } W^2 = 4q - 2, \\ A &= U - W, \quad \text{Tr } A = 0, \quad \text{Tr } A^2 = 8q - 4 - 2 \text{Tr}(UW). \end{aligned} \quad (\text{AT20a})$$

Thus the equal mass $2q$ is an operator trace, and its multiplicity-two subspace of dimension $q - 1$ is retained in each rank calculation.

The identity $M_* C = \Delta M_* = \Delta M_*^*$ proves that C is self-adjoint in $M_*(t)$. The exact factorization $M_*(t) = M_*(0)((1-t)I + tD_*)$ gives $C = ((1-t)I + tD_*)^{-1}(D_* - I)$. A basis of $M_*(0)$ -orthonormal eigenvectors of D_* remains orthogonal in $M_*(t)$, with positive squared lengths $1 - t + tb_j$, proving that the displayed $c_j(t)$ are its eigenvalues. Differentiation in b_j gives $(1 - t + tb_j)^{-2} > 0$, so they retain their order.

For any $M_*(t)$ -orthogonal projection P of rank r , choose an orthonormal eigenbasis e_j of C . Its diagonal values $a_j = \langle e_j, P e_j \rangle$ lie in $[0, 1]$ and sum to r . Thus $\text{Tr}(PC) = \sum_j c_j a_j$ lies between the sum of the smallest r and the sum of the largest r eigenvalues: moving a fixed amount of mass from a smaller coefficient to a larger coefficient increases the sum, until these two extrema are obtained. This also proves the assertion when there are ties. Either $Z = U$ or $Z = W$ is the sum of its range projection of rank $q + 1$ and its eigenvalue-two projection of rank $q - 1$. Applying the just-proved inequality gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AT20b})$$

Subtracting the extrema for the two original operators, retaining both sums before their common middle term cancels, proves $|\operatorname{Tr}(AC)| \leq S_{\text{AW}}(t)$.

Both ranges have dimension $q + 1$ in dimension $2q + 1$; hence $s(t) \geq 1$. Let E_L be the orthogonal projection onto their intersection. Each of U, W dominates its range projection, since its nonzero eigenvalues are 1, 2; that projection dominates E_L . Therefore $U - E_L, W - E_L$ are positive, of trace $2q - s(t)$, and

$$\|A\|_1 \leq 4q - 2s(t), \quad \|A\|_1 \leq \sqrt{(2q + 1) \operatorname{Tr} A^2}. \quad (\text{AT20c})$$

The first inequality is the trace-norm triangle inequality applied to their difference. The second is Cauchy-Schwarz on all $2q + 1$ real eigenvalues of A . Subtracting the midpoint $a = (c_1 + c_n)/2$ from C leaves $\operatorname{Tr}(AC)$ unchanged by $\operatorname{Tr} A = 0$. In an orthonormal eigenbasis f_j of A with eigenvalues α_j , $|\operatorname{Tr}(A(C - aI))| \leq \sum_j |\alpha_j| |\langle f_j, (C - aI)f_j \rangle| \leq d(t) \|A\|_1/2$. Together with (AT20a)–(AT20c), this proves the full pointwise bound

$$|\operatorname{Tr}((U - W)C)| \leq \min\{S_{\text{AW}}(t), S_{\text{SP}}(t)\}. \quad (\text{AT20})$$

No pairwise commutation of U, W, C has been used. The Gram inverses and projectors are smooth on the compact interval; their ranks and intersection dimensions are Borel, since ranks are specified by vanishing and nonvanishing minors. All terms are bounded, hence the displayed minimum is integrable. Integrating (AT20) in (AT15) gives $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}$ for these two positive forms, even without moment structure. On the actual measures, RMT11 retains the two full q -angle lists of the crossed pairs $(q - 1, 2q)$ and $(q, 2q - 1)$, including the designated zero angle, and proves $|\mathcal{B}'| \leq A_{h,k}$. Its strict positivity proof applies here: if $\mathcal{B} = 0$, the first pair has identical minimum sections, putting 1 in the orthogonal complement of $\chi\mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \sum_j \bar{\chi}_j(k - S)^j$ then gives $0 = \int \chi\chi^{\#k} dm_t = \int |\chi|^2 dm_t$, contradicting the positive original Gram of χ . Thus $\mathcal{B}(t) > 0$. RMT11 follows from its complete original angle proof and concavity of $\sqrt{1 - e^{-z}}$ on the retained a_q -entry scalar list. Applying all five inequalities to this same derivative proves $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k}$. Compactness and the positive continuous $\mathcal{B}$ make every denominator in (AT18b) nonzero and bounded away from zero.

For each retained relative eigenvalue,

$$\int_0^1 c_j(t) dt = \log b_j, \quad \int_0^1 d(t) dt = \log(b_n/b_1) = \log \kappa_*. \quad (\text{AT21})$$

This follows by differentiating $\log(1 - t + tb_j)$; it includes $b_j = 1$, whose integrand is zero. Thus $\int S_{\text{AW}} = \mathcal{L}_q(D_*)$ and $\mathcal{J}_{h,k} \leq \int S_{\text{SP}}$. Each ratio in $\mathcal{L}_q$ is at most κ_* ; the retained total coefficient is $1 + 2(q - 1) = 2q - 1$. This proves all inequalities. Finally the absolute branches of (AT19) follow by applying both signs of (AT18), and its constructed signed branches follow from (AT19bb) proved below, to the exact equality $\mathcal{B}_{h,k}^{\text{ar}} = \mathcal{B}_{h,k}^{\Gamma} + \Delta\mathcal{B}$. $\square$

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact end-point forms as well. Here $\mathbf{M}_z = M_*(t)|_{t=z}$, $\mathbf{C}_z = C(t)|_{t=z}$, $\mathbf{A}_z = (U(t) - W(t))|_{t=z}$, and $\mathbf{D} = D_*$. Thus the RMT/ISM coefficient map is the identity with $x_{\text{RMT}} = z = t$, and its $\mathcal{B}$ is (AT14). Write $\mathbf{D} = (\mathbf{M}_0)^{-1}\mathbf{M}_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let $\mathbf{A}_z$ be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1 - z) + z(b_1 + b_{2q+1})}{2}, & H_z^\circ &= I - \frac{(1 - z)I + z\mathbf{D}}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (\mathbf{D} - I), & R_z^{[\ell_N]} &= (H_z^\circ)^{\ell_N+1} \mathbf{C}_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\operatorname{Tr} R_z^{[\ell_N]}}{2q + 1} I. \end{aligned} \quad (\text{AT19ba})$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1 - z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \tag{AT19bb}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q - 6)} \longrightarrow 0. \tag{AT19bc}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

The same actual-moment calculation also propagates to the nonlinear endpoint coordinate. Set $\mathcal{H}_{a_q}(B) = 2 \text{arcosh}(e^{B/(2a_q)})$ and $H_0 = \mathcal{H}_{a_q}(\mathcal{B}_{h,k}^\Gamma)$. Its exact derivative is $\mathcal{H}_{a_q}'(B) = 1/(a_q \sqrt{1 - e^{-B/a_q}})$ and its continuous inverse is $2a_q \log \cosh(z/2)$ for $z \geq 0$. The positivity just proved permits the chain rule throughout this fixed path. Dividing the five derivative bounds by their positive factor and integrating gives $|\mathcal{H}_{a_q}(\mathcal{B}_{h,k}^{\text{ar}}) - H_0| \leq \mathcal{I}_{h,k} \leq \log \kappa_*$. Consequently the existing interval (AT19) has the simultaneous strengthening

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{ar}} &\geq \max \left\{ 0, \mathcal{B}_{h,k}^\Gamma - \mathcal{J}_{h,k}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{h,k}\}}{2}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}_{h,k}^{\text{ar}} &\leq \min \left\{ \mathcal{B}_{h,k}^\Gamma + \mathcal{J}_{h,k}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{h,k}}{2}, \mathcal{B}_{h,k}^\Gamma + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned} \tag{AT19a}$$

The inverse and nonlinear branches are RMT16–RMT18; the signed branches are RMT19–RMT22 and the full local calculation (AT19bb), on the forms already specified in AT; the scalar continuous value at zero bounds an actual strictly positive endpoint.

When $M_*(1) = cM_*(0)$, every $b_j = c$, so $d = 0$ and $\mathcal{J}_{h,k} = 0$. The original quotient determinants are $(1 - t + tc)^q V_N(0)$, whose four factors cancel exactly in (AT14). This retains both original masses. For an invertible coefficient arrow $F : H' \rightarrow H$, let $M'_* = F^* M_* F$ and transport every subspace by F^{-1} . The projector formula gives $P'_N = F^{-1} P_N F$, $Q'_N = F^{-1} Q_N F$, $D'_* = F^{-1} D_* F$, and $C' = F^{-1} C F$. Their spectra, intersection dimensions, $\text{Tr}(UW)$, the signed trace and $\mathcal{J}_{h,k}$ consequently agree under this exact map.

For the original reflection-stable quartet with displacement $\delta > 0$ and multiplicity m , retain $q = [1 + k(m - 1)](k + 1)^2$. Its supplied Gamma theorem gives $\mathcal{B}_{h,k}^\Gamma \geq 4q \log(\delta k / (2\sqrt{5}))$. Applying (AT19)

with its actual sign proves

$$\begin{aligned}
\mathcal{B}_{h,k}^{\text{ar}} &\geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \mathcal{J}_{h,k}^{(5)} \\
&\geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \mathcal{L}_q(D_*) \\
&\geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - (2q-1) \log \kappa_*, \\
\log \kappa_* &\geq \frac{4q \log(\delta k/(2\sqrt{5})) - \mathcal{B}_{h,k}^{\text{ar}}}{2q-1}.
\end{aligned} \tag{AT22}$$

The signed branch also yields $\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log(\delta k/(2\sqrt{5})) + j_{\ell_N} - e_{\ell_N}$: substitute the same Gamma lower bound into $\mathcal{B}_{h,k}^{\text{ar}} \geq \mathcal{B}_{h,k}^{\Gamma} + j_{\ell_N} - e_{\ell_N}$. This branch and (AT22) are both retained by their maximum. The last denominator is positive for every $q \geq 1$. This is the same actual packet application, with its earlier necessary condition updated to the stronger proved finite control. No existence of such a quartet or uniform growth bound is assumed. The full signed formulas (AT15)–(AT17), all source primitives and the retained cohomology observations remain the objects whose moments and actual overlaps enter the bound.

82.5 Source dependencies and precise status

The complete retained Tau Base note supplies (AT1), its tensor pushforward, right adjoint and supported maps. The original arithmetic-input and cyclic-sum proofs supply (AT2), full packet units and the theta primitives. The Gamma/source-density notes supply the actual measures and finite moments in (AT4)–(AT5). The new Deligne-bound note supplies the full quartet Gamma lower bound used only in (AT22). Each is preserved in the accompanying source closure, with its actual edition and hash.

The new proved calculations here are (AT6)–(AT21) on those unchanged input objects, and their substitution (AT22). They give an analytic calculation on the canonical finite complex mapping to $A_{\tau,k}(\mathcal{T}^{\boxtimes k})$. They neither replace that pushforward by a generic stalk nor deduce a verdict about the entire tau-based program from an auxiliary model.

83 Exact identification of the retained AT extraction hash

This additive record identifies the processing frame of a retained metadata hash. The entire accepted AT proof source and reader, including the original preamble and all 23 equation tags, remain unchanged.

83.0.1 The exact identified value

The value in the sealed evidence/BODY_EXTRACTION.json field `pre_layout_reader_body_sha256` is 42b97470bfe0f67e0a0247e5cefc95609c487b8ff20c275457aeabc02b7b706. It identifies the complete final *post-layout* reader body in LF encoding, with 17149 bytes. Replacing each CRLF in the 17558-byte accepted reader by LF gives this value exactly. The retained field name and its explanatory note name the wrong processing stage. The four exact frames are:

Complete body frame	Bytes	SHA-256
pre layout LF	17113	5ccb346f20b37ea30a40491a9c6ec84e7121869816efbb08107ae8f64cc9116
pre layout CRLF	17520	3ed5d0dbedbcffdaa8949a4dc6108e99546acd3023c32c4785388860a5d1815

post layout LF	17149	42b97470bfe0f67e0a0247e5cefc95609c487b8ff20c275457aeebc02b7b706
post layout CRLF	17558	9fa19a509edf91f5aa0a9cb4762485449196d0b1ff521f2445f791fa3290cc75

83.0.2 Complete forward and inverse reconstruction

Read the pinned original source, extraction record and ten-entry layout record. The exact original body begins with the 13-byte sequence whose base64 representation is `ClxtYWtldGl0bGUKCg==`. Remove exactly that title prefix; no proof prose is removed. The remaining complete LF body has 17113 bytes. Replacing each LF by CRLF gives its 17520-byte CRLF representation.

Apply all ten ordered old/new pairs from the sealed layout map. In the CRLF representation, convert every LF in both sides of each pair to CRLF. Each old payload occurs exactly once at its recorded byte span. The final LF and CRLF results are the complete final bodies identified above. The full payloads, source spans and before/after hashes for every step are retained in `CLARIFICATION.json`.

For the inverse, start with the accepted CRLF reader and reverse all ten pairs in reverse order, replacing each unique new payload by the complete old payload. This recovers the entire pre-layout CRLF body. Replace each CRLF by LF and restore the exact removed title prefix. Then concatenate the original preamble, the literal begin-document boundary, the restored complete body, the literal end-document boundary and the original postamble. The result is every byte of the pinned original source, whose SHA-256 is `24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae`. The JSON record gives exact zero-based byte intervals and hashes for all five original-file components, and retains the complete original preamble and postamble literally. The source itself retains every proof paragraph.

83.0.3 Complete byte-verification program

The following standard-library program checks the five sealed input pins, the exact source spans and every intermediate hash in both representations, the ten reverse transformations and all original-file components. It writes no source file, imports no mathematical program and performs no document build. Invoke it with `--artifact-root` naming the sealed artifact directory. Its complete JSON recipe accompanies this file.

```

"""Verify the AT clarification using bytes only; no TeX or mathematical code runs."""
import argparse
import base64
import hashlib
import json
from pathlib import Path

def require(value, message):
    if not value:
        raise ValueError(message)

def pin(data):
    return {'bytes': len(data), 'sha256': hashlib.sha256(data).hexdigest()}

def check(data, expected, label):
    require(pin(data) == {k: expected[k] for k in ('bytes', 'sha256')}, label)

def decode(value):
    return base64.b64decode(value, validate=True)

def verify(artifact_root, recipe):
    root = Path(artifact_root).resolve()
    inputs = {}
    for label, spec in recipe['sealed_inputs'].items():

```

```

    path = (root / spec['path']).resolve()
    require(path.is_relative_to(root), 'Input leaves the named artifact root')
    data = path.read_bytes()
    check(data, spec, label + ': sealed input pin')
    inputs[label] = data
    original = inputs['original_source']
    reader = inputs['accepted_reader']
    extraction = json.loads(inputs['body_extraction'])
    layout = json.loads(inputs['layout_map'])
    original_body = extraction['original_body'].encode('utf-8')
    prefix = decode(recipe['body_prefix_removed_base64'])
    require(original_body.startswith(prefix), 'Exact original title prefix')
    pre_lf = original_body[len(prefix):]
    require(b'\n' not in pre_lf, 'The retained original body uses LF')
    pre_crlf = pre_lf.replace(b'\n', b'\r\n')
    check(pre_lf, recipe['frames']['pre_layout_LF'], 'Complete pre-layout LF body')
    check(pre_crlf, recipe['frames']['pre_layout_CRLF'], 'Complete pre-layout CRLF body')
    lf, crlf = pre_lf, pre_crlf
    require(len(layout['changes']) == len(recipe['layout_steps']) == 10, 'All ten layout edits')
    for i, (source_change, step) in enumerate(zip(layout['changes'], recipe['layout_steps']), 1):
        require(step['step'] == i, 'Layout order')
        old, new = (source_change[k].encode('utf-8') for k in ('old', 'new'))
        require(old == decode(step['old_LF_base64']) and new == decode(step['new_LF_base64']),
            'Complete original layout payloads')
        old_crlf, new_crlf = (s.replace(b'\n', b'\r\n') for s in (old, new))
        require(lf.count(old) == crlf.count(old_crlf) == 1, 'Unique layout source spans')
        require(lf.index(old) == step['source_span_LF']['start'] and
            crlf.index(old_crlf) == step['source_span_CRLF']['start'], 'Exact layout span starts')
        check(lf, step['before_LF'], 'Intermediate LF source')
        check(crlf, step['before_CRLF'], 'Intermediate CRLF source')
        lf, crlf = lf.replace(old, new), crlf.replace(old_crlf, new_crlf)
        check(lf, step['after_LF'], 'Intermediate LF target')
        check(crlf, step['after_CRLF'], 'Intermediate CRLF target')
    check(lf, recipe['frames']['post_layout_LF'], 'Complete final LF body')
    check(crlf, recipe['frames']['post_layout_CRLF'], 'Complete final CRLF body')
    require(crlf == reader, 'Final CRLF body is every accepted reader byte')
    require(reader.replace(b'\r\n', b'\n') == lf, 'Exact final reader LF frame')
    require(hashlib.sha256(lf).hexdigest() == extraction['pre_layout_reader_body_sha256'],
        'The legacy-named field identifies final LF bytes')
    backwards = reader
    for step in reversed(recipe['layout_steps']):
        old = decode(step['old_LF_base64']).replace(b'\n', b'\r\n')
        new = decode(step['new_LF_base64']).replace(b'\n', b'\r\n')
        require(backwards.count(new) == 1, 'Unique reverse layout target')
        backwards = backwards.replace(new, old)
    require(backwards == pre_crlf, 'All ten inverse transformations recover pre-layout CRLF bytes')
    inverse_body = prefix + backwards.replace(b'\r\n', b'\n')
    require(inverse_body == original_body, 'Full original body including removed title prefix')
    parts = [extraction['preamble'].encode('utf-8'), b'\\begin{document}', inverse_body,
        b'\\end{document}', extraction['postamble'].encode('utf-8')]
    cursor = 0
    for component, data in zip(recipe['original_file_components'], parts):
        require(component['start'] == cursor and component['end'] == cursor + len(data),
            'Exact source component offsets')
        check(data, component, 'Exact source component pin')
        require(original[cursor:cursor + len(data)] == data, 'Exact source component bytes')
        cursor += len(data)
    require(cursor == len(original) and b''.join(parts) == original,
        'Complete original preamble, boundaries, body and postamble reconstructed')
    return {
        'status': 'verified-exact-current-AT-source-reconstruction',
        'legacy_field_identified_as': 'post-layout complete reader body in LF encoding',
        'all_ten_layout_steps_and_inverses_verified': True,
        'full_original_file_restored_byte_for_byte': True,
        'original_file': pin(original), 'accepted_reader': pin(reader),

```

```

    'frames': recipe['frames'], 'mathematical_code_executed': False,
    'TeX_or_other_build_executed': False, 'sealed_inputs_modified': False,
}

def main():
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument('--artifact-root', required=True)
    parser.add_argument('--recipe', default=str(Path(__file__).with_name('CLARIFICATION.json')))
    args = parser.parse_args()
    recipe = json.loads(Path(args.recipe).read_bytes())
    print(json.dumps(verify(args.artifact_root, recipe), indent=2))

if __name__ == '__main__':
    main()

```

84 Exact relation-window spectral transport on the original tau complex

This calculation continues arithmetic determinant transport on its original finite source over the absolute tau base. It computes the full spectra of the two retained window operators, constructs an isometry intertwining them in the actual source metric, and strengthens the signed determinant bound. The source measures, their masses, the cyclic polynomial, the Taylor unit, and the cohomology observation remain fixed.

84.1 The original source and its four windows

Use a full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and the original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Let $E = \mathbb{C}[S]/(\chi)$, $E_h = \mathbb{C}[s]/(h)$, and $v_h = j_h(g/h) \in E_h^\times$. The original tensor-sum observation is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1. \quad (\text{AW1})$$

The original section $\sigma_h : E_h \hookrightarrow Q = \mathcal{B}/\Theta V$ gives the separate injection $\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}$. Here $Q^{\otimes k}$ is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau} = \{\tau, 1\}$; the one-factor complex is $[V \oplus V \xrightarrow{\Theta(\phi-\psi)} \mathcal{B}]$. The complete Tau Base and cyclic source proofs accompany the preceding arithmetic determinant transport calculation (AT).

For $N \geq q - 1$, retain the increasing-power spaces and maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad \mathcal{P}_{-1} = 0. \quad (\text{AW2})$$

The common source is $H = \mathcal{P}_{2q}$, of dimension $n = 2q + 1$. Its original coordinate on the integration line is $S = k/2 + iu$. Set $v_N(u) = (1, S(u), \dots, S(u)^N)$ and retain the two densities

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, & c_a &= 2^{1-2a}\Gamma(2a). \end{aligned} \quad (\text{AW3})$$

Their actual masses are μ_h^k and $(2\pi)^{k/2}$. For the real parameter $0 \leq t \leq 1$, put $m_t = (1-t)m_0 + tm_1$, $M_N(t) = \int v_N^* v_N m_t$, and $M(t) = M_{2q}(t)$. The source density proofs give all finite moments and positivity almost everywhere, so all these finite Grams are positive definite. We use the original form $\langle x, y \rangle_t = x^* M(t) y$ on H .

Let $P_N(t)$ be the orthogonal projection onto $\mathcal{P}_N \subset H$, and $Q_N(t)$ the projection onto $\chi\mathcal{P}_{N-q} \subset H$. Set $Q_{q-1} = 0$. For the coefficient inclusion $I_N : \mathcal{P}_N \rightarrow H$, their fully typed coefficient formulas are

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M, \\ Q_N &= I_N B_N (B_N^* M_N B_N)^{-1} B_N^* I_N^* M. \end{aligned} \quad (\text{AW4})$$

The zero relation domain at $N = q - 1$ contributes the zero operator. Let $G_N(t)$ be the minimum-lift Gram on E from (AW2) and $V_N(t) = \det G_N(t)$. Retain

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}, \\ C(t) &= M(t)^{-1} (M(1) - M(0)), \\ U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{AW5})$$

The AT proof, included in full, establishes $V_N = \det M_N / \det(B_N^* M_N B_N)$ with the original orientation $(-1)^{q(N-q+1)}$, and

$$\Delta \mathcal{B} = \mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \quad (\text{AW6})$$

Thus the following computation evaluates the operators in the already proved signed arithmetic correction.

84.2 Full spectra and an explicit source isometry

Theorem 84.1. *In the original $M(t)$ metric, both $U(t)$ and $W(t)$ have spectrum*

$$0 \text{ with multiplicity } q, \quad 1 \text{ with multiplicity } 2, \quad 2 \text{ with multiplicity } q - 1. \quad (\text{AW7})$$

The last multiplicity is zero when $q = 1$. There is an explicitly constructed smooth $M(t)$ -unitary $T(t) : H \rightarrow H$ such that $U(t) = T(t)W(t)T(t)^{-1}$.

Proof. Every subspace in either of the following flags is the literal displayed polynomial subspace, with its original coefficient inclusion. For the relation flag use

$$F_0 = \chi\mathcal{P}_0 \subset F_1 = \chi\mathcal{P}_{q-1} \subset F_2 = \chi\mathcal{P}_q \subset H. \quad (\text{AW8})$$

Their dimensions are 1, q , $q+1$, since multiplication by χ is injective. When $q = 1$, $F_0 = F_1$ is retained as that equality. Projections of nested subspaces commute and their products equal the projection of the smaller subspace: each larger space is the orthogonal direct sum of the smaller space and its orthogonal complement within the larger space. On the original orthogonal decomposition $F_0 \oplus (F_1 \cap F_0^\perp) \oplus (F_2 \cap F_1^\perp) \oplus F_2^\perp$, the operator $U = Q_{F_1} + Q_{F_2} - Q_{F_0}$ acts respectively by 1, 2, 1, 0. The dimensions are respectively 1, $q - 1$, 1, q .

For W use $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q} = H$. On the four corresponding orthogonal pieces, W acts by 0, 1, 2, 1, with dimensions q , 1, $q - 1$, 1. This proves (AW7), including the repeated middle subspaces at $q = 1$.

For an explicit intertwiner retain every Gram factor as follows. For any specified ordered independent list $z_0, \dots, z_d$, apply the recursion

$$y_j = z_j - \sum_{i < j} e_i \langle e_i, z_j \rangle_t, \quad a_j = (y_j^* M(t) y_j)^{1/2} > 0, \quad e_j = y_j / a_j. \quad (\text{AW9})$$

The positive real square root fixes its phase. No source density or mass is changed; each a_j is the norm in the original form. Independence proves $y_j \neq 0$ at every step, so this recursion is defined and smooth on the entire parameter interval.

First use $z_j = S^j$, $0 \leq j \leq 2q$, obtaining $p_0, \dots, p_{2q}$. Next use $z_j = \chi S^j$, $0 \leq j \leq q$, obtaining $r_0, \dots, r_q$. These lie in the successive relation spaces in (AW8). Finally let $R_{2q}(t) : E \rightarrow H$ be the original minimum section from AT6 and apply (AW9) to $R_{2q}(t)[1], \dots, R_{2q}(t)[S^{q-1}]$, obtaining $c_0, \dots, c_{q-1}$. The relation orthogonality in AT6 puts this entire list in $F_2^\perp$. Since $J_{2q}R_{2q} = I_E$, the list is independent and spans that q -dimensional complement.

Define T on the first orthonormal basis by

$$\begin{aligned} Tp_j &= c_j & (0 \leq j \leq q-1), \\ Tp_q &= r_0, & Tp_{2q} &= r_q, \\ Tp_{q+j} &= r_j & (1 \leq j \leq q-1). \end{aligned} \tag{AW10}$$

The middle range is empty for $q = 1$, with p_q and p_{2q} still distinct. The target list is an orthonormal basis. Thus this linear map is invertible and $T^*MT = M$. The spectral actions already calculated agree on each indicated basis vector, proving $UT = TW$. The recursion, the smooth minimum section and the positive norms prove the asserted smoothness. This completes the construction. $\square$

The exact isometry gives a commutator form of the signed integrand:

$$\text{Tr}((U - W)C) = \text{Tr}(WT^{-1}[C, T]), \quad [C, T] = CT - TC. \tag{AW11}$$

Indeed cyclicity moves T in $\text{Tr}(TWT^{-1}C)$ to the right, and subtraction then gives the displayed product. This T is an isometry of the original source; its cohomology observation is computed, rather than silently assigned an intertwining property. The unchanged arithmetic observation $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ satisfies

$$\mathcal{O}T = \sigma_h^{\otimes k} \eta J_{2q} T, \quad \mathcal{O}T - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} (T - I), \quad \ker(\mathcal{O}T) = T^{-1}F_2. \tag{AW12}$$

The last identity follows from both injections in (AW1) and $\ker J_{2q} = F_2$. By (AW10), $T^{-1}F_2$ is exactly $\text{span}\{p_q, \dots, p_{2q}\} = \mathcal{P}_{q-1}^\perp$ in H . At any fixed nonempty support label A , lift every displayed linear map by $(A, x) \mapsto (A, fx)$ and send τ to τ . Consequently the supported-zero inverse image of the lifted $\mathcal{O}T$ at A is the full set $\{A\} \times \mathcal{P}_{q-1}^\perp$. The Taylor unit in (AW1) and the defect in (AW12) remain present.

84.3 The spectral bound with every eigenvalue retained

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues, counted with multiplicity, of $D = M(0)^{-1}M(1)$ on its original $M(0)$ Hilbert space. All are positive. Define

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \kappa(D) = b_n/b_1. \tag{AW13}$$

For $q = 1$ the sum is empty and $\mathcal{L}_1(D) = \log(b_3/b_1)$.

Retain also the actual polynomial-space overlap. At each t , let $s(t) = \dim(\text{ran } U \cap \text{ran } W)$ and $d(t) = c_n(t) - c_1(t)$. With the actual $c_j(t)$ of (AW17), set

$$\begin{aligned} S_{\text{AW}}(t) &= c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j), \\ S_{\text{SP}}(t) &= \frac{d(t)}{2} \min \left\{ 4q - 2s(t), \sqrt{(2q+1)(8q-4-2\text{Tr}(UW))} \right\}, \\ \mathcal{J}_{h,k} &= \int_0^1 \min \{ S_{\text{AW}}(t), S_{\text{SP}}(t) \} dt. \end{aligned} \tag{AW13a}$$

The overlap refinement from SP8–SP10 is transported to this earlier calculation by the exact arrows $x = t, y = u, B_N^{\text{SP}} = I_N B_N^{\text{AW}}$. Indeed $M_N = I_N^* M I_N$ gives $(B_N^{\text{SP}})^* M B_N^{\text{SP}} = B_N^* M_N B_N$, so its projectors are precisely (AW4), and $(\mathcal{R}^{\text{SP}}, \mathcal{P}^{\text{SP}}, T^{\text{SP}}) = (U, W, C)$. The original Gamma densities agree because $r_{1/4}^{*k} = c_{1/4}^k r_{k/4} / c_{k/4}$ with $c_{1/4} = \sqrt{2\pi}$, exactly (AW3). The arithmetic densities already coincide as w_h^{*k} . Thus this comparison preserves the moments, masses, polynomial coefficients and all four endpoint signs.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(t) &= \frac{d(t)}{2} \|U(t) - W(t)\|_1, \\ \Sigma(t) &= \text{Tr}(C(t)^2) - \frac{(\text{Tr } C(t))^2}{2q+1}, \\ S_{\text{HS}}(t) &= \sqrt{\Sigma(t) \text{Tr}((U(t) - W(t))^2)}. \end{aligned} \tag{AW13c}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(t))I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(t)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(t)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

The original-moment angle calculation supplies a further simultaneous bound at this same claim site. Put $a_q = 2q - 1$ and

$$\begin{aligned} A_{h,k}(t) &= a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}} d(t), \\ \mathcal{J}_{h,k}^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(t), S_{\text{tr}}(t), S_{\text{SP}}(t), S_{\text{HS}}(t), A_{h,k}(t)\} dt, \\ \mathcal{I}_{h,k} &= \int_0^1 \min \left\{ d(t), \frac{S_{\text{AW}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{SP}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{HS}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}} \right\} dt. \end{aligned} \tag{AW13b}$$

These specialize RMT1–RMT18 through $a_{\text{RMT}} = a_q, x = t, C_x = C(t)$ and $(U_x, W_x) = (U(t), W(t))$. The density and relation arrows were proved in (AW3)–(AW4) and (AW13a), so every source moment and original cohomology map agrees. This application uses the actual moment measures (AW3). It does not impose moment structure on an arbitrary positive coefficient Gram.

Theorem 84.2. *On the unchanged arithmetic source,*

$$|\Delta \mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k} \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa(D). \tag{AW14}$$

The previously proved coefficient $2q - 1$ and full relative-spectrum bound are retained. The pointwise minimum additionally preserves the actual polynomial-space overlap, and is at most $\int_0^1 S_{\text{SP}}(t) dt$.

Proof. We first prove the needed finite spectral trace inequality. Let C be self-adjoint in a finite Hilbert space with eigenvalues $c_1 \leq \dots \leq c_n$. If P is an orthogonal projection of rank r , let e_j be an orthonormal eigenbasis of C and $d_j = \langle e_j, P e_j \rangle$. Then $0 \leq d_j \leq 1$ and $\sum_j d_j = r$, since $d_j = \|P e_j\|^2$

and $\text{Tr } P = r$. Thus $\text{Tr}(PC) = \sum_j c_j d_j$. Its upper bound by $\sum_{j=n-r+1}^n c_j$ follows by subtracting the two expressions: the mass $\sum_{j \leq n-r} d_j$ equals $\sum_{j > n-r} (1 - d_j)$, and every coefficient in the latter group is at least every coefficient in the former. Pairing the same nonnegative mass, or bounding both groups by c_{n-r+1} , proves the inequality. Applying this argument to $-C$ proves the lower bound $\sum_{j=1}^r c_j$. This proof includes repeated eigenvalues and the ranks zero and n .

For an operator Z with spectrum (AW7), its spectral projection A_Z onto eigenvalues 1, 2 has rank $q + 1$, and its spectral projection B_Z onto eigenvalue 2 has rank $q - 1$. The exact decomposition is $Z = A_Z + B_Z$. Applying the proved projection bounds gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AW15})$$

Each bound is attained for an abstract operator of this spectrum by assigning its eigenspaces to the appropriate eigenvectors of C . This attainment statement is only about the displayed spectral constraint; it assigns no freely chosen orientation to the actual polynomial relation spaces. Both U and W satisfy (AW15). Subtract their bounds and cancel the common c_{q+1} term only after writing both complete sums:

$$|\text{Tr}((U - W)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{AW16})$$

No mutual commutation of U, W, C is used.

The range dimensions in (AW7) give $s(t) \geq 1$. Let E_L be the orthogonal projector onto their intersection. Each $Z = U, W$ is at least its range projection, since its nonzero eigenvalues are 1, 2, and therefore $Z - E_L$ is positive of trace $2q - s(t)$. Writing $A = U - W = (U - E_L) - (W - E_L)$ proves

$$\|A\|_1 \leq 4q - 2s(t), \quad \text{Tr } A = 0, \quad \text{Tr } A^2 = 8q - 4 - 2 \text{Tr}(UW), \quad \|A\|_1 \leq \sqrt{(2q + 1) \text{Tr } A^2}. \quad (\text{AW16a})$$

The square identity uses $\text{Tr } U^2 = \text{Tr } W^2 = 4q - 2$ and cyclicity. The last inequality is Cauchy-Schwarz on every real eigenvalue of the self-adjoint A . Subtracting $(c_1 + c_n)I/2$ from C does not change its trace pairing with A . In an orthonormal eigenbasis of A , that pairing is bounded in absolute value by $d(t)\|A\|_1/2$. This proves $|\text{Tr}((U - W)C)| \leq S_{\text{SP}}(t)$ in addition to (AW16), and hence bounds it by their pointwise minimum. Smooth positive Grams give bounded continuous eigenvalues and overlaps; $s(t)$ is Borel because its rank strata are specified by matrix minors. Therefore the minimum is integrable. The signed identity (AW6) now proves $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}$ and $\mathcal{J}_{h,k} \leq \int S_{\text{SP}}$. For the original moment path, the exact original-source identification above also applies RMT11-RMT12: its full crossed angle lists give $|\mathcal{B}'| \leq A_{h,k}$, and $\mathcal{B} > 0$ follows because otherwise $1 \perp \chi \mathcal{P}_q$ would force $\int |\chi|^2 m_t = 0$ by the reflected polynomial $\chi^{\#k}(S) = \sum_j \bar{\chi}_j(k - S)^j$. Taking the pointwise minimum of this additional proved inequality with (AW16), (AW16a), and both complete bounds in (AW13c) gives $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)}$. The complete proof of the original angle estimate and its exact chain-rule continuation is retained in EP and RMT; it uses AW's earlier projection identities, not the final claim (AW14).

For the actual interpolated metric, $C(t) = ((1-t)I + tD)^{-1}(D - I)$, so its eigenvalues in increasing order are

$$c_j(t) = \frac{b_j - 1}{1 - t + tb_j}. \quad (\text{AW17})$$

The denominator is positive and the derivative with respect to b_j is $(1 - t + tb_j)^{-2} > 0$. Thus the ordering is valid for the entire closed interval. Each $c_j(t)$ is the derivative of $\log(1 - t + tb_j)$ and has integral $\log b_j$. Integrating (AW16) proves $\mathcal{J}_{h,k} \leq \mathcal{L}_q(D)$; the preceding pointwise-minimum argument proves the first inequality of (AW14). Every ratio in (AW13) lies between 1 and b_n/b_1 , so retaining its coefficient gives $1 + 2(q - 1) = 2q - 1$ times $\log \kappa(D)$ for the second bound. This proves the theorem for all $q \geq 1$. $\square$

All bounds retain the literal source masses. If $M(1) = cM(0)$, every $b_j = c$, so (AW13) is zero. Directly each original quotient determinant is multiplied at time t by $(1-t+tc)^q$, and the four displayed factors in (AW5) cancel. For a fixed invertible source-coordinate arrow F , the transported forms are $M' = F^*MF$, all subspaces have their F^{-1} images, and $D' = F^{-1}DF$. Hence the eigenvalues, (AW6) and (AW13) are unchanged under this explicit map.

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Here $M_z = M(t)|_{t=z}$, $C_z = C(t)|_{t=z}$ and $A_z = (U(t) - W(t))|_{t=z}$, with $D = D$. The exact RMT/ISM source arrow is the identity, and $\mathcal{B}$ is the function (AW5). Write $D = (M_0)^{-1}M_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let A_z be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, & H_z^\circ &= I - \frac{(1-z)I + zD}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (D - I), & R_z^{[\ell_N]} &= (H_z^\circ)^{\ell_N+1} C_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \tag{AW17aa}$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1-z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \tag{AW17ab}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q-6)} \longrightarrow 0. \tag{AW17ac}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

The original quartet Gamma theorem, when applied to the actual full quartet packet with displacement $\delta > 0$ and common order m , retains $q = [1 + k(m-1)](k+1)^2$ and gives

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) + \max\{-\mathcal{J}_{h,k}^{(5)}, j_{\ell_N} - e_{\ell_N}\}. \tag{AW18}$$

This substitutes the complete supplied Gamma theorem into the lower branches of (AW14) and the constructed signed inequality (AW17ab), then takes their maximum. It retains the stronger actual overlap bound, and also implies the earlier lower bounds with $-\mathcal{L}_q(D)$ and $-(2q-1) \log \kappa(D)$ respectively. It proves no uniform estimate for the eigenvalues b_j , and asserts no existence of an off-line packet. The new computed objects are the exact window spectra, their source isometry, its arithmetic observation and commutator, and the strengthened finite bound.

85 The one-dimensional quotient: exact angle and nonlinear transport

When $q = 1$, the same four endpoints are 0, 1, 1, 2. Their middle factors cancel in the specified ratio, leaving $\mathcal{B}(t) = \log(V_0(t)/V_2(t))$ with both original masses still present. Write $\chi = S - a$. The actual operators now are $U = Q_2$ and $W = I - P_0$. If $\mathcal{N} = (\chi\mathcal{P}_1)^\perp$ and $P_{\mathcal{N}} = I - Q_2$, then $U - W = P_0 - P_{\mathcal{N}}$. Both projections have rank one.

Let $e = 1 \in H$, $z = R_2(t)[1]$, and retain their actual norms $\langle e, e \rangle_t = V_0(t)$ and $\langle z, z \rangle_t = V_2(t)$. The source splitting gives $z = P_{\mathcal{N}}e$, since $J_2e = 1$. Thus $\langle e, z \rangle_t = V_2(t)$, and the angle between the two lines has the exact value

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)}. \quad (\text{AW19})$$

Both Grams are positive, so the cosine is positive. It is strictly less than one for the actual measure (AW3). Otherwise $e \in \mathcal{N}$, implying $\int \chi(S)m_t = 0$ and $\int S\chi(S)m_t = 0$. On the unchanged line $\bar{S} = k - S$, and consequently $|\chi(S)|^2 = ((k - \bar{a}) - S)\chi(S)$. The two equalities would force $\int |\chi(S)|^2 m_t = 0$, contrary to the positive Gram of this nonzero polynomial. Hence $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the whole closed interval.

For completeness, take the unit vector $p = e/\sqrt{V_0}$ and write $z/\sqrt{V_2} = \cos \theta p + \sin \theta r$, where r is the resulting unit vector orthogonal to p . The phase is already fixed since $\langle e, z \rangle_t = V_2 > 0$. In this orthonormal pair the exact difference of projections is

$$(P_0 - P_{\mathcal{N}})|_{\text{span}(p,r)} = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & -\sin^2 \theta \end{pmatrix}. \quad (\text{AW20})$$

Its trace is zero and its determinant is $-\sin^2 \theta$. It vanishes on the remaining one-dimensional orthogonal complement. Thus its full eigenvalues are $\sin \theta, 0, -\sin \theta$. Applying the proved rank-one spectral trace bounds to its positive and negative eigenprojections yields

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{AW21})$$

This retains the actual alignment of the two lines.

The function $\Phi(B) = 2 \operatorname{arcosh}(e^{B/2})$ for $B > 0$ has derivative $\Phi'(B) = 1/\sqrt{1 - e^{-B}}$, by differentiating the inverse of the hyperbolic cosine on its positive branch. The preceding strict positivity on the compact interval justifies its smooth composition with $\mathcal{B}(t)$. Divide (AW21) by its positive square-root factor. The four independent generic inequalities (AW16), (AW16a), and the trace-norm and Hilbert–Schmidt inequalities (AW13c) can all be divided by that same factor because $q = 1$ gives $a_q = 1$ and $\mathcal{B} > 0$. Their pointwise minimum integrates to $\mathcal{I}_{h,k}$ in (AW13b); integration of (AW17) bounds it by $\log(b_3/b_1)$. The exact nonlinear endpoint bound is

$$\left| 2 \operatorname{arcosh}(e^{B(1)/2}) - 2 \operatorname{arcosh}(e^{B(0)/2}) \right| \leq \mathcal{I}_{h,k} \leq \log(b_3/b_1). \quad (\text{AW22})$$

Combining (AW22) with the signed interval (AW17ab), with the original values $A_0 = \operatorname{arcosh}(e^{B(0)/2})$ and $L = \mathcal{I}_{h,k}/2 \leq \frac{1}{2} \log(b_3/b_1)$, monotonicity of $2 \log \cosh A$ for $A \geq 0$ gives

$$\begin{aligned} \mathcal{B}(1) &\geq \max\{2 \log \cosh(\max\{0, A_0 - L\}), \mathcal{B}(0) + j_{\ell_N} - e_{\ell_N}\}, \\ \mathcal{B}(1) &\leq \min\{2 \log \cosh(A_0 + L), \mathcal{B}(0) + j_{\ell_N} + e_{\ell_N}\}. \end{aligned} \quad (\text{AW23})$$

This last lower bound can be zero although the actual $\mathcal{B}(1)$ is strictly positive, as proved above. Every step concerns the original $q = 1$ quotient and its degree-two source. No extension of that dimension restriction to a quartet quotient is asserted.

85.1 Complete proof dependencies

The complete AT calculation accompanies this note, together with its 29 complete source dependencies and independent proof review. Those sources prove the original pushforward, tensor signs, full Taylor-unit injections, theta primitive, positive arithmetic and Gamma moments, and the quartet application. Every new assertion (AW7)–(AW23) is proved above on those unchanged finite objects. A separate independent review supplies the spectral calculation and its edge-case checks.

86 Complete arithmetic cohomology over the tau-base

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3

Complete source. sources/next_edition/actual_tau_conversions_v1/sources/tau_base/NOTE.md

Source SHA-256. d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3

86.0.1 Arithmetic cohomology over the τ -base

86.0.2 Scope and result

This note changes the geometric base of the existing arithmetic theta model to a one-point absolute base with distinguished absorbing element τ . It does **not** start with a variety over a finite field and then relabel its coefficients.

The main calculation is the right adjoint to the actual global-section functor on this τ -based model. A projective resolution computes the arithmetic cohomology; a dual injective complex computes its right-adjoint object. These are then linked to the existing finite Mellin-jet quotients, including the full nilpotent jets and the original e/τ distinction.

This is an algebraic, finite-chart realization. No assertion is made that the chart model is an arithmetic scheme, that its topology already supplies all prime correspondences, that the right-adjoint object is an involutive Verdier duality on every sheaf, or that the calculation proves a weight estimate. Each operation used below has its stated category. In particular the algebraic dual is not an assertion about the continuous dual of the Schwartz quotient.

The supplied SplitZero results are the source for the scalar operations, support fibres and transports. The new constructions and proofs below are calculations in this note; they have not been Lean-formalized.

86.0.3 1. An actual base whose structural zero is τ

Let

$$\mathbf{F}_{1,\tau} = \{\tau, 1\}$$

be the pointed multiplicative monoid with identity 1 and absorbing element τ . As a blueprint, its only additional zero relation is that the empty sum represents τ . No additive equation $1 + 1 = 1$ is imposed. Its spectrum $\mathfrak{b}_\tau$ has the one prime $\{\tau\}$.

For every nonzero commutative ring R , retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet. \quad (\text{TauBase 1})$$

Let B_R be its associated blueprint: the original multiplicative pointed monoid, with all its original finite additive relations. In particular retain $[\tau] = 0$, in agreement with the presentation correction in the supplied formalization note. There is a blueprint morphism

$$\mathcal{J}_R : \mathbf{F}_{1,\tau} \longrightarrow B_R, \quad \tau \mapsto \tau, \quad 1 \mapsto 1_R^\bullet. \quad (\text{TauBase 2})$$

On spectra it gives a genuine structural morphism

$$a_R : \operatorname{Spec} B_R \longrightarrow \mathfrak{b}_\tau. \quad (\text{TauBase 3})$$

Every prime has inverse image $\{\tau\}$ under (2). Thus every arithmetic point lies over the single base point, by a specific inverse-image operation on prime ideals.

The arithmetic quotient stays attached:

$$p_R : G(R) \longrightarrow R, \quad p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r. \quad (\text{TauBase 4})$$

For a domain,

$$P_\tau = \{\tau\}, \quad P_0 = \{\tau, e_R\}, \quad \operatorname{Spec} R \xrightarrow{i_R} \operatorname{Spec}_{\text{sr}} G(R), \quad \mathfrak{p} \mapsto \mathfrak{p} \sqcup \{\tau\}. \quad (\text{TauBase 5})$$

Here $i_R(\operatorname{Spec} R) = V(e_R) = \overline{\{P_0\}}$. These maps carry the source's infinite quotient when $R = \mathbb{Z}$. They do not assign an additional finite norm to either bottom ideal.

There is a second, different-direction morphism in the same diagram:

$$\operatorname{Spec} \mathbb{B} \xrightarrow{\operatorname{Spec} \chi_R} \operatorname{Spec}_{\text{sr}} G(R) \xrightarrow{a_R} \mathfrak{b}_\tau, \quad \chi_R(\tau) = 0, \quad \chi_R(r^\bullet) = 1. \quad (\text{TauBase 6})$$

The first arrow lands at P_τ . The original localization calculation is

$$G(R)[e_R^{-1}] \cong \mathbb{B}. \quad (\text{TauBase 7})$$

Indeed invertibility and $e_R^2 = e_R$ force $e_R = 1$, and $e_R r^\bullet = e_R$ then forces every supported scalar to equal 1. External absence remains distinct. We use (3), not localization (7), for the change of base. The arithmetic structure and its quotient remain upstream of (6).

For $R = \mathbb{Z}$, the source's infinite target is retained in the square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \hookrightarrow & \mathbb{C}. \end{array} \quad (\text{TauBase 8})$$

Here $j : \mathbb{Z} \hookrightarrow \mathbb{C}$ is the specified unital embedding, and $G(j)$ is its split lift.

86.0.4 2. The arithmetic theta chart space over that base

Fix $E = \mathbb{C}$. Let P be the four-point poset

$$+ < \eta < \sigma, \quad - < \eta < \sigma, \quad (\text{TauBase 9})$$

with $+$ and $-$ incomparable. Give P the upper-set topology. Its proper nonempty opens are

$$\{\sigma\}, \quad \{\eta, \sigma\}, \quad \{+, \eta, \sigma\}, \quad \{-, \eta, \sigma\}.$$

The structural sheaf $\mathcal{S}$ has value $G(E)$ on every open containing an arithmetic chart point and value $\mathbb{B}$ on $\{\sigma\}$. Restriction to σ is χ_E ; other nontrivial restrictions are identities. Compatible supported local amplitudes glue, and compatibility at σ distinguishes an absent family from a supported one. This proves the sheaf assertion directly.

Together with (2), this defines a space with a sheaf of blueprints over $\mathfrak{b}_\tau$. There is also the explicit structural morphism to $\operatorname{Spec}_{\text{sr}} G(E)$, sending $+, -, \eta$ to P_0 and σ to P_τ , followed by the morphism induced by $G(j)$ to $\operatorname{Spec}_{\text{sr}} G(\mathbb{Z})$. On arithmetic stalks its map is $G(\mathbb{Q}) \rightarrow G(E)$; at σ it is $\mathbb{B} \rightarrow \mathbb{B}$. Thus the infinite quotient is still in a commuting structural diagram. This particular chart model maps to the arithmetic generic point; its finite-prime geometry is not being claimed to cover the closed prime fibres. Arithmetic prime information in the present realization enters through the theta map and its Euler/explicit formula, not through extra chart points invented here.

The coefficient spaces are exactly

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad (\text{TauBase 10})$$

and

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}. \quad (\text{TauBase 11})$$

Define

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad JF(x) = x^{-1}F(1/x). \quad (\text{TauBase 12})$$

The retained Poisson identity gives

$$J\Theta\psi = \Theta\widehat{\psi}. \quad (\text{TauBase 13})$$

The arithmetic coefficient sheaf is the covariant diagram

$$\mathcal{T}_+ = V, \quad \mathcal{T}_- = V, \quad \mathcal{T}_\eta = \mathcal{B}, \quad \mathcal{T}_\sigma = 0, \quad r_+ = \Theta, \quad r_- = J\Theta = \Theta\mathcal{F}. \quad (\text{TauBase 14})$$

This use of a zero vector space at σ will not identify the supported zero with τ . It is reconstructed as the support skeleton in the next paragraph.

All original two-leg support faces Set $L = \mathcal{P}(\{+, -\})$. For nonempty $A \subseteq \{+, -\}$, define a sheaf $\mathcal{T}_A$ by

$$(\mathcal{T}_A)_+ = \begin{cases} V & + \in A \\ 0 & + \notin A, \end{cases} \quad (\mathcal{T}_A)_- = \begin{cases} V & - \in A \\ 0 & - \notin A, \end{cases} \quad (\mathcal{T}_A)_\eta = \mathcal{B}, \quad (\mathcal{T}_A)_\sigma = 0. \quad (\text{TauBase 15})$$

The nonzero restriction maps are those in (14); put $\mathcal{T}_\emptyset = 0$. Transports for $A \subseteq A'$ are coordinate inclusions at $+$ and $-$ and the identity at η . These form a strict diagram.

Reconstruct the supplied SplitZero semimodule sheaf

$$\widetilde{\mathcal{T}} = \{\tau\} \sqcup \bigsqcup_{A \neq \emptyset} \{A\} \times \mathcal{T}_A. \quad (\text{TauBase 16})$$

At σ its stalk is the semilattice L , with Boolean scalar action. At an arithmetic chart the scalar action is by $G(E)$. Restrictions to σ are semilinear along χ_E . In particular $e_E(A, v) = (A, 0)$, which restricts to A at σ , while τ restricts to the empty label. The original distinction is present at the new base boundary.

The additive module in degree zero is exactly the original two-leg carrier $(V^\tau)^2$: a missing leg is absent, and a supported zero in that leg is present. Formula (15) records a missing leg as a zero coefficient space **with its separate mask A retained**. Thus no new support is inserted into the original missing leg.

86.0.5 3. Compute global cohomology over the τ -base

We now use the explicitly declared linear-fibre category

$$\mathcal{A} = \text{Fun}(P, \text{Vect}_E).$$

This is a realization category of the marked blueprint space, not an asserted category of étale sheaves on a new arithmetic scheme. Its passage to the original supported sheaves is (15)–(16), with kernels and quotients reconstructed fibrewise.

For $x \in P$, let P_x be the projective diagram

$$P_x(y) = \begin{cases} E & x \leq y \\ 0 & \text{otherwise.} \end{cases} \quad \text{Hom}_{\mathcal{A}}(P_x, F) = F_x. \quad (\text{TauBase 17})$$

The constant diagram $\mathbf{E}$ has the exact projective resolution

$$0 \longrightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \mathbf{E} \longrightarrow 0. \quad (\text{TauBase 18})$$

At $+$ and $-$ the last map is identity. At η and σ it is addition with kernel the displayed anti-diagonal. This checks every stalk and every arrow.

Consequently the base pushforward is

$$\mathbf{A}_\tau(F) := R\Gamma(P, F) \simeq [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta], \quad \text{degrees } 0, 1. \quad (\text{TauBase 19})$$

This is global cohomology over the base point, rather than restriction to the added generic point. The typed comparison to the latter is

$$\text{res}_\sigma : \mathbf{A}_\tau(F) \longrightarrow F_\sigma[0]. \quad (\text{TauBase 20})$$

Two chain representatives of (20) in degree zero are $(x, y) \mapsto r_{+\sigma}x$ and $(x, y) \mapsto r_{-\sigma}y$. Their difference is

$$r_{\eta\sigma}(r_+x - r_-y),$$

so they are chain homotopic, with homotopy $r_{\eta\sigma} : F_\eta \rightarrow F_\sigma$. Thus (20) is canonical in the derived category. On the actual joint arithmetic sheaf,

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \widehat{\psi})} \mathcal{B}], \quad (\text{TauBase 21})$$

$$H^0 \mathbf{A}_\tau(\mathcal{T}) \cong V, \quad \psi \mapsto (\widehat{\psi}, \psi), \quad H^1 \mathbf{A}_\tau(\mathcal{T}) = Q := \mathcal{B}/\Theta V. \quad (\text{TauBase 22})$$

Injectivity of Θ follows from its Mellin transform on $\Re s > 1$, the nonvanishing Euler product there, and Mellin uniqueness. This is the existing source proof, not an assumption about zeros in the critical strip.

The actual supported differential is

$$\widetilde{d}(\phi, \psi) = (A, \Theta p(\phi) - J\Theta p(\psi)), \quad (\text{TauBase 23})$$

where A is the set of present legs; the absent pair maps to τ . Its cohomology uses

$$Z_{\text{sp}}^i = \text{Eq}(d, \epsilon d), \quad H_{\text{sp}}^i = \text{coeq}(b, \epsilon b), \quad \epsilon(x) = ex. \quad (\text{TauBase 24})$$

For every nonempty A , $H^1(\mathcal{T}_A) = Q$. For a single-leg A , $H^0 = 0$; for the joint A , $H^0 = V$. Therefore

$$\widetilde{H}_\tau^1 = \{\tau\} \sqcup \bigsqcup_{A \neq \emptyset} \{A\} \times Q. \quad (\text{TauBase 25})$$

Restriction (20) sends each of these arithmetic classes to the supported zero at its own label in the σ skeleton. Their global cohomology classes in (25) remain present. This is the exact answer to what remains when all geometric points lie over the τ -base.

86.0.6 4. Compute the right adjoint on this base

Deligne defines a dualizing object through a right adjoint and then proves weight bounds. Here we compute the right adjoint to our stated functor (19), rather than importing a dualizing complex from a finite-field variety.

For $x \in P$ and a vector space W , define

$$I_x(W)(y) = \begin{cases} W & y \leq x \\ 0 & \text{otherwise.} \end{cases} \quad \text{Hom}_{\mathcal{A}}(F, I_x(W)) = \text{Hom}_E(F_x, W). \quad (\text{TauBase 26})$$

Because evaluation at x is exact and W is an injective E -module, $I_x(W)$ is injective. The right-adjoint object is

$$\boxed{K_\tau(W) = [I_\eta(W) \xrightarrow{(+\text{res}, -\text{res})} I_+(W) \oplus I_-(W)], \quad \text{degrees } -1, 0.} \quad (\text{TauBase 27})$$

For every sheaf F ,

$$\begin{aligned} R\text{Hom}_A(F, K_\tau(W)) &= [\text{Hom}(F_\eta, W) \longrightarrow \text{Hom}(F_+, W) \oplus \text{Hom}(F_-, W)] \\ &\cong R\text{Hom}_E(A_\tau(F), W). \end{aligned} \quad (\text{TauBase 28})$$

The differential is $\ell \mapsto (\ell r_+, -\ell r_-)$. This proves the adjunction, with its signs. For a bounded complex W , totalize the same formula with its original differential.

Let $S_\eta W$ denote the diagram with W at η and zero at the other three points. Calculation of the four stalk complexes gives

$$K_\tau(W)_+ = [W \xrightarrow{1} W], \quad K_\tau(W)_- = [W \xrightarrow{-1} W], \quad K_\tau(W)_\eta = W[1], \quad K_\tau(W)_\sigma = 0. \quad (\text{TauBase 29})$$

Thus

$$\boxed{K_\tau(W) \simeq S_\eta W[1].} \quad (\text{TauBase 30})$$

The symbol $[1]$ means W is in cohomological degree -1 . It is obtained from (18) and (27). The functor is a right adjoint to $R\Gamma$ on this model; no assertion that it is the extraordinary pullback for an unconstructed proper arithmetic morphism is made. In particular, an involutive local Verdier duality or a finite-field purity theorem has not been obtained just from (28).

Apply (27) componentwise to support diagrams. Every zero coefficient stalk reconstructs as its labelled zero, including at σ . Dual transports are actual precomposition maps

$$\rho_{AB}^\vee : \text{Hom}(H_B, W) \longrightarrow \text{Hom}(H_A, W), \quad \ell \mapsto \ell \rho_{AB}. \quad (\text{TauBase 31})$$

They therefore have index category L^{op} ; the label arrows have not been reversed without retaining the precomposition operation.

86.0.7 5. Retain the actual characteristic-zero arithmetic action

For a positive real a , use

$$U_a F(x) = F(x/a), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{TauBase 32})$$

Then

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = a U_{1/a} \widehat{\phi}. \quad (\text{TauBase 33})$$

The action on the two source charts is consequently

$$(\phi, \psi) \mapsto (U_a \phi, a U_{1/a} \psi), \quad F \mapsto U_a F \text{ at } \eta. \quad (\text{TauBase 34})$$

Using the same U_a on both source charts would fail the required intertwining. Direct calculation gives

$$U_a J = a J U_{1/a}, \quad d(U_a \phi, a U_{1/a} \psi) = U_a d(\phi, \psi). \quad (\text{TauBase 35})$$

It fixes every support mask. At σ it fixes the support skeleton. The new base remains the same point; the induced action on its pushforward remains the arithmetic action (33).

The source's moment map gives the two coefficient lines

$$\mathcal{L}_0 = E \text{ with action } 1, \quad \mathcal{L}_1 = E \text{ with action } a, \quad m(\phi) = (\phi(0), \int_{\mathbb{R}} \phi(x) dx), \quad (\text{TauBase 36})$$

$$mU_a = \text{diag}(1, a)m.$$

Both choices $K_\tau(\mathcal{L}_0)$ and $K_\tau(\mathcal{L}_1)$ exist. The latter produces, on dual functionals, the action

$$(U_a \ell)(v) = a \ell(U_a^{-1} v). \quad (\text{TauBase 37})$$

The factor a is carried from the actual second moment representation. It is not set to one or inferred from the cardinality of a base point.

86.0.8 6. An actual morphism from arithmetic cohomology to its τ -base dual

Put $g = 2\xi$. Retain a finite reflection-stable set Z of actual zeros, with actual multiplicities. Define

$$A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho / (g), \quad (J_Z F)_\rho = \sum_{j < m_\rho} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j. \quad (\text{TauBase 38})$$

The source identity $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ entire, proves that the maps

$$\mathcal{T} \longrightarrow S_\eta A_Z, \quad 0 \text{ on } +, -, \sigma, \quad J_Z \text{ on } \eta, \quad (\text{TauBase 39})$$

are a sheaf morphism. Applying A_τ gives precisely the already-specified map $Q \rightarrow A_Z$ in degree one. This finite jet map is onto; the source proof is independence of the finite family $e^{\rho y} y^j / j!$ on any nonempty interval, followed by a compactly supported smooth right inverse.

On A_Z , the scaling action is

$$U_a|_\rho = a^\rho \sum_{j < m_\rho} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho z^j = z^{j+1}. \quad (\text{TauBase 40})$$

Using the reflected germ map

$$f^\dagger(s) = \overline{f(1-\bar{s})}, \quad \sum c_j z_{i_\rho}^j \mapsto \sum (-1)^j \bar{c}_j z_\rho^j, \quad (\text{TauBase 41})$$

retain the perfect finite residue pairing

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger(s)h(s)}{g(s)} ds. \quad (\text{TauBase 42})$$

It has $R_Z(U_a f, U_a h) = a R_Z(f, h)$. Its raw skew-Hermitian orientation is retained; no multiplication by a phase is inserted to call it positive.

Precomposing the dual map of (39) with (42) gives an explicit injective linear morphism from the conjugate finite packet:

$$\boxed{\overline{A_Z} \longrightarrow H^{-1} R \text{Hom}_{\mathcal{A}}(\mathcal{T}, K_\tau(\mathcal{L}_1)), \quad \bar{f} \longmapsto \left([F] \longmapsto \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger(s) \mathcal{M}F(s)}{g(s)} ds \right)}. \quad (\text{TauBase 43})$$

The target is canonically $\text{Hom}_E(Q, \mathcal{L}_1)$ by (28). Well-definedness follows because replacing F by $F + \Theta\phi$ adds the holomorphic residue expression $f^\dagger H_\phi$. Injectivity follows from the surjectivity of J_Z and perfectness of (42). Formula (37) and covariance of (42) prove equivariance.

Thus the duality here is a morphism **to the right-adjoint object computed on the τ -base**, not merely a pairing written separately from a cohomological model.

The trace contraction is the specified endomorphism

$$J_g : A_Z \rightarrow A_Z, \quad [h] \mapsto [g'h], \quad (\text{TauBase 44})$$

$$R_Z(f, J_g h) = \sum_{\rho} m_{\rho} \overline{f(1 - \bar{\rho})} h(\rho) = \text{Tr}(M_{f^{\dagger}h}|A_Z). \quad (\text{TauBase 45})$$

At a zero of order m , write the actual $g = u(z)z^m$. Then $g' \equiv mu(z)z^{m-1} \pmod{z^m}$. The radical of (45) consists of the jets with zero residue value. They map to e under the original internal quotient, while (42) retains their full pairing before that observation.

All maps in (38)–(45) lift to supported fibres. At label A the map is $(A, [F]) \mapsto (A, J_Z F)$. A zero functional obtained in (43) or (45) is the supported zero functional at that label, not τ . For inclusions of admitted source labels the resulting squares commute, and the dual arrows are the precomposition arrows (31).

86.0.9 7. The tensor-amplification operation on the changed base

For the joint arithmetic fibre, take products of the actual chart space P over the underlying point of $\mathfrak{b}_{\tau}$, and external products of its coefficient sheaf. This is the ordinary product of the declared finite topological models with external coefficient tensors. It is not a claim that its structure-sheaf product has already been identified with the schematic square of the full arithmetic space.

The projective resolution for P^r is the external tensor product of (18). Its differential on homogeneous factors is

$$d(x_1 \otimes \cdots \otimes x_r) = \sum_{j=1}^r (-1)^{\sum_{k < j} |x_k|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_r. \quad (\text{TauBase 46})$$

Because the tensor product is over the field E , this yields the explicit Künneth map and, at the top degree,

$$\boxed{H^r \mathbf{A}_{\tau, r}(\mathcal{T}^{\boxtimes r}) \cong Q^{\boxtimes_{E^r}}.} \quad (\text{TauBase 47})$$

Indeed the numerator of top cohomology is $\mathcal{B}^{\boxtimes r}$; its relations are the sum of all subspaces obtained by putting ΘV in at least one factor. The quotient is exactly the displayed tensor quotient. The joint-support evaluation is a stated functor from the full retained support diagram. Other original support fibres are not removed; their cochain complexes and transports remain in that diagram.

The top-degree kernel relation is again quotiented internally in its supported scalar fibre. Pure external absence is absorbing. A supported tensor whose amplitude becomes a tensor relation maps to the supported zero of that tensor fibre. We do not identify a coefficient external tensor with the Cartesian product of support labels; such a tensor has its specified absorbing basepoint and balancing relations.

The dual right-adjoint object on the product is computed from the dual tensor resolution:

$$K_{\tau, r}(W) \simeq S_{(\eta, \dots, \eta)} W[r]. \quad (\text{TauBase 48})$$

For $r = 2$, the degree -2 term and both degree -1 terms occur in that resolution with the signs from (46). The shift $[2]$ is obtained by this actual product calculation, not inserted into (30).

For a residue-value eigenvector at an actual zero ρ , the action in (47) on its r -fold tensor is

$$a^{r\rho}. \quad (\text{TauBase 49})$$

In particular the exact modulus defect relative to r copies of $\mathcal{L}_1^{1/2}$, expressed without rescaling the action, is

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re \rho - 1) \log a \quad (a > 1). \quad (\text{TauBase 50})$$

This performs the amplification operation on the actual arithmetic spectral quotient over the new base. It does not prove a bound on the left-hand side. Deligne's geometric upper estimate is additional information in his setting; it has not been derived for (47).

86.0.10 8. What a map to the base does, and what deleting a spectral packet does

The structural map (3) sends every point to the base point. The derived pushforward (19) carries its coefficient relations and arithmetic operators along with it. Restriction (20), by contrast, has value the support skeleton on our sheaf. These are two explicitly compared operations on the same model.

For clarity about the proposed “exceptions land in τ ,” let W and W' be ordinary vector spaces and use their single-active-fibre split lifts. If a $G(E)$ -linear map

$$f : W^\tau \rightarrow (W')^\tau$$

sends one supported vector v to τ , then it sends every supported vector to τ . Proof: $f(0_W^\bullet) = f(ev) = ef(v) = \tau$. For every w , $ef(w) = f(ew) = \tau$. In $(W')^\tau$, the only element whose e -multiple is τ is τ itself. This result is for this specified target; it is not asserted for arbitrary semimodules with a nontrivial bottom group fibre.

A labelled projection can delete a whole packet while retaining other labels, but it has a calculable cohomological contribution. For a reflection-stable finite set Z split into C and B , the actual projection is

$$A_Z \twoheadrightarrow A_C, \quad \ker = A_B. \quad (\text{TauBase 51})$$

The difference of convolution traces is

$$\text{Tr}(H_h|A_Z) - \text{Tr}(H_h|A_C) = \sum_{\rho \in B} m_\rho \mathcal{M}h(\rho). \quad (\text{TauBase 52})$$

This is nonzero for some admissible compactly supported smooth h whenever B is nonempty, by finite Mellin interpolation. Its split value for that test is a supported nonzero number. The cone of the projection $A_Z[-1] \rightarrow A_C[-1]$ has $H^0 = A_B$; that is the precise cohomological trace lost by deletion.

Consequently a structural projection of the entire geometry to the τ -base does not delete a packet. A packet deletion is (51), with the separate kernel and trace (52). Establishing that all actual exceptional contributions are absent requires this kernel or its full trace contribution to vanish by arithmetic geometry, not merely by relabelling its support.

86.0.11 9. Result and next mathematically defined target

The completed construction in this note is

$$\tilde{\mathcal{T}} \text{ over } \mathfrak{b}_\tau \xrightarrow{A_\tau} \tilde{H}_\tau^1 \xrightarrow{J_Z} A_Z^\tau,$$

with the computed adjoint object $K_\tau(\mathcal{L}_1)$, the dual morphism (43), the trace contraction (45), and all tensor powers (47). It is a characteristic-zero scaling calculation, not a purity calculation on a finite-field curve.

The arithmetic quotient $G(\mathbb{Z}) \rightarrow \mathbb{Z}$ is kept in the structural map (8). The current theta chart model sees its generic arithmetic image; a global prime-sensitive geometric realization with a Lefschetz theorem and two-sided weight estimates is not obtained from this finite chart topology alone. The specific remaining quantity in this model is the tensor-growth defect (50), or equivalently the sign of the full trace pairing (45) after its global arithmetic realization. Neither is set to zero in the note.

The new work is the actual change of base, the computed right adjoint, and the cohomological placement of the finite arithmetic duality maps on that base. It narrows the task to a property of these explicit objects rather than repeating the assertion that a shared support should imply purity.

86.0.12 Sources read for this calculation

1. `globalization_note.tex`: lines 432–563 (localizations and stalks), 673–751 (contracted monoid algebra and blueprint remark). The separate supplied Lean-summary note provides the correction `[tau]=0` and the implemented support-fibre operations. Those source results are not claimed as new.
2. Retained `Split_Support_Actual_Comparison/CONTINUATION.tex`: test spaces, Fourier and theta maps, the arithmetic quotient and finite jet realization. This is an earlier research derivation, not an independent verification of every global statement in its package.

3. The supplied French Weil II TeX excerpts, original §6.2.1–6.2.7 and §3.3: the right-adjoint definition of a dualizing complex, opposite dual-weight bounds, proper pushforward, and the arithmetic-base twist/degree equation. The excerpts’ historical-transcription status remains recorded. Reading these windows is not a claim to read or audit the whole paper in this turn.
4. O. Lorscheid, *The geometry of blueprints. Part I* (arXiv:1103.1745): external framework for the declared blueprint base. The finite-chart model and adjoint calculation above are proved here; they are not attributed to that paper.

86.0.13 Checks

The accompanying executable checks finite instances of the stated poset resolution, dual complex, internal support differential, action compatibility, tensor differentials and trace deletion. They use explicit exceptions rather than optimizable assert statements. They are regression checks of these identities, not a formal proof of RH, of Deligne’s theorem, or of a global arithmetic realization.

87 Complete Gamma-reference lower bound and arithmetic determinant comparison

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 830fbfe19d688b1e138b41ea4a784c662629dfec71812627b66927f91ce468f

Complete source. sources/next_edition/actual_tau_conversions_v1/sources/gamma_review/deligne_bound_gamma_independent_review_20260913.md

Source SHA-256. 830fbfe19d688b1e138b41ea4a784c662629dfec71812627b66927f91ce468f

87.0.1 Independent proof audit of the Gamma-reference lower bound

Date: 13 September 2026. Status: the Gamma theorem and the signed arithmetic comparison are accepted with the domains proved below. This review gives the complete calculation for delivered equations (28)–(43) and (51)–(56). Its appended Gamma derivation is part of this proof body.

87.0.2 Sources and reading scope

Paths below are relative to the mathematical workspace. The delivered files have not been changed.

Source	Bytes	SHA-256	Reading scope
output/split_zero_rh_t andem_2026-09-12/ sources/web_deligne_ bound_audit_delivery /Tau_Deligne_Bound _Audit/NOTE.tex The same delivery’s verify.py	35701 11159	bfe63094ec360b6059 f2936e5e456c6c0fd0 8db144bbabc29621a5 27b89c13c6 fb39ffeeec2de6002c d477e92ab2db1e52ed dfdd945f8596f15702 63cf9c2cb0	Complete note Complete code; no execution of this old suite is claimed here
The same delivery’s SOURCES.json	3582	6edb542a72730aa6ba 05db9f37700b65a50e 345e96fdc5f48bf623 e9e47b2286	Complete source record
output/split_zero_rh_t andem_2026-09-12/te x/arithmetic_input.tex	14644	a50b0fb587b644c4ea 94dba45b2b423d038f 2bfbe7fef188543939 ba850e58d2	Complete A1–A19 and proofs

Source	Bytes	SHA-256	Reading scope
output/split_zero_rh_t andem_2026-09-12/te x/toda_cv_exact_brid ge.tex	45603	c9629b11d4b4a173dd 52bdf75fafb437883a 85aba05e598c637f39 0386d45512	Lines 682–1098: separate first degree, phase term, spectral projection, consecutive estimates, and scope
output/split_zero_rh_t andem_2026-09-12/ sources/web_gamma_ convolution_delivery /Tau_Gamma_Convolu tion_Descent/NOTE.te x	34859	7cf984d87de405b815 250b44454c4ee27c8a b46574da245b377ebc 6e49327fa2	Opening and Sections 2–6.1: literal measure, sum coordinate, amplitude, conditional multiplier and bounds
work/deligne_bound _gamma_generating_ subreview_20260913 .md	11155	d4dfa59e923e7c7223 b31396e84dc17b595f 2fdd5eeddbadb63f3d 11aec971cb	Complete independent Gamma derivation; appended verbatim below

This is an all-degree written proof. No fresh Lean execution, complete Deligne scan audit, interval certification of arithmetic moments, or remote publication is claimed by this review. The purity/operator branch is being audited separately.

87.0.3 1. Literal norms and their coordinate map

The complete Beta/Fourier calculation in the appended proof derives, for $a > 0$,

$$r_a(x) = \frac{|\Gamma(a + ix/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \quad \int e^{iyx} r_a(x) dx = c_a (\cosh y)^{-2a}. \quad (\text{GR1})$$

In particular c_a is the original mass. The literal substitution $t = e^{2y}$ in the Beta integral gives the factor 2^{1-2a} . Exponential moments exist for $|\theta| < \pi/2$, and the same calculation gives $c_a (\cos \theta)^{-2a}$. Fourier uniqueness for finite measures then gives

$$r_a^{*k}(x) = \frac{c_a^k}{c_{ka}} r_{ka}(x). \quad (\text{GR2})$$

The coordinate is the original sum $x = x_1 + \cdots + x_k$. At $a = 1/4$, $c_{1/4} = \sqrt{2\pi}$, so the mass is $(2\pi)^{k/2}$.

The printed generating function, with branches equal to one at zero, is

$$F_a(z, x) = (1 - iz)^{-a+ix/2} (1 + iz)^{-a-ix/2} = \sum_{n \geq 0} b_n^{(a)}(x) \frac{z^n}{n!}. \quad (\text{GR3})$$

For real small z , it is $(1 + z^2)^{-a} \exp(x \arctan z)$. Thus $b_1 = x$, with the positive sign. The appended proof shows

$$\int F_a(z, x) F_a(w, x) r_a(x) dx = c_a (1 - zw)^{-2a}. \quad (\text{GR4})$$

All coefficient extractions are justified by exponential moments on a larger neighborhood. Thus the real, monic polynomials have orthogonal norms $c_a n! (2a)_n$. Their recurrence follows from $(1 + z^2) \partial_z F_a = (x - 2az) F_a$:

$$b_{n+1}^{(a)} = x b_n^{(a)} - n(n + 2a - 1) b_{n-1}^{(a)}, \quad b_0 = 1, \quad b_1 = x. \quad (\text{GR5})$$

Keep $S = k/2 + ix$ and define

$$p_n(S) = i^n b_n^{(k/4)} ((S - k/2)/i). \quad (\text{GR6})$$

The leading coefficient is $i^n i^{-n} = 1$. On the integration line the factor i^n has squared modulus one. Substituting in GR5 and using $i^2 = -1$ gives, with the actual original Gamma sum measure,

$$\begin{aligned}\omega_{k,n}^\Gamma &= \int |p_n(k/2 + ix)|^2 r_{1/4}^{*k}(x) dx = (2\pi)^{k/2} n! (k/2)_n, \\ Sp_n &= p_{n+1} + \frac{k}{2} p_n - n(n + k/2 - 1) p_{n-1}.\end{aligned}\tag{GR7}$$

At $n = 0$ the last term is absent. Thus delivered (28), (29), and (35) have the correct masses, monic convention, and negative recurrence sign.

87.0.4 2. The complete cyclic polynomial for every multiplicity

Write

$$h_*(S) = \prod_{\epsilon, \eta \in \{\pm 1\}} (S - 1/2 - \epsilon/4 - i\eta).\tag{GR8}$$

For $Z = S - 1/2$, multiplying conjugate factors gives $h_* = Z^4 + (15/8)Z^2 + 289/256$. It is monic, real, and fixed by $S \mapsto 1 - S$.

Fix integers $m \geq 1, k \geq 1$. Consider the full tensor algebra of k copies of $\mathbb{C}[S]/(h_*^m)$, the sum multiplication operator, and cyclic vector $1^{\otimes k}$. Chinese remainders decompose each factor into four local algebras $\mathbb{C}[n]/(n^m)$. Every tensor occupation therefore has local algebra

$$\mathbb{C}[n_1, \dots, n_k]/(n_1^m, \dots, n_k^m)$$

and sum operator equal to its center plus $n_1 + \dots + n_k$. Its power of degree $k(m-1) + 1$ vanishes because every monomial then has some exponent at least m . Its power of degree $k(m-1)$ is exactly

$$\frac{(k(m-1))!}{((m-1)!)^k} \prod_{j=1}^k n_j^{m-1} \neq 0.\tag{GR9}$$

For $m = 1$ this is the zeroth power 1, giving the same length-one result. The coefficient is nonzero in characteristic zero. The cyclic vector projects to the unit in every local algebra, so this full nilpotency length also occurs in its cyclic minimal polynomial.

For margins $0 \leq a, b \leq k$, the center is

$$\lambda_{a,b} = k/2 + (2a - k)/4 + i(2b - k).\tag{GR10}$$

All margin pairs occur. Indeed, $t = \max(0, a + b - k)$ gives nonnegative occupations $t, a - t, b - t, k - a - b + t$, with total k and the required two margins. Distinct margins have distinct centers because both coordinate coefficients are nonzero. Repeated occurrences of the same center take the maximum of their identical nilpotency lengths in the minimal polynomial, not their sum. Therefore exactly

$$\ell_k = 1 + k(m-1), \quad \chi_k(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^{\ell_k}, \quad q_k = \ell_k(k+1)^2\tag{GR11}$$

is the cyclic minimal polynomial. Each primary summand in $\mathbb{C}[S]/(\chi_k)$ has dimension and block length ℓ_k . This proves delivered (30), including all $m > 1$.

87.0.5 3. Canonical quotient metric and the complete boundary identity

Fix k, m and abbreviate $q = q_k, \chi = \chi_k, c = k/2$. Use the ordered remainder basis $1, S, \dots, S^{q-1}$ of $E = \mathbb{C}[S]/(\chi)$, and let $A : E \rightarrow E$ be multiplication by S . Write $b_n = [p_n]_\chi$.

For $N \geq q - 1$, the first q monic remainder columns are a basis. In the source basis $p_0, \dots, p_N$, the source Gram is $O_N = \text{diag}(\omega_0, \dots, \omega_N)$, and the remainder matrix is $J_N = [b_0 \cdots b_N]$. Hence

$$K_N = J_N O_N^{-1} J_N^* = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n} > 0. \quad (\text{GR12})$$

The coefficient matrix of the canonical minimum section is

$$C_N = O_N^{-1} J_N^* K_N^{-1}. \quad (\text{GR13})$$

One has $J_N C_N = I_E$. If $z \in \ker J_N$, then $z^* O_N C_N v = z^* J_N^* K_N^{-1} v = 0$. Every other representative is $C_N v + z$, whose squared norm is $\|C_N v\|^2 + \|z\|^2$. Thus the actual minimum quotient Gram and volume are

$$G_N = K_N^{-1}, \quad V_N = \det G_N > 0. \quad (\text{GR14})$$

Put $\alpha_n = n(n + k/2 - 1)$. GR7 gives $\omega_n/\omega_{n-1} = \alpha_n$, and reduction gives $Ab_n = b_{n+1} + cb_n - \alpha_n b_{n-1}$. Substitution in $AK_N + K_N A^* - kK_N$ cancels each interior pair: the term $(b_{n+1}b_n^* + b_n b_{n+1}^*)/\omega_n$ is cancelled by the next negative term because $\alpha_{n+1}/\omega_{n+1} = 1/\omega_n$. No lower term occurs at zero. Therefore

$$AK_N + K_N A^* - kK_N = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{GR15})$$

Multiplication by G_N on both sides gives

$$W_N := A^* G_N + G_N A - kG_N = G_N \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N} G_N. \quad (\text{GR16})$$

Thus $H_N = G_N^{-1} W_N$ is self-adjoint in the original G_N inner product and has rank at most two. Its trace is $2\Re \text{Tr } A - kq = 0$: reflection $(a, b) \leftrightarrow (k - a, b)$ pairs real parts with sum k , retaining equal full multiplicities. Its possible nonzero eigenvalues are $+\epsilon_N, -\epsilon_N$, with $\epsilon_N \geq 0$.

To preserve the whole complex cross term, apply the exact isometry $G_N^{1/2} : (E, G_N) \rightarrow (\mathbb{C}^q, I)$. Put $u = G_N^{1/2} b_N, v = G_N^{1/2} b_{N+1}$, and $z_N = u^* v = b_N^* G_N b_{N+1}$. The conjugated matrix is $(vu^* + uv^*)/\omega_N$. Its zero trace says $\Re z_N = 0$. Direct multiplication gives trace of its square equal to

$$\frac{2\|u\|^2\|v\|^2 + 2\Re(z_N^2)}{\omega_N^2} = \frac{2(\|u\|^2\|v\|^2 - |z_N|^2)}{\omega_N^2}.$$

This trace is also $2\epsilon_N^2$, including the zero-rank case. Consequently

$$\epsilon_N^2 = \frac{(b_N^* G_N b_N)(b_{N+1}^* G_N b_{N+1}) - |z_N|^2}{\omega_N^2}. \quad (\text{GR17})$$

The isometry does not assign a different metric to either source. The phase term remains present before every ensuing inequality.

87.0.6 4. The reflected generalized trace and its exact count

Let $E_{>} \subset E$ be the sum of all primary spaces with $\Re \lambda > k/2$, retaining every power in GR11. It is A -invariant. Let $P_{>}$ be its G_N -orthogonal projection. In $E_{>} \oplus E_{>}^{\perp G_N}$, the lower-left block of A vanishes. Taking the first diagonal block of $A^{\dagger G_N} + A - kI$ proves

$$\text{Tr}(P_{>} H_N) = 2\Re \text{Tr}(A|_{E_{>}}) - k \dim E_{>}. \quad (\text{GR18})$$

If $\epsilon_N > 0$, choose unit eigenvectors u_+, u_- at its two nonzero eigenvalues. Its spectral expansion gives $\text{Tr}(P_{>} H_N) = \epsilon_N \|P_{>} u_+\|^2 - \epsilon_N \|P_{>} u_-\|^2 \leq \epsilon_N$. If $\epsilon_N = 0$, the same trace is zero. This uses the actual orthogonal projection and does not silently identify it with a generally nonorthogonal algebraic spectral projector.

The exact count is

$$\sum_{a > k/2} (2a - k) = \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{GR19})$$

For $k = 2r$ it is $2 + 4 + \cdots + 2r = r(r+1)$; for $k = 2r+1$ it is $1 + 3 + \cdots + (2r+1) = (r+1)^2$. There are $k+1$ choices of imaginary margin and multiplicity ℓ_k at each center. Thus

$$L_k := \text{Tr}(P_{>} \mathbf{H}_N) = \frac{1}{2} \ell_k (k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_N. \quad (\text{GR20})$$

The two parity formulas also prove $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$, so

$$L_k \geq kq_k/8 > 0. \quad (\text{GR21})$$

The trace retains the full generalized dimensions for every m .

87.0.7 5. Consecutive degrees and the separate first degree

For $N \geq q$, the update is $K_N = K_{N-1} + b_N b_N^* / \omega_N$. If $t = b_N^* G_{N-1} b_N / \omega_N \geq 0$, the determinant lemma gives $\delta_N := V_N / V_{N-1} = (1+t)^{-1} \in (0, 1]$. Multiplying the rank-one inverse formula verifies $b_N^* G_N b_N = \omega_N t / (1+t)$. The next determinant update uses this same G_N . Hence

$$\begin{aligned} b_N^* G_N b_N &= \omega_N (1 - \delta_N), \\ b_{N+1}^* G_N b_{N+1} &= \omega_{N+1} (\delta_{N+1}^{-1} - 1). \end{aligned} \quad (\text{GR22})$$

Substitution in GR17 yields

$$\epsilon_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1) - \frac{|z_N|^2}{\omega_N^2}. \quad (\text{GR23})$$

Because $0 \leq 1 - \delta_N \leq 1$ and $0 \leq \delta_{N+1}^{-1} - 1 \leq \delta_{N+1}^{-1}$, this gives

$$\epsilon_N^2 \leq \frac{\omega_{N+1}}{\omega_N} \frac{V_N}{V_{N+1}} = \frac{\Lambda_{N+1}}{\Lambda_N}, \quad \Lambda_N = \omega_N / V_N. \quad (\text{GR24})$$

At $N = q-1$, do not invert a preceding rank-deficient Gram. Expand the original monic relation

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \nu_0 = \|\chi\|_\Gamma^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j. \quad (\text{GR25})$$

The degree- $(q-1)$ remainder map is an isomorphism, so its canonical metric satisfies $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Reducing GR25 gives $b_q = -\sum_{j < q} \gamma_j b_j$. Therefore

$$\begin{aligned} b_{q-1}^* G_{q-1} b_{q-1} &= \omega_{q-1}, \\ b_q^* G_{q-1} b_q &= \nu_0 - \omega_q, \\ z_{q-1} &= -\gamma_{q-1} \omega_{q-1}, \\ V_q / V_{q-1} &= \omega_q / \nu_0. \end{aligned} \quad (\text{GR26})$$

The final equality is the forward determinant lemma at K_q . Substitution in GR17 proves the stronger exact endpoint identity

$$\begin{aligned} \epsilon_{q-1}^2 &= \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} \\ &= \frac{\nu_0 - \omega_q}{\omega_{q-1}} - |\gamma_{q-1}|^2 \\ &\leq \frac{\nu_0}{\omega_{q-1}} = \frac{\Lambda_q}{\Lambda_{q-1}}. \end{aligned} \quad (\text{GR27})$$

Thus delivered (41) is valid. No G_{q-2} , K_{q-2}^{-1} , or V_{q-2} is used. The endpoint phase term is present before the inequality.

87.0.8 6. Both windows and the literal constant

For either $i = q - 1$ or $i = q$, let $j = i + q$. For each $n = i, \dots, j - 1$, GR20 gives $L_k^2 \leq \epsilon_n^2$, and GR24 or its one initial instance GR27 gives $\epsilon_n^2 \leq \Lambda_{n+1}/\Lambda_n$. Multiplication of these q inequalities telescopes:

$$L_k^{2q} \leq \frac{\Lambda_j}{\Lambda_i} = \frac{\omega_j}{\omega_i} \frac{V_i}{V_j}. \quad (\text{GR28})$$

The indices in ω_j/ω_i are $n = i + 1, \dots, j$, all at most $2q$. Also $k \leq q$ by GR11. Every positive recurrence factor therefore satisfies

$$n(n + k/2 - 1) \leq 2q(2q + q/2 - 1) \leq 5q^2. \quad (\text{GR29})$$

Thus, cancelling the common mass only in the displayed ratio,

$$\frac{\omega_j}{\omega_i} = \prod_{n=i+1}^j n(n + k/2 - 1) \leq (5q^2)^q. \quad (\text{GR30})$$

Combining GR21, GR28, and GR30 yields, at each of the two windows,

$$\frac{V_i}{V_j} \geq \left(\frac{k}{8\sqrt{5}} \right)^{2q}, \quad \log(V_i/V_j) \geq 2q \log \left(\frac{k}{8\sqrt{5}} \right). \quad (\text{GR31})$$

Their final degrees are respectively $2q - 1$ and $2q$. Addition proves

$$\boxed{\mathcal{B}_k^\Gamma = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}} \geq 4q_k \log \left(\frac{k}{8\sqrt{5}} \right)}. \quad (\text{GR32})$$

When the right side is negative the inequality remains valid; the separate monotonicity $V_i/V_j \geq 1$ does not conflict with it. For $k > 1$, division by $q_k \log k$ gives $4 - 4 \log(8\sqrt{5})/\log k$, tending to four. This proves the claimed liminf for every fixed $m \geq 1$.

For centers $1/2 \pm \delta \pm i\gamma$ with fixed $\delta > 0, \gamma \neq 0$, the same occupation calculation is valid because both margins remain distinct. Its trace is $L = 2\delta\ell_k(k+1)\lfloor(k+1)^2/4\rfloor \geq \delta kq/2$. The identical two windows therefore give

$$\mathcal{B}_{h,k}^\Gamma \geq 4q_k \log \left(\frac{\delta k}{2\sqrt{5}} \right). \quad (\text{GR33})$$

This proves delivered (55) for its stipulated quartet geometry. It does not assert that an off-line zeta quartet exists.

The optional Gaussian comparison also checks exactly. Its mass and monic norms are 7^k and $7^k k^n n!$, so the ratio at index $n \leq 2q$ is $kn \leq 2kq$. The same $L \geq kq/8$ gives each window at least $q \log(kq/128)$. At $m = 1$, $q = (k+1)^2$, making this $3q \log k + q(2 \log(1 + 1/k) - \log 128)$. The combined liminf is at least six. The mass remains in each norm and cancels only from the ratio. This calibration does not change either the Gamma source or the original zeta source.

87.0.9 7. The actual arithmetic source and the exact determinant correction

Now fix an actual finite packet h of zeros of $g = 2\xi$ with its complete selected orders. Retain its actual cyclic polynomial $\chi = \chi_{h,k}$, quotient E , and full observation

$$\begin{aligned} \mathcal{V}_{h,k}P &= P(D_1 + \dots + D_k)F_h^{\otimes k}, & \mathcal{M}F_h &= g/h, \\ J^{(k)}\mathcal{V}_{h,k} &= \eta_{h,k}\pi_\chi, & \eta_{h,k}[P] &= v_h^{\otimes k}P(A_h^{(k)})1, & v_h &= j_h(g/h). \end{aligned} \quad (\text{GR34})$$

The actual amplitude and scalar norm observation are

$$\begin{aligned}
A_h(t) &= \frac{(g/h)(1/2 + it)}{\Gamma(1/4 + it/2)}, & B_h(t) &= |A_h(t)|^2, \\
w_h(t) &= B_h(t)r_{1/4}(t), & B_{h;k}(u) &= C_\Sigma(B_h^{\otimes k})(u), \\
m_{h,k}(u) &= r_{1/4,k}(u)B_{h;k}(u).
\end{aligned} \tag{GR35}$$

Here C_Σ is the original conditional integral on $u = t_1 + \cdots + t_k$, with denominator the actual fibre mass $r_{1/4,k}(u)$. Fubini proves the last equality by testing against an arbitrary nonnegative measurable function of that sum. The full complex amplitude multiplies the Gamma tensor amplitude before taking its squared norm; GR35 therefore does not replace that source vector or its phase by a real scalar.

At every degree $N \geq q - 1$, use the same monomial coefficient space $\mathcal{P}_N$, remainder map J_N , and relation injection

$$B_N : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad Q \longmapsto \chi Q. \tag{GR36}$$

At $N = q - 1$, the domain is zero. Let $M_N^\Gamma, M_N^{\text{ar}}$ be the original two source Grams. Each is positive: Gamma has positive density; the nonzero entire function g/h vanishes at a discrete subset of the line, so no nonzero polynomial has zero arithmetic norm. The convolution retains positivity. Every moment is finite in the original source; for a quartet the proved $B_{h;k} \leq C_h^k$ and Gamma moments also prove finiteness.

The coefficient identity $I_N : P \mapsto P$, from $(\mathcal{P}_N, M_N^\Gamma)$ to $(\mathcal{P}_N, M_N^{\text{ar}})$, intertwines both J_N and B_N . Its induced quotient map is the identity on E with the two stated metrics.

Let R_N^Γ be the Gamma minimum section. Every arithmetic representative of a class v is uniquely $R_N^\Gamma v + B_N z$. Orthogonality to the arithmetic relation subspace gives $(B_N^* M_N^{\text{ar}} B_N)z = -B_N^* M_N^{\text{ar}} R_N^\Gamma v$. The relation Gram is positive on its actual domain. Thus

$$R_N^{\text{ar}} = R_N^\Gamma - B_N(B_N^* M_N^{\text{ar}} B_N)^{-1} B_N^* M_N^{\text{ar}} R_N^\Gamma. \tag{GR37}$$

At zero relation dimension, the inverse is the unique automorphism of the zero space; the composite is the zero map $E \rightarrow \mathcal{P}_N$. Applying the actual source map $\mathcal{V}_{h,k}$ to the difference gives an original relation and its theta primitive, while GR34 keeps the full arithmetic jet unchanged. This proves delivered (53) with its exact source, codomain, and original unit.

There is a full determinant formula. Define

$$L_N : E \oplus \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad ([P], Q) \longmapsto \text{rem}_\chi(P) + \chi Q. \tag{GR38}$$

In the ordered monomial bases, the first columns are $1, S, \dots, S^{q-1}$; each later χS^j has coefficient one at degree $q + j$ and zero above. The matrix is triangular with determinant one. Its inverse is monic polynomial division into the remainder and quotient. For either source Gram M , $L_N^* M L_N$ has lower-right block $H = B_N^* M B_N$. The triangular change of coefficients determined by GR37 completes the square, with determinant one, and its other diagonal block is the minimum quotient Gram G . Hence

$$\det M = (\det G)(\det H), \quad V_N = \frac{\det M_N}{\det(B_N^* M_N B_N)}. \tag{GR39}$$

At $N = q - 1$, the relation determinant is the empty determinant one. Therefore the positive quotient-volume correction $T_{k,N} = V_{k,N}^{\text{ar}}/V_{k,N}^\Gamma$ is exactly

$$T_{k,N} = \frac{\det M_N^{\text{ar}} \det(B_N^* M_N^\Gamma B_N)}{\det M_N^\Gamma \det(B_N^* M_N^{\text{ar}} B_N)}. \tag{GR40}$$

Taking the four literal endpoint ratios gives

$$\mathcal{B}_{h,k}^{\text{ar}} - \mathcal{B}_{h,k}^\Gamma = \log T_{k,q-1} + \log T_{k,q} - \log T_{k,2q-1} - \log T_{k,2q}. \tag{GR41}$$

This proves delivered (54). The final two minus signs are essential. An individual upper bound on each positive $T_{k,N}$ provides no lower bound for those denominator factors. On the reference itself, $B_{h;k} = 1$ is realized, all $T_{k,N} = 1$, and the correction is zero while GR32 holds. Positivity and the assertion that the multiplier is bounded, without other restrictions, include this reference calibration.

For the stipulated actual off-line quartet displacement $\delta > 0$, an arithmetic upper estimate $(4 - \varepsilon)q \log k$, together with the proved GR33, would force exactly

$$\log \frac{T_{k,q-1}T_{k,q}}{T_{k,2q-1}T_{k,2q}} \leq -\varepsilon q \log k - 4q \log \left(\frac{\delta}{2\sqrt{5}} \right). \quad (\text{GR42})$$

The last term supplies the complete constant represented by the delivered $O_h(q)$. This is the necessary consequence of the displayed estimates, not an assertion that the arithmetic upper estimate has been proved.

87.0.10 8. What this establishes about the broader program

For any monic test polynomial h_0 with centers in the open critical strip, the original theta source proves the exact quotient square

$$\begin{array}{ccc} V/h_0(D)V & \xrightarrow{\Theta} & \mathcal{B}/h_0(D)\mathcal{B} \\ j_{h_0}H \downarrow \wr & & \wr \downarrow j_{h_0}\mathcal{M} \\ E_{h_0} & \xrightarrow{\times j_{h_0}g} & E_{h_0}. \end{array} \quad (\text{GR43})$$

These are the full Taylor-jet maps. The original Euler inverses prove their kernels to be $h_0(D)V$ and $h_0(D)\mathcal{B}$; the seed $H_{P(D)\phi_*} = P$ gives source surjectivity, and the inverse Gram of the independent functions $y^j e^{\rho y}/j!$ against an interval bump gives target surjectivity. The literal identity $\mathcal{M}\Theta\phi = gH_\phi$ proves commutativity.

The associated exact sequence is

$$0 \longrightarrow Q[h_0(D)] \longrightarrow E_{h_0} \xrightarrow{\times j_{h_0}g} E_{h_0} \longrightarrow Q/h_0(D)Q \longrightarrow 0. \quad (\text{GR44})$$

Its first arrow has an explicit inverse on the kernel. If $h_0(D)[F] = 0$, write $h_0(D)F = \Theta\phi$, uniquely by theta injectivity, and send the class to $j_{h_0}H_\phi$. A change by a theta relation changes ϕ by $h_0(D)V$, proving well-definedness. Zero image permits $\phi = h_0(D)\psi$; injectivity of $h_0(D)$ on $\mathcal{B}$ gives $F = \Theta\psi$. Conversely, a polynomial representative P killed by $j_{h_0}g$ gives the class of $S_{h_0}^{\mathcal{B}}\Theta(P(D)\phi_*)$, defined by the original Euler inverse and the range identity. Changing P by h_0T changes this by $\Theta(T(D)\phi_*)$, so its class is independent and maps back to P . Quotienting the square proves the final cokernel statement.

For an actual full packet h , $j_h g = 0$, g/h is entire, and $j_h(g/h)$ is a unit because the exact zero orders were used. These are additional properties of the original source. They have not been established for h_* , and none is assumed in GR8–GR32. Choosing an invertible observation on a calibration model preserves its feasible coefficient sets; it does not identify that observation with the actual Taylor unit.

On each fixed support fibre, all the linear arrows above lift by $(\ell, v) \mapsto (\ell, fv)$, with $\tau \mapsto \tau$. The preimage of the supported zero is exactly $\{(\ell, v) : v \in \ker f\}$. In particular the support lift records the actual kernel of multiplication by $j_{h_0}g$; it does not replace that map by zero.

The Gamma obstruction invalidates the universal sub-threshold estimate using only the generic Gamma source and displayed quotient identities. When the same auxiliary purity is available for these calibration polynomials, adjoining that fact cannot restore this universal implication. The original arithmetic restrictions, full theta primitive, and signed correction GR41 remain present through the proved maps GR34–GR44. No result here states that all F1 geometry or split-base constructions fail, that the desired estimate for the actual arithmetic source is false, or that an off-critical zero of 2ξ exists.

87.0.11 9. Audit findings and verification scope

No correction is required to the Gamma constants, monic recurrence, full multiplicity q_k , reflected rank-two trace, first admitted degree, two consecutive windows, the $8\sqrt{5}$ constant, or the signs of the arithmetic correction. GR27 and GR40 expand portions of the supplied argument into explicit identities.

The delivered verifier's complete canonical-volume calculations are at $m = 1, k = 1$. Its separate repeated-primary check verifies a nonzero nilpotent with square zero. The all- m , all- k statement rests on the written proof GR9–GR33. Its negative-control flag rejects the scalar equality $3/4 = 1/2$; that flag alone does not replay or certify the full auxiliary-purity argument. These scope limits do not affect the proof above.

87.0.12 Appendix: Complete independent Gamma derivation

The following complete proof is appended from the pinned generating-function subreview. It preserves its equation labels G1–G28 and original measures.

87.0.13 Independent Gamma generating-function calculation

Date: 2026-09-13. Scope: the Gamma source, its exact mass and convolution, and the polynomial generating function and recurrence in equations (28), (29), (31), and (35) of the delivered `Tau_Deligne_Bound_Audit/NO TE.tex`. The source SHA-256 is `bfe63094ec360b6059f2936e5e456c6c0fd08db144bbabc29621a527b89c13c6`. The source remains unchanged. The quotient, trace, and determinant estimates are reviewed separately.

Finding. The printed generating function has linear coefficient $+x$, and its exponential generating coefficients are literally monic polynomials in x . The printed substitution produces literally monic polynomials in S , and produces the negative last term of recurrence (35). The constants and convolution mass in (28)–(29) are also correct. No sign or mass correction is required for these formulas.

The proof below retains the original density, coordinate $S = k/2 + ix$, positive integer k , and all factors of 2, i , and π . Euler's Beta integral is recorded in [DLMF 5.12.1](#) and [DLMF 5.12.3](#). The Fourier, convolution, and polynomial calculations here are derived explicitly from that integral.

87.0.14 1. Fourier transform of the unchanged Gamma density

Let $a > 0$, and define

$$r_a(x) = \frac{|\Gamma(a + ix/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \quad x \in \mathbb{R}. \quad (\text{G1})$$

We use the convention

$$\mathcal{F}_+ f(t) = \int_{\mathbb{R}} f(x) e^{itx} dx. \quad (\text{G2})$$

The Beta identity with arguments $a + iu, a - iu$, followed by $v = e^y$, gives

$$\begin{aligned} \frac{\Gamma(a + iu)\Gamma(a - iu)}{\Gamma(2a)} &= \int_0^\infty \frac{v^{a+iu-1}}{(1+v)^{2a}} dv \\ &= \int_{\mathbb{R}} \frac{e^{ay}}{(1+e^y)^{2a}} e^{iuy} dy \\ &= \int_{\mathbb{R}} f_a(y) e^{iuy} dy, \quad f_a(y) = (2 \cosh(y/2))^{-2a}. \end{aligned} \quad (\text{G3})$$

The function f_a and all its derivatives decay exponentially as $y \rightarrow \pm\infty$. In particular, $f_a, f_a'' \in L^1(\mathbb{R})$; integration by parts twice shows that its Fourier transform is $O((1 + |u|)^{-2})$. Fourier inversion therefore applies as an absolutely convergent integral. Write the left side of (G3) as $F_a(u)$. Then

$$f_a(y) = \frac{1}{2\pi} \int_{\mathbb{R}} F_a(u) e^{-iuy} du. \quad (\text{G4})$$

Since $r_a(x) = \Gamma(2a)F_a(x/2)/(2\pi)$, the change of variables $x = 2u$ in the original density gives

$$\begin{aligned} \mathcal{F}_+ r_a(t) &= \frac{\Gamma(2a)}{\pi} \int_{\mathbb{R}} F_a(u) e^{2itu} du \\ &= 2\Gamma(2a)f_a(-2t) \\ &= 2\Gamma(2a)(2 \cosh t)^{-2a} = c_a(\cosh t)^{-2a}. \end{aligned} \quad (\text{G5})$$

Every integral used in (G5) is absolutely convergent. In particular, its value at $t = 0$ proves the original mass

$$\int_{\mathbb{R}} r_a(x) dx = c_a. \quad (\text{G6})$$

For later coefficient extraction we also prove the exponential moments, rather than assuming them. The expression $f_a(z) = (2 \cosh(z/2))^{-2a}$ has the analytic branch positive on the real axis throughout $|\operatorname{Im} z| < \pi$. For $|\sigma| < \pi$,

$$|\cosh((y + i\sigma)/2)|^2 = \sinh^2(y/2) + \cos^2(\sigma/2) \geq \cos^2(\sigma/2) \cosh^2(y/2). \quad (\text{G7})$$

Consequently $f_a(y + i\sigma)$ is integrable in y , uniformly when σ ranges in a closed smaller strip. The vertical sides of the contour rectangle in (G3) tend to zero because f_a decays exponentially in the real direction. Shifting the contour upward for $u \geq 0$ and downward for $u < 0$ therefore proves, for each $0 < \sigma < \pi$,

$$|F_a(u)| \leq e^{-\sigma|u|} \int_{\mathbb{R}} |f_a(y + i\sigma)| dy. \quad (\text{G8})$$

For $u < 0$ the integral along $-i\sigma$ has the same bound by conjugation. Inserting $u = x/2$ proves that $r_a(x)$ is bounded by a constant times $e^{-\sigma|x|/2}$. Thus every exponential moment of order strictly less than $\pi/2$ is finite. Both sides of (G5) extend holomorphically to $|\operatorname{Im} t| < \pi/2$; the right side uses the branch equal to the positive real value on the real axis. Equality on the real axis and the identity theorem imply

$$\int_{\mathbb{R}} e^{vx} r_a(x) dx = c_a (\cos v)^{-2a}, \quad v \in \mathbb{R}, \quad |v| < \pi/2. \quad (\text{G9})$$

This is the exact moment generating function of the density with mass c_a ; no mass has been divided out.

87.0.15 2. The k -fold convolution and its mass

Fix the original value $a = 1/4$. Since $\Gamma(1/2) = \sqrt{\pi}$, (G1) gives

$$c_{1/4} = 2^{1/2} \Gamma(1/2) = \sqrt{2\pi}. \quad (\text{G10})$$

For every positive integer k , the convolution theorem for the integrable functions in (G1) and formula (G5) give

$$\begin{aligned} \mathcal{F}_+(r_{1/4}^{*k})(t) &= c_{1/4}^k (\cosh t)^{-k/2} \\ &= \mathcal{F}_+ \left(\frac{c_{1/4}^k}{c_{k/4}} r_{k/4} \right) (t). \end{aligned} \quad (\text{G11})$$

Fourier uniqueness proves equality almost everywhere. The densities here are continuous: each r_a is continuous, bounded, and integrable, and convolution of an integrable function with a bounded uniformly continuous function is continuous. The required uniform continuity of r_a follows either from (G4) and the exponential estimate (G8), or directly by dominated differentiation there. Hence the equality holds at every real x :

$$r_{1/4,k}(x) = r_{1/4}^{*k}(x) = \frac{c_{1/4}^k}{c_{k/4}} \frac{|\Gamma(k/4 + ix/2)|^2}{2\pi}. \quad (\text{G12})$$

At $t = 0$, or by Tonelli's theorem in the original k variables,

$$\int_{\mathbb{R}} r_{1/4,k}(x) dx = c_{1/4}^k = (2\pi)^{k/2}. \quad (\text{G13})$$

These are exactly equations (28)–(29) of the delivered note, including the coefficient $c_{1/4}^k / c_{k/4}$.

87.0.16 3. The printed generating function has $+x$, not $-x$

The printed function is, for z near zero,

$$F_a(z, x) = (1 - iz)^{-a+ix/2} (1 + iz)^{-a-ix/2} = \sum_{n=0}^{\infty} b_n^{(a)}(x) \frac{z^n}{n!}. \quad (\text{G14})$$

Use the logarithms vanishing at $z = 0$. Their exact derivatives give

$$\begin{aligned}\partial_z \log F_a(z, x) &= \frac{(-a + ix/2)(-i)}{1 - iz} + \frac{(-a - ix/2)i}{1 + iz} \\ &= \frac{x - 2az}{1 + z^2}.\end{aligned}\tag{G15}$$

In particular $F_a(0, x) = 1$ and $\partial_z F_a(0, x) = x$. For real z near zero the same branches give the exact expression

$$F_a(z, x) = (1 + z^2)^{-a} \exp(x \arctan z).\tag{G16}$$

The sign in the exponential is positive because

$$\log(1 - iz) - \log(1 + iz) = -2i \arctan z, \quad \frac{ix}{2}(-2i \arctan z) = x \arctan z.\tag{G17}$$

Multiply the differential equation (G15) by F_a and compare the coefficient of $z^n/n!$. For $n \geq 1$, the coefficient contributed by $z^2 \partial_z F_a$ is $n(n-1)b_{n-1}^{(a)}$, and that contributed by $2azF_a$ is $2anb_{n-1}^{(a)}$. Therefore

$$b_0^{(a)} = 1, \quad b_1^{(a)} = x, \quad b_{n+1}^{(a)}(x) = xb_n^{(a)}(x) - n(n+2a-1)b_{n-1}^{(a)}(x).\tag{G18}$$

For $n = 0$, the same equation reads $b_1^{(a)} = xb_0^{(a)}$, with the last term omitted. Induction in (G18) proves that $b_n^{(a)} \in \mathbb{R}[x]$ has degree n and leading coefficient exactly 1. For example, the first values are

$$\begin{aligned}b_0^{(a)} &= 1, & b_1^{(a)} &= x, \\ b_2^{(a)} &= x^2 - 2a, & b_3^{(a)} &= x^3 - (6a+2)x.\end{aligned}\tag{G19}$$

Thus no replacement of z , x , or either exponent is needed to obtain monicity.

87.0.17 4. Exact orthogonality and norms

Take real z, w in a neighborhood of zero sufficiently small that $|\arctan z + \arctan w| < \pi/2$. Equations (G9) and (G16) give

$$\begin{aligned}\int_{\mathbb{R}} F_a(z, x) F_a(w, x) r_a(x) dx &= c_a (1 + z^2)^{-a} (1 + w^2)^{-a} (\cos(\arctan z + \arctan w))^{-2a} \\ &= c_a (1 - zw)^{-2a}.\end{aligned}\tag{G20}$$

The last equality uses the exact identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}.\tag{G21}$$

On a sufficiently small closed complex bidisc around $z = w = 0$, the absolute value of the integrand in (G20) is bounded by a constant times $r_a(x)e^{v_0|x|}$ for some $v_0 < \pi/2$. This follows directly from the analytic logarithms in (G14). That dominating function is integrable by (G8). Consequently the integral is holomorphic on the open bidisc. The real identity (G20) extends there by applying the one-variable identity theorem first in z , then in w . Differentiation under the integral at zero is justified by Cauchy's formula and the same domination.

The coefficient of $z^n w^m$ on the left side of (G20) is

$$\frac{1}{n!m!} \int_{\mathbb{R}} b_n^{(a)}(x) b_m^{(a)}(x) r_a(x) dx.$$

The binomial series on the right side is $c_a \sum_{j \geq 0} (2a)_j (zw)^j / j!$. Comparing coefficients yields

$$\int_{\mathbb{R}} b_n^{(a)}(x) b_m^{(a)}(x) r_a(x) dx = \mathbf{1}_{n=m} c_a n! (2a)_n.\tag{G22}$$

The polynomials have real coefficients, so (G22) is also their Hermitian orthogonality relation. In particular it is the norm of the original monic polynomial, with mass c_a retained.

87.0.18 5. Exact transport to $S = k/2 + ix$

Set $a = k/4$ and $c = k/2$. Define the polynomial in the original complex variable S by

$$p_n(S) = i^n b_n^{(k/4)}((S - c)/i). \quad (\text{G23})$$

The coefficient of S^n is $i^n i^{-n} = 1$; hence this is literally monic in S . The lower coefficients are obtained by the displayed affine substitution, with no change to the source norm or its mass. On the source line, $S = c + ix$, equation (G23) gives $p_n(c + ix) = i^n b_n^{(k/4)}(x)$. Therefore (G12) and (G22) imply

$$\begin{aligned} \int_{\mathbb{R}} \overline{p_n(c + ix)} p_m(c + ix) r_{1/4, k}(x) dx &= \frac{c_{1/4}^k}{c_{k/4}} (-i)^n i^m \int_{\mathbb{R}} b_n^{(k/4)}(x) b_m^{(k/4)}(x) r_{k/4}(x) dx \\ &= \mathbf{1}_{n=m} (2\pi)^{k/2} n! (k/2)_n. \end{aligned} \quad (\text{G24})$$

In the surviving term $m = n$, the phase is $(-i)^n i^n = 1$. This proves the exact norm $\omega_{k,n}^\Gamma$ printed in (35).

To transport the recurrence, multiply (G18), in its form $xb_n = b_{n+1} + n(n + 2a - 1)b_{n-1}$, by i^{n+1} . Since $S - c = ix$, the result is

$$\begin{aligned} (S - c)p_n(S) &= p_{n+1}(S) + n(n + k/2 - 1)i^{n+1}b_{n-1}^{(k/4)}((S - c)/i) \\ &= p_{n+1}(S) - n(n + k/2 - 1)p_{n-1}(S). \end{aligned} \quad (\text{G25})$$

The minus sign is exactly $i^{n+1}/i^{n-1} = i^2 = -1$. Restoring $c = k/2$ proves

$$Sp_n = p_{n+1} + \frac{k}{2}p_n - n(n + k/2 - 1)p_{n-1}, \quad \omega_{k,n}^\Gamma = (2\pi)^{k/2} n! (k/2)_n. \quad (\text{G26})$$

The last term is omitted at $n = 0$. The first polynomials in the original coordinate are

$$p_0 = 1, \quad p_1 = S - k/2, \quad p_2 = (S - k/2)^2 + k/2, \quad p_3 = (S - k/2)^3 + (3k/2 + 2)(S - k/2). \quad (\text{G27})$$

Finally, consecutive norm ratios retain the same constants before their displayed cancellation:

$$\frac{\omega_{k,n}^\Gamma}{\omega_{k,n-1}^\Gamma} = \frac{(2\pi)^{k/2} n! (k/2)_n}{(2\pi)^{k/2} (n-1)! (k/2)_{n-1}} = n(n + k/2 - 1), \quad n \geq 1. \quad (\text{G28})$$

This supplies precisely the coefficient needed for the interior cancellation following equation (36). The derivation establishes the Gamma formulas and their original polynomial coordinates; it does not by itself assert the later determinant bound or a comparison with the arithmetic theta source.

88 Complete Deligne comparison and its actual tau-base cohomology

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 8f5676711115ecbe926e825a98f498af74d7685d3777608c6b1b0429af3e74f1

Complete source. `sources/next_edition/actual_tau_conversions_v1/sources/purity_review/deligne_bound_purity_independent_review_20260913.md`

Source SHA-256. 8f5676711115ecbe926e825a98f498af74d7685d3777608c6b1b0429af3e74f1

88.0.1 Independent audit of the Deligne bound comparison

Date: 13 September 2026. This is a complete written review of the operator, coefficient, period, purity, and critical-kernel parts of the delivered Tau_Deligne_Bound_Audit/NOTE.tex. The complete note was read, including its Gamma argument; independent verification of the quantitative Gamma estimate belongs to the separate Gamma review. This document supplies full proofs for its new comparison statements. It preserves the delivered note unchanged.

88.0.2 Finding and precise scope

The delivered note proves a failure of a particular implication from auxiliary exponential Frobenius purity to the original arithmetic operator estimate. Its rank-one and reflection-stable counterexamples survive the review below. They do not prove failure of the Split-Zero or broader F1 geometry program. Their test polynomials are not known to be factors of the actual function $g = 2\xi$, and the exact arithmetic comparison records that difference by multiplication by $j_h g$.

Two narrow corrections are required in the exposition. The rank and trace statements attached to the Artin-Schreier equation must specify the nontrivial additive-character sheaf, or equivalently its character summand in the cover. The whole affine cover also has a degree-two compact-support contribution. Furthermore, vanishing of $j_{h_0} g$ and entire divisibility by h_0 do not require that every selected zero order be its full actual order; the extra full-order condition makes $j_{h_0}(g/h_0)$ a unit. Neither correction invalidates the exhibited operator counterexamples.

The strongest comparisons proved here include a complete classification of the possible pure-target intertwiners, with nilpotent lengths and exponential collisions retained; a universal observable quotient for each specified target Frobenius; and a tensor construction from the original arithmetic packet whose finite-field realization has the correct tensor weight k . The latter has an explicit injective complex period map retaining the original Taylor unit and sum action. Its existence prevents a failure of the simpler rank- q_k , weight-one auxiliary inference from being mistaken for failure of every geometric comparison.

88.0.3 1. The actual coefficient complex and its recovered operator

Retain the original monic annihilator

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q > 0, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S,$$

and the specified potential

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS. \quad (\text{DP.1})$$

Let $B = \mathbb{C}[u, t]$. The two-term complex in degrees zero and one is

$$\mathcal{C} = [B[S] \xrightarrow{D} B[S]dS], \quad D(P) = (u\partial_S P + (\chi - t)P)dS. \quad (\text{DP.2})$$

For a nonzero polynomial P of degree n and leading coefficient b , the term of degree $n+q$ in $D(P)/dS$ is exactly bS^{n+q} . Every other term has smaller degree. Thus D is injective. For a polynomial Q of degree at least q , subtract the image under D/dS of its leading coefficient times $S^{\deg Q - q}$. The degree strictly decreases. The procedure terminates and writes $Q = D(P)/dS + R$ with $\deg R < q$. Uniqueness follows because a nonzero image has degree at least q . Consequently

$$B[S] = \text{im}(D/dS) \oplus \bigoplus_{b=0}^{q-1} BS^b, \quad H^0(\mathcal{C}) = 0, \quad \mathcal{H} = H^1(\mathcal{C}) \cong B^q. \quad (\text{DP.3})$$

This proves freeness over the displayed coefficient ring and its compatibility with every coefficient specialization. The special fibre at $(u, t) = (0, 0)$ is exactly E . No repeated root is removed.

On $u \neq 0$, put $\nabla_t = \partial_t - S/u$ on polynomial coefficients in both terms. Directly,

$$[\partial_t, D/dS] = -1, \quad [-S/u, u\partial_S] = 1, \quad [-S/u, \chi - t] = 0.$$

Their sum is zero, so ∇_t descends to $\mathcal{H}[u^{-1}]$. In the fixed increasing-power frame, reduction of $S \cdot S^{q-1} = S^q$ uses only $D(1) = (\chi - t)dS$; therefore the matrix is

$$\nabla_t = \partial_t - A(t)/u, \quad A(t) = A + t\mathcal{R}, \quad \mathcal{R} = 1_E \otimes \ell, \quad \ell[P] = [S^{q-1}] \text{rem}_\chi P. \quad (\text{DP.4})$$

The operator $-u\nabla_t = -u\partial_t + A(t)$ preserves $(u, t)\mathcal{H}$: its derivative on tv contributes $-uv$, which belongs to that ideal. Hence it induces precisely A on the special fibre. This is a regular operator specialization even though ∇_t itself has a pole. Multiplication by S on arbitrary representatives does not separately define an algebra structure on $\mathcal{H}$; the displayed connection and fixed-frame endomorphism give its exact replacement.

88.0.4 2. Rank-one periods, orientation, and the finite-field trace

For $\chi(S) = S - \lambda$, with $\lambda = 3/4$, the coefficient ring $\mathbb{Z}[1/2]$ contains every coefficient. Let $u > 0$, put $b = \lambda + t$, and first take b real. On the upward contour $S = ix$, $-\infty < x < \infty$,

$$\Pi_{\text{up}}(u, t) = i \int_{\mathbb{R}} e^{-x^2/(2u)} e^{-ibx/u} dx. \quad (\text{DP.5})$$

Write the real integral as $I(b)$. Differentiation is justified by a Gaussian bound. Integration by parts, whose endpoint terms vanish, gives

$$0 = \int_{\mathbb{R}} \partial_x (e^{-x^2/(2u)} e^{-ibx/u}) dx = -u^{-1} \int_{\mathbb{R}} x e^{-x^2/(2u)} e^{-ibx/u} dx - ibu^{-1} I(b).$$

Thus $I'(b) = -bI(b)/u$. The Gaussian value $I(0) = \sqrt{2\pi u}$, obtained by squaring the integral and using polar coordinates, yields

$$\Pi_{\text{up}} = i\sqrt{2\pi u} e^{-b^2/(2u)}, \quad -u \frac{\partial_t \Pi_{\text{up}}}{\Pi_{\text{up}}} = b, \quad |\Pi_{\text{up}}|^2 = 2\pi u e^{-b^2/u}. \quad (\text{DP.6})$$

Both sides of the unsquared period identity are entire in b : compact sets of complex b still admit an integrable Gaussian majorant. Hence the unsquared identity extends to complex t ; the last real norm formula is asserted only for real b .

For the inherited ray order, $d = 2$, $\theta_0 = \pi/2$, $\theta_1 = 3\pi/2$, and $\Gamma_1 = -\ell_0 + \ell_1$. Its path travels from $+i\infty$ to zero and then to $-i\infty$. Thus $\Gamma_{\text{up}} = -\Gamma_1$ as oriented chains and $\Pi_{\text{up}} = -\Pi_1$. The unsquared sign in the delivered note is correct.

Now fix a finite field $\mathbb{F}_Q$ of odd characteristic, a nontrivial complex additive character ψ , $u \in \mathbb{F}_Q^\times$, and $b \in \mathbb{F}_Q$. Translation $x \mapsto x + b$ is a bijection and gives

$$\sum_x \psi((x^2/2 - bx)/u) = \psi(-b^2/(2u)) G_Q(u), \quad G_Q(u) = \sum_x \psi(x^2/(2u)). \quad (\text{DP.7})$$

The character takes values in roots of unity, so its conjugate at y is its value at $-y$. The invertible coordinate change

$$(x, y) \mapsto (r, s) = (x - y, x + y), \quad (r, s) \mapsto ((s + r)/2, (s - r)/2)$$

therefore gives

$$|G_Q(u)|^2 = \sum_{r,s} \psi(rs/(2u)) = Q. \quad (\text{DP.8})$$

Indeed the inner sum is Q at $r = 0$. For $r \neq 0$, choose a with $\psi(ra/(2u)) \neq 1$; shifting s by a multiplies that inner sum by a number unequal to one, so the inner sum vanishes.

The cohomological object for this sum is

$$H_c^1(\mathbb{A}_{\mathbb{F}_Q}^1, \mathcal{L}_\psi((S^2/2 - bS)/u)). \quad (\text{DP.9})$$

With the geometric-Frobenius convention in the source, its trace is the negative of DP.7. The degree-two character-polynomial theorem makes DP.9 one-dimensional and all other compact cohomology of this **character sheaf** zero. Thus the sole Frobenius eigenvalue has modulus $\sqrt{Q}$, including its trace sign.

For precision about the cover, fix $\Psi : \mathbb{F}_p \rightarrow \overline{\mathbb{Q}}_\ell^\times$ and write a nontrivial ψ as $\Psi \circ \text{Tr}_{\mathbb{F}_Q/\mathbb{F}_p}(a \cdot)$, $a \neq 0$. The cover used for this summand is $Y^p - Y = af(S)$; alternatively fix $a = 1$ and the corresponding character. The finite étale pushforward from that cover decomposes by the character idempotents of its translation group $\mathbb{F}_p$. Its trivial summand is the constant sheaf on $\mathbb{A}^1$, whose H_c^2 is $\overline{\mathbb{Q}}_\ell(-1)$. Every nontrivial character summand has the type DP.9. Consequently the whole cover has an additional degree-two contribution, whereas DP.9 has the rank and alternating-trace sign used in the note. This is the required clarification of its equations (14) and (22)–(23).

Finally, on the recovered complex one-dimensional source $A = 3/4$, any positive scalar Gram G gives

$$A^*G + GA - G = \frac{1}{2}G. \quad (\text{DP.10})$$

This identity and DP.8 coexist in the same coefficient family. Hence the indicated coefficient specialization and period identity do not imply a vanishing arithmetic weight-one defect.

88.0.5 3. Primary-source hypotheses and the reflection-stable example

The local original Deligne scan was inspected visually on complete printed pages 215 and 216, including the theorem, its character-sheaf definition, and its family argument. The complete relevant S20 French text and the dated primary correction were also read. The primary bibliographic page was checked online: [Deligne, La conjecture de Weil II](#).

The theorem used is Weil II (3.7.2.3): for a degree- d polynomial in n variables, $p \nmid d$, and smooth leading projective hypersurface, the associated nontrivial additive-character sheaf has compact cohomology concentrated in degree n , of dimension $(d-1)^n$, pure of weight n . The parameter sheaves are lisse by (3.7.3). Its propagation argument uses mixedness as well as lissity in (1.8.12). In (3.7.4) the final reference is Roman **I.8.11**. This audit does not claim a fresh proof of Deligne's theorem or a visual rereading of all Weil II pages.

For DP.1, $n = 1$, $d = q + 1$, and $u \neq 0$, the leading coefficient is exactly $1/(du)$. In $\mathbb{P}^0$, its zero scheme is empty because that coefficient is invertible. The empty scheme is smooth. Taking $p > d$ is stronger than the required $p \nmid d$ and retains all denominators. The lower critical points may collide; they do not alter this leading homogeneous hypothesis. Thus the stated theorem applies after each specified good coefficient reduction.

The proposed quartet polynomial is correct. With $x = S - 1/2$, its four factors have roots $x = \pm 1/4 \pm i$. Pairing conjugates at each real part gives

$$\begin{aligned} h_*(S) &= ((x - 1/4)^2 + 1)((x + 1/4)^2 + 1) \\ &= (x^2 + 17/16)^2 - x^2/4 \\ &= x^4 + (15/8)x^2 + 289/256. \end{aligned} \quad (\text{DP.11})$$

It is monic over $\mathbb{Z}[1/2]$, real, and invariant under $S \mapsto 1 - S$. Its roots have real parts $1/4, 3/4$, including both signs of their nonzero imaginary parts. Its degree-five potential has leading coefficient $1/5$. For all specified good reductions with $p > 5$, the previous theorem applies to its character sheaf and yields weight one. Thus reflection symmetry does not restore the failed operator implication. The statement does not assert that h_* divides 2ξ .

The exact translation proof in the retained source was also read in full. For $\chi_a(S) = \chi(S - a)$, $\Phi_a(S) = \Phi(S - a) - \Phi(-a)$, and $T_a P(X) = P(X + a)$, direct substitution gives the chain isomorphism and

$$T_a A_a T_a^{-1} = A + aI, \quad \Pi_a = f_a \Pi C_a, \quad f_a = e^{(-\Phi(-a) - ta)/u}. \quad (\text{DP.12})$$

The fixed-frame matrix $(C_a)_{rb} = \binom{b}{r} a^{b-r}$ for $r \leq b$ has inverse C_{-a} . The scalar is nonvanishing on $u \neq 0$ and is retained. For a holomorphic determinant-line metric multiplied by $|f_a|^{2p}$, the curvature change is $2p \partial \bar{\partial} \text{Re} \log f_a = 0$ locally; logarithm branches differ by constants and give the same derivative. The corresponding additive trace defect changes by $2p \text{Re} a$. The reflection-preserving DP.11 shows that the broader counterexample does not rely on changing the original reflection centre by DP.12.

88.0.6 4. Classification of all intertwiners to a specified pure target

Fix $Q > 1$, $a = \log Q > 0$, and a finite-dimensional complex vector space V with an invertible operator F . Let the source be the **original** cyclic algebra $E = \mathbb{C}[S]/(\chi)$, $A = M_S$, and $U = e^{aA}$. We classify

$$\mathcal{T} = \{T \in \text{Hom}_{\mathbb{C}}(E, V) : TU = FT\}. \quad (\text{DP.13})$$

Write $\chi = \prod_{\lambda} (S - \lambda)^{r_{\lambda}}$. The Chinese-remainder map, with the full powers, gives

$$E = \bigoplus_{\lambda} E_{\lambda}, \quad E_{\lambda} = \mathbb{C}[\epsilon_{\lambda}]/(\epsilon_{\lambda}^{r_{\lambda}}), \quad A|_{E_{\lambda}} = \lambda I + N_{\lambda}, \quad N_{\lambda} = M_{\epsilon_{\lambda}}. \quad (\text{DP.14})$$

Its inverse exists by Bezout: for each factor, the product of the other factors is a unit modulo that factor; multiply its inverse there and sum the resulting mutually orthogonal idempotent lifts. This retains every nilpotent power.

Let $V_{\mu} = \ker(F - \mu I)^{s_{\mu}}$ be a full primary summand of F , with s_{μ} any exponent at least its longest Jordan length. When $\mu = e^{a\lambda}$, define the nilpotent polynomial

$$L_{\mu} = \frac{1}{a} \sum_{j=1}^{s_{\mu}-1} \frac{(-1)^{j+1}}{j} (\mu^{-1}F - I)^j \quad \text{on } V_{\mu}. \quad (\text{DP.15})$$

The expression is independent of enlarging s_{μ} , because additional powers vanish. The formal identities $\log(\exp z) = z$ and $\exp(\log(1+z)) = 1+z$, valid in the formal power-series ring, become finite polynomial identities after substituting a nilpotent. Therefore

$$e^{aL_{\mu}} = \mu^{-1}F, \quad \frac{1}{a} \log(e^{aN_{\lambda}}) = N_{\lambda}. \quad (\text{DP.16})$$

The source satisfies $(U - e^{a\lambda})^{r_{\lambda}} = 0$ on E_{λ} . Intertwining carries this equation to $(F - e^{a\lambda})^{r_{\lambda}}T = 0$. Projection to any different target primary summand gives zero because $F - e^{a\lambda}$ is invertible there. Hence $T(E_{\lambda}) \subseteq V_{e^{a\lambda}}$, interpreted as zero if that primary summand is absent. Applying the logarithm polynomial to the intertwining equation proves

$$TN_{\lambda} = L_{e^{a\lambda}}T. \quad (\text{DP.17})$$

Conversely DP.17 implies the exponential intertwining by DP.16. Since 1_{λ} generates the source block under N_{λ} , every such map is determined by

$$v_{\lambda} = T(1_{\lambda}) \in \ker L_{e^{a\lambda}}^{r_{\lambda}}, \quad T\left(\sum_{j=0}^{r_{\lambda}-1} c_j \epsilon_{\lambda}^j\right) = \sum_{j=0}^{r_{\lambda}-1} c_j L_{e^{a\lambda}}^j v_{\lambda}. \quad (\text{DP.18})$$

These are inverse constructions. Consequently

$$\mathcal{T} \cong \bigoplus_{\lambda} \ker L_{e^{a\lambda}}^{r_{\lambda}}. \quad (\text{DP.19})$$

No injectivity of $\lambda \mapsto e^{a\lambda}$ is assumed. If two roots differ by $2\pi i n/a$, their source blocks have the same exponential eigenvalue and map independently into the same target primary space. Their images can overlap. Formula DP.18 retains those overlaps and therefore does not incorrectly identify the total kernel with the sum of the individual block kernels for an arbitrary fixed T .

If F is pure of weight k in its specified complex realization, every μ has modulus $Q^{k/2}$. For $\text{Re } \lambda \neq k/2$, $|e^{a\lambda}| = Q^{\text{Re } \lambda}$ differs, so that target primary space is absent. It follows that

$$E_{\text{off}} := \bigoplus_{\text{Re } \lambda \neq k/2} E_{\lambda} \subseteq \ker T \quad (T \in \mathcal{T}). \quad (\text{DP.20})$$

This proves the delivered note's off-critical annihilation including every generalized vector. Purity alone does not assert semisimplicity of F ; DP.15–DP.19 therefore retain all target nilpotents as well.

There is also an exact quotient visible to **all** such maps. Let d_μ be the largest Jordan length of F on V_μ , and set $d_\mu = 0$ when that summand is absent. The nilpotent L_μ has the same Jordan lengths as $F - \mu I$, because it is the latter times an invertible polynomial with nonzero constant coefficient $1/(a\mu)$. Put $b_\lambda = \min(r_\lambda, d_{e^{a\lambda}})$. Then

$$K_F := \bigcap_{T \in \mathcal{T}} \ker T = \bigoplus_{\lambda} \begin{pmatrix} \epsilon_\lambda^{b_\lambda} \\ \epsilon_\lambda^{r_\lambda} \end{pmatrix}. \quad (\text{DP.21})$$

To prove this, DP.18 makes every map kill $\epsilon_\lambda^{b_\lambda}$. Conversely choose a target Jordan chain of length $d = d_{e^{a\lambda}} > 0$. If $d \geq r_\lambda$, choose v of nilpotent length exactly r_λ within that chain; if $d < r_\lambda$, take a top vector of length d . The vectors $v, Lv, \dots, L^{b_\lambda-1}v$ are independent, so the map defined by this v has kernel exactly the displayed ideal in that source block. Set the other source-block maps to zero. This also proves DP.21 in the presence of exponential collisions. If $d = 0$, the displayed ideal is the entire source block.

The quotient has a canonical faithful realization in a pure target without choosing individual maps:

$$\text{ev} : E \longrightarrow \text{Hom}_{\mathbb{C}}(\mathcal{T}, V), \quad \text{ev}(x)(T) = T(x), \quad \ker \text{ev} = K_F. \quad (\text{DP.22})$$

The operator on its target is $F_{\text{out}}\phi = F \circ \phi$. For every T , $\text{ev}(Ux)(T) = T U x = F T x$, hence $\text{ev} U = F_{\text{out}} \text{ev}$. The target is a direct sum of $\dim \mathcal{T}$ copies of (V, F) , so it is pure of the same weight. Thus DP.22 induces an injection of E/K_F onto its actual image. This construction is the maximal quotient detected by all maps of the required type to the specified F . It supplies a complete typed relation rather than replacing a missing faithful map by an assertion of unrelatedness.

In particular, even a critical block can be invisible when its exponential eigenvalue is absent from F . A semisimple target has $d_\mu = 1$, so all higher critical nilpotents are killed by DP.21. A nonsemisimple pure target can preserve them up to its actual Jordan length. No reduction to simple zeros is justified by purity alone.

88.0.7 5. A weight-matched tensor construction from the original arithmetic packet

Let h be the original finite packet polynomial, with its full actual zero orders, and retain

$$E_h = \mathbb{C}[s]/(h), \quad A_h = M_s, \quad v_h = j_h(g/h), \quad g = 2\xi. \quad (\text{DP.23})$$

The original packet uses the product of its monic linear powers, so h is monic in this construction. If an earlier presentation carried a nonzero scalar leading factor, the ideal map is multiplication by that recorded scalar and the factor in g/h must be retained; no such factor is set to one here. Complete zero orders make v_h a unit in the actual finite algebra. Its inverse is $j_h(h/g)$, with the quotient analytically filled at every selected zero.

For the fixed tensor degree k , let

$$\mathcal{E}_k = E_h^{\otimes k}, \quad \mathcal{A}_k = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A_h \otimes I^{\otimes(k-j)}, \quad v_k = 1_h^{\otimes k}. \quad (\text{DP.24})$$

Let $\chi_{h,k}$ be the original minimal annihilator of the cyclic vector v_k under $\mathcal{A}_k$. This is the sum polynomial already used in the source, with all its multiplicities. Polynomial evaluation gives the injective map

$$\iota_k : E_{h,k} = \mathbb{C}[S]/(\chi_{h,k}) \longrightarrow \mathcal{E}_k, \quad [P] \longmapsto P(\mathcal{A}_k)v_k. \quad (\text{DP.25})$$

Indeed its kernel in $\mathbb{C}[S]$ is the ideal generated by that minimal annihilator, proving injectivity on the displayed quotient. Multiplication by S gives $\mathcal{A}_k \iota_k = \iota_k A_k$, where $A_k = M_S$. Let $U_k = M_{v_h}^{\otimes k}$. Each tensor factor of this unit is a polynomial in the corresponding A_h ; hence U_k commutes with $\mathcal{A}_k$. It is invertible with inverse $M_{j_h(h/g)}^{\otimes k}$. Therefore

$$\eta_{h,k} = U_k \iota_k, \quad \mathcal{A}_k \eta_{h,k} = \eta_{h,k} A_k, \quad \eta_{h,k} \text{ is injective.} \quad (\text{DP.26})$$

This is precisely the original arithmetic inclusion. It retains the full Taylor unit rather than replacing it by a basis convention. At the source level the already defined original map satisfies

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_{\chi_{h,k}}, \quad \mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad \mathcal{M}F_h = g/h. \quad (\text{DP.27})$$

In particular the arithmetic source being embedded is the original one.

Apply DP.1–DP.4 to the original degree- d_h polynomial h , with independent parameters $t_1, \dots, t_k$ and the same nonzero parameter u . Tensor the resulting coefficient complexes over $\mathbb{C}[u, t_1, \dots, t_k]$. The total differential has the usual signs: on a homogeneous tensor it differentiates the j -th factor with sign $(-1)^{e_1 + \cdots + e_{j-1}}$, where $e_i \in \{0, 1\}$ are the preceding cochain degrees. The direct decomposition in DP.3 contracts each factor onto its free degree-one remainder module: its other summand is an isomorphism from its degree-zero module onto its own differential image. Tensoring these decompositions leaves exactly the tensor of the degree-one remainders in degree k ; every summand containing one of the isomorphism complexes is contracted using its inverse, with the same total-complex signs. Thus the tensor cohomology is free of rank d_h^k , concentrated in degree k , and its special fibre is $\mathcal{E}_k$.

The parameter connections commute on distinct factors. Their recovered sum is

$$\mathfrak{A} = -u \sum_{j=1}^k \nabla_{t_j} = -u \sum_{j=1}^k \partial_{t_j} + \sum_{j=1}^k A_h(t_j)^{[j]}, \quad \mathfrak{A} \bmod (u, t_1, \dots, t_k) = \mathcal{A}_k. \quad (\text{DP.28})$$

Here the bracket means insertion in that tensor factor. This is the sum, with no division by k .

Let $\Pi_h(u, t_j)$ be the full period matrix in the original ray ordering, including every unsquared phase. Fix $u \neq 0$ in a sector with the stated choice of $\arg u$; the labelled rays have angles $\theta_r = (\arg u + \pi)/(d_h + 1) + 2\pi r/(d_h + 1)$, and $\Gamma_r = -\ell_0 + \ell_r$, $1 \leq r \leq d_h$. Each t_j is an arbitrary complex parameter. Its invertibility follows directly as follows. At the monomial potential the (r, b) period is a nonzero Gamma/ray factor times $\zeta^{r(b+1)} - 1$, with $\zeta^{d_h+1} = 1$. A column dependence would give a polynomial of degree at most d_h vanishing at all $d_h + 1$ roots of unity, hence every coefficient is zero. Along a straight coefficient path the periods satisfy $\Pi' = \Pi B$, because integration of the image of DP.2 is an exact derivative and the endpoints vanish. Uniform domination by a negative leading exponential times a polynomial justifies each derivative. The adjugate determinant identity yields $(\det \Pi)' = (\text{Tr } B) \det \Pi$, so its initially nonzero determinant never vanishes. This argument includes repeated roots.

Consequently the map

$$\mathcal{P}_k(u, \mathbf{t}) = \Pi_h(u, t_1) \otimes \cdots \otimes \Pi_h(u, t_k) : \mathcal{E}_k \longrightarrow \mathbb{C}^{d_h^k} \quad (\text{DP.29})$$

is invertible, with inverse the tensor of the exact inverses. The original arithmetic cyclic source therefore has the explicit injective rectangular period map

$$P_{h,k}(u, \mathbf{t}) = \mathcal{P}_k(u, \mathbf{t})\eta_{h,k} : E_{h,k} \hookrightarrow \mathbb{C}^{d_h^k}. \quad (\text{DP.30})$$

Holding $\eta_{h,k}$ fixed in its original marked coordinates, the individual period equations and DP.26 give the exact first-order identity at $\mathbf{t} = 0$:

$$-u \sum_{j=1}^k \partial_{t_j} P_{h,k}(u, 0) = \mathcal{P}_k(u, 0)\mathcal{A}_k\eta_{h,k} = P_{h,k}(u, 0)\mathcal{A}_k. \quad (\text{DP.31})$$

The nilpotent blocks are retained because DP.30 is injective and DP.31 intertwines the displayed recovered action. No invariance of its image under all deformed $A_h(t_j)$, and no missing higher-order period identity, is assumed.

For any original finite-cutoff positive Gram $G_{k,N}$, put $P = P_{h,k}(u, 0)$ and $L = (P^*P)^{-1}P^*$. Then $LP = I$, and the concrete positive form on $\text{im } P$ is the restriction of $L^*G_{k,N}L$. Its pullback is

$$P^*L^*G_{k,N}LP = G_{k,N}. \quad (\text{DP.32})$$

The action on that image is PA_kL , and its control form pulls back to $A_k^*G_{k,N} + G_{k,N}A_k - kG_{k,N}$. Thus every original mass, source norm, and generalized defect is preserved by an actual metric-and-action map. The ambient form may be degenerate outside the image; no positive extension or orthogonal-complement invariance is asserted.

There is a finite coefficient model retaining this injection. Form the finitely generated subring of $\mathbb{C}$ containing the coefficients of h , $\chi_{h,k}$, v_h , its inverse, and the matrices in DP.25–DP.26, together with all denominators

$1/(d_h + 1)!$. Since $\eta_{h,k}$ has full column rank, choose one of its nonzero square maximal minors Δ and adjoin Δ^{-1} . Call the resulting ring $R_{h,k}$. The selected rows give a left inverse by their adjugate divided by Δ . Thus $\eta_{h,k}$ is a split injection over this very ring, and its action identity holds there because every entry is an element of its subring in $\mathbb{C}$ and the equality holds in $\mathbb{C}$.

For each specified reduction $R_{h,k} \rightarrow \mathbb{F}_Q$ with $p > d_h + 1$, the same matrices remain a split injection and intertwiner of the reduced coefficient algebras. Its reduced cyclic image is therefore isomorphic, through those matrices, to the literal quotient $\mathbb{F}_Q[S]/(\overline{\chi}_{h,k})$. Every nilpotent power of that reduced image is the corresponding power in this quotient, including any root collisions after reduction. The nonzero minor used for this particular finite construction has been inverted. This does not claim that every prime is allowed or that one localization works for unbounded k .

On the finite-field branch, form the **external product** of the k original one-variable nontrivial additive-character sheaves. The Künneth map for compact support identifies its degree- k cohomology, with product-coordinate orientation, with

$$\bigotimes_{j=1}^k H_c^1(\mathbb{A}_{\mathbb{F}_Q}^1, \mathcal{L}_\psi(\Phi_{h,t_j}/u)). \quad (\text{DP.33})$$

Each factor is concentrated in degree one and pure of weight one by the checked theorem. Thus DP.33 is concentrated in degree k , has rank d_h^k , and is pure of weight k : its Frobenius is the tensor product, whose eigenvalues are all products of the factor eigenvalues. To see the last assertion including nonsemisimple factors, triangularize each complex coefficient realization; the tensor basis makes the tensor product triangular, with precisely those products on its diagonal. Each product has modulus $(\sqrt{Q})^k$. The alternating trace sign is $(-1)^k$, agreeing with the product of the k one-variable negative trace signs.

The retained construction therefore has a weight-matched common coefficient span, with DP.28–DP.32 on its complex side and DP.33 on its finite-field side. The reduced finite coefficient injection is not, by its existence, a map into the separate ℓ -adic cohomology. A specified operator-compatible comparison with that target is governed exactly by DP.13–DP.22. This locates the remaining comparison inside an explicit surviving geometric construction, preserving the arithmetic input and tensor degree throughout.

88.0.8 5A. Placement inside the original cohomology program over the privileged tau point

The preceding tensor construction is now placed in the actual base program, using the complete original `Tau_Base_Cohomology_2026-09-12/NOTE.md`, read in full. It does not replace that base by a finite field.

Retain the original pointed blueprint

$$\mathbf{F}_{1,\tau} = \{\tau, 1\}, \quad \mathbf{b}_\tau = \text{Spec } \mathbf{F}_{1,\tau}, \quad j_R : \mathbf{F}_{1,\tau} \rightarrow B_R, \quad \tau \mapsto \tau, \quad 1 \mapsto 1_R^\bullet. \quad (\text{DB.1})$$

Its additive zero relation is that the empty sum represents τ ; no Boolean equation $1 + 1 = 1$ is introduced into this base. For each original nonzero ring R , B_R is the blueprint of $G(R) = \{\tau\} \sqcup R^\bullet$ with its original additive relations. A prime of B_R contains τ and does not contain the multiplicative identity. Its inverse image under j_R is therefore exactly $\{\tau\}$. This proves that the structural spectrum map lands at the single privileged point through the actual prime inverse-image map. The arithmetic map $p_R(\tau) = 0$, $p_R(r^\bullet) = r$, and the Boolean support map $b_R(\tau) = 0$, $b_R(r^\bullet) = 1$, remain attached. Their pair is injective, since support separates τ from every represented element and amplitude distinguishes represented elements. The map here is DB.1; the separate localization that forces $e_R = 1$ is not performed.

For a coefficient map $\varphi : R \rightarrow R'$, its split lift satisfies

$$G(\varphi)j_R = j_{R'}, \quad p_{R'}G(\varphi) = \varphi p_R, \quad b_{R'}G(\varphi) = b_R. \quad (\text{DB.2})$$

Each equality follows on τ and on each $r^\bullet$, the two exhaustive kinds of elements. Apply DB.2 to the maps from $R_{h,k}$ in DP.26–DP.33, to the complex coefficient realization, and to every specified finite-field reduction. All these diagrams remain over the same $\mathbf{b}_\tau$.

There is an exact categorical version for the linear fibres. For an R -module M , set $M^\tau = \{\tau\} \sqcup M^\bullet$, with

$$v^\bullet + w^\bullet = (v + w)^\bullet, \quad r^\bullet v^\bullet = (rv)^\bullet, \quad \tau + z = z, \quad \tau z = \tau, \quad r^\bullet \tau = \tau. \quad (\text{DB.3})$$

For an R -linear map $f : M \rightarrow N$, its lift sends τ to τ and $v^\bullet$ to $f(v)^\bullet$. This is $G(R)$ -linear by the two ordinary linearity identities and the rules for τ . Conversely, a $G(R)$ -linear map between these objects which preserves represented support restricts to a map $f : M \rightarrow N$. Its additive and scalar identities are exactly the original R -linear identities. Moreover its value on $0_M^\bullet = e_R v^\bullet$ is $0_N^\bullet$. Thus this gives a fully faithful equivalence with the category consisting of these split module objects and support-preserving maps.

In that category the zero morphism is the supported-zero map $v^\bullet \mapsto 0_N^\bullet$, with $\tau \mapsto \tau$. The zero object is $\{\tau, 0^\bullet\}$. These assertions follow either by the fully faithful correspondence or directly from DB.3: each zero-object arrow preserving support is forced, and composition through it gives exactly that supported-zero map. Consequently the original equation $d^{i+1}d^i = 0$ is carried to the correct supported-zero composition, preserving the separate absorbing element. This supplies the precise linear-fibre category in which the original support cohomology is reconstructed. With the several masks of the source, apply it at each admitted mask and retain all inclusion transports; do not tensor the masks as if they were unrecorded vector coordinates.

Now retain the actual chart poset

$$+ < \eta < \sigma, \quad - < \eta < \sigma, \quad \mathcal{A} = \text{Fun}(P, \text{Vect}_{\mathbb{C}}), \quad (\text{DB.4})$$

and the actual coefficient sheaf

$$\mathcal{T}_+ = \mathcal{T}_- = V, \quad \mathcal{T}_\eta = \mathcal{B}, \quad \mathcal{T}_\sigma = 0, \quad r_+ = \Theta, \quad r_- = \Theta\mathcal{F} = J\Theta. \quad (\text{DB.5})$$

The J in DB.5 is the original reflection $JF(x) = x^{-1}F(1/x)$; the finite jet map below is written J_Z . Let $P_x(y) = \mathbb{C}$ for $x \leq y$ and zero otherwise. Evaluation gives $\text{Hom}(P_x, F) = F_x$, hence these diagrams are projective. The sequence

$$0 \rightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \mathbb{C} \rightarrow 0 \quad (\text{DB.6})$$

is exact: at $+$ and $-$ its last map is the identity, and at η and σ its kernel is exactly the displayed anti-diagonal. Each map commutes with the poset arrows because its entries are the same scalar identity at all nonzero stalks. Applying $\text{Hom}(-, F)$ proves

$$\mathbf{A}_\tau(F) = R\Gamma(P, F) \simeq [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta]. \quad (\text{DB.7})$$

On DB.5 this is the original two-leg theta complex. Its degree-zero kernel consists of $(\hat{\psi}, \psi)$, since Θ is injective, and its degree-one quotient is Q . This identifies the actual arithmetic cohomology over the privileged base point. Restriction to the σ stalk is a separate map: its two degree-zero representatives $r_{+\sigma}x$ and $r_{-\sigma}y$ differ by $r_{\eta\sigma}(r_+x - r_-y)$, so the homotopy is exactly $r_{\eta\sigma}$. This proves its derived comparison to DB.7 without replacing global cohomology by the support stalk.

Define $I_x(W)(y) = W$ for $y \leq x$ and zero otherwise. The identity $\text{Hom}(F, I_x W) = \text{Hom}(F_x, W)$ and exactness of evaluation show that each is injective over $\mathbb{C}$. Dualizing DB.6 gives

$$K_\tau(W) = [I_\eta W \xrightarrow{(+\text{res}, -\text{res})} I_+ W \oplus I_- W] \quad \text{in degrees } -1, 0. \quad (\text{DB.8})$$

Applying $\text{Hom}(F, -)$ gives the explicit differential $\ell \mapsto (\ell r_+, -\ell r_-)$, which is the dual of DB.7. This proves its stated adjunction to $\mathbf{A}_\tau$. The stalk complexes at $+$, $-$ are $[W \xrightarrow{1} W]$ and $[W \xrightarrow{-1} W]$, so they are acyclic; at η the sole term is W in degree -1 , and at σ all terms vanish. Thus $K_\tau(W) \simeq S_\eta W[1]$, with the exact inherited shift.

Tensor DB.6 over $\mathbb{C}$ in the fixed factor order to resolve the constant diagram on P^k . Its differential is the signed sum with sign $(-1)^{\sum_{i < j} e_i}$. In top degree the actual coefficient numerator is $\mathcal{B}^{\otimes k}$, and the boundary subspace is the sum of tensors with ΘV in at least one factor. Quotienting one factor at a time proves, by the defining relations of the tensor quotient,

$$H^k \mathbf{A}_{\tau, k}(\mathcal{T}^{\boxtimes k}) = Q^{\otimes k}, \quad K_{\tau, k}(W) \simeq S_{(\eta, \dots, \eta)} W[k]. \quad (\text{DB.9})$$

The second identity follows by the same dual tensor resolution: any stalk with at least one $+$ or $-$ factor contains its signed two-term identity contraction, any σ factor gives zero, and the all- η stalk has exactly W in degree $-k$. This proves the product right-adjoint calculation with its signs and no deleted cohomological shift.

Use the actual full-jet map $J_Z : \mathcal{B} \rightarrow E_h$. Since $\mathcal{M}\Theta\phi = gH_\phi$, its jets vanish through every full selected zero order; only holomorphy in neighborhoods of those selected zeros in the strip is required here. The map with component J_Z at η and zero at the other stalks is therefore a sheaf map $\mathcal{T} \rightarrow S_\eta E_h$. Its tensor product gives the actual top-cohomology arrow

$$\mathcal{J}_k : H^k \mathbf{A}_{\tau,k}(\mathcal{T}^{\boxtimes k}) = Q^{\otimes k} \longrightarrow \mathcal{E}_k, \quad \mathcal{J}_k = \overline{\mathcal{J}}_Z^{\otimes k}. \quad (\text{DB.10})$$

It is surjective because $\overline{\mathcal{J}}_Z \sigma_Z = I$ in the original full-order packet; tensoring that explicit right inverse gives a right inverse of DB.10. Compose the other way with the original cyclic inclusion to obtain the arithmetic split pair

$$\sigma_Z^{\otimes k} \eta_{h,k} : E_{h,k} \hookrightarrow Q^{\otimes k}, \quad \mathcal{J}_k \sigma_Z^{\otimes k} \eta_{h,k} = \eta_{h,k}. \quad (\text{DB.11})$$

Thus DP.30 is the period realization of the actual cyclic submodule of this tau-base cohomology: the map is $\mathcal{P}_k \mathcal{J}_k$ restricted through DB.11. The units in $\eta_{h,k}$, the full jets, and the original cochain representative boundaries are all present. On top cohomology the scaling action is $U_a^{\otimes k}$; its exact intertwining through DB.10 and DB.11 is $e^{(\log a) \mathcal{A}_k} \eta_{h,k} = \eta_{h,k} e^{(\log a) A_k}$. This follows from DP.26 by the convergent exponential series, or by its finite nilpotent expansions on each block.

The original right-adjoint dual injection also extends to this cyclic module. For the original reflection-stable packet Z , let R_Z be the full residue form from the tau-base note,

$$R_Z(f, v) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger(s) v(s)}{g(s)} ds, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}. \quad (\text{DB.12})$$

Its local denominator is the actual $g = z^m u(z)$, $u(0) \neq 0$. Multiplication by u^{-1} is an invertible triangular Taylor map; extracting the coefficient of z^{m-1} pairs powers z^j, z^{m-1-j} with nonzero anti-diagonal. Reflection sends the opposite block's z^j to $(-1)^j \bar{z}^j$ in coefficients. Hence this sesquilinear form is nondegenerate on the full finite direct sum, retaining all nilpotents and the exact reflection signs. Tensor its numerical values in the fixed order to get a nondegenerate sesquilinear pairing $R_Z^{\otimes k}$ on $\mathcal{E}_k$; the tensor Gram is the tensor of invertible Grams and has the tensor of their inverses. No phase is inserted to claim positivity. In particular the skew-Hermitian parity is retained as $(-1)^k$ under conjugate exchange.

Let $\mathcal{L}_k = \mathcal{L}_1^{\otimes k}$ be the original moment line, with scaling action a^k . Define

$$\begin{aligned} \mathcal{D}_{h,k} : \overline{E_{h,k}} &\longrightarrow H^{-k} R \text{Hom}_{\mathcal{A}(P^k)}(\mathcal{T}^{\boxtimes k}, K_{\tau,k}(\mathcal{L}_k)) \\ &\cong \text{Hom}_{\mathbb{C}}(Q^{\otimes k}, \mathcal{L}_k), \\ \mathcal{D}_{h,k}(\bar{x})(q) &= R_Z^{\otimes k}(\eta_{h,k} x, \mathcal{J}_k q). \end{aligned} \quad (\text{DB.13})$$

The displayed cohomological identification follows from the adjunction DB.8–DB.9: over the field, taking the algebraic dual is exact, so degree $-k$ is the dual of top degree k . The formula is linear in $\bar{x}$. It is well-defined on $Q^{\otimes k}$ because DB.10 is, including every theta relation. If it vanishes for all q , surjectivity of DB.10 and nondegeneracy of the tensor residue form imply $\eta_{h,k} x = 0$; injectivity of DP.26 then implies $x = 0$. Thus DB.13 is an actual injection to the computed tau-base dual object.

For real $a > 0$, the original scaling multiplier satisfies $(a^s)^\dagger = a^{1-s}$. Multiplying the two factors in DB.12 therefore gives exactly a , so $R_Z(U_a f, U_a v) = a R_Z(f, v)$, including the full Taylor expansions. Tensoring gives the factor a^k . On the algebraic dual the action is $(U_a \ell)(q) = a^k \ell(U_a^{-\otimes k} q)$. It follows by direct substitution in DB.13 that this injection is equivariant with the conjugate of the original cyclic scaling action. This proves the tensor counterpart of the original tau-base note's (43), with $K_{\tau,k}$ and the correct moment line, rather than leaving the pairing outside the cohomological model.

Finally, each nonempty original support mask retains its copy of the coefficient complex and each transport retains its stated linear map. Reconstructing them by DB.3 gives the supported degree-one object $\{\tau\} \sqcup \coprod_{A \neq \emptyset} \{A\} \times Q$ and the original tensor supported cohomology. All maps DB.10–DB.13 preserve the admitted mask; a zero amplitude maps to that mask's supported zero and external absence maps to τ . The dual mask arrows are precomposition in the opposite mask category, as in the original note. This completes the placement of the weight-matched auxiliary tensor comparison **inside** the actual split cohomology program over $\mathfrak{b}_\tau$.

88.0.9 6. Original critical-line comparison and the retained kernel

The complete local proof of the critical-only comparison and of the arbitrary-polynomial theta sequence is supplied in the companion `deligne_bound_critical_kernel_subreview_20260913.md`. Its sources were read in full independently. The exact sequence from the original theta complex is

$$0 \longrightarrow Q[h_0(D)] \longrightarrow E_{h_0} \xrightarrow{\times j_{h_0} g} E_{h_0} \longrightarrow Q/h_0(D)Q \longrightarrow 0. \quad (\text{DP.34})$$

The first arrow sends a class represented by F , with $h_0(D)F = \Theta\phi$, to $j_{h_0}H_\phi$. It is independent of the representative because changing F by $\Theta\psi$ changes ϕ by $h_0(D)\psi$. The source and target Euler inverse formulas in `arithmetic_input.tex` show that this arrow identifies $Q[h_0(D)]$ with the kernel of the displayed multiplication. The target jet map identifies its cokernel with $Q/h_0(D)Q$. This retains the exact theta source, not an arbitrary test spectral model.

At a selected centre, write $\epsilon = s - \rho$, $h_0 = \epsilon^n a$, $g = \epsilon^m u$, with $a(\rho)u(\rho) \neq 0$, and $r = \min(n, m)$; when $g(\rho) \neq 0$, set $m = r = 0$. In $\mathbb{C}[\epsilon]/\epsilon^n$, multiplication by $j_{h_0}g$ has kernel $(\epsilon^{n-r})/(\epsilon^n)$ and cokernel $\mathbb{C}[\epsilon]/\epsilon^r$. Indeed the factor u is invertible and multiplication by ϵ^m has precisely these kernel and image ideals. These formulas prove that vanishing occurs whenever $n \leq m$, including undercounted orders. Exact full order $n = m$ makes $g/h_0 = u/a$ a unit; it is that unit which appears in DP.23 and the full packet retraction. The precise local map from the kernel to the quotient, through the original cohomology, is proved in the companion, so that no equality between two different jet presentations is presumed here.

For a full actual packet Z , the existing Connes comparison has

$$T_Z = \Gamma\sigma_Z : E_Z \longrightarrow \mathcal{S}(\mathbb{R})/\overline{\zeta(1/2 + it)\mathcal{S}(\mathbb{R})}, \quad \ker T_Z = E_{Z, \text{off}}. \quad (\text{DP.35})$$

The full higher-jet inverse on the selected critical blocks identifies its actual image with $E_{Z, \text{crit}}$. Thus DP.35 yields an exact equivariant sequence

$$0 \longrightarrow E_{Z, \text{off}} \longrightarrow E_Z \xrightarrow{T_Z} \text{im } T_Z \longrightarrow 0, \quad (\text{DP.36})$$

with the canonical primary splitting of its source. Its two-term comparison complex $[E_Z \xrightarrow{T_Z} \text{im } T_Z]$, in degrees zero and one, has degree-zero cohomology $E_{Z, \text{off}}$ and degree-one cohomology zero. The inclusion of the kernel in degree zero is therefore a quasi-isomorphism. If the full target is retained instead, degree-one cohomology is its quotient by $\text{im } T_Z$; it must not be silently set to zero. This is an exact derived record of everything killed by the critical comparison.

For any receiving vector-space fibre labelled by λ , the split lift of a linear map is

$$\tilde{T}(\tau) = \tau, \quad \tilde{T}(\lambda, v) = (\lambda, Tv). \quad (\text{DP.37})$$

The preimages are exactly

$$\tilde{T}^{-1}\{(\lambda, 0)\} = \{(\lambda, v) : v \in \ker T\}, \quad \tilde{T}^{-1}\{\tau\} = \{\tau\}. \quad (\text{DP.38})$$

Consequently DP.35 retains its full off-critical kernel as supported source vectors. The support observation distinguishes those vectors from external absence; it does not turn an annihilating linear map into an injection. At the same time DP.36 gives the precise surviving morphism and the complex that records the lost blocks. That is a structural continuation of the program, not a refutation of it.

88.0.10 7. Acceptance and the next exact arithmetic calculation

Equations (12)–(13), (17)–(24), (26)–(27), and (44)–(50) of the delivered note have the stated mathematical scope after the character-summand clarification. The note's full-order sentence following (57) needs the distinction proved in DP.34. Its quantitative Gamma estimate is covered by the separate independent Gamma review; this document does not convert a full reading of that section into an independent numerical or all- k certificate.

The note's period congruence is exact: with a common invertible Π , $\tilde{G} = \Pi^{-*}G\Pi^{-1}$ and $\tilde{W} = \Pi^{-*}W\Pi^{-1}$ give $\tilde{G}^{-1}\tilde{W} = \Pi(G^{-1}W)\Pi^{-1}$ by direct multiplication. In particular every generalized eigenvalue and every common-index determinant ratio is retained. This is consistent with the faithful tensor map DP.30 and the original theta relation DP.27.

The actual arithmetic multiplier still enters the two source metrics through their identity coefficient map and their different least-norm sections. Its signed endpoint determinant correction in delivered (54) remains an exact quantity in the program. The present audit neither assumes that correction is small nor replaces its calculation by a condition. It establishes the full comparison diagram that remains valid while that source-specific calculation proceeds.

The phrase “the F1 geometry program has failed” does not follow from this package. A defensible conclusion is: the displayed auxiliary purity, by itself and through the existing maps, does not force the desired original theta-source estimate. The exact failures and surviving maps have now been calculated, including a weight- k tensor construction and the complete target-dependent comparison kernel.

88.0.11 Source pins and actual reading record

The companion JSON records SHA-256, byte counts, and reading scope for every selected source, this review, and the independent critical-kernel proof. The new delivered note was read completely. `arithmetic_input.tex`, `Exponential_Translation_Comparison.tex`, and the primary correction were read completely. The SGA trace-period note was read from Sections 6 through 12, including all coefficient, period, tensor-weight, and metric formulas used above. The original Deligne scan was visually inspected on complete printed pages 215–216 only. The retained S20 text is a location aid; the original scan supplies the theorem typography.

No Lean invocation, remote mutation, whole-library audit, or numerical certification is claimed in this review.

89 The original tau-base weight line in the complete exponential reflection flow

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. `bfd95736800fa643796fd9b083a6eaf1d2c5ba2af62989450f31378aefb873ca`

Complete source. `sources/next_edition/actual_tau_conversions_v1/sources/f1_reflection/F1_REFLECTION_WEIGHT_TRANSPORT.md`

Source SHA-256. `bfd95736800fa643796fd9b083a6eaf1d2c5ba2af62989450f31378aefb873ca`

89.0.1 The original tau-base weight line in the exponential reflection flow

13 September 2026. This is a calculation on the existing Split-Zero arithmetic construction, not a replacement finite-field model and not a weight estimate. It joins the original moment line in the tau-base right adjoint to an exact reflection and flow on the already constructed polynomial-exponential family. The finite transport identity extends through the special fibre even though the corresponding contour gauge contains an essential exponential in $1/u$.

89.0.2 1. Retained source and conventions

The controlling source is `Tau_Base_Cohomology_2026-09-12/NOTE.md`, SHA256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`. Its whole body was read. Sections 3–7 supply, respectively,

$$A_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}], \quad H^1 = Q = \mathcal{B}/\Theta V, \quad K_\tau(W) \simeq S_\eta W[1]. \quad (\text{RW1})$$

The original action is $U_p F(x) = F(x/p)$. Its two source legs are $(U_p \phi, pU_{1/p} \psi)$, not two copies of U_p . Its Mellin multiplier is p^s . The actual second moment $\int \phi(x) dx$ has multiplier p ; its k -fold tensor is the specified line L_k with multiplier p^k . The dual action is consequently p^k times precomposition by U_p^{-1} . All these maps fix support masks and carry supported zero to supported zero, with external absence tau carried to tau.

Use an actual full quartet packet and its original cyclic tensor module,

$$E = \mathbb{C}[S]/(\chi), \quad q = \deg \chi, \quad A = M_S, \quad S = s_1 + \cdots + s_k. \quad (\text{RW2})$$

Here the retained quartet and its complete multiplicities give

$$\bar{\chi} = \chi, \quad \chi(k - S) = \varepsilon \chi(S), \quad \varepsilon = (-1)^q. \quad (\text{RW3})$$

Indeed complex conjugation and reflection permute the tensor-sum roots with their original multiplicities; the leading coefficient of the reflected monic degree- q polynomial is $(-1)^q$. This statement is about those actual symmetry-stable packets. It does not invent an off-line zero or discard a nilpotent direction. The arithmetic inclusion remains $\eta[P] = v_h^{\otimes k} P(A_h^{(k)})1$. In particular its full Taylor unit is not absorbed into a scalar or removed from the comparison.

The existing exponential family uses the antiderivative with $\Phi(0) = 0$,

$$\Phi' = \chi, \quad \Phi_t(S) = \Phi(S) - tS, \quad D_{u,t} = u\partial_S + \chi(S) - t, \quad \mathcal{H} = \text{coker } D_{u,t}. \quad (\text{RW4})$$

Leading-term division makes H free of rank q over $\mathbb{C}[u, t]$, with basis $1, S, \dots, S^{q-1}$. In that basis the already proved connection is

$$\nabla_t = \partial_t - \frac{A(t)}{u}, \quad A(t) = A + t\mathcal{R}, \quad \mathcal{R} = 1 \otimes \ell_{q-1}. \quad (\text{RW5})$$

The matrix $A(t)$ is multiplication by S in $\mathbb{C}[S]/(\chi - t)$ on these representatives. Multiplication by an arbitrary S -polynomial is not thereby asserted to descend to the full u -deformed differential cokernel. The connection and the chain maps below supply the actual descended operators.

89.0.3 2. Reflection is a cochain map of the actual family

Let $RP(S) = P(k - S)$ and

$$r(u, t) = (-\varepsilon u, \varepsilon t) = (u', t'). \quad (\text{RW6})$$

The matrix R on degree-below- q representatives has the explicit entries

$$R_{ij} = \begin{cases} \binom{j}{i} k^{j-i} (-1)^i & i \leq j, \\ 0 & i > j, \end{cases} \quad R^2 = I. \quad (\text{RW7})$$

For every polynomial P , differentiation and (RW3) give

$$D_{u', t'}(\varepsilon RP) = R(D_{u, t}P). \quad (\text{RW8})$$

In detail, the derivative term on the left is $(-\varepsilon u)\varepsilon(-RP') = uRP'$; its remaining term is $\varepsilon(\chi - \varepsilon t)RP = (\varepsilon\chi - t)RP$, which is exactly the reflected right side. Thus degree-zero map εR and degree-one map R give a cochain isomorphism, with its inverse given by the same reflection. This extends over $(u, t) = (0, 0)$, where it is the original polynomial reflection on E . Adding coefficient conjugation gives the actual dagger operation, with parameter map $(u, t) \mapsto (-\varepsilon \bar{u}, \varepsilon \bar{t})$.

On the companion matrices the same calculation is

$$RA(t)R = kI - A(\varepsilon t). \quad (\text{RW9})$$

This follows either by reducing $(k - S)P(k - S)$ modulo $\chi - \varepsilon t$, or by using the algebra isomorphism defined by (RW3). All entries and all repeated-root blocks remain; no diagonalization is used.

89.0.4 3. The weight twist is the original potential's constant, not a choice

Integrating (RW3) with the original endpoint gives

$$\Phi(k - S) = \Phi(k) - \varepsilon \Phi(S),$$

$$\boxed{\frac{\Phi_t(k - S)}{u} = \frac{\Phi_{\varepsilon t}(S)}{-\varepsilon u} + \frac{\Phi(k) - kt}{u}.} \quad (\text{RW10})$$

Neither $\Phi(k)$ nor k is set to zero. Pull the target connection back along r , keeping u fixed while differentiating t . Since $dt' = \varepsilon dt$,

$$r^*\nabla_t = \partial_t + \frac{A(\varepsilon t)}{u}, \quad (r^*\nabla_t)(Rv) - R\nabla_tv = \frac{k}{u}Rv. \quad (\text{RW11})$$

Consequently the exact horizontal map on $u \neq 0$ is

$$\boxed{\mathcal{F}(u, t) = \exp\left(\frac{\Phi(k) - kt}{u}\right) R.} \quad (\text{RW12})$$

Its logarithmic t -derivative is $-k/u$, so substituting (RW11) proves $(r^*\nabla)(\mathcal{F}v) = \mathcal{F}\nabla v$. This uses the full constant supplied by the original potential. It does not select a new norm on H .

For an oriented rapid-decay contour Γ , its reflected contour is the pushforward $\Gamma' = k - \Gamma$ with that orientation. The substitution $S_{\text{old}} = k - S_{\text{new}}$ has $dS_{\text{old}} = -dS_{\text{new}}$. Therefore the actual integrals satisfy

$$\int_{\Gamma} e^{\Phi_t(S)/u} P(S) dS = -\exp\left(\frac{\Phi(k) - kt}{u}\right) \int_{\Gamma'} e^{\Phi_{t'}(S)/u'} (RP)(S) dS. \quad (\text{RW13})$$

The minus sign is the one-form pullback. The constant exponential is retained. The exponential identity also carries the original decay along Γ to Γ' , so this substitution does not postulate a different contour class.

89.0.5 4. A regular finite flow identity through the special fibre

The scaling calculation computes the regular lift of forward theta dilation from $L = u\partial_t - S$. Its coefficient transport C_a is characterized by

$$\frac{d}{da} C_a(u, t) = -C_a(u, t) A(t + ua), \quad C_0 = I. \quad (\text{RW14})$$

This linear matrix equation, with polynomial coefficients, has a unique entire solution in a , depending holomorphically on u, t . Its local power series recursion is explicit. It obeys

$$C_{a+b}(u, t) = C_a(u, t) C_b(u, t + ua), \quad C_a^{-1}(u, t) = C_{-a}(u, t + ua). \quad (\text{RW15})$$

To prove the product law, differentiate its right side in b and compare its initial value at $b=0$ with C_{a+b} in (RW14). Uniqueness gives equality. In particular this finite transport is invertible even at $u=0$; there $C_a(0, t) = \exp(-aA(t))$.

Its reflection is the new exact formula

$$\boxed{RC_a(u, t)R = e^{-ka} C_{-a}(-\varepsilon u, \varepsilon t).} \quad (\text{RW16})$$

Proof: write the left side as B_a . By (RW9),

$$B'_a = -kB_a + B_a A(\varepsilon t + \varepsilon ua).$$

For $u' = -\varepsilon u$, $t' = \varepsilon t$, the derivative of $C_{-a}(u', t')$ is $C_{-a}(u', t') A(t' - u'a)$. Thus the right side of (RW16) obeys this same equation and has the same initial value I . This proves (RW16) for all parameters, without dividing by u . It is not an attempt to evaluate the singular exponential (RW12) at $u=0$.

Along $t \mapsto t + ua$, the ratio of the two full scalar factors in (RW12) is exactly $\exp(-ka)$. At $a = \log p$, forward theta dilation has moment line p^{-k} ; inverse dilation U_p has moment line p^k . Thus the very same factor appears both in the actual tau-base adjunction and in the exact exponential-family reflection flow. It is not a finite-field cardinality assigned to the absolute base point.

Specializing (RW16) gives the full arithmetic identity

$$Rp^{-A}R = p^{-k}p^A, \quad Rp^A R = p^k p^{-A}. \quad (\text{RW17})$$

Every block is retained:

$$p^{\pm A}|_{\lambda} = p^{\pm \lambda} \sum_{j=0}^{\ell_{\lambda}-1} \frac{(\pm \log p)^j}{j!} N_{\lambda}^j. \quad (\text{RW18})$$

89.0.6 5. The computed adjoint transport and its exact evaluation

On coefficient sections the scaling operator is semilinear: $T_a v(t) = C_a(u, t)v(t + ua)$. The k -fold moment line has $T_a^L z(t) = \exp(-ka)z(t + ua)$. Therefore the actual twisted dual is

$$\boxed{T_a^{\vee} \ell(t) = e^{-ka} C_a(u, t)^{-T} \ell(t + ua).} \quad (\text{RW19})$$

Substitution proves, with no omitted scalar,

$$(T_a^{\vee} \ell)(t)^T (T_a v)(t) = e^{-ka} \ell(t + ua)^T v(t + ua). \quad (\text{RW20})$$

The product rule (RW15) proves the composition law for (RW19). Its infinitesimal generator is

$$u \partial_t + A(t)^T - kI. \quad (\text{RW21})$$

At the actual special fibre and inverse dilation it becomes $p^k(p^A)^{-T}$. This is precisely the action obtained by applying the right adjoint $K_{(\tau, k)}(L_k)$ in the original source, then precomposing finite jet evaluation. In the conjugate packet, the source's dagger and residue pairing give its specified linear map to this twisted dual. No Hermitian positivity has been inferred from this bilinear calculation.

For completeness, the existing constant residue matrix

$$S_{ij} = [S^{-1}] \frac{S^{i+j}}{\chi(S) - t} = [S^{-1}] \frac{S^{i+j}}{\chi(S)} \quad (0 \leq i, j < q) \quad (\text{RW22})$$

has determinant $(-1)^{q(q-1)/2}$ and satisfies $A(t)^T S = SA(t)$. Its t -independence follows because the difference of the two fractions starts at degree at most S^{-2} . With $J = R^T S$, the computed two-parameter pairing identity is

$$\boxed{A(t)^T J + JA(\varepsilon t) = kJ.} \quad (\text{RW23})$$

Indeed transpose $RA(t) = (kI - A(\varepsilon t))R$ and apply the residue self-adjointness of $A(\varepsilon t)$. At $t=0$ this recovers the additive weight relation. For the actual arithmetic residue with denominator g , one must additionally retain the existing full unit $j_h(g/h)$ and the actual arithmetic inclusion eta ; the polynomial χ residue is not silently substituted for that pairing. No extension of that arithmetic unit as a globally invertible operator on every deformed fibre is claimed.

89.0.7 6. Position in Deligne's argument and in the current program

The printed primary text of Weil II, pp. 178, 202, 203 and 206, was read in the supplied S20 source-language records together with the existing primary-scan correction. In (3.3.5), the dual degree and Tate twist give $-n + (2N - i) - 2N = -n - i$. In (3.3.6), opposite bounds then constrain the image of the actual compact-support comparison. Page 203 (3.2.13) tensors the same eigenvalue with itself; its square is not replaced by an unspecified tensor constituent.

Our computation supplies a concrete part of the analogous tau-base diagram: the original moment-weight dual action has been transported through the exact exponential deformation and its reflection, including the full special-fibre blocks. It has not supplied Deligne's geometric upper weight bound for the theta quotient. The canonical arithmetic metrics G_N and their original relation volumes remain unchanged.

The next use of this result is to transport the original residue-dual map and its arithmetic norm comparison along this explicit flow; the factor and parameter shift are now calculated, so they need not be guessed or replaced by a generic purity assertion. The external-product construction and signed determinant calculation are the continuations on those two sides.

89.0.8 Provenance and verification scope

- Full controlling tau-base note as pinned above; no whole-paper Lean claim.
- Existing exponential delivery Tau_Deligne_Exponential_Comparison/RESEARCH_NOTE.md, section 7 (37)–(42), read in full for the constant residue duality; DS66–69 supply the previously constructed family and original theta map.
- The scaling calculation supplies the independent chain-level lift underlying (RW14). Formulas (RW6)–(RW13), (RW16), and their joining to the actual weight line are derived here, with their complete algebra above.
- Exact finite fixtures are supporting checks, not replacements for these general proofs. No new Lean execution, RH proof, or remote publication is claimed by this note.

90 Complete independent tau-base arithmetic determinant transport proof

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. ce6b9fbdad43e5233b76756f5cbbbadf6aca0a33db62437f7caf0b90321683bb

Complete source. sources/next_edition/actual_tau_conversions_v1/sources/determinant_review/tau_arithmetic_determinant_transport_independent_review_20260913.md

Source SHA-256. ce6b9fbdad43e5233b76756f5cbbbadf6aca0a33db62437f7caf0b90321683bb

90.0.1 Independent proof review: arithmetic determinant transport

Date: 13 September 2026. Review status: the mathematical statements AT1–AT22, including AT2a, are accepted on the corrected source pinned below. The first-pass proof-order issue in the explanation of AT6 has been repaired. This review retains the original Tau Base cohomology, arithmetic theta source, full Taylor unit, tensor sum coordinate, and both reference and arithmetic masses.

90.0.2 1. Sources and scope

The complete first-pass root source was read at SHA-256 41c7b45a0a94f0e65d422b213a4bcbddedbf983ab5127da6095e449e001fef98. Its corrected proof paragraph was then read completely. The accepted source is work/tau_arithmetic_determinant_transport_20260913.tex, 17706 bytes, SHA-256 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae. No displayed formula changed between those editions.

The original dependencies checked directly were:

Source relative to output/split_zero_rh_t andem_2026-09-12	Bytes	SHA-256	Scope
sources/Tau_Base_Cohomology_2026-09-12/NOTE.tex	28353	e7a494d79e01b774ad18045b1e244b28738ddf7b85aec31c58724e2fe35c3aba	Lines 225–408 and 672–755: coefficient sheaf, support faces, projective resolution, actual cohomology, tensor signs and top quotient
sources/web_cyclic_sum_delivery/Tau_Cyclic_Sum_Control/NOTE.tex	41804	2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c	Lines 108–183 and 300–349: source, full unit, exact cyclic map and theta primitive

Source relative to output/split_zero_rh_t andem_2026-09-12	Bytes	SHA-256	Scope
tex/arithmetic_input.t ex	14644	a50b0fb587b644c4ea 94dba45b2b423d038f 2bfbe7fef188543939 ba850e58d2	Complete A1–A19, already fully read in this audit

The Gamma norms, all-multiplicity quartet theorem, and exact original arithmetic comparison were independently proved in `work/deligne_bound_gamma_independent_review_20260913.md`, SHA-256 57069ebc4b4eab844d18d3bca9b4b7bd26cda138d9497fe3c126ee65bdfb9186. Its complete 72-equation proof remains a dependency of the application AT22. The section and variation proof for the present transport is given completely in RV6–RV11 below.

The exact finite fixtures described at the end exercise these transport formulas. They are calibration measures for the formulas; they do not assert values of arithmetic zeta moments or the existence of a selected zeta packet. No old suite, Lean process, or remote mutation is part of this review.

90.0.3 2. Original cohomology and the two separately typed injections

On the four-point chart $+, - < \eta < \sigma$, the original coefficient diagram is

$$(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0), \quad r_+ = \Theta, \quad r_- = J\Theta = \Theta\mathcal{F}. \quad (\text{RV1})$$

Here V is the original even Schwartz space with its value and integral zero, $\mathcal{B}$ is the original strong Schwartz space on the multiplicative line, $JF(x) = x^{-1}F(1/x)$, and $\mathcal{F}$ is the original Fourier transform. The diagram category is the declared linear-fibre realization of the Tau Base chart. At each chart point, the projective resolution

$$0 \rightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \mathbf{C} \rightarrow 0 \quad (\text{RV2})$$

is exact: at $+$ and $-$, its last map is identity; at η and σ , its kernel is the displayed anti-diagonal. Applying its Hom complex to RV1 gives

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}]. \quad (\text{RV3})$$

Its degree-one image is exactly ΘV , so its degree-one cohomology is the original $Q = \mathcal{B}/\Theta V$. Tensoring the resolution uses the literal differential sign $(-1)^{\sum_{j < i} |x_j|}$. At top degree the source is $\mathcal{B}^{\otimes k}$, and the relation subspace is the sum of all tensors with ΘV in one factor. Tensoring the quotient maps over the field therefore identifies the top cohomology with $Q^{\otimes k}$. This proves the use of AT1 in the specified category, without replacing the pushforward by evaluation at the zero coefficient stalk σ .

For an actual full packet h , retain $E_h = \mathbb{C}[s]/(h)$, $v_h = j_h(g/h) \in E_h^\times$, $g = 2\xi$, and the actual F_h with Mellin transform g/h . The polynomial tensor algebra is $E_h^{\otimes k}$, its operator is multiplication by $S = s_1 + \cdots + s_k$, and its cyclic annihilator is the original $\chi = \chi_{h,k}$.

The injective algebra map $\alpha : E = \mathbb{C}[S]/(\chi) \rightarrow E_h^{\otimes k}$ sends $[P]$ to $P(A_h^{(k)})1$. Its kernel before quotienting is exactly the defining cyclic annihilator, proving injectivity. Multiplication by the full tensor unit has inverse multiplication by its inverse. Hence the first injection is

$$\eta = M_{v_h^{\otimes k}} \alpha : E \hookrightarrow E_h^{\otimes k}. \quad (\text{RV4})$$

It is a module map; multiplication by a unit is not being assigned the type of a unital algebra map. The original single-packet section $\sigma_h : E_h \hookrightarrow Q$ has a left inverse induced by its full arithmetic jet observation. Tensoring this left inverse proves that $\sigma_h^{\otimes k}$ remains injective. The separate cohomology injection is consequently

$$\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}. \quad (\text{RV5})$$

The two target spaces in RV4 and RV5 are carried by the explicit arrow $\sigma_h^{\otimes k}$; neither is substituted for the other. Applying the original source map $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$ gives exactly $J^{(k)}\mathcal{V}P = \eta[P]$ and $q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta[P]$, as in AT2 and AT2a.

Every relation $\chi(S)P(S)$ has its original theta primitive. Indeed, the cyclic annihilator says it belongs to the tensor ideal $(h(s_1), \dots, h(s_k))$. Fixed-order monic division writes it as $\sum_i h(s_i)P_i(\mathbf{s})$. In factor i , replace F_h by ϕ_* , apply $P_i(D_1, \dots, D_k)$, and assign the primitive sign $(-1)^{i-1}$. The tensor differential supplies a second $(-1)^{i-1}$; their product is $+1$, and $h(D)F_h = \Theta\phi_*$ recovers the desired source relation. This checks that every variation below retains an actual top cochain boundary.

90.0.4 3. Finite source, zero relation space, and variations

Let $q = \deg \chi \geq 1$. For $N \geq q - 1$, use the original exact sequence

$$0 \rightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \rightarrow 0 \quad (\text{RV6})$$

in increasing monomial frames, with the fixed remainder section s_N . Its coefficient maps do not depend on the real interpolation parameter t . The densities are precisely $m_0 = r_{1/4,k}$, of mass $(2\pi)^{k/2}$, and $m_1 = w_h^{*k}$, of mass μ_h^k , in $S = k/2 + iu$. Neither mass is changed.

Each source density is positive almost everywhere and has moments of every finite degree. For the arithmetic density, the zeros of the nonzero entire function g/h on a real line are discrete; at $k \geq 2$, its convolution preserves almost-everywhere positivity. A nonzero polynomial has only finitely many zeros on this line, so the two source Grams are positive definite. Their convex interpolation $M_N(t)$ remains positive definite on $0 \leq t \leq 1$. At a fixed finite degree, its eigenvalues have a positive lower bound on this compact interval. The relation $\text{Gram } H_N = B_N^* M_N B_N$ is positive on its actual domain by injectivity of B_N . These facts justify all finite differentiations, including a small open neighborhood of the interval.

The minimum section is $R = s - BH^{-1}B^*Ms$. Orthogonality gives $JR = I$, $B^*MR = 0$, and $G = R^*MR > 0$. To prove the inverse expression without presupposing it, decompose every vector $p \in \mathcal{P}_N$ uniquely as $By + RJp$. Then

$$R^*Mp = GJp, \quad MR = J^*G, \quad (JM^{-1}J^*)G = I. \quad (\text{RV7})$$

The last equality follows from $JR = I$; it proves $G = (JM^{-1}J^*)^{-1}$. This is the noncircular derivation now in the corrected AT6 proof.

The determinant-line sign is also retained. With $r = N - q + 1$, the increasing-degree matrix $[s \ B]$ is triangular with determinant one because χ is monic. Thus $[B \ s]$ has determinant $(-1)^{qr}$. The actual line map wedges the r relation vectors before the q quotient lifts. Replacing a lift by a relation leaves this wedge unchanged. Its Gram block Schur complement is G , so

$$\det H \det G = |(-1)^{qr}|^2 \det M = \det M, \quad V_N = \det G = \det M / \det H. \quad (\text{RV8})$$

At $r = 0$, the domain $\mathcal{P}_{-1}$ is zero, its determinant is one, and its unique endomorphism is invertible as a map from the zero space to itself. Every composite through that space has the zero map of its stated outer domain and codomain. The formula gives $R = s$, $G = s^*Ms$, and $V_N = \det M$; no earlier singular quotient Gram is inverted.

Differentiation of $JR = I$ shows that $\dot{R}$ lies in $\text{im } B$. Differentiation of $B^*MR = 0$ fixes its coefficient. Thus

$$\dot{R} = -BH^{-1}B^*\Delta MR, \quad \dot{G} = R^*\Delta MR, \quad \ddot{G} = -2R^*\Delta MBH^{-1}B^*\Delta MR. \quad (\text{RV9})$$

The two terms involving $\dot{R}$ vanish in the first derivative of R^*MR by orthogonality. In the next derivative they are the two equal displayed negative terms because ΔM is Hermitian. These identities prove AT7 and show that G is concave as an original Hermitian form.

Taking determinant derivatives in RV8, or directly in G , gives

$$\frac{d}{dt} \log V_N = \text{Tr}(M^{-1}\Delta M) - \text{Tr}(H^{-1}B^*\Delta MB) = \text{Tr}(G^{-1}R^*\Delta MR). \quad (\text{RV10})$$

The second derivative is exactly

$$\frac{d^2}{dt^2} \log V_N = -2\|H^{-1/2}B^* \Delta M R G^{-1/2}\|_{\text{HS}}^2 - \|G^{-1/2}\dot{G}G^{-1/2}\|_{\text{HS}}^2. \quad (\text{RV11})$$

Each square is for the indicated exact isometry of the original source forms. No term is removed in this identity. It proves AT8 and AT9, including the empty first square when $r = 0$.

Applying $\mathcal{V}$ to the relation-valued $\dot{R}$ gives the primitive constructed after RV5. The jet $J^{(k)}\mathcal{V}R = \eta$ and cohomology class $\sigma_h^{\otimes k}\eta$ are constant in t . On a fixed nonempty support label A , every such linear map lifts as $(A, v) \mapsto (A, fv)$, with $\tau \mapsto \tau$. Thus a nonzero relation vector can map to the supported zero while its energy in RV9–RV11 is retained. Its preimage is the whole source kernel on that label, not only the zero vector.

90.0.5 4. The exact common-source projection trace

Set $N_* = 2q$, and let $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$ be the literal coefficient inclusion. In the common metric $M_* = M_{N_*}(t)$, define

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M_*, \\ Q_N &= I_N B_N H_N^{-1} B_N^* I_N^* M_*, \\ C &= M_*^{-1} \Delta M_*. \end{aligned} \quad (\text{RV12})$$

The first two are orthogonal projections with images $I_N \mathcal{P}_N$ and $I_N \text{im } B_N$. Substitution uses $I_N^* M_* I_N = M_N$ to verify idempotence and M_* -self-adjointness. Since the relation image lies in the polynomial image, $P_N Q_N = Q_N P_N = Q_N$. Thus $P_N - Q_N$ is the orthogonal projection onto $I_N R_N(E)$, with rank and trace q .

The cyclic trace computation has no omitted metric factor:

$$\begin{aligned} \text{Tr}(P_N C) &= \text{Tr}(M_N^{-1} I_N^* \Delta M_* I_N) = \text{Tr}(M_N^{-1} \Delta M_N), \\ \text{Tr}(Q_N C) &= \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N). \end{aligned} \quad (\text{RV13})$$

Combining these equalities with RV10 proves AT12.

For $i_0 = q - 1, i_1 = q, j_0 = 2q - 1, j_1 = 2q$, set

$$U = (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \quad W = (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \quad (\text{RV14})$$

For nested subspaces $X \subset Y$ of one Hilbert space, their orthogonal projections satisfy $P_X P_Y = P_Y P_X = P_X$; hence $P_Y - P_X$ is the projection onto $Y \cap X^\perp$. Each difference in RV14 is therefore a positive projection of rank q : the polynomial dimensions differ by q ; the relation dimensions $N - q + 1$, including zero at $q - 1$, also differ by q . Consequently

$$U \succeq 0, \quad W \succeq 0, \quad \text{Tr } U = \text{Tr } W = 2q. \quad (\text{RV15})$$

Their sums need not be idempotent, their two summands need not have orthogonal images, and no such assertion enters RV15. Subtracting the four exact traces in RV13 gives

$$\frac{d}{dt} \log \frac{V_{i_0} V_{i_1}}{V_{j_0} V_{j_1}} = \text{Tr}((U - W)C). \quad (\text{RV16})$$

All inverses are smooth and bounded on the parameter interval. Integrating proves AT15 with the exact arithmetic correction $\Delta \mathcal{B}$. The individual concavity in RV11 gives no sign to this four-term difference; RV16 is its complete signed expression.

90.0.6 5. Pointwise expression and arithmetic density zeros

The two orthogonal source images $\text{im } B_N$ and $\text{im } R_N$ decompose the whole finite polynomial space. Their projection formulas imply

$$M_N^{-1} = B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*. \quad (\text{RV17})$$

For example, multiplication on the right by M_N acts as identity separately on each of those two images, by orthogonality and their own Gram identities; it therefore acts as identity on their direct sum.

For $v_N(u) = (1, S(u), \dots, S(u)^N)$, evaluation of RV17 gives the exact nonnegative function

$$\begin{aligned} d_N(t, u) &= v_N M_N^{-1} v_N^* - |\chi(S(u))|^2 v_{N-q} H_N^{-1} v_{N-q}^* \\ &= v_N R_N G_N^{-1} R_N^* v_N^* \geq 0. \end{aligned} \quad (\text{RV18})$$

At $N = q - 1$ the relation term is absent and defined as zero. The factor $|\chi|^2$ follows from the actual evaluation identity $v_N B_N = \chi(S(u)) v_{N-q}$; it is not a replacement of the relation polynomial or its powers.

Integration against m_t yields $\int d_N m_t = \text{Tr}(G_N^{-1} R_N^* M_N R_N) = q$. Integration against Δm , using the original moment matrices, gives

$$\frac{d}{dt} \log V_N = \int d_N(t, u) \Delta m(u) du. \quad (\text{RV19})$$

Subtracting at the four endpoints and integrating t is exactly AT17. To justify both steps without any quotient of densities, note that $d_N(t, u)$ is a fixed finite-degree polynomial in $S(u)$ and $\overline{S(u)}$, whose coefficients are bounded for $0 \leq t \leq 1$. Its absolute value is bounded by a fixed constant times $1 + |u|^{2N^*}$. The inequality $|\Delta m| \leq m_0 + m_1$ and the two actual moment bounds prove absolute integrability on $[0, 1] \times \mathbb{R}$. Fubini applies. At a zero of m_1 , this expression is still an ordinary polynomial times a finite signed density; no division by m_1 appears. This proves the claimed zero-set and unbounded-ratio domains.

90.0.7 6. Spectral width and the condition number

The operator $D_* = M_*(0)^{-1} M_*(1)$ is positive and self-adjoint for the original $M_*(0)$ metric. Let $b_{\min}, b_{\max} > 0$ be its extreme eigenvalues. Factor the interpolated Gram in its stated matrix order:

$$M_*(t) = M_*(0)((1-t)I + tD_*), \quad C(t) = ((1-t)I + tD_*)^{-1}(D_* - I). \quad (\text{RV20})$$

No commutation of the two original Gram matrices is required for this factorization. The matrix on the right of $C(t)$ is a rational function of the single diagonalizable positive operator D_* . Its eigenvalue on a b -eigenvector is

$$c_b(t) = \frac{b-1}{1-t+tb}. \quad (\text{RV21})$$

The denominator is positive throughout the closed interval. Differentiating with respect to b gives $(1-t+tb)^{-2} > 0$, so the extrema are attained at the same $b_{\min}, b_{\max}$ for every t . Repeated eigenvalues cause no exception. The operator $C(t)$ is self-adjoint in $M_*(t)$ because $M_*(t)C(t) = \Delta M_*$ is Hermitian.

For any positive self-adjoint Z on that same Hilbert space,

$$c_{\min} \text{Tr } Z \leq \text{Tr}(ZC) \leq c_{\max} \text{Tr } Z.$$

Indeed $\text{Tr}(Z(c_{\max}I - C))$ is nonnegative: by cyclicity it is the trace of the positive matrix $Z^{1/2}(c_{\max}I - C)Z^{1/2}$. The lower bound is proved the same way. This does not require Z to commute with C . Applying it to RV15 and subtracting gives

$$|\text{Tr}((U - W)C)| \leq 2q(c_{\max} - c_{\min}). \quad (\text{RV22})$$

For $b > 0$, $c_b(t)$ is the derivative of $\log(1-t+tb)$. For $b = 1$, both are respectively zero and the constant zero logarithm, so the same integral gives $\log b = 0$. Hence

$$\int_0^1 (c_{\max} - c_{\min}) dt = \log b_{\max} - \log b_{\min} = \log \kappa_*, \quad \kappa_* = b_{\max}/b_{\min}. \quad (\text{RV23})$$

The triangle inequality for the real integral of RV16, followed by RV22–RV23, proves

$$|\Delta \mathcal{B}| \leq 2q \log \kappa_*. \quad (\text{RV24})$$

This confirms AT18–AT21 with the factor $2q$. The source data at $N_* = 2q$ are actual finite Gram data. No bound uniform in packet or tensor degree, and no pointwise upper or lower bound for m_1/m_0 , is used.

90.0.8 7. Mass and frame covariance

If $M_*(1) = cM_*(0)$ with its literal positive scalar c , every degree form is $(1-t+tc)M_N(0)$, and every minimum section remains the same. Its quotient form is $(1-t+tc)G_N(0)$, so its determinant is $(1-t+tc)^q V_N(0)$. The two numerator and two denominator factors in $\mathcal{B}(t)$ cancel exactly. Thus $\Delta\mathcal{B} = 0$, also implied by $\kappa_* = 1$. The separate original masses and determinant factors remain those displayed in these formulas.

For a precise source-frame arrow, let F carry new common-source coefficients to old coefficients. Then $M'_* = F^*M_*F$, $\Delta M'_* = F^*\Delta M_*F$, $D'_* = F^{-1}D_*F$, $C' = F^{-1}CF$, and every projection transforms as $P' = F^{-1}PF$ onto the correspondingly transported subspace. Its inclusion matrix is transported as $I'_N = F^{-1}I_N$ if the internal frame of $\mathcal{P}_N$ is unchanged. Products and traces are then identical by conjugacy. This includes nonunitary complex frames and frames mixing the displayed common-source degree coordinates, because the actual degree subspaces are transported with them.

Independently, let a fixed $Y : E_{\text{new}} \rightarrow E_{\text{old}}$ change the common quotient frame. Then $R'_N = R_N Y$, $G'_N = Y^*G_N Y$, and $V'_N = |\det Y|^2 V_N$, with the same factor for every N, t . That factor cancels from the four endpoint ratios. These explicit frame maps prove the covariance paragraph following AT21; they do not change the original source norms.

90.0.9 8. Original quartet application and scope

For the stipulated actual reflection-stable quartet $1/2 \pm \delta \pm i\gamma$, with $\delta > 0$, nonzero imaginary displacement, and complete common order m , the independent Gamma theorem retains

$$q = [1 + k(m-1)](k+1)^2, \quad \mathcal{B}_{h,k}^\Gamma \geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right).$$

The exact arithmetic/reference comparison RV24 therefore yields

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) - 2q \log \kappa_*. \quad (\text{RV25})$$

Since $2q > 0$, moving terms and dividing gives

$$\log \kappa_* \geq 2 \log \left(\frac{\delta k}{2\sqrt{5}} \right) - \frac{\mathcal{B}_{h,k}^{\text{ar}}}{2q}. \quad (\text{RV26})$$

Both are exactly AT22. They require no existence claim for such a packet. If the fixed packet restriction is invoked, its g/h , Taylor unit, theta primitive, and finite jet injection remain the original ones in RV4–RV6. In particular this is an analytic comparison on the original finite Tau Base complex with its action-compatible arithmetic observation, and does not replace it by a freely chosen polynomial.

The result computes a finite signed arithmetic term and bounds its magnitude using an actual relative Gram spectrum. It neither proves an arithmetic sub-threshold estimate nor draws a conclusion that the entire Tau Base or F1 geometry program has failed.

90.0.10 9. Exact finite fixtures and final review record

The independent fixture source and its exact JSON receipt are `work/tau_arithmetic_determinant_transport_exact_fixtures_20260913.py` and the same basename with `.json`. The calibration retains the literal original Gamma coordinate $S = 1/2 + iu$ and mass $\sqrt{2\pi}$, takes the test relation $\chi = S^2 - S + 3/16$, and compares that Gamma density with $(u^2 + u/3 + 2)m_0(u)$. This latter multiplier is positive since it is $(u + 1/6)^2 + 71/36$, and it is unbounded. Its mass is $5\sqrt{2\pi}/2$. The test relation is explicitly a calibration polynomial, not a zero-packet assertion.

The fixture tests the first empty relation domain and all four endpoints 1, 2, 3, 4 on cutoff 4. It checks the exact minimum section, relation orthogonality, full inverse splitting, Schur determinants, common-source traces, both positive-sum traces, and the signed four-endpoint identity at $t = 0, 1/3, 1$. It explicitly verifies that each sum U, W is not an idempotent and that neither commutes with C in this calibration, exercising two assumptions deliberately absent from the proof.

It also differentiates the exact inverse-kernel construction at the first nonzero relation space and checks both matrix variations and both nonzero logarithmic curvature losses. Omitting either loss is detected by a

nonzero exact residual. A second fixture rescales the source mass by $7/3$, checks every quotient scales by that literal scalar and the four-volume correction is zero, and verifies that the relative source operator is $(7/3)I$. A complex nonunitary frame change additionally verifies the exact similarity of D_* .

One initial harness comparison encountered an unevaluated symbolic zero written $0\sqrt{2}$. Its exact expression evaluator was repaired; no mathematical identity, hypothesis, or expected value was changed. The actual completed run and source hashes are recorded in the attached receipt. These finite checks supplement the written proof RV1–RV26; they do not prove its analytic moment facts by sampling.

The completed run passed 98 exact checks. Its fixture source SHA-256 is 8f29a721b8409708e3cc6e3b469cf35429743b3e19150fba467636d98085a497, and its JSON output SHA-256 is 367bb7b3c59e092e6b9ea9e444739d83bcf69701e0dd45545dc595f4bbfd67b6. It gives the exact nonzero correction

$$\exp(\Delta\mathcal{B}) = \frac{66206077200175080056805947}{54881827734195887080729255}. \quad (\text{RV27})$$

At $t = 1/3$, the positive relation contribution and the positive metric contribution whose negatives occur in RV11 are, respectively,

$$\frac{77593974624}{617198528867}, \quad \frac{88134062040620154}{21159417165147361}. \quad (\text{RV28})$$

Thus this execution checks the two distinct curvature terms with nonzero exact values. No rounding or interval approximation enters these fixture results.

The AT6 proof-order issue was resolved in the accepted source. No further mathematical correction was found. The complete written calculation in RV1–RV28 and the exact fixture receipt support the mathematical acceptance recorded here.

91 Actual tau-base scaling and its complete exponential cohomology flow

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 8becda9f286ae37357589af9f85b5520a5981d4689a4d07c0e290e5265ee2304

Complete source. `sources/next_edition/actual_tau_conversions_v2/sources/f1_scaling/ACTUAL_TAU_SCALING_AND_EXPONENTIAL_FLOW.md`

Source SHA-256. 8becda9f286ae37357589af9f85b5520a5981d4689a4d07c0e290e5265ee2304

91.0.1 Actual tau-base scaling and its exponential cohomology flow

Independent local calculation, 2026-09-13. The starting point is the specified $\mathbf{F}_{1,\tau}$ -based arithmetic cohomology, not a replacement finite-field geometry or localization at the generic support point. The construction below applies its actual theta action, full jet maps, right-adjoint pairing, and polynomial-exponential deformation. All source support masks, scalar factors, Taylor units, and multiplicities are retained.

91.0.2 1. Outcome: the existing arithmetic action has a concrete lift

The source already defines $U_p F(x) = F(x/p)$ over the absolute tau-base. It induces p^A on the full finite arithmetic quotient. The forward dilation $L_p F(x) = F(px) = U_{1/p} F(x)$ induces p^{-A} . Their exact finite block formulas, theta-boundary corrections, and tau-base dual maps are given below.

More than a fixed-basis operator is available on the existing exponential deformation. Write

$$D_{u,t} = u\partial_S + \chi(S) - t, \quad L = u\partial_t - S.$$

Then $[D_{u,t}, L] = 0$. The formal cochain automorphism

$$\exp(aL)P(u, t, S) = \exp(-aS)P(u, t + ua, S)$$

preserves the special-fibre ideal (u, t) and induces $\exp(-aA)$ there. On its finite free cohomology it has an entire, explicitly computed matrix $C_a(u, t)$; for $u \neq 0$ the actual oriented period matrices give

$$C_a(u, t) = \Pi(u, t)^{-1} \Pi(u, t + ua).$$

At $u = t = 0$, this is exactly p^{-A} when $a = \log p$. The inverse parameter $a = -\log p$ realizes the source's p^A . Thus the arithmetic scaling acts through the actual cohomological family with its full nilpotent coefficients. Formal polynomial complexes, evaluated finite matrices, and nonpolynomial contour integrands are distinguished below; no nonpolynomial expression is silently treated as a polynomial map.

The same reduced algebraic connection also supplies an explicit characteristic- r power operator, its r -curvature. The finite-field Artin-Schreier Frobenius and its character-twist map are computed in Section 7 as separate operations on the same potential data. This is a positive comparison of actual maps, not a generic failure argument about the tau-base program.

91.0.3 2. Dilation on the original source and both tau-base legs

Fix $p > 0$, $a = \log p$, and use the positive real logarithm. The source spaces are exactly those of arithmetic_inpt.tex, A1, and Tau_Base_Cohomology/NOTE.md, equations (10)–(14):

$$V = \{\text{even Schwartz } \phi : \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$B = \{F : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for every } b \in \mathbb{Z} \text{ and } j \geq 0\}, \text{ with } D = -x\partial_x,$$

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), MF(s) = \int_0^\infty F(x) x^s dx/x.$$

Let $L_p\phi(x) = \phi(px)$ and $L_p F(x) = F(px)$. Dilation preserves evenness, Schwartz regularity, and the two source moments:

$$(L_p\phi)(0) = \phi(0), \int L_p\phi = p^{-1} \int \phi.$$

For the target seminorms, $\sup x^b |D^j L_p F(x)| = p^{-b} \sup y^b |D^j F(y)|$. Thus these are continuous invertible source maps, with inverse $L_{1/p}$. Termwise theta evaluation and change of Mellin variable give

$$\Theta L_p = L_p \Theta, D L_p = L_p D, M L_p F(s) = p^{-s} M F(s).$$

The theta identity uses the original convergent series, including its factor two. The Mellin identity holds on the entire transform of B ; the source identity $H_{L_p\phi}(s) = p^{-s} H_\phi(s)$ follows on its specified strip from the same change of variables. Hence ΘV is stable and there is an induced invertible action on $Q = B/\Theta V$. The group law is literal: $L_p L_q = L_{pq}$.

The Fourier transform satisfies $\text{Fourier}(L_p\phi) = p^{-1} L_{1/p} \text{Fourier}(\phi)$. The tau-base joint pushforward is the two-leg complex

$$A_\tau(T) = [V_+ \oplus V_- \rightarrow B], d(\phi, \psi) = \Theta(\phi - \text{Fourier}(\psi)).$$

Its correct forward dilation is

$$(\phi, \psi) \rightarrow (L_p\phi, p^{-1} L_{1/p}\psi), F \rightarrow L_p F.$$

Indeed $\text{Fourier}(p^{-1} L_{1/p}\psi) = L_p \text{Fourier}(\psi)$, proving the cochain equality with both signs and the factor p^{-1} . On the original convention $U_p = L_{1/p}$, this becomes

$$(\phi, \psi) \rightarrow (U_p\phi, p U_{1/p}\psi), F \rightarrow U_p F,$$

exactly tau-base equations (32)–(35). The joint degree-zero cycle $(\text{Fourier}(\psi), \psi)$ is sent to the cycle indexed by $p U_{1/p}\psi$. The single-leg degree-zero groups stay zero.

These are actual sheaf automorphisms on the four-point chart model: the displayed maps act at $+$, $-$, η , and the identity acts on every support mask at σ . They commute with restriction, with support inclusions, and with the projective-resolution differential (18). Consequently they induce the stated action on $\text{R}\Gamma(P, T)$ over the actual one-point $\mathbf{F}_{1,\tau}$ base. No localization $G(R)[e^{-1}] = B_{\text{boolean}}$ is performed. The infinite scalar map $G(\mathbb{Z}) \rightarrow \mathbb{Z}$ remains in the original structural square.

91.0.4 3. Full jets, the Taylor unit, and an explicit source boundary

Full finite action For the original full-order packet polynomial h and full jet map $J_Z F = j_h M F$, let $A = M_s$ on $E_h = \mathbb{C}[s]/(h)$. Taylor expansion at a root ρ of its actual order m_ρ gives

$$j_\rho(p^{-s}) = p^{-\rho} \sum_{j=0}^{m_\rho-1} (-\log p)^j z_\rho^j / j!.$$

Therefore

$$J_Z L_p = F_p^- J_Z, F_p^- = p^{-A} = \exp(-(\log p)A).$$

On the block where $A = \rho I + N_\rho$, the entire matrix is

$$F_p^-|_\rho = p^{-\rho} \sum_{j=0}^{m_\rho-1} (-\log p)^j N_\rho^j / j!.$$

The inverse is $F_p^+ = p^A$ with all signs positive in the nilpotent sum. No Jordan level or exponential phase is dropped. Equivalently there is a unique degree-below-deg h Hermite remainder whose full jets are those of p^{-s} , and multiplication by that remainder is the same matrix. This is a linear module automorphism, not a claim that multiplying every class by a unit is a unital ring homomorphism.

The finite source/target square A8 has arrows $\times j_h g$. Multiplication by $j_h(p^{-s})$ commutes with this arrow. Dilation also commutes with $h(D)$ and its unique inverse on its range. Hence every map, kernel, and quotient in A7–10 is equivariant. In particular the original packet summand $Q[h(D)]$ and its complementary $h(D)Q$ remain distinct invariant subspaces with the same original projector.

Write $v = g/h$, $v = j_h v$, $\epsilon = v^{-1} = j_h(h/g)$. These are the full original Taylor units, not just their residue values. The specified representative is

$$R_{\text{ref}}x = S_h^B \Theta(R_Z(x)(D)\phi_*), R_Z(x) = \text{rem}_h(\epsilon x).$$

Since $j_h(p^{-s})$ commutes with both v and ϵ , the same unit remains in this representation after either dilation. In the cyclic tensor source the already specified inclusion

$$\eta[P] = v^{\otimes k} P(A_k)1, A_k = \sum_i A_i,$$

obeys $\exp(-aA_k)\eta = \eta \exp(-aA)$, where A is the original cyclic sum action on $\mathbb{C}[S]/(\chi_{h,k})$. This follows directly by commuting multiplication operators. The existing equivariant cyclic retraction has the same exponential intertwining. This is only the inherited tensor-source action, not a second construction of an external-product exponential family.

The boundary correction is an actual element of V A11 supplies the unchanged finite-rank defect

$$DR_{\text{ref}} - R_{\text{ref}}A = \Theta(\phi_*)\ell, \ell(x) = [s^{\deg h-1}]R_Z(x).$$

Define the finite-interval vector-valued integral

$$K_a x = - \int_0^a L_{\exp(a-r)} \phi_* \cdot \ell(\exp(-rA)x) dr.$$

Every integrand is in the original V ; dependence on r is smooth in its Schwartz topology, so the integral is in V , including for negative a with the usual oriented integral. Differentiate $L_{\exp(a-r)} R_{\text{ref}} \exp(-rA)$ with respect to r . Since the derivative of $L_{\exp(a-r)}$ is $DL_{\exp(a-r)}$, one obtains $L_{\exp(a-r)}(DR_{\text{ref}} - R_{\text{ref}}A) \exp(-rA)$. Integration from zero to a gives the exact sign and boundary formula

$$L_p R_{\text{ref}} - R_{\text{ref}} p^{-A} = \Theta K_{\log p}.$$

Its Mellin expression retains the entire unit and polynomial remainder:

$$H_{K_a x}(s) = [\exp(-as)R_Z(x)(s) - R_Z(\exp(-aA)x)(s)]/h(s).$$

The numerator has all zero jets required by h , because $R_Z = \text{rem}_h(\epsilon \times -)$ and the two finite multipliers commute. The expression is thus holomorphic at every selected root, with its full order. The integral, rather than divisibility alone, establishes that it is the transform of an admitted original V element.

In the exact extension coordinates $B_Z = \Theta V + R_{\text{ref}}E$, dilation is

$$(v, x) \rightarrow (L_p v + K_a x, p^{-A} x).$$

The complete cocycle identity is $K_{a+b} = L_{\exp a} K_b + K_a \exp(-bA)$. It follows by composing the displayed source action, or by splitting its integral. Thus the equivariant roof A13 is equivariant for this actual multiplicative group, not merely for its infinitesimal generator.

For any already specified representative r_N with $D^{(k)}r_N - r_N A = dK_N$, the identical finite-interval argument gives

$$\begin{aligned} L_p^{\otimes k} r_N - r_N p^{-A} \\ = -d \int_0^{\log p} L_{\exp(\log p - r)} K_N \exp(-rA) dr. \end{aligned}$$

Here dilation is the chain action on every term containing K_N ; its commutation with d is the tensor cochain rule. The correction remains an original relation with its specified primitive. The finite representative image need not be invariant as a subspace of the source.

91.0.5 4. Tau-base right adjoint and the existing holonomy observer

The tau-base note computes, with its original signs,

$$K_\tau(W) = [I_\eta(W) \rightarrow I_+(W) \oplus I_-(W)],$$

$$\text{in degrees } -1, 0, \text{ and } \text{RHom}(T, K_\tau(W)) = \text{RHom}(A_\tau(T), W).$$

The source's second-moment line L_1 has U_p action p , since $\int U_p \phi = p \int \phi$. Naturality of that right-adjoint

calculation gives the exact action on a dual functional

$$(U_p \ell)(v) = p \ell(U_p^{-1} v).$$

For forward dilation it is $p^{-1} \ell(L_p^{-1} v)$. This factor comes from the retained moment representation, not the cardinality of the absolute base. On finite coefficient duals the inverse-dilation action is p^{I-A^T} ; the forward action is $p^{-(I-A^T)}$.

The actual injection from the conjugate finite packet into the degree-minus-one right-adjoint cohomology is tau-base equation (43):

$$\bar{f} \mapsto [F] \mapsto \sum_{\rho} \text{Res}_{\rho} f^{\dagger}(s) M F(s) ds/g(s),$$

where $f^{\dagger}(s) = \overline{f(1-\bar{s})}$ retains the Taylor coefficient signs $(-1)^j$. Since $(p^s)^{\dagger} = p^{1-s}$,

$$R_Z(p^A f, p^A h) = p R_Z(f, h).$$

This proves equivariance with the just-computed dual action, at every nilpotent order. For forward dilation the factor is p^{-1} . Changing a source representative by $\Theta \phi$ adds a holomorphic residue, exactly as in the original right-adjoint construction.

The Taylor unit is visible in the residue join to the exponential special fibre. In the monic packet case $\chi = h$, let S_{χ} denote the full bilinear polynomial residue pairing with denominator χ . Then

$$R_Z(f, h) = S_{\chi}(f^{\dagger}, \epsilon h), \epsilon = j_h(h/g).$$

Indeed $1/g = (1/h)(1/v)$ locally, and multiplying by the complete jet ϵ leaves the same residues. Replacing this unit by its constant term would generally change the pairing on higher jets; no such replacement occurs here. The exponential special-fibre coefficient identification therefore receives the same finite packet and this exact unit-bearing pairing.

There is also a direct join to the already written sector holonomy. The original $L^2(dx)$ norm obeys $\|U_p F\|^2 = p \|F\|^2$. Hence $p^{-1/2} U_p$ is the normalized source dilation, and on an actual eigenvector at ρ its finite jet value is $p^{\rho-1/2}$. Including the source's retained unit phase ω gives exactly

$$z_{p,\omega}(\rho) = \omega p^{\rho-1/2}$$

in `split_zero_carriers.tex`, not merely an unrelated scalar analogy. Its full block extension is

$$\omega p^{\rho-1/2} \sum_{j < m_{\rho}} (\log p)^j N_{\rho}^j / j!.$$

For the original k-fold source the literal norm factor is p^k , and the corresponding normalization is $p^{-k/2} U_p^{\otimes k}$. This records an exact source isometry and its algebraic quotient map; it does not set an original finite canonical Gram equal to a new unitarizing metric. The source relation correction in Section 3 stays part of the comparison.

91.0.6 5. Exact cochain scaling flow on the original exponential family

Keep the original $\chi(S)$, all its coefficients, and the potential

$$\Phi_t(S) = S^{q+1}/(q+1) + \sum_{b=0}^{q-1} c_b S^{b+1}/(b+1) - tS.$$

DS66-69 and the original exponential note use the complex

$$C_{u,t} = [\mathbb{C}[u, t, S] \xrightarrow{D_{u,t}} \mathbb{C}[u, t, S] dS], D_{u,t} P = u P_S + (\chi - t) P.$$

Its differential is injective and has the unique degree-below-q remainder, by its monic leading term. Thus its first cohomology is free with the unchanged basis $1, S, \dots, S^{q-1}$, and its special fibre $u = t = 0$ is exactly $E = \mathbb{C}[S]/(\chi)$, with full multiplicities.

Define $L = u \partial_t - M_S$ on the coefficient in each cochain degree. The individual commutators are

$$[D, u \partial_t] = u I, [D, -M_S] = -u I.$$

They cancel, so $DL = LD$. In the formal a -adic completion over characteristic zero this proves that $\exp(aL)$ is a cochain automorphism with inverse $\exp(-aL)$. Since $u \partial_t$ commutes with M_S , each coefficient of its formal series is exactly that of

$$\exp(-aS) P(u, t + ua, S).$$

At every fixed order in a , only finitely many polynomial operations occur. The differential and this formula are therefore well defined on the formal completion; no analytic function has been substituted into the uncompleted polynomial ring. The map is semilinear for the base translation $t \rightarrow t + ua$, while u is fixed. The monic direct-sum reduction commutes coefficientwise with this formal completion, so its cohomology is the corresponding completed free rank-q module; no unproved interchange with an infinite quotient is involved.

The ideal (u, t) is stable: $L(uP)$ is a multiple of u , and $L(tP) = uP + tuP_t - tSP$ lies in (u, t) . Thus formal specialization gives $\exp(-aS)$ on the polynomial quotient, or precisely $\exp(-aA)$ on its finite coefficient module.

Entire finite-cohomology evaluation and the exact periods The descended generator in the retained basis is

$$L_H = u\partial_t - A(t), A(t) = A + tR, R = e_0 e_{q-1}^T.$$

For all complex u, t, a , define $C_a(u, t)$ by the finite matrix ODE

$$\partial_a C_a = -C_a A(t + ua), C_0 = I.$$

Its solution is entire in these parameters. One direct verification is the iterated-integral series: on any compact parameter set the coefficient matrix is bounded along the segment from 0 to a ; the n -th term is bounded by $(|a|M)^n/n!$. The series converges locally uniformly and differentiates to the ODE. Uniqueness follows by the same integral equation. Solving the reverse ODE gives its inverse, so this entire matrix is nonsingular.

The induced action on coefficient sections is

$$v(u, t) \rightarrow C_a(u, t)v(u, t + ua).$$

To check the generator, the ODE and parameter differentiation give $\partial_a C_a = u\partial_t C_a - A(t)C_a$; equivalently this follows from uniqueness of the differential equation for the flow. At $a = 0$ the derivative of the displayed section is exactly $u\partial_t v - A(t)v$. Uniqueness therefore identifies its Taylor series with the formal $\exp(aL_H)$ induced above. Alternatively for $u \neq 0$, the following period formula proves both differential identities directly, and their entire extension proves them at $u = 0$.

XD12 gives $\partial_t \Pi(u, t) = -\Pi(u, t)A(t)/u$. The actual periods are invertible, with their original ray orientations and monomial factors, by XD19–21. Therefore

$$\Pi(u, t)^{-1} \Pi(u, t + ua)$$

satisfies precisely the defining ODE and initial condition. This proves the formula in Section 1 and retains every period coefficient. It also gives the exact cocycle, first for $u \neq 0$ and then by entire continuation,

$$C_a(u, t)C_b(u, t + ua) = C_{a+b}(u, t).$$

At $u = 0$ the ODE is constant, so

$$C_a(0, t) = \exp(-aA(t)), C_{\log p}(0, 0) = p^{-A}.$$

In particular evaluation at $a = \log p$ is legitimate for this entire finite matrix, although the corresponding $\exp(-aS)$ is not a polynomial cochain map after numerical evaluation. This distinction is not removable by writing an exponential symbol alone.

At $u \neq 0$, an equally explicit contour identity is

$$\begin{aligned} & \int_{\Gamma} \exp(\Phi_t(S)/u) \exp(-aS) P(S) dS \\ &= \int_{\Gamma} \exp(\Phi_{t+ua}(S)/u) P(S) dS. \end{aligned}$$

Both integrals converge for every complex a : the original leading negative degree- $q + 1 \geq 2$ ray term dominates the additional linear exponent. Furthermore

$$D_{u,t}(\exp(-aS)P) = \exp(-aS)D_{u,t+ua}P.$$

This identity has the correct source direction: the multiplier takes relations at $t + ua$ to analytic integrands at t , not polynomial relations in an unchanged fibre. On the finite cohomology the actual map from the fibre at $t + ua$ to the fibre at t is $C_a(u, t)$. No quotient by all entire functions is used or asserted.

The distinction from a frozen endomorphism is exactly evaluable:

$$C_a(u, t) = I - aA(t) + a^2(A(t)^2 - uR)/2 + O(a^3).$$

The second derivative follows by differentiating its ODE once. Thus the first correction to $\exp(-aA(t))$ is $-a^2uR/2$. The residue map here is exactly the original coefficient map $R = e_0 e_{q-1}^T$, including $q = 1$; it is not discarded.

Fixed-fibre source-to-period action For any one already specified (u, t) with $u \neq 0$, the original coefficient comparison $E \rightarrow H_{u,t}$ and XD22 also transport the fixed original theta action to the concrete period-coordinate matrix

$$F_{p;u,t}^{\text{per}} = \Pi(u, t)p^{-A}\Pi(u, t)^{-1}.$$

Indeed $\Pi j_E L_p = F_{p;u,t}^{\text{per}} \Pi j_E$ on the admitted original test-function domain, using its full equivariant jet observation. The kernel is exactly $\ker j_E$; on the canonical representative image the inverse comparison is

$r_N \Pi^{-1}$. The difference of source representatives is the theta boundary computed in Section 3. This fixed-fibre conjugate and the base-translating $C_a(u, t)$ are both actual maps, with different domains. Their coincidence at the coefficient special fibre does not identify them elsewhere.

91.0.7 6. The existing logarithmic comparison receives the same action

DS31 defines $F_A = \sum_{j=0}^{q-1} A^j E_0$, with its actual invariant lattice $M_A = F_A O + EO(-D)$. Every polynomial in A , and therefore its finite-matrix exponential, preserves F_A and induces an operator on E/F_A . Hence the same $p^{\pm A}$ acts on the computed complex and its cohomology:

$$H^0(K_A) = F_A, \text{ action } p^{\pm A}|_{F_A};$$

$$H^2(K_A) = (E/F_A) \otimes T, \text{ action } p^{\pm \bar{A}} \otimes I_T.$$

The dual action for ordinary evaluation is the inverse transpose; the tau-base moment-line choice multiplies it by the retained p (or p^k on the inherited tensor moment line). Thus DS34's annihilator lattice and the full boundary quotient also carry these actions. The trace across the two retained groups is the trace of the original finite action, by a basis adapted to F_A .

For the global algebraic power-map discussion in this paragraph only, additionally restrict p to a positive integer; it need not be prime. On the characteristic-zero comparison curve, the separately existing parameter power map $z \rightarrow z^p$ then has the different explicit action $dz/z \rightarrow pdz/z$, and pulls divisor depth one to the integer depth p . It acts on the curve coordinate z , not the Mellin coordinate S or the original positive variable x . No global algebraic power map $z \rightarrow z^p$ is asserted for nonintegral real p . The theta dilations, full-jet actions $p^{\pm A}$, and finite exponential flow at $a = \log p$ retain their stated domains for every real $p > 0$; the integer restriction applies only to this separate curve map and its divisor-depth pullback.

91.0.8 7. Actual finite-field power maps of this same potential

This section applies the already specified coefficient realization in the original exponential note, equations (47)–(49), retaining its privileged characteristic-zero tau-base branch. Use r for residue characteristic to distinguish it from the positive theta-dilation parameter p , and put $Q = r^f$. The source ring is

$$R_0 = \mathbb{Z}[c_0, \dots, c_{q-1}, 1/(q+1)!] \subset \mathbb{C}.$$

Under any of the original specified maps $R_0 \rightarrow \mathbb{F}_Q$ with $r > q + 1$, the same polynomial differential and coefficient basis base-change literally. This is a map from R_0 , not a field map from $\mathbb{C}$. No value of the transcendental parameter $\log p$ is silently assigned a residue-field value.

A computed characteristic- r power operator from the connection Over the reduced coefficient ring, the polynomial derivation obeys $\partial_t^r = 0$: the r -th derivative of every monomial contains a product of r consecutive integers. Since $u\partial_t$ and M_S commute, their characteristic- r binomial expansion gives

$$(u\partial_t - M_S)^r = -M_{S^r}.$$

Moreover $[D_{u,t}, M_{S^r}] = r u M_{S^{r-1}} = 0$. Thus multiplication by S^r , unlike multiplication by S at nonzero u , genuinely descends to the reduced polynomial cohomology. This gives the following completely specified matrix in its original basis:

$$\Psi_r(\partial_t) = -u^{-r}(\text{red}_{u,t}(S^{r+b}))_{0 \leq b < q}.$$

The notation means column b is the indicated remainder vector. It is the r -curvature of $\nabla_t = \partial_t - S/u$, because the restricted r -th power of the base derivation is zero. Multiplying by $-u^r$ gives a regular coefficient endomorphism which at $u = t = 0$ is exactly A^r in the reduced full quotient. Every coefficient is obtained by the same monic reduction algorithm. This is a computed power map supplied by the actual deformation, without attempting to reduce an infinite exponential series with factorial denominators in characteristic r .

The cover Frobenius and the character-twist comparison For the source's specified $u_0 \neq 0, t_0$ in $\mathbb{F}_Q$, write

$$f_t(S) = (\Phi(S) - t_0 S)/u_0.$$

Its actual Artin–Schreier cover is

$$Y^r - Y = f_t(S).$$

The arithmetic Q -power operation on its geometric coordinates is $(S, Y) \rightarrow (S^Q, Y^Q)$. It preserves the cover because coefficients are in $\mathbb{F}_Q$ and $(Y^Q)^r - Y^Q = (Y^r - Y)^Q = f_t(S)^Q = f_t(S^Q)$. It commutes with

every deck transformation $Y \rightarrow Y + c$, $c \in \mathbb{F}_r$. The chosen additive-character eigensheaf is consequently stable. Geometric Frobenius uses the inverse Galois action convention, as fixed in the original DT.12–13 trace statement.

With that convention, its exact trace on the existing compact-support degree-one cohomology is

$$\mathrm{Tr}(\mathrm{Frob}_Q^n | H_c^1) = - \sum_{s \in \mathbb{F}_{Q^n}} \psi(\mathrm{Tr}_{\mathbb{F}_{Q^n}/\mathbb{F}_r} f_t(s)).$$

This follows from the trace formula with its degree-one sign and the concentration theorem already applied in the original source. Its degree is $q + 1$, the leading coefficient $1/((q + 1)u_0)$ is a unit, and the leading projective zero locus in $\mathbb{P}^0$ is empty and smooth. The retained Deligne theorem therefore applies to these actual fibres: dimension q and weight one, including repeated critical roots. This is the finite-field application already present in the program, not a claim that such a realization has failed.

The exact finite-field operation corresponding to the exponential linear multiplier is also calculable. For a specified scalar b in the residue field,

$$f_{t_0+u_0b}(S) = f_{t_0}(S) - bS,$$

so the actual sheaves obey

$$L_\psi(f_{t_0+u_0b}) = L_\psi(f_{t_0}) \otimes L_\psi(-bS).$$

The cover proof is addition: if $Y_1^r - Y_1 = f_t(S)$ and $Y_2^r - Y_2 = -bS$, then $Y_1 + Y_2$ satisfies the equation with right side $f_t - bS$. On character eigenspaces, addition tensors the character lines. This is the precise analogue of $\exp(\Phi_t/u) \exp(-aS) = \exp(\Phi_{t+ua}/u)$: it is parameter translation with a linear character twist, not the Q-power Frobenius operation on the cover.

The existing DT.12 comparison under coordinate translation $S = X + b$ is a further, different map and retains its original constant line $L_\psi((- \Phi(-b) - t_0b)/u_0)$. Its Frobenius scalar is the exact root of unity in DT.13, not a real theta-dilation mass. Thus the coefficient translation, linear character twist, r-curvature, and cover Frobenius have separately computed coordinates. The matrix $p^{\pm A}$ remains the action from the original theta sheaf over the tau-base throughout this comparison.

91.0.9 8. Support reconstruction at every arrow

At every retained nonempty support mask Λ , every displayed linear/cochain map f has the original lift

$$(\Lambda, v) \rightarrow (\Lambda, f(v)), \tau \rightarrow \tau.$$

For invertible dilations the inverse image of the labelled zero is the labelled zero; for the quotient observation it is its full displayed kernel at the same label. Scalar coefficient specialization uses $G(R_0) \rightarrow G(\mathbb{F}_Q)$ and carries a represented coefficient specializing to zero to e , while external τ stays τ . Dual label arrows are the original precomposition maps in the opposite support-index category, as in tau-base equation (31).

Consequently the calculation composes the actual diagram

τ -based theta sheaf $\rightarrow R\Gamma$ over $\tau \rightarrow$ full finite jets

$\rightarrow$ exponential special fibre $\rightarrow$ finite exponential cohomology / periods,

together with its computed tau-base right-adjoint morphism. It does not replace global cohomology over the tau-base by restriction to the support skeleton, or replace the full packet by reduced values.

91.0.10 9. Source pins and exact read scope

Read in full before completing this calculation:

- output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md: SHA-256 D03E71AD18F187BD89880CB8CA6FC9C5D6F260B3DE8E8E5702C27DB97E7B63C3; source equations (18)–(22), (27)–(43), (47)–(50) anchor the actual base, pushforward, adjoint, scaling, and residue morphism.
- output/split_zero_rh_tandem_2026-09-12/tex/arithmic_input.tex: SHA-256 A50B0FB587B644C4EA94DBA45B2B423D038F2BFBE7FEF188543939BA850E58D2; A1 at line 43, A7–10 at 186–231, A11–13 at 265–295.
- work/deligne_split_sidebar_20260913.tex: SHA-256 8601C882177ECBF64A374BDED1641573F7CBBA4D950DD8786E23D15B64307384; DS31–34 at 496–531, DS44 at 665, DS66–69 at 1014–1078.
- work/deligne_exponential_determinant_extension_20260913.tex: SHA-256 4A6B6C237CA3CF F29ABCEBA6CB5F8285E080B080C5F923F0E09D6F130A30DFE8; XD12 at 131, XD19–22 at 208–246. Its determinant proof is used only to justify the actual period inverse, not to audit a signed Gamma

allowance or claim a new upper bound.

- output/split_zero_rh_tandem_2026-09-12/sources/web_deligne_exponential_delivery/Tau_Deligne_Exponential_Comparison/RESEARCH_NOTE.md: SHA-256 C2246C9F6046347BBD246890DB88B1676B1A53487D66583A13B5883528BC0402; equations (5), (24)–(36j.1), (37)–(49) provide the original unit-bearing cyclic inclusion, polynomial cohomology, connection, periods, residue pairing, coefficient realization, and cover.
- output/split_zero_rh_tandem_2026-09-12/tex/deligne_translation_bridge.tex: SHA-256 7DC6C73941A3EA30B141E7FB1110D6FB7B688AF8A9C79A996ACCE31F9086611D; DT.12–13 at lines 247–255 give the original character convention.
- output/split_zero_rh_tandem_2026-09-12/tex/deligne_primary_source_correction.tex: SHA-256 B12BF0C1FF1175DE1099075F0DA66CF51996ACC43B2ACC5D0ED01523DC05C001; the corrected source-theorem application begins at line 58. The historical citation correction supersedes the older mismatch report preserved in the original exponential delivery.

Additionally read the exact carrier definitions and the complete sector/holonomy/positive-moment section of output/split_zero_rh_tandem_2026-09-12/tex/split_zero_carriers.tex, SHA-256 49ED53151068F726E5E392A36D076C95E4E050E163C204D85FD3B179A4A25805; the sector-to-holonomy proposition is at lines 197–218.

No new Lean execution, signed Gamma audit, or claim of a tensor-degree upper estimate accompanies this calculation.

91.0.11 10. Local correction record

2026-09-13: a transcript audit identified that Section 6’s global algebraic curve map $z \rightarrow z^p$ inherited the arbitrary-real $p > 0$ convention from Section 2 without stating its additional integer restriction. The power-map paragraph now explicitly assumes positive integer p on the characteristic-zero comparison curve. Its affected unnumbered formulas are $z \rightarrow z^p$, $dz/z \rightarrow pdz/z$, and the pullback of divisor depth one to depth p . No primeness assumption is needed.

The preceding real-parameter theta dilation, full-jet and moment-line actions, and the entire finite exponential flow are unchanged. The residue-characteristic r and finite-field Q operations of Section 7 are also unchanged. Only this local parameter-domain correction and this transparent record were added; no publication payload or source-owner file was modified.

Pre-correction SHA-256 of this artifact: 209385837d3fbd72b1f0d1164c9f6b261f312243f4427befe7148729f934029.

92 Complete relation-window spectral proof and literal-flag strictness

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 06e611a52b369d927b90a0dbc065283ce118402905cea046fe028c81f507d96d

Complete source. sources/next_edition/actual_tau_conversions_v2/sources/window_review/tau_relation_window_spectral_review_20260913.md

Source SHA-256. 06e611a52b369d927b90a0dbc065283ce118402905cea046fe028c81f507d96d

92.0.1 Exact spectra and trace spread of the original four relation windows

13 September 2026. Independent mathematical continuation of the original arithmetic determinant transport. This note leaves the sealed transport TeX and its PDF unchanged. It derives the complete spectra of the actual window operators, proves the spectral trace extremum directly, and integrates it against the original relative Gram eigenvalues. The packet, tensor degree, polynomial coordinate, measures, and all four cutoff indices remain fixed.

92.0.2 1. Original common source and projections

Fix the original actual packet h , tensor degree $k \geq 1$, and its literal cyclic annihilator $\chi = \chi_{h,k}$, of degree $q \geq 1$. Keep the common coefficient space

$$\mathcal{H} = \mathcal{P}_{2q} = \mathbb{C}[S]_{\leq 2q}, \quad \dim \mathcal{H} = n = 2q + 1, \quad (\text{WS1})$$

with its original increasing-power frame. Let M_0 and M_1 be the two actual positive source Grams at this cutoff, retaining the Gamma mass and arithmetic mass. Set

$$M(t) = (1-t)M_0 + tM_1, \quad \Delta M = M_1 - M_0, \quad D = M_0^{-1}M_1, \quad C(t) = M(t)^{-1}\Delta M, \quad 0 \leq t \leq 1. \quad (\text{WS2})$$

No rescaling of either Gram is performed. Positivity of $M(t)$ follows because its quadratic form on a nonzero vector is a convex combination of two strictly positive numbers.

Write $P_N(t)$ for the $M(t)$ -orthogonal projection onto the original coefficient subspace $\mathcal{P}_N$. Write $Q_N(t)$ for the same-metric orthogonal projection onto

$$\mathcal{R}_N = \chi \mathcal{P}_{N-q}, \quad \mathcal{P}_{-1} = 0, \quad N \geq q-1. \quad (\text{WS3})$$

Multiplication by the nonzero polynomial χ is injective, so $\dim \mathcal{R}_N = N - q + 1$. In particular $Q_{q-1} = 0$. These are the original relation images, with every power of χ and every finite coefficient retained.

The two actual operators are

$$\begin{aligned} U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q) = Q_{2q-1} + Q_{2q} - Q_q, \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{WS4})$$

All projections in WS4 are computed in the one space WS1 with the same form $M(t)$. Every equation below is pointwise in this retained parameter unless an integral is displayed.

For completeness, if $E \subseteq F$ are subspaces and P_E, P_F their orthogonal projections for that form, then $P_F P_E = P_E$ because $P_E v \in F$. Also $P_E P_F = P_E$, because $v - P_F v \in F^\perp \subseteq E^\perp$. Hence $P_F - P_E$ is self-adjoint, idempotent, and has image $F \cap E^\perp$. This proves every nested-projection decomposition used below; none of them assumes unrelated subspaces are orthogonal.

92.0.3 2. The exact relation and polynomial window spectra

The original relation spaces have the nested flag

$$\mathcal{R}_q = \chi \mathcal{P}_0 \subseteq \mathcal{R}_{2q-1} = \chi \mathcal{P}_{q-1} \subseteq \mathcal{R}_{2q} = \chi \mathcal{P}_q, \quad (\text{WS5})$$

of dimensions $1, q, q+1$. For $q = 1$, the first two spaces coincide; the following middle complement is then the zero space. Define

$$\begin{aligned} L_{U,0} &= \mathcal{R}_{2q}^\perp, & \dim L_{U,0} &= q, \\ L_{U,1a} &= \mathcal{R}_q, & \dim L_{U,1a} &= 1, \\ L_{U,2} &= \mathcal{R}_{2q-1} \cap \mathcal{R}_q^\perp, & \dim L_{U,2} &= q-1, \\ L_{U,1b} &= \mathcal{R}_{2q} \cap \mathcal{R}_{2q-1}^\perp, & \dim L_{U,1b} &= 1. \end{aligned} \quad (\text{WS6})$$

The four spaces are mutually orthogonal and their dimensions add to $2q+1$. The flag decomposition proves that they span $\mathcal{H}$. On them the three summands in U act respectively as follows:

	$L_{U,0}$	$L_{U,1a}$	$L_{U,2}$	$L_{U,1b}$
Q_{2q-1}	0	1	1	0
Q_{2q}	0	1	1	1
Q_q	0	1	0	0
U	0	1	2	1

(WS7)

Every entry is a restriction of the actual projection, not an assigned new operator.

The original polynomial flag is

$$\mathcal{P}_{q-1} \subseteq \mathcal{P}_q \subseteq \mathcal{P}_{2q-1} \subseteq \mathcal{P}_{2q} = \mathcal{H}. \quad (\text{WS8})$$

Its dimensions are $q, q+1, 2q, 2q+1$; the middle two coincide for $q = 1$. The corresponding four orthogonal pieces are

$$\begin{aligned} L_{W,0} &= \mathcal{P}_{q-1}, & \dim L_{W,0} &= q, \\ L_{W,1a} &= \mathcal{P}_q \cap \mathcal{P}_{q-1}^\perp, & \dim L_{W,1a} &= 1, \\ L_{W,2} &= \mathcal{P}_{2q-1} \cap \mathcal{P}_q^\perp, & \dim L_{W,2} &= q-1, \\ L_{W,1b} &= \mathcal{H} \cap \mathcal{P}_{2q-1}^\perp, & \dim L_{W,1b} &= 1. \end{aligned} \quad (\text{WS9})$$

On these pieces $P_{2q-1} - P_{q-1}$ has respective values $0, 1, 1, 0$, while $I - P_q = P_{2q} - P_q$ has respective values $0, 0, 1, 1$. Therefore W acts by $0, 1, 2, 1$.

Combining WS6–WS9 proves the complete spectra, including multiplicities:

$$\boxed{\text{spec } U = \text{spec } W = \{0^{[q]}, 1^{[2]}, 2^{[q-1]}\}.} \quad (\text{WS10})$$

Both traces are $2 + 2(q-1) = 2q$. Both eigenvalue-two spaces have dimension zero when $q = 1$; no nonexistent vector or inverse is used.

Equivalently, introduce the actual spectral projections

$$\begin{aligned} E_{U,1} &= Q_{2q}, & E_{U,2} &= Q_{2q-1} - Q_q, \\ E_{W,1} &= I - P_{q-1}, & E_{W,2} &= P_{2q-1} - P_q. \end{aligned} \quad (\text{WS11})$$

They satisfy $E_{X,2} \leq E_{X,1}$, with ranks $q-1$ and $q+1$, respectively, and

$$U = E_{U,1} + E_{U,2}, \quad W = E_{W,1} + E_{W,2}. \quad (\text{WS12})$$

This is also a useful direct explanation of the repeated eigenvalues in WS10.

92.0.4 3. Direct proof of the spectral trace extremum

Let C be any self-adjoint endomorphism of the same finite-dimensional Hilbert space, with eigenvalues

$$c_1 \leq c_2 \leq \cdots \leq c_n, \quad n = 2q + 1. \quad (\text{WS13})$$

All repetitions are retained. By the finite spectral theorem there is an orthonormal eigenbasis $e_1, \dots, e_n$ for this **same** source form. This use of coordinates is an isometry of the original form, not a change of any vector's norm or any scalar mass.

For an orthogonal projection E of rank r , put $w_i = \langle e_i, Ee_i \rangle = \|Ee_i\|^2$. Then

$$0 \leq w_i \leq 1, \quad \sum_i w_i = r, \quad \text{Tr}(EC) = \sum_i c_i w_i. \quad (\text{WS14})$$

For $0 < r < n$, set $a = c_{n-r+1}$. The difference between the largest- r eigenvalue sum and the last expression is exactly

$$\begin{aligned} \sum_{i=n-r+1}^n c_i - \sum_{i=1}^n c_i w_i &= \sum_{i=n-r+1}^n (c_i - a)(1 - w_i) \\ &\quad + \sum_{i=1}^{n-r} (a - c_i)w_i \geq 0. \end{aligned} \quad (\text{WS15})$$

To verify equality, expand the right side; its coefficient of a is $-r + \sum_{i=n-r+1}^n w_i + \sum_{i=1}^{n-r} w_i = 0$, leaving the displayed difference. Each remaining summand is nonnegative because of the sorted eigenvalues and WS14.

Applying this calculation to $-C$ gives the lower bound. When $r = 0$ the trace and both sums are zero; when $r = n$, $E = I$ and both sides equal $\sum c_i$. Thus in every case

$$\sum_{i=1}^r c_i \leq \text{Tr}(EC) \leq \sum_{i=n-r+1}^n c_i. \quad (\text{WS16})$$

This proof requires no strict eigenvalue inequality. It also gives the precise equality condition. At the upper endpoint, E is identity on all eigenspaces with eigenvalues strictly above the cutoff c_{n-r+1} , zero on all eigenspaces strictly below it, and may select the needed rank within the cutoff eigenspace. Indeed a strict coefficient in WS15 forces $w_i = 1$ or $w_i = 0$, and those identities imply $Ee_i = e_i$ or $Ee_i = 0$ by orthogonal projection. The lower endpoint has the reversed condition. Repeated cutoff values therefore allow their full indicated subspace choices rather than requiring a distinguished eigenbasis there.

Apply WS16 to each projection in WS11. Their ranks give, for $X = U$ or W ,

$$L_q(c) \leq \text{Tr}(XC) \leq H_q(c), \quad (\text{WS17})$$

where the exact endpoints are

$$\begin{aligned} L_q(c) &= 2 \sum_{j=1}^{q-1} c_j + c_q + c_{q+1}, \\ H_q(c) &= c_{q+1} + c_{q+2} + 2 \sum_{j=q+3}^{2q+1} c_j. \end{aligned} \quad (\text{WS18})$$

The sums of length $q - 1$ are empty for $q = 1$. The upper bound is the sum of the largest $q + 1$ eigenvalues and the largest $q - 1$ eigenvalues; the lower bound is the corresponding sum of the smallest eigenvalues. They are attained by nested spectral subspaces of these ranks, so the rank bounds are simultaneously consistent with the nested form WS12.

Subtracting WS17 for U and W , in both orders, proves

$$\begin{aligned} |\text{Tr}((U - W)C)| &\leq H_q(c) - L_q(c) \\ &= c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \end{aligned} \quad (\text{WS19})$$

No commutation between U , W , and C has been assumed. All traces are real: for self-adjoint X , C , conjugating $\text{Tr}(XC)$ gives $\text{Tr}(CX) = \text{Tr}(XC)$.

The bound in WS19 is the optimal one from the separate spectra WS10 and WS13. To prove this assertion explicitly, take the two endomorphisms whose diagonal entries in the eigenbasis of C are the list

$$(0, \dots, 0, 1, 1, 2, \dots, 2) \quad \text{and its reverse.} \quad (\text{WS20})$$

Their spectra are precisely WS10. The first attains $H_q(c)$, the second $L_q(c)$, and their difference attains the positive right side of WS19. Exchanging them attains the negative endpoint. This remains true for repeated c_i , with equality inside any repeated block understood through WS15. It proves a sharp statement about the indicated unitary orbits; additional alignment information from the literal polynomial and relation flags remains available and can improve a particular instance.

92.0.5 4. Integration along the unchanged arithmetic Gram path

The original $D = M_0^{-1}M_1$ is positive and self-adjoint for M_0 , because $M_0D = M_1$ is Hermitian and its quadratic form is positive. Thus it is diagonalizable with positive eigenvalues

$$0 < b_1 \leq \dots \leq b_{2q+1}, \quad \kappa = b_{2q+1}/b_1. \quad (\text{WS21})$$

Factor WS2 in its actual order:

$$M(t) = M_0((1-t)I + tD), \quad \Delta M = M_0(D - I), \quad C(t) = ((1-t)I + tD)^{-1}(D - I). \quad (\text{WS22})$$

The denominator is invertible because each of its eigenvalues $1 - t + tb_i$ is positive. Also $M(t)C(t) = \Delta M$ is Hermitian, proving self-adjointness for the exact source metric used in WS4. Functional calculus on the diagonalization of D gives its eigenvalues

$$c_i(t) = \frac{b_i - 1}{1 - t + tb_i}. \quad (\text{WS23})$$

The derivative of $(b-1)/(1-t+tb)$ with respect to b is exactly $(1-t+tb)^{-2} > 0$. Hence the same ordered list WS21 supplies the ordered list WS13 at every t , including $t = 0, 1$, repeated eigenvalues, and values $b_i = 1$. No eigenvalue branch relabelling is needed.

The original determinant transport proves, with these exact projections,

$$\Delta \mathcal{B} = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}} = \int_0^1 \text{Tr}((U(t) - W(t))C(t))dt, \quad T_N = V_N(1)/V_N(0). \quad (\text{WS24})$$

Here V_N is the determinant of the original least-norm quotient Gram, so WS24 is the signed arithmetic/reference endpoint correction. The identity was read in the complete sealed source; this review supplies the new spectral refinement. Its integrand is continuous: projection coefficients are rational expressions in positive finite Grams and their positive relation restrictions, so their inverses remain continuous on the compact path.

The elementary derivative

$$\frac{d}{dt} \log(1 - t + tb_i) = c_i(t)$$

holds also when $b_i = 1$, where both sides are zero. Thus $\int_0^1 c_i(t)dt = \log b_i$. Apply WS19 at every t , integrate, and use the triangle inequality in WS24. This gives the complete ordered-eigenvalue estimate

$$|\Delta \mathcal{B}| \leq \mathcal{S}_q(b) := \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}. \quad (\text{WS25})$$

All ratios in WS25 are at least one and are literal ratios of eigenvalues of the original relative Gram. Every such ratio is at most κ . The total positive coefficient count is $1 + 2(q-1) = 2q-1$. Therefore

$$|\Delta \mathcal{B}| \leq \mathcal{S}_q(b) \leq (2q-1) \log \kappa. \quad (\text{WS26})$$

For $q = 1$, the statement is exactly $|\Delta \mathcal{B}| \leq \log(b_3/b_1)$; there is no sum and no factor two. If all b_i equal a positive scalar c , then WS23 makes $C(t)$ a scalar multiple of the identity and the traces of U, W cancel. Equations WS25–WS26 give zero as well. The original common scalar mass remains present in the individual Grams; it cancels only in this signed ratio.

The coefficient $2q-1$ cannot be decreased using only the separate spectra of U, W and the extremes of D . For any retained numbers $0 < a < d$, choose the ordered data

$$b_1 = \cdots = b_q = a, \quad b_{q+1} = c \in [a, d], \quad b_{q+2} = \cdots = b_{2q+1} = d. \quad (\text{WS27})$$

Then every ratio in WS25 is d/a , so $\mathcal{S}_q(b) = (2q-1) \log(d/a)$. Choose the two commuting spectral flags WS20 in a fixed M_0 -orthogonal eigenbasis of D . Since $M(t)$ only changes each basis vector's squared norm by the positive factor $1 - t + tb_i$, these coordinate projections stay orthogonal for every original t in this abstract comparison. They attain WS19 with the same sign at every t , hence its integrated value is exactly WS25 and WS26. This is an equality construction within the stated spectral-data class, preserving arbitrary positive a, d . It does not identify arbitrary spectral flags with the actual multiplication-by- χ flag.

92.0.6 5. What the literal polynomial flag retains beyond the spectra

The original relation and polynomial flags impose the exact incidence

$$\chi\mathcal{P}_{q-1} \cap \mathcal{P}_{q-1} = 0. \quad (\text{WS28})$$

A nonzero product χP has degree $q + \deg P \geq q$, whereas every polynomial on the right has degree at most $q - 1$; this proves WS28. In particular independent opposite eigenvalue alignments need not be attainable by these specific subspaces.

For example, assume C has strictly increasing eigenvalues and $q \geq 2$. Equality at the positive endpoint of WS19 forces U to have its eigenvalue-two space on the largest $q - 1$ eigenspaces of C , and forces W to vanish on its largest q eigenspaces, by the equality criterion in WS15. Hence $L_{U,2} \subseteq L_{W,0} = \mathcal{P}_{q-1}$. But $L_{U,2} \subseteq \chi\mathcal{P}_{q-1}$ and has positive dimension $q - 1$, contradicting WS28. At the negative endpoint the same argument uses the smallest eigenspaces: minimizing U puts its eigenvalue-two space in the smallest $q - 1$ eigenspaces, while maximizing W makes those eigenspaces part of its kernel. Thus that endpoint is also impossible for these literal flags and simple C .

At $q = 1$, the actual formulas reduce to

$$U = Q_2 = I - P_{\mathcal{R}_2^\perp}, \quad W = I - P_0, \quad U - W = P_0 - P_{\mathcal{R}_2^\perp}. \quad (\text{WS29})$$

For simple C , a positive equality in WS19 makes $\ker U = \mathcal{R}_2^\perp$ the lowest eigenline and $\ker W = P_0$ the highest eigenline. They are orthogonal, which would force $P_0 \subseteq \mathcal{R}_2 = \chi\mathcal{P}_1$. This is impossible because the nonzero constant polynomial is not divisible by the degree-one χ . The negative endpoint is excluded by exchanging the lowest and highest eigenlines. The strictness statement concerns nonconstant simple spectral data; repeated cutoff eigenspaces have precisely the freedom retained in WS15.

These incidence calculations identify an exact additional relation that the source-specific unitary, commutator, and full-jet observations can use. The spectral estimate WS25 remains valid for the original flags, while its equality statement is explicitly scoped. The signed determinant correction itself remains WS24, with all four original endpoints and all relation primitives present.

92.0.7 6. The original one-dimensional quotient gives a nonlinear bound

This section concerns exactly $q = 1$, with the original common source $\mathcal{P}_2$, original vertical line $S = k/2 + iu$, original positive measures m_t , and original degree-one relation polynomial $\chi = S - a$. It makes no assertion that the same two-line formula applies in higher quotient dimension.

The four cutoff indices become 0, 1, 1, 2, so the two copies of V_1 cancel in this specified ratio and

$$\mathcal{B}(t) = \log \frac{V_0(t)}{V_2(t)}. \quad (\text{WS30})$$

Let $e = 1$ be the original constant polynomial, $\mathcal{N} = (\chi\mathcal{P}_1)^\perp$, and $z = R_2(t)[1]$ its canonical minimum-norm representative. Every polynomial with remainder one differs from e by an element of $\chi\mathcal{P}_1$, so orthogonal minimization gives $z = P_{\mathcal{N}}e$. The norms are the unchanged quotient values

$$\langle e, e \rangle_t = V_0(t), \quad \langle z, z \rangle_t = V_2(t), \quad \langle e, z \rangle_t = V_2(t) > 0. \quad (\text{WS31})$$

The last equality follows by the orthogonal splitting of e ; its sign and phase are fixed. Positivity of V_2 follows independently from $J_2 z = 1$, which makes $z \neq 0$. Hence the angle between the two lines has

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)} > 0. \quad (\text{WS32})$$

This ratio is strictly below one for the actual source measures. Indeed equality would give $e \in \mathcal{N}$, hence both $\int \chi(S)m_t = 0$ and $\int S\chi(S)m_t = 0$. On the retained vertical line, $\bar{S} = k - S$, so

$$|\chi(S)|^2 = ((k - \bar{a}) - S)\chi(S). \quad (\text{WS33})$$

The two vanishing integrals would force $\int |\chi(S)|^2 m_t = 0$. This contradicts positive definiteness of the original Gram on the nonzero polynomial χ ; alternatively positivity almost everywhere of m_t and the finite

zero set of χ give the contradiction directly. Thus $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the entire compact interval. This argument uses the actual source line and its measure, not an arbitrary coefficient Gram calibration.

Put $p = e/\sqrt{V_0}$ and $n_0 = z/\sqrt{V_2}$. Their inner product is the positive real number $c = \sqrt{V_2/V_0}$. Set $s = \sqrt{1 - c^2} > 0$, and $r = (n_0 - cp)/s$. Directly $\langle p, r \rangle_t = 0$ and $\|r\|_t = 1$. These are explicit coordinates with the original norms displayed, so no Gram factor or mass is removed. In the basis (p, r) , the difference in WS29 is

$$P_0 - P_{\mathcal{N}} = \begin{pmatrix} s^2 & -sc \\ -sc & -s^2 \end{pmatrix} \quad \text{on } \text{span}(p, r), \quad (\text{WS34})$$

and is zero on its orthogonal complement. The displayed matrix has trace zero and determinant $-s^2$, so the full three eigenvalues are $s, 0, -s$. This proves the exact alignment-dependent spectrum

$$\text{spec}(U - W) = \left\{ \sqrt{1 - e^{-\mathcal{B}(t)}}, 0, -\sqrt{1 - e^{-\mathcal{B}(t)}} \right\}. \quad (\text{WS35})$$

Write its positive and negative rank-one spectral projections as E_+ and E_- . Then $U - W = s(E_+ - E_-)$. Applying the already proved rank-one bound WS16 to their two traces gives

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{WS36})$$

Here $\dot{\mathcal{B}}$ is the original derivative in WS24; the equality identifying it uses all four original window signs even though their scalar endpoint expression has the cancellation in WS30.

For real $B > 0$, let $\Phi(B) = 2 \text{arcosh}(e^{B/2})$, using the nonnegative real inverse of cosh. Differentiating $\cosh(\Phi(B)/2) = e^{B/2}$ gives

$$\Phi'(B) = \frac{e^{B/2}}{\sqrt{e^B - 1}} = \frac{1}{\sqrt{1 - e^{-B}}}. \quad (\text{WS37})$$

Since $\mathcal{B}(t)$ is continuous and strictly positive on the compact interval, it has a positive minimum; every composition and division in WS37 is therefore defined and smooth on this whole path. Multiplying WS36 by WS37 yields $|(\Phi \circ \mathcal{B})'| \leq c_3 - c_1$. Integrating WS23 gives the exact nonlinear endpoint control

$$\left| 2 \text{arcosh}(e^{\mathcal{B}(1)/2}) - 2 \text{arcosh}(e^{\mathcal{B}(0)/2}) \right| \leq \log \frac{b_3}{b_1}. \quad (\text{WS38})$$

Equivalently, put $A_0 = \text{arcosh}(e^{\mathcal{B}(0)/2})$ and $L = \frac{1}{2} \log(b_3/b_1)$. Then $\text{arcosh}(e^{\mathcal{B}(1)/2})$ lies between $\max(0, A_0 - L)$ and $A_0 + L$. The function $A \mapsto 2 \log \cosh A$ is increasing for $A \geq 0$, because its derivative is $2 \tanh A \geq 0$. Hence

$$2 \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2 \log \cosh(A_0 + L). \quad (\text{WS39})$$

The lower bound may be zero although the actual value was proved strictly positive. A scalar relative Gram gives $L = 0$ and the exact equality of the two original endpoint values, in agreement with WS25. This is an additional analytic calculation on the actual $q = 1$ source, retaining its canonical line angle.

92.0.8 7. Complete independent review of the root AW1-AW23 source

The complete tau_relation_window_spectral_transport_20260913.tex was read and checked at SHA-256 c79e76c087efd59871e3234ff44fb18964515dbd34f0318dffba6b3f1440f6ec. No mathematical correction is requested at that pin.

The following checks cover the new assertions rather than only their final inequalities.

1. **Original objects and types.** AW1-AW6 retain the exact arithmetic Taylor unit, two distinct injected observations into the full tensor jet module and into $Q^{\otimes k}$, the common $2q$ coefficient cutoff, the full masses, and the same-metric projection formulas. The correction sign agrees with the four terms of the sealed AT15, whose source pin is 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae.

2. **Smooth actual isometry.** The Gram recursion AW9 retains every positive norm a_j . Linear independence of each original list makes every residual nonzero; positive square roots are smooth along the entire compact path. The canonical quotient list lies in $(\chi\mathcal{P}_q)^\perp$ because of the original minimum-section orthogonality and spans it because $J_{2q}R_{2q} = I$. Thus AW10 maps an actual orthonormal basis to another one, with matching eigenvalues from WS7 and WS9. It proves $T^*MT = M$ and $UT = TW$, including the empty middle list at $q = 1$.
3. **Commutator and unchanged cohomology observation.** Finite trace cyclicity gives $\text{Tr}((TWT^{-1} - W)C) = \text{Tr}(WT^{-1}(CT - TC))$, with precisely the displayed sign. The original observation has kernel $\chi\mathcal{P}_q$ because its two later maps are injective. Therefore $\ker(\mathcal{O}T) = T^{-1}(\chi\mathcal{P}_q)$. The explicit AW10 basis assignment identifies this with $\text{span}(p_q, \dots, p_{2q}) = \mathcal{P}_{q-1}^\perp$. The full observation defect $\sigma_h^{\otimes k} \eta J_{2q}(T - I)$ is retained, so this isometry is not assigned an unwritten identity on arithmetic classes. The support lift preserves the label and gives exactly that supported-zero inverse image.
4. **Repeated spectra and integration.** AW13–AW17 agree with the independently derived WS14–WS26. The zero rank at $q = 1$, repeated Gram eigenvalues, values $b_i = 1$, and the full coefficient $2q - 1$ are all covered. The claimed sharpness is expressly restricted to the separate spectral constraint; its text does not freely orient the original polynomial relations. AW18 is a correct substitution of the supplied Gamma lower estimate into this new finite-source bound, with the actual quartet restriction retained.
5. **One-dimensional continuation.** AW19–AW23 are independently proved in full by WS30–WS39 above. The vertical-line identity, positive phase of $\langle e, z \rangle$, exact two-dimensional projection matrix, remaining zero eigenvalue, positive derivative divisor, and both real hyperbolic-function branches are correct. The factor on the right of AW22 is $\log(b_3/b_1)$, while AW23 correctly uses half that number in its cosh arguments. The calculation remains explicitly restricted to $q = 1$.

The whole AW source is accepted at the exact pin above. The original finite Gram data remain the input to the proved estimates; no uniform bound in k has been presumed.

92.0.9 8. Source and verification scope

The complete sealed tau_arithmetic_determinant_transport_20260913.tex was read. Its AT11–AT15 fix the original common Hilbert space, projection conventions, correction sign, and relative Gram path used here. This note proves WS5–WS39 directly from those data and the actual $q = 1$ measures; it does not alter the sealed source or its PDF. The complete root AW1–AW23 source was also read and accepted as recorded above. These complete written proofs constitute the closed mathematical review; no supplementary source is needed for any acceptance recorded here.

No external trace extremum theorem was used: WS14–WS16 are its direct finite-dimensional proof, including repeated eigenvalues. No Lean execution, remote mutation, numerical-only proof, source-multiplier boundedness hypothesis, or unproved asymptotic estimate is asserted.

93 The complete original arithmetic Taylor-unit formal gauge

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. b0d60706b6250e4c8c0ede3db0b847852af20299076518ebd76bc43775ad3e9c

Complete source. sources/next_edition/unit_gauge_conversions_v1/sources/f1_unit_gauge/ACTUAL_TAYLOR_UNIT_FORMAL_GAUGE.md

Source SHA-256. b0d60706b6250e4c8c0ede3db0b847852af20299076518ebd76bc43775ad3e9c

93.0.1 The original arithmetic Taylor unit as a formal geometric gauge

13 September 2026. This calculation continues the original cohomology over the absolute tau base. It retains the actual unit $j_h(g/h)$ in the exponential deformation, constructs every localization and inverse map, and proves that the newly introduced pole quotient is contractible in the specified formal neighbourhood. It does not replace that unit by a constant, replace an arithmetic metric, or identify formal and analytic cohomology.

93.0.2 1. Fixed arithmetic input

Use the retained theta complex and its actual finite packet:

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad g = 2\xi, \quad h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}. \quad (\text{UG1})$$

Every selected order is its full original zero order. Set $d = \deg h$, $E_h = \mathbb{C}[s]/(h)$, and retain

$$v_h = j_h(g/h), \quad \nu(s) = \text{rem}_h(v_h), \quad \epsilon_h = v_h^{-1} = j_h(h/g). \quad (\text{UG2})$$

Here rem_h in the middle denotes the unique polynomial of degree less than d with those full Hermite jets, not a remainder of an entire function obtained by dropping its Taylor derivatives. The actual g/h is nonzero at every selected root, so ν and h are relatively prime. Polynomial Euclidean division gives coefficients a, b with $ah + b\nu = 1$. These are facts about the retained packet. No additional zero of g is postulated.

The original arithmetic source section and cyclic tensor inclusion stay

$$\begin{aligned} J_Z R_{\text{ref}} &= I, \quad \mathcal{M} R_{\text{ref}} x = (g/h) \text{rem}_h(\epsilon_h x), \\ \eta_{h,k}[P] &= v_h^{\otimes k} P(A_h^{(k)}) 1, \quad A_h^{(k)} = \sum_{i=1}^k A_i. \end{aligned} \quad (\text{UG3})$$

The full inverse unit remains in the source section, and the full direct unit remains in eta. They are not mutually cancelled across different arrows or used to change the original theta norm.

93.0.3 2. The actual completed rings and pole quotient

Write s for the single-factor polynomial coordinate and (u, t) for the existing deformation parameters. The two rings used below are

$$\widehat{B} = \mathbb{C}[s][[u, t]], \quad \widehat{B}_\nu = \mathbb{C}[s, \nu(s)^{-1}][[u, t]]. \quad (\text{UG4})$$

They are coefficientwise formal power series. In the second ring the power of ν in a denominator may grow with the (u, t) coefficient. We do not silently replace it by a localization allowing only one uniform denominator for the whole series.

Let

$$M = \mathbb{C}[s, \nu^{-1}]/\mathbb{C}[s].$$

Coefficientwise inclusion and quotient give the exact sequence

$$0 \longrightarrow \widehat{B} \xrightarrow{i} \widehat{B}_\nu \xrightarrow{\pi} M[[u, t]] \longrightarrow 0. \quad (\text{UG5})$$

An explicit section σ of π is the proper rational part at infinity. Divide each rational function P/ν^n into its polynomial part and a remainder with numerator degree strictly smaller than $\deg(\nu^n)$. The proper part depends only on the class modulo polynomials: a rational function both polynomial and vanishing at infinity is zero. This defines σ independently of the presentation with denominator ν^n . Apply it coefficient by coefficient. If ν is a nonzero constant, M is zero, π and σ are zero, and all the formulas below remain valid.

93.0.4 3. Multiplication by h is invertible on the pole quotient only

Multiplication by h on M is injective. If hP/ν^n is polynomial, relative primality forces ν^n to divide P , so the original class is zero. It is surjective: choose a_n, b_n with

$$a_n h + b_n \nu^n = 1.$$

For $[P/\nu^n]$, the class $[a_n P/\nu^n]$ maps to it, since the difference is the polynomial $-b_n P$. Injectivity shows that this inverse does not depend on the Bezout representative. Denote this exact inverse by H .

$$H = (M_h|_M)^{-1} : M \longrightarrow M. \quad (\text{UG6})$$

This inverse is **not** an inverse to h on E_h : there multiplication by h is still zero with all original jets present. The two operators occur on the different, explicitly related terms of (UG5).

Differentiation preserves both rational functions with poles among zeros of ν and polynomials; it therefore induces ∂_s on M . Neither this derivative nor H is asserted to commute with the other.

93.0.5 4. An explicit contracting homotopy for the additional pole complex

The original formal exponential differential is

$$D = u\partial_s + h(s) - t, \quad \widehat{C} = [\widehat{B} \xrightarrow{D} \widehat{B} ds], \quad \widehat{C}_\nu = [\widehat{B}_\nu \xrightarrow{D} \widehat{B}_\nu ds]. \quad (\text{UG7})$$

It respects (UG5). On its pole quotient put $E = u\partial_s - t$ and

$$J = D_M^{-1} = \sum_{n=0}^{\infty} (-HE)^n H. \quad (\text{UG8})$$

The order of these operators is part of the formula. The operator HE raises (u, t) -adic order by at least one. Thus each fixed coefficient of (UG8) contains only finitely many terms; this is a defined continuous operator on $M[[u, t]]$. Because $D_M = M_h(I + HE)$, multiplication of finite partial sums gives a remainder of arbitrarily high parameter order. Completeness yields both $JD_M = I$ and $D_M J = I$.

It follows that the two-term quotient complex

$$[M[[u, t]] \xrightarrow{D_M} M[[u, t]] ds] \quad (\text{UG9})$$

is contractible, with degree-minus-one homotopy J . This conclusion uses the displayed inverse, not a claim that a common support label makes a kernel vanish. In particular the entire arithmetic quotient of the original complex is retained.

There is an explicit deformation retraction, not only a cohomology count. Set

$$K = \sigma J \pi : \widehat{C}_\nu^1 \longrightarrow \widehat{C}_\nu^0, \\ r^0 = I - KD = I - \sigma \pi, \quad r^1 = I - DK. \quad (\text{UG10})$$

Each r has its image in the corresponding original polynomial-series space: for r^0 this follows from its definition, and $\pi r^1 = \pi - D_M J \pi = 0$. Direct multiplication gives

$$r^1 D = D r^0, \quad r i = I_{\widehat{C}}, \quad i r = I_{\widehat{C}_\nu} - (DK + KD). \quad (\text{UG11})$$

In this last expression the degree-one component is DK and the degree-zero component is KD ; no sum of incompatible-degree maps is intended. Since $\pi i = 0$, one also has $K i = 0$. Formula (UG10) is a coefficientwise evaluable inverse to localization in the derived category. It retains the explicit pole complex and contracts it.

For comparison, the map from the mapping cone of i to (UG9) sends its localized component to its quotient and the other component to zero. The kernel is the cone of the identity on the original complex, which contracts directly. Thus the mapping-cone account and (UG10) give the same vanishing of this additional pole contribution.

93.0.6 5. The original Taylor unit supplies a gauge isomorphism

Multiplication by ν is invertible on $\widehat{B}_\nu$. Compute its conjugation of the unchanged differential:

$$D^\nu = \nu D \nu^{-1} = u\partial_s + h - t - u \frac{\nu'}{\nu}. \quad (\text{UG12})$$

Indeed $\partial_s(P/\nu) = P'/\nu - P\nu'/\nu^2$. Consequently

$$D^\nu(\nu P) = \nu DP$$

for every localized formal coefficient. Multiplication by ν in both degrees is an exact cochain isomorphism

$$\widehat{C}_\nu \xrightarrow{\times \nu} \widehat{C}_\nu^\nu = [\widehat{B}_\nu \xrightarrow{D^\nu} \widehat{B}_\nu ds]. \quad (\text{UG13})$$

Its inverse is multiplication by the same full rational ν^{-1} . The new rational derivative term is written explicitly, with its sign and coefficient ν ; it is not dropped or called a change of coordinates.

Combining (UG11) and (UG13) provides the geometric arrow

$$\boxed{\widehat{C} \xrightarrow{i} \widehat{C}_\nu \xrightarrow{\times \nu} \widehat{C}_\nu^\nu} \quad (\text{UG14})$$

It is a quasi-isomorphism of formal complexes. The original special fibre is $[\mathbb{C}[s] \xrightarrow{h} \mathbb{C}[s] ds]$, with $H_1=E_h$ and $H_0=0$. The localized special fibre has the same cohomology, because ν is invertible modulo h . Under those original coefficient identifications, the special-fibre map (UG14) is exactly

$$E_h \xrightarrow{\times \nu_h} E_h. \quad (\text{UG15})$$

At this fibre $D^\nu = h$, but the cochain map is still multiplication by ν . In particular a basis change using $\nu, \nu s, \dots$ must retain its transition matrix to the original monomial frame; otherwise it would conceal the actual arithmetic unit in (UG15).

93.0.7 6. The same arithmetic scaling flow commutes with this gauge

The previously constructed regular lift of theta dilation is

$$L = u\partial_t - s, \quad [D, L] = 0, \quad \exp(aL)P = e^{-as}P(u, t + ua, s) \quad (\text{UG16})$$

in the a -adic cochain completion. Because ν depends only on s ,

$$[L, M_\nu] = [L, M_{\nu^{-1}}] = 0, \quad [D^\nu, L] = \nu[D, L]\nu^{-1} = 0. \quad (\text{UG17})$$

Thus (UG14) intertwines the **same** formal scaling operator L and its a -adic exponential. No modified scaling generator or replacement moment line is needed. The proper-rational retraction need not itself commute strictly with L : its exact chain homotopies remain available; the quasi-isomorphism i and the gauge do commute strictly.

In the target frame represented by νs^j , the induced finite flow matrix is the already calculated C_a . In the special-fibre original frame, the transition is the full multiplication matrix M_{ν_h} . It commutes with $p^{\pm A_h}$ because both are multiplication operators on the same full algebra, not because its higher jets have been ignored. The finite entire evaluation of C_a for $a = \log p$ is as proved in the scaling note; no convergence of the rational formal gauge in arbitrary analytic (u, t) neighbourhoods is inferred here.

93.0.8 7. The tensor inclusion is recovered without changing the sum module

Apply the construction to each original h factor, using its coordinate s_i and its deformation parameters. External tensor maps have degree zero, so tensoring (UG14) introduces no new Koszul sign in those maps; the differentials keep the original tensor signs. More precisely use independent parameters (u_i, t_i) and the coefficientwise completion defined by the cofinal rectangular truncations $(u_i, t_i)^{n_i+1} = 0$ for every factor i . Each fixed multi-index receives only finitely many terms from the homotopies (UG10).

There is an explicit tensor homotopy, so no inverse-limit exactness assertion is needed. Set $P_i = i_i r_i$. On homogeneous tensors the j -th term of the tensor homotopy is

$$(-1)^{|x_1|+\dots+|x_{j-1}|} P_1 x_1 \otimes \dots \otimes P_{j-1} x_{j-1} \otimes K_j x_j \otimes x_{j+1} \otimes \dots \otimes x_k.$$

Summing these terms gives K_{total} . The differential's original Koszul rule cancels its cross terms and gives $dK_{\text{total}} + K_{\text{total}}d = I - P_1 \otimes \dots \otimes P_k$, by telescoping. The other retraction identity follows by tensoring

$r_i i_i = I$. These are coefficientwise identities on every rectangular truncation and hence on the specified completed coefficient space.

The special-fibre map is

$$\mathcal{E}_k = E_h^{\otimes k} \xrightarrow{M_{v_h}^{\otimes k}} \mathcal{E}_k. \quad (\text{UG18})$$

Compose it with the original cyclic inclusion

$$E_{h,k} \longrightarrow \mathcal{E}_k, \quad [P] \longmapsto P(A_h^{(k)})1.$$

The resulting map is exactly eta in (UG3). This proof does not assume that the full tensor unit is a polynomial in the single sum coordinate S ; different occupancies with the same sum remain in the original tensor object. The original cyclic annihilator and its full jet lengths are unchanged. The original section into $Q^{\otimes k}$ and the tau-base pushforward therefore receive the very same map eta.

For any fixed admitted support mask, every map above has its existing split lift $(\text{mask}, v) \mapsto (\text{mask}, f(v))$, with tau going to tau. A formal coefficient specializing to zero goes to that mask's e. The structural map over $\mathbf{F}_{1,\tau}$ is retained; neither localization at e nor restriction of global cohomology to the sigma skeleton has been used.

93.0.9 8. The residue-dual arrow retains the inverse unit as well

For this section use the original reflection-stable packet of the tau-base residue construction. Then dagger is an endomorphism of its full jet space. The localization and gauge proofs above did not need reflection; this specifies the original domain of the residue-dual arrow, rather than silently applying it to an arbitrary packet.

Let the original polynomial residue pairing be

$$\mathcal{S}_h(f, v) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f(s)v(s)}{h(s)} ds.$$

On full jets, the arithmetic residue is exactly

$$R_Z(f, v) = \mathcal{S}_h(f^\dagger, \epsilon_h v), \quad \epsilon_h = j_h(h/g) = v_h^{-1}. \quad (\text{UG19})$$

To prove it, write $g = h(g/h)$ in each germ. The inverse of g/h agrees with epsilon to every order needed for the residue, so replacing that inverse by epsilon changes the integrand by a holomorphic germ. The original dagger carries its full alternating Taylor signs.

The dual of (UG15) is its actual transpose, and the inverse-dual transport uses $M_{v_h}^{-T}$. After tensoring and composing the source's residue-dual arrow to $K_{\tau,k}(L_k)$, (UG19) retains the same inverse units that already occur there. The weight factor p^k is still the original moment representation. Thus the formal gauge joins the direct unit-bearing inclusion and its specified residue dual, rather than discarding one of their arithmetic factors.

93.0.10 9. Exact scope for the ongoing geometric argument

This is a formal-neighbourhood computation over the original base program. It establishes an explicit quasi-isomorphism carrying the actual Taylor unit, its localization cone and contraction, its commuting arithmetic scaling, and its tensor and dual maps.

The Neumann series was justified in the (u, t) -adic topology only. Specializing it at a nonzero complex u is not an authorized inference from that proof. Such an analytic comparison would require its actual continuation and contour/pole contribution to be calculated. The original canonical theta metrics and their source/relation volumes remain the analytic quantities in the separate continuation; this note neither chooses a new metric nor asserts a weight bound.

The completed result improves the concrete program nonetheless: the original arithmetic unit is now present in a proved formal geometric map through the same deformation, with its exact extra term and contractible pole quotient, rather than being left outside that map.

93.0.11 Source and verification record

The original tau-base note was read completely in the preceding turn, SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. The full current tex/arithmetic_input.tex was also read completely; A7–13 give the full unit, actual source section, quotient square and equivariant roof used here. The original exponential family and its constant residue duality are those of the retained exponential note, not a new arbitrary replacement. The adjoining accepted scaling and reflection notes supply the exact flow and moment-weight transport.

All localization, contraction and gauge claims above have their elementary proofs in this note. Finite algebra fixtures are supporting checks and do not certify actual zero data, all analytic coefficients, or an RH statement. No new Lean or public upload is claimed here.

94 Complete independent review of the actual Taylor-unit formal gauge

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 5cc96c73e8fabda87d87a5a58184f059361a38210a9696cf498caba911423316

Complete source. sources/next_edition/unit_gauge_conversions_v1/sources/f1_unit_gauge_review/INDEPENDENT_REVIEW.md

Source SHA-256. 5cc96c73e8fabda87d87a5a58184f059361a38210a9696cf498caba911423316

94.0.1 Independent review of the actual-unit formal gauge

Date: 2026-09-13. Read ACTUAL_TAYLOR_UNIT_FORMAL_GAUGE.md completely, UG1–19 and its scope/source paragraphs. The original tau-base note, arithmetic_input.tex, and original exponential comparison were read in full in the adjoining independent scaling work. This review does not edit the source note, run its fixtures, or claim a Lean or analytic-convergence result.

94.0.2 Disposition and one domain clarification

Accept UG1–18 and the formal construction. The pole inverse has the correct order, the coefficientwise completions are essential and correctly specified, and the explicit retract proves the claimed quasi-isomorphism. The tensor map retains the complete direct Taylor units and recovers the original eta without an assumption that their tensor product is a function of the sum coordinate alone.

UG19's residue identity is correct on the original reflection-stable packet used in the tau-base residue construction. That domain should be explicit here: UG1 otherwise admits any full-order selected packet, while dagger is not automatically a well-defined endomorphism of its quotient. The two equivalent precise typings are:

- use the already retained reflection-stable packet, so dagger maps E_h to E_h ; or
- for arbitrary h , put $h^\dagger(s) = (-1)^{d_h} \overline{h(1 - \bar{s})}$ (the monic reflected polynomial) and take the first argument from $E_{h^\dagger}$, so its dagger lies in E_h .

No change to the full-order arithmetic source or formal gauge is needed. This is the exact domain of the existing residue arrow, not a new analytic hypothesis. The issue and the tensor-completion convention below were recorded before writing this review.

94.0.3 Coefficientwise quotient and its inverse

Let $A = \mathbb{C}[s]$, $A_\nu = \mathbb{C}[s, \nu^{-1}]$, and $M = A_\nu / A$. A rational function has a unique decomposition into a polynomial and a proper rational function; two proper representatives differing by a polynomial are equal. Thus the stated section sigma is well defined, including when different denominator powers represent the same rational function. Applying this section to each (u, t) coefficient proves surjectivity in UG5 without assuming a uniform denominator. The constant-nu case gives $M = 0$.

Since $\gcd(h, \nu) = 1$, hP/ν^n being polynomial forces $\nu^n \mid P$. Conversely $a_n h + b_n \nu^n = 1$ gives an inverse class $[a_n P/\nu^n]$. These prove injectivity and surjectivity of M_h on M , and uniqueness makes its inverse H independent of all Bezout choices. This inverse acts on the additional pole quotient, not on E_h . Differentiation descends to M , but no commutation with H is needed.

Writing $E = u\partial_s - t$, the exact factorization is

$$D_M = M_h(I + HE).$$

Thus the inverse must be $(I + HE)^{-1}H$, which is precisely UG8. The powers of HE raise parameter order; at any fixed coefficient only finitely many terms contribute. Both inverse identities follow by finite geometric sums modulo each parameter power, then coefficientwise completion. This does not use a norm-convergent Neumann series or reorder H and the derivative.

94.0.4 Explicit retraction and formal finite cohomology

Using $\pi D = D_M \pi$ and $JD_M = D_M J = I$, one gets

$$KD = \sigma\pi, r^0 = I - \sigma\pi, \pi r^1 = 0.$$

The remaining formulas follow by multiplication: $r^1 D = Dr^0$, $ri = I$, and $ir = I - dK - Kd$. The degree-zero and degree-one occurrences of K are correctly typed. Thus no assumption about exactness of a completed cohomology functor is replacing the displayed deformation retraction. The mapping-cone description has the same quotient and the contractible identity-cone kernel.

The completed original polynomial complex also retains its expected finite coefficient basis. At the lowest parameter order, divide by h ; the $u\partial_s - t$ term only affects higher orders. Recursive monic division therefore constructs the unique degree-below- d formal remainder and its formal preimage, coefficient by coefficient. The lowest-order nonzero term of a putative kernel would be killed by the injective polynomial multiplier h , so the differential remains injective. Hence its cohomology is the coefficientwise completed free rank- d module. Unbounded s -degrees or pole orders across formal coefficients are allowed exactly as stated in UG4.

94.0.5 Gauge, special fibre, and scaling

Direct Leibniz differentiation gives

$$\nu D\nu^{-1} = u\partial_s + h - t - uv'/\nu.$$

Both the minus sign and factor u are correct. Multiplication by ν in both degrees is an isomorphism only on the explicitly localized complex, with inverse the same rational ν^{-1} . At $u = t = 0$, both target differentials are h , but this cochain map still multiplies by ν . Its induced map under the original quotient identification is therefore exactly multiplication by ν_h , not identity.

For the composite to the gauge target, an explicit inverse up to homotopy is $rM_{\nu^{-1}}$; its homotopy is $M_{\nu}KM_{\nu^{-1}}$. This also supplies a direct way to tensor the composite itself.

The operator $L = u\partial_t - s$ commutes with multiplication by ν and its inverse, because these depend only on s . Therefore its commutation with D transfers to D^ν with no additional term. The inclusions and gauge map strictly intertwine L and its formal a -adic exponential. It is correct not to demand that the particular proper-part retraction strictly intertwine L . The coefficient matrix in the transported frame νs^j is consequently the same formal C_a ; an entire evaluation of that finite matrix does not claim analytic convergence of the rational formal gauge. The note keeps these domains distinct.

94.0.6 Precise tensor-completion interpretation and homotopy

The tensor statement is valid with independent parameter pairs and the completed tensor product defined by the cofinal rectangular truncations $(u_i, t_i)^{n_i}$ in the finitely many factors. Each such quotient is the ordinary tensor product over $\mathbb{C}$ of its factor truncations. These rectangular filtrations are cofinal with total parameter-adic truncation, so they give the same coefficientwise formal product. Calling every total-degree truncation a literal tensor product would be inaccurate; the rectangular convention is the precise one under which UG7's tensor paragraph operates.

There is no need to infer inverse-limit exactness abstractly. Let $P_i = i_i r_i$ and use the source's homotopies K_i . For two factors set

$$K_{\text{tot}} = K_1 \otimes I + P_1 \otimes K_2,$$

with the graded convention $(P_1 \otimes K_2)(x \otimes y) = (-1)^{\deg x} P_1 x \otimes K_2 y$. Since P is a cochain map, direct expansion gives

$$dK_{\text{tot}} + K_{\text{tot}}d = (I - P_1) \otimes I + P_1 \otimes (I - P_2) = I - P_1 \otimes P_2.$$

For k factors the sum has $P_1, \dots, P_{j-1}$ before K_j and identities after it, with these same tensor signs. The terms telescope to $I - \bigotimes_i P_i$. All maps preserve parameter order, so this identity holds at each compatible rectangular truncation and hence coefficientwise in the completed product. The gauge-conjugated homotopies give the same statement for the targets with D_i^ν .

Specialization yields $M_{v_h}^{\otimes k}$ on the whole original tensor packet. Composing with $[P] \mapsto P(\sum A_i)1$ is exactly eta. All same-sum occupancies remain in the tensor module and no cyclic polynomial formula for the tensor unit is used.

94.0.7 Residue scope, support, and limits of this review

With the dagger domain fixed as above, $g = h(g/h)$ proves UG19: the complete inverse Taylor unit differs from the inverse germ by a multiple of h , so the residue difference is holomorphic. Its transpose and inverse-transpose transports keep the actual source inverse units; they do not assert a positive pairing or a globally invertible extension on every analytic deformed fibre.

Every operation is on the declared linear coefficient fibre, followed by its support-preserving lift. Inverting ν is not inverting the supported zero e . The formal pole contraction does not identify represented zero with τ or pass from τ -base pushforward to the generic skeleton. No specialization of UG8 at a nonzero complex u , no analytic period assertion, no metric estimate, and no tensor-growth bound is certified by this review. The affirmative result is the explicit formal quasi-isomorphism with its actual arithmetic unit and all the displayed kernels retained.

94.0.8 Final acceptance of the revised note

The preceding domain-clarification discussion is the historical review record, not an outstanding objection. On 2026-09-13 I reread the revised Section 7 and Section 8, including the full signed tensor homotopy and the explicit original reflection-stable packet restriction. Both comments are resolved in the text. In the tensor discussion above, “UG7’s tensor paragraph” means Section 7, not the equation labelled UG7.

The exact accepted source artifact is ACTUAL_TAYLOR_UNIT_FORMAL_GAUGE.md, SHA-256:

9ded2486160b76bf65f92d44b73cdd0c59a3ca7e5334f88682e10592d5a45b2f.

Final disposition: accept UG1–19 and the stated formal construction, with no outstanding mathematical correction. Independent parameter pairs and cofinal rectangular truncations make every tensor identity coefficientwise well defined; the displayed Koszul signs give $dK_{\text{total}} + K_{\text{total}}d = I - \bigotimes_i P_i$. Section 8 now has exactly the domain on which the original dagger and residue-dual arrow are defined, while the preceding gauge proof retains its greater generality. No assumption of analytic convergence, positivity, or global invertibility of the original arithmetic residue unit is added by this acceptance.

This final seal is a proof review. The separately reported exact fixtures and negative controls were not rerun by this reviewer.

95 Complete arithmetic determinant variations and their original determinant lines

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. dee20e249e980fdf0251513c3ab5d504e1a17e6d408c3e7ad69d08816a6f115e

Complete source. sources/next_edition/actual_tau_variation_v1/sources/variation_review/tau_arithmetic_determinant_variation_subreview_20260913.md

Source SHA-256. dee20e249e980fdf0251513c3ab5d504e1a17e6d408c3e7ad69d08816a6f115e

95.0.1 Independent verification of the arithmetic determinant variations

Date: 2026-09-13. Reviewed source: tau_arithmetic_determinant_transport_20260913.tex. Final accepted source SHA-256: 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae. Scope: AT3–AT10, including the literal section, inverse quotient Gram, determinant-line orientation, and full first and second variations. This subreview does not claim a review of the later four-window estimate or the analytic source construction.

The displayed formulas AT6–AT10 are correct. The first source, SHA-256 41c7b45a0a94f0e65d422b213a4bcbddedbf983ab5127da6095e449e001fef98, asserted that $M_N^{-1}J_N^*G_N$ had remainder I_E before establishing that fact. The author has now inserted the exact decomposition proof. I read the corrected paragraph, verified its final source hash above, and accept the repair. The observation is resolved, and no displayed formula changed. Section 3 supplies the complete independent derivation without assuming $(J_N M_N^{-1} J_N^*)G_N = I_E$.

95.0.2 1. Original finite spaces and their types

Fix the actual $h, k, \chi = \chi_{h,k}$ of the source, with $k \geq 1$, monic χ of degree $q \geq 1$, and $N \geq q - 1$. Set

$$U = \mathcal{P}_{N-q}, \quad P = \mathcal{P}_N, \quad E = \mathbb{C}[S]/(\chi), \quad r = N - q + 1. \quad (\text{AV1})$$

At $N = q - 1$, the convention is $U = \mathcal{P}_{-1} = \{0\}$, so $r = 0$. The source spaces carry their increasing-power coefficient frames, and E has its increasing-power remainder frame. The fixed maps are

$$B : U \longrightarrow P, \quad y \longmapsto \chi y, \quad J : P \longrightarrow E, \quad p \longmapsto [p]_\chi, \quad s : E \longrightarrow P, \quad (\text{AV2})$$

where s chooses the polynomial of degree less than q . Polynomial division by the original monic χ proves

$$JB = 0, \quad Js = I_E, \quad \ker J = \text{im } B, \quad P = \text{im } B \oplus \text{im } s. \quad (\text{AV3})$$

In particular B is injective. The coefficient y in $p = By + sJp$ is the actual polynomial quotient in division by χ .

All stars below are conjugate transpose in these fixed frames. Let

$$M(t) = M_N(t) = M(0) + t\Delta M, \quad \Delta M = M(1) - M(0), \quad H(t) = B^*M(t)B. \quad (\text{AV4})$$

These are the source's matrices from $m_t = (1 - t)m_0 + tm_1$ on the original line $S = k/2 + iu$; no scalar mass is divided out. Their entries have the original finite moments. For nonzero $p \in P$, polynomial evaluation vanishes at only finitely many real u , whereas m_t is positive almost everywhere, so $p^*M(t)p > 0$. Thus $M(t)$ is positive definite on $0 \leq t \leq 1$. For nonzero $y \in U$, injectivity of B gives $y^*H(t)y = (By)^*M(t)(By) > 0$.

The lower bound

$$M(t) \succeq \min\{\lambda_{\min}(M(0)), \lambda_{\min}(M(1))\}I_P$$

holds on the compact interval. Continuity gives a real open neighborhood of this interval on which $M(t)$ remains positive definite. The corresponding assertion holds for $H(t)$ when $r > 0$. All inverse matrices and sections below are smooth there. This uses finite matrices only.

95.0.3 2. The actual section and its minimum property

Every lift of $x \in E$ is uniquely $sx + By$, with $y \in U$. Define

$$d = H^{-1}B^*Ms : E \longrightarrow U, \quad R = s - Bd : E \longrightarrow P. \quad (\text{AV5})$$

For $x \in E$ and $y \in U$, direct expansion gives

$$\begin{aligned} (sx + By)^*M(sx + By) &= x^*(s^*Ms - s^*MBH^{-1}B^*Ms)x \\ &\quad + (y + dx)^*H(y + dx). \end{aligned} \quad (\text{AV6})$$

Here H^{-1} is Hermitian, and $d^*Hd = s^*MBH^{-1}B^*Ms$. Thus the unique least-norm lift is Rx , attained at $y = -dx$. Direct substitution gives

$$JR = I_E, \quad B^*MR = 0, \quad G := R^*MR = s^*Ms - s^*MBH^{-1}B^*Ms. \quad (\text{AV7})$$

The map R is injective because $JR = I_E$. Therefore G is positive definite, so its inverse exists before using the inverse quotient formula.

95.0.4 3. Inverse quotient formula without circularity

For every $p \in P$, the vector $p - RJp$ lies in $\ker J$. Exactness and injectivity of B therefore give a unique $y \in U$ with

$$p = By + RJp. \quad (\text{AV8})$$

Multiplication by R^*M , using $R^*MB = (B^*MR)^* = 0$, gives

$$R^*Mp = GJp \quad \text{for every } p \in P, \quad R^*M = GJ, \quad MR = J^*G. \quad (\text{AV9})$$

The last equality is the conjugate transpose of the preceding equality, since M, G are Hermitian. Multiplying by M^{-1} , then composing with J , proves

$$R = M^{-1}J^*G, \quad I_E = JR = (JM^{-1}J^*)G. \quad (\text{AV10})$$

The matrix $K = JM^{-1}J^*$ is positive definite: J^* is injective because J is surjective, and for $x \neq 0$

$$x^*Kx = (J^*x)^*M^{-1}(J^*x) > 0. \quad (\text{AV11})$$

Consequently $G = K^{-1}$. This proves the inverse quotient identity from the literal section and orthogonality, without presupposing its consequence about the remainder.

95.0.5 4. Determinant-line sign and metric

Let $y_1, \dots, y_r$ be the increasing-power frame of U , and $x_1, \dots, x_q$ the increasing-power remainder frame of E . In the increasing-power frame of P ,

$$[s \ B] = \begin{pmatrix} I_q & T \\ 0 & U_\chi \end{pmatrix}, \quad \det U_\chi = 1. \quad (\text{AV12})$$

Indeed, relation column j , starting with $j = 0$, is $S^j\chi(S)$ and has highest term S^{q+j} of coefficient one. In the bottom rows corresponding to $S^q, \dots, S^N$, no entry occurs below that diagonal. Hence U_χ is upper triangular with all diagonal entries one, proving $\det[s \ B] = 1$.

Moving the first q columns past the last r columns uses exactly qr transpositions. The relation-first frame therefore has

$$L = [B \ s], \quad \det L = (-1)^{qr}. \quad (\text{AV13})$$

The determinant-line map is literally

$$\det U \otimes \det E \longrightarrow \det P, \\ (y_1 \wedge \dots \wedge y_r) \otimes (x_1 \wedge \dots \wedge x_q) \longmapsto By_1 \wedge \dots \wedge By_r \wedge sx_1 \wedge \dots \wedge sx_q. \quad (\text{AV14})$$

The wedge is nonzero by (AV3), so this is an isomorphism of one-dimensional lines. Replacing sx_j by $sx_j + Bz_j$ leaves it unchanged: every term in the difference contains $r+1$ vectors in the r -dimensional space $\text{im } B$. This establishes section independence, while preserving the displayed frame sign.

The change from s to $R = s - Bd$ is

$$[B \ R] = [B \ s] \begin{pmatrix} I_U & -d \\ 0 & I_E \end{pmatrix}, \quad [B \ R]^*M[B \ R] = \begin{pmatrix} H & 0 \\ 0 & G \end{pmatrix}. \quad (\text{AV15})$$

The right factor in the first equality has determinant one. The second equality is exactly (AV7). Taking determinants retains the orientation scalar explicitly:

$$\det H \det G = \overline{(-1)^{qr}} \det M (-1)^{qr} = |(-1)^{qr}|^2 \det M = \det M. \quad (\text{AV16})$$

The squared norm of a wedge is its Gram determinant. Thus (AV16) also proves that (AV14), using the relation metric H and quotient metric G , is the product metric map. The orientation remains (AV13); its squared absolute value is one in the metric equation. In particular

$$V(t) = \det G(t) = \frac{\det M(t)}{\det H(t)}. \quad (\text{AV17})$$

All original source masses remain in these matrices and determinants.

95.0.6 5. First variation of the section and Gram

The maps B, J, s are fixed in t . From $H = B^*MB$ and $H^{-1}H = I_U$,

$$\dot{H} = B^*\Delta MB, \quad \frac{d}{dt}H^{-1} = -H^{-1}\dot{H}H^{-1}. \quad (\text{AV18})$$

Differentiating $d = H^{-1}B^*Ms$ gives

$$\begin{aligned} \dot{d} &= -H^{-1}(B^*\Delta MB)H^{-1}B^*Ms + H^{-1}B^*\Delta Ms \\ &= H^{-1}B^*\Delta M(s - Bd) = H^{-1}B^*\Delta MR. \end{aligned} \quad (\text{AV19})$$

Consequently, as maps on the original spaces,

$$\dot{R} = -BH^{-1}B^*\Delta MR : E \longrightarrow P, \quad \text{im } \dot{R} \subseteq \text{im } B, \quad J\dot{R} = 0. \quad (\text{AV20})$$

Differentiating $G = R^*MR$ gives all three terms

$$\dot{G} = \dot{R}^*MR + R^*\Delta MR + R^*M\dot{R}. \quad (\text{AV21})$$

The first and third vanish by the relation-valued derivative and $B^*MR = 0$. Therefore

$$\dot{G} = R^*\Delta MR. \quad (\text{AV22})$$

This is a Hermitian matrix on the unchanged remainder space E .

95.0.7 6. Second variation of the quotient Gram

The matrix ΔM is fixed and Hermitian, so differentiating (AV22) gives

$$\ddot{G} = \dot{R}^*\Delta MR + R^*\Delta M\dot{R} = -2R^*\Delta MBH^{-1}B^*\Delta MR. \quad (\text{AV23})$$

The two terms are equal because H^{-1} and ΔM are Hermitian. Their sign is established on each actual $x \in E$ by

$$x^*\ddot{G}x = -2(B^*\Delta MRx)^*H^{-1}(B^*\Delta MRx) = -2\|H^{-1/2}B^*\Delta MRx\|^2 \leq 0. \quad (\text{AV24})$$

Thus the factor 2 in AT7 retains both differentiated occurrences of R .

95.0.8 7. Both logarithmic derivative formulas

For any positive definite differentiable finite matrix A , multilinearity of the determinant gives $\det(I + \varepsilon D) = 1 + \varepsilon \operatorname{Tr} D + O(\varepsilon^2)$: the terms choosing exactly one perturbed column sum to the diagonal trace. Applying this to $\det(A + \varepsilon \dot{A}) = \det A \det(I + \varepsilon A^{-1} \dot{A})$ therefore proves

$$\frac{d}{dt} \log \det A = \operatorname{Tr}(A^{-1} \dot{A}). \quad (\text{AV25})$$

Apply (AV25) to (AV17), and separately to G using (AV22):

$$\begin{aligned} \frac{d}{dt} \log V &= \operatorname{Tr}(M^{-1} \Delta M) - \operatorname{Tr}(H^{-1} B^* \Delta M B) \\ &= \operatorname{Tr}(G^{-1} R^* \Delta M R). \end{aligned} \quad (\text{AV26})$$

The traces are on P, U, E , respectively. The equality follows from the proved determinant identity, so it requires no unproved rearrangement across different domains.

Differentiating $G^{-1} G = I_E$ gives $\dot{G}^{-1} = -G^{-1} \dot{G} G^{-1}$. A second differentiation of the last expression in (AV26), followed by (AV23), now retains both terms:

$$\begin{aligned} \frac{d^2}{dt^2} \log V &= \operatorname{Tr}(G^{-1} \ddot{G}) - \operatorname{Tr}(G^{-1} \dot{G} G^{-1} \dot{G}) \\ &= -2 \operatorname{Tr}(G^{-1} R^* \Delta M B H^{-1} B^* \Delta M R) - \operatorname{Tr}(G^{-1} \dot{G} G^{-1} \dot{G}). \end{aligned} \quad (\text{AV27})$$

Their exact norm realizations are

$$\begin{aligned} \operatorname{Tr}(G^{-1} R^* \Delta M B H^{-1} B^* \Delta M R) &= \|H^{-1/2} B^* \Delta M R G^{-1/2}\|_{\text{HS}}^2, \\ \operatorname{Tr}(G^{-1} \dot{G} G^{-1} \dot{G}) &= \|G^{-1/2} \dot{G} G^{-1/2}\|_{\text{HS}}^2. \end{aligned} \quad (\text{AV28})$$

For the first, multiply $X^* X$ with $X = H^{-1/2} B^* \Delta M R G^{-1/2} : E \rightarrow U$, and take its trace on E . For the second, $Y = G^{-1/2} \dot{G} G^{-1/2}$ is Hermitian, so $\operatorname{Tr}(Y^* Y) = \operatorname{Tr}(Y^2)$. Cyclicity of the trace on E gives exactly the two expressions in (AV28). Formula (AV27) is consequently $-2\|X\|_{\text{HS}}^2 - \|Y\|_{\text{HS}}^2 \leq 0$.

Differentiating the first line of (AV26) also gives the compatible full source/relation expression

$$\frac{d^2}{dt^2} \log V = -\operatorname{Tr}(M^{-1} \Delta M M^{-1} \Delta M) + \operatorname{Tr}(H^{-1} \dot{H} H^{-1} \dot{H}). \quad (\text{AV29})$$

Equations (AV27)–(AV28) prove why this particular difference is nonpositive for the exact quotient.

95.0.9 8. Empty relation domain and equality

When $N = q - 1$, $U = \{0\}$, $P = \mathcal{P}_{q-1}$, and J, s identify the same increasing-power frame. The only endomorphism of U is both its zero and identity map; it is invertible on U , with itself as inverse. Its determinant is 1, and its trace is 0. The maps $B : U \rightarrow P$ and $B^* : P \rightarrow U$ are the corresponding zero maps. Every composition through U is the correctly typed zero map on its stated exterior domain and codomain. Thus

$$R = s, \quad \dot{R} = 0, \quad G = s^* M s, \quad \dot{G} = s^* \Delta M s, \quad \ddot{G} = 0. \quad (\text{AV30})$$

The determinant-line map uses the empty wedge $1 \in \det U = \mathbb{C}$; its sign is $(-1)^{q-0} = 1$. Formula (AV17) is $\det G = \det M$ in the stated frames. The logarithmic second derivative still retains the inverse-Gram term:

$$\frac{d^2}{dt^2} \log V = -\|G^{-1/2} \dot{G} G^{-1/2}\|_{\text{HS}}^2. \quad (\text{AV31})$$

For either $r = 0$ or $r > 0$, both terms in (AV28) vanish exactly when $B^* \Delta M R = 0$ and $R^* \Delta M R = 0$. The matrix $[B \ R]$ is invertible by (AV8), so its conjugate transpose is invertible. These equations are therefore equivalent to

$$\Delta M R = 0 : E \longrightarrow P. \quad (\text{AV32})$$

Conversely this identity makes both terms vanish directly. Thus equality in the second-derivative inequality is characterized by the actual Gram increment annihilating the current canonical section, including the empty relation case.

The corrected source's proof now contains the noncircular step (AV8)–(AV10). AT6–AT10 are accepted on the final source pin above; the first-pass proof-order observation is resolved.

96 All-degree angle transport on the original arithmetic four-volume

The original finite quotient has two pairs of minimum-section subspaces. This calculation identifies their complete principal angles with the energies of their actual relation-valued differences. Their determinant is the original arithmetic four-volume. The original vertical-line moment identity then proves strict positivity of that four-volume and gives a nonlinear endpoint bound in every quotient dimension.

96.1 Fixed arithmetic source, maps and forms

Fix the original full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and its original cyclic polynomial $\chi = \chi_{h,k}$, monic of degree $q \geq 1$. Keep $E = \mathbb{C}[S]/(\chi)$ in its increasing-power remainder frame. The full Taylor unit $v_h = j_h(g/h)$ occurs in the unchanged injections

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1, \quad \sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}, \quad (\text{AGT1})$$

where $E_h = \mathbb{C}[s]/(h)$ and $Q = \mathcal{B}/\Theta V$. The latter is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau}$. All these maps and the original theta primitives are supplied, with their complete proofs, in the retained AT and AW dependency packages.

The common polynomial source is $H = \mathcal{P}_{2q}$, with coordinate $S = k/2 + iu$ on the original integration line. Its original measures are

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a} \Gamma(2a). \end{aligned} \quad (\text{AGT2})$$

Their full masses $(2\pi)^{k/2}$ and μ_h^k remain present. Set $m_t = (1-t)m_0 + tm_1$, $0 \leq t \leq 1$. In the original coefficient frame write $M(t)$ for its Gram on H , and $\langle x, y \rangle_t = x^* M(t) y$. The complete source-density proofs give all finite moments and positivity almost everywhere. Therefore this form is positive on every nonzero polynomial, for every t in the closed interval.

For each $q-1 \leq N \leq 2q$, use the original exact sequence $0 \rightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} E \rightarrow 0$, with $\mathcal{P}_{-1} = 0$. Let $R_N(t) : E \hookrightarrow \mathcal{P}_N \subset H$ be its unique minimum-norm section, and retain

$$G_N(t) = R_N(t)^* M(t) R_N(t), \quad V_N(t) = \det G_N(t). \quad (\text{AGT3})$$

The inclusions into H are part of this formula. Write P_N, Q_N for the orthogonal projections in H onto $\mathcal{P}_N$ and $\chi \mathcal{P}_{N-q}$ respectively, and $Z_N = P_N - Q_N$. Then Z_N projects onto $R_N E$ and has rank q .

The existing complete determinant-variation proof gives

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}, \\ C(t) &= M(t)^{-1} (M(1) - M(0)), \\ \dot{\mathcal{B}}(t) &= \text{Tr}((Z_{q-1} - Z_{2q} + Z_q - Z_{2q-1}) C(t)). \end{aligned} \quad (\text{AGT4})$$

This groups the same four signed terms into two pairs; all four original cutoff indices and quotient Grams remain fixed.

96.2 Relation energy, Gram ratios and the full principal angles

Theorem 96.1. For $q - 1 \leq i \leq j \leq 2q$, define the actual difference of minimum sections $D_{ij} = R_i - R_j : E \rightarrow H$. Then

$$\begin{aligned} D_{ij}E &\subset \chi\mathcal{P}_{j-q}, & R_j &= Z_j R_i, \\ G_i - G_j &= D_{ij}^* M D_{ij}, & K_{ij} &= G_i^{-1/2} G_j G_i^{-1/2}, \\ I - K_{ij} &= G_i^{-1/2} D_{ij}^* M D_{ij} G_i^{-1/2}, & 0 &\prec K_{ij} \preceq I. \end{aligned} \quad (\text{AGT5})$$

The eigenvalues of K_{ij} , including multiplicities, are $\cos^2 \theta_{ij,a}$ for the full principal-angle list between $R_i E$ and $R_j E$, with $0 \leq \theta_{ij,a} < \pi/2$ and $1 \leq a \leq q$. Consequently

$$\log(V_i/V_j) = \sum_{a=1}^q -\log \cos^2 \theta_{ij,a}. \quad (\text{AGT6})$$

The nonzero eigenvalues of $Z_i - Z_j$ are exactly the pairs $\sin \theta_{ij,a}, -\sin \theta_{ij,a}$ for the nonzero angles; the remaining eigenvalues are zero.

Proof. Both sections have the same remainder, so their difference lies in the displayed relation space. The vector $R_j x$ is orthogonal to that entire space. Also $R_i x \in \mathcal{P}_j$. Thus the orthogonal splitting of $R_i x$ into its relation and minimum-lift components is $D_{ij}x + R_j x$, proving $R_j = Z_j R_i$ and the Gram identity. Positivity of G_j and this nonnegative difference prove the last inequalities in (AGT5). Each square root is the unique positive Hermitian square root on the displayed original coefficient space; the order of its products is retained.

For the principal-angle calculation put $U_i = R_i G_i^{-1/2}$ and $U_j = R_j G_j^{-1/2}$. They are isometries from the displayed Euclidean coefficient space to the two original source subspaces: $U_i^* M U_i = I$ and $U_j^* M U_j = I$. Their full cross matrix is

$$U_j^* M U_i = G_j^{-1/2} R_j^* M R_i G_i^{-1/2} = G_j^{1/2} G_i^{-1/2}. \quad (\text{AGT7})$$

The equality uses $R_j^* M D_{ij} = 0$. Its squared singular values are therefore the eigenvalues of K_{ij} , with no commutation of the two Grams presumed. These singular values are positive and at most one, defining the angles just stated. Determinants of the displayed positive matrices give (AGT6), keeping all original determinant factors.

Here is also a direct proof of the projection-difference spectrum. Choose singular-vector bases e_a in $R_i E$ and f_a in $R_j E$ for (AGT7), so their cross inner products are $\langle f_a, e_b \rangle_t = c_a \delta_{ab}$, with $c_a = \cos \theta_{ij,a} > 0$. For $c_a = 1$, the equality of norms gives $e_a = f_a$. For $c_a < 1$, put $s_a = \sqrt{1 - c_a^2}$ and $r_a = (f_a - c_a e_a)/s_a$. Substitution in these inner products proves that all resulting pairs (e_a, r_a) are orthonormal, orthogonal to the other pairs and to the common vectors. On each pair the exact matrix is

$$(Z_i - Z_j)|_{\text{span}(e_a, r_a)} = \begin{pmatrix} s_a^2 & -s_a c_a \\ -s_a c_a & -s_a^2 \end{pmatrix}. \quad (\text{AGT8})$$

It has trace zero and determinant $-s_a^2$, hence eigenvalues $s_a, -s_a$. The difference vanishes on the common vectors and on the orthogonal complement of the two subspaces' sum. This proves the full spectrum, including all zero-angle cases. $\square$

The difference D_{ij} in (AGT5) is an original polynomial relation. The original source map $\mathcal{V}$ therefore gives its literal theta primitive, while its full-jet and arithmetic cohomology observations are zero. Precisely, $J_j D_{ij} = 0$, $\eta J_j D_{ij} = 0$ and $\sigma_h^{\otimes k} \eta J_j D_{ij} = 0$; nevertheless its source energy is exactly $G_i - G_j$ in (AGT5). At each nonempty original support label A , the corresponding linear maps lift as $(A, x) \mapsto (A, fx)$, with $\tau \mapsto \tau$. The boundary then has a supported-zero class at that same label, and (AGT5) retains its entire preimage energy.

96.3 The actual four-volume has at most $2q - 1$ nonzero angles

Apply the theorem to the two pairs $(q - 1, 2q)$ and $(q, 2q - 1)$. For the second pair both q -dimensional minimum-section subspaces lie in

$$\mathcal{P}_{2q-1} \cap (\chi\mathcal{P}_0)^\perp, \quad \dim(\mathcal{P}_{2q-1} \cap (\chi\mathcal{P}_0)^\perp) = 2q - 1. \quad (\text{AGT9})$$

Indeed both are orthogonal to $\chi\mathcal{P}_0$, a one-dimensional subspace of $\mathcal{P}_{2q-1}$. Their intersection thus has dimension at least one, by the dimension formula for a sum of subspaces. Each common unit vector gives singular value one in the principal-angle calculation, so this pair has at least one exact zero angle. When $q = 1$ its two cutoff indices coincide, and its single angle is exactly zero.

Retain both full lists of q angles. Their combined logarithmic cost from (AGT6) is exactly $\mathcal{B}(t)$. Remove one of the proved zero contributions only in the following scalar list, and write the remaining $m = 2q - 1$ entries as

$$x_\ell(t) = -\log \cos^2 \theta_\ell(t) \geq 0, \quad \sum_{\ell=1}^m x_\ell(t) = \mathcal{B}(t), \quad \sin \theta_\ell(t) = \sqrt{1 - e^{-x_\ell(t)}}. \quad (\text{AGT10})$$

Other zero angles remain in the list. The zero-angle eigenspaces and the full source maps were not removed from their original spaces. No differentiability of a chosen eigenvector or ordered angle branch will be needed.

Let $c_{\min}(t), c_{\max}(t)$ be the extreme eigenvalues of $C(t)$ for $M(t)$. A self-adjoint projection difference with the spectrum in (AGT8) is a sum of its positive and negative rank-one eigenprojections with coefficients $s_a, -s_a$. Each eigenprojection's trace against $C(t)$ is the Rayleigh quotient of $C(t)$ at its unit vector and lies between these two extremes. Subtracting each pair gives an absolute value at most $s_a(c_{\max} - c_{\min})$. Apply this to both terms of (AGT4), retaining their signs, to obtain

$$|\dot{\mathcal{B}}(t)| \leq \left(\sum_{\ell=1}^m \sqrt{1 - e^{-x_\ell(t)}} \right) (c_{\max}(t) - c_{\min}(t)). \quad (\text{AGT11})$$

This first estimate retains all principal angles through the actual source relation energies in (AGT5).

96.4 The moment identity forces strict positivity

Write the original polynomial as $\chi(S) = \sum_{j=0}^q a_j S^j$, $a_q = 1$. The antilinear algebra involution on $\mathbb{C}[S]$ defined by

$$\chi^{\#_k}(S) = \sum_{j=0}^q \overline{a_j}(k - S)^j \quad (\text{AGT12})$$

satisfies $\chi^{\#_k}(S) = \overline{\chi(S)}$ on the original line, because $\overline{S} = k - S$. Its degree is q and its leading coefficient is $(-1)^q$. Applying it twice restores each coefficient and the variable since k is real.

Each term in (AGT10) is nonnegative, so $\mathcal{B}(t) \geq 0$. If it were zero, all angles of the first pair would be zero, hence $R_{q-1}E = R_{2q}E$. The first space is $\mathcal{P}_{q-1}$; the second is $(\chi\mathcal{P}_q)^\perp$ in H . It would follow that 1 is orthogonal to $\chi\mathcal{P}_q$. But $\chi^{\#_k} \in \mathcal{P}_q$, and therefore this orthogonality would imply

$$0 = \langle 1, \chi\chi^{\#_k} \rangle_t = \int_{\mathbb{R}} |\chi(k/2 + iu)|^2 m_t(u) du > 0. \quad (\text{AGT13})$$

The last strict inequality is positivity of the original form on the nonzero polynomial χ . This contradiction proves $\mathcal{B}(t) > 0$ at every $t \in [0, 1]$. Thus the strict positivity follows from an exact identity in the actual moment form; an arbitrary positive coefficient form has not been substituted for that form.

96.5 A nonlinear endpoint estimate in every degree

Theorem 96.2. Put $m = 2q - 1$, let $b_{\min}, b_{\max}$ be the extreme eigenvalues of the original relative Gram $M(0)^{-1}M(1)$, and set $\kappa = b_{\max}/b_{\min}$. Then the original two endpoint four-volumes satisfy

$$\left| 2 \operatorname{arcosh}\left(e^{\mathcal{B}(1)/(2m)}\right) - 2 \operatorname{arcosh}\left(e^{\mathcal{B}(0)/(2m)}\right) \right| \leq \log \kappa. \quad (\text{AGT14})$$

In particular this holds for every actual full quartet quotient, with its unchanged value of q .

Proof. The function $f(x) = \sqrt{1 - e^{-x}}$ is continuous for $x \geq 0$, with $f(0) = 0$. For $x > 0$ its two derivatives give

$$f'(x) = \frac{e^{-x}}{2\sqrt{1 - e^{-x}}}, \quad f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} < 0. \quad (\text{AGT15})$$

It is consequently concave on the whole closed half-line, by continuity at zero. More explicitly, its decreasing derivative shows that its tangent at any positive a is above its graph on $(0, \infty)$, and continuity extends this inequality to zero. Choose $a = \mathcal{B}(t)/m > 0$, proved admissible in (AGT13), and sum those tangent inequalities at the entries in (AGT10). Their linear terms cancel exactly, giving

$$\sum_{\ell=1}^m f(x_{\ell}(t)) \leq m\sqrt{1 - e^{-\mathcal{B}(t)/m}}, \quad |\dot{\mathcal{B}}(t)| \leq m\sqrt{1 - e^{-\mathcal{B}(t)/m}}(c_{\max}(t) - c_{\min}(t)). \quad (\text{AGT16})$$

This argument allows every remaining zero angle without differentiating it.

The positive finite Grams and their sections are smooth through the whole compact interpolation interval. Therefore $\mathcal{B}$ is smooth and, by (AGT13), has a positive minimum there. The real function $\Phi_m(B) = 2 \operatorname{arcosh}(e^{B/(2m)})$, using the nonnegative real inverse of the hyperbolic cosine, satisfies

$$\Phi'_m(B) = \frac{1}{m\sqrt{1 - e^{-B/m}}}, \quad B > 0. \quad (\text{AGT17})$$

One obtains this by differentiating $\cosh(\Phi_m(B)/2) = e^{B/(2m)}$. Multiply (AGT16) by this positive derivative.

To integrate the remaining spectral width, set $D = M(0)^{-1}M(1)$. It is positive and self-adjoint for the original $M(0)$ metric, and $C(t) = ((1 - t)I + tD)^{-1}(D - I)$. Each relative eigenvalue $b > 0$ gives $c_b(t) = (b - 1)/(1 - t + tb)$. Its derivative with respect to b is $(1 - t + tb)^{-2} > 0$, so the same extreme eigenvalues give the extremes for all t . Direct integration of the logarithmic derivative yields

$$\int_0^1 (c_{\max}(t) - c_{\min}(t)) dt = \int_0^1 \frac{d}{dt} \log \frac{1 - t + tb_{\max}}{1 - t + tb_{\min}} dt = \log \kappa. \quad (\text{AGT18})$$

Integrating $|(\Phi_m \circ \mathcal{B})'| \leq c_{\max} - c_{\min}$ proves (AGT14), including repeated relative eigenvalues. $\square$

An explicit interval for the original arithmetic endpoint is obtained without an implicit inversion. Put $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/(2m)})$ and $L = \frac{1}{2} \log \kappa$. Since $2m \log \cosh A$ is increasing on $A \geq 0$, (AGT14) is equivalent to

$$2m \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2m \log \cosh(A_0 + L). \quad (\text{AGT19})$$

The lower endpoint can be zero although the actual $\mathcal{B}(1)$ was proved strictly positive. Also $\Phi'_m(B) > 1/m$ for finite positive B , so (AGT14) implies $|\mathcal{B}(1) - \mathcal{B}(0)| \leq m \log \kappa$, with strict inequality in this last comparison whenever the two four-volumes differ. The full-relative-spectrum bound in AW13–AW14 remains available as a second, separately proved bound; none of its eigenvalues is discarded by the exact identities (AGT5)–(AGT11).

When $q = 1, m = 1$ and the second pair is identical. Equations (AGT14)–(AGT19) recover AW19–AW23 with exactly the original constant and mass factors. If the endpoint forms differ by a positive scalar only, then $\kappa = 1$; every quotient determinant has the same literal scalar to the power q , and the four-volume is unchanged, as the new inequality also proves.

96.6 Complete source and mathematical scope

The full AT and AW source packages accompany this calculation. Their complete proofs establish the original tau-base observation, tensor units, theta primitives, source densities, minimum sections and signed derivative (AGT4). The new principal-angle and moment calculations above retain those objects and prove (AGT5)–(AGT19) in every quotient dimension. The actual endpoint measures enter both the relative Gram D and the strict positivity argument. No upper estimate uniform in k for that relative Gram is assumed or claimed. The new bound supplies an explicit nonlinear constraint on the original arithmetic four-volume, with the complete source relation energies still present.

97 Complete four-window spectral sharpness on the original polynomial flags

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 11c26c3bcfce135e1882190af9dc3a8f1e4e6487fd282cc252fa441fdf90651c

Complete source. `sources/next_edition/actual_tau_four_window_v1/sources/four_window_sharpness/tau_four_window_spectral_sharpness_20260913.md`

Source SHA-256. 11c26c3bcfce135e1882190af9dc3a8f1e4e6487fd282cc252fa441fdf90651c

97.0.1 Four-window projection spectra and the exact polynomial-flag sharpness calculation

13 September 2026. Independent written calculation for the common finite source in the arithmetic determinant transport. The original monic polynomial, coefficient frame, relation maps, and masses remain explicit.

The source work/tau_arithmetic_determinant_transport_20260913.tex was consulted through its first 190 lines, including AT1–AT10. Its SHA-256 at consultation was 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae. The present calculation proves the projection statements below directly. It does not edit that source, run Lean, or assert that an arbitrary positive coefficient Gram is an arithmetic moment Gram.

97.0.2 1. Original flag spaces and exact spectra

Fix the original monic polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$. Work on

$$\mathcal{H} = \mathcal{P}_{2q} = \mathbb{C}[S]_{\leq 2q}, \quad n = 2q + 1, \quad \langle x, y \rangle_M = x^* M y, \quad M > 0. \quad (\text{FS1})$$

The inner product is conjugate-linear in its first slot. Let P_N be the M -orthogonal projection of $\mathcal{H}$ onto $\mathcal{P}_N$. Let Q_N be the M -orthogonal projection onto the literal relation space $\chi \mathcal{P}_{N-q}$, with $Q_{q-1} = 0$. In particular $P_{2q} = I$. Define

$$U = Q_{2q-1} + Q_{2q} - Q_q, \quad W = P_{2q-1} - P_{q-1} + P_{2q} - P_q, \quad D = U - W. \quad (\text{FS2})$$

All projections in each one of the two displayed flags commute: if $A \subset B$, their orthogonal projections satisfy $P_A P_B = P_B P_A = P_A$, as follows by decomposing into $A \oplus (B \cap A^\perp) \oplus B^\perp$.

Write

$$B_0 = \mathbb{C}\chi, \quad B_1 = \chi \mathcal{P}_{q-1}, \quad B_2 = \chi \mathcal{P}_q. \quad (\text{FS3})$$

Their dimensions are 1, q , $q + 1$, respectively, because multiplication by the nonzero polynomial χ is injective. The exact eigenspaces of U are

eigenvalue	eigenspace	dimension
0	$B_2^\perp$	q
1	$B_0 \oplus (B_2 \cap B_1^\perp)$	2
2	$B_1 \cap B_0^\perp$	$q - 1$.

(FS4)

Every direct sum in (FS4) is orthogonal. Applying the three projections of (FS2) to each of these spaces gives the stated eigenvalue, and the dimensions sum to $2q + 1$. For $q = 1$, $B_0 = B_1$, the last space is zero, and $U = Q_2$.

The exact eigenspaces of W are

eigenvalue	eigenspace	dimension	
0	$\mathcal{P}_{q-1}$	q	(FS5)
1	$(\mathcal{P}_q \cap \mathcal{P}_{q-1}^\perp) \oplus \mathcal{P}_{2q-1}^\perp$	2	
2	$\mathcal{P}_{2q-1} \cap \mathcal{P}_q^\perp$	$q - 1$.	

This follows by the same nested decomposition of the degree flag. Thus both operators have precisely the spectrum

$$0^{[q]}, \quad 1^{[2]}, \quad 2^{[q-1]}, \quad \text{Tr } U = \text{Tr } W = 2q, \quad \text{Tr } D = 0. \quad (\text{FS6})$$

The notation $a^{[r]}$ means an eigenvalue a of multiplicity r . No claim that the spectrum of their difference is fixed by these two individual spectra is made.

97.0.3 2. Best bound from the two individual spectra alone

Let C be any M -self-adjoint endomorphism with eigenvalues

$$c_1 \leq c_2 \leq \cdots \leq c_{2q+1}. \quad (\text{FS7})$$

For a rank- r orthogonal projection E , choose an orthonormal eigenbasis v_i of C , and put $a_i = \langle v_i, E v_i \rangle$. Then $0 \leq a_i \leq 1$, $\sum a_i = r$, and

$$\text{Tr}(EC) = \sum_i c_i a_i.$$

Transferring any positive weight from a lower coefficient to an available higher coefficient does not decrease this sum. Repeating gives the exact two-sided inequality

$$\sum_{i=1}^r c_i \leq \text{Tr}(EC) \leq \sum_{i=n-r+1}^n c_i. \quad (\text{FS8})$$

If the eigenvalues of C are distinct, equality on either side forces the corresponding spectral projection: a non-extremal weight transfer would otherwise strictly improve the sum, and a diagonal projection value zero or one forces the corresponding basis vector into its kernel or image.

For any self-adjoint operator with spectrum (FS6), its spectral projections satisfy

$$U = E_{\{U \geq 1\}} + E_{\{U=2\}}, \quad \text{rank } E_{\{U \geq 1\}} = q + 1, \quad \text{rank } E_{\{U=2\}} = q - 1. \quad (\text{FS9})$$

Applying (FS8) twice gives

$$\begin{aligned} \text{Tr}(UC) &\leq c_{q+1} + c_{q+2} + 2 \sum_{i=q+3}^{2q+1} c_i, \\ \text{Tr}(WC) &\geq c_q + c_{q+1} + 2 \sum_{i=1}^{q-1} c_i. \end{aligned} \quad (\text{FS10})$$

Subtracting, and then applying the same reasoning with U, W interchanged, proves

$$|\text{Tr}((U - W)C)| \leq B_q(c) := c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{FS11})$$

For $q = 1$, the sum is empty and the right side is exactly $c_3 - c_1$.

This is the best bound that uses only the individual spectra. To prove sharpness in that class, use the eigenbasis of C , put

$$u_i = \begin{cases} 0 & i \leq q, \\ 1 & i = q+1, q+2, \\ 2 & i \geq q+3, \end{cases} \quad U_* = \text{diag}(u_1, \dots, u_n), \quad W_* = \text{diag}(u_n, \dots, u_1). \quad (\text{FS12})$$

Both have spectrum (FS6), and direct multiplication gives $\text{Tr}((U_* - W_*)C) = B_q(c)$. Swapping them gives the negative equality. Their difference has the ordered spectrum

$$\lambda^* = (-2^{[q-1]}, -1, 0, 1, 2^{[q-1]}). \quad (\text{FS13})$$

These are exact independent-unitary-orbit equality fixtures. They do not yet assert realization by the two literal polynomial flags in (FS1)–(FS3).

97.0.4 3. Why the literal polynomial flags cannot attain nontrivial equality

Two exact polynomial identities govern the flag positions:

$$\mathcal{P}_{q-1} \cap B_2 = 0, \quad \mathcal{P}_q + B_2 = \mathcal{P}_{2q}. \quad (\text{FS14})$$

The first follows because a nonzero multiple of a degree- q polynomial has degree at least q . For the second, $B_2 \cap \mathcal{P}_q = \mathbb{C}\chi$: a nonconstant multiplier has degree above q . Thus the sum has dimension $(q+1) + (q+1) - 1 = 2q+1$, and is the whole space.

Let $U_2 = \ker(U - 2I)$, $W_2 = \ker(W - 2I)$, $U_0 = \ker U$, and $W_0 = \ker W$. Equations (FS4)–(FS5) and (FS14) imply

$$U_2 \cap W_0 = 0, \quad W_2 \cap U_0 = 0. \quad (\text{FS15})$$

For the first, $U_2 \subset B_1 \subset B_2$ whereas $W_0 = \mathcal{P}_{q-1}$. For the second, a vector in $W_2 \cap U_0$ is orthogonal to both $\mathcal{P}_q$ and B_2 ; their sum is the whole space by (FS14).

Here is the requested simple-spectrum obstruction. If equality held in the positive version of (FS11) with distinct c_i , both bounds in (FS10) would have to be equalities. The $U = 2$ eigenspace would be the span of the highest $q-1$ eigenvectors of C , and the kernel of W would be the span of the highest q . For $q \geq 2$, this contradicts the first equation in (FS15). For $q = 1$, the image of U would contain the highest eigenline, which is $\ker W = \mathcal{P}_0$. That would put the constant polynomial in $B_2 = \chi\mathcal{P}_1$, contrary to (FS14). The negative equality is treated by applying the same argument to $-C$.

In fact the inequality is strict for every nonscalar C , even when its spectrum has repeated eigenvalues. A full proof follows, retaining the spectral gaps rather than assuming they are nonzero individually.

For $1 \leq r < n$, let a_r be the sum of the largest r entries in (FS13). Applying (FS8) to U and W , with a rank- r projection E as the variable, gives

$$\text{Tr}(DE) \leq \sum_{i=n-r+1}^n u_i - \sum_{i=1}^r u_i = a_r. \quad (\text{FS16})$$

Equality is impossible in every nontrivial rank:

- For $1 \leq r \leq q-1$, equality requires $\text{im } E \subset U_2$ and $\text{im } E \subset W_0$, contradicting (FS15).
- For $r = q$ and $q \geq 2$, minimization of $\text{Tr}(WE)$ forces $\text{im } E = W_0$; maximization of $\text{Tr}(UE)$ forces that image to contain all of U_2 . Again (FS15) excludes it. If $q = r = 1$, the same equality instead forces $\mathcal{P}_0 \subset B_2$, excluded by (FS14).
- For $r = q+1$, the maximum of $\text{Tr}(UE)$ forces $\text{im } E = B_2$, since this is its complete positive spectral space. The minimum of $\text{Tr}(WE)$ forces $\text{im } E$ to contain $W_0 = \mathcal{P}_{q-1}$. Their intersection in (FS14) is zero, so this is impossible.
- For $q+2 \leq r \leq 2q$, use the complementary projection $I - E$, whose rank is between one and $q-1$. Since $\text{Tr } D = 0$, equality would require $\text{im}(I - E) \subset W_2 \cap U_0$, excluded by (FS15).

These equality conditions use only the spectral levels zero, one, and two of U, W ; their eigenspace descriptions follow by the same projection-weight argument proving (FS8).

The set of rank- r orthogonal projections is compact in finite dimension: it is closed and bounded in the matrix space by the equations $E^* = E$, $E^2 = E$, and $\text{Tr } E = r$. Hence the maximum in (FS16) is attained, and its failure to equal a_r proves strict inequality for that maximum. If $d_1 \leq \dots \leq d_n$ are the actual eigenvalues of D , (FS8) applied to D identifies this maximum. Therefore

$$\sum_{i=n-r+1}^n d_i < a_r \quad (1 \leq r < n), \quad \sum_i d_i = 0. \quad (\text{FS17})$$

For completeness, trace rearrangement for two arbitrary self-adjoint matrices follows by the same projection argument. In an eigenbasis of C , write

$$C = c_1 I + \sum_{r=1}^{n-1} (c_{n-r+1} - c_{n-r}) E_r,$$

where E_r projects onto its highest r eigenvectors, with any fixed choice inside a repeated eigenspace. The trace of the scalar term against D is zero, so (FS17) gives

$$\begin{aligned} \text{Tr}(DC) &\leq \sum_{r=1}^{n-1} (c_{n-r+1} - c_{n-r}) \sum_{i=n-r+1}^n d_i \\ &< \sum_{r=1}^{n-1} (c_{n-r+1} - c_{n-r}) a_r = B_q(c) \end{aligned} \quad (\text{FS18})$$

whenever C is nonscalar. At least one spectral gap is then positive, so the second inequality is strict. Applying the same proof to $-C$ gives

$|\text{Tr}(DC)| < B_q(c) \quad \text{for every nonscalar } C \text{ and every fixed } M > 0 \text{ on the literal polynomial flags.}$

(FS19)

For scalar C , both sides of (FS11) are zero and equality holds. This completely identifies actual finite positive-metric equality, while allowing the sharp supremum described next.

97.0.5 4. Explicit positive metrics approaching the bound on the unchanged polynomial flags

Keep the same original monic χ . The following ordered list is a basis of $\mathcal{P}_{2q}$:

$$b_i = S^i \quad (0 \leq i < q), \quad b_q = \chi, \quad b_{q+j} = \chi S^j \quad (1 \leq j \leq q). \quad (\text{FS20})$$

Its elements have strictly increasing degrees with leading coefficient one. Thus their first $N+1$ spans equal the original $\mathcal{P}_N$. Neither χ nor its coefficients have been changed.

Choose any retained positive mass $\mu > 0$ and any $\epsilon > 0$. Let $e_0, \dots, e_{2q}$ be the standard coordinate vectors, and define the actual linear isomorphism

$$\begin{aligned} T_\epsilon b_i &= e_i & (0 \leq i \leq q), \\ T_\epsilon b_{q+j} &= e_{q+j} + \epsilon e_{q+j} & (1 \leq j \leq q). \end{aligned} \quad (\text{FS21})$$

Its diagonal in the ordered basis is 1 on the first $q+1$ entries and ϵ on the last q , so its determinant is ϵ^q . Its inverse sends

$$e_i \mapsto b_i \quad (0 \leq i \leq q), \quad e_{q+j} \mapsto \epsilon^{-1}(b_{q+j} - b_{j-1}). \quad (\text{FS22})$$

Prescribe the positive coefficient metric by

$$\langle P, R \rangle_{M_\epsilon} = \mu \langle T_\epsilon P, T_\epsilon R \rangle_{\text{Euclidean}}. \quad (\text{FS23})$$

The map $\sqrt{\mu} T_\epsilon$ is its displayed isometry to the Euclidean space. In the original polynomial coefficients $P = \sum x_i b_i$, its complete norm is

$$\|P\|_{M_\epsilon}^2 = \mu \left(|x_q|^2 + \sum_{j=1}^q |x_{j-1} + x_{q+j}|^2 + \epsilon^2 \sum_{j=1}^q |x_{q+j}|^2 \right). \quad (\text{FS24})$$

In particular $\|1\|^2 = \|\chi\|^2 = \mu$, $\|\chi S^j\|^2 = \mu(1 + \epsilon^2)$, and $\langle S^{j-1}, \chi S^j \rangle = \mu$. Every unlisted cross pairing between the distinct two-coordinate blocks and b_q is zero. Each two-coordinate block has Gram

$$\mu \begin{pmatrix} 1 & 1 \\ 1 & 1 + \epsilon^2 \end{pmatrix}, \quad (\text{FS25})$$

whose determinant is $\mu^2 \epsilon^2 > 0$. The total Gram determinant in (FS20) is exactly $\mu^{2q+1} \epsilon^{2q}$. As (FS20) has determinant one relative to the increasing monomial basis, that is also the determinant there. The mass μ , all cross terms, and all powers of ϵ remain present.

Because (FS21) is triangular by degree, every transformed degree flag is the standard coordinate flag. Hence W , in these isometric coordinates, has value zero on $e_0, \dots, e_{q-1}$, value one on e_q, e_{2q} , and value two on $e_{q+1}, \dots, e_{2q-1}$. Put

$$v_j = \frac{e_{j-1} + \epsilon e_{q+j}}{\sqrt{1 + \epsilon^2}} \quad (1 \leq j \leq q). \quad (\text{FS26})$$

These vectors are orthonormal and orthogonal to e_q . The transformed relation flag has $B_0 = \mathbb{C}e_q$, $B_1 = \text{span}(e_q, v_1, \dots, v_{q-1})$, and $B_2 = \text{span}(e_q, v_1, \dots, v_q)$. Thus the exact transformed operator is

$$U = |e_q\rangle\langle e_q| + 2 \sum_{j=1}^{q-1} |v_j\rangle\langle v_j| + |v_q\rangle\langle v_q|. \quad (\text{FS27})$$

Here the normalization in (FS26) states an orthonormal coordinate map inside the specified metric; the original norms remain (FS24)–(FS25).

97.0.6 5. Fixed-spectrum self-adjoint control and its exact trace

For the prescribed ordered numbers in (FS7), define a diagonal operator C_0 in the transformed coordinates by

$$\begin{aligned} C_0 e_{j-1} &= c_{2q+2-j} e_{j-1}, & C_0 e_{q+j} &= c_j e_{q+j} & (1 \leq j < q), \\ C_0 e_{q-1} &= c_{q+2} e_{q-1}, & C_0 e_{2q} &= c_q e_{2q}, & C_0 e_q &= c_{q+1} e_q. \end{aligned} \quad (\text{FS28})$$

This lists each eigenvalue exactly once. Define $C_\epsilon = T_\epsilon^{-1} C_0 T_\epsilon$, an endomorphism of the original polynomial space. Equation (FS23) proves it is M_ϵ -self-adjoint with exactly those eigenvalues. Its polynomial-coordinate formula is also explicit. Set

$$a_j = c_{2q+2-j}, \quad b'_j = c_j \quad (1 \leq j < q), \quad a_q = c_{q+2}, \quad b'_q = c_q.$$

Then

$$\begin{aligned} C_\epsilon b_{j-1} &= a_j b_{j-1}, \\ C_\epsilon b_{q+j} &= b'_j b_{q+j} + (a_j - b'_j) b_{j-1} \quad (1 \leq j \leq q), \\ C_\epsilon b_q &= c_{q+1} b_q. \end{aligned} \quad (\text{FS29})$$

In fact this endomorphism in the original basis is independent of ϵ . Its self-adjointness for every metric (FS25) can be verified by multiplying each block: the product of its Gram with its matrix is $\mu \begin{pmatrix} a_j & a_j \\ a_j & a_j + \epsilon^2 b'_j \end{pmatrix}$, which is Hermitian.

Using (FS26)–(FS28), each of the first $q - 1$ two-coordinate blocks contributes

$$2 \frac{a_j + \epsilon^2 b'_j}{1 + \epsilon^2} - 2b'_j = \frac{2(a_j - b'_j)}{1 + \epsilon^2},$$

and the last contributes $(a_q - b'_q)/(1 + \epsilon^2)$. The e_q terms cancel with their equal coefficient one. Therefore the complete trace is

$$\boxed{\text{Tr}((U_\epsilon - W_\epsilon)C_\epsilon) = \frac{B_q(c)}{1 + \epsilon^2}.} \quad (\text{FS30})$$

This proves that (FS11) is a sharp supremum even with the original polynomial and relation flags held fixed, when arbitrary positive coefficient metrics are allowed to vary. For nonscalar C , it is never an attained maximum at a finite positive metric by (FS19). The limit $\epsilon = 0$ has Gram determinant zero by (FS25), so it is not an admissible equality metric.

The case $q = 1$ is present literally: $b_0 = 1, b_1 = \chi, b_2 = \chi S, T_\epsilon b_2 = e_0 + \epsilon e_2, W = I - |e_0\rangle\langle e_0|, U = |e_1\rangle\langle e_1| + |v_1\rangle\langle v_1|$, and (FS30) is $(c_3 - c_1)/(1 + \epsilon^2)$.

The difference operator in this family has the exact spectrum

$$\left(-\frac{2}{\sqrt{1 + \epsilon^2}}^{[q-1]}, -\frac{1}{\sqrt{1 + \epsilon^2}}, 0, \frac{1}{\sqrt{1 + \epsilon^2}}, \frac{2}{\sqrt{1 + \epsilon^2}}^{[q-1]} \right). \quad (\text{FS31})$$

To verify it, each first pair in (FS27) gives twice the difference of the rank-one projection onto v_j and that onto e_{q+j} . Its matrix has trace zero and determinant $-1/(1 + \epsilon^2)$, so its eigenvalues are $\pm 1/\sqrt{1 + \epsilon^2}$, before multiplication by two. The last pair has the same matrix without the factor two. The common vector e_q is in the zero eigenspace. This also shows directly how the extremal vector (FS13) is approached.

If it is desired that the chosen C_ϵ be an actual logarithmic derivative of a positive metric path, retain the same coefficient frame and define

$$M_\epsilon(t) = e^{tC_\epsilon^*/2} M_\epsilon e^{tC_\epsilon/2} = M_\epsilon e^{tC_\epsilon}. \quad (\text{FS32})$$

The first expression is positive for every real t . The equality follows from $C_\epsilon^* M_\epsilon = M_\epsilon C_\epsilon$, term by term in the exponential series. Differentiation gives $M_\epsilon(t)^{-1} \dot{M}_\epsilon(t) = C_\epsilon$, with no rescaling of its eigenvalues. A locally affine path $M_\epsilon + tM_\epsilon C_\epsilon$ gives the same derivative at zero and is positive whenever all $1 + tc_i$ are positive.

97.0.7 6. The moment-Gram class remains a stricter class

The original arithmetic and Gamma norms have the additional form

$$\langle P, R \rangle_m = \int_{\mathbb{R}} \overline{P(k/2 + iu)} R(k/2 + iu) m(u) du. \quad (\text{FS33})$$

For all polynomials for which the displayed products belong to the finite space, the literal vertical coordinate forces

$$\langle SP, R \rangle_m + \langle P, SR \rangle_m = k \langle P, R \rangle_m. \quad (\text{FS34})$$

Indeed the integrands on the left add the factor $\overline{k/2 + iu} + (k/2 + iu) = k$. The original mass is the value at $P = R = 1$. These relations are extra linear constraints on the coefficient Gram; positive definiteness by itself does not supply them.

The family (FS24) does not satisfy these moment constraints. Set $P = \chi, R = 1$; then SP has degree $q + 1 \leq 2q$. The exact family gives

$$\langle \chi, 1 \rangle_{M_\epsilon} = 0, \quad \langle S\chi, 1 \rangle_{M_\epsilon} = \mu. \quad (\text{FS35})$$

If $q \geq 2$, $S \in \mathcal{P}_{q-1}$ and its image under (FS21) is a coordinate vector orthogonal to $e_q = T_\epsilon \chi$; hence $\langle \chi, S \rangle = 0$. If $q = 1$, monicity writes $S = \chi - \chi(0)$, and $\langle \chi, S \rangle = \mu$. Thus the left side of (FS34) is μ for $q \geq 2$, and 2μ for $q = 1$, whereas its right side is zero. Since $\mu > 0$, neither case can be a moment Gram of the form (FS33), for any real vertical centre.

Consequently the family proves sharpness for arbitrary finite positive coefficient metrics and for the literal polynomial flags. It does not prove sharpness within the original arithmetic or Gamma moment-Gram class. The exact map (FS23) and the explicit violated relation (FS34) identify the relation between those classes, while preserving the original polynomial and its full relation space. No estimate on the original source multiplier, and no arithmetic equality fixture, is inferred from the larger class.

98 Complete independent four-window sharpness proof and fixed-metric optimum

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 63fdcf0cc874e32cba395d2f2f2a6a5aab16cfa36c4a282708c4ec092041c954

Complete source. sources/next_edition/actual_tau_four_window_v1/sources/four_window_sharpness_review/tau_four_window_spectral_sharpness_independent_review_20260913.md

Source SHA-256. 63fdcf0cc874e32cba395d2f2f2a6a5aab16cfa36c4a282708c4ec092041c954

98.0.1 Independent proof review of four-window spectral sharpness

Date: 13 September 2026. Accepted source: work/tau_four_window_spectral_sharpness_20260913.md, 18872 bytes, SHA-256 11c26c3bcfce135e1882190af9dc3a8f1e4e6487fd282cc252fa441fdf90651c. The complete FS1–FS35 proof was read. The projection and strictness arguments, the explicit coefficient-metric family, its control and difference spectra, and its stated arithmetic scope are accepted without correction.

The source relies on the finite spaces of work/tau_arithmetic_determinant_transport_20260913.tex, 17706 bytes, SHA-256 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae. Those exact spaces, original source maps, and determinant transport were independently verified in work/tau_arithmetic_determinant_transport_independent_review_20260913.md, SHA-256 ce6b9fbdad43e5233b76756f5cbbbadf6aca0a33db62437f7caf0b90321683bb.

No closed AW, WS, or AT source or PDF was changed. This review is an additional complete proof body. It includes an exact refinement of the supplied family: its fixed-metric optimum over the prescribed control spectrum is computed explicitly below. No old test suite, Lean process, numerical moment calculation, or remote operation was used.

98.0.2 1. Literal flags, dimensions, and two exact individual spectra

Fix the original monic polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$. Set $n = 2q + 1$, $\mathcal{H} = \mathcal{P}_{2q}$, and retain an arbitrary positive coefficient form M , with inner product x^*My . All projections and complements in this review use that same form. Let P_N project onto $\mathcal{P}_N$, and Q_N onto $\chi\mathcal{P}_{N-q}$, with $Q_{q-1} = 0$. Define

$$U = Q_{2q-1} + Q_{2q} - Q_q, \quad W = P_{2q-1} - P_{q-1} + I - P_q, \quad D = U - W. \quad (\text{SR1})$$

For nested subspaces $X \subset Y$, their orthogonal projections have products $P_X P_Y = P_Y P_X = P_X$. Indeed the orthogonal decomposition $X \oplus (Y \cap X^\perp) \oplus Y^\perp$ gives their diagonal actions. Thus each of the individual degree and relation flags is a commuting projection flag.

Set $B_0 = \mathbb{C}\chi$, $B_1 = \chi\mathcal{P}_{q-1}$, $B_2 = \chi\mathcal{P}_q$, of dimensions $1, q, q + 1$. These dimensions use injectivity of multiplication by the original nonzero polynomial. The relation flag decomposes $\mathcal{H}$ orthogonally into B_0 , $B_1 \cap B_0^\perp$, $B_2 \cap B_1^\perp$, and $B_2^\perp$. Substitution in SR1 shows that U has respective values $1, 2, 1, 0$. The degree flag similarly decomposes into $\mathcal{P}_{q-1}$, $\mathcal{P}_q \cap \mathcal{P}_{q-1}^\perp$, $\mathcal{P}_{2q-1} \cap \mathcal{P}_q^\perp$, and $\mathcal{P}_{2q-1}^\perp$; W has respective values $0, 1, 2, 1$. Consequently both operators have the precise spectrum

$$0^{[q]}, \quad 1^{[2]}, \quad 2^{[q-1]}, \quad \text{Tr } U = \text{Tr } W = 2q, \quad \text{Tr } D = 0. \quad (\text{SR2})$$

For $q = 1$, the multiplicity of two is zero; $B_0 = B_1$, and $U = Q_2$. These are sums of projections with the indicated weights; for $q > 1$ neither is an idempotent.

98.0.3 2. Trace rearrangement with all equality conditions needed later

Let A be self-adjoint with ordered eigenvalues $a_1 \leq \dots \leq a_n$. For a rank- r orthogonal projection E , choose an orthonormal eigenbasis v_i , and write $e_i = \langle v_i, Ev_i \rangle = \|Ev_i\|^2$. Then $0 \leq e_i \leq 1$, $\sum_i e_i = r$, and $\text{Tr}(AE) = \sum_i a_i e_i$.

The largest possible sum of these weights is the sum of the largest r eigenvalues. To see this directly, choose a threshold a_{n-r+1} . The mass on indices strictly below the threshold has nonpositive contribution after subtracting that threshold times r , and the mass above it contributes at most the sum of their full coefficients. Assigning weight one at the top r indices reaches the resulting bound. The analogous argument at the bottom gives

$$\sum_{i=1}^r a_i \leq \text{Tr}(AE) \leq \sum_{i=n-r+1}^n a_i. \quad (\text{SR3})$$

Equality on the upper side forces E to contain every eigenspace with eigenvalue strictly above the cutoff and be orthogonal to every eigenspace strictly below it. A diagonal value one gives $\|(I - E)v_i\|^2 = 0$; a value zero gives $Ev_i = 0$. The only free subspace is inside the eigenspace at the cutoff. The lower side has the reversed conditions. Thus this proof retains arbitrary repeated eigenvalues.

For a self-adjoint C with ordered eigenvalues $c_1 \leq \dots \leq c_n$, decompose U as its positive spectral projection plus its level-two projection, of ranks $q + 1$ and $q - 1$. Applying SR3 and doing the same for W yields

$$\begin{aligned} \text{Tr}(UC) &\leq c_{q+1} + c_{q+2} + 2 \sum_{i=q+3}^{2q+1} c_i, \\ \text{Tr}(WC) &\geq c_q + c_{q+1} + 2 \sum_{i=1}^{q-1} c_i. \end{aligned} \quad (\text{SR4})$$

Subtracting, and also reversing the two extrema, proves

$$|\text{Tr}(DC)| \leq B_q(c), \quad B_q(c) = c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{SR5})$$

For $q = 1$, this is $c_3 - c_1$. If only the two individual spectra in SR2 are prescribed, equality is attained by diagonal operators whose ordered entries are $u = (0^{[q]}, 1, 1, 2^{[q-1]})$ and its reversal, in an eigenbasis of C . Their difference has ordered eigenvalues

$$\lambda^* = (-2^{[q-1]}, -1, 0, 1, 2^{[q-1]}). \quad (\text{SR6})$$

Direct multiplication of the diagonal entries gives precisely SR5 with positive sign; exchanging the two operators gives the negative sign. This proves the claimed sharp value for that class of two independent spectral orbits.

98.0.4 3. The polynomial intersections exclude every nontrivial rank equality

The two needed positions of the literal polynomial subspaces follow solely from degrees:

$$\mathcal{P}_{q-1} \cap B_2 = 0, \quad \mathcal{P}_q + B_2 = \mathcal{H}. \quad (\text{SR7})$$

A nonzero multiple of χ has degree at least q , proving the first. In the second, $B_2 \cap \mathcal{P}_q = \mathbb{C}\chi$, because multiplication by a nonconstant polynomial has degree exceeding q . Its sum therefore has dimension $(q + 1) + (q + 1) - 1 = 2q + 1$, proving equality with the whole space.

Let $U_j = \ker(U - jI)$, $W_j = \ker(W - jI)$. The eigenspaces computed in Section 1 and SR7 imply

$$U_2 \cap W_0 = 0, \quad W_2 \cap U_0 = 0. \quad (\text{SR8})$$

The first uses $U_2 \subset B_2$ and $W_0 = \mathcal{P}_{q-1}$. For the second, a vector in $W_2 \cap U_0$ is orthogonal to both $\mathcal{P}_q$ and B_2 , hence to their whole sum.

Let a_r be the sum of the largest r entries of SR6, for $1 \leq r < n$. Applying SR3 separately to U and W gives, for every rank- r orthogonal projection E ,

$$\mathrm{Tr}(DE) \leq \sum_{i=n-r+1}^n u_i - \sum_{i=1}^r u_i = a_r. \quad (\text{SR9})$$

Equality in this subtraction requires equality in both component bounds; their two deficits are nonnegative. The exact equality conditions from SR3 now give the following exhaustive cases.

For $1 \leq r \leq q-1$, the maximum for U is $2r$ and the minimum for W is zero. Equality therefore requires $\mathrm{im} E \subset U_2 \cap W_0$, excluded by SR8.

For $r = q$ and $q \geq 2$, minimization for W forces $\mathrm{im} E = W_0$, because that kernel already has dimension q . Maximization for U forces the image to contain its entire level-two space of dimension $q-1 > 0$. This contradicts $U_2 \cap W_0 = 0$. At $q = r = 1$, the maximum for U instead says $\mathrm{im} E \subset B_2$, while the minimum for W says $\mathrm{im} E = \mathcal{P}_0$; SR7 excludes this case as well.

For $r = q+1$, the maximum for U forces the image to equal its entire positive spectral space B_2 : its dimension is exactly $q+1$. The minimum for W forces the image to contain $W_0 = \mathcal{P}_{q-1}$. Since $q \geq 1$, that subspace is nonzero, contradicting SR7.

For $q+2 \leq r \leq 2q$, put $F = I - E$, of rank $s = n - r$ with $1 \leq s \leq q-1$. Since $\mathrm{Tr} D = 0$ and the spectrum SR6 is invariant under negation, $a_r = a_s = 2s$. Equality in SR9 would force $\mathrm{Tr}(DF) = -2s$. The lower bound for U and upper bound for W then require $\mathrm{im} F \subset U_0 \cap W_2$, again excluded by SR8.

These arguments cover every rank including $q = 1$, and use the actual repeated eigenspaces of U, W ; they do not suppose that C has simple spectrum.

For a fixed positive M , the rank- r orthogonal projections form a compact set. Conjugation by the explicit isometry $M^{1/2}$ identifies it with the Euclidean self-adjoint idempotents of trace r , a closed bounded set in finite matrix dimension. Hence the continuous functional $\mathrm{Tr}(DE)$ attains its maximum. Since equality in SR9 is impossible, if $d_1 \leq \dots \leq d_n$ are the actual eigenvalues of D , SR3 applied to D gives

$$\sum_{i=n-r+1}^n d_i < a_r \quad (1 \leq r < n), \quad \sum_i d_i = 0. \quad (\text{SR10})$$

98.0.5 4. Strictness for every nonscalar control, with repeated eigenvalues retained

Choose one orthonormal eigenbasis of C , in any order inside a repeated eigenspace, and let E_r project onto its last r vectors. The exact expansion is

$$C = c_1 I + \sum_{r=1}^{n-1} (c_{n-r+1} - c_{n-r}) E_r. \quad (\text{SR11})$$

All coefficients are nonnegative. The scalar term has zero trace against D . Applying SR10 term by term therefore yields

$$\mathrm{Tr}(DC) < \sum_{r=1}^{n-1} (c_{n-r+1} - c_{n-r}) a_r = B_q(c) \quad (\text{SR12})$$

whenever C is nonscalar: at least one spectral gap is then strictly positive, and its corresponding inequality in SR10 is strict. The last equality follows by telescoping with the ordered zero-sum vector SR6, or by multiplying its entries with c_i . Applying the same argument to $-C$ proves the negative bound; its ordered eigenvalues are $-c_n, \dots, -c_1$, and the symmetry of SR6 gives the same $B_q(c)$. Hence

$$|\mathrm{Tr}(DC)| < B_q(c) \quad \text{for every nonscalar } C \text{ and every fixed } M > 0. \quad (\text{SR13})$$

For scalar C , both sides are zero. Every a_r is strictly positive, so SR11 also proves $B_q(c) > 0$ for every nonscalar C . This completes the repeated-spectrum part of FS19.

The finite strictness has an exact quantitative record: define

$$g_r(M) = a_r - \sum_{i=n-r+1}^n d_i > 0. \quad (\text{SR14})$$

Before making SR12 strict, its positive-side deficit satisfies

$$B_q(c) - \text{Tr}(DC) \geq \sum_{r=1}^{n-1} (c_{n-r+1} - c_{n-r})g_r(M) > 0. \quad (\text{SR15})$$

For the negative side use the expansion of $-C$ and its own ordered gaps against the same $g_r(M)$. This retains every spectral gap and proves exactly where strictness comes from. It supplies no uniform lower bound for those positive deficits as M varies.

98.0.6 5. The complete metric family on the unchanged polynomial flag

Retain every coefficient of the original χ . The ordered polynomials

$$b_i = S^i \ (0 \leq i < q), \quad b_q = \chi, \quad b_{q+j} = \chi S^j \ (1 \leq j \leq q) \quad (\text{SR16})$$

have increasing degrees and leading coefficient one. Their coordinate matrix relative to the monomial frame is triangular with determinant one, and each initial segment spans the original degree space.

Fix $\mu > 0, \epsilon > 0$. Define $T_\epsilon : \mathcal{H} \rightarrow \mathbb{C}^n$ by

$$T_\epsilon b_i = e_i \ (0 \leq i \leq q), \quad T_\epsilon b_{q+j} = e_{j-1} + \epsilon e_{q+j} \ (1 \leq j \leq q). \quad (\text{SR17})$$

It has determinant ϵ^q , and its inverse sends e_i to b_i for $i \leq q$, and e_{q+j} to $\epsilon^{-1}(b_{q+j} - b_{j-1})$. Substituting each basis vector into these two formulas verifies both inverse compositions.

Set $M_\epsilon = \mu T_\epsilon^* T_\epsilon$ as a form on the original polynomial coefficients. The explicit isometry to the Euclidean target is $\sqrt{\mu} T_\epsilon$. For $P = \sum_i x_i b_i$, its full norm is

$$\|P\|_{M_\epsilon}^2 = \mu \left(|x_q|^2 + \sum_{j=1}^q |x_{j-1} + x_{q+j}|^2 + \epsilon^2 \sum_{j=1}^q |x_{q+j}|^2 \right). \quad (\text{SR18})$$

The cross term in each two-coordinate block is retained. Each such block is $\mu \begin{pmatrix} 1 & 1 \\ 1 & 1 + \epsilon^2 \end{pmatrix}$, of determinant $\mu^2 \epsilon^2$, and the remaining b_q block is μ . Therefore

$$\det M_\epsilon = \mu^{2q+1} \epsilon^{2q}, \quad \|1\|^2 = \|\chi\|^2 = \mu, \quad \|\chi S^j\|^2 = \mu(1 + \epsilon^2), \quad \langle S^{j-1}, \chi S^j \rangle = \mu. \quad (\text{SR19})$$

The determinant is the same in the original monomial frame, by the unit determinant of SR16. All unlisted cross terms between different paired blocks or the b_q block are zero.

The map SR17 transports the degree flag to the standard coordinate flag. Thus W has values zero on $e_0, \dots, e_{q-1}$, one on e_q, e_{2q} , and two on $e_{q+1}, \dots, e_{2q-1}$. Define the orthonormal vectors

$$v_j = \frac{e_{j-1} + \epsilon e_{q+j}}{\sqrt{1 + \epsilon^2}} \quad (1 \leq j \leq q). \quad (\text{SR20})$$

They are orthogonal to one another and to e_q . The transformed relation spaces are spanned by e_q , then $e_q, v_1, \dots, v_{q-1}$, then $e_q, v_1, \dots, v_q$. Their projections therefore give

$$U = |e_q\rangle\langle e_q| + 2 \sum_{j=1}^{q-1} |v_j\rangle\langle v_j| + |v_q\rangle\langle v_q|. \quad (\text{SR21})$$

The unit vectors in SR20 describe the coordinate isometry; the original masses and norms remain SR18–SR19.

98.0.7 6. Prescribed spectrum, exact trace, and full difference spectrum

For prescribed ordered real numbers $c_1, \dots, c_n$, set

$$a_j = c_{2q+2-j}, \quad b'_j = c_j \quad (1 \leq j < q), \quad a_q = c_{q+2}, \quad b'_q = c_q. \quad (\text{SR22})$$

Give e_{j-1} eigenvalue a_j , e_{q+j} eigenvalue b'_j , and e_q eigenvalue c_{q+1} . This diagonal C_0 uses every prescribed eigenvalue exactly once. Its pullback $C_\epsilon = T_\epsilon^{-1}C_0T_\epsilon$ has, on each ordered pair (b_{j-1}, b_{q+j}) , matrix

$$\begin{pmatrix} a_j & a_j - b'_j \\ 0 & b'_j \end{pmatrix}. \quad (\text{SR23})$$

In particular its entries are independent of ϵ . Multiplication by the original Gram block gives

$$\mu \begin{pmatrix} 1 & 1 \\ 1 & 1 + \epsilon^2 \end{pmatrix} \begin{pmatrix} a_j & a_j - b'_j \\ 0 & b'_j \end{pmatrix} = \mu \begin{pmatrix} a_j & a_j \\ a_j & a_j + \epsilon^2 b'_j \end{pmatrix}, \quad (\text{SR24})$$

which is Hermitian because a_j, b'_j are real. This proves self-adjointness in every M_ϵ , retaining all masses and cross terms.

In the isometric coordinates, each paired block of D is

$$D_j = \frac{w_j}{1 + \epsilon^2} \begin{pmatrix} 1 & \epsilon \\ \epsilon & -1 \end{pmatrix}, \quad w_j = 2 \ (j < q), \quad w_q = 1. \quad (\text{SR25})$$

The common coordinate e_q has value zero. The diagonal C_0 therefore contributes $w_j(a_j - b'_j)/(1 + \epsilon^2)$ to the trace of each block. Summing proves

$$\text{Tr}(DC_\epsilon) = \frac{B_q(c)}{1 + \epsilon^2}. \quad (\text{SR26})$$

The unweighted matrix in SR25 has trace zero and determinant $-1/(1 + \epsilon^2)$. Its two eigenvalues are $\pm 1/\sqrt{1 + \epsilon^2}$. Including the weights gives the exact spectrum

$$\text{Spec } D = \left(-\frac{2}{\sqrt{1 + \epsilon^2}}^{[q-1]}, -\frac{1}{\sqrt{1 + \epsilon^2}}, 0, \frac{1}{\sqrt{1 + \epsilon^2}}, \frac{2}{\sqrt{1 + \epsilon^2}}^{[q-1]} \right). \quad (\text{SR27})$$

Superscripts in square brackets denote multiplicities, as in SR2, rather than powers. The case $q = 1$ simply has one paired block of weight one and one zero coordinate; SR26 is $(c_3 - c_1)/(1 + \epsilon^2)$.

For any nonscalar prescribed spectrum, SR26 tends to $B_q(c)$ as ϵ decreases to zero through positive values. Yet $\det M_\epsilon$ tends to zero by SR19, so the limiting form is outside the positive class. Together with SR13, this proves a sharp supremum and excludes an attained nontrivial maximum within the literal polynomial flags.

98.0.8 7. Additional exact fixed-metric optimum for this family

The source's chosen control in SR23 is independent of ϵ , and its trace is exactly SR26. The full difference spectrum SR27 also permits computing the optimum over all controls with the prescribed spectrum at any fixed $\epsilon > 0$.

Set $\rho = \sqrt{1 + \epsilon^2}$. On the j -th pair, define

$$f_j^+ = \frac{(\rho + 1)e_{j-1} + \epsilon e_{q+j}}{\sqrt{2\rho(\rho + 1)}}, \quad f_j^- = \frac{-\epsilon e_{j-1} + (\rho + 1)e_{q+j}}{\sqrt{2\rho(\rho + 1)}}. \quad (\text{SR28})$$

Their inner product is zero, and their squared norms are one because $(\rho + 1)^2 + \epsilon^2 = 2\rho(\rho + 1)$. Multiplication by the unscaled matrix $\begin{pmatrix} 1 & \epsilon \\ \epsilon & -1 \end{pmatrix}$ gives eigenvalues $+\rho$ and $-\rho$, respectively. Hence $D_j f_j^\pm = \pm(w_j/\rho) f_j^\pm$.

Define $\tilde{C}_0$ to assign a_j to f_j^+ , b'_j to f_j^- , and c_{q+1} to e_q , and pull it back to the original polynomial space by $T_\epsilon^{-1}\tilde{C}_0T_\epsilon$. This is a fully specified M_ϵ -self-adjoint operator with the required eigenvalue multiset. Its trace is

$$\text{Tr}(D\tilde{C}_\epsilon) = \frac{1}{\rho} \sum_j w_j(a_j - b'_j) = \frac{B_q(c)}{\sqrt{1 + \epsilon^2}}. \quad (\text{SR29})$$

For any other control with that spectrum, the spectral projection expansion SR11 and SR3 applied to D bound its trace above by the ordered pairing of its eigenvalues with SR27, which is the right side of SR29. Reversing the control assignment gives the corresponding negative bound. Thus the exact maximum of the absolute trace, at this fixed M_ϵ , is $B_q(c)/\sqrt{1 + \epsilon^2}$.

This additional formula preserves the correctness of the source's particular choice SR26. It computes the optimization left available when its prescribed spectrum is allowed a different self-adjoint realization. It remains in the same positive coefficient-metric family. Its full Ky Fan deficits are $g_r(M_\epsilon) = a_r(1 - 1/\sqrt{1 + \epsilon^2})$, by SR14 and SR27, where a_r here is the reference sum from SR9.

98.0.9 8. Metric-path realization and its literal mass

The source's fixed polynomial-coordinate operator C_ϵ satisfies $C_\epsilon^* M_\epsilon = M_\epsilon C_\epsilon$. Therefore

$$M_\epsilon(t) = e^{tC_\epsilon^*/2} M_\epsilon e^{tC_\epsilon/2} = M_\epsilon e^{tC_\epsilon} \quad (\text{SR30})$$

is positive for every real t . The equality follows by the exponential series and the stated intertwining. Differentiating gives $M_\epsilon(t)^{-1} \dot{M}_\epsilon(t) = C_\epsilon$. Thus the operator is an actual logarithmic derivative, with the same eigenvalues, on an explicit positive coefficient-metric path.

For the locally affine path $M_\epsilon + tM_\epsilon C_\epsilon$, positivity is equivalent to $1 + tc_i > 0$ for all prescribed eigenvalues, after the same explicit isometry. Its logarithmic derivative at zero is again C_ϵ .

For clarity, the fixed mass statement $\|1\|^2 = \mu$ holds across the parameter family $\epsilon > 0$ at $t = 0$. The path's original constant polynomial has eigenvalue c_n : this is $a_1 = c_{2q+1}$ for $q \geq 2$, and $a_q = c_3$ at $q = 1$. Consequently its literal norm along SR30 is μe^{tc_n} , and along the affine path is $\mu(1 + tc_n)$. No claim that those paths preserve mass in t is needed or made; these formulas give the actual factors.

98.0.10 9. Exact moment obstruction and the retained arithmetic scope

Put $N = 2q$. Let $\mathcal{M}_N$ be the real vector space of finite real signed measures ν satisfying $\int (1 + |u|^{2N}) d|\nu|(u) < \infty$. On a vertical line $S = k/2 + iu$, the real-linear moment map $\mathfrak{M}_k : \mathcal{M}_N \rightarrow \text{Herm}(\mathcal{P}_N)$ is

$$\mathfrak{M}_k(\nu)(P, R) = \int \overline{P(k/2 + iu)} R(k/2 + iu) d\nu(u). \quad (\text{SR31})$$

For a nonnegative measure the form is positive semidefinite, and it is positive definite when no nonzero polynomial in the source vanishes almost everywhere. The original positive densities satisfy this condition. For $P, R \in \mathcal{P}_{N-1}$, evaluation gives the real-linear constraint map $\mathcal{L}_k : \text{Herm}(\mathcal{P}_N) \rightarrow \text{Herm}(\mathcal{P}_{N-1})$,

$$\mathcal{L}_k(M)(P, R) := M(SP, R) + M(P, SR) - kM(P, R) = 0 \quad \text{when } M = \mathfrak{M}_k(\nu). \quad (\text{SR32})$$

Indeed $\overline{S(u)} + S(u) = k$ pointwise, so the three integrands cancel. In monomial coefficients, this linear map has entries $M_{i+1,j} + M_{i,j+1} - kM_{ij}$ wherever both shifted indices occur in the finite domain. Thus SR31 has image in the kernel of the explicitly typed linear map $\mathcal{L}_k$. This supplies the exact relation between moment forms and general coefficient forms.

In the constructed family take $P = \chi, R = 1$. All products belong to $\mathcal{P}_{2q}$, because $q + 1 \leq 2q$. Equations SR17–SR19 give

$$\langle \chi, 1 \rangle = 0, \quad \langle S\chi, 1 \rangle = \mu. \quad (\text{SR33})$$

For $q \geq 2$, $S = b_1$ is in the lower degree flag and is orthogonal to $b_q = \chi$, so $\langle \chi, S \rangle = 0$. At $q = 1$, writing the original monic polynomial as $\chi = S + \chi(0)$ gives $S = \chi - \chi(0)$, hence $\langle \chi, S \rangle = \mu$. The convention is conjugate-linear in the first slot and linear in the second; the coefficient $-\chi(0)$ multiplies the zero cross pairing, so no unretained conjugation occurs. Therefore

$$\mathcal{L}_k(M_\epsilon)(\chi, 1) = \begin{cases} \mu, & q \geq 2, \\ 2\mu, & q = 1. \end{cases} \quad (\text{SR34})$$

This is nonzero for every $\epsilon > 0, \mu > 0$, and the subtracted term is zero for every real center parameter k . Thus this family violates a necessary moment identity for every real vertical line, not only the particular arithmetic center.

The original Gamma and arithmetic source forms remain in the image of their actual measures under SR31, with masses $(2\pi)^{k/2}$ and μ_h^k , and with the original χ, η , and $\sigma_h^{\otimes k} \eta$ from the accepted AT source. The constructed M_ϵ is linked to its explicit coordinate model by SR17–SR19 and has the computed nonzero obstruction SR34. It therefore proves sharpness in the larger class of positive coefficient forms while proving no sharpness assertion for the original arithmetic moment class.

For each actual fixed positive moment Gram, the strict inequality SR13 still applies because its literal flags are the same and it belongs to the positive coefficient class. The approach-to-equality family supplied here cannot establish that the actual moment class approaches the same supremum: its obstruction SR34 persists at every finite parameter. No conclusion about a uniform arithmetic improvement or about the failure of the F1 or Tau Base program is drawn from this calibration.

98.0.11 10. Final findings

The strictness proof is valid for every nonscalar control, including arbitrary repeated spectra and $q = 1$. Both middle-rank equality cases and the complementary high-rank case are explicitly excluded by the two original polynomial intersections. The family preserves χ , its coefficient frame and relation flags, its base mass μ , every cross term, and all powers of ϵ . Its trace and full difference spectrum are correct. Its violation of the moment identity is exactly μ or 2μ on the stated pair.

The supplementary independent calculation of FS20–FS35 agreed with these block formulas and required no correction. SR28–SR29 additionally compute the exact prescribed-spectrum optimum for each fixed metric in this family. All these conclusions are add-only; the previously sealed AT proof and artifacts remain unchanged.

99 Complete analytic pole classes, connection, periods and endpoint transport

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 6e8bba584c5788e0d1ea29127ace2d89d9a65a2e111da01df5870d9c9b57d1e7

Complete source. `sources/next_edition/analytic_pole_conversions_v2/sources/analytic_pole/ANALYTIC_POLE_RESIDUE_TRANSPORT.md`

Source SHA-256. 6e8bba584c5788e0d1ea29127ace2d89d9a65a2e111da01df5870d9c9b57d1e7

Current mathematical continuation. APU1–23 below are inserted into the complete original proof at the AP13–AP14 transition. Their source is `sources/current_boundary/fragments/analytic_pole_u_connection.tex`, SHA-256 1b42c530ccee6fd019b8536170277e3909ba43bfe4e3b2fefab3f7b9fc666c05. The preceding source record identifies the preserved historical body; the exact current body and reversible wrapper mapping are pinned by the boundary patch manifest.

99.0.1 Actual Taylor-unit gauge: analytic pole classes and their connection

13 September 2026. This continues UG1–19 on the original arithmetic packet. The accompanying derivation supplied the exact weighted-residue sequence and the connection on simple-pole lifts. This note proves those constructions in full, calculates their entire finite scaling transport and dual, and compares them with the already proved formal contraction. It does not substitute a new source, a new metric, a simple-root packet, or an assumed arithmetic bound.

99.0.2 1. Original input and the fixed nonzero-parameter complex

Keep the monic polynomial h of degree $d > 0$ with every selected actual zero order, $g = 2\xi$, and the unique polynomial ν of degree less than d representing the complete Taylor unit $v_h = j_h(g/h)$. Thus $\gcd(h, \nu) = 1$, and no factor of h or Taylor coefficient of its unit has been removed. Let P_ν be the set of distinct zeros of ν , and r its cardinality; $r = 0$ when ν is constant. The multiplicities of ν are retained in ν itself throughout the gauge.

Fix the original primitive Φ with $\Phi' = h$ and its specified additive constant. For $u \in \mathbb{C}^*$ and $t \in \mathbb{C}$ define

$$\begin{aligned} D &= u \frac{d}{ds} + h(s) - t, \\ w(s; u, t) &= \exp((\Phi(s) - ts)/u), \\ B &= \mathbb{C}[s], \quad B_{\text{loc}} = \mathbb{C}[s, \nu^{-1}], \quad M = B_{\text{loc}}/B. \end{aligned} \tag{AP1}$$

Use complexes $[B \xrightarrow{D} B ds]$ and $[B_{\text{loc}} \xrightarrow{D} B_{\text{loc}} ds]$ in degrees 0, 1. The equation $wDF = u(wF)'$ is literal, with the constant in Φ retained. Although w is entire and nowhere zero in s , it is not asserted rational. The quotient M consists exactly of finite sums of principal parts at P_ν : partial fractions gives existence and uniqueness, with the polynomial part discarded only in this explicitly specified quotient.

99.0.3 2. The entire additional pole cohomology is calculated

Define, for $[F] \in M$,

$$\text{Res}_w[F] = \left(\text{Res}_{s=p} w(s)F(s) ds \right)_{p \in P_\nu}. \quad (\text{AP2})$$

Adding a polynomial changes none of these residues. Also $\text{Res}_w D = 0$ by AP1, since a derivative of a meromorphic Laurent series has zero residue.

The exact sequence is

$$0 \longrightarrow M \xrightarrow{D} M \xrightarrow{\text{Res}_w} \mathbb{C}^{P_\nu} \longrightarrow 0. \quad (\text{AP3})$$

Here is a proof of every assertion. If F has pole order $n \geq 1$ at p and leading term $c(s-p)^{-n}$, DF has leading pole term $-unc(s-p)^{-n-1}$. Multiplication by $h-t$ has pole order at most n , so it cannot cancel this term. Thus $D[F] = 0$ in M implies that F has no poles, and $[F] = 0$. This proves injectivity.

For each p , the rational function $1/(w(p)(s-p))$ has residue vector equal to the p -th coordinate vector. It is analytic at every other point of P_ν . This proves surjectivity without suppressing the factor $w(p)$.

Suppose $\text{Res}_w[F] = 0$. In a disk at p containing no other point of P_ν , the Laurent series of wF has finitely many negative terms and no $(s-p)^{-1}$ term. Integrating term by term therefore gives a meromorphic primitive H_p , with no logarithm. Put $Q_p = H_p/(uw)$. Then $DQ_p = F$ locally. Let Q be the rational function obtained by summing the finite principal parts of the Q_p at all p . The difference $Q - Q_p$ is holomorphic at p , so $DQ - F$ is holomorphic there. This rational function has no other possible finite poles, hence is a polynomial. Therefore $D[Q] = [F]$ in M , which proves exactness in the middle. Changing any integration constant changes only a holomorphic local function and does not change this construction of the principal part.

All higher pole orders are explicitly observed, rather than ignored:

$$\begin{aligned} \text{Res}_p \frac{w(s)}{(s-p)^n} ds &= \frac{w^{(n-1)}(p)}{(n-1)!}, \\ D(s-p)^{-n} &= -un(s-p)^{-n-1} \\ &+ \sum_{j=0}^d \frac{h^{(j)}(p)}{j!} (s-p)^{j-n} - t(s-p)^{-n}. \end{aligned} \quad (\text{AP4})$$

These formulas give the triangular reduction of each higher pole while retaining every derivative of h and every derivative of w that contributes. They explain why there is one quotient coordinate per distinct pole point; they do not replace h by its radical or discard its nilpotent zero jets.

99.0.4 3. The original arithmetic classes inject into the analytic extension

The polynomial complex has $H^0 = 0$ and H^1 of dimension d , with basis $1, s, \dots, s^{d-1}$. Indeed, for a nonzero polynomial of degree n , the highest term of DF has degree $n + d$, with unchanged leading coefficient. Successive subtraction of such leading terms reduces every polynomial uniquely to degree less than d . Write this unchanged remainder module as H_{pol} .

The same highest-pole argument shows H^0 of the localized complex is zero: a rational solution of $DF = 0$ has no finite pole, and then the preceding polynomial argument applies. The short exact sequence of the three complexes, together with AP3, gives

$$0 \longrightarrow H_{\text{pol}} \xrightarrow{i} H_{\text{loc}} \xrightarrow{\text{Res}_w} \mathbb{C}^{P_\nu} \longrightarrow 0. \quad (\text{AP5})$$

One can check injectivity without a long exact sequence. If a polynomial F equals DQ for rational Q , the leading-pole calculation forces Q to be a polynomial. Thus no original polynomial cohomology class becomes a boundary. The kernel and surjectivity assertions follow from AP3 by subtracting DQ and retaining the resulting polynomial.

An explicit basis of H_{loc} is

$$1, s, \dots, s^{d-1}, \quad r_p = \frac{1}{s-p} \quad \text{for } p \in P_\nu. \quad (\text{AP6})$$

For existence, subtract $\sum_p (\text{Res}_w F)_p r_p / w(p)$. The remainder has zero weighted residues, and AP3 reduces it modulo D to a polynomial, then to degree less than d . For independence, applying AP2 to a linear combination forces every pole coefficient to vanish because $w(p) \neq 0$; injectivity in AP5 then gives independence of the polynomial coefficients. Hence $\dim H_{\text{loc}} = d + r$. The original d -dimensional arithmetic fibre, with all repeated roots of h , is present as the explicitly injected summand of this vector-space basis.

99.0.5 4. Actual parameter connection and both explicit residue frames

The differential operator $L = u \, d/dt - s$ commutes with D :

$$\left[u \frac{d}{dt}, D \right] = -u, \quad \left[-s, u \frac{d}{ds} \right] = u, \quad [-s, h - t] = 0. \quad (\text{AP7})$$

It therefore descends as a connection operator on the analytic family over $\mathbb{C}^* \times \mathbb{C}$, not as an unsupported multiplication endomorphism on the fixed de Rham quotient. Let $A(t)$ be the original companion matrix of $h - t$ in the polynomial frame, and let e_0 be its constant-polynomial column. Set

$$P = \text{diag}(p)_{p \in P_\nu}, \quad B_0 = (e_0 \quad \cdots \quad e_0), \quad (\text{AP8})$$

$$A_{\text{ext}}(t) = \begin{bmatrix} A(t) & B_0 \\ 0 & P \end{bmatrix}.$$

Here each pole basis vector satisfies the exact identity

$$L r_p = -\frac{s}{s - p} = -p r_p - 1. \quad (\text{AP9})$$

For the last polynomial vector, reduction of s^d uses $D(1) = h - t$, so the polynomial block is exactly the already specified $A(t)$, without a derivative correction. Thus in the full frame AP6 the connection is

$$L = u \frac{d}{dt} - A_{\text{ext}}(t). \quad (\text{AP10})$$

All coefficients depend holomorphically on (u, t) for $u \neq 0$. In fact A_{ext} itself has no singular u coefficient; this statement about the matrix does not identify its rank- $(d + r)$ frame with the formal cohomology at $u = 0$.

Put $W = \text{diag}(w(p; u, t))$. The residue map in AP6 is the rectangular matrix

$$\text{Res}_w = \begin{bmatrix} 0 & W \end{bmatrix}, \quad u \frac{dW}{dt} = -PW. \quad (\text{AP11})$$

For parameter-dependent representatives direct differentiation proves

$$u \frac{d}{dt} \text{Res}_w[F] = \text{Res}_w \left[\left(u \frac{d}{dt} - s \right) F \right]. \quad (\text{AP12})$$

Thus the quotient connection in weighted-residue coordinates is $u \, d/dt$. The ordinary simple-pole coefficients instead have $u \, d/dt - P$. Their exact transition is multiplication by W , not deletion of an exponential phase.

For completeness use the second frame $r_p/w(p)$. Its residue coordinates are the standard coordinate vectors, and

$$L \left(\frac{r_p}{w(p)} \right) = -\frac{1}{w(p)}. \quad (\text{AP13})$$

The pole diagonal becomes zero while its coupling to the original polynomial class is $-e_0/w(p)$. This is the same connection under the explicit diagonal frame map $\text{diag}(I_d, W^{-1})$; no factor has been set to one in the original frame.

99.0.6 The complete two-parameter connection on the original analytic pole extension

This calculation continues AP1–26 with its original monic h of degree $d > 0$, primitive $\Phi' = h$, full Taylor representative ν of $j_h(g/h)$, and localization $\mathbb{C}[s, \nu^{-1}]$. The polynomial ν and its complete multiplicities are fixed; its distinct pole points $P_\nu = \{p : \nu(p) = 0\}$ do not vary with u or t . Write $r = \#P_\nu$, $n = d + 1$, and

$$f(s, t) = \Phi(s) - ts, \quad D = u\partial_s + h(s) - t, \quad w(s; u, t) = e^{f(s, t)/u}. \quad (\text{APU1})$$

All formulas below are on $S = \mathbb{C}_u^\times \times \mathbb{C}_t$. The original coefficient frame in AP6 is

$$(1, s, \dots, s^{d-1}, (r_p)_{p \in P_\nu}), \quad r_p = (s - p)^{-1}. \quad (\text{APU2})$$

The free polynomial submodule, the localized module, and their quotient are the actual cohomology modules in AP5. In particular there is no new source frame that discards their extension.

Degree-correct chain connections. On both the polynomial and localized complexes, in degree $j = 0, 1$, define

$$\nabla_u^j = \partial_u - \frac{f}{u^2} + \frac{1-j}{u}, \quad \nabla_t^j = \partial_t - \frac{s}{u}. \quad (\text{APU3})$$

Multiplication by f and differentiation in the parameters preserve the stated localization, since all poles are fixed. The two connections are actual cochain maps:

$$\nabla_u^1 D = D \nabla_u^0, \quad \nabla_t^1 D = D \nabla_t^0. \quad (\text{APU4})$$

Indeed $\partial_u D - D \partial_u = \partial_s, [-f/u^2, D] = f_s/u = (h-t)/u$, and $D/u = \partial_s + (h-t)/u$; their three terms cancel in the first identity. For the second, $[\partial_t, D] = -1$ and $[-s/u, D] = 1$. The original degree-zero $1/u$ term is essential in the first cancellation. Their curvature vanishes in each degree:

$$[\nabla_u^j, \nabla_t^j] = 0. \quad (\text{APU5})$$

The only nonzero terms are $[\partial_u, -s/u] = s/u^2$ and $[-f/u^2, \partial_t] = f_t/u^2 = -s/u^2$. The scalar $(1-j)/u$ commutes with ∇_t^j . Thus the descended connection is flat on the original two-parameter family, with no commutation assertion inferred from pointwise matrix coincidence.

Every coefficient of the u -connection. For $0 \leq b < d$, perform the original monic polynomial division

$$f(s, t)s^b = (h(s) - t)q_b(s, t) + c_b(s, t), \quad \deg c_b < d.$$

Here $\deg q_b \leq b + 1$, so $\deg q'_b \leq b < d$. Let $C(t)$ have coefficient column c_b and $B(t)$ have coefficient column q'_b , both in the original ascending polynomial frame. Subtraction of Dq_b gives the exact differential remainder $[fs^b] = [c_b - uq'_b]$. Therefore the polynomial block is

$$\Omega_{\text{pol}} = -\frac{C(t)}{u^2} + \frac{B(t)}{u}. \quad (\text{APU6})$$

This is the full polynomial connection from the marked-family calculation, with its derivative correction retained.

For a fixed $p \in P_\nu$, set

$$f_p(t) = f(p, t), \quad q_p(s, t) = \frac{f(s, t) - f(p, t)}{s - p}, \quad Q_p(s, t) = q_p(s, t) - \frac{h(s) - t}{n}. \quad (\text{APU7})$$

Since h is monic of degree d , the leading coefficient of Φ is $1/n$. Thus q_p has degree d and leading coefficient $1/n$; its ordinary quotient by $h-t$ is precisely the constant $1/n$, and Q_p has degree less than d . The derivative of this quotient is zero. Subtracting $D(1/n)$ therefore proves the full differential-remainder equality $[q_p] = [Q_p]$, at every $u \neq 0$ and every t . Now the literal rational identity $fr_p = q_p + f_p r_p$ gives

$$\nabla_u^1[r_p] = -\frac{[Q_p]}{u^2} - \frac{f_p[r_p]}{u^2}. \quad (\text{APU8})$$

No higher-pole or derivative term is omitted: fixed p implies $\partial_u r_p = \partial_t r_p = 0$, and the only polynomial quotient just computed is constant.

Let $Q(t)$ be the d -by- r matrix of these exact Q_p columns and $F(t) = \text{diag}(f_p(t))$. Retain $A(t)$, $B_0 = (e_0, \dots, e_0)$, and $P = \text{diag}(p)$ from AP8. The complete coefficient connections in the frame APU2 are

$$\begin{aligned} \nabla_u &= \partial_u + \Omega_u, & \Omega_u &= \begin{bmatrix} -C(t)/u^2 + B(t)/u & -Q(t)/u^2 \\ 0 & -F(t)/u^2 \end{bmatrix}, \\ \nabla_t &= \partial_t + \Omega_t, & \Omega_t &= -\frac{1}{u} \begin{bmatrix} A(t) & B_0 \\ 0 & P \end{bmatrix}. \end{aligned} \quad (\text{APU9})$$

The second formula is exactly AP10 divided by u , not a change of its connection convention.

The off-diagonal flatness can also be checked directly in these original coefficient matrices. First,

$$\partial_t Q_p = -\frac{d}{n} e_0, \quad B(t) e_0 = \frac{1}{n} e_0.$$

The first identity differentiates APU7, using $\partial_t q_p = -1$; the second uses the constant derivative of the leading linear quotient of f by $h - t$. Also reducing

$$(s - p)Q_p = f - f_p - \frac{(s - p)(h - t)}{n}$$

modulo the original polynomial $h - t$ proves

$$(A(t) - pI_d)Q_p = C(t)e_0 - f_p e_0. \quad (\text{APU10})$$

Consequently the upper-right curvature block equals

$$\frac{B_0 + \partial_t Q - BB_0}{u^2} + \frac{CB_0 + QP - AQ - B_0 F}{u^3} = 0.$$

The lower-right curvature is zero because $F_t = -P$; the polynomial block is flat by APU4–5. This proves the matrix curvature identity $\partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] = 0$ with every extension column retained.

Weighted residues, oriented loops, and their exact horizontal maps. Preserve separately the two actual matrices

$$W_{\text{res}}(u, t) = \text{diag}(e^{f_p(t)/u}), \quad W_{\text{loop}}(u, t) = (2\pi i)W_{\text{res}}(u, t). \quad (\text{APU11})$$

The first is the matrix W in AP11: its rectangular map $\mathcal{R} = [0 \ W_{\text{res}}]$ is the weighted residue, with no $2\pi i$ in its definition. The second gives the positively oriented small-loop period map $\mathcal{L} = [0 \ W_{\text{loop}}] = (2\pi i)\mathcal{R}$. Direct differentiation gives

$$\partial_u W_{\text{res}} = -FW_{\text{res}}/u^2, \quad \partial_t W_{\text{res}} = -PW_{\text{res}}/u, \quad \partial_u \mathcal{R} = \mathcal{R}\Omega_u, \quad \partial_t \mathcal{R} = \mathcal{R}\Omega_t. \quad (\text{APU12})$$

The same equations hold for $\mathcal{L}$. Equivalently, for every parameter-dependent rational representative $a(s; u, t)$,

$$\partial_u \text{Res}_w[a] = \text{Res}_w[\nabla_u^1 a], \quad \partial_t \text{Res}_w[a] = \text{Res}_w[\nabla_t^1 a].$$

To prove these equalities, use $\partial_u w = -(f/u^2)w$ and $\partial_t w = -(s/u)w$ in its convergent Laurent expansion on a circle containing that fixed pole and no other pole. Parameter differentiation and the finite contour integral commute. These identities descend because both residue maps kill D -boundaries by AP1. Thus AP5 is an exact sequence of the actual flat connections in both parameters: its polynomial inclusion is horizontal and its weighted-residue quotient has the ordinary trivial connection.

The explicit AP13 frame comparison remains available: replace only the pole columns by $r_p/w(p)$, through the displayed matrix $\text{diag}(I_d, W_{\text{res}}^{-1})$. In that frame their exact derivatives are

$$\nabla_u[r_p/w(p)] = -[Q_p]/(u^2 w(p)), \quad \nabla_t[r_p/w(p)] = -e_0/(uw(p)). \quad (\text{APU13})$$

The pole diagonal terms cancel against derivatives of the displayed inverse weights. The off-diagonal polynomial extension remains, with its full inverse exponential in both directions.

For the unchanged rapid-decay contours and positively oriented small loops of AP19–19b, their complete period matrix satisfies

$$\partial_u \Pi_{\text{ext}} = \Pi_{\text{ext}} \Omega_u, \quad \partial_t \Pi_{\text{ext}} = \Pi_{\text{ext}} \Omega_t. \quad (\text{APU14})$$

Here the contours can be held fixed locally in an admitted u -sector: smaller compact subsectors retain the same decaying ends; their finite detours avoid the fixed poles. The negative leading exponential dominates its polynomial derivatives and simple-pole factors uniformly on compact parameter subsets. This proves differentiation under the integral. The differential relations integrate to zero at the decaying ends or around the closed loops, so APU6–8 give the stated matrices. The lower period block is exactly W_{loop} , and the determinant remains the full AP19a expression

$$\det \Pi_{\text{ext}} = \det \Pi_{\text{pol}} (2\pi i)^r \exp \left(\frac{\sum_{p \in P_\nu} f_p(t)}{u} \right). \quad (\text{APU15})$$

In particular its u -derivative is fixed by $\partial_u \det \Pi_{\text{ext}} = (\text{Tr } \Omega_u) \det \Pi_{\text{ext}}$; the polynomial trace, every pole phase, and the $2\pi i$ factors retain their original values.

The full Taylor gauge and its dual commute with the new connection. The original cochain isomorphism multiplies both degrees by the complete polynomial ν , with inverse ν^{-1} , and has

$$D^\nu = \nu D \nu^{-1} = u \partial_s + h - t - u \nu' / \nu.$$

Because ν is fixed in both parameters, it commutes with ∇_u^j and ∇_t^j in APU3. Conjugating APU4–5 therefore proves the same chain and flatness identities for D^ν . In the actual transported frame

$$(\nu, \nu s, \dots, \nu s^{d-1}, (\nu/(s-p))_{p \in P_\nu})$$

the matrices are exactly APU9. The target weight is w/ν , and

$$(w/\nu)(\nu a) = wa, \quad (w/\nu)D^\nu a = u((w/\nu)a)'. \quad (\text{APU16})$$

These prove preservation of every old contour and every loop period, including all multiplicities of ν . If $\nu/(s-p)$ has no rational pole, its weighted form still equals $w/(s-p)$. The loop class and its original weighted pole are thereby retained.

For the ordinary linear dual, the two connections are $\partial_u - \Omega_u^T$ and $\partial_t - \Omega_t^T$. The injection of the weighted-residue dual is $\mathcal{R}^T$. Transposing APU12 gives

$$(\partial_u - \Omega_u^T)(\mathcal{R}^T a) = \mathcal{R}^T \partial_u a, \quad (\partial_t - \Omega_t^T)(\mathcal{R}^T a) = \mathcal{R}^T \partial_t a. \quad (\text{APU17})$$

Thus its full two-parameter horizontality is proved, with no positive pairing inferred from an algebraic dual. The dual of the Taylor gauge is the inverse transpose of its actual multiplication map, preserving the inverse unit.

The exact endpoint arrow to the original arithmetic source. The singular u -connection is specified on $u \neq 0$. Its comparison with the original marked endpoint uses the actual arrow already calculated in AP23–26. Since $\gcd(h, \nu) = 1$, each retained pole satisfies $h(p) \neq 0$. Set

$$R_p(s) = -\frac{h(s) - h(p)}{(s-p)h(p)}, \quad Q_0 = (R_p)_{p \in P_\nu}, \quad J_0 = [I_d \ Q_0]. \quad (\text{APU18})$$

The polynomial identity $(s-p)R_p = 1 - h(s)/h(p)$ proves that R_p is the exact image of r_p in $E_h = \mathbb{C}[s]/h$. Thus J_0 sends every pole coordinate and every polynomial coordinate to its original marked class, with all jets of h retained. It has the exact kernel $\{(-Q_0 c, c) : c \in \mathbb{C}^r\}$ and satisfies

$$J_0 A_{\text{ext}}(0) = A(0) J_0. \quad (\text{APU19})$$

These identities follow by multiplying the preceding polynomial inverse by s , exactly as in AP25. No value of the singular matrix Ω_u at $u = 0$ is assigned by this endpoint arrow.

For completeness its original theta-source representative is also explicit. Retain the original even Schwartz source V , rapid-decay target $\mathcal{B}$, theta map $\Theta \phi(x) = 2 \sum_{a \geq 1} \phi(ax)$, Euler operator $\mathcal{D} = -x \partial_x$, and seed

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M} \Theta \phi_* = g.$$

The original Euler inverse $F_h \in \mathcal{B}$ satisfies $h(\mathcal{D})F_h = \Theta \phi_*$ and $\mathcal{M} F_h = g/h$. The polynomial chain map is the original

$$P \longmapsto P(\mathcal{D})\phi_*, \quad P ds \longmapsto P(\mathcal{D})F_h, \quad \Theta P(\mathcal{D})\phi_* = (hP)(\mathcal{D})F_h. \quad (\text{APU20})$$

Consequently the coefficient map

$$\Psi_h : E_h \longrightarrow Q_{\text{ar}} = \mathcal{B}/\Theta V, \quad [P] \longmapsto [P(\mathcal{D})F_h] \quad (\text{APU21})$$

is well defined: replacing P by $P + hT$ changes its source representative by $\Theta T(\mathcal{D})\phi_*$. The original Mellin-jet map sends it to $j_h(g/h)[P] = [\nu][P]$. Since $[\nu]$ is the full invertible Taylor class, this proves injectivity on the original packet. Equivalently, with the retained corrected section $\sigma_h = \Psi_h[\nu]^{-1}$, it is $\Psi_h = \sigma_h[\nu]$. The complete endpoint-source arrow is therefore

$$\mathbb{C}^{d+r} \xrightarrow{J_0} E_h \xrightarrow{[\nu]} E_h \xrightarrow{\sigma_h} Q_{\text{ar}}, \quad v \longmapsto [(J_0 v)(\mathcal{D})F_h]. \quad (\text{APU22})$$

All inverse Taylor coefficients are those of the original $[\nu]^{-1}$; they have not been replaced by a constant. Since multiplication by $[\nu]$ commutes with the original $A(0)$, APU19 intertwines the displayed endpoint action with this actual theta-source action. On the transported gauge fibre the coefficient is $[\nu]J_0 v$, whose image under σ_h is the same source vector. This is the precise unit and endpoint comparison for the original rational pole extension.

Backward propagation through the entire finite scaling transport. Retain AP15–19b’s exact transport, for every complex a ,

$$C_{\text{ext},a}(u, t) = \Pi_{\text{ext}}(u, t)^{-1} \Pi_{\text{ext}}(u, t + ua), \quad \partial_a C_{\text{ext},a} = -C_{\text{ext},a} A_{\text{ext}}(t + ua).$$

Its invertible diagonal blocks and full upper-right integral are the ones in AP15; this paragraph propagates APU9 through that already existing finite map. Put $t_a = t + ua$. At fixed a , the exact two parameter derivatives are

$$\begin{aligned} \partial_u C_{\text{ext},a} &= -\Omega_u(u, t) C_{\text{ext},a} \\ &\quad + C_{\text{ext},a} (\Omega_u(u, t_a) + a\Omega_t(u, t_a)), \\ \partial_t C_{\text{ext},a} &= -\Omega_t(u, t) C_{\text{ext},a} + C_{\text{ext},a} \Omega_t(u, t_a). \end{aligned} \quad (\text{APU23})$$

To prove every term, differentiation of $\Pi_{\text{ext}}^{-1} \Pi_{\text{ext}} = I$ and APU14 gives

$$\partial_u \Pi_{\text{ext}}^{-1} = -\Omega_u \Pi_{\text{ext}}^{-1}, \quad \partial_t \Pi_{\text{ext}}^{-1} = -\Omega_t \Pi_{\text{ext}}^{-1}.$$

The derivative of the shifted factor is, by the ordinary chain rule and APU14,

$$\partial_u (\Pi_{\text{ext}}(u, t + ua)) = \Pi_{\text{ext}}(u, t + ua) (\Omega_u(u, t + ua) + a\Omega_t(u, t + ua)),$$

whereas its t -derivative is $\Pi_{\text{ext}}(u, t + ua) \Omega_t(u, t + ua)$. Multiplying these inverse and shifted derivatives proves APU23. In particular the $a\Omega_t$ term is the derivative of the actual shifted parameter and cannot be omitted. Equivalently, under the precise base map $F_a : S \rightarrow S$, $F_a(u, t) = (u, t + ua)$, the pulled-back connection has coefficients $\Omega_u(u, t_a) + a\Omega_t(u, t_a)$ and $\Omega_t(u, t_a)$; the matrix $C_{\text{ext},a}$ is the horizontal map from that pullback to the unshifted coefficient bundle, by APU23.

The original weighted-residue identity AP18 uses the rectangular matrix $\mathcal{R} = [0 \ W_{\text{res}}]$:

$$\mathcal{R}(u, t) C_{\text{ext},a}(u, t) = \mathcal{R}(u, t + ua).$$

It remains horizontal in both parameters. In fact its left-hand u -derivative is, by APU12 and APU23,

$$\begin{aligned} \mathcal{R}(u, t) \Omega_u(u, t) C_{\text{ext},a} + \mathcal{R}(u, t) \partial_u C_{\text{ext},a} \\ = \mathcal{R}(u, t + ua) (\Omega_u(u, t + ua) + a\Omega_t(u, t + ua)), \end{aligned}$$

which is the exact u -derivative of the right side by APU12 and the same chain rule. The t -derivative is $\mathcal{R}(u, t + ua) \Omega_t(u, t + ua)$, by the two cancelling unshifted terms in the identical calculation. Multiplication by the constant $2\pi i$ proves these same compatibilities for the positively oriented loop-period map $\mathcal{L}$. Thus the finite scaling, its full extension integral, both parameter derivatives, and both residue conventions all remain in the same original commuting diagrams.

For constant ν , the set P_ν is empty and every additional block, residue coordinate and loop row in APU7–23 is the unique empty map; the polynomial connection remains unchanged. For a nonempty pole set, all coefficient-face maps act on these matrices slot by slot. Constant zero insertion commutes with both parameter derivatives and every displayed coefficient map; hence the independent original support labels and their represented zeros are retained by the entire two-parameter calculation.

99.0.7 5. Entire finite scaling and its retained extension term

Let $C_a(u, t)$ be the original polynomial scaling transport, satisfying

$$\frac{dC_a}{da} = -C_a A(t + ua), \quad C_0 = I_d. \quad (\text{AP14})$$

It is the previously proved entire solution, including its full higher terms. Define an integral along the straight segment from 0 to a in the complex a -plane (equivalently the primitive of its entire integrand):

$$\begin{aligned} K_a &= - \int_0^a C_b(u, t) B_0 \exp(-(a-b)P) db, \\ C_{\text{ext},a} &= \begin{bmatrix} C_a & K_a \\ 0 & \exp(-aP) \end{bmatrix}. \end{aligned} \quad (\text{AP15})$$

Differentiating the integral gives $K'_a = -C_a B_0 - K_a P$ and $K_0 = 0$. Consequently

$$\frac{dC_{\text{ext},a}}{da} = -C_{\text{ext},a} A_{\text{ext}}(t + ua), \quad C_{\text{ext},0} = I_{d+r}. \quad (\text{AP16})$$

Existence and uniqueness for this finite polynomial-coefficient matrix ODE prove that these are the actual entire finite connection transports. The block integral retains the coupling into the original polynomial class. Both diagonal blocks are invertible, and the inverse has upper-right block $-C_a^{-1} K_a \exp(aP)$. The groupoid law is

$$C_{\text{ext},a}(u, t) C_{\text{ext},b}(u, t + ua) = C_{\text{ext},a+b}(u, t). \quad (\text{AP17})$$

It follows either by the shifted ODE and uniqueness, or from the commuting operator's formal flow $\exp(aL)F = e^{-as}F(u, t + ua, s)$. The latter is coefficientwise in a when F is rational; no claim that $e^{-as}F$ is itself a rational function at a nonzero complex a is needed. The finite matrix ODE supplies its entire cohomological continuation.

The residue quotient has the exact transport identity

$$\begin{bmatrix} 0 & W(u, t) \end{bmatrix} C_{\text{ext},a}(u, t) = \begin{bmatrix} 0 & W(u, t + ua) \end{bmatrix}, \quad (\text{AP18})$$

because $w(p; u, t + ua) = w(p; u, t)e^{-ap}$. In weighted residues the entire action is parameter translation; in the unchanged simple-pole frame the factor e^{-ap} remains. The exponentials and their polynomial extension K_a are both displayed, including the case when p is not real.

99.0.8 6. Small-loop periods, the actual unit gauge and dual maps

For a positively oriented small circle γ_p enclosing only p , the exact additional period is

$$\int_{\gamma_p} wF ds = 2\pi i (\text{Res}_w F)_p. \quad (\text{AP19})$$

Thus the extra periods are concrete loops, with their orientation and $2\pi i$ factor retained. Every original polynomial image has zero such loop period. For any original rapid-decay contour avoiding P_ν , the incoming polynomial period is still its old integral, with its original ray orientation and phase.

Choose the finite parts of the original d rapid-decay contours to avoid every point of P_ν , including zero when zero belongs to that set. Such detours leave their polynomial integrals unchanged, since those integrands are entire. Different detours for a rational integrand can add the small loops AP19; those are retained rows, not discarded contour ambiguities. At infinity w has the same decay on the specified rays as in the original polynomial period proof, and the simple-pole functions are $O(1/s)$, so all new integrals converge. On compact parameter subsets in an admitted u -sector and bounded t , the negative leading exponential dominates the lower terms and justifies differentiation.

In the unchanged AP6 frame these d contours followed by the r small loops give the full period matrix

$$\Pi_{\text{ext}} = \begin{bmatrix} \Pi_{\text{pol}} & V_{\text{pole}} \\ 0 & (2\pi i)W \end{bmatrix}, \quad (\text{AP19a})$$

$$\det \Pi_{\text{ext}} = \det \Pi_{\text{pol}} (2\pi i)^r \prod_p w(p).$$

Π_{pol} is the original full polynomial period matrix, whose nonzero determinant was proved by the monomial ray matrix and coefficient-connection transport. Hence AP19a is invertible: it realizes all $d+r$ analytic classes and all their loop periods. Integrals of D -exact forms vanish at the decaying endpoints or on closed loops. Differentiating with t therefore gives

$$\frac{d\Pi_{\text{ext}}}{dt} = -\Pi_{\text{ext}} A_{\text{ext}}(t)/u, \quad (\text{AP19b})$$

$$C_{\text{ext},a} = \Pi_{\text{ext}}(u, t)^{-1} \Pi_{\text{ext}}(u, t + ua).$$

The second identity follows from the same ODE and initial value as AP16, for fixed nonzero u in the admitted sector. No period matrix is evaluated at $u = 0$.

On the same localized complex define the actual gauge differential

$$D^\nu = \nu D \nu^{-1} = u \frac{d}{ds} + h - t - u \frac{\nu'}{\nu}. \quad (\text{AP20})$$

The cochain map is multiplication by ν in both degrees. Its inverse is multiplication by ν^{-1} in B_{loc} . The target weight is w/ν , and the identities

$$\begin{aligned} (w/\nu) D^\nu F &= u((w/\nu)F)', \\ (w/\nu)(\nu F) &= wF \end{aligned} \quad (\text{AP21})$$

show exact equality of every small-loop and every old rapid-decay period under this map. The target extra class $\nu/(s-p)$ need not have a pole as a rational function; its weighted form $(w/\nu)\nu/(s-p) = w/(s-p)$ retains the pole. The original image is ν times H_{pol} , not the unmodified polynomial frame.

The ordinary algebraic dual of AP5 is the exact sequence

$$0 \longrightarrow (\mathbb{C}^{P_\nu})^* \xrightarrow{\text{Res}_w^T} H_{\text{loc}}^* \xrightarrow{i^T} H_{\text{pol}}^* \longrightarrow 0. \quad (\text{AP22})$$

In AP6 the injected column for α is $(0, W^T \alpha)$. The dual connection is $\frac{d}{dt} + A_{\text{ext}}(t)^T/u$. Its action on that column is $(0, W^T \alpha')$: the derivative $-PW/u$ cancels PW/u , and the upper block is zero. Thus the loop-functional injection is explicitly horizontal for the trivial weighted-residue dual. Under the gauge the dual is the inverse transpose of multiplication by ν . This statement concerns the ordinary linear dual; the previous τ -base dagger and moment-line dual retain their separate, already calculated conjugation and weight-line maps. No positive pairing is inferred from AP22, and no original inverse Taylor unit is replaced by ν on a dual.

99.0.9 7. Exact relation to the formal contraction, including the endpoint

UG uses $\mathbb{C}[s][[u, t]]$ and $\mathbb{C}[s, \nu^{-1}][[u, t]]$, with coefficientwise completion. On the added pole module h is invertible because $\gcd(h, \nu) = 1$. Writing H for this inverse and $E = u \frac{d}{ds} - t$, its exact formal inverse is the ordered sum $\sum_{n \geq 0} (-HE)^n H$. That is the contraction proved in UG. AP3 instead fixes $u \neq 0$ first and calculates a cokernel of dimension r , nonzero when $r > 0$. For constant ν , $r = 0$ and the additional pole complex and its residue map are zero; localization adds no analytic or formal classes in that case.

The two operations are related by the original rational complexes and the localization arrow, but the formal completion has no evaluation homomorphism at an arbitrary nonzero complex u . A formal coefficient sequence may diverge there. Thus the previously displayed formal contraction is not evaluated to claim that AP3's residue classes vanish. AP3 gives exactly the missing cokernel.

There is a concrete endpoint calculation. For p in P_ν , $\gcd(h, \nu) = 1$ gives $h(p) \neq 0$. At $u = t = 0$ the class of $1/(s-p)$ modulo h is the original polynomial class

$$\left[\frac{1}{s-p} \right] = - \left[\frac{h(s) - h(p)}{(s-p)h(p)} \right] \quad \text{in } \mathbb{C}[s]/(h). \quad (\text{AP23})$$

Indeed multiplying the displayed polynomial by $s - p$ gives $1 - h(s)/h(p)$. This is the literal inverse of $s - p$ in the original algebra, with every jet of h retained. For formal u, t before the endpoint, $D(1/(s - p))$ gives

$$(h(p) - t) \left[\frac{1}{s - p} \right] = u \left[\frac{1}{(s - p)^2} \right] - \left[\frac{h(s) - h(p)}{s - p} \right]. \quad (\text{AP24})$$

The successive higher-pole relations AP4 and ordered formal inverse supply the remaining u, t coefficients. AP23 is not a rank- $(d + r)$ basis at $u = 0$. It explicitly describes how these rational lifts are represented in the original rank- d formal fibre.

The weighted residue factors also retain their parameter behavior: $w(p) = \exp((\Phi(p) - tp)/u)$. At fixed t with $\Phi(p) - tp \neq 0$ this has an essential singularity at $u = 0$. If that numerator vanishes, that particular value is one; its exact derivative is always $u d_t w(p) = -p w(p)$. For $p = 0$, $w(0)$ is independent of t : it is identically one when $\Phi(0) = 0$, and is $\exp(\Phi(0)/u)$ otherwise. Thus no universal singularity or t -dependence is asserted for every residue coordinate. Even when $w(p) = 1$, AP4 on higher pole orders uses $w^{(n)}(p)$, whose highest inverse- u term is $w(p)(h(p) - t)^n/u^n$. Because $h(p) \neq 0$, these contain unbounded inverse- u powers as the pole order increases near $t = 0$. Consequently when $r > 0$, AP2 does not define a coefficientwise continuous residue map on UG's completed pole module. For $r = 0$ its zero map is of course continuous. The obstruction to evaluating the nonzero-pole formal contraction is not replaced by a false assertion that every individual loop factor is singular.

There is also an exact map for the entire displayed endpoint matrix. Let Q_0 have p -th column the polynomial in AP23, expressed in the original degree- d frame, and put $J_0 = [I_d \ Q_0]$. Direct multiplication of AP23 gives

$$\begin{aligned} A Q_0 &= B_0 + Q_0 P, \\ J_0 A_{\text{ext}}(0) &= A J_0, \\ \ker J_0 &= \{(-Q_0 c, c) : c \in \mathbb{C}^r\}. \end{aligned} \quad (\text{AP25})$$

The map $c \mapsto (-Q_0 c, c)$ carries P to $A_{\text{ext}}(0)$, by the first identity. Thus the endpoint has an explicit invariant r -dimensional kernel and the original arithmetic quotient, rather than an unexplained rank comparison. Entire matrix exponentiation yields

$$J_0 C_{\text{ext},a}(0, 0) = \exp(-aA) J_0. \quad (\text{AP26})$$

Here $C_{\text{ext},a}(0, 0) = \exp(-aA_{\text{ext}}(0))$ is the matrix specialization of AP16; J_0 then maps it to actual formal-fibre cohomology. Multiplication by the full unit v_h after J_0 gives the gauge endpoint. It commutes with A since that unit is an element of the original algebra, so the same action diagram holds. The endpoint comparison therefore retains the full unit as well as all jets.

Completed scope: the original arithmetic polynomial cohomology injects into the analytic localization. The complete two-parameter connection, weighted-residue and oriented-loop maps, finite-scaling pullback including its shifted endpoint derivative, full Taylor gauge, dual and original theta-source endpoint have now been proved in APU1–23. All original additional pole classes and their finite-scaling extension integral remain in the same maps. The formal and analytic constructions retain their exact common maps and different coefficient operations. This does not supply a uniform θ -norm bound or an arithmetic Frobenius comparison not constructed in these equations.

99.0.10 Provenance and verification scope

Input proof: ACTUAL_TAYLOR_UNIT_FORMAL_GAUGE.md, SHA256 9ded2486160b76bf65f92d44b73cdd0c59a3ca7e5334f88682e10592d5a45b2f. Original scaling proof after the accepted integer-power-map clarification: ../f1_scaling_frobenius/ACTUAL_TAU_SCALING_AND_EXPONENTIAL_FLOW.md, SHA256 4e4cb66e3cbbff6379ae389ddaaad65045ac0ed380f17a87ac7c4241c45e6c08. The accompanying derivation supplied AP3's local-primitive construction and AP8–AP13's exact connection in the preceding calculation. This note supplies the full written continuation, AP15–AP18's entire extension transport, AP22's explicit dual and AP23–AP24's endpoint comparison. The independent reviewer supplied the determinant extension AP19a and the endpoint intertwiner AP25–AP26; the text incorporates their full proofs and includes the review's corrected $p = 0$ residue-factor exception explicitly. No new Lean execution or certified arithmetic-zero computation is claimed.

100 Complete independent review of analytic pole-residue transport

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 7205cfbac7a37856a04ef5f7a9a579d8dcdf8c77dab6e6f86cb4f72d44f7c730

Complete source. `sources/next_edition/analytic_pole_conversions_v2/sources/analytic_pole_review/ANALYTIC_POLE_INDEPENDENT_REVIEW.md`

Source SHA-256. 7205cfbac7a37856a04ef5f7a9a579d8dcdf8c77dab6e6f86cb4f72d44f7c730

100.0.1 Independent review of analytic pole-residue transport

Date: 2026-09-13. This reviewer read the complete ANALYTIC_POLE_RESIDUE_TRANSPORT.md, originally AP1–24, its unnumbered assertions, and its provenance and scope paragraphs. The revised AP19a–b, AP25–26, and their complete surrounding proofs were then read as well. The formal-unit gauge and the original scaling proof were independently reviewed or derived in the adjoining work. This review changes neither the source proof nor its sources, and does not certify an arithmetic-zero computation, a metric estimate, an analytic evaluation of the formal inverse, or a Lean result.

100.0.2 Disposition and the one local edge case

Accept AP1–26, including AP19a–b, and their proofs. The weighted-residue exact sequence, the coupling into polynomial cohomology, the off-diagonal transport integral, the actual unit gauge, and the ordinary dual have the correct domains, signs, and matrix order. Neither repeated roots of h nor the multiplicities of ν need to be removed.

One unnumbered sentence in Section 7 was reported for correction before sealing this review. The claim that the full residue exponential and its t -derivative still contain t/u when its exponent vanishes requires an exception for a pole point $p = 0$. The exact derivative is $u\partial_t w(p) = -pw(p)$. If $p = 0$ and $\Phi(0) = 0$, then $w(0; u, t) = 1$ identically and its t -derivative is zero. If $p = 0$ and $\Phi(0)$ is nonzero, it is independent of t and has an essential singularity only in u . For p nonzero, the displayed two-parameter exponential retains its t/u dependence. This comment concerns only that local prose assertion; AP11, AP18, and all the numbered formulas already have the correct factors.

100.0.3 Pole quotient: all maps and exactness

For fixed u nonzero, the quotient $M = \mathbb{C}[s, \nu^{-1}]/\mathbb{C}[s]$ is the vector space of finite principal parts at the distinct roots of ν . Although ν can have repeated roots, localization allows every finite pole order at each such point. Polynomial differences are killed only in this quotient. Multiplication by the entire nonvanishing weight w does not define a rational multiplier, and the proof does not use it as one: its uses are local meromorphic products and contour integrands.

The residue functional descends to M because w times a polynomial is holomorphic at all the designated points. The identity $wDQ = u(wQ)'$ proves that its composite with D is zero, retaining u . If a principal part of Q at p starts with $c(s - p)^{-n}$, its derivative contributes the uncancellable highest pole $-unc(s - p)^{-n-1}$. Thus D is injective on M .

For a zero-residue F , each Laurent series wF has a meromorphic primitive H_p without a logarithm. The function $H_p/(uw)$ is meromorphic locally and satisfies $DQ_p = F$. Taking its finite principal part, and summing over the finitely many points, gives a rational Q with only allowed poles. The difference $DQ - F$ is rational and holomorphic at every finite point, hence polynomial. No global analytic primitive or rationality of w is assumed. A change of integration constant contributes a holomorphic germ because w never vanishes, so the principal part does not change.

The explicit lift $1/(w(p)(s - p))$ has weighted-residue vector exactly the p -th coordinate vector, including its nonzero phase and mass. This proves surjectivity. These arguments establish all of AP3. AP4 is the literal Taylor expansion of h and the Laurent-residue formula for w , so its factorials, derivative order, t term, and leading sign are correct.

100.0.4 Polynomial injection and the displayed basis

For nonzero polynomial Q of degree n , DQ has degree $n + d$, since h is monic and $d > 0$. Therefore D is injective on polynomials, and recursive leading-term subtraction gives a unique remainder of degree below d . For a rational Q with polynomial DQ , the preceding highest-pole argument first forces Q to be polynomial. Thus localization kills no original polynomial cohomology class.

AP3 now gives AP5 with the indicated kernel and surjection. Subtracting $\sum_p (\text{Res}_w F)_p / (w(p)(s - p))$ removes all weighted residues; reduction then leaves exactly a degree-below- d polynomial. Conversely, applying the residues to a linear relation among AP6's vectors kills every pole coefficient, and polynomial injectivity kills the rest. The dimension is therefore $d + r$ for u nonzero, with the original rank- d object explicitly injected. These statements include ν constant, when $r = 0$ and the additional maps and blocks have zero size. No squarefreeness of h is used.

100.0.5 Connection, frames, and entire transport

AP7's two nonzero commutators are $-u$ and $+u$, so L commutes with D . Its action on the last polynomial basis vector uses only $D(1) = h - t$; it is the companion block $A(t)$. On a simple pole, $-s/(s - p) = -1 - p/(s - p)$. Consequently the upper-right block of A_{ext} is the constant column e_0 repeated once per pole, with the positive sign in A_{ext} and the negative sign in L . All of AP8–10 follows without asserting that multiplication by s alone descends on the quotient.

The residue row is $[0, W]$, and $uW_t = -PW$. Thus $\text{Res}_w L = u\partial_t \text{Res}_w$, both by direct differentiation and by multiplying the block matrices. The normalized vector $r_p/w(p)$ has $L(r_p/w(p)) = -1/w(p)$: the two terms with p cancel exactly. AP13's diagonal frame change is therefore correct, without replacing the original W by identity.

For the source-to-target flow from the fibre at $t + ua$ to the fibre at t , the ODE must be $C'_{\text{ext},a} = -C_{\text{ext},a}A_{\text{ext}}(t + ua)$. Its upper-right block therefore obeys $K'_a = -C_a B_0 - K_a P$, with zero initial value. The integral in AP15 solves exactly this equation; in particular its C_b is on the left of B_0 and its final exponential is on the right. The integrand is entire in the integration variable and the finite matrix ODE gives the asserted entire continuation. The inverse block, shifted groupoid law AP17, and weighted-residue identity AP18 follow with the stated order and signs. Since W never vanishes for u nonzero, none of these operations silently discards a residue coordinate.

The formal flow on rational representatives is only coefficientwise in a . The note correctly obtains nonzero- a transport from finite matrices rather than treating $\exp(-as)$ as a rational cochain multiplier. The ODE matrix can extend to $u = 0$ without its $d + r$ representatives becoming a cohomology basis at that endpoint.

100.0.6 Contours, actual gauge, and ordinary dual

Counterclockwise small loops give exactly $2\pi i \text{Res}_w$, not the normalized residue alone. The original rapid-decay contours have their finite portions routed away from the designated poles, including zero if necessary. Polynomial periods are unchanged because their integrands are entire; different rational routings can change periods by precisely the retained small-loop periods. The infinite rays are unchanged, so their original rapid decay dominates the rational factors. Local uniform domination on the admitted u -sector and bounded parameter sets justifies the asserted differentiation in t .

In these actual contours the extended period matrix has the block form of AP19a. Its determinant is $\det(\Pi_{\text{pol}})(2\pi i)^r \prod_p w(p)$, with no missing factor or sign for the stated counterclockwise loops. The original nonzero polynomial determinant and nonvanishing of each $w(p)$ prove invertibility. Boundaries integrate to zero along these contours, so the connection calculation gives $\partial_t \Pi_{\text{ext}} = -\Pi_{\text{ext}} A_{\text{ext}}/u$. Consequently $\Pi_{\text{ext}}(u, t)^{-1} \Pi_{\text{ext}}(u, t + ua)$ solves AP16 with the correct initial value. This proves AP19b for fixed nonzero u and does not evaluate any period matrix at $u = 0$.

Leibniz differentiation verifies $D^\nu = u\partial_s + h - t - w'/\nu$. Multiplication by ν in both degrees is an isomorphism on the explicitly localized complexes. The equalities in AP21 prove exact equality of weighted periods, including all multiplicities in ν and the original constant in Φ . It is correct that the target representative $\nu/(s - p)$ can be polynomial: its weighted form still has the relevant simple pole. The image of the old polynomial subspace is represented by ν times that frame, not by a silently unchanged polynomial frame in the new differential.

Dualizing the finite-dimensional exact sequence gives AP22 with reversed arrows. The injected dual col-

umn is $(0, W^T \alpha)$. Acting by $\partial_t + A_{\text{ext}}^T/u$ gives $(0, W^T \alpha')$, since $W_t^T = -W^T P/u$ cancels the lower diagonal block, and the upper component remains zero. The inverse transpose is the correct forward transport of functionals under the gauge. This is an ordinary linear-dual statement; it neither supplies a Hermitian positivity assertion nor replaces the original τ -base conjugation or moment-line twist.

100.0.7 Formal endpoint and limits of the result

The coefficientwise formal inverse of the earlier note is not evaluated at a nonzero u here. The distinction is essential: AP3 directly computes its nonzero- u residue cokernel, whereas the formal pole complex is contractible because h is invertible on M and the perturbation raises parameter order. These are different coefficient operations on the same original rational complex, not contradictory dimension assertions.

For p a root of ν , $h(p)$ is nonzero. Multiplying AP23's displayed polynomial by $s - p$ gives $1 - h(s)/h(p)$, proving its exact inverse identity in the full quotient, including repeated roots. Directly setting the cohomology class of $D(1/(s - p))$ to zero gives AP24 with its positive u term and negative divided-polynomial term.

In particular, if Q_0 has these AP23 polynomials as columns, then $AQ_0 = B_0 + Q_0P$. The specialization map from the $d + r$ displayed lifts to the actual rank- d endpoint is $[I, Q_0]$, and it intertwines $A_{\text{ext}}(0)$ with A . This confirms the note's endpoint distinction without assigning $d + r$ independent classes to the formal special fibre.

The affirmative result is the fully computed analytic extension, its weighted residues, connection transport, actual-unit gauge, and linear dual, together with their precise original polynomial subobject. No uniform metric bound or unconstructed Frobenius identification follows from this review.

100.0.8 Revised endpoint and empty-pole clarification

The revised AP25–26 include the preceding explicit endpoint check. The kernel of $[I, Q_0]$ is precisely $\{(-Q_0c, c)\}$, and $A_{\text{ext}}(0)(-Q_0c, c) = (-Q_0Pc, Pc)$. Its dimension is r , including the zero-dimensional case, and the induced quotient action is the original A . At $u = t = 0$ the matrix ODE has constant coefficient $A_{\text{ext}}(0)$, so exponentiating the intertwiner proves AP26. The original full unit multiplier commutes with A on the full algebra, which also verifies the stated gauge endpoint without dropping any Taylor coefficient.

The corrected $p = 0$ discussion separates all exceptional cases accurately. For each existing pole p , the highest inverse- u term in $w^{(n)}(p)/w(p)$ is $(h(p) - t)^n/u^n$; its coefficient is nonzero at $t = 0$ because $h(p)$ is nonzero. In the usual u -adic scalar topology, one can see the noncontinuity explicitly even after dividing out $w(p)$: the source sequence $u^n/(s - p)^{2n+1}$ tends to zero, whereas its normalized residue at $t = 0$ has u -valuation $-n$, with nonzero leading coefficient $h(p)^{2n}/(2n)!$. Thus the weighted-residue operation is not a coefficientwise continuous extension to the earlier formal pole module. No claim that $w(p)$ itself is always singular is required.

The last argument presupposes at least one pole. A final local wording comment therefore asked that Section 7's "nonzero cokernel" and noncontinuous-residue assertions be explicitly qualified by $r > 0$. For ν constant, the pole module, residue cokernel, and residue map are all zero; the zero map is continuous. This is the complete degenerate case, not an exception to AP3 or any matrix formula.

100.0.9 Final acceptance seal

On 2026-09-13 I reread the revised pole-factor discussion and both explicit $r = 0$ qualifications. The $p = 0$ and empty-pole comments above are preserved as the historical review record; neither remains an outstanding correction.

The exact accepted proof is ANALYTIC_POLE_RESIDUE_TRANSPORT.md, SHA-256:

357923aead35c77f2fa98598fbc4ed8bc1f7cd6ebc04760da504a0ded765a652.

Final disposition: accept AP1–26, including AP19a–b and all accompanying scope statements, with no outstanding mathematical correction. This is a complete independent written-proof review of the residue exact sequence, full connection and period transport, actual-unit gauge, ordinary dual, and original formal endpoint. No numerical fixture, arithmetic-zero certificate, Lean replay, positive metric estimate, or additional Frobenius comparison is certified by this seal.

101 Complete all-degree principal-angle proof and literal tensor theta primitive

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. b0634428d69f80dedd4f68766ef9e6468c390eaac86dfe58b08c21c8fb896e03

Complete source. sources/next_edition/all_degree_review_conversions_v1/sources/all_degree_angle_review/tau_all_degree_angle_independent_review_20260913.md

Source SHA-256. b0634428d69f80dedd4f68766ef9e6468c390eaac86dfe58b08c21c8fb896e03

101.0.1 All-degree principal angles of the original theta quotient sections

13 September 2026. This is a complete independent derivation of the nonlinear arithmetic determinant comparison in every quotient degree. All spaces are the original polynomial spaces, all projections use the actual common interpolated source Gram, and every change of representative is exhibited as a theta boundary. The earlier sealed AT and AW sources are unchanged.

101.0.2 1. Fixed original objects and the canonical projection relation

Retain a full actual finite zero packet h for $g = 2\xi$, an integer $k \geq 1$, its original tensor-sum annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and

$$E = \mathbb{C}[S]/(\chi), \quad H = \mathcal{P}_{2q}, \quad \mathcal{P}_N = \mathbb{C}[S]_{\leq N}, \quad \mathcal{P}_{-1} = 0. \quad (\text{AA1})$$

Use the original increasing-power remainder frame of E and the original source coordinate $S = k/2 + iu$. The two source densities are exactly

$$\begin{aligned} m_1(u) &= w_h^k(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a}\Gamma(2a). \end{aligned} \quad (\text{AA2})$$

Their literal masses μ_h^k and $(2\pi)^{k/2}$ remain in all Grams. Put $m_t = (1-t)m_0 + tm_1$, $0 \leq t \leq 1$, and let $M(t)$ be its common Gram on H . The source proofs establish positivity almost everywhere and every polynomial moment for the two densities. Therefore $M(t)$ and every restricted Gram below are positive definite. Inner products are conjugate-linear in their first argument, with $\langle x, y \rangle_t = x^* M(t) y$.

For $q-1 \leq N \leq 2q$, retain the exact sequence

$$0 \rightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \chi} \mathcal{P}_N \xrightarrow{J_N} E \rightarrow 0. \quad (\text{AA3})$$

Let P_N be the common-metric orthogonal projection onto $\mathcal{P}_N$, and Q_N the projection onto $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$. In particular $Q_{q-1} = 0$. Since $\mathcal{R}_N \subseteq \mathcal{P}_N$, the two projections commute and $P_N Q_N = Q_N P_N = Q_N$. Hence

$$N_N := P_N - Q_N \quad \text{is the orthogonal projection onto} \quad K_N = \mathcal{P}_N \cap \mathcal{R}_N^\perp. \quad (\text{AA4})$$

Polynomial division gives $\dim K_N = q$. The original minimum section $R_N : E \rightarrow \mathcal{P}_N \subset H$ satisfies

$$J_N R_N = I_E, \quad \text{im } R_N = K_N, \quad G_N = R_N^* M(t) R_N > 0, \quad V_N = \det G_N. \quad (\text{AA5})$$

These objects retain their dependence on t . We suppress it only inside pointwise equations.

For any two cutoffs $q-1 \leq i \leq j \leq 2q$, both sections have remainder equal to their input. Thus $R_i - R_j$ has image in $\mathcal{R}_j$. Meanwhile R_j lies in K_j , orthogonal to that image. Orthogonal projection of R_i onto K_j consequently gives the exact relation

$$R_j = N_j R_i, \quad R_i^* M R_j = G_j, \quad G_i - G_j = (R_i - R_j)^* M (R_i - R_j) \succeq 0. \quad (\text{AA6})$$

No equality between the different canonical sections is assumed. The same formula is valid at $i = j$, with zero difference.

Let $H_j = B_j^* M B_j$, with every map understood through its original coefficient inclusion in H . The relation coefficient and its full energy are

$$\begin{aligned} X_{ij} &= H_j^{-1} B_j^* M R_i : E \rightarrow \mathcal{P}_{j-q}, \\ R_i - R_j &= B_j X_{ij}, \quad G_i - G_j = X_{ij}^* H_j X_{ij}. \end{aligned} \quad (\text{AA7})$$

Indeed $Q_j R_i = B_j H_j^{-1} B_j^* M R_i$, and the splitting in $\mathcal{P}_j$ shows that this is $R_i - R_j$. If the relation domain is zero, the formula uses its unique zero map and no inverse on a nonzero space. This is the actual source relation energy that will become the squared sine data.

101.0.3 2. Finite principal-angle calculation with all Gram factors

All square roots in this section are the positive Hermitian square roots in the fixed coefficient frame. Define the isometric frames

$$F_i = R_i G_i^{-1/2} : \mathbb{C}^q \rightarrow (H, M), \quad F_j = R_j G_j^{-1/2} : \mathbb{C}^q \rightarrow (H, M). \quad (\text{AA8})$$

They satisfy $F_i^* M F_i = F_j^* M F_j = I$; this equality retains the original Grams rather than assigning new source norms. Their exact cross matrix is

$$Z_{ij} = F_i^* M F_j = G_i^{-1/2} G_j G_j^{-1/2} = G_i^{-1/2} G_j^{1/2}. \quad (\text{AA9})$$

The order of its factors matters. Multiplication by its actual adjoint gives

$$A_{ij} := Z_{ij} Z_{ij}^* = G_i^{-1/2} G_j G_j^{-1/2}, \quad 0 < A_{ij} \preceq I. \quad (\text{AA10})$$

Its positivity follows from $G_j > 0$, and its upper bound from AA6. Also $I - A_{ij} = G_i^{-1/2} X_{ij}^* H_j X_{ij} G_i^{-1/2}$, retaining the complete source boundary energy.

Let $v_1, \dots, v_q$ be an orthonormal eigenbasis of A_{ij} , with eigenvalues $c_1^2, \dots, c_q^2$ in $(0, 1]$, including every multiplicity. Define

$$x_a = F_i v_a \in K_i, \quad y_a = c_a^{-1} N_j x_a \in K_j, \quad 1 \leq a \leq q. \quad (\text{AA11})$$

There is no division by zero because of AA10. Since $F_i^* M N_j F_i = Z_{ij} Z_{ij}^* = A_{ij}$, direct calculation gives

$$\langle x_a, x_b \rangle_t = \delta_{ab}, \quad \langle y_a, y_b \rangle_t = \delta_{ab}, \quad \langle x_a, y_b \rangle_t = c_b \delta_{ab}. \quad (\text{AA12})$$

For example the second equality follows from $\langle N_j x_a, N_j x_b \rangle_t = \langle x_a, N_j x_b \rangle_t = c_b^2 \delta_{ab}$, divided by $c_a c_b$. The y_a form a basis of K_j because they are q orthonormal vectors in that space.

Define $\theta_a \in [0, \pi/2)$ by $c_a = \cos \theta_a$. These are the complete principal angles of the two actual section spaces, as witnessed by the paired frames AA11–AA12. Their squared cosines are exactly the eigenvalues of AA10, with no interchange of noncommuting square roots.

If $c_a = 1$, then the orthogonal projection $N_j x_a$ has the full norm of x_a , so $x_a \in K_j$ and $y_a = x_a$. Conversely every $x \in K_i \cap K_j$ is fixed by $N_i N_j$ on K_i , and its coefficient vector is an eigenvector of AA10 of eigenvalue one. Hence

$$\#\{a : \theta_a = 0\} = \dim(K_i \cap K_j), \quad (\text{AA13})$$

including all repeated eigenvalues of one.

For every a with $s_a = \sin \theta_a > 0$, put $z_a = (y_a - c_a x_a)/s_a$. Equations AA12 imply that the vectors z_a are orthonormal, orthogonal to every x_b , and that different two-dimensional spaces $\text{span}(x_a, z_a)$ are orthogonal. They are also orthogonal to every common vector with $c_b = 1$. On each such plane, the difference of the original projections is exactly

$$(N_i - N_j)|_{\text{span}(x_a, z_a)} = \begin{pmatrix} s_a^2 & -s_a c_a \\ -s_a c_a & -s_a^2 \end{pmatrix}. \quad (\text{AA14})$$

This follows because N_i is projection onto x_a on that plane, while N_j is projection onto $y_a = c_a x_a + s_a z_a$. The trace is zero and the determinant is $-s_a^2$, giving eigenvalues $+s_a, -s_a$. On the common vectors and on $(K_i + K_j)^\perp$, the difference is zero. Thus if $r = \#\{a : \theta_a > 0\}$, its full spectrum on H is

$$\text{spec}(N_i - N_j) = \{+\sin \theta_a, -\sin \theta_a : \theta_a > 0\} \sqcup \{0^{[\dim H - 2r]}\}. \quad (\text{AA15})$$

This proves the complete finite angle-to-projection spectral relation, including zero angles, repeated positive angles, and the remaining zero eigenvalues.

Finally, taking determinants of AA10 gives

$$\prod_{a=1}^q \cos^2 \theta_a = \frac{V_j}{V_i}, \quad \log \frac{V_i}{V_j} = \sum_{a=1}^q -\log \cos^2 \theta_a \geq 0. \quad (\text{AA16})$$

Every determinant is computed in the unchanged remainder frame. In particular this identity retains the full original quotient metric, not merely its eigenvalue locations.

101.0.4 3. The two original cutoff pairs and their forced common direction

Use the original grouped pairs

$$(i_0, j_1) = (q-1, 2q), \quad (i_1, j_0) = (q, 2q-1). \quad (\text{AA17})$$

The second pair is ordered for $q \geq 2$; for $q = 1$ its two indices both equal one. Apply AA11–AA16 to both pairs, retaining all $2q$ angles, and define the original endpoint quantity

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t)V_q(t)}{V_{2q-1}(t)V_{2q}(t)} \\ &= \log \frac{V_{q-1}(t)}{V_{2q}(t)} + \log \frac{V_q(t)}{V_{2q-1}(t)} = \sum_{a=1}^{2q} x_a(t), \quad x_a = -\log \cos^2 \theta_a \geq 0. \end{aligned} \quad (\text{AA18})$$

At least one angle in the second pair is exactly zero. To prove this from the original spaces, both K_q and K_{2q-1} lie in

$$L = \mathcal{P}_{2q-1} \cap (\chi \mathcal{P}_0)^\perp, \quad \dim L = 2q - 1. \quad (\text{AA19})$$

They lie in that hyperplane because each is orthogonal to its own relation space, which contains $\chi \mathcal{P}_0$. The latter is a nonzero line inside $\mathcal{P}_{2q-1}$, so it cuts that $2q$ -dimensional space by exactly one dimension in a positive form. Since each K has dimension q , $\dim(K_q \cap K_{2q-1}) = 2q - \dim(K_q + K_{2q-1}) \geq 1$. Now AA13 proves the claim. At $q = 1$, the two one-dimensional spaces are identical, in agreement with their identical cutoffs.

Keep the complete angle list. At each t , remove one of these forced entries $x_a = 0$ only for the purpose of writing the following finite sum, and put

$$m = 2q - 1. \quad (\text{AA20})$$

The remaining m entries include any further zero angles and still sum to $\mathcal{B}(t)$. No continuous choice of the removed entry or of individual eigenvectors is needed: the next inequality is a pointwise symmetric inequality in this entire list. The smooth quantity differentiated below is the original log determinant, not individual angle branches.

101.0.5 4. Direct trace bound and finite concavity calculation

Define the unchanged relative operators

$$D = M(0)^{-1}M(1), \quad C(t) = M(t)^{-1}(M(1) - M(0)) = ((1-t)I + tD)^{-1}(D - I). \quad (\text{AA21})$$

The last identity follows by factoring $M(t) = M(0)((1-t)I + tD)$ in its actual order. The operator D is positive self-adjoint for $M(0)$, and $C(t)$ is self-adjoint for $M(t)$ because $M(t)C(t) = M(1) - M(0)$ is Hermitian. Let its extreme eigenvalues be $c_{\min}(t)$, $c_{\max}(t)$, and set $w(t) = c_{\max}(t) - c_{\min}(t) \geq 0$.

For any unit vector v , expansion in an orthonormal eigenbasis of $C(t)$ gives $\langle v, C(t)v \rangle_t = \sum c_j |v_j|^2$, between those two extremes. Apply this fact to the positive and negative eigenvectors in AA15. Each pair contributes at most $\sin \theta_a w(t)$ in absolute value. Summing and using the triangle inequality proves directly

$$|\mathrm{Tr}((N_i - N_j)C(t))| \leq \left(\sum_{a=1}^q \sin \theta_a \right) w(t). \quad (\text{AA22})$$

No external trace extremum theorem, or commutation with the section projections, is used.

The original determinant variation is $\frac{d}{dt} \log V_N = \mathrm{Tr}(N_N C(t))$. Its four original signs give exactly

$$\dot{\mathcal{B}} = \mathrm{Tr}((N_{q-1} - N_{2q} + N_q - N_{2q-1})C(t)). \quad (\text{AA23})$$

It follows either by differentiating the finite determinant ratio from the exact sequence, or from the retained AT12. Apply AA22 to the two pairs in AA17. Using AA18 and AA20 yields the full angle-profile bound

$$|\dot{\mathcal{B}}(t)| \leq \left(\sum_{a=1}^{2q} \sin \theta_a(t) \right) w(t) = \left(\sum_{a=1}^m \sqrt{1 - e^{-x_a(t)}} \right) w(t). \quad (\text{AA24})$$

For $x > 0$, direct differentiation of $f(x) = \sqrt{1 - e^{-x}}$ gives

$$f'(x) = \frac{e^{-x}}{2\sqrt{1 - e^{-x}}}, \quad f''(x) = -\frac{e^{-x}(2 - e^{-x})}{4(1 - e^{-x})^{3/2}} < 0. \quad (\text{AA25})$$

Thus its derivative decreases, so its secant slopes decrease, proving concavity on $(0, \infty)$. Iterating the two-point concavity inequality proves the finite equal-weight inequality $m^{-1} \sum f(y_a) \leq f(m^{-1} \sum y_a)$ for positive y_a . For nonnegative x_a , apply it to $y_a = x_a + \varepsilon$ and let $\varepsilon \downarrow 0$; continuity at zero, with $f(0) = 0$, gives the same result. Therefore AA24 implies

$$|\dot{\mathcal{B}}(t)| \leq m \sqrt{1 - e^{-\mathcal{B}(t)/m}} w(t), \quad m = 2q - 1. \quad (\text{AA26})$$

At $m = 1$ the finite concavity step is an equality. Additional zero angles in higher degree are still present in AA24; AA26 is the stated consequence using only their sum and the guaranteed one zero angle.

101.0.6 5. Strict positivity from the actual vertical-line moment source

Every term in AA18 is nonnegative. Suppose that $\mathcal{B}(t) = 0$ at one value of t . Then every angle of the first pair is zero, so AA13 makes its two q -dimensional spaces equal:

$$K_{q-1} = K_{2q}. \quad (\text{AA27})$$

The first is $\mathcal{P}_{q-1}$, because its relation domain is zero. It contains the original constant polynomial one. The second is orthogonal to $\chi \mathcal{P}_q$. Thus $1 \perp \chi \mathcal{P}_q$.

Write the original polynomial in full as $\chi(S) = \sum_{j=0}^q a_j S^j$, $a_q = 1$, and define its actual reflected-conjugate polynomial

$$\chi_k^\#(S) = \sum_{j=0}^q \overline{a_j}(k - S)^j. \quad (\text{AA28})$$

Its degree is q , with leading coefficient $(-1)^q \overline{a_q}$; this sign is retained. On the unchanged integration line, $\overline{S} = k - S$, and so

$$\chi_k^\#(S) = \overline{\chi(\overline{S})}, \quad \chi(S) \chi_k^\#(S) = |\chi(S)|^2. \quad (\text{AA29})$$

The polynomial product belongs to the literal relation subspace $\chi \mathcal{P}_q$. Orthogonality of one to that space would give

$$0 = \langle 1, \chi \chi_k^\# \rangle_t = \int_{\mathbb{R}} |\chi(k/2 + iu)|^2 m_t(u) du > 0, \quad (\text{AA30})$$

a contradiction. The strict inequality is the positive original Gram of the nonzero polynomial χ , and all integrals exist by the retained moments. This argument uses the actual source measure and vertical-line multiplication identity. It is not a claim about every abstract positive coefficient metric. We have proved

$$\mathcal{B}(t) > 0 \quad (0 \leq t \leq 1). \quad (\text{AA31})$$

The finite positive Grams and their inverses are smooth near the compact path, so $\mathcal{B}$ is smooth and attains a strictly positive minimum on it.

101.0.7 6. Nonlinear endpoint transport in every quotient degree

Let the full ordered positive spectrum of the original relative Gram be

$$0 < b_1 \leq \cdots \leq b_{2q+1}, \quad \kappa = b_{2q+1}/b_1. \quad (\text{AA32})$$

From AA21 its eigenvalues are $c_j(t) = (b_j - 1)/(1 - t + tb_j)$. Differentiating with respect to b_j gives $(1 - t + tb_j)^{-2} > 0$, so the same extreme indices apply for every t , including repeated values and values equal to one. Since each c_j is the derivative of $\log(1 - t + tb_j)$,

$$\int_0^1 w(t) dt = \log b_{2q+1} - \log b_1 = \log \kappa. \quad (\text{AA33})$$

For the fixed integer $m = 2q - 1$, define on $B > 0$

$$\Phi_m(B) = 2 \operatorname{arcosh}(e^{B/(2m)}), \quad \Phi'_m(B) = \frac{1}{m\sqrt{1 - e^{-B/m}}}. \quad (\text{AA34})$$

The derivative follows by differentiating $\cosh(\Phi_m(B)/2) = e^{B/(2m)}$, using the nonnegative real inverse branch of $\cosh$. The square root divisor is strictly positive on the original path by AA31. Multiply AA26 by AA34 and integrate the absolute derivative. Equation AA33 gives the all-degree nonlinear comparison

$$\left| 2 \operatorname{arcosh}(e^{\mathcal{B}(1)/(2m)}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/(2m)}) \right| \leq \log \kappa, \quad m = 2q - 1. \quad (\text{AA35})$$

The original values $\mathcal{B}(0) = \mathcal{B}_{h,k}^\Gamma$ and $\mathcal{B}(1) = \mathcal{B}_{h,k}^{\text{ar}}$ are kept. No monotonicity of $\mathcal{B}(t)$, individual angles, or individual eigenvectors was assumed.

An equivalent endpoint interval retains both branches explicitly. Set

$$A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/(2m)}), \quad L = \frac{1}{2} \log \kappa. \quad (\text{AA36})$$

The endpoint inverse-cosh value lies in $[\max(0, A_0 - L), A_0 + L]$. Since $A \mapsto 2m \log \cosh A$ has derivative $2m \tanh A \geq 0$ for $A \geq 0$,

$$2m \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2m \log \cosh(A_0 + L). \quad (\text{AA37})$$

The lower bound may be zero while the actual endpoint is strictly positive. When $q = 1, m = 1$, the second pair has its one zero angle and AA35 is the previously proved one-dimensional formula. For any higher q , including the actual quartet dimension, the same proof applies. It establishes no uniform-in- k bound for κ .

The new estimate implies the earlier bound $|\mathcal{B}(1) - \mathcal{B}(0)| \leq m \log \kappa$: the derivative of the inverse function $B = 2m \log \cosh(\Phi/2)$ is $m \tanh(\Phi/2) \leq m$. Integrating that derivative between the two endpoint Φ values and using AA35 proves the assertion. The angle-profile inequality AA24 remains a still finer pointwise estimate, retaining every individual original Gram ratio.

101.0.8 7. The literal tensor theta boundary of the angle energy

This section supplies the source primitive behind AA7. Keep the original one-leg complex $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, with $D = -x\partial_x$, and the original functions satisfying

$$\mathcal{M}F_h = g/h, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad h(D)F_h = \Theta\phi_*. \quad (\text{AA38})$$

The last identity follows by Mellin transformation: both sides transform to the exact g , and Mellin uniqueness on these source spaces identifies them. The original source intertwining $D\Theta = \Theta D$ lets all polynomial differential operators below act on the complex. Its inclusion in the original two-leg complex sends ϕ to $(\phi, 0)$ in degree zero and is identity in degree one, so the same primitive maps into the actual tau-base complex with its retained two-leg signs.

In the coefficient ring $\mathbb{C}[s_1, \dots, s_k]$, the cyclic definition of χ gives

$$\chi(s_1 + \dots + s_k) \in (h(s_1), \dots, h(s_k)). \quad (\text{AA39})$$

Indeed the tensor algebra $E_h^{\otimes k}$ is this polynomial ring modulo that ideal, and its unit is the cyclic vector used to define χ . Evaluating χ on the sum operator and that unit is zero by its actual annihilator definition.

The ideal expression can be chosen by a fully specified finite division. Start with $F_0 = \chi(s_1 + \dots + s_k)$; for $j = 1, \dots, k$, divide F_{j-1} by the literal monic $h(s_j)$, viewing the other variables as coefficients:

$$F_{j-1} = h(s_j)L_j + F_j, \quad \deg_{s_j} F_j < \deg h. \quad (\text{AA40})$$

Later divisions do not increase the already restricted degrees in earlier variables, because $h(s_j)$ has complex constant coefficients in those variables. Thus the final F_k lies in the span of the tensor remainder monomials, which are a basis of the quotient algebra. Its image in that algebra is zero by AA39, so $F_k = 0$. Summing the exact division identities proves

$$\chi(s_1 + \dots + s_k) = \sum_{j=1}^k h(s_j)L_j(s_1, \dots, s_k). \quad (\text{AA41})$$

Every original coefficient of h and χ remains in these finite quotient polynomials. This construction does not replace χ by a product of simple factors or discard any nilpotent multiplicity.

For a polynomial P in the original sum variable, define the degree- $k - 1$ tensor cochain

$$\begin{aligned} \mathcal{K}_\chi(P) &= \sum_{j=1}^k (-1)^{j-1} L_j(D_1, \dots, D_k) P(D_1 + \dots + D_k) \\ &\quad (F_h \otimes \dots \otimes F_h \otimes \phi_* \otimes F_h \otimes \dots \otimes F_h), \end{aligned} \quad (\text{AA42})$$

where the j -th position is ϕ_* in the degree-zero space V , and every other position is the displayed F_h in degree one. There are $j - 1$ degree-one factors before that position. The tensor differential therefore acts on it with sign $(-1)^{j-1}$; this cancels the explicit coefficient sign in AA42. It acts trivially on the other degree-one factors. By AA38, by commutation of the polynomial operators with the differential, and then by AA41, one obtains exactly

$$d\mathcal{K}_\chi(P) = P(D_1 + \dots + D_k)\chi(D_1 + \dots + D_k)F_h^{\otimes k} = \mathcal{V}_{h,k}(\chi P). \quad (\text{AA43})$$

This proves the primitive with its full tensor signs. In particular, for $x \in E$, regard $X_{ij}x$ from AA7 as its literal polynomial in S . Then

$$\mathcal{V}_{h,k}(R_i x - R_j x) = d\mathcal{K}_\chi(X_{ij}x), \quad G_i - G_j = X_{ij}^* H_j X_{ij}. \quad (\text{AA44})$$

Thus the angle energy is the Gram energy of this exact original relation, while its cohomology class is zero. The unchanged arithmetic observation still satisfies $J^{(k)}\mathcal{V}_{h,k}R_N = \eta_{h,k}$, with $\eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1$ and the full unit $v_h = j_h(g/h)$. Its further cohomology image is $\sigma_h^{\otimes k}\eta_{h,k}$ in the original $Q^{\otimes k}$. No source relation is dropped merely because those observations kill it.

At every nonempty original support label A , lift the displayed coefficient maps by $(A, v) \mapsto (A, fv)$, and send external absence τ to τ . The class of AA44 is the supported zero at the same label; its full source primitive and energy remain as written. This preserves the original cohomology over the tau base throughout the all-degree estimate.

101.0.9 8. Full review of the root AGT1–AGT19 source

The complete corrected tau_all_degree_angle_transport_20260913.tex was read at SHA-256 cc0c0c008b0532c5f120d1cea76b94e2d08153c87a694ce18775a5189c72d5f8. It is accepted without a further mathematical correction. The initial source-cutoff sentence now states $q - 1 \leq N \leq 2q$, making its inclusion $\mathcal{P}_N \subseteq H = \mathcal{P}_{2q}$ correct on the entire declared range.

The full acceptance covers the following calculations.

1. AGT1–AGT4 retain the actual packet, full Taylor unit, original source and cohomology observations, both full masses, original vertical line, positive finite Grams, and all four signed determinant terms. The bounded source codomain is now exact.
2. AGT5–AGT8 agree with the independent derivation AA6–AA16. In the root source the cross matrix is written in the opposite orientation, $U_j^* M U_i = G_j^{1/2} G_i^{-1/2}$; its adjoint-times-itself is precisely $G_i^{-1/2} G_j G_i^{-1/2}$. Thus both presentations give the same squared singular values without commuting the square-root factors. The displayed two-dimensional projection block, all unit-cosine common vectors, and the remaining zero spectrum are correct. The primitive behind its relation-valued difference is supplied explicitly by AA38–AA44 above.
3. AGT9–AGT11 retain both full angle lists and remove only one exactly proved zero scalar contribution. The common ambient space has dimension $2q - 1$, including $q = 1$. The trace bound is the sum of the paired positive and negative Rayleigh bounds, so it needs no mutual commutation and introduces no extra factor two.
4. AGT12–AGT13 use the precise involution with variable $k - S$ and conjugated coefficients. Its leading sign $(-1)^q$, its square, and its equality to complex conjugation on the original line are correct. The resulting integral is the full positive norm of the nonzero original polynomial χ , proving strict positivity on the actual moment path.
5. AGT14–AGT19 have the exact constant $m = 2q - 1$. The derivative and tangent-line proof include the remaining zero-angle values by continuity and the actual positive mean $\mathcal{B}/m$. The factor $1/m$ in the derivative of Φ_m cancels the factor m in the angle estimate. The integrated width equals the full original $\log \kappa$, including repeated eigenvalues and endpoint eigenvalues equal to one. Both inverse hyperbolic branches and the explicit endpoint interval are correct. The source states the valid all-degree application, including actual quartet quotients, and preserves the absence of a new uniform estimate for κ .

The separate complete PA1–PA29 derivation in tau_principal_angle_volume_independent_review_20260913.md, SHA-256 5f6a75afdbdf4dcd81781a336702ea6d6545fd8fd0a9b2e014a605e414b3198, was read in full. Its principal bases, projection spectra, positivity argument, and integrated constant agree with AA1–AA37. This provides an additional independent derivation, with its source scope retained; no finite numerical test is being substituted for these proofs.

101.0.10 9. Reading, source pins, and mathematical scope

The complete sealed AT source tau_arithmetic_determinant_transport_20260913.tex at SHA-256 24cf7ce5be6a43cc656641b516185df10b2e4d7824600ece657120d1c549deae and the complete AW source tau_relation_window_spectral_transport_20260913.tex at SHA-256 c79e76c087efd59871e3234ff44fb18964515dbd34f0318dffba6b3f1440f6ec were read in the preceding independent calculation. Their original projection and moment conventions are retained here. The full original arithmetic_input.tex and Tau_Base_Cohomology/NOTE.md were read earlier in this same audit and supply the stated theta complex, finite-packet unit, and tau-base tensor realization. AA6–AA44 provide the new finite angle, analytic transport, and explicit primitive proofs.

The companion receipt pins the final sources and the complete root-source acceptance above. These are closed written proofs and do not depend on a forthcoming supplement. No numerical or Lean execution, source normalization, remote mutation, or unproved uniform eigenvalue bound is claimed. The all-degree conclusion applies to the actual original moment Grams and includes every admissible quotient degree $q \geq 1$.

102 Complete independent principal-angle volume proof and integrated metric comparison

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 5f6a75afdbdf4dcd81781a336702ea6d6545fd8fd0a9b2e014a605e414b3198

Complete source. sources/next_edition/all_degree_review_conversions_v1/sources/principal_angle_volume_review/tau_principal_angle_volume_independent_review_20260913.md

Source SHA-256. 5f6a75afdbdf4dcd81781a336702ea6d6545fd8fd0a9b2e014a605e414b3198

102.0.1 Independent principal-angle calculation for the arithmetic four-volume

13 September 2026. This is a separate written derivation of the proposed canonical-section angle formula and its integrated metric comparison. No sealed source is edited.

102.0.2 1. Canonical sections in one original Hilbert space

Fix the original monic polynomial χ of degree $q \geq 1$, the original real weight k , and the literal finite polynomial space

$$\mathcal{H} = \mathcal{P}_{2q}, \quad E = \mathbb{C}[S]/(\chi), \quad J : \mathcal{H} \longrightarrow E \quad (\text{PA1})$$

with its fixed coefficient basis and remainder map. Let $M(t) > 0$ be a continuously differentiable Gram on this common space. All orthogonal projections and norms below use that Gram at the same parameter value. The adjoint in the coefficient formula R^*MR is the ordinary conjugate transpose in the retained bases; no coefficient basis is changed implicitly.

For $q - 1 \leq N \leq 2q$, let P_N project onto $\mathcal{P}_N$, and let Q_N project onto the original relation space $\chi\mathcal{P}_{N-q}$, with $Q_{q-1} = 0$. Since the latter is a subspace of the former, their projections commute and satisfy $P_N Q_N = Q_N P_N = Q_N$. Therefore

$$N_N = P_N - Q_N \quad (\text{PA2})$$

is exactly the orthogonal projection onto $K_N = \mathcal{P}_N \cap (\chi\mathcal{P}_{N-q})^\perp$. The original least-norm remainder section $R_N : E \rightarrow \mathcal{P}_N$ has $JR_N = I_E$, image K_N , and positive Gram

$$G_N = R_N^* M R_N > 0. \quad (\text{PA3})$$

These assertions follow from the direct orthogonal decomposition of the surjective remainder sequence; its kernel is multiplication by the unchanged monic χ .

If $i \leq j$, both $R_i x$ and $R_j x$ are in $\mathcal{P}_j$ and have remainder x . Their difference belongs to $\chi\mathcal{P}_{j-q}$, which is orthogonal to K_j . Hence

$$\boxed{R_j = N_j R_i}, \quad R_i^* M R_j = G_j, \quad G_i - G_j = (R_i - R_j)^* M (R_i - R_j) \succeq 0. \quad (\text{PA4})$$

The second identity follows by writing $R_i = R_j + (R_i - R_j)$; its last summand is orthogonal to $R_j E$. The same expansion gives the last identity with all cross terms accounted for.

102.0.3 2. Exact principal angles and the square-root order

Define the actual isometries

$$V_i = R_i G_i^{-1/2}, \quad V_j = R_j G_j^{-1/2}. \quad (\text{PA5})$$

Their domains have the usual coefficient inner product; their targets carry M . Thus $V_i^* M V_i = V_j^* M V_j = I$. Their cross-Gram and its two products, in their exact noncommutative orders, are

$$\begin{aligned}
V_i^* M V_j &= G_i^{-1/2} G_j^{1/2}, \\
(V_i^* M V_j)(V_j^* M V_i) &= G_i^{-1/2} G_j G_i^{-1/2}, \\
(V_j^* M V_i)(V_i^* M V_j) &= G_j^{1/2} G_i^{-1} G_j^{1/2}.
\end{aligned} \tag{PA6}$$

The first follows from (PA4), with $G_j G_j^{-1/2} = G_j^{1/2}$ in the adjacent factors. No commutation between G_i and G_j is used.

The singular values of the first matrix in (PA6) are the cosines of the principal angles between K_i and K_j . This can be seen directly. Choose an orthonormal eigenbasis a_ℓ of $V_i^* M N_j V_i = G_i^{-1/2} G_j G_i^{-1/2}$, and let its eigenvalues be c_ℓ^2 . By (PA3)–(PA4), $0 < c_\ell \leq 1$. Put

$$x_\ell = V_i a_\ell, \quad y_\ell = c_\ell^{-1} N_j x_\ell. \tag{PA7}$$

The x_ℓ form an orthonormal basis of K_i . The projection identity gives $\langle y_\ell, y_r \rangle = \delta_{\ell r}$ and $\langle x_\ell, y_r \rangle = c_r \delta_{\ell r}$. Since the c_ℓ values are positive, the y_ℓ form an orthonormal basis of K_j . These are principal bases. Set $\theta_\ell = \arccos c_\ell \in [0, \pi/2)$. Consequently

$$\text{spec}(G_i^{-1/2} G_j G_i^{-1/2}) = \{\cos^2 \theta_\ell : 1 \leq \ell \leq q\}. \tag{PA8}$$

The multiplicity of $\cos \theta = 1$ is exactly $\dim(K_i \cap K_j)$. Indeed equality $\|N_j x\| = \|x\|$ for an orthogonal projection is equivalent to $x \in K_j$; apply this on $x \in K_i$. The same multiplicity is $\dim \ker(R_i - R_j)$: a common vector has the same remainder in both sections, and conversely equal section values define a vector in the intersection. Thus all unit cosines are retained, including multiplicities arising from coinciding section subspaces.

Taking the determinant of the actual positive matrix in (PA8) gives

$$\log \frac{\det G_i}{\det G_j} = \sum_{\ell=1}^q -\log \cos^2 \theta_\ell. \tag{PA9}$$

All terms are finite and nonnegative, because the original section Grams are positive definite.

102.0.4 3. The two pairs and the forced unit cosine

Use exactly the pairs

$$(i_1, j_1) = (q-1, 2q), \quad (i_2, j_2) = (q, 2q-1). \tag{PA10}$$

For the second pair, both section spaces are contained in

$$\mathcal{L} = \mathcal{P}_{2q-1} \cap (\mathbb{C}\chi)^\perp, \quad \dim \mathcal{L} = 2q-1. \tag{PA11}$$

For K_q , this follows from its defining orthogonality to χ ; for K_{2q-1} , its relation space contains χ , so it has the same orthogonality. The dimension in (PA11) is exact because $\chi \in \mathcal{P}_{2q-1}$ is nonzero and its norm is positive. Both section spaces have dimension q . The identity $\dim(A \cap B) = \dim A + \dim B - \dim(A + B)$ therefore gives an intersection of dimension at least one. By (PA8), at least one angle in the second pair has cosine one. For $q = 1$, the second pair is literally $(1, 1)$, so its sole angle is zero and the whole pair contributes zero.

The original four-volume is precisely

$$\begin{aligned}
\mathcal{B}(t) &= \log \frac{\det G_{q-1}(t) \det G_q(t)}{\det G_{2q-1}(t) \det G_{2q}(t)} \\
&= \sum_{a=1}^2 \sum_{\ell=1}^q -\log \cos^2 \theta_{a,\ell}(t).
\end{aligned} \tag{PA12}$$

Remove one of the proved zero contributions from the second pair. There remain exactly $m = 2q - 1$ nonnegative slots, with every additional zero slot kept. This removal changes neither the value nor either

bound below. It is a pointwise assertion; no differentiability of a chosen angle enumeration at eigenvalue crossings is required.

102.0.5 4. Full projection-difference spectrum and derivative estimate

For a positive angle in (PA7), define $z_\ell = (y_\ell - c_\ell x_\ell) / \sin \theta_\ell$. The principal-basis identities show that the z_ℓ are orthonormal and orthogonal to K_i , and that distinct positive-angle two-planes are orthogonal. In the ordered basis (x_ℓ, z_ℓ) , the two projections and their difference have the exact matrices

$$N_i = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad N_j = \begin{pmatrix} c_\ell^2 & c_\ell s_\ell \\ c_\ell s_\ell & s_\ell^2 \end{pmatrix}, \quad N_i - N_j = \begin{pmatrix} s_\ell^2 & -c_\ell s_\ell \\ -c_\ell s_\ell & -s_\ell^2 \end{pmatrix}, \quad (\text{PA13})$$

where $s_\ell = \sin \theta_\ell$. The last matrix has trace zero and determinant $-s_\ell^2$, hence eigenvalues $+s_\ell, -s_\ell$. A zero-angle common vector has zero difference, and the orthogonal complement of the two spaces also has zero difference. Thus these are the complete nonzero eigenvalues with multiplicity. In particular

$$\text{Tr}(N_i - N_j)_+ = \text{Tr}(N_i - N_j)_- = \sum_{\ell=1}^q \sin \theta_\ell. \quad (\text{PA14})$$

Here the positive and negative parts are positive operators, and the latter is the negative of the operator on its negative spectral subspace.

Let $C = M^{-1}\dot{M}$, which is self-adjoint for the same M . Let $\omega(C) = \lambda_{\max}(C) - \lambda_{\min}(C)$. For any trace-zero self-adjoint D , subtracting the scalar $\lambda_{\min}(C)I$ leaves its trace pairing unchanged. The inequalities $0 \preceq C - \lambda_{\min}(C)I \preceq \omega(C)I$, applied to the two positive parts of D , give

$$|\text{Tr}(DC)| \leq \omega(C) \text{Tr} D_+ \quad \text{if } \text{Tr} D = 0. \quad (\text{PA15})$$

To verify the inequality without a commutation assumption, the trace of a product of two positive operators is nonnegative: it equals the trace of their positive conjugation $A^{1/2}BA^{1/2}$. Hence each of the two positive-part trace terms lies between zero and $\omega(C) \text{Tr} D_+$, and their difference has the asserted absolute bound.

The differentiability of the canonical section follows from its explicit positive-Gram inverse formula. Because $JR_N = I$, its derivative is a relation vector. Orthogonality to that relation space cancels the two section-derivative cross terms, giving

$$\dot{G}_N = R_N^* \dot{M} R_N, \quad \frac{d}{dt} \log \det G_N = \text{Tr}(N_N C), \quad N_N = R_N G_N^{-1} R_N^* M. \quad (\text{PA16})$$

The last formula is the orthogonal projection onto $R_N E$, since it is the identity there and annihilates its orthogonal complement. Cyclicity of the finite trace then proves the middle identity.

Using the two pairs (PA10), (PA12), (PA14)–(PA16), and the triangle inequality gives

$$|\mathcal{B}'(t)| \leq \omega(C(t)) \sum_{a=1}^2 \sum_{\ell=1}^q \sin \theta_{a,\ell}(t). \quad (\text{PA17})$$

This bound does not presume that the two projection differences commute, or that either one commutes with C .

For each of the m retained slots, set $b_r = -\log \cos^2 \theta_r \geq 0$. Then $\sum b_r = \mathcal{B}$ and $\sin \theta_r = f(b_r)$, where

$$f(b) = \sqrt{1 - e^{-b}}. \quad (\text{PA18})$$

It is continuous at zero and strictly concave on $(0, \infty)$, because

$$f''(b) = -\frac{e^{-b}}{2\sqrt{1 - e^{-b}}} - \frac{e^{-2b}}{4(1 - e^{-b})^{3/2}} < 0.$$

Concavity extends to the endpoint by continuity: apply the interior inequality to all arguments increased by a positive number and let that number decrease to zero. Jensen's inequality, which follows by iterating the defining two-point concavity inequality, therefore gives

$$\sum_{r=1}^m f(b_r) \leq m f(\mathcal{B}/m).$$

Combining this with (PA17) proves the proposed estimate with its exact constant:

$$\boxed{|\mathcal{B}'| \leq m \sqrt{1 - e^{-\mathcal{B}/m}} \omega(M^{-1} \dot{M}), \quad m = 2q - 1.} \quad (\text{PA19})$$

102.0.6 5. Strict positivity for the actual moment Grams

The original arithmetic and Gamma norms have the common vertical coordinate $S = k/2 + iu$ and the form

$$\langle P, R \rangle_{M(t)} = \int_{\mathbb{R}} \overline{P(k/2 + iu)} R(k/2 + iu) m_t(u) du. \quad (\text{PA20})$$

The original densities and their convex interpolation have the required finite moments and positive definite polynomial Grams. Write the unchanged polynomial as $\chi(S) = \sum_{j=0}^q a_j S^j$, retaining its coefficients, and define the exact reflected polynomial

$$\chi^\#(S) = \sum_{j=0}^q \overline{a_j} (k - S)^j. \quad (\text{PA21})$$

On the original vertical line, $k - S = \overline{S}$, so $\chi^\#(S) = \overline{\chi(\overline{S})}$. It has degree q , with leading coefficient $(-1)^q \overline{a_q}$; that sign is not removed. Hence $\chi \chi^\# \in \chi \mathcal{P}_q$ and

$$\langle 1, \chi \chi^\# \rangle_{M(t)} = \int |\chi(k/2 + iu)|^2 m_t(u) du = \|\chi\|_{M(t)}^2 > 0. \quad (\text{PA22})$$

All integrals are within the specified moment degree $2q$. If the first pair in (PA10) had every cosine equal to one, its two q -dimensional spaces would coincide. Its first space is exactly $\mathcal{P}_{q-1}$, and its second is $(\chi \mathcal{P}_q)^\perp$. The constant polynomial would then be orthogonal to $\chi \chi^\#$, contradicting (PA22). Thus at least one angle in the first pair is positive, and

$$\boxed{\mathcal{B}(t) > 0} \quad (\text{PA23})$$

for every actual moment Gram in this path. This proves strict positivity without a presumption about the zero locations or a discarded relation primitive. It also verifies the needed endpoint at $q = 1$, where the second pair contributes exactly zero.

102.0.7 6. Integrated arcosh comparison on the actual arithmetic/reference path

For $b > 0$, define

$$F_m(b) = 2 \operatorname{arcosh}(e^{b/(2m)}), \quad F'_m(b) = \frac{1}{m \sqrt{1 - e^{-b/m}}}. \quad (\text{PA24})$$

The derivative is obtained by differentiating the displayed exponential and arcosh; no rescaling of $\mathcal{B}$ or its original mass is performed. By (PA23), the chain rule applies along the entire actual path. Equation (PA19) gives

$$\left| \frac{d}{dt} F_m(\mathcal{B}(t)) \right| \leq \omega(M(t)^{-1} \dot{M}(t)). \quad (\text{PA25})$$

Now take exactly the convex source-metric interpolation $M(t) = (1 - t)M_\Gamma + tM_{\text{ar}}$, $0 \leq t \leq 1$, inherited from the two actual densities in (PA20). Put

$$A = M_\Gamma^{-1/2} M_{\text{ar}} M_\Gamma^{-1/2} > 0, \quad 0 < \lambda_1 \leq \dots \leq \lambda_{2q+1}, \quad \kappa = \lambda_{2q+1}/\lambda_1. \quad (\text{PA26})$$

This is the complete common-source relative Gram, not a quotient-metric substitute. Factoring $M(t) = M_\Gamma^{1/2}((1-t)I + tA)M_\Gamma^{1/2}$ shows that $M(t)^{-1}\dot{M}(t)$ is similar to $((1-t)I + tA)^{-1}(A - I)$. Its eigenvalues are

$$c_j(t) = \frac{\lambda_j - 1}{1 + t(\lambda_j - 1)}. \quad (\text{PA27})$$

Every denominator is positive. For fixed t , the derivative of this scalar expression with respect to λ is $(1 + t(\lambda - 1))^{-2} > 0$, so its largest and smallest values have the same indices as the original λ_j . Therefore

$$\begin{aligned} \int_0^1 \omega(M(t)^{-1}\dot{M}(t)) dt &= \int_0^1 \left(\frac{\lambda_{2q+1} - 1}{1 + t(\lambda_{2q+1} - 1)} - \frac{\lambda_1 - 1}{1 + t(\lambda_1 - 1)} \right) dt \\ &= \log \lambda_{2q+1} - \log \lambda_1 = \log \kappa. \end{aligned} \quad (\text{PA28})$$

The identity also includes any $\lambda_j = 1$, whose integrand is zero, and cases in which both extreme eigenvalues are above or below one.

Integrating (PA25) and using (PA28) gives precisely

$$\left| 2 \operatorname{arcosh}\left(e^{\mathcal{B}_{\text{ar}}/(2m)}\right) - 2 \operatorname{arcosh}\left(e^{\mathcal{B}_\Gamma/(2m)}\right) \right| \leq \log \kappa, \quad m = 2q - 1. \quad (\text{PA29})$$

Every endpoint volume is the original four-volume in (PA12). All section angles are strictly below $\pi/2$ because every finite quotient Gram is positive definite. The forced unit-cosine slot is retained through the exact common-space argument (PA11). For $q = 1$, $m = 1$ and the formula uses the single nontrivial pair. If $\kappa = 1$, the two source Grams differ by one positive scalar; all quotient Grams acquire that same scalar and their four-volume ratio is unchanged, consistently giving zero on both sides of (PA29).

The proposed formulas are accepted. This calculation proves their exact finite metric geometry and their application to the original positive moment interpolation. It supplies no bound on κ beyond its computed source definition, and no extra arithmetic conclusion is assigned to that quantity.

103 Complete independent acceptance proof of the literal tensor theta primitive

Source role. Complete independently accepted supporting calculation, with every original prose and mathematical payload retained and exact delimiter-transcription inverses.

Exact revision. a78384fe3a1cff8398713f89181c38c62351315d6f32aca314d357e457938de4

Complete source. sources/next_edition/all_degree_review_conversions_v1/sources/angle_primitive_acceptance/tau_all_degree_angle_primitive_acceptance_20260913.md

Source SHA-256. a78384fe3a1cff8398713f89181c38c62351315d6f32aca314d357e457938de4

103.0.1 Acceptance addendum: the full all-degree angle derivation and theta primitive

13 September 2026. The complete work/tau_all_degree_angle_independent_review_20260913.md, equations AA1–AA44 and all prose, was read for this addendum. Accepted source SHA-256: b0634428d69f80dedd4f68766ef9e6468c390eaac86dfe58b08c21c8fb896e03.

The original output/split_zero_rh_tandem_2026-09-12/tex/arithmetic_input.tex was previously read completely and its current pin was checked again: a50b0fb587b644c4ea94dba45b2b423d038f2bfbfe7fef188543939ba850e58d2. It supplies the unchanged spaces, Euler inverses, original Gaussian, theta factor, and full finite-packet unit used below. The earlier independent PA1–PA29 calculation is tau_principal_angle_volume_independent_review_20260913.md, SHA-256 5f6a75afdbdfd4dcd81781a336702ea6d6545fd8fd0a9b2e014a605e414b3198.

AA1–AA37 agree with that independent principal-angle derivation. The relation coefficient in AA7 keeps its actual domain; the positive square roots in AA9–AA10 occur in the correct order; AA13 and AA15 retain every unit cosine and all remaining zero eigenvalues. The second pair's common hyperplane gives the exact one-angle reduction to $m = 2q - 1$, including its identical-cutoff case when $q = 1$. The strict moment

positivity in AA28–AA31 retains the leading sign of the reflected polynomial. AA33 integrates the actual full-source spectral width, and the two inverse-cosh branches in AA36–AA37 have the stated domains. No further correction is required in those calculations.

Section 7 was checked separately as follows.

1. The identity $h(D)F_h = \Theta\phi_*$ in AA38 has exactly the original Mellin transform $g = 2\xi$ on both sides. The source statement $H_{\phi_*} = 1$ gives $\mathcal{M}\Theta\phi_* = g$, while multiplication by the original h cancels the displayed quotient g/h . Mellin uniqueness in the stated source space therefore proves the identity. There is no extra Taylor-unit factor in this identity: the full unit belongs to the later arithmetic jet observation, where the review retains it.
2. In the tensor algebra, the sum operator is multiplication by $s_1 + \cdots + s_k$, and its cyclic vector is the literal unit. Thus the defining annihilation equation is the vanishing of the actual element $\chi(s_1 + \cdots + s_k)$ modulo the ideal generated by the original $h(s_j)$. This proves AA39, including repeated roots and the full cyclic multiplicities.
3. Division in AA40 is valid over the polynomial coefficient ring in the other variables because each divisor $h(s_j)$ is monic. Its division operations multiply earlier-variable coefficients only by the complex coefficients of that divisor and add or subtract such coefficients. Therefore later divisions cannot raise a previously restricted degree in an earlier variable. At the final step every variable degree is less than $\deg h$. The corresponding tensor monomials are a basis of the quotient: one-variable monic remainders form a basis, and their external products form the tensor basis. A remainder in their span with zero quotient image is consequently the zero polynomial. Telescoping the actual division identities gives AA41 with no leftover term.
4. In the j -th summand of AA42, the j -th tensor factor is ϕ_* in degree zero. The preceding $j - 1$ factors are all F_h in degree one. Thus the only nonzero term of the tensor differential carries precisely the sign $(-1)^{j-1}$; it cancels the displayed coefficient sign in that summand. Every other factor is in the top degree of the two-term complex, where the differential is zero. All D_j have cohomological degree zero and commute with the differential because $D\Theta = \Theta D$. Applying the differential therefore gives exactly

$$\sum_{j=1}^k L_j(D_1, \dots, D_k) P(D_1 + \cdots + D_k) h(D_j) F_h^{\otimes k}.$$

AA41 turns this sum into $P(\sum D_j)\chi(\sum D_j)F_h^{\otimes k}$, which is the unchanged $\mathcal{V}_{h,k}(\chi P)$. This proves AA43 with every tensor sign and the original theta factor retained. For $k = 1$, the same division gives $L_1 = 1$ because the cyclic annihilator is the original monic h , and the formula reduces to the one-leg identity without any tensor sign.

5. Substituting the literal polynomial $P = X_{ij}x$ from AA7 proves the first equation of AA44. Its second equation is the independently checked orthogonal source energy. That energy uses the specified common interpolated coefficient Gram $M(t)$; at $t = 1$ it is the ordinary arithmetic $\mathcal{V}_{h,k}$ norm, and along the path it is the stated arithmetic/reference interpolated norm. Neither equality identifies those two norms. The actual theta primitive exists as the displayed source cochain at every parameter value, independently of which of those specified norms is being evaluated.
6. Inclusion of the one-leg complex into the first leg of the two-leg complex commutes with the differential: the second coefficient is zero, so the original difference differential is still Θ on that image. The supported reconstruction retains its admitted mask, sends represented zero amplitudes to supported zeros, and sends external absence to τ . Consequently this ordinary coefficient inclusion does not require identifying a missing support leg with a new represented amplitude. The explicit primitive and its full energy remain attached before taking the supported quotient.

The full AA1–AA44 review is accepted at the pin above, including its added deterministic division and theta primitive. No mathematical correction is requested. This addendum records a written proof review; it makes no claim of a new finite test, Lean execution, remote operation, or fresh reading of the separate root AGT file. No sealed source was changed.

104 Which completion retains the arithmetic jets?

This section extends the Hilbert/jet calculation of the September 12 orthogonal-boundary note. All changes of topology are specified maps on the same arithmetic functions. The theta relation space,

the operator $D = -x\partial_x$, and $g = 2\xi$ are retained.

104.1 Logarithmic coordinates and the original measure

Let $\mathcal{B}, V, \Theta$ have the definitions in Section 6. Put $\mathcal{H} = L^2(\mathbb{R}_+, dx)$. The map

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{L}^{-1}f(x) = x^{-1/2}f(\log x)$$

is unitary from $\mathcal{H}$ to $L^2(\mathbb{R}, dy)$, since $dx = e^y dy$. Direct differentiation gives

$$\mathcal{L}D\mathcal{L}^{-1} = \frac{1}{2} - \partial_y, \quad \mathcal{M}F(\rho) = \int_{\mathbb{R}} (\mathcal{L}F)(y)e^{(\rho-1/2)y} dy. \quad (\text{JT1})$$

On $\mathcal{H}$, D denotes the closed maximal distributional operator which $\mathcal{L}$ identifies with $1/2 - \partial_y$ on the Sobolev domain $H^1(\mathbb{R})$. For an integer $r \geq 0$, let $\mathcal{H}^{(r)}$ be the domain of the first r powers of D in $\mathcal{H}$, with norm $\|F\|_r^2 = \sum_{k=0}^r \|D^k F\|_{\mathcal{H}}^2$. The Mellin-line transform makes this the norm with density $w_r(t) = \sum_{k=0}^r |1/2 + it|^{2k}$ relative to $dt/(2\pi)$.

Theorem 104.1 (Finite differential graph norms and the complete jet graph). *Let Z be a nonempty finite packet of actual nontrivial zeros with full multiplicities, $E_Z = \mathbb{C}[t]/h_Z$, and $J_Z : \mathcal{B} \rightarrow E_Z$ the complete Mellin-jet map. For every finite r ,*

$$\overline{\text{span}\{D^j f_0 : j \geq 0\}}^{\mathcal{H}^{(r)}} = \mathcal{H}^{(r)}, \quad f_0 = \Theta\phi_*, \quad (\text{JT2})$$

$$\overline{\{(F, J_Z F) : F \in \mathcal{B}\}}^{\mathcal{H}^{(r)} \oplus E_Z} = \mathcal{H}^{(r)} \oplus E_Z. \quad (\text{JT3})$$

Thus J_Z is not closable in this norm. The completed theta-boundary graph is $\mathcal{H}^{(r)} \oplus 0$; its quotient in (JT3) is E_Z through $[(F, u)] \mapsto u$.

Proof. For each $0 < c < \pi/4$, the actual source gives $|g(1/2 + it)| \leq C_c e^{-c|t|}$. Indeed $e^{z/2}f_0(e^z)$ is holomorphic in $|\text{Im } z| < \pi/4$: its Gaussian series and derivatives converge uniformly on compact subsets since $\text{Re}(e^{2z}) > 0$. On smaller horizontal strips it decays faster than every exponential at the positive end. Poisson gives $Jf_0 = f_0$, so analytic continuation gives $e^{z/2}f_0(e^z) = e^{-z/2}f_0(e^{-z})$; this controls the negative end. Shift the Fourier contour to height c or $-c$. The vertical integrals vanish by that decay, proving the bound.

The measure $d\mu_r = w_r(t)|g(1/2 + it)|^2 dt/(2\pi)$ has a positive exponential moment. If $u \in L^2(\mu_r)$ is orthogonal to every polynomial, $d\nu = \bar{u} d\mu_r$ is a finite complex measure with a smaller positive exponential moment, by Cauchy–Schwarz. Its Fourier transform is analytic in a strip, and every derivative at zero vanishes. Analytic continuation makes it zero there. Fourier uniqueness gives $\nu = 0$, hence $u = 0$. For this uniqueness step, convolve with Gaussian approximate identities: each convolution is in L^2 , since $\|\nu * \gamma_\epsilon\|_2 \leq \|\nu\|_{\text{TV}} \|\gamma_\epsilon\|_2$; its transform is zero, so Plancherel makes it zero, and weak convergence then gives $\nu = 0$. Polynomials are therefore dense in $L^2(\mu_r)$. Multiplication by $g(1/2 + it)$ is an isometry onto $L^2(w_r dt/(2\pi))$: divide almost everywhere, since this nonzero analytic function vanishes on a discrete, measure-zero set. The Mellin transform of $D^j f_0$ is $(1/2 + it)^j g(1/2 + it)$, proving (JT2).

Every theta polynomial has zero J_Z -jet. Thus the closure in (JT3) contains $(h, 0)$ for every h . The actual packet section $R_{\text{ref}} : E_Z \rightarrow \mathcal{B}$ satisfies $J_Z R_{\text{ref}} = 1$. Approximate $R_{\text{ref}} u$ in $\mathcal{H}^{(r)}$ by theta polynomials $p_n(D)f_0$. Then $(R_{\text{ref}} u - p_n(D)f_0, u) \rightarrow (0, u)$. This proves (JT3). For $u \neq 0$ it prevents closability. The boundary and quotient statements follow from the coordinate projections. $\square$

Corollary 104.2 (Differential on the completed graph). *The continuous map*

$$(D, A) : \mathcal{H}^{(r+1)} \oplus E_Z \longrightarrow \mathcal{H}^{(r)} \oplus E_Z, \quad (F, u) \mapsto (DF, Au) \quad (\text{JT4})$$

extends the original $(F, J_Z F) \mapsto (DF, J_Z DF)$. On the quotient by the boundary closure it is exactly $A = M_t$.

Proof. The norm definitions bound $D : \mathcal{H}^{(r+1)} \rightarrow \mathcal{H}^{(r)}$; the jet identity $J_Z D = A J_Z$ follows by retaining the complete Taylor polynomial of $s\mathcal{M}F(s)$. Apply (JT3). $\square$

Its split observation records $(e_{\mathcal{H}}, u^\bullet)$ when the Hilbert component tends to its supported zero and the jet survives. The projections compare this point to $e_{\mathcal{H}}$ and to $u^\bullet$; they do not identify it with (τ, τ) .

104.2 Exponential weights and exact jet norms

For $a \geq 0$, retain the original functions and introduce

$$\mathcal{H}_a = L^2(\mathbb{R}_+, e^{2a|\log x|} dx), \quad \|F\|_a^2 = \int_{\mathbb{R}} |\mathcal{L}F(y)|^2 e^{2a|y|} dy. \quad (\text{JT5})$$

The identity inclusions $\mathcal{B} \hookrightarrow \mathcal{H}_a \hookrightarrow \mathcal{H}_0$ are continuous. Source seminorms with powers exceeding $a + 1/2$ control both ends. Smooth functions compactly supported in log coordinates prove density.

Theorem 104.3 (Exact threshold, including the boundary). *Put $\ell_{\rho,k}(F) = (\mathcal{M}F)^{(k)}(\rho)/k!$ and $b = \operatorname{Re} \rho - 1/2$, with $k \geq 0$. The functional on $\mathcal{B}$ extends continuously to $\mathcal{H}_a$ if and only if $|b| < a$. Its exact squared norm is*

$$\|\ell_{\rho,k}\|^2 = \frac{(2k)!}{(k!)^2} \left(\frac{1}{[2(a-b)]^{2k+1}} + \frac{1}{[2(a+b)]^{2k+1}} \right). \quad (\text{JT6})$$

Proof. Inner products are conjugate-linear in their first argument. The Riesz vector in log coordinates must be

$$r_{\rho,k}(y) = \frac{y^k}{k!} e^{(\bar{\rho}-1/2)y} e^{-2a|y|}.$$

Its norm squared is $(k!)^{-2} \int y^{2k} e^{2by-2a|y|} dy$. The two half-lines converge exactly when $|b| < a$; their gamma integrals give (JT6). If $|b| \geq a$, testing any putative Riesz representative against compactly supported smooth functions forces it to be this same non-square-integrable density, a contradiction. The equality case is included: one end has no exponential decay. $\square$

Theorem 104.4 (Complete confluent Cauchy Gram). *For distinct centres ρ_i with $|\operatorname{Re} \rho_i - 1/2| < a$ and orders $0 \leq k < m_i$, let $J_a : \mathcal{H}_a \rightarrow \mathbb{C}^d$ be their Taylor-jet map. In the original Taylor coordinates its Gram $C_a = J_a J_a^*$ is*

$$(C_a)_{(i,k),(j,l)} = \frac{(k+l)!}{k!l!} \left(\frac{1}{(2a-z_{ij})^{k+l+1}} + \frac{(-1)^{k+l}}{(2a+z_{ij})^{k+l+1}} \right), \quad z_{ij} = \rho_i + \bar{\rho}_j - 1. \quad (\text{JT7})$$

It is positive definite. The exact right inverse $S_a = J_a^ C_a^{-1}$ satisfies*

$$J_a S_a = 1, \quad \|S_a u\|_a^2 = u^* C_a^{-1} u, \quad \|F\|_a^2 = \|(1 - S_a J_a)F\|_a^2 + (J_a F)^* C_a^{-1} (J_a F). \quad (\text{JT8})$$

Proof. The entry is $\langle r_{\rho_i,k}, r_{\rho_j,l} \rangle_a$; integrating $y^{k+l} e^{z_{ij}y-2a|y|} / (k!l!)$ on both half-lines proves (JT7) with its stated sign. The vectors are independent. After removing their common nonzero factor, a dependence is an exponential polynomial $\sum_i P_i(y) e^{\bar{\rho}_i y} = 0$. For each i , apply $\prod_{j \neq i} (\partial_y - \bar{\rho}_j)^{m_j}$. On the i -th polynomial factor every nonzero constant plus derivative is invertible on the finite-degree polynomial space; hence $P_i = 0$. Thus $C_a > 0$. Now $S_a J_a$ is the orthogonal projection onto $\operatorname{range} J_a^* = (\ker J_a)^\perp$. Multiplying the specified matrices proves all of (JT8). $\square$

For an actual packet the unchanged theta subspace lies in $\ker J_a$. Consequently there is an onto continuous quotient map

$$\mathcal{H}_a / \overline{\Theta V}^{\mathcal{H}_a} \longrightarrow \mathbb{C}^d, \quad [F] \mapsto J_a F, \quad \ker = (\ker J_a) / \overline{\Theta V}^{\mathcal{H}_a}. \quad (\text{JT9})$$

Its surjectivity follows from (JT8). Define the retained coordinate map

$$T : E_Z \longrightarrow \mathbb{C}^d, \quad T[p] = (p^{(k)}(\rho_i)/k!)_{i,k}.$$

Then $J_a|_{\mathcal{B}} = TJ_Z$, and the exact representative difference is $R_{\text{ref}}T^{-1} - S_a : \mathbb{C}^d \rightarrow \ker J_a$. Here R_{ref} is the original source map; S_a is a Hilbert map, with no claim it lies in $\mathcal{B}$. In E_Z coordinates the observation Gram is $C_E = T^{-1}C_a(T^{-1})^*$ and the right inverse is S_aT .

104.3 The energy term carried by the weighted topology

Theorem 104.5 (Weighted boundary identity on the actual representatives). *On $\mathcal{B}$, with $(\Sigma H)(x) = \text{sgn}(\log x)H(x)$,*

$$\langle DF, H \rangle_a + \langle F, DH \rangle_a - \langle F, H \rangle_a = 2a \langle F, \Sigma H \rangle_a. \quad (\text{JT10})$$

*For an actual packet representative $R : E_Z \rightarrow \mathcal{B}$ with $DR - RA = B = \Theta k$, put $G_a = R^*R$, $T_a = R^*\Sigma R$. Adjoints use exactly $\mathcal{H}_a$. Then*

$$A^*G_a + G_aA - G_a = 2aT_a - (B^*R + R^*B), \quad -G_a \leq T_a \leq G_a. \quad (\text{JT11})$$

For $Au = \rho u \neq 0$,

$$2 \operatorname{Re} \rho - 1 = 2a \frac{u^*T_a u}{u^*G_a u} - 2 \operatorname{Re} \frac{\langle Ru, Bu \rangle_a}{\|Ru\|_a^2}. \quad (\text{JT12})$$

Proof. In log coordinates integrate $-(\bar{f}h)'e^{2a|y|}$ on the two half-lines. Source decay kills the terms at infinity; those at zero cancel because the weight is continuous there. The weak derivative of the weight is $2a \operatorname{sgn}(y)e^{2a|y|}$. This proves (JT10) with its positive sign. Substitute $DR = RA + B$ and move both boundary terms to obtain (JT11). Multiplication by sign is a self-adjoint contraction, proving the form bounds. Evaluation on u proves (JT12). $\square$

At $a = 0$, (JT11) is the source boundary-control formula. At $a > 0$, the topology making a jet continuous has its explicitly retained energy term $2aT_a$. Off-line jets cannot remain continuous as $a \downarrow 0$. For line-centred jets, all $a > 0$ work but (JT6) diverges as $a \downarrow 0$. These are exact comparisons on the same Mellin map and differential, without a weight theorem assumed.

104.4 The tesserine coordinate in the actual weight form

Retain $c_{-+} = 4(\beta - 1/2)$, $c_{--} = 4i\gamma$ for $\rho = \beta + i\gamma$. For $p > 1$ and $|\psi| = 1$,

$$z = \psi \exp\left(\frac{\log p}{4}(c_{-+} + c_{--})\right), \quad \frac{u^*(A^*G + GA - G)u}{u^*G u} = \frac{c_{-+}}{2}, \quad |b| < a \iff |c_{-+}| < 4a. \quad (\text{JT13})$$

The first equality is exactly $z = \psi p^{\rho-1/2}$; the second is evaluation on $Au = \rho u$. Thus the original sector, local holonomy radius, arithmetic weight form and jet-continuity strip have specified comparison maps. The scalar evaluations do not replace the original matrix or its support-labelled source.

Part II

Complete counterfactual, mixed-support and boundary calculations

105 The object, its actual elements, and its assembly

Put $U = \{s : 0 < \Re s < 1, \Re s \neq 1/2\}$ and retain the original $g = 2\xi$, source V , target $\mathcal{B}$, operator $D = -x\partial_x$, and map Θ defined in full in the following construction. Let $\mathcal{P}_U$ consist of all monic polynomials whose roots lie in U , including 1. The construction starts with actual functions:

$$\mathcal{B}_U = \{F \in \mathcal{B} : \exists p \in \mathcal{P}_U, p(D)F \in \Theta V\}, \quad C_U = [V \xrightarrow{\Theta} \mathcal{B}_U], \quad \mathfrak{T}_U = \bigoplus_{k \geq 0} C_U^{\otimes k}. \quad (\text{T0.1})$$

The differential, multiplication, maps, topologies, full local coordinates and every finite presentation of this object are proved below. In particular, (T0.1) specifies a subspace of the original functions and a differential graded algebra, rather than introducing free numerical variables to which necessary conditions are assigned. An element is a finite sum of finite tensors of actual original source functions and target functions satisfying the displayed annihilator equation. Its differential applies the original theta map with the original Koszul signs.

The complete object includes its maps into the original two-leg tau diagram and its arithmetic, geometric and norm observations. We specify exactly what one object with these maps means. Let

$$C_\tau^{\text{raw}} = [V_+ \oplus V_- \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \widehat{\psi})} \mathcal{B}], \quad i : C_U \longrightarrow C_\tau^{\text{raw}}, \quad (\text{T0.2})$$

where $i^0 v = (v/2, -\widehat{v}/2)$ and $i^1 F = F$. The minus leg has the action $t = 1 - D$, the plus and target legs have $t = D$, and Fourier uses $e^{-2\pi i x v}$. Thus i is an injective cochain map: its differential on v is $\Theta(v/2 + \widehat{v}/2) = \Theta v$, and Fourier's Euler identity proves its $A = \mathbb{C}[t]$ linearity. Tensoring these literal inclusions over $\mathbb{C}$ gives the injective homomorphism of differential graded algebras

$$T(i) : \mathfrak{T}_U \hookrightarrow T_{\mathbb{C}}(C_\tau^{\text{raw}}). \quad (\text{T0.3})$$

Injectivity follows because every inclusion of vector spaces admits a linear retraction: extend a basis, project onto the original basis, and tensor the retractions. This is a proof of injectivity, not a claim that these auxiliary retractions preserve the Euler action.

The arithmetic sheaf observation is the actual balanced diagram

$$\mathcal{A}_U \xrightarrow{a} C_\tau \xrightarrow{c} C_g = [\mathcal{O}_U \xrightarrow{g} \mathcal{O}_U], \quad C_{\ker} \hookrightarrow C_\tau, \quad c \circ C_{\ker} = 0, \quad (\text{T0.4})$$

with the all-polynomial complexes, all arrows, and their exact cohomology constructed in (CAU.1)–(CAU.33). Here balancing means sheafification of $\mathcal{O}_U \otimes_A$; it is a specified observation of the original source. Independent tensor variables always tensor over $\mathbb{C}$, as in (T0.1), not over A or a single analytic stalk.

The connection from (T0.1) to this diagram is also explicit. At a polynomial p , a degree-one cycle of the original complex

$$[V \xrightarrow{(\Theta, p(D))} \mathcal{B} \oplus V \xrightarrow{(p(D), -\Theta)} \mathcal{B}]$$

is (F, ϕ) with $p(D)F = \Theta\phi$. Send it to $F \in \mathcal{B}_U$. Its boundaries map to ΘV . Under the transitions $p \mid P$, the first coordinate F is unchanged. Consequently its induced map to $H^1(C_U)$ is the identity on

$$T_U Q = \bigcup_{p \in \mathcal{P}_U} \ker(p(D) : Q \rightarrow Q), \quad Q = \mathcal{B}/\Theta V. \quad (\text{T0.5})$$

Composition with (TO.2), balancing, and Mellin is exactly the map in (TO.4), as is checked on these representatives in CAU.

Define $\mathbf{C}_U$ to be the following single represented diagram. Start with the source vertices and arrows (TO.1)–(TO.4), including the all-polynomial vertices and transitions of CAU. Adjoin the finite polynomial source, full tensor jet, conormal, cyclic and metric observation vertices and their arrows in (AG19)–(AG38) and the subsequent exact continuation of those formulas. Common source vertices are identified by the literal function equalities in (AG12), (AG20), (AG22) and (AG26), and by the representative map preceding (TO.5). Retain the proper source-label and support arrows of (CAU.28)–(CAU.32).

The original all-packet observation also retains $\Omega = \{0 < \Re s < 1\}$ and $L = \Omega \setminus U$. For $X = \Omega, L$ put $\mathcal{B}_X = q^{-1}(T_X Q)$ using the same all-polynomial torsion definition as (TO.5), and $C_X = [V \rightarrow \mathcal{B}_X]$. Include the actual cochain inclusions $C_U \rightarrow C_\Omega \rightarrow C_+ = [V \rightarrow \mathcal{B}]$, with identity in degree zero and the original inclusions in degree one. The full calculation (MCF41)–(MCF41g) proves $T_\Omega Q = T_U Q \oplus T_L Q$ by its exact Bezout maps, and proves the cochain quotient and all proper-source comparisons. Thus critical and offcritical full packets occur on their specified domains while (TO.7) retains its offcritical detector.

For every actual full-order packet h in Ω , every $n \geq 1$, and every coefficient mask $A \subseteq I_n = \{0, \dots, n-1\}$, also adjoin the original independent carrier $D_{E_h, n} = G(E_h)^n$, every labelled face $E_h[A] = \bigoplus_{i \in A} E_h$, and the source faces $C_\Omega[A]$ and $C_{W, S}^\Omega[A]$, together with their original U -subcomplexes and their maps into $C_{W, S}[A]$. The section for a packet rooted in U lands in C_U 's degree-one term; a general packet in Ω uses the degree-one term of C_Ω , as proved in (MCF41e). Here $E_h = \mathbb{C}[x]/(h)$; the scalar ring $\mathbb{C}[x]$ acts on C_U by $x \mapsto D$ and is identified with the earlier Euler ring $\mathbb{C}[t]$ by $x \mapsto t$ at that Euler-ring vertex. The coefficient variable t in the negacyclic observation $E_h[t]/(t^n + 1)$ is a separate tagged coordinate from x and from the earlier Euler-ring variable bearing the same glyph. The additional generators are the amplitude and complete-mask maps, negacyclic product, synchronization and localization map and its specified nonunital section, coefficient extraction and insertion, signed coefficient Frobenius, and every zero-insertion arrow between faces. Include the coordinatewise original quotient, full-unit packet section, divisor, arithmetic action and theta-boundary homotopies, together with the strict free-resolution realization which can be balanced over $\mathbb{C}[x]$. Their complete formulas and proofs are the mixed-support and cochain attachments included below. In particular, the finite section s_h remains a complex-linear observation; it is not silently tensored over $\mathbb{C}[x]$ as a module morphism. The nonempty original leg label $\emptyset \neq S \subseteq \{+, -\}$ and the coefficient mask A remain independent indices. Every coefficient face, including its represented zero and the empty face, remains a vertex; no colimit over the face-filling arrows is taken.

Include also the original quadratic mixed carrier, its exact synchronization quotient and prime branches, and the original higher lattice zero fibres with their specified threshold and localization maps, as constructed in the accompanying full-carrier proofs. Retain also the original chain-to-ideal identifications $j \mapsto p^{n-j}(\mathbb{Z}/p^n\mathbb{Z})$ and $j \mapsto \epsilon^{n-j}k[\epsilon]/(\epsilon^n)$. For the power-indexed ideals $I_a = (p^a)$ or $I_a = (\epsilon^a)$, $0 \leq a \leq n$, their separate product is $I_a I_b = I_{\min(n, a+b)}$, in addition to their stated sum and intersection operations. These ideal-product degrees remain distinct from the spectral nilpotent filtration below. Their connections to the independent coefficient carrier are those explicit maps. The nilpotent filtrations, their graded quotients, and the additional Tate-scaling operators of (MW1)–(MW22) are further observation vertices and arrows. The reconstruction identity (MW19) retains the unchanged original arithmetic dilation at the same vertex. Include the original marked-product family with coefficient differentials $u\partial_{s_i} + h(s_i) - t_i$, its ordered exterior complex, its marked-fibre theta map, the specified finite-field coefficient realizations and their geometric sheaves, and its original complex connection and period observations. The marked-product boundary calculation below constructs these maps, the full u -connection and its period determinant. Its coefficient specializations have their stated source and target; the represented category imposes only the proved comparison relations. Adjoin also the four original common-source projections and signed trace obser-

vations of (SP1)–(SP13). Close all arrows under composition. Close the complex-linear amplitude complexes under their specified finite ordered tensor products over $\mathbb{C}$ and Koszul permutations, and the ringed-space complexes under their specified external tensor products and base maps. For the mixed semiring carriers and support spaces retain only their explicitly defined products and the proved coefficient-face tensor attachments. No complex tensor product or Koszul action is inferred on a semiring carrier, prime branch or lattice-zero set from its underlying-set observation.

This specifies a set-sized indexing category and one representation of it. For its complex and sheaf vertices the target is the category of ringed spaces with the specified module complexes and base-map/semilinear cochain morphisms defined in CAU. External tensor words carry product bases. Its polynomial and Hilbert vertices have their displayed vector-space domains and linear maps; forms and scalar norm values are typed observations. A common target for all evaluations is the category of their underlying sets, with each original structure retained in its vertex label and each displayed function as an arrow. This set-valued representation with marked structural data does not declare a nonlinear norm evaluation to be a linear or cochain morphism. Concretely, form the free path category on the stated vertices and arrows, then divide its paths by precisely their proved composition, tensor-interchange and common-source equalities. Assign each vertex its displayed actual space and each arrow its displayed map. Every defining relation holds as an equality of those maps by the proofs cited above; hence the assignment factors through the path relations and is a well-defined functor. This functor is $\mathbf{C}_U$. Its explicit generating source chain is $C_U \rightarrow C_\Omega \rightarrow C_+$. The offcritical tensor sources come from C_U , all nontrivial full-packet tensor sources from C_Ω , and the ambient source observations retain C_+ . The graded cochain attachment (CFA.38) also retains the actual Hom-complex vertices and their differential. A proved homotopy has the relation $f - g = dH + Hd$ with its specified degree-minus-one map H ; it does not impose the false equality $f = g$ on the original degree-zero paths.

The universal assertion for this assembly concerns the actual maps: any compatible realization of these same generating spaces and maps extends uniquely to the represented category, by sending a path to the composite of its maps and a tensor word to the specified tensor product. Well-definedness is precisely the stated path relations; uniqueness follows because every arrow is generated in this manner. No colimit over quotient observations is taken. Such a colimit would identify classes with images in zero targets and erase the very fibres the programme constructs. Only the directed inclusion colimits explicitly proved in CAU and AG are used.

106 The absolute base and the two distinct zero fibres

The base of this represented object is the constructed pointed monoid blueprint $\mathfrak{b}_\tau = \text{Spec}\{\tau, 1\}$, with absorbing τ and empty-sum relation $\emptyset \equiv \tau$. At a commutative coefficient ring R , its split carrier is $G(R) = \{\tau\} \sqcup R^\bullet$, with $e_R = 0_R^\bullet \neq \tau$. Supported addition and multiplication are the operations in R , while τ is the additive identity and multiplicative absorber. At a labelled linear arrow $f : E \rightarrow F$ the lifted arrow is

$$\tau \longmapsto \tau, \quad (\lambda, x) \longmapsto (\lambda', f(x)). \quad (\text{T0.6})$$

For a linear arrow its additivity and supported scalar compatibility follow by applying the linear identities for f ; compositions give the lift of the composite. An element killed by an arrow lands at $(\lambda', 0)$ and retains support. For the conjugate-linear reflected-conjugate operator and for conjugation, the scalar is conjugated in the semilinear rule proved in (AG38). The Fourier reflection retains its complex-linear map and the base involution $s \mapsto 1 - s$. For multiplication the rule is $x^\bullet y^\bullet = (xy)^\bullet$, with absorbing τ ; the product identities prove compatibility. These rules give the pointed structural arrows of $\mathbf{C}_U$. Numerical norm evaluations remain the specified observations and are not asserted additive.

For independent coefficients, the coordinatewise lift of a linear map preserves every coefficient mask, including coordinates with zero image. Face filling is a different specified arrow: for $A \subseteq$

B , zero insertion sends the zero e_A of the labelled additive face group to e_B . If A is empty and B is nonempty, it sends the all-absent coefficient tuple to a supported-zero tuple. It is therefore not declared to preserve the global absent point. Both maps occur in the represented underlying-set category with their respective structural types. The Cartesian product semiring has intersection of coefficient supports under multiplication; the negacyclic semiring has their cyclic sumset. These are two marked structures with their actual formulas, not an identification of their products. The negacyclic product is an arrow of the finite algebra observation. The cochain tensor algebra uses its original concatenation into the next tensor degree; no degree-zero cochain multiplication $C_U \otimes C_U \rightarrow C_U$ is inferred from that finite product.

For the scalar carrier $G(R)$, the amplitude/support map is injective: support zero identifies τ , while support one and amplitude identify $r^\bullet$. The scalar maps continue to be $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ and $\mathbb{Z} \rightarrow \mathbb{C}$. The lower arithmetic ring is the entire infinite ring $\mathbb{Z}$. Applying (T0.6) to the four point chart $+$, $- < \eta < \sigma$ leaves its original central amplitude $\mathcal{B}$, its two theta legs, and its zero σ amplitude. It does not identify relative cohomology with the zero amplitude at σ .

107 Exactly what survives and how every offcritical zero enters

The complete calculation gives an isomorphism with an explicit inverse on original functions:

$$H^1(C_U) = T_U Q \xrightarrow{[F]} [\mathcal{M}F] \sim \bigoplus_{\substack{\rho \in U \\ g(\rho)=0}} \mathcal{O}_{\mathbb{C},\rho}/(g). \quad (\text{T0.7})$$

For finite support Z , put $h = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$ with each actual full order. A prescribed full jet α has the unique remainder polynomial P satisfying $[P]_\rho = [(g/h)^{-1} \alpha_\rho]$; its preimage is

$$[F_\alpha], \quad F_\alpha = S_h \Theta(P(D)\phi_*), \quad \mathcal{M}F_\alpha = (g/h)P. \quad (\text{T0.8})$$

Here S_h is the iterated original Euler inverse and $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. CAU proves that every inverse acts on its required moment-zero domain, that the result belongs to the original $\mathcal{B}$, and that changes of representatives are actual theta boundaries. The complete unit g/h is retained in (T0.8).

The local element 1 at any actual offcritical zero is nonzero modulo (g) . Formula (T0.8) therefore gives a nonzero arithmetic class, not merely a parameter bearing that zero's name. Conversely, the original finite-jet kernel calculation proves that a nonzero class in (T0.5) has a root of g in its annihilator support. Thus this construction is exhaustive in both directions. The exclusion of zeros outside the open strip is the original arithmetic input for $g = 2\xi$; within that input, vanishing of $H^1(C_U)$ excludes every counterexample to RH.

The entire object is retained when this marked cohomology vanishes: its theta relations, scalar unit, diagonal tau source and meromorphic comparison kernel remain actual parts of the represented diagram. The precise arithmetic detector is (T0.7), not an assertion that all those background spaces vanish. Its tensor powers are

$$H^j(C_U^{\otimes k}) = \begin{cases} (T_U Q)^{\otimes k}, & j = k, \\ 0, & j \neq k. \end{cases} \quad (\text{T0.9})$$

AG proves this with the full explicit tensor contraction. For a nonzero class u , a functional taking u to 1 proves $u^{\otimes k} \neq 0$. A contradiction to any universally forced component of $\mathbf{C}_U$ therefore excludes its originating offcritical class. The following amplification calculation uses these exact nonvanishing maps.

108 The norm observations are values on this same object

All Hilbert, phase and finite quotient forms in the calculation come from evaluating the original functions represented above. For a full packet and polynomial P they begin with the identity

$$\|P(D_1, \dots, D_k)F_h^{\otimes k}\|_{L^2(d^k x)}^2 = \int_{\mathbb{R}^k} |P(1/2+it_1, \dots, 1/2+it_k)|^2 \prod_i \frac{|(g/h)(1/2+it_i)|^2}{2\pi} d\mathbf{t}. \quad (\text{T0.10})$$

Successive monic division gives their relation spaces and the theta primitive (AG21). Orthogonal minimization within precisely these finite relation spaces defines their matrices G , their canonical lifts and their volume determinants. These quantities are thereby coupled through the same F_h ; none is an independently adjustable entry in the construction.

Conormal logarithmic derivatives and other observations made before the theta quotient retain their target in the full ambient tensor source of (T0.3). Closure of $\mathcal{B}_U$ under those extra operations is not presumed. Hilbert completions likewise remain observations with their proved arrows. In particular the FKG calculation gives actual nonzero observations of canonical lifts whose Hilbert norms tend to zero; it proves their exact algebraic quotient and their Hausdorff reduction separately. Replacing (T0.7) by that Hausdorff reduction is not an arrow that preserves the arithmetic detector.

Every original coefficient, conormal, phase, control and volume restriction is thus a proved identity or inequality on these specified vertices and arrows. The complete proof inputs are included later in this manuscript. The next calculation tests their joint quantitative content against the surviving tensor class, including all error terms.

109 One anchored offcritical assembly of the original tau complex

We construct a single diagram from the original source complexes, with a marked arithmetic image and with its other fibres still attached. Its index set does not require naming an actual zero. In particular, the construction makes sense with no decision about the existence of an offcritical zero. The analytic input consists of the exact original calculations (BK.1)–(BK.24), the finite derived comparison (DP.1)–(DP.8), and the original two-leg maps (DC.17)–(DC.23).

109.1 The source-indexed complex and its canonical directed limit

Keep $A = \mathbb{C}[t]$, $D = -x\partial_x$, the original even Schwartz space V with $\phi(0) = 0$ and $\int \phi = 0$, and the original rapid-at-both-ends space $\mathcal{B}$ of (BK.1). All tensor products in this section are algebraic. The operators and entire functions are

$$\begin{aligned} \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), & \mathcal{M}F(s) &= \int_0^\infty F(x)x^s \frac{dx}{x}, \\ Q &= \mathcal{B}/\Theta V, & g(s) &= 2\xi(s), & \mathcal{M}\Theta\phi &= gH_\phi. \end{aligned} \quad (\text{CAU.1})$$

Here $g = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $H_\phi = 2\pi^{s/2}M_+\phi/(s(s-1)\Gamma(s/2))$ with its original entire continuation. The Fourier transform uses $e^{-2\pi i x \nu}$. Set

$$U = \{s \in \mathbb{C} : 0 < \Re s < 1, \quad \Re s \neq \tfrac{1}{2}\}, \quad \mathcal{P}_U = \{p \in \mathbb{C}[t] : p \text{ monic and every root lies in } U\}. \quad (\text{CAU.2})$$

The constant polynomial 1 belongs to this set. Order it by divisibility. It is directed because pP is a common upper bound of p and P . No condition $p \mid g$ is imposed on this index set.

For every $p \in \mathcal{P}_U$ define the following original-source complex in degrees zero, one, and two:

$$\mathcal{K}_p = [V \xrightarrow{v \mapsto (\Theta v, p(D)v)} \mathcal{B} \oplus V \xrightarrow{(F, \phi) \mapsto p(D)F - \Theta\phi} \mathcal{B}]. \quad (\text{CAU.3})$$

The differential squares to zero because $D\Theta = \Theta D$. For $P = rp$, the transition $\iota_{p,P}$ is the identity on degree zero and has the formulas

$$(F, \phi) \mapsto (F, r(D)\phi), \quad F \mapsto r(D)F \quad (\text{CAU.4})$$

in degrees one and two. Applying both differentials verifies the cochain identity, and $\iota_{P,R}\iota_{p,P} = \iota_{p,R}$ because the polynomial multipliers multiply. Thus (CAU.3)–(CAU.4) are one specified functor from the directed category $\mathcal{P}_U$ to complexes of A -modules, not a set of requested properties.

The original Θ is injective and every nonzero $p(D)$ is injective on $\mathcal{B}$ and V . The exact cohomology maps are

$$H^0 \mathcal{K}_p = 0, \quad H^1 \mathcal{K}_p \xrightarrow{\sim} Q[p], \quad [(F, \phi)] \mapsto [F], \quad H^2 \mathcal{K}_p \xrightarrow{\sim} Q/pQ. \quad (\text{CAU.5})$$

Indeed a cycle has $p(D)F = \Theta\phi$, exactly the torsion condition. The source ϕ is unique. If $F = \Theta v$, injectivity of Θ gives $\phi = p(D)v$, so the cycle is the displayed boundary. Conversely every torsion class has such a representative and gives a cycle. The degree-two quotient is $\mathcal{B}/(p(D)\mathcal{B} + \Theta V)$, which is exactly Q/pQ . Under (CAU.5), (CAU.4) induces the literal inclusion $Q[p] \subset Q[P]$ in degree one, and $[q] \mapsto [r(D)q]$ in degree two.

Define the actual directed colimit complex

$$\mathcal{K}_U = \operatorname{colim}_{p \in \mathcal{P}_U} \mathcal{K}_p, \quad T_U Q = \bigcup_{p \in \mathcal{P}_U} Q[p]. \quad (\text{CAU.6})$$

Its universal property is explicit: a cochain map $\mathcal{K}_U \rightarrow L$ is uniquely the same as cochain maps $f_p : \mathcal{K}_p \rightarrow L$ satisfying $f_p \iota_{p,P} = f_P$. To construct it, represent a colimit vector by $x \in \mathcal{K}_p$ and send it to $f_p(x)$. The common-upper-bound relation makes this independent of the representative, and the cochain assertion follows at that stage. Conversely restriction gives the compatible family. Filtered colimits of vector spaces are exact: an element and any relation involving it occur at finitely many stages, and a common upper stage witnesses the required kernel or image identity. This also proves that cohomology commutes with this colimit.

In fact

$$H^1 \mathcal{K}_U = T_U Q, \quad H^i \mathcal{K}_U = 0 \quad (i \neq 1). \quad (\text{CAU.7})$$

Only degree two remains to prove. The original finite-jet calculation (BK.8)–(BK.11) identifies Q/pQ with $A/(p, j_p g)$ by the Mellin jet. At a root ρ of p of order n , let $m = \operatorname{ord}_\rho g$, allowing $m = 0$. The local factor of this cokernel is $\mathbb{C}[z]/(z^{\min(n,m)})$, with $z = s - \rho$; this records the actual unit of g in the Mellin map, although the ideal alone is generated by this power. For $P = p^L$, the transition on degree two is multiplication by p^{L-1} . At ρ this multiplier has zero order $(L-1)n$, while the target cokernel is killed by $z^{\min(Ln,m)}$. Choose L so that $(L-1)n \geq m$ at every one of the finitely many roots with $m > 0$. Every local component then maps to zero. Components with $m = 0$ were zero already. Thus each degree-two class is killed at a later stage, proving (CAU.7). This argument never assumes that $p(D)$ is onto its original test space.

109.2 The exact Mellin identification, including every actual zero

Let $\mathcal{N}_U = (\mathcal{O}_{\mathbb{C}}/(g))|_U$. The notation $\Gamma_c(U, \mathcal{N}_U)$ means sheaf sections with compact support in U . The zeros of the nonzero entire function g form a discrete closed set. Each compact subset of U meets that set in finitely many points. A local section at one of these points extends uniquely by zero outside a sufficiently small disc containing no other zero: on the punctured disc g is a unit and the quotient sheaf is zero. These extensions glue, since their restrictions agree on overlaps. Consequently there is the canonical identification

$$\Gamma_c(U, \mathcal{N}_U) = \bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_{\mathbb{C},\rho}/(g). \quad (\text{CAU.8})$$

The direct sum specifies finite support and keeps the full stalk quotient.

The original Mellin map gives a canonical isomorphism

$$\mathfrak{m} : T_U Q \xrightarrow{\sim} \Gamma_c(U, \mathcal{N}_U), \quad [F] \longmapsto ([\mathcal{M}F]_\rho)_\rho. \quad (\text{CAU.9})$$

We prove all assertions, rather than presupposing a spectral identification. If $p(D)F = \Theta\phi$, then

$$p(s)\mathcal{M}F(s) = g(s)H_\phi(s). \quad (\text{CAU.10})$$

Away from the roots of p , this says that $\mathcal{M}F$ is locally a multiple of g . Hence the proposed section has finite support in U . Changing F by an original theta boundary changes its Mellin transform by gH , so the map is well-defined and A -linear.

For injectivity, suppose the section in (CAU.9) is zero. At a root ρ of p of order n , write $p = a(z)z^n$ and $g = u(z)z^m$, with $a(0)u(0) \neq 0$, allowing $m = 0$. The vanishing section says that $\mathcal{M}F = gB$ for an analytic germ B if $m > 0$; it says the same when $m = 0$ because g is then a unit. Equation (CAU.10) and cancellation in the local analytic domain give $H_\phi = pB$. Thus every jet of H_ϕ of order less than the corresponding multiplicity in p is zero. The proved original range identity (BK.8) gives $\phi = p(D)\psi$ for $\psi \in V$. Then $p(D)(F - \Theta\psi) = 0$, and injectivity on $\mathcal{B}$ gives $F = \Theta\psi$. This proves injectivity.

For surjectivity, take a finitely supported section α of (CAU.8), with support Z , and put

$$p_Z(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad m_\rho = \text{ord}_\rho g, \quad k_Z(s) = g(s)/p_Z(s). \quad (\text{CAU.11})$$

The quotient k_Z is entire, and its germ at each $\rho \in Z$ is a unit. The function g , every multiplicity, and this entire quotient are retained. Choose the unique polynomial P of degree less than $\deg p_Z$ whose jets satisfy

$$[P]_\rho = [k_Z^{-1}\alpha]_\rho \quad \text{in } \mathcal{O}_{\mathbb{C},\rho}/(s - \rho)^{m_\rho}. \quad (\text{CAU.12})$$

Existence and uniqueness follow from the polynomial Chinese remainder map: relatively prime factors have Bezout polynomials, whose products construct the orthogonal idempotents; multiplication by these idempotents and summation realizes any prescribed finite set of Taylor polynomials. The kernel is (p_Z) , and degree reduction gives the unique representative.

Use the original vector $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, for which $H_{P(D)}\phi_* = P$. The Mellin transform of $\Theta(P(D)\phi_*)$ is gP and vanishes to every order specified by p_Z . The exact original range identity for $\mathcal{B}$ in (BK.8) therefore supplies its unique inverse under $p_Z(D)$:

$$F_\alpha = S_{p_Z}\Theta(P(D)\phi_*) \in \mathcal{B}, \quad p_Z(D)F_\alpha = \Theta(P(D)\phi_*), \quad \mathcal{M}F_\alpha = k_Z P. \quad (\text{CAU.13})$$

Here S_{p_Z} is the composite of the original full-plane Euler inverses at the roots with their full multiplicities. Uniqueness makes the composite independent of the order of those factors. Formula (CAU.12) gives $[\mathcal{M}F_\alpha]_\rho = \alpha_\rho$. Away from Z the quotient by p_Z is a unit at every zero of g , so the other stalk classes are zero. This proves surjectivity. The empty support gives the zero class. Any alternative polynomial lift differing by $p_Z R$ changes F_α by $\Theta(R(D)\phi_*)$, so its class in Q is unchanged. Thus (CAU.13) constructs the inverse to (CAU.9) on the original functions and relations.

In particular, for every actual $\rho \in U$ the element 1 of $\mathcal{O}_{\mathbb{C},\rho}/(g)$ has a nonzero preimage in $T_U Q$. Indeed g vanishes there, so $1 \notin (g)$, and (CAU.9) is injective. Every nilpotent coefficient at that zero is carried by the same inverse. This exhaustiveness assertion is a proved map at every actual zero; it is not an assertion that such a ρ exists.

109.3 The anchored sheaf diagram and its retained diagonal and kernel

Sheafify $\mathcal{O}_U \otimes_A$ applied termwise to (CAU.6), and denote the resulting complex by $\mathcal{A}_U$. Its stalks are the corresponding tensor products because tensor products commute with filtered colimits. Every

analytic stalk is torsion-free over the PID A , hence flat by the directed-union-of-finite-free-modules proof in (BK.12). Therefore

$$H^1 \mathcal{A}_U = \mathcal{O}_U \otimes_A T_U Q, \quad H^i \mathcal{A}_U = 0 \quad (i \neq 1). \quad (\text{CAU.14})$$

The natural sheaf map induced by Mellin identifies this degree-one sheaf with $\mathcal{N}_U$. To check it exactly, (CAU.8)–(CAU.9) identify $T_U Q$ as an A -module with the direct sum of $A/(t - \rho)^{m_\rho}$. At a point s_0 , tensoring a summand gives $\mathcal{O}_{s_0}/(s - \rho)^{m_\rho}$ through the original map $a \otimes [P(t)] \mapsto [aP(s)]$. It is zero if $s_0 \neq \rho$, and is the full quotient if $s_0 = \rho$. The Mellin identification retains the unit k_Z used in (CAU.12)–(CAU.13), so this is the actual germ of the map to $\mathcal{O}_{s_0}/(g)$. It is an isomorphism at every stalk and hence an isomorphism of sheaves.

Use the original two-leg complex, with $t = D$ on V_+ and $t = 1 - D$ on V_- , and its analytic comparison:

$$C_\tau = [\mathcal{V}_+ \oplus \mathcal{V}_- \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \hat{\psi})} \mathcal{B}], \quad C_g = [\mathcal{O}_U \xrightarrow{g} \mathcal{O}_U]. \quad (\text{CAU.15})$$

Their terms are in degrees zero and one and are restricted to U . The comparison $c : C_\tau \rightarrow C_g$ sends (ϕ, ψ) to $H_{\phi - \hat{\psi}}$ and sends F to $\mathcal{M}F$; on balanced tensors these formulas are multiplied by their analytic coefficient. The identity in (CAU.1) proves it is a cochain map.

There is a cochain map from every $\mathcal{K}_p$ to the original unbalanced joint complex, compatible with (CAU.4), given by

$$v \mapsto \left(\frac{1}{2}v, -\frac{1}{2}\hat{v}\right), \quad (F, \phi) \mapsto F, \quad F \text{ in degree two} \mapsto 0. \quad (\text{CAU.16})$$

The differential of the first image is exactly Θv because Fourier squares to the identity on even functions. The half factors in (CAU.16) are part of this displayed map; no amplitude is discarded. The map is compatible with every transition because their first degree-one coordinate is unchanged. Taking the colimit and balancing therefore gives $a : \mathcal{A}_U \rightarrow C_\tau$.

Write $C_{\ker} = \ker c$. As proved by the explicit lifts in (DC.20)–(DC.22), there is a short exact sequence of the original complexes $0 \rightarrow C_{\ker} \rightarrow C_\tau \xrightarrow{c} C_g \rightarrow 0$. The single anchored diagram we use is

$$\mathfrak{T}_U^{\text{coh}} := \left(\mathcal{A}_U \xrightarrow{a} C_\tau \xrightarrow{c} C_g, \quad C_{\ker} \hookrightarrow C_\tau, \quad \{\mathcal{O}_U \otimes_A \mathcal{K}_p \rightarrow \mathcal{A}_U\}_{p \in \mathcal{P}_U} \right). \quad (\text{CAU.17})$$

The tensor notation in the final family means its sheafification. This is a representation of the complex-linear path category of the following fully specified quiver with relations: vertices A, C, G, R and one vertex P_p for every $p \in \mathcal{P}_U$; arrows $P_p \rightarrow P_P$ for $p \mid P$, arrows $P_p \rightarrow A$, and $A \rightarrow C \rightarrow G, R \rightarrow C$. Relations identify composites of divisibility arrows, require $P_p \rightarrow A$ to be compatible with them, and require $R \rightarrow C \rightarrow G = 0$. The representation assigns respectively $\mathcal{A}_U, C_\tau, C_g, C_{\ker}$ and the displayed balanced source complexes, with precisely the maps above. The quiver is a set-sized category, since $\mathcal{P}_U$ is a set. Its colimit vertex has the proved universal property following (CAU.6). This specifies one object in a diagram category and its actual entries and arrows, rather than an object defined by satisfying a list of conditions.

The composite ca is a quasi-isomorphism. Both its source and target have cohomology only in degree one, and its degree-one map is the isomorphism (CAU.14) to $\mathcal{N}_U$. With the original notation $\mathcal{M} = \mathcal{O}_U \otimes_A Q, \beta : \mathcal{M} \rightarrow \mathcal{N}_U$, and $\mathcal{K} = \ker \beta$, its degree-one factorization is

$$\mathcal{N}_U \xrightarrow{j} \mathcal{M} \xrightarrow{\beta} \mathcal{N}_U, \quad \beta j = 1. \quad (\text{CAU.18})$$

It is the canonical original inclusion of the full torsion summand. Indeed each local class in its image is annihilated by some polynomial, whereas every nonzero polynomial is invertible on $\mathcal{K}$ by BK. Thus it has zero $\mathcal{K}$ component, and β is an isomorphism on that torsion summand. This identifies the actual map induced by (CAU.16), not just an abstract summand of equal length.

For clarity the entire middle term is retained in (CAU.17). On the original joint degree-zero amplitude make the reversible change

$$u = \phi - \hat{\psi}, \quad v = \frac{1}{2}(\phi + \hat{\psi}), \quad \phi = v + \frac{1}{2}u, \quad \psi = \hat{v} - \frac{1}{2}\hat{u}. \quad (\text{CAU.19})$$

Its differential is Θu ; the v component is the full diagonal. Together with the original canonical splitting $\mathcal{M} = \mathcal{N}_U \oplus \mathcal{K}$, it proves

$$C_\tau \simeq \mathcal{V}_+[0] \oplus \mathcal{N}_U[-1] \oplus \mathcal{K}[-1], \quad C_{\ker} \simeq \mathcal{V}_+[0] \oplus \mathcal{K}[-1]. \quad (\text{CAU.20})$$

These are derived identifications of the explicitly retained complexes. The marked arithmetic image of (CAU.17) is $\text{im } H^1(a)$, which by (CAU.18) is $j\mathcal{N}_U$. Its support is therefore exactly the actual zero divisor of g in U , with its scheme multiplicities. The support of the entire middle term is not being substituted for this specified arithmetic support. In particular the nonzero meromorphic kernel and the diagonal are present even at points where this arithmetic image is zero.

109.4 Full finite observations and all tensor jets of the same diagram

For each $p \in \mathcal{P}_U$ let $E_p = A/(p)$ and let j_p denote full Taylor reduction. The original Mellin maps define the following cochain map, with every factor of g retained:

$$\mathcal{K}_p \longrightarrow [\mathcal{O}_U \xrightarrow{(g,p)} \mathcal{O}_U^2 \xrightarrow{(p,-g)} \mathcal{O}_U], \quad v \mapsto H_v, \quad (F, \phi) \mapsto (\mathcal{M}F, H_\phi), \quad F \mapsto \mathcal{M}F. \quad (\text{CAU.21})$$

On balanced tensors it is $\mathcal{O}_U$ -linear. Applying either differential gives the same expression by (CAU.1) and $\mathcal{M}p(D)F = p(s)\mathcal{M}F$. It induces the original maps to the kernel and cokernel of multiplication by $j_p g$ on E_p ; equivalently its analytic degree-one target is $\mathcal{N}_U[p]$. The local conversion between these two presentations, for $p = a(z)z^n$ and $g = u(z)z^m$, is

$$[v] \longmapsto [(g/p)v], \quad z^{\max(n-m,0)}\mathcal{O}/z^n\mathcal{O} \xrightarrow{\sim} z^{\max(m-n,0)}\mathcal{O}/z^m\mathcal{O}, \quad (\text{CAU.22})$$

with the exact multiplier $(u/a)z^{m-n}$ and inverse $(a/u)z^{n-m}$ on those specified powers. This is a well-defined isomorphism because changing a representative by p changes the product by g , and the source kernel condition makes the product holomorphic. It includes zero factors when $m = 0$ and includes every repeated-zero order. Thus every finite factor of the all-polynomial index is computed, including those which observe no actual zero. In fact (CAU.21), after balancing, is a quasi-isomorphism at every p : the degree-one map is (CAU.22), the degree-two map is the same Mellin identification $\mathcal{O}/(p, g)$, and both degree-zero groups vanish. Flatness and (CAU.5) justify these cohomology calculations for the original balanced source, so no chain-level kernel has been assumed zero.

For an actual finite monic divisor $h \mid g$ with roots in U , the finite packet is the prescribed quotient $A_h = A/(h) \simeq \Gamma(U, \mathcal{O}_U/(h))$. It is obtained from the marked image by the reduction $\mathcal{N}_U \rightarrow \mathcal{O}_U/(h)$ and from the original source by $[F] \mapsto j_h \mathcal{M}F$. Retain the entire quotient $k = g/h$; its germ at a root of h is a unit exactly when h contains the full multiplicity of g there. Both degrees of its derived observation are retained by

$$E_h^{\text{free}} = [\mathcal{O}_U \xrightarrow{h} \mathcal{O}_U] \quad \text{in degrees } -1, 0, \quad - \otimes_{\mathcal{O}_U} E_h^{\text{free}}. \quad (\text{CAU.23})$$

This is a tensor operation on every vertex and arrow of (CAU.17), so it is an observation of the single diagram. On its analytic vertex C_g the total complex in degrees $-1, 0, 1$ is

$$\mathcal{O}_U \xrightarrow{a \mapsto (ha, ga)} \mathcal{O}_U^2 \xrightarrow{(b, c) \mapsto gb - hc} \mathcal{O}_U. \quad (\text{CAU.24})$$

It maps to $[\mathcal{O}_U/(h) \xrightarrow{0} \mathcal{O}_U/(h)]$ in degrees zero and one by $(b, c) \mapsto [b]$ and $d \mapsto [d]$. This is a quasi-isomorphism because it tensors the free complex C_g with the quasi-isomorphism $E_h^{\text{free}} \rightarrow \mathcal{O}_U/(h)$; alternatively its degree-zero cycles are exactly (b, kb) , modulo (ha, ga) , and its degree-one quotient is $\mathcal{O}/(g, h) = \mathcal{O}/(h)$. Under the map to the torsion in $\mathcal{N}_U$ its degree-zero class is $[b] \mapsto [kb]$. The multiplier $k = g/h$ is not dropped.

On the full joint vertex the same observation also retains its diagonal. By (BK.8)–(BK.9), $\mathcal{V}_+/h\mathcal{V}_+ \simeq \mathcal{O}_U/(h)$ through H , and h is injective on $\mathcal{V}_+$. Using (CAU.20) and the contractibility of $[\mathcal{K} \xrightarrow{h} \mathcal{K}]$ therefore gives

$$H^0(C_\tau \otimes E_h^{\text{free}}) \simeq (\mathcal{O}_U/(h))^2, \quad H^1(C_\tau \otimes E_h^{\text{free}}) \simeq \mathcal{O}_U/(h). \quad (\text{CAU.25})$$

In the original plus/minus jet coordinates the degree-zero comparison is $(\alpha, \beta) \mapsto \alpha - \beta$; its kernel is the full diagonal (α, α) . Thus the finite observation does not confuse the joint source with the one-leg complex.

For $h \mid H$, the quotient transition of these observations is the cochain map $E_H^{\text{free}} \rightarrow E_h^{\text{free}}$ equal to H/h in degree -1 and the identity in degree zero. Its cochain identity is $h(H/h) = H$. All transitions of the total diagram are obtained by tensoring this exact map.

To specify all endpoints of these arrows, first adjoin to the seed diagram a vertex (v, h) with value $D_v \otimes_{\mathcal{O}_U} E_h^{\text{free}}$ for every existing vertex v of value D_v and every admitted h . Adjoin the maps $f \otimes 1$ for each original arrow f , the transition maps $1 \otimes (E_H^{\text{free}} \rightarrow E_h^{\text{free}})$, and their cohomology vertices and induced arrows, including the finite quotient vertices in (CAU.24)–(CAU.25). Their relations are exactly the commuting squares supplied by tensor functoriality and by cohomology. This internal tensor operation over $\mathcal{O}_U$ is an explicitly adjoined observation; it is not identified with the external tensor operation in the next paragraph.

Finally close this represented diagram under finite ordered external tensor products over $\mathbb{C}$, with the specified symmetry maps and Koszul signs. The ambient category is graded by the nonnegative integer r : degree- r values are sheaf complexes on U^r , with the coefficient sheaf equal to the tensor product over $\mathbb{C}$ of the pulled-back coefficient sheaves; degree zero is the point with unit $\mathbb{C}$. Permutations use their actual coordinate permutations on U^r . This is a single canonical enlargement: its vertices are finite words of existing vertices, including the empty word; arrows are generated by tensoring the existing arrows with identities, by permutations, and by the displayed finite-observation maps. Their relations are composition, identity, tensor interchange, and the Koszul permutation relations. The value of a word is the literal external tensor product of its original complexes; its differential is

$$d(x_1 \otimes \cdots \otimes x_r) = \sum_{i=1}^r (-1)^{\sum_{j<i} |x_j|} x_1 \otimes \cdots \otimes dx_i \otimes \cdots \otimes x_r. \quad (\text{CAU.26})$$

This construction has the explicit free symmetric monoidal universal property: a tensor-compatible extension of the original represented diagram must have these values and arrows on every word, and the displayed relations make that extension unique. Pairwise interchange of the two differential terms proves $d^2 = 0$ with these signs. For finite packets $h_1, \dots, h_r$, their tensor observation is exactly

$$A_{h_1} \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} A_{h_r} = \frac{\mathbb{C}[s_1, \dots, s_r]}{(h_1(s_1), \dots, h_r(s_r))}. \quad (\text{CAU.27})$$

Successive monic division in each variable proves the isomorphism: the monomials with $0 \leq j_i < \deg h_i$ give bases on both sides and the map takes the tensor basis to that basis. At specified actual centres it retains all jets $0 \leq j_i < \text{ord}_{\rho_i} h_i$; when each selected divisor contains the full multiplicity, these are exactly all jets $0 \leq j_i < m_{\rho_i}$. It retains every one-variable multiplier $g(s_i)/h_i(s_i)$ from (CAU.24). The total derived observation is the tensor of the two-term observations, so every degree and every tensor nilpotent survives in its prescribed summand. Tensoring over a field is exact: a basis extends a basis of a subspace and splits every vector-space short exact sequence. This proves the cohomological tensor assertion without an unproved flatness claim about a completed analytic tensor product. The tensor sheaves here are sheaves of modules over the literal tensor product of the pulled-back analytic coefficient sheaves; no formal or analytic completion of the original source is implicit.

109.5 All proper source and original leg-support fibres, and the exact actions

The diagram has its original label enlargement as part of the same construction. For every D -invariant subspace $W \subset V$ and every nonempty original leg mask $S \subset \{+, -\}$, include the original complex $C_{W,S}$ with central term $\mathcal{B}$, plus source W if $+$ $\in S$, minus source $\widehat{W}$ if $- \in S$, and differential $\Theta\phi - \Theta\widehat{\psi}$ on the present sources. Give the two sources the actions D and $1 - D$, respectively. Balancing and restriction to U are the same operations used above. For $W \subset W'$ and $S \subset S'$, inclusion on each present source and the identity on $\mathcal{B}$ is a cochain map; the differentials agree term by term. These maps compose literally and are included as arrows in the represented category of (CAU.17). The map to C_g is the original signed H map in degree zero and Mellin in degree one, so these arrows commute with that comparison.

For the joint mask, the change (CAU.19) has $u, v \in W$ and proves $H^0 C_{W,\{+,-\}} = W$ and $H^1 C_{W,\{+,-\}} = Q_W$, where $Q_W = \mathcal{B}/\Theta W$. Each one-leg mask has zero degree-zero cohomology and the same Q_W in degree one; on the minus leg the actual differential is $-\Theta\widehat{\psi}$. The proper-label transition retains the exact original sequence

$$0 \longrightarrow V/W \xrightarrow{[\phi] \mapsto [\Theta\phi]} Q_W \longrightarrow Q \longrightarrow 0. \quad (\text{CAU.28})$$

Injectivity of Θ and the quotient definitions prove exactness; flatness retains it after balancing. Thus every additional proper-label class and its specified vanishing under transition remain in this single diagram. No field-vector assertion for its larger kernel is substituted for the proved assertion about the full source V .

On a support fibre the total carrier is the original disjoint union with external absence. Every displayed linear map f lifts as

$$\tau \mapsto \tau, \quad (W, S, x) \mapsto (W', S', f(x)), \quad e_{W,S} = (W, S, 0). \quad (\text{CAU.29})$$

For a fixed source and target label its supported-zero fibre is exactly $\{(W, S, x) : f(x) = 0\}$, while only external τ maps to external τ . Composition of these lifts follows from composition of the linear maps. Thus the supported part has the actual Grothendieck category of elements: at a fixed cochain degree and open set its objects are the labelled amplitude elements, and its arrows are exactly the represented maps carrying one element to the other. Forgetting the amplitude gives the label projection on this supported part. The single external τ is adjoined to its carrier outside that projection, with the pointed actions in (CAU.29). It is not identified with a labelled object (W, S, τ) of a category of elements. No zero fibre has been replaced by that external point.

These are the original leg-mask fibres. The independent coefficient slots of the original PS construction have an additional index $A \subseteq I_n$, retained jointly with (W, S) . Their face complex is $C_{W,S}[A] = \bigoplus_{i \in A} C_{W,S}$, with every arrow and differential acting in each active coordinate. Consequently the transition to the full source has kernel $(V/W)^A$: injectivity, image, and surjectivity in (CAU.28) are checked in each coordinate. Zero insertion between coefficient faces commutes with every linear cochain arrow because a new coordinate has value zero on both composites. It fills a labelled supported zero and does not change the independent outer leg label S . The mixed cochain attachment below proves the complete packet-section, boundary, balancing and tensor comparisons on these faces. Their disjoint labelled carrier and synchronization are additional represented vertices; the single synchronized lift $G(E^n)$ is not used in place of the independent carrier $G(E)^n$.

The symmetries and actions are also actual arrows. To type their base maps precisely, the enlargement with these arrows uses the category whose objects are ringed spaces with their specified module complexes and whose morphisms consist of a specified base map and the indicated semilinear cochain map. Composition uses the composite base map and the composite semilinear map. Ordinary arrows of (CAU.17) have the identity base map; reflection and conjugation have their displayed base maps. This is the Grothendieck extension of that represented diagram by the following actual action maps. Dilation $R_a F(x) = F(x/a)$, $a > 0$, sends the label W to $R_a W$; on the plus source

and target it acts by R_a , and on the minus source it acts by $aR_{1/a}$. The latter lies in $\widehat{R_a W} = aR_{1/a}\widehat{W}$, by direct Fourier substitution. It commutes with the differentials and Mellin takes it to a^s . It acts on (CAU.3) by R_a in each original coordinate and hence commutes with its polynomial transitions and with (CAU.16).

For the reflection $r(s) = 1 - s$ put $JF(x) = x^{-1}F(1/x)$ and $p^r(s) = (-1)^{\deg p}p(1 - s)$, which is again monic in $\mathcal{P}_U$. The r -semilinear chain map from the (W, S) vertex to $(\widehat{W}, \text{swap } S)$ sends

$$(\phi, \psi) \mapsto (-\psi, -\phi), \quad F \mapsto JF. \quad (\text{CAU.30})$$

The theta identity $J\Theta = \Theta^\wedge$ proves the chain identity with the displayed signs, and its square is the identity. On (CAU.3), writing $\epsilon_p = (-1)^{\deg p}$, the corresponding map to $\mathcal{K}_{p^r}$ is

$$v \mapsto \widehat{v}, \quad (F, \phi) \mapsto (JF, \epsilon_p \widehat{\phi}), \quad F \mapsto \epsilon_p JF. \quad (\text{CAU.31})$$

Indeed $\widehat{p(D)v} = p(1 - D)\widehat{v} = \epsilon_p p^r(D)\widehat{v}$ and $p^r(D)JF = \epsilon_p Jp(D)F$. These equalities verify both differentials. For $P = rp$, one has $r^r = \epsilon_r r(1 - s)$ and $\epsilon_P = \epsilon_r \epsilon_p$, which proves compatibility of (CAU.31) with (CAU.4). Formula (CAU.16) intertwines (CAU.31) with (CAU.30) exactly. In (CAU.19) coordinates reflection acts by $(u, v) \mapsto (\widehat{u}, -\widehat{v})$, retaining the diagonal.

On the adjoined finite observations the required cochain map is $r^* E_h^{\text{free}} \rightarrow E_{h^r}^{\text{free}}$, with multiplier $\epsilon_h = (-1)^{\deg h}$ in degree -1 and multiplier 1 in degree zero; indeed $h^r = \epsilon_h h(1 - s)$. On (CAU.24) this sends its three coordinates to $\epsilon_h r(a)$, $(r(b), \epsilon_h r(c))$, and $r(d)$, respectively. Substitution into both differentials proves the cochain identity. For $h \mid H$, the identity $(H/h)^r = \epsilon_{H/h}(H/h)(1 - s)$ and $\epsilon_H = \epsilon_h \epsilon_{H/h}$ proves compatibility with the quotient transition of the observation complexes. Thus the reflection on every observed vertex is typed by its pullback and retains the signs forced by monic polynomial reflection.

Complex conjugation acts on the original functions and all coefficients, takes $p(s)$ to $\overline{p(\overline{s})}$, and commutes with the differentials because the theta sums, D , and g have the corresponding real structure. It commutes with Fourier on the even source spaces. Together with (CAU.30) it gives the original reflected-conjugate operation $s \mapsto 1 - \overline{s}$ on the packet. Direct substitution proves

$$JR_a J = aR_{1/a}, \quad r R_a = aR_{1/a} r \quad (\text{CAU.32})$$

on the labelled chain maps, with the scalar a acting on every term. These are the actual semidirect relations, not a commuting-action replacement. All analytic actions agree because $a^{1-s} = a(a^{-1})^s$ and $g(1 - s) = g(s)$.

At an actual zero ρ of order m_ρ the finite arithmetic action obtained from this single diagram is precisely

$$a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho[v] = [(s - \rho)v] \text{ in } \mathcal{O}_\rho/(g). \quad (\text{CAU.33})$$

It is multiplication by the Taylor expansion of a^s modulo the full original ideal, so every coefficient and nilpotent order is specified. The finite residue and its weight-one dual action are the original maps (OCQ.26)–(OCQ.30) applied to these same factors; their source inclusion is (CAU.18), and their tensor extensions are fixed by (CAU.26). Thus (CAU.17), enlarged by its explicitly defined label, action, and tensor arrows, is one anchored original-programme object. Its marked arithmetic support is exactly $\mathcal{N}_U$, and (CAU.13) provides a nonzero original-source lift of every nonzero class of each actual offcritical packet. The construction asserts neither that this support is nonempty nor that it is empty.

109.6 The exact attachment to the single arithmetic tensor algebra

The arithmetic tensor algebra (AG14) is attached to (CAU.17) by an explicit source-level map, with no claim that all operations on the ambient test space preserve its arithmetic subspace. Retain its

original definitions

$$\mathcal{B}_U = q^{-1}(T_U Q) \subset \mathcal{B}, \quad C_U = [V \xrightarrow{\Theta} \mathcal{B}_U], \quad \mathfrak{T}_U = \bigoplus_{k \geq 0} C_U^{\otimes_{\mathbb{C}} k}. \quad (\text{CAU.34})$$

These agree exactly with (AG2) and (AG14), since $[F] \in Q[p]$ means $p(D)F \in \Theta V$. Each term has the original topology as a subspace when a topological observation is taken; the displayed DGA itself uses the algebraic tensors specified in AG.

It is essential not to project all of $\mathcal{K}_p^1$ into $\mathcal{B}_U$: an arbitrary F in that degree need not have a torsion class. The exact connecting complex is the good truncation

$$Z_p = \{(F, \phi) : p(D)F = \Theta\phi\}, \quad \tau_{\leq 1}\mathcal{K}_p = [V \xrightarrow{(\Theta, p(D))} Z_p]. \quad (\text{CAU.35})$$

The map $Z_p \rightarrow q^{-1}(Q[p])$, $(F, \phi) \mapsto F$, is a linear isomorphism: existence of ϕ is the defining condition on its target and uniqueness follows from injectivity of Θ . It intertwines the displayed differential with Θ . The transitions (CAU.4) preserve it and become literal inclusions of these target subspaces. Directed exactness consequently proves the cochain isomorphism and the anchored zigzag

$$\mathcal{K}_U \longleftarrow \operatorname{colim}_p \tau_{\leq 1}\mathcal{K}_p \xrightarrow{\sim} C_U \xrightarrow{\sigma} C_\tau^{\text{orig}}, \quad \sigma^0(v) = (\tfrac{1}{2}v, -\tfrac{1}{2}\widehat{v}), \quad \sigma^1(F) = F. \quad (\text{CAU.36})$$

Here C_τ^{orig} is the original unbalanced joint complex; its balancing is (CAU.15). The left arrow is the inclusion of the good truncation into the directed complex and is a quasi-isomorphism by (CAU.7). The middle arrow is the actual cochain isomorphism just proved, since the union of $q^{-1}(Q[p])$ is $\mathcal{B}_U$. The right arrow is injective in both degrees and is a cochain map by (CAU.16). The two maps from the middle complex into the full tau complex agree literally. Balancing preserves this zigzag and its quasi-isomorphism because all the analytic stalks are flat over A . Thus it is the exact source-level attachment of AG's arithmetic complex to the arithmetic vertex of (CAU.17).

Tensoring the right arrow and taking the direct sum gives one injective unital differential graded algebra map

$$\mathfrak{T}_U \hookrightarrow T(C_\tau^{\text{orig}}) := \bigoplus_{k \geq 0} (C_\tau^{\text{orig}})^{\otimes_{\mathbb{C}} k}, \quad x_1 \otimes \cdots \otimes x_k \mapsto \sigma(x_1) \otimes \cdots \otimes \sigma(x_k). \quad (\text{CAU.37})$$

Its multiplicativity follows from concatenation, its compatibility with the differential follows term by term from (CAU.26), and its injectivity follows by extending bases over $\mathbb{C}$ in every cochain degree before tensoring. The degree-zero tensor unit maps to the same unit. It has the tensor-algebra universal property: every cochain map from C_U into the underlying complex of a unital DGA extends uniquely to a unital DGA map by multiplying the images of the tensor factors, since the Leibniz identity is exactly the signed formula (CAU.26). In particular (CAU.37) is the unique such extension of the specified original source inclusion σ .

Adjoining (CAU.36)–(CAU.37) as arrows to the same represented diagram identifies its arithmetic tensor vertex with the already constructed single DGA (AG14). All its finite packet lifts in (AG6)–(AG13) have the same Mellin image as (CAU.12)–(CAU.13); equality of their classes follows from the injectivity proved in (CAU.9). The ambient vertex retains the full $\mathcal{B}$ and both original tau legs. Accordingly the original logarithmic-derivative and conormal observations remain attached to their actual ambient domains. No closure of $\mathcal{B}_U$ under those operations is asserted, and no observation is silently retyped as an endomorphism of the arithmetic tensor algebra.

110 The single arithmetic carrier and its divisor–tensor gluing

Put $U = \{s \in \mathbb{C} : 0 < \Re s < 1, \Re s \neq 1/2\}$. The original objects are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \\ D &= -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \rightarrow Q, \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}. \end{aligned}$$

The original theta and Euler inverse calculations (A1)–(A9) prove that Θ is injective, commutes with D , and that every nonzero polynomial $p(D)$ is injective on $\mathcal{B}$. For roots of p in the open strip they also prove

$$p(D)\mathcal{B} = \ker(j_p \mathcal{M}), \quad p(D)V = \ker(j_p H), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad g = 2\xi, \quad H_{P(D)\phi_*} = P, \quad (\text{AG1})$$

where $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. The maps j_p in this formula are the complete Taylor jets, including every nilpotent coefficient through the indicated multiplicity. The following construction takes place inside these original test function spaces. Their topologies are the displayed Schwartz topologies and their inherited subspace topologies.

110.1 A definition that does not presuppose a list of zeros

Let $\mathcal{P}_U$ be the multiplicative set of monic polynomials all of whose roots lie in U , including 1. Define the actual spaces

$$\begin{aligned} T_U Q &= \{u \in Q : p(D)u = 0 \text{ for some } p \in \mathcal{P}_U\}, \\ \mathcal{B}_U &= \{F \in \mathcal{B} : p(D)F \in \Theta V \text{ for some } p \in \mathcal{P}_U\}, \\ C_U &= [V \xrightarrow{\Theta} \mathcal{B}_U] \subseteq [V \xrightarrow{\Theta} \mathcal{B}]. \end{aligned} \quad (\text{AG2})$$

The complex has its original degrees zero and one. Products of annihilating polynomials show that both displayed sets are vector spaces and D -modules. For example, if $p(D)u = 0$ and $r(D)v = 0$, then $(pr)(D)(au + bv) = 0$; D commutes with both polynomials. The inverse image under q of $T_U Q$ is exactly $\mathcal{B}_U$. Consequently

$$H^0(C_U) = 0, \quad H^1(C_U) = T_U Q, \quad 0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_U \xrightarrow{q} T_U Q \rightarrow 0 \quad (\text{AG3})$$

is the actual exact sequence, with the original arrows.

Here is its precise finite-packet presentation. For any $p \in \mathcal{P}_U$, the map

$$Q[p(D)] \longrightarrow \ker(M_{j_p g} : E_p \rightarrow E_p), \quad [F] \longmapsto j_p H_\phi, \quad p(D)F = \Theta\phi, \quad E_p = \mathbb{C}[s]/(p), \quad (\text{AG4})$$

is a $\mathbb{C}[s]$ -linear isomorphism. To check every arrow, ϕ is unique by injectivity of Θ . Replacing F by $F + \Theta\psi$ replaces ϕ by $\phi + p(D)\psi$, whose p -jet changes by zero by (AG1). The target condition follows from $j_p \mathcal{M}(p(D)F) = 0$. A zero source jet gives $\phi = p(D)\psi$ by (AG1); injectivity of $p(D)$ then gives $F = \Theta\psi$. Conversely a target vector has a polynomial representative P , and $j_p(gP) = 0$ allows the unique inverse $F = S_p^{\mathcal{B}} \Theta(P(D)\phi_*)$. Changing P by pR changes this inverse by $\Theta(R(D)\phi_*)$. These operations are inverse and commute with D , again by its injectivity and (AG1).

At a root ρ of p of order n , use the literal coordinate $z = s - \rho$. If g has order $m \geq 0$ there, its full local jet has the form $z^m u(z)$, $u(0) \neq 0$, with the convention that it is zero modulo z^n when $m \geq n$. The inverse of the unit is the finite geometric expansion of its nilpotent part. Thus

$$\ker(M_g : \mathbb{C}[z]/z^n \rightarrow \mathbb{C}[z]/z^n) = z^{\max(n-m, 0)} \mathbb{C}[z]/z^n. \quad (\text{AG5})$$

This kernel is zero when $m = 0$, and is killed by z^m when $m > 0$. Chinese remainders and (AG4) imply that every $u \in T_U Q$ is killed by the polynomial consisting of the actual roots of g in its finite annihilator, with their full orders. Adding the finite orbits under conjugation and $s \mapsto 1 - s$ still gives an annihilator whose roots lie in U .

Let $\mathcal{D}_U$ denote these finite invariant full-order packet polynomials, with 1 included. Divisibility makes it a directed set: the least common multiple is the polynomial on the union of the two supports, retaining each full order once. This directed set is now a presentation of (AG2), rather than its definition.

110.2 The exact divisor arrows and their arithmetic units

For $h \in \mathcal{D}_U$, $h \neq 1$, retain

$$\begin{aligned} E_h &= \mathbb{C}[s]/h, & d_h &= \deg h, & A_h &= M_s, \\ v_h &= g/h, & v_h &= j_h v_h, & \varepsilon_h &= v_h^{-1}, \\ F_h &= S_h^{\mathcal{B}} \Theta \phi_*, & \mathcal{T}_h P &= P(D)F_h. \end{aligned} \quad (\text{AG6})$$

Thus $h(D)F_h = \Theta \phi_*$ and $\mathcal{M}F_h = v_h$. The notation F_h denotes this actual inverse function, not an unconstructed functional calculus. Complete root orders imply $v_h(\rho) \neq 0$ on the packet, proving the asserted unit property. Write r_h for the unique remainder representative of degree $< d_h$, and define

$$s_h = \mathcal{T}_h r_h M_{\varepsilon_h}, \quad J_h = j_h \mathcal{M}, \quad \sigma_h = q s_h. \quad (\text{AG7})$$

Then $J_h s_h = I$, $J_h \Theta = 0$, and

$$h(D)s_h u = \Theta(r_h(\varepsilon_h u)(D)\phi_*), \quad Ds_h - s_h A_h = \Theta \phi_* \ell_h, \quad \ell_h(u) = [s^{d_h-1}]r_h(\varepsilon_h u). \quad (\text{AG8})$$

For the second identity, the difference $sr_h(\varepsilon_h u) - r_h(\varepsilon_h A_h u)$ is divisible by h , has degree at most d_h , and has quotient exactly $\ell_h(u)$. Applying $\mathcal{T}_h$ proves it.

The image of σ_h is exactly $Q[h(D)]$. In fact (AG4) identifies the latter with E_h , since $j_h g = 0$. Applying its map to $\sigma_h u$ gives $\varepsilon_h u$ by the first identity of (AG8), an invertible map of E_h . Injectivity of σ_h also follows from $J_h s_h = I$ and $J_h \Theta = 0$. Hence

$$\mathcal{B}_U = \Theta V + \bigcup_{h \in \mathcal{D}_U} s_h E_h, \quad T_U Q = \varinjlim_{h \in \mathcal{D}_U} (E_h, \iota_{hH}), \quad (\text{AG9})$$

with the following explicit transition arrows. The $h = 1$ terms mean $E_1 = 0$, $s_1 = 0$, and ΘV ; no inverse unit is required in the zero space.

For $h \mid H$, put $b = H/h$; b and h are coprime. Let

$$\pi_{Hh} : E_H \twoheadrightarrow E_h, \quad [P]_H \mapsto [P]_h, \quad \iota_{hH} : E_h \hookrightarrow E_H \quad (\text{AG10})$$

be respectively restriction of jets and insertion of zero jets at the added centres. Chinese remainder division proves $\pi \iota = I$, $\iota \pi = M_{e_{hH}}$, where e_{hH} is the idempotent equal to one on the old local factors and zero on the added ones. Both maps commute with multiplication by s . The map π is a unital algebra map; ι preserves products and is $\mathbb{C}[s]$ -linear, with $\iota(1) = e_{hH}$. Define the different, raw numerator map

$$\mu_b : E_h \rightarrow E_H, \quad [P]_h \mapsto [bP]_H, \quad \mu_b = \iota M_{[b]_h}, \quad \iota = \mu_b M_{[b]_h}^{-1}. \quad (\text{AG11})$$

These identities follow on every old factor by multiplication by the full unit $[b]_h$, and on every added factor by its full-order zero. They prove well-definedness and injectivity of μ_b .

The analytic and unit identities are exactly

$$\begin{aligned} v_h &= bv_H, & b(D)F_H &= F_h, & \mathcal{T}_H(bP) &= \mathcal{T}_hP, \\ \pi v_H &= [b]_h^{-1}v_h, & \pi\varepsilon_H &= [b]_h\varepsilon_h, & J_H\mathcal{T}_hP &= \iota J_h\mathcal{T}_hP, \\ s_H\iota &= s_h, & \sigma_H\iota &= \sigma_h, & \ell_H\iota &= \ell_h. \end{aligned} \quad (\text{AG12})$$

The function identity follows by applying the injective $h(D)$ to $b(D)F_H$ and F_h . The jet identity holds at the old centres by equality of the entire functions, and at added centres because v_h vanishes there to their full orders. For the section identity, $br_h(\varepsilon_h u)$ has degree $< d_H$ and its product with v_H has complete H -jet ιu . Since v_H is a unit, this is precisely $r_H(\varepsilon_H \iota u)$. The last identity follows either from monicity of b and its leading coefficient, or by subtracting (AG8) for these equal sections. These arguments also prove the composition laws $\iota_{HK}\iota_{hH} = \iota_{hK}$ and $\mu_{K/H}\mu_{H/h} = \mu_{K/h}$.

In particular the directed limit in (AG9) is explicitly

$$E_U := \bigoplus_{\substack{\rho \in U \\ g(\rho)=0}} \mathbb{C}[z_\rho]/z_\rho^{m_\rho}, \quad s \cdot u_\rho = (\rho + z_\rho)u_\rho, \quad \sigma_U : E_U \xrightarrow{\sim} T_U Q. \quad (\text{AG13})$$

All vectors have finite support. The direct sum is closed under coordinatewise products, but has no prescribed global unit when the set of roots is infinite. No nilpotent coordinate is removed. The map $s_U : E_U \rightarrow \mathcal{B}_U$ is defined by the coherent s_h ; the formula $\mathcal{B}_U = \Theta V \oplus s_U E_U$ is an equality of vector spaces proved by $J_h s_h = I$. It is not asserted to be a D -equivariant splitting.

110.3 One differential graded arithmetic object

Define the algebraic differential graded tensor algebra

$$\mathfrak{T}_U = \bigoplus_{k \geq 0} C_U^{\otimes k}, \quad C_U^{\otimes 0} = \mathbb{C}, \quad (x_1 \otimes \cdots \otimes x_k)(y_1 \otimes \cdots \otimes y_l) = x_1 \otimes \cdots \otimes x_k \otimes y_1 \otimes \cdots \otimes y_l. \quad (\text{AG14})$$

On homogeneous tensors retain the literal differential

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_{i=1}^k (-1)^{\sum_{j < i} |x_j|} x_1 \otimes \cdots \otimes dx_i \otimes \cdots \otimes x_k. \quad (\text{AG15})$$

Every $|x_i|$ is its original cochain degree, zero or one. The two orders of applying the differential at distinct positions have opposite signs, and $d^2 x_i = 0$, proving $d^2 = 0$. Separating this sum at the concatenation point proves $d(xy) = (dx)y + (-1)^{|x|}x(dy)$. Concatenation is associative on pure tensors and hence by linearity. Thus (AG14) is one actual differential graded algebra embedded in the corresponding tensor algebra of the original theta complex. The operator $D_{\text{tot}} = \sum_i D_i$ commutes with d and is a derivation for concatenation. It is zero on the length-zero term.

Every tensor uses finitely many entries, so (AG9) and a common packet containing their supports prove

$$C_U^{\otimes k} = \varinjlim_{h \in \mathcal{D}_U} [V \rightarrow \Theta V + s_h E_h]^{\otimes k}. \quad (\text{AG16})$$

These are literal inclusions into the same tensor test-function complex, rather than isomorphisms of unspecified carriers.

There is a direct proof of its complete cohomology. For $F = \Theta\phi + s_U u$, define $tF = \phi$ and $pF = s_U u$; on degree zero put $p = 0$. Uniqueness of the decomposition makes these linear maps well defined. Then $dt + td = I - p$, with t zero on degree zero, and p is a cochain projection onto $s_U E_U$ in degree one. On a length- k tensor set

$$t_k = \sum_{i=1}^k p^{\otimes(i-1)} \otimes t \otimes I^{\otimes(k-i)}, \quad (\text{AG17})$$

where the action of the degree- -1 map t includes the sign $(-1)^{\sum_{j < i} |x_j|}$ on each homogeneous input. Computing $dt_k + t_k d$ cancels every term involving different positions, by precisely these signs. The remaining terms are $\sum_i p^{\otimes(i-1)} \otimes (I - p) \otimes I^{\otimes(k-i)} = I - p^{\otimes k}$ by telescoping. Consequently, in each length k ,

$$H^j(C_U^{\otimes k}) = \begin{cases} (T_U Q)^{\otimes k}, & j = k, \\ 0, & j \neq k, \end{cases} \quad H^\bullet(\mathfrak{T}_U) = \bigoplus_{k \geq 0} (T_U Q)^{\otimes k} \quad (\text{AG18})$$

with concatenation and the degree k retained. A cocycle has only finitely many length components, so cohomology commutes with this direct sum directly. The projection in (AG18) is the original $q^{\otimes k}$ and is D_{tot} -equivariant. The contraction t_k is only the explicit complex-linear proof of its quasi-isomorphism. It introduces no equivariant section into $\mathcal{B}$: such a section would send any h -torsion vector to an element killed by the injective $h(D)$ and hence to zero.

110.4 The finite original source and metric diagrams

For $k \geq 1$ and $M \geq k(d_h - 1)$, put

$$\begin{aligned} \mathcal{P}_M^{(k)} &= \mathbb{C}[s_1, \dots, s_k]_{\leq M}, \quad I_{h,k} = (h(s_1), \dots, h(s_k)), \quad E_h^{(k)} = E_h^{\otimes k}, \quad U_{h,k} = v_h^{\otimes k}, \\ \mathcal{T}_h^{(k)} P &= P(D_1, \dots, D_k) F_h^{\otimes k}, \quad J_{h,k,M} P = U_{h,k}[P]_{I_{h,k}}. \end{aligned} \quad (\text{AG19})$$

Thus the following entire diagram has its given source, quotient, and original cohomology arrow:

$$\begin{array}{ccccc} \mathcal{P}_M^{(k)} & \xrightarrow{\mathcal{T}_h^{(k)}} & \mathcal{B}_U^{\otimes k} & \xrightarrow{q^{\otimes k}} & (T_U Q)^{\otimes k} \\ J_{h,k,M} \downarrow & & & & \uparrow \sigma_h^{\otimes k} \\ E_h^{(k)} & \xrightarrow{I} & E_h^{(k)} & \xrightarrow{I} & E_h^{(k)}. \end{array} \quad (\text{AG20})$$

To prove commutativity, divide P successively by the monic $h(s_i)$; its remainder is represented by a linear combination of separate-variable monomials of degree $< d_h$ in each variable. The difference from the numerator of $s_h^{\otimes k}(U_{h,k}[P])$ belongs to $I_{h,k}$. Every relation $\sum_i h(s_i)P_i$ has the explicit primitive

$$\sum_{i=1}^k (-1)^{i-1} P_i(\mathbf{D})(F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}). \quad (\text{AG21})$$

The $i-1$ preceding factors have degree one; (AG15) supplies the second sign $(-1)^{i-1}$. Its differential is therefore exactly $\mathcal{T}_h^{(k)}(\sum_i h(s_i)P_i)$. Conversely its complete jet vanishes precisely when $P \in I_{h,k}$, since $U_{h,k}$ is a unit. This proves all the kernel assertions used by the diagram. Monic division lowers total degree, so at degree M the relation coefficients have degree at most $M - d_h$.

For $h \mid H$ and $b = H/h$, put $B_b(\mathbf{s}) = \prod_i b(s_i)$ and $N = M + k \deg b$. The diagram of divisor arrows is exactly

$$\begin{array}{ccc} \mathcal{P}_M^{(k)} & \xrightarrow{\times B_b} & \mathcal{P}_N^{(k)} \\ J_{h,k,M} \downarrow & & \downarrow J_{H,k,N} \\ E_h^{(k)} & \xrightarrow{\iota_{hH}^{\otimes k}} & E_H^{(k)}, \end{array} \quad \mathcal{T}_H^{(k)}(B_b P) = \mathcal{T}_h^{(k)} P. \quad (\text{AG22})$$

This follows first on pure monomials from (AG12), and then on all polynomials by linearity. It sends the smaller relation space into the larger one and hence agrees with the actual cochain inclusion.

All norms in this diagram are values of the original functions. Mellin–Fourier Plancherel in $x = e^y$ gives exactly

$$\|\mathcal{T}_h^{(k)} P\|_{L^2(d^k x)}^2 = \int_{\mathbb{R}^k} |P(1/2 + it_1, \dots, 1/2 + it_k)|^2 \prod_{i=1}^k w_h(t_i) dt, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}. \quad (\text{AG23})$$

The one-variable normalization is seen directly from $F(x) \mapsto e^{y/2}F(e^y)$, which is unitary for dx and dy , followed by Fourier measure $dt/(2\pi)$. Applying it in every variable proves (AG23). The entire multiplier $\prod_i v_h(s_i)$ is nonzero, so $\mathcal{T}_h^{(k)}$ is injective: a zero Mellin product on an open set makes the polynomial zero there and hence everywhere. Thus the Gram matrix in (AG23) is positive definite. The actual mass is $\mu_h^k, \mu_h = \int w_h > 0$. The numerator map in (AG22) is an isometry for these specified source forms, because its actual functions are equal. It does not state that the bare coordinate insertion preserves Euclidean metrics or every minimized quotient metric.

For completeness the exact minimized comparison is as follows. Let $\mathcal{L}_{h,k,M} = \mathcal{T}_h^{(k)}(I_{h,k} \cap \mathcal{P}_M^{(k)})$, with its inherited Hilbert inner product, and let $P_{h,k,M}$ be its orthogonal projection. Then

$$\begin{aligned} R_{h,k,M} &= (I - P_{h,k,M})s_h^{\otimes k}, \quad G_{h,k,M} = R_{h,k,M}^* R_{h,k,M}, \\ W_{h,k,M} &= (A_h^{(k)})^* G_{h,k,M} + G_{h,k,M} A_h^{(k)} - k G_{h,k,M}. \end{aligned} \quad (\text{AG24})$$

Indeed every source lift of a fixed jet differs from $s_h^{\otimes k}$ by exactly $\mathcal{L}_{h,k,M}$; orthogonal projection is its unique minimum. The section has numerator of total degree at most $k(d_h - 1)$, so it belongs to the declared source. Surjectivity follows from this section. With $\iota_k = \iota_{hH}^{\otimes k}$ and the degrees in (AG22), set

$$\mathcal{F} = \mathcal{L}_{H,k,N} \cap \mathcal{L}_{h,k,M}^\perp, \quad Z = P_{\mathcal{F}} R_{h,k,M}.$$

The inclusion of relation spaces in (AG22) implies that the difference of their projections is precisely $P_{\mathcal{F}}$. Their common section is $s_H^{\otimes k} \iota_k = s_h^{\otimes k}$ by (AG12). Therefore

$$\begin{aligned} R_{H,k,N} \iota_k &= R_{h,k,M} - Z, \\ \iota_k^* G_{H,k,N} \iota_k &= G_{h,k,M} - Z^* Z, \\ \iota_k^* W_{H,k,N} \iota_k &= W_{h,k,M} - ((A_h^{(k)})^* Z^* Z + Z^* Z A_h^{(k)} - k Z^* Z). \end{aligned} \quad (\text{AG25})$$

Orthogonality proves the second line; $A_H^{(k)} \iota_k = \iota_k A_h^{(k)}$ and expansion prove the third. When $k = 1$ and $M = n + d_h$, both relation spaces are $\Theta \text{span}\{\phi_*, D\phi_*, \dots, D^n \phi_*\}$. Then $Z = 0$, so (AG25) is the exact one-leg metric and control compression equality. In general the displayed $Z^* Z$ is retained.

Tensor concatenation also retains its source norm exactly. For $P \in \mathcal{P}_M^{(k)}$ and $Q \in \mathcal{P}_L^{(l)}$, set $P \boxtimes Q = P(s_1, \dots, s_k)Q(s_{k+1}, \dots, s_{k+l})$. Then

$$\begin{aligned} \mathcal{T}_h^{(k+l)}(P \boxtimes Q) &= \mathcal{T}_h^{(k)}P \otimes \mathcal{T}_h^{(l)}Q, \\ J_{h,k+l,M+L}(P \boxtimes Q) &= J_{h,k,M}P \otimes J_{h,l,L}Q. \end{aligned} \quad (\text{AG26})$$

Fubini gives the product of the two squared norms; polarization gives the full tensor inner product. Thus the map on the Hilbert tensor of these finite polynomial sources is isometric onto its specified product subspace of $\mathcal{P}_{M+L}^{(k+l)}$. Different packets are first inserted into their least common multiple by (AG22); equality of the actual functions makes the result independent of this presentation. Multiplication of the factors $b(s_i)$ proves that (AG22) and (AG26) commute, including their degree shifts. The symmetric coefficient projection is the literal average $\frac{1}{k!} \sum_{\sigma \in S_k} \sigma$. At top cochain degree its realization is the signed cochain average $\frac{1}{k!} \sum_{\sigma} \text{sgn}(\sigma) T_{\sigma}$, since interchanging degree-one factors contributes the other factor $\text{sgn}(\sigma)$. Concatenation followed by this specified projection supplies the symmetric product; no isometry of that projection is presumed.

110.5 The cyclic sources and every conormal depth in the same diagram

Retain the original finite local algebras

$$B_{h,k,r} = \mathbb{C}[\mathbf{s}]/I_{h,k}^r, \quad A_{h,k,r} = B_{h,k,r}^{S_k}, \quad U_{h,k,r} = [\prod_i v_h(s_i)]_{I_{h,k}^r}, \quad r \geq 1. \quad (\text{AG27})$$

At an ordered root tuple ρ , with multiplicities m_i , their local presentation is

$$\mathbb{C}[z_1, \dots, z_k]/(z_1^{m_1}, \dots, z_k^{m_k})^r.$$

The basis consists exactly of $z^{\mathbf{a}}$ with $\sum_i [a_i/m_i] \leq r-1$: ideal membership is equivalent to choosing r generator factors whose exponents are bounded by $\mathbf{a}$. The highest surviving total degree is $D_\rho + (r-1)M_\rho$, where $D_\rho = \sum_i (m_i-1)$ and $M_\rho = \max_i m_i$. Put every one of the $r-1$ allowed blocks at an index attaining the maximum and retain each remainder exponent m_i-1 to attain it. Its multinomial coefficient in the corresponding power of $\sum_i z_i$ is nonzero. Thus the exact sum nilpotency order is $1 + D_\rho + (r-1)M_\rho$. The separate-variable Chinese remainder theorem proves the stated local presentations: the local ideals are pairwise comaximal, and their products, intersections, and r th powers agree.

Consequently, with $S = \sum_i s_i$,

$$\begin{aligned} \ell_{h,k,\lambda,r} &= \max_{\sum_i \rho_i = \lambda} \{1 + D_\rho + (r-1)M_\rho\}, \\ \chi_{h,k,r}(T) &= \prod_{\lambda} (T - \lambda)^{\ell_{h,k,\lambda,r}}, \quad C_{h,k,r} = \mathbb{C}[T]/\chi_{h,k,r}, \\ \alpha_{h,k,r} : C_{h,k,r} &\hookrightarrow A_{h,k,r}, \quad [P] \mapsto [P(S)], \quad \eta_{h,k,r} = M_{U_{h,k,r}} \alpha_{h,k,r}. \end{aligned} \quad (\text{AG28})$$

Intersecting the local annihilator ideals takes the maximum at each collided sum, proving that α is an injective unital algebra map. The full Taylor unit U is invertible on every local factor, so η is an injective $\mathbb{C}[T]$ -module map. It is not called a unital algebra map. At $r = 1$ its analytic source is the exact restriction

$$\mathcal{V}_{h,k}P = \mathcal{T}_h^{(k)}P(S), \quad J_h^{\otimes k} \mathcal{V}_{h,k}P = \eta_{h,k,1}[P], \quad \|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 w_h^{*k}(u) du. \quad (\text{AG29})$$

The last equality is (AG23) with $u = \sum_i t_i$, eliminating one coordinate with absolute Jacobian one, which is exactly the definition of the convolution. Higher r in (AG27) are full Taylor observations of this same analytic Mellin multiplier; they are not new cohomology quotients killing the original theta relations, since those relations need only vanish to depth one. The truncation maps record their further jets.

For fixed h , tensor regrouping has an exact all-depth form:

$$\begin{aligned} \Delta_{k,l,r} : C_{h,k+l,r} &\hookrightarrow \prod_{a=0}^{r-1} (C_{h,k,a+1} \otimes C_{h,l,r-a}), \\ [P(T)] &\longmapsto ([P(T_1 + T_2)])_{a=0}^{r-1}. \end{aligned} \quad (\text{AG30})$$

Each component is well defined: a product of r generators from the two blocks has either at least $a+1$ first-block generators or at least $r-a$ second-block generators. Hence $I_{h,k+l}^r \subseteq I_{h,k}^{a+1} + I_{h,l}^{r-a}$, giving an algebra quotient to the corresponding two-block algebra. The scalar map in (AG30) factors through its cyclic subalgebras. On a pair of local sum factors the nilpotency order is the sum of their orders minus one: expand $(z+w)^n$ in $\mathbb{C}[z,w]/(z^A, w^B)$ and retain $\binom{A+B-2}{A-1} z^{A-1} w^{B-1} \neq 0$. The maximal order at a total sum in all components of (AG30) is therefore the maximum, over tuples and a , of

$$1 + D_{\text{first}} + D_{\text{second}} + aM_{\text{first}} + (r-1-a)M_{\text{second}}.$$

The maximum over a is attained at an endpoint and equals $1 + D_{\text{total}} + (r-1) \max(M_{\text{first}}, M_{\text{second}})$. This is exactly (AG28), proving that the kernel before taking the quotient is $(\chi_{h,k+l,r})$ and thus proving injectivity. For $r = 1$ the product has one factor and gives the familiar injection $C_{h,k+l,1} \hookrightarrow C_{h,k,1} \otimes C_{h,l,1}$. In every component the full unit maps to $U_{h,k,a+1} \otimes U_{h,l,r-a}$. Accordingly

$$\Xi_a \eta_{h,k+l,r} = (\eta_{h,k,a+1} \otimes \eta_{h,l,r-a}) \Delta_a, \quad (\text{AG31})$$

where Ξ_a is the just-proved quotient of the full joint algebras. Grouping three blocks gives the components indexed by $a + b + c = r - 1$ and evaluation at $T_1 + T_2 + T_3$ in either order. This proves associativity of these gluing maps with all depths and all their nilpotent coefficients retained.

For divisor inclusion, the local old factors of $B_{H,k,r}$ are exactly those of $B_{h,k,r}$: the extra $b(s_i)$ are units there, so the generated ideals and all their powers coincide. Restriction π_r and zero insertion ι_r therefore give

$$\pi_r \iota_r = I, \quad \iota_r \pi_r = M_{e_r}, \quad \iota_r U_{h,k,r} = e_r B_b U_{H,k,r}, \quad \pi_r U_{H,k,r} = (\pi_r B_b)^{-1} U_{h,k,r}. \quad (\text{AG32})$$

They commute with the depth truncations. The e_r are the idempotents equal to one on tuples entirely in the old support and zero on every other tuple; permutations preserve this set, so these maps also preserve the invariant algebras. The exact polynomial map through depth r is

$$\mu_{b,r} : B_{h,k,r} \hookrightarrow B_{H,k,r}, \quad [P] \mapsto [B_b^r P], \quad \mu_{b,r} = \iota_r M_{[B_b^r]}, \quad (\text{AG33})$$

where the multiplier on the right is taken in $B_{h,k,r}$. To prove well-definedness on generators, for $\sum_i a_i = r$ one has $B_b^r \prod_i h(s_i)^{a_i} = \prod_i H(s_i)^{a_i} \prod_i b(s_i)^{r-a_i} \in I_{H,k}^r$. On an added tuple at least one $b(s_i)$ belongs to the corresponding local generator ideal, so B_b^r vanishes modulo its r th power. On old tuples it is a unit. These facts prove (AG33), injectivity, and image $e_r B_{H,k,r}$. The complete arithmetic identity is

$$U_{H,k,r} \mu_{b,r} [P] = \iota_r [B_b^{r-1} U_{h,k,r} P], \quad \mathcal{T}_H^{(k)}(B_b^r P) = \mathcal{T}_h^{(k)}(B_b^{r-1} P). \quad (\text{AG34})$$

It follows from (AG32) on old factors and from zero on added ones; the analytic identity follows directly from $v_h = b v_H$. The analytic equality in (AG34) is asserted for each specified polynomial P before passage to a finite quotient. Its two functions are not defined from $[P]$ alone; changing the polynomial representative retains the corresponding actual source relation. Under truncation from $r + 1$ to r , the raw map satisfies $\text{red } \mu_{b,r+1} = \mu_{b,r} M_{[B_b]} \text{red}$. Thus its extra multiplier is explicitly part of the depth diagram. In particular the one-power physical map must not be declared to descend at every depth: for $k = 1$ and a proper inclusion, $bh^r = Hh^{r-1}$ is not divisible by H^r for $r > 1$, since b and h are coprime. This tests the claim on the actual packet polynomials themselves.

At depth one the physical divisor image of a cyclic vector is the precise vector

$$\iota_1 \eta_{h,k,1} [P] = e_1 B_b U_{H,k,1} [P(S)] \quad \text{in } E_H^{\otimes k}. \quad (\text{AG35})$$

This is always the common-carrier arrow, including collisions of old and added sum eigenvalues. It belongs to the larger cyclic module $\eta_{H,k,1} C_{H,k,1}$ exactly when $e_1 B_b [P(S)] \in \alpha_{H,k,1} C_{H,k,1}$, since U_H is a unit. No such membership is imposed by the gluing. An exact common-source pullback, available without imposing it, is

$$\mathcal{Z}_{hH,k} = \{(x, y) \in C_{h,k,1} \oplus C_{H,k,1} : \iota_1 \eta_{h,k,1} x = \eta_{H,k,1} y\}. \quad (\text{AG36})$$

Both projections are injective by injectivity of the displayed maps; their images are exactly the cyclic vectors whose specified physical jets agree. This states the full agreement relation inside (AG20), rather than substituting an unsupported scalar transition. The CRT restriction and insertion maps compose by their explicit local-factor formulas. They commute with the block quotients in (AG31), because restriction or insertion of old tuples and truncation of their local monomial ideals commute coordinate by coordinate.

There is also a scalar restriction in the reverse direction at every depth. Every old tuple occurs in the larger packet with unchanged orders, so $\chi_{h,k,r} \mid \chi_{H,k,r}$. Let $\pi_C : C_{H,k,r} \twoheadrightarrow C_{h,k,r}$ be polynomial remainder. Evaluation and (AG32) give the exact squares

$$\pi_r \alpha_{H,k,r} = \alpha_{h,k,r} \pi_C, \quad M_{[B_b]_{I_{h,k}^r}} \pi_r \eta_{H,k,r} = \eta_{h,k,r} \pi_C. \quad (\text{AG36a})$$

Thus even this backwards arithmetic observation retains its full extra multiplier.

The forward cyclic image admits complete complementary coordinates, with no membership assumption in (AG35). Here is the original full-unit retraction, with its proof. On each sum factor of $A_{H,k,r}$, let $N = S - \lambda$, $\ell = \ell_{H,k,\lambda,r}$, and choose a tuple and a highest surviving monomial z^{a^*} attaining degree $\ell - 1$, as in the proof of (AG28). On this specific local factor let θ extract its coefficient. For $X = T - \lambda$, define

$$\Pi_\lambda^0(v) = \sum_{j=0}^{\ell-1} \theta(N^{\ell-1-j}v)X^j, \quad p_\lambda(X) = \Pi_\lambda^0(e_\lambda(S)U_{H,k,r}), \quad \Pi_\lambda = p_\lambda^{-1}\Pi_\lambda^0. \quad (\text{AG36b})$$

Here e_λ is the complete scalar CRT idempotent and the polynomials on the right are reduced modulo X^ℓ . Since $N^\ell = 0$, coefficient comparison gives $\Pi_\lambda^0(Nv) = X\Pi_\lambda^0(v)$. Moreover

$$p_\lambda(0) = \left(\prod_i v_H(\rho_i) \right) \frac{(\ell-1)!}{\prod_i a_i^{*!}} \neq 0.$$

In fact terms of positive total degree in the unit would exceed the maximal surviving degree of the chosen local factor, while the remaining multinomial coefficient is the one displayed. This proves invertibility by the finite geometric series and shows that $\Pi_\lambda\eta(P) = P$ on each local scalar factor. Combining these maps by scalar CRT gives the exact $\mathbb{C}[T]$ -linear retraction $\Pi_H : A_{H,k,r} \rightarrow C_{H,k,r}$. Put $K_H = \ker \Pi_H$ and $Q_H = I - \eta_H\Pi_H$. Direct multiplication gives the inverse module isomorphisms

$$A_{H,k,r} \longleftrightarrow C_{H,k,r} \oplus K_H, \quad v \longmapsto (\Pi_H v, Q_H v), \quad (c, z) \longmapsto \eta_H c + z. \quad (\text{AG36c})$$

They are the retained original coefficient changes; no orthogonality for the arithmetic form is inferred. The complete forward insertion is consequently the injection

$$(X_{hH}, Y_{hH}) : C_{h,k,r} \hookrightarrow C_{H,k,r} \oplus K_H, \quad (\text{AG36d})$$

$$X_{hH} = \Pi_H \iota_r \eta_h, \quad Y_{hH} = Q_H \iota_r \eta_h, \quad \iota_r \eta_h = \eta_H X_{hH} + Y_{hH}.$$

The last equality follows from (AG36c); if both coordinates vanish, injectivity of $\iota_r \eta_h$ forces the input to vanish. Its image is wholly cyclic exactly when $Y_{hH} = 0$. Under another divisor insertion the whole map is transformed by (AG36c), so both coordinates, including the cross maps from K_H , remain present.

For clarity the raw polynomial arrows also have their exact compatibility with the distributed-depth quotients. Write $B_b = B_1 B_2$ for its first and second blocks. On their stated finite quotients,

$$\Xi_a^H \mu_{b,r}^{(k+l)} = (\mu_{b,a+1}^{(k)} \otimes \mu_{b,r-a}^{(l)}) M_{B_1^{r-a-1} B_2^a} \Xi_a^h. \quad (\text{AG36e})$$

Both sides multiply a polynomial representative by $B_1^r B_2^r$ before the same block reduction; this proves equality and retains every additional power. The arithmetic identity (AG34) supplies its corresponding actual source-function equality.

110.6 Reflection and the retained split zero

For an invariant packet use the original involutions $P^\dagger(s) = \overline{P(1-\bar{s})}$ and $j_{\mathcal{B}}F(x) = x^{-1}\overline{F(1/x)}$. The latter is antiunitary for dx by substitution $y = 1/x$. Its Mellin transform is $(\mathcal{M}F)^\dagger$ by the same substitution, and $Dj_{\mathcal{B}} = j_{\mathcal{B}}(1-D)$ by differentiation. Since $h^\dagger = (-1)^{d_h}h$ and $g^\dagger = g$, $v_h^\dagger = (-1)^{d_h}v_h$. The full jet argument and degree bound used in (AG12) now give $j_{\mathcal{B}}s_h = s_h j_h$. The CRT maps commute with reflection, since old support and added support are separately invariant. The raw numerator map carries the exact sign

$$(\mu_{b,r}P)^\dagger = (-1)^{kr(d_H-d_h)}\mu_{b,r}(P^\dagger), \quad (\text{AG37})$$

as follows from $B_b^\dagger = (-1)^{k(d_H - d_h)} B_b$. Thus no sign has been removed from the original arithmetic unit. These involutions, with reversal $S \mapsto k - \bar{S}$, intertwine the divisor CRT, scalar-evaluation and depth-truncation arrows just constructed. In the complementary coordinates (AG36c) their exact action is $(c, z) \mapsto (\Pi_H j(\eta_H c + z), Q_H j(\eta_H c + z))$. No commutation of the chosen coefficient extractor with reflection is assumed. Before any parity consequence, the full-unit map has the exact law $j_A \eta_{h,k,r} = (-1)^{kd_h} \eta_{h,k,r} j_C$, because $U_{h,k,r}^\dagger = (-1)^{kd_h} U_{h,k,r}$. For the present off-critical invariant packets d_h is even: $\rho \mapsto 1 - \bar{\rho}$ has no fixed centre in U and pairs equal full multiplicities. The sign is therefore one here.

Finally apply the original split lift to every proved linear arrow. For a vector space X put $G(X) = \{\tau\} \sqcup \{x^\bullet : x \in X\}$, $e_X = 0_X^\bullet$, with τ the additive identity and $x^\bullet + y^\bullet = (x + y)^\bullet$. For $L : X \rightarrow Y$, its specified lift is

$$G(L)(\tau) = \tau, \quad G(L)(x^\bullet) = (Lx)^\bullet. \quad (\text{AG38})$$

Checking two supported inputs proves additivity and scalar compatibility; a case involving τ follows from the given identity and absorbing rules. Thus composition and every commutative diagram above lift exactly. Products lift by $x^\bullet y^\bullet = (xy)^\bullet$ and make τ absorbing; the same check applies to concatenation and to the multiplicative arrows of their proved types. A retained support label is carried by $(\lambda, x) \mapsto (\lambda, Lx)$ at each such arrow. In particular the differential of (AG21) remains the represented theta boundary $\mathcal{T}_h^{(k)}(\sum_i h(s_i) P_i)$; it is the subsequent map $G(q^{\otimes k})$ that sends this boundary to the supported zero e . Every inserted zero jet remains a supported zero. External absence is always τ . These lifts produce the split form of the single actual object (AG14), with its divisor, tensor, cohomology, full-jet and metric observation diagrams intact. The conjugate-linear reflection uses its corresponding semilinear lift, with $G(j)(c^\bullet x^\bullet) = \bar{c}^\bullet G(j)(x^\bullet)$. The square of the lifted differential is the lift of the zero linear map: it sends a supported input to the supported zero in the indicated degree, while fixing τ . This is the original split-zero identity, not a replacement of e by τ .

All positive matrices, convolution densities, minimum representatives and control forms appearing here are defined by (AG19)–(AG25) from the fixed $g = 2\xi$ and its original test functions. The construction does not assign independent values to these quantities. It supplies the exact common arithmetic carrier on which their previously proved analytic and control formulas are evaluated.

111 Full mixed carriers, their faces, prime branches, and depth maps

MFC.1. Conventions and the original split scalar. All rings in this attachment are commutative with identity and are nonzero. All semiring homomorphisms preserve the specified zero and unit unless a different category is explicitly named. A semiring ideal contains zero, is closed under addition, and absorbs multiplication by arbitrary elements; no subtractivity requirement is imposed. A prime ideal is a proper ideal whose complement is multiplicatively closed. The length of a strict prime chain is its number of strict inclusions. For a ring R , write

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad 1_{G(R)} = 1_R^\bullet. \quad (\text{MFC.1})$$

The union is disjoint. Its operations are

$$r^\bullet + s^\bullet = (r + s)^\bullet, \quad r^\bullet s^\bullet = (rs)^\bullet, \quad \tau_R + x = x, \quad \tau_R x = \tau_R. \quad (\text{MFC.2})$$

In particular, a cancellation of supported amplitudes produces e_R , not τ_R . These operations make a commutative semiring. On supported entries the laws are the ring laws. In a sum an occurrence of τ_R can be deleted, and a product with an occurrence of τ_R equals τ_R ; these observations prove both associativities in every other case. For distributivity the new cases are $\tau_R(x + y) = \tau_R = \tau_R x + \tau_R y$ and $x(\tau_R + y) = xy = x\tau_R + xy$. The amplitude and support maps

$$\begin{aligned} p_R : G(R) &\longrightarrow R, & p_R(\tau_R) &= 0_R, & p_R(r^\bullet) &= r, \\ \chi_R : G(R) &\longrightarrow \mathbb{B}, & \chi_R(\tau_R) &= 0, & \chi_R(r^\bullet) &= 1 \end{aligned} \quad (\text{MFC.3})$$

are unital semiring homomorphisms; here $\mathbb{B} = \{0, 1\}$ has join as addition and meet as multiplication. Indeed (MFC.2) proves the amplitude identities, and a sum is absent exactly when both terms are absent, whereas a product is absent exactly when at least one factor is absent.

MFC.2. Independent Cartesian faces and their entire fibres. Let I be a nonempty finite set and R_i a nonzero ring for each $i \in I$. Put

$$P_I = \prod_{i \in I} G(R_i), \quad R_I = \prod_{i \in I} R_i, \quad p = \prod_i p_{R_i}, \quad s(x) = \{i : x_i \neq \tau_{R_i}\}. \quad (\text{MFC.4})$$

Both operations of P_I are coordinatewise. The map $s : P_I \rightarrow \mathcal{P}(I)$ is a unital semiring homomorphism for union and intersection on $\mathcal{P}(I)$, by the last two assertions of MFC.1 in each coordinate. Its exact face over $A \subseteq I$ is

$$s^{-1}(A) = \prod_{i \in A} \{r_i^\bullet : r_i \in R_i\} \times \prod_{i \notin A} \{\tau_{R_i}\}. \quad (\text{MFC.5})$$

The ordering of factors is the original ordering indexed by I ; the displayed expression specifies each coordinate rather than changing that ordering. For $r = (r_i) \in R_I$, put $J(r) = \{i : r_i \neq 0_{R_i}\}$. For each $A \supseteq J(r)$ define

$$x(r, A)_i = \begin{cases} r_i^\bullet & i \in A, \\ \tau_{R_i} & i \notin A. \end{cases} \quad p^{-1}(r) = \{x(r, A) : J(r) \subseteq A \subseteq I\}. \quad (\text{MFC.6})$$

To prove equality, an absent coordinate of any preimage must have zero amplitude; hence its mask contains $J(r)$. At an active coordinate its amplitude forces the unique supported entry $r_i^\bullet$. Conversely each displayed tuple has amplitude r . Distinct masks give distinct tuples, including at zero amplitude. Consequently

$$|p^{-1}(r)| = 2^{|I| - |J(r)|}, \quad (p, s) : P_I \hookrightarrow R_I \times \mathcal{P}(I) \text{ has image } \{(r, A) : J(r) \subseteq A\}. \quad (\text{MFC.7})$$

In particular the amplitude-zero fibre has all $2^{|I|}$ faces. These statements retain supported zeros on every proper face.

MFC.3. Synchronized inclusion and the precise failed collapse. The original synchronized inclusion is

$$j : G(R_I) \hookrightarrow P_I, \quad j(\tau_{R_i}) = (\tau_{R_i})_i, \quad j(r^\bullet) = (r_i^\bullet)_i. \quad (\text{MFC.8})$$

It is injective: the absent image differs from every fully supported image, and amplitudes distinguish fully supported images. Applying (MFC.2) coordinatewise proves preservation of sum, product, zero, and unit. Its image is exactly $s^{-1}(\{\emptyset, I\})$. Thus the square

$$\begin{array}{ccc} G(R_I) & \xrightarrow{j} & P_I \\ \chi_{R_I} \downarrow & & \downarrow s \\ \mathbb{B} & \xrightarrow{\delta} & \mathcal{P}(I) \end{array} \quad \delta(0) = \emptyset, \quad \delta(1) = I \quad (\text{MFC.9})$$

is a pullback of semirings. Explicitly, an element of the fibre product is a pair (ϵ, x) with mask empty when $\epsilon = 0$ and full when $\epsilon = 1$. In the first case it comes uniquely from τ_{R_i} ; in the second case its amplitudes give its unique preimage $r^\bullet$. This bijection respects all operations by (MFC.8), proving the asserted universal property. Also $pj = p_{R_I}$.

For $I = \{1, 2\}$, there is no semiring homomorphism

$$q : P_I \longrightarrow G(R_1 \times R_2), \quad p_{R_1 \times R_2} q = p, \quad q(x) \text{ supported whenever } s(x) \neq \emptyset. \quad (\text{MFC.10})$$

Indeed let $x = (1_{R_1}^\bullet, \tau_{R_2})$ and $y = (\tau_{R_1}, 1_{R_2}^\bullet)$. Their product is the specified zero of P_I . The stipulated images are $(1, 0)^\bullet$ and $(0, 1)^\bullet$, whose product is $e_{R_1 \times R_2}$, distinct from $q(0_{P_I}) = \tau_{R_1 \times R_2}$. This contradicts multiplicativity and zero preservation. The statement concerns exactly the map in (MFC.10); it leaves the proven maps j, p, s intact.

MFC.4. The original quadratic convolution carrier. Fix a nonzero ring A , set $S = G(A)$, $u = (-1_A)^\bullet$, and put

$$D_A = S \oplus S, \quad (a, b) \star (c, d) = (ac + ubd, ad + bc), \quad \mathbf{0} = (\tau_A, \tau_A), \quad X_1 = (1_A^\bullet, \tau_A). \quad (\text{MFC.11})$$

Addition remains coordinatewise. In this formula a, b, c, d lie in S , so every sum and product on the right is the original split operation. Set

$$B = A[t]/(t^2 + 1), \quad X_a = (a^\bullet, \tau_A), \quad Y_b = (\tau_A, b^\bullet), \quad Z_{a+tb} = (a^\bullet, b^\bullet). \quad (\text{MFC.12})$$

Division by the monic polynomial $t^2 + 1$ gives every element of B a unique expression $a + tb$. For completeness, repeated subtraction of the leading monomial times $t^2 + 1$ reduces a polynomial to degree at most one. A nonzero multiple of this monic polynomial has degree at least two, since its leading coefficient equals the nonzero leading coefficient of the multiplier. Therefore two reduced expressions cannot differ by a nonzero multiple. In particular $A \hookrightarrow B$, $B \neq 0$, and t is a unit with inverse $-t$. No assumption on the characteristic or on the invertibility of 2 is used.

The multiplication in (MFC.11) is associative. For $(a, b), (c, d), (h, k) \in S^2$, the first coordinate in either association is

$$ach + ubdh + uadk + ubck, \quad \text{and the second is} \quad ack + ubdk + adh + bch. \quad (\text{MFC.13})$$

This follows by distributing each of the two products; commutativity of S puts the resulting four terms in the displayed order. Commutativity follows directly from (MFC.11). Distributing its two coordinates proves distributivity over coordinatewise addition. Substitution gives the stated zero and unit, so D_A is a commutative semiring. Substitution also gives every nonempty-face product:

$$\begin{aligned} X_a X_c &= X_{ac}, & X_a Y_b &= Y_{ab}, & Y_b Y_d &= X_{-bd}, \\ X_a Z_z &= Z_{az}, & Y_b Z_z &= Z_{tbz}, & Z_z Z_w &= Z_{zw}. \end{aligned} \quad (\text{MFC.14})$$

For example, for $z = c + td$, $Y_b Z_z$ has coordinates $(-bd)^\bullet, (bc)^\bullet$, which are exactly those of tbz . Products of two full faces similarly have coordinates $(ac - bd)^\bullet, (ad + bc)^\bullet$. The four elements $\mathbf{0}, X_0, Y_0, Z_0$ are distinct. Unlike $P_{\{1,2\}}$, the carrier D_A uses convolution multiplication; (MFC.14) specifies the difference without identifying the two semiring structures.

MFC.5. Four support fibres and the scalar character. Write $\mathbb{B}[C_2]$ for the semiring of pairs of bits with coordinatewise join and

$$(p, q)(r, s) = (pr \vee qs, ps \vee qr). \quad X : D_A \longrightarrow \mathbb{B}[C_2], \quad X(a, b) = (\chi_A(a), \chi_A(b)). \quad (\text{MFC.15})$$

Here pr means Boolean meet. Applying χ_A to (MFC.11), and using $\chi_A(u) = 1$, proves the multiplication formula; addition, zero, and unit are also preserved. The support semiring laws can alternatively be verified by distributing Boolean products as in (MFC.13). The fibres over $(0, 0), (1, 0), (0, 1), (1, 1)$ are respectively $\{\mathbf{0}\}, \{X_a : a \in A\}, \{Y_b : b \in A\}, \{Z_z : z \in B\}$. Every one retains its zero-amplitude entry. The amplitude map $\pi : D_A \rightarrow B$, obtained by replacing absent coordinates by zero, preserves all operations by (MFC.11). The map (π, X) is injective: support determines which coordinates are absent, and the unique coordinates of $\pi(x)$ determine all active entries.

There is exactly one unital semiring homomorphism $f : D_A \rightarrow \mathbb{B}$: it sends $\mathbf{0}$ to 0 and every nonempty face to 1. To prove uniqueness, $f(X_1) = 1$ and $X_1 + X_{-1} = X_0$ imply $f(X_0) = 1$. The equality $X_0 X_a = X_0$ then forces $f(X_a) = 1$ for every a . Next $Y_1^2 = X_{-1}$ forces $f(Y_1) = 1$, and $Y_a = X_a Y_1$ forces $f(Y_a) = 1$. Finally $Z_{a+tb} = X_a + Y_b$ forces $f(Z_{a+tb}) = 1$. Existence follows because a sum or product of two nonempty faces is nonempty, as follows respectively from coordinate addition and (MFC.14); the cases with $\mathbf{0}$ are the zero laws. Thus scalar support is an exact further observation of the four fibres, and does not replace (MFC.15).

MFC.6. The quadratic synchronization quotient and all fibres. Define

$$q_D : D_A \longrightarrow G(B), \quad \mathbf{0} \longmapsto \tau_B, \quad X_a \longmapsto a^\bullet, \quad Y_b \longmapsto (tb)^\bullet, \quad Z_z \longmapsto z^\bullet. \quad (\text{MFC.16})$$

This is a surjective unital semiring homomorphism. Surjectivity follows from the Z -face and $\mathbf{0}$. A sum of two nonempty faces is nonempty and has amplitude equal to the sum in B . The same assertion for multiplication follows from the six products in (MFC.14). Thus both operations are preserved, even when the resulting amplitude is zero. Operations with $\mathbf{0}$ obey the zero laws, and $q_D(X_1) = 1_B^\bullet$. Its complete fibre list, in the unique coordinates $z = a + tb$, is

target	fibre	
τ_B	$\{\mathbf{0}\}$	
e_B	$\{X_0, Y_0, Z_0\}$	
$(a + tb)^\bullet, a \neq 0, b \neq 0$	$\{Z_{a+tb}\}$	(MFC.17)
$a^\bullet, a \neq 0$	$\{X_a, Z_a\}$	
$(tb)^\bullet, b \neq 0$	$\{Y_b, Z_{tb}\}$	

Indeed an X -preimage requires second coordinate zero, a Y -preimage requires first coordinate zero, and the Z -preimage always exists uniquely. These alternatives exhaust the four faces.

The kernel congruence of q_D is generated by $X_0 \sim Y_0$. Adding X_0 and then Y_0 to this relation shows $X_0 \sim Z_0 \sim Y_0$. Adding X_a to $X_0 \sim Y_0$ gives $X_a \sim Z_a$; adding Y_b gives $Z_{tb} \sim Y_b$. Hence each fibre in (MFC.17) is identified. Conversely the kernel of the proved homomorphism q_D is a congruence containing the generator, so no generated identification can connect distinct fibres. This proves the equality of congruences.

MFC.7. The full section, its unit, and the localization. There is a zero-preserving additive and multiplicative section

$$\iota : G(B) \longrightarrow D_A, \quad \iota(\tau_B) = \mathbf{0}, \quad \iota(z^\bullet) = Z_z, \quad q_D \iota = \text{id}_{G(B)}. \quad (\text{MFC.18})$$

Addition within the Z -face is $Z_z + Z_w = Z_{z+w}$; multiplication is (MFC.14), and $\mathbf{0}$ obeys the required zero laws. These prove every asserted property. This section is not unital as a map to D_A , since $\iota(1_B^\bullet) = Z_1 = (1_A^\bullet, 0_A^\bullet) \neq X_1$.

In fact no unital semiring section σ of q_D exists. The equality $1_B^\bullet + e_B = 1_B^\bullet$ forces $X_1 + \sigma(e_B) = X_1$. Among the three possibilities in the e_B -fibre, only X_0 satisfies this equality: adding Y_0 or Z_0 gives Z_1 . Thus $\sigma(e_B) = X_0$. The $t^\bullet$ -fibre is $\{Y_1, Z_t\}$, using $1_A \neq 0_A$. But $e_B t^\bullet = e_B$, while $X_0 Y_1 = Y_0 \neq X_0$ and $X_0 Z_t = Z_0 \neq X_0$. Both possibilities contradict multiplicativity.

The element Z_1 is idempotent, and its corner is

$$Z_1 D_A = \{\mathbf{0}\} \sqcup \{Z_z : z \in B\}, \quad D_A[Z_1^{-1}] \cong Z_1 D_A \cong G(B). \quad (\text{MFC.19})$$

The corner has unit Z_1 . The map $d \mapsto Z_1 d$ is a unital homomorphism to that corner: idempotence gives $(Z_1 d)(Z_1 d') = Z_1 dd'$, and distributivity gives additivity. It equals ιq_D with this codomain, by (MFC.14). To prove the localization universal property, let $f : D_A \rightarrow T$ be a unital homomorphism for which $f(Z_1)$ is invertible. An invertible idempotent is 1_T , since multiplying $f(Z_1)^2 = f(Z_1)$ by its inverse gives that equality. Therefore $f(d) = f(Z_1 d)$. Restricting f to the corner gives the unique unital homomorphism through which f factors; it is unital because $f(Z_1) = 1_T$, and uniqueness follows from surjectivity of $d \mapsto Z_1 d$. This proves (MFC.19), retaining both the full fibres of q_D and the exact category of its section.

MFC.8. All ordinary semiring ideals of D_A . For ring ideals $H \triangleleft B, J \triangleleft A$, define

$$\begin{aligned} M_H &= \{\mathbf{0}\} \cup \{Z_z : z \in H\}, & M &= M_B, \\ I_{J,H} &= M_H \cup \{X_a : a \in J\} \cup \{Y_a : a \in J\}, & JB &\subseteq H. \end{aligned} \quad (\text{MFC.20})$$

Here JB is the ideal of B generated by the image of J . Every ideal of D_A is exactly one of

$$\{\mathbf{0}\}, \quad M_H, \quad I_{J,H} \quad (H \triangleleft B, J \triangleleft A, JB \subseteq H). \quad (\text{MFC.21})$$

We prove the statement for the ordinary ideal convention in MFC.1. Let $\mathcal{I}$ be such an ideal. If it has no X - or Y -entry and is not $\{\mathbf{0}\}$, it contains some Z_z . The set $H = \{z : Z_z \in \mathcal{I}\}$ is nonempty. Multiplication by Z_0 shows $0 \in H$, multiplication by Z_{-1} gives additive inverses, addition gives closure under sums, and multiplication by Z_w for arbitrary $w \in B$ gives absorption. Hence H is a ring ideal and $\mathcal{I} = M_H$.

If $\mathcal{I}$ has an X - or Y -entry, put $J_X = \{a : X_a \in \mathcal{I}\}$ and $J_Y = \{a : Y_a \in \mathcal{I}\}$. The identities $X_a Y_1 = Y_a$ and $Y_a Y_{-1} = X_a$ give $J_X = J_Y =: J$, and this common set is nonempty. Multiplication of an X -entry by X_0 puts 0 in J ; addition, multiplication by X_{-1} , and multiplication by arbitrary X_r prove respectively closure under sums, additive inverses, and absorption by A . Thus $J \triangleleft A$. Multiplying an X -entry by Z_0 puts Z_0 in $\mathcal{I}$, and the preceding Z -face argument proves that $H = \{z : Z_z \in \mathcal{I}\}$ is a ring ideal of B . For $a \in J, w \in B$, $X_a Z_w = Z_{aw}$ puts aw in H ; closure under finite sums proves $JB \subseteq H$. The definitions now give $\mathcal{I} = I_{J,H}$.

Conversely M_H is closed under addition, and every nonempty face multiplied by $Z_h, h \in H$, remains a Z -entry with amplitude in H , by (MFC.14). The absent face multiplies it to $\mathbf{0}$, which is also in M_H . Thus M_H is an ideal. For $I_{J,H}$, same-face sums stay in J or H . Cross-face sums have the precise forms

$$X_a + Y_b = Z_{a+tb}, \quad X_a + Z_h = Z_{a+h}, \quad Y_b + Z_h = Z_{tb+h}. \quad (\text{MFC.22})$$

When $a, b \in J$ and $h \in H$, these amplitudes lie in H because $JB \subseteq H$. Multiplication of an X - or Y -entry of $I_{J,H}$ by an X - or Y -entry of D_A has coefficient in J , and its product with a Z -entry has amplitude in JB . A Z -entry of $I_{J,H}$ multiplied by any face remains in M_H . Products with $\mathbf{0}$ are $\mathbf{0}$. This verifies absorption in every case. Each listed family is therefore an ideal. Their descriptions are unique by reading the amplitudes in each face. The three families are disjoint: M_H contains Z_0 but no X - or Y -entry, while $I_{J,H}$ contains X_0, Y_0, Z_0 .

Their complete inclusion rules, aside from the least ideal $\{\mathbf{0}\}$, are

$$\begin{aligned} M_H \subseteq M_{H'} &\iff H \subseteq H', \\ M_H \subseteq I_{J',H'} &\iff H \subseteq H', \\ I_{J,H} \subseteq I_{J',H'} &\iff J \subseteq J' \text{ and } H \subseteq H'. \end{aligned} \quad (\text{MFC.23})$$

No $I_{J,H}$ is contained in an $M_{H'}$, since it contains X_0 . Each equivalence follows by inspecting the relevant faces in (MFC.20), which are disjoint.

MFC.9. All prime branches and their exact inclusions. For $\mathfrak{q} \in \text{Spec } B$ and $\mathfrak{p} \in \text{Spec } A$, put

$$P_\tau = \{\mathbf{0}\}, \quad Q_\mathfrak{q} = I_{\mathfrak{q} \cap A, \mathfrak{q}}, \quad T_\mathfrak{p} = I_{\mathfrak{p}, B}. \quad \text{Spec } D_A = \{P_\tau, M\} \sqcup \{Q_\mathfrak{q}\} \sqcup \{T_\mathfrak{p}\}. \quad (\text{MFC.24})$$

We prove exhaustion and primality, rather than imposing a subtractive prime convention. The complement of P_τ is closed under multiplication by (MFC.14). The complement of M consists of all X - and Y -entries, and is also closed under multiplication. Both ideals are proper, so both are prime. If $H \subsetneq B$, choose $z \notin H$. Then $X_0, Z_z \notin M_H$ but $X_0 Z_z = Z_0 \in M_H$; thus M_H is not prime.

Suppose $I_{J,H}$ is prime and $H \subsetneq B$. The rule $Z_z Z_w = Z_{zw}$ shows that H is prime in B . The condition $JB \subseteq H$ gives $J \subseteq H \cap A$. If $a \in H \cap A$, then $X_a Z_1 = Z_a \in I_{J,H}$ while $Z_1 \notin I_{J,H}$, so primality gives $a \in J$. Thus $I_{J,H} = Q_H$. Conversely take a prime $\mathfrak{q}$ in B , and let $\mathfrak{p} = \mathfrak{q} \cap A$. This contraction is proper and prime: $1 \notin \mathfrak{q}$, and $ab \in \mathfrak{q}$ forces $a \in \mathfrak{q}$ or $b \in \mathfrak{q}$. Two X - or Y -entries outside $Q_\mathfrak{q}$ have coefficients outside $\mathfrak{p}$, so their product coefficient, with either sign, remains outside $\mathfrak{p}$. An X_a outside this ideal and a Z_z outside it have $a, z \notin \mathfrak{q}$, so $az \notin \mathfrak{q}$. For $Y_a Z_z$, the same holds for taz , since t is a unit and cannot lie in a proper prime. Two Z -entries outside the ideal have product outside by primality of $\mathfrak{q}$. These are all products in its complement, proving primality.

If $H = B$, the ideal $I_{J,B}$ is proper precisely when $J \subsetneq A$. Its complement consists of X_a, Y_a with $a \notin J$. It is multiplicatively closed when J is prime, by the first row of (MFC.14). Conversely, if it is multiplicatively closed and $ab \in J$, the product $X_a X_b$ belongs to the ideal, so at least one factor does, proving that J is prime. This gives exactly the T_p -family. Together with (MFC.21), it proves (MFC.24).

The complete prime inclusion rules are

$$\begin{aligned} P_\tau \subsetneq M, \quad P_\tau \subsetneq Q_q, \quad P_\tau \subsetneq T_p, \quad M \subsetneq T_p, \\ Q_q \subseteq Q_{q'} \iff q \subseteq q', \quad T_p \subseteq T_{p'} \iff p \subseteq p', \\ Q_q \subseteq T_p \iff q \cap A \subseteq p. \end{aligned} \tag{MFC.25}$$

Every inclusion of a Q -prime in a T -prime is strict, because Z_1 lies only in the latter. The prime M is incomparable with each Q_q : $Z_1 \in M \setminus Q_q$ and $X_0 \in Q_q \setminus M$. No T -prime is contained in a Q -prime, because it contains Z_1 ; no T -prime is contained in M , because it contains X_0 . These assertions and (MFC.25) follow directly from (MFC.23), and exhaust the possible pairs of families. The ordinary ideal convention matters here. Each M_H is non-subtractive: $Z_0 \in M_H$ and $Z_0 + X_0 = Z_0 \in M_H$, whereas $X_0 \notin M_H$. Each T_p is likewise non-subtractive: $Z_0, Z_0 + X_1 = Z_1 \in T_p$, whereas $X_1 \notin T_p$ because p is proper. Thus restricting to subtractive primes would remove original prime branches occurring in (MFC.24).

MFC.10. The dimension calculation, including incomparability. If $A \neq 0$ and its Krull dimension is $\dim A = d < \infty$, then

$$\dim D_A = d + 2. \tag{MFC.26}$$

Here dimension means the ideal-theoretic prime-chain dimension of MFC.1. We give the integral-extension argument needed for the upper bound in full. The ring $B = A[t]/(t^2 + 1)$ is integral over its embedded copy of A , since every $b = a + tc$ satisfies

$$b^2 - 2ab + (a^2 + c^2) = 0. \tag{MFC.27}$$

Suppose primes $q \subsetneq q'$ of B had the same contraction p in A . Then $R = A/p$ embeds in the domain $C = B/q$, and C is integral over R : reduce any monic equation over A modulo these ideals. The nonzero prime $\tau = q'/q$ of C has zero contraction in R . Let $S = R \setminus \{0\}$ and $K = S^{-1}R$. The localization $S^{-1}C$ is a domain, and every c/s in it is integral over the field K . Indeed, dividing a monic equation for c by the corresponding power of s produces a monic equation for c/s with coefficients in K . A domain integral over a field is a field: for a nonzero element v , choose a monic relation of least degree over that field. Its constant coefficient is nonzero, because otherwise division by v in the domain would give a monic relation of lower degree. Rearranging the relation expresses v^{-1} as a polynomial in v divided by that nonzero constant coefficient. Thus $S^{-1}C$ is a field. Since $\tau \cap S = \emptyset$, its localization is a proper prime ideal of this field: if $1 = c/s$ with $c \in \tau$, $s \in S$, clearing the equality gives $v(c - s) = 0$ for some $v \in S$, and the domain property implies $s = c \in \tau$, a contradiction. It follows that $S^{-1}\tau = 0$. But a nonzero $c \in \tau$ remains nonzero after localization, since $sc = 0$, $s \in S$, is impossible in the domain. This contradiction proves that strict prime inclusions in B have strict contractions in A .

The ring $A \neq 0$ has a maximal proper ideal: the union of a chain of proper ideals is proper, since otherwise it contains 1 at some stage, and Zorn's lemma applies. A maximal ideal is prime because its quotient is a field: if the class of a is nonzero, maximality of the ideal generated by that class supplies its multiplicative inverse. Thus $d \geq 0$, and a chain of length d exists. Indeed the supremum of a nonempty set of nonnegative integer lengths, when finite, is attained. Choose $p_0 \subsetneq \cdots \subsetneq p_d$. Then (MFC.25) gives the strict chain

$$P_\tau \subsetneq M \subsetneq T_{p_0} \subsetneq \cdots \subsetneq T_{p_d}, \tag{MFC.28}$$

of length $d + 2$.

For the upper bound, it suffices to bound every finite strict chain, since an infinite chain would contain finite chains of arbitrarily large length. Adjoin P_τ at its beginning if it is missing. By (MFC.25), a chain containing M contains no Q -prime, and after M contains only a T -chain. If that T -chain has ℓ entries, then $\ell - 1 \leq d$, and the full chain has length at most $\ell + 1 \leq d + 2$. A chain with no Q - or T -entries has length at most one. A chain with Q -entries and no M has a Q -chain of $k \geq 1$ entries followed by a T -chain of $\ell \geq 0$ entries. The contractions of the Q -entries are strictly increasing by the preceding proof. If $\ell \geq 1$, they are followed by the strictly increasing T -indices, with at most one equality, at the junction. Therefore this chain in $\text{Spec } A$ has at least $k + \ell - 1$ distinct entries, so $k + \ell - 2 \leq d$. The original chain, including P_τ , has length $k + \ell \leq d + 2$. If $\ell = 0$, then $k - 1 \leq d$ and its length is $k \leq d + 1$. A chain having T -entries but neither Q -entries nor M has length at most $d + 1$. These cases exhaust (MFC.24), proving the upper bound and therefore (MFC.26).

MFC.11. Higher distributive-lattice zero fibres. Let L be a bounded distributive lattice, with bottom 0_L and top 1_L , and let $R \neq 0$ be a ring. The original carrier is

$$\begin{aligned} G_L(R) &= \{(0_R, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subseteq R \times L, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \\ (r, \lambda)(s, \mu) &= (rs, \lambda \wedge \mu), \\ 0_{G_L(R)} &= (0_R, 0_L), \quad 1_{G_L(R)} = (1_R, 1_L). \end{aligned} \tag{MFC.29}$$

This is a union inside a product; the two descriptions of $(0_R, 1_L)$ denote one element. A nonzero amplitude is permitted only at top support. The operations are closed. If $r + s \neq 0$, at least one summand has nonzero amplitude and hence top support, so the sum has top support. If $rs \neq 0$, both amplitudes are nonzero and both supports are top. Zero-amplitude results allow any support. Associativity and commutativity follow in each coordinate from the ring and lattice laws. Distributivity follows from ring distributivity and distributivity of meet over join in L . The displayed zero and unit satisfy their laws in each coordinate. Thus (MFC.29) is a semiring.

The projections

$$p_L : G_L(R) \rightarrow R, \quad (r, \lambda) \mapsto r, \quad \chi_L : G_L(R) \rightarrow L, \quad (r, \lambda) \mapsto \lambda, \quad p_L^{-1}(0_R) = \{(0_R, \lambda) : \lambda \in L\} \tag{MFC.30}$$

are unital semiring homomorphisms, where L has join and meet as its operations. The last fibre is exactly the entire original lattice. Moreover p_L is the universal unital homomorphism from $G_L(R)$ to a ring. To prove this, write $z_\lambda = (0_R, \lambda)$ and $e = (0_R, 1_L)$. For any homomorphism $f : G_L(R) \rightarrow T$ to a ring, $e + z_\lambda = e$ gives $f(z_\lambda) = 0_T$ by additive cancellation in T . The function $\bar{f} : R \rightarrow T, r \mapsto f(r, 1_L)$, therefore preserves zero, addition, multiplication, and unit, so is a ring homomorphism. Every element of $G_L(R)$ is a z_λ or a $(r, 1_L)$, proving $f = \bar{f}p_L$, and the surjectivity of p_L proves uniqueness. If L is the one-element lattice, the carrier is R itself; the distinct absent and supported-zero elements occur precisely when $0_L \neq 1_L$.

MFC.12. All lattice-localization fibres and the universal map. For every $\lambda \in L$, the element z_λ is idempotent and

$$z_\lambda G_L(R) = \{z_\nu : \nu \leq \lambda\}, \quad G_L(R)[z_\lambda^{-1}] \cong \downarrow \lambda, \quad \ell_\lambda : G_L(R) \rightarrow \downarrow \lambda, \quad (r, \mu) \mapsto \lambda \wedge \mu. \tag{MFC.31}$$

Here $\downarrow \lambda$ has join and meet, bottom 0_L , and unit λ . Multiplying in (MFC.29) gives $z_\lambda(r, \mu) = z_{\lambda \wedge \mu}$; every $\nu \leq \lambda$ occurs by taking $(0_R, \nu)$. This proves the corner equality and its identification with $\downarrow \lambda$, including its unit. Distributivity of L proves that ℓ_λ preserves addition, and associativity and idempotence of meet prove that it preserves multiplication. It sends the original zero to 0_L , the original unit to λ , and z_λ to that unit. For a unital map f making z_λ invertible, idempotence forces $f(z_\lambda) = 1$. Hence $f(x) = f(z_\lambda x)$, and restriction to the corner gives its unique factorization through

ℓ_λ . This is the required localization universal property. At $\lambda = 0_L$ the localization is the one-element semiring; the universal statement allows that zero semiring. Its exact fibres are

$$\ell_\lambda^{-1}(\nu) = \{(r, \mu) \in G_L(R) : \lambda \wedge \mu = \nu\}, \quad \nu \leq \lambda. \quad (\text{MFC.32})$$

Thus every top-supported amplitude, including every nonzero amplitude, maps to the unit λ of the localized interval. The formula specifies every erased amplitude and every identified support stratum.

MFC.13. The higher-zero carrier inside the full independent cube. Take $R = R_I$ and $L = \mathcal{P}(I)$. Define

$$J : G_{\mathcal{P}(I)}(R_I) \hookrightarrow P_I, \quad J(r, I) = (r^\bullet)_i, \quad J(0, A) = x(0, A). \quad (\text{MFC.33})$$

The two formulas agree at $(0, I)$. Two full-support entries add and multiply by the ring operations in every coordinate, so J preserves their sum and product. Two zero-amplitude entries have support union under addition and intersection under multiplication, by (MFC.5), again as in (MFC.29). A full entry plus a zero-amplitude entry stays full with the original amplitude; a full entry multiplied by a zero-amplitude entry gives the same zero-amplitude face, because supported multiplication by zero gives supported zero in exactly the active coordinates of that face. These cases exhaust all pairs and prove preservation of addition and multiplication. The bottom entry maps to the all-absent zero and $(1, I)$ to the unit. Recovering amplitude and support proves injectivity. Its exact image is all full-support tuples together with all zero-amplitude faces. Proper faces with nonzero amplitude are the additional elements of P_I . The synchronized inclusion factors as

$$G(R_I) \xrightarrow{k} G_{\mathcal{P}(I)}(R_I) \xrightarrow{J} P_I, \quad k(\tau_{R_I}) = (0, \emptyset), \quad k(r^\bullet) = (r, I), \quad Jk = j. \quad (\text{MFC.34})$$

The formulas prove that k is injective and unital, using (MFC.2) and (MFC.29). Both maps also commute with amplitude and support, since those coordinates are the coordinates used in their definitions.

MFC.14. Every original chain stratum and each threshold fibre. Let $n \geq 1$ and $C_n = \{0 < 1 < \dots < n\}$. In $G_{C_n}(R)$ put

$$\begin{aligned} e &= (0_R, n), & \tau_i &= (0_R, n - i) \quad (1 \leq i \leq n), \\ \tau_0 &= e & \text{when indexing formulas,} \\ \tau_i + \tau_j &= \tau_{\min(i, j)}, & \tau_i \tau_j &= \tau_{\max(i, j)}. \end{aligned} \quad (\text{MFC.35})$$

These equalities follow from join = max and meet = min in (MFC.29). For every $r \in R$, $(r, n) + \tau_i = (r, n)$ and $(r, n)\tau_i = \tau_i$. The carrier consists of the full ring copy $\{(r, n) : r \in R\}$, with its zero e , and the n further distinct entries $\tau_1, \dots, \tau_n$. In particular its amplitude-zero fibre has $n + 1$ entries.

For $1 \leq k \leq n$, the original threshold and induced map are

$$\begin{aligned} \alpha_k : C_n &\rightarrow \mathbb{B}, & \alpha_k(j) &= \begin{cases} 0 & j < k, \\ 1 & j \geq k, \end{cases} \\ T_k : G_{C_n}(R) &\rightarrow G(R), & T_k(r, n) &= r^\bullet, & T_k(0_R, j) &= \begin{cases} \tau_R & j < k, \\ e_R & j \geq k. \end{cases} \end{aligned} \quad (\text{MFC.36})$$

The formulas agree at $(0_R, n)$. The threshold sends bottom to zero and top to one. The maximum of two indices is at least k precisely when at least one is, and the minimum is at least k precisely when both are. Thus it preserves join and meet. For two zero-amplitude entries this proves both laws for T_k . Two full entries use the original ring operations and stay full, even upon amplitude cancellation. A full entry plus any zero-amplitude entry stays full; its image plus either τ_R or e_R is

the same supported image. A full entry times $(0_R, j)$ equals $(0_R, j)$; its supported image times the prescribed image of that entry is respectively τ_R or e_R . These are all cases, proving that T_k is a unital semiring homomorphism. Its entire fibres are

$$\begin{aligned} T_k^{-1}(\tau_R) &= \{(0_R, j) : 0 \leq j < k\}, \\ T_k^{-1}(e_R) &= \{(0_R, j) : k \leq j \leq n\}, \\ T_k^{-1}(r^\bullet) &= \{(r, n)\} \quad (r \neq 0_R). \end{aligned} \quad (\text{MFC.37})$$

They have respectively $k, n - k + 1, 1$ entries. These formulas retain the exact depth lost under every binary threshold.

MFC.15. The all-ideal chain for the original two arithmetic rings. Let p be a prime integer, let k now denote a field, and let $n \geq 1$. Define the two rings and their ideals by

$$R_{p,n} = \mathbb{Z}/p^n\mathbb{Z}, \quad I_a = p^a R_{p,n}, \quad R_{k,n} = k[\epsilon]/(\epsilon^n), \quad J_a = \epsilon^a R_{k,n}, \quad 0 \leq a \leq n. \quad (\text{MFC.38})$$

In particular $I_0 = R_{p,n}$, $I_n = (0)$, $J_0 = R_{k,n}$, and $J_n = (0)$. We prove that these are all ideals, not a selected subfamily.

For a nonzero element of $R_{p,n}$, take its representative between 1 and $p^n - 1$. Its exact power of p is p^a with $0 \leq a < n$; thus the element is $p^a u$ with u prime to p . The class of u is a unit: the Euclidean algorithm applied to u, p^n , whose greatest common divisor is 1, supplies integers v, w with $uv + p^n w = 1$. Let $\mathfrak{a}$ be a nonzero ideal, and choose among its nonzero elements one with least exponent a . Multiplying this element by u^{-1} puts p^a in $\mathfrak{a}$, so $I_a \subseteq \mathfrak{a}$. Every other nonzero element has exponent at least a and is therefore a multiple of p^a ; zero also lies in I_a . Hence $\mathfrak{a} = I_a$. The zero ideal is I_n . The ideals are distinct: for $a < n$, an equality $p^a = p^{a+1}b$ modulo p^n would imply that p^{n-a} divides $1 - pb$, impossible upon reduction modulo p .

Every element of $R_{k,n}$ has a unique polynomial representative of degree less than n , by division by the monic polynomial ϵ^n . A nonzero representative with least nonzero degree a is $\epsilon^a u$, where $u = u_0(1 + v)$, $u_0 \in k^\times$, and $v \in (\epsilon)$. Since $v^n = 0$,

$$u^{-1} = u_0^{-1} \sum_{j=0}^{n-1} (-v)^j \quad (\text{MFC.39})$$

is an inverse: multiplication by $1 + v$ gives $1 - (-v)^n = 1$. In a nonzero ideal choose an element with least such degree a . Multiplication by u^{-1} puts ϵ^a in the ideal, while every other element has degree at least a and is divisible by ϵ^a . Thus the ideal is J_a ; the zero ideal is J_n . Uniqueness of the degree-less-than- n representative shows that $\epsilon^a \notin J_{a+1}$ for $a < n$, proving that these $n + 1$ ideals are distinct.

In both rings the ideals are nested by reversed exponent. Consequently their sums and intersections are

$$I_a + I_b = I_{\min(a,b)}, \quad I_a \cap I_b = I_{\max(a,b)}, \quad J_a + J_b = J_{\min(a,b)}, \quad J_a \cap J_b = J_{\max(a,b)}. \quad (\text{MFC.40})$$

For instance, if $a \leq b$, then $I_b \subseteq I_a$, so the sum is I_a and the intersection is I_b ; the other cases are the same two inclusions with indices interchanged. It follows that the explicit bijections

$$C_n \xrightarrow{\theta_p} \text{Id}(R_{p,n}), \quad j \mapsto I_{n-j}, \quad C_n \xrightarrow{\theta_k} \text{Id}(R_{k,n}), \quad j \mapsto J_{n-j} \quad (\text{MFC.41})$$

preserve bottom, top, join, and meet, with ideal sum as join and ideal intersection as meet. Surjectivity here is the all-ideal statement just proved; injectivity is the strictness proved above. Thus these are isomorphisms of the bounded-lattice semirings, preserving every stratum.

MFC.16. The separate, complete ideal-product law. The very same ideal carriers also retain ordinary ideal multiplication, with its different operation:

$$\begin{aligned} I_a I_b &= I_{\min(n, a+b)}, & J_a J_b &= J_{\min(n, a+b)}, \\ V_n &= (\{0, \dots, n\}, \min, (a, b) \mapsto \min(n, a+b), 0_{V_n} = n, 1_{V_n} = 0). \end{aligned} \quad (\text{MFC.42})$$

Indeed the product of the principal ideals generated by x, y is generated by xy : every finite sum of products $xr \cdot ys$ is a multiple of xy , and xy itself is one of the generating products. Applying this fact to p^a, p^b , and to ϵ^a, ϵ^b , proves the displayed laws, including the vanishing for $a + b \geq n$. The maps $a \mapsto I_a$ and $a \mapsto J_a$ are therefore bijective semiring homomorphisms from V_n to the respective all-ideal carriers with sum and ordinary ideal product. For completeness, associativity of the operation on exponents follows from $\min(n, \min(n, a+b) + c) = \min(n, a+b+c)$; its unit and absorbing zero are 0 and n . The identity

$$\min(n, a + \min(b, c)) = \min(\min(n, a+b), \min(n, a+c)) \quad (\text{MFC.43})$$

proves distributivity over the addition $\min$. This also checks all semiring laws directly on V_n .

Under the order-preserving coordinates of (MFC.41), ordinary ideal multiplication is the operation

$$j \odot \ell = \max(0, j + \ell - n) \quad \text{on } C_n, \quad (\text{MFC.44})$$

because $n - \min(n, (n-j) + (n-\ell)) = \max(0, j + \ell - n)$. It is distinct from the meet $\min(j, \ell)$ when $n \geq 2$: for $j = \ell = n-1$, their values are $\max(0, n-2)$ and $n-1$. Equivalently, $I_1^2 = I_2$ whereas $I_1 \cap I_1 = I_1$, with the convention $I_2 = 0$ when $n = 2$, and the same distinction holds for J_1 . The identity map on either all-ideal carrier preserves the carrier, order, zero, unit, and addition between these two presentations, but does not in general preserve multiplication. Equations (MFC.40)–(MFC.44) give its exact transported operations. No identification of the two underlying arithmetic rings is asserted.

MFC.17. Coefficient transport, support preservation, and prime maps. The preceding constructions are accompanied by their original coefficient maps; they are not merely lists of properties. For a unital ring homomorphism $f : R \rightarrow R'$, define

$$G(f)(\tau_R) = \tau_{R'}, \quad G(f)(r^\bullet) = f(r)^\bullet. \quad (\text{MFC.45})$$

Equation (MFC.2) proves that this is a unital semiring homomorphism, including when a nonzero r has $f(r) = 0$: that entry is still supported. For a family $f_i : R_i \rightarrow R'_i$, the product map $\prod_i G(f_i)$ preserves each mask exactly, has amplitude $\prod_i f_i$, and commutes with j and J in (MFC.8) and (MFC.33). Each assertion follows by substituting into one coordinate of the displayed formulas; inactive coordinates stay absent and active ones stay supported. Thus the support square and every independent face remain part of the transported diagram.

For $f : A \rightarrow A'$, let $\bar{f} : A[t]/(t^2 + 1) \rightarrow A'[t]/(t^2 + 1)$ be $\bar{f}(a + tb) = f(a) + tf(b)$. This is well-defined and a unital ring homomorphism: applying f to the two coefficients of a product $(ac - bd) + t(ad + bc)$ gives exactly the corresponding product of the two images. Define

$$D(f)(a, b) = (G(f)(a), G(f)(b)), \quad q_{D_{A'}} D(f) = G(\bar{f}) q_{D_A}, \quad D(f) \iota_A = \iota_{A'} G(\bar{f}). \quad (\text{MFC.46})$$

The equality $G(f)(u) = (-1_{A'})^\bullet$ and (MFC.11) prove that $D(f)$ is a unital semiring homomorphism. It sends X_a, Y_b, Z_z respectively to $X_{f(a)}, Y_{f(b)}, Z_{\bar{f}(z)}$, so it preserves each of the four faces exactly and proves both commuting identities in (MFC.46). It sends Z_1 to Z_1 , hence also induces the corner and localization map already represented by $G(\bar{f})$.

For ideals in the primed rings its inverse-image formulas are

$$\begin{aligned} D(f)^{-1}(M_{H'}) &= M_{\bar{f}^{-1}(H')}, \\ D(f)^{-1}(I_{J', H'}) &= I_{f^{-1}(J'), \bar{f}^{-1}(H')}, \\ D(f)^{-1}(P_\tau) &= P_\tau, & D(f)^{-1}(M) &= M, \\ D(f)^{-1}(Q_{q'}) &= Q_{\bar{f}^{-1}(q')}, & D(f)^{-1}(T_{p'}) &= T_{f^{-1}(p')}. \end{aligned} \quad (\text{MFC.47})$$

The first two formulas follow by the exact face images, and the required condition on the second ideal pair holds: if $a \in f^{-1}(J')$ and $b \in B$, then $\bar{f}(ab) = f(a)\bar{f}(b) \in J'B' \subseteq H'$. Closure under sums gives $f^{-1}(J')B \subseteq \bar{f}^{-1}(H')$. The two constant-family formulas follow because nonempty faces and full faces remain in precisely those classes. For the Q -formula the contraction identity $\bar{f}^{-1}(q') \cap A = f^{-1}(q' \cap A')$ proves agreement of both indices. Inverse images of proper prime ideals under unital ring maps are proper primes, because they omit the unit and satisfy the prime product implication. This proves the typing and the final two formulas. These maps are continuous on ideal-theoretic spectra: for any ideal $\mathcal{I} \subseteq D_A$, the inverse image, under prime contraction, of the closed set of primes containing $\mathcal{I}$ is the set of primes of $D_{A'}$ containing the ideal generated by $D(f)(\mathcal{I})$. The two containment conditions are equivalent by the definition of inverse image.

Finally, for a bounded-lattice homomorphism $\beta : L \rightarrow L'$ preserving bottom, top, joins, and meets and a unital ring homomorphism $f : R \rightarrow R'$, the map

$$G_\beta(f) : G_L(R) \rightarrow G_{L'}(R'), \quad (r, \lambda) \mapsto (f(r), \beta(\lambda)) \quad (\text{MFC.48})$$

is well-defined: a nonzero target amplitude comes from a nonzero source amplitude, hence from $\lambda = 1_L$, whose image is $1_{L'}$. The two operations in (MFC.29) and the hypotheses on f, β prove preservation of sum, product, zero, and unit. It commutes with amplitude and support via f, β , and sends z_λ to $z_{\beta(\lambda)}$. Its localization transport is the map $\downarrow \lambda \rightarrow \downarrow \beta(\lambda), \nu \mapsto \beta(\nu)$, because

$$\beta(\ell_\lambda(r, \mu)) = \beta(\lambda \wedge \mu) = \beta(\lambda) \wedge \beta(\mu) = \ell_{\beta(\lambda)}(G_\beta(f)(r, \mu)). \quad (\text{MFC.49})$$

It preserves the interval unit and bottom. The threshold maps of MFC.14 are exactly this construction for $f = \text{id}_R$ and $\beta = \alpha_k$, after the isomorphism $G_{\mathbb{B}}(R) \cong G(R)$ sending $(0_R, 0)$ to τ_R and $(r, 1)$ to $r^\bullet$. Substitution in (MFC.29) and (MFC.2) proves that this last bijection is a unital semiring isomorphism. All maps in this attachment preserve their stated source indices and codomains. Their exact fibres identify the information forgotten by a specified observation while retaining the full carriers to which the observation is attached.

112 All independent coefficient faces attached to the original theta source

This construction attaches the independent coefficient masks of the original mixed-support algebra to the actual theta complexes. It retains the original outer source label and leg mask independently of the coefficient mask. A further slot-local leg diagram and its exact comparison with that outer diagram are constructed explicitly. The source-level maps, their cohomology, and the full arithmetic boundary of the finite section are computed below. Every tensor operation already present in the ambient programme keeps its original tensor variables; the finite direct sums in this section are explicitly coefficient-slot observations.

112.1 The original source and its full finite arithmetic sections

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+) : \sup_{x>0} x^b |D^j F(x)| < \infty \quad (b \in \mathbb{Z}, j \geq 0)\}, \\ D &= -x\partial_x, \quad \hat{\phi}(\nu) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \nu} dx, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q. \end{aligned} \quad (\text{MCF1})$$

The letter q in (MCF1) denotes the quotient map; a coefficient Frobenius prime power will be denoted q_0 below. Put

$$U = \{s : 0 < \Re s < 1, \Re s \neq \tfrac{1}{2}\}, \quad T_U Q = \bigcup_p \ker(p(D) : Q \rightarrow Q), \quad (MCF2)$$

$$\mathcal{B}_U = q^{-1}(T_U Q), \quad C_U = [V \xrightarrow{\Theta} \mathcal{B}_U],$$

where p ranges over all monic polynomials with roots in U , including 1. The original source calculation proves injectivity of Θ , $D\Theta = \Theta D$, and injectivity of every nonzero polynomial $p(D)$ on V and $\mathcal{B}$. We use its exact range identities

$$p(D)V = \ker(j_p H), \quad p(D)\mathcal{B} = \ker(j_p \mathcal{M}),$$

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}, \quad \mathcal{M}\Theta\phi = gH_\phi, \quad g = 2\xi, \quad (MCF3)$$

$$H_{P(D)\phi_*} = P, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}.$$

These are the original arithmetic identities (AG1) and (CAU1), with their full-plane Euler inverses. The attachment below uses their stated domains; it does not introduce a different test-function completion. For example, the last identity follows from

$$M_+\phi_*(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad H_\phi = \frac{2\pi^{s/2}M_+\phi}{s(s-1)\Gamma(s/2)}.$$

The displayed Mellin formula is obtained by integrating the two Gaussian monomials and using $\Gamma(z+1) = z\Gamma(z)$; the apparent exceptional values are handled by the original entire continuation. The moments of ϕ_* are zero because $\phi_*(0) = 0$ and $4\pi^2 \int x^4 e^{-\pi x^2} dx - 6\pi \int x^2 e^{-\pi x^2} dx = 3 - 3 = 0$.

For a nonempty finite full-order packet in U , write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad m_\rho = \text{ord}_\rho g, \quad d_h = \deg h, \quad (MCF4)$$

$$E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1},$$

$$F_h = S_h^{\mathcal{B}} \Theta \phi_*, \quad P_h(u) = r_h(\varepsilon_h u), \quad s_h u = P_h(u)(D)F_h, \quad \sigma_h = q s_h.$$

Here r_h is the unique polynomial representative of degree less than d_h , and $S_h^{\mathcal{B}}$ is the inverse of $h(D)$ on the exact range in (MCF3). Every actual multiplicity is retained, so v_h is nonzero at each packet centre and v_h is a unit in the full local algebra. Every packet index and directed limit in this module also includes the empty packet $h = 1$, with $F_1 = \Theta\phi_*$, $E_1 = 0$, $s_1 = \sigma_1 = \ell_1 = 0$, and $\mathcal{B}_1 = \Theta V$. Its maps into other packet stages are zero on E_1 and the original inclusion on ΘV . No inverse unit at this zero-vector-space stage is used. The definition of F_1 is forced by $1(D)F_1 = \Theta\phi_*$ and gives $\mathcal{M}F_1 = g$. Thus the directed systems are defined even when there is no nonempty packet in U . The equalities

$$h(D)F_h = \Theta\phi_*, \quad \mathcal{M}F_h = v_h, \quad J_h s_h = 1, \quad J_h = j_h \mathcal{M}, \quad J_h \Theta = 0,$$

$$h(D)s_h u = \Theta(P_h(u)(D)\phi_*), \quad (MCF5)$$

$$Ds_h - s_h A_h = \Theta\phi_* \ell_h, \quad \ell_h(u) = [s^{d_h-1}]P_h(u)$$

have the original source types. To verify the last equality in full, $sP_h(u) - P_h(A_h u)$ has zero h -jet, degree at most d_h , and leading coefficient $\ell_h(u)$. Monicity gives $sP_h(u) - P_h(A_h u) = h\ell_h(u)$. Apply this polynomial in D to F_h and use the first equality. The other identities follow directly from (MCF3)–(MCF4) and retain the entire multiplier v_h .

For completeness the finite arithmetic image is exactly $\sigma_h E_h = Q[h(D)]$. If $h(D)F = \Theta\phi$, the assignment $[F] \mapsto j_h H_\phi$ is well-defined because replacing F by $F + \Theta\psi$ replaces ϕ by $\phi + h(D)\psi$. A zero jet makes $\phi = h(D)\psi$ by (MCF3), and injectivity of $h(D)$ gives $F = \Theta\psi$. Every jet represented by a polynomial P is obtained from $S_h^{\mathcal{B}} \Theta(P(D)\phi_*)$, since $j_h(gP) = 0$. Thus this assignment is an isomorphism. Applied to $s_h u$ it gives $\varepsilon_h u$ by (MCF5), an invertible change of full jet coordinates. In particular σ_h is injective and has the asserted image.

112.2 The independent coefficient face category and its exact quotient

Fix $n \geq 1$ and $I_n = \{0, \dots, n-1\}$. For every subset $A \subseteq I_n$, including the empty subset, put

$$C_U[A] = \bigoplus_{i \in A} C_U = [V^A \xrightarrow{(\phi_i) \mapsto (\Theta \phi_i)} \mathcal{B}_U^A], \quad E_h[A] = E_h^A. \quad (\text{MCF6})$$

Empty direct sums are zero vector spaces. If $A \subseteq B$, define z_{AB} by the identity in each old coordinate and by the typed zero in every coordinate of $B \setminus A$, in both degrees and in $E_h[A]$. Then $dz_{AB} = z_{AB}d$ coordinate by coordinate, and $z_{BC}z_{AB} = z_{AC}$. This is a functor from the inclusion category $\mathcal{P}(I_n)$ to complexes. The complete face diagram is retained; no colimit is taken that would identify its different masks.

For a complex vector space X , set $G(X) = \{\tau\} \sqcup \{x^\bullet : x \in X\}$, with the original supported addition and scalar action. The degree- j mixed carrier is

$$\mathcal{D}_n^j = \prod_{i \in I_n} G(C_U^j) \simeq \prod_{A \subseteq I_n} \{A\} \times (C_U^j)^A. \quad (\text{MCF7})$$

The empty term on the right is one point, corresponding to the all-absent tuple. The bijection sends each present coordinate to its original amplitude and records every present zero in A . Its inverse puts those amplitudes in the coordinates of A as supported values and puts τ in the other coordinates. Thus all 2^n masks, including the proper nonempty faces, are present. A face arrow z_{AB} changes the recorded mask from A to B and fills the new coordinates with supported zeros. It is an arrow between labelled faces, not an identification of them.

The differential preserves the original mask of its input and applies Θ independently in every present coordinate. A source vector annihilated by its amplitude differential reaches the supported zero at the same mask. In the internal split cycle and boundary construction, $\epsilon(A, x) = (A, 0)$ and $Z = \text{Eq}(d, \epsilon d)$; boundary relations are the coequalizer of b and ϵb within each labelled fibre. Consequently the exact cohomology and quotient are

$$\begin{aligned} H^0(C_U[A]) &= 0, & H^1(C_U[A]) &= (T_U Q)^A, \\ \mathcal{D}_n^1 &\longrightarrow \prod_i G(T_U Q), & F_i^\bullet &\longmapsto (qF_i)^\bullet, \quad \tau \longmapsto \tau. \end{aligned} \quad (\text{MCF8})$$

Indeed Θ is injective in every active coordinate, and the image in degree one is exactly $(\Theta V)^A$. Two amplitudes at mask A are identified precisely when their coordinate differences belong to $(\Theta V)^A$. The fibre over the supported zero at A is $\{A\} \times (\Theta V)^A$; it is not combined with any other mask. More generally, choosing any lift F of an amplitude in $(T_U Q)^A$ gives its entire fibre as $F + (\Theta V)^A$ at that same mask.

The full finite arithmetic attachment is

$$\begin{array}{ccccc} E_h[A] & \xrightarrow{s_h^A} & \mathcal{B}_U^A & \xrightarrow{q^A} & (T_U Q)^A \\ z_{AB} \downarrow & & z_{AB} \downarrow & & \downarrow z_{AB} \\ E_h[B] & \xrightarrow{s_h^B} & \mathcal{B}_U^B & \xrightarrow{q^B} & (T_U Q)^B. \end{array} \quad (\text{MCF9})$$

All squares commute because every displayed map is applied independently to the unchanged old coordinates and is linear on the inserted zeros. Every $\sigma_h^A = q^A s_h^A$ is injective by (MCF5). In every active coordinate its exact Euler boundary is

$$(D^A s_h^A - s_h^A A_h^A)(u_i)_{i \in A} = (\Theta \phi_* \ell_h(u_i))_{i \in A}. \quad (\text{MCF10})$$

This is an equality of actual functions, whose primitive is $(\phi_* \ell_h(u_i))_{i \in A} \in V^A$. After the original quotient it proves $D^A \sigma_h^A = \sigma_h^A A_h^A$. Thus every proper coefficient face carries the same full arithmetic boundary calculation, with its mask still specified.

112.3 Independent slot-local leg presentations

There is also a slot-local copy of the original two-leg presentation at every active coefficient. This auxiliary indexing does not replace the original outer leg label: its exact attachment, including empty coefficient support, is (MCF34)–(MCF40) below. Its objects are

$$\lambda = (A, (S_i)_{i \in A}), \quad A \subseteq I_n, \quad \emptyset \neq S_i \subseteq \{+, -\}. \quad (\text{MCF11})$$

For empty A the family is empty. Order these objects by $\lambda \leq \lambda'$ when $A \subseteq A'$ and $S_i \subseteq S'_i$ for every $i \in A$. Thus each coefficient slot has four states: absent, plus only, minus only, or both. There are exactly 4^n slot-local masks.

At λ put

$$\begin{aligned} C_\lambda^0 &= \bigoplus_{i \in A} \left(\bigoplus_{+ \in S_i} V_+ \oplus \bigoplus_{- \in S_i} V_- \right), \\ C_\lambda^1 &= \mathcal{B}_U^A, \quad d_\lambda(\phi_i, \psi_i)_i = (\Theta\phi_i - \Theta\widehat{\psi}_i)_i. \end{aligned} \quad (\text{MCF12})$$

A coordinate of a missing leg in the displayed formula is its typed zero; that convention evaluates the formula without adjoining that leg to the mask. The plus and target actions of $t \in \mathbb{C}[t]$ are D ; the minus action is $1 - D$. The Fourier identity $\widehat{D}\phi = (1 - D)\widehat{\phi}$ proves that this is a complex of $\mathbb{C}[t]$ -modules. For $\lambda \leq \lambda'$, retain every old source leg and coefficient, insert typed zeros in the new legs and slots, and use zero insertion in degree one. Both differentials then agree term by term, proving a strictly composing cochain map. The actual supported arrow changes the label from λ to λ' and retains every newly supported zero as such.

The natural comparison with (MCF6) is the cochain map

$$p_\lambda : C_\lambda \longrightarrow C_U[A], \quad p_\lambda^0(\phi_i, \psi_i)_i = (\phi_i - \widehat{\psi}_i)_i, \quad p_\lambda^1 = 1. \quad (\text{MCF13})$$

It commutes with every zero-insertion transition. In a plus-only slot its degree-zero map is the identity, in a minus-only slot it is $-\widehat{}$, and in a joint slot it has the exact inverse coordinates

$$u = \phi - \widehat{\psi}, \quad v = \frac{1}{2}(\phi + \widehat{\psi}), \quad \phi = v + \frac{1}{2}u, \quad \psi = \widehat{v} - \frac{1}{2}\widehat{u}. \quad (\text{MCF14})$$

Fourier squares to the identity on V , so every displayed map and inverse has its stated domain. Write $J(\lambda) = \{i \in A : S_i = \{+, -\}\}$. The exact kernel and cohomology are

$$\begin{aligned} 0 \longrightarrow V^{J(\lambda)}[0] &\xrightarrow{\delta_\lambda} C_\lambda \xrightarrow{p_\lambda} C_U[A] \longrightarrow 0, \\ \delta(v)_i &= (v_i, \widehat{v}_i) \quad (i \in J(\lambda)), \\ H^0(C_\lambda) &= V^{J(\lambda)}, \quad H^1(C_\lambda) = (T_U Q)^A. \end{aligned} \quad (\text{MCF15})$$

Surjectivity of p^0 follows from the single-leg formulas and (MCF14). Its kernel in a joint slot is exactly the displayed diagonal, and its degree-one kernel is zero. Its differential on (u, v) is Θu . Injectivity of Θ now proves the cohomology statement directly. Thus the two-leg diagonal remains present at every relevant independent coefficient; it is not removed when the arithmetic degree-one quotient is observed.

The slot-local cohomology carrier is the following. The separate outer label is retained in (MCF34).

$$\widetilde{H}_n^1 = \{\tau_n\} \sqcup \coprod_{\substack{\lambda=(A,S) \\ A \neq \emptyset}} \{\lambda\} \times (T_U Q)^A. \quad (\text{MCF16})$$

Forgetting only the leg masks gives a pointed map to $\prod_i G(T_U Q)$. Over a specified tuple of coefficient mask A it has exactly $3^{|A|}$ labelled preimages, one for every nonempty S_i ; the amplitudes are fixed.

Over the all-absent tuple its fibre is the single τ_n . A specified section takes the joint leg mask at every active coefficient. These formulas prove the exact relationship between the combined support diagram and the coefficient-mask observation; neither is silently used in place of the other. The maps s_h^A and σ_h^A of (MCF9) give their source and arithmetic amplitudes at every λ , with the same recorded coefficient and leg masks.

One can also lift each original one-leg complex into its selected leg presentation. At a single slot the degree-zero section of p_λ is

$$j_+(u) = u, \quad j_-(u) = -\widehat{u}, \quad j_{+-}(u) = (u/2, -\widehat{u}/2), \quad (\text{MCF17})$$

and degree one is the identity. They are $\mathbb{C}[t]$ -linear cochain maps by the Fourier identity. For the two inclusions into the joint slot their precise difference is

$$\text{inc}_+ j_+ - j_{+-} = \delta(u/2), \quad \text{inc}_- j_- - j_{+-} = -\delta(u/2) \quad \text{in degree zero,} \quad (\text{MCF18})$$

and zero in degree one. These retain the original diagonal; a claim of strict naturality for these three chosen sections would omit it. There is also an explicit complex-linear homotopy. The coherent original sections give $\mathcal{B}_U = \Theta V \oplus s_U E_U$, with $E_U = \varinjlim_h E_h$; define $t_U(\Theta\phi + s_U u) = \phi$. Then $t_U\Theta = 1$. The degree-minus-one maps $H_\pm(F) = \pm\delta(t_U F)/2$ obey $dH_\pm = 0$ and $H_\pm\Theta = \pm\delta/2$, so $dH_\pm + H_\pm d$ equals (MCF18). This is the exact homotopy of the chosen sections. No equivariance of t_U or $H_\pm$ is asserted. Existence of the coherent s_U and its original unit is checked in the divisor calculation below.

112.4 Arithmetic dilation on every face and its full source boundary

For $a > 0$ write $\mathcal{R}_a F(x) = F(x/a)$, with the same formula on V . Changing variables shows that it preserves the original spaces, preserves the value-zero condition, and multiplies the source integral by a . Direct substitution gives

$$\Theta\mathcal{R}_a = \mathcal{R}_a\Theta, \quad \widehat{\mathcal{R}_a\phi} = a\mathcal{R}_{1/a}\widehat{\phi}, \quad \mathcal{M}\mathcal{R}_a F = a^s \mathcal{M}F. \quad (\text{MCF19})$$

It commutes with D and hence preserves $T_U Q$ and its inverse image $\mathcal{B}_U$. On every combined face its action is

$$(\phi_i, \psi_i)_i \mapsto (\mathcal{R}_a\phi_i, a\mathcal{R}_{1/a}\psi_i)_i, \quad F_i \mapsto \mathcal{R}_a F_i. \quad (\text{MCF20})$$

The coefficient and leg masks are unchanged. The Fourier formula in (MCF19) makes the two differentials commute exactly. These actions also commute with every face inclusion and with p_λ , since

$$\mathcal{R}_a\phi - a\widehat{\mathcal{R}_{1/a}\psi} = \mathcal{R}_a(\phi - \widehat{\psi}).$$

On the diagonal the action is $v \mapsto \mathcal{R}_a v$, because its minus coordinate becomes $a\mathcal{R}_{1/a}\widehat{v} = \widehat{\mathcal{R}_a v}$. Thus (MCF15) is an equivariant exact sequence.

The finite arithmetic action is

$$A_{a,h} = M_{j_h(a^s)} : E_h \longrightarrow E_h, \quad A_{a,h}|_\rho = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho = M_{s-\rho}. \quad (\text{MCF21})$$

The complete nilpotent Taylor polynomial is displayed. Mellin and (MCF19) give $J_h \mathcal{R}_a = A_{a,h} J_h$. Dilation preserves $Q[h(D)]$, and J_h is the inverse of σ_h there by (MCF5). Therefore $\widehat{\mathcal{R}_a\sigma_h} = \sigma_h A_{a,h}$, an exact equality on the original arithmetic quotient.

The source defect has a full original primitive, rather than an assumed equivariant lift. Define

$$\begin{aligned} \chi_{a,h}(u) &= \mathcal{R}_a(P_h(u)(D)\phi_*) - P_h(A_{a,h}u)(D)\phi_* \in V, \\ \Psi_{a,h}(u) &= S_h^V \chi_{a,h}(u), \end{aligned} \quad (\text{MCF22})$$

where S_h^V is the inverse of $h(D)$ on $\ker(j_h H)$ from (MCF3). This inverse is applied on its proved domain: the Mellin calculation gives

$$H_{\chi_{a,h}(u)} = a^s P_h(u) - P_h(A_{a,h}u), \quad j_h H_{\chi_{a,h}(u)} = j_h(a^s) \varepsilon_h u - \varepsilon_h A_{a,h}u = 0. \quad (\text{MCF23})$$

The complete $\varepsilon_h = j_h(g/h)^{-1}$ occurs in both terms. Applying $h(D)$ to the two sides and using (MCF5), (MCF19), and $h(D)\Psi = \chi$, gives

$$\mathcal{R}_a s_h u - s_h A_{a,h}u = \Theta \Psi_{a,h}(u). \quad (\text{MCF24})$$

Indeed the difference after application of $h(D)$ is zero, and its injectivity on $\mathcal{B}$ proves the displayed equality. In every coefficient face this equality is applied separately to each u_i . In a combined leg presentation an actual degree-zero primitive is $j_{S_i}(\Psi_{a,h}(u_i))$ from (MCF17). Its differential is precisely the corresponding coordinate of (MCF24). The alternative single-leg and joint-leg primitives differ by the diagonal terms of (MCF18), which remain recorded in (MCF15).

The primitives themselves have the exact composition law

$$\Psi_{ab,h}(u) = \mathcal{R}_a \Psi_{b,h}(u) + \Psi_{a,h}(A_{b,h}u). \quad (\text{MCF25})$$

Expand $\mathcal{R}_{ab}s_h - s_h A_{a,h}A_{b,h}$ as $\mathcal{R}_a(\mathcal{R}_b s_h - s_h A_{b,h}) + (\mathcal{R}_a s_h - s_h A_{a,h})A_{b,h}$, apply (MCF24), and cancel the injective Θ . This proves the law with the original dilation order, source action, and full arithmetic multiplier.

112.5 Packet enlargement with every coefficient face retained

For full-order packets $h \mid H$, put $b = H/h$. Their supports differ by whole local factors, so b and h are coprime. Let $\iota_{hH} : E_h \rightarrow E_H$ insert the old full jets and zero jets at every added centre. CRT gives this injective $\mathbb{C}[s]$ -linear map and the restriction left inverse. At old centres $v_H = b^{-1}v_h$ and $\varepsilon_H = b\varepsilon_h$. At added centres b vanishes to the full required order. Hence

$$r_H(\varepsilon_H \iota_{hH} u) = b r_h(\varepsilon_h u), \quad b(D)F_H = F_h, \quad s_H \iota_{hH} = s_h. \quad (\text{MCF26})$$

When $h = 1, b = H$ and the middle identity is exactly $H(D)F_H = \Theta \phi_* = F_1$; the first and third identities have zero domain and are zero-map identities. The highest-coefficient, action, and primitive identities below also have zero domain there. For a nonempty h , the first identity holds in every full local factor and both polynomials have degree less than $\deg H$; uniqueness of the remainder proves it. For the second identity, apply $h(D)$ and use the injectivity in (MCF3). Together they prove the third. Monicity of b preserves the highest coefficient of the remainders in the first identity, giving

$$\ell_H \iota_{hH} = \ell_h, \quad A_{a,H} \iota_{hH} = \iota_{hH} A_{a,h}, \quad \Psi_{a,H} \iota_{hH} = \Psi_{a,h}. \quad (\text{MCF27})$$

The action identity follows in the unchanged old jets and the zero added jets. The primitive identity follows from (MCF24), the section identity, and injectivity of Θ . All statements hold coordinatewise in $E_h[A]$ and commute with every coefficient and combined-leg face map. Thus the finite face attachments glue by literal source equalities. Their directed union supplies the coherent s_U used in (MCF18). The decomposition $\mathcal{B}_U = \Theta V \oplus s_U E_U$ follows from surjectivity onto $T_U Q$ and injectivity of each σ_h ; every vector and every finite relation occurs in a common full packet. The original all-polynomial calculation (CAU5)–(CAU13) proves that these packets exhaust $T_U Q$, retaining every local zero order.

112.6 The original negacyclic masks and coefficient Frobenius

At its finite arithmetic-algebra vertex, the complete independent carrier is the original

$$D_{E_h,n} = G(E_h)^n, \quad B_{h,n} = E_h[t]/(t^n + 1), \quad \pi_n(c) = \sum_{i=0}^{n-1} p(c_i)t^i. \quad (\text{MCF28})$$

The variable t here is the additional coefficient variable of this finite observation. The spectral variable in E_h is still s . The negacyclic product, its original unit, and its mask law are

$$(c \star d)_k = \bigoplus_{i+j=k} c_i d_j \oplus \bigoplus_{i+j=k+n} (-1)^{\bullet} c_i d_j, \quad 1_D = (1^{\bullet}, \tau, \dots, \tau), \quad (\text{MCF29})$$

$$\text{mask}(c + d) = \text{mask}(c) \cup \text{mask}(d), \quad \text{mask}(c \star d) = \text{mask}(c) + \text{mask}(d) \pmod{n}.$$

For the product, an output is supported exactly when at least one pair of supported input indices sums to it. Wrap signs are supported and a nonempty sum of supported summands remains supported even when its coefficient cancels. This proves the stated law with every mixed face. Monic remainder in t gives the unique coefficients of $\pi_n(c)$; the mask then distinguishes each coefficient zero from absence. Thus (mask, π_n) is injective. Its facewise source and cohomology attachment is exactly (MCF9), or (MCF16) when leg masks are also retained. The product in (MCF29) belongs to this finite algebra vertex. The source maps s_h^A have their stated linear types; no unproved ring multiplication on the original source is inserted.

Let q_0 be an odd prime power with $\gcd(q_0, n) = 1$, and put $\kappa(i) = q_0 i \bmod n$ and $\epsilon_i = (-1)^{\lfloor q_0 i/n \rfloor}$. The coefficient Frobenius lift on the entire independent object is

$$c_i \mapsto \epsilon_i^{\bullet} c_i \quad \text{in slot } \kappa(i), \quad A \mapsto \kappa(A). \quad (\text{MCF30})$$

An absent coordinate stays absent. On a combined label it sends S_i to $S'_{\kappa(i)} = S_i$ and multiplies every present source-leg and target amplitude in that coefficient by ϵ_i . It commutes with the differential because it uses the same coefficient sign in each term of that differential. It commutes with face inclusion, the maps p_λ , s_h^A , σ_h^A , and all arithmetic dilations by coordinate permutation and complex linearity.

On the integral coefficient ring $\mathbb{Z}[t]/(t^n + 1)$ this is the matrix of $t \mapsto t^{q_0}$. It is well-defined since $(t^{q_0})^n = -1$, and choosing r with $q_0^r \equiv 1 \pmod{2n}$ proves its r th power is the identity. Reduction to $\mathbb{F}_{q_0}$ is exactly coefficient Frobenius; extension to $\mathbb{C}$ retains the same integral signed-permutation matrix. On supported monomials, reducing the exponent before or after substitution gives the same wrap sign, since q_0 is odd. Distributivity and the absent cases therefore prove multiplicativity on (MCF29), before synchronization. In particular its support action is the exact permutation in (MCF30), rather than an observation of one global Boolean support bit.

The synchronization arrow is also retained explicitly:

$$E_n = (1^{\bullet}, e, \dots, e), \quad E_n D_{E_h, n} \simeq G(B_{h, n}), \quad D_{E_h, n} [E_n^{-1}] \simeq E_n D_{E_h, n}. \quad (\text{MCF31})$$

Its multiplication fills every missing coordinate of an active input with a supported zero while leaving all ordinary coefficients unchanged; every output has a supported summand using any chosen active input coordinate. Hence $E_n^2 = E_n$. A unital map making this idempotent invertible sends it to 1 and identifies c with $E_n c$, proving the localization universal property. If $f = \sum f_i t^i$ and $J(f) = \{i : f_i \neq 0\}$, its fibre over $f^{\bullet}$ consists of one original vector for each nonempty mask A with $J(f) \subseteq A \subseteq I_n$. The fibre over τ is the all-absent vector alone. In particular a supported zero has $2^n - 1$ distinct independent-mask preimages. These exact fibres remain attached; the synchronized observation does not replace (MCF28).

Finally the original arithmetic coefficients have an exact retract

$$\iota_n : E_h \longrightarrow B_{h, n}, \quad f \mapsto f, \quad r_n : B_{h, n} \longrightarrow E_h, \quad r_n(b) = \frac{1}{n} \text{Tr}_{B_{h, n}/E_h}(M_b), \quad r_n \iota_n = 1. \quad (\text{MCF32})$$

In the basis $1, t, \dots, t^{n-1}$, multiplication by t^i has no diagonal entries for $1 \leq i < n$, while multiplication by b_0 has n diagonal entries b_0 . Thus $r_n(\sum b_i t^i) = b_0$ and its kernel is exactly $\bigoplus_{i=1}^{n-1} E_h t^i$. Writing Ψ_{q_0} for (MCF30) on coefficients, it fixes ι_n and preserves r_n . The coefficientwise extension of $A_{a, h}$ commutes with it and intertwines both arrows in (MCF32). On a coefficient eigenspace of

eigenvalue ω , define the combined operator $T_{a,q_0} = A_{a,h}\Psi_{q_0} = \Psi_{q_0}A_{a,h}$ on $B_{h,n}$. Its actual full local action is

$$\omega a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{MCF33})$$

Every ω is a root of unity by the proved finite order, and the original arithmetic action is recovered on the constant-coefficient retract. This completes the attachment of independent mixed coefficient support to the original theta source, its combined leg diagram, and its actual arithmetic operators. All masks, source diagonals, full units, nilpotents, and comparison fibres have their displayed maps.

112.7 The independent outer source label, including empty coefficient support

The original outer source label is (W, S) , with $W \subseteq V$ a complex linear D -invariant subspace and $\emptyset \neq S \subseteq \{+, -\}$. It is independent of the coefficient mask $A \subseteq I_n$. Denote the set of those subspaces W by $\mathcal{W}_D$. Its exact coefficient diagram is

$$\begin{aligned} \mathcal{J}_n^{\text{out}} &= \{(W, S; A) : W \in \mathcal{W}_D, \emptyset \neq S \subseteq \{+, -\}, A \subseteq I_n\}, \\ C_{W,S;A}^0 &= \left(\bigoplus_{+ \in S} W \oplus \bigoplus_{- \in S} \widehat{W} \right)^A, \quad C_{W,S;A}^1 = \mathcal{B}_U^A, \\ d_{W,S;A}(\phi_i, \psi_i)_i &= (\Theta\phi_i - \Theta\psi_i)_i. \end{aligned} \quad (\text{MCF34})$$

The minus leg is exactly $\widehat{W}$; it is not replaced by W unless the two spaces are equal. Its action is $1 - D$, preserving that space by the Fourier identity. The target restriction $\mathcal{B}_U \subseteq \mathcal{B}$ is an injective cochain map to the original outer complex, since $\Theta W \subseteq \Theta V \subseteq \mathcal{B}_U$. For $W \subseteq W', S \subseteq S'$ and $A \subseteq A'$, the transition includes the old source subspaces and coordinates and inserts the typed zeros in new legs and coefficient slots. The degree-one transition is zero insertion. These are cochain maps, by substitution in the displayed differential, and compose literally.

At $A = \emptyset$ this is a zero complex *at its retained outer label* (W, S) . Its single coefficient value has empty coefficient support; the outer leg mask is still S . In the represented carrier one retains all triples $(W, S; A, x)$ and separately adjoins a global external point τ_{out} . In particular

$$(W, S; \emptyset, 0) \neq \tau_{\text{out}}, \quad (W, S; A, 0) \neq (W, S; A', 0) \quad \text{if } A \neq A'. \quad (\text{MCF35})$$

These are explicit distinct elements of the disjoint labelled carrier; they do not assert a nonzero amplitude in a zero complex. Taking a colimit over coefficient masks would identify these zero insertions with their images in the terminal full face. That colimit is not part of this construction.

Define $Q_{W,U} = \mathcal{B}_U / \Theta W$ and $\eta_W : V/W \rightarrow Q_{W,U}$ by $[v] \mapsto [\Theta v]_W$. The complete arithmetic quotient and kernel are

$$\begin{aligned} 0 \longrightarrow (V/W)^A &\xrightarrow{\eta_W^A} Q_{W,U}^A \xrightarrow{\pi_W^A} (T_U Q)^A \longrightarrow 0, \\ H^0(C_{W,S;A}) &= \begin{cases} W^A, & S = \{+, -\}, \\ 0, & |S| = 1, \end{cases} \quad H^1(C_{W,S;A}) = Q_{W,U}^A. \end{aligned} \quad (\text{MCF36})$$

For the kernel sequence, injectivity of Θ proves that $\Theta v \in \Theta W$ exactly when $v \in W$. A class of $\mathcal{B}_U / \Theta W$ maps to zero modulo ΘV exactly when it is represented by Θv for some $v \in V$. Surjectivity follows from the definition of $\mathcal{B}_U$. Applying this proof independently in the coordinates proves the exact sequence. For the cohomology calculation, a joint slot has the reversible coordinates (MCF14) with $u, v \in W$, since $\psi \in \widehat{W}$. Its differential is Θu and its diagonal is W . A single-leg differential is injective with image ΘW . This proves every case, including the zero coefficient vector spaces at empty A . For $W \subseteq W'$ the transition on degree one has kernel exactly $(W'/W)^A$ under the same original map $[v] \mapsto [\Theta v]_W$. The outer-label kernel is thus fully retained.

The exact relation to the slot-local diagram is an enlarged index

$$(W, S; A, (T_i)_{i \in A}), \quad \emptyset \neq T_i \subseteq S. \quad (\text{MCF37})$$

Its source uses W on the present plus legs and $\widehat{W}$ on the present minus legs, according to T_i , and its target is $\mathcal{B}_U^A$. Every differential is the original signed theta differential in that coordinate. The outer diagram (MCF34) embeds by $T_i = S$ for every $i \in A$, with the identity on its coefficient spaces. This embedding remains injective on the index when A is empty because W and S are retained separately. Forgetting W, S and including $W, \widehat{W}$ into V gives the explicit cochain comparison to (MCF12); its target map is the identity on $\mathcal{B}_U^A$. Its induced map on H^1 is precisely π_W^A in (MCF36), and its induced map on H^0 is the diagonal inclusion $W^{\{i:T_i=\{+,-\}\}} \subseteq V^{\{i:T_i=\{+,-\}\}}$. For fixed $W = V$, forgetting only the outer S has exactly three labelled preimages over the empty slot-local mask, two when the union of the nonempty T_i is a single leg, and one when that union contains both legs. Indeed these are exactly the nonempty subsets $S \subseteq \{+, -\}$ containing $\bigcup_i T_i$. This computes the lost outer information of that specified forgetting map; its empty coefficient value is not identified with global absence in (MCF35).

There is a useful original-outer-source retract, which also checks the empty coefficient endpoint without changing the label. At a fixed (W, S) , write $C_{W,S}^U$ for its one-coefficient original complex with central term $\mathcal{B}_U$, and take the constant coefficient map

$$c_0 : C_{W,S;A} \longrightarrow C_{W,S}^U, \quad c_0(x) = \begin{cases} x_0, & 0 \in A, \\ 0, & 0 \notin A, \end{cases} \quad i_0 : C_{W,S}^U \longrightarrow C_{W,S;\{0\}}, \quad x \longmapsto (x)_0. \quad (\text{MCF38})$$

Here i_0 includes the original outer complex as the coefficient-face vertex $A = \{0\}$ in the full represented diagram. This notation does not assert an inclusion into every other face. Each c_0 is a cochain map and commutes with all coefficient zero insertions, since a newly inserted coordinate has zero amplitude. On the represented carrier it sends $(W, S; \emptyset, 0)$ to the supported outer zero at (W, S) . On global τ_{out} the pointed extension has value τ_{out} . Its composite with the selected i_0 is the identity on the original outer coefficient amplitude and label. The coefficient Frobenius fixes index zero and its sign is one, so $c_0 \Psi_{q_0} = c_0$. This proves an actual outer-label-preserving comparison, rather than identifying the original outer label with a slot-local mask.

The finite arithmetic section at a proper outer source is $\sigma_{h,W}(u) = [s_h u]_W \in Q_{W,U}$. It is injective because $\pi_W \sigma_{h,W} = \sigma_h$ and σ_h is injective. Its exact retained defects are

$$\begin{aligned} D\sigma_{h,W} - \sigma_{h,W} A_h &= \eta_W([\phi_* \ell_h(\cdot)]_W), \\ \overline{\mathcal{R}}_a^{W \rightarrow \mathcal{R}_a W} \sigma_{h,W} - \sigma_{h,\mathcal{R}_a W} A_{a,h} &= \eta_{\mathcal{R}_a W}([\Psi_{a,h}(\cdot)]_{\mathcal{R}_a W}). \end{aligned} \quad (\text{MCF39})$$

The first is (MCF5) in the quotient by ΘW and the second is (MCF24) in the quotient by $\Theta \mathcal{R}_a W$. Dilation transports the outer label to $\mathcal{R}_a W$, as required by (MCF19); it need not fix an arbitrary D -invariant W . On the minus source it maps $\widehat{W}$ to $a\mathcal{R}_{1/a}\widehat{W} = \overline{\mathcal{R}_a W}$. This proves that every arrow in (MCF39) has the stated domain and codomain. The original primitives belong to V ; their possibly nonzero classes in V/W or $V/\mathcal{R}_a W$ are retained in (MCF36). No primitive in V has been declared to be a boundary in a smaller source W . For the full outer source $W = V$, both kernel classes are zero and (MCF39) recovers the proved full-source intertwining.

Apply (MCF39) coordinatewise on every A , keeping all outer and slot-local labels. Packet inclusion, zero insertion, and coefficient Frobenius commute with these defects by (MCF27), linearity and the permutation (MCF30). In particular the complete retained data form the commuting represented diagram

$$\{C_{W,S;A}\}_{W,S,A} \longrightarrow \{Q_{W,U}^A[-1]\}_{W,S,A} \xrightarrow{\{\pi_W^A\}} \{(T_U Q)^A[-1]\}_{W,S,A}, \quad \ker \pi_W^A = \eta_W^A((V/W)^A), \quad (\text{MCF40})$$

with the enriched slot-local faces and their inclusion and forgetting maps specified in (MCF37). The first arrow is the actual cochain map to $Q_{W,U}^A[-1]$ given by zero in degree zero and the original quotient in degree one. Its exact kernel complex is also retained:

$$K_{W,S;A} = [C_{W,S;A}^0 \xrightarrow{d} (\Theta W)^A], \quad 0 \longrightarrow K_{W,S;A} \longrightarrow C_{W,S;A} \longrightarrow Q_{W,U}^A[-1] \longrightarrow 0. \quad (\text{MCF40a})$$

Degree-wise kernel computation proves this short exact sequence. The single-leg differential onto ΘW is an isomorphism, and the joint differential in (MCF14) has kernel the full W diagonal. Thus $K_{W,S;A}$ retains exactly the degree-zero diagonal specified in (MCF36), including its zero-insertion and source-inclusion maps. The original outer source, every coefficient mask including the empty one, every present source leg, and the exact arithmetic comparison kernel are simultaneously specified by this diagram.

112.8 Exact enlargement from offcritical support to all nontrivial packets

The offcritical region U throughout the preceding construction remains unchanged. A packet containing critical-line zeros has a larger original source domain, which we now attach by its exact inclusions. Put

$$\begin{aligned} \Omega &= \{s \in \mathbb{C} : 0 < \Re s < 1\}, & L &= \Omega \setminus U = \{s : \Re s = \tfrac{1}{2}\}, \\ T_X Q &= \bigcup_{p \in \mathcal{P}_X} \ker(p(D) : Q \rightarrow Q), & \mathcal{B}_X &= q^{-1}(T_X Q), & C_X &= [V \xrightarrow{\Theta} \mathcal{B}_X] \quad (X = U, \Omega, L), \\ \mathcal{P}_X &= \{p \text{ monic} : \text{every root of } p \text{ belongs to } X\}, & C_+ &= [V \xrightarrow{\Theta} \mathcal{B}]. \end{aligned} \quad (\text{MCF41})$$

The polynomial 1 is in every $\mathcal{P}_X$. The notation C_L denotes an algebraic source complex; no openness of the line L or sheaf restriction to an open L is required. The same polynomial range identities (MCF3) hold for every root in the open strip Ω , including its critical line.

The exact source inclusions, with their original degree-zero maps, are

$$C_U[A] \xrightarrow{(1, \text{inc})} C_\Omega[A] \xrightarrow{(1, \text{inc})} C_+[A]. \quad (\text{MCF41a})$$

Indeed $\mathcal{P}_U \subseteq \mathcal{P}_\Omega$ gives $T_U Q \subseteq T_\Omega Q$ and then $\mathcal{B}_U \subseteq \mathcal{B}_\Omega \subseteq \mathcal{B}$. All differentials are the same original Θ and the degree-zero maps are the identity on V^A , proving both cochain identities. Injectivity holds in each degree. Cohomology in degree zero is zero by injectivity of Θ , and the induced degree-one sequence is the literal sequence of submodules $(T_U Q)^A \hookrightarrow (T_\Omega Q)^A \hookrightarrow Q^A$.

Its full quotient has the exact arithmetic meaning

$$T_\Omega Q = T_U Q \oplus T_L Q, \quad T_\Omega Q / T_U Q \xrightarrow{\sim} T_L Q. \quad (\text{MCF41b})$$

Here is a direct proof without presuming a list of existing zeros. For $u \in T_\Omega Q$, choose an annihilator p in $\mathcal{P}_\Omega$ and factor it as $p = p_U p_L$, retaining every multiplicity, with roots of p_U in U and roots of p_L in L . Either factor can be 1. The factors have disjoint roots, so there are polynomials a, b with $ap_U + bp_L = 1$. Then

$$u_U = b(D)p_L(D)u, \quad u_L = a(D)p_U(D)u, \quad u = u_U + u_L, \quad (\text{MCF41c})$$

and multiplication shows $p_U(D)u_U = p_L(D)u_L = 0$. If a vector is killed by a polynomial rooted in U and by a polynomial rooted in L , the same Bezout identity makes that vector zero. Thus the two submodules intersect in zero and the decomposition is unique, independent of the chosen annihilator and Bezout polynomials. This proves (MCF41b), including its inverse: include $T_L Q$ in $T_\Omega Q$ and take its class modulo $T_U Q$.

There is an exact original Mellin description of this quotient as well:

$$T_X Q \xrightarrow{\sim} \bigoplus_{\substack{\rho \in X \\ g(\rho)=0}} \mathcal{O}_\rho / (g), \quad [F] \mapsto ([\mathcal{M}F]_\rho)_\rho, \quad X = U, L, \Omega. \quad (\text{MCF41d})$$

To check the map, write $p(D)F = \Theta\phi$. Then $p(s)\mathcal{M}F(s) = g(s)H_\phi(s)$, so the Mellin class is zero at every zero of g outside the finite root set of p . Replacing F by a theta boundary changes its Mellin transform by gH and does not change a local class. For injectivity, suppose all local classes at roots of p vanish. At a root with g nonzero, g is a local unit; at a root of g , the vanishing class says that $\mathcal{M}F$ is a multiple of g . In either case write $\mathcal{M}F = gB$ locally. Cancellation in the local analytic domain gives $H_\phi = pB$, so $j_p H_\phi = 0$ at every required order. By (MCF3), $\phi = p(D)\psi$ with $\psi \in V$. Injectivity of $p(D)$ on $\mathcal{B}$ then yields $F = \Theta\psi$.

For surjectivity, take a finitely supported family of full local jets α_ρ at actual zeros in X . For its finite support Z put $h = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$ with the complete actual orders and choose the unique polynomial P of degree less than $\deg h$ with $j_h P = j_h (g/h)^{-1} \alpha$. The complete unit exists at each specified centre and CRT supplies that polynomial. The original Euler range identity gives

$$F_\alpha = S_h^{\mathcal{B}} \Theta(P(D)\phi_*), \quad \mathcal{M}F_\alpha = (g/h)P, \quad h(D)F_\alpha \in \Theta V. \quad (\text{MCF41e})$$

These equations give the specified full jets and zero classes at all other zeros of g . The empty family uses the stage $h = 1$ and the zero class. Changing P by hR changes F_α by the original boundary $\Theta(R(D)\phi_*)$. This proves both inverse identities in (MCF41d), retaining the entire g/h , every coordinate $s - \rho$, and every multiplicity. Under this isomorphism the decomposition (MCF41b) is exactly zero insertion and restriction to the disjoint offcritical and critical-line sets of full local factors.

Degree-wise quotients of the actual complexes now give

$$\begin{aligned} 0 \longrightarrow C_U[A] \longrightarrow C_\Omega[A] \longrightarrow (T_\Omega Q / T_U Q)^A[-1] \longrightarrow 0, \\ 0 \longrightarrow C_\Omega[A] \longrightarrow C_+[A] \longrightarrow (Q / T_\Omega Q)^A[-1] \longrightarrow 0. \end{aligned} \quad (\text{MCF41f})$$

Their degree-zero quotients vanish because both inclusions are the identity on V^A . For the first degree-one quotient, the map $\mathcal{B}_\Omega / \mathcal{B}_U \rightarrow T_\Omega Q / T_U Q$ induced by q is onto, and has zero kernel because $qF \in T_U Q$ is exactly $F \in \mathcal{B}_U$. The second quotient is proved by the identical calculation with $\mathcal{B} / \mathcal{B}_\Omega$ and $Q / T_\Omega Q$. These prove the displayed short exact sequences on the original source domains. In particular a nonzero critical local factor survives in the first quotient; it has not been discarded by working with the original offcritical subcomplex.

For every actual nontrivial full packet h in Ω , including a critical pair or a mixture of critical and offcritical centres, the formulas (MCF4)–(MCF5) and (MCF22)–(MCF27) have their unchanged values and their source now lies in $\mathcal{B}_\Omega$ because $h(D)s_h u \in \Theta V$. Their proofs used (MCF3) in the open strip and the full local unit; they did not use $\Re \rho \neq 1/2$. Thus they apply to that precise larger domain. Their restriction to packets rooted in U agrees literally with the earlier formulas through (MCF41a). Every packet system includes $h = 1$ with $E_1 = 0$; no nonempty offcritical packet is assumed to make the enlarged construction or its embeddings.

Finally retain every coefficient and outer label in this enlargement. Replacing only the central term of (MCF34) by $\mathcal{B}_X^A$ defines $C_{W,S;A}^X$ and $Q_{W,X} = \mathcal{B}_X / \Theta W$. The maps from $X = U$ to $X = \Omega$ are the identity on the original source legs and the inclusion on the central term, and their degree-one quotient is again $(T_\Omega Q / T_U Q)^A$:

$$0 \longrightarrow Q_{W,U}^A \longrightarrow Q_{W,\Omega}^A \longrightarrow (T_\Omega Q / T_U Q)^A \longrightarrow 0. \quad (\text{MCF41g})$$

Indeed $\Theta W \subseteq \mathcal{B}_U$; quotienting the two nested central subspaces by that same subspace proves injectivity and the stated quotient. The proper-source kernel $(V/W)^A$ of (MCF36) remains attached

on each side. Empty coefficient support still retains (W, S) as in (MCF35). The enlarged maps commute with zero insertion, the specified slot-local comparisons, dilation and its proper-source transport, and coefficient Frobenius, since their source maps are identities and their central maps are the original coordinatewise inclusions. This attaches all actual nontrivial packets while preserving the original offcritical construction and its full comparison quotient.

112.9 Literal comparison with the earlier finite source and its cocycle

The earlier source (TC1) writes the spectral polynomial coordinate as t and the same residue representative as $R_Z(u)$; the coordinate isomorphism sends t to s , with $h_Z(t)$ sent to $h(s)$, every power $(t-\rho)^r$ sent to $(s-\rho)^r$, and the full unit unchanged. Consequently $R_Z(u)$ is $P_h(u)$ under this specified isomorphism. For the actual source functions, not only their quotient classes,

$$R_{\text{ref}} = s_h : E_h \longrightarrow \mathcal{B}_X, \quad X = U \text{ or } \Omega \text{ as the packet requires.} \quad (\text{MCF42})$$

Indeed applying $h(D)$ to either representative gives $\Theta(P_h(u)(D)\phi_*)$. The difference lies in $\mathcal{B}$ and is killed by the injective $h(D)$, hence vanishes. This proves the asserted equality with the original units and domains.

Put $t = \log a$, retaining its orientation when $a < 1$, and define

$$c_{t,h}(u) = \int_0^t \mathcal{R}_{e^{t-r}\phi_*} \ell_h(e^{rA_h}u) dr : E_h \longrightarrow V. \quad (\text{MCF43})$$

The integral converges in the original Schwartz topology. In fact on the compact interval joining 0 and t , every derivative of $\mathcal{R}_{e^{t-r}\phi_*}$ is a continuously varying rescaled derivative of the same Schwartz function, with the scaling parameter in a compact subset of $(0, \infty)$. For every pair N, k , its seminorm $\sup_x (1 + |x|)^N |\partial_x^k(\cdot)|$ is uniformly bounded there. The finite matrix e^{rA_h} is smooth and bounded there too. Riemann sums therefore converge in every seminorm in the complete Schwartz space; parity and the two zero moments pass to the limit by continuity, so the limit is in V .

Differentiate the actual $\mathcal{B}$ -valued function $\mathcal{R}_{e^{t-r}s_h} e^{rA_h}u$. Its derivative is $-\mathcal{R}_{e^{t-r}}(Ds_h - s_h A_h) e^{rA_h}u$. The same seminorm argument on $\mathcal{B}$, followed by (MCF5), allows the fundamental theorem of calculus. It gives

$$\Theta_{c_{\log a,h}} = \mathcal{R}_a s_h - s_h A_{a,h}, \quad \Psi_{a,h} = c_{\log a,h}. \quad (\text{MCF44})$$

The second equality follows by comparing with (MCF24) and canceling the injective map $\Theta : V \rightarrow \mathcal{B}$. Thus the previous oriented integral and the Euler-inverse primitive are the same map. Their packet and coefficient-face naturality are literal, by (MCF27) and coordinatewise integration. The joint primitive is $(c/2, -\widehat{c}/2)$; the difference between the two single-leg primitives $(c, 0)$ and $(0, -\widehat{c})$ is the full cycle $(c, \widehat{c})$. No diagonal is discarded by this comparison.

The earlier dual two-leg complex is obtained by applying algebraic $\text{Hom}_{\mathbb{C}}(-, \mathbb{C})$, so the map induced by $C_U \rightarrow C_+$ is restriction $\mathcal{B}^\vee \rightarrow \mathcal{B}_U^\vee$ in degree -1 and identity on both $V^\vee$ source coordinates in degree zero. Its differential, in the same order of legs, is

$$\ell \longmapsto (\ell\Theta, -\ell\Theta\mathcal{F}), \quad \mathcal{F}\phi = \widehat{\phi}. \quad (\text{MCF45})$$

Because $\Theta V \subseteq \mathcal{B}_U$, restriction followed by this differential equals the differential followed by the identity. Thus this is an actual cochain map. It induces restriction $Q^\vee \rightarrow (T_U Q)^\vee$ on degree -1 cohomology. At a joint source the two diagonal parameters $v \mapsto (v, \mathcal{F}v)$ and $w \mapsto (\mathcal{F}w, w)$ agree under $w = \mathcal{F}v$. Define $\mathcal{F}^\vee : V^\vee \rightarrow V^\vee$ by $\mathcal{F}^\vee(\ell) = \ell \circ \mathcal{F}$; the maps $\pi_v, \pi_w : V^\vee \oplus V^\vee \rightarrow V^\vee$ are

$$\pi_v(\alpha, \beta) = \alpha + \beta\mathcal{F}, \quad \pi_w(\alpha, \beta) = \alpha\mathcal{F} + \beta, \quad \pi_v = \mathcal{F}^\vee \circ \pi_w. \quad (\text{MCF46})$$

These identities follow by evaluation on the displayed diagonals and $\mathcal{F}^2 = 1$. The original actions intertwine because $\mathcal{F}\mathcal{R}_a = a\mathcal{R}_{1/a}\mathcal{F}$. For a proper source W , dualizing $W \hookrightarrow V$ gives restriction of diagonal functionals to W . For coefficient faces, dualizing zero insertion gives coordinate restriction. These are algebraic dual statements; no continuous splitting for arbitrary nonclosed W is asserted.

112.10 The shifted coefficient quotient on the original theta complexes

Fix one of the actual full packets h in Ω and a member ρ with $m = m_\rho \geq 2$. The calculation is indexed by precisely those members; it requires no assertion that the indexing set is nonempty. The coordinate isomorphism from the x presentation in (MW23) to this s presentation sends x to s and $e_\rho(x - \rho)^r$ to $e_\rho(s - \rho)^r$, with coefficient one and inverse $s \mapsto x$. It carries the full defining polynomial, its CRT idempotents, and every retained jet order. Use the original local subspace $E_\rho = e_\rho E_h$, $J_\rho = (s - \rho)^2 E_\rho$, and $C_\rho = E_\rho / J_\rho$ of (MW23). Write σ_ρ for the restriction of the original injective $\sigma_h : E_h \rightarrow Q$. Define actual subspaces of the original central source by

$$\mathcal{B}_\rho = q^{-1}(\sigma_\rho E_\rho), \quad \mathcal{B}_{J_\rho} = q^{-1}(\sigma_\rho J_\rho), \quad \Theta V \subseteq \mathcal{B}_{J_\rho} \subseteq \mathcal{B}_\rho \subseteq \mathcal{B}_\Omega. \quad (\text{MCF47})$$

The last inclusion follows because every displayed quotient class lies in the $h(D)$ -torsion, and h is rooted in Ω . The literal direct-sum equalities are

$$\mathcal{B}_\rho = \Theta V \oplus s_h E_\rho, \quad \mathcal{B}_{J_\rho} = \Theta V \oplus s_h J_\rho. \quad (\text{MCF48})$$

For the first, if $qF = \sigma_\rho x$, then $F - s_h x \in \Theta V$; injectivity of σ_ρ makes this x unique. If $\Theta v + s_h x = 0$, quotienting gives $x = 0$, then injectivity of Θ gives $v = 0$. The same argument with $x \in J_\rho$ proves the second equality. Every representative here is the original $s_h x = P_h(x)(D)F_h$ with the full unit $j_h(g/h)^{-1}$, including its components at all selected centres.

There is an actual short exact sequence of source complexes

$$0 \longrightarrow [V \xrightarrow{\Theta} \mathcal{B}_{J_\rho}] \xrightarrow{(1, \text{inc})} [V \xrightarrow{\Theta} \mathcal{B}_\rho] \longrightarrow C_\rho[-1] \longrightarrow 0. \quad (\text{MCF49})$$

The final map is zero in degree zero. In degree one it sends F to $[x]$, where the unique $x \in E_\rho$ satisfies $qF = \sigma_\rho x$. It is linear, vanishes on ΘV , is onto because $s_h x$ is a representative of every $[x]$, and has kernel exactly $\mathcal{B}_{J_\rho}$. These assertions prove both the cochain identity and degreewise exactness. Its inverse on the degree-one quotient sends $[x]$ to $[s_h x] \in \mathcal{B}_\rho / \mathcal{B}_{J_\rho}$; replacing x by $x + j$, $j \in J_\rho$, changes its representative by the actual element $s_h j \in \mathcal{B}_{J_\rho}$. Both inverse identities follow from (MCF48). Thus the length-two quotient used in (MW24)–(MW27) and (MRE34)–(MRE42) has this explicit original-source realization.

The arithmetic actions in this exact sequence also have their original types. Both E_ρ and J_ρ are invariant under $A_h, N_\rho = A_h - \rho I$ on E_ρ , and each $A_{a,h}$. The full-source intertwining (MCF5), (MCF21) makes $D, D - \rho$, and $\mathcal{R}_a$ preserve the corresponding spaces in (MCF47). Their maps on degree zero are respectively $D, D - \rho$, and $\mathcal{R}_a$ on V , and their maps on degree one use the same original operators on functions. Since each commutes with Θ , they are cochain maps. Under the final map in (MCF49) they give exactly the quotient $\bar{A}_h, \bar{N}_\rho$, and $\bar{A}_{a,h}$ of (MW25), with every surviving coefficient retained. The full higher powers remain in the original source operators and in the subcomplex on the left.

The induced filtration is realized before taking cohomology. For every integer j , set

$$\begin{aligned} \mathcal{B}_{\rho,j} &= q^{-1}(\sigma_\rho(M_j E_\rho)), & \mathcal{B}_{J_\rho,j} &= q^{-1}(\sigma_\rho(J_\rho \cap M_j E_\rho)) = \mathcal{B}_{J_\rho} \cap \mathcal{B}_{\rho,j}, \\ 0 \longrightarrow [V \xrightarrow{\Theta} \mathcal{B}_{J_\rho,j}] &\longrightarrow [V \xrightarrow{\Theta} \mathcal{B}_{\rho,j}] \longrightarrow \bar{M}_j C_\rho[-1] \longrightarrow 0. \end{aligned} \quad (\text{MCF50})$$

The intersection equality uses injectivity of σ_ρ . For the exact sequence, the last map of (MCF49) sends $\mathcal{B}_{\rho,j}$ onto $\pi_\rho(M_j E_\rho)$, because the representatives $s_h x$ exist for every $x \in M_j E_\rho$. Its kernel is precisely the displayed intersection. The degree-zero map remains identity and the last differential is zero, proving every case. Consequently the quotient filtration is exactly (MW24), with degrees $m - 1$ and $m - 3$, hence with its retained shift $m - 2$ relative to the displayed length-two centered filtration.

For all sufficiently negative j , the two central terms in (MCF50) equal ΘV . Their source complexes then remain $[V \xrightarrow{\Theta} \Theta V]$, with its actual invertible differential; no assertion that these source complexes are literally zero is made. They have zero cohomology because Θ is injective and onto its

displayed image. At every step the original D preserves $\mathcal{B}_{\rho,j}$, and $D - \rho$ sends it into $\mathcal{B}_{\rho,j-2}$ since its quotient action is precisely N_ρ . This proves the filtered source-map types as well as the induced filtered cohomology identities.

For each proper original $W \subseteq V$, replace the degree-zero term in (MCF49)–(MCF50) by the outer source $\bigoplus_{+ \in S} W \oplus \bigoplus_{- \in S} \widehat{W}$ with the original signed theta differential. Use that identical source in the left and middle complexes. The same proof of degreewise exactness applies, and in degree one their individual cohomology groups have the exact sequences

$$\begin{aligned} 0 \longrightarrow V/W &\longrightarrow \mathcal{B}_{\rho,j}/\Theta W \longrightarrow M_j E_\rho \longrightarrow 0, \\ 0 \longrightarrow V/W &\longrightarrow \mathcal{B}_{J_\rho,j}/\Theta W \longrightarrow J_\rho \cap M_j E_\rho \longrightarrow 0. \end{aligned} \tag{MCF51}$$

In both lines $[v] \mapsto [\Theta v]_W$ is the original kernel map, and the last map is $\sigma_\rho^{-1}q$ on its indicated image. Injectivity, surjectivity, and the kernel computation are the same as in (D11). Thus the low-filtration degree-one cohomology is V/W at a proper source, while the joint degree-zero diagonal is W ; both remain present. For D use $DW \subseteq W'$ and the specified source codomain W' as in (D14). For $D - \rho$ use $(D - \rho)W \subseteq W'$; its minus source operator is $1 - D - \rho$, because $\mathcal{F}(D - \rho) = (1 - D - \rho)\mathcal{F}$. The choice $W' = W + DW$ admits both maps. Dilation uses $\mathcal{R}_a W$. Apply the same proof independently at every coefficient mask, including its retained empty outer label. Zero insertion and coefficient Frobenius commute with the displayed quotient maps by linearity and their original signed coordinate permutation. This transports the strict shifted coefficient quotient to the original mixed source system without removing its actual comparison kernels.

113 The empty-packet endpoint of the exact source diagram

The original mixed coefficient attachment includes the empty packet $h = 1$, with $E_1 = 0$ and $\mathcal{B}_1 = \Theta V$. Its section, arithmetic-action and primitive maps on E_1 are the unique zero maps. To include this endpoint in the named source functions of (MCF4)–(MCF5) and (MCF26), define

$$F_1 = \Theta\phi_*, \quad 1(D)F_1 = \Theta\phi_*, \quad \mathcal{M}F_1 = g. \tag{EPE1}$$

These identities follow from the original theta seed equation; the first is forced by the second because $1(D)$ is the identity. For a nonempty full packet H , the defining equation $H(D)F_H = \Theta\phi_*$ consequently gives

$$(H/1)(D)F_H = F_1. \tag{EPE2}$$

For $H = 1$ this is the identity on the same source function. Every section, action and primitive compatibility with domain E_1 is the equality of the unique zero maps. Unit inverses and the nonempty local jet and polynomial-remainder formulas are used only at nonempty stages. Thus no unit inverse in the zero-vector-space stage is needed. The source inclusion at this stage is exactly $\Theta V \hookrightarrow \mathcal{B}_H$, not a new packet class. This makes the original empty endpoint explicit without assuming that the offcritical packet category has a nonempty member.

The historical MCF bytes are preserved in this reader. Equations (EPE1)–(EPE2) supply their explicit endpoint convention, also accepted by the author of the coordinating source.

114 The exact original cochain attachment on every independent coefficient face

This attachment supplies the cochain, source-section, balancing, and tensor arrows for the independent coefficient faces of the original mixed carrier. The coefficient mask and the original two-leg mask are separate indices. The finite mixed semiring multiplication remains the original algebra observation; the original source spaces acquire no additional multiplication. All tensor products below

are algebraic unless a topology is explicitly specified. We retain the original maps of (AG1)–(AG18), their analytic comparison in (CAU.9)–(CAU.20), and their two-leg source attachment in (CAU.34)–(CAU.37).

114.1 Original spaces, independent faces, and their exact cochain maps

Retain

$$\begin{aligned}
 V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \quad \phi(0) = 0, \quad \int_{\mathbb{R}} \phi(x) dx = 0\}, \\
 \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \quad (b \in \mathbb{Z}, j \geq 0)\}, \\
 D &= -x\partial_x, \\
 \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \\
 Q &= \mathcal{B}/\Theta V, \quad q : \mathcal{B} \rightarrow Q, \quad U = \{s : 0 < \Re s < 1, \Re s \neq \tfrac{1}{2}\}, \\
 T_U Q &= \{u \in Q : p(D)u = 0 \text{ for some } p \in \mathcal{P}_U\}, \\
 \mathcal{B}_U &= q^{-1}(T_U Q), \quad C_U = [V \xrightarrow{\Theta} \mathcal{B}_U].
 \end{aligned} \tag{CFA.1}$$

Here $\mathcal{P}_U$ is the set of monic polynomials all of whose roots lie in U , including 1. The last complex is in degrees zero and one. Products of annihilating polynomials prove that $T_U Q$ and $\mathcal{B}_U$ are vector spaces stable under D . The original theta map is injective. Its image lies in $\mathcal{B}_U$, and the definition of q proves the actual sequence

$$0 \longrightarrow V \xrightarrow{\Theta} \mathcal{B}_U \xrightarrow{q} T_U Q \longrightarrow 0. \tag{CFA.2}$$

We use $A_{\text{sp}} = \mathbb{C}[x]$ for spectral polynomials, with x acting by D . This is the original Euler polynomial algebra $\mathbb{C}[t]$ under its explicit isomorphism $t \mapsto x$. The cyclic coefficient variable t in the separate finite algebra $E_h[t]/(t^n + 1)$ is never substituted for the spectral variable x .

Fix $n \geq 1$, put $I_n = \{0, \dots, n-1\}$, and let $A \subseteq I_n$ be any coefficient mask, including the empty mask. For a vector space M put $M[A] = \bigoplus_{i \in A} M$. Its labelled carrier element $(A, (m_i))$ means the tuple with m_i in active slot i and external τ in each inactive slot. Thus the zero of this additive face group is e_A , supported exactly on A ; the empty face has the single wholly absent coefficient tuple. For $A \subseteq B \subseteq I_n$ define

$$z_A^B : M[A] \longrightarrow M[B], \quad (z_A^B m)_i = \begin{cases} m_i, & i \in A, \\ 0_M, & i \in B \setminus A. \end{cases} \tag{CFA.3}$$

This is linear. For $A \subseteq B \subseteq C$, each coordinate of $z_B^C z_A^B$ is its original value if it lies in A and is zero otherwise, proving $z_B^C z_A^B = z_A^C$. On labelled carriers (CFA.3) sends e_A to e_B . In particular, when A is empty and B is nonempty it sends the wholly absent coefficient tuple to a supported zero tuple. It is a support-changing map of face groups, not a globally τ -preserving pointed lift.

For any linear map $f : M \rightarrow N$, define $f[A]$ by applying f in each active coordinate. On coordinates in A , both composites below are $f(m_i)$; on coordinates in $B \setminus A$, both are zero because $f(0_M) = 0_N$. Thus

$$f[B] z_A^B = z_A^B f[A]. \tag{CFA.4}$$

This proves naturality on every face, including the empty one. The separate pointed coordinate map $\prod_{i \in I_n} G(f)$ preserves the mask exactly: it sends $m_i^\bullet$ to $f(m_i)^\bullet$, even when $f(m_i) = 0$, and sends an absent coordinate to an absent coordinate. Its inverse image of the supported zero on face A is exactly $(\ker f)[A]$. These statements apply in the actual scalar categories of q , the Mellin maps, the finite sections, the quotient inclusions, and the analytic comparison maps below.

The face diagram has a terminal vertex I_n . Its colimit as a diagram of vector spaces is $M[I_n] = M^n$: every compatible family of maps is uniquely its map from this terminal vertex. Such a colimit identifies each zero face with its full-face image. The independent mixed carrier is instead the labelled disjoint union of the faces. Its simultaneous amplitude and mask determine every coordinate, since they distinguish supported zero from absence. No colimit over the arrows (CFA.3) is taken in this attachment.

The coefficient mask is independent of the original source-leg mask $S \subseteq \{+, -\}$, with S nonempty. For a D -invariant original subspace $W \subseteq V$, give the plus leg W the action D and the minus leg $\widehat{W}$ the action $1 - D$, where the Fourier kernel is $e^{-2\pi i x \nu}$. The original complex $C_{W,S}$ has central term $\mathcal{B}$ in degree one, the indicated present legs in degree zero, and differential $d(\phi, \psi) = \Theta\phi - \Theta\widehat{\psi}$ with only present terms included. Define its coefficient face by

$$C_{W,S}[A] = \bigoplus_{i \in A} C_{W,S}, \quad C_U[A] = [V[A] \xrightarrow{\Theta[A]} \mathcal{B}_U[A]]. \quad (\text{CFA.5})$$

The differential acts independently on every active coordinate. Hence it commutes with (CFA.3); the same coordinate check proves that all source inclusions $W \subseteq W'$ and leg inclusions $S \subseteq S'$ commute with these face maps. The source inclusion is inclusion on each present degree-zero leg and the identity on the central term. Its cochain identity is the equality of the indicated signed theta terms before and after inclusion.

For the joint leg mask, set $u = \phi - \widehat{\psi}$ and $v = (\phi + \widehat{\psi})/2$. Then $\phi = v + u/2$ and $\psi = \widehat{v} - \widehat{u}/2$, and the differential is Θu . Injectivity of Θ gives a full diagonal copy of W in degree zero and $Q_W = \mathcal{B}/\Theta W$ in degree one. For a plus-only leg injectivity of Θ gives zero in degree zero; for a minus-only leg the Fourier isomorphism and the original minus sign give the same conclusion. In all cases the image is ΘW . Taking active coordinates proves

$$H^1(C_{W,S}[A]) = Q_W[A], \quad H^0(C_{W,S}[A]) = \begin{cases} W[A], & S = \{+, -\}, \\ 0, & |S| = 1. \end{cases} \quad (\text{CFA.6})$$

For (CFA.5)'s arithmetic complex, taking coordinates in (CFA.2) gives

$$0 \rightarrow V[A] \xrightarrow{\Theta[A]} \mathcal{B}_U[A] \xrightarrow{q[A]} (T_U Q)[A] \rightarrow 0. \quad (\text{CFA.7})$$

Every vector in the last kernel has unique original source coordinates under Θ ; this proves exactness directly. The boundary relation on face A is therefore exactly $(\Theta V)[A]$.

For the independent split carrier, the coordinatewise lift of the differential squares to e_A on an element of face A , because the ordinary differential squares to zero without changing the face. Even outside the nonzero ordinary degrees, the split carrier of the zero amplitude space is $G(0)^n = \{\tau, e\}^n$. The supported cycle condition is the equality of the differential with the supported-zero map on that face; it is precisely the ordinary linear cycle condition. The boundary coequalizer identifies exactly pairs whose difference is an ordinary boundary within that same face. Thus (CFA.6)–(CFA.7) compute the supported cohomology without replacing the independent face skeleton by one globally absent zero object.

114.2 The complete original section and its Euler boundary

Let $h(x) = \prod_{\rho \in Z} (x - \rho)^{m_\rho}$ be any nonempty actual finite full-order packet with $Z \subset U$, where $m_\rho = \text{ord}_\rho g$ and

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad E_h = A_{\text{sp}}/(h), \quad d_h = \deg h, \quad A_h = M_x : E_h \rightarrow E_h. \quad (\text{CFA.8})$$

Every original nilpotent coefficient through order $m_\rho - 1$ is retained. The statements on an empty packet use $E_1 = 0$ and zero maps, without asking for an inverse unit in a zero vector space. For a nonempty packet define the original maps of (AG6)–(AG8):

$$\begin{aligned} v_h(s) &= g(s)/h(s), & v_h &= j_h v_h \in E_h, & \varepsilon_h &= v_h^{-1}, \\ \phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & H_{P(D)\phi_*} &= P, & F_h &= S_h^{\mathcal{B}} \Theta \phi_*, \\ h(D)F_h &= \Theta \phi_*, & \mathcal{M}F_h &= v_h, & P_h(u) &= r_h(\varepsilon_h u), & s_h u &= P_h(u)(D)F_h, \\ J_h &= j_h \mathcal{M}, & \sigma_h &= q s_h. \end{aligned} \quad (\text{CFA.9})$$

Here $r_h : E_h \rightarrow \mathbb{C}[x]_{<d_h}$ is the unique monic remainder representative and j_h retains all the specified Taylor jets. The original Euler range identity supplies $S_h^{\mathcal{B}}$ on this input because g has the required zeros. Injectivity of $h(D)$ on $\mathcal{B}$ makes the inverse unique. Full orders give $v_h(\rho) \neq 0$ at every retained centre, so ε_h is the inverse of the full unit jet, including its nilpotent coefficients.

Taking Mellin transforms gives $\mathcal{M}(s_h u) = v_h P_h(u)$, so its h -jet is $v_h \varepsilon_h u = u$. Therefore $J_h s_h = 1$ and s_h is injective. Since $h(D)s_h u = \Theta(P_h(u)(D)\phi_*)$, every image belongs to $\mathcal{B}_U$, with the specified original boundary. Define

$$\ell_h(u) = [x^{d_h-1}]P_h(u), \quad b_h(u) = \phi_* \ell_h(u). \quad Ds_h - s_h A_h = \Theta b_h. \quad (\text{CFA.10})$$

To prove the last identity, the polynomial $xP_h(u) - P_h(A_h u)$ has zero h -remainder and degree at most d_h . Its leading coefficient is $\ell_h(u)$, because the second polynomial has degree less than d_h . Since h is monic, this polynomial is exactly $h\ell_h(u)$. Apply it at D to F_h and use (CFA.9). This proves the entire coefficient, sign, and source boundary.

Applying these maps in every active coordinate gives

$$\begin{aligned} J_h[A]s_h[A] &= 1, & q[A]s_h[A] &= \sigma_h[A], \\ D[A]s_h[A] - s_h[A]A_h[A] &= \Theta[A]b_h[A], & b_h[A](u_i) &= (\phi_* \ell_h(u_i))_{i \in A}. \end{aligned} \quad (\text{CFA.11})$$

The proof is the coordinate identity (CFA.10), and (CFA.4) proves naturality of every map here under every face insertion. Regard $E_h[A]$ as a complex concentrated in degree one. Its map $s_h[A]$ to $C_U[A]$ is a cochain map, since the target differential out of degree one is zero. The map $b_h[A]$ has degree minus one, and (CFA.11) says exactly that the Euler composites differ by the cochain homotopy $db_h[A] + b_h[A]d = db_h[A]$. After $q[A]$ the difference vanishes, proving exact spectral equivariance of $\sigma_h[A]$. Before the quotient the section is a complex-linear observation and is not asserted to be a ring map or an A_{sp} -linear map.

114.3 Full-unit divisor transport on every face

For full packet divisors $h \mid H$, write $b = H/h$; their added and old zero sets are disjoint, so b and h are relatively prime. Retain the restriction and arithmetic zero-extension maps

$$\pi_{Hh} : E_H \rightarrow E_h, \quad \iota_{hH} : E_h \rightarrow E_H, \quad \pi \iota = 1, \quad \iota \pi = M_{e_{hH}}, \quad (\text{CFA.12})$$

where e_{hH} is one on old local factors and zero on the added ones. The polynomial Chinese remainder theorem gives these exact maps: Bezout for h, b constructs the complementary idempotents, and multiplication by them inserts the prescribed old or new jets. Thus π is unital, whereas ι preserves products and sends 1 to e_{hH} . Both are spectral-polynomial linear. The raw numerator map is a different map:

$$\mu_b : [P]_h \mapsto [bP]_H, \quad \mu_b = \iota M_{[b]_h}, \quad \iota = \mu_b M_{[b]_h}^{-1}. \quad (\text{CFA.13})$$

The value on each old factor is multiplication by its full unit $[b]_h$, and the value on each added factor is zero to the full order. These facts prove both equations and well-definedness.

The following are literal original function and polynomial identities:

$$\begin{aligned} v_h &= bv_H, & b(D)F_H &= F_h, & \pi\varepsilon_H &= [b]_h\varepsilon_h, \\ r_H(\varepsilon_H\iota u) &= bP_h(u), & s_H\iota &= s_h, & \sigma_H\iota &= \sigma_h, & \ell_H\iota &= \ell_h. \end{aligned} \quad (\text{CFA.14})$$

For the second function identity, applying $h(D)$ to both sides gives $\Theta\phi_*$, and injectivity of $h(D)$ proves equality. The inverse-unit equation follows by restricting $v_h = bv_H$ and taking the inverse in the old finite quotient. The polynomial $bP_h(u)$ has degree less than $\deg H$. On old factors, $v_H bP_h(u)$ has jet u ; on new factors it has zero jet because $v_H b = v_h$ vanishes to their full orders. Thus its full jet is ιu . Multiplication by the invertible v_H proves that this is the required unique remainder of $\varepsilon_H\iota u$. Applying that polynomial at D to F_H proves the section identity; applying q proves the quotient identity. Monicity of b identifies the top coefficient of $bP_h(u)$ with that of $P_h(u)$, proving the identity for ℓ .

On each coefficient face, (CFA.14) holds in every active coordinate. Equation (CFA.4) then proves all divisor/face squares commute. For a chain of full divisors, restriction and zero insertion compose because their values agree at every local factor; the raw numerator maps compose by multiplication of their literal polynomials. None of these equalities replaces μ_b by ι without its unit. The independent split lift $G(\iota)^n$ preserves every coefficient mask, and its finite mixed unit has image $(e_{hH}^\bullet, \tau, \dots, \tau)$, so it is not falsely declared to preserve the larger packet's unit.

114.4 The exact dilation homotopy on the original source

Let $R_a F(x) = F(x/a)$ for $a > 0$ on both original spaces. Positive dilation preserves evenness and the value at zero, and it multiplies the integral by a ; hence it preserves V . It preserves $\mathcal{B}$ by substituting in each Euler seminorm. Direct substitution in the theta sum gives $\Theta R_a = R_a \Theta$, and in the Mellin integral gives $\mathcal{M} R_a F = a^s \mathcal{M} F$. Write $L = \log a$ and $A_{h,t} = \exp(tA_h)$, so $A_{h,L}$ is the actual finite multiplication by $j_h(a^s)$. Define

$$B_{h,a}(u) = \int_0^L R_{e^{L-t}} \phi_* \ell_h(A_{h,t}u) dt \quad \text{in } V. \quad (\text{CFA.15})$$

This integral is taken with its indicated orientation also for $L < 0$. It is an original Schwartz-space integral. On the compact interval between 0 and L , the dilation parameters range in a compact subset of $(0, \infty)$. For a Schwartz seminorm $p_{r,j}(\phi) = \sup_x |x^r \phi^{(j)}(x)|$, substitution gives $p_{r,j}(R_c \phi) = c^{r-j} p_{r,j}(\phi)$. The same identity with the Euler derivatives of ϕ controls derivatives in t . The finite-dimensional function $\ell_h(A_{h,t}u)$ is smooth. Thus Riemann sums converge in every Schwartz seminorm, defining the integral in the complete Schwartz space. Each integrand has value zero at zero and integral zero; these are continuous functionals on that space, so the result belongs to V .

Differentiation in the same seminorms gives

$$\frac{d}{dt} (R_{e^{L-t}} s_h A_{h,t} u) = -R_{e^{L-t}} (Ds_h - s_h A_h) A_{h,t} u. \quad (\text{CFA.16})$$

Indeed the derivative of $R_{e^{L-t}}$ is $-DR_{e^{L-t}}$, and the derivative of $A_{h,t}$ is $A_h A_{h,t}$. The two terms give the displayed difference. Integrate from 0 to L and use (CFA.10) and $\Theta R_c = R_c \Theta$ to obtain exactly

$$R_a s_h - s_h A_{h,L} = \Theta B_{h,a}. \quad (\text{CFA.17})$$

The commutation with the integral follows from the original continuous theta map, or by its convergent original seminorm estimates on these compact families. This formula includes $a = 1$, when both sides are zero. It is a source-level cochain homotopy, not an assertion that s_h intertwines dilation on its nose. Coordinatewise application and (CFA.4) give this same homotopy naturally on every

face. Applying q gives the exact action relation $\overline{R}_a \sigma_h = \sigma_h A_{h,L}$. All multiplicities are retained: on the ρ factor this action is

$$a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho = M_{x-\rho}. \quad (\text{CFA.18})$$

It is the full Taylor expansion of a^s modulo the actual factor. The divisor identities (CFA.14), their commutation with A_h , and the coordinate formula for the integral also give $B_{H,a} \iota_{hH} = B_{h,a}$. Thus the full source homotopies, not only their quotients, have the exact divisor/face compatibility.

114.5 A strict cochain map that can actually be balanced

The section s_h is not A_{sp} -linear for any nonempty packet. For example take $u = v_h$, so $P_h(u) = 1$ and $h(D)s_h u = \Theta \phi_* \neq 0$, although $hu = 0$ in E_h . Here $\phi_* \neq 0$ and injectivity of Θ prove the nonvanishing. Thus $O \otimes_{A_{\text{sp}}} s_h$ is not a defined tensor arrow. There is a strict replacement with the same original cohomology map. Put

$$e_h = r_h(\varepsilon_h), \quad F'_h = e_h(D)F_h, \quad \phi'_h = e_h(D)\phi_*, \quad P_h^{\text{free}} = [A_{\text{sp}} \xrightarrow{h} A_{\text{sp}}] \quad (\text{CFA.19})$$

with degrees zero and one. Define

$$\kappa_h : P_h^{\text{free}} \longrightarrow C_U, \quad \kappa_h^0(P) = P(D)\phi'_h, \quad \kappa_h^1(P) = P(D)F'_h. \quad (\text{CFA.20})$$

Both maps are A_{sp} -linear because the polynomial actions commute. Moreover $h(D)F'_h = \Theta \phi'_h$, so $\kappa_h^1(hP) = \Theta \kappa_h^0(P)$, proving the cochain identity. For every polynomial P , retain the exact polynomial quotient

$$R_h(P) = \frac{Pe_h - r_h(\varepsilon_h[P]_h)}{h}, \quad \kappa_h^1(P) - s_h([P]_h) = \Theta(R_h(P)(D)\phi_*). \quad (\text{CFA.21})$$

The numerator has zero h -remainder, so the quotient is a polynomial. Applying the numerator at D to F_h and using (CFA.9) proves the displayed equality. The degree-one cohomology of the free source is E_h , and (CFA.21) proves that its induced map is exactly σ_h . Thus no new spectral injection or replacement unit has been substituted.

For every face, take the direct sum of (CFA.20). It commutes with zero insertion in both degrees by (CFA.4) and can be balanced at every analytic stalk $O = \mathcal{O}_{\mathbb{C},s_0}$ by the actual map $x \mapsto s$:

$$[O[A] \xrightarrow{h(s)} O[A]] \xrightarrow{O \otimes \kappa_h[A]} [(O \otimes_{A_{\text{sp}}} V)[A] \xrightarrow{\Theta[A]} (O \otimes_{A_{\text{sp}}} \mathcal{B}_U)[A]]. \quad (\text{CFA.22})$$

The formula is balanced because (CFA.20) is polynomial-linear. The stalk O is torsion-free over the PID A_{sp} and hence flat: it is the directed union of its finite generated torsion-free submodules, each free over the PID; tensoring by free modules and taking filtered colimits preserve exactness. Thus its degree-one map is the actual balanced version of $\sigma_h[A]$. The finite-dimensional section $s_h[A]$ remains a complex-linear observation with (CFA.11) and (CFA.17) as its full source homotopies. This keeps both descriptions connected by (CFA.21), rather than dropping the section when passing to the strict balanced map.

The divisor arrows between these strict free vertices must also be specified. For nonempty full divisors $h \mid H = bh$, put $a_{hH} = r_h([b]_h^{-1})$ and define

$$L_{hH} : P_h^{\text{free}} \longrightarrow P_H^{\text{free}}, \quad L_{hH}^0(P) = a_{hH}P, \quad L_{hH}^1(P) = ba_{hH}P. \quad (\text{CFA.22a})$$

Its cochain identity is $Ha_{hH} = ba_{hH}h$. On degree-one cohomology, the polynomial ba_{hH} equals one at the old factors and vanishes at the added factors, so this map induces exactly ι_{hH} . The original sections commute strictly by (CFA.14), whereas their free resolutions have the explicit homotopy

$$w_{hH} = \frac{a_{hH}e_H - e_h}{h} \in \mathbb{C}[x], \quad K_{hH}^1(P) = P(D)w_{hH}(D)\phi_*, \quad K_{hH}^0 = 0, \quad (\text{CFA.22b})$$

$$\kappa_H L_{hH} - \kappa_h = dK_{hH} + K_{hH}d.$$

The polynomial exists because $[e_H]_h = [b]_h \varepsilon_h$ and $[a_{hH}b]_h = 1$, so its numerator has zero h -remainder. In degree zero the difference is $P(D)(a_{hH}e_H - e_h)(D)\phi_* = K_{hH}^1(hP)$. In degree one, using $b(D)F_H = F_h$, the difference is $P(D)(a_{hH}e_H - e_h)(D)F_h = \Theta(P(D)w_{hH}(D)\phi_*)$. These are exactly the two components of (CFA.22b), proving its sign and degree.

For $h \mid H \mid J$, put $\eta = (a_{hH}a_{HJ} - a_{hJ})/h$. Its numerator vanishes modulo h : both factors give the inverse of J/h there. Thus η is a polynomial, and the degree-minus-one map $T^1 : P_h^{\text{free},1} \rightarrow P_J^{\text{free},0}$ given by multiplication by η satisfies

$$L_{HJ}L_{hH} - L_{hJ} = dT + Td. \quad (\text{CFA.22c})$$

In degree zero its right side is multiplication by $h\eta$, and in degree one it is multiplication by $J\eta$; these are precisely the differences of (CFA.22a)'s two multipliers. Consequently free-resolution divisor composition is retained with its exact homotopy, not asserted to be strict in these remainder coordinates. All multipliers here are polynomial-linear, and the homotopies in (CFA.22b)–(CFA.22c) are A_{sp} -linear. Their direct sums commute with every face insertion and can be balanced just as (CFA.20). The attachment homotopies have the exact further composition identity

$$K_{HJ}L_{hH} + K_{hH} = K_{hJ} + \kappa_J T. \quad (\text{CFA.22d})$$

On a degree-one input its left multiplier on ϕ_* is $ba_{hH}w_{HJ} + w_{hH}$, while the right multiplier is $w_{hJ} + \eta e_J$. Substituting their displayed polynomial definitions makes both equal to $(a_{hH}a_{HJ}e_J - e_h)/h$. All other components are zero by degree. This proves compatibility of the attachment with the composition homotopy itself, with its actual coefficients. The empty finite packet remains its specified zero amplitude observation; its independent split carrier still has every face of $G(0)^n$. The inverse-unit free vertices here are indexed by nonempty packets.

114.6 Every proper source retains its residual class

For any proper D -invariant $W \subseteq V$ the original exact sequence is

$$0 \rightarrow V/W \xrightarrow{[\phi] \mapsto [\Theta\phi]_W} Q_W \rightarrow Q \rightarrow 0, \quad Q_W = \mathcal{B}/\Theta W. \quad (\text{CFA.23})$$

The first map is well-defined. If $\Theta\phi \in \Theta W$, injectivity of Θ gives $\phi \in W$, proving injectivity. The quotient definitions identify its image with the last kernel and prove surjectivity of the last arrow. Taking active coordinates retains the kernel $(V/W)[A]$. For $W \subseteq W' \subseteq V$, the quotient $V/W \rightarrow V/W'$ has kernel W'/W , and the quotient $Q_W \rightarrow Q_{W'}$ has kernel identified with W'/W by $[\phi] \mapsto [\Theta\phi]_W$. The coordinate maps commute with all face arrows by their explicit zero-insertion formula.

Let $s_{h,W}(u) = [s_h u]_W$ as a complex-linear amplitude map to Q_W . Equation (CFA.10) has the exact residual form

$$Ds_{h,W} - s_{h,W}A_h = (u \mapsto [\Theta\phi_*\ell_h(u)]_W), \quad (\text{CFA.24})$$

which corresponds under the first map of (CFA.23) to $[\phi_*\ell_h(u)] \in V/W$. It is zero exactly when that source correction lies in W . Likewise (CFA.17) gives a precise residual class $[B_{h,a}(u)] \in V/W$ when its two target functions are compared in Q_W . Dilation as an action on source-labelled quotients actually goes from Q_W to $Q_{R_a W}$, since $R_a \Theta W = \Theta R_a W$; no invariance of W under all dilations is assumed. In that target the exact comparison is

$$\overline{R_a} s_{h,W}(u) - s_{h,R_a W}(A_{h,L}u) = [\Theta B_{h,a}(u)]_{R_a W}, \quad (\text{CFA.25})$$

whose residual source class lies in $V/R_a W$. Applying the transition to the full source kills these residual classes by (CFA.23). Their coefficientwise versions lie in $(V/W)[A]$ or $(V/R_a W)[A]$ with all labels retained. The strict cochain map (CFA.20) has degree-zero source in V ; it is not silently retyped as a map to a proper source not containing its source vectors. Thus the full-source equivariance statements do not erase any proper-label obstruction.

114.7 Naturality to the independent mixed observation vertices

Only the cochain transport of the original mixed observation is needed here. On the independent carrier $G(M)^n$ define the filling map syn_M to keep the wholly absent tuple absent and otherwise make every slot present with its original amplitude. For any linear $f : M \rightarrow N$, one has

$$G(f)^n \text{syn}_M = \text{syn}_N G(f)^n. \quad (\text{CFA.26})$$

If the input is wholly absent, both sides are wholly absent. Otherwise its mask remains nonempty after $G(f)^n$, even when all amplitudes vanish. Both sides then have all coordinates supported with coefficients $f(m_i)$, taking $m_i = 0$ at a formerly absent slot. This proves the equality with its exact empty and zero cases. It applies to the original section, quotient, divisor, and analytic comparison maps in their declared linear categories.

At the finite PS algebra $D_{E_h, n}$ this filling map is its proved synchronization by $E_{n, h} = (1^\bullet, e, \dots, e)$. Under the nonunital divisor map it has image $(e_{h, H}^\bullet, e, \dots, e)$, which need not equal $E_{n, H}$. Thus the synchronization square for this map is justified by (CFA.26), not by applying a unital localization universal property to a nonunital arrow. Its synchronized section has the actual image of the unit $E_{n, h}$, not the original mixed unit $(1^\bullet, \tau, \dots, \tau)$.

The PS arithmetic retraction is taken after synchronization. To verify its coordinate formula, in $B_n = E_h[t]/(t^n + 1)$ multiplication by t^j for $1 \leq j < n$ has no diagonal entry in the original basis $1, t, \dots, t^{n-1}$, because it shifts indices by the nonzero class j modulo n . Multiplication by the constant coefficient b_0 has n diagonal entries b_0 . Hence

$$r_n \left(\sum b_i t^i \right) = \frac{1}{n} \text{Tr}_{B_n/E_h} (M_{\sum b_i t^i}) = b_0. \quad (\text{CFA.27})$$

Its independent-carrier composite sends an empty mask to τ and every nonempty mask to $p(b_0)^\bullet$, including supported zero when coefficient slot zero was absent. Raw projection to slot zero instead returns τ on that latter face. These are distinct arrows. The PS constant section is carried to $(f^\bullet, e, \dots, e)$; (CFA.27) sends it back to $f^\bullet$. Consequently the section and retraction remain exactly the original arithmetic retract on all coefficient faces, with the factor $1/n$ and the changed masks retained.

For the original coefficient Frobenius, let q_0 be an odd prime power with $\gcd(q_0, n) = 1$. Its integral signed permutation has

$$t^i \mapsto (-1)^{\lfloor q_0 i / n \rfloor} t^{q_0 i \bmod n}. \quad (\text{CFA.28})$$

It takes coefficient mask A to $q_0 A$ modulo n . Applying any complex-linear map to a coordinate commutes with its specified sign, and permuting positions commutes with applying the same map in each position. This proves exact compatibility of (CFA.28) with the sections, source differentials, quotient maps, boundary homotopies, and face transitions. The original nilpotent arithmetic action (CFA.18) instead acts within each face.

For completeness its compatibility with the finite product has a different type from a semiring automorphism. Write $\star$ for the original PS negacyclic convolution, retaining every minus sign for a wrap and its relation $t^n + 1 = 0$. For $a, c > 0$ the coefficient elements satisfy $j_h(a^s)j_h(c^s) = j_h((ac)^s)$. Every convolution summand therefore gives

$$(A_{h, \log a} b) \star (A_{h, \log c} d) = A_{h, \log(ac)} (b \star d). \quad (\text{CFA.29})$$

The scalar elements commute with every coefficient and with the wrap sign, so summing proves the equality, including supported cancellations and absent slots. In particular the diagonal action with parameter a has target action with parameter a^2 , not a . The finite convolution is an algebra observation in degree zero. The packet source in C_U is in cohomological degree one; a degree-zero chain map $C_U \otimes C_U \rightarrow C_U$ induces zero on its degree-two cohomology because $H^2(C_U) = 0$. It therefore cannot realize the nonzero finite packet product on those degree-one source classes. The original DGA product of (AG14) is concatenation into tensor length two and degree two. The attachment keeps those two differently typed products as their actual arrows.

114.8 The full coherent and tensor comparison on all faces

The original Mellin isomorphism, with its original inverse, is

$$m : T_U Q \xrightarrow{\sim} E_U := \bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_\rho / (g), \quad [F] \mapsto ([MF]_\rho)_\rho. \quad (\text{CFA.30})$$

Its complete original-function proof is (CAU.9)–(CAU.13). For clarity, its coordinate inverse on a full common packet h sends α to the class of

$$S_h^{\mathcal{B}} \Theta(r_h(\varepsilon_h \alpha)(D)\phi_*), \quad MF_\alpha = (g/h)r_h(\varepsilon_h \alpha). \quad (\text{CFA.31})$$

The Euler inverse exists because this theta input has all required moment-zero jets. Its Mellin identity gives exactly α at each retained centre, using the complete inverse unit. A different polynomial representative differs by hR and changes the source by $\Theta(R(D)\phi_*)$. If a torsion representative has zero full local Mellin class, its equation $pMF = gH_\phi$ forces every p -jet of H_ϕ to vanish. The original range identity gives $\phi = p(D)\psi$, and injectivity of $p(D)$ on $\mathcal{B}$ gives $F = \Theta\psi$. This proves both inverse identities used here and does not assume the existence of an offcritical zero.

On face A , apply these maps to each active coordinate. Its finitely many entries have altogether finite spectral support, so one common full packet contains all of them. Formula (CFA.14) proves that enlarging this common packet changes none of the original functions in the sections. Thus (CFA.30) gives the exact face isomorphism $(T_U Q)[A] \simeq E_U[A]$, naturally under all face maps by (CFA.4). An active zero entry remains active at its supported zero. At every analytic stalk with $s_0 \in U$, balancing and the canonical original maps j and β give

$$M[A] = N[A] \oplus K[A], \quad j[A] : N[A] \hookrightarrow M[A], \quad \beta[A]j[A] = 1, \quad N = O/(g). \quad (\text{CFA.32})$$

The coordinate formulas for j, β prove these identities; the full-source kernel K and the joint-leg diagonal of (CFA.6) remain separate attached factors. No field-vector property of K is asserted for a proper-source kernel from (CFA.23).

For a finite tensor word retain the tuple of coefficient masks $(A_1, \dots, A_k)$ and every original source-leg label. Tensor the already defined maps over $\mathbb{C}$, with the original differential

$$d(x_1 \otimes \dots \otimes x_k) = \sum_{i=1}^k (-1)^{\sum_{j<i} |x_j|} x_1 \otimes \dots \otimes dx_i \otimes \dots \otimes x_k. \quad (\text{CFA.33})$$

The two orders of differentiating distinct positions have opposite signs and cancel, while the same-position terms vanish because $d^2 x_i = 0$. Thus this is the original cochain tensor differential. An equality of the already defined maps holds on each factor of a pure tensor and therefore on their tensor product; pure tensors span, so every face, divisor, quotient, and section square remains an exact square. A tensor of injective complex-linear sections is injective: extend a basis of each image to a basis of its target, obtaining a linear retraction in each factor, and tensor those retractions.

Tensor source homotopies have the exact signs as well. If f_i, g_i are degree-zero cochain maps and H_i has degree minus one with $f_i - g_i = dH_i + H_i d$, then the tensor homotopy is

$$H = \sum_{i=1}^k g_1 \otimes \dots \otimes g_{i-1} \otimes H_i \otimes f_{i+1} \otimes \dots \otimes f_k. \quad (\text{CFA.34})$$

On a homogeneous input the i th summand includes the sign $(-1)^{\sum_{j<i} |x_j|}$ forced by the degree of H_i . Computing $dH + Hd$ cancels the different-position terms with the same signs as (CFA.33). The remaining terms are $\sum_i g_1 \otimes \dots \otimes g_{i-1} \otimes (f_i - g_i) \otimes f_{i+1} \otimes \dots \otimes f_k$; the middle products telescope to $\bigotimes_i f_i - \bigotimes_i g_i$. This proves the full tensor transport of the Euler and dilation homotopies, without asserting a strict section intertwining that the original source does not have.

The finite tensor packet on a word $h_1, \dots, h_k$ is exactly

$$\bigotimes_{i=1}^k E_{h_i} \simeq \frac{\mathbb{C}[x_1, \dots, x_k]}{(h_1(x_1), \dots, h_k(x_k))}. \quad (\text{CFA.35})$$

Successive monic division gives a basis with exponent of x_i strictly less than $\deg h_i$, and the tensor map takes the tensor of the one-variable remainder bases to precisely that basis. This proves the isomorphism and retains all local exponents $0 \leq j_i < m_{\rho_i}$. On each independent face of that word, (CFA.31) acts in each original coefficient coordinate, so its unit is the tensor of the literal full inverse unit jets. No nilpotent coordinate or coefficient mask is replaced by one common support bit.

114.9 Finite-stage continuity and the attachment category

Every fixed-packet section has a direct original seminorm bound. Choose a finite basis u_j of E_h , write $F_j = s_h u_j$, and let p be any original seminorm of $\mathcal{B}$. Linearity gives

$$p \left(s_h \sum_j c_j u_j \right) \leq \sum_j |c_j| p(F_j). \quad (\text{CFA.36})$$

Thus s_h is continuous in its actual finite-dimensional topology. Continuity of its Mellin inverse on the image can also be seen directly. For ρ of real part σ , choose an integer $N > |\sigma|$, put $p_+(F) = \sup_{x>0} x^N |F(x)|$ and $p_-(F) = \sup_{x>0} x^{-N} |F(x)|$. Splitting the integral at one gives

$$\left| \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} \right| \leq \frac{p_+(F)}{(N - \sigma)^{j+1}} + \frac{p_-(F)}{(N + \sigma)^{j+1}}. \quad (\text{CFA.37})$$

Indeed $\int_1^\infty x^{\sigma-N} (\log x)^j dx/x = j!/(N - \sigma)^{j+1}$ by $x = e^y$, and $\int_0^1 x^{\sigma+N} |\log x|^j dx/x = j!/(N + \sigma)^{j+1}$ by $x = e^{-y}$. The same integrable bounds justify differentiation at the fixed jet. Every fixed finite Mellin observation is therefore continuous. Together with $J_h s_h = 1$ this proves that the finite section is a topological embedding onto its image. The quotient map q is continuous in the original quotient topology, so the induced section σ_h is continuous; its inverse J_h descends continuously through q since $J_h \Theta = 0$.

Finite zero insertions and fixed-face maps are finite linear maps. For the labelled disjoint-union topology, addition has output mask $A \cup B$, and negacyclic multiplication has output mask $A + B$ in $\mathbb{Z}/n\mathbb{Z}$. A nonempty supported sum remains supported even when its amplitude cancels, so these output masks are fixed on a pair of input faces. On that pair the amplitude formulas are respectively linear and polynomial finite-dimensional maps, proving continuity there; the disjoint-union topology proves continuity on the whole finite mixed carrier. These are fixed-packet, fixed- n facts. Formula (CFA.36) does not give a uniform bound as the packet or n varies, and it does not identify the topology of an infinite section s_U with a direct-limit topology. No such additional continuity or estimate is asserted by this attachment.

Finally, the attachment is one specified extension of the represented object. Its generating indices are (W, S, h, n, A) with independent S and A , together with the actual full-source and empty-packet vertices. Its values are the face complexes, strict free complexes, finite observation modules, and the original mixed carriers above. For the cochain part use the graded complex-linear path category: degree-zero generators are the displayed cochain maps, and the specified homotopies are degree-minus-one linear generators. The target Hom complexes have their actual differential

$$\partial f = d_{\text{target}} f - (-1)^{|f|} f d_{\text{source}}. \quad (\text{CFA.38})$$

Direct expansion using $d^2 = 0$ gives $\partial^2 f = 0$, and expansion of both sides gives $\partial(fg) = (\partial f)g + (-1)^{|f|} f(\partial g)$. In particular a cochain map has $\partial f = 0$, while a homotopy has $\partial H = dH + H d$.

Impose the proved coordinate and linear composition relations, including $f - g = \partial H$ with the displayed generator H . This does not identify the two unmodified cochain paths f and g . Sending each generator to its actual graded map and each linear combination of paths to the same linear combination of composites is well-defined because precisely these identities were proved above. The differential formula proves compatibility with the graded structure.

Include all tensor words with (CFA.33) and their Koszul permutations. The finite mixed carriers, face-filling maps, and algebra observations are also retained with their independently specified types in the underlying-set representation of TO; no non-cochain observation is declared a degree-zero chain map. Thus this extends the same TO/CAU representation by the original coefficient faces and their exact graded cochain comparison. No colimit discards their labels and no supported zero is identified with global absence. The full ambient $\mathcal{B}$, original two legs, and proper-source residual kernels remain attached.

115 The nilpotent filtration on every independent arithmetic face

This construction applies Deligne's nilpotent-filtration linear algebra to the actual arithmetic packets and to every independent coefficient face. It does not assert an arithmetic purity estimate. Every formula below retains the original arithmetic dilation, its scalar eigenvalue, and all its nilpotent terms. The additional grading operator is recorded as an additional operator.

115.1 Original packets and the filtration

Let Z be a finite set of distinct zeros of the original function $g(s) = 2\xi(s)$, with their full orders $m_\rho \geq 1$. Put

$$h_Z(x) = \prod_{\rho \in Z} (x - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[x]/(h_Z), \quad E_\rho = e_\rho E_Z. \quad (\text{MW1})$$

Here e_ρ is the Chinese-remainder idempotent, and $u_\rho = e_\rho(x - \rho)$ is an element of this algebra, with local unit e_ρ . Write $v_{\rho,r} = e_\rho(x - \rho)^r$, for $0 \leq r < m_\rho$. These vectors are the original local jet basis. The nilpotent operator N_ρ is multiplication by u_ρ :

$$N_\rho v_{\rho,r} = v_{\rho,r+1} \quad (r < m_\rho - 1), \quad N_\rho v_{\rho,m_\rho-1} = 0. \quad (\text{MW2})$$

For an integer j , define the increasing filtration

$$M_j E_\rho = \text{span}_{\mathbb{C}}\{v_{\rho,r} : m_\rho - 1 - 2r \leq j\}, \quad M_j E_Z = \bigoplus_{\rho \in Z} M_j E_\rho. \quad (\text{MW3})$$

In particular a zero vector in $M_j E_\rho$ remains the zero of its specified coefficient face when this filtration is split-lifted.

Filtration calculation. Multiplication by u_ρ lowers the displayed index by exactly two. Thus $N_\rho M_j \subseteq M_{j-2}$. For $k \geq 0$, the only way $\text{Gr}_k^M E_\rho$ is nonzero is

$$k = m_\rho - 1 - 2r \geq 0, \quad 0 \leq r < m_\rho.$$

It is then one dimensional with basis the class of $v_{\rho,r}$. The map induced by N_ρ^k sends this basis vector to the class of $v_{\rho,r+k}$, whose index is $-k$ and whose exponent $r + k = m_\rho - 1 - r$ lies between zero and $m_\rho - 1$. Its coefficient is exactly one. Thus

$$N_\rho^k : \text{Gr}_k^M E_\rho \xrightarrow{\sim} \text{Gr}_{-k}^M E_\rho. \quad (\text{MW4})$$

When the source graded space is zero, the target is zero by the same index calculation. Direct sums prove the corresponding assertion for $N = \bigoplus_{\rho} N_{\rho}$ on E_Z . This is precisely the centered nilpotent filtration characterized in Weil II, Proposition 1.6.1.

There is an intrinsic formula which also proves functoriality without choosing a new jet basis:

$$M_j V = \sum_{i \geq 0} \left(\ker N^{i+1} \cap \operatorname{im} N^{\max(0, i-j)} \right), \quad V = E_Z. \quad (\text{MW5})$$

To verify it on a block of length m , both spaces in each intersection are terminal spans of the original basis. Their intersection starts at exponent $\max(0, m-1-i, i-j)$. The sum over i starts at the smallest such exponent, namely $\max(0, -j, \lceil (m-1-j)/2 \rceil)$; an exponent at least m denotes the zero space. When the extra term $-j$ exceeds the other terms, $j < 1-m$ and both spans are zero. Otherwise the starting exponents agree with MW3. Kernels and images commute with the finite direct-sum decomposition, proving MW5. A linear map commuting with the nilpotent operators carries each kernel and each image into the corresponding kernel and image. It therefore preserves MW5.

115.2 Independent masks and the arithmetic action

For $n \geq 1$, retain the independent carrier

$$D_{E_Z, n} = G(E_Z)^n, \quad G(E_Z) = \{\tau\} \sqcup \{b^{\bullet} : b \in E_Z\}. \quad (\text{MW6})$$

For each $A \subseteq I_n = \{0, \dots, n-1\}$, its face is $\{A\} \times E_Z^A$, with the empty face a single absent point. The filtration on the entire carrier is the collection

$$\widetilde{M}_j D_{E_Z, n} = \{\tau\} \sqcup \coprod_{\emptyset \neq A \subseteq I_n} \{A\} \times (M_j E_Z)^A. \quad (\text{MW7})$$

This notation specifies additive faces and their filtrations. No assertion that each $\widetilde{M}_j$ is a unital sub-semiring is made. Every present face has its own zero $(A, 0)$ in every filtration step; the filtration does not turn any of these zeros into absence. The zero-insertion maps for $A \subseteq B$ preserve MW7 because they insert the original zero of $M_j E_Z$.

The original arithmetic operator, for real $a > 0$, is multiplication by the full jet of $a^s = \exp(s \log a)$. On a local block it is

$$A_a|_{E_{\rho}} = a^{\rho} \sum_{r=0}^{m_{\rho}-1} \frac{(\log a)^r}{r!} N_{\rho}^r. \quad (\text{MW8})$$

It preserves MW3, and its induced action on every nonzero graded piece of that block is exactly a^{ρ} , since all terms with $r > 0$ lower the filtration index. Its inverse is $A_{a^{-1}}$. Thus the split lift preserves every independent mask, including the zero in each active slot.

For an odd prime power q with $\gcd(q, n) = 1$, retain the coefficient Frobenius from the original negacyclic algebra:

$$\Psi_q(v t^i) = (-1)^{\lfloor qi/n \rfloor} v t^{qi \bmod n}, \quad B_n = E_Z[t]/(t^n + 1). \quad (\text{MW9})$$

It sends A to qA and leaves the jet index unchanged. Consequently it preserves the full filtered face diagram, with its specified permutation of face labels and its original signs. It commutes with N and A_a . Define the combined operator, at its full linear coefficient-space type, by

$$\begin{aligned} B_{\rho, n} &= E_{\rho}[t]/(t^n + 1), & M_j B_{\rho, n} &= \bigoplus_{i=0}^{n-1} (M_j E_{\rho}) t^i, \\ T_{a, q} &= A_a \Psi_q = \Psi_q A_a : B_n \longrightarrow B_n, \\ T_{a, q}(v_{\rho, r} t^i) &= (-1)^{\lfloor qi/n \rfloor} a^{\rho} \sum_{\ell=0}^{m_{\rho}-1-r} \frac{(\log a)^{\ell}}{\ell!} v_{\rho, r+\ell} t^{qi \bmod n}. \end{aligned} \quad (\text{MW9a})$$

Here A_a and N act coefficientwise, and B_n is the direct sum of the $B_{\rho,n}$. Applying MW8 to MW9 gives every coefficient in MW9a; the two orders agree because the signed permutation acts only on the coefficient-slot index. The split lift of $T_{a,q}$ sends the face labelled A to the face labelled qA , with exactly the displayed signs. Eigenspaces below are taken in the full linear space $B_{\rho,n}$; an individual moving face need not be invariant.

Since q is a unit modulo $2n$, choose $h \geq 1$ with $q^h \equiv 1 \pmod{2n}$. Substitution sends t^i to $t^{q^h i} = t^i$, so $\Psi_q^h = I$. The polynomial $X^h - 1$ has distinct complex roots, hence Ψ_q is diagonalizable and its eigenvalues ω satisfy $\omega^h = 1$. On each nonzero graded part of its ω -eigenspace, the three separate operators therefore have eigenvalues

$$\mathrm{Gr}^M(A_a) : a^\rho, \quad \mathrm{Gr}^M(\Psi_q) : \omega, \quad \mathrm{Gr}^M(T_{a,q}) : \omega a^\rho, \quad |\omega| = 1. \quad (\text{MW10})$$

The eigenspace decomposition respects the filtration. Its projection onto the ω -eigenspace is

$$P_\omega = h^{-1} \sum_{d=0}^{h-1} \omega^{-d} \Psi_q^d.$$

On an eigenspace with eigenvalue ω' , this sum is one if $\omega' = \omega$ and zero otherwise, by the finite geometric-sum identity and $(\omega'/\omega)^h = 1$. Every summand preserves every filtration step, so the projection does too. Thus the asserted graded eigenspaces and the restrictions of A_a are defined. MW10 computes the combined operator's modulus as $a^{\mathrm{Re} \rho}$; no bound on $\mathrm{Re} \rho$ has been inserted.

115.3 Tensor products with every coefficient and sign retained

The following proof supplies the tensor compatibility on these blocks, including its constants. On a block of length m , define auxiliary operators

$$Hv_r = (m - 1 - 2r)v_r, \quad Lv_r = v_{r+1}, \quad Ev_r = r(m - r)v_{r-1}. \quad (\text{MW11})$$

The formulas at the endpoints use $v_{-1} = v_m = 0$; L is precisely the original nilpotent N_ρ . Direct subtraction on each basis vector gives

$$[H, L] = -2L, \quad [H, E] = 2E, \quad [E, L] = H. \quad (\text{MW12})$$

For example the coefficient in the last commutator is $(r+1)(m-r-1) - r(m-r) = m-1-2r$. For this algebraic proof alone equip this basis with the auxiliary positive Hermitian form having squared lengths

$$c_r = \frac{r!(m-1)!}{(m-1-r)!}, \quad c_0 = 1. \quad (\text{MW13})$$

This is an explicitly specified additional form, not a replacement for a source norm or an arithmetic positivity assertion. The identity $c_{r+1} = (r+1)(m-r-1)c_r$ proves $L^* = E$, and $H^* = H$.

On any finite tensor product of blocks, retain the original tensor basis and define $L_{\mathrm{tot}}, E_{\mathrm{tot}}, H_{\mathrm{tot}}$ as the sums of these operators acting in the individual factors. Operators on different factors commute, so MW12 still holds. With the product form from MW13, $L_{\mathrm{tot}}^* = E_{\mathrm{tot}}$ and $H_{\mathrm{tot}}^* = H_{\mathrm{tot}}$.

Here is a direct proof of the needed graded isomorphisms. Choose a nonzero vector v of maximal H_{tot} -eigenvalue d . Then $E_{\mathrm{tot}}v = 0$, because E_{tot} raises that eigenvalue by two. Induction using MW12 gives

$$E_{\mathrm{tot}}L_{\mathrm{tot}}^r v = r(d-r+1)L_{\mathrm{tot}}^{r-1}v. \quad (\text{MW14})$$

The operator L_{tot} is nilpotent. Let ℓ be the largest exponent for which $L_{\mathrm{tot}}^\ell v \neq 0$. Applying MW14 at $r = \ell + 1$ gives $(\ell+1)(d-\ell)L_{\mathrm{tot}}^\ell v = 0$, hence $d = \ell$. Thus d is a nonnegative integer and the chain has weights $d, d-2, \dots, -d$. Its span is invariant under $L_{\mathrm{tot}}, E_{\mathrm{tot}}, H_{\mathrm{tot}}$. Its orthogonal complement is invariant too, because the first two are adjoints and the third is self-adjoint. Induction on dimension

decomposes the entire tensor space into such chains. On each chain L_{tot}^k identifies weight k with weight $-k$ exactly as in MW4. Therefore the filtration by sums of H_{tot} -eigenspaces of index at most j is the centered nilpotent filtration. In the original tensor coordinates it is

$$M_j \left(\bigotimes_{\nu=1}^k E_{\rho_\nu} \right) = \sum_{j_1+\dots+j_k \leq j} M_{j_1} E_{\rho_1} \otimes \dots \otimes M_{j_k} E_{\rho_k}. \quad (\text{MW15})$$

This also proves the precise relation to Weil II 1.6.9 on these actual blocks. Tensoring labelled faces retains the ordered tuple of their masks; MW15 operates on their amplitude fibres.

The original arithmetic tensor operator is, without dropping any term,

$$\bigotimes_{\nu=1}^k A_a|_{E_{\rho_\nu}} = a^{\rho_1+\dots+\rho_k} \exp \left((\log a) \sum_{\nu=1}^k N_{\rho_\nu} \right). \quad (\text{MW16})$$

The exponential is finite and the equality follows by expanding the product of commuting finite exponentials. On the M -graded pieces its eigenvalue is $a^{\rho_1+\dots+\rho_k}$. For identical packets its modulus is $a^{k \operatorname{Re} \rho}$. In particular the original arithmetic weight survives this complete nilpotent-filtration construction.

For the coefficient spaces, take $n_\nu \geq 1$, a common odd prime power q coprime to every n_ν , and $B_\nu = E_{\rho_\nu}[t_\nu]/(t_\nu^{n_\nu} + 1)$. Write N_ν for the nilpotent in factor ν , extended by identities to the other factors. Define $A_a^\otimes = \bigotimes_\nu A_a|_{B_\nu}$, $\Psi_q^\otimes = \bigotimes_\nu \Psi_q|_{B_\nu}$, and $T_{a,q}^\otimes = \bigotimes_\nu T_{a,q}|_{B_\nu}$. Then

$$\begin{aligned} T_{a,q}^\otimes &= A_a^\otimes \Psi_q^\otimes = a^{\sum_\nu \rho_\nu} \exp \left((\log a) \sum_\nu N_\nu \right) \Psi_q^\otimes, \\ T_{a,q}^\otimes \left(\bigotimes_\nu v_{\rho_\nu, r_\nu} t_\nu^{i_\nu} \right) &= a^{\sum_\nu \rho_\nu} \left(\prod_\nu (-1)^{\lfloor q i_\nu / n_\nu \rfloor} \right) \\ &\quad \times \sum_{0 \leq \ell_\nu \leq m_{\rho_\nu} - 1 - r_\nu \text{ for all } \nu} \left(\prod_\nu \frac{(\log a)^{\ell_\nu}}{\ell_\nu!} \right) \bigotimes_\nu v_{\rho_\nu, r_\nu + \ell_\nu} t_\nu^{q i_\nu \bmod n_\nu}. \end{aligned} \quad (\text{MW16a})$$

Expanding the factor formulas MW9a proves the second line, including every sign and every factorial. The first line follows by grouping the commuting operators on different tensor factors, and then using MW16. On the tensor of coefficient eigenspaces with eigenvalues ω_ν , the three graded eigenvalues are respectively

$$A_a^\otimes : a^{\sum_\nu \rho_\nu}, \quad \Psi_q^\otimes : \prod_\nu \omega_\nu, \quad T_{a,q}^\otimes : \left(\prod_\nu \omega_\nu \right) a^{\sum_\nu \rho_\nu}. \quad (\text{MW16b})$$

The positive nilpotent powers in MW16a decrease total degree by at least two, which proves this graded formula from the complete operator formula. Its combined modulus is $a^{\sum_\nu \operatorname{Re} \rho_\nu}$. The coefficient permutation sends the ordered tuple $(A_\nu)_\nu$ to $(qA_\nu)_\nu$.

115.4 The exact Tate-scaling comparison

Deligne's local-monodromy operator has type $N : V(1) \rightarrow V$. For geometric Frobenius over $\mathbb{F}_q$, the Tate line has Frobenius eigenvalue q^{-1} . After recording a generator of that line, equivariance has the precise matrix form

$$FNF^{-1} = q^{-1}N. \quad (\text{MW17})$$

For the original arithmetic action MW8 one instead calculates $A_a N_\rho A_a^{-1} = N_\rho$. The exact relation between these operator presentations can be constructed on the same jet space. For real $q > 1$ define

$$\Gamma_{q,\rho} v_{\rho,r} = q^{(m_\rho-1)/2-r} v_{\rho,r}, \quad F_{q,\rho} = q^\rho \Gamma_{q,\rho}. \quad (\text{MW18})$$

Both are invertible and retain every original basis vector. The quotient of successive diagonal entries is q^{-1} , so

$$F_{q,\rho} N_\rho F_{q,\rho}^{-1} = q^{-1} N_\rho, \quad A_q = F_{q,\rho} \Gamma_{q,\rho}^{-1} \exp((\log q) N_\rho). \quad (\text{MW19})$$

The second equality is obtained by multiplying the first two diagonal operators, giving exactly $q^\rho I$, and then retaining the complete exponential MW8. It proves reconstruction of the original arithmetic action from the added operators. Moreover, if $U(z) = \exp(z N_\rho)$, then

$$F_{q,\rho} U(z) F_{q,\rho}^{-1} = U(q^{-1} z). \quad (\text{MW20})$$

Thus these explicit matrices define a complex representation of the group generated by the additive one-parameter operators $U(z)$ and an invertible element acting on the parameter by $z \mapsto q^{-1} z$. This statement is at the displayed complex-linear type; it does not assert that these matrices arise from an arithmetic lisse sheaf.

On the graded piece of index $j = m_\rho - 1 - 2r$, the eigenvalue of the added $F_{q,\rho}$ is $q^{\rho+j/2}$, with

$$2 \log_q |q^{\rho+j/2}| = 2 \operatorname{Re} \rho + j. \quad (\text{MW21})$$

This gives exactly the index shift $\beta + j$ in Weil II 1.8.4, with the explicitly computed $\beta = 2 \operatorname{Re} \rho$. It does not determine β . The two-sided boundary estimate in Deligne's theorem comes from the pure lisse source, the trace and Euler-product bound, and its tensor and dual arguments. MW18 constructs the matrix relation and MW21 computes every graded modulus; neither computation supplies that geometric input. All operators MW18 act coefficientwise on MW7, preserve its masks, and commute with the coefficient permutation MW9.

For admissible odd prime powers q as in MW9, retaining the separate coefficient operator also gives the exact reconstruction and its tensor version

$$\begin{aligned} T_{q,q}|_{B_{\rho,n}} &= F_{q,\rho} \Gamma_{q,\rho}^{-1} \exp((\log q) N_\rho) \Psi_q, \\ T_{q,q}^\otimes &= F_q^\otimes (\Gamma_q^\otimes)^{-1} \exp\left((\log q) \sum_\nu N_\nu\right) \Psi_q^\otimes, \\ F_q^\otimes &= \bigotimes_\nu F_{q,\rho_\nu}, \quad \Gamma_q^\otimes = \bigotimes_\nu \Gamma_{q,\rho_\nu}. \end{aligned} \quad (\text{MW21a})$$

The first two factors in the first line multiply to $q^\rho I$; in the tensor line they multiply to $q^{\sum_\nu \rho_\nu} I$. This proves both identities in their displayed order. On total degree j , the added $F_q^\otimes$ acts by $q^{\sum_\nu \rho_\nu + j/2}$. Its combination with $\Psi_q^\otimes$ acts by $(\prod_\nu \omega_\nu) q^{\sum_\nu \rho_\nu + j/2}$ on the indicated coefficient eigenspaces. These are added-operator eigenvalues; the unchanged arithmetic and combined eigenvalues remain MW16b.

For completeness, duality at the vector-space level also retains its signs. On $V^\vee$, use $N^\vee = -N^\top$ and $M_j V^\vee = (M_{-j-1} V)^\perp$. The dual of a basis vector of index d has index $-d$, and $N^\vee$ lowers that index by two with coefficient -1 . Its iterated graded isomorphisms have the nonzero coefficient $(-1)^k$. The original contragredient arithmetic action is

$$(A_a^{-1})^\top = a^{-\rho} \exp(-(\log a) N^\top), \quad (\text{MW22})$$

so its graded eigenvalue is exactly $a^{-\rho}$. On the full coefficient dual put $\Psi_q^\vee = (\Psi_q^{-1})^\top$ and $T_{a,q}^\vee = (T_{a,q}^{-1})^\top$. Commutativity in MW9a and transposition of the inverse product give

$$T_{a,q}^\vee = (A_a^{-1})^\top (\Psi_q^{-1})^\top. \quad (\text{MW22a})$$

The dual of the ω -coefficient eigenspace has $\Psi_q^\vee$ -eigenvalue ω^{-1} : evaluation gives $(\Psi_q^\vee f)(v) = f(\Psi_q^{-1} v) = \omega^{-1} f(v)$. Thus the three dual graded eigenvalues are $a^{-\rho}$, ω^{-1} , and $\omega^{-1} a^{-\rho}$, respectively. These formulas establish the tensor, dual and Tate-scaling comparisons on every arithmetic amplitude fibre while retaining the separate mixed-support labels.

115.5 The actual length-two quotient retains its induced degrees

Define $Z_{\geq 2} = \{\rho \in Z : m_\rho \geq 2\}$. For each member of this specified subset put $m = m_\rho$, $b = m - 2$, and

$$\begin{aligned} J_\rho &= u_\rho^2 E_\rho = \text{span}\{v_{\rho,r} : 2 \leq r < m\}, & C_\rho &= E_\rho / J_\rho, & \pi_\rho &: E_\rho \longrightarrow C_\rho, \\ w_0 &= \pi_\rho(v_{\rho,0}), & w_1 &= \pi_\rho(v_{\rho,1}), & \phi_\rho &: C_\rho \longrightarrow V_2 = \mathbb{C}[u]/(u^2), & \phi_\rho(c_0 w_0 + c_1 w_1) &= c_0 + c_1 u. \end{aligned} \quad (\text{MW23})$$

Here $u = x - \rho$ is the same local coordinate: the map sends the original local unit to 1 and the original first power to u . The basis multiplication satisfies $v_{\rho,r} v_{\rho,s} = v_{\rho,r+s}$ for $r + s < m$, and zero otherwise. Therefore J_ρ is an ideal, w_0 is the quotient unit, $w_1^2 = 0$, and ϕ_ρ is an algebra isomorphism with inverse $c_0 + c_1 u \mapsto c_0 w_0 + c_1 w_1$. For $m = 2$, the displayed span for J_ρ is zero. This construction is indexed by $Z_{\geq 2}$; it does not assert that this subset has a member. Raising an exponent preserves J_ρ , so N_ρ induces the operator $\bar{N}$ on C_ρ .

Give J_ρ the subspace filtration and C_ρ the quotient filtration of MW3. The latter is exactly

$$\begin{aligned} \bar{M}_j C_\rho = \pi_\rho(M_j E_\rho) &= \begin{cases} 0 & j < m - 3, \\ \mathbb{C}w_1 & m - 3 \leq j < m - 1, \\ C_\rho & j \geq m - 1, \end{cases} & (\text{MW24}) \\ \phi_\rho(\bar{M}_j C_\rho) &= M_{j-b} V_2. \end{aligned}$$

Indeed, all original powers of exponent at least two vanish under π_ρ ; the two surviving powers have their original degrees $m - 1$ and $m - 3$. Substituting $b = m - 2$ into the length-two degrees 1, -1 proves the second line. For every j the kernel of $M_j E_\rho \rightarrow \bar{M}_j C_\rho$ is $J_\rho \cap M_j E_\rho$, and its image is the whole indicated quotient step. Hence $0 \rightarrow J_\rho \cap M_j E_\rho \rightarrow M_j E_\rho \rightarrow \bar{M}_j C_\rho \rightarrow 0$ is exact, proving strictness of both maps. The quotient filtration is centered at b : its only nonidentity graded isomorphism is $\bar{N} : \text{Gr}_{b+1}^{\bar{M}} C_\rho \rightarrow \text{Gr}_{b-1}^{\bar{M}} C_\rho$, sending the class of w_0 to that of w_1 with coefficient one. For $k = 0$ the centered-at- b map is the identity, and for $k > 1$ both graded spaces are zero. If $m > 2$, this filtration is not centered at zero: Gr_{m-1} is nonzero whereas $\text{Gr}_{-(m-1)}$ is zero. MW24 retains the quotient's actual degrees in every case.

The ideal J_ρ is invariant under N_ρ , every A_a , and the added diagonal operators $\Gamma_{q,\rho}, F_{q,\rho}$. For N_ρ this follows by raising the exponent; for A_a it follows term by term in MW8; for the diagonals each basis line is invariant. Their induced quotient operators consequently satisfy

$$\begin{aligned} \bar{N}w_0 &= w_1, & \bar{N}w_1 &= 0, & \bar{A}_a w_0 &= a^\rho(w_0 + (\log a)w_1), & \bar{A}_a w_1 &= a^\rho w_1, \\ \bar{\Gamma}_q w_r &= q^{(m-1)/2-r} w_r, & \bar{F}_q w_r &= q^{\rho+(m-1)/2-r} w_r & (r = 0, 1), & & & (\text{MW25}) \\ \phi_\rho \bar{N} &= N_2 \phi_\rho, & \phi_\rho \bar{A}_a &= A_{a,2} \phi_\rho, & \phi_\rho \bar{\Gamma}_q &= q^{b/2} \Gamma_{q,2} \phi_\rho, & \phi_\rho \bar{F}_q &= q^{b/2} F_{q,2} \phi_\rho. \end{aligned}$$

The subscript 2 specifies the fixed length-two operators $N_2(1) = u$, $N_2(u) = 0$, $A_{a,2} = a^\rho(I + (\log a)N_2)$, $\Gamma_{q,2}(1) = q^{1/2}$, $\Gamma_{q,2}(u) = q^{-1/2}u$, and $F_{q,2} = q^\rho \Gamma_{q,2}$. Every line of MW25 follows by applying both sides to w_0, w_1 . In particular, the higher terms of MW8 are killed by this specified quotient map; they have not been discarded from the original packet.

An exact filtered operator isomorphism records both the degree and diagonal scalar shifts. Let $D_b = \mathbb{C}d_b$ have filtration $(D_b)_j = 0$ for $j < b$ and $(D_b)_j = D_b$ for $j \geq b$, with $N_{D_b} = 0$, $A_{a,D_b} = I$, and $\Gamma_{q,D_b} = F_{q,D_b} = q^{b/2}I$. Use sum filtrations, tensor-sum nilpotents, and tensor-product invertible operators. Then

$$\begin{aligned} \theta_\rho : C_\rho &\xrightarrow{\sim} V_2 \otimes D_b, & w_r &\longmapsto u^r \otimes d_b & (r = 0, 1), & & & (\text{MW26}) \\ \theta_\rho(\bar{M}_j C_\rho) &= M_{j-b} V_2 \otimes D_b, & \bar{A}_q &= \bar{F}_q \bar{\Gamma}_q^{-1} (I + (\log q) \bar{N}). \end{aligned}$$

The inverse sends $u^r \otimes d_b$ to w_r , so the displayed filtration equality proves strictness in both directions. MW25 proves intertwining for A, Γ, F , and its coefficient-one nilpotent relation proves

intertwining for N . In the reconstruction, $\overline{F}_q \overline{\Gamma}_q^{-1} = q^\rho I$, so the degree shift has changed no arithmetic scalar or logarithmic coefficient. The original auxiliary H in MW11 descends to $H_2 + bI$ under ϕ_ρ , where $H_2(1) = 1$ and $H_2(u) = -u$, by its two diagonal entries. For $m > 2$, the original auxiliary E does not descend: $E(v_{\rho,2}) = 2(m-2)v_{\rho,1} \notin J_\rho$. Thus MW26 is the explicitly proved filtered isomorphism for N, A, Γ, F , with the exact additional relation for H .

The kernel also has a specified shifted comparison. For $m > 2$, let $V_{m-2} = \mathbb{C}[u]/(u^{m-2})$ use the original powers, the centered filtration of a length- $m-2$ block, and the operators MW8 and MW18 with the same ρ . The linear map $\alpha_\rho : V_{m-2} \rightarrow J_\rho$, $\alpha_\rho(u^r) = v_{\rho,r+2}$, is bijective, with inverse given by removing these specified two powers. It is a module comparison for the displayed nilpotent and arithmetic operators. Its complete filtration and diagonal relations are

$$\begin{aligned} \alpha_\rho(M_{j+2}V_{m-2}) &= J_\rho \cap M_j E_\rho, & N_\rho \alpha_\rho &= \alpha_\rho N_{m-2}, & A_a \alpha_\rho &= \alpha_\rho A_{a,m-2}, \\ \Gamma_{q,\rho} \alpha_\rho &= q^{-1} \alpha_\rho \Gamma_{q,m-2}, & F_{q,\rho} \alpha_\rho &= q^{-1} \alpha_\rho F_{q,m-2}. \end{aligned} \quad (\text{MW26a})$$

The source degree of u^r is $m-3-2r$; the image's degree is $m-1-2(r+2) = (m-3-2r)-2$, proving the first relation at every step. Raising the exponent proves the nilpotent relation including the last zero vector; expanding MW8 proves the arithmetic relation with all its coefficients. The diagonal exponent on the image is $(m-1)/2 - (r+2) = ((m-3)/2 - r) - 1$, which proves both remaining relations. Consequently the kernel, quotient, and their exact sequence all retain their actual induced shifts. For $m = 2$ the kernel is zero as already specified.

The same quotient is available coefficientwise. Its map on each face is

$$(A, (x_i)_{i \in A}) \mapsto (A, (\pi_\rho x_i)_{i \in A}).$$

Even a coefficient in J_ρ becomes a present zero in the same slot. Surjectivity on each filtered face follows by choosing, independently in each slot, a preimage in $M_j E_\rho$ from MW24. Zero insertion commutes with this map because $\pi_\rho(0) = 0$. On the full coefficient space it intertwines Ψ_q since the map is linear and sends $(-1)^{\lfloor qi/n \rfloor} x$ to that same sign times $\pi_\rho(x)$; MW25 then proves intertwining for $T_{a,q}$. For a tensor of such quotient blocks the exact comparison is

$$\begin{aligned} \bigotimes_{\nu} C_{\rho_{\nu}} &\xrightarrow{\bigotimes_{\nu} \theta_{\rho_{\nu}}} \left(\bigotimes_{\nu} V_{2,\rho_{\nu}} \right) \otimes \left(\bigotimes_{\nu} D_{m_{\rho_{\nu}}-2} \right), \\ \text{total degree shift } \sum_{\nu} (m_{\rho_{\nu}} - 2), &\quad \text{added } F \text{ and } \Gamma \text{ factor } q^{\frac{1}{2} \sum_{\nu} (m_{\rho_{\nu}} - 2)}. \end{aligned} \quad (\text{MW27})$$

The tensor factors are rearranged by the explicit ordinary vector-space map $\bigotimes_{\nu} (u^{r_{\nu}} \otimes d_{b_{\nu}}) \mapsto (\bigotimes_{\nu} u^{r_{\nu}}) \otimes (\bigotimes_{\nu} d_{b_{\nu}})$, with coefficient one. Degrees add and diagonal eigenvalues multiply, which proves both formulas and their inverse on every tensor basis vector. The tensor quotient is strict: a source tensor basis vector survives exactly when each exponent is zero or one, and the degree of every surviving vector is unchanged. Thus each filtered target step is the image of the same source step. Its kernel is spanned by tensors with at least one exponent at least two, giving the complete filtered kernel as their intersection with that step. The unchanged arithmetic eigenvalues and the signed coefficient products are precisely MW16a–MW16b, with the nilpotent powers transported through these actual quotient maps. The strict length-two subobject and quotient maps are developed with these induced degrees in MRE34–MRE42 below.

Status of the geometric control. The construction proves MW1–MW27, including the lettered formulas, on the original finite packets and the specified quotients. It explicitly supplies the nilpotent filtration and its tensor and dual compatibility, and it reconstructs the unchanged arithmetic operator from a Tate-scaling operator. No theorem here identifies the resulting complex representation with a pure lisse sheaf over the absolute τ -base. In particular it proves no numerical restriction on $2 \operatorname{Re} \rho$. The relevant further comparison is with the actual hypotheses and proof of Weil II 1.8.4–1.8.5 and the compact-support pushforward estimate 3.3.1; the finite-poset section functor by itself has not supplied those estimates.

116 Strict filtered control of the original length-two jet extension

This calculation constructs the induced filtrations and exact operator maps on a specified arithmetic jet extension. It includes the uniqueness proof for the centered nilpotent filtration, the translation to a specified constituent degree, and the relative filtration on this extension. The original arithmetic action and the added Tate-scaling action are both retained as separate, explicitly related operators.

116.1 Uniqueness of the centered nilpotent filtration

Let V be a finite-dimensional complex vector space and N a nilpotent endomorphism. A finite increasing filtration in this paragraph means subspaces $M_j V$, indexed by $j \in \mathbb{Z}$, which are zero for all sufficiently negative j and equal to V for all sufficiently positive j . Suppose it has the centered nilpotent properties

$$NM_j V \subseteq M_{j-2} V, \quad N^k : \mathrm{Gr}_k^M V \xrightarrow{\sim} \mathrm{Gr}_{-k}^M V \quad (k \geq 0). \quad (\text{MRE1})$$

Here and below an isomorphism from the zero vector space to itself is included in the assertion.

We prove that these properties determine M uniquely. Choose $d \geq 0$ with $N^{d+1} = 0$. For $k > d$, the isomorphism in MRE1 is the zero map; its source and target therefore vanish. Exhaustiveness and finiteness give $M_d V = V$ and $M_{-d-1} V = 0$. The isomorphism for $k = d$ is induced by N^d from $V/M_{d-1} V$ to $M_{-d} V$. Consequently

$$M_{d-1} V = \ker N^d, \quad M_{-d} V = \mathrm{im} N^d. \quad (\text{MRE2})$$

For $d = 0$, this says that the only possibly nonzero graded piece is degree zero, and determines the filtration completely.

For $d \geq 1$, the subspace $\mathrm{im} N^d$ lies in $\ker N^d$, because $N^{2d} = 0$. The induced operator on

$$K_d = \ker N^d / \mathrm{im} N^d \quad (\text{MRE3})$$

satisfies $N^d = 0$. The filtration induced on K_d has the same graded pieces as V for $-d < j < d$, and zero graded pieces outside this range. Thus MRE1 descends to this quotient: for $0 \leq k < d$ its maps are the original graded maps, and for $k \geq d$ both sides vanish. Induction on d determines the filtration on K_d . Every intermediate step on V is its inverse image under the quotient $\ker N^d \rightarrow K_d$; MRE2 fixes the endpoints. This proves uniqueness.

For the original length- m jet basis v_r , $0 \leq r < m$, with $Nv_r = v_{r+1}$ and $Nv_{m-1} = 0$, existence is explicit:

$$M_j V = \mathrm{span}\{v_r : m - 1 - 2r \leq j\}. \quad (\text{MRE4})$$

The operator lowers the displayed index by two. When index $k \geq 0$ occurs, its basis vector has exponent $r = (m - 1 - k)/2$; applying N^k gives the index- $-k$ vector v_{r+k} , with coefficient one and $r + k = m - 1 - r$. If index k is absent, index $-k$ is absent as well. This proves MRE1. Taking direct sums proves existence on any finite packet of these original blocks. The preceding uniqueness proof identifies this explicit filtration with the centered nilpotent filtration.

116.2 Translation to a specified constituent degree

For an integer b , put a given original block in the one-step filtration

$$W_i^{[b]} V = \begin{cases} 0 & i < b, \\ V & i \geq b, \end{cases} \quad M_j^{[b]} V = M_{j-b} V. \quad (\text{MRE5})$$

The relative nilpotent-filtration identities for this specified $W^{[b]}$ are

$$NM_j^{[b]} \subseteq M_{j-2}^{[b]}, \quad N^k : \mathrm{Gr}_{i+k}^{M^{[b]}} \mathrm{Gr}_i^{W^{[b]}} V \xrightarrow{\sim} \mathrm{Gr}_{i-k}^{M^{[b]}} \mathrm{Gr}_i^{W^{[b]}} V. \quad (\text{MRE6})$$

For $i \neq b$, both sides of the second map vanish. For $i = b$, that map is exactly the already proved $N^k : \text{Gr}_k^M V \rightarrow \text{Gr}_{-k}^M V$. The first inclusion follows directly by shifting both indices in MRE1. This proves the relative-filtration translation for every integer b . The construction of these filtrations assigns the displayed degree; it does not change or determine any arithmetic eigenvalue.

116.3 The original extension and its arithmetic action

Fix $\rho \in \mathbb{C}$, retain the local coordinate $u = x - \rho$, and define the length-two jet algebra and its nilpotent operator by

$$V_2 = \mathbb{C}[u]/(u^2), \quad v_0 = 1, \quad v_1 = u, \quad Nv_0 = v_1, \quad Nv_1 = 0. \quad (\text{MRE7})$$

In this entire calculation these are the fixed original basis vectors and the coefficient of Nv_0 remains exactly one. For arbitrary ρ , MRE7 specifies this displayed local algebra; it asserts no zero or repeated zero of $g(s) = 2\xi(s)$. The exact comparison with any actual packet member of multiplicity at least two is proved in MRE34–MRE42. That comparison retains the quotient filtration induced from the full block, together with its degree translation and both added diagonal factors. Let $K = \mathbb{C}u \subset V_2$, let $Q = V_2/K$, and write $\bar{1}$ for the class of 1 in Q . The maps

$$i : K \longrightarrow V_2, \quad i(cu) = cu, \quad p : V_2 \longrightarrow Q, \quad p(a + bu) = a\bar{1} \quad (\text{MRE8})$$

give the exact sequence

$$0 \longrightarrow K \xrightarrow{i} V_2 \xrightarrow{p} Q \longrightarrow 0. \quad (\text{MRE9})$$

Indeed, i is injective, p is surjective, and $\ker p = \mathbb{C}u = \text{im } i$. The restricted operator on K and the induced operator on Q are both zero.

For real $a > 0$, retain the original arithmetic action

$$A_{a,2} = a^\rho (I + (\log a)N), \quad A_{a,K} = a^\rho I_K, \quad A_{a,Q} = a^\rho I_Q, \quad a^\rho = \exp(\rho \log a). \quad (\text{MRE10})$$

Since $N^2 = 0$, this is exactly the complete original finite exponential. It gives

$$A_{a,2}(c + du) = a^\rho (c + (d + c \log a)u). \quad (\text{MRE11})$$

The restriction to K and the induced quotient action are therefore the displayed scalars. In particular

$$A_{a,2}i = iA_{a,K}, \quad pA_{a,2} = A_{a,Q}p. \quad (\text{MRE12})$$

All maps in MRE9 preserve these original arithmetic actions.

116.4 Induced subobject and quotient filtrations

The centered filtration MRE4 on V_2 is

$$M_j V_2 = \begin{cases} 0 & j < -1, \\ K & -1 \leq j < 1, \\ V_2 & j \geq 1. \end{cases} \quad (\text{MRE13})$$

Give the subobject and quotient their actual induced filtrations:

$$M_j^K = K \cap M_j V_2 = \begin{cases} 0 & j < -1, \\ K & j \geq -1, \end{cases} \quad M_j^Q = p(M_j V_2) = \begin{cases} 0 & j < 1, \\ Q & j \geq 1. \end{cases} \quad (\text{MRE14})$$

Let $C_j K$ and $C_j Q$ denote the centered filtrations of the zero nilpotent on these one-dimensional spaces: each is zero for $j < 0$ and the whole space for $j \geq 0$. Then the precise shifts are

$$M_j^K = C_{j+1}K, \quad M_j^Q = C_{j-1}Q. \quad (\text{MRE15})$$

Thus the subobject retains degree -1 , the quotient retains degree $+1$, and the middle object's filtration remains MRE13.

For every integer j , the sequence

$$0 \longrightarrow M_j^K \xrightarrow{i} M_j V_2 \xrightarrow{p} M_j^Q \longrightarrow 0 \quad (\text{MRE16})$$

is exact. For $j < -1$, all three spaces are zero. For $-1 \leq j < 1$, it is $0 \rightarrow K \xrightarrow{1} K \rightarrow 0 \rightarrow 0$. For $j \geq 1$, it is MRE9. This exhaustive calculation also proves strictness: the image of the filtered inclusion is exactly $\text{im } i \cap M_j V_2$, and $p(M_j V_2) = M_j^Q$. On associated graded, the inclusion identifies the two degree- -1 spaces, the quotient identifies the two degree- $+1$ spaces, and all other graded spaces vanish.

The relation to the separately centered one-dimensional filtrations is also explicit. Their step -1 is zero, whereas $K \cap M_{-1} V_2 = K$. Therefore the original inclusion with the separately centered filtration is filtration preserving but is not strict. Formula MRE15 supplies the exact shifted subobject and quotient filtrations which restore the strict filtered exact sequence MRE16.

116.5 Added Frobenius and the exact shifted intertwiners

For real $q > 1$, use the additional diagonal operators

$$\Gamma_{q,2} v_0 = q^{1/2} v_0, \quad \Gamma_{q,2} v_1 = q^{-1/2} v_1, \quad F_{q,2} = q^\rho \Gamma_{q,2}. \quad (\text{MRE17})$$

Their induced operators on MRE9 are

$$\begin{aligned} \Gamma_{q,K} &= q^{-1/2} I_K, & F_{q,K} &= q^{\rho-1/2} I_K, \\ \Gamma_{q,Q} &= q^{1/2} I_Q, & F_{q,Q} &= q^{\rho+1/2} I_Q. \end{aligned} \quad (\text{MRE18})$$

Every displayed operator preserves the indicated filtration. Directly on u and $\bar{1}$, the inclusion and quotient satisfy

$$F_{q,2} i = i F_{q,K}, \quad p F_{q,2} = F_{q,Q} p, \quad \Gamma_{q,2} i = i \Gamma_{q,K}, \quad p \Gamma_{q,2} = \Gamma_{q,Q} p. \quad (\text{MRE19})$$

This proves exact filtered intertwining with the unchanged extension maps. If $F_{q,K}^{\text{cen}} = q^\rho I_K$ and $F_{q,Q}^{\text{cen}} = q^\rho I_Q$ are the operators obtained by assigning the one-dimensional spaces their separate centered degree zero, the same equations read

$$F_{q,2} i = q^{-1/2} i F_{q,K}^{\text{cen}}, \quad p F_{q,2} = q^{1/2} F_{q,Q}^{\text{cen}} p. \quad (\text{MRE20})$$

The degree shifts and the scalar factors are therefore exactly the same correction, in filtration and operator form respectively.

The original arithmetic operators are reconstructed without replacing their eigenvalues:

$$\begin{aligned} A_{q,2} &= F_{q,2} \Gamma_{q,2}^{-1} (I + (\log q) N), \\ A_{q,K} &= F_{q,K} \Gamma_{q,K}^{-1} = q^\rho I_K, \\ A_{q,Q} &= F_{q,Q} \Gamma_{q,Q}^{-1} = q^\rho I_Q. \end{aligned} \quad (\text{MRE21})$$

The first line has the displayed multiplication order. In all three lines $F \Gamma^{-1}$ is exactly q^ρ times the identity. Thus the shifted added eigenvalues retain, through these exact formulas, the original arithmetic scalar and its complete nilpotent term.

The added graded moduli are fully determined:

$$2 \log_q |F_{q,K}| = 2 \operatorname{Re} \rho - 1, \quad 2 \log_q |F_{q,Q}| = 2 \operatorname{Re} \rho + 1. \quad (\text{MRE22})$$

Here $|F|$ means the modulus of the displayed one-dimensional eigenvalue. Equation MRE22 records the exact shift; it imposes no restriction on $\operatorname{Re} \rho$.

116.6 The nilpotent link and its Tate-typed isomorphism

Since N vanishes on K and maps V_2 into K , it factors uniquely through the quotient as

$$N = i \bar{N} p, \quad \bar{N} : Q \xrightarrow{\sim} K, \quad \bar{N}(\bar{1}) = u. \quad (\text{MRE23})$$

The factorization follows by applying both sides to $a + bu$, where both sides give au . The factor map is bijective because it sends the specified basis vector to the specified basis vector with coefficient one. It lowers the induced filtration degree by exactly two. It commutes with the original arithmetic scalars and satisfies the exact added-operator relation

$$A_{a,K} \bar{N} = \bar{N} A_{a,Q}, \quad F_{q,K} \bar{N} = q^{-1} \bar{N} F_{q,Q}. \quad (\text{MRE24})$$

The second equality follows from $q^{\rho-1/2} = q^{-1} q^{\rho+1/2}$.

Define the explicit one-dimensional complex character $T_q = \mathbb{C}t$ with its chosen generator t , filtration concentrated in degree -2 , and operators

$$F_{q,T} t = q^{-1} t, \quad \Gamma_{q,T} t = q^{-1} t, \quad A_{a,T} t = t \quad (a > 0). \quad (\text{MRE25})$$

The added eigenvalue is the geometric-Frobenius Tate character. At this stage T_q is the displayed complex filtered line; no arithmetic sheaf realization is required for the following proved matrix map. Give tensor products the sum filtration and tensor product operators. Then $Q \otimes T_q$ is concentrated in degree $1 - 2 = -1$, with added eigenvalue $q^{\rho+1/2} q^{-1} = q^{\rho-1/2}$ and original arithmetic eigenvalue a^ρ . Therefore

$$\nu : Q \otimes T_q \xrightarrow{\sim} K, \quad \nu(\bar{1} \otimes t) = u \quad (\text{MRE26})$$

is a strict filtered isomorphism intertwining F_q , Γ_q , and every A_a . Its inverse sends u to $\bar{1} \otimes t$, so these claims include both inverse maps. Formula MRE26 is the exact Tate-typed form of the original nilpotent link MRE23.

The map N itself has the same strict filtered type. On $V_2 \otimes T_q$, define $\mathcal{N}(v \otimes t) = Nv$. The domain filtration is $M_j(V_2 \otimes T_q) = M_{j+2} V_2 \otimes T_q$. For every j ,

$$N(M_{j+2} V_2) = \text{im } N \cap M_j V_2. \quad (\text{MRE27})$$

For $j < -1$, both sides vanish: the domain step is at most K , which N kills. For $j \geq -1$, the domain step is V_2 , its image is K , and the right side is also K . This proves strictness. The relation $F_{q,2} N F_{q,2}^{-1} = q^{-1} N$, checked on v_0, v_1 , proves F_q -equivariance of $\mathcal{N}$. The corresponding relation for Γ_q proves its equivariance, and $A_{a,2} N = N A_{a,2}$ proves equivariance for every original arithmetic action.

116.7 The explicit relative filtration on this extension

Retain the extension filtration with its two induced constituent degrees:

$$W_j V_2 = \begin{cases} 0 & j < -1, \\ K & -1 \leq j < 1, \\ V_2 & j \geq 1. \end{cases} \quad (\text{MRE28})$$

This is a specified filtered extension, with degree- -1 constituent K and degree- $+1$ constituent Q . Set $R_j V_2 = M_j V_2$, with M as in MRE13. Then $R = W$ as displayed filtrations, and R is the relative nilpotent filtration for this specified W .

Here is the complete verification. Multiplication by u gives $N R_j \subseteq R_{j-2}$. The only nonzero W -graded pieces are $\text{Gr}_{-1}^W V_2 = K$ and $\text{Gr}_1^W V_2 = Q$. The operator induced by N on each of these pieces

is zero. The filtration induced by R on the first piece is concentrated in degree -1 , and on the second in degree $+1$, as proved in MRE14. Consequently, for every integer i and every $k \geq 0$,

$$N^k : \mathrm{Gr}_{i+k}^R \mathrm{Gr}_i^W V_2 \xrightarrow{\sim} \mathrm{Gr}_{i-k}^R \mathrm{Gr}_i^W V_2. \quad (\text{MRE29})$$

For $k = 0$, this is the identity. For $k > 0$, both source and target are zero, including when $i = -1$ or $i = 1$. For other i , the W -graded piece is itself zero. This proves every case of the relative-filtration identities.

In this two-dimensional extension uniqueness can also be proved directly. Any relative filtration R' induces the degree- -1 one-step filtration on K , and the degree- $+1$ one-step filtration on Q , because the induced nilpotent on each is zero and the relative graded isomorphisms force all other degrees to vanish. If $j < -1$, then $R'_j \cap K = 0$ and its image in Q is zero, so $R'_j = 0$. If $-1 \leq j < 1$, it contains K and its image in Q is zero, so $R'_j = K$. If $j \geq 1$, it contains K and surjects onto Q , so it equals V_2 . Thus $R' = R$, with every step determined.

This proves a relative filtration on the displayed original extension. Its additional eigenvalue shifts are MRE18 and MRE22. The proof assigns no independent geometric purity assertion to these constituents.

116.8 The original extension class is retained

Every linear section $s : Q \rightarrow V_2$ of p has the form

$$s(\bar{1}) = 1 + cu \quad (c \in \mathbb{C}). \quad (\text{MRE30})$$

For each such section,

$$Ns(\bar{1}) = u, \quad sN_Q(\bar{1}) = 0. \quad (\text{MRE31})$$

Hence no section intertwines the nilpotent operators: MRE9 is a non-split extension of nilpotent representations. The fixed nonzero map $\bar{N}$ of MRE23 records its extension coupling, and the Tate-typed isomorphism MRE26 retains that same coefficient-one coupling.

The arithmetic non-splitting is equally explicit. For every $a > 0$,

$$(A_{a,2}s - sA_{a,Q})(\bar{1}) = a^\rho(\log a)u. \quad (\text{MRE32})$$

The scalar is nonzero when $a \neq 1$, because $a^\rho \neq 0$ and $\log a \neq 0$. Thus no section intertwines even one fixed original arithmetic operator with $a \neq 1$, and no section intertwines the whole arithmetic family. The strict filtered exact sequence has therefore preserved the complete original arithmetic extension.

For the added diagonal operator alone, $s_0(\bar{1}) = 1$ is a filtered F_q -equivariant section. It is the only such section: equivariance for MRE30 forces $cq^{\rho-1/2} = cq^{\rho+1/2}$, and $q > 1$ forces $c = 0$. Equations MRE31–MRE32 calculate precisely how this additional diagonal splitting fails to split the original nilpotent and arithmetic actions. All three operator structures, their maps, and their difference are explicit on the original basis.

Their interaction also has an exact formula. Conjugating MRE10 gives

$$F_{q,2}A_{a,2}F_{q,2}^{-1} = a^\rho(I + q^{-1}(\log a)N), \quad F_{q,2}A_{a,2}F_{q,2}^{-1} - A_{a,2} = a^\rho(q^{-1} - 1)(\log a)N. \quad (\text{MRE33})$$

Every coefficient follows from $F_{q,2}NF_{q,2}^{-1} = q^{-1}N$. For $a \neq 1$, the last operator is nonzero, so the two retained actions do not commute. Equation MRE33 records their full interaction; no commuting-action assumption has been used in the filtered exact sequence or its twisted nilpotent maps.

116.9 Transport from the full original packet to its actual quotient

We give the exact downstream comparison, including the subobject, quotient, and all operators. In the original packet MW1 let $Z_{\geq 2} = \{\rho \in Z : m_\rho \geq 2\}$. The following construction is indexed by this subset and makes no assertion that it is nonempty. For each $\rho \in Z_{\geq 2}$ write $m = m_\rho$, $b = m - 2$, and retain the original $v_{\rho,r} = e_\rho(x - \rho)^r$. Define

$$\begin{aligned} J_\rho &= \text{span}\{v_{\rho,r} : 2 \leq r < m\}, \quad C_\rho = E_\rho/J_\rho, \quad \pi_\rho \left(\sum_{r=0}^{m-1} c_r v_{\rho,r} \right) = c_0 w_0 + c_1 w_1, \\ K_\rho &= \mathbb{C}w_1, \quad Q_\rho = C_\rho/K_\rho, \quad i_\rho(cw_1) = cw_1, \quad p_\rho(c_0 w_0 + c_1 w_1) = c_0[w_0], \\ \phi_\rho &: C_\rho \longrightarrow V_2, \quad \phi_\rho(c_0 w_0 + c_1 w_1) = c_0 + c_1 u. \end{aligned} \tag{MRE34}$$

Here $w_r = \pi_\rho(v_{\rho,r})$, $r = 0, 1$, and the coordinate $u = x - \rho$ is fixed as in MRE7. Multiplication of original powers shows that $J_\rho = u_\rho^2 E_\rho$ is an ideal and that w_0 is the quotient unit with $w_1^2 = 0$. The map ϕ_ρ is an algebra isomorphism: its inverse sends $c_0 + c_1 u$ to $c_0 w_0 + c_1 w_1$, and the multiplication rules on the two indicated bases agree. The maps i_ρ, p_ρ give an exact sequence, since i_ρ is injective, p_ρ is surjective, and $\ker p_\rho = \text{im } i_\rho = K_\rho$.

The filtration on C_ρ is the quotient of the full original MW3 filtration. Its two degrees are retained explicitly:

$$\begin{aligned} \overline{M}_j C_\rho &= \pi_\rho(M_j E_\rho) = \begin{cases} 0 & j < b - 1, \\ K_\rho & b - 1 \leq j < b + 1, \\ C_\rho & j \geq b + 1, \end{cases} \quad \phi_\rho(\overline{M}_j C_\rho) = M_{j-b} V_2, \\ \overline{M}_j K_\rho &= K_\rho \cap \overline{M}_j C_\rho = \begin{cases} 0 & j < b - 1, \\ K_\rho & j \geq b - 1, \end{cases} \quad \overline{M}_j Q_\rho = p_\rho(\overline{M}_j C_\rho) = \begin{cases} 0 & j < b + 1, \\ Q_\rho & j \geq b + 1, \end{cases} \\ 0 &\longrightarrow \overline{M}_j K_\rho \xrightarrow{i_\rho} \overline{M}_j C_\rho \xrightarrow{p_\rho} \overline{M}_j Q_\rho \longrightarrow 0. \end{aligned} \tag{MRE35}$$

Only the exponents zero and one survive π_ρ ; MW3 places them in degrees $m - 1 = b + 1$ and $m - 3 = b - 1$. This proves the first line and, by intersection and image, the second. The last line is zero in every position for $j < b - 1$; for $b - 1 \leq j < b + 1$, its only nonzero map is the identity on K_ρ ; for $j \geq b + 1$, it is the full exact sequence just proved. These three cases prove exactness and strictness. In terms of the separately centered degree-zero line filtrations C_j , the actual shifts are $\overline{M}_j K_\rho = C_{j-b+1} K_\rho$ and $\overline{M}_j Q_\rho = C_{j-b-1} Q_\rho$. The same coordinate calculation proves that $0 \rightarrow J_\rho \cap M_j E_\rho \rightarrow M_j E_\rho \xrightarrow{\pi_\rho} \overline{M}_j C_\rho \rightarrow 0$ is exact: its kernel is the displayed intersection and its image is the defined quotient step. Thus the map from the actual packet is strict before the subobject/quotient extension is formed.

All retained operators preserve J_ρ : the nilpotent raises exponents, each term of the original finite exponential raises exponents, and the added diagonals preserve every original basis line. Denote their induced operators on C_ρ by bars. Their entire formulas and their restriction and quotient values are

$$\begin{aligned} \overline{N}w_0 &= w_1, \quad \overline{N}w_1 = 0, \quad \overline{A}_a(c_0 w_0 + c_1 w_1) = a^\rho(c_0 w_0 + (c_1 + c_0 \log a)w_1), \\ \overline{\Gamma}_q w_r &= q^{(b+1)/2-r} w_r, \quad \overline{F}_q w_r = q^{\rho+(b+1)/2-r} w_r \quad (r = 0, 1), \\ \begin{array}{c|ccc} & A_a & \Gamma_q & F_q \\ \hline K_\rho & a^\rho I & q^{(b-1)/2} I & q^{\rho+(b-1)/2} I \\ Q_\rho & a^\rho I & q^{(b+1)/2} I & q^{\rho+(b+1)/2} I \end{array} \\ \overline{A}_q &= \overline{F}_q \overline{\Gamma}_q^{-1} (I + (\log q) \overline{N}), \\ 2 \log_q |F_{q,K_\rho}| &= 2 \text{Re } \rho + b - 1, \quad 2 \log_q |F_{q,Q_\rho}| = 2 \text{Re } \rho + b + 1. \end{aligned} \tag{MRE36}$$

In the full source exponential, terms with nilpotent power at least two have image in J_ρ ; applying π_ρ to each term therefore proves the first line with exactly its coefficient $a^\rho \log a$. The source diagonal

exponents are $(m-1)/2 - r = (b+1)/2 - r$, proving the second line. Restriction to w_1 and passage to $[w_0]$ prove the table. For each $O = A_a, \Gamma_q, F_q$, applying to w_1 proves $\overline{O}i_\rho = i_\rho O_{K_\rho}$, and applying to both w_0, w_1 proves $p_\rho \overline{O} = O_{Q_\rho} p_\rho$. For the nilpotent these maps intertwine its zero restriction and quotient. The product $\overline{F}_q \overline{\Gamma}_q^{-1}$ is $q^\rho I$, proving reconstruction in the displayed order. The two real logarithms follow from $|q^{\rho+c}| = q^{\operatorname{Re} \rho + c}$ for real c .

For a precise filtered comparison of entire exact sequences, take the line $D_b = \mathbb{C}d_b$ concentrated in degree b , with $N_{D_b} = 0$, $A_{a,D_b} = I$, and $\Gamma_{q,D_b} = F_{q,D_b} = q^{b/2}I$. This is a specified complex filtered line for every integer $b = m-2$; its definition does not assert a geometric realization. Give tensors sum filtrations, tensor-sum nilpotents, and tensor-product invertible operators. The comparison maps are

$$\begin{aligned} \theta_C : C_\rho &\xrightarrow{\sim} V_2 \otimes D_b, & w_0 &\mapsto 1 \otimes d_b, & w_1 &\mapsto u \otimes d_b, \\ \theta_K : K_\rho &\xrightarrow{\sim} K \otimes D_b, & w_1 &\mapsto u \otimes d_b, & \theta_Q : Q_\rho &\xrightarrow{\sim} Q \otimes D_b, & [w_0] &\mapsto \overline{1} \otimes d_b, \\ \theta_C i_\rho &= (i \otimes I) \theta_K, & (p \otimes I) \theta_C &= \theta_Q p_\rho, \\ \phi_\rho \overline{A}_a &= A_{a,2} \phi_\rho, & \phi_\rho \overline{\Gamma}_q &= q^{b/2} \Gamma_{q,2} \phi_\rho, & \phi_\rho \overline{F}_q &= q^{b/2} F_{q,2} \phi_\rho. \end{aligned} \quad (\text{MRE37})$$

The inverse maps send the displayed target basis vectors to their displayed sources. The filtration step on each tensor is the original MRE13–MRE14 step at $j-b$, tensored with D_b . Equation MRE35 therefore proves strictness in both directions for each comparison map. The two diagram equations hold on w_1 and on w_0, w_1 , respectively, proving commutativity of the whole exact-sequence comparison. The last three equations follow by applying both sides to w_0, w_1 , using MRE17 and MRE36. The same calculation intertwines the nilpotents with coefficient one. Thus all maps in the sequence are intertwined by the specified θ 's for N, A, Γ, F . For $m=2, b=0$ and these are exactly the fixed length-two maps. For $m>2$, ϕ_ρ with separately centered target filtration is filtration preserving because $j-b \leq j$, but non-strict: at $j=1$ its image of $\overline{M}_1 C_\rho$ omits 1, whereas $M_1 V_2 = V_2$. The scalar $q^{b/2}$ in both added operators and the degree shift b are retained in MRE37.

The actual nilpotent link is

$$\begin{aligned} \overline{N} &= i_\rho \eta_\rho p_\rho, & \eta_\rho : Q_\rho &\xrightarrow{\sim} K_\rho, & \eta_\rho([w_0]) &= w_1, \\ \nu_\rho : Q_\rho \otimes T_q &\xrightarrow{\sim} K_\rho, & \nu_\rho([w_0] \otimes t) &= w_1, \\ F_{q,K_\rho} \eta_\rho &= q^{-1} \eta_\rho F_{q,Q_\rho}, & \Gamma_{q,K_\rho} \eta_\rho &= q^{-1} \eta_\rho \Gamma_{q,Q_\rho}, & A_{a,K_\rho} \eta_\rho &= \eta_\rho A_{a,Q_\rho}. \end{aligned} \quad (\text{MRE38})$$

Here T_q is exactly MRE25, with degree -2 , both added scalars q^{-1} , and arithmetic scalar one. Applying the factorization to $c_0 w_0 + c_1 w_1$ gives $c_0 w_1$ on both sides. In ν_ρ , the domain and target both have degree $(b+1)-2 = b-1$, added F -eigenvalue $q^{\rho+(b+1)/2} q^{-1} = q^{\rho+(b-1)/2}$, and the analogous Γ -eigenvalue with ρ omitted. Their original arithmetic eigenvalues are both a^ρ . Its inverse is $w_1 \mapsto [w_0] \otimes t$. These computations prove strict filtered equivariance, its inverse, and all three identities in the last line. The comparison with MRE26 sends $(\overline{1} \otimes d_b) \otimes t$ to $(\overline{1} \otimes t) \otimes d_b$ and then to $u \otimes d_b$; the specified tensor rearrangement has coefficient one, so this is exactly $\theta_K \nu_\rho$.

The whole quotient has the corresponding strict map

$$\begin{aligned} \mathcal{N}_\rho : C_\rho \otimes T_q &\longrightarrow C_\rho, & z \otimes t &\longmapsto \overline{N} z, \\ \overline{N}(\overline{M}_{j+2} C_\rho) &= K_\rho \cap \overline{M}_j C_\rho, & \pi_\rho \mathcal{N}_{E_\rho} &= \mathcal{N}_\rho(\pi_\rho \otimes I), & \mathcal{N}_{E_\rho}(v \otimes t) &= N_\rho v. \end{aligned} \quad (\text{MRE39})$$

For $j < b-1$, the domain step in the middle equality is at most K_ρ , which $\overline{N}$ kills, and the right side is zero. For $j \geq b-1$, that domain step is C_ρ , its image is K_ρ , and the right side is K_ρ . This proves strictness at every integer. The quotient diagonal entries give $\overline{F}_q \overline{N} \overline{F}_q^{-1} = q^{-1} \overline{N}$ and the identical relation for $\overline{\Gamma}_q$; the full arithmetic formula gives $\overline{A}_a \overline{N} = \overline{N} \overline{A}_a$. These prove equivariance with the three operators on the Tate line. The last equality follows on $v \otimes t$ from the already proved identity $\pi_\rho N_\rho = \overline{N} \pi_\rho$.

The relative-filtration comparison shifts both filtrations of MRE28–MRE29. Define

$$\begin{aligned} W_i^{(\rho)} C_\rho &= \phi_\rho^{-1}(W_{i-b} V_2), & R_j^{(\rho)} C_\rho &= \phi_\rho^{-1}(R_{j-b} V_2), & W^{(\rho)} &= R^{(\rho)} = \overline{M}, \\ \overline{N}^k : \mathrm{Gr}_{i+k}^{R^{(\rho)}} \mathrm{Gr}_i^{W^{(\rho)}} C_\rho &\xrightarrow{\sim} \mathrm{Gr}_{i-k}^{R^{(\rho)}} \mathrm{Gr}_i^{W^{(\rho)}} C_\rho \quad (i \in \mathbb{Z}, k \geq 0). \end{aligned} \quad (\text{MRE40})$$

Multiplication by the local coordinate lowers the displayed degree by two. The only nonzero $W^{(\rho)}$ -graded spaces are K_ρ in degree $b-1$ and Q_ρ in degree $b+1$. The induced nilpotents on them are zero, and the induced $R^{(\rho)}$ -filtrations are concentrated in their respective degrees by MRE35. Thus for $k=0$ the displayed map is the identity, and for $k>0$ both spaces are zero. The comparison ϕ_ρ identifies each iterated graded space with $\mathrm{Gr}_{i-b\pm k}^R \mathrm{Gr}_{i-b}^W V_2$; hence these are exactly the MRE29 maps with i replaced by $i-b$. Any other relative filtration must induce degree $b-1$ on K_ρ and degree $b+1$ on Q_ρ , because for a zero nilpotent the relative isomorphisms force all other graded pieces to vanish. Below $b-1$ its intersection with K_ρ and image in Q_ρ are zero, so the step is zero. Between $b-1$ and $b+1$ it contains K_ρ with zero quotient image, so it equals K_ρ . From $b+1$ onward it contains K_ρ and surjects onto Q_ρ , so it equals C_ρ . This proves uniqueness with all induced degrees intact. The displayed $W^{(\rho)}$ is also the quotient of the specified full-block filtration $W_j^{\text{full}} E_\rho = M_j E_\rho$, since its quotient steps are MRE35. This is the specified filtration on the original coefficient packet, with its exact quotient map. For the separately specified one-step filtration on C_ρ concentrated in degree b , MRE5–MRE6 likewise identify $\overline{M}$ as its relative filtration: the only distance-one map sends $[w_0]$ to $[w_1]$ with coefficient one, distance zero gives the identity, and every distance at least two has both graded spaces zero.

The extension discrepancies also survive without any factor change. Every linear section $s_c : Q_\rho \rightarrow C_\rho$ of p_ρ sends $[w_0]$ to $w_0 + cw_1$. Direct substitution gives

$$\begin{aligned} (\overline{N} s_c - s_c N_{Q_\rho})([w_0]) &= w_1, & (\overline{A}_a s_c - s_c A_{a, Q_\rho})([w_0]) &= a^\rho (\log a) w_1, \\ (\overline{F}_q s_c - s_c F_{q, Q_\rho})([w_0]) &= c(q^{\rho+(b-1)/2} - q^{\rho+(b+1)/2}) w_1, \\ \overline{F}_q \overline{A}_a \overline{F}_q^{-1} - \overline{A}_a &= a^\rho (q^{-1} - 1) (\log a) \overline{N}. \end{aligned} \quad (\text{MRE41})$$

The first two differences are independent of c ; the second is nonzero for $a \neq 1$. Thus no section splits either the nilpotent or an original arithmetic operator with $a \neq 1$. The third difference vanishes only for $c=0$, since its parenthesized factor is $q^{\rho+(b-1)/2}(1-q) \neq 0$. The section s_0 preserves the filtration because its only source degree is $b+1$, when the target is already the full space. It therefore gives exactly the added diagonal splitting. Conjugating the full arithmetic formula in MRE36 by $\overline{F}_q$ and using its q^{-1} nilpotent relation proves the last line with its unchanged scalar. These are the transported MRE30–MRE33 calculations on the actual quotient.

Finally the transport is compatible with every independent slot and with the combined operator of MW9a. For an admissible odd prime power q coprime to n , extend π_ρ linearly to the coefficient algebra and put $\overline{T}_{a,q} = \overline{A}_a \Psi_q$. On the fixed bases,

$$\begin{aligned} \overline{T}_{a,q}(w_r t^i) &= (-1)^{\lfloor qi/n \rfloor} a^\rho \sum_{\ell=0}^{1-r} \frac{(\log a)^\ell}{\ell!} w_{r+\ell} t^{qi \bmod n}, \quad r=0,1, \\ \pi_\rho T_{a,q} &= \overline{T}_{a,q} \pi_\rho, & \bigotimes_{\nu} \theta_{C, \rho_\nu} : \bigotimes_{\nu} C_{\rho_\nu} &\xrightarrow{\sim} \left(\bigotimes_{\nu} V_{2, \rho_\nu} \right) \otimes \left(\bigotimes_{\nu} D_{b_\nu} \right), \\ b_\nu &= m_{\rho_\nu} - 2, & \text{degree translation } \sum_{\nu} b_\nu, & \text{added } F, \Gamma \text{ factor } q^{\frac{1}{2} \sum_{\nu} b_\nu}. \end{aligned} \quad (\text{MRE42})$$

Linearity of π_ρ preserves the displayed sign, and applying it to the full source finite Taylor sum keeps exactly the terms shown. This proves both the first line and the intertwining identity. On a face the map retains the mask and sends each coefficient to its quotient; a vanished coefficient remains the zero of that same present slot. Zero insertion commutes with π_ρ , since it is linear. For the tensor

comparison, send each displayed original tensor basis vector to $(\bigotimes_{\nu} u^{r_{\nu}}) \otimes (\bigotimes_{\nu} d_{b_{\nu}})$, with the inverse prescribed by the same basis pairs. Every filtration degree is translated by the sum $\sum_{\nu} b_{\nu}$, and both added diagonal factors multiply to $q^{\frac{1}{2} \sum_{\nu} b_{\nu}}$. This proves strictness and operator compatibility of this full tensor isomorphism. On coefficient eigenspaces with eigenvalues ω_{ν} , the original arithmetic graded eigenvalue is $a^{\sum_{\nu} \rho_{\nu}}$, the coefficient eigenvalue is $\prod_{\nu} \omega_{\nu}$, and their combined eigenvalue is $(\prod_{\nu} \omega_{\nu}) a^{\sum_{\nu} \rho_{\nu}}$, exactly as in MW16b. Positive nilpotent powers lower total degree by two per power, so none changes these graded eigenvalues; every such power is still retained in the exact tensor formula MW16a.

Proved control and continuing scope. The calculation constructs the strict filtered exact sequence on the length-two arithmetic extension, its exact shifted added eigenvalues, the Tate-typed nilpotent isomorphism, and its relative filtration. Every original arithmetic scalar and nilpotent coefficient survives. These results supply this local extension control within the program. MRE34–MRE42 supply its strict comparison with every specified actual length-two packet quotient, including the original induced degree translation, both added diagonal factors, and the full arithmetic and coefficient operators. The arbitrary parameter calculation MRE7–MRE33 makes no zero-multiplicity assertion. They do not determine $2 \operatorname{Re} \rho$, establish a pure lisse source, or supply the analytic upper estimate used in Weil II 1.8.1. Those numerical and geometric controls remain separate constructions to be carried out on the original arithmetic sources.

117 The same arithmetic discrepancy in the mixed filtration and the source boundary

The following calculation connects the independent mixed faces, the nilpotent filtration and the original phase observation on the same arithmetic class. All maps used here are constructed in the preceding attachments; this section computes their composite on the full packet.

Fix an actual zero $\rho = 1/2 + \delta + i\gamma$ in the open strip, with $0 < \delta < 1/2$ and $\gamma > 0$, its full order m , and a full packet h containing it. The other offcritical quadrants map to this quadrant by the original reflection and conjugation arrows of (CAU.30)–(CAU.32), retaining their full local jets. Retain the original CRT idempotent and terminal jet

$$v_{\rho} = e_{\rho}(s - \rho)^{m-1} \in E_h, \quad u_{\rho} = \sigma_h v_{\rho} \in T_U Q, \quad 0 < \delta < 1/2, \quad \gamma > 0. \quad (\text{MAI1})$$

The coefficient of $(s - \rho)^{m-1}$ in this local jet is one; therefore $v_{\rho} \neq 0$. The full-unit section satisfies $J_h s_h = 1$, so σ_h is injective and $u_{\rho} \neq 0$. The nilpotent kills v_{ρ} , and the full original exponential therefore gives

$$A_a v_{\rho} = a^{\rho} v_{\rho}, \quad \overline{\mathcal{R}}_a u_{\rho} = a^{\rho} u_{\rho} \quad (a > 0). \quad (\text{MAI2})$$

The second identity uses the actual theta-boundary homotopy, not an equivariance assertion for the section before quotienting.

For $n \geq 1$, in the coefficient algebra $B_{h,n} = E_h[t]/(t^n + 1)$, the original insertion and trace retraction satisfy

$$E_h \xrightarrow{\iota_n} B_{h,n} \xrightarrow{r_n} E_h, \quad r_n \iota_n = 1, \quad \Psi_{q_0} \iota_n = \iota_n, \quad r_n \Psi_{q_0} = r_n. \quad (\text{MAI3})$$

Here q_0 is an odd prime power coprime to n , and r_n is $1/n$ times the trace of multiplication. The complete coefficient computation is $r_n(\sum b_i t^i) = b_0$. Thus $\iota_n v_{\rho}$ is nonzero, fixed by coefficient Frobenius, and has arithmetic eigenvalue a^{ρ} . It is represented on the unsynchronized face $\{0\}$ by its supported constant coefficient and absent remaining slots. Synchronization also observes it on the full face. Both vertices and their connecting arrow are retained; (MAI3) identifies their unchanged arithmetic amplitude.

Tensor each of these maps over $\mathbb{C}$ on its amplitude spaces. To prove nonvanishing explicitly, choose a linear functional ℓ with $\ell(v_\rho) = 1$. Then $\ell^{\otimes k}(v_\rho^{\otimes k}) = 1$; similarly the tensor retraction $r_n^{\otimes k}$ carries $(\iota_n v_\rho)^{\otimes k}$ to $v_\rho^{\otimes k}$. The injective original cohomology map carries the latter to $u_\rho^{\otimes k} \neq 0$. These tensor arrows retain the ordered tuple of coefficient-face labels. Their arithmetic eigenvalue is $a^{k\rho}$, and hence

$$\log\left(\frac{|a^{k\rho}|^2}{a^k}\right) = 2k\delta \log a. \quad (\text{MAI4})$$

Now retain the entire tensor of the original length- m local blocks, not only this terminal line. In the original jet basis write $N_{\text{tot}} = \sum_{\nu=1}^k N_\nu$ and use the filtration proved in (MW15). On every nonzero graded piece Gr_j^M the added Tate-scaling operator $F_q^{\otimes k}$, for real $q > 1$, acts by $q^{k\rho+j/2}$. Indeed on a basis tensor of indices $j_1, \dots, j_k$ its scalar is the product $\prod_\nu q^{\rho+j_\nu/2}$, and their sum is j . The coefficient operator on a retained eigenspace contributes a root of unity ω . Every eigenvalue on that graded eigenspace is therefore

$$\lambda_{k,j,\omega} = \omega q^{k\rho+j/2}, \quad 2\log_q |\lambda_{k,j,\omega}| - (k+j) = 2k\delta. \quad (\text{MAI5})$$

The equality uses $|\omega| = 1$ and the original real logarithm of q . It holds for every chosen nonzero graded piece, including $j = 0$. For the tensor of the terminal lines in (MAI1), the index is exactly $j = -k(m-1)$ and $\omega = 1$. Thus this computation includes a specified nonzero line in the arithmetic retract.

The exact reconstruction from (MW19) also tensors without commuting operators in an unproved order:

$$A_q^{\otimes k} = F_q^{\otimes k} (\Gamma_q^{-1})^{\otimes k} \exp((\log q) N_{\text{tot}}). \quad (\text{MAI6})$$

Each tensor factor has that ordered identity. Operators in different factors commute, so multiplying their finite exponentials gives the displayed sum in the exponent. On the graded piece, the exponential induces the identity and $(\Gamma_q^{-1})^{\otimes k}$ contributes $q^{-j/2}$. The resulting arithmetic eigenvalue is exactly $q^{k\rho}$. This proves the map between the added Tate-scaling observation and the original arithmetic action, including every nilpotent term. Integer monodromy indices alone do not change its real exponent.

For its original source-norm observation, take the reflected quartet packet h_0 of (PAM.1), with $\delta > 0$, and the complete cyclic tensor factor of (PAM.2)–(PAM.5). Its nonzero terminal eigenvector v_k has eigenvalue $k\rho$. Its exact relation to the tensor in (MAI4) includes a nonzero scalar. Put $K = k(m-1)$. The extreme tensor centre $k\rho$ can arise only by choosing ρ in every factor of that quartet. On this block, the multinomial expansion gives $(N_1 + \dots + N_k)^K = K! / ((m-1)!)^k \prod_\nu N_\nu^{m-1}$. Every other term vanishes because some exponent is at least m . The complete unit of (PAM.3) acts on this terminal monomial by its nonzero scalar value. Consequently

$$\eta_k v_k = c_{h_0,k} v_\rho^{\otimes k}, \quad \mathcal{I}_k v_k = c_{h_0,k} u_\rho^{\otimes k}, \quad c_{h_0,k} = \frac{(k(m-1))!}{((m-1)!)^k} ((g/h_0)(\rho))^k \neq 0. \quad (\text{MAI6a})$$

Here v_ρ in the first equality is taken in E_{h_0} . If the initially chosen h differs from h_0 , use the common full packet $H = \text{lcm}(h, h_0)$. Both terminal jets have the same image in E_H , and the literal identities $\sigma_H \iota_h H = \sigma_h$ and $\sigma_H \iota_{h_0} H = \sigma_{h_0}$ identify their original arithmetic classes. This proves the second equality with the original u_ρ of (MAI1), retaining the entire coefficient. For this phase observation take $k \geq 2$, $L > 0$, $0 \leq \theta < 2\pi$, and an admitted polynomial cutoff $N \geq q_k - 1$, with $q_k = (1 + k(m-1))(k+1)^2$. Write $R_k = R_{N,\theta}$ for the canonical lift on this specified source of (PAM.9)–(PAM.13). The source lift $R_k v_k$ is nonzero because its original cohomology observation is injective. With the exact twisted phase domain and its generator $D_{L,\theta} = -\partial_r + k/2$, let

$$B_k = D_{L,\theta} R_k - R_k A_k. \quad (\text{MAI7})$$

Every term belongs to its proved domain. Put $f = R_k v_k$. The spectral expansion of the skew-adjoint derivative on that domain gives frequencies $u_n = (2\pi n + \theta)/L$ and

$$\frac{\|B_k v_k\|^2}{\|R_k v_k\|^2} = k^2 \delta^2 + \frac{\sum_n (u_n - k\gamma)^2 \|f_n\|^2}{\sum_n \|f_n\|^2} \geq k^2 \delta^2. \quad (\text{MAI8})$$

To verify the equality, apply $D_{L,\theta} - k\rho$ to each orthogonal frequency component. Its multiplier is $-k\delta + i(u_n - k\gamma)$, whose squared modulus is $k^2 \delta^2 + (u_n - k\gamma)^2$. Parseval and the domain condition justify both sums. The denominator is positive because $f \neq 0$.

Equations (MAI4), (MAI5) and (MAI8) are thus connected observations of the same original arithmetic displacement. The first follows through the coefficient-pure retract; the second computes it after the complete monodromy grading; the third measures its necessary original theta-boundary energy. In particular the fully evaluated phase calculation does not erase it: at the first cutoff the boundary is the nonzero residue map of (PSC.50)–(PSC.64), even when the two section gluing maps agree. The arithmetic four-window identities in (AAM.19)–(AAM.30) retain that same boundary scale and all its inverse Gram factors.

Deligne's final strict-weight exclusion in Weil II applies to geometric constituents whose integer weights and strict upper bound have both been proved: the complete calculation is recorded in the following dyadic-bootstrap chapter. In (MAI5) the integer j has been computed, whereas the central real weight is the unchanged $2\Re\rho$. Calling that central weight one would replace the explicitly computed scalar rather than deduce its value. The present calculation identifies its exact location in the assembled mixed object and in the original boundary operator; it proves no disappearance of the marked cohomology $T_U Q$.

Abstract

For the original monic packet polynomial, retaining all selected zero multiplicities, we calculate the missing u -connection of its polynomial exponential complex in the original monomial lattice. The connection has an explicit second-order coefficient, a first-order coefficient of trace $d/2$, and a degree-correct chain realization. We calculate its full tensor version and the exact period determinant, including its gamma factors and ordered contour phase. We also apply Deligne's lissity and local-monodromy theorems to the actual finite-field family on $u \neq 0$, obtaining a genuine mixed boundary extension there. The original theta source, its Taylor unit, and its metric enter through explicit maps. The exact single-primary branch now includes its full period matrix, boundary monodromy, finite cover, source-unit maps and finite-field character decomposition through every ordered tensor degree, together with the exact original coefficient lattice and its specialization maps. The general multi-primary sector transitions and original arithmetic uniform four-volume estimate remain continuing calculations on the retained objects.

118 Fixed objects and original source maps

Let Z be the original nonempty finite set of selected zeros of $g = 2\xi$, with the complete selected orders m_ρ , and put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho} = s^d + \sum_{b=0}^{d-1} h_b s^b, \quad d = \sum_{\rho \in Z} m_\rho \geq 1. \quad (1)$$

The primitive and its constant are fixed as

$$\Phi_h(s) = \frac{s^{d+1}}{d+1} + \sum_{b=0}^{d-1} \frac{h_b s^{b+1}}{b+1}, \quad \Phi_h(0) = 0, \quad f_t(s) = \Phi_h(s) - ts. \quad (2)$$

Retain the original source spaces, Fourier convention, and operators

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^m F(x)| < \infty \text{ for all } b \in \mathbb{Z}, m \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \\ \widehat{\phi}(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx. \end{aligned} \quad (3)$$

The seed is

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g. \quad (4)$$

The inherited Euler inverses supply the original $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h = g/h$. Define

$$E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = j_h v_h \in E_h^\times, \quad \alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h. \quad (5)$$

The original identities are

$$\Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h \mathcal{T}_h(P) = v_h[P]_h. \quad (6)$$

Thus $[\mathbb{C}[s] \xrightarrow{h} \mathbb{C}[s]ds]$ maps to $[V \xrightarrow{\Theta} \mathcal{B}]$ by $(\alpha_h, \mathcal{T}_h)$. Its degree-one map is exactly $\sigma_h M_{v_h}$ into $Q = \mathcal{B}/\Theta V$. The linear source section is not asserted to be a ring homomorphism. Tensoring these chain maps preserves the full factor $v_h^{\otimes k}$ and the original ordered exterior signs.

119 The full lattice and its marked fibre

For $R_k = \mathbb{C}[u, t_1, \dots, t_k]$ set

$$\mathcal{C}_k^j = R_k[s_1, \dots, s_k] \otimes \bigwedge^j (ds_1, \dots, ds_k), \quad \mathcal{D} = \sum_{i=1}^k L_i ds_i \wedge (-), \quad L_i = u\partial_{s_i} + h(s_i) - t_i. \quad (7)$$

The L_i commute, so the exterior signs give $\mathcal{D}^2 = 0$.

Proposition 119.1 (The original complete lattice). *The complex (7) has cohomology only in degree k , with*

$$\mathcal{H}_k := H^k(\mathcal{C}_k) = \bigoplus_{0 \leq a_i < d} R_k[s_1^{a_1} \cdots s_k^{a_k} ds_1 \wedge \cdots \wedge ds_k]. \quad (8)$$

Specialization at $u = t_1 = \cdots = t_k = 0$ identifies this displayed module with $E_h^{\otimes k}$, retaining every primary jet.

Proof. In one factor $L = u\partial_s + h - t$ increases the degree of a nonzero polynomial by exactly d and retains its leading coefficient. Consequently L is injective. Subtracting $L(as^j)$ at each highest degree gives a remainder of degree below d . A polynomial of degree below d in the image of L must be zero by the same degree identity. This proves the direct module decomposition

$$R[s] = L(R[s]) \oplus R[s]_{<d}, \quad R = \mathbb{C}[u, t].$$

The inverse of L on its image is R -linear and contracts the image two-term complex. Tensor the direct decompositions in the specified factor order. Every tensor summand with a contractible factor contracts by that factor's homotopy with the sign contributed by preceding degrees; the cross terms cancel by the tensor differential rule. Only the top remainder summand remains. The splitting is over R_k , so it remains a splitting after the stated coefficient specialization. $\square$

The family is a literal deformation of the original finite polynomial source images: with $V_* = \alpha_h(\mathbb{C}[s])$ and $\mathcal{B}_h = \mathcal{T}_h(\mathbb{C}[s])$, its one-factor differential is

$$d_{u,t}^\theta = \Theta|_{V_*} + u\mathcal{T}_h\partial_s\alpha_h^{-1} - t\mathcal{T}_h\alpha_h^{-1}. \quad (9)$$

The inverse maps here are defined on the displayed injective images. Equation (6) gives $d_{u,t}^\theta\alpha_h = \mathcal{T}_h L$. The derivative term is also the original function-space commutator

$$\mathcal{T}_h(P') = (\log x)P(D)F_h - P(D)((\log x)F_h),$$

because $[\log x, D] = 1$. In Mellin coordinates this subtraction retains and cancels the two equal $v_h'P$ terms, leaving $v_h P'$.

If χ is the minimal polynomial of $S = \sum_i s_i$ on $E_h^{\otimes k}$, the weighted cyclic injection is

$$\eta : \mathbb{C}[S]/\chi \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P\left(\sum_i s_i\right)1. \quad (10)$$

For fixed-order monic division $\chi(S)P(S) = \sum_i h(s_i)Q_i(\mathbf{s})$, the ordered primitive $\sum_i (-1)^{i-1} Q_i ds_1 \wedge \cdots \widehat{ds_i} \cdots \wedge ds_k$ has image $\sum_i (h_i Q_i - t_i Q_i + u \partial_i Q_i)$. It follows that

$$[\chi(S)P(S)]_{\mathcal{H}_k} = \sum_i [t_i Q_i - u \partial_{s_i} Q_i]_{\mathcal{H}_k}. \quad (11)$$

The quotient by the cyclic image and this parameter defect remain in the full family.

120 A genuine finite-field mixed extension

Choose the same finitely generated coefficient ring $R_0 \subset \mathbb{C}$ containing the coefficients of h , the matrices of the complete v_h and its inverse, and $1/(d+1)!$. Fix a specified finite-field point $R_0 \rightarrow \kappa = \mathbb{F}_Q$ of characteristic $p > d+1$. For the rest of this section polynomial coefficients mean their images under this specified map. Let

$$\begin{aligned} S &= \mathbb{G}_{m_u} \times \mathbb{A}_{\mathbf{t}}^k, & \pi : \mathbb{A}_{\mathbf{s}}^k \times S &\longrightarrow S, \\ \mathcal{L} &= \mathcal{L}_\psi \left(\frac{\sum_i \Phi_h(s_i) - \sum_i t_i s_i}{u} \right), & \mathcal{F} &= R^k \pi_! \mathcal{L}. \end{aligned} \quad (12)$$

Here ψ is a fixed nontrivial additive character with values in the chosen ℓ -adic coefficient field, $\ell \neq p$.

Proposition 120.1 (The actual open family and boundary stalk). *The sheaf $\mathcal{F}$ in (12) is lisse, of rank d^k , and pointwise pure of weight k ; all $R^j \pi_! \mathcal{L}$ with $j \neq k$ vanish. Fix any finite-field parameter value $\mathbf{t} = \mathbf{t}_0$, after a residue-field extension when necessary, and write $\mathcal{F}_{\mathbf{t}_0}$ for its pullback to $\mathbb{G}_m$. For $j : \mathbb{G}_m \hookrightarrow \mathbb{A}_u^1$ and $i : \{0\} \hookrightarrow \mathbb{A}_u^1$ there is an exact sequence*

$$0 \longrightarrow j_! \mathcal{F}_{\mathbf{t}_0} \longrightarrow j_* \mathcal{F}_{\mathbf{t}_0} \longrightarrow i_*(V^I) \longrightarrow 0, \quad (13)$$

where V is the local geometric generic representation and I its inertia group. The boundary stalk V^I is mixed of weights at most k .

Proof. The highest homogeneous phase is $\sum_i s_i^{d+1}/((d+1)u)$. Its partial derivatives s_i^d/u cannot vanish simultaneously at a projective point. Therefore the explicit coefficient map from S to the universal degree- $(d+1)$ polynomial parameter space lands entirely in Deligne's good open. Weil II, Lemma 3.7.3, proves lissity of the universal proper-support cohomology sheaves there. Proper-support base change pulls these sheaves to (12). Statement 3.7.2.3, with polynomial degree $d+1$ and affine dimension k , proves the asserted concentration, rank, and weight. This check uses no simplicity assumption on finite critical roots.

The stalk of $j_*\mathcal{F}_{t_0}$ at zero is V^I ; the stalk of $j_!\mathcal{F}_{t_0}$ there is zero. On the open the map is the identity, proving (13) on every geometric stalk. The extension by zero is mixed of weight k .

On an open finite-index inertia subgroup with unipotent action, let N be the actual logarithmic monodromy operator and M its monodromy filtration. Weil II 1.8.4 applies to the lisse pointwise pure $\mathcal{F}_{t_0}$ and proves that $\mathrm{Gr}_r^M V$ is pure of weight $k + r$. By 1.8.8, V^I is obtained from $\ker N$ by finite-group invariants; the filtration it inherits has no graded pieces with $r > 0$. Its nonzero graded pieces therefore have weights at most k . No graded rank or numerical matrix for N is inserted in this argument. $\square$

This supplies an actual use of Deligne's mixed-boundary machinery on the finite-field branch. The coefficient specialization of (8), and the theta source map (6), are separately explicit. A comparison from their marked complex fibre to the local representation in (13) has not been proved by these constructions. The complex boundary data relevant to such a comparison are calculated next, with all original coefficients retained.

121 Exact division and the full boundary connection

For $0 \leq b < d$ perform monic division in the original variable s :

$$f_t(s)s^b = (h(s) - t)q_b(s, t) + r_b(s, t), \quad \deg_s r_b < d, \quad \deg_s q_b \leq b + 1 \leq d. \quad (14)$$

Let $C_h(t)$ and $B_h(t)$ be the matrices whose b th columns are the coefficients of r_b and $\partial_s q_b$, respectively, in the ascending monomial basis $1, s, \dots, s^{d-1}$.

Lemma 121.1 (Exact differential remainder). *In the one-factor top lattice,*

$$[f_t s^b] = [r_b - u \partial_s q_b]. \quad (15)$$

There is no further reduction of the displayed quotient derivative.

Proof. Subtract $Lq_b = (h - t)q_b + u \partial_s q_b$ from (14). The derivative has degree below d , so the right side is already in the retained remainder basis. $\square$

Theorem 121.2 (Degree-correct flat connection). *On the one-factor complex localized at u , the degree-one and degree-zero connections in the u direction are*

$$\nabla_u^1 = \partial_u - \frac{f_t}{u^2}, \quad \nabla_u^0 = \partial_u - \frac{f_t}{u^2} + \frac{1}{u}. \quad (16)$$

The induced top connection in the original monomial lattice is exactly

$$\boxed{\nabla_u = \partial_u - \frac{C_h(t)}{u^2} + \frac{B_h(t)}{u}}. \quad (17)$$

On the ordered k -factor complex the degree- j operator and its top cohomology operator are

$$\nabla_u^{(j)} = \partial_u - \frac{\sum_i f_{t_i}(s_i)}{u^2} + \frac{k - j}{u}, \quad (18)$$

$$\boxed{\nabla_u} = \left[\partial_u - \frac{\sum_i C_{h,i}(t_i)}{u^2} + \frac{\sum_i B_{h,i}(t_i)}{u} \right]. \quad (19)$$

These commute with the existing parameter connections $\nabla_{t_i} = \partial_{t_i} - s_i/u$ and are compatible with the original complex differential.

Proof. The commutators on polynomial coefficients are

$$[L, \partial_u] = -\partial_s, \quad [L, -f_t/u^2] = -(h-t)/u,$$

so $[L, \nabla_u^1] = -L/u$ and $\nabla_u^1 L = L(\nabla_u^1 + 1/u) = L\nabla_u^0$. This proves the chain identity, including the degree-zero correction. The differential remainder (15) proves (17).

For (18), the commutator with each L_i is $-L_i/u$, and the scalar difference between adjacent exterior degrees contributes $+L_i/u$. Their cancellation proves chain compatibility in the original order. Top cohomology and (15) give (19). Finally, differentiating $-s_i/u$ in u gives $+s_i/u^2$, whereas the commutator of $-\sum f_{t_i}/u^2$ with ∂_{t_i} is $-s_i/u^2$. All other terms commute. This proves the asserted flatness. $\square$

Proposition 121.3 (The exact first-order trace). *$B_h(t)$ is independent of t . In the unchanged ascending monomial basis it is upper triangular and satisfies*

$$(B_h)_{bb} = \frac{b+1}{d+1}, \quad \boxed{\text{Tr } B_h = \frac{d}{2}}. \quad (20)$$

Proof. In (14), compare coefficients of degrees greater than d . The perturbation $-ts^{b+1}$ and the product $-tq_b$ both have degree at most d . All coefficients of q_b except possibly its constant coefficient are therefore determined independently of t . Differentiation removes that constant. The leading term of q_b is $s^{b+1}/(d+1)$, so its derivative has degree b and leading coefficient $(b+1)/(d+1)$. Summing those diagonal entries proves (20). $\square$

The matrix B_h alone is not asserted to be the local monodromy residue: the order-two coefficient $C_h(t)$ is present and generally does not commute with B_h . Equations (17) and (19) retain their exact relationship.

122 The full primary jets at the boundary

Proposition 122.1 (Leading coefficient on every original primary block). *Retain the Chinese-remainder summands $E_\rho = \mathbb{C}[s]/(s-\rho)^{m_\rho}$ and their projectors. At $t=0$,*

$$C_h(0)|_{E_\rho} = \Phi_h(\rho)I_{E_\rho}, \quad A_h|_{E_\rho} = \rho I_{E_\rho} + N_\rho, \quad \dim E_\rho = m_\rho. \quad (21)$$

For the ordered tuple $(\rho_1, \dots, \rho_k)$ the leading matrix in (19) is scalar $\sum_i \Phi_h(\rho_i)$ on a block of dimension $\prod_i m_{\rho_i}$.

Proof. Write $h(s) = (s-\rho)^{m_\rho} a_\rho(s)$, where $a_\rho(\rho) \neq 0$. Integration of its Taylor polynomial, with the original constant retained, gives the exact identity

$$\Phi_h(s) - \Phi_h(\rho) = \sum_{j \geq 0} \frac{a_\rho^{(j)}(\rho)}{j!} \frac{(s-\rho)^{m_\rho+j+1}}{m_\rho+j+1}. \quad (22)$$

The sum is finite. Every term is divisible by $(s-\rho)^{m_\rho+1}$. The matrix $C_h(0)$ is multiplication by $[\Phi_h]_h$, by its definition in (14); hence its restriction is the first expression in (21). Multiplication by s remains $\rho + N_\rho$ on that same full module. The tensor statement follows by applying the sum of these matrices to the ordered tensor projectors. $\square$

Equal sums of critical values retain their separate original projectors and source labels. The scalar exponential factors for two tuples have the exact modulus ratio

$$\exp\left(\text{Re} \frac{\sum_i \Phi_h(\rho_i) - \sum_i \Phi_h(\rho'_i)}{u}\right). \quad (23)$$

For a nonzero difference, equal modulus occurs on $\arg u = \arg(\text{difference}) + \pi/2 \pmod{\pi}$. These are candidate Stokes directions of the displayed exponential factors. Equation (23) supplies no assertion that a particular Stokes entry is nonzero, and does not replace the calculation of the matrices for (19).

123 Exact ordered period determinant

Keep the original rays and contours

$$\vartheta_j = \frac{\arg u + \pi}{d+1} + \frac{2\pi j}{d+1}, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq d), \quad (\Pi_h)_{jb} = \int_{\Gamma_j} e^{f_t(s)/u} s^b ds. \quad (24)$$

The leading radial exponent is $-r^{d+1}/((d+1)|u|)$; it proves absolute convergence and coefficient differentiation on compact parameter sets. Contours on nearby parameters deform within the same decaying sectors, with their original orientations. Every formula below uses a fixed retained branch for u and the ordered rays.

Theorem 123.1 (Full period determinant, including its constant). *Let $c_h(t) = \text{Tr } C_h(t)$, and put*

$$\begin{aligned} \zeta &= e^{2\pi i/(d+1)}, & \beta_b &= \frac{b+1}{d+1}, \\ W_{jb} &= \zeta^{j(b+1)} - 1 \quad (1 \leq j \leq d, 0 \leq b < d), \\ K_d &= (d+1)^{-d/2} e^{\pi i d/2} \left(\prod_{b=0}^{d-1} \Gamma(\beta_b) \right) \det W. \end{aligned} \quad (25)$$

Then $K_d \neq 0$ and

$$\boxed{\det \Pi_h(u, t) = K_d u^{d/2} e^{c_h(t)/u}.} \quad (26)$$

For the ordered external product $\Pi_k = \bigotimes_i \Pi_h(u, t_i)$,

$$\boxed{\det \Pi_k = K_d^{kd^{k-1}} u^{kd^k/2} \exp\left(\frac{d^{k-1} \sum_i c_h(t_i)}{u}\right).} \quad (27)$$

Both identities retain all lower coefficients of h and extend across critical-root collisions.

Proof. Differentiating (24) in u inserts $-f_t/u^2$. The exact reduction (15) gives

$$\partial_u \Pi_h = \Pi_h \left(-\frac{C_h(t)}{u^2} + \frac{B_h}{u} \right).$$

The determinant equation, valid as the exterior-power equation even before invertibility is known, is

$$\partial_u \det \Pi_h = \left(-\frac{c_h(t)}{u^2} + \frac{d}{2u} \right) \det \Pi_h. \quad (28)$$

Thus the remaining factor after division by $u^{d/2} e^{c_h(t)/u}$ is independent of u on a contour branch.

We prove its independence of t and all lower coefficients without deleting their effects. On the dense open of simple roots of $h - t$,

$$c_h(t) = \sum_{h(\rho)=t} f_t(\rho), \quad \partial_t c_h(t) = - \sum_{h(\rho)=t} \rho = -\text{Tr } A_h(t).$$

The terms containing derivatives of the roots vanish because $f_t'(\rho) = 0$. Both sides are polynomial expressions in the coefficients and t , so this identity holds across collisions too. The period equation $\partial_t \Pi_h = \Pi_h(-A_h(t)/u)$ cancels the derivative of $e^{c_h(t)/u}$, proving that the remaining factor is independent of t .

Differentiation in h_b inserts $s^{b+1}/((b+1)u)$. For a column s^j , its numerator has degree $j+b+1 \leq 2d-1$. In division by $h-t$ the quotient derivative has degree at most $j+b-d < j$. Consequently the differential-reduction correction has zero trace. The determinant logarithmic derivative in h_b is $\text{Tr } M_{s^{b+1}}/((b+1)u)$. On the simple-root open this equals $(\partial_{h_b} c_h(t))/u$, since again the root derivative is multiplied by $f'_t(\rho) = 0$. Polynomial continuation proves the identity for all coefficients. Therefore the remaining factor has zero derivative in every lower coefficient.

It remains to compute that factor at $h = s^d, t = 0$, with the original contours. Substitution $v = r^{d+1}/((d+1)|u|)$ on each ray gives

$$(\Pi_h)_{jb} = [(d+1)u]^{\beta_b} e^{\pi i \beta_b} \frac{\Gamma(\beta_b)}{d+1} (\zeta^{j(b+1)} - 1). \quad (29)$$

The sum of the β_b is $d/2$, so taking determinants gives exactly (25) and (26). For invertibility of W , suppose $\sum_{r=1}^d a_r(z^r - 1)$ vanishes at the d nontrivial roots of $z^{d+1} = 1$. It also vanishes at $z = 1$. Its degree is at most d , so it is identically zero and every $a_r = 0$. All gamma factors are nonzero, proving $K_d \neq 0$.

The periods and their parameter derivatives are entire in the lower coefficients on compact sets by the fixed leading radial decay. This justifies the coefficient continuation used above and proves the identities also at collisions. Finally, $\det(A \otimes B) = (\det A)^{\dim B} (\det B)^{\dim A}$ in the retained ordered product bases gives (27). $\square$

124 Exact comparison with the original source metric

The period Gram determinant following from (27) is

$$\det(\Pi_k^* \Pi_k) = |K_d|^{2kd^{k-1}} |u|^{kd^k} \exp\left(2d^{k-1} \text{Re} \frac{\sum_i c_h(t_i)}{u}\right). \quad (30)$$

Retain, independently, the actual one-factor source Gram

$$(G_\theta)_{ab} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{(1/2 + iy)^a} (1/2 + iy)^b |v_h(1/2 + iy)|^2 dy, \quad 0 \leq a, b < d. \quad (31)$$

This is the Gram of $\mathcal{T}_h(s^b) = D^b F_h$ in the original $L^2(dx)$ source norm by Mellin–Plancherel. Its complete multiplier is $v_h = g/h$. It is positive definite: a zero norm forces the continuous source function $\mathcal{T}_h(P)$ to vanish; its Mellin transform Pv_h is then zero, and the nonzero entire v_h forces $P = 0$. The original ordered tensor source Gram is $G_\theta^{\otimes k}$.

The exact comparison of the two finite forms is the endomorphism

$$T_{\text{per}} = (G_\theta^{\otimes k})^{-1} (\Pi_k^* \Pi_k). \quad (32)$$

It is self-adjoint and positive in the original source metric, since

$$\langle v, T_{\text{per}} w \rangle_{G_\theta^{\otimes k}} = v^* \Pi_k^* \Pi_k w, \quad \langle v, T_{\text{per}} v \rangle_{G_\theta^{\otimes k}} = \|\Pi_k v\|^2 > 0 \quad (v \neq 0).$$

Its determinant is exactly (30) divided by $(\det G_\theta)^{kd^{k-1}}$. This preserves both metrics rather than identifying them.

On the weighted cyclic image (10), the period form is $\eta^* \Pi_k^* \Pi_k \eta$. Its relation to the original cyclic source metric retains the rectangular map $\Pi_k \eta$, the complement of η , and the transverse map

$$E \xrightarrow{\eta} E_h^{\otimes k} \xrightarrow{\sum_i \mathcal{R}_{h,i}} E_h^{\otimes k} \longrightarrow E_h^{\otimes k} / \eta E.$$

The full determinant (30) does not determine the singular values of this rectangular restriction. The displayed maps are the exact relationship; no original relation norm has been set to zero.

125 Compatibility with independently retained mixed faces

For an original coefficient mask A , retain its A -indexed coefficient complex and the original outer theta-leg label. Let j_A^B be the specified constant zero-insertion linear map for $A \subseteq B$. It inserts represented zero in each added slot. The preceding complex differential and connection act with the same coefficients in each slot. Therefore

$$j_A^B \mathcal{D}_A = \mathcal{D}_B j_A^B, \quad j_A^B \nabla_A = \nabla_B j_A^B. \quad (33)$$

These equations hold componentwise on retained coordinates; on the inserted coordinates each derivative and matrix coefficient sends zero to zero. Since j_A^B is constant in all parameters, differentiation commutes with it as well.

On the carrier retaining its mask, a vector killed by an observation therefore remains the receiving represented zero with that mask. The empty coefficient face maps to the zero vector in the receiving face under zero insertion; this face-group map is not a τ -preserving pointed lift. A nonempty outer theta-leg label remains present when the coefficient mask is empty. No colimit over the entire mask poset is taken, since its terminal full face would discard the independent face labels.

The scalar structural specialization is still

$$\mathbb{F}_{1,\tau} \longrightarrow G(\mathbb{C}[u, \mathbf{t}]) \xrightarrow{G(\text{ev}_0)} G(\mathbb{C}).$$

It sends represented u, t_i to represented zero e , and τ to τ . The original absolute point, its source, and the coefficient specialization are retained. Equation (33) is an explicit compatibility with the mixed face maps; it supplies no claim that a Boolean mask itself is a Deligne weight.

126 Completed single-primary boundary calculation and the continuing mixed-support calculation

The full single-primary calculation SP.1–36 and SPF.1–29, included in this edition as *The actual single-primary marked family and its complete boundary monodromy* and its finite-field companion, sharpens the preceding connection and determinant calculations on their exact primary subcase. The formulas in this section are reused from that complete proof, with the original source, constant, unit and basis retained. They replace the earlier statement that this subcase's local inertia and periods still required calculation.

126.1 The exact primary specialization, connection and period matrix

Choose an actual nontrivial zero ρ of $g = 2\xi$, with its complete order $m \geq 1$. In this subsection fix $t = 0$ and put

$$\begin{aligned} n &= m + 1, & h(s) &= (s - \rho)^m, & y &= s - \rho, \\ \Phi_h(s) &= \frac{(s - \rho)^n - (-\rho)^n}{n} = \frac{y^n}{n} + c, & c &= -\frac{(-\rho)^n}{n}. \end{aligned}$$

Thus $\Phi_h(0) = 0$ and $\Phi'_h = h$; the constant c has its original value. In the ordered rows $e_s = (1, s, \dots, s^{m-1})$ and $e_y = (1, y, \dots, y^{m-1})$, the exact coefficient map is

$$e_s = e_y P, \quad P_{ab} = \binom{b}{a} \rho^{b-a} \quad (a \leq b), \quad (P^{-1})_{ab} = \binom{b}{a} (-\rho)^{b-a} \quad (a \leq b),$$

with entries zero when $a > b$. Substitution in the binomial theorem proves both maps and their inverse relation; $\det P = 1$.

The original complex has differential $L_u = u\partial_s + (s - \rho)^m$. Its leading-degree term gives the free degree-one remainder module $\bigoplus_{b=0}^{m-1} \mathbb{C}[u][s^b ds]$, whose marked fibre is $\mathbb{C}[s]/h ds$, with the full primary order. In these unchanged coefficient columns the exact division and its derivative are

$$\Phi_h(s)s^b = h(s)\frac{(s - \rho)s^b}{n} + cs^b, \quad \left(\frac{(s - \rho)s^b}{n}\right)' = \frac{(b + 1)s^b - b\rho s^{b-1}}{n}.$$

Subtracting $L_u((s - \rho)s^b/n)$ proves

$$C_s = cI_m, \quad (B_s)_{bb} = \frac{b + 1}{n}, \quad (B_s)_{b-1,b} = -\frac{b\rho}{n}, \quad \nabla_u = \partial_u - \frac{cI_m}{u^2} + \frac{B_s}{u}.$$

All other entries of B_s vanish. The degree-zero lift is still $\partial_u - \Phi_h/u^2 + 1/u$; it intertwines L_u with the degree-one lift $\partial_u - \Phi_h/u^2$. Writing $\beta_a = (a + 1)/n$ gives the exact conjugation

$$PB_sP^{-1} = \text{diag}(\beta_0, \dots, \beta_{m-1}), \quad \text{Tr } B_s = m/2.$$

These are SP.1–9, including the original coefficient and chain maps.

On a fixed branch of $\log u$, retain the ordered outward rays of angles $(\arg u + \pi)/n + 2\pi j/n$ and their differences $\Gamma_j = -\ell_0 + \ell_j$, $1 \leq j \leq m$. Set

$$\zeta = e^{2\pi i/n}, \quad W_{ja} = \zeta^{j(a+1)} - 1, \quad d_a = n^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a), \quad D(u) = \text{diag}(d_a u^{\beta_a}).$$

The complete original period matrix, with original monomial columns, is

$$\Pi_s(u) = \left(\int_{\Gamma_j} e^{\Phi_h(s)/u} s^b ds \right)_{jb} = e^{c/u} W D(u) P.$$

SP.10–14 prove this equality on the original contours: translating their junction by ρ leaves connector integrals tending to zero by $\text{Re}(\Phi_h/u) = -R^n/(n|u|) + O(R^{n-1}) + O(1)$; the finite junction paths cancel with their original orientations. The radial substitution then gives exactly the d_a above. The matrix W is invertible by the degree- $(n - 1)$ polynomial root argument in SP.14. In particular,

$$\det \Pi_s = K_m u^{m/2} e^{mc/u}, \quad K_m = n^{-m/2} e^{\pi i m/2} \left(\prod_{a=0}^{m-1} \Gamma(\beta_a) \right) \det W \neq 0.$$

Thus the preceding determinant has its complete matrix refinement on this precise subcase, with all Gamma factors, contour phases, Pascal coefficients and the original scalar retained.

126.2 The computed boundary and its exact marked-source maps

The transported cycles satisfy $\Gamma_j \mapsto \Gamma_{j+1} - \Gamma_1$ for $j < m$ and $\Gamma_m \mapsto -\Gamma_1$. Consequently SP.16–21 give

$$M_{\text{cyc}} = W \text{diag}(\zeta, \zeta^2, \dots, \zeta^m) W^{-1}, \quad M_{\text{cyc}}^n = I, \quad \ker(M_{\text{cyc}} - I) = 0.$$

The horizontal coefficient convention is determined by the exact fundamental matrix

$$H_s(u) = P^{-1} e^{-c/u} \text{diag}(u^{-\beta_a}).$$

It has inverse eigencharacters, with the same zero invariant space. Under $u = v^n$, the displayed connection becomes

$$\partial_v - \frac{nc}{v^{n+1}} I_m + \frac{1}{v} \text{diag}(1, \dots, m).$$

The meromorphic change $x_y = \text{diag}(v^{-1}, \dots, v^{-m})z$ removes precisely these integer residues and retains the common irregular factor e^{-c/v^n} . This exact fundamental solution has identity monodromy after the cover; its unipotent logarithm is $N_{\text{boundary}} = 0$. Its sector restrictions agree on

overlaps with the same branch, so there are no additional Stokes factors in this one-critical-value calculation. Both ordinary invariant stalks and the derived local invariant and coinvariant groups vanish before the cover, since $M_{\text{cyc}} - I$ is invertible.

On the same marked fibre let J_m be multiplication by y , with $(J_m)_{a+1,a} = 1$. The original multiplication and its precise relation to the boundary connection are

$$A_s = P^{-1}(\rho I_m + J_m)P, \quad N_{\text{ar}} = P^{-1}J_mP, \quad [B_s, A_s] = \frac{1}{n}N_{\text{ar}}, \quad [\nabla_u, A_s] = \frac{N_{\text{ar}}}{nu}.$$

With $D_0 = \text{diag}(d_a)$ and $K_0 = WD_0J_mD_0^{-1}W^{-1}$, the complete period transport is

$$\Pi_s A_s \Pi_s^{-1} = \rho I_m + u^{1/n} K_0, \quad M_{\text{cyc}} K_0 M_{\text{cyc}}^{-1} = \zeta K_0.$$

The coefficient calculation on each monomial proves these commutators and conjugations, including the last nilpotent column. They retain the original arithmetic nilpotent through its exact nonhorizontal map; see SP.22–24.

For the entire quotient $v_h = g/h$, retain every coefficient

$$a_j = \frac{v_h^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!}, \quad 0 \leq j < m, \quad a_0 \neq 0.$$

Its marked multiplication and inverse are

$$U_s = P^{-1} \left(\sum_{j=0}^{m-1} a_j J_m^j \right) P, \quad U_s^{-1} = P^{-1} \left(\sum_{j=0}^{m-1} b_j J_m^j \right) P,$$

$$b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}.$$

Multiplication of these finite series proves their inverse relation, using $J_m^m = 0$. Its complete period and connection images are

$$\Pi_s U_s \Pi_s^{-1} = \sum_{j=0}^{m-1} a_j u^{j/n} K_0^j, \quad [\nabla_u, U_s] = \frac{1}{nu} P^{-1} \left(\sum_{j=1}^{m-1} j a_j J_m^j \right) P.$$

These retain the full Taylor unit, rather than only its leading coefficient.

The original Euler/theta square remains, with $D = -x\partial_x$ and $\mathcal{M}F_h = v_h$,

$$\alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \quad \Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h\mathcal{T}_h(P) = U_s[P]_s.$$

Here $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, $\Theta\phi = 2\sum_{r \geq 1} \phi(rx)$, and $\mathcal{M}\Theta\phi_* = g$, with the original function spaces and Mellin convention. The degree-one map is the retained $\sigma_h U_s : E_h \hookrightarrow Q_{\text{ar}}$. The exact translated source vector is $\mathcal{T}_h(y^b) = (D - \rho)^b F_h$, and at general u the source differential is the specified $\Theta + u\mathcal{T}_h\partial_s\alpha_h^{-1}$ on $\alpha_h(\mathbb{C}[s])$. For $v_{\text{pol}} = \sum_{j < m} a_j (s - \rho)^j$, the original differential gives the full representative defect

$$v_{\text{pol}} L_u Q = L_u(v_{\text{pol}} Q) - u v'_{\text{pol}} Q.$$

If $v_{\text{pol}} p = hq + r$ for an original remainder p , its differential remainder is $r - uq'$, whereas the constant coefficient extension U_s returns r . SP.25–31 prove these identities and retain precisely the derivative term needed away from $u = 0$.

The exact finite-cover lattice map is also available:

$$e^{-c/v^n} \Pi_s(v^n) = WD_0 \text{diag}(v, v^2, \dots, v^m)P.$$

After the two displayed constant invertible changes, its cokernel is $\bigoplus_{a=0}^{m-1} \mathbb{C}\{v\}/(v^{a+1})$, of length $m(m+1)/2$. Its image lattice has rank- m fibre modulo v , while its inclusion in the ambient constant lattice induces zero modulo v . This is the complete lattice map between the retained marked fibre and the computed local boundary; the factored exponential remains an irregular factor. See SP.32–33.

In the unchanged source metric of the preceding calculation, the full matrix refinement is

$$T_{\text{per}}(u) = G_{\theta}^{-1} \Pi_s(u)^* \Pi_s(u) = e^{2 \operatorname{Re}(c/u)} G_{\theta}^{-1} P^* D(u)^* W^* W D(u) P.$$

It is positive and self-adjoint in G_{θ} , with the complete multiplier g/h in that Gram. For $N \geq m-1$, the enlarged polynomial source has the exact period factorization and kernel

$$\operatorname{Per}_N = \Pi_s J_{u,N}, \quad \ker J_{u,N} = L_u(\mathbb{C}[s]_{\leq N-m}).$$

Thus its period Gram has rank m ; on the displayed remainder frame it is positive definite. SP.34–36 prove this by the original Mellin source and the endpoint calculation for $u d(e^{\Phi_h/u} Q)$. The original relation space and source norm therefore remain in subsequent restriction calculations.

126.3 The exact finite-field boundary on the same primary phase

For the finite-field statement retain the specified coefficient homomorphism $R \rightarrow k_0 = \mathbb{F}_{Q_0}$, with $p > n$, including $\rho, 1/n!$, and every coefficient of the full Taylor unit and its inverse. Use the explicit constant extension

$$k = \mathbb{F}_Q, \quad Q = Q_0^d, \quad d = \operatorname{ord}_{(\mathbb{Z}/n\mathbb{Z})^\times}(Q_0),$$

so $n \mid Q-1$. Fix $\ell \neq p$, geometric Frobenius, and the nontrivial additive character ψ obtained from the specified one on k_0 by the field trace. SPF.1–29 compute the actual sheaf $\mathcal{F} = R^1 \pi_{s!} \mathcal{L}_{\psi}(\Phi_h(s)/u)$. With

$$X_n = \{\chi : k^\times \rightarrow \overline{\mathbb{Q}}_{\ell}^\times : \chi^n = 1\}, \quad G_k(\chi, \psi) = \sum_{z \in k^\times} \chi(z) \psi(z),$$

and $\mathcal{A}_{\gamma}$ denoting the geometrically constant line with geometric Frobenius eigenvalue γ , the exact parameter-sheaf isomorphism is

$$\mathcal{F} \simeq \mathcal{L}_{\psi}(c/u) \otimes \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{A}_{-\chi(n)G_k(\chi, \psi)} \otimes \mathcal{K}_{\chi}(u).$$

The Kummer convention has trace χ . SPF.6–19 prove the isomorphism using the original translation, the finite power map $y \mapsto y^n$, its character projectors including the zero stalk, and compact-support base change. The minus sign is the degree-one cohomological sign; $\chi(n)$ is the original coefficient factor. Both it and the common Artin–Schreier factor remain.

If the specialized $c \neq 0$, the common wild character has break one, so

$$V^{I_0} = 0, \quad \operatorname{Swan}_0(\mathcal{F}) = m.$$

If $c = 0$, the nontrivial tame Kummer characters instead give

$$V^{I_0} = 0, \quad \operatorname{Swan}_0(\mathcal{F}) = 0.$$

Writing $\psi(x) = \vartheta(\operatorname{Tr}_{k/\mathbb{F}_p}(bx))$, the explicit cover $u = v^n$, followed when needed by $z^p - z = bc/v^n$, kills all geometric inertia. Its restricted unipotent logarithm is zero, while its restricted invariant space is the full rank m . SPF.22–27 prove the conductor, original invariant space and finite-cover statements by the exact local ramification calculation. Finally SPF.28–29 prove $|G_k(\chi, \psi)|^2 = Q$ directly for every nontrivial χ ; all local eigenvalues consequently have the original weight one. These are completed local and weight calculations for the specified sheaf and coefficient specialization.

126.4 Continuation through the full mixed-support family

The preceding complete primary calculation now supplies actual boundary matrices, period maps, conductors, unit defects and source forms to the earlier family. The original independent coefficient faces, outer theta-leg labels, proper-source kernels and structural specialization remain those already defined above; every constant coefficient map retains the corresponding represented zero and its label.

For several distinct primary critical values, the continuing calculation includes the sector transitions and local extensions between the computed blocks, with their exact maps through the retained mixed faces. The single-primary tensor calculation now gives the following complete continuation on the original parameter line.

The one-factor calculation gives the original finite-field sheaf's exact decomposition, weight one, invariant-space vanishing and zero finite-cover logarithm. The ordered tensor calculation now carries that result through every positive tensor order on the same parameter line. Keeping $n = m + 1$, the original phase has constant kc , and the character on an ordered monomial form is $\zeta^{\sum_i (b_i + 1)}$, where $0 \leq b_i < m$ and $\zeta = e^{2\pi i/n}$. The complete sum of these characters gives the exact ordinary complex invariant dimension

$$d_{0,k} = \frac{1}{n} \sum_{j=0}^{n-1} \left(\sum_{a=1}^{n-1} \zeta^{ja} \right)^k = \frac{m^k + m(-1)^k}{m+1}. \quad (\text{SPBT.1})$$

The inner sum is m for $j = 0$ and -1 otherwise, proving the equality. It is zero at $k = 1$; for $m \geq 2$ it is nonzero at every $k \geq 2$, and for $m = 1$ it is one at even k and zero at odd k .

More precisely, let E_0 be the diagonal mask selecting $\sum_i (b_i + 1) \equiv 0 \pmod{n}$, let $P_k = P^{\otimes k}$ be the original Pascal coordinate map, and let U_k be the complete original source-unit matrix. The retained tensor period matrix Π_k , with all its Gamma constants and original contours, gives

$$Q_0 = P_k^{-1} E_0 P_k, \quad \widehat{Q}_0 = U_k Q_0 U_k^{-1}, \quad \mathcal{P}_0 \Pi_k U_k^{-1} = \Pi_k U_k^{-1} \widehat{Q}_0. \quad (\text{SPBT.2})$$

The diagonal power matrix in the exact tensor period formula commutes with E_0 , so multiplication of those matrices proves this identity. Thus $\widehat{Q}_0$ is the projector on the actual original-source jet, and its period observation is $\Pi_k U_k^{-1}$; neither unit nor inverse is removed. The terminal monomial $\prod_i (s_i - \rho)^{m-1}$ has character ζ^{-k} , and hence survives exactly when $n \mid k$. The original sum multiplication still acts on that same vector by $k\rho$, since multiplication by each $s_i - \rho$ kills it. Its dilation eigenvalue remains $a^{k\rho}$.

On the actual finite-field coefficient stage with $p > m + 1$, the entire ordered sheaf is the direct sum of the original Gauss lines for the ordered character tuples, tensored with the unchanged additive factor $\mathcal{L}_\psi(kc/u)$. The exact original-inertia projector is the tame average multiplied by its wild average; the latter is one when $kc = 0$, zero otherwise. Consequently

$$\dim V_k^{I_0} = \begin{cases} d_{0,k}, & kc = 0, \\ 0, & kc \neq 0, \end{cases} \quad \text{Swan}_0(V_k) = \begin{cases} 0, & kc = 0, \\ m^k, & kc \neq 0. \end{cases} \quad (\text{SPBT.3})$$

The wild average is $p^{-1} \sum_{a \in \mathbb{F}_p} \vartheta(-ka)$ when the original $c \neq 0$, which proves the two cases; for $c = 0$ the original additive torsor is trivial. The Swan value follows because every nontrivial $1/u$ Artin-Schreier character has break one, shared by all m^k ordered summands. Each nonzero invariant orbit block has its full product of the original negative Gauss constants and has weight k . The terminal character survives precisely for $n \mid k$ and $kc = 0$, hence for $pn \mid k$ in the branch $c \neq 0$.

The coefficient lattice now records the same original cyclic map as $T^r \mapsto r!UV_r$. In the actual coefficient localization $\mathcal{S} = A_{\ker \alpha}$ of the given coefficient datum, its image L_U lies in its saturation Λ_U , with the exact quotient $\Lambda_U/L_U \simeq \bigoplus_r \mathcal{S}/(p^{v_p(r!)}).$ The quotient of the full tensor target also retains

its free complementary summand, as proved in (BPC.5) and (BPC.9). The complete proof in (BPC.1)–(BPC.14) identifies its specialized kernel (T^p), including all the original unit coefficients and the Tor connecting map. The unequal-order and residual-collision forms are (PC30a)–(PC30g). These statements apply only on their actual coefficient-map domains; the phase denominators and the additional CRT denominators have the explicit extension tests in (BTF.9) and the preceding coefficient-domain calculation. The common cyclotomic projector maps in (BTF.6) relate the coefficient characters to the actual Gauss-line representation through their two base changes. They preserve the distinct coefficient and sheaf scalar targets. Thus one-factor vanishing is retained as its exact $k = 1$ case, and the tensor invariants, original arithmetic action and coefficient specialization are carried forward with their proved maps.

The continuing induced relation-form calculation on the weighted cyclic image retains the parameter defect (11), the transverse map, its rectangular period restriction, and all original source cross-pairings. The explicit primary formula for T_{per} above can now enter that calculation, while its complete coefficient Gram and restriction maps remain. An arithmetic Frobenius comparison and the uniform four-volume estimate still require their own calculations on those original objects; the completed primary boundary result has not performed those additional steps.

Source and verification record

The original packet-family inputs are those of *Tau Marked Product Four Endpoint*, 13 September 2026, NOTE.tex, read in full, with SHA256

40749e2a9d599a322e7b125eb43ad0878
a066a51b61da52cc3bb2ffb94cbd841.

The relevant primary source is Pierre Deligne, *La conjecture de Weil II*, Publications mathématiques de l’IHÉS 52 (1980), 137–252: 1.8.4, 1.8.8, 3.3.1–3.3.6, 3.7.2.3, 3.7.3, 6.1.13 and 6.2.1–6.2.6. The local typed text of those statements and surrounding proofs was read for this audit.

The companion exact symbolic checker verifies differential reduction, flatness, $\partial_t B_h = 0$, the trace and diagonal of B_h , the critical-value trace derivative, and complete jet divisibility for four polynomial fixtures of degrees 1 through 4, including repeated roots. All passed. These fixtures are explicitly not asserted to be arithmetic zero packets; the proofs here apply to the original h .

127 Joint trace control on the original source and relation spaces

Retain $h, k, \chi = \chi_{h,k}$ from the marked-product note. Here $q = \deg \chi \geq 1$ denotes the polynomial degree. Use the original coordinate $S = k/2 + iy$, the densities $m = w_h^{*k}$ and $r = r_{1/4}^{*k}$ with their literal masses, and $\mu_x = ((1-x)r + xm) dy$ for $0 \leq x \leq 1$. The common source is $H = \mathcal{P}_{2q}$ in the retained monomial basis. Its positive definite Gram and its original metric derivative are

$$M_x = M_0 + x\dot{M}, \quad M_0 = M_{2q}^\Gamma, \quad \dot{M} = M_{2q}^{\text{ar}} - M_{2q}^\Gamma, \quad T_x = M_x^{-1}\dot{M}. \quad (\text{SP1})$$

The equality $M_x T_x = \dot{M} = \dot{M}^*$ proves that T_x is self-adjoint in the M_x inner product. Its exact relation to the positive M_0 -self-adjoint operator $R = M_0^{-1}M_1$ is $T_x = (I + x(R - I))^{-1}(R - I)$, obtained by factoring $M_x = M_0(I + x(R - I))$. Thus its eigenvalues are $(r - 1)/(1 + x(r - 1))$ for the positive eigenvalues r of R , with their actual signs retained.

The precise arrow to the earlier AW source keeps H and every coefficient fixed, sends t_{AW} to x and u_{AW} to y , and sends the relation map by $B_N^{\text{SP}} = I_N B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow H$. The Gamma convolution constant is $c_a = 2^{1-2a}\Gamma(2a)$, so

$$r_{1/4}^{*k}(y) = \frac{c_{1/4}^k |\Gamma(k/4 + iy/2)|^2}{c_{k/4} 2\pi}, \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{SP1a})$$

This is exactly AW3, including its mass $(2\pi)^{k/2}$; both arithmetic densities are w_h^{*k} with mass μ_h^k . Thus $M_x = M_{AW}(x)$, and $(B_N^{\text{SP}})^* M_x B_N^{\text{SP}} = (B_N^{\text{AW}})^* M_N B_N^{\text{AW}}$ with $M_N = I_N^* M_x I_N$.

127.1 All four endpoints as operators on one source

Let $I_N : \mathcal{P}_N \rightarrow H$ be literal coefficient inclusion, and let $B_N : \mathcal{P}_{N-q} \rightarrow H$ be multiplication by the original monic χ , followed by coefficient inclusion. Define

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{SP2})$$

The second formula is used for $N \geq q$; its Gram is invertible because multiplication by χ is injective on polynomials. Direct multiplication proves that these are the M_x -orthogonal projections onto the original source and relation subspaces. Within either family the subspaces are nested, so their orthogonal projections commute and their product is the smaller projection. Indeed the smaller subspace, its orthogonal complement inside the larger one, and the orthogonal complement of the larger one give three invariant summands on which this statement is immediate.

Let $V_N(x)$ be the original canonical quotient volume. In the fixed quotient-lift and monic-relation coordinates, block determinant gives

$$\log V_N(x) = \log \det(I_N^* M_x I_N) - \log \det(B_N^* M_x B_N). \quad (\text{SP3})$$

The relation determinant is one when its dimension is zero. The fixed monic change-of-basis determinant has modulus one, so SP3 retains every volume factor. Differentiation and cyclicity of finite matrix trace give

$$(\log V_N)' = \text{Tr}(T_x(P_N - Q_N)). \quad (\text{SP4})$$

For example $\text{Tr}(T_x P_N) = \text{Tr}((I_N^* M_x I_N)^{-1} I_N^* \dot{M} I_N)$; the relation formula uses precisely the same calculation. Set

$$F(x) = \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \quad \Delta_{h,k} = F(1) - F(0), \quad (\text{SP5})$$

and define two positive operators on this same source:

$$\mathcal{R}_x = Q_{2q-1} + Q_{2q} - Q_q, \quad \mathcal{P}_x = (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q), \quad \mathcal{A}_x = \mathcal{R}_x - \mathcal{P}_x. \quad (\text{SP6})$$

The four original signs in SP4 give

$$F'(x) = \text{Tr}(T_x \mathcal{A}_x). \quad (\text{SP7})$$

For the original row $v = v_{2q}(y)$, the density identity has the exact metric factor

$$\mathbf{R}_x(y) - \mathbf{P}_x(y) = v \mathcal{A}_x M_x^{-1} v^*.$$

Indeed $P_N M_x^{-1} = I_N(I_N^* M_x I_N)^{-1} I_N^*$, and the same identity holds with B_N, Q_N . This recovers the marked-product kernels and proves the common-source matrix comparison.

127.2 Ranks, multiplicities and the retained overlap

The following complete spectra and the final condition-number coefficient were already established in AW7 and AW14. They are retained here with the exact source correspondence above; the actual cross-trace and intersection terms supply the added refinement.

The nested relation spaces in SP6 have dimensions 1, q , $q+1$. On the first space $\mathcal{R}_x$ has eigenvalue one; on the next orthogonal increment it has eigenvalue two with multiplicity $q-1$; on the last

increment it has eigenvalue one; elsewhere it is zero. The nested source increments in $\mathcal{P}_x$ give the same list. When $q = 1$, the multiplicity $q - 1$ is zero. Consequently

$$\begin{aligned} \text{rank } \mathcal{R}_x &= \text{rank } \mathcal{P}_x = q + 1, & \text{Tr } \mathcal{R}_x &= \text{Tr } \mathcal{P}_x = 2q, \\ \text{Tr } \mathcal{R}_x^2 &= \text{Tr } \mathcal{P}_x^2 = 4q - 2, & \text{Tr } \mathcal{A}_x &= 0, \\ \text{Tr } \mathcal{A}_x^2 &= 8q - 4 - 2 \text{Tr}(\mathcal{R}_x \mathcal{P}_x). \end{aligned} \quad (\text{SP8})$$

The equal kernel masses $2q$ are traces. The two positive operators have rank $q + 1$, with their multiplicity-two directions retained.

The overlap in SP8 is at least one. Each positive operator is at least the orthogonal projection onto its range, since its nonzero eigenvalues are one and two. The two ranges each have dimension $q + 1$ in dimension $2q + 1$, so their intersection has dimension at least one. Choose a unit vector in this intersection and complete it to an orthonormal basis of the first range. The trace of the product of the two range projections is the sum of squared norms of the second projection on this basis, hence at least one. For positive operators, replacing one factor by a larger positive factor cannot decrease the trace of the product: the difference is $\text{Tr}(B^{1/2}(A' - A)B^{1/2}) \geq 0$. Applying this twice proves the stated bound for the actual operators.

127.3 A bound that uses cancellation before estimating

Write $d_x = \lambda_{\max}(T_x) - \lambda_{\min}(T_x)$. Since $\text{Tr } \mathcal{A}_x = 0$, for every real c

$$F'(x) = \text{Tr}((T_x - cI)\mathcal{A}_x). \quad (\text{SP9})$$

Choose c to be the midpoint of the two extreme eigenvalues. In an M_x -orthonormal eigenbasis e_j of $\mathcal{A}_x$, with real eigenvalues α_j , one obtains

$$|F'(x)| \leq \sum_j |\alpha_j| |\langle e_j, (T_x - cI)e_j \rangle_{M_x}| \leq \frac{1}{2}d_x \sum_j |\alpha_j|.$$

This proves the trace-norm bound directly. Let L_x be the intersection of the two ranges in SP6, with $s_x = \dim L_x \geq 1$, and let E_{L_x} be its orthogonal projection in M_x . Each of $\mathcal{R}_x, \mathcal{P}_x$ is at least E_{L_x} . Thus $\mathcal{A}_x = (\mathcal{R}_x - E_{L_x}) - (\mathcal{P}_x - E_{L_x})$ is a difference of two positive operators of trace $2q - s_x$. The triangle inequality for trace norm gives $\|\mathcal{A}_x\|_1 \leq 4q - 2s_x \leq 4q - 2$. Cauchy-Schwarz on its at most $2q + 1$ eigenvalues gives the refinement

$$|F'(x)| \leq \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q + 1)(8q - 4 - 2 \text{Tr}(\mathcal{R}_x \mathcal{P}_x))} \right\}. \quad (\text{SP10})$$

The overlap is calculated from the unchanged source and relation Grams in SP2. Every cross-pairing remains in that expression.

The complete earlier AW13–AW16 bound applies to this same source through $(U, W, C) = (\mathcal{R}_x, \mathcal{P}_x, T_x)$. Write $c_1(x) \leq \dots \leq c_{2q+1}(x)$ for all eigenvalues of T_x in M_x and define

$$S_{\text{AW}}(x) = c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \quad S_{\text{SP}}(x) = \text{the right-hand side of (SP10)}. \quad (\text{SP10a})$$

Here is the complete retained spectral proof. For an orthogonal projection P of rank r , its diagonal values a_j in an orthonormal eigenbasis of T_x satisfy $0 \leq a_j \leq 1$ and $\sum_j a_j = r$. Thus $\text{Tr}(PT_x) = \sum_j c_j a_j$ is at least the sum of the smallest r and at most the sum of the largest r eigenvalues: moving a fixed diagonal mass from a smaller coefficient to a larger one increases this sum. Either $Z = \mathcal{R}_x$

or $Z = \mathcal{P}_x$ equals its range projection, of rank $q + 1$, plus its eigenvalue-two projection, of rank $q - 1$. Therefore

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZT_x) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{SP10b})$$

Subtracting the two extrema with their exact multiplicities gives $|F'(x)| \leq S_{\text{AW}}(x)$. Combining this bound with SP10 before integration proves

$$|F'(x)| \leq \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\}. \quad (\text{SP10c})$$

Positive smooth Gram matrices make the eigenvalues and overlaps bounded on $[0, 1]$; s_x is Borel since its rank strata are given by matrix minors. Hence the right side is integrable. The original source isometry of AW10 also yields the exact commutator identity $F' = \text{Tr}(W(T^{\text{iso}})^{-1}[C, T^{\text{iso}}])$. It follows from $U = T^{\text{iso}}W(T^{\text{iso}})^{-1}$ and cyclicity, with T^{iso} the explicit isometry and $C = T_x$ the metric derivative. Its full original observation remains $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}$ and has exact defect $\mathcal{O}T^{\text{iso}} - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}(T^{\text{iso}} - I)$ and kernel $\ker(\mathcal{O}T^{\text{iso}}) = (T^{\text{iso}})^{-1}(\chi\mathcal{P}_q)$. The defect follows by expansion and the kernel follows from injectivity of $\sigma_h^{\otimes k} \eta$ and $\ker J_{2q} = \chi\mathcal{P}_q$. Thus no additional cohomology-intertwining property of the metric isometry is used.

There is a closed integrated bound. Set $R = M_0^{-1}M_{2q}^{\text{ar}}$. It is positive and self-adjoint in the M_0 inner product, since $\langle v, Rv \rangle_{M_0} = \langle v, v \rangle_{M_{2q}^{\text{ar}}} > 0$. Let its smallest and largest eigenvalues be $r_{\min}, r_{\max} > 0$. The literal factorization of M_x gives

$$T_x = (I + x(R - I))^{-1}(R - I), \quad d_x = \frac{r_{\max} - 1}{1 + x(r_{\max} - 1)} - \frac{r_{\min} - 1}{1 + x(r_{\min} - 1)}. \quad (\text{SP11})$$

For fixed x , the function $(r - 1)/(1 + x(r - 1))$ is increasing on $r > 0$, its derivative being $(1 + x(r - 1))^{-2} > 0$. This proves the extreme-eigenvalue formula even though the inner product varies. Integration, including the identically zero integrand when an eigenvalue equals one, yields

$$\int_0^1 d_x dx = \log(r_{\max}/r_{\min}). \quad (\text{SP12})$$

Write $r_1 \leq \dots \leq r_{2q+1}$ for all positive eigenvalues of R , and retain

$$\mathcal{L}_q(R) = \log \frac{r_{q+2}}{r_q} + 2 \sum_{j=1}^{q-1} \log \frac{r_{2q+2-j}}{r_j}.$$

The ordered form of SP11 gives $c_j(x) = (r_j - 1)/(1 + x(r_j - 1))$ and $\int_0^1 c_j(x) dx = \log r_j$. Therefore SP7–SP12 and the full spectral proof above give both original-source controls

$$\boxed{|\Delta_{h,k}| \leq \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx \leq \mathcal{L}_q(R) \leq (2q - 1) \log \frac{r_{\max}}{r_{\min}},} \quad (\text{SP13})$$

$$|\Delta_{h,k}| \leq \frac{1}{2} \int_0^1 d_x \|\mathcal{R}_x - \mathcal{P}_x\|_1 dx \leq (2q - 1) \log \frac{r_{\max}}{r_{\min}}.$$

The first integral is also at most $\int_0^1 S_{\text{SP}}(x) dx$. Every logarithmic ratio in $\mathcal{L}_q$ is bounded by $\log(r_{\max}/r_{\min})$, with coefficient total $1 + 2(q - 1) = 2q - 1$; this proves the last inequality explicitly. The two displayed integral bounds each remain available: no ordering between the exact trace norm and the spectral refinement is assumed. When the Grams are proportional, every c_j agrees, both bounds are zero, and the correction is exactly zero without rescaling their original masses.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(x) &= \frac{d_x}{2} \|\mathcal{R}_x - \mathcal{P}_x\|_1, \\ \Sigma(x) &= \text{Tr}(T_x^2) - \frac{(\text{Tr } T_x)^2}{2q+1}, \\ S_{\text{HS}}(x) &= \sqrt{\Sigma(x) \text{Tr}((\mathcal{R}_x - \mathcal{P}_x)^2)}. \end{aligned} \tag{SP13d}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(T_x)/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(x)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(x)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

On the original moment source (SP1a), the later angle calculation propagates further at this endpoint. With $a_q = 2q - 1$ define

$$\begin{aligned} A(x) &= a_q \sqrt{1 - e^{-F(x)/a_q}} d_x, \\ J^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{tr}}(x), S_{\text{SP}}(x), S_{\text{HS}}(x), A(x)\} dx, \\ I &= \int_0^1 \min \left\{ d_x, \frac{S_{\text{AW}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \frac{S_{\text{SP}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \frac{S_{\text{HS}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}} \right\} dx. \end{aligned} \tag{SP13a}$$

The exact map to RMT is $a_{\text{RMT}} = a_q$, $F = \mathcal{B}$, $T_x = C_x$ and $(\mathcal{R}_x, \mathcal{P}_x) = (U_x, W_x)$, with the same original moment measures by SP1a. RMT11–RMT12 proves $F > 0$ and $|F'| \leq A$: its full two angle lists keep the designated zero angle, and vanishing F would imply $1 \perp \chi \mathcal{P}_q$, contradicting $\int |\chi|^2 d\mu_x > 0$ through $\chi^{\#k}(S) = \sum_j \overline{\chi_j}(k-S)^j$. Thus the two-bound proof above remains valid for general positive forms, while on these actual moment forms its displayed first bound in SP13 strengthens to

$$|\Delta_{h,k}| \leq J^{(5)} \leq \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx. \tag{SP13b}$$

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Use $\mathbf{M}_z = M_x|_{x=z}$, $\mathbf{C}_z = T_x|_{x=z}$, $\mathbf{A}_z = (\mathcal{R}_x - \mathcal{P}_x)|_{x=z}$, and $\mathbf{D} = R$. The same-source RMT/ISM function $\mathcal{B}$ is exactly F in SP5. Write $\mathbf{D} = (\mathbf{M}_0)^{-1} \mathbf{M}_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let $\mathbf{A}_z$ be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, \quad H_z^\circ = I - \frac{(1-z)I + z\mathbf{D}}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (\mathbf{D} - I), \quad R_z^{[\ell_N]} = (H_z^\circ)^{\ell_N+1} \mathbf{C}_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \tag{SP13ea}$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1 - z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \tag{SP13eb}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q - 6)} \longrightarrow 0. \tag{SP13ec}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

For $H_0 = 2 \text{arccosh}(e^{F(0)/(2a_q)})$, the derivative $[2 \text{arccosh}(e^{B/(2a_q)})]' = 1/(a_q \sqrt{1 - e^{-B/a_q}})$ and its inverse $2a_q \log \cosh(z/2)$ give the actual endpoint interval

$$\begin{aligned} F(1) &\geq \max \left\{ 0, F(0) - J^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - I\}}{2}, F(0) + j_{\ell_N} - e_{\ell_N} \right\}, \\ F(1) &\leq \min \left\{ F(0) + J^{(5)}, 2a_q \log \cosh \frac{H_0 + I}{2}, F(0) + j_{\ell_N} + e_{\ell_N} \right\}, \quad I \leq \log(r_{\max}/r_{\min}). \end{aligned} \tag{SP13c}$$

The nonlinear branches follow by dividing all five derivative bounds by the same positive factor, integrating, and applying the increasing inverse. Intersecting with the constructed signed interval (SP13eb) gives the additional signed branches; compactness and $F > 0$ justify every denominator. It is precisely RMT13–RMT22 on the stated unchanged original source, including the full local signed construction (SP13ea)–(SP13ec).

SP1–SP13 give finite two-sided control on every original packet. They do not bound the displayed condition number or projection overlap along the growing arithmetic family. The next calculation can use these exact quantities, their original moment formulae, and the newly retained singular-boundary geometry; none has been replaced by an assumed asymptotic estimate or an RH verdict.

128 Combining the existing spectral and overlap controls on the original source

This attachment records the exact comparison already established in the coordinating source intake. The complete earlier AW proof is retained in this reader; SP reproduces its spectra and final condition-number bound and adds the stated overlap refinement. Neither bound is discarded.

Fix the actual full packet h , tensor order $k \geq 1$, and its original cyclic polynomial χ of degree $q \geq 1$. The common coefficient space is $H = \mathcal{P}_{2q}$. Its integration coordinate is $S = k/2 + iy$. Write

$x \in [0, 1]$ for metric interpolation and reserve u for the geometric deformation parameter of BC. The exact AW/SP coordinate map is

$$y = u_{AW}, \quad x = t_{AW}, \quad H_{AW} \xrightarrow{\text{id}} H_{SP}. \quad (\text{ACM1})$$

This identity leaves every polynomial coefficient unchanged. The arithmetic density is w_h^{*k} in both calculations. For the reference density, the original constant $c_a = 2^{1-2a}\Gamma(2a)$ gives $c_{1/4} = \sqrt{2\pi}$, so the exact convolution formula is

$$r_{1/4}^{*k}(y) = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}(y) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}. \quad (\text{ACM2})$$

Consequently the densities, their masses and all monomial integrals coincide. In particular $M_{AW}(x) = M_{SP}(x) = M_x$ on H , and $M_{N,AW} = I_N^* M_x I_N$ on the original subspace $I_N : \mathcal{P}_N \hookrightarrow H$.

AW's relation map has codomain $\mathcal{P}_N$, whereas SP incorporates its inclusion into H . Their literal equality is

$$B_N^{SP} = I_N B_N^{AW}. \quad (\text{ACM3})$$

Substitution into the orthogonal projector formula gives

$$\begin{aligned} P_N &= I_N (I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= I_N B_N^{AW} ((B_N^{AW})^* I_N^* M_x I_N B_N^{AW})^{-1} (B_N^{AW})^* I_N^* M_x. \end{aligned} \quad (\text{ACM4})$$

The two definitions of each projector therefore agree, including $Q_{q-1} = 0$ for the zero relation domain. Put

$$\begin{aligned} U_x &= Q_{2q-1} + Q_{2q} - Q_q = \mathcal{R}_x^{SP}, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q) = \mathcal{P}_x^{SP}, \\ C_x &= M_x^{-1} \dot{M}_x = T_x^{SP}. \end{aligned} \quad (\text{ACM5})$$

These are AW's original U, W, C . Its independently constructed source isometry T_x^{iso} satisfies

$$(T_x^{\text{iso}})^* M_x T_x^{\text{iso}} = M_x, \quad U_x = T_x^{\text{iso}} W_x (T_x^{\text{iso}})^{-1}. \quad (\text{ACM6})$$

Thus cyclicity of the same finite trace proves

$$\text{Tr}((U_x - W_x)C_x) = \text{Tr}(W_x(T_x^{\text{iso}})^{-1}[C_x, T_x^{\text{iso}}]). \quad (\text{ACM7})$$

The original observation is still $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}$. Its exact difference and kernel after the isometry remain

$$\begin{aligned} \mathcal{O} T_x^{\text{iso}} - \mathcal{O} &= \sigma_h^{\otimes k} \eta J_{2q} (T_x^{\text{iso}} - I), \\ \ker(\mathcal{O} T_x^{\text{iso}}) &= (T_x^{\text{iso}})^{-1} (\chi \mathcal{P}_q). \end{aligned} \quad (\text{ACM8})$$

The first equality is expansion. The second follows because $\sigma_h^{\otimes k}$ and η are the original injections and $\ker J_{2q} = \chi \mathcal{P}_q$. This retains the actual cohomology defect of the isometry rather than assigning it an intertwining property.

Let $F(x)$ be the original logarithmic four-endpoint ratio and $\Delta_{h,k} = F(1) - F(0)$. Its signed identity is $F'(x) = \text{Tr}((U_x - W_x)C_x)$ by [AW6], or equivalently (SP4)–(SP5) and (ACM5). Both U_x, W_x have the complete spectrum 0 with multiplicity q , 1 with multiplicity 2, and 2 with multiplicity $q - 1$. All statements below also cover $q = 1$.

Write $c_1(x) \leq \dots \leq c_{2q+1}(x)$ for the eigenvalues of C_x in the original M_x metric and $d_x = c_{2q+1}(x) - c_1(x)$. Let $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two already proved pointwise bounds, on this identical source, are

$$\begin{aligned} S_{AW}(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ S_{SP}(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}. \end{aligned} \quad (\text{ACM9})$$

For completeness, their simultaneous application follows directly as follows. An orthogonal projection P of rank r satisfies $\sum_{j=1}^r c_j \leq \text{Tr}(PC_x) \leq \sum_{j=2q+2-r}^{2q+1} c_j$: in an orthonormal eigenbasis of C_x , its diagonal entries lie in $[0, 1]$ and have sum r , so moving their mass to the lowest or highest eigenvalues proves each inequality. Each of U_x, W_x is the sum of its range projection of rank $q + 1$ and its eigenvalue-two projection of rank $q - 1$. Apply these bounds to both sums and subtract their extrema to obtain $|F'(x)| \leq S_{\text{AW}}(x)$, with every multiplicity retained.

The range dimensions give $s_x \geq 1$. If E_x projects onto the intersection, then $U_x - E_x$ and $W_x - E_x$ are positive, each of trace $2q - s_x$. Therefore $\|U_x - W_x\|_1 \leq 4q - 2s_x$. The same complete spectra give

$$\text{Tr}(U_x - W_x) = 0, \quad \text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{ACM10})$$

Cauchy-Schwarz on all $2q + 1$ real eigenvalues gives the other trace-norm bound in (ACM9). Subtract the midpoint $(c_1 + c_{2q+1})I/2$ from C_x in the trace pairing; the first identity in (ACM10) leaves F' unchanged. Diagonalizing $U_x - W_x$ then bounds the pairing by $d_x \|U_x - W_x\|_1/2$. This proves $|F'(x)| \leq S_{\text{SP}}(x)$ in the same original metric.

Every subsequent bound is now evaluated on this identical path, including the centered Hilbert-Schmidt, angle and signed finite-residual controls. Retain the original monomial row $v(y) = (1, S, \dots, S^{2q})$, $S = k/2 + iy$, in the same H , and the original interpolation measure $d\mu_x(y) = ((1-x)r_{1/4}^{*k}(y) + xw_h^{*k}(y)) dy$. For compact notation put $n = 2q + 1$, $a_q = 2q - 1$, and define

$$\begin{aligned} T_q(x) &= \frac{1}{2} d_x \|U_x - W_x\|_{1, M_x}, \\ S_q(x) &= \sqrt{\left(\text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{n} \right) \text{Tr}((U_x - W_x)^2)}, \\ z_x &= a_q \sqrt{1 - e^{-F(x)/a_q}}, \quad A_q(x) = z_x d_x, \\ J_q &= \int_0^1 \min\{S_{\text{AW}}(x), T_q(x), S_{\text{SP}}(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\{d_x, S_{\text{AW}}(x)/z_x, T_q(x)/z_x, S_{\text{SP}}(x)/z_x, S_q(x)/z_x\} dx. \end{aligned} \quad (\text{ACM11a})$$

These symbols are exactly $E_q, T_q, O_q, S_q, A_q, J_q, I_q$ of (RMT9)-(RMT13), with $E_q = S_{\text{AW}}$ and $O_q = S_{\text{SP}}$. Equations (ACM1)-(ACM5) prove the operator and measure identities needed for that substitution. In particular they retain the actual inverse metric in the signed density $v(U_x - W_x)M_x^{-1}v^*$.

Here is the proof of the additional terms at this earlier use site. Trace zero in (ACM10) allows subtraction of $\text{Tr } C_x/n$ from C_x ; Hilbert-Schmidt Cauchy-Schwarz gives $|F'| \leq S_q$. The midpoint argument already proved gives $|F'| \leq T_q \leq S_{\text{SP}}$. Retain both complete q -entry angle lists of the minimum-section pairs $(q - 1, 2q)$ and $(q, 2q - 1)$. The second pair has a designated zero angle by the original ambient dimension. Each additional zero remains in its list. The exact projection-difference spectra give $|F'| \leq d_x \sum \sin \theta$ over the two lists. Their cosine-square determinants give $F = \sum -\log \cos^2 \theta$; applying concavity of $t \mapsto \sqrt{1 - e^{-t}}$ to the remaining $2q - 1$ entries gives $|F'| \leq A_q$, exactly as proved with every multiplicity in (RMT11). The designated zero still contributes its displayed zero to both scalar sums.

The denominator in (ACM11a) is strictly positive on this actual path. Indeed equality $F = 0$ makes both crossed minimum sections equal: their Gram difference is the squared norm of their difference, and every relative eigenvalue is at least one. Equality of their determinants makes every such eigenvalue one. In particular the constant remainder 1 would be orthogonal to $\chi \mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \overline{\chi(k - \bar{S})}$ has degree q and equals $\overline{\chi(S)}$ on $S = k/2 + iy$. That orthogonality would imply $\int |\chi|^2 d\mu_x = 0$, contradicting the positive Gram of the nonzero polynomial χ . This proves $F(x) > 0$ without inserting a new hypothesis. Compactness gives a positive minimum. All remaining functions are continuous except the intersection dimension, whose rank strata are Borel sets defined by matrix minors. Thus every displayed integral is finite.

The connection to the retained (J_*, I_*) is an exact identity, not an additional choice of estimate. Put $\Pi_N = P_N - Q_N$, the orthogonal projection onto the original minimum-section space at cutoff N . Substitution in (ACM5), including $Q_{q-1} = 0$, gives

$$U_x - W_x = (\Pi_{q-1} - \Pi_{2q}) + (\Pi_q - \Pi_{2q-1}).$$

For each pair of rank- q projections in H of dimension $2q + 1$, the complete difference spectrum consists of the positive and negative sine of each of its q principal angles and one additional ambient zero. All zero angles still contribute both of their zero entries. The triangle inequality for the trace norm therefore gives

$$\frac{1}{2} \|U_x - W_x\|_{1, M_x} \leq \sum_{j=1}^q \sin \theta_j + \sum_{j=1}^q \sin \varphi_j \leq a_q \sqrt{1 - e^{-F(x)/a_q}} = z_x. \quad (\text{ACM11b})$$

Here (θ_j) and (φ_j) are respectively the two complete angle lists just used; $\varphi_1 = 0$ is the designated forced zero. The last inequality is the same concavity calculation, and no other zero entry is removed. Consequently $T_q \leq A_q$ and the already proved $T_q \leq S_{SP}$ gives, pointwise and then by integration,

$$\begin{aligned} J_q &= \int_0^1 \min\{S_{AW}, T_q, S_q\} dx = J_*, \\ I_q &= \int_0^1 \frac{\min\{S_{AW}, T_q, S_q\}}{z_x} dx = I_*. \end{aligned} \quad (\text{ACM11c})$$

These are precisely the three terms in (AT21a)–(AT21c), through the already proved identical coefficients, measures and operators. The exact path dictionary is $B(x) = F(x)$, $a = a_q$ and $f_a(B(x)) = z_x$; it preserves the original initial value and every measure factor. For I_q , the omitted entry d_x is at least T_q/z_x by (ACM11b); the omitted overlap term is also at least T_q/z_x . Thus both equalities retain the same original nonlinear denominator.

Both bounds are finite and measurable on $[0, 1]$. Indeed the positive Grams and their inverses vary smoothly, the eigenvalues vary continuously, and the dimension s_x is a Borel function: in any fixed coefficient frame it is determined by the ranks of matrices of continuous functions, whose rank strata are given by vanishing and nonvanishing minors. Thus the triangle inequality for integration gives the combined, original-source control

$$|\Delta_{h,k}| \leq J_* \leq \underbrace{\int_0^1 \min\{S_{AW}(x), \frac{1}{2}d_x \|U_x - W_x\|_{1, M_x}\} dx}_{\mathfrak{C}_*} \leq \underbrace{\int_0^1 \min\{S_{AW}(x), S_{SP}(x)\} dx}_{\mathfrak{C}_*^{\text{ov}}}. \quad (\text{ACM11})$$

The first bound preserves the exact trace norm already present in (SP13); its proof above gives $d_x \|U_x - W_x\|_{1, M_x} / 2 \leq S_{SP}(x)$. The identity maps (ACM1)–(ACM5) identify these quantities with (AT18a), including the source density and its mass. If $b_1 \leq \dots \leq b_{2q+1}$ are the positive eigenvalues of $D = M_0^{-1}M_1$ in M_0 , the original interpolation gives

$$c_j(x) = \frac{b_j - 1}{1 - x + xb_j}, \quad \int_0^1 c_j(x) dx = \log b_j. \quad (\text{ACM12})$$

The order is preserved since differentiation of the displayed fraction with respect to b_j gives $(1 - x + xb_j)^{-2} > 0$. Integration of (ACM9) consequently yields

$$\begin{aligned} |\Delta_{h,k}| &\leq J_* \leq \mathfrak{C}_* \leq \mathfrak{C}_*^{\text{ov}} \leq \min \left\{ \mathcal{L}_q(D), \int_0^1 S_{SP}(x) dx \right\}, \\ \mathcal{L}_q(D) &= \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}. \end{aligned} \quad (\text{ACM13})$$

The same pointwise minimum also retains the individually integrated trace, centered variance and angle bounds. Termwise integration in (ACM12), with the original order and all multiplicities, proves

$$J_q \leq \min \left\{ \mathcal{L}_q(D), \int_0^1 T_q(x) dx, \int_0^1 S_{\text{SP}}(x) dx, \int_0^1 S_q(x) dx, \int_0^1 A_q(x) dx \right\} \quad (\text{ACM13a})$$

$$\mathcal{L}_q(D) \leq (2q-1) \log(b_{2q+1}/b_1).$$

The last inequality bounds each of the $1 + 2(q-1)$ logarithmic factors in the unchanged (ACM13) by $\log(b_{2q+1}/b_1)$. It is therefore valid also for $q = 1$, where the original sum is empty.

The empty sum at $q = 1$ is zero. No universal ordering of the two refinements is used: each bounds the same signed integrand, hence their pointwise minimum does. Their full original arithmetic moments, relative eigenvalues and overlap remain the actual terms to be evaluated along the growing packet family.

The actual moment interpolation above also satisfies the complete proof (AT21a)–(AT21e). With $a = 2q - 1$ its recursive additive minimum also retains the independently proved centered HS term S_{HS} in (AT21a), so $J_* \leq \mathfrak{C}_*$. The exact trace norm is at most both the overlap and full angle bounds. The nonlinear radius $I_* = \int_0^1 \min(S_{\text{spec}}, S_{\text{tr}}, S_{\text{HS}}) / [a \sqrt{1 - e^{-B(x)/a}}] dx$ is defined on the same strictly positive four-volume path, with $I_* \leq \log \kappa$. Thus the current endpoint consequence here is the simultaneous interval $\mathfrak{b}_*^- \leq B(1) \leq \mathfrak{b}_*^+$ of (AT21e), not only its additive corollary. The identity of the coefficient maps and original measures proved above verifies its hypotheses; no purity or tensor-uniform bound is supplied by this finite transport.

The following signed construction further intersects this same retained additive/nonlinear interval. By (ACM11c) its (J_q, I_q) are literally (J_*, I_*) ; hence no earlier endpoint or overlap bound is discarded.

The nonlinear coordinate also receives every one of these controls. For $B > 0$ put

$$\mathcal{H}_{a_q}(B) = 2 \operatorname{arcosh}(e^{B/(2a_q)}), \quad \Psi_{a_q}(t) = 2a_q \log \cosh(t/2) \quad (t \geq 0). \quad (\text{ACM14})$$

Their two inverse composites on the positive half-lines are identities. Direct differentiation gives

$$\mathcal{H}'_{a_q}(B) = \frac{1}{a_q \sqrt{1 - e^{-B/a_q}}}.$$

Dividing the five proved derivative bounds by z_x and integrating therefore gives $|\mathcal{H}_{a_q}(F(1)) - \mathcal{H}_{a_q}(F(0))| \leq I_q$. Moreover $I_q \leq \int dx = \log(b_n/b_1)$.

For every integer $p \geq 0$ retain the following specified finite signed construction, on the same source operators:

$$\begin{aligned} B_x^\circ &= (1-x)I + xD, & \alpha_x &= \tfrac{1}{2}(2(1-x) + x(b_1 + b_n)), & H_x^\circ &= I - B_x^\circ/\alpha_x, \\ C_x^{[p]} &= \alpha_x^{-1} \sum_{r=0}^p (H_x^\circ)^r (D - I), \\ R_x^{[p]} &= (H_x^\circ)^{p+1} C_x, & E_x^{[p]} &= R_x^{[p]} - \operatorname{Tr}(R_x^{[p]})I/n, \\ j_p &= \int_0^1 \operatorname{Tr}(C_x^{[p]}(U_x - W_x)) dx, \\ e_p &= \int_0^1 \sqrt{\operatorname{Tr}((E_x^{[p]})^2) \operatorname{Tr}((U_x - W_x)^2)} dx. \end{aligned} \quad (\text{ACM15})$$

The finite geometric identity proves $C_x - C_x^{[p]} = R_x^{[p]}$. Every factor is a real function of D , hence self-adjoint in M_x . Trace zero removes the centered scalar, so Hilbert–Schmidt Cauchy–Schwarz proves $j_p - e_p \leq \Delta_{h,k} \leq j_p + e_p$. Writing $\vartheta = (b_n - b_1)/(b_n + b_1) < 1$ and $K_D = \sum_{j=1}^n (|b_j - 1|/\min\{1, b_j\})^2$,

the full residual eigenvalues give $\text{Tr}((E_x^{[p]})^2) \leq \vartheta^{2p+2} K_D$. Both U_x, W_x dominate their range projections, whose intersection has dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$. Equation (ACM10) then gives $e_p \leq \vartheta^{p+1} \sqrt{K_D(8q-6)} \rightarrow 0$. This retains the exact signed center and its controlled residual, with no assumption of uniformity in the tensor order.

Put $F_0 = F(0)$ and $H_0 = \mathcal{H}_{a_q}(F_0)$. For every $p \geq 0$ define the actual simultaneous endpoint functions

$$\begin{aligned}\beta_p^- &= \max\{0, F_0 - J_q, \Psi_{a_q}(\max\{0, H_0 - I_q\}), F_0 + j_p - e_p\}, \\ \beta_p^+ &= \min\{F_0 + J_q, \Psi_{a_q}(H_0 + I_q), F_0 + j_p + e_p\}.\end{aligned}\tag{ACM16}$$

The preceding additive, nonlinear and signed inequalities prove $\beta_p^- \leq F(1) \leq \beta_p^+$ by taking their intersection. Both original angle lists, source masses, relation maps and arithmetic Taylor unit remain fixed. At $D = cI$ all diameter and residual terms vanish: each quotient volume is multiplied by the same scalar power q , which cancels with the four signs. The formulas then give $F(1) = F_0$ exactly. Numerical integration of a signed center must retain its own integration error; (ACM15) specifies exact integrals.

128.1 The literal graph augmentation on this finite coefficient source

The following additional calculation uses the full quartet domain of PGB.1: $\gamma > 2, 0 < \delta < 1/2, m, k \geq 1$. It leaves the preceding Gamma-to-arithmetic formulas on their entire stated domain. Here χ is the literal monic polynomial

$$\chi(S) = \prod_{a,b=0}^k (S - k/2 - (2a - k)\delta - i(2b - k)\gamma)^{1+k(m-1)}.\tag{ACM17}$$

Thus (HT2) and PGB.1 have identical roots, multiplicities, leading coefficient, increasing remainder frame and remainder map; their coefficient identification is the identity, including every nilpotent primary coordinate. Retain $g = 2\xi, v_h = g/h, \Phi' = h, \Phi(0) = 0, s_i = 1/2 + it_i, \Sigma = \sum_i s_i$ and $\tau_k = \sum_i \Phi(s_i)$. The integration measure in the full source is $\prod_i dt_i/(2\pi)$, and

$$\begin{aligned}(\mathcal{V}P)(\mathbf{t}) &= \left(\prod_i v_h(s_i)\right) P(\Sigma), \\ L_{\text{obs}}x &= \left(\prod_i v_h(s_i)\right) \sum_{b=0}^{q-1} x_b \text{rem}_{h(s_1), \dots, h(s_k)}(\tau_k \Sigma^b), \\ D_{\text{def}}P &= \tau_k \mathcal{V}P - L_{\text{obs}} J_{2q} P, \quad H = \mathcal{P}_{2q}, \quad M^{\text{ar}} = \mathcal{V}^* \mathcal{V}.\end{aligned}\tag{ACM18}$$

The symbols L_{obs} and D_{def} denote exactly L and D_{2q} of PGB.2; neither is the circle circumference or the positive relative matrix below. PGB.2 and its leading-coefficient argument prove that these columns lie in the stated Hilbert source and that D_{def} is injective. In particular every following finite Gram exists and is positive. On the unchanged coefficient space define the actual affine path

$$M_x^{\text{gr}} = M^{\text{ar}} + x D_{\text{def}}^* D_{\text{def}}, \quad 0 \leq x \leq 1, \quad \mathcal{V}_x^{\text{gr}} P = (\mathcal{V}P, \sqrt{x} D_{\text{def}} P).\tag{ACM19}$$

Its Gram is exactly M_x^{gr} , by the orthogonal direct-sum norm; no mixed term was omitted. At $x = 1$ this is the unchanged graph of PGB.2. The restriction to $\mathcal{P}_N$ uses the literal I_N and has defect $D_{\text{def}} I_N = D_N$: monic remainders satisfy $J_{2q} I_N = J_N$, proving this equality directly in (ACM18). Its relation columns are $I_N B_N^{\text{AW}}$, exactly as in (ACM3).

Use (ACM4)–(ACM5) with this new, explicitly specified form and put $U_x^{\text{gr}} = Q_{2q-1} + Q_{2q} - Q_q$ and $W_x^{\text{gr}} = P_{2q-1} - P_{q-1} + P_{2q} - P_q$. Let $G_N(x)$ denote the quotient Gram of its restriction and set

$$\begin{aligned}F_{\text{gr}}(x) &= \log \det G_{q-1}(x) + \log \det G_q(x) - \log \det G_{2q-1}(x) - \log \det G_{2q}(x), \\ C_x^{\text{gr}} &= (M_x^{\text{gr}})^{-1} D_{\text{def}}^* D_{\text{def}}, \quad A_x^{\text{gr}} = U_x^{\text{gr}} - W_x^{\text{gr}}, \\ \Gamma_k^{\text{gr}} &= F_{\text{gr}}(1) - F_{\text{gr}}(0) = \hat{\mathcal{B}}_k - \mathcal{B}_k.\end{aligned}\tag{ACM20}$$

Here $\widehat{\mathcal{B}}_k$ is the four-endpoint logarithm of $\widehat{G}_N$ in PGB.26, and $\mathcal{B}_k$ is that of the original arithmetic source. To prove the derivative at this use site, the minimum section $R_N(x)$ satisfies $J_N R_N = I$ and $B_N^* M_N(x) R_N = 0$. Differentiating the first equality puts $\dot{R}_N$ in the relation space. The two cross terms in $d(R_N^* M_N R_N)/dx$ therefore vanish by the second equality, and $\dot{G}_N = R_N^* \dot{M}_N R_N$. Jacobi's determinant formula and finite trace cyclicity give $d \log \det G_N/dx = \text{Tr}((P_N - Q_N)C_x^{\text{gr}})$. Taking the four signs proves

$$F'_{\text{gr}}(x) = \text{Tr}(A_x^{\text{gr}} C_x^{\text{gr}}). \quad (\text{ACM21})$$

The flag inclusions imply that each difference of nested orthogonal projections is an orthogonal projection. In particular Q_q has rank one and is contained in both Q_{2q-1} and Q_{2q} . The original dimensions give the complete spectra $0^{[q]}, 1^{[2]}, 2^{[q-1]}$ for $U_x^{\text{gr}}, W_x^{\text{gr}}$. Their traces agree, so $\text{Tr} A_x^{\text{gr}} = 0$ and

$$\text{Tr}((A_x^{\text{gr}})^2) = 8q - 4 - 2 \text{Tr}(U_x^{\text{gr}} W_x^{\text{gr}}). \quad (\text{ACM22})$$

These arguments use positive finite forms and the same flags; they do not use a scalar moment representation of the graph norm. Hence the proofs at (ACM9)–(ACM11a) apply to their spectral, trace-norm, overlap and centered Hilbert–Schmidt terms. Explicitly let $c_1^{\text{gr}} \leq \dots \leq c_n^{\text{gr}}$ be the full eigenvalue list of C_x^{gr} , $n = 2q + 1$, and let $d_x^{\text{gr}} = c_n^{\text{gr}} - c_1^{\text{gr}}$ and $s_x^{\text{gr}} = \dim(\text{ran } U_x^{\text{gr}} \cap \text{ran } W_x^{\text{gr}})$. Set

$$\begin{aligned} E_{\text{gr}}(x) &= c_{q+2}^{\text{gr}} - c_q^{\text{gr}} + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}^{\text{gr}} - c_j^{\text{gr}}), \\ T_{\text{gr}}(x) &= \frac{1}{2} d_x^{\text{gr}} \|A_x^{\text{gr}}\|_{1, M_x^{\text{gr}}}, \\ O_{\text{gr}}(x) &= \frac{1}{2} d_x^{\text{gr}} \min \left\{ 4q - 2s_x^{\text{gr}}, \sqrt{n \text{Tr}((A_x^{\text{gr}})^2)} \right\}, \\ S_{\text{gr}}(x) &= \sqrt{\left(\text{Tr}((C_x^{\text{gr}})^2) - \frac{(\text{Tr} C_x^{\text{gr}})^2}{n} \right) \text{Tr}((A_x^{\text{gr}})^2)}, \\ J_{\text{gr}} &= \int_0^1 \min\{E_{\text{gr}}, T_{\text{gr}}, O_{\text{gr}}, S_{\text{gr}}\} dx. \end{aligned} \quad (\text{ACM23})$$

Rank-stratum measurability and continuity of the other entries prove integrability exactly as before. The four pointwise trace proofs give $|\Gamma_k^{\text{gr}}| \leq J_{\text{gr}}$.

For a specified integer $p \geq 0$, this same path also has an actual signed finite approximation. Put

$$\begin{aligned} D^{\text{gr}} &= (M^{\text{ar}})^{-1} (M^{\text{ar}} + D_{\text{def}}^* D_{\text{def}}), \quad 1 < b_1^{\text{gr}} \leq \dots \leq b_n^{\text{gr}}, \\ B_x^{\text{gr}} &= (1-x)I + xD^{\text{gr}}, \quad \alpha_x^{\text{gr}} = 1-x + \frac{x}{2}(b_1^{\text{gr}} + b_n^{\text{gr}}), \\ H_x^{\text{gr}} &= I - B_x^{\text{gr}}/\alpha_x^{\text{gr}}, \quad C_x^{[p],\text{gr}} = (\alpha_x^{\text{gr}})^{-1} \sum_{r=0}^p (H_x^{\text{gr}})^r (D^{\text{gr}} - I), \\ R_x^{[p],\text{gr}} &= (H_x^{\text{gr}})^{p+1} C_x^{\text{gr}}, \quad E_x^{[p],\text{gr}} = R_x^{[p],\text{gr}} - \text{Tr}(R_x^{[p],\text{gr}})I/n, \\ j_{p,\text{gr}} &= \int_0^1 \text{Tr}(C_x^{[p],\text{gr}} A_x^{\text{gr}}) dx, \\ e_{p,\text{gr}} &= \int_0^1 \sqrt{\text{Tr}((E_x^{[p],\text{gr}})^2) \text{Tr}((A_x^{\text{gr}})^2)} dx. \end{aligned} \quad (\text{ACM24})$$

The strict lower bound for b_1^{gr} follows from injectivity of D_{def} in this finite dimension. The finite geometric identity gives $C_x^{\text{gr}} - C_x^{[p],\text{gr}} = R_x^{[p],\text{gr}}$. All these factors are real functions of the same D^{gr} , so they are self-adjoint for M_x^{gr} . Pairing the last identity with the trace-zero A_x^{gr} and applying Hilbert–Schmidt Cauchy–Schwarz proves

$$j_{p,\text{gr}} - e_{p,\text{gr}} \leq \Gamma_k^{\text{gr}} \leq j_{p,\text{gr}} + e_{p,\text{gr}}. \quad (\text{ACM25})$$

For completeness set

$$\vartheta_{\text{gr}} = \frac{b_n^{\text{gr}} - b_1^{\text{gr}}}{b_n^{\text{gr}} + b_1^{\text{gr}}}, \quad K_{\text{gr}} = \sum_{j=1}^n \left(\frac{|b_j^{\text{gr}} - 1|}{\min\{1, b_j^{\text{gr}}\}} \right)^2.$$

The eigenvalue formula $c_j^{\text{gr}}(x) = (b_j^{\text{gr}} - 1)/(1 - x + xb_j^{\text{gr}})$ shows $\|H_x^{\text{gr}}\| \leq \vartheta_{\text{gr}} < 1$ and $\text{Tr}((E_x^{[p],\text{gr}})^2) \leq \vartheta_{\text{gr}}^{2p+2} K_{\text{gr}}$. The range intersection has dimension at least one. Since each of $U_x^{\text{gr}}, W_x^{\text{gr}}$ dominates its range projection, the positive-product trace inequality gives $\text{Tr}(U_x^{\text{gr}} W_x^{\text{gr}}) \geq 1$. Together with (ACM22), this proves

$$e_{p,\text{gr}} \leq \vartheta_{\text{gr}}^{p+1} \sqrt{K_{\text{gr}}(8q - 6)} \rightarrow 0. \quad (\text{ACM26})$$

Thus the two actual signed endpoints

$$g_{p,\text{gr}}^- := \max\{-J_{\text{gr}}, j_{p,\text{gr}} - e_{p,\text{gr}}\}, \quad g_{p,\text{gr}}^+ := \min\{J_{\text{gr}}, j_{p,\text{gr}} + e_{p,\text{gr}}\} \quad (\text{ACM27})$$

satisfy $g_{p,\text{gr}}^- \leq \Gamma_k^{\text{gr}} \leq g_{p,\text{gr}}^+$. This calculation uses no division by the angle coordinate of the new graph form. The moment-path nonlinear formulas (ACM14)–(ACM16) retain their original scope. All integrals in (ACM23)–(ACM27) are exact integrals of the displayed finite matrices; numerical evaluation must add its own certified integration error.

128.2 The complete original relation rows in the graph endpoint control

Keep exactly the domain and coefficient identity of (ACM17)–(ACM20). The symbols in this paragraph include a superscript *rel* so that the monic relation polynomial is never confused with a consecutive quotient-volume ratio. They are the literal objects of (PGRT.2)–(PGRT.8):

$$\begin{aligned} r_k(u) &= \int_{\sum t_i=u} (1 + |\tau_k|^2) \prod_i w_h(t_i), \quad F_b = \text{rem}_{h(s_1), \dots, h(s_k)}(\tau_k \Sigma^b), \\ z_b(u) &= \int_{\sum t_i=u} \overline{\tau_k} F_b \prod_i w_h(t_i), \quad g_x = z(u)x/r_k(u), \quad F^0 x = (sx)(k/2 + iu) - g_x(u), \\ d_j^{\text{rel}} &= \chi p_{j+1-q}, \quad a_j^{\text{rel}} = \int |d_j^{\text{rel}}(k/2 + iu)|^2 r_k(u) du > 0, \\ (\alpha_j^{\text{rel}})_b &= \int \overline{d_j^{\text{rel}}(k/2 + iu)} ((k/2 + iu)^b r_k(u) - z_b(u)) du, \quad j \geq q-1, \quad 0 \leq b < q. \end{aligned} \quad (\text{ACM28})$$

Here $s : E \rightarrow \mathcal{P}_{q-1}$ is the original monic remainder section, p_n is the unique monic polynomial of degree n orthogonal to lower polynomials for $|\chi|^2 r_k du$, and the fibre integral eliminates t_k with its original Lebesgue measure. For $k = 1$ it is evaluation. The constant factors are $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$ throughout. Each coefficient is absolutely integrable by the proved polynomial exponential moments and Cauchy–Schwarz applied to F^0 . Expanding the full polynomials in these integrals gives the finite products of the original one-factor moments in (PGS.35) and (PGS.38), without dropping their conjugate or cross terms. The coefficients of p_n are obtained from its positive finite lower-polynomial Gram system. This is the exact finite arithmetic evaluation of each row.

The complete weighted density proof (PGRT.3)–(PGRT.6) gives

$$\widehat{G}_N - \Omega_k = \sum_{j=N}^{\infty} (a_j^{\text{rel}})^{-1} (\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}, \quad \Omega_k = K_{\text{low}} + D_{\text{mix}}^* D_{\text{mix}} > 0. \quad (\text{ACM29})$$

The series converges in the original finite matrix operator norm. Specifically, the relation columns are a complete orthogonal family in $L^2(r_k du)$; the coefficient of $F^0 x$ on its j th column is $(a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} x$. Removing the first $N - q + 1$ columns is precisely minimization over the original relations $\chi \mathcal{P}_{N-q}$.

Parseval gives the displayed quadratic forms, and polarization gives their matrices. The trace of the remaining positive tail is the sum of the squared tails of the q fixed frame vectors, hence tends to zero and bounds its operator norm. The positive completed term and its mixed realization are the same as (HC22), with full proof in PGMB.

For $q - 1 \leq j \leq 2q - 1$ put

$$\begin{aligned} l_j^{\text{rel}} &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \widehat{G}_{q-1}^{-1} (\alpha_j^{\text{rel}})^*, \\ u_j^{\text{rel}} &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \widehat{G}_{2q}^{-1} (\alpha_j^{\text{rel}})^*, \\ Z_{2q}^{\text{mix}} x &= (D_{\text{mix}} x, f_{2q} x), \\ u_j^{\text{rel}} &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \\ &\quad - (a_j^{\text{rel}})^{-1} \left\| (I + Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (Z_{2q}^{\text{mix}})^*)^{-1/2} Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \right\|^2. \end{aligned} \tag{ACM30}$$

The domain of Z_{2q}^{mix} is the original E , and its ordered target is the mixed weighted Hilbert space of (HC22) followed by $L^2(r_k du)$. In particular it retains the finite tail as well as the mixed component. To prove the subtraction, write $K = K_{\text{low}}$ and $Z = Z_{2q}^{\text{mix}}$. The identity $\widehat{G}_{2q} = K + Z^* Z$ gives

$$(K + Z^* Z)^{-1} = K^{-1} - K^{-1} Z^* (I + Z K^{-1} Z^*)^{-1} Z K^{-1}.$$

Multiplication by $K + Z^* Z$ cancels the leftover factor $Z^* Z K^{-1} - Z^* (I + Z K^{-1} Z^*) (I + Z K^{-1} Z^*)^{-1} Z K^{-1}$. The target inverse exists because it is the identity off the finite-dimensional range of Z and is positive with eigenvalues at least one on that range. Pairing with the specified row proves (ACM30). If $k \geq 2$ and $\alpha_j^{\text{rel}} \neq 0$, the subtracted squared norm is strictly positive: D_{mix} is injective by the full fixed-sum calculation PGM.6–9, and the displayed inverse square root has zero kernel. For $k = 1$ its mixed component is zero but its finite tail is retained. A zero row gives zero in both terms. These statements retain the original strictness domain and the whole negative term.

Use the endpoint weights $\omega_{q-1} = \omega_{2q-1} = 1$ and $\omega_j = 2$ for $q \leq j \leq 2q - 2$, and define

$$\begin{aligned} L_k^{\text{rel}} &= \sum_{j=q-1}^{2q-1} \omega_j \log(1 + l_j^{\text{rel}}), \quad U_{k,2q}^{\text{rel}} = \sum_{j=q-1}^{2q-1} \omega_j \log(1 + u_j^{\text{rel}}), \\ L_k^{\text{rel}} &\leq \widehat{\mathcal{B}}_k \leq U_{k,2q}^{\text{rel}}, \\ L_k^{\text{rel}} - g_{s,\text{gr}}^+ &\leq \mathcal{B}_k \leq U_{k,2q}^{\text{rel}} - g_{s,\text{gr}}^- \quad (s \geq 0). \end{aligned} \tag{ACM31}$$

To verify every index, (ACM29) gives $\widehat{G}_j = \widehat{G}_{j+1} + (a_j^{\text{rel}})^{-1} (\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}$. The rank-one determinant identity therefore gives the j th decrement $\log(1 + (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \widehat{G}_{j+1}^{-1} (\alpha_j^{\text{rel}})^*)$. Since $\widehat{G}_{2q} \preceq \widehat{G}_{j+1} \preceq \widehat{G}_{q-1}$, inverse order bounds its argument between $1 + l_j^{\text{rel}}$ and $1 + u_j^{\text{rel}}$. Inverse order follows by congruence with the smaller positive form, inversion of its positive eigenvalues, and reversal of that congruence. Summing the decrements from $q - 1$ to $2q - 2$ and from q to $2q - 1$ gives precisely the displayed weights. The full quartet in (ACM17) has $q \geq 4$, so the listed endpoints and interior have these literal ranges. Finally (ACM20) says $\mathcal{B}_k = \widehat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$. Subtracting (ACM27) yields the last line, with its ordered negative $g_{s,\text{gr}}^-$ term intact. This is a finite relation-row bound on the same arithmetic scalar; it does not replace the exact value of $\widehat{\mathcal{B}}_k$ when that value is retained.

All retained relation correlations in the same inverse. The complete calculation PGRC.1–18 now sharpens the row enclosure (ACM31) by evaluating its full cross terms. Keep $H = \widehat{G}_{2q}$, $I = \{q - 1, \dots, 2q - 1\}$ and put $C_{ij} = \alpha_i^{\text{rel}} H^{-1} (\alpha_j^{\text{rel}})^*$. For an ordered subset $J \subset I$ let A_J have the

corresponding original rows, let $D_J = \text{diag}(a_i^{\text{rel}} : i \in J)$, and define

$$\begin{aligned} H_J &= H + A_J^* D_J^{-1} A_J, \quad B_J = D_J + C_{JJ}, \\ V_J : (E, H_J) &\longrightarrow (E, H) \oplus (\mathbb{C}^J, D_J^{-1}), \quad V_J x = (x, A_J x), \\ V_J^\dagger(y, z) &= H_J^{-1}(Hy + A_J^* D_J^{-1} z), \\ V_J^\dagger V_J &= I_E, \quad \ker V_J = 0, \quad \ker V_J^\dagger = \{(y, z) : Hy + A_J^* D_J^{-1} z = 0\}. \end{aligned} \tag{ACM32}$$

The norm expansion proves the Gram H_J and the adjoint; multiplication proves the inverse on the image and both kernel descriptions. Empty subsets mean the zero-dimensional coordinate space and its unique empty inverses. For every subset, direct multiplication also proves $H_J^{-1} = H^{-1} - H^{-1} A_J^* B_J^{-1} A_J H^{-1}$: the extra middle factor is $D_J^{-1} - B_J^{-1} - D_J^{-1} C_{JJ} B_J^{-1} = 0$ because $I + D_J^{-1} C_{JJ} = D_J^{-1} B_J$.

Let $J_j = \{j+1, \dots, 2q-1\}$. For $J \subset J_j$ put $r_j(J) = C_{jJ} B_J^{-1} C_{Jj}$ and $v_j(J) = (C_{jj} - r_j(J))/a_j^{\text{rel}}$. The inverse identity proves $v_j(J) = \alpha_j^{\text{rel}} H_J^{-1} (\alpha_j^{\text{rel}})^* / a_j^{\text{rel}} \geq 0$. Telescoping (ACM29) identifies $H_{J_j} = \widehat{G}_{j+1}$. Thus the exact decrement parameter is

$$\eta_j = \frac{C_{jj} - C_{jJ_j} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j j}}{a_j^{\text{rel}}}. \tag{ACM33}$$

For disjoint $J, K \subset J_j$, keep the remaining block and row

$$\begin{aligned} S_{K|J} &= D_K + C_{KK} - C_{KJ} B_J^{-1} C_{JK} = D_K + A_K H_J^{-1} A_K^* \succeq D_K > 0, \\ e_{j,K|J} &= C_{jK} - C_{jJ} B_J^{-1} C_{JK} = \alpha_j^{\text{rel}} H_J^{-1} A_K^*, \\ r_j(J \cup K) - r_j(J) &= e_{j,K|J} S_{K|J}^{-1} e_{j,K|J}^*, \\ 0 \leq r_j(J \cup K) - r_j(J) &\leq \sum_{i \in K} |(e_{j,K|J})_i|^2 / a_i^{\text{rel}}. \end{aligned} \tag{ACM34}$$

Substitution of the proved inverse gives the first two lines. For interleaved disjoint subsets, first use the exact coordinate map $P_{J,K} : \mathbb{C}^{J \cup K} \rightarrow \mathbb{C}^J \oplus \mathbb{C}^K$, $z \mapsto (z|_J, z|_K)$, with inverse reassembling the coordinates in their inherited order. This is an isometry for the retained diagonal lengths: its two squared norms sum to the original sum over $J \cup K$. On coefficient matrices it is a permutation, so $P_{J,K} (D_{J \cup K} + C_{J \cup K, J \cup K}) P_{J,K}^{-1}$ is exactly the displayed two-block matrix. Conjugating the inverse and applying the same restriction to the row leaves the scalar $r_j(J \cup K)$ unchanged. To prove the third line, solve the resulting two block equations $B_J x + C_{JK} y = C_{Jj}$ and $C_{KJ} x + (D_K + C_{KK}) y = C_{Kj}$. The first gives $x = B_J^{-1} (C_{Jj} - C_{JK} y)$ and the second then gives $y = S_{K|J}^{-1} e_{j,K|J}^*$. Pairing with (C_{jJ}, C_{jK}) proves exactly the displayed quadratic correction. Inverse order from $S_{K|J} \succeq D_K$ gives the final inequality. In particular all cross subtractions are retained even when they vanish.

Choose actual finite subsets $J(j) \subset J_j$, let $K(j) = J_j \setminus J(j)$, and set

$$\begin{aligned} \bar{\eta}_j &= v_j(J(j)), \quad \underline{\eta}_j = \max \left\{ l_j^{\text{rel}}, v_j(J(j)) - \sum_{i \in K(j)} \frac{|(e_{j,K(j)|J(j)})_i|^2}{a_j^{\text{rel}} a_i^{\text{rel}}} \right\}, \\ L_k^{\text{corr}}(J) &= \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \underline{\eta}_j), \quad U_k^{\text{corr}}(J) = \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \bar{\eta}_j), \\ L_k^{\text{rel}} &\leq L_k^{\text{corr}}(J) \leq \widehat{B}_k \leq U_k^{\text{corr}}(J) \leq U_{k,2q}^{\text{rel}}. \end{aligned} \tag{ACM35}$$

The last line of (ACM34) with $J \cup K = J_j$ proves $0 \leq \underline{\eta}_j \leq \eta_j \leq \bar{\eta}_j$. Moreover $\underline{\eta}_j \geq l_j^{\text{rel}}$ and $\bar{\eta}_j \leq C_{jj}/a_j^{\text{rel}} = u_j^{\text{rel}}$. Increasing logarithms and the unchanged endpoint weights prove all the

inequalities in (ACM35). If every $J(j) = J_j$, the remaining block is empty and both new bounds equal the exact graph volume. Insertion of one further index b subtracts exactly

$$\frac{|C_{jb} - C_{jJ}B_J^{-1}C_{Jb}|^2}{a_j^{\text{rel}}(a_b^{\text{rel}} + C_{bb} - C_{bJ}B_J^{-1}C_{Jb})}$$

from $v_j(J)$, by (ACM34). Its denominator is positive. Thus successive upper bounds decrease and become exact after at most q later-row insertions for each row; retaining the maximum of all previous valid lower bounds makes that retained sequence increasing as well.

In particular the exact $a_j^{\text{rel}}\eta_j$ is the nonnegative $\alpha_j^{\text{rel}}K_{\text{low}}^{-1}(\alpha_j^{\text{rel}})^*$ minus the full squared norm in (ACM30) minus $C_{jJ_j}(D_{J_j} + C_{J_jJ_j})^{-1}C_{J_jj}$. The latter vanishes exactly when $C_{J_jj} = 0$, because its inverse matrix is positive. This proves that both negative corrections occur in the same original inverse quantity. Their signs and equality cases are retained. The matrix C in the correlation subtraction is computed using $\hat{G}_{2q}^{-1}$, with its full mixed term; it is never replaced there by K_{low}^{-1} .

At every independent coefficient face $(W, S; A)$ the coefficient inclusions and zero entries act componentwise on these same linear source maps. Retain the original source F_h with $\mathcal{M}F_h = g/h$, $D_i = -x_i\partial_{x_i}$, $h(D)F_h = \Theta\phi_*$, and $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$ from (ISM1). The full Taylor unit in that observation is unchanged. The original minimum-section difference is a relation χQ ; successive monic division constructs $\chi(\sum s_i)Q(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$. Its original primitive is $\sum_i (-1)^{i-1}Q_i(D_1, \dots, D_k)(F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)})$. The tensor differential contributes the same sign and therefore recovers the relation. The differential operators commute with the original tensor differential, and $h(D_i)F_h = \Theta\phi_*$, proving that its image is exactly $\mathcal{V}(\chi Q)$. For the original two-leg complex, use the degree-zero map $j_+(v) = (v, 0)$ and degree-one identity into $[V \oplus \hat{V} \rightarrow \mathcal{B}]$, whose differential is $\Theta(\phi - \mathcal{F}^{-1}\hat{\psi})$. The tensor cochain map preserves every sign and gives the same top boundary. The one-factor two-leg differential has the entire diagonal kernel $(v, \mathcal{F}v)$: its zero equation is equivalent to $\phi = \mathcal{F}^{-1}\hat{\psi}$ by injectivity of Θ , and every such diagonal pair has zero differential. When the two compared top representatives lie in an admitted proper subcomplex $\mathbf{C}_{W,S;A}^\bullet$ of the fixed full coefficient carrier, their residual is the class in $(d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_{W,S;A}^k)/d\mathbf{C}_{W,S;A}^{k-1}$, with its complete two-leg primitive retained. The inclusion-induced map sends a numerator representative to its class in $\mathbf{C}_{W,S;A}^k/d\mathbf{C}_{W,S;A}^{k-1}$. Its image is exactly the kernel of the map to the full top quotient: mapping to zero there means being a full boundary. Its inverse uses that same representative, and both are well-defined because their denominator is the identical $d\mathbf{C}_{W,S;A}^{k-1}$. This proves the stated kernel isomorphism, as in the complete coefficient and source maps (ISM38)–(ISM43). Consequently none of the metric estimates deletes this residual, the full diagonal degree-zero source, or an empty mask's outer label.

129 The actual single-primary marked family and its complete boundary monodromy

This calculation specializes the original marked polynomial family to a single selected zero with its complete order. The parameter t is fixed at its original value zero. The variable s , its ascending monomial basis, the primitive with value zero at $s = 0$, and the original theta source are retained. A translation below is a displayed coordinate isomorphism on that same family; its constant term and contour transport are part of the calculation.

129.1 The actual packet and the exact Pascal map

Let ρ be a selected actual nontrivial zero of $g = 2\xi$, of complete order $m \geq 1$. Put

$$n = m + 1, \quad h(s) = (s - \rho)^m, \quad E_h = \mathbb{C}[s]/((s - \rho)^m), \quad A_s = M_s : E_h \rightarrow E_h. \quad (\text{SP.1})$$

The original polynomial primitive is exactly

$$\Phi_h(s) = \frac{(s - \rho)^n - (-\rho)^n}{n} = \frac{y^n}{n} + c, \quad y = s - \rho, \quad c = -\frac{(-\rho)^n}{n}. \quad (\text{SP.2})$$

Differentiation gives $\Phi'_h = h$, and substitution $s = 0$ gives $\Phi_h(0) = 0$. Thus its constant has not been reassigned. Since a nontrivial zero ρ is nonzero, this complex number c is nonzero; later coefficient specializations may send it to zero and are treated separately.

Write $e_s = (1, s, \dots, s^{m-1})$ and $e_y = (1, y, \dots, y^{m-1})$ as ordered rows. The coefficient map from the former basis to the latter is the explicit Pascal matrix

$$e_s = e_y P, \quad P_{ab} = \begin{cases} \binom{b}{a} \rho^{b-a}, & 0 \leq a \leq b < m, \\ 0, & a > b, \end{cases} \quad (P^{-1})_{ab} = \begin{cases} \binom{b}{a} (-\rho)^{b-a}, & a \leq b, \\ 0, & a > b. \end{cases} \quad (\text{SP.3})$$

The binomial theorem proves the first identity. Substituting $s = y + \rho$ and $y = s - \rho$ in succession gives the identity on every monomial, proving the inverse formula. Both matrices have diagonal entries one and determinant one. If x_s is an original coefficient column, its new coordinates are exactly $x_y = P x_s$.

The polynomial-exponential complex at $t = 0$ is

$$\mathbb{C}[u, s] \xrightarrow{L_u} \mathbb{C}[u, s] ds, \quad L_u = u \partial_s + (s - \rho)^m. \quad (\text{SP.4})$$

Under the same translation $ds = dy$ and $L_u = u \partial_y + y^m$. For a nonzero polynomial Q , the degree of $L_u Q$ in s is exactly $\deg Q + m$, with the same leading coefficient. Successive subtraction of its leading monomials therefore gives a unique remainder of degree below m . The complex is injective in degree zero and its degree-one cohomology is the free module

$$\mathcal{H} = \bigoplus_{b=0}^{m-1} \mathbb{C}[u] [s^b ds], \quad \mathcal{H}/u\mathcal{H} = E_h ds. \quad (\text{SP.5})$$

The unique-remainder proof also verifies the equality after specialization: at $u = 0$ the differential is multiplication by the original h , retaining its complete primary order.

129.2 The two boundary coefficients in the unchanged basis

The BC connection is defined by monic division of $\Phi_h(s)s^b$ by $h(s)$. Here that division is explicit:

$$\Phi_h(s)s^b = h(s) \frac{(s - \rho)s^b}{n} + cs^b, \quad q_b(s) = \frac{(s - \rho)s^b}{n}, \quad r_b(s) = cs^b. \quad (\text{SP.6})$$

For $0 \leq b < m$, its quotient derivative is already a remainder:

$$q'_b(s) = \frac{(b+1)s^b - b\rho s^{b-1}}{n}, \quad (\text{SP.7})$$

where the second term is zero for $b = 0$. Subtracting $L_u q_b$ in (SP.6) proves $[\Phi_h s^b] = [cs^b - u q'_b]$ in (SP.5). Thus the exact coefficient matrices and connection are

$$C_s = cI_m, \quad (B_s)_{bb} = \frac{b+1}{n}, \quad (B_s)_{b-1,b} = -\frac{b\rho}{n} \quad (1 \leq b < m), \quad \nabla_u = \partial_u - \frac{cI_m}{u^2} + \frac{B_s}{u}, \quad (\text{SP.8})$$

and all other entries of B_s are zero. The induced degree-one formula comes from the original chain connection $\nabla_u^1 = \partial_u - \Phi_h/u^2$ and $\nabla_u^0 = \partial_u - \Phi_h/u^2 + 1/u$. Indeed $[L_u, \partial_u] = -\partial_s$ and $[L_u, -\Phi_h/u^2] = -h/u$, so $\nabla_u^1 L_u = L_u \nabla_u^0$. This proves the connection on the stated complex, with the degree-zero correction and the marked scalar retained.

In the y -coordinates, the same division of $(y^n/n + c)y^a$ gives the quotient y^{a+1}/n . Consequently

$$PB_sP^{-1} = \text{diag}(\beta_0, \dots, \beta_{m-1}), \quad \beta_a = \frac{a+1}{n}, \quad PC_sP^{-1} = cI_m, \quad \text{Tr } B_s = \frac{m}{2}. \quad (\text{SP.9})$$

This also proves the matrix conjugation directly, without replacing the original matrix by one having the same eigenvalues.

129.3 The original ordered contours and their exact translation

Fix a branch $\log u = \log |u| + i \arg u$ on a simply connected sector of $\mathbb{C}^\times$. Put

$$\vartheta_j = \frac{\arg u + \pi}{n} + \frac{2\pi j}{n}, \quad \ell_j = \{re^{i\vartheta_j} : 0 \leq r < \infty\}, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq m), \quad (\text{SP.10})$$

with each ℓ_j oriented toward infinity. These are the original ordered contours in the s -plane. Their periods, in the original monomial columns, are

$$(\Pi_s(u))_{jb} = \int_{\Gamma_j} e^{\Phi_h(s)/u} s^b ds, \quad 1 \leq j \leq m, \quad 0 \leq b < m. \quad (\text{SP.11})$$

We prove the translation needed to evaluate these exact contours. For $a \in [0, 1]$, translate the broken contour by $a\rho$. Its two ends remain in the same ordered decay sectors, while its junction moves from 0 to $a\rho$. Truncate its two rays at radius R . Cauchy's theorem applies to the quadrilateral swept by each finite ray because the integrand is entire in s . The two junction paths have opposite orientations in $-\ell_0 + \ell_j$ and cancel. Each remaining connector has length at most $|\rho|$ and lies at $s = Re^{i\vartheta_l} + a\rho$, $l = 0, j$. Uniformly for $a \in [0, 1]$,

$$\text{Re} \frac{\Phi_h(s)}{u} = -\frac{R^n}{n|u|} + O(R^{n-1}) + O(1). \quad (\text{SP.12})$$

This follows by expanding $(Re^{i\vartheta_l} + (a-1)\rho)^n$, whose leading term divided by nu has the displayed negative real value. The polynomial factor is $O((1+R)^b)$; the connector integrals therefore tend to zero as $R \rightarrow \infty$. The same estimate proves absolute convergence of all rays. It follows that each period over its original Γ_j equals the period over $\rho + \Gamma_j$. This is the exact sector-preserving contour deformation, with the original orientation and junction cancellation.

On the latter contour set $s = \rho + y$. The binomial expansion is exactly (SP.3), while (SP.2) retains the factor $e^{c/u}$. Let

$$\zeta = e^{2\pi i/n}, \quad W_{ja} = \zeta^{j(a+1)} - 1, \quad d_a = n^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a), \quad D(u) = \text{diag}(d_a u^{\beta_a})_{a=0}^{m-1}. \quad (\text{SP.13})$$

On the j -th y -ray, the substitution $v = r^n/(n|u|)$, with its specified positive real radial variable, gives

$$\int_{\ell_j} e^{y^n/(nu)} y^a dy = n^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a) u^{\beta_a} \zeta^{j(a+1)}.$$

Subtracting the $j = 0$ ray proves the complete ordered period matrix in the original columns:

$$\Pi_s(u) = e^{c/u} W D(u) P. \quad (\text{SP.14})$$

Every factor in this equality has been specified, including the original primitive constant, the gamma constants, all powers of ρ , and the ordered contour phase.

For completeness, W is invertible. If its columns have a linear relation with coefficients $a_1, \dots, a_m$, then $\sum_{r=1}^m a_r (z^r - 1)$ vanishes at every nontrivial n -th root of unity. It also vanishes at 1. Its degree is at most $m = n - 1$; division by the degree- n polynomial $z^n - 1$, or the elementary bound on the

number of roots of a nonzero polynomial, makes it the zero polynomial. All its coefficients a_r are zero. The gamma factors in (SP.13) are positive-real gamma values times nonzero constants, since $0 < \beta_a < 1$. Thus (SP.14) is invertible for every $u \neq 0$. Its determinant is exactly

$$\det \Pi_s(u) = K_m u^{m/2} e^{mc/u}, \quad K_m = n^{-m/2} e^{\pi i m/2} \left(\prod_{a=0}^{m-1} \Gamma(\beta_a) \right) \det W \neq 0. \quad (\text{SP.15})$$

Here $\sum_a \beta_a = m/2$ and $\det P = 1$ prove every factor. This is the single-primary value of the original BC determinant, with $\text{Tr } C_s = mc$.

129.4 Complete monodromy, its finite cover, and the boundary invariants

Under one positive circuit of $u = 0$, the ordered decay rays move ℓ_j to ℓ_{j+1} , with indices modulo n . Thus their transported relative cycles satisfy

$$\Gamma_j \mapsto \begin{cases} \Gamma_{j+1} - \Gamma_1, & 1 \leq j < m, \\ -\Gamma_1, & j = m. \end{cases} \quad (\text{SP.16})$$

The equality follows by writing each as $-\ell_1 + \ell_{j+1}$ and subtracting the defining $-\ell_0 + \ell_1$. It remains valid for $m = 1$, when the second case is the single map $\Gamma_1 \mapsto -\Gamma_1$. Let M_{cyc} be the matrix whose rows record (SP.16). The constant exponential is single-valued on $\mathbb{C}^\times$. Equation (SP.14) therefore gives

$$\Pi_s^{\text{cont}} = M_{\text{cyc}} \Pi_s, \quad M_{\text{cyc}} = W \text{diag}(\zeta, \zeta^2, \dots, \zeta^m) W^{-1}, \quad M_{\text{cyc}}^n = I, \quad \ker(M_{\text{cyc}} - I) = 0. \quad (\text{SP.17})$$

For example the row difference in (SP.16), applied to the column $W_{\bullet a}$, is $\zeta^{a+1}(\zeta^{j(a+1)} - 1)$, proving the stated diagonalization directly. None of $\zeta, \dots, \zeta^m$ equals one.

The signs for horizontal coefficient solutions can also be specified without an implicit duality convention. The connection (SP.8) acts on coefficient columns. Its horizontal equation is $x'_s = (cI/u^2 - B_s/u)x_s$, whose fundamental matrix is

$$H_s(u) = P^{-1} e^{-c/u} \text{diag}(u^{-\beta_a})_{a=0}^{m-1}. \quad (\text{SP.18})$$

Differentiate it and use (SP.9) to verify the equation. Its coefficient monodromy is conjugate to $\text{diag}(\zeta^{-1}, \dots, \zeta^{-m})$. This is the inverse convention to the period-cycle action in (SP.17). Its eigenvalue set is the same set of all nontrivial n -th roots, so it too has zero invariants.

On the finite ramified cover $u = v^n$, the pulled-back connection in the y -basis is exactly

$$\partial_v - \frac{nc}{v^{n+1}} I_m + \frac{1}{v} \text{diag}(1, 2, \dots, m). \quad (\text{SP.19})$$

The meromorphic basis change $x_y = \text{diag}(v^{-1}, \dots, v^{-m})z$ cancels precisely the integer diagonal term and leaves $\partial_v - nc v^{-n-1} I_m$. Its horizontal factor remains e^{-c/v^n} . It is not a meromorphic gauge removing that irregular factor when $c \neq 0$. Analytic continuation of the displayed fundamental matrix around $v = 0$ is, nevertheless, the identity. Thus the actual finite-cover unipotent monodromy logarithm is exactly

$$N_{\text{boundary}} = 0. \quad (\text{SP.20})$$

There is no uncomputed sector transition in this one-critical-value calculation: (SP.18) is an exact diagonal fundamental solution on every branch, and after (SP.19) all columns have the same explicit scalar exponential and integer powers. On two overlapping sectors with the same branch they are identical. This proves the absence of additional Stokes factors for this specified family; it is not a claim about a packet with several distinct critical values.

Let $j : \Delta^\times \hookrightarrow \Delta$ be a small complex disc punctured at zero, and let $\mathcal{V}_{\text{an}}$ denote either the period local system or its horizontal dual, with the convention just specified. Its stalk at zero under ordinary j_* is the invariant subspace: a local section is a vector unchanged by a circuit of the puncture. Equation (SP.17) or (SP.18) proves

$$(j_* \mathcal{V}_{\text{an}})_0 = 0, \quad (j! \mathcal{V}_{\text{an}})_0 = 0. \quad (\text{SP.21})$$

Moreover the circle cochain complex is $[\mathbb{C}^m \xrightarrow{M-I} \mathbb{C}^m]$; cutting the circle at one point and identifying its endpoints by M gives exactly this cellular differential. Since $M - I$ is invertible, both its kernel and cokernel are zero. Thus the derived local invariant and coinvariant groups also vanish. This statement concerns the local system specified above and does not replace the marked algebraic fibre $\mathcal{H}/u\mathcal{H}$ in (SP.5).

129.5 The same full marked fibre retains the arithmetic nilpotent

In the y -basis, let J_m be the matrix of multiplication by y on $\mathbb{C}[y]/y^m$; explicitly $(J_m)_{a+1,a} = 1$ for $0 \leq a < m-1$, with all other entries zero. Then

$$A_s = P^{-1}(\rho I_m + J_m)P, \quad N_{\text{ar}} := A_s - \rho I_m = P^{-1}J_mP, \quad J_m^m = 0, \quad J_m^{m-1} \neq 0. \quad (\text{SP.22})$$

For $m = 1$, the last statement means $J_1^0 = I_1 \neq 0$ and $J_1 = 0$; its nilpotency index is one. These identities follow by applying multiplication to each of the ordered monomials; no monodromy identification is used. They reconstruct the full original companion multiplication matrix, since the polynomial identity $(s - \rho)^m = 0$ determines its last column and (SP.3) its complete change of coordinates.

The relation between this arithmetic action and the boundary connection is also explicit. From $\beta_{a+1} - \beta_a = 1/n$,

$$[B_s, A_s] = \frac{1}{n} N_{\text{ar}}, \quad [\nabla_u, A_s] = \frac{N_{\text{ar}}}{nu}. \quad (\text{SP.23})$$

The first identity is verified on every y^a , including the last one where $J_m y^{m-1} = 0$, and then conjugated by P . The scalar $C_s = cI$ commutes with A_s , and A_s is constant in u , proving the second. Thus for $m > 1$ this marked arithmetic action is not a horizontal endomorphism of the displayed boundary connection. The exact nonhorizontal term has been computed, rather than discarding the action.

It can be transported through the actual period isomorphism. Set $D_0 = \text{diag}(d_a)$, with the constants in (SP.13), and

$$K_0 = W D_0 J_m D_0^{-1} W^{-1}.$$

The scalar exponential cancels by its actual inverse in conjugation, and (SP.14) gives

$$\Pi_s A_s \Pi_s^{-1} = \rho I_m + u^{1/n} K_0, \quad K_0^m = 0, \quad K_0^{m-1} \neq 0, \quad M_{\text{cyc}} K_0 M_{\text{cyc}}^{-1} = \zeta K_0. \quad (\text{SP.24})$$

Indeed the $(a+1, a)$ entry of $\text{diag}(u^\beta) J_m \text{diag}(u^{-\beta})$ is $u^{1/n}$. The same calculation with ζ^{a+1} proves the last identity. Differentiating (SP.24) retains the exact $1/(nu)$ commutator in (SP.23). This is the concrete connection between the nonzero arithmetic nilpotent and the finite boundary monodromy, on the same rank- m space at each $u \neq 0$.

129.6 The complete Taylor unit and the original theta source

The entire quotient and its full Taylor coefficients are

$$v_h(s) = \frac{g(s)}{(s - \rho)^m}, \quad a_j = \frac{v_h^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!} \quad (0 \leq j < m), \quad a_0 \neq 0. \quad (\text{SP.25})$$

The equality follows by dividing the convergent Taylor series of g , whose first nonzero term is of order m . In the unchanged marked fibre its multiplication matrix is

$$U_s = P^{-1} \left(\sum_{j=0}^{m-1} a_j J_m^j \right) P, \quad U_s^{-1} = P^{-1} \left(\sum_{j=0}^{m-1} b_j J_m^j \right) P, \quad b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}. \quad (\text{SP.26})$$

Multiplying the two finite series proves the inverse: the constant coefficient is one and every coefficient of J_m^r , $1 \leq r < m$, is zero by the displayed recursion. Terms of degree at least m vanish by $J_m^m = 0$. All coefficients a_j , including their derivative constants, remain in the same matrix. When $u \neq 0$, the occurrence of this same U_s on the displayed remainder lattice means its constant coefficient extension in that specified basis. It is not an assertion of an unrecorded multiplication operation on twisted polynomial cohomology. The exact multiplication defect is calculated below.

Retain the original function spaces $V, \mathcal{B}$, Euler operator $D = -x\partial_x$, Fourier convention, theta map $\Theta\phi(x) = 2 \sum_{r \geq 1} \phi(rx)$, and seed

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g, \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

Specifically V is the even Schwartz subspace with $\phi(0) = \int \phi = 0$, and $\mathcal{B}$ consists of smooth functions on $\mathbb{R}_{>0}$ with finite seminorms $\sup_{x>0} x^b |D^r F(x)|$ for every integer b and every nonnegative integer r . The original iterated Euler inverse gives $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$. The actual polynomial maps are

$$\alpha_h(P) = P(D)\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \quad \Theta\alpha_h(P) = \mathcal{T}_h(hP), \quad J_h \mathcal{T}_h(P) = U_s[P]_s. \quad (\text{SP.27})$$

Here $J_h = j_h \mathcal{M}$ and $[P]_s$ is the original remainder coordinate column. These are the inherited exact Euler/theta identities in original marked equations (18)–(19) and BC's source square. In particular $[\mathbb{C}[s] \xrightarrow{h} \mathbb{C}[s]ds]$ maps to $[V \xrightarrow{\Theta} \mathcal{B}]$ by $(\alpha_h, \mathcal{T}_h)$. Its degree-one injection is $\sigma_h U_s : E_h \hookrightarrow Q_{\text{ar}} = \mathcal{B}/\Theta V$, where σ_h is the original corrected section.

The coordinate change is realized on these exact source vectors:

$$\mathcal{T}_h(y^b) = (D - \rho)^b F_h, \quad (\mathcal{T}_h(s^b))_{b=0}^{m-1} = (\mathcal{T}_h(y^a))_{a=0}^{m-1} P. \quad (\text{SP.28})$$

The binomial theorem for the commuting scalar ρ and operator D proves both identities. Thus the Pascal map, full nilpotent and Taylor unit have explicit source representatives on the same theta complex. At general u , the original source deformation is still $\Theta + u\mathcal{T}_h\partial_s\alpha_h^{-1}$ on $\alpha_h(\mathbb{C}[s])$; its square with (SP.4) commutes. There is no assertion that the unit-valued coefficient map turns this deformed differential into the undeformed theta map.

Under the period map, (SP.26) has the exact image

$$\Pi_s U_s \Pi_s^{-1} = \sum_{j=0}^{m-1} a_j u^{j/n} K_0^j, \quad [\nabla_u, U_s] = \frac{1}{nu} P^{-1} \left(\sum_{j=1}^{m-1} j a_j J_m^j \right) P. \quad (\text{SP.29})$$

The first follows by taking powers in (SP.24); the second uses $[B_y, J_m^j] = (j/n) J_m^j$, proved by induction from (SP.23). These formulas specify exactly how every Taylor coefficient is transported and why its arithmetic multiplication is not silently identified with a constant flat monodromy operator.

More explicitly put $v_{\text{pol}}(s) = \sum_{j=0}^{m-1} a_j (s - \rho)^j$, the exact original Taylor remainder of v_h . For every polynomial $Q(s)$, the Leibniz rule in the original differential gives

$$v_{\text{pol}} L_u Q = L_u(v_{\text{pol}} Q) - uv'_{\text{pol}} Q. \quad (\text{SP.30})$$

Indeed the $h v_{\text{pol}} Q$ terms agree and the derivative of $v_{\text{pol}} Q$ contributes exactly the additional $uv'_{\text{pol}} Q$. At the marked fibre $u = 0$ this is ordinary multiplication of the full class $v_h \in E_h^\times$, with inverse

(SP.26). Away from that fibre it records precisely the derivative term needed to compare multiplication on representatives with the specified coefficient extension U_s . For a remainder polynomial p of degree less than m , divide $v_{\text{pol}}p = hq + r$ monically. The degree of q is at most $m - 2$, with $q = 0$ when the product has degree less than m . Consequently q' already has degree below m , and

$$[v_{\text{pol}}p]_{\mathcal{H}_u} = [r - uq']_{\mathcal{H}_u}. \quad (\text{SP.31})$$

Subtracting $L_u q = hq + uq'$ proves the identity with its specified correction. The constant matrix U_s gives the coefficient r ; (SP.31) computes exactly its difference from multiplying these representatives and then taking the differential remainder. No term involving v'_h or its Taylor coefficients has been deleted.

129.7 The finite-cover lattice and the original source metric

The marked lattice also has an explicit image on the cover $u = v^n$. After retaining and factoring its common scalar e^{c/v^n} in the period expression, its columns are

$$e^{-c/v^n} \Pi_s(v^n) = WD_0 \text{diag}(v, v^2, \dots, v^m)P. \quad (\text{SP.32})$$

This is an equality on the punctured disc that also defines a holomorphic matrix on the whole v -disc. The constant invertible matrices WD_0, P identify its image lattice with $\bigoplus_{a=0}^{m-1} v^{a+1} \mathbb{C}\{v\}$. Its cokernel in the displayed ambient holomorphic free module is consequently

$$\bigoplus_{a=0}^{m-1} \mathbb{C}\{v\}/(v^{a+1}), \quad \text{length} = \sum_{a=0}^{m-1} (a+1) = \frac{m(m+1)}{2}. \quad (\text{SP.33})$$

Both assertions follow by multiplying by those constant inverses and computing the image and quotient in each one-dimensional coordinate. The image lattice itself has a rank- m quotient by multiplication by v , as does the pulled-back marked lattice. Its inclusion into the ambient constant lattice induces zero modulo v , since every column contains a positive power of v . Thus the vanishing boundary invariants and the nonzero marked fibre have their exact lattice map and cokernel recorded; neither is used to erase the other's coefficients. The scalar exponential factored in (SP.32) remains an irregular factor and is not being declared a meromorphic gauge.

In the original source norm, retain the positive matrix

$$(G_\theta)_{ab} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{(1/2 + it)}^a (1/2 + it)^b |v_h(1/2 + it)|^2 dt, \quad 0 \leq a, b < m. \quad (\text{SP.34})$$

Mellin–Plancherel identifies this with the Gram of $\mathcal{T}_h(s^b)$ in the actual $L^2(dx)$ source. The source moments are finite by the inherited arithmetic source estimates. A zero polynomial-vector norm would make its entire Mellin transform Pv_h zero; since $v_h \neq 0$, this forces $P = 0$. Hence G_θ is positive definite, with its original mass and full multiplier retained. The actual comparison endomorphism is

$$T_{\text{per}}(u) = G_\theta^{-1} \Pi_s(u)^* \Pi_s(u) = e^{2\text{Re}(c/u)} G_\theta^{-1} P^* D(u)^* W^* W D(u) P. \quad (\text{SP.35})$$

Its positivity and self-adjointness are in G_θ : $x^* G_\theta T_{\text{per}} x = \|\Pi_s x\|^2 > 0$ for $x \neq 0$. The ratio of its largest and smallest eigenvalues is unchanged by its explicit positive scalar $e^{2\text{Re}(c/u)}$. Every other factor in (SP.35), including the original Pascal matrix and G_θ , remains. In particular the scalar cancellation alone gives no estimate for that ratio.

Fix $u \neq 0$. For a larger polynomial source $\mathbb{C}[s]_{\leq N}$, $N \geq m - 1$, let $J_{u,N}$ be the unique differential remainder for (SP.4). It is onto the same m -dimensional remainder module and has exact kernel

$$\ker J_{u,N} = L_u(\mathbb{C}[s]_{\leq N-m}), \quad \text{Per}_N = \Pi_s(u) J_{u,N}. \quad (\text{SP.36})$$

When $N = m - 1$ the relation space is zero. The degree argument for (SP.5) proves the kernel and surjectivity. For the period factorization, each polynomial differential relation has integral

$$\int_{\Gamma_j} e^{\Phi_h/u} L_u Q ds = u \int_{\Gamma_j} d(e^{\Phi_h/u} Q) = 0;$$

the endpoint values vanish by the same polynomial decay estimate (SP.12), and the junction endpoints cancel with the original orientations. Thus the period Gram at degree N has rank exactly m , with kernel (SP.36). Its restriction to the displayed m -dimensional remainder frame is positive definite. It is not a positive definite Gram on an enlarged source with more columns.

129.8 Support labels and the scope of the computed boundary

All constant coefficient maps above act at each original nonbottom support label by $(A, x) \mapsto (A, fx)$, with external $\tau \mapsto \tau$. The Pascal matrix, its inverse, the finite Taylor-unit matrix and its inverse preserve the labelled coefficient space. A killed coefficient is its represented zero at that label. The coefficient specialization at $u = 0, t = 0$ sends represented parameter values to the represented zero e , as in the original marked structural map $\mathbb{F}_{1,\tau} \rightarrow G(\mathbb{C}[u, t]) \rightarrow G(\mathbb{C})$. It does not send the marked primary fibre to external absence.

Equations (SP.17)–(SP.21) compute the actual one-factor boundary monodromy, invariant stalk and finite-cover nilpotent logarithm. Equations (SP.22)–(SP.33) reconstruct the original arithmetic action, full Taylor unit, theta representatives and marked lattice through explicit maps. They do not identify N_{boundary} with N_{ar} . These conclusions are restricted to this complete single-primary packet at $t = 0$; tensor products may multiply its characters to the trivial character, and no assertion of vanishing tensor invariants is made.

130 The finite-field sheaf of the single-primary phase

130.1 Specified coefficient specialization and all constants

Let the specified single primary polynomial be

$$h(s) = (s - \rho)^m, \quad m \geq 1, \quad n = m + 1. \quad (\text{SPF.1})$$

Retain the given coefficient ring R , including ρ , $1/n!$, and every coefficient of the full Taylor class $v_h = j_h(g/h)$ and its inverse in the retained m -dimensional remainder basis. The finite-field datum for this calculation is the specified ring homomorphism $\alpha : R \rightarrow k_0 = \mathbf{F}_{Q_0}$, of characteristic $p > n$. All coefficients, including the unit and its inverse, are mapped by this same homomorphism. We continue to write ρ for $\alpha(\rho)$. At $t = 0$, the specified primitive with value zero at $s = 0$ is exactly

$$\Phi(s) = \frac{(s - \rho)^n - (-\rho)^n}{n}, \quad \Phi'(s) = h(s), \quad c = -\frac{(-\rho)^n}{n}. \quad (\text{SPF.2})$$

In particular, the constant c is retained throughout. The only constant-field extension we require is

$$k = \mathbf{F}_Q, \quad Q = Q_0^d, \quad d = \text{ord}_{(\mathbf{Z}/n\mathbf{Z})^\times}(Q_0). \quad (\text{SPF.3})$$

Thus $n \mid Q - 1$. This is an explicit finite extension because $\gcd(Q_0, n) = 1$. All original coefficients are mapped by $R \xrightarrow{\alpha} k_0 \hookrightarrow k$.

Fix $\ell \neq p$, work over $\overline{\mathbf{Q}}_\ell$, and fix a nontrivial additive character $\psi_0 : k_0 \rightarrow \overline{\mathbf{Q}}_\ell^\times$. On k use $\psi = \psi_0 \circ \text{Tr}_{k/k_0}$. For a further extension $k_r = \mathbf{F}_{Q^r}$, write

$$\psi_r = \psi \circ \text{Tr}_{k_r/k}, \quad \chi_r = \chi \circ \text{N}_{k_r/k}. \quad (\text{SPF.4})$$

Every Frobenius below is geometric Frobenius, the inverse of $x \mapsto x^{Q^r}$ at a k_r -rational point. Define $\mathcal{L}_\psi(f)$ by the Artin–Schreier torsor convention whose trace there is $\psi_r(f)$. Define the multiplicative Kummer sheaf $\mathcal{K}_\chi$ by the convention whose trace there is $\chi_r(x)$. These conventions fix both inverse characters and cohomological signs.

Let

$$\begin{aligned} \pi_s : \mathbf{A}_s^1 \times \mathbf{G}_{m,u} &\longrightarrow \mathbf{G}_{m,u}, \\ \mathcal{F}_0 &= R^1 \pi_{s!} \mathcal{L}_{\psi_0}(\Phi(s)/u) \quad \text{over } k_0, \\ \mathcal{F} &= (\mathcal{F}_0)_k = R^1 \pi_{s!} \mathcal{L}_\psi(\Phi(s)/u) \quad \text{over } k. \end{aligned} \quad (\text{SPF.5})$$

The equality is compact-support base change. Our computation concerns this precise sheaf. It does not replace the specified phase by an expression involving g/h , and does not remove that unit from the original coefficient ring.

The source isomorphism

$$T : \mathbf{A}_y^1 \times \mathbf{G}_{m,u} \longrightarrow \mathbf{A}_s^1 \times \mathbf{G}_{m,u}, \quad T(y, u) = (y + \rho, u), \quad T^{-1}(s, u) = (s - \rho, u) \quad (\text{SPF.6})$$

fixes the coefficient ring and satisfies $\pi_s T = \pi_y$. It preserves compact supports because it is an isomorphism. For any original coefficient function $v(s)$, the transported function is exactly $v(y + \rho)$; for an invertible such function the inverse is exactly $v^{-1}(y + \rho)$. Direct substitution into (SPF.2), and the additive torsor tensor-product law, give

$$T^* \mathcal{L}_\psi(\Phi(s)/u) = \mathcal{L}_\psi(y^n/(nu)) \otimes \pi_y^* \mathcal{L}_\psi(c/u). \quad (\text{SPF.7})$$

The tensor-product law follows by adding two torsor coordinates: if $z_i^p - z_i = f_i$, their sum satisfies the equation for $f_1 + f_2$; the associated one-dimensional character representations multiply. Projection formula therefore yields

$$\mathcal{F} = \mathcal{L}_\psi(c/u) \otimes R^1 \pi_{y!} \mathcal{L}_\psi(y^n/(nu)). \quad (\text{SPF.8})$$

130.2 The actual power-map splitting, including its zero stalk

Put $X_n = \{\chi : k^\times \rightarrow \overline{\mathbf{Q}}_\ell^\times : \chi^n = 1\}$, so $\#X_n = n$, and let $q : \mathbf{A}_y^1 \rightarrow \mathbf{A}_z^1$ be $z = y^n$. Let $j : \mathbf{G}_{m,z} \hookrightarrow \mathbf{A}_z^1$. Then the following is an isomorphism of sheaves over k , not merely an equality of trace functions:

$$q_* \overline{\mathbf{Q}}_\ell \simeq \overline{\mathbf{Q}}_{\ell, \mathbf{A}_z^1} \oplus \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} j_! \mathcal{K}_\chi. \quad (\text{SPF.9})$$

Here is the explicit splitting. The deck group on $\mathbf{G}_m$ is $\mu_n(k)$. On the stalk with basis e_r , indexed by $r^n = z$, set $D_\zeta e_r = e_{\zeta r}$. For $\chi \in X_n$ there is a unique character $\lambda_\chi : \mu_n(k) \rightarrow \overline{\mathbf{Q}}_\ell^\times$ with

$$\lambda_\chi(x^{(Q-1)/n}) = \chi(x) \quad (x \in k^\times), \quad e_\chi = \frac{1}{n} \sum_{\zeta \in \mu_n(k)} \lambda_\chi(\zeta) D_\zeta. \quad (\text{SPF.10})$$

Uniqueness and existence follow because $x \mapsto x^{(Q-1)/n}$ surjects onto $\mu_n(k)$, with kernel the subgroup of n th powers, on which χ is trivial. Character orthogonality gives $e_\chi e_{\chi'} = \delta_{\chi, \chi'} e_\chi$ and $\sum_{\chi \in X_n} e_\chi = 1$. Its image is the λ_χ^{-1} -eigenline of the operators D_ζ . For $z = x \in k_r^\times$, a chosen root has $r^{Q^r} = x^{(Q^r-1)/n} r$; geometric Frobenius consequently acts on this eigenline by $\lambda_\chi(x^{(Q^r-1)/n}) = \chi_r(x)$. It is therefore the stated Kummer summand with the stated descent convention.

At $z = 0$, the geometric fibre of q has one point. Its constant-sheaf stalk has trivial deck action. Thus e_χ has zero image there for $\chi \neq 1$, and the trivial projector has a one-dimensional stalk. The unit map from the constant sheaf identifies the latter summand with the constant sheaf on $\mathbf{A}_z^1$. Each other summand is canonically $j_! \mathcal{K}_\chi$: the adjunction map is an isomorphism at every geometric stalk, on $\mathbf{G}_m$ and at zero. Finally, q is finite, so its higher direct images on this sheaf vanish; the argument has neither suppressed the ramified fibre nor added a term at it.

130.3 Compact-support cohomology and the negative Gauss line

For a nonzero scalar a , write

$$\mathcal{E}_{\chi,a} = \mathcal{K}_{\chi}(z) \otimes \mathcal{L}_{\psi}(az) \quad \text{on } \mathbf{G}_{m,z}, \quad H_{\chi} = H_c^1(\mathbf{G}_{m,\bar{k}}, \mathcal{E}_{\chi,1}). \quad (\text{SPF.11})$$

For $\chi \neq 1$ one has

$$H_c^i(\mathbf{G}_{m,\bar{k}}, \mathcal{E}_{\chi,a}) = 0 \ (i \neq 1), \quad \dim H_{\chi} = 1. \quad (\text{SPF.12})$$

We give the complete application of the curve formulas. A lisse sheaf on the connected, nonproper curve $\mathbf{G}_m$ has no nonzero compactly supported section: a nonzero locally constant section has support the whole curve. Thus $H_c^0 = 0$. Poincaré duality identifies H_c^2 with the Tate-twisted dual of the global invariants of the dual sheaf. Those invariants are zero, since the Artin–Schreier factor at infinity has nontrivial wild inertia and the Kummer factor is tame. The Swan conductors at zero and infinity are respectively 0 and 1. The second value also follows from the direct local calculation below, with the local parameter z^{-1} and nonzero numerator a . The Euler–Poincaré formula for this rank-one sheaf gives

$$\chi_c(\mathbf{G}_m, \mathcal{E}_{\chi,a}) = 1 \cdot \chi_c(\mathbf{G}_m, \bar{\mathbf{Q}}_{\ell}) - \text{Swan}_0 - \text{Swan}_{\infty} = 0 - 0 - 1 = -1. \quad (\text{SPF.13})$$

Together with the two vanishing groups and the cohomological dimension bound for a curve, this proves (SPF.12). The formulas used here, including their proofs and geometric Frobenius convention, are in Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Sections 2.1–2.3, especially 2.3.1–2.3.3.

The trivial summand of (SPF.9) instead contributes

$$H_c^i(\mathbf{A}_k^1, \mathcal{L}_{\psi}(az)) = 0 \quad \text{for every } i, \ a \neq 0. \quad (\text{SPF.14})$$

Indeed $H_c^0 = 0$ for the same support reason, $H_c^2 = 0$ by its nontrivial wild inertia at infinity and duality, and Euler–Poincaré gives $1 - 1 = 0$, hence $H_c^1 = 0$. This proves why the final rank is $n - 1$, with no trivial character contribution.

Set

$$G_k(\chi, \psi) = \sum_{z \in k^{\times}} \chi(z) \psi(z). \quad (\text{SPF.15})$$

The trace formula, with (SPF.12), gives the exact one-dimensional geometric Frobenius action

$$\text{Fr}_Q | H_{\chi} = -G_k(\chi, \psi). \quad (\text{SPF.16})$$

The minus sign is the cohomological sign in degree one. There is no Tate twist in this identity.

This computation is an isomorphism of the entire parameter sheaf. On $\mathbf{G}_{m,z} \times \mathbf{G}_{m,a}$, use the source isomorphism $(z, a) \mapsto (w = az, a)$, whose inverse is $(w, a) \mapsto (w/a, a)$. The multiplicative Kummer torsor law gives

$$\mathcal{K}_{\chi}(w/a) \otimes \mathcal{L}_{\psi}(w) \simeq \text{pr}_w^*(\mathcal{K}_{\chi}(w) \otimes \mathcal{L}_{\psi}(w)) \otimes \text{pr}_a^* \mathcal{K}_{\chi^{-1}}(a). \quad (\text{SPF.17})$$

This law follows by multiplying or dividing the root coordinates of Kummer torsors; the associated character representations multiply or invert accordingly. The isomorphism preserves compact supports in every fibre, so projection formula and base change identify the parameter sheaf with $H_{\chi} \otimes \mathcal{K}_{\chi^{-1}}(a)$. Here H_{χ} is the pullback from $\text{Spec } k$ of the one-dimensional Galois module in (SPF.16); it is geometrically constant.

130.4 The exact decomposition on the original parameter

For a nonzero scalar γ , denote by $\mathcal{A}_\gamma$ the geometrically constant one-dimensional sheaf with geometric Fr_Q -eigenvalue γ . In its use below γ is the actual eigenvalue in (SPF.16), or that eigenvalue times a root of unity, so this notation denotes an existing continuous Galois representation. Combining the preceding source maps with $a = 1/(nu)$ gives

$$\mathcal{F} \simeq \mathcal{L}_\psi(c/u) \otimes \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{A}_{-G_k(\chi, \psi)} \otimes \mathcal{K}_{\chi^{-1}}(1/(nu)). \quad (\text{SPF.18})$$

In particular, $\mathcal{F}$ is lisse of rank $n - 1 = m$. The inverse Kummer character here is essential. Since inversion twice returns χ and multiplication by the constant n has trace $\chi(n)$, the same isomorphism on $\mathbf{G}_{m,u}$ is exactly

$$\mathcal{F} \simeq \mathcal{L}_\psi(c/u) \otimes \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{A}_{-\chi(n)G_k(\chi, \psi)} \otimes \mathcal{K}_\chi(u). \quad (\text{SPF.19})$$

Neither the factor n nor the additive factor c/u has been dropped.

For completeness this determines every finite-extension trace. Applying the trace formula to the base change of the one-dimensional group H_χ proves directly

$$-G_{k_r}(\chi_r, \psi_r) = (-G_k(\chi, \psi))^r. \quad (\text{SPF.20})$$

Thus the usual lifting sign is derived from this actual group, without an independent convention for a normalized Gauss sum. For $u \in k_r^\times$,

$$\begin{aligned} \text{Tr}(\text{Fr}_{Q^r, u} | \mathcal{F}_{\bar{u}}) &= - \sum_{s \in k_r} \psi_r(\Phi(s)/u) \\ &= -\psi_r(c/u) \sum_{\substack{\chi \in X_n \\ \chi \neq 1}} \chi_r(nu) G_{k_r}(\chi_r, \psi_r) \\ &= \psi_r(c/u) \sum_{\substack{\chi \in X_n \\ \chi \neq 1}} (-\chi(n)G_k(\chi, \psi))^r \chi_r(u). \end{aligned} \quad (\text{SPF.21})$$

To verify the middle equality also by finite sums, translate s to $y = s - \rho$. The number of n th roots of $z \in k_r^\times$ is $\sum_{\chi \in X_n} \chi_r(z)$. There is one root at zero. The trivial-character part is $1 + \sum_{z \neq 0} \psi_r(az) = 0$. For each other character the change of variable $w = az$ gives $\chi_r(a)^{-1} G_{k_r}(\chi_r, \psi_r)$; here $a^{-1} = nu$. In the last equality, $\chi_r(n) = \chi(n)^r$, since $n \in k^\times$ and its extension-field norm is n^r .

130.5 Inertia at zero, including a zero constant specialization

Let I_0 denote geometric inertia over $\bar{k}((u))$, let P_0 be its wild subgroup, and let $V = \mathcal{F}_{\bar{\eta}}$. Write $\psi(x) = \vartheta(\text{Tr}_{k/\mathbf{F}_p}(bx))$, where $\vartheta : \mathbf{F}_p \rightarrow \bar{\mathbf{Q}}_\ell^\times$ is a fixed nontrivial additive character and $b \in k^\times$. Such a unique b exists by nondegeneracy of the finite-field trace pairing. Then $\mathcal{L}_\psi(c/u) = \mathcal{L}_\vartheta(bc/u)$. Put $d_0 = bc$.

First suppose $c \neq 0$. The Artin–Schreier equation

$$z^p - z = d_0/u \quad (\text{SPF.22})$$

defines a degree- p extension of $\bar{k}((u))$. Indeed $f^p - f = d_0/u$ has no Laurent-series solution: if f has a pole of order $r > 0$, then $f^p - f$ has pole order pr , whereas the right side has order one; if f has no pole, neither does $f^p - f$. An Artin–Schreier polynomial of prime degree with no root is irreducible,

since its roots form one translate of $\mathbf{F}_p$ and its Galois group is a subgroup of that group, hence has order either one or p .

Let v_E be its integer-valued normalized valuation, and write $v_E(u) = e$ for the ramification index. The equation forces $v_E(z) < 0$, hence $p v_E(z) = -e$. Thus $p \mid e$; since $1 \leq e \leq [E : \bar{k}((u))] = p$, one has $e = p$ and $v_E(z) = -1$. The extension is therefore totally ramified and z^{-1} is a uniformizer. For $\sigma_a(z) = z + a$, $a \in \mathbf{F}_p^\times$,

$$v_E(\sigma_a(z^{-1}) - z^{-1}) = v_E\left(\frac{-a}{z(z+a)}\right) = 2. \quad (\text{SPF.23})$$

The lower ramification groups are consequently $G_0 = G_1 = \mathbf{F}_p$, $G_2 = 1$; their unique positive upper break is one. Every nontrivial character of this group has Swan conductor one. In particular the character ε_{d_0} supplied by $\mathcal{L}_{\vartheta}(d_0/u)$ is nontrivial on P_0 . All geometrically constant factors of (SPF.19) are trivial on I_0 , and all Kummer factors are trivial on P_0 . Thus

$$V|_{P_0} \simeq \varepsilon_{d_0}^{\oplus(n-1)}, \quad V^{P_0} = 0, \quad V^{I_0} = 0, \quad \text{Swan}_0(\mathcal{F}) = n - 1 \quad (c \neq 0). \quad (\text{SPF.24})$$

The first vanishing follows explicitly by choosing $\sigma \in P_0$ with $\varepsilon_{d_0}(\sigma) \neq 1$: an invariant vector v would satisfy $(\varepsilon_{d_0}(\sigma) - 1)v = 0$, hence $v = 0$. Swan conductors add under direct sum.

Now suppose $c = 0$. This occurs exactly when $\rho = 0$ in the specified residue field, since $n \neq 0$ there. The additive factor is the constant sheaf, and

$$V|_{I_0} \simeq \bigoplus_{\substack{\chi \in X_n \\ \chi \neq 1}} \mathcal{K}_\chi|_{I_0}, \quad V^{I_0} = 0, \quad \text{Swan}_0(\mathcal{F}) = 0 \quad (c = 0). \quad (\text{SPF.25})$$

Each displayed character is nontrivial on tame inertia: the totally ramified Kummer cover $u = v^n$ has deck group μ_n , and its summand is the nontrivial λ_χ^{-1} -character proved in (SPF.10). An invariant in a direct sum of these eigenlines has invariant components in each, hence is zero. Thus this specialization does not produce an invariant line.

Here is one finite cover that kills all geometric inertia in either case and keeps track of the additive factor:

$$u = v^n, \quad z^p - z = \frac{bc}{v^n}. \quad (\text{SPF.26})$$

Both maps are finite etale on the punctured u -line or its pullback: their derivatives there are $nv^{n-1} \neq 0$ and $-1 \neq 0$. The pullback of $\mathcal{K}_\chi(u)$ is $\mathcal{K}_{\chi^n}(v)$, hence constant. The pullback of the Artin-Schreier factor is trivialized by the displayed torsor itself. For $c = 0$, the first map alone suffices and the second is a disjoint trivial torsor. For $c \neq 0$, the second map remains a nontrivial degree- p extension after the first: its right side has pole order n , and $p \nmid n$, so the preceding pole-order argument applies. The composite has degree np . Constants $\mathcal{A}_{-\chi(n)G_k}$ can retain arithmetic Frobenius action; they have no geometric inertia action.

On the resulting finite-index subgroup of I_0 , every monodromy matrix is the identity. Its unipotent logarithm is therefore exactly

$$N = \log(1) = 0. \quad (\text{SPF.27})$$

Equivalently, the original inertia representation has finite image and is a direct sum of finite-order characters, so its unipotent part is already the identity. Formula (SPF.27) and the vanishing $V^{I_0} = 0$ hold simultaneously. Invariants refer to the original group I_0 ; after the cover its restricted representation is trivial, so the new inertia invariants are the whole rank- m vector space.

130.6 Purity proved by the exact Gauss calculation

For every complex embedding of the cyclotomic field of the character values, and every nontrivial χ ,

$$|G_k(\chi, \psi)|^2 = \sum_{t \in k^\times} \chi(t) \sum_{y \in k^\times} \psi((t-1)y) = (Q-1) - \sum_{\substack{t \in k^\times \\ t \neq 1}} \chi(t) = Q. \quad (\text{SPF.28})$$

The first equality expands the square of the sum, uses the complex conjugates $\overline{\chi(y)} = \chi(y)^{-1}$ and $\overline{\psi(y)} = \psi(-y)$, and sets $x = ty$. The inner sum is $Q - 1$ at $t = 1$ and -1 otherwise, by nontrivial additive-character orthogonality. Multiplicative-character orthogonality gives $\sum_{t \neq 0} \chi(t) = 0$, so the sum over $t \neq 1$ is -1 . This proves every equality, with the original unscaled Gauss sum.

Every local eigenvalue in (SPF.21) is

$$\psi_r(c/u) \left(-\chi(n) G_k(\chi, \psi) \right)^r \chi_r(u), \quad \left| \psi_r(c/u) \left(-\chi(n) G_k(\chi, \psi) \right)^r \chi_r(u) \right| = Q^{r/2}. \quad (\text{SPF.29})$$

All character factors have absolute value one, and (SPF.28) gives the last equality. Thus the specified sheaf is pointwise pure of weight one. No division by $Q^{1/2}$, half Tate twist, or change of the original phase was made. Over k_0 the same purity follows as well: for a closed point of degree e , pass to a constant extension in which its Frobenius has a positive power covered by (SPF.29). If its original eigenvalue is β and that power is r , the computed equality $|\beta^r| = Q_0^{er/2}$ implies $|\beta| = Q_0^{e/2}$. The inertia calculation is geometric and is unchanged by constant-field extension.

This establishes the exact finite-field decomposition, purity, invariant-space vanishing, and finite-cover logarithm of the sheaf in (SPF.5). The finite-field parameter and its Frobenius eigenvalues have been constructed here. No complex off-critical zeta zero or identification of such a zero with one of these finite-field Frobenius eigenvalues is constructed by this calculation.

Primary sources for the cohomological foundations. The compact-support curve formulas and geometric Frobenius convention used above are Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups* (Annals of Mathematics Studies 116), Sections 2.1–2.3; the unscaled Gauss convention is Section 4.0. The Artin–Schreier torsor construction is the exact sequence in the Stacks Project, Section 59.63, tag 0A3J; the Kummer torsor construction is Section 59.28, tag 03PK. All specializations, source maps, power-map projectors, Gauss factors, and local ramification computations needed for this particular sheaf have been proved above.

131 Ordered tensor primary packets: invariant characters and the actual arithmetic action

This calculation takes the ordered tensor product of the complete single-primary marked family, with a common parameter u and with every original t -parameter fixed at zero. It does not pass to a symmetric quotient. Its source is the full packet attached to a selected actual zero when such a packet is given; it asserts the existence of no zero off the critical line. The constant in the original primitive, the original monomial coordinates, and the full source unit are retained.

131.1 The original ordered family and its period matrix

Let ρ be the selected zero of $g = 2\xi$ of complete order $m \geq 1$, put $N = m + 1$, and let $k \geq 1$ be an integer. For $1 \leq i \leq k$, put $y_i = s_i - \rho$. The precise phase and marked algebra are

$$\begin{aligned} h(s) &= (s - \rho)^m, & c &= -\frac{(-\rho)^N}{N}, \\ \Phi_k(s_1, \dots, s_k) &= \sum_{i=1}^k \frac{(s_i - \rho)^N - (-\rho)^N}{N} = \sum_{i=1}^k \frac{y_i^N}{N} + kc, \\ E_k &= \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)). \end{aligned} \quad (\text{TP.1})$$

Each primitive is zero at its original base point $s_i = 0$. Index every ordered basis lexicographically by $\mathbf{b} = (b_1, \dots, b_k) \in \{0, \dots, m-1\}^k$. Write $s^{\mathbf{b}} = \prod_i s_i^{b_i}$, $y^{\mathbf{b}} = \prod_i y_i^{b_i}$, $|\mathbf{b}| = \sum_i b_i$, and $L(\mathbf{b}) = |\mathbf{b}| + k$. The unchanged top differential is $ds_1 \wedge \dots \wedge ds_k = dy_1 \wedge \dots \wedge dy_k$.

The one-factor Pascal matrix and its ordered tensor power are

$$P_{ab} = \begin{cases} \binom{b}{a} \rho^{b-a} & a \leq b, \\ 0 & a > b, \end{cases} \quad (P^{-1})_{ab} = \begin{cases} \binom{b}{a} (-\rho)^{b-a} & a \leq b, \\ 0 & a > b, \end{cases} \quad P_k = P^{\otimes k}. \quad (\text{TP.2})$$

Here all one-factor indices run from zero to $m - 1$. The binomial theorem gives $e_s = e_y P_k$, where e_s, e_y are the ordered rows of monomials. Substitution in both directions proves the inverse formula, and $\det P = 1$. Coefficient columns therefore satisfy $x_y = P_k x_s$, with no change of the polynomial object.

The ordered exponential complex is the tensor product over $\mathbb{C}[u]$ of the two-term complexes with differential

$$L_i = u \partial_{s_i} + h(s_i). \quad (\text{TP.3})$$

Equivalently its total differential on polynomial forms is $u d + d\Phi_k \wedge$, with the usual ordered wedge signs. The operators L_i commute for different indices, proving that the square of this differential is zero. In one variable the leading degree of $L_i Q$, for $Q \neq 0$, is $\deg Q + m$, with the same leading coefficient. Monic subtraction therefore gives a unique remainder of degree below m . This decomposes the two-term complex as a contractible complex mapping isomorphically onto its image, plus a free $\mathbb{C}[u]$ -module of rank m in degree one. To see contractibility explicitly, send an image element $L_i Q$ back to its unique Q , and send its remainder to zero. Tensoring a contracting homotopy with the identity contracts every summand having a contractible factor; the total-complex signs cancel the differentials in the other factors. Consequently all cohomology vanishes except in degree k , where

$$\mathcal{H}_k = \bigoplus_{\mathbf{b}} \mathbb{C}[u][s^{\mathbf{b}} ds_1 \wedge \cdots \wedge ds_k], \quad \mathcal{H}_k / u \mathcal{H}_k = E_k ds_1 \wedge \cdots \wedge ds_k. \quad (\text{TP.4})$$

In particular its rank is m^k , and every primary order is retained.

Fix a branch of $u^{1/N}$ on a sector of $u \neq 0$, let $\zeta = e^{2\pi i/N}$, and for $0 \leq a < m$ define

$$\beta_a = \frac{a+1}{N}, \quad d_a = N^{\beta_a-1} e^{\pi i \beta_a} \Gamma(\beta_a), \quad W_{ja} = \zeta^{j(a+1)} - 1 \quad (1 \leq j \leq m), \quad D(u) = \text{diag}(d_a u^{\beta_a}). \quad (\text{TP.5})$$

Use the original rays at angles $(\arg u + \pi)/N + 2\pi j/N$, oriented outwards, and the original relative contour $\Gamma_j = -\ell_0 + \ell_j$. The ordered product contour is $\Gamma_{j_1} \times \cdots \times \Gamma_{j_k}$, with its product orientation. There is no permutation of its factors or top form. The full original period matrix is

$$\Pi_k(u) = e^{kc/u} W_k D_k(u) P_k, \quad W_k = W^{\otimes k}, \quad D_k(u) = D(u)^{\otimes k}. \quad (\text{TP.6})$$

Here is a direct verification retaining the original contours. In each variable, translate its contour to $\rho + \Gamma_j$. Cauchy's theorem on the finite quadrilateral swept by each ray applies to the entire integrand. The junction connectors cancel between the negative and positive rays. At their remote endpoints $s = R e^{i\vartheta} + a\rho$, $0 \leq a \leq 1$, the exponent has real part $-R^N/(N|u|) + O(R^{N-1}) + O(1)$, uniformly in a . Each connector has length at most $|\rho|$; its polynomially weighted integral tends to zero. Absolute integrability on each ray follows from the same estimate. Translation is thus a proved contour equality, preserving the scalar $e^{c/u}$. On the translated ray the substitution $v = r^N/(N|u|)$ gives precisely $d_a u^{\beta_a} \zeta^{j(a+1)}$; subtracting the zeroth ray gives the entry of $W D(u)$. The Pascal expansion gives the factor P . Absolute convergence permits Fubini on every product contour and proves (TP.6), including its orientation and all constants.

The matrix W is invertible: a relation among its columns gives a polynomial $\sum_{r=1}^m a_r (z^r - 1)$ of degree at most $m = N - 1$, vanishing at every N -th root of unity, including one. A nonzero polynomial cannot have more roots than its degree, so each $a_r = 0$. Every $d_a \neq 0$, since $0 < \beta_a < 1$. Thus (TP.6) is invertible for every $u \neq 0$. For reference its determinant, without discarding its scalar, is

$$\det \Pi_k(u) = K_m^{km^{k-1}} u^{km^k/2} e^{km^k c/u}, \quad K_m = N^{-m/2} e^{\pi i m/2} \left(\prod_{a=0}^{m-1} \Gamma(\beta_a) \right) \det W. \quad (\text{TP.7})$$

Indeed $\sum_a \beta_a = m/2$, and $\det(X \otimes Y) = (\det X)^{\dim Y} (\det Y)^{\dim X}$. The latter identity follows first on triangular matrices by multiplying their diagonal entries and then on all complex matrices by triangularizing them separately. Applying it repeatedly proves the displayed exponent km^{k-1} , also when $m = 1$.

131.2 Full monodromy and the exact character projectors

Let B_s have column b equal to $((b+1)s^b - b\rho s^{b-1})/N$ in the original basis, and let $B_k = \sum_i I^{\otimes(i-1)} \otimes B_s \otimes I^{\otimes(k-i)}$. The exact connection on (TP.4) is

$$\nabla_u = \partial_u - \frac{kc}{u^2}I + \frac{B_k}{u}, \quad P_k B_k P_k^{-1} = \text{diag}(L(\mathbf{b})/N). \quad (\text{TP.8})$$

For one factor, divide $\Phi(s)s^b$ monically by $h(s)$: the quotient is $(s - \rho)s^b/N$ and the remainder is cs^b . Subtracting its L_u -image gives $[\Phi s^b] = [cs^b - u((s - \rho)s^b/N)']$. This is exactly $\partial_u - \Phi/u^2$ on cohomology and gives the stated column. The binomial coordinate map makes this column diagonal with entries β_a . Summing the tensor-factor connections proves (TP.8).

On one positive circuit of $u = 0$, each relative contour changes by $\Gamma_j \mapsto \Gamma_{j+1} - \Gamma_1$ for $j < m$, and $\Gamma_m \mapsto -\Gamma_1$. This follows directly by advancing each ray index by one modulo N . Applying this to the columns of W gives eigenvalue ζ^{a+1} . Consequently the full period-cycle monodromy is

$$\Pi_k^{\text{cont}} = M_k \Pi_k, \quad M_k = W_k \text{diag}(\zeta^{L(\mathbf{b})}) W_k^{-1}, \quad M_k^N = I. \quad (\text{TP.9})$$

The scalar $e^{kc/u}$ is single-valued and has not been removed. The horizontal coefficient fundamental matrix is $P_k^{-1} e^{-kc/u} \text{diag}(u^{-L(\mathbf{b})/N})$, as follows by differentiation using (TP.8). Its monodromy has the inverse eigenvalues. Both conventions have the same invariant indices, namely $N \mid L(\mathbf{b})$.

For a residue $r \in \mathbb{Z}/N\mathbb{Z}$, define the diagonal mask E_r on the y -basis by $E_r y^{\mathbf{b}} = \mathbf{1}_{L(\mathbf{b}) \equiv r} y^{\mathbf{b}}$. The period-space and marked-space projectors are, respectively,

$$\begin{aligned} \mathcal{P}_r &= \frac{1}{N} \sum_{j=0}^{N-1} \zeta^{-rj} M_k^j = W_k E_r W_k^{-1}, \\ Q_r &= P_k^{-1} E_r P_k, \quad \mathcal{P}_r \Pi_k = \Pi_k Q_r. \end{aligned} \quad (\text{TP.10})$$

The geometric sum of N -th roots is N when its exponent is divisible by N and zero otherwise; this proves the first equality and the identities $Q_r Q_t = \mathbf{1}_{r=t} Q_r$, $\sum_r Q_r = I$, and the same identities for $\mathcal{P}_r$. The diagonal matrix $D_k(u)$ commutes with every E_r , proving the last equality. In particular the pulled-back projector is constant in u . These are algebraic projectors; no orthogonality in the original theta-source metric is assumed.

Their dimensions are exactly

$$d_r = \text{rank } Q_r = \begin{cases} (m^k + m(-1)^k)/N, & r = 0, \\ (m^k - (-1)^k)/N, & r \neq 0. \end{cases} \quad (\text{TP.11})$$

In fact the character sum giving the number of ordered tuples is $N^{-1} \sum_{j=0}^{N-1} \zeta^{-rj} (\sum_{l=1}^{N-1} \zeta^{jl})^k$. Its inner sum is m at $j = 0$ and -1 otherwise. The sum of the remaining outer characters is m for $r = 0$ and -1 for $r \neq 0$. This proves (TP.11) as an actual integer count, including all small values of m, k .

The ordinary invariant stalk of this punctured-disc local system is $\ker(M_k - I) = \text{im } \mathcal{P}_0$, of dimension d_0 . Its coinvariant space is identified with that same summand by $\mathcal{P}_0$: on every other summand $M_k - I$ is invertible. Thus both circle cohomology groups have dimension d_0 . On the cover $u = v^N$, (TP.8) becomes

$$\partial_v - \frac{Nkc}{v^{N+1}}I + \frac{1}{v} \text{diag}(L(\mathbf{b})). \quad (\text{TP.12})$$

The basis change by $\text{diag}(v^{-L(\mathbf{b})})$ removes precisely the integer residue. The remaining exact horizontal scalar is e^{-kc/v^N} . It is not a meromorphic gauge removing the irregular term when $kc \neq 0$. Nevertheless all displayed solutions are single-valued on this cover, so its analytic monodromy is the identity and its unipotent logarithm is zero. The exact diagonal solution leaves no additional sector transition to determine for this tensor of the one-critical-value family. The invariant and coinvariant statements above refer to the ordinary local system; they do not impose an unmentioned growth condition at the puncture.

131.3 The arithmetic action moves the invariant character

Let J be the matrix of multiplication by y on $\mathbb{C}[y]/y^m$: $J_{a+1,a} = 1$, all other entries zero. Put $J_i = I^{\otimes(i-1)} \otimes J \otimes I^{\otimes(k-i)}$, $J_{\text{tot}} = \sum_i J_i$, and

$$R_k = P_k^{-1} J_{\text{tot}} P_k, \quad A_k = M_{s_1+\dots+s_k} = k\rho I + R_k. \quad (\text{TP.13})$$

This is multiplication in the original algebra (TP.1), since $s_i = \rho + y_i$. No geometric nilpotent replaces R_k . Put $K = k(m-1)$. The commuting nilpotents have the exact multinomial powers

$$J_{\text{tot}}^r y^{\mathbf{b}} = \sum_{\substack{\alpha \in \mathbb{Z}_{\geq 0}^k, |\alpha|=r \\ b_i + \alpha_i \leq m-1 \ (1 \leq i \leq k)}} \frac{r!}{\prod_i \alpha_i!} y^{\mathbf{b}+\alpha}, \quad R_k^{K+1} = 0, \quad R_k^K \neq 0. \quad (\text{TP.14})$$

Expansion of the commuting operators proves the sum. A term with an exponent at least m vanishes in (TP.1), giving its exact index restrictions. At $r = K$, the input one has nonzero top coefficient $K!/((m-1)!)^k$; at larger r no tuple survives. This also proves the conventions when $K = 0$: $R_k^0 = I \neq 0$ and $R_k = 0$.

Define $D_0 = \text{diag}(d_a)$ and $K_0 = W D_0 J D_0^{-1} W^{-1}$, and let K_{tot} be the sum of its ordered factor placements. The actual period transport of the arithmetic action is

$$\Pi_k A_k \Pi_k^{-1} = k\rho I + u^{1/N} K_{\text{tot}}, \quad M_k K_{\text{tot}} M_k^{-1} = \zeta K_{\text{tot}}. \quad (\text{TP.15})$$

Conjugating J by $\text{diag}(u^{\beta_a})$ multiplies each nonzero entry by $u^{\beta_{a+1}-\beta_a} = u^{1/N}$; the tensor placements give the first identity. Replacing these powers by ζ^{a+1} gives the second. The scalar $e^{kc/u}$ cancels by its displayed inverse in this conjugation. Equivalently the exact connection commutator is

$$[B_k, A_k] = R_k/N, \quad [\nabla_u, A_k] = R_k/(Nu). \quad (\text{TP.16})$$

These follow on each monomial from the difference of its two residue eigenvalues, including each truncated top term, and then from (TP.8). Thus the arithmetic action is explicitly related to the boundary connection, with its precise nonhorizontal term.

Multiplication by a surviving y_i increases L by one. For all residues and integers $r \geq 0$, this gives

$$\begin{aligned} Q_t R_k^r &= R_k^r Q_{t-r}, & [Q_0, A_k] &= R_k(Q_{-1} - Q_0), \\ Q_0 A_k Q_0 &= k\rho Q_0, & (I - Q_0) A_k Q_0 &= R_k Q_0 = Q_1 R_k Q_0. \end{aligned} \quad (\text{TP.17})$$

All subscripts are taken modulo N . The first identity holds on every basis monomial, including zero output terms. The other identities follow from it and from $N \geq 2$. They calculate both the compression and the outgoing map. In particular a scalar compression is not an assertion that the invariant plane is preserved.

For a precise nonvanishing statement put

$$d_{\min} = N \lceil k/N \rceil - k \in \{0, \dots, N-1\}. \quad (\text{TP.18})$$

The possible degrees $|\mathbf{b}|$ fill every integer from zero to K : distribute any such integer successively into the k slots, never exceeding their capacity $m-1$. Thus $Q_0 \neq 0$ exactly when $d_{\min} \leq K$. Moreover

$R_k Q_0 \neq 0$ exactly when $d_{\min} < K$. If strict inequality holds, a tuple of that degree has an unfilled slot, and (TP.14) with $r = 1$ gives nonzero output with no cancellation, since its distinct monomials have positive integer coefficients. If equality holds, the only possible invariant tuple is the top one and every J_i kills it. This proves both directions.

For $k = 1$ we have $d_{\min} = m > K = m - 1$, so there are no invariants. For $m = 1$, $R_k = 0$ and the unique tensor character is invariant exactly when k is even. For every $m \geq 2$, $k \geq 2$, instead,

$$Q_0 \neq 0, \quad R_k Q_0 \neq 0. \quad (\text{TP.19})$$

Indeed at $k = 2$, $d_{\min} = m - 1 < 2m - 2 = K$. For $k \geq 3$, $K = k(m - 1) \geq m + 1 = N$, while $d_{\min} \leq N - 1$. This proves (TP.19), including $m = 2$, $k = 3$, the smallest equality case for the capacity bound.

131.4 The complete compressed dilation, including every return power

Work on $V_0 = \text{im } Q_0$. If this space is zero, all the following compressed maps are the unique zero maps. Suppose it is nonzero and put

$$q_* = \left\lfloor \frac{K - d_{\min}}{N} \right\rfloor = \lfloor km/N \rfloor - \lceil k/N \rceil, \quad T = Q_0 R_k^N Q_0|_{V_0}. \quad (\text{TP.20})$$

Since R_k^N commutes with Q_0 , $T^q = Q_0 R_k^{qN} Q_0|_{V_0}$. For every $0 \leq q \leq q_*$ this map is nonzero; for $q > q_*$ it is zero. To prove nonvanishing, choose an invariant tuple of minimum degree. Its remaining capacities sum to $K - d_{\min}$, so an increment tuple of size qN can be distributed within them. Its term in (TP.14) is nonzero and cannot cancel. For vanishing, every input has degree at least $d_{\min}$, so all output terms exceed capacity when $qN > K - d_{\min}$. Thus the nilpotency index of T is exactly $q_* + 1$, with the identity zeroth power when $q_* = 0$.

For $a > 0$, define the original ordered dilation on functions by simultaneous replacement $x_i \mapsto x_i/a$. Its generator is $\sum_i D_i$, with $D_i = -x_i \partial_{x_i}$, and hence its action on this marked packet is a^{A_k} . Write $z = \log a$. The complete compression is the finite expression

$$Q_0 a^{A_k} Q_0|_{V_0} = a^{k\rho} \sum_{q=0}^{q_*} \frac{z^{qN}}{(qN)!} T^q. \quad (\text{TP.21})$$

Indeed $a^{A_k} = a^{k\rho} \sum_{r=0}^K z^r R_k^r / r!$ by (TP.14). The first identity in (TP.17) kills its compression unless $N \mid r$, and (TP.20) determines exactly which surviving powers are nonzero. Equivalently, the sum without its scalar is the restriction of $N^{-1} \sum_{j=0}^{N-1} \exp(\zeta^j z R_k) Q_0$; the same root-of-unity sum proves this equality. In the original y -monomial coordinates its exact entry, from invariant $\mathbf{b}$ to invariant $\mathbf{d}$, is

$$\begin{cases} a^{k\rho} \frac{z^{|\mathbf{d}|-|\mathbf{b}|}}{\prod_i (d_i - b_i)!}, & d_i \geq b_i \text{ for all } i, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{TP.22})$$

The two invariant conditions make the exponent divisible by N . Formula (TP.14) cancels its total factorial against that of the exponential and proves (TP.22). Conjugation by the explicit columns of P_k^{-1} returns this same matrix to the original s -basis.

In particular $k = 2$ always has $q_* = 0$, so (TP.21) is scalar even though (TP.19) proves nonzero outgoing arithmetic action when $m \geq 2$. Conversely nonconstant returns do occur: for $m = 2$, $N = 3$, $k = 3$, V_0 is spanned in the y -basis by 1 and $y_1 y_2 y_3$, and $R_k^3(1) = 6y_1 y_2 y_3$. The compression is exactly $a^{3\rho}(I + z^3 T/6)$, with that nonzero T . When $q_* \geq 1$, the compressed maps are not a one-parameter representation: after removing the scalar, their derivative at $z = 0$ is zero but their coefficient of z^N is the nonzero $T/N!$. A differentiable matrix-valued function F with $F(0) = I$ and

$F(z + w) = F(z)F(w)$ satisfies $F'(z) = F(z)F'(0)$ by differentiating in w ; zero derivative at zero would make it constant. This proves the asserted failure without identifying compression with a quotient action.

131.5 The full Taylor unit and its exact original-source projector

Retain the entire quotient $v_h = g/h$ and all coefficients of its marked Taylor class:

$$a_j = \frac{v_h^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!} \quad (0 \leq j < m), \quad a_0 \neq 0, \quad U_s = P^{-1} \left(\sum_{j=0}^{m-1} a_j J^j \right) P, \quad U_k = U_s^{\otimes k}. \quad (\text{TP.23})$$

The Taylor expansion at the complete-order zero proves the derivative equality. The inverse coefficients are $b_0 = a_0^{-1}$ and $b_r = -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}$; multiplying the finite series modulo J^m proves the inverse. Thus (TP.23) is invertible and keeps every coefficient. It commutes with A_k , since all its factors are polynomials in the commuting multiplication matrices.

For a multi-index $\mathbf{j} \in \{0, \dots, m-1\}^k$, write $a_{\mathbf{j}} = \prod_i a_{j_i}$ and let $R_i = P_k^{-1} J_i P_k$, so $R_k = \sum_i R_i$. The full graded unit, its period transport and its connection defect are

$$\begin{aligned} U_k &= \sum_{\mathbf{j}} a_{\mathbf{j}} \prod_i R_i^{j_i}, \\ \Pi_k U_k \Pi_k^{-1} &= \sum_{\mathbf{j}} a_{\mathbf{j}} u^{|\mathbf{j}|/N} \prod_i K_i^{j_i}, \\ [\nabla_u, U_k] &= \frac{1}{Nu} \sum_{\mathbf{j}} |\mathbf{j}| a_{\mathbf{j}} \prod_i R_i^{j_i}. \end{aligned} \quad (\text{TP.24})$$

Here K_i is the corresponding placement of K_0 . The first equality expands the tensor product, the second uses (TP.15) factor by factor, and the third follows from the residue difference $|\mathbf{j}|/N$ on each surviving monomial. Zero terms at the truncation boundary satisfy the identities as well.

The actual source maps can now be specified. On the original one-variable spaces retain $D = -x\partial_x$, $\Theta\phi(x) = 2 \sum_{r \geq 1} \phi(rx)$, $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, and the original F_h with Mellin transform v_h . Here the Mellin convention is $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$. The source V is the even Schwartz subspace with $\phi(0) = \int \phi = 0$; the target $\mathcal{B}$ has all seminorms $\sup_{x>0} x^b |D^r F(x)|$ finite for every integer b and nonnegative integer r . The inherited original source identity is $\mathcal{M}\Theta\phi_* = g$, and the exact Euler inverse constructs this $F_h \in \mathcal{B}$. Define the actual finite-packet maps on multivariate polynomials by

$$\begin{aligned} \mathcal{T}_k(f) &= f(D_1, \dots, D_k) \prod_{i=1}^k F_h(x_i), \\ \alpha_k(f) &= f(D_1, \dots, D_k) \prod_{i=1}^k \phi_*(x_i), \\ \mathcal{J}_k \mathcal{T}_k(f) &= U_k[f]_s. \end{aligned} \quad (\text{TP.25})$$

Here $\mathcal{J}_k$ means the multivariable Mellin transform followed by reduction modulo all $h(s_i)$, and $[f]_s$ is the unchanged remainder column. Every displayed source lies in the original smooth product source; it is an explicit finite sum of tensor functions, not an assertion that the whole completed source equals its algebraic tensor product. Integration by parts gives $\mathcal{M}(D_i F) = s_i \mathcal{M}F$; all endpoint terms vanish by the stated seminorms. Fubini then gives $\mathcal{M}_k \mathcal{T}_k(f) = f(s) \prod_i v_h(s_i)$. Its full jet is exactly the last equality of (TP.25). The factorwise identities $\Theta\alpha_h(f) = \mathcal{T}_h(hf)$ give the ordered tensor source square with every usual total-complex sign. The corrected one-factor section σ_h therefore gives the

actual degree- k injection $\sigma_h^{\otimes k} U_k$ on this finite tensor packet. This retains the original full unit; it is not set equal to the identity.

In particular, if x_s is a marked coefficient column and $q = U_k x_s$ is its actual theta-jet coordinate, the period observation of that source is $\Pi_k U_k^{-1} q$. Its exact invariant projector is

$$\widehat{Q}_0 = U_k Q_0 U_k^{-1}, \quad \mathcal{P}_0 \Pi_k U_k^{-1} = \Pi_k U_k^{-1} \widehat{Q}_0, \quad [\widehat{Q}_0, A_k] = U_k [Q_0, A_k] U_k^{-1}. \quad (\text{TP.26})$$

The first two formulas follow by substituting $q = U_k x_s$ into (TP.10). The last uses the proved commutation of U_k with A_k . Thus the actual invariant source plane is $U_k \text{im } Q_0$, with the same exact arithmetic leakage conjugated through its original unit. On the cohomology image the corresponding projector is $\sigma_h^{\otimes k} \widehat{Q}_0 (\sigma_h^{\otimes k})^{-1}$, where the inverse is only on that injective image. Its maps are fully specified.

For comparison, when $V_0 \neq 0$, compressing the coefficient matrix U_k by Q_0 is another concrete operation. Its full expression on V_0 is

$$Q_0 U_k Q_0|_{V_0} = a_0^k I_{V_0} + Z, \quad Z = \sum_{\substack{\mathbf{j} \neq 0 \\ N || \mathbf{j}}} a_{\mathbf{j}} Q_0 \prod_i R_i^{j_i} Q_0 \Big|_{V_0}, \quad Z^{q_*+1} = 0. \quad (\text{TP.27})$$

Every omitted term changes the character and so is killed by (TP.10); every term of Z raises degree by at least N . The degree bound defining q_* proves its stated vanishing power, without requiring any a_j other than a_0 to be nonzero. Its exact inverse is $a_0^{-k} \sum_{r=0}^{q_*} (-a_0^{-k} Z)^r$. Multiplying this finite geometric series proves the claim. This inverse is not substituted for the full U_k^{-1} in (TP.26).

Finally the difference between marked coefficient multiplication and multiplication of twisted representatives is also explicit. Put $v_{\text{pol}}(s) = \sum_{j=0}^{m-1} a_j (s - \rho)^j$ and $V_{\text{pol}} = \prod_i v_{\text{pol}}(s_i)$. For every polynomial f , the actual differential obeys

$$V_{\text{pol}} L_i f = L_i (V_{\text{pol}} f) - u(\partial_{s_i} V_{\text{pol}}) f. \quad (\text{TP.28})$$

This is the Leibniz rule, with the full derivative term. To compute its complete remainder, divide in one variable $v_{\text{pol}} s^b = h q_b + r_b$, with $\deg r_b < m$. Let H_s have column b the original coordinate column of q'_b . Since $\deg q_b \leq m - 2$, q'_b is already a remainder. Subtracting $L_u q_b = h q_b + u q'_b$ gives $[v_{\text{pol}} s^b] = [r_b - u q'_b]$. For explicit coefficients in the y -basis,

$$(H_y)_{ab} = \begin{cases} (a+1)a_{a+m+1-b}, & 0 \leq a+m+1-b < m, \\ 0, & \text{otherwise,} \end{cases} \quad H_s = P^{-1} H_y P, \quad (\text{TP.29})$$

$$\text{Rem}_u(V_{\text{pol}} \cdot) = (U_s - u H_s)^{\otimes k}.$$

on the retained original remainder frame. Indeed the quotient of $v_{\text{pol}} y^b$ has terms $a_j y^{j+b-m}$ exactly when $j+b \geq m$; differentiating gives the stated matrix entry. The tensor formula follows by replacing the factors successively; their variables are distinct and their top-form order stays fixed. This proves the exact full representative correction on the same frame. It does not assert that multiplication descends on arbitrary twisted-cohomology representatives.

131.6 The original terminal class on its exact invariant subsequence

The complete tensor primary algebra contains its original terminal jet

$$\tau_k = \prod_{i=1}^k y_i^{m-1}, \quad R_k \tau_k = 0, \quad A_k \tau_k = k \rho \tau_k, \quad U_k \tau_k = a_0^k \tau_k. \quad (\text{TP.30})$$

Its displayed coefficient in the retained basis is one, so it is nonzero. Each multiplication by y_i kills it, proving the next two statements by (TP.13). Every positive Taylor term in (TP.23) also kills it, proving the last statement with its full scalar. Since $L(m-1, \dots, m-1) = km \equiv -k \pmod{N}$,

$$Q_0 \tau_k = \widehat{Q}_0 \tau_k = \begin{cases} \tau_k, & N \mid k, \\ 0, & N \nmid k. \end{cases} \quad (\text{TP.31})$$

The equality for Q_0 is its diagonal-mask definition, and that for $\widehat{Q}_0$ follows from $U_k^{-1} \tau_k = a_0^{-k} \tau_k$ in (TP.26). Thus on the infinite subsequence $k = N\ell, \ell \geq 1$, the same nonzero arithmetic terminal class lies in the actual boundary-invariant plane, in both marked and original jet coordinates.

Its complete period vector is

$$\Pi_k \tau_k = e^{kc/u} d_{m-1}^k u^{km/N} (W_{\bullet, m-1})^{\otimes k}, \quad d_{m-1} = N^{-1/N} e^{\pi i m/N} \Gamma(m/N), \quad W_{j, m-1} = \zeta^{-j} - 1. \quad (\text{TP.32})$$

This is column $(m-1, \dots, m-1)$ of (TP.6) in its specified y -coordinates. Every entry $W_{j, m-1}$ is nonzero, so the vector remains nonzero for $u \neq 0$. Its period-cycle monodromy is multiplication by $\zeta^{km} = \zeta^{-k}$, agreeing with (TP.31). For the original corrected-section class $u_\rho^{\otimes k} = \sigma_h^{\otimes k} \tau_k$, the period observation defined in (TP.26) is precisely $a_0^{-k} \Pi_k \tau_k$, not the vector with this scalar omitted.

For the comparison with the original MAI/PAM counterfactual branch, retain its selected $\rho = 1/2 + \delta + i\gamma$, with $0 < \delta < 1/2$ and $\gamma > 0$, reached by its original reflection and conjugation arrows. Let h_0 be its original reflected quartet packet containing the same order- m zero, put $w = h_0/h$, and retain its original CRT idempotent e_ρ . The map $E_h \rightarrow e_\rho E_{h_0}$, $[f] \mapsto e_\rho[f]$, is an isomorphism with inverse reduction modulo h : the idempotent is one on this primary factor and zero on all the other factors. This proves both composites directly on every CRT component. Moreover $w(\rho) \neq 0$, and the exact unit relation on this factor is $(g/h_0) = v_h/w$; the inverse of the class of w is its unique truncated Taylor inverse. Thus this local passage retains the whole original unit, not only its constant coefficient.

Write $K = k(m-1)$ as before. The original amplified terminal coefficient is

$$c_{h_0, k} = \frac{K!}{((m-1)!)^k} ((g/h_0)(\rho))^k \neq 0, \quad \eta_k v_k = c_{h_0, k} \tau_k, \quad \mathcal{I}_k v_k = c_{h_0, k} u_\rho^{\otimes k}. \quad (\text{TP.33})$$

Here $v_k, \eta_k, \mathcal{I}_k$ are the original cyclic terminal vector, coefficient map and cohomology map of MAI6a. The selected quartet corner $k\rho$ arises only by choosing ρ in every tensor factor, as in that construction. The first equality for that map is also its direct local calculation: (TP.14) at $r = K$ leaves precisely $K!/((m-1)!)^k \prod_i y_i^{m-1}$; the original complete unit acts on that terminal monomial by $((g/h_0)(\rho))^k$. All other Taylor terms vanish because they exceed a retained primary order. The original corrected-section transition $\sigma_{h_0} \iota_{h, h_0} = \sigma_h$ carries this identity to the second cohomology equality. Thus the original source period observation of this specified amplified vector is exactly

$$c_{h_0, k} a_0^{-k} \Pi_k \tau_k = \frac{K!}{((m-1)!)^k} \left(\frac{(g/h_0)(\rho)}{(g/h)(\rho)} \right)^k e^{kc/u} d_{m-1}^k u^{km/N} (W_{\bullet, m-1})^{\otimes k}. \quad (\text{TP.34})$$

Every factor is retained and nonzero over $\mathbb{C}$. The ratio of unit values is $w(\rho)^{-1}$, as follows from the full unit relation above. No rescaling of the original vector was used to get (TP.34).

For $a > 0$, (TP.30) gives $a^{A_k} \tau_k = a^{k\rho} \tau_k$; the original source action has the same equality after applying its established theta-boundary homotopy and passing to cohomology. If $\delta = \text{Re } \rho - 1/2$, its exact discrepancy is

$$\log \left(\frac{|a^{k\rho}|^2}{a^k} \right) = 2k\delta \log a. \quad (\text{TP.35})$$

This follows by taking the real part of $k\rho \log a$. On $k = N\ell$, it holds on the same nonzero original terminal class inside the boundary-invariant plane. The invariant projection therefore does not eliminate that discrepancy. This calculation supplies no bound on it by a constant independent of k .

131.7 The exact rank of arithmetic departure from the invariant plane

Let $E_{k,d}$ be the span of $y^{\mathbf{b}}$ with $|\mathbf{b}| = d$, and define $h_d = \dim E_{k,d}$, with $h_d = 0$ outside $0 \leq d \leq K$. The exact coefficient count is

$$\sum_{d=0}^K h_d t^d = (1 + t + \cdots + t^{m-1})^k. \quad (\text{TP.36})$$

Expanding the product proves it by the ordered monomial basis. To compute the rank of J_{tot} on these levels, define on one factor $Fy^b = b(m-b)y^{b-1}$, with $F1 = 0$, and use the positive diagonal form with $\|y^b\|^2 = b!(m-1)!/(m-1-b)!$. The ratio of consecutive diagonal entries is $(b+1)(m-1-b)$; hence the adjoint of J is exactly F . This form is used only to prove a basis-independent algebraic rank. It is not substituted for the original theta-source norm.

Use the tensor product of these forms and put $F_{\text{tot}} = \sum_i F_i$. Direct evaluation on each one-factor monomial gives $[F, J]y^b = (m-1-2b)y^b$. Operators in different factors commute, so on degree d this proves

$$\|J_{\text{tot}}x\|^2 - \|F_{\text{tot}}x\|^2 = (K-2d)\|x\|^2 \quad (x \in E_{k,d}). \quad (\text{TP.37})$$

If $d < K/2$, the identity makes J_{tot} injective on $E_{k,d}$. If $d+1 > K/2$, its reversed identity makes F_{tot} injective on $E_{k,d+1}$, so its adjoint $J_{\text{tot}} : E_{k,d} \rightarrow E_{k,d+1}$ is surjective: $(\text{im } J_{\text{tot}})^\perp = \ker(F_{\text{tot}}|_{E_{k,d+1}}) = 0$. Every integer d satisfies one or both of these inequalities, including the central degree. It follows that

$$\text{rank}(J_{\text{tot}} : E_{k,d} \rightarrow E_{k,d+1}) = \min(h_d, h_{d+1}). \quad (\text{TP.38})$$

No unimodality assumption is used; the necessary dimension inequalities follow from the proved injectivity or surjectivity.

The invariant degrees have $d+k \equiv 0 \pmod{N}$. Their outgoing images have distinct degrees, so the direct-sum decomposition and (TP.38) give the full departure rank and its exact kernel:

$$\begin{aligned} \text{rank}(R_k Q_0) &= \sum_{\substack{0 \leq d \leq K \\ N|d+k}} \min(h_d, h_{d+1}), \\ \dim(\ker R_k \cap \text{im } Q_0) &= \sum_{\substack{0 \leq d \leq K \\ N|d+k}} \max(h_d - h_{d+1}, 0). \end{aligned} \quad (\text{TP.39})$$

Conjugation by P_k , then by the original U_k in (TP.26), preserves these ranks and kernels on the actual marked and original-source invariant planes.

For example, when $k=2, m \geq 2$, the sole invariant degree is $m-1$, and the exact kernel is the line

$$\mathbb{C}\omega_m, \quad \omega_m = \sum_{i=0}^{m-1} (-1)^i y_1^i y_2^{m-1-i}, \quad (y_1 + y_2)\omega_m = 0 \quad \text{in } E_2. \quad (\text{TP.40})$$

The interior coefficients cancel pairwise, and the two end terms have exponent m , so vanish. For a general input $\sum_i c_i y_1^i y_2^{m-1-i}$, the output coefficients are $c_{j-1} + c_j, 1 \leq j < m$. Its kernel therefore has exactly the stated alternating coefficients and dimension one, and its rank is $m-1$. This is a nonzero arithmetic eigenclass of eigenvalue 2ρ already lying in the boundary-invariant plane. It coexists with the nonzero departure on the remaining $m-1$ directions and with the scalar compressed dilation of (TP.21).

Every constant coefficient map above acts at each original nonbottom split-support label by $(A, x) \mapsto (A, fx)$, and fixes the external absence symbol τ . Its killed coefficient remains the represented zero at that label. The marked specialization $u=0$ is the original represented-zero specialization; none of these projector computations replaces it by external absence. Equations (TP.10), (TP.17), (TP.21) and (TP.26) specify the exact maps relating boundary invariants, the full arithmetic action, its dilation and the original source. The dimension of that invariant space alone supplies no conclusion about the location of the selected zero.

132 Ordered tensors of the actual single-primary finite-field sheaf

132.1 The original coefficient map and the common parameter

Retain the specified coefficient map $R \rightarrow k_0 = \mathbf{F}_{Q_0}$ of characteristic p , the full Taylor class $v_h = j_h(g/h)$ and its inverse in the retained m -dimensional remainder basis, and every other original coefficient. Write

$$N = m + 1 \geq 2, \quad p > N, \quad h(s) = (s - \rho)^m, \quad \Phi(s) = \frac{(s - \rho)^N - (-\rho)^N}{N}, \quad c = -\frac{(-\rho)^N}{N}. \quad (\text{TPF.1})$$

The same coefficient map supplies ρ, c, N^{-1} . In particular, no coefficient of an infinite analytic function is being adjoined in place of its specified full Taylor class. Set

$$d = \text{ord}_{(\mathbf{Z}/N\mathbf{Z})^\times}(Q_0), \quad L = \mathbf{F}_Q, \quad Q = Q_0^d, \quad N \mid Q - 1. \quad (\text{TPF.2})$$

This is defined since $p > N$ implies $\gcd(Q_0, N) = 1$. Fix $\ell \neq p$, coefficients $\overline{\mathbf{Q}}_\ell$, and a nontrivial additive character $\psi_0 : k_0 \rightarrow \overline{\mathbf{Q}}_\ell^\times$. Put $\psi = \psi_0 \circ \text{Tr}_{L/k_0}$. Every Frobenius below is geometric Frobenius. Kummer sheaves $\mathcal{K}_\chi$ have trace χ on $L^\times$; over finite extensions their characters extend by norm. Artin–Schreier characters extend by trace.

Let $\pi : \mathbf{A}_s^1 \times \mathbf{G}_{m,u} \rightarrow \mathbf{G}_{m,u}$ be projection and retain

$$\mathcal{F}_0 = R^1 \pi_! \mathcal{L}_{\psi_0}(\Phi(s)/u), \quad \mathcal{T}_{0,k} = \mathcal{F}_0^{\otimes k}, \quad k \geq 1. \quad (\text{TPF.3})$$

The positive integer k has no restriction relative to p or N . The tensor is ordered and taken on this one common u -line. No symmetric, alternating, or sum-coordinate quotient is taken. Write $\mathcal{F} = (\mathcal{F}_0)_L$ and $\mathcal{T}_k = (\mathcal{T}_{0,k})_L$. The complete single-primary calculation SPF gives

$$\mathcal{F} \simeq \mathcal{L}_\psi(c/u) \otimes \bigoplus_{\substack{\chi \in X_N \\ \chi \neq 1}} \mathcal{A}_{a_\chi} \otimes \mathcal{K}_\chi(u), \quad X_N = \{\chi : L^\times \rightarrow \overline{\mathbf{Q}}_\ell^\times : \chi^N = 1\}, \quad (\text{TPF.4})$$

where all constants are exactly

$$G_L(\chi, \psi) = \sum_{z \in L^\times} \chi(z) \psi(z), \quad a_\chi = -\chi(N) G_L(\chi, \psi). \quad (\text{TPF.5})$$

Here $\mathcal{A}_a$ is the geometrically constant line with geometric Fr_Q -eigenvalue a . SPF proves this isomorphism using the finite power-map idempotents, the support-preserving translation $s = y + \rho$, and $w = z/(Nu)$. Its minus sign comes from compact cohomology in degree one. In particular, all $R^i \pi_!$ vanish for $i \neq 1$. We use that proved single-primary result without changing its source or phase.

The relative compact-support Kunneth map consequently gives the precise source realization

$$\mathcal{T}_{0,k} \simeq R^k \Pi_! \mathcal{L}_{\psi_0} \left(\frac{\sum_{i=1}^k \Phi(s_i)}{u} \right), \quad \Pi : \mathbf{A}_{s_1, \dots, s_k}^k \times \mathbf{G}_{m,u} \longrightarrow \mathbf{G}_{m,u}. \quad (\text{TPF.6})$$

Indeed the derived relative Kunneth isomorphism tensors the k complexes $R\pi_! \mathcal{L}_{\psi_0}(\Phi/u)$, each concentrated in degree one; their tensor is concentrated in degree k . The external product of the additive sheaves is the sheaf of the sum of their functions, by addition of Artin–Schreier torsor coordinates. No factors are reordered, so no permutation sign is inserted. The source theorem is SGA 4, Expose XVII, 5.4.3 and its finite-family version 5.4.4.1: apply it at the finite coefficient levels of these lisse ℓ -adic sheaves and then pass to the ℓ -adic system and invert ℓ . Here every projection is compactifiable by $\mathbf{P}^1 \times \mathbf{G}_m \rightarrow \mathbf{G}_m$; the product compactification is $(\mathbf{P}^1)^k \times \mathbf{G}_m$. The coordinate isomorphism on this same source is

$$(y_1, \dots, y_k, u) \longmapsto (y_1 + \rho, \dots, y_k + \rho, u), \quad \frac{\sum_i \Phi(s_i)}{u} = \frac{\sum_i y_i^N}{Nu} + \frac{kc}{u}. \quad (\text{TPF.7})$$

Its inverse subtracts ρ in every coordinate, and it preserves compact supports. Each original coefficient function is pulled back by this literal substitution, and its inverse is pulled back by the same map.

132.2 Every ordered character tuple and its exact constant

Define

$$\begin{aligned} \Omega_k &= (X_N \setminus \{1\})^k, & r_k &= (N-1)^k, \\ \eta_{\chi} &= \prod_{i=1}^k \chi_i, & A_{\chi} &= \prod_{i=1}^k (-\chi_i(N) G_L(\chi_i, \psi)), & w_k &= kc \in k_0. \end{aligned} \quad (\text{TPF.8})$$

Thus k is reduced in characteristic p only when it multiplies the field element c . Its integer value still counts all k tensor factors and remains in every cohomological sign. Distributing the ordered tensor in (TPF.4), and using the additive and multiplicative torsor tensor laws, proves

$$\boxed{\mathcal{T}_k \simeq \mathcal{L}_{\psi}(w_k/u) \otimes \bigoplus_{\chi \in \Omega_k} \mathcal{A}_{A_{\chi}} \otimes \mathcal{K}_{\eta_{\chi}}(u).} \quad (\text{TPF.9})$$

Different ordered tuples remain different summands, even when their product characters or constants agree. This sheaf is lisse of rank r_k .

For $L_a = \mathbf{F}_{Q^a}$, put $\psi_a = \psi \circ \text{Tr}_{L_a/L}$ and $\chi_{i,a} = \chi_i \circ N_{L_a/L}$. At $u \in L_a^{\times}$, the eigenvalue on the tuple summand is exactly

$$\psi_a(w_k/u) A_{\chi}^a \prod_i \chi_{i,a}(u). \quad (\text{TPF.10})$$

SPF proves $-G_{L_a}(\chi_{i,a}, \psi_a) = (-G_L(\chi_i, \psi))^a$ by its one-dimensional compact cohomology. Also $\chi_{i,a}(N) = \chi_i(N)^a$. These two equalities prove the stated extension-field constant, without replacing a negative Gauss sum by the negative of its a th power. The trace identity on the actual ordered source is consequently

$$\text{Tr}(\text{Fr}_{Q^a, u} | \mathcal{T}_k) = (-1)^k \sum_{(s_1, \dots, s_k) \in L_a^k} \psi_a \left(\frac{\sum_i \Phi(s_i)}{u} \right). \quad (\text{TPF.11})$$

To check its sign directly, the original sum factors into k one-variable sums, each the negative of the trace on $\mathcal{F}$. Multiplying gives precisely $(-1)^k$.

132.3 Original inertia, its tame projector, and multiplicities

Let I_0 be geometric inertia over $\bar{k}_0((u))$, and put $V_k = (\mathcal{T}_{0,k})_{\bar{\eta}}$. Choose $\vartheta : \mathbf{F}_p \rightarrow \bar{\mathbf{Q}}_{\ell}^{\times}$ nontrivial and the unique $b \in k_0^{\times}$ satisfying

$$\psi_0(x) = \vartheta(\text{Tr}_{k_0/\mathbf{F}_p}(bx)). \quad (\text{TPF.12})$$

Nondegeneracy of the finite-field trace pairing supplies this unique b ; after extension to L the same b represents ψ .

First let $c \neq 0$. Keep the original cover for $\mathcal{F}$, before taking a tensor:

$$u = v^N, \quad z^p - z = \frac{bc}{v^N}. \quad (\text{TPF.13})$$

The Kummer degree is N . Over $\bar{k}_0((v))$ the Artin–Schreier polynomial has no root, since its right side has pole order N , prime to p , while a pole of $f^p - f$ has order divisible by p . Its degree is therefore p . The total extension has degree Np , and the Np displayed automorphisms

$$(\zeta, a) : (v, z) \mapsto (\zeta v, z + a), \quad \zeta \in \mu_N, a \in \mathbf{F}_p \quad (\text{TPF.14})$$

commute and exhaust its Galois group $\Gamma = \mu_N \times \mathbf{F}_p$. This is a quotient of the original geometric inertia: the Kummer part is totally ramified, and if e is the ramification index of the Artin–Schreier extension over $\bar{k}_0((v))$, its defining equation gives $p v_E(z) = -Ne$. Since $\gcd(p, N) = 1$ and $1 \leq e \leq p$, it follows that $e = p$.

For $\chi \in X_N$, let $\lambda_\chi : \mu_N(L) \rightarrow \overline{\mathbf{Q}}_\ell^\times$ be uniquely specified by $\lambda_\chi(x^{(Q-1)/N}) = \chi(x)$. Under the geometric Frobenius convention fixed above, the exact action on the tuple line in (TPF.9) is

$$\rho_k(\zeta, a)|_\chi = \lambda_{\eta_\chi}(\zeta)^{-1} \vartheta(-ka). \quad (\text{TPF.15})$$

The Kummer inverse is the power-map projector convention of SPF: with $D_\zeta e_r = e_{\zeta r}$, its χ -projector is $N^{-1} \sum_\zeta \lambda_\chi(\zeta) D_\zeta$, whose image has eigencharacter λ_χ^{-1} . For the Artin-Schreier torsor $z^p - z = f$, the projector $p^{-1} \sum_a \vartheta(a) D_a$ has eigencharacter ϑ^{-1} . At a finite-field point geometric Frobenius translates the torsor coordinate by $-\text{Tr}(f)$, giving trace $\vartheta(\text{Tr}(f))$. Thus its action is $\vartheta(-a)$, and k ordered tensor factors give $\vartheta(-ka)$, proving every sign in (TPF.15).

When $c = 0$, the original additive torsor is trivial. The geometric inertia action instead factors through the Kummer quotient μ_N , with the exact action $\lambda_{\eta_\chi}^{-1}$; there is no nontrivial wild factor.

Let T_ζ denote the action of $(\zeta, 0)$ in (TPF.15), or its Kummer action when $c = 0$. Its exact tame projector is

$$P_{0,k}^{\text{tame}} = \frac{1}{N} \sum_{\zeta \in \mu_N} T_\zeta, \quad P_{0,k}^{\text{tame}}|_\chi = \begin{cases} 1, & \eta_\chi = 1, \\ 0, & \eta_\chi \neq 1. \end{cases} \quad (\text{TPF.16})$$

Character orthogonality proves the second formula and idempotence. In the case $c \neq 0$, the true original inertia projector is

$$\begin{aligned} P_{I_0,k} &= \frac{1}{Np} \sum_{\zeta \in \mu_N} \sum_{a \in \mathbf{F}_p} \rho_k(\zeta, a) \\ &= P_{0,k}^{\text{tame}} \left(\frac{1}{p} \sum_{a \in \mathbf{F}_p} \vartheta(-ka) \right) = \begin{cases} P_{0,k}^{\text{tame}}, & p \mid k, \\ 0, & p \nmid k. \end{cases} \end{aligned} \quad (\text{TPF.17})$$

Division by Np occurs in the characteristic-zero coefficient field, regardless of the prime ℓ . For $c = 0$, the true projector is $P_{I_0,k} = P_{0,k}^{\text{tame}}$. This calculates it inside the original representation; the tame projector alone is not substituted for full inertia when its wild character is nontrivial.

Here are all tame multiplicities. Identify the cyclic group X_N with the additive group $\mathbf{Z}/N\mathbf{Z}$ using a chosen generator. A tuple is an ordered sequence $(j_1, \dots, j_k)$, each $1 \leq j_i < N$. For $t \in \mathbf{Z}/N\mathbf{Z}$, let $D_{k,t}$ count those with $\sum_i j_i = t$. Choose any primitive complex N th root ω . The elementary root-of-unity filter yields

$$\begin{aligned} D_{k,t} &= \frac{1}{N} \sum_{a=0}^{N-1} \omega^{-at} \left(\sum_{j=1}^{N-1} \omega^{aj} \right)^k \\ &= \begin{cases} \frac{(N-1)^k + (N-1)(-1)^k}{N}, & t = 0, \\ \frac{(N-1)^k - (-1)^k}{N}, & t \neq 0. \end{cases} \end{aligned} \quad (\text{TPF.18})$$

The filter follows by summing $N^{-1} \sum_a \omega^{a(\sum j_i - t)}$, which is one when the congruence holds and zero otherwise. At $a = 0$ the inner sum is $N - 1$; for every other a it is -1 . Finally $\sum_{a=1}^{N-1} \omega^{-at}$ is $N - 1$ for $t = 0$ and -1 otherwise. This proves the count, including all repeated tuple occurrences.

Put $d_{0,k} = D_{k,0}$. Since a field has no nonzero nilpotents and $N \neq 0$, $c = 0$ holds exactly when $\rho = 0$ after the given specialization. Consequently

$$w_k = 0 \iff (c = 0 \text{ or } p \mid k), \quad \boxed{\dim V_k^{I_0} = \begin{cases} d_{0,k} = \frac{(N-1)^k + (N-1)(-1)^k}{N}, & w_k = 0, \\ 0, & w_k \neq 0. \end{cases}} \quad (\text{TPF.19})$$

This allows $d_{0,k} = 0$: for $k = 1$ it is zero, and for $N = 2$ it is zero at odd k , one at even k . For $N \geq 3$ and $k \geq 2$, both formulas in (TPF.18) are strictly positive: at even k , $(N - 1)^k > 1$; at odd $k \geq 3$, $(N - 1)^k > N - 1$.

The action in (TPF.15) specifies the entire original inertia representation, including its kernel. For clarity its image order can also be read off. Define

$$h_k = \begin{cases} 1, & N = 2 \text{ and } k \text{ even,} \\ N, & \text{otherwise.} \end{cases} \quad (\text{TPF.20})$$

The tame image has order h_k . Indeed for $N = 2$ there is only one character, whose k th power has that order. For $N \geq 3$, $k = 1$ the list includes a primitive character; for $N \geq 3$, $k \geq 2$ every product character occurs by (TPF.18). In either case the tame kernel is trivial. When $w_k \neq 0$, the wild image has order p , and no nontrivial tame scalar can cancel a nontrivial wild scalar: their orders divide the coprime integers N and p . Thus the full image has order $h_k p$ in that case, and order h_k when $w_k = 0$.

132.4 Swan conductor and the finite-cover logarithm

If $w_k \neq 0$, the nontrivial character $\mathcal{L}_\psi(w_k/u)$ has Swan conductor one. One may verify it directly over $\bar{k}_0((u))$: in $z^p - z = bw_k/u$, the ramification index is p by the valuation argument above, $v_E(z) = -1$, and

$$v_E((z + a)^{-1} - z^{-1}) = 2 \quad (a \in \mathbf{F}_p^\times). \quad (\text{TPF.21})$$

Thus its lower ramification groups are $G_0 = G_1 = \mathbf{F}_p$, $G_2 = 1$, giving one positive break of value one. Tensoring by a tame character does not change this wild restriction. Every tuple line in (TPF.9) has this same wild character. If $w_k = 0$, every tuple line is tame. Additivity of Swan under direct sum therefore proves

$$\boxed{\text{Swan}_0(\mathcal{T}_k) = \begin{cases} 0, & w_k = 0, \\ (N - 1)^k, & w_k \neq 0. \end{cases}} \quad (\text{TPF.22})$$

The finite cover

$$u = v^N, \quad z_k^p - z_k = \frac{bk c}{v^N} \quad (\text{TPF.23})$$

trivializes all geometric inertia of $\mathcal{T}_k$. The first map kills every product Kummer character. The second trivializes the displayed additive factor by its own torsor. If $w_k = 0$, it is a trivial disjoint torsor and the first map alone suffices. If $w_k \neq 0$, its degree is p , as its pole order N is prime to p . When $c \neq 0$ and $p \nmid k$, it is explicitly the original cover (TPF.13) with $z_k = kz$; this equality follows from $k^p = k$ in $\mathbf{F}_p$. When $p \mid k$, the original cover still kills the representation, and its wild deck subgroup already acts trivially by (TPF.15).

On the subgroup defined by this finite cover every monodromy matrix is the identity. Hence its unipotent logarithm is exactly

$$N_{\text{mon},k} = 0. \quad (\text{TPF.24})$$

There is no conflict with any of the invariant dimensions in (TPF.19). Those concern the original inertia group; after the displayed cover, the restricted representation has the full r_k -dimensional invariant space.

132.5 Frobenius on the invariant space and exact descent

Let $j : \mathbf{G}_m \hookrightarrow \mathbf{A}_u^1$. The ordinary invariant stalk $(j_* \mathcal{T}_{0,k})_{\bar{0}} = V_k^{I_0}$: sections over the punctured strict henselian neighborhood are precisely the invariant vectors of its geometric generic stalk. If $w_k \neq 0$ this stalk is zero. For $w_k = 0$, set

$$\Omega_k^0 = \{\chi \in \Omega_k : \eta_\chi = 1\}. \quad (\text{TPF.25})$$

Over L , the exact Frobenius module is

$$V_k^{I_0} \simeq \bigoplus_{\chi \in \Omega_k^0} \overline{\mathbf{Q}}_\ell(A_\chi), \quad \det(1 - T \operatorname{Fr}_Q | V_k^{I_0}) = \prod_{\chi \in \Omega_k^0} (1 - A_\chi T). \quad (\text{TPF.26})$$

Each surviving summand is the constant sheaf $\mathcal{A}_{A_\chi}$: its product Kummer character and its additive factor are both trivial. This proves the formula for the boundary stalk, as well as for its constant extension to the u -line.

Over k_0 , Frobenius need not preserve the individual tuple lines. Let F_0 denote its geometric Frobenius action. Its exact permutation is

$$F_0 : \chi \mapsto \chi^{Q_0} = (\chi_1^{Q_0}, \dots, \chi_k^{Q_0}). \quad (\text{TPF.27})$$

To prove its direction, over the splitting field the deck operator satisfies $F_0 D_\zeta F_0^{-1} = D_{\zeta^{Q_0^{-1}}}$; the exponent denotes the inverse modulo N . Conjugating the exact projector gives

$$F_0 \left(\frac{1}{N} \sum_{\zeta} \lambda_\chi(\zeta) D_\zeta \right) F_0^{-1} = \frac{1}{N} \sum_{\eta} \lambda_\chi(\eta^{Q_0}) D_\eta = e_{\chi^{Q_0}}. \quad (\text{TPF.28})$$

Tensoring this identity in the retained order proves (TPF.27). The coefficients of the projectors are in the fixed ℓ -adic coefficient field; the conjugation changes their deck operators, not their coefficient values. The subset Ω_k^0 is stable because taking a Q_0 th power preserves the trivial product character.

For each orbit $\mathcal{O}$ in Ω_k^0 , choose a tuple χ in it and let its exact orbit length be e . Then $e \mid d$. Put $L_e = \mathbf{F}_{Q_0^e} \subset L$. Each χ_i uniquely descends by norm to a nontrivial character $\kappa_{i,e} : L_e^\times \rightarrow \overline{\mathbf{Q}}_\ell^\times$:

$$\chi_i = \kappa_{i,e} \circ \mathbf{N}_{L/L_e}, \quad \psi_e = \psi_0 \circ \operatorname{Tr}_{L_e/k_0}. \quad (\text{TPF.29})$$

Indeed $\chi_i^{Q_0^e} = \chi_i$, so its order divides $Q_0^e - 1$. If g generates $L^\times$, the kernel of this norm is generated by $g^{Q_0^e - 1}$, on which χ_i is trivial. The norm is surjective, proving existence, uniqueness, and nontriviality of $\kappa_{i,e}$. Also $\prod_i \kappa_{i,e} = 1$, again by surjectivity of the norm.

It is not necessary that all N th roots of unity belong to L_e . Here is the exact source map giving each individual descent. Put $r_i = \operatorname{ord}(\chi_i)$. Then $r_i \mid N$ and $r_i \mid Q_0^e - 1$, so $\mu_{r_i} \subset L_e$. Factor the original power map by

$$\mathbf{A}_y^1 \longrightarrow \mathbf{A}_x^1 \longrightarrow \mathbf{A}_z^1, \quad x = y^{N/r_i}, \quad z = x^{r_i} = y^N.$$

Pullback of functions on geometric fibre roots identifies the direct image for $x^{r_i} = z$ with the μ_{N/r_i} -invariants of the original $y^N = z$ direct image. For $z \neq 0$, the quotient of its root set by μ_{N/r_i} is exactly the root set of $x^{r_i} = z$, and invariant functions are exactly the functions constant on these orbits. At zero both geometric fibres have one point and the map is the identity. The sheaf map is therefore an isomorphism onto those invariants at every geometric stalk.

The character λ_{χ_i} factors through $\mu_N \rightarrow \mu_{r_i}$, $\zeta \mapsto \zeta^{N/r_i}$, as $\lambda_{\kappa_{i,e}}$. Indeed for $a \in L^\times$,

$$\lambda_{\kappa_{i,e}}(a^{(Q-1)/r_i}) = \kappa_{i,e}(a^{(Q-1)/(Q_0^e-1)}) = \chi_i(a) = \lambda_{\chi_i}(a^{(Q-1)/N}),$$

and $a \mapsto a^{(Q-1)/N}$ surjects onto μ_N . Thus the descended individual source summand is precisely the trace- $\kappa_{i,e}$ Kummer line, including its zero stalk. The phase on that source remains $z/(Nu) + c/u$, with the original denominator N , not r_i . Applying the same source isomorphism and trace formula as in SPF over L_e therefore computes the cycle scalar exactly:

$$\Lambda_{\mathcal{O}} = \prod_{i=1}^k (-\kappa_{i,e}(N) G_{L_e}(\kappa_{i,e}, \psi_e)). \quad (\text{TPF.30})$$

Every individual factor, including N and the minus sign, is retained. More explicitly F_0^e acts by this product on the selected tuple line: at $u = 1$ all its Kummer values are one, while the product of its additive values is $\psi_e(kc) = 1$, since $w_k = 0$. The tuple product is a constant sheaf, so this is its action at the invariant boundary stalk too. This proves the scalar over its actual orbit field; it has not been selected as an arbitrary root of an eigenvalue over L .

Choose $v_0 \neq 0$ in the tuple line and set $v_j = F_0^j v_0$ for $0 \leq j < e$. These vectors are in distinct direct-summand lines, so form a basis for the orbit block. In this basis $F_0 v_j = v_{j+1}$ for $j < e - 1$, and $F_0 v_{e-1} = \Lambda_{\mathcal{O}} v_0$. Consequently

$$\begin{aligned} \det(T - F_0 \mid V_k^{I_0}) &= \prod_{\mathcal{O} \subset \Omega_k^0} (T^{|\mathcal{O}|} - \Lambda_{\mathcal{O}}), \\ \det(1 - TF_0 \mid V_k^{I_0}) &= \prod_{\mathcal{O} \subset \Omega_k^0} (1 - \Lambda_{\mathcal{O}} T^{|\mathcal{O}|}). \end{aligned} \tag{TPF.31}$$

These expressions concern $w_k = 0$; when $w_k \neq 0$ both determinants on the zero space are one. Choosing another representative does not change $\Lambda_{\mathcal{O}}$, because F_0^e commutes with F_0 and hence has the same scalar on every line of that orbit. Formula (TPF.31) thus proves descent without asserting diagonal Frobenius in the original tuple coordinates over k_0 .

132.6 Exact weight and the two cancellation cases

The elementary unscaled Gauss calculation in SPF gives, for every complex embedding,

$$|G_L(\chi, \psi)| = Q^{1/2}, \quad |A_{\chi}| = Q^{k/2}. \tag{TPF.32}$$

In (TPF.10) all other factors are roots of unity. At a point of degree a over L , every eigenvalue therefore has absolute value $Q^{ak/2}$. This proves pointwise purity of weight k for the actual ordered tensor. It descends to k_0 : a positive power of any original Frobenius is covered by the calculation over L , and $|\beta^a| = Q_0^{fak/2}$ forces $|\beta| = Q_0^{fk/2}$ at an original point of degree f .

When $w_k = 0$, the same calculation over L_e in (TPF.30) gives

$$|\Lambda_{\mathcal{O}}| = Q_0^{ek/2}, \quad |\beta| = Q_0^{k/2} \quad \text{for every root } \beta^e = \Lambda_{\mathcal{O}}. \tag{TPF.33}$$

Thus every nonzero invariant block has weight k on $\text{Spec } k_0$, including the blocks that occur because $p \mid k$. If $c = 0$, the original single-factor wild character was already trivial. If $c \neq 0$ and $p \mid k$, it becomes trivial by the exact tensor character $\vartheta(-ka)$. Both branches have the same tame multiplicity formula, but the displayed maps account for their different origins. No complex arithmetic zeta zero or comparison of such a zero with these finite-field Frobenius eigenvalues is constructed by this calculation.

132.7 Exact counts of the arithmetic orbit blocks

For $f \mid d$, retain the full greatest common divisor and define

$$g_f = \gcd(N, Q_0^f - 1), \quad D(f) = \frac{(g_f - 1)^k + (g_f - 1)(-1)^k}{g_f}. \tag{TPF.34}$$

The characters fixed by exponent Q_0^f form a cyclic subgroup of X_N of order g_f . A fixed ordered tuple has all entries in this subgroup. Applying the root-of-unity count (TPF.18) inside that subgroup shows that exactly $D(f)$ tuples in Ω_k^0 are fixed. When $g_f = 1$ there are no allowed nontrivial entries

and the formula is zero, since $k \geq 1$. If $C(e)$ denotes the number of orbits in Ω_k^0 of exact length $e \mid d$, then

$$D(f) = \sum_{e \mid f} e C(e), \quad C(e) = \frac{1}{e} \sum_{f \mid e} \mu(e/f) D(f). \quad (\text{TPF.35})$$

The first identity counts the elements of every orbit fixed by F_0^f . For the second, multiply that identity by $\mu(e/f)$ and sum over $f \mid e$. The coefficient of $r C(r)$ is $\sum_{r \mid f \mid e} \mu(e/f)$, which is one for $r = e$ and otherwise zero: the latter sum factors as $\prod_{q \mid e/r} (1-1)$ over its distinct prime divisors. This proves the orbit count by integer arithmetic. These orbit numbers count the nonzero boundary blocks exactly when $w_k = 0$; when $w_k \neq 0$ the boundary stalk is zero by (TPF.19).

132.8 The terminal character, its orbit, and the original scalar

Fix a generator of the abstract cyclic deck group of order N , and identify its chosen complex N th root of unity and its finite-field root in $\mu_N(L)$ with a chosen N th root in the ℓ -adic coefficient field. These are explicit identifications of finite cyclic groups. Let $\chi_* \in X_N$ be the primitive character whose λ_{χ_*} is the resulting tautological character.

The direction of the deck action matters. For $\delta_\zeta(y) = \zeta y$, direct pullback gives

$$\delta_\zeta^*(y^b dy) = \zeta^{b+1} y^b dy, \quad \delta_\zeta^* e_r = e_{\zeta^{-1}r}, \quad D_\zeta = (\delta_\zeta^{-1})^*. \quad (\text{TPF.36})$$

The middle equality follows by evaluating the indicator function of the point r at ζy . Thus the finite SPF summand labelled χ has D_ζ -eigenvalue $\lambda_\chi(\zeta)^{-1}$ but ordinary deck-pullback eigenvalue $\lambda_\chi(\zeta)$. Comparing the pullback characters with the pullback characters, rather than with the inverse action D_ζ , gives the exact character-label map

$$b \mapsto \chi_b = \chi_*^{b+1}, \quad \chi_{\text{ter}} = \chi_*^m = \chi_*^{-1}, \quad \text{ord}(\chi_{\text{ter}}) = N. \quad (\text{TPF.37})$$

Here the terminal coefficient is y^{m-1} , and its form is $y^{m-1} dy$, so $b = m - 1 = N - 2$. The ordered terminal form has deck-pullback character $\zeta^{km} = \zeta^{-k}$, precisely as the product χ_{ter}^k records. This proves a map of the finite deck-character index sets. It does not assert a de Rham-étale comparison isomorphism or identify a complex arithmetic operator with finite-field Frobenius.

The repeated terminal tuple is $\chi_{\text{ter}} = (\chi_{\text{ter}}, \dots, \chi_{\text{ter}})$. Its actual finite-field summand on the same u -line is exactly

$$\mathcal{L}_\psi(kc/u) \otimes \mathcal{A}_{[-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k} \otimes \mathcal{K}_{\chi_{\text{ter}}^k}(u). \quad (\text{TPF.38})$$

Its original-inertia invariants have dimension

$$\begin{cases} 1, & N \mid k \text{ and } kc = 0, & c = 0 : \text{ the nonzero case is } N \mid k, \\ 0, & \text{otherwise.} & c \neq 0 : \text{ the nonzero case is } pN \mid k. \end{cases} \quad (\text{TPF.39})$$

Indeed its tame product is trivial exactly when $N \mid k$, since χ_{ter} is primitive. Its wild character is trivial exactly when $kc = 0$. In the second branch $p \mid k$ and $N \mid k$ are equivalent to $pN \mid k$, since $\gcd(p, N) = 1$.

This single terminal line need not descend to k_0 . Its exact arithmetic orbit consists of the d distinct ordered tuples

$$\left(\chi_{\text{ter}}^{Q_0^j}, \dots, \chi_{\text{ter}}^{Q_0^j} \right), \quad 0 \leq j < d.$$

Its length is exactly d : equality of two tuple entries requires equality of their primitive characters, hence $Q_0^e = 1 \bmod N$, whose least positive solution is d . If the criterion in (TPF.39) holds, all these lines are invariant and the descended terminal-orbit block has dimension d , with

$$\begin{aligned} \Lambda_{\text{ter}} &= [-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k, \\ \det(T - F_0 \mid \text{terminal orbit}) &= T^d - \Lambda_{\text{ter}}, \\ \det(1 - TF_0 \mid \text{terminal orbit}) &= 1 - \Lambda_{\text{ter}} T^d. \end{aligned} \quad (\text{TPF.40})$$

This is (TPF.30)–(TPF.31) with its actual orbit field L . Every root has absolute value $Q_0^{k/2}$. If the criterion fails, the invariant part of this orbit is zero. No Frobenius eigenline over k_0 is inferred from a single summand over L .

Finally retain the original scalar multiplying the terminal coefficient in the full marked packet h_0 , evaluated on a retained stage where its original source coefficients belong to the domain of the actual map α :

$$K = k(m-1), \quad u_0 = (g/h_0)(\rho), \quad C_{h_0,k} = \frac{K!}{((m-1)!)^k} u_0^k. \quad (\text{TPF.41})$$

Here K is an integer, and u_0 is the specified marked coefficient in that actual coefficient model. This calculation does not extend α to a larger cyclic CRT stage by assigning values to new denominators. If such a stage inverts p , no extension of α to it exists: applying a purported extension to $1 = p p^{-1}$ would give $1 = 0$. The absence of a stage specialization is not a vanishing coefficient specialization. Writing a bar for that same specialization gives the exact image

$$\overline{C}_{h_0,k} = \overline{K!} (\overline{(m-1)!})^{-k} \overline{u_0^k}, \quad \overline{C}_{h_0,k} \neq 0 \iff K < p \text{ and } \overline{u_0} \neq 0. \quad (\text{TPF.42})$$

The denominator is nonzero because $m-1 = N-2 < p$. The numerator $K!$ is nonzero modulo p exactly when $K < p$, and a positive power of a field element is nonzero exactly when that element is nonzero. These facts prove the equivalence. When the original unit class and its inverse are retained at this marked point, their product is one after evaluation at ρ and after specialization; hence $\overline{u_0} \neq 0$ in that specified model. Even there the factorial can vanish.

In particular, when $m \geq 2$ and $p \mid k$, one has $K = k(m-1) \geq p$, so $\overline{C}_{h_0,k} = 0$. Thus the $c \neq 0$ terminal-invariant degrees $pN \mid k$, on a marked stage admitting the specified α , have zero specialized original scalar for $m \geq 2$, although their sheaf summand and invariant orbit block computed above are nonzero. For $m = 1$, $K = 0$ and the scalar is exactly u_0^k , so only its original coefficient specialization can make it zero. For $c = 0$, the tame condition $N \mid k$ and the separate scalar condition $k(m-1) < p$, $\overline{u_0} \neq 0$ are both retained.

The last scalar calculation has a precise coefficient-module target. The original translation and coefficient specialization give

$$\begin{aligned} R[y_1, \dots, y_k]/(y_1^m, \dots, y_k^m) &\longrightarrow \overline{E}_k = k_0[y_1, \dots, y_k]/(y_1^m, \dots, y_k^m), \\ \sum_a q_a \prod_i y_i^{a_i} &\longmapsto \sum_a \alpha(q_a) \prod_i y_i^{a_i}, \\ \overline{\tau}_{\text{ter}} = \prod_i y_i^{m-1} &\neq 0, \quad \overline{C}_{h_0,k} \overline{\tau}_{\text{ter}} = 0 \iff \overline{C}_{h_0,k} = 0. \end{aligned} \quad (\text{TPF.43})$$

Successive division by the monic polynomials y_i^m gives the unique remainder basis $\prod_i y_i^{a_i}$, $0 \leq a_i < m$, so $\overline{\tau}_{\text{ter}}$ is one of its nonzero basis vectors and the last equivalence follows. All nilpotent directions of this coefficient algebra are retained. These scalar statements are in characteristic p . There is no unital field map from k_0 to $\overline{\mathbf{Q}}_\ell$, since such a map would send $p \cdot 1 = 0$ to the nonzero element $p \cdot 1$. Thus no multiplication of an ℓ -adic Gauss line by $\overline{C}_{h_0,k}$ is being asserted. The deck-character map (TPF.37) does not by itself transport coefficient vectors to etale cohomology; the finite-field sheaf is the independently specified compact-support sheaf and does not depend on this scalar.

132.9 The shared integral deck-character model and its two base changes

There is an exact common integral representation behind the character correspondence, which does not require a map between fields of different characteristics. Let

$$\begin{aligned} B &= \mathbf{Z}[\zeta_N, 1/N], \quad C_N = \langle \gamma : \gamma^N = 1 \rangle, \\ e_j &= \frac{1}{N} \sum_{a=0}^{N-1} \zeta_N^{-ja} \gamma^a \in B[C_N] \quad (0 \leq j < N). \end{aligned} \quad (\text{TPF.44})$$

Here ζ_N is a fixed primitive root in the cyclotomic ring. Finite geometric sums give

$$e_i e_j = \delta_{ij} e_j, \quad \sum_{j=0}^{N-1} e_j = 1, \quad \gamma e_j = \zeta_N^j e_j. \quad (\text{TPF.45})$$

For example the coefficient convolution in $e_i e_j$ contains $\sum_a \zeta_N^{a(j-i)}$, which is N for $i = j$ and zero otherwise. The sum of the e_j has coefficient one at γ^0 and zero elsewhere, and reindexing $a + 1$ proves the last identity.

The corresponding free module of polynomial forms with the original degree retained is

$$M_B = B[y] dy / y^m B[y] dy, \quad \gamma(y^b dy) = \zeta_N^{b+1} y^b dy \quad (0 \leq b < m). \quad (\text{TPF.46})$$

It is the quotient of polynomial one-forms by y^m times that free module, not the module of Kahler differentials of the truncated algebra. Its basis is $dy, y dy, \dots, y^{m-1} dy$. Thus the exact $B[C_N]$ -linear isomorphism

$$\bigoplus_{j=1}^{N-1} B e_j \longrightarrow M_B, \quad e_j \longmapsto y^{j-1} dy \quad (\text{TPF.47})$$

follows from (TPF.45) and the displayed bases. In an ordered k -fold tensor, $e_{j_1} \otimes \dots \otimes e_{j_k}$ therefore projects onto the corresponding ordered monomial-form basis line. The simultaneous e_0 -projector selects $\sum_i j_i = 0 \bmod N$, exactly the count in (TPF.18). The terminal line is $(B e_m)^{\otimes k}$.

Choose a primitive root $\zeta_L \in L$ and the primitive ℓ -adic root ζ_ℓ already used in (TPF.37). The two actual ring maps are

$$\iota_p : B \rightarrow L, \quad \zeta_N \mapsto \zeta_L, \quad \iota_\ell : B \rightarrow \overline{\mathbf{Q}}_\ell, \quad \zeta_N \mapsto \zeta_\ell. \quad (\text{TPF.48})$$

They are well defined: $p \nmid N$, $N \mid Q - 1$, and both selected roots are primitive N th roots. The first base change of (TPF.47) is the concrete isomorphism with $L[y] dy / y^m L[y] dy$, under which e_j is the projector onto $L y^{j-1} dy$. The coefficient-map connection is also exact: $R \otimes_{\mathbf{Z}} B \rightarrow L$ sends $r \otimes b'$ to $\alpha(r) \iota_p(b')$. Tensoring its polynomial-form module gives this same characteristic- p module. This does not introduce new inverses from a larger marked stage.

For the second base change, let W_ℓ be the fibre at $u = 1$ of the actual untwisted sheaf $\mathcal{L}_\psi(c/u)^\vee \otimes \mathcal{F}$. Give it the source deck-pullback action $\gamma = D_{\zeta_L}^{-1}$. The image of the same integral projector is

$$\iota_\ell(e_j) = \frac{1}{N} \sum_{a=0}^{N-1} \zeta_\ell^{-ja} D_{\zeta_L}^{-a} = \frac{1}{N} \sum_{a=0}^{N-1} \zeta_\ell^{ja} D_{\zeta_L}^a = e_{\chi_*^j}^{\text{SPF}}. \quad (\text{TPF.49})$$

The second equality reindexes $a \mapsto -a$. The third is precisely the original SPF power-map projector, with $\lambda_{\chi_*^j}(\zeta_L) = \zeta_\ell^j$. Thus it projects onto the actual one-dimensional Gauss line $H_j = e_{\chi_*^j}^{\text{SPF}} W_\ell$, whose geometric Frobenius over L is

$$-\chi_*^j(N) G_L(\chi_*^j, \psi).$$

Keep that entire line, including its Frobenius, rather than choosing a numerical basis for it. If H_j^{triv} denotes the same vector space and Frobenius with only its C_N -action replaced by the trivial action, there is the canonical isotypic reconstruction below. Give each first character-module factor the identity splitting-field Frobenius:

$$\bigoplus_{j=1}^{N-1} (B e_j \otimes_{B, \iota_\ell} \overline{\mathbf{Q}}_\ell) \otimes_{\overline{\mathbf{Q}}_\ell} H_j^{\text{triv}} \longrightarrow W_\ell, \quad (\text{TPF.50})$$

$$(a e_j) \otimes h \longmapsto a h.$$

It is an isomorphism: on the j th summand it is the scalar multiplication isomorphism onto H_j , and the SPF projectors are complete and orthogonal. It preserves the deck action by (TPF.45) and

preserves the splitting-field Frobenius by the retained action on H_j^{triv} . No Frobenius action on a characteristic- p coefficient vector has been assumed.

Tensoring these actual maps in the fixed order gives the shared terminal-character projector on both sides: $(e_m)^{\otimes k}$ maps to the characteristic- p line $L(\prod_i y_i^{m-1} dy_i)$ and to the ordered Gauss line $H_m^{\otimes k}$. The full finite-field sheaf reinstates its already specified additive factor $\mathcal{L}_\psi(kc/u)$; the common integral model supplies the finite μ_N representation, not an unconstructed universal additive sheaf. Formulas (TPF.44)–(TPF.50) are the exact common projector and base-change correspondence. They retain the coefficient scalar specialization in (TPF.43) and the independently nonzero ℓ -adic sheaf lines as their stated, different targets.

132.10 The multiplicity-one case and its exact scalar comparison

When $m = 1$, one has $N = 2$, $p > 2$, $d = 1$, and $L = k_0 = \mathbf{F}_Q$, because every odd Q_0 satisfies $Q_0 \equiv 1 \pmod{2}$. There is just the nontrivial quadratic character χ , and it is the terminal character. Retain the original constants

$$c = -\rho^2/2, \quad \gamma_\chi = -\chi(2)G_L(\chi, \psi). \quad (\text{TPF.51})$$

Their exact Gauss-square identity is

$$\gamma_\chi^2 = \chi(-1)Q. \quad (\text{TPF.52})$$

To prove it, take a complex embedding of the cyclotomic field of character values. Quadratic-character values are real, so changing z to $-z$ in the conjugate Gauss sum gives $\overline{G_L(\chi, \psi)} = \chi(-1)G_L(\chi, \psi)$. The proved absolute-square identity $G_L \overline{G_L} = Q$ implies $G_L^2 = \chi(-1)Q$, since $\chi(-1)^2 = 1$. Also $\chi(2)^2 = 1$, proving (TPF.52) with the original minus sign and the factor 2 retained. Both sides lie in the same cyclotomic number field; injectivity of the chosen embedding proves that algebraic identity there and hence in its chosen ℓ -adic realization.

For $c \neq 0$, every positive terminal-invariant degree is exactly $k = 2pr$, $r \geq 1$. Its one-dimensional original-base invariant Frobenius is

$$\gamma_\chi^k = (\chi(-1)Q)^{pr}, \quad |\gamma_\chi^k| = Q^{k/2}. \quad (\text{TPF.53})$$

Its additive factor is exactly $\mathcal{L}_\psi(2prc/u) = \mathcal{L}_\psi(0)$; its Kummer factor is $\mathcal{K}_{\chi^{2pr}} = \mathcal{K}_1$. Thus the line is an actual nonzero invariant line of weight k . Here $K = k(m-1) = 0$ and the original marked coefficient is exactly u_0^k . On an actual coefficient stage retaining the inverse of u_0 , its reduction is nonzero. There is no factorial loss. For $c = 0$, every even k is tame invariant and the same formula reads $\gamma_\chi^k = (\chi(-1)Q)^{k/2}$; no divisibility by p is needed.

There is also an exact comparison calculation for any chosen map between the corresponding one-dimensional complex spaces, without asserting a canonical comparison. The actual arithmetic primary algebra is $\mathbf{C}[s]/(s - \rho)$. In its ordered k -fold tensor, the original sum operator $S = \sum_i s_i$ is multiplication by $k\rho$. For $a > 0$, its dilation operator is therefore

$$A_a = \exp((\log a)S) = a^{k\rho} \text{id}, \quad a^{k\rho} = \exp(k\rho \log a). \quad (\text{TPF.54})$$

Choose a complex embedding of the cyclotomic number field containing γ_χ , and let L_G be the resulting one-dimensional complex Frobenius model, with $F_G = \gamma_\chi^k \text{id}$. This takes a number-field model of the computed eigenvalue to $\mathbf{C}$, not a field map from k_0 to $\mathbf{C}$. For every nonzero complex-linear map $T : L_G \rightarrow L_{\text{arith}}$, scalar multiplication proves

$$\begin{aligned} A_a T &= \frac{a^{k\rho}}{\gamma_\chi^k} T F_G, \\ A_a T - T F_G &= (a^{k\rho} - \gamma_\chi^k) T, \quad \left| \frac{a^{k\rho}}{\gamma_\chi^k} \right| = \frac{a^{k \operatorname{Re} \rho}}{Q^{k/2}}. \end{aligned} \quad (\text{TPF.55})$$

For example such maps exist by sending a nonzero basis vector to a nonzero basis vector; changing either basis multiplies the matrix of T by a nonzero scalar and leaves the displayed ratio unchanged. At $a = Q$ the ratio has modulus $Q^{k(\operatorname{Re} \rho - 1/2)}$. This is the exact intertwining defect for every such map. Existence of a nonzero map does not turn this computed ratio into one, and no identification of the two operators has been assumed.

Sources and dependencies. The sole specialized input is the proved SPF.1–29 single-primary sheaf calculation, with its exact power-map projectors, cohomology, Gauss signs, and local inertia. The foundational curve and trace formulas are in Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Sections 2.0–2.3 and 4.0–4.3. The torsor constructions are the Stacks Project, tags 03PK and 0A3J. The relative compact-support Kunneth map in (TPF.6) is SGA 4, Expose XVII, Theorem 5.4.3, applied to the common base $\mathbf{G}_{m,u}$. All tensor multiplicities, original inertia projectors, cancellation cases, descent permutations, and orbit scalars are proved here.

133 The retained primary cyclic lattice at its specified coefficient prime

This calculation uses the ordered primary tensor and full Taylor unit of TP.1 and TP.23–25. It computes its original cyclic map over the specified coefficient prime, including the map before reduction, its primitive containing lattice, its cokernel, and the exact reduction sequence. The support carrier used below is the original G_{C_n} of MFC.29 and MFC.35–44.

133.1 The actual coefficient localization and the unchanged coordinates

Use the single-primary coefficient specialization already specified in SPF.1. Denote its retained coefficient subring of $\mathbb{C}$ by A , its map by $\alpha : A \rightarrow k_0$, and its characteristic by the prime p . Retain every coefficient of the full Taylor class and its inverse, and put

$$\mathfrak{q} = \ker \alpha, \quad S = A_{\mathfrak{q}}, \quad \tilde{\alpha} : S \longrightarrow k_0, \quad \tilde{\alpha}(a/b) = \alpha(a)/\alpha(b). \quad (\text{TCL.1})$$

Thus $b \notin \mathfrak{q}$ in this formula. The image of α is a finite subring of a field. For each nonzero element of that image, multiplication on the finite image is injective and therefore surjective; its inverse lies in the image. The image is consequently a field, proving that $\mathfrak{q}$ is maximal. The fraction formula is independent of representatives: equality of two fractions gives equality after cross multiplication in the domain A , and applying α and dividing by the nonzero denominator images gives the asserted equality. It preserves the ring operations by the same calculation. The map extends the specified map, rather than selecting a new one.

Since $A \subset \mathbb{C}$, both A and its localization S are domains of characteristic zero. The element p is a nonzero nonunit: its image is zero whereas the image of the unit is one. Every integer prime to p is outside $\mathfrak{q}$ and is a unit in S . In particular $\mathbb{Z}_{(p)} \rightarrow S$ is a well-defined injective scalar map. No assertion that S is a discrete valuation ring, a principal ideal domain, or a ring with all ideals in a chain is made or used. The algebra below uses $p > m$; the retained SPF specialization has the stronger $p > m + 1$, as required there by its phase denominators.

Keep the selected primary coordinate ρ , its full order $m \geq 1$, and the ordered number of factors $k \geq 1$. Put

$$\begin{aligned} K &= k(m-1), & y_i &= s_i - \rho, & E_S &= S[s_1, \dots, s_k] / ((s_i - \rho)^m)_{i=1}^k = S[y_1, \dots, y_k] / (y_i^m)_{i=1}^k, \\ J &= \sum_{i=1}^k y_i, & M &= \sum_{i=1}^k s_i = k\rho + J, & \mathcal{C}_S &= S[T] / (T^{K+1}), & \mathcal{C}_S^{\text{orig}} &= S[Z] / ((Z - k\rho)^{K+1}). \end{aligned} \quad (\text{TCL.2})$$

The two displayed presentations of E_S are identified by the mutually inverse substitutions $s_i = y_i + \rho$ and $y_i = s_i - \rho$. The cyclic-source coordinate map $\mathcal{C}_S^{\text{orig}} \rightarrow \mathcal{C}_S$ is $Z \mapsto k\rho + T$, with inverse $T \mapsto Z - k\rho$. These substitutions respect the defining ideals and their composites are the identity on each generator. They preserve the original objects and will be recorded explicitly in the original monomial matrices.

Let $\mathcal{B} = \{0, \dots, m-1\}^k$, ordered lexicographically. Repeated division by the monic relations y_i^m proves that $\{y^{\mathbf{b}} : \mathbf{b} \in \mathcal{B}\}$ is an S -basis: deleting monomials with any exponent at least m gives a remainder, and the relation ideal is spanned by precisely the deleted monomials, so no relation between retained monomials is possible. Every surviving degree is at most K ; hence $J^{K+1} = 0$.

Write the retained one-factor unit and its retained inverse as

$$\begin{aligned} v(y) &= \sum_{j=0}^{m-1} a_j y^j, & a_j &= \frac{(g/h)^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!}, & h &= (s - \rho)^m, \\ w(y) &= \sum_{j=0}^{m-1} b_j y^j, & b_0 &= a_0^{-1}, & b_r &= -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}, \\ U &= \prod_{i=1}^k v(y_i), & U^{-1} &= \prod_{i=1}^k w(y_i) \quad \text{in } E_S. \end{aligned} \tag{TCL.3}$$

These are the actual retained coefficients, all lying in A by its specified construction. The derivative identity follows by expanding $g = (s - \rho)^m (g/h)$ at ρ . The recurrence makes the constant coefficient of vw equal to one and its coefficients of degrees $1, \dots, m-1$ equal to zero. Thus $vw = 1$ modulo y^m , proving the inverse in every factor and then in the tensor. In particular multiplication by U is an actual S -linear automorphism of E_S commuting with multiplication by J and M . No coefficient of this unit has been discarded.

The unchanged unweighted cyclic ring map and the original unit-weighted cyclic coefficient map are respectively

$$\begin{aligned} f : \mathcal{C}_S &\longrightarrow E_S, & f(T^r) &= J^r, \\ f_U : \mathcal{C}_S &\longrightarrow E_S, & f_U(T^r) &= U J^r, \\ f_U^{\text{orig}} : \mathcal{C}_S^{\text{orig}} &\longrightarrow E_S, & f_U^{\text{orig}}(Z^j) &= U M^j. \end{aligned} \tag{TCL.4}$$

The relation $J^{K+1} = 0$ proves that all maps are well-defined. The last two are linear maps, and are maps of modules over the cyclic source ring acting on the target by $T \mapsto J$, respectively $Z \mapsto M$. They are not asserted to be unital ring maps: their value at one is the retained U . Their exact intertwining identities are

$$f_U(Tc) = J f_U(c), \quad f_U^{\text{orig}}(Zc) = M f_U^{\text{orig}}(c). \tag{TCL.5}$$

Both follow by multiplication in the commutative algebra E_S . This is the coefficient realization of the original source map $\mathcal{J}_k \mathcal{T}_k$ in TP.25, restricted to polynomials in the sum of its unchanged s_i coordinates.

133.2 The primitive lattice and the exact factorial inclusion

For $0 \leq r \leq K$, define

$$\mathcal{B}_r = \{\mathbf{b} \in \mathcal{B} : |\mathbf{b}| = r\}, \quad \mathbf{b}! = \prod_{i=1}^k b_i!, \quad V_r = \sum_{\mathbf{b} \in \mathcal{B}_r} \frac{y^{\mathbf{b}}}{\mathbf{b}!}, \quad V_{K+1} = 0. \tag{TCL.6}$$

Every denominator is a unit because $b_i < m < p$. Each set $\mathcal{B}_r$ is nonempty. For $m > 1$ a specified member $\mathbf{b}^*(r)$ is obtained recursively by $b_i^*(r) = \min(m-1, r - \sum_{j < i} b_j^*(r))$. The remaining degree

is nonnegative at each step and is zero after k steps because $r \leq k(m-1)$. For $m = 1$ there is only $r = 0$, with $\mathbf{b}^*(0) = (0, \dots, 0)$.

The multinomial expansion and direct multiplication give

$$J^r = r!V_r \quad (0 \leq r \leq K), \quad JV_r = (r+1)V_{r+1} \quad (0 \leq r \leq K). \quad (\text{TCL.7})$$

To prove the first formula, expand the product of r copies of $\sum_i y_i$. For a prescribed exponent vector $\mathbf{b}$ there are $r!/\mathbf{b}!$ assignments of its r ordered factors to the variables. Exponents at least m give zero in the quotient. This leaves exactly the displayed coefficients. For the second formula, the coefficient of a retained $y^{\mathbf{b}}$ of degree $r+1$ is $\sum_{i:b_i>0} 1/(b_i-1)! \prod_{j \neq i} 1/b_j! = (\sum_i b_i)/\mathbf{b}! = (r+1)/\mathbf{b}!$. For $r = K$ every resulting monomial is zero, which agrees with the convention for V_{K+1} .

Define coefficient functionals, Λ containing lattice and its projection by

$$\begin{aligned} \lambda_r(x) &= \mathbf{b}^*(r)! [y^{\mathbf{b}^*(r)}]x, \\ \Lambda &= \bigoplus_{r=0}^K SV_r \subset E_S, \quad \pi(x) = \sum_{r=0}^K \lambda_r(x)V_r, \\ F &= \text{span}_S\{y^{\mathbf{b}} : \mathbf{b} \notin \{\mathbf{b}^*(0), \dots, \mathbf{b}^*(K)\}\}. \end{aligned} \quad (\text{TCL.8})$$

Distinct V_r have distinct degrees, and the pivot coefficient of V_r is the unit $1/\mathbf{b}^*(r)!$. Therefore $\lambda_r(V_t) = \delta_{rt}$, proving both the direct sum and $\pi^2 = \pi$. Subtracting $\pi(x)$ deletes exactly all pivot coordinates, so $x - \pi(x) \in F$. Conversely all pivot coordinates vanish on F ; hence $\pi(F) = 0$. We have proved the explicit splitting

$$E_S = \Lambda \oplus F, \quad \ker \pi = F. \quad (\text{TCL.9})$$

In particular each V_r is primitive in the sense that it is part of the displayed basis of a direct summand, with no division by $r!$ in the lattice construction.

Write $L = f(\mathcal{C}_S)$. Equations (TCL.7)–(TCL.9) imply

$$\begin{aligned} L &= \bigoplus_{r=0}^K r!SV_r \subset \Lambda, \quad \Lambda/L \simeq \bigoplus_{r=0}^K S/(r!), \\ W_r &= UV_r, \quad \Lambda_U = U\Lambda, \quad L_U = f_U(\mathcal{C}_S) = UL, \quad F_U = UF, \\ E_S &= \Lambda_U \oplus F_U, \quad \pi_U = U\pi U^{-1}, \quad L_U = \bigoplus_{r=0}^K r!SW_r. \end{aligned} \quad (\text{TCL.10})$$

Since S is a characteristic-zero domain, each $r!$ is nonzero and multiplication by it is injective. Thus f and f_U are injective. The quotient isomorphism sends $\sum_r c_r V_r$ to $(c_r \bmod r!)_r$; its kernel and surjectivity follow coordinate by coordinate. Applying the automorphism U proves every transported identity in the display. There is no claim that $U\Lambda = \Lambda$; it is the pair of lattices $L \subset \Lambda$ that is transported.

Let $Q = \text{Frac}(S)$. The exact saturation statement is

$$(L_U \otimes_S Q) \cap E_S = \Lambda_U \quad \text{inside } E_S \otimes_S Q. \quad (\text{TCL.11})$$

Indeed the nonzero $r!$ are invertible in Q , so $L_U \otimes Q = \Lambda_U \otimes Q$. In the splitting $E_S = \Lambda_U \oplus F_U$, an integral element belonging to this rational subspace has zero F_U -component, because the free module F_U injects into $F_U \otimes Q$. This proves the stated intersection, including both inclusions.

Both lattices are stable under J . On their indicated bases, the full action and its covariant inclusion are

$$\begin{aligned} JW_r &= (r+1)W_{r+1}, \quad J(r!W_r) = (r+1)!W_{r+1}, \\ \mathcal{C}_S &\xrightarrow{f_U} \Lambda_U, \quad T^r \mapsto r!W_r, \quad f_U \circ M_T = M_J \circ f_U. \end{aligned} \quad (\text{TCL.12})$$

At $r = K$ both actions are zero. Commutation of U and J and (TCL.7) prove all entries. This displays the actual source multiplication and the containing-lattice multiplication together with their exact intertwining map.

For clarity, the same map in all original monomial coordinates is

$$\begin{aligned} P_{ab} &= \begin{cases} \binom{b}{a} \rho^{b-a} & a \leq b, \\ 0 & a > b, \end{cases} \quad (0 \leq a, b < m), \quad P_k = P^{\otimes k}, \\ B_{rj} &= \begin{cases} \binom{j}{r} (k\rho)^{j-r} & r \leq j, \\ 0 & r > j, \end{cases} \quad (0 \leq r, j \leq K), \\ \mathbf{V}_{\mathbf{b}r} &= \begin{cases} 1/\mathbf{b}! & |\mathbf{b}| = r, \\ 0 & \text{otherwise,} \end{cases} \quad \Delta = \text{diag}(0!, 1!, \dots, K!), \\ [f_U^{\text{orig}}]_{(s^{\mathbf{b}}), (Z^j)} &= P_k^{-1} [U]_{(y^{\mathbf{b}})} \mathbf{V} \Delta B. \end{aligned} \tag{TCL.13}$$

The binomial theorem gives the two translation matrices P_k and B . Their inverses replace ρ and $k\rho$ by their negatives, respectively, as verified by the inverse substitutions above; their diagonal entries are one. The j -th column on the right is consequently the unchanged polynomial $U \sum_{r=0}^j \binom{j}{r} (k\rho)^{j-r} r! V_r = U(k\rho + J)^j$ written in the original $s^{\mathbf{b}}$ -basis. This proves the matrix formula without replacing the original source map or its unit.

133.3 Exact integer depths, quotient coordinates, and reduction

For each integer $r \geq 0$ define

$$d_r = v_p(r!) = \sum_{a \geq 1} \left\lfloor \frac{r}{p^a} \right\rfloor, \quad r! = p^{d_r} \varepsilon_r, \quad \varepsilon_r \in \mathbb{Z}_{(p)}^\times \subset S^\times. \tag{TCL.14}$$

Here $0! = 1$, and the sum is finite because its terms vanish when $p^a > r$. Each integer j between one and r contributes one factor of p for each positive a with $p^a \mid j$. Interchanging these two finite counts yields $\sum_a \#\{1 \leq j \leq r : p^a \mid j\} = \sum_a \lfloor r/p^a \rfloor$, proving the valuation formula. The remaining integer factor is prime to p , hence is a unit in S . Therefore $(r!) = p^{d_r} S$ exactly. The depth cannot increase in S : an equality $p^d = p^{d+1} s$ would permit cancellation in the domain and give $1 = ps$, contradicting $\tilde{\alpha}(1) = 1$ and $\tilde{\alpha}(p) = 0$. This argument proves strictness of every consecutive pair of principal p -power ideals, without assigning a valuation to arbitrary elements of S .

Set $C = \Lambda_U/L_U$. The following are actual quotient maps:

$$\begin{aligned} q_r : E_S &\longrightarrow S/(p^{d_r}), \quad q_r(x) = \lambda_r(U^{-1}x) \bmod p^{d_r}, \\ C &\xrightarrow{\sim} \bigoplus_{r=0}^K S/(p^{d_r}), \quad \left[\sum_r c_r W_r \right] \longmapsto (c_r \bmod p^{d_r})_r, \\ E_S/L_U &\xrightarrow{\sim} F_U \oplus \bigoplus_{r=0}^K S/(p^{d_r}). \end{aligned} \tag{TCL.15}$$

The convention is $S/(p^0) = 0$. Each q_r is linear, vanishes on F_U and on L_U , and takes W_t to δ_{rt} in its indicated target. Equations (TCL.9)–(TCL.10) prove the two isomorphisms and all these claims. For the coordinate class $\nu_r = [W_r] \in C$, its exact annihilator in S is $p^{d_r} S$, including the unit ideal when $\nu_r = 0$ and $d_r = 0$.

Let B denote either S/pS with its quotient map, or the specified target k_0 with the map $\tilde{\alpha}$. In both cases $B \neq 0$ and has characteristic p . Append a bar to each base-changed module, element and map. Equations (TCL.8)–(TCL.10) base change directly, because their projections and inverse changes of

basis have their coefficients in S . Thus $\overline{W}_0, \dots, \overline{W}_K$ remain a basis of the direct summand $\overline{\Lambda}_U \subset \overline{E}$. The original map becomes

$$\begin{aligned} \overline{f}_U(\overline{T}^r) &= \begin{cases} r! \overline{W}_r & 0 \leq r < p, \\ 0 & p \leq r \leq K, \end{cases} \\ \ker \overline{f}_U &= \bigoplus_{r=p}^K B\overline{T}^r = (\overline{T}^p), \\ \operatorname{im} \overline{f}_U &= \bigoplus_{r=0}^{\min(K, p-1)} B\overline{W}_r. \end{aligned} \quad (\text{TCL.16})$$

An empty direct sum is zero, and the ideal $(\overline{T}^p)$ is zero when $p > K$. For $r < p$, every factor of $r!$ is a nonzero element of the prime field and therefore a unit in B . For $r \geq p$ the factorial has a factor p and maps to zero. Applying these facts to the independent $\overline{W}_r$ proves exactly the displayed kernel and image.

The nilpotency indices of the indicated actions are

module and action	nilpotency index
$\overline{\mathcal{C}}_S = B[T]/(T^{K+1}), M_T$	$K + 1$
$\operatorname{im} \overline{f}_U, M_{\overline{J}}$	$\min(p, K + 1)$
$\overline{E}, M_{\overline{J}}$	$\min(p, K + 1)$
$\overline{\Lambda}_U, M_{\overline{J}}$	$\min(p, K + 1).$

(TCL.17)

The nilpotency index is the least positive integer whose power is zero; thus it is one for the zero action on a nonzero module. In the first row, $T^K \neq 0$ in its free monomial basis and $T^{K+1} = 0$. On $\overline{E}$, the binomial coefficients $\binom{p}{i}$ for $0 < i < p$ are divisible by p : their factorial expression has one factor of p in its numerator and none in its denominator. The commuting-variable binomial formula therefore gives $\overline{J}^p = \sum_i \overline{y}_i^p = 0$, since $p > m$. Also $\overline{J}^{K+1} = 0$. If $d = \min(p - 1, K)$, then $\overline{J}^d = \overline{d}! \overline{V}_d \neq 0$, by its unit pivot coordinate. This proves the third row. Its application to $\overline{W}_0 = \overline{U}$ gives $\overline{J}^d \overline{W}_0 = \overline{d}! \overline{W}_d \neq 0$, proving the second and fourth rows together with their inherited upper bounds. In particular the abstract reduced cyclic source has not acquired the smaller index: its exact map to the target has lost the directions in (TCL.16).

The reduced containing lattice has the more precise chain decomposition

$$\overline{\Lambda}_U = \bigoplus_{\substack{j \geq 0 \\ jp \leq K}} \bigoplus_{t=0}^{\min(p-1, K-jp)} B e_{j,t}, \quad e_{j,t} = \left(\prod_{a=1}^t \overline{jp+a} \right) \overline{W}_{jp+t}. \quad (\text{TCL.18})$$

The empty product is one. All displayed products are units because $1 \leq a \leq p - 1$. These are therefore changes of basis on disjoint intervals of the basis $\overline{W}_r$. Formula (TCL.12) gives $\overline{J} e_{j,t} = e_{j,t+1}$ except at the last position of a displayed interval. There it is zero, either because the next coefficient is $\overline{(j+1)p} = 0$, or because the next index is $K + 1$. This proves each full chain, including every chain that is absent from the image in (TCL.16).

The lost directions are also the exact connecting image in the base-change sequence

$$\begin{aligned} 0 &\longrightarrow \mathcal{C}_S \xrightarrow{\overline{f}_U} \overline{\Lambda}_U \longrightarrow C \longrightarrow 0, \\ 0 &\longrightarrow \operatorname{Tor}_1^S(C, B) \xrightarrow{\partial} \overline{\mathcal{C}}_S \xrightarrow{\overline{f}_U} \overline{\Lambda}_U \longrightarrow C \otimes_S B \longrightarrow 0, \\ \operatorname{Tor}_1^S(C, B) &= \bigoplus_{r=p}^K B \epsilon_r, \quad \partial(\epsilon_r) = \overline{T}^r, \quad C \otimes_S B \simeq \bigoplus_{r=p}^K B[\overline{W}_r]. \end{aligned} \quad (\text{TCL.19})$$

Here Tor_1 can be computed directly, without a flatness assertion about B . The first line is a free resolution of C of length one, whose differential in the displayed bases is Δ . Tensoring this two-term complex with B gives differential $\overline{\Delta}$. By definition its degree-one homology is $\operatorname{Tor}_1^S(C, B)$. The

factorial calculation in (TCL.16) gives its kernel with the displayed basis and gives its cokernel with the displayed basis. The degree-one differential was chosen as $+f_U$, so the map of the kernel into the degree-one free module is precisely $\epsilon_r \mapsto \bar{T}^r$, with no additional sign. This proves the entire sequence and its maps explicitly.

For the top original vector the formula, including its full scalar, is

$$f_U(T^K) = \frac{K!}{((m-1)!)^k} a_0^k \prod_{i=1}^k y_i^{m-1}. \quad (\text{TCL.20})$$

Only $\mathbf{b} = (m-1, \dots, m-1)$ has total degree K , so (TCL.7) gives its multinomial coefficient. Every nonconstant term of U annihilates that monomial; its constant coefficient is exactly a_0^k , proving the formula. Its scalar is nonzero in the actual characteristic-zero domain, and its exact factorial depth is d_K , because $(m-1)!$ and a_0 are units. The displayed vector maps to zero under the specified characteristic p coefficient map exactly when $K \geq p$. This statement concerns the already specified local single-primary source. It does not extend that specialization to a larger coefficient ring containing additional inverse CRT denominators. A ring in which p is a unit cannot map unitaly to characteristic p , since applying such a map to $pp^{-1} = 1$ would give $0 = 1$.

133.4 The exact original chain-support map for these depths

Fix an integer $L \geq \max(1, d_K)$ and put

$$\begin{aligned} R_{p,L} &= \mathbb{Z}/p^L\mathbb{Z}, & S_L &= S/p^L S, \\ \iota_L : R_{p,L} &\longrightarrow S_L, & [z] &\longmapsto [z], \\ I_a &= p^a R_{p,L}, & H_a &= p^a S_L \quad (0 \leq a \leq L). \end{aligned} \quad (\text{TCL.21})$$

The scalar map is well-defined and unital. It is injective. Indeed, if an integer z not divisible by p^L has image zero, write $z = p^a u$ with $0 \leq a < L$ and u prime to p . Then $p^a u = p^L s$ in S . Cancellation and inversion of u imply $1 = p^{L-a} s u^{-1}$, contradicting the specified residue map. Thus precisely $p^L \mathbb{Z}$ maps to zero, proving injectivity. The same argument proves $p^a \notin p^{a+1} S_L$ for $a < L$: a purported equality lifts to $p^a = p^{a+1} s + p^L t$, and cancellation gives $1 = ps + p^{L-a} t$, again impossible in the residue field. Consequently all H_a are distinct.

For completeness, every ideal of $R_{p,L}$ is some I_a . For a nonzero ideal, choose a nonzero element with least exponent of p among its integer representatives. It has the form $p^a u$ with u a unit modulo p^L , by the integer Euclidean algorithm. Multiplying by u^{-1} puts p^a in the ideal, and all its other elements have exponent at least a by choice, proving that the ideal is I_a . The zero ideal is I_L . This argument is not applied to S_L : there we retain exactly the principal subchain $\{H_a\}$.

Extension and contraction on this actual subchain are

$$\iota_L(I_a)S_L = H_a, \quad \iota_L^{-1}(H_a) = I_a, \quad (0 \leq a \leq L). \quad (\text{TCL.22})$$

Extension follows because the image of the principal generator is p^a . For contraction, an integer of exponent $b < a$ belonging to $p^a S_L$ would give $p^b u = p^a s + p^L t$. Cancelling p^b contradicts the nonzero residue of the unit u . Thus only the integers divisible by p^a contract, and they all do, proving the second formula.

The original ideal laws are retained exactly:

$$\begin{aligned} I_a + I_b &= I_{\min(a,b)}, & I_a \cap I_b &= I_{\max(a,b)}, & I_a I_b &= I_{\min(L,a+b)}, \\ H_a + H_b &= H_{\min(a,b)}, & H_a \cap H_b &= H_{\max(a,b)}, & H_a H_b &= H_{\min(L,a+b)}. \end{aligned} \quad (\text{TCL.23})$$

For sum and intersection the proof is inclusion: when $a \leq b$, the ideal indexed by b is contained in the one indexed by a . Their sum is the larger and their intersection the smaller. For product, every

sum of products of multiples of p^a and p^b is a multiple of p^{a+b} , and the generator p^{a+b} itself is such a product. It becomes zero for $a + b \geq L$. These arguments apply in both displayed rings and prove all six laws, including the boundary indices.

Let $C_L = \{0 < 1 < \dots < L\}$. The original lattice map and its actual coefficient extension are

$$\theta_p : C_L \longrightarrow \text{Id}(R_{p,L}), \quad j \longmapsto I_{L-j}, \quad \theta_S : C_L \longrightarrow \{H_a : 0 \leq a \leq L\}, \quad j \longmapsto H_{L-j}. \quad (\text{TCL.24})$$

They are bijections by the distinctness and classification just proved. Each takes bottom to the zero ideal, top to the unit ideal, maximum to ideal sum and minimum to ideal intersection, by (TCL.23). Thus they are bounded-lattice isomorphisms onto their stated codomains. Extension $I_a \mapsto H_a$ commutes with these two maps and preserves all three ideal operations of (TCL.23). In the j -coordinates, ordinary ideal multiplication is $j \odot t = \max(0, j+t-L)$, since its exponent is $\min(L, (L-j)+(L-t))$. It is not identified with lattice meet: for $L \geq 2, j = t = L-1$ gives respective values $L-2$ and $L-1$.

For either coefficient ring $B = R_{p,L}$ or $B = S_L$, use precisely the original split-support carrier

$$\begin{aligned} G_{C_L}(B) &= \{(0_B, j) : 0 \leq j \leq L\} \cup \{(b, L) : b \in B\}, \\ (b, j) + (c, t) &= (b + c, \max(j, t)), \quad (b, j)(c, t) = (bc, \min(j, t)), \\ \tau_a &= (0_B, L - a) \quad (0 \leq a \leq L), \quad e = \tau_0 = (0_B, L), \quad \tau_L = (0_B, 0). \end{aligned} \quad (\text{TCL.25})$$

The union is inside $B \times C_L$, so the two occurrences of $(0_B, L)$ give one element. A nonzero amplitude occurs only at top support. A nonzero sum has a nonzero summand, so has top support; a nonzero product has both amplitudes nonzero, so has top support. This proves closure. Associativity and commutativity hold in each coordinate. Distributivity follows from ring distributivity and $\min(j, \max(t, v)) = \max(\min(j, t), \min(j, v))$, verified by ordering t, v . The zero is τ_L , and the unit is $(1_B, L)$. These facts prove the stated semiring structure directly.

The exact support-preserving coefficient morphism is

$$\begin{aligned} G_{C_L}(\iota_L) : G_{C_L}(R_{p,L}) &\longrightarrow G_{C_L}(S_L), \quad (b, j) \longmapsto (\iota_L(b), j), \\ \chi_B : G_{C_L}(B) &\longrightarrow C_L, \quad (b, j) \longmapsto j, \\ \theta_S \chi_{S_L} G_{C_L}(\iota_L) &= \text{ext}_{\iota_L} \theta_p \chi_{R_{p,L}}. \end{aligned} \quad (\text{TCL.26})$$

The first map is well-defined because nonzero target amplitude has nonzero source amplitude and hence top support. Applying the ring homomorphism to amplitudes and leaving the support coordinate fixed proves preservation of both operations, zero and unit. Its injectivity follows from injectivity of ι_L and the unchanged support coordinate. The support projection preserves the two operations by their defining formulas. The last equality follows on every entry (b, j) from $I_{L-j}S_L = H_{L-j}$; it is an equality into the principal ideal subchain with ideal sum and ideal intersection. In particular each τ_a maps to the distinct τ_a , then to I_a or H_a . The supported zero e remains distinct from the absent zero τ_L .

Finally the connection of this support diagram to the actual cyclic quotient is not an identification of a torsion module with a semiring. The precise coordinate and annihilator maps are

$$\begin{aligned} C &\simeq \bigoplus_{r=0}^K S/(p^{d_r}), \quad p^L C = 0, \\ \text{Ann}_{S_L}(\nu_r) &= H_{d_r} = \theta_S(L - d_r) = \theta_S \chi_{S_L}(\tau_{d_r}), \\ q_r(x) &= \lambda_r(U^{-1}x) \bmod p^{d_r}. \end{aligned} \quad (\text{TCL.27})$$

The first and last lines are the proved maps of (TCL.15), with p^L annihilating every summand because $d_r \leq L$. An element of S_L kills the unit class of the r -th summand exactly when a representative in S lies in $p^{d_r}S$; this proves the annihilator formula and all its equalities using (TCL.24)–(TCL.26). Thus the depths of the original factorial inclusion have explicit labels in the original chain-support carrier, with their arithmetic ideal product as well as their lattice sum and intersection still recorded. The entry τ_{d_r} labels the annihilator; it does not replace the vector ν_r . In particular, when $d_r = L$ the vector ν_r is the nonzero unit class in S_L , whereas its annihilator is the zero ideal labelled by τ_L .

133.5 The retained boundary-character labels and the full-stage obstruction

The action on the lost source directions is also exact. For $K \geq p$, the kernel in (TCL.16), equivalently the connecting image of the Tor group in (TCL.19), has the ordered basis $\bar{T}^p, \dots, \bar{T}^K$, with

$$M_T \bar{T}^r = \bar{T}^{r+1} \ (p \leq r < K), \quad M_T \bar{T}^K = 0, \quad \text{ind}(M_T|_{\ker \bar{f}_U}) = K - p + 1. \quad (\text{TCL.28})$$

These equalities follow in the free monomial source basis. In particular the $(K - p)$ -th power takes $\bar{T}^p$ to the nonzero $\bar{T}^K$, and the next power is zero on every basis element. This index can exceed p . It is an action on the exact kernel in the source, not a subspace of the ambient target. Thus it is compatible with $\bar{J}^p = 0$ on $\bar{E}$: the original map is zero on every one of these kernel vectors.

Put $N = m + 1$. For a residue class $c \in \mathbb{Z}/N\mathbb{Z}$ define the original coefficient character masks in y -coordinates and their full-unit transports by

$$\begin{aligned} Q_c(y^{\mathbf{b}}) &= \mathbf{1}_{\{|\mathbf{b}|+k \equiv c \pmod{N}\}} y^{\mathbf{b}}, & \hat{Q}_c &= U Q_c U^{-1} : E_S \longrightarrow E_S, \\ D_c(T^r) &= \mathbf{1}_{\{r+k \equiv c \pmod{N}\}} T^r & \text{on } \mathcal{C}_S. \end{aligned} \quad (\text{TCL.29})$$

Each diagonal entry defining Q_c, D_c is zero or one in S , so these maps exist over this exact coefficient ring without adjoining roots of unity. On every indicated basis vector exactly one of the masks is one. Hence each family consists of pairwise orthogonal algebraic idempotents summing to the identity; conjugation by the actual U proves the same assertion for $\hat{Q}_c$. The word orthogonal here means that distinct idempotents compose to zero; no inner product or orthogonal projection on a metric space is introduced.

The full original boundary-character comparison is

$$\begin{aligned} Q_c V_r &= \mathbf{1}_{\{r+k \equiv c\}} V_r, & \hat{Q}_c W_r &= \mathbf{1}_{\{r+k \equiv c\}} W_r, \\ \hat{Q}_c f_U &= f_U D_c, & \hat{Q}_c J &= J \hat{Q}_{c-1}, & D_c M_T &= M_T D_{c-1}. \end{aligned} \quad (\text{TCL.30})$$

Every summand of V_r has degree r , proving the first equality, and conjugation proves the second. Apply these formulas to $f_U(T^r) = r!W_r$ to prove the third equality on a source basis. Multiplication by each y_i raises degree by one for every surviving monomial and gives zero at the truncation boundary. It follows on the complete monomial basis that $Q_c J = J Q_{c-1}$. Conjugation and $UJ = JU$ prove the fourth identity; the identical degree argument on T^r proves the last. These are covariant character shifts, not a claim that a fixed character mask commutes with J .

Over $\mathbb{C}$, this is exactly the coefficient mask of the boundary monodromy in TP.9–10: that period monodromy has eigenvalue $\exp(2\pi i(|\mathbf{b}| + k)/N)$ on its period character indexed by $\mathbf{b}$. In the unchanged original $s^{\mathbf{b}}$ -coordinates the coefficient mask is $P_k^{-1} Q_c P_k$, and the source mask is its conjugate by $P_k^{-1}[U]_{(y^{\mathbf{b}})} P_k$. The complex period matrix is $\Pi_k = e^{kc_0/u} W_k D_k(u) P_k$, with the original phase constant here denoted $c_0 = -(-\rho)^N/N$, and $D_k(u)$ diagonal in these same $\mathbf{b}$ indices. Consequently

$$\mathcal{P}_c \Pi_k = \Pi_k (P_k^{-1} Q_c P_k), \quad \mathcal{P}_c \Pi_k [U]_{(s^{\mathbf{b}})}^{-1} = \Pi_k [U]_{(s^{\mathbf{b}})}^{-1} [\hat{Q}_c]_{(s^{\mathbf{b}})}, \quad \mathcal{P}_c = W_k Q_c W_k^{-1}. \quad (\text{TCL.31})$$

Indeed Q_c commutes with the diagonal $D_k(u)$, which proves the first equality by multiplication, and substitution of the conjugation formula for $\hat{Q}_c$ proves the second. The common scalar $e^{kc_0/u}$, the original P_k and the entire U are present in these identities.

The masks preserve both L_U and Λ_U , as follows from their diagonal action on $r!W_r$ and W_r . They therefore act on the quotient C , and the exact free resolution in (TCL.19) commutes with D_c on its source and $\hat{Q}_c$ on its target. Its base change and its kernel inherit precisely those same masks. Thus the simultaneously retained depth and character of every quotient coordinate and lost direction are

$$\begin{aligned} \nu_r &= [W_r] : \quad \text{Ann}_{S_L}(\nu_r) = H_{d_r}, & \hat{Q}_c \nu_r &= \mathbf{1}_{\{r+k \equiv c\}} \nu_r, \\ \epsilon_r &\in \text{Tor}_1^S(C, B) \ (r \geq p) : & \bar{D}_c \partial \epsilon_r &= \mathbf{1}_{\{r+k \equiv c\}} \bar{T}^r. \end{aligned} \quad (\text{TCL.32})$$

The first line combines the proved annihilator in (TCL.27) with the induced quotient map. The second follows from the explicit connecting map $\partial\epsilon_r = \overline{T}'$. This proves both labels on the actual maps, including the zero coordinates when $d_r = 0$. It does not assert an inertia-invariant description of a different sheaf after reduction.

There is an exact obstruction to carrying this local specialization through the full amplified quartet CRT stage. In the counterfactual quartet of PAM.1–4 retain $\rho = \frac{1}{2} + \delta + i\gamma$, with $0 < \delta < \frac{1}{2}$, $\gamma > 0$, and take $k \geq p$. Two actual ordered tensor centres in that construction are

$$\lambda_0 = k\rho, \quad \lambda_1 = (k-p)\rho + p(1-\bar{\rho}), \quad \lambda_0 - \lambda_1 = p(\rho + \bar{\rho} - 1) = 2p\delta \neq 0. \quad (\text{TCL.33})$$

The first chooses ρ in every factor; the second chooses $1-\bar{\rho}$ in exactly p factors and ρ in the other $k-p$ factors. Both choices are permitted in the retained quartet, and the difference follows by subtraction. In the grid coordinates of PAM.2 they have indices respectively $(a, b) = (k, k)$ and $(k-p, k)$.

Let $A_{\text{CRT}} \subset \mathbb{C}$ be the actual coefficient ring carrying those quartet coordinates and every coefficient of the retained terminal CRT idempotent polynomial representative $e(Z) = \sum_{r=0}^d e_r Z^r$ at λ_0 . By its specified CRT residues, $e(\lambda_0) = 1$ and $e(\lambda_1) = 0$. Define the following element of this same ring:

$$\mathcal{D} = \sum_{r=1}^d e_r \sum_{j=0}^{r-1} \lambda_0^{r-1-j} \lambda_1^j \in A_{\text{CRT}}, \quad 1 = p(\rho + \bar{\rho} - 1)\mathcal{D}, \quad p^{-1} = (\rho + \bar{\rho} - 1)\mathcal{D} \in A_{\text{CRT}}. \quad (\text{TCL.34})$$

For every $r \geq 1$, multiplying the inner sum by $\lambda_0 - \lambda_1$ cancels all intermediate monomials and leaves $\lambda_0^r - \lambda_1^r$. After multiplication by e_r and summation the result is $e(\lambda_0) - e(\lambda_1) = 1$. Substitution of (TCL.33) proves the second equality and then the displayed inverse of p . This proof uses the original CRT polynomial itself and does not depend on a selected formula for its coefficients or on an assumed list of inverted pairwise differences. Since p is a unit, no prime ideal of A_{CRT} lies over (p) , and there is no unital characteristic- p specialization of that complete stage. A prime containing p would contain 1, and any such specialization would send the proved identity $pp^{-1} = 1$ to $0 = 1$.

Thus for $k \geq p$ the actual local prime used in (TCL.1) does not extend to this full quartet stage with its retained terminal CRT idempotent. This is a proved obstruction for that specified stage, not a statement that the local and global characteristic-zero objects are unrelated. Their local cyclic coefficient map is still exactly (TCL.4), their unchanged monomial translation is (TCL.13), and the characteristic-zero terminal coefficient is (TCL.20). The local characteristic- p quotient and its lost directions are the maps (TCL.16)–(TCL.19); a missing extension of their coefficient map is not the image of a global vector under a nonexistent specialization.

More generally let $A_{\text{CRT},k} \subset \mathbb{C}$ denote the retained coefficient ring of the complete split corner at degree k . The argument (TCL.33)–(TCL.34), applied to every rational prime $q \leq k$ in place of p , proves

$$q^{-1} \in A_{\text{CRT},k} \quad (q \text{ prime}, q \leq k). \quad (\text{TCL.35})$$

Indeed the second centre uses q reflected factors, which are available for every such q , and the same retained idempotent separates it from the terminal centre. Its divided difference gives the inverse without changing the idempotent. For a directed coefficient ring containing all these actual split stages one may take the explicit subring

$$A_{\text{CRT},\infty} = \bigcup_{n \geq 2} \mathbb{Z}[A_{\text{CRT},2}, \dots, A_{\text{CRT},n}] \subset \mathbb{C}, \quad \mathbb{Q} \subset A_{\text{CRT},\infty}. \quad (\text{TCL.36})$$

The finite generated-stage subrings in this union form an increasing sequence, so the union is a subring containing every stage with its actual coefficient maps. For any prime q , the stage $k = q$ supplies q^{-1} by (TCL.35). Factoring a nonzero integer into primes then puts its reciprocal in the union, proving the asserted copy of $\mathbb{Q}$. Every residue field of this union therefore has characteristic zero: a characteristic- q ring homomorphism would send the unit identity $qq^{-1} = 1$ to $0 = 1$. Equivalently

the scalar morphism $\text{Spec } A_{\text{CRT},\infty} \rightarrow \text{Spec } \mathbb{Z}$ has no point over any (q) . This follows directly by contraction of primes and the proved invertibility of every q . These conclusions concern the full split corner diagram with its retained CRT idempotents. They do not remove the finite single-primary models or the local maps (TCL.1)–(TCL.32).

There are also exact surviving character-zero vectors in the reduced local cyclic image. Take $m \geq 2$ and $k = p\ell$ with $\ell \geq 1$, so $K = k(m-1) \geq p$. Its final nonzero cyclic image basis vector and its unchanged characteristic-zero precursor satisfy

$$\begin{aligned} \bar{J}\bar{W}_{p-1} &= 0, & \bar{W}_{p-1} &\neq 0, & JW_{p-1} &= pW_p \neq 0, \\ \bar{Q}_0 \bar{W}_{p-1} &= \bar{W}_{p-1} &\iff & p(\ell+1) \equiv 1 \pmod{N}, & N &= m+1. \end{aligned} \quad (\text{TCL.37})$$

The first line follows from (TCL.12), the unit pivot after reduction, and the nonzero scalar p on the free characteristic-zero containing lattice. The vector belongs to the original reduced cyclic image because $\bar{f}_U(\bar{T}^{p-1}) = \overline{(p-1)!} \bar{W}_{p-1}$, whose scalar is a unit. Formula (TCL.30) gives the second line as $p-1+k \equiv 0 \pmod{N}$, and substitution of $k = p\ell$ gives the stated congruence. In the retained coefficient datum $p > N$, so p is invertible modulo N . Choosing the single residue class $\ell \equiv p^{-1} - 1 \pmod{N}$ and then its infinitely many positive representatives proves that the displayed character-zero case occurs for infinitely many tensor orders $k = p\ell$. These vectors are killed by J only after base change, as their exact characteristic-zero image pW_p shows. This is a calculation in the local coefficient module with its retained character masks, not an identification with a finite-field sheaf vector or a recovery of the original terminal jet.

Every tensor coefficient, the original monomial translations, the full unit and inverse, and the specified coefficient map occur in the displayed morphisms. The characteristic- p kernel in (TCL.16) is a statement about these coefficient modules. It neither sets an integer scalar to zero in $\mathbb{Q}_\ell$ nor identifies this reduction with the inertia or Frobenius action on the finite-field sheaf. No verdict about the location of an actual zero of ξ follows from this lattice calculation alone.

134 Original cyclic coefficients and the tensor boundary maps

134.1 The cyclic exponent with its retained coefficient map

The characteristic-zero nilpotent exponent in the cyclic-sum calculation has the following full coefficient form. This gives the original map before specialization, its saturation, its precise cokernel, and the specialized map. In particular the characteristic-zero injection and the specialized injection are no longer treated as the same assertion.

Let ρ be the retained primary centre, let $m \geq 1$ be its complete order, let $k \geq 1$, and put $K = k(m-1)$. Retain a coefficient subring $A \subset \mathbb{C}$ containing ρ , its already specified unital map $\alpha : A \rightarrow k_0$ to a finite field of characteristic $p > m$, and every coefficient of the original full unit and its inverse. The finite-field phase used below requires the stronger $p > m+1$. Since the image of α is a finite subring of a field, multiplication by any nonzero element of that image is an injective self-map of a finite set, hence a surjective map. The image is therefore a field and $\mathfrak{q} = \ker \alpha$ is maximal. Put $\mathcal{S} = A_{\mathfrak{q}}$. The extension $\tilde{\alpha} : \mathcal{S} \rightarrow k_0$ sends a/b to $\alpha(a)/\alpha(b)$; cross multiplication proves independence of the chosen fraction. The ring $\mathcal{S}$ is a characteristic-zero domain, p is a nonzero nonunit, and every integer prime to p is a unit. These statements use this actual coefficient map; they make no discrete-valuation or principal-ideal assumption on $\mathcal{S}$.

The coefficient datum and the phase datum have separate explicit domains. If A_{coef} carries the original centre and the full unit and inverse, the coefficient maps below are defined over A_{coef} . Adjoining the original phase requires the map $A_{\text{coef}} \rightarrow A_{\text{coef}}[1/(m+1)]$; a source presentation which also retains $1/(m+1)!$ uses that larger, explicitly given localization. A coefficient map extends to either localization exactly when its inverted elements have nonzero images: the extension must send each inverse to the inverse of its image, and that formula proves existence and uniqueness in the

nonzero case. Thus the factorial calculation with $p > m$ does not itself construct a phase specialization. For $m = 1, p = 2$, its coefficient map, when given on A_{coef} , sends the one-dimensional cyclic image by multiplication by the retained unit $\bar{a}_0^k$, which is nonzero; $K = 0$ and the nilpotent has index one. This map has no extension to $A_{\text{coef}}[1/2]$, since the identity $2(1/2) = 1$ would acquire images $0 = 1$. Every finite-field phase statement below uses its actual stronger datum $p > m + 1$, with all other retained denominator exclusions.

The original objects and the two explicit coordinate substitutions are

$$\begin{aligned} E_{\mathcal{S}} &= \mathcal{S}[s_1, \dots, s_k] / ((s_i - \rho)^m)_{i=1}^k \simeq \mathcal{S}[y_1, \dots, y_k] / (y_i^m)_{i=1}^k, & y_i &= s_i - \rho, \\ J &= \sum_i y_i, & M &= \sum_i s_i = k\rho + J, \\ \mathcal{C}_{\mathcal{S}}^{\text{orig}} &= \mathcal{S}[Z] / ((Z - k\rho)^{K+1}) \simeq \mathcal{C}_{\mathcal{S}} = \mathcal{S}[T] / (T^{K+1}), & T &= Z - k\rho. \end{aligned} \quad (\text{BPC.1})$$

The inverse substitutions are $s_i = y_i + \rho$ and $Z = T + k\rho$; both respect the displayed ideals and their composites fix every generator. Successive division by the monic relations y_i^m proves that the ordered monomials $y^{\mathbf{b}}, 0 \leq b_i < m$, are an $\mathcal{S}$ -basis.

Write $h = (s - \rho)^m, g = 2\xi$, and retain

$$\begin{aligned} v(y) &= \sum_{j=0}^{m-1} a_j y^j, & a_j &= \frac{(g/h)^{(j)}(\rho)}{j!} = \frac{g^{(m+j)}(\rho)}{(m+j)!}, \\ w(y) &= \sum_{j=0}^{m-1} b_j y^j, & b_0 &= a_0^{-1}, & b_r &= -a_0^{-1} \sum_{j=1}^r a_j b_{r-j}, \\ U &= \prod_i v(y_i), & U^{-1} &= \prod_i w(y_i). \end{aligned} \quad (\text{BPC.2})$$

All these coefficients belong to the specified A . Expanding $g = (s - \rho)^m(g/h)$ proves the derivative identity. The recurrence makes the constant coefficient of vw equal to one and all its positive coefficients of degree below m zero; thus it proves the stated inverse. Multiplication by U is an actual $\mathcal{S}$ -linear automorphism of $E_{\mathcal{S}}$, commuting with J and M . The same proof applies to the full unit induced by a retained quartet packet on this primary factor, using that unit's original coefficients throughout.

The unchanged cyclic ring map and the actual unit-weighted module map are

$$f : \mathcal{C}_{\mathcal{S}} \longrightarrow E_{\mathcal{S}}, \quad T^r \longmapsto J^r, \quad f_U : \mathcal{C}_{\mathcal{S}} \longrightarrow E_{\mathcal{S}}, \quad T^r \longmapsto UJ^r, \quad f_U^{\text{orig}}(Z^j) = UM^j. \quad (\text{BPC.3})$$

All surviving degrees in $E_{\mathcal{S}}$ are at most K , so $J^{K+1} = 0$ and these maps are well defined. The weighted map is a map of modules for $T \mapsto J$, or for $Z \mapsto M$ in the original coordinates: $f_U(Tc) = Jf_U(c)$ and $f_U^{\text{orig}}(Zc) = Mf_U^{\text{orig}}(c)$. Its value at one is the actual U . These formulas also give its type when $U \neq 1$; they do not require it to be a unital ring homomorphism.

For $0 \leq r \leq K$ set

$$V_r = \sum_{\substack{0 \leq b_i < m \\ \sum_i b_i = r}} \frac{y^{\mathbf{b}}}{\prod_i b_i!}, \quad W_r = UV_r, \quad V_{K+1} = W_{K+1} = 0. \quad (\text{BPC.4})$$

Every denominator here is a unit because $b_i < m < p$. For each $r \leq K$, a tuple of this degree exists: fill the slots successively up to capacity $m - 1$. Select that tuple $\mathbf{b}^*(r)$ and put $\lambda_r(x) = \mathbf{b}^*(r)! [y^{\mathbf{b}^*(r)}]x$. Then $\lambda_r(V_t) = \delta_{rt}$. The coefficient projection $\pi(x) = \sum_r \lambda_r(x) V_r$ deletes every selected pivot coordinate from $x - \pi(x)$. Consequently, with F the span of all unselected monomials, the maps just given prove

$$E_{\mathcal{S}} = \Lambda \oplus F, \quad \Lambda = \bigoplus_{r=0}^K \mathcal{S}V_r, \quad E_{\mathcal{S}} = \Lambda_U \oplus UF, \quad \Lambda_U = U\Lambda = \bigoplus_{r=0}^K \mathcal{S}W_r, \quad \pi_U = U\pi U^{-1}. \quad (\text{BPC.5})$$

The sum defining each V_r is retained in full; its selected pivot only supplies this explicit splitting.

The multinomial expansion and a direct coefficient calculation give

$$J^r = r!V_r, \quad JV_r = (r+1)V_{r+1}, \quad f_U(T^r) = r!W_r, \quad JW_r = (r+1)W_{r+1}. \quad (\text{BPC.6})$$

For the first formula the coefficient of $y^{\mathbf{b}}$ is the number $r!/\mathbf{b}!$ of its ordered assignments among the r factors. Terms exceeding a primary order vanish in the defining quotient. For the second formula the coefficient at degree $r+1$ is $\sum_{i:b_i>0} b_i/\mathbf{b}! = (r+1)/\mathbf{b}!$. The last two formulas follow by multiplying by the retained U . At degree K there is exactly one tuple, so

$$J^K = \frac{K!}{((m-1)!)^k} \prod_i y_i^{m-1} \neq 0, \quad J^{K+1} = 0, \quad f_U(T^K) = \frac{K!}{((m-1)!)^k} a_0^k \prod_i y_i^{m-1}. \quad (\text{BPC.7})$$

Its coefficient is nonzero in the characteristic-zero domain $\mathcal{S}$. Every positive Taylor term of U kills the top monomial, which proves the last equality with its full original scalar. Thus the old characteristic-zero exponent is exactly $K+1 = 1 + k(m-1)$, and f_U is injective: its columns are the independent W_r multiplied by the nonzero $r!$. This includes $m=1$: then $K=0$, the zeroth power is the identity and the zero nilpotent has index one.

In the original coordinates the same map has matrix

$$[f_U^{\text{orig}}]_{(s^{\mathbf{b}}), (Z^j)} = P_k^{-1}[U]_{(y^{\mathbf{b}})} \mathbf{V} \Delta B, \quad \Delta = \text{diag}(0!, 1!, \dots, K!), \quad (\text{BPC.8})$$

where $P_k = P^{\otimes k}$, $P_{ab} = \binom{b}{a} \rho^{b-a}$ for $a \leq b$ and zero otherwise, $B_{rj} = \binom{j}{r} (k\rho)^{j-r}$ for $r \leq j$ and zero otherwise, and $\mathbf{V}_{\mathbf{b}r} = 1/\mathbf{b}!$ for $|\mathbf{b}| = r$, zero otherwise. The inverses of P and B replace ρ and $k\rho$ by their negatives, as follows by the inverse substitutions in (BPC.1). The j -th column on the right is the original polynomial $U \sum_r \binom{j}{r} (k\rho)^{j-r} r! V_r = U M^j$, proving the formula and retaining every original coordinate.

Write $L_U = f_U(\mathcal{C}_{\mathcal{S}})$ and $d_r = v_p(r!) = \sum_{a \geq 1} \lfloor r/p^a \rfloor$. Counting, for every integer from one to r , each of its factors of p proves the sum formula. The complementary factor $\varepsilon_r = r!/p^{d_r}$ is prime to p , so is a unit in $\mathcal{S}$. We obtain the actual inclusion and quotient

$$L_U = \bigoplus_{r=0}^K r! \mathcal{S} W_r \subset \Lambda_U, \quad C_U := \Lambda_U / L_U \simeq \bigoplus_{r=0}^K \mathcal{S} / (p^{d_r}), \quad \left[\sum_r c_r W_r \right] \mapsto (c_r \bmod p^{d_r})_r. \quad (\text{BPC.9})$$

The kernel and surjectivity follow coordinate by coordinate from the displayed basis; the convention $\mathcal{S}/(p^0) = 0$ is used. The extension of these quotient coordinates to $E_{\mathcal{S}}$ is $x \mapsto \lambda_r(U^{-1}x) \bmod p^{d_r}$, and its kernel also retains the complementary summand UF . In particular $E_{\mathcal{S}}/L_U \simeq UF \oplus C_U$. Over $Q = \text{Frac } \mathcal{S}$ every $r!$ is invertible. The splitting (BPC.5) therefore proves the precise saturation

$$(L_U \otimes_{\mathcal{S}} Q) \cap E_{\mathcal{S}} = \Lambda_U \quad \text{inside } E_{\mathcal{S}} \otimes_{\mathcal{S}} Q. \quad (\text{BPC.10})$$

Indeed an integral element in the rational span of Λ_U has zero UF -component, since that free module injects into its scalar extension. This proves both inclusions without dividing an original vector by its factorial in $E_{\mathcal{S}}$.

Now take either $B = \mathcal{S}/p\mathcal{S}$ or the specified finite-field target $B = k_0$, using the actual coefficient map, and put a bar on each base change. The splitting and its inverse in (BPC.5) base change because all their coefficients lie in $\mathcal{S}$. Thus every $\overline{W}_r$ remains a distinct basis vector of a direct summand. The exact specialized map is

$$\overline{f}_U(\overline{T}^r) = \begin{cases} \overline{r!} \overline{W}_r, & r < p, \\ 0, & r \geq p, \end{cases} \quad \ker \overline{f}_U = (\overline{T}^p) = \bigoplus_{r=p}^K B \overline{T}^r, \quad \text{im } \overline{f}_U = \bigoplus_{r=0}^{\min(K, p-1)} B \overline{W}_r. \quad (\text{BPC.11})$$

The direct sum is empty when $K < p$. The factorial is a unit for $r < p$, and zero for $r \geq p$; the independent basis proves every claim. The unchanged abstract source $B[T]/T^{K+1}$ consequently

has nilpotency index $K + 1$, whereas multiplication by $\bar{J}$ on its actual image, on $\bar{\Lambda}_U$, and on $\bar{E}_{\mathcal{J}}$ has index

$$\min(p, K + 1). \quad (\text{BPC.12})$$

For the upper bound, in characteristic p the intermediate binomial coefficients vanish, so $\bar{J}^p = \sum_i \bar{y}_i^p = 0$, while $\bar{J}^{K+1} = 0$ still holds. For the lower bound put $d = \min(p - 1, K)$. Then $\bar{J}^d \bar{W}_0 = \bar{d}! \bar{W}_d \neq 0$. This vector is in the image, which proves all three lower bounds. All additional chains of the containing lattice are retained by

$$e_{j,t} = \left(\prod_{a=1}^t \bar{J}^{p+a} \right) \bar{W}_{jp+t}, \quad 0 \leq jp \leq K, \quad 0 \leq t \leq \min(p - 1, K - jp). \quad (\text{BPC.13})$$

Each coefficient here is a unit. Equation (BPC.6) makes $\bar{J}e_{j,t} = e_{j,t+1}$ until the last position; the next coefficient is zero modulo p , or the next degree is $K + 1$. These are all the reduced lattice chains.

The coefficient loss has the exact homological map

$$\begin{aligned} 0 &\longrightarrow \mathcal{C}_{\mathcal{J}} \xrightarrow{f_U} \Lambda_U \longrightarrow C_U \longrightarrow 0, \\ 0 &\longrightarrow \text{Tor}_1^{\mathcal{J}}(C_U, B) \xrightarrow{\partial} \bar{\mathcal{C}}_{\mathcal{J}} \xrightarrow{\bar{f}_U} \bar{\Lambda}_U \longrightarrow C_U \otimes_{\mathcal{J}} B \longrightarrow 0, \\ \text{Tor}_1^{\mathcal{J}}(C_U, B) &= \bigoplus_{r=p}^K B\epsilon_r, \quad \partial\epsilon_r = \bar{T}^r, \quad C_U \otimes_{\mathcal{J}} B = \bigoplus_{r=p}^K B[\bar{W}_r]. \end{aligned} \quad (\text{BPC.14})$$

The first line is an explicit free resolution whose matrix is Δ in (BPC.8). Tensoring that complex with B gives the diagonal matrix $\bar{\Delta}$. Its kernel and cokernel are exactly (BPC.11), proving the last two lines and the sign of ∂ , which is the inclusion into the degree-one free module for differential $+f_U$. When $K \geq p$, multiplication by $\bar{T}$ on this kernel has index $K - p + 1$: its $(K - p)$ -th power sends $\bar{T}^p$ to the nonzero $\bar{T}^K$. The original source, actual image, containing lattice, and connecting image are thus all present with their proved maps.

134.2 The same cyclic map carries the tensor boundary projector

Put $N = m + 1$ and retain the primitive fixed at the original point $s = 0$:

$$\Phi(s) = \frac{(s - \rho)^N - (-\rho)^N}{N}, \quad c = -\frac{(-\rho)^N}{N}, \quad \sum_i \Phi(s_i) = \sum_i \frac{y_i^N}{N} + kc. \quad (\text{BTP.1})$$

The ordered tensor complex has the commuting differentials $u\partial_{s_i} + h(s_i)$ and the original form $ds_1 \wedge \cdots \wedge ds_k$. The one-factor monic remainder decomposition, tensored with its explicit contracting homotopy, concentrates its cohomology in degree k , with the free basis $s^{\mathbf{b}} ds_1 \wedge \cdots \wedge ds_k, 0 \leq b_i < m$. No permutation quotient is taken.

For $\zeta = e^{2\pi i/N}$ and the fixed branch of $u^{1/N}$, keep

$$\begin{aligned} \beta_b &= (b + 1)/N, & d_b &= N^{\beta_b - 1} e^{\pi i \beta_b} \Gamma(\beta_b), \\ W_{jb} &= \zeta^{j(b+1)} - 1, & D(u) &= \text{diag}(d_b u^{\beta_b}), \\ \Pi_k(u) &= e^{kc/u} W^{\otimes k} D(u)^{\otimes k} P_k. \end{aligned} \quad (\text{BTP.2})$$

Here $1 \leq j \leq m, 0 \leq b < m$, and the rows are the ordered products of the original relative contours $\Gamma_j = -\ell_0 + \ell_j$, whose outward rays have angles $(\arg u + \pi)/N + 2\pi j/N$. The one-variable substitution $t = r^N/(N|u|)$ gives precisely $d_b u^{\beta_b}$ on the zeroth ray and its factor $\zeta^{j(b+1)}$ on the j -th ray. Their oriented difference gives W_{jb} . Translation to $y = s - \rho$ is the original one-factor contour equality: the remote connector integral tends to zero by the bound $\exp(-R^N/(N|u|) + O(R^{N-1}))$ times a polynomial. It leaves the factor $e^{c/u}$ intact. Absolute integrability permits Fubini on the ordered product

contours, and the binomial substitution gives P_k , proving (BTP.2). The polynomial $\sum_{j=1}^{N-1} a_j(z^j - 1)$ cannot vanish at all N -th roots unless every coefficient is zero; this proves W invertible. Also $d_b \neq 0$, so every $\Pi_k(u)$ is invertible for $u \neq 0$.

In y -coordinates let

$$E_r y^{\mathbf{b}} = \mathbf{1}_{\{|\mathbf{b}|+k \equiv r \pmod{N}\}} y^{\mathbf{b}}, \quad Q_r = P_k^{-1} E_r P_k, \quad \hat{Q}_r = U_k Q_r U_k^{-1}, \quad U_k = [U]_{(s^{\mathbf{b}})}. \quad (\text{BTP.3})$$

These diagonal zero-one masks exist over the exact retained coefficient ring. The period monodromy and its projectors are

$$M_k = W^{\otimes k} \text{diag}(\zeta^{|\mathbf{b}|+k})(W^{\otimes k})^{-1}, \quad \mathcal{P}_r = \frac{1}{N} \sum_{j=0}^{N-1} \zeta^{-rj} M_k^j, \quad \mathcal{P}_r \Pi_k = \Pi_k Q_r. \quad (\text{BTP.4})$$

One positive circuit multiplies the $\mathbf{b}$ -th diagonal entry in (BTP.2) by $\zeta^{|\mathbf{b}|+k}$; its scalar exponential is single-valued. This proves the monodromy formula. The finite geometric sum is one at matching character and zero otherwise, proving the projectors and their intertwining identity. The horizontal coefficient convention has the inverse monodromy, with the same invariant index set.

The source-period map is exactly $\Pi_k U_k^{-1}$. Indeed the original source has Mellin transform $f(s_1, \dots, s_k) \prod_i (g/h)(s_i)$, whose full jet is $U_k[f]_{s_i}$; integration by parts for $D_i = -x_i \partial_{x_i}$ gives multiplication by s_i with vanishing endpoint terms in the original source space. It follows that

$$\mathcal{P}_r \Pi_k U_k^{-1} = \Pi_k U_k^{-1} \hat{Q}_r. \quad (\text{BTP.5})$$

This is the exact source comparison, including the entire unit and its inverse. It does not replace the source norm by the auxiliary period coordinate norm.

The invariant dimension is

$$d_{0,k} = \frac{m^k + m(-1)^k}{m+1}. \quad (\text{BTP.6})$$

To prove it, filter the ordered tuples by their total character: $N^{-1} \sum_{j=0}^{N-1} (\sum_{a=1}^{N-1} \zeta^{ja})^k$. The inner sum is m for $j = 0$ and -1 otherwise, giving the formula. Thus the one-factor vanishing remains true at $k = 1$; tensor invariants appear for every $m \geq 2, k \geq 2$, and for $m = 1$ precisely when k is even. These are the ordinary local-system invariants at the original puncture.

The original arithmetic matrix is $A_k = k\rho I + R_k$, where $R_k = P_k^{-1}[J]P_k$. Each surviving multiplication by y_i increases the boundary character by one. Therefore, on every monomial,

$$\begin{aligned} Q_r R_k^j &= R_k^j Q_{r-j}, & [Q_0, A_k] &= R_k(Q_{-1} - Q_0), \\ Q_0 A_k Q_0 &= k\rho Q_0, & (I - Q_0) A_k Q_0 &= R_k Q_0. \end{aligned} \quad (\text{BTP.7})$$

Terms that exceed a primary order are zero on both sides. Conjugation by the commuting U_k gives the identical formulas on the actual source invariant plane. In particular the scalar compression retains the explicit outgoing map.

For $\text{im } Q_0 \neq 0$, set $d_{\min} = N \lceil k/N \rceil - k$ and $q_* = \lfloor (K - d_{\min})/N \rfloor$. Put $R_0 = Q_0 R_k^N Q_0|_{\text{im } Q_0}$. Then the complete dilation on this plane is

$$Q_0 a^{A_k} Q_0|_{\text{im } Q_0} = a^{k\rho} \sum_{j=0}^{q_*} \frac{(\log a)^{jN}}{(jN)!} R_0^j, \quad R_0^{q_*} \neq 0, \quad R_0^{q_*+1} = 0. \quad (\text{BTP.8})$$

Expand the original finite exponential through degree K . Equation (BTP.7) removes exactly the powers whose degree is not divisible by N . Every invariant input has degree at least $d_{\min}$, proving the upper cutoff. At that minimum degree, distribute any increment $jN \leq K - d_{\min}$ among the remaining slot capacities. Its multinomial coefficient is positive and distinct monomials cannot cancel,

proving the stated nonzero powers. The source compression is the conjugate of (BTP.8) by the full U_k .

The original terminal monomial $\tau_k = \prod_i y_i^{m-1}$ satisfies

$$A_k \tau_k = k \rho \tau_k, \quad U_k \tau_k = a_0^k \tau_k, \quad Q_0 \tau_k = \widehat{Q}_0 \tau_k = \mathbf{1}_{\{N|k\}} \tau_k. \quad (\text{BTP.9})$$

Multiplication by every y_i kills this top monomial, proving the first two formulas. Its character is $km \equiv -k \pmod{N}$, proving the third, including the unit conjugation. Its exact period is

$$\Pi_k \tau_k = e^{kc/u} \left(N^{-1/N} e^{\pi i m/N} \Gamma(m/N) \right)^k u^{km/N} (\zeta^{-j} - 1)_{j=1}^m \otimes k. \quad (\text{BTP.10})$$

This is its column in (BTP.2); all its factors are nonzero. The corresponding actual corrected-section class has observation $a_0^{-k} \Pi_k \tau_k$. For the original amplified quartet vector the retained coefficient is $C_{h_0,k} = K!((m-1)!)^{-k}((g/h_0)(\rho))^k$, so its observation is exactly $C_{h_0,k} a_0^{-k} \Pi_k \tau_k$. Multiplication by (g/h_0) on the terminal monomial has this scalar because all its positive Taylor terms vanish; this is also (BPC.7) for that retained unit. Therefore, on the infinite subsequence $N \mid k$, the same original terminal arithmetic eigenclass lies in the boundary invariant plane. Its original dilation still satisfies

$$\log \left(\frac{|a^{k\rho}|^2}{a^k} \right) = 2k(\operatorname{Re} \rho - 1/2) \log a. \quad (\text{BTP.11})$$

Taking the real part of $k\rho \log a$ proves the identity. No invariant-plane projection has altered this arithmetic scalar.

134.3 Finite-field boundary and the actual coefficient-specialization restriction

Use the already specified finite-field map with $p > N$, and write $k_0 = \mathbb{F}_{Q_0}$, $L = \mathbb{F}_{Q_0^d}$, $d = \operatorname{ord}_{(\mathbb{Z}/N\mathbb{Z})^\times}(Q_0)$. Thus L contains μ_N . Retain $\psi = \psi_0 \circ \operatorname{Tr}_{L/k_0}$, geometric Frobenius, the original compact-support one-factor sheaf, and every one-factor Gauss constant. The already proved SPF decomposition, distributed over the ordered tensor factors on the common u -line, gives the exact isomorphism

$$\mathcal{F}_L^{\otimes k} \simeq \mathcal{L}_\psi(kc/u) \otimes \bigoplus_{\substack{\mathbf{x}=(\chi_1, \dots, \chi_k) \\ \chi_i^N=1, \chi_i \neq 1}} \mathcal{A}_{A_{\mathbf{x}}} \otimes \mathcal{K}_{\prod_i \chi_i}(u), \quad A_{\mathbf{x}} = \prod_i \left(-\chi_i(N) \sum_{z \in L^\times} \chi_i(z) \psi(z) \right). \quad (\text{BTF.1})$$

The torsor tensor laws add the Artin–Schreier phases and multiply the Kummer characters. Distribution of tensor over direct sum retains every ordered tuple, even if different tuples have equal product characters. These facts prove the displayed map from the established one-factor map. The compact-support Kunnet map places it in degree k on the original ordered product source, since each one-factor complex is concentrated in degree one. No factor permutation changes its sign.

For $c \neq 0$, choose the original additive character $\psi_0(x) = \vartheta(\operatorname{Tr}_{k_0/\mathbb{F}_p}(bx))$. The cover $u = v^N$, $z^p - z = bc/v^N$ has deck group $\mu_N \times \mathbb{F}_p$. Its degree is Np : the second right side has pole order N prime to p , whereas a pole of $f^p - f$ has order divisible by p , so it is a nontrivial degree- p extension. The displayed commuting deck maps $(v, z) \mapsto (\zeta v, z + a)$ then exhaust that degree. On the tuple line the original action is

$$(\zeta, a) \mapsto \lambda_{\prod_i \chi_i}(\zeta)^{-1} \vartheta(-ka), \quad (\text{BTF.2})$$

where $\lambda_{\chi}(x^{(Q-1)/N}) = \chi(x)$ and $Q = Q_0^d$. These are the original SPF inverse deck-action conventions, tensored without changing them. Averaging over the complete finite inertia image in the characteristic-zero ℓ -adic coefficient field proves

$$P_{I_0,k} = \begin{cases} P_{0,k}^{\text{tame}}, & kc = 0, \\ 0, & kc \neq 0, \end{cases} \quad \dim V_k^{I_0} = \begin{cases} \frac{m^k + m(-1)^k}{m+1}, & kc = 0, \\ 0, & kc \neq 0. \end{cases} \quad (\text{BTF.3})$$

The tame average is one exactly when $\prod_i \chi_i = 1$. The wild average $p^{-1} \sum_a \vartheta(-ka)$ is one for $p \mid k$ and zero otherwise. For $c = 0$ the original additive torsor is trivial, so only the tame average is present. Since $kc = 0$ is equivalent to $c = 0$ or $p \mid k$, these computations prove the full case distinction. The tuple count is the same geometric sum proved in (BTP.6).

For $kc \neq 0$, the Artin-Schreier factor of $z^p - z = bkc/u$ has a single break one: z^{-1} is a uniformizer and its difference under $z \mapsto z + a$, $a \neq 0$, has valuation two. Every tuple line has that same wild factor, while Kummer twists are tame. Therefore

$$\text{Swan}_0(\mathcal{F}_L^{\otimes k}) = \begin{cases} 0, & kc = 0, \\ m^k, & kc \neq 0, \end{cases} \quad N_{\text{mon},k} = 0 \text{ after } u = v^N, \quad z_k^p - z_k = bkc/v^N. \quad (\text{BTF.4})$$

The cover trivializes every geometric character, so its monodromy is the identity and its logarithm is zero. This leaves the original-inertia invariant space in (BTF.3) unchanged as a separately specified stalk over the original base.

The exact boundary Frobenius for $kc = 0$ over L is the direct sum of the scalars A_χ with $\prod_i \chi_i = 1$. Over k_0 , geometric Frobenius sends χ to χ^{Q_0} : conjugating the SPF deck projector by Frobenius replaces its deck argument by $\zeta^{Q_0^{-1}}$, and reindexing yields that character power. For an orbit $\mathcal{O}$ of length e , each χ_i descends uniquely by norm to $\kappa_{i,e}$ over $L_e = \mathbb{F}_{Q_0^e}$. Indeed its order divides $Q_0^e - 1$, so it kills the norm kernel; norm surjectivity proves uniqueness. Retaining the original denominator N , the one-factor SPF calculation over this orbit field gives

$$\Lambda_{\mathcal{O}} = \prod_i (-\kappa_{i,e}(N) G_{L_e}(\kappa_{i,e}, \psi_0 \circ \text{Tr}_{L_e/k_0})) , \quad \det(1 - tF_0 \mid V_k^{I_0}) = \prod_{\mathcal{O}} (1 - \Lambda_{\mathcal{O}} t^e). \quad (\text{BTF.5})$$

For the descent map when L_e does not contain all N -th roots, factor the source power map as $y \mapsto x = y^{N/r_i} \mapsto z = x^{r_i} = y^N$, where $r_i = \text{ord } \chi_i$. Functions constant on the μ_{N/r_i} -orbits are exactly functions on the intermediate root set; the same map is the identity at the zero fibre. This identifies the descended $\kappa_{i,e}$ -summand of the original sheaf, whose phase is still $z/(Nu) + c/u$. Thus the displayed scalar is the actual F_0^e -action. A basis $v, F_0 v, \dots, F_0^{e-1} v$ of distinct tuple lines gives the factor $1 - \Lambda_{\mathcal{O}} t^e$. The unscaled Gauss absolute-square calculation $|G_{L_e}|^2 = Q_0^e$ therefore gives $|\Lambda_{\mathcal{O}}| = Q_0^{ek/2}$ and weight k for every nonzero boundary orbit block.

To specify the comparison of characteristic types, put $B_0 = \mathbb{Z}[\zeta_N, 1/N]$ and

$$e_j = \frac{1}{N} \sum_{a=0}^{N-1} \zeta_N^{-ja} \gamma^a \in B_0[C_N], \quad M_{B_0} = B_0[y]dy/y^m B_0[y]dy, \quad e_j \longmapsto y^{j-1} dy \quad (1 \leq j < N). \quad (\text{BTF.6})$$

Give γ the action $\gamma(y^b dy) = \zeta_N^{b+1} y^b dy$. The finite geometric sum proves that the e_j are complete orthogonal idempotents, and the displayed basis map is an isomorphism $\bigoplus_{j=1}^{N-1} B_0 e_j \simeq M_{B_0}$. The actual maps $B_0 \rightarrow L$ and $B_0 \rightarrow \overline{\mathbb{Q}_\ell}$ send ζ_N to the respective chosen primitive roots. They give, respectively, the coefficient monomial-form projector and the actual SPF Gauss-line projector: the latter is $N^{-1} \sum_a \zeta_\ell^{-ja} D_\zeta^{-a} = N^{-1} \sum_a \zeta_\ell^{ja} D_\zeta^a$. If H_j is that complete Gauss line, retain its Frobenius and reconstruct its isotypic component by $(ae_j) \otimes h \mapsto ah$ from $(B_0 e_j \otimes \overline{\mathbb{Q}_\ell}) \otimes H_j^{\text{triv}}$, where only the deck action on the second factor is made trivial. The scalar map is bijective on each summand and preserves both deck action and the retained splitting-field Frobenius, proving the comparison. Ordered tensor products of these maps retain every tuple. This is the common integral representation and its two proved base changes; it introduces no field map from characteristic p to $\overline{\mathbb{Q}_\ell}$.

The terminal form has deck-pullback character $\zeta^m = \zeta^{-1}$ in each factor; let its primitive Kummer character be χ_{ter} . Its ordered sheaf line is consequently

$$\mathcal{L}_\psi(kc/u) \otimes \mathcal{A}_{[-\chi_{\text{ter}}(N) G_L(\chi_{\text{ter}}, \psi)]^k} \otimes \mathcal{K}_{\chi_{\text{ter}}^k}(u). \quad (\text{BTF.7})$$

It has invariants exactly for $N \mid k$ and $kc = 0$; this means $N \mid k$ if $c = 0$, and $pN \mid k$ if $c \neq 0$. Its original base orbit has length d , because its character is primitive, with polynomial

$1 - [-\chi_{\text{ter}}(N)G_L(\chi_{\text{ter}}, \psi)]^k t^d$. These claims follow from (BTF.2)–(BTF.5) without assigning a finite-field Frobenius to an original complex arithmetic scalar.

On a coefficient stage admitting the specified map, the original amplified coefficient instead has the literal coefficient image

$$\overline{C}_{h_0,k} = K! ((m-1)!)^{-k} \overline{u}_0^k, \quad u_0 = (g/h_0)(\rho), \quad \overline{C}_{h_0,k} \neq 0 \iff K < p \text{ and } \overline{u}_0 \neq 0. \quad (\text{BTF.8})$$

The denominator is a unit since $m-1 < p$, while $K!$ vanishes exactly for $K \geq p$. Retaining the inverse of the original full unit ensures $\overline{u}_0 \neq 0$. The target is the coefficient algebra $k_0[y_1, \dots, y_k]/(y_i^m)$, with coefficient map obtained by applying α to every coefficient in its unchanged monomial basis. Its terminal basis vector remains nonzero; the original multiple vanishes precisely as in (BTF.8). This is the map (BPC.11). It does not multiply a characteristic- p scalar into the characteristic-zero Gauss line (BTF.7); their common projector maps are instead (BTF.6).

For $m \geq 2$, the wild-cancellation orders $p \mid k$ have $K \geq p$, so the specialized original terminal multiple vanishes on every stage admitting this local coefficient map. The sheaf invariant lines calculated above remain as specified by (BTF.7). For $m = 1$, $K = 0$ and $C_{h_0,k} = u_0^k$, there is no factorial loss. Here $N = 2$ and the quadratic Gauss constant $\gamma_\chi = -\chi(2)G_L(\chi, \psi)$ satisfies $\gamma_\chi^2 = \chi(-1)Q$. Indeed complex conjugation gives $\overline{G} = \chi(-1)G$, while $G\overline{G} = Q$, which proves the algebraic equality in its cyclotomic number field and hence in its ℓ -adic realization. Thus at $k = 2pr$ with $c \neq 0$, the exact invariant eigenvalue is $(\chi(-1)Q)^{pr}$, and the coefficient image is the retained nonzero $\overline{u}_0^k$.

Finally, the local coefficient map does not silently extend through new CRT denominators. In the original off-critical quartet branch $\rho = 1/2 + \delta + i\gamma$, $\delta > 0$, the complete degree- k split corner has centres $\lambda_0 = k\rho$ and $\lambda_1 = (k-p)\rho + p(1-\bar{\rho})$ whenever $k \geq p$. Retain its actual terminal CRT polynomial $e(Z) = \sum_r e_r Z^r$ with $e(\lambda_0) = 1$, $e(\lambda_1) = 0$. In its own coefficient ring set

$$\mathcal{D} = \sum_{r \geq 1} e_r \sum_{j=0}^{r-1} \lambda_0^{r-1-j} \lambda_1^j, \quad 1 = p(\rho + \bar{\rho} - 1)\mathcal{D}, \quad p^{-1} = (\rho + \bar{\rho} - 1)\mathcal{D}. \quad (\text{BTF.9})$$

The divided difference telescopes term by term: $(\lambda_0 - \lambda_1)\mathcal{D} = e(\lambda_0) - e(\lambda_1) = 1$. Since $\lambda_0 - \lambda_1 = p(\rho + \bar{\rho} - 1)$, this proves all the formulas. Hence that full stage inverts p and has no unital characteristic- p specialization. Replacing p by any rational prime $q \leq k$ proves that the degree- k split corner inverts every such prime. The directed coefficient union of all these retained stages therefore contains $\mathbb{Q}$: each reciprocal prime occurs at its corresponding stage. The explicit local maps (BPC.3), (BPC.8), (BPC.11) and (BPC.14) continue to relate the local characteristic-zero source, its lattice and its valid coefficient specialization. The absence of an extension of that specialization is not the zero image of a vector under a nonexistent map.

The depth and boundary character can be carried together on those same local maps. Define $D_c T^r = \mathbf{1}_{\{r+k \equiv c \pmod{N}\}} T^r$ and use the full-unit character masks $\widehat{Q}_c$ in (BTP.3). Every term in V_r has degree r , so

$$\widehat{Q}_c f_U = f_U D_c, \quad \widehat{Q}_c W_r = \mathbf{1}_{\{r+k \equiv c\}} W_r, \quad \text{Ann}_{\mathcal{S}}([W_r]) = p^{d_r} \mathcal{S}. \quad (\text{BTF.10})$$

These masks preserve both lattices and their free resolution, so the Tor connecting map in (BPC.14) has the same character: $\overline{D}_c \partial \epsilon_r = \mathbf{1}_{\{r+k \equiv c\}} \overline{T}^r$. For $k = p\ell$, $m \geq 2$, its surviving reduced image vector $\overline{W}_{p-1}$ has the exact calculation

$$\begin{aligned} \overline{J} \overline{W}_{p-1} &= 0, \quad \overline{W}_{p-1} \neq 0, \quad J W_{p-1} = p W_p \neq 0, \\ \widetilde{\overline{Q}}_0 \overline{W}_{p-1} &= \overline{W}_{p-1} \iff p(\ell+1) \equiv 1 \pmod{N}. \end{aligned} \quad (\text{BTF.11})$$

Equation (BPC.6) and the primitive pivot prove the first three assertions. Its degree is $p-1$, so (BTF.10) gives the congruence. Since $p > N$, it selects an infinite positive residue class of ℓ . The precursor $p W_p$ is retained, so this new reduced kernel direction has not been identified with the

original complex terminal eigenclass. Every coefficient map above acts at each existing nonbottom split-support label without changing that label; a killed coefficient remains its represented zero, and the external absent symbol is fixed separately. Thus this update propagates the arithmetic exponent, tensor boundary projection and coefficient depth on the same original maps.

135 The actual amplified class and the exact phase exclusion error

The calculation here identifies the precise quantity in the proved holonomy and quotient comparisons that would have to decrease after division by tensor degree. It uses the actual divisor of $g = 2\xi$, the original tensor source, the full Taylor unit, and the original phase generator. Its estimates are calculations on those objects. The tensor amplification itself and its nonzero arithmetic observation are retained throughout.

135.1 The full amplified primary factor and its nonzero observation

Retain an actual off-line zero $\rho = \frac{1}{2} + \delta + i\gamma$ of $g = 2\xi$, of full order m , where $0 < \delta < \frac{1}{2}$ and $\gamma > 0$, and its full reflected quartet divisor

$$h(s) = \prod_{\epsilon, \zeta \in \{-1, 1\}} (s - \frac{1}{2} - \epsilon\delta - i\zeta\gamma)^m. \quad (\text{PAM.1})$$

For $k \geq 2$ put

$$\begin{aligned} c_k &= k/2, & d_k &= 1 + k(m-1), & q_k &= d_k(k+1)^2, \\ \chi_k(S) &= \prod_{a, b=0}^k (S - c_k - (2a - k)\delta - i(2b - k)\gamma)^{d_k}, \\ E_k &= \mathbb{C}[S]/(\chi_k), & A_k &= M_S, & \lambda_k &= k\rho = c_k + k\delta + ik\gamma. \end{aligned} \quad (\text{PAM.2})$$

These are precisely (HT1)–(HT2). The full original injection is

$$\eta_k[P] = v_h^{\otimes k} P(A_h^{(k)})1, \quad v_h = j_h(g/h) \in (\mathbb{C}[s]/h)^\times, \quad \mathcal{I}_k = \sigma_h^{\otimes k} \eta_k : E_k \hookrightarrow Q_{\text{ar}}^{\otimes k}, \quad (\text{PAM.3})$$

where $Q_{\text{ar}} = \mathcal{B}/\Theta V$ is the original arithmetic cohomology. The tensor-sum operator is $A_h^{(k)} = \sum_{j=1}^k A_{h,j}$, and the original source section σ_h and jet remainder j_h have their unchanged meanings.

The factor of E_k at λ_k is exactly

$$E_{k, \lambda_k} = e_{\lambda_k} E_k \simeq \mathbb{C}[z]/(z^{d_k}), \quad A_k|_{E_{k, \lambda_k}} = \lambda_k I + N_k, \quad N_k = M_z. \quad (\text{PAM.4})$$

Here e_{λ_k} is the Chinese remainder idempotent equal to one modulo $(S - \lambda_k)^{d_k}$ and zero modulo every other factor. Distinct pairs (a, b) have distinct complex centers since $\delta, \gamma > 0$; hence these factors are relatively prime and the idempotent exists by polynomial Bezout identities. In particular $v_k = e_{\lambda_k} (S - \lambda_k)^{d_k-1}$ is a nonzero eigenvector with eigenvalue λ_k . Its arithmetic class $\mathcal{I}_k v_k$ is nonzero by (PAM.3).

For explicit multiplicity preservation, on the tensor product of the ρ -primary factors, write its nilpotents as $z_1, \dots, z_k$, each with $z_j^m = 0$. Their sum has

$$(z_1 + \dots + z_k)^{k(m-1)} = \frac{(k(m-1))!}{((m-1)!)^k} z_1^{m-1} \dots z_k^{m-1} \neq 0, \quad (z_1 + \dots + z_k)^{k(m-1)+1} = 0. \quad (\text{PAM.5})$$

The first equality follows from the multinomial expansion: every surviving exponent is at most $m-1$ and their sum is $k(m-1)$, so they all equal $m-1$. Its coefficient is a nonzero complex integer. In the next power at least one exponent is at least m . Multiplication by the retained unit $v_h^{\otimes k}$ preserves both equalities and nonvanishing. Thus the eigenvector in (PAM.4) and every lower jet lie in the actual amplified factor; no reduced-root replacement is involved.

135.2 The exact phase generator and its boundary on that class

Use the original density

$$w_h(u) = \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \int_{\mathbb{R}} m_{h,k}(u) du = \mu_h^k, \quad \mu_h = \int_{\mathbb{R}} w_h(u) du. \quad (\text{PAM.6})$$

For $L > 0, 0 \leq \theta < 2\pi$, put

$$u_n = \frac{2\pi n + \theta}{L}, \quad \nu_{L,\theta} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}. \quad (\text{PAM.7})$$

The original half-density map and twisted periodization in (HT4)–(HT5) give the exact polynomial source $\mathcal{V}_{L,\theta} = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}$. Its inner product is integration of $P(c_k + iu)Q(c_k + iu)$ against (PAM.7). The relative-variable space is $\mathcal{K}_k = L^2(\mathbb{R}^{k-1}, dz)$, and the phase Hilbert space and generator are

$$\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K}_k), \quad D_{L,\theta} = -\partial_r + c_k, \quad (\text{PAM.8})$$

$$\text{Dom}(D_{L,\theta}) = \{f \in H^1([0, L]; \mathcal{K}_k) : f(L) = e^{-i\theta} f(0)\}.$$

Integration by parts cancels the two endpoint products, since both vectors have the same multiplier of modulus one. Consequently $-\partial_r$ is skew-adjoint on this domain, and $D_{L,\theta} + D_{L,\theta}^* = kI$. The Fourier modes $L^{-1/2}e^{-iu_n r}$ give its eigenvalues $c_k + iu_n$, acting identically on $\mathcal{K}_k$.

For an admitted degree $N \geq q_k - 1$, retain the original Gram $M_{N,\theta}$, remainder $J_N : \mathbb{C}[S]_{\leq N} \rightarrow E_k$, minimum coefficient section and arithmetic metric

$$C_{N,\theta} = M_{N,\theta}^{-1} J_N^* G_{N,\theta}, \quad G_{N,\theta} = (J_N M_{N,\theta}^{-1} J_N^*)^{-1}, \quad R_{N,\theta} = \mathcal{V}_{L,\theta} C_{N,\theta}. \quad (\text{PAM.9})$$

All these inverse matrices exist by the positive lattice weights and the surjectivity of literal remainder. They obey $J_N C_{N,\theta} = I$ and $R_{N,\theta}^* R_{N,\theta} = G_{N,\theta} > 0$. In particular $r_k = R_{N,\theta} v_k \neq 0$.

The exact generator comparison is the original polynomial division

$$S C_{N,\theta} - C_{N,\theta} A_k = \chi_k Q_{N,\theta}, \quad \mathcal{B}_{N,\theta} = \mathcal{V}_{L,\theta}(\chi_k Q_{N,\theta}) = D_{L,\theta} R_{N,\theta} - R_{N,\theta} A_k. \quad (\text{PAM.10})$$

It is relation-valued because applying J_{N+1} to the first difference gives zero. Its coefficient quotient by the monic χ_k is unique. Its original theta primitive, with its signs, is already computed in (HT13): if tensor division gives $\chi_k Q = \sum_j h(s_j) Q_j(\mathbf{s})$, it is

$$\sum_{j=1}^k (-1)^{j-1} Q_j(D_1, \dots, D_k) (F_h^{\otimes(j-1)} \otimes \phi_h \otimes F_h^{\otimes(k-j)}). \quad (\text{PAM.11})$$

Here $h(D)F_h = \Theta \phi_h$; F_h has degree one and ϕ_h degree zero. Applying the tensor differential adds $(-1)^{j-1}$, canceling the displayed sign and producing exactly the boundary of (PAM.10) after the same half-density and phase maps.

Apply (PAM.10) to v_k . The cross term between a real scalar and a skew-adjoint operator has real part zero. Parseval therefore proves the exact identity

$$\begin{aligned} \frac{\|\mathcal{B}_{N,\theta} v_k\|^2}{\|R_{N,\theta} v_k\|^2} &= (k\delta)^2 + \frac{\|(-\partial_r - ik\gamma) R_{N,\theta} v_k\|^2}{\|R_{N,\theta} v_k\|^2} \\ &= (k\delta)^2 + \frac{\sum_{n \in \mathbb{Z}} (u_n - k\gamma)^2 \|\widehat{r}_k(n)\|_{\mathcal{K}_k}^2}{\sum_{n \in \mathbb{Z}} \|\widehat{r}_k(n)\|_{\mathcal{K}_k}^2}. \end{aligned} \quad (\text{PAM.12})$$

Indeed $\mathcal{B} v_k = (D_{L,\theta} - \lambda_k) r_k$, and this operator is $-k\delta + (-\partial_r - ik\gamma)$. Every source vector belongs to its domain by the original weighted smoothness estimates; the Fourier identities therefore hold with convergent squared sums. In particular the error forced on this actual eigenclass obeys

$$\frac{\|\mathcal{B}_{N,\theta} v_k\|}{\|R_{N,\theta} v_k\|} \geq \sqrt{k^2 \delta^2 + \text{dist}(k\gamma, \{u_n : n \in \mathbb{Z}\})^2} \geq k\delta. \quad (\text{PAM.13})$$

Changing θ can place a lattice point exactly at $k\gamma$; this removes that particular lattice-distance term and leaves the term $k\delta$ unchanged. Thus lattice refinement or choice of phase does not divide the real displacement by k .

135.3 The full-jet inverse of the boundary comparison

For the entire primary factor (PAM.4), the relation and source are connected by an exact finite resolvent formula. Put $\mathcal{R} = R_{N,\theta}|_{E_{k,\lambda_k}}$ and $\mathcal{B} = \mathcal{B}_{N,\theta}|_{E_{k,\lambda_k}}$. The resolvent $(D_{L,\theta} - \lambda_k)^{-1}$ exists and satisfies

$$\|(D_{L,\theta} - \lambda_k)^{-1}\|_{\text{op}} = \frac{1}{\sqrt{k^2\delta^2 + \text{dist}(k\gamma, \{u_n\})^2}} \leq \frac{1}{k\delta}. \quad (\text{PAM.14})$$

Its exact Fourier multiplier is $(-k\delta + i(u_n - k\gamma))^{-1}$. Its absolute values have the displayed maximum. The inverse maps into the stated H^1 domain: the multipliers $u_n/(-k\delta + i(u_n - k\gamma))$ are bounded, so the Fourier derivative has finite squared norm.

Writing (PAM.10) on the full block gives $(D_{L,\theta} - \lambda_k)\mathcal{R} - \mathcal{R}N_k = \mathcal{B}$. Multiply by the resolvent, substitute the resulting expression for $\mathcal{R}$ repeatedly, and use $N_k^{d_k} = 0$. This gives the exact morphism, with every nilpotent term retained,

$$\mathcal{R} = \sum_{j=0}^{d_k-1} (D_{L,\theta} - \lambda_k)^{-j-1} \mathcal{B} N_k^j. \quad (\text{PAM.15})$$

All factors have specified domains: N_k^j acts on the finite primary algebra, $\mathcal{B}$ maps it to $\mathcal{H}_{L,\theta}$, and the resolvent maps that Hilbert space to itself and into the generator domain. Finite iteration is therefore legitimate. Conversely multiplying this finite sum by $D_{L,\theta} - \lambda_k$ and subtracting its right product with N_k cancels every term except $\mathcal{B}$. It is consequently the unique solution: the same iteration with zero $\mathcal{B}$ makes any homogeneous solution zero. In particular the already nonzero arithmetic primary block cannot have zero generator boundary on this phase Hilbert space.

135.4 The scaling of the proved global phase controls

For brevity in this subsection write $G = G_{N,\theta}$, $\mathcal{B} = \mathcal{B}_{N,\theta}$ and $q = q_k$. The proved rank-two matrix and its exact positive spectral floor (HT8)–(HT12) are

$$H = G^{-1/2}(A_k^*G + GA_k - kG)G^{-1/2}, \quad \text{spec}(H) = \{\varepsilon, -\varepsilon, 0, \dots, 0\},$$

$$\varepsilon \geq \mathcal{L}_{h,k} = 2\delta d_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{PAM.16})$$

These statements include every phase and every admitted degree. Their original energy identity is

$$\|\mathcal{B}G^{-1/2}\|_{\text{HS}}^2 = \frac{\varepsilon^2}{2} + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_{\perp}\|_{\text{HS}}^2, \quad H = -(Z + Z^*), \quad (\text{PAM.17})$$

where $U = RG^{-1/2}$, $T = \mathcal{B}G^{-1/2}$, $Z = U^*T$, and $T_{\perp} = (I - UU^*)T$, with $U^*U = I$. Thus both actual norm choices for a possible exclusion error have the exact lower estimates

$$\|\mathcal{B}G^{-1/2}\|_{\text{op}} \geq \frac{\varepsilon}{2} \geq \frac{\mathcal{L}_{h,k}}{2}, \quad \|\mathcal{B}G^{-1/2}\|_{\text{HS}} \geq \frac{\mathcal{L}_{h,k}}{\sqrt{2}}. \quad (\text{PAM.18})$$

The first follows from $\varepsilon = \|H\|_{\text{op}} \leq 2\|Z\|_{\text{op}} \leq 2\|T\|_{\text{op}}$; the second is immediate from (PAM.17).

The precise scaling, including the full multiplicity, is

$$\frac{\mathcal{L}_{h,k}}{kq_k} = \frac{2\delta \lfloor (k+1)^2/4 \rfloor}{k(k+1)} = \begin{cases} \frac{\delta(k+2)}{2(k+1)}, & k \text{ even}, \\ \frac{\delta(k+1)}{2k}, & k \text{ odd}. \end{cases} \rightarrow \frac{\delta}{2}. \quad (\text{PAM.19})$$

The equalities follow respectively from $\lfloor (k+1)^2/4 \rfloor = k(k+2)/4$ and $(k+1)^2/4$. In particular even the rank-divided operator lower bound satisfies

$$\frac{\|\mathcal{B}G^{-1/2}\|_{\text{op}}}{kq_k} \geq \frac{\mathcal{L}_{h,k}}{2kq_k}, \quad \liminf_{k \rightarrow \infty} \frac{\|\mathcal{B}_{N_k, \theta_k} G_{N_k, \theta_k}^{-1/2}\|_{\text{op}}}{kq_k} \geq \frac{\delta}{4}, \quad (\text{PAM.20})$$

where the limit inferior allows any admitted degrees N_k , periods $L_k > 0$, and phases θ_k . The Hilbert-Schmidt lower bound divided by kq_k tends to $\delta/(2\sqrt{2})$. Without division by q_k , the proved lower-bound functions for either norm divided by k diverge: $q_k = (k+1)^2$ for $m = 1$, and $q_k = (1+k(m-1))(k+1)^2$ for $m > 1$. For the root-mean-square quantity $\|\mathcal{B}G^{-1/2}\|_{\text{HS}}/\sqrt{q_k}$, its proved lower bound divided by k likewise diverges at least as $\delta\sqrt{q_k}/(2\sqrt{2})$. These claims concern the explicitly displayed lower-bound functions; they do not assign unproved asymptotic equalities to the actual energies.

The available upper comparison (HC12) retains two additional actual source quantities:

$$\|\mathcal{B}G_N^{-1/2}\|_{\text{HS}} \leq (1 + \sqrt{\beta_N})\sqrt{J_N^D}, \quad J_N^D = \|D_{L, \theta} R_N G_N^{-1/2}\|_{\text{HS}}^2, \quad \beta_N = \frac{V_N}{V_{N+1}} \geq 1. \quad (\text{PAM.21})$$

Here the equality for β_N is the proved single-rank update (HC12a), and $V_N = \det G_N$. Integration by parts in (PAM.8) gives the exact identity

$$J_N^D = c_k^2 q_k + \|\partial_r R_N G_N^{-1/2}\|_{\text{HS}}^2 \geq \frac{k^2 q_k}{4}. \quad (\text{PAM.22})$$

Consequently the numerical right-hand side of (PAM.21), divided by k , is at least $\sqrt{q_k}$; it is not an $o(k)$ exclusion error. After dividing that upper expression by kq_k , its elementary floor $1/\sqrt{q_k}$ does tend to zero, but (PAM.18)–(PAM.19), combined with the same true upper inequality, give instead

$$\frac{(1 + \sqrt{\beta_N})\sqrt{J_N^D}}{kq_k} \geq \frac{\mathcal{L}_{h,k}}{\sqrt{2} kq_k}, \quad (\text{PAM.23})$$

whose right side tends to $\delta/(2\sqrt{2}) > 0$. Thus the actual derivative and contraction factors in the already proved upper expression cannot simultaneously create a vanishing error at this divisor. This is an exact consequence of their proved relation, not permission to choose either factor freely.

135.5 What the computed quotient repairs do to the amplified generator

The QT construction can be performed on the identical finite phase Gram at $N = 2q_k$: (PAM.7) has all positive polynomial moments, so all the Gram-Schmidt lists and projection inverses defining the original four windows and their isometry remain well defined. Write $F = \chi_k \mathbb{C}[S]_{\leq q_k}$, $K = F^\perp$, $\Pi = P_K$, $Q = P_F$, and $C = C_{2q_k, \theta} : E_k \rightarrow K$. Here $H_{\text{pol}} = \mathbb{C}[S]_{\leq 2q_k}$ is the polynomial source, and $R = \mathcal{V}_{L, \theta} C$. The repaired source map and its exact cochain homotopy are

$$X = QT + \Pi, \quad JX = J, \quad X - I = BK_X, \quad K_X = (B^* M B)^{-1} B^* M (T - I), \quad B = \times \chi_k. \quad (\text{PAM.24})$$

These are the explicit QT.7–QT.8 and QT.27–QT.28 formulas on that original phase form. Its new representative is

$$R_X = R + \mathcal{H}, \quad \mathcal{H} = \mathcal{V}_{L, \theta}(BK_X C), \quad \mathcal{B}_X = \mathcal{B} + (D_{L, \theta} \mathcal{H} - \mathcal{H} A_k). \quad (\text{PAM.25})$$

The first equality follows by applying $\mathcal{V}_{L, \theta}$ to $XC = C + BK_X C$; substituting it into $\mathcal{B}_X = DR_X - R_X A_k$ proves the second with both signs. The same full-unit observation is unchanged: $\mathcal{I}_k JXC = \mathcal{I}_k$. The square-zero defect in QT.19 concerns the four-window operators U, W ; (PAM.25) is its separately calculated relation to the actual generator A_k . No equality between those two operators has been assigned.

The exact norm of this new representative is

$$G_X = R_X^* R_X = 2G - GG_{\text{low}}^{-1}G, \quad G \preceq G_X \preceq 2G, \quad G_{\text{low}} = G_{q_k-1, \theta}. \quad (\text{PAM.26})$$

Indeed $XC = C + QTC$ has orthogonal summands; the isometry of T and the full metric formula QT.12 give $(QTC)^*M(QTC) = G - GG_{\text{low}}^{-1}G$. Minimum-norm monotonicity gives $0 \preceq GG_{\text{low}}^{-1}G \preceq G$, proving the bounds. Thus $R_X v_k \neq 0$, and the direct calculation (PAM.12) applies to $R_X, \mathcal{B}_X$ with its own unaltered Gram G_X . It again gives $\|\mathcal{B}_X v_k\|/\|R_X v_k\| \geq k\delta$. The exact closest observation-preserving unitary in VR12 is $V_{\text{obs}} = I_K \oplus \text{polar}(T|_{F \rightarrow F})$. It fixes $C(E_k) \subset K$ pointwise, so

$$\mathcal{V}_{L, \theta} V_{\text{obs}} C = R, \quad D_{L, \theta} \mathcal{V}_{L, \theta} V_{\text{obs}} C - \mathcal{V}_{L, \theta} V_{\text{obs}} C A_k = \mathcal{B}. \quad (\text{PAM.27})$$

This is an exact equality of the original source representatives and their boundaries, not only an equality of cohomology classes.

Finally, the QT source-variation error is $e(t)^2 = -\frac{1}{2} \text{Tr}(G^{-1} \ddot{G}) = \text{Tr}(G^{-1} \dot{C}^* M \dot{C})$. It measures change of the representative with an interpolated source metric. On the constant path equal to the same actual phase Gram, $\dot{C} = 0$, its signed correction is zero, and its bound QT.24 is exactly zero. The generator boundary (PAM.10) and its strictly positive error (PAM.13) are unchanged on that constant path. This explicitly computes why that signed source-variation error cannot be substituted for the generator error in (PAM.13). The original Gamma-to-arithmetic interpolation retains its own nonconstant error and its exact QT bounds.

135.6 The exact cancellation inside the original derivative energy

Retain the objects and hypotheses of (PAM.1)–(PAM.10). In this subsection fix k, L, θ , put $c = c_k = k/2$, $q = q_k$, $A = A_k$, and use this same phase on every degree-dependent quantity. Write $D = D_{L, \theta} = -\partial_r + cI$ on the domain (PAM.8). The boundary in (PAM.10) has the exact second expression

$$\mathcal{B}_N = DR_N - R_N A = (D - cI)R_N - R_N(A - cI). \quad (\text{PSC.1})$$

Indeed the two new scalar terms are $-cR_N + cR_N = 0$. The original operators D, A , their scalar $c = k/2$, and the original boundary are all retained in this equality.

Define the actual two derivative energies and the original one-degree contraction by

$$\begin{aligned} J_N^\partial &= \|(D - cI)R_N G_N^{-1/2}\|_{\text{HS}}^2, & J_N^D &= \|DR_N G_N^{-1/2}\|_{\text{HS}}^2, \\ \mathcal{E}_N &= \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}}^2, & \beta_N &= \lambda_{\max}(G_{N+1}^{-1/2} G_N G_{N+1}^{-1/2}) = \frac{V_N}{V_{N+1}} \geq 1, \quad V_N = \det G_N. \end{aligned} \quad (\text{PSC.2})$$

The last equality is the original rank-one update (HC12a), on this same phase measure. Since $D - cI = -\partial_r$ is skew-adjoint on the stated twisted domain, its scalar cross term has real part zero. Applying that equality to each column of $R_N G_N^{-1/2}$ gives

$$J_N^\partial = J_N^D - c^2 q > 0. \quad (\text{PSC.3})$$

Here the strict positivity also follows directly from the source. A nonzero column is the periodized image of a nonzero polynomial P , since $J_N C_{N, \theta} = I_E$. The norm of its derivative is the sum of the positive lattice weights times $u_n^2 |P(c + iu_n)|^2$. There are infinitely many nonzero u_n , and a nonzero polynomial cannot vanish at all their distinct values $c + iu_n$. Thus each nonzero source column has positive derivative norm. No lattice term, including a possible $u_n = 0$, has been deleted.

The vector $(D - cI)R_N x$ is an original degree- $(N + 1)$ representative of $(A - cI)x$. Indeed its coefficient polynomial is $(S - c)C_{N, \theta} x$, and its full remainder is $(A - cI)x$. The degree- $(N + 1)$ minimum therefore proves the Hermitian inequality

$$(A - cI)^* G_{N+1} (A - cI) \preceq R_N^* (D - cI)^* (D - cI) R_N. \quad (\text{PSC.4})$$

This is an inequality between the original forms on E ; the source map, the full remainder, and its unit-valued arithmetic inclusion in (PAM.3) have not changed. From $G_N \preceq \beta_N G_{N+1}$, sandwiching (PSC.4) by $G_N^{-1/2}$ and taking traces gives

$$\|R_N(A - cI)G_N^{-1/2}\|_{\text{HS}}^2 \leq \beta_N J_N^\partial.$$

The triangle inequality in (PSC.1), combined with (HT14), now gives the sharper actual energy bound

$$\frac{\mathcal{L}_{h,k}^2}{2} \leq \mathcal{E}_N \leq (1 + \sqrt{\beta_N})^2 (J_N^\partial - c^2 q) = (1 + \sqrt{\beta_N})^2 J_N^\partial. \quad (\text{PSC.5})$$

The improvement over (HC12) is the exact cancelling contribution $c^2 q$ in the same measured derivative energy. It is not a change of metric or a choice of a new representative.

The contraction identity $e_N = \log \beta_N$, with $e_N = \log(V_N/V_{N+1})$, then implies

$$e_N \geq 2 \log \max \left\{ 1, \frac{\mathcal{L}_{h,k}}{\sqrt{2J_N^\partial}} - 1 \right\}, \quad (\text{PSC.6})$$

$$\mathcal{B}_{k,\theta} \geq 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}_{h,k}}{\sqrt{2J_N^\partial}} - 1 \right\}, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q-2).$$

For the first line, divide (PSC.5) by $J_N^\partial > 0$, take nonnegative square roots, and use $\sqrt{\beta_N} \geq 1$. For the second, put $\mathcal{B}_{k,\theta} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$; its original four endpoint logarithms telescope to $\mathcal{B}_{k,\theta} = \sum_N w_N e_N$. Thus the strengthened energy enters exactly the same windows.

135.7 A finite moment upper expression for the same generator error

Put $P_N = \mathbb{C}[S]_{\leq N}$. Let $\mathbb{L}_N : P_N \rightarrow P_{N+1}$ be the literal coefficient map $P(S) \mapsto (S - c)P(S)$. Define

$$W_N = \mathbb{L}_N^* M_{N+1,\theta} \mathbb{L}_N, \quad \Gamma_N = \lambda_{\max}(M_{N,\theta}^{-1/2} W_N M_{N,\theta}^{-1/2}),$$

$$P^* W_N P = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} u_n^2 m_{h,k}(u_n) |P(c + iu_n)|^2, \quad 0 < \Gamma_N < \infty. \quad (\text{PSC.7})$$

The quadratic-form identity follows from $S - c = iu$ on the original sampled line. Its strict positivity is the same nonvanishing-polynomial argument used for (PSC.3). All moments through degree $2N+2$ are finite by the original source estimates; the finite positive matrix $M_{N,\theta}$ has an inverse, proving finiteness of Γ_N . The full factor $2\pi/L$ and the original mass of $m_{h,k}$ remain in both matrices.

From $W_N \preceq \Gamma_N M_{N,\theta}$ and $C_{N,\theta}^* M_{N,\theta} C_{N,\theta} = G_N$, one gets

$$\|(D - cI)R_N G_N^{-1/2}\|_{\text{op}} \leq \sqrt{\Gamma_N}, \quad J_N^\partial \leq q\Gamma_N.$$

Keeping (PSC.4) as a Hermitian inequality, rather than only taking its trace, likewise gives $\|R_N(A - cI)G_N^{-1/2}\|_{\text{op}} \leq \sqrt{\beta_N \Gamma_N}$. Consequently

$$\frac{\mathcal{L}_{h,k}}{2} \leq \|\mathcal{B}_N G_N^{-1/2}\|_{\text{op}} \leq (1 + \sqrt{\beta_N}) \sqrt{\Gamma_N},$$

$$\frac{\mathcal{L}_{h,k}}{\sqrt{2}} \leq \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}} \leq (1 + \sqrt{\beta_N}) \sqrt{J_N^\partial}, \quad (\text{PSC.8})$$

$$\liminf_{k \rightarrow \infty} \frac{(1 + \sqrt{\beta_{N_k}}) \sqrt{\Gamma_{N_k}}}{k q_k} \geq \frac{\delta}{4}, \quad \liminf_{k \rightarrow \infty} \frac{(1 + \sqrt{\beta_{N_k}}) \sqrt{J_{N_k}^\partial}}{k q_k} \geq \frac{\delta}{2\sqrt{2}}.$$

The first lower bound is (PAM.18); the second is (HT14). The two limit inferior statements hold for every choice of admitted integers $N_k \geq q_k - 1$, periods $L_k > 0$, and phases $0 \leq \theta_k < 2\pi$, with every quantity on its indicated same phase. They follow from the pointwise bounds and the exact parity-dependent limit (PAM.19). In particular neither computed upper expression is $o(k)$: each divided by k tends to infinity at least as its displayed positive multiple of q_k . This statement concerns the upper expressions actually linked to the original generator by (PSC.1)–(PSC.7).

135.8 Original spaces and the actual source at both degrees

Fix a finite original packet h , a tensor order $k \geq 2$, and the exact original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q > 0$. Retain

$$E = \mathbb{C}[S]/(\chi), \quad A = M_S : E \longrightarrow E, \quad P_N = \mathbb{C}[S]_{\leq N}, \quad N \geq q - 1, \quad J_N = \pi_\chi|_{P_N}. \quad (\text{PSC.9})$$

The source and its half-density version are precisely

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad \mathcal{M}F_h = g/h, \quad \Psi_N = \mathcal{U}_k \mathcal{V}|_{P_N}. \quad (\text{PSC.10})$$

Here $D_i = -x_i \partial_{x_i}$, $r = \log x_k$, $z_i = \log x_i - \log x_k$, and

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r).$$

The target is $L^2(\mathbb{R}, dr; \mathcal{K})$, with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$. Its relative variables are retained. The original full-jet map is the injection

$$\eta_{H8}[P] = j_h(g/h)^{\otimes k} P(A_h^{(k)})1, \quad J^{(k)} \mathcal{V}P = \eta_{H8} \pi_\chi P. \quad (\text{PSC.11})$$

Here $E_h = \mathbb{C}[s]/(h)$, $A_h = M_s : E_h \rightarrow E_h$, and $A_h^{(k)} = \sum_{i=1}^k A_{h,i}$ acts on $E_h^{\otimes k}$. Its restriction to the displayed cyclic image agrees with the original $A = M_S$. The subscript merely identifies the original map called η in H8; the scalar source error below is the η of H32. No unit factor is removed from (PSC.11).

Write $\iota_N : P_N \hookrightarrow P_{N+1}$ for inclusion and $S_N : P_N \rightarrow P_{N+1}$, $P \mapsto SP$, for multiplication before reduction. These have different codomains from A , and satisfy

$$J_{N+1} S_N = A J_N, \quad J_{N+1} \iota_N = J_N, \quad \partial_r \Psi_N = \Psi_{N+1} ((k/2) \iota_N - S_N). \quad (\text{PSC.12})$$

The last formula follows by applying the exact H10 identity $\mathcal{U}_k D^{(k)} = (-\partial_r + k/2) \mathcal{U}_k$ to each polynomial column. It is why the comparison below uses a source bound at degree $N + 1$.

For $L > 0$ define

$$W_{N,\theta} = \mathcal{Z}_{L,\theta} \Psi_N, \quad M_N = \Psi_N^* \Psi_N, \quad M_{N,\theta} = W_{N,\theta}^* W_{N,\theta}. \quad (\text{PSC.13})$$

The phase source is in $\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K})$. Its generator is

$$D_{L,\theta} = -\partial_r + k/2, \quad \text{Dom } D_{L,\theta} = \{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}.$$

All columns used here belong to this domain. The original weighted source bounds justify the periodization and its r -derivative. In particular no derivative in θ is involved when $D_{L,\theta}$ is applied to a coefficient map depending on θ .

The complete phase transform gives

$$\int_0^{2\pi} M_{N,\theta} \frac{d\theta}{2\pi} = M_N. \quad (\text{PSC.14})$$

135.9 The exact relation map between the two sections

Let $B_N : \mathbb{C}^{N-q+1} \hookrightarrow P_N$ be the coefficient matrix of literal multiplication by χ , with zero-dimensional domain if $N = q - 1$. Thus $\text{im } B_N = \chi P_{N-q} = \ker J_N$. Keep the original sections

$$\begin{aligned} G &= (J_N M_N^{-1} J_N^*)^{-1}, & C &= M_N^{-1} J_N^* G, \\ G_\theta &= (J_N M_{N,\theta}^{-1} J_N^*)^{-1}, & C_\theta &= M_{N,\theta}^{-1} J_N^* G_\theta. \end{aligned} \quad (\text{PSC.15})$$

Since $J_N C = J_N C_\theta = I_E$, the difference $Z_\theta := C - C_\theta : E \rightarrow P_N$ has image in $\ker J_N$. Hence there is a unique polynomial coefficient map $X_\theta : E \rightarrow P_{N-q}$ such that

$$Z_\theta = \chi X_\theta = B_N X_\theta, \quad X_\theta = (B_N^* M_{N,\theta} B_N)^{-1} B_N^* M_{N,\theta} C \quad (\text{PSC.16})$$

in the fixed coefficient bases. An inverse on a zero-dimensional space in this notation is the unique empty map.

Indeed $B_N^* M_{N,\theta} C_\theta = 0$, so multiplying the first identity by $B_N^* M_{N,\theta}$ proves the second. This is precisely the literal original theta-relation difference from H26.

Define maps $E \rightarrow \mathcal{H}_{L,\theta}$ by

$$R_\theta^c = W_{N,\theta} C, \quad R_\theta^p = W_{N,\theta} C_\theta, \quad L_\theta = W_{N,\theta} Z_\theta = R_\theta^c - R_\theta^p. \quad (\text{PSC.17})$$

The superscripts label the common section and phase minimum. The orthogonality from (PSC.15) is the exact map identity $(R_\theta^p)^* L_\theta = 0$. Therefore

$$\begin{aligned} \mathcal{D} &:= \int_0^{2\pi} L_\theta^* L_\theta \frac{d\theta}{2\pi} = \int_0^{2\pi} X_\theta^* B_N^* M_{N,\theta} B_N X_\theta \frac{d\theta}{2\pi}, \\ G &= \overline{G} + \mathcal{D}, \quad \overline{G} := \int_0^{2\pi} G_\theta \frac{d\theta}{2\pi}. \end{aligned} \quad (\text{PSC.18})$$

For clarity, the proof is an expansion: $(R_\theta^c)^* R_\theta^c = G_\theta + L_\theta^* L_\theta$. Integration and (PSC.14) give $C^* M_N C = G$, hence (PSC.18). No cancellation of the norm of a relation has been made.

135.10 Exact generator difference, with its literal primitive

Define the common and phase generator boundaries by

$$\mathcal{B}_\theta^c = D_{L,\theta} R_\theta^c - R_\theta^c A, \quad \mathcal{B}_\theta^p = D_{L,\theta} R_\theta^p - R_\theta^p A. \quad (\text{PSC.19})$$

Both are maps on the original E ; $\mathcal{B}_\theta^p$ is H54. At polynomial level put

$$T_c = S_N C - \iota_N C A, \quad T_{p,\theta} = S_N C_\theta - \iota_N C_\theta A : E \longrightarrow P_{N+1}.$$

Applying J_{N+1} gives zero, so unique monic division yields $T_c = \chi Q_c$ and $T_{p,\theta} = \chi Q_{p,\theta}$, where both $Q_c, Q_{p,\theta}$ map to P_{N+1-q} . Using (PSC.16), their exact difference is

$$Q_c - Q_{p,\theta} = S X_\theta - X_\theta A. \quad (\text{PSC.20})$$

Here $S X_\theta : E \rightarrow P_{N-q+1}$ raises degree before reduction, whereas $X_\theta A$ is composition on E . The calculation is

$$T_c - T_{p,\theta} = S(C - C_\theta) - (C - C_\theta)A = \chi(S X_\theta - X_\theta A);$$

injectivity of multiplication by the nonzero polynomial χ proves (PSC.20).

After applying the same source and phase maps, the full comparison is

$$\begin{aligned}
\Delta \mathcal{B}_\theta &:= \mathcal{B}_\theta^c - \mathcal{B}_\theta^p \\
&= W_{N+1,\theta} (S_N Z_\theta - \iota_N Z_\theta A) \\
&= \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}(\chi(SX_\theta - X_\theta A)) \\
&= D_{L,\theta} L_\theta - L_\theta A \\
&= -\partial_r L_\theta - L_\theta (A - (k/2)I_E).
\end{aligned} \tag{PSC.21}$$

The final line expands the stated generator without changing either A or its marked arithmetic value.

There is an explicit original tensor primitive for every displayed relation. If $Y(S)$ is any of X_θ , Q_c , $Q_{p,\theta}$, or $SX_\theta - X_\theta A$, tensor monic division gives polynomial coefficient maps Y_i with

$$\chi(S)Y(S) = \sum_{i=1}^k h(s_i)Y_i(s_1, \dots, s_k).$$

This holds because the left side has zero image in the full tensor jet algebra, not because the tensor relation ideal was replaced by (χ) . With the original relation $h(D)F_h = \Theta\phi_h$, a degree- $(k-1)$ primitive is

$$\sum_{i=1}^k (-1)^{i-1} Y_i(D_1, \dots, D_k) \left(F_h^{\otimes(i-1)} \otimes \phi_h \otimes F_h^{\otimes(k-i)} \right). \tag{PSC.22}$$

The differential crosses $i-1$ degree-one factors before acting on ϕ_h , supplying a second $(-1)^{i-1}$. The two signs multiply to $+1$, so its differential is exactly $\mathcal{V}(\chi Y)$. Applying $\mathcal{U}_k$ and $\mathcal{Z}_{L,\theta}$ gives the maps in (PSC.17), (PSC.19), and (PSC.21).

135.11 Exact energy identities before any inequality

Introduce the matrices on the original coefficient space E

$$\mathcal{E} = \int (\partial_r L_\theta)^* (\partial_r L_\theta) \frac{d\theta}{2\pi}, \quad \mathcal{K} = \int (\partial_r L_\theta)^* L_\theta \frac{d\theta}{2\pi}. \tag{PSC.23}$$

All integrals in this note range over $0 \leq \theta < 2\pi$. Expanding the last line of (PSC.21), with both signs retained, gives

$$\begin{aligned}
\mathcal{E}_\Delta &:= \int (\Delta \mathcal{B}_\theta)^* \Delta \mathcal{B}_\theta \frac{d\theta}{2\pi} \\
&= \mathcal{E} + (A^* - (k/2)I_E) \mathcal{D} (A - (k/2)I_E) \\
&\quad + \mathcal{K} (A - (k/2)I_E) + (A^* - (k/2)I_E) \mathcal{K}^*.
\end{aligned} \tag{PSC.24}$$

It also has the original finite-moment formula

$$\mathcal{E}_\Delta = \int (S_N Z_\theta - \iota_N Z_\theta A)^* M_{N+1,\theta} (S_N Z_\theta - \iota_N Z_\theta A) \frac{d\theta}{2\pi}. \tag{PSC.25}$$

For the full boundaries, the exact comparison includes its cross term:

$$\begin{aligned}
T_c^* M_{N+1} T_c &= \int (\mathcal{B}_\theta^c)^* \mathcal{B}_\theta^c \frac{d\theta}{2\pi} \\
&= \int (\mathcal{B}_\theta^p)^* \mathcal{B}_\theta^p \frac{d\theta}{2\pi} + \mathcal{E}_\Delta \\
&\quad + \int [(\mathcal{B}_\theta^p)^* \Delta \mathcal{B}_\theta + (\Delta \mathcal{B}_\theta)^* \mathcal{B}_\theta^p] \frac{d\theta}{2\pi}.
\end{aligned} \tag{PSC.26}$$

The first equality is (PSC.14) at degree $N + 1$, because T_c is independent of phase. The remaining equality is expansion of $\mathcal{B}_\theta^c = \mathcal{B}_\theta^p + \Delta \mathcal{B}_\theta$. Orthogonality of R_θ^p and L_θ does not delete the last integral in (PSC.26).

There is also an exact comparison for the adjoint generator forms. Set $\mathfrak{H}(M) = A^*M + MA - kM$, for a Hermitian form M on the same E . Then

$$\boxed{\mathfrak{H}(G) - \mathfrak{H}(\overline{G}) = A^* \mathcal{D} + \mathcal{D} A - k \mathcal{D}.} \quad (\text{PSC.27})$$

This follows by substituting (PSC.18) into the left side. It concerns the difference of the two forms; it does not assert that either full form is small.

135.12 Actual source derivative moments and a uniform comparison

The source Gram for its derivative is the finite, explicitly specified matrix

$$\begin{aligned} Q_N &:= (\partial_r \Psi_N)^* (\partial_r \Psi_N) \\ &= ((k/2)\iota_N - S_N)^* M_{N+1} ((k/2)\iota_N - S_N). \end{aligned} \quad (\text{PSC.28})$$

In the original coefficient basis $P_a(S)$, its entries are exactly

$$\begin{aligned} (Q_N)_{ab} &= \int_{\mathbb{R}} u^2 \overline{P_a(k/2 + iu)} P_b(k/2 + iu) m_{h,k}(u) du, \\ m_{h,k} &= w_h^{*k}, \quad w_h(u) = \frac{|(g/h)(1/2 + iu)|^2}{2\pi}. \end{aligned} \quad (\text{PSC.29})$$

Indeed the derivative has Fourier multiplier $-iu$, while (H17) and the continuous source norm retain precisely $m_{h,k}$. Its squared absolute value is u^2 , proving every entry in (PSC.29). The full vector-valued source remains in (PSC.28); (PSC.29) evaluates its Gram.

Define the exact finite number

$$d_N = \sup_{v \neq 0} \frac{v^* Q_N v}{v^* M_N v}. \quad (\text{PSC.30})$$

It exists because $M_N \succ 0$, Q_N is finite, and the finite-dimensional unit sphere for M_N is compact. Thus $Q_N \preceq d_N M_N$ is a proved statement, not an additional arithmetic hypothesis. H50's original derivative matrix satisfies

$$Q_N \preceq H_{1,N} = \int_{\mathbb{R}} (1 + r^2) (\partial_r \Psi_N)^* (\partial_r \Psi_N) dr. \quad (\text{PSC.31})$$

Consequently the original $h_{1,N}$ of H51 can be used in every occurrence of d_N below if its bound has been supplied. Formula (PSC.30), using (PSC.28)–(PSC.29), is the sharper exact source moment used here.

Choose $a > 0$ and retain the original two weighted source Grams at degree $N + 1$:

$$M_{a,\pm,N+1} = (e^{\pm ar} \Psi_{N+1})^* (e^{\pm ar} \Psi_{N+1}).$$

The weighted column maps take the finite coefficient space P_{N+1} into the same $L^2(\mathbb{R}, dr; \mathcal{K})$; their squared weights are the original coordinate factors $x_k^{\pm 2a}$. Define their actual finite cost

$$\kappa_{a,N+1} = \sup_{v \neq 0} \frac{v^* (M_{a,+,N+1} + M_{a,-,N+1}) v}{v^* M_{N+1} v}, \quad \eta = \frac{\kappa_{a,N+1}}{e^{aL} - 1} < 1. \quad (\text{PSC.32})$$

The original weighted Schwartz estimates make both weighted Grams finite for the chosen finite source. H41 therefore gives, with no new phase hypothesis,

$$(1 - \eta) M_{N+1} \preceq M_{N+1,\theta} \preceq (1 + \eta) M_{N+1}. \quad (\text{PSC.33})$$

Restricting gives the same comparison at degree N . H33, whose hypotheses are now supplied by the actual source, gives

$$0 \preceq \mathcal{D} \preceq \eta^2 G. \quad (\text{PSC.34})$$

For each $x \in E$, apply (PSC.33) to $((k/2)\iota_N - S_N)Z_\theta x$, then (PSC.30), then the lower degree- N bound:

$$\begin{aligned} \|\partial_r L_\theta x\|^2 &\leq (1 + \eta)x^* Z_\theta^* Q_N Z_\theta x \\ &\leq (1 + \eta)d_N x^* Z_\theta^* M_N Z_\theta x \\ &\leq \frac{1 + \eta}{1 - \eta} d_N x^* Z_\theta^* M_{N,\theta} Z_\theta x \\ &= \frac{1 + \eta}{1 - \eta} d_N \|L_\theta x\|^2. \end{aligned} \quad (\text{PSC.35})$$

Periodization commutes with the r -derivative on these source columns, so the first line uses the actual derivative in (PSC.23). Integrating (PSC.35) and using (PSC.34) proves

$$0 \preceq \mathcal{E} \preceq \frac{1 + \eta}{1 - \eta} d_N \mathcal{D} \preceq \frac{1 + \eta}{1 - \eta} d_N \eta^2 G. \quad (\text{PSC.36})$$

135.13 Generator-difference bounds in the original quotient metric

For a family $Y_\theta : E \rightarrow \mathcal{H}_{L,\theta}$, define the Hilbert–Schmidt norm using the retained G by

$$\|Y\|_{2,G}^2 := \int \text{Tr}(G^{-1} Y_\theta^* Y_\theta) \frac{d\theta}{2\pi}. \quad (\text{PSC.37})$$

Equivalently this is the sum of $\|Y_\theta e_j\|^2$ over any G -orthonormal basis (e_j) , integrated in the original phase measure. This definition does not rescale either the source mass or the arithmetic metric.

Keep the exactly computed coefficient quantities

$$a_{\text{HS}}^2 = \text{Tr}(G^{-1}(A^* - (k/2)I_E)G(A - (k/2)I_E)), \quad a_{\text{op}} = \|A - (k/2)I_E\|_G. \quad (\text{PSC.38})$$

They are computed from the actual multiplication matrix and original minimum metric. In particular no diagonalizability or removal of nilpotent parts is used.

Equations (PSC.34), (PSC.36), and (PSC.21) imply

$$\boxed{\begin{aligned} \|L\|_{2,G} &\leq \eta\sqrt{q}, \\ \|\Delta\mathcal{B}\|_{2,G} &\leq \eta \left(\sqrt{\frac{1 + \eta}{1 - \eta} d_N q} + a_{\text{HS}} \right). \end{aligned}} \quad (\text{PSC.39})$$

To prove the second statement, the triangle inequality in the Hilbert space of phase families gives

$$\|\Delta\mathcal{B}\|_{2,G} \leq \|\partial_r L\|_{2,G} + \|L(A - (k/2)I_E)\|_{2,G}.$$

The first squared norm is $\text{Tr}(G^{-1}\mathcal{E}) \leq \eta^2(1 + \eta)(1 - \eta)^{-1}d_N q$. The second squared norm is

$$\text{Tr}(G^{-1}(A^* - (k/2)I_E)\mathcal{D}(A - (k/2)I_E)) \leq \eta^2 a_{\text{HS}}^2,$$

which proves (PSC.39). The first statement is $\text{Tr}(G^{-1}\mathcal{D}) \leq \eta^2 q$.

For every real $t > 0$, a matrix form of the same estimate is

$$\mathcal{E}_\Delta \preceq \eta^2 \left[(1 + t) \frac{1 + \eta}{1 - \eta} d_N G + (1 + t^{-1})(A^* - (k/2)I_E)G(A - (k/2)I_E) \right]. \quad (\text{PSC.40})$$

For each vector x , use $2 \text{Re}\langle u, v \rangle \leq t\|u\|^2 + t^{-1}\|v\|^2$ in the squared norm of (PSC.21), integrate, and then apply (PSC.34) and (PSC.36). This proves all quadratic forms in (PSC.40).

The adjoint-form comparison (PSC.27) has a quadratic error:

$$\left\| G^{-1/2}(\mathfrak{H}(G) - \mathfrak{H}(\overline{G}))G^{-1/2} \right\|_{\text{op}} \leq 2\eta^2 a_{\text{op}}. \quad (\text{PSC.41})$$

Indeed let $D = G^{-1}\mathcal{D}$; it is positive self-adjoint for G , and $\|D\|_G \leq \eta^2$. Multiplication of (PSC.27) by G^{-1} gives

$$(A - (k/2)I_E)^{\dagger_G} D + D(A - (k/2)I_E),$$

where †_G is the metric adjoint. Submultiplicativity proves (PSC.41). The statement is in the original G -metric; it is not an unqualified Euclidean bound on the coefficient matrix in (PSC.27).

135.14 An explicit length choice makes both comparison errors small

Keep a fixed original packet h and let k vary, with any finite cutoff $N_k \geq q_k - 1$. In the following formulas all matrices are the actual source matrices for that k, N_k . Set

$$\eta_k = \frac{1}{(k+1)(1 + \sqrt{q_k} + \sqrt{3d_{N_k}q_k + a_{\text{HS},k}})}, \quad L_k = \frac{1}{a} \log \left(1 + \frac{\kappa_{a,N_k+1}}{\eta_k} \right). \quad (\text{PSC.42})$$

Every input in these formulas is finite, and the weighted Gram cost is strictly positive. Thus both parameters are finite and positive. Substitution gives

$$\frac{\kappa_{a,N_k+1}}{e^{aL_k} - 1} = \eta_k.$$

Also $0 < \eta_k < 1/2$, so $(1+\eta_k)/(1-\eta_k) \leq 3$. Thus (PSC.39) proves the explicit simultaneous estimates

$$\boxed{\|L\|_{2,G} \leq \frac{1}{k+1}, \quad \|\mathcal{B}^c - \mathcal{B}^p\|_{2,G} \leq \frac{1}{k+1}.} \quad (\text{PSC.43})$$

For example, the first bound follows by inserting the numerator $\sqrt{q_k}$ into the positive denominator in (PSC.42); the second follows by inserting $\sqrt{3d_{N_k}q_k + a_{\text{HS},k}}$. Both errors are therefore $o(k)$, indeed they tend to zero.

No uniform estimate on the growth of κ_{a,N_k+1} , d_{N_k} , or $a_{\text{HS},k}$ was claimed. The length in (PSC.42) explicitly pays their full cost. This is an existence and exact-formula comparison at each finite k, N_k , not a certified numerical evaluation of an unevaluated arithmetic Gram.

135.15 The complete quartet boundary cannot satisfy the same estimate

Now remain inside the counterfactual branch of the original program: a root $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 0$, of its full order m , together with its reflected and conjugate roots, is selected. Retain HT1–HT2 exactly:

$$\ell_k = 1 + k(m-1), \quad q_k = \ell_k(k+1)^2, \quad \mathcal{L}_{h,k} = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{PSC.44})$$

The full original cyclic polynomial is

$$\chi_{h,k}(S) = \prod_{a_0, b_0=0}^k (S - k/2 - (2a_0 - k)\delta - i(2b_0 - k)\gamma)^{\ell_k}.$$

No order in this polynomial is reduced.

The independently proved HT14–HT15 give, for every phase, every length, and every $N_k \geq q_k - 1$,

$$\text{Tr}(G_{\theta}^{-1}(\mathcal{B}_{\theta}^p)^* \mathcal{B}_{\theta}^p) \geq \frac{\mathcal{L}_{h,k}^2}{2} \geq \frac{\delta^2 k^2 q_k^2}{8}. \quad (\text{PSC.45})$$

Those arithmetic lower bounds are used here, not reproved.

To place (PSC.45) in precisely the same metric as (PSC.43), first note that the source bound (PSC.33) gives

$$(1 - \eta_k)G \preceq G_\theta \preceq (1 + \eta_k)G. \quad (\text{PSC.46})$$

Indeed the minimum over $J_N P = x$ of each quadratic source form is its quotient form $x^* G_\theta x$, respectively $x^* G x$; minimizing the two inequalities proves both statements. The lower inequality implies $G^{-1} \succeq (1 - \eta_k)G_\theta^{-1}$. Taking its trace against the positive matrix $(\mathcal{B}_\theta^p)^* \mathcal{B}_\theta^p$, and then integrating (PSC.45), gives

$$\|\mathcal{B}^p\|_{2,G} \geq \sqrt{1 - \eta_k} \frac{\mathcal{L}_{h,k}}{\sqrt{2}} \geq \frac{\sqrt{1 - \eta_k} \delta k q_k}{2\sqrt{2}}. \quad (\text{PSC.47})$$

The same lower bound reaches the common section with its exact comparison cost. The reverse triangle inequality applied to (PSC.21) and (PSC.43) gives

$$\boxed{\begin{aligned} \|\mathcal{B}^c\|_{2,G} &= (\text{Tr}(G^{-1} T_c^* M_{N_k+1} T_c))^{1/2} \\ &\geq \sqrt{1 - \eta_k} \frac{\mathcal{L}_{h,k}}{\sqrt{2}} - \frac{1}{k+1}. \end{aligned}} \quad (\text{PSC.48})$$

The equality is the exact all-phase isometry (PSC.26), so the common boundary in this formula is the original source boundary.

In particular (PSC.42) gives $\eta_k \rightarrow 0$, and $q_k \geq 1$, whence

$$\liminf_{k \rightarrow \infty} \frac{\|\mathcal{B}^c\|_{2,G}}{k q_k} \geq \frac{\delta}{2\sqrt{2}}, \quad \liminf_{k \rightarrow \infty} \frac{\|\mathcal{B}^p\|_{2,G}}{k q_k} \geq \frac{\delta}{2\sqrt{2}}. \quad (\text{PSC.49})$$

Thus neither full generator boundary has $o(k)$ norm, even after reporting its norm per quotient dimension. The actual phasewise metric version (PSC.45) is uniform in L without any length restriction.

Equations (PSC.43), (PSC.47), and (PSC.48) coexist because (PSC.43) makes the two large boundaries close. It does not set either boundary equal to zero. The morphism that relates these statements is the exact original relation map (PSC.20)–(PSC.22).

135.16 At the minimal cutoff the gluing error is identically zero

There is a useful exact case requiring no length choice or limiting statement. Set $N = q - 1$. Then

$$J_{q-1} : P_{q-1} \longrightarrow E$$

is an isomorphism: monic division gives a unique remainder of degree at most $q - 1$ for every class, and a nonzero multiple of a degree- q monic polynomial cannot have degree at most $q - 1$.

Every section in (PSC.15) must consequently equal the unique inverse, so

$$C = C_\theta = J_{q-1}^{-1}, \quad Z_\theta = X_\theta = L_\theta = 0, \quad \mathcal{D} = \mathcal{E}_\Delta = 0 \quad (\text{PSC.50})$$

for every θ and $L > 0$.

Define the original H57 residue coefficient

$$\ell : E \longrightarrow \mathbb{C}, \quad \ell([P]) = [S^{q-1}] \text{rem}_\chi P. \quad (\text{PSC.51})$$

This is nonzero since $\ell([S^{q-1}]) = 1$. For a remainder P of degree at most $q - 1$, the polynomial SP has degree at most q . Its monic division by χ has constant quotient exactly $[S^{q-1}]P$. Its remainder is $CA[P]$. Therefore the complete coefficient identity is

$$S_{q-1}C - \iota_{q-1}CA = \chi \ell. \quad (\text{PSC.52})$$

It follows that the persistent boundary is the explicit rank-one map

$$\boxed{\mathcal{B}_\theta^c = \mathcal{B}_\theta^p = f_{\chi,\theta} \otimes \ell, \quad f_{\chi,\theta} := \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}_\chi.} \quad (\text{PSC.53})$$

It factors through exactly the same scalar functional as the original residue constituent $\mathcal{R} = 1_E \otimes \ell$ in H57. In (PSC.53) the receiving vector is the actual source relation $f_{\chi,\theta}$, and its tensor theta primitive is (PSC.22) with $Y = 1$.

This vector is nonzero. Its exact phase norm is

$$\|f_{\chi,\theta}\|^2 = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k} \left(\frac{2\pi n + \theta}{L} \right) \left| \chi \left(k/2 + i \frac{2\pi n + \theta}{L} \right) \right|^2 > 0. \quad (\text{PSC.54})$$

All weights are positive for $k \geq 2$, as proved for the actual source in HT6. The nonzero polynomial χ has only finitely many distinct roots; it cannot vanish on every point of the infinite distinct sampled lattice. At least one summand is therefore positive and all are nonnegative. Convergence follows from the original source bounds.

Thus (PSC.53) has rank exactly one and the full generator energy is exactly

$$\text{Tr}(G_\theta^{-1}(\mathcal{B}_\theta^p)^* \mathcal{B}_\theta^p) = \|f_{\chi,\theta}\|^2 \ell G_\theta^{-1} \ell^* > 0. \quad (\text{PSC.55})$$

To verify this equation, the Gram of $f_{\chi,\theta} \otimes \ell$ is $\|f_{\chi,\theta}\|^2 \ell^* \ell$; cyclicity of the finite matrix trace gives (PSC.55). Positive definiteness of G_θ^{-1} and $\ell \neq 0$ prove strict positivity. Under the selected quartet, HT14 supplies the quantitative lower bound (PSC.45) to this exact expression.

Hence even perfect gluing, $\mathcal{D} = 0$, is already compatible with the full positive generator defect in the original finite quotient at its first cutoff. This is an exact calculation on the counterfactual packet, not an asserted RH counterexample.

135.16.1 Exact first-cutoff energy, including the monic residual

In this paragraph fix one phase and $N = q - 1$, and write $R = R_\theta^p = R_\theta^c$, $G_0 = G_\theta$, $f = f_{\chi,\theta}$. Thus G_0 denotes this phase metric, not the common metric G in (PSC.15). Put

$$U = R G_0^{-1/2}, \quad r^* = \ell G_0^{-1/2}, \quad T = \mathcal{B}_\theta^p G_0^{-1/2} = f r^*, \quad z = U^* f, \quad \alpha = r^* z. \quad (\text{PSC.56})$$

Here $U^* U = I_E$. The exact integration-by-parts identity H55 gives

$$H_0 = G_0^{-1/2} (A^* G_0 + G_0 A - k G_0) G_0^{-1/2} = -(z r^* + r z^*). \quad (\text{PSC.57})$$

For the complete quartet, $\text{Tr } H_0 = 0$; therefore $\text{Re } \alpha = 0$. HT9 identifies its two nonzero eigenvalues as $\varepsilon_\theta, -\varepsilon_\theta$, with $\varepsilon_\theta \geq \mathcal{L}_{h,k}$.

Multiplying out the square in (PSC.57) and taking its trace gives

$$2\varepsilon_\theta^2 = 2\|z\|^2 \|r\|^2 + 2\text{Re}(\alpha^2) = 2\|z\|^2 \|r\|^2 - 2(\text{Im } \alpha)^2.$$

The decomposition $f = U z + (I - U U^*) f$ is orthogonal. Since $T = f r^*$ is rank one, both its squared operator norm and squared Hilbert–Schmidt norm equal $\|f\|^2 \|r\|^2$. Consequently

$$\boxed{\|T\|_{\text{op}}^2 = \|T\|_{\text{HS}}^2 = \varepsilon_\theta^2 + (\text{Im } \alpha)^2 + \|r\|^2 \|(I - U U^*) f\|^2.} \quad (\text{PSC.58})$$

In particular the projection term is present.

Let p_j be the actual monic orthogonal polynomials for the same original phase measure

$$d\nu_{L,\theta}(u) = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}, \quad u_n = (2\pi n + \theta)/L, \quad S = k/2 + iu,$$

and let $\omega_j = \int |p_j(k/2 + iu)|^2 d\nu_{L,\theta}(u) > 0$. The range of U is precisely $W_{q-1,\theta}P_{q-1}$, since $C = J_{q-1}^{-1}$ is an isomorphism. Both χ and p_q are monic of degree q , so $\chi - p_q \in P_{q-1}$, while $W_{q,\theta}p_q$ is orthogonal to that range. It follows that

$$(I - UU^*)f = W_{q,\theta}p_q, \quad \|(I - UU^*)f\|^2 = \omega_q. \quad (\text{PSC.59})$$

On the orthogonal basis $p_0, \dots, p_{q-1}$ of P_{q-1} , the functional ℓ is zero on p_j for $j < q-1$ and equals one on p_{q-1} . Its squared dual norm is therefore exactly $1/\omega_{q-1}$. The identification with the phase quotient is the isometry $C : (E, G_0) \rightarrow (P_{q-1}, M_{q-1,\theta})$. Thus

$$\|r\|^2 = \ell G_0^{-1} \ell^* = \frac{1}{\omega_{q-1}}, \quad \|T\|_{\text{op}}^2 = \|T\|_{\text{HS}}^2 = \varepsilon_\theta^2 + (\text{Im } \alpha)^2 + \frac{\omega_q}{\omega_{q-1}} > \varepsilon_\theta^2 \geq \mathcal{L}_{h,k}^2. \quad (\text{PSC.60})$$

This is a strictly stronger first-cutoff floor, using the original monic source moments and preserving their mass.

135.16.2 A full-jet eigenvector and its exact arithmetic boundary ratio

Let λ be any root of χ of its complete order d_λ , and write

$$\psi_\lambda(S) = \frac{\chi(S)}{(S - \lambda)^{d_\lambda}}, \quad \psi_\lambda(\lambda) \neq 0, \quad v_\lambda = \left[\frac{\chi(S)}{(S - \lambda)\psi_\lambda(\lambda)} \right] \in E. \quad (\text{PSC.61})$$

The displayed representative has degree $q-1$. It satisfies $(A - \lambda I_E)v_\lambda = 0$, since multiplication by $S - \lambda$ produces the multiple $\chi/\psi_\lambda(\lambda)$. At the full local factor $\mathbb{C}[\epsilon]/(\epsilon^{d_\lambda})$, its value is

$$\epsilon^{d_\lambda-1} \frac{\psi_\lambda(\lambda + \epsilon)}{\psi_\lambda(\lambda)} = \epsilon^{d_\lambda-1}.$$

At every other full primary factor its value is zero. Thus this is a specified nonzero vector in the full original jet algebra, with every local order retained. Its leading coefficient is

$$\ell(v_\lambda) = \frac{1}{\psi_\lambda(\lambda)}. \quad (\text{PSC.62})$$

The exact two squared norms before any cancellation of a coefficient are

$$\begin{aligned} \|\mathcal{B}_\theta^p v_\lambda\|^2 &= \frac{1}{|\psi_\lambda(\lambda)|^2} \int |\chi(k/2 + iu)|^2 d\nu_{L,\theta}(u), \\ \|Rv_\lambda\|^2 &= \frac{1}{|\psi_\lambda(\lambda)|^2} \int \left| \frac{\chi(k/2 + iu)}{k/2 + iu - \lambda} \right|^2 d\nu_{L,\theta}(u). \end{aligned} \quad (\text{PSC.63})$$

The quotient inside the second integral is the polynomial $\chi(S)/(S - \lambda)$, including at any sampled root. In particular no singular point is omitted. The denominator is strictly positive by the infinite positive lattice argument used in (PSC.54).

For the extreme complete primary factor $\lambda = k/2 + k\delta + ik\gamma$, with $d_\lambda = \ell_k$, division of the two equations in (PSC.63) gives the exact identity

$$\frac{\|\mathcal{B}_\theta^p v_\lambda\|^2}{\|Rv_\lambda\|^2} = k^2 \delta^2 + \frac{\int (u - k\gamma)^2 \left| \frac{\chi(k/2 + iu)}{k/2 + iu - \lambda} \right|^2 d\nu_{L,\theta}(u)}{\int \left| \frac{\chi(k/2 + iu)}{k/2 + iu - \lambda} \right|^2 d\nu_{L,\theta}(u)} > k^2 \delta^2. \quad (\text{PSC.64})$$

Indeed on the original sampled line, $|k/2 + iu - \lambda|^2 = k^2 \delta^2 + (u - k\gamma)^2$. Multiplying this equality by $|\chi(k/2 + iu)/(k/2 + iu - \lambda)|^2$, integrating, and using (PSC.63) proves the formula. The last numerator is positive: at most one lattice point has $u = k\gamma$, and the nonzero polynomial in the integrand cannot vanish at all remaining infinitely many points. Both the nonzero factor $\psi_\lambda(\lambda)$ and the full arithmetic unit in (PSC.11) were retained in the actual vector and observation maps before the equality of ratios was calculated.

135.17 Split-base typing, provenance, and independent check

Each displayed linear coefficient map acts on a fixed original support label by

$$(\lambda, v) \mapsto (\lambda, fv), \quad \tau \mapsto \tau.$$

If $fv = 0$, the target is the represented zero at that label. For instance the full-jet observation of $\mathcal{V}(\chi Y)$ is zero by (PSC.11), although its source representative and norm remain in (PSC.22), (PSC.26), and (PSC.55). This is the original map to the receiving coefficient zero; it does not send represented data to external absence. Arithmetic scalars remain under $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$, the marked $a = 0$ coefficient remains fixed, and neither the phase θ nor the length L translates A , g , or χ .

135.18 The actual match to the amplification calculation

The amplified nonzero arithmetic class is supplied by (PAM.3)–(PAM.5), with eigenvalue $k\rho$. The phase Hilbert space supplies a precisely critical-line generator, by (PAM.8). The exact map between them is (PAM.9)–(PAM.11), and on the amplified primary factor its full inverse relation is (PAM.15). The real displacement is $k\delta$; the actual comparison error on its eigenvector is the explicitly computed norm (PAM.12), at least $k\delta$. The available canonical global controls grow at least as the functions in (PAM.18)–(PAM.23), and the class-preserving repairs preserve this generator obstruction by (PAM.25)–(PAM.27). Consequently none of these proved holonomy or repaired-source estimates is an exclusion of the amplified class with error $o(k)$. They give the explicit positive boundary energies and exact maps displayed above. These calculations do not assert that a new opposing estimate is impossible, or that an off-line zeta packet exists.

136 The actual arithmetic amplification and its proved error scales

This calculation uses one fixed original counterfactual anchor $\rho = 1/2 + \delta + i\gamma$, of full order m in $g = 2\xi$, with $0 < \delta < 1/2$ and $\gamma > 0$. Its entire quotient and full Taylor unit are retained throughout:

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - 1/2 - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad v_h = j_h(v_h), \quad w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}. \quad (\text{AAM.1})$$

The original tensor source, its arithmetic injection and the finite source degrees provide the following simultaneous data for every integer $k \geq 3$:

$$\begin{aligned} c &= k/2, \quad \ell_k = 1 + k(m-1), \quad q_k = \ell_k(k+1)^2, \quad f_k = \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \\ \lambda_{a,b} &= c + (2a-k)\delta + i(2b-k)\gamma \quad (0 \leq a, b \leq k), \\ \chi_k(S) &= \prod_{a,b=0}^k (S - \lambda_{a,b})^{\ell_k}, \quad E_k = \mathbb{C}[S]/(\chi_k), \quad A_k^{\text{cyc}} = M_S, \\ m_{h,k} &= w_h^{*k}, \quad \mathcal{I}_k = \sigma_h^{\otimes k} \eta_{h,k} : E_k \hookrightarrow Q^{\otimes k}, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_h^{(k)})1. \end{aligned} \quad (\text{AAM.2})$$

The superscript on A_k^{cyc} distinguishes this operator from the scalar upper-envelope constant A_k used in (TW8). No coordinate or operator has been multiplied by this scalar. The full local nilpotent sum has exact index ℓ_k : its power $k(m-1)$ has coefficient $(k(m-1))!/((m-1)!)^k \neq 0$ on the product of the k top local monomials, whereas the next power is zero. Every real and imaginary sign count occurs in the ordered tensor product, and $\delta\gamma \neq 0$ makes the displayed sums distinct. Thus the stated complete cyclic annihilator and its dimension are exactly those in (HT2), with no local directions omitted.

For $N \geq q_k - 1$, the original exact sequence is

$$0 \longrightarrow \mathcal{P}_{N-q_k} \xrightarrow{\times \chi_k} \mathcal{P}_N \xrightarrow{J_{k,N}} E_k \longrightarrow 0, \quad \mathcal{P}_N = \mathbb{C}[S]_{\leq N}. \quad (\text{AAM.3})$$

The source norm is the integral of $|P(c + iu)|^2 m_{h,k}(u)$. The relation source is zero when $N = q_k - 1$. The unique canonical minimum lift $R_{k,N} : E_k \rightarrow \mathcal{P}_N$ satisfies $J_{k,N} R_{k,N} = I$ and $R_{k,N}^* H_{k,N} R_{k,N} = G_{k,N} > 0$, in the original remainder frame, and $V_{k,N} = \det G_{k,N}$. These facts follow by orthogonal projection onto the kernel in the finite positive-definite source form, as proved in (CA6).

136.1 The invariant that the tensor construction amplifies

A nonzero eigenvector v of A_h at ρ has nonzero ordered tensor power: a linear functional taking v to one takes $v^{\otimes k}$ to one. The sum generator acts on that tensor by $k\rho$, whose real displacement from c is exactly $k\delta$. The complete positive-half-plane trace uses all primary spaces, not only this vector. Its original value is

$$\mathcal{L}_k = 2\delta \ell_k(k+1)f_k = 2 \sum_{a > k/2} \sum_{b=0}^k \ell_k(2a-k)\delta. \quad (\text{AAM.4})$$

To evaluate the sum, for $k = 2r$ its positive real coefficients sum to $2(1 + \dots + r) = r(r+1)$; for $k = 2r-1$ they sum to $1 + 3 + \dots + (2r-1) = r^2$. These are exactly f_k . In particular the following are exact finite equalities:

$$\frac{\mathcal{L}_k}{kq_k} = \begin{cases} \frac{\delta}{2}(1 + 1/k), & k \text{ odd}, \\ \frac{\delta}{2}(1 + 1/(k+1)), & k \text{ even}, \end{cases} \quad \mathcal{L}_k \geq \frac{\delta k q_k}{2}, \quad \frac{\mathcal{L}_k}{kq_k} \longrightarrow \frac{\delta}{2}. \quad (\text{AAM.5})$$

Here is the exact finite control to which that trace applies. Suppress k in the subscripts while keeping it fixed, and put $A = A_k^{\text{cyc}}$. The Hermitian matrix in the original quotient metric is

$$\mathbf{H}_N = G_N^{-1/2} (A^* G_N + G_N A - k G_N) G_N^{-1/2}. \quad (\text{AAM.6})$$

Write p_n for the original monic orthogonal polynomials and ω_n for their squared norms, and $b_n = [p_n]_{\chi_k}$. The monic recurrence in the exact sum coordinate is $S p_n = p_{n+1} + (c + i\beta_n)p_n - (\omega_n/\omega_{n-1})p_{n-1}$ with real β_n . Inserting it in $K_N = \sum_{n=0}^N b_n b_n^*/\omega_n$ cancels every internal adjacent term and gives

$$A K_N + K_N A^* - k K_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}, \quad K_N = G_N^{-1}. \quad (\text{AAM.7})$$

Congruence by $G_N^{1/2}$ identifies its left side with (AAM.6). It has rank at most two. Its trace is $2\Re \text{Tr } A - kq_k = 0$ by the complete root grid. It therefore has eigenvalues $\varepsilon_{k,N}$, $-\varepsilon_{k,N}$ and zeros, with $\varepsilon_{k,N} \geq 0$.

Let E_+ be the direct sum of the complete primary spaces with $a > k/2$, and let P_+ be the orthogonal projection onto $G_N^{1/2} E_+$ in the displayed Euclidean coordinates. This subspace is invariant under $G_N^{1/2} A G_N^{-1/2}$. Taking the trace in an orthonormal basis of that invariant subspace gives $\text{Tr}(P_+ \mathbf{H}_N) = \mathcal{L}_k$. For a Hermitian matrix with spectrum $\varepsilon, -\varepsilon, 0, \dots, 0$, any orthogonal projection has trace pairing at most ε : its diagonal coefficients in an eigenbasis lie between zero and one. Thus

$$\boxed{\varepsilon_{k,N} = \|\mathbf{H}_N\| \geq \mathcal{L}_k, \quad \frac{\varepsilon_{k,N}}{kq_k} \geq \frac{\mathcal{L}_k}{kq_k} \geq \frac{\delta}{2}.} \quad (\text{AAM.8})$$

This is the common normalized invariant at every admitted source degree. It is also the phasewise invariant in (HT9), with the phase metric retained there. Neither a source norm nor an unnormalized matrix can be substituted into the numerator of (AAM.8).

136.2 The exact original window map into that control

In this subsection $q = q_k$, and every measure is the unperiodized original $m_{h,k}$. Define the original four-volume and norm term

$$\mathcal{B}_k = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}}, \quad \Omega_k = \log \frac{\omega_{2q-1}\omega_{2q}}{\omega_{q-1}\omega_q}. \quad (\text{AAM.9})$$

Retain $d_j = V_j/V_{j-1}$ and the complete contraction penalties

$$P_- = -\log(1 - d_{2q-1}) - 2 \sum_{j=q}^{2q-2} \log(1 - d_j), \quad (\text{AAM.10})$$

$$P_+ = -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - d_j), \quad P_k = P_- + P_+.$$

Here $0 < d_j < 1$ on these degrees by the original consecutive identities and the positive trace floor. The phase is the original real number ϕ_N specified by $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. Let $w_{q-1} = w_{2q-1} = 1$ and $w_N = 2$ for the intervening degrees; their sum is $2q$. The original first-step and regular products (CJ.9)–(CJ.13) prove exactly

$$\mathcal{B}_k = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_{k,N}^2 + \phi_N^2) - \Omega_k + P_k. \quad (\text{AAM.11})$$

There is no inverse at degree $q - 2$: the first product has one terminal contraction and the squared interior contractions displayed in P_- .

The following quantities extract the same control from that identity, retaining its entire phase and contraction data:

$$\Phi_k = \sum_N w_N \log(1 + \phi_N^2/\varepsilon_{k,N}^2) \geq 0, \quad \bar{\varepsilon}_k = \prod_N \varepsilon_{k,N}^{w_N/(2q)}, \quad (\text{AAM.12})$$

$$\mathbf{n}_k = e^{\Omega_k/(4q)}, \quad \mathbf{c}_k = e^{(\mathcal{B}_k - P_k - \Phi_k)/(4q)}.$$

Every denominator is positive by (AAM.8). Splitting each logarithm in (AAM.11) proves the exact multiplicative correspondence

$$\boxed{\bar{\varepsilon}_k = \mathbf{n}_k \mathbf{c}_k, \quad \mathcal{L}_k \leq \min_{q-1 \leq N \leq 2q-1} \varepsilon_{k,N} \leq \bar{\varepsilon}_k.} \quad (\text{AAM.13})$$

In particular the exact scalar exclusion threshold for the given window is

$$\mathcal{B}_k^{\text{crit}} = 4q \log \mathcal{L}_k - \Omega_k, \quad \mathcal{B}_k - \mathcal{B}_k^{\text{crit}} = P_k + \sum_N w_N \log \frac{\varepsilon_{k,N}^2 + \phi_N^2}{\mathcal{L}_k^2} \geq 0. \quad (\text{AAM.14})$$

More precisely, $\mathcal{B}_k - P_k - \Phi_k \geq \mathcal{B}_k^{\text{crit}}$. This follows from (AAM.13), so it retains rather than deletes the actual phases.

136.3 Uniform norm errors and their precise amplified rates

We use the literal constants already proved in (TW2), (TW7)–(TW8):

$$\alpha = \pi/2, \quad p = 8m - 9/2, \quad B = 42 + 8m, \quad M = B/2,$$

$$\vartheta_h = \int_{-1}^1 w_h(u) du, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha u} w_h(u) du,$$

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_\Gamma 2^M}, \quad A_k = \frac{C_h}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \quad (\text{AAM.15})$$

$$D_N = [1 + 4(N + M)^2]^M, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}.$$

Here C_h is the full boundary-tilt constant of (BT12), and c_h is the original positive convolution lower-envelope constant of (AU.10). Their original constructions, compact contributions and full masses remain in the retained chapters. No new zeta bound is assumed in using them. The proved all-polynomial estimate is $a_k H_N^\sigma / D_N \preceq H_{k,N} \preceq A_k H_N^\sigma$, where $\sigma(u) = |\Gamma(1/4 + iu/2)|^2 / (2\pi)$ has its original mass $\sqrt{2\pi}$. Minimizing on monic affine fibres gives $a_k \gamma_n / D_n \leq \omega_n \leq A_k \gamma_n$, with $\gamma_n = \sqrt{2\pi}(2n)!/4^n$.

Define the exact nonnegative norm-window error and common-source comparison error by

$$\mathcal{E}_k = 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q, \quad \Lambda_k = \log(A_k/a_k) + \log D_{2q}. \quad (\text{AAM.16})$$

The symbol $\mathcal{E}_k$ here denotes exactly E_k in (TW20), and is distinct from the quotient vector space E_k in (AAM.2). The two original norm windows (TW16)–(TW19) imply

$$\mathfrak{n}_k/q \longrightarrow 4/e, \quad \Omega_k \leq 4q \log(4q/e) + \mathcal{E}_k, \quad \mathcal{E}_k/q \longrightarrow 0. \quad (\text{AAM.17})$$

For completeness the first claim uses the separate limits $q^{-1}(\omega_{2q}/\omega_q)^{1/(2q)} \rightarrow 4/e$ and $q^{-1}(\omega_{2q-1}/\omega_{q-1})^{1/(2q)} \rightarrow 4/e$; their positive geometric mean is exactly $\mathfrak{n}_k/q$. The upper logarithmic estimate is the sum of the two original upper norm comparisons and the exact factorial inequality (TW18).

The exponential dependence of the scalar constants can be calculated more precisely than a big-O estimate. Put

$$d_h = \alpha + \log(M_\alpha/\vartheta_h). \quad (\text{AAM.18})$$

The actual w_h is even and positive almost everywhere, so $M_\alpha = \int \cosh(\alpha u) w_h(u) du > \vartheta_h$: the integral outside $[-1, 1]$ is strictly positive and on that interval $\cosh(\alpha u) \geq 1$. Hence $d_h > \alpha > 0$. Taking logarithms of (AAM.15) gives the exact identity

$$\begin{aligned} \log(A_k/a_k) &= (k-1) \log M_\alpha - (k-3) \log \vartheta_h + \alpha(k-3) + (p+1) \log k + B \log(k-2) \\ &+ \log \frac{C_h C_\Gamma 2^M}{c_\Gamma c_h} = k d_h + (p+1+B) \log k + O_h(1). \end{aligned} \quad (\text{AAM.19})$$

The last remainder is bounded because $\log(k-2) - \log k = \log(1-2/k)$ is bounded for $k \geq 3$. Since $(k+1)^2 \leq q_k \leq mk(k+1)^2$ and $\log D_{2q} = O_h(\log q)$, these equalities prove

$$\frac{\Lambda_k}{k} \longrightarrow d_h, \quad \frac{\mathcal{E}_k}{k} \longrightarrow 2d_h, \quad \frac{\mathcal{E}_k}{\mathcal{L}_k} \longrightarrow 0, \quad \frac{(2q_k-1)\Lambda_k}{\mathcal{L}_k} \longrightarrow \frac{4d_h}{\delta} > 0. \quad (\text{AAM.20})$$

Thus the small norm error and the four-volume comparison error have different rates on the original amplified scale, with their exact dimension factors displayed.

Equations (AAM.5), (AAM.13), and (AAM.17) now give the particularly direct quantitative correspondence

$$\frac{\mathfrak{n}_k}{\mathcal{L}_k} \sim \frac{8}{e\delta k} \longrightarrow 0, \quad \mathfrak{c}_k \geq \frac{\mathcal{L}_k}{\mathfrak{n}_k}, \quad \liminf_{k \rightarrow \infty} \frac{\mathfrak{c}_k}{k} \geq \frac{e\delta}{8}. \quad (\text{AAM.21})$$

The vanishing first ratio is therefore not an upper estimate on $\bar{\varepsilon}_k/\mathcal{L}_k$. The exact second factor in the original source identity has at least the stated compensating growth. This proves the relationship between the small norm-window factor and the retained arithmetic control, instead of merely declaring them to be different quantities.

The exact threshold itself has the computed asymptotic

$$\frac{\mathcal{B}_k^{\text{crit}}}{q_k} - 4 \log k \longrightarrow 4 \log(e\delta/8), \quad \frac{\mathcal{B}_k^{\text{crit}}}{\mathcal{L}_k} \longrightarrow 0. \quad (\text{AAM.22})$$

Indeed $\mathcal{B}_k^{\text{crit}}/(4q) = \log(\mathcal{L}_k/\mathfrak{n}_k)$ and (AAM.21) evaluates this logarithm. The actual finite lower estimate in (CA20) follows as well:

$$\mathcal{B}_k \geq 4q_k \log(\delta e k/8) - \mathcal{E}_k. \quad (\text{AAM.23})$$

All further contraction, phase, control and trace savings in (TW20) remain nonnegative. Its final trace ratio is exactly $2\mathcal{L}_k/(\delta k q_k) = 1 + 1/k$ for odd k and $1 + 1/(k+1)$ for even k , by (AAM.5).

136.4 The available upper functions on the identical window

The full finite reference number $\mathcal{B}_k^\sigma$ is evaluated with the original χ_k by (CA18)–(CA19); in particular it is nonnegative. On the common degree- $2q$ source let $D_* = (H_{2q}^\sigma)^{-1} H_{k,2q}$ have ordered positive eigenvalues $b_1 \leq \dots \leq b_{2q+1}$ and put

$$\begin{aligned} L_q(D_*) &= \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \\ \widehat{\Phi}_q(B) &= 2 \operatorname{arcosh}(e^{B/(4q-2)}), \quad \widehat{\Psi}_q(t) = (4q-2) \log \cosh(t/2), \\ \widehat{A}_\sigma &= \widehat{\Phi}_q(\mathcal{B}_k^\sigma), \quad U_k^{\operatorname{spec}} = \mathcal{B}_k^\sigma + L_q(D_*), \quad U_k^{\operatorname{ang}} = \widehat{\Psi}_q(\widehat{A}_\sigma + \Lambda_k). \end{aligned} \tag{AAM.24}$$

These are the actual full-spectrum and angle bounds proved in (CA14), (CA17b). They give $\mathcal{B}_k \leq \min(U_k^{\operatorname{spec}}, U_k^{\operatorname{ang}})$ and $L_q(D_*) \leq (2q-1)\Lambda_k$. The full spectrum in $L_q(D_*)$ is retained; its value has not been replaced by the scalar envelope when taking the first minimum.

The precise asymptotic cost of the displayed angle upper function can be calculated even without an asymptotic for $\mathcal{B}_k^\sigma$ at the moving endpoints. For $t, x \geq 0$,

$$\frac{e^x}{2} \leq \frac{\cosh(t+x)}{\cosh t} \leq e^x.$$

The upper inequality follows by expanding both hyperbolic cosines into exponentials; the lower follows from $\cosh(t+x) \geq e^{t+x}/2$ and $\cosh t \leq e^t$. Use $t = \widehat{A}_\sigma/2$ and $x = \Lambda_k/2$, and use the inverse relation between $\widehat{\Phi}_q$ and $\widehat{\Psi}_q$. It gives

$$\begin{aligned} (2q-1)\Lambda_k - (4q-2) \log 2 &\leq U_k^{\operatorname{ang}} - \mathcal{B}_k^\sigma \leq (2q-1)\Lambda_k, \\ \frac{U_k^{\operatorname{ang}} - \mathcal{B}_k^\sigma}{\mathcal{L}_k} &\longrightarrow \frac{4d_h}{\delta}. \end{aligned} \tag{AAM.25}$$

The discarded constant term divided by $\mathcal{L}_k$ tends to zero by (AAM.5). This is a statement about the proved upper function; it is not an asymptotic assertion about the signed actual difference.

For an independent upper function retain exactly the original (HC8) constants. Set $R_0 = \sqrt{\delta^2 + \gamma^2}$, choose the original $0 < b < \pi/2$, and put

$$D = \max(1, 2/b, 2R_0), \quad a = \alpha D + \log 256, \quad X = \max(1, M_h(b), M_h(-b)), \quad K = \max(1, \vartheta_h^{-1}),$$

$$C_{\operatorname{env}} = \max(1, 2/(3c_h D)), \quad M_h(t) = \int e^{tu} w_h(u) du,$$

$$\widehat{\mathcal{U}}_k = 2aq^2 + 2 \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq + 2Bq \log(1 + (D+1)q) + 2q \log C_{\operatorname{env}} + q \log 20. \tag{AAM.26}$$

All these constants are fixed by the original anchor, and the retained proof gives $\mathcal{B}_k \leq \widehat{\mathcal{U}}_k$. Since $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$, direct division yields

$$\frac{\widehat{\mathcal{U}}_k}{q_k^2} \longrightarrow 2a, \quad \frac{\widehat{\mathcal{U}}_k}{\mathcal{L}_k} \sim \frac{4a}{\delta} \frac{q_k}{k} \longrightarrow \infty. \tag{AAM.27}$$

Combine these proved bounds on the identical actual scalar by defining

$$U_k^{\operatorname{all}} = \min\{\widehat{\mathcal{U}}_k, U_k^{\operatorname{spec}}, U_k^{\operatorname{ang}}\}. \tag{AAM.28}$$

The following are exact finite comparisons, obtained by substituting $\mathcal{B}_k \leq U_k^{\operatorname{all}}$ into (AAM.12)–

(AAM.14):

$$\begin{aligned}
1 \leq \frac{\bar{\varepsilon}_k}{\mathcal{L}_k} &\leq \exp\left(\frac{U_k^{\text{all}} - \mathcal{B}_k^{\text{crit}} - P_k - \Phi_k}{4q_k}\right) \\
&\leq \frac{4q_k}{e\mathcal{L}_k} \exp\left(\frac{U_k^{\text{all}} + \mathcal{E}_k}{4q_k}\right), \\
1 \leq \frac{\varepsilon_{k,N}}{\mathcal{L}_k} &\leq \exp\left(\frac{U_k^{\text{all}} - \mathcal{B}_k^{\text{crit}} - P_k}{2w_N}\right) \quad (q_k - 1 \leq N \leq 2q_k - 1).
\end{aligned} \tag{AAM.29}$$

For the individual bound, isolate its one term in the nonnegative sum in (AAM.14), and use $\log((\varepsilon_{k,N}^2 + \phi_N^2)/\mathcal{L}_k^2) \geq 2\log(\varepsilon_{k,N}/\mathcal{L}_k)$. These formulas retain every known finite input. They do not supply an upper ratio below one: their exponent with the actual data is nonnegative by (AAM.14). The coarser scalar function $\hat{\mathcal{U}}_k$ gives

$$\frac{\hat{\mathcal{U}}_k - \mathcal{B}_k^{\text{crit}}}{4q_k} = \frac{a}{2}q_k + O_h(k + \log q_k), \tag{AAM.30}$$

which follows from (AAM.22) and (AAM.26), and grows without bound. The angle upper function has the strictly positive $\mathcal{L}_k$ -scale overhead (AAM.25). The full-spectrum expression U_k^{spec} remains the explicitly defined original finite matrix calculation in (AAM.24); none of (CA)–(TW) evaluates its moving- k spectrum by a smaller scalar error. In particular the proved subquadratic comparison $(\mathcal{B}_k - \mathcal{B}_k^\sigma)/q_k^2 \rightarrow 0$ does not give a sublinear upper bound for the normalized control (AAM.8).

136.5 What the fixed-packet degree limit preserves

For each fixed k , the full-jet theorem (FPK.23)–(FPK.29) and the actual comparison (LC.26)–(LC.28) give the exact leading coefficient

$$\begin{aligned}
\lim_{N \rightarrow \infty} \frac{\log V_{k,N}^{\text{ar}}}{\log N} &= -\mathcal{L}_k, \quad \log V_{k,N}^\sigma = -\mathcal{L}_k \log N - \mathcal{B}_{\chi_k} \log \log N + C_{\sigma,k} + o(1), \\
\log V_{k,N}^\Gamma &= -\mathcal{L}_k \log N - \mathcal{B}_{\chi_k} \log \log N + C_{\Gamma,k} + o(1), \\
\mathcal{B}_{\chi_k} &= \begin{cases} (k+1)\ell_k^2, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}
\end{aligned} \tag{AAM.31}$$

The original tensor-Gamma reference in the third line has mass $(2\pi)^{k/2}$; its full constant $C_{\Gamma,k}$ is the one in (FPK.29). The fixed comparison in the second line has mass $\sqrt{2\pi}$. The original remainder and full unit maps are the same for these statements. Thus, for $r_{k,N} = V_{k,N}^{\text{ar}}/V_{k,N}^\Gamma$,

$$\frac{\log r_{k,N}}{\log N} \rightarrow 0. \tag{AAM.32}$$

Equivalently, for every positive ϵ and this fixed k , all sufficiently large N satisfy $N^{-\epsilon} \leq r_{k,N} \leq N^\epsilon$. The threshold here depends on the complete fixed k -packet, and is not specified by the moving endpoints $q_k, 2q_k$.

There are useful exact consequences without exchanging limits. For fixed $t > 1$, (AAM.31) proves

$$\frac{V_{k,N}^\sigma}{V_{k,[tN]}^\sigma} \rightarrow t^{\mathcal{L}_k}, \quad \frac{V_{k,N}^\Gamma}{V_{k,[tN]}^\Gamma} \rightarrow t^{\mathcal{L}_k}, \quad \frac{\log(V_{k,N}^{\text{ar}}/V_{k,[tN]}^{\text{ar}})}{\log N} \rightarrow 0. \tag{AAM.33}$$

The first two assertions follow by subtraction of their complete asymptotics: the constants cancel, the logarithmic ratios tend to $\log t$, the log log differences tend to zero, and both remainders tend to zero. For the third, divide each of the two actual logarithms by its own degree logarithm and use

$\log \lfloor tN \rfloor / \log N \rightarrow 1$. It does not prove that the actual ratio tends to $t^{\mathcal{L}_k}$. For a fixed power $t > 1$ the same argument with $\lfloor N^t \rfloor$ instead gives the nonzero actual exponent

$$\frac{\log(V_{k,N}^{\text{ar}}/V_{k,\lfloor N^t \rfloor}^{\text{ar}})}{\log N} \rightarrow (t-1)\mathcal{L}_k. \quad (\text{AAM.34})$$

These are degree limits on a fixed original quotient; neither proof substitutes a growing k into a fixed- k error term.

Finally the full operator result of the Hilbert-completion chapter is $\|G_{k,N}^{\text{ar}}\| \rightarrow 0$ for each fixed k , while $\mathcal{O}_{k,N}R_{k,N} = \mathcal{I}_k \neq 0$. Its exact effect on the control matrix can be computed. Put $W_{k,N} = A^*G_{k,N} + G_{k,N}A - kG_{k,N}$. Then

$$\|W_{k,N}\| \leq 2\|A - cI\| \|G_{k,N}\| \rightarrow 0, \quad G_{k,N}^{-1/2}W_{k,N}G_{k,N}^{-1/2} = H_N, \quad \|H_N\| \geq \mathcal{L}_k. \quad (\text{AAM.35})$$

The first inequality expands $W = (A - cI)^*G + G(A - cI)$ and applies submultiplicativity in the original fixed frame. The second is its literal congruence (AAM.6). There is no bounded inverse factor in that congruence: $G_{k,N} \rightarrow 0$ forces $\|G_{k,N}^{-1/2}\| \rightarrow \infty$. In the actual source norm the normalized lift remains exactly isometric:

$$(R_{k,N}G_{k,N}^{-1/2})^*H_{k,N}(R_{k,N}G_{k,N}^{-1/2}) = I_{E_k}, \quad \|R_{k,N}G_{k,N}^{-1/2}\| = 1, \quad \|R_{k,N}G_{k,N}^{-1/2}\|_{\text{HS}}^2 = q_k. \quad (\text{AAM.36})$$

This is direct substitution of $R_{k,N}^*H_{k,N}R_{k,N} = G_{k,N}$. Thus the original vanishing representatives, their nonzero arithmetic classes, and the amplified normalized boundary are connected by the explicit maps (AAM.35)–(AAM.36). Their Hilbert reduction is the proved quotient map killing the dense relation space; it is not an exclusion of the nonzero original tensor class.

In Deligne’s paragraph 7.3 the tensor eigenvalue and the proved exclusion band belong to the same amplified object, and its exponent error stays constant while the displacement grows. In the present original arithmetic data the corresponding displacement survives as $k\delta$ and the full trace as $\mathcal{L}_k$. The exact upper comparisons currently proved here are (AAM.24)–(AAM.30) on the window scalar and (AAM.29) on its normalized boundary. Their rates are calculated above: the small norm-window factor is paired with the forced growing quotient factor (AAM.21), the scalar comparison overhead has the positive amplified rate (AAM.20), (AAM.25), and the fixed-packet Gram limit retains the normalizing inverses (AAM.35). These statements prove the exact arithmetic match of the available elements; they do not insert Deligne’s finite-field exclusion theorem into the original source.

137 Deligne’s final amplification: the exact contradiction

The particular argument requested here is in Pierre Deligne, *La conjecture de Weil I* (1974), Lemma 7.1 and paragraph 7.3, printed pages 298 and 301. It is not a quotation from *Weil II* (1980). The [original paper](#) is the primary source; the [English translation](#) was checked against it. The source input is the proved estimate below and the Frobenius-equivariant Künneth isomorphism. The subsequent calculation spells out their contradiction mechanism.

137.1 The proved bound being reused

Let $q > 1$ be the cardinality of the finite base field and let ℓ be a prime different from its characteristic. For a smooth, projective, geometrically irreducible variety $Y_0/\mathbb{F}_q$ of **even** dimension D , put $Y = Y_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}$. Deligne’s Lemma 7.1 establishes that each eigenvalue β of geometric Frobenius on $H^D(Y, \mathbb{Q}_\ell)$ is algebraic and that every complex conjugate satisfies

$$q^{D/2-1/2} \leq |\beta| \leq q^{D/2+1/2}. \quad (\text{DA1})$$

This already excludes everything outside a band of width one in the exponent. The additive error $1/2$ is the same for every such variety and every even dimension. There is no unrecorded constant depending on its Betti numbers in (DA1).

137.2 Constructing the object to which the earlier exclusion applies

Take a smooth, projective, geometrically irreducible $X_0/\mathbb{F}_q$ of dimension d . Let X be its geometric base change and F_X^* its geometric Frobenius action. Take an eigenvalue α in the middle group $H^d(X, \mathbb{Q}_\ell)$, and an eigenvector $v \neq 0$ after extending coefficients to $\overline{\mathbb{Q}_\ell}$.

For every positive even integer k , form the actual product variety

$$X_0^k = \underbrace{X_0 \times_{\mathbb{F}_q} \cdots \times_{\mathbb{F}_q} X_0}_{k \text{ factors}}.$$

It is again smooth, projective and geometrically irreducible. Its dimension is kd , which is even. The field cardinality stays q . The Frobenius-equivariant Künneth map has the precise form

$$\bigoplus_{i_1 + \cdots + i_k = kd} \bigotimes_{j=1}^k H^{i_j}(X, \overline{\mathbb{Q}_\ell}) \xrightarrow{\sim} H^{kd}(X^k, \overline{\mathbb{Q}_\ell}),$$

$$x_1 \otimes \cdots \otimes x_k \mapsto \text{pr}_1^* x_1 \smile \cdots \smile \text{pr}_k^* x_k. \quad (\text{DA2})$$

In particular, the summand with $i_1 = \cdots = i_k = d$ injects into the middle cohomology of the product. The tensor $v^{\otimes k}$ is nonzero: choose a linear functional λ with $\lambda(v) = 1$; then $\lambda^{\otimes k}(v^{\otimes k}) = 1$. It stays nonzero under the isomorphism (DA2). Frobenius acts on it by

$$(F_X^*)^{\otimes k} v^{\otimes k} = (\alpha v) \otimes \cdots \otimes (\alpha v) = \alpha^k v^{\otimes k}. \quad (\text{DA3})$$

Thus the construction retains the offending eigenvalue as α^k in exactly the cohomological degree covered by (DA1). No semisimplicity assertion is needed. Nor is a symmetric or exterior quotient taken: even when d is odd, the external tensor of v with itself in different factors survives.

Algebraicity is not an extra assumption: applying (DA1) to X_0^2 shows α^2 is algebraic; then α is algebraic over the algebraic extension containing α^2 . Every complex embedding of $\mathbb{Q}(\alpha)$ restricts to one on $\mathbb{Q}(\alpha^k)$, so the following inequalities hold for each fixed complex conjugate of α .

137.3 Apply the earlier bound to the newly constructed product

Substitute $Y_0 = X_0^k$, $D = kd$ and $\beta = \alpha^k$ into (DA1):

$$q^{kd/2-1/2} \leq |\alpha^k| \leq q^{kd/2+1/2}. \quad (\text{DA4})$$

Taking positive real k th roots, and using $|\alpha^k| = |\alpha|^k$, gives

$$q^{d/2-1/(2k)} \leq |\alpha| \leq q^{d/2+1/(2k)} \quad (k = 2, 4, 6, \dots). \quad (\text{DA5})$$

The expected exponent $kd/2$ and the actual exponent of $|\alpha|^k$ both scale by k . The already-proved error remains $1/2$. This is precisely why its contribution after taking the root is $1/(2k)$.

137.4 The finite contradiction, without relying on informal limiting language

For a fixed complex conjugate define

$$\varepsilon = \log_q |\alpha| - \frac{d}{2}.$$

Suppose $\varepsilon \neq 0$. Choose the explicit even positive integer

$$k = 2 \left(\left\lfloor \frac{1}{4|\varepsilon|} \right\rfloor + 1 \right). \quad (\text{DA6})$$

Then $k > 1/(2|\varepsilon|)$, hence $k|\varepsilon| > 1/2$.

If $\varepsilon > 0$, (DA3) supplies an actual eigenvalue with

$$|\alpha^k| = q^{kd/2+k\varepsilon} > q^{kd/2+1/2},$$

contradicting the upper bound in (DA4). If $\varepsilon < 0$, it supplies

$$|\alpha^k| = q^{kd/2-k|\varepsilon|} < q^{kd/2-1/2},$$

contradicting the lower bound in (DA4). Thus $\varepsilon = 0$ and

$$\boxed{|\alpha| = q^{d/2}}. \quad (\text{DA7})$$

An arbitrarily small fixed deviation is enough. A smaller deviation merely requires a larger finite product. The exact boundary $\varepsilon = 0$ is retained and satisfies every bound. This closes the middle-degree argument; the cited paper's paragraph 7.2 then uses weak Lefschetz, Poincaré duality, components and finite field extensions to cover the other degrees.

137.5 Why repeating a bound is not sufficient by itself

In exponent coordinates the entire product family is the intersection

$$\bigcap_{k \in 2\mathbb{N}_{>0}} \left[-\frac{1}{2k}, \frac{1}{2k} \right] = \{0\}. \quad (\text{DA8})$$

To prove the equality, zero belongs to every interval; a nonzero real number is excluded by the explicit k in (DA6). This is an exhaustive exclusion, not a computation for finitely many chosen dimensions.

Both quantitative and geometric ingredients are essential to the calculation. The same eigenclass maps to a nonzero tensor class; here the exclusion for the resulting class has error exactly $1/2$. With a nonnegative error e_k , the corresponding calculation would produce $|\varepsilon| \leq e_k/k$. The resulting intersection is the singleton zero exactly when $\inf_{k \in 2\mathbb{N}_{>0}} e_k/k = 0$. Sublinear error $e_k = o(k)$ is sufficient; convergence along every even integer is not necessary. If e_k were a fixed positive multiple of k , that calculation would leave a positive-width band. No such proportional error is present in (DA4).

Extending the base field alone gives a different calculation. Take d even so that (DA1) applies directly. Replacing $\mathbb{F}_q$ by $\mathbb{F}_{q^r}$ replaces α by α^r but leaves dimension d unchanged. The same weak bound would read

$$(q^r)^{d/2-1/2} \leq |\alpha|^r \leq (q^r)^{d/2+1/2}.$$

Its r th root is the original weak bound; the error does not shrink. The product construction keeps q fixed while increasing dimension and tensor exponent together.

137.6 The logical pattern recorded for the actual Split-Zero object

The pattern is: an offending class survives an exact amplification map; the amplified class belongs to a previously bounded object; its fixed nonzero displacement grows linearly with tensor degree; a proved exclusion has smaller growth; a finite tensor degree contradicts that exclusion.

In the actual programme the additive tensor generator is $A_h^{(k)} = \sum_j A_{h,j}$. A nonzero eigenvector with eigenvalue $\rho = 1/2 + \delta + i\gamma$ has nonzero tensor power with eigenvalue $k\rho$. Its real displacement from $k/2$ is exactly $k\delta$. This statement follows by applying each summand to the original tensor and by the same linear-functional test proving nonvanishing. The complete local nilpotent sum remains $\sum_j N_j$; on a full order- m block its top nonzero power is $k(m-1)$ with coefficient $(k(m-1))!/((m-1)!)^k$ on $\prod_j N_j^{m-1}$. Consequently no multiplicity is discarded in making this comparison.

Deligne's numerical bound is a theorem about his finite-field cohomology. It is not inserted into the Split-Zero tensor source as an assumption. The present construction keeps every already-calculated original arithmetic bound and its actual error term on the same amplified source. Their compatibility or contradiction must be calculated there. The finite contradiction (DA6) records exactly what a successful amplification/exclusion calculation accomplishes.

138 The matching passage in Weil II

The local S20 text identifies a second, particularly close match to the recollection about a bound involving a minus sign and two: *Weil II*, Lemma 1.8.1, its formulas 1.8.1.1–1.8.1.2, and the explicit return to Corollary 1.4.3. The controlling text is [current English S20 article \(source locator\)](#), lines 1073–1095, checked against the current French transcription. The [published paper](#) is Deligne, *La conjecture de Weil II*, IHÉS 52 (1980), 137–252.

Let $j : U_0 \hookrightarrow X_0$ be an open immersion of smooth geometrically irreducible curves over $\mathbb{F}_q$, with finite complement S_0 . Fix $\ell \neq \text{char } \mathbb{F}_q$ and the coefficient embedding $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ in Deligne's convention. Let $\mathcal{F}_0$ be lisse on U_0 and pointwise ι -pure of weight β . For $N(x) = q^{\deg x}$ use the exact convention

$$w_{N(x)}(\alpha) = 2 \log_{N(x)} |\iota\alpha|.$$

After making X_0 affine, the cohomological identity is

$$\begin{aligned} \prod_{x \in |U_0|} \iota \det(1 - F_x t^{\deg x}, \mathcal{F}_0)^{-1} & \prod_{x \in |S_0|} \iota \det(1 - F_x t^{\deg x}, j_* \mathcal{F}_0)^{-1} \\ &= \frac{\iota \det(1 - Ft, H_c^1(X, j_* \mathcal{F}))}{\iota \det(1 - Ft, H_c^2(X, j_* \mathcal{F}))}. \end{aligned} \quad (\text{DII1})$$

The open-set Euler product is holomorphic and nonzero in $|t| < q^{-(\beta+2)/2}$. Corollary 1.4.3 excludes poles of the right side there. The remaining finite product has numerator one, so no denominator root can cancel inside that product. Dividing (DII1) by its nonzero open-set factor therefore excludes each boundary local pole in that disc.

For an eigenvalue α in a boundary factor, a root of $1 - \iota\alpha t^{\deg x}$ has modulus $|\iota\alpha|^{-1/\deg x}$. Pole exclusion gives

$$|\iota\alpha|^{-1/\deg x} \geq q^{-(\beta+2)/2}, \quad |\iota\alpha| \leq N(x)^{(\beta+2)/2}, \quad w_{N(x)}(\alpha) \leq \beta + 2. \quad (\text{DII2})$$

Now take tensor powers. Write $W = \mathcal{F}_{\bar{\eta}}$ for the geometric generic fibre near x , and I_x for its local inertia group. The boundary stalk is $(j_* \mathcal{F})_{\bar{x}} = W^{I_x}$. The relevant map is literally

$$(W^{I_x})^{\otimes k} \hookrightarrow (W^{\otimes k})^{I_x}, \quad v_1 \otimes \cdots \otimes v_k \longmapsto v_1 \otimes \cdots \otimes v_k. \quad (\text{DII3})$$

The tensor inclusion is injective by a basis extension. Its image is invariant because inertia fixes every factor; it commutes with Frobenius because Frobenius normalizes inertia. Consequently an eigenvector $v \neq 0$ at the boundary has a nonzero image $v^{\otimes k}$ with eigenvalue α^k . The sheaf $\mathcal{F}_0^{\otimes k}$ is pointwise pure of weight $k\beta$: eigenvalues in a tensor product are products of eigenvalues, so their complex moduli multiply. Applying the already-proved (DII2) to it yields

$$k w_{N(x)}(\alpha) = w_{N(x)}(\alpha^k) \leq k\beta + 2, \quad w_{N(x)}(\alpha) \leq \beta + \frac{2}{k}. \quad (\text{DII4})$$

A violation $w_{N(x)}(\alpha) = \beta + \varepsilon$ with $\varepsilon > 0$ contradicts (DII4) at the finite integer $k = \lfloor 2/\varepsilon \rfloor + 1$. Thus $w_{N(x)}(\alpha) \leq \beta$. This lemma gives an upper bound at the boundary; it does not assert equality of all boundary weights.

This is the exact pattern requested: first prove a larger forbidden region by returning to an earlier cohomological exclusion; then map the same offending class injectively into tensor powers until it would enter that forbidden region. The earlier exclusion is reapplied to an object proved to retain the offending eigenvalue.

The older English math-mode file has an incorrect H^0 denominator in its transcription of 1.8.1.1. The current S20 text has H_c^2 , as used in (DII1). The historical file is retained unchanged and its status is recorded in the source index.

139 The actual final bootstrap in Weil II

The proof of the curve theorem in *Weil II*, 3.2.3–3.2.15, uses a more specific mechanism than tensor amplification alone. A local calculation makes the weights of every relevant nonconstant constituent integral. An earlier analytic theorem excludes weight 2 and all larger weights. Consequently these integral weights are at most 1. The remaining constituents are geometrically constant on the projective line, where their first cohomology vanishes. These two exclusions make a product-square argument improve a universal error e to $e/2$. The proof below retains the schemes, coefficient sheaves, boundary terms, maps, twists, and indices which make that improvement possible.

The primary source is Pierre Deligne, *La conjecture de Weil II*, Publications mathématiques de l’IHÉS 52 (1980), 137–252: the introduction, printed pp. 140–141; 1.8.4, pp. 175–176; 2.2.10, p. 196; 3.1, pp. 197–200; 3.2, pp. 200–204; and 3.3, pp. 204–207. All numbered results cited in this section are Deligne’s published results unless a displayed tag begins with WDB. The local current S20 English and French texts were read alongside the controlling published French scan. This section is an exposition of that proof and its exact algebraic steps, not a claim that a construction over the absolute base has already acquired the finite-field sheaves in this proof.

139.1 Conventions, original weight, and the universal assertion

Fix a finite field $\mathbb{F}_q$, an algebraic closure $\overline{\mathbb{F}}_q$, a prime $\ell \neq \text{char } \mathbb{F}_q$, and Deligne’s coefficient isomorphism $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$. For a scheme or sheaf with subscript 0, omission of that subscript means extension to $\overline{\mathbb{F}}_q$. Use geometric Frobenius and the weight convention

$$w_q(\alpha) = 2 \log_q |\iota \alpha|, \quad w_q(\alpha^d) = w_q(\alpha), \quad w_q(\alpha^r) = r w_q(\alpha). \quad (\text{WDB1})$$

A Tate twist (a) multiplies Frobenius eigenvalues by q^{-a} and lowers their weights by $2a$.

Let U_0 be a smooth curve and let $\mathcal{F}_0$ be a lisse pointwise ι -pure sheaf of weight β on U_0 . Deligne proves the universal assertions

$$\mathbf{B}_k : \quad w_q(\alpha) \leq \beta + 1 + 2^{-k} \quad \text{for every eigenvalue of } F \text{ on } H_c^1(U, \mathcal{F}), \quad k = 0, 1, 2, \dots \quad (\text{WDB2})$$

The quantifiers range over all such curves, sheaves, weights, and finite fields. This universality is used when the induction is applied to new coefficient sheaves on the base of a pencil, rather than only to the original $\mathcal{F}_0$.

Here is the precise relation to the weight-zero formulation of 3.2.4. Choose $c \in \overline{\mathbb{Q}}_\ell^*$ with $\iota(c) = q^{-\beta/2}$, and let $\mathcal{L}_c$ be the constant geometric Weil sheaf whose arithmetic datum is Frobenius c . Deligne's twisting operation 1.2.7 gives

$$\mathcal{F}_0^{[0]} = \mathcal{F}_0 \otimes \mathcal{L}_c, \quad H_c^1(U, \mathcal{F}^{[0]}) = H_c^1(U, \mathcal{F}) \otimes \mathcal{L}_c, \quad \alpha^{[0]} = c\alpha, \quad w_q(\alpha^{[0]}) = w_q(\alpha) - \beta. \quad (\text{WDB3})$$

Thus the original weight and eigenvalue are recovered by tensoring with $\mathcal{L}_{c^{-1}}$ and multiplying by c^{-1} . We use $\mathcal{F}_0^{[0]}$, not a change in the meaning of $\mathcal{F}_0$, for the auxiliary weight-zero proof. This twist is made in the Weil-sheaf setting allowed by Deligne; it is not an assertion that every such character extends to an ℓ -adically continuous character of the profinite arithmetic Galois group.

We will repeatedly use the following elementary linear algebra. If a linear automorphism preserves a finite filtration, a basis adapted to the filtration makes its matrix block triangular. Its characteristic polynomial is the product of the characteristic polynomials on the successive quotients. Hence a uniform upper weight bound is inherited by subspaces, quotients, and finite extensions. For a convergent Frobenius-equivariant spectral sequence, each term on a later page is a subquotient of an earlier term; the abutment has a filtration with the E_∞ terms as successive quotients. These facts prove the weight comparisons below without assuming spectral-sequence degeneration or Frobenius semisimplicity.

139.2 The two earlier inputs: a strict bound and local integral shifts

Deligne's 1.8.2 supplies the initial estimate $w_q(\alpha) \leq \beta + 2$, hence B_0 . For a smooth affine curve it applies directly with the open immersion in 1.8.2 taken to be the identity: the Euler product and the H_c^2 denominator have no zero in $|t| < q^{-(\beta+2)/2}$, so the numerator has none there. For a projective component, remove a closed point and use the surjection from the compactly supported first cohomology of the remaining affine curve, proved explicitly in (WDB7)–(WDB8) below. Connected components are handled by a finite extension and (WDB1).

The stronger earlier result used in the induction is 2.2.10:

$$\alpha \in \text{Spec}(F; H_c^1(U, \mathcal{F})) \implies w_q(\alpha) < \beta + 2. \quad (\text{WDB4})$$

The inequality is strict. Deligne proves it in Section 2 using the Hadamard-de la Vallée-Poussin method and the cohomological expression for the relevant L -function. After weight-zero twisting and reduction to an irreducible representation, 2.2.9 excludes zeros on the boundary line $\text{Re } s = 1$; 2.2.8 and 2.2.9.1 identify this with the absence of the corresponding numerator roots on $|t| = q^{-1}$. The possible geometrically constant character is accounted for by the simple pole stated there, rather than discarded. Its pole order equals the order of the H_c^2 denominator zero, so a numerator zero at that point would contradict the exact simple pole and is excluded as well. Since a reciprocal root $t = \iota(\alpha)^{-1}$ has modulus $q^{-w_q(\alpha)/2}$, the absence of numerator roots for $|t| \leq q^{-1}$ is exactly $w_q(\alpha) < 2$ in weight zero. This is an earlier proved input in the order of the paper; it is not deduced by assuming the conclusion of 3.2.3.

The second earlier result is local monodromy purity, 1.8.4. If a lisse sheaf on a punctured smooth curve is pointwise ι -pure of weight b , and M is the centered local monodromy filtration on its geometric generic stalk V , then

$$\text{Gr}_j^M V \text{ is } \iota\text{-pure of weight } b + j, \quad j \in \mathbb{Z}. \quad (\text{WDB5})$$

In particular every Frobenius eigenvalue of the local Weil representation has weight in $b + \mathbb{Z}$. Under unipotent inertia the graded pieces have trivial inertia: writing $\rho(\sigma) = \exp(t_\ell(\sigma)N)$, the operator N lowers M by two, so its exponential acts as the identity on every Gr_j^M .

The weight shift in (WDB5) is itself proved in 1.8.4, not read off from the existence of a nilpotent operator. To recall its numerical content, twist to weight zero and use a finite covering so that all inertia is unipotent. Then $\ker N = V^I$, and 1.8.1 says its eigenvalues have modulus at most 1. For a primitive bottom piece $P_{-j}V$, Deligne's tensor decomposition 1.6.14.4 makes

$$P_{-j}V \otimes P_{-j}V(-j) \text{ a direct summand of } P_0(V \otimes V). \quad (\text{WDB6})$$

If α occurs in $P_{-j}V$, the resulting eigenvalue $\alpha^2 q^j$ therefore has modulus at most 1, giving $|\iota\alpha| \leq q^{-j/2}$. Applying the same argument to the dual primitive piece, whose corresponding eigenvalue is $\alpha^{-1} q^{-j}$ by 1.6.14.5, gives $|\iota\alpha| \geq q^{-j/2}$. Equality follows. The primitive decomposition 1.6.14.3 and the Tate shifts on its strings give (WDB5). Thus the integral shifts later used by the pencil argument have already been tied to the actual local Weil action.

139.3 The exact open-extension map used by the induction

Let $j : W_0 \hookrightarrow C_0$ be a dense open immersion of curves, with finite complement, and let $\mathcal{K}_0$ be a constructible sheaf on C_0 . Extension by zero gives the short exact sequence

$$0 \longrightarrow j_! j^* \mathcal{K}_0 \longrightarrow \mathcal{K}_0 \longrightarrow \mathcal{K}_0 / j_! j^* \mathcal{K}_0 \longrightarrow 0. \quad (\text{WDB7})$$

The quotient is supported on finitely many points. Over the algebraic closure its positive-degree compactly supported cohomology vanishes: each geometric point has only degree-zero cohomology and a finite disjoint union has the direct sum of those cohomologies. The long exact sequence of (WDB7) consequently gives the Frobenius-equivariant surjection

$$H_c^1(W, j^* \mathcal{K}) \twoheadrightarrow H_c^1(C, \mathcal{K}). \quad (\text{WDB8})$$

If W_0 is smooth and $j^* \mathcal{K}_0$ is lisse and pointwise pure of weight b , assertion B_k applied to the source proves that every eigenvalue of the target has weight at most $b + 1 + 2^{-k}$. This is precisely 3.2.5(k). No purity of the possibly singular boundary stalks of $\mathcal{K}_0$ is assumed in this step.

139.4 The product and pencil, with all three reduction hypotheses

Assume B_k . Initially impose the three hypotheses of 3.2.6 on the auxiliary sheaf $\mathcal{F}_0^{[0]}$.

- (A) If X_0 is the smooth projective compactification of U_0 , the local monodromy of $\mathcal{F}^{[0]}$ at $X - U$ is tame and unipotent.
- (B) Put $S_0 = X_0 \times X_0$, $V_0 = U_0 \times U_0$, and $D_0 = S_0 - V_0$. Choose a projective embedding and a hyperplane pencil with axis A_0 satisfying 3.1.1(A)–(D): the axis is transverse to S and disjoint from D ; the only singularities of the pencil on S are ordinary quadratic points outside D ; its only singularities on the normalization of D are ordinary quadratic points away from the preimages of the nodes of D ; and each exceptional fibre contains exactly one exceptional point.
- (C) These exceptional points are defined over $\mathbb{F}_q$.

Write $P_0 = A_0^* \simeq \mathbb{P}_{\mathbb{F}_q}^1$ for the parameter line of the pencil, and use the incidence surface

$$\tilde{S}_0 = \{(s, t) \in S_0 \times P_0 : s \in H_t\}, \quad \pi : \tilde{S}_0 \longrightarrow S_0, \quad \bar{f} : \tilde{S}_0 \longrightarrow P_0. \quad (\text{WDB9})$$

It is the blow-up at the finite set $S_0 \cap A_0$. Put $\tilde{V}_0 = \pi^{-1}(V_0)$, let $\tilde{j} : \tilde{V}_0 \hookrightarrow \tilde{S}_0$ be the open immersion, and set $f = \tilde{f}\tilde{j}$. The actual coefficient sheaf is

$$\mathcal{G}_0 = \text{pr}_1^* \mathcal{F}_0^{[0]} \otimes \text{pr}_2^* \mathcal{F}_0^{[0]} \quad \text{on } V_0. \quad (\text{WDB10})$$

Both tensor factors are retained. In terms of the original sheaf this is $(\mathcal{F}_0 \boxtimes \mathcal{F}_0) \otimes \mathcal{L}_{c^2}$, so its original weight 2β has been recorded by an explicit invertible twist. Its auxiliary weight is zero.

Let $T_0 \subset P_0$ be the finite set of exceptional values, $W_0 = P_0 - T_0$, and $w : W_0 \hookrightarrow P_0$. Define

$$\mathcal{R}_0^b = R^b f_! \pi^* \mathcal{G}_0 = R^b \tilde{f}_* j_! \pi^* \mathcal{G}_0. \quad (\text{WDB11})$$

These sheaves are lisse over W_0 . The compact-support Leray spectral sequence is Frobenius-equivariant:

$$E_2^{ab} = H^a(P, \mathcal{R}^b) \implies H_c^{a+b}(\tilde{V}, \pi^* \mathcal{G}). \quad (\text{WDB12})$$

Since both the base and the fibres have dimension one, total degree two has only the terms $(a, b) = (0, 2), (1, 1), (2, 0)$.

139.5 Why the middle coefficient sheaf has pure constituents

Let $x \in |W_0|$. Denote the proper fibre of $\tilde{f}$ by $\tilde{S}_x$, its open part by Y_x , and the open immersion by $j_x : Y_x \hookrightarrow \tilde{S}_x$. Proper base change applied to (WDB11) gives the exact stalk identification

$$(\mathcal{R}_0^b)_{\bar{x}} \simeq H^b(\tilde{S}_{\bar{x}}, j_{x!} \pi^* \mathcal{G}) \simeq H_c^b(Y_{\bar{x}}, \pi^* \mathcal{G}). \quad (\text{WDB13})$$

These are identifications of representations of geometric Frobenius over $\kappa(x)$. The open fibre is a smooth curve for $x \in W_0$; it need not be connected.

There is a notation defect in the published display 3.2.7.1, printed p. 202: it writes ordinary $H^1(Y, \pi^* \mathcal{H})$ for an open fibre while the left side is $R^1 f_!$. The controlling French scan and both current S20 transcriptions show the same display. Equation (WDB13) states the corrected support convention and its base-change proof. Ordinary cohomology of the open fibre is not used in place of compactly supported cohomology.

Set $\mathcal{H}_0 = \mathcal{G}_0 \oplus \mathcal{G}_0^\vee$. At a closed point the eigenvalues of $\mathcal{G}_0$ have ι -modulus one. The images of the eigenvalues of its dual are their inverses, hence their complex conjugates. Therefore the characteristic polynomial of the direct sum has real coefficients, and $\mathcal{H}_0$ is pointwise pure of weight zero and ι -real. Deligne's 3.2.1–3.2.2 applied to the smooth fibre in (WDB13) shows that $w^* R^1 f_! \pi^* \mathcal{H}_0$ is ι -real. The sheaf $w^* \mathcal{R}_0^1$ is an actual direct summand of it because pullback and $R^1 f_!$ preserve finite direct sums. The purity criterion 1.5.1 now gives a finite filtration of $w^* \mathcal{R}_0^1$ whose successive quotients are lisse and pointwise ι -pure. This proves 3.2.7 with its coefficient and support maps specified.

To make this use of 3.2.1 explicit, let $\mathcal{E}_0$ be pointwise ι -pure of weight b and ι -real on a smooth geometrically connected curve C_0 . Put $P_a(t) = \iota \det(1 - Ft, H_c^a(C, \mathcal{E}))$. Grothendieck's formula says that $P_1/(P_0 P_2)$ has real Taylor coefficients and hence is a real rational function. The roots of P_2 all have modulus $q^{-(b+2)/2}$ by 1.4.3; those of P_0 , if any, have modulus $q^{-b/2}$; and (WDB4) excludes roots of P_1 on the former circle. Consequently the poles on that circle, with their multiplicities, are exactly the roots of P_2 . They are stable under complex conjugation, so P_2 has real coefficients and constant term one. If C is affine, $P_0 = 1$. If it is projective, the correct scalar duality is

$$H^0(C, \mathcal{E})^\vee \simeq H_c^2(C, \mathcal{E}^\vee(1)). \quad (\text{WDB13a})$$

The coefficient sheaf on the right is again ι -real and pure, so the just-proved degree-two argument and inversion of eigenvalues show that P_0 is real. It follows that P_1 is real. If geometric components are permuted in a cycle of length d , their cohomological characteristic polynomial is $\det(1 - F^d t^d, H_c^a(C', \mathcal{E}))$ for one component C' ; the preceding argument over $\mathbb{F}_{q^d}$ makes this polynomial real, proving the extension 3.2.2.

There is a second printed notation defect in the proof of 3.2.1, p. 200: it writes $\mathcal{E}^\vee(-1)$ in place of the $\mathcal{E}^\vee(1)$ in (WDB13a). The printed group equals $H^0(C, \mathcal{E})^\vee(-2)$, so it still implies the required reality after the real scalar factor q^2 is retained. Formula (WDB13a) records the exact untwisted duality used here; the later 3.2.15 itself has the correct (1) twist.

139.6 Vanishing cycles detect the nonconstant weights

The calculation in 3.1 treats exactly three exceptional geometries: an ordinary node of the fibre away from D , tangency of the fibre to one smooth branch of D , and passage of the fibre through a node of D . Let $\bar{t} \in T$ and $\bar{x}$ be its unique exceptional point. The vanishing-cycle sheaves are supported at $\bar{x}$ and vanish in every degree other than one. If $\varepsilon(B)$ is the one-dimensional sign representation of the two branches or two local sheets, the calculation is

$$\begin{aligned} \text{node outside } D : \quad & \Phi_{\bar{x}}^1 = \mathcal{G}_{\bar{x}}(-1) \otimes \varepsilon(B), \\ \text{tangency to } D : \quad & \text{Gr}^a \Phi_{\bar{x}}^1 = \text{Gr}^a \mathcal{G}_{\bar{x}}^{\text{loc}} \otimes \varepsilon(B), \\ \text{node of } D : \quad & \text{Gr}^a \Phi_{\bar{x}}^1 = \text{Gr}^a \mathcal{G}_{\bar{x}}^{\text{loc}} \otimes \varepsilon(B). \end{aligned} \tag{WDB14}$$

Here the latter two expressions mean the stalks of the constant extensions of the local graded sheaves of 3.1.2; they do not mean that the original sheaf was defined at a deleted boundary point. For the first case the Picard–Lefschetz vanishing cycle supplies the twist (-1) . For tangency, apply vanishing cycles to $0 \rightarrow j_! \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_{\ell D} \rightarrow 0$: the constant ambient term has no vanishing cycles, which shifts the divisor computation by one. At a node of D , the corresponding two-step exact sequence through the normalization of D shifts the computation by two and gives the same degree-one sign representation. These are 3.1.3.1–3.1.5.4.

For each original boundary point of $X - U$, (WDB5) applied to $\mathcal{F}_0^{[0]}$ gives integral local weights. Hypothesis (A) makes its graded sheaves locally constant across the puncture. Taking their external tensor product gives the local filtration of $\mathcal{G}_0$ used in (WDB14); its weights are sums of integers. At an interior point the weight is zero. The sign representation has Frobenius eigenvalue 1 or -1 and weight zero, and (-1) adds two to the weight. Every case of (WDB14) therefore proves

$$\text{all Frobenius eigenvalues of } \Phi_{\bar{x}}^1 \text{ have integral weights.} \tag{WDB15}$$

The proof retains the distinction between the three geometric supports while obtaining the same integrality conclusion from them.

Let $\bar{\eta}$ be the local geometric generic point at $\bar{t}$. The vanishing of Φ^a for $a \neq 1$ gives the exact sequence 3.2.9.1

$$0 \rightarrow \mathcal{R}_{\bar{t}}^1 \rightarrow \mathcal{R}_{\bar{\eta}}^1 \rightarrow \Phi_{\bar{x}}^1 \rightarrow \mathcal{R}_{\bar{t}}^2 \rightarrow \mathcal{R}_{\bar{\eta}}^2 \rightarrow 0. \tag{WDB16}$$

In particular $\mathcal{R}^1$ has no sections supported at $\bar{t}$. Varying $\bar{t}$ gives the injective sheaf map $\mathcal{R}_0^1 \hookrightarrow w_* w^* \mathcal{R}_0^1$ of 3.2.8.

Choose the finite pure-constituent filtration G on $w^* \mathcal{R}_0^1$ obtained above and extend it by intersection:

$$G^a \mathcal{R}_0^1 = \mathcal{R}_0^1 \cap w_* G^a(w^* \mathcal{R}_0^1), \quad \mathcal{K}_{a,0} = \text{Gr}_G^a \mathcal{R}_0^1. \tag{WDB17}$$

The intersection is in $w_* w^* \mathcal{R}_0^1$. Write b_a for the pure weight of $w^* \mathcal{K}_{a,0}$. At $\bar{t}$, this is the filtration induced by intersection with the subspace in (WDB16); consequently taking its associated graded sequence gives

$$0 \rightarrow (\mathcal{K}_{a,0})_{\bar{t}} \rightarrow (\mathcal{K}_{a,0})_{\bar{\eta}} \rightarrow A_{a,\bar{t}} \rightarrow 0, \quad A_{a,\bar{t}} \text{ a subquotient of } \Phi_{\bar{x}}^1. \tag{WDB18}$$

For completeness, this subquotient statement is elementary for an inclusion $E \hookrightarrow V$ and a filtration of V : filter E by $E \cap G^a V$ and V/E by the images of $G^a V$. The associated graded sequence is exact, so the cokernel in degree a is a graded piece of V/E . Here V/E injects into $\Phi_{\bar{x}}^1$ by (WDB16).

If $A_{a,\bar{t}} \neq 0$, select any one of its Frobenius eigenvalues λ . As a subquotient of the vanishing cycle it has an integral weight by (WDB15). As a quotient of the local Weil stalk of the pure sheaf $w^* \mathcal{K}_{a,0}$, its weight lies in $b_a + \mathbb{Z}$ by (WDB5). Thus

$$A_{a,\bar{t}} \neq 0 \implies w_{N(t)}(\lambda) \in \mathbb{Z} \cap (b_a + \mathbb{Z}) \implies b_a \in \mathbb{Z}. \tag{WDB19}$$

If instead every $A_{a,\bar{t}}$ vanishes, (WDB18) identifies the special stalk with the entire generic stalk at every boundary point. The image of specialization is inertia invariant, so inertia acts trivially and the sheaf is lisse across every such point. It is therefore lisse on $P \simeq \mathbb{P}^1$; its geometric monodromy is trivial because $\mathbb{P}^1$ over an algebraically closed field is simply connected. It follows that $\mathcal{K}_a$ is geometrically constant. Frobenius on its constant fibre can still be nontrivial. We have proved the exact alternative of 3.2.9:

$$\mathcal{K}_a \text{ is geometrically constant on } P, \quad \text{or} \quad b_a \in \mathbb{Z}. \quad (\text{WDB20})$$

The pure weight b_a can be measured at any closed point of W_0 . There its eigenvalues are among those of (WDB13) with $b = 1$. Applying the strict bound (WDB4) to the weight-zero coefficient sheaf on that smooth fibre proves $b_a < 2$, which is 3.2.10. Combining this with (WDB20) gives

$$\mathcal{K}_a \text{ not geometrically constant} \implies b_a \in \mathbb{Z}, \quad b_a < 2, \implies b_a \leq 1. \quad (\text{WDB21})$$

This is the decisive return to the earlier exclusion. A weak bound $b_a \leq 2$ would leave the integer 2 available and would not start the improvement from $k = 0$.

139.7 Leray transfers the surviving error only once

If $\mathcal{K}_a$ is geometrically constant with fibre L_a , then

$$H^1(P, \mathcal{K}_a) = H^1(\mathbb{P}^1, \overline{\mathbb{Q}}_\ell) \otimes L_a = 0. \quad (\text{WDB22})$$

The equality is Frobenius-equivariant and holds for every Frobenius action on L_a . No numerical weight bound for L_a is required. In the other case, (WDB8), B_k , and (WDB21) give

$$\alpha \in \text{Spec}(F; H^1(P, \mathcal{K}_a)) \implies w_q(\alpha) \leq b_a + 1 + 2^{-k} \leq 2 + 2^{-k}. \quad (\text{WDB23})$$

The long exact sequences of the finite filtration (WDB17) and the block-triangular argument following (WDB3) now give

$$w_q(\alpha) \leq 2 + 2^{-k} \quad \text{on } E_2^{11} = H^1(P, \mathcal{R}^1). \quad (\text{WDB24})$$

For the two outer terms, 3.2.12 uses the elementary degree-zero and degree-two cohomology of curves from 1.4, not the theorem being proved. On any fibre, H_c^0 is a space of sections of the weight-zero lisse sheaf and its eigenvalues have weight zero. For H_c^2 , remove the finitely many singular points and pass to smooth connected components. Formula 1.4.1.3 identifies the top cohomology with geometric coinvariants of a stalk, followed by the twist (-1) ; its eigenvalues have weight two. If components are permuted, a finite extension makes them individually defined, and the weight identity (WDB1) descends the same bound. Hence the stalk weights of $\mathcal{R}_0^0$ and $\mathcal{R}_0^2$ are respectively at most 0 and 2.

Applying the same elementary curve statements on the base proves

$$w_q(\alpha) \leq 2 \quad \text{on } E_2^{02} = H^0(P, \mathcal{R}^2) \text{ and } E_2^{20} = H^2(P, \mathcal{R}^0). \quad (\text{WDB25})$$

For the first assertion, a nonzero global eigensection is nonzero at some closed-point stalk; Frobenius there acts by the corresponding power of its global eigenvalue. For the second, removal of finitely many points does not change H_c^2 ; on a dense lisse open, 1.4.1.3 expresses it as coinvariants of a stalk with weights at most zero, twisted by (-1) . This adds exactly two. These arguments also cover constructible sheaves on the base with singular boundary stalks. The β in the printed explanation of 3.2.12 is specialized here to the auxiliary weight zero of $\mathcal{G}_0$.

Equations (WDB24)–(WDB25) and the spectral sequence (WDB12) prove

$$w_q(\mu) \leq 2 + 2^{-k} \quad \text{for every eigenvalue on } H_c^2(\tilde{V}, \pi^* \mathcal{G}). \quad (\text{WDB26})$$

Only one error 2^{-k} has appeared: it enters through the base cohomology in (WDB23). The fibre coefficient weight has already been reduced to at most one by the separate integer and strict-bound argument. Applying B_k twice without that reduction would instead give $2 + 2 \cdot 2^{-k}$ and no improvement after taking a square root.

139.8 The nonzero class survives the two exact injections

Künneth gives a Frobenius-equivariant direct-sum decomposition in degree two. Its $(1, 1)$ summand provides the first map below; the second is pullback along the blow-up:

$$\begin{aligned} H_c^1(U, \mathcal{F}^{[0]}) \otimes H_c^1(U, \mathcal{F}^{[0]}) &\hookrightarrow H_c^2(V, \mathcal{G}) \\ &\hookrightarrow H_c^2(\tilde{V}, \pi^* \mathcal{G}). \end{aligned} \quad (\text{WDB27})$$

The second injection is 3.1.1.4. More precisely 3.1.1.3 identifies

$$H_c^2(\tilde{V}, \pi^* \mathcal{G}) \simeq H_c^2(V, \mathcal{G}) \oplus H^0(V \cap A, \mathcal{G})(-1), \quad (\text{WDB28})$$

with pullback as the inclusion of the first summand. The center is disjoint from D by (B), so these are the actual blow-up centers in the open surface. Equivalently the Poincaré-dual transpose described in 3.1.1.4 supplies a retraction. The extra summand in (WDB28) is retained rather than identified with zero.

If $v \neq 0$ is an eigenvector of F with eigenvalue $\alpha^{[0]}$ in the original degree-one space, choose a linear functional λ with $\lambda(v) = 1$. Then $(\lambda \otimes \lambda)(v \otimes v) = 1$, so $v \otimes v \neq 0$. Both maps in (WDB27) are injective, hence their composite carries it to a nonzero eigenvector with eigenvalue $(\alpha^{[0]})^2$. No exterior square is being taken: the product is on two copies of the curve, so the odd cohomological degree does not force this tensor to vanish. Invertibility and the existence of an eigenvector hold over $\overline{\mathbb{Q}}_\ell$ without any diagonalizability assumption.

Applying (WDB26) to this specific image gives

$$\begin{aligned} 2(w_q(\alpha) - \beta) &= w_q((c\alpha)^2) \leq 2 + 2^{-k}, \\ w_q(\alpha) &\leq \beta + 1 + 2^{-(k+1)}. \end{aligned} \quad (\text{WDB29})$$

This proves B_{k+1} under (A)–(C), with the original eigenvalue and original coefficient weight restored exactly.

139.9 Removing the reduction hypotheses preserves the offending eigenvalue

For a finite extension $\mathbb{F}_{q^d}/\mathbb{F}_q$, the geometric cohomology space is unchanged and F is replaced by F^d . Thus an eigenvalue α is replaced by α^d , and (WDB1) proves equality of its old and new weights. A bound over the extension gives the identical bound over the original field. Deligne uses this operation in 3.2.14 to define over a finite field an embedding and a general-position pencil that exist over the algebraic closure, and then to make their finitely many exceptional points rational. This removes (B) and (C). It does not itself divide any exponent error by d .

To remove (A), the monodromy theorem supplies a finite surjective cover $g : U'_0 \rightarrow U_0$, with U'_0 smooth, for which $g^* \mathcal{F}_0^{[0]}$ has tame unipotent boundary monodromy. Purity is preserved by pullback: if x' lies over x with residue degree e , its Frobenius eigenvalues are the e -th powers of those at x , and $N(x') = N(x)^e$. The exact map used to retain cohomology is

$$g^* : H_c^a(U, \mathcal{F}^{[0]}) \hookrightarrow H_c^a(U', g^* \mathcal{F}^{[0]}). \quad (\text{WDB30})$$

Here is a trace construction which includes the multiplicities at ramification points. On a connected smooth target curve the finite surjective map is flat of constant degree $d_g > 0$: its finite local modules are torsion-free over the discrete valuation rings of the target and hence free. At a geometric point u , the trace of the constant coefficient sheaf $E = \overline{\mathbb{Q}}_\ell$ is

$$\text{Tr}_{g,u} : \bigoplus_{u'|u} E \longrightarrow E, \quad (a_{u'}) \longmapsto \sum_{u'|u} \text{length}(\mathcal{O}_{g^{-1}(u),u'}) a_{u'}. \quad (\text{WDB30a})$$

The sum of these lengths is d_g . Compatibility with restriction follows by decomposing the inverse image of an étale neighborhood into open and closed pieces: each piece is finite flat and its rank is locally constant. Thus (WDB30a) is a sheaf morphism $g_*E \rightarrow E$, with the unit followed by trace equal to $d_g \text{id}$. The projection formula tensors this identity with $\mathcal{F}^{[0]}$. Finite maps are proper and their geometric fibres have no higher étale cohomology, so $Rg_* = g_* = g_!$. Taking compact-support cohomology gives the map (WDB30) with retraction $d_g^{-1} \text{Tr}_g$. The coefficient field has characteristic zero, so the inverse d_g^{-1} exists even if ℓ divides d_g . The formula includes inseparable lengths as well. Because these maps are Frobenius-equivariant, they retain the same eigenvalue. Connected components can be treated after a finite base extension and descended by (WDB1). The universal B_{k+1} follows.

139.10 A finite contradiction, followed by the exact dual lower bound

Induction now proves (WDB2) for every integer $k \geq 0$. Suppose an actual eigenvalue had $w_q(\alpha) = \beta + 1 + \varepsilon$ with $\varepsilon > 0$. The specific finite integer

$$K = \max\{0, \lfloor \log_2(1/\varepsilon) \rfloor + 1\} \quad \text{satisfies} \quad 2^{-K} < \varepsilon. \quad (\text{WDB31})$$

Indeed if $\varepsilon > 1$, $K = 0$ works; otherwise $K > \log_2(1/\varepsilon)$. At this finite stage (WDB2) would say $\beta + 1 + \varepsilon \leq \beta + 1 + 2^{-K}$, a contradiction. Consequently

$$w_q(\alpha) \leq \beta + 1 \quad \text{on } H_c^1(U, \mathcal{F}). \quad (\text{WDB32})$$

This is the dyadic 2^{-k} bootstrap in Weil II, not the $1/k$ error obtained from arbitrarily large products in the earlier proof of Weil I. Both retain the offending eigenvalue, but their geometric estimates and induction parameters differ.

Let now $j : U_0 \hookrightarrow X_0$ with X_0 smooth projective. Equation (WDB8) transfers (WDB32) to $H^1(X, j_*\mathcal{F})$. Deligne's duality in 3.2.15 is the perfect Frobenius-equivariant pairing with values in the trivial coefficient field

$$H^1(X, j_*\mathcal{F}) \otimes H^1(X, j_*\mathcal{F}^\vee(1)) \longrightarrow \overline{\mathbb{Q}}_\ell. \quad (\text{WDB33})$$

Its perfectness on these actual image groups can also be seen directly from Poincaré duality on U . The compact-to-ordinary maps for $\mathcal{F}$ and $\mathcal{F}^\vee(1)$ are transposes, up to the degree-one cup-product sign. The kernel of either transpose is the annihilator of the other image. Quotienting the compactly supported groups by these kernels therefore induces a nondegenerate pairing on the two images. Their identification with the two j_* cohomology groups gives (WDB33), including its scalar target. The dual coefficient sheaf has weight $-\beta - 2$. For an eigenvalue α of the first factor, α^{-1} occurs in the second, whether or not Frobenius is semisimple. Applying the already proved upper bound to that factor gives

$$-w_q(\alpha) = w_q(\alpha^{-1}) \leq (-\beta - 2) + 1 = -\beta - 1. \quad (\text{WDB34})$$

Together with the upper bound this proves $w_q(\alpha) = \beta + 1$. In degrees zero and two, the invariant and coinvariant descriptions of 1.4, with the Tate twist (-1) in degree two, give weights β and $\beta + 2$. Thus all three degrees give 3.2.3:

$$w_q(\alpha) = \beta + i \quad \text{on } H^i(X, j_*\mathcal{F}). \quad (\text{WDB35})$$

For degree one this group is also the image of $H_c^1(U, \mathcal{F}) \rightarrow H^1(U, \mathcal{F})$, as stated at the start of 3.2.3. Purity is being asserted for this actual image; the mixed boundary part of all of $H_c^1(U, \mathcal{F})$ has not been declared pure.

139.11 The mixed-sheaf theorem retains and controls the boundary

Theorem 3.3.1 concerns a morphism $f : X \rightarrow Y$ of schemes of finite type over $\mathbb{Z}$, with ℓ invertible as in Deligne's conventions. It asserts that a mixed constructible sheaf $\mathcal{F}$ of weights at most n has $R^i f_! \mathcal{F}$ mixed of weights at most $n + i$. Deligne proves it by reduction to the curve theorem together with the local boundary calculation. The reductions are by exact sequences in the sheaf; decomposition into an open set and its closed complement; base change on the target; and the Frobenius-equivariant Leray spectral sequence for a composition. For the sheaf reduction involving a kernel, the preceding term in the long exact sequence has bound $n + i - 1$, which is at most $n + i$; thus that reduction also preserves the claimed bound. Relative dimension zero contributes only degree zero and preserves the pointwise weight. Universal homeomorphisms preserve the *étale* theory. Stratification and composition reduce to relative curves and then to a lisse pure coefficient sheaf. These are the explicit reductions 3.3.1(a)–(f).

On the generic relative curve one removes finitely many singular points, takes a finite covering with tame ramification at infinity, and uses the trace splitting of its coefficient sheaf. A purely inseparable field extension is allowed in making the smooth compactification. Deligne then spreads this finite geometric data over a neighborhood of the generic point of the reduced target. Reductions 3.3.1(c),(e) make this neighborhood the whole reduced target and allow the smooth reduced target used in Deligne's relative boundary calculation; the target reduction handles its complement. The resulting relative geometry is

$$\begin{aligned} X &\xrightarrow{j} \overline{X} \xleftarrow{i} D, & f &= \bar{f}j, \\ \bar{f} : \overline{X} &\longrightarrow Y \text{ smooth projective of relative dimension } 1, & & \\ D &\longrightarrow Y \text{ finite étale.} & & \end{aligned} \tag{WDB36}$$

Here $\mathcal{F}$ is lisse pure of weight n on X and tamely ramified along D . There is an actual short exact sequence on the compactification

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* B \rightarrow 0, \quad B = i^* j_* \mathcal{F}. \tag{WDB37}$$

The boundary stalk is inertia invariants, not the entire generic stalk. By 1.8.6–1.8.8 its induced monodromy filtration satisfies

$$\mathrm{Gr}_a^M B = 0 \ (a > 0), \quad \mathrm{Gr}_a^M B \text{ pure of weight } n + a \ (a \leq 0). \tag{WDB38}$$

The signs here follow from $V^I \subset \ker N \subset M_0 V$; when a finite residual inertia group remains, its invariant functor is exact in characteristic zero and retains the same upper indices. These sheaves and their filtrations commute with passage to the fibres in (WDB36).

Applying 3.2.3 on each fibre, and for every coefficient embedding ι , proves that $R^a \bar{f}_* j_* \mathcal{F}$ is pointwise pure of weight $n + a$. Because $D \rightarrow Y$ is finite, its higher direct images vanish and its direct image carries the finite filtration (WDB38) to weights at most n . The long exact sequence of (WDB37) contains, in degrees zero and one,

$$\begin{aligned} 0 \rightarrow f_! \mathcal{F} &\rightarrow \bar{f}_* j_* \mathcal{F} \rightarrow (\bar{f}i)_* B \rightarrow R^1 f_! \mathcal{F} \\ &\rightarrow R^1 \bar{f}_* j_* \mathcal{F} \rightarrow 0, \end{aligned} \tag{WDB39}$$

and in degrees $a \geq 2$ gives $R^a f_! \mathcal{F} \simeq R^a \bar{f}_* j_* \mathcal{F}$. Thus degree zero has weights at most n ; degree one is an extension of the pure sheaf $R^1 \bar{f}_* j_* \mathcal{F}$, of weight $n + 1$, by the boundary quotient $\mathrm{coker}(\bar{f}_* j_* \mathcal{F} \rightarrow (\bar{f}i)_* B)$, of weights at most n ; and degree two has weight $n + 2$. This proves the desired upper bound in the reduced relative-curve case, and the exact reductions give 3.3.1. Boundary contributions have been included through (WDB37)–(WDB39), with their weights and maps specified.

For a finite-type $\mathbb{F}_q$ -scheme this gives 3.3.4: $H_c^i(X, \mathcal{F})$ has weights at most $n + i$ when $\mathcal{F}_0$ is mixed of weights at most n . If in addition X_0 is smooth of pure dimension d and $\mathcal{F}_0$ is lisse and mixed of

weights at least n , then $\mathcal{F}_0^\vee$ has weights at most $-n$. Apply the upper bound to the exact Poincaré dual:

$$H^i(X, \mathcal{F})^\vee \simeq H_c^{2d-i}(X, \mathcal{F}^\vee)(d), \quad -n + (2d - i) - 2d = -n - i. \quad (\text{WDB40})$$

Duality reverses weights, giving the lower bound $n + i$ for $H^i(X, \mathcal{F})$, which is 3.3.5. For a lisse pure sheaf of weight n , the natural map from compactly supported to ordinary cohomology is Frobenius-equivariant, and its image is both a quotient of a space of weights at most $n + i$ and a subspace of one of weights at least $n + i$. The characteristic-polynomial argument after (WDB3) then proves the exact conclusion 3.3.6:

$$\text{im}(H_c^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})) \quad \text{is pure of weight } n + i. \quad (\text{WDB41})$$

The original compact-support and ordinary cohomology groups are preserved in this statement; purity is proved for their specified common image.

139.12 The exact contradiction mechanism available for comparison

The decisive numerical chain in this proof is

$$\begin{aligned} \text{nonzero local boundary quotient} &\xrightarrow{(\text{WDB14})-(\text{WDB19})} b_a \in \mathbb{Z}, \\ \{b_a \in \mathbb{Z}, b_a < 2\} &\implies b_a \leq 1, \\ \text{all local boundary quotients zero} &\xrightarrow{(\text{WDB18})-(\text{WDB22})} H^1(P, \mathcal{K}_a) = 0, \\ \mathbf{B}_k &\xrightarrow{(\text{WDB23})-(\text{WDB26})} w_q(\mu) \leq 2 + 2^{-k}, \\ \alpha^{[0]} &\xrightarrow{(\text{WDB27})} (\alpha^{[0]})^2 \implies \mathbf{B}_{k+1}. \end{aligned} \quad (\text{WDB42})$$

The word “boundary” in this chain names the actual specialization-cokernel map (WDB18) into vanishing cycles, and the constant alternative has the actual cohomology-vanishing map (WDB22). The improvement is proved only after both alternatives have been treated. The finite integer (WDB31) then excludes every strict upper violation, and (WDB33)–(WDB34) exclude every strict lower violation of the pure image weight. Equations (WDB37)–(WDB41) extend this calculation to mixed coefficients without deleting their boundary-supported pieces.

140 What has been reconstructed, and what has been decided

This manuscript reconstructs the actual programme in the shared conversation [Branch · Compile Navier Stokes Literature](#), together with the two supplied continuation instructions. Its question is the counterfactual to RH inside that programme: take a genuine off-critical zero of its original arithmetic function, and carry its complete cohomological packet through the constructed maps. The question is not answered by assigning arbitrary roots to an auxiliary polynomial.

The recovered source does construct split support, an absolute base, coefficient Frobenius, coefficient purity, arithmetic cohomology over that base, its right adjoint, actual theta representatives, tensor and symmetric pushforwards, conormal maps, exponential periods, and finite holonomy comparisons. These constructions are used below. The source’s final auxiliary counterexample does not establish that RH is false, or that the whole programme fails. This continuation also establishes neither an RH contradiction nor an off-critical zeta zero. Its new results are exact comparisons, explicit retained kernels, and quantitative restrictions on an actual putative zero. Each is proved in the subsequent chapters.

140.1 Complete source coverage and its limits

The saved public response contains 5,350 mapping nodes. Following every parent from the declared current node to the root produces exactly the same 5,350 nodes in exactly the published linear order. There are 5,349 message records, of which 67 are user messages and 148 are readable assistant prose messages. All 215 readable user/assistant records were read in five complete contiguous assignments. The complete user history was also read by the integrating auditor. The original node identifiers, message hashes, verbatim bodies, assignment coverage and independent reviews are included in the source repository. A second public capture has exactly the same 5,349 message identifiers and body hashes. Differences in transient HTML metadata do not change any message body.

The share explicitly redacts tool payloads and does not embed the bodies of every historical attachment. Those absent contents are not counted as read. Original local proof files are retained separately and are cited as such. The reconstruction uses the actual visible messages and inspected proof files; it does not infer hidden mathematics from a redacted tool call.

The source begins with Navier–Stokes and thermodynamic calculations. It does not begin with the eventual tau-base construction. In particular the early function denoted $e(y)$ and early time or dual-trace coordinates denoted τ are not retrospectively identified with the later supported zero and absolute base. The user’s negative-polynomial detour is expressly withdrawn at U0010. The later user instruction at U0043 demands the tau base itself, and U0046 accepts the corrected arithmetic construction of A1523. These corrections determine the programme continued here.

140.2 Source order and the surviving mathematical content

Source	Constructed object	Exact role in this continuation
U0001–15	Heat/NS test maps and the arithmetic seed	The full early audit retains its integral identities and constants. The test-field realization is kinematic with its specified force; it does not prove that an arbitrary fixed Navier–Stokes evolution supplies arithmetic positivity.
U0016–28	Split support, origin fibres, arithmetic amplitude, mixed-support synchronization, co-efficient Frobenius	The packet chapter uses the actual localization and its integral coefficient action, and constructs a trace retract retaining the arithmetic eigenclass.
U0029–40	Original theta quotient, source labels, raw and balanced Mellin comparison, finite full jets	BK, DP and PL compute the previously retained balanced kernel and every smaller-label finite defect on these spaces.
U0041–45	Absolute base $\mathfrak{b}_\tau$, marked chart, A_τ , restriction and K_τ	The companion tau chapter computes restriction’s complete homotopy fibre and its adjoint, including the nonzero arithmetic quotient at supported zero.
U0046–54	Continuous theta retraction, tensor and symmetric source, relative sum algebra, Fisher and conormal maps	The scaling chapter calculates the original cocycle and boundary pairing. The conormal chapter retains all local factors and computes the intrinsic symmetric trace, including negative directions within one exact sum fibre.
U0055–66	Original cyclic interpolation, volume windows, exponential periods, residue, periodization, all-phase recovery	The holonomy and volume chapters calculate what the original off-line packet forces into the boundary and compare the proved actual upper and lower estimates.

Source	Constructed object	Exact role in this continuation
U0067, A2611	Auxiliary exponential/purity and Gamma-reference counterexamples	These refute specified reduced implications. Their polynomials are not asserted to divide 2ξ ; their arithmetic correction is retained, rather than set to zero.

Every row has a complete source-pinned audit in `audit/`; the table is a navigation device, not a substitute for those proofs. The original 821-page source cut, including its editable proof dependencies, is retained in `original_programme/`. The later original holonomy and audit messages occur in the complete saved transcript. None of the new results is attributed to a historical source that only supplied its starting objects.

141 The original absolute base and the arithmetic scalar square

For a commutative ring R , retain the disjoint union

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

Addition has identity τ , with $r^\bullet \oplus s^\bullet = (r + s)^\bullet$; multiplication has identity $1_R^\bullet$, has absorbing element τ , and satisfies $r^\bullet s^\bullet = (rs)^\bullet$. The laws follow directly from the ring laws when all inputs are supported; cases containing τ use its stated identity or absorption law. In particular $e \neq \tau$, $e \oplus e = e$, and $er^\bullet = e$ for every supported scalar.

Let $\mathbb{B} = \{0, 1\}$ have addition OR and multiplication AND. The amplitude and support observations are the unital semiring maps

$$p_R : G(R) \longrightarrow R, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r, \quad \chi_R : G(R) \longrightarrow \mathbb{B}, \quad \chi_R(\tau) = 0, \quad \chi_R(r^\bullet) = 1. \quad (\text{R.1})$$

To verify additivity of χ_R , two supported summands have a supported sum even if their amplitudes cancel. Their OR is still one. The cases with an absent input follow from the additive identity. Multiplicativity is the corresponding support conjunction. The pair (p_R, χ_R) is injective: support zero identifies τ uniquely, whereas support one leaves the amplitude to identify $r^\bullet$ uniquely. Thus the zero amplitude at supported zero retains an observable absent from the ordinary ring quotient.

For a unital ring map $f : R \rightarrow S$, the map $G(f)$ fixes absence and sends $r^\bullet$ to $f(r)^\bullet$. The operation rules prove it is a unital semiring map, and $p_S G(f) = f p_R$ on both sorts of input. In particular the actual scalar square is

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(\iota)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{\iota} & \mathbb{C}. \end{array} \quad (\text{R.2})$$

The lower-left arithmetic target is the full infinite ring $\mathbb{Z}$. No finite characteristic or Boolean quotient replaces it in (R.2).

The original pointed-monoid blueprint base has elements $\{\tau, 1\}$, absorbing τ , and the empty-sum relation $\emptyset \equiv \tau$. It does not impose $1 + 1 = 1$. Its original structural map sends these elements to absence and the unit in $G(R)$. This defines the chart's structure over $\mathfrak{b}_\tau$. The support map in (R.1), whose codomain does impose Boolean addition, is a further map, not that structural map. The stated operations prove the two maps are compatible; their different additive relations are not identified.

For a complex linear map $f : E \rightarrow F$ at an admitted support label λ , its split lift sends (λ, v) to $(\lambda, f v)$ and external τ to external τ . It is additive because f is, and commutes with supported scalar multiplication for the same reason. If $f v = 0$, its value is $(\lambda, 0)$. This proves the support rule

used by every linear complex, quotient and kernel below. Between labels, the same statement uses the explicitly specified linear transport; the PL chapter calculates a transport that kills a nonzero relation class to supported zero, without deleting its source label.

142 The actual cohomological route of a hypothetical zero

The full-space definitions of $V, \mathcal{B}, \Theta, D, \mathcal{M}$ are given in BK.1 and proved against the original source in the companion chapters. In particular

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}\Theta\phi = gH_\phi.$$

The exact arithmetic object is $Q = \mathcal{B}/\Theta V$, with its original locally convex quotient topology. The later global theta retraction proves this quotient Hausdorff in that topology; it does not substitute an L^2 completion for Q .

On the original four-point chart $+, - < \eta < \sigma$, the arithmetic diagram is

$$\mathcal{T} = (V, V, \mathcal{B}, 0; \Theta, \Theta\mathcal{F}, 0).$$

The two-leg complex in degrees 0, 1 is

$$A_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \hat{\psi})} \mathcal{B}]. \quad (\text{R.3})$$

Its full H^1 is Q . The derived restriction comparison of the diagram is $A_\tau(\mathcal{T}) \rightarrow \mathcal{T}_\sigma[0] = 0$; its homotopy fibre is (R.3) itself. The companion tau proof computes the projective resolution, both representatives of restriction, their homotopy, the fibre, its right adjoint, and the induced residue map. Thus nonzero classes sent to supported zero at sigma remain in the actual relative arithmetic cohomology. There is no inference here that the zero stalk makes Q vanish.

The counterfactual studied throughout the calculations is the existence of an actual nontrivial zero

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 0,$$

with its full positive order m . The original conjugation and functional equation give the complete reflection-conjugation quartet of the same order. Choose that full packet and retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - \frac{1}{2} - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad v_h = j_h(v_h). \quad (\text{R.4})$$

Here division is entire because these are stipulated actual zeros, and v_h is a unit because their complete orders were used. Its entire Taylor class, including every nonconstant coefficient, is retained. This is the temporary counterfactual input to deductions, not evidence that any such zero exists.

The original arithmetic jet sequence, proved in the retained source and extended to every nonzero polynomial in BK, is

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0, \quad E_h = \mathbb{C}[s]/(h). \quad (\text{R.5})$$

For the actual complete packet, $j_h g = 0$. Its finite torsion class is therefore a full copy of E_h , and the Mellin observation on that copy includes the invertible factor v_h rather than dropping it. The original representative map uses the inverse unit to obtain $J_h \sigma_h = 1$. The Duhamel chapter gives its source representatives, its nonsplit source extension and its full scaling action explicitly.

At each local zero its action under $U_a F(x) = F(x/a)$, $a > 0$, is

$$a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{R.6})$$

This is not replaced by its reduced eigenvalue. BK and DP prove that the full-source balanced Mellin kernel cannot contain, cancel or hide this finite primary block. PL proves that proper source labels have additional finite kernel classes, even at parameters that are not zeros of g , and calculates their exact transition to the full source. This is why the source labels cannot be suppressed when using (R.5).

The mixed-support chapter next carries (R.4)–(R.6) through the actual coefficient synchronization. Its trace retract proves that the original arithmetic eigenclass remains nonzero. The complex-linear lift of the integral coefficient Frobenius adds roots of unity to the spectrum, and has a fixed coefficient line on which the arithmetic eigenvalue is unchanged. All tensor amplifications in this statement are over $\mathbb{C}$, as in the original external product; tensoring the local top nilpotent over E_h instead would give a different map and could annihilate it.

The tensor programme also contains more than its cyclic sum variable. The full conormal and symmetric chapters preserve the relative occupation algebra, its local lengths, original cochain signs, residue/Jacobian comparison and source representatives. In particular two full symmetric-square occupation factors can both have sum eigenvalue 1 while the difference of their idempotents has intrinsic arithmetic trace $-2m^2$. The detailed proof identifies the actual mixed derivative jet that detects this trace. Purity of a sum eigenvalue alone therefore cannot erase these retained relative directions.

Finally the actual holonomy comparison retains the common source generated by v_h and its tensor convolution. Its canonical section has a computed relation-valued action defect. The holonomy chapter proves a lower bound for its full original boundary energy at every phase and every admitted finite degree, and a spectral-distance lower bound for the associated Laplacian boundary. The concluding volume calculation compares those restrictions with the proved scalar arithmetic upper estimate. That comparison, rather than an auxiliary Gamma example, determines the remaining numerical range obtained by this continuation.

143 What the purity construction does, and the precise remaining calculation

The original finite coefficient algebra has proved weight-zero coefficient purity. The later finite-field exponential family has its proved weight-one cohomology under its stated good-reduction hypotheses; its external k -fold product retains degree k and the corresponding weight. The original source computes specialization and period maps relating its full arithmetic packet to those families. These are actual constructions. The supplied audits retain their periods, connections, cohomological degrees and kernels, rather than calling the families unrelated to the arithmetic source.

The operational numerical use of the Deligne-style amplification in this programme is to control the original arithmetic action or its equivalent original source volume defect uniformly as tensor degree grows. A period congruence transports both the metric and its action defect. It does not set the defect to zero. For the following formula let E be the retained finite tensor coefficient space (or a specified invariant constituent), let $A_k : E \rightarrow E$ be the restriction of the sum operator, and let $R : E \rightarrow L^2((\mathbb{R}_{>0})^k, dx_1 \cdots dx_k)$ be the original source section. Its columns are in $\mathcal{B}^{\otimes k}$, so $D^{(k)} = \sum_i (-x_i \partial_{x_i})$ acts on them. Put $G = R^* R$ and $\mathcal{B} = D^{(k)} R - R A_k$ as maps on this exact E . The symbol $\mathcal{B}$ in this formula denotes the boundary map; the function space with the same inherited symbol is distinguished by its domain and codomain. The exact boundary formula, proved on the original $dx_1 \cdots dx_k$ source in the following chapters, is

$$A_k^* G + G A_k - kG = -(R^* \mathcal{B} + \mathcal{B}^* R), \quad \mathcal{B} = D^{(k)} R - R A_k. \quad (\text{R.7})$$

The source relation producing $\mathcal{B}$ is explicitly retained. Neither its zero cohomology class nor its zero observation at sigma sets either cross-pairing in (R.7) to zero. This continuation computes the contribution and gives a strictly positive counterfactual lower bound on its original energy.

For the actual four endpoints take $k \geq 2$ and the original cyclic sum quotient of length $q = (1 + k(m - 1))(k + 1)^2$. In that quotient let $V_{k,N}^\nu = \det G_{k,N}^\nu > 0$, $N \geq q - 1$, be the determinant of the canonical source Gram for the specified measure $\nu \in \{\text{ar}, \Gamma\}$, as defined in the holonomy and volume chapters. Define $\mathcal{B}_{h,k}^\nu = \log(V_{k,q-1}^\nu/V_{k,2q-1}^\nu) + \log(V_{k,q}^\nu/V_{k,2q}^\nu)$. Substituting the positive volume ratios at all four endpoints gives the exact retained comparison

$$\mathcal{B}_{h,k}^{\text{ar}} = \mathcal{B}_{h,k}^\Gamma + \log \frac{T_{k,q-1} T_{k,q}}{T_{k,2q-1} T_{k,2q}}, \quad T_{k,N} = V_{k,N}^{\text{ar}}/V_{k,N}^\Gamma. \quad (\text{R.8})$$

The final auxiliary Gamma counterexample retains the reference volumes but does not calculate the actual signed correction in (R.8). The new concluding calculation bounds the actual four-window quantity on both sides, retaining this exact signed term. The coarse scalar upper and lower bound functions are compatible; the refined common-source interval retains unevaluated quantities, as detailed in the final combined restriction. No actual off-line zero is certified by the coarse compatibility, and no RH contradiction follows from those bounds.

Consequently the remaining issue after the mathematics below is the stronger original arithmetic control of (R.7), or of the joint signed term in (R.8), that would close the amplification argument. This is a precise limit of the results obtained, not an assumption used to prove them. The manuscript supplies the new exact maps and inequalities in full and retains the complete original sources for further calculation.

144 1. Objects, provenance, and cochain conventions

The complete successor *A global retraction of the original theta complex over the tau base*, retained as `Tau_Global_Comparison_2026-09-12/RESEARCH_NOTE.md`, was also read completely. Its SHA256 is 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859. Equations (6.12a)–(6.12i) below reproduce its constructive retraction and evaluate the previously general closure kernel for this exact source.

This calculation continues equations (18)–(48) of *Arithmetic cohomology over the tau-base*. The complete source `NOTE.md` has SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. Its complete retained dependency sources/`Split_Support_Actual_Comparison__CONTINUATION.tex` has SHA256 98e8aaa63b1f7ee4661c043cd985ca274646b749f36d23d2ec46cbc4b0737dc5. Both were read completely for this calculation. The category, original theta source, masks, analytic topology, multiplicities, and actions below are those sources' objects.

Let $E = \mathbb{C}$, let P have order $+$ $< \eta < \sigma$ and $-$ $< \eta < \sigma$, and let $\mathcal{A} = \text{Fun}(P, \text{Vect}_E)$. A diagram $F \in \mathcal{A}$ is specified by

$$U = F_+, \quad V_- = F_-, \quad M = F_\eta, \quad S = F_\sigma, \quad r_+ : U \rightarrow M, \quad r_- : V_- \rightarrow M, \quad t : M \rightarrow S. \quad (1.1)$$

Its remaining two arrows are the composites tr_+ and tr_- . The symbol V_- here denotes a general stalk; V in Section 5 denotes the original even Schwartz source.

Use cohomological complexes. For a chain map $f : C \rightarrow D$, define

$$\text{Cone}(f)^n = D^n \oplus C^{n+1}, \quad d(y, x) = (d_D y + f x, -d_C x), \quad d_{C[k]} = (-1)^k d_C. \quad (1.2)$$

After ordering its summands as $C^n \oplus D^{n-1}$, the homotopy fibre is

$$\text{Fib}(f) = \text{Cone}(f)[-1], \quad d(c, y) = (d_C c, -d_D y - f c). \quad (1.3)$$

For internal Hom use

$$d_{\text{Hom}} h = d_W h - (-1)^{|h|} h d_C. \quad (1.4)$$

These conventions, including the minus sign in the second component of (1.3), remain fixed throughout.

145 2. The original pushforward and the complete restriction fibre

Define $P_x(y) = E$ for $x \leq y$, and zero otherwise, with identity arrows between nonzero stalks. A natural map $P_x \rightarrow F$ is uniquely specified by the image of $1 \in P_x(x)$, so $\text{Hom}(P_x, F) = F_x$. Evaluation is exact, hence P_x is projective. The sequence

$$0 \longrightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \mathbf{E} \longrightarrow 0 \quad (2.1)$$

is exact: at $+$ and $-$ its last arrow is identity; at η, σ its kernel is the displayed anti-diagonal; all restriction squares commute. Consequently a representative of the source functor is

$$C(F) = A_\tau(F) = [U \oplus V_- \xrightarrow{\delta} M], \quad \delta(u, v) = r_+u - r_-v, \quad (2.2)$$

in degrees 0, 1. With (1.4), applying internal Hom to (2.1) initially gives differential $-\delta$; the explicit isomorphism to (2.2) is identity in degree zero and minus identity in degree one. This is a change between two stated cochain presentations, not a modification of the arithmetic differential in (2.2).

The two representatives of derived restriction are

$$f_+(u, v) = tr_+u, \quad f_-(u, v) = tr_-v, \quad C(F) \longrightarrow S[0]. \quad (2.3)$$

They are chain maps because the target has only degree zero. Their difference is $t\delta$. Thus $h^1 = t : M \rightarrow S$ satisfies $f_+ - f_- = d_S h + h d_C$. In particular both represent the same natural derived morphism.

Using f_+ , the complete fibre is

$$B_\sigma(F) = [U \oplus V_- \xrightarrow{b} M \oplus S], \quad b(u, v) = (r_+u - r_-v, -tr_+u). \quad (2.4)$$

The cone itself has degrees $-1, 0$ and differential

$$U \oplus V_- \longrightarrow S \oplus M, \quad (u, v) \longmapsto (tr_+u, -r_+u + r_-v). \quad (2.5)$$

No stalk or quotient has been dropped in either presentation.

Put

$$I = \text{im } r_+ \cap \text{im } r_- \subset M. \quad (2.6)$$

Then the complete cohomology of (2.4) is

$$H^0 B_\sigma(F) = \{(u, v) : r_+u = r_-v \in \ker t\}, \quad (2.7)$$

$$H^1 B_\sigma(F) = \frac{M \oplus S}{\{(r_+u - r_-v, -tr_+u) : u \in U, v \in V_-\}}, \quad (2.8)$$

and all its other cohomology groups vanish. There are exact sequences

$$0 \longrightarrow \ker r_+ \oplus \ker r_- \longrightarrow H^0 B_\sigma(F) \xrightarrow{(u, v) \mapsto r_+u} I \cap \ker t \longrightarrow 0, \quad (2.9)$$

$$0 \longrightarrow S/t(I) \xrightarrow{[s] \mapsto [(0, s)]} H^1 B_\sigma(F) \xrightarrow{[(m, s)] \mapsto [m]} M/(\text{im } r_+ + \text{im } r_-) \longrightarrow 0. \quad (2.10)$$

For (2.9), every element of the displayed intersection has a preimage under each leg, and the kernel is exactly the indicated direct sum. For (2.10), the last map is well defined and surjective. A class in its kernel has $m = \delta(u, v)$; subtracting $b(u, v)$ replaces it by $(0, s + tr_+u)$. The class of $(0, s)$ vanishes exactly when there exist u, v with $r_+u = r_-v$ and $s = -tr_+u$, which is precisely $s \in t(I)$. This proves the sequence, including the sign in its presentation. These are exact sequences of specified objects, without a chosen splitting.

Equivalently the restriction long exact sequence is

$$0 \rightarrow H^0 B_\sigma(F) \rightarrow \ker \delta \xrightarrow{f_+} S \xrightarrow{s \mapsto [(0, -s)]} H^1 B_\sigma(F) \rightarrow \operatorname{coker} \delta \rightarrow 0. \quad (2.11)$$

Its signs agree with (1.2): when the fibre-to-source arrow is the positive projection, the rotated distinguished triangle has boundary arrow minus the inclusion of S in $\operatorname{Cone}(f) = B_\sigma(F)[1]$. This accounts for the minus in (2.11); the independently specified exact presentation (2.10) uses the positive inclusion. The only nonzero possible cone cohomology is $H^{-1} \operatorname{Cone}(f) = H^0 B_\sigma(F)$ and $H^0 \operatorname{Cone}(f) = H^1 B_\sigma(F)$.

The two representative fibres are related by the chain isomorphism

$$B_{\sigma,+}(F) \longrightarrow B_{\sigma,-}(F), \quad \Phi^0 = 1, \quad \Phi^1(m, s) = (m, s + tm), \quad (2.12)$$

whose inverse subtracts tm . Indeed $-f_+ + t\delta = -f_-$. Thus the fibre calculation does not depend on a privileged choice of the plus leg.

146 3. Naturality and the right-adjoint comparison

A morphism $a : F \rightarrow F'$ consists of maps $a_+, a_-, a_\eta, a_\sigma$ commuting with r_+, r_-, t . The fibre map is exactly

$$B_\sigma(a)^0 = a_+ \oplus a_-, \quad B_\sigma(a)^1 = a_\eta \oplus a_\sigma. \quad (3.1)$$

For example its second differential component is $-a_\sigma t r_+ u = -t' r'_+ a_+ u$; the first follows from the two other commutative squares. Formula (3.1) also commutes with (2.12), because $a_\sigma t = t' a_\eta$. This proves the naturality of every kernel, quotient, exact sequence, and representative comparison above.

For a vector space W , let $I_x(W)(y) = W$ for $y \leq x$, and zero otherwise. A natural map $F \rightarrow I_x(W)$ is uniquely determined by its component $F_x \rightarrow W$; the other nonzero components are precomposition with $F_y \rightarrow F_x$. Hence

$$\operatorname{Hom}_{\mathcal{A}}(F, I_x(W)) = \operatorname{Hom}_E(F_x, W). \quad (3.2)$$

Every vector space is injective over E : a linear map on a subspace extends after extending a basis. Therefore the right side is exact contravariantly in F , and $I_x(W)$ is injective.

Let

$$K_\tau(W) = [I_\eta(W) \xrightarrow{(+\operatorname{res}, -\operatorname{res})} I_+(W) \oplus I_-(W)] \quad (3.3)$$

have degrees $-1, 0$. It is a bounded injective complex. Applying (3.2) gives

$$R \operatorname{Hom}_{\mathcal{A}}(F, K_\tau(W)) = [\operatorname{Hom}(M, W) \xrightarrow{\ell \mapsto (\ell r_+, -\ell r_-)} \operatorname{Hom}(U, W) \oplus \operatorname{Hom}(V_-, W)]. \quad (3.4)$$

The internal Hom of (2.2) into $W[0]$ has exactly this differential: the element ℓ has degree -1 , so (1.4) gives $d\ell = +\ell\delta$. In particular the signs in the original single-factor adjoint and residue arrow require no repair.

For completeness the adjunction holds for bounded complexes, with all totalization signs specified. Define

$$C(F)^n = (F_+^n \oplus F_-^n) \oplus F_\eta^{n-1}, \quad d(b, m) = (d_F b, \delta b - d_F m), \quad (3.5)$$

and $K_\tau(W) = K_\tau(E) \otimes W$, with tensor differential. A degree n map $F \rightarrow K_\tau(W)$ consists of $b_p : F_+^p \oplus F_-^p \rightarrow W^{p+n}$ and $a_p : F_\eta^p \rightarrow W^{p+n+1}$. The isomorphism to $\operatorname{Hom}^n(C(F), W)$ keeps b_p and takes

$$\ell_p = (-1)^{n+1} a_p. \quad (3.6)$$

On the source differential these components become

$$a'_p = -d_W a_p - (-1)^n a_{p+1} d_F, \quad b'_p = d_W b_p + a_p \delta - (-1)^n b_{p+1} d_F. \quad (3.7)$$

On the target they become

$$\ell'_p = d_W \ell_p + (-1)^n \ell_{p+1} d_F, \quad b'_p = d_W b_p - (-1)^n (b_{p+1} d_F + \ell_p \delta). \quad (3.8)$$

Substitution of (3.6) gives equality in both components. Bounded injective targets compute derived Hom, and (2.1), with the degree n multiplier $(-1)^n$ on its eta component, gives (3.5) from its projective resolution. This proves the bounded derived adjunction as an actual natural chain isomorphism.

The stalk complexes of (3.3) are $[W \xrightarrow{1} W]$ at $+$, $[W \xrightarrow{-1} W]$ at $-$, $W[1]$ at η , and zero at σ . The stalkwise kernel diagram is $S_\eta W$, supported only at η . Its inclusion in degree -1 is a natural quasi-isomorphism

$$S_\eta W[1] \longrightarrow K_\tau(W). \quad (3.9)$$

Since every point of P lies below σ , $I_\sigma(W)$ is the constant diagram W . It represents the dual of the sigma stalk. The actual adjoint of restriction is

$$\alpha_+ : I_\sigma(W) \longrightarrow K_\tau(W), \quad (3.10)$$

given in degree zero by restriction into $I_+(W)$, and zero into $I_-(W)$. The alternative α_- restricts into $I_-(W)$. Their difference is $d_K h$, where $h : I_\sigma(W) \rightarrow I_\eta(W)$ is restriction in degree -1 . Applying Hom to (3.10) sends $w : S \rightarrow W$ to $(wtr_+, 0)$, which is exactly precomposition with f_+ . This identifies the morphism, rather than only comparing its endpoints.

147 4. The dual of the restriction fibre and its nonsplit extension

Define the bounded injective complex

$$R_\tau(W) = [I_\eta(W) \oplus I_\sigma(W) \xrightarrow{d_R} I_+(W) \oplus I_-(W)], \quad (4.1)$$

in degrees $-1, 0$, where

$$d_R(\ell, w) = (\ell \text{res}_+ - w \text{res}_{\sigma,+}, -\ell \text{res}_-). \quad (4.2)$$

Here restriction symbols denote the natural maps of injective diagrams. Formula (3.2) makes its Hom differential

$$(\ell, w) \longmapsto (\ell r_+ - wtr_+, -\ell r_-). \quad (4.3)$$

This is precisely precomposition with the two components $(\delta, -f_+)$ of (2.4), with the positive sign supplied by degree -1 in (1.4). Therefore

$$\boxed{R \text{Hom}_A(F, R_\tau(W)) \cong R \text{Hom}_E(B_\sigma(F), W).} \quad (4.4)$$

Equivalently $R_\tau(W) = \text{Cone}(-\alpha_+)$. Multiplying its $I_\sigma(W)$ cone summand by -1 identifies this presentation with $\text{Cone}(\alpha_+)$; (4.2) fixes which presentation is being used in (4.4).

At $+$, the differential is $(\ell, w) \mapsto \ell - w$, with kernel (w, w) . At $-$, it is $(\ell, w) \mapsto -\ell$, with kernel $(0, w)$. At η it has domain $W \oplus W$ and zero target, and at σ it has domain W and zero target. All degree-zero cokernels vanish. Consequently, if $D(W)$ has stalks

$$D(W)_+ = W, \quad D(W)_- = W, \quad D(W)_\eta = W \oplus W, \quad D(W)_\sigma = W, \quad (4.5)$$

and arrows

$$w \longmapsto (w, w) \quad (+ \rightarrow \eta), \quad w \longmapsto (0, w) \quad (- \rightarrow \eta), \quad (u, w) \longmapsto w \quad (\eta \rightarrow \sigma), \quad (4.6)$$

the kernel inclusion is a natural quasi-isomorphism $D(W)[1] \rightarrow R_\tau(W)$.

There is an exact sequence of diagrams

$$0 \longrightarrow S_\eta W \xrightarrow{w \mapsto (u,0)} D(W) \xrightarrow{\text{second coordinate}} I_\sigma(W) \longrightarrow 0. \quad (4.7)$$

At $+$, $-$, σ the last map is identity under (4.5); at η its kernel is the first coordinate. Naturality follows directly from (4.6). For $W \neq 0$, the extension has no natural splitting. Indeed a splitting is identity at the plus and minus stalks, since the quotient there is identity. Commutation along $+\rightarrow\eta$ forces its η component to be $w \mapsto (w, w)$, whereas commutation along $-\rightarrow\eta$ forces $w \mapsto (0, w)$. They disagree on every nonzero $w \in W$. This proves nonsplitting explicitly.

In the original case $F_\sigma = 0$, every term $\text{Hom}(F, I_\sigma(W)) = \text{Hom}(0, W)$ is zero. Thus the inclusion $K_\tau(W) \rightarrow R_\tau(W)$ induces an actual equality of the represented Hom complexes, as well as an isomorphism in the derived category. The nonzero original global arithmetic classes are therefore all in the relative fibre, and their original adjoint is precisely the restriction of (4.4) to these objects.

148 5. Calculation on the original theta source and every mask

Retain the exact original spaces and maps

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad (5.1)$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad (5.2)$$

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad J\Theta = \Theta\mathcal{F}. \quad (5.3)$$

The last identity is the retained Poisson identity. Explicitly, the full Poisson equation is $\sum_n \phi(nx) = x^{-1} \sum_n \widehat{\phi}(n/x)$; the two zero terms vanish by (5.1), giving (5.3). At infinity every differentiated summand is bounded by a constant times $(|n|x)^{-N}$ for any chosen $N > 1$; summing $2 \sum_{n \geq 1} n^{-N}$ proves rapid decay. Applying (5.3) proves arbitrary power decay at zero, with all derivatives, so $\Theta : V \rightarrow \mathcal{B}$. The Fourier transform preserves (5.1) because it exchanges the value and integral conditions, and its square is reflection, which is identity on even functions.

The map Θ is injective. For $\Re s = c > 1$, absolute convergence gives

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)\mathcal{M}\phi(s), \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}. \quad (5.4)$$

For absolute convergence, the integral of $|\phi(nx)|x^{c-1}$ is $n^{-c} \int_0^\infty |\phi(u)|u^{c-1}du$, and both the integral and $\sum n^{-c}$ are finite. The Euler product is nonzero for $c > 1$, as follows from absolute convergence of its logarithm. If $\Theta\phi = 0$, (5.4) makes $\mathcal{M}\phi(c+it) = 0$ for every real t . This is the Fourier transform, with its frequency sign explicitly reversed, of the integrable function $y \mapsto e^{cy}\phi(e^y)$. Fourier uniqueness gives zero almost everywhere and continuity gives zero everywhere. Evenness then gives $\phi = 0$ on the full real line.

The joint sheaf is

$$\mathcal{T} = (V, V, \mathcal{B}, 0; \Theta, \Theta\mathcal{F}, 0), \quad C(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\widehat{\psi})} \mathcal{B}]. \quad (5.5)$$

Its restriction map is zero on both chain representatives, so

$$B_\sigma(\mathcal{T}) = C(\mathcal{T}), \quad \text{Cone}(\text{res}_\sigma) = C(\mathcal{T})[1]. \quad (5.6)$$

Injectivity just proved gives

$$H^0 B_\sigma(\mathcal{T}) = \{(\widehat{\psi}, \psi) : \psi \in V\}, \quad H^1 B_\sigma(\mathcal{T}) = Q := \mathcal{B}/\Theta V. \quad (5.7)$$

The quotient here is the algebraic quotient with its original quotient locally convex topology. The later global retraction source proves that this image is closed; its exact retraction and the resulting zero closure kernel are included after (6.12) below.

Let $L = \mathcal{P}(\{+, -\})$. For each nonempty mask $A \in L$, retain V at each admitted leg and the zero coefficient space at a missing leg, retain $\mathcal{B}$ at eta and zero at sigma, and retain the restrictions in (5.5). The empty mask has the zero diagram. Every nonempty A has restriction fibre exactly $C(\mathcal{T}_A)$; its differential image is ΘV , since both Θ and $\Theta\mathcal{F}$ have that image. A one-leg differential is injective, with its literal plus or minus sign retained. It follows that

$$H^0 B_\sigma(\mathcal{T}_A) = \begin{cases} V, & A = \{+, -\}, \\ 0, & |A| = 1, \end{cases} \quad H^1 B_\sigma(\mathcal{T}_A) = Q \quad (A \neq \emptyset). \quad (5.8)$$

For mask inclusions the eta map is identity and leg maps are coordinate inclusions, hence the induced H^1 map is exactly identity on Q . This explicitly proves compatibility with every original mask transport.

Reconstruction gives

$$\tilde{H}^1 B_\sigma = \{\tau\} \sqcup \bigsqcup_{\emptyset \neq A \subseteq \{+, -\}} \{A\} \times Q. \quad (5.9)$$

The scalar $e = 0_E^\bullet \in G(E)$ sends (A, q) to $(A, 0)$. The scalar τ sends it to the absent element. Linear restriction to the zero coefficient at sigma sends q to zero, but its supported reconstruction is $(A, q) \mapsto (A, 0)$, preserving A . No class of (5.9) becomes absent merely because its coefficient restricts to zero. The same statement applies to (2.4), (3.4), and (4.4) fibrewise, with dual label arrows given by actual precomposition and hence indexed by L^{op} .

149 6. Finite jets, their full kernel, and the retained topology

Put $g = 2\xi$, where

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (6.1)$$

The source's full-plane multiplier identity is

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_\phi = \frac{\mathcal{M}\phi}{\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)} \in \mathcal{O}(\mathbb{C}). \quad (6.2)$$

Here is the full continuation used in this identity. Write the Taylor expansion of the even Schwartz function at zero as $\phi(x) = \sum_{j=0}^N a_{2j}x^{2j} + O(x^{2N+2})$. Subtract this finite sum in its Mellin integral over $(0, 1)$. The remainder is holomorphic for $\Re s > -2N - 2$, while the subtracted terms have integrals $a_{2j}/(s+2j)$. The integral over $(1, \infty)$ is entire by Schwartz decay. Since N is arbitrary, $\mathcal{M}\phi$ has meromorphic continuation with only possible simple poles at the nonpositive even integers. The reciprocal $1/\Gamma(s/2)$ has a simple zero at each such integer and cancels each possible pole. At $s = 0$, the condition $a_0 = \phi(0) = 0$ removes the Mellin pole, and $1/\Gamma(s/2)$ cancels the explicit factor s in the denominator of H_ϕ . At $s = 1$, evenness and the integral condition give $\mathcal{M}\phi(1) = \frac{1}{2} \int_{\mathbb{R}} \phi = 0$, cancelling $s - 1$. No other denominator zero occurs, since the gamma function has no zeros. Thus the displayed H_ϕ is entire. Formula (5.4) proves (6.2) first on $\Re s > 1$, and analytic continuation proves it everywhere, with the original factor $g = 2\xi$ retained.

Let Z be a finite set of actual zeros of g with their actual orders m_ρ , and write

$$z_\rho = s - \rho, \quad g(s) = u_\rho(z_\rho)z_\rho^{m_\rho}, \quad u_\rho(0) \neq 0, \quad A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho/(g). \quad (6.3)$$

The germ coordinate z_ρ , the full jet length, and the actual unit u_ρ are retained. The jet arrow is

$$J_Z : \mathcal{B} \longrightarrow A_Z, \quad (J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j. \quad (6.4)$$

Equation (6.2) proves $J_Z \Theta = 0$, so it induces $q_Z : Q \rightarrow A_Z$.

Here is a full finite interpolation proof, with the original dx Riesz density preserved. Index jets by $\alpha = (\rho, j)$, and fix any $R > 0$ and $0 < R' < R$. Let

$$\eta_{R'}(y) = \begin{cases} \exp[-1/((R')^2 - y^2)], & |y| < R', \\ 0, & |y| \geq R', \end{cases} \quad r_\alpha(x) = x^{\bar{\rho}-1}(\log x)^j/j!. \quad (6.5)$$

Set

$$B_{\alpha\beta} = \frac{1}{j!k!} \int_{-R'}^{R'} \eta_{R'}(y) e^{(\rho+\bar{\sigma}-1)y} y^{j+k} dy, \quad \beta = (\sigma, k). \quad (6.6)$$

This is $\ell_\alpha(\eta_{R'}(\log x)r_\beta(x))$, where ℓ_α is the corresponding component of (6.4). It is Hermitian. For a column c , its quadratic form is

$$\bar{c}^\top B c = \int_{-R'}^{R'} \eta_{R'}(y) e^{-y} \left| \sum_\alpha \bar{c}_\alpha e^{\rho y} y^j / j! \right|^2 dy. \quad (6.7)$$

The exponential polynomials are independent: for a fixed exponent ρ , apply $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{m_\sigma}$ to a vanishing linear combination. It kills every other exponent block, and acts on the remaining block by $e^{\rho y}$ times $\prod_{\sigma \neq \rho} (\partial_y + \rho - \sigma)^{m_\sigma}$. Each factor on finite-degree polynomials is invertible, since it is a nonzero constant plus a nilpotent differentiation operator, with inverse given by the terminating geometric series. Hence that polynomial block is zero. This holds for every ρ , proving independence. Therefore (6.7) is strictly positive for $c \neq 0$, so B is invertible.

For $v \in A_Z$, the function

$$s_Z(v)(x) = \eta_{R'}(\log x) \sum_\beta (B^{-1}v)_\beta r_\beta(x) \quad (6.8)$$

is smooth, belongs to $\mathcal{B}$, has support in $[e^{-R'}, e^{R'}] \subset (e^{-R}, e^R)$, and satisfies $J_Z s_Z(v) = v$ by matrix multiplication. Thus (6.4) and q_Z are surjective, with an explicitly calculated finite-dimensional continuous section of (6.4).

For continuity, choose an integer $N > |\Re \rho|$ and put $p_N(F) = \max(\sup x^N |F(x)|, \sup x^{-N} |F(x)|)$. Splitting the integral at one gives the exact bound

$$|\ell_{\rho,j}(F)| \leq p_N(F) \left((N + \Re \rho)^{-j-1} + (N - \Re \rho)^{-j-1} \right). \quad (6.9)$$

Indeed the two integrals of $|\log x|^j/j!$ against $x^{N+\Re \rho} dx/x$ and $x^{-N+\Re \rho} dx/x$ are the two displayed terms. This also proves that the Mellin jet derivatives used here are well-defined by dominated differentiation.

Define the actual retained kernel spaces, using $\mathcal{K}_Z$ to avoid changing the different space named $\mathcal{B}_Z$ in the source's finite arithmetic extension:

$$\mathcal{K}_Z = \ker J_Z, \quad Q_Z^0 = \mathcal{K}_Z / \Theta V. \quad (6.10)$$

They give exact sequences

$$0 \longrightarrow \mathcal{K}_Z \longrightarrow \mathcal{B} \xrightarrow{J_Z} A_Z \longrightarrow 0, \quad \boxed{0 \longrightarrow Q_Z^0 \longrightarrow Q \xrightarrow{q_Z} A_Z \longrightarrow 0.} \quad (6.11)$$

The first is split by (6.8); the second is split as a topological vector-space sequence by the induced section. No equivariance of this section is asserted: only the arrows (6.11) are canonical scaling morphisms.

To verify the topology assertion, the map $(F_0, v) \mapsto F_0 + s_Z(v)$ is a continuous linear bijection $\mathcal{K}_Z \oplus A_Z \rightarrow \mathcal{B}$, with continuous inverse $F \mapsto (F - s_Z J_Z F, J_Z F)$. Since $\Theta V \subset \mathcal{K}_Z$, passing to the

quotient gives the claimed topological direct sum. By (6.9), $\mathcal{K}_Z$ is closed, so $\overline{\Theta V} \subset \mathcal{K}_Z$. The same proof gives

$$0 \rightarrow \mathcal{K}_Z / \overline{\Theta V} \rightarrow Q^H := \mathcal{B} / \overline{\Theta V} \rightarrow A_Z \rightarrow 0, \quad \ker(Q \rightarrow Q^H) = \overline{\Theta V} / \Theta V. \quad (6.12)$$

Thus every finite jet survives both specified quotients, with the closure kernel written explicitly. No new global topology has been substituted.

For this actual source the kernel in (6.12) is zero. We include the later source's constructive reason, retaining its auxiliary cutoffs and contraction constant. Let μ be the arithmetic Möbius function, and define on $x \neq 0$

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|). \quad (6.12a)$$

For $j \geq 0$, write $x^j \partial_x^j = P_j(D)$, where $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, and let $C_{N,j}(F)$ be the sum of $p_{N,k}(F) = \sup x^N |D^k F|$ weighted by the absolute coefficients of P_j . For each integer $N > 1$, termwise differentiation gives

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j} \quad (x > 0). \quad (6.12b)$$

Uniform convergence away from zero justifies each derivative. On the original theta image the double sum is absolutely convergent, bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which converges by Schwartz decay and $d(k) \leq k$. The identity $\sum_{n|k} \mu(n) = 1$ for $k = 1$, zero otherwise, therefore proves

$$\mathcal{U}\Theta\phi = \phi, \quad \mathcal{U}J\Theta\phi = \hat{\phi} \quad (x \neq 0). \quad (6.12c)$$

Choose the original type of real even compactly supported smooth cutoffs $0 \leq \alpha, \beta \leq 1$, each equal to one near zero, and set on $L^2(\mathbb{R}, dx)$

$$T = M_\alpha \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad c = \|T\|, \quad YF = (1 - \alpha) \mathcal{U}F + \alpha \mathcal{F}^{-1}((1 - \beta) \mathcal{U}JF). \quad (6.12d)$$

The kernel of T is $\alpha(x)\beta(x-y)$, with squared L^2 norm $\|\alpha\|_2^2 \|\beta\|_2^2$, so T is compact. Both its factors are contractions. If $c = 1$, compactness supplies a norm-attaining unit vector f ; equality in both contraction inequalities forces $\hat{f}$ to have compact support where $\beta = 1$ and f to have compact support where $\alpha = 1$. Compact Fourier support makes the inverse Fourier integral entire, because the compactly supported L^2 transform is L^1 and allows all complex derivatives under its integral. This entire function vanishes on a real interval outside the support of f , hence vanishes identically, a contradiction. Therefore $c < 1$, and

$$(1 - T)^{-1} = \sum_{n \geq 0} T^n, \quad \|(1 - T)^{-1}\| \leq (1 - c)^{-1}. \quad (6.12e)$$

The cut-off exterior maps in Y are continuous into even Schwartz functions by (6.12b); their values are extended by zero where the cutoff vanishes near zero. The operator $T : L^2 \rightarrow \mathcal{S}$ is continuous: compact Fourier support bounds every derivative of $\mathcal{F}^{-1} M_\beta \mathcal{F} f$ by Cauchy-Schwarz, and multiplication by smooth compactly supported α controls every Schwartz seminorm. Thus $\Phi = (1 - T)^{-1} Y$ is Schwartz-valued and continuous: the equation $\Phi F = YF + T\Phi F$ bounds each Schwartz seminorm p by

$$p(\Phi F) \leq p(YF) + C_p (1 - c)^{-1} \|YF\|_2. \quad (6.12f)$$

It preserves evenness. With $\gamma_0 = e^{-\pi x^2}$ and $\gamma_2 = 2\pi x^2 e^{-\pi x^2}$, whose value/integral columns are $(1, 1)$, $(0, 1)$, define $P_V : \mathcal{S}(\mathbb{R})^{\text{even}} \rightarrow V$ by

$$P_V \phi = \phi - \phi(0) \gamma_0 - \left(\int_{\mathbb{R}} \phi - \phi(0) \right) \gamma_2, \quad \Lambda = P_V (1 - T)^{-1} Y. \quad (6.12g)$$

This continuous projection has image V and is identity on V . On $\Theta\phi$, (6.12c) gives $Y\Theta\phi = (1 - T)\phi$, so

$$\Lambda\Theta = 1_V, \quad K = 1_{\mathcal{B}} - \Theta\Lambda, \quad K^2 = K, \quad \ker K = \Theta V. \quad (6.12h)$$

The last equality follows in both directions by direct substitution. Since K is continuous, its kernel is closed. Therefore $\overline{\Theta V} = \Theta V$, $Q = Q^H$, and the closure kernel in (6.12) is exactly zero. The section

$$s : Q \rightarrow \mathcal{B}, \quad s[F] = KF \quad (6.12i)$$

is well-defined and continuous in the quotient topology, satisfies $qs = 1$ and $\Lambda s = 0$, and gives the topological decomposition $\mathcal{B} \cong V \oplus Q$ by $F \mapsto (\Lambda F, qF)$. Every cutoff and the factor $(1 - c)^{-1}$ remain in this construction. It establishes closedness of the theta image, and makes no assertion that the different kernel Q_Z^0 in (6.11) vanishes.

Let $\mathcal{T}_Z$ have the same source legs and sigma stalk as $\mathcal{T}$, eta stalk $\mathcal{K}_Z$, and the original restrictions. Then

$$0 \rightarrow \mathcal{T}_Z \rightarrow \mathcal{T} \xrightarrow{J_Z} S_\eta A_Z \rightarrow 0 \quad (6.13)$$

is exact at each stalk; its eta surjectivity is (6.8). Applying the explicit complexes gives a termwise exact sequence

$$0 \rightarrow C(\mathcal{T}_Z) \rightarrow C(\mathcal{T}) \rightarrow A_Z[-1] \rightarrow 0. \quad (6.14)$$

The kernel inclusion into the homotopy fibre of the last map is a quasi-isomorphism. This follows directly by quotienting that inclusion: the quotient fibre is the fibre of the identity of $A_Z[-1]$, whose two identical terms have an invertible differential and zero cohomology. Therefore the full cohomology of this jet-comparison fibre is

$$H^0 = V \quad \text{via } \psi \mapsto (\hat{\psi}, \psi), \quad H^1 = Q_Z^0, \quad (6.15)$$

and nothing else. On a nonempty one-leg mask $H^0 = 0$ and $H^1 = Q_Z^0$; the same proof applies to each exact mask diagram. On $S_\eta A_Z$ itself, sigma restriction is zero, its restriction fibre is exactly $A_Z[-1]$, and its restriction cone is $A_Z[0]$.

For disjoint finite packets Z, W of actual zeros, simultaneous application of (6.8) to $Z \cup W$ with prescribed coordinates $(0, w)$ proves the further exact surjection

$$Q_Z^0 \longrightarrow A_W. \quad (6.16)$$

Consequently the kernel in (6.11) retains all these explicitly reachable omitted packets. It has not been set to zero.

150 7. Original scaling, generator, and dual actions

For $a > 0$, retain $U_a F(x) = F(x/a)$. Substitution $x = ay$ gives

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = aU_{1/a} \hat{\phi}. \quad (7.1)$$

These operators preserve the original test spaces: dilation rescales each defining Schwartz or $\mathcal{B}$ seminorm by finite constants, preserves evenness and the value-zero condition, and multiplies the integral by a . The source sheaf action is exactly

$$U_a^T : (\phi, \psi, F) \longmapsto (U_a \phi, aU_{1/a} \psi, U_a F). \quad (7.2)$$

The identities $\Theta U_a = U_a \Theta$ and (7.1) give

$$d(U_a \phi, aU_{1/a} \psi) = U_a \Theta(\phi - \hat{\psi}). \quad (7.3)$$

Thus (7.2) is a diagram automorphism and (3.1) proves equivariance of every restriction-fibre map. On $H^0 = V$ in the parameter ψ of (5.7), its action is $\psi \mapsto aU_{1/a}\psi$; on Q it is $[F] \mapsto [U_a F]$. Every support mask is fixed.

The infinitesimal generator for $a = e^t$ at $t = 0$ is $D = -x\partial_x$ on the plus and eta stalks, and $1 - D$ on the minus stalk. Differentiating (7.1) gives the original relation $\mathcal{F}D = (1 - D)\mathcal{F}$. Integration by parts, with zero boundary terms because $F \in \mathcal{B}$, gives $\mathcal{M}(DF) = s\mathcal{M}F$.

On the full local jet algebra,

$$U_a|_\rho = a^\rho \exp((\log a)N_\rho) = a^\rho \sum_{k=0}^{m_\rho-1} \frac{(\log a)^k}{k!} N_\rho^k, \quad N_\rho(z_\rho^j) = z_\rho^{j+1}, \quad (7.4)$$

where $N_\rho(z_\rho^{m_\rho-1}) = 0$. The generator is $\rho I + N_\rho$. This follows by multiplying the jet by the exact germ $a^{\rho+z_\rho}$, so (6.4), (6.11), (6.13), and (6.14) intertwine it; in particular $\mathcal{K}_Z$ and Q_Z^0 are stable under scaling. Nilpotent directions have not been replaced by their trace.

Let $\mathcal{L}_1 = E$ with action $w \mapsto aw$, the original integral-moment line. On every dual object the action is the induced Hom action

$$(U_a \ell)(v) = a\ell(U_{1/a}v). \quad (7.5)$$

The formula is forced by the evaluation map $\ell \otimes v \mapsto \ell(v) \in \mathcal{L}_1$: applying (7.5) and U_a to both inputs multiplies the result by a . Since all maps in Sections 3 and 4 are defined by precomposition and restriction, this proves their equivariance with (7.5), including every contravariant mask transport.

The known continuous section (6.12i) has an exact, retained scaling defect. Denote the induced action on Q by $\bar{U}_a$, and define

$$k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\bar{U}_a = \Theta k_a. \quad (7.6)$$

To prove the second equality, use $1_{\mathcal{B}} = sq + \Theta\Lambda$ on $U_a s$, and $qU_a s = \bar{U}_a$. The group identity then gives, without dropping a boundary term,

$$k_{ab} = U_a k_b + k_a \bar{U}_b, \quad k_1 = 0. \quad (7.7)$$

Indeed expand $U_{ab}s - s\bar{U}_{ab}$ by first applying the equation for b , then that for a , and cancel the injective Θ . These maps are continuous by their formulas. Thus the known topological splitting is accompanied by the full original source-valued action defect.

151 8. Full finite residue arrow and its chain degree

The symmetry of the original $g = 2\xi$ used here follows directly from the original Gaussian Poisson formula, with no assumption on its zeros. Put $\psi_0(t) = \sum_{n \geq 1} e^{-\pi n^2 t}$. Poisson gives $\psi_0(t) = t^{-1/2} \psi_0(1/t) + \frac{1}{2}(t^{-1/2} - 1)$. For $\Re s > 1$, termwise integration gives $\pi^{-s/2} \Gamma(s/2) \zeta(s) = \int_0^\infty \psi_0(t) t^{s/2} dt/t$. Splitting at one and substituting $t = 1/v$ in the reflected term therefore yields the entire continuation

$$g(s) = 1 + s(s-1) \int_1^\infty \psi_0(t) (t^{s/2} + t^{(1-s)/2}) \frac{dt}{t}. \quad (8.0)$$

The constant is exactly one: the remaining elementary integral is $1/(s-1) - 1/s = 1/[s(s-1)]$. The bound $\psi_0(t) \leq e^{-\pi t}/(1 - e^{-\pi t})$ for $t \geq 1$ proves locally uniform convergence with all s -derivatives. The expression is invariant under $s \mapsto 1-s$, and its real coefficients give conjugation symmetry. Consequently $g^\dagger = g$, and g is not the zero function (for example its continuation has $g(0) = 1$).

Assume Z is reflection-stable, meaning the finite chosen set satisfies $\iota Z = Z$ for $\iota\rho = 1 - \bar{\rho}$; this selects an actual finite divisor, without a hypothesis on the location of its points. The completed function has $g^\dagger = g$ for $f^\dagger(s) = \overline{f(1 - \bar{s})}$. In original coordinates,

$$f_\rho^\dagger(z_\rho) = \sum_j (-1)^j \overline{f_{\iota\rho, j} z_\rho^j}, \quad u_\rho(z) = (-1)^{m_\rho} \overline{u_{\iota\rho}(-\bar{z})}. \quad (8.1)$$

The residue pairing, with its raw orientation, is

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger(s)h(s)}{g(s)} ds. \quad (8.2)$$

Changing any representative by a multiple of g adds a holomorphic differential, so this is well-defined. At ρ , if $u_\rho(z)^{-1} = \sum_{k \geq 0} c_{\rho,k} z^k$, its literal formula is

$$R_Z(f, h) = \sum_{\rho} \sum_{i,j=0}^{m_\rho-1} (-1)^i \overline{f_{\iota\rho,i}} h_{\rho,j} c_{\rho,m_\rho-1-i-j}, \quad c_{\rho,k} = 0 \ (k < 0). \quad (8.3)$$

The matrix $G_{ij} = c_{\rho,m_\rho-1-i-j}$ has determinant $(-1)^{m_\rho(m_\rho-1)/2} u_\rho(0)^{-m_\rho} \neq 0$: reversing the columns makes it triangular with diagonal $u_\rho(0)^{-1}$, and reversal has the displayed sign. Multiplication of the i -th row by $(-1)^i$, together with the invertible reflection-permutation of blocks, preserves invertibility. Hence R_Z is perfect, including all nilpotent jets and actual analytic units.

For a meromorphic function k , expansion in z_ρ gives

$$\text{Res}_\rho k^\dagger(s) ds = -\overline{\text{Res}_{\iota\rho} k(s)} ds. \quad (8.4)$$

Only its coefficient of z^{-1} contributes, and reflection multiplies that coefficient by $(-1)^{-1} = -1$. Apply (8.4) to $k = u^\dagger f/g$, sum over the permuted divisor, and use $g^\dagger = g$. The result is

$$R_Z(f, h) = -\overline{R_Z(h, f)}. \quad (8.5)$$

The raw pairing is therefore skew-Hermitian. No phase has been inserted.

The identity $(a^s)^\dagger = a^{1-s}$ gives

$$R_Z(U_a f, U_a h) = a R_Z(f, h). \quad (8.6)$$

Thus the following is an injective linear equivariant map from the conjugate vector space:

$$\overline{A_Z} \longrightarrow H^{-1} R \text{Hom}_{\mathcal{A}}(\mathcal{T}, K_\tau(\mathcal{L}_1)), \quad \bar{f} \longmapsto \ell_f, \quad \ell_f(F) = R_Z(f, J_Z F). \quad (8.7)$$

Here ℓ_f is literally the degree -1 cochain on $\mathcal{B}$ in (3.4). It is closed, since (6.2) gives $\ell_f \Theta = 0$, and $\ell_f \Theta \mathcal{F} = 0$. There is no degree -2 term, so this cocycle represents zero exactly when it is the zero functional. Surjectivity of J_Z and perfectness of (8.2) then prove injectivity. Formula (8.6), evaluated on $J_Z U_{1/a} F$, proves $\ell_{U_a f}(F) = a \ell_f(U_{1/a} F)$, exactly (7.5). Bound (6.9) also proves that all these functionals are continuous on the original quotient topology, even though the adjunction (3.4) uses the full algebraic dual. Thus (8.7) exhibits a specified subspace of continuous functionals inside that full dual, by an actual injection.

The contraction $J_g : h \mapsto g' h$ retains its local form

$$g' = m_\rho u_\rho(z) z^{m_\rho-1} + u'_\rho(z) z^{m_\rho} \equiv m_\rho u_\rho(z) z^{m_\rho-1} \pmod{z^{m_\rho}}. \quad (8.8)$$

Using $g'/g = m_\rho/z + u'_\rho/u_\rho$ gives

$$R_Z(f, J_g h) = \sum_{\rho} m_\rho \overline{f(\iota\rho)} h(\rho) = \text{Tr}(M_{f^\dagger h} | A_Z). \quad (8.9)$$

The trace equality follows because multiplication by a germ in the ordered basis $1, z, \dots, z^{m-1}$ is triangular with its value repeated m times on the diagonal. The radical in either variable of (8.9) is exactly the subspace of all packets whose constant term is zero: the formula annihilates that subspace, and arbitrary constant terms in the reflected block detect every nonzero remaining value. The differentiated symmetry is $(g')^\dagger = -g'$, which together with (8.5) also confirms the Hermitian orientation of (8.9). No positivity of the raw residue form was used or asserted.

Every arrow in (8.7) is reconstructed at its original label A . A zero functional is $(A, 0)$, and the absent input alone is sent to τ . Precomposition with the mask inclusions commutes with the residue formula because H^1 transport in (5.8) is identity. This completes the labelled adjoint arrow with its degree, sign, topology, and action.

152 9. Tensor compatibility, including the dual Koszul sign

All coefficient tensors in this section are the original algebraic tensors over E ; no unproved completed tensor topology is substituted. For a product P^r , tensoring the stalkwise exact resolution (2.1) produces a finite projective resolution of the constant diagram. To see exactness without assuming a tensor theorem, at each stalk split an exact finite complex into its cohomology and elementary two-term identity complexes by choosing complements to boundaries and cycles. Tensoring an elementary identity complex with any complex is contractible, by its usual contracting homotopy tensored with identity. Only the tensor of the degree-zero constant cohomology remains. Each external product of the P_x 's represents evaluation at the corresponding product point and is projective. This proves the claimed resolution and yields the natural cochain identity

$$A_{\tau,r}(F_1 \boxtimes \cdots \boxtimes F_r) \simeq C(F_1) \otimes \cdots \otimes C(F_r), \quad (9.1)$$

with the full differential

$$d(x_1 \otimes \cdots \otimes x_r) = \sum_{j=1}^r (-1)^{\sum_{k < j} |x_k|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_r. \quad (9.2)$$

The standard cross product of cohomology classes is an isomorphism because the same split-complex argument kills all summands with a contractible factor. The proof uses a splitting to verify that the canonical cross product is bijective; it does not make that map dependent on the splitting.

Product restriction is the tensor $f_1 \otimes \cdots \otimes f_r$ to $S_1 \otimes \cdots \otimes S_r[0]$. Changes from plus to minus representatives are compatible: for $g_i = f_{i,-}$ and $f_i = f_{i,+}$, a homotopy is

$$H = \sum_{j=1}^r g_1 \otimes \cdots \otimes g_{j-1} \otimes h_j \otimes f_{j+1} \otimes \cdots \otimes f_r, \quad (9.3)$$

where tensor maps act with the Koszul sign and $h_j^1 = t_j$. The differential of each tensor summand is the same tensor with $d_{\text{Hom}} h_j = f_j - g_j$; every other factor is a degree-zero chain map. Summing telescopes to $\otimes f_i - \otimes g_i$, proving the homotopy identity.

For the original sheaves and all nonempty masks, every $S_i = 0$. Thus product restriction is zero and

$$B_{\sigma^r}(\mathcal{T}_{A_1} \boxtimes \cdots \boxtimes \mathcal{T}_{A_r}) = \bigotimes_{i=1}^r C(\mathcal{T}_{A_i}) = \bigotimes_{i=1}^r B_{\sigma}(\mathcal{T}_{A_i}). \quad (9.4)$$

The complete cohomology is the graded tensor product of the two groups in (5.8). Explicitly its degree n is the direct sum over subsets $I \subseteq \{1, \dots, r\}$, $|I| = n$, with Q in factors indexed by I and $H^0 C(\mathcal{T}_{A_i})$ in the other factors. In particular

$$\boxed{H^r B_{\sigma^r}(\mathcal{T}^{\boxtimes r}) = Q^{\otimes r}.} \quad (9.5)$$

The numerator is $\mathcal{B}^{\otimes r}$, and its denominator is exactly

$$\sum_{j=1}^r \mathcal{B}^{\otimes(j-1)} \otimes \Theta V \otimes \mathcal{B}^{\otimes(r-j)}. \quad (9.6)$$

Signs in (9.2) change none of these subspaces but remain in the differential itself. The quotient map to $Q^{\otimes r}$ has kernel (9.6), by successive tensoring of the quotient presentation over the field.

For general diagrams one cannot replace the fibre of a tensor product by the tensor of fibres. The exact replacement can itself be computed. For composable chain maps $X \xrightarrow{f} Y \xrightarrow{g} Z$, put $B(f) = \text{Fib}(f)$ with (1.3). There are chain maps

$$u : B(f) \rightarrow B(gf), \quad (x, y) \mapsto (x, gy), \quad v : B(gf) \rightarrow B(g), \quad (x, z) \mapsto (fx, z). \quad (9.7)$$

Their composite has nullhomotopy $H(x, y) = (-y, 0)$, since $dH + Hd = (fx, gy)$. A surjective chain map

$$\text{Cone}(u) \longrightarrow B(g), \quad (x, z; x', y') \longmapsto (fx - y', z) \quad (9.8)$$

has kernel $z = 0, y' = fx$, with remaining coordinates $x \in X^n, x' \in X^{n+1}$ and differential $(x, x') \mapsto (dx + x', -dx')$. The degree -1 map $(x, x') \mapsto (0, x)$ contracts this kernel. Thus (9.8) is a quasi-isomorphism and proves the fibre triangle explicitly. Apply this with $f = f_1 \otimes 1_{C_2}$ and $g = 1_{S_1} \otimes f_2$. The resulting triangle is

$$B(f_1) \otimes C_2 \longrightarrow B(f_1 \otimes f_2) \longrightarrow S_1 \otimes B(f_2) \longrightarrow (B(f_1) \otimes C_2)[1]. \quad (9.9)$$

The first fibre identification follows immediately from (1.3) and the tensor differential; in the second, S_1 is in degree zero. This proves the general tensor relation and explains precisely why it reduces to (9.4) for the original source.

The product right adjoint is

$$K_{\tau,r}(W) = K_{\tau}(E)^{\boxtimes r} \otimes W, \quad K_{\tau,r}(W) \simeq S_{(\eta, \dots, \eta)} W[r]. \quad (9.10)$$

Its terms are finite direct sums of $I_{(x_1, \dots, x_r)}(W)$, hence injective. Its only cohomology follows by the same stalkwise contraction proof used for (3.9) and (9.1). No assertion that an infinite-dimensional algebraic dual commutes with tensor products is used.

The adjunction on the product has a sign that must be stated. In total degree $-p$, multiply the coefficient functional arising from $K_{\tau}(E)^{\boxtimes r}$ by

$$\epsilon_p = (-1)^{p(p-1)/2} \quad (9.11)$$

to identify it with $\text{Hom}^{-p}(\bigotimes C(F_i), W)$. To check this, an index set I of p eta factors contributes to degree $-p$. Removing $j \in I$ in d_K has sign $(-1)^{|I \cap \{1, \dots, j-1\}|}$. The corresponding incoming cochain differential in (9.2) has the same sign, while its internal Hom differential has the additional factor $(-1)^{p+1}$ from (1.4). The equality $\epsilon_{p-1} = (-1)^{p+1} \epsilon_p$ proves commutation in every component. This verifies the sign on every product differential.

In top degree the identification is therefore

$$H^{-r} R \text{Hom}_{\mathcal{A}(P^r)}(\mathcal{T}^{\boxtimes r}, K_{\tau,r}(W)) \cong \text{Hom}_E(Q^{\boxtimes r}, W), \quad (9.12)$$

with the coefficient multiplier ϵ_r . In particular, for $r = 2$ it is -1 . If a product residue arrow is written on the right of (9.12) as the literal functional

$$([F_1] \otimes \cdots \otimes [F_r]) \longmapsto \prod_{i=1}^r R_Z(f_i, J_Z F_i), \quad (9.13)$$

its coefficient representative in the unaltered tensor complex $K_{\tau,r}$ is ϵ_r times that product. This is the cohomological Koszul sign, not a phase inserted into (8.2) and not a modification of its positivity properties.

Tensoring (6.11) gives the exact surjection $Q^{\boxtimes r} \rightarrow A_Z^{\boxtimes r}$, with kernel

$$\sum_{j=1}^r Q^{\otimes(j-1)} \otimes Q_Z^0 \otimes Q^{\otimes(r-j)}. \quad (9.14)$$

This follows by successively quotienting each factor; a vector-space complement verifies equality of the kernel, without imposing an equivariant complement. The fibre of the cochain jet map

$$C(\mathcal{T})^{\otimes r} \longrightarrow A_Z^{\otimes r}[-r] \quad (9.15)$$

has the same cohomology as $C(\mathcal{T})^{\otimes r}$ in degrees below r , has (9.14) in degree r , and zero above r . Indeed the target has only degree r , its degree- r map is the proved surjection, and the explicit fibre long exact sequence then has exactly these kernels and cokernels.

The action in all tensor expressions is the tensor of the original (7.2) and (7.4). On a specified packet tuple $(\rho_1, \dots, \rho_r)$ it is

$$a^{\rho_1 + \dots + \rho_r} \exp \left((\log a) \sum_{j=1}^r 1 \otimes \dots \otimes N_{\rho_j} \otimes \dots \otimes 1 \right). \quad (9.16)$$

The summands commute, so their exponential is exactly the product of the terminating exponentials in (7.4). On the target line $\mathcal{L}_1^{\otimes r}$ the action is a^r ; (8.6) proves covariance of (9.13) by this exact factor. Formula (9.16) retains every nilpotent term, not only its eigenvalue. Support reconstruction is performed on each original admitted tensor fibre and its balancing relations: a zero tensor amplitude remains the supported zero of that fibre, whereas an absent tensor input is absorbing. The proof never identifies the coefficient tensor with a Cartesian product of label sets.

There is also the source's proved passage to completed projective tensor products, now obtained from the retained continuous contraction instead of an assumed exactness of completion. For the joint complex $C = C(\mathcal{T})$, let $H_0 = V[0] \oplus Q[-1]$, with zero differential, and put

$$p^0(\phi, \psi) = \psi, \quad p^1 = q, \quad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s, \quad H^1(F) = (\Lambda F, 0). \quad (9.17)$$

Then $pi = 1$ and $1 - ip = dH + Hd$: in degree zero the latter equation is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$, obtained from $\Lambda\Theta = 1$; in degree one it is $1 - sq = \Theta\Lambda$. All the maps are continuous in the already specified topologies. If $P = ip$, the tensor homotopy

$$H_r = \sum_{j=1}^r P^{\otimes(j-1)} \otimes H \otimes 1^{\otimes(r-j)} \quad (9.18)$$

uses the sign $(-1)^{\sum_{k < j} |x_k|}$ on homogeneous inputs. The differential telescopes to $1 - P^{\otimes r}$, since $d_{\text{Hom}} H = 1 - P$, while $P, 1$ are chain maps of degree zero. Every finite tensor map is continuous for the projective tensor topology and therefore extends uniquely to its completion; the identity extends by density. The completed complex is thus continuously homotopy equivalent to $H_0^{\widehat{\otimes} \pi^r}$. Its degree n cohomology is the direct sum of ordered completed tensors with Q in n slots and V in the others, and in particular its top cohomology is $Q^{\widehat{\otimes} \pi^r}$. The comparison from (9.5) is the canonical completion map on these tensors. For an individual one-leg mask replace H_0 by $Q[-1]$, keep $p^1 = q, i^1 = s$, and take $H^1 = \Lambda$ on the plus leg or $H^1 = -\mathcal{F}\Lambda$ on the minus leg. On the latter the differential is $-\Theta\mathcal{F}$, so $Hd = \mathcal{F}\Lambda\Theta\mathcal{F} = 1_V$ and $dH = \Theta\Lambda = 1 - sq$; the plus leg has the same identities with its positive signs. Thus the same tensor proof applies to every mask. This specifies the exact completed continuation while retaining the algebraic tensor calculation and the action defect (7.6).

153 10. Exact consequences for continuation

The original global arithmetic object, its generic-boundary restriction, and its right adjoint are related by the explicit maps (2.3), (3.10), and (4.4), and the nonsplit extension (4.7). On the original theta sheaf, the restriction fibre is the original cochain complex itself, with every supported copy of Q retained. Finite jets give the exact kernel sequence (6.11), its sheaf and fibre realizations (6.13)–(6.15), and the injective continuous residue arrow into the original adjoint (8.7). Tensor products give (9.5), (9.14), and the full action (9.16), with the independently checked Koszul sign (9.11).

These are completed calculations on the original $\Theta, V, \mathcal{B}, \mathcal{A}_\tau, K_\tau$ and $g = 2\xi$. No vanishing of Q , of the finite-jet kernel $Q_{\mathbb{Z}}^0$, or of an arithmetic boundary contribution was used to obtain them. The calculations establish exact comparison morphisms and retain their kernels; they supply no upper estimate for the original tensor action beyond the identities proved here.

154 Finite dilation of the original arithmetic packet over the tau base

154.1 0. Scope and exact source correspondence

This calculation continues the original two-leg theta complex. The retained infinitesimal identity is (A11) of `arithmetic_input.tex`; the absolute-base pushforward, its two-leg action, and its tensor differential are (19), (21), (34), and (46) of *Arithmetic cohomology over the tau-base*. The full global retraction and its cocycle were already constructed in transcript turn **A1694**, node `1fa0ebbc-f569-40c1-b7ac-b523da19a5f0`, chain position 3576, and in *Tau_Global_Comparison*, equations (16), (19), (29)–(35), and (41). Those results are retained as prior work, not reported as newly discovered here.

The additional calculation is the explicit finite-packet dilation cocycle for every positive real dilation, its full Taylor-unit and nilpotent coefficients, convergent source-space integral, incomplete-gamma evaluation, exact finite/global change of representatives, two-leg homotopies, and tensor and boundary formulas. These formulas perform the group-level continuation of the existing infinitesimal extension. They do not assign positivity to a representative pairing.

The source bytes used are pinned at the end. Every integral on the positive half-line keeps the displayed measure. In particular the Hilbert pairing below uses dx , and its dilation mass is a , not 1. The notation $t = \log a$ is a real time parameter; the polynomial coordinate in $\mathbb{C}[X]/(h)$ is written X to avoid confusing it with that time. Thus $A = M_X$ here is exactly the source operator denoted M_t .

154.2 1. The original spaces, maps, and topology

Work over $\mathbb{C}$. Let

$$V = \left\{ \phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \quad \phi(0) = 0, \quad \int_{\mathbb{R}} \phi(x) dx = 0 \right\},$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0) \right\}, \quad D = -x\partial_x.$$

Use the Schwartz seminorms

$$q_{N,k}(\phi) = \sup_{x \in \mathbb{R}} (1 + |x|)^N |\partial_x^k \phi(x)|, \quad N, k \geq 0.$$

The space V is a closed subspace of the complete Schwartz space: parity, value at zero, and integration are continuous. For example $|\int \phi| \leq q_{2,0}(\phi) \int_{\mathbb{R}} (1 + |x|)^{-2} dx$. The space $\mathcal{B}$ is complete because a Cauchy sequence in all $p_{b,k}$ converges with all derivatives on compact subsets after $x = e^y$; the weighted uniform bounds pass to its limit. These observations justify limits of Riemann sums in the actual spaces below.

Retain

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad JF(x) = x^{-1} F(1/x),$$

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

Schwartz estimates, followed by the original Poisson identity, give

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta \widehat{\phi} = J\Theta \phi, \quad D\Theta = \Theta D.$$

Here is the continuity argument. At $x \geq 1$, each $D^k \phi(nx)$ is bounded by a finite sum of Schwartz seminorms times $(nx)^{-M}$, where M can be any sufficiently large positive integer. Summing n^{-M} gives every required bound at infinity. The Fourier transform preserves Schwartz seminorms continuously, preserves evenness, and interchanges the two vanishing moments. Poisson summation therefore gives $\Theta \phi(x) = x^{-1} \Theta \widehat{\phi}(1/x)$; applying D^k and the preceding bounds gives every required estimate at zero. Each resulting estimate involves finitely many source seminorms. Termwise differentiation is justified uniformly on every $x \geq \delta > 0$ by the same summable majorants.

The theta map is injective. For fixed $x > 0$, absolute convergence permits

$$\frac{1}{2} \sum_{m \geq 1} \mu(m) \Theta \phi(mx) = \sum_{r \geq 1} \left(\sum_{m|r} \mu(m) \right) \phi(rx) = \phi(x).$$

Indeed the rearranged absolute sum is bounded by $\sum_{r \geq 1} d(r) |\phi(rx)| < \infty$, since $d(r) \leq r$ and ϕ is Schwartz. Thus no theta relation is lost when its source is recovered.

For $a > 0$ define $U_a F(x) = F(x/a)$, and use the same formula on V . Direct substitutions give

$$p_{b,k}(U_a F) = a^b p_{b,k}(F), \quad q_{N,k}(U_a \phi) \leq a^{-k} \max(1, a)^N q_{N,k}(\phi), \quad (1)$$

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = a U_{1/a} \widehat{\phi}, \quad \Theta U_a = U_a \Theta. \quad (2)$$

The two moments of $U_a \phi$ are $\phi(0)$ and $a \int \phi$, so V is invariant. Differentiating the defining formula gives

$$\frac{d}{dt} U_{e^t} \phi = D U_{e^t} \phi = U_{e^t} D \phi.$$

This is a derivative in every Schwartz seminorm. To verify it, apply the ordinary Taylor formula with integral remainder to $U_{e^{t+h}} \phi$; its second derivative is $U_{e^{t+h}} D^2 \phi$, uniformly bounded on each compact time interval by (1). The same argument, using the exact first identity in (1), proves smoothness in $\mathcal{B}$. Repeating with $D^j \phi$ proves derivatives of all orders. In particular this is a smooth group action on the retained Frechet spaces.

The four-point source poset is

$$+ < \eta < \sigma, \quad - < \eta < \sigma.$$

For the sheaf with stalks $V, V, \mathcal{B}, 0$ and restriction maps $\Theta, \Theta \mathcal{F}$, the source projective resolution computes

$$C = \mathbf{A}_\tau(\mathcal{T}) = \left[V \oplus V \xrightarrow{d} \mathcal{B} \right], \quad d(v, w) = \Theta(v - \widehat{w}), \quad |V \oplus V| = 0, |\mathcal{B}| = 1. \quad (3)$$

Its actual chain action is

$$\mathcal{U}_a^0(v, w) = (U_a v, a U_{1/a} w), \quad \mathcal{U}_a^1 F = U_a F. \quad (4)$$

Equation (2) proves $d\mathcal{U}_a^0 = \mathcal{U}_a^1 d$. On even Schwartz functions $\mathcal{F}^2 = 1$, so the exact degree-zero cycle inclusion is

$$i_0(w) = (\widehat{w}, w), \quad \mathcal{U}_a i_0(w) = i_0(a U_{1/a} w). \quad (5)$$

It follows from theta injectivity that $H^0(C) = i_0(V)$ and $H^1(C) = Q = \mathcal{B}/\Theta V$. These are the original global cohomology groups over the absolute tau base.

154.3 2. The finite arithmetic packet and its unchanged Taylor unit

Retain the original Gaussian source

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0 = \Theta\phi_*.$$

It has value zero at zero, and direct Gaussian integration gives zero integral. More precisely, with $M_+\phi(s) = \int_0^\infty \phi(x)x^s dx/x$,

$$M_+\phi_*(s) = \pi^{-s/2}(2\Gamma(s/2+2) - 3\Gamma(s/2+1)) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2). \quad (6)$$

Consequently the entire function $g = \mathcal{M}f_0$ agrees, first by absolutely convergent integration in a right half-plane and then by continuation, with the original function

$$g(s) = 2\zeta(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Entireness of the Mellin transform of $f_0 \in \mathcal{B}$ follows directly from the bounds in $\mathcal{B}$. For $0 < \operatorname{Re} s < 1$, set

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}M_+\phi(s).$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s). \quad (7)$$

For completeness, $M_+\phi$ is holomorphic on $\operatorname{Re} s > -2$, since smooth evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$. In $\operatorname{Re} s > 1$, interchanging its theta series and Mellin integral gives $\mathcal{M}\Theta\phi = 2\zeta(s)M_+\phi(s)$. Continue to the indicated strip to obtain the first equality in (7). Integrating $D\phi$ by parts proves $H_{D\phi} = sH_\phi$; the endpoint term $[x^s\phi(x)]_0^\infty$ vanishes on that strip. Equation (6) then proves the other two equalities.

Fix the original nonempty finite packet Z of actual nontrivial zeros, each with its complete order m_ρ , and put

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad d_h = \deg h, \quad E = \mathbb{C}[X]/(h), \quad A = M_X,$$

$$v(s) = g(s)/h(s), \quad \varepsilon = j_h(h/g) \in E^\times.$$

The quotient h/g here is taken as a germ at each selected zero; these germs are holomorphic and nonzero because the complete multiplicity has been retained. The local unit is **not** replaced by 1. For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree less than d_h representing εu , and put

$$\ell(u) = [X^{d_h-1}]R_Z(u). \quad (8)$$

We record the inverse used to define the source section, rather than taking an unproved inverse as an extra hypothesis. On $F \in \mathcal{B}$ satisfying $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y)y^{\rho-1} dy.$$

Differentiation gives $(D - \rho)S_\rho F = F$. The moment condition makes the upper integral the negative lower integral. If $\sigma = \operatorname{Re} \rho$, the upper estimate, for $N > \sigma$, is

$$|S_\rho F(x)| \leq \frac{p_{N,0}(F)}{N - \sigma} x^{-N}.$$

The lower estimate, for $N + \sigma > 0$, is

$$|S_\rho F(x)| \leq \frac{p_{-N,0}(F)}{N + \sigma} x^N.$$

Using the upper estimate on $x \geq 1$ and the lower estimate on $x \leq 1$ gives every weighted bound. The differential identity expresses $D^k S_\rho F$ as $\rho^k S_\rho F$ plus a finite linear combination of $D^j F$, proving all derivative bounds and continuity. The homogeneous solution $cx^{-\rho}$ cannot belong to $\mathcal{B}$ unless $c = 0$, so the inverse is unique. Integration by parts gives

$$\mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho},$$

with the removable value filled in. Iterating this construction, removing one vanishing factor at a time, defines the continuous inverse S_h of $h(D)$ on $\ker(j_h \mathcal{M})$. The operation preserves the remaining zeros and lowers the order at the chosen zero by exactly one at each step. Conversely any $h(D)F$ has those vanishing jets. This proves

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D) \text{ is injective.} \quad (9)$$

The section in the question is therefore the actual well-defined map

$$R = R_{\text{ref}} : E \longrightarrow \mathcal{B}, \quad Ru = S_h \Theta(R_Z(u)(D)\phi_*). \quad (10)$$

Indeed its input has Mellin transform $gR_Z(u)$, which vanishes to all the required orders. Equations (7) and (9) give

$$\mathcal{M}Ru = vR_Z(u), \quad J_Z R = 1_E, \quad J_Z = j_h \mathcal{M}, \quad J_Z \Theta = 0. \quad (11)$$

The Mellin-jet map is continuous: split its integral at 1, dominate the logarithmic factors at zero by a sufficiently negative weight and those at infinity by a sufficiently positive weight. This yields a finite sum of seminorm bounds for each derivative at each ρ . In (11), $j_h(v)\varepsilon = 1$ proves the middle identity. Hence R and $\sigma = qR : E \rightarrow Q$ are injective, since J_Z descends to Q .

The infinitesimal boundary is precisely the retained one:

$$DR - RA = \Theta\phi_*\ell = f_0\ell. \quad (12)$$

To prove it without losing its coefficient, the polynomial $XR_Z(u) - R_Z(Au)$ vanishes modulo h and has degree at most d_h . Since h is monic, its quotient by h is exactly $\ell(u)$. Apply $h(D)$ to (12), use (10), $D\Theta = \Theta D$, and that polynomial identity. Injectivity of $h(D)$ from (9) proves (12). For a single simple zero, $R_Z(u) = u/g'(\rho)$ and $\ell(u) = u/(2\xi'(\rho))$, with the original factor of two.

154.4 3. The finite dilation cocycle as an actual convergent source integral

For every real t , define $T_t = e^{tA}$ and

$$c_t(u) = \int_0^t U_{e^{t-r}} \phi_* \ell(T_r u) dr \in V. \quad (13)$$

If $t < 0$, $\int_0^t = -\int_t^0$; no orientation is changed. Write $c_a = c_{\log a}$ and $a^A = T_{\log a}$ when using the multiplicative parameter.

Convergence and continuity. Fix any norm on the finite-dimensional coefficient space, solely for estimates, and its induced operator norms. On the oriented interval between 0 and t , both $|r|$ and $|t - r|$ are at most $|t|$. Equation (1) and $\|T_r\| \leq e^{|r|\|A\|}$ yield

$$q_{N,k}(c_t u) \leq |t| e^{(N+k+\|A\|)|t|} q_{N,k}(\phi_*) \|\ell\| \|u\|. \quad (14)$$

The integrand is continuous in every seminorm. Its Riemann sums are Cauchy in every seminorm by uniform continuity on the compact interval. Completeness of V proves their common limit exists in V , and the inequality follows first for sums and then for their limit. Thus the integral is an actual source vector with the two original zero moments and even Schwartz regularity. It is not merely an integral in a weak dual or in L^2 .

Smoothness in t follows either by the differentiable compact-interval rule justified by the same estimates for all time derivatives, or by writing (13) as $t \int_0^1 U_{e^{t(1-r)}} \phi_* \ell(T_r u) dr$. This latter formula works without a sign convention change for negative t . Every derivative has a finite sum of terms involving $D^j \phi_*$, powers of A , and bounded polynomial factors in r . Inequality (14) with those finitely many seminorms proves local uniform convergence of the differentiated Riemann sums. Therefore $t \mapsto c_t$ is smooth as a map to continuous linear maps $E \rightarrow V$, with their bounded-set topology. Finite dimensionality of E makes the preceding uniform estimates on its unit ball sufficient for this statement.

The exact dilation identity. For all $t \in \mathbb{R}$,

$$\boxed{U_{e^t} R - R T_t = \Theta c_t.} \quad (15)$$

For fixed u , differentiate in the Frechet space $\mathcal{B}$:

$$\frac{d}{dr} (U_{e^{t-r}} R T_r u) = -U_{e^{t-r}} (DR - RA) T_r u = -\Theta U_{e^{t-r}} \phi_* \ell(T_r u).$$

The fundamental theorem of calculus follows in every defining seminorm from the same Riemann-sum argument; integrating with the indicated orientation proves (15), including $t < 0$. Continuity of Θ permits moving it through the integral.

Cocycle and derivative. Directly splitting the defect in (15) gives

$$\boxed{c_{t+s} = U_{e^t} c_s + c_t T_s, \quad c_0 = 0, \quad c_{-t} = -U_{e^{-t}} c_t T_{-t}.} \quad (16)$$

Indeed applying Θ to the asserted first equality yields the identity obtained by inserting $U_{e^t} R T_s$ in $U_{e^{t+s}} R - R T_{t+s}$. Injectivity of Θ proves equality in the actual source V . The other two statements follow by setting $t = s = 0$ and $s = -t$. Differentiating (13), or the cocycle in either variable, gives the two equal expressions

$$\boxed{\partial_t c_t = D c_t + \phi_* \ell T_t = U_{e^t} \phi_* \ell + c_t A, \quad \partial_t c_t|_{t=0} = \phi_* \ell.} \quad (17)$$

All derivatives are in V . In multiplicative notation (16) is

$$c_{ab} = U_a \circ c_b + c_a \circ b^A,$$

where the final product means the map $c_a \circ b^A$; to remove any typographical ambiguity it can also be written $c_{ab}(u) = U_a(c_b(u)) + c_a(b^A u)$.

The space $\mathcal{B}_Z = \Theta V + RE$ is a direct topological sum with coordinates $(\phi, u) \mapsto \Theta \phi + Ru$. Its inverse on that subspace is $(\Theta^{-1}(F - RJ_Z F), J_Z F)$; continuity of the inverse theta map is also proved in Section 6 below. Equations (15) and (16) integrate the source block operator to the exact group representation

$$U_a|_{\mathcal{B}_Z} = \begin{pmatrix} U_a|_V & c_a \\ 0 & a^A \end{pmatrix}. \quad (18)$$

This is the explicit group-level form of the already known extension (A12). In particular (15) proves the true quotient intertwining $\overline{U}_a \sigma = \sigma a^A$.

154.5 4. All nilpotents, the Mellin multiplier, and an incomplete-gamma evaluation

The Chinese remainder decomposition is

$$E = \bigoplus_{\rho \in Z} \mathbb{C}[z_\rho]/(z_\rho^{m_\rho}), \quad A|_\rho = \rho I + N_\rho,$$

where $N_\rho z_\rho^j = z_\rho^{j+1}$ and $N_\rho^{m_\rho} = 0$. Denote the summand projector by P_ρ , extending N_ρ by zero on other summands. These are the actual full local algebras. They can be constructed explicitly: if $H_\rho(X) = h(X)/(X - \rho)^{m_\rho}$, then P_ρ is multiplication by the remainder modulo h of

$$H_\rho(X) \sum_{j=0}^{m_\rho-1} \frac{(H_\rho^{-1})^{(j)}(\rho)}{j!} (X - \rho)^j.$$

At every other centre this has all required zero jets; at ρ it has jet 1. This verifies all its projector identities by the Chinese remainder map.

The action and the entire boundary integral therefore contain the full expressions

$$T_t = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho, \quad (19)$$

$$c_t u = \sum_{\rho \in Z} \sum_{j=0}^{m_\rho-1} \ell(N_\rho^j P_\rho u) \int_0^t U_{e^{t-r}\phi_*} e^{r\rho} \frac{r^j}{j!} dr. \quad (20)$$

The unit entering every coefficient is still $\varepsilon = j_h(h/g)$ through (8). If the packet has one zero of order m , and $\varepsilon = \sum_{i < m} \varepsilon_i z^i$, $u = \sum_{i < m} u_i z^i$, then $\ell(u) = \sum_{i=0}^{m-1} \varepsilon_i u_{m-1-i}$. This follows because the top coefficient of a polynomial of degree less than m is the same in the X and $z = X - \rho$ bases. Thus even the functional ℓ retains the lower coefficients of the Taylor unit, rather than only its nonzero constant value.

There is a closed expression for the integral kernels in (20). For $x > 0$ let

$$K_{\rho,t}(x) = \int_0^t e^{r\rho} \phi_*(x e^{r-t}) dr.$$

Define the oriented incomplete-gamma interval by

$$\Gamma_{[L,H]}(w) = \int_L^H y^{w-1} e^{-y} dy \quad (L, H > 0),$$

where $y^{w-1} = \exp((w-1) \log y)$ uses the real logarithm. Then substitution first of $y = x e^{r-t}$ and then of $z = \pi y^2$ gives

$$K_{\rho,t}(x) = e^{t\rho} x^{-\rho} \pi^{-\rho/2} [2\Gamma_{[\pi x^2 e^{-2t}, \pi x^2]}(\rho/2 + 2) - 3\Gamma_{[\pi x^2 e^{-2t}, \pi x^2]}(\rho/2 + 1)]. \quad (21)$$

Both substitutions preserve orientation when $t < 0$. The coefficients 2 and -3 come respectively from $4\pi^2 x^4$ and $-6\pi x^2$, including $dy/y = dz/(2z)$. No Gaussian constant is removed.

Differentiation under the compact oriented r -integral now gives the finite formula

$$c_t u = \sum_{\rho \in Z} \sum_{j=0}^{m_\rho-1} \frac{\ell(N_\rho^j P_\rho u)}{j!} \partial_\rho^j K_{\rho,t}. \quad (22)$$

The derivative here acts on the complex parameter in the entire kernel $K_{\rho,t}$ before evaluation at the indicated centre, not on the arithmetic zero or on the coefficient $\ell(N_\rho^j P_\rho u)$. The formula on $x > 0$ is extended evenly to $\mathbb{R}$, with its smooth value at 0 supplied by the integral defining K . That integral and Section 3 prove smoothness and both zero moments, so no regularity is inferred from cancelling the separate singular-looking factors in (21).

The full Mellin multiplier of the source boundary is also explicit:

$$H_{c_t u}(s) = \int_0^t e^{(t-r)s} \ell(T_r u) dr = \frac{e^{ts} R_Z(u)(s) - R_Z(T_t u)(s)}{h(s)}. \quad (23)$$

For the first equality in the original strip, use (7), $H_{U_a \phi_*} = a^s$, and the continuous source integral. For the second, differentiate $e^{(t-r)s} R_Z(T_r u)(s)$ in r . The polynomial identity proving (12) says its derivative is $-e^{(t-r)s} h(s) \ell(T_r u)$. Integration proves (23) wherever $h(s) \neq 0$. The numerator has every required vanishing jet: in the local algebra its jet is $e^{tA} \varepsilon u - \varepsilon T_t u = 0$. Hence the quotient in (23) has removable singularities at every chosen zero and is entire. This entire expression extends the original source multiplier; it is not a replacement of h or g .

For every integer $n \geq 0$ its actual Taylor coefficient at any complex centre s_0 is

$$\frac{H_{c_t u}^{(n)}(s_0)}{n!} = \int_0^t e^{(t-r)s_0} \frac{(t-r)^n}{n!} \ell(T_r u) dr. \quad (24)$$

Inserting (19) gives every mixed coefficient in $(t-r)^n r^j / (n!j!)$ and keeps all nilpotent terms. This is also a direct convergence proof for every jet, since the parameter interval is compact.

154.6 5. Actual two-leg homotopies and the support-face comparison

Regard $E[-1]$ as the complex with E in degree 1 and zero elsewhere. The finite section defines a degree-zero cochain map $r : E[-1] \rightarrow C$, with $r^1 = R$. The maps $\mathcal{U}_a r$ and ra^A have homotopies

$$\begin{aligned} H_a^+ : E[-1] &\longrightarrow C[-1], & (H_a^+)^1 u &= (c_a u, 0), \\ H_a^- : E[-1] &\longrightarrow C[-1], & (H_a^-)^1 u &= (0, -\widehat{c_a u}). \end{aligned} \quad (25)$$

Here the notation records maps of cohomological degree -1 ; the source only has degree 1. In ordinary component notation the exact identity is

$$\mathcal{U}_a r - ra^A = dH_a^\pm + H_a^\pm d_{E[-1]} = dH_a^\pm. \quad (26)$$

Indeed $d(c_a u, 0) = \Theta c_a u$, whereas $d(0, -\widehat{c_a u}) = \Theta \mathcal{F}^2 c_a u = \Theta c_a u$. This verifies the minus sign using the original differential. Equation (2) also gives the homotopy cocycle itself:

$$H_{ab}^\pm = \mathcal{U}_a^0 H_b^\pm + H_a^\pm b^A. \quad (27)$$

No extra Fourier scalar is inserted: $\widehat{U_a c} = aU_{1/a}\widehat{c}$ is exactly the second-leg action in (4).

The original support masks are $\lambda \in \{\{+\}, \{-\}, \{+, -\}\}$. The fibre C_λ^0 consists of the legs present in λ , $C_\lambda^1 = \mathcal{B}$, with the same differential. The $+$ homotopy in (25) is defined on $\{+\}$ and on the joint face. The $-$ homotopy is defined on $\{-\}$ and on the joint face. On the joint face their difference is the precisely retained cycle

$$H_a^+ - H_a^- = (c_a, \widehat{c}_a) = i_0(\widehat{c}_a). \quad (28)$$

It lies in $H^0(C)$ by (5) and theta injectivity. There is no term in degree -1 of C , so a nonzero such cycle is not a boundary in C .

The two singleton homotopies are forced. On $\{+\}$, injectivity of theta forces the first one; on $\{-\}$ it forces the second. Consequently a single homotopy that is strictly natural for both singleton-to-joint inclusions would require $(c_a, 0) = (0, -\widehat{c}_a)$, which is equivalent to $c_a = 0$. This is the exact scope of the support-naturality obstruction; it does not remove the homotopies in (25), their difference (28), or the resulting quotient action.

For completeness, c_a as a linear map is nonzero for every $a \neq 1$. If $c_a = 0$, (15) would make the nonzero finite-dimensional subspace RE invariant under U_a , with invertible action a^A . It therefore has a nonzero eigenvector $F \in \mathcal{B}$ with $U_a F = \lambda F$, $\lambda \neq 0$. Replacing a by a^{-1} if needed, take $a > 1$. Choose $x_0 > 0$ with $F(x_0) \neq 0$. Iteration gives $F(a^n x_0) = \lambda^{-n} F(x_0)$ for $n \geq 0$. The bound $p_{N,0}(F) < \infty$ implies $a^{nN} |\lambda|^{-n} |F(x_0)| \leq x_0^{-N} p_{N,0}(F)$. Choose N with $a^N > |\lambda|$ and let $n \rightarrow \infty$ to obtain a contradiction. Thus no such finite invariant subspace exists. This extends the previously proved failure of a strict equivariant section by giving its exact nonzero finite-dilation defect; the failure itself was already present in (A12).

At a fixed admitted mask the split lift of every displayed map is

$$\widetilde{f}(\lambda, u) = (\lambda, f(u)), \quad \widetilde{f}(\tau) = \tau.$$

An amplitude $f(u) = 0$ has value $(\lambda, 0)$, the supported zero, including at $a = 1$. All homotopy identities are interpreted using addition and the supported scalar $(-1)^\bullet$ in that fibre. No absent leg is inserted by the face-specific formulas.

The known synchronization comparison can be retained explicitly when a common joint target is desired. Send every active degree-one mask to $(\{+, -\}, F)$ and send the degree-zero pair to its joint-supported pair with the same two amplitudes, putting a supported zero in a formerly missing coordinate. Denote this operation by τ . It is the source map $(a, b) \mapsto (a + eb, b + ea)$, because multiplication by e gives a supported zero at a present coordinate. It preserves the differential, fixes joint masks, and is idempotent. The pair $(\text{supp } x, \tau x)$ recovers the original element by selecting the originally present coordinates, proving injectivity of that comparison. After this explicit synchronization, (25)–(28) hold in the common joint fibre. The original mask must stay attached to the comparison; forgetting it is not part of the homotopy calculation.

One can also see the extra degree-zero group in an exact equivariant roof. Let $C_Z = [V \oplus V \rightarrow \mathcal{B}_Z]$ with the same differential, and define

$$p_Z^0(v, w) = w, \quad p_Z^1 = J_Z.$$

This is a cochain map to $V[0] \oplus E[-1]$. It is equivariant for the action $aU_{1/a}$ on V and a^A on E , and induces isomorphisms on both cohomology groups: the degree-zero assertion follows from (5), while the degree-one assertion follows from the direct sum $\mathcal{B}_Z = \Theta V \oplus RE$ and (11). Thus the exact two-leg version of the source roof is

$$V[0] \oplus E[-1] \xleftarrow{p_Z} C_Z \longrightarrow C.$$

Its ordinary cochain section is $i_Z^0(w) = (\widehat{w}, w)$ and $i_Z^1 = R$. If $F = \Theta\phi + Ru$, the map $h_Z^1(F) = (\phi, 0)$ obeys $1 - i_Z p_Z = dh_Z + h_Z d$: in degree zero it returns $(v - \widehat{w}, 0)$ and in degree one it returns $\Theta\phi$. Its failure of equivariance in degree one is exactly (25). This proves the replacement on the entire original two-leg complex, retaining its $V[0]$ summand.

154.6.1 The transported minus face has an explicitly nonzero extension component

The preceding roof also computes what happens to the original support arrows when the action is retained. Work over $\mathbb{C}[X]$, with $X = D$ on $\mathcal{B}_Z$, $X = A$ on E , and $X = 1 - D$ on the degree-zero second-leg module, denoted V_- . Let $r(X) = R_Z(1)$ and $\psi = r(D)\phi_*$. Then

$$h(D)R(1) = \Theta\psi.$$

The class of ψ in $V/h(D)V$ is nonzero: applying $j_h H$ gives $j_h r = \varepsilon \neq 0$, whereas $j_h H$ vanishes on $h(D)V$ by (7). The Fourier map carries this quotient isomorphically onto $V/h(1 - D)V$, since differentiating (2) gives $\mathcal{F}D = (1 - D)\mathcal{F}$.

For the minus-face extension the injection is $i_- = -\Theta\mathcal{F}$. Its extension representative, with the convention $h(D)R(1) = i_-(w)$, is therefore

$$w = -\widehat{\psi} \quad \text{in } V/h(1 - D)V.$$

The component of the minus support arrow in the $V_-[0]$ summand of the joint roof has the opposite sign to this extension representative. Here is the exact shifted-resolution calculation. The free cochain resolution of E in degrees $-1, 0$ has differential h . We use $(C[n])^j = C^{j+n}$ and $d_{C[n]} = (-1)^n d_C$; after shifting by -1 , that resolution is in degrees $0, 1$ with differential $-h$. Its degree-one lift to the minus-face complex sends 1 to $R(1)$, extended $\mathbb{C}[X]$ -linearly. Its degree-zero lift must send 1 to $-w$, because $i_-(-w) = -h(D)R(1)$. Projection to $V_-[0]$ is identity in degree zero. Thus the induced derived map $E[-1] \rightarrow V_-[0]$ represents exactly

$$[-w] = [\widehat{\psi}] \quad \text{in} \quad \text{Ext}_{\mathbb{C}[X]}^1(E, V_-) \cong V/h(1 - D)V. \quad (28a)$$

The isomorphism here follows by applying $\text{Hom}_{\mathbb{C}[X]}(-, V_-)$ to that two-term free resolution; its cokernel is the stated quotient. The class (28a) is nonzero by the preceding source-jet calculation. The same arrow has identity as its $E[-1]$ component. By contrast, the plus support arrow has zero $V_-[0]$ component and identity $E[-1]$ component, because its degree-zero image has second coordinate zero. Consequently the two support arrows in the equivariant derived category do not both become inclusion of the $E[-1]$ summand: the exact difference carries (28a), with the original Taylor unit. The strict original cochain support diagram remains available before passage through the roof.

154.7 6. The exact bridge to the already constructed global section

We recall the global section in a form that makes its use here mathematically explicit. Choose the same fixed even real cutoffs $\alpha, \beta \in C_c^\infty(\mathbb{R})$, with $0 \leq \alpha, \beta \leq 1$, equal to 1 near zero. For $x \neq 0$ define

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|).$$

If $P_j(D) = (-1)^j D(D+1) \cdots (D+j-1) = x^j \partial_x^j$, and $C_{N,j}(F)$ is the sum of $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then for $x > 0$, $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (29)$$

Each summand is bounded by $\frac{1}{2}C_{N,j}(F)n^{-N}x^{-N-j}$; summation proves the estimate and local uniform convergence of every derivative. Thus the cutoff exterior functions $(1 - \alpha)\mathcal{U}F$ and $(1 - \beta)\mathcal{U}JF$, extended by zero near zero, are continuous Schwartz-valued maps. The theta inversion in Section 1 and Poisson give $\mathcal{U}\Theta\phi = \phi$ and $\mathcal{U}J\Theta\phi = \widehat{\phi}$ away from zero.

On $L^2(\mathbb{R}, dx)$ define

$$T = M_\alpha \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad YF = (1 - \alpha)\mathcal{U}F + \alpha \mathcal{F}^{-1}((1 - \beta)\mathcal{U}JF).$$

The kernel of T is $\alpha(x)\check{\beta}(x - y)$; its squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$, so T is compact. Its norm c is strictly less than 1. Otherwise compactness would give a unit vector attaining norm 1. Equality in the two contraction inequalities forces its Fourier transform to be supported where $\beta = 1$, and then the vector itself to be supported where $\alpha = 1$. A compactly supported Fourier transform in L^2 is in L^1 , and its inverse Fourier transform extends to an entire function by its finite-interval integral. That entire function would vanish on a real open interval because of the compact spatial support, hence vanish identically, a contradiction.

Thus $(1 - T)^{-1} = \sum_{n \geq 0} T^n$ in operator norm, with bound $(1 - c)^{-1}$. The operator $T : L^2 \rightarrow \mathcal{S}$ is continuous: Cauchy-Schwarz on the compact Fourier support bounds every derivative of $\mathcal{F}^{-1}(\beta\widehat{f})$ by a constant times $\|f\|_2$, and multiplication by α gives compact spatial support and every Schwartz seminorm. It follows that $\Phi = (1 - T)^{-1}Y$ is Schwartz-valued and continuous; the explicit identity $\Phi = Y + T\Phi$ gives

$$q_{N,k}(\Phi F) \leq q_{N,k}(YF) + \frac{C_{N,k}}{1 - c} \|YF\|_2.$$

No uniform lower bound for $1 - c$ as cutoffs vary is used. On a theta input, $Y\Theta\phi = (1 - T)\phi$, so $\Phi\Theta\phi = \phi$. The range of Φ is the even Schwartz space $\mathcal{S}(\mathbb{R})^{\text{even}}$. To check this exact domain, $\mathcal{U}F$ is even because its defining series uses $|x|$, and the same is true of $\mathcal{U}JF$. The cutoffs α, β are even, and Fourier transformation and inverse Fourier transformation preserve evenness, so YF is even. Reflection $\phi(x) \mapsto \phi(-x)$ commutes with T ; hence it commutes with every partial sum of the norm-convergent inverse $(1 - T)^{-1}$ and with its limit. Thus ΦF is even as an L^2 vector, and its Schwartz representative is even pointwise by continuity. With the original Gaussian moment corrections, define the map on this precise domain by

$$P_V\phi = \phi - \phi(0)e^{-\pi x^2} - \left(\int_{\mathbb{R}} \phi - \phi(0) \right) 2\pi x^2 e^{-\pi x^2}, \quad \Lambda = P_V\Phi.$$

The moment columns of the two displayed Gaussians are $(1, 1)$ and $(0, 1)$ by direct Gaussian integration. The two subtractions therefore set the value at zero and the integral to zero, while preserving evenness. On V both correction coefficients vanish. Consequently $P_V : \mathcal{S}(\mathbb{R})^{\text{even}} \rightarrow V$ is a continuous projection and $P_V|_V = 1$. This is its stated domain; no projection of the entire Schwartz space onto the even subspace is asserted by that formula. It follows that

$$\Lambda\Theta = 1_V, \quad K = 1 - \Theta\Lambda, \quad s : Q \rightarrow \mathcal{B}, \quad s[F] = KF. \quad (30)$$

This proves all needed existence and continuity claims for the particular global construction used in A1694. Since $\ker K = \Theta V$, that image is closed. The expression for s is well-defined, $qs = 1$, $\Lambda s = 0$, and it is continuous by the definition of the quotient topology. In particular Θ^{-1} on its image is Λ , verifying the topological inverse asserted after (17).

The already existing global cocycle is $k_a = \Lambda U_a s$, with

$$U_a s - s\overline{U}_a = \Theta k_a.$$

This follows by applying $1 = K + \Theta\Lambda$ to $U_a s$. The finite/global comparison is now fully determined, with no new choice of arithmetic unit:

$$b_Z = \Lambda R : E \rightarrow V, \quad R^s = s\sigma = R - \Theta b_Z. \quad (31)$$

Define $c_a^s = k_a \sigma$. Substitution in (15) yields

$$\boxed{c_a^s = c_a - U_a b_Z + b_Z a^A.} \quad (32)$$

To verify the equality in V , use $\overline{U}_a \sigma = \sigma a^A$ from (18) to compute

$$\begin{aligned} U_a R^s - R^s a^A &= U_a R - R a^A - \Theta U_a b_Z + \Theta b_Z a^A \\ &= \Theta(c_a - U_a b_Z + b_Z a^A) = \Theta k_a \sigma. \end{aligned}$$

Theta injectivity proves (32). Its derivative is the exact infinitesimal bridge

$$\boxed{k_D \sigma = \phi_* \ell - D b_Z + b_Z A, \quad k_D = \Lambda D s.} \quad (33)$$

All terms are defined continuous maps; differentiation is allowed by Section 1. This proves that the explicitly evaluated Duhamel integral and the transcript's global continuous cocycle represent the same actual arithmetic extension, through the displayed coboundary.

The full extension class is unchanged. Applying $h(D)$ to (31) gives

$$h(D)R^s(u) = \Theta(R_Z(u)(D)\phi_* - h(D)b_Z u).$$

Modulo $h(D)V$, its source class equals that of $R_Z(u)(D)\phi_*$. Under $j_h H$ its value is exactly εu , by (7). There is no quotient operation that discards the nonconstant coefficients of ε .

At cochain level the two finite representative maps r^s and r have the difference

$$r^s - r = dL^\pm, \quad L^+(u) = (-b_Z u, 0), \quad L^-(u) = (0, \widehat{b_Z u}). \quad (34)$$

The minus-face sign follows by applying $d(0, \widehat{b}) = -\Theta b$. Equations (25), (32), and (34) give the exact homotopy comparison

$$H_a^{s,\pm} = H_a^\pm + \mathcal{U}_a^0 L^\pm - L^\pm a^A.$$

These identities retain the joint degree-zero difference as in (28), rather than asserting strict agreement of the two face homotopies.

154.8 7. Tensor powers with every cohomological sign and jet coefficient

Let $k \geq 1$. On the original ordered tensor complex $C^{\otimes k}$ use

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_{j=1}^k (-1)^{\sum_{i<j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k. \quad (35)$$

This is exactly the tau-base product resolution differential. It squares to zero because the two orders of applying differentials in different factors have opposite signs; a repeated differential in one factor is zero.

Write $f_a = U_a R : E \rightarrow C^1$, $g_a = R a^A : E \rightarrow C^1$, and $h_a = H_a^\pm : E \rightarrow C^0$ for any admitted choice of face in each factor. The explicit degree- -1 homotopy on the top packet is

$$\mathcal{H}_{a,k}(u_1 \otimes \cdots \otimes u_k) = \sum_{j=1}^k (-1)^{j-1} f_a u_1 \otimes \cdots \otimes f_a u_{j-1} \otimes h_a u_j \otimes g_a u_{j+1} \otimes \cdots \otimes g_a u_k. \quad (36)$$

In this evaluated formula the factor $(-1)^{j-1}$ is displayed exactly once. If the same formula is instead written as a tensor of graded linear maps, it is the Koszul evaluation sign of the degree -1 map in slot j crossing the preceding $j-1$ inputs, each of degree 1; it must not be inserted a second time.

Only the slot containing $h_a u_j$ contributes to the differential, because the other slots are already in top degree. Its differential sign from (35) is $(-1)^{j-1}$, which cancels the prefactor. Since $dh_a = f_a - g_a$, the result is the telescoping identity

$$d\mathcal{H}_{a,k} = (U_a R)^{\otimes k} - (R a^A)^{\otimes k}. \quad (37)$$

For $k=2$ this reads $\mathcal{H}_{a,2} = h_a \otimes g_a - f_a \otimes h_a$ on evaluated inputs, and its differential is $(f_a - g_a) \otimes g_a + f_a \otimes (f_a - g_a)$. This displays the original minus sign without relying on an implicit convention.

Define

$$R_k = R^{\otimes k}, \quad A_k = \sum_{j=1}^k 1^{\otimes(j-1)} \otimes A \otimes 1^{\otimes(k-j)}, \quad T_{t,k} = T_t^{\otimes k} = e^{tA_k}.$$

For the ordered local block $(\rho_1, \dots, \rho_k)$ the action is exactly

$$T_{t,k} = e^{t \sum_i \rho_i} \sum_{0 \leq j_i < m_{\rho_i}} \frac{t^{j_1 + \cdots + j_k}}{j_1! \cdots j_k!} N_{\rho_1}^{j_1} \otimes \cdots \otimes N_{\rho_k}^{j_k}. \quad (38)$$

The formula follows by multiplying (19); the nilpotents in different factors commute, and no term has been discarded. The representative is still the tensor of the maps using ε in (10), hence the coefficient unit is $\varepsilon^{\otimes k}$ in the ordered local algebra.

For the multivariable Mellin transform $\mathcal{M}_k$ with measure $\prod_i dx_i/x_i$, the top boundary $B_{a,k} = d\mathcal{H}_{a,k}$ has, on a pure coefficient tensor, the exact expression

$$\begin{aligned} \mathcal{M}_k B_{a,k}(\mathbf{s}) &= \prod_{i=1}^k [a^{s_i} v(s_i) R_Z(u_i)(s_i)] \\ &\quad - \prod_{i=1}^k [v(s_i) R_Z(a^A u_i)(s_i)]. \end{aligned} \quad (39)$$

All integrals factor by absolute convergence of the Schwartz Mellin integrals. Applying the full packet jet in each factor gives zero, as also follows from (37). The homotopy (36) exhibits the actual tensor relation giving that zero. It is not inferred from a dimension count or an assumed vanishing tensor kernel.

Every map in (36) is continuous. More explicitly, for seminorms $p_1, \dots, p_k$ on the appropriate target factors, the projective tensor seminorm of one summand on a pure input is bounded by the product of the corresponding operator bounds from (1), (10), and (14), times $\prod_i \|u_i\|$. Sum over the k terms and use the defining infimum for the projective seminorm on arbitrary finite sums. Thus all maps extend to completed projective tensor products, and (37) persists by density and continuity. The coefficient domain $E^{\otimes k}$ is finite dimensional, but this continuity statement identifies its targets in the same completed topology used by the original global construction.

The support rule is applied to each admitted factor before the tensor balancing quotient: an external absent factor is absorbing, while a zero amplitude in an active factor remains its supported tensor

zero. The chosen face in that factor determines which of (25) occurs. Changing that face replaces its homotopy by (28); substitution into (36) gives an explicit tensor of the retained H^0 cycle with the other degree-one factors. No identification of external tensor support with a Cartesian product of labels is needed or asserted.

This tensor homotopy difference can be detected without discarding a possible lower-degree boundary. Define cochain maps $\pi_0 : C \rightarrow V[0]$ by $\pi_0^0(v, w) = w$, $\pi_0^1 = 0$, and $\pi_1 : C \rightarrow Q[-1]$ by $\pi_1^0 = 0$, $\pi_1^1 = q$. In the difference between the all-plus and all-minus versions of (36), apply π_0 in slot j and π_1 in the others. Every summand but the j -th vanishes. The result is

$$(-1)^{j-1}(\sigma a^A)^{\otimes(j-1)} \otimes \widehat{c}_a \otimes (\sigma a^A)^{\otimes(k-j)}.$$

It is a nonzero map whenever $a \neq 1$: σa^A is injective by (11), $\widehat{c}_a$ is nonzero by Section 5, and a tensor of nonzero vectors over $\mathbb{C}$ is nonzero, as witnessed by a tensor of linear functionals nonzero on those vectors. The target complex has zero differential, so the detected class cannot be a boundary. This proves the exact survival of the homotopy difference in degree $k - 1$, where one factor carries the original $H^0(C)$ and the others carry Q .

154.9 8. The complete dx boundary pairing, its derivative, and tensor mass

The inclusion $\mathcal{B} \hookrightarrow L^2(\mathbb{R}_{>0}, dx)$ is continuous. For example on $(0, 1)$ use $|F(x)| \leq p_{-1,0}(F)x$, and on $(1, \infty)$ use $|F(x)| \leq p_{1,0}(F)x^{-1}$. Both squared majorants are integrable. Use the inner product conjugate-linear in its first argument. Direct substitution, retaining dx , gives

$$\langle U_a F, U_a G \rangle = a \langle F, G \rangle. \quad (40)$$

Let $G = R^* R$ in the retained coefficient basis and let $B_a = \Theta c_a : E \rightarrow L^2(dx)$. Since $J_Z R = 1$, G is positive definite. Equation (15), $U_a R = R a^A + B_a$, gives the exact finite boundary identity

$$\boxed{(a^A)^* G a^A - a G = -(U_a R)^* B_a - B_a^* (U_a R) + B_a^* B_a.} \quad (41)$$

On vectors u, w , its right side is

$$-\langle U_a R u, \Theta c_a w \rangle - \langle \Theta c_a u, U_a R w \rangle + \langle \Theta c_a u, \Theta c_a w \rangle.$$

To check the signs, solve $R a^A = U_a R - B_a$ and expand its squared pairing. Equation (40) supplies $a G$. Equivalently the right side is $-(R a^A)^* B_a - B_a^* R a^A - B_a^* B_a$; these two forms agree by (15). Thus the positive sign in the version using $U_a R$ is forced by the expansion, not a sign choice.

Every term is finite and depends smoothly on a , by the Frechet bounds already proved and the continuous Hilbert inclusion. For a full generalized packet vector u , the left side is evaluated using (19), retaining every power of $\log a$. For a true eigenvector $Au = \rho u$ it is $(a^{2\operatorname{Re} \rho} - a) u^* G u$.

The infinitesimal identity obtained by differentiating (41) at $t = \log a = 0$ is

$$W = A^* G + G A - G = -R^* f_0 \ell - \ell^* f_0^* R. \quad (42)$$

This agrees exactly with (12) and the original integration by parts. Indeed for $F, H \in \mathcal{B}$,

$$\langle DF, H \rangle + \langle F, DH \rangle = - \int_0^\infty x (\overline{F} H)' dx = \langle F, H \rangle,$$

since the endpoint term $[x \overline{F}(x) H(x)]_0^\infty$ vanishes by the displayed weighted bounds. This independently verifies the mass term $-G$ and the two negative boundary cross terms in (42).

The finite defect is exactly the integrated infinitesimal one:

$$T_t^* G T_t - e^t G = \int_0^t e^{t-r} T_r^* W T_r dr. \quad (43)$$

Set $F(t) = T_t^* G T_t - e^t G$. Direct differentiation gives $F'(t) = F(t) + T_t^* W T_t$ and $F(0) = 0$. Multiplying by e^{-t} and integrating proves (43) with the original orientation for $t < 0$. This is a finite-dimensional exact matrix integration, valid with all the nilpotents in (19).

For the global representative R^s from (31), replace R by R^s and c_a by the fully computed c_a^s in (32). The finite expansion (41) then holds with

$$G^s = (R - \Theta b_Z)^*(R - \Theta b_Z), \quad \Theta c_a^s = \Theta(c_a - U_a b_Z + b_Z a^A).$$

Its infinitesimal boundary is the changed map

$$B_D^s = \Theta k_D \sigma = \Theta(\phi_* \ell - D b_Z + b_Z A). \quad (43a)$$

Indeed differentiating the explicit $c_{e^t}^s$ at $t = 0$ gives $k_D \sigma$ by (33), and direct differentiation of $R^s = R - \Theta b_Z$ gives $DR^s - R^s A = B_D^s$ by (12). Apply the same dx integration-by-parts identity to $R^s u, R^s w$ for arbitrary coefficient vectors. Substituting $DR^s = R^s A + B_D^s$ and moving the two boundary terms to the other side proves exactly

$$W^s = A^* G^s + G^s A - G^s = -(R^s)^* B_D^s - (B_D^s)^* R^s. \quad (43b)$$

Thus the factor $f_0 \ell$ in (42) is replaced by B_D^s , not retained as the old infinitesimal source boundary. Finally differentiating $T_t^* G^s T_t - e^t G^s$ as in the proof of (43) gives $F_s'(t) = F_s(t) + T_t^* W^s T_t$, $F_s(0) = 0$, and therefore

$$T_t^* G^s T_t - e^t G^s = \int_0^t e^{t-r} T_r^* W^s T_r dr. \quad (43c)$$

This is exactly the restriction of the A1694 global pairing identity to the actual packet σE . It proves the finite/global relationship between both the cocycle and its boundary pairings, without assuming an unproved vanishing of either cross term.

For k factors use the Hilbert pairing with measure $dx_1 \cdots dx_k$. Fubini gives its value on pure tensors as the product of the original pairings, hence

$$\langle U_a^{\otimes k} F, U_a^{\otimes k} H \rangle = a^k \langle F, H \rangle.$$

Let $G_k = G^{\otimes k}$, $B_{a,k}$ be the actual boundary (37), and $\mathcal{R}_a = (U_a R)^{\otimes k}$. Then the exact tensor version of (41) is

$$(a^A)^{* \otimes k} G_k (a^A)^{\otimes k} - a^k G_k = -\mathcal{R}_a^* B_{a,k} - B_{a,k}^* \mathcal{R}_a + B_{a,k}^* B_{a,k}. \quad (44)$$

Indeed $R_k(a^A)^{\otimes k} = \mathcal{R}_a - B_{a,k}$ by (37), and expansion proves the formula. Its derivative has the exact ordered sum

$$W_k = A_k^* G_k + G_k A_k - k G_k = \sum_{j=1}^k G^{\otimes(j-1)} \otimes W \otimes G^{\otimes(k-j)}, \quad (45)$$

because each of the k copies of the term $-G$ in (42) contributes one copy of $-G_k$. The finite integral is

$$T_{t,k}^* G_k T_{t,k} - e^{kt} G_k = \int_0^t e^{k(t-r)} T_{r,k}^* W_k T_{r,k} dr. \quad (46)$$

The proof differentiates the left side exactly as in (43), with mass exponent k . Equations (38) and (46) retain all tensor nilpotents. They are actual formulas for the original representative boundary contribution on the original tau-base tensor complex.

154.10 9. Source pins and status of the calculation

The complete input files read for this calculation were:

- `arithmetic_input.tex` (original September 12 cumulative source), SHA256 a50b0fb587b644c4ea94dba45b2b423d038f2bfbe7fef188543939ba850e58d2.
- `Tau_Base_Cohomology, NOTE.md`, SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3.
- Original transcript turn A1694.md, SHA256 a87bff4960c0a1742734e9e711780e0533ea76ba6f46ceab1fea115a2fd66f18; its original locator is recorded in Section 0.
- `Tau_Global_Comparison, NOTE.tex`, SHA256 50284fcebade0dc859ac398a00839a29e15be60c008e02314ac1f35c31950ed01; its equations are cited at the beginning.

The completed new formulas are (13)–(24), their exact finite/global comparison (31)–(34), the evaluated tensor homotopy and jets (36)–(39), and the integrated finite and tensor boundary expressions (43)–(46). Sections 1, 2, 5, 6, and the unintegrated boundary pairing explicitly reprove the source interfaces needed to type and verify those formulas. The global retraction, global cocycle, and support comparison in A1694 remain credited to that earlier calculation.

The resulting boundary terms have been computed, including their orientations, topologies, full arithmetic coefficients, and product masses. None has been set to zero by calling it a theta boundary. No estimate forcing the sign or vanishing of the original global weight defect is asserted by these identities. That precise limit on the conclusion does not delete the displayed maps, the full quotient Q , or the original tau-base construction.

155 The original balanced Mellin kernel: full polynomial comparison and meromorphic remainder

The authoritative shared transcript constructs the raw Mellin realization in A1167, retains its kernel complex in A1361, and replaces it by the $\mathbb{C}[t]$ -balanced realization in A1427. This section computes the polynomial action on the remaining kernel on those original objects. It uses no replacement arithmetic function and no auxiliary zero packet.

155.1 Original objects and full-plane Euler inverses

Retain, exactly as in A1167 and A1427,

$$\begin{aligned}
 V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\
 \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\
 \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}, \quad (\text{BK.1}) \\
 g(s) &= 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \\
 H_\phi(s) &= \frac{2\pi^{s/2}M_+\phi(s)}{s(s-1)\Gamma(s/2)}.
 \end{aligned}$$

The original Fourier convention is $\widehat{\phi}(\nu) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \nu} dx$. The original theta construction proves

$$\mathcal{M}\Theta\phi = gH_\phi, \quad \Theta D = D\Theta, \quad \widehat{D\phi} = (1-D)\widehat{\phi}, \quad H_{D\phi}(s) = sH_\phi(s), \quad H_{\widehat{\phi}}(s) = H_\phi(1-s). \quad (\text{BK.2})$$

The last equality follows directly from Poisson's identity $\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x)$, Mellin substitution, and $g(1-s) = g(s)$; cancellation holds on the nonempty open set $g \neq 0$ and then throughout the

plane by holomorphy. Here H_ϕ is entire: Taylor subtraction of the even smooth function at zero gives only simple possible poles of $M_+\phi$ at the negative even integers; the corresponding zeros of $1/\Gamma(s/2)$ cancel them. At $s = 0$, $M_+\phi$ is holomorphic because $\phi(0) = 0$, and $s\Gamma(s/2)$ tends to 2. At $s = 1$, $M_+\phi(1) = 0$ is exactly the retained integral condition. In particular

$$H_\phi(1) = 2(M_+\phi)'(1). \quad (\text{BK.3})$$

The topology and all these maps are the original Schwartz ones.

For every $\rho \in \mathbb{C}$ one has the exact range formulas

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)V = \ker(H_{(\cdot)}(\rho)). \quad (\text{BK.4})$$

Both operators are injective. We now prove the full-plane source formula, including the endpoint $\rho = 1$ that cannot be handled by dividing by $1 - \rho$.

On the target zero-Mellin domain the inverse is the original integral

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy. \quad (\text{BK.5})$$

Differentiating gives $(D - \rho)S_\rho F = F$. Since the total Mellin moment is zero, the same expression is $-x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ near zero. For $\sigma = \Re \rho$, an upper bound $|F(y)| \leq C_N y^{-N}$ yields $|S_\rho F(x)| \leq C_N x^{-N}/(N - \sigma)$ when $N > \sigma$. A lower bound $|F(y)| \leq C'_N y^N$ yields $|S_\rho F(x)| \leq C'_N x^N/(N + \sigma)$ when $N + \sigma > 0$. The differential identity controls each further Euler derivative. Thus every original seminorm is finite and the inverse is continuous on the indicated kernel. The only homogeneous solution is $cx^{-\rho}$, which belongs to $\mathcal{B}$ only when $c = 0$. Integration by parts proves $\mathcal{M}((D - \rho)F) = (s - \rho)\mathcal{M}F$, so the range is exactly the first kernel in (BK.4).

For the source, first let $\Re \rho > 0$ and $H_\phi(\rho) = 0$. For $\rho \neq 1$, the multiplier of $M_+\phi(\rho)$ in (BK.1) is finite and nonzero, so this Mellin moment vanishes. For $\rho = 1$ the same moment already vanishes by the definition of V . The integral (BK.5) is consequently also

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du. \quad (\text{BK.6})$$

For every ordinary derivative of order k , differentiating this integral inserts $u^k \phi^{(k)}(xu)$. Uniform domination on bounded x intervals follows from $\Re \rho + k > 0$. Therefore the result extends smoothly and evenly across zero with value zero there. The upper-tail expression gives Schwartz decay, with every derivative, at infinity. If $\rho \neq 1$, integration of $(D - \rho)S_\rho \phi = \phi$ over $\mathbb{R}$ gives

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi = \int_{\mathbb{R}} \phi = 0.$$

If $\rho = 1$, integration by parts in a right half-plane gives

$$M_+(S_1 \phi)(s) = \frac{M_+\phi(s)}{s - 1}.$$

Taking its removable value at 1 and using (BK.3) gives $\int_{\mathbb{R}} S_1 \phi = 2(M_+\phi)'(1) = H_\phi(1) = 0$. Thus the original source integral condition is retained at every ρ with positive real part.

If $\Re \rho \leq 0$, put $\lambda = 1 - \rho$, so $\Re \lambda \geq 1$. Fourier preserves V , and (BK.2) gives $H_{\widehat{\phi}}(\lambda) = H_\phi(\rho) = 0$. The exact inverse is

$$S_\rho^V \phi = -\widehat{S_\lambda \widehat{\phi}}. \quad (\text{BK.7})$$

On even functions Fourier squares to the identity. The intertwiner in (BK.2) therefore gives $(D - \rho)(-\widehat{S_\lambda \widehat{\phi}}) = \phi$, with every source condition preserved. For positive real part the inverse is (BK.6). The homogeneous solution cannot be Schwartz at infinity unless zero, so the inverse is unique. Identity (BK.2) proves the reverse containment in the second formula of (BK.4). This completes the full-plane calculation without excluding either endpoint or negative even Taylor coordinate.

155.2 Every polynomial kernel and cokernel is computed on the actual theta sequence

Set $A = \mathbb{C}[t]$, with t acting as the original D , and $Q = \mathcal{B}/\Theta V$. For a nonconstant monic polynomial

$$h(t) = \prod_{\rho} (t - \rho)^{n_{\rho}}, \quad E_h = A/(h),$$

let j_h retain all Taylor jets of orders less than n_{ρ} . Iterating (BK.4) gives

$$h(D)V = \ker(j_h H), \quad h(D)\mathcal{B} = \ker(j_h \mathcal{M}). \quad (\text{BK.8})$$

Indeed after dividing by $D - \rho$, the Mellin multiplier or H is divided by $s - \rho$ by (BK.2); the remaining zero orders are exactly the orders needed at the next division. Injectivity proves that the iterated inverse is independent of the chosen ordering.

The source jet map is onto because the original vector

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \quad \text{satisfies} \quad H_{P(D)\phi_*} = P. \quad (\text{BK.9})$$

The target jet map is onto because its functionals on compactly supported smooth logarithmic functions are the independent functions $y^j e^{\rho y} / j!$. For precision, a dependence vanishing on an interval extends to an entire exponential polynomial. Applying $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{n_{\sigma}}$ removes the other summands; its restriction to $e^{\rho y}$ times polynomials of degree less than n_{ρ} is invertible, since it is a product of nonzero constants plus nilpotent derivatives. Then $(\partial_y - \rho)$ repeatedly separates the polynomial coefficients. All coefficients vanish. If the finite vector-valued jet map were not onto, a nonzero linear functional on its missing quotient would give such a dependence, a contradiction.

Consequently the quotient square and exact theta sequence are

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{M_{j_h g}} & E_h, \end{array} \quad (\text{BK.10})$$

$$0 \longrightarrow Q[h(D)] \xrightarrow{\delta_h} E_h \xrightarrow{M_{j_h g}} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{BK.11})$$

These identities now hold for every polynomial h , without a strip restriction. The map δ_h sends $[F]$ to $j_h H_{\phi}$ where $h(D)F = \Theta\phi$. The source ϕ is unique because Θ is injective. Changing F by a theta relation changes ϕ by an $h(D)$ multiple, so the map is well-defined. If its value is zero, (BK.8) writes $\phi = h(D)\psi$ and injectivity of $h(D)$ on $\mathcal{B}$ gives $F = \Theta\psi$. A vector in the kernel of $M_{j_h g}$ is represented by a polynomial P ; the zero-jet condition permits applying the actual inverse of $h(D)$ to $\Theta(P(D)\phi_*)$, producing its preimage. Quotienting the right target in (BK.10) proves the last cokernel. This proves the entire sequence with its original maps.

155.3 The balanced kernel has invertible polynomial action

Fix any analytic stalk $O = \mathcal{O}_{\mathbb{C}, s_0}$, with t acting on O by multiplication by s . Write

$$M = O \otimes_A Q, \quad N = O/(g), \quad b : M \rightarrow N, \quad b(a \otimes [F]) = [a\mathcal{M}F], \quad K = \ker b. \quad (\text{BK.12})$$

The map is balanced by (BK.2). The ring O is torsion-free over the PID A : a nonzero polynomial times a holomorphic germ is zero only when that germ is zero. It is therefore flat. One direct justification of this familiar PID fact is to write a torsion-free module as the directed union of its finitely generated submodules; each such submodule is free by the PID structure theorem. Tensoring with a free module preserves injections, and filtered colimits preserve exactness, proving flatness.

The map b is onto. In fact the original test function

$$F_0(x) = \exp[-(\log x)^2], \quad H_0(s) = \mathcal{M}F_0(s) = \sqrt{\pi} \exp(s^2/4) \quad (\text{BK.13})$$

belongs to $\mathcal{B}$ and H_0 is an invertible germ at every point. Euler derivatives of F_0 are polynomials in $\log x$ times F_0 , and $x^b F_0$ is bounded for every integer b . Its Mellin integral becomes $\int_{\mathbb{R}} e^{-y^2+sy} dy$; real completion of the square gives the formula for real s , and entire continuation gives it for every complex s . Thus $b(aH_0^{-1} \otimes [F_0]) = [a]$, proving surjectivity.

For every nonzero polynomial h , the two induced maps are isomorphisms:

$$b : M[h] \xrightarrow{\sim} N[h], \quad \bar{b} : M/hM \xrightarrow{\sim} N/hN. \quad (\text{BK.14})$$

For a nonzero constant both statements are immediate. For a nonconstant polynomial flatness gives $M[h] = O \otimes_A Q[h(D)]$ and $M/hM = O \otimes_A (Q/h(D)Q)$. Also $O \otimes_A E_h = O/(h)$ by the explicit map $a \otimes [P(t)] \mapsto [aP(s)]$, with inverse $[a] \mapsto a \otimes 1$. The cokernel assertion in (BK.11) consequently gives

$$M/hM \cong O/(h, g) = N/hN,$$

and the isomorphism is precisely the Mellin map $\bar{b}$.

For the kernel map retain the actual local units. Put $z = s - s_0$ and write

$$h(s) = a(z)z^n, \quad g(s) = u(z)z^m, \quad a(0)u(0) \neq 0, \quad n, m \geq 0. \quad (\text{BK.15})$$

Under (BK.11), a local vector v in $\ker(g : O/(h) \rightarrow O/(h))$ corresponds to a torsion class with $h\mathcal{M}F = gH_\phi$, and its image under b is exactly

$$v \mapsto \left[\frac{g}{h} v \right] \in N[h]. \quad (\text{BK.16})$$

This is well-defined: changing v by h times a germ changes the displayed product by g times a germ. The kernel condition makes the product holomorphic. Explicitly its domain is $z^{\max(n-m, 0)} O / z^n O$ and its target is $z^{\max(m-n, 0)} O / z^m O$. Formula (BK.16) is multiplication by the actual unit u/a , together with the power z^{m-n} . It sends precisely the indicated source powers to the indicated target powers and has inverse multiplication by a/u and the power z^{n-m} . It is an isomorphism, including $m = 0$ or $n = 0$ when both spaces are zero. This proves the first map in (BK.14) without dropping a local factor of g or a repeated-zero order.

Apply the kernel-cokernel exact sequence for multiplication by h to $0 \rightarrow K \rightarrow M \xrightarrow{b} N \rightarrow 0$. It is

$$0 \rightarrow K[h] \rightarrow M[h] \rightarrow N[h] \rightarrow K/hK \rightarrow M/hM \rightarrow N/hN \rightarrow 0. \quad (\text{BK.17})$$

Both maps in (BK.14) are isomorphisms, so $K[h] = 0$ and $K/hK = 0$. We have proved

$$\boxed{h : K \xrightarrow{\sim} K \quad \text{for every nonzero } h \in \mathbb{C}[t].} \quad (\text{BK.18})$$

This is a computed property of the original balanced kernel.

It has a stronger local form. Every nonzero holomorphic germ is a unit times z^r . Units already act bijectively, and (BK.18) applied to $(t - s_0)^r$ makes z^r bijective. Hence K has a unique module structure over the meromorphic germ field $\text{Frac } O$, obtained by taking these actual inverses:

$$\frac{a}{b} \cdot k = a (b|_K)^{-1} k. \quad (\text{BK.19})$$

If $a/b = c/d$, the equality $ad = bc$ and invertibility of bd show that the two prescribed actions agree. The field laws follow by multiplying through their nonzero denominators.

155.4 Canonical finite spectral part and an explicit nonzero kernel class

At a point where $m = \text{ord}_{s_0} g > 0$, (BK.14) for $h = (t - s_0)^m$ gives a unique inverse $j : N \xrightarrow{\sim} M[z^m]$ of b on that kernel. Therefore

$$M = M[z^m] \oplus K, \quad x = j(bx) + (x - j(bx)), \quad M[z^m] \cong O/(g). \quad (\text{BK.20})$$

The first summand is all the O -torsion of M . Indeed every torsion element is killed by a power of z . For larger powers than m , (BK.14) still gives the same unique preimage of each element of N , because K has no z -torsion. When $m = 0$, $N = 0$ and the formula is $M = K$. All the full nilpotent jets of an actual zero occur in the first summand of (BK.20).

The kernel K is nonzero. The only classical zero-set input needed for the following assertion is the already-retained Hardy theorem: there are distinct $T_j \rightarrow +\infty$ with $g(1/2 + iT_j) = 0$. The cumulative reader uses this exact input in `theta_gamma_reference.tex`, (TG.26), and `sum_connection_stieltjes.tex`, (FC.21), with Hardy's original 1914 source. No assumption of RH is used.

Let $q_0 = [F_0] \in Q$. It is A -nontorsion. If $P(D)F_0 = \Theta\phi$ with a polynomial P , then (BK.2) and (BK.13) give

$$P(s)\sqrt{\pi}e^{s^2/4} = g(s)H_\phi(s).$$

At every $1/2 + iT_j$, the nonzero exponential forces $P(1/2 + iT_j) = 0$. A nonzero polynomial cannot have infinitely many distinct roots, so $P = 0$. Thus $A \rightarrow Q, P \mapsto P(D)q_0$, is injective. Flatness now proves that $O \rightarrow M, a \mapsto a \otimes q_0$, is injective. In particular

$$0 \neq g \otimes q_0 \in K. \quad (\text{BK.21})$$

The corresponding cycle is explicit on the original balanced complex. Put $b_* = \Theta\phi_*$, so $\mathcal{M}b_* = g$. In $O \otimes_A \mathcal{B}$, set

$$Z_0 = g \otimes F_0 - H_0 \otimes b_*. \quad \beta Z_0 = gH_0 - H_0g = 0. \quad (\text{BK.22})$$

Its degree-one class is exactly the nonzero class in (BK.21), since its second term is the original theta boundary. For the original invariant source label $W_* = \mathbb{C}[D]\phi_*$, the map $O \otimes_A W_* \rightarrow O$ induced by H is an isomorphism by (BK.9). Thus the kernel complex at that label has degree-zero term zero, and (BK.22) is an explicitly nonzero degree-one kernel cycle. Passing to the full label V preserves its nonzero class, as proved by the A -nontorsion argument. This resolves the status left open in A1427: balancing removes the earlier displayed spurious relation, but its full degree-one kernel is not zero.

This statement does not create a spectral obstruction to RH. For every finite-dimensional A -module E , let p_E be the characteristic polynomial of its t action. Cayley–Hamilton gives $p_E E = 0$. If $u : E \rightarrow K$ is A -linear, then $p_E u(E) = 0$ and injectivity in (BK.18) gives $u = 0$. If $v : K \rightarrow E$ is A -linear, every $k \in K$ is $p_E k'$ for some k' , and $v(k) = p_E v(k') = 0$. Consequently

$$\text{Hom}_A(E, K) = 0 = \text{Hom}_A(K, E) \quad (\dim_{\mathbb{C}} E < \infty). \quad (\text{BK.23})$$

The entire kernel has no finite-dimensional invariant subspace and no nonzero finite-dimensional equivariant quotient. This does not assert the same for arbitrary subquotients of its proper submodules. Its calculated contribution to every full finite spectral jet observation is zero, whereas (BK.20) proves that the actual finite spectral torsion survives bijectively.

Finally every construction occurs in the original supported quotient. The structure assertions (BK.18)–(BK.23) concern the full source label V used in (BK.12). They are not asserted for an arbitrary proper D -invariant source label. For such a label $W \subset V$, the exact comparison that must be retained is

$$0 \longrightarrow V/W \xrightarrow{[\phi] \mapsto [\Theta\phi]} Q_W \longrightarrow Q \longrightarrow 0, \quad Q_W = \mathcal{B}/\Theta W. \quad (\text{BK.24})$$

Injectivity of Θ proves the first map injective, and the quotient definitions prove its image is the last map's kernel. All maps commute with D when W is invariant. Tensoring this sequence with the

flat $\mathcal{O}$ retains the entire kernel $\mathcal{O} \otimes_A (V/W)$, which can have additional finite spectral torsion before transport to the full label. No such proper-label torsion is discarded by the scope of the structure theorem. The inclusion of labels $W_* \hookrightarrow V$ carries $\mathcal{B}/\Theta W_* \rightarrow \mathcal{B}/\Theta V$ with the original kernel $\Theta V/\Theta W_*$. Balancing imposes the explicit linear relations $sa \otimes v - a \otimes Dv$ within each such label. A linear map L at a fixed label ℓ lifts as $(\ell, x) \mapsto (\ell, Lx)$ and sends external τ to external τ . Addition joins labels and applies their given linear transports; the displayed formulas commute with those transports. Therefore (BK.22) is sent to the supported zero $(\ell, 0)$ by Mellin evaluation, while its nonzero kernel class remains at that label. Equations (BK.18)–(BK.24), including every actual nilpotent jet and the proper-label transition kernel, describe precisely which full-source kernel assertions survive the complete original quotient diagram.

156 The exact derived comparison at every finite spectral thickening

This calculation uses the balanced comparison of the original theta complex, as constructed in the shared conversation at A1427, and the polynomial-invertibility theorem proved in the accompanying balanced-kernel chapter. It uses no replacement polynomial in place of an actual zeta packet. Write $\mathcal{O} = \mathcal{O}_{\mathbb{C}, s_0}$ for the local ring of holomorphic germs at the specified spectral coordinate s_0 , $A = \mathbb{C}[t]$, $M = \mathcal{O} \otimes_A Q$, and

$$Q = \mathcal{B}/\Theta V, \quad g = 2\xi, \quad \beta : M \longrightarrow \mathcal{O}/(g), \quad \beta(a \otimes [F]) = [a\mathcal{M}F].$$

The A -action is exactly $tF = DF = -x\partial_x F$ on the original function space and $ta = sa$ on $\mathcal{O}$. Let $K = \ker \beta$. The preceding calculation on these original spaces proves that

$$0 \longrightarrow K \longrightarrow M \xrightarrow{\beta} \mathcal{O}/(g) \longrightarrow 0 \quad \text{is exact,} \quad (s - s_0) : K \xrightarrow{\sim} K. \quad (\text{DP.1})$$

In particular this is not an assumption of injectivity of β ; the preceding chapter exhibits an explicit nonzero element of K .

Proposition 156.1 (Derived finite spectral observations). *For every nonzero holomorphic germ $h \in \mathcal{O}$, the original comparison induces a quasi-isomorphism*

$$M \otimes_{\mathcal{O}}^{\mathbf{L}} \mathcal{O}/(h) \xrightarrow{\beta \otimes 1} (\mathcal{O}/(g)) \otimes_{\mathcal{O}}^{\mathbf{L}} \mathcal{O}/(h). \quad (\text{DP.2})$$

Both the degree-minus-one kernel and the degree-zero quotient are retained. The map commutes with multiplication by every holomorphic germ, in particular the original action a^s for every real $a > 0$.

Proof. Write the specified germ as $h = (s - s_0)^n u$, where $n \geq 0$ is its actual vanishing order and $u(s_0) \neq 0$. This identity retains the entire unit u , not merely its value at s_0 . Multiplication by u is invertible on every $\mathcal{O}$ -module. By (DP.1), multiplication by $(s - s_0)^n$ is invertible on K . Thus the literal multiplication map $h : K \rightarrow K$ is invertible.

The integral domain $\mathcal{O}$ has the free resolution

$$0 \longrightarrow \mathcal{O} \xrightarrow{\times h} \mathcal{O} \longrightarrow \mathcal{O}/(h) \longrightarrow 0.$$

Place its two free terms in degrees $-1, 0$. Tensoring it with the three modules in (DP.1) gives a short exact sequence of complexes

$$0 \longrightarrow [K \xrightarrow{h} K] \longrightarrow [M \xrightarrow{h} M] \longrightarrow [\mathcal{O}/(g) \xrightarrow{h} \mathcal{O}/(g)] \longrightarrow 0. \quad (\text{DP.3})$$

The first complex has zero cohomology in both degrees because its differential is invertible. The cohomology exact sequence of (DP.3) therefore proves (DP.2), including the degree -1 terms $\ker(h : M \rightarrow M)$ and $\ker(h : \mathcal{O}/(g) \rightarrow \mathcal{O}/(g))$. Its degree-zero terms are the actual quotients by h . All maps in (DP.3) are multiplication or $\mathcal{O}$ -linear maps; they commute with multiplication by every germ. This proves the claimed action compatibility without removing a nilpotent term. $\square$

At an actual zero ρ of complete order m_ρ , retain $g(s) = (s - \rho)^{m_\rho} v_\rho(s)$ with its nonvanishing full unit. For the exact thickening $h = (s - \rho)^n$, the right complex in (DP.3) has

$$H^{-1} = \{z \in \mathcal{O}_\rho/(g) : (s - \rho)^n z = 0\}, \quad H^0 = \mathcal{O}_\rho/(g, (s - \rho)^n). \quad (\text{DP.4})$$

When $n \geq m_\rho$, its differential is zero, so both terms are the entire length- m_ρ arithmetic local algebra. When $0 < n < m_\rho$, its kernel is exactly $(s - \rho)^{m_\rho - n} \mathcal{O}_\rho/(g)$ and its quotient is $\mathcal{O}_\rho/((s - \rho)^n)$, each of length n . These statements follow by division by the retained unit v_ρ , which is an isomorphism of the specified ideals and does not change the action. On every retained full block the action is still

$$a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho[z^j] = [z^{j+1}], \quad z = s - \rho. \quad (\text{DP.5})$$

Here is the exact bridge to the original dilation, rather than only to scalar multiplication on a balanced module. Define $R_a F(x) = F(x/a)$ and $R_a \phi(x) = \phi(x/a)$ for $a > 0$. These maps preserve the original spaces, commute with D , and $\Theta R_a = R_a \Theta$ term by term. Substitution $x = ay$ gives $\mathcal{M}(R_a F)(s) = a^s \mathcal{M}F(s)$; hence the induced $\mathcal{O}$ -linear map $1 \otimes R_a$ on M satisfies $\beta(1 \otimes R_a) = a^s \beta$. It preserves $M[h]$ and hM , since it commutes with h . The proved comparison on kernels and quotients therefore transports its action to literal multiplication by a^s , giving (DP.5). This does not assert equality of $1 \otimes R_a$ and scalar a^s on the entire balanced module: their difference has image in K , which is retained.

For the counterfactual under examination, ρ is an actual off-critical nontrivial zero of g ; its existence is the temporary premise, not a conclusion. Equations (DP.2)–(DP.5) prove that its complete finite block persists through this balanced comparison, even after taking every finite derived spectral thickening. It cannot be an element of the nonzero kernel K , since multiplication by its annihilating polynomial is invertible on K .

On each of the three original nonempty two-leg masks whose admitted theta image is the full ΘV , these complexes have the split lifts $(\lambda, x) \mapsto (\lambda, f(x))$, and external τ maps to external τ . A zero in (DP.3) is the supported zero at that same mask. This assertion uses the full admitted source V . The arithmetic spectral coordinate s_0 and the absolute base point τ are connected by this labelled comparison; they are not identified by it. Smaller admitted sources have the following additional, explicitly computed finite spectral data.

Proposition 156.2 (All finite defects of a smaller source label). *Let $W \subseteq V$ be an original D -invariant admitted source, and retain $Q_W = \mathcal{B}/\Theta W$. Put*

$$M_W = \mathcal{O} \otimes_A Q_W, \quad L_W = \mathcal{O} \otimes_A (V/W), \quad \beta_W(a \otimes [F]_W) = [a \mathcal{M}F], \quad K_W = \ker \beta_W.$$

Then the source inclusion induces the exact sequences

$$0 \longrightarrow L_W \xrightarrow{[\phi] \mapsto [\Theta \phi]} M_W \xrightarrow{\pi_W} M \longrightarrow 0, \quad 0 \longrightarrow L_W \longrightarrow K_W \longrightarrow K \longrightarrow 0. \quad (\text{DP.6})$$

For every nonzero germ h , their induced maps give

$$L_W[h] \xrightarrow{\sim} K_W[h], \quad L_W/hL_W \xrightarrow{\sim} K_W/hK_W. \quad (\text{DP.7})$$

More fully, inclusion is a quasi-isomorphism

$$[L_W \xrightarrow{h} L_W] \xrightarrow{\sim} [K_W \xrightarrow{h} K_W]. \quad (\text{DP.8})$$

Proof. Injectivity of the original $\Theta : V \rightarrow \mathcal{B}$ gives the exact A -module sequence $0 \rightarrow V/W \xrightarrow{\Theta} Q_W \rightarrow Q \rightarrow 0$. The flatness proved in the balanced-kernel chapter yields the first sequence in (DP.6). The equality $\beta_W = \beta \pi_W$ follows directly from the retained Mellin formula. Every element

of L_W maps to zero under β_W , since $\mathcal{M}\Theta\phi = gH_\phi$. If $k \in K$, lift it to $x \in M_W$ using the surjection π_W ; then $\beta_W x = \beta k = 0$. This proves the second sequence in (DP.6), including surjectivity of its last map. Tensor its terms with the same two-term free resolution of $\mathcal{O}/(h)$ used in (DP.3). The quotient complex is $[K \xrightarrow{h} K]$, whose differential is invertible by (DP.1) and its unit-factor argument. Its cohomology vanishes. The resulting exact cohomology sequence proves both (DP.7) and (DP.8). Every arrow comes from the actual source inclusion, and commutes with the given actions. $\square$

Thus a finite spectral class in a smaller label's Mellin kernel is precisely a finite spectral class in its retained relation quotient V/W . It vanishes upon passing to the full arithmetic source V , by (DP.6). This is an actual transition map, not a deletion of the smaller label or its extension. The next chapter constructs such a nonzero class explicitly from the original dilates of ϕ_* , so that the restriction to the full source in (DP.1)–(DP.5) is also tested by a concrete example. At a smaller label only D -invariance has been imposed. Accordingly (DP.6)–(DP.8) commute with the D and holomorphic-scalar actions; an original dilation R_a acts between the labels W and $R_a W$, and is an endomorphism at W only when $R_a W = W$. This follows from the exact map $[F]_W \mapsto [R_a F]_{R_a W}$ and $\Theta R_a = R_a \Theta$.

157 An actual proper source label with a finite spectral kernel

The full-source calculation (BK.18) applies to $Q = \mathcal{B}/\Theta V$. It does not assert the same result for $Q_W = \mathcal{B}/\Theta W$ at an arbitrary proper, D -invariant source label $W \subset V$. The following construction proves the distinction and its exact transition map on the original source functions. It does not replace $g = 2\xi$ by a model function and does not assume an off-critical zero.

Retain $A = \mathbb{C}[t]$, $t = D = -x\partial_x$, and the original source vector and its literal dilation

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \psi(x) = \phi_*(2x). \quad (\text{PL.1})$$

Both functions belong to V : dilation preserves evenness, Schwartz decay, and value zero at the origin, while $\int_{\mathbb{R}} \psi(x) dx = \frac{1}{2} \int_{\mathbb{R}} \phi_*(y) dy = 0$. The change of variable $y = 2x$ in the original positive Mellin integral gives

$$M_+ \psi(s) = 2^{-s} M_+ \phi_*(s), \quad H_\psi(s) = 2^{-s}, \quad \mathcal{M}\Theta\psi(s) = g(s)2^{-s}. \quad (\text{PL.2})$$

The first equality holds in its initial half-plane of absolute convergence; the entire continuation of H proved in (BK.1)–(BK.3) extends the second equality to the full plane. Here $2^{-s} = \exp(-s \log 2)$ uses the positive real logarithm of the actual dilation constant 2.

Define the original invariant source label

$$W = A\phi_* + A\psi \subset V. \quad (\text{PL.3})$$

This is a free rank-two A -module on the two displayed vectors. Indeed an equality $P(D)\phi_* + Q(D)\psi = 0$, for polynomials $P, Q \in \mathbb{C}[t]$, implies by (BK.2) and (PL.2)

$$P(s) + Q(s)2^{-s} = 0 \quad (s \in \mathbb{C}).$$

As the real variable s tends to positive infinity, $Q(s)2^{-s}$ tends to zero. Thus $P(s)$ tends to zero. A nonzero complex polynomial cannot tend to zero along that ray: division by its highest real power gives its nonzero leading coefficient. Consequently $P = 0$, and the nonvanishing exponential forces $Q = 0$. This also proves the exponential-polynomial independence used below.

Fix any $\rho \in \mathbb{C}$ and retain its exact nonzero scalar $c_\rho = 2^{-\rho}$. Set

$$\phi_\rho = \psi - c_\rho \phi_* \in W, \quad H_{\phi_\rho}(s) = 2^{-s} - c_\rho. \quad (\text{PL.4})$$

The last entire function vanishes at $s = \rho$. The already-proved full-plane inverse (BK.4), including its endpoint and Fourier-reflected cases, therefore gives the unique actual vector

$$\chi_\rho = S_\rho^V \phi_\rho \in V, \quad (D - \rho)\chi_\rho = \phi_\rho, \quad H_{\chi_\rho}(s) = \frac{2^{-s} - 2^{-\rho}}{s - \rho}. \quad (\text{PL.5})$$

Its displayed quotient is entire with its retained value $H_{\chi_\rho}(\rho) = -(\log 2)2^{-\rho} \neq 0$. The vector is specified without an implicit choice: for $\text{Re } \rho > 0$ its positive-half-line formula is

$$\chi_\rho(x) = x^{-\rho} \int_x^\infty \phi_\rho(y) y^{\rho-1} dy = - \int_0^1 u^{\rho-1} \phi_\rho(xu) du, \quad (\text{PL.6})$$

extended evenly; for $\text{Re } \rho \leq 0$ it is $-\widehat{S_{1-\rho}\phi_\rho}$, with the same positive-half-line integral for $S_{1-\rho}$. The proof in (BK.4)–(BK.7) establishes all original source conditions for these formulas.

The vector χ_ρ is not in W . Suppose otherwise and write $\chi_\rho = P(D)\phi_* + Q(D)\psi$. Multiplying the equality of their H transforms by $s - \rho$ gives the entire identity

$$((s - \rho)P(s) + 2^{-\rho}) + ((s - \rho)Q(s) - 1)2^{-s} = 0. \quad (\text{PL.7})$$

The proved exponential-polynomial independence makes both polynomial coefficients identically zero. Evaluating the first at $s = \rho$ would give $2^{-\rho} = 0$, a contradiction. This proves $\chi_\rho \notin W$, so in particular $W \neq V$.

Now take the actual theta representative and its original supported quotient class:

$$F_\rho = \Theta\chi_\rho \in \mathcal{B}, \quad q_\rho = [F_\rho]_W \in Q_W = \mathcal{B}/\Theta W. \quad (\text{PL.8})$$

It is nonzero because $F_\rho = \Theta w, w \in W$, would imply $\chi_\rho = w$ by injectivity of the original theta map. Nevertheless

$$(t - \rho)q_\rho = 0, \quad \mathcal{M}F_\rho(s) = g(s) \frac{2^{-s} - 2^{-\rho}}{s - \rho}. \quad (\text{PL.9})$$

The first equality follows from $(D - \rho)F_\rho = \Theta\phi_\rho \in \Theta W$; the second is the original identity $\mathcal{M}\Theta = gH$ applied to χ_ρ . In fact $Aq_\rho \cong A/(t - \rho)$. For any polynomial P , division by $t - \rho$ gives $P(t)q_\rho = P(\rho)q_\rho$. Since $q_\rho \neq 0$, this vanishes exactly when $P(\rho) = 0$.

Let $O = \mathcal{O}_{\mathbb{C}, \rho}$, $z = s - \rho$, and retain the actual proper-label comparison

$$M_W = O \otimes_A Q_W, \quad b_W : M_W \longrightarrow O/(g), \quad b_W(a \otimes [F]_W) = [a\mathcal{M}F], \quad K_W = \ker b_W. \quad (\text{PL.10})$$

It is well-defined because $\mathcal{M}\Theta w = gH_w$ for every $w \in W$; it is balanced because $\mathcal{M}DF = s\mathcal{M}F$. It is onto by the original F_0 and its invertible transform H_0 in (BK.13): $b_W(aH_0^{-1} \otimes [F_0]_W) = [a]$. The inclusion $Aq_\rho \hookrightarrow Q_W$ remains injective after tensoring with the flat A -module O . Since

$$O \otimes_A Aq_\rho \cong O/(z) \neq 0,$$

the element $k_\rho = 1 \otimes q_\rho$ is nonzero in M_W . Its original image (PL.9) is divisible by g , giving

$$0 \neq k_\rho \in K_W, \quad zk_\rho = 0. \quad (\text{PL.11})$$

Thus K_W has a nonzero one-dimensional spectral submodule on an actual proper source label containing the original arithmetic generator ϕ_* . Invariance of W and inclusion of that generator do not imply the polynomial-invertibility theorem for K_W .

We record the exact morphism that transports this class to the proved full-source comparison. Inclusion of source labels gives

$$0 \longrightarrow V/W \xrightarrow{[v] \mapsto [\Theta v]_W} Q_W \xrightarrow{[F]_W \mapsto [F]_V} Q \longrightarrow 0. \quad (\text{PL.12})$$

The first arrow is injective by theta injectivity. The final map is onto, and its kernel consists exactly of those representatives in ΘV , modulo ΘW . These statements prove exactness and identify its kernel with the original V/W , without changing the source quotient. The arrows commute with D because theta does. Flatness therefore gives the actual balanced exact sequence

$$0 \longrightarrow O \otimes_A (V/W) \longrightarrow M_W \xrightarrow{\pi} M \longrightarrow 0, \quad M = O \otimes_A Q, \quad b_W = b\pi. \quad (\text{PL.13})$$

Restricting π to the indicated kernels gives

$$0 \longrightarrow O \otimes_A (V/W) \longrightarrow K_W \xrightarrow{\pi} K \longrightarrow 0. \quad (\text{PL.14})$$

Indeed the first term is already in K_W since its Mellin values are multiples of g . Every element of K has a lift in M_W by (PL.13), and every such lift is in K_W because $b_W = b\pi$. This proves surjectivity of the last arrow and identifies its kernel. The explicit k_ρ is the image of $1 \otimes [\chi_\rho]$ in the left term of (PL.14), and its image in K is exactly zero. Its finite spectral contribution is therefore removed by the specified enlargement from W to V . It does not contradict the polynomial invertibility of the full-source K .

This difference is also detected by the finite derived observations. Tensor the free resolution $[O \xrightarrow{z} O]$ with the proper-label exact sequence $0 \rightarrow K_W \rightarrow M_W \xrightarrow{b_W} O/(g) \rightarrow 0$. It produces the short exact sequence of complexes

$$0 \rightarrow [K_W \xrightarrow{z} K_W] \rightarrow [M_W \xrightarrow{z} M_W] \rightarrow [O/(g) \xrightarrow{z} O/(g)] \rightarrow 0. \quad (\text{PL.15})$$

The first complex has nonzero degree-minus-one cohomology, since $k_\rho \in \ker(z : K_W \rightarrow K_W) \setminus \{0\}$. Equivalently the degree-minus-one map on the last two complexes has the nonzero kernel containing k_ρ . Thus $b_W \otimes_O^L O/(z)$ is not a quasi-isomorphism. The finite derived comparison (DP.2) is therefore a theorem for its stated full-source Q and its specified copies in the supported construction, not for arbitrary proper source labels.

Every element above retains its original label. The source-label transport carries $[\Theta\chi_\rho]_W$ to the supported zero in the V fibre; it does not send that element to the external absence point τ . Equations (PL.12)–(PL.15) prove the exact relation between the proper-label finite spectral kernel and the full-source meromorphic kernel, including the class that the transition kills. Nothing in the construction asserts $g(\rho) = 0$; it works at every specified ρ . If ρ is an actual off-critical zero, its arithmetic block in $O/(g)$ remains the one computed in (DP.4), and the proper-label class (PL.11) is the separately computed kernel of the displayed source-label transition.

158 The actual arithmetic packet through mixed-support synchronization

This calculation starts with the mixed-support construction in original shared-thread record A0961, node a7bbd422-311c-4380-ba6c-a4bdd61fce2f, chain 2032, and its full authored proof in original record A0953, node 0d2934ea-14ac-48c1-a1f3-cd918c1f5fe5, chain 1987. It does not replace the actual arithmetic function by a symmetric test polynomial. The polynomial below is the full divisor polynomial of an actual packet in the counterfactual under study.

Let Z be a nonempty finite set of actual nontrivial zeros of the original completed function $g(s) = 2\xi(s)$, with their full orders m_ρ . Retain

$$h_Z(x) = \prod_{\rho \in Z} (x - \rho)^{m_\rho}, \quad d = \sum_{\rho \in Z} m_\rho, \quad E_Z = \mathbb{C}[x]/(h_Z(x)).$$

The variable x is the original spectral variable. The additional variable t below is the cyclic coefficient coordinate; it is not substituted for x . For $a > 0$ let

$$A_a = M_{j_{h_Z}(a^s)} : E_Z \longrightarrow E_Z.$$

Here $a^s = \exp(s \log a)$ uses the real logarithm and j_{h_Z} retains all Taylor coefficients at the actual zero centres. On the summand

$$E_\rho = \mathbb{C}[z_\rho]/(z_\rho^{m_\rho}), \quad x = \rho + z_\rho,$$

this is exactly

$$A_a|_{E_\rho} = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho = M_{z_\rho}. \quad (34)$$

Thus no multiplicity or nilpotent term is omitted.

158.1 The original mixed-support object, with its unit

Fix $n \geq 1$. Form the source object $D_{E_Z, n} = G(E_Z)^n$ with coordinatewise addition and the source product

$$(b \star c)_k = \bigoplus_{i+j=k} b_i c_j \oplus \bigoplus_{i+j=k+n} (-1)^{\bullet} b_i c_j, \quad 0 \leq k < n.$$

An empty sum is τ , and the original unit is $(1^\bullet, \tau, \dots, \tau)$. Write $e = 0^\bullet_{E_Z}$. The exact amplitude target is

$$B_n = E_Z[t]/(t^n + 1), \quad \pi_n(b) = \sum_{i=0}^{n-1} p(b_i) t^i.$$

The support mask of b has its i th entry equal to one precisely when $b_i \neq \tau$. The pair consisting of the mask and $\pi_n(b)$ is injective: the monic polynomial gives the free E_Z -basis $1, t, \dots, t^{n-1}$, and the mask then distinguishes e from τ in each zero coefficient.

Let

$$E_n = (1^\bullet, e, \dots, e).$$

Multiplication by E_n sends the wholly absent input to itself and every active input to the fully supported vector with the same ordinary coefficients. To check the support assertion, fix an active input coefficient at index j . At every output index k the summand using the E_n coefficient at $k - j$ modulo n is supported; all wrap factors are supported. Thus every output coordinate is supported, including a coordinate whose resulting amplitude is zero. The amplitude assertion follows from $\pi_n(E_n) = 1$. It follows at once that $E_n^2 = E_n$ and

$$D_{E_Z, n}[E_n^{-1}] \simeq E_n D_{E_Z, n} \simeq G(B_n). \quad (35)$$

The image has unit E_n . For completeness, a unital semiring map making E_n invertible sends it to 1 by idempotence and therefore sends b and $E_n b$ to the same value. This is the localization universal property. The inclusion of the image into the original unsynchronized object is not falsely declared to preserve the original unit.

158.2 A proved split injection and its exact retraction

Define

$$\iota_n : E_Z \longrightarrow B_n, \quad f \longmapsto f, \quad r_n : B_n \longrightarrow E_Z, \quad b \longmapsto \frac{1}{n} \text{Tr}_{B_n/E_Z}(M_b).$$

The first is a unital ring homomorphism. The second is an E_Z -linear map of modules; it is not asserted to be multiplicative. In the basis $1, t, \dots, t^{n-1}$, multiplication by t^k has no diagonal entries for $1 \leq k < n$, since it shifts every basis index by the nonzero class $k \pmod{n}$. Its wrap entries carry the original minus signs. Multiplication by $b_0 \in E_Z$ is the scalar matrix with n diagonal entries b_0 . Consequently

$$\text{Tr}_{B_n/E_Z} \left(M_{\sum_{i=0}^{n-1} b_i t^i} \right) = n b_0, \quad r_n \left(\sum_{i=0}^{n-1} b_i t^i \right) = b_0, \quad r_n \iota_n = 1_{E_Z}. \quad (36)$$

In particular the trace factor n and its inverse in the retraction are both retained. No arithmetic trace form has been rescaled.

For a linear map $T : M \rightarrow N$ over E_Z , its split lift is $T^\tau(m^\bullet) = T(m)^\bullet$ and $T^\tau(\tau) = \tau$. It is $G(E_Z)$ -linear: addition of supported inputs is the original module addition, scalar multiplication is the original module action, and the cases containing an absent scalar or input are absorbed by τ . A zero result is $(0_N)^\bullet$, not τ . Thus (36) lifts to an actual retract

$$E_Z^\tau \xrightarrow{\iota_n^\tau} G(B_n) \xrightarrow{r_n^\tau} E_Z^\tau, \quad r_n^\tau \iota_n^\tau = 1. \quad (37)$$

Through (35), the first map is the fully supported mixed vector $(f^\bullet, e, \dots, e)$ for supported f , and the wholly absent vector for τ . Therefore every nonzero amplitude packet class remains nonzero through this specific mixed-support realization and localization.

Write $\sigma_Z : E_Z \hookrightarrow Q$ and $J_Z : Q \rightarrow E_Z$ for the original arithmetic packet injection and observation, whose proved identity is $J_Z \sigma_Z = 1_{E_Z}$. Their composite with the maps just computed is

$$\sigma_Z r_n : B_n \longrightarrow Q, \quad (\sigma_Z r_n) \iota_n = \sigma_Z.$$

This is a composite of the established maps, not a new assertion that the spectral coefficients are unrelated to arithmetic cohomology.

The scaling intertwiner used here comes from the original source. Retain $Q = \mathcal{B}/\Theta V$, $D = -x\partial_x$, and $\Theta\phi(x) = \sum_{j \in \mathbb{Z} \setminus \{0\}} \phi(jx)$. On both original test spaces define

$$(\mathcal{R}_a F)(x) = F(x/a), \quad a > 0.$$

Positive dilation preserves even Schwartz functions and their two zero moments: the value at zero is unchanged and the integral is multiplied by a . Direct substitution in the theta sum proves $\Theta \mathcal{R}_a = \mathcal{R}_a \Theta$. Thus $\mathcal{R}_a$ induces the original action $\bar{\mathcal{R}}_a$ on Q . Substitution $x = ay$ in the Mellin integral gives

$$\mathcal{M}(\mathcal{R}_a F)(s) = a^s \mathcal{M}F(s), \quad J_Z \bar{\mathcal{R}}_a = A_a J_Z.$$

The source's complete finite-packet theorem identifies $\sigma_Z(E_Z) = Q[h_Z(D)]$ and makes J_Z its inverse there. Since $\mathcal{R}_a D = D \mathcal{R}_a$, that subspace is invariant. Applying the bijection J_Z to its two vectors therefore proves

$$\bar{\mathcal{R}}_a \sigma_Z = \sigma_Z A_a.$$

This carries the full factor a^ρ and every Taylor coefficient in (34) from the actual theta quotient.

158.3 The exact coefficient Frobenius comparison

Let q be an odd prime power with $\gcd(q, n) = 1$. On the integral coefficient algebra

$$C_{n, \mathbb{Z}} = \mathbb{Z}[t]/(t^n + 1)$$

define the unital algebra map

$$\Psi_q(t) = t^q, \quad \Psi_q(c) = c \quad (c \in \mathbb{Z}).$$

It is well-defined since $(t^q)^n = (-1)^q = -1$. Because $\gcd(q, 2n) = 1$, choose $r > 0$ with $q^r \equiv 1 \pmod{2n}$. The identity $t^{2n} = 1$ then gives $\Psi_q^r = 1$, proving it is an automorphism. Its matrix in the original coefficient basis is the signed permutation

$$t^i \longmapsto (-1)^{\lfloor qi/n \rfloor} t^{qi \bmod n}. \quad (38)$$

Reduction of this integral map to $\mathbb{F}_q$ is exactly the source's q -power Frobenius on $\mathbb{F}_q[t]/(t^n + 1)$, because every scalar of $\mathbb{F}_q$ has q th power equal to itself. Its extension to $\mathbb{C}$ is a coefficient-linear lift

of that integral matrix. It is not the map $c \mapsto c^q$ on complex scalars. These two specializations are linked by the displayed integral map; no ring homomorphism $\mathbb{F}_q \rightarrow \mathbb{C}$ is asserted.

There is also a map on the full unsynchronized mixed-support object: send coefficient b_i to $((-1)^\bullet)^{\lfloor qi/n \rfloor} b_i$ in slot $qi \bmod n$, fixing E_Z scalars and preserving absent inputs. It is additive. On two supported monomials, reducing the exponent before or after the map gives the same wrap factor because $((-1)^\bullet)^q = (-1)^\bullet$; the absent cases are absorbed, and distributivity proves the product identity for arbitrary vectors. Hence this is a semiring morphism, fixes E_n , preserves every mask up to the stated permutation, and induces Ψ_q through (35).

Extend Ψ_q E_Z -linearly to B_n . It fixes the subspace $\iota_n(E_Z)$ pointwise. It preserves r_n : formula (38) moves no index $i \in \{1, \dots, n-1\}$ to zero modulo n , and fixes index zero. Hence

$$\Psi_q \iota_n = \iota_n, \quad r_n \Psi_q = r_n. \quad (39)$$

Every eigenvalue of the complex coefficient operator Ψ_q is a root of unity, because $\Psi_q^r = 1$. Over $\mathbb{C}$ the polynomial $X^r - 1$ has no repeated roots, so the operator is diagonalizable. This is the exact finite coefficient purity calculation transported through its integral matrix.

158.4 The original arithmetic action, full jets, and modulus

Extend A_a to B_n by coefficient multiplication:

$$A_a^B \left(\sum b_i t^i \right) = \sum A_a(b_i) t^i.$$

Since A_a is multiplication by $j_{h_Z}(a^s) \in E_Z$, this commutes with the E_Z -linear automorphism Ψ_q . The maps in (36) intertwine the arithmetic action exactly:

$$A_a^B \iota_n = \iota_n A_a, \quad r_n A_a^B = A_a r_n.$$

Consequently they also intertwine A_a with $A_a^B \Psi_q$ because of (39). On the coefficient eigenspace of Ψ_q with eigenvalue ω , the original ρ -packet action is

$$\omega a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (40)$$

The entire nilpotent polynomial is retained. For $a \neq 1$, its nilpotent part has the same nilpotence index as N_ρ :

$$\exp((\log a) N_\rho) - 1 = N_\rho \left((\log a) I + \sum_{j=2}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^{j-1} \right),$$

and the factor in parentheses is invertible because its constant coefficient is nonzero. The expression also states the case $m_\rho = 1$ directly.

Let e_ρ be the full Chinese-remainder idempotent in E_Z . The vector

$$v_\rho = e_\rho(x - \rho)^{m_\rho-1}$$

is nonzero: in its ρ -local quotient its coefficient of $z_\rho^{m_\rho-1}$ is one. It is killed by N_ρ and satisfies $A_a v_\rho = a^\rho v_\rho$. The vector $\iota_n(v_\rho)$ is nonzero by the retraction, lies in the fixed coefficient line of Ψ_q , and is therefore an eigenvector of $A_a^B \Psi_q$ with the unchanged eigenvalue a^ρ .

Suppose now the actual counterfactual supplies $\rho = \frac{1}{2} + \delta + i\gamma$, $0 < \delta < \frac{1}{2}$. For every $a > 1$,

$$\frac{|a^\rho|^2}{a} = a^{2\delta} > 1, \quad \log \frac{|a^\rho|^2}{a} = 2\delta \log a. \quad (41)$$

For the other coefficient eigenspaces in (40), $|\omega| = 1$ gives exactly the same modulus. At $a = q$, these eigenvalues are $q^\rho \omega$. The coefficient pure spectrum therefore preserves the actual arithmetic weight defect in this specified construction.

Take every tensor power in this paragraph over $\mathbb{C}$. The vector $\iota_n(v_\rho)^{\otimes_{\mathbb{C}} k}$ is nonzero because $(r_n)^{\otimes_{\mathbb{C}} k}$ sends it to $v_\rho^{\otimes_{\mathbb{C}} k}$. The latter is nonzero: a complex linear functional taking v_ρ to 1 has k -fold complex tensor power taking that tensor to 1. Its eigenvalue under $(A_a^B \Psi_q)^{\otimes_{\mathbb{C}} k}$ is $a^{k\rho}$, and

$$\log \frac{|a^{k\rho}|^2}{a^k} = 2k\delta \log a.$$

This is an exact input to the user's selected Deligne amplification argument. The computation does not import a cohomological upper estimate from another object. It identifies which actual nonzero class and which unchanged eigenvalue such an estimate must control.

158.5 The trace obstruction also survives the pure coefficient factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$ and under conjugation, with the original multiplicities. Retain the actual derivative-residue trace contraction

$$\mathcal{T}_Z(f, u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho).$$

This formula is the contraction of the original residue pairing using g' ; the factors in $g = 2\xi$ are not modified here. In a genuine off-line quartet let m be its common multiplicity and set $f = e_\rho - e_{1-\bar{\rho}}$. The two nonzero evaluations contribute $-m$ each, so $\mathcal{T}_Z(f, f) = -2m$.

On $C_n = \mathbb{C}[t]/(t^n + 1)$ retain the positive form

$$H_n(c, d) = \sum_{j=0}^{n-1} \overline{c(r_j)} d(r_j), \quad r_j = \exp\left(\frac{(2j+1)\pi i}{n}\right).$$

The distinct roots make evaluation an isomorphism $C_n \simeq \mathbb{C}^n$, proving positive definiteness. The operator Ψ_q permutes the evaluations, so it preserves H_n . The original constant vector has $H_n(1, 1) = n$. Under $B_n \simeq E_Z \otimes_{\mathbb{C}} C_n$ the tensor form therefore satisfies

$$(\mathcal{T}_Z \otimes_{\mathbb{C}} H_n)(\iota_n f, \iota_n f) = -2mn < 0.$$

This proves the actual negative trace vector survives this exact positive coefficient attachment, with the factor n retained. It does not assert that the global arithmetic polarization is automatically this tensor form. Any different global comparison has its own map and must be calculated on this surviving input.

159 The multiplicity boundary floor

This section proves an exact lower bound on the relative control defect of the theta-generated representatives. The bound counts every retained zero with its complete multiplicity. The source constructions are A1644, lines 2576–2718, and A1771, lines 4477–4835, in the authoritative `audit_segment_U0041_U0054.md`. The rank-two identity itself is A1771, lines 4688–4725. The invariant trace and the invariant-subspace argument below are new deductions from those source constructions.

All matrix coordinates on the packet algebra are the original polynomial coordinates. The generator is $D = -x\partial_x$, the arithmetic multiplier is $g = 2\xi$, and the Hilbert-space measure is dx on the positive half-line. The local extension unit, the unscaled orthogonal vectors, their norm squares, and all nilpotent jets are retained.

159.1 The original source, packet algebra, and arithmetic section

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(s) = 2\xi(s),$$

with the entire continuations understood at the apparent singularities. Let

$$\begin{aligned} V &= \left\{ \phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0 \right\}, \\ \mathcal{B} &= \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0 \right\}, \\ D &= -x\partial_x, \quad (\Theta\phi)(x) = \sum_{a \neq 0} \phi(ax) = 2 \sum_{a \geq 1} \phi(ax). \end{aligned} \quad (\text{BF.01})$$

Here $\mathcal{S}(\mathbb{R})$ is the Schwartz space, and the summation indices a in Θ are integers. The symbol $\mathcal{B}$ denotes exactly the strong-Schwartz target specified in A1694, lines 3049–3073. Put

$$Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H}_+ = L^2(\mathbb{R}_{>0}, dx).$$

The scalar product convention is

$$\langle F, H \rangle_+ = \int_0^\infty \overline{F(x)} H(x) dx,$$

antilinear in the first argument. Adjoint signs on maps into $\mathcal{H}_+$ use this scalar product and the standard scalar product in the stated polynomial coordinates on their finite-dimensional domains.

Retain the actual source function and its derivative sequence:

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \\ f_0 &= \Theta\phi_*, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta\phi_j = D^j f_0 \quad (j \geq 0), \\ (\mathcal{M}F)(s) &= \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \mathcal{M}f_0 = g. \end{aligned} \quad (\text{BF.02})$$

The source identity $\mathcal{M}(D^j f_0)(s) = s^j g(s)$, with these exact constants, is A1644, lines 2566–2574, and A1771, lines 4479–4529. The commutation $D\Theta\phi = \Theta D\phi$ uses the same generator on the source and target. Differentiation of the defining series is justified on each compact subinterval of $(0, \infty)$ by the Schwartz estimates.

Let Z be a nonempty finite set of actual zeros of g . For $\rho \in Z$, let $m_\rho \geq 1$ be its complete order of vanishing. Define

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z = \sum_{\rho \in Z} m_\rho, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t : E_Z \longrightarrow E_Z. \quad (\text{BF.03})$$

The ordered coordinates on E_Z are the classes of $1, t, \dots, t^{d-1}$. Multiplication by t is not replaced by a shifted generator. The empty packet is treated separately at the end.

For a function holomorphic near every member of Z , j_{h_Z} means its complete collection of jets of orders $0, \dots, m_\rho - 1$, identified with E_Z by the Chinese remainder isomorphism. In particular, the finite Mellin-jet map is

$$J_Z : \mathcal{B} \longrightarrow E_Z, \quad J_Z F = j_{h_Z}(\mathcal{M}F).$$

In the local coordinate $z = s - \rho$, the ρ -component is

$$\sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z^j \pmod{z^{m_\rho}}.$$

The map retains the factorials and the full jet length.

The actual extension unit is

$$\varepsilon_Z = (j_{h_Z}(g/h_Z))^{-1} = j_{h_Z}(h_Z/g) \in E_Z^\times. \quad (\text{BF.04})$$

To see why it is a unit, write

$$g(\rho + z) = z^{m_\rho} w_\rho(z), \quad w_\rho(0) \neq 0.$$

The polynomial $h_Z(\rho + z)$ has precisely the same zero order. Thus g/h_Z and h_Z/g are mutually inverse holomorphic units near every selected zero. Equation (BF.04) keeps their complete jets. This finite-algebra unit is distinct from the supported-zero scalar e of the original split-support construction.

For $u \in E_Z$, let $R_Z(u)$ be the unique polynomial of degree less than d representing $\varepsilon_Z u$, and define the extension row

$$\ell_Z : E_Z \longrightarrow \mathbb{C}, \quad \ell_Z(u) = [t^{d-1}]R_Z(u). \quad (\text{BF.05})$$

The established arithmetic section in A1644, lines 2594–2718, and A1771, lines 4499–4510, is the map

$$s_Z : E_Z \longrightarrow \mathcal{B}$$

with the identities

$$J_Z s_Z = I_{E_Z}, \quad J_Z \Theta = 0, \quad q s_Z = \sigma_Z : E_Z \hookrightarrow Q, \quad D s_Z - s_Z A = f_0 \ell_Z. \quad (\text{BF.06})$$

These are the inherited source identities used below, with the actual source section and its analytic construction. For a one-point packet at a simple zero, their coefficient is exactly

$$\ell_Z(1) = \frac{1}{g'(\rho)} = \frac{1}{2\xi'(\rho)}.$$

No replacement of this coefficient by a unit-free scalar is made.

159.2 Theta relation spaces and the exact rank-two boundary

Define

$$V_n = \text{span}_{\mathbb{C}}\{\phi_0, \dots, \phi_n\}, \quad L_n = \Theta V_n = \text{span}_{\mathbb{C}}\{f_0, \dots, f_n\}, \quad n \geq 0. \quad (\text{BF.07})$$

The vectors f_j are linearly independent. Indeed, a finite dependence has Mellin transform $g(s)P(s) = 0$, where $P(s) = \sum_j c_j s^j$. Since g is not identically zero, it is nonzero on a nonempty open set. Therefore P vanishes on that open set and is the zero polynomial. Every coefficient c_j is zero.

Let $F_n : \mathbb{C}^{n+1} \rightarrow \mathcal{H}_+$ have columns $f_0, \dots, f_n$, and set

$$H_n = F_n^* F_n, \quad P_n = F_n H_n^{-1} F_n^*, \quad R_n = (I - P_n) s_Z, \quad G_n = R_n^* R_n. \quad (\text{BF.08})$$

Independence gives $H_n \succ 0$. Direct multiplication gives $P_n^2 = P_n = P_n^*$, and its range is L_n ; it is the actual orthogonal projection onto the specified finite theta relation space. The expression

$$b_n(u) = \sum_{j=0}^n (H_n^{-1} F_n^* s_Z u)_j \phi_j$$

defines a map $b_n : E_Z \rightarrow V_n$ satisfying $P_n s_Z = \Theta b_n$. Consequently, by (BF.06),

$$q R_n = \sigma_Z, \quad J_Z R_n = I_{E_Z}, \quad G_n = s_Z^* s_Z - s_Z^* F_n H_n^{-1} F_n^* s_Z \succ 0. \quad (\text{BF.09})$$

For the strict inequality, $v^* G_n v = \|R_n v\|_+^2 = 0$ implies $R_n v = 0$ and hence $v = J_Z R_n v = 0$. Thus all inverses of G_n used below are inverses of strictly positive matrices at the stated finite level.

The original theta boundary and the associated control matrix are

$$\begin{aligned} B_n &= DR_n - R_n A = \Theta K_n, \\ K_n &= \phi_* \ell_Z - D_V b_n + b_n A, \\ W_n &= A^* G_n + G_n A - G_n. \end{aligned} \tag{BF.10}$$

Here D_V is D on V . The boundary formula follows by substituting $R_n = s_Z - \Theta b_n$ into (BF.06). It uses only original theta boundaries.

For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle_+ + \langle F, DH \rangle_+ = - \left[x \overline{F(x)} H(x) \right]_0^\infty + \langle F, H \rangle_+ = \langle F, H \rangle_+. \tag{BF.11}$$

The defining estimates of $\mathcal{B}$ make the boundary expression zero at both endpoints. Applying (BF.11) to columns of R_n , and substituting $DR_n = R_n A + B_n$, proves

$$W_n = -(R_n^* B_n + B_n^* R_n). \tag{BF.12}$$

The coefficient of G_n in (BF.10) is exactly one because the measure in (BF.11) is dx . Even extension to the full line has squared norm twice this norm; it is not silently substituted here.

Construct the unscaled orthogonal vectors

$$u_0 = f_0, \quad u_n = (I - P_{n-1})f_n \quad (n \geq 1), \quad \mathfrak{h}_n = \langle u_n, u_n \rangle_+ > 0, \quad r_n = \mathfrak{h}_n^{-1} u_n^* s_Z : E_Z \rightarrow \mathbb{C}. \tag{BF.13}$$

The operator $u_n^* : \mathcal{H}_+ \rightarrow \mathbb{C}$ sends F to $\langle u_n, F \rangle_+$. Each u_n has leading coefficient one on f_n ; none is divided by its norm. Since the f_j are real functions, their Gram matrices are real, and each u_n is real.

Degree and orthogonality determine the recurrence

$$\begin{aligned} Du_0 &= u_1 + \frac{1}{2}u_0, \\ Du_n &= u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1). \end{aligned} \tag{BF.14}$$

Here is the complete coefficient calculation. As $u_n = f_n + \sum_{j < n} c_j f_j$, Du_n belongs to L_{n+1} and has coefficient one on u_{n+1} . For $j \leq n-2$, (BF.11) gives

$$\langle u_j, Du_n \rangle_+ = \langle u_j, u_n \rangle_+ - \langle Du_j, u_n \rangle_+ = 0,$$

because $Du_j \in L_{j+1} \subseteq L_{n-1}$. Reality and (BF.11) give

$$2\langle u_n, Du_n \rangle_+ = \mathfrak{h}_n,$$

so the full diagonal coefficient is $1/2$, not merely its real part. For $n \geq 1$,

$$\langle u_{n-1}, Du_n \rangle_+ = -\langle Du_{n-1}, u_n \rangle_+ = -\mathfrak{h}_n.$$

Dividing only this coefficient by the corresponding original norm square $\mathfrak{h}_{n-1}$ yields the last term of (BF.14).

The section coefficients satisfy

$$P_n s_Z = \sum_{j=0}^n u_j r_j, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} r_{n+1} = r_n \left(\frac{1}{2}I - A \right) + r_{n-1} - \mathbf{1}_{\{n=0\}} \ell_Z, \quad r_{-1} = 0. \tag{BF.15}$$

The indicator $\mathbf{1}_{\{n=0\}}$ equals one for $n = 0$ and zero otherwise. To check the row equation, pair $Ds_Z = s_Z A + u_0 \ell_Z$ against u_n . The right side is

$$\mathfrak{h}_n r_n A + \mathbf{1}_{\{n=0\}} \mathfrak{h}_0 \ell_Z.$$

By (BF.11) and (BF.14), the left side is

$$\frac{1}{2} \mathfrak{h}_n r_n - \mathfrak{h}_{n+1} r_{n+1} + \mathfrak{h}_n r_{n-1},$$

where the last term is zero for $n = 0$. Rearrangement gives (BF.15). The initial extension functional is therefore fixed by the original arithmetic section.

We now obtain the source's rank-two formula with its exact sign. Since

$$B_n = f_0 \ell_Z - DP_n s_Z + P_n s_Z A,$$

the first and third terms have image in L_n , orthogonal to the image of R_n . Equation (BF.12) becomes

$$W_n = R_n^* DP_n s_Z + (DP_n s_Z)^* R_n.$$

Equations (BF.14) and (BF.15) give

$$(I - P_n) DP_n s_Z = u_{n+1} r_n, \quad u_{n+1}^* R_n = u_{n+1}^* s_Z = \mathfrak{h}_{n+1} r_{n+1}.$$

It follows that

$$W_n = \mathfrak{h}_{n+1} (r_{n+1}^* r_n + r_n^* r_{n+1}). \quad (\text{BF.16})$$

Also $R_{n+1} = R_n - u_{n+1} r_{n+1}$, with R_{n+1} orthogonal to u_{n+1} . Taking Gram matrices proves the exact decrement

$$G_n - G_{n+1} = \mathfrak{h}_{n+1} r_{n+1}^* r_{n+1} \succeq 0. \quad (\text{BF.17})$$

These formulas are A1771, lines 4688–4734. The rank bound in (BF.16) concerns the boundary control form on E_Z ; the dimension d , and the full operator A , have not been reduced to two dimensions.

For clarity, the retained source-support transition is also exact. Write $Q_{V_n} = \mathcal{B}/\Theta V_n = \mathcal{B}/L_n$. Since $D\Theta V_n \subseteq \Theta V_{n+1}$, differentiation gives a well-defined map

$$\overline{D}_n : Q_{V_n} \longrightarrow Q_{V_{n+1}}, \quad [F] \longmapsto [DF].$$

Before that transition, the boundary class is

$$[B_n v]_{Q_{V_n}} = -[u_{n+1}]_{Q_{V_n}} r_n(v).$$

The layer map

$$\Theta V_{n+1}/\Theta V_n \longrightarrow L_{n+1} \cap L_n^\perp, \quad [z] \longmapsto (I - P_n)z$$

is well-defined and injective because its kernel in L_{n+1} is L_n . It is surjective because every vector in the displayed orthogonal complement is fixed by $I - P_n$. The subsequent support transport makes this particular class the supported zero in $Q_{V_{n+1}}$. Its earlier pairing in (BF.16) is retained. These are the actual maps of A1771, lines 4736–4773.

159.3 Relative control spectrum and its invariant trace

For $G_n \succ 0$, define its least two-sided control excess by

$$\epsilon_n = \inf\{\epsilon \geq 0 : -\epsilon G_n \preceq W_n \preceq \epsilon G_n\}. \quad (\text{BF.18})$$

Here $X \preceq Y$ means that $Y - X$ is a positive semidefinite Hermitian matrix. The operator $G_n^{-1}W_n$ is self-adjoint in the positive inner product $\langle u, v \rangle_{G_n} = u^* G_n v$:

$$\langle u, G_n^{-1}W_n v \rangle_{G_n} = u^* W_n v = \langle G_n^{-1}W_n u, v \rangle_{G_n}.$$

Thus ϵ_n is the largest absolute value of its eigenvalues. This uses the actual metric and operator and makes no assumption that A is normal for this metric.

In the unchanged coordinates, set

$$\mathbf{a}_n = r_n G_n^{-1} r_n^*, \quad \mathbf{b}_n = r_{n+1} G_n^{-1} r_{n+1}^*, \quad \mathbf{c}_n = r_n G_n^{-1} r_{n+1}^*. \quad (\text{BF.19})$$

The exact maps

$$\begin{aligned} \mathcal{T}_n : E_Z &\longrightarrow \mathbb{C}^2, & \mathcal{T}_n v &= (r_n v, r_{n+1} v), \\ \mathcal{U}_n : \mathbb{C}^2 &\longrightarrow E_Z, \\ \mathcal{U}_n(a, b) &= \mathfrak{h}_{n+1} (G_n^{-1} r_{n+1}^* a + G_n^{-1} r_n^* b) \end{aligned} \quad (\text{BF.20})$$

satisfy

$$\mathcal{U}_n \mathcal{T}_n = G_n^{-1} W_n, \quad \mathcal{T}_n \mathcal{U}_n = \mathfrak{h}_{n+1} \begin{pmatrix} \mathbf{c}_n & \mathbf{a}_n \\ \mathbf{b}_n & \overline{\mathbf{c}_n} \end{pmatrix}.$$

For every nonzero eigenvalue, applying $\mathcal{T}_n$ and $\mathcal{U}_n$ gives mutually inverse eigenspace maps up to that nonzero scalar. Therefore the nonzero eigenvalues of the two composites agree. Cauchy-Schwarz for the positive form on row covectors induced by G_n^{-1} gives $\mathbf{a}_n \mathbf{b}_n \geq |\mathbf{c}_n|^2$. The two roots of the displayed 2-by-2 characteristic polynomial are consequently

$$\lambda_{n,\pm} = \mathfrak{h}_{n+1} \left(\operatorname{Re} \mathbf{c}_n \pm \sqrt{\mathbf{a}_n \mathbf{b}_n - (\operatorname{Im} \mathbf{c}_n)^2} \right), \quad \lambda_{n,+} \geq 0 \geq \lambda_{n,-}. \quad (\text{BF.21})$$

Zero roots are included in this notation even when the corresponding eigenspace does not occur in E_Z . There are no other nonzero eigenvalues by the factorization (BF.20). Thus

$$\epsilon_n = \mathfrak{h}_{n+1} \left(|\operatorname{Re} \mathbf{c}_n| + \sqrt{\mathbf{a}_n \mathbf{b}_n - (\operatorname{Im} \mathbf{c}_n)^2} \right). \quad (\text{BF.22})$$

This is the exact expression in A1771, lines 4775–4821. In particular, G_n^{-1} remains present in every relative coefficient.

The trace of this control operator is independent of the chosen positive representative metric. Cyclicity of finite-dimensional trace and (BF.10) give

$$\begin{aligned} \operatorname{tr}(G_n^{-1} W_n) &= \operatorname{tr}(G_n^{-1} A^* G_n) + \operatorname{tr} A - d \\ &= 2 \operatorname{Re} \operatorname{tr} A - d \\ &= \sum_{\rho \in Z} m_\rho (2 \operatorname{Re} \rho - 1). \end{aligned} \quad (\text{BF.23})$$

For the final equality, the Chinese remainder decomposition preserves the local coordinates $z = t - \rho$. On the length- m_ρ summand, multiplication by t is $\rho I + M_z$. In the ordered basis $1, z, \dots, z^{m_\rho-1}$, M_z has zero diagonal and trace zero. Thus its full trace contribution is $m_\rho \rho$. No generalized eigenvector has been deleted.

Taking the trace directly in (BF.16), and using (BF.19), yields the exact row constraint

$$\boxed{2 \mathfrak{h}_{n+1} \operatorname{Re} \mathbf{c}_n = \sum_{\rho \in Z} m_\rho (2 \operatorname{Re} \rho - 1).} \quad (\text{BF.24})$$

Define the reflection $\iota\rho = 1 - \bar{\rho}$. A reflection-stable packet means that $\iota Z = Z$ and that $m_{\iota\rho} = m_\rho$ at every member. For such a packet the contributions of every two-point orbit in (BF.23) cancel. At a fixed point of ι , $\operatorname{Re} \rho = 1/2$, so its contribution is zero. Since $\mathfrak{h}_{n+1} > 0$,

$$\begin{aligned} \operatorname{tr}(G_n^{-1}W_n) &= 0, & \operatorname{Re} \mathbf{c}_n &= 0, \\ \lambda_{n,+} &= \epsilon_n = -\lambda_{n,-}, \\ \epsilon_n &= \mathfrak{h}_{n+1} \sqrt{\mathbf{a}_n \mathbf{b}_n - (\operatorname{Im} \mathbf{c}_n)^2}. \end{aligned} \tag{BF.25}$$

If $W_n \neq 0$, the two nonzero eigenvalues are ϵ_n and $-\epsilon_n$, each once. If $\epsilon_n = 0$, the self-adjoint operator $G_n^{-1}W_n$ has only the zero eigenvalue and hence is zero. Trace cancellation alone does not set the matrix W_n to zero.

159.4 The invariant-subspace projection argument

The next estimates hold for an arbitrary actual packet Z ; reflection stability will be used only for their final combination. Define

$$Z_+ = \{\rho \in Z : \operatorname{Re} \rho > 1/2\}, \quad Z_- = \{\rho \in Z : \operatorname{Re} \rho < 1/2\}.$$

Let

$$E_+ = \bigoplus_{\rho \in Z_+} \mathbb{C}[z]/(z^{m_\rho}), \quad E_- = \bigoplus_{\rho \in Z_-} \mathbb{C}[z]/(z^{m_\rho}). \tag{BF.26}$$

The inclusions $I_\pm : E_\pm \rightarrow E_Z$ are the actual Chinese remainder inclusions, not coordinate projections chosen to make A normal. Explicitly, put $h_\rho = (t - \rho)^{m_\rho}$ and $H_\rho = h_Z/h_\rho$. Since H_ρ and h_ρ are coprime, choose $b_\rho(t)$ with $H_\rho b_\rho \equiv 1 \pmod{h_\rho}$, and let e_ρ be the class of $H_\rho b_\rho$ in E_Z . It has residue one at the ρ -summand and zero at all other summands. The maps in (BF.26) are

$$I_\pm((u_\rho)_\rho) = \sum_{\rho \in Z_\pm} e_\rho u_\rho(t - \rho).$$

Changing the representative b_ρ changes nothing in E_Z . All original polynomial coordinates and every full local length remain in these maps.

Let $A_\pm$ be multiplication by $\rho + z$ on each local summand of $E_\pm$. The intertwining identities are

$$AI_\pm = I_\pm A_\pm. \tag{BF.27}$$

In particular the images are invariant generalized eigenspaces, with their nilpotent parts unchanged.

Fix $n \geq 0$. For a nonzero $E_\pm$, define

$$G_\pm = I_\pm^* G_n I_\pm, \quad P_\pm = I_\pm G_\pm^{-1} I_\pm^* G_n. \tag{BF.28}$$

The notation $G_\pm, P_\pm$ in this subsection refers to this fixed value of n . Since $I_\pm$ is injective and $G_n \succ 0$, $G_\pm \succ 0$. Multiplication gives $P_\pm^2 = P_\pm$, $P_\pm I_\pm = I_\pm$, and

$$P_\pm^* G_n = G_n P_\pm.$$

Thus $P_\pm$ is the G_n -orthogonal projection onto $\operatorname{im} I_\pm$. When $E_\pm = \{0\}$, define $P_\pm = 0$ directly; no inverse of a nonpositive-dimensional Gram matrix is needed.

For nonzero $E_\pm$, the restricted defect is exact. By (BF.27),

$$I_\pm^* W_n I_\pm = A_\pm^* G_\pm + G_\pm A_\pm - G_\pm. \tag{BF.29}$$

For nonzero $E_\pm$, cyclicity of the trace of the rectangular composites, followed by (BF.29), gives

$$\begin{aligned}\mathrm{tr}(P_\pm G_n^{-1} W_n) &= \mathrm{tr}(G_\pm^{-1} I_\pm^* W_n I_\pm) \\ &= 2 \operatorname{Re} \mathrm{tr} A_\pm - \dim E_\pm \\ &= 2 \sum_{\rho \in Z_\pm} m_\rho (\operatorname{Re} \rho - 1/2).\end{aligned}\tag{BF.30}$$

For $E_\pm = \{0\}$, the first and last expressions in (BF.30) are both zero and remain equal; the inverse-containing expression is not invoked. The last trace evaluation again uses $\rho I + M_z$ on each full local summand.

The rank-two factorization gives a stronger estimate for these compressed traces than a generic rank bound would give. Indeed, the Hermitian form $v^* W_n v$ is the pullback under $v \mapsto (r_n v, r_{n+1} v)$ of

$$\mathfrak{h}_{n+1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.\tag{BF.31}$$

This form has one positive and one negative direction. To verify the inherited inertia bound directly, a positive definite subspace for W_n must map injectively under $v \mapsto (r_n v, r_{n+1} v)$; otherwise its kernel would contain a nonzero vector on which the form is zero. Its image is a positive definite subspace of (BF.31), so has dimension at most one. The same reasoning applies to a negative definite subspace. Therefore $G_n^{-1} W_n$, self-adjoint in the G_n metric, has at most one strictly positive and at most one strictly negative eigenvalue.

Use $\lambda_{n,+}$ and $\lambda_{n,-}$ from (BF.21). Let Π_+ and Π_- be their G_n -orthogonal spectral projections when the corresponding eigenvalue is strictly positive or strictly negative. When a sign is absent, set both its eigenvalue and its projection equal to zero. The finite-dimensional spectral theorem gives

$$G_n^{-1} W_n = \lambda_{n,+} \Pi_+ + \lambda_{n,-} \Pi_-, \quad \operatorname{rank} \Pi_\pm \leq 1.\tag{BF.32}$$

All remaining spectral directions have eigenvalue zero.

Let P be any G_n -orthogonal projection. For either of the nonzero projections $\Pi_\pm$, $\Pi_\pm P \Pi_\pm$ acts on a one-dimensional subspace as a positive contraction. Positivity follows because

$$\langle x, \Pi_\pm P \Pi_\pm x \rangle_{G_n} = \langle P \Pi_\pm x, P \Pi_\pm x \rangle_{G_n} \geq 0,$$

and the upper bound follows from $\|P \Pi_\pm x\|_{G_n} \leq \|\Pi_\pm x\|_{G_n}$. Its eigenvalue belongs to $[0, 1]$. Cyclicity and $\Pi_\pm^2 = \Pi_\pm$ then prove

$$0 \leq \operatorname{tr}(P \Pi_\pm) = \operatorname{tr}(\Pi_\pm P \Pi_\pm) \leq 1.\tag{BF.33}$$

The equality and both inequalities also hold when $\Pi_\pm = 0$. Combining (BF.32) and (BF.33) gives

$$\lambda_{n,-} \leq \operatorname{tr}(P G_n^{-1} W_n) \leq \lambda_{n,+}.\tag{BF.34}$$

For example the upper bound is $\lambda_{n,+} \operatorname{tr}(P \Pi_+) + \lambda_{n,-} \operatorname{tr}(P \Pi_-) \leq \lambda_{n,+}$; the lower bound follows with the two signs interchanged.

Apply (BF.34) to the actual projections P_+ and P_- in (BF.30). For every packet and every $n \geq 0$, this proves

$$\boxed{\begin{aligned}\lambda_{n,+} &\geq 2 \sum_{\rho \in Z_+} m_\rho (\operatorname{Re} \rho - 1/2), \\ -\lambda_{n,-} &\geq 2 \sum_{\rho \in Z_-} m_\rho (1/2 - \operatorname{Re} \rho).\end{aligned}}\tag{BF.35}$$

These estimates do not require reflection stability or semisimplicity.

For a reflection-stable packet, reflection bijects Z_+ and Z_- , preserves multiplicity, and negates $\operatorname{Re} \rho - 1/2$. Thus the two positive distance sums in (BF.35) are equal. Equation (BF.25) identifies the

two control eigenvalues with ϵ_n and $-\epsilon_n$. Their combination is the claimed multiplicity boundary floor:

$$\begin{aligned} \epsilon_n &= \mathfrak{h}_{n+1} \sqrt{\mathfrak{a}_n \mathfrak{b}_n - (\operatorname{Im} c_n)^2} \\ &\geq \sum_{\rho \in Z} m_\rho \left| \operatorname{Re} \rho - \frac{1}{2} \right| \quad (n \geq 0). \end{aligned} \quad (\text{BF.36})$$

The restriction traces in (BF.30) show explicitly why complete multiplicities occur. The proof includes every generalized eigenspace and uses its induced metric. No normality of A , deletion of nilpotent jets, or positivity of the arithmetic Weil form has entered.

159.5 Exact consequences for a putative off-line zero

An individual eigenvector gives a useful comparison. If $0 \neq v \in E_Z$ satisfies $Av = \rho v$, then directly from (BF.10),

$$v^* W_n v = (2 \operatorname{Re} \rho - 1) v^* G_n v. \quad (\text{BF.37})$$

Such a vector exists in every local summand: the class $z^{m_\rho-1}$, embedded by e_ρ , is nonzero and is annihilated by $A - \rho I$. By (BF.18), (BF.37) gives $\epsilon_n \geq |2 \operatorname{Re} \rho - 1|$. Equation (BF.36) includes the additional constraints from all other roots and their multiplicities.

For a reflected off-line pair $\{\rho, 1 - \bar{\rho}\}$, with complete multiplicity m at both points, put $\delta = \operatorname{Re} \rho - 1/2 \neq 0$. The two points are distinct, and their contributions to the sum in (BF.36) give

$$\epsilon_n \geq 2m|\delta| = m|2 \operatorname{Re} \rho - 1| \quad (n \geq 0). \quad (\text{BF.38})$$

The real structure and functional equation of the retained xi-function are $g(\bar{s}) = \overline{g(s)}$ and $g(1-s) = g(s)$. Composition of a local Taylor expansion with conjugation or the affine map $s \mapsto 1-s$ preserves its first nonzero degree, so these symmetries preserve the complete vanishing multiplicities. Consequently, when the four points $\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}$ are distinct, they form a reflection-stable packet with the same complete multiplicity m . Their exact bound is

$$\epsilon_n \geq 4m|\delta| = 2m|2 \operatorname{Re} \rho - 1| \quad (n \geq 0). \quad (\text{BF.39})$$

For a collapsed symmetry orbit, only the actual distinct points with their actual multiplicities are counted. Formula (BF.38) already treats the two-point reflection orbit.

In particular, a reflection-stable actual packet containing an off-line root forces a strictly positive right side of (BF.36) at every finite level. Then W_n is nonzero, indefinite, and of rank exactly two, with generalized eigenvalues $\epsilon_n, -\epsilon_n$. The strict positivity follows from the packet sum; both signs and exact rank follow from (BF.25) and (BF.16). For an arbitrary packet without reflection stability, the valid statements are instead the separate inequalities (BF.35); its trace need not vanish.

This lower bound is on the relative expression containing G_n^{-1} . The absolute monotonicity (BF.17) does not supply an upper bound on that inverse-weighted expression. The homotopy freedom in A1771, lines 4432–4475, does not alter this conclusion. Retain the Fourier transform

$$(\mathcal{F}\phi)(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

It restricts to an isomorphism $V \rightarrow V$: it preserves evenness and the Schwartz space, and interchanges evaluation at zero with the integral, so preserves the two zero moments in (BF.01). Its inverse on V is the inverse Fourier transform with the same convention. For the two-leg differential and its cycle map,

$$\begin{aligned} d_{\text{leg}} : V \oplus V &\longrightarrow \mathcal{B}, & d_{\text{leg}}(v, w) &= \Theta(v - \mathcal{F}w), \\ \gamma : V &\longrightarrow V \oplus V, & \gamma(v) &= (v, \mathcal{F}^{-1}v), \end{aligned}$$

one has $d_{\text{leg}}\gamma = 0$ by direct substitution. For a linear map $L : E_Z \rightarrow V$, adding γL to a primitive therefore leaves B_n unchanged, and hence leaves W_n unchanged for fixed R_n . The support transition described above likewise records where the boundary becomes a supported zero while retaining the pairing that produced (BF.16).

There is also an exact tensor obstruction independent of the rank-two argument. For an integer $k \geq 1$, keep

$$A_k = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(k-j)} \quad \text{on } E_Z^{\otimes k}. \quad (\text{BF.40})$$

For any positive definite metric H on this same tensor space, set $W_{k,H} = A_k^* H + H A_k - kH$. The nonzero tensor $v^{\otimes k}$ of the eigenvector above satisfies $A_k v^{\otimes k} = k \rho v^{\otimes k}$. Consequently

$$\frac{(v^{\otimes k})^* W_{k,H} v^{\otimes k}}{(v^{\otimes k})^* H v^{\otimes k}} = k(2 \operatorname{Re} \rho - 1). \quad (\text{BF.41})$$

The denominator is strictly positive. Thus the least two-sided excess for $H, W_{k,H}$ is at least $k|2 \operatorname{Re} \rho - 1|$. In particular this applies to the actual orbit metric of A1721, lines 4102–4139,

$$H_{k,T} = \int_0^T e^{-kt} \exp(tA_k)^* G_n^{\otimes k} \exp(tA_k) dt, \quad T > 0.$$

It is positive definite because the integrand has positive quadratic form on every nonzero vector at every t ; $\exp(tA_k)$ is invertible. The same time parameter is used in every factor, and the full weight e^{-kt} remains. Changing T cannot change the Rayleigh value (BF.41).

For the empty packet, $h_Z = 1$, $E_Z = \{0\}$, and every packet sum is zero. Define its control excess to be zero. Then the boundary-floor assertion is the equality $0 = 0$. The extension row $[t^{d-1}]R_Z$ and the positive-dimensional Gram inverses above are not invoked in this empty case.

Equations (BF.35) and (BF.36) are lower bounds proved on the actual arithmetic objects. They neither assert an off-line zero nor provide the arithmetic upper estimate needed to exclude one. The source explicitly retains that missing estimate in A1721, line 4259, and A1771, line 4890. Finite regression checks, the stated integer-sum tail bound, and absolute Gram decrements do not supply it.

160 Arithmetic symmetric frontier cost and determinant growth

This section proves finite restrictions on the spectral positions of the original arithmetic packet. It retains the actual theta source, the measure $|2\xi/h|^2 dy/(2\pi)$, all zero multiplicities, and the unscaled orbit coordinates. No purity assertion is used. Stars denote conjugate transpose in the displayed finite coefficient bases, and Hilbert adjoint for maps into the source Hilbert space. For Hermitian matrices, $U \preceq V$ means that $V - U$ is positive semidefinite. The one-factor quotient coefficient basis is always $[1]_h, [t]_h, \dots, [t^{d-1}]_h$, with its specified tensor and orbit inclusions below.

Authoritative source pins. The retained construction is in the original messages A1829 and A1859 of `audit_segment_U0041_U0054.md`. The line numbers here are literal lines in that source, whose SHA-256 is recorded in the accompanying audit. A1829, lines 4956–5071, specifies the source, the arithmetic function, the full-jet map, and its exact polynomial kernel; lines 5073–5169 give the canonical minimum and its metric; lines 5383–5561 give the joint tensor source and actual next relation layer. A1859, lines 5693–5874, gives the unscaled polynomial coordinates, the relation isomorphism, and the exact metric update; lines 5876–6040 give the recurrence and finite lowering bound; lines 6042–6197 give the signed symmetric summand and its unscaled orbit factors. The aggregate inequalities proved below are new consequences of those retained objects.

160.1 The retained arithmetic source and its canonical finite metric

Keep the original operator, theta map, and arithmetic function

$$\begin{aligned} D &= -x\partial_x, & \Theta\phi(x) &= \sum_{m \neq 0} \phi(mx), \\ \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & f_0 &= \Theta\phi_*, \\ g(s) &= \mathcal{M}f_0(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \end{aligned} \quad (\text{FD.1})$$

Here $\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1}dx$, on its original source domain. Choose a nonempty finite packet of actual zeros and retain the full order m_ρ at every selected centre:

$$\begin{aligned} h(t) &= \prod_{\rho \in Z} (t - \rho)^{m_\rho}, & d &= \deg h, \\ E_h &= \mathbb{C}[t]/(h), & v_h &= [g/h]_h \in E_h^\times. \end{aligned} \quad (\text{FD.2})$$

The bracket $[\cdot]_h$ means the complete local Taylor jets modulo h ; it includes every nilpotent jet at a multiple zero. Since each selected order is the full order of that zero, g/h is nonzero at every selected centre, proving that v_h is a unit in the finite quotient. Let A_h be multiplication by $[t]_h$ on E_h .

The retained actual source inverse supplies F_h with

$$h(D)F_h = f_0, \quad \mathcal{M}F_h = g/h, \quad \mathcal{T}_h(P) = P(D)F_h. \quad (\text{FD.3})$$

Its exact identities, retained from the original source construction, are

$$\mathcal{M}\mathcal{T}_h(P) = (g/h)P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (\text{FD.4})$$

Here $J_h F = [\mathcal{M}F]_h$ is the full Mellin-jet map on the original analytic source. The second identity gives an actual theta primitive, and the last identity includes the arithmetic coefficient v_h . The map $\mathcal{T}_h$ is linear and injective: a zero image has identically zero Mellin transform $(g/h)P$, which forces $P = 0$.

The Hilbert observation remains $\mathcal{H} = L^2((0, \infty), dx)$. The exact change of variables and Fourier map

$$F \longmapsto (y \longmapsto e^{y/2}F(e^y)) \longmapsto \left(t \longmapsto \int_{\mathbb{R}} e^{y/2}F(e^y)e^{ity} dy \right)$$

have target measure $dt/(2\pi)$ and composite $F \mapsto \mathcal{M}F(1/2 + it)$. Consequently the arithmetic polynomial measure is exactly

$$d\nu_h(y) = \left| \frac{g(\frac{1}{2} + iy)}{h(\frac{1}{2} + iy)} \right|^2 \frac{dy}{2\pi}. \quad (\text{FD.5})$$

At selected zeros on the integration line, the quotient is evaluated as the entire function g/h . The finiteness of every polynomial moment can also be checked directly from the retained factors. If $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$, its alternating coefficient partial sums have absolute value at most one. Partial summation gives $|\eta(s)| \leq |s|/\Re s$ for $\Re s > 0$. The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$, first obtained by splitting the absolutely convergent series and then continued analytically, implies

$$|\zeta(\tfrac{1}{2} + iy)| \leq \frac{2|\frac{1}{2} + iy|}{\sqrt{2} - 1},$$

because $|1 - 2^{1/2 - iy}| \geq \sqrt{2} - 1$. Fix $0 < \theta < \pi/2$. Rotating the Euler gamma integral through angle $\text{sgn}(y)\theta$ gives

$$|\Gamma(\tfrac{1}{4} + iy/2)| \leq \Gamma(\tfrac{1}{4})(\cos \theta)^{-1/4} e^{-\theta|y|/2}.$$

For $y = 0$, the same bound follows from the unrotated integral. The rotation follows by integrating $z^{1/4+iy/2-1}e^{-z}$ around a truncated sector using its continuous logarithm there. The small circular arc tends to zero as its radius to the power $1/4$; the large arc tends to zero because $\cos \theta > 0$. Taking absolute values on the rotated ray yields the displayed bound. Retaining the factors in (FD.1) therefore gives

$$|g(\tfrac{1}{2} + iy)|^2 \leq \frac{4\pi^{-1/2}\Gamma(\frac{1}{4})^2}{(\sqrt{2}-1)^2\sqrt{\cos \theta}} (\tfrac{1}{4} + y^2)^3 e^{-\theta|y|}.$$

Monicity of the fixed degree- d polynomial h gives $|h(\frac{1}{2} + iy)| \geq c_h(1 + |y|)^d$ for some $c_h > 0$ outside a sufficiently large interval: divide its value by $(\frac{1}{2} + iy)^d$, whose quotient tends to one. On the remaining compact interval, the entire function g/h is bounded. These estimates prove finiteness of every moment of ν_h . The measure is positive away from a discrete set. Therefore the norm of a nonzero polynomial is strictly positive: zero norm would make the polynomial vanish almost everywhere on a line and hence identically.

Fix $k \geq 1$, and let

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, & A_k &= \sum_{j=1}^k A_{h,j}, & D^{(k)} &= \sum_{j=1}^k D_j, \\ \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, & \mathcal{T}^{(k)}(P) &= P(D_1, \dots, D_k) F_h^{\otimes k}, \\ \mathcal{J}(P) &= v_h^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \tag{FD.6}$$

All tensor generators in this section are additive sums, as displayed in (FD.6). The multiplicative symmetric power of a linear map is a different construction and is not used here.

For $M \geq k(d-1)$, the map $\mathcal{J}$ on $\mathcal{P}_M^{(k)}$ is onto: each reduced monomial has each coordinate degree below d , hence total degree at most $k(d-1)$, and multiplication by $v_h^{\otimes k}$ is invertible. Its kernel and its actual theta image are

$$\mathcal{I}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)} \mathcal{I}_M. \tag{FD.7}$$

Successive monic division in the variables expresses every polynomial in the kernel as $\sum_j h(t_j) P_j$. Under $\mathcal{T}^{(k)}$, each summand is an actual theta boundary in factor j , by (FD.4). Its cochain primitive has coefficient $(-1)^{j-1}$; the tensor differential has the same coefficient, so their product gives the displayed positive summand $+h(t_j) P_j$.

Let $R_M : E^{(k)} \rightarrow \mathcal{T}^{(k)} \mathcal{P}_M^{(k)}$ be the unique least-norm lift of the full jet, using the product measure induced by (FD.5), and write $G_M = R_M^* R_M$, $\mathcal{C}_M = G_M^{-1}$. The existence, uniqueness, and inverse are also obtained directly below. The minimum is orthogonal to the specified $\mathcal{B}_M$, and its full jet is identity. Its arithmetic class is that of the original product source lift with the specified jet, since all admissible changes are the actual boundaries in (FD.7).

Start with the monic, unscaled orthogonal polynomials and their actual norms

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathbb{C}[t]_{\leq n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \tag{FD.8}$$

For a multi-index $\alpha \in \mathbb{N}_0^k$, set

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_{j=1}^k \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}(p_\alpha). \tag{FD.9}$$

Let T_M have columns $\mathcal{T}^{(k)} p_\alpha$ for $|\alpha| \leq M$, let $J_M = [z_\alpha]_{|\alpha| \leq M}$, and let $\Omega_{\leq M} = \text{diag}_{|\alpha| \leq M} \omega_\alpha$. Product orthogonality gives $T_M^* T_M = \Omega_{\leq M}$. Surjectivity of J_M implies

$$\begin{aligned} \mathcal{C}_M &= J_M \Omega_{\leq M}^{-1} J_M^* = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha} > 0, \\ G_M &= \mathcal{C}_M^{-1}, \quad R_M = T_M \Omega_{\leq M}^{-1} J_M^* G_M. \end{aligned} \tag{FD.10}$$

Indeed $J_M \Omega_{\leq M}^{-1} J_M^* G_M = I$. Any other coefficient vector with jet u is $\Omega_{\leq M}^{-1} J_M^* G_M u + w$, where $J_M w = 0$. Its cross term with the first vector is $w^* J_M^* G_M u = 0$, so its squared norm is exactly $u^* G_M u + w^* \Omega_{\leq M} w$. This proves the unique minimum, its orthogonality to $\mathcal{B}_M$, and $R_M^* R_M = G_M$.

160.2 The actual next relation and its recurrence bound

For $|\beta| = M + 1$, define the frontier matrices

$$\Omega = \text{diag}_{|\beta|=M+1} \omega_\beta, \quad F = [z_\beta]_{|\beta|=M+1}, \quad T_+ = [\mathcal{T}^{(k)} p_\beta]_{|\beta|=M+1}. \quad (\text{FD.11})$$

There are $r = \binom{M+k}{k-1}$ columns. They are orthogonal to the entire degree- M source image. If $P_{\mathcal{B}_M}$ denotes orthogonal projection onto the actual boundary space, put

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M}) \mathcal{B}_{M+1}, \quad b_M = T_+ - R_M F : \mathbb{C}^r \longrightarrow \mathcal{E}_M. \quad (\text{FD.12})$$

The full jet of b_M is $F - F = 0$, and its two terms are orthogonal to $\mathcal{B}_M$, so its range is contained in $\mathcal{E}_M$. Its highest orthogonal polynomial component is T_+ , proving injectivity. Conversely, extract the highest component of a member of $\mathcal{E}_M$ and subtract its corresponding b_M -image. The remainder is simultaneously in $\mathcal{B}_M$ and orthogonal to $\mathcal{B}_M$, and is therefore zero. This proves that b_M is an isomorphism and constructs its inverse.

The exact Gram and cross-pairing are

$$H_M^{\text{rel}} := b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (\text{FD.13})$$

The superscript distinguishes this relation Gram from the Hermitian control matrix introduced later. The projection of R_M onto the new layer and the updated representative are

$$Y_M = -b_M (H_M^{\text{rel}})^{-1} F^* G_M, \quad R_{M+1} = R_M - Y_M. \quad (\text{FD.14})$$

These corrections are actual boundaries. In particular, the full jet and the original arithmetic class stay fixed. The quotient-layer map

$$\mathcal{B}_{M+1}/\mathcal{B}_M \longrightarrow \mathcal{E}_M, \quad [z] \longmapsto (1 - P_{\mathcal{B}_M})z$$

is an isomorphism, whose inverse is the inclusion followed by quotient. Its further map into the quotient by $\mathcal{B}_{M+1}$ is zero. Write λ_M for the specified original joint support label recording the relation space $\mathcal{B}_M$, and write 0 for the zero amplitude in that support fibre. On the original supported reconstruction this is $(\lambda_M, u) \mapsto (\lambda_{M+1}, 0)$. For $u \neq 0$ the preceding class is nonzero. The final class is the supported zero of that fibre; it is distinct from τ , the original external-absence element. The notation $e = 0^\bullet$ denotes the nonabsent represented scalar zero in the original scalar reconstruction. Thus the relation Gram in (FD.13) measures the actual relations becoming represented zero at the next support.

Orthogonal projection and the finite sum in (FD.10) give

$$G_{M+1} = G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \quad (\text{FD.15})$$

$$\mathcal{C}_{M+1} - \mathcal{C}_M = F \Omega^{-1} F^*.$$

The second equality also follows simply by adding the degree- $M + 1$ terms to (FD.10); it does not require a limiting argument.

Set $p_{-1} = 0$. Multiplication by t on the integration line has adjoint multiplication by $1 - t$. Orthogonality and monicity therefore give

$$tp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad a_n = \omega_n / \omega_{n-1} \ (n \geq 1), \quad a_0 = 0, \quad (\text{FD.16})$$

$$b_n = \frac{\int (\frac{1}{2} + iy) |p_n(\frac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, \quad \Re b_n = \frac{1}{2}.$$

For $n \geq 1$, pairing against a polynomial of degree below $n - 1$ in the expansion of tp_n vanishes by the adjoint identity. Pairing against p_{n-1} gives $-\omega_n/\omega_{n-1}$, and pairing against p_n gives the displayed b_n . For $n = 0$, monicity gives $tp_0 = p_1 + b_0p_0$, and the term $a_0p_{-1} = 0$ is exactly the declared zero term. This proves every coefficient in the recurrence. The minus sign and the possibly nonzero imaginary part of b_n are retained.

Define

$$w_\beta = \sum_{j:\beta_j>0} a_{\beta_j} z_{\beta-e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (\text{FD.17})$$

Here e_j is the standard multi-index vector; it is not the supported scalar $e = 0^\bullet$. The full arithmetic jet is a module map, so $\mathcal{J}((\sum_j t_j)P) = A_k \mathcal{J}(P)$. Inserting (FD.16) into (FD.10) proves

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*. \quad (\text{FD.18})$$

To verify the cancellation explicitly, for every internal edge $\alpha \rightarrow \alpha + e_j$, the raising term has coefficient $1/\omega_\alpha$, while the lowering term has coefficient $-a_{\alpha_j+1}/\omega_{\alpha+e_j} = -1/\omega_\alpha$. Their adjoints cancel as well. The diagonal contribution is $2 \sum_j \Re b_{\alpha_j} = k$. The only uncanceled edges have $|\alpha| = M$ and $|\alpha + e_j| = M + 1$, and their coefficients are exactly those in the right side of (FD.18).

For completeness, the arithmetic derivative defect has the same actual relation coordinates. Set

$$B_M^{\text{der}} = D^{(k)} R_M - R_M A_k, \quad C_M^{\text{rel}} = (1 - P_{\mathcal{B}_M}) B_M^{\text{der}}.$$

Its full jet is zero, and its highest polynomial component is $T_+ \Omega^{-1} E_+^* G_M$. The latter assertion follows from the coefficient of p_β in $(\sum_j t_j) R_M$: it is $\sum_{j:\beta_j>0} z_{\beta-e_j}^* G_M / \omega_{\beta-e_j} = w_\beta^* G_M / \omega_\beta$. The layer isomorphism (FD.12) therefore proves

$$\begin{aligned} C_M^{\text{rel}} &= b_M \Omega^{-1} E_+^* G_M, \\ W_M &:= A_k^* G_M + G_M A_k - k G_M \\ &= G_M \mathcal{Z}_M G_M \\ &= -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M). \end{aligned} \quad (\text{FD.19})$$

The last equality follows by substituting (FD.13) and (FD.14); in particular its minus sign is retained.

The recurrence constant is

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}) > 0. \quad (\text{FD.20})$$

Strict positivity follows from $a_{n+1} > 0$ for every $n \geq 0$. We now prove its complete lowering estimate.

For coefficient calculations only, use the exact coordinate map that divides the basis vector p_α by $\sqrt{\omega_\alpha}$. Its inverse multiplies by $\sqrt{\omega_\alpha}$; the original polynomials, their norms, and the measure are unchanged. The degree- (-1) coefficient space is by definition $\{0\}$; in particular every lowering map from degree zero has that zero codomain. This retains the admitted endpoint $d = 1$, $M = 0$. In these coefficient spaces let R_j be the raising map $e_\alpha \mapsto \sqrt{a_{\alpha_j+1}} e_{\alpha+e_j}$, and let R_j^* be its lowering adjoint. Put

$$Q_M^{\text{jet}} = [z_\alpha / \sqrt{\omega_\alpha}]_{|\alpha|=M}, \quad B^{\text{inc}} = (\sum_j R_j)^*|_{M+1 \rightarrow M}.$$

The retained coefficients in (FD.17) give exactly $E_+ \Omega^{-1/2} = Q_M^{\text{jet}} B^{\text{inc}}$. Raising and lowering in distinct coordinates commute. Hence, on the degree- M coefficient space,

$$B^{\text{inc}} (B^{\text{inc}})^* = \text{diag}_{|\alpha|=M} \sum_j (a_{\alpha_j+1} - a_{\alpha_j}) + L^* L, \quad L = \sum_j R_j^*|_{M \rightarrow M-1}.$$

For every coefficient vector c , Cauchy-Schwarz gives $\|Lc\|^2 \leq k \sum_j \|R_j^* c\|^2$. Thus

$$L^* L \preceq k \text{diag}_{|\alpha|=M} \sum_j a_{\alpha_j}, \quad B^{\text{inc}} (B^{\text{inc}})^* \preceq \Gamma_{k,M} I.$$

Multiplication by Q_M^{jet} and its adjoint, followed by inclusion of the positive degree- M sum in (FD.10), proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Q_M^{\text{jet}} (Q_M^{\text{jet}})^* \preceq \Gamma_{k,M} \mathcal{C}_M. \quad (\text{FD.21})$$

160.3 Exact symmetric restriction with every orbit factor

Let P_π be ordinary factor permutation. The relevant module is

$$E_k^{\text{sym}} = (E_h^{\otimes k})^{S_k}, \quad N = \dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}.$$

It is the actual top-degree summand selected on the original cochain tensor product by $S_k = (1/k!) \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, where T_π is permutation with its Koszul sign. In top degree $T_\pi = \text{sgn}(\pi) P_\pi$, so the two signs cancel. The numerator squares to $k!$ times itself, proving the idempotent with its literal coefficient $1/k!$. In particular, every repeated tensor $v^{\otimes k}$ remains in this summand.

Use the quotient basis formed by the unscaled orbit sums of $[t^{a_1}] \otimes \cdots \otimes [t^{a_k}]$, $0 \leq a_i < d$. Let I_k be its coefficient inclusion in the full tensor basis. If n_j counts factors with exponent j , its orbit size is $k! / \prod_{j=0}^{d-1} n_j!$. The orbit supports are disjoint, so

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_{j=0}^{d-1} n_j!} \right), \quad L_k = D_k^{-1} I_k^*, \quad L_k I_k = I. \quad (\text{FD.22})$$

The projector onto invariant coefficient vectors is $\Pi_k = I_k L_k$. Permutations preserve the source, the relation ideal, the product measure, and $v_h^{\otimes k}$. Uniqueness of the minimum therefore gives equivariance of R_M , and consequently of G_M and $\mathcal{C}_M$. The exact restricted matrices are

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad \mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (\text{FD.23})$$

The first two matrices are inverse. For example,

$$G_M^{\text{sym}} \mathcal{C}_M^{\text{sym}} = I_k^* G_M \Pi_k \mathcal{C}_M L_k^* = I_k^* \Pi_k L_k^* = I,$$

because both G_M and $\mathcal{C}_M$ commute with Π_k , their product is identity, $\Pi_k L_k^* = L_k^*$, and $I_k^* L_k^* = (L_k I_k)^* = I$.

In the source frontier let Q_+ be the coefficient inclusion matrix whose columns are the 0/1 orbit indicators in the ordered basis $(p_\beta)_{|\beta|=M+1}$. Thus each column represents the unscaled orbit sum of those p_β . If $m_j(\beta)$ counts the entries of β equal to j , the corresponding source orbit norm is exactly

$$\omega_{[\beta]} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!} \prod_{i=1}^k \omega_{\beta_i}. \quad (\text{FD.24})$$

This is a sum of equal squared norms of pairwise orthogonal distinct product polynomials. Define the actual restricted frontier matrices

$$\Omega_s = Q_+^* \Omega Q_+, \quad F_s = L_k F Q_+, \quad E_s = L_k E_+ Q_+. \quad (\text{FD.25})$$

Thus Ω_s is the positive diagonal matrix with entries (FD.24); none of its orbit factors are suppressed.

The source invariant projector is $\Pi_+ = Q_+ (Q_+^* Q_+)^{-1} Q_+^*$. Since Ω is constant on each orbit, it commutes with Π_+ , and direct calculation in one orbit gives

$$Q_+ \Omega_s^{-1} Q_+^* = \Omega^{-1} \Pi_+.$$

Indeed, on an orbit of size m and weight ω , the left side is the all-ones matrix divided by $m\omega$; this is precisely $\omega^{-1} \Pi_+$ there. Equivariance implies $L_k F = L_k F \Pi_+$ and $L_k E_+ = L_k E_+ \Pi_+$: averaging any

source vector before applying an equivariant map has the same invariant component as applying the map first. Congruence of (FD.15), (FD.18), and (FD.21) therefore gives, with

$$\begin{aligned} \mathcal{C} &= \mathcal{C}_M^{\text{sym}}, \quad G = G_M^{\text{sym}} = \mathcal{C}^{-1}, \quad A = A_k^{\text{sym}}, \\ \Delta\mathcal{C} &= \mathcal{C}_{M+1}^{\text{sym}} - \mathcal{C}_M^{\text{sym}}, \quad \mathcal{Z} = AC + CA^* - k\mathcal{C}, \end{aligned} \quad (\text{FD.26})$$

the exact formulas

$$\begin{aligned} \Delta\mathcal{C} &= F_s \Omega_s^{-1} F_s^*, \\ \mathcal{Z} &= F_s \Omega_s^{-1} E_s^* + E_s \Omega_s^{-1} F_s^*, \\ E_s \Omega_s^{-1} E_s^* &\preceq \Gamma_{k,M} \mathcal{C}. \end{aligned} \quad (\text{FD.27})$$

For the operator term, equivariance also gives $L_k A_k = A_k^{\text{sym}} L_k$, proving that the congruence of $A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k\mathcal{C}_M$ is exactly the $\mathcal{Z}$ in (FD.26).

160.4 The aggregate spectral-position restriction

Positive square roots below are the unique positive square roots of the retained positive matrices. They are used as explicit comparison maps; their inverses are retained, and neither the source measure nor the arithmetic metric is redefined. Set

$$\begin{aligned} X &= G^{1/2} F_s \Omega_s^{-1/2}, \quad Y = G^{1/2} E_s \Omega_s^{-1/2}, \\ K = XX^* &= G^{1/2} \Delta\mathcal{C} G^{1/2}, \quad H = G^{1/2} \mathcal{Z} G^{1/2} = XY^* + YX^*. \end{aligned} \quad (\text{FD.28})$$

The last inequality of (FD.27) gives $YY^* \preceq \Gamma_{k,M} I$, and hence $Y^*Y \preceq \Gamma_{k,M} I$ on its own coefficient space. Here the latter statement follows from $\|Y\|_{\text{op}}^2 = \|YY^*\|_{\text{op}}$, or directly from the singular values. For the Frobenius norm $\|T\|_F^2 = \text{tr}(T^*T)$,

$$\|XY^*\|_F^2 = \text{tr}(X^*X Y^*Y) \leq \Gamma_{k,M} \text{tr}(X^*X).$$

The norm of a matrix equals the norm of its adjoint, so the triangle inequality gives the proved upper estimate

$$\|H\|_F^2 \leq 4\Gamma_{k,M} \text{tr} K. \quad (\text{FD.29})$$

List the roots of h , repeated with their original algebraic multiplicities, as $\rho_1, \dots, \rho_d$, and put

$$\delta_i = \Re \rho_i - \frac{1}{2}, \quad \eta_{\mathbf{n}} = \sum_{i=1}^d n_i \rho_i, \quad \mathbf{n} \in \mathbb{N}_0^d, \quad |\mathbf{n}| = k. \quad (\text{FD.30})$$

The eigenvalues of A , with algebraic multiplicity, are exactly the $\eta_{\mathbf{n}}$. To prove this while retaining nilpotents, triangularize A_h by induction using an eigenvector and the induced operator on the quotient. In a basis with $A_h e_j = \rho_j e_j + \sum_{i < j} c_{ij} e_i$, applying the additive generator to a symmetric monomial of occupations $\mathbf{n}$ gives diagonal coefficient $\sum_j n_j \rho_j$. Every other term replaces an index j by $i < j$, reducing the sum of the indices. Ordering monomials by that sum, and resolving ties arbitrarily, makes the operator triangular with precisely the displayed diagonal. Each weak composition contributes once; equal eigenvalues remain repeated with their algebraic multiplicities. The triangular off-diagonal entries are retained, including those caused by nontrivial Jordan blocks.

Let $B = G^{1/2} A G^{-1/2}$. Substitution into (FD.26) gives $H = B + B^* - kI$. Unitary Schur triangularization gives $T = U^* B U$, upper triangular with diagonal entries $\eta_{\mathbf{n}}$. Such a triangularization is obtained inductively by taking a unit eigenvector as the first basis vector and applying the induction to the compression on its orthogonal complement. Unitary conjugation preserves the Frobenius norm. Its diagonal and off-diagonal entries give the exact identity

$$\|H\|_F^2 = \sum_{|\mathbf{n}|=k} (2\Re \eta_{\mathbf{n}} - k)^2 + 2 \sum_{i < j} |T_{ij}|^2 \geq 4 \sum_{|\mathbf{n}|=k} \left(\sum_i n_i \delta_i \right)^2. \quad (\text{FD.31})$$

For $i < j$, the two corresponding entries of $T + T^* - kI$ are T_{ij} and $\overline{T_{ij}}$; this proves the factor two. Thus the retained nilpotent and nonnormal contributions cannot weaken the displayed lower bound.

We evaluate its sum exactly. For $d \geq 2$, the ordinary generating functions for weak compositions give

$$\begin{aligned} \sum_{|\mathbf{n}|=k} n_1 &= \binom{k+d-1}{d}, \\ \sum_{|\mathbf{n}|=k} n_1(n_1-1) &= 2 \binom{k+d-1}{d+1}, \\ \sum_{|\mathbf{n}|=k} n_1 n_2 &= \binom{k+d-1}{d+1}. \end{aligned} \tag{FD.32}$$

Indeed their generating functions are respectively

$$z(1-z)^{-d-1}, \quad 2z^2(1-z)^{-d-2}, \quad z^2(1-z)^{-d-2}.$$

These follow by differentiating the appropriate geometric-series factors. Extracting the coefficient of z^k gives (FD.32). Write $a = \binom{k+d-1}{d}$ and $b = \binom{k+d-1}{d+1}$. Then the square expands as

$$(a+2b) \sum_i \delta_i^2 + 2b \sum_{i < j} \delta_i \delta_j = (a+b) \sum_i \delta_i^2 + b \left(\sum_i \delta_i \right)^2.$$

Pascal's identity therefore proves

$$\begin{aligned} \mathcal{S}_{h,k} &:= \sum_{|\mathbf{n}|=k} \left(\sum_i n_i \delta_i \right)^2 \\ &= \binom{k+d}{d+1} \sum_i \delta_i^2 + \binom{k+d-1}{d+1} \left(\sum_i \delta_i \right)^2. \end{aligned} \tag{FD.33}$$

Binomial coefficients with smaller upper than nonnegative lower index are zero. For $d = 1$, direct evaluation gives $k^2 \delta_1^2$; the right side of (FD.33) is $[(\binom{k+1}{2} + \binom{k}{2})] \delta_1^2 = k^2 \delta_1^2$, so the same formula holds in that case.

Combining (FD.29), (FD.31), and (FD.33) proves the aggregate arithmetic restriction

$$\boxed{\Gamma_{k,M} \operatorname{tr}(G_M^{\operatorname{sym}} (\mathcal{C}_{M+1}^{\operatorname{sym}} - \mathcal{C}_M^{\operatorname{sym}})) \geq \mathcal{S}_{h,k}.} \tag{FD.34}$$

The trace in this formula is exactly $\operatorname{tr} K$, by cyclicity. All its matrices were defined from the actual measure, complete arithmetic jet unit, and unscaled orbit norms.

160.5 Coordinate invariance and determinant growth

The comparison maps have an exact coordinate-invariance statement. For invertible changes S in quotient coordinates and R in frontier coordinates, set

$$\begin{aligned} A' &= SAS^{-1}, & C' &= SCS^*, & G' &= S^{-*}GS^{-1}, \\ F'_s &= SF_sR, & E'_s &= SE_sR, & \Omega'_s &= R^*\Omega_sR. \end{aligned} \tag{FD.35}$$

Here $S^{-*} = (S^{-1})^*$. Cancellation of R, R^{-1} in the factorizations proves $\Delta C' = S \Delta C S^*$, and substitution in the generator identity gives $\mathcal{Z}' = S \mathcal{Z} S^*$. The lowering inequality is carried through the same congruence. Direct cyclic cancellation and determinant multiplicativity give

$$\begin{aligned} \operatorname{tr}(G' \Delta C') &= \operatorname{tr}(G \Delta C), \\ \frac{\det(C' + \Delta C')}{\det C'} &= \frac{\det(C + \Delta C)}{\det C}. \end{aligned} \tag{FD.36}$$

Furthermore $U_S = (\mathcal{C}')^{-1/2} S \mathcal{C}^{1/2}$ satisfies $U_S U_S^* = I$, and hence is unitary, with $H' = U_S H U_S^*$. This follows by substituting $\mathcal{C}' = S \mathcal{C} S^*$ and $\mathcal{Z}' = S \mathcal{Z} S^*$. Thus the exact trace, determinant ratio, and Frobenius quantity are preserved by the comparison-coordinate maps; no arithmetic normalization has been imposed.

The exact update and (FD.28) give

$$\frac{\det \mathcal{C}_{M+1}^{\text{sym}}}{\det \mathcal{C}_M^{\text{sym}}} = \det(I + K). \quad (\text{FD.37})$$

Indeed congruence by $\mathcal{C}^{-1/2} = G^{1/2}$ takes $\mathcal{C} + \Delta \mathcal{C}$ to $I + K$, and its determinant multiplier is $1/\det \mathcal{C}$. If $\kappa_1, \dots, \kappa_N \geq 0$ are the eigenvalues of K , expanding $\prod_j (1 + \kappa_j)$ gives $\det(I + K) \geq 1 + \sum_j \kappa_j = 1 + \text{tr } K$. Since $\Gamma_{k,M} > 0$ was proved in (FD.20), (FD.34) now proves

$$\frac{\det \mathcal{C}_{M+1}^{\text{sym}}}{\det \mathcal{C}_M^{\text{sym}}} \geq 1 + \frac{\mathcal{S}_{h,k}}{\Gamma_{k,M}}. \quad (\text{FD.38})$$

For any integers $M_1 > M_0 \geq k(d-1)$, multiplying the finitely many positive inequalities (FD.38) telescopes to

$$\frac{\det \mathcal{C}_{M_1}^{\text{sym}}}{\det \mathcal{C}_{M_0}^{\text{sym}}} \geq \prod_{M=M_0}^{M_1-1} \left(1 + \frac{\mathcal{S}_{h,k}}{\Gamma_{k,M}} \right). \quad (\text{FD.39})$$

This is a finite product over the specified actual support degrees. It requires no limit or unproved asymptotic estimate.

For a reflection-stable packet, the original reflection maps each zero ρ to $1 - \bar{\rho}$, retaining its full order. Each paired displacement δ_i is sent to $-\delta_i$, so $\sum_i \delta_i = 0$. Hence the exact coefficient becomes

$$\mathcal{S}_{h,k} = \binom{k+d}{d+1} \sum_{i=1}^d (\Re \rho_i - \frac{1}{2})^2. \quad (\text{FD.40})$$

In particular, the presence of any actual off-axis zero makes this coefficient strictly positive. It forces a positive relative frontier increment and the determinant lower bound at every admitted support degree, within the actual symmetric summand.

160.6 The exact signed eigenline witness

Let $v \neq 0$ be an actual eigenvector of A_h , with $A_h v = \rho v$. In the unscaled symmetric quotient coordinates put $u = L_k(v^{\otimes k})$, so $I_k u = v^{\otimes k}$. This vector is nonzero: choose a linear functional ℓ with $\ell v = 1$; then $\ell^{\otimes k}(v^{\otimes k}) = 1$. The additive generator satisfies $Au = k\rho u$. With the exact positive number $q = u^* G u$, put

$$x = \Omega_s^{-1/2} F_s^* G u, \quad y = \Omega_s^{-1/2} E_s^* G u. \quad (\text{FD.41})$$

The symmetric restriction of (FD.19) is $W = A^* G + G A - kG = G \mathcal{Z} G$. On the eigenvector,

$$u^* W u = k(2\Re \rho - 1)q.$$

On the other hand, (FD.27) and (FD.41) give $u^* G \mathcal{Z} G u = x^* y + y^* x = 2\Re(x^* y)$. Equating these two exact expressions proves the signed identity

$$\Re(x^* y) = k(\Re \rho - \frac{1}{2})q. \quad (\text{FD.42})$$

Thus this correlation has the retained sign of the actual real-part displacement. No absolute-value estimate was used in obtaining it.

Define the two directional costs and the actual symmetric maximum cost

$$\alpha = \frac{\|x\|^2}{q}, \quad \beta = \frac{\|y\|^2}{q}, \quad \lambda_{k,M}^{\text{sym}} = \max_{w \neq 0} \frac{w^* \Delta \mathcal{C} w}{w^* \mathcal{C} w} = \lambda_{\max}(K). \quad (\text{FD.43})$$

The last equality follows by the invertible substitution $w = \mathcal{C}^{-1/2}z$ in the Rayleigh quotient. The lowering estimate in (FD.27), evaluated at Gu , gives $\beta \leq \Gamma_{k,M}$, because $(Gu)^* \mathcal{C}(Gu) = q$. The same substitution into the maximum defining $\lambda_{k,M}^{\text{sym}}$ gives $\alpha \leq \lambda_{k,M}^{\text{sym}}$. Cauchy–Schwarz applied to (FD.42) gives

$$\alpha\beta \geq k^2(\Re\rho - \tfrac{1}{2})^2.$$

Consequently the actual arithmetic costs satisfy

$$\boxed{\Gamma_{k,M} \lambda_{k,M}^{\text{sym}} \geq k^2(\Re\rho - \tfrac{1}{2})^2.} \quad (\text{FD.44})$$

Every term in the signed identity and this bound is attached to the actual next original relation through (FD.12), (FD.19), and the exact symmetric restrictions.

Exact scope of the resulting restriction. Equations (FD.34)–(FD.40) and (FD.42)–(FD.44) prove the stated frontier and determinant restrictions for the original arithmetic family, including all multiplicities and nilpotent actions. The source explicitly records the unestimated quantity: A1829, line 5580, states that no uniform sublinear excess was established; A1859, line 6013, states that the finite bound’s constants were not bounded sublinearly in tensor degree; A1859, line 6327, states that the uniform sublinear estimate for the actual arithmetic family was not proved. No upper estimate incompatible with (FD.38) or (FD.44) is asserted here. These are proved necessary restrictions on an actual RH counterexample, not an assertion that their arithmetic values cannot satisfy them.

161 The original conormal map, the arithmetic Jacobian, and complete symmetric trace fibres

161.1 Source pin and the unchanged arithmetic objects

The inherited source is the authoritative shared transcript, A1912 (chain 3986, segment lines 6410–7016) and A1959 (chain 4075, lines 7086–7810), together with the signed chain projector in A1859 (chain 3887, lines 6042–6098). These locators refer to the complete segment U0041–U0054; all 43 message bodies have been matched against the original transcript records. The calculations below extend the source’s actual morphisms. Their arithmetic coefficient object is the same one over the constructed absolute base $\mathfrak{b}_\tau = \text{Spec}\{\tau, 1\}$, with its structural square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

Here j is the original unital inclusion, $G(R) = \{\tau\} \sqcup \{a^\bullet : a \in R\}$, $e_R = 0_R^\bullet$, and $p_R(\tau) = 0$, $p_R(a^\bullet) = a$. A linear map is lifted by $a^\bullet \mapsto f(a)^\bullet$, $\tau \mapsto \tau$. Every coefficient map below has this lift on its original support fibre.

Retain the actual entire function

$$g(s) = 2\xi(s), \quad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with the entire continuation in the inherited arithmetic construction. Its source is

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\eta) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \eta} dx,$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^a F(x)| < \infty \text{ for every } b \in \mathbb{Z}, a \geq 0 \right\},$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x.$$

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The degrees of C_+ are zero and one. The original seed is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g.$$

For a nonempty finite actual zero packet Z , with its complete positive multiplicities m_ρ , write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad r = |Z|, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h.$$

The full CRT idempotents have an explicit polynomial construction. For each centre put $h_\rho = h/(s - \rho)^{m_\rho}$ and $q_\rho(z) = h_\rho(\rho + z)/h_\rho(\rho) - 1$. Since $q_\rho(0) = 0$, the polynomial class

$$a_\rho(\rho + z) = \frac{1}{h_\rho(\rho)} \sum_{j=0}^{m_\rho-1} (-q_\rho(z))^j \mod z^{m_\rho}$$

is the inverse of h_ρ modulo $(s - \rho)^{m_\rho}$: multiplication of the finite geometric sum leaves $1 - (-q_\rho)^{m_\rho}$, whose final term vanishes modulo z^{m_ρ} . Then $e_\rho = [h_\rho a_\rho]_h$ equals one on the ρ factor and zero on every other factor. Pairwise coprimality of the powers $(s - \rho)^{m_\rho}$ implies that a polynomial divisible by each is divisible by their product h . It follows that $\sum_\rho e_\rho = 1$, $e_\rho e_\sigma = 0$ for $\rho \neq \sigma$, and $e_\rho^2 = e_\rho$. These identities prove the stated CRT inverse by summing e_ρ times each local representative. The inherited arithmetic realization constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$, and its literal polynomial source map is

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h, \quad v_h = j_h(v_h) \in E_h^\times.$$

Here j_h takes full Taylor germs modulo the factors of h , with the inverse Chinese-remainder map into E_h , and $J_h F = j_h(\mathcal{M}F)$. These identities, with their analytic domains, are the inherited source pins, not new purity assumptions. Their tensor products use the original order of variables and the cochain differential signs. The source identity $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ holomorphic at the actual packet, proves $J_h \Theta\phi = 0$, including every prescribed multiplicity. Thus J_h induces $\bar{J}_h : Q \rightarrow E_h$. Put $\varepsilon_h = v_h^{-1}$, and use its fixed polynomial representative in the original coordinates. The map

$$\sigma_h : E_h \longrightarrow Q, \quad [P]_h \longmapsto [\mathcal{T}_h(\varepsilon_h P)]_{\Theta V}$$

is well-defined: changing P by hA , or the representative of ε_h by hB , changes its source by an explicitly displayed original theta relation. The identity $\bar{J}_h \sigma_h = 1_{E_h}$ follows by multiplication by $v_h \varepsilon_h = 1$, so σ_h is injective. Its tensor powers and the signed symmetric projector below retain these finite arithmetic classes and the same left inverses. For $k \geq 1$, put

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)), \quad E = \mathcal{P}/I = E_h^{\otimes k}, \quad E^{[2]} = \mathcal{P}/I^2,$$

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}, \quad J^{(k)} = J_h^{\otimes k}, \quad U(\mathbf{s}) = \prod_{i=1}^k v_h(s_i).$$

The tensor expressions here act on finite sums of the displayed source functions, so all stated Mellin derivatives have those original rapid-decrease domains. In coefficient multiplication U means its full class modulo I ; in a differentiated formula before reduction it means the displayed entire function. Its full class modulo I^2 is denoted $\hat{U}$. These classes retain every Taylor coefficient allowed by their respective ideals.

161.2 The exact conormal domain and its arithmetic observation

The ring $\mathcal{P}$ is free over $\mathbb{C}[h(s_1), \dots, h(s_k)]$, with basis s^α , $0 \leq \alpha_i < d$. Successive monic division in each variable proves spanning. Independence follows by taking a largest multidegree: the leading monomials $s^{d\beta+\alpha}$, with $0 \leq \alpha_i < d$, are distinct for different pairs (β, α) . The same leading-term argument proves that the $h(s_i)$ are algebraically independent. Modulo I^2 , precisely the terms of degree zero or one in these generators survive. This proves the exact sequence and coordinate map

$$0 \longrightarrow I/I^2 \longrightarrow E^{[2]} \xrightarrow{\pi} E \longrightarrow 0, \quad \iota_N : E^k \xrightarrow{\sim} I/I^2, \quad (a_i) \mapsto \left[\sum_i h(s_i) \tilde{a}_i \right].$$

Changing a lift $\tilde{a}_i$ by an element of I changes the sum by an element of I^2 , proving well-definedness directly.

The original relative-sum derivative and logarithmic source observable are

$$\partial_S^{\text{rel}} = \frac{1}{k} \sum_i \partial_{s_i}, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i.$$

Leibniz's rule gives $\partial_S^{\text{rel}} I^2 \subseteq I$, hence defines $\delta_S : E^{[2]} \rightarrow E$. On the conormal domain its exact row is

$$\delta_S \iota_N(a_1, \dots, a_k) = \frac{1}{k} \sum_i h'(s_i) a_i.$$

Indeed differentiation of a lifted coefficient leaves a factor $h(s_i)$ and therefore belongs to I . In subsequent rank formulas δ_S denotes this restricted conormal map; its domain is $I/I^2 \cong E^k$, not all of $E^{[2]}$. For an actual original relation $P = \sum_i h(s_i) P_i$, differentiation of its Mellin transform yields

$$J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P = \frac{U}{k} \sum_i h'(s_i) [P_i]_I.$$

The terms differentiating U or P_i still contain $h(s_i)$, so they vanish only at this displayed final quotient. In the split lift the quotient $G(\pi)$ sends a supported conormal vector to e_E , whereas the lifted coefficient derivative sends it to $(\delta_S n)^\bullet$. Both send external absence to τ . This is the original map through which the subsequent calculation observes a theta relation before it becomes its receiving supported zero.

For a reflection-stable packet the involution is $f^\dagger(s) = \overline{f(1 - \bar{s}_1, \dots, 1 - \bar{s}_k)}$. The ordered arithmetic residue form used below is explicitly

$$\mathcal{R}_k(f, v) = \sum_{(\rho_1, \dots, \rho_k) \in Z^k} \text{Res}_{s_1=\rho_1} \cdots \text{Res}_{s_k=\rho_k} \frac{f^\dagger(s) v(s)}{\prod_{i=1}^k g(s_i)} ds_1 \cdots ds_k.$$

The order of the iterated residues is the displayed original order; there is no extra Koszul factor in this definition. To check independence of representatives, changing one input by $h(s_i)a$ cancels that factor against $g(s_i) = h(s_i)v_h(s_i)$; the resulting expression is holomorphic in the i -th variable at every corresponding centre and its residue is zero. The same statement for the first input uses $h^\dagger = (-1)^d h$.

Its nondegeneracy retains every local unit. At a centre ρ , write $z = s - \rho$ and $g(\rho + z) = z^{m_\rho} u_\rho(z)$, $u_\rho(0) \neq 0$. The local residue is the coefficient of $z^{m_\rho-1}$ in $f^\dagger(z)v(z)/u_\rho(z)$. Multiplication by u_ρ^{-1} is invertible in $\mathbb{C}[z]/(z^{m_\rho})$. The coefficient pairing of the basis $1, z, \dots, z^{m_\rho-1}$ with itself has entries 1 when the two exponents sum to $m_\rho - 1$, and 0 otherwise; its anti-diagonal determinant is nonzero. The reflected conjugate germ is an invertible conjugate-linear map and only pairs matched centres.

The CRT decomposition and tensor product of these finite perfect pairings prove nondegeneracy of $\mathcal{R}_k$. For any finite coefficient algebra A , $M_a : A \rightarrow A$ denotes multiplication by a , and $\text{Tr}_A M_a$ is its ordinary complex-linear trace. The inertia (n_+, n_-, n_0) of a Hermitian form means the dimensions of its positive, negative and zero subspaces in a diagonalization by congruence; the zero dimension is the dimension of its radical.

161.3 The entire conormal image of an actual zero packet

Let Z be the finite actual packet in the transcript, with all its distinct centres and full positive multiplicities. For the residue and Hermitian trace calculations retain the transcript's reflection stability: $\iota(\rho) = 1 - \bar{\rho}$ permutes Z and $m_{\iota\rho} = m_\rho$. Its conjugate-linear involution is $f^\dagger(s) = \overline{f(1 - \bar{s})}$; since $h^\dagger = (-1)^d h$, it descends to E_h and componentwise to its tensor powers. The rank calculations alone do not need this involution. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \sum_{\rho \in Z} m_\rho, \quad r = |Z|, \quad E_h = \mathbb{C}[s]/(h).$$

No factor of h is removed. Denote by e_ρ the full Chinese-remainder idempotent of E_h . The exact CRT map sends a polynomial class to its classes modulo $(s - \rho)^{m_\rho}$; its inverse is multiplication by these idempotents followed by summation. At the rho component use the named local coordinate $z = s - \rho$. Let

$$h_\rho(s) = h(s)/(s - \rho)^{m_\rho}.$$

Its value $h_\rho(\rho)$ is nonzero. Before reduction the derivative is

$$h'(\rho + z) = m_\rho z^{m_\rho-1} h_\rho(\rho + z) + z^{m_\rho} h'_\rho(\rho + z).$$

In $\mathbb{C}[z]/(z^{m_\rho})$, the second displayed term maps to zero and the first equals $m_\rho h_\rho(\rho) z^{m_\rho-1}$. This explicitly records the killed term and every original scalar. Therefore the ideal (h') on this factor is the line generated by $z^{m_\rho-1}$. For $m_\rho = 1$ this line is the whole one-dimensional factor. We have proved

$$\dim(h')E_h = r, \quad \dim E_h/(h') = d - r. \quad (\text{C1})$$

For the ordered k -fold packet put $E = E_h^{\otimes k}$. The image of δ_S is exactly

$$\mathcal{J} = (h'(s_1), \dots, h'(s_k))E. \quad (\text{C2})$$

Containment follows from its displayed row. For the reverse inclusion, a term $h'(s_i)b$ is the image of the vector with i th entry kb and all other entries zero. This retains the original $1/k$ instead of dropping it. There is the exact quotient morphism

$$E/\mathcal{J} \xrightarrow{\sim} (E_h/(h'))^{\otimes k}, \quad [a_1 \otimes \dots \otimes a_k] \mapsto [a_1] \otimes \dots \otimes [a_k]. \quad (\text{C3})$$

To prove its kernel, choose any basis of $(h')E_h$ and extend it to a basis of E_h . The tensor basis decomposes into those tensors with at least one ideal-basis factor and those with none. The former span exactly the sum of the k ideals in (C2); the latter project bijectively onto a basis of the target. The basis is used only to prove the named quotient map, and it does not replace the arithmetic Gram or the original polynomial coordinates. Equations (C1)–(C3) imply

$$\text{rank } \delta_S = d^k - (d - r)^k, \quad \dim \ker \delta_S = (k - 1)d^k + (d - r)^k. \quad (\text{C4})$$

The full local version, for an ordered tuple $(\rho_1, \dots, \rho_k)$, has image dimension

$$\prod_i m_{\rho_i} - \prod_i (m_{\rho_i} - 1) \quad (\text{C5})$$

inside the unchanged local algebra $\mathbb{C}[z_1, \dots, z_k]/(z_i^{m_{\rho_i}} : 1 \leq i \leq k)$. Indeed the quotient in (C3) replaces each local vector-space dimension by $m_{\rho_i} - 1$. This includes every repeated-root direction rather than passing to residue values.

The actual coefficient U in A1959 is an invertible element of E . Multiplication by U is therefore an automorphism with explicit inverse multiplication by U^{-1} . Since an ideal is invariant under multiplication by a unit, the actual arithmetic observation $U\delta_S$ has precisely the same image (C2) and kernel as δ_S . These are rank calculations for the original arithmetic source relation map, including its unit, not merely ranks of an unrelated polynomial map.

161.4 The full mixed derivative and its arithmetic trace

Let

$$J_k = \prod_{i=1}^k g'(s_i) \in E, \quad g = 2\xi.$$

A1959 lines 7743–7764 gives the exact source identity

$$\left[\partial_{s_1} \cdots \partial_{s_k} \left(U P \prod_i h(s_i) \right) \right]_I = J_k [P]_I. \quad (\text{C6})$$

The product rule proves this without suppressing terms: every term in which some derivative misses its corresponding h retains an undifferentiated $h(s_i)$ and therefore belongs to I ; exactly the one term hitting every h survives. Its coefficient is $U \prod_i h'(s_i) = \prod_i g'(s_i)$ in E because $g' = h'(g/h) + h(g/h)'$. The term involving h has thus been recorded before the quotient.

The actual unit $v_h = g/h$ is nonzero at every centre. On the rho factor,

$$g'(\rho + z) = m_\rho u_\rho(0) z^{m_\rho - 1} \pmod{z^{m_\rho}}, \quad u_\rho(z) = g(\rho + z)/z^{m_\rho}, \quad u_\rho(0) \neq 0. \quad (\text{C7})$$

Thus $M_{g'}$ has rank one at each centre, and the tensor-product multiplication map in (C6) has

$$\text{rank } M_{J_k} = r^k, \quad \dim \ker M_{J_k} = d^k - r^k. \quad (\text{C8})$$

Its image on an ordered tuple is the line with the original coefficient

$$\left(\prod_i m_{\rho_i} u_{\rho_i}(0) \right) z_1^{m_{\rho_1} - 1} \cdots z_k^{m_{\rho_k} - 1}. \quad (\text{C9})$$

The map may land in the nilradical when a multiplicity exceeds one, but the original perfect residue pairing still detects (C9). Concretely, the ordered residue contraction is

$$\mathcal{R}_k(f, J_k u) = \sum_{\rho \in Z^k} \left(\prod_i m_{\rho_i} \right) \overline{f(1 - \bar{\rho})} u(\rho). \quad (\text{C10})$$

Here $\mathcal{R}_k$ means the ordinary ordered iterated residue in the fixed order $s_1, \dots, s_k$, with no additional Koszul factor. Equation (C10) is repeated application of the actual simple-pole residue of g'/g ,

whose coefficient is the full m_ρ . It is also the ordinary multiplication trace on the full ordered algebra. Formula (C10) proves the exact map between the conormal/Jacobian observation and the trace; no assertion that a nilpotent is irrelevant is used. The cohomological supertrace has its separate factor $(-1)^k$. A differently signed tensor evaluation must be related by its explicit sign multiplication; no automatic cancellation is claimed here.

161.5 Restriction to the actual signed symmetric cochain image

The original idempotent in A1859 lines 6054–6078 is

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi, \quad T_\pi = \text{sgn}(\pi) P_\pi \text{ in top degree.}$$

Its top-degree action is therefore the ordinary Reynolds projector $R_k = \frac{1}{k!} \sum_{\pi \in S_k} P_\pi$. We use the exact image $B_k = (E_h^{\otimes k})^{S_k}$, including full nilpotents. Simultaneous permutation of variables and conormal indices makes δ_S equivariant. Averaging a preimage proves that its invariant image is $\mathcal{J}^{S_k}$. This argument uses the same $1/k!$ as the actual chain idempotent. Taking invariants in (C3) is exact: if an invariant quotient vector has a lift, averaging that lift gives an invariant lift. It follows that

$$\text{rank}(\delta_S \text{ on invariant conormals}) = \binom{d+k-1}{k} - \binom{d-r+k-1}{k}. \quad (\text{C11})$$

Here $\binom{k-1}{k} = 0$ when $d-r=0$, since the k th symmetric power of the zero vector space is zero for $k \geq 1$. To verify the dimensions, a basis of a t -dimensional vector space gives an unscaled orbit-sum basis of its k th tensor invariants indexed by nonnegative t -tuples summing to k ; placing k identical marks among t slots gives $\binom{t+k-1}{k}$. This is a vector-space basis calculation, not a metric replacement.

The invariant conormal domain has dimension

$$d \binom{d+k-2}{k-1}. \quad (\text{C12})$$

Indeed an invariant tuple of k coefficients is determined by its k th coefficient, which must be invariant under the permutations of the first $k-1$ tensor positions. The other coefficients are recovered by permutation; independence of the chosen permutation is exactly this stabilizer invariance. This constructs the isomorphism to $E_h \otimes \text{Sym}^{k-1}(E_h)$ and proves (C12). Subtracting (C11) gives the complete invariant kernel dimension.

The mixed Jacobian is permutation invariant. Its image is the k th tensor power of the r -dimensional image $(g')E_h$; the same averaging proves surjectivity after taking invariants. Therefore

$$\text{rank}(M_{J_k}|_{B_k}) = \binom{r+k-1}{k}. \quad (\text{C13})$$

These are exact maps on the original signed symmetric cochain image, not on the unsigned cochain average that kills repeated degree-one vectors.

161.6 Intrinsic symmetric trace: the exact comparison of its multiplicities

It is necessary to keep both trace maps, with their actual comparison. Let an occupation vector be $\mathbf{n} = (n_\rho)_{\rho \in Z}$, with every $n_\rho \geq 0$ and $\sum_{\rho \in Z} n_\rho = k$. Define $e_{\mathbf{n}}$ to be the unscaled sum of the full ordered CRT idempotents whose ordered tuples have those occupations. Products of distinct summands vanish, so $e_{\mathbf{n}}^2 = e_{\mathbf{n}}$, and these are the full orthogonal idempotents of B_k . The associated local factor is the invariant tensor product of the original local algebras. Its dimension is

$$D_n = \prod_{\rho \in Z} \binom{m_\rho + n_\rho - 1}{n_\rho}. \quad (\text{C14})$$

For proof, one chosen ordering has stabilizer $\prod_{\rho \in Z} S_{n_\rho}$. An invariant tuple of local components is determined by that component and its stabilizer invariance. The latter is $\bigotimes_{\rho \in Z} ((\mathbb{C}[z]/(z^{m_\rho}))^{\otimes n_\rho})^{S_{n_\rho}}$. Applying the orbit-sum dimension argument separately proves (C14), and also constructs the isomorphism without changing local variables.

The component has nilpotent ideal of zero constant terms: in the full ordered local algebra the ideal $(z_1, \dots, z_k)$ has a power equal to zero, hence so does its intersection with the invariant factor. Every element with nonzero constant is invertible by the finite geometric series in this ideal. Thus this is a local algebra, with residue field $\mathbb{C}$ and dimension (C14). Multiplication by $a = c + n$, n nilpotent, has trace $D_n c$: multiplication by n is nilpotent, hence has zero trace (triangularize its successive kernels). We obtain the intrinsic symmetric trace form

$$\mathcal{T}_k^{\text{sym}}(f, u) = \text{Tr}_{B_k} M_{f^\dagger u} = \sum_n D_n \overline{f(\iota n)} u(n), \quad \iota \rho = 1 - \bar{\rho}. \quad (\text{C15})$$

Here $\iota n = (n_{\iota \rho})_{\rho \in Z}$, the induced permutation of the original occupations. It is also, by the image/kernel decomposition of the Reynolds projector,

$$\mathcal{T}_k^{\text{sym}}(f, u) = \text{Tr}_{E_h^{\otimes k}} (R_k M_{f^\dagger u}). \quad (\text{C16})$$

Indeed $M_{f^\dagger u}$ commutes with R_k , and $R_k M$ equals M on its image B_k and zero on its kernel. This is the exact operator joining the ordered tensor trace and the actual symmetric cohomology trace.

The unprojected ordered trace in (C10), restricted to invariant inputs, has the different weight

$$O_n = \frac{k!}{\prod_\rho n_\rho!} \prod_\rho m_\rho^{n_\rho}.$$

The exact further map between the two arithmetic forms is multiplication by

$$B = \sum_n \frac{D_n}{O_n} e_n, \quad \mathcal{T}_k^{\text{sym}}(f, u) = \mathcal{R}_k(f, J_k B u). \quad (\text{C17})$$

Every ratio is a specified positive nonzero rational number, and the inverse replaces each ratio by its reciprocal. The equation follows by comparing the two literal finite sums. Thus the symmetric intrinsic trace is reached by the same actual theta mixed derivative, followed by its explicitly displayed coefficient map B . No orbit factor or local multiplicity has been normalized away.

For precision, this contraction has the exact induced isomorphism

$$B_k / \text{rad } \mathcal{T}_k^{\text{sym}} \xrightarrow{\sim} \text{im}(M_{J_k}|_{B_k}), \quad [u] \longmapsto J_k B u. \quad (\text{C17a})$$

The local formula (C7) shows that multiplication by the full mixed Jacobian kills exactly those classes whose constant term is zero on every ordered tuple. Restricting to invariants gives exactly the nilpotent ideal of B_k , already computed from its local factors. Multiplication by B preserves that ideal and is invertible. Its kernel is therefore precisely the trace radical, and surjectivity onto the stated invariant image follows from invertibility of B . The image here has dimension (C13); it is the invariant part of the ordered image (C8), not the full ordered image. This proves (C17a), with its exact kernel, rather than identifying its two spaces without a map.

An explicit original source representative for its value is obtained as follows. Choose the unique full CRT polynomial representative P of Bu , of degree less than d in each of the unchanged variables;

for invariant u this representative is invariant by uniqueness and the permutation invariance of h . Then

$$F_{u,k} := \left(\prod_{i=1}^k \log x_i \right) P(D_1, \dots, D_k) (\Theta \phi_*)^{\otimes k} \quad \text{satisfies} \quad J^{(k)} F_{u,k} = J_k B u. \quad (\text{C17b})$$

Indeed the undifferentiated source on the right has Mellin transform $P \prod_i g(s_i) = U P \prod_i h(s_i)$. Each log multiplication differentiates its specified Mellin variable, so (C6) gives precisely (C17b). These source functions and their logarithmic multiples remain in the given rapidly decreasing test spaces by the bounds defining those spaces. The undifferentiated expression is an original theta tensor boundary, for example using the first source leg ϕ_* with the other theta legs and the same polynomial operator. The mixed derivative is taken before that relation is quotiented. For $m > 1$ its observed jet may be trace-radical, even while its residue contraction in (C17) is nonzero; (C17a) and (C17b) retain exactly which map is being used.

161.7 Consequence for the actual RH counterfactual, with every multiplicity

Suppose the counterfactual supplies an actual quartet

$$Z = \{ \tfrac{1}{2} + \delta + i\gamma, \tfrac{1}{2} - \delta + i\gamma, \tfrac{1}{2} + \delta - i\gamma, \tfrac{1}{2} - \delta - i\gamma \}, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 0,$$

with the common full multiplicity m supplied by the original reflection and conjugation. The existence assumption in this paragraph is exactly the RH counterfactual. The statements below compute its consequences on the original objects. Put the first and second centres in one reflection pair, and the third and fourth in the other. Then $d=4m$ and $r=4$, so

$$\text{rank } \delta_S = (4m)^k - (4(m-1))^k, \quad \text{rank } M_{J_k} = 4^k, \quad (\text{C18})$$

and on the actual signed symmetric image,

$$\text{rank } \delta_S^{\text{sym}} = \binom{4m+k-1}{k} - \binom{4(m-1)+k-1}{k}, \quad \text{rank } M_{J_k}^{\text{sym}} = \binom{k+3}{3}. \quad (\text{C19})$$

The trace radical is exactly the nilpotent ideal. To prove this, a zero-constant element is orthogonal to everything by (C15). Conversely, if its value at occupation n is nonzero, pairing against $e_{\iota n}$ gives the nonzero scalar $D_{\iota n}$ times that value or its conjugate. The radical therefore has dimension

$$\binom{4m+k-1}{k} - \binom{k+3}{3}. \quad (\text{C20})$$

On a two-element reflection orbit of occupations n and ιn , the literal Gram in the idempotent basis is

$$\begin{pmatrix} 0 & D_n \\ D_n & 0 \end{pmatrix}, \quad D_n = \prod_{i=1}^4 \binom{m+n_i-1}{n_i} > 0,$$

since matched multiplicities make $D_n = D_{\iota n}$. The vectors $e_n + e_{\iota n}$ and $e_n - e_{\iota n}$ have values $+2D_n$ and $-2D_n$, respectively. The basis and both factors two have been retained. A fixed occupation gives the positive one-dimensional Gram D_n .

A fixed occupation must have $n_1 = n_2$ and $n_3 = n_4$. There are no such occupations for odd k . For $k = 2l$, choosing $n_1 = n_2 = a$ and $n_3 = n_4 = l - a$, with $0 \leq a \leq l$, gives exactly $l+1$. Let

$$b_k = \binom{k+3}{3}, \quad f_k = \begin{cases} 0 & k \text{ odd,} \\ k/2 + 1 & k \text{ even.} \end{cases}$$

Counting each nonfixed orbit twice proves the complete inertia, including the entire radical:

$$\boxed{\begin{aligned} n_+(\mathcal{T}_k^{\text{sym}}) &= (b_k + f_k)/2, \\ n_-(\mathcal{T}_k^{\text{sym}}) &= (b_k - f_k)/2, \\ n_0(\mathcal{T}_k^{\text{sym}}) &= \binom{4m+k-1}{k} - b_k. \end{aligned}} \quad (\text{C21})$$

These negative directions are nonradical. Each of their negative trace values is the pairing with the explicit original theta mixed-derivative observation $J_k B f$ in (C6) and (C17). This states the exact map: it does not assert that an idempotent f is itself in the Jacobian image, which would fail for repeated roots. The class f survives the nilradical quotient and the actual signed symmetric projector; its mixed-derivative observation remains available before the corresponding source relation is quotiented to its supported zero. Formula (C21) supplies a growing family of exact nonzero arithmetic directions that every claimed purity comparison for the RH counterfactual must control. It establishes no off-line zeta zero and no unconditional positivity contradiction: neither assertion occurs in the authoritative source, and neither was used in this calculation.

161.8 The relative fibres at the expected sum weight still retain negative directions

The sum pushforward in A1912 retains the occupation idempotents even when their sum eigenvalues coincide. On the quartet label ordering just specified, its eigenvalue at occupation n is exactly

$$\lambda_n = \frac{k}{2} + \delta(n_1 - n_2 + n_3 - n_4) + i\gamma(n_1 + n_2 - n_3 - n_4). \quad (\text{C22})$$

To verify this with nilpotents, restrict multiplication by $S = \sum_i s_i$ to one ordered CRT factor, where it is the displayed scalar plus multiplication by $\sum_i z_i$. The latter has a power equal to zero, since all monomials of total degree above $k(m-1)$ vanish. Passing to stabilizer invariants retains this action and its scalar; it does not set the nilpotent to zero. This proves the full generalized eigenvalue statement on the occupation factor.

For $k = 2l$, let B_{bal} be the direct summand defined by the sum of all full idempotents with $n_1 + n_3 = l$ and $n_2 + n_4 = l$. It is a specified algebra direct summand of B_k , invariant under both reflection and the sum operator. Its eigenvalues all have real part $k/2$ by (C22). There are exactly $(l+1)^2$ such occupations: independently distribute l among the two right centres and l among the two left centres. Exactly $l+1$ of them are reflection-fixed, as in (C21). Its trace form therefore has

$$n_-(\mathcal{T}_k^{\text{sym}}|_{B_{\text{bal}}}) = \frac{l(l+1)}{2}, \quad n_+(\mathcal{T}_k^{\text{sym}}|_{B_{\text{bal}}}) = \frac{(l+1)(l+2)}{2}, \quad (\text{C23})$$

and radical dimension $\sum_{n \text{ balanced}} D_n - (l+1)^2$, with the same literal weights (C14). This is proved by the same explicit two-idempotent Gram blocks, now on this exact invariant direct summand. It is not a new positivity assumption.

Already at $k=2$ the two distinct occupations $(1, 0, 0, 1)$ and $(0, 1, 1, 0)$ both have the actual sum eigenvalue 1. Their full idempotents remain different, and

$$\mathcal{T}_2^{\text{sym}}(e_{(1,0,0,1)} - e_{(0,1,1,0)}, e_{(1,0,0,1)} - e_{(0,1,1,0)}) = -2m^2. \quad (\text{C24})$$

For $m > 1$ the local nilpotent sum operators remain attached; no assertion that the idempotents are eigenvectors is needed. Equation (C24) is exactly why the original program's relative algebra cannot be replaced by the list of sums alone. It is an actual counterfactual consequence inside A1912's constructed pushforward. It does not say that an upper bound on the entire tensor image would be insufficient: the same whole image also retains repeated extreme eigenlines of real part $k(1/2 + \delta)$. Rather, it specifies the additional negative relative directions that remain even in its expected-real-part summand, with their exact source and trace maps (C17a)–(C17b).

The negative count can be located on each exact sum fibre. For an integer q with $-l \leq q \leq l$, the fibre whose eigenvalue is $l + 2i\gamma q$ consists of the full occupation factors

$$\mathbf{n} = (a, b, l - a, l - b), \quad 0 \leq a, b \leq l, \quad a + b = l + q.$$

There are $N_q = l + 1 - |q|$ such pairs: when $q \geq 0$, a runs from q to l ; when $q \leq 0$, a runs from 0 to $l+q$. Reflection exchanges a and b , so its fixed count is $f_q = 1$ if $l + q$ is even and zero otherwise. This also equals one precisely when N_q is odd. The exact nonradical inertia of that particular sum fibre is consequently

$$n_- = \left\lfloor \frac{l+1-|q|}{2} \right\rfloor, \quad n_+ = \left\lceil \frac{l+1-|q|}{2} \right\rceil, \quad (\text{C25})$$

with its unchanged nilpotent radical of dimension $\sum_{a+b=l+q} D_{(a,b,l-a,l-b)} - N_q$. The proof is the full Gram-block argument above restricted by the literal spectral projector which sums these occupation idempotents. Thus the persistent negative directions are established inside individual exact sum fibres, not only in a union of fibres with the same real part.

162 The off-line counterfactual inside the original holonomy construction

This calculation uses the original theta source in the final analytic continuation of the programme. In particular, the scalar measure below is computed from g/h with $g = 2\xi$; it is not a Gamma reference measure. The source equations being continued are (H5)–(H17), (H24)–(H28), and (H53)–(H55) of *All-holonomy descent of the original theta quotient*. The complete source record and locators are recorded in the accompanying audit. The conclusions concern the stated counterfactual and do not assert the existence of an off-line zero.

162.1 The retained arithmetic objects and every phase

Suppose $1/2 + \delta + i\gamma$ is a zero of g of its full order m , where $0 < \delta < 1/2$ and $\gamma > 0$. Reflection and conjugation give the complete quartet, with the same order. Retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - 1/2 - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}. \quad (\text{HT1})$$

The division is entire and its full jet $v_h = j_h(v_h)$ is a unit. For an integer $k \geq 2$, put

$$c = k/2, \quad \ell_k = 1 + k(m-1), \quad q = \ell_k(k+1)^2, \quad m_{h,k} = w_h^{*k},$$

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^{\ell_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \quad (\text{HT2})$$

These are the original cyclic sum quotient, its full multiplicities, and its multiplication operator. For completeness, in each tensor local factor the nilpotent sum is $n_1 + \cdots + n_k$ with $n_i^m = 0$. Its power

$k(m-1)$ has nonzero coefficient $(k(m-1))!/((m-1)!)^k$ on $n_1^{m-1} \cdots n_k^{m-1}$, whereas the next power vanishes. Thus its nilpotency index is exactly ℓ_k . Every pair (a, b) is realized by choosing independently the real and imaginary signs in the k tensor factors. Distinct pairs give distinct complex sums, so the annihilator in the sum variable is exactly χ .

The original analytic construction provides F_h with Mellin transform v_h , and the original source and jet maps are

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{HT3})$$

Here η is an injective complex linear module map. Its unit factor is retained. The norm of this source is exactly $\int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du$. The half-density map is

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad r = \log x_k, \quad z_i = \log x_i - \log x_k. \quad (\text{HT4})$$

It maps into $L^2(\mathbb{R}, dr; \mathcal{K})$ with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$ and intertwines $D_1 + \cdots + D_k$ with $-\partial_r + c$. No relative variable is removed.

Fix any $L > 0$ and $\theta \in [0, 2\pi)$. The source of phase θ is

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{j \in \mathbb{Z}} e^{ij\theta} \psi(r + jL), \quad \psi(r + L) = e^{-i\theta} \psi(r) \quad \text{in the target domain.} \quad (\text{HT5})$$

On the original finite source, its Gram form is exactly the polynomial norm for the measure

$$\nu_{L,\theta} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}, \quad u_n = \frac{2\pi n + \theta}{L}, \quad S = c + iu. \quad (\text{HT6})$$

The factor $2\pi/L$ is unchanged. The measure has positive mass at every u_n . Indeed the nonzero entire function v_h has only isolated zeros on the line; hence $w_h > 0$ almost everywhere. For every u , the product $w_h(v)w_h(u-v)$ is positive almost everywhere in v , so its integral is positive. Iterated convolution preserves strict positivity. The original weighted Schwartz estimates give convergence and all moments in (HT6). A nonzero polynomial cannot vanish at every lattice point; consequently every finite polynomial Gram matrix is positive definite.

For $N \geq q-1$, let $P_N = \mathbb{C}[S]_{\leq N}$ and $J_N = \pi_\chi|_{P_N}$. With the fixed coefficient bases define

$$M_{N,\theta} = (\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N})^*(\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N}), \quad K = J_N M_{N,\theta}^{-1} J_N^*, \quad G = K^{-1},$$

$$C_{N,\theta} = M_{N,\theta}^{-1} J_N^* G, \quad R = \mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V} C_{N,\theta}. \quad (\text{HT7})$$

Thus $J_N C_{N,\theta} = I_E$ and $R^* R = G$. All inverses in (HT7) exist: $M_{N,\theta}$ is positive definite and J_N is surjective. No inverse at degree $q-2$ is used.

162.2 A uniform phasewise lower bound for the original control

Define

$$H = G^{-1/2}(A^*G + GA - kG)G^{-1/2}, \quad \mathcal{L}_{h,k} = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{HT8})$$

For every $k \geq 2$, $L > 0$, $\theta \in [0, 2\pi)$, and $N \geq q-1$, the nonzero eigenvalues of H are ε , $-\varepsilon$, and

$$\boxed{\varepsilon \geq \mathcal{L}_{h,k} > 0, \quad \frac{\mathcal{L}_{h,k}}{q} \geq \frac{\delta k}{2}.} \quad (\text{HT9})$$

Proof. Let $p_j(S)$ be the monic orthogonal polynomials for (HT6), with squared norms $\omega_j > 0$, and let $b_j = [p_j] \in E$. In the real coordinate u , write $p_j(c + iu) = i^j r_j(u)$, where r_j is real monic. Orthogonality and the symmetry of multiplication by u show

$$ur_j = r_{j+1} + a_j r_j + \frac{\omega_j}{\omega_{j-1}} r_{j-1}, \quad a_j \in \mathbb{R},$$

with the last term omitted at $j = 0$. To verify that no earlier term occurs, pair ur_j with r_i , $i \leq j - 2$, and move u to r_i ; its degree is at most $j - 1$. Pairing with r_{j-1} gives precisely ω_j/ω_{j-1} . In the S coordinate this is

$$Sp_j = p_{j+1} + (c + ia_j)p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}. \quad (\text{HT10})$$

Since the source Gram is diagonal in this basis, $K = \sum_{j=0}^N b_j b_j^* / \omega_j$. Apply A on the left and A^* on the right and use (HT10). The imaginary diagonal terms cancel, and each adjacent term except the last cancels its predecessor. This gives, including $N = q - 1$,

$$AK + KA^* - kK = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{HT11})$$

Congruence by $G^{1/2}$ identifies the left side with H . Thus its rank is at most two. Also $\text{Tr } H = 2 \text{Re Tr } A - kq = 0$ by the complete grid (HT2).

Let E_+ be the direct sum of the complete primary spaces with $a > k/2$, and let P be the orthogonal projection onto $G^{1/2}E_+$ in the standard Hermitian space. The latter subspace is invariant under $G^{1/2}AG^{-1/2}$. Taking traces in an orthonormal basis of this subspace gives

$$\text{Tr}(PH) = 2 \sum_{a > k/2} \sum_{b=0}^k \ell_k(2a - k) \delta = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = \mathcal{L}_{h,k}. \quad (\text{HT12})$$

The finite sum is r^2 if $k = 2r - 1$, and $r(r+1)$ if $k = 2r$. A Hermitian rank-at-most-two matrix of trace zero has eigenvalues ε , $-\varepsilon$ and zeroes. For every orthogonal projection, $\text{Tr}(PH) \leq \varepsilon$: in an eigenbasis the positive contribution is at most ε and the negative one is nonpositive. Equation (HT12) proves the first inequality of (HT9), and $4\lfloor (k+1)^2/4 \rfloor \geq k(k+1)$ proves the second. $\square$

162.3 The exact energy forced into the retained theta boundary

In $\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K})$ put $D_{L,\theta} = -\partial_r + c$, with domain $\{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}$. The domain of its square is $\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), f'(L) = e^{-i\theta} f'(0)\}$. The original smooth periodized source columns belong to these domains. Monic division in the original coefficient source gives the unique polynomial map $Q_{N,\theta}$ with

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}.$$

Indeed applying J_{N+1} gives zero. Define the actual boundary

$$\mathcal{B} = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}(\chi Q_{N,\theta}) = D_{L,\theta} R - RA. \quad (\text{HT13})$$

This retains its theta primitive. Explicitly, tensor monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}),$$

because its image in the full tensor jet algebra vanishes. In the original theta complex, ϕ_h has degree zero, F_h has degree one, and $h(D)F_h = \Theta\phi_h$. The degree- $(k-1)$ tensor primitive is exactly

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_h \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes a further sign $(-1)^{i-1}$ when it acts on ϕ_h , because the preceding $i - 1$ factors have degree one. The two signs cancel, so its differential is precisely $\mathcal{V}(\chi Q)$. Applying the same half-density and phase maps gives precisely (HT13).

Set

$$U = RG^{-1/2}, \quad T = \mathcal{B}G^{-1/2}, \quad Z = U^*T, \quad T_\perp = (I - UU^*)T.$$

Then the following exact energy identity, with no omitted source term, holds:

$$\boxed{\text{Tr}(G^{-1}\mathcal{B}^*\mathcal{B}) = \frac{\varepsilon^2}{2} + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2 \geq \frac{\mathcal{L}_{h,k}^2}{2} \geq \frac{\delta^2 k^2 q^2}{8}.} \quad (\text{HT14})$$

All quantities on the left are calculated from g/h and its original theta representatives. In particular the same estimate holds after integration with the original Fourier inversion measure:

$$\int_0^{2\pi} \text{Tr}(G_{N,\theta}^{-1}\mathcal{B}_{N,\theta}^*\mathcal{B}_{N,\theta}) \frac{d\theta}{2\pi} \geq \frac{\mathcal{L}_{h,k}^2}{2}. \quad (\text{HT15})$$

Proof. The two arguments in integration by parts have the same boundary multiplier $e^{-i\theta}$, so the endpoint products cancel. Hence $D_{L,\theta}^* + D_{L,\theta} = 2cI$ on the stated domain. Inserting $D_{L,\theta}R = RA + \mathcal{B}$ and $R^*R = G$ gives

$$A^*G + GA - kG = -(R^*\mathcal{B} + \mathcal{B}^*R), \quad H = -Z - Z^*.$$

Since $U^*U = I$, the decomposition $T = UZ + T_\perp$ is orthogonal in Hilbert–Schmidt norm. The Hermitian and skew-Hermitian parts of Z are orthogonal for the real Hilbert–Schmidt scalar product. Therefore

$$\|T\|_{\text{HS}}^2 = \frac{1}{4}\|H\|_{\text{HS}}^2 + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2.$$

By (HT9), $\|H\|_{\text{HS}}^2 = 2\varepsilon^2$, which proves (HT14). For fixed k, L, N the weighted source sums and their derivatives converge uniformly in θ . Thus $M_{N,\theta}$ is continuous. Its smallest eigenvalue is positive on the compact phase circle, so its inverse, $G_{N,\theta}$, and the boundary columns are continuous there. Integration of (HT14) proves (HT15). $\square$

162.4 The full lattice test for the original Laplacian boundary

Put $\mathcal{L}_{L,\theta} = D_{L,\theta}^2 - kD_{L,\theta} = \partial_r^2 - c^2$. Direct substitution of (HT13) gives the exact map

$$\mathcal{L}_{L,\theta}R - R(A^2 - kA) = (D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A. \quad (\text{HT16})$$

If $Av = \lambda v$ with $v \neq 0$, write $\lambda = c + x + iy$. Then

$$\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq \min_{n \in \mathbb{Z}} \sqrt{(u_n^2 + x^2 - y^2)^2 + 4x^2y^2}. \quad (\text{HT17})$$

In particular every phase and every admitted degree satisfy, on an eigenvector for $\lambda = k(1/2 + \delta + i\gamma)$,

$$\boxed{\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq 2k^2\delta\gamma.} \quad (\text{HT18})$$

For an eigenvalue with $x \neq 0$ and $y = 0$, (HT17) also gives the positive lower bound $x^2 + \min_n u_n^2$, even though its imaginary part is zero.

Proof. The Fourier basis $L^{-1/2}e^{-iu_nr}$ diagonalizes the quasi-periodic Laplacian with eigenvalues $-u_n^2 - c^2$, each acting identically on $\mathcal{K}$. The source vectors lie in its domain by the original weighted differentiability estimates. Writing Rv in that basis and using Parseval gives

$$\|(\mathcal{L}_{L,\theta} - (\lambda^2 - k\lambda))Rv\|^2 = \sum_n ((u_n^2 + x^2 - y^2)^2 + 4x^2y^2) \|\widehat{Rv}(n)\|_{\mathcal{K}}^2.$$

The lattice is discrete and the expression tends to infinity as $|n| \rightarrow \infty$, so its infimum is a minimum. Also R is injective because $R^*R = G \succ 0$. Divide the displayed identity by $\|Rv\|^2$; (HT16) proves (HT17). The extreme primary factor in (HT2) is nonzero and has an eigenvector, so substituting $x = k\delta$, $y = k\gamma$ proves (HT18). If $y = 0$, the square root in (HT17) is $u_n^2 + x^2$. $\square$

162.5 What these computations decide

The original off-line counterfactual survives passage to every finite holonomy quotient only with the explicit boundary energy (HT14) and Laplacian residual (HT17). These quantities belong to the original arithmetic source. The statement is compatible with the exact all-phase identity $\int M_{N,\theta} d\theta/(2\pi) = M_N$ and with the original small gluing-norm estimate. That estimate concerns $C_N - C_{N,\theta}$; (HT13) applies S and subtracts right multiplication by A , preserving a different, explicitly displayed source quantity. There is no claim here that averaging the $G_{N,\theta}$ preserves the rank-two identity (HT11): the lower bound (HT15) integrates the proved phasewise energies themselves.

The transcript's threshold-four volume lower bound was already present before its final Deligne comparison. Its last counterexamples retain many generic identities but do not satisfy the actual divisor condition $h \mid 2\xi$. Consequently they disprove the reduced generic inference stated there; they do not disprove the entire arithmetic programme or provide an RH counterexample. Equations (HT9), (HT14), and (HT17) give further quantified consequences within the original programme without replacing its theta metric or its represented relations.

163 Actual bounds for the retained four-window costs

Continue with (HT1)–(HT18), now with $k \geq 3$. All unadorned source norms and quotient volumes in this section are those of the original $m_{h,k}(u) du$, with $g = 2\xi$ and the complete hypothetical quartet. Write $\mathcal{L} = \mathcal{L}_{h,k}$ and retain $q = \ell_k(k+1)^2$. The threshold-four lower bound is inherited from *Consecutive identities and the first window*, (CJ.9)–(CJ.19); it is already explicitly present in the recovered transcript at lines 4202–4228. It is not a new claim of the present continuation.

163.1 Every term of the inherited lower bound

Put $d_j = V_j/V_{j-1}$ for $j \geq q$. For the two original windows $I_- = (q-1, 2q-1)$ and $I_+ = (q, 2q)$, retain the contraction penalties

$$\begin{aligned} P_- &= -\log(1 - d_{2q-1}) - 2 \sum_{j=q}^{2q-2} \log(1 - d_j), \\ P_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - d_j). \end{aligned} \tag{HC1}$$

The source identities prove $0 < d_j < 1$ in these ranges. The real phase ϕ_N is defined by the retained source pairing $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. The first window uses its original boundary identity; no inverse at degree $q-2$ occurs. Define weights

$$w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q-2), \quad \sum_{N=q-1}^{2q-1} w_N = 2q.$$

Adding the two exact source products gives

$$\mathcal{B}_k := \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + P_- + P_+. \quad (\text{HC2})$$

Here every $\varepsilon_N \geq \mathcal{L}$, including the first degree.

For clarity, retain all constants in the inherited comparable-window bound. With $\alpha = \pi/2$, $B = 42 + 8m$, and $\vartheta_h = \int_{-1}^1 w_h$, put

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ X &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{X}/b, \\ C_h^{\text{win}} &= 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2, \quad D_h = \frac{\delta}{2C_h^{\text{win}}}. \end{aligned} \quad (\text{HC3})$$

Here $0 < b < \pi/2$, and $c_h > 0$ is the original proved convolution constant. The source proves $(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{win}} n$ for $n \geq k \geq 3$, $n/2 \leq r \leq 2n$. Thus it applies to both literal windows, and $0 < D_h < 1/108$.

The exact residual in (HC2) has the following five nonnegative pieces:

$$\begin{aligned} \mathcal{R}_k &= P_- + P_+ + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) \\ &\quad + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} \\ &\quad + 4q \log(C_h^{\text{win}} q) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + 4q \log \frac{2\mathcal{L}}{\delta k q}. \end{aligned} \quad (\text{HC4})$$

Both sums use exactly $q-1 \leq N \leq 2q-1$. Nonnegativity follows, respectively, from $0 < d_j < 1$, the squared phases, (HT9), the two comparable-window estimates, and $\mathcal{L}/q \geq \delta k/2$. Expanding each logarithm in (HC4), with $D_h = \delta/(2C_h^{\text{win}})$, proves the exact identity

$$\boxed{\mathcal{B}_k = 4q \log(D_h k) + \mathcal{R}_k, \quad \mathcal{R}_k \geq 0.} \quad (\text{HC5})$$

This retains, rather than discards, all the contraction, phase, norm, and trace savings in the original lower bound.

163.2 A proved upper bound for these same four original volumes

The original convolution estimate and Laplace moments are

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bu} w_h(u) du.$$

Write $\mu_h = \int_{\mathbb{R}} w_h(u) du$ for the original source mass. Put

$$\begin{aligned} R_0 &= \sqrt{\delta^2 + \gamma^2}, \quad D = \max(1, 2/b, 2R_0), \quad T = Dq, \quad r = kR_0/T \leq 1/2, \\ a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B}, \\ M_k &= M_h(b)^k + M_h(-b)^k, \quad U_k(T) = \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{HC6})$$

The capital D in the interval scale is distinct from the inherited lower constant D_h and from the source differential. All masses, coordinates, and root multiplicities remain as in (HT1)–(HT7). One has the actual finite Hermitian comparisons

$$G_{q-1} \preceq C_{2q-1} G_{2q-1}, \quad G_q \preceq G_{q-1} \preceq C_{2q} G_{2q}.$$

Consequently, with the exact ratio

$$\rho_q = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2,$$

$$\boxed{0 \leq \mathcal{B}_k \leq q \log C_{2q-1} + q \log C_{2q} = 2q \log C_{2q-1} + q \log \rho_q.} \quad (\text{HC7})$$

Proof of the degree extension and its maps. Use the exact substitution $f(x) = P(c + iTx)$ with inverse $P(S) = f((S-c)/(iT))$. Its source measure is $Tm_{h,k}(Tx) dx$, with mass μ_h^k , and its monic relation is $\chi_T(x) = (iT)^{-q} \chi(c + iTx)$. The roots $z_1, \dots, z_q$, listed with their repetitions, have modulus at most r . The raw jet of order d acquires exactly $(iT)^d$; the quotient determinant acquires $T^{-q(q-1)}$. This common factor cancels in each displayed quotient-volume ratio.

The Legendre expansion on $[-1, 1]$ gives $\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}$ for $\deg f \leq N$. Indeed Rodrigues' polynomial has coefficient norm at most $\binom{2j}{j} \leq 4^j$ and squared integral $2/(2j+1)$. Pairing the expansion with this polynomial bounds its coefficient by $\sqrt{(2j+1)/2} \|f\|_2$; summing the geometric series gives exactly A_N in (HC6).

Write $\prod_a (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j$. Geometric-series multiplication gives $|h_j| \leq \binom{q+j-1}{j} r^j$ and $\|\chi_T\|_{\text{coeff},1} \leq (1+r)^q$. For every integer $s \geq 0$, monic division of x^{q+s} has quotient $\sum_{j=0}^s h_j x^{s-j}$. The inverse-series identity cancels all other terms of degree at least q . Removing the leading coefficient one therefore gives

$$\|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} \leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq B_q(r).$$

The bound is valid also for $s = q$, the extra degree required here. At degrees below q the remainder is the original monomial, whose coefficient norm is one. Linearity gives the same remainder-map bound for every degree at most $2q$.

For a remainder v of degree below q ,

$$\int |v(u/T)|^2 m_{h,k}(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2.$$

This follows by bounding each monomial by $\max(1, |u/T|^{q-1})$ and using $\int u^{2q-2} m_{h,k} \leq (2q-2)! b^{-(2q-2)} M_k$. Restriction of the original norm to $[-T, T]$ gives the lower bound $T a_k(T) \|f\|_{L^2([-1,1])}^2$. Combining these three estimates proves

$$\|\text{rem}_\chi P\|_{H_{q-1}}^2 \leq C_N \|P\|_{H_N}^2 \quad (N = 2q-1, 2q).$$

The map is the literal remainder representative of the original jet. Apply it to the canonical least lift of $x \in E$ at degree N . Its norm at degree $q-1$ is at least $x^* G_{q-1} x$; this proves the Hermitian comparisons preceding (HC7). Inclusion of polynomial degrees gives $G_q \preceq G_{q-1}$ and monotonicity of all volumes. Taking determinants proves (HC7). Applying the same remainder inequality to any nonzero polynomial below degree q proves $C_N \geq 1$ without an additional assumption. $\square$

For an entirely scalar upper bound retain the original constants

$$K = \max(1, \vartheta_h^{-1}), \quad C_{\text{env}} = \max(1, 2/(3c_h D)), \quad a = \alpha D + \log 256,$$

and the original expression (AU.31)

$$\mathcal{U}_k = aq^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}.$$

Then the proved four-window upper bound is

$$\boxed{\mathcal{B}_k \leq \widehat{\mathcal{U}}_k := 2\mathcal{U}_k + q \log 20.} \quad (\text{HC8})$$

To verify the constants, $bT \geq 2q$ implies the factorial ratio in (HC6) is at most one; also $\mu_h \leq X$ and $M_k \leq 2X^k$, so $U_k(T) \leq 3X^k$. Directly from (HC6),

$$A_{2q-1}^2 \leq (2q/9)256^q, \quad A_{2q}^2 \leq (40q/9)256^q.$$

The logarithmic derivative $2/(1-r^2) \leq 8/3$ on $[0, 1/2]$ gives $\log B_q(r)^2 \leq 16kR_0/(3D)$. Substitution into C_N yields $q \log C_{2q-1} \leq \mathcal{U}_k$ and $q \log C_{2q} \leq \mathcal{U}_k + q \log 20$. Adding proves (HC8). This argument does not use the false inequality $\rho_2 < 20$; in the actual quartet domain $q \geq 16$ anyway.

163.3 The exact compatible interval for the signed arithmetic correction

Let V_N^Γ denote the inherited Gamma-reference quotient volume for the *same* original polynomial χ , and define

$$T_N = V_N/V_N^\Gamma, \quad \mathcal{B}_k^\Gamma = \log \frac{V_{q-1}^\Gamma V_q^\Gamma}{V_{2q-1}^\Gamma V_{2q}^\Gamma}, \quad \Delta_k^\Gamma = \log \frac{T_{q-1} T_q}{T_{2q-1} T_{2q}}.$$

Cancellation of the four literal volume factors gives $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$. No auxiliary polynomial is introduced and no sign is imposed on Δ_k^Γ . From (HC5) and (HC8) the actual retained residuals must belong to the following explicit intervals. Let $\mathfrak{C}_*$ be (AT18a) on the same arithmetic and tensor Gamma moment matrices. Its proof uses the original full packet and applies to exactly these four determinants, so $|\Delta_k^\Gamma| \leq J_* \leq \mathfrak{C}_*$. Write $T_k = 4q \log(D_h k)$ for this displayed scalar only:

$$\boxed{\max\{0, -T_k, \mathfrak{b}_*^- - T_k\} \leq \mathcal{R}_k \leq \min\{\widehat{\mathcal{U}}_k, \mathfrak{b}_*^+\} - T_k,} \quad (\text{HC9})$$

$$\boxed{\max\{\mathfrak{d}_*^-, \max\{0, 4q \log(D_h k)\} - \mathcal{B}_k^\Gamma\} \leq \Delta_k^\Gamma \leq \min\{\mathfrak{d}_*^+, \widehat{\mathcal{U}}_k - \mathcal{B}_k^\Gamma\}.} \quad (\text{HC10})$$

The new entries follow by substituting $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$ and $\mathcal{R}_k = \mathcal{B}_k - T_k$ into (AT18)–(AT19). Here $\mathfrak{b}_*^\pm, \mathfrak{d}_*^\pm$ are the simultaneous original Gamma endpoints (AT21e); its complete actual moment proof supplies both the additive and nonlinear coordinate. Their intersection width has not been evaluated for an actual offcritical packet. Removing the simultaneous (AT21e) endpoint entries $\mathfrak{b}_*^\pm$ and $\mathfrak{d}_*^\pm$ gives the preceding coarse intervals, whose width is strictly positive for every $k \geq 3$. Indeed all nonquadratic terms of $\mathcal{U}_k$ are nonnegative, $a > 1$, $q \geq (k+1)^2 > 2k$, and $D_h < 1$. Thus

$$\widehat{\mathcal{U}}_k \geq 2aq^2 > 4qk \geq \max\{0, 4q \log(D_h k)\}.$$

Moreover, keeping the fixed packet constants, each lower-order term divided by q^2 tends to zero, so

$$\lim_{k \rightarrow \infty} \frac{\widehat{\mathcal{U}}_k - \max\{0, 4q \log(D_h k)\}}{q^2} = 2(\alpha D + \log 256) > 0. \quad (\text{HC11})$$

This proves numerical compatibility of those coarse bound functions. It does not prove compatibility of the refined (HC9)–(HC10), which retain the source-dependent (AT21e) endpoints computed from J_* and I_* . It does not construct a packet realizing an arbitrarily chosen point of either interval, and it does not assert the existence of the hypothetical zero. In particular the interval may lie entirely at negative values of Δ_k^Γ ; determinant-ratio positivity does not imply positivity of its signed four-window logarithm.

The constructed signed refinement on this identical source. The exact coordinate map to (ACM1)–(ACM5) is $y = u$, $x = t$, with $S = k/2 + iu$ and every ascending coefficient unchanged. The reference density (ACM2) is exactly the tensor Gamma density of (AT4), including its mass $(\sqrt{2\pi})^k$; the arithmetic density is the original $m_{h,k} = w_h^{*k}$. The relation map in the common source is $I_N B_N^{\text{AW}}$ in both presentations. Consequently the source Grams, their quotient minima, and all four signed determinants agree. In particular $F(0) = \mathcal{B}_k^\Gamma$ and $F(1) = \mathcal{B}_k$ for the F of (ACM14)–(ACM16).

Here is the exact dictionary for all pointwise controls:

$$(S_{\text{AW}}, T_q, S_{\text{SP}}, S_q, A_q) = (S_{\text{spec}}, S_{\text{tr}}, S_{\text{ov}}, S_{\text{HS}}, S_{\text{ang}}), \quad \mathcal{H}_{2q-1} = H_{2q-1}.$$

The original crossed-pair proof (AT21b) gives $S_{\text{tr}} \leq S_{\text{sin}} \leq S_{\text{ang}}$, and the original overlap proof gives $S_{\text{tr}} \leq S_{\text{ov}}$. Therefore the full pointwise minima in (ACM11a) and (AT21a)–(AT21c) are identical, as are those minima divided by $(2q-1)\sqrt{1-e^{-F(x)/(2q-1)}}$. Thus $J_q = J_*$ and $I_q = I_*$ on this actual path. These equalities retain both complete angle lists, including their designated and additional zero entries. They compare the calculated expressions without removing any source vector or changing either endpoint form.

For every finite-series degree $p \geq 0$, let $j_{p,\Gamma}$, $e_{p,\Gamma}$ be the exact signed center and residual constructed in (ACM15) on this Gamma-to-arithmetic path. Their actual finite geometric identity and centered Hilbert–Schmidt proof give $j_{p,\Gamma} - e_{p,\Gamma} \leq \Delta_k^\Gamma \leq j_{p,\Gamma} + e_{p,\Gamma}$. The explicit residual in (ACM15) tends to zero for this fixed source pair, with every eigenvalue and the factor $8q-6$ retained. Substituting the preceding dictionary into (ACM16) gives exactly

$$\begin{aligned} \beta_{p,\Gamma}^- &= \max\{\mathfrak{b}_*^-, \mathcal{B}_k^\Gamma + j_{p,\Gamma} - e_{p,\Gamma}\}, \\ \beta_{p,\Gamma}^+ &= \min\{\mathfrak{b}_*^+, \mathcal{B}_k^\Gamma + j_{p,\Gamma} + e_{p,\Gamma}\}, \\ \mathfrak{b}_*^- &\leq \beta_{p,\Gamma}^- \leq \mathcal{B}_k \leq \beta_{p,\Gamma}^+ \leq \mathfrak{b}_*^+. \end{aligned} \tag{HC9a}$$

Every lower entry is at most the same actual $\mathcal{B}_k$ and every upper entry is at least it, proving the displayed intersection and its order. Their signed entries also prove $0 \leq \beta_{p,\Gamma}^+ - \beta_{p,\Gamma}^- \leq 2e_{p,\Gamma}$. Indeed the upper endpoint is at most $\mathcal{B}_k^\Gamma + j_{p,\Gamma} + e_{p,\Gamma}$ and the lower endpoint is at least $\mathcal{B}_k^\Gamma + j_{p,\Gamma} - e_{p,\Gamma}$. Thus both endpoints converge to the actual $\mathcal{B}_k$ on this fixed source as $p \rightarrow \infty$; no nesting in p is required. Inserting (HC5) and subtracting the same original Gamma value yields the stronger intervals

$$\begin{aligned} \max\{0, -T_k, \beta_{p,\Gamma}^- - T_k\} &\leq \mathcal{R}_k \leq \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\} - T_k, \\ \max\{0, T_k, \beta_{p,\Gamma}^- - T_k\} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \\ &\leq \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\} - \mathcal{B}_k^\Gamma. \end{aligned} \tag{HC10a}$$

The first line uses $\mathcal{R}_k = \mathcal{B}_k - T_k \geq 0$, and the last two use $\Delta_k^\Gamma = \mathcal{B}_k - \mathcal{B}_k^\Gamma$. The nesting in (HC9a), together with $\mathfrak{d}_*^\pm = \mathfrak{b}_*^\pm - \mathcal{B}_k^\Gamma$, proves that these intervals lie inside (HC9)–(HC10). Those original bounds and their full proofs are consequently retained corollaries. Removing all source-dependent endpoint entries gives the original coarse intervals in (HC11); its positive width is not an evaluation of the refined width. All signed centers here are the specified exact integrals. Any numerical evaluation must retain its integration error; fixed-source convergence in p supplies no uniform estimate in k .

The graph calculation carried back into these arithmetic intervals. On the additional full-quartet domain $\gamma > 2$ of (ACM17)–(ACM27), retain $k \geq 3$ as in this section. The coefficients, polynomial χ , original $v_h = 2\xi/h$ and quotient maps are identified there literally with (HT1)–(HT3). The graph computation therefore controls these same arithmetic quantities by $\mathcal{B}_k = \widehat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$. For any specified integers $p, s \geq 0$, define

$$\begin{aligned} b_{p,s}^{\text{gr},-} &= \max\{\beta_{p,\Gamma}^-, \widehat{\mathcal{B}}_k - g_{s,\text{gr}}^+\}, \\ b_{p,s}^{\text{gr},+} &= \min\{\beta_{p,\Gamma}^+, \widehat{\mathcal{B}}_k - g_{s,\text{gr}}^-\}. \end{aligned} \tag{HC9g}$$

The $\beta_{p,\Gamma}^\pm$ retain the original Gamma-to-arithmetic path; $g_{s,\text{gr}}^\pm$ are the separately computed arithmetic-to-graph endpoints (ACM27). Subtracting the two bounds on Γ_k^{gr} from this same $\hat{\mathcal{B}}_k$ gives

$$\beta_{p,\Gamma}^- \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq \beta_{p,\Gamma}^+.$$

Inserting this exact intersection at the use sites (HC9)–(HC10) and keeping their original $T_k = 4q \log(D_h k)$ proves

$$\begin{aligned} \max\{0, -T_k, b_{p,s}^{\text{gr},-} - T_k\} &\leq \mathcal{R}_k \leq \min\{\hat{\mathcal{U}}_k, b_{p,s}^{\text{gr},+}\} - T_k, \\ \max\{0, T_k, b_{p,s}^{\text{gr},-}\} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \\ &\leq \min\{\hat{\mathcal{U}}_k, b_{p,s}^{\text{gr},+}\} - \mathcal{B}_k^\Gamma. \end{aligned} \tag{HC10g}$$

Indeed $\mathcal{R}_k = \mathcal{B}_k - T_k \geq 0$ by (HC5), $\mathcal{B}_k \geq 0$ by (HC7), and $\Delta_k^\Gamma = \mathcal{B}_k - \mathcal{B}_k^\Gamma$ by its definition. Subtracting these unchanged constants gives every entry and proves nesting into the preceding intervals. The additional interval width obeys

$$0 \leq b_{p,s}^{\text{gr},+} - b_{p,s}^{\text{gr},-} \leq \min\{2e_{p,\Gamma}, 2e_{s,\text{gr}}\}. \tag{HC10h}$$

For the first bound, (ACM16) places the two Gamma endpoints inside the interval centered at $\mathcal{B}_k^\Gamma + j_{p,\Gamma}$ with radius $e_{p,\Gamma}$. For the second, (ACM27) places the graph-corrected endpoints inside the interval centered at $\hat{\mathcal{B}}_k - j_{s,\text{gr}}$ with radius $e_{s,\text{gr}}$. Taking an intersection cannot increase either width. These are actual finite-source enclosures. Their fixed-source residual limits do not estimate the separate limit in the growing tensor order. All bounds preceding this paragraph retain their original domain $\gamma > 0$; (HC9g)–(HC10h) assert the additional calculated refinement on the explicitly stated graph-source domain.

The original relation-row enclosure at these same use sites. Retain every equation (HC9g)–(HC10h), and keep its domain $k \geq 3$, $\gamma > 2$, $0 < \delta < 1/2$, $m \geq 1$. Equations (ACM28)–(ACM31) compute the additional lower and upper values $L_k^{\text{rel}}, U_{k,2q}^{\text{rel}}$ from the full monic relation rows and the original finite arithmetic graph Grams. Here d_j^{rel} is a relation polynomial as defined there; the original $d_j = V_j/V_{j-1}$ of (HC1) remains its stated scalar. Likewise the form $T_N^{\text{rel}} := f_N^* f_N$ below retains its superscript and does not rename either $T_N = V_N/V_N^\Gamma$ or the scalar $T_k = 4q \log(D_h k)$. With $g_s^\pm = g_{s,\text{gr}}^\pm$ exactly as in (ACM27), put

$$\begin{aligned} b_{p,s}^{\text{rel},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{rel}} - g_{s,\text{gr}}^+\}, \\ b_{p,s}^{\text{rel},+} &= \min\{\beta_{p,\Gamma}^+, U_{k,2q}^{\text{rel}} - g_{s,\text{gr}}^-\}, \\ \beta_{p,\Gamma}^- &\leq b_{p,s}^{\text{rel},-} \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq b_{p,s}^{\text{rel},+} \leq \beta_{p,\Gamma}^+. \end{aligned} \tag{HC9r}$$

The coefficient identity (ACM17)–(ACM20) identifies the graph and arithmetic remainder frames by the literal identity, including the full primary nilpotents. Consequently (ACM31) bounds the same $\mathcal{B}_k$ as (HC9a). Taking their interval intersection proves the displayed relation enclosure. For its comparison with (HC9g), $L_k^{\text{rel}} \leq \hat{\mathcal{B}}_k$ gives $\max\{\beta^-, L_k^{\text{rel}} - g^+\} \leq \max\{\beta^-, \hat{\mathcal{B}}_k - g^+\}$, and $\hat{\mathcal{B}}_k \leq U_{k,2q}^{\text{rel}}$ gives the corresponding order for the two upper minima. These two elementary monotonicities prove every remaining order in (HC9r). The exact graph-volume interval is contained in the relation-row interval; its intersection with the latter is itself. Thus (HC9g) retains its full exact input, while (HC9r) gives the separately calculated finite row enclosure for that input.

At the two receiving residuals define

$$\begin{aligned} L_{k;p,s}^{\text{ar,rel}} &= \max\{0, T_k, b_{p,s}^{\text{rel},-}\}, \\ U_{k;p,s}^{\text{ar,rel}} &= \min\{\hat{\mathcal{U}}_k, b_{p,s}^{\text{rel},+}\}, \\ L_{k;p,s}^{\text{ar,rel}} - T_k &\leq \mathcal{R}_k \leq U_{k;p,s}^{\text{ar,rel}} - T_k, \\ L_{k;p,s}^{\text{ar,rel}} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \leq U_{k;p,s}^{\text{ar,rel}} - \mathcal{B}_k^\Gamma. \end{aligned} \tag{HC10r}$$

To prove this complete substitution, (HC5) gives $\mathcal{B}_k \geq T_k$ and $\mathcal{R}_k = \mathcal{B}_k - T_k \geq 0$, (HC7)–(HC8) give $0 \leq \mathcal{B}_k \leq \hat{\mathcal{U}}_k$, and (HC9r) gives its third lower and second upper entries. Their intersection contains this same scalar. The two maps are the translations $x \mapsto x - T_k$ and $x \mapsto x - \mathcal{B}_k^\Gamma$, with inverse maps adding the displayed unchanged constant. They are order-preserving bijections of the real line, so they preserve interval widths and give the two complete receiving intervals. The first lower endpoint is exactly $\max\{0, -T_k, b_{p,s}^{\text{rel},-} - T_k\}$. No sign is assigned to $g_{s,\text{gr}}^-$, to T_k , or to Δ_k^Γ before these explicit operations. Monotonicity of the maximum and minimum also proves that (HC10g) is contained in (HC10r), which is contained in (HC10a).

The relation-row width has the precise bound

$$\begin{aligned} 0 \leq b_{p,s}^{\text{rel},+} - b_{p,s}^{\text{rel},-} &\leq \min\{2e_{p,\Gamma}, U_{k,2q}^{\text{rel}} - L_k^{\text{rel}} + g_{s,\text{gr}}^+ - g_{s,\text{gr}}^-\} \\ &\leq \min\{2e_{p,\Gamma}, U_{k,2q}^{\text{rel}} - L_k^{\text{rel}} + 2e_{s,\text{gr}}\}. \end{aligned} \quad (\text{HC10s})$$

The first lower sign follows because the interval contains $\mathcal{B}_k$. Its Gamma width is at most $2e_{p,\Gamma}$ by (HC9a). Its other interval has lower endpoint $L_k^{\text{rel}} - g^+$ and upper endpoint $U_{k,2q}^{\text{rel}} - g^-$, so direct subtraction gives the second entry. Equation (ACM27) bounds $g^+ - g^-$ by $2e_{s,\text{gr}}$. Taking an intersection cannot increase either of these two widths, which proves the stated order. The complete negative mixed squared norm in (ACM30) reduces the row upper comparison relative to its K_{low}^{-1} term. It does not alter the exact-graph nesting just proved. All fixed-source limits in p and s retain their stated fixed h, k domain.

The full correlation calculation in the retained signed intervals. Keep the exact row subsets $J(j)$ of (ACM35), on the same receiving domain as (HC9r). Their calculated bounds now give

$$\begin{aligned} b_{p,s}^{\text{corr},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{corr}}(J) - g_{s,\text{gr}}^+\}, \\ b_{p,s}^{\text{corr},+} &= \min\{\beta_{p,\Gamma}^+, U_k^{\text{corr}}(J) - g_{s,\text{gr}}^-\}, \\ b_{p,s}^{\text{rel},-} &\leq b_{p,s}^{\text{corr},-} \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq b_{p,s}^{\text{corr},+} \leq b_{p,s}^{\text{rel},+}. \end{aligned} \quad (\text{HC9rc})$$

The exact identity $\mathcal{B}_k = \hat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$ and the proved bounds (ACM27), (ACM35) show that the interval contains $\mathcal{B}_k$. Comparing the arguments of the lower maximum and upper minimum, using each inequality in (ACM35), proves the complete order in the last line. All later rows included gives $L_k^{\text{corr}} = U_k^{\text{corr}} = \hat{\mathcal{B}}_k$, so both endpoints become exactly (HC9g). For any specified subsets their complete width obeys

$$\begin{aligned} 0 \leq b_{p,s}^{\text{corr},+} - b_{p,s}^{\text{corr},-} &\leq \min\{2e_{p,\Gamma}, U_k^{\text{corr}}(J) - L_k^{\text{corr}}(J) + g_{s,\text{gr}}^+ - g_{s,\text{gr}}^-\} \\ &\leq \min\{2e_{p,\Gamma}, U_k^{\text{corr}}(J) - L_k^{\text{corr}}(J) + 2e_{s,\text{gr}}\}. \end{aligned} \quad (\text{HC10rcw})$$

The Gamma interval lies in its radius- $e_{p,\Gamma}$ signed interval, and the graph correction lies in its radius- $e_{s,\text{gr}}$ signed interval by their actual finite remainder proofs. Taking an intersection cannot increase either width. Subtracting the displayed graph lower endpoint from its upper endpoint gives the middle expression, proving every term of this bound.

The complete residual substitution at (HC9)–(HC10) is

$$\begin{aligned} L_{k;p,s}^{\text{ar},\text{corr}} &= \max\{0, T_k, b_{p,s}^{\text{corr},-}\}, \\ U_{k;p,s}^{\text{ar},\text{corr}} &= \min\{\hat{\mathcal{U}}_k, b_{p,s}^{\text{corr},+}\}, \\ L_{k;p,s}^{\text{ar},\text{corr}} - T_k &\leq \mathcal{R}_k \leq U_{k;p,s}^{\text{ar},\text{corr}} - T_k, \\ L_{k;p,s}^{\text{ar},\text{corr}} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \leq U_{k;p,s}^{\text{ar},\text{corr}} - \mathcal{B}_k^\Gamma. \end{aligned} \quad (\text{HC10rc})$$

Indeed intersect (HC9rc) with the original inequalities $\mathcal{B}_k \geq 0$, $\mathcal{B}_k \geq T_k$ and $\mathcal{B}_k \leq \hat{\mathcal{U}}_k$ proved in (HC5), (HC7), (HC8). The order-preserving translations already proved in (HC10r) give the two displayed receiving intervals with their exact inverse additions. Their widths do not exceed (HC10rcw).

The inclusion of intervals is now exact-graph inside correlation inside relation-row inside Gamma; monotonicity of the maximum, minimum and the two translations proves the same inclusion at both residuals. All original lower-bound pieces and all negative signed corrections remain present.

163.4 The holonomy energy is tied to the same endpoint contractions

Fix an actual holonomy θ and period L . In this paragraph all metrics, lifts and volumes carry that same phase. Put

$$e_N = \log(V_N/V_{N+1}), \quad \beta_N = \lambda_{\max}(G_{N+1}^{-1/2}G_N G_{N+1}^{-1/2}),$$

$$J_N^D = \|D_{L,\theta}R_N G_N^{-1/2}\|_{\text{HS}}^2, \quad \mathcal{E}_N = \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}}^2.$$

These use original source moments through degree $2N + 2$. They obey

$$\frac{\mathcal{L}^2}{2} \leq \mathcal{E}_N \leq (1 + \sqrt{\beta_N})^2 J_N^D, \quad e_N \geq 2 \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}. \quad (\text{HC12})$$

The denominator is positive: integration by parts on the stated twisted domain gives $\|D_{L,\theta}f\|^2 = c^2\|f\|^2 + \|\partial_r f\|^2$, whence $J_N^D \geq c^2 q > 0$.

To prove the upper inequality, $D_{L,\theta}R_N v$ is an original degree- $(N + 1)$ representative of Av . The same quotient minimum therefore gives

$$A^*G_{N+1}A \preceq R_N^*D_{L,\theta}^*D_{L,\theta}R_N.$$

Use $G_N \preceq \beta_N G_{N+1}$, sandwich by $G_N^{-1/2}$, and take traces. This gives $\|R_N A G_N^{-1/2}\|_{\text{HS}}^2 \leq \beta_N J_N^D$. The triangle inequality in $\mathcal{B}_N = D_{L,\theta}R_N - R_N A$ proves the upper inequality in (HC12); (HT14) proves the lower. All eigenvalues in the definition of β_N are at least one by degree monotonicity. In fact the original rank-one kernel update proves the exact identity

$$e_N = \log \beta_N, \quad \beta_N = 1 + \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}}. \quad (\text{HC12a})$$

Here b_{N+1} and ω_{N+1} are the remainder column and monic squared norm for this same original phase measure. Indeed $K_{N+1} = K_N + b_{N+1}b_{N+1}^*/\omega_{N+1}$, so

$$K_N^{-1/2}K_{N+1}K_N^{-1/2} = I + vv^*, \quad v = K_N^{-1/2}b_{N+1}/\sqrt{\omega_{N+1}}.$$

The relative matrix defining β_N is $K_{N+1}^{1/2}K_N^{-1}K_{N+1}^{1/2} = X^*X$ for the invertible $X = K_N^{-1/2}K_{N+1}^{1/2}$. The displayed matrix $I + vv^*$ is XX^* , and $X(X^*X)X^{-1} = XX^*$ proves they have identical eigenvalues. These are 1 with multiplicity at least $q - 1$ and $1 + \|v\|^2$. Taking the determinant gives $V_N/V_{N+1} = \det K_{N+1}/\det K_N = 1 + \|v\|^2 = \beta_N$, which proves (HC12a), including a zero update if one occurs. Rearranging the two energy bounds now proves the second inequality of (HC12) with this exact endpoint identity.

The same weights as in (HC2) reconstruct the actual phase volume:

$$\mathcal{B}_{k,\theta} = \sum_{N=q-1}^{2q-1} w_N e_N \geq 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}. \quad (\text{HC13})$$

Thus a small original derivative-source energy at a degree has the explicit endpoint contraction cost (HC12); that cost enters the same four-window budget, with its actual multiplicity.

Finally the supplied holonomy construction allows any fixed $0 < \eta < 1$ by choosing the original period from its computed weighted source comparison at degree $2q + 2$. It proves simultaneously

$(1 - \eta)G_N \preceq G_{N,\theta} \preceq (1 + \eta)G_N$ at all these degrees. For completeness, apply the comparison before quotient minimization on the common coefficient space $H = \mathcal{P}_{2q}$. The supplied weighted source estimate, written explicitly in (PSC.33), is $(1 - \eta)M_{2q} \preceq M_{2q,\theta} \preceq (1 + \eta)M_{2q}$. Set $H_\theta(t) = (1 - t)M_{2q} + tM_{2q,\theta}$, $C^\theta = H_\theta(t)^{-1}(M_{2q,\theta} - M_{2q})$, and define P_N^θ, Q_N^θ by (AT11) with $M_* = H_\theta(t)$ and the unchanged inclusions I_N, B_N . Then $U^\theta = Q_{2q-1}^\theta + Q_{2q}^\theta - Q_q^\theta$ and $W^\theta = P_{2q-1}^\theta - P_{q-1}^\theta + P_{2q}^\theta - P_q^\theta$. Substitution proves all projector identities and the exact signed trace formula (AT15) on this distinct interpolation. Both positive forms have the same original remainder fibres. Thus (AT18a)–(AT21) defines its own control $\mathfrak{C}_\theta$ and proves

$$\begin{aligned} |\mathcal{B}_{k,\theta} - \mathcal{B}_k| &\leq J_\theta \leq \mathfrak{C}_\theta := \int_0^1 \min\{S_{\text{spec}}^\theta(t), \tfrac{1}{2}d_\theta(t)\|U^\theta - W^\theta\|_{1,H_\theta(t)}\} dt \\ &\leq (2q - 1) \log \frac{1 + \eta}{1 - \eta}. \end{aligned} \quad (\text{HC14})$$

Indeed every eigenvalue of $M_{2q}^{-1}M_{2q,\theta}$ lies in $[1 - \eta, 1 + \eta]$ by the source sandwich, and the full spectra and trace norms in the finite AT proof are on the displayed polynomial flags. This is an explicit source comparison; taking determinants only after quotient minimization gives the preceding $2q$ allowance as a coarser consequence. The interpolated forms are not identified with either Gamma reference. On these same phase-source operators define S_{HS}^θ by the exact centered trace formula (AT21a), and $J_\theta = \int_0^1 \min\{S_{\text{spec}}^\theta, S_{\text{tr}}^\theta, S_{\text{HS}}^\theta\} dt$. The same finite Hermitian Cauchy–Schwarz proof gives the first entry in (HC14); hence $J_\theta \leq \mathfrak{C}_\theta$. The positive atomic phase measure also meets the moment hypothesis proved above. Define I_θ by (AT21a) on the exact path $H_\theta(t)$; its original $B_\theta(t)$ is strictly positive by the same nonzero moment $\int |\chi|^2 d\mu_t$. Put

$$\begin{aligned} \mathfrak{U}_{\text{ar}} &= \min\{\widehat{\mathcal{U}}_k, \mathfrak{b}_*^+\}, \\ \mathfrak{U}_\theta &= \min\{\mathfrak{U}_{\text{ar}} + J_\theta, \Psi_a(H_a(\mathfrak{U}_{\text{ar}}) + I_\theta)\}, \quad a = 2q - 1. \end{aligned} \quad (\text{HC14a})$$

Then (HC13)’s right side is at most $\mathfrak{U}_\theta$. Indeed $0 < \mathcal{B}_k \leq \mathfrak{U}_{\text{ar}}$ by the original upper estimates. Apply (AT21e) on $H_\theta(t)$, then use monotonicity of $\Psi_a \circ (H_a + I_\theta)$ in its positive initial value. This proves both upper entries and their minimum. This substitution uses (AT19) on the original arithmetic form before the explicit source-phase trace calculation; it does not identify the sampled phase metric with the original one.

The original source-phase map and its signed refinement. We now refine the same bound (HC14a), retaining its original $J_\theta, I_\theta, \mathfrak{C}_\theta$ and both upper entries. Let $I_{2q}^{2q+2} : \mathcal{P}_{2q} \hookrightarrow \mathcal{P}_{2q+2}$ be literal coefficient inclusion. The two original forms on $H = \mathcal{P}_{2q}$ are

$$\begin{aligned} M_0^\theta &= (I_{2q}^{2q+2})^* M_{2q+2} I_{2q}^{2q+2} = \int_{\mathbb{R}} v(u)^* v(u) m_{h,k}(u) du, \\ M_1^\theta &= (I_{2q}^{2q+2})^* M_{2q+2,\theta} I_{2q}^{2q+2} = \frac{2\pi}{L} \sum_{r \in \mathbb{Z}} m_{h,k}(u_r) v(u_r)^* v(u_r), \\ v(u) &= (1, c + iu, \dots, (c + iu)^{2q}), \quad c = k/2, \quad u_r = (2\pi r + \theta)/L. \end{aligned} \quad (\text{HC14b})$$

By (HT3)–(HT7), these are exactly the squared norms of $\mathcal{U}_k \mathcal{V}|_H$ and $\mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}|_H$. Here $\mathcal{U}_k$ is the half-density map (HT4), whose source and target include every relative variable; the earlier scalar upper bound with this notation is still the expression in (HC8). The original relative variables, full Taylor unit, coordinate, phase, and factor $2\pi/L$ are present in these two source maps. To specify the period cost without colliding with $a = 2q - 1$, write $a_{\text{wt}} > 0$ for the weighted-source exponent of (PSC.32). Its actual degree- $2q + 2$ cost is

$$\kappa_{a_{\text{wt}}, 2q+2} = \sup_{v \neq 0} \frac{v^*(M_{a_{\text{wt}}, +, 2q+2} + M_{a_{\text{wt}}, -, 2q+2})v}{v^* M_{2q+2} v}.$$

The weighted Grams retain the squared weights $x_k^{\pm 2a_{\text{wt}}}$. Retain the originally fixed period L whenever $\kappa_{a_{\text{wt}}, 2q+2}/(e^{a_{\text{wt}}L} - 1) \leq \eta$. One exact admissible choice is $L = a_{\text{wt}}^{-1} \log(1 + \kappa_{a_{\text{wt}}, 2q+2}/\eta)$; the comparison below holds for every retained period satisfying the displayed inequality. Equation (PSC.33), restricted by I_{2q}^{2q+2} , proves $(1 - \eta)M_0^\theta \preceq M_1^\theta \preceq (1 + \eta)M_0^\theta$. For $q - 1 \leq N \leq 2q$ the remainder J_N is surjective; taking the two infima over its identical nonempty fibre $J_N P = v$ also proves the quotient sandwiches used above.

Put $M_x^\theta = (1 - x)M_0^\theta + xM_1^\theta$ for $0 \leq x \leq 1$. At $x = t$ this is precisely $H_\theta(t)$ in (HC14). Use $I_N : \mathcal{P}_N \rightarrow H$ and the unpadded original relation $B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N, P \mapsto \chi P$. Its included map is $B_N^\theta = I_N B_N^{\text{AW}}$, and the exact projections and operators are

$$\begin{aligned} P_N^\theta &= I_N(I_N^* M_x^\theta I_N)^{-1} I_N^* M_x^\theta, \\ Q_N^\theta &= B_N^\theta((B_N^\theta)^* M_x^\theta B_N^\theta)^{-1} (B_N^\theta)^* M_x^\theta, \\ U_x^\theta &= Q_{2q-1}^\theta + Q_{2q}^\theta - Q_q^\theta, \\ W_x^\theta &= P_{2q-1}^\theta - P_{q-1}^\theta + P_{2q}^\theta - P_q^\theta, \\ C_x^\theta &= (M_x^\theta)^{-1}(M_1^\theta - M_0^\theta), \quad D_\theta = (M_0^\theta)^{-1}M_1^\theta. \end{aligned} \tag{HC14c}$$

At $N = q - 1$ the relation domain is zero and the second formula is the unique zero map $Q_{q-1}^\theta = 0$. These are exactly (ACM4)–(ACM5) after the stated substitution, and exactly the projections already used in (HC14). The source and relation ranges are the same nested polynomial flags, so both U_x^θ, W_x^θ have full spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. Differentiating the actual canonical section as in (ISM18) and taking the four determinant signs gives

$$F'_\theta(x) = \text{Tr}(C_x^\theta(U_x^\theta - W_x^\theta)), \quad F_\theta(0) = \mathcal{B}_k, \quad F_\theta(1) = \mathcal{B}_{k,\theta}.$$

The polynomial χ has positive norm for the continuous measure and for the infinite positive lattice measure (HT6). Their convex mixture has the same property. The original $\chi^{\#_k}$ proof (AT21a)–(AT21c) therefore gives $F_\theta(x) > 0$ on this closed path. In particular every nonlinear denominator is bounded away from zero for this fixed source comparison.

Apply the pointwise dictionary preceding (HC9a) on this path. The additional trace inequalities $S_{\text{tr}}^\theta \leq S_{\text{ov}}^\theta$ and $S_{\text{tr}}^\theta \leq S_{\text{sin}}^\theta \leq S_{\text{ang}}^\theta$ prove that the complete (ACM11a) minima are precisely the J_θ, I_θ already defined above. In particular both complete angle lists, the overlap majorant and the centered Hilbert–Schmidt expression are retained in these identities. Let $j_{p,\theta}, e_{p,\theta}$ be the (ACM15) construction with

$$(M_x, C_x, U_x, W_x, D, F(x)) \mapsto (M_x^\theta, C_x^\theta, U_x^\theta, W_x^\theta, D_\theta, F_\theta(x)).$$

The integer $p \geq 0$ is its finite-series degree, while the original period is still L . If $b_1^\theta \leq \dots \leq b_n^\theta$ are every eigenvalue of D_θ in M_0^θ , $n = 2q + 1$, write

$$\vartheta_\theta = \frac{b_n^\theta - b_1^\theta}{b_n^\theta + b_1^\theta}, \quad K_{D_\theta} = \sum_{j=1}^n \left(\frac{|b_j^\theta - 1|}{\min\{1, b_j^\theta\}} \right)^2.$$

The finite identity $C_x^\theta - (C_x^\theta)^{[p]} = (H_x^{\circ,\theta})^{p+1} C_x^\theta$ is the identical (ACM15) geometric sum on these specified operators. Subtracting the trace mean of this residual leaves its pairing with $U_x^\theta - W_x^\theta$ unchanged. All its factors are real functions of D_θ , hence self-adjoint in M_x^θ . The Hilbert–Schmidt inequality therefore gives

$$\begin{aligned} j_{p,\theta} - e_{p,\theta} &\leq \mathcal{B}_{k,\theta} - \mathcal{B}_k \leq j_{p,\theta} + e_{p,\theta}, \\ 0 \leq e_{p,\theta} &\leq \vartheta_\theta^{p+1} \sqrt{K_{D_\theta}(8q - 6)} \xrightarrow{p \rightarrow \infty} 0. \end{aligned} \tag{HC14d}$$

For completeness the first factor bounds every residual eigenvalue because $\|H_x^{\circ,\theta}\|_{M_x^\theta} \leq \vartheta_\theta < 1$ and $|c_j^\theta(x)| \leq |b_j^\theta - 1|/\min\{1, b_j^\theta\}$. Centering subtracts the nonnegative square of the trace mean. Both

positive operators dominate their range projections, whose ranges intersect in dimension at least one. Thus $\text{Tr}(U_x^\theta W_x^\theta) \geq 1$, and the full spectra give $\text{Tr}((U_x^\theta - W_x^\theta)^2) = 8q - 4 - 2\text{Tr}(U_x^\theta W_x^\theta) \leq 8q - 6$. This proves the displayed constant on the original metric.

The source sandwich places every b_j^θ in $[1 - \eta, 1 + \eta]$. It consequently gives $\vartheta_\theta \leq \eta$ and $K_{D_\theta} \leq (2q+1)\eta^2/(1-\eta)^2$: the first follows by increasing the larger endpoint and decreasing the smaller one in their positive ratio, and each summand of the second has numerator at most η^2 and denominator at least $(1 - \eta)^2$. Hence the explicit further bound is

$$e_{p,\theta} \leq \frac{\eta^{p+2}}{1 - \eta} \sqrt{(2q+1)(8q-6)}.$$

This holds for every phase at each retained period satisfying the displayed original degree- $2q + 2$ source-cost inequality, retaining the complete $q = [1 + k(m-1)](k+1)^2$ dependence. Retaining their entire ordered list, the same signed derivative and the exact derivative of H_a give

$$\begin{aligned} |\mathcal{B}_{k,\theta} - \mathcal{B}_k| &\leq J_\theta \leq \min\{\mathfrak{C}_\theta, \mathcal{L}_q(D_\theta)\} \leq (2q-1) \log \frac{1+\eta}{1-\eta}, \\ |H_{2q-1}(\mathcal{B}_{k,\theta}) - H_{2q-1}(\mathcal{B}_k)| &\leq I_\theta \leq \log \frac{1+\eta}{1-\eta}, \\ \mathcal{L}_q(D_\theta) &= \log \frac{b_{q+2}^\theta}{b_q^\theta} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}^\theta}{b_j^\theta}. \end{aligned} \tag{HC14e}$$

Indeed integration of each ordered $c_j^\theta(x) = (b_j^\theta - 1)/(1 - x + xb_j^\theta)$ gives $\log b_j^\theta$. The total coefficient of the resulting logarithmic ratios is $1 + 2(q-1) = 2q-1$. Dividing the original pointwise minimum by $(2q-1)\sqrt{1 - e^{-F_\theta(x)/(2q-1)}}$ and integrating proves the second line. These are the same additive and nonlinear bounds as (HC14)–(HC14a), now accompanied by the constructed signed center (HC14d).

For any integers $p, r \geq 0$, define

$$\begin{aligned} U_{k,p}^{\text{ar}} &= \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\}, \\ U_{k,\theta;p,r} &= \min\{U_{k,p}^{\text{ar}} + J_\theta, \Psi_{2q-1}(H_{2q-1}(U_{k,p}^{\text{ar}}) + I_\theta), \\ &\quad U_{k,p}^{\text{ar}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \tag{HC14f}$$

The actual $\mathcal{B}_k$ is positive and at most $U_{k,p}^{\text{ar}}$ by (HC9a)–(HC10a). The additive and signed expressions increase with this initial scalar. For the fixed computed I_θ , both H_{2q-1} and Ψ_{2q-1} are increasing on the displayed domains. Applying the three proved phase upper bounds and substituting this initial upper value proves

$$2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} \leq \mathcal{B}_{k,\theta} \leq U_{k,\theta;p,r} \leq \mathfrak{U}_\theta. \tag{HC13a}$$

For the final inequality (HC9a) gives $U_{k,p}^{\text{ar}} \leq \mathfrak{U}_{\text{ar}}$, and monotonicity of the first two upper expressions bounds their minimum by the original (HC14a). The third entry can only lower it. This proof holds the actual phase-path integrals fixed during the scalar comparison; it does not replace their source path. The weights w_N , derivative-source energies J_N^D , and every boundary term in (HC12)–(HC13) remain in their original phase metric. No fixed-source residual limit is promoted to a uniform estimate in the growing tensor order.

The source comparison also retains its exact cochain map. For two minimum sections ρ_0, ρ_1 on this common source, $(\rho_1 - \rho_0)v = \chi Q_v$ by monic division. Successive division in $s_1, \dots, s_k$ gives the original polynomials $Q_{v,i}$ with $\chi(\sum s_i)Q_v(\sum s_i) = \sum_i h(s_i)Q_{v,i}(\mathbf{s})$. With the original $h(D)F_h = \Theta\phi_*$, the actual primitive is

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes the same $(-1)^{i-1}$, so $dK_v = \mathcal{V}((\rho_1 - \rho_0)v)$, as in (ISM38)–(ISM41). For the two-leg complex with differential $(\phi, \hat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\hat{\psi})$, the degree-zero map $j_+(\phi) = (\phi, 0)$ and degree-one identity form a cochain map because their differential is exactly $\Theta\phi$. Tensor these specified maps and use the original coefficient insertions into the fixed full carrier of (ISM42). This defines $\tilde{K}_v$ and preserves its displayed top differential. At a declared support $\lambda = (W, S; A)$, apply these maps coordinatewise. With the original coefficient injections ι_i , the exact top maps and their common inverse-image domain are

$$F_{a,A} : E^{\oplus A} \longrightarrow \mathcal{C}_{\text{full}}^k, \quad F_{a,A}((v_i)_{i \in A}) = \sum_{i \in A} \iota_i \mathcal{V} \rho_a v_i \quad (a = 0, 1),$$

$$E_\lambda = F_{0,A}^{-1}(\mathcal{C}_\lambda^k) \cap F_{1,A}^{-1}(\mathcal{C}_\lambda^k).$$

This is a linear subspace. For $\mathbf{v} = (v_i)_{i \in A} \in E_\lambda$, put $\tilde{K}_\mathbf{v} = \sum_{i \in A} \iota_i j_+^{\otimes k} K_{v_i}$, where the tensor cochain map has the degree-one identities already specified. Its top differential is $F_{1,A}(\mathbf{v}) - F_{0,A}(\mathbf{v})$, which lies in $\mathcal{C}_\lambda^k$ and is a boundary in the full carrier. The exact residual map on this specified domain is

$$E_\lambda \longrightarrow \frac{d\mathcal{C}_{\text{full}}^{k-1} \cap \mathcal{C}_\lambda^k}{d\mathcal{C}_\lambda^{k-1}}, \quad \mathbf{v} \longmapsto [d\tilde{K}_\mathbf{v}]_\lambda.$$

Its inclusion-induced map to the kernel of $\mathcal{C}_\lambda^k / d\mathcal{C}_\lambda^{k-1} \rightarrow \mathcal{C}_{\text{full}}^k / d\mathcal{C}_{\text{full}}^{k-1}$ and its inverse both use the same representative, proving the exact isomorphism (ISM42). The full two-leg degree-zero diagonal and the source images are those of (ISM41)–(ISM43). For independent coefficient masks the linear maps act in each active coordinate; zero insertion commutes because every new coordinate maps 0 to 0 with its original outer label. This calculation retains the proper-source residual and the empty mask's outer label throughout.

The graph refinement in this original phase-energy budget. Retain the domain $\gamma > 2$ of (HC9g), the actual original period and phase above, and every previously calculated $J_\theta, I_\theta, j_{r,\theta}, e_{r,\theta}$. For $p, s, r \geq 0$ put

$$\begin{aligned} U_{k;p,s}^{\text{ar,gr}} &= \min\{\hat{\mathcal{U}}_k, b_{p,s}^{\text{gr,+}}\}, \\ U_{k,\theta;p,s,r}^{\text{gr}} &= \min\{U_{k;p,s}^{\text{ar,gr}} + J_\theta, \\ &\quad \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k;p,s}^{\text{ar,gr}}) + I_\theta), \\ &\quad U_{k;p,s}^{\text{ar,gr}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \tag{HC14g}$$

The original arithmetic moment proof at (ACM11a) gives strict positivity of $\mathcal{B}_k$; (HC10g) and (HC9g) then give $0 < \mathcal{B}_k \leq U_{k;p,s}^{\text{ar,gr}} \leq U_{k,p}^{\text{ar}}$. For each of the three fixed phase controls the upper expression is increasing in its initial scalar: the first and third are affine of slope one, and the second is the composite of the two increasing functions already differentiated in (ACM14). Applying their actual phase inequalities and this scalar comparison proves the propagated bound at (HC13a):

$$\begin{aligned} 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} &\leq \mathcal{B}_{k,\theta} \\ &\leq U_{k,\theta;p,s,r}^{\text{gr}} \leq U_{k,\theta;p,r}. \end{aligned} \tag{HC13g}$$

The original w_N, J_N^D , phase source, twisted derivative domain and boundary residual in (HC12)–(HC13) are unchanged. The graph data enter through the proved signed bound for the initial arithmetic volume. The inequality does not identify the two-component graph source with the phase source.

The relation-row interval propagated into the phase budget. Keep the original phase, circumference, source-path matrices and the exact integrals $J_\theta, I_\theta, j_{r,\theta}, e_{r,\theta}$ from (HC14b)–(HC14g). Use $U_{k;p,s}^{\text{ar,rel}}$ defined in (HC10r), and define the complete three-entry upper value

$$U_{k,\theta;p,s,r}^{\text{rel}} = \min\{U_{k;p,s}^{\text{ar,rel}} + J_\theta, \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k;p,s}^{\text{ar,rel}}) + I_\theta), U_{k;p,s}^{\text{ar,rel}} + j_{r,\theta} + e_{r,\theta}\}. \quad (\text{HC14r})$$

By (HC9r)–(HC10r) and the strict original-moment positivity,

$$0 < \mathcal{B}_k \leq U_{k;p,s}^{\text{ar,gr}} \leq U_{k;p,s}^{\text{ar,rel}} \leq U_{k,p}^{\text{ar}}.$$

For this fixed phase path the first and third functions of an initial scalar x in (HC14r) are affine of slope one. The second is the composite of increasing $\mathcal{H}_{2q-1}$ on $x > 0$ and increasing Ψ_{2q-1} on its nonnegative argument; their derivatives and inverse domains were computed in (ACM14). The actual phase inequalities (HC14d)–(HC14e) use $x = \mathcal{B}_k$. Substituting the larger scalar upper endpoints and applying each monotonicity proves

$$2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} \leq \mathcal{B}_{k,\theta} \leq U_{k,\theta;p,s,r}^{\text{gr}} \leq U_{k,\theta;p,s,r}^{\text{rel}} \leq U_{k,\theta;p,r}. \quad (\text{HC13r})$$

The left inequality is the unchanged original phase-energy identity (HC13), with the complete multiplicities, twisted derivative domain and boundary terms. The comparison holds the actual path controls fixed; the scalar substitution does not change their integration path or measure. It also retains the entire cochain map and proper-source residual following (HC13a): the only new maps here are the two real translations in (HC10r) and these three functions of their proved arithmetic upper endpoint. The exact-graph phase bound (HC13g) remains the smaller one at the same p, s, r , as the displayed inequalities prove.

The correlation interval in the three original phase functions. With the same fixed original phase controls as (HC14r), set

$$U_{k,\theta;p,s,r}^{\text{corr}} = \min\{U_{k;p,s}^{\text{ar,corr}} + J_\theta, \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k;p,s}^{\text{ar,corr}}) + I_\theta), U_{k;p,s}^{\text{ar,corr}} + j_{r,\theta} + e_{r,\theta}\}. \quad (\text{HC14rc})$$

Equation (HC9rc) and the arithmetic minimum in (HC10rc) give $0 < \mathcal{B}_k \leq U_{k;p,s}^{\text{ar,gr}} \leq U_{k;p,s}^{\text{ar,corr}} \leq U_{k;p,s}^{\text{ar,rel}}$. The three increasing functions proved in (HC14r) therefore give

$$2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} \leq \mathcal{B}_{k,\theta} \leq U_{k,\theta;p,s,r}^{\text{gr}} \leq U_{k,\theta;p,s,r}^{\text{corr}} \leq U_{k,\theta;p,s,r}^{\text{rel}} \leq U_{k,\theta;p,r}. \quad (\text{HC13rc})$$

This proves the full propagation into the original phase-energy budget by applying each actual phase inequality before taking their minimum. The source-path integrals, circumference condition, twisted derivative domain, complete weights, boundary terms and proper-source cochain residual are those already proved above. Including all later rows makes the correlation upper bound equal to the exact-graph bound (HC14g) at the same p, s, r . No tensor uniform rate is inferred from this finite exactness statement.

For the averaged original phase metrics the already proved stronger comparison $(1 - \eta^2)G_N \preceq \overline{G}_N \preceq G_N$ gives instead $|\overline{\mathcal{B}}_k - \mathcal{B}_k| \leq 2q \log(1/(1 - \eta^2))$. These changes are of order q for a fixed η , whereas the compatible width (HC11) has positive quadratic order. The exact Laplacian residual (HT17) remains present at every phase. No displayed upper estimate for its norm, or for the original derivative moments in (HC12), contradicts these lower bounds.

The graph-to-average cost and its complete mixed remainder. On the same graph-source domain $\gamma > 2$, fix one original circumference for all four endpoints. The inverse-form statements below use the already proved degree- $2q + 2$ original source comparison above. For the graph map L_{obs} of (ACM18), write L_{circ} for this circumference to avoid a collision. At each $N \in (q - 1, q, 2q - 1, 2q)$ use the original arithmetic minimum section C_N and the relation map B_N of PGB.3. Put $H_N = B_N^* M_N B_N$, $T_N^{\text{def}} = D_N B_N$, $H_{+,N} = H_N + (T_N^{\text{def}})^* T_N^{\text{def}}$, and retain

$$Q_{D,N} = I - T_N^{\text{def}} H_{+,N}^{-1} (T_N^{\text{def}})^*, \quad \mathcal{E}_N^{\text{aug}} = C_N^* D_N^* Q_{D,N} D_N C_N, \quad \mathcal{D}_N^{\text{hol}} = G_N - \overline{G}_N. \quad (\text{HC16})$$

For zero relation dimension the maps through that domain are zero and $Q_{D,N} = I$. PGB.3–PGB.10 proves, by the displayed completed squares and the original phase section difference,

$$\begin{aligned} \widehat{G}_N &= G_N + \mathcal{E}_N^{\text{aug}}, \quad \mathcal{E}_N^{\text{aug}} > 0, \\ \mathcal{D}_N^{\text{hol}} &= \int (C_N - C_{N,\theta})^* M_{N,\theta} (C_N - C_{N,\theta}) \frac{d\theta}{2\pi} \succeq 0, \\ \widehat{G}_N - \overline{G}_N &= \mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}}. \end{aligned} \quad (\text{HC17})$$

The exact map whose Gram is the last difference is

$$x \mapsto \left(Q_{D,N}^{1/2} D_N C_N x, [\theta \mapsto (C_N - C_{N,\theta}) x] \right) \in \mathcal{H}_k \oplus L^2(\Theta; \text{im } B_N, M_{N,\theta} d\theta / (2\pi)). \quad (\text{HC18})$$

Its squared norm is (HC17), and its kernel is zero by the injectivity of $D_N C_N$ proved from the full leading coefficient in PGB.6. The cochain map inducing the identity contraction $(E, \widehat{G}_N) \rightarrow (E, \overline{G}_N)$ sends each polynomial and each relation to its constant phase field. It commutes with the inclusion differential and preserves J_N ; its norm identity is $\int p^* M_{N,\theta} p d\theta / (2\pi) = p^* M_N p \leq p^* \widehat{M}_N p$. Thus the finite comparison is accompanied by its original source map and its complete nonzero metric loss.

To carry this calculation into the four-volume use site define the exact individual determinant costs

$$\begin{aligned} a_N^{\text{aug}} &= \log \frac{\det \widehat{G}_N}{\det G_N} = \log \det(I + G_N^{-1/2} \mathcal{E}_N^{\text{aug}} G_N^{-1/2}) > 0, \\ h_N^{\text{hol}} &= \log \frac{\det G_N}{\det \overline{G}_N} = \log \det(I + \overline{G}_N^{-1/2} \mathcal{D}_N^{\text{hol}} \overline{G}_N^{-1/2}) \geq 0, \\ a_N^{\text{av}} &= \log \frac{\det \widehat{G}_N}{\det \overline{G}_N} = a_N^{\text{aug}} + h_N^{\text{hol}}. \end{aligned} \quad (\text{HC19})$$

These equalities follow by determinant multiplication; no commutation of their positive matrices is required. List the four endpoints as $(N_1, N_2, N_3, N_4) = (q - 1, q, 2q - 1, 2q)$ with $\epsilon = (1, 1, -1, -1)$, retaining repeated entries if they occur. The exact signed identities are

$$\begin{aligned} \Gamma_k^{\text{gr}} &= \sum_{j=1}^4 \epsilon_j a_{N_j}^{\text{aug}}, \\ H_k^{\text{hol}} &:= \sum_{j=1}^4 \epsilon_j h_{N_j}^{\text{hol}} = \mathcal{B}_k - \overline{\mathcal{B}}_k, \\ \widehat{\mathcal{B}}_k - \overline{\mathcal{B}}_k &= \Gamma_k^{\text{gr}} + H_k^{\text{hol}} = \sum_{j=1}^4 \epsilon_j a_{N_j}^{\text{av}}. \end{aligned} \quad (\text{HC20})$$

For the original source sandwich chosen above, $(1 - \eta^2)G_N \preceq \overline{G}_N \preceq G_N$ implies $0 \leq h_N^{\text{hol}} \leq q \log(1/(1 - \eta^2))$. There are two positive and two negative signs, so

$$|H_k^{\text{hol}}| \leq 2q \log \frac{1}{1 - \eta^2}, \quad \left| \widehat{\mathcal{B}}_k - \overline{\mathcal{B}}_k - \Gamma_k^{\text{gr}} \right| \leq 2q \log \frac{1}{1 - \eta^2}. \quad (\text{HC21})$$

This proves exactly where the earlier order- q allowance enters. The augmentation's signed term is retained through (ACM23)–(ACM27), rather than assigned that allowance. The separate positivity of the four costs in (HC19) does not give a sign to their complete signed sum.

The graph volume appearing in (HC9g) also has its full mixed-source expression. Retain $\Phi(1/2 + it) = A + iq\Phi(t)$, the density r_k , function $g_x = z(u)x/r_k(u)$, and maps $Y_\perp$, $\mathcal{J}_k$, $\widehat{R}_N$ exactly as in PGB.14–PGB.22. In particular $\Omega_k = Y_\perp^* Y_\perp > 0$ and $f_N x = \iota \widehat{R}_N x - g_x$, with $\iota P(u) = P(k/2 + iu)$. The complete squared-norm identity PGMB.5 is

$$\begin{aligned} \widehat{G}_N &= K_{\text{low}} + (\mathcal{D}_k^{\text{mix}})^* \mathcal{D}_k^{\text{mix}} + f_N^* f_N = \Omega_k + f_N^* f_N, \\ (\mathcal{D}_k^{\text{mix}} x)(\mathbf{t}) &= \frac{\overline{\tau}_k F_x(\mathbf{s})}{1 + |\tau_k|^2} - g_x(u). \end{aligned} \quad (\text{HC22})$$

The target norm of the second line is the full weighted space $L^2((1 + |\tau_k|^2) \prod_i w_h(t_i) dt_i)$. Thus it retains $1 + k^2 A^2 + (\sum_i q\Phi(t_i))^2$, the conjugate phase and every tensor variable. PGMB.2 evaluates its kernel: it is zero for $k \geq 2$, while the map vanishes for $k = 1$. Its first term has the original realization $L_{\text{low}} x = (\prod_i v_h(s_i)) F_x / \sqrt{1 + |\tau_k|^2}$, with Gram K_{low} . The map $x \mapsto (L_{\text{low}} x, \mathcal{D}_k^{\text{mix}} x, f_N x)$ has precisely the Gram $\widehat{G}_N$ in (HC22), proving the source realization of every term.

Define its finite positive tail costs by

$$t_N = \log \det(I + \Omega_k^{-1/2} f_N^* f_N \Omega_k^{-1/2}). \quad (\text{HC23})$$

Equation (HC22) gives $\log \det \widehat{G}_N = \log \det \Omega_k + t_N$. The minimum over the nested original relation spaces proves $\widehat{G}_{N+1} \preceq \widehat{G}_N$, hence $t_{N+1} \leq t_N$; determinant monotonicity follows by congruence with the smaller positive form and its eigenvalues. PGB.22's affine approximation proves $f_N^* f_N \rightarrow 0$ at fixed h, k , so $t_N \downarrow 0$ in that limit. The actual four endpoints therefore give

$$\begin{aligned} \widehat{\mathcal{B}}_k &= (t_{q-1} - t_{2q-1}) + (t_q - t_{2q}), \\ 0 \leq \widehat{\mathcal{B}}_k &\leq t_{q-1} + t_q, \\ \mathcal{B}_k &= t_{q-1} + t_q - t_{2q-1} - t_{2q} - \Gamma_k^{\text{gr}}. \end{aligned} \quad (\text{HC24})$$

Substitution into (HC9g), (HC10g) and then (HC14g) is the explicit original-source tail calculation for their graph entry. In particular it retains all four tails and the signed augmentation correction. Although the common $\log \det \Omega_k$ cancels with $1 + 1 - 1 - 1 = 0$, that full matrix remains in all four congruences (HC23). Its strictly positive mixed contribution at every $k \geq 2$ has not been removed. The fixed- k degree limit of t_N does not supply a rate for these four growing endpoints as k varies.

Finally the completed receiving maps remain explicit. PGB.20–PGB.21 gives $\Xi(f, x) = \mathcal{J}_k f - Y_\perp x$ and $j_\infty \Xi(f, x) = x$, where $j_\infty = -\Omega_k^{-1} Y_\perp^*$ and $\ker j_\infty = \mathcal{J}_k L^2(r_k du)$. The first-coordinate projection to the original unweighted completed source is $\pi_1 \Xi(f, x) = \mathcal{J}_k^0 I_{r,m}(f + g_x)$. It maps the closed graph relation into the original closed unweighted relation, inducing $(E, \Omega_k) \rightarrow 0$ with full kernel E , exactly as proved in PGB.21b. On the graph quotient, PGMB.7–PGMB.8 instead gives $\mathfrak{C}_k \Xi(f, x) = (L_{\text{low}} x, \mathcal{D}_k^{\text{mix}} x)$ with kernel $\mathcal{J}_k L^2(r_k du)$, and, for $k \geq 2$,

$$(\Omega_k - K_{\text{low}})^{-1} (\mathcal{D}_k^{\text{mix}})^* \text{pr}_2 \mathfrak{C}_k = j_\infty. \quad (\text{HC25})$$

The equality follows by the exact Gram in (HC22). These are the maps from the same completed source to its unweighted and mixed targets, with their complete kernels. On each fixed coefficient mask they act coordinatewise; represented zero and an empty present tuple keep their original label, and external absence keeps its singleton inverse image. The finite graph-to-average contraction (HC18) and these completed maps use their stated different domains. Every proper-source residual retained above continues to use its original inverse-image domain and boundary quotient; none of these Hilbert quotient calculations declares that residual zero.

163.5 The sharper finite comparison retains its exact error

The original lower-envelope measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du.$$

Let V_N^ν be its quotient volume for the identical full remainder map. The actual finite upper expression for the four-window volume is

$$\mathcal{U}_{\nu,k} = \log(V_{q-1} V_q) - \log(V_{2q-1}^\nu V_{2q}^\nu),$$

with the exact nonnegative error

$$\boxed{\mathcal{U}_{\nu,k} - \mathcal{B}_k = \log \frac{V_{2q-1}}{V_{2q-1}^\nu} + \log \frac{V_{2q}}{V_{2q}^\nu} \geq 0.} \quad (\text{HC15})$$

Indeed $m_{h,k}(u) du \geq d\nu_k$ bounds every source polynomial quadratic form. Minimizing on each identical affine remainder fibre gives $G_N \succeq G_N^\nu \succ 0$. Conjugation by $(G_N^\nu)^{-1/2}$ shows that each volume ratio in (HC15) is at least one. Cancellation proves the equality. The low-degree factors require only original moments through $2q$, and the comparison moments in the last factors go through $4q$. These are well-defined finite expressions, with the actual full jet map and source mass. The supplied proof records do not evaluate their errors at the hypothetical arithmetic quartet or bound them at a rate opposing (HC5). Retaining this sharper finite expression therefore supplies no extra contradictory inequality.

164 The same arithmetic source in every bound

Retain the actual quartet, $g = 2\xi$, h , $v_h = g/h$, δ , γ , and full order m from (HT1). For $k \geq 3$, retain $q = [1 + k(m-1)](k+1)^2$ and the exact cyclic sum polynomial of (HT2). All determinant factors below are taken in its original increasing-power remainder frame. They are positive because the original common source Grams are positive definite and the remainder maps are onto.

Use the original arithmetic measure $m_{h,k} = w_h^{*k}$ and its fixed Gamma reference from (AT4), with their unchanged masses. Set

$$B_{\text{ar}} = \log \frac{V_{q-1}^{\text{ar}} V_q^{\text{ar}}}{V_{2q-1}^{\text{ar}} V_{2q}^{\text{ar}}}, \quad B_\Gamma = \log \frac{V_{q-1}^\Gamma V_q^\Gamma}{V_{2q-1}^\Gamma V_{2q}^\Gamma}, \quad \Delta = B_{\text{ar}} - B_\Gamma. \quad (\text{CR1})$$

In the fixed degree- $2q$ source define the positive relative operator $D_* = M_{2q}(0)^{-1} M_{2q}(1)$. This symbol is not the Euler differential. Let $b_1 \leq \dots \leq b_{2q+1}$ be all its eigenvalues with their full multiplicities, and put

$$L_q = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}. \quad (\text{CR2})$$

Every eigenvalue here is computed from the two actual moment matrices on their common retained polynomial source. There is no free replacement by a diagonal arithmetic metric. The diagonalization used in the AW proof is the spectral theorem applied to that specified positive relative operator, and its proof transports the original form.

Let D_h and $\hat{\mathcal{U}}_k$ have their full definitions in (HC3), (HC6) and (HC8). Let $\mathcal{U}_{\nu,k}$ be the sharper finite lower-envelope comparison in (HC15), including its two actual low-degree determinants and two envelope determinants. Then the following combined restriction follows on the original objects:

$$\max\{0, 4q \log(D_h k), B_\Gamma - L_q\} \leq B_{\text{ar}} \leq \min\{\hat{\mathcal{U}}_k, \mathcal{U}_{\nu,k}, B_\Gamma + L_q\}. \quad (\text{CR3})$$

For the signed arithmetic correction itself this is

$$\max\{-B_\Gamma, 4q \log(D_h k) - B_\Gamma, -L_q\} \leq \Delta \leq \min\{\hat{\mathcal{U}}_k - B_\Gamma, \mathcal{U}_{\nu,k} - B_\Gamma, L_q\}. \quad (\text{CR4})$$

Proof. Polynomial-degree inclusion decreases the minimum quotient Gram in the positive-semidefinite order, so each original volume ratio in (CR1) is at least one. This proves $B_{\text{ar}} \geq 0$. The full inherited two-window identity and the new retained residual calculation give $B_{\text{ar}} \geq 4q \log(D_h k)$ in (HC5). The actual remainder-map estimates prove $B_{\text{ar}} \leq \widehat{U}_k$ in (HC8), while comparison on the identical affine remainder fibres proves $B_{\text{ar}} \leq \mathcal{U}_{\nu,k}$ in (HC15). The computed full window spectrum and the actual relative moment eigenvalues give $|\Delta| \leq L_q$ in (AW14). Substituting (CR1) into this last inequality gives the remaining lower and upper entries of (CR3). Taking the maximum of the three proved lower bounds and minimum of the three proved upper bounds proves (CR3), with no independent choice of their source data. Subtracting the same literal B_{Γ} everywhere gives (CR4). $\square$

The signed value appearing in these inequalities has also been calculated as an exact integral, not only bounded:

$$\begin{aligned} \Delta &= \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt \\ &= \int_0^1 \text{Tr}(W(t)T(t)^{-1}[C(t), T(t)]) dt. \end{aligned} \tag{CR5}$$

Here U, W are the two original relation and polynomial window operators (AT13); $C = M_{2q}(t)^{-1}(M_{2q}(1) - M_{2q}(0))$; and T is the actual smooth source isometry constructed in (AW9)–(AW10). The first equality is (AT15), whose proof retains all four determinant orientations and both relation spaces. The second is (AW11), proved by inserting $U = TWT^{-1}$ and cyclically moving T in the finite trace. The source isometry does not preserve the arithmetic observation: its exact observation defect and full kernel are (AW12), and a nonzero witness is proved below. Thus (CR5) has not silently replaced the arithmetic map by an isometry that preserves different cohomology classes.

In particular every putative actual quartet must obey the additional explicit spectral compatibility inequality

$$L_q \geq \max\{0, B_{\Gamma} - \widehat{U}_k, B_{\Gamma} - \mathcal{U}_{\nu,k}, \max(0, 4q \log(D_h k)) - B_{\Gamma}\}. \tag{CR6}$$

Indeed the first two nonzero entries follow by combining $B_{\Gamma} - L_q \leq B_{\text{ar}}$ with the two upper estimates. The last entry follows from $\max(0, 4q \log(D_h k)) \leq B_{\text{ar}} \leq B_{\Gamma} + L_q$. All logarithmic ratios defining L_q are nonnegative, proving its lower bound by zero as well. This proves (CR6) with the actual relative spectrum, not a freely chosen condition number.

165 An explicit original relation that the source isometry does not preserve

Here the arithmetic observation is the exact map $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ in (AW12), and $T(t) : H \rightarrow H$ is the constructed source isometry. Write the original monic polynomial as $\chi(S) = \sum_{j=0}^q a_j S^j$ and define its reflected conjugate on this fixed sum line by $\chi^{\dagger}(S) = \sum_{j=0}^q \overline{a_j}(k-S)^j$. The actual vector

$$v(S) = \chi(S)\chi^{\dagger}(S) \in \mathcal{P}_{2q} \quad \text{satisfies} \quad \mathcal{O}v = 0, \quad \mathcal{O}T(t)v \neq 0 \quad (0 \leq t \leq 1). \tag{CR7}$$

To prove this, $v \in \chi\mathcal{P}_q = \ker J_{2q}$ proves the first equality using the original observation map. On the unchanged line $S = k/2 + iu$ one has $k - S = \overline{S}$, so

$$\langle 1, v \rangle_t = \int_{\mathbb{R}} |\chi(k/2 + iu)|^2 m_t(u) du > 0.$$

The integral is finite by the original moment bounds, and positive because χ is a nonzero polynomial and m_t is positive almost everywhere. Therefore $v \notin \mathcal{P}_{q-1}^{\perp}$. Equation (AW12), proved by the explicit isometry on the two full flags, gives $\ker(\mathcal{O}T) = \mathcal{P}_{q-1}^{\perp}$, establishing the second assertion of (CR7).

On a supported fibre the first value is its supported zero, whereas the second is a nonzero arithmetic coefficient at that same label. Neither value is external τ . Thus the exact source isometry and its calculated commutator are available, but that particular isometry does not descend through the original relation quotient. This obstruction is proved on the same actual polynomial and measure used in (CR1)–(CR6).

166 Status at the preceding proof increment

The later chapters in this cumulative reader sharpen the arithmetic norm, volume, and coherent comparisons. The following records the scope of the preceding bounds, all of which remain valid.

The coarse actual upper and lower bound functions have strictly positive separation for every $k \geq 3$, as proved in (HC9)–(HC11). Keeping the packet constants fixed, their separation divided by q^2 tends to $2((\pi/2)D + \log 256) > 0$, with D exactly the original interval scale in (HC6). Thus those proved estimates give no contradiction. The more informative interval (CR3) retains actual unevaluated relative moment eigenvalues and the sharper envelope determinants. This calculation does not claim to have verified a nonempty instance of that refined interval for any off-line zeta packet: no such packet has been certified.

The strongest original-object restrictions proved here are simultaneous. The actual packet survives the full balanced finite comparison, the mixed-support coefficient retract, and the original tensor action. Its proper-label kernel contribution has the exact transition of DP and PL. Its full relative symmetric trace has the calculated negative directions and original derivative representatives in C. Its canonical source at every phase has the boundary energy and Laplacian residual in HT. Its phase derivative energy contributes to the same endpoint contraction costs by (HC12)–(HC14). Its original arithmetic correction is the computed common-source integral (CR5), bounded by the actual relative spectrum in (CR2)–(CR6).

These conclusions eliminate particular ways an off-line packet might have disappeared from the programme, and quantify the cost of its remaining original source representatives. They do not eliminate all such packets. Establishing the requested contradiction would require an opposing estimate on these original arithmetic quantities, proved from the same g/h and theta relations. Establishing RH false would require an actual off-critical zero of g , which has not been obtained. Neither outcome is asserted by this manuscript. The full calculations, source evidence and corrected comparisons are retained for continuation.

167 The original coherent quotient and the derived classes it does not remove

This section continues the actual balanced realization in A1427 and the absolute pointed-base comparison in A1523. The balanced-kernel chapter, (BK.1)–(BK.24), proves the original analytic input used below, including the full-plane Euler inverse and all source-label transition maps. The derived-packet chapter, (DP.1)–(DP.8), retains both degrees of every finite spectral thickening. We compute a different observation here: maps from the same balanced object into coherent analytic modules, and then derived maps into those modules. The distinction is established by explicit morphisms and resolutions.

167.1 The original sheaf and its canonical local summands

Retain the original definitions

$$\begin{aligned}
 V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \quad \phi(0) = 0, \quad \int_{\mathbb{R}} \phi(x) dx = 0\}, \\
 \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\
 \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad Q = \mathcal{B}/\Theta V, \quad A = \mathbb{C}[t], \quad (\text{OCQ.1}) \\
 g(s) &= 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad H_\phi(s) = \frac{2\pi^{s/2}M_+\phi(s)}{s(s-1)\Gamma(s/2)}.
 \end{aligned}$$

Here t acts by D on the original function spaces and by multiplication by s on analytic functions. The expression defining H_ϕ is its entire continuation, with every removable point treated in (BK.1)–(BK.8). In particular $\mathcal{M}\Theta\phi = gH_\phi$ and $\mathcal{M}DF = s\mathcal{M}F$ on the original spaces. The tensor products and module Hom groups below are algebraic, as in the displayed balanced construction in A1427. The original Schwartz spaces remain the source of their specified elements and maps.

Let $X = \mathbb{C}$ with its analytic structure. Define $\mathcal{M}$ to be the sheafification of the presheaf $U \mapsto \mathcal{O}_X(U) \otimes_A Q$, and put $\mathcal{N} = \mathcal{O}_X/(g)$. The presheaf maps

$$a \otimes [F] \longmapsto [a\mathcal{M}F]$$

are balanced, respect restrictions, and vanish on the original theta boundaries by (OCQ.1). They therefore define an actual sheaf morphism

$$\beta : \mathcal{M} \longrightarrow \mathcal{N}, \quad \mathcal{K} = \ker \beta. \quad (\text{OCQ.2})$$

Tensor products commute with filtered colimits, so its stalk at s_0 is the original map

$$\begin{aligned}
 O &= \mathcal{O}_{X,s_0}, \quad z = s - s_0, \quad F_{\text{mer}} = \text{Frac } O = O[1/z], \\
 M = O \otimes_A Q &\xrightarrow{b} N = O/(g), \quad b(a \otimes [F]) = [a\mathcal{M}F], \quad K = \ker b.
 \end{aligned} \quad (\text{OCQ.3})$$

The letter F_{mer} denotes meromorphic germs and does not denote a test function or the formal Laurent-series field. In these fixed coordinates $O = \mathbb{C}\{z\}$.

For completeness, surjectivity is visible on an original global test:

$$F_0(x) = e^{-(\log x)^2}, \quad H_0(s) = \mathcal{M}F_0(s) = \sqrt{\pi}e^{s^2/4}, \quad q_0 = [F_0]. \quad (\text{OCQ.4})$$

Euler derivatives and multiplication by x^b preserve the required Gaussian bounds, so $F_0 \in \mathcal{B}$. Substitution $x = e^y$ evaluates the Mellin integral, first for real s by completing the square and then everywhere by entire continuation. Since H_0 is nowhere zero, $a \mapsto aH_0^{-1} \otimes q_0$ lifts any local section $[a]$ of $\mathcal{N}$. Thus (OCQ.2) is a sheaf epimorphism.

The proved original comparison (BK.14)–(BK.19) says that every nonzero polynomial acts bijectively on K . Since every nonzero element of O is a unit times a power of z , it follows that K is a vector space over F_{mer} , with the uniquely forced action $(a/c)k = a(c|_K)^{-1}k$. It is nonzero at every stalk. The explicit witness is

$$k_0 = g \otimes q_0 \neq 0 \quad \text{in } K. \quad (\text{OCQ.5})$$

Here the proof of nonvanishing is exactly the proof retained in (BK.21): the classical infinitude of distinct actual zeros of g forces $P = 0$ in any identity $P(s)H_0(s) = g(s)H_\phi(s)$; hence $Aq_0 \simeq A$, and flatness of the torsion-free A -module O gives $Oq_0 \simeq O$. The classical zero-set input is the retained Hardy theorem and its original source identified in that chapter. Neither RH nor an off-line zero is assumed. The structure and factorization arguments below only require the computed vector-space property; the assertions of noncoherence and nonzero derived classes also use this proved witness.

Retain the full local factor $g(s) = u(z)z^m$ with $u(0) \neq 0$ and $m = \text{ord}_{s_0} g$. There is an explicit canonical decomposition. For $x \in M$, $gx \in K$, so define

$$p_K(x) = (g|_K)^{-1}(gx), \quad p_T(x) = x - p_K(x), \quad T = M[g]. \quad (\text{OCQ.6})$$

These are O -linear, p_K is the identity on K , and $gp_T(x) = 0$. The intersection $T \cap K$ is zero because g is invertible on K . The map $b|_T$ is injective and is surjective because $b(p_T x) = b(x)$. Consequently, with $j = (b|_T)^{-1}$,

$$M = T \oplus K, \quad T \xrightarrow[b]{\sim} O/(g), \quad x = j(bx) + p_K x. \quad (\text{OCQ.7})$$

An O -torsion element of M has zero K component, so T is the entire torsion submodule. This includes $m = 0$, when $N = T = 0$. Formula (OCQ.6) retains u ; no replacement of the function g has been made.

167.2 Every ordinary coherent observation factors through the actual zero scheme

Call an O -module divisible when multiplication by every nonzero element of O is surjective, and call a module reduced when it has no nonzero divisible submodule. Over this ring divisibility is equivalent to surjectivity of multiplication by z . Every finitely generated O -module E is reduced. To see this without an assumption on a submodule, the elementary divisor decomposition gives $E \simeq O^r \oplus \bigoplus_{j=1}^d O/(z^{n_j})$. This decomposition follows by reducing a finite presentation matrix over the discrete valuation ring: choose a nonzero entry of least valuation, use its divisibility into all entries to clear its row and column by invertible operations, and repeat on the remaining finite matrix. The resulting diagonal entries are units times powers of z ; unit row operations remove the units. The presentation exists because every nonzero ideal of O contains an element of smallest valuation and is generated by it. More explicitly, for a submodule of O^n , its projection to the first coordinate is a principal ideal. Choose one lift of its generator; subtracting its multiples leaves a submodule of O^{n-1} . Induction on n proves finite generation of every such submodule and hence of the kernel of a finite free surjection. The decomposition proves

$$\bigcap_{n \geq 0} z^n E = 0, \quad (\text{OCQ.8})$$

since a holomorphic germ divisible by every power is zero, and sufficiently large powers kill each torsion summand. Every divisible submodule is contained in this intersection.

An O -linear image of K is divisible. Therefore every map $M \rightarrow E$ with E reduced vanishes on K and factors uniquely through b . For a finite module this yields, naturally in E ,

$$b^* : \text{Hom}_O(N, E) \xrightarrow{\sim} \text{Hom}_O(M, E), \quad \text{Hom}_O(M, E) \simeq E[g]. \quad (\text{OCQ.9})$$

The last map sends f to $f(j([1]))$. Its inverse sends $e \in E[g]$ to $x \mapsto ae$ whenever $b(x) = [a]$; changing a by gc does not change the answer.

The reduced targets form the exact largest class with this property. Indeed, if an arbitrary target E contains a nonzero divisible submodule D , choose $d_0 \neq 0$ and $d_{n+1} \in D$ with $zd_{n+1} = d_n$. The formula

$$\alpha : F_{\text{mer}} \longrightarrow D, \quad \alpha(az^{-n}) = ad_n \quad (\text{OCQ.10})$$

is well-defined: equality $az^{-n} = bz^{-m}$, with $m \geq n$, means $b = z^{m-n}a$, while $z^{m-n}d_m = d_n$. It is additive, O -linear, and nonzero at 1. Choose an F_{mer} -basis of K containing k_0 , and let $\ell : K \rightarrow F_{\text{mer}}$ be its coordinate projection with $\ell(k_0) = 1$. Then $\alpha \ell p_K : M \rightarrow E$ is nonzero on K , so it does not factor through b . The choices are explicit choices of a divisible chain and a vector-space complement, not canonical maps.

Also K is the maximal divisible submodule of M : its image under b in the finite module N must be zero. The universal property implies

$$K = \bigcap_{\substack{E \text{ finite over } O \\ f: M \rightarrow E}} \ker f. \quad (\text{OCQ.11})$$

The inclusion from left to right is (OCQ.9), and the reverse inclusion uses $f = b$ itself.

Neither K nor M is finitely generated at any analytic stalk. Otherwise (OCQ.8), applied respectively to K or M , contradicts $0 \neq k_0 \in \bigcap_n z^n K \subset \bigcap_n z^n M$. Hence $\mathcal{K}$ and $\mathcal{M}$ are nowhere coherent, although $\mathcal{N}$ is coherent. For a coherent analytic sheaf $\mathcal{E}$, any actual sheaf morphism $\mathcal{M}|_U \rightarrow \mathcal{E}|_U$ vanishes on $\mathcal{K}|_U$ by (OCQ.9) on stalks. The sheaf quotient property gives a unique factorization on U , compatible on overlapping opens. We obtain a proved sheaf isomorphism

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{N}, \mathcal{E}) \xrightarrow{\sim} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{M}, \mathcal{E}). \quad (\text{OCQ.12})$$

Thus β is the universal coherent quotient of this original object. This statement neither identifies $\mathcal{M}$ with its quotient nor asserts existence of such a quotient for arbitrary analytic sheaves.

Every coherent quotient of M is consequently a quotient of $O/(g)$. At the fixed stalk its kernel lifts to an ideal I of O containing $(g) = (z^m)$. The valuation argument above gives $I = (z^r)$ for exactly one integer $0 \leq r \leq m$. Conversely every such ideal defines a quotient. The full map and its removed part are

$$M \xrightarrow{b} O/(g) \longrightarrow O/(z^r), \quad \ker(O/(g) \rightarrow O/(z^r)) = (z^r)/(g). \quad (\text{OCQ.13})$$

This records all $m-r$ removed nilpotent levels and retains the actual ideal (g) . Globally these kernels glue exactly to ideal sheaves between (g) and $\mathcal{O}_X$, or equivalently to effective subdivisors of the actual divisor of g . In particular the ordinary analytic duals are

$$\text{Hom}_O(M, O) = 0, \quad \text{Hom}_O(K, O) = 0, \quad M^{\vee\vee} = 0. \quad (\text{OCQ.14})$$

The next calculation constructs the derived comparison that (OCQ.14) does not address.

167.3 A complete local derived calculation

All Ext groups in this subsection are groups of modules over the specified analytic stalk $O = \mathbb{C}\{z\}$. Put $\widehat{E} = \varprojlim_n E/z^n E$ for a finite O -module E . Equation (OCQ.8) embeds E into its completion. The original source is not replaced by a completed module.

The meromorphic germ field has the following explicit two-term free resolution. With $P_0 = P_1 = \bigoplus_{n \geq 0} Oe_n$, set

$$0 \longrightarrow P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} F_{\text{mer}} \longrightarrow 0, \quad d(e_n) = e_n - ze_{n+1}, \quad \epsilon(e_n) = z^{-n}. \quad (\text{OCQ.15})$$

Surjectivity follows from the definition of $O[1/z]$. If a finitely supported vector has last nonzero coefficient a_m , the coefficient of e_{m+1} in its image under d is $-za_m \neq 0$; hence d is injective. Modulo dP_1 one has $e_n = z^{m-n}e_m$ for $n \leq m$. Therefore a vector $\sum_{n=0}^m a_n e_n$ in $\ker \epsilon$ is congruent to $(\sum_{n=0}^m z^{m-n} a_n) e_m = 0$. This proves exactness with the displayed maps.

Applying $\text{Hom}_O(-, E)$ gives the product map

$$\Delta : \prod_{n \geq 0} E \longrightarrow \prod_{n \geq 0} E, \quad \Delta(a)_n = a_n - za_{n+1}, \quad \text{Ext}_O^1(F_{\text{mer}}, E) = \text{coker } \Delta. \quad (\text{OCQ.16})$$

The kernel is $\text{Hom}_O(F_{\text{mer}}, E) = 0$ by (OCQ.8), and all higher Ext groups vanish by this free resolution. Define the convergent-in-the-completion sum

$$S : \prod_{n \geq 0} E \longrightarrow \widehat{E}/E, \quad S(c) = \left[\sum_{n \geq 0} z^n c_n \right]. \quad (\text{OCQ.17})$$

Every summand of index at least n lies in $z^n \widehat{E}$, so the sum is well-defined. Finite partial sums for $c = \Delta a$ are $a_0 - z^{N+1} a_{N+1}$; the second term tends to zero in the completion. Thus $S\Delta = 0$.

For $E = O$, suppose $S(c) = 0$ and write $a = \sum_{n \geq 0} z^n c_n \in O$. Define

$$a_n = z^{-n} \left(a - \sum_{j < n} z^j c_j \right). \quad (\text{OCQ.18})$$

The parenthesis is an analytic germ whose formal Taylor coefficients below degree n vanish. Its analytic zero order is consequently at least n , so $a_n \in O$. Subtraction gives $a_n - z a_{n+1} = c_n$, proving $\ker S = \text{im } \Delta$. Given an arbitrary formal series $\sum b_n z^n$, choose the constant germs $c_n = b_n$; this proves surjectivity of S . The argument applies componentwise to finite free modules. For a finite torsion module T killed by z^q , one has $\widehat{T} = T$ and the explicit formula

$$a_n = \sum_{j=0}^{q-1} z^j c_{n+j}, \quad a_n - z a_{n+1} = c_n - z^q c_{n+q} = c_n \quad (\text{OCQ.19})$$

proves surjectivity of Δ . The finite-module decomposition used in (OCQ.8) completes the proof for every finite E . Since the map S itself commutes with every homomorphism $E \rightarrow E'$, the resulting isomorphism is natural, independently of the decomposition used to verify it:

$$\text{Ext}_O^1(F_{\text{mer}}, E) \xrightarrow[S]{} \widehat{E}/E. \quad (\text{OCQ.20})$$

Choose an F_{mer} -basis B of K . The direct sum over B of (OCQ.15) is a free resolution of K . Applying Hom gives products of (OCQ.16). Products of these surjective image-to-cokernel maps are surjective, and their kernels are the corresponding products of images; thus their cokernel is the product of the individual cokernels. Consequently

$$\text{Hom}_O(K, E) = 0, \quad \text{Ext}_O^1(K, E) \simeq \prod_{b \in B} (\widehat{E}/E), \quad \text{Ext}_O^i(K, E) = 0 \quad (i \geq 2). \quad (\text{OCQ.21})$$

Only the product identification depends on the chosen basis. The canonical summands in (OCQ.7), together with the resolution $0 \rightarrow O \xrightarrow{g} O \rightarrow N \rightarrow 0$, now give

$$\begin{aligned} \text{Hom}_O(M, E) &\simeq E[g], \\ \text{Ext}_O^1(M, E) &\simeq E/gE \oplus \text{Ext}_O^1(K, E), \\ \text{Ext}_O^i(M, E) &= 0 \quad (i \geq 2). \end{aligned} \quad (\text{OCQ.22})$$

The first summand in degree one is exactly the map b^* from $\text{Ext}_O^1(N, E)$.

These statements determine the exact scope for all bounded coherent derived targets. If $C \in D_{\text{fg}}^b(O)$, then

$$\begin{aligned} R\text{Hom}_O(N, C) &\xrightarrow{b^*} R\text{Hom}_O(M, C) \text{ is a quasi-isomorphism} \\ \iff H^q(C) &\text{ has finite length for every } q. \end{aligned} \quad (\text{OCQ.23})$$

Indeed, the two-term projective resolution of K and the finite cohomological filtration of C give the hyper-Ext spectral sequence with terms $\text{Ext}_O^p(K, H^q(C))$. The column $p = 0$ is zero by (OCQ.21), columns $p \geq 2$ are zero by the resolution, and no differential enters or leaves the remaining column. The finite filtration therefore has precisely one possible nonzero graded piece in each total degree, giving

$$H^n R\text{Hom}_O(K, C) \simeq \text{Ext}_O^1(K, H^{n-1}(C)). \quad (\text{OCQ.24})$$

For a finite module E the quotient $\widehat{E}/E$ is zero exactly when its free rank is zero: its torsion part contributes zero, and $\mathbb{C}[[z]]/\mathbb{C}\{z\} \neq 0$, as witnessed by the divergent formal series $\sum_{n \geq 0} n! z^n$. The

ratio of successive absolute terms at any fixed $z \neq 0$ is $(n+1)|z|$, so that series cannot converge in any nonzero disc. Since B is nonempty by (OCQ.5), (OCQ.21), (OCQ.24), and the splitting of M prove both directions of (OCQ.23).

There is also a concrete nonsplit extension, rather than only an abstract dimension assertion. Put $\omega = O ds$ and use (OCQ.15). Define $\varphi : P_1 \rightarrow \omega$ by $\varphi(e_n) = n! ds$, and set

$$E_! = \frac{\omega \oplus P_0}{\{(-\varphi(v), d(v)) : v \in P_1\}}. \quad 0 \longrightarrow \omega \longrightarrow E_! \longrightarrow F_{\text{mer}} \longrightarrow 0. \quad (\text{OCQ.25})$$

The last map sends (w, p) to $\epsilon(p)$. Injectivity of the first map follows from injectivity of d ; if $\epsilon(p) = 0$, write $p = dv$, so the class of (w, p) is the class of $(w + \varphi(v), 0)$. This proves exactness. Writing $\bar{e}_n$ for the class of e_n , the relations are $\bar{e}_n - z\bar{e}_{n+1} = n! ds$. A splitting would give compatible lifts $\bar{e}_n - a_n ds$ and hence $a_n - za_{n+1} = n!$. Equations (OCQ.17)–(OCQ.20) would then make $\sum n!z^n$ convergent, a contradiction. Thus (OCQ.25) is nonsplit. Pulling it back along $\ell_{p_K} : M \rightarrow F_{\text{mer}}$ from (OCQ.10) gives a class in $\text{Ext}_O^1(M, \omega)$ whose restriction to the embedded line $F_{\text{mer}}k_0$ is (OCQ.25). It is therefore nonzero on K and does not come from $\text{Ext}_O^1(N, \omega)$.

Accordingly the universal ordinary coherent quotient (OCQ.12) does not extend to all derived coherent targets. In particular $\omega[1]$ detects the explicit additional class. This coexists with the exact derived tensor comparison (DP.2): for every nonzero germ h , $K \otimes_O^L O/(h)$ is the contractible two-term complex $[K \xrightarrow{h} K]$. The two calculations concern explicitly different functors. No identification of stalks of sheaf Ext with Ext groups of these noncoherent stalks is asserted or needed.

167.4 The residue morphism into the original pointed-base dual

The finite analytic summand has a direct relation to the actual A1523 duality. The resolution using multiplication by the actual g gives

$$\text{Ext}_O^1(N, \omega) \simeq \omega/g\omega \xrightarrow{\sim} g^{-1}\omega/\omega, \quad [v ds] \longmapsto [v ds/g]. \quad (\text{OCQ.26})$$

The second map is well-defined and invertible by multiplication by g . There is a canonical complex-linear residue map

$$g^{-1}\omega/\omega \longrightarrow \text{Hom}_{\mathbb{C}}(N, \mathbb{C}), \quad [\eta] \longmapsto ([v] \mapsto \text{Res}_{s=s_0}(v\eta)). \quad (\text{OCQ.27})$$

Adding a holomorphic form or changing v by a multiple of g changes the integrand by a holomorphic form, so the map is well-defined. It is perfect: with $g = u(z)z^m$, the basis $1, z, \dots, z^{m-1}$ is dual to the classes $u(z)z^{m-1-j} ds/g = z^{-j-1} ds$, for $0 \leq j < m$. Their residue matrix is the identity. This calculation retains the entire analytic unit u in the representative before cancellation. For $m = 0$ both sides are zero.

Let Z be an actual finite set of zeros stable under $\rho \mapsto \rho^\dagger = 1 - \bar{\rho}$, and retain all multiplicities $m_\rho = \text{ord}_\rho g$ in $A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_{X,\rho}/(g)$. The Mellin map from the original quotient is $J_Z : Q \rightarrow A_Z$, $J_Z[F] = ([\mathcal{M}F]_\rho)_\rho$. Its surjectivity is the original finite Mellin-jet statement retained in A1523, with complete interpolation proof in (BK.9)–(BK.11); no arbitrary surrogate packet is used. For $f \in A_Z$, use the reflected germ $\overline{f^\dagger}(s) = \overline{f(1 - \bar{s})}$ at the corresponding paired centre. The original identities $g(1 - s) = g(s)$ and $\overline{g(\bar{s})} = g(s)$ give $g^\dagger = g$, so this operation preserves the actual ideals and is well-defined on the full packets. Equations (OCQ.26)–(OCQ.27) give the exact composite

$$\overline{A_Z} \longrightarrow \bigoplus_{\rho \in Z} \text{Ext}_{\mathcal{O}_{X,\rho}}^1(\mathcal{O}_{X,\rho}/(g), \mathcal{O}_{X,\rho} ds) \longrightarrow \text{Hom}_{\mathbb{C}}(Q, \mathbb{C}), \quad (\text{OCQ.28})$$

where its value at f is

$$[F] \longmapsto \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) \mathcal{M}F(s)}{g(s)} ds. \quad (\text{OCQ.29})$$

The first map sends f to $[f^\dagger ds/g]$; it is a complex-linear isomorphism from the conjugate vector space. The second map is injective by the perfectness in (OCQ.27) and surjectivity of J_Z . Thus (OCQ.28) is precisely the finite residue injection retained in A1523, after equipping its target with the original one-dimensional dilation character $\mathcal{L}_1$.

To check that character and every sign, let $R_a F(x) = F(x/a)$ for $a > 0$. Direct Mellin substitution gives $\mathcal{M}R_a F(s) = a^s \mathcal{M}F(s)$. The original dual action on $\text{Hom}_{\mathbb{C}}(Q, \mathcal{L}_1)$ is $(U_a \lambda)(q) = a\lambda(R_{1/a} q)$. In (OCQ.29) its effect is multiplication of the differential representative by a^{1-s} . On the source packet, $(a^s f)^\dagger = a^{1-s} f^\dagger$. Thus the displayed injection is equivariant, with the exact character a and the original reflection. Inserting g' gives the original trace form, since $g'/g = m_\rho/(s - \rho) + u'/u$ at each centre:

$$\sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) g'(s) v(s)}{g(s)} ds = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} v(\rho). \quad (\text{OCQ.30})$$

This proves the exact morphism from the finite coherent dual summand to the actual pointed-base comparison; the additional classes in (OCQ.25) do not replace or delete that summand.

The A1523 target is the complex-linear right adjoint for the original four-point amplitude diagram. Explicitly its points satisfy $+ < \eta < \sigma$ and $- < \eta < \sigma$; the amplitudes are $\mathcal{T}_+ = \mathcal{T}_- = V$, $\mathcal{T}_\eta = \mathcal{B}$, $\mathcal{T}_\sigma = 0$, with restrictions Θ and $J\Theta = \hat{\Theta}$, where $JF(x) = x^{-1}F(1/x)$ and the Fourier kernel is $e^{-2\pi i x \nu}$. Its cohomology complex is $[V \oplus V \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \hat{\psi})} \mathcal{B}]$, in degrees zero and one, so $H^1 = Q$. The right adjoint computed there satisfies $H^{-1}R \text{Hom}(\mathcal{T}, K_\tau(\mathcal{L}_1)) \simeq \text{Hom}_{\mathbb{C}}(Q, \mathcal{L}_1)$. Consequently ordinary O -dual vanishing in (OCQ.14) cannot be inserted as a vanishing statement about that different, explicitly related right-adjoint target.

167.5 Proper labels and the precise consequence for purity comparison

For every original D -invariant source $W \subset V$, keep

$$Q_W = \mathcal{B}/\Theta W, \quad M_W = O \otimes_A Q_W, \quad L_W = O \otimes_A (V/W), \quad \pi_W : M_W \rightarrow M, \quad b_W = b\pi_W. \quad (\text{OCQ.31})$$

The inclusion $L_W \rightarrow M_W$ sends $a \otimes [\phi]$ to $a \otimes [\Theta\phi]$. Injectivity of the original Θ and flatness of O over A prove

$$0 \rightarrow L_W \rightarrow M_W \xrightarrow{\pi_W} M \rightarrow 0, \quad 0 \rightarrow L_W \rightarrow \ker b_W \rightarrow K \rightarrow 0. \quad (\text{OCQ.32})$$

For a finite target E , a map $f : M_W \rightarrow E$ factors through b_W if and only if $f|_{L_W} = 0$. The reverse direction is proved by first descending along the first quotient in (OCQ.32) and then applying (OCQ.9); the forward direction follows because $b_W L_W = 0$. More precisely,

$$0 \rightarrow \text{Hom}_O(N, E) \xrightarrow{b_W^*} \text{Hom}_O(M_W, E) \rightarrow \text{Hom}_O(L_W, E) \xrightarrow{\delta} \text{Ext}_O^1(M, E) \quad (\text{OCQ.33})$$

is exact. Here $\delta(\alpha)$ is the pushout of the first sequence in (OCQ.32) by $\alpha : L_W \rightarrow E$: its middle module is $(E \oplus M_W)/\{(-\alpha(l), l) : l \in L_W\}$. This extension splits exactly when α extends to M_W . Indeed an extension f defines a retraction by $[e, x] \mapsto e + f(x)$, and any retraction gives $f(x) = [0, x]$ followed by that retraction, with $f|_{L_W} = \alpha$. This proves exactness at the last displayed Hom term, while the previous factorization argument proves exactness at the others.

If $W \subset W' \subset V$, the source-quotient maps give the commuting diagram of (OCQ.32), with $L_W \rightarrow L_{W'}$ induced by $V/W \rightarrow V/W'$ and $M_W \rightarrow M_{W'}$ induced by $Q_W \rightarrow Q_{W'}$. Their respective kernels are both $O \otimes_A (W'/W)$, identified by $[\phi] \mapsto [\Theta\phi]$. This follows by tensoring the two exact original quotient sequences. Thus the coherent observations of a proper source retain exactly their extendable observations on L_W and the obstruction map δ ; they are not replaced by the universal property for the full source. For example, the actual proper-label classes constructed in (PL.8)–(PL.15) lie in L_W and can have finite spectral torsion that vanishes only under this specified transition.

All these maps act inside the original support fibres. For any fixed nonempty support label S , the split-zero extension of a linear map $f : E \rightarrow E'$ is

$$\tau \mapsto \tau, \quad (S, v) \mapsto (S, f(v)), \quad e_S = (S, 0). \quad (\text{OCQ.34})$$

Its inverse image of e_S is the full supported kernel $\{(S, v) : v \in \ker f\}$, and its inverse image of τ is only $\{\tau\}$. This follows directly from the disjoint union defining the split-zero object. Formula (OCQ.34) applies to β , to each label transition, and to every quotient in (OCQ.13); no nonempty kernel fibre is identified with external absence.

At an actual zero ρ of order m , scalar multiplication by a^s on N is $a^\rho \exp((\log a)z)$ modulo the full ideal (g) . For the quotient of depth r in (OCQ.13), the removed module $(z^r)/(g)$ has the ordered basis $z^r, \dots, z^{m-1}$, and the action is triangular with diagonal a^ρ . Therefore its dimension and trace are exactly

$$\dim_{\mathbb{C}}(z^r)/(g) = m - r, \quad \text{Tr}(a^s; (z^r)/(g)) = (m - r)a^\rho. \quad (\text{OCQ.35})$$

Every partial or total deletion of an actual local packet is consequently recorded by a nonzero finite coherent kernel and by this exact change in the original trace, including all multiplicities. The dilation on the full balanced module remains the original $1 \otimes R_a$, whose difference from scalar a^s has image in K as proved in (DP.4)–(DP.6). Formula (OCQ.35) is asserted on the actual analytic packet and its specified quotient, where the actions agree.

The obtained consequence for the tau purity comparison is exact. The full original quotient has the actual zero scheme as its universal ordinary coherent quotient; retaining that universal map retains every actual off-line packet if such a packet exists. Proper source labels retain the additional maps and kernels in (OCQ.32)–(OCQ.33). Derived coherent targets of positive rank additionally see the explicit kernel extension (OCQ.25), whereas every finite-length derived target has exactly the comparison (OCQ.23). The original finite residue injection is the composite (OCQ.28), with its computed action and trace. None of these proved maps supplies a positive metric or a weight bound on the actual packet; no numerical location constraint has been deduced from coherence. The original purity question therefore continues with these full morphisms and obstructions retained, rather than with a claimed identification of the full analytic object and its coherent quotient.

168 The canonical derived dual and the retained original tau complex

This continuation uses the actual sheaf morphism (OCQ.2), constructed from the original theta quotient in A1427, and the original four-point amplitude complex in A1523. All completed objects below occur as explicit targets in a derived calculation. The original analytic sheaf and the original Schwartz spaces are unchanged.

168.1 The global splitting is an actual sheaf map

Keep $X = \mathbb{C}$ with coordinate s , $g = 2\xi$, and

$$0 \longrightarrow \mathcal{K} \xrightarrow{\iota} \mathcal{M} \xrightarrow{\beta} \mathcal{N} \longrightarrow 0, \quad \mathcal{N} = \mathcal{O}_X/(g). \quad (\text{DC.1})$$

These are precisely the sheaves of (OCQ.1)–(OCQ.3). At every analytic stalk the original calculation makes every nonzero germ invertible on $\mathcal{K}_{s_0}$. In particular multiplication by the same entire function g is an isomorphism of the sheaf $\mathcal{K}$: a morphism of sheaves is an isomorphism when its maps on all stalks are isomorphisms. The inverse is unique and commutes with restrictions.

Since $g\mathcal{N} = 0$, multiplication by g on $\mathcal{M}$ factors uniquely through a sheaf map $u_g : \mathcal{M} \rightarrow \mathcal{K}$ with $\iota u_g = g \text{id}_{\mathcal{M}}$. Define

$$r = (g|_{\mathcal{K}})^{-1} u_g : \mathcal{M} \longrightarrow \mathcal{K}, \quad P = \text{id}_{\mathcal{M}} - \iota r. \quad (\text{DC.2})$$

One has $r\iota = \text{id}_{\mathcal{K}}$, $rP = 0$, $P^2 = P$, $\beta P = \beta$, and $gP = 0$, by substituting (DC.2). The intersection $\mathcal{M}[g] \cap \mathcal{K}$ is zero, because g acts injectively on $\mathcal{K}$. Thus β restricts to an isomorphism $\mathcal{M}[g] \rightarrow \mathcal{N}$. Its inverse $j : \mathcal{N} \rightarrow \mathcal{M}$ satisfies $j\beta = P$, and gives the canonical sheaf isomorphisms

$$(\beta, r) : \mathcal{M} \xrightarrow{\sim} \mathcal{N} \oplus \mathcal{K}, \quad (n, k) \mapsto jn + \iota k. \quad (\text{DC.3})$$

Both composites are the identity by the displayed projection identities. No local vector-space basis or choices of representatives enter these maps. In particular the stalk decomposition (OCQ.7) genuinely glues.

Let $\omega_X = \mathcal{O}_X ds$. Applying the derived internal Hom functor to the finite direct sum (DC.3) gives

$$R\mathcal{H}om(\mathcal{M}, \omega_X) \simeq R\mathcal{H}om(\mathcal{N}, \omega_X) \oplus R\mathcal{H}om(\mathcal{K}, \omega_X). \quad (\text{DC.4})$$

The original exact locally free resolution $[\mathcal{O}_X \xrightarrow{g} \mathcal{O}_X] \rightarrow \mathcal{N}$, with its free terms in degrees $-1, 0$, computes the first summand. Multiplication by g is injective on ω_X , so

$$R\mathcal{H}om(\mathcal{N}, \omega_X) \simeq (\omega_X/g\omega_X)[-1]. \quad (\text{DC.5})$$

The injection of (DC.5) into (DC.4) is β^* and its retraction is j^* ; their composite is the identity since $\beta j = 1$. This is a global derived summand of the actual source. It makes no identification between stalks of the second internal Hom in (DC.4) and Ext groups computed from its noncoherent stalks.

168.2 A canonical local dual without a choice of meromorphic basis

Fix the original analytic coordinate s_0 , and retain

$$\begin{aligned} O = \mathcal{O}_{X, s_0} &= \mathbb{C}\{z\}, \quad z = s - s_0, \quad F = \text{Frac } O = O[1/z], \\ \omega &= O ds, \quad \widehat{\omega} = \mathbb{C}[[z]] ds, \quad C_\omega = \widehat{\omega}/\omega. \end{aligned} \quad (\text{DC.6})$$

Thus F consists of convergent meromorphic germs, and is not the formal Laurent-series field. The injective inclusion $\omega \hookrightarrow \widehat{\omega}$ is the Taylor-series map. Its quotient is nonzero: $\sum_{n \geq 0} n! z^n ds$ is a formal series, while its terms fail to tend to zero at every fixed $z \neq 0$. It is therefore not an analytic germ.

Multiplication by z is bijective on C_ω . To prove surjectivity, subtract the constant coefficient of a formal representative and divide the remaining formal series by z . The removed constant lies in ω . For injectivity, if $z\widehat{f} ds \in \omega$, the analytic germ on the left has zero constant term and can be divided analytically by z ; the uniqueness of its Taylor series gives $\widehat{f} ds \in \omega$. Every nonzero analytic germ is a unit times a power of z , so C_ω is an F -vector space with the uniquely determined action by inverses of these multiplications.

For every F -vector space L , viewed as an O -module, one has

$$\text{Hom}_O(L, \widehat{\omega}) = 0, \quad \text{Ext}_O^i(L, \widehat{\omega}) = 0 \quad (i \geq 1). \quad (\text{DC.7})$$

The Hom assertion follows from $f(l) \in \bigcap_n z^n \widehat{\omega} = 0$, since $l = z^n l_n$. For the Ext assertion first use the exact free resolution (OCQ.15) of F . Its Hom differential on the products of $\widehat{\omega}$ is $\Delta(a)_n = a_n - z a_{n+1}$. Given an arbitrary sequence (c_n) in $\widehat{\omega}$, set

$$a_n = \sum_{j \geq 0} z^j c_{n+j} \in \widehat{\omega}. \quad (\text{DC.8})$$

For each fixed coefficient only finitely many summands contribute, so this sum is well-defined in the specified completion. Subtracting $z a_{n+1}$ cancels every term except c_n . Thus Δ is onto. The resolution has length one, so all higher Ext groups vanish. A direct sum of these free resolutions over an F -basis of L is a free resolution of L ; its Hom differential is the product of the preceding surjections, hence is onto. This proves (DC.7) for arbitrary L . The basis is used only to prove this vanishing assertion.

An O -linear map between F -vector spaces is F -linear: applying the map to $z(z^{-1}l) = l$ and using invertibility of z in the target proves compatibility with z^{-1} , and compatibility with all meromorphic fractions follows. Consequently the exact completion sequence gives a canonical isomorphism, natural in L ,

$$0 \longrightarrow \omega \longrightarrow \widehat{\omega} \longrightarrow C_\omega \longrightarrow 0, \quad \partial_L : \operatorname{Hom}_F(L, C_\omega) \xrightarrow{\sim} \operatorname{Ext}_O^1(L, \omega). \quad (\text{DC.9})$$

Indeed its Hom–Ext sequence has zero groups adjacent to this map by (DC.7). The map ∂_L sends f to the literal pullback extension

$$E_f = \{(l, \widehat{v}) \in L \oplus \widehat{\omega} : f(l) = [\widehat{v}]\}, \quad 0 \rightarrow \omega \xrightarrow{v \mapsto (0, v)} E_f \xrightarrow{(l, \widehat{v}) \mapsto l} L \rightarrow 0. \quad (\text{DC.10})$$

Surjectivity of the last arrow follows from surjectivity onto C_ω , and its kernel is the displayed copy of ω . The extension is split exactly when f lifts to an O -linear map $L \rightarrow \widehat{\omega}$; by (DC.7), this occurs exactly when $f = 0$. Thus the canonical description detects every such nonzero map without choosing a basis.

For $L = K = \mathcal{K}_{s_0}$, evaluation at a chosen F -basis identifies the left side of (DC.9) with $\prod_B C_\omega$. This recovers the original product calculation (OCQ.21), with its full differential form factor. The isomorphism (DC.9) itself is independent of that choice.

168.3 The natural local derived bidual retains every kernel class

Define the local derived dual by

$$D_\omega L = R\operatorname{Hom}_O(L, \omega[1]). \quad (\text{DC.11})$$

Here complexes use cohomological grading; $\omega[1]$ has ω in degree -1 . For an F -vector space L , its free resolution has length one, its ordinary Hom into ω is zero, and (DC.9) computes its Ext in degree one. Therefore

$$D_\omega L \simeq \operatorname{Hom}_F(L, C_\omega)[0]. \quad (\text{DC.12})$$

The module on the right is again an F -vector space, by pointwise scalar multiplication. Applying the same calculation a second time yields

$$D_\omega^2 L \simeq \operatorname{Hom}_F(\operatorname{Hom}_F(L, C_\omega), C_\omega)[0]. \quad (\text{DC.13})$$

Under these canonical identifications the natural derived bidual map is exactly

$$\operatorname{ev}_L : L \longrightarrow D_\omega^2 L, \quad l \longmapsto (f \longmapsto f(l)). \quad (\text{DC.14})$$

To check the map and its sign, represent $\omega[1]$ by

$$J = [\widehat{\omega} \xrightarrow{-\pi} C_\omega], \quad \deg \widehat{\omega} = -1, \quad \deg C_\omega = 0. \quad (\text{DC.15})$$

The quasi-isomorphism $\omega[1] \rightarrow J$ is the positive Taylor inclusion in degree -1 . Both terms of J are Hom-acyclic for an F -vector source: this is (DC.7) for $\widehat{\omega}$; for C_ω its telescope differential is onto by the recursion $a_0 = 0$, $a_{n+1} = z^{-1}(a_n - c_n)$, and the same direct-sum resolution proves higher vanishing. Thus $\operatorname{Hom}_O(L, J)$ computes $D_\omega L$ and has only the term $\operatorname{Hom}_F(L, C_\omega)$ in degree zero. The identical reasoning applies to that F -vector space as source. Evaluation into J consequently computes the derived bidual map. Its dg sign is $(-1)^{|l||f|} = 1$, since both l and f have degree zero. It is the displayed positive evaluation. Equivalently, comparison of J with an injective resolution gives quasi-isomorphisms on these two Hom complexes by the proved acyclicity, and carries this evaluation to the natural derived one.

The map (DC.14) is injective. If $l \neq 0$, extend l to an F -basis of L , and take its coordinate functional $\lambda : L \rightarrow F$ with $\lambda(l) = 1$. Choose the nonzero class $c = [\sum n! z^n ds] \in C_\omega$. The F -linear map

$f(x) = \lambda(x)c$ satisfies $f(l) = c \neq 0$, so $\text{ev}_L(l) \neq 0$. This proves injectivity of the canonical map; the auxiliary basis and nonzero c only verify it. No assertion of surjectivity is made.

In particular every nonzero class in the original K survives its natural local derived bidual. This includes the original class $k_0 = g \otimes [F_0]$ of (OCQ.5). The original ordinary dual vanishing $\text{Hom}_O(M, \omega) = 0$ therefore cannot be used to discard this kernel from the derived comparison. The finite summand is retained simultaneously. Represent N by $P = [O \xrightarrow{g} O]$ in degrees $-1, 0$. Its dual is $[\omega \xrightarrow{g} \omega]$ in the same degrees: the Hom differential on a degree-minus-one map is $-(-1)^{-1}fg = fg$. Dualizing again gives precisely $[O \xrightarrow{g} O]$. In each term the evaluation sign is positive, since the possible degree pairs are $(-1, 0)$ and $(0, -1)$. Finite free evaluation is the identity in the displayed bases. Consequently $D_\omega N = \omega/g\omega$ in degree zero and $N \xrightarrow{\sim} D_\omega^2 N$. Thus the canonical splitting $M = N \oplus K$ makes its natural derived bidual injective, with the entire finite arithmetic summand mapped isomorphically.

168.4 The exact global target of the local Ext calculation

Let i_{s_0} denote the inclusion of the one-point ringed space with ring O into $(X, \mathcal{O}_X)$. For an O -module E , its direct image has sections E on an open containing s_0 and zero on an open not containing it, with the analytic action through germs. This functor is exact. The exact stalk functor is its left adjoint, as one checks by restricting a sheaf map to germs, so the direct image preserves injective modules. Applying it to an injective resolution proves the exact derived adjunction

$$R\text{Hom}_{\mathcal{O}_X}(\mathcal{M}, i_{s_0*}\omega) \simeq R\text{Hom}_O(M, \omega). \quad (\text{DC.16})$$

The same proof on every open gives the internal version with i_{s_0*} on the right. This places the explicit local extension (DC.10) in a genuine sheaf-derived target. That target $i_{s_0*}\omega$ is not coherent on X : a coherent sheaf supported at a point has a stalk killed by a power of z , whereas no such power kills ω . Formula (DC.16) does not replace the coherent target ω_X by a claimed equality. The genuine coherent sheaf target retains the globally split summand (DC.5), independently of this skyscraper realization.

168.5 The original tau complex, its dilation, and its retained degree zero

The source in A1523 has the two incoming amplitude legs V , the central amplitude $\mathcal{B}$, restrictions Θ and $\widehat{\Theta}$, and its cohomology complex $[V \oplus V \rightarrow \mathcal{B}]$ with differential $(\phi, \psi) \mapsto \Theta(\phi - \widehat{\psi})$. For its balanced action give the plus leg V_+ the action $t = D$, and the minus leg V_- the action $t = 1 - D$. With the original Fourier kernel $e^{-2\pi i x \nu}$, $\widehat{D}\widehat{\psi} = (1 - D)\widehat{\psi}$ and Fourier squares to the identity on these even functions. Therefore Fourier is the exact A -linear isomorphism $V_- \rightarrow V_+$.

Write $\mathcal{V}_+$, $\mathcal{V}_-$, and $\mathcal{B}$ for the sheafifications of the corresponding $\mathcal{O}_X \otimes_A$ presheaves. Flatness of every analytic stalk over A proves that

$$C_A = [\mathcal{V}_+ \xrightarrow{\Theta} \mathcal{B}], \quad C_\tau = [\mathcal{V}_+ \oplus \mathcal{V}_- \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \widehat{\psi})} \mathcal{B}] \quad (\text{DC.17})$$

have their indicated original differentials, with terms in degrees zero and one. There is an explicit cochain isomorphism

$$C_\tau \xrightarrow{\sim} C_A \oplus \mathcal{V}_+[0], \quad (\phi, \psi) \mapsto (\phi - \widehat{\psi}, \widehat{\psi}) \quad (\text{DC.18})$$

in degree zero and the identity on $\mathcal{B}$ in degree one. Its inverse sends (u, v) to $(u + v, \widehat{v})$. Applying the two differentials gives Θu in both directions, proving the cochain assertion. The quotient map in degree one is a quasi-isomorphism $C_A \rightarrow \mathcal{M}[-1]$, because the original Θ is injective and balancing is flat. Thus (DC.3) gives the canonical derived decomposition

$$C_\tau \simeq \mathcal{V}_+[0] \oplus \mathcal{N}[-1] \oplus \mathcal{K}[-1]. \quad (\text{DC.19})$$

The degree-zero diagonal source is retained. On the actual degree-one tau cohomology, (DC.19) is exactly the split comparison $\mathcal{M} = \mathcal{N} \oplus \mathcal{K}$.

The comparison to the analytic complex $C_g = [\mathcal{O}_X \xrightarrow{g} \mathcal{O}_X]$ is also an actual cochain map: on C_A it sends $a \otimes \phi$ to aH_ϕ and $a \otimes F$ to $a\mathcal{M}F$. The original identity $\mathcal{M}\Theta = gH$ proves that it commutes with the differentials. On C_τ compose it with the first projection in (DC.18). The diagonal source is exactly the degree-zero kernel of that first projection. The full composite has the larger kernel computed next; no injectivity of H is inferred. Its induced degree-one map is precisely β .

Write $H : \mathcal{V}_+ \rightarrow \mathcal{O}_X$ and $\text{Mel} : \mathcal{B} \rightarrow \mathcal{O}_X$ for the two displayed maps, and put

$$A_0 = \ker H, \quad A_1 = \ker \text{Mel}. \quad (\text{DC.20})$$

Both maps are onto as sheaf maps. For H use the original $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$ with $H_{\phi_*} = 1$; the section a lifts to $a \otimes \phi_*$. For Mel use $aH_0^{-1} \otimes F_0$ with the original nowhere-zero $H_0 = \sqrt{\pi}e^{s^2/4}$. In (DC.18) coordinates the full kernel complex and its actual short exact sequence are therefore

$$C_{\ker} = [A_0 \oplus \mathcal{V}_+ \xrightarrow{(u,v) \mapsto \Theta u} A_1], \quad 0 \rightarrow C_{\ker} \rightarrow C_\tau \rightarrow C_g \rightarrow 0. \quad (\text{DC.21})$$

The inclusion $\Theta A_0 \subset A_1$ follows from $\text{Mel}\Theta = gH$. Since Θ is injective, $H^0 C_{\ker} = \mathcal{V}_+$, its diagonal factor. The map $A_1 \rightarrow \mathcal{K}$ sends a target vector to its original theta-quotient class. It is onto locally: if a representative F has $\text{Mel}(F) = ga$, subtract $\Theta(a \otimes \phi_*)$ to obtain a representative in A_1 . If a vector in A_1 is $\Theta\phi$, then $gH(\phi) = 0$; injectivity of g on $\mathcal{O}_X$ gives $\phi \in A_0$. Thus its kernel is exactly ΘA_0 . This proves

$$H^0 C_{\ker} = \mathcal{V}_+, \quad H^1 C_{\ker} = A_1 / \Theta A_0 \xrightarrow{\sim} \mathcal{K}, \quad C_{\ker} \simeq \mathcal{V}_+[0] \oplus \mathcal{K}[-1]. \quad (\text{DC.22})$$

For the last assertion project the explicit complex onto its diagonal factor and take the quotient of its remaining degree-one term. The other degree-zero differential is injective, so the resulting map is a quasi-isomorphism.

Finally the original dilation $R_a F(x) = F(x/a)$ acts as R_a on $\mathcal{B}$ and V_+ and as $aR_{1/a}$ on V_- . Direct Fourier substitution gives $\widehat{aR_{1/a}\psi} = R_a\widehat{\psi}$, so (DC.18) is equivariant with these exact actions. The Mellin substitution $x = ay$ gives $\mathcal{M}R_a F = a^s \mathcal{M}F$ and $H_{R_a\phi} = a^s H_\phi$. Hence the cochain map to C_g intertwines these actions with multiplication by a^s on both analytic terms. Dilation commutes with g and preserves $\mathcal{K}$, so it commutes with the projections (DC.2); in particular $R_a j = j a^s$ on $\mathcal{N}$.

At an actual zero ρ with its full order m and full local unit $g(s) = u(s - \rho)(s - \rho)^m$, this action is exactly

$$a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho[v] = (s - \rho)[v] \quad \text{in } \mathcal{O}/(g). \quad (\text{DC.23})$$

No nilpotent term is removed. The finite coherent dual summand (DC.5) passes by the original maps $[v ds] \mapsto [v ds/g]$ and residue to the original tau dual target exactly as proved in (OCQ.26)–(OCQ.30). Its dual dilation is $\lambda \mapsto (q \mapsto a\lambda(R_{1/a}q))$; the differential representative is multiplied by a^{1-s} , including the retained one-dimensional character.

For a proper admitted source W , the entire relation quotient $\mathcal{O}_X \otimes_A (V/W)$ and its source-label transition remain the ones in (OCQ.31)–(OCQ.33). No invertibility on that larger proper-label kernel is used in (DC.1)–(DC.23). The diagonal degree-zero summand in (DC.19) belongs to the joint face $S = \{+, -\}$. Under (DC.18), the plus-only source is exactly the subspace $\{(u, 0)\}$, and the minus-only source is exactly $\{(-v, v)\}$. On these subspaces the differential is respectively Θu and $-\Theta v$, so each one-leg complex has $H^0 = 0$ and $H^1 = \mathcal{M}$. These are the specified restrictions of the cochain maps; no joint-face diagonal is added to a one-leg face. On each such original support fibre the restricted linear maps lift by $(S, v) \mapsto (S, f(v))$ and external $\tau \mapsto \tau$, as in (OCQ.34). Thus a killed vector lands at its supported zero. These exact cochain, coherent, residue, and derived-dual maps retain the actual counterfactual packet throughout; they do not supply a new weight bound or assert the existence of such a zero.

169 Exact arithmetic quotient transport and its unavoidable window defect

This calculation uses the literal four windows of (AT13) and (AW5), the source isometry constructed by (AW9)–(AW10), and the arithmetic observation of (AT2a). The principal-angle calculations (EW.25)–(EW.35c) remain available; the new calculation concerns the arithmetic observation of a repaired source map and the exact operator left in its intertwining identity.

169.1 The retained source, forms, and arithmetic maps

Fix an actual full packet polynomial h of $g = 2\xi$, an integer $k \geq 1$, and the original cyclic annihilator $\chi = \chi_{h,k}$, monic of degree $q \geq 1$. Retain

$$E = \mathbb{C}[S]/(\chi), \quad E_h = \mathbb{C}[s]/(h), \quad H = \mathbb{C}[S]_{\leq 2q}, \quad S(u) = k/2 + iu. \quad (\text{QT.1})$$

The increasing-power remainder coordinates of E are fixed. The actual arithmetic injection, including its full Taylor unit, is

$$\begin{aligned} \eta : E &\longrightarrow E_h^{\otimes k}, & \eta[P] &= v_h^{\otimes k} P(A_h^{(k)})1, \\ v_h &= j_h(g/h), & Q_{\text{ar}} &= \mathcal{B}/\Theta V, \\ \mathcal{I} &= \sigma_h^{\otimes k} \eta : E \hookrightarrow Q_{\text{ar}}^{\otimes k}, & \mathcal{O} = \mathcal{I}J : H &\longrightarrow Q_{\text{ar}}^{\otimes k}. \end{aligned} \quad (\text{QT.2})$$

Here $A_h^{(k)}$ is the original tensor-sum multiplication operator, J is literal remainder modulo χ , j_h takes the full Taylor remainder modulo h , and $\sigma_h : E_h \hookrightarrow Q_{\text{ar}}$ is the original section. Both injections in (QT.2) are the source constructions of (AT2)–(AT2a), not new choices. In particular no nilpotent Taylor coefficient of h , χ , or v_h is discarded. The target is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau} = \{\tau, 1\}$, with one-factor complex $[V \oplus V \longrightarrow \mathcal{B}]$ and differential $(\phi, \psi) \mapsto \Theta(\phi - \hat{\psi})$.

Use precisely the two densities

$$\begin{aligned} m_1 &= w_h^{*k}, & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a} \Gamma(2a), \\ m_t &= (1-t)m_0 + tm_1, \quad 0 \leq t \leq 1, & v(u) &= (1, S(u), \dots, S(u)^{2q}), \\ M(t) &= \int_{\mathbb{R}} v(u)^* v(u) m_t(u) du. \end{aligned} \quad (\text{QT.3})$$

These densities have the positive almost-everywhere property and all finite polynomial moments proved in the supplied AT sources. Their masses are retained: $\int m_t = (1-t)(2\pi)^{k/2} + t\mu_h^k$, where $\mu_h = \int w_h$. All orthogonality, operator adjoints, singular values, and Hilbert–Schmidt norms below use the positive form $\langle x, y \rangle_t = x^* M(t) y$. For operators on H , write $Z^\sharp = M^{-1} Z^* M$ and $\|Z\|_{\text{HS},t}^2 = \text{Tr}(Z^\sharp Z)$. For maps between its orthogonal subspaces, the same notation uses their restricted forms. A star on a coefficient matrix means its ordinary conjugate transpose.

At a fixed t , put

$$\begin{aligned} F &= \chi \mathbb{C}[S]_{\leq q} = \ker J, & K &= F^\perp, & L &= \mathbb{C}[S]_{\leq q-1}, \\ Q &= P_F, & \Pi &= I - Q = P_K, & P &= P_L, \\ s : E &\longrightarrow L \subset H, & s[P] &= \text{the degree} < q \text{ remainder}, \\ R &= R_{2q}(t) = \Pi s, & G &= R^* M R, & G_\ell &= s^* M s. \end{aligned} \quad (\text{QT.4})$$

The symbol P_X denotes the orthogonal projection onto the specified subspace X . Monic division gives the literal algebraic direct sum $H = L \oplus F$; it does not declare these two subspaces orthogonal. The minimum-section formula (AT6) proves $JR = I_E$ and $\Pi = RJ$. Thus $R : (E, G) \rightarrow K$ is an isometry with inverse $J|_K$. The low-degree Gram is $G_\ell = G_{q-1}(t)$.

Retain U, W, T exactly as in (AW5)–(AW10). Those source formulas prove

$$\begin{aligned} T^\sharp T = TT^\sharp = I, \quad UT = TW, \quad TL = K, \quad T(L^\perp) = F, \\ U|_K = 0, \quad I_F \preceq U_F := U|_F \preceq 2I_F. \end{aligned} \quad (\text{QT.5})$$

The eigenvalues of U_F are 1, 1 and $q - 1$ copies of 2. For $q = 1$, they are just 1, 1. In the orthogonal decomposition $H = K \oplus F$, define its fully typed blocks by

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a : K \rightarrow K, \quad b : F \rightarrow K, \quad c : K \rightarrow F, \quad d : F \rightarrow F. \quad (\text{QT.6})$$

In particular $\Pi T = TP$.

169.2 An invertible repair preserving every arithmetic coefficient

Both square diagonal blocks in (QT.6) are invertible. Indeed, if $ax = 0$ for $x \in K$, then $Tx \in F = T(L^\perp)$, so $x \in K \cap L^\perp = (F + L)^\perp = 0$. If $dy = 0$ for $y \in F$, then $Ty \in K = TL$, so $y \in F \cap L = 0$. These are injective endomorphisms of finite spaces, hence bijective.

Define

$$X = QT + \Pi = \begin{pmatrix} I_K & 0 \\ c & d \end{pmatrix}, \quad X^{-1} = \begin{pmatrix} I_K & 0 \\ -d^{-1}c & d^{-1} \end{pmatrix}. \quad (\text{QT.7})$$

Multiplying these displayed block matrices gives both inverse identities. Since $JQ = 0$ and $J\Pi = J$, they obey

$$JX = J, \quad JX^{-1} = J, \quad \mathcal{O}X = \mathcal{O}, \quad \mathcal{O}X^{-1} = \mathcal{O}. \quad (\text{QT.8})$$

These are equalities in the original target, before changing its presentation. For every original coefficient operator B on that target for which the composition is defined, $B\mathcal{O}X = B\mathcal{O}$. This statement does not require B to preserve the cyclic subspace $\mathcal{I}(E)$. On that cyclic subspace the induced map of X is exactly I_E ; it therefore commutes with every multiplication $f(A_\chi)$, where A_χ is multiplication by S on E and $f \in \mathbb{C}[S]$. The complete arithmetic unit in (QT.2) remains in both sides. At any nonempty original support label A , the lift is $(A, x) \mapsto (A, Xx)$, and $\tau \mapsto \tau$. Consequently the two supported arithmetic coefficients coincide at that label, including supported zero; no supported zero is replaced by τ .

Moreover, X is the unique linear map nearest to T , in the original Hilbert–Schmidt norm, among all maps Z satisfying $\mathcal{O}Z = \mathcal{O}$. Indeed injectivity of $\mathcal{I}$ gives $JZ = J$, equivalently $\Pi Z = \Pi$. Such a Z has top block $(I_K, 0)$, whereas its lower block is unconstrained. Orthogonality of the two output summands gives the exact identity

$$\|Z - T\|_{\text{HS},t}^2 = \|I_K - a\|_{\text{HS},t}^2 + \|b\|_{\text{HS},t}^2 + \|\mathcal{Q}(Z - T)\|_{\text{HS},t}^2. \quad (\text{QT.9})$$

Its last summand vanishes precisely for $Z = X$. Unitarity gives $\|a\|_{\text{HS},t}^2 + \|b\|_{\text{HS},t}^2 = q$, and hence

$$\|X - T\|_{\text{HS},t}^2 = 2q - 2 \operatorname{Re} \operatorname{Tr}_K a. \quad (\text{QT.10})$$

Every map and inverse constructed here is smooth in t : the source Grams, their positive inverses, and the fixed-phase Gram–Schmidt construction of T are smooth on the compact parameter interval.

169.3 The exact quotient contraction and its original metric

For comparison, the unique nearest linear map to T that merely preserves F is

$$Y = T - \Pi T Q = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}. \quad (\text{QT.11})$$

The same orthogonal-block minimization proves this assertion: preserving F forces precisely the upper-right block to be zero. It is invertible since a, d are. Its quotient action is the explicit invertible map $\mathfrak{a} = JTR : E \rightarrow E$.

Here its entire original-metric contraction can be calculated. Put $\mathfrak{w} = JTs : E \rightarrow E$. Because $Ts(E) = K$, $Ts = R\mathfrak{w}$; the isometry of T therefore gives $\mathfrak{w}^*G\mathfrak{w} = G_\ell$. Also $s^*MR = G$, since $s - R$ is relation-valued and orthogonal to $R(E)$. The projection formula onto $L = s(E)$ is $P = sG_\ell^{-1}s^*M$, so $PR = sG_\ell^{-1}G$. Using $\Pi T = TP$ yields all of the following exact identities:

$$\mathfrak{a} = \mathfrak{w}G_\ell^{-1}G, \quad \mathfrak{a}^*G\mathfrak{a} = GG_\ell^{-1}G, \quad \mathfrak{a}^\sharp G\mathfrak{a} = G_\ell^{-1}G, \quad \mathfrak{a}^\sharp G = G^{-1}\mathfrak{a}^*G. \quad (\text{QT.12})$$

No basis diagonalization or deletion of multiplicities entered these identities. In particular

$$\begin{aligned} \ell(t) &:= \|b\|_{\text{HS},t}^2 = \|c\|_{\text{HS},t}^2 = q - \text{Tr}(G_\ell^{-1}G), \\ |\det X|^2 &= |\det d|^2 = \det(\mathfrak{a}^\sharp \mathfrak{a}) = \frac{\det G}{\det G_\ell}. \end{aligned} \quad (\text{QT.13})$$

For the first line, the diagonal blocks of $TT^\sharp = I$ and $T^\sharp T = I$ respectively give $aa^\sharp + bb^\sharp = I_K$ and $a^\sharp a + c^\sharp c = I_K$. Take traces and then use (QT.12). For the second line, triangularity of X gives $\det X = \det d$. The identities $dd^\sharp = I_F - cc^\sharp$ and $a^\sharp a = I_K - c^\sharp c$ show that their determinants coincide: the nonzero eigenvalues of $cc^\sharp$ and $c^\sharp c$, with their multiplicities, coincide, because for any eigenvalue $\lambda > 0$, c and $\lambda^{-1}c^\sharp$ are inverse maps between the corresponding eigenspaces. The extra dimension of F contributes the eigenvalue one after subtracting from its identity. Finally take determinants in (QT.12).

The repair's source-energy change is also exact, for every $z \in H$:

$$\|Xz\|_t^2 - \|z\|_t^2 = \|\Pi z\|_t^2 - \|\Pi Tz\|_t^2. \quad (\text{QT.14})$$

Indeed $Xz = QTz + \Pi z$ has orthogonal summands, whereas the isometry gives $\|z\|_t^2 = \|QTz\|_t^2 + \|\Pi Tz\|_t^2$. Thus its nonunitarity is a calculated source energy, while its arithmetic class is unchanged by (QT.8).

169.4 A positive original-moment obstruction and a sharp repair minimum

Write $\chi(S) = \sum_{j=0}^q \alpha_j S^j$, and put

$$\chi^\dagger(S) = \sum_{j=0}^q \overline{\alpha_j}(k - S)^j, \quad v = \chi\chi^\dagger \in F, \quad \beta(t) = \frac{\left(\int_{\mathbb{R}} |\chi(S(u))|^2 m_t(u) du\right)^2}{\left(\int_{\mathbb{R}} m_t(u) du\right) \left(\int_{\mathbb{R}} |\chi(S(u))|^4 m_t(u) du\right)}. \quad (\text{QT.15})$$

These are the literal reflected polynomial and relation of (CR7). All displayed integrals are finite. Both denominator factors and the numerator are strictly positive, because a nonzero polynomial has finitely many zeros on the line and $m_t > 0$ almost everywhere. On that line $k - S = \overline{S}$, so $\langle 1, v \rangle_t = \int |\chi|^2 m_t > 0$. As $1 \in L$, orthogonal projection followed by Cauchy-Schwarz gives

$$\|Pv\|_t^2 \geq \frac{|\langle 1, v \rangle_t|^2}{\|1\|_t^2}.$$

Since $v \in F$ and $bv = \Pi T v = T P v$, isometry of T implies $\|bv\|_t = \|Pv\|_t$. The operator norm of b is at most its Hilbert–Schmidt norm. Dividing the preceding inequality by $\|v\|_t^2 = \int |\chi|^4 m_t$ proves the explicit floor

$$0 < \beta(t) \leq \ell(t). \quad (\text{QT.16})$$

It also proves $L \neq K$, directly on the original density.

The two retained windows have a nonzero intertwining defect after any arithmetic-class-preserving repair. More precisely,

$$UX - XW = -\Pi W, \quad \min_{OZ=O} \|UZ - ZW\|_{\text{HS},t}^2 = \|\Pi W\|_{\text{HS},t}^2 =: \nu(t)^2. \quad (\text{QT.17})$$

To prove the first equality, $UQ = U$, $U\Pi = 0$, and $UT = TW$ give $UX = TW$. Since $\Pi TW = \Pi UT = 0$, one also has $XW = QTW + \Pi W = TW + \Pi W$. For any competing Z , (QT.8)’s argument gives $\Pi Z = \Pi$. Then $\Pi(UZ - ZW) = -\Pi W$, since $\Pi U = 0$. Orthogonality of output components proves its squared norm is at least $\|\Pi W\|_{\text{HS},t}^2$, and X attains it.

This minimum has its complete arithmetic quotient expression:

$$\begin{aligned} \nu(t)^2 &= \text{Tr}(G^{-1} R^* M W^2 R) = \text{Tr}_K(c^\sharp U_F^2 c), \\ 0 < \beta(t) &\leq \ell(t) \leq \nu(t)^2 \leq 4\ell(t). \end{aligned} \quad (\text{QT.18})$$

For the first trace, use $\Pi = R G^{-1} R^* M$, which is the orthogonal projection formula from (QT.4), and $\|\Pi W\|_{\text{HS},t}^2 = \text{Tr}(\Pi W^2)$. For the second, $W = T^\sharp U T$, hence the K, K block of $W^2 = T^\sharp U^2 T$ is $c^\sharp U_F^2 c$. Finally (QT.5) gives $I_F \preceq U_F^2 \preceq 4I_F$; taking traces after the congruence by c and using (QT.13), (QT.16) proves every inequality.

In fact there is no invertible F -preserving operator Z with $UZ = ZW$, even if its induced quotient map is allowed to change. An invertible such Z has $Z^{-1}F = F$. The purported intertwining and $\text{im } U = F$ would give $\text{im } W = Z^{-1}F = F$. But (AW7)–(AW10) give $\text{im } W = L^\perp$. Thus $F = L^\perp$ and $K = L$, contradicting the original positive pairing just proved. This obstruction does not stop the construction: (QT.7), (QT.17), and the next subsection give an invertible exact quotient repair, its attained optimal defect, and its full effect on the signed arithmetic calculation.

169.5 Square-zero conjugation error and the signed arithmetic bound

The repaired intertwining has an especially explicit conjugation form:

$$XW X^{-1} = U + \mathcal{N}, \quad \mathcal{N} = \Pi W X^{-1} = \begin{pmatrix} 0 & c^\sharp U_F \\ 0 & 0 \end{pmatrix}, \quad \mathcal{N}^2 = 0, \quad \|\mathcal{N}\|_{\text{HS},t}^2 = \nu(t)^2. \quad (\text{QT.19})$$

Indeed $W = T^\sharp U T$ gives its complete two-by-two blocks

$$W = \begin{pmatrix} c^\sharp U_F c & c^\sharp U_F d \\ d^\sharp U_F c & d^\sharp U_F d \end{pmatrix}.$$

Multiply its top row by X^{-1} in (QT.7): the first entry is $c^\sharp U_F c - c^\sharp U_F d d^{-1} c = 0$, and the second is $c^\sharp U_F d d^{-1} = c^\sharp U_F$. This proves the displayed matrix, its square-zero property, and its norm equality by (QT.18). Its range is in K and it vanishes on K ; its nonzero arithmetic output on suitable relation vectors has not been suppressed.

Write

$$\mathcal{C}(t) = M(t)^{-1}(M(1) - M(0)), \quad \mathcal{E}(t) = \text{Re Tr}(W X^{-1}[\mathcal{C}, X]), \quad [\mathcal{C}, X] = \mathcal{C}X - X\mathcal{C}. \quad (\text{QT.20})$$

The original four-window value is $\mathcal{B}(t) = \log(V_{q-1}(t)V_q(t)/(V_{2q-1}(t)V_{2q}(t)))$, where $V_j = \det G_j$ uses the unchanged minimum quotient Grams. The exact signed formula (AT15) now becomes

$$\begin{aligned} \dot{\mathcal{B}}(t) &= \mathcal{E}(t) - \text{Re Tr}(\mathcal{N}\mathcal{C}), \\ \Delta \mathcal{B} &= \int_0^1 \mathcal{E}(t) dt - \int_0^1 \text{Re Tr}(\mathcal{N}(t)\mathcal{C}(t)) dt. \end{aligned} \quad (\text{QT.21})$$

For the first identity, substitute $U = XW X^{-1} - \mathcal{N}$ from (QT.19) into $\dot{\mathcal{B}} = \text{Tr}((U - W)\mathcal{C})$. Cyclicity of the finite trace transforms its first two terms into $\text{Tr}(W X^{-1}[\mathcal{C}, X])$. The left-hand side is real, so taking real parts gives exactly (QT.21). Integrating its smooth functions proves the second identity. This repaired commutator preserves $\mathcal{O}$ exactly; the explicit square-zero contribution is the remaining correction.

It is controlled by a particular part of the original arithmetic source energy, rather than by its whole spectral spread. Set

$$e(t)^2 := \text{Tr}\left(G^{-1}\dot{R}^* M \dot{R}\right) = -\frac{1}{2} \text{Tr}(G^{-1}\ddot{G}) \geq 0. \quad (\text{QT.22})$$

Here all dots differentiate the same source interpolation (QT.3). For completeness the AT variation identity gives, with $B : \mathbb{C}[S]_{\leq q} \rightarrow H$ multiplication by χ ,

$$\dot{R} = -B(B^* M B)^{-1} B^*(M(1) - M(0))R = -\mathcal{Q}C R.$$

Substituting this formula into $\dot{R}^* M \dot{R}$ cancels the middle factor $B^* M B$, giving

$$\dot{R}^* M \dot{R} = R^*(M(1) - M(0))B(B^* M B)^{-1} B^*(M(1) - M(0))R.$$

The fully proved second variation (AT7) is minus twice this matrix, which proves (QT.22). Moreover, under the isometry $R : (E, G) \rightarrow K$, $\mathcal{Q}C|_K$ has coefficient map $\mathcal{Q}C R = -\dot{R}$. Thus

$$\|\mathcal{Q}C\Pi\|_{\text{HS},t} = e(t). \quad (\text{QT.23})$$

Only this off-diagonal metric-variation block contributes to $\text{Tr}(\mathcal{N}\mathcal{C})$, since (QT.19) has a single upper-right block. In any orthonormal bases of K, F , the trace is the sum of entrywise products of those two blocks. Cauchy-Schwarz on that finite list, then (QT.19), (QT.23), proves the new quantitative transport estimate

$$|\dot{\mathcal{B}}(t) - \mathcal{E}(t)| \leq \nu(t)e(t) \leq 2\sqrt{\ell(t)}e(t). \quad (\text{QT.24})$$

In particular the error vanishes exactly when $\text{Re Tr}(\mathcal{N}\mathcal{C}) = 0$, and the sufficient condition $e(t) = 0$ does not require the whole relative source operator to be scalar.

Every quantity is continuous on the compact interval. Define the fully specified original-source quantities

$$\mathfrak{E} = \int_0^1 \mathcal{E}(t) dt, \quad \mathfrak{F} = \int_0^1 \nu(t)e(t) dt.$$

Integration, followed by Cauchy-Schwarz for real functions of t , gives

$$|\Delta\mathcal{B} - \mathfrak{E}| \leq \mathfrak{F} \leq \left(\int_0^1 \nu(t)^2 dt\right)^{1/2} \left(-\frac{1}{2} \int_0^1 \text{Tr}(G^{-1}\ddot{G}) dt\right)^{1/2}. \quad (\text{QT.25})$$

This is an interval centered at the actual class-preserving commutator, with an explicitly calculated relation-speed error. It supplements, rather than assumes a sign for, the original signed correction.

To combine with the fully retained AW spectrum, let $b_1 \leq \dots \leq b_{2q+1}$ be the positive eigenvalues of the actual relative source operator $M(0)^{-1}M(1)$, and put

$$\mathfrak{L}_q = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}.$$

The sum is empty when $q = 1$. The original proof (AW14) and (QT.25) give simultaneously

$$\max\{-\mathfrak{L}_q, \mathfrak{E} - \mathfrak{F}\} \leq \Delta\mathcal{B} \leq \min\{\mathfrak{L}_q, \mathfrak{E} + \mathfrak{F}\}. \quad (\text{QT.26})$$

This follows by intersecting two intervals both proved for the same unchanged signed value; the second interval keeps the measured quotient-transport defect and the original source-variation energy. None of these expressions treats the arithmetic density or its full Taylor unit as a freely selectable coefficient.

If $M(1) = c_0 M(0)$ with $c_0 > 0$, the source mass retains that literal factor, $\mathcal{C}(t)$ is scalar, and R is constant. Thus $\mathcal{E} = e = \mathfrak{F} = 0$, and (QT.21) gives $\Delta\mathcal{B} = 0$ exactly. The strictly positive obstruction $\nu(t)^2$ to a class-preserving window intertwiner still exists; its contraction with this scalar metric variation is zero. This illustrates why the nonzero source obstruction alone is not a proof of an RH contradiction. Equations (QT.17)–(QT.26) compute its exact contribution on the retained arithmetic programme.

169.6 The literal finite relation complex and its computed homotopy

The repair also acts on the original finite cochain complex, not just on its quotient. Put the two terms in degrees $-1, 0$:

$$\mathcal{C}^{-1} = \mathbb{C}[S]_{\leq q}, \quad \mathcal{C}^0 = H, \quad \partial = B = \times\chi, \quad H_F = B^*MB.$$

These are exactly the $N = 2q$ terms of (AT3), before taking cohomology. Multiplication by the literal monic χ is injective, so H_F is positive definite. Define the three actual coefficient maps

$$\begin{aligned} X_{\text{deg}}^{-1} &:= H_F^{-1}B^*MTB : \mathcal{C}^{-1} \longrightarrow \mathcal{C}^{-1}, & X_{\text{deg}}^0 &:= X : \mathcal{C}^0 \longrightarrow \mathcal{C}^0, \\ K_X &:= H_F^{-1}B^*M(T - I) : \mathcal{C}^0 \longrightarrow \mathcal{C}^{-1}. \end{aligned} \quad (\text{QT.27})$$

The superscript -1 in X_{deg}^{-1} labels cochain degree; it does not denote the inverse operator. The latter exists, as the calculation below also proves. Substituting the original projection $Q = BH_F^{-1}B^*M$ gives every chain and homotopy identity:

$$X_{\text{deg}}^0 B = BX_{\text{deg}}^{-1}, \quad X_{\text{deg}}^0 - I_H = BK_X, \quad X_{\text{deg}}^{-1} - I_{\mathcal{C}^{-1}} = K_X B. \quad (\text{QT.28})$$

For the first equality, $\Pi B = 0$, hence $XB = QT B = BH_F^{-1}B^*MTB$. For the second, $X - I = QT + \Pi - I = Q(T - I) = BK_X$. For the third use $H_F = B^*MB$ directly in (QT.27). These three substitutions prove that the displayed two-term map is a cochain map and that K_X is a cochain homotopy from its identity to that map, with the displayed signs and degrees. Moreover $X(F) = F$ by (QT.7), and $X|_F = d$ is invertible. Since $B : \mathcal{C}^{-1} \rightarrow F$ is an isomorphism, the first equality identifies $X_{\text{deg}}^{-1} = B^{-1}(X|_F)B$, proving its invertibility without using an inverse on the infinite source.

In particular, for every polynomial vector $z \in H$, the change of its original source representative is the literal relation $\chi K_X z$, with its full coefficients given by (QT.27). Composing the original source comparison $\mathcal{V}$ of (AT2) gives $\mathcal{V}Xz - \mathcal{V}z = \mathcal{V}(\chi K_X z)$. The comparison identifies the latter's arithmetic class with $\mathcal{I}J(\chi K_X z) = 0$, using the same full unit $v_h^{\otimes k}$ of (QT.2). Thus the computed finite homotopy and the equality in the original arithmetic target agree under the already constructed source map; no new choice of cohomology section is made here. Every map in (QT.27)–(QT.28) lifts at a nonempty original support label by retaining that label and its displayed coefficient map, with $\tau \mapsto \tau$. Both sides of the homotopy equations therefore agree at each such supported fibre, including its supported zero. This is the exact morphism connecting the repaired source calculation to the relation complex used by the programme.

170 The closest unitary preserving the actual arithmetic observation

The source isometry in AW9–AW10 has a unique closest unitary preserving the original arithmetic observation. This calculation constructs that unitary and proves its exact source-metric cost. It also

computes the coefficient defect of the triangular relation-preserving repair and the positive triangular quotient block forced by the prescribed Gram–Schmidt phases. Every formula retains the original polynomial, arithmetic measure, Taylor unit, relation quotient, and absolute τ base.

170.1 Original objects and forms

Fix $g = 2\xi$, its actual full packet h , the full coefficient algebra $E_h = \mathbb{C}[s]/(h)$, a positive integer k , the exact monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $t \in [0, 1]$. Retain

$$H = \mathbb{C}[S]_{\leq 2q}, \quad E = \mathbb{C}[S]/(\chi), \quad J = J_{2q} : H \longrightarrow E.$$

Here J is literal remainder in the original increasing-power frame. The original arithmetic observation is

$$\mathcal{O} = \iota J, \quad \iota = \sigma_h^{\otimes k} \eta, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1, \quad v_h = j_h(g/h) \in E_h^\times.$$

Both η and ι are the original injections; thus $\ker \mathcal{O} = \ker J$. Every Taylor coefficient of the unit remains present.

Write $v_{2q}(u) = (1, S(u), \dots, S(u)^{2q})$. Give H its actual form

$$\langle x, y \rangle_t = x^* M(t) y, \quad M(t) = \int_{\mathbb{R}} v_{2q}(u)^* v_{2q}(u) m_t(u) du, \quad S = k/2 + iu,$$

$$m_t = (1 - t)m_0 + tm_1, \quad m_1 = w_h^{*k}, \quad w_h(u) = \frac{|(g/h)(1/2 + iu)|^2}{2\pi},$$

$$m_0(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, \quad c_a = 2^{1-2a} \Gamma(2a).$$

With $\mu_h = \int_{\mathbb{R}} w_h(u) du$, the original masses are μ_h^k and $(2\pi)^{k/2}$. Adjoints denoted star below use this form and the induced subspace forms. In the fixed coefficient frame, $Z^* = M^{-1} Z^* M$. Define $\|Z\|_{\text{HS},t}^2 = \text{Tr}(Z^* Z)$.

Set

$$F = \chi \mathbb{C}[S]_{\leq q} = \ker J, \quad K = F^\perp, \quad L = \mathbb{C}[S]_{\leq q-1}, \quad Q = Q_F, \quad \Pi = I - Q. \quad (\text{VR1})$$

Here Q_F is the orthogonal projection onto F in the actual form. Their dimensions are $q + 1, q, q$ respectively. Polynomial division proves $H = L \oplus F$ algebraically: their intersection is zero by degree and their dimensions add to $2q + 1$. No orthogonality is assigned to that direct sum.

Let $s : E \rightarrow L$ be the remainder section, $R = R_{2q} : E \rightarrow K$ the original minimum section, and $G = G_{2q} = R^* M R$. AT6 gives

$$JR = I_E, \quad RJ = \Pi, \quad R^* M p = G J p \quad (p \in H). \quad (\text{VR2})$$

Indeed RJ and Π have image K , kernel F , and equal the identity on K . For the last identity decompose $p = R J p + Q p$ and use $R(E) \perp F$. Thus R is an isometry from the original G -space E to K . Write $G_0 = s^* M s = G_{q-1}$; this subscript denotes the low section, not parameter value zero.

The prescribed AW isometry satisfies

$$T^* T = T T^* = I, \quad T L = K, \quad T L^\perp = F, \quad U = T W T^*, \quad (\text{VR3})$$

with exactly the original window operators U, W . Relative to $H = K \oplus F$ set

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A = \Pi T|_K, \quad B = \Pi T|_F, \quad C = Q T|_K, \quad D = Q T|_F. \quad (\text{VR4})$$

Here C is the block $K \rightarrow F$, not the derivative operator of AT/AW.

170.2 Invertible blocks and strict leakage

Multiplication of the two unitarity identities gives

$$\begin{aligned} A^*A + C^*C &= I_K, & AA^* + BB^* &= I_K, \\ D^*D + B^*B &= I_F, & DD^* + CC^* &= I_F, \\ A^*B + C^*D &= 0, & AC^* + BD^* &= 0. \end{aligned} \quad (\text{VR5})$$

Both A, D are invertible. If $Ax = 0$ with $x \in K$, then $Tx \in F$, so $x \in K \cap T^{-1}F = F^\perp \cap L^\perp = (F + L)^\perp = 0$. If $Dy = 0$ with $y \in F$, then $Ty \in K$, so $y \in F \cap T^{-1}K = F \cap L = 0$. Both maps are square maps on their displayed spaces, proving invertibility.

Their singular values lie in $(0, 1]$. Define $|A| = (A^*A)^{1/2}$ and $\text{polar}(A) = A|A|^{-1}$; both are defined because A is invertible. Taking traces in VR5 gives

$$\ell := \|B\|_{\text{HS},t}^2 = \|C\|_{\text{HS},t}^2 = q - \text{Tr}(A^*A). \quad (\text{VR6})$$

In fact $\ell > 0$ on the actual source. For $\chi(S) = \sum_{j=0}^q a_j S^j$, define $\chi^\dagger(S) = \sum_{j=0}^q \overline{a_j}(k - S)^j$. Then $v = \chi\chi^\dagger \in F$ and

$$\langle 1, v \rangle_t = \int_{\mathbb{R}} |\chi(k/2 + iu)|^2 m_t(u) du > 0. \quad (\text{VR7})$$

On the unchanged line $k - S = \overline{S}$, proving the equality. The moment bounds give finiteness; the nonzero polynomial χ has only finitely many zeros on that line, and m_t is positive almost everywhere, giving strict positivity. Thus $1 \in L$ is not in K , so $K \neq L$. If $\ell = 0$ then $B = 0$, so $T(F) = F$ by equal finite dimensions. But $T^{-1}F = L^\perp$ by VR3; hence $F = L^\perp$ and $K = L$, a contradiction.

If $s_1, \dots, s_q$ are all singular values of A , those of D are exactly

$$1, s_1, \dots, s_q. \quad (\text{VR8})$$

For any $\lambda > 0$ the maps B^* and $\lambda^{-1}B$ identify the λ -eigenspaces of BB^* and B^*B . Their nonzero spectra therefore coincide, including multiplicities; the respective dimensions add one zero eigenvalue to the latter. Subtracting from one in $AA^* = I - BB^*$ and $D^*D = I - B^*B$, then taking positive square roots, proves VR8, including rank-deficient B .

170.3 Closest repairs

The unique closest linear map to T under the exact constraint $JX = J$ is

$$X = QT + \Pi = \begin{pmatrix} I_K & 0 \\ C & D \end{pmatrix}. \quad (\text{VR9})$$

Indeed $JY = J$ is equivalent, by multiplication by R , to $\Pi Y = \Pi$. It fixes the upper blocks of Y to $I_K, 0$ and leaves its lower blocks free. The four block spaces are orthogonal in Hilbert–Schmidt inner product, so the unique minimizers of the remaining block errors are C, D . Thus

$$\begin{aligned} d_{\text{lin}}^2 &:= \min_{JY=J} \|Y - T\|_{\text{HS},t}^2 \\ &= \|I_K - A\|_{\text{HS},t}^2 + \|B\|_{\text{HS},t}^2 = 2q - 2 \text{Re Tr } A. \end{aligned} \quad (\text{VR10})$$

The last equality expands the square and uses $AA^* + BB^* = I_K$.

The complete class of exact-observation-preserving unitaries is

$$\{V : V^*V = VV^* = I, JV = J\} = \{I_K \oplus V_F : V_F \text{ unitary on } F\}. \quad (\text{VR11})$$

For $JV = J$ implies $V(F) \subset F$, hence $V(F) = F$. Unitarity then preserves K . For $x \in K, Vx - x \in K$ and $J(Vx - x) = 0$, hence $Vx = x$. Conversely every displayed block map has $JV = J$.

Since D is invertible, $|D| = (D^*D)^{1/2}$ is positive definite and $V_D = D|D|^{-1}$ is unitary on F . The unique closest member of VR11 is

$$V_{\text{obs}} = \Pi + V_D Q = I_K \oplus V_D. \quad (\text{VR12})$$

Only the lower diagonal block varies in the distance, so one maximizes $\text{Re Tr}(V_F^* D)$. Write $Z = V_F^* V_D$, a unitary, and diagonalize $|D|$ in an orthonormal basis with positive eigenvalues d_j . The expression is $\sum_j d_j \text{Re } Z_{jj} \leq \sum_j d_j$. Equality forces $Z_{jj} = 1$ for every j . The unit length of each column then makes its other entries zero, so $Z = I$. This proves attainment and uniqueness.

The exact minimum is

$$\begin{aligned} d_{\text{unit}}^2 &:= \min_{\substack{V^* V = I \\ J V = J}} \|V - T\|_{\text{HS},t}^2 \\ &= d_{\text{lin}}^2 + \|C\|_{\text{HS},t}^2 + \|V_D - D\|_{\text{HS},t}^2 \\ &= d_{\text{lin}}^2 + 2 \text{Tr}_F(I_F - |D|) \\ &= 2q - 2 \text{Re Tr } A + 2q - 2 \text{Tr}_K |A|. \end{aligned} \quad (\text{VR13})$$

The second line adds orthogonal block errors beyond X . For the third, $\|C\|^2 = \text{Tr}(I_F - |D|^2)$ and $\|V_D - D\|^2 = \text{Tr}((I_F - |D|)^2)$. Adding these polynomials gives the stated expression; VR8 gives the last line. Consequently

$$\ell \leq d_{\text{unit}}^2 - d_{\text{lin}}^2 = 2 \sum_{j=1}^q (1 - s_j) \leq 2\ell, \quad d_{\text{unit}}^2 \geq 2\ell > 0. \quad (\text{VR14})$$

Each inequality follows termwise from $1 - s_j^2 \leq 2(1 - s_j) \leq 2(1 - s_j^2)$, with $\ell = \sum_j (1 - s_j^2)$, and from $d_{\text{lin}}^2 \geq \ell$.

If only preservation of F is imposed on the unitary, its two diagonal blocks may vary independently. The same positive-weight trace argument proves that the unique closest one is

$$V_{\text{rel}} = \text{polar}(A) \oplus \text{polar}(D), \quad \|V_{\text{rel}} - T\|_{\text{HS},t}^2 = 4(q - \text{Tr } |A|). \quad (\text{VR15})$$

Its quotient action is the unitary corresponding to $\text{polar}(A)$, not necessarily the identity. The exact extra cost of the entire original observation is

$$d_{\text{unit}}^2 - \|V_{\text{rel}} - T\|_{\text{HS},t}^2 = 2(\text{Tr } |A| - \text{Re Tr } A) \geq 0. \quad (\text{VR16})$$

The last inequality follows from the preceding trace maximization with the identity as a candidate unitary against A .

170.4 AW's specified phases and a determinant bound for the repair cost

This calculation uses the prescribed Gram–Schmidt recursion. Write the positive definite original Grams as

$$G_0 = L_0^* L_0, \quad G = L_G^* L_G, \quad (\text{VR17})$$

with upper triangular Cholesky factors having positive real diagonals. These factors follow directly by the Gram–Schmidt recursion on the ordered remainder monomials: the change matrix is upper triangular with positive diagonal and its orthonormality equation gives VR17. The successive positive norms uniquely determine every entry, proving uniqueness as well.

The prescribed orthonormal frames of L and K are

$$p = sL_0^{-1} : \mathbb{C}^q \longrightarrow L, \quad c = RL_G^{-1} : \mathbb{C}^q \longrightarrow K. \quad (\text{VR18})$$

The preceding triangular uniqueness verifies every Gram–Schmidt phase, and AW10 gives $Tp = c$.

Let $a = JTR : E \rightarrow E$. The matrix of the block A in the K -orthonormal frame c is

$$A_c = c^* T c = p^* c = L_0^{-*} G L_G^{-1} = L_0^{-*} L_G^*. \quad (\text{VR19})$$

Here $c^* T = p^*$ follows from $T^* c = p$, and $s^* M R = G$ follows from VR2. Thus A_c is lower triangular with positive real diagonal entries

$$\lambda_j = (L_G)_{jj} / (L_0)_{jj} \in (0, 1] \quad (1 \leq j \leq q). \quad (\text{VR20})$$

Their upper bound follows because they are eigenvalues of the contraction A . Hence $\text{Tr } A$ is positive real, and

$$\det a = \det A_c = \frac{\det L_G}{\det L_0} = \sqrt{\frac{\det G}{\det G_0}} > 0. \quad (\text{VR21})$$

The square root is positive, forced by the actual AW phases. Since L_G identifies the original remainder frame with the c -frame, its full original-frame matrix is

$$a = L_G^{-1} A_c L_G = L_G^{-1} L_0^{-*} G. \quad (\text{VR22})$$

Define the actual two-endpoint quantity

$$\beta = \log \frac{\det G_0}{\det G} = \log \frac{V_{q-1}(t)}{V_{2q}(t)}. \quad (\text{VR23})$$

It is strictly positive. Indeed least-norm minimization gives $G \leq G_0$ and $G_0 - G = s^* M Q s$. Equality would imply $Q s = 0$, hence $L = K$, contradicting VR7. Thus $G_0^{-1/2} G G_0^{-1/2}$ has all eigenvalues in $(0, 1]$ with at least one smaller than one; its determinant is strictly smaller than one.

The positive λ_j have product $e^{-\beta/2}$, as do the positive singular values s_j . The arithmetic-geometric mean inequality gives $\text{Tr } A \geq q e^{-\beta/(2q)}$ and $\text{Tr } |A| \geq q e^{-\beta/(2q)}$. To recall a proof without an extra hypothesis, among nonnegative q -tuples $(z_1, \dots, z_q)$ with a fixed sum the product has a maximum by compactness. Since the target product is positive, zero entries are excluded at the maximum. Replacing two unequal entries x, y by their common mean increases their product by $(x-y)^2/4$ while preserving the other factors and the sum. Thus every maximizing entry is equal, proving $\prod_{j=1}^q z_j \leq (q^{-1} \sum_{j=1}^q z_j)^q$. Applying this to the two actual positive lists proves

$$\begin{aligned} 0 < d_{\text{lin}}^2 &= 2(q - \text{Tr } A) \leq 2q(1 - e^{-\beta/(2q)}), \\ 2\ell &\leq d_{\text{unit}}^2 = 4q - 2 \text{Tr } A - 2 \text{Tr } |A| \leq 4q(1 - e^{-\beta/(2q)}). \end{aligned} \quad (\text{VR24})$$

These estimates retain the prescribed AW phase. An abstract unitary intertwiner with arbitrary phase on its zero eigenspace could change $\text{Tr } A$ while retaining its singular values; VR19 removes that ambiguity for the actual T .

For $q = 1$, A is the positive scalar $\sqrt{G/G_0}$, and the formulas reduce exactly to $d_{\text{lin}}^2 = 2(1 - \sqrt{G/G_0})$ and $d_{\text{unit}}^2 = 4(1 - \sqrt{G/G_0})$, retaining the original norms in AW19.

170.5 Full coefficient action and the exact defect

For $f \in E$ let $m_f : E \rightarrow E$ be multiplication by the full residue class f , including all nilpotents. The original full-jet action obeys

$$\eta m_f = f(A_h^{(k)}) \eta. \quad (\text{VR25})$$

Both sides on $[P]$ equal the full coefficient $v_h^{\otimes k} f(A_h^{(k)}) P(A_h^{(k)}) 1$; the Taylor unit commutes with these polynomial actions in the same commutative tensor coefficient algebra. The annihilating relation χ makes the expression independent of representatives.

The relation-preserving triangular repair

$$Y_{\text{rel}} = T - \Pi T Q = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} \quad (\text{VR26})$$

is invertible because A, D are. It has the exact quotient and arithmetic actions

$$JY_{\text{rel}} = aJ, \quad \mathcal{O}Y_{\text{rel}} = \iota aJ. \quad (\text{VR27})$$

Its value on $R(E)$ is TR , so its induced quotient map is $a = JTR$; its value on F stays in F . This proves the first identity on the direct sum $R(E) \oplus F$, and the second follows by ι .

The coefficient-equivariance defect is exactly

$$\eta(am_f - m_fa) = \eta am_f - f(A_h^{(k)})\eta a. \quad (\text{VR28})$$

It vanishes for every f if and only if a is multiplication by $a(1)$: commutation gives $a(f) = a(m_f 1) = m_fa(1) = fa(1)$, and multiplication by a fixed element conversely commutes with every m_f . Since E is generated by $[S]$, it is equivalent to the single matrix identity

$$am_{[S]} - m_{[S]}a = 0. \quad (\text{VR29})$$

Commutation with the generator implies commutation with its powers by induction and with every polynomial by linearity. Thus VR22 and the literal full companion multiplication matrix compute the entire defect, without replacing E by its reduced root algebra. No unsupported assertion that this actual commutator vanishes or is nonzero is made.

For a fully typed source formula define $\rho_f = Rm_f J : H \rightarrow H$. Then

$$J[T, \rho_f]R = am_f - m_fa. \quad (\text{VR30})$$

This follows by substitution of $JR = I$. Its identity element acts by $\rho_1 = \Pi$, so this extension by zero is not asserted to be a unital action on all H . On the canonical image K it is precisely the unital original coefficient action.

The closest observation-preserving unitary satisfies

$$JV_{\text{obs}} = J, \quad \mathcal{O}V_{\text{obs}} = \mathcal{O}, \quad V_{\text{obs}}R = R, \quad V_{\text{obs}}\rho_f = \rho_f V_{\text{obs}} = \rho_f \quad (f \in E). \quad (\text{VR31})$$

Its identity block on K proves the first three equalities; their substitution in $Rm_f J$ proves the last. It therefore preserves the full arithmetic coefficient action on the original canonical quotient, with every nilpotent and the full Taylor unit. There is no claim of commutation for an undefined degree-truncated source multiplication.

The contraction of a has the exact original G -energy

$$\|x\|_G^2 - \|ax\|_G^2 = \|CRx\|_t^2 = \|QTRx\|_t^2 \quad (x \in E). \quad (\text{VR32})$$

Indeed $\Pi TRx = Rax$ and $QTRx = CRx$ are orthogonal, T is an isometry, and R is the original G -isometry. Thus, in the original remainder frame,

$$G - a^*Ga = R^*T^*MQTR \succeq 0, \quad \ell = \text{Tr}(G^{-1}(G - a^*Ga)). \quad (\text{VR33})$$

The first identity is the just-proved norm identity with $Q^*Q = Q$; the trace equality transports C^*C by R .

170.6 No invertible relation-preserving exact intertwiner

There is no invertible $Z : H \rightarrow H$ with $Z(F) \subset F$ and $UZ = ZW$. For invertibility gives $Z(F) = F$ and $Z^{-1}(F) = F$. Since $\text{im } U = F$ from AW7–AW8, $ZW(H) = UZ(H) \subset F$, so $W(H) \subset Z^{-1}(F) = F$. But AW7 on the original polynomial flag gives $\text{im } W = L^\perp$, which has the same dimension $q + 1$ as F . Hence $F = L^\perp$ and $K = L$, contradicting VR7. This proof does not assume Z is unitary.

Consequently exact window intertwining on this actual source cannot coexist with an invertible map preserving the original relation quotient. The explicit VR9, VR12 and VR26 maps instead compute each available correction, its full quotient action, its original-metric energy, and the exact coefficient defect.

At each fixed nonempty support label A_0 all displayed linear maps lift by $(A_0, x) \mapsto (A_0, Zx)$, with external τ sent to τ . Their zero outputs remain supported zero coefficients at A_0 . In particular $\mathcal{O}V_{\text{obs}} = \mathcal{O}$ preserves the supported arithmetic observation and does not identify a supported zero with the external basepoint.

171 Actual boundary tilts and uniform convolution tails

This calculation retains the original arithmetic density and the complete hypothetical actual quartet. It does not replace that density with a reference measure. The estimates do not assume RH or a pointwise lower bound for zeta.

171.1 1. Objects and the exact retained expression

Fix $0 < \delta < 1/2$, $\gamma > 0$, and an integer $m \geq 1$, and suppose the selected full-order quartet is a quartet of zeros of the original $g = 2\xi$. Its monic polynomial is

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - \tfrac{1}{2} - \epsilon\delta - i\eta\gamma)^m, \quad \deg h = 4m. \quad (\text{BT1})$$

The entire function g/h , its original Taylor unit, source maps, and measure remain unchanged:

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad \alpha = \frac{\pi}{2}, \quad R_0 = \sqrt{\delta^2 + \gamma^2}. \quad (\text{BT2})$$

The original completion formula gives, exactly,

$$\begin{aligned} w_h(t) &= \frac{(t^2 + 1/4)^2}{\sqrt{\pi}} \frac{|\zeta(1/2 + it)|^2}{|h(1/2 + it)|^2} \sigma(t), \\ \sigma(t) &= \frac{|\Gamma(1/4 + it/2)|^2}{2\pi}, \\ |h(1/2 + it)|^2 &= [((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2)]^{2m}. \end{aligned} \quad (\text{BT3})$$

The two factors with each fixed imaginary sign contribute a minus sign before multiplication, so their product is the positive expression in the last line. Every denominator in (BT3) is positive. Conjugation of the original g and the real coefficients of h give $w_h(-t) = w_h(t)$.

171.2 2. A centered partial-sum bound from the original floor formula

For every integer $N \geq 1$, initially for $\Re s > 1$,

$$\zeta(s) = \sum_{n=1}^N n^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty \{x\} x^{-s-1} dx. \quad (\text{BT4})$$

To verify (BT4), integrate the constant value $\lfloor x \rfloor$ on each interval $[n, n+1)$, or telescope integration by parts of the absolutely convergent tail. The last integral is absolutely and locally uniformly convergent for $\Re s > 0$, so the identity continues to that half-plane with $s \neq 1$.

Let $B_1(x) = \{x\} - 1/2$, so $|B_1(x)| \leq 1/2$. The identity $\int_N^\infty x^{-s-1} dx = N^{-s}/s$ gives exactly

$$\zeta(s) = \sum_{n=1}^{N-1} n^{-s} + \frac{1}{2}N^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty B_1(x)x^{-s-1} dx. \quad (\text{BT5})$$

No asymptotic remainder is substituted here.

On $s = 1/2 + it$, put $\varrho = \sqrt{t^2 + 1/4}$, which is both $|s|$ and $|s-1|$. For $N \geq 2$,

$$\sum_{n=1}^{N-1} n^{-1/2} + \frac{1}{2}N^{-1/2} \leq 2\sqrt{N-1} - 1 + \frac{1}{2\sqrt{N}} \leq 2\sqrt{N} - 1.$$

The first inequality follows from the integral bound for the decreasing function $x^{-1/2}$. For the second, $2(\sqrt{N} - \sqrt{N-1}) = 2/(\sqrt{N} + \sqrt{N-1}) \geq 1/\sqrt{N}$. At $N = 1$, the same asserted inequality holds directly, with left side $1/2$. The absolute integral in (BT5), before its multiplier s , is at most $\frac{1}{2} \int_N^\infty x^{-3/2} dx = N^{-1/2}$. Therefore

$$|\zeta(1/2 + it)| \leq 2\sqrt{N} - 1 + \frac{\sqrt{N}}{\varrho} + \frac{\varrho}{\sqrt{N}} \quad (t \in \mathbb{R}, N \geq 1). \quad (\text{BT6})$$

The infimum over these explicitly specified integer choices is also a bound.

For $x = |t| \geq 1$, choose $N = \lfloor x \rfloor$. Then

$$x/2 \leq N \leq x, \quad \varrho \leq \frac{\sqrt{5}}{2}x, \quad \frac{\sqrt{N}}{\varrho} \leq x^{-1/2}.$$

Since $-1 + x^{-1/2} \leq 0$, (BT6) yields

$$|\zeta(1/2 + it)| \leq Z_* \sqrt{|t|} \quad (|t| \geq 1), \quad Z_* = 2 + \sqrt{5/2}. \quad (\text{BT7})$$

This improves the polynomial exponent in the uncentered estimate used in TG.15. It is not a claim that (BT7) is the strongest known zeta bound. On the whole real line the original $N = 1$ uncentered formula gives independently

$$|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4}. \quad (\text{BT8})$$

171.3 3. Explicit two-sided Gamma bounds

The complete Binet calculation included below proves

$$c_\Gamma e^{-\alpha|t|}(1+|t|)^{-1/2} \leq \sigma(t) \leq C_\Gamma e^{-\alpha|t|}(1+|t|)^{-1/2} \quad (t \in \mathbb{R}), \quad (\text{BT9})$$

where

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{BT10})$$

For completeness, its exact starting identity for $\Re z > 0$ is

$$\log \Gamma z = (z - \tfrac{1}{2}) \log z - z + \tfrac{1}{2} \log(2\pi) + R(z), \quad R(z) = \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}.$$

The logarithmic branch is the one real on the positive real axis. The kernel is $(\coth x - 1/x)/(4x)$, with $x = u/2 > 0$. Define

$$F(x) = x \cosh x - \sinh x, \quad G(x) = (x^2 + 3) \sinh x - 3x \cosh x.$$

Both vanish at zero, and $F'(x) = x \sinh x \geq 0$, $G'(x) = xF(x) \geq 0$. Dividing their nonnegative values by the appropriate positive factors gives $0 \leq \coth x - 1/x \leq x/3$. Thus the kernel lies between zero and $1/12$. At $z = 1/4 + it/2$, integration of $e^{-u/4}/12$ proves $|R(z)| \leq 1/3$.

Twice the real part of Binet's identity, with $x = |t|$, gives exactly

$$\sigma(t) = |z|^{-1/2} \exp[-x \arctan(2x) - 1/2 + 2\Re R(z)].$$

For $x > 0$,

$$-x \arctan(2x) - \frac{1}{2} = -\alpha x + x \arctan(1/(2x)) - \frac{1}{2} \leq -\alpha x,$$

because $\arctan v \leq v$ for $v \geq 0$. At $x = 0$, the same inequality is $-1/2 \leq 0$. The lower bound uses $\arctan(2x) \leq \pi/2$. Hence the full exponent is between $-\alpha x - 7/6$ and $-\alpha x + 2/3$, including zero.

Finally $|z|^2 = 1/16 + x^2/4$, and the exact squared differences

$$\frac{(1+x)^2}{4} - |z|^2 = \frac{3+8x}{16}, \quad |z|^2 - \frac{(1+x)^2}{20} = \frac{(4x-1)^2}{80}$$

give

$$\sqrt{2}(1+x)^{-1/2} \leq |z|^{-1/2} \leq \sqrt{2\sqrt{5}}(1+x)^{-1/2}.$$

These prove (BT9)–(BT10) with no unproved Stirling remainder and no unquantified compact Gamma minimum.

171.4 4. A global explicit actual boundary-tilt bound

Set

$$p = 8m - \frac{9}{2}, \quad \beta = p - \frac{1}{2} = 8m - 5, \quad T_h = \max(1, 2R_0). \quad (\text{BT11})$$

Then $p \geq 7/2 > 1$ and $\beta > 0$. Define

$$\begin{aligned} C_{\text{comp}} &= \frac{C_\Gamma}{\sqrt{\pi}} (T_h^2 + 1/4)^2 [1 + 2\sqrt{T_h^2 + 1/4}]^2 \delta^{-8m} (1 + T_h)^\beta, \\ C_{\text{tail}} &= \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} \left(\frac{4}{3}\right)^{4m} \left(1 + \frac{1}{T_h}\right)^\beta, \\ C_h^{\text{tilt}} &= \max(C_{\text{comp}}, C_{\text{tail}}). \end{aligned} \quad (\text{BT12})$$

These constants are explicit in the fixed quartet and elementary constants. None is an unevaluated zeta supremum, and none changes the source mass.

For $|t| \leq T_h$, (BT3) has $|h(1/2 + it)|^2 \geq \delta^{8m}$. Apply (BT8) and the upper inequality of (BT9). Multiplying by $e^{\alpha|t|}(1+|t|)^p$ leaves the factor $(1+|t|)^\beta$; bounding $|t|$ by T_h proves the constant C_{comp} .

For $x = |t| \geq T_h$, retain the exact identity

$$((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2) = (t^2 - R_0^2)^2 + 4\delta^2 t^2 \geq \frac{9}{16} t^4. \quad (\text{BT13})$$

The last inequality uses $t^2 \geq 4R_0^2$. Thus

$$|h(1/2 + it)|^2 \geq (3/4)^{4m} x^{8m}, \quad (t^2 + 1/4)^2 \leq (25/16)x^4.$$

Applying (BT7) and (BT9) to (BT3) gives

$$e^{\alpha x} w_h(t) \leq \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} \left(\frac{4}{3}\right)^{4m} x^{5-8m} (1+x)^{-1/2}.$$

Multiplication by $(1+x)^p$ leaves $((1+x)/x)^\beta \leq (1+1/T_h)^\beta$. This proves the global bound

$$0 \leq e^{\alpha|t|} w_h(t) \leq C_h^{\text{tilt}} (1+|t|)^{-p} \quad (t \in \mathbb{R}). \quad (\text{BT14})$$

The exact expression (BT3), including its actual zeta numerator and complete quartet denominator, is the object bounded throughout.

171.5 5. Boundary moments and finite derivative orders

Define the original two boundary tilts and their actual masses:

$$f_{\pm}(t) = e^{\pm\alpha t} w_h(t), \quad M_{\pm} = \int_{\mathbb{R}} f_{\pm}(t) dt. \quad (\text{BT15})$$

The nonzero entire function g/h has only isolated zeros on the integration line, so both masses are positive. Evenness gives $M_+ = M_- =: M_{\alpha}$. Integration of (BT14) proves

$$0 < M_{\alpha} = M_h(\alpha) = M_h(-\alpha) \leq \frac{2C_h^{\text{tilt}}}{p-1} < \infty. \quad (\text{BT16})$$

For every real $j \geq 0$ with $j < p-1$, the inequality $|t|^j \leq (1+|t|)^j$ gives

$$\int_{\mathbb{R}} |t|^j e^{\pm\alpha t} w_h(t) dt \leq \frac{2C_h^{\text{tilt}}}{p-j-1}. \quad (\text{BT17})$$

More precisely, for $R \geq 0$, the part $|t| \geq R$ is at most

$$\frac{2C_h^{\text{tilt}}}{p-j-1} (1+R)^{j-p+1}. \quad (\text{BT18})$$

Thus every integer order $0 \leq j \leq 8m-6$ is absolutely integrable at both boundary tilts. For a simple quartet, this includes orders zero, one and two.

For the actual Laplace function $M_h(z) = \int e^{zt} w_h(t) dt$, differentiation in $|\Re z| < \alpha$ gives $\int t^j e^{zt} w_h(t) dt$. For each stated order, (BT14) dominates the absolute integrand uniformly on $|\Re z| \leq \alpha$ by the integrable function $C_h^{\text{tilt}} |t|^j (1+|t|)^{-p}$. Dominated convergence therefore extends those derivatives continuously to the closed strip with exactly the boundary moments above. This does not assert holomorphic continuation through the boundary or finiteness of every endpoint derivative.

171.6 6. A uniform convolution envelope with the literal tilted mass

If a nonnegative integrable function f has actual mass $M > 0$ and $f(x) \leq C(1+|x|)^{-p}$, then for every integer $k \geq 1$,

$$f^{*k}(u) \leq kCM^{k-1}(1+|u|/k)^{-p}. \quad (\text{BT19})$$

For $k=1$ this is the hypothesis. In the convolution integral with $x_1 + \cdots + x_k = u$, at least one index has $|x_i| \geq |u|/k$. Bound its integrand by the sum of the k restrictions to these events. On the event for index i , the factor $f(x_i)$ is at most $C(1+|u|/k)^{-p}$. Parametrize the convolution hyperplane by the other $k-1$ coordinates; the change from any such convolution parametrization to another has absolute determinant one. Dropping the event now leaves the full product integral M^{k-1} . Summing proves (BT19), without a probability normalization.

Multiplication inside the original convolution gives exactly

$$f_{\pm}^{*k}(u) = e^{\pm\alpha u} m_{h,k}(u), \quad m_{h,k} = w_h^{*k}. \quad (\text{BT20})$$

Use the plus tilt for $u \geq 0$, the minus tilt for $u < 0$, and their same actual mass M_{α} . The result is

$$m_{h,k}(u) \leq kC_h^{\text{tilt}} M_{\alpha}^{k-1} e^{-\alpha|u|} (1+|u|/k)^{-p} \quad (k \geq 1, u \in \mathbb{R}). \quad (\text{BT21})$$

All dependence on k is displayed. The constants $C_h^{\text{tilt}}, p, M_{\alpha}$ belong to fixed h .

For every integer $n \geq 0$ with $2n > p - 1$, use $(1 + u/k)^{-p} \leq k^p u^{-p}$ for $u > 0$. The Gamma integral gives

$$\int_{\mathbb{R}} u^{2n} m_{h,k}(u) du \leq 2C_h^{\text{tilt}} k^{p+1} M_\alpha^{k-1} \frac{\Gamma(2n - p + 1)}{\alpha^{2n-p+1}}. \quad (\text{BT22})$$

The condition $2n > p - 1$ is precisely what makes this comparison integrable at zero. The actual raw moments exist also when the condition fails, but (BT22) is not used there.

For the retained source comparison, one direct corollary is

$$m_{h,k}(u) \leq b_k \sigma(u), \quad b_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1} > 0. \quad (\text{BT23})$$

Indeed $1 + |u|/k \geq (1 + |u|)/k$; (BT21) and the lower Gamma bound leave the additional factor $(1 + |u|)^{-p+1/2} \leq 1$. This is a proved identity-map norm comparison; it does not assert equality of the measures.

172 Internal derivation of the exact Binet identity

This note proves the exact identity used in the actual boundary-tilt calculation from the defining Gamma integral, elementary integration, locally uniform products, and the Beta substitution. No Stirling formula, Gamma asymptotic, or cited Binet identity is assumed.

All complex powers of positive real integration variables use their real logarithms. The notation $\text{Log } z$ on $\Re z > 0$ denotes the principal holomorphic logarithm. The holomorphic logarithm of Gamma will be constructed below and will be real on the positive real axis.

172.1 1. Defining integrals and the finite Euler formula

For $\Re z > 0$, define

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx. \quad (\text{BI1})$$

This is absolutely convergent. On a compact set with $0 < a \leq \Re z \leq b$, its integrand and each of its first finitely many derivatives in z are bounded in absolute value by

$$(x^{a-1} + x^{b-1})(1 + |\log x|^j) e^{-x},$$

with the appropriate fixed integer j . This function is integrable at zero and infinity. Differentiation under the integral is therefore valid, and Γ is holomorphic on the right half-plane. Integration by parts gives

$$\Gamma(z + 1) = z\Gamma(z), \quad \Gamma(1) = 1, \quad (\text{BI2})$$

because $x^z e^{-x}$ tends to zero at both endpoints for $\Re z > 0$.

For $\Re z, \Re w > 0$, let

$$B(z, w) = \int_0^1 t^{z-1} (1-t)^{w-1} dt. \quad (\text{BI3})$$

For an integer $N \geq 1$, integration by parts gives

$$B(z, N + 1) = \frac{N}{z} B(z + 1, N).$$

Iterating, and using $B(z + N, 1) = 1/(z + N)$, proves the exact finite formula

$$B(z, N + 1) = \frac{N!}{z(z + 1) \cdots (z + N)}. \quad (\text{BI4})$$

In particular none of these denominators vanishes in the right half-plane.

Set

$$E_N(z) = N^z B(z, N+1) = \frac{N^z N!}{z(z+1) \cdots (z+N)}, \quad N \geq 1. \quad (\text{BI5})$$

The substitution $t = x/N$ gives

$$E_N(z) = \int_0^N x^{z-1} (1 - x/N)^N dx. \quad (\text{BI6})$$

For $0 \leq x < N$, $\log(1 - x/N) \leq -x/N$, so

$$0 \leq (1 - x/N)^N \leq e^{-x}.$$

For each fixed $x > 0$, the expression on the left tends to e^{-x} , with the indicator of $x < N$ understood.

On a compact set as above, therefore

$$\sup_z |E_N(z) - \Gamma(z)| \leq \int_0^\infty (x^{a-1} + x^{b-1}) \cdot |(1 - x/N)^N \mathbf{1}_{\{x < N\}} - e^{-x}| dx \rightarrow 0$$

by dominated convergence. The same argument with an additional factor $|\log x|^j$, for $j = 1, 2$, proves locally uniform convergence of those two derivatives as well. Thus the Euler limit and the derivative limits are established directly from their integrals.

172.2 2. Nonvanishing and the holomorphic Gamma logarithm

Write $H_N = \sum_{j=1}^N 1/j$. The real numbers

$$\gamma_N = H_N - \log N$$

decrease and are bounded below by zero. Indeed

$$\gamma_{N+1} - \gamma_N = \frac{1}{N+1} - \log(1 + 1/N) < 0$$

by integrating $1/x > 1/(N+1)$ on $[N, N+1)$, while $H_N \geq \int_1^{N+1} dx/x = \log(N+1)$. Let $\gamma_E = \lim_N \gamma_N$.

For $\Re z > 0$, every $1 + z/j$ has positive real part. Its principal logarithm is holomorphic. Define

$$Q_N(z) = \text{Log } z + z(H_N - \log N) + \sum_{j=1}^N (\text{Log}(1 + z/j) - z/j). \quad (\text{BI7})$$

Exponentiating this literal sum and using (BI5) gives exactly

$$e^{Q_N(z)} = z N^{-z} \prod_{j=1}^N (1 + z/j) = \frac{1}{E_N(z)}. \quad (\text{BI8})$$

The logarithmic series in (BI7) converges locally uniformly. To prove this explicitly, for $|w| \leq 1/2$,

$$\text{Log}(1 + w) - w = - \int_0^1 \frac{tw^2}{1 + tw} dt,$$

whence $|\operatorname{Log}(1+w) - w| \leq |w|^2$. On a compact set with $|z| \leq R$, this bounds the j -th summand by R^2/j^2 whenever $j \geq 2R$. The finite initial part is holomorphic and the summable bound proves uniform convergence of the tail. Hence

$$Q(z) = \operatorname{Log} z + \gamma_E z + \sum_{j=1}^{\infty} (\operatorname{Log}(1+z/j) - z/j) \quad (\text{BI9})$$

is holomorphic and $Q_N \rightarrow Q$ locally uniformly.

Taking limits in $E_N e^{Q_N} = 1$, using Section 1, proves

$$\Gamma(z) e^{Q(z)} = 1. \quad (\text{BI10})$$

This proves that Gamma is nowhere zero on the entire right half-plane. It also constructs the required holomorphic logarithm:

$$\mathcal{G}(z) := -Q(z), \quad e^{\mathcal{G}(z)} = \Gamma(z). \quad (\text{BI11})$$

For $x > 0$, every term in $Q(x)$ is real and the defining Gamma integral is positive. Thus $\mathcal{G}(x)$ is precisely the real logarithm of $\Gamma(x)$; there is no unspecified additive $2\pi i$ constant.

The second derivatives of the series in (BI9) converge locally uniformly, since $\sum_j |z+j|^{-2}$ does. Termwise differentiation gives

$$\mathcal{G}''(z) = \sum_{j=0}^{\infty} \frac{1}{(z+j)^2} \quad (\Re z > 0). \quad (\text{BI12})$$

For justification without assuming a derivative-series theorem, on any slightly larger compact subset of the same half-plane the series converges uniformly; Cauchy's integral formula on small circles then gives uniform convergence of its derivatives. Direct estimates of the derivatives of its summands give the same conclusion.

172.3 3. Beta product, duplication, and the factor $\sqrt{\pi}$

The absolutely convergent product of the two defining Gamma integrals admits the change of variables

$$x = rt, \quad y = r(1-t), \quad r > 0, \quad 0 < t < 1.$$

Its Jacobian matrix has determinant $-r$, so the positive integration Jacobian is r . Consequently

$$\Gamma(z)\Gamma(w) = \Gamma(z+w)B(z, w) \quad (\Re z, \Re w > 0). \quad (\text{BI13})$$

All powers here are powers of positive real bases. Fubini and the change of variables are justified by the same integrals with real parts of the exponents.

In $B(z, z)$, first substitute $t = (1+u)/2$, and then use symmetry and $v = u^2$. Every factor is retained:

$$\begin{aligned} B(z, z) &= 2^{1-2z} \int_{-1}^1 (1-u^2)^{z-1} du \\ &= 2^{2-2z} \int_0^1 (1-u^2)^{z-1} du \\ &= 2^{1-2z} B(1/2, z). \end{aligned} \quad (\text{BI14})$$

Finally $x = t^2$ in (BI1) gives $\Gamma(1/2) = 2 \int_0^\infty e^{-t^2} dt$. The latter is the positive integral over the whole real line. Its square is $\int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$. Polar coordinates, whose positive Jacobian is r , evaluate this square as

$$2\pi \int_0^\infty r e^{-r^2} dr = \pi.$$

Thus $\Gamma(1/2) = \sqrt{\pi}$. Inserting (BI13) into (BI14), and using the nonvanishing established in (BI10), proves

$$\boxed{\Gamma(z)\Gamma(z+1/2) = 2^{1-2z}\sqrt{\pi}\Gamma(2z) \quad (\Re z > 0).} \quad (\text{BI15})$$

On the positive real axis all these factors are positive, so their real logarithms obey the corresponding additive identity without any branch ambiguity.

172.4 4. The bounded integral remainder

For $u > 0$, define

$$K(u) = \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{1}{u}. \quad (\text{BI16})$$

We prove its exact bounds rather than assuming a Bernoulli expansion. With $x = u/2 > 0$,

$$K(u) = \frac{\coth x - 1/x}{4x}.$$

The two functions

$$F_0(x) = x \cosh x - \sinh x, \quad F_1(x) = (x^2 + 3) \sinh x - 3x \cosh x$$

vanish at zero and satisfy $F'_0(x) = x \sinh x \geq 0$ and $F'_1(x) = xF_0(x) \geq 0$. Dividing their nonnegative values by positive factors gives

$$0 \leq \coth x - \frac{1}{x} \leq \frac{x}{3}.$$

Therefore

$$\boxed{0 \leq K(u) \leq \frac{1}{12} \quad (u > 0).} \quad (\text{BI17})$$

Define the actual remainder candidate

$$R(z) = \int_0^\infty e^{-zu} K(u) du, \quad \Re z > 0. \quad (\text{BI18})$$

It is absolutely convergent and holomorphic. Differentiating j times is justified on every compact set with $\Re z \geq a > 0$ by $u^j e^{-au}/12$. In particular

$$|R(z)| \leq \frac{1}{12\Re z}, \quad 0 \leq R(x) \leq \frac{1}{12x} \quad (x > 0). \quad (\text{BI19})$$

172.5 5. Equality of second derivatives

Put $c = \frac{1}{2} \log(2\pi)$, and define

$$\mathcal{F}(z) = (z - \tfrac{1}{2}) \text{Log } z - z + c + R(z). \quad (\text{BI20})$$

Differentiation and the exact algebraic identity

$$u^2 K(u) = \frac{u}{e^u - 1} - 1 + \frac{u}{2}$$

give

$$\begin{aligned} \mathcal{F}''(z) &= \frac{1}{z} + \frac{1}{2z^2} + \int_0^\infty e^{-zu} \left(\frac{u}{e^u - 1} - 1 + \frac{u}{2} \right) du \\ &= \frac{1}{z^2} + \int_0^\infty \frac{ue^{-zu}}{e^u - 1} du. \end{aligned} \quad (\text{BI21})$$

Here $\int_0^\infty e^{-zu} du = 1/z$ and $\int_0^\infty ue^{-zu} du = 1/z^2$ follow by elementary integration on $\Re z > 0$; all cancellation factors are displayed.

The geometric series $1/(e^u - 1) = \sum_{j=1}^\infty e^{-ju}$ is valid for $u > 0$. Its absolute integral after multiplication by ue^{-zu} is bounded by the same expression with $\Re z$, which is integrable at zero and infinity. Hence termwise integration is valid, and

$$\mathcal{F}''(z) = \frac{1}{z^2} + \sum_{j=1}^\infty \int_0^\infty ue^{-(z+j)u} du = \sum_{j=0}^\infty \frac{1}{(z+j)^2} = \mathcal{G}''(z). \quad (\text{BI22})$$

The right half-plane is connected, so the holomorphic function $\mathcal{F} - \mathcal{G}$ has zero second derivative and consequently equals $az + b$ there, for constants $a, b \in \mathbb{C}$.

172.6 6. The recurrence fixes the linear constant

For positive real x , (BI2) and the real-axis branch in (BI11) give

$$\mathcal{G}(x+1) - \mathcal{G}(x) = \log x.$$

The corresponding exact difference of the explicit function (BI20) is

$$\mathcal{F}(x+1) - \mathcal{F}(x) - \log x = (x + \tfrac{1}{2}) \log(1 + 1/x) - 1 + R(x+1) - R(x). \quad (\text{BI23})$$

The left side equals a . The elementary integral bounds

$$\frac{y}{1+y} \leq \log(1+y) \leq y \quad (y > 0)$$

show that $x \log(1 + 1/x) \rightarrow 1$ and $\frac{1}{2} \log(1 + 1/x) \rightarrow 0$. The remainder bound (BI19) makes the remaining difference tend to zero. Thus $a = 0$, and $\mathcal{F} - \mathcal{G} = b$ throughout the right half-plane.

172.7 7. Duplication fixes the additive constant

For $x > 0$, the positive-real logarithm of (BI15) is

$$\mathcal{G}(x) + \mathcal{G}(x + 1/2) - \mathcal{G}(2x) = (1 - 2x) \log 2 + \tfrac{1}{2} \log \pi. \quad (\text{BI24})$$

Since $\mathcal{F} = \mathcal{G} + b$, the difference between the same expression using $\mathcal{F}$ and the right side of (BI24) equals exactly $b + b - b = b$.

Insert (BI20) without dropping any constant. The terms containing $\log x$ cancel, and the resulting difference is

$$b = x \log(1 + 1/(2x)) - \tfrac{1}{2} + (c - \tfrac{1}{2} \log(2\pi)) + R(x) + R(x + 1/2) - R(2x). \quad (\text{BI25})$$

Our displayed $c = \frac{1}{2} \log(2\pi)$ cancels exactly. The same elementary logarithm bounds show $x \log(1 + 1/(2x)) \rightarrow 1/2$, and each remainder tends to zero by (BI19). Thus $b = 0$.

Combining Sections 5–7 proves the complete exact identity

$$\boxed{\log \Gamma(z) = (z - \tfrac{1}{2}) \text{Log } z - z + \tfrac{1}{2} \log(2\pi) + \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}, \quad \Re z > 0.} \quad (\text{BI26})$$

The logarithm on the left is the uniquely specified holomorphic branch from (BI11). The integral is absolutely convergent, and its bound and all differentiations have been justified. This is the identity used by the actual boundary-tilt proof.

173 Uniform original source comparison and the two growing norm windows

Retain an actual complete off-line quartet $1/2 \pm \delta \pm i\gamma$ of common order m , with $0 < \delta < 1/2$ and $\gamma > 0$. The polynomial h , the entire quotient $v_h = (2\xi)/h$, its full Taylor unit, and the original source are unchanged. Set

$$w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \mu_h = \int_{\mathbb{R}} w_h(u) du, \quad \alpha = \pi/2. \quad (\text{TW1})$$

The proved actual boundary-tilt calculation supplies

$$p = 8m - \frac{9}{2} > 1, \quad e^{\alpha|u|} w_h(u) \leq C_h(1 + |u|)^{-p}, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha u} w_h(u) du \in (0, \infty). \quad (\text{TW2})$$

Here C_h denotes exactly the explicit constant C_h^{tilt} of (BT12), with its original compact and tail expressions. Its proof retains the exact factor $-\pi^{-1/4}(u^2 + 1/4)e^{-iu \log \pi/2} \zeta(1/2 + iu)$ and the full quartet denominator. In particular (TW2) does not use a positive lower bound for ζ or an assumption about its zero locations.

The comparison measure used below is the fixed original one-factor Gamma measure

$$d\sigma(u) = \frac{|\Gamma(1/4 + iu/2)|^2}{2\pi} du, \quad \int d\sigma = \sqrt{2\pi}. \quad (\text{TW3})$$

Its proved two-sided boundary bounds are

$$c_\Gamma e^{-\alpha|u|}(1 + |u|)^{-1/2} \leq \frac{d\sigma}{du} \leq C_\Gamma e^{-\alpha|u|}(1 + |u|)^{-1/2}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{TW4})$$

This fixed measure has mass $\sqrt{2\pi}$ for every k . It is distinct from the k -fold Gamma convolution of mass $(2\pi)^{k/2}$ in (AT4). Both comparisons, with their exact connecting constants, will be retained.

173.1 A uniform convolution bound at the actual exponential boundary

For every integer $k \geq 1$ and every $u \in \mathbb{R}$,

$$m_{h,k}(u) \leq kC_h M_\alpha^{k-1} (1 + |u|/k)^{-p} e^{-\alpha|u|}. \quad (\text{TW5})$$

To prove this, put $f_+(x) = e^{\alpha x} w_h(x)$. Its mass is exactly M_α , and $f_+(x) \leq C_h(1 + |x|)^{-p}$. The original product integral gives $f_+^{*k}(u) = e^{\alpha u} m_{h,k}(u)$. On the hyperplane $x_1 + \cdots + x_k = u$, at least one coordinate satisfies $|x_i| \geq |u|/k$. Split this integral into the union of those k sets. On its i th set the i th factor is at most $C_h(1 + |u|/k)^{-p}$. Eliminate $x_i = u - \sum_{j \neq i} x_j$; the change from the original convolution coordinates has absolute Jacobian one. Integrate the other $k - 1$ nonnegative factors over all their original domains; the resulting bound is $C_h(1 + |u|/k)^{-p} M_\alpha^{k-1}$. Summing proves the bound with $e^{-\alpha u}$. Conjugation of the original real-entire 2ξ and of the complete quartet polynomial proves $w_h(-x) = w_h(x)$. Thus $f_-(x) = e^{-\alpha x} w_h(x)$ has the same mass, and the same argument with f_- proves the bound with $e^{\alpha u}$. Taking the smaller of these two proved expressions gives exactly (TW5). This argument also covers $k = 1$ directly. Tonelli's theorem applies because all integrands are nonnegative and the displayed masses are finite.

For integers n with $2n > p - 1$, this yields the literal raw moment bound

$$\int_{\mathbb{R}} u^{2n} m_{h,k}(u) du \leq 2C_h k^{p+1} M_\alpha^{k-1} \frac{\Gamma(2n - p + 1)}{\alpha^{2n-p+1}}. \quad (\text{TW6})$$

Indeed $(1 + u/k)^{-p} \leq k^p u^{-p}$ for $u > 0$, and substitution $y = \alpha u$ evaluates the remaining positive half-line integral. No arithmetic density has been assigned the value of a Gamma density.

173.2 Pointwise comparison of the original convolution

For $k \geq 3$, retain the proved original convolution lower envelope

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (k-2+|u|)^{-B}, \quad B = 42 + 8m, \quad \vartheta_h = \int_{-1}^1 w_h(u) du > 0. \quad (\text{TW7})$$

Its complete proof is the original arithmetic envelope (AU.10), also retained as (CJ.15); it concerns this same g/h convolution. Define the following explicit constants, without altering either measure:

$$M = B/2 = 21 + 4m, \\ a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_\Gamma 2^{B/2}}, \quad A_k = \frac{C_h}{c_\Gamma} k^{p+1} M_\alpha^{k-1}. \quad (\text{TW8})$$

Then

$$\boxed{a_k (1+u^2)^{-M} \frac{d\sigma}{du} \leq m_{h,k}(u) \leq A_k \frac{d\sigma}{du}}. \quad (\text{TW9})$$

For the upper bound, $1 + |u|/k \geq (1 + |u|)/k$ in (TW5), and the lower inequality of (TW4) leaves the factor $(1 + |u|)^{-(p-1/2)} \leq 1$. This gives precisely A_k . For the lower bound, $k-2 \geq 1$ gives $k-2+|u| \leq (k-2)(1+|u|)$. The upper inequality of (TW4) implies $e^{-\alpha|u|} \geq C_\Gamma^{-1} d\sigma/du$. Finally $(1+|u|)^B \leq 2^{B/2} (1+u^2)^{B/2}$. Substitution of these three inequalities in (TW7) proves its lower counterpart in (TW9), with its displayed constant and direction.

173.3 Full finite polynomial forms, with the original coordinate maps

Let b_j be the fixed Gamma monic polynomials in u . The original generating-integral calculation (TG.7)–(TG.14) proves

$$\gamma_j = \|b_j\|_\sigma^2 = \sqrt{2} j! \Gamma(j+1/2) = \sqrt{2\pi} \frac{(2j)!}{4^j}, \quad ub_j = b_{j+1} + j(j-1/2)b_{j-1}. \quad (\text{TW10})$$

The last term is absent at $j = 0$. The second equality follows by induction from $\Gamma(1/2) = \sqrt{\pi}$ and the Gamma recurrence; no asymptotic formula is used there.

For $N \geq 0$ put

$$D_N = [1 + 4(N+M)^2]^M.$$

Every polynomial P of degree at most N satisfies

$$\int |P(u)|^2 (1+u^2)^M d\sigma(u) \leq D_N \int |P(u)|^2 d\sigma(u). \quad (\text{TW11})$$

Here is the full multiplication estimate. The orthonormal vectors $e_j = b_j/\sqrt{\gamma_j}$ obey

$$ue_j = \sqrt{(j+1)(j+1/2)} e_{j+1} + \sqrt{j(j-1/2)} e_{j-1}.$$

On degrees at most N , the first weighted shift has norm at most $N+1$ and the second has norm at most N ; this follows by taking squared norms of their orthogonal images term by term. The triangle inequality therefore gives $\|uP\|_\sigma \leq 2(N+1)\|P\|_\sigma$. After each multiplication the degree increases by one. Iteration gives $\|u^j P\|_\sigma \leq [2(N+M)]^j \|P\|_\sigma$ for $0 \leq j \leq M$. Expand $(1+u^2)^M$, integrate its nonnegative summands, and use the binomial theorem; the result is exactly (TW11).

Apply Cauchy–Schwarz with the two functions $|P|(1+u^2)^{-M/2}$ and $|P|(1+u^2)^{M/2}$ in $L^2(\sigma)$:

$$\left(\int |P|^2 d\sigma \right)^2 \leq \left(\int |P|^2 (1+u^2)^{-M} d\sigma \right) \left(\int |P|^2 (1+u^2)^M d\sigma \right).$$

For $P \neq 0$, divide by its positive σ norm and apply (TW11). For $P = 0$ the resulting inequality is immediate. Combining with (TW9) proves the full finite-form comparison

$$\boxed{\frac{a_k}{D_N} \int |P|^2 d\sigma \leq \int |P|^2 m_{h,k}(u) du \leq A_k \int |P|^2 d\sigma \quad (\deg P \leq N).} \quad (\text{TW12})$$

It holds for every complex coefficient vector, not just monic polynomials. In particular $A_k D_N / a_k \geq 1$ by applying (TW12) to any nonzero P .

The original sum coordinate is $S = c + iu$, $c = k/2$. Its exact algebra map is $\Psi_k : \mathbb{C}[S] \rightarrow \mathbb{C}[u]$, $\Psi_k P(u) = P(c + iu)$, with inverse $f \mapsto f((S - c)/i)$. In degree at most N its coefficient matrix has diagonal $1, i, \dots, i^N$. Thus (TW12) is a congruence statement in the original S coordinates as well. On the monic degree- n affine space, $P \mapsto i^{-n} \Psi_k P$ is a bijection to the monic u polynomials and preserves the norm because $|i^{-n}| = 1$. No degree-dependent phase is asserted to be a single algebra map. Consequently, writing $H_{k,N}$ for the actual original polynomial Gram and $H_N^{\sigma,k}$ for the fixed σ Gram evaluated at $S = c + iu$, the fully typed matrix form is

$$\boxed{(a_k/D_N) H_N^{\sigma,k} \preceq H_{k,N} \preceq A_k H_N^{\sigma,k}.} \quad (\text{TW13})$$

The identity on the original polynomial source commutes with remainder modulo the same $\chi_{h,k}$ and hence with its full unitful jet observation. Neither this comparison nor the coordinate congruence changes $\chi_{h,k}$, S , its full jets, or the original source norm.

173.4 Uniform arithmetic monic norms

Let $\omega_{h,k,n}$ be the minimum original squared norm of a monic degree- n polynomial. Minimizing (TW12) on that same affine space, using (TW10) for its fixed comparison minimum, gives

$$\boxed{\frac{a_k \gamma_n}{D_n} \leq \omega_{h,k,n} \leq A_k \gamma_n, \quad k \geq 3, n \geq 0.} \quad (\text{TW14})$$

The upper bound may equivalently be verified on the explicit trial polynomial b_n ; the lower bound applies to every monic polynomial. Thus no minimizer was transferred between different metrics without the stated inequality.

Here and below constants in O_h may depend on the fixed complete quartet and the fixed proven envelope constants, but not on k, n . Taking logarithms of the exact constants in (TW8) gives

$$\log a_k = (k-3)(\log \vartheta_h - \alpha) - B \log(k-2) + \log c_h - \log C_\Gamma - (B/2) \log 2,$$

$$\log A_k = (k-1) \log M_\alpha + (p+1) \log k + \log(C_h/c_\Gamma).$$

Also $\log D_n = M \log(1 + 4(n+M)^2) = O_h(1 + \log(n+1))$. These displayed equalities prove, uniformly for $k \geq 3, n \geq 0$,

$$\boxed{\log \omega_{h,k,n} = \log \left(\sqrt{2\pi} \frac{(2n)!}{4^n} \right) + O_h(k + \log(n+1)).} \quad (\text{TW15})$$

This is a high-degree norm estimate for the actual arithmetic convolution. It is not an assertion that consecutive norm ratios have an asymptotic; the error in (TW15) does not justify that differentiation in n .

The same all-polynomial comparison also controls the original raw even moments $I_{k,2n} = \int_{\mathbb{R}} u^{2n} m_{h,k}(u) du$. For $k \geq 3$ and $n \geq 1$ it gives the explicit two-sided bounds

$$\boxed{\frac{a_k \sqrt{2} c_\Gamma e^{-\alpha} (2n-1)!}{D_n \alpha^{2n}} \leq I_{k,2n} \leq 2 A_k C_\Gamma \frac{(2n)!}{\alpha^{2n+1}}.} \quad (\text{TW15a})$$

For the lower comparison moment, restrict (TW4) to $u \geq 1$ and use $(1+u)^{-1/2} \geq 1/(\sqrt{2}u)$. Both half-lines give $\int u^{2n} d\sigma \geq \sqrt{2}c_\Gamma \alpha^{-2n} \Gamma(2n, \alpha)$. Repeated integration by parts gives $\Gamma(2n, \alpha) = (2n-1)!e^{-\alpha} \sum_{j=0}^{2n-1} \alpha^j/j! \geq (2n-1)!e^{-\alpha}$. For the upper comparison moment, drop the factor $(1+|u|)^{-1/2} \leq 1$ in (TW4) and evaluate the Gamma integral. Applying (TW12) to the original degree- n polynomial u^n proves (TW15a). Its logarithms give, uniformly on these stated degrees,

$$\log I_{k,2n} = \log((2n)!\alpha^{-2n}) + O_h(k + \log(n+1)). \quad (\text{TW15b})$$

The full one-factor zeta numerator and actual tilted mass in (TW2) remain in a_k, A_k ; (TW15a) is a bound on the original moments.

173.5 The two original windows with full finite errors

Put $q = [1 + k(m-1)](k+1)^2$. For $n = q-1$ or $n = q$, division of (TW14) at the two exact degrees gives

$$\log \frac{\gamma_{n+q}}{\gamma_n} - \log \frac{A_k D_{n+q}}{a_k} \leq \log \frac{\omega_{h,k,n+q}}{\omega_{h,k,n}} \leq \log \frac{\gamma_{n+q}}{\gamma_n} + \log \frac{A_k D_n}{a_k}. \quad (\text{TW16})$$

All logarithmic error bounds here are nonnegative by (TW12). The exact factorial formula (TW10) and monotonicity of $\log x$ give

$$\log \frac{\gamma_{2q}}{\gamma_q} = 2q \log(4q/e) + \eta_q, \quad 0 \leq \eta_q \leq \log 2. \quad (\text{TW17})$$

Indeed the logarithm of $(4q)!/(2q)!$ is $\sum_{j=2q+1}^{4q} \log j$. Its excess over $\int_{2q}^{4q} \log x dx$ is nonnegative and at most $\sum_{j=2q+1}^{4q} (\log j - \log(j-1)) = \log 2$. Subtracting $q \log 4$ gives precisely (TW17). The other original window satisfies the exact identity

$$\frac{\gamma_{2q-1}}{\gamma_{q-1}} = \frac{\gamma_{2q}}{\gamma_q} \frac{q-1/2}{4q-1}, \quad \frac{1}{6} \leq \frac{q-1/2}{4q-1} < \frac{1}{4}.$$

It follows in particular that

$$4q \log(4q/e) - \log 6 \leq \log \frac{\gamma_{2q-1} \gamma_{2q}}{\gamma_{q-1} \gamma_q} \leq 4q \log(4q/e). \quad (\text{TW18})$$

For fixed h , one has $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Thus each finite error in (TW16), divided by $2q$, tends to zero. Equations (TW16)–(TW18) prove both limits on their stated windows:

$$\lim_{k \rightarrow \infty} \frac{1}{q} \left(\frac{\omega_{h,k,2q}}{\omega_{h,k,q}} \right)^{1/(2q)} = \lim_{k \rightarrow \infty} \frac{1}{q} \left(\frac{\omega_{h,k,2q-1}}{\omega_{h,k,q-1}} \right)^{1/(2q)} = \frac{4}{e}. \quad (\text{TW19})$$

No limit with an unspecified ratio n/q is asserted.

173.6 The resulting actual four-window lower estimate

Retain the original relative controls ε_N , phases ϕ_N , and contraction penalties P_- , P_+ of (HC1)–(HC2). Their exact first boundary and regular products, inherited from (CJ.9)–(CJ.13), give

$$\mathcal{B}_k = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{h,k,2q-1} \omega_{h,k,2q}}{\omega_{h,k,q-1} \omega_{h,k,q}} + P_- + P_+,$$

where the endpoint weights are $1, 2, \dots, 2, 1$. Set

$$E_k = 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q \geq 0, \quad E_k = O_h(k + \log q).$$

Adding the upper sides of (TW16) and applying (TW18) gives the actual sum-norm upper bound $4q \log(4q/e) + E_k$. Consequently, with the full original trace $\mathcal{L} = 2\delta\ell_k(k+1)\lfloor(k+1)^2/4\rfloor$,

$$\begin{aligned} \mathcal{B}_k &\geq 4q \log(\delta ek/8) - E_k + P_- + P_+ \\ &\quad + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} + 4q \log \frac{2\mathcal{L}}{\delta kq}. \end{aligned} \quad (\text{TW20})$$

All terms after $-E_k$ are the displayed original nonnegative costs. The sums retain both endpoint degrees and all their multiplicities. To verify the constant, $4q \log \mathcal{L} - 4q \log(4q/e)$ equals $4q \log(\delta ek/8) + 4q \log(2\mathcal{L}/(\delta kq))$. This proves the identity used in the bound with no rescaling of S . Since $E_k/q \rightarrow 0$, the quantified refinement includes

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_k - 4q \log k}{q} \geq 4 \log(\delta e/8). \quad (\text{TW21})$$

The leading threshold four is inherited. The new content is the actual boundary-tilt comparison, its uniform finite-degree errors, and the sharper literal window constant. These lower estimates alone do not assign a sign to the signed arithmetic correction in (AT14)–(AT17).

174 Fixed-packet asymptotics of the complete original Gamma jet determinant

Fix the original positive integer k , put $c = k/2$, and retain the actual monic cyclic annihilator $\chi = \chi_{h,k}$, with all its distinct roots $\lambda_1, \dots, \lambda_d$ and their complete multiplicities $m_1, \dots, m_d$. Write $q = \sum_i m_i$. The finite-packet hypothesis only specifies these original objects; no new arithmetic zero or replacement annihilator enters this section. The same computation applies to every specified finite jet observation on the given source line. This last fact will be used to transport the original observation through a literal comparison-measure dilation.

The source line and comparison density are exactly

$$S = c + iu, \quad \sigma(u) = \frac{|\Gamma(1/4 + iu/2)|^2}{2\pi}. \quad (\text{FPK.1})$$

Use the original monic polynomials and squared norms

$$p_n(S) = i^n b_n((S - c)/i), \quad \gamma_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2} n! \Gamma(n + 1/2). \quad (\text{FPK.2})$$

The generating-integral calculation in the retained Gamma chapter proves their orthogonality for (FPK.1), including these constants and phases. We calculate the full confluent kernel, not just the diagonal scalar evaluation kernel.

For $f^{[j]} = f^{(j)}/j!$, let E_χ be the original Taylor-coordinate map from the increasing-power remainder frame:

$$(E_\chi)_{(i,j),r} = \begin{cases} \binom{r}{j} \lambda_i^{r-j}, & r \geq j, \\ 0, & r < j, \end{cases} \quad 0 \leq j < m_i, \quad 0 \leq r < q. \quad (\text{FPK.3})$$

Define

$$(\mathbf{K}_N)_{(i,j),(a,l)} = \sum_{n=0}^N \frac{p_n^{[j]}(\lambda_i) \overline{p_n^{[l]}(\lambda_a)}}{\gamma_n} \quad (N \geq q - 1). \quad (\text{FPK.4})$$

The minimum-norm quotient Gram and its original determinant are

$$G_N^\sigma = E_\chi^* \mathbf{K}_N^{-1} E_\chi, \quad V_N^\sigma = \det G_N^\sigma = \frac{|\det E_\chi|^2}{\det \mathbf{K}_N}. \quad (\text{FPK.5})$$

Indeed the normalized polynomial basis makes the source Gram identity, so the Gram of the onto jet observation is (FPK.4). Minimizing a Euclidean norm subject to that observation gives its inverse Gram. The first q monic polynomials span all remainders, proving that this Gram is positive definite. Transport by the actual map (FPK.3) gives (FPK.5). Thus V_N means the determinant of the Gram, as in (AT6), and does not mean its square root.

The exact determinant of the displayed coordinate map is

$$\det E_\chi = \prod_{i < a} (\lambda_a - \lambda_i)^{m_i m_a}. \quad (\text{FPK.6})$$

Here root blocks and derivatives are ordered as in (FPK.3). To prove the constant, replace the m_i evaluations in block i by evaluations at $\lambda_i + \epsilon x_{i,j}$ with fixed distinct $x_{i,j}$. Expand each row into its Taylor series and divide its block determinant by $\epsilon^{m_i(m_i-1)/2} \prod_{j < l} (x_{i,l} - x_{i,j})$. Multilinearity gives the determinant of the normalized derivative rows in the limit. The ordinary Vandermonde formula on all the evaluation points gives precisely the right side of (FPK.6); the within-block factors cancel the displayed divisors. This also proves that no factorial or orientation factor is missing.

174.1 The two singularities with the original coordinate and constants

Put $w = S - c$. Substituting iz in the retained generating function $(1 + z^2)^{-1/4} \exp(u \arctan z)$ gives the exact identity

$$\sum_{n=0}^{\infty} \frac{p_n(c+w)}{n!} z^n = (1-z)^{-a(w)} (1+z)^{-b(w)}, \quad a(w) = \frac{1}{4} + \frac{w}{2}, \quad b(w) = \frac{1}{4} - \frac{w}{2}. \quad (\text{FPK.7})$$

All powers use the branches equal to one at $z = 0$. Define the entire coefficient functions

$$C_+(w) = \frac{2^{-1/2+w/2}}{\Gamma(1/4 + w/2)}, \quad C_-(w) = \frac{2^{-1/2-w/2}}{\Gamma(1/4 - w/2)}. \quad (\text{FPK.8})$$

For every compact set of w and each fixed derivative order j , the normalized polynomials $e_n(w) = p_n(c+w)/\sqrt{\gamma_n}$ satisfy

$$e_n(w) = C_+(w)n^{w/2-1/2} + (-1)^n C_-(w)n^{-w/2-1/2} + R_n(w), \quad (\text{FPK.9})$$

$$|\partial_w^j R_n(w)| \leq C_j(1 + \log n)^j \left(n^{\Re w/2-3/2} + n^{-\Re w/2-3/2} + n^{-J} \right), \quad n \geq 2. \quad (\text{FPK.10})$$

The positive integer J can be chosen as large as needed, with the constant depending on the compact set, j , and J .

Here is a coefficient proof of these estimates. It is included so no unproved asymptotic theorem about the orthogonal polynomials is used. Near $z = 1$, expand the other factor in (FPK.7):

$$(1+z)^{-b} = 2^{-b} \sum_{r=0}^{L-1} \frac{(b)_r}{2^r r!} (1-z)^r + (1-z)^L A_L(z, w), \quad (\text{FPK.11})$$

where A_L is analytic near that point, uniformly with each specified parameter derivative on a compact parameter set. Near $z = -1$ use the same expansion with a, b exchanged and z replaced by $-z$. Subtract from (FPK.7) both finite lists of singular terms. Choose L larger than the maximum real part of a, b plus any specified integer $J_0 + 2$. The remainder and its first J_0 angular derivatives on the unit circle then extend continuously across both 1 and -1 . Indeed its nonanalytic local remainder is a power of real part greater than $J_0 + 1$, times an analytic bounded factor; parameter derivatives only add fixed powers of its logarithm. The singular terms subtracted at the other point are analytic there. Away from these two points everything is analytic across the circle.

The resulting boundary function is C^{J_0} and periodic. Integrating its Fourier coefficient by parts J_0 times bounds that coefficient by Cn^{-J_0} , uniformly for all the specified parameter derivatives. Its nonnegative Fourier coefficients are its Taylor coefficients: apply Cauchy's formula on circles of radius less than one and use uniform boundary convergence. Each subtracted term has its exact coefficient

$$[z^n](1-z)^{-a+r} = \frac{\Gamma(n+a-r)}{\Gamma(a-r)\Gamma(n+1)}, \quad (\text{FPK.12})$$

with analytic values at nonpositive integral $a-r$; the corresponding coefficient at -1 has the additional factor $(-1)^n$.

For completeness, uniformly on a compact complex set of α ,

$$\frac{\Gamma(n+\alpha)}{\Gamma(n+1)} = n^{\alpha-1}(1+O(n^{-1})), \quad (\text{FPK.13})$$

with differentiated estimates after retaining the derivatives of $n^{\alpha-1}$. One direct proof first arranges $\Re\alpha < 1$ by the gamma recurrence. The beta integral is

$$\frac{\Gamma(n+\alpha)}{\Gamma(n+1)} = \frac{n^{\alpha-1}}{\Gamma(1-\alpha)} \int_0^n (1-y/n)^{n+\alpha-1} y^{-\alpha} dy.$$

On $0 \leq y \leq n/2$, the Taylor estimate for $\log(1-y/n)$, with its integral remainder, bounds the difference from e^{-y} in the integral by $Cn^{-1}e^{-y/2}(1+y)^2y^{-\Re\alpha}$. One may first use $0 \leq y \leq n/4$ for this estimate and absorb $n/4 \leq y \leq n/2$ into its exponentially small bound. On $n/2 \leq y \leq n$, integrate the exponentially small factor $(1-y/n)^{n+\Re\alpha-1}$ directly. The same estimates hold after each fixed parameter derivative, with additional integrable powers of $|\log y|$ and $|\log(1-y/n)|$. The limiting integral is $\Gamma(1-\alpha)$; its reciprocal is bounded on the chosen compact set. This proves (FPK.13) and its parameter derivatives. Recurrence back to the original α multiplies by a fixed finite product of $n+\alpha-r$, yielding the asserted formula on every compact set.

In (FPK.12), the $r=0$ terms therefore have coefficients

$$2^{-b}n^{a-1}/\Gamma(a), \quad (-1)^n 2^{-a}n^{b-1}/\Gamma(b),$$

each with a relative error of order $1/n$ interpreted analytically when a reciprocal gamma factor vanishes. The terms $r \geq 1$ are of the next indicated order or smaller. The exact normalization (FPK.2) multiplies these coefficients by

$$\frac{n!}{\sqrt{\gamma_n}} = 2^{-1/4} \left(\frac{\Gamma(n+1)}{\Gamma(n+1/2)} \right)^{1/2} = 2^{-1/4} n^{1/4} (1+O(n^{-1})).$$

This proves (FPK.8)–(FPK.10), including the original factors of two and every derivative required for the full jet calculation.

174.2 The complete off-critical blocks

For $w_i = \lambda_i - c$, put $\delta_i = \Re w_i$. When $\delta_i \neq 0$, set $\epsilon_i = \text{sgn } \delta_i$. The dominant coefficient $C_{\epsilon_i}(w)$ is nonzero on a neighborhood of w_i , because the real part of its gamma argument at w_i is $1/4 + |\delta_i|/2 > 0$. On that full local jet block define $T_{N,i}$ to be multiplication by the analytic germ

$$\left(C_{\epsilon_i}(w) N^{\epsilon_i w/2} \right)^{-1}, \quad \det T_{N,i} = \left(C_{\epsilon_i}(w_i) N^{\epsilon_i w_i/2} \right)^{-m_i}. \quad (\text{FPK.14})$$

This retains all the derivatives of C_{ϵ_i} and the triangular logarithmic jet terms. It is a specified invertible coordinate map used to compute the determinant, and will be undone.

Apply it to the complete Taylor column of e_n at w_i . The dominant component in position j becomes

$$v_{N,n;i,j} = \vartheta_{i,n} n^{-1/2} (n/N)^{\epsilon_i w_i/2} \frac{(\epsilon_i \log(n/N)/2)^j}{j!}, \quad \vartheta_{i,n} = \begin{cases} 1, & \epsilon_i = 1, \\ (-1)^n, & \epsilon_i = -1. \end{cases} \quad (\text{FPK.15})$$

The difference between the normalized exact row and (FPK.15) has squared-sum norm tending to zero. To verify that claim without neglecting small indices, the nondominant term and each of its fixed derivatives have squared-sum bound

$$CN^{-|\delta_i|}(1 + \log N)^{2j} \sum_{n=1}^N n^{-1-|\delta_i|}(1 + \log n)^{2j} = o(1).$$

For the remainder (FPK.10), multiplication and differentiation in (FPK.14) give a bound by a constant times a fixed power of $\log N$ times

$$N^{-|\delta_i|} \sum_{n=2}^N n^{|\delta_i|-3} + N^{-|\delta_i|} \sum_{n=2}^N n^{-1-|\delta_i|}.$$

It tends to zero: the first term is $O(N^{-2})$ if $|\delta_i| > 2$, $O(N^{-2} \log N)$ if it equals two, and $O(N^{-|\delta_i|})$ otherwise. Choose J in (FPK.10) sufficiently large for the remaining rapidly decaying term. The finitely many initial columns obey the same vanishing conclusion directly, because their normalization is $O(N^{-|\delta_i|/2}(\log N)^j)$.

For two blocks in the same half-plane, the Gram of (FPK.15) converges by an integrable Riemann sum to

$$(\mathbf{H}_\epsilon)_{(i,j),(a,l)} = \int_0^1 x^{\epsilon(w_i + \bar{w}_a)/2 - 1} \frac{(\epsilon \log x/2)^{j+l}}{j!l!} dx = \frac{(-\epsilon/2)^{j+l}(j+l)!}{j!l! [\epsilon(w_i + \bar{w}_a)/2]^{j+l+1}}. \quad (\text{FPK.16})$$

Here $\epsilon_i = \epsilon_a = \epsilon$, and the denominator has positive real part. Convergence at zero follows, for every fixed power of $\log x$, from that positive real part. For a direct Riemann-sum justification, split at any $b > 0$; on $[b, 1]$ the integrand is smooth, and on $(0, b)$ its absolute sums and integrals are bounded by $Cb^\alpha(1 + |\log b|^{j+l})$, with some $\alpha > 0$, which tends to zero.

The matrix $\mathbf{H}_\epsilon$ is positive definite. It is the Gram of the complete list $x^{\epsilon w_i/2 - 1/2}(\epsilon \log x/2)^j/j!$ in $L^2(0, 1)$. An equality between these functions, after $x = e^{-y}$ and division by the common nonzero factor $e^{y/2}$, gives an exponential-polynomial equality on $y > 0$. Apply the product of $(\partial_y + \epsilon w_a/2)^{m_a}$ over $a \neq i$. It removes the other blocks and acts on the remaining polynomial by a product of nonzero constants plus nilpotent differentiation. That product is invertible on the specified finite polynomial space. Every remaining polynomial is therefore zero, proving independence.

For two opposite half-planes the Gram has instead an additional $(-1)^n$. Its absolute integrable Riemann envelope is the same with real exponent $(|\delta_i| + |\delta_a|)/2 > 0$ at zero. On $[b, 1]$, pairing successive indices proves that its alternating Riemann sum tends to zero by uniform continuity. The absolute bound on $(0, b)$ just proved then gives a zero limit for the whole sum. Thus the two half-plane blocks decouple in this exact limit; no entry of the original finite kernel has been set to zero.

174.3 Critical full jets and all mixed blocks

For $\delta_i = 0$, put $L_N = \log N$ and use the diagonal jet map

$$(\mathbf{T}_{N,i})_{j,j} = L_N^{-j-1/2}, \quad \det \mathbf{T}_{N,i} = L_N^{-m_i^2/2}. \quad (\text{FPK.17})$$

In a differentiated dominant term (FPK.9), the part in which all j derivatives hit the power of n is

$$\frac{(\log n)^j}{2^j j! \sqrt{n}} \left(C_+(w_i) n^{w_i/2} + (-1)^{n+j} C_-(w_i) n^{-w_i/2} \right). \quad (\text{FPK.18})$$

Every other differentiated term has a smaller power of $\log n$. After multiplication by (FPK.17), its squared-sum norm tends to zero, since $\sum_{n \leq N} n^{-1}(\log n)^r = O(1 + L_N^{r+1})$. The remainder in (FPK.10) and every fixed initial column vanish in the same normalized squared sum.

At the same critical point, the nonalternating products in (FPK.18) give the limit

$$(\mathbf{H}_i^0)_{j,l} = \frac{|C_+(w_i)|^2 + (-1)^{j+l}|C_-(w_i)|^2}{2^{j+l}j!!(j+l+1)} = \frac{|C_+(w_i)|^2}{2^{j+l}j!!} \int_{-1}^1 t^{j+l} dt. \quad (\text{FPK.19})$$

The last equality uses the exact relation $C_-(w_i) = \overline{C_+(w_i)}$ for purely imaginary w_i . The coefficient is nonzero, and (FPK.19) is a positive multiple of the Gram of independent real monomials of degrees below m_i . It is therefore positive definite, including every nilpotent direction.

For different critical points, a nonalternating product has $\sum_{n \leq N} n^{-1+i\theta}(\log n)^r$ with $\theta \neq 0$. Its difference from the corresponding integral is bounded uniformly in N : the derivative of $x^{-1+i\theta}(\log x)^r$ is absolutely integrable on $[1, \infty)$. Substitution $x = e^t$ and integration by parts bound that integral by $O(1 + L_N^r)$. Division by L_N^{r+1} gives zero. Alternating products are bounded uniformly before this division, by pairing successive terms and the same integrable derivative estimate. This includes the alternating products within a single critical block. Hence all different critical blocks have zero cross limit.

Every mixed off-critical/critical block also has zero limit. Here is a bound that does not require any oscillatory cancellation. The normalized off-critical row (FPK.15) has a limiting squared mass distribution on $x = n/N \in (0, 1)$ with integrable density $x^{|\delta_i|-1}$ times a fixed power of $\log x$. For any fixed $R > 0$, its mass in $n \leq Ne^{-R}$ has limiting upper bound equal to the integral of that density on $(0, e^{-R})$, which tends to zero as R grows. On the complementary indices, the squared mass of a normalized critical row is $O(R/L_N)$: use $(\log n)^{2j} \leq L_N^{2j}$ and $\sum_{Ne^{-R} < n \leq N} 1/n \leq R + o(1)$. Both rows have bounded total squared norm by their already-proved diagonal limits. Cauchy-Schwarz on the two index ranges, first letting N tend to infinity and then R tend to infinity, proves that their inner product tends to zero. The vanishing remainder rows do not change this conclusion.

174.4 The entire determinant and the original quotient volume

Let $\mathbf{T}_N$ be the block diagonal combination of (FPK.14) and (FPK.17), with any fixed root-block ordering. The preceding proofs establish the full finite-matrix limit

$$\mathbf{T}_N \mathbf{K}_N \mathbf{T}_N^* \longrightarrow \mathbf{H} := \mathbf{H}_+ \oplus \mathbf{H}_- \oplus \bigoplus_{\delta_i=0} \mathbf{H}_i^0 > 0. \quad (\text{FPK.20})$$

Absent blocks are omitted and their determinants are one. Put

$$\mathcal{A}_\chi = \sum_i m_i |\Re \lambda_i - c|, \quad \mathcal{B}_\chi = \sum_{\Re \lambda_i = c} m_i^2, \quad (\text{FPK.21})$$

$$\mathcal{C}_\chi = (\det \mathbf{H}) \prod_{\delta_i \neq 0} |C_{\epsilon_i}(\lambda_i - c)|^{2m_i} > 0. \quad (\text{FPK.22})$$

All entries defining this constant have been given in (FPK.8), (FPK.16), and (FPK.19), with their actual coordinates and factors.

Theorem 174.1 (Complete fixed-Gamma jet determinant). *For the original fixed source line, the entire original finite jet observation, and $N \rightarrow \infty$ through integers,*

$$\det \mathbf{K}_N = \mathcal{C}_\chi N^{\mathcal{A}_\chi} (\log N)^{\mathcal{B}_\chi} (1 + o(1)), \quad (\text{FPK.23})$$

$$V_N^\sigma = \frac{|\det \mathbf{E}_\chi|^2}{\mathcal{C}_\chi} N^{-\mathcal{A}_\chi} (\log N)^{-\mathcal{B}_\chi} (1 + o(1)), \quad (\text{FPK.24})$$

and therefore

$$\log V_N^\sigma = -\mathcal{A}_\chi \log N - \mathcal{B}_\chi \log \log N + \log \frac{|\det \mathbf{E}_\chi|^2}{\mathcal{C}_\chi} + o(1). \quad (\text{FPK.25})$$

Proof. Determinant is a polynomial in the matrix entries, so (FPK.20) gives $\det(\mathbf{T}_N \mathbf{K}_N \mathbf{T}_N^*) \rightarrow \det \mathbf{H} > 0$. The literal determinant factors of the coordinate maps are

$$|\det \mathbf{T}_N|^2 = N^{-\mathcal{A}_\chi} (\log N)^{-\mathcal{B}_\chi} \prod_{\delta_i \neq 0} |C_{\epsilon_i}(w_i)|^{-2m_i}.$$

Undoing this congruence proves (FPK.23). The exact original coordinate formula (FPK.5) proves (FPK.24). Its positive constant and positive determinants allow taking real logarithms, which gives (FPK.25). $\square$

The original arithmetic observation remains $\sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}$, with $\eta_{h,k}$ containing its complete Taylor unit $v_h^{\otimes k}$. No factor in this injection was changed in forming the comparison norm on the original remainder coordinate E . If that coordinate is transported by a specified fixed invertible unit matrix U , the quotient Gram becomes $U^* G_N^\sigma U$ and its determinant is multiplied by the exact fixed factor $|\det U|^2$. Thus a unit-containing alternative frame retains the same two exponents and its own explicitly transported constant; the unit is never set to one.

This theorem is a fixed- k , fixed- χ statement. It provides no uniform error as the arithmetic packet or k changes, and does not assign a sign to the original signed four-window correction. Its assertions are exact asymptotic statements for the stated full kernel and quotient determinant, retaining every original jet.

174.5 The original tensor-Gamma reference with its full mass

The original reference in (AT4), on this same line $S = k/2 + iu$, is

$$m_0(u) = \frac{(2\pi)^{k/2}}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad c_a = 2^{1-2a} \Gamma(2a). \quad (\text{FPK.26})$$

Its mass is exactly $(2\pi)^{k/2}$; this scalar is retained. The original generating-integral calculation gives its monic polynomials $p_{k,n}(S) = i^n b_n^{(k/4)} ((S - c)/i)$ and squared norms

$$\gamma_{k,n} = (2\pi)^{k/2} \frac{n! \Gamma(n + k/2)}{\Gamma(k/2)}. \quad (\text{FPK.27})$$

Their generating function in (FPK.7) has $1/4$ replaced by $k/4$. The coefficient proof (FPK.11)–(FPK.13), with this fixed positive parameter, gives the normalized expansion (FPK.9)–(FPK.10) with the exact coefficients

$$C_{\pm,k}(w) = (2\pi)^{-k/4} \sqrt{\Gamma(k/2)} \frac{2^{-k/4 \pm w/2}}{\Gamma(k/4 \pm w/2)}. \quad (\text{FPK.28})$$

To verify the unchanged power of n , a singular coefficient is $n^{k/4 \pm w/2 - 1}$, while division of $n!$ by the square root of (FPK.27) contributes $n^{1/2 - k/4}$ and the displayed prefactor. Their product is $n^{\pm w/2 - 1/2}$. The next-order coefficient and every derivative have exactly the bounds already used above. The dominant coefficients remain nonzero, since their gamma arguments have positive real part. At critical points the two coefficients are conjugate. Thus every full-block estimate, positive Gram limit, and determinant argument (FPK.14)–(FPK.25) applies with (FPK.28), keeping the original root coordinates and the same E_χ . In particular

$$\log V_N^{m_0} = -\mathcal{A}_\chi \log N - \mathcal{B}_\chi \log \log N + \log \frac{|\det E_\chi|^2}{C_{\chi,k}} + o(1), \quad (\text{FPK.29})$$

where $C_{\chi,k} > 0$ is given by the literal constant formula (FPK.22) with every $C_\pm$ replaced by $C_{\pm,k}$, including their occurrences in the critical blocks (FPK.19). The off-critical Cauchy blocks (FPK.16) remain unchanged. This proves the corresponding fixed-packet result for the programme's original reference, with its entire original mass.

175 A fixed denominator in the Gamma weight

Throughout this section the original real number $c = k/2$, the original coordinate S , and the actual monic polynomial

$$\chi(S) = \prod_{\lambda \in \Lambda} (S - \lambda)^{m_\lambda}, \quad q = \sum_{\lambda \in \Lambda} m_\lambda \geq 1,$$

are fixed. Distinct elements of Λ denote distinct roots, and every m_λ is the full multiplicity in χ . No root is discarded. Set

$$\sigma(u) = \frac{|\Gamma(1/4 + iu/2)|^2}{2\pi}, \quad w_M(u) = \frac{\sigma(u)}{(1 + u^2)^M}, \quad M \in \mathbb{Z}_{\geq 0}. \quad (\text{LC.1})$$

For a positive weight w with finite moments, let

$$\|p\|_w^2 = \int_{\mathbb{R}} |p(c + iu)|^2 w(u) du, \quad p \in \mathcal{P}_N = \mathbb{C}[S]_{\leq N}. \quad (\text{LC.2})$$

For $N \geq q - 1$ and a remainder $a(S) = \sum_{j=0}^{q-1} a_j S^j$, put

$$\|[a]\|_{w,N,\chi}^2 = \min_{\substack{p \in \mathcal{P}_N \\ p-a \in (\chi)}} \|p\|_w^2 = \mathbf{a}^* G_N(w, \chi) \mathbf{a}, \quad V_N(w, \chi) = \det G_N(w, \chi). \quad (\text{LC.3})$$

The minimum exists uniquely because it is the minimum of a positive definite quadratic form on a nonempty affine subspace of a finite dimensional space. The form is positive definite since a polynomial vanishing almost everywhere on $c + i\mathbb{R}$ vanishes identically. In particular, V_N retains the original Gram determinant convention.

175.1 Explicit comparison of the weights

The exact Gamma envelope, including $u = 0$, is

$$c_\Gamma e^{-\alpha|u|} (1 + |u|)^{-1/2} \leq \sigma(u) \leq C_\Gamma e^{-\alpha|u|} (1 + |u|)^{-1/2}, \quad (\text{LC.4})$$

$$\alpha = \frac{\pi}{2}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}.$$

For completeness, its proof is recalled. Binet's exact identity on $\Re z > 0$, with the logarithm real on positive reals, is

$$\log \Gamma(z) = (z - \tfrac{1}{2}) \log z - z + \tfrac{1}{2} \log(2\pi) + R(z), \quad R(z) = \int_0^\infty e^{-zt} \frac{1}{t} \left(\frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2} \right) dt. \quad (\text{LC.5})$$

This identity is given in NIST DLMF, equation 5.9.10₂, https://dlmf.nist.gov/5.9.E10_2. Writing $x = t/2$ in the kernel, its value is $(\coth x - 1/x)/(4x)$. The functions $F(x) = x \cosh x - \sinh x$ and $H(x) = (x^2 + 3) \sinh x - 3x \cosh x$ vanish at zero and satisfy $F'(x) = x \sinh x \geq 0$ and $H'(x) = xF(x) \geq 0$ for $x \geq 0$. Dividing by the positive denominators $x \sinh x$ and $3x \sinh x$ gives $0 \leq \coth x - 1/x \leq x/3$ for $x > 0$. Thus the kernel in (LC.5) lies between 0 and $1/12$, and $|R(z)| \leq 1/(12\Re z)$.

At the retained argument $z = 1/4 + iu/2$, put $x = |u|$ and $\rho = |z| = (1/16 + x^2/4)^{1/2}$. Twice the real part of (LC.5) gives exactly

$$\sigma(u) = \rho^{-1/2} \exp(-x \arctan(2x) - \tfrac{1}{2} + 2\Re R(z)). \quad (\text{LC.6})$$

The remainder bound is $|R(z)| \leq 1/3$. For $x \geq 0$ the exponent lies between $-\alpha x - 7/6$ and $-\alpha x + 2/3$: the lower inequality uses $\arctan(2x) \leq \pi/2$, while for $x > 0$ the upper inequality follows from

$\arctan(2x) = \pi/2 - \arctan(1/(2x))$ and $\arctan y \leq y$ for $y \geq 0$. At $x = 0$ the upper inequality follows directly from $-1/2 \leq 0$. The exact squared differences

$$\frac{(1+x)^2}{4} - \rho^2 = \frac{3+8x}{16}, \quad \rho^2 - \frac{(1+x)^2}{20} = \frac{(4x-1)^2}{80}$$

give $(1+x)/(2\sqrt{5}) \leq \rho \leq (1+x)/2$. Combining these inequalities with (LC.6) proves (LC.4), including the origin.

Fix $r > 1$ and put $\beta = \alpha(r-1) > 0$. Define the finite explicit constant

$$B_{r,M} = \frac{C_\Gamma}{c_\Gamma} \sum_{j=0}^{2M} \binom{2M}{j} \frac{j!}{\beta^j}. \quad (\text{LC.7})$$

For $x = |u| \geq 0$, the Gamma envelope gives

$$\frac{\sigma(ru)}{\sigma(u)} \leq \frac{C_\Gamma}{c_\Gamma} e^{-\beta x} \left(\frac{1+x}{1+rx} \right)^{1/2} \leq \frac{C_\Gamma}{c_\Gamma} e^{-\beta x}.$$

For each integer $j \geq 0$, the exponential series implies $e^{\beta x} \geq (\beta x)^j / j!$, so that $x^j e^{-\beta x} \leq j! \beta^{-j}$, with the case $j = 0$ read directly. Since $(1+x^2)^M \leq (1+x)^{2M}$, expansion of the latter polynomial now proves

$$B_{r,M}^{-1} \sigma(ru) \leq w_M(u) \leq \sigma(u) \quad (u \in \mathbb{R}). \quad (\text{LC.8})$$

Every comparison is global, and r is fixed independently of N .

175.2 The exact morphism on the original jet constraints

Order the roots once and use that ordering throughout. For a polynomial p , write its divided jets as

$$J_\chi p = \left(\frac{p^{(j)}(\lambda)}{j!} \right)_{\lambda \in \Lambda, 0 \leq j < m_\lambda}.$$

Let E_χ be the matrix of J_χ on the original remainder frame $1, S, \dots, S^{q-1}$. This matrix is invertible: a remainder in its kernel has a zero of order at least m_λ at every λ , is consequently divisible by χ , and has degree less than q , hence is zero. For all polynomials p, a , the same multiplicity criterion proves

$$p - a \in (\chi) \iff J_\chi p = J_\chi a. \quad (\text{LC.9})$$

For the comparison parameter $r > 0$, define the polynomial bijection

$$(T_r p)(S) = p \left(c + \frac{S-c}{r} \right), \quad \zeta_\lambda(r) = c + r(\lambda - c), \quad D_r = \text{diag}_{\lambda,j}(r^j). \quad (\text{LC.10})$$

Its inverse is $T_{1/r}$. The locations $\zeta_\lambda(r)$ are pairwise distinct, because $r > 0$. Write J_r for the divided-jet map of the same multiplicities at these locations. The chain rule, with all constants retained, gives

$$\frac{(T_r p)^{(j)}(\zeta_\lambda(r))}{j!} = r^{-j} \frac{p^{(j)}(\lambda)}{j!}, \quad J_r T_r = D_r^{-1} J_\chi. \quad (\text{LC.11})$$

The literal substitution $v = ru$ in the integral on the original line gives

$$\|p\|_{\sigma(r \cdot)}^2 = r^{-1} \int_{\mathbb{R}} \left| p \left(c + \frac{iv}{r} \right) \right|^2 \sigma(v) dv = r^{-1} \|T_r p\|_{\sigma}^2. \quad (\text{LC.12})$$

Equations (LC.11)–(LC.12) are the exact relation between the comparison measure and the original observation. In particular, the original condition $p \equiv a \pmod{\chi}$ becomes the condition $J_r(T_r p) =$

$D_r^{-1}E_\chi \mathbf{a}$; it has not been discarded or replaced by an independent arithmetic condition. The symbols $\zeta_\lambda(r)$ describe these transported evaluation functionals only.

Let $K_N^\sigma(r)$ be the Gram matrix of all these divided-jet functionals on $(\mathcal{P}_N, \|\cdot\|_\sigma)$. More explicitly, for any orthonormal basis $e_0, \dots, e_N$ of this space,

$$[K_N^\sigma(r)]_{(\lambda,j),(\mu,l)} = \sum_{n=0}^N \frac{e_n^{(j)}(\zeta_\lambda(r))}{j!} \overline{\frac{e_n^{(l)}(\zeta_\mu(r))}{l!}}. \quad (\text{LC.13})$$

The map J_r is onto, since its restriction to $\mathcal{P}_{q-1}$ is injective by the same multiplicity proof as for E_χ and the two spaces have dimension q . Hence $K_N^\sigma(r)$ is positive definite. For clarity, if L is the q by $N+1$ matrix of J_r in an orthonormal basis, then $K_N^\sigma(r) = LL^*$. For any required jet vector y , the vector $b_0 = L^*(LL^*)^{-1}y$ satisfies $Lb_0 = y$. Every other solution is $b_0 + h$ with $Lh = 0$, and $h^*b_0 = h^*L^*(LL^*)^{-1}y = (Lh)^*(LL^*)^{-1}y = 0$. Thus the minimum squared norm is $y^*(K_N^\sigma(r))^{-1}y$. Applying this identity to the exact original constraints yields

$$G_N(\sigma(r\cdot), \chi) = r^{-1}E_\chi^* D_r^{-1} (K_N^\sigma(r))^{-1} D_r^{-1} E_\chi. \quad (\text{LC.14})$$

Since $\det D_r = r^{\sum_\lambda m_\lambda(m_\lambda-1)/2}$,

$$V_N(\sigma(r\cdot), \chi) = r^{-\sum_\lambda m_\lambda^2} \frac{|\det E_\chi|^2}{\det K_N^\sigma(r)}. \quad (\text{LC.15})$$

This is an identity in the original remainder frame, with the original χ on its left and the full transported jet matrix on its right.

There is an equivalent check on the remainder determinant. On $\mathcal{P}_{q-1}$ the coefficient matrix C_r of T_r is upper triangular with entries

$$(C_r)_{l,j} = \binom{j}{l} c^{j-l} (1 - r^{-1})^{j-l} r^{-l} \quad (0 \leq l \leq j < q), \quad \det C_r = r^{-q(q-1)/2}. \quad (\text{LC.16})$$

Let E_r be the divided-jet matrix at $\zeta_\lambda(r)$ on this displayed polynomial frame. Equation (LC.11) reads $E_r C_r = D_r^{-1} E_\chi$. Consequently

$$G_N(\sigma(r\cdot), \chi) = r^{-1} C_r^* [E_r^* (K_N^\sigma(r))^{-1} E_r] C_r,$$

and taking determinants gives the total frame factor r^{-q^2} . This explicitly agrees with (LC.15); no modulus, Jacobian, or derivative factor has been absorbed into a normalization.

175.3 Passage to the asymptotic coefficient

Write

$$A_\chi = \sum_{\lambda \in \Lambda} m_\lambda |\Re \lambda - c|, \quad B_\chi = \sum_{\substack{\lambda \in \Lambda \\ \Re \lambda = c}} m_\lambda^2. \quad (\text{LC.17})$$

The full confluent Gamma-kernel theorem (Theorem 174.1) proved in the preceding module, applied to the particular fixed functionals in (LC.13), gives

$$\log \det K_N^\sigma(r) = r A_\chi \log N + B_\chi \log \log N + O_{r,\chi}(1). \quad (\text{LC.18})$$

Indeed, $\Re \zeta_\lambda(r) - c = r(\Re \lambda - c)$, and the roots on the line are exactly those original roots with $\Re \lambda = c$, with all their multiplicities unchanged. This use of the Gamma theorem is for each fixed $r > 0$; no uniformity in a moving parameter is asserted or needed. Equation (LC.15) therefore proves

$$\log V_N(\sigma(r\cdot), \chi) = -r A_\chi \log N - B_\chi \log \log N + O_{r,\chi}(1). \quad (\text{LC.19})$$

We next pass the pointwise inequalities to the fixed quotient without altering the admissible polynomial space. If $b w_1 \leq w_2 \leq d w_3$ pointwise with $b, d > 0$, then for every polynomial p the same inequalities hold for the squared norms. Taking the infimum over the identical set $\{p \in \mathcal{P}_N : p - a \in (\chi)\}$ gives

$$b \mathbf{a}^* G_N(w_1, \chi) \mathbf{a} \leq \mathbf{a}^* G_N(w_2, \chi) \mathbf{a} \leq d \mathbf{a}^* G_N(w_3, \chi) \mathbf{a}.$$

Thus these are inequalities of positive definite Hermitian matrices. If $0 < H \leq G$, then $H^{-1/2} G H^{-1/2} \geq I$; all its eigenvalues are at least one, and $\det G \geq \det H$. This proves, in particular,

$$B_{r,M}^{-q} V_N(\sigma(r \cdot), \chi) \leq V_N(w_M, \chi) \leq V_N(\sigma, \chi). \quad (\text{LC.20})$$

Using (LC.19) at r and at 1, we obtain, for each fixed $r > 1$,

$$-r A_\chi \log N - B_\chi \log \log N + O_{r,M,\chi}(1) \leq \log V_N(w_M, \chi) \leq -A_\chi \log N - B_\chi \log \log N + O_\chi(1). \quad (\text{LC.21})$$

Dividing by $\log N$ for $N > 1$ and taking lower and upper limits gives

$$-r A_\chi \leq \liminf_{N \rightarrow \infty} \frac{\log V_N(w_M, \chi)}{\log N} \leq \limsup_{N \rightarrow \infty} \frac{\log V_N(w_M, \chi)}{\log N} \leq -A_\chi.$$

Here $A_\chi \geq 0$. The lower inequality holds separately for every fixed $r > 1$. Taking its supremum as r decreases to 1 proves the exact limit

$$\lim_{N \rightarrow \infty} \frac{\log V_N(w_M, \chi)}{\log N} = - \sum_{\lambda \in \Lambda} m_\lambda |\Re \lambda - c|. \quad (\text{LC.22})$$

This order of limits makes the possible growth of $B_{r,M}$ near $r = 1$ irrelevant: the constant is fixed before N tends to infinity.

If every original root lies on $\Re S = c$, then $A_\chi = 0$. Already a single fixed choice, for example $r = 2$, makes the two sides of (LC.21) equal up to bounded errors. Hence in this case

$$\log V_N(w_M, \chi) = - \left(\sum_{\lambda \in \Lambda} m_\lambda^2 \right) \log \log N + O_{M,\chi}(1). \quad (\text{LC.23})$$

For $A_\chi > 0$, equation (LC.22) is the conclusion proved by this comparison; it does not assert a bounded remainder at the $\log \log N$ scale.

175.4 Application to the actual arithmetic weight

Retain the actual weight $m_{h,k}$ and the original polynomial χ from the arithmetic construction. The proved arithmetic-tail envelope for fixed $k \geq 3$ is

$$a_k \frac{\sigma(u)}{(1+u^2)^M} \leq m_{h,k}(u) \leq A_k \sigma(u), \quad u \in \mathbb{R}, \quad (\text{LC.24})$$

where a_k, A_k are the positive constants calculated there and M is its fixed integer exponent. Applying the same quotient order argument gives

$$a_k^q V_N(w_M, \chi) \leq V_N(m_{h,k}, \chi) \leq A_k^q V_N(\sigma, \chi). \quad (\text{LC.25})$$

The constants k, q, a_k, A_k, M , every root, and every root multiplicity are fixed in this limit. Equations (LC.22) and (LC.19) at $r = 1$ now prove

$$\lim_{N \rightarrow \infty} \frac{\log V_N(m_{h,k}, \chi)}{\log N} = - \sum_{\lambda \in \Lambda} m_\lambda |\Re \lambda - k/2|. \quad (\text{LC.26})$$

If all original roots lie on $\Re S = k/2$, the stronger conclusion is

$$\log V_N(m_{h,k}, \chi) = - \left(\sum_{\lambda \in \Lambda} m_\lambda^2 \right) \log \log N + O_{h,k,\chi}(1). \quad (\text{LC.27})$$

These statements concern the actual fixed quotient. They provide no uniform estimate when the arithmetic polynomial or k changes with N , and they do not determine a signed difference of growing-window quotient determinants. All such parameters remain in their original domains.

Finally retain the original tensor-Gamma density m_0 of (AT4), with its original mass $(2\pi)^{k/2}$. Its proved complete asymptotic (FPK.29) has the same leading coefficient $-A_\chi$. Hence the actual end-point quotient ratio satisfies

$$\lim_{N \rightarrow \infty} \frac{1}{\log N} \log \frac{V_N(m_{h,k}, \chi)}{V_N(m_0, \chi)} = 0. \quad (\text{LC.28})$$

If all original roots lie on $\Re S = k/2$, (LC.27) and (FPK.29) give the stronger bounded logarithmic ratio, equivalently positive upper and lower bounds for this ratio for all sufficiently large N . Both conclusions are for the original fixed h, k, χ . No uniformity needed for a varying-packet four-window expression is asserted by (LC.28).

176 The original arithmetic classes under Hilbert completion

Fix the original complete counterfactual quartet h of (HT1), the integer $k \geq 3$, and its complete cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$. Every parameter remains fixed as $N \rightarrow \infty$. Retain

$$\begin{aligned} c &= k/2, \quad \mathcal{P} = \mathbb{C}[S], \quad E = \mathcal{P}/\chi\mathcal{P}, \quad J : \mathcal{P} \longrightarrow E, \\ \mathcal{H}_{h,k} &= L^2(\mathbb{R}, m_{h,k}(u) du), \quad \iota : \mathcal{P} \longrightarrow \mathcal{H}_{h,k}, \quad (\iota P)(u) = P(c + iu). \end{aligned} \quad (\text{FKG.1})$$

The map ι is injective, since the original density is positive almost everywhere and a nonzero polynomial has finitely many zeros on the source line. We write $\mathcal{P}$ and $\chi\mathcal{P}$ for these actual embedded subspaces when taking Hilbert-space quotients. All quotient arrows below are complex linear. Multiplication by S on the full Hilbert space has its unbounded-operator domain; no everywhere-defined polynomial-module structure on that space is assumed.

176.1 Vanishing of the entire quotient Gram

Put $w_i = \lambda_i - c$ and $\delta_i = \Re w_i$ as in (FPK.14). For a noncritical block, write $a_i(w) = C_{\epsilon_i}(w)^{-1}$, where $\epsilon_i = \text{sgn } \delta_i$. In normalized Taylor coordinates the complete triangular matrix in (FPK.14) has entries

$$(\mathsf{T}_{N,i})_{j,l} = \begin{cases} N^{-\epsilon_i w_i/2} \sum_{r=0}^{j-l} a_i^{[r]}(w_i) \frac{(-\epsilon_i \log N/2)^{j-l-r}}{(j-l-r)!}, & l \leq j, \\ 0, & l > j. \end{cases} \quad (\text{FKG.2})$$

This follows by the product rule for Taylor coefficients of $a_i(w) \exp(-\epsilon_i w \log N/2)$; thus no derivative of the original gamma unit has been dropped. For fixed roots and multiplicities,

$$\|\mathsf{T}_{N,i}\| \leq C_i N^{-|\delta_i|/2} (1 + \log N)^{m_i-1} \longrightarrow 0 \quad (\delta_i \neq 0). \quad (\text{FKG.3})$$

For a critical block, (FPK.17) gives $\|\mathsf{T}_{N,i}\| = (\log N)^{-1/2} \rightarrow 0$ for $N \geq 3$. There are finitely many blocks, so $\|\mathsf{T}_N\| \rightarrow 0$.

Let $A_N = \mathsf{T}_N \mathsf{K}_N \mathsf{T}_N^*$. The complete matrix limit (FPK.20) gives $A_N \rightarrow \mathsf{H} > 0$. If h_0 is the smallest eigenvalue of H , then $A_N \geq (h_0/2)I$ for all sufficiently large N : for each unit vector the difference of

the quadratic forms is at most $\|A_N - H\|$. In particular $\|A_N^{-1}\| \leq 2/h_0$. Undoing the exact congruence gives

$$\begin{aligned} K_N^{-1} &= T_N^* A_N^{-1} T_N, \\ G_N^\sigma &= E_\chi^* T_N^* A_N^{-1} T_N E_\chi, \\ \|G_N^\sigma\| &\leq \frac{2\|E_\chi\|^2}{h_0} \|T_N\|^2 \longrightarrow 0. \end{aligned} \tag{FKG.4}$$

All norms on E here use the original increasing-power remainder frame. In particular this is vanishing of the entire positive operator, not merely of its determinant. The explicit bounds (FKG.3) also give

$$\|G_N^\sigma\| \leq C_\chi \left(\frac{\mathbf{1}_{\{\exists i: \delta_i=0\}}}{\log N} + \sum_{\delta_i \neq 0} N^{-|\delta_i|} (1 + \log N)^{2m_i-2} \right) \tag{FKG.5}$$

for all sufficiently large N , with one finite constant for the fixed original observation.

Write $R_N^{\text{ar}} : E \rightarrow \mathcal{P}_N$ for the canonical least-norm polynomial lift in the original arithmetic form (AT6). It is the unique minimizer on each affine fibre $J_N P = x$: orthogonal projection onto $\chi \mathcal{P}_{N-q}$ in the finite positive-definite polynomial space proves existence and uniqueness. Its source Gram is G_N^{ar} , so

$$J_N R_N^{\text{ar}} = I_E, \quad \|\iota R_N^{\text{ar}} x\|_{\mathcal{H}_{h,k}}^2 = x^* G_N^{\text{ar}} x. \tag{FKG.6}$$

The proved actual envelope $m_{h,k} \leq A_k \sigma$ in (LC.24) applies to every polynomial in the same affine fibre. Taking the minimum first for the arithmetic norm and then comparing with the minimizing gamma lift proves

$$0 < G_N^{\text{ar}} \leq A_k G_N^\sigma, \quad \|G_N^{\text{ar}}\| \rightarrow 0, \quad \|\iota R_N^{\text{ar}}\|_{E \rightarrow \mathcal{H}_{h,k}} \rightarrow 0. \tag{FKG.7}$$

For the middle assertion a positive matrix has operator norm equal to the supremum of its unit-vector quadratic forms; the last assertion follows from (FKG.6). The constant A_k is the original arithmetic upper-comparison constant, including its full scalar mass.

176.2 Exact arithmetic observations and theta primitives

The original cohomology observation and its injection remain

$$\begin{aligned} \mathcal{O} &= q^{(k)} \mathcal{V} : \mathcal{P} \longrightarrow Q^{\otimes k}, \quad \mathcal{I} = \sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}, \\ \mathcal{V} P &= P(D_1 + \cdots + D_k) F_h^{\otimes k}, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_h^{(k)}) 1, \\ v_h &= j_h(g/h), \quad g = 2\xi, \quad D = -x\partial_x, \quad \mathcal{O} = \mathcal{I}J. \end{aligned} \tag{FKG.8}$$

These are precisely (AT2)–(AT2a), with the complete invertible Taylor unit v_h retained. Consequently for every $x \neq 0$ and every $N \geq q-1$,

$$\mathcal{O}(R_N^{\text{ar}} x) = \mathcal{I}x \neq 0, \quad \|\iota R_N^{\text{ar}} x\|_{\mathcal{H}_{h,k}} \longrightarrow 0. \tag{FKG.9}$$

This comparison also retains the exact original tensor-source norm: Mellin Plancherel on each variable gives

$$\|\mathcal{V} P\|_{L^2((0,\infty)^k, dx_1 \cdots dx_k)}^2 = \int_{\mathbb{R}^k} |P(k/2 + i \sum t_i)|^2 \prod_i w_h(t_i) dt = \|\iota P\|_{\mathcal{H}_{h,k}}^2. \tag{FKG.10}$$

Indeed the Mellin transform of the source is $P(\sum s_i) \prod_i v_h(s_i)$, the k Plancherel factors $(2\pi)^{-k}$ are exactly those in $\prod_i w_h(t_i)$, and the change $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. The final equality is the defining convolution. Thus (FKG.9) quantifies the original source representatives themselves.

For $q - 1 \leq a < b$, monic division gives a unique polynomial $\kappa_x \in \mathcal{P}_{b-q}$ satisfying

$$R_a^{\text{ar}}x - R_b^{\text{ar}}x = \chi\kappa_x. \quad (\text{FKG.11})$$

The inclusions into $\mathcal{P}_b$ are the literal coefficient inclusions. Both remainders are x , so their difference is divisible by χ ; the degree bound and uniqueness follow from monicity. Division also proves linearity in x . To retain the entire original boundary, successive monic division in $s_1, \dots, s_k$ gives specific polynomials Q_i with

$$\chi(\sum s_i) = \sum_{i=1}^k h(s_i)Q_i(\mathbf{s}). \quad (\text{FKG.12})$$

The final multivariable remainder is zero because the monomials of degree less than $\deg h$ in each variable form a basis of $(\mathbb{C}[s]/h)^{\otimes k}$ and the original χ annihilates the sum operator on this algebra. Using the original $h(D)F_h = \Theta\phi_0$, where $\phi_0 = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}$, form the cochain

$$\mathcal{P}_\chi(\kappa_x) = \sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D})\kappa_x(\sum_j D_j)(F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{FKG.13})$$

It has degree $k - 1$ and

$$d_{\text{tot}}\mathcal{P}_\chi(\kappa_x) = \mathcal{V}(R_a^{\text{ar}}x - R_b^{\text{ar}}x). \quad (\text{FKG.14})$$

At position i , the $i - 1$ preceding degree-one factors contribute $(-1)^{i-1}$ to the tensor differential, cancelling the displayed sign. The relation $\Theta\phi_0 = h(D_i)F_h$ then makes the sum equal to (FKG.12) applied to the original tensor source. Polynomial Euler operators commute with this differential by the original identity $D\Theta = \Theta D$. This proves (FKG.14) term by term. It is the full primitive (EW.47)–(EW.48), with the actual canonical arithmetic lifts substituted.

176.3 The exact algebraic quotient and its Hausdorff reduction

Polynomials are dense in $\mathcal{H}_{h,k}$. Here the exponential moment needed for the complete density theorem (ND.11)–(ND.12) follows in the original sum coordinate: for $0 < \eta < \pi/2$, Tonelli's theorem and $|\sum t_i| \leq \sum |t_i|$ give

$$\int_{\mathbb{R}} e^{\eta|u|} m_{h,k}(u) du \leq \left(\int_{\mathbb{R}} e^{\eta|t|} w_h(t) dt \right)^k < \infty. \quad (\text{FKG.15})$$

Finiteness follows directly from (ND.3), choosing $\eta < 2b < \pi/2$. To specify the density argument in this one-dimensional application, if f is orthogonal to all polynomials, the finite complex measure $d\nu = \bar{f} m_{h,k} du$ has an exponential moment of any order less than $\eta/2$, by Cauchy–Schwarz and (FKG.15). Its Fourier transform is holomorphic on the corresponding strip and all its derivatives at zero vanish, because polynomials in u are exactly polynomials in $S = c + iu$ under this invertible affine coordinate map. Its Taylor series and then the identity theorem make the transform zero on the strip. The Gaussian convolution and approximate-identity proof (ND.12) then gives $\nu = 0$ and $f = 0$ almost everywhere. Taking orthogonal complements proves the asserted density, including the original measure and mass.

For any $P \in \mathcal{P}$ and $N \geq \max(q - 1, \deg P)$, put

$$B_N(P) = P - R_N^{\text{ar}}JP \in \chi\mathcal{P}. \quad \|\iota(P - B_N(P))\|_{\mathcal{H}_{h,k}} \longrightarrow 0. \quad (\text{FKG.16})$$

The first assertion follows by taking its remainder; the second is (FKG.7) at the fixed vector JP . Thus the closure of $\chi\mathcal{P}$ contains $\mathcal{P}$, whose density was just proved. Consequently

$$\overline{\chi\mathcal{P}}^{\mathcal{H}_{h,k}} = \mathcal{H}_{h,k}. \quad (\text{FKG.17})$$

Every approximating relation in (FKG.16) has a literal polynomial quotient by χ , and hence the primitive (FKG.13) as well. The isometry (FKG.10) also extends to the exact original tensor-source completion:

$$\begin{aligned} \overline{\mathcal{V}} : \mathcal{H}_{h,k} &\xrightarrow{\cong} \overline{\mathcal{V}(\mathcal{P})}^{L^2((0,\infty)^k)}, \\ \overline{\mathcal{V}(\chi\mathcal{P})}^{L^2((0,\infty)^k)} &= \overline{\mathcal{V}(\mathcal{P})}^{L^2((0,\infty)^k)}. \end{aligned} \quad (\text{FKG.17a})$$

Indeed polynomial approximants form Cauchy sequences in the original tensor norm by (FKG.10); completeness defines the extension, and that norm equality proves its independence of the approximants and its isometry. Its range is closed by completeness and contains the dense polynomial source, hence is the displayed closure. Applying this isometry to (FKG.17) proves the second equality. Every vector before closure on its left is the boundary (FKG.14) for its literal polynomial quotient. No full test-space cohomology is quotiented in forming these two specified Hilbert closures.

There is nevertheless an injective map of the unchanged algebraic quotient into the algebraic Hilbert-space quotient, in the exact sequence

$$\begin{aligned} 0 \longrightarrow E &\xrightarrow{j} \mathcal{H}_{h,k}/\chi\mathcal{P} \xrightarrow{\pi} \mathcal{H}_{h,k}/\mathcal{P} \longrightarrow 0, \\ j([P]) &= \iota P + \chi\mathcal{P}, \quad \pi(f + \chi\mathcal{P}) = f + \mathcal{P}. \end{aligned} \quad (\text{FKG.18})$$

The first map is independent of the polynomial representative. Its kernel is zero because ι is injective and membership in the embedded $\chi\mathcal{P}$ means literal polynomial divisibility. The second map is well-defined and surjective since $\chi\mathcal{P} \subset \mathcal{P}$; its kernel consists exactly of the cosets represented by polynomials, which is the image of j . This proves exactness at every term without a topological closure.

With the quotient seminorm inherited from the original Hilbert norm, the Hausdorff reduction is instead the specific map

$$\begin{aligned} \mathcal{H}_{h,k}/\chi\mathcal{P} &\longrightarrow \mathcal{H}_{h,k}/\overline{\chi\mathcal{P}} = 0, \\ f + \chi\mathcal{P} &\longmapsto f + \overline{\chi\mathcal{P}}. \end{aligned} \quad (\text{FKG.19})$$

Indeed the seminorm of any coset is its distance to the dense subspace $\chi\mathcal{P}$, hence zero. In particular (FKG.19) kills the injected algebraic copy $j(E)$, although every nonzero $x \in E$ retains the nonzero arithmetic class $\mathcal{I}x$ in (FKG.8). This is an exact comparison between these two quotients.

One can also compute the entire closure of the original remainder graph. Give E its finite-dimensional topology in its original frame. For any $f \in \mathcal{H}_{h,k}$ and $x \in E$, choose polynomials $P_j \rightarrow f$ by density. For each fixed j , (FKG.7) permits an $N_j \geq \max(q-1, \deg P_j)$ such that $\|\iota R_{N_j}^{\text{ar}}(x - JP_j)\| < 1/j$. Then

$$\tilde{P}_j = P_j + R_{N_j}^{\text{ar}}(x - JP_j), \quad J\tilde{P}_j = x, \quad \iota\tilde{P}_j \longrightarrow f. \quad (\text{FKG.20})$$

It follows that the graph of $J : \mathcal{P} \subset \mathcal{H}_{h,k} \rightarrow E$ has closure $\mathcal{H}_{h,k} \times E$. In particular this densely defined observation is not closable when $q \geq 1$: (FKG.9) gives sequences tending to zero with any prescribed nonzero remainder. This precisely states why a continuous extension of the same nonzero finite observation to this Hilbert completion is impossible.

At an original nonempty support label α , each linear comparison just written lifts as $(\alpha, v) \mapsto (\alpha, Lv)$ and sends the external point τ to τ . Thus the killed amplitude in (FKG.19) maps to $(\alpha, 0)$, and the exact boundary (FKG.14) has that same supported zero observation. The label remains specified. These maps quantify this particular Hilbert completion; they do not identify its zero Hausdorff quotient with the entire original arithmetic cohomology or alter any original support restriction.

176.4 The coefficient for the complete counterfactual quartet

In the actual off-line counterfactual (HT1)–(HT2), retain $\rho = 1/2 + \delta + i\gamma$ with $0 < \delta < 1/2, \gamma > 0$, and its full order m . Put $\ell_k = 1 + k(m-1)$. The distinct cyclic roots and their complete multiplicities

are exactly

$$\lambda_{a,b} = k/2 + (2a - k)\delta + i(2b - k)\gamma, \quad 0 \leq a, b \leq k, \quad m_{a,b} = \ell_k. \quad (\text{FKG.21})$$

For completeness, in each ordered tensor-local factor the nilpotent sum $n_1 + \cdots + n_k$, with $n_i^m = 0$, has nonzero top coefficient $(k(m-1))!/((m-1)!)^k$ at total degree $k(m-1)$ and its next power vanishes. Its index is therefore ℓ_k . The real and imaginary signs may be chosen independently in each tensor factor, so every pair (a, b) occurs. Since $\delta\gamma \neq 0$, different pairs give different sums. The full annihilator is consequently the product over (FKG.21), each to power ℓ_k , with $q = \ell_k(k+1)^2$ as originally specified.

The exponents of the full gamma determinant therefore evaluate to

$$\begin{aligned} \mathcal{A}_\chi &= \delta\ell_k(k+1) \sum_{a=0}^k |2a - k| = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = \mathcal{L}_{h,k}, \\ \mathcal{B}_\chi &= \begin{cases} (k+1)\ell_k^2, & k \text{ even,} \\ 0, & k \text{ odd.} \end{cases} \end{aligned} \quad (\text{FKG.22})$$

To verify the finite sum, pair a with $k-a$. For $k = 2r$ the sum of its positive terms is $2(1 + \cdots + r) = r(r+1)$; for $k = 2r-1$ it is $1 + 3 + \cdots + (2r-1) = r^2$. Doubling gives $2\lfloor (k+1)^2/4 \rfloor$ in both cases. A root is critical exactly when $2a = k$, which occurs only for even k , with its $k+1$ distinct imaginary coordinates and multiplicity ℓ_k . This proves the second exponent with all nilpotent jets included. The equality with $\mathcal{L}_{h,k}$ uses its exact definition in (HT8), not a change of normalization.

In particular (LC.26) gives, on the actual unchanged arithmetic source for every fixed $k \geq 3$,

$$\lim_{N \rightarrow \infty} \frac{\log V_N^{\text{ar}}}{\log N} = -2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{FKG.23})$$

The value of $\mathcal{B}_\chi$ in (FKG.22) is the proved gamma $\log \log N$ coefficient. For a mixed critical and non-critical arithmetic packet, the envelope proof establishes (FKG.23), and does not establish its next $\log \log N$ coefficient. None of (FKG.3)–(FKG.23) supplies an error uniform in growing k , q , or the packet. Their precise conclusions concern fixed original objects and the displayed exact maps.

177 The actual arithmetic form passes to every original quotient

Retain $g = 2\xi$, the full actual hypothetical quartet polynomial h , its complete common order m , and $v_h = g/h$. For $k \geq 3$ put $c = k/2$, $\ell = 1 + k(m-1)$, and $q = \ell(k+1)^2$. The original cyclic polynomial is

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a - k)\delta - i(2b - k)\gamma)^\ell. \quad (\text{CA1})$$

Thus $E = \mathbb{C}[S]/(\chi)$ has the increasing-power frame of length q . No distinct factors or nilpotent orders are merged. The original observation is $\mathcal{O}_N = \sigma_h^{\otimes k} \eta J_N : \mathcal{P}_N \rightarrow Q_{\text{ar}}^{\otimes k}$, with $Q_{\text{ar}} = \mathcal{B}/\Theta V$, where J_N is monic remainder and $\eta : E \hookrightarrow E_h^{\otimes k}$, $\eta[P] = v_h^{\otimes k} P(A_h^{(k)})1$ retains the full Taylor unit $v_h = j_h(g/h)$. For $N \geq q-1$ its original source and relation sequence is

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{CA2})$$

The empty relation space at $N = q-1$ is zero. All comparison maps below are the identity on $\mathcal{P}_N$ and E in this sequence; they therefore commute with $\mathcal{O}_N$ and retain its supported label.

Let $m_{h,k} = w_h^{*k}$, with $w_h(u) = |v_h(1/2 + iu)|^2/(2\pi)$, and let

$$\sigma(u) = \frac{|\Gamma(1/4 + iu/2)|^2}{2\pi}, \quad H_N^\sigma = \int v_N(u)^* v_N(u) \sigma(u) du, \quad v_N(u) = (1, c + iu, \dots, (c + iu)^N). \quad (\text{CA3})$$

The masses are respectively μ_h^k and $\sqrt{2\pi}$. In particular this fixed one-factor comparison is different from the inherited k -fold Gamma convolution, whose mass is $(2\pi)^{k/2}$. Write H_N^{ar} for the same moment matrix with density $m_{h,k}$. The preceding boundary-tilt calculation proves positive explicit a_k, A_k and a fixed integer M depending on h , with

$$\frac{a_k \sigma(u)}{(1+u^2)^M} \leq m_{h,k}(u) \leq A_k \sigma(u), \quad D_N = [1 + 4(N+M)^2]^M, \quad c_{k,N} = a_k/D_N, \quad (\text{CA4})$$

and proves, by its finite Gamma recurrence and weighted Cauchy argument,

$$c_{k,N} H_N^\sigma \preceq H_N^{\text{ar}} \preceq A_k H_N^\sigma. \quad (\text{CA5})$$

The complete definitions and proofs of these constants and estimates are in the preceding chapter. Equations (CA4)–(CA5) invoke those proved arithmetic estimates; they add no assumption about a zeta lower bound or about the sign of a determinant correction.

For $\star \in \{\text{ar}, \sigma\}$ retain the exact matrices

$$K_N^\star = J_N (H_N^\star)^{-1} J_N^\star, \quad G_N^\star = (K_N^\star)^{-1}, \quad R_N^\star = (H_N^\star)^{-1} J_N^\star G_N^\star, \quad V_N^\star = \det G_N^\star. \quad (\text{CA6})$$

Positivity and surjectivity give positive invertible matrices $H_N^\star, K_N^\star, G_N^\star$ of their displayed square dimensions. The map $R_N^\star : E \rightarrow \mathcal{P}_N$ is the displayed right inverse, which is rectangular when $N > q - 1$. Direct multiplication gives $J_N R_N^\star = I$ and $B_N^\star H_N^\star R_N^\star = 0$. Every lift of $x \in E$ is uniquely $R_N^\star x + B_N y$, with squared norm $x^\star G_N^\star x + y^\star B_N^\star H_N^\star B_N y$. Thus $G_N^\star$ is the minimum on the same original affine remainder fibre. Applying (CA5) to each point of that fibre and taking its infimum proves

$$c_{k,N} G_N^\sigma \preceq G_N^{\text{ar}} \preceq A_k G_N^\sigma, \quad c_{k,N}^q V_N^\sigma \leq V_N^{\text{ar}} \leq A_k^q V_N^\sigma. \quad (\text{CA7})$$

For the determinant claim, conjugate a positive comparison by the inverse square root of its smaller form; all relative eigenvalues are at least one, so their product is at least one. This also proves the claim when the forms do not commute.

There is an exact original cochain connecting the two representatives. Use the original two-term complex $[V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one and the polynomial source map $\mathcal{V}_{h,k} : \mathbb{C}[S] \rightarrow \mathcal{B}^{\otimes k}$, $P \mapsto P(\sum_j D_j) F_h^{\otimes k}$. Since $J_N(R_N^{\text{ar}} - R_N^\sigma)x = 0$, monic division gives the unique $\kappa_x \in \mathcal{P}_{N-q}$ with

$$(R_N^{\text{ar}} - R_N^\sigma)x = \chi \kappa_x. \quad (\text{CA8})$$

Its coefficients depend linearly on x . Use the original polynomials Q_i from successive tensor division, with $\chi(\sum s_j) = \sum_i h(s_i) Q_i(\mathbf{s})$, and the original F_h satisfying $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. Then

$$\mathfrak{p}_x = \sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) \kappa_x(\sum_j D_j) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}) \quad (\text{CA9})$$

has degree $k - 1$ in the original tensor theta complex and satisfies

$$d_{\text{tot}} \mathfrak{p}_x = \mathcal{V}_{h,k} (R_N^{\text{ar}} - R_N^\sigma)x. \quad (\text{CA10})$$

Indeed the $i - 1$ preceding factors have degree one, so the differential contributes $(-1)^{i-1}$, cancelling the written sign. Its remaining factor is $\Theta\phi_* = h(D_i)F_h$; the displayed polynomial identity then proves (CA10) term by term. This is the same explicit primitive as inherited EW.46–48 applied to the new representative difference. It maps to the supported zero in the original labelled fibre; external τ is unchanged.

178 A finite computable comparison for the four arithmetic volumes

Define on the four original endpoints

$$B_{\text{ar}} = \log \frac{V_{q-1}^{\text{ar}} V_q^{\text{ar}}}{V_{2q-1}^{\text{ar}} V_{2q}^{\text{ar}}}, \quad B_{\sigma} = \log \frac{V_{q-1}^{\sigma} V_q^{\sigma}}{V_{2q-1}^{\sigma} V_{2q}^{\sigma}}, \quad \Delta_{\sigma} = B_{\text{ar}} - B_{\sigma}. \quad (\text{CA11})$$

All are in the original quotient frame. Quotient minima decrease with source degree, hence both B 's are nonnegative. Applying (CA7) at the four endpoints gives the asymmetric exact scalar enclosures

$$-q \log \frac{A_k^2}{c_{k,q-1} c_{k,q}} \leq \Delta_{\sigma} \leq q \log \frac{A_k^2}{c_{k,2q-1} c_{k,2q}}. \quad (\text{CA12})$$

For example the upper numerator is at most $A_k^{2q} V_{q-1}^{\sigma} V_q^{\sigma}$, and the lower denominator is at least $(c_{k,2q-1} c_{k,2q})^q V_{2q-1}^{\sigma} V_{2q}^{\sigma}$. The opposite choices prove the lower bound without imposing a sign on Δ_{σ} .

For the stronger common-source comparison let

$$D_* = (H_{2q}^{\sigma})^{-1} H_{2q}^{\text{ar}}, \quad b_1 \leq \dots \leq b_{2q+1}, \quad \Lambda_k = \log(A_k D_{2q}/a_k). \quad (\text{CA13})$$

The b_j are all positive eigenvalues of this specified relative operator, including every multiplicity. The Rayleigh quotient in (CA5) gives $c_{k,2q} \leq b_j \leq A_k$ for each j ; in particular $\log(b_{2q+1}/b_1) \leq \Lambda_k$ and $\Lambda_k \geq 0$. The unchanged source flags and their minimum relation maps have the AW.7 spectrum $0^q, 1^2, 2^{q-1}$ along $H(t) = (1-t)H_{2q}^{\sigma} + tH_{2q}^{\text{ar}}$. Applying its fully proved spectral trace calculation gives

$$|\Delta_{\sigma}| \leq L_q(D_*) := \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1)\Lambda_k. \quad (\text{CA14})$$

This application changes the endpoint form to (CA3) on the same polynomial and relation spaces; all hypotheses of the finite AW proof hold by (CA3)–(CA6). Its explicit conjugation is a spectral calculation of those forms, not an assertion that an arbitrary isometry preserves their arithmetic observation.

The stronger nonlinear angle estimate is already proved in the original source EW.25–35c for every packet dimension and every window length. For these endpoints its $s = \min(q, q) = q$. To retain its attribution and its exact effect, define

$$\Phi_q(B) = 2 \operatorname{arcosh}(e^{B/(4q)}), \quad \Psi_q(t) = 4q \log \cosh(t/2) \quad (t \geq 0), \quad A_{\sigma} = \Phi_q(B_{\sigma}). \quad (\text{CA15})$$

They are mutually inverse increasing continuous functions on $[0, \infty)$. That inherited theorem gives

$$|\Phi_q(B_{\text{ar}}) - \Phi_q(B_{\sigma})| \leq \log(b_{2q+1}/b_1) \leq \Lambda_k. \quad (\text{CA16})$$

Equivalently it supplies

$$\Psi_q(\max(0, A_{\sigma} - \Lambda_k)) \leq B_{\text{ar}} \leq \Psi_q(A_{\sigma} + \Lambda_k). \quad (\text{CA17})$$

For clarity its exact derivative mechanism is as follows. The two pairs of original canonical-lift projections have at most $2q$ nonzero angle losses $x_i = -\log \sigma_i^2$, with $\sum_i x_i = B(t)$. Their trace norms and concavity of $\sqrt{1-e^{-x}}$ give $|B'(t)| \leq 2q o(t) \sqrt{1-e^{-B(t)/(2q)}}$, where $o(t) = \lambda_{\max}(H(t)^{-1} H') - \lambda_{\min}(H(t)^{-1} H')$. At $B > 0$, division is differentiation of Φ_q , and replacing B by $B + \varepsilon$ justifies integration through $B = 0$. The exact eigenvalues $(b_j - 1)/(1 - t + tb_j)$ integrate to $\log b_j$, proving (CA16). This recalls the proved inherited argument; it does not claim it as a new generalization of the one-dimensional AW angle result.

The accompanying calculation AGT, from the coordinating programme, sharpens this inherited method by pairing the same endpoints as $(q-1, 2q)$ and $(q, 2q-1)$. The second pair lies in the actual $(2q-1)$ -dimensional space $\mathcal{P}_{2q-1} \cap (\chi\mathcal{P}_0)^\perp$; its two q -dimensional section images therefore have a common direction. One of its angles is exactly zero. There are consequently at most $2q-1$ nonzero angle contributions in the same four-volume, rather than the $2q$ allowed in the equal-gap pairing. Its full proof is included in this reader. Put

$$\widehat{\Phi}_q(B) = 2 \operatorname{arcosh}(e^{B/(4q-2)}), \quad \widehat{\Psi}_q(t) = (4q-2) \log \cosh(t/2), \quad \widehat{A}_\sigma = \widehat{\Phi}_q(B_\sigma). \quad (\text{CA17a})$$

Apply the same finite angle sum to (CA3)–(CA6), whose positive moment form and relation flags meet all the proved hypotheses of AGT. The concavity argument just displayed now has $2q-1$ terms. Its derivative bound is $(2q-1)o(t)\sqrt{1-e^{-B(t)/(2q-1)}}$; its factor multiplying $o(t)$ is exactly $1/\widehat{\Phi}'_q(B(t))$. Integration therefore proves

$$\widehat{\Psi}_q(\max(0, \widehat{A}_\sigma - \Lambda_k)) \leq B_{\text{ar}} \leq \widehat{\Psi}_q(\widehat{A}_\sigma + \Lambda_k). \quad (\text{CA17b})$$

The strict positive moment $\int |\chi|^2 m_t > 0$ used in AGT holds for this interpolation as well, so its differentiation is valid throughout the compact interval. This states the precise improvement over EW; the general angle mechanism and its original theta primitive remain attributed to their earlier source.

The reference number B_σ itself is a fully specified finite calculation with the identical original χ . Define real monic $b_n^{(1/4)}(u)$ by

$$b_0 = 1, \quad b_1 = u, \quad b_{n+1} = ub_n - n(n-1/2)b_{n-1}, \quad \gamma_n = \sqrt{2\pi}(2n)!/4^n, \quad p_n(S) = i^n b_n^{(1/4)}((S-c)/i). \quad (\text{CA18})$$

The fixed Gamma integral proves that p_n is monic in the original S coordinate and has squared norm γ_n . Put $z_n = [p_n] \bmod \chi$ in the length- q remainder frame for every $n \geq 0$. Orthogonal expansion of (CA6) gives, exactly,

$$K_N^\sigma = \sum_{n=0}^N \frac{z_n z_n^*}{\gamma_n}, \quad B_\sigma = \log \frac{\det K_{2q-1}^\sigma \det K_{2q}^\sigma}{\det K_{q-1}^\sigma \det K_q^\sigma}. \quad (\text{CA19})$$

To prove the kernel formula, the monic polynomial columns form an invertible triangular matrix Z with diagonal one and $Z^* H_N^\sigma Z = \operatorname{diag}(\gamma_0, \dots, \gamma_N)$. Thus $(H_N^\sigma)^{-1} = Z \operatorname{diag}(\gamma_n^{-1}) Z^*$. Multiplying by J_N and J_N^* gives the displayed sum, and the determinant identity follows from $V_N = 1/\det K_N$. This retains the original phases, centre, degree, unit, and mass.

179 The strengthened restriction and its exact reach

The preceding actual-tail chapter proves the original arithmetic lower bound

$$B_{\text{ar}} \geq 4q \log(\delta ek/8) - E_k, \quad E_k = 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q \geq 0. \quad (\text{CA20})$$

It is obtained from the inherited CJ identity, the full-multiplicity trace floor, and the newly proved norm bounds. Its exact nonnegative contraction, phase, control and trace residuals are retained in that chapter. Equations (CA14), (CA17b), and (CA20) give simultaneously

$$\begin{aligned} & \max\{0, 4q \log(\delta ek/8) - E_k, B_\sigma - L_q(D_*), \\ & \quad \widehat{\Psi}_q(\max(0, \widehat{A}_\sigma - \Lambda_k))\} \\ & \leq B_{\text{ar}} \leq \min\{B_\sigma + L_q(D_*), \widehat{\Psi}_q(\widehat{A}_\sigma + \Lambda_k)\}. \end{aligned} \quad (\text{CA21})$$

Each entry bounds the same original scalar (CA11); taking the indicated maximum and minimum therefore proves (CA21). The earlier original upper bounds and full-spectrum old-Gamma bounds remain simultaneous bounds and may also be inserted in this minimum. They are not discarded.

For the fixed hypothetical actual packet the proved constants give $\log A_k - \log a_k = O_h(k + \log k)$ and hence

$$E_k = O_h(k + \log q) = o(q), \quad \Lambda_k = O_h(k + \log q), \quad |B_{\text{ar}} - B_{\sigma}| = O_h(qk + q \log q) = o(q^2). \quad (\text{CA22})$$

Indeed $q = \ell(k + 1)^2 \geq (k + 1)^2$, while $\ell \leq mk$ for $k \geq 1$, so $\log q = O_h(\log k)$, $k/q \rightarrow 0$, and $\log q/q \rightarrow 0$. Also $\log D_{2q} = M \log[1 + 4(2q + M)^2] = O_h(\log q)$. Substitution in (CA14) and (CA20) proves every assertion in (CA22). In particular the actual arithmetic four-volume loss and the explicitly specified fixed-Gamma four-volume loss have the same leading quadratic scale:

$$\frac{B_{\text{ar}} - B_{\sigma}}{q^2} \rightarrow 0. \quad (\text{CA23})$$

This statement specifies their difference; it asserts neither separate limit exists without its calculation.

The improved exact lower restriction is

$$\liminf_{k \rightarrow \infty} \left(\frac{B_{\text{ar}}}{q} - 4 \log k \right) \geq 4 \log(\delta e/8). \quad (\text{CA24})$$

It follows by dividing (CA20) by q and using $E_k/q \rightarrow 0$. The coefficient four was already present in CJ; the literal constant and vanishing per-dimension error here follow from the new arithmetic tail estimate. Since the error bound in (CA14) has order qk , it is larger than the $q \log k$ lower scale in (CA20). Thus (CA23) alone does not establish an opposing estimate on that smaller scale. No actual off-critical zero is constructed by any comparison here, and no contradiction to its counterfactual is asserted. What is proved is the sharper arithmetic restriction (CA21), the exact cochain map (CA8)–(CA10), and the subquadratic comparison (CA23) on the unchanged original source.

180 The original convolution reference has a compulsory spectral spread

The old k -fold Gamma comparison remains part of the programme and has the explicit density and monic squared norms

$$m_k^{\Gamma}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad \gamma_{k,n}^{\Gamma} = (2\pi)^{k/2} n! \frac{\Gamma(n + k/2)}{\Gamma(k/2)}. \quad (\text{CA25})$$

These are the inherited generating-integral identities with their literal mass. On its original degree- $2q$ space put $D_{\text{old}} = (H_{2q}^{\Gamma})^{-1} H_{2q}^{\text{ar}}$ and let κ_{old} be the ratio of its largest to smallest eigenvalues. The preceding actual estimate forces

$$\kappa_{\text{old}} \geq \max \left\{ 1, \frac{\mu_h^k}{\sqrt{2} A_k \Gamma(k/2)} \frac{\Gamma(2q + k/2)}{\Gamma(2q + 1/2)} \right\}. \quad (\text{CA26})$$

To prove it without changing a source frame, let b_- , b_+ be these two extreme eigenvalues. The constant polynomial gives $b_+ \geq \mu_h^k / (2\pi)^{k/2}$. The full form comparison defining b_- gives $\omega_{h,k,n} \geq b_- \gamma_{k,n}^{\Gamma}$ on every monic affine fibre with $n \leq 2q$; take its infimum to see this directly. Thus $b_- \leq \omega_{h,k,2q} / \gamma_{k,2q}^{\Gamma}$. The proved TW.14 inequality further gives $\omega_{h,k,2q} \leq A_k \gamma_{2q}^{\Gamma}$. Combining these inequalities and substituting (CA18), (CA25) proves (CA26), including the factor $\Gamma(1/2)/\sqrt{2\pi} = 1/\sqrt{2}$. The independent lower bound one is the ordered positive-spectrum inequality.

Here the size of the compulsory spread can also be proved explicitly. For integers $n \geq 1$, $k \geq 1$ one has

$$n^{(k-1)/2} \leq \frac{\Gamma(n + k/2)}{\Gamma(n + 1/2)} \leq (n + k/2)^{(k-1)/2}. \quad (\text{CA27})$$

For odd $k = 2r + 1$, the ratio is the exact product $\prod_{j=0}^{r-1} (n + 1/2 + j)$, and bounding each factor proves it. For even $k = 2r$, separate the product $\prod_{j=1}^{r-1} (n + j)$ and the factor $\Gamma(n + 1)/\Gamma(n + 1/2)$. The Gamma integral and Cauchy–Schwarz give $\Gamma((x + y)/2)^2 \leq \Gamma(x)\Gamma(y)$ for $x, y > 0$: apply that inequality to the functions $t^{(x-1)/2}e^{-t/2}$ and $t^{(y-1)/2}e^{-t/2}$. Using $(x, y) = (n, n + 1)$ and $(n + 1/2, n + 3/2)$, and using the Gamma recurrence in both cases, yields

$$\sqrt{n} \leq \frac{\Gamma(n + 1)}{\Gamma(n + 1/2)} \leq \sqrt{n + 1/2}.$$

Every factor is positive, so multiplication proves (CA27) also for even k . Empty products in the cases $k = 1, 2$ are one. In particular the error in replacing the logarithm of its ratio by $((k - 1)/2) \log n$ lies between zero and $((k - 1)/2) \log(1 + k/(2n)) \leq k^2/(4n)$.

The actual A_k in TW.8 satisfies $\log A_k = O_h(k + \log k)$. Also $\log \Gamma(k/2) = (k/2) \log k + O(k)$ for integer $k \rightarrow \infty$. For the latter claim no complex asymptotic is needed: bounding the sum $\log(r!)$ between its two adjacent integrals of $\log x$ gives $\log(r!) = r \log r - r + O(\log(r + 1))$. For even k apply this to $\Gamma(k/2) = (k/2 - 1)!$; for odd $k = 2r + 1$ use the exact identity $\Gamma(r + 1/2) = \sqrt{\pi}(2r)!/(4^r r!)$ and the same two bounds. Both substitutions give the stated $O(k)$ form. Taking logarithms in (CA26), using the lower side of (CA27) at $n = 2q$, and retaining the fixed original mass, proves

$$\log \kappa_{\text{old}} \geq \frac{k - 1}{2} \log(2q) - \frac{k}{2} \log k + O_h(k + \log k). \quad (\text{CA28})$$

Here the meaning of the signed O_h term is a function bounded in absolute value by a fixed multiple of $k + \log k$, as follows from the preceding explicit formula. This inequality yields

$$\liminf_{k \rightarrow \infty} \frac{\log \kappa_{\text{old}}}{k \log k} \geq \begin{cases} \frac{1}{2}, & m = 1, \\ 1, & m \geq 2. \end{cases} \quad (\text{CA29})$$

Indeed when $m = 1$, $q = (k + 1)^2$ gives $\log(2q) = 2 \log k + O(1)$. When $m \geq 2$ is fixed, $q = [1 + k(m - 1)](k + 1)^2$ gives $\log(2q) = 3 \log k + O_h(1)$. Substitution in (CA28), division by $k \log k$, and passage to the lower limit prove both cases.

This is a constraint on the actual relative moment operator, not on freely chosen positive matrices. It shows that the old convolution reference cannot have a uniformly bounded condition number along these original amplified sources. In contrast, (CA13) proves the upper bound $\log \kappa(D_*) = O_h(k + \log k)$ for the fixed reference, with the exact identity comparison through (CA2). Both source masses and both reference maps remain explicit. A condition-number lower bound does not fix the orientation of the original window projections in its eigenvectors; consequently (CA29) assigns no sign to the original signed four-volume correction.

181 The certified theta moments as observations of the total original object

The two certified matrices attach to the same original theta seed and the same finite arithmetic packets in $\mathbf{C}_U$. The global moment sum includes critical and offcritical packets in the ambient $Q = \mathcal{B}/\Theta V$ of (T0.5). Its offcritical part is exactly the already constructed $T_U Q$ of (T0.7), not a replacement for that carrier. We construct the attachment and calculate what the finite certificate excludes.

181.1 From the original theta boundary to the finite moment matrices

Retain the original operators and seed, with every scalar unchanged:

$$\begin{aligned} D &= -x\partial_x, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V, \quad f_0 = \Theta\phi_* = 2 \sum_{n \geq 1} \phi_*(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x)x^s \frac{dx}{x}, \quad \mathcal{M}f_0 = g = 2\xi, \quad \psi(y) = e^{y/2}f_0(e^y). \end{aligned} \quad (\text{HCA.1})$$

The complete theta calculation (WBR1)–(WBR8) gives $\psi(-y) = \psi(y) > 0$ and, for $y \geq 0$, $\psi(y) \leq C_\theta e^{9y/2 - \pi e^{2y}}$ with $C_\theta = 8\pi^2 \sum_{n \geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. It follows by differentiation under this integrable majorant that

$$M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy = g^{(2j)}(1/2), \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad F(w) = \sum_{j \geq 0} a_j w^j, \quad F(z^2) = g(1/2 + iz). \quad (\text{HCA.2})$$

The map $z \mapsto z^2$ and the even entire Taylor series define F without choosing a square-root branch. Its mass is exactly $a_0 = M_0 = g(1/2) > 0$.

The original moment observation on a general $F_1 \in \mathcal{B}$ is the linear map

$$F_1 \mapsto \left(\int_{\mathbb{R}} y^{2j} e^{y/2} F_1(e^y) dy \right)_{j=0}^{32}. \quad (\text{HCA.3})$$

The two-ended defining bounds of $\mathcal{B}$ dominate each integrand by an integrable exponential times a polynomial, so every component is defined. At the actual boundary $f_0 = \Theta\phi_*$, this map has first component $M_0 > 0$. Thus it is an observation of the actual degree-one source functions, and does not factor through their theta quotient: factoring would send the class of f_0 to zero. This explicit map is permitted by the marked source vertices of (T0.1)–(T0.4), which retain ϕ_* , f_0 and their differential.

On the open set of coefficient vectors with $a_0 \neq 0$, define the finite rational recurrence

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0} \quad (0 \leq n \leq 31), \quad H_{15}^{(\nu)} = (b_{r+s+\nu})_{0 \leq r, s \leq 15}, \quad \nu = 0, 1. \quad (\text{HCA.4})$$

Multiplication of $-F'/F = \sum_{n \geq 0} b_n w^n$ by F proves each coefficient formula. Thus the source observation (HCA.3), the factorial/sign map (HCA.2), the recurrence and the two matrix assemblies form a single explicit composite. The recurrence is a rational observation of its coefficients; it is not asserted to be a linear or cochain map. This is exactly the typed convention for numerical observations in (T0.4)–(T0.6).

181.2 The reciprocal action on every full original packet

Let Z be any finite actual zero packet in the open strip, stable under reflection and conjugation, with all full orders m_ρ . Critical points are allowed in this ambient packet. Put

$$\begin{aligned} h(s) &= \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A_h = M_s, \quad v_h = g/h, \\ v_h &= j_h(v_h), \quad \varepsilon_h = v_h^{-1}, \quad R_h(u) = r_h(\varepsilon_h u), \\ s_h u &= S_h \Theta(R_h(u)(D)\phi_*), \quad \sigma_h u = [s_h u] \in Q, \quad J_h = j_h \mathcal{M}. \end{aligned} \quad (\text{HCA.5})$$

Here r_h is the unique degree-below-deg h remainder and S_h is the original iterated Euler inverse on zero full Mellin jets. These are precisely (AG6)–(AG8) and (WBR22)–(WBR25): the workbench's η_h is the same element as ε_h here. In particular

$$\mathcal{M}s_h u = v_h R_h(u), \quad J_h s_h = I, \quad J_h \Theta = 0, \quad D_Q \sigma_h = \sigma_h A_h. \quad (\text{HCA.6})$$

The first equality follows by dividing $gR_h(u)$ by h ; the second is $j_h(v_h)\varepsilon_h = 1$ with the entire Taylor unit retained. Thus σ_h is injective. Polynomial division gives $Ds_h - s_h A_h = \Theta\phi_*[s^{\deg h-1}]R_h$, proving the last equality before passing to the quotient, including its boundary. The range and inverse proofs used here apply to every root in the open strip and therefore include the ambient critical packets.

Since $g(1/2) > 0$, the polynomial $s - 1/2$ is a unit on every E_h . Define the actual finite-packet operator

$$T_h = -(A_h - \tfrac{1}{2}I)^{-2}, \quad \beta_\rho = -\frac{1}{(\rho - 1/2)^2}. \quad (\text{HCA.7})$$

No inverse on the whole quotient Q is required. On the full ρ -block, write $A_h = \rho I + N_\rho$ and $\lambda_\rho = \rho - 1/2$. The finite coordinate maps are

$$\begin{aligned} T_h &= \sum_{j=0}^{m_\rho-1} (-1)^{j+1} (j+1) \lambda_\rho^{-j-2} N_\rho^j, \quad Y_\rho = T_h - \beta_\rho I, \\ N_\rho &= \lambda_\rho \sum_{j=1}^{m_\rho-1} \binom{-1/2}{j} (Y_\rho/\beta_\rho)^j. \end{aligned} \quad (\text{HCA.8})$$

Multiplication of the first polynomial by $(\lambda_\rho + N_\rho)^2$ gives -1 modulo $N_\rho^{m_\rho}$. For the inverse, the element with constant term one solving $U^2(1 + Y_\rho/\beta_\rho) = 1$ is unique by successive coefficient comparison, whose leading scalar is two. The binomial polynomial and $1 + N_\rho/\lambda_\rho$ both solve it. This proves (HCA.8) with every nilpotent coefficient and with no analytic branch choice.

For $h \mid H$ with full orders, let ι_{hH} insert zero jets at added centres, let π_{Hh} restrict to old jets, and let $e_{hH} = \iota_{hH}(1)$ be the complete old-packet idempotent. The original arrows (AG10)–(AG12) give

$$\begin{aligned} \pi\iota &= I, \quad \iota\pi = M_{e_{hH}}, \quad s_H\iota = s_h, \quad \sigma_H\iota = \sigma_h, \\ \pi T_H &= T_h\pi, \quad T_H\iota = \iota T_h. \end{aligned} \quad (\text{HCA.9})$$

To prove the last two identities, the maps commute with $A - 1/2$; its invertibility on both domains makes them commute with its inverse, and then with its inverse square. Thus the reciprocal action is natural for the actual divisor arrows, including their full units and literal representatives. Through σ_h it acts on the actual finite image in Q .

The representative of each polynomial of this action is also fixed. Let $t_h(s) = r_h(-(s - 1/2)^{-2})$, let $p_h(s) = r_h(P(t_h(s)))$, and put

$$K_{P,h}(u; s) = \frac{p_h(s)R_h(u)(s) - R_h(P(T_h)u)(s)}{h(s)}. \quad (\text{HCA.10})$$

The numerator is zero modulo h because multiplication by the unit ε_h commutes with $P(T_h)$; thus the quotient is a literal polynomial. Mellin transformation gives

$$p_h(D)s_h u - s_h P(T_h)u = \Theta(K_{P,h}(u; D)\phi_*). \quad (\text{HCA.11})$$

Both sides have transform $g(s)K_{P,h}(u; s)$, so Mellin injectivity proves this equality. This is the full original theta boundary, not only equality of its classes. Tensoring (HCA.11) with the other source factors gives its boundaries in (TO.3) with the original tensor differential and its Koszul signs.

The Mellin isomorphism of (TO.7), (CAU.9) sends $\sigma_h u$ on its offcritical part to precisely its prescribed local jets u_ρ . Equations (HCA.7)–(HCA.8) therefore give the following endomorphism of the already constructed offcritical direct sum:

$$\bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_{\mathbb{C},\rho}/(g) \longrightarrow \bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_{\mathbb{C},\rho}/(g), \quad (u_\rho)_\rho \longmapsto (-(s - 1/2)^{-2}u_\rho)_\rho. \quad (\text{HCA.12})$$

Each denominator is a local analytic unit. Finite support is unchanged. Thus this is exactly the same reciprocal action as the one defined in the ambient finite images in Q .

181.3 The full half-trace and the exact global moment limit

For the packet Z , let $\mathcal{B}_Z$ be its distinct reciprocal nodes. Each fibre of $\rho \mapsto \beta_\rho$ is the pair $\rho, 1 - \rho$, since equality of nonzero reciprocal squares is equality up to sign. Reflection gives the same full order m_β at both points. Define

$$f_Z(X) = \prod_{\beta \in \mathcal{B}_Z} (X - \beta)^{m_\beta}, \quad \mathcal{A}_Z = \mathbb{C}[X]/(f_Z). \quad (\text{HCA.13})$$

The inverse local map (HCA.8) proves that the kernel of $P \mapsto P(T_h)$ is exactly (f_Z) . On choosing one point in each reflected pair it also gives the full module isomorphism $E_h \simeq \mathcal{A}_Z \oplus \mathcal{A}_Z$. A different choice interchanges the corresponding two factors; both remain present. Triangular multiplication in each local basis gives, for $\nu = 0, 1$,

$$\mathcal{L}_{Z,\nu}(PQ) := \frac{1}{2} \text{Tr}_{E_h} (T_h^{\nu+1} P(T_h) Q(T_h)) = \sum_{\beta \in \mathcal{B}_Z} m_\beta \beta^{\nu+1} P(\beta) Q(\beta). \quad (\text{HCA.14})$$

Indeed every diagonal entry on a length- m_β block equals $\beta^{\nu+1} P(\beta) Q(\beta)$, and there are two reflected blocks. This proves the factor $1/2$ and the retained multiplicity. The radical of this bilinear form on $\mathcal{A}_Z$ is exactly its nilradical $(\text{rad } f_Z)/(f_Z)$: vanishing at every node is sufficient by the sum, and Lagrange interpolation at a node where a value is nonzero proves necessity, since $m_\beta \beta^{\nu+1} \neq 0$. Thus (HCA.8) retains all local jets before the trace and (HCA.14) specifies its exact scalar kernel.

Packet growth has a specific trace law. From (HCA.9) and the idempotent splitting of the larger packet,

$$\begin{aligned} \frac{1}{2} \text{Tr}_{E_H} (M_{e_{hH}} T_H^{\nu+1} P(T_H) Q(T_H)) &= \mathcal{L}_{Z,\nu}(PQ), \\ \mathcal{L}_{Z',\nu}(PQ) &= \mathcal{L}_{Z,\nu}(PQ) + \mathcal{L}_{Z' \setminus Z,\nu}(PQ) \quad (Z \subseteq Z'). \end{aligned} \quad (\text{HCA.15})$$

The first is a trace on the old direct summand, conjugate by ι and π to the original endomorphism. The second is additivity of trace on the two complementary full summands. Thus uncompressed traces on growing packets are not asserted constant. Their precise change is the added original packet.

The growth calculation (WBR9) and the complete product proof (GF1)–(GF13) for this same F give distinct nonzero reciprocal nodes β_j , their full orders m_j , and

$$\sum_j m_j |\beta_j| < \infty, \quad F(w) = a_0 \prod_j (1 - \beta_j w)^{m_j}, \quad b_n = \sum_j m_j \beta_j^{n+1}. \quad (\text{HCA.16})$$

That proof applies the explicit bound $\log M_F(R) = O(\sqrt{R} \log R)$ with exponent $3/4 < 1$. Its circle-mean calculation bounds the zero count by $CR^{3/4}$; dyadic annuli then prove reciprocal summability. The uniformly convergent product has exactly the original zeros. Dividing F by it gives a nonvanishing entire function e^H . The selected circles in (GF7)–(GF9) bound $\Re H$ by $CR^{3/4} \log R$; the circle Fourier formula (GF10)–(GF11) forces every positive Taylor coefficient of H to vanish. Evaluation at zero fixes the remaining constant to a_0 . Thus no unproved factorization input, exponential factor, or change of mass enters (HCA.16). The complete product proof is included with the attachment. The nodes are bounded, since absolute summability excludes infinitely many nodes outside any disc about zero. Hence, for any fixed P, Q , the sum in (HCA.14) over all nodes is absolutely convergent, also for $\nu = 1$. Exhausting by finite full reflection- and conjugation-stable packets proves the exact limit

$$\begin{aligned} \mathcal{L}_\nu(PQ) &= \lim_Z \mathcal{L}_{Z,\nu}(PQ) = \sum_j m_j \beta_j^{\nu+1} P(\beta_j) Q(\beta_j), \\ \mathcal{L}_\nu(X^r X^s) &= b_{r+s+\nu}, \quad \mathcal{Q}_\nu(P) := \mathcal{L}_\nu(P^2) = c^\top H_{15}^{(\nu)} c \quad \left(P(X) = \sum_{r=0}^{15} c_r X^r \right). \end{aligned} \quad (\text{HCA.17})$$

Absolute convergence proves independence of the exhaustion and justifies the finite polynomial expansion. For real coefficients the value is real by conjugation of the full node set. The square in this identity is the algebraic square $P(\beta)^2$.

Equations (HCA.3)–(HCA.4) and (HCA.5)–(HCA.17) are therefore two equal evaluations of the same original data: the first through the theta integral and coefficient recurrence, the second through the full packet action and absolutely convergent half-trace. Their equality is proved by (HCA.16), not imposed as a new relation between independently adjustable matrices. Adjoining these specified observation vertices and the displayed maps to the path representation $\mathbf{C}_U$ of (TO.4) gives one represented extension: every path is evaluated by composition of the actual maps, and (HCA.9), (HCA.11), (HCA.15), (HCA.17) prove the new path relations. The old represented diagram is retained through its identity inclusion. No quotient identifies its offcritical component with a scalar or with a zero target.

In particular the global trace splits into the absolutely convergent critical and offcritical sums. The critical-only projection of (WBR37)–(WBR40) kills every full offcritical packet but need not kill its trace: its reciprocal primary idempotent e_β has $\text{Tr}_{\mathcal{A}_Z} M_{X^{\nu+1}e_\beta} = m_\beta \beta^{\nu+1} \neq 0$. Thus the full functional has this explicit omitted summand under critical projection. At the split support level all linear maps above send $(\mathfrak{a}, u)$ to $(\mathfrak{a}, Lu)$ and τ to τ ; a killed class lands at $(\mathfrak{a}, 0)$. The moment and quadratic evaluations remain the marked observations in their actual linear or polynomial types.

181.4 The certified finite restriction on this attached observation

The certified theta calculation (TC1)–(TC28) encloses exactly the moments (HCA.2). It retains theta terms $1 \leq n < 20$ on $0 \leq y \leq 4$, and, for $J = 0, 2, \dots, 64$, adds the proved nonnegative tails

$$\frac{16\pi^2 4^{J+1} e^{18} 20^4 e^{-400\pi}}{1 - e^{4/20-40\pi}}, \quad (HCA.18)$$

$$\frac{16\pi^2 4^J e^{18-\pi e^8}}{(1 - e^{4-2\pi})(2\pi e^8 - 9/2 - J/4)}.$$

These are exactly (TC5) at $N = 20, L = 4$, with N the first omitted integer. All denominators are strictly positive as proved there. The finite integrands, quadrature subdivisions and dyadic interval endpoints are preserved in the certificate. The interval recurrence (TC11)–(TC13) encloses the exact coefficients in (HCA.4) and their $LDL^\top$ decompositions in the unchanged coordinate order. Every one of its thirty-two pivot intervals has a positive lower endpoint, proving

$$H_{15}^{(0)} > 0, \quad H_{15}^{(1)} > 0. \quad (HCA.19)$$

The certificate is the actual file `certificate_dimension16_1024.json`, with SHA-256 869ebacd3f82e46c3fe5fac1f07bcbaae7f4c333e1a5ba2e0ffcb1f91884a0fc. Its exact endpoint checks and the full analytic tail and validated quadrature proof accompany this statement.

The exact rational endpoint check also gives a uniform quantitative margin in the original monomial coefficient frame. Write the exact decompositions as $H_{15}^{(\nu)} = L_\nu D_\nu L_\nu^\top$. The sum of the absolute upper endpoint bounds in every off-diagonal row and column of $L_\nu - I$ is strictly less than $1/64$. Every diagonal pivot of D_0 is greater than $6 \cdot 10^{-107}$, and every pivot of D_1 is greater than $7 \cdot 10^{-111}$. These comparisons use exact dyadic fractions in the certificate. For a matrix E , Cauchy–Schwarz with row weights proves $\|Ex\|_2^2 \leq \|E\|_\infty \|E\|_1 \|x\|_2^2$: bound the square of row i by $(\sum_j |E_{ij}|)(\sum_j |E_{ij}| |x_j|^2)$ and sum in i . Thus $\|L_\nu - I\|_2 < 1/64$ and $\|L_\nu^\top x\|_2 > (63/64)\|x\|_2$ for $x \neq 0$. Substitution in the exact decomposition proves

$$H_{15}^{(\nu)} \succ \gamma_\nu I_{16}, \quad \gamma_0 = (63/64)^2 6 \cdot 10^{-107}, \quad \gamma_1 = (63/64)^2 7 \cdot 10^{-111}. \quad (HCA.19a)$$

No polynomial basis or mass was changed in obtaining this bound.

Consequently, every nonzero real polynomial of degree at most fifteen satisfies $Q_\nu(P) > 0$ for both $\nu = 0, 1$. Padding its coefficient vector with zeros identifies its form with the corresponding leading principal submatrix; positivity of (HCA.19) proves the assertion. More generally, on the explicitly specified finite cone

$$\mathcal{C}_{15} = \left\{ \sum_j A_j(X)^2 + X \sum_j B_j(X)^2 : A_j, B_j \in \mathbb{R}[X]_{\leq 15}, \text{ finite sums} \right\}, \quad (\text{HCA.20})$$

one has $\mathcal{L}_0(C) > 0$ for every nonzero C in that cone. Indeed it is the sum of the positive forms in (HCA.19), and a nonzero polynomial represented in the cone has at least one nonzero summand. This uses the shifted matrix through $\mathcal{L}_0(XB^2) = Q_1(B)$; no characterization of larger polynomial cones or positivity at higher degrees is presumed.

181.5 The precise exclusion for an offcritical separator

Take any actual offcritical pair $\beta, \bar{\beta}$ of multiplicity m in (HCA.16). In original zero coordinates it is

$$\beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}, \quad \rho = 1/2 + \delta + i\gamma. \quad (\text{HCA.21})$$

It is nonreal because $\delta\gamma \neq 0$. Put $r = |\beta|/2$, let $\mathcal{A}$ be the set of all other distinct nodes of modulus at least r , and define

$$B_\beta(X) = \prod_{\alpha \in \mathcal{A}} (X - \alpha), \quad B_0 = \prod_{\alpha \in \mathcal{A}} (r + |\alpha|), \quad a = \#\mathcal{A}. \quad (\text{HCA.22})$$

The set is finite by (HCA.16), conjugation stable, and excludes $\beta, \bar{\beta}$, so B_β is real and $B_\beta(\beta) \neq 0$. For $\nu = 0, 1$ choose either c_ν with $\beta^{\nu+1}c_\nu^2 = -1$, and retain the actual constants

$$C_\nu = \frac{|c_\nu|B_0}{|B_\beta(\beta)|} \left(1 + \frac{|\Re\beta| + r}{|\Im\beta|} \right), \quad T_\nu = \sum_{|\beta_j| < r} m_j |\beta_j|^{\nu+1}. \quad (\text{HCA.23})$$

Both tails converge and $T_1 \leq rT_0$. Define for every integer $N \geq 0$

$$\begin{aligned} w_{\nu,N} &= \frac{c_\nu}{\beta^N B_\beta(\beta)}, \quad v_{\nu,N} = \frac{\Im w_{\nu,N}}{\Im \beta}, \quad u_{\nu,N} = \Re w_{\nu,N} - (\Re \beta) v_{\nu,N}, \\ P_{\nu,N}(X) &= X^N B_\beta(X) (u_{\nu,N} + v_{\nu,N} X), \quad D_{\nu,N} = \deg P_{\nu,N} \leq N + a + 1. \end{aligned} \quad (\text{HCA.24})$$

This is precisely (TC24) for $\nu = 0$. The shifted version uses the original shifted form and its exponent $\nu + 1 = 2$. The polynomial is real and nonzero, because its value at β is $c_\nu \neq 0$. Its actual degree is $N + a + 1$ when $v_{\nu,N} \neq 0$, and $N + a$ otherwise. Evaluation proves $P_{\nu,N}(\beta) = c_\nu$ and conjugate equality at $\bar{\beta}$. Hence its selected two contributions to the actual scalar form are exactly $-2m$.

Every other node of modulus at least r is a zero of the polynomial. For $|X| \leq r$, the elementary estimates $|B_\beta(X)| \leq B_0$ and $|u_{\nu,N} + v_{\nu,N}X| \leq |w_{\nu,N}| + (|\Re\beta| + r)|w_{\nu,N}|/|\Im\beta|$ give $|P_{\nu,N}(X)| \leq C_\nu 2^{-N}$. Absolute summation proves the exact identity and tail bound

$$\begin{aligned} Q_\nu(P_{\nu,N}) &= -2m + \mathcal{T}_{\nu,N}, \quad \mathcal{T}_{\nu,N} = \sum_{|\beta_j| < r} m_j \beta_j^{\nu+1} P_{\nu,N}(\beta_j)^2 \in \mathbb{R}, \\ |\mathcal{T}_{\nu,N}| &\leq C_\nu^2 T_\nu 4^{-N}. \end{aligned} \quad (\text{HCA.25})$$

No tail is assigned a sign before applying the certificate. If $D_{\nu,N} \leq 15$, the nonzero polynomial in (HCA.24) is within (HCA.19), and therefore

$$\boxed{\mathcal{T}_{\nu,N} = 2m + Q_\nu(P_{\nu,N}) > 2m, \quad C_\nu^2 T_\nu 4^{-N} > 2m.} \quad (\text{HCA.26})$$

This is the finite exclusion on the actual packet together with its entire reciprocal tail. In particular, when $a \leq 14$, choosing $N = 14 - a$ makes its degree at most fifteen and gives

$$C_\nu^2 T_\nu > 2m 4^{14-a} \quad (\nu = 0, 1). \quad (\text{HCA.27})$$

If the affine factor has zero leading coefficient, the exact degree condition in (HCA.26) additionally retains that case. There is also an explicit strictly positive margin. For a polynomial $P(X) = \sum_{j=0}^D p_j X^j$ with $D \leq 15$, Cauchy-Schwarz gives $|P(\beta)|^2 \leq (\sum_j |p_j|^2)(\sum_{j=0}^D |\beta|^{2j})$. For $P = P_{\nu,N}$ its value is c_ν and $|c_\nu|^2 = |\beta|^{-\nu-1}$. Applying (HCA.19a) to its coefficient vector padded with zeros strengthens (HCA.26) to

$$C_\nu^2 T_\nu 4^{-N} \geq \mathcal{T}_{\nu,N} > 2m + \frac{\gamma_\nu |\beta|^{-\nu-1}}{\sum_{j=0}^{D_{\nu,N}} |\beta|^{2j}} \quad (D_{\nu,N} \leq 15). \quad (\text{HCA.27a})$$

The inequality on the left uses the actual tail's strict positivity already proved in (HCA.26), followed by its absolute upper bound.

The shifted certificate also tests the identical original polynomial $P_{0,N}$, without changing c_0 or any coefficient. Its two selected terms in $\mathcal{Q}_1$ are $m\beta^2 c_0^2 + m\bar{\beta}^2 \bar{c}_0^2 = -2m\Re\beta$. The absolute value of its remaining sum is at most $C_0^2 T_1 4^{-N} \leq r C_0^2 T_0 4^{-N}$. If $D_{0,N} \leq 15$ and p is its original coefficient vector padded with zeros through degree fifteen, (HCA.19a) consequently proves the simultaneous exact bounds

$$\begin{aligned} C_0^2 T_0 4^{-N} &> \max \left\{ 2m + \gamma_0 \|p\|_2^2, \frac{2m\Re\beta + \gamma_1 \|p\|_2^2}{r} \right\} \\ &> \max \left\{ 2m, \frac{4m\Re\beta}{|\beta|} \right\}. \end{aligned} \quad (\text{HCA.27b})$$

The second entry remains valid even when its numerator is negative: the absolute upper tail bound is still an upper bound for the signed tail. Both positive coefficient margins are strict. Substitution $\|p\|_2^2 \geq |\beta|^{-1} / \sum_{j=0}^{D_{0,N}} |\beta|^{2j}$ from the preceding evaluation argument retains an entirely explicit margin in both entries of the first maximum. The elementary shifted entry improves $2m$ exactly when $\Re\beta/|\beta| > 1/2$. In the original coordinates (HCA.21) that inequality is $\gamma^2 > 3\delta^2$, since $\Re\beta/|\beta| = (\gamma^2 - \delta^2)/(\gamma^2 + \delta^2)$. This additional comparison uses the same original unshifted separator, as well as the separately constructed shifted family.

There is a further comparison between the two separately phased families. Set $U_{\nu,N} = C_\nu^2 T_\nu 4^{-N}$. The exact modulus $|c_\nu|^2 = |\beta|^{-\nu-1}$ in (HCA.23) gives $C_1^2 = C_0^2/|\beta|$. Since $T_1 \leq rT_0$ and $r = |\beta|/2$, one obtains $U_{1,N} \leq U_{0,N}/2$. For $D_{1,N} \leq 15$, apply (HCA.19a) to the original coefficient vector p_1 of $P_{1,N}$, padded through degree fifteen, and use (HCA.25). The resulting exact inequalities are

$$\begin{aligned} U_{0,N} &\geq 2U_{1,N} > 4m + 2\gamma_1 \|p_1\|_2^2 \geq 4m + \frac{2\gamma_1 |\beta|^{-2}}{\sum_{j=0}^{D_{1,N}} |\beta|^{2j}}, \\ a \leq 14 &\implies C_0^2 T_0 > 4m 4^{14-a} = m 4^{15-a}. \end{aligned} \quad (\text{HCA.27c})$$

Here the last line chooses $N = 14 - a$, so that $D_{1,N} \leq 15$; the coefficient estimate is the same evaluation argument as in (HCA.27a). This estimate uses the shifted polynomial $P_{1,N}$ to bound the original unshifted tail constant. It is distinct from the same-polynomial comparison (HCA.27b), which retains $P_{0,N}$.

For the original unshifted quantitative separator of (TC23), put

$$N_* = \begin{cases} 0, & T_0 = 0, \\ \max \left\{ 0, 1 + \left\lfloor \frac{\log(C_0^2 T_0/m)}{\log 4} \right\rfloor \right\}, & T_0 > 0. \end{cases} \quad (\text{HCA.28})$$

Its definition gives $C_0^2 T_0 4^{-N_*} \leq m$ and thus $\mathcal{Q}_0(P_{0,N_*}) \leq -m < 0$ by (HCA.25). Comparison with (HCA.19) proves the concrete restriction

$$\deg P_{0,N_*} \geq 16, \quad N_* + a + 1 \geq 16. \quad (\text{HCA.29})$$

The shifted comparison strengthens the nominal bound for the same original threshold. If $a \leq 14$, (HCA.27c) implies $T_0 > 0$ and $\log(C_0^2 T_0 / m) / \log 4 > 15 - a$. Because $15 - a$ is an integer, its floor is at least $15 - a$, and (HCA.28) gives

$$a \leq 14 \implies N_* \geq 16 - a, \quad N_* + a + 1 \geq 17. \quad (\text{HCA.29a})$$

This is a bound on the nominal degree. Its actual degree remains $N_* + a + 1$ if $v_{0,N_*} \neq 0$ and $N_* + a$ otherwise; the latter case can still have degree sixteen. There is a further exact boundary case: if $a \leq 15$ and $v_{1,15-a} = 0$, the shifted polynomial with $N = 15 - a$ has degree exactly fifteen. The first line of (HCA.27c) then gives $C_0^2 T_0 > m 4^{16-a}$, and the identical integer-floor argument yields $N_* \geq 17 - a$. No assertion on a matrix of degree sixteen is needed for either conclusion.

The rational rounding of (TC25) keeps the actual quadratic form strictly negative and has degree at most $N_* + a + 1$. Its actual degree is therefore at least sixteen as well. Coefficient rounding cannot produce a smaller-degree negative witness against the certified matrices. The same conclusion holds for any rational negative polynomial in either shifted family, by (HCA.19).

The full finite packet alone may already contribute $-2m$ under the exact half-trace (HCA.14); its remaining original arithmetic nodes still enter (HCA.17). The certificate therefore excludes the low-degree global separators specified above, rather than deleting that offcritical packet from $T_U Q$. It imposes no positivity assertion on H_{16} or any larger matrix. Every source block, its full reciprocal coordinate, its scalar radical, its receiving supported zero and its exact ambient trace remain the explicitly constructed parts of this attachment.

182 Original theta Hankel witness: independent proof and full packet transport

This is an additive mathematical review of the complete supplied NOTE.md, equations (1)–(23), and its complete LaTeX counterpart. Both original files remain byte-for-byte unchanged. The proof of the Hankel criterion and the infinite-tail separator is accepted after the explicit symbol repair below. This proof supplies explicit growth constants, the inverse local coordinate, the complete finite algebra map, and the original theta-boundary representative of that map. The source note’s numerical results retain their exploratory status.

The source defines $u_N = \Re w_N - (\Re \beta) v_N$, but its boxed equation (16) uses the undefined symbol ν_N . The exact repair is the identity

$$\nu_N := u_N = \Re w_N - (\Re \beta) v_N.$$

This occurs at NOTE.md lines 212/216 and NOTE.tex lines 350/354. The user’s pasted narrative already uses u_N in both places. Equation (WBR16) below writes that complete definition and the polynomial using the same u_N . No value or coefficient is changed.

182.1 WBR1. Source and scalar factors

Let

$$D = -x \partial_x, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx,$$

$$V = \{ \phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0 \},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\},$$

$$\Theta : V \rightarrow \mathcal{B}, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{WBR1})$$

The quotient is the original $Q = \mathcal{B}/\Theta V$, with its quotient topology. Its split coefficient set is $G(R) = \{\tau\} \sqcup R^\bullet$, with represented zero $e_R = 0_R^\bullet$. A map on a coefficient fibre sends (ℓ, v) to (ℓ, fv) and fixes external absence τ . Thus $(\ell, 0)$ keeps the source label.

Put

$$q_0 = e^{-\pi x^2}, \quad \phi_* = (D^2 - D)q_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0 = \Theta \phi_*. \quad (\text{WBR2})$$

Direct differentiation gives $Dq_0 = 2\pi x^2 q_0$ and $D^2 q_0 = (4\pi^2 x^4 - 4\pi x^2)q_0$. The function ϕ_* is even and vanishes at zero. Integration by parts gives $\int Df = \int f$ on Schwartz functions, so $\int \phi_* = 0$. Hence $\phi_* \in V$. Fourier differentiation and integration by parts give $\mathcal{F}D = (1 - D)\mathcal{F}$. Since $\mathcal{F}q_0 = q_0$, this proves $\mathcal{F}\phi_* = \phi_*$. Poisson summation, with zero origin terms on both sides, gives

$$f_0(x) = x^{-1} f_0(1/x). \quad (\text{WBR3})$$

For $x \geq 1$, the series and its derivatives decay faster than every power. Reflection proves the same assertion at zero, so $f_0 \in \mathcal{B}$.

For $\Re s > 1$, absolute convergence allows termwise integration:

$$\begin{aligned} \mathcal{M}f_0(s) &= 2\zeta(s) \int_0^\infty (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2} x^{s-1} dx \\ &= 2\zeta(s) \pi^{-s/2} \{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} \\ &= s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) = 2\xi(s). \end{aligned} \quad (\text{WBR4})$$

The two-ended decay makes the Mellin transform entire. Thus this identity also supplies the entire continuation of the completed zeta expression. Write $g = 2\xi$, retaining its actual value at $1/2$. The conventional factor in ξ is given by [DLMF 25.4.4](#).

182.2 WBR2. Positive density and explicit growth

Define

$$\psi(y) = e^{y/2} f_0(e^y), \quad C_\theta = 8\pi^2 \sum_{n=1}^\infty n^4 e^{-\pi(n^2-1)}. \quad (\text{WBR5})$$

The constant is finite. Reflection gives $\psi(-y) = \psi(y)$. For $y \geq 0$, each theta summand is strictly positive because $2\pi n^2 e^{2y} > 3$. Dropping its negative term only for an upper bound, while retaining the complete summand in the definition, yields

$$0 < \psi(y) \leq 8\pi^2 e^{9y/2} \sum_{n \geq 1} n^4 e^{-\pi n^2 e^{2y}} \leq C_\theta e^{9y/2 - \pi e^{2y}}. \quad (\text{WBR6})$$

Reflection gives the same estimate with $|y|$ on the whole line. For any compact set of s , this dominates every $|y|^j \psi(y) e^{(\Re s - 1/2)y}$ by an integrable function. Therefore

$$g(s) = \int_{\mathbb{R}} \psi(y) e^{(s-1/2)y} dy, \quad M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy = g^{(2j)}(1/2). \quad (\text{WBR7})$$

In particular $g(s) > 0$ for every real s , without numerical input.

The entire function $G(z) = g(1/2 + iz)$ is even. Its Taylor series gives

$$G(z) = \sum_{j \geq 0} \frac{(-1)^j M_{2j}}{(2j)!} z^{2j} = F(z^2), \quad F(w) = \sum_{j \geq 0} a_j w^j, \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}. \quad (\text{WBR8})$$

The map $z \mapsto z^2$ sends the circle of radius $\sqrt{R}$ onto the circle of radius R , so the convergent Taylor series proves that F is entire and $M_F(R) = M_G(\sqrt{R})$. No square-root branch is chosen. Its unchanged mass is $F(0) = a_0 = g(1/2) > 0$.

Set $a(R) = 9/4 + \sqrt{R}/2$. Substituting $t = e^{2y}$, with $dy = dt/(2t)$, into the majorant gives

$$\begin{aligned} M_F(R) &\leq 2C_\theta \int_0^\infty e^{(9/2 + \sqrt{R})y - \pi e^{2y}} dy \\ &= C_\theta \int_1^\infty t^{a(R)-1} e^{-\pi t} dt \\ &\leq C_\theta \pi^{-a(R)} \Gamma(a(R)) \\ &\leq 2C_\theta \pi^{-a(R)} \left(\frac{2(a(R) - 1)}{e} \right)^{a(R)-1}. \end{aligned} \tag{WBR9}$$

For the last inequality, maximize $t^{a-1}e^{-t/2}$ at $2(a-1)$ and integrate the remaining $e^{-t/2}$. Here $a(R) > 1$. Thus $\log M_F(R) = O(\sqrt{R} \log R)$, and the order is at most $1/2$.

182.3 WBR3. Reciprocal zeros and coefficient map

Use the standard genus-zero case of Hadamard factorization in this precise form: an entire function of finite order strictly below one, nonzero at zero, is its value at zero times the product $1 - w/\alpha$ over its zeros with multiplicity; the reciprocal absolute zero values are summable, and no nonconstant exponential factor occurs. WBR2 verifies its hypotheses for the actual F .

Let α_j be its distinct zeros, m_j their orders, and $\beta_j = 1/\alpha_j$. Then

$$F(w) = a_0 \prod_j (1 - \beta_j w)^{m_j}, \quad S_\beta = \sum_j m_j |\beta_j| < \infty. \tag{WBR10}$$

The reciprocal zeros are nonzero and have only zero as a possible accumulation point. On a disk where all $|\beta_j w| < 1/2$, the logarithmic derivative is dominated by $2 \sum m_j |\beta_j|$. Geometric expansion and absolute convergence give

$$-\frac{F'(w)}{F(w)} = \sum_j \frac{m_j \beta_j}{1 - \beta_j w} = \sum_{n \geq 0} b_n w^n, \quad b_n = \sum_j m_j \beta_j^{n+1}. \tag{WBR11}$$

Multiplying by F and equating coefficients, with a_0 retained, gives

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \tag{WBR12}$$

These b_n are real because all a_j are real.

At a zero $\rho = 1/2 + \delta + i\gamma$, the exact coordinates are

$$z = -i(\rho - 1/2) = \gamma - i\delta, \quad \alpha = z^2, \quad \beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \tag{WBR13}$$

Since $g(1/2) > 0$, $z \neq 0$, so $z \mapsto z^2$ has nonzero derivative and preserves the local zero order. The two reflected points $\rho, 1 - \rho$ supply one zero α of that order. Conjugation preserves the orders and sends α, β to their conjugates. For real v , $F(-v^2) = g(1/2 - v) > 0$. Thus no nonpositive real zero occurs, and RH is equivalent to every β_j being positive real. An off-line zero has $\delta \neq 0$, while positivity of g on the real axis gives $\gamma \neq 0$; its reciprocal coordinate is therefore nonreal by (WBR13).

182.4 WBR4. Finite criterion and full infinite-tail proof

For $P(X) = \sum_{r=0}^d c_r X^r$, with real coefficients, put

$$H_d = (b_{r+s})_{0 \leq r, s \leq d}, \quad \mathcal{L}(P^2) = c^T H_d c = \sum_j m_j \beta_j P(\beta_j)^2. \quad (\text{WBR14})$$

The reciprocal zeros together with zero form a compact set. Thus boundedness of P and (WBR10) prove absolute convergence. Conjugate terms make the sum real. Under RH its terms are nonnegative, proving $H_d \succeq 0$. The square retained in this formula is the algebraic $P(\beta)^2$.

For the converse, choose a nonreal reciprocal zero β of order m , and its conjugate. Set $r = |\beta|/2$. Among the finitely many distinct reciprocal zeros of modulus at least r , remove these two and call the remaining set $\mathcal{A}$. Define

$$B(X) = \prod_{\gamma \in \mathcal{A}} (X - \gamma), \quad B_0 = \prod_{\gamma \in \mathcal{A}} (r + |\gamma|). \quad (\text{WBR15})$$

The empty products are one. Conjugation permutes $\mathcal{A}$, so B is real and $B(\beta) \neq 0$. Choose either root c of $\beta c^2 = -1$. For an integer $N \geq 0$, put

$$w_N = \frac{c}{\beta^N B(\beta)}, \quad v_N = \frac{\Im w_N}{\Im \beta}, \quad u_N = \Re w_N - (\Re \beta) v_N, \quad P_N(X) = X^N B(X)(u_N + v_N X). \quad (\text{WBR16})$$

The coefficients are real, $u_N + v_N \beta = w_N$, and hence $P_N(\beta) = c$ and $P_N(\bar{\beta}) = \bar{c}$. The selected contributions total $-2m$. Every other point of modulus at least r contributes zero, regardless of its multiplicity.

The remaining infinite tail has the explicit constants

$$C = \frac{|c|B_0}{|B(\beta)|} \left(1 + \frac{|\Re \beta| + r}{|\Im \beta|} \right), \quad T = \sum_{|\beta_j| < r} m_j |\beta_j|. \quad (\text{WBR17})$$

Indeed $|v_N| \leq |w_N|/|\Im \beta|$ and $|u_N| \leq |w_N|(1 + |\Re \beta|/|\Im \beta|)$. For $|X| \leq r$, $|B(X)| \leq B_0$, giving $|P_N(X)| \leq C 2^{-N}$. Therefore

$$\left| \sum_{|\beta_j| < r} m_j \beta_j P_N(\beta_j)^2 \right| \leq C^2 T 4^{-N}, \quad \mathcal{L}(P_N^2) \leq -2m + C^2 T 4^{-N}. \quad (\text{WBR18})$$

The tail is real by conjugation, which justifies the ordered inequality. If $T = 0$, take $N = 0$. If $T > 0$, the explicit choice

$$N = \max \left(0, 1 + \left\lfloor \log_4 \left(\frac{C^2 T}{2m} \right) \right\rfloor \right) \quad (\text{WBR19})$$

satisfies $4^N > C^2 T / (2m)$. This yields a finite negative direction of degree at most $N + |\mathcal{A}| + 1$, proving

$$\text{RH} \iff H_d \succeq 0 \quad \text{for every } d \geq 0. \quad (\text{WBR20})$$

Every actual zero and every multiplicity is included in this proof. The degree depends on the specified pair and tail; no uniform degree or actual off-line zero is thereby supplied.

For a strict negative vector c , set $M = \max_{i,j} |(H_d)_{ij}|$ and $n = d + 1$. If $\|\tilde{c} - c\|_\infty \leq \varepsilon$, expansion gives

$$|\tilde{c}^T H_d \tilde{c} - c^T H_d c| \leq M(2\|c\|_1 n \varepsilon + n^2 \varepsilon^2). \quad (\text{WBR21})$$

If the original value is $-\mu < 0$, choose ε making the right side less than μ , then choose rational coefficients within that distance. This proves the rational-witness assertion with the original matrix fixed. The shifted family has quadratic form $\sum m_j \beta_j^2 P(\beta_j)^2$, and is also nonnegative under RH.

182.5 WBR5. Actual packet and its complete representative

Let Z be a nonempty finite set of actual nontrivial zeros of g , stable under reflection and conjugation, with every full order retained. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d_h = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad A_h = M_s, \quad \nu_h = g/h, \quad \eta_h = j_h(1/\nu_h). \quad (\text{WBR22})$$

The map j_h keeps all Taylor coefficients through orders $m_\rho - 1$. The element η_h is a unit because every order in h is the full order in g . Let $R_Z(u)$ be the unique polynomial of degree below d_h representing $\eta_h u$.

The original Euler inverse on a zero Mellin value is

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{WBR23})$$

The zero Mellin value makes the upper integral the negative lower integral. At infinity the bound $F(x) \leq Cx^{-N}$ yields $C/(N - \Re \rho)$; at zero $F(x) \leq C'x^N$ yields $C'/(N + \Re \rho)$, when the denominators are positive. These inequalities, and $DS_\rho F = \rho S_\rho F + F$, prove every required seminorm bound. The homogeneous function $x^{-\rho}$ is excluded by the two-ended decay, so the inverse is unique and $D - \rho$ is injective. Iteration gives S_h on the vanishing full Mellin jets, independent of inversion order by uniqueness.

The actual section and observation are

$$R_{\text{ref}}u = S_h \Theta(R_Z(u)(D)\phi_*), \quad \sigma_h u = [R_{\text{ref}}u] \in Q, \quad J_h F = j_h \mathcal{M}F, \\ \mathcal{M}R_{\text{ref}}u = \nu_h R_Z(u), \quad J_h R_{\text{ref}} = 1_{E_h}, \quad J_h \Theta = 0. \quad (\text{WBR24})$$

The transform $gR_Z(u)$ has all the zero jets needed by S_h . The section identity is $j_h \nu_h \eta_h = 1$. Thus σ_h is injective. Polynomial division gives

$$DR_{\text{ref}} - R_{\text{ref}}A_h = f_0 \ell_h, \quad \ell_h(u) = [s^{d_h-1}]R_Z(u). \quad (\text{WBR25})$$

Indeed $sR_Z(u) - R_Z(su) = h(s)\ell_h(u)$; the two sides of (WBR25) have equal Mellin transforms. Mellin injectivity follows from Fourier injectivity in logarithmic coordinates. Passing to the original quotient proves $D_Q \sigma_h = \sigma_h A_h$.

The source sequence cited by the package is

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{WBR26})$$

Its complete proof, including the source Euler inverse, full-jet surjectivity and the actual arrows, is included verbatim in dependencies/arithmetic_input.tex, (A5)–(A10). Equations (WBR23)–(WBR25) also prove directly every packet inclusion and operator identity used by this witness, with the full η_h .

182.6 WBR6. Reciprocal action, inverse jet, and literal theta boundary

Since $g(1/2) > 0$, $A_h - \frac{1}{2}I$ is invertible. Define on this finite packet

$$T_h = -(A_h - \frac{1}{2}I)^{-2}. \quad (\text{WBR27})$$

Through σ_h , this is also an endomorphism of the actual finite image in Q . The construction has this precise domain; no inverse on unrelated quotient classes is required.

At a primary block write $\lambda_\rho = \rho - 1/2$ and $A_h = \rho I + N_\rho$, with full nilpotent length m_ρ . The exact finite expansion is

$$T_h = \sum_{j=0}^{m_\rho-1} (-1)^{j+1} (j+1) \lambda_\rho^{-j-2} N_\rho^j. \quad (\text{WBR28})$$

This is the product $-\lambda_\rho^{-2}(1 + N_\rho/\lambda_\rho)^{-2}$. Multiplying the series by $(\lambda_\rho + N_\rho)^2$ gives constant -1 and zero at every positive degree below m_ρ , which proves the identity in the truncated algebra.

Put $\beta_\rho = -\lambda_\rho^{-2}$ and $Y_\rho = T_h - \beta_\rho I$. The coefficient of N_ρ is $2\lambda_\rho^{-3} \neq 0$. The complete inverse coordinate is

$$N_\rho = \lambda_\rho \sum_{j=1}^{m_\rho-1} \binom{-1/2}{j} (Y_\rho/\beta_\rho)^j. \quad (\text{WBR29})$$

This is a finite algebra identity without any analytic branch choice. In the truncated algebra the element U with constant term one and $U^2(1 + Y_\rho/\beta_\rho) = 1$ is unique: each successive coefficient has leading scalar 2. The displayed binomial coefficients solve these recursive equations. The element $1 + N_\rho/\lambda_\rho$ solves the same equation since $T_h/\beta_\rho = (1 + N_\rho/\lambda_\rho)^{-2}$, proving (WBR29). For $m_\rho = 1$, the positive-degree sums are empty and $N_\rho = 0$. Thus every original local length is retained.

There is a literal original-source representative of this action. Let $t_h(s)$ be the unique polynomial of degree below d_h representing $-(s - 1/2)^{-2}$ in E_h . For a polynomial P , let $p_h(s)$ be the degree-below- d_h representative of $P(t_h(s))$. Define

$$K_{P,h}(u; s) = \frac{p_h(s)R_Z(u)(s) - R_Z(P(T_h)u)(s)}{h(s)}. \quad (\text{WBR30})$$

The numerator vanishes modulo h because multiplication by η_h commutes with $P(T_h)$. Thus the quotient is a polynomial in the original variable s . Mellin transformation gives the exact original boundary

$$p_h(D)R_{\text{ref}}u - R_{\text{ref}}P(T_h)u = \Theta(K_{P,h}(u; D)\phi_*). \quad (\text{WBR31})$$

Both transforms are $gK_{P,h}(u; s)$, proving the formula and every sign. It specifies the difference of representatives before passing to Q , and proves the finite packet intertwining there. The support lift keeps the label of that theta boundary fixed.

182.7 WBR7. Full finite algebra, half-trace, and scalar radical

Let $\mathcal{B}_Z$ be the set of distinct reciprocal coordinates in this packet. A fibre of $\rho \mapsto \beta_\rho$ consists exactly of the distinct pair $\rho, 1 - \rho$: equality of the reciprocal squares gives equality or negation of $\rho - 1/2$. Their orders agree by reflection. Call this common order m_β , and put

$$f_Z(X) = \prod_{\beta \in \mathcal{B}_Z} (X - \beta)^{m_\beta}, \quad \mathcal{A}_Z = \mathbb{C}[X]/(f_Z). \quad (\text{WBR32})$$

The map $P \mapsto P(T_h)$ has kernel exactly (f_Z) . On each primary block, the inverse (WBR29) identifies the local algebra with $\mathbb{C}[Y]/Y^{m_\rho}$. Consequently the action vanishes precisely when every Taylor coefficient of P at β_ρ below that order is zero. Intersecting the coprime primary ideals proves the stated kernel and the embedding $\mathcal{A}_Z = \mathbb{C}[T_h] \subset \text{End}(E_h)$.

Choose one point of each reflected pair to write coordinates. Applying (WBR29) on both primary blocks and then Chinese remainders gives an explicit $\mathcal{A}_Z$ -module isomorphism

$$E_h \simeq \mathcal{A}_Z \oplus \mathcal{A}_Z. \quad (\text{WBR33})$$

The action of X on each copy is multiplication by X . Both reflected blocks remain in this display. Changing the coordinate choice swaps the corresponding two components without altering E_h or T_h .

On the local basis $1, Y, \dots, Y^{m_\beta-1}$, multiplication by $Q(\beta+Y)$ is triangular with constant diagonal $Q(\beta)$. Therefore

$$\begin{aligned} \frac{1}{2} \operatorname{Tr}_{E_h} Q(T_h) &= \operatorname{Tr}_{\mathcal{A}_Z} M_Q = \sum_{\beta \in \mathcal{B}_Z} m_\beta Q(\beta), \\ \frac{1}{2} \operatorname{Tr}_{E_h} (T_h P(T_h)^2) &= \sum_{\beta \in \mathcal{B}_Z} m_\beta \beta P(\beta)^2. \end{aligned} \quad (\text{WBR34})$$

This proves the original factor $1/2$, the full multiplicity and the source-to-scalar map. Exhaustion by finite stable packets converges to (WBR14) by (WBR10).

The precise scalar information loss is also computable. On $\mathcal{A}_Z$, define the bilinear form

$$\mathfrak{b}_Z(U, V) = \operatorname{Tr}_{\mathcal{A}_Z} M_{XUV} = \sum_{\beta} m_\beta \beta U(\beta) V(\beta). \quad (\text{WBR35})$$

Its radical is exactly

$$\mathfrak{n}_Z = \{U : U(\beta) = 0 \text{ for every } \beta\} = (\operatorname{rad} f_Z)/(f_Z). \quad (\text{WBR36})$$

One inclusion follows from (WBR35). If $U(\beta_0) \neq 0$, choose a polynomial V equal to one at β_0 and zero at every other node, by Lagrange interpolation. Then the bilinear value is $m_{\beta_0} \beta_0 U(\beta_0) \neq 0$, proving the other inclusion. Thus the scalar form factors through the exact quotient $\mathcal{A}_Z \rightarrow \prod_{\beta} \mathbb{C}$, whose kernel is the displayed nilradical. Before that quotient, (WBR28)–(WBR33) retain every nilpotent coefficient; the trace weights retain every algebraic multiplicity. This specifies the full map relating the local algebra to its scalar observation.

182.8 WBR8. Exact critical-line comparison

The inherited source map is

$$\Gamma : Q \longrightarrow Q_{\text{crit}} = \mathcal{S}(\mathbb{R})/\overline{\zeta(1/2+it)\mathcal{S}(\mathbb{R})}, \quad \Gamma[F] = [\mathcal{M}F(1/2+it)]. \quad (\text{WBR37})$$

Its construction and range are included whole in dependencies/HOCHSCHILD_COMPARISON.md, (H16)–(H25). For $\Re \rho \neq 1/2$, multiplication by $(1/2+it-\rho)^{-1}$ is a continuous Schwartz multiplier, preserves the displayed relation and its closure, and inverts $M_{1/2+it-\rho}$ on Q_{crit} . Intertwining by D therefore sends every full generalized ρ -block to zero under $\Gamma\sigma_h$.

At a critical zero $\rho = 1/2 + i\gamma$ of full order m , the continuous jet map

$$[f] \longmapsto \sum_{j=0}^{m-1} \frac{i^{-j} f^{(j)}(\gamma)}{j!} \epsilon^j \quad (\text{WBR38})$$

vanishes on the relation and its closure. On $\Gamma[R_{\text{ref}}u]$, it equals $j_\rho u$: each t -derivative gives the factor i^j in the Mellin derivative. Together these jet maps invert every critical block. Thus

$$\ker(\Gamma\sigma_h) = E_{h,\text{off}}. \quad (\text{WBR39})$$

In the reciprocal algebra, the corresponding actual map is the Chinese-remainder projection

$$\pi_{\text{crit}} : \mathcal{A}_Z \longrightarrow \mathcal{A}_{Z,\text{crit}}, \quad \ker \pi_{\text{crit}} = \mathcal{A}_{Z,\text{off}}. \quad (\text{WBR40})$$

The functional (WBR34) decomposes into its critical and off-critical summands. If an off-critical node β occurs, its primary idempotent e_β , equal to one on its entire local block and zero on the others, belongs to $\ker \pi_{\text{crit}}$, yet has $\operatorname{Tr}_{\mathcal{A}_Z} M_{Xe_\beta} = m_\beta \beta \neq 0$. Hence in that case the full functional does not factor through π_{crit} . Its exact omitted part is the displayed weighted trace on $\mathcal{A}_{Z,\text{off}}$. On real polynomials the two-value interpolation (WBR16) selects the conjugate pair and kills every other finite packet node, giving the value $-2m$; the actual infinite arithmetic tail is controlled by (WBR18).

All these maps lift with the source label fixed. Thus an off-critical vector maps to $(\ell, 0)$ under the split lift of $\Gamma\sigma_h$, while remaining the input (ℓ, v) of the full source operator and its trace. The exact maps (WBR29)–(WBR40) relate the arithmetic source, the scalar witness and the receiving quotient.

182.9 WBR9. Computation and integration record

The original integral in the numerical script is

$$M_{2j} = 4 \sum_{n \geq 1} \int_0^\infty y^{2j} (4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}} dy. \quad (\text{WBR41})$$

The factor four is the two in Θ and the two half-lines under reflection. The supplied script retains these constants. It uses finite theta sums and the interval $0 \leq y \leq 4$. Its two high-precision runs have no validated enclosure of the finite quadrature or omitted tails.

The finite independent replay verifies the saved decimal matrices by exact rational LDL arithmetic. Those matrices are positive through dimension 16 in both shifts. This is a theorem about the specified decimal rational matrices. The map from those matrices to actual H_d requires coefficient enclosures; none is supplied by agreement between two floating runs.

The supplied verifier also permits negative shift values, using Python negative indices, and the coefficient recurrence uses floating division when passed integer inputs. These narrow input-contract defects do not affect the archived default runs or the mathematical recurrence. The additive strict verifier accepts explicit exact rationals, validates nonnegative integral indices and interval order, and leaves the archive unchanged. The full replay report, exact tests and machine receipts accompany this proof.

The source NOTE equations (1)–(23), with the explicit $\nu_N = u_N$ symbol repair above, are accepted with their stated standard Hadamard input. For cumulative integration, include its entire original LaTeX, this entire proof, the complete finite replay and both inherited dependency files. Place the reciprocal action after the original arithmetic packet, and its Hankel scalar after the original theta moments. Its source class remains available to the ongoing metric and absolute-base geometric calculations.

Classical context was checked directly at [Zhang’s author record](#). This lies in the established power-sum approach. The current proof makes no novelty claim for that general approach and does not depend on an uninspected theorem from that paper.

183 Complete product proof for the original theta entire function

This appendix proves the genus-zero factorization used in the counterfactual workbench. It is a proof of the classical special case, included to make the application’s analytic dependencies explicit.

183.1 A sublinear growth bound and its zeros

Let F be entire, $F(0) \neq 0$, and suppose that for some $0 < \alpha < 1$ and constants $C, R_0 > 0$,

$$\log \max_{|z| \leq R} |F(z)| \leq CR^\alpha \quad (R \geq R_0). \quad (\text{GF1})$$

The workbench’s original F satisfies this with $\alpha = 3/4$: its explicitly proved bound $\log M_F(R) = O(\sqrt{R} \log R)$ implies (GF1) because $\log R/R^{1/4}$ is bounded for $R \geq 1$. Let the nonzero zeros be a_j , repeated with multiplicity, and let $n(r)$ count those with $|a_j| \leq r$.

We first prove the required counting formula. Choose a circle $|z| = R$ containing no zero. There are finitely many zeros in its closed disk, since otherwise an accumulation point would force the nonzero entire function to vanish identically. Divide F by their linear factors; the resulting holomorphic nonvanishing function on a neighborhood of the disk has a holomorphic logarithm there. Indeed its logarithmic derivative is holomorphic on the disk and has a primitive, as follows by termwise integration of its Taylor series on concentric smaller disks. The exponential of that primitive has the

same logarithmic derivative; their quotient is constant, giving the logarithm after choosing the constant at zero. The real part of this logarithm has its value at zero as its circle mean, by integrating its Taylor series. For $|a| < R$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |Re^{it} - a| dt = \log R. \quad (\text{GF2})$$

To see this, expand $\log(1 - ae^{-it}/R) = -\sum_{\ell \geq 1} (a/R)^\ell e^{-i\ell t}/\ell$, which converges uniformly on the circle; each term has zero mean. Taking the logarithm of the removed factors now gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log |F(Re^{it})| dt - \log |F(0)| = \sum_{|a_j| < R} \log \frac{R}{|a_j|}. \quad (\text{GF3})$$

For any sufficiently large r , choose $R \in (2r, 3r)$ containing no zero. All terms are nonnegative and every zero with $|a_j| \leq r$ contributes at least $\log 2$. Equations (GF1)–(GF3) therefore imply $n(r) \leq C_1 r^\alpha$, for a fixed positive C_1 and all sufficiently large r .

Choose a positive r_0 smaller than the modulus of every zero; such a radius exists since $F(0) \neq 0$. After enlarging C_1 , the inequality $n(r) \leq C_1 r^\alpha$ holds for every $r \geq r_0$. The sum over successive dyadic annuli gives

$$\sum_j \frac{1}{|a_j|} \leq \sum_{\ell \geq 0} \frac{n(2^{\ell+1}r_0)}{2^\ell r_0} \leq C_1 2^\alpha r_0^{\alpha-1} \sum_{\ell \geq 0} 2^{(\alpha-1)\ell} < \infty. \quad (\text{GF4})$$

The inequalities can overcount zeros and still give the stated upper bound. If the zero set is empty, every sum is zero.

183.2 The product and a lower bound on selected circles

Define the literal product

$$P(z) = \prod_j (1 - z/a_j). \quad (\text{GF5})$$

On a fixed disk $|z| \leq A$, all sufficiently late factors have $|z/a_j| \leq 1/2$. The power series logarithm satisfies $|\log(1 - z/a_j)| \leq 2|z/a_j|$, so (GF4) makes its tail uniformly convergent there. The finite preceding factors and that exponential tail define an entire function, locally uniformly independent of the ordering. Its zeros and their multiplicities are exactly the a_j , and $P(0) = 1$. Thus

$$H(z) = F(z)/P(z) \quad (\text{GF6})$$

extends through every zero to a nonvanishing entire function. An entire primitive of H'/H , obtained by integrating its entire power series, gives an entire logarithm h with $H = e^h$ after adjusting its constant.

We prove h constant without assuming any growth bound for $1/P$ away from the following selected circles. Let t be sufficiently large, put $N_t = n(4t)$, and set $\delta_t = t/[4(N_t + 1)]$. Delete from $[t, 2t]$ each open interval of radius δ_t about a number $|a_j| \leq 4t$. Their total length is at most $2N_t\delta_t < t/2$. Hence a radius $R_t \in [t, 2t]$ remains with $|R_t - |a_j|| \geq \delta_t$ for every such zero. For $|z| = R_t$,

$$|1 - z/a_j| \geq \frac{\delta_t}{4t} = \frac{1}{16(N_t + 1)} \quad (|a_j| \leq 4t). \quad (\text{GF7})$$

For the remaining zeros, $|z/a_j| \leq 1/2$, and the same logarithm estimate used for (GF5) gives $\log |1 - z/a_j| \geq -2|z|/|a_j|$. Another dyadic estimate, retaining only $|a_j| > 4t$, gives

$$\sum_{|a_j| > 4t} \frac{1}{|a_j|} \leq \sum_{\ell \geq 0} \frac{n(2^{\ell+1}4t)}{2^\ell 4t} \leq C_2 t^{\alpha-1}, \quad (\text{GF8})$$

where $C_2 = C_1 2^\alpha 4^{\alpha-1} / (1 - 2^{\alpha-1})$ is positive and fixed. Summing the logarithms of (GF7) and the uniformly convergent tail proves, on the entire selected circle,

$$\log |P(z)| \geq -N_t \log(16(N_t + 1)) - 2R_t C_2 t^{\alpha-1} \geq -C_3 t^\alpha \log t \quad (\text{GF9})$$

for another fixed constant $C_3 > 0$ and all large t . This estimate keeps every zero, including repetitions.

183.3 The zero-free factor is constant

Combining (GF1), (GF6), and (GF9) shows that $\Re h(z) \leq C_4 t^\alpha \log t$ on $|z| = R_t$. Write $h(z) = \sum_{\ell \geq 0} d_\ell z^\ell$ and put $B_t = C_4 t^\alpha \log t$, enlarged if necessary so that the bound holds on every selected circle under consideration. The Taylor series, uniformly convergent on a circle, gives

$$d_\ell R_t^\ell = \frac{1}{\pi} \int_0^{2\pi} \Re h(R_t e^{i\theta}) e^{-i\ell\theta} d\theta \quad (\ell \geq 1). \quad (\text{GF10})$$

The integral of the constant B_t against $e^{-i\ell\theta}$ is zero. Because $B_t - \Re h \geq 0$ on the circle, its mean and the mean-value identity yield

$$|d_\ell| R_t^\ell \leq \frac{1}{\pi} \int_0^{2\pi} (B_t - \Re h(R_t e^{i\theta})) d\theta = 2(B_t - \Re h(0)). \quad (\text{GF11})$$

Since $R_t \geq t$ and $\alpha < 1$, the right side divided by R_t^ℓ tends to zero for every integer $\ell \geq 1$. Thus every such d_ℓ is zero. It follows that H is constant. Evaluating at zero now proves the full factorization

$$F(z) = F(0) \prod_j (1 - z/a_j). \quad (\text{GF12})$$

This also covers finitely many zeros and the empty zero set.

183.4 Logarithmic coefficients with their full multiplicities

Group equal zeros in (GF12), writing distinct reciprocals $\beta_j = a_j^{-1}$ and multiplicities m_j . Equation (GF4) becomes $\sum_j m_j |\beta_j| < \infty$. Their supremum is finite. Choose a sufficiently small positive r such that $r|\beta_j| \leq 1/2$ for all j . The derivative series $\sum_j m_j \beta_j / (1 - \beta_j z)$ converges uniformly on $|z| \leq r$, by domination with $2 \sum_j m_j |\beta_j|$. It is the negative logarithmic derivative of (GF12), by termwise differentiation of its locally uniformly convergent logarithm, or by integrating the uniformly convergent derivative series and matching its value at zero. Expanding each denominator and interchanging absolutely convergent sums gives

$$-\frac{F'(z)}{F(z)} = \sum_j \frac{m_j \beta_j}{1 - \beta_j z} = \sum_{n \geq 0} \left(\sum_j m_j \beta_j^{n+1} \right) z^n. \quad (\text{GF13})$$

Every series interchange is justified by $\sum_j m_j |\beta_j| / (1 - r|\beta_j|) < \infty$. Equations (GF12)–(GF13) prove exactly the product and power-sum input used in the workbench and (TC16), with the unchanged $F(0) = g(1/2)$.

This calculation continues the supplied *RH Counterfactual Workbench*. It keeps its original $g = 2\xi$, theta source, Fourier coordinate, unscaled mass, reciprocal-zero multiplicities, and monomial matrices. The accompanying workbench gives the full arithmetic packet and support comparison. Here the infinite theta integral receives explicit truncation bounds, and the negative-witness construction receives quantitative rational approximation and finite interval verification. The computational certificate uses these bounds to prove positivity of both original matrices of dimension 16. This finite result has the same matrix definitions as the workbench's exploratory calculation.

184 The original integral and its two omitted regions

Retain

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \psi(y) = e^{y/2} f_0(e^y). \quad (\text{TC1})$$

The original reflection and Mellin calculation give $\psi(-y) = \psi(y)$ and $g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy$. For every even integer $J \geq 0$, write $M_J = \int_{\mathbb{R}} y^J \psi(y) dy$. Substitution of (TC1), with its theta factor two and reflection factor two, gives

$$M_J = \int_0^\infty I_J(y) dy, \quad I_J(y) = 4y^J \sum_{n \geq 1} \left(4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2} \right) e^{-\pi n^2 e^{2y}}. \quad (\text{TC2})$$

At $y \geq 0$, each summand is nonnegative; it is strictly positive at $y > 0$, because $2\pi n^2 e^{2y} - 3 > 0$. This also proves positivity of every M_J once convergence is established below.

For an integer $N \geq 1$ and a real $L > 0$, retain exactly the terms $1 \leq n < N$ on $0 \leq y \leq L$. Denote their integral by $M_J^{N,L}$. Thus N is the first omitted integer. Set

$$r_N = e^{4/N - 2\pi N}, \quad r_1 = e^{4 - 2\pi}, \quad \lambda_J(L) = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{TC3})$$

Both r_N and r_1 lie strictly between zero and one.

Theorem 184.1 (Two explicit positive tails). *For $\lambda_J(L) > 0$ the full original moment satisfies*

$$0 \leq M_J - M_J^{N,L} \leq T_{J,N,L} + Y_{J,L}, \quad (\text{TC4})$$

where every factor is retained in

$$T_{J,N,L} = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - r_N}, \quad Y_{J,L} = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - r_1) \lambda_J(L)}. \quad (\text{TC5})$$

Proof. For $n = N + \ell$, $\ell \geq 0$, use $(N + \ell)^4 \leq N^4 e^{4\ell/N}$, which follows from $\log(1 + x) \leq x$ for $x \geq 0$, and $(N + \ell)^2 \geq N^2 + 2N\ell$. For $y \geq 0$ these imply

$$\sum_{n \geq N} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{N^4 e^{-\pi N^2 e^{2y}}}{1 - e^{4/N - 2\pi N e^{2y}}} \leq \frac{N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TC6})$$

The negative term in each parenthesis of (TC2) can be dropped for an upper bound. On $[0, L]$ one has $y^J \leq L^J$ and $e^{9y/2} \leq e^{9L/2}$. Integrating that bound over an interval of length L proves the first expression in (TC5).

For the second region, apply the first bound in (TC6) with $N = 1$, retaining its numerator's dependence on y and bounding its denominator below by $1 - r_1$. Consequently

$$0 \leq I_J(y) \leq \frac{16\pi^2}{1 - r_1} y^J e^{9y/2 - \pi e^{2y}} \quad (y \geq 0). \quad (\text{TC7})$$

Write $y = L + v$ with $v \geq 0$. The elementary inequalities $(L + v)^J \leq L^J e^{Jv/L}$ and $e^{2(L+v)} \geq e^{2L}(1 + 2v)$ yield

$$y^J e^{9y/2 - \pi e^{2y}} \leq L^J e^{9L/2 - \pi e^{2L}} e^{-\lambda_J(L)v}. \quad (\text{TC8})$$

Its integral for $v \geq 0$ is the numerator divided by $\lambda_J(L)$, proving the second bound. The two omitted regions are disjoint: the first has $0 \leq y \leq L$ and $n \geq N$; the second has $y > L$ and every $n \geq 1$. Every omitted summand is nonnegative. Tonelli's theorem therefore justifies all exchanges before convergence is known; the finite upper bounds then prove convergence and (TC4). $\square$

For every fixed J , the bounds tend to zero by first increasing L and then increasing N . Indeed (TC8) is dominated by the negative double exponential as $L \rightarrow \infty$, while for fixed L the factor $N^4 e^{-\pi N^2}$ tends to zero and $r_N \rightarrow 0$. The finite integrand

$$y^J e^{y/2} \sum_{n=1}^{N-1} (16x_n(y)^2 - 24x_n(y)) e^{-x_n(y)}, \quad x_n(y) = \pi n^2 e^{2y}, \quad (\text{TC9})$$

is entire as a function of the original complex coordinate y . Expanding x_n shows that it is exactly the retained integrand of (TC2); no scalar factor or negative term has been changed. This entire extension supplies the analyticity condition for validated complex-ball quadrature on the finite real intervals.

185 Exact propagation to the original Hankel matrices

Define, in the original coefficient convention,

$$a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad F(w) = \sum_{j \geq 0} a_j w^j, \quad -\frac{F'(w)}{F(w)} = \sum_{n \geq 0} b_n w^n. \quad (\text{TC10})$$

The original entire-function comparison is $F(z^2) = g(1/2 + iz)$, and $a_0 = M_0 > 0$. Multiplication of the last identity by F and extraction of the coefficient of w^n prove the complete recurrence

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{TC11})$$

In particular the original mass remains in every division by a_0 . The two real symmetric matrices of dimension $d+1$ are

$$H_d = (b_{r+s})_{0 \leq r, s \leq d}, \quad H_d^{(1)} = (b_{r+s+1})_{0 \leq r, s \leq d}. \quad (\text{TC12})$$

Proposition 185.1 (Finite enclosures and signs). *Suppose that closed rational intervals enclose the finite integrals and upper bounds in (TC4)–(TC5). Adding the interval $[0, T_{J,N,L} + Y_{J,L}]$ with outward upper endpoints gives rational enclosures of the full moments. Applying (TC10)–(TC11) by interval arithmetic, with an enclosure of a_0 having positive lower endpoint, gives enclosures of every required b_n . An interval LDL^T recursion whose every pivot has positive lower endpoint proves positive definiteness of the original matrix in (TC12).*

Proof. Addition, subtraction, multiplication and division by an interval excluding zero each contain the exact operation on contained arguments. This follows for multiplication from the extrema of a bilinear function on a rectangle; division reduces to multiplication by the interval reciprocal. Induction in (TC11) proves the assertion. No independence of the interval arguments is required for enclosure.

For a symmetric matrix A , the exact recursion in the unchanged coordinate order is

$$D_j = A_{jj} - \sum_{k < j} L_{jk}^2 D_k, \quad (\text{TC13})$$

$$L_{jj} = 1, \quad L_{ij} = \frac{A_{ij} - \sum_{k < j} L_{ik} L_{jk} D_k}{D_j} \quad (i > j), \quad L_{ij} = 0 \quad (i < j).$$

If the interval pivots have positive lower endpoints, induction shows that every exact D_j is positive and every division is defined. Formula (TC13) makes $A_{ij} = \sum_k L_{ik} D_k L_{jk}$ for all $i \geq j$; symmetry gives the remaining entries. Thus $A = LDL^T$. The triangular L is invertible with determinant one. For every nonzero real vector v , $v^T A v = \sum_j D_j ((L^T v)_j)^2 > 0$. The same equality for complex vectors uses the conjugate transpose and squared moduli and proves Hermitian positive definiteness. $\square$

For a specified rational vector $c = (c_0, \dots, c_d)$ let $\gamma_n = \sum_{r+s=n} c_r c_s \in \mathbb{Q}$. The exact signed witness is

$$c^\top H_d c = \sum_{n=0}^{2d} \gamma_n b_n. \quad (\text{TC14})$$

If $b_n \in [\ell_n, u_n]$, its upper endpoint is

$$U(c) = \sum_{\gamma_n \geq 0} \gamma_n u_n + \sum_{\gamma_n < 0} \gamma_n \ell_n. \quad (\text{TC15})$$

Every comparison here is rational. A strict inequality $U(c) < 0$ is a finite negative certificate. If $\widehat{b}_n$ have errors $|b_n - \widehat{b}_n| \leq \epsilon_n$, the exact error in (TC14) is at most $\sum_n |\gamma_n| \epsilon_n$, by the triangle inequality. These statements include all mixed matrix entries.

186 Quantitative rational form of the counterfactual separator

We record the full argument to show precisely how a negative counterfactual form becomes a rational certificate. The entire factorization in the supplied workbench gives the distinct reciprocal zeros β_j of F , with positive integer multiplicities m_j , and

$$\sum_j m_j |\beta_j| < \infty, \quad b_n = \sum_j m_j \beta_j^{n+1}. \quad (\text{TC16})$$

No reciprocal zero is zero. The multiset is stable under complex conjugation. The actual map from a zeta zero $\rho = 1/2 + \delta + i\gamma$ is

$$\beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \quad (\text{TC17})$$

The reflected pair $\rho, 1 - \rho$ corresponds to a single zero of F with the same multiplicity. Both primary blocks remain in the original packet before taking its half-trace.

Let

$$S_* = \sum_j m_j |\beta_j|, \quad R_* = \max\{1, \sup_j |\beta_j|\} < \infty, \quad \|P\|_{R_*} = \sum_{r=0}^{\deg P} |p_r| R_*^r \quad (\text{TC18})$$

for $P(X) = \sum p_r X^r$. This coefficient norm is an explicit error-control map; it changes neither P nor the original Hankel form. Define

$$\mathcal{Q}(P) = \sum_j m_j \beta_j P(\beta_j)^2 = p^\top H_{\deg P} p. \quad (\text{TC19})$$

Absolute convergence follows from $|P(\beta_j)| \leq \|P\|_{R_*}$ and (TC16). The sum is real for real P because conjugate terms pair, and real reciprocal terms are real. The equality with the finite matrix follows by expanding the finite polynomial product and using (TC16).

Lemma 186.1 (Coefficient perturbation with the full reciprocal tail). *For real polynomials P, E one has*

$$|\mathcal{Q}(P + E) - \mathcal{Q}(P)| \leq S_* \|E\|_{R_*} (2\|P\|_{R_*} + \|E\|_{R_*}). \quad (\text{TC20})$$

Proof. The polynomial identity $(P + E)^2 - P^2 = 2PE + E^2$ holds at every β_j including every multiplicity. Take absolute values term by term in the absolutely convergent series (TC19), bound $|P(\beta_j)|$ and $|E(\beta_j)|$ by (TC18), and sum $m_j |\beta_j|$. This gives (TC20). $\square$

Theorem 186.2 (Explicit finite degree and rational coefficient accuracy). *Choose any nonreal reciprocal zero β and its conjugate of multiplicity m . Put $r = |\beta|/2$. Let $\mathcal{A}$ be the finite set of all other distinct reciprocal zeros of modulus at least r , and set*

$$B(X) = \prod_{\alpha \in \mathcal{A}} (X - \alpha), \quad B_0 = \prod_{\alpha \in \mathcal{A}} (r + |\alpha|), \quad T_* = \sum_{|\beta_j| < r} m_j |\beta_j|. \quad (\text{TC21})$$

Choose either $c \in \mathbb{C}$ satisfying $\beta c^2 = -1$ and define

$$C_* = \frac{|c|B_0}{|B(\beta)|} \left(1 + \frac{|\Re\beta| + r}{|\Im\beta|} \right). \quad (\text{TC22})$$

If $T_* = 0$, take $N_* = 0$. Otherwise take the explicit integer

$$N_* = \max \left\{ 0, 1 + \left\lfloor \frac{\log(C_*^2 T_*/m)}{\log 4} \right\rfloor \right\}. \quad (\text{TC23})$$

For any integer $N \geq N_*$ put

$$w_N = \frac{c}{\beta^N B(\beta)}, \quad v_N = \frac{\Im w_N}{\Im \beta}, \quad u_N = \Re w_N - (\Re \beta) v_N, \quad P_N(X) = X^N B(X)(u_N + v_N X). \quad (\text{TC24})$$

Then $\mathcal{Q}(P_N) \leq -m$. A rational polynomial $\tilde{P}_N$ of degree at most $D = N + \#\mathcal{A} + 1$ can be obtained by rounding each of the $D + 1$ original coefficients to a nearest multiple of $1/H$, with either choice allowed at a tie, for any positive integer H with

$$\frac{\sum_{r'=0}^D R_*^{r'}}{2H} \leq \min \left\{ 1, \frac{m}{2S_*(2\|P_N\|_{R_*} + 1)} \right\}. \quad (\text{TC25})$$

Every such rounding satisfies $\mathcal{Q}(\tilde{P}_N) \leq -m/2 < 0$.

Proof. Absolute summability in (TC16) makes $\mathcal{A}$ finite. It is conjugation stable; hence B has real coefficients, $B(\beta) \neq 0$, and u_N, v_N are real. Evaluation of their formulas gives $u_N + v_N \beta = w_N$ and therefore $P_N(\beta) = c$ and $P_N(\bar{\beta}) = \bar{c}$. The selected two terms of (TC19) sum to $-2m$ exactly. All other terms with modulus at least r vanish.

For $|X| \leq r$, one has $|B(X)| \leq B_0$ and

$$|u_N + v_N X| \leq |w_N| \left(1 + \frac{|\Re\beta| + r}{|\Im\beta|} \right).$$

Consequently $|P_N(X)| \leq C_* 2^{-N}$. The absolute sum of all remaining terms is at most $C_*^2 T_* 4^{-N}$, so

$$\mathcal{Q}(P_N) \leq -2m + C_*^2 T_* 4^{-N}. \quad (\text{TC26})$$

The chosen integer (TC23) makes the last summand at most m ; if $T_* = 0$ it is zero. This proves the first assertion. The coefficient rounding error $E = \tilde{P}_N - P_N$ has each coefficient at most $1/(2H)$ in absolute value. Thus $\|E\|_{R_*} \leq \sum_{r'=0}^D R_*^{r'}/(2H)$. The bound in (TC25) and (TC20) make the change in the quadratic form at most $m/2$, proving the strict negative result. Every original coefficient slot is included in the error bound. A nonzero coefficient may round to zero; an original zero coefficient remains an exact zero. $\square$

All constants in this counterfactual degree bound retain their dependence on the chosen actual zero and on the full reciprocal set. It supplies no universal dimension for an off-critical witness. Once a rational polynomial is specified, however, (TC14)–(TC15) require only finitely many original moments. As their enclosing intervals shrink, their propagated intervals shrink to the exact coefficients: (TC11) is a finite composition of continuous rational operations with denominator $a_0 > 0$. The strict margin in (TC25) therefore eventually yields a strictly negative rational upper bound in (TC15).

187 The completed finite certificate

The supplied computation uses $N = 20$, $L = 4$, and $J = 0, 2, \dots, 64$. It keeps all nineteen original theta terms on each of the eight intervals $[j/2, (j+1)/2]$, $0 \leq j < 8$. The omitted two regions are enclosed by (TC5). For these parameters every $\lambda_J(L)$ is positive: already $\pi > 3$ and $e^8 > 1 + 8 + 8^2/2 = 41$ give $2\pi e^8 - 9/2 - J/4 > 246 - 9/2 - 16 > 0$. The actual computation retains ball enclosures of the complete expressions, rather than replacing them by that rough inequality.

The finite integrations use python-flint 0.9.0's validated `acb.integral`, at 1024 bits and target 850 bits. The callback is exactly (TC9); its entire expression satisfies the integration routine's analytic-ball contract. Each returned real interval is enlarged by the positive tails. All endpoints are exported as exact dyadic rational numerators and denominators. The complete a_j, b_n, L_{ij}, D_j enclosures and the individual integration intervals are retained in the machine-readable certificate. All thirty-two computed pivots have strictly positive lower endpoints, proving

$$H_{15} > 0, \quad H_{15}^{(1)} > 0. \quad (\text{TC27})$$

Every smaller leading principal matrix inherits positivity. For orientation, the readable enclosures include

$$\det H_1 \in [1.9493355548191422147398782191721395 \times 10^{-9} \pm 5.60 \times 10^{-45}]. \quad (\text{TC28})$$

The actual certificate contains the more precise exact rational endpoints. Its last pivots are positive intervals centered at $6.1698420811114408502314906319786538 \times 10^{-107}$ and $7.1026100619440799659087832255229554 \times 10^{-111}$, respectively. The original a_0 remains centered at $0.99424155637662821982554747937079544$. These values are given to identify the checked objects; the strict signs are decided by the enclosing rational endpoints.

The complete workbench proves that positivity for every unshifted dimension is equivalent to RH. Equation (TC27) settles the displayed finite dimensions and makes no assertion about larger dimensions. The packet trace still evaluates the retained arithmetic source before any critical-only comparison, and its support maps retain τ and every receiving supported zero exactly as in the included packet proofs.

References

- [1] *RH Counterfactual Workbench*, original supplied package, 13 September 2026, complete note and calculation sources.
- [2] The python-flint developers, *acb: complex ball arithmetic and validated integration*, <https://python-flint.readthedocs.io/en/latest/acb.html>.
- [3] R. Zhang, *On Power Sums of Positive Numbers*, <https://arxiv.org/abs/1510.03420>.

Original marked-product note: literal source and reading corrections

The following appendix preserves the complete original body, including its abstract and scope statements. Only its document wrapper, `\maketitle` and `\tableofcontents` controls are removed here. The separate `MARKED_PRODUCT_ORIGINAL_NOTE.tex` is byte exact. The original frequency symbol is y in $S = k/2 + iy$; the deformation parameter is u . In the original analytic displays, the measure written du after defining a density in y is read as dy , and the frequency arguments written $c + iu$ are read as

$c + iy$. These are distinct original variables, and this correction does not specialize or replace either one.

The sentence following original equation (64) overstates positivity of the relative derivative. For the literal positive source Grams $M(0)$ and $M(1)$, the identities

$$M(0)(M(0)^{-1}M(1)) = M(1), \quad M(0)(M(0)^{-1}(M(1) - M(0))) = M(1) - M(0)$$

prove that the first endomorphism is positive and self-adjoint in the $M(0)$ metric and that the second is self-adjoint. The second need not be positive: when $M(1) = M(0)/2$, it equals $-I/2$. Its original operator norm and the resulting midpoint enclosure remain meaningful without the extra positivity assertion.

The two nonnegative source/relation operators whose densities occur in the signed four-endpoint cancellation each have trace $2q$. Their rank is $q + 1$, with spectrum $1, 2, \dots, 2, 1$ on their range (the multiplicity of 2 is $q - 1$). The earlier AW7 proves this from the actual nested source and relation projections; SP6–SP8 reproduce that spectrum on the identical operators proved above. Thus a kernel mass $2q$ is read as that trace, not as rank $2q$. The complete exact cancellation is retained in SP1–SP13. Its final bound is the earlier AW14, whose full-spectrum refinement AW13 is also included; the additional overlap and common-range refinement is SP8–SP10.

The marked τ -base, weight-matched external products,
and the signed arithmetic four-endpoint correction

Original abstract. We continue the original marked Split-Zero construction, retaining its absolute pointed base, supported zero, amplitude and support observations, full theta complex, and pre-quotient norms. We construct the external product of the original packet’s polynomial-exponential complex. Its top cohomology has rank d^k , its special fibre is the full tensor packet, and its diagonal parameter operator specializes to the original sum action. Its finite-field realization has weight k under Deligne’s actual hypotheses. An explicit coefficient and source diagram connects this family to the original theta complex; no Frobenius intertwiner is assumed. We then calculate the signed arithmetic/Gamma correction simultaneously at all four original endpoints. The result is a single integral of the actual multiplier against a signed difference of two equal-rank source/relation projection densities. Its differentiated sections are original relations, their complete quadratic contribution is calculated, and a finite quadrature error bound is proved. A closed formula for the first conjugate-pair calculation retains the exact cancellation $|h|^2 w_h = |2\xi|^2/(2\pi)$. These results do not prove the remaining uniform upper bound.

188 Scope and fixed inputs

The earlier audit constrains particular auxiliary implications. It is not a no-go theorem for geometry over $\mathbb{F}_1$ or for this marked τ -base. We retain its valid examples and its exact arithmetic/Gamma decomposition. We do not use an arbitrary test polynomial as an arithmetic zero packet. Everywhere below, an arithmetic packet means a nonempty finite set of actual nontrivial zeros of $g = 2\xi$, with each selected zero included to its complete order.

The current formal integration distinguishes an original relation module from the larger kernel of a finite observation, and distinguishes a source-section defect from a coefficient-action defect. Those distinctions are used below. Analytic period invertibility and the analytic theta constructions remain written results, not assertions of a new Lean run.

189 The original absolute base and its actual fibres

For a commutative ring R , retain the semiring

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad \tau + x = x, \quad r^\bullet + s^\bullet = (r + s)^\bullet, \quad \tau x = \tau, \quad r^\bullet s^\bullet = (rs)^\bullet. \quad (1)$$

The multiplicative unit is $1_R^\bullet$, including when R is the zero ring. Define

$$p_R(\tau) = 0, \quad p_R(r^\bullet) = r, \quad b_R(\tau) = 0, \quad b_R(r^\bullet) = 1 \in \mathbb{B}. \quad (2)$$

Both are semiring maps. The pair (p_R, b_R) is injective: support distinguishes τ from every represented element, and amplitude distinguishes represented elements. In particular e_R has observations $(0, 1)$, while τ has observations $(0, 0)$.

Let $\mathbb{F}_{1,\tau} = \{\tau_0, 1_0\}$ be the pointed commutative monoid. Its finite-word preaddition identifies precisely words having the same number of 1_0 terms; τ_0 terms can be deleted. Thus $[] \sim [\tau_0]$, but $[1_0, 1_0] \not\sim [1_0]$. There is a unique pointed multiplicative structural map

$$j_R : \mathbb{F}_{1,\tau} \longrightarrow G(R), \quad \tau_0 \mapsto \tau, \quad 1_0 \mapsto 1_R^\bullet. \quad (3)$$

It preserves the preaddition because the image of a word is its integer number of units in $G(R)$. Its base spectrum $\mathfrak{b}_\tau$ has exactly one multiplicative prime, $\{\tau_0\}$.

We work in marked monoidal spaces carrying their original semiring sheaf and the two observations (2), and in their specified coefficient-support diagrams. The structural maps are local pointed-monoid maps. This declares the category actually used; it does not identify a finite chart with a finite-type arithmetic scheme or assume a general six-functor formalism for semirings.

For a nonzero integral domain R , the semiring prime ideals are

$$P_\tau = \{\tau\}, \quad P_{\mathfrak{p}} = \{\tau\} \sqcup \mathfrak{p}^\bullet \quad (\mathfrak{p} \in \operatorname{Spec} R). \quad (4)$$

Indeed, an ideal containing any $a^\bullet$ contains $e_R = a^\bullet + (-a)^\bullet$. Its represented part is a ring ideal, and primality reduces to ring primality. The remaining ideal $\{\tau\}$ is prime since a product of represented elements is represented. Thus

$$P_e = \{\tau, e_R\}, \quad V(e_R) \simeq \operatorname{Spec} R, \quad D(e_R) = \{P_\tau\}. \quad (5)$$

Here P_e is the supported-zero prime, P_τ is a point in the enlarged source, and the unique point of $\mathfrak{b}_\tau$ is the structural target. Their connecting maps are the closed inclusion induced by p_R , the inclusion of $D(e_R)$, and the structural map $a_R : X_R \rightarrow \mathfrak{b}_\tau$.

One explicit sheaf presentation is obtained by adjoining P_τ to every nonempty open of $\operatorname{Spec} R$. On $U \cup \{P_\tau\}$ its value is $G(\mathcal{O}_R(U))$, and on $\{P_\tau\}$ its value is $G(0) \simeq \mathbb{B}$. The restriction to that point is b_R . Compatible sections on a cover have the same support because all opens meet at P_τ ; represented amplitudes then glue by the ring sheaf, and absent sections glue to τ . This proves the sheaf property. It agrees with the displayed localizations on principal opens, including localization at e_R .

Consequently the direct image and its further restriction are

$$(a_{R*} \mathcal{S}_R)(\mathfrak{b}_\tau) = G(R) \xrightarrow{b_R} G(R)[e_R^{-1}] \simeq \mathbb{B}. \quad (6)$$

We retain the left-hand object. The absolute structural fibre is the whole marked source; it is not replaced by the Boolean stalk. The original arithmetic square is

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(\iota)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{\iota} & \mathbb{C}. \end{array} \quad (7)$$

Its integer target is infinite. For every ring ideal I , the square with $G(R/I)$ commutes; $G(R \rightarrow R/I)^{-1}(e_{R/I}) = I^\bullet$ and its inverse image of τ is $\{\tau\}$. In particular the ordinary finite ideal norms $|R/I|$ are retained by the arithmetic observation; no finite Euler factor is assigned to the infinite quotient by the zero ideal.

For a coefficient vector space V , write $S(V) = \{\tau\} \sqcup V^\bullet$. A linear map lifts by $\tau \mapsto \tau$ and $v^\bullet \mapsto f(v)^\bullet$. For an original join-support diagram, the same formula is applied fibrewise and uses its specified transport maps. On fixed nonbottom labels,

$$\tilde{f}^{-1}(e) = (\ker f)^\bullet, \quad \tilde{f}^{-1}(\tau) = \{\tau\}. \quad (8)$$

For general support-changing maps, this nonabsence assertion is made only when the receiving label is nonbottom. The current reconstruction allows nontrivial bottom coefficient fibres; that case is not silently treated as the disjoint single-fibre model.

190 The theta complex over the marked point

Retain the original Fréchet spaces and seminorms

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^m F(x)| < \infty \text{ for every } b \in \mathbb{Z}, m \geq 0\}, \quad D = -x\partial_x. \quad (9)$$

The Fourier and Mellin maps are

$$\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (10)$$

The original source proofs establish continuity and injectivity of Θ , $D\Theta = \Theta D$, and

$$\Theta\hat{\phi}(x) = x^{-1}\Theta\phi(1/x), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (11)$$

Here

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad H_{\phi_*} = 1. \quad (12)$$

Every constant in (10)–(12) is retained.

The four-point chart P has $+\langle \eta < \sigma$ and $-\langle \eta < \sigma$. Its coefficient sheaf has values $V, V, \mathcal{B}, 0$, restrictions $\Theta, \Theta\mathcal{F}$ into η , and zero restrictions into σ . Its structural map to $X_{\mathbb{C}}$ sends the first three points to P_e and σ to P_τ , then to $\mathfrak{b}_\tau$. The coefficient scalar restriction at σ is the specified $G(\mathbb{C}) \rightarrow \mathbb{B}$; reconstructing the zero coefficient fibre leaves its supported zero and absence.

The projective-diagram resolution

$$0 \rightarrow P_\eta \xrightarrow{x \mapsto (x, -x)} P_+ \oplus P_- \xrightarrow{(x, y) \mapsto x+y} \underline{\mathbb{C}} \rightarrow 0 \quad (13)$$

is exact at every chart point. Applying the sheaf Hom functor gives the actual global-coefficient complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta(v-\hat{w})} \mathcal{B}]. \quad (14)$$

The map $(v, w) \mapsto (v - \hat{w}, w)$ with inverse $(a, w) \mapsto (a + \hat{w}, w)$ identifies it with $C_+ \oplus V[0]$, where $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Thus $H^1(C_+) = Q = \mathcal{B}/\Theta V$, while the entire additional degree-zero cycle is retained in (14). This is a coefficient-complex calculation over the marked base, not a replacement of that base by $\text{Spec } \mathbb{C}$.

190.1 Actual packets and the finite observation kernel

For $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, with the stipulated full orders, set

$$E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times. \quad (15)$$

The original Euler inverse

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy \quad (16)$$

on its zero-Mellin-value domain is continuous in the stated spaces. Its iterations supply $F_h = S_h(\Theta\phi_*) \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$. They give the actual square and exact sequence

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h, \end{array} \quad 0 \rightarrow Q[h(D)] \rightarrow E_h \xrightarrow{\times j_h g} E_h \rightarrow Q/h(D)Q \rightarrow 0. \quad (17)$$

For the actual packet $j_h g = 0$ and v_h is a unit. The canonical finite observation $J_h F = j_h \mathcal{M}F$ and source section

$$r_h u = \mathcal{T}_h(\text{rem}_h(v_h^{-1}u)), \quad \mathcal{T}_h P = P(D)F_h, \quad \alpha_h P = P(D)\phi_* \quad (18)$$

satisfy

$$\mathcal{T}_h(hP) = \Theta\alpha_h P, \quad J_h \mathcal{T}_h P = v_h[P]_h, \quad J_h r_h = 1, \quad J_h \Theta = 0. \quad (19)$$

Thus the coefficient complex $[\mathbb{C}[s] \xrightarrow{h} \mathbb{C}[s]ds]$ maps to the original C_+ by $(\alpha_h, \mathcal{T}_h)$. Its H^1 map is $\sigma_h M_{v_h} : E_h \hookrightarrow Q$, where $\sigma_h = [r_h]$. Its cone retains $Q/\sigma_h E_h$ in degree one. On the entire source the observation kernel remains $\ker J_h/\Theta V$; equality with zero is asserted only on the stated finite packet image.

191 The complete dilation and relation action

For $a > 0$ retain $U_a F(x) = F(x/a)$. Mellin transformation gives $\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s)$ and $U_a \Theta = \Theta U_a$. Fourier transformation gives

$$\widehat{U_a \phi} = a U_{1/a} \widehat{\phi}. \quad (20)$$

Therefore the two-leg action is $(U_a, aU_{1/a})$ and its generators are $(D, 1 - D)$. On E_h the exact action is multiplication by $j_h(a^s)$; on a primary block it is

$$a^{A_h}|_{E_\rho} = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (21)$$

The full v_h commutes with this multiplication. Thus $J_h U_a = a^{A_h} J_h$ and the finite inclusion into Q intertwines the same action.

There is a source primitive for the representative defect. Put $P_u = \text{rem}_h(v_h^{-1}u)$ and $\tilde{P}_u = \text{rem}_h(j_h(a^s)P_u)$. Then

$$K_a u = S_h^V (U_a \alpha_h P_u - \alpha_h \tilde{P}_u), \quad U_a r_h - r_h a^{A_h} = \Theta K_a. \quad (22)$$

The input to S_h^V has every required zero jet by construction. Applying the injective $h(D)$ proves the second equation. Injectivity of Θ also proves

$$K_{ab} = U_a K_b + K_a b^{A_h}. \quad (23)$$

Polynomial source spans are preserved by D , not generally by every U_a ; (22) is the explicit map into the full original source which supplies the latter comparison.

On the ordered k -fold tensor complex, $D^{(k)} = \sum_i D_i$, the finite action is $A_k = \sum_i A_{h,i}$, and

$$a^{A_k} = \bigotimes_{i=1}^k a^{A_{h,i}}. \quad (24)$$

This retains every term in the exponential of $N_1 + \cdots + N_k$. A primitive for the tensor difference is the sum whose i -th term is

$$(-1)^{i-1} (U_a r_h)^{\otimes(i-1)} \otimes K_a \otimes (r_h a^{A_h})^{\otimes(k-i)}. \quad (25)$$

The tensor differential contributes the second $(-1)^{i-1}$, giving the positive telescoping difference. All primitives remain in their original tensor support labels, with the specified join when terms are added.

192 The weight-matched external family over the same base

Write $d = \deg h$ and $h(s) = s^d + \sum_{b < d} h_b s^b$. Define

$$\Phi_h(s) = \frac{s^{d+1}}{d+1} + \sum_{b=0}^{d-1} \frac{h_b s^{b+1}}{b+1}. \quad (26)$$

Its constant is fixed to zero; $\Phi'_h = h$. Let $R = \mathbb{C}[u, t_1, \dots, t_k]$. On $R[s_1, \dots, s_k] \otimes \bigwedge^\bullet(ds_1, \dots, ds_k)$ define

$$\mathcal{D} = \sum_{i=1}^k (u \partial_{s_i} + h(s_i) - t_i) ds_i \wedge (-). \quad (27)$$

The coefficient operators commute, and the wedge signs give $\mathcal{D}^2 = 0$.

Theorem 192.1 (The full external-product fibre). *The complex (27) has cohomology only in degree k . That cohomology is free over R with the retained basis $s_1^{a_1} \cdots s_k^{a_k} ds_1 \wedge \cdots \wedge ds_k$, $0 \leq a_i < d$, and has rank d^k . At the coefficient specialization $u = t_1 = \cdots = t_k = 0$, it is $E_h^{\otimes k}$ with its full nilpotent structure.*

Proof. In one variable, $(u \partial_s + h - t)P$ has degree $\deg P + d$ and the same leading coefficient as P . It is injective, and successive leading-term subtraction gives a unique remainder of degree below d . This splits the one-factor complex, as an R -module complex, into its free remainder in degree one and a contractible pair. Tensor those explicit splittings. Every summand having a contractible factor is contractible, with the usual tensor-homotopy signs. The sole remaining summand is the stated tensor remainder in degree k . Since all splittings are over R , the result survives the displayed coefficient specialization, without a derived-fibre Tor term. $\square$

This family has an explicit original-source realization. The coefficient maps α_h and $\mathcal{T}_h$ are injective; denote their images by V_* and $\mathcal{B}_h$. Transport the one-variable family differential to them:

$$d_{u,t}^\theta = \Theta|_{V_*} + u \mathcal{T}_h \partial_s \alpha_h^{-1} - t \mathcal{T}_h \alpha_h^{-1} : V_*[u, t] \rightarrow \mathcal{B}_h[u, t]. \quad (28)$$

Then $d_{u,t}^\theta \alpha_h = \mathcal{T}_h(u \partial_s + h - t)$. Thus (28), not an asserted unidentified complex, is a deformation of the original polynomial subcomplex. The derivative is also realized by the original logarithmic multiplier:

$$\mathcal{T}_h(P') = (\log x)P(D)F_h - P(D)((\log x)F_h). \quad (29)$$

This follows from $[\log x, D] = 1$. Its Mellin proof retains both terms $v'_h P + v_h P'$; the $v'_h P$ term is not dropped.

Tensor (28) in the original order and then include at the marked specialization in $C_+^{\otimes k}$. The resulting cohomology map is

$$E_h^{\otimes k} \xrightarrow{M_{v_h^{\otimes k}}} E_h^{\otimes k} \xrightarrow{\sigma_h^{\otimes k}} Q^{\otimes k}. \quad (30)$$

Over $\mathbb{C}$ these are literal algebraic tensor complexes. On the inherited completed projective tensor realization, the same maps apply to the finite packet and its continuous sections. A new exactness theorem for arbitrary completed tensor products is not assumed.

The scalar structural maps and specialization are

$$\mathbb{F}_{1,\tau} \rightarrow G(R), \quad G(R) \xrightarrow{G(\text{ev}_0)} G(\mathbb{C}). \quad (31)$$

They commute with (7). The images of the represented u, t_i are e , while τ maps to τ . The original source chart, the family, and the marked fibre remain over the same unique point of $\mathfrak{b}_\tau$. Evaluation in (31) is a coefficient specialization over that point; it does not identify the absolute base with an extra complex parameter value.

192.1 The cyclic inclusion and its complete family defect

Let $\chi = \chi_{h,k}$ be the minimal polynomial of the sum on $E_h^{\otimes k}$ and let $E = \mathbb{C}[S]/\chi$. The algebra injection $\alpha[P] = P(\sum_i s_i)1$ and weighted module injection

$$\eta = M_{v_h^{\otimes k}} \alpha : E \hookrightarrow E_h^{\otimes k} \quad (32)$$

are retained. Their sum actions intertwine. The quotient $E_h^{\otimes k}/\eta E$ and its induced marked action remain; the word “cyclic” is not a deletion of this quotient.

Fixed-order monic division gives, for each $P(S)$,

$$\chi(S)P(S) = \sum_{i=1}^k h(s_i)Q_i(\mathbf{s}), \quad S = \sum_i s_i. \quad (33)$$

At the special fibre its actual primitive is $\sum_i (-1)^{i-1} Q_i ds_1 \wedge \cdots \widehat{ds_i} \cdots \wedge ds_k$. At general parameters (27) gives instead the exact equation

$$[\chi(S)P(S)]_{\mathcal{H}} = \sum_i [t_i Q_i - u \partial_{s_i} Q_i]_{\mathcal{H}}. \quad (34)$$

Thus the original cyclic annihilator is a marked-fibre relation. The full free family retains the correction (34), rather than declaring that annihilator constant throughout the family.

193 Deligne’s step, with the actual parameter and action maps

For $u \neq 0$, the one-factor connection is

$$\nabla_{t_i} = \partial_{t_i} - \frac{A_{h,i}(t_i)}{u}, \quad A_h(t) = A_h + t\mathcal{R}_h, \quad \mathcal{R}_h = 1 \otimes [S^{d-1}] \text{rem}_h. \quad (35)$$

On the diagonal $t_i = t$, the operator is

$$\nabla_t = \partial_t - \frac{A_k + t \sum_i \mathcal{R}_{h,i}}{u}, \quad (-u \nabla_t) \bmod (u, t) = A_k. \quad (36)$$

The coefficient commutators prove descent of (35): $[u\partial_s + h - t, \partial_t] = 1$ and $[u\partial_s + h - t, -s/u] = -1$. Formula (36) is an operator on the full tensor coefficient module; on ηE its transverse term is the explicitly retained composite

$$E \xrightarrow{\eta} E_h^{\otimes k} \xrightarrow{\sum_i \mathcal{R}_{h,i}} E_h^{\otimes k} \xrightarrow{\pi_\eta} E_h^{\otimes k} / \eta E. \quad (37)$$

It is not presumed zero.

Let $R_0 \subset \mathbb{C}$ be the finitely generated coefficient ring containing the coefficients of h , the finitely many coefficients of v_h and its inverse in the retained remainder basis, and $1/(d+1)!$. Thus the finite Taylor-unit multiplication and its inverse also specialize through this same coefficient model; no finite-field interpretation of the entire analytic function g is asserted. At a specified finite-field specialization of characteristic $p > d+1$, and $u \neq 0$, take

$$\mathcal{L}_\psi \left(\frac{\sum_i \Phi_h(s_i) - \sum_i t_i s_i}{u} \right) \quad \text{on } \mathbb{A}^k. \quad (38)$$

Its highest homogeneous part is $\sum_i s_i^{d+1}/((d+1)u)$. The projective hypersurface is smooth: its partial derivatives are s_i^d/u , which cannot vanish simultaneously at a projective point. Deligne's exponential-sum theorem, Weil II 3.7.2.3, therefore gives compact-support cohomology only in degree k , of dimension d^k , pure of weight k . Equivalently, the actual external-product sheaf and compact Künneth map give the tensor product of the k degree-one, weight-one factors. Repeated critical roots do not remove smoothness of the leading homogeneous part.

This uses the product of the original degree- d packet family. The rank- q , weight-one exponential family constructed from the sum annihilator has a different specified map and is not substituted for (38). The current correction resolves the citation as Roman I.8.11 to Weil I, retains strict < 2 , and retains lisse/mixed hypotheses for propagation. The algebraic fibre (31) is not automatically a lisse fibre of the $u \neq 0$ exponential system.

For completeness the complex periods use the original ordered rays

$$\begin{aligned} \vartheta_j &= \frac{\arg u + \pi}{d+1} + \frac{2\pi j}{d+1}, \quad \Gamma_j = -\ell_0 + \ell_j, \quad 1 \leq j \leq d, \\ (\Pi_h)_{jb}(u, t) &= \int_{\Gamma_j} e^{(\Phi_h(s) - ts)/u} s^b ds, \quad 0 \leq b < d. \end{aligned} \quad (39)$$

The leading exponent is $-r^{d+1}/((d+1)|u|)$, proving absolute convergence on those rays and differentiation on compact parameter sets. At the monomial member the matrix is a nonzero gamma-column factor times $(\exp(2\pi i j(b+1)/(d+1)) - 1)_{jb}$. A polynomial of degree at most d vanishing at all $d+1$ roots of unity proves that last matrix invertible. Coefficient differentiation and unique differential remainder yield a matrix equation $\Pi' = \Pi B$ along a coefficient path; hence $(\det \Pi)' = \text{Tr}(B) \det \Pi$, proving invertibility at every parameter, including collisions.

The ordered external period map is exactly

$$\Pi_k(u, \mathbf{t}) = \bigotimes_i \Pi_h(u, t_i) : \mathcal{H}_{u, \mathbf{t}}^k \xrightarrow{\sim} \mathbb{C}^{d^k}. \quad (40)$$

Fubini proves the product formula with the ordered product contours and $ds_1 \wedge \cdots \wedge ds_k$. The cyclic map is the actual rectangular map $\Pi_k \eta$. On its image it has the inverse induced by $\eta^{-1} \Pi_k^{-1}$. On the full source its observation kernel is unchanged by invertibility of Π_k .

In particular the complete base diagram is the split lift of

$$\mathcal{H}^k/(u, \mathbf{t}) \mathcal{H}^k \simeq E_h^{\otimes k} \xrightarrow{\sigma_h^{\otimes k} M_{v_h^{\otimes k}}} Q^{\otimes k}, \quad E \xrightarrow{\eta} E_h^{\otimes k} \xrightarrow{\Pi_k} \mathbb{C}^{d^k}. \quad (41)$$

Every arrow is defined above, and (21), (24), (35)–(37) specify its action. The finite-field branch of (38) has its own Frobenius on that realization. No faithful Frobenius comparison with a^{A_k} is added to (41).

194 The original boundary pairing, before its supported quotient

For the canonical cyclic section $r_N = \mathcal{V}C_N : E \rightarrow \mathcal{B}^{\widehat{\otimes} k}$, retain

$$J^{(k)}r_N = \eta, \quad [r_N] = \sigma_h^{\otimes k}\eta, \quad D^{(k)}r_N - r_N A = dK_N, \quad G_N = r_N^* r_N. \quad (42)$$

Here C_N is the coefficient minimum for the actual source norm. Integration by parts, including the vanishing endpoint $[x\overline{F}H]_0^\infty$, proves

$$W_N = A^*G_N + G_N A - kG_N = -r_N^* dK_N - (dK_N)^* r_N. \quad (43)$$

For $a = e^t$ the original dilation defect is the convergent finite-interval source integral

$$U_a^{\otimes k} r_N - r_N a^A = d \int_0^t U_{e^{t-v}}^{\otimes k} K_N e^{vA} dv. \quad (44)$$

Differentiation of $U_{e^{t-v}} r_N e^{vA}$ verifies the sign and both endpoints. All factors in e^{vA} on every nilpotent block remain as in (21).

The new tensor family uses (42) through the finite source deformation $D_{\mathbf{t}} = D^{(k)} + \sum_i t_i r_{\text{full}} \mathcal{R}_{h,i} j_{\text{full}}$ on the full packet presentation. Its finite observation is $A_k + \sum_i t_i \mathcal{R}_{h,i}$; its representative theta primitive is unchanged. For a cyclic restriction the additional term (37) remains in the full quotient. On changing packet or support, the current integration's exact identity is used:

$$D(h)\widehat{a}_i - \widehat{a}_j D(h) = \Delta_h a_i J_i + r_j(E(h)a_i - a_j E(h))J_i. \quad (45)$$

Thus neither equal finite jets nor a coefficient basis change is used to manufacture source naturality.

The arithmetic content of the relation norm is more explicit than positivity. From (33), let $s_l = 1/2 + it_l$ and put

$$Z_i(P; \mathbf{t}) = g(s_i) \prod_{l \neq i} v_h(s_l) Q_i(P; \mathbf{s}).$$

Then full Mellin–Plancherel gives

$$\|\mathcal{V}(\chi P)\|^2 = \frac{1}{(2\pi)^k} \sum_{i,j=1}^k \int_{\mathbb{R}^k} \overline{Z_i(P; \mathbf{t})} Z_j(P; \mathbf{t}) d\mathbf{t}. \quad (46)$$

For two different inputs use the corresponding polarized formula. The terms $i \neq j$ are retained. This formula uses the actual identities $h v_h = g$ and $j_h g = 0$. It does not replace g/h by a generic bounded function. All its integrals converge, since each displayed multiplier is the Mellin transform of its specified original theta-source function.

195 One common source for the signed arithmetic/Gamma correction

Use the original sum-frequency coordinate $S = k/2 + iy$ in the analytic calculation. The Gamma reference is the retained density

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}. \quad (47)$$

We use $\lambda = 1/4$ for the original reference, keeping $\int r_{1/4,k} = (2\pi)^{k/2}$. An explicit half-line reference vector is

$$F_\lambda^\Gamma(x) = 2x^{2\lambda-1/2} e^{-x^2}, \quad \mathcal{M}F_\lambda^\Gamma(1/2 + it) = \Gamma(\lambda + it/2), \quad \|F_\lambda^\Gamma\|_{L^2(dx)}^2 = c_\lambda. \quad (47a)$$

Substitution $y = x^2$ proves the Mellin identity and the norm, including their factors two. It is an auxiliary weighted L^2 reference, mapped to the same polynomial coefficient source; it is not declared an element of the original strong-Schwartz space. Its polynomial-dilation columns have the weighted $x^{\pm 2a}$ norms needed for holonomy comparison whenever $0 < a < 2\lambda$. The actual density is

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad m(y) = w_h^{*k}(y), \quad r(y) = r_{1/4,k}(y). \quad (48)$$

No mass is set to one. The ratio $b(y) = m(y)/r(y)$ is the actual arithmetic multiplier. For $k = 1$ its isolated zeros are retained; for $k \geq 2$ its convolution is positive everywhere.

Let $c = k/2$ and use the common source $\mathcal{P}_{2q}$, $q = \deg \chi$. For $0 \leq x \leq 1$ set

$$d\mu_x(y) = ((1-x)r(y) + xm(y))du, \quad \dot{\mu}(y) = m(y) - r(y). \quad (49)$$

This is the Gram of the explicit map

$$P \mapsto (\sqrt{1-x} P(c + iu), \sqrt{x} \mathcal{V}P) \quad (50)$$

into the direct sum of the Gamma reference Hilbert space and the original arithmetic source Hilbert space. Their original norms are used. All finite source Grams $M_N(x)$ are positive definite, including the endpoints: a nonzero polynomial cannot vanish on a positive-measure part of either original measure.

Every quotient map $J_N : \mathcal{P}_N \rightarrow E$ and every relation inclusion $B_N = M_\chi : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ is unchanged as x varies. Define

$$G_N(x) = (J_N M_N(x)^{-1} J_N^*)^{-1}, \quad C_N(x) = M_N(x)^{-1} J_N^* G_N(x), \quad V_N(x) = \det G_N(x). \quad (51)$$

At $x = 0, 1$ these are precisely the reference and arithmetic canonical objects, not just matrices chosen to have their dimensions.

The underlying coefficient complex $[B_N \hookrightarrow \mathcal{P}_N]$ over $\mathfrak{b}_\tau$ is constant in (49). Its degree-one image is always the same supported E . The Hermitian forms in (50) are varying observations of that common source; they do not change the cochain relations.

196 An exact signed four-endpoint calculation

Let $v_N(y)$ be the row $(1, c + iu, \dots, (c + iu)^N)$. Define the source and original-relation projection densities

$$K_N(x, y) = v_N(y) M_N(x)^{-1} v_N(y)^*, \\ K_N^{\text{rel}}(x, y) = v_N(y) B_N (B_N^* M_N(x) B_N)^{-1} B_N^* v_N(y)^*, \quad (52)$$

with the second expression zero at $N = q - 1$. Their residual is

$$L_N(x, y) = K_N - K_N^{\text{rel}} = v_N C_N G_N^{-1} C_N^* v_N^* \geq 0. \quad (53)$$

The inverse identity in (53) follows by block inversion in the retained basis of quotient lifts followed by the relation columns. In that ordering the monic relation matrix has determinant one. Passing to the conventional relation-first determinant order would contribute $(-1)^{q(N-q+1)}$; its phase is not a change of metric.

Theorem 196.1 (Signed arithmetic correction on a common source). *Define*

$$F(x) = \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \quad \Delta_{h,k} = F(1) - F(0).$$

Then

$$\Delta_{h,k} = \int_0^1 \int_{\mathbb{R}} \dot{\mu}(y) (L_{q-1} + L_q - L_{2q-1} - L_{2q})(x, y) dy dx. \quad (54)$$

This is exactly

$$\mathcal{B}_{h,k}^{\text{ar}} - \mathcal{B}_{h,k}^{\Gamma} = \log \frac{T_{k,q-1} T_{k,q}}{T_{k,2q-1} T_{k,2q}}, \quad T_{k,N} = V_N(1)/V_N(0). \quad (55)$$

Proof. Differentiate $J_N C_N = I$ and $B_N^* M_N C_N = 0$. Since C'_N lies in the original kernel of J_N ,

$$C'_N = -B_N (B_N^* M_N B_N)^{-1} B_N^* \dot{M}_N C_N. \quad (56)$$

All derivatives are in the fixed original coordinates. Orthogonality eliminates both derivative-section cross terms in $G_N = C_N^* \dot{M}_N C_N$, so

$$G'_N = C_N^* \dot{M}_N C_N, \quad (\log V_N)' = \text{Tr}(G_N^{-1} C_N^* \dot{M}_N C_N) = \int \dot{\mu} L_N. \quad (57)$$

The four finite matrices are smooth on the compact interval and have positive minimum eigenvalues; their polynomial integrands have the moments supplied by (48). This justifies the finite differentiation and integrations. Add the four expressions with their displayed signs and integrate in x . $\square$

The derivative section (56) is not merely killed by the finite observation: it is explicitly in $\chi \mathcal{P}_{N-q}$. Equation (33) supplies its theta primitive. The lift sends this derivative relation to the receiving supported zero while retaining its original source norm.

196.1 The equal-rank cancellation is calculated before estimation

Let $K_n^\chi(x, y)$ denote the ordinary polynomial kernel of degree at most n for the measure $|\chi(c + iu)|^2 d\mu_x$. Set

$$\begin{aligned} \mathbf{R}_x(y) &= |\chi(c + iu)|^2 (K_{q-1}^\chi + K_q^\chi - K_0^\chi)(x, y), \\ \mathbf{P}_x(y) &= (K_{2q-1} - K_{q-1} + K_{2q} - K_q)(x, y). \end{aligned} \quad (58)$$

Both are nonnegative. Differences of consecutive polynomial-kernel projections are positive, so the first is the sum of the degree-0 through degree- $(q-1)$ relation kernel and the degree-1 through degree- q relation kernel. Their exact integrals are

$$\int \mathbf{R}_x d\mu_x = 2q, \quad \int \mathbf{P}_x d\mu_x = 2q. \quad (59)$$

Substitution in (53) proves, without estimating the terms independently,

$$L_{q-1} + L_q - L_{2q-1} - L_{2q} = \mathbf{R}_x - \mathbf{P}_x. \quad (60)$$

In particular every constant contrast cancels from the signed derivative. More explicitly, put $f_x = (m - r)/((1 - x)r + xm)$ where defined. The endpoint values at isolated zeros are read in the integrable products, not by an assumed bounded pointwise logarithmic derivative. Then

$$F'(x) = \frac{1}{2q} \iint (f_x(y) - f_x(v)) \mathbf{R}_x(y) \mathbf{P}_x(v) d\mu_x(y) d\mu_x(v). \quad (61)$$

The factor $1/(2q)$ follows from the two actual masses in (59); neither source measure is rescaled. Formulas (46), (54), and (58) calculate this signed term using the original g/h , its convolution, and the relation multiplication by g .

197 The retained quadratic term and a finite upper certificate

Set $H_N = B_N^* M_N B_N$. A second derivative of (57) gives the exact matrix equation

$$\boxed{G_N'' = -2C_N^* \dot{M}_N B_N H_N^{-1} B_N^* \dot{M}_N C_N.} \quad (62)$$

Thus

$$(\log V_N)'' = -\text{Tr}((G_N^{-1} G_N')^2) - 2\text{Tr}(G_N^{-1} C_N^* \dot{M}_N B_N H_N^{-1} B_N^* \dot{M}_N C_N). \quad (63)$$

Both terms are nonnegative before their minus signs. The second is the squared norm of the literal original relation correction (56). The signed four-endpoint sum contains all four occurrences with their original signs; separate concavity does not assert a sign for that sum.

There is a fully finite error bound for calculating (54), including all inverses. Work in the original top-source metric $M_{2q}(0)$. Let

$$\delta = \|M_{2q}(0)^{-1}(M_{2q}(1) - M_{2q}(0))\|_{M_{2q}(0)}, \quad a = \min\{1, \lambda_{\min}(M_{2q}(0)^{-1}M_{2q}(1))\} > 0. \quad (64)$$

The indicated endomorphism is positive self-adjoint for the stated metric; (64) does not choose a new metric. Every source and relation restriction inherits $M(x) \succeq aM(0)$ and relative derivative norm at most δ/a .

For a fixed full-column-rank inclusion U , its Gram $H(x) = U^* M(x) U$ is affine in x , and

$$\frac{d^3}{dx^3} \log \det H(x) = 2\text{Tr}((H(x)^{-1} H'(x))^3). \quad (65)$$

The four source ranks sum to $6q + 2$, and their four relation ranks sum to $2q + 2$. Therefore

$$|F'''(x)| \leq 2(8q + 4)(\delta/a)^3. \quad (66)$$

Composite midpoint quadrature applied to F' proves the explicit two-sided enclosure

$$\boxed{\left| \Delta_{h,k} - \frac{1}{m} \sum_{j=0}^{m-1} F'\left(\frac{j+1/2}{m}\right) \right| \leq \frac{8q+4}{12m^2} (\delta/a)^3.} \quad (67)$$

The proof of the midpoint constant is the second-order Taylor integral remainder on each interval of length $1/m$: its absolute integral is at most $\sup |F'''|/(24m^3)$, then sum the m intervals. Arithmetic moment enclosures must still be propagated into (64) and the sampled traces. Formula (67) is a finite analytic certificate, not a uniform asymptotic bound on its constants.

198 All inputs are the actual one-factor arithmetic moments

Define without rescaling

$$\mu_{h,n} = \frac{1}{2\pi} \int_{\mathbb{R}} t^n |g(1/2 + it)/h(1/2 + it)|^2 dt. \quad (68)$$

Every selected apparent singularity is filled by the entire quotient g/h , using the stipulated full multiplicity. Exponential Gamma decay and polynomial bounds for ζ give convergence of every finite moment. For the sum density,

$$\int y^n m_{h,k}(y) dy = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i \mu_{h,n_i}. \quad (69)$$

The source entries in the retained S -basis are

$$(M_N^{\text{ar}})_{ab} = \sum_{r=0}^a \sum_{s=0}^b \binom{a}{r} \binom{b}{s} c^{a+b-r-s} (-i)^r i^s \int y^{r+s} m_{h,k}(y) dy. \quad (70)$$

Relation matrices are $B_N^* M_N^{\text{ar}} B_N$, with B_N the actual coefficient matrix of multiplication by χ . The four endpoints use the single moment list through degree $4q$. Equation (46) is the additional, arithmetic source identity behind those relation entries; their interpretation is not supplied by generic positivity.

The Gamma input has the exact Laplace transform

$$\int e^{\theta y} r_{1/4,k}(y) dy = (2\pi)^{k/2} (\cos \theta)^{-k/2}, \quad |\theta| < \pi/2. \quad (71)$$

Its derivatives supply the other entries in the same common matrix. Thus (54)–(67) are executable using one original arithmetic moment list, the same χ and unit, and exact Gamma moments.

198.1 A closed first-pair calculation with the original relation cancellation

Take $k = 1$ and an actual simple conjugate pair $1/2 \pm i\gamma$. Then

$$h(S) = (S - 1/2)^2 + \gamma^2, \quad |h(1/2 + it)|^2 w_h(t) = w_1(t) = |g(1/2 + it)|^2 / (2\pi). \quad (72)$$

Let $a_j = \mu_{h,2j}$, $0 \leq j \leq 4$, and $b_j = \mu_{1,2j}$, $0 \leq j \leq 2$. They satisfy the exact arithmetic identities

$$b_0 = \gamma^4 a_0 - 2\gamma^2 a_1 + a_2, \quad b_1 = \gamma^4 a_1 - 2\gamma^2 a_2 + a_3, \quad b_2 = \gamma^4 a_2 - 2\gamma^2 a_3 + a_4. \quad (73)$$

Put

$$E_3 = \det \begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_3 \\ a_2 & a_3 & a_4 \end{pmatrix}.$$

Parity separates the source Hankel matrices into their even and odd blocks. Their Schur quotient by the original relation gives

$$\exp(\mathcal{B}_{h,1}^{\text{ar}}) = \frac{a_0 a_1^2 b_1^2 (b_0 b_2 - b_1^2)}{E_3 (a_1 a_3 - a_2^2)^2}. \quad (74)$$

For example the four individual volumes are

$$V_1 = a_0 a_1, \quad V_2 = \frac{a_1(a_0 a_2 - a_1^2)}{b_0}, \quad V_3 = \frac{(a_0 a_2 - a_1^2)(a_1 a_3 - a_2^2)}{b_0 b_1}, \quad (75)$$

$$V_4 = \frac{E_3(a_1 a_3 - a_2^2)}{b_1(b_0 b_2 - b_1^2)}.$$

Substitution proves (74), including every mass factor. The Gamma comparison uses the same h and (73), but the actual reference moment list is

$$(a_0^\Gamma, a_1^\Gamma, a_2^\Gamma, a_3^\Gamma, a_4^\Gamma) = \sqrt{2\pi} \left(1, \frac{1}{2}, \frac{7}{4}, \frac{139}{8}, \frac{5473}{16} \right). \quad (76)$$

The difference of the logarithms of (74) for these two inputs is the original signed four-endpoint correction. The supplied numerical script evaluates (68) for an approximation to the first actual pair, keeping the removable value and the unmodified 2ξ . It is a numerical evaluation, not an interval proof of the root, its simplicity, the infinite tail, or the final sign. A number is not promoted from an unfinished earlier run into a proved result.

199 Holonomy recovery and the exact original metric correction

The retained all-holonomy transform is

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad \widehat{\mathcal{Z}_{L,\theta}\psi}(n) = L^{-1/2} \widehat{\psi}((2\pi n + \theta)/L). \quad (77)$$

It uses the original half-density map and all relative variables. Fourier inversion in a gives an exact unitary direct integral, hence the original source mean $\int M_\theta d\theta/(2\pi) = M$. The zero mode is retained.

For the same finite J and relation columns B , the existing sections obey

$$C - C_\theta = BX_\theta, \quad G = \overline{G} + \int X_\theta^* B^* M_\theta B X_\theta \frac{d\theta}{2\pi}. \quad (78)$$

Thus no interchange of quotient minimization with phase averaging is made in (54). At each interpolation value x , the same formula applies to the direct-sum source (50). Its discrepancy is again an explicit original relation. If the already calculated source-relative holonomy errors are at most $\eta < 1$, the established quadratic comparison gives $(1 - \eta^2)G_N(x) \preceq \overline{G}_N(x) \preceq G_N(x)$. Applying determinant bounds at both endpoints of (54) supplies a total signed-correction error at most

$$4q \log \frac{1}{1 - \eta^2}. \quad (79)$$

This is a proved finite transport allowance once the original weighted-source constants and sampling errors have been evaluated. It is not the missing asymptotic estimate. Completing a fixed-phase Hilbert quotient first remains a further map with kernel $(p_{L,\theta})/(\chi)$; that operation is not used here.

200 What is newly established and what remains

The marked-base and scalar inputs are retained, and the new constructions are (27)–(41), the original-source relation norm (46), the signed common-source formula (54)–(61), its complete differentiated relation cost (62)–(63), the finite enclosure (67), and the arithmetic first-pair formula (72)–(76). They use actual packets rather than arbitrary polynomials as arithmetic data. The Gamma model remains a specified norm comparison of the same coefficient source.

Deligne’s theorem now applies to a concrete external family with degree, rank, and finite-field weight matched to the original tensor packet. Its special-fibre arrow and the cyclic transverse correction are calculated. This is an additional geometric construction over the original τ -base, not a deduction of arithmetic purity from the word “absolute” or from auxiliary finite-field purity.

The retained audit shows that auxiliary purity alone is insufficient; it does not establish failure of the marked programme. The original endpoint identity is still

$$\mathcal{B}_{h,k}^{\text{ar}} = \mathcal{B}_{h,k}^\Gamma + \Delta_{h,k}. \quad (80)$$

The signed term is now a single explicit arithmetic source/relation integral and admits finite error control. What remains unproved is a uniform upper estimate on (80) below the inherited off-line quartet threshold $4q_k \log k$ along the required growing packet/tensor family, or an independently proved faithful action-and-metric transfer producing that estimate. Neither a numerical first-pair calculation nor the finite-field weight- k statement is represented as that result.

Source and execution scope

The owner’s uploaded `arithmetic_input.tex`, `coherent_tensor_integration.tex`, `toda_cv_exact_bridge.tex`, SGA trace/period note, marked-base recovery, holonomy note, and

dated Deligne primary-source correction supply the specified inherited inputs. The repo's `tau-split-integration/RESEARCH_NOTE.md` supplies the current source/coefficient-defect distinction. The current work preserves those meanings and does not claim a new Lean or full-primary-source audit.

Deligne, *La conjecture de Weil II*, Publ. Math. IHES 52 (1980), especially 3.3.6 and 3.7.2–3.7.4, supplies the cited finite-field statements at their actual hypotheses. Standard finite-dimensional Schur complements, determinant differentiation, tensor-complex signs, and polynomial moment facts are proved here at the displayed map types; no general novelty claim is made for them.

The package's symbolic checks are finite regression tests. Numerical quadrature is separately labelled and is not an interval certificate. The execution record, rather than this prose, reports which supplied scripts actually ran.

Part III

Complete source, dual and transcript calculations

201 Verified typed maps in the original τ -base construction

This is an independent source-map verification for the transcript continuation audit. It reads the entire primary note and the entire canonical source route. It does not claim to audit the whole shared transcript or the analytic arithmetic estimates. It does not duplicate the homotopy-fibre/right-adjoint continuation assigned elsewhere.

201.1 Source pins and exact scope

1. output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md, 25,512 bytes, SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. All 611 lines read. Equation locators below refer to this body.
2. work/f1_geometry_current_source_route_20260913.md, SHA-256 a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74. Entire body read. Its dependencies were not thereby deemed independently reread.
3. output/split_zero_rh_tandem_2026-09-12/tex/support_diagrams.tex, SHA-256 7586ac17247456a55bdacfcf8b46c196c29eafd3b57846a71792e918704e069b. Entire body read to check the kernel types and support-index transport. Its labels D1–D8 are used below.

The primary note explicitly works in the algebraic category

$$\mathcal{A} = \text{Fun}(P, \text{Vect}_E), \quad E = \mathbb{C}, \quad + < \eta < \sigma, \quad - < \eta < \sigma.$$

The topology is the upper-set topology; diagrams are covariant. The calculations below keep that category, its arrows, and its cohomological degrees. The source does not replace the Schwartz topology by a theorem about its continuous dual: it calculates an algebraic right adjoint, and says so.

201.2 1. The structural map does not perform Boolean localization

Primary equations (2)–(3), lines 33–45, use the blueprint map

$$j_R : \mathbf{F}_{1,\tau} = \{\tau, 1\} \longrightarrow B_R, \quad j_R(\tau) = \tau, \quad j_R(1) = 1_R^\bullet.$$

The only additional additive relation imposed by the source blueprint is that its empty sum represents τ . The target retains that relation, its own addition relations, and its original multiplicative monoid, so the displayed map preserves all source relations. Every proper prime contains τ and excludes the unit. Its inverse image is consequently exactly $\{\tau\}$. This proves the structural map to the one-point base.

Primary equations (6)–(7), lines 64–80, concern a separate map $\chi_R : G(R) \rightarrow \mathbb{B}$, with $\chi_R(\tau) = 0$, $\chi_R(r^\bullet) = 1$, and localization at $e = 0_R^\bullet$. Inverting the idempotent e gives $e=1$; then $er^\bullet = e$ gives $r^\bullet = 1$. No cancellation with τ occurs. The source already proves this localization, and does not use it as the definition of the structural base pushforward.

The canonical route supplies the exact comparison for every $G(R)$ -semimodule M :

$$\Phi : \mathbb{B} \otimes_{G(R)} M \longrightarrow eM, \quad b \otimes m \longmapsto b \cdot em.$$

Here eM is a Boolean semimodule under its original addition. For a supported scalar a , $ea = e$, so both sides of the balancing relation map to $b \cdot em$; for $a=\tau$ both map to the global zero. Additivity in m follows from distributivity of e . Additivity in b uses $em + em = em$. Its inverse is

$$\Psi : eM \longrightarrow \mathbb{B} \otimes_{G(R)} M, \quad l \longmapsto 1 \otimes l.$$

Indeed $\Phi\Psi(l) = el = l$, and $\Psi\Phi(1 \otimes m) = 1 \otimes em = \chi_R(e) \otimes m = 1 \otimes m$; tensors with Boolean coefficient 0 vanish. Both formulas commute with every split-linear map. The comparison is therefore a natural isomorphism, not a claimed lack of relation between the two constructions.

201.3 2. The global pushforward and its canonical derived restriction

Primary equations (17)–(20), lines 200–242, are correct with their displayed domains and signs.

For x in P , $P_x(y) = E$ when $x \leq y$, and 0 otherwise, with identity maps on nonzero arrows. A natural transformation $P_x \rightarrow F$ is determined by the image v of 1 at x ; at y above x its value must be $r_{xy}v$. Conversely those values are natural. Thus

$$\mathrm{Hom}_{\mathcal{A}}(P_x, F) = F_x.$$

Evaluation is exact because kernels and cokernels in the diagram category are computed at each point. This proves projectivity of P_x .

The source resolution is

$$0 \longrightarrow P_\eta \xrightarrow{\delta} P_+ \oplus P_- \xrightarrow{\varepsilon} \mathbf{E} \longrightarrow 0, \quad \delta(v) = (v, -v), \quad \varepsilon(u, v) = u + v.$$

At $+$ its sequence is $0 \rightarrow 0 \rightarrow E \xrightarrow{1} E \rightarrow 0$, and at $-$ it is the same sequence in the second summand. At η and σ its sequence is $0 \rightarrow E \rightarrow E^2 \rightarrow E \rightarrow 0$, with kernel of addition exactly $\{(v, -v) : v \in E\}$. Restriction maps commute with both displayed maps because each nonzero component is an identity. This checks exactness and naturality on every point and arrow.

Compatible global sections of F are determined by $(u, v) \in F_+ \oplus F_-$ with $r_+u = r_-v$. Their η -value is this common vector and their σ -value is its image under $r_{\eta\sigma}$. Hence $\Gamma(P, F) = \mathrm{Hom}(\mathbf{E}, F)$, and the resolution computes

$$A_\tau(F) = R\Gamma(P, F) \simeq C_F, \quad C_F^0 = F_+ \oplus F_-, \quad C_F^1 = F_\eta, \quad d_F(u, v) = r_+u - r_-v.$$

The convention is the displayed resolution coboundary by precomposition. The degrees and the differential are those of the source, not an identification with F_σ .

Put $r = r_{\eta\sigma}$. The two maps $c_+, c_- : C_F \rightarrow F_\sigma[0]$ have degree-zero parts

$$c_+^0(u, v) = rr_+u, \quad c_-^0(u, v) = rr_-v,$$

and degree-one parts 0. They are chain maps because the target has no term in degree 1. With degree -1 homotopy h defined by $h^1 = r : F_\eta \rightarrow F_\sigma$,

$$(c_+ - c_-)^0 = h^1 d_F, \quad (c_+ - c_-)^1 = 0 = d_{F_\sigma} h^1.$$

They therefore give a single canonical derived morphism $\mathrm{res}_\sigma : A_\tau(F) \rightarrow F_\sigma[0]$. Its action on a global compatible section is its actual σ -value. This proves the comparison, including its sign, without substituting the target for the source.

201.4 3. The actual arithmetic cohomology and its three support fibres

Primary equations (10)–(15), (21)–(25), retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx,$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0 \right\},$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad J\Theta = \Theta\mathcal{F}.$$

Its joint diagram is $(V, V, \mathcal{B}, 0)$, with differential

$$d(\phi, \psi) = \Theta\phi - J\Theta\psi = \Theta(\phi - \widehat{\psi}).$$

Consequently its image is exactly ΘV : one inclusion follows from Fourier invariance of V , and the reverse inclusion follows by using $(\phi, 0)$. Fourier invariance follows from evenness and $\mathcal{F}^2\phi(x) = \phi(-x)$, which exchanges the two vanishing moments and gives $\mathcal{F}^2 = 1$ on V . Using the source's established injectivity of Θ gives

$$\ker d = \{(\widehat{\psi}, \psi) : \psi \in V\}, \quad H^1 C_{\mathcal{T}} = Q = \mathcal{B}/\Theta V.$$

This verifier does not replace the source's Mellin/Poisson proof of injectivity by an independent analytic claim. The algebraic identification just made uses that named source result at its exact place.

For the $+$ singleton mask the differential is Θ . For the $-$ singleton mask it is $-J\Theta = -\Theta\mathcal{F}$. Each is injective, and each has image ΘV , so its H^0 is 0 and its H^1 is Q . The inclusions from either singleton to the joint mask are coordinate inclusions in degree 0 and identity in degree 1; they are chain maps since the joint differential restricts to the indicated signed singleton differential. Their induced H^1 maps are the identity of Q . The empty support diagram is zero.

Thus the entire H^1 support diagram has zero bottom fibre and Q at all three nonempty masks, with identity transitions among nonempty comparable masks. Its reconstruction is exactly

$$M = \widetilde{H}_{\mathcal{T}}^1 = \{\tau\} \sqcup \coprod_{\emptyset \neq A \subseteq \{+, -\}} \{A\} \times Q.$$

The linear map induced by res_{σ} on H^1 is $Q \rightarrow H^1(0[0]) = 0$ at each nonempty mask. Reconstructing that same-mask diagram map gives

$$\epsilon_M : M \longrightarrow eM, \quad \tau \longmapsto \tau, \quad (A, q) \longmapsto (A, 0).$$

This is the precise typed reading of the source's statement immediately after (25), at lines 283–288. It is not a degree-one map to the unshifted linear stalk treated as if that stalk had nonzero H^1 . The supported skeleton is obtained by the explicit support reconstruction. Its fibres are

$$\epsilon_M^{-1}(\tau) = \{\tau\}, \quad \epsilon_M^{-1}(A, 0) = \{A\} \times Q \quad (A \neq \emptyset).$$

Every arithmetic amplitude remains in its recorded fibre. The projection and its section $(A, 0) \mapsto (A, 0)$ prove the relation to the boundary skeleton exactly.

201.5 4. The right adjoint is computed algebraically on that same chart

Primary equations (26)–(30), lines 294–347, give the correct right-adjoint object.

Define $I_x(W)(y) = W$ if $y \leq x$ and 0 otherwise. A map $F \rightarrow I_x(W)$ is uniquely determined by $\ell : F_x \rightarrow W$: its y -component for $y \leq x$ is ℓr_{yx} , and its other components are 0. These components are natural on every arrow. Therefore

$$\mathrm{Hom}_{\mathcal{A}}(F, I_x(W)) = \mathrm{Hom}_E(F_x, W).$$

Every vector space is injective: a linear map defined on a subspace extends after extending a basis of that subspace to a basis of the whole space. Evaluation at x is exact, so $I_x(W)$ is injective in $\mathcal{A}$.

The source complex is

$$K_\tau(W)^{-1} = I_\eta(W), \quad K_\tau(W)^0 = I_+(W) \oplus I_-(W), \quad d_K = (+\mathrm{res}, -\mathrm{res}).$$

A bounded complex of injectives calculates the derived Hom from a sheaf here. Evaluation of its Hom complex gives

$$R\mathrm{Hom}_{\mathcal{A}}(F, K_\tau(W)) = \left[\mathrm{Hom}_E(F_\eta, W) \xrightarrow{\ell \mapsto (\ell r_+, -\ell r_-)} \mathrm{Hom}_E(F_+, W) \oplus \mathrm{Hom}_E(F_-, W) \right],$$

in degrees -1 and 0 . The ordinary Hom complex from C_F to $W[0]$ has exactly these terms. For a degree -1 map ℓ its differential is $-(-1)^{-1} \ell d_F = \ell d_F$, giving exactly the displayed pair. This proves the chain isomorphism in (28), naturally in F and W , and hence the stated adjunction.

For bounded coefficient complexes the same result follows by taking the finite total complexes, preserving the original differential of W . An explicit convention is

$$K_\tau(W)^n = I_\eta(W^{n+1}) \oplus I_+(W^n) \oplus I_-(W^n), \quad d(s, u, v) = (-d_W s, \mathrm{res}_+ s + d_W u, -\mathrm{res}_- s + d_W v).$$

Its square is zero: the first component is $d_W^2 s$, and each other component cancels its two restriction-of- $d_W s$ terms, leaving $d_W^2 u$ or $d_W^2 v$. This is the signed totalization of the displayed two-term adjoint.

At $+$, $-$, η , σ , respectively, the W -vector-space stalk complexes are

$$[W \xrightarrow{1} W], \quad [W \xrightarrow{-1} W], \quad W[1], \quad 0.$$

The inclusion $S_\eta W[1] \rightarrow K_\tau(W)$, identity at η and zero at the other points, is natural: at each arrow entering η its source is zero; at arrows leaving η its target is zero; all other squares have a zero entry on the source side. It induces an isomorphism in cohomology at every stalk, since the $\pm$ complexes have invertible differentials. Hence

$$K_\tau(W) \simeq S_\eta W[1]$$

by an explicit quasi-isomorphism. In particular $K_\tau(W)_\sigma = 0$ does not imply that the right adjoint is zero, nor that its Hom from the arithmetic sheaf is zero. Its degree -1 Hom cohomology is

$$\{\ell : \mathcal{B} \rightarrow W : \ell \Theta V = 0\} \cong \mathrm{Hom}_E(Q, W).$$

The inverse correspondence sends a quotient functional $\lambda : Q \rightarrow W$ to $\lambda \circ q$; the forward correspondence sends ℓ to $[F] \mapsto \ell(F)$. Equality of values on representatives is exactly $\ell \Theta V = 0$.

This verifies the target of the full finite-jet/residue morphism (43), whose source has been retained as $\overline{A_Z}$ and whose codomain is $H^{-1} R\mathrm{Hom}_{\mathcal{A}}(\mathcal{T}, K_\tau(\mathcal{L}_1))$. The source's surjective $J_Z : Q \rightarrow A_Z$ and perfect finite residue pairing are the specified reasons its precomposition map is injective. Neither a continuous-dual identification nor involutive Verdier duality is supplied by this algebraic adjunction.

201.6 5. Exact categorical and all-support kernels

Use D1–D8 with $M = \mathcal{T}(L, V)$, $N = \mathcal{T}(L', W)$, and a split-linear map

$$F = \mathcal{T}(\alpha, f), \quad F(l, x) = (\alpha(l), f_l x).$$

The target's global zero is $(\perp', 0)$, while its support skeleton consists of all $(k, 0)$. Consequently

$$F(l, x) = 0_N \iff \alpha(l) = \perp' \text{ and } f_l x = 0,$$

$$F(l, x) \in eN \iff f_l x = 0.$$

Writing $L_0 = \alpha^{-1}(\perp')$, these equalities give

$$\ker_{G(R)} F = \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}),$$

$$F^{-1}(eN) = \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN.$$

The original transition maps restrict to these kernels: naturality gives $f_k \rho_{lk} x = \rho'_{\alpha(l), \alpha(k)} f_l x = 0$. Also L_0 is closed under joins because α preserves joins. Therefore both displayed carriers are sub-semimodules. Any map with the corresponding vanishing property has its image in the indicated carrier and hence factors uniquely through the inclusion. This proves the two universal properties, not just set identities. Their exact comparison is the inclusion induced by $L_0 \hookrightarrow L$.

For ϵ_M above, α is the identity and every fibre map is the zero map $Q \rightarrow 0$. Its bottom fibre is zero. Thus

$$\ker_{G(E)} \epsilon_M = \{\tau\}, \quad M \times_{eM} eM = M.$$

A trivial categorical kernel here cannot establish disappearance of supported arithmetic cohomology: the second displayed object is the entire original arithmetic carrier. The formula states the scope of the trivial-kernel result exactly.

The quotient q at a fixed nonempty mask has source $\mathcal{B}$, boundary subspace ΘV , and target Q . Its supported form sends (A, F) to $(A, [F])$. In the all-support sense its kernel fibre is ΘV , while every boundary maps to $(A, 0)$, never τ . This follows from $[F] = 0 \iff F \in \Theta V$. More generally, a cochain map has

$$\ker H(f) = \{z \in \ker d_C : fz \in \text{im } d_D\} / \text{im } d_C.$$

The map sends the class of z to its cohomology class. It is well-defined because boundaries map to boundaries, onto because every cohomology class has a cycle representative, and injective because the zero class is precisely a boundary. This is D7, and reconstructing it at every label gives the all-support object in D8.

201.7 6. Dual support arrows: the exact reversal and its limit

Primary equation (31), lines 349–356, retains the correct map for every inclusion $A \subseteq B$:

$$\rho_{AB}^\vee : \text{Hom}_E(H_B, W) \longrightarrow \text{Hom}_E(H_A, W), \quad \ell \longmapsto \ell \rho_{AB}.$$

For $A \subseteq B \subseteq C$, $\rho_{AB}^\vee(\rho_{BC}^\vee \ell) = \ell \rho_{BC} \rho_{AB} = \ell \rho_{AC}$, proving functoriality on L^{op} . The identity arrow acts as the identity. Thus the exact dual-index diagram exists.

It is not a covariant diagram on the original inclusion-ordered L with the same arrow directions. The source does not assert that it is. A further reconstruction into split semimodules must declare its index semilattice. For the finite Boolean lattice in this source, L^{op} itself has join given by intersection and bottom $\{+, -\}$. Applying D1–D5 to this opposite lattice is valid and keeps the dual arrows, but its bottom fibre is then $\text{Hom}_E(H_{\{+, -\}}, W)$, not the original empty-mask fibre. In degree 1 this is $\text{Hom}_E(Q, W)$. D1–D5 explicitly allow such nonzero bottom fibres. Thus an argument that applies a theorem requiring a singleton bottom fibre to this reconstructed dual has changed a hypothesis and must calculate the actual bottom fibre. The original dual map (43), formed label by label with precomposition, does not require that unsupported identification.

201.8 7. A completed exact jet map that makes the eigenline wording precise

Primary (40), (44)–(45), and (49) retain full nilpotent jets. The phrase “residue-value eigenvector” in (49) is read through the following exact maps, so it does not discard the nilpotents of (40).

At an actual zero ρ of multiplicity m , keep the original coordinate $z = s - \rho$ and factor

$$g(\rho + z) = u(z)z^m, \quad u(0) \neq 0.$$

Then $A_\rho = \mathcal{O}_\rho/(g) = \mathcal{O}_\rho/(z^m)$, with the identity on germs giving the latter equality of ideals because u is a unit. Its reduction map is

$$q_\rho : A_\rho \longrightarrow E, \quad [h] \longmapsto h(\rho).$$

The endomorphism J_g of (44) factors as an actual quotient followed by an injection:

$$A_\rho \xrightarrow{q_\rho} E \xrightarrow{t_\rho} A_\rho, \quad t_\rho(c) = c[g'], \quad J_g = t_\rho q_\rho.$$

Indeed

$$g' = u'(z)z^m + mu(z)z^{m-1} \equiv mu(0)z^{m-1} \pmod{z^m}.$$

Writing $h = h(0) + z\tilde{h}$, multiplication by $[g']$ kills the second summand, so $[g'h] = h(0)[g']$, proving factorization. Since $mu(0) \neq 0$, $[g'] \neq 0$; therefore t_ρ is injective. Its image is exactly

$$\text{Ann}_{A_\rho}(z) = Ez^{m-1}.$$

To verify this equality, multiply $\sum_{j=0}^{m-1} c_j z^j$ by z . The product is zero modulo z^m exactly when $c_0 = \dots = c_{m-2} = 0$. This includes $m=1$, when the annihilator is all of $A_\rho = E$. Consequently

$$\ker J_g = \ker q_\rho = zA_\rho, \quad \text{im } J_g = \text{Ann}_{A_\rho}(z).$$

The actual scaling operator remains

$$U_a = a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} N^j, \quad N[z^j] = [z^{j+1}], \quad a > 0.$$

Reduction gives $q_\rho U_a = a^\rho q_\rho$, since every $j>0$ term has zero constant coefficient. Multiplication of $t_\rho(c)$, which lies in Ez^{m-1} , by a $j>0$ term is zero, so $U_a t_\rho(c) = t_\rho(a^\rho c)$. Both maps are equivariant. Thus the residue-value representation and its socle image inside the full jet are linked by the explicit nonzero map t_ρ . When $m > 1$, the constant lift 1 of a residue value is not generally an eigenvector of U_a for $a \neq 1$, because its z coefficient is $a^\rho \log a \neq 0$. The socle image $t_\rho(1)$ is an actual eigenvector. No full jet was replaced by its reduction in proving that fact.

The tensor of r copies of this explicit eigenvector is nonzero in $A_\rho^{\otimes Er}$: the coefficient of $(z^{m-1})^{\otimes r}$ is $(mu(0))^r \neq 0$. Its action on that finite tensor quotient is exactly $a^{r\rho}$. Its precise relation to primary (47) is the equivariant surjection

$$Q^{\otimes Er} \xrightarrow{(\text{pr}_\rho J_Z)^{\otimes r}} A_\rho^{\otimes Er}.$$

Here Z contains ρ , $J_Z : Q \rightarrow A_Z$ is the source’s full jet quotient, and pr_ρ is its block projection. Each map is equivariant by the full jet scaling formula (40). The first is surjective by the source’s jet-surjectivity result, the block projection is surjective, and their tensor is surjective because each simple target tensor has a tensor of preimages and such tensors span the target. This proves the exact relation to $Q^{\otimes r}$.

The quotient-and-socle argument of this subsection, taken alone, does not establish an equivariant lift to Q . The original arithmetic section supplies that additional map: Subsection 201.10 below retains its full Taylor unit and proves its scaling equivariance, giving the actual eigenline inside Q

and its tensor inside $Q^{\otimes r}$. Thus this is a completed source lift, with the quotient-only argument and the additional original section both specified. On the finite quotient the tensor modulus expression is

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2 \operatorname{Re} \rho - 1) \log a \quad (a > 1),$$

with its original action and coefficient line. This is an identity on the actual eigenline embedded in $A_\rho^{\otimes r}$, which is a quotient of the tensor cohomology in (47); it proves no upper bound on that identity. At every nonempty supported mask, q_ρ , t_ρ , and J_g lift by retaining the mask, so a killed nilpotent maps to the supported zero of its mask.

201.9 Audit uses

The source already contains the structural base, global pushforward, derived restriction, right adjoint, full supported Q , dual-index arrows, and the finite-jet morphism into that right adjoint. A passage that calls any of these missing must be tested against the equation locators above. The algebraic calculation does not supply analytic bounds; that limitation has the exact scope stated in the primary note. A boundary-stalk vanishing or a trivial categorical kernel does not reach the original supported Q , because the explicit maps above retain the whole Q fibre. A duality argument must retain the dual arrow direction, its coefficient line, and its actual algebraic category. A tensor argument can use the exact socle embedding into the finite tensor quotient above while retaining every nilpotent in that ambient jet object; it must retain the displayed quotient map from the full tensor cohomology. Subsection 201.10 constructs the source eigenvector using the original arithmetic section and its proved finite scaling cocycle, rather than inferring it from the quotient alone.

201.10 8. The original arithmetic section completes the source eigenline lift

The section used here is already constructed in the retained arithmetic source, equations (A7)–(A11) of output/split_zero_rh_tandem_2026-09-12/tex/arithmetic_input.tex, SHA-256 a50b0fb587b644c4ea94dba45b2b423d038f2bfbef7fef188543939ba850e58d2. The group-level source integral is proved in Sections 2–3, equations (9)–(19), of output/tau_f1_transcript_audit_2026-09-13/proofs/theta_scaling_cocycle.md, SHA-256 5aae80114cbf3cb0c3bec25e4e655b37117a932c1a92132161c162393b7ea377. We retain their maps and give the source-lift proof explicitly. The earlier quotient calculation did not prove this additional map, but the map is present in the original construction and is completed here at the full scaling-group level.

The original packet and its unit. Let Z be a nonempty finite set of actual nontrivial zeros of the original $g = 2\xi$, with each complete multiplicity m_ρ , and set

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[X]/(h), \quad A = M_X.$$

The letter X here is the original polynomial variable denoted t in (A10)–(A11); the notation E_h distinguishes that source's finite packet E from the ground field $E = \mathbb{C}$ used earlier in this verification. The complex Mellin coordinate remains s . At each selected zero, the quotient

$$v(s) = g(s)/h(s), \quad \varepsilon = j_h(h/g) \in E_h$$

is holomorphic and nonzero. Indeed, in the original coordinate $z = s - \rho$, both g and h have order exactly m_ρ , so their quotient has a nonzero constant coefficient. A truncated power series with nonzero constant coefficient is invertible by recursively solving for its inverse coefficients. The Chinese-remainder decomposition therefore makes ε a unit, with $j_h(v)\varepsilon = 1$. No coefficient of this unit is replaced by 1. For $u \in E_h$, let $R_Z(u)$ be the unique polynomial of degree less than d representing εu . Polynomial division by the monic h proves this existence and uniqueness.

The retained source is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP.$$

Its moments and this Mellin identity are the proved original (A3), with the same Fourier convention and both factors of two retained. Write

$$D = -x\partial_x, \quad p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)|, \quad b \in \mathbb{Z}, \quad k \geq 0.$$

These are the defining seminorms of the original $\mathcal{B}$. The operator inverse is the actual inverse

$$S_h^{\mathcal{B}} : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B} \quad \text{of} \quad h(D) : \mathcal{B} \longrightarrow \ker(j_h \mathcal{M}).$$

Equations (A5)–(A7), or equation (9) in the finite-cocycle chapter, prove its range, injectivity, and continuity. The explicit one-factor inverse used in that proof is

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad \mathcal{M}F(\rho) = 0.$$

It satisfies $(D-\rho)S_\rho F = F$. The zero moment writes its upper integral as the negative lower integral; these two expressions give, respectively, all decay bounds at infinity and zero. More explicitly, for $N > \operatorname{Re} \rho$ and $N + \operatorname{Re} \rho > 0$,

$$|S_\rho F(x)| \leq \frac{p_{N,0}(F)}{N - \operatorname{Re} \rho} x^{-N}, \quad |S_\rho F(x)| \leq \frac{p_{-N,0}(F)}{N + \operatorname{Re} \rho} x^N.$$

Using the first bound for $x \geq 1$, the second for $x \leq 1$, and $DS_\rho F = \rho S_\rho F + F$ proves every defining derivative seminorm bound. The homogeneous solution $cx^{-\rho}$ belongs to $\mathcal{B}$ only when $c = 0$. Integration by parts gives $\mathcal{M}S_\rho F(s) = \mathcal{M}F(s)/(s - \rho)$, with the removable value filled in. Iteration removes one of the prescribed zero factors at each step and preserves every other vanishing order. This proves the stated inverse on the complete jet kernel and injectivity of $h(D)$, without assuming an unproved operator range.

Define the original section exactly as in (A10):

$$R_{\text{ref}} u = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M}F, \quad q : \mathcal{B} \longrightarrow Q = \mathcal{B}/\Theta V, \quad \sigma = qR_{\text{ref}}.$$

The input to $S_h^{\mathcal{B}}$ has Mellin transform $gR_Z(u)$, which has all required zero jets, so this definition has the stated domain for every u . The inverse formula gives

$$\mathcal{M}R_{\text{ref}} u = vR_Z(u), \quad J_Z R_{\text{ref}} u = j_h(v)\varepsilon u = u.$$

For $\phi \in V$, the original Mellin identity $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ holomorphic at the selected zeros, gives $J_Z\Theta\phi = 0$. Thus J_Z factors as $\overline{J}_Z q$, and

$$\overline{J}_Z \sigma = 1_{E_h}, \quad h(D)\sigma(u) = q\Theta(R_Z(u)(D)\phi_*) = 0.$$

The displayed left inverse proves that $\sigma : E_h \hookrightarrow Q[h(D)]$ is injective. This is an actual section inside the arithmetic cohomology, not a section inferred from finite-dimensional quotient data alone.

All positive dilations act equivariantly after the original boundary. Set $\ell(u) = [X^{d-1}]R_Z(u)$. Since multiplication by X commutes with the unit ε , the polynomial $XR_Z(u) - R_Z(Au)$ is divisible by h . Its degree is at most d ; its leading coefficient is $\ell(u)$. Hence the exact polynomial identity is

$$XR_Z(u) - R_Z(Au) = h(X)\ell(u).$$

Applying the defining inverse, or multiplying by the injective $h(D)$, gives the original boundary coefficient (A11):

$$DR_{\text{ref}} - R_{\text{ref}}A = \Theta\phi_*\ell.$$

For a single simple zero this coefficient is $\ell(u) = u/g'(\rho) = u/(2\xi'(\rho))$; its arithmetic unit and factor of two remain present.

For $a > 0$ let $\lambda = \log a$, $T_r = e^{rA}$, and retain $U_a F(x) = F(x/a)$. Define

$$c_a(u) = \int_0^\lambda U_{e^{\lambda-r}} \phi_* \ell(T_r u) dr \in V.$$

When $\lambda < 0$ this is the oriented integral $\int_0^\lambda = -\int_\lambda^0$. No orientation changes are hidden. Fix any norm on the finite-dimensional E_h , only for this convergence estimate, with the induced operator norms for A and ℓ . For the original Schwartz seminorms $q_{N,k}(\phi) = \sup_x (1 + |x|)^N |\partial_x^k \phi(x)|$,

$$q_{N,k}(U_b \phi) \leq b^{-k} \max(1, b)^N q_{N,k}(\phi).$$

On the interval between 0 and λ , this inequality and $\|T_r\| \leq e^{|r|\|A\|}$ give

$$q_{N,k}(c_a(u)) \leq |\lambda| e^{(N+k+\|A\|)|\lambda|} q_{N,k}(\phi_*) \|\ell\| \|u\|.$$

The integrand is continuous in every seminorm. Its Riemann sums converge in the complete closed Schwartz subspace V : uniform continuity on the compact parameter interval makes them Cauchy in each seminorm, and their limits retain evenness and both zero moments. Differentiating dilation is legitimate in every seminorm, as the difference quotient's integral remainder is bounded by finitely many seminorms of $D^2\phi$ on that compact interval. The same argument applies in $\mathcal{B}$, using $p_{b,k}(U_a F) = a^b p_{b,k}(F)$. Consequently

$$\frac{d}{dr}(U_{e^{\lambda-r}} R_{\text{ref}} T_r u) = -U_{e^{\lambda-r}} (DR_{\text{ref}} - R_{\text{ref}}A) T_r u = -\Theta U_{e^{\lambda-r}} \phi_* \ell(T_r u).$$

Integrating in the actual $\mathcal{B}$ seminorms and using continuity of Θ proves, for every $a > 0$,

$$U_a R_{\text{ref}} - R_{\text{ref}} a^A = \Theta c_a, \quad a^A = e^{(\log a)A}.$$

This is precisely equation (15) of the finite-cocycle calculation. The original relation space is invariant because $U_a \Theta = \Theta U_a$, so U_a induces $\overline{U}_a$ on Q . Applying q to the identity gives the required full intertwining

$$\boxed{\overline{U}_a \sigma = \sigma a^A \quad \text{for every } a > 0.}$$

Inserting $U_a R_{\text{ref}} a^A$ into the defect for ab and using injectivity of Θ also gives the retained cocycle law $c_{ab} = U_a \circ c_b + c_a \circ b^A$. This proves group equivariance of the section in Q ; it does not assert strict equivariance of R_{ref} in $\mathcal{B}$. Indeed the displayed boundary is exactly what connects those two maps.

The exact full-jet insertion and the source eigenline. The identity on germs gives the Chinese-remainder isomorphism

$$E_h \xrightarrow{\sim} \bigoplus_{\nu \in Z} A_\nu, \quad A_\nu = \mathcal{O}_\nu / (g) = \mathcal{O}_\nu / ((s - \nu)^{m_\nu}).$$

At each centre the ideals generated by h and g both equal the displayed power ideal, since their remaining factors are units. For a fixed $\rho \in Z$, let $\iota_\rho : A_\rho \hookrightarrow E_h$ be the inverse Chinese-remainder map with all other blocks zero. An explicit formula, retaining all Taylor coefficients, uses $H_\rho(X) = h(X)/(X - \rho)^{m_\rho}$: for a germ representative f ,

$$\iota_\rho[f] = \left[H_\rho(X) \sum_{j=0}^{m_\rho-1} \frac{1}{j!} \frac{d^j}{dX^j} \left(\frac{f(X)}{H_\rho(X)} \right) \right]_{X=\rho} (X - \rho)^j \Big|_h.$$

At ρ its jet is f ; at every other selected centre the factor H_ρ gives all required zero jets. A change of representative f by a germ of order at least m_ρ leaves the displayed derivatives unchanged. These facts prove that this is the exact well-defined insertion, inverse to its block projection. In each block,

$$A|_\nu = \nu I + N_\nu, \quad a^A|_\nu = a^\nu \sum_{j=0}^{m_\nu-1} \frac{(\log a)^j}{j!} N_\nu^j.$$

Thus ι_ρ intertwines the full block scaling; it does not replace that block by its residue-value quotient.

Retain the preceding subsection's actual factorization $g(\rho + z) = u_\rho(z)z^m$, $m = m_\rho$, and put

$$t_\rho : \mathbb{C} \longrightarrow A_\rho, \quad t_\rho(c) = c[g'] = cmu_\rho(0)z^{m-1}.$$

Its domain has action $c \mapsto a^\rho c$. The previous calculation proves that t_ρ is an injective intertwiner, since $mu_\rho(0) \neq 0$ and $N_\rho z^{m-1} = 0$. Therefore the complete original source lift is

$$L_\rho = \sigma \iota_\rho t_\rho : (\mathbb{C}, a^\rho) \hookrightarrow (Q, \bar{U}_a), \quad w_\rho = L_\rho(1).$$

Its equivariance follows by composing the three proved intertwiners. Its nonvanishing follows without any choice of a quotient representative: $\bar{J}_Z w_\rho = \iota_\rho[g'] \neq 0$. For an explicit left inverse, let pr_ρ denote the block projection and set

$$\Lambda_\rho(q_0) = \frac{[z^{m-1}] \text{pr}_\rho \bar{J}_Z(q_0)}{mu_\rho(0)}.$$

Then $\Lambda_\rho L_\rho = 1_{\mathbb{C}}$. This linear left inverse is used to certify injectivity; no equivariance of Λ_ρ on the whole Q is asserted. In particular, for $m > 1$ it is not residue evaluation, which would vanish on the socle. The construction keeps precisely the nilpotent coefficient required for a nonzero lift.

The full tensor lift in the original top cohomology. For every integer $r \geq 1$, tensor over $\mathbb{C}$ to obtain

$$L_\rho^{\otimes r} = \sigma^{\otimes r} \iota_\rho^{\otimes r} t_\rho^{\otimes r} : \mathbb{C}^{\otimes r} \hookrightarrow Q^{\otimes r}, \quad w_\rho^{\otimes r} = L_\rho^{\otimes r}(1^{\otimes r}).$$

The map $\Lambda_\rho^{\otimes r}$ is a left inverse, since the tensor of the identities $\Lambda_\rho L_\rho = 1$ is the identity. Thus this is a nonzero tensor in the source cohomology, not only in its finite quotient. Composing the exact scaling intertwiners gives

$$\bar{U}_a^{\otimes r} w_\rho^{\otimes r} = a^{r\rho} w_\rho^{\otimes r} \quad (a > 0).$$

The top-degree comparison in primary (47) identifies this tensor with an actual class of $H^r \mathbf{A}_{\tau,r}(\mathcal{T}^{\boxtimes r})$. Explicitly, put

$$b_\rho = R_{\text{ref}} \iota_\rho[g'] \in \mathcal{B}.$$

Its representative in the original tensor complex is $b_\rho^{\otimes r} \in \mathcal{B}^{\otimes r}$. Top boundaries are sums of tensors with a ΘV factor. The map $J_Z^{\otimes r}$ kills every such tensor, whereas it sends this representative to $(\iota_\rho[g'])^{\otimes r} \neq 0$. This also proves its nonzero top cohomology class directly in the original complex.

The group action on this representative has the original source boundary and tensor signs. Put $c = c_a(\iota_\rho[g'])$. Then $U_a b_\rho - a^\rho b_\rho = \Theta c$. In the joint two-leg complex, $d(c, 0) = \Theta c$. With each $\mathcal{B}$ factor in degree one, form the degree $r - 1$ tensor

$$H_{a,r} = \sum_{j=1}^r (-1)^{j-1} (U_a b_\rho)^{\otimes(j-1)} \otimes (c, 0) \otimes (a^\rho b_\rho)^{\otimes(r-j)}.$$

Only the degree-zero factor contributes to its differential, with sign $(-1)^{j-1}$ from its preceding degree-one factors. It cancels the displayed coefficient. The resulting sum telescopes, giving

$$dH_{a,r} = (U_a b_\rho)^{\otimes r} - (a^\rho b_\rho)^{\otimes r}.$$

This is the cochain-level proof of the stated tensor eigencharacter on the original complex, retaining its source boundary rather than postulating a strict invariant representative.

Finally, the source character now has the exact original modulus identity

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2 \operatorname{Re} \rho - 1) \log a \quad (a > 1)$$

on the nonzero class $w_\rho^{\otimes r}$ itself. The actual moment line still has character a ; no norm or mass has been changed. This identity supplies the source eigenline requested by the tensor construction. It does not supply an upper estimate for its modulus. At each original nonempty support mask, σ , ι_ρ , and t_ρ lift by keeping that mask and its labelled zero. Their cohomological lifts commute with the identity transports among the nonempty Q -fibres; external τ remains external absence. The joint tensor class above therefore belongs to the retained supported arithmetic cohomology over the same absolute base.

202 Scope and source pins

This calculation completes the specific promise in transcript A1545, node bacc895-0427-42ee-b6f6-4c61752dfde2, to compute “the full τ -chart dual, including bottom fibers created by reversed support arrows.” It uses the actual four-point chart, coefficient spaces, differential, and right adjoint of the retained *Arithmetic cohomology over the τ -base*, NOTE.md, SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e5702c27db97e7b63c3. The complete transcript segment U0041–U0054 has SHA-256 e79b46722d9a2ee95b68ae6512f37a19fd41e4b3ecc4e53583cd357d0442c251. The reconstruction convention allowing a nonzero bottom fibre is the fully read `tex/support_diagrams.tex`, equations (D1)–(D8). No generic-stalk restriction replaces global cohomology here.

203 1. Original objects, signs, and both index sets

Retain $E = \mathbb{C}$, the even Schwartz space

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx,$$

and

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}.$$

Write $\mathcal{F} : V \rightarrow V$ for this Fourier transform; on this even space $\mathcal{F}^{-1} = \mathcal{F}$. Retain

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad J\Theta = \Theta\mathcal{F}, \quad Q = \mathcal{B}/\Theta V,$$

and denote the quotient by $q : \mathcal{B} \rightarrow Q$. The source proves injectivity of Θ from its Mellin identity in $\Re s > 1$; no zero in the critical strip is assumed absent.

The geometric index poset remains $P = \{+, -, \eta, \sigma\}$, with $+ < \eta < \sigma$ and $- < \eta < \sigma$. The support index poset is separately

$$L = \{\emptyset, +, -, J_0\}, \quad J_0 = \{+, -\},$$

ordered by inclusion. The symbol J_0 denotes a support mask; J above denotes the original inversion operator. At active mask A , the coefficient diagram $\mathcal{T}_A$ retains precisely the source legs belonging to A , has overlap $\mathcal{B}$, and has zero vector stalk at σ . At the empty mask it is the zero diagram. The source computes

$$\mathbf{A}_\tau(F) = R\Gamma(P, F) \simeq [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta], \quad K_\tau(W) = [I_\eta(W) \longrightarrow I_+(W) \oplus I_-(W)],$$

with degrees 0, 1 and $-1, 0$, respectively, and differential signs $+$, $-$. Its proved adjunction gives

$$D_A(W) := R\mathrm{Hom}_{\mathrm{Fun}(P, \mathrm{Vect}_E)}(\mathcal{T}_A, K_\tau(W)) \cong R\mathrm{Hom}_E(\mathbf{A}_\tau(\mathcal{T}_A), W).$$

Here W is an ordinary coefficient vector space, and $X^\vee = \mathrm{Hom}_E(X, W)$ throughout the algebraic calculation.

204 2. The complete complexes and every support arrow

The following are literal representatives, not just their cohomology:

A	D_A^{-1}	D_A^0	δ_A
$\emptyset$	0	0	0
$+$	$\mathcal{B}^\vee$	$V^\vee$	$\ell \mapsto \ell\Theta$
$-$	$\mathcal{B}^\vee$	$V^\vee$	$\ell \mapsto -\ell\Theta\mathcal{F}$
J_0	$\mathcal{B}^\vee$	$V^\vee \oplus V^\vee$	$\ell \mapsto (\ell\Theta, -\ell\Theta\mathcal{F})$.

For $A \subseteq A'$, the arrow $D_{A'} \rightarrow D_A$ is precomposition with the original coefficient inclusion. On degree -1 it is the identity between active masks. On degree zero the arrows $J_0 \rightarrow +$ and $J_0 \rightarrow -$ are respectively

$$(a, b) \mapsto a, \quad (a, b) \mapsto b.$$

Both leaf-to-empty arrows are zero, as is the direct joint-to-empty arrow. Applying either coordinate projection to $\delta_{J_0}\ell$ gives the corresponding displayed leaf differential; the maps therefore commute with the differentials. Both composites from joint to empty are zero. These checks prove all naturality and composition identities in $\mathrm{Fun}(L^{\mathrm{op}}, \mathrm{Ch}(E))$.

205 3. Cohomology, with an explicit quotient and inverse

For every active mask, the degree- -1 kernel is

$$\{\ell \in \mathcal{B}^\vee : \ell\Theta = 0\} \xleftarrow{q^\vee} Q^\vee, \quad q^\vee(\lambda) = \lambda q.$$

The inverse sends ℓ to $[F] \mapsto \ell(F)$. It is well-defined precisely because $\ell\Theta = 0$. The Fourier isomorphism makes the condition on the minus face equivalent to the same equation. All active transition maps on this kernel are the identity in these $Q^\vee$ coordinates.

On either single leg, the degree-zero cohomology is zero. Indeed any linear functional on V extends to $\mathcal{B}$ through the injective map Θ . More concretely, the global retraction constructed in transcript A1694, fully reproved with explicit fixed cutoffs and estimates in Appendix A below, is a continuous linear map $\Lambda : \mathcal{B} \rightarrow V$ with $\Lambda\Theta = 1$. Then $a\Lambda$ lifts $a \in V^\vee$ on the plus face and $-b\mathcal{F}^{-1}\Lambda$ lifts $b \in V^\vee$ on the minus face.

At the joint mask define

$$\pi_0 : V^\vee \oplus V^\vee \longrightarrow V^\vee, \quad \pi_0(a, b) = a\mathcal{F} + b.$$

This evaluates the functional on the original cycle $\psi \mapsto (\mathcal{F}\psi, \psi)$. The composite $\pi_0\delta_{J_0}$ is zero. Conversely, if $a\mathcal{F} + b = 0$, then $b = -a\mathcal{F}$, and

$$\delta_{J_0}(a\Lambda) = (a, -a\mathcal{F}) = (a, b).$$

The quotient map is surjective, with representative $c \mapsto (0, c)$. Its inverse on the quotient sends c to the class of $(0, c)$: subtracting this representative from (a, b) gives

$$(a, b) - (0, a\mathcal{F} + b) = (a, -a\mathcal{F}) = \delta_{J_0}(a\Lambda).$$

This proves both inverse identities. Consequently the entire cohomology diagram is

$$\begin{array}{c|ccc} A & J_0 & + & - & \emptyset \\ \hline H^{-1}D_A & Q^\vee & Q^\vee & Q^\vee & 0 \\ H^0D_A & V^\vee & 0 & 0 & 0. \end{array}$$

The arrows in degree -1 are identity from joint to either leaf and zero to the empty mask. Every arrow in degree zero is zero. No degree-zero joint class is discarded.

In particular $V^\vee \neq 0$ when $W = E$. A concrete continuous functional is $\phi \mapsto \phi(1)$. It is nonzero on

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V.$$

The function is even, Schwartz, and vanishes at zero. Integration by parts gives $\int x^4 e^{-\pi x^2} dx = \frac{3}{2\pi} \int x^2 e^{-\pi x^2} dx$, proving its integral is zero; its value at one is $2\pi(2\pi - 3)e^{-\pi} \neq 0$.

206 4. What the reversed bottom actually contains

In L^{op} the bottom is J_0 , the join operation is original intersection, and the top is the original empty mask. Thus the dual's bottom cohomology fibres are exactly $Q^\vee$ in degree -1 and $V^\vee$ in degree zero. In particular the latter is nonzero for $W = E$. The rule “every bottom fibre is zero” is therefore incompatible with this calculation.

The source's full reconstruction retains these fibres. For either cohomology diagram $H^i D$, it is

$$M^i = \coprod_{A \in L^{\text{op}}} H^i D_A,$$

with

$$(A, u) + (B, v) = (A \cap B, \rho_{A, A \cap B} u + \rho_{B, A \cap B} v), \quad r^\bullet(A, u) = (A, ru), \quad \tau(A, u) = (J_0, 0).$$

The global additive zero is $(J_0, 0)$, and $e(A, u) = (A, 0)$. Associativity transports three amplitudes to $A \cap B \cap C$; equality of the two iterated transports proves the two sums agree. The zero law uses the unique map from the bottom and its zero value. Linearity of transitions proves distributivity, and the fibrewise vector-space scalar laws prove the remaining semimodule laws. This is the exact reconstruction, including nonzero vectors in the fibre of its global zero under multiplication by e .

If an additional external-absence label is desired while retaining every original reversed label, it can also be constructed explicitly: adjoin a new bottom $\perp_\dagger < L^{\text{op}}$, put zero fibre there, and retain all old fibres and arrows. The new external zero is $(\perp_\dagger, 0)$. The label map sending $\perp_\dagger$ and J_0 both to J_0 , and fixing every other label, preserves joins and bottom. With identity maps on old fibres and the zero map on the new fibre it reconstructs a surjective split-linear comparison to M^i . Its only additional identification is between the new external zero and the zero vector at the old bottom J_0 . It does not remove any nonzero bottom vector. This proves the exact relation between the two declared conventions; it does not silently impose a zero bottom fibre on the original reversed diagram.

207 5. Strict natural splitting fails, but the derived splitting has a concrete roof

Let $C = V^\vee$, let Z_C be the diagram supported at the bottom J_0 with value C , and let H be the degree- -1 cohomology diagram just calculated. The maps $q^\vee : H[1] \rightarrow D$ and $\pi : D \rightarrow Z_C[0]$, where π is π_0 only at the joint mask in degree zero, are natural cochain maps.

There is no strict natural cochain section $s : Z_C[0] \rightarrow D$ of π when $C \neq 0$. At the joint mask write $s(c) = (a(c), b(c))$. Naturality to the plus leaf forces $a(c) = 0$, since the source at that leaf is zero.

Naturality to the minus leaf forces $b(c) = 0$. Hence $\pi s = 0$, contradicting $\pi s = 1_C$. This is exactly a strict-section obstruction.

It is not a derived nonsplitting statement. Here is the actual surviving derived map. Write $b = J_0, x = +, y = -, t = \emptyset$ in the reversed diamond. For a vertex v , define the projective diagram

$$P_v(C)(w) = \begin{cases} C, & v \leq w, \\ 0, & v \not\leq w. \end{cases}$$

The augmented resolution of Z_C is

$$\mathcal{R}_C : P_t(C) \xrightarrow{c \mapsto (c, -c)} P_x(C) \oplus P_y(C) \xrightarrow{(a,b) \mapsto a+b} P_b(C) \xrightarrow{\epsilon} Z_C,$$

with degrees $-2, -1, 0$ before the augmentation. At b , the augmentation is identity. At either leaf the preceding map is identity and the augmented stalk is zero. At t , the stalk sequence is $0 \rightarrow C \rightarrow C^2 \rightarrow C \rightarrow 0$, with the displayed anti-diagonal and sum. This proves exactness stalkwise and on all arrows. Each $P_v(C)$ is projective because its Hom functor is $\text{Hom}_E(C, -_v)$, an exact functor over a field. Thus $\epsilon : \mathcal{R}_C \rightarrow Z_C[0]$ is a quasi-isomorphism.

Define $f : \mathcal{R}_C \rightarrow D$ as follows. In degree zero it is determined by

$$f_b^0(c) = (0, c), \quad f_x^0(c) = 0, \quad f_y^0(c) = c, \quad f_t^0(c) = 0.$$

In degree -1 , its map from $P_x(C)$ is zero, and its map from $P_y(C)$ is determined at y by

$$f_y^{-1}(c) = -c\mathcal{F}^{-1}\Lambda \in \mathcal{B}^\vee.$$

At t it is zero; in degree -2 it is zero. These maps are natural by their projective-source definitions and the zero target at t . At y , the chain equation is

$$\delta_y(-c\mathcal{F}^{-1}\Lambda) = c\mathcal{F}^{-1}\Lambda\Theta\mathcal{F} = c.$$

At x both sides are zero, and at t both sides are zero. These verify every chain equation. Finally $\pi f = \epsilon$, so

$$Z_C[0] \xleftarrow[\simeq]{\epsilon} \mathcal{R}_C \xrightarrow{f} D$$

is an explicit right inverse of π in the derived category. The combined cochain map

$$H[1] \oplus \mathcal{R}_C \longrightarrow D, \quad (h, r) \longmapsto q^\vee h + f(r),$$

induces identity on degree -1 cohomology and the displayed isomorphism on degree zero cohomology. It is therefore a quasi-isomorphism. This proves the full derived relationship without replacing the original support arrows by equalities.

208 6. The original action and the actual defect of this roof

For $a > 0$, retain $U_a F(x) = F(x/a)$. On the source plus and minus legs the actions are U_a and $aU_{1/a}$. On $\mathcal{B}$ the action is U_a . Let $W = \mathcal{L}_w$ carry a^w , where the source supplies $w = 0, 1$. Then on the displayed dual complex the action is

$$\ell \longmapsto a^w \ell U_{1/a}, \quad (\alpha, \beta) \longmapsto (a^w \alpha U_{1/a}, a^{w-1} \beta U_a).$$

The source identities $\Theta U_a = U_a \Theta$ and $\mathcal{F} U_a = aU_{1/a} \mathcal{F}$ verify directly that these operators commute with every differential and support arrow. On cohomology they give

$$H^{-1} : \lambda \longmapsto a^w \lambda \bar{U}_{1/a}, \quad H^0 : c \longmapsto a^{w-1} c U_a.$$

For $w = 1$, these are the original weight-one dual action on $Q^\vee$ and the explicit action $c \mapsto cU_a$ on the retained bottom $V^\vee$.

The roof above need not be strictly equivariant because Λ is not asserted equivariant. Its degree-zero component is equivariant. At the minus generator of degree -1 , its exact failure is

$$U_a^D f_y^{-1}(c) - f_y^{-1}(U_a^C c) = a^w c \mathcal{F}^{-1}(U_{1/a} \Lambda - \Lambda U_{1/a}).$$

Here the first $U_{1/a}$ in parentheses acts on V , and the second on $\mathcal{B}$. The formula follows by substitution and $U_a \mathcal{F}^{-1} = a \mathcal{F}^{-1} U_{1/a}$. Its precomposition with Θ is zero, using $\Lambda \Theta = 1$ and the theta intertwining. It consequently factors through the exact injection $q^\vee : Q^\vee \rightarrow \mathcal{B}^\vee$, with functional

$$[F] \mapsto a^w c \mathcal{F}^{-1}(U_{1/a} \Lambda - \Lambda U_{1/a}) F.$$

This is the computed relation between the original action, the chosen derived roof, and the retained $Q^\vee$ fibre. No equivariant splitting is inferred by erasing this term. In this statement each projective $P_v(C)$ carries the displayed coefficient action $c \mapsto a^{w-1} c U_a$ at every nonzero stalk, and its restriction maps are identities; its resolution maps therefore intertwine the actions.

The derived class of this defect is also computable. Applying $\text{Hom}(-, H)$ to the displayed projective resolution gives the complex

$$\text{Hom}_E(C, Q^\vee) \xrightarrow{\ell \mapsto (\ell, \ell)} \text{Hom}_E(C, Q^\vee)^2 \longrightarrow 0.$$

The last term is zero because $H_t = 0$. Thus

$$\text{Ext}^1(Z_C, H) \cong \text{Hom}_E(C, Q^\vee), \quad [(u, v)] \mapsto v - u, \quad d \mapsto [(0, d)].$$

The two maps are inverse: their first composite is identity, and subtracting $(0, v - u)$ from (u, v) gives the diagonal pair (u, u) . The action defect of the chosen roof has zero plus component and minus component $q^\vee d_a$, where

$$d_a(c)([F]) = a^w c \mathcal{F}^{-1}(U_{1/a} \Lambda - \Lambda U_{1/a}) F.$$

It therefore has the exact Ext class d_a . Equivalently, a nullhomotopy would be a degree- -1 map from $P_b(C)$ to D , represented by the same functional on both leaves; its degree-zero equation forces that functional to annihilate ΘV . The leaf equations require its two copies to equal $(0, q^\vee d_a)$. Such a map exists exactly when $d_a = 0$. This proves that this particular derived section intertwines the action exactly when the displayed class vanishes; it does not assert that no other equivariant section can exist.

The class is the transpose of the original global theta cocycle by an exact formula. Put

$$s[F] = (1 - \Theta \Lambda) F, \quad k_b = \Lambda U_b s.$$

Every F has its original decomposition $F = \Theta \Lambda F + s q F$. Substitution, with $\Lambda \Theta = 1$ and $U_b \Theta = \Theta U_b$, gives

$$U_b \Lambda - \Lambda U_b = -k_b q.$$

Consequently the same class is

$$\boxed{d_a(c) = -a^w c \mathcal{F}^{-1} k_{1/a} : Q \longrightarrow W.}$$

The minus sign, Fourier inverse, reciprocal parameter, and coefficient action are retained. The reversed-support defect and the source's scaling boundary are thus connected by a proved morphism.

209 7. Continuity and the finite residue inclusion remain attached

For $W = E$, all the formulas also hold for continuous duals with their strong topologies. The maps $\Theta, \mathcal{F}, q$ and the source's explicit Λ are continuous, and $\Lambda\Theta = 1$ splits the original topological sequence. Thus the plus and minus lifts used above are continuous; all quotient and inverse maps are continuous on the displayed finite products. Precomposition by a continuous linear map is strongly continuous because it takes bounded subsets to bounded subsets. This verifies the asserted topology of every displayed dual transport. The derived roof assertion in Section 5 is in the declared algebraic diagram category; no undeclared derived category of all locally convex spaces is used.

The explicit inverse pair on strong duals is

$$\mathcal{B}'_b \longrightarrow V'_b \oplus Q'_b, \quad \ell \longmapsto (\ell\Theta, \ell s), \quad (\alpha, \lambda) \longmapsto \alpha\Lambda + \lambda q.$$

Its composites are identity by $\Lambda\Theta = 1, qs = 1, \Lambda s = 0, q\Theta = 0$, and $\Theta\Lambda + sq = 1$. Strong continuity follows directly from $p_B(T'\ell) = p_{T(B)}(\ell)$ for a bounded subset B and continuous linear T . In particular q' identifies Q'_b with the annihilator of ΘV , with continuous inverse $\ell \mapsto \ell s$. The joint relation subspace is the closed kernel of π_0 , and $c \mapsto [(0, c)]$ is its continuous quotient inverse.

For the original reflection-stable finite arithmetic packet A_Z , keep $g = 2\xi$, every actual multiplicity, the complete jet map $J_Z : Q \rightarrow A_Z$, and the full residue form

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s)h(s)}{g(s)} ds, \quad f^\dagger(s) = \overline{f(1-\bar{s})}.$$

The source proves its perfectness, J_Z 's surjectivity and continuity, and its covariance with $\mathcal{L}_1$. Therefore the map

$$\overline{A_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto ([F] \mapsto R_Z(f, J_Z[F]))$$

is the same injective continuous residue map at each active vertex of $H^{-1}D$. The active support arrows are identities, so its three naturality squares commute exactly; at the empty vertex both maps are zero. Every nilpotent remains in the full A_Z and residue pairing. The new bottom $H^0 = V'_b$, the action defect of Section 6, and the original arithmetic residue inclusion now coexist in one completely specified reversed support calculation.

For the continuous-jet assertion one can retain an explicit estimate. Write $r = \Re \rho$ and choose an integer $n > |r|$. Differentiating the Mellin integral and splitting it at one gives, for every $j \geq 0$,

$$|(\mathcal{M}F)^{(j)}(\rho)| \leq j! \left(\frac{p_{n,0}(F)}{(n-r)^{j+1}} + \frac{p_{-n,0}(F)}{(n+r)^{j+1}} \right).$$

The two integrals used here are respectively $\int_1^\infty x^{r-n}(\log x)^j dx/x = j!/(n-r)^{j+1}$ and $\int_0^1 x^{r+n}|\log x|^j dx/x = j!/(n+r)^{j+1}$. They follow from substitution $y = |\log x|$ and j integrations by parts. Thus every derivative in the finite jet is continuous, with its actual order retained. The map descends by the original quotient topology since it annihilates ΘV .

The finished result is the explicit dual diagram, its nonzero bottom degree-zero fibre, the precise strict-section obstruction, the constructed derived roof that survives that obstruction, and its full scaling defect. None of these computations proves a theta weight bound, removes a finite-prime trace term, or replaces A_τ by restriction to σ .

210 Appendix A: complete retraction on the original theta source

210.1 A.1. Original spaces and the fixed auxiliary functions

Work over $\mathbb{C}$, with Fourier convention

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

Retain the original spaces, with their indicated seminorm topologies:

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \widehat{\phi}(0) = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x.$$

Use Schwartz seminorms $\sigma_{m,j}(u) = \sup_{x \in \mathbb{R}} (1 + |x|)^m |u^{(j)}(x)|$. The Fourier transform preserves the even Schwartz space and satisfies $\mathcal{F}^2 \phi(x) = \phi(-x)$. Consequently it preserves V , since $\widehat{\phi}(0) = \int \phi$ and $\int \widehat{\phi} = \phi(0)$.

Fix the following explicit smooth functions for the entire construction:

$$h(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases} \quad \chi(x) = \frac{h(1/16 - x^2)}{h(1/16 - x^2) + h(x^2 - 1/64)}, \quad \alpha = \beta = \chi.$$

All derivatives of h at zero vanish: each derivative for $t > 0$ is a polynomial in $1/t$ times $e^{-1/t}$, which tends to zero. The denominator defining χ never vanishes, since its two arguments cannot both be nonpositive. Thus χ is real, even, smooth, lies in $[0, 1]$, equals one for $|x| \leq 1/8$, and vanishes for $|x| \geq 1/4$. In particular

$$\|\chi\|_2^2 \leq \frac{1}{2}.$$

These are choices of reconstruction data on the original coordinates. The spaces, Fourier convention and arithmetic theta map remain exactly those displayed here.

210.2 A.2. Theta continuity and recovery on both exterior regions

The original maps are

$$\Theta \phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \quad JF(x) = x^{-1} F(1/x).$$

Poisson summation gives $\Theta \mathcal{F} = J\Theta$ on V . For this identity, periodize the Schwartz function $t \mapsto \phi(xt)$; its Fourier coefficients are $x^{-1} \widehat{\phi}(n/x)$, obtained by integrating over all translated unit intervals. Both the periodization and its Fourier series converge absolutely with derivatives. Evaluation at zero gives $\sum_n \phi(nx) = x^{-1} \sum_n \widehat{\phi}(n/x)$. The two zero-index terms vanish by the original moments.

The theta map is continuous into $\mathcal{B}$. Indeed, for $x \geq 1$, an integer $N > 1$, and $s_{N,k}(\phi) = \sup_{t>0} t^N |D^k \phi(t)|$,

$$|D^k \Theta \phi(x)| \leq 2\zeta(N) s_{N,k}(\phi) x^{-N}.$$

Each $s_{N,k}$ is bounded by finitely many Schwartz seminorms. Choosing $N \geq b$ bounds the part of $p_{b,k}$ on $x \geq 1$. The identities

$$D^k J = J(1 - D)^k, \quad p_{b,k}(JF) \leq \sum_{j=0}^k \binom{k}{j} p_{1-b,j}(F)$$

follow by differentiating $x^{-1} F(1/x)$. For $x \leq 1$, apply the preceding theta estimate to $\widehat{\phi}$ at $1/x$; choosing also $N \geq 1 - b$ bounds this part by $2\zeta(N) \sum_{j=0}^k \binom{k}{j} s_{N,j}(\widehat{\phi})$. This proves every required bound and also continuity of J .

For $x \neq 0$, define the actual arithmetic inverse series

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|).$$

Here μ is the ordinary Möbius function; the factor $1/2$ inverts the sum over both nonzero integer signs. Put

$$P_j(T) = (-1)^j T(T+1) \cdots (T+j-1), \quad P_0 = 1, \quad C_{N,j}(F) = \sum_{k=0}^j |[T^k]P_j(T)| p_{N,k}(F).$$

The identity $x^j \partial_x^j = P_j(D)$ follows by induction. Therefore, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{\zeta(N)}{2} C_{N,j}(F) x^{-N-j}. \quad (1)$$

The same absolute bound holds for negative x . It proves locally uniform convergence of every differentiated series away from zero.

For $\phi \in V$, the divisor rearrangement is legitimate: for fixed $x > 0$, $\sum_{m,n \geq 1} |\phi(mnx)|$ is finite, by a Schwartz bound of degree $N > 1$ and $\sum_{m,n} (mn)^{-N} = \zeta(N)^2$. Thus

$$\mathcal{U}\Theta\phi(x) = \sum_{k \geq 1} \left(\sum_{d|k} \mu(d) \right) \phi(kx) = \phi(x).$$

The divisor sum is one for $k = 1$ and zero otherwise: prime factorization gives the product $\prod_{p|k} (1 - 1)$ when $k > 1$. Evenness gives recovery on negative x , and Poisson summation gives the second exterior recovery:

$$\boxed{\mathcal{U}\Theta\phi = \phi, \quad \mathcal{U}J\Theta\phi = \hat{\phi} \quad (x \neq 0).} \quad (2)$$

In particular Θ is injective, including at zero by continuity. No information about zeros of zeta in a critical strip enters this proof.

210.3 A.3. Fixed contraction and full Schwartz estimates

On $L^2(\mathbb{R}, dx)$, set

$$A = M_\chi, \quad B = \mathcal{F}^{-1} M_\chi \mathcal{F}, \quad T = AB.$$

Its integral kernel is $\chi(x)\check{\chi}(x-y)$. Plancherel and the change of variable $z = x - y$ give

$$\|T\|^2 \leq \|T\|_{\text{HS}}^2 = \|\chi\|_2^2 \|\check{\chi}\|_2^2 = \|\chi\|_2^4 \leq \frac{1}{4}.$$

Consequently the operator-norm convergent series supplies the actual inverse:

$$\boxed{(1 - T)^{-1} = \sum_{n=0}^{\infty} T^n, \quad \|(1 - T)^{-1}\| \leq 2.} \quad (3)$$

Multiplying its partial sums by $1 - T$ leaves $1 - T^{n+1}$, which tends to the identity, proving both inverse identities.

Define exterior functions, extended by zero near the origin, by

$$a_F = (1 - \chi)\mathcal{U}F, \quad b_F = (1 - \chi)\mathcal{U}JF, \quad YF = a_F + \chi\mathcal{F}^{-1}b_F.$$

Here are explicit bounds verifying all their Schwartz seminorms. Write $c_0 = 1$ and $c_r = \|\chi^{(r)}\|_\infty$ for $r \geq 1$. Leibniz's rule and (1), with $N = m + 2$, give

$$\sigma_{m,j}(a_F) \leq \frac{\zeta(m+2)}{2} 9^m \sum_{r=0}^j \binom{j}{r} c_r 8^{2+j-r} C_{m+2,j-r}(F). \quad (4)$$

Indeed every nonzero differentiated term occurs at $|x| \geq 1/8$, where $1 + |x| \leq 9|x|$ and $|x|^{-2-j+r} \leq 8^{2+j-r}$. The same bound holds for b_F with JF in place of F ; the displayed bounds for J express it in finitely many original $p_{b,k}(F)$. Smoothness across the plateau endpoints follows from smoothness of the cutoffs and their identically zero interior factors.

Fourier inversion is continuous on these Schwartz seminorms. Explicitly, integration by parts gives

$$\sigma_{m,j}(\mathcal{F}^{-1}u) \leq \sum_{r=0}^m \binom{m}{r} (2\pi)^{j-r} \|\partial_\xi^r(\xi^j u)\|_1.$$

Each integral on the right is bounded by a finite sum of Schwartz seminorms, using Leibniz's rule and $\int_{\mathbb{R}} (1 + |\xi|)^{-2} d\xi = 2$. Together with (4), this proves that $Y : \mathcal{B} \rightarrow \mathcal{S}^{\text{even}}$ is continuous.

The operator T also maps L^2 continuously to Schwartz space. Put

$$b_j = \|(2\pi\xi)^j \chi(\xi)\|_2 \leq \frac{(\pi/2)^j}{\sqrt{2}}, \quad A_{m,j} = (5/4)^m \sum_{r=0}^j \binom{j}{r} \|\chi^{(r)}\|_\infty b_{j-r}.$$

Cauchy-Schwarz in the Fourier integral bounds $\|\partial^j Bf\|_\infty$ by $b_j \|f\|_2$. Leibniz's rule and the support of χ therefore give

$$\sigma_{m,j}(Tf) \leq A_{m,j} \|f\|_2.$$

Set $\Phi F = (1 - T)^{-1} YF$, initially in L^2 . The equation $\Phi F = YF + T\Phi F$ proves that it is Schwartz, and supplies the explicit continuity estimate

$$\sigma_{m,j}(\Phi F) \leq \sigma_{m,j}(YF) + 2A_{m,j} \|YF\|_2. \quad (5)$$

The final L^2 norm is bounded by $\sqrt{2} \sigma_{1,0}(YF)$. Every operator preserves parity; hence ΦF is even. Equations (2) give

$$Y\Theta\phi = (1 - A)\phi + A(1 - B)\phi = (1 - T)\phi, \quad \Phi\Theta\phi = \phi. \quad (6)$$

210.4 A.4. The original moment correction and exact splitting

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int_{\mathbb{R}} \phi)$ are $(1, 1)$ and $(0, 1)$. For the integral constants, the square of the Gaussian integral is the planar polar integral $2\pi \int_0^\infty r e^{-\pi r^2} dr = 1$. Integrating $\partial_x(xe^{-\pi x^2})$ then gives $2\pi \int x^2 e^{-\pi x^2} dx = 1$. Thus the continuous projection

$$P_V u = u - u(0)\gamma_0 - \left(\int_{\mathbb{R}} u - u(0) \right) \gamma_2$$

has both original moments zero and is identity on V . Continuity follows from $|u(0)| \leq \sigma_{0,0}(u)$ and $|\int u| \leq 2\sigma_{2,0}(u)$. The required map is therefore

$$\boxed{\Lambda = P_V(1 - T)^{-1}Y : \mathcal{B} \longrightarrow V, \quad \Lambda\Theta = 1_V.} \quad (7)$$

All constants and seminorm dependence have been supplied in (1)–(5). For the original quotient $q : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V$, put $K = 1 - \Theta\Lambda$. Equation (7) gives $K^2 = K$, $\ker K = \Theta V$, and $\text{im } K = \ker \Lambda$. The image ΘV is closed because it is the kernel of continuous K . The continuous map K descends by the quotient topology to

$$s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0, \quad \Theta\Lambda + sq = 1.$$

These equations prove the continuous inverse pair

$$V \oplus Q \longrightarrow \mathcal{B}, \quad (\phi, u) \longmapsto \Theta\phi + su, \quad F \longmapsto (\Lambda F, qF).$$

Finally, retain $U_a F(x) = F(x/a)$. It is continuous because $p_{b,k}(U_a F) = a^b p_{b,k}(F)$; its source action preserves V and obeys $U_a \Theta = \Theta U_a$. It therefore induces $\overline{U}_a$ on Q . Put $k_a = \Lambda U_a s$. The decomposition of $U_a s$ gives

$$U_a s = \Theta k_a + s \overline{U}_a, \quad k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0.$$

For the middle identity, substitute the first identity with parameter b into $\Lambda U_a U_b s$, and use $\Lambda U_a \Theta = U_a$ and $\Lambda U_a s = k_a$. Writing $F = \Theta \Lambda F + sqF$ also proves

$$U_a \Lambda - \Lambda U_a = -k_a q.$$

Consequently the reversed-support dual roof's action defect is exactly

$$\boxed{d_a(c) = -a^w c \mathcal{F}^{-1} k_{1/a}.$$

This connects the completed retraction to the original action on the full Q , retaining its Fourier inverse, weight factor, reciprocal parameter and sign. The calculation does not set this cocycle to zero.

This calculation completes the operator comparison left unspecified in shared transcript A0961, node a7bbd422-311c-4380-ba6c-a4bdd61fce2f, §7. It uses the original arithmetic objects of *Arithmetic cohomology over the tau-base*, equations (10)–(14), (19), (27), and (38)–(45). It preserves the actual function $g = 2\xi$, every local unit, multiplicity, coordinate, reflected-germ sign and support mask. The new grading is an explicitly constructed additional action. It is not identified with arithmetic Frobenius or with the original Mellin scaling.

210.5 1. Original arithmetic objects and analytic identities

Write $E = \mathbb{C}$ and retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The target is the original Fréchet space

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)|, \quad D = -x\partial_x.$$

Its Mellin transform is

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

For $F \in \mathcal{B}$ this is entire. Indeed, for any compact set of s and any derivative order, choose integers b_- below every $\Re s$ in that compact set and b_+ above every $\Re s$. On $(0, 1)$ and $(1, \infty)$ respectively, the bounds $|F(x)| \leq p_{b_-,0}(F)x^{-b_-}$ and $|F(x)| \leq p_{b_+,0}(F)x^{-b_+}$ give integrable dominating functions even after multiplication by any power of $|\log x|$. Differentiation under the integral is therefore justified.

Retain the two original maps

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{J}F(x) = x^{-1}F(1/x).$$

For every $\phi \in V$, $\Theta\phi \in \mathcal{B}$. At infinity this follows by termwise Euler differentiation and the Schwartz estimate, choosing a decay exponent greater than one so that its sum over n converges. At zero Poisson summation, with both original vanishing moments, gives

$$\Theta\phi(x) = x^{-1} \Theta \widehat{\phi}(1/x).$$

The same infinity estimates for $\widehat{\phi}$ apply after differentiating this identity; the resulting finite sums of Euler derivatives still decay faster than every power at zero. All sums and differentiated sums converge normally on compact positive intervals. Fourier transformation preserves V , and

$$\mathcal{J}\Theta\psi = \Theta\widehat{\psi}.$$

Thus the original two-leg cochain complex is

$$C_\theta = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(\phi, \psi) = \Theta\phi - \mathcal{J}\Theta\psi = \Theta(\phi - \widehat{\psi}),$$

in degrees 0, 1. Its degree-one quotient is $Q = \mathcal{B}/\Theta V$.

For clarity, the exact Mellin factor used below can be proved without replacing g by its leading local monomial. Put

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The entire continuation and reflection identity needed here follow directly from the Gaussian theta integral. With

$$\vartheta(x) = 2 \sum_{n \geq 1} e^{-\pi n^2 x^2},$$

Poisson summation gives $\vartheta(x) = x^{-1}\vartheta(1/x) + x^{-1} - 1$. Mellin integration for $\Re s > 1$, followed by splitting at 1 and substituting $x \mapsto 1/x$ in the lower integral, yields

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \vartheta(x)(x^s + x^{1-s}) \frac{dx}{x}.$$

The latter integral is entire by Gaussian decay. Consequently

$$g(s) = 1 + s(s-1) \int_1^\infty \vartheta(x)(x^s + x^{1-s}) \frac{dx}{x}, \quad g(1-s) = g(s), \quad \overline{g(\bar{s})} = g(s).$$

In particular g is not zero identically and satisfies $g^\dagger = g$ for the reflection below.

For $\phi \in V$, absolute convergence for $\Re s > 1$ gives the exact factorization

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)\mathcal{M}\phi(s) = g(s)H_\phi(s), \quad H_\phi(s) = \frac{2\pi^{s/2}\mathcal{M}\phi(s)}{s(s-1)\Gamma(s/2)}.$$

Here $\mathcal{M}\phi$ uses the positive half-line. To verify that H_ϕ is entire, split its integral at 1 and subtract finitely many terms of the even Taylor expansion of ϕ at zero. Since $\phi(0) = 0$, its only possible Mellin poles are simple poles at $s = -2, -4, \dots$, with the corresponding even Taylor coefficients as residues. Each is canceled by the simple zero of $1/\Gamma(s/2)$. At $s = 0$ the same reciprocal gamma factor cancels the displayed factor s . At $s = 1$, $\mathcal{M}\phi(1) = \frac{1}{2} \int_{\mathbb{R}} \phi = \frac{1}{2} \widehat{\phi}(0) = 0$ cancels $s-1$. This proves the asserted entire continuation. The displayed identity then holds everywhere by continuation. No vanishing at an arithmetic zero was assumed in this argument.

210.6 2. Full actual jet packets, reflection, and residue pairing

Fix any finite set Z of actual zeros of this g which is stable under

$$\iota\rho = 1 - \bar{\rho}.$$

If Z is empty, set $A_Z = 0$, $J_Z = 0$, $S = 0$, and $L_q = 1$; all packet identities below are the unique identities on the zero space, and no minimum or maximum over Z is taken. For the remaining construction assume Z is nonempty. Retain the actual order $m_\rho \geq 1$ of each zero and the coordinate $z_\rho = s - \rho$. Write the actual germ, with its unit intact, as

$$g_\rho(z) = g(\rho + z) = u_\rho(z)z^{m_\rho}, \quad u_\rho(0) \neq 0.$$

The coefficient algebra is

$$A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho / (g),$$

where $\mathcal{O}_\rho$ is the ring of holomorphic germs at ρ . Since multiplication by the actual unit u_ρ is invertible, the ideal (g_ρ) equals (z^{m_ρ}) . This equality identifies classes with unique truncated Taylor coordinates; it does not replace the denominator g_ρ in a residue. Define

$$(J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j.$$

The preceding factorization proves $J_Z \Theta = 0$, so J_Z induces $\bar{J}_Z : Q \rightarrow A_Z$.

Reflection is the conjugate-linear involution

$$(f^\dagger)_\rho(z) = \overline{f_{\iota\rho}(-\bar{z})} = \sum_{j \geq 0} (-1)^j \overline{f_{\iota\rho,j}} z^j.$$

Because $g^\dagger = g$, reflection preserves the defining ideals, $m_{\iota\rho} = m_\rho$, and the actual units satisfy

$$u_\rho(z) = (-1)^{m_\rho} \overline{u_{\iota\rho}(-\bar{z})}.$$

Retain the original residue form, conjugate-linear in its first variable and linear in its second:

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_{z=0} \frac{(f^\dagger)_\rho(z) h_\rho(z)}{u_\rho(z) z^{m_\rho}} dz.$$

Changing either representative by a multiple of its actual g changes the integrand by a holomorphic germ, so this form is well-defined. It is perfect: if a nonzero component of h starts with $h_k z^k$, choose $(f^\dagger)_\rho = z^{m_\rho-1-k}$ and all other components zero. Its residue is $h_k/u_\rho(0) \neq 0$. Reflection is a bijection, so that choice of $f^\dagger$ comes from an f . Finite dimension then gives nondegeneracy in both variables.

The form is **skew-Hermitian**, with no phase change. In fact for a meromorphic germ $a(z) = \sum a_j z^j$, reflection changes its residue coefficient to $-\bar{a}_{-1}$, because $(-1)^{-1} = -1$. Applying that identity, relabeling ρ by $\iota\rho$, and using $g^\dagger = g$ gives

$$R_Z(h, f) = -\overline{R_Z(f, h)}.$$

This sign will not be replaced by a positivity assertion.

210.7 3. The exact additional grading and its inverse

For each real $q > 0$ define componentwise

$$(C_q f)_\rho(z) = f_\rho(z/q) \pmod{g_\rho(z)}.$$

This is an algebra automorphism of the original quotient. To check well-definedness with the original unit retained, compute

$$\frac{g_\rho(z/q)}{g_\rho(z)} = q^{-m_\rho} \frac{u_\rho(z/q)}{u_\rho(z)}.$$

The right side is a holomorphic unit at zero. Thus substitution sends the ideal (g_ρ) onto itself. Its inverse is

$$(C_q^{-1} f)_\rho(z) = f_\rho(qz),$$

as is seen directly by composing the substitutions. In particular

$$C_p C_q = C_{pq}, \quad C_1 = 1, \quad C_q(z_\rho^j) = q^{-j} z_\rho^j.$$

Since q is real and positive, substitution commutes with the reflected conjugation:

$$(C_q f)^\dagger = C_q(f^\dagger).$$

The separate packet labels ρ , local coordinates, multiplicities, and positive-degree jets have all been retained.

On a component let

$$N_\rho f = z_\rho f, \quad U_a|_\rho = a^\rho e^{(\log a)N_\rho}, \quad a > 0, \quad a^\rho = \exp(\rho \log a),$$

with the real logarithm of the original parameter a . These are exactly the jet operators induced by the original scaling $U_a F(x) = F(x/a)$. Multiplication by $a^{\rho+z}$ follows from a change of variables in the original Mellin integral. No centered or normalized scaling has been substituted.

Direct multiplication proves

$$U_a N_\rho U_a^{-1} = N_\rho, \quad C_q N_\rho C_q^{-1} = q^{-1} N_\rho.$$

Conjugating the full exponential gives the exact surviving comparison

$$C_q U_a|_\rho C_q^{-1} = a^\rho \exp\left(\frac{\log a}{q} N_\rho\right) = a^{\rho(1-1/q)} U_{a^{1/q}}|_\rho.$$

Every exponential is its full finite sum through degree $m_\rho - 1$. The factor $a^{\rho(1-1/q)}$ remains part of the formula. This is a relation between two distinct actions on the same actual arithmetic packet. The conjugated operator $C_q U_a C_q^{-1}$ still commutes with N_ρ , because its displayed formula is a polynomial in N_ρ times a scalar. It is C_q itself, not that conjugated arithmetic operator, which scales N_ρ by q^{-1} .

210.8 4. Exact obstruction to a literal simultaneous identification

Let $N = \bigoplus N_\rho$ and $U_a = \bigoplus U_a|_\rho$. Fix $q > 0$, $q \neq 1$, and retain any target vector space W with invertible F and an operator N_D satisfying the exact Deligne-type relation

$$F N_D F^{-1} = q^{-1} N_D.$$

Every linear map $T : A_Z \rightarrow W$ intertwining both proposed operators,

$$T U_a = F T, \quad T N = N_D T,$$

annihilates $N A_Z$. Indeed $F^{-1} T = T U_a^{-1}$, so

$$F N_D F^{-1} T = F T N U_a^{-1} = T U_a N U_a^{-1} = T N = N_D T.$$

The left side is also $q^{-1} N_D T$. Since $1 - q^{-1} \neq 0$, $N_D T = T N = 0$. Hence T factors uniquely through

$$A_Z \twoheadrightarrow A_Z / N A_Z = \bigoplus_{\rho \in Z} \mathbb{C},$$

the actual residue-value observation. An injective such T therefore exists only when every $N_\rho = 0$, which is equivalent to every $m_\rho = 1$.

This proves precisely the obstruction to the **literal simultaneous identification** of original Mellin scaling and the same full jet nilpotent with that pair. It does not forbid other functors or additional actions. There is a concrete surviving quotient map: evaluation at $z = 0$ intertwines U_a with $\text{diag}(a^\rho)$ and sends N to zero. On that quotient the target relation holds with $N_D = 0$, and it imposes no condition on $\Re \rho$. Alternatively, C_q retains all nilpotents and satisfies the target conjugacy relation, with its distinct semidirect interaction with U_a proved above.

210.9 5. Residue Jacobian, adjoint, and trace contraction

Define the actual unit in each component

$$\kappa_{q,\rho}(z) = q^{1-m_\rho} \frac{u_\rho(z)}{u_\rho(qz)}.$$

Changing the local coordinate of integration by $z = qw$ in the residue gives, with its Jacobian retained,

$$\begin{aligned} R_Z(C_q f, C_q h) &= \sum_\rho q \operatorname{Res}_{w=0} \frac{(f^\dagger)_\rho(w) h_\rho(w)}{g_\rho(qw)} dw \\ &= \sum_\rho \operatorname{Res}_{w=0} \frac{(f^\dagger)_\rho(w) \kappa_{q,\rho}(w) h_\rho(w)}{g_\rho(w)} dw. \end{aligned}$$

Thus

$$\boxed{R_Z(C_q f, C_q h) = R_Z(f, M_{\kappa_q} h).}$$

In particular the factor is q^{1-m_ρ} , not q^{-m_ρ} , and the quotient of the full local units remains. The unit relation from reflection proves, without dropping its sign,

$$(\kappa_q^\dagger)_\rho(z) = q^{1-m_\rho} \frac{(-1)^{m_\rho} u_\rho(z)}{(-1)^{m_\rho} u_\rho(qz)} = \kappa_{q,\rho}(z).$$

Put

$$D_q = M_{\kappa_q} C_q^{-1}.$$

Replacing the second $C_q h$ in the residue formula by an arbitrary h proves

$$R_Z(C_q f, h) = R_Z(f, D_q h).$$

Skew-Hermitian symmetry then gives the equally important other-side formula

$$R_Z(D_q f, h) = R_Z(f, C_q h).$$

For example, apply the first formula to (h, f) and take negative complex conjugates; no phase is needed. These are the exact residue adjoint identities.

Retain the original trace contraction

$$J_g = M_{g'} : A_Z \rightarrow A_Z.$$

The actual derivative is $g'_\rho(z) = m_\rho u_\rho(z) z^{m_\rho-1} + u'_\rho(z) z^{m_\rho}$. Modulo its original ideal $(g_\rho) = (z^{m_\rho})$ this gives

$$\begin{aligned} \kappa_{q,\rho}(z) g'_\rho(qz) &\equiv q^{1-m_\rho} \frac{u_\rho(z)}{u_\rho(qz)} m_\rho u_\rho(qz) q^{m_\rho-1} z^{m_\rho-1} \\ &= m_\rho u_\rho(z) z^{m_\rho-1} \equiv g'_\rho(z). \end{aligned}$$

Consequently

$$\boxed{D_q J_g C_q = J_g.}$$

The contracted form is therefore invariant:

$$R_Z(C_q f, J_g C_q h) = R_Z(f, J_g h).$$

Its exact value is

$$R_Z(f, J_g h) = \sum_{\rho \in Z} m_\rho \overline{f_{\iota\rho}(0)} h_\rho(0) = \operatorname{Tr}(M_{f^\dagger h} | A_Z).$$

For the first equality, $g'_\rho/g_\rho = m_\rho/z + u'_\rho/u_\rho$ and the second term is holomorphic. For the second, multiplication by a germ on the basis $1, z, \dots, z^{m_\rho-1}$ is triangular with its constant coefficient on all m_ρ diagonal entries. The radical of this contracted form is exactly NA_Z ; the full residue form is perfect before contraction. C_q is invertible and preserves that radical. No nilpotent or multiplicity has been discarded.

210.10 6. A proved continuous section on the original theta target

We now construct a section of J_Z in the actual space $\mathcal{B}$, not in a substitute polynomial test family. Index the Taylor coordinates by

$$\Lambda = \{(\rho, j) : \rho \in Z, 0 \leq j < m_\rho\}, \quad M = |\Lambda|.$$

Choose fixed real numbers $\alpha < \beta$ and retain the bump

$$\chi(y) = \begin{cases} \exp\left(-\frac{1}{(y-\alpha)(\beta-y)}\right), & \alpha < y < \beta, \\ 0, & \text{otherwise.} \end{cases}$$

It is smooth and all its endpoint derivatives vanish. For $i = (\rho, j)$ let

$$w_i(y) = e^{\rho y} \frac{y^j}{j!}, \quad H_{ik} = \int_{\mathbb{R}} \chi(y) w_i(y) \overline{w_k(y)} dy.$$

The matrix H is invertible. Here is the required independence proof. An identically zero linear combination of the w_i on a nonempty real interval has the form $\sum_{\rho} e^{\rho y} P_{\rho}(y) = 0$ with $\deg P_{\rho} < m_{\rho}$. By analyticity it vanishes everywhere. Fix ρ and apply

$$\prod_{\sigma \in Z, \sigma \neq \rho} (\partial_y - \sigma)^{m_{\sigma}}.$$

All other summands vanish. The remaining equation is

$$e^{\rho y} \prod_{\sigma \neq \rho} (\partial_y + \rho - \sigma)^{m_{\sigma}} P_{\rho}(y) = 0.$$

On polynomials of degree below m_{ρ} , each factor $\partial_y + c$ with $c \neq 0$ is invertible: its inverse is the finite sum

$$c^{-1} \sum_{r=0}^{m_{\rho}-1} (-c^{-1} \partial_y)^r.$$

Thus $P_{\rho} = 0$. Doing this for every ρ proves independence. For any nonzero column v ,

$$v^* H v = \int \chi(y) \left| \sum_i \overline{v_i} w_i(y) \right|^2 dy > 0,$$

because the combination cannot vanish on the whole interval where $\chi > 0$. Hence H is positive definite and invertible.

Define functions in the original positive variable by

$$F_k(x) = \chi(\log x) \sum_{\ell \in \Lambda} \overline{w_{\ell}(\log x)} (H^{-1})_{\ell k}.$$

Each is smooth and supported in $[e^{\alpha}, e^{\beta}]$, with all derivatives zero at its endpoints, so $F_k \in \mathcal{B}$. Direct substitution $y = \log x$ gives

$$\frac{(\mathcal{M}F_k)^{(j)}(\rho)}{j!} = \int \chi(y) w_{(\rho, j)}(y) \sum_{\ell} \overline{w_{\ell}(y)} (H^{-1})_{\ell k} dy = \delta_{(\rho, j), k}.$$

Thus the explicit finite-rank map

$$S : A_Z \rightarrow \mathcal{B}, \quad S\left(\sum_i c_i z_i^{j_i}\right) = \sum_i c_i F_i, \quad J_Z S = 1_{A_Z},$$

is a continuous linear section. The notation in the sum means the specified Taylor coordinates in their separate ρ -components.

For explicit continuity of the observation, choose integers

$$b_- < \min_{\rho \in Z} \Re \rho, \quad b_+ > \max_{\rho \in Z} \Re \rho.$$

For each $i = (\rho, j)$, direct integration of $|\log x|^j$ on the two half-lines proves

$$|(J_Z F)_{\rho, j}| \leq \frac{p_{b_-, 0}(F)}{(\Re \rho - b_-)^{j+1}} + \frac{p_{b_+, 0}(F)}{(b_+ - \Re \rho)^{j+1}}.$$

The $j!$ in the jet definition exactly cancels the $j!$ in those elementary integrals. All constants and original seminorms remain explicit. In particular J_Z and S are continuous, J_Z is surjective, and $\bar{J}_Z$ on Q is surjective. No assertion about closedness of ΘV or completeness of its quotient was needed.

210.11 7. An exact action on the original four-point theta sheaf

Define on $\mathcal{B}$

$$L_q = 1 + S(C_q - 1)J_Z.$$

Then

$$J_Z L_q = C_q J_Z, \quad L_q|_{\ker J_Z} = 1, \quad L_q \Theta = \Theta.$$

The identity $J_Z S = 1$ shows by expansion that

$$L_p L_q = 1 + S(C_p C_q - 1)J_Z = L_{pq}, \quad L_q^{-1} = 1 + S(C_q^{-1} - 1)J_Z.$$

These are exact inverses on the full original function space, not just on the observed packet.

Keep the original four-point upper-set space

$$+ < \eta < \sigma, \quad - < \eta < \sigma,$$

with coefficient sheaf $\mathcal{T}_+ = \mathcal{T}_- = V$, $\mathcal{T}_\eta = \mathcal{B}$, $\mathcal{T}_\sigma = 0$, and restriction maps Θ and $\mathcal{J}\Theta$. Define a sheaf automorphism to be the identity at $+$ and $-$, L_q at η , and the unique map on the zero vector space at σ . This is a sheaf morphism because L_q fixes both ΘV and $\mathcal{J}\Theta V = \Theta V$ pointwise. On derived global sections it is precisely

$$\begin{array}{ccc} V \oplus V & \xrightarrow{\Theta - \mathcal{J}\Theta} & \mathcal{B} \\ \downarrow 1 & & \downarrow L_q \\ V \oplus V & \xrightarrow{\Theta - \mathcal{J}\Theta} & \mathcal{B}. \end{array}$$

The signs and the original two source legs are unchanged.

Consequently L_q descends to an automorphism $\bar{L}_q$ of the exact quotient Q . It induces the identity on degree-zero cohomology and satisfies

$$\bar{J}_Z \bar{L}_q = C_q \bar{J}_Z.$$

Equip Q here with the quotient topology of the original $\mathcal{B}$. Both $\bar{L}_q$ and its inverse are continuous by the quotient universal property. This does not assert that Q is Hausdorff or that ΘV is closed. It is natural for the retained inclusions of nonempty support masks when the same S is used on their common $\mathcal{B}$ stalk. Indeed each $\mathcal{T}_A$ has its original active source legs, this same target, and the corresponding restrictions. The source identities and common L_q commute with every such inclusion. The split reconstruction acts by

$$\tau \mapsto \tau, \quad (A, F) \mapsto (A, L_q F).$$

It fixes $(A, 0)$ and keeps the label A . Thus neither synchronization of masks nor conversion of a supported zero into τ has occurred. At σ the reconstructed support skeleton is fixed.

The action has concrete analytic bounds. In the Taylor coordinate basis C_q is diagonal, so

$$p_{b,k}(L_q F) \leq p_{b,k}(F) + \sum_{\rho \in \mathbb{Z}} \sum_{j=0}^{m_\rho-1} |q^{-j} - 1| p_{b,k}(F_{\rho,j}) |(J_Z F)_{\rho,j}|.$$

Combining this with the preceding observation estimate proves continuity in every original Fréchet seminorm. For $q \geq 1$, all $|q^{-j} - 1| \leq 1$, so the displayed family of bounds is uniform in $q \geq 1$. For its inverse replace these factors by $|q^j - 1|$; it grows at most with the explicitly retained maximum degree $\max_\rho(m_\rho - 1)$. Nothing here bounds the original U_a in terms of $a^{1/2}$: L_q is the new finite-packet grading, and its inverse need not remain bounded as q grows.

The construction is noncanonical in its chosen bump and section. If S' is a second section and $K = S' - S$, then $J_Z K = 0$ and

$$L'_q - L_q = K(C_q - 1)J_Z.$$

Its image is in $\ker J_Z$, it vanishes on $\ker J_Z$, and both choices induce the same C_q on A_Z . This formula records the exact dependence instead of asserting a canonical global arithmetic action.

210.12 8. Compatibility with the original right-adjoint dual map

For the declared diagram category on the four-point space, the original calculation is

$$A_\tau(F) = [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta], \quad K_\tau(W) = [I_\eta(W) \xrightarrow{(+\text{res}, -\text{res})} I_+(W) \oplus I_-(W)]$$

in degrees 0, 1 and $-1, 0$ respectively. Here $I_x(W)(y) = W$ for $y \leq x$ and zero otherwise. Evaluation gives $\text{Hom}(F, I_x(W)) = \text{Hom}(F_x, W)$; applying this to the two displayed terms proves the right-adjoint identity with differential $\ell \mapsto (\ell r_+, -\ell r_-)$. Therefore

$$H^{-1} R \text{Hom}(\mathcal{T}, K_\tau(\mathcal{L}_1)) = \text{Hom}_E(Q, \mathcal{L}_1),$$

where the original moment line $\mathcal{L}_1$ has the original scaling action a . Its underlying vector space is E ; no action of the new grading on it is being inferred from arithmetic purity.

The original finite-packet morphism is the linear map from the conjugate vector space

$$\iota_Z : \overline{A_Z} \longrightarrow \text{Hom}_E(Q, \mathcal{L}_1), \quad \iota_Z(\bar{f})([F]) = R_Z(f, J_Z F).$$

It is well-defined because $J_Z \Theta = 0$, and injective by the just-proved surjectivity of J_Z and perfection of R_Z . Precomposition with $\bar{L}_q$ has the exact effect

$$\begin{aligned} (\bar{L}_q^* \iota_Z(\bar{f}))([F]) &= R_Z(f, J_Z L_q F) \\ &= R_Z(f, C_q J_Z F) \\ &= R_Z(D_q f, J_Z F). \end{aligned}$$

Thus the diagram into the **same** computed right-adjoint object satisfies

$$\boxed{\bar{L}_q^* \iota_Z(\bar{f}) = \iota_Z(\overline{D_q f}), \quad D_q = M_{\kappa_q} C_q^{-1}.}$$

Every arrow is now calculated on the original theta source and the full actual arithmetic jets. This is not merely a pairing on an unrelated finite algebra. All dual support arrows remain precomposition, and the zero functional at a present mask remains its supported zero functional.

210.13 9. Exact scope of the completed comparison

The new automorphism C_q retains the actual zero ρ , its complete local algebra, and its unit. It provides the conjugacy $N \mapsto q^{-1}N$. Its continuous lift L_q acts on the original theta sheaf and fixes the original relation space ΘV pointwise; its finite-packet effect and dual effect are the explicit diagrams above. These are completed maps with full proofs and analytic seminorm control.

The actual arithmetic scaling remains $U_a = a^\rho e^{(\log a)N}$. It commutes with N and has the exact semidirect relation with C_q . The grading eigenvalues are q^{-j} on the original Taylor coordinates. They do not contain a bound on $\Re \rho$. On simple packets C_q is the identity for every q , while U_a still has eigenvalue a^ρ .

More generally, multiplying C_q on a component by any nonzero scalar c leaves its conjugacy with N unchanged and changes all its eigenvalues to cq^{-j} . The residue formula and its full unit, not that conjugacy alone, state the actual additional information carried by our chosen C_q . Consequently this completed comparison supplies neither a canonical arithmetic Frobenius nor an absolute arithmetic weight estimate. The original tensor-growth calculation over A_τ remains the original problem; the new maps give a precise continuation instead of a false literal identification or an unsupported claim of irrelevance.

210.14 Provenance

Primary source: `output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md`, SHA-256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`. Canonical route: `work/f1_geometry_current_source_route_20260913.md`, SHA-256 `a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74`. Exact transcript node and its scope are recorded at the beginning. The independent algebra review is retained in `GERM_PAIRING_CHECK.md`. No Lean run, remote publication or shared-master edit is claimed.

211 Completed Mellin-realization calculations on the original theta source

These calculations close two finite algebraic gaps identified by the fully read transcript section U0029–U0040. They use the original theta source, its Fourier convention, its differentiated Gaussian, its complete two-leg support, and the original generator $g = 2\xi$. They do not replace that source by an arbitrary polynomial family. They concern Mellin realization, not the separately assigned homotopy fibre of restriction to σ .

211.1 Retained objects and source equations

Let $E = \mathbb{C}$ and retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx,$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Set

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The primary Tau Base NOTE.md equations (10)–(14),(21)–(22),(38) and the actual comparison source retain the identities

$$J\Theta = \Theta\mathcal{F}, \quad \mathcal{F}D = (1 - D)\mathcal{F}, \quad \Theta D = D\Theta, \quad \mathcal{M}(DF) = s\mathcal{M}F.$$

The fixed source function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \widehat{\phi_*} = \phi_*,$$

and

$$b_* = \Theta\phi_*, \quad \mathcal{M}b_* = g, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The entire function g is not zero. For example, the retained theta-boundary calculation gives $g(0)=1$. The arithmetic quotient is $Q = \mathcal{B}/\Theta V$. The original two-leg complex is

$$C^\bullet = [V \oplus V \xrightarrow{\Theta\phi - J\Theta\psi} \mathcal{B}], \quad C^0 = V \oplus V, \quad C^1 = \mathcal{B}.$$

Its H^0 is the actual Fourier graph V and its H^1 is Q . A single-leg complex has $H^0 = 0$; it is not the same complex.

All sheaf arguments below are stalkwise over $R = \mathcal{O}_{\mathbb{C}, s_0}$, with its original coordinate s , and hence glue. This local analytic ring is an integral domain. A nonzero entire function has a nonzero germ at every point by the identity theorem, even if its value at that point is zero.

211.2 Exact Fourier calculation for the retained theta source

We prove the Fourier self-duality used in the preceding definition, with the original convention and every coefficient. Put $u(x) = e^{-\pi x^2}$. Its integral is one. Indeed, its positive integral I satisfies, by Tonelli's theorem and the polar-coordinate Jacobian,

$$I^2 = \int_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^\infty e^{-\pi r^2} r dr d\vartheta = 2\pi \frac{1}{2\pi} = 1,$$

so $I = 1$.

Differentiation under the Fourier integral is justified by the integrable majorants $(2\pi|x|)^k u(x)$, for each fixed derivative order k . Since $u' = -2\pi x u$, integration by parts gives

$$\begin{aligned} \frac{d}{d\xi} \widehat{u}(\xi) &= \int_{\mathbb{R}} (-2\pi i x) u(x) e^{-2\pi i x \xi} dx \\ &= i \int_{\mathbb{R}} u'(x) e^{-2\pi i x \xi} dx \\ &= i(2\pi i \xi) \widehat{u}(\xi) = -2\pi \xi \widehat{u}(\xi). \end{aligned}$$

The boundary term is zero because $u(x)$ tends to zero at both ends of the real line. The differential equation, together with $\widehat{u}(0) = 1$, gives

$$\widehat{u}(\xi) = e^{-\pi \xi^2}.$$

For every Schwartz function f , the same differentiation and integration by parts prove the typed operator identities

$$\mathcal{F}(xf) = \frac{i}{2\pi} \partial_\xi \widehat{f}, \quad \mathcal{F}(f') = 2\pi i \xi \widehat{f}.$$

The Gaussian derivatives are

$$\begin{aligned} u''(\xi) &= (4\pi^2 \xi^2 - 2\pi) u(\xi), \\ u^{(4)}(\xi) &= (16\pi^4 \xi^4 - 48\pi^3 \xi^2 + 12\pi^2) u(\xi). \end{aligned}$$

Consequently,

$$\begin{aligned} \mathcal{F}(x^2 u)(\xi) &= -\frac{1}{4\pi^2} u''(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) u(\xi), \\ \mathcal{F}(x^4 u)(\xi) &= \frac{1}{16\pi^4} u^{(4)}(\xi) = \left(\xi^4 - \frac{3}{\pi} \xi^2 + \frac{3}{4\pi^2} \right) u(\xi). \end{aligned}$$

Apply these formulas to the original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)u$:

$$\begin{aligned}\widehat{\phi_*}(\xi) &= \left[4\pi^2 \left(\xi^4 - \frac{3}{\pi} \xi^2 + \frac{3}{4\pi^2} \right) - 6\pi \left(\frac{1}{2\pi} - \xi^2 \right) \right] u(\xi) \\ &= (4\pi^2 \xi^4 - 12\pi \xi^2 + 3 - 3 + 6\pi \xi^2) u(\xi) \\ &= (4\pi^2 \xi^4 - 6\pi \xi^2) e^{-\pi \xi^2} = \phi_*(\xi).\end{aligned}$$

This establishes the literal equality $\widehat{\phi_*} = \phi_*$. The function is even and Schwartz, $\phi_*(0) = 0$, and $\widehat{\phi_*}(0) = \phi_*(0) = 0$; hence it belongs to the exact source space V .

The same calculation also verifies the source operator expression and the sign of its Fourier intertwiner. Since

$$Du = 2\pi x^2 u, \quad D^2 u = (-4\pi x^2 + 4\pi^2 x^4)u,$$

we have

$$D(D-1)u = (4\pi^2 x^4 - 6\pi x^2)u = \phi_*.$$

For every Schwartz f ,

$$\begin{aligned}\widehat{Df}(\xi) &= -\mathcal{F}(xf')(\xi) = -\frac{i}{2\pi} \partial_\xi (2\pi i \xi \widehat{f}(\xi)) \\ &= \widehat{f}(\xi) + \xi \partial_\xi \widehat{f}(\xi) = (1 - D_\xi) \widehat{f}(\xi).\end{aligned}$$

Here $D_\xi = -\xi \partial_\xi$, with the same sign as the original operator. This proves $\mathcal{F}D = (1 - D)\mathcal{F}$ on its stated Schwartz domain. Moreover, D preserves V : it preserves evenness and the Schwartz class, $(D\phi)(0) = 0$, and

$$\int_{\mathbb{R}} D\phi(x) dx = -[x\phi(x)]_{-\infty}^{\infty} + \int_{\mathbb{R}} \phi(x) dx = 0 \quad (\phi \in V).$$

The boundary term vanishes by Schwartz decay. Thus every $D^m \phi_*$ and every $(1 - D)^m \phi_*$ occurring in the two-leg calculation remains in the original domain.

The Mellin constant can also be obtained directly from the same function. For $\Re s > 0$, substitution $t = \pi x^2$ and $\Gamma(z+1) = z\Gamma(z)$ give

$$\begin{aligned}M_+ \phi_*(s) &= \int_0^\infty (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2} x^s \frac{dx}{x} \\ &= \pi^{-s/2} \left(2\Gamma\left(\frac{s}{2} + 2\right) - 3\Gamma\left(\frac{s}{2} + 1\right) \right) \\ &= \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \left[2\frac{s}{2} \left(\frac{s}{2} + 1\right) - 3\frac{s}{2} \right] \\ &= \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right).\end{aligned}$$

The recurrence follows by integration by parts in the defining Gamma integral on this half-plane; its boundary terms vanish at zero and infinity. For $\Re s > 1$, the full nonzero-integer theta sum is absolutely integrable term by term, because

$$\sum_{n \neq 0} \int_0^\infty |\phi_*(nx)| x^{\Re s} \frac{dx}{x} = 2 \sum_{n \geq 1} n^{-\Re s} \int_0^\infty |\phi_*(y)| y^{\Re s} \frac{dy}{y} < \infty.$$

Therefore

$$\begin{aligned}\mathcal{M}\Theta\phi_*(s) &= 2\zeta(s)M_+\phi_*(s) \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s) = g(s).\end{aligned}$$

This retains both the factor two from the full integer sum and the factor one half in the source Mellin integral. The source's entire continuation identifies these same functions away from this initial half-plane; no division by g at an arithmetic zero has occurred.

211.3 The raw two-leg syzygy at every source order

For each integer $m \geq 1$, retain the actual functions

$$b_{+,m} = \Theta D^m \phi_* = D^m b_*, \quad b_{-,m} = J \Theta D^m \phi_* = (1 - D)^m b_*.$$

Their Mellin transforms are exactly

$$\mathcal{M}b_{+,m} = s^m g, \quad \mathcal{M}b_{-,m} = (1 - s)^m g.$$

Thus the same source input $D^m \phi_*$ on the two original charts gives these two images. No independent spectral function has been chosen.

Put $a_m(s) = s^m$ and $v_m(s) = (1 - s)^m$. Define

$$\begin{aligned} \delta_m : R^2 &\longrightarrow R \otimes_E \mathcal{B}, & (f, h) &\longmapsto f \otimes b_{+,m} - h \otimes b_{-,m}, \\ \beta : R \otimes_E \mathcal{B} &\longrightarrow R, & f \otimes F &\longmapsto f \mathcal{M}F. \end{aligned}$$

These give the literal commutative comparison

$$\begin{array}{ccc} R^2 & \xrightarrow{\delta_m} & R \otimes_E \mathcal{B} \\ 1 \downarrow & & \downarrow \beta \\ R^2 & \xrightarrow{(f,h) \mapsto g(a_m f - v_m h)} & R. \end{array}$$

First, $b_{+,m}$ and $b_{-,m}$ are E-linearly independent. A relation $cb_{+,m} + db_{-,m} = 0$ gives $g(s)(cs^m + d(1 - s)^m) = 0$. On the nonempty open set on which g does not vanish the polynomial factor vanishes, so that polynomial is identically zero. Evaluation at $s=0$ gives $d=0$; evaluation at $s=1$ gives $c=0$. Tensoring this two-dimensional E-subspace with R is injective because vector spaces split over E . Therefore δ_m is injective.

Here are the exact polynomial coefficients used for its composite:

$$A_m(s) = \sum_{k=0}^{m-1} \binom{m+k-1}{k} s^k, \quad B_m(s) = \frac{1 - A_m(s)(1 - s)^m}{s^m}.$$

The power series $(1 - s)^{-m}$ has coefficients $\binom{m+k-1}{k}$, so its truncation A_m gives a numerator divisible by s^m . Thus B_m is a polynomial and

$$B_m a_m + A_m v_m = 1.$$

The germ g is a non-zero-divisor. Consequently $\beta \delta_m(f, h) = 0$ is equivalent to $a_m f = v_m h$. Set $r = B_m h + A_m f$. Then

$$v_m r = B_m v_m h + A_m v_m f = B_m a_m f + A_m v_m f = f,$$

$$a_m r = B_m a_m h + A_m a_m f = B_m a_m h + A_m v_m h = h.$$

Conversely $(f, h) = (v_m r, a_m r)$ has $a_m f - v_m h = 0$. The parametrization is injective because its two coefficients generate the unit ideal. Hence

$$\boxed{\ker \delta_m = 0, \quad \ker(\beta \delta_m) = R(v_m, a_m).}$$

In particular

$$\boxed{z_m = v_m \otimes b_{+,m} - a_m \otimes b_{-,m} \neq 0, \quad \beta z_m = 0.}$$

This proves the exact all-order extension of A1361's $m=1$ witness.

Its support is not optional. Let $L = \mathcal{P}(\{+, -\})$, with empty mask corresponding to external absence. On the $+$ face δ_m is $f \mapsto f \otimes b_{+,m}$; on the $-$ face it is $h \mapsto -h \otimes b_{-,m}$; on the joint face it is the map above. All maps retain their masks. The raw maps on either singleton face, and their Mellin composites, are injective because $b_{\pm,m}$ and ga_m, gv_m respectively are nonzero. The new syzygy occurs only in the joint face. Its lifted image is

$$(\{+, -\}, z_m) \mapsto (\{+, -\}, 0),$$

not τ . At $m=0$ the two source images coincide, and the raw map already has kernel $R(1,1)$; thus $m=0$ is not falsely included in the new-relation assertion.

This result excludes a quasi-isomorphism of these particular raw complexes: the first H^0 is zero while the second H^0 is R . It says nothing by itself about nonvanishing of the full realized H^1 kernel; that different calculation is given in Section 211.5.

211.4 Operator balancing removes the raw tensor relation, and retains honest joint cycles

Let $A = E[t]$. On $\mathcal{B}$ let t act by D , and on R let t act by multiplication by s . The quotient map

$$\pi_{\text{bal}} : R \otimes_E \mathcal{B} \longrightarrow R \otimes_A \mathcal{B}$$

imposes the R -linear span of the actual relations

$$sh \otimes F - h \otimes DF.$$

These are Mellin-null because $\mathcal{M}(DF) = s\mathcal{M}F$.

For every $j \geq 1$, the exact telescoping identity is

$$h \otimes D^j b_* - h s^j \otimes b_* = \sum_{k=0}^{j-1} \left(h s^k \otimes D^{j-k} b_* - h s^{k+1} \otimes D^{j-k-1} b_* \right).$$

Each summand is the negative of a displayed elementary balancing relation. Applying this identity to all coefficients of the original polynomials t^m and $(1-t)^m$ proves

$$\pi_{\text{bal}}(h \otimes P(D)b_*) = hP(s) \otimes b_* \quad (P \in E[t]).$$

It follows without removing any sign that

$$\pi_{\text{bal}}(z_m) = v_m a_m \otimes b_* - a_m v_m \otimes b_* = 0.$$

More explicitly, with $\Delta_P = 1 \otimes P(D)b_* - P(s) \otimes b_*$,

$$z_m = v_m \Delta_{t^m} - a_m \Delta_{(1-t)^m}.$$

The coefficients of $(1-t)^m$ remain $(-1)^j \binom{m}{j}$.

Balancing is not a reason to erase genuine degree-zero cycles in the original two-leg source. The exact saturated source calculation makes this precise. On the plus leg let t act by D ; on the minus leg let t act by $1 - D$. These are the infinitesimal actions induced by primary (34), since the minus scaling is $aU_{1/a}$. Define

$$\begin{aligned} i_{+,m} : A &\longrightarrow V, & P &\longmapsto P(D)D^m \phi_*, \\ i_{-,m} : A &\longrightarrow V, & Q &\longmapsto Q(1-D)D^m \phi_*. \end{aligned}$$

They are injections. Indeed, Θ applied to the first has Mellin transform $P(s)s^m g$; $J\Theta$ applied to the second has transform $Q(s)(1-s)^m g$, and polynomial multiplication by either nonzero entire function

is injective. Let Ab_* be the cyclic A -submodule of $\mathcal{B}$ generated by b_* . It too is free rank one because $P(D)b_* = 0$ implies $P(s)g(s)=0$ and therefore $P=0$.

The original two-leg differential restricted to these saturated subspaces is therefore exactly

$$C_m^{\text{cyc}} = \left[A^2 \xrightarrow{(P,Q) \mapsto (t^m P - (1-t)^m Q)b_*} Ab_* \right].$$

Fourier supplies the minus term:

$$J\Theta(Q(1-D)D^m\phi_*) = Q(D)(1-D)^m b_*.$$

The same Bezout polynomials show the differential is onto and that its kernel is $A((1-t)^m, t^m)$. Hence

$$\boxed{H^0(C_m^{\text{cyc}}) = A((1-t)^m, t^m), \quad H^1(C_m^{\text{cyc}}) = 0.}$$

This cycle is an actual Fourier-compatible two-leg cycle before Mellin realization. It is not a spurious raw tensor relation.

The A -module R is torsion-free: multiplication by a nonzero polynomial $P(s)$ is injective in the analytic domain R . A torsion-free module over a PID is flat. One direct proof writes it as the directed union of its finitely generated submodules; those submodules are finitely generated torsion-free modules over a PID and hence free, and tensoring commutes with directed colimits. A filtered colimit of exact sequences of modules is exact. Thus R is flat, and tensoring the displayed exact cyclic sequence gives

$$H^0(R \otimes_A C_m^{\text{cyc}}) = R((1-s)^m, s^m), \quad H^1 = 0.$$

No joint H^0 has been lost.

For completeness, its analytic comparison has $\beta_{\text{cyc}}(h \otimes b_*) = hg$ and identity in degree zero:

$$[R^2 \xrightarrow{a_m f - v_m h} Rb_*] \longrightarrow [R^2 \xrightarrow{g(a_m f - v_m h)} R].$$

The degree-one map $\times g$ is injective with cokernel $R/(g)$, and the degree-zero map is identity. Hence there is an exact sequence of these cochain complexes whose quotient is $R/(g)$ concentrated in degree one. The induced map on H^0 is identity on the displayed R -syzygy; the analytic target has $H^1 = R/(g)$. This calculation locates precisely what changes for this saturated cyclic subcomplex.

The cyclic subcomplex is not the full theta complex. Its inclusion is the literal pair $(i_{+,m}, i_{-,m})$ in degree zero and $Ab_* \hookrightarrow \mathcal{B}$ in degree one, and its differential is the restriction just proved. Extending these maps to the corresponding supported faces retains all masks. No equality with the full balanced kernel has been inferred.

211.5 An explicit free submodule in the raw realization kernel

This makes A1427's nonvanishing statement constructive with the actual finite-jet maps.

Let Z be any finite set of actual zeros of g , with their actual positive multiplicities m_ρ . Retain the complete jet space

$$A_Z = \bigoplus_{\rho \in Z} E[z_\rho]/(z_\rho^{m_\rho}), \quad z_\rho = s - \rho,$$

and the primary map

$$J_Z : Q \longrightarrow A_Z, \quad [F] \longmapsto \left(\sum_{j < m_\rho} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j \right)_\rho.$$

The source's smooth compactly supported right inverse can be written in its original measure. Fix $0 < R' < R_0$, let

$$\eta(y) = \begin{cases} \exp(-1/(R'^2 - y^2)), & |y| < R', \\ 0, & |y| \geq R', \end{cases}$$

and for $\alpha = (\rho, j)$ put

$$r_\alpha(x) = x^{\bar{\rho}-1} \frac{(\log x)^j}{j!},$$

$$B_{\alpha\beta} = \frac{1}{j!k!} \int_{-R'}^{R'} \eta(y) e^{(\rho+\bar{\sigma}-1)y} y^{j+k} dy \quad (\beta = (\sigma, k)).$$

For a coefficient vector c ,

$$c^* B c = \int_{-R'}^{R'} \eta(y) e^{-y} \left| \sum_{\alpha} \bar{c}_{\alpha} e^{\rho y} \frac{y^j}{j!} \right|^2 dy.$$

The exponential polynomials with distinct exponents and all orders below the retained multiplicities are linearly independent on every open interval. To verify this, a vanishing sum is an entire function of y and vanishes identically. For a selected exponent ρ , applying $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{m_{\sigma}}$ kills the other blocks. On $e^{\rho y} P_{\rho}(y)$, each factor becomes $(\partial_y + \rho - \sigma)^{m_{\sigma}}$ on the polynomial P_{ρ} ; its constant term is nonzero, and in the degree-ordered polynomial basis its matrix is triangular with nonzero diagonal. Its product is therefore invertible. Thus $P_{\rho} = 0$ for each ρ . Since η is positive on the interval, $c^* B c = 0$ forces $c=0$. Hence B is positive definite and invertible.

For $v \in A_Z$ define

$$F_v(x) = \eta(\log x) \sum_{\beta} (B^{-1}v)_{\beta} r_{\beta}(x).$$

It lies in $C_c^{\infty}((e^{-R_0}, e^{R_0})) \subset \mathcal{B}$. Substituting $x = e^y$ in each Mellin derivative gives $J_Z[F_v] = v$, with exactly the displayed factorials and exponent -1 . Thus

$$s_Z : A_Z \longrightarrow Q, \quad v \longmapsto [F_v]$$

is an explicit E -linear section of J_Z . It is not asserted to be equivariant under scaling; that additional property is neither needed nor inferred.

At each analytic stalk retain

$$\bar{\beta} : R \otimes_E Q \longrightarrow R/(g), \quad h \otimes [F] \longmapsto [h M F], \quad K_{\text{raw}} = \ker \bar{\beta}.$$

Define

$$\boxed{\iota_Z : R \otimes_E A_Z \longrightarrow K_{\text{raw}}, \quad h \otimes v \longmapsto g h \otimes s_Z(v).}$$

The image is in the kernel because its Mellin evaluation is a multiple of g . If $\iota_Z(w) = 0$, applying $1 \otimes J_Z$ gives $g w = 0$ in $R \otimes_E A_Z$. This latter module is free of rank $\sum_{\rho \in Z} m_{\rho}$, and g is a non-zero-divisor. Thus $w=0$. Therefore ι_Z is an injective R -linear map.

This gives an explicit free rank- $\sum_{\rho \in Z} m_{\rho}$ submodule of the raw kernel for every nonempty retained packet Z . The map is natural under stalk restriction for the fixed smooth section. Every derivative direction below each actual multiplicity appears in its domain. In particular the proof does not merely cite existence of a Hamel basis for Q or reduce the packet to its values.

At each original nonempty support mask L , lift the map as

$$(L, w) \longmapsto (L, \iota_Z w),$$

with $\tau \mapsto \tau$. It remains injective, retains its labelled zero, and commutes with the identity transports among the primary nonempty H^1 fibres. The target is the all-support realization-kernel fibre, not an identification with external absence.

Neither this free raw submodule nor the balancing calculation proves that the global balanced kernel is nonzero or zero. They prove the exact raw obstruction and the exact repair of its exhibited operator relation. The surviving full map is

$$\mathcal{O} \otimes_{E[t]} Q \longrightarrow \mathcal{O}/(g),$$

together with its actual kernel and the original strong-Schwartz/quotient topology. It must be connected through the original A_τ and K_τ ; the present calculations have made no replacement of those objects.

211.6 Source pins and transcript locators

The complete primary source was read: Tau_Base_Cohomology_2026-09-12/NOTE.md, SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. The canonical route was read completely, SHA-256 a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74.

The all-order calculation answers A1361 (node c9e551ee-298f-438b-9bdc-2e23c2c0613a), source Section 1, and A1427 (node d7e752ce-cb39-4130-b089-b20333037990), source Section 4, while retaining A1304's actual two-leg source. The smooth section retains A1167 (node 157eecd7f96-4ec5-96bd-6dd376276ea6), source Section 5, with its original measure, factorials, and full jet orders. These are exact transcript UUID locators; line endings of the readable extraction were repaired during the audit without changing message text.

These are written proofs. No Lean session, remote write, numerical zero computation, or independently rerun historical checker is claimed.

212 Exact weighted theta-source constants

This calculation finishes the finite moment reduction left explicit in U0065/A2497 and U0066/A2559 of the retained transcript. It calculates the weighted-source Grams controlling the circle length, and the derivative Grams controlling its Fourier cutoff, on the original theta source. It supplies no asymptotic bound on their growing-degree inverses.

212.1 Original objects and domains

Retain the source's actual finite packet polynomial $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, with the full selected orders, $g = 2\xi$, and its specified inverse Mellin representative $F_h \in \mathcal{B}$, where

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}.$$

Write $v_h(s) = \mathcal{M}F_h(s) = g(s)/h(s)$, $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$, and $D = -x\partial_x$. For $k \geq 1$, set

$$c = k/2, \quad D^{(k)} = \sum_{j=1}^k D_j, \quad \mathcal{V}P = P(D^{(k)})F_h^{\otimes k}, \quad S = \sum_{j=1}^k s_j. \quad (1)$$

The polynomial source of degree at most N is $\mathcal{P}_N$, with the original basis $1, S, \dots, S^N$. Its arithmetic quotient map is the unchanged remainder map $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[S]/(\chi_{h,k})$. Its full observation is

$$J^{(k)}\mathcal{V} = \eta_{h,k}J_N, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad v_h = j_h(g/h). \quad (2)$$

No coefficient of this unit is changed by the calculations below.

The original half-density coordinates are

$$r = \log x_k, \quad z_i = \log x_i - \log x_k \ (i < k), \quad (3)$$

and

$$(\mathcal{U}_k F)(r, \mathbf{z}) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r). \quad (4)$$

The coordinate inverse in (3) has determinant one in logarithmic coordinates. Consequently (4) is an isometry from $L^2((0, \infty)^k, d^k x)$ to $L^2(\mathbb{R}, dr; L^2(\mathbb{R}^{k-1}, d\mathbf{z}))$, and $\partial_r \mathcal{U}_k = -\mathcal{U}_k(D^{(k)} - c)$. These statements follow by substitution in the measure and by the chain rule, respectively; for $k = 1$ the relative space is $\mathbb{C}$.

212.2 Weighted lines and exact finite moments

For every real a , define

$$w_a(t) = \frac{|v_h(1/2 + a + it)|^2}{2\pi}, \quad \mu_j(a) = \int_{\mathbb{R}} t^j w_a(t) dt, \quad \mathcal{M}_a(z) = \sum_{j \geq 0} \mu_j(a) \frac{z^j}{j!}. \quad (5)$$

The last expression is a formal series: each coefficient below uses only finitely many of its entries. No analytic convergence assertion about that series is needed.

All the integrals in (5) are finite. Indeed, for any real σ and nonnegative integers b, j , $y^b \partial_y^j (e^{\sigma y} F_h(e^y))$ decays exponentially at both ends. To check this, expand the derivative into finitely many terms $e^{\sigma y} D^l F_h(e^y)$, use a positive integer exponent larger than σ at $+\infty$, and an integer exponent less than σ at $-\infty$ in the definition of $\mathcal{B}$. Any polynomial in y is absorbed by the resulting strict exponential margins. Thus $e^{\sigma y} F_h(e^y)$ is Schwartz. Its Fourier transform with kernel e^{ity} is exactly $v_h(\sigma + it)$, which is therefore Schwartz on that line.

Let

$$m_{k,a} = w_0^{*(k-1)} * w_a, \quad \mu_j^{(k,a)} = j! [z^j] \mathcal{M}_0(z)^{k-1} \mathcal{M}_a(z), \quad (6)$$

where the exponent zero in the case $k = 1$ means the identity for convolution. Expansion of $(t_1 + \dots + t_k)^j$, with Fubini justified by absolute polynomial moments, proves $\mu_j^{(k,a)} = \int t^j m_{k,a}(t) dt$.

Theorem 212.1 (The actual weighted source Gram). For $P, Q \in \mathcal{P}_N$, the original weighted observation is

$$\begin{aligned} M_{k,N}(a)(P, Q) &:= \int_{(0, \infty)^k} x_k^{2a} \overline{\mathcal{V}P(\mathbf{x})} \mathcal{V}Q(\mathbf{x}) d^k x \\ &= \int_{\mathbb{R}} \overline{P(c + a + iu)} Q(c + a + iu) m_{k,a}(u) du. \end{aligned} \quad (7)$$

In the unchanged basis $1, S, \dots, S^N$, its entries are exactly

$$[M_{k,N}(a)]_{b,d} = \sum_{r=0}^b \sum_{s=0}^d \binom{b}{r} \binom{d}{s} (c+a)^{b+d-r-s} (-i)^r i^s (r+s)! [z^{r+s}] \mathcal{M}_0^{k-1} \mathcal{M}_a. \quad (8)$$

In particular these matrices require only $\mu_j(0), \mu_j(a)$ for $0 \leq j \leq 2N$. Their masses are

$$[M_{k,N}(a)]_{0,0} = \mu_0(0)^{k-1} \mu_0(a). \quad (9)$$

Proof. Put $y_j = \log x_j$, use $\sigma_j = 1/2$ for $j < k$ and $\sigma_k = 1/2 + a$, and apply ordinary Fourier Plancherel to $e^{\sum_j \sigma_j y_j} \mathcal{V}P(e^{\mathbf{y}})$. Integration by parts has no boundary term by the estimates just proved. It sends each D_j to multiplication by $s_j = \sigma_j + it_j$. Hence its Fourier transform is

$$P(c + a + i \sum_j t_j) \prod_{j < k} v_h(1/2 + it_j) v_h(1/2 + a + it_k).$$

The factor $(2\pi)^{-k}$ in Plancherel is exactly the product of the $(2\pi)^{-1}$ factors in (5). Expanding the complex inner product and integrating over the sum coordinate proves (7). The binomial formula for each original monomial gives (8), including both phases. Formula (9) is its constant entry. In particular the shift $c+a$ has not been replaced by c , and the sum has not been replaced by an average. $\square$

Each matrix $M_{k,N}(a)$ is positive definite. The entire nonzero Mellin function v_h has isolated zeros on each vertical line, so $w_a > 0$ almost everywhere. For $k \geq 2$, a convolution in (6) is positive at every real sum: for almost every choice of the first $k-1$ variables all factors of its integrand are positive. For $k=1$, the almost-everywhere positivity suffices. A nonzero polynomial has only finitely many zeros on the integration line in (7), so its squared norm is strictly positive.

Let $M = M_{k,N}(0)$. The constants required by A2497 and A2559 are therefore the explicitly defined finite-matrix values

$$\kappa_{a,N} = \lambda_{\max}(M^{-1}(M_{k,N}(a) + M_{k,N}(-a))), \quad a > 0, \quad (10)$$

where eigenvalues are understood in the positive inner product M . In particular there is a fully specified upper value avoiding an additional extreme-eigenvalue calculation:

$$0 < \kappa_{a,N} \leq \mathfrak{k}_{a,N} := \text{Tr}(M^{-1}(M_{k,N}(a) + M_{k,N}(-a))). \quad (11)$$

To prove (11), conjugate by $M^{1/2}$; the resulting matrix is positive, and each eigenvalue is at most the sum of its eigenvalues. The congruence is used to prove the inequality, not to replace the source basis or its metric. With $\kappa_{\pm} = \lambda_{\max}(M^{-1}M_{k,N}(\pm a))$, the same argument gives the individual upper values $\text{Tr}(M^{-1}M_{k,N}(\pm a))$.

212.3 Derivative and Fourier-cutoff Grams, including the cross terms

Write

$$a_h(t) = v_h(1/2 + it)/\sqrt{2\pi}, \quad n(t) = |a'_h(t)|^2, \quad b(t) = \overline{a'_h(t)}a_h(t). \quad (12)$$

All three measures $w_0(t)dt, n(t)dt, b(t)dt$ have finite absolute polynomial moments by the same Schwartz argument. The last is a complex measure; it is not replaced by a positive one. Define

$$\begin{aligned} m_k &= w_0^{*k}, & n_k &= w_0^{*(k-1)} * n, & b_k &= w_0^{*(k-1)} * b, \\ \mathcal{N}(z) &= \sum_{j \geq 0} \int t^j n(t) dt \, z^j / j!, & \mathcal{C}(z) &= \sum_{j \geq 0} \int t^j b(t) dt \, z^j / j!. \end{aligned} \quad (13)$$

Let $p \geq 1$, $R_P(S) = (S - c)^p P(S)$, and let primes on R_P mean polynomial derivatives in the unchanged coordinate S . The derivative Gram in A2497/A2559 is

$$H_{p,N}(P, Q) = \int_{\mathbb{R} \times \mathbb{R}^{k-1}} (1 + r^2) \overline{\partial_r^p \mathcal{U}_k} \mathcal{V} P \, \partial_r^p \mathcal{U}_k \mathcal{V} Q \, dr \, d\mathbf{z}. \quad (14)$$

Theorem 212.2 (Complete derivative Gram). With all polynomials in the following formula evaluated at $c + iu$,

$$\begin{aligned} H_{p,N}(P, Q) &= \int_{\mathbb{R}} (\overline{R_P} R_Q (m_k + n_k) + \overline{R'_P} R'_Q m_k \\ &\quad + i \overline{R_P} R'_Q b_k - i \overline{R'_P} R_Q \overline{b_k})(u) \, du. \end{aligned} \quad (15)$$

Every entry is a finite expression in the moments encoded by $\mathcal{M}_0^k$, $\mathcal{M}_0^{k-1} \mathcal{N}$, and $\mathcal{M}_0^{k-1} \mathcal{C}$, through degree $2(N + p)$.

Proof. The chain rule in (4) gives $\partial_r^p \mathcal{U}_k \mathcal{V} P = (-1)^p \mathcal{U}_k \mathcal{V} R_P$. The two occurrences of $(-1)^p$ cancel in (14). The term without r^2 is (7) at $a = 0$, with polynomial R_P . For the second term return to the independent logarithmic coordinates $y_1, \dots, y_k$; its multiplier is exactly $r = y_k$. For the Fourier kernel $e^{i \sum t_j y_j}$, multiplication by y_k transforms to $-i \partial_{t_k}$. The factor $-i$ cancels against its conjugate in the inner product. After retaining the Plancherel constants in the product of the a_h 's, the differentiated last factor is

$$a'_h(t_k) R_P(c + i \sum_j t_j) + i a_h(t_k) R'_P(c + i \sum_j t_j).$$

Multiplication by its conjugate for P and by the displayed expression for Q gives, respectively, the n_k term, the derivative m_k term, and the cross terms $+i \overline{R_P} R'_Q b_k$ and $-i \overline{R'_P} R_Q \overline{b_k}$. Integrating the other variables gives (15). Expansion in powers of the sum and Fubini prove the stated coefficient products. The largest degree occurs in $\overline{R_P} R_Q$ and is at most $2(N + p)$. $\square$

212.4 A finite choice of comparison parameters and its exact scope

Set

$$\mathfrak{h}_{p,N} = \text{Tr}(M^{-1} H_{p,N}) \geq 0. \quad (16)$$

Both sides in (16) are defined on the original coefficient space. The same positive-matrix argument used for (11) gives $H_{p,N} \preceq \mathfrak{h}_{p,N} M$. Here are the explicit parameters obtained from these completed calculations. For prescribed positive error contributions η_1, η_2 , choose

$$L \geq a^{-1} \log(1 + \mathfrak{k}_{a,N}/\eta_1), \quad (17)$$

and, for the twisted lattice used by all scalar holonomy phases, take an integer

$$J \geq \max \left\{ 2, 1 + \frac{L}{2\pi} \left(\frac{\mathfrak{h}_{p,N}}{(2p-1)\eta_2} \right)^{1/(2p-1)} \right\}. \quad (18)$$

The ordinary lattice can omit the extra one; it has denominator J instead of $J - 1$.

For completeness, the source estimates underlying this parameter choice have direct proofs. Write $\Psi = \mathcal{U}_k \mathcal{V}$ and $\mathcal{C}(s) = \int \Psi(r)^* \Psi(r+s) dr$. Weighted Cauchy–Schwarz gives, for $s > 0$,

$$|v^* \mathcal{C}(s) v| \leq e^{-as} \sqrt{v^* M_{k,N}(-a) v \, v^* M_{k,N}(a) v}.$$

The negative shifts exchange the two weights. Unfolding one period gives the Gram difference as the sum of the nonzero correlations at shifts mL ; holonomy only multiplies each by a unit complex number. Summing the geometric series and using $2\sqrt{xy} \leq x + y$ proves its absolute quadratic-form bound $(M_{k,N}(a) + M_{k,N}(-a))/(e^{aL} - 1)$. This is at most $\eta_1 M$ by (11) and (17).

Integration by parts p times in the retained Fourier transform, then Cauchy–Schwarz with $\int (1 + r^2)^{-1} dr = \pi$, gives $\|\widehat{\Psi} v(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For the ordinary lattice, summing $|n| > J$ and bounding $\sum_{n>J} n^{-2p} \leq J^{1-2p}/(2p-1)$ gives

$$0 \preceq M(L) - M(L, J) \preceq \frac{1}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1} H_{p,N}. \quad (19)$$

The constants arise from π/L in each Fourier coefficient and the two lattice directions; thus the factor in (19) is literal. For phases $0 \leq \theta < 2\pi$, $|n + \theta/(2\pi)| \geq |n| - 1$ for the omitted indices. Repeating the integral bound replaces J by $J - 1$, and (18) makes this at most $\eta_2 M$. With $\eta_1 + \eta_2 < 1$, the resulting finite source Gram lies between $(1 - \eta_1 - \eta_2)M$ and $(1 + \eta_1 + \eta_2)M$.

For this quotient conclusion put $q = \deg \chi_{h,k}$ and take $N \geq q - 1$, so the unchanged remainder map is onto E . Taking the infimum over the unchanged affine sets $\{P : J_N P = v\}$ gives the same inequalities for the canonical quotient Grams. Their sections differ by the original $\ker J_N = \chi_{h,k} \mathcal{P}_{N-q}$,

with $\mathcal{P}_{-1} = 0$, hence by the original theta relations under (1)–(2). At a retained support label that correction maps to its supported zero; it never becomes external absence.

These estimates compute the finite error parameters from the actual one-factor shifted Mellin moments and their full derivative cross terms. They do not bound $\mathfrak{k}_{a,N}$ or $\mathfrak{h}_{p,N}$ uniformly as k, N grow. For the original four endpoints, one may use $N = 2q_k + 2$ in (17)–(18), exactly as in A2497; all smaller polynomial spaces are restrictions of that single source. The inverse M^{-1} and every original mass remain visible, so this result cannot be mistaken for the missing sub-threshold determinant estimate.

Finally, the weighted observations are norms of the already specified representatives, not new maps identifying the arithmetic cohomology with the Boolean stalk. In the primary Tau Base source the cohomology is $A_\tau = R\Gamma(P, -)$, its right adjoint is $K_\tau(W) \simeq S_\eta W[1]$, and its actual top tensor quotient is $Q^{\otimes k}$. Equations (1)–(2) remain the same finite source and full arithmetic observation into that construction. Multiplying the source norm by x_k^{2a} has not been asserted to descend as an endomorphism of $Q^{\otimes k}$.

213 Circle resolvent with all arithmetic jets

U0065/A2470 promised a resolvent derived from theta boundaries. The subsequent A2497 and its retained note calculate the first- and second-order source defects. This note completes the resolvent calculation from those exact defects, including the resonant case. It makes no purity inference from the circle spectrum.

213.1 Original operator and its domain

Fix $k \geq 1$, $L > 0$, $c = k/2$, and the relative Hilbert space $\mathcal{K} = L^2(\mathbb{R}^{k-1})$; for $k = 1$ use $\mathcal{K} = \mathbb{C}$. The scalar holonomy phase is $0 \leq \theta < 2\pi$. Let

$$\mathcal{H} = L^2([0, L]; \mathcal{K}), \quad \mathcal{D} = -\partial_r + c, \quad \text{Dom } \mathcal{D} = \{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}. \quad (1)$$

The original orthonormal scalar functions are $e_n(r) = L^{-1/2} e^{-i\omega_n r}$, $\omega_n = (2\pi n + \theta)/L$. Fourier series, including $n = 0$, identify (1) with the diagonal

$$\mathcal{D}(e_n \otimes v) = (c + i\omega_n) e_n \otimes v, \quad \sum_n (1 + \omega_n^2) \|f_n\|_{\mathcal{K}}^2 < \infty. \quad (2)$$

In particular (1) is closed and normal: the adjoint of the diagonal has conjugate entries and the same domain. Its spectrum is exactly $\{c + i\omega_n : n \in \mathbb{Z}\}$. To check the last assertion, each listed value has its nonzero eigenvectors, while at any other complex number z the inverse diagonal is bounded and maps onto the stated domain. Its exact operator norm is

$$\|T(z)\| = \delta(z)^{-1}, \quad T(z) = (\mathcal{D} - z)^{-1}, \quad \delta(z) = \min_{n \in \mathbb{Z}} |c + i\omega_n - z| > 0. \quad (3)$$

The minimum exists because the distances tend to infinity. Equality in the norm follows by taking any unit vector in a minimizing mode.

Retain the actual cyclic packet $E = \mathbb{C}[S]/(\chi_{h,k})$, $A = M_S$, and the phase-periodized canonical representative $R : E \rightarrow \mathcal{H}$. Its source is still $P(D_1 + \cdots + D_k)F_h^{\otimes k}$, with $\mathcal{M}F_h = 2\xi/h$, and full arithmetic observation $v_h^{\otimes k} P(A_h^{(k)})1$. Use an admitted finite degree $N \geq \deg \chi_{h,k} - 1$. For this source comparison choose the circle length by the proved formula (17) of `shifted_mellin_source_constants.tex`, with $a = 1$ and $\eta_1 = 1/2$: thus $L \geq \log(1 + 2\mathfrak{k}_{1,N})$. Its phase source Gram is at least one half of the positive original Gram. Periodization is therefore injective on the admitted polynomial

source; composing with the canonical section proves that R is injective. It has values in every twisted Sobolev domain, and the actual theta boundary is

$$B = \mathcal{D}R - RA. \quad (4)$$

It is the scalar-holonomy periodization of the retained source vector $D^{(k)}r_N - r_N A = dK_N^{\text{prim}}$, not the zero vector obtained after applying its arithmetic quotient.

213.2 The resolvent comparison and the complete nonresonant packet

For z outside both spectra, the exact comparison is

$$T(z)R - R(A - z)^{-1} = -T(z)B(A - z)^{-1}. \quad (5)$$

Indeed, (4) says $(\mathcal{D} - z)R = R(A - z) + B$. Multiplying on the left by $T(z)$ and on the right by $(A - z)^{-1}$ gives (5), including its minus sign. All products are defined because E is finite dimensional and $R(E) \subset \text{Dom } \mathcal{D}$.

For a complete primary block $E_\lambda \subset E$, put $N_\lambda = (A - \lambda)|_{E_\lambda}$ and retain its full length r , so $N_\lambda^r = 0$. Restrict (4) to that block. If $\lambda \notin \text{Spec } \mathcal{D}$, it gives $R_\lambda = T(\lambda)R_\lambda N_\lambda + T(\lambda)B_\lambda$. Substitute this expression successively r times. Its last term contains N_λ^r and vanishes, giving the finite formula

$$R_\lambda = \sum_{j=0}^{r-1} T(\lambda)^{j+1} B_\lambda N_\lambda^j. \quad (6)$$

No jet was discarded: all r powers occur. Conversely the right side satisfies (4) on E_λ , by telescoping after multiplication by $\mathcal{D} - \lambda$; the last nilpotent term is zero. It is the unique such solution, because the same iteration applied to a homogeneous solution makes that solution zero. For the original finite coefficient norm, (3) also proves

$$\|R_\lambda\| \leq \sum_{j=0}^{r-1} \delta(\lambda)^{-j-1} \|B_\lambda N_\lambda^j\|. \quad (7)$$

Thus the displayed map is a calculated reconstruction from the retained boundary, not a vanishing assertion.

In particular, for an actual eigenvector $0 \neq v \in E_\lambda$ with $N_\lambda v = 0$,

$$Rv = T(\lambda)Bv, \quad \frac{\|Bv\|}{\|Rv\|} \geq \delta(\lambda) \geq |\Re \lambda - k/2|. \quad (8)$$

The denominator is positive by the injectivity of the canonical source. The first inequality follows either from (3) or directly from (2). Every off-weight primary block is nonresonant, regardless of its imaginary part. Hence making its quotient boundary zero does not make its source boundary zero.

213.3 The resonant block with its full kernel and nilpotents

Suppose $\lambda = c + i\omega_{n_0}$. Let P be the orthogonal projection onto $e_{n_0} \otimes \mathcal{K}$, and write $Q = I - P$. On $Q\mathcal{H}$, the inverse

$$T_Q = (\mathcal{D} - \lambda)^{-1}|_{Q\mathcal{H}} \quad \text{has norm } L/(2\pi). \quad (9)$$

This follows from the nonzero distances $|\omega_n - \omega_{n_0}| = 2\pi|n - n_0|/L$. Projection of (4) and the same nilpotent iteration now yield

$$QR_\lambda = \sum_{j=0}^{r-1} T_Q^{j+1} QB_\lambda N_\lambda^j, \quad PB_\lambda = -PR_\lambda N_\lambda. \quad (10)$$

These are all components of the comparison; (9) has not been used on the resonant mode.

For clarity, write $\chi_{h,k} = (S - \lambda)^r h_\lambda$. The two displayed factors are relatively prime. Choose polynomials a_λ, b_λ with $a_\lambda(S - \lambda)^r + b_\lambda h_\lambda = 1$, and set $\varepsilon_\lambda = [b_\lambda h_\lambda]_{\chi_{h,k}}$. Its residues are one in the λ -local factor and zero in every other primary factor. The Chinese remainder map and its inverse therefore identify $E_\lambda = \varepsilon_\lambda E$ with $\mathbb{C}[S - \lambda]/((S - \lambda)^r)$. The actual embedded increasing power basis is $e_j = \varepsilon_\lambda(S - \lambda)^j$. Thus $N_\lambda e_j = e_{j+1}$ for $j < r - 1$, and $N_\lambda e_{r-1} = 0$. Write $X = PR_\lambda, Y = PB_\lambda$. Equation (10) says exactly

$$Y e_{r-1} = 0, \quad X e_{j+1} = -Y e_j \quad (0 \leq j < r - 1), \quad X e_0 \text{ is free in } P\mathcal{H}. \quad (11)$$

Necessity follows by applying $Y = -X N_\lambda$ to every basis vector. Conversely the values in (11) define a linear map satisfying that equation. Hence (11) proves both sufficiency and the complete freedom, with no assumed semisimplicity. The actual source supplies its particular $X e_0$. A squarefree sampled-value quotient would retain only a further observation of this data; it is not the map in (10)–(11).

213.4 The original Laplacian, including its two boundary terms

Set

$$\mathcal{L} = \mathcal{D}^2 - k\mathcal{D} = \partial_r^2 - k^2/4, \quad A^{\text{lap}} = A^2 - kA, \quad C = (\mathcal{D} - k)B + BA. \quad (12)$$

The domain of $\mathcal{L}$ is exactly

$$\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), f'(L) = e^{-i\theta} f'(0)\},$$

equivalently the Fourier Sobolev domain of order two. Its eigenvalues are exactly $-\omega_n^2 - k^2/4$; the zeroth Fourier index remains included. Applying $\mathcal{D}$ to (4) and subtracting k times (4) proves

$$\mathcal{L}R - RA^{\text{lap}} = C. \quad (13)$$

In particular both terms of C in (12) are necessary. For z outside both relevant spectra, exactly the proof of (5) gives

$$(\mathcal{L} - z)^{-1}R - R(A^{\text{lap}} - z)^{-1} = -(\mathcal{L} - z)^{-1}C(A^{\text{lap}} - z)^{-1}. \quad (14)$$

On E_λ , retain

$$\mu = \lambda^2 - k\lambda, \quad M_\lambda = (2\lambda - k)N_\lambda + N_\lambda^2, \quad M_\lambda^r = 0. \quad (15)$$

If $\mu \notin \text{Spec } \mathcal{L}$, (13) gives the complete finite reconstruction

$$R_\lambda = \sum_{j=0}^{r-1} (\mathcal{L} - \mu)^{-j-1} C_\lambda M_\lambda^j. \quad (16)$$

The proof is the same finite substitution as (6). Formula (15) retains the exceptional coefficient $2\lambda - k$, including the case where it is zero, and retains the N_λ^2 term; it does not replace the original primary block by a simple root.

The linear source maps in (4)–(16) lift at each original support label as $(\ell, v) \mapsto (\ell, fv)$, with $\tau \mapsto \tau$. A source theta boundary has a supported-zero image under the original arithmetic quotient and retains its Hilbert vector in (6) or (10). Its necessary nonvanishing on each nonresonant eigenvector is proved by (8); the resonant boundary can vanish, with its full freedom given by (10)–(11). This resolves the promised resolvent calculation while proving its precise scope: a negative circle Laplacian controls its own resolvent and leaves the displayed arithmetic boundary terms. No bound for the four arithmetic quotient volumes follows by deleting those terms.

214 Exact fluid-cutoff transport into the retained theta complex

This is a completed calculation prompted by A0618, node ba85139d-2899-4a3b-b05e-9f30a7341c82, chain 1146, transcript-segment lines 2698–2818. That answer constructed an admissible cutoff test but left its equivariance and quotient class uncalculated. We calculate the exact scaling morphism and its image in the original theta complex; we do not assert a sign for its arithmetic pairing or replace the original fluid by a self-similar ansatz.

The arithmetic target is the original `Tau_Base_Cohomology_2026-09-12/NOTE.md`, SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3: equations (10)–(14), (19)–(22), (32)–(45). All spaces and actions below retain those definitions. The factor connecting the early centered test with this target is an explicit invertible multiplication map, not an identification of their coordinates.

214.1 1. Original spaces and the cutoff map

Fix viscosity $\nu > 0$, a time interval I , and a smooth real vector field u on $\mathbb{R}^3 \times I$. Define $e_\theta = (-\sin \theta, \cos \theta, 0)$ and retain the **unaveraged** angular-momentum integral

$$a_u(r, t) = \int_0^{2\pi} r e_\theta \cdot u(r \cos \theta, r \sin \theta, 0, t) d\theta.$$

Let $\chi(r, t)$ be a smooth cutoff whose radial support is compact in $(0, \infty)$, locally uniformly in t . Set

$$D = r\partial_r, \quad b(r, t) = \chi(r, t)a_u(r, t), \quad g(r, t) = (D^2 - \tfrac{1}{4})b(r, t).$$

For every fixed t , $g_t \in C_c^\infty(\mathbb{R}_{>0})$. Define the original centered admissible space

$$\mathcal{T} = \{g \in C_c^\infty(\mathbb{R}_{>0}) : \int_0^\infty g(r)r^{1/2} dr/r = \int_0^\infty g(r)r^{-1/2} dr/r = 0\}.$$

The Mellin transform used in the tau source is $\mathcal{M}h(s) = \int_0^\infty h(r)r^s dr/r$. The support of b_t lies in some $[\alpha, \beta] \subset (0, \infty)$; it and every derivative vanish near both integration endpoints. Therefore, for every complex s ,

$$\mathcal{M}(Db)(s) = [b(r)r^s]_0^\infty - s\mathcal{M}b(s) = -s\mathcal{M}b(s),$$

and a second integration by parts gives

$$\mathcal{M}g(s) = (s^2 - \tfrac{1}{4})\mathcal{M}b(s). \quad (1)$$

At $s = \pm \frac{1}{2}$, this proves both admissibility equations, with all boundary terms zero for the stated support reason. Smooth parameter dependence follows by differentiation under the integral on each common compact support interval.

214.2 2. The explicit map to the original uncentered theta target

Retain the tau source's space

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{r>0} r^j |(-r\partial_r)^k F(r)| < \infty \text{ for all } j \in \mathbb{Z}, k \geq 0\}.$$

Multiplication defines a linear injection

$$\mathcal{C} : \mathcal{T} \longrightarrow \mathcal{B}, \quad (\mathcal{C}g)(r) = r^{-1/2}g(r). \quad (2)$$

Its image lies in $C_c^\infty \subset \mathcal{B}$, and its inverse on that image is multiplication by $r^{1/2}$. No endpoint extension is used. For $F = \mathcal{C}g$, equation (1) gives the exact entire identity

$$\boxed{\mathcal{M}F(s) = s(s-1)\mathcal{M}b(s - \tfrac{1}{2})}. \quad (3)$$

Thus the early two centered moment cancellations become the uncentered zeros at $s = 0, 1$, through the stated multiplication map.

The test space retains its usual inductive-limit topology over compact support intervals. On each such interval the multiplier $r^{-1/2}$ and all its derivatives are bounded; the product rule therefore bounds every displayed $\mathcal{B}$ seminorm by finitely many test-function seminorms. Thus (2) is continuous into $\mathcal{B}$. The quotient and adjoint calculations below retain the source's algebraic quotient and algebraic dual; they make no assertion that ΘV is closed or that this dual is the continuous dual of a completed quotient.

Retain the actual source spaces V , its Fourier transform with exponent $-2\pi i x \xi$, $\Theta\phi(r) = \sum_{n \neq 0} \phi(nr)$, $JF(r) = r^{-1}F(1/r)$, and

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(\phi, \psi) = \Theta\phi - J\Theta\psi, \quad Q = \mathcal{B}/\Theta V.$$

Formula (2) followed by the quotient map gives a concrete linear map $\mathcal{T} \rightarrow H^1 \mathbf{A}_\tau(\mathcal{T}) = Q$. It does not say that every input class is nonzero. Equation (3), and the full jet formula below, compute its actual finite arithmetic observations without making that assumption.

214.2.1 The early theta background and its exact factor of two

A0598 used $\mathcal{E}\phi(r) = r^{1/2} \sum_{n \geq 1} \phi(nr)$, whereas the retained tau source uses $\Theta\phi(r) = \sum_{n \neq 0} \phi(nr)$. For an even $\phi \in V$, pairing the terms $n, -n$ gives the exact identity

$$\mathcal{C}(\mathcal{E}\phi) = \frac{1}{2}\Theta\phi. \quad (3a)$$

Here multiplication $\mathcal{C}$ extends to the displayed smooth functions by its same formula; this sentence does not assert that the uncut $\mathcal{E}\phi$ belongs to the compactly supported domain $\mathcal{T}$. On every compact positive r -interval the Schwartz bounds permit termwise differentiation of the theta sum, giving $D\Theta\phi = \Theta(D\phi)$. The operator D preserves even Schwartz functions and $\phi(0) = 0$; integration by parts gives $\int_{\mathbb{R}} D\phi = -\int_{\mathbb{R}} \phi = 0$. It therefore preserves V . Direct differentiation of $r^{1/2}h$ gives

$$r^{-1/2}(D^2 - \frac{1}{4})(r^{1/2}h) = (D^2 + D)h.$$

Consequently the uncut background maps to the original theta relation

$$\mathcal{C}(D^2 - \frac{1}{4})\mathcal{E}\phi = \frac{1}{2}\Theta((D^2 + D)\phi) \in \Theta V. \quad (3b)$$

For a compact cutoff χ , the exact image instead is

$$\begin{aligned} \mathcal{C}(D^2 - \frac{1}{4})(\chi\mathcal{E}\phi) &= \frac{1}{2}\chi\Theta((D^2 + D)\phi) + (D\chi)\Theta(D\phi) \\ &\quad + \frac{1}{2}(D^2\chi + D\chi)\Theta\phi. \end{aligned} \quad (3c)$$

The first term still contains its cutoff; membership of the uncut theta image does not erase that factor. Thus (3c) retains all three pieces of the actual early cutoff background in the original Q , without assuming that any one belongs to the relation space.

214.3 3. Constant parabolic scaling, with the same viscosity

For a constant $\lambda > 0$, put

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t), \quad f_\lambda(x, t) = \lambda^3 f(\lambda x, \lambda^2 t), \quad I_\lambda = \{t : \lambda^2 t \in I\}.$$

Differentiating each component gives

$$\partial_t u_\lambda = \lambda^3 \partial_t u, \quad (u_\lambda \cdot \nabla) u_\lambda = \lambda^3 (u \cdot \nabla) u, \quad \nabla p_\lambda = \lambda^3 \nabla p, \quad \Delta u_\lambda = \lambda^3 \Delta u, \quad \nabla \cdot u_\lambda = \lambda^2 \nabla \cdot u,$$

where the right sides are evaluated at $(\lambda x, \lambda^2 t)$. Consequently the forced equation $\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f$ is carried to that same equation with exactly the same ν . This is a map between fields at different coordinates, not an assertion that one fixed solution obeys a self-similarity identity.

Set $\chi_\lambda(r, t) = \chi(\lambda r, \lambda^2 t)$. Substitution in the unaveraged angular integral gives

$$a_{u_\lambda}(r, t) = \int_0^{2\pi} (\lambda r) e_\theta \cdot u(\lambda r \cos \theta, \lambda r \sin \theta, 0, \lambda^2 t) d\theta = a_u(\lambda r, \lambda^2 t).$$

The operator D commutes with $R_\lambda h(r) = h(\lambda r)$. Thus

$$b_\lambda(r, t) = b(\lambda r, \lambda^2 t), \quad g_\lambda(r, t) = g(\lambda r, \lambda^2 t). \quad (4)$$

At corresponding times the theta target satisfies

$$\boxed{F_\lambda = \lambda^{1/2} U_{1/\lambda} F, \quad \mathcal{M}F_\lambda(s) = \lambda^{1/2-s} \mathcal{M}F(s),} \quad (5)$$

where the source's action remains $U_a F(r) = F(r/a)$. Indeed $r^{-1/2} g(\lambda r) = \lambda^{1/2} (\lambda r)^{-1/2} g(\lambda r)$.

Equation (5) lifts to the **original two-leg chain map**

$$(\phi, \psi) \mapsto (\lambda^{1/2} U_{1/\lambda} \phi, \lambda^{-1/2} U_\lambda \psi), \quad F \mapsto \lambda^{1/2} U_{1/\lambda} F. \quad (6)$$

The first leg follows from $\Theta U_a = U_a \Theta$. On the second leg the source identity $U_a J = a J U_{1/a}$ gives $\lambda^{1/2} U_{1/\lambda} J \Theta = \lambda^{-1/2} J \Theta U_\lambda$. Both are therefore precisely the two terms of d , with the original minus sign retained. Each scaling preserves V , since the zero value and zero integral remain zero, and fixes every support mask. It consequently descends to Q . At σ the coefficient map remains the zero map on the zero coefficient space, with the original labelled support skeleton fixed; no class is deleted by this comparison.

214.4 4. Full actual-zero jets and the right-adjoint dual

Let $G(s) = 2\xi(s)$, and retain a finite reflection-stable set Z of actual zeros, with actual orders m_ρ . Write $z_\rho = s - \rho$, $A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho / (G)$, and let N_ρ denote multiplication by z_ρ . The finite observation is the source's map

$$(J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j.$$

Equation (3) supplies every coefficient of this polynomial from the original cutoff b . In particular no unproved nonvanishing or simplicity assertion enters its definition. Expanding the exponential multiplier in (5) gives

$$\boxed{(J_Z F_\lambda)_\rho = \lambda^{1/2-\rho} \sum_{k=0}^{m_\rho-1} \frac{(-\log \lambda)^k}{k!} N_\rho^k (J_Z F)_\rho.} \quad (7)$$

Equivalently, coefficient j is

$$\lambda^{1/2-\rho} \sum_{k=0}^j \frac{(-\log \lambda)^k}{k!} \frac{(\mathcal{M}F)^{(j-k)}(\rho)}{(j-k)!}.$$

This retains all nilpotent jets, not only the residue-value eigenline.

The source computes $K_\tau(\mathcal{L}_1) = [I_\eta(\mathcal{L}_1) \rightarrow I_+(\mathcal{L}_1) \oplus I_-(\mathcal{L}_1)]$ in degrees $-1, 0$, with differential $(+\text{res}, -\text{res})$, and $H^{-1} R \text{Hom}(\mathcal{T}, K_\tau(\mathcal{L}_1)) = \text{Hom}(Q, \mathcal{L}_1)$. Thus the retained residue injection evaluates the computed class by

$$\bar{h} \mapsto \ell_h, \quad \ell_h([F]) = \sum_{\rho \in Z} \text{Res}_\rho \frac{h^\dagger(s) \mathcal{M}F(s)}{G(s)} ds, \quad h^\dagger(s) = \overline{h(1-\bar{s})}. \quad (8)$$

For a scaling packet $T_\lambda h(s) = \lambda^{1/2-s} h(s)$ in each local algebra,

$$(T_\lambda h)^\dagger(s) = \lambda^{s-1/2} h^\dagger(s).$$

Multiplying by the multiplier in (5) cancels these two displayed factors exactly. Hence the original residue pairing and its trace contraction are invariant:

$$R_Z(T_\lambda h, T_\lambda k) = R_Z(h, k), \quad R_Z(T_\lambda h, G' T_\lambda k) = R_Z(h, G' k). \quad (9)$$

The derivative multiplier G' commutes with T_λ , since both act by multiplication in the same local algebra. No phase changes the source's raw skew-Hermitian residue orientation; $\mathcal{L}_1$ has not been replaced by a different coefficient line. In particular (9) concerns the explicitly displayed transport $T_\lambda = \lambda^{1/2} U_{1/\lambda}$. The original uncentered action still has $R_Z(U_a h, U_a k) = a R_Z(h, k)$; here its factor $a = 1/\lambda$ cancels the two real scalar factors $\lambda^{1/2}$ carried by the two inputs.

214.5 5. The unchanged source Weil scalar is invariant under this transport

Retain exactly

$$(g_1 * g_2)(x) = \int_0^\infty g_1(y) g_2(x/y) dy/y, \quad g^*(x) = \overline{g(x^{-1})}.$$

Then

$$(g * g^*)(x) = \int_0^\infty g(y) \overline{g(y/x)} dy/y.$$

For $g_\lambda = R_\lambda g$, substitute $v = \lambda y$ to obtain

$$(g_\lambda * g_\lambda^*)(x) = \int_0^\infty g(v) \overline{g(v/x)} dv/v = (g * g^*)(x). \quad (10)$$

This is equality of the entire test function, before any arithmetic functional is applied. In particular it retains every prime term and the complete archimedean subtraction in

$$W_p(f) = (\log p) \sum_{m \geq 1} p^{-m/2} (f(p^m) + f(p^{-m})),$$

$$W_{\mathbb{R}}(f) = (\log 4\pi + \gamma) f(1) + \int_1^\infty (f(x) + f(x^{-1}) - 2x^{-1/2} f(1)) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}.$$

Therefore the source scalar $\sum_v W_v(g * g^*)$, whose required sign in A0618 is nonpositive under RH, is unchanged by (4) at corresponding times. No estimate, omitted boundary term, discarded prime or replacement norm proves this: equation (10) does.

214.6 6. Time-dependent concentration and its exact extra force

Let $\lambda(t) > 0$ and $s(t) \in I$ be smooth. Define

$$u^\#(x, t) = \lambda(t) u(\lambda(t)x, s(t)), \quad p^\#(x, t) = \lambda(t)^2 p(\lambda(t)x, s(t)),$$

and transport χ by the same arguments. Equations (4)–(10) still hold instantaneously with source time $s(t)$. They do not by themselves claim a fluid-equation intertwiner for this varying parameter.

Write $X = \lambda(t)x$ and $\mathcal{N}(u, p) = (u \cdot \nabla_X)u + \nabla_X p - \nu \Delta_X u$. Direct differentiation, using the original equation $\partial_s u + \mathcal{N}(u, p) = f$, yields the complete transformed force

$$\begin{aligned} & \partial_t u^\# + (u^\# \cdot \nabla_x) u^\# + \nabla_x p^\# - \nu \Delta_x u^\# \\ &= \lambda s' f(X, s) + (\lambda^3 - \lambda s') \mathcal{N}(u, p)(X, s) \\ &+ \frac{\lambda'}{\lambda} (u^\# + x \cdot \nabla_x u^\#). \end{aligned} \quad (11)$$

Indeed $\partial_t u^\# = (\lambda'/\lambda)(u^\# + x \cdot \nabla_x u^\#) + \lambda s' \partial_s u$, whereas each nonlinear, pressure and viscosity term carries λ^3 . This proves (11) term by term. Divergence is $\lambda^2 \nabla_X \cdot u$, so incompressibility is retained. If $s' = \lambda^2$, the middle term cancels and the remaining force is

$$\lambda^3 f(X, s) + \frac{\lambda'}{\lambda}(u^\# + x \cdot \nabla_x u^\#). \quad (12)$$

The second summand is a calculated force contribution; its endpoint regularity cannot be inferred from the original force's regularity.

For the test and theta amplitude, direct differentiation gives

$$\begin{aligned} \partial_t g^\# &= \frac{\lambda'}{\lambda} Dg^\# + s' R_\lambda \partial_s g, \\ \partial_t F^\# &= \frac{\lambda'}{\lambda} (D + \tfrac{1}{2}) F^\# + s' \lambda^{1/2} U_{1/\lambda} \partial_s F. \end{aligned} \quad (13)$$

Writing w for the Mellin variable in this sentence, to distinguish it from the source-time function $s(t)$, the first transformed generator is exactly $(\lambda'/\lambda)(\frac{1}{2} - w)$; on the ρ -jet it is $(\lambda'/\lambda)((\frac{1}{2} - \rho)I - N_\rho)$. Equation (10) shows at each time that the transported Weil scalar equals the source scalar at $s(t)$. Thus parameter variation in λ alone contributes zero to that scalar, whereas changes in the actual source profile or a cutoff not transported by this formula need their own calculation.

214.7 Proven scope

We have completed the exact transport of A0618's cutoff construction into the original A_τ complex, Q , the full actual-zero jets and the retained right-adjoint dual. Pure transported parabolic concentration preserves the entire autocorrelation and the source Weil scalar; this does not rule out effects from the actual NS solution's changing anisotropic profile, interactions or cutoff. The explicit extra force in (11)–(12) shows why a time-dependent coordinate transport cannot be called a new smooth-forced solution without calculating that term. Neither a vanished supported quotient kernel nor a global weight estimate has been assumed.

215 Finished calculations on the retained physical and arithmetic formulas

These calculations use the objects and constants displayed in A0656 and A0706. They close bounded algebraic omissions. They do not identify the modular surface, the physical wedge, or the fluid/gravity expansion with the later split-base cohomology by assumption. The source coordinates below are LF coordinates in the pinned transcript segment.

215.1 1. The modular scattering coefficient retains every zero multiplicity

A0656, lines 2964–3096, displays

$$S(k) = \pi^{-2ik} \frac{\Gamma(\frac{1}{2} + ik)}{\Gamma(\frac{1}{2} - ik)} \frac{\zeta(1 + 2ik)}{\zeta(1 - 2ik)}, \quad E_* = \frac{\hbar^2}{2mR^2} > 0, \quad \mathcal{E}(k) = E_*(k^2 + \tfrac{1}{4}).$$

Take any nontrivial zeta zero in the specified lower half-plane, with its actual multiplicity:

$$\rho = a - i\gamma, \quad 0 < a < 1, \quad \gamma > 0, \quad q = \text{ord}_{s=\rho} \zeta(s) \geq 1.$$

Solving the denominator equation exactly gives

$$k_\rho = \frac{\gamma}{2} - \frac{i(1-a)}{2}, \quad 1 - 2ik_\rho = \rho, \quad 1 + 2ik_\rho = 2 - \rho, \quad \frac{1}{2} - ik_\rho = \frac{\rho}{2}, \quad \frac{1}{2} + ik_\rho = 1 - \frac{\rho}{2}.$$

Both Gamma arguments have positive real part. Gamma is finite and nonzero there. Also $\Re(2 - \rho) = 2 - a > 1$, where the absolutely convergent Euler product is nonzero. Consequently the analytic numerator

$$N(k) = \pi^{-2ik} \frac{\Gamma(\frac{1}{2} + ik)}{\Gamma(\frac{1}{2} - ik)} \zeta(1 + 2ik)$$

is nonzero at k_ρ . Taylor expansion at the actual zero gives

$$\zeta(1 - 2ik) = \frac{\zeta^{(q)}(\rho)}{q!} (-2i)^q (k - k_\rho)^q + O((k - k_\rho)^{q+1}).$$

Thus the pole order is exactly q , and its leading Laurent coefficient is exactly

$$a_{-q} = \frac{q! N(k_\rho)}{(-2i)^q \zeta^{(q)}(\rho)} \neq 0.$$

The map to energy has derivative $2E_* k_\rho \neq 0$, so it is locally invertible at this pole. The energy-plane pole has the same order and leading coefficient

$$b_{-q} = a_{-q} (2E_* k_\rho)^q.$$

Direct squaring, without removing either real correction, gives

$$\mathcal{E}_\rho = E_* \left[\frac{\gamma^2}{4} - \frac{(1-a)^2}{4} + \frac{1}{4} \right] - i \frac{E_* \gamma (1-a)}{2}.$$

Let $\sum_{j=1}^q b_{-j} (\mathcal{E} - \mathcal{E}_\rho)^{-j}$ be the complete principal part. The positively oriented local contour contribution with the explicitly specified convention $(2\pi i)^{-1} \oint e^{-i\mathcal{E}t/\hbar} \mathfrak{S}(k(\mathcal{E})) d\mathcal{E}$ equals

$$e^{-i\mathcal{E}_\rho t/\hbar} \sum_{j=1}^q \frac{b_{-j}}{(j-1)!} \left(-\frac{it}{\hbar} \right)^{j-1}.$$

Indeed, expand the exponential in powers of $\mathcal{E} - \mathcal{E}_\rho$; its coefficient of power $j - 1$ is the displayed factor, which is exactly the residue paired with the order- j pole. A lower-half-plane contour used in a global inverse transform may carry a different overall orientation; the convention above fixes this sign and makes no unproved contour deformation. Every multiple zero contributes the complete polynomial of degree $q - 1$, with nonzero top coefficient, multiplying the decaying exponential. Its exponential amplitude factor is

$$\exp \left[-\frac{E_* \gamma (1-a)t}{2\hbar} \right].$$

The source's simple-pole width $\Gamma_\rho = E_* \gamma (1-a)$ is therefore retained, and the previously undisplayed multiplicities are now explicit. This calculation concerns the principal part of the continued scattering coefficient. It does not assert a complete time-domain expansion for every wavepacket, or convert poles of a continued coefficient into nonreal eigenvalues of a self-adjoint Hamiltonian. Those claims require a specified inversion contour, packet class, and bounds on the remaining integral.

215.2 2. Channel redistribution is exact and need not be positive

A0656, lines 3104–3137, uses a Hilbert space $\mathcal{H}$, a unitary $U : \mathcal{H} \rightarrow \mathcal{H}$, and an orthogonal projection $P = P^* = P^2$. For every $\psi \in \mathcal{H}$,

$$\langle \psi, (U^*PU - P)\psi \rangle = \langle U\psi, PU\psi \rangle - \langle \psi, P\psi \rangle = \|PU\psi\|^2 - \|P\psi\|^2.$$

Moreover,

$$\frac{1}{2}U^*[2P - I, U] = \frac{1}{2}U^*((2P - I)U - U(2P - I)) = U^*PU - P.$$

No domain issue occurs: both operators are bounded. Unitarity does not imply positivity of this operator. In $\mathbb{C}^2$ with its usual inner product, take

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$U^*PU - P = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The two coordinate unit vectors have expectations -1 and $+1$. This retains the precise useful interpretation of a change in the selected observable. Any use of this identity to prove an arithmetic positivity statement still requires the actual arithmetic subspace and pairing; the identity by itself supplies no such sign.

215.3 3. Equal marginal histories leave a phase family even after cross-energy measurement

A0706, lines 3835–3901, displays a thermofield vector and the partial-trace map. Work on the exact countable energy eigenbasis appearing there. Put

$$p_n = \frac{e^{-\beta E_n}}{Z(\beta)}, \quad \rho_\beta = \sum_n p_n |n\rangle\langle n|, \quad \sum_n p_n = 1.$$

For every real phase sequence $\theta = (\theta_n)$, the norm-convergent vector

$$|\Psi_\theta\rangle = \sum_n \sqrt{p_n} e^{i\theta_n} |n\rangle_L |n\rangle_R$$

has norm one because its summands are mutually orthogonal. In the partial trace of $|\Psi_\theta\rangle\langle\Psi_\theta|$, the right basis inner product is $\langle m|n\rangle = \delta_{mn}$; hence both marginal density operators are exactly ρ_β . The product density $\rho_\beta \otimes \rho_\beta$ has the same marginals. This proves the source's many-to-one statement.

When $\sum_n p_n E_n^2 < \infty$, diagonal summation gives

$$\langle H_L H_R \rangle_{\Psi_\theta} = \sum_n p_n E_n^2, \quad \langle \delta H_L \delta H_R \rangle_{\Psi_\theta} = \sum_n p_n E_n^2 - \left(\sum_n p_n E_n \right)^2.$$

This value is independent of all phases. It is zero in the product state after centering because the expectation factors. Thus the particular cross-energy observable in A0706 distinguishes the displayed product state from a nonzero-variance thermofield vector, but it does not recover the phase pattern inside the entire pure-state family.

An explicit bounded cross observable that does retain this missing phase is

$$O_{mn} = (|m\rangle\langle n|)_L \otimes (|m\rangle\langle n|)_R, \quad m \neq n.$$

Direct action on the series gives

$$\langle \Psi_\theta | O_{mn} | \Psi_\theta \rangle = \sqrt{p_m p_n} e^{i(\theta_n - \theta_m)}.$$

Its Hermitian real and imaginary parts measure respectively the cosine and sine of this relative phase, with the exact factor $\sqrt{p_m p_n}$. This is a concrete continuation beyond the source's single cross-energy observable.

For completeness, partial trace after arbitrary local unitary driving obeys

$$\text{Tr}_R[(U_L \otimes U_R)\rho(U_L^* \otimes U_R^*)] = U_L(\text{Tr}_R \rho)U_L^*.$$

To prove it, pair both sides with any bounded A_L ; cyclicity of the trace replaces $(A_L \otimes I)$ by $(U_L^* A_L U_L \otimes I)$, and the defining property of partial trace gives equality of all such expectations. The same argument applies on the right. Therefore equal initial marginals give equal histories for the same specified local unitary drives. This is an obstruction to reconstructing the joint state from those histories alone. It does not obstruct a reconstruction that includes the displayed cross observables, a joint evolution law, or an independently specified joint state.

215.4 4. The arithmetic occupation map, modular clock, and trace-class boundary

A0706, lines 3903–3980, gives the BC Hamiltonian

$$H|n\rangle = E_* \log n |n\rangle, \quad E_* > 0.$$

Let $\mathfrak{P}$ denote the primes. The bosonic Fock occupation basis over $\ell^2(\mathfrak{P})$ is indexed by sequences $m = (m_p)_{p \in \mathfrak{P}}$ of nonnegative integers of finite total sum. Such a sequence has finite support. Unique factorization is a bijection

$$n \longleftrightarrow (v_p(n))_{p \in \mathfrak{P}}, \quad (m_p)_p \longmapsto \prod_p p^{m_p}.$$

Sending one orthonormal basis to the other gives a surjective isometry $\mathcal{U}$, hence a unitary. On its full diagonal-operator domain,

$$\mathcal{U}H\mathcal{U}^* = \sum_p E_* \log p N_p.$$

The equality of domains is explicit: a Fock coefficient sequence c_m belongs to this domain exactly when

$$\sum_m \left(\sum_p E_* m_p \log p \right)^2 |c_m|^2 < \infty,$$

which is the image of $\sum_n (E_* \log n)^2 |c_n|^2 < \infty$. Thus this is an operator equality, including its domain, rather than a formal equality of symbols.

Write $\sigma = \beta E_*$. The positive trace is

$$\text{Tr}(e^{-\beta H}) = \sum_{n \geq 1} n^{-\sigma}.$$

It is finite exactly when $\sigma > 1$, by comparison with $\int_1^\infty x^{-\sigma} dx$; for $\sigma \leq 0$ the terms do not even tend to zero. For any finite prime set F , summing the independent geometric occupation series gives

$$\sum_{\text{supp}(m) \subset F} \prod_{p \in F} p^{-\sigma m_p} = \prod_{p \in F} (1 - p^{-\sigma})^{-1} \quad (\sigma > 0).$$

Increasing F through all primes and using nonnegative monotone sums recovers $\sum_n n^{-\sigma}$. Hence the full Gibbs density and doubled vector exist in the displayed Hilbert-space representation exactly for $\beta E_* > 1$. In that range,

$$|\text{TFD}_\beta\rangle = \zeta(\beta E_*)^{-1/2} \sum_{n \geq 1} n^{-\beta E_*/2} |n\rangle_L |n\rangle_R,$$

and its prime factors are exactly those displayed in A0706. Every energy moment used in the preceding calculation is finite because $\sum_n n^{-\sigma} (\log n)^j < \infty$ for every fixed nonnegative integer j and $\sigma > 1$, as follows from the same integral comparison after $x = e^y$.

The frequency dictionary is $\omega_p = E_* \log p / \hbar$. Substitution into the source's thermal-coefficient match gives

$$\beta E_* \log p = \frac{2\pi c \omega_p}{a} \iff \beta = \frac{2\pi c}{\hbar a}.$$

No prime-dependent choice remains. Combining this equality with the actual trace-class domain yields the precise domain of this global vector correspondence:

$$0 < a < \frac{2\pi c E_*}{\hbar}.$$

At the upper endpoint, $\beta E_* = 1$, the vector normalization diverges through the harmonic series, although every individual prime's geometric factor is finite. This is an omitted global boundary in the retained response. It is a limitation of this trace-class arithmetic representation, not a theorem that an arbitrary Rindler observer lacks a KMS state.

The modular sign and proper clock also agree exactly with the source convention. On the thermofield representation the modular operator for the left algebra is

$$\Delta = \rho_\beta \otimes \rho_\beta^{-1}.$$

For a bounded left observable, the convention used in A0706 gives

$$\Delta^{-is} (A \otimes I) \Delta^{is} = (e^{i\beta s H} A e^{-i\beta s H}) \otimes I.$$

The scalar partition factors cancel. Physical Heisenberg time is $\alpha_t(A) = e^{itH/\hbar} A e^{-itH/\hbar}$, so $t = \beta \hbar s = 2\pi c s / a$, exactly the source relation $s = a\tau / (2\pi c)$. This proves a thermal representation and clock correspondence. It does not construct a spatial observable net, a relativistic stress tensor, or a cooling generator.

215.5 5. The force pullback has a real terminal-control issue

A0706, lines 3660–3697, constructs a one-form through the actual preterminal fluid flow:

$$A_t = -(\Phi_t^{-1})^* \int_0^t \Phi_s^* f_s ds, \quad h_{00} = -2v^j A_j.$$

Pulling back by Φ_t , differentiating, and using the chain rule for a transported one-form gives

$$\Phi_t^*[(\partial_t + \mathcal{L}_v)A_t] = -\Phi_t^* f_t, \quad \partial_t A_i + v^j \partial_j A_i + A_j \partial_i v^j = -f_i.$$

Therefore the boundary-force expression is exactly

$$\frac{1}{2} \partial_i h_{00} - \partial_t A_i + v^j (\partial_i A_j - \partial_j A_i) = -\partial_t A_i - v^j \partial_j A_i - A_j \partial_i v^j = f_i.$$

In Lagrangian coordinates the full derivative factor is

$$A_t(\Phi_t(x)) = -D\Phi_t(x)^{-T} \int_0^t D\Phi_s(x)^T f_s(\Phi_s(x)) ds.$$

This identifies exactly what must be computed on the source's actual flow, rather than silently inferring terminal regularity from f .

The general implication from terminal-flat forcing to smooth coefficients is false even among smooth preterminal forced incompressible NS solutions. The following explicit comparison tests that implication without replacing the source's finite-energy field. Retain arbitrary viscosity $\nu > 0$, source time $0 \leq t < 1$, and put

$$u = 1 - t, \quad g(t) = e^{-1/u^2}, \quad v(t, x) = \left(-\frac{x_1}{u}, \frac{x_2}{u}, 0 \right), \quad p(t, x) = g(t)x_1 - \frac{x_2^2}{u^2}, \quad f = g(t) dx^1.$$

The force extends smoothly by zero to $t = 1$, with all terminal time derivatives zero: every derivative is a polynomial in u^{-1} times e^{-1/u^2} , which tends to zero. The velocity has divergence $-1/u + 1/u = 0$ and zero Laplacian. Its first material-acceleration component is zero and its second is $2x_2/u^2$; adding $\nabla p = (g(t), -2x_2/u^2, 0)$ gives exactly f . Thus

$$\partial_t v + v \cdot \nabla v - \nu \Delta v + \nabla p = (g(t), 0, 0)$$

for every retained $\nu > 0$.

The flow and pulled-back force are

$$\Phi_t(x) = (ux_1, u^{-1}x_2, x_3), \quad \Phi_s^* f_s = (1-s)g(s) dx^1.$$

Writing $I(t) = \int_0^t (1-s)g(s) ds$, the prescribed construction yields

$$A_t = -\frac{I(t)}{u} dx^1, \quad h_{00} = -\frac{2I(t)x_1}{u^2}.$$

Here $I(t)$ tends to the strictly positive finite number $\int_0^1 (1-s)e^{-1/(1-s)^2} ds$, so these coefficients do not extend smoothly to the endpoint. This example is not compactly supported and does not have finite total kinetic energy; it therefore does not settle the actual NS manuscript's terminal behavior. It proves exactly that the displayed preterminal force-map identity alone cannot make that inference. The strongest continuation remains evaluating its full flow derivatives and the gravitational corrections on the actual retained NS profile.

216 Exact source-viscosity and cutoff-clock morphism

The retained A0656 states a viscosity-one NS stage (LF2866). A0706 displays cutoff viscosity r_c (LF3571–3577) and later the coordinate embedding $\tau = t$ with unchanged spatial coordinates (LF3611–3627). The retained response does not display the equation-preserving conversion between those parameters. The linked full research package was not acquired by this bounded worker; this finding is about the retained response.

Keep the original source viscosity $\nu_s > 0$, source coordinates (t, x) , and the complete forced source equations

$$\partial_t u_i + u^j \partial_j u_i - \nu_s \Delta u_i + \partial_i p = f_i, \quad \partial_i u^i = 0.$$

Keep the target cutoff parameter $r_c > 0$ and its coordinates (τ, x) . Set

$$b = \frac{r_c}{\nu_s}, \quad t = b\tau, \quad v_i(\tau, x) = b u_i(b\tau, x),$$

$$P(\tau, x) = b^2 p(b\tau, x), \quad f_i^{\text{cut}}(\tau, x) = b^2 f_i(b\tau, x).$$

Here b has the time-coordinate conversion units required by the displayed equations; no equality of source and cutoff clocks is presumed. The map on points is

$$j_c^{\nu_s, r_c} : (t, x) \mapsto \left(\tau = \frac{\nu_s}{r_c} t, r = r_c, x \right).$$

Direct differentiation gives

$$\partial_\tau v_i = b^2 \partial_t u_i, \quad v^j \partial_j v_i = b^2 u^j \partial_j u_i, \quad r_c \Delta v_i = r_c b \Delta u_i = b^2 \nu_s \Delta u_i, \quad \partial_i P = b^2 \partial_i p.$$

All right-hand source fields are evaluated at $(b\tau, x)$. Consequently

$$\partial_\tau v_i + v^j \partial_j v_i - r_c \Delta v_i + \partial_i P = f_i^{\text{cut}}, \quad \partial_i v^i = b \partial_i u^i = 0.$$

The map is invertible: $\tau = t/b$, $u_i(t, x) = b^{-1} v_i(t/b, x)$, $p(t, x) = b^{-2} P(t/b, x)$, and $f_i(t, x) = b^{-2} f_i^{\text{cut}}(t/b, x)$. This is an exact bijection between the displayed forced-equation data, with the original viscosity retained. It is not a modification of the source solution.

The source's cutoff proper clock is $dt_c = \sqrt{r_c} d\tau/c$. With origins chosen at source $t = 0$, the same morphism gives

$$t_c = \frac{\nu_s}{c\sqrt{r_c}} t.$$

The original source endpoint ($t = 1, x = 0$) is therefore sent, under this particular same-spatial-coordinate map, to

$$\left(\tau = \frac{\nu_s}{r_c}, r = r_c, x = 0 \right), \quad t_c = \frac{\nu_s}{c\sqrt{r_c}}.$$

The simple identification $\tau = t$, $v = u$, $P = p$, and unchanged spatial coordinates intertwines these equations only in the parameter-matching case $r_c = \nu_s$: otherwise the target residual differs by $(\nu_s - r_c) \Delta u_i$. One may instead use a different explicitly typed spatial-coordinate conversion, but it must be written and its action on the source support, wavelengths, force, and norms must be carried through.

For this exact map, the source spatial support and wave numbers at corresponding times are unchanged, whereas

$$\int |v(\tau, x)|^2 dx = b^2 \int |u(b\tau, x)|^2 dx,$$

$$\int_{\tau_0}^{\tau_1} \int |\nabla v|^2 dx d\tau = b \int_{b\tau_0}^{b\tau_1} \int |\nabla u|^2 dx dt.$$

These follow from the pointwise factor b in v and the exact change $dt = b d\tau$. The displayed Brown–York coefficient therefore becomes

$$[\mathfrak{B}_2(v, P)]_{\tau\tau} = \frac{c^4 b^2}{16\pi G_5 \sqrt{r_c}} |u(b\tau, x)|^2.$$

Finally, this change of clock does not remove the actual higher-derivative problem. At the corresponding source Fourier wave number k , the magnitude ratio of the displayed correction $-\frac{3}{2}r_c^2 \Delta^2 v_i$ to $-r_c \Delta v_i$ remains exactly $\frac{3}{2}r_c |k|^2$. Its growth must still be calculated on the source’s actual decreasing wavelengths.

217 Complete mathematical continuation ledger

This volume preserves the complete passage audits, including their historical repairs and exact scope statements. The separate proof volume contains the newly completed calculations. Long coverage tables have been expanded into labelled row entries without dropping cells.

218 Overview

219 Mathematical continuation audit of the original tau-base programme

This audit concerns the complete published conversation at <https://chatgpt.com/share/6aa6bc6e-98cc-83eb-9343-e811c26d6631>. It evaluates calculations and their exact scope. The starting mathematical object for continuation is the construction explicitly accepted at U0046: the original split cohomology over the absolute base b_τ , with its actual pushforward A_τ , right adjoint K_τ , theta source, full finite jets and tensor operations.

The public source was fully recovered and its ordering verified. All 67 user turns and 148 readable assistant prose records were read in five contiguous segments. The complete 5,350-node source, including other assistant and tool records, is retained separately. The share redacts 2,600 tool payloads; the audit does not claim to recover those hidden contents. Source-file proofs are checked against separately retained files, rather than inferred from redacted tool responses.

The transcript contains historical scope substitutions and unfinished calculations, but it also contains substantial later repairs. In particular, the early generic comparison, selected Jordan model, and finite-field illustration precede the accepted tau-base construction. The later source must not be used to suggest that it already existed when those earlier replies were written. Conversely, once A1523 constructs the actual tau-base model and U0046 accepts it, that model is the object every subsequent continuation must preserve.

The main accepted programme locators are:

- U0043, node d908924e-830b-46ad-8d9c-f2b78a536989: the request to use tau as the absolute base, rather than merely repeat the finite-field result in split notation.
- A1523, node b670b6c0-a774-4398-b978-9c9779fca379: the original absolute base, $A_\tau = R\Gamma(P, -)$, separate derived restriction to σ , supported Q , computed K_τ , finite residue arrow and tensor pushforward.
- U0046, node 0af0ed52-c7ab-45fb-85a5-1911193398cb: explicit acceptance of that context and instruction to continue using the resulting objects.
- A1694, node 1fa0ebbc-f569-40c1-b7ac-b523da19a5f0: the actual continuous theta retraction and global representative section, including its nonzero scaling cocycle and complete boundary-pairing expression.
- U0067 and A2611, final node bdd08a3e-5f07-47a3-8892-35c4d52aff5a: the user's alternative of proving the estimate or demonstrating a failed inference, followed by a valid localized auxiliary-purity and Gamma-reference counterexample. Its proven scope is not failure of the original tau-base programme.

The passage ledger contains 115 findings across the five contiguous reading segments. It distinguishes an unfinished calculation, a proved obstruction to a specified map, a later repaired defect, and a user-withdrawn detour. A finite identity with a stated unproved uniform estimate is recorded as a completed finite calculation and an open estimate. It is not classified as an abandoned calculation merely because the estimate is difficult.

The accompanying 13-chapter proof volume completes selected omitted maps and calculations on the original objects, including the two coordinated determinant and spectral-window continuations. It retains the separate support masks, all multiplicities and nilpotents, the actual Taylor units, the original Fourier and Mellin coordinates, and the original source norms. The calculations do not assert an RH proof or assume that an arithmetic kernel vanishes. The supplementary continuation starts from the completed signed arithmetic trace. Its postscript records the subsequently completed analytic pole extension, full period determinant and endpoint quotient; the actual density/spectrum estimate, the quantitative theta-metric contribution of that comparison, and the global balanced ker-

nel retain their specified open scope.

The standalone repository preserves the entire published source, complete full-proof dependencies, mathematical review receipts, and reproducible extraction and build scripts. Its original source snapshots remain byte-preserving. Publication fragments repair rendering and explicitly record the two mathematical typing clarifications: the retraction projection has even Schwartz domain, and changing to the global representative also changes the source boundary in the metric identity. The original arithmetic section is used to prove the full eigenline lift into $Q^{\otimes r}$; a quotient-only argument is not left mislabeled as the strongest available result.

220 Acquisition and complete source coverage

221 Shared transcript acquisition and coverage

The public share is <https://chatgpt.com/share/6aa6bc6e-98cc-83eb-9343-e811c26d6631>. It was acquired as an HTTP 200 HTML response on 13 September 2026 at 15:15:44 UTC, after the ordinary web fetch timed out and two browser acquisition attempts failed. Those earlier failures were not treated as evidence of unavailability.

The raw response is `share_page.html`: 14,471,434 bytes, SHA256 4875e00a46a064b9f466b638ffffcf02c8bef64b756e33f128ace7674adf042e. The extractor parses JSON string arguments of the embedded React Router stream and resolves its reference table. It does not execute page JavaScript. The original reference table is also retained as `share_stream.json`.

The decoded `share_conversation.json` has 5,350 mapping nodes. Its `linear_conversation` has exactly 5,350 nodes. Following every `parent` link from current node `bdd08a3e-5f07-47a3-8892-35c4d52aff5a` to the root, reversing that chain, and comparing the complete identifier lists proves exact equality with the supplied linear order. Neither comparison has a missing identifier. The source stream explicitly closes.

There are 5,349 nonempty message records: 67 user, 2,611 assistant, 2,654 tool and 17 system. The assistant count includes tool calls, thought records, progress messages and final responses. The original empty root node is retained in the decoded conversation. The source gives backing conversation `6aa092d6-e978-83ed-92cc-9c30f5d91ca6`.

`TRANSCRIPT_FULL.md` preserves every message role and channel in that exact order. `TRANSCRIPT_USER_ASSISTANT.md` retains both those roles, including assistant tool calls. `TRANSCRIPT_VISIBLE.md` is the 215-record readable user/prose view used for the passage audit. `transcript_records.json` retains metadata and content types; the original decoded conversation retains complete content objects. Each string text part is unchanged; the readable extraction joins multiple parts with a newline. Non-text content is rendered as JSON rather than guessed from filenames. The mathematical Markdown is retained literally.

The locator `U0043`, for example, means the forty-third user message, and `A1523` the 1,523rd assistant message including intervening tool-call records. These are extraction locators, not invented timestamps. Every heading also contains the exact original node UUID and chronological chain ordinal. Individual user and assistant records are retained under `turns/`.

The share itself replaces 2,600 tool responses by the literal text “The output of this plugin was redacted.” This audit has acquired those records completely as published; it has not recovered their hidden private payloads. Linked sandbox files and attachment bytes are not contained merely because the conversation mentions them. Retained local mathematical sources are pinned separately. `U0065` has empty textual content; its attachment metadata remains in the decoded source. No attached mathematics is inferred from that empty text.

Complete visible reading was divided into five contiguous segments, `U0001–U0015`, `U0016–U0028`, `U0029–U0040`, `U0041–U0054` and `U0055–U0067`. Each segment includes its assistant responses up to the next segment’s first user message. This partition covers all 67 user turns and all 148 readable assistant prose records. Early Navier–Stokes reporting and a user-withdrawn negative-blow-up detour are classified explicitly rather than used to redefine the later accepted tau-base objective.

The decoded JSON, original source bytes and coverage manifest are the provenance record. Claims about a source-file proof are based on a separate pinned file read; a redacted tool message alone does not prove that file’s contents or that a claimed test was run.

222 Early context: U0001–U0015

223 Complete early-context audit: U0001–U0015

Every retained turn in this segment is covered: **51 nodes, comprising 15 user turns and 36 assistant turns**, across delivered source lines **1–4074**. The full read was collective, divided into three disjoint contiguous ranges. Exact node IDs and chain positions were checked again against the delivered source.

Current source SHA256: 1e2669766857411354807efbacafc1add6157e28c2d5454ae210c8c83f0ebc2f. The former read-source SHA256 was 2dc59dcf167759e7198afc474d356285d293a1e7d187a3329f7ce9f7a349ecc5; the parent corrected Windows newline translation without changing message text or UUIDs. The physical reader originally read PowerShell logical lines 2819–4083, covering all delivered LF lines 2819–4074.

223.1 Findings that matter for the current tau continuation

1. **The negative detour was expressly withdrawn.** A0473 completes the negative polynomial-branch calculation requested near U0007/U0008; U0010 asks to omit it and use the released fluid proof. Its unconstructed finite-energy extension is not an active redo target.
2. **The exact arithmetic sign was repaired, not left unresolved.** A0618 gives the source scalar $\text{sum_vW_v}(g^*g^*)$, its nonpositive RH direction, its original factor $1/2$, its actual local-factor unitary, and $Q_{\text{previous}} = -\text{sum_vW_v}$. The previous finite positive values were never positive witnesses for the source scalar.
3. **The actual cutoff map had an unfinished action comparison.** A0618 proves admissibility for $(D_r^2 - 1/4)(\text{chia_u})$ but stops before proving its arithmetic action or class. The completed supplementary appendix below now provides the original theta map, full nilpotent jets, right-adjoint residue evaluation and exact scaling behavior.
4. **Pure transported dilation has a precisely scoped obstruction.** It leaves the entire test autocorrelation unchanged, hence leaves every original prime and archimedean contribution unchanged. This does not describe the full actual NS evolution, which has changing profiles, interactions and cutoffs. Its time-dependent transport creates an explicitly computed additional force.
5. **The finite heat model and numerical packet search have limited scope.** A0598 labels its five-state example as auxiliary and its 18-packet arithmetic search as uncertified. Neither closes the original analytic calculation or rejects its remaining directions.
6. **A0706 materially repairs the earlier physical substitution.** A0656 gives exact modular scattering but does not complete the requested NS/horizon chain. A0706 subsequently supplies several explicit maps. The remaining physical endpoint and full arithmetic-kernel realization gaps do not obstruct the later $A_{\text{tau}}/K_{\text{tau}}$ construction.

The late accepted program remains $A_{\text{tau}} = R\Gamma(P, -)$, its separate res_sigma , K_{tau} , and the actual theta quotient with all support labels. None of the early finite-family, compact-error, physical-clock or endpoint arguments proves failure of that entire program.

223.2 Completed calculation supplied for integration

[Full proof in Markdown](#) and [editable LaTeX fragment](#) give the exact map from the A0618 test to the original theta complex. The proof includes the factor $1/2$ between early Ephi and Theta, all cutoff derivative terms, unchanged physical viscosity, the unequal actions on the two theta legs, every actual-zero nilpotent jet, the K_{tau} residue evaluation, unchanged autocorrelation, and the complete force produced by a varying scale.

The [independent review](#) read the complete proof and complete Tau Base source. It found no algebraic correction. Its two notation/scope clarifications were applied. The integration owner separately read and checked the proof.

Additional bounded contextual calculations are retained in [the opening audit](#), [the physical calculations](#), and [the viscosity/clock map](#). These remain subordinate to the late tau construction.

223.3 Passage-by-passage ledgers

- [Opening U0001–U0007 ledger](#): 28 nodes, 8 passage entries, all lines 1–623. Its boundary-pending strain exchange is resolved by A0473 and U0010 as stated above.
- [Central U0008–U0012 ledger](#): 15 nodes, 12 passage entries, all lines 624–2818. Every quotation was checked against the exact retained node.
- [Physical U0013–U0015 ledger](#): 8 nodes, 14 passage entries, all delivered lines 2819–4074.

The complete structured union is [audit.json](#); [read_coverage.json](#) records provenance and all node classifications. Historical source pins inside the worker files remain as read receipts; the current source hash above controls delivered-source lookup.

223.4 Complete coverage table

Complete table entries (51 rows).

Entry 1.

Turn: U0001

Exact node: dac6bebe-48e1-465f-a4b0-5f89a3607966

Chain: 2

Lines: 1–4

Classification: unrelated context

Entry 2.

Turn: A0004

Exact node: 02b0ce59-201b-4bc3-96da-e2882088188d

Chain: 11

Lines: 5–8

Classification: unrelated context

Entry 3.

Turn: A0013

Exact node: 68fc959f-e66b-4d65-8905-f37a91bd0865

Chain: 30

Lines: 9–12

Classification: unrelated context

Entry 4.

Turn: A0035

Exact node: 47fe5f30-0886-41ac-9f29-95606b12fb9d

Chain: 65

Lines: 13–16

Classification: unrelated context

Entry 5.**Turn:** A0066**Exact node:** fd131720-edb7-4c97-91ea-6829415d0891**Chain:** 113**Lines:** 17-20**Classification:** unrelated context**Entry 6.****Turn:** A0109**Exact node:** e8ac4297-9029-4e26-9cc8-678ef4b7e646**Chain:** 187**Lines:** 21-24**Classification:** unrelated context**Entry 7.****Turn:** A0120**Exact node:** deac9312-3427-4078-9917-0c31d13c7677**Chain:** 217**Lines:** 25-67**Classification:** relevant precursor**Entry 8.****Turn:** U0002**Exact node:** 3d720ce3-ac0c-4160-943b-c520b73310f2**Chain:** 218**Lines:** 68-73**Classification:** unrelated context**Entry 9.****Turn:** A0121**Exact node:** bb27dcf5-f8ed-48cc-aaec-c6d6a15f8b93**Chain:** 220**Lines:** 74-77**Classification:** unrelated context**Entry 10.****Turn:** U0003**Exact node:** 457946ec-6fb0-4742-8e69-1539cebc3424**Chain:** 226**Lines:** 78-81**Classification:** relevant precursor**Entry 11.****Turn:** A0130**Exact node:** 25b4c507-60a2-4de0-9bf0-2b2bdd584a19**Chain:** 235**Lines:** 82-85**Classification:** relevant precursor

Entry 12.**Turn:** A0164**Exact node:** b7efc463-95f2-4da9-81a1-fb6aedebfccb**Chain:** 292**Lines:** 86–89**Classification:** unrelated context**Entry 13.****Turn:** A0197**Exact node:** d898a7d2-dd7f-445c-956b-af0a672db8bb**Chain:** 346**Lines:** 90–93**Classification:** relevant precursor**Entry 14.****Turn:** A0224**Exact node:** 53788422-8f29-4060-9fc2-65a2bf1b1077**Chain:** 399**Lines:** 94–441**Classification:** relevant precursor**Entry 15.****Turn:** U0004**Exact node:** 4e24bdf7-f83a-4bb5-a736-987c66df71e8**Chain:** 400**Lines:** 442–445**Classification:** relevant precursor**Entry 16.****Turn:** A0226**Exact node:** cf86f088-05ad-491e-a927-9b4e6dbd8855**Chain:** 403**Lines:** 446–449**Classification:** relevant precursor**Entry 17.****Turn:** U0005**Exact node:** 3d3f6197-c4e4-43b1-aa22-a6b36207dda0**Chain:** 406**Lines:** 450–453**Classification:** unrelated context**Entry 18.****Turn:** U0006**Exact node:** b37c6fd5-dd0c-4bfb-a22c-2429afe90f22**Chain:** 414**Lines:** 454–457**Classification:** unrelated context

Entry 19.**Turn:** A0235**Exact node:** cb370134-32f5-4d74-b99d-839bb81c6b48**Chain:** 420**Lines:** 458–461**Classification:** unrelated context**Entry 20.****Turn:** A0236**Exact node:** 67662ac1-3c95-4d5e-b1c0-1271e7781966**Chain:** 421**Lines:** 462–465**Classification:** unrelated context**Entry 21.****Turn:** A0288**Exact node:** 47b323eb-3a93-412e-8b9d-9daa5ef064ee**Chain:** 503**Lines:** 466–469**Classification:** unrelated context**Entry 22.****Turn:** A0289**Exact node:** a4aa3670-4083-48ad-b9ac-cf2479047d0b**Chain:** 504**Lines:** 470–473**Classification:** unrelated context**Entry 23.****Turn:** A0364**Exact node:** c0902270-3f97-4b6b-90c3-2aeb109a137b**Chain:** 627**Lines:** 474–477**Classification:** relevant precursor**Entry 24.****Turn:** A0365**Exact node:** c9c094ab-7a2a-4a39-8f14-31fd78aad17c**Chain:** 628**Lines:** 478–481**Classification:** relevant precursor**Entry 25.****Turn:** A0407**Exact node:** 20549590-6782-4c11-97ef-47be2af361fb**Chain:** 727**Lines:** 482–611**Classification:** relevant precursor

Entry 26.**Turn:** U0007**Exact node:** db5368dd-de50-4c4c-8d0b-2e6a3bead0bb**Chain:** 728**Lines:** 612–615**Classification:** relevant precursor; answered by A0473, then route withdrawn by U0010**Entry 27.****Turn:** A0408**Exact node:** 5ce20ff2-2066-4a9f-9918-0955f6b863f5**Chain:** 730**Lines:** 616–619**Classification:** relevant precursor; answered by A0473, then route withdrawn by U0010**Entry 28.****Turn:** A0437**Exact node:** 3597cc38-5eda-4031-8799-ee39f36b9a33**Chain:** 772**Lines:** 620–623**Classification:** relevant precursor; answered by A0473, then route withdrawn by U0010**Entry 29.****Turn:** U0008**Exact node:** 752013ce-3c9a-429b-b2cf-17c1552c2bb9**Chain:** 789**Lines:** 624–627**Classification:** superseded request: withdrawn explicitly by U0010**Entry 30.****Turn:** U0009**Exact node:** 98acef2e-4181-44ff-b1bd-ecf367fac17b**Chain:** 790**Lines:** 628–632**Classification:** feedback record: superseded negative-branch exchange; historical evidence only**Entry 31.****Turn:** A0446**Exact node:** 72567af7-46dc-41de-8447-8db089d11675**Chain:** 792**Lines:** 633–636**Classification:** interim plan in superseded negative-branch exchange**Entry 32.****Turn:** A0466**Exact node:** c2f9ec84-e542-4f1c-81ef-6734728c9418**Chain:** 821**Lines:** 637–640**Classification:** interim result in superseded negative-branch exchange

Entry 33.**Turn:** A0473**Exact node:** 55647499 - 2804 - 4e35 - ad71 - 91297fcd648e**Chain:** 831**Lines:** 641–1010**Classification:** completed polynomial calculation; proposed NS extension superseded by U0010**Entry 34.****Turn:** U0010**Exact node:** d6dd8d5d - d324 - 42f5 - 8ea9 - f72c609c3ecb**Chain:** 832**Lines:** 1011–1014**Classification:** explicit scope change: withdraw negative detour; retain workbench integration**Entry 35.****Turn:** A0474**Exact node:** 6fa597d0 - e8e2 - 4817 - aae5 - 509f92e6c36d**Chain:** 834**Lines:** 1015–1018**Classification:** interim integration plan**Entry 36.****Turn:** A0533**Exact node:** 3df45a90 - 4fcc - 443f - 9e7f - 3fb0aaac8b87**Chain:** 936**Lines:** 1019–1022**Classification:** interim relevant arithmetic-kernel-jet provenance**Entry 37.****Turn:** A0546**Exact node:** 3933b69b - 4530 - 4ad1 - 8212 - 80bd14ef2157**Chain:** 980**Lines:** 1023–1123**Classification:** relevant arithmetic quotient precursor; other fluid/gauge exposition context**Entry 38.****Turn:** U0011**Exact node:** a0d9bfa2 - 9ee6 - 4d7f - a9ab - e1dd7b561632**Chain:** 981**Lines:** 1124–1127**Classification:** relevant original heat/BCM/arithmetic mechanism request**Entry 39.****Turn:** A0548**Exact node:** 377718cf - 398c - 4cf9 - b148 - 034d28cc99e0**Chain:** 987**Lines:** 1128–1131**Classification:** interim plan for exact thermal/cooling/quartet bridge

Entry 40.**Turn:** A0574**Exact node:** fbfa075e-f262-4061-a646-ee8689822cec**Chain:** 1026**Lines:** 1132–1135**Classification:** interim claim of four-channel construction**Entry 41.****Turn:** A0598**Exact node:** 571d627f-3a00-4eb7-b59c-e6c007ca58b0**Chain:** 1089**Lines:** 1136–1994**Classification:** substantive relevant bridge, scoped obstruction, finite model and uncertified arithmetic test**Entry 42.****Turn:** U0012**Exact node:** a4c4bad7-0fae-4cda-a7c4-9f569e70bebb**Chain:** 1090**Lines:** 1995–1998**Classification:** controlling precision and full-Shimura correction request**Entry 43.****Turn:** A0618**Exact node:** ba85139d-2899-4a3b-b05e-9f30a7341c82**Chain:** 1146**Lines:** 1999–2818**Classification:** substantive corrected sign and typed cooling/cutoff maps; genuine unfinished analytic transfer**Entry 44.****Turn:** U0013**Exact node:** 94eb43d9-2a32-48be-8e40-d65160f3def6**Chain:** 1147**Lines:** 2819–2824**Classification:** controlling_request: Physical interpretation, not merely crossed product**Entry 45.****Turn:** U0014**Exact node:** 9ae478a7-c6c1-4b1f-9057-7723ae70b11f**Chain:** 1161**Lines:** 2825–2830**Classification:** controlling_request: Observer/modular/KMS/Rindler/horizon chain; excludes resonance substitution**Entry 46.****Turn:** A0626**Exact node:** c7b2e42f-53e5-428a-af50-68e2a7918456**Chain:** 1163**Lines:** 2831–2834

Classification: progress_promise_partly_fulfilled: Detector, horizon stress, and backreaction; later A0706 supplies several maps

Entry 47.

Turn: A0638

Exact node: f8c34278-7bce-4b37-9d29-0297bd9ee405

Chain: 1180

Lines: 2835–2838

Classification: superseded_proposed_branch: Mass inflation candidate not calculated in retained range

Entry 48.

Turn: U0015

Exact node: 8cfab729-2a37-4ebc-9be9-7319066d3775

Chain: 1185

Lines: 2839–2851

Classification: controlling_request: NS/ER=EPR physical chains and precision requirements

Entry 49.

Turn: A0656

Exact node: 0fcbfffe-d051-4d02-a7ca-b03970ed734a

Chain: 1229

Lines: 2852–3158

Classification: scope_substitution_with_retained_results: Exact modular scattering calculation; no actual NS-to-scattering bridge

Entry 50.

Turn: A0665

Exact node: 9478295f-156e-451a-9c64-a6b781125868

Chain: 1243

Lines: 3159–3162

Classification: progress_promise_fulfilled: Boundary forcing and joint-observable maps appear in A0706

Entry 51.

Turn: A0706

Exact node: 15a7724d-c82b-4704-9e93-932256ec5c6c

Chain: 1336

Lines: 3163–4074

Classification: substantive_repair_with_limited_obstructions: Explicit physical maps; full endpoint and arithmetic realization remain uncalculated

223.5 Acquisition and audit limits

The source acquisition itself belongs to the parent task. This lane read every assigned retained user/assistant text, including displays and original order; it did not silently substitute a latest ZIP. Historical linked sandbox notebooks and external articles were not all independently acquired here. When an early response reports proofs inside those packages, the ledger preserves that distinction. Newly completed proofs are identified separately. No motive judgement, remote publication, shared-master edit or Lean run occurred.

224 Opening passage ledger: U0001–U0007

224.1 Opening-context coverage and mathematical continuation audit

224.1.1 Scope and receipt

The assigned source is `output/tau_f1_transcript_audit_2026-09-13/sources/audit_segment_U0001_U0015.md`. Its SHA-256 at the read was `2dc59dcf167759e7198afc474d356285d293a1e7d187a3329f7ce9f7a349ecc5`. Read coverage is **every line 1–623 inclusive**, in three independently displayed complete ranges 1–210, 211–420, and 421–623. An initial combined display was truncated; it was not relied on for coverage. The retained LF extraction `assigned_transcript_lines_0001_0623.md` has SHA-256 `930d93dce77dc69e3254ee f95592463ab44676f30809ee909c66b38f930767b5`.

There are **28 turn nodes: seven user nodes and 21 assistant nodes**. All seven user turns are preserved verbatim in `user_inputs_verbatim.md`. The companion JSON supplies exact UUIDs, chain indices, inclusive line bounds, classes, and passage records. Line numbers below refer to the original assigned source, not this report.

This bounded read audits the mathematical antecedents of the later split-base/theta/Deligne construction. It does not independently investigate the fluid-equation announcements or controversy, retrieve the packages cited inside the transcript, or independently certify claims whose proofs are only said to be in those packages. In particular, wording such as “the notebook proves” below describes what the transcript reports; it is not a receipt for reading that notebook.

No definition or calculation of $A_\tau = R\Gamma(P, -)$, res_σ , K_τ , or the original theta complex occurs in lines 1–623. The references to arithmetic observables and RH at A0224 do not supply maps into those later objects. No failure in this segment establishes failure of that original program.

224.1.2 Complete turn coverage

“Unrelated context” means outside the later tau program’s mathematical scope, including records that remain useful as provenance. “Relevant precursor” means it supplies a mathematical construction, a sign choice, an explicit restriction of an obstruction, or a user instruction to preserve full-strength relationships. “Superseded calculation” is reserved for a calculation actually replaced within this bounded segment; no complete mathematical calculation here meets that criterion. Public-record framing that changed is not classified as a superseded mathematical calculation.

Complete table entries (28 rows).

Entry 1.

Locator: U0001

UUID: `dac6bebe-48e1-465f-a4b0-5f89a3607966`

Chain: 2

Lines: 1–4

Class: unrelated context

Coverage finding: Literature/archive request for the fluid reconstruction.

Entry 2.

Locator: A0004

UUID: `02b0ce59-201b-4bc3-96da-e2882088188d`

Chain: 11

Lines: 5–8

Class: unrelated context

Coverage finding: Sourcing plan.

Entry 3.

Locator: A0013

UUID: 68fc959f-e66b-4d65-8905-f37a91bd0865

Chain: 30

Lines: 9–12

Class: unrelated context

Coverage finding: Dated archive plan.

Entry 4.

Locator: A0035

UUID: 47fe5f30-0886-41ac-9f29-95606b12fb9d

Chain: 65

Lines: 13–16

Class: unrelated context

Coverage finding: Reported forcing specifications; no tau construction.

Entry 5.

Locator: A0066

UUID: fd131720-edb7-4c97-91ea-6829415d0891

Chain: 113

Lines: 17–20

Class: unrelated context

Coverage finding: Viscosity distinction.

Entry 6.

Locator: A0109

UUID: e8ac4297-9029-4e26-9cc8-678ef4b7e646

Chain: 187

Lines: 21–24

Class: unrelated context

Coverage finding: Preliminary fractional parameter obstruction report.

Entry 7.

Locator: A0120

UUID: deac9312-3427-4078-9917-0c31d13c7677

Chain: 217

Lines: 25–67

Class: relevant precursor

Coverage finding: Explicitly confines failed parameter extension to one construction; acquisition limits disclosed.

Entry 8.

Locator: U0002

UUID: 3d720ce3-ac0c-4160-943b-c520b73310f2

Chain: 218

Lines: 68–73

Class: unrelated context

Coverage finding: Public exposition and coherent fluid reconstruction request.

Entry 9.

Locator: A0121

UUID: bb27dcf5-f8ed-48cc-aaec-c6d6a15f8b93

Chain: 220

Lines: 74–77

Class: unrelated context

Coverage finding: Editorial verification plan.

Entry 10.

Locator: U0003

UUID: 457946ec-6fb0-4742-8e69-1539cebc3424

Chain: 226

Lines: 78–81

Class: relevant precursor

Coverage finding: Explicit request to retain the reconstruction accurately at full strength.

Entry 11.

Locator: A0130

UUID: 25b4c507-60a2-4de0-9bf0-2b2bdd584a19

Chain: 235

Lines: 82–85

Class: relevant precursor

Coverage finding: Promises full-strength presentation; does not itself calculate.

Entry 12.

Locator: A0164

UUID: b7efc463-95f2-4da9-81a1-fb6aede bfc cb

Chain: 292

Lines: 86–89

Class: unrelated context

Coverage finding: Public-record commentary.

Entry 13.

Locator: A0197

UUID: d898a7d2-dd7f-445c-956b-af0a672db8bb

Chain: 346

Lines: 90–93

Class: relevant precursor

Coverage finding: Reports pressure-only obstruction, subsequently spelled out in A0224.

Entry 14.

Locator: A0224

UUID: 53788422-8f29-4060-9fc2-65a2bf1b1077

Chain: 399

Lines: 94–441

Class: relevant precursor

Coverage finding: Exact physical bridge formulas, signs, finite-sector scope and disclosed

infinite-stage gap.

Entry 15.

Locator: U0004

UUID: 4e24bdf7 - f83a - 4bb5 - a736 - 987c66df71e8

Chain: 400

Lines: 442–445

Class: relevant precursor

Coverage finding: Asks for demonstrative relationships to the original S6 research and continued mathematics.

Entry 16.

Locator: A0226

UUID: cf86f088 - 05ad - 491e - a927 - 9b4e6dbd8855

Chain: 403

Lines: 446–449

Class: relevant precursor

Coverage finding: Promises clarification of S6 relationships; no computation in the message.

Entry 17.

Locator: U0005

UUID: 3d3f6197 - c4e4 - 43b1 - aa22 - a6b36207dda0

Chain: 406

Lines: 450–453

Class: unrelated context

Coverage finding: Speech-to-text fragment corrected by U0006; no mathematical content.

Entry 18.

Locator: U0006

UUID: b37c6fd5 - dd0c - 4bfb - a22c - 2429afe90f22

Chain: 414

Lines: 454–457

Class: unrelated context

Coverage finding: Corrected author-opinion wording; no mathematical content.

Entry 19.

Locator: A0235

UUID: cb370134 - 32f5 - 4d74 - b99d - 839bb81c6b48

Chain: 420

Lines: 458–461

Class: unrelated context

Coverage finding: Editorial continuation plan.

Entry 20.

Locator: A0236

UUID: 67662ac1 - 3c95 - 4d5e - b1c0 - 1271e7781966

Chain: 421

Lines: 462–465

Class: unrelated context

Coverage finding: Public-record update.

Entry 21.

Locator: A0288

UUID: 47b323eb-3a93-412e-8b9d-9daa5ef064ee

Chain: 503

Lines: 466–469

Class: unrelated context

Coverage finding: Repository discovery report.

Entry 22.

Locator: A0289

UUID: a4aa3670-4083-48ad-b9ac-cf2479047d0b

Chain: 504

Lines: 470–473

Class: unrelated context

Coverage finding: Repository claim report.

Entry 23.

Locator: A0364

UUID: c0902270-3f97-4b6b-90c3-2aeb109a137b

Chain: 627

Lines: 474–477

Class: relevant precursor

Coverage finding: Reports cross-shell Beltrami work and accumulating forcing pulses.

Entry 24.

Locator: A0365

UUID: c9c094ab-7a2a-4a39-8f14-31fd78aad17c

Chain: 628

Lines: 478–481

Class: relevant precursor

Coverage finding: Reports transformed-diffusion calculation; preserves distinction between volume and metric.

Entry 25.

Locator: A0407

UUID: 20549590-6782-4c11-97ef-47be2af361fb

Chain: 727

Lines: 482–611

Class: relevant precursor

Coverage finding: Reports a specific polynomial-family obstruction and an explicit all-order pulse construction.

Entry 26.

Locator: U0007

UUID: db5368dd-de50-4c4c-8d0b-2e6a3bead0bb

Chain: 728

Lines: 612–615

Class: relevant precursor

Coverage finding: Introduces the signed “negative” blowup direction.

Entry 27.

Locator: A0408

UUID: 5ce20ff2-2066-4a9f-9918-0955f6b863f5

Chain: 730

Lines: 616–619

Class: relevant precursor

Coverage finding: Promises negative-strain, pressure and norm comparison.

Entry 28.

Locator: A0437

UUID: 3597cc38-5eda-4031-8799-ee39f36b9a33

Chain: 772

Lines: 620–623

Class: relevant precursor

Coverage finding: Interim negative-strain conclusion at the assigned boundary; continuation is outside this range.

224.1.3 Passage ledger

NS-01: a specific obstruction already has an explicit scope **Locator:** A0120, chain 217, lines 52–60; A0224, chain 399, lines 339–365. **Request:** U0001 and U0003 seek the actual reconstruction at full supported strength. **Short passage:** “not an impossibility result for a new construction.”

Objects and calculation reported: $\Lambda^\alpha = (-\Delta)^{\alpha/2}$; the particular fractional-cascade parameter selection has, at $\alpha = 2$, $R = 1/\sqrt{7}$, $s = 2 - \sqrt{7}$, and $\alpha - a = 2 - 2/\sqrt{7}$. Thus $R < 1$, $s < 0$, and $\alpha - a > 0$ by $1 < \sqrt{7} < 3$. The displayed values support failure of that selected scale hierarchy and sign budget. The imported analytic estimates are explicitly not reproved in the message.

Unfinished calculation: The transcript explicitly leaves a different common classical-viscosity cascade sequence unconstructed. The parameter audit alone is not such a construction. **Scope:** This is a legitimate particular-family obstruction, with no map to the later tau construction. **Next derivation:** If this fluid lane is ever resumed, retain its exact five force-error classes and derive their exponents from the original estimates for one common sequence. This is not a priority redo item for the present tau task, and the bounded segment does not justify replacing it with another polynomial family.

NS-02: the unchanged-velocity obstruction does not cover corrections **Locator:** A0224, chain 399, lines 205–272; continuation acknowledgement at A0407, chain 727, line 582. **Request:** U0003 asks for the attempted reconstruction “(acurately in full strength)”. **Short passage:** “It does **not** rule out a genuinely different, viscosity-dependent construction.”

Objects and calculation reported: An Euler solution u, p_E, f_E , the same velocity in Navier-Stokes, and $p_{NS} = p_E + q$ give $f_{NS} = f_E - \nu \Delta u + \nabla q$, hence $\text{curl } f_{NS} = \text{curl } f_E - \nu \Delta \omega$. The quantitative interpolation estimate is reported with $E^{-4/5} \|\omega\|_\infty^{9/5}$; its full constant and analytic proof are in a named attachment, not in this excerpt.

Unfinished calculation: No infinite corrected solution is supplied here. **Scope:** The same-velocity obstruction is explicitly narrow. **Completed local continuation:** Section C1 below derives the complete correction map with all mixed and quadratic terms, proving exactly which terms a viscosity-dependent correction introduces. A0407 says its attachment also retains these terms; this local derivation is not represented as recovering or auditing that unseen attachment.

NS-03: the positive swirl residual is retained, not discarded **Locator:** A0224, chain 399, lines 314–337. **Request:** U0002–U0003 ask for coherent, full-strength reconstruction. **Short passage:** “That positive gradient-square term is part of the **exact viscous calculation**.”

Objects and calculation reported: $h = u^\phi/r$, $h^2 = c + \mu\theta > 0$, and the displayed residual

$$\frac{\nu\mu^2|\nabla\theta|_Q^2}{4(c + \mu\theta)^{3/2}}.$$

The coordinate diffusion operator and the other geometric terms are not written out in this bounded excerpt.

Unfinished calculation: Uniform terminal control of this term while the proposed cascade amplifies remains open in the exposition. **Scope:** This is a retained cost of the particular physical bridge; no conclusion about a tau-supported kernel, theta cohomology, weight transfer, or an arithmetic mass follows. **Completed local sign calculation:** Section C2 derives the displayed sign and coefficient without deleting lower-order diffusion drift. The exact three-dimensional geometric terms still require the named original source `swirl_carrier_transfer.tex`, and are not invented here.

NS-04: coordinate preservation is related to, but does not imply, metric preservation **Locator:** A0224, chain 399, lines 367–402; A0365, chain 628, lines 478–481. **Request:** U0004 explicitly asks “this is how it relates” concerning the original S6 research. **Short passage:** “It computes the full material diffusion operator”.

Objects and calculation reported:

$$(\Delta f) \circ X_t = \operatorname{div}_\xi(K_t \nabla_\xi(f \circ X_t)), \quad K_t = (DX_t)^{-1}(DX_t)^{-T}.$$

Along the stated original branch $t \uparrow 1/8$, the reported eigenvalue asymptotics retain the constants 64/121, 121/337, 337/64 and powers +3, 0, −3 of $1 - 8t$. Their product of leading factors is

$$(64/121)(121/337)(337/64) = 1,$$

which is consistent with determinant-one diffusion. This is not an independent proof of the asymptotics: the map X_t and coefficients needed for that proof are absent from the excerpt.

Unfinished calculation: The excerpt reports exact coordinate transfer and loss of uniform ellipticity, not a completed energy-admissible singular limit. **Scope:** It does not infer that the coordinate objects are unrelated. It supplies their actual operator map. It cannot be exported as a failure of the later A_τ construction. **Next derivation:** A source-level audit of these constants must start with the original X_t in `material_generator_transfer.tex` and `material_endpoint.tex`; the excerpt does not authorize substitution by a generic map.

NS-05: finite-sector physical maps are positive results with a defined range **Locator:** A0224, chain 399, lines 404–418; A0364, chain 627, lines 474–477. **Request:** U0003–U0004 seek all established relationships at full strength. **Short passage:** “A Bost–Connes heat observable produces genuine real Navier–Stokes shears.”

Objects and calculation reported: A Bost–Connes heat observable, an arithmetic projection, a logarithmic-derivative map to Burgers dynamics, a divergence-free completion, a homogeneous gauge/scalar oscillator, a two-mode periodic Beltrami sector, and a Wilson ground-state transformation. The full state spaces, mode vectors and force maps live in `arithmetic_fluid_bridge.tex` and `s6_dynamics_bridge.tex`; they are named but not reproduced here. The transcript does not claim an RH or continuum Yang–Mills proof.

Unfinished calculation: The fixed-curl sector does not itself supply the infinite interacting cascade. A0364 reports that cross-shell interactions were subsequently calculated; the interim report

alone does not expose those formulas. **Scope:** It would be incorrect to infer “no relationship” or “every nonlinear interaction vanishes.” The sector calculation only removes the nonlinearity within one curl eigenvalue. **Completed local continuation:** Section C3 proves the exact cross-eigenvalue residual $(\lambda - \mu)v \times w$, retaining its sign. Its vanishing or pressure exactness for particular original modes needs those actual modes, so no fabricated original-mode conclusion is made.

NS-06: the polynomial energy limit is one tested limiting procedure **Locator:** A0407, chain 727, lines 508–545. **Request:** U0004 asks for continued demonstrative relationships to the original research. **Short passage:** “It is **not** a theorem excluding all S6-related approaches”.

Objects and calculation reported: The specified localized original polynomial vector field W , its rescaled velocity U , $R \geq 2$, a uniform $\|U\|_2 \leq E$, and the two displayed bounds

$$\|W\|_{L^2(B_R)}^2 \geq \frac{24\pi}{12155} R^{19}, \quad \sup_{t,x} |\operatorname{curl} f| \geq 18\nu \left(\frac{24\pi}{12155} \right)^4 \frac{R^{76}}{E^8}.$$

The exact source vector field and rescaling are absent from the message, so the numerical constants are retained as transcript-reported bounds, not independently certified here.

Unfinished calculation: Corrections or other limits are not constructed in this passage. **Scope:** The message explicitly preserves valid finite-stage forced solutions and excludes only the proposed uniformly regular-force limit. This is not evidence of abandonment or of general program failure. **Next derivation:** Any reconsideration must use the original field and rescaling to calculate the full corrected force. Nothing here reaches A_τ , K_τ , or the original theta source.

NS-07: the common-sequence force-extension step is actually carried out **Locator:** A0407, chain 727, lines 547–582; earlier gap at A0224, lines 420–430. **Short passage:** “**One common sequence works for every finite derivative order.**”

Objects and calculation reported: Disjoint time pulses accumulating at $T = 1$, $\delta_n = 2^{-n-1}$, $\ell_n = e^{-n^2}$, $\epsilon_n = e^{-n^3}$, and fixed compactly supported smooth profiles. Every fixed (m, β) has the exact exponent

$$-n^3 + |\beta|n^2 + m(n+1) \log 2 \longrightarrow -\infty.$$

This follows by dividing the exponent by n^3 : the quotient tends to -1 . Thus the same displayed sequence works for every fixed derivative order; it is not a separate sequence selected for each derivative.

Unfinished calculation: Coupling that forcing to unbounded fluid amplification despite viscosity remains explicitly uncomputed. This is a substantive unresolved fluid calculation, but it supplies no interruption in the later tau program. **Scope:** The excerpt fills an earlier force-extension step rather than abandoning it. **Next derivation:** Retain the exact original velocity residual and derive whether it obeys these simultaneous pulse bounds; arbitrary prescribed smooth pulses cannot stand in for that residual.

NS-08: signed negative blowup is pending at the coverage boundary **Locator:** U0007, chain 728, lines 612–615; A0408, chain 730, lines 616–619; A0437, chain 772, lines 620–623. **Exact user request:** “Now let’s assume that it blows up by going to negative. That it just blows a hole in the bottom.” **Short assistant passage:** “negative strain diverging, with incompressibility forcing compensating stretching elsewhere.”

Objects and calculation present: These turns choose a direction of divergence and mention pressure and norm comparisons. They do not yet display the strain matrix, state-space map, solution or pressure. **Unfinished within the assigned boundary:** The displayed calculation is absent by line 623. **Classification:** Boundary-pending, not demonstrated abandonment. The next source reader must inspect U0008 onward before concluding that the web session dropped the request.

Completed local calculation: Section C4 proves the exact trace-zero sign relation and constructs an explicit negative-strain solution with its pressure and exact energy divergence. It is contextual mathematical completion, not a substitute for the later split-zero or theta state space. It does not show that this example was the one used by the web session after line 623.

224.1.4 Complete local calculations

These calculations are proved directly from their stated equations. They do not depend on the unseen fluid attachments and do not assert a new theorem about the later tau construction.

C1. The complete Euler-to-Navier-Stokes correction map Let $u, w : I \times \Omega \rightarrow \mathbb{R}^3$ be smooth vector fields, $p_E, q : I \times \Omega \rightarrow \mathbb{R}$ smooth functions, and $f_E : I \times \Omega \rightarrow \mathbb{R}^3$ a smooth force on an open time interval I and spatial domain Ω . Suppose

$$\partial_t u + (u \cdot \nabla)u + \nabla p_E = f_E, \quad \nabla \cdot u = 0.$$

Fix the original positive viscosity ν . Set $v = u + w$, $p_{NS} = p_E + q$. Then $\nabla \cdot v = 0$ holds exactly when $\nabla \cdot w = 0$. For such w , the force making v, p_{NS} solve

$$\partial_t v + (v \cdot \nabla)v + \nabla p_{NS} = \nu \Delta v + f_{NS}$$

is exactly

$$f_{NS} = f_E + \partial_t w + (u \cdot \nabla)w + (w \cdot \nabla)u + (w \cdot \nabla)w - \nu \Delta u - \nu \Delta w + \nabla q.$$

Indeed, expand $(u + w) \cdot \nabla(u + w)$ into its four terms and $\Delta(u + w)$ into its two terms, then substitute the original Euler equation. Every displayed term follows with its original sign. Conversely, substituting this force cancels these terms and recovers the Navier-Stokes equation. Hence the formula is necessary and sufficient, not merely an example of a possible force.

Taking curl gives

$$\text{curl } f_{NS} = \text{curl } f_E - \nu \Delta \omega + \text{curl}(\partial_t w + (u \cdot \nabla)w + (w \cdot \nabla)u + (w \cdot \nabla)w - \nu \Delta w), \quad \omega = \text{curl } u.$$

Curl commutes with Δ for smooth fields and annihilates ∇q . The final curl term is zero when $w = 0$; no identity forces it to vanish for arbitrary divergence-free w . Thus the obstruction based only on $-\nu \Delta \omega$ is precisely an unchanged-velocity obstruction, as the transcript already states.

C2. The retained square-root diffusion cost, including drift Let θ be a smooth real scalar on $I \times \Omega$, let c, μ be fixed real constants with $c + \mu\theta > 0$, and set $h = (c + \mu\theta)^{1/2}$. Let

$$D = \partial_t + b^i \partial_i, \quad L = Q^{ij} \partial_i \partial_j + d^i \partial_i$$

with smooth coefficients, symmetric positive-semidefinite Q , and the usual repeated-index summation. This includes a divergence-form diffusion after its coefficient derivatives are retained in the first-order drift d^i . Define $|\nabla \theta|_Q^2 = Q^{ij} \partial_i \theta \partial_j \theta$. Differentiation gives

$$h'(\theta) = \frac{\mu}{2h}, \quad h''(\theta) = -\frac{\mu^2}{4h^3}.$$

Since D is first order, $Dh = h'D\theta$. Expanding each second derivative gives

$$Lh = h' L\theta + h'' |\nabla \theta|_Q^2.$$

The d^i -terms are included in $h' L \theta$; none are discarded. Therefore, for the same positive viscosity ν ,

$$(D - \nu L)h = \frac{\mu}{2h}(D - \nu L)\theta + \frac{\nu\mu^2}{4h^3}|\nabla\theta|_Q^2.$$

The last term is nonnegative, has the denominator $(c + \mu\theta)^{3/2}$, and retains the exact coefficient $1/4$. This proves the sign in A0224. It does not estimate the other geometric terms in the full swirl equation, which are not supplied in this source segment.

C3. Fixed-curl linearity and its exact cross-sector residual Use the standard orientation on $\mathbb{R}^3$, $(a \times b)_i = \varepsilon_{ijk}a_jb_k$, and $(\text{curl } a)_i = \varepsilon_{ijk}\partial_ja_k$. For any smooth vector field u ,

$$(u \cdot \nabla)u = \nabla(|u|^2/2) - u \times \text{curl } u.$$

For completeness, contraction of the two epsilon symbols gives

$$(u \times \text{curl } u)_i = u_j\partial_iu_j - u_j\partial_ju_i = \partial_i(|u|^2/2) - (u \cdot \nabla)u_i,$$

which proves the identity with the displayed sign.

If $\nabla \cdot u = 0$ and $\text{curl } u = \lambda u$ for a constant real λ , then $u \times \text{curl } u = 0$, and

$$\Delta u = \nabla(\nabla \cdot u) - \text{curl } \text{curl } u = -\lambda^2 u.$$

Consequently the pressure choice $p = -|u|^2/2$ makes the incompressible Navier–Stokes equation equivalent, in this sector, to

$$\partial_t u = -\nu\lambda^2 u + f.$$

This calculation applies at each time to any smooth time-dependent field remaining in that fixed curl eigenspace; it neither asserts that an arbitrary force preserves the eigenspace nor supplies a mode basis.

Now retain two original smooth divergence-free fields v, w with $\text{curl } v = \lambda v$, $\text{curl } w = \mu w$, where λ, μ are real constants. Set $u = v + w$. Then

$$u \times \text{curl } u = (v + w) \times (\lambda v + \mu w) = (\mu - \lambda)v \times w,$$

because $v \times v = w \times w = 0$ and $w \times v = -v \times w$. Hence

$$(u \cdot \nabla)u = \nabla(|u|^2/2) + (\lambda - \mu)v \times w.$$

The exact residual after the same pressure choice is $(\lambda - \mu)v \times w$. To absorb it by an additional pressure requires that this actual residual be a gradient on the original domain; that is not established merely by the two curl-eigenvalue equations. No particular source mode pair is inserted or replaced here. This gives the explicit relationship between the linear fixed sector and possible cross-sector nonlinear interactions.

C4. Negative strain, compensating stretch, pressure and the physical domain For a smooth incompressible real velocity u , define the symmetric strain matrix

$$S = \frac{\nabla u + (\nabla u)^T}{2}.$$

Its trace is $\nabla \cdot u = 0$. At any fixed point, write its ordered real eigenvalues as $\lambda_1 \leq \lambda_2 \leq \lambda_3$. If $\lambda_1 < 0$, then

$$\lambda_2 + \lambda_3 = -\lambda_1, \quad \lambda_3 \geq -\frac{\lambda_1}{2} > 0,$$

since $\lambda_2 \leq \lambda_3$. Similarly, if $\lambda_3 > 0$, then $\lambda_1 \leq -\lambda_3/2$, since $\lambda_1 \leq \lambda_2$. These inequalities prove the precise relation between negative compression and positive stretching; the two signs cannot be replaced by a scalar norm without losing directional information.

Here is a complete example for the physical space $\mathbb{R}^3$, a fixed finite time T , and $t < T$:

$$a(t) = \frac{1}{T-t}, \quad A(t) = \text{diag}(-2a(t), a(t), a(t)),$$

$$u(t, x) = A(t)x, \quad p(t, x) = -\frac{|x|^2}{(T-t)^2}, \quad f(t, x) = 0.$$

It solves the unforced incompressible Navier–Stokes equation for every fixed $\nu > 0$. Indeed, $\nabla \cdot u = \text{tr } A = 0$, $\Delta u = 0$, $\partial_t u = A'x$, and $(u \cdot \nabla)u = A^2x$. Since $a' = a^2$,

$$A' + A^2 = \text{diag}(2a^2, 2a^2, 2a^2) = 2a^2 I, \quad \nabla p = -2a^2 x.$$

Their sum is zero, exactly as required. The strain is $S = A$ and its eigenvalues are

$$-\frac{2}{T-t}, \quad \frac{1}{T-t}, \quad \frac{1}{T-t}.$$

Thus the negative eigenvalue diverges to $-\infty$ with precisely compensating positive eigenvalues. The curl vanishes identically because A is symmetric. This is a direct example of negative-strain divergence without vorticity divergence.

The spatial energy cannot be omitted. Rotational symmetry gives $\int_{B_R} x_i^2 dx = 4\pi R^5/15$. Therefore

$$\int_{B_R} |u(t, x)|^2 dx = \frac{(4 + 1 + 1) 4\pi R^5}{15(T-t)^2} = \frac{8\pi R^5}{5(T-t)^2}.$$

For every $t < T$, the expression diverges as $R \rightarrow \infty$, so $u(t, \cdot) \notin L^2(\mathbb{R}^3)$. It is also not periodic: for a lattice translation Le_i , its increment is $LA(t)e_i \neq 0$. The example therefore establishes the signed local mechanism and an exact solution, while failing the finite-energy and periodic state spaces named by the original fluid target.

The displayed pressure tends to $-\infty$ for every fixed $x \neq 0$. Adding an arbitrary time-only function $C(t)$ leaves ∇p and the equation unchanged, so pressure values alone do not fix an invariant blowup criterion. The Hessian $\nabla^2 p = -2(T-t)^{-2}I$ is unchanged by this gauge freedom. These exact statements complete the elementary negative-strain comparison requested in U0007 without claiming that pressure values prove the original finite-energy blowup target.

224.1.5 Integration conclusion and next action

Within the assigned range there is **no demonstrated interruption of the later tau program**. The principal surviving content is the requirement, already respected in several passages, to retain exact maps and costs and to limit a counterexample to its actual family. The source explicitly reports substantive finite-stage physical bridges, acknowledges the absent infinite-stage coupling, and later completes one all-order force-extension step.

Carry NS-08 forward to the reader of U0008 onward before deciding whether the negative-direction calculation was abandoned. Use C1–C4 as optional fully proved contextual calculations. Do not insert them as replacements for the original A_τ , K_τ , supported H^1 , Mellin jets, tensor push-forward, or analytic theta estimates. The later source-specific audit determines the actual tau redo priorities.

225 Central passage ledger: U0008–U0012

225.1 Early-context local audit: U0008–U0012

Source SHA256: 1e2669766857411354807efbacafc1add6157e28c2d5454ae210c8c83f0ebc2f. Every line 624–2818 was read in the ten complete contiguous chunks recorded in `audit_local.json`. The retained early response text, not inaccessible historical sandbox notebooks, is the evidence audited here. The entire later Tau Base NOTE.md was also read to anchor the current continuation.

The central unfinished early calculation is A0618's actual cutoff-to-arithmetic action and survival calculation. The accompanying complete proof now supplies its exact dilation map into the original theta complex and full jets. This is supplementary context: the late accepted absolute-tau program remains the primary task. U0010 expressly withdraws the negative branch, so no redo prompt should revive it.

225.1.1 Complete local turn coverage

Complete table entries (15 rows).

Entry 1.

Turn: U0008

Node: 752013ce-3c9a-429b-b2cf-17c1552c2bb9

Chain: 789

Lines: 624–627

Classification: superseded request: withdrawn explicitly by U0010

Entry 2.

Turn: U0009

Node: 98acef2e-4181-44ff-b1bd-ecf367fac17b

Chain: 790

Lines: 628–632

Classification: feedback record: superseded negative-branch exchange; historical evidence only

Entry 3.

Turn: A0446

Node: 72567af7-46dc-41de-8447-8db089d11675

Chain: 792

Lines: 633–636

Classification: interim plan in superseded negative-branch exchange

Entry 4.

Turn: A0466

Node: c2f9ec84-e542-4f1c-81ef-6734728c9418

Chain: 821

Lines: 637–640

Classification: interim result in superseded negative-branch exchange

Entry 5.

Turn: A0473

Node: 55647499-2804-4e35-ad71-91297fcd648e

Chain: 831

Lines: 641–1010

Classification: completed polynomial calculation; proposed NS extension superseded by U0010

Entry 6.

Turn: U0010

Node: d6dd8d5d-d324-42f5-8ea9-f72c609c3ecb

Chain: 832

Lines: 1011–1014

Classification: explicit scope change: withdraw negative detour; retain workbench integration

Entry 7.

Turn: A0474

Node: 6fa597d0-e8e2-4817-aae5-509f92e6c36d

Chain: 834

Lines: 1015–1018

Classification: interim integration plan

Entry 8.

Turn: A0533

Node: 3df45a90-4fcc-443f-9e7f-3fb0aaac8b87

Chain: 936

Lines: 1019–1022

Classification: interim relevant arithmetic-kernel-jet provenance

Entry 9.

Turn: A0546

Node: 3933b69b-4530-4ad1-8212-80bd14ef2157

Chain: 980

Lines: 1023–1123

Classification: relevant arithmetic quotient precursor; other fluid/gauge exposition context

Entry 10.

Turn: U0011

Node: a0d9bfa2-9ee6-4d7f-a9ab-e1dd7b561632

Chain: 981

Lines: 1124–1127

Classification: relevant original heat/BCM/arithmetic mechanism request

Entry 11.

Turn: A0548

Node: 377718cf-398c-4cf9-b148-034d28cc99e0

Chain: 987

Lines: 1128–1131

Classification: interim plan for exact thermal/cooling/quartet bridge

Entry 12.

Turn: A0574

Node: fbfa075e-f262-4061-a646-ee8689822cec

Chain: 1026

Lines: 1132–1135

Classification: interim claim of four-channel construction

Entry 13.

Turn: A0598

Node: 571d627f-3a00-4eb7-b59c-e6c007ca58b0

Chain: 1089

Lines: 1136–1994

Classification: substantive relevant bridge, scoped obstruction, finite model and uncertified arithmetic test

Entry 14.

Turn: U0012

Node: a4c4bad7-0fae-4cda-a7c4-9f569e70bebb

Chain: 1090

Lines: 1995–1998

Classification: controlling precision and full-Shimura correction request

Entry 15.

Turn: A0618

Node: ba85139d-2899-4a3b-b05e-9f30a7341c82

Chain: 1146

Lines: 1999–2818

Classification: substantive corrected sign and typed cooling/cutoff maps; genuine unfinished analytic transfer

225.1.2 EC08: A0473, lines 973–1009

Node 55647499-2804-4e35-ad71-91297fcd648e, chain 831. Status: **superseded by explicit user instruction.**

Its trajectory travels to increasingly negative spatial infinity, where its coefficients and derivatives are already large.

User request. U0008 asked whether the original $-1/4$ fibre carries negative blowup; U0010 subsequently says to omit this negative detour.

Original objects and maps. Original polynomial F with $\det DF = -2$, $U = 2W_1 + 6F_3W_2$, $\gamma_{\min}$, Euclidean strain tensor; no tau cohomology is present.

What was actually proved. The response computes explicit finite-time inverse escape and signed strain asymptotics for a polynomial vector field. It distinguishes spatial escape from blowup at a finite spatial point.

What was left uncalculated. The bounded-region finite-energy, endpoint-smooth-force extension is not established, but U0010 withdraws this as the requested route.

Exact scope. Its scope is one polynomial lift and its physical extension. It cannot obstruct the later theta/absolute-base construction.

Concrete continuation. Do not revive the discarded negative detour. Retain the signed polynomial calculations only as historical context; continue the released-fluid and later tau source.

225.1.3 EC10: A0546, lines 1074–1090

Node 3933b69b-4530-4ad1-8212-80bd14ef2157, chain 980. Status: **relevant retained precursor.**

The distinction is the **order of operations**: lifting into an actual test function and then quotienting is not the same map as quotienting first and taking a jet.

User request. U0010 asks to integrate the actual workbench additions; U0011 asks for more than a generic non-self-adjointness statement.

Original objects and maps. Four-jet lift L_k into original arithmetic tests followed by quotient; $[T]=0$, $[(\log x)T]$ nonzero under the imported trace-ideal hypotheses.

What was actually proved. The response retains the source-dependent nonzero map and explicitly marks its hypotheses as imported. It does not equate every jet construction.

What was left uncalculated. Its full pairing sign and finite quadrature/rounding certification remain uncalculated in this response; the linked kernel paper itself is not independently re-audited by this early-context lane.

Exact scope. Nonzero class is not a signed arithmetic witness. Conversely failure to sign the pairing does not erase the proved nonzero map.

Concrete continuation. Use the original later full Mellin-jet map J_Z and K_{τ} residue injection, retaining multiplicities and supported zeros; calculate the actual trace contraction and global theta relation, without changing to a residue-only packet.

225.1.4 EC11A: A0598, lines 1153–1231

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **completed local calculation with limited scope**.

An RH counterexample would have to emerge from the remaining, globally structured part of the completed Mellin transform—not merely reproduce this leading heat divergence.

User request. U0011 asks for the exact image of the fluid/heat origin in BCM and the arithmetic spectrum.

Original objects and maps. $N|n\rangle = n|n\rangle$, $H_{BC} = \log N$, heat semigroup $\exp(-\pi \times N^2)$, Gamma-Mellin transform and the Poisson theta expansion.

What was actually proved. The Gamma integral and the small-time term $1/(\beta-1)$ explicitly connect the arithmetic heat endpoint to the BC Gibbs pole.

What was left uncalculated. This local leading-term computation does not compute the global remainder that contains the original theta quotient and zeros.

Exact scope. It proves the scope of the leading heat singularity; it is not a proof that the global theta program cannot yield further arithmetic information.

Concrete continuation. Keep both endpoint terms and the complete Poisson theta remainder when passing to the original B/Θ V. The late accepted tau construction, not this leading thermal pole, is the continuation object.

225.1.5 EC11B: A0598, lines 1274–1312

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **completed supplementary calculation in this audit**.

The important missing identification is whether an NS-derived object actually represents that character in the arithmetic quotient.

User request. U0011 requests a concrete thermodynamic and spectral map, with the blowing-up source retained.

Original objects and maps. Logarithmic scale $s = -\log(1-t)$, centered character $\Lambda^{(\rho-1/2)}$, and source quotient.

What was actually proved. The growth/chirp coordinate identity is exact, but it does not identify a specific source test or quotient class.

What was left uncalculated. An actual equivariant map from the retained cutoff test to the uncentered theta quotient and its full jets is absent here.

Exact scope. A missing identification is not nonexistence; it specifies the morphism to calculate.

Concrete continuation. Completed in NS_CUTOFF_THETA_SCALING.md equations (1)–(9): $Cg = r^{(-1/2)}g$, $MF(s) = s(s-1)Mb(s-1/2)$, original two-leg chain map, actual-zero full jet multiplier and right-adjoint residue evaluation.

225.1.6 EC11C: A0598, lines 1428–1520

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **legitimate scoped obstruction; sign presentation repaired by A0618.**

We cannot replace that weight by an arbitrary rank-one projector onto a negative direction.

User request. U0011 asks how the original absorption operator could register a quartet/cancellation mechanism; U0012 requires its exact original sign.

Original objects and maps. $P = \text{diag}(0, I)$, $U = [[A, B], [C, D]]$, $D_{\text{script}} = P - UPU$, *arithmetic multipliers theta(gg*)* on the source trace domain.

What was actually proved. The full block form gives $||By||^2 - ||Cx||^2 - 2\text{Re}\langle Cx, Dy \rangle$. The response correctly limits the compact perturbation example and restricts arithmetic tests to allowed convolution weights.

What was left uncalculated. It does not compute a signed bound for the actual local-factor U and the full allowed class. The initial sign-reversed scalar is later explicitly related to the original source scalar by A0618.

Exact scope. Compact leakage invalidates an inference from compactness alone to positivity. It does not prove the opposite sign for the actual arithmetic trace, nor global failure of any tau-base argument.

Concrete continuation. Resume the actual semilocal local-factor operator, original W scalar and trace-class domain. Retain the C^*D cross term and full archimedean subtraction. Do not use arbitrary projector tests as a substitute.

225.1.7 EC11D: A0598, lines 1522–1672

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **auxiliary model with explicitly limited scope.**

It does not assume that interpolation eliminates all other zeros.

User request. U0011 asks for the quartet mechanism and an actual arithmetic ansatz rather than a verbal RH criterion.

Original objects and maps. Actual functional-equation quartet symmetry; finite interpolation $a=1, b=-1$; auxiliary five-state $F_{\eta} = 3 + 2\exp(\eta)\cos z$; actual Riemann heat kernel separately.

What was actually proved. The quartet contribution is $4\text{Re}(a \text{ conjugate}(b))$; the finite thermal system explicitly shows positive equilibrium states can coexist with off-real observable zeros. The response labels the five-state model as distinct.

What was left uncalculated. The finite interpolation has no global rest-of-zero-set control. The auxiliary model provides no analytic estimate for the original Riemann kernel.

Exact scope. This is an auxiliary example, not itself a false proof. It would become a scope substitution only if used to close or replace the original arithmetic calculation.

Concrete continuation. Retain the original Theta and full analytic source. Use finite jets only as exact observations of Q , with all omitted spectral and boundary contributions calculated rather than declared negligible.

225.1.8 EC11E: A0598, lines 1686–1755

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **unproved sufficient criterion; not a conclusion about the later tau program.**

Proving that the bound persists for the **actual** Riemann heat family is the part not established here.

User request. U0011 requested the actual heat mechanism and attempted proof rather than a restated criterion.

Original objects and maps. $\text{Xi_eta}(s)=8H_eta(-2i(s-1/2))$, PDE $\text{partial_eta Xi}=(1/4)\text{partial_s}^2\text{Xi}$; adjacent gap d and exterior zero interaction.

What was actually proved. The coordinate PDE and gap identity $(d^2)\text{prime}=8-4d^2 \sum \text{exterior } 1/((b-x_j)(a-x_j))$ retain the exterior terms. Positive η breaks the raw termwise Dirichlet expansion, while the completed kernel still defines the family.

What was left uncalculated. The response supplies a sufficient inequality with $\theta < 2$ and an initial gap inequality; it does not prove either for the actual family over the needed interval.

Exact scope. The displayed conditional collision mechanism is not a completed calculation for the Riemann kernel. Divergence of the raw Dirichlet series does not invalidate the completed kernel.

Concrete continuation. Do not promote the sufficient inequalities to proved input. Under the later accepted tau program, continue its original theta analytic estimates; revisit this heat route only as a specifically requested auxiliary calculation, retaining the completed kernel.

225.1.9 EC11F: A0598, lines 1757–1883

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **unfinished certification and transfer; no false witness claimed.**

The calculation uses floating-point quadrature, not certified intervals.

User request. U0011 requests an arithmetic realization of the concentration/cross-term mechanism.

Original objects and maps. Original bump χ_L , E_{phi} , two reflected chirped packets, pole-cancelling $D_y^2 - 1/4$, five-dimensional restriction of the unchanged Weil form, Schur complement $q_0 - b^T A^{-1} b$.

What was actually proved. The test has exact compact support and pole cancellation; the response retains prime powers, archimedean subtraction and the exact tail. It reports 18 floating-point searches and positive minima, expressly without certification.

What was left uncalculated. All finite quadrature and roundoff enclosures are absent. The minimum establishes neither global positivity nor a negative witness and is not shown to arise dynamically from the NS pulses.

Exact scope. Finite positive searches cannot reject the full original test space; numerical convergence is not proof at a small eigenvalue.

Concrete continuation. If this exact packet calculation is reused, recover its original code/parameters and enclose every error. For the tau continuation, keep the full original Theta rather than replacing the source with this finite packet family.

225.1.10 EC11G: A0598, lines 1885–1905

Node 571d627f - 3a00 - 4eb7 - b59c - e6c007ca58b0, chain 1089. Status: **retained nonzero map with unfinished pairing.**

it does not yet determine the sign of its Weil pairing.

User request. U0011 asks what the existing arithmetic work actually contributes.

Original objects and maps. The logarithmic multiplier on the retained theta radical, the original trace-ideal lift and quotient pairing; weighted ambient L^2 norm separately.

What was actually proved. The response retains that log multiplication differentiates the Mellin transform and can remove a zero order, unlike multiplicative translations or log derivatives that preserve the zeta factor.

What was left uncalculated. The actual pairing on that surviving class is not signed, and no continuous global pairing transfer is calculated here.

Exact scope. Small ambient norm or an uncomputed sign cannot be used to annihilate the quotient class.

Concrete continuation. Use the complete later residue injection into $\text{Hom}(Q, L1)$ and trace contraction G_{prime} , with all nilpotents and labels. The new appendix (3a)–(3c) also computes the exact early E-to-Theta factor $1/2$ and every cutoff product term.

225.1.11 EC12A: A0618, lines 1999–2348

Node ba85139d - 2899 - 4a3b - b05e - 9f30a7341c82, chain 1146. Status: **completed repair; retain as authoritative early sign dictionary.**

The positive numbers previously reported were values of Q_{previous} . They were not positive values of equation (20)'s left-hand side, and were not counterexamples.

User request. U0012 requires exact signs, typed morphisms, and proof instead of separation rhetoric.

Original objects and maps. Source scalar $\sum_v W_v(gg) \leq 0$, exact factor $1/2$ trace $M_f U^*[2P - I, U]$, original local-factor U , full admissibility; BC groupoid, Haar state and σ_i .

What was actually proved. A0618 explicitly gives $Q_{\text{previous}} = -\sum_v W_v$, computes the opposite block form without changing source sign, and constructs KMS_1 by the groupoid/Haar measure identity without an RH assumption.

What was left uncalculated. Actual signed arithmetic control is still missing; the sign and KMS dependency themselves are repaired rather than abandoned.

Exact scope. The existence of KMS_1 is not an RH conclusion, and this is established through the construction. The sign correction does not turn any previous positive Q_{previous} value into a counterexample.

Concrete continuation. Preserve the exact A0618 source scalar and convolution convention in every later comparison. Do not reopen the resolved sign discrepancy or treat KMS-state existence as the desired weight theorem.

225.1.12 EC12B: A0618, lines 2352–2696

Node ba85139d - 2899 - 4a3b - b05e - 9f30a7341c82, chain 1146. Status: **completed typed maps and strictly scoped quotient obstruction.**

At this arrow, β is unchanged.

User request. U0012 requests the full Shimura version and an exact morphism after every non-identity claim.

Original objects and maps. Full datum (G, X, V, M, L, K, K_M) , quotient stack, determinant d_ϕ , level multiplicity m_K , operator-field cooling Π , dual scaling θ_λ and cyclic cokernel q_β .

What was actually proved. The response supplies the same- t dynamics inclusion of rank-one BC and computes $\Pi_\epsilon(H)(\theta_\lambda X) = \Pi_\epsilon(H + \log \lambda I)(X)$. It retains the operator field and possible Dixmier–Douady twisting. $q_\beta \delta_\beta, 0=0$ is the exact cokernel factorization identity.

What was left uncalculated. A specific NS input is not shown to lie outside the cooling image, and no global Shimura arithmetic pairing is supplied for every possible datum.

Exact scope. Only maps factoring through the specified cooling image have zero quotient. No assertion that every map factors that way is justified. This cannot erase the distinct Q or supported H_1 of the later A_τ construction.

Concrete continuation. Compute the actual factorization or nonfactorization on the original source. In the late τ program use its derived res_σ comparison and homotopy fibre, rather than substituting the generic stalk or an unrelated cooling cokernel.

225.1.13 EC12C: A0618, lines 2698–2818

Node ba85139d - 2899 - 4a3b - b05e - 9f30a7341c82, chain 1146. Status: **partly completed here; actual original-source analytic estimate remains.**

Its value has not been proved positive. Its factorization through—or survival modulo—the cooling image has not been proved either.

User request. U0011/U0012 request the actual source-to-arithmetic transfer, with unchanged viscosity, signs, domains, constants and action.

Original objects and maps. Unaveraged angular integral a_u ; moving compact cutoff χ ; $T_\chi(u) = (D_r^2 - 1/4)(\chi a_u)$; original NS equation and full W scalar.

What was actually proved. Admissibility is proved by two integrations by parts, all four radial product terms and the full time derivative are retained, and the arithmetic composition is typed.

What was left uncalculated. The displayed answer stops at an unspecified I and equations to prove: its physical/arithmetic action, quotient survival and signed scalar are not completed.

Exact scope. This identifies a genuine unfinished calculation. It does not establish impossibility, and the later τ model supplies a specific quotient/adjoint target that must be retained.

Concrete continuation. Completed in `NS_CUTOFF_THETA_SCALING.md/.tex`: full constant and variable scaling, retained viscosity, original two-leg action, all Mellin jets, residue dual, exact scalar invariance, and extra time-dependent force. The remaining actual-source estimate concerns changing anisotropic profile and cutoff, not pure transported dilation.

226 Physical passage ledger: U0013–U0015

226.1 Retained physical bridge and modular-scattering precursor audit

Source: `output/tau_fl_transcript_audit_2026-09-13/sources/audit_segment_U0001_U0015.md`, SHA256 `1e2669766857411354807efbacafcladd6157e28c2d5454ae210c8c83f0ebc2f`.

Full coverage: LF lines 2819–4074, inclusive. The original full-read source SHA256 was `2dc59dcf167759e7198afc474d356285d293a1e7d187a3329f7ce9f7a349ecc5`; at that read, these were PowerShell Get-Content logical lines 2819–4083 because U0015 contained repeated isolated CR characters. The current source pin above incorporates only the subsequent newline correction. PB03 and PB10 were revalidated as exact substrings against that corrected source. Every line was read. Every retained turn in that range is classified below. Node IDs and chain positions are stable locators; LF line numbers match `rg - n`.

This lane audits the retained transcript, not the motives of its author. The external papers and linked `Horizon_NS_Entanglement` package were not acquired in this bounded lane. A0706's reported 33 symbolic groups and 53 scalar identities are recorded as a transcript claim, not an independent rerun. New calculations are given fully in `CALCULATIONS.md` and `VISCOSITY_MAP.md`.

226.1.1 Complete turn coverage

Complete table entries (8 rows).

Entry 1.

Turn: U0013

Exact node ID: `94eb43d9-2a32-48be-8e40-d65160f3def6`

Chain: 1147

LF range: 2819–2824

Classification: Controlling physical-interpretation request; rejects a merely algebraic label.

Entry 2.

Turn: U0014

Exact node ID: `9ae478a7-c6c1-4b1f-9057-7723ae70b11f`

Chain: 1161

LF range: 2825–2830

Classification: Detailed observer/modular/KMS/Rindler/horizon chain, explicit instruction not to substitute a resonance description.

Entry 3.

Turn: A0626

Exact node ID: `c7b2e42f-53e5-428a-af50-68e2a7918456`

Chain: 1163

LF range: 2831–2834

Classification: Progress promise: detector observables, horizon stress, gravitational backreaction. No completed derivation in this turn. Partly fulfilled by A0706.

Entry 4.

Turn: A0638

Exact node ID: `f8c34278-7bce-4b37-9d29-0297bd9ee405`

Chain: 1180

LF range: 2835–2838

Classification: Proposed mass-inflation comparison. No Cauchy-horizon/mass-inflation derivation appears later in this retained range; superseded by the more explicit horizon-fluid and world-sheet examples.

Entry 5.

Turn: U0015

Exact node ID: 8cfab729 - 2a37 - 4ebc - 9be9 - 7319066d3775

Chain: 1185

LF range: 2839–2851

Classification: Controlling NS/ER=EPR research-program request and precision requirements.

Entry 6.

Turn: A0656

Exact node ID: 0fcbfffe - d051 - 4d02 - a7ca - b03970ed734a

Chain: 1229

LF range: 2852–3158

Classification: Substantial modular-scattering calculation, but a scope substitution relative to U0014/U0015. Retain its exact arithmetic maps as precursor results.

Entry 7.

Turn: A0665

Exact node ID: 9478295f - 156e - 451a - 9c64 - a6b781125868

Chain: 1243

LF range: 3159–3162

Classification: Progress promise of boundary forcing and cross-horizon observables. Fulfilled by explicit maps in A0706.

Entry 8.

Turn: A0706

Exact node ID: 15a7724d - c82b - 4704 - 9e93 - 932256ec5c6c

Chain: 1336

LF range: 3163–4074

Classification: Substantive repair: explicit physical maps, several legitimate limited obstructions, plus unresolved analytic endpoint and arithmetic-realization work.

There are no wholly unrelated retained turns in this range. The mass-inflation proposal is a superseded proposed branch, not an established obstruction or a completed calculation.

226.1.2 Passage ledger

PB01 — The requested physical chain was replaced in A0656, then substantially restored Locators and short passages. U0013, lines 2821–2823: “the question was what is this PHYSICALLY”. U0014, lines 2827–2829: “do not say resonance” and a demand to follow modular time, uniformly accelerated observables, horizon thermality, and possible singular geometry. U0015, lines 2841–2850, adds both NS/horizon and entanglement/bridge chains and requests coordinate-level maps. A0656 opens, lines 2854–2858, “the useful picture is an open resonator”.

Requested calculation. Carry the concrete thermal and NS objects through both physical chains, with state space, maps, dynamics, constants, and observables. **What A0656 does.** Introduces the hyperbolic modular surface, a particle Hamiltonian, a cusp channel, and its scattering coefficient. **What is missing at that turn.** No map from the source NS state/flow or BC equilibrium system into that particle experiment, and no gravitational endpoint. The NS comparison is expressly an analogy.

Scope. The scattering model is a valid separate arithmetic realization on the displayed formulas; it cannot substitute for the requested horizon construction. **Continuation.** Retain A0656's pole calculation, while resuming the maps A0706 actually supplies. A0706 is a real subsequent repair and must not be omitted from the audit.

PB02 — A0656 calculates a zero-to-pole map but only displays the simple-pole time response

Locator. A0656, lines 2964–3096; “For a simple resonance” at LF3043. **Objects.** The exact $S(k)$, $\rho = a - i\gamma$, $k_\rho = \gamma/2 - i(1 - a)/2$, $E_* = \hbar^2/(2mR^2)$, and $\mathcal{E}(k) = E_*(k^2 + 1/4)$. **Proved in the response.** Pole location, absence of numerator cancellation, complex energy, and a simple-pole width; the response explicitly avoids claiming a universal single exponential. **Missing.** Actual zero multiplicity and all polynomial time factors of a repeated pole; a full packet inversion with the remaining contour contribution. **Finished here.** CALCULATIONS.md §1 computes the exact pole order, leading coefficient in both variables, and every principal-part residue factor. **Scope.** These poles neither prove runaway real-time energy nor disprove the later tau construction. The remaining packet estimate is an analytic calculation on a specified packet and contour, not an RH reformulation.

PB03 — The projected-intensity identity is useful but supplies no positivity theorem

Locator. A0656, lines 3104–3137; “outgoing intensity in the selected channel minus incoming intensity in that channel”. **Objects.** U , orthogonal P , $U^*PU - P$, and the original Hilbert norm. **Proved.** The difference-of-squared-projection-norms identity and the commutator identity. **Missing.** An intertwiner from these chosen channel observables to the original arithmetic pairing; no sign on the relevant original subspace is derived. **Finished here.** CALCULATIONS.md §2 proves both identities and gives an exact two-dimensional sign test. **Scope.** This disproves only the inference that unitarity alone makes the redistribution operator positive. It says nothing against a positivity theorem using additional arithmetic structure. **Next derivation.** Identify the actual arithmetic test space, pairing, and action before transferring channel positivity or almost-null behavior.

PB04 — The observer-to-wedge arrow is a specialization, not a theorem about every entangling subsystem

Locator. U0014, line 2829, asserts a broad observer prescription. A0706, lines 3231–3331, begins its worked case with $\iota_W : \mathcal{A}(W_R) \hookrightarrow \mathcal{A}(\mathbb{M}^{1,3})$, $\omega_R = \omega_0 \circ \iota_W$, and the accelerated trajectory. **Proved or displayed.** An explicit wedge-algebra state restriction, the stated modular-sign convention, the boost parameter, proper-time conversion, and detector detailed balance. **Missing.** A construction assigning the required wedge/local net/state to a general arithmetic or entangling subsystem. **Scope.** The Bisognano–Wichmann identification applies in its stated wedge-field setting; the source never proves the broad first arrow for every subsystem. The explicit restriction is valuable and survives this scope limit. **Next derivation.** Construct the actual arithmetic representation and a compatible spacetime local net, then prove the modular group intertwinement there. Prime occupation matching alone is insufficient to perform this arrow.

PB05 — Static-observer divergence is computed; a source-driven gravitational singularity is not

Locator. A0626, lines 2831–2833, promises a freely falling test. A0638, lines 2835–2837, proposes mass inflation. A0706, lines 3333–3397, computes the Schwarzschild radial substitution, acceleration temperature, Tolman temperature, and finite horizon curvature. **Proved or displayed.** The coordinate relation, finite- ρ metric corrections, $T_{\text{acceleration}}/T_{\text{local}} = r_h^2/r^2$, and the distinction between a static-observer limit and finite horizon curvature. **Missing.** No coupling of the source concentrating NS/arithmetic state to Einstein evolution, no reconstructed stress tensor along a freely falling trajectory, and no mass-inflation calculation. **Scope.** Finite Schwarzschild horizon curvature rules out reading that particular static-observer temperature divergence as a curvature singularity. It does not rule out curvature growth in a different sourced geometry. **Next derivation.** Carry the actual force/stress data into a controlled bulk solution and evaluate invariant curvature and freely

falling measurements. The mass-inflation candidate is lower priority than the explicit fluid/gravity construction that was retained.

PB06 — The accelerating-string pullback is an actual bridge example with a restricted input **Locator.** A0706, lines 3399–3483, the embedding $x^2 + z^2 - (X^0)^2 = b^2$ and induced metric. **Requested calculation.** Identify the extra dimension and physical objects in an ER=EPR example. **Proved or displayed.** The right-exterior pullback, endpoint proper clock, worldsheet temperature $\hbar c / (2\pi k_B b)$, and scalar curvature $-2/R^2$, with the heavy-probe and holographic domain stated. **Missing.** A map from the actual NS singular state or arithmetic state into this gauge/string configuration. **Scope.** This is a substantive retained geometric example, not a proof of a universal ER=EPR identification or a disproof of the requested route. **Next derivation.** Supply the joint boundary state and its cross observables for the proposed source, and show how the source enters the string/bulk equations without assuming the missing state dictionary.

PB07 — The source-viscosity-to-cutoff-clock morphism is absent from the retained response **Locator.** A0656, LF2866, explicitly names a viscosity-one stage. A0706, LF3571–3577, retains target viscosity r_c ; LF3611–3627 embeds by $\tau = t$ with unchanged spatial coordinates. **Proved or displayed.** The target proper clock and $\nu_{\text{proper}} = c\sqrt{r_c}$. **Missing.** The actual equation-preserving parameter dictionary from the source viscosity to this target and the induced endpoint/norm factors. **Finished here.** VISCOSITY_MAP.md proves the invertible map $b = r_c/\nu_s$, $t = b\tau$, $v = bu$, $P = b^2 p$, $f^{\text{cut}} = b^2 f$, its domain, endpoint image, kinetic and dissipation factors, and Brown-York coefficient. **Scope.** The direct unchanged-clock identification only intertwines the displayed equations when $r_c = \nu_s$. This is a repair of a missing parameter morphism, not an obstruction to the fluid/gravity program. The linked full note may contain additional dictionary data; this bounded audit makes no claim about unread artifact contents.

PB08 — The boundary-force construction is exact on each preterminal interval and needs the actual endpoint calculation **Locator.** A0706, LF3660–3697; “does not assert an uncomputed smooth terminal metric”. **Objects.** $A_t = -(\Phi_t^{-1})^* \int_0^t \Phi_s^* f_s ds$, $h_{00} = -2v^j A_j$, the actual source flow, and the forcing equation. **Proved.** The coordinate force identity, retaining the transport and one-form derivative terms. **Missing.** Terminal control of $D\Phi_t^{-T} \int_0^t D\Phi_s^T f_s \circ \Phi_s ds$, its spatial/time derivatives, and the resulting metric-source coefficients on the actual NS field. **Finished here.** CALCULATIONS.md §5 expands the exact derivative factors and constructs an explicit smooth preterminal forced incompressible NS comparison with terminal-flat force and divergent coefficients. **Scope.** The comparison is not finite-energy or compactly supported, so it does not decide the manuscript’s profile. It proves only that the general smooth-force inference is invalid. **Next derivation.** Calculate these factors for the actual retained field, retaining all support and logarithmic corrections, and feed them into the chosen gravitational construction.

PB09 — The higher-derivative ratio is a real obstruction to uniform truncation, not to a full endpoint **Locator.** A0706, LF3805–3831; “which gravitational approximation has stopped controlling the answer”. **Objects.** Leading $-r_c \Delta v_i$, actual correction $-\frac{3}{2} r_c^2 \Delta^2 v_i$, and the source pulse wavelengths with their retained support/logarithmic factors. **Proved.** For a fixed Fourier component, the magnitude ratio is exactly $\frac{3}{2} r_c |k|^2$; this grows on decreasing source wavelengths. **Missing.** The complete higher-order/source-dependent cancellation or domination calculation, controlled re-summation/full Einstein solution, and endpoint observables. **Scope.** A single unbounded relative coefficient rules out asserting uniform control from the displayed finite derivative truncation. It does not establish nonexistence of an exact bulk solution or settle its curvature. **Next derivation.** Use the full retained correction equation and actual source pulses; calculate a remainder bound or solve

the relevant bulk equations with the original forcing. The exact clock morphism in PB07 does not remove this ratio.

PB10 — Partial trace proves only nonuniqueness from local histories; stronger joint data are explicit **Locator.** A0706, LF3835–3901; “The map from the joint state to local stress data is demonstrably many-to-one”. **Objects.** $m(\rho) = (\text{Tr}_R \rho, \text{Tr}_L \rho)$, TFD density, product Gibbs density, cross-energy covariance, and local drives. **Proved.** Identical marginals and identical one-sided driven histories coexist with different cross-energy covariance. **Missing.** A selected joint state and joint evolution for the actual source; a construction of geometry from those data. **Finished here.** CALCULATIONS.md §3 exhibits the entire thermofield phase family with the same marginals and the same cross-energy covariance, and gives explicit bounded cross observables recovering its relative phases. **Scope.** The obstruction reaches reconstruction from one-sided data alone. It does not establish unrelatedness of NS and entanglement, and it does not prevent reconstruction after adding sufficient joint observables. **Next derivation.** Calculate the actual cross-observable state and its dynamical relation to the source singular profile rather than treating local stress as a complete joint-state invariant.

PB11 — BC prime occupations and modular clocks match, but the global trace-class boundary must be retained **Locator.** A0706, LF3903–3972; “A spatial field net and its stress tensor are additional physical data”. **Objects.** $\ell^2(\mathbb{N})$, prime bosonic Fock occupations, $E_* \log n, \omega_p = E_* \log p/\hbar$, the doubled thermal vector, and the exact coefficient match. **Proved or displayed.** The occupation bijection and Hamiltonian dictionary; partition function stated in its trace-class range. **Missing.** Operator domains, the simultaneous global normalization boundary after imposing the Rindler temperature match, and the spatial/local stress realization. **Finished here.** CALCULATIONS.md §4 proves the full unitary operator/domain equality, prime product convergence, energy moments, modular sign/time map, and the exact global range $\beta E_* > 1$, equivalently $0 < a < 2\pi c E_*/\hbar$ for this correspondence. **Scope.** Failure of this particular global Gibbs normalization does not establish absence of all KMS states or a physical cooling singularity. **Next derivation.** Construct the relevant field net and full arithmetic observable correspondence on its actual state domain; retain the semigroup and multiplicities of any BCM extension instead of replacing them by the elementary ideal model.

PB12 — Equilibrium data do not determine cooling dynamics **Locator.** A0706, LF3974–3980; $\dot{n} = -\Gamma(n - n_\beta)$. **Proved.** Every $\Gamma > 0$ gives the same equilibrium occupation but a different relaxation time: direct solution is $n(t) = n_\beta + (n(0) - n_\beta)e^{-\Gamma t}$. **Missing.** A source-selected dynamical generator, coupling, and energy exchange for the proposed arithmetic physical system. **Scope.** This excludes inferring a cooling deadline from KMS populations alone, not constructing a cooling model from additional dynamics. **Next derivation.** Specify and calculate the actual generator transported by the physical/arithmetic map; pulse choices in a fixed scattering experiment do not alter the underlying zeta zeros.

PB13 — The Schwarzian diagonal identity is a local bridge; the full arithmetic kernel has not been realized **Locator.** A0706, LF3982–4058; “Identifying the full arithmetic kernel with a particular spacetime ray map remains a separate construction”. **Objects.** The exact diagonal kernel subtraction, $\{F, u\}$, central charge c_{CFT} , the null time coordinates, power $-\hbar c_{\text{CFT}}\{F, u\}/(24\pi)$, and entropy $-c_{\text{CFT}} \log F'/12$. **Proved or displayed.** The local diagonal identity and the direct entropy/flux/horizon-exponential derivative calculations. **Missing.** A full arithmetic kernel-to-ray-map morphism, including off-diagonal equality, orientation $F' > 0$, global state/boundary conditions, and transport of the arithmetic action into physical evolution. **Scope.** Sharing a local invariant neither supplies the full kernel identity nor proves incompatibility. **Next derivation.** Work on the

actual retained arithmetic kernel and solve/test the full pullback equation with the original variables and boundary data. This has not yet become a proof about the later theta source.

PB14 — No retained physical obstruction reaches A_{τ}/K_{τ} Locator. Entire assigned range, especially A0706's final LF4068–4074. **Evidence.** The range contains modular-surface scattering, BC occupation dynamics, and physical constructions, but it does not define or calculate the later $A_{\tau} = R\Gamma(P, -)$, K_{τ} , their restriction/homotopy-fibre maps, or the original theta action. **Scope.** None of the demonstrated many-to-one maps, finite-truncation limits, or trace-class boundaries proves failure of the split-base/privileged-tau program. **Integration.** Preserve the exact scattering multiplicities, units, force-map caution, and state/action questions as context. The main redo prompt must still resume the actual later A_{τ} , K_{τ} , and theta calculations, with the parent's later-turn locators. Replacing that work with the physical examples in this range would repeat the earlier scope substitution.

226.1.3 Priority for integration

1. Preserve A0706's completed maps and its correctly scoped limitations; do not label the entire physical response an abandonment.
2. Carry the exact multiplicities, operator domains, trace-class boundary, phase-sensitive joint observables, and viscosity/clock map from the new proof files into the relevant passage audit.
3. Mark the actual forced gravitational endpoint, arithmetic local-net/ray-map realization, and full packet estimates as calculations not completed in this retained range. These are concrete original-object continuations, not substitute assumptions.
4. Keep this physical precursor secondary to the current original theta/split-base continuation. This lane supplies no blanket program-failure verdict and no general ER=EPR theorem.

227 Formation of the original base: U0016–U0028

228 Formation of the original tau-base program: passage audit U0016–U0028

This audit records calculations and their scope, without judging motives. It reads the entire assigned transcript segment and the complete primary Tau_Base note and canonical source route. Six substantive answers and sixteen progress nodes cover thirteen user turns. The primary Tau_Base construction is a later successor: it determines the continuation available now, not what an earlier answer could already have read.

228.1 Coverage and source pins

- output/tau_fl_transcript_audit_2026-09-13/sources/audit_segment_U0016_U0028.md — 199,384 bytes; SHA-256 9705e32233d967f4abc6d522d40cfddc20a9f94c057d47410b37a3ee9e065298.
- output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md — 25,512 bytes; SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3.
- work/fl_geometry_current_source_route_20260913.md — 15,362 bytes; SHA-256 a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74.

All 198,330 decoded source characters (198,330 after CRLF-to-LF conversion) were read through the last A0963 node. The contiguous reading windows and all node IDs are retained in AUDIT.json. Exact user turns are separately copied to SEGMENT_USER_INPUTS_VERBATIM.md.

Historical cited attachments and test counts are not independently verified by this lane. A0952/A0961 cite Deligne witnesses; this audit assesses what those transcript passages calculate and claim, while the root task owns independent Deligne/purity verification. U0028 remains active at the segment boundary.

228.2 Status of every user turn

Complete table entries (13 rows).

Entry 1.

User turn / node: U0016 / 7d76e278-ae0c-43db-ab75-81ea7405723a

Response nodes: A0707, A0720, A0751, A0756

Classification and evidence: **advanced-partial; universal transfer unfinished.** Full sequential reconstruction, exact ideal/character/Mellin maps and genuine bounded obstructions are delivered. §9 assumes the decisive universal transfer/decay. U0016 did request a conditional sketch component, so that component is valid but not completion of the requested full mapping.

Entry 2.

User turn / node: U0017 / 9dd06d78-abdb-4213-a643-73780d99ad1f

Response nodes: A0758

Classification and evidence: **superseded by further user steering.** Only a progress acknowledgment follows before U0018. Its correction—one support beneath distinct origins—remains active and must not be dropped.

Entry 3.**User turn / node:** U0018 / a4f2753d-2aa4-4c11-b394-3b77c1640c80**Response nodes:** A0781, A0788

Classification and evidence: **advanced side and all-degree algebra; main transfer unfinished.** The all-n holonomy construction and specifically requested ES/Jordan integration are substantive. The answer explicitly lacks identification of its chosen H with actual arithmetic zeros; U0019 repeats the main task.

Entry 4.**User turn / node:** U0019 / 844aed0c-72f8-4ac2-b351-5ad9ef346c7a**Response nodes:** A0790

Classification and evidence: **interrupted by explicit source-reading prerequisite.** A0790 states an intent to isolate the central implication; U0020 immediately requires reading the split-zero work. No substantive calculation answers U0019 in that interval.

Entry 5.**User turn / node:** U0020 / b3ff5366-d7c7-46ec-9f49-951d2f29177f**Response nodes:** A0796, A0805

Classification and evidence: **source-reading progress; scope widened by U0021.** A0805 reports reading 956 lines of globalization_note.tex and one localization theorem. This audit does not independently verify the historical reading claim. U0021 requires additional current GitHub material.

Entry 6.**User turn / node:** U0021 / 1f65ae67-a886-4ef1-9e54-082d7c928fe4**Response nodes:** A0812, A0837, A0844

Classification and evidence: **source repairs and typed maps advanced; integration unfinished.** A0844 explicitly repairs repeated-adjunction, bare- C and all-observer substitutions, giving exact replacement maps. The requested original cohomological argument is still not executed, leading to U0022/U0023.

Entry 7.**User turn / node:** U0022 / b48b70f3-9b82-48fc-b7b9-d3e6fd428c42**Response nodes:** A0862

Classification and evidence: **advanced original-object calculation; weight inference still uncalculated.** The ideal absorption, infinite norm, exact-kernel quotient and all-character weight calculation are real results. Their scope does not prove purity, and the response ends at a quoted positivity criterion.

Entry 8.**User turn / node:** U0023 / c1f5c457-5da2-439d-b133-0a4b032008ed**Response nodes:** Merged into subsequent steering

Classification and evidence: **merged into U0024; substantive response delayed.** The user makes the Deligne-at-tau construction explicit and requires reintegration. U0024 immediately states two routes; A0863 responds to the combined request. Later A0952/A0961 execute only part of that task.

Entry 9.**User turn / node:** U0024 / 65b8bdef-be04-4ac6-a7f1-9bb25a8f74f4**Response nodes:** A0863

Classification and evidence: **progress then source provenance restart.** The two routes are acknowledged. U0025 challenges Deligne reading and U0026 restarts the prompt; A0952/A0961

later give source comparison and finite coefficient/NS calculations but not arithmetic purity.

Entry 10.

User turn / node: U0025 / bd3fe5a5-4f4c-4f92-ae76-5fc2f58512d1

Response nodes: Merged into subsequent steering

Classification and evidence: **source claim challenged; later retracted.** U0025 is followed by U0026 before an assistant node. A0877 and A0952 explicitly retract the unestablished reading claim; this is source-provenance evidence, not a motive finding.

Entry 11.

User turn / node: U0026 / 0cda31d8-8a8d-428c-b551-77999e854978

Response nodes: A0877

Classification and evidence: **restart accepted; method refined by U0027.** A0877 promises to locate the actual attachment and use its precise theorems. U0027 then requires TeX-only access.

Entry 12.

User turn / node: U0027 / b48fac36-14fb-4a9b-a6fe-7d6fe8b94886

Response nodes: A0882, A0934, A0937, A0952, A0961

Classification and evidence: **source route repaired; mixed-support calculations advanced; arithmetic transfer unfinished.** A0952 reports scan checks despite TeX-only instruction; A0961 explicitly uses historical TeX witnesses and preserves discrepancies. Mixed-double localization/Frobenius and arbitrary monodromy blocks advance the algebra. The finite H0 spectrum is not linked to original theta H1.

Entry 13.

User turn / node: U0028 / 9d5e82ea-d670-4145-b5d3-c22d8948133e

Response nodes: A0963

Classification and evidence: **active at assigned segment boundary.** Only a progress node is included after U0028. The response continues in the next segment, so no session-wide abandonment conclusion is drawn here.

228.3 Passage-by-passage findings

Priorities: 0 = essential original-program continuation; 1 = integrate or preserve exact scope; 2 = useful bounded source result. A finding that a result is limited is not a claim that the program fails. Each cited line is a one-based line of the retained segment.

228.3.1 BF01 — A0756 §§2.2–3.3

User U0016 (7d76e278-ae0c-43db-ab75-81ea7405723a); assistant A0756 (7feaa63b-bf8a-46d2-bffc-4ab530615f9d), chain 1444; passage line 341. Priority 1.

The finite-norm ideal correspondence is multiplicative:

Status: advanced; retain exact arithmetic maps.

Requested calculation. Recover the Dedekind system and all character sectors after adjoining the common split support; map the complete arithmetic picture.

Original objects and maps. $S=G(O_K)$, $e=0_R$, $I_J=JU\{\tau\}$, p_R , π_J , $N(I_J)=N(J)$; the ideal-basis unitary V , inertia projectors E_I , ramification-complete group-algebra Euler product.

What was actually proved. The answer gives exact norm/ideal multiplication, intertwines the log-norm Hamiltonians including their domains, gives finite character Fourier inversion, Mellin

transform $\Gamma(s)L(s,\chi)$, and repairs omitted ramified local factors. The $p=2$ factor for $Q(i)$ remains explicit.

What remained uncalculated. Those maps do not yet place the character spectral quotient in a cohomology over the absolute base, or calculate an arithmetic weight bound. This is an incomplete larger task, not an error in the retained ideal calculation.

Exact obstruction scope. No obstruction to the original program is established by arithmetic preservation; these maps are usable inputs.

Concrete surviving derivation. Attach the preserved square $G(Z) \rightarrow G(C)$, p_Z and p_C to the actual tau-base structural map and theta complex, as the later Tau_Base (2)–(8), (14), (19) do. Preserve inertia and omitted-prime data for any character generalization.

228.3.2 BF02 — A0756 §4; corrected by A0844 §1

User U0016 (7d76e278-ae0c-43db-ab75-81ea7405723a); assistant A0756 (7feaa63b-bf8a-46d2-bffc-4ab530615f9d), chain 1444; passage line 743. Priority 1.

The support-chain operations are

Status: scope substitution, explicitly repaired later.

Requested calculation. Relate one split support to the full family of distinct coordinate origins, higher nullity operations, winding and characters.

Original objects and maps. The answer uses $G_{\{C_n\}}(R[G])$, chain support max/min, and $c_n(a,j)=(a,\min(j,n-1))$. The later source correction retains interlevel maps $i_n:L_n \rightarrow L_{\{n+1\}}$ and $\text{Orig}_i(y)=i^{\{-1\}}(y)$.

What was actually proved. It proves amplitude preservation for the displayed support-chain collapse. That result is about that operation and its specified carrier; it cannot identify every origin-and-operation raise with iteration of zero adjunction.

What remained uncalculated. The actual interlevel nullity operation and its typed comparison with cyclic/character constructions were not computed. A0844 explicitly acknowledges the substitution, and U0022 clarifies that repeated realizations of the same bottom support and higher nullity operations must both remain.

Exact obstruction scope. The collapse result does not reach the full origin ladder. Equal empty origin fibres do not identify their domain points or higher operations.

Concrete surviving derivation. Retain fixed scalar $G(R)$, the full interlevel maps and their origin fibres. In the current tau model keep all masks $A \subseteq \{+,-\}$, the original T_A and their transports; compute any higher operation on those diagrams rather than replace it by c_n .

Later repair or steering: A0844 §§1–3; U0022.

228.3.3 BF03 — A0756 §4.1; corrected by A0844 §4

User U0016 (7d76e278-ae0c-43db-ab75-81ea7405723a); assistant A0756 (7feaa63b-bf8a-46d2-bffc-4ab530615f9d), chain 1444; passage line 823. Priority 1.

Complex-valued observations necessarily forget the extra zero

Status: overbroad heading; precise theorem retained and scope repaired.

Requested calculation. Carry the added supported zero into the arithmetic and operator geometry without erasing it by an inappropriate observation.

Original objects and maps. Additive unital semiring maps $G(R) \rightarrow B$ with B a ring factor through p_R ; multiplicative representations of the pointed monoid instead factor through the contracted monoid algebra.

What was actually proved. The displayed factorization theorem is correct for additive semiring maps into rings. A0844 provides the inverse of the contracted-monoid decomposition and $T(n) = \text{diag}(1, n)$, $T(\tau) = 0$, which retains $T(e) = \text{diag}(1, 0)$.

What remained uncalculated. The heading and associated interpretation exceeded the theorem type; no all-observer vanishing was proved. The central idempotent representation was initially left outside the purported complete map.

Exact obstruction scope. The ring-reflection theorem obstructs support detection only for its stated additive maps. It does not kill the multiplicative linearization or the supported cohomology fibres.

Concrete surviving derivation. Use both p_R and χ_R and keep their types. For current Q and its dual, retain supported zero functionals at every label and the actual precomposition arrows of Tau_Base (31), (43); do not infer τ from a zero complex value.

Later repair or steering: A0844 §4.

228.3.4 BF04 — A0756 §§5.1–5.4

User U0016 (7d76e278 - ae0c - 43db - ab75 - 81ea7405723a); assistant A0756 (7feaa63b - bf8a - 46d2 - bff c - 4ab530615f9d), chain 1444; passage line 934. Priority 2.

This is the concrete content of the winding/character connection.

Status: advanced; finite observation scope subsequently enlarged.

Requested calculation. Count full integer winding and map it to arithmetic holonomies, rather than treating a return of phase as loss of all winding information.

Original objects and maps. $Z \rightarrow G$, $n \mapsto \text{Frob}_p^n$, followed by χ ; the exponential covering and lifted angle; the K4 quartet coefficients $c_{\{++\}} = 2, c_{\{+-\}} = 0, c_{\{-+\}} = 4(\beta - 1/2), c_{\{-\}} = 4i\gamma$.

What was actually proved. The displayed maps retain the prime-orbit character value $\chi(\text{Frob}_p)^n$ and recover the actual real displacement from the quartet. The exponential cover of $C^\times$ has no fibre over zero, precisely for that cover.

What remained uncalculated. The requested fixed-origin full integer-action realization was not yet built in this answer. A0788 later builds it on a lifted angular space and Jordan tangent data, but still does not identify the chosen action with the actual arithmetic spectral action.

Exact obstruction scope. The empty fibre of $\exp$ over zero is a property of that particular covering. It is not an obstruction to an origin blow-up, an action groupoid at a fixed origin, or the original support base.

Concrete surviving derivation. Keep the full-turn $T_n(H)$ representation before finite character observation, then instantiate H by the original arithmetic U_a on a finite full jet packet and prove the resulting determinant/trace maps with all nilpotents.

Later repair or steering: A0788 §§1–4.

228.3.5 BF05 — A0756 §§6.2–7.2

User U0016 (7d76e278 - ae0c - 43db - ab75 - 81ea7405723a); assistant A0756 (7feaa63b - bf8a - 46d2 - bff c - 4ab530615f9d), chain 1444; passage line 1184. Priority 2.

smooth bulk extension exists for every

Status: legitimate calculated obstruction, with limited scope.

Requested calculation. Test whether a coherent higher-dimensional cancellation forces the real arithmetic period defect to vanish.

Original objects and maps. $\alpha_a = d\Phi + a d\ell$ on the circle of length L ; disk extension $(aL/2\pi)(X dY - Y dX)$; paired holonomy $\text{diag}(e^{-aL}, e^{aL})$; hyperbolic anomaly envelope $A \oplus \hat{A}$.

What was actually proved. The disk curvature integral equals aL . Flat extension of the specified meridional period forces $a=0$, but smooth curved extension exists for every a . A determinant-one pair preserves an indefinite pairing at arbitrary a ; its positive invariant form would force $a=0$. The finite hyperbolic completion is independent of the spectral zero.

What remained uncalculated. No map from the actual arithmetic theta source to the flat extension or positive form was constructed. Universal abstract cancellation alone does not supply it.

Exact obstruction scope. These are exact counterexamples to the stated implication from mere smooth completion, determinant cancellation, or an abstract finite envelope. They do not disprove a more constrained arithmetic construction over τ .

Concrete surviving derivation. Compute the actual arithmetic residue/trace pairing and its analytic source, preserving the raw orientation, moment twist and boundary terms. Use Tau_Base (38)–(45) rather than declaring an arbitrary completion to be its duality.

228.3.6 BF06 — A0756 §8

User U0016 (7d76e278-ae0c-43db-ab75-81ea7405723a); assistant A0756 (7feaa63b-bf8a-46d2-bffc-4ab530615f9d), chain 1444; passage line 1411. Priority 2.

Smooth flatness does not supply that map by itself.

Status: exact regularity distinction; transfer unfinished.

Requested calculation. Map the NS all-order residual cancellation into an arithmetic analytic identity while preserving the original singular velocity and profiles.

Original objects and maps. The explicit approach map $\beta(q, X, \eta, \theta)$, its $q^{-1/2-h}$ velocity factor, residual flatness, and the comparison germ $\exp(-1/q^2)$.

What was actually proved. The answer retains the unbounded velocity factor and identifies the weighted profile as a different observable. The flat smooth example proves that all derivatives vanishing at an endpoint do not imply the holomorphic identity theorem without a function-space transfer.

What remained uncalculated. The actual map on profiles, pressure corrections, moment conditions and residuals into the arithmetic Schwartz/Mellin spaces is not calculated.

Exact obstruction scope. Only a direct inference from smooth flatness to holomorphic vanishing is blocked. The evidence does not rule out a new controlled transformation.

Concrete surviving derivation. If this historical route is resumed, specify and calculate the map from the original NS correction histories into the original V and B seminorms, including every q -dependent weight and endpoint term. The root task now prioritizes the original theta cohomology route.

228.3.7 BF07 — A0756 §§9–9.2

User U0016 (7d76e278-ae0c-43db-ab75-81ea7405723a); assistant A0756 (7feaa63b-bf8a-46d2-bffc-4ab530615f9d), chain 1444; passage line 1451. Priority 0.

Suppose a construction assigns, to every

Status: valid conditional sketch component; requested construction interrupted.

Requested calculation. First reconstruct the whole proof sketch, then actually map its stages and prove the proposed transfer for all arithmetic sectors.

Original objects and maps. Actual zeros (χ, ρ) , $a_{\{\chi, \rho\}} = \text{Re} \rho - 1/2$, circles of fixed positive $L_{\{\chi, \rho\}}$, $\alpha_{\{j+1\}} - \alpha_j = d\varphi_j$, and an assumed L1 decay of α_j .

What was actually proved. The period-preservation argument is correct: $|a|L$ is bounded by the assumed L1 norm, hence vanishes when that norm tends to zero. U0016 explicitly asked for a sketch of what would follow if all steps were mapped, so this is responsive to that portion of the request.

What remained uncalculated. The transfer $T_{\{\chi, \rho\}}$ from actual NS correction histories, universal coverage of actual zeros, and the required decay are all introduced as assumptions. They are the substantive missing calculation, not conclusions of the theorem.

Exact obstruction scope. No obstruction is proved against the complete arithmetic program. The theorem only shows what its added hypotheses would imply; it cannot be counted as completion of those hypotheses.

Concrete surviving derivation. At the current successor, resume the actually constructed $A_{\tau}(T)$, $K_{\tau}(L1)$, $Q \rightarrow A_Z$ and dual injection. Derive an actual bound on the original theta source instead of replacing it with period decay or a positivity assumption.

228.3.8 BF08 — A0756 §10.1

User U0016 (7d76e278-ae0c-43db-ab75-81ea7405723a); assistant A0756 (7feaa63b-bf8a-46d2-bffc-4ab530615f9d), chain 1444; passage line 1654. Priority 1.

This is **not** a counterexample to GRH:

Status: legitimate bounded model test; no broad program verdict.

Requested calculation. Determine what original arithmetic rigidity follows from support, symmetry and abstract anomaly cancellation.

Original objects and maps. $P_{a,b}(s) = ((s - 1/2 - a)^2 + b^2)((s - 1/2 + a)^2 + b^2)$, $0 < a < 1/2$, $b > 0$, with the original four roots and symmetries.

What was actually proved. The polynomial has the two indicated reflection symmetries and four off-line roots. It can be assigned the general support labels and abstract envelope used in the answer. This disproves only an implication based solely on that common structure.

What remained uncalculated. No morphism identifying this polynomial family with the original arithmetic theta image, its gamma factor, trace formula or adjoint was supplied.

Exact obstruction scope. The answer expressly limits the example to non-arithmetic amplitudes. Preserve that limitation: it cannot establish failure of the tau-base program or replace its arithmetic function.

Concrete surviving derivation. Retain $g=2\xi$, the actual Θ , exact Mellin image, and A_{τ}/K_{τ} . Any subsequent surrogate bound must be transferred through a proved map before its failure is attributed to the original construction.

228.3.9 BF09 — A0788 §§2,6

User U0018 (a4f2753d-2aa4-4c11-b394-3b77c1640c80); assistant A0788 (c65ef26a-1cb7-418f-ba78-f620261edde9), chain 1521; passage line 1937. Priority 1.

All levels compose explicitly

Status: genuine all-degree advance; arithmetic instantiation incomplete.

Requested calculation. Give the full n -level cyclic/holonomy ladder, not merely its second or third example, and map its Mellin and arithmetic operations.

Original objects and maps. $T_n(H)(v_0, \dots, v_{n-1}) = (Hv_{n-1}, v_0, \dots, v_{n-2})$; $P_{\{m,n\}}(e_a \otimes e_b \otimes v) = e_{\{a+mb\}} \otimes v$; $\nu_{\{H,Q\}} = \sum a_k \delta_{\{Q^k\}}$.

What was actually proved. The answer proves $T_n(H)^n = \text{diag}(H)$, determinant $\det(I - zT_n(H)) = \det(I - z^n H)$, trace vanishing unless n divides the power, and the all-degree composition law. The pushforward $t \mapsto t^n$ gives spectral dilation with the displayed Mellin factor $1/n$ for functions. It also retains inertia invariants and nonabelian multiplicities.

What remained uncalculated. The chosen H in the main example is not derived from the actual arithmetic zero cohomology. The local finite permutation determinant and global spectral zero packet have not been joined by a map.

Exact obstruction scope. This is substantial completion of an all-degree algebra subtask; it does not by itself furnish the global GRH-sensitive invariant requested repeatedly.

Concrete surviving derivation. Apply the same proved T_n construction to $U_a|A_Z = a^\rho \exp(\log(a)N_\rho)$ on the actual full jet quotient, and compare its trace/determinant to the arithmetic theta morphism. Keep n , every multiplicity, all powers of N and the original a .

228.3.10 BF10 — A0788 §§3–5,8

User U0018 (a4f2753d - 2aa4 - 4c11 - b394 - 3b77c1640c80); assistant A0788 (c65ef26a - 1cb7 - 418f - ba78 - f620261edde9), chain 1521; passage line 2690. Priority 1.

It does not identify the chosen Jordan holonomy with a particular arithmetic zero

Status: authorized side construction advanced; main transfer left unfinished.

Requested calculation. Map the three-coordinate/Jordan/ES component as part of the entire arithmetic program; U0018 also explicitly requested ES repository integration.

Original objects and maps. $J = H_3(O)$, marked frame, Cayley–Dickson coordinates, $H = G_{\{(3+4i)/5\}C}$, $T_n(H)$, exact defect $\Delta(q) = -1$ and rational ES chart.

What was actually proved. The response gives an all-integer faithful automorphism construction, an ordered-frame tangent map, retained cubic coupling $2\text{Re}((xy)z)$, and the reversible non-diagonal ES chart preserving $\Delta = -1$. It expressly admits the missing arithmetic-zero identification.

What remained uncalculated. The construction does not calculate the requested universal interlevel arithmetic realization or show that this specific holonomy is the one on the original zeta zero space. U0019 repeats the still-open initial task; A0844 later retracts counting these examples as execution of the whole program.

Exact obstruction scope. No failure of the full program follows. The example shows that a fixed projected origin can retain winding; that success is compatible with the missing arithmetic identification.

Concrete surviving derivation. Keep the proved ES and Jordan maps as their own results. Resume the common-support arithmetic cohomology and derive its actual action, rather than selecting another illustrative holonomy.

Later repair or steering: A0844 §§1,7.

228.3.11 BF11 — A0844 §§1–3,7

User U0021 (1f65ae67 - a886 - 4ef1 - 9e54 - 082d7c928fe4); assistant A0844 (2aeb23d1 - e3be - 4204 - 9bc2 - 3abe87a3cbcd), chain 1712; passage line 2807. Priority 1.

My error was to substitute

Status: source repair completed; program continuation still absent.

Requested calculation. Read the actual current globalization/nullity sources, correct previous mistakes, and integrate those corrections into the original task.

Original objects and maps. Free zero-adjunction $Z \rightarrow U$ over a fixed zero-forgotten input; common scalar $S_0 = G(R)$; $A_d = R[x_1, \dots, x_d]$; $j_d^*: A_{d+1} \rightarrow A_d$ and ε_d .

What was actually proved. The answer supplies the unique-isomorphism argument from the universal property, uniqueness/naturality of the Boolean character, the fully faithful comparison for common-scalar algebra maps, and the commuting coordinate-origin diagram. It retracts the bare-C test and repeated-adjunction replacement.

What remained uncalculated. Source correction is not the requested cohomological realization or purity estimate. U0022 correctly insists that bottom uniqueness and higher nullity coexist; both claims must be propagated into subsequent work rather than alternately substituted.

Exact obstruction scope. Uniqueness over a fixed universal input is proved. Equality of dimensions, empty fibres, or Boolean observations does not identify arbitrary higher objects; conversely their distinction is no obstruction to a common base.

Concrete surviving derivation. The exact present continuation is the four-point P with all support fibres T_A over b_τ , the structural map, A_τ and res_σ comparison. No fresh bottom support should be invented for each mask or degree.

228.3.12 BF12 — A0862 §§1–2

User U0022 (b48b70f3-9b82-48fc-b7b9-d3e6fd428c42); assistant A0862 (b6c3dc97-18dd-487e-bb70-933fab060c64), chain 1745; passage line 3387. Priority 1.

There is a direct contradiction proving that the supported-zero ideal cannot receive a finite positive multiplicative norm

Status: completed original-object norm rigidity; scope exact.

Requested calculation. Keep bottom-support uniqueness while using the infinite quotient to calculate a real rigidity statement relevant to the whole arithmetic system.

Original objects and maps. $S = G(O_K)$, e-ideal $\mathfrak{e} = \{\tau, e\}$, supported ideal monoid $\{I_J\}$, $\mathfrak{e} I_J = \mathfrak{e}$; norm map to $[1, \infty]$.

What was actually proved. The absorption argument proves uniqueness in that specified ideal monoid. Since $N(2R) = 2^{[K:Q]} > 1$, a positive finite extension q would obey $q = 2^{[K:Q]} q$; hence the multiplicative extension has $N(\mathfrak{e}) = \infty$. The converse assignment and multiplication are checked.

What remained uncalculated. This exact result does not compute the sign or growth of arithmetic zero cohomology. The infinite norm alone does not define a geometric proper direct image or identify every possible F1 object.

Exact obstruction scope. The contradiction excludes finite positive norms with those operations and retained ordinary norms. It neither disproves nor proves the original purity program.

Concrete surviving derivation. Use the preserved arithmetic square as structural input to b_τ . Retain the source infinity value and prove how arithmetic operators act on $A_\tau(T)$; do not replace the norm by a finite mass or derive weights from its cardinality.

228.3.13 BF13 — A0862 §3

User U0022 (b48b70f3-9b82-48fc-b7b9-d3e6fd428c42); assistant A0862 (b6c3dc97-18dd-487e-bb70-933fab060c64), chain 1745; passage line 3682. Priority 0.

The contradiction concerns finiteness while preserving the specified arithmetic kernel

Status: completed quotient classification at exact zero fibre.

Requested calculation. Compute what a finite quotient at the distinguished support would do while preserving the original arithmetic.

Original objects and maps. $q:G(O_K)\rightarrow T$ unital, zero fibre exactly $\epsilon=\{\tau,e\}$; $p:G(O_K)\rightarrow O_K$; $\chi:G(O_K)\rightarrow B$ with zero fibre $\{\tau\}$.

What was actually proved. The restriction of q to O_K is a ring map, is injective by the exact zero fibre, and is surjective; therefore $T\cong O_K$ and is infinite. Finite arithmetic quotients instead have enlarged zero fibre I_J ; the finite Boolean observation has a different zero fibre and collapses all supported amplitudes.

What remained uncalculated. No uniqueness theorem for all F1 candidates or all infinite quotient cardinalities follows. No spectrum/weight transfer is calculated.

Exact obstruction scope. This excludes a finite quotient with precisely that kernel. It does not identify the structural base map with the Boolean generic-stalk restriction.

Concrete surviving derivation. Keep the later source's opposite-direction maps $j_{\text{math}_R}:F_{\{1,\tau\}}\rightarrow B_R$ and $\chi_R:G(R)\rightarrow B$. The first defines the absolute base; the second defines res_σ . Calculate their actual derived comparison, not an identity between them.

228.3.14 BF14 — A0862 §§4–5

User U0022 (b48b70f3-9b82-48fc-b7b9-d3e6fd428c42); assistant A0862 (b6c3dc97-18dd-487e-bb70-933fab060c64), chain 1745; passage line 3835. Priority 0.

It does not yet determine the sign of that full trace pairing.

Status: all-character support calculation advanced; analytic task interrupted.

Requested calculation. Propagate norm rigidity through every character and use it to study the full arithmetic rigidity mechanism.

Original objects and maps. $w_{\{\chi,s\}}(I_J)=\chi(J)N(J)^{-s}$; $\epsilon I_J=\epsilon$; $C[I_m\cup\{\epsilon\}]\cong C\times C[I_m]$; the retained arithmetic trace representation W .

What was actually proved. For $\text{Re } s>0$, a weight with $|\chi(J)N(J)^{-s}|<1$ forces $w(\epsilon)=0$. The direct-product algebra preserves the support idempotent and entire ordinary ideal algebra. This proves the exact zero contribution of that idempotent to those particular weights.

What remained uncalculated. The response then quotes the GRH-equivalent trace-positivity criterion and stops. It has not derived the full trace from the theta source, analyzed its sign, or shown that the kernel carrying actual spectral classes vanishes.

Exact obstruction scope. Zero weight on the absorbing ideal is not zero cohomology or zero spectral trace on the original quotient Q . The later source explicitly retains $H_1(T_A)=Q$ for every nonempty mask.

Concrete surviving derivation. Compute the full Mellin-jet trace contraction $R_Z(f,j_g h)=\sum m_\rho \text{conjugate}(f(1-\bar{\rho}))h(\rho)$ within $K_\tau(L_1)$, with the full g -unit and supported zero functionals. Any vanishing of a larger kernel must be proved by its actual test traces.

228.3.15 BF15 — A0952 opening; corrected reading route A0961

User U0027 (b48fac36-14fb-4a9b-a6fe-7d6fe8b94886); assistant A0952 (50a6a43b-ec70-40dc-b17f-864477ddcff4), chain 1985; passage line 3897. Priority 1.

I checked eight of those pages against the original French scan.

Status: source provenance repaired in stages; requested TeX-only route not followed by this account.

Requested calculation. U0025 challenged the claimed reading; U0026 required restart; U0027 explicitly required unpacking and reading LaTeX, with no PDF/OCR.

Original objects and maps. The nested six-part S20 archive, English/French historical TeX witnesses, their $j_!/j_*$ and dual-weight sign discrepancies; original scan checks are reported by the assistant.

What was actually proved. A0877 and A0952 retract the unestablished reading claim. A0952 reports a bounded 32-page text reading and eight scan checks rather than a full-paper audit. A0961 then explicitly uses the retained TeX and documents witness disagreements.

What remained uncalculated. This segment itself cannot verify every historical source-access claim or every attached package. A0952's reported scan checks do not comply with the immediately preceding TeX-only instruction. A0961 improves the source route but its reading remains section-bounded.

Exact obstruction scope. A source-reading defect is not a mathematical counterexample. It limits what the reported comparison establishes and requires exact witness pins; it does not license stopping the original program.

Concrete surviving derivation. Cite the actual retained French theorem text and exact ranges for the signs/maps used, keep historical English discrepancies visible, and calculate the comparison on the original tau objects. Do not repeat the source-reading cycle after sufficient source evidence is acquired.

Later repair or steering: A0961 opening and §§5.2–5.4.

228.3.16 BF16 — A0952 §§1–2

User U0023 (c1f5c457-5da2-439d-b133-0a4b032008ed); assistant A0952 (50a6a43b-e70-40dc-b17f-864477ddcfff4), chain 1985; passage line 4140. Priority 0.

a functor identifying your full mixed-double multiplication with Deligne's sheaf operations has not been constructed

Status: relevant source correspondence advanced; actual functor unfinished.

Requested calculation. Map the complete mixed-support structure into the analogous Deligne argument over one privileged base while retaining all directions.

Original objects and maps. Deligne common intersection E , Kummer cover X_m , $F[E]$, commuting N_i , successive specialization isomorphism; split mixed double D_A with X_a, Y_b, Z_z and four masks.

What was actually proved. The answer identifies the actual multi-direction specialization and coherence theorem. It gives mixed-double multiplication, amplitude r_D , Boolean group-semiring support map, and subset-to-bitvector bijection. This is more informative than treating all directions as fresh scalar zeros.

What remained uncalculated. The only explicit cross-comparison here is the incidence-label bijection. It is not a functor carrying sheaves, mixed-double multiplication, monodromy, derived global sections, or the arithmetic action.

Exact obstruction scope. No obstruction to a full functor is established. A failure to have constructed it cannot be promoted to nonexistence or to failure of the tau-base program.

Concrete surviving derivation. The later actual realization supplies T_A on P , reconstructed SplitZero sheaf, global functor A_τ and K_τ . Use those concrete objects as the domain of the comparison, preserving each mask and precomposition arrow, then calculate monodromy/scaling compatibility rather than identifying labels with weights.

228.3.17 BF17 — A0952 §§3,5

User U0023 (c1f5c457-5da2-439d-b133-0a4b032008ed); assistant A0952 (50a6a43b-e70-40dc-b17f-864477ddc ff4), chain 1985; passage line 4352. Priority 0.

I have not established that your split-zero construction realizes these sheaf categories

Status: Deligne mechanism identified; original-object calculation deferred.

Requested calculation. Develop the analogous cohomological proof over τ , including duality, mixed support, specialization and weight transfer.

Original objects and maps. $c_i : H_c^i \rightarrow H^i$, image I^i ; upper bound $n+i$, lower bound from $F^\vee$ and twist (d) ; $K_X = \text{Ra}^! Q_L$; proper potentially-pure specialization in §6.2.9.

What was actually proved. The answer preserves the two bounds on one image, the $2d-i$ index, both dual signs and twists, and the invariant-cycle target $H^*(X_{\text{bar}\eta}, K)^{\text{Gal}}$. It recognizes that singularity need not destroy duality in the stated rational-homology-manifold setting.

What remained uncalculated. No c_i for the actual theta sheaf, no extraordinary pullback or involutive duality on the original arithmetic category, and no two-sided arithmetic weight bounds are constructed in this response. The potentially-pure hypothesis of the quoted specialization theorem is not proved for the proposed system.

Exact obstruction scope. The quoted theorem proves results for its declared proper arithmetic setting. A common base point or infinite quotient does not itself satisfy those hypotheses. Conversely, the lack of that literal category identification does not disprove an analogous construction.

Concrete surviving derivation. Compute the actual A_τ right adjoint K_τ , its unit/counit and comparison with restriction; use the established finite-jet injection into its derived Hom. Distinguish this proved adjunction from unproved involutive local duality and calculate the surviving analytic bound directly.

228.3.18 BF18 — A0952 §4

User U0023 (c1f5c457-5da2-439d-b133-0a4b032008ed); assistant A0952 (50a6a43b-e70-40dc-b17f-864477ddc ff4), chain 1985; passage line 4295. Priority 0.

tensor amplification, geometric control, and an error bound that decreases under iteration

Status: actual Deligne amplification identified; arithmetic bound not calculated.

Requested calculation. Reproduce the proof mechanism on the original split cohomology at the privileged τ base, rather than only describe purity.

Original objects and maps. Deligne bound $w_q(\alpha) \leq 1 + 2^k(-k)$, the tensor-square inclusion and Lefschetz-pencil estimate. The successor Tau_Base uses P^r , signed tensor differential, $Q^\vee \otimes r$ and $a^r(r\rho)$.

What was actually proved. The answer correctly identifies how a fixed positive excess contradicts a bound with shrinking error. A0961 further retains the even tensor powers, nonnegative traces, determinant pole argument and $r+1/k$ estimate.

What remained uncalculated. The geometric upper estimate has not been derived for the original characteristic-zero theta source. In the successor source the amplification itself is completed but equation (50) explicitly leaves its bound unproved.

Exact obstruction scope. Failure to have that estimate is an unfinished calculation, not evidence that every possible τ -base argument fails. A bound tested on a later auxiliary polynomial family has its own scope.

Concrete surviving derivation. Use the already proved top tensor $Q^\wedge \otimes r$ and $K_{\{\tau, r\}} = S_{(\eta, \dots, \eta)} W[r]$, retaining the original Koszul signs. Derive an actual bound on $r(2\text{Re}\rho - 1)\log a$ from the original theta relations and analytic boundary terms; do not assume sublinear growth.

Later repair or steering: $\text{Tau_Base } \S 7$ completes amplification, not its bound.

228.3.19 BF19 — A0961 §§1–3

User U0023 (c1f5c457 - 5da2 - 439d - b133 - 0a4b032008ed); assistant A0961 (a7bbd422 - 311c - 4380 - ba6c - a4bdd61fce2f), chain 2032; passage line 4500. Priority 1.

the idempotent synchronizes support and preserves amplitudes.

Status: completed mixed-object localization; discarded support observation must remain explicit.

Requested calculation. Perform an actual operation on the original mixed support and compare what it preserves with the proposed purity mechanism.

Original objects and maps. $D_A = G(A)^\wedge 2$, $E = (1, e)$, $D_A[E^\wedge(-1)] \cong G(A[t]/(t^2+1))$; (Chi, π) injective; higher $D_{\{A, n\}}$, $E_n = (1, e, \dots, e)$.

What was actually proved. The answer proves the multiplication formulas, the synchronization localization universal property, and injectivity of the joint support/amplitude observation. E sends every nonempty mask to the synchronized face while preserving the amplitude. The full n -formulas retain $u = -1$ and all wrap exponents.

What remained uncalculated. Localization forgets which nonempty face the input occupied. Its coefficient-amplitude identity does not calculate a cohomological image, trace sign or weight inequality. No kernel/cone of this observation on the original theta complex is computed here.

Exact obstruction scope. This is a real morphism with a calculable information loss. It neither deletes arbitrary spectral packets nor forces their vanishing. It is not the same as the structural pushforward A_τ .

Concrete surviving derivation. Carry the maps on the complete mask diagram T_A , record the nonempty-mask saturation separately from coefficient maps, and compute induced kernel/cokernel or cone. Keep A_τ and res_σ distinct as in the successor (19)–(25).

228.3.20 BF20 — A0961 §4, completed purity calculation

User U0023 (c1f5c457 - 5da2 - 439d - b133 - 0a4b032008ed); assistant A0961 (a7bbd422 - 311c - 4380 - ba6c - a4bdd61fce2f), chain 2032; passage line 4690. Priority 0.

This gives **weight-zero purity of this coefficient object**.

Status: valid coefficient calculation; no arithmetic spectral transfer.

Requested calculation. Obtain a Frobenius-like arithmetic spectrum and prove the requested purity for the original zeta cohomology.

Original objects and maps. $B_{\{A, n\}} = A[t]/(t^{n+1})$; reductions $O_K \rightarrow k_p$; Φ_q with the exact coefficient sign $u^{\lfloor q_i/n \rfloor}$; $Y_n = \text{Spec}(F_q[t]/(t^{n+1}))$, $\gcd(q, 2n) = 1$.

What was actually proved. The response computes the affine permutation $j \mapsto qj + (q-1)/2 \pmod n$, geometric Frobenius as its inverse, and $\det(1 - \text{TF}|H_0) = \prod (1 - T^{d_j})$; thus this finite reduced coefficient spectrum has roots of unity. It explicitly retains primes dividing $2n$ outside the calculation and supported residue zeros under reduction.

What remained uncalculated. There is no map from $Q=B/\Theta V$, its H1 scaling representation or full Mellin jets to this finite H0 permutation representation which identifies the desired weight. The degree change $1 \rightarrow 0$ and coefficient change $C \rightarrow F_q$ are not bridged.

Exact obstruction scope. The stated weight-zero result is correctly scoped to Y_n . It cannot prove purity of the original theta spectrum, and a failure of a later t^{n+1} variant cannot disprove it. This family must not replace $g=2\xi$.

Concrete surviving derivation. Return to the original $T, A_{\tau}(T)$ and full actual zero packets. If using a finite-prime model, construct its integral lattice and actual specialization/trace map; otherwise calculate the characteristic-zero scaling bound directly. No unital map $C \rightarrow F_q$ can supply coefficient reduction, since $p \cdot 1$ is invertible in C and zero in F_q .

228.3.21 BF21 — A0961 §6

User U0024 (65b8bdef - be04 - 4ac6 - a7f1 - 9bb25a8f74f4); assistant A0961 (a7bbd422 - 311c - 4380 - ba6c - a4bdd61fce2f), chain 2032; passage line 4906. Priority 1.

The force is translated too; this does not claim different locations with an unchanged prescribed force.

Status: legitimate exact NS symmetry obstruction.

Requested calculation. Test the claim that the original cancellation can occur only at the origin and hence forces a unique arithmetic support.

Original objects and maps. $R_{\nu}(u,p) = \partial_{\tau}u + (u \cdot \nabla)u - \nu \Delta u + \nabla p$; T_b on the full (u,p,f) ; disjoint translates with all supports retained.

What was actually proved. The exact translation commutes with the full residual, divergence and spatial norms. Under the stated compact/disjoint supports the nonlinear cross terms vanish and squared energies add. It transports the marked center while translating the force.

What remained uncalculated. The answer does not determine all possible blow-up locations for a fixed original force. It also supplies no map from the NS triple to the unique arithmetic support or the theta cohomology.

Exact obstruction scope. It refutes universal coordinate-origin uniqueness for the translation-stable class of full triples, not a fixed-force uniqueness claim or the existence of an intrinsically marked support in another model.

Concrete surviving derivation. Retain the proved symmetry and specify whether the proposed arithmetic marking is invariant, equivariant, or tied to a fixed force. Any renewed NS transfer must carry the full force and residual; translating only the velocity cannot serve as that map.

228.3.22 BF22 — A0961 §7

User U0024 (65b8bdef - be04 - 4ac6 - a7f1 - 9bb25a8f74f4); assistant A0961 (a7bbd422 - 311c - 4380 - ba6c - a4bdd61fce2f), chain 2032; passage line 5010. Priority 0.

Unipotent behavior organizes the boundary grades; it does not erase them or fix their center by itself.

Status: completed arbitrary-length block calculation; literal action comparison missing.

Requested calculation. Relate unipotent support and the complete higher-direction ladder to the arithmetic weight needed in the proposed proof.

Original objects and maps. $Nv_0=0, Nv_j=v_{\{j-1\}}, Fv_j=cq^jv_j$; $FN F^{(-1)} = q^{(-1)}N$; grade $2j-d$, center $\gamma+d$. Original $\tau_{\text{Base}}(40)$ instead has $Nz^j = z^{j+1}, U_a = a^{\rho} \exp(\log(a)N)$.

What was actually proved. The answer retains all block lengths $d+1$, the actual monodromy twist and the free center scalar c . Multiplication of F by q leaves monodromy and relative filtration unchanged while shifting absolute weights; hence this monodromy data alone does not fix the center.

What remained uncalculated. No map from the original theta action or NS correction operator to this F, N pair is calculated. In particular the original scaling commutes with its jet N , whereas this F scales N by $q^{(-1)}$.

Exact obstruction scope. The result obstructs an inference of absolute weight from unipotence alone. It is not a broad obstruction to adding the missing arithmetic dynamics or to the original tau cohomology.

Concrete surviving derivation. Calculate the exact full-jet comparison: $U_a N U_a^{(-1)} = N$. A simultaneous intertwiner with the displayed Deligne pair kills N . On the original germ $g = u(z)z^m$, the separately defined $C_q f(z) = f(z/q)$ retains the unit ratio $q^{(-m)}u(z/q)/u(z)$, gives $C_q N C_q^{(-1)} = q^{(-1)}N$ and $C_q U_a C_q^{(-1)} = a^{\rho(1-1/q)} U_{a^{1/q}}$. This is a surviving semidirect comparison, not an arithmetic weight theorem.

228.3.23 BF23 — A0963 closing progress node

User U0028 (9d5e82ea - d670 - 4145 - b5d3 - c22d8948133e); assistant A0963 (6c82ae69 - e025 - 419b - ac36 - a20d2cf62f4c), chain 2036; passage line 5036. Priority 0.

I'm continuing the global construction

Status: open segment boundary; continuation owned by later transcript.

Requested calculation. Continue mapping all original split-zero and Spec Z objects into the analogous Weil II proof; do the work before judging the program.

Original objects and maps. The user explicitly selects the privileged absolute base, the whole supported spectrum above it, original arithmetic preservation, and the analogue of the proof rather than the literal finite-field theorem.

What was actually proved. This node states the intended next work on sheaves, duality and arithmetic trace. It contains no completed calculation.

What remained uncalculated. The substantive answer to U0028 lies beyond this assigned segment. It is not legitimate to call the absence of that answer inside the segment an abandonment of the full session.

Exact obstruction scope. No obstruction or program verdict occurs in this closing node.

Concrete surviving derivation. Carry U0028 as an active request into the following segment. The current continuation must start from the later Tau_Base construction rather than restart the source-discovery and finite-coefficient detour.

228.4 Exact continuation route already supplied by the later source

The absolute base is the pointed blueprint $F_{\{1, \tau\}} = \{\tau, 1\}$, with structural map induced by j_{math_R} , not the Boolean localization at the generic stalk. The original four-point upper-set space has $+\langle \eta \rangle \sigma$ and $-\langle \eta \rangle \sigma$. Its fibre category computes $A_{\tau}(F) = [F_+ \oplus F_- \rightarrow F_{\eta}]$ with differential $r_+ - r_-$. The separate $\text{res}_{\sigma}: A_{\tau}(F) \rightarrow F_{\sigma}$ has two degree-zero representatives whose difference is $r_{\{\eta \sigma\}}(r_+ - r_-)$, retaining the chain homotopy.

For the original theta sheaf T , the complex is $[V \oplus V \rightarrow B]$, $d(\varphi, \psi) = \Theta(\varphi - \hat{\psi})$. Its H_0 is V by $\psi \mapsto (\hat{\psi}, \psi)$; $H_1 = Q = B/\Theta V$. Every nonempty mask has the same H_1 coefficient Q ; single-leg H_0 is zero while joint-mask H_0 is V . The split lift retains tau separately from each labelled supported zero. The derived comparison with σ does not delete these classes.

$K_{\tau}(W)=[I_{\eta}(W)\rightarrow I_{+}(W)\oplus I_{-}(W)]$ in degrees $-1,0$ has $(+res,-res)$, and is quasi-isomorphic to $S_{\eta}W[1]$. The original source action is $(\varphi,\psi)\mapsto(U_a\varphi,aU_{\{1/a\}}\psi)$, with $U_aF(x)=F(x/a)$; both legs and the factor a are required. The L1 moment representation supplies the twist in the dual action.

The actual finite quotient $Q\rightarrow A_Z$ retains $g=2\xi$ and all local multiplicities. The residue form with reflected-germ signs $(-1)^j$ gives the injective map of the conjugate full packet into $\text{Hom}(Q,L1)=H^*(-1)\text{RHom}(T,K_{\tau}(L1))$. The trace contraction uses multiplication by the actual g' and is not the perfect residue form. Product charts retain every Koszul sign and yield $Q^{\wedge}\otimes r$ and $S_{(\eta,\dots,\eta)}W[r]$. These constructions are completed mathematics; the arithmetic growth estimate is the remaining calculation, not an assumed theorem.

228.5 Prioritized redo instructions for integration

1. Resume those actual A_{τ} and K_{τ} objects. Carry all masks and maps; do not replace the pushforward by σ restriction, the origin ladder by repeated adjunction, or the arithmetic quotient by the finite t^{n+1} coefficient example.
2. Complete the derived restriction fibre/adjoint comparison and its induced supported cohomology maps. Every supported class that maps to a labelled zero must retain its exact source fibre and kernel contribution.
3. Put the full original Mellin-jet injection, raw residue orientation, trace contraction and actual two-leg scaling into that comparison. Do the analytic theta calculation, including endpoint terms and gamma arithmetic, that a tensor estimate would require.
4. Preserve every legitimate obstruction above at its proved scope. In particular the non-arithmetic quartet test, smooth curved filling and finite H_0 purity result cannot determine the fate of the original theta H_1 program.
5. If comparing full jets to Deligne monodromy, calculate the exact action relation recorded in BF22. A literal identification silently kills nilpotents; the semidirect coordinate-dilation comparison survives with its stated extra action and original unit.
6. Carry U0028 into the following transcript segment. Do not count this segment boundary as an abandoned answer, and do not repeat source-reading apologies after the relevant source route has been acquired.

228.6 Completed tractable calculation delivered to root

BF22 isolates an algebraic calculation on the original finite jet packet, rather than a new polynomial family. The original U_a commutes with multiplication by z . The separately constructed $C_q:f(z)\mapsto f(z/q)$ respects the exact ideal (g) because $g(z/q)/g(z)=q^{(-m)}u(z/q)/u(z)$ is a unit. It is invertible, with $C_q^{(-1)}f(z)=f(qz)$. Thus $C_qNC_q^{(-1)}=q^{(-1)}N$ and $C_qU_aC_q^{(-1)}=a^{\rho(1-1/q)}U_{a^{1/q}}$. The comparison retains the same ρ , z , m and unit u . It does not identify C_q with the original arithmetic action.

The complete new proof is GERM_GRADING_AND_THETA_LIFT.md, with standalone TeX and an inclusion fragment alongside it. It also computes the full reflected residue Jacobian and adjoint, preserves the exact g -prime contraction, constructs an explicit compact-support Mellin right inverse by an invertible Gram matrix, and lifts C_q to a continuous automorphism $L_q=I+S(C_q-I)_Z$ of the ORIGINAL two-leg theta sheaf, with uniform forward seminorm bounds for $q\geq 1$. The lift fixes Theta V and all masks and has a proved comparison into the existing K_{τ} dual. Its noncanonical section-dependence and inverse growth are recorded. GERM_PAIRING_CHECK.md is an independent full algebra review; its eight exact fixtures and mutation controls are auxiliary verification.

No shared master, remote service or Lean process was changed by this lane.

229 Typed route: U0029–U0040

230 Passage-by-passage typed-route audit: U0029–U0040

This is a mathematical continuation audit of the full assigned transcript segment. It makes no inference about the model’s motives. Exact short passages and UUIDs below are checked against the final source extraction. Later corrections are part of each finding.

230.1 Coverage and source provenance

The assigned segment contains 12 user turns and 22 assistant turns, including progress replies, in exact retained order. All were read. Final extraction: 159,410 bytes, SHA-256 `a6574b90b21cd3e46533cc9aee7dc3ef8874dfa7a0860df93cc458c1bf1500e9`. The first full reading used the pre-newline-repair hash recorded in the JSON; root subsequently removed doubled CR/LF translation without changing message text. Quotes were revalidated after that repair.

The complete primary Tau Base NOTE.md (SHA-256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`) and complete canonical source route (SHA-256 `a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74`) were read. Dependencies named in those sources are not thereby claimed as freshly read in full. The companion source-map verifier separately read `support_diagrams.tex` in full and supplied its proofs and source pins.

The ledger’s ‘actual result’ states the scope of the displayed argument, independently checking its elementary typed algebra. This audit does not claim a fresh independent verification of all cited external books, every historical package, or historical test counts. Root owns the separate analytic and replay lanes.

230.2 Controlling continuation

The current primary construction is the structural blueprint map $F1, \tau \rightarrow B_R$ and its spectrum map to $b\tau$; the four-point upper-set chart; $A\tau = R\Gamma(P, -)$; the distinct derived restriction res_σ ; the complete supported H^1 fibres carrying Q ; $K\tau \simeq S_\eta W[1]$; the full-jet map $Q \rightarrow A_Z$; the residue injection into $R\text{Hom}(T, K\tau(L1))$; and actual tensor powers $Q \otimes r$. They are equations (1)–(50) of the primary source, not missing hypotheticalal objects.

The earlier transcript predates that completed primary note. Its missing-arrow statements must be updated using the later exact constructions rather than repeated as the present state. In particular Boolean generic-stalk restriction is a proved comparison with $A\tau$, not its definition or replacement.

An additional typed comparison is now completed: the finite-quotient argument alone places the socle eigenline in $A\rho$, but original `arithmetic_input.tex` (A10)–(A11) already supplies $\text{sigma} = qR_ref: E_h \rightarrow Q[h(D)]$, with $Jbar_Z$ $\text{sigma} = \text{id}$ and the complete unit $\text{epsilon} = j_h(h/g)$. The proved finite cocycle $U_aR_ref - R_ref$ $a^A = \text{Theta } c_a$ establishes equivariance for every positive a . The full source lift is therefore $L\rho = \text{sigma } \text{iota } \rho$ $t\rho: C \rightarrow Q$, and its tensor is a nonzero equivariant injection into $Q \otimes r$. Section 8 of the companion source-map proof gives the complete original construction, source estimates, retained boundary, full jet insertion, explicit left inverse, and tensor homotopy signs. The earlier quotient argument’s limited scope is retained; the source eigenline lift is no longer described as missing. The upper weight estimate remains a different calculation.

230.3 Every user-response episode

230.3.1 U0029 — 53352f69-6c38-4cab-a6aa-c225647ebe9a

A1007 announces the concrete sheaf lift; A1021 performs multiplication, gluing, support-prime restriction, nonsplitting, and the supported-null-cycle construction. T01 records its exact scope.

230.3.2 U0030 — 2e3cac15-6cbf-4fbd-b826-ceed04867399

A1022/A1032 announce the adelic reduction and retention of global norm errors; A1070 constructs HN/spectral filtration lifts, Rees defects, and slope diagnostics. The arithmetic theta specialization remains missing (T02–T04).

230.3.3 U0031 — fd1d4ef4-537b-4bca-8d5b-0cb82a441737

A1071/A1098/A1106 announce the Rees amplification and trace work; A1140 supplies the uniform-family formulas and explicitly admits the absent global cohomological identification. T05–T06 preserve this limit.

230.3.4 U0032 — f4609209-f2cd-4dbc-8ec2-a8995aedc998

A1142/A1155 explicitly turn to the actual theta source; A1167 supplies the real extension-by-zero, raw analytic realization, smooth full-jet right inverse, and quotient Gram metric (T07–T09). This partly repairs U0031.

230.3.5 U0033 — 622780af-97a6-4c56-987c-135df8409461

A1169 promises original-congruence propagation; A1206 supplies the internal coequalizer, relation-labelled quotients, transition kernels, actual differentiated theta endpoint families, and support-preserving operators (T10–T11).

230.3.6 U0034 — b5ee7e11-a089-4433-af7a-ed73bbc5bcf

A1210 announces reconciliation with the spectral paper; A1235 changes the primary target to unshifted Weil positivity, computes its exact shift correction, quotient correspondence, and residual/gap comparison, then leaves the central analytic term unestimated (T12–T15).

230.3.7 U0035 — 7b9f7612-3a57-4687-9bfb-39d1d933baf8

This action-continuity correction has no separate substantive assistant reply before U0036. It jointly controls A1280, which acknowledges and repairs the program displacement (T16).

230.3.8 U0036 — d6542f7d-b917-4de5-b35d-07062f16e9c1

A1280 restores cohomology/action/Lefschetz order, constructs correspondence saturation and full-jet traces, and retains moment endpoints and the explicit-formula terms. It explicitly leaves infinite cohomological trace and geometric endpoint-degree identifications unproved (T17–T18).

230.3.9 U0037 — 8b2110e7-195b-42d0-8c86-7277004e35fd

This demand to apply the program is followed by U0038 before a substantive reply. A1304 responds to both with actual mixed-source endpoint-image and residue calculations; T19 records the result and overstatement.

230.3.10 U0038 — 4ab2ff8f-915a-417d-876f-7294ed662b24

A1304 computes all masks, source-induced c_m , endpoint duality, and the J_g trace contraction. Its statement that Deligne had been applied is narrower in its own caveat and later corrected in A1361/A1427 (T20–T21).

230.3.11 U0039 — 71c1d5ee-668e-4eea-ac21-d844702248ea

A1306 announces the requested audit; A1361 supplies concrete raw-realization, Frobenius, cohomological-degree, residue-orientation and checker corrections, and preserves their limited scope (T22–T26).

230.3.12 U0040 — a10456e9-7f72-4c81-bc00-118fcd575269

A1362 announces source/F1/Deligne checking; A1427 supplies actual blueprint/prime/completion maps, proves raw-kernel nonvanishing, constructs balanced realization and finite-jet isomorphisms, and accurately limits its Deligne application status (T27–T30).

230.4 Exact passage ledger**230.4.1 T01: completed exact replacement; no obstruction to original program**

User: U0029 (53352f69-6c38-4cab-a6aa-c225647ebe9a, chain 2057). **Reply:** A1021 (cf9ad54a-2906-4d48-9cbd-b9e9677bb362, chain 2138).

Its failure to admit that natural section is information about the extension itself. It is not a reason to discard the extension.

Requested calculation. When a map fails, construct the typed relation and strongest surviving replacement.

Original objects and maps. $H_G(Q) \rightarrow H_Q$, archimedean ℓ_1 -bounded Γ -objects, split sheaf pull-back, added support prime σ .

Actual result and proof scope. The fold on $(1/2, -1/2)$ obstructs a pointed natural section. The pullback sheaf itself exists, glues, retains τ/e , and maps to the original arithmetic sheaf.

Left uncalculated. This episode defines a supported-null cycle object but computes neither its higher derived cohomology nor a weight bound.

Exact obstruction scope. Only a section of this particular pointed natural projection is excluded. The sheaf extension survives.

Later completion or correction. A1206 computes the internal coequalizer; A1280 installs correspondence action; the later primary τ -base note computes $A\tau$ and $K\tau$.

Concrete surviving derivation. Use the existing support-reconstructed theta sheaf (primary (14)–(25)) and its actual derived pushforward, retaining the pullback rather than trying to split it.

230.4.2 T02: correct general construction; arithmetic specialization left uncalculated

User: U0030 (2e3cac15-6cbf-4fbd-b826-ceed04867399, chain 2139). **Reply:** A1070 (86b375dc-bd7b-494b-b050-0d8680eadae0, chain 2308).

For a compatible supported graded algebra

Requested calculation. Push the richer supported adelic construction toward Weil II/RH.

Original objects and maps. Adelic bundle E with actual place norms, HN filtration, spectral norm, supported graded algebra $M_\bullet \rightarrow E_\bullet$, Rees modules.

Actual result and proof scope. Pullback filtrations retain the full amplitude-zero fibre; the two norm assignments are connected by the displayed error bound; multiplicativity is proved for the stated graded input.

Left uncalculated. No actual adelic norm family or graded comparison is supplied for the original theta complex Q . The phrase introduces an input instead of deriving that input from the source.

Exact obstruction scope. HN and spectral norms may be different; the recorded inequalities do not authorize replacing the original metric or transferring their bounds to Q .

Later completion or correction. A1167 supplies a genuine finite theta-quotient Gram metric but explicitly leaves its adelic realization unbuilt; A1361 audits the missing HN/spectral intertwiner.

Concrete surviving derivation. Attach each chosen norm to a concrete finite quotient of $A_T(T)$, retain its exact comparison with the original theta measure, and calculate the corresponding map height before using Chen–Moriwaki.

230.4.3 T03: legitimate measured obstruction; vanishing unproved for arithmetic map

User: U0030 (2e3cac15-6cbf-4fbd-b826-ceed04867399, chain 2139). **Reply:** A1070 (86b375dc-bd7b-494b-b050-0d8680eadae0, chain 2308).

It replaces the unproved assertion that every such comparison is automatically strict.

Requested calculation. Construct the supported comparison relevant to purity.

Original objects and maps. $c: (V, F) \rightarrow (W, G)$, $I = \text{im } c$, quotient/subspace filtrations $F_q \subset F_s$, $D_c = \text{Rees}_{F_s}(I) / \text{Rees}_{F_q}(I)$.

Actual result and proof scope. For integer jumps D_c is finite T -power torsion and $\text{length}(D_c) = \sum \mu - \sum \lambda$; the special-fibre Tor sequence retains its kernel and cokernel. For the actual spectral-weight filtration, an equivariant map is strict by generalized eigenspaces.

Left uncalculated. The transcript has not identified this c and filtration with the arithmetic theta comparison or proved their strictness.

Exact obstruction scope. The rank-one source-weight 0/target-weight 2 example disproves automatic strictness of arbitrary filtered maps only.

Later completion or correction. A1140 amplifies this exact defect; A1361 explicitly retains separate HN and spectral filtrations.

Concrete surviving derivation. Compute the induced map on the finite Mellin-jet quotients of the original cohomology together with its actual source/target filtration. If the filtration comes from scaling, prove equivariance using the full nilpotent operator before computing D_c .

230.4.4 T04: complete diagnostic; different example family is not arithmetic proof

User: U0030 (2e3cac15-6cbf-4fbd-b826-ceed04867399, chain 2139). **Reply:** A1070 (86b375dc-bd7b-494b-b050-0d8680eadae0, chain 2308).

The appropriate replacement for a mean-only invariant is now explicit

Requested calculation. Obtain the purity information of the arithmetic construction.

Original objects and maps. $C_V(c) = \sum \max(\lambda_j - c, 0)$, endomorphism norm, weighted Gauss metric on $H^0(P^1, \mathcal{O}(n))$.

Actual result and proof scope. Distributional second derivative recovers all slopes; $\deg + \text{End}(E_n) = |\alpha - \beta| n(n+1)(n+2)/6$ while the Hilbert–Samuel leading coefficient is $\alpha + \beta$.

Left uncalculated. No map identifies this chosen P^1 metric family with the original theta cohomology or gives its arithmetic weight distribution.

Exact obstruction scope. The example shows only that a mean invariant does not determine all weights. It is not a counterexample to τ -base purity.

Later completion or correction. A1361 preserves these calculations but removes any unsupported transfer to the arithmetic comparison.

Concrete surviving derivation. Retain the diagnostic formulas, apply them only after the concrete arithmetic filtration and operator have been constructed, and do not substitute the P^1 family for T .

230.4.5 T05: amplification completed; requested estimate transfer still missing

User: U0031 (fd1d4ef4-537b-4bca-8d5b-0cb82a441737, chain 2309). **Reply:** A1140 (4096bcde-e267-4ebe-8148-43bb2424da1a, chain 2445).

Their casting construction supplies the relevant kind of quantitative estimate

Requested calculation. Continue the calculation through tensor powers, duality, and the arithmetic trace.

Original objects and maps. $LQ \subset LP$, D_c , $\text{Sym}^m(I)$, actual Smith exponents.

Actual result and proof scope. $\text{length}(D_{\text{Sym}^m}) = \text{binom}(m+r-1, r) \text{length}(D_c)$; any bound on this very family sublinear per tensor degree would force $D_c = 0$.

Left uncalculated. No bound for this very comparison family is calculated. A sufficient inequality remains an unproved input.

Exact obstruction scope. The exact amplification law is valid. The law alone gives no vanishing and no weight estimate on $A_\tau(T)$.

Later completion or correction. The primary note (46)–(50) later computes actual Q tensor powers and their growth defect; that is the surviving original family.

Concrete surviving derivation. Apply tensor amplification to $Q \otimes r$ and $K_{\tau, r}$ from primary (47)–(48), and derive an analytic upper estimate from the original theta relations rather than import the casting bound by resemblance.

230.4.6 T06: explicit substitute family; gap acknowledged and later partly repaired

User: U0031 (fd1d4ef4-537b-4bca-8d5b-0cb82a441737, chain 2309). **Reply:** A1140 (4096bcde-e267-4ebe-8148-43bb2424da1a, chain 2445).

The uniform comparison family is constructed and its scalar trace identified.

Requested calculation. Connect the actual supported comparison to arithmetic cohomology and trace.

Original objects and maps. Uniform lattices $k[T] \otimes V \subset T^{\wedge}(-a)k[T] \otimes V$, shifted completed function $\Lambda(s - a\epsilon)/\Lambda(s)$.

Actual result and proof scope. The grading derivative yields the defect trace; the chosen uniform family produces $-a\Lambda'/\Lambda$, with discriminant, primes, gamma factors and poles retained.

Left uncalculated. This construction chooses a uniform shift rather than deriving the global cohomological comparison; it does not compute the theta relation kernel or its weights.

Exact obstruction scope. It proves the scalar trace of that uniform family. Equality of scalar traces is not an isomorphism of the original comparison complexes.

Later completion or correction. U0032 explicitly says ‘So no map yet’; A1167 constructs the actual theta/extension-by-zero/Mellin map. Do not retain T06 as an entirely unremedied absence of any map.

Concrete surviving derivation. Use the actual source-induced arrows of A1167 and primary (19),(39),(43). Preserve any uniform-family calculation only with its typed comparison to those objects.

230.4.7 T07: actual extension-by-zero calculation completed

User: U0032 (f4609209 - f2cd - 4dbc - 8ec2 - a8995aedc998, chain 2446). **Reply:** A1167 (157eecd - 7f96 - 4ec5 - 96bd - 6dd376276ea6, chain 2520).

is the identity on

Requested calculation. Construct the missing actual map.

Original objects and maps. $C_+ = [V \rightarrow \Theta B]$, $C = [V \oplus V \rightarrow \Theta(\varphi - F\psi) B]$, $c_0(\varphi) = (\varphi, 0)$, $c_1 = \text{identity}$.

Actual result and proof scope. The chart extension-by-zero map induces identity on Q ; the source coordinate map $(\varphi, \psi) \mapsto (\varphi - F\psi, \psi)$ splits the joint complex as C_+ plus V in degree zero.

Left uncalculated. The displayed chart map is not identified with global arithmetic $f! \rightarrow f^*$. Its trivial H_1 comparison supplies no weight estimates.

Exact obstruction scope. An exact chart calculation, not the missing global geometric purity theorem. The nonzero H_0 joint summand must remain when passing back to $A_\tau(T)$.

Later completion or correction. Primary (18)–(22) incorporates this same joint theta complex in the four-point τ -base model.

Concrete surviving derivation. Carry the explicit joint complex through A_τ and K_τ ; retain $H_0 \cong V$ and the endpoint moment representations without renaming the latter as geometric H_0/H_2 .

230.4.8 T08: kernel formula correct; cohomological effect initially under-carried

User: U0032 (f4609209 - f2cd - 4dbc - 8ec2 - a8995aedc998, chain 2446). **Reply:** A1167 (157eecd - 7f96 - 4ec5 - 96bd - 6dd376276ea6, chain 2520).

Its kernel is calculated:

Requested calculation. Transport the actual theta quotient into the arithmetic divisor.

Original objects and maps. $\alpha: 0 \otimes V \rightarrow 0$, $\beta: 0 \otimes B \rightarrow 0$, $\delta = 1 \otimes \Theta$, $\beta\delta = g\alpha$, $g = 2\xi$.

Actual result and proof scope. The sheafified image ideal is (g) , α and β are sheaf-surjective, and raw realization $0 \otimes Q \rightarrow 0/(g)$ has kernel $\ker \beta / \delta \ker \alpha$.

Left uncalculated. The formula is not a proof that the kernel vanishes, and the raw tensor map is not shown to preserve full cohomology. ‘Image ideal’ concerns analytic O -linear sheaf generation, not equality of bare global Mellin images.

Exact obstruction scope. Only the quotient and its specified kernel are proved. Neither finite-jet surjectivity nor the generator φ^* erases that kernel.

Later completion or correction. A1361 gives the kernel complex and explicit raw syzygy; A1427 proves the raw kernel nonzero and supplies operator-balanced realization.

Concrete surviving derivation. Retain both raw and balanced realization maps and calculate their kernels as modules and complexes. The attached completed calculations give all-order syzygies and explicit free submodules of the raw kernel.

230.4.9 T09: finite quotient metric fully calculated; global metric bound not calculated

User: U0032 (f4609209-f2cd-4dbc-8ec2-a8995aedc998, chain 2446). **Reply:** A1167 (157eecd-f7f96-4ec5-96bd-6dd376276ea6, chain 2520).

The infimum over smooth compactly supported exact lifts is nevertheless the same

Requested calculation. Compute weight control on the actual source.

Original objects and maps. $H_R = L_2((e^{-R}, e^R), dx)$, finite Mellin-jet map, Gram matrix A, smooth bump right inverse.

Actual result and proof scope. All finite jets are surjective; A is positive definite; least-lift squared norm is $v^* A^{-1} v$; evaluation norm growth is $|\text{Re } \rho - 1/2|$. A correction by the smooth right inverse proves the same exact-lift infimum for C_c^∞ .

Left uncalculated. No global arithmetic comparison estimate bounds that growth or supplies an adelic metric inducing this exact norm.

Exact obstruction scope. A positive quotient metric always exists on finite observations; it is not the Weil trace form and does not establish spectral purity.

Later completion or correction. A1235 retains the unshifted form correction; A1361 warns against HN/spectral transfer.

Concrete surviving derivation. Keep dx , the $x^{(1/2)}$ map and all Gram coefficients. Estimate the actual theta relations in this norm, including boundary terms, rather than change measure or metric to force boundedness.

230.4.10 T10: requested internal quotient and transport kernels completed

User: U0033 (622780af-97a6-4c56-987c-135df8409461, chain 2521). **Reply:** A1206 (ea7fad65-aa59-4169-8bbd-5e73a631d3ae, chain 2613).

The quotient can now be taken in the original split-zero semimodule category

Requested calculation. Propagate the original split-zero quotient through every typed map.

Original objects and maps. $f=(\varphi, f_l): M \rightarrow N$, $B_k(f) = \sum_{\{\varphi(l) \leq k\}} \rho f_l(M_l)$, $\text{coeq}(f, \varepsilon f)$.

Actual result and proof scope. The original congruence yields coproduct $N_k/B_k(f)$, universal against arbitrary split-linear targets; finite relation labels give Q_W and transport kernel $\Theta W'/\Theta W \cong W'/W$.

Left uncalculated. Retaining these internal fibres does not itself calculate weight estimates, but that was not claimed as the new theorem.

Exact obstruction scope. This is substantive propagation rather than a label appended to an already collapsed ordinary quotient.

Later completion or correction. Canonical source E D6–D8 and primary (24)–(25) retain this result with categorical-kernel/all-support distinction.

Concrete surviving derivation. Use the same universal quotient for cohomology and every later realization, keeping bottom fibres and support maps explicit. Do not redo already proved quotient algebra as the purported missing purity calculation.

230.4.11 T11: source-induced endpoint-jet family completed; scope must remain local

User: U0033 (622780af - 97a6 - 4c56 - 987c - 135df8409461, chain 2521). **Reply:** A1206 (ea7fad65 - aa59 - 4169 - 8bbd - 5e73a631d3ae, chain 2613).

origin information: it consists of

Requested calculation. Track higher supported relation data rather than discard it.

Original objects and maps. $\varphi_j = D^j \varphi^*$, W_m, N , $I_W = (gs^m)$, $D_m = O/(gs^m)$.

Actual result and proof scope. The endpoint kernel $O/(s^m)$ embeds by $\times g$; $g(0)=1$ is computed from the actual theta boundary; Bezout coefficients give the split decomposition over a common Boolean support. Fourier supplies $s(1-s)$ with all signs.

Left uncalculated. No purity bound follows from these additional endpoint jets, and the finite subspaces are not silently D-invariant.

Exact obstruction scope. The extra endpoint factors are caused by restricting theta relations. They are not new arithmetic zeros of ξ .

Later completion or correction. A1304 builds the reflected comparison; A1361 corrects its cohomological placement; A1427 enlarges finite labels to D-saturation with its kernel.

Concrete surviving derivation. Keep the endpoint exact sequences and Fourier transport, then use the actual full two-leg source when computing A_τ . The attached all-m calculation identifies the exact realization syzygy rather than introduce another polynomial family.

230.4.12 T12: scope displacement, explicitly conceded later

User: U0034 (b5ee7e11 - a089 - 4433 - af7a - ede73bbc5bcf, chain 2614). **Reply:** A1235 (ca9499fc - 3b1b - 49a8 - 8705 - 047fc4fec20e, chain 2661).

The programme needs to return to one RH-level target: positivity of the unshifted Weil form

Requested calculation. Return to Weil II on the user's program and clarify the route.

Original objects and maps. Finite spectral-triple unshifted form QW , ε shift, finite ground-line quotient, theta truncation.

Actual result and proof scope. The reply gives genuine finite comparison formulas and sufficient lower-bound criteria.

Left uncalculated. It promotes a downstream RH-equivalent criterion to the primary task instead of completing the requested cohomology/Lefschetz/Deligne construction.

Exact obstruction scope. The positivity criterion is mathematically relevant but does not replace the requested construction over the τ -base.

Later completion or correction. A1280 explicitly states that the lower bound 'should not have displaced' cohomology, action, and duality.

Concrete surviving derivation. Resume primary $A_\tau = R\Gamma$, K_τ , Mellin-jet injection and actual tensor powers. Use positivity as a consequence to be proved for their trace, not a generic replacement deliverable.

230.4.13 T13: legitimate exact descent obstruction

User: U0034 (b5ee7e11-a089-4433-af7a-ed73bbc5bcf, chain 2614). **Reply:** A1235 (ca9499fc-3b1b-49a8-8705-047fc4fec20e, chain 2661).

the unshifted form descends through this particular quotient precisely when

Requested calculation. Carry the original quotient into the cited spectral construction.

Original objects and maps. $A, \varepsilon = \min$ eigenvalue, $B = A - \varepsilon I$, ground line $L = C_V$, quotient q .

Actual result and proof scope. $QW(f, g) = \text{Bbar}(qf, qg) + \varepsilon \langle f, g \rangle$. The unshifted form descends through E/L iff $\varepsilon = 0$; the shifted form descends, and its distribution is $\Psi - \varepsilon \delta 1$.

Left uncalculated. The missing estimate is control of the original ε term, not proof that the shifted form is positive.

Exact obstruction scope. This excludes descent through that ground-line quotient for $\varepsilon \neq 0$ only. It does not exclude all supported comparisons or all τ -base duality.

Later completion or correction. A1361 retains this exact correction instead of a blanket rejection of the finite model.

Concrete surviving derivation. Retain the direct form identity or the two terms as structured data and calculate their exact pullback along the original theta map. Do not identify the shifted positive form with the raw residue/Weil trace pairing.

230.4.14 T14: typed correspondence completed; estimate abandoned at explicit term

User: U0034 (b5ee7e11-a089-4433-af7a-ed73bbc5bcf, chain 2614). **Reply:** A1235 (ca9499fc-3b1b-49a8-8705-047fc4fec20e, chain 2661).

Those kernels are the retained discrepancy

Requested calculation. Integrate arithmetic theta relations with the finite spectral quotient.

Original objects and maps. $T = \Pi_{NP} \lambda C$, $R = T \ominus W$, $L = C_V$, common quotient $E/(R+L)$, linear relation $C_{W, \lambda, N}$.

Actual result and proof scope. Both quotient kernels $L/(L \cap R)$, $R/(R \cap L)$ are explicit; the latter is the multivalued output fibre of the theta-to-ground-line relation.

Left uncalculated. R is not computed for all original W , and the residual-to-gap term remains unestimated uniformly.

Exact obstruction scope. Failure of a single-valued map to E/L is measured by $R/(R \cap L)$; the exact correspondence survives.

Later completion or correction. A1280 restores the cohomological route; the raw finite relation can remain a diagnostic.

Concrete surviving derivation. Compute R and the unshifted form on actual theta-generated finite subspaces, retaining its kernel. Any residual estimate must include the same support relations and all endpoint terms.

230.4.15 T15: explicit incomplete calculation, not a finished bound

User: U0034 (b5ee7e11-a089-4433-af7a-ed73bbc5bcf, chain 2614). **Reply:** A1235 (ca9499fc-3b1b-49a8-8705-047fc4fec20e, chain 2661).

The uniform residual-to-gap estimate is the nontrivial term left

Requested calculation. Use Weil II to make actual progress on the arithmetic program.

Original objects and maps. $u = \text{IIP } K$, v lowest eigenvector; spectral gap $g_{\lambda, N}$, theta tail and Fourier projection tail.

Actual result and proof scope. A concrete complex-strip error bound is derived with all three terms: theta tail, Fourier projection tail, and residual/gap.

Left uncalculated. No uniform estimate for $\|(A - \epsilon I)u\|/\text{gap}$ is carried out; small Rayleigh quotient controls only the upper side of the minimum.

Exact obstruction scope. The inequality locates an analytic problem; it neither proves convergence nor proves the desired lower bound.

Later completion or correction. U0035 demands calculations rather than proposals; U0036 redirects to traces and cohomology; A1280 accepts the correction.

Concrete surviving derivation. For the exact original theta source, calculate residual and boundary contributions before invoking a gap argument. Root's original-source analytic lane should supply that computation rather than this audit duplicate it.

230.4.16 T16: instruction incorporated after consecutive user steering

User: U0035 (7b9f7612 - 3a57 - 4687 - 9bfb - 39d1d933baf8, chain 2662). **Reply:** A1280 (3c7dd1ba - 95eb - 4919 - b2db - efef2948ddd1, chain 2749).

The lower bound I named is a downstream criterion

Requested calculation. Do requested mathematics rather than merely say it should be done.

Original objects and maps. User U0035 followed by U0036; no substantive assistant response between them in this retained segment.

Actual result and proof scope. A1280 acknowledges the displacement and performs convolution-action/finite-trace/endpoint-action calculations.

Left uncalculated. Its global cohomological trace and purity remain incomplete; the acknowledgement does not complete them.

Exact obstruction scope. This user directive applies to continued mathematical work, not just wording or a redo prompt.

Later completion or correction. A1304 and A1427 execute further concrete algebraic repairs.

Concrete surviving derivation. Continue actual calculations on retained A_{τ}/K_{τ} ; supplementary redo instructions do not satisfy the instruction by themselves.

230.4.17 T17: actual action repaired by enlargement; change of label category must stay visible

User: U0036 (d6542f7d - b917 - 4de5 - b35d - 07062f16e9c1, chain 2679). **Reply:** A1280 (3c7dd1ba - 95eb - 4919 - b2db - efef2948ddd1, chain 2749).

its source relations must be invariant

Requested calculation. Construct cohomologies and Lefschetz traces.

Original objects and maps. $Q_W = B/\Theta W$, convolution H_h , $W^A = W + \text{span}\{h^*\varphi\}$.

Actual result and proof scope. Convolution commutes with Θ , saturation is idempotent and join-compatible, and the map to invariant relations has kernel $\Theta W^A/\Theta W$.

Left uncalculated. The new saturated label is usually not finite-dimensional. It is a transition into a larger invariant-subspace category, not an endomorphism of every original finite fibre.

Exact obstruction scope. Non-invariance obstructs only the claimed fixed- W endomorphism. The saturated action and moving-label map both survive.

Later completion or correction. A1361 explicitly corrects treating saturation as another finite label. Primary (34)–(35) supplies the full two-leg action without a label substitution.

Concrete surviving derivation. Use $(U_a, aU_1/a)$ on $V \oplus V$ and U_a on B . For finite subspaces keep either the moving label or the explicit saturation kernel, and retain the original absent-leg masks.

230.4.18 T18: finite trace and arithmetic distribution completed; global trace not established

User: U0036 (d6542f7d - b917 - 4de5 - b35d - 07062f16e9c1, chain 2679). **Reply:** A1280 (3c7dd1ba - 95eb - 4919 - b2db - efeb2948ddd1, chain 2749).

not asserted to be an ordinary trace of every infinite semimodule quotient

Requested calculation. Apply cohomological Lefschetz machinery to the original object.

Original objects and maps. Finite A_Z , multiplication by Mellin h , E_0/E_1 moment lines, spectral sum, prime and archimedean distribution.

Actual result and proof scope. The full-jet multiplication trace is $\sum \rho Mh(\rho)$; moment actions give $Mh(0)$, $Mh(1)$; compact smooth test decay gives the stated spectral distribution, using the cited explicit formula.

Left uncalculated. The reply does not prove that this distribution is an ordinary/nuclear trace on the full supported theta quotient or calculate a global realization kernel contribution. E_0/E_1 are not yet constructed as geometric cohomological degrees 0 and 2.

Exact obstruction scope. All finite traces may be correct while the infinite cohomological trace identification still requires a topology and exact trace theorem.

Later completion or correction. A1361 records this gap explicitly; primary (43) retains an algebraic dual map and deliberately avoids asserting continuous/global duality.

Concrete surviving derivation. Specify and prove continuity/nuclearity or a regularized trace construction on the original strong-Schwartz quotient, track the realization kernel, and preserve all prime, gamma, moment, and boundary terms.

230.4.19 T19: response to consecutive U0037/U0038; actual algebra executed, Deligne application overstated

User: U0037 (8b2110e7 - 195b - 42d0 - 8c86 - 7277004e35fd, chain 2750). **Reply:** A1304 (7fd7ecd5 - 0f3d - 4673 - 94dc - 9cda82bee9ed, chain 2801).

The concrete application is a

Requested calculation. Apply the requested machinery to the program.

Original objects and maps. Original mixed double $G(R)^2$, all three masks, $d_m = (gs^m p(a) - g(1-s)^m p(b))$, c_m .

Actual result and proof scope. The three support fibres, endpoint transport kernels, joint cycles, and analytic image are computed using explicit Bezout coefficients and actual differentiated theta inputs.

Left uncalculated. The analytic realization creates relations and changes cohomology; that effect is not carried through here. The reply does not construct the global Deligne comparison.

Exact obstruction scope. These are valid source-induced local algebraic calculations. They are not a proof of weight bounds for the global original construction.

Later completion or correction. A1361 identifies the raw syzygy and image-versus-cohomology issue; A1427 supplies operator balancing and corrects the claim of applying Deligne.

Concrete surviving derivation. Retain the d_m comparison as an analytic image of actual theta source subcomplexes, with the exact comparison cone. The attached all- m proof computes that map and its support faces.

230.4.20 T20: perfect residue duality calculated; transfer to right adjoint not yet present here

User: U0038 (4ab2ff8f-915a-417d-876f-7294ed662b24, chain 2753). **Reply:** A1304 (7fd7ecd5-0f3d-4673-94dc-9cda82bee9ed, chain 2801).

Now use the duality operation needed by the Deligne route

Requested calculation. Use both Deligne and split zero in the actual program.

Original objects and maps. $C_m \rightarrow O_m$, finite reflected packets, $\text{Ext}_1(-, R \text{ ds})$, $R_m, \mathbb{Z}$, L_1 action.

Actual result and proof scope. The residue pairing is perfect, $\text{im } c_m = (\ker c_m)^\perp$, endpoint jets have the displayed anti-diagonal signs, and covariance has factor a .

Left uncalculated. A local Ext residue pairing is not yet identified with the right adjoint of the program's global pushforward in this episode.

Exact obstruction scope. Perfect duality and weight-one covariance alone do not imply positivity. This is a limited obstruction, not program failure.

Later completion or correction. A1361 proves skew-Hermitian orientation and formal counterexample; the primary (26)–(43) later computes $K\tau$ and gives an explicit injection into its algebraic dual.

Concrete surviving derivation. Use primary (43) to place this finite residue duality in $\text{RHom}(T, K\tau(L_1))$; carry its continuous/topological status separately and calculate its arithmetic estimate.

230.4.21 T21: trace contraction completed; full nilpotents retained before contraction

User: U0038 (4ab2ff8f-915a-417d-876f-7294ed662b24, chain 2753). **Reply:** A1304 (7fd7ecd5-0f3d-4673-94dc-9cda82bee9ed, chain 2801).

specific morphism whose pairing is the actual finite Weil spectral pairing

Requested calculation. Connect the actual duality and arithmetic trace.

Original objects and maps. $J_g = [\times g']$, $P_m = [\times (1-s)^m g']$, raw residue R_Z , trace Q_Z .

Actual result and proof scope. $R_Z(f, J_g h) = \sum m \rho \text{ conjugate}(f(\iota \rho)) h(\rho) = \text{Tr } M_{f \uparrow h}$; at multiplicity r , J_g has kernel (z) and image (z^{r-1}) .

Left uncalculated. The finite trace contraction is not an estimate, and the radical does not prove supported jet fibres are absent.

Exact obstruction scope. Scalar trace forgets nilpotent off-diagonal data through an explicit endomorphism. The perfect residue pairing still sees it.

Later completion or correction. A1361 keeps the orientation and generator-change factors; primary (42)–(45) integrates this same operation.

Concrete surviving derivation. Carry R_Z and J_g as distinct typed maps into $K\tau$, preserving $g=2$ ξ and the full unit $u_\rho(z)$. Use the trace contraction only for the observable it computes.

230.4.22 T22: legitimate narrow obstruction, explicit scope retained

User: U0039 (71c1d5ee-668e-4eea-ac21-d844702248ea, chain 2802). **Reply:** A1361 (c9e551ee-298f-438b-9bdc-2e23c2c0613a, chain 2922).

It does not, by itself, prove that this final kernel is nonzero.

Requested calculation. Audit and repair roughly 25 turns without normalizing or terminating a lane.

Original objects and maps. Raw $\delta: O^2 \rightarrow O \otimes B$ from $b+ = \Theta D\varphi$, $b- = \Theta(1-D)\varphi$, β evaluation.

Actual result and proof scope. δ is injective while $\ker \beta\delta = O(1-s, s)$; $z = (1-s) \otimes b+ - s \otimes b-$ is nonzero and maps to zero. For the full source z lies in $\delta \ker \alpha$ and therefore does not prove the final H1 kernel nonzero.

Left uncalculated. The full raw H1 kernel remains unresolved at this point.

Exact obstruction scope. The obstruction excludes the raw restricted comparison being a quasi-isomorphism. It does not establish no other realization or no τ -base program.

Later completion or correction. A1427 proves raw H1 kernel nonzero by a different argument and identifies z as an operator-balancing relation.

Concrete surviving derivation. Keep both calculations and their distinct scopes. The attached all-order proof extends z to every $m \geq 1$ without changing masks or g .

230.4.23 T23: corrected cohomological placement

User: U0039 (71c1d5ee-668e-4eea-ac21-d844702248ea, chain 2802). **Reply:** A1361 (c9e551ee-298f-438b-9bdc-2e23c2c0613a, chain 2922).

is **the image**, not either cohomology group.

Requested calculation. Make all preceding work self-consistent.

Original objects and maps. Two-term $[C_m \rightarrow O_m]$, kernel $E_{0,m}$, image I , cokernel $E_{1,m}$.

Actual result and proof scope. $H_0 = E_{0,m}$, $H_1 = E_{1,m}$; I is the comparison image and cancels from this complex's supertrace. Dilation supertrace is $m(1-a)$.

Left uncalculated. No actual global comparison producing I as a middle image with both bounds is constructed in this episode.

Exact obstruction scope. A correct kernel-image-cokernel exact sequence cannot be relabelled as $H_1 = I$.

Later completion or correction. Primary (21)–(22) really has $H_1 = Q$ and (39) induces $Q \rightarrow A_Z$; this is the strongest later exact cohomological placement.

Concrete surviving derivation. Retain the two endpoint short exact sequences as data over A_τ , then use the actual Q cohomology for the spectral packet map. Never take the arithmetic trace of I as the supertrace of $[C_m \rightarrow O_m]$.

230.4.24 T24: legitimate formal counterexample; no failure verdict for original arithmetic program

User: U0039 (71c1d5ee-668e-4eea-ac21-d844702248ea, chain 2802). **Reply:** A1361 (c9e551ee-298f-438b-9bdc-2e23c2c0613a, chain 2922).

perfect duality and weight-one covariance alone do not force positivity

Requested calculation. Give exact scope and constructive successor for every obstruction.

Original objects and maps. Actual raw R_Z, J_g ; auxiliary $g_0=(s-1/4)(s-3/4)$, $U=\text{diag}(2,8)$.

Actual result and proof scope. The auxiliary model has R skew-Hermitian, $Q=RJ=[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}]$, $U*QU=16Q$ and negative value on $(1,-1)$.

Left uncalculated. It does not compute the additional arithmetic property of the actual theta image which would exclude such a pairing.

Exact obstruction scope. Only the universal implication ‘perfect residue duality plus covariance implies positivity’ fails. ξ , its theta source, and τ -base $A\tau/K\tau$ have not been disproved.

Later completion or correction. A1427 explicitly retains richer arithmetic geometry as the continuing route; primary (50) states the actual unproved tensor defect.

Concrete surviving derivation. Use the counterexample as a mutation control on any proposed proof, then calculate the original theta relation estimates. Do not replace ξ by this auxiliary polynomial as the main research object.

230.4.25 T25: actual supported discrepancy completed

User: U0039 (71c1d5ee-668e-4eea-ac21-d844702248ea, chain 2802). **Reply:** A1361 (c9e551ee-298f-438b-9bdc-2e23c2c0613a, chain 2922).

the two Frobenius operations differ

Requested calculation. Carry exact maps when operations cannot be identified.

Original objects and maps. Φ_q additive Frobenius lift and $P_q=x*q$ in $D_{A,n}$; $A=F_5, n=3, x=(1,1, \tau)$.

Actual result and proof scope. $\Phi_5(x)=(1,\tau,4)$, $x*5=(1,e,4)$; $\pi\Phi_q=F_q\pi=\pi P_q$ and (P_q, Φ_q) maps to the arithmetic fibre product in pointed multiplicative monoids.

Left uncalculated. No global Frobenius action on the richer arithmetic cohomology is built from that coefficient comparison.

Exact obstruction scope. The two supported operations differ; they remain explicitly related over the arithmetic quotient. Finite étale amplitude purity cannot be assigned to both.

Later completion or correction. A1427 supplies a root-cover coefficient square but carefully limits its geometric claims.

Concrete surviving derivation. Keep the fibre-product comparison when using finite-characteristic coefficients, while retaining the actual characteristic-zero scaling action on $A\tau$. Do not conflate the two.

230.4.26 T26: verification overstatement found and repaired

User: U0039 (71c1d5ee-668e-4eea-ac21-d844702248ea, chain 2802). **Reply:** A1361 (c9e551ee-298f-438b-9bdc-2e23c2c0613a, chain 2922).

identical normal and optimized output demonstrated equivalent checking was not justified

Requested calculation. Audit rigor and prior calculations.

Original objects and maps. Two prior checkers, 33 assert sites, normal/optimized executions.

Actual result and proof scope. The transcript reports hardened copies and deliberate failing guards in both modes.

Left uncalculated. Passing finite checks remains distinct from proof of the analytic/global claims; this audit has not independently rerun those old packages.

Exact obstruction scope. The defect concerns the optimized checker claim, not all mathematical formulas in those packages.

Later completion or correction. Root has an independent current-audit replay lane; this segment audit intentionally does not duplicate it.

Concrete surviving derivation. Preserve source hashes and use explicit exception/mutation controls for new computational checks; cite full written arguments for infinite analytic statements.

230.4.27 T27: requested F1 morphisms substantially supplied

User: U0040 (a10456e9-7f72-4c81-bc00-118fcd575269, chain 2923). **Reply:** A1427 (d7e752ce-cb39-4130-b089-b20333037990, chain 3056).

The full square is the object to retain

Requested calculation. Supply exact maps, audit Deligne reading, check globalization primality and F1 connection.

Original objects and maps. $P\tau \subset P0 \subset Pp$, p_R , Spec inclusions, $B_{bare} \rightarrow B_R$, contracted monoid ring, Boolean nonunital section.

Actual result and proof scope. The two primality statements, arithmetic closed subspace $V(e)$, distinct generic localizations, blueprint completion/recovery, nonunital Boolean retraction, and all-rank monomial automorphism sequence are explicitly connected.

Left uncalculated. No literal six-item specification was located, so certification against all six was not performed. This is an acquisition/specification limit, not evidence the F1 program fails.

Exact obstruction scope. Unital Boolean inclusion failure only requires retaining the nonunital idempotent retraction or genuine pointed blueprint base. It does not erase the absolute base.

Later completion or correction. Primary (1)–(8) constructs $F1, \tau$ and structural map to $b\tau$; canonical route proves $B \otimes_G(R)M \cong eM$ alongside, not instead of, $A\tau$.

Concrete surviving derivation. Use the actual structural blueprint morphism and cohomological pushforward. Never substitute the generic σ Boolean stalk for the τ -base or claim the source omitted (0).

230.4.28 T28: raw-kernel nonvanishing completed; balanced successor only partly calculated

User: U0040 (a10456e9-7f72-4c81-bc00-118fcd575269, chain 2923). **Reply:** A1427 (d7e752ce-cb39-4130-b089-b20333037990, chain 3056).

The raw tensor realization really does have a nonzero kernel.

Requested calculation. Replace every nonidentity warning with a typed map and continuation.

Original objects and maps. $Q = B/\Theta V$, raw $0 \otimes C Q \rightarrow 0/(g)$, $g=2\xi$; balanced $0 \otimes C[t]Q$.

Actual result and proof scope. The nonzero free raw source contains $g(0 \otimes Q)$ in the kernel. Operator balancing imposes $sh \otimes v - h \otimes Dv$, factors the raw map and kills the earlier explicit z .

Left uncalculated. K_{bal}^1 is still not computed globally. The nonzero raw kernel argument does not apply automatically after balancing.

Exact obstruction scope. The raw scalar extension is not faithful to the torsion divisor realization. This excludes that raw map being an isomorphism, while the operator-balanced route survives.

Later completion or correction. Primary (39),(43) gives actual finite quotients and dual injections without assuming either global realization is an isomorphism.

Concrete surviving derivation. Compute K_{bal} on the original module with its topology and action. The attached explicit free finite-packet embedding makes the raw-kernel theorem constructive and preserves every finite jet.

230.4.29 T29: finite realization completed; finite-to-global extrapolation forbidden

User: U0040 (a10456e9-7f72-4c81-bc00-118fcd575269, chain 2923). **Reply:** A1427 (d7e752ce-cb39-4130-b089-b20333037990, chain 3056).

The finite-jet realization is now an actual isomorphism

Requested calculation. Construct the missing typed morphisms.

Original objects and maps. $A = C[t]$, $Q_{\rho,m} = A/(t-\rho)^m$, $0 \otimes A Q_{\rho,m} \rightarrow 0/(s-\rho)^m$.

Actual result and proof scope. $h \otimes P \mapsto hP(s)$ has inverse $[h] \mapsto h \otimes 1$; 0 is torsion-free over PID A and flat. The single-source complex $[0 \otimes A V \rightarrow 0 \otimes A B]$ has $H_0 = 0$.

Left uncalculated. The complete Q is not proved to be the direct sum/limit of these finite jet modules with an appropriate topology. Its balanced realization kernel remains.

Exact obstruction scope. Matching every finite jet does not prove global quasi-isomorphism. The single-source $H_0 = 0$ must not replace the actual joint $A\tau$ complex, whose $H_0 = V$.

Later completion or correction. Primary (21)–(22),(38)–(43) preserves this distinction and uses finite packet maps with exact scope.

Concrete surviving derivation. Retain the honest joint H_0 and all support masks. The attached saturated two-leg calculation proves its surviving syzygy explicitly rather than declaring all H_0 removed by balancing.

230.4.30 T30: accurate application limit; source audit delivered, missing geometry not calculated

User: U0040 (a10456e9-7f72-4c81-bc00-118fcd575269, chain 2923). **Reply:** A1427 (d7e752ce-cb39-4130-b089-b20333037990, chain 3056).

A verified application of its purity theorem to the full split-support arithmetic object was not constructed.

Requested calculation. Audit what Deligne was actually read and push the object through the method.

Original objects and maps. Recorded Weil II windows, local root covers $G(A) \rightarrow G(B)$, tame coefficient maps, analytic c_m , intended global cohomological comparison.

Actual result and proof scope. The reply distinguishes actual fresh windows from earlier reading claims, and constructs the coefficient square with supported residue zeros and commuting root-of-unity automorphisms.

Left uncalculated. The square does not construct tame lisse sheaves with the required weights or the global arithmetic $f! \rightarrow f^*$. Calling local analytic c_m an application of Deligne had been too strong.

Exact obstruction scope. The theorem's hypotheses are not established for that local comparison. This does not disprove an analogous τ -base argument; the later primary explicitly computes its own $A\tau$ and $K\tau$.

Later completion or correction. Use the later complete primary as the present starting point, so 'construct a right adjoint' is no longer left as a generic demand.

Concrete surviving derivation. Continue the exact original A_τ/K_τ method: prove the analytic bound on its theta-source relations and calculate all boundary/kernel terms. Keep Deligne's original argument as a method to reproduce, not a theorem whose hypotheses may be assumed.

230.5 Completed work delivered with this audit

COMPLETED_MELLIN_CALCULATIONS.md proves the raw two-leg syzygy for every original differentiated-theta order $m \geq 1$, retains each mask, supplies the exact operator-balancing relation, and calculates the honest cycles in the saturated original two-leg source. It also uses the actual smooth full-jet right inverse to inject a free rank- $\Sigma m\rho$ module into the raw Mellin-realization kernel. These results neither assume vanishing of the balanced kernel nor replace the theta source.

source_maps/VERIFIED_SOURCE_MAPS.md proves the exact structural/base-restriction relation, A_τ and K_τ maps, all-support versus categorical kernels, and the opposite dual-index arrows. It additionally gives the explicit equivariant factorization $A_\rho \rightarrow E \rightarrow A_\rho$ of J_g , with the retained coefficient $m\rho u_\rho(0)$ and the actual socle eigenline inside the full nilpotent packet.

230.6 Prioritized continuation for integration

1. Integrate the later primary A_τ/K_τ construction into the historical findings first. The missing right adjoint and finite residue-to-dual arrow are now completed; no generic-stalk substitute is needed.
2. Carry the raw and operator-balanced Mellin maps as separate exact comparisons. The raw kernel is provably nonzero; the balanced kernel requires actual calculation on Q with its topology and action. The attached all-order calculations give explicit subcomplexes and kernel elements.
3. Calculate the original theta relation estimates with all boundary terms, masses, Fourier/scaling factors, and the retained unshifted trace correction. The unproved uniform residual/gap term and unsupported norm transfer are identifiable interruptions, not complete results.
4. Establish the trace on a specified infinite arithmetic realization and account for its kernel/cone contribution. Finite full-jet traces and an absolutely convergent spectral distribution alone do not provide that theorem.
5. Retain valid obstructions at their exact scope: pointed-section nonsplitting, filtered-map non-strictness, unshifted-form descent, raw realization non-quasi-isomorphism, and formal covariance without positivity. Each has an explicit surviving map in this ledger. None proves failure of the entire τ -base program.

These are integration priorities for the original program. They are not conditional theorems in which a missing bound or vanishing is assumed.

230.7 Retained-turn index

Complete table entries (34 rows).

Entry 1.

Locator: U0029

UUID: 53352f69-6c38-4cab-a6aa-c225647ebe9a

Chain: 2057

Disposition: user directive read verbatim

Entry 2.**Locator:** A1007**UUID:** 52a96f2c-82c8-4b11-8b84-892d97e0bfa3**Chain:** 2102**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 3.****Locator:** A1021**UUID:** cf9ad54a-2906-4d48-9cbd-b9e9677bb362**Chain:** 2138**Disposition:** substantive reply audited in records T01**Entry 4.****Locator:** U0030**UUID:** 2e3cac15-6cbf-4fbd-b826-ceed04867399**Chain:** 2139**Disposition:** user directive read verbatim**Entry 5.****Locator:** A1022**UUID:** 6a6b4385-d69e-4e8e-996d-fed8f95a0447**Chain:** 2141**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 6.****Locator:** A1032**UUID:** 882cb536-7647-4572-a180-f9208370725d**Chain:** 2154**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 7.****Locator:** A1070**UUID:** 86b375dc-bd7b-494b-b050-0d8680eadae0**Chain:** 2308**Disposition:** substantive reply audited in records T02, T03, T04**Entry 8.****Locator:** U0031**UUID:** fd1d4ef4-537b-4bca-8d5b-0cb82a441737**Chain:** 2309**Disposition:** user directive read verbatim**Entry 9.****Locator:** A1071**UUID:** c3c6b5c5-2cd8-4ceb-91a8-e063cbd72b7a**Chain:** 2311**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem

Entry 10.**Locator:** A1098**UUID:** 7c3e6ce4-6d6c-4697-b468-e830ec4607bd**Chain:** 2348**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 11.****Locator:** A1106**UUID:** 14112bb8-e64e-41ed-ab82-630beeb21d6f**Chain:** 2362**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 12.****Locator:** A1140**UUID:** 4096bcde-e267-4ebe-8148-43bb2424da1a**Chain:** 2445**Disposition:** substantive reply audited in records T05, T06**Entry 13.****Locator:** U0032**UUID:** f4609209-f2cd-4dbc-8ec2-a8995aedc998**Chain:** 2446**Disposition:** user directive read verbatim**Entry 14.****Locator:** A1142**UUID:** 5e7f991c-86d7-42ce-a2af-223291387447**Chain:** 2449**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 15.****Locator:** A1155**UUID:** 733dc023-e98b-4fc4-9855-15de54e1438e**Chain:** 2467**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 16.****Locator:** A1167**UUID:** 157eecdf-7f96-4ec5-96bd-6dd376276ea6**Chain:** 2520**Disposition:** substantive reply audited in records T07, T08, T09**Entry 17.****Locator:** U0033**UUID:** 622780af-97a6-4c56-987c-135df8409461**Chain:** 2521**Disposition:** user directive read verbatim

Entry 18.**Locator:** A1169**UUID:** 45c55f8c-4af1-4c8c-981a-7f90d9eaa7e7**Chain:** 2524**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 19.****Locator:** A1206**UUID:** ea7fad65-aa59-4169-8bbd-5e73a631d3ae**Chain:** 2613**Disposition:** substantive reply audited in records T10, T11**Entry 20.****Locator:** U0034**UUID:** b5ee7e11-a089-4433-af7a-ed73bbc5bcf**Chain:** 2614**Disposition:** user directive read verbatim**Entry 21.****Locator:** A1210**UUID:** bc9240bf-7f10-4885-b8b7-97225b1ba74d**Chain:** 2620**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 22.****Locator:** A1235**UUID:** ca9499fc-3b1b-49a8-8705-047fc4fec20e**Chain:** 2661**Disposition:** substantive reply audited in records T12, T13, T14, T15**Entry 23.****Locator:** U0035**UUID:** 7b9f7612-3a57-4687-9bfb-39d1d933baf8**Chain:** 2662**Disposition:** user directive read verbatim**Entry 24.****Locator:** U0036**UUID:** d6542f7d-b917-4de5-b35d-07062f16e9c1**Chain:** 2679**Disposition:** user directive read verbatim**Entry 25.****Locator:** A1280**UUID:** 3c7dd1ba-95eb-4919-b2db-efeb2948ddd1**Chain:** 2749**Disposition:** substantive reply audited in records T16, T17, T18

Entry 26.**Locator:** U0037**UUID:** 8b2110e7 - 195b - 42d0 - 8c86 - 7277004e35fd**Chain:** 2750**Disposition:** user directive read verbatim**Entry 27.****Locator:** U0038**UUID:** 4ab2ff8f - 915a - 417d - 876f - 7294ed662b24**Chain:** 2753**Disposition:** user directive read verbatim**Entry 28.****Locator:** A1304**UUID:** 7fd7ecd5 - 0f3d - 4673 - 94dc - 9cda82bee9ed**Chain:** 2801**Disposition:** substantive reply audited in records T19, T20, T21**Entry 29.****Locator:** U0039**UUID:** 71c1d5ee - 668e - 4eea - ac21 - d844702248ea**Chain:** 2802**Disposition:** user directive read verbatim**Entry 30.****Locator:** A1306**UUID:** 26f39b92 - 21f6 - 48be - 91f8 - d30a0e06723b**Chain:** 2805**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem**Entry 31.****Locator:** A1361**UUID:** c9e551ee - 298f - 438b - 9bdc - 2e23c2c0613a**Chain:** 2922**Disposition:** substantive reply audited in records T22, T23, T24, T25, T26**Entry 32.****Locator:** U0040**UUID:** a10456e9 - 7f72 - 4c81 - bc00 - 118fcd575269**Chain:** 2923**Disposition:** user directive read verbatim**Entry 33.****Locator:** A1362**UUID:** e5721b92 - 7c4b - 4be6 - 9100 - d6e7fd61fe5c**Chain:** 2926**Disposition:** progress statement; classified in its controlling user episode; not treated as an independent completed theorem

Entry 34.**Locator:** A1427**UUID:** d7e752ce - cb39 - 4130 - b089 - b20333037990**Chain:** 3056**Disposition:** substantive reply audited in records T27, T28, T29, T30

231 Actual tau cohomology: U0041–U0054

232 Passage audit: actual tau cohomology and continuation, U0041–U0054

The entire 7,811-line segment and complete primary Tau Base NOTE were read. The current segment SHA-256 is e79b46722d9a2ee95b68ae6512f37a19fd41e4b3ecc4e53583cd357d0442c251. All 14 user inputs, progress responses and substantive answers are represented below. The extraction owner’s newline-only repair preserves the UUID locators and exact quotations. The full user text is preserved in the JSON and separate provenance file.

The accepted original construction is A1523, explicitly accepted by U0046. The concrete A1545 promise to calculate the full reversed-support dual is completed in FULL_SUPPORT_DUAL.md/.tex with its standalone retraction supplement. The repeated remaining quantitative endpoint is the uniform arithmetic control, not a missing definition of A_{tau} or K_{tau} .

232.1 AT41: U0041 → A1458

User node b85d7358 - ede3 - 4c0e - ab3f - 45b067391b12, chain 3057, source line 1. Completed literal support propagation; arithmetic transfer not computed.

Retained turns: U0041 (b85d7358 - ede3 - 4c0e - ab3f - 45b067391b12), A1429 (0db9d22c - 28a2 - 4f93 - a2f8 - fd09c4ecb11d), A1458 (5ae3c5b9 - 7f5c - 46a6 - 92c6 - 09933220f7aa).

Exact passage, U0041 node b85d7358 - ede3 - 4c0e - ab3f - 45b067391b12, line 3:

Try to get hardcore typed morphisms without simplification that you propagate through each and every equation, while tracking the interesting bits continually.

Exact passage, A1458 node 5ae3c5b9 - 7f5c - 46a6 - 92c6 - 09933220f7aa, line 818:

No equivalence between that entire analytic complex and the geometric complexes above is asserted here.

Requested calculation. Carry the original infinite quotient and split support through the stated Deligne operations by exact typed maps.

Actually proved. The answer constructs the marked geometry $(\widehat{X}, \mathcal{S}_X, \mathbf{p}_X)$, its induced site, support diagrams and original coequalizer cohomology. It retains $G(\mathcal{O}_K) \rightarrow \mathcal{O}_K$, the closed arithmetic subspace $V(e)$, the cycle formula for support-comparison kernels, extension by absence versus the retained boundary skeleton, dual precomposition, all tensor signs and the e -seeded Lefschetz sum, including an empty fixed-point set. Its finite-field comparison image is proved under Deligne’s explicitly stated smooth/lisse/pure hypotheses; its final support projection kernel is not set to zero.

Left uncalculated. No source-sensitive morphism transfers the finite-field weight estimates to the original characteristic-zero theta cohomology. The retained map $\mathcal{O} \otimes_{\mathbb{C}[t]} Q \rightarrow \mathcal{O}/(2\xi)$ is not proved to be an equivalence to the finite-field comparison complexes.

Exact obstruction scope. The finite-field theorem is a legitimate scoped calculation, not a proof or disproof of the original absolute-base program. No kernel vanishing or broad nonrelationship is proved.

Later completion. U0043 rejects stopping at a finite-field lift; A1523 constructs the actual absolute-base $A_{\text{tau}}/K_{\text{tau}}$ system.

Concrete continuation. Retain these support operations but resume A1523’s accepted absolute-base theta cohomology; do not repeat a renamed finite-field theorem.

232.2 AT42: U0042 → A1474

User node 40ac47e6-d648-406d-91e5-085f3e6ecb67, chain 3130, source line 822. Completed finite-packet nullity and signature calculation; absolute-base estimate remains.

Retained turns: U0042 (40ac47e6-d648-406d-91e5-085f3e6ecb67), A1474 (32b7efb8-d9e3-4f15-9dd5-e15827f2241d).

Exact passage, A1474 node 32b7efb8-d9e3-4f15-9dd5-e15827f2241d, line 1136:

This is a **newly stated test of a particular trace-theoretic relation**, not a redefinition of your existing quotient.

Exact passage, A1474 node 32b7efb8-d9e3-4f15-9dd5-e15827f2241d, line 1267:

The sharpened question is whether the arithmetic e -null fibre is a radical through which the pairing descends.

Requested calculation. Determine the individual mathematical roles of tau and e in testing the arithmetic pairing and RH.

Actually proved. The response proves pairing descent exactly when its restricted map to the dual vanishes. For the actual packet it computes

$$Q_Z(f, h) = \sum_{\rho} m_{\rho} \overline{f(1 - \bar{\rho})} h(\rho), \quad \text{Rad } Q_Z = \{f : f(\rho) = 0 \forall \rho\}.$$

It retains all local units in the full residue pairing, gives its Jacobian contraction to this trace form, and calculates every off-line reflected block's isotropic coordinate vectors, nonzero cross pairing and positive/negative signature. The trace radical quotient legitimately forgets nilpotent amplitudes while retaining zeros and multiplicities. Synchronization preserves the arithmetic amplitude and hence cannot erase a diagnostic value.

Left uncalculated. The finite nullity criterion is not the cohomological arithmetic argument forcing positivity or radicality. No absolute-base weight estimate is calculated.

Exact obstruction scope. Failure of a proposed self-null congruence is an exact obstruction to that quotient, not to the already valid original coequalizer or the whole tau-base program. A negative auxiliary finite model is not an RH counterexample.

Later completion. A1523 supplies the original-base adjoint and residue injection; A1644 supplies the arithmetic extension unit and equivariant finite trace.

Concrete continuation. Use the computed residue/Jacobian map in the actual $A_{\text{tau}}/K_{\text{tau}}$ continuation. Prove the analytic control of that image; do not replace it by the already established RH criterion.

232.3 AT43: U0043 → A1523

User node d908924e-830b-46ad-8d9c-f2b78a536989, chain 3157, source line 1271. Requested change to absolute tau base completed and accepted.

Retained turns: U0043 (d908924e-830b-46ad-8d9c-f2b78a536989), U0044 (457e7cf8-ce51-4d20-afe4-e998185d2e17), U0045 (c7241c17-6d01-4d7f-bd87-78977a059471), A1481 (5d7a3844-3f4c-4652-a5d9-cd4d586e4c75), A1520 (3f9d0b49-19dc-44cc-87a7-f139138cd6b9), A1523 (b670b6c0-a774-4398-b978-9c9779fca379).

Exact passage, U0043 node d908924e-830b-46ad-8d9c-f2b78a536989, line 1273:

and then use his machinery instead over the other thing, like tau instead of E, which was the intent, for the record. That's what I wanted you to do.

Exact passage, A1523 node b670b6c0 - a774 - 4398 - b978 - 9c9779fca379 , line 1957:

The construction uses the structural morphism a_R , keeping p_R attached. It does not replace the arithmetic object by its Boolean localization.

Exact passage, A1523 node b670b6c0 - a774 - 4398 - b978 - 9c9779fca379 , line 2478:

The remaining issue is **a weight or positivity theorem for this new-base arithmetic realization**.

Requested calculation. Change the actual base to tau, instead of repeating the finite-field theorem with split notation, and perform the Deligne-like constructions on that object.

Actually proved. A1520 first supplies a pointed-monoidal structural map, the mixed-boundary contraction with stalk $G(A)^2 \rightarrow A^2$, and boundary cohomology Q^2 , retaining mixed multiplication and synchronization. Its product is expressly topological, not asserted to be the universal semiring-scheme product. A1523 then gives the canonical retained blueprint construction: the structural map $F_{1,\tau} \rightarrow B_R$, four-point original theta chart, projective resolution computing $A_\tau = R\Gamma(P, -)$, separate restriction to sigma with its chain homotopy, signed right-adjoint complex $K_\tau(W) \simeq S_\eta W[1]$, and full reversed precomposition arrows. It retains the source actions $U_a, aU_{1/a}$, the arithmetic weight-one moment line, all Mellin jets and the residue injection into the computed K_tau dual. It proves top tensor cohomology $Q^{\otimes r}$, product adjoint, and exact finite-packet deletion cone and trace.

Left uncalculated. No weight estimate is proved from the tensor action alone. An involutive global Verdier duality and prime-sensitive Lefschetz geometry are not constructed merely by the structural map.

Exact obstruction scope. The direct image A_τ is not Boolean generic restriction. Its explicitly compared restriction retains the labelled support skeleton while the global Q classes survive. The single-active-fibre lemma about mapping a supported vector to tau does not apply unchanged to arbitrary nonzero-bottom semimodules. A1520's Q^2 is the pushforward of two pulled-back coefficient copies at a contracted boundary; A1523's Q is the original joint chart's global cohomology. They are different declared functors, not contradictory values of one functor.

Later completion. U0046 expressly says this is now the correct context and orders continuation using these objects. A1644-A1959 genuinely continue the original theta source; no whole-program failure is established in this segment.

Concrete continuation. Resume canonical A1523/retained NOTE objects, preserving its comparison to sigma and all supports. Complete the original arithmetic estimate, not a generic-stalk replacement or an unrelated polynomial family.

232.4 AT44: U0044 $\rightarrow$ A1523

User node 457e7cf8 - ce51 - 4d20 - afe4 - e998185d2e17 , chain 3159, source line 1275. Reinforcement of U0043; completed by shared A1523 response.

Retained turns: U0044 (457e7cf8 - ce51 - 4d20 - afe4 - e998185d2e17) .

Exact passage, U0044 node 457e7cf8 - ce51 - 4d20 - afe4 - e998185d2e17 , line 1277:

which is thus what you will in fact do

Requested calculation. Carry out the absolute-tau-base construction ordered in U0043.

Actually proved. This has no separate assistant response before U0045. The complete shared response is audited at AT43.

Left uncalculated. Same subsequent arithmetic estimate as AT43; no additional mathematical calculation is introduced by this reinforcement.

Exact obstruction scope. Do not invent a separate abandoned episode from adjacent user reinforcement.

Later completion. A1520 and A1523 supply the construction, followed by explicit user acceptance U0046.

Concrete continuation. Use AT43 continuation, retaining this input in provenance.

232.5 AT45: U0045 → A1523

User node c7241c17-6d01-4d7f-bd87-78977a059471, chain 3161, source line 1279. Urgency reinforcement of U0043; completed by shared A1523 response.

Retained turns: U0045 (c7241c17-6d01-4d7f-bd87-78977a059471) .

Exact passage, U0045 node c7241c17-6d01-4d7f-bd87-78977a059471, line 1281:

nnow

Requested calculation. Execute the already ordered absolute-base construction immediately.

Actually proved. A1481 progress and A1520/A1523 are the shared response; their mathematics is audited at AT43.

Left uncalculated. Same later quantitative estimate as AT43.

Exact obstruction scope. No new mathematical hypothesis or alternative construction is requested.

Later completion. U0046 explicitly accepts A1523's context.

Concrete continuation. Use AT43 continuation, retaining this input in provenance.

232.6 AT46: U0046 → progress responses

User node 0af0ed52-c7ab-45fb-85a5-1911193398cb, chain 3260, source line 2482. Progress-only interval; explicit full reversed-support dual promise completed by this audit.

Retained turns: U0046 (0af0ed52-c7ab-45fb-85a5-1911193398cb), A1525 (d07bf9b2-4bbf-4450-b206-9df18980b10d), A1533 (48573948-14de-4d6d-a29b-2e291b816c22), A1545 (bacc895-0427-42ee-b6f6-4c61752dfde2) .

Exact passage, U0046 node 0af0ed52-c7ab-45fb-85a5-1911193398cb, line 2484:

you started on the DELINGE program in the correct context nowadays

Exact passage, U0046 node 0af0ed52-c7ab-45fb-85a5-1911193398cb, line 2484:

continue using the objects as the results so far allo

Exact passage, A1525 node d07bf9b2-4bbf-4450-b206-9df18980b10d, line 2488:

then I'll extend the comparison kernel while retaining e , τ , and support transports.

Exact passage, A1545 node bacc895-0427-42ee-b6f6-4c61752dfde2, line 2496:

including bottom fibers created by reversed support arrows.

Requested calculation. Publish the existing work as GitHub merge requests and continue the now-accepted original tau-base program using the constructed objects.

Actually proved. The retained assistant responses are progress messages about integration and continuation. No substantive final mathematical calculation intervenes before U0047 'cont'. A1545 specifically promises the full tau-chart dual, including bottom fibres created by reversing support arrows.

Left uncalculated. The full L^{op} cohomology diagram and its nonzero new bottom fibre are not written in this interval. Later A1644/A1694 retain joint H0 and single-fibre duals but do not explicitly complete the whole reversed diagram.

Exact obstruction scope. This is a precise deferred mathematical promise, not evidence about motives. Historical PR status in these messages is not independently verified remote state in this lane.

Later completion. Completed here in FULL_SUPPORT_DUAL.md/.tex, with a full standalone retraction supplement.

Concrete continuation. Use the completed explicit complexes and precomposition maps: $H^{-1} = Q^{\vee}$ on all active faces and $H^0 = V^{\vee}$ on the original joint mask, now the bottom. Retain the strict-natural-section obstruction, its actual surviving derived roof, and the action defect $d_a(c) = -a^w c \mathcal{F}^{-1} k_{1/a}$. Do not impose zero bottom fibres.

232.7 AT47: U0047 → A1644

User node b6dc9d48-0230-480b-8688-c4532c9a9796, chain 3344, source line 2498. Completed original theta extension, support-coherent homotopy and finite cochain trace.

Retained turns: U0047 (b6dc9d48-0230-480b-8688-c4532c9a9796), A1580 (ee1160cc-4814-4d26-9f63-5802e7202b95), A1600 (dc3d2a73-4433-4fb7-b21d-1b912836fa18), A1601 (4071829e-6607-477c-970f-6b4149337366), A1644 (d448b4f7-00f5-449f-a2b0-43ebd22cc66f).

Exact passage, A1600 node dc3d2a73-4433-4fb7-b21d-1b912836fa18, line 2508:

Next I'm computing its compact-support-to-ordinary cohomology map.

Exact passage, A1601 node 4071829e-6607-477c-970f-6b4149337366, line 2512:

although it vanishes in cohomology, its extension class remains part of the comparison.

Exact passage, A1644 node d448b4f7-00f5-449f-a2b0-43ebd22cc66f, line 3015:

They do not yet establish the global weight estimate.

Requested calculation. Continue the accepted $A_{\text{tau}}/K_{\text{tau}}$ construction and its original comparison.

Actually proved. The response retains $C_+ = [V \rightarrow \mathcal{B}]$, the full joint comparison, original $D = -x\partial_x$, Gaussian source and $g = 2\xi$. The actual section uses the full arithmetic unit $\varepsilon_Z = j_{h_Z}(h_Z/g)$. It computes the theta extension class, polynomial-action defect, rank-one infinitesimal boundary and full dilation cocycle. At a simple zero the coefficient is exactly $1/(2\xi'(\rho))$. The two homotopies differ by the retained joint H0 cycle $(b, \widehat{b})$; synchronization supplies a common supported target without killing it. The same unit enters residue duality and the Jacobian trace. The non-A-linear section is replaced by the strict equivariant roof $E_Z[-1] \leftarrow C_Z \rightarrow C_+$. The finite-rank cochain trace keeps its degree-one minus sign and every tensor sign.

Left uncalculated. The global topological section, continuous-dual splitting and quantitative estimate are not yet supplied here. A finite packet roof is not by itself every missing prime-sensitive compact-support theorem.

Exact obstruction scope. A nonzero extension obstructs a strict equivariant representative section, not the displayed equivariant derived inclusion. The exact one-leg-to-joint comparison is present in the primary dependency and fully calculated again in the later-control supplement; its H1 map is identity and its complementary $V[0]$ survives.

Later completion. A1694 completes the global continuous section, transpose and completed tensor contractions. A1771 later computes the complete homotopy family.

Concrete continuation. Retain the extension unit and actual roof through the global control calculation. Do not drop its boundary pairings or treat focus on C_+ as deletion of the joint H0 complement.

232.8 AT48: U0048 $\rightarrow$ A1694

User node 29b1576a-8ac7-421a-8b66-cbd70eea994d, chain 3469, source line 3017. Completed global analytic retraction and Hausdorff-kernel calculation; full quantitative boundary estimate remains.

Retained turns: U0048 (29b1576a-8ac7-421a-8b66-cbd70eea994d), A1694 (1fa0ebbc-f569-40c1-b7ac-b523da19a5f0).

Exact passage, A1694 node 1fa0ebbc-f569-40c1-b7ac-b523da19a5f0, line 3287:

Their kernels are not identified with the now-vanishing Hausdorff-comparison kernel.

Exact passage, A1694 node 1fa0ebbc-f569-40c1-b7ac-b523da19a5f0, line 3575:

All three boundary and cross-pairing terms remain.

Exact passage, A1694 node 1fa0ebbc-f569-40c1-b7ac-b523da19a5f0, line 3592:

No global positivity or purity conclusion is being claimed from the retraction alone.

Requested calculation. Continue pushing the original theta program.

Actually proved. A1694 constructs the continuous left inverse $\Lambda : \mathcal{B} \rightarrow V$ using the original $\frac{1}{2} \sum \mu(n)F(n|x|)$, both Fourier-related exterior regions, fixed cutoff operator $T = A_\alpha B_\beta$, the retained inverse bound $1/(1 - \|T\|)$, and both original moments. It proves Schwartz regularity and $\Lambda\Theta = 1$, hence closed theta image, Hausdorff Q, the section s , and $\mathcal{B} \cong V \oplus Q$. It retains the different spectral and balanced analytic kernels and the exact smaller-source sequence with V/W . The full global cocycle $k_a = \Lambda U_a s$, change-of-section coboundary, old extension unit and trace are preserved. Transposition supplies continuous duals; completed tensor contractions retain all signs and joint degree-zero terms.

Left uncalculated. The full boundary expression

$$\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a\mathcal{H}_s(u, v) = -\langle U_a s u, \Theta k_a v \rangle - \langle \Theta k_a u, U_a s v \rangle + \langle \Theta k_a u, \Theta k_a v \rangle$$

is computed but not bounded uniformly. The auxiliary positive form is not the arithmetic Weil form. Closedness of Theta V in the original Frechet topology does not prove vanishing of the remaining spectral or balanced analytic kernels.

Exact obstruction scope. This is a real global analytic proof and a specific vanishing comparison kernel. Neither a uniform cutoff gap for changing cutoffs nor vanishing Hilbert pairing of theta boundaries is asserted. No purity follows from the global section alone.

Later completion. A1721 expands cutoff/Fourier/moment commutators and creates source matrices. A1771-A1959 further refine the original arithmetic control; none completes uniform sublinear tensor excess within this segment.

Concrete continuation. Use those later canonical representatives and retain all cross terms and inverse conditioning. The standalone RETRACTION_PROOF.md now supplies the exact Lambda used by the full-support dual completion.

233 U0049-U0054: complete later-control audit

234 Later control continuation audit: U0049-U0054

234.1 Scope, coverage and source pins

This bounded audit read every line 3596-7811 of output/tau_f1_transcript_audit_2026-09-13/sources/audit_segment_U0041_U0054.md, in order, including every retained user and assistant message. Segment SHA256: e79b46722d9a2ee95b68ae6512f37a19fd41e4b3ecc4e53583cd357d0442c251. Locators below are one-based lines in that file, not lines in an inferred ZIP. The complete output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md was also read. Its verified SHA256 is d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3.

Supplementary checking read the sum-connection NOTE.tex at output/split_zero_rh_tandem_2026-09-12/sources/web_sum_connection_delivery/Tau_Sum_Connection_Control/NOTE.tex, lines 176-287, 391-421 and 659-694. This is a bounded proof-window check of the Fisher identity, original relation derivative and explicitly stated transfer gap; it is not a claim to have audited that entire package or executed its regression suite. Earlier retraction, inverse-range and packet-section results cited by these episodes remain dependency inputs for the parent audit.

The original object is the four-point chart with $+$, $-$ $\leq \eta < \sigma$, the functor

$$A_\tau(F) = [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta]$$

in degrees zero and one, its separately computed restriction to σ , and

$$K_\tau(W) = [I_\eta(W) \longrightarrow I_+(W) \oplus I_-(W)] \simeq S_\eta W[1].$$

Tau Base (21)-(25), (27)-(31), (38)-(48) retain the two-leg complex, supported copies of $\mathbb{Q}$, the adjoint, residue jets and tensor products. No message in the audited six episodes replaces this push-forward with the generic stalk or proves failure of the entire original programme. Later concentration on the one-leg complex has an exact relation to the joint complex, calculated below.

234.2 Episode ledger

234.2.1 LC49: U0049 $\rightarrow$ A1721, source boundary and control machinery

- User, line 3596, node a9162d93-9cf1-4d45-b6f1-9fd92ae43471, chain 3577: “then let us make the machinery for the deligne like control”.
- Progress A1696, line 3600, node 63828eab-7482-48e8-93d0-4fc0f615a74e, chain 3580.
- Final A1721, line 3604, node 40ab7ee4-4dcd-409c-9e0d-f211bc6aeccec, chain 3634; response ends at line 4262.

The requested calculation is source-side Deligne-like norm, duality and tensor control on the original arithmetic classes. The response expands the cutoff/moment commutator of the inherited continuous retraction, constructs finite representatives $R=s_Z\text{-}\Theta_{\text{tab}}$ with exactly unchanged $qR=\sigma_Z$ and $J_ZR=1$, and computes

$$B = DR - RA = \Theta K, \quad G = R^*R > 0, \quad W = A^*G + GA - G = -(R^*B + B^*R).$$

At line 3905 it explicitly says: “The boundary expression $[x\overline{F}H]_0^\infty$ vanishes by their stated decay.” This is an analytic endpoint boundary justified by the original test space, not a claim that the cohomological relation B has zero Hilbert pairing. Indeed the following subsection explicitly retains $\langle Rv, Bv \rangle$ and the difference of the two theta primitives in H^0 .

The residue calculation $A^*S+SA=S$ gives the exact metric transfer

$$G^D = S^*G^{-1}S, \quad W^D = -S^*G^{-1}WG^{-1}S.$$

This proves upper-to-lower transport, retaining the original weight-one action. It does not itself supply an upper arithmetic estimate. The tensor computation retains W_n as the sum of all n factor contributions, hence a copied error costs `nepsilon`. Its orbit-Gram construction has an exact endpoint identity; the endpoint has not been shown to be small on the actual arithmetic family.

The fixed cutoff example $r=s_0=1/4$, the nilpotent block and the off-line diagonal block are explicitly auxiliary calibrations. Their role is stated; the text does not identify them with actual zero packets or change $g=2xi$ to those examples. The malformed opening display $A^*GA+(\text{not used})$ is a presentation defect, immediately replaced by the correct infinitesimal identity, not a substantive mathematical obstruction.

Actual stopping point. Line 4259: “The uniform sublinear error estimate for the **actual arithmetic tensor family** is not proved here.” The answer builds the machinery the user requested; it does not finish the subsequent bound. The matrix certificate interface certifies supplied matrices with supplied error enclosures; no arithmetic integral or uniform family has been enclosed by that interface here.

Later completion. A1771 fixes the actual theta derivative filtration and derives its rank-two defect. A1829 supplies canonical full-jet interpolation, same-metric reflection and joint tensor corrections. Thus the later work does not merely leave the machinery at the orbit-metric stage.

Concrete continuation. Use the canonical representatives from A1829, calculate their arithmetic full-jet moment matrices and signed boundary forms with the inverse interpolation matrix retained, then estimate their relative extremum as k, M vary. Do not infer the missing smallness from $qB=0$, the fixed-cutoff inverse bound, or a calibration certificate.

234.2.2 LC50: U0050 → A1771, homotopy freedom and the original theta recurrence

- User, line 4263, node 95ddc7db-1037-4e47-b839-13eaece6695, chain 3635: “The other session is working on this. It may not have everything that it needs, but it’s still trying to work. Consider corrections if they are required and continue.”
- Progress A1722, line 4267, node ce1a8f2e-af6c-4022-8160-af9a12024a25, chain 3637; A1735, line 4271, node 3e321a1f-89f3-42a8-86a4-d6e948dde8b2, chain 3656.
- Final A1771, line 4275, node 552c8e45-2599-46da-8b95-214fa4d5ce0c, chain 3721; response ends at line 4891.

The response checks both homotopy equations and gives the entire continuous family

$$H_\alpha(b) = (\kappa b + \alpha(qb), \mathcal{F}^{-1}\alpha(qb)), \quad \alpha \in \text{Hom}_{\text{cont}}(Q, V).$$

It lifts the family through support synchronization and retains $\text{Hom}(H^1, H^0)$; line 4430 states: “synchronization supplies the common target; it does not remove the H^0 -class.” The zero composite $L \rightarrow \gamma L \rightarrow d\gamma L = 0$ establishes exactly why changing this homotopy cannot change the source boundary or control matrix. It then continues through the effective operation $R_T = R + \text{Theta} T$, with $K_T = K + D_V T - T A$.

The calculation fixes the actual theta seed $\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$, $f_j = D^j \text{Theta} \phi_*$, $L_n = \text{span}(f_0, \dots, f_n)$, and the orthogonal representative $R_n = (1 - P_n) s_Z$. Every unscaled orthogonal norm $\text{mathfrak{h}}_n = \|u_n\|^2$ remains. Integration by parts gives the three-term recurrence, including the negative coefficient $-\text{mathfrak{h}}_n / \text{mathfrak{h}}_{n-1}$, and then

$$W_n = \mathfrak{h}_{n+1}(r_{n+1}^* r_n + r_n^* r_{n+1}), \quad G_n - G_{n+1} = \mathfrak{h}_{n+1} r_{n+1}^* r_{n+1}.$$

The derivative is typed $Q_{\{V_n\}} \rightarrow Q_{\{V_{n+1}\}}$. Its nonzero class at the old relation level is explicitly $-[u_{n+1}] r_n$, and the next transport makes it that receiving fibre’s supported zero. The response proves the relation-layer isomorphism rather than dismissing this as zero after quotient.

The least relative excess is reduced to an explicit two-by-two spectral calculation using $r_n G_n^{-1} r_n^*$, $r_{n+1} G_{n+1}^{-1} r_{n+1}^*$ and the mixed coefficient. Line 4821 stresses that “the inverse G_n^{-1} remains”. This is an exact finite reduction, not the evaluation or asymptotic control of those three quantities for arbitrary actual packets.

Numerical and boundary scope. The actual first four seed norms are evaluated; a double theta sum is integrated only after moving $(0, 1)$ to $(1, \text{infinity})$ by the original inversion. The displayed integer-tail estimate is explicitly distinguished at line 4882 from validated rounding and inverse-matrix errors. It is not a certificate for a nonempty packet’s relative excess.

Actual stopping point. Line 4890: “A uniform small arithmetic excess has not yet been proved.” This is successful continuation through homotopy classification and an exact source recurrence, followed by a still-open quantitative calculation. The obstruction only says homotopy changes cannot change B ; it does not say original theta-boundary changes, or the programme, fail.

Later completion and next derivation. A1829 supplies the full-jet inverse-moment metric and its convergent tail, making the earlier inverse-conditioning issue explicit. The next work is the arithmetic asymptotic or certified finite evaluation of that retained two-row expression, not a new homotopy parameter search.

234.2.3 LC51: U0051 $\rightarrow$ A1829, canonical full-jet moment problem and joint tensors

- User, line 4892, node e8523a76-cc80-41ad-a0f0-f5ed9ad6ff3b, chain 3722: “The other session is continuing to try and formalize. This can happen, of course, while you also continue to work on pushing this through the program further. So continue doing that as you are.”
- Progress A1773, line 4896, node 11e00aa0-8036-4382-a795-16e531c812af, chain 3725; A1779, line 4900, node 560d9cac-da7f-4022-9dcc-22095069129f, chain 3732.
- Final A1829, line 4904, node aff5580e-a451-48c3-85f4-f19487a5d35e, chain 3826; response ends at line 5583.

The response acknowledges the previous metric-freedom calculation, then fixes the original L_n instead of optimizing unrestricted metrics. Its exact source remains $T_h(P) = P(D) F_h$, $MF_h = g/h$, with $T_h(hP) = \text{Theta}(P(D) \phi_*)$ and the arithmetic jet unit $\text{upsilon}_h = j_h(g/h)$. The polynomial family is therefore an explicitly mapped presentation of the original quotient, not a replacement family. The full-jet kernel is precisely hP_n in the one-factor source, and the original

relation ideal in the tensor source.

It proves the constrained minimum formula

$$d\nu_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad G_{h,n} = (J_{h,n+d} M_{h,n+d}^{-1} J_{h,n+d}^*)^{-1}.$$

The packet inclusions retain their arithmetic units and every nilpotent jet; the exact multiplication $P \rightarrow (H/h)P$ is an isometry between the two specified measures. The reflection acts on the same canonical metric and pairs the generalized control eigenvalues with their negatives. This completes a compatibility issue left by transferring an upper bound to a different dual metric in A1721; it still supplies no small upper bound.

The density calculation proves the original $D^{\wedge n} \mathbf{f}_0$ family dense in $L^2(dx)$, hence $G_{\{h, n\}} \rightarrow 0$, but simultaneously proves every nonzero dual row diverges against $G_{\{h, n\}}^{-1}$. The exact Hilbert quotient map is $Q \rightarrow H/\text{closure}(\Theta \tau V) = 0$; this map is explicitly distinguished from the original Frechet quotient Q . Its supported lift sends a class to $\mathbf{e}$, not to τ . This is an exact obstruction to reading arithmetic purity off absolute Hilbert decay, with the original class and its jet still present. It is not failure of A_τ or K_τ .

The joint total-degree source and ideal are explicitly constructed. The next relation layer has dimension $\text{binom}(M+k, k-1)$, and the actual maps $Y_{\{k, M\}}, C_{\{k, M\}}$ give

$$W_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}.$$

All tensor primitive signs are retained. This is a real change to admitted original boundaries, not the unsupported inference that a smaller Gram matrix has smaller relative defect.

Actual stopping point. Line 5580: “No uniform sublinear excess has been established in this continuation.” The eigenline diagnostic remains exactly $k(2\text{Re}\rho - 1)$ independently of the chosen positive metric. The source equations and moment problem are completed; the arithmetic extremal estimate is not.

Later completion and next derivation. A1859 gives finite highest-layer coordinates, a proved finite bound and a signed symmetry restriction. The surviving task is bounding the actual recurrence coefficients and full-jet interpolation increments together; independent positivity of the measure, shrinking Gram matrices, or the bare existence of the moment inverse are insufficient.

234.2.4 LC52: U0052 $\rightarrow$ A1859, finite bound and signed symmetric summand

- User, line 5584, node af10d798-2756-445d-9e9b-5c7b064cade9, chain 3827: “Yeah, let’s push that.”
- Progress A1831, line 5588, node fe850773-525d-45f2-a728-1fbb3b854c75, chain 3830.
- Final A1859, line 5592, node f39f3d53-73e5-40f4-bdf0-25c37492dba8, chain 3887; response ends at line 6332.

The response constructs and inverts the map $\mathbf{b}_M = \mathbf{T}_+ - \mathbf{R}_M \mathbf{F}$ from the next homogeneous polynomial space to the actual next theta-relation layer. It computes the exact Gram $\Omega_{\mathbf{G}_M} + \mathbf{F}^* \mathbf{G}_M$, retains every raw norm and actual measure mass, and proves the Woodbury metric update. The inverse metric obeys

$$\mathcal{C}_{M+1} = \mathcal{C}_M + \mathbf{F} \Omega^{-1} \mathbf{F}^*, \quad \mathcal{Z}_M = \mathbf{A}_k \mathcal{C}_M + \mathcal{C}_M \mathbf{A}_k^* - k \mathcal{C}_M = \mathbf{F} \Omega^{-1} \mathbf{E}_+^* + \mathbf{E}_+ \Omega^{-1} \mathbf{F}^*.$$

The recurrence retains $\text{Reb}_n = 1/2$ and potentially nonzero Imb_n ; it does not silently impose an even measure for an arbitrary reflection-stable packet. It proves the finite upper and lower estimate $\epsilon_{\{k, M\}} \leq 2\sqrt{\Gamma_{\{k, M\}} \lambda_{\{k, M\}}}$. Line 6013 explicitly limits

this: “Its constants have not yet been bounded sublinearly in tensor degree.” The sharper signed finite matrix is retained after Cauchy–Schwarz, so discarded phases are not silently identified with the original form.

The signed cochain idempotent $(1/k!) \sum \text{sign}(\pi) T_\pi$ cancels the Koszul sign in top degree and selects ordinary symmetric tensors. It keeps every repeated eigenvector $v^{\{\text{tensork}\}}$, unscaled orbit sums, all factorial orbit masses, and the complement with its exact trace. The smaller module is a computed direct summand, not a deletion of off-line spectral witnesses.

The Gaussian repeated-root comparison is explicitly a calibration. The arithmetic measure remains $|2xi/h|^2 dt/(2\pi)$ throughout the proposed application; the literal Gaussian mass $\sqrt{2\pi}$ is retained in the test. The response also reports a smaller-degree example where enlarging the correction space increases the relative excess. This is a legitimate counterexample to a naive monotonicity claim, not to all possible arithmetic asymptotics.

Actual stopping point. Line 6327 again states that uniform sublinear excess for the actual arithmetic family “has not been proved here.” The finite bound meets part of the requested push; its constants and signed cancellation have not been estimated in the required limit.

Later completion and next derivation. A1912 transforms this same matrix problem to a sum-coordinate matrix weight, preserving the relative fibres. One must estimate the actual signed frontier matrix or the combined arithmetic Γ_λ cost after that exact transformation. The Gaussian improvement cannot supply that estimate.

234.2.5 LC53: U0053 → A1912, exact sum descent but no improvement of the constant

- User, line 6333, node bbd82b42 - aa50 - 4eeb - abf3 - 27b7320953d3, chain 3888: “Okay, yeah, continue.”
- Progress A1860, line 6337, node 275efe72 - c7eb - 4763 - 82d1 - 9bf3ce0e3186, chain 3890.
- Final A1912, line 6341, node 64c5b883 - 17ba - 441a - b5c9 - bfa6ff1ef787, chain 3986; response ends at line 7043.

The response performs an exact change of presentation using $S = \sum s_i$, not a replacement of g . In the symmetric square it keeps $\Delta = (s_1 - s_2)^2$, proves the invariant quotient $(H_0, \Delta H_1)$ including collisions, and supplies its original theta primitives with both tensor signs. The real change of variables retains Jacobian $1/2$ and the relation $\Delta = -v^2$ in every matrix-weight entry.

The scalar convolution is only the $(0, 0)$ entry. Every relative-coordinate moment remains in the full positive matrix $\text{mathsf{W}}(u)$. The triangular square-completion map is invertible, but its coefficient $m_{0^{\{-1\}}b}$ need not be polynomial. The response explicitly keeps the exact image of the original filtered polynomials instead of enlarging to arbitrary sections. This avoids an otherwise serious source substitution.

The coordinate congruences $M' = T^*MT$, $J' = U^{\{-1\}}JT$, $G' = U^*GU$, $W' = U^*WU$, $R' = RU$ prove that the generalized control constants are unchanged. Line 6788 says precisely: “The new operation has not manufactured a smaller constant.” This is an explicit preservation theorem, not the bound requested by the wider programme.

The finite sum-line module dual is computed by the $SI - A$ resolution, retaining the full resolvent and every nilpotent ladder. Its scalar trace is a further observation. The displayed $\text{RHom}_{\{C[S]\}}(B_k, -)$ belongs to a restriction-of-scalars category; it is not identified in this episode with the original chart functor K_τ . Tau Base already supplies a finite-jet morphism to the K_τ dual. Reattaching the new dual through that original finite-jet morphism is the concrete task, rather than calling the two right adjoints identical or unrelated.

Actual stopping point. Line 7038: “A uniform sublinear tensor excess has not yet been estab-

lished.” The response leaves explicit arithmetic convolution mass, relative covariance and full-jet interpolation as the unestimated quantities. The finite pushforward’s lack of an identified Deligne Lefschetz-pencil realization is stated with exact scope; no general failure theorem follows.

Later completion and next derivation. A1959 differentiates the actual sum amplitude, proves a mass-retaining scalar contraction, computes the full matrix connection and normal part, and adds the missing frame derivative $C^{\{-1\}}C'$. Those are genuine later completions of part of the analytic work. The still-missing estimate is on the original degree-constrained relative image, not on the unrestricted direct-sum space produced by pointwise square completion.

234.2.6 LC54: U0054 → A1959, real Fisher estimate; missing transfer to the arithmetic extremal form

- User, line 7044, node 9cde09a7-2e48-4d0c-aa8b-4d54a316f9b9, chain 3987: “ok the formalizing is in progress - you work on this”.
- Progress A1914, line 7048, node c84c5062-8d5b-45fb-8eea-c0f4c924908e, chain 3990; A1936, line 7052, node 951e58ad-9a13-43d3-847c-6b57a8616eef, chain 4024.
- Final A1959, line 7056, node ac351c92-ca3e-47a6-9014-85f239a2760e, chain 4075; response ends with the segment at line 7811.

The source amplitude is the actual $a_h(t) = (g/h) (1/2 + it) / \sqrt{2\pi i}$, with unchanged mass $\mu_h = |F_h|^2$. Reflection gives fixed real or imaginary phase, justifying the exact Fisher identity without requiring a bounded logarithmic derivative at zeros. In centered sum coordinates the Jacobian is one. The response proves

$$\mathcal{I}_{h,k} + 4 \int \|n_k(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h.$$

The proof retains all diagonal terms, computes all distinct-index cross terms as $\mu_h^{k-2} |\text{intconjugate}(a_h) a_h'|^2 = 0$, and retains the normal residual by orthogonal decomposition. The endpoint zero is justified by Schwartz decay after the exact Mellin–Fourier transform. The mass is never set to one. This is a completed analytic calculation on the actual arithmetic seed family.

Line 7333 states: “The scalar calculation is only the constant relative column.” The full relative family then has exact matrices $\text{mathsf{fW}} = j^* j$, $\text{mathsf{fB}} = j^* j'$, $\text{mathsf{fT}} = (j')^* j'$, connection $\text{Gamma} = \text{mathsf{fW}}^{\{-1\}} \text{mathsf{fB}}$, normal map $N = (1 - \text{Pi}) j'$, and full energy

$$\|\partial_u(j_u c)\|^2 = (c' + \Gamma c)^* \text{W} (c' + \Gamma c) + c^* N^* N c.$$

The connection formula correctly retains the extra $C^{\{-1\}}C'$ under a changing frame and the potentially noncommuting product $\text{mathsf{fW}}' \text{mathsf{fW}}^{\{-1\}} \text{mathsf{fW}}'$. Neither matrix term is bounded on the whole degree-constrained source in this response.

The exact observable dictionary is

$$\mathcal{U}_k D^{(k)} = (k/2 + iu) \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \quad \mathcal{L}_k = \frac{1}{k} \sum \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -1.$$

Thus the new differential estimate concerns a conjugate observable, not the original generator defect W_M . The response does not conflate them; it exhibits their exact commutator and filtered graph. It also calculates the response of actual theta relations: $J_h(\log x \text{Thetaphi}) = j_h(g') j_h(H_{\text{phi}})$, the complete conormal thickening P/I^2 , its derivative map, the local arithmetic unit correction, and higher relation layers. These carry the original multiplicities and lead to the same residue trace contraction as Tau Base (44)–(45).

Actual stopping point. Line 7790: “It is not yet a uniform sublinear bound for every relative polynomial direction in W_M .” This statement has exact mathematical content. Neither the full-matrix energy identity nor the conormal map supplies a quantitative comparison between all admissible coefficient functions and the inverse full-jet interpolation metric. The only displayed numerical arithmetic evaluation is for $h=1$; line 7807 expressly says it is not “a substitute for a nonempty arithmetic packet.”

Later completion. None occurs inside this audited segment. The supplementary source windows confirm that the final note itself states the same transfer gap. Any later completion must be matched to its actual subsequent turns; it cannot be inferred from a later ZIP title.

Concrete continuation. Retain the exact least-norm coefficient lift $C_{\{k, M\}} = M_{\{k, M\}}^{\{-1\}} J_{\{k, M\}}^* G_{\{k, M\}}$. Substitute its actual coefficients into both terms of the full derivative energy, retaining the normal matrix and the frame derivative. Evaluate the corresponding finite quadratic forms and the original signed $W_{\{k, M\}}$ in the same quotient coordinates; control the change between them using the original theta boundary equation. Track $\mu_h^{\{k-1\}}$, all packet-unit jets, the polynomial degree, tensor degree and inverse interpolation matrix. An estimate of $I_{\{h, k\}}$ alone, or a proof that the two differential pieces are retained, does not complete this calculation.

234.3 Two completed source-comparison calculations for the audit

234.3.1 A. The one-leg complex is an explicit equivariant retract of the original joint push-forward

Let F denote the original even Fourier involution on V , so $F^2=1$. In the original coordinates put

$$C_{\tau} = [V \oplus V \xrightarrow{\Theta(\phi - \mathcal{F}\psi)} \mathcal{B}], \quad C_{+} = [V \xrightarrow{\Theta} \mathcal{B}].$$

Define degree-zero and degree-one maps

$$i^0(v) = (v, 0), \quad i^1(F) = F; \quad p^0(\phi, \psi) = \phi - \mathcal{F}\psi, \quad p^1(F) = F.$$

The equations $d_{\tau} i^0 = i^1 \Theta$ and $\Theta p^0 = p^1 d_{\tau}$ follow directly from the displayed original differential; therefore these are chain maps, and $p i = 1$. Their complementary map in degree zero is

$$(1 - ip)^0(\phi, \psi) = (\mathcal{F}\psi, \psi),$$

and in degree one is zero. Its image is the full original kernel, not a vanished subspace. Explicit mutually inverse chain maps are

$$C_{\tau} \longrightarrow C_{+} \oplus V[0], \quad (\phi, \psi) \longmapsto (\phi - \mathcal{F}\psi, \psi),$$

$$C_{+} \oplus V[0] \longrightarrow C_{\tau}, \quad (v, w) \longmapsto (v + \mathcal{F}w, w),$$

with the identity on the degree-one copy of $\mathcal{B}$. Every component is continuous in the original Schwartz topology. Under Tau Base (34), the joint action is $(U_{\phi}, aU_{\{1/a\}\psi})$. Since $\mathcal{F}(aU_{\{1/a\}\psi}) = U_{\mathcal{F}\psi}$, the first factor has exactly U_a , and the retained second factor has exactly $aU_{\{1/a\}}$. Thus this is an equivariant chain splitting with the full action, not a replacement by identical actions on both legs.

In the split reconstruction, the actual map from the single + support fibre to the joint fibre inserts the supported zero in the minus leg. It is the existing support transport and preserves the nonempty

mask. External absence maps to external absence. No assertion that this linear direct-sum description erases other support fibres is made.

Consequently every one-leg $H^1=Q$ calculation has the identity comparison to the original $H^1 A_{\tau}(T)=Q$, while the original $H^0=V$ remains in the explicit complementary complex. Applying the original adjunction to this actual split complex gives

$$R\mathrm{Hom}_E(C_{\tau}, W) \cong R\mathrm{Hom}_E(C_{+}, W) \oplus \mathrm{Hom}_E(V, W)[0].$$

In degree minus one the comparison is precisely $\mathrm{Hom}(Q, W)$, hence the finite-jet dual maps in these episodes land in the same original $H^{-1}R\mathrm{Hom}_A(T, K_{\tau}(W))$ from Tau Base (28),(43). This proves the relevant source and adjoint comparison rather than calling the two presentations unrelated.

234.3.2 B. The announced global logarithmic-relation map and its exact full-jet image

Use the original $LF(x)=(\log x)F(x)$ on $\mathbf{mathscr{B}}$. This is a continuous endomorphism: differentiating LF finitely many times produces the original derivatives of F times $\log x$ or constants; for every integer weight b , $|\log x| x^b$ is bounded by a constant times $x^{b-1}+x^{b+1}$. The seminorms in Tau Base (11) therefore control all resulting terms. Define

$$\nu : V \longrightarrow Q, \quad \nu(\phi) = q(L\Theta\phi).$$

This is a continuous linear map using the original quotient topology. In original unshifted coordinates,

$$U_a L = (L - \log a)U_a, \quad U_a \Theta = \Theta U_a, \quad q\Theta = 0.$$

For each ϕ_i their direct substitution gives

$$U_a \nu(\phi) = qU_a L\Theta\phi = qL\Theta U_a\phi - (\log a)q\Theta U_a\phi = \nu(U_a\phi).$$

Similarly $DL=LD-1$ and $D\Theta=\Theta D_V$ imply $D_Q\nu=\nu D_V$. Thus A1936's announced global map has a complete equivariance proof without discarding the commutator; the commutator term is an actual original theta relation before its quotient.

For a finite packet $h=\mathrm{prod}(s-\rho)^{\{m_{\rho}\}}$ with complete actual orders, the original Mellin identity gives

$$J_h \nu(\phi) = j_h(g')j_h(H_{\phi}).$$

For every polynomial P , the actual source $\phi=P(D_V)\phi_*$ lies in V , since differentiation preserves both zero moments, and $H_{\phi}=P$. Polynomial reduction is onto E_h ; hence the exact image, not just an inclusion, is

$$\boxed{J_h \nu(V) = j_h(g')E_h.}$$

At a selected root write the original germ as $g=u_{\rho}(z)z^{\{m_{\rho}\}}$, with the original nonzero unit u_{ρ} . Differentiation gives

$$g' = m_{\rho}u_{\rho}(z)z^{m_{\rho}-1} + u'_{\rho}(z)z^{m_{\rho}} \equiv m_{\rho}u_{\rho}(0)z^{m_{\rho}-1} \pmod{z^{m_{\rho}}}.$$

Therefore multiplication by g' has image the full one-dimensional socle $Cz^{\{m_{\rho}-1\}}$ in each local factor. Its kernel is (z) in that factor, including the case $m_{\rho}=1$, when $(z)=0$ and the image

is the whole factor. On the complete packet its rank is exactly the number of distinct selected roots and its kernel dimension is $\text{sum_rho}(m_rho - 1)$. No nilpotent jet has been declared absent: the derivative reads its socle through a fully specified contraction.

Finally Tau Base (42) gives, for this exact image,

$$R_Z(f, J_h \nu(P(D_V)\phi_*)) = \sum_{\rho \in Z} \text{Res}_\rho f^\dagger P \frac{g'}{g} ds = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho).$$

The second equality follows from $g' / g = m_rho / z + u_rho' / u_rho$; the unit term is holomorphic and has zero residue. This completes the global equivariance and the exact arithmetic image behind the final episode's trace formula. It does not establish a positivity or tensor-growth bound.

234.4 Integration judgment and priority

These six episodes contain substantial mathematical continuation, including analytic work on the original theta source. Their repeated endpoint is an uncompleted arithmetic uniform estimate. A passage-by-passage audit should identify that endpoint without relabelling legitimate computations or precisely scoped obstructions as abandonment of all mathematics.

The main redo request should resume the final filtered matrix-weight comparison, with A_tau , K_tau , the actual $g=2 \times i$, the arithmetic unit, all local jets, supported relation layers and original actions attached. The following shortcuts are specifically excluded by the calculations already present: treating a theta boundary as an orthogonal vector; choosing an arbitrary favourable metric; reading purity from $G_n - > 0$; promoting an auxiliary Gaussian example; dropping the complement of the signed symmetric projector; replacing the full relative matrix by its scalar convolution entry; assigning unit mass; or replacing the conjugate logarithmic observable by $D^\wedge(k)$.

The tractable comparison calculations above can be integrated immediately. The harder continuation is the explicit all-direction arithmetic interpolation estimate left at A1959, not a new generic RH equivalence or an unspecified purity assumption.

235 Late analytic control: U0055–U0067

236 Late mathematical continuation audit: U0055–U0067

The late segment contains substantial completed mathematics and an unresolved arithmetic estimate. It does not support a blanket finding of abandoned work or a collapse of the original global τ -base cohomology to the Boolean stalk. The exact counterexamples in the final response refute a particular auxiliary-purity inference. They do not refute the complete original construction.

236.1 Reading and evidence scope

All supplied segment text and both required primary notes were read completely. Quotes below are exact substring-validated against current segment. Claims about earlier deliverables are credited at their supplied written-proof scope, not presented as a fresh independent audit of every predecessor source. Parent separately owns gamma/purity replay and integration.

The source UUID and ordered U/A locator are primary. The source owner corrected only newline serialization after the full read; hashes below pin the current serialization. Every quoted passage is checked as an exact substring against it. No remote/master/Lean work was performed in this lane.

Public locators identify the cited sources; the byte counts and hashes below remain those of the original source witnesses. The first locator is a source record, not the private conversation.

- provenance/public_reference_locators/audit_segment_U0055_U0067.md; 267375 bytes; SHA-256 1c1965d5497705ed262a98cf1054736543427ba75569ed2c1a1d9a448ba60cbc. Read: Complete U0055–U0067 segment, sequentially, including every mathematical display and progress node.
- sources/current_cohorts/snapshots/transcript/sources/Tau_Base_Cohomology_NOTE.md; 25512 bytes; SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. Read: Complete file
- sources/current_cohorts/snapshots/transcript/sources/Tau_Deligne_Bound_Audit_NOTE.tex; 35701 bytes; SHA-256 bfe63094ec360b6059f2936e5e456c6c0fd08db144bbabc29621a527b89c13c6. Read: Complete file
- sources/web_periodized_source_delivery/Tau_Periodized_Source_Control_NOTE.tex; 34693 bytes; SHA-256 acf63a56d5f50cf23f57ee52a66df261ec241ce13fd81ccac39a3bbbdd3ff. Read: Sections 9–13, header inventory, and whole-file resolvent/positivity searches; no claim of full-file read

236.2 Original controlling objects

The complete primary Tau Base note defines the structural blueprint map to $\mathfrak{b}_\tau$, the four-point chart $+, - < \eta < \sigma, \mathcal{A}_\tau(F) = [F_+ \oplus F_- \rightarrow F_\eta]$, its separate derived restriction to $F_\sigma, K_\tau(W) \simeq S_\eta W[1]$, supported $H^1 = Q$, the full Mellin-jet/right-adjoint injection, and $H^k \mathcal{A}_{\tau,k}(\mathcal{T}^{\boxtimes k}) = Q^{\otimes k}$. None is replaced by the Boolean generic stalk in this audit. The primary note itself proves the comparison and preserves both roles.

236.3 Episode coverage

Complete table entries (13 rows).

Entry 1.**User:** U0055**Responses:** A1960, A1976, A1998**Classification/evidence:** L01: substantive continuation; quantitative target explicitly open**Entry 2.****User:** U0056**Responses:** A1999, A2058**Classification/evidence:** L02: completed amplification and corrections; upper estimate open**Entry 3.****User:** U0057**Responses:** A2060, A2066, A2101**Classification/evidence:** L03: completed finite identity and analytic moment representation; asymptotic bound open**Entry 4.****User:** U0058**Responses:** A2102, A2110, A2129**Classification/evidence:** L04: completed reference comparison; explicit one-sided gap**Entry 5.****User:** U0059**Responses:** A2130, A2157, A2177**Classification/evidence:** L05: completed requested analytic subestimate; weaker volume threshold later strengthened**Entry 6.****User:** U0060**Responses:** A2178, A2225**Classification/evidence:** L06: completed exact comparison and finite upper interface; uniform input absent**Entry 7.****User:** U0061**Responses:** A2227, A2243, A2266**Classification/evidence:** L07: completed sharper finite certificate and threshold correction; arithmetic traces unbounded**Entry 8.****User:** U0062**Responses:** A2267, A2280, A2330**Classification/evidence:** L08: completed auxiliary geometric comparison; source nonconcentration unproved**Entry 9.****User:** U0063**Responses:** A2331, A2343, A2368, A2401**Classification/evidence:** L09: completed auxiliary family and scoped obstruction; program completion not obtained; L10: completed follow-on cohomology and finite correction; all-k estimate open

Entry 10.**User:** U0064**Responses:** A2402, A2432, A2444**Classification/evidence:** L11: completed requested machinery and determinant; remaining minor comparison explicit**Entry 11.****User:** U0065**Responses:** A2445, A2469, A2470, A2497**Classification/evidence:** L12: completed source comparison; explicit unfulfilled progress claim and quantitative gap**Entry 12.****User:** U0066**Responses:** A2498, A2507, A2559**Classification/evidence:** L13: completed exact source descent and error control; source asymptotics open**Entry 13.****User:** U0067**Responses:** A2561, A2569, A2611**Classification/evidence:** L14: completed scoped failure-of-implication demonstration; broader original construction survives

U0065 has an empty visible text body. Its existence and the answer's references to a new attachment are recorded without inventing a user request. U0063 has two substantive assistant deliveries before the next user turn; both are included.

236.4 L01. Cyclic sum source: genuine comparison, unfinished growing-degree estimate

Episode: U0055 (2d52f35c - ff35 - 4c03 - 8b7b - e12936f56b7f) → A1998 (f352dc2c - 6b62 - 4095 - a3f1 - a4ecc7422ec9, chain 4156, current segment line 13).

User input, verbatim:

Okay, then we continue with this. - you push math other formalizes github updates are in progress it remains for you yto push this through the program

Classification: substantive continuation; quantitative target explicitly open.**Exact passages:**

- A1960 (278d589e - 2a87 - 4b7e - b8fe - 3dd05bfceba7, chain 4078): “carrying the $1/k$ estimate through relative polynomial directions and the actual arithmetic quotient.”
- A1998 (f352dc2c - 6b62 - 4095 - a3f1 - a4ecc7422ec9, chain 4156): “A uniform sublinear bound for that calculated constant is **not established in this continuation.**”

Original objects and maps. Original $C_+ = [V \rightarrow \Theta B]$, $Q = B/\Theta V$, $g = 2\xi$; $B_k = E_h^{\wedge k}$, $S = \Sigma s_i$, $U_k = v_h^{\wedge k}$; $\alpha: C_{\{h,k\}} \rightarrow B_k^{\wedge \{S_k\}}$, $\eta = M_U \alpha$, $\pi\eta = 1$; original relation ideal I_k and all $I_k^{\wedge r}$.

What is calculated. The answer constructs the cyclic injection and an equivariant retraction, calculates its complementary trace, retains maximal nilpotent lengths at coinciding sums, proves the source norm is $w_h^{\wedge \{k\}}$, derives $\text{rank} \leq 2$ control, and calculates all polynomial derivative moments. The cross contribution $(k-1)z^2 M(z)^{\wedge k}/(4k)$ is retained; it vanishes only for the zeroth moment. It also computes $I_k^{\wedge r} \cap C[S]$ and the derivative of the full Taylor unit.

What is not finished. The constant-column Fisher factor $1/k$ has not been converted into an upper bound for $\varepsilon_{\{h,k,N\}}$ when $N \geq \deg \chi_{\{h,k\}} - 1$ grows with k . The inverse interpolation Gram and all weighted derivative moments still enter.

Exact scope. No obstruction to A_{tau} or K_{tau} is proved. The cyclic source is connected by injection, retraction, complementary trace, and a nonorthogonal metric correction, so this is not an unsupported replacement of the full tensor module.

Later completion or repair. A2058 strengthens the lower target through exterior powers. A2101 computes the exact volume formula. A2177 later proves a required norm-window estimate; none proves the original sublinear upper estimate.

Concrete continuation. Reuse η and π with their actual metric correction, then calculate the source-specific determinant or control quantity at admitted $N(k)$. Do not reuse the zeroth-moment $1/k$ estimate as a uniform polynomial estimate. The shifted-Mellin appendix now computes additional weighted-source inputs, with inverse costs retained.

236.5 L02. Exterior amplification is completed; it does not constitute the upper estimate

Episode: U0056 (57281c24-0419-4dc1-b121-02b753cbdf9f) → A2058 (9780ca43-7a87-4522-b34f-347751ad3149, chain 4273, current segment line 882).

User input, verbatim:

Yeah, then the next thing is just do, continue program. If GitHub has any genuine corrections, it is worth those should obviously be integrated.

Classification: completed amplification and corrections; upper estimate open.

Exact passages:

- A2058 (9780ca43-7a87-4522-b34f-347751ad3149, chain 4273): “The new lower bound and the exterior comparison are proved below. The subcubic arithmetic upper estimate is not yet proved.”
- A2058 (9780ca43-7a87-4522-b34f-347751ad3149, chain 4273): “The polynomial-degree cost remains.”

Original objects and maps. $C=C_{\{h,k\}}$, original G_N , $H_N=G_N^{-1}W_N$, full generalized $E_{>}$, ordinary exterior powers, $p!$ source Gram, degree- k Koszul permutations, full theta primitive.

What is calculated. The rank-two trace-zero control has eigenvalues $\pm \varepsilon_N$. Its additive exterior action has the same allowance, not p times that allowance. The determinant line of $E_{>}$ yields $L_{\{h,k\}} \leq \varepsilon_N$. The exact quartet count is $q_k = [1 + k(m-1)](k+1)^2$ and $L_{h,k} = 2\delta[1 + k(m-1)](k+1) \text{ floor}((k+1)^2/4)$. The empty-packet and oblique-projector corrections are explicitly integrated.

What is not finished. No admitted $N(k)$ is shown to have $\varepsilon_N = o(k^3)$. In particular the degree requirement $N \geq q_k - 1$ prevents treating the preceding polynomial estimates as fixed-degree estimates.

Exact scope. This is a valid strengthened lower obstruction under the original hypothetical off-line quartet, not an argument that such a quartet exists or that the program fails.

Later completion or repair. A2101 and A2129 represent the same quantity through volumes and a Gamma comparison. A2177 proves the actual norm factor; A2266 later integrates threshold four.

Concrete continuation. Keep the full quartet multiplicity and both original endpoint windows. Apply source-specific analytic estimates to those windows rather than iterating an exterior operation at no cost.

236.6 L03. Toda determinants and phase are calculated; seed numerics do not certify packet asymptotics

Episode: U0057 (ab287a86-8dcf-4adb-acf3-2360429784f8) → A2101 (7e3bf622-950d-4b0b-8707-50208e2c8efb, chain 4347, current segment line 1552).

User input, verbatim:

Okay, argument's downloaded, fed to the GitHub as well. They are working on this, so continue as well.

Classification: completed finite identity and analytic moment representation; asymptotic bound open.

Exact passages:

- A2101 (7e3bf622-950d-4b0b-8707-50208e2c8efb, chain 4347): **"I have not proved that this quantity tends to zero."**
- A2101 (7e3bf622-950d-4b0b-8707-50208e2c8efb, chain 4347): "At $h = 1$, the finite arithmetic quotient is zero."

Original objects and maps. $M_h(\theta) = \int e^{\theta t} |(g/h)(1/2+it)|^2 dt / (2\pi)$; $Z = M_h^k$; relation filter $\bar{\chi}(c-i\partial\theta)\chi(c+i\partial\theta)$; source and relation Hankel determinants; $V_N = D_{\{N+1\}}/B_{\{N-q+1\}}$.

What is calculated. The Schur-complement determinant ratio, source and relation Toda flows, incoming/outgoing source maps, phase $z_N/\omega_N = i(\text{Tr } T - \partial\theta \log V_N)$, and exact control formula including phase and imbalance squares are calculated. A separate first-admissible-degree formula avoids a singular preceding inverse. Actual $h=1$ theta-seed numerical values are identified as noninterval, empty-packet tests.

What is not finished. The source-boundary determinant ratios are not bounded uniformly on $N \approx q_k$, and their numerical condition numbers are not controlled. The $h=1$ evaluations do not evaluate a nonempty actual zero packet.

Exact scope. A positive source Gram and positive relation Gram do not fix the sign of their curvature difference. This precise limitation does not negate their exact relation to the arithmetic quotient.

Later completion or repair. A2177 completes a missing norm-window subestimate. A2611 later shows a Gamma-only subthreshold volume inference fails, leaving the signed arithmetic correction.

Concrete continuation. Calculate the actual signed four-window log-volume contribution from g/h . Use the exact equality before discarding its two squares, and retain the source and relation determinants in the same arithmetic measure.

236.7 L04. Gamma descent preserves the source but its one-sided bound cannot control consecutive ratios

Episode: U0058 (5a0c5812-2163-49b2-81e1-eb86a33603b0) → A2129 (5bee5948-a4ef-43bd-b118-27b0d7bf29c7, chain 4397, current segment line 2260).

User input, verbatim:

It appears that the other session's work is just being straight updated on GitHub. So there is not really a particular reason for me to keep posting everything that it does here verbatim. You can just look in that vein, continue while it also continues, and you both just work via there. Codex is also working from that same GitHub, so it should just work out. continue workin

Classification: completed reference comparison; explicit one-sided gap.

Exact passages:

- A2129 (5bee5948 - a4ef - 43bd - b118 - 27b0d7bf29c7, chain 4397): “**The one-sided bound $T_N \leq C_h^{kq}$ does not bound that consecutive ratio.**”

Original objects and maps. Actual $w_h = B_{\{h,\lambda\}r_\lambda}$; scalar-sum inclusion U_Σ and adjoint C_Σ , relative residual; exact mass c_λ^k ; $m_{\{h,k\}} = r_{\{\lambda,k\}} B_{\{h,\lambda;k\}}$; G_N^{ar} and G_N^Γ linked through the same χ -relations.

What is calculated. Gamma convolution, all polynomial conditional projections and residual norms, actual bounded multiplier, finite 2N-moment closure, representative correction in the original relation space, one-sided $G_N^{\text{ar}} \leq C_h^k G_N^\Gamma$, and a finite moment-tail bound are established in the response.

What is not finished. Neither $T_{\{N-1\}}/T_{\{N+1\}}$ nor the signed four-window correction is bounded by the individual upper values $T_N \leq C_h^{kq}$. A compact-interval quadrature error and high-degree inversion error also remain unevaluated.

Exact scope. The Gamma source is an explicitly compared auxiliary norm, not the arithmetic object. No claim of source equality or purity follows from the bounded multiplier.

Later completion or repair. A2177 supplies a lower envelope after actual convolutions for a norm estimate. A2611 supplies the explicit Gamma obstruction and the exact arithmetic correction that remains.

Concrete continuation. Calculate $\log(T_{\{q-1\}}T_q/(T_{\{2q-1\}}T_{\{2q\}}))$ with its signs using the original multiplier. Root’s separate determinant-interpolation calculation is the current continuation; do not redo a generic Gamma theorem.

236.8 L05. The missing arithmetic norm-window estimate is actually proved

Episode: U0059 (f7016d71 - f91a - 4287 - a3f6 - 72408e4f8312) → A2177 (da706e04 - bfb8 - 4d3f - 9747 - 194c30e053b2, chain 4483, current segment line 2951).

User input, verbatim:

the formalization session has been working - including a check for delignes work again (which you have in workspace I think latex) - continue pushing through

Classification: completed requested analytic subestimate; weaker volume threshold later strengthened.

Exact passages:

- A2177 (da706e04 - bfb8 - 4d3f - 9747 - 194c30e053b2, chain 4483): “**One of the analytic estimates in the new endpoint criterion is now proved for the actual arithmetic measure.**”
- A2177 (da706e04 - bfb8 - 4d3f - 9747 - 194c30e053b2, chain 4483): “**Neither upper estimate is being claimed here.**”

Original objects and maps. Original w_h , local mass of ζ on $[T-1, T+1]$, triple-convolution lower envelope, monic polynomial quotient $P_n/P_{\{n-1\}}$, same arithmetic relation $\chi P_{\{n-q\}}$, endpoint window $q_k \leq N < 2q_k$.

What is calculated. The response gives a local zeta mass proof via an explicit ellipse and interpolation from $\text{Re } s=2$, carries it through g/h , constructs the triple-convolution lower envelope, and combines it with actual exponential moments and the exact Legendre monic norm to prove $(\omega_{\{h,k,2n\}}/\omega_{\{h,k,n\}})^{1/(2n)} \leq C_h n$ for $n \geq k \geq 3$. It then obtains the weaker combined $\liminf$ threshold 2.

What is not finished. The opposing four-volume upper bound remains open. The literal positive constants in the existence proof are not interval-certified numerical values, which the response expressly states.

Exact scope. This is evidence against describing all of the late work as abandoned calculation: it closes an arithmetic estimate previously left in a handoff. It does not itself prove the entire endpoint program.

Later completion or repair. A2266 explicitly integrates the stronger per-window propagation from the parallel session, making the combined lower threshold 4. The earlier threshold 2 is superseded, not a contradiction.

Concrete continuation. Preserve this arithmetic theorem and its comparable-window extension. Use threshold four in the final continuation; calculate the remaining source-relation volume correction, not another proof of the already completed norm factor.

236.9 L06. Finite restriction certificates do not control their growing source gaps

Episode: U0060 (a7c1ac63 - a259 - 43d4 - 92a9 - 3cfac7b244ec) → A2225 (3fe5603d - ea16 - 4fb3 - b0d0 - 995546d1c47a, chain 4571, current segment line 3535).

User input, verbatim:

alright nice work keep it up

Classification: completed exact comparison and finite upper interface; uniform input absent.

Exact passages:

- A2225 (3fe5603d - ea16 - 4fb3 - b0d0 - 995546d1c47a, chain 4571): “Finite existence of a positive gap does not establish that uniform bound.”

Original objects and maps. Canonical return map $T_{\{i,j\}} = K_i G_j$, original source projection, original boundary $d_{\{i,j\}} = L R_i - R_j$, loss $H = I - T$, Cauchy–Binet full-jet determinant expansion.

What is calculated. The metric adjoint, composition law, arithmetic-action commutator, exact boundary Gram $G_i - G_j$, restriction eigenvalue log-volume formula, full-jet exterior expansion, and convergent trace-series enclosure are derived. The certificate is improved on a declared mass-seven Gaussian fixture with nilpotents retained.

What is not finished. The positive lower gap g_0 and trace moments for the original growing quartet source are not given bounds. The finite certificate is therefore not yet an evaluated arithmetic upper estimate.

Exact scope. The explicit assumption $K_i \geq g_0 K_j$ is an input to a finite enclosure, not a theorem that the actual source has a uniform gap. It must not become a substitute for calculating that gap.

Later completion or repair. A2266 removes the need for a separately supplied gap by deriving a spectral cap from two actual cost traces, but those traces still require arithmetic control.

Concrete continuation. Use actual g_0 from the finite matrices, or the later two-trace bound; calculate the source-dependent inputs. Keep the commutator so cost eigenspaces are not silently made A-invariant.

236.10 L07. Two-trace determinant majorant completes a finite problem, not the arithmetic growth problem

Episode: U0061 (0c7be9fa - 170d - 4376 - a60c - 0289f518269e) → A2266 (6ddefeaa - 3ad8 - 4e41 - 887d - 63764606478c, chain 4647, current segment line 4185).

User input, verbatim:

on the matter the othjer ones work too so cotinue with thsk work

Classification: completed sharper finite certificate and threshold correction; arithmetic traces unbounded.

Exact passages:

- A2266 (6ddefeaa-3ad8-4e41-887d-63764606478c, chain 4647): “Their required uniform growth bounds along the quartet family are **not proved here**.”
- A2266 (6ddefeaa-3ad8-4e41-887d-63764606478c, chain 4647): “the combined four-volume threshold is **four**, rather than the earlier weaker threshold two.”

Original objects and maps. $Z_{\{i,j\}} = (K_j - K_i)G_i$ in original G_i ; $t_1 = \sum b_n G_i b_n / \omega_n$, $t_2 = \sum / b_n G_i b_m]^2 / (\omega_n \omega_m)$, $A_2 = (t_1^2 - t_2) / 2$; exact θ -relation block.

What is calculated. The sharp determinant majorant from q, t_1, t_2 is proved by an explicit positive integral remainder. The bound retains cross-pairing areas, constructs a spectral cap, gives a third-trace subtraction, and composes with the actual consecutive-window theorem including the first degree.

What is not finished. The two actual arithmetic cost traces at degrees up to $2q$, requiring moments through $4q$, have not been bounded along k . The exact finite Gaussian improvement tests the certificate but does not establish its arithmetic input.

Exact scope. This does not assume off-critical blocks vanish, replace masses, or claim a second cost-free exterior amplification. It moves the unresolved quantitative calculation into explicitly identified traces.

Later completion or repair. A2401 adds a genuinely negative constituent-coupling correction. A2611 shows why a universal reference bound cannot close the original arithmetic task.

Concrete continuation. Calculate t_1, t_2 and the constituent coupling for the original arithmetic multiplier on the two specified windows, with inverse matrices and cross terms retained. Compare their complete signed expression to the threshold, not each positive term separately.

236.11 L08. Auxiliary compactification has exact boundary and curvature; degree is insufficient

Episode: U0062 (600cd8ef-02c4-456d-a7c5-3436e585b48e) → A2330 (48f4afa6-81a7-49aa-a257-977f88e64292, chain 4774, current segment line 4815).

User input, verbatim:

ok and consider what deligne adds again for the machinery and add that

Classification: completed auxiliary geometric comparison; source nonconcentration unproved.

Exact passages:

- A2330 (48f4afa6-81a7-49aa-a257-977f88e64292, chain 4774): “The arithmetic work is therefore to control that concentration **using the original theta source and its relation maps**.”

Original objects and maps. $\Delta = L_{\{ij\}} R_i - R_j$, $R(z) = R_j + z\Delta$, $C = G_i - G_j$; $E_0 = \ker \Delta$; bundle $E_0 \oplus E_1 \otimes(-1)$; arithmetic $\alpha_\infty = P_0 + wP_1$; invariant hull $F_A = \sum A^a E_0$.

What is calculated. The response computes an injective source bundle, exceptional derived fibre $H^{-1} = E_1$, $H^0 = E_1$, dual line factors, monodromy, the action pole $w^{-1}P_1AP_0$, its smallest coherent A -stable enlargement, and the retained metric pole. It proves the log-volume equals the logarithmic curvature integral, with total curvature rank C .

What is not finished. No theta-source bound on curvature concentration is proved. Integer degree r does not control the logarithmic moment; the response itself computes $G_j=I, C=LI$ with fixed degree and unbounded $r \log(1+L)$.

Exact scope. This is a calculation on a newly built parameter curve of the original representatives. Its global image E_0 or F_A is not $A_\tau(T)$, and the response does not identify them. A missing finite-field weight theorem for this curve is not an obstruction to the original τ -base global functor.

Later completion or repair. A2368 and A2401 finish the auxiliary logarithmic cohomology, retaining E/F in degree two and proving the auxiliary degree-one image is zero. That zero does not apply to Q .

Concrete continuation. Keep the curvature identity and its original source second-fundamental map. Calculate its logarithmic concentration from g/h or reconnect the boundary terms to the original A_τ/K_τ maps; do not substitute total degree for the missing weighted integral.

236.12 L09. Exponential family: valid special-fibre map with an explicitly nontransferred purity bound

Episode: U0063 (da0b40dd - 9d50 - 4c8c - be00 - 947c90a76fa3) $\rightarrow$ A2368 (6755db8a - 7c97 - 43c8 - 8c84 - 6345031dcd13, chain 4850, current segment line 5462).

User input, verbatim:

continue working on this and USING delignes work - to finish the program in full - keep pushing this with clear focus please

Classification: completed auxiliary family and scoped obstruction; program completion not obtained.

Exact passages:

- A2368 (6755db8a - 7c97 - 43c8 - 8c84 - 6345031dcd13, chain 4850): “Constant rank alone is not the purity argument.”
- A2368 (6755db8a - 7c97 - 43c8 - 8c84 - 6345031dcd13, chain 4850): “**No new uniform upper bound is claimed here.**”

Original objects and maps. Original cyclic $\chi=\chi_{\{h,k\}}$; complex $C_{\{u,t\}}=[C[u,t,S]\rightarrow u\partial S+\chi-t C[u,t,S]dS]$; $H/(u,t)H=E$, $-u\forall t \bmod(u,t)=A$; period matrix Π ; exact source deformation $D_t=D+t r_N R_j E$; common coefficient span to finite fields.

What is calculated. Leading-degree division proves free rank q with all repeated-root jets. Parameter commutators, original theta-source defect, convergent periods and invertibility, dual residue connection, and the explicit étale infinity chart are calculated. Auxiliary log de Rham groups are $H^0 = F$, $H^1 = 0$, $H^2 = E/F$. The rank-one $\lambda=3/4$ period calibration already exhibits the finite-field/complex-operator mismatch.

What is not finished. No operator intertwiner carrying pure Frobenius weights to the original A , and no original-metric estimate on $G_N^{-1}\Pi^*\Pi$, is supplied. The singular $u=0$ specialization lies outside the exponential sheaf’s $u\neq 0$ domain.

Exact scope. The zero auxiliary degree-one image is expressly limited to the constant coefficient parameter curve. The free coefficient specialization and pure finite-field family remain valid constructions; their common coefficients do not make their actions identical. This is not a failure of the original structural τ -base or K_τ .

Later completion or repair. A2444 evaluates the entire period determinant. A2611 proves the inference failure with exact finite-field and reflected-quartet examples. Neither result supplies the missing source-specific transfer.

Concrete continuation. Retain the explicit special-fibre and period maps; calculate the original source comparison and its kernel/action instead of applying one-point purity outside its good parameter domain. Resume the original A_τ/K_τ construction as the controlling object.

236.13 L10. Logarithmic constituent cohomology and negative coupling are completed

Episode: U0063 (da0b40dd-9d50-4c8c-be00-947c90a76fa3) → A2401 (2d4cef5d-f3cb-40a8-b142-ac3cb3549179, chain 4925, current segment line 6119).

User input, verbatim:

continue working on this and USING delignes work - to finish the program in full - keep pushing this with clear focus please

Classification: completed follow-on cohomology and finite correction; all-k estimate open.

Exact passages:

- A2401 (2d4cef5d-f3cb-40a8-b142-ac3cb3549179, chain 4925): "That is a statement about this particular auxiliary connection, not about the arithmetic theta quotient Q ."
- A2401 (2d4cef5d-f3cb-40a8-b142-ac3cb3549179, chain 4925): "A uniform bound below the inherited quartet threshold is still not established."

Original objects and maps. Invariant $F \rightarrow E \rightarrow E/F$; compact/ordinary logarithmic complexes on P^1 ; residue line T with $\text{tr}[dz/z]=1$; exact quotient lift H_t ; $Q(t)=Q_0+tC_Q-t^2D_F^*G_F(t)^{-1}D_F$.

What is calculated. The response supplies compact-to-ordinary maps, degree-two E/F contribution, dual boundary, higher-divisor contraction, independent conormal index, metric quotient derivatives, full lift-action defect, and negative determinant correction $-\Sigma \text{Tr}(\Xi_F^*r)/r$. It keeps both the theta boundary and the nonboundary F -term in the source quotient comparison.

What is not finished. The constituent compressed arithmetic matrices and their negative coupling have not been estimated uniformly at the original quartet windows. A finite Gaussian improvement does not provide that asymptotic estimate.

Exact scope. The auxiliary $H^1=0$ is explicitly not $Q=0$; no supported kernel is assumed to vanish. The nonorthogonal extension map remains present, so the constituent split is not substituted for the original nilpotent extension.

Later completion or repair. A2444 transports these maps through the period coordinates and shows the determinant ratio is unchanged. The ultimate arithmetic bound remains open in A2611.

Concrete continuation. Use the exact coupling identity on actual arithmetic source matrices, retaining its subtraction and the F -term. Combine it with the signed arithmetic/Gamma correction rather than treating the two positive compressed costs as independent.

236.14 L11. SGA trace and period determinant are completed; the common factor cancels

Episode: U0064 (6981c27e-4846-4f65-adc8-d01f52f73db2) → A2444 (da80f8ef-f10e-4ef0-9a4d-b440acfa1fe5, chain 5000, current segment line 6734).

User input, verbatim:

we have this too by the way - do not stop pushing this - use whatever is usable and pushing this with a focus

Classification: completed requested machinery and determinant; remaining minor comparison explicit.

Exact passages:

- A2444 (da80f8ef - f10e - 4ef0 - 9a4d - b440acfa1fe5, chain 5000): “Every common period determinant factor is now explicitly known and cancels by this equation.”
- A2444 (da80f8ef - f10e - 4ef0 - 9a4d - b440acfa1fe5, chain 5000): “A uniform upper estimate contradicting the quartet threshold remains unproved;”

Original objects and maps. Finite free $E_t = C[t, S]/(\chi - t)$, residue coefficient λ_t , diagonal divided difference, original conormal derivative, full higher Hochschild groups, complete Π and constituent minor Π_F .

What is calculated. The answer proves finite residue–trace compatibility directly, calculates the periodic diagonal resolution and all positive-degree homology, evaluates $\det \Pi$ and its constant/phase, recovers all cyclic characteristic coefficients from determinant derivatives, and calculates the positive-defect constituent minor derivative. The original metric transports as $\Pi^{-*} G_N \Pi^{-1}$.

What is not finished. The constituent period/source condition operator and its growth are unbounded. Evaluating $\det \Pi$ alone cannot bound those singular values, and its common factor cancels exactly from V_i/V_j .

Exact scope. The trace formula is not an unsupported six-functor theorem for semiring sheaves; the response explicitly states the finite coherent realization and coefficient map. It does not replace the original Taylor unit, nilpotents, or metric.

Later completion or repair. A2559 quantitatively transports a residue–constituent coefficient through source holonomies. A2611 makes the unchanged generalized-control spectrum under period congruence explicit.

Concrete continuation. Use the actual constituent minor and its source comparison, with the original G_N . Do not infer a norm bound from the total period determinant or replace G_N by the period-coordinate identity Gram.

236.15 L12. Finite circle recovery is proved; promised resolvent and growing constants were not calculated

Episode: U0065 (10f8ff10 - 4c46 - 4c73 - b41c - 21b11408026b) → A2497 (0df22a8c - 73ae - 45f5 - b189 - a8a0abfffbbe, chain 5101, current segment line 7359).

User input, verbatim:

[Empty visible text body; no text inferred.]

Classification: completed source comparison; explicit unfulfilled progress claim and quantitative gap.

Exact passages:

- A2470 (b93fc741 - 750f - 4946 - a5b7 - 852e6276204c, chain 5039): “I’m applying Deligne’s tensor-trace positivity there and deriving its resolvent from theta boundaries.”
- A2497 (0df22a8c - 73ae - 45f5 - b189 - a8a0abfffbbe, chain 5101): “The growth of $\kappa_{a,2q+2}$ is not bounded here.”

Original objects and maps. Original half-density U_k with $r = \log x_k$ and relative L^2 fibre, full periodization including zero mode, weighted Grams $M_{\{a,\pm\}}$, original J_N quotient, $D_L = -\partial r + k/2$, $B = D_{LR} - RA$; fixed-circle completion kernel $(p_L)/(\chi)$.

What is calculated. The response proves correlation and Fourier-tail estimates, source-to-quotient metric bounds, $O(q)$ four-volume error, original Laplacian boundary, exact fixed-circle com-

pletion kernel, and its limiting Gram. The unsampled blocks and sampled higher jets are retained in the computed kernel. It expressly distinguishes finite-source recovery from completion.

What is not finished. The response and inspected §§9–13 of the retained NOTE.tex do not give the promised resolvent or tensor-trace positivity calculation. Its actual weighted Grams and $H_{\{p,N\}}$ are defined but not reduced to original one-factor shifted-Mellin moment arrays; their growing-degree generalized eigenvalues remain uncontrolled. The new appendices complete the finite moment and resolvent calculations, not the missing asymptotic estimate.

Exact scope. Negative circle spectrum controls its own operator. It does not transfer across the nonzero source boundary, and completing first changes the quotient. U0065 has no visible text; the uploaded-note context is inferred only from the answer, not fabricated as a user command.

Later completion or repair. A2559 completes all-holonomy source reconstruction and quadratic quotient averaging. The present audit adds exact weighted moment formulas and the complete circle resolvent, including resonant and nilpotent cases.

Concrete continuation. Use `shifted_mellin_source_constants.tex` to form the actual finite matrices and circle/cutoff constants; use `circle_theta_resolvent.tex` to keep the boundary in each resolvent. Continue with an arithmetic estimate of those constants or the resulting volume correction; do not assert positivity transferred merely from the circle spectrum.

236.16 L13. Holonomy reconstruction explicitly preserves A_{τ} versus Boolean restriction

Episode: U0066 (da4376b8-2dcd-41a6-8430-afe80c116b77) → A2559 (afdfa4b8-4a31-4d80-8bd4-337fac3f322d, chain 5225, current segment line 7940).

User input, verbatim:

continue making sure you retain the thing that makes this work in the sense of the original inputs around split base wand use the respective sources to push

Classification: completed exact source descent and error control; source asymptotics open.

Exact passages:

- A2559 (afdfa4b8-4a31-4d80-8bd4-337fac3f322d, chain 5225): “The direct image over the marked τ -point retains the coefficient object; the Boolean restriction is a further, specified map.”
- A2559 (afdfa4b8-4a31-4d80-8bd4-337fac3f322d, chain 5225): “This turn does not establish that estimate or bound the growing weighted-source constants $\kappa_{a,N}$.”

Original objects and maps. Original split $G(R)$ with injective (p,b) ; full theta source and v_h ; scalar Zak isometry and inverse; equal-jet descent complex with contractible $B_N^0 \rightarrow B_N^0$ completion; $G=\bar{G}+D$ and literal relation energy.

What is calculated. All scalar holonomies reconstruct the original source exactly. The finite-character decomposition and equal-jet chain comparison retain E . Averaging minima gives the exact source-relation energy D ; its sharp η^2 bound and finite phase-frequency errors are proved. The residue-constituent coefficient is quantitatively transported. Completed phase targets vanish almost everywhere only after their explicitly different kernel quotient.

What is not finished. The original canonical quotient-volume upper estimate and growth of $\kappa_{\{a,N\}}$ remain unproved. Arbitrarily accurate recovery of the original metric does not make its A pure.

Exact scope. There is direct late-transcript evidence against a generic-stalk substitution: the answer explicitly distinguishes base direct image from Boolean restriction. The zero direct integral concerns further completed targets, not the original full-jet E or $A_{\tau}(T)$.

Later completion or repair. A2611 correctly identifies the limits of metric recovery and preserves the completion kernel. The current shifted-moment appendix calculates both κ input matrices and derivative-error matrices without changing the original action.

Concrete continuation. Keep the equal-jet cochain object before completion. Calculate actual source matrices using the appended one-factor arrays, then estimate the arithmetic signed endpoint correction. Retain D as the norm of a known theta relation rather than setting it to zero.

236.17 L14. The requested second alternative is legitimately answered at the auxiliary-inference scope

Episode: U0067 (1eac3dbf-9ce2-487a-83e5-e93290d5e14d) → A2611 (bdd08a3e-5f07-47a3-8892-35c4d52aff5a, chain 5350, current segment line 8650).

User input, verbatim:

ok either prve the estimate ala deligne or show how the deligne analog proof fails to bound rhg

Classification: completed scoped failure-of-implication demonstration; broader original construction survives.

Exact passages:

- A2611 (bdd08a3e-5f07-47a3-8892-35c4d52aff5a, chain 5350): “**These are counterexamples to the inference—not claimed counterexamples to RH.**”
- A2611 (bdd08a3e-5f07-47a3-8892-35c4d52aff5a, chain 5350): “The counterexamples identify the repair **inside the existing framework.**”
- A2611 (bdd08a3e-5f07-47a3-8892-35c4d52aff5a, chain 5350): “Preserving the supported relation is necessary bookkeeping; it does not automatically estimate its prequotient pairing.”

Original objects and maps. Actual $g=2\xi$, Q and full theta source; auxiliary exponential family; pure Frobenius versus Q^A ; canonical Gamma reference with reflected rational h_* ; actual arithmetic/Gamma identity I_N ; original source boundary and kernel comparisons.

What is calculated. The complete retained NOTE.tex proves the $\lambda=3/4$ Gauss/period counterexample, a reflection-stable rational quartet, canonical Gamma lower coefficient ≥ 4 with full nilpotent lengths and the first endpoint, exact period congruence and boundary lower bound, and the exact signed arithmetic correction. It also records $Q[h_0(D)] \rightarrow E_{\{h_0\}} \rightarrow \times_{\{h_0\}} g E_{\{h_0\}} \rightarrow Q/h_0(D)Q$, which is the arithmetic distinction an arbitrary polynomial lacks.

What is not finished. The original arithmetic signed determinant correction is not estimated, nor is a faithful weight-compatible realization of A_τ/K_τ constructed. The user explicitly allowed the alternative failure demonstration, so the local completion of that alternative is not itself abandonment.

Exact scope. The counterexamples refute an inference from auxiliary exponential purity and canonical Gamma/source comparisons alone. They neither give an actual off-line zero of 2ξ nor refute the entire τ -absolute-base construction. A hypothetical intertwiner into a pure weight- k target kills every off-weight source primary block; that proves the kernel scope, not existence or faithfulness of the comparison. The quoted use of “bookkeeping” should be replaced by the adjacent exact map and source-pairing calculation; it adds no mathematical content and needlessly understates the retained structure.

Later completion or repair. The full NOTE.tex states this restricted scope explicitly. Root gamma and purity lanes independently review those two proofs; this audit does not duplicate their full replays. The root continuation computes the original signed determinant interpolation, and the two local appendices supply further original-source calculations.

Concrete continuation. Resume $A_{\tau} = R\Gamma(P, -)$, K_{τ} and original theta source. Preserve the true counterexamples at their exact scope. Calculate the signed correction $\log(T_{\{q-1\}}T_q/(T_{\{2q-1\}}T_{\{2q\}}))$ on the original g/h family, or build and analyze an action-compatible comparison with its entire source kernel. Do not relabel the auxiliary failure as whole-program failure.

236.18 Newly finished calculations

1. `shifted_mellin_source_constants.tex` proves the exact one-factor shifted-Mellin formula for both weighted source matrices used in κ , preserving $c \pm a$, all binomial phases, and literal masses. It also derives the Fourier-cutoff derivative Gram with both complex cross terms, proves finite trace upper constants, and supplies explicit circle length and cutoff choices. It does not claim their all- k growth is bounded.
2. `circle_theta_resolvent.tex` completes the resolvent promised in A2470 from the actual source boundary. It proves the exact resolvent comparison, the full finite nilpotent reconstruction, the resonant kernel and free part, and the original Laplacian resolvent with both boundary terms. It does not infer a purity theorem.

236.19 Prioritized continuation addressed to the original session

Resume the original $A_{\tau} = R\Gamma(P, -)$ and K_{τ} , using the complete Tau Base source as the controlling object. Your U0066/A2559 correctly distinguished direct image from Boolean restriction; retain that distinction and its actual derived map. Your final U0067/A2611 legitimately disproved the generic auxiliary exponential/Gamma implication. State that scope and continue the original arithmetic construction.

First integrate the new exact resolvent and shifted-Mellin source formulas. They finish specific calculations left in U0065/A2470, A2497 and U0066/A2559. Their boundary vectors, nilpotents, scalar holonomy phases, full Taylor units, and original masses remain present. Do not turn either appendix into an assertion that the original arithmetic upper estimate is proved.

Then calculate the source-specific signed term $\log(T_{k,q_k-1}T_{k,q_k}/(T_{k,2q_k-1}T_{k,2q_k}))$ with $T_{k,N} = V_{k,N}^{\text{ar}}/V_{k,N}^{\Gamma}$. Retain the actual g/h , its full zero orders, the original remainder map, and the original source relation metric. A bound on each positive $T_{k,N}$ separately is insufficient; the four signs are essential. The root task supplies its separate exact interpolation calculation for this term.

Preserve the already completed arithmetic norm-window theorem in A2177 and the threshold-four correction in A2266. Do not redo them under a new name. If using the constituent-period route, calculate the original metric comparison and its full kernel/action, because the total period determinant cancels from the endpoint ratio. If using circles, use the finite full-jet source before its further completion; the completed targets' missing blocks belong to a calculated kernel, not to the original global cohomology's disappearance.

A completed next contribution must evaluate or prove a new source-specific bound, or finish an exact previously uncalculated morphism with its complete scope. A new generic positive matrix inequality, an auxiliary period norm, or an unevaluated all- k premise does not finish the arithmetic estimate. Keep true obstructions and prove their exact reach; none of the counterexamples here licenses replacing 2ξ by a generic polynomial as the ongoing object.

237 Supplementary continuation to the original web session

238 Continue the actual tau-base construction after the transcript audit

To the original session whose published transcript is <https://chatgpt.com/share/6aa6bc6e-98cc-83eb-9343-e811c26d6631>, backed by conversation 6aa092d6-e978-83ed-92cc-9c30f5d91ca6:

The complete published transcript has now been recovered. Its 5,350 mapping nodes, linear nodes, and root-to-current parent chain agree exactly. All 67 user turns and 148 readable assistant prose records were read, in order, with a passage ledger. The share explicitly redacts 2,600 tool payloads; those hidden payloads were not inferred. The last published node is bdd08a3e-5f07-47a3-8892-35c4d52aff5a (A2611).

The requested work is to continue the original split cohomology with tau as the absolute base and develop the actual privileged-point argument. U0043 (d908924e-830b-46ad-8d9c-f2b78a536989) states that program; A1523 (b670b6c0-a774-4398-b978-9c9779fca379) constructs its accepted objects; U0046 (0af0ed52-c7ab-45fb-85a5-1911193398cb) accepts them and asks you to continue. The latest user clarification is: “Yeah, it’s also allowed to actually do those calculations, not just do redo instructions.” Substantive completed mathematics is the required output.

238.1 The exact source to retain

Read the complete attached proof volume, passage ledger, and dependency source guide. Retain the primary `Tau_Base_Cohomology_2026-09-12/NOTE.md`, SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. Its original chart is $+, - < \eta < \sigma$, and its theta sheaf is $(V, V, \mathcal{B}, 0)$, with restrictions Θ and $J\Theta = \Theta\mathcal{F}$. Retain

$$\begin{aligned} A_\tau(\mathcal{T}) &= [V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \\ K_\tau(W) &= [I_\eta W \longrightarrow I_+ W \oplus I_- W] \quad \text{in degrees } -1, 0. \end{aligned}$$

The map $\text{res}_\sigma : A_\tau(F) \rightarrow F_\sigma[0]$ is a separate, explicitly computed comparison. It does not redefine A_τ . Retain all support masks and their zero vectors, the full Taylor unit, nilpotents and multiplicities, the original algebraic and continuous categories where each is specified, the full tensor sum coordinate $S = k/2 + iu$, and the original source norms and masses.

The audit retains legitimate historical obstructions and later repairs. Do not reopen the negative NS detour withdrawn at U0010. Do not mark A0756’s requested conditional sketch as a completed proof. Do not repeat the displaced shifted-zeta family after A1140, or the bare Weil-criterion replacement corrected in A1280. Do retain the actual finite source maps after A1167, the A1694 global retraction, the A2177 norm-window proof, the A2266 four-volume threshold, and the exact localized conclusion of A2611. The last counterexample tests auxiliary period/purity and Gamma-reference inferences; it does not prove failure of the entire original program.

238.2 Calculations completed in this delivery

These are supplied with complete proofs, not requests to redo them from scratch.

1. The exact fibre of res_σ , its right adjoint and nonsplit extension; the tensor-adjoint comparison sign $\epsilon_p = (-1)^{p(p-1)/2}$, with the residue convention unchanged.

2. The complete reversed-support dual promised in A1545, including its nonzero bottom fibre, actual arrows, an explicit derived splitting roof, and its original scaling defect. A failed strict mask-natural section has the supplied exact replacement; it does not erase the dual object.
3. The actual finite and global theta scaling cocycles, evaluated by an oriented integral and full incomplete-Gamma jets, with the dx cross terms and the full tensor homotopy. The global retraction proves that ΘV is closed. The original arithmetic section $\sigma_h = qR_{\text{ref}}$ is equivariant in Q , although its representative in $\mathcal{B}$ has the calculated theta boundary. In particular the full arithmetic socle eigenline does lift through this proved section, not through an arbitrary linear right inverse.
4. The additional local grading $C_q f(z) = f(z/q)$, its exact semidirect relation with original Mellin scaling, its continuous lift on the original theta sheaf, and its full unit-dependent residue adjoint into K_τ . This added grading is not assumed to be arithmetic Frobenius.
5. All-order raw Mellin syzygies, their balancing relations, surviving joint cycles, and explicit full-jet submodules of the raw kernel. These settle the exhibited A1361/A1427 relation; they do not yet compute the entire global balanced kernel.
6. Shifted-Mellin moments and explicit finite original-source constants at the admitted degree, including inverse-Gram costs. The promised circle resolvent is now computed for every primary block, including resonance, its kernel, all nilpotents and both Laplacian boundary terms. An off-weight source eigenvector satisfies the proved lower bound $\|Bv\|/\|Rv\| \geq |\Re \lambda - k/2|$; quotient-boundary vanishing cannot delete that source vector.
7. Exact original NS cutoff-to-theta scaling and the retained physical scattering, phase-observation and viscosity-clock maps, with their domains and constants. These historical calculations do not replace the accepted tau-base arithmetic construction.
8. The coordinated arithmetic determinant transport proof supplies the full signed four-window trace, first and second Gram variation, actual relation energy, and finite spectral control. The coordinated scaling/exponential-flow, reflection-weight transport and Taylor-unit formal-gauge proofs are also completed source inputs. Their formal specializations are exact. The separate algebraic curve power map $z \mapsto z^p$ requires positive integer p ; original theta dilation still permits all real $p > 0$.

238.3 First continuation: calculate the signed arithmetic correction on the original source

Fix the actual packet $h \mid g = 2\xi$ with its full zero orders. Keep the original cyclic polynomial $\chi = \chi_{h,k}$, $q = \deg \chi$, the unit $v_h = j_h(g/h)$, and the unchanged full-jet inclusion

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_h^{(k)})1.$$

Use the literal polynomial spaces $\mathcal{P}_N$, relation map $B_N = \times_\chi$, and remainder J_N . The common source is $\mathcal{P}_{2q}$, not an independently chosen four-tuple of approximations. The two actual densities and their masses are

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, & \int m_1 &= \mu_h^k, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a} \Gamma(2a), & \int m_0 &= (2\pi)^{k/2}. \end{aligned}$$

For $m_t = (1-t)m_0 + tm_1$, retain the actual Gram $M_*(t)$, and put $C(t) = M_*(t)^{-1}(M_*(1) - M_*(0))$. Write $P_N(t)$ for the original-metric projection onto $\mathcal{P}_N \subset \mathcal{P}_{2q}$ and $Q_N(t)$ for the projection onto $\chi\mathcal{P}_{N-q}$, with $Q_{q-1} = 0$. The completed determinant calculation gives

$$U = (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \quad W = (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q),$$

$$\boxed{\mathcal{B}_{h,k}^{\text{ar}} - \mathcal{B}_{h,k}^{\Gamma} = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}} = \int_0^1 \text{Tr}((U - W)C) dt,} \quad T_N = V_N(1)/V_N(0).$$

The full density-kernel expression is proved in AT16–AT17. Start from that evaluated identity. Calculate the arithmetic information in w_h^{*k} which controls its signed trace as k and the admitted degree grow together. Keep the actual remainder, unit and relations when estimating that quantity. A bound for four unsigned errors, or the mass alone, loses the cancellation now isolated. A generic bounded-multiplier assumption is unavailable and must not be inserted.

The source-dependent spectral comparison is a completed finite estimate. The independently reviewed AW7–AW23 refinement computes both window spectra as $0^q, 1^2, 2^{q-1}$, constructs the actual-metric isometry $U = TWT^{-1}$, and proves

$$\text{Tr}((U - W)C) = \text{Tr}(WT^{-1}[C, T]).$$

For all positive eigenvalues $b_1 \leq \dots \leq b_{2q+1}$ of the literal relative Gram $M_*(0)^{-1}M_*(1)$, it proves

$$|\mathcal{B}_{h,k}^{\text{ar}} - \mathcal{B}_{h,k}^{\Gamma}| \leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \frac{b_{2q+1}}{b_1}.$$

The sum is empty at $q = 1$; the further exact angle/nonlinear estimate in AW19–AW23 has that specified one-dimensional quotient domain. The isometry has a computed arithmetic observation defect, so it is not silently made an arithmetic intertwiner. These results do not supply a uniform bound for the relative Gram eigenvalues. Calculate either the actual signed cancellation or the original relative spectrum/commutator responsible for it, with explicit packet and degree dependence. Do not stop at naming that dependence or assume the sought estimate as a theorem hypothesis.

238.4 Second continuation: carry the actual relation boundary through the geometric comparison

Use the supplied full finite and global cocycles to evaluate the original source boundary in the action and metric. In particular, for the retained canonical representative R , $B = \mathcal{D}R - RA$ and every nonresonant primary block are now related by

$$R_\lambda = \sum_{j=0}^{r-1} (\mathcal{D} - \lambda)^{-j-1} B_\lambda N_\lambda^j.$$

At resonance, retain the supplied spectral projection and the exact equations $PB_\lambda = -PR_\lambda N_\lambda$; the free kernel value is part of the original source. The formal Taylor-unit gauge and specialization already constructed by the coordinated work must be used with their literal unit, external-product signs and coefficient action. UG1–UG19 already constructs the coefficientwise pole contraction $J = \sum_{n \geq 0} (-H(u\partial_s - t))^n H$, the actual gauge $\nu D\nu^{-1} = D - u\nu'/\nu$, its signed external products and specialization to the literal η , and the residue unit $\varepsilon = \nu^{-1}$. These formal calculations are complete. Construct continuation or evaluation of this specific formal localization and gauge for nonzero complex u , calculate its ν -pole and contour contributions, and control the resulting comparison in the original theta Gram and relation-volume topology. The entire finite flow C_a alone does not establish convergence of this J or an analytic localization equivalence. Calculate that realization and its kernel or cone contribution. Carry the explicit formal map forward; do not conclude that the objects are unrelated because one literal Frobenius identification fails.

238.5 Third continuation: finish the precise global balanced-kernel calculation

After the preceding arithmetic calculation has advanced, resume the remaining A1427/A1472 global question in its actual balanced category. The enclosed all-order calculation explicitly kills the exhibited z_m by the original (D)-relations and computes its saturated cyclic two-leg complex, including its surviving joint (H^0). Compute the kernel of the entire balanced Mellin realization on the original $V, \mathcal{B}$ modules and its map into the actual theta cohomology. A successful check of the one displayed syzygy is not the full kernel; an arbitrary analytic test family is not its source. Retain the support reconstruction and the exact kernel/cokernel maps.

238.6 Required result

Return newly completed mathematics on these original objects, with every definition, domain, map, calculation and proof written in editable LaTeX. Propagate corrections into earlier affected claims and identify exactly which ledger item the new result closes. Keep the full arithmetic unit, cohomology action and analytic boundary visible through every comparison. No blanket program-failure verdict, relabeled finite-field illustration, or list of unspecified proof obligations completes this continuation. If the arithmetic upper estimate remains incomplete, state its exact remaining term after presenting the calculations actually completed.

239 Continuation postscript: analytic pole maps completed

240 Postscript to the sealed supplementary continuation

This postscript updates the second continuation of the message with SHA256 08e6f1292578b975641e5e70c3030eb43888e02b78f3ce929445d045410aa795 . The first signed arithmetic-trace calculation and the third global balanced-kernel calculation retain their stated scope.

The coordinated proof [Analytic pole residue transport](#), SHA256 357923aead35c77f2fa98598fbc4ed8bc1f7cd6ebc04760da504a0ded765a652 , has now completed AP1–AP26. The audit owner read that entire source. Do not repeat its pole, contour, finite connection or endpoint computations as unperformed work.

On the actual localization $\mathbb{C}[s, \nu^{-1}]$, at fixed $u \neq 0$, the proof retains every original zero order of h and every Taylor coefficient of $\nu = \text{rem}_h j_h(g/h)$. It computes the exact weighted-residue extension

$$0 \longrightarrow H_{\text{pol}} \longrightarrow H_{\text{loc}} \xrightarrow{\text{Res}_w} \mathbb{C}^{P_\nu} \longrightarrow 0, \quad w = \exp((\Phi - ts)/u).$$

It provides all higher-pole reductions, the full block connection, the entire off-diagonal scaling integral, the full period determinant with the original rapid-decay contours and positively oriented $2\pi i$ small-loop factors, and the exact gauge and linear dual. Its explicit endpoint arrow $J_0 = [I_d \ Q_0]$ satisfies

$$J_0 A_{\text{ext}}(0) = A J_0, \quad \ker J_0 = \{(-Q_0 c, c) : c \in \mathbb{C}^{P_\nu}\}, \quad J_0 C_{\text{ext},a}(0, 0) = e^{-aA} J_0.$$

The unit after that arrow remains the full v_h . The cases of a constant unit and a pole at zero are treated explicitly. The formal contraction is not evaluated at nonzero u ; the analytic pole cokernel and the actual endpoint map provide the calculated comparison.

Continue with the quantitative term these completed maps leave: carry this explicit original polynomial inclusion, pole extension, gauge, period matrix and endpoint quotient into the original theta-source Gram and relation-volume calculation. Compute the norm or metric contribution of the retained extension and its endpoint kernel in that specified source topology, and its contribution to the signed arithmetic correction. AP1–AP26 supplies no uniform theta-norm estimate. The remaining request is that quantitative calculation on the original source, not another construction of the already-proved residue extension or a claim that its formal counterpart can be evaluated away from the formal fibre.

The full AP source is included in the attachment repository; this postscript does not substitute for its proofs.

Part IV**Complete original-source analytic transport and moment certification**

241 Which analytic pole classes belong to the original theta source?

13 September 2026. This is the next calculation after AP1–26. It keeps the original one-factor arithmetic source, its Mellin measure and mass, and computes both its Hilbert admission and its strong-Schwartz admission. The resulting complexes, their residue images, their exact source maps and their theta-boundary defects are displayed. No freely chosen metric or hypothetical zero replaces the arithmetic input.

241.1 1. Original source, fixed parameters, and the exact norm

Retain the full actual packet h of degree $d > 0$, $g = 2\xi$, the entire function $v_h = g/h$, and the original F_h in the strong Schwartz source B with Mellin transform v_h . Retain the complete Taylor-unit polynomial ν representing $v_h = j_h(g/h)$; $\gcd(h, \nu) = 1$. Let P_ν be its distinct roots. At every $p \in P_\nu$ put $m_p = \text{ord}_p v_h$, a finite nonnegative integer. Since $h(p) \neq 0$, this is exactly $\text{ord}_p g$, not a multiplicity of ν or a presumed simple zero.

Use s for the Mellin coordinate and y for its critical-line imaginary part. The independent geometric parameters are $u \in \mathbb{C}^*$ and $t \in \mathbb{C}$. Distinguish

$$D_x = -x \frac{d}{dx}, \quad D_{\text{exp}} = u \frac{d}{ds} + h(s) - t, \quad L = u \frac{d}{dt} - s. \quad (\text{TA1})$$

On rational coefficients $R \in \mathbb{C}[s, \nu^{-1}]$, the stipulated original Hilbert norm is

$$\|V_{h,\text{rat}}R\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |v_h(1/2 + iy)R(1/2 + iy)|^2 dy. \quad (\text{TA2})$$

When finite, inverse Mellin on this line defines $V_{h,\text{rat}}R$ in $L^2(\mathbb{R}_+, dx)$. Indeed logarithmic coordinates $x = e^z$ turn $F(x)$ into $e^{z/2}F(e^z)$, whose Fourier Plancherel identity is precisely TA2. No total mass is divided out. For polynomial R this is exactly $R(D_x)F_h$, the original source map.

241.2 2. Complete Hilbert admission, including zeros on the observation line

Partition P_ν into $O = \{p : \text{Re } p \neq 1/2\}$ and $C = \{p : \text{Re } p = 1/2\}$. Let $\text{PP}_p^{\leq n}$ be the span of $(s-p)^{-1}, \dots, (s-p)^{-n}$, and PP_p the union over n . The zero bound $n = 0$ means the zero space, not a discarded supported point.

The rational coefficients of finite TA2 norm are exactly

$$A^1 = \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq m_p}. \quad (\text{TA3})$$

To prove necessity at $p = 1/2 + iy_p$, write $v_h(s) = (s-p)^{m_p}a(s)$, where $a(p) \neq 0$, and $R(s) = (s-p)^{-n}c(s)$, where $c(p) \neq 0$. Along the line, the squared modulus is bounded above and below by positive constants times $|y - y_p|^{2(m_p-n)}$ near p . Its integral is finite exactly when $2(m_p - n) > -1$, or, because the orders are integers, $n \leq m_p$. The leading coefficient cannot be cancelled by principal parts at another point.

There are no local line singularities from O . Outside a fixed compact interval, any rational R has at most polynomial growth. The actual Mellin transform v_h is rapidly decreasing on every vertical strip, as follows from F_h in B ; equivalently its gamma decay gives more than the required line decay. Thus every coefficient described in TA3 has finite norm, proving sufficiency.

The exact domain making D_{exp} an admitted differential is

$$\begin{aligned}
A^0 &= D_{\exp}^{-1}(A^1) \\
&= \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq \max(m_p-1, 0)}.
\end{aligned} \tag{TA4}$$

For a pole of order $n \geq 1$, $D_{\exp} R$ has the uncancellable leading pole $-unc(p)(s-p)^{-n-1}$. Hence at a line point its image is admitted exactly when $n+1 \leq m_p$. A polynomial has polynomial image and is always admitted. This proves TA4 and also $A^0 \subset A^1$ without any extra domain assumption.

The two-term complex $[A^0 \xrightarrow{D_{\exp}} A^1 ds]$ is consequently the original norm-admitted subcomplex of AP's analytic localization at fixed $u \neq 0$. This is a graph-domain calculation, not an assertion that all finite-norm coefficients form a differential-invariant space in both degrees.

241.3 3. Its entire cohomology and exact inclusion into the analytic family

Put $S = O \cup \{p \in C : m_p \geq 1\}$ and $Z = \{p \in C : m_p = 0\}$. Use the actual weight $w = \exp((\Phi(s) - ts)/u)$, with $\Phi' = h$ and its original constant. AP's exact residue map restricts to give

$$0 \longrightarrow H_{\text{pol}} \longrightarrow H^1(A) \xrightarrow{\text{Res}_w} \mathbb{C}^S \longrightarrow 0, \tag{TA5}$$

$$0 \longrightarrow H^1(A) \longrightarrow H_{\text{loc}} \xrightarrow{\text{Res}_w|_Z} \mathbb{C}^Z \longrightarrow 0. \tag{TA6}$$

Here $H^0(A) = 0$, and a basis of $H^1(A)$ is the original polynomial basis $1, s, \dots, s^{d-1}$, followed by $1/(s-p)$ for $p \in S$.

For full proof, the highest-pole argument makes $D_{\exp}$ injective on rational functions, followed by its leading-degree argument on polynomials. If an admitted F equals $D_{\exp} Q$ in the larger localized complex, TA4 forces Q to belong to A^0 . Thus the induced map on H^1 has no kernel. Subtract from an admitted F its weighted residues using $(\text{Res}_w F)_p / (w(p)(s-p))$ for $p \in S$. At a point with bound m_p , AP's log-free local primitive then has pole order at most $m_p - 1$. Its sum of principal parts belongs to A^0 . Subtracting its $D_{\exp}$ image leaves a polynomial and proves the kernel assertion in TA5. Those simple-pole lifts prove surjectivity. Independence follows by residues and then polynomial cohomology. Finally AP's full residue decomposition shows that precisely the classes with zero Z -residues have such representatives, proving TA6.

All original m_p occur in the two chain spaces and in their local primitive maps. One coordinate per admitted point is the computed cohomology, not a replacement of the original zero orders by one.

Because L does not increase finite pole order and commutes with $D_{\exp}$, it preserves A^0 and A^1 . Its connection on TA5 is exactly the submatrix of AP:

$$L = u \frac{d}{dt} - \begin{bmatrix} A(t) & B_S \\ 0 & \text{diag}(p)_{p \in S} \end{bmatrix}, \tag{TA7}$$

where each column of B_S is the original constant-polynomial column. Its weighted-residue factors $w(p)$, its entire upper-right scaling integral and its oriented $2\pi i$ loop periods are the corresponding AP formulas restricted to S . No new connection or residue scaling is chosen.

241.4 4. Strong-Schwartz admission is an actual full-zero packet construction

The original source is

$$B = \left\{ F \text{ smooth on } \mathbb{R}_+ : \begin{array}{l} \sup_{x>0} x^b |D_x^j F(x)| < \infty \\ \text{for every integer } b \text{ and } j \geq 0 \end{array} \right\}. \tag{TA8}$$

Its Mellin transforms are entire: for s in a compact set the bounds at zero and infinity dominate the integral and all its logarithmic derivatives. Therefore $V_{h,\text{rat}}R$ can belong to B only when every finite pole of $v_h R$ is removable, at all $p \in P_\nu$, not just on the observation line. Agreement of its entire Mellin transform with $v_h R$ on a nonsingular segment of the critical line extends to the connected punctured plane by the identity theorem, which proves the necessity of the removable-pole assertion.

Define the actual polynomials

$$\begin{aligned}
 S_B &= \{p \in P_\nu : m_p > 0\}, \\
 \kappa(s) &= \prod_{p \in S_B} (s - p)^{m_p}, \\
 \kappa_0(s) &= \prod_{p \in S_B} (s - p)^{m_p - 1}, \\
 b(s) &= \kappa / \kappa_0 = \prod_{p \in S_B} (s - p), \\
 c(s) &= \sum_{p \in S_B} (m_p - 1)b(s)/(s - p), \\
 H(s) &= h(s)\kappa(s).
 \end{aligned} \tag{TA9}$$

Empty products equal one and the empty sum is zero. These are products of the specified monic factors, not renormalizations of an arbitrary leading coefficient. Since $\gcd(h, \nu) = 1$, the roots of h and κ are disjoint. H therefore selects only actual zeros of g , retaining their complete orders.

All entire-admitted rational coefficients are exactly $\kappa^{-1}\mathbb{C}[s]$. This follows either from partial fractions with the full order bounds, or by multiplying by κ : the resulting rational function has no finite poles and is polynomial. Conversely P/κ has at most the prescribed orders.

Sufficiency for the original B follows through its existing Euler inverses, without a new space-identification assumption. Apply the original S_ρ inverses successively to $\Theta\phi_*$ through every full factor of H . At each step the remaining Mellin zero supplies the required kernel condition. Their already proved source map gives $F_H \in B$ with Mellin transform g/H . Therefore

$$\begin{aligned}
 V_{h,\text{rat}}(P/\kappa) &= P(D_x)F_H, \\
 M(P(D_x)F_H) &= gP/H = v_h P/\kappa.
 \end{aligned} \tag{TA10}$$

Every term on the right is an original strong-Schwartz theta-source vector. This proves the converse and the exact source realization, not merely finite norm. The differential graph domain of this strong-source degree-one space is $\kappa_0^{-1}\mathbb{C}[s]$, by the same highest-pole calculation as TA4 at every p .

Thus the strong-source complex is the explicit subcomplex

$$[\kappa_0^{-1}\mathbb{C}[s] \xrightarrow{D_{\text{exp}}} \kappa^{-1}\mathbb{C}[s] ds]. \tag{TA11}$$

It injects into the Hilbert-admitted and full analytic cohomologies with no extra kernel, by the same D_{exp} -preimage proof. Its residue image is exactly $\mathbb{C}^{S_B}$. In particular a ν -root off the critical line with $m_p = 0$ is admitted by the Hilbert norm but not by B ; its exact omitted coordinate is its AP residue.

241.5 5. Polynomial coordinates, source maps and every retained boundary defect

Use the literal maps $Q \mapsto Q/\kappa_0$ in degree zero and $P \mapsto P/\kappa$ in degree one. They carry TA11 to a polynomial complex with differential

$$\begin{aligned} E(Q) &= \kappa D_{\exp}(Q/\kappa_0) \\ &= ubQ' + [b(h-t) - uc]Q. \end{aligned} \quad (\text{TA12})$$

The quotient $\kappa'_0/\kappa_0 = \sum_p(m_p - 1)/(s - p)$ proves this identity with all orders and the negative derivative term retained. Since b and h are the specified monic polynomials, the leading term of $E(Q)$ has degree $\deg Q + d + |S_B|$ and unchanged leading coefficient. Every other term has lower degree. Thus E is injective and its cokernel has the unique polynomial basis of degrees less than $d + |S_B|$, by terminating leading-term reduction.

The original polynomial exponential complex maps into it by the exact square

$$\begin{array}{ccc} \mathbb{C}[s] & \xrightarrow{D_{\exp}} & \mathbb{C}[s] \\ \text{multiplication } \kappa_0 \downarrow & & \downarrow \text{multiplication } \kappa \\ \mathbb{C}[s] & \xrightarrow{E} & \mathbb{C}[s] \end{array} \quad (\text{TA13})$$

$$E(\kappa_0 Q) = \kappa D_{\exp} Q.$$

Leibniz differentiation proves this directly, or use the denominator maps. The original arithmetic source is carried identically, since $V_{h,\text{rat}}((\kappa P)/\kappa) = P(D_x)F_h$. No map here sends supported zero to τ or changes the base point; lift each displayed linear map on its original mask.

There is also an exact arithmetic-quotient map, with its boundary value fully retained. Write $E_H = \mathbb{C}[s]/(H)$, $v_H = j_H(g/H)$, and use the existing full packet section $\sigma_H : E_H \rightarrow Q = B/\Theta V$. For the original full jet J_H ,

$$\begin{aligned} J_H V_{h,\text{rat}}(P/\kappa) &= v_H[P]_H, \\ qV_{h,\text{rat}}(P/\kappa) &= \sigma_H(v_H[P]_H). \end{aligned} \quad (\text{TA14})$$

To justify the second equation, $H(D_x)P(D_x)F_H = \Theta(P(D_x)\phi_*)$ by Mellin identity and injectivity, so its Q -class lies in $Q[H(D_x)]$. On that actual submodule the original full-jet map and σ_H are inverse. Thus the displayed equation follows from the first one, which is the literal product of Taylor series. Restriction to the original h -block gives

$$j_h(\kappa v_H P) = v_h[P]_h. \quad (\text{TA15})$$

At each added κ -block, κP vanishes through its full order. The old source inclusion therefore retains its actual full-unit h -component and has zero added components, not a hidden mixture of old and new source classes.

For a geometric degree-zero numerator Q the observed boundary is exactly

$$qV_{h,\text{rat}}(E(Q)/\kappa) = \sigma_H(v_H[E(Q)]_H). \quad (\text{TA16})$$

It need not be zero. TA16 is the calculated defect preventing an unproved descent from geometric de Rham cohomology to the original theta quotient. The cochain source map, its full-unit jet value and its entire residual boundary have all been supplied; no false cohomology intertwiner replaces them.

At the fixed-domain specialization $u = t = 0$, $E = bh$. The quotient of the original arithmetic E_H by its ideal $(bh)/(H)$, transported by the unit v_H , is exactly the geometric polynomial quotient $\mathbb{C}[s]/(bh)$. At an added root p this ideal is $(s - p)/(s - p)^{m_p}$; at an original h -root its component is zero. This computes the complete endpoint kernel, including every removed higher jet, instead of identifying the two quotients outright. For $m_p = 1$ that component is zero. Its rank count is $d + |S_B|$, with no deletion from the larger original E_H before the specified quotient map.

The fixed pair TA11 was derived for $u \neq 0$. At $u = 0$, recomputing the maximal graph domain allows all coefficients in $\kappa^{-1}\mathbb{C}[s]$, since $h - t$ has no pole; that is a different degree-zero domain.

At $u = t = 0$ its localized cohomology has rank d . The fixed-domain specialization just calculated has rank $d + |S_B|$. Both domains and the exact endpoint quotient are explicit; no flatness of the recomputed graph domain is asserted.

241.6 6. The actual theta Gram and finite source cost

For the literal denominator frame TA10, TA2 becomes the exact identity

$$\begin{aligned} \|V_{h,\text{rat}}(P/\kappa)\|^2 &= \int_{\mathbb{R}} |P(1/2 + iy)|^2 w_H(y) dy, \\ w_H(y) &= |(g/H)(1/2 + iy)|^2 / (2\pi), \quad \int w_H = \mu_H. \end{aligned} \quad (\text{TA17})$$

All removable values on the line are filled by the original entire g/H . This is the already specified arithmetic convolution seed for the actual extended packet H , with its own full mass μ_H , not a unit-mass replacement. For any finite numerator degree N , the exact Gram is

$$(M_{H,N})_{ij} = \int_{\mathbb{R}} \overline{(1/2 + iy)^i} (1/2 + iy)^j w_H(y) dy, \quad 0 \leq i, j \leq N. \quad (\text{TA18})$$

It is positive definite: a nonzero polynomial is nonzero outside a finite set, and the nonzero entire g/H is nonzero almost everywhere on the line. All moments converge by the original rapid vertical decay. In an arbitrary specified finite rational basis R_j , the unchanged norm instead has matrix $\int \overline{R_i} R_j w_h$; TA17–18 are its exact denominator-coordinate pullback, not a change of norm.

Let E_N be the coefficient matrix of TA12 from degrees $\leq N$ to degrees $\leq N + d + |S_B|$. The full source cost of the actual geometric boundary is

$$\|V_{h,\text{rat}}(E(Q)/\kappa)\|^2 = Q^* E_N^* M_{H,N+d+|S_B|} E_N Q. \quad (\text{TA19})$$

This includes u, t , every m_p through c , and all cross terms. Its arithmetic class is TA16. If G_H is an existing canonical quotient Gram at an admitted cutoff and T_N is the literal matrix $Q \mapsto v_H[E(Q)]_H$, the quotient cost is exactly $Q^* T_N^* G_H T_N Q$. No new quotient metric is selected. No uniform bound on these actual matrices is inferred from their positivity.

Completed result: all analytic pole classes have their exact original-Hilbert admission test; those with genuine strong-Schwartz source representatives are realized by an actual enlarged full-zero packet. Their complex, residues, source Gram and arithmetic boundary defect are explicitly calculated.

241.7 7. The connection in the same arithmetic numerator frame

Write $r = |S_B|$ and $q = d + r$. In the degree-one numerator frame $f_j = s^j/\kappa$, $0 \leq j < q$, the actual connection is

$$\begin{aligned} L &= u \frac{d}{dt} - A_{\text{adm}}(u, t), \\ A_{\text{adm}} &= \text{Companion}(\chi_{\text{adm}}), \\ \chi_{\text{adm}}(u, t; s) &= b(s)(h(s) - t) - uc(s). \end{aligned} \quad (\text{TA20})$$

Here Companion means the column-convention matrix taking e_j to e_{j+1} for $j < q - 1$ and taking e_{q-1} to minus the coefficient column of the lower-degree part of the displayed monic polynomial. This specifies every matrix entry without a basis convention left implicit.

Indeed $[L, E] = 0$ on parameter-dependent polynomial numerators: the term $[u d/dt, E] = -ub$ cancels $[-s, ub d/ds] = ub$, and all remaining commutators vanish. Thus L descends through the

parameter-dependent cokernel. Its value on f_j is $-s^{j+1}/\kappa$. Only the last basis vector requires reduction, and the exact reduction relation is $E(1) = \chi_{\text{adm}}$. This proves TA20. It does not assert that multiplication by s alone descends on the fixed quotient by the image of the differential operator E . Nor is χ_{adm} being declared an annihilator made of new actual zeros of g . It is the calculated connection matrix in the already admitted arithmetic frame.

Here is its exact relation to AP's polynomial-plus-simple-pole frame. Use that frame restricted to S_B , and denote its connection matrix by $T_B = \begin{bmatrix} A(t) & B_{S_B} \\ 0 & \text{diag}(p) \end{bmatrix}$. Let $G(u, t)$ be the matrix whose j th column is the AP coordinates of $[s^j/\kappa]$. Its residue-block entries are

$$\begin{aligned} G_{(p,j)} &= \text{Res}_p(w(s)s^j/\kappa(s) ds)/w(p) \\ &= \frac{\left(\frac{d}{ds}\right)^{m_p-1} [w(s)s^j/\kappa_p(s)] \Big|_{s=p}}{(m_p-1)!w(p)}, \\ \kappa_p(s) &= \kappa(s)/(s-p)^{m_p}. \end{aligned} \tag{TA21}$$

All m_p derivatives, the original exponential weight, and its value $w(p)$ are retained. To compute the polynomial-block entries, subtract the simple poles with these coefficients from f_j . The result has zero weighted residue at every admitted point. Take AP's log-free local primitives, keep their finite principal parts, and subtract their D_{exp} images. This leaves a polynomial. Reduce it to degree less than d with the original leading-term reduction for D_{exp} . These coefficients are exactly the remaining entries of G . This is a finite formula for each column, not a choice of an unspecified comparison map.

Both sets are proved bases of the same admitted cohomology, so G is invertible. The primitive calculations only divide by nonzero $u, w(p)$, nonzero leading coefficients, and fixed nonzero differences of distinct points. Their finite coefficients are therefore holomorphic for $u \neq 0$ and all t . Applying L to each column, with its parameter dependence included, gives the exact identity and finite scaling transport

$$\begin{aligned} uG_t &= T_B G - G A_{\text{adm}}, \\ A_{\text{adm}} &= G^{-1} T_B G - uG^{-1} G_t, \\ C_{\text{adm}}(a; t) &= G(t)^{-1} C_B(a; t) G(t + ua). \end{aligned} \tag{TA22}$$

The fixed u is suppressed only in the last line's arguments. For proof of that line, differentiate its right side with respect to a , use $C'_B(a; t) = -C_B(a; t)T_B(t + ua)$, and then the first identity of TA22. The derivative is $-C_{\text{adm}}(a; t)A_{\text{adm}}(t + ua)$ and the value at $a = 0$ is I . The uniqueness of this finite linear differential equation proves the transport formula. Both endpoint frames and the derivative gauge term are present; a pointwise similarity would omit an actual term.

These same numerator vectors are the actual source vectors $D_x^j F_H$ by TA10, so their original norm is the explicitly calculated $M_{H,q-1}$. The cohomological frame change G , however, also used D_{exp} primitives. Those primitive differences have exactly the source cost TA19 and the full arithmetic image TA16. A norm isometry between arbitrary reduced representatives is not silently inferred from a cohomology change of basis. Thus TA20–22 supply the concrete arithmetic-frame connection, its exact comparison with AP, and the already computed source-level discrepancy.

241.8 8. Exact recovery of the old canonical quotient metric

The comparison reaches the original quotient metric as well as individual source vectors. Put $K = \deg \kappa$ and $D = \deg H = d + K$. Use the literal injection

$$i_\kappa : \mathbb{C}[s]/h \longrightarrow \mathbb{C}[s]/H, \quad [P]_h \longmapsto [\kappa P]_H.$$

It is well-defined because $H = h\kappa$ and injective because the polynomial ring has no zero divisors. Its image is the ideal $(\kappa)/(H)$. At the old h -block it multiplies by the actual nonzero Taylor unit $j_h(\kappa)$; at every new block it is zero through that block's full order. Composing with v_H yields precisely the original arithmetic unit v_h by TA15, not a newly chosen identification of the h -component.

Let $R_{H,N}^{\text{coef}}$ be the existing unique least-source-norm representative for coefficient classes in $\mathbb{C}[s]/H$ among polynomials of degree at most N ; let $G_{H,N}^{\text{coef}}$ be its Gram. Define $R_{h,N-K}^{\text{coef}}$ and $G_{h,N-K}^{\text{coef}}$ in precisely the same way using the original weight w_h . Take $N \geq D - 1$ so both quotient maps are onto and both metrics are defined. Then, as maps and forms,

$$\begin{aligned} R_{H,N}^{\text{coef}} i_\kappa &= M_\kappa R_{h,N-K}^{\text{coef}}, \\ i_\kappa^* G_{H,N}^{\text{coef}} i_\kappa &= G_{h,N-K}^{\text{coef}}. \end{aligned} \tag{TA23}$$

where M_κ is literal polynomial multiplication. These are the original least-norm constructions, not newly selected quotient metrics.

For proof, a degree-at-most- N representative Q of $i_\kappa[P]_h$ has $Q - \kappa P = HT$ for some polynomial T . Thus $Q = \kappa R$ with $[R]_h = [P]_h$ and $\deg R \leq N - K$. Conversely every such R gives an allowed Q . This is a bijection between the two complete affine spaces of representatives, including all their relation directions. Their norms agree exactly:

$$\begin{aligned} \int |\kappa(s)R(s)|^2 w_H(y) dy &= \int |R(s)|^2 w_h(y) dy, \quad s = 1/2 + iy, \\ w_h(y) &= |\kappa(1/2 + iy)|^2 w_H(y). \end{aligned} \tag{TA24}$$

The last identity is $g/h = \kappa g/H$, with the removable line values filled by those entire functions. It retains both unnormalized masses; neither mass is asserted equal to the other. Finite-dimensional positive definiteness gives a unique minimizer in each affine space. The norm bijection therefore proves the first identity of TA23, and polarization proves its full Hermitian Gram identity, including cross terms.

Under the actual source maps the two minimizing polynomials give the identical strong-Schwartz function:

$$(R_{H,N}^{\text{coef}} i_\kappa [P])(D_x)F_H = (R_{h,N-K}^{\text{coef}} [P])(D_x)F_h,$$

because $\kappa(D_x)F_H = F_h$ follows from the original Mellin transforms. Thus TA23 is a calculated isometric inclusion of the original arithmetic quotient observation into the extended-packet observation with the exact degree shift K . It is stronger than a bound inferred from positivity. For precision about full-unit coordinates, write $G_{H,N}^{\text{jet}}$ for the same canonical observation expressed in physical jet coordinates $J_H = v_H \pi_H$. Then

$$\begin{aligned} G_{H,N}^{\text{coef}} &= M_{v_H}^* G_{H,N}^{\text{jet}} M_{v_H}, \\ j &= M_{v_H} i_\kappa M_{v_h}^{-1}, \\ j^* G_{H,N}^{\text{jet}} j &= G_{h,N-K}^{\text{jet}}. \end{aligned}$$

Here M_v denotes multiplication by the indicated complete Taylor unit, not its leading coefficient. TA15 proves that j is exactly the identity on the original h -block and zero on each added full κ -block. These equations specify the equivalent metric statement in the physical jet convention used for the arithmetic cost in TA19. No unit-coordinate change is left implicit or omitted from a Gram pullback.

TA23 does not identify the extra quotient directions or their mixed Gram entries with old directions. Those remain in the actual extended metric; the geometric-boundary image TA16 and its source cost TA19 remain unchanged.

241.9 Sources and verification scope

Root reread A1–A5 of the current `tex/arithmetic_input.tex`, including the definition of B , $g = 2\xi$, $\Theta\phi_*$, and the original Euler inverse. The full A1–A19 source and its full-order section were read previously. AP1–26 is the accepted analytic input, SHA256 357923aead35c77f2fa98598fbc4ed8bc1f7cd6ebc04760da504a0ded765a652. The independent collaborator derived and checked TA3–TA7, the global graph domain and TA12–TA16 while root wrote their complete proofs and source-cost formulas here. This note claims written mathematics, not new Lean execution, numerical zero certification, a uniform estimate, or remote publication.

Independent calculation and review of theta-admitted pole complexes

Date: 2026-09-13. This records the independent calculation supplied while the root proof THETA_ADMITTED_POLE_COMPLEX.md was being written. A final review seal, if appended below, identifies the complete root version read and accepted. No source-owner or publication file is edited here. There is no numerical, arithmetic-zero, Lean, or quotient-positivity certificate.

1. Exact admission for the stipulated critical-line norm

Keep the original entire function $v_h = g/h$, with h containing the full selected zero orders. Its norm on a rational coefficient F is exactly

$$\int_{\mathbb{R}} \left| v_h \left(\frac{1}{2} + iy \right) F \left(\frac{1}{2} + iy \right) \right|^2 \frac{dy}{2\pi}.$$

The parameter y on this line is distinct from the deformation parameter t in $D = u\partial_s + h - t$. Throughout the pole-complex calculation u is fixed and nonzero. No phase, cancellation, or mass in this integral is replaced by a discrete or diagonal surrogate.

Let P be the distinct roots of the actual Hermite polynomial ν , and let O and C be its off-line and critical-line subsets. At $p \in C$ put $m_p = \text{ord}_p(v_h)$. If F has an actual pole of order n at p , the weighted integrand is a nonzero smooth factor times $|y - y_p|^{2(m_p - n)}$ locally. Its integrability is equivalent to $2(m_p - n) > -1$, hence exactly to $n \leq m_p$, since both orders are integers. Apparent terms whose leading coefficients cancel must first be combined into F 's actual Laurent order.

There is no obstruction at infinity for a rational F . This can be checked from the original source without a new zeta estimate: A1–A3 give $g = \text{Mellin}(\Theta\phi_*)$ with $\Theta\phi_*$ in the strong-Schwartz space. Its Mellin restriction to any fixed vertical line is a Schwartz Fourier transform after $x = \exp(r)$. Division by h preserves this rapid decay at infinity, and the selected finite singularities are removable. Rational functions and their derivatives grow at most polynomially away from their finite poles. Off-line poles therefore impose no further condition for this line norm. Indeed every admitted weighted restriction $v_h F$ is Schwartz on the line, not merely square-integrable.

This statement proves exactly line-norm admission. It does not turn a meromorphic off-line Mellin multiplier into an entire Mellin transform.

2. The maximal two-term norm-admitted complex

Write PP_p for finite principal parts at p , and $\text{PP}_p^{\leq j} = 0$ for $j \leq 0$. The degree-one admitted rational space and its degree-zero graph domain are

$$A^1 = \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq m_p},$$

$$A^0 = \mathbb{C}[s] + \bigoplus_{p \in O} \text{PP}_p + \bigoplus_{p \in C} \text{PP}_p^{\leq \max(m_p - 1, 0)}.$$

All sums here are direct as polynomial-plus-principal-part vector spaces. The complex is $[A^0 \xrightarrow{D} A^1 ds]$. It is the maximal such subcomplex of the analytic localization with the stipulated degree-one admission:

$$A^0 = D^{-1}(A^1) \text{ inside } \mathbb{C}[s, \nu^{-1}].$$

Indeed, at a pole of order $n \geq 1$ in Q , the highest pole of DQ has order $n + 1$ and coefficient $-un$ times the leading coefficient of Q . The multiplier $h - t$ cannot cancel it. Consequently DQ is admitted at a critical point exactly when Q has no pole or has order at most $m_p - 1$. Such a Q is automatically

in A^1 as well. This proves the asserted maximal graph domain rather than assuming D preserves the original degree-one space.

Let $S = O \cup \{p \in C : m_p \geq 1\}$ and $Z = \{p \in C : m_p = 0\}$. The weighted analytic residue of AP, with $w = \exp((\Phi - ts)/u)$, has exactly the image $\mathbb{C}^S$ on the admitted complex. Its coordinates on Z vanish identically. For every $p \in S$, the admitted lift $1/(w(p)(s - p))$ gives the p -th coordinate vector, with its actual weight retained.

If an admitted degree-one F has zero weighted residues, the AP local primitive construction gives a rational Q with $DQ - F$ polynomial. At a critical point, Q has pole order at most one less than F , so Q belongs to A^0 ; at every off-line point no additional restriction is imposed. Thus the same proof gives the exact sequence

$$0 \longrightarrow H_{\text{pol}} \longrightarrow H^1(A) \longrightarrow \mathbb{C}^S \longrightarrow 0.$$

A basis is $1, s, \dots, s^{d-1}$ followed by $1/(s - p)$ for $p \in S$. In particular $\dim H^1(A) = d + |S|$ and $H^0(A) = 0$. Full multiplicities m_p remain in the cochain spaces and their differentials; the single residue coordinate per admitted point is a proved cohomological quotient.

3. Exact map to AP and absence of an additional kernel

If an admitted F is DQ in the full analytic localization, the graph-domain identity forces Q into A^0 . Therefore the induced cohomology map is injective: restricting degree zero in this maximal way creates no extra kernel. Its image is exactly the classes whose weighted residues on Z vanish. In full, the actual map fits into

$$0 \longrightarrow H^1(A) \longrightarrow H_{\text{loc}} \longrightarrow \mathbb{C}^Z \longrightarrow 0,$$

where the last map is the corresponding coordinate restriction of AP's weighted-residue map. This also accounts for all cokernel coordinates.

There is a direct quotient-complex check. At critical p with $m_p \geq 1$, the quotient complex is

$$\text{PP}_p / \text{PP}_p^{\leq m_p - 1} \xrightarrow{D} \text{PP}_p / \text{PP}_p^{\leq m_p}.$$

Its differential is bijective: remove the highest target pole with the unique predecessor having one lower order, and continue downward; all remaining target orders at most m_p are zero in this quotient. Injectivity is the same leading-pole argument. At $m_p = 0$ the quotient is the full AP principal-part complex with its one-dimensional residue cokernel. At off-line points the quotient is zero. Hence this computation gives exactly the preceding short exact sequence and no hidden additional classes.

The operator $L = u\partial_t - s$ preserves both A^0 and A^1 and commutes with D . Multiplication by s does not raise any finite pole order, and t differentiation does not alter these fixed pole bounds. The admitted connection, weighted-residue transition, and scaling matrix are therefore AP's matrices with precisely the pole rows and columns in S retained: $\begin{bmatrix} A(t) & B_S \\ 0 & P_S \end{bmatrix}$, W_S , and their displayed off-diagonal integral. This is an invariant subfamily, not a chosen replacement operator.

4. Entire-Mellin admission and the actual strong-Schwartz source

For this stronger admission put $m_p = \text{ord}_p(v_h)$ at every p in P , not just on the critical line. Since $\gcd(h, \nu) = 1$, h does not vanish at these points and this equals $\text{ord}_p(g)$. Let

$$\begin{aligned}
k &= \prod_{m_p > 0} (s - p)^{m_p}, \\
k_0 &= \prod_{m_p > 0} (s - p)^{m_p - 1}, \\
b = k/k_0 &= \prod_{m_p > 0} (s - p).
\end{aligned}$$

The empty products are one. The rational coefficients F for which $v_h F$ is entire are exactly $k^{-1}\mathbb{C}[s]$. Their graph domain under D for u nonzero is $k_0^{-1}\mathbb{C}[s]$, by the same leading-pole argument. The inclusion into the line-norm admitted complex is literal. Its image on cohomology retains just the residues at points with $m_p > 0$, and it is injective by the same maximal-domain argument. Thus the difference from the line-norm domain consists exactly of off-line points where $m_p = 0$.

This stronger space has an exact realization in the original source, not only a formal entire-function criterion. Set $H = hk$. Every root of H is an actual g -zero with its full order, and its two factors have disjoint root sets. The original repeated-root division construction gives $F_H = S_H \Theta \phi_*$ in the strong-Schwartz space, with Mellin transform g/H . Consequently

$$V(P/k) = P(D_x)F_H, \text{ where } D_x = -x\partial_x,$$

has Mellin transform $gP/H = v_h P/k$. This proves sufficiency by the original source construction. Necessity follows because the Mellin transform of a strong-Schwartz source is entire: every pole of $v_h F$ must cancel. The coefficient derivative ∂_s in D is not D_x ; the latter corresponds to multiplication by s under Mellin transform.

The full source jet map is

$$J_H V(P/k) = v_H[P]_H,$$

with the actual complete unit $v_H = j_H(g/H)$. On the old polynomial inclusion the target numerator is kP ; restricting to the original h packet gives $j_h(kv_H P) = v_h[P]$. At newly added roots kP vanishes to the full corresponding orders. These identities retain all original jets and the full Taylor units; they do not replace them by leading values.

5. Polynomial coordinates for the stronger complex

Under the degree-zero and degree-one denominator identifications, put

$$c = bk'_0/k_0 = \sum_{m_p > 0} (m_p - 1)b/(s - p).$$

This is a polynomial, including when some or all m_p equal one. The exact polynomial differential is

$$E = ub\partial_s + b(h - t) - uc,$$

because $kD(P/k_0) = E(P)$. The old polynomial complex embeds by multiplication by k_0 in degree zero and k in degree one:

$$E(k_0 P) = kD(P).$$

If $q = d + \deg b$, the leading term of $E(P)$ for nonzero polynomial P has degree $\deg P + q$ and unchanged leading coefficient. The derivative and c terms have strictly smaller degree. Hence E is injective, its cohomology is free on the degree-below- q remainder basis at fixed parameters, and its dimension q agrees with the exact residue sequence.

The commuting connection in this numerator basis is also explicit. $[u\partial_t - s, E] = 0$, and the last basis column is reduced using $E(1) = b(h - t) - uc$. Thus its matrix is

$$u\partial_t - \text{Companion}(b(h - t) - uc).$$

It need not be pointwise similar to AP's polynomial-plus-simple-poles block: their parameter-dependent frame transition is a connection gauge, whose derivative term must be included. In particular c records the full multiplicities without changing the original arithmetic packet H .

These coefficient/source maps are degreewise. An exponential boundary $E(P)$ need not be an arithmetic theta boundary; its exact arithmetic jet observation is $\nu_H[E(P)]_H$. No descent through the arithmetic quotient is asserted by merely writing a strong-Schwartz representative. Likewise the fixed pair with denominators k_0, k is the maximal graph domain only when u is nonzero. Setting $u = 0$ in its matrix is not the same as recomputing the maximal graph domain, which then admits the full orders m_p in degree zero as well.

Source and verification scope

The original source identities A1–A3 were reread in `arithmetic_input.tex`. Its previously recorded SHA-256 is `a50b0fb587b644c4ea94dba45b2b423d038f2bfbe7fef188543939ba850e58d2`. The repeated-root primitive, full-unit section, and exact original source were read completely in the prior independent scaling work. The accepted AP proof supplies the already independently checked local weighted-residue primitive and entire transport, with no change of arithmetic input.

All cohomology assertions above are algebraic. Finite norm of the displayed representatives does not by itself prove closedness of the boundary range or a nondegenerate induced quotient norm. This review neither chooses a new metric nor asserts such a conclusion.

Full read of the root TA1–19 proof

The complete root TA1–19 note was read after it became available, including its source, endpoint, and cost paragraphs. No mathematical correction was required in that version. In addition to the independent calculations above, the following root extensions were checked directly.

First, the original source identification descends exactly to its stated arithmetic-quotient value. Multiplying $P(D_x)F_H$ by $H(D_x)$ gives $\Theta(P(D_x)\phi_*)$, so its Q -class belongs to the actual H -torsion submodule. The original packet section is inverse to the full jet map there. This proves the root's $qV(P/\kappa) = \sigma_H(\nu_H[P]_H)$ with the complete unit. Applying the same identity to $E(Q)$ proves the exact nonzero boundary observation; no assertion of geometric-boundary descent is needed or made.

Second, at the fixed-pair endpoint $u = t = 0$ the geometric polynomial differential is multiplication by bh . Since bh divides H , the map $\mathbb{C}[s]/H \rightarrow \mathbb{C}[s]/(bh)$ is onto with kernel $(bh)/(H)$. At an old h -root its kernel is zero, and at an added p it is exactly $(s - p)/(s - p)^{m_p}$. The kernel dimension is $\sum_{p \in S_B} (m_p - 1)$, and the quotient dimension is $d + |S_B|$. Multiplication by the actual unit preserves this ideal and gives the stated transported comparison. Recomputing the graph domain at $u = t = 0$ instead uses κ in both degrees and produces the rank- d quotient. This verifies both endpoint statements with their distinct domains.

Third, TA17–19 are exact pullbacks of the retained source norm. For numerator P the weight is $|g/H|^2/(2\pi)$, with the original unrenormalized mass. Its Gram is positive on every finite polynomial coefficient space because g/H is a nonzero entire function and a nonzero polynomial has only finitely many zeros. All moments converge by the original source's rapid vertical decay. For a fixed matrix E_N , expanding the original quadratic form gives precisely $Q^*E_N^*M_HE_NQ$, including every cross term. Composing instead with the existing arithmetic quotient map and its specified G_H gives $Q^*T_N^*G_HT_NQ$. These equalities are changes of coordinates in given forms; they do not define a new quotient norm or infer a uniform estimate from positive definiteness.

Complete review of TA20–22 and the frame transport

The full new Section 7 was read, including the explicitly stated column companion convention, the finite primitive construction of G , its holomorphic parameter domain, and the source-discrepancy paragraph. Its formulas agree exactly with the independent connection calculation.

For each numerator vector s^j/κ , the simple-pole AP coefficient at p is its weighted residue divided by $w(p)$. Expanding κ at its full order m_p gives exactly TA21's derivative of order $m_p - 1$ and factorial $(m_p - 1)!$. The complementary factor κ_p is nonzero at p . After subtracting those simple poles, AP's primitive reduces the remaining zero-residue rational coefficient within the admitted degree-zero bound. Polynomial leading-term reduction then supplies its unique old-polynomial coordinates. This proves that the displayed finite algorithm computes G with every required unit, derivative, and residue value retained.

The two bases have already been proved independent and spanning, hence G is invertible without a separate determinant assumption. The finite algorithm has only the stated nonzero denominators on u nonzero and all t ; a value $h(p) - t = 0$ does not introduce an unlisted division. Thus the asserted holomorphic parameter domain is correct, including the empty-pole case in which G is the identity.

The frame orientation is AP coordinates = G times numerator coordinates. Applying L gives $uG_t - T_B G = -GA_{\text{adm}}$, which is TA22 with exactly the negative derivative term in $G^{-1}T_B G - uG^{-1}G_t$. Differentiating $G(t)^{-1}C_B(a; t)G(t + ua)$ therefore yields $-C_{\text{adm}}(a; t)A_{\text{adm}}(u, t + ua)$, with initial value identity. This checks the matrix order, signs, source and target parameter fibres, and both endpoint frame factors. No pointwise similarity is substituted for the connection gauge.

The representative differences used to construct this cohomological frame change are actual D_{exp} primitives in the strong-source graph domain. Their source images consequently have precisely the already computed TA19 cost and TA16 arithmetic boundary value. Keeping this discrepancy prevents an unjustified source-norm isometry or arithmetic cohomology descent from being inferred from G .

Final independent acceptance

The complete root proof TA1–22, including all unnumbered scope statements and the final source paragraph, has now been read and checked. The exact accepted artifact is THETA_ADMITTED_POLE_COMPLEX.md, SHA-256:

3f9a79e61de1f38bc0944be4e43b1ef20f02f99b55637f3afa8b773fb38cf8ec.

Final disposition: accept TA1–22 with no outstanding mathematical correction. This is a written-proof review of the norm and strong-source admission, both maximal graph complexes, exact residue images and injections, full-packet source realization, retained arithmetic boundary defect, endpoint quotient, actual Gram/cost pullbacks, and connection frame transport. No numerical fixture, arithmetic-zero evaluation, Lean replay, uniform metric bound, or new Frobenius identification is certified by this acceptance.

Additive review of the exact canonical-metric recovery, TA23–24

The complete Section 8 was subsequently read, including its explicit coefficient-versus-physical-jet conventions. The preceding TA1–22 seal and its hash remain a historical acceptance of that earlier version. The following checks concern the additive TA23–24 continuation.

Set $K = \deg \kappa$, so K includes every full added zero order, not merely the number of distinct added points. With $D = \deg H = d + K$, the stated cutoff $N \geq D - 1$ gives $N - K \geq d - 1$. Both coefficient quotient observations are therefore onto, including the endpoint cutoffs and the empty-added- packet case $K = 0$. Multiplication by κ is injective from $\mathbb{C}[s]/h$ to $\mathbb{C}[s]/H$ because $H = h\kappa$ in the polynomial integral domain.

Fix a coefficient class $[P]_h$. A representative Q of its image satisfies $Q - \kappa P = HT$ if and only if $Q = \kappa R$ with $[R]_h = [P]_h$. Because multiplication by the nonzero degree- K polynomial κ shifts the degree of every nonzero R by exactly K , the restriction $\deg Q \leq N$ is exactly $\deg R \leq N - K$. Zero is included separately with no difficulty. This proves a bijection of the whole affine representative spaces, not only an injection of selected representatives or a comparison inequality.

The pointwise identity $v_H \kappa = v_h$ gives equality of the original norms for each pair of corresponding representatives, with their original measure and neither source mass renormalized. Positive definiteness of the finite coefficient Grams gives unique minimizers; the bijection therefore proves the representative identity in TA23. Polarization proves its full Hermitian-form identity, retaining all mixed terms among old quotient directions. The actual source vectors coincide because $\kappa(D_x)F_H = F_h$ by Mellin injectivity. Thus there is no unrecorded source representative or cutoff defect.

The addendum correctly distinguishes G^{coef} from the physical-jet G^{jet} used in TA19. For $U_H = M_{v_H}$ and $U_h = M_{v_h}$, the identities

$$\begin{aligned} G_H^{\text{coef}} &= U_H^* G_H^{\text{jet}} U_H, \\ j &= U_H i_\kappa U_h^{-1} \end{aligned}$$

give $j^* G_H^{\text{jet}} j = G_h^{\text{jet}}$ at the stated respective cutoffs. On the original h -block, $v_H \kappa = v_h$ proves that j is identity. On every added block it is zero through the full order because κ vanishes to that order and both units are genuine full Taylor units. This is precisely the original packet's extension by zero in physical jets, not the bare coefficient multiplication map silently reused in a different coordinate convention.

No orthogonality to the added directions follows or is claimed. Their mutual and mixed Gram entries remain in the extended metric. TA16's arithmetic boundary value and TA19's full cost are unchanged. Therefore the addendum establishes an exact restriction of the existing canonical metrics, without selecting a new metric or deriving a uniform estimate.

Final acceptance of TA1–24

The complete added Section 8 and its final unit-coordinate clarification have been read and checked. The exact accepted proof is `THETA_ADMITTED_POLE_COMPLEX.md`, SHA-256:

830e9a254465434fd61cb16212544f004ddf54a5adc72554bccabe217b045e5a.

Final disposition: accept TA1–24 and their stated scope, with no outstanding mathematical correction. This supersedes only the version pin of the earlier TA1–22 acceptance, which remains recorded above. The present addition is a written-proof review of exact cutoff, minimizer, source-map, and full-unit Gram identities; it includes no numerical replay, Lean execution, arithmetic-zero evaluation, or uniform metric-bound claim.

242 Counterfactual RH search on the original split-supported theta input

242.1 Scope

The preceding Deligne-bound audit is not a no-go theorem for geometry over F1, for the monoidal base tau, or for a future purity construction using the full arithmetic input. Its stated conclusion concerns a particular inference from an auxiliary exponential family's Frobenius purity to a bound for the original theta metric. Its examples do not assert that their chosen polynomials divide the original function $g = 2\xi$.

This continuation takes the counterexample search as an independent research objective. It proves a finite-sign certificate for a counterexample to RH, directly from derivatives of the original theta function. It also proves that an actual counterexample would eventually supply such a certificate. The finite calculation reported below found no negative witness. No counterexample to RH is claimed.

The power-sum/moment method belongs to established zero-positivity methods; see Zhang (2015). The derivation below gives all its maps, signs, and constants for the present source, rather than claiming a new general RH criterion.

242.2 1. Original input and support

Retain

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r,$$

and the coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \longrightarrow & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \longrightarrow & \mathbb{C}. \end{array}$$

The support observation sends τ to Boolean zero and every represented coefficient to Boolean one. On the original support-labelled coefficient fibres, a linear map lifts as $(\ell, v) \mapsto (\ell, fv)$, with external absence fixed. A killed coefficient therefore reaches $(\ell, 0)$, not external absence. No conclusion about the sign of an arithmetic functional follows just from that support value.

Use the unchanged source

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

$$f_0 = \Theta\phi_*, \quad \mathcal{M}f_0(s) = g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

These are the arithmetic-input equations (A1)–(A3), also identified with the specified Hochschild trace in HOCHSCHILD_COMPARISON.md, equations (H4)–(H10). Define the explicit logarithmic-coordinate function

$$\psi(y) = e^{y/2} f_0(e^y). \tag{1}$$

Mellin substitution gives

$$g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy. \tag{2}$$

The original reflection $f_0(x) = x^{-1}f_0(1/x)$ gives $\psi(-y) = \psi(y)$. This reflection also follows by applying the source Fourier identity to ϕ_* : Fourier sends D to $1 - D$, so it fixes $(D^2 - D)e^{-\pi x^2} = \phi_*$.

For $x \geq 1$, every summand in $f_0(x)$ is positive, since

$$4\pi^2 n^4 x^4 - 6\pi n^2 x^2 = 2\pi n^2 x^2 (2\pi n^2 x^2 - 3) > 0.$$

Reflection proves positivity for all $x > 0$. The Gaussian series also gives a constant C such that

$$0 < \psi(y) \leq C \exp(9|y|/2 - \pi e^{2|y|}). \quad (3)$$

For $y \geq 0$, bound the convergent sum of $n^4 e^{-\pi(n^2-1)e^{2y}}$ by its value at $y = 0$; reflection supplies the other half-line. Thus all differentiations and all moments below converge absolutely.

242.3 2. Coefficients with the original mass retained

Put

$$M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy = g^{(2j)}(1/2), \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}. \quad (4)$$

Define the entire function F by its specified coefficients:

$$F(w) = \sum_{j \geq 0} a_j w^j, \quad F(z^2) = g(1/2 + iz), \quad a_0 = g(1/2) > 0. \quad (5)$$

No square-root branch is part of the definition, and a_0 is not assigned the value one. The map $z \mapsto z^2$ identifies the two reflected coordinates only at this scalar-function observation; the original packet modules are retained in Section 5.

Define b_n by the analytic logarithmic derivative near zero:

$$-\frac{F'(w)}{F(w)} = \sum_{n \geq 0} b_n w^n. \quad (6)$$

The complete finite coefficient map is

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (7)$$

It follows by multiplying (6) by F and extracting the coefficient of w^n . For example,

$$b_0 = -\frac{a_1}{a_0}, \quad b_1 = \frac{a_1^2}{a_0^2} - \frac{2a_2}{a_0}, \quad b_2 = -\frac{a_1^3}{a_0^3} + \frac{3a_1 a_2}{a_0^2} - \frac{3a_3}{a_0}. \quad (8)$$

This operation need not preserve positivity of arbitrary coefficient sequences. The matrices used below are not positive by their definition.

242.4 3. A finite certificate contradicting RH

Let α_j be the distinct zeros of F , with their multiplicities m_j , and put $\beta_j = \alpha_j^{-1}$. These symbols are used to prove the test, not supplied to its numerical implementation.

Estimate (3), followed by $t = e^{2|y|}$, bounds the maximum modulus of g by an exponential of $O(R \log R)$. The maximum modulus of F at radius R is consequently bounded by an exponential of $O(\sqrt{R} \log R)$. Its order is at most $1/2 < 1$. Genus-zero Hadamard factorization therefore gives

$$F(w) = a_0 \prod_j (1 - \beta_j w)^{m_j}, \quad \sum_j m_j |\beta_j| < \infty. \quad (9)$$

There is no nonconstant exponential factor of integer degree at this order. Differentiating locally uniformly away from zeros proves

$$b_n = \sum_j m_j \beta_j^{n+1}. \quad (10)$$

The exact coordinate of a zeta zero $\rho = 1/2 + \delta + i\gamma$ is

$$z = -i(\rho - 1/2) = \gamma - i\delta, \quad \alpha = z^2 = \gamma^2 - \delta^2 - 2i\gamma\delta,$$

$$\beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \quad (11)$$

The two reflected roots $\rho, 1 - \rho$ give one zero α of the same multiplicity in F . Conjugation gives the conjugate α and β .

For every real s , (1)–(3) give $g(s) = \int \psi(y) e^{(s-1/2)y} dy > 0$. Hence F has no nonpositive real zeros: $F(-v^2) = g(1/2 - v) > 0$. Thus RH is equivalent to every β_j being positive real.

For $d \geq 0$, define the actual Hankel coefficient matrix

$$H_d = (b_{r+s})_{0 \leq r, s \leq d}. \quad (12)$$

For $P(X) = \sum_{r=0}^d c_r X^r$ with real coefficients, the exact value is

$$c^T H_d c = \sum_j m_j \beta_j^2 P(\beta_j)^2. \quad (13)$$

Absolute convergence follows from (9) and boundedness of the polynomial on the compact set of reciprocal zeros and zero. Conjugate pairs make (13) real.

Under RH, each term is nonnegative. Therefore

$$c^T H_d c < 0 \implies \text{RH is false.} \quad (14)$$

The certificate may use $c \in \mathbb{Q}^{d+1}$: a strict negative value persists under sufficiently small rational coefficient perturbations. With validated enclosures $b_n \in [\underline{b}_n, \bar{b}_n]$, exact rational interval arithmetic on $\sum c_r c_s b_{r+s}$ is enough to certify that its upper endpoint is negative.

A second necessary family is $H_d^{(1)} = (b_{r+s+1})$. Under RH its quadratic form is $\sum_j m_j \beta_j^2 P(\beta_j)^2 \geq 0$. We tested both, though the unshifted family alone is complete by the next construction.

242.5 4. Work through the off-line counterfactual: construct its negative direction

Assume for this argument that RH is false. Equation (11) supplies a nonreal reciprocal zero β and its conjugate, each of multiplicity m . They are nonreal because g has no real zeros, so both γ and δ are nonzero.

Put $r = |\beta|/2$. There are finitely many distinct reciprocal zeros of modulus at least r . Remove $\beta, \bar{\beta}$ from that finite set and call the remaining set $\mathcal{A}$. It is closed under conjugation. Define the real polynomial

$$B(X) = \prod_{\gamma \in \mathcal{A}} (X - \gamma). \quad (15)$$

Choose $c \in \mathbb{C}$ satisfying $\beta c^2 = -1$; either of the two choices works. For an integer $N \geq 0$, put

$$w_N = \frac{c}{\beta^N B(\beta)}, \quad v_N = \frac{\operatorname{Im} w_N}{\operatorname{Im} \beta}, \quad u_N = \operatorname{Re} w_N - \operatorname{Re} \beta v_N,$$

$$\boxed{P_N(X) = X^N B(X)(\nu_N + v_N X).} \quad (16)$$

The coefficients are real. At the two target points,

$$P_N(\beta) = c, \quad P_N(\bar{\beta}) = \bar{c}.$$

At every other point of modulus at least r , P_N vanishes. The two target contributions to (13) are exactly

$$m\beta c^2 + m\bar{\beta} \bar{c}^2 = -2m. \quad (17)$$

Here is an explicit bound on everything else. Let

$$B_0 = \prod_{\gamma \in \mathcal{A}} (r + |\gamma|), \quad C = \frac{|c|B_0}{|B(\beta)|} \left(1 + \frac{|\operatorname{Re} \beta| + r}{|\operatorname{Im} \beta|} \right),$$

$$T = \sum_{|\beta_j| < r} m_j |\beta_j| < \infty.$$

For $|X| \leq r$, the coefficient definitions imply

$$|P_N(X)| \leq C \left(\frac{r}{|\beta|} \right)^N = C 2^{-N}.$$

Consequently the complete, real quadratic form satisfies

$$\boxed{\mathcal{L}(P_N^2) := \sum_j m_j \beta_j P_N(\beta_j)^2 \leq -2m + C^2 T 4^{-N}.} \quad (18)$$

For all sufficiently large N the right side is strictly negative. Every omitted reciprocal zero is included in the tail estimate, and every multiplicity is retained. This proves

$$\boxed{\text{RH} \iff H_d \succeq 0 \text{ for every } d \geq 0.} \quad (19)$$

It also explains the limitations of a finite search. The degree needed in (16) depends on the location of the off-line pair, the other larger reciprocal zeros, and the tail. The proof does not say that a counterexample must be visible in a small matrix. The polynomial (16) is a counterfactual construction; no actual off-line pair has been used to instantiate it.

242.6 5. Connect the witness to the original finite packet, including its jets

For a finite conjugation- and reflection-stable packet of actual zeros, retain the original inclusion

$$\sigma_h : E_h = \mathbb{C}[s]/(h) \hookrightarrow Q, \quad D_Q \sigma_h = \sigma_h A_h.$$

The original arithmetic input constructs this through the kernel of multiplication by $j_h g$. An arbitrary polynomial example is not automatically such a packet.

Because $g(1/2) > 0$, the operator $A_h - \frac{1}{2}I$ is invertible on every actual packet. Define the specified rational functional calculus

$$T_h = -(A_h - \frac{1}{2}I)^{-2}, \quad \mathbb{C}[X] \longrightarrow \text{End}_{\mathbb{C}}(E_h), \quad P \longmapsto P(T_h). \quad (20)$$

On a full local block $A_h = \rho I + N_\rho$, $N_\rho^m = 0$, the matrix is exactly

$$T_h = \sum_{j=0}^{m-1} (-1)^{j+1} (j+1) (\rho - 1/2)^{-j-2} N_\rho^j. \quad (21)$$

The coefficient of N_ρ is $2(\rho - 1/2)^{-3} \neq 0$, so this local coordinate change retains the nilpotent length. The two reflected primary modules can have the same T_h eigenvalue, but they remain separate in E_h .

The packet contribution to (13) is

$$\mathcal{L}_h(P^2) = \frac{1}{2} \text{Tr}_{E_h}(T_h P(T_h)^2). \quad (22)$$

The factor $1/2$ counts each reflected pair once, matching (5). The trace records algebraic multiplicities, while the full operators and their nilpotent coefficients remain upstream of the trace map. Absolute convergence in (9) makes the full functional the limit over packets exhausting the actual zero set.

The existing critical-line comparison satisfies

$$\ker(\Gamma \sigma_h) = E_{h,\text{off}}.$$

Equation (22) is evaluated on the source packet before applying that map. An off-critical class maps to $(\ell, 0)$ under the split lift of Γ , but is not deleted from the source on which the trace and negative witness are evaluated. This is why the search does not obtain a false positive or a false proof by restricting to the critical-only target.

242.7 6. The performed search uses the original theta seed, not zero ordinates

The numerical calculation uses (4), in the explicit form

$$M_{2j} = 4 \sum_{n \geq 1} \int_0^\infty y^{2j} \left(4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2} \right) e^{-\pi n^2 e^{2y}} dy. \quad (23)$$

The factor four consists of the source theta factor two and the reflection factor two. The implementation computes $M_0, \dots, M_{64}$, forms (7), and tests both matrix families through dimension 16 by an unpivoted LDL^T calculation in their original monomial coordinates. No rescaling or preconditioning is used. No zero ordinate is supplied.

Two runs were completed:

Working decimal precision	Gauss–Legendre nodes	Retained theta terms	y interval	Outcome
120	384	12	[0,4]	All 16 pivots positive numerically, in both families
180	768	14	[0,4]	Same outcome

Selected values are

$$a_0 = 0.9942415563766282198255474793707954396 \dots,$$

$$b_0 = 0.0231049931154189707889338104303390140 \dots,$$

$$b_1 = 0.00003717259928526968616486626247174057845 \dots,$$

$$b_2 = 0.0000001441739314009732796953815560948209 \dots,$$

$$\det H_1 = b_0 b_2 - b_1^2 = 1.949335554819142214739878219172139506 \times 10^{-9} \dots$$

The largest relative difference between the two runs' stored pivot values is below 3×10^{-79} . Those stored values contain 80 significant digits, so the comparison does not support a stronger precision claim. The scripts report the input choices and complete pivot sequences.

These are exploratory multiple-precision quadratures, not interval-certified integrations. A failed attempt to install an interval-arithmetic backend did not produce an interval certificate; only the documented mpmath computations are reported. Positive numerical pivots do not prove the corresponding signs, and a negative numerical pivot would require certified analytic enclosures before it could count as a disproof. No negative numerical pivot was found.

242.8 7. Finite verification and the next decisive object

Six standard-library exact rational checks passed in normal and optimized Python. They verify the logarithmic recurrence, positive-node identity, an explicit nonreal-pair negative control, and a rational interval evaluator that cannot call an interval containing zero strictly negative. The negative control is expressly an auxiliary polynomial, not an actual zeta packet.

The interval evaluator's contract is limited: it checks the quadratic form from supplied valid intervals. Establishing that those intervals enclose the actual b_n remains an analytic task. No Lean certificate is claimed.

The decisive counterexample object in this branch is now a finite tuple

$$(d, c_0, \dots, c_d, [b_0, \bar{b}_0], \dots, [b_{2d}, \bar{b}_{2d}])$$

with rational c_r , rigorously justified coefficient enclosures from the original theta input, and a strictly negative upper bound for $\sum_{r,s} c_r c_s b_{r+s}$. Equation (14) proves exactly what that tuple would establish. Equation (18) proves that every actual off-line counterexample has a finite witness of this type at some degree.

The calculation performed here did not find that tuple. The failure of the previous auxiliary-purity inference is not used as evidence that one exists. Geometry over the original absolute base remains a

separate possible source of further constraints; this note establishes no impossibility theorem about it.

242.9 Source and execution record

- Original arithmetic source: supplied `arithmetic_input.tex`, equations (A1)–(A10), including the actual multiplication-by- $j_h g$ kernel sequence.
- Original support reconstruction and internal homology: supplied SplitZero reconstruction record, Sections 1–4. The source-labelled zero and the external absent value are kept distinct throughout.
- Original critical comparison: supplied `HOCHSCHILD_COMPARISON.md`, equations (H16)–(H25). The visible text supplies the exact off-critical kernel and critical-jet retractions.
- Prior scope: supplied `Tau_Deligne_Bound_Audit/NOTE.tex`, “Result and scope”. It explicitly does not refute the split-zero construction or establish falsity of the arithmetic estimate for 2ξ .
- Classical context: R. Zhang, *On Power Sums of Positive Numbers*, arXiv:1510.03420 (2015). Its abstract documents moment-theoretic positivity criteria for genus-zero entire functions and applications to RH. The concrete criterion and separation proof above are proved here, not attributed as a verbatim theorem from an uninspected full text.
- Classical background: Hadamard factorization for entire functions of order less than one; the functional equation and completed-zeta notation are also recorded in NIST DLMF Section 25.4. The source Gaussian estimate above supplies the order bound needed for this application.
- Computation: mpmath 1.3.0. Results are in `probe_120dps.json` and `probe_180dps.json`; reproduction script is `search_theta_moments.py`. Exact finite checks and the interval-input verifier are in `exact_witness_checks.py`.

No GitHub file, branch, comment, or remote publication was changed in this continuation.

243 Certified original-theta Hankel calculation

This calculation certifies strict positive definiteness of both original matrices

$$H_{15} = (b_{r+s})_{0 \leq r, s \leq 15}, \quad H_{15}^{(1)} = (b_{r+s+1})_{0 \leq r, s \leq 15}. \quad (\text{CH.1})$$

Their coefficients are computed from the original $g = 2\xi$ theta source. Every one of their 32 leading principal minors is certified positive. The coordinates, original mass, theta terms and logarithmic coefficient recurrence are retained. No zero ordinate is supplied to the calculation.

243.1 1. Original integral and coefficients

Use the original functions

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \psi(y) = e^{y/2} f_0(e^y). \quad (\text{CH.2})$$

For completeness, write $\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x^2}$ and $D = -x\partial_x$. Termwise differentiation on compact positive x -intervals gives $f_0 = (D^2 - D)\vartheta$. The original Gaussian Poisson identity is $\vartheta(x) = x^{-1}\vartheta(1/x)$. For the exact reflection operator $(\mathcal{R}v)(x) = x^{-1}v(1/x)$, differentiation gives $D\mathcal{R} = \mathcal{R}(1 - D)$, and hence $(D^2 - D)\mathcal{R} = \mathcal{R}(D^2 - D)$. Applying this identity to ϑ proves $f_0(x) = x^{-1}f_0(1/x)$ and therefore $\psi(-y) = \psi(y)$. All powers of x are retained in this map. Consequently, for every even nonnegative integer J ,

$$\begin{aligned} M_J &= \int_{\mathbb{R}} y^J \psi(y) dy \\ &= \int_0^\infty y^J \sum_{n \geq 1} \left(16\pi^2 n^4 e^{9y/2} - 24\pi n^2 e^{5y/2} \right) e^{-\pi n^2 e^{2y}} dy. \end{aligned} \quad (\text{CH.3})$$

Both factors two, from theta summation and reflection, are present. Every summand in (CH.3) is nonnegative for $y \geq 0$, and is strictly positive for $y > 0$: its non-polynomial coefficient factors as

$$8\pi n^2 e^{5y/2} (2\pi n^2 e^{2y} - 3) e^{-\pi n^2 e^{2y}} > 0. \quad (\text{CH.4})$$

For $J = 0$, positivity holds at $y = 0$ as well.

The Mellin substitution $x = e^y$, including $dx = e^y dy$, gives

$$g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy, \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad F(w) = \sum_{j \geq 0} a_j w^j. \quad (\text{CH.5})$$

The bounds proved below supply absolute integrability after multiplication by every power of y , and by $e^{A|y|}$ for each fixed A : replace $9/2$ by $9/2 + A$ in (CH.11) and choose a sufficiently large positive L , so its decay constant is positive. Reflection gives the same estimate on the other half-line. Thus dominated differentiation on compact z -sets is valid and proves $F(z^2) = g(1/2 + iz)$ with the displayed coefficients. In particular the original $a_0 = M_0 = g(1/2)$ is strictly positive. Multiplication of the convergent power series $-F'(w)/F(w) = \sum_{n \geq 0} b_n w^n$ by $F(w)$ gives

$$a_0 b_n + \sum_{j=1}^n a_j b_{n-j} = -(n+1)a_{n+1}, \quad b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{CH.6})$$

This is the exact coefficient map used by the program. The denominator is the computed interval for the original a_0 . It is never replaced by one.

243.2 2. Explicit positive omitted regions

Fix $N = 20$ and $L = 4$. The finite integral retains precisely $1 \leq n \leq 19$ and $0 \leq y \leq 4$. Its complement is the disjoint union of the omitted terms $n \geq 20$ on $0 \leq y \leq 4$ and every $n \geq 1$ on $y > 4$. Endpoints have measure zero.

For $n = N + \ell$, $\ell \geq 0$, the elementary inequality $\log(1 + t) \leq t$ for $t \geq 0$, proved by differentiating $t - \log(1 + t)$, implies

$$(N + \ell)^4 \leq N^4 e^{4\ell/N}, \quad (N + \ell)^2 \geq N^2 + 2N\ell. \quad (\text{CH.7})$$

Hence, for $y \geq 0$,

$$\sum_{n \geq N} n^4 e^{-\pi n^2 e^{2y}} \leq N^4 e^{-\pi N^2} \sum_{\ell \geq 0} e^{(4/N - 2\pi N)\ell} = \frac{N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}}. \quad (\text{CH.8})$$

The exponent $4/N - 2\pi N$ is negative for every integer $N \geq 1$. Using (CH.4), and bounding the positive difference in (CH.3) by its first positive term, the first omitted region lies in $[0, T_{J,n}]$, where

$$T_{J,n} = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}}. \quad (\text{CH.9})$$

Indeed $y^J \leq L^J$ and $e^{9y/2} \leq e^{9L/2}$ on the finite interval, whose length contributes the displayed factor L . The deliberately larger bound L^{J+1} is retained.

For the second omitted region the same argument, with $N = 1$ but retaining the y -dependent first exponential, gives

$$\sum_{n \geq 1} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{e^{-\pi e^{2y}}}{1 - e^{4 - 2\pi}}, \quad y \geq 0. \quad (\text{CH.10})$$

Set $h_J(y) = y^J e^{9y/2 - \pi e^{2y}}$ on $y \geq L > 0$. Its logarithmic derivative is

$$\frac{h'_J(y)}{h_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq -\alpha_J, \quad \alpha_J = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{CH.11})$$

For $0 \leq J \leq 64$, $N = 20$, $L = 4$, all α_J are strictly positive. For example, $\pi > 3$ and $e^8 > 1 + 8 + 8^2/2 = 41$ give $\alpha_J > 246 - 9/2 - 16 > 0$. Integration of (CH.11) gives $h_J(y) \leq h_J(L) e^{-\alpha_J(y-L)}$, and then

$$T_{J,y} = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4 - 2\pi})(2\pi e^{2L} - 9/2 - J/L)} \quad (\text{CH.12})$$

bounds the second omitted region. Therefore the complete exact moment satisfies

$$M_J \in I_{J,\text{finite}} + [0, T_{J,n} + T_{J,y}]. \quad (\text{CH.13})$$

The constants and signs of both original terms survive (CH.3) and the finite integration. The positive upper majorant is used only for the two explicitly omitted regions.

243.3 3. Validated finite integration and its exact export

The program uses python-flint 0.9.0, linked to FLINT 3.6.0, with 1024-bit arithmetic. For each even $J = 0, 2, \dots, 64$, it evaluates

$$z^J e^{z/2} \sum_{n=1}^{19} (16(\pi n^2 e^{2z})^2 - 24\pi n^2 e^{2z}) e^{-\pi n^2 e^{2z}}. \quad (\text{CH.14})$$

This is algebraically identical to the finite integrand in (CH.3). It is an entire complex function: all powers are nonnegative integer powers, and every remaining operation is addition, multiplication or the entire exponential. It has no branch cut, pole, absolute value, conjugation or piecewise real continuation.

The `acb.integral` callback receives a Boolean `analytic`. The official contract requires the callback to establish analyticity on the input complex ball when this flag is true. Formula (CH.14) is entire, so the same formula is valid for both values of the flag on every ball. This is the reason that the implementation may ignore the flag. It is not assuming that analyticity along the real segment alone suffices.

The finite interval is divided at the exact rationals $0, 1/2, 1, \dots, 7/2, 4$. Each of its eight integrals is evaluated with relative and absolute tolerance goals 2^{-850} , evaluation limit 200000 and depth limit 50. These controls affect effort and enclosure width; the final proof uses the returned enclosures themselves. Every returned value must be finite and its imaginary part must contain zero. Its real interval therefore encloses the real integral. The eight real intervals are added with outward rounding.

The two tail formulas are evaluated with Arb arithmetic, after verifying strict positivity of each denominator interval. If their sum has upper endpoint U_J , the program forms the interval $\text{hull}(0, U_J)$, and adds it to the real finite interval. This proves (CH.13) with the actually stored enclosure.

For every reported Arb ball, `lower()` and `upper()` produce exact dyadic endpoints rounded respectively downward and upward. `man_exp()` writes each as $m2^e$, with integers m, e . The export writes that same rational as numerator $m2^e$ and denominator one if $e \geq 0$, and numerator m , denominator 2^{-e} otherwise. No decimal conversion intervenes. Every decimal display field is for reading only. The lower and upper integer fractions are the authoritative certificate. The program raises the Python integer-string limit to accommodate the large dyadic denominator of (CH.12); this changes only serialization capacity.

243.4 4. Interval coefficient and LDL proof

Arb interval operations enclose every real operation on represented inputs. Induction in (CH.6), beginning with $a_0 > 0$, therefore proves that each stored a_j encloses the exact coefficient in (CH.5) and each stored b_n encloses the exact logarithmic coefficient in (CH.6). The 33 moments determine $a_0, \dots, a_{32}$ and $b_0, \dots, b_{31}$, exactly the data required by (CH.1).

For each shift $\epsilon = 0, 1$, put $A_{ij} = b_{i+j+\epsilon}$. In the original ordered monomial basis, define the unit lower triangular matrix L and diagonal $D = \text{diag}(d_0, \dots, d_{15})$ recursively by

$$d_j = A_{jj} - \sum_{k < j} L_{jk}^2 d_k, \quad L_{ij} = \frac{A_{ij} - \sum_{k < j} L_{ik} L_{jk} d_k}{d_j} \quad (i > j). \quad (\text{CH.15})$$

The implementation checks that the entire computed interval for d_j is strictly positive before every division. Induction on j proves that the stored interval values contain the exact values of (CH.15). The equations for the diagonal and lower triangular entries are exactly the equations of $A = LDL^T$; symmetry supplies the upper triangular entries. Since L is invertible with determinant one,

$$c^T A c = \sum_{j=0}^{15} d_j (L^T c)_j^2 > 0 \quad \text{for every } c \in \mathbb{R}^{16} \setminus \{0\}. \quad (\text{CH.16})$$

For each leading size r , restriction to the first r coordinates and triangularity give

$$\det A_{[0, r-1]} = \prod_{j=0}^{r-1} d_j > 0. \quad (\text{CH.17})$$

The exact interval computation certifies all 32 pivots as positive. No pivoting, coordinate rescaling, diagonal congruence or replacement of the source mass occurs.

243.5 5. Recorded values and finite conclusion

The authoritative file `certificate_dimension16_1024.json` stores all 264 finite integral pieces, their real and imaginary intervals, all 66 tail bounds, all 33 complete moments, all coefficients, and both complete lower triangular factor intervals. The run took about 16.14 seconds on the recorded host. This duration measures one reproducible computation, rather than a mathematical assumption.

For orientation the original mass and first coefficients begin

$$\begin{aligned} a_0 &= 0.99424155637662821982554747937079544 \dots, \\ b_0 &= 0.023104993115418970788933810430339014 \dots, \\ b_1 &= 3.7172599285269686164866262471740578 \dots \times 10^{-5}, \\ b_2 &= 1.4417393140097327969538155609482091 \dots \times 10^{-7}. \end{aligned} \tag{CH.18}$$

The last pivots and full determinants are enclosed within the following simple rational intervals:

$$\begin{array}{c|c|c} \epsilon & d_{15} & \det H_{15}^{(\epsilon)} \\ \hline 0 & (6 \cdot 10^{-107}, 7 \cdot 10^{-107}) & (9 \cdot 10^{-822}, 10 \cdot 10^{-822}) \\ 1 & (7 \cdot 10^{-111}, 8 \cdot 10^{-111}) & (6 \cdot 10^{-878}, 7 \cdot 10^{-878}) \end{array} \tag{CH.19}$$

The exact stored rational endpoints are much narrower; the respective last pivots retain 656 and 648 relative accuracy bits.

Thus this calculation establishes strict positivity of the two specified dimension-16 matrices and every leading truncation. Its quantified conclusion concerns these finite matrices. It supplies certified original-theta inputs in place of the Workbench’s unvalidated quadrature values at these dimensions.

243.6 Reproduction

Run `certify_theta_hankel.py` with arguments `--dimension 16 --precision 1024 --target-bits 850 --output certificate_dimension16_1024.json` using `python-flint 0.9.0`. The defaults retain $N = 20$, $L = 4$. The file `certificate_dimension2_256.json` records a separate smaller run with dimension two, precision 256 and tolerance exponent 200.

The official integration and real-endpoint API contracts are documented at:

- [python-flint complex integration](#)
- [python-flint real ball endpoints](#)

The local `API_CONTRACT.txt` records the installed method docstrings used when implementing the computation.

244 Independent theta-tail proof and certificate code review

Date: 2026-09-13.

This review concerns the exact original theta integrand, its omitted tails, and the complete code path that produces the dimension-16 logarithmic Hankel certificate. It retains every coefficient in the integrand. The parent is independently replaying the rational interval arithmetic; this review supplies the analytic proof and an independent complete source read.

244.1 1. Exact objects and retained integral

Let J be a nonnegative integer, N a positive integer, and $L > 0$. For $n \geq 1$ and real $y \geq 0$, set

$$f_{J,n}(y) = y^J \left(16\pi^2 n^4 e^{9y/2} - 24\pi n^2 e^{5y/2} \right) e^{-\pi n^2 e^{2y}}, \quad F_J(y) = \sum_{n=1}^{\infty} f_{J,n}(y). \quad (\text{TR1})$$

When $J = 0$, the factor y^0 is the constant function 1, including at zero. The intended integral and retained part are

$$I_J = \int_0^{\infty} F_J(y) dy, \quad I_{J,N,L}^{\text{fin}} = \int_0^L \sum_{n=1}^{N-1} f_{J,n}(y) dy. \quad (\text{TR2})$$

For $N = 1$, the finite sum is empty and its integral is zero. The actual run uses $N = 20$, $L = 4$, and $J = 0, 2, \dots, 64$; it therefore retains precisely $n = 1, \dots, 19$.

244.2 2. Positivity and the exact majorant

The identity

$$16\pi^2 n^4 e^{9y/2} - 24\pi n^2 e^{5y/2} = 8\pi n^2 e^{5y/2} (2\pi n^2 e^{2y} - 3) \quad (\text{TR3})$$

proves that the parenthesis in (TR1) is strictly positive, since $2\pi n^2 e^{2y} - 3 \geq 2\pi - 3 > 0$. Thus

$$0 \leq f_{J,n}(y) \leq 16\pi^2 y^J e^{9y/2} n^4 e^{-\pi n^2 e^{2y}}. \quad (\text{TR4})$$

The upper estimate discards a nonnegative subtracted term only in the estimate. Both terms remain in the actual finite integral.

For every $N \geq 1$,

$$0 < r_N := e^{4/N - 2\pi N} \leq r_* := e^{4 - 2\pi} < 1. \quad (\text{TR5})$$

Here $4/N \leq 4$, $N \geq 1$, and $\pi > 3$.

244.3 3. Omitted summands on the retained interval

Put $a_n = n^4 e^{-\pi n^2}$. Its exact consecutive-term ratio is

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n} \right)^4 e^{-\pi(2n+1)}. \quad (\text{TR6})$$

For $x \geq 0$, $\log(1+x) \leq x$: the difference $x - \log(1+x)$ is zero at zero and has derivative $x/(1+x) \geq 0$. Consequently, for $n \geq N$,

$$\begin{aligned} \log \frac{a_{n+1}}{a_n} &= 4 \log(1 + 1/n) - 2\pi n - \pi \\ &\leq 4/n - 2\pi n - \pi \\ &\leq 4/N - 2\pi N - \pi \leq 4/N - 2\pi N = \log r_N. \end{aligned} \quad (\text{TR7})$$

It follows by induction that $a_{N+j} \leq a_N r_N^j$ for every integer $j \geq 0$, and hence

$$\sum_{n=N}^{\infty} n^4 e^{-\pi n^2} \leq \frac{N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TR8})$$

On $0 \leq y \leq L$, use $y^J \leq L^J$, $e^{9y/2} \leq e^{9L/2}$, and $e^{-\pi n^2 e^{2y}} \leq e^{-\pi n^2}$. Equations (TR4) and (TR8) then give

$$\begin{aligned} 0 \leq E_{J,N,L}^{(n)} &:= \int_0^L \sum_{n=N}^{\infty} f_{J,n}(y) dy \\ &\leq \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N-2\pi N}} =: T_n(J, N, L). \end{aligned} \quad (\text{TR9})$$

The extra negative term $-\pi$ in (TR7) was discarded in the direction that enlarges the upper bound. This proves the proposed expression with its stated denominator.

The same argument with $N = 1$ gives a summable uniform majorant on each compact interval $[0, L]$. The series defining F_J is therefore uniformly convergent there and continuous. Its summands are nonnegative, so monotone convergence also justifies the sum-integral interchanges above and below.

244.4 4. The complete integration tail

For fixed $y \geq 0$, write $a_n(y) = n^4 e^{-\pi n^2 e^{2y}}$. Then

$$\begin{aligned} \log \frac{a_{n+1}(y)}{a_n(y)} &= 4 \log(1 + 1/n) - \pi(2n+1)e^{2y} \\ &\leq 4/n - 2\pi n e^{2y} - \pi e^{2y} \\ &\leq 4 - 2\pi = \log r_*. \end{aligned} \quad (\text{TR10})$$

The last inequality uses $1/n \leq 1$, $n e^{2y} \geq 1$, and $-\pi e^{2y} \leq 0$. Since $a_1(y) = e^{-\pi e^{2y}}$, the geometric-series estimate is

$$\sum_{n=1}^{\infty} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{e^{-\pi e^{2y}}}{1 - e^{4-2\pi}}. \quad (\text{TR11})$$

Therefore

$$0 \leq F_J(y) \leq \frac{16\pi^2}{1 - e^{4-2\pi}} y^J e^{9y/2 - \pi e^{2y}}. \quad (\text{TR12})$$

Define

$$A_{J,L} = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{TR13})$$

When $A_{J,L} > 0$, the positive function $q_J(y) = y^J e^{9y/2 - \pi e^{2y}}$, for $y \geq L > 0$, satisfies

$$\frac{q'_J(y)}{q_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq \frac{J}{L} + \frac{9}{2} - 2\pi e^{2L} = -A_{J,L}. \quad (\text{TR14})$$

The first summand is identically zero for $J = 0$. Integrating this differential inequality gives $q_J(y) \leq q_J(L) e^{-A_{J,L}(y-L)}$. Thus

$$\begin{aligned} 0 \leq E_{J,L}^{(y)} &:= \int_L^{\infty} \sum_{n=1}^{\infty} f_{J,n}(y) dy \\ &\leq \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4-2\pi})(2\pi e^{2L} - 9/2 - J/L)} =: T_y(J, L). \end{aligned} \quad (\text{TR15})$$

This integral includes every $n \geq 1$, including all omitted summands $n \geq N$ beyond L .

244.5 5. The exact remainder and interval addition

The omitted domain is the disjoint union, up to the measure-zero endpoint slice, of

$$\{(n, y) : n \geq N, 0 \leq y < L\} \quad \text{and} \quad \{(n, y) : n \geq 1, L \leq y < \infty\}. \quad (\text{TR16})$$

The two integrals are nonnegative and finite by (TR9) and (TR15). They therefore give the exact identity and enclosure

$$I_J = I_{J,N,L}^{\text{fin}} + E_{J,N,L}^{(n)} + E_{J,L}^{(y)}, \quad I_J - I_{J,N,L}^{\text{fin}} \in [0, T_n + T_y]. \quad (\text{TR17})$$

There is no double counting of $n \geq N$ beyond L : that region occurs only in $E^{(y)}$. If $[a, b]$ encloses the finite integral and $\bar{T} \geq T_n + T_y$, then

$$I_J \in [a, b] + [0, \bar{T}] = [a, b + \bar{T}]. \quad (\text{TR18})$$

The entire interval $[0, \bar{T}]$, rather than a tail midpoint, is what must be added.

For the actual parameters the formulas are

$$\begin{aligned} T_n(J, 20, 4) &= \frac{16\pi^2 4^{J+1} e^{18} 20^4 e^{-400\pi}}{1 - e^{1/5-40\pi}}, \\ T_y(J, 4) &= \frac{16\pi^2 4^J e^{18-\pi e^8}}{(1 - e^{4-2\pi})(2\pi e^8 - 9/2 - J/4)}. \end{aligned} \quad (\text{TR19})$$

All denominator signs are proved exactly for $0 \leq J \leq 64$:

$$1/5 - 40\pi < -599/5 < 0, \quad 4 - 2\pi < -2 < 0, \quad (\text{TR20})$$

and $e^8 > 1 + 8 = 9$ gives

$$2\pi e^8 - \frac{9}{2} - \frac{J}{4} > 6 \cdot 9 - \frac{9}{2} - 16 = \frac{67}{2} > 0. \quad (\text{TR21})$$

Thus the claimed tail hypotheses are verified for every actual moment in the run, without a floating-point sign assumption.

244.6 6. Entire finite integrand and its literal code expression

For each nonnegative integer J , the function

$$z \mapsto z^J \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{9z/2} - 24\pi n^2 e^{5z/2}) e^{-\pi n^2 e^{2z}} \quad (\text{TR22})$$

is entire on $\mathbb{C}$: the only operations are a polynomial, exponentials and their compositions, and finite sums and products. The code sets $x = \pi n^2 e^{2z}$ and evaluates

$$z^J e^{z/2} \sum_{n=1}^{N-1} (16x^2 - 24x) e^{-x}. \quad (\text{TR23})$$

The exact identities

$$16x^2 e^{z/2} = 16\pi^2 n^4 e^{9z/2}, \quad 24x e^{z/2} = 24\pi n^2 e^{5z/2} \quad (\text{TR24})$$

prove equality of (TR23) and (TR22). This rewrite retains the subtracted term and all original coefficients.

The installed python-flint 0.9.0 documentation for `acb.integral` was read directly. Its callback receives a complex ball and an analytic flag. When that flag is true, the callback must establish analyticity on the ball or return a nonfinite ball. The entire function (TR22) meets this requirement on every complex ball, so ignoring that flag in this particular callback is justified by the proof above. There are no fractional powers, logarithms, branches, or pole-containing denominators in the callback.

The code uses `arb.pi()` and integer arithmetic for its coefficients. Every division appearing in the callback or integration endpoints is performed on an `Arb/Acb` object and an exact integer. The finite path is partitioned into the eight consecutive exact intervals $[j/2, (j+1)/2]$, $j = 0, \dots, 7$. Their union is exactly $[0, 4]$. The code rejects nonfinite integrals and any reported imaginary enclosure excluding zero, then sums the complex balls. Since the exact integral is real, its real-part enclosure contains (TR2). The tolerance is an exact dyadic 2^{-850} ; the working precision is 1024 bits. The elapsed-time floats do not enter the mathematics.

244.7 7. Complete tail-code and exact-export review

The function `positive_tails` computes precisely (TR9), (TR15), and (TR13), with outward-rounded `Arb` operations. It explicitly rejects any run that fails to certify $1 - r_N > 0$, $1 - r_* > 0$, or $A_{J,L} > 0$. It uses the original factor $16\pi^2$ in each tail.

The function `compute_moment` takes the outward upper endpoint of the sum of the two tail balls, forms the ball union of that endpoint with zero, and adds that union to the real finite-integral ball. An enclosing ball union can slightly enlarge the endpoint interval, but it contains the whole interval required by (TR18). It does not replace the nonnegative tail interval with its midpoint. The final strict-positive comparison certifies positivity of the whole resulting moment ball.

The installed `arb.lower` and `arb.upper` documentation was also read directly. These methods return exact finite floating-point numbers rounded respectively toward negative and positive infinity at the active precision. The installed `arb.man_exp` documentation says it requires an exact finite input and returns an integer mantissa m and integer exponent e . Therefore the exported endpoint is exactly

$$m2^e = \begin{cases} (m2^e)/1, & e \geq 0, \\ m/(2^{-e}), & e < 0. \end{cases} \quad (\text{TR25})$$

This is precisely the case distinction in `exact_dyadic`. Python integers and their decimal strings retain these numerators and denominators exactly. They need not be reduced to lowest terms to represent the same rational endpoint. The enclosing real interval exported by `enclosure` consequently retains a certified lower and upper rational bound; its shorter decimal display is additional presentation data.

The helper `from_enclosure` parses both endpoint fractions through `Arb` and unions their enclosures. This remains sound if rounding enlarges them. It is unused in the original integration and LDL run and therefore does not participate in the output's arithmetic chain.

244.8 8. Original logarithmic coefficient recurrence

Write $M_{2j} = I_{2j}$ for the original moments defined by (TR2). Define the exact formal series

$$A(z) = \sum_{j \geq 0} a_j z^j, \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad B(z) = -\frac{A'(z)}{A(z)} = \sum_{n \geq 0} b_n z^n. \quad (\text{TR26})$$

The nonzero constant term $a_0 = I_0 > 0$ is proved by (TR1), (TR3), and integration over any interval of positive length. Thus the formal division in (TR26) is uniquely defined, and no convergence

assumption is needed to compute its finite jet. The identity $A(z)B(z) = -A'(z)$, in degree n , is

$$a_0 b_n + \sum_{j=1}^n a_j b_{n-j} = -(n+1)a_{n+1}. \quad (\text{TR27})$$

Solving for b_n gives exactly the recurrence in `logarithmic_coefficients`:

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{TR28})$$

The factorials and signs are Python integers; the divisions and sums occur in Arb arithmetic. Starting from enclosing moment balls, induction on n proves that the computed a_j and b_n balls contain the exact coefficients (TR26). The denominator ball is certified positive before division. Correlations between occurrences of earlier balls can enlarge the resulting intervals but cannot invalidate their containment.

For dimension $d = 16$, the loop integrates $J = 0, 2, \dots, 4d$, giving $2d + 1 = 33$ moments and $a_0, \dots, a_{32}$. The recurrence produces $b_0, \dots, b_{31}$. Both Hankel matrices

$$H^{(r)} = (b_{i+j+r})_{0 \leq i, j < 16}, \quad r \in \{0, 1\}, \quad (\text{TR29})$$

are therefore completely specified: their largest needed indices are 30 and 31. The code uses these exact indices. It introduces no rescaling of the coefficient sequence or matrix.

244.9 9. Unpivoted LDL proof and the meaning of positivity

Fix $r \in \{0, 1\}$. For the exact real symmetric matrix $H^{(r)}$, let L be lower triangular with diagonal entries 1 and define, in increasing order of j ,

$$\begin{aligned} d_j &= H_{jj}^{(r)} - \sum_{k=0}^{j-1} L_{jk}^2 d_k, \\ L_{ij} &= \frac{H_{ij}^{(r)} - \sum_{k=0}^{j-1} L_{ik} L_{jk} d_k}{d_j}, \quad i > j. \end{aligned} \quad (\text{TR30})$$

The code implements these exact ordered recurrences. At step j , every lower entry used in the sums was computed at an earlier step. Induction on j proves that the stored balls enclose the exact values: the initial coefficient balls enclose $H^{(r)}$; sums and products preserve enclosure; and every divisor interval is certified strictly positive before its use. Hence each exact $d_j > 0$.

To establish the matrix identity, for $i = j$, the first recurrence in (TR30) gives $(LDL^T)_{jj} = H_{jj}^{(r)}$; for $i > j$, the second gives $(LDL^T)_{ij} = H_{ij}^{(r)}$. Symmetry supplies the entries $i < j$. Thus

$$H^{(r)} = L \operatorname{diag}(d_0, \dots, d_{15}) L^T. \quad (\text{TR31})$$

Since L is unit lower triangular it is invertible. For every nonzero $v \in \mathbb{R}^{16}$, the vector $L^T v$ is nonzero, and

$$v^T H^{(r)} v = \sum_{j=0}^{15} d_j (L^T v)_j^2 > 0. \quad (\text{TR32})$$

This proves positive definiteness of each original unscaled matrix, once all 16 pivot intervals are positive. Restricting (TR31) to the first s indices gives the exact leading determinant

$$\det H_{[0,s]}^{(r)} = \prod_{j=0}^{s-1} d_j, \quad 1 \leq s \leq 16, \quad (\text{TR33})$$

which is the interval product saved in each row. The unsuccessful branch reports `positivity_not_certified` rather than asserting that the matrix is indefinite. A finite useful certificate is produced only after all divisions used in it have positive divisor intervals.

244.10 10. Reviewed source and observed certificate

The complete 163-line source was read, including argument validation, partial-file writes, numerical functions, serialization, final status, and both matrix calls. The file reviewed is `certify_theta_hankel.py`, SHA-256:

a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950

The inspected certificate is `certificate_dimension16_1024.json`, SHA-256:

869ebacd3f82e46c3fe5fac1f07bcbaae7f4c333e1a5ba2e0ffcb1f91884a0fc

Its `source_sha256` equals the independently hashed reviewed source. It records python-flint 0.9.0, 1024-bit working precision, 850 target bits, dimension 16, $N = 20$, and $L = 4$. It contains all 33 required moments, eight finite-integration pieces per moment, 33 coefficient a intervals, 32 coefficient b intervals, and 16 successful pivot rows for each $r = 0, 1$. Its final status is `positive_definite_certified`.

The certificate's displayed dimension-16 endpoints, quoted here only as orientation while exact dyadic endpoints remain authoritative, are approximately:

Matrix	Last pivot		Full determinant
$H^{(0)}$	6.1698420811114408502314906319786538 ×	10^{-107}	9.2735841166129968999872729909592731 ×
			10^{-822}
$H^{(1)}$	7.1026100619440799659087832255229554 ×	10^{-111}	6.3606810193647631787476650057683745 ×
			10^{-878}

244.11 11. Acceptance and scope of the completed proof

Both proposed tail formulas are valid with every original constant. The analytic callback meets its contract, the retained summand matches the original expression exactly, the code adds the required nonnegative tail interval, the endpoint export is exact rational, the logarithmic recurrence has the correct signs and indices, and the LDL recurrences certify the original two matrices. This review found no formula/code mismatch and no correction necessary for the inspected dimension-16 run.

The exact objects certified are the moments (TR2), the finite logarithmic jet (TR26), and the two finite matrices (TR29). Their positive definiteness gives every leading determinant through dimension 16 by (TR33). Identifying a separately defined zeta function with A , or passing from these finite matrices to an assertion covering every dimension, requires the corresponding exact map and argument in the surrounding research; neither further statement is silently inserted into this certificate. The completed tail and finite-matrix calculations themselves use no zero ordinates, off-line zero assumption, or unproved infinite-dimensional positivity assertion.

245 Independent verification of the positive theta tails

Date: 2026-09-13. The complete file `theta_certified_hankel_20260913/certify_theta_hankel.py` was read. Its SHA-256 is `a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950`. This review verifies the two formulas in `positive_tails` and the finite integration callback's analyticity. It does not rerun integration or review the separately assigned logarithmic recurrence, dyadic export, and LDL calculation.

Finding: both tail bounds are valid upper bounds with the printed constants, and the omitted integrand is nonnegative. For $N = 20, L = 4, J \in \{0, 2, \dots, 64\}$, all three denominator signs are strictly positive. No correction is required.

245.1 1. Exact integrand and the disjoint omitted regions

For $y \geq 0, n \geq 1$, and nonnegative integer J , write

$$x_n(y) = \pi n^2 e^{2y}, \quad f_{J,n}(y) = y^J e^{y/2} (16x_n(y)^2 - 24x_n(y)) e^{-x_n(y)}. \quad (\text{TT1})$$

For $J = 0$, the polynomial y^0 is identically one, including at zero. The finite callback integrates $\sum_{n=1}^{N-1} f_{J,n}$ on $[0, L]$. Its omitted part is exactly

$$\int_0^L \sum_{n=N}^{\infty} f_{J,n}(y) dy + \int_L^{\infty} \sum_{n=1}^{\infty} f_{J,n}(y) dy. \quad (\text{TT2})$$

The shared boundary has measure zero. There is no extra multiplicity or factor of two in this decomposition of the literal half-line integral.

Since $x_n(y) \geq \pi > 3/2$,

$$0 < 16x_n(y)^2 - 24x_n(y) \leq 16x_n(y)^2.$$

Consequently

$$0 \leq f_{J,n}(y) \leq 16\pi^2 n^4 y^J \exp\left(\frac{9}{2}y - \pi n^2 e^{2y}\right). \quad (\text{TT3})$$

The factor $9/2$ is the exact sum $4 + 1/2$ coming from $x_n(y)^2$ and $e^{y/2}$. The negative $-24x_n$ is retained in the finite callback; dropping it only increases the omitted-tail majorant.

245.2 2. Omitted n -sum on $0 \leq y \leq L$

On this interval, $y^J \leq L^J$, $e^{9y/2} \leq e^{9L/2}$, and $e^{-\pi n^2 e^{2y}} \leq e^{-\pi n^2}$. Therefore the first term in (TT2) is at most

$$16\pi^2 L^{J+1} e^{9L/2} \sum_{n=N}^{\infty} n^4 e^{-\pi n^2}. \quad (\text{TT4})$$

For $a_n = n^4 e^{-\pi n^2}$, the exact ratio and its majorant satisfy, for every $n \geq N$,

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= (1 + 1/n)^4 e^{-\pi(2n+1)} \\ &\leq e^{4/n - \pi(2n+1)} \leq e^{4/N - 2\pi N} =: \rho_n. \end{aligned} \quad (\text{TT5})$$

The first inequality uses $\log(1 + 1/n) \leq 1/n$. The second drops the additional negative term $-\pi$ and uses $1/n \leq 1/N$ and $n \geq N$. If $\rho_n < 1$, induction gives $a_{N+j} \leq a_N \rho_n^j$, hence

$$0 \leq \int_0^L \sum_{n=N}^{\infty} f_{J,n}(y) dy \leq \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N-2\pi N}} =: T_n. \quad (\text{TT6})$$

This is exactly the returned expression tn , including its first omitted index N and denominator.

245.3 3. Omitted y -interval, including every $n \geq 1$

For fixed $y \geq 0$, let $a_n(y) = n^4 e^{-\pi n^2 e^{2y}}$. For every $n \geq 1$,

$$\begin{aligned} \frac{a_{n+1}(y)}{a_n(y)} &= (1 + 1/n)^4 e^{-\pi(2n+1)e^{2y}} \\ &\leq e^{4/n-2\pi n e^{2y}} \leq e^{4-2\pi e^{2y}} \leq e^{4-2\pi} =: \rho_y. \end{aligned} \quad (\text{TT7})$$

Thus

$$\sum_{n=1}^{\infty} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{e^{-\pi e^{2y}}}{1 - \rho_y}.$$

Combining this with (TT3), the second omitted integral in (TT2) is at most

$$\frac{16\pi^2}{1 - \rho_y} \int_L^{\infty} F_J(y) dy, \quad F_J(y) = y^J e^{9y/2 - \pi e^{2y}}. \quad (\text{TT8})$$

For $L > 0$, F_J is positive on $[L, \infty)$. Its logarithmic derivative is exactly

$$\frac{F'_J(y)}{F_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq \frac{J}{L} + \frac{9}{2} - 2\pi e^{2L} = -\delta, \quad \delta := 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{TT9})$$

Both inequalities have the displayed direction because $J \geq 0$, $y \geq L > 0$, and the exponential is increasing. When $\delta > 0$, integrating (TT9) from L to y gives $F_J(y) \leq F_J(L) e^{-\delta(y-L)}$. Integrating this inequality proves

$$0 \leq \int_L^{\infty} \sum_{n=1}^{\infty} f_{J,n}(y) dy \leq \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4-2\pi})(2\pi e^{2L} - 9/2 - J/L)} =: T_y. \quad (\text{TT10})$$

This is exactly ty and its returned decay denominator. The constant geometric majorant ρ_y is deliberately larger than the y -dependent ratio; using it cannot underestimate the tail.

The bounds also justify integrability of the positive series. Tonelli's theorem permits the nonnegative sums and integrals throughout; (TT6) and (TT10), together with the finite continuous integral, show their total is finite.

245.4 4. Denominator signs at the actual parameters

For $N = 20$, $L = 4$, and every stated $J \leq 64$, the elementary inequalities $\pi > 3$ and $e^8 > 1 + 8 = 9$ give

$$\begin{aligned} \frac{4}{N} - 2\pi N &< \frac{1}{5} - 120 = -\frac{599}{5}, \\ 4 - 2\pi &< -2, \\ \delta = 2\pi e^8 - \frac{9}{2} - \frac{J}{4} &> 54 - \frac{9}{2} - 16 = \frac{67}{2} > 0. \end{aligned} \quad (\text{TT11})$$

Hence $1 - \rho_n > 1 - e^{-599/5} > 0$, $1 - \rho_y > 1 - e^{-2} > 0$, and $\delta > 67/2$. These are proved lower bounds for the exact denominators at every required even J ; evenness is not needed for the estimates.

The code independently requires the Arb comparisons for those signs to succeed. All quantities in T_n, T_y are nonnegative, and the omitted part lies in $[0, T_n + T_y]$. Therefore using the upper endpoint of the computed enclosure of $tn+ty$ and taking its interval hull with zero gives the correct one-sided tail enclosure. This conclusion uses the printed real interval operations and the exact inequalities above; it does not assert that this review executed those operations.

245.5 5. Callback analyticity

The finite callback, as a function of complex y , is

$$y^J e^{y/2} \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{4y} - 24\pi n^2 e^{2y}) \exp(-\pi n^2 e^{2y}). \quad (\text{TT12})$$

Every power y^J has a nonnegative integer exponent. All other operations are finite sums, products, and compositions of exponential functions. The function is therefore entire, including at $y = 0$ and for $J = 0$. It uses no conjugation, absolute value, logarithm, division by y , or noninteger power. Accordingly the `analytic=True` callback branch requires no restriction of its complex domain.

The value returned for a complex ball encloses the expression using the real Arb enclosure of π . For each fixed coefficient value within that enclosure, the same finite expression is entire; in particular the target with the exact constant π is entire. Possible nonfinite numerical enclosures do not create an analytic singularity, and the caller separately rejects a nonfinite integration result.

The program partitions $[0, 4]$ into the eight exact half-unit subintervals with endpoints $j/2, (j + 1)/2$. The omitted regions and the denominators proved above match that finite integration interval and the loop $n=1, \dots, 19$ exactly. This review accepts the tail formulas and callback analyticity without rerunning that integration.

246 Independent review of the theta error bounds and finite rational witness

Date: 2026-09-13. The complete source tau_theta_hankel_error_control_20260913.tex, TC1–TC28, was read at SHA-256 55f7a9cce6f22b11a30321a7ba726f300a990e78c39780a43b7fdd692c7935e5. This review independently checks the tail estimates, finite algebra, explicit degree cutoff, coefficient rounding, and interval detection of a negative witness. The integration was not rerun, and the source was not edited.

Finding. The mathematical formulas TC1–TC26 are consistent with the retained source identities, and TC23 and TC25 give the claimed margins, including their endpoint cases. The finite positive-definiteness implication in TC27 follows from the stated exact pivot certificates; their complete numerical endpoint replay is the separate parent review. The displayed TC28 decimal enclosure matches its display string in certificate_dimension16_1024.json.

One prose correction was reported: “No coefficient is removed by the rounding rule” can be read as asserting that a nonzero coefficient never rounds to zero, which is false. The precise statement is: every original coefficient slot is included in the error estimate; nearest-grid rounding may produce zero. The theorem should explicitly specify rounding to a nearest multiple of $1/H$, with either choice allowed at a tie. These changes clarify the already used bound $1/(2H)$; no displayed inequality needs alteration.

246.1 1. Retained integral and complete positive tail bounds

The review uses the workbench’s original reflection and Mellin identities stated before TC2, with $g = 2\xi$. There are two distinct factors of two: one in $f_0 = 2 \sum \phi_*$, and one from reflection of the even moment. Their product gives the factor four in TC2. Expanding

$$y^J e^{y/2} (16x_n^2 - 24x_n) e^{-x_n}, \quad x_n = \pi n^2 e^{2y},$$

gives exactly

$$4y^J (4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}}. \quad (\text{TR1})$$

Thus TC2 and the callback TC9 agree with their original constants.

For $y \geq 0, n \geq 1, x_n \geq \pi > 3/2$, so

$$0 \leq y^J e^{y/2} (16x_n^2 - 24x_n) e^{-x_n} \leq 16\pi^2 n^4 y^J e^{9y/2 - \pi n^2 e^{2y}}. \quad (\text{TR2})$$

For the polynomial y^0 , its value at zero is one. For $J > 0$, the vanishing factor at zero creates no exception to the inequality.

For $n = N + \ell, \ell \geq 0$, the elementary bounds $(N + \ell)^4 \leq N^4 e^{4\ell/N}$ and $(N + \ell)^2 \geq N^2 + 2N\ell$ imply

$$\sum_{n=N}^{\infty} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{N^4 e^{-\pi N^2 e^{2y}}}{1 - e^{4/N - 2\pi N e^{2y}}} \leq \frac{N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TR3})$$

Here $r_N = e^{4/N - 2\pi N} \in (0, 1)$: $4/N - 2\pi N \leq 4 - 2\pi < 0$ for $N \geq 1$. Multiplication by (TR2), bounding $y^J \leq L^J$ and $e^{9y/2} \leq e^{9L/2}$, and integrating over an interval of length L prove exactly

$$T_{J,N,L} = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TR4})$$

For the omitted interval $y > L$, use (TR3) with $N = 1$, retaining its exponential numerator. This gives

$$I_J(y) \leq \frac{16\pi^2}{1 - r_1} y^J e^{9y/2 - \pi e^{2y}}, \quad r_1 = e^{4 - 2\pi}. \quad (\text{TR5})$$

For $y = L + v, v \geq 0$, the inequalities $(L + v)^J \leq L^J e^{Jv/L}$ and $e^{2(L+v)} \geq e^{2L}(1 + 2v)$ yield

$$y^J e^{9y/2 - \pi e^{2y}} \leq L^J e^{9L/2 - \pi e^{2L}} e^{-(2\pi e^{2L} - 9/2 - J/L)v}. \quad (\text{TR6})$$

Integrating when $\lambda_J(L) = 2\pi e^{2L} - 9/2 - J/L > 0$ proves exactly

$$Y_{J,L} = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - r_1)\lambda_J(L)}. \quad (\text{TR7})$$

The first tail covers $0 \leq y \leq L, n \geq N$, and the second covers $y > L, n \geq 1$; their interiors are disjoint. The positive integrands permit Tonelli before integrability is established. The finite bounds then prove convergence and TC4.

At $N = 20, L = 4, J \leq 64$, the exact denominator signs follow already from $\pi > 3, e^8 > 41$:

$$\frac{4}{N} - 2\pi N < -\frac{599}{5}, \quad 4 - 2\pi < -2, \quad \lambda_J(4) > 246 - \frac{9}{2} - 16 = \frac{451}{2} > 0. \quad (\text{TR8})$$

Thus the quoted actual-parameter sign check is correct.

For fixed J , first choose L large enough to make $Y_{J,L}$ arbitrarily small; its double-exponential decay dominates the polynomial and exponential factors. For that fixed L , increase N to make $T_{J,N,L}$ arbitrarily small. This is a sequential choice of L , then N , not a claim that the first tail tends to zero with L while N remains fixed.

The finite integrand is entire in complex y : y^J has a nonnegative integer exponent, and all remaining operations are finite sums, products, and compositions of exponentials. There are no branch cuts, absolute values, or conjugations in the callback. Its analyticity therefore meets the recorded analytic-ball callback contract, including at $y = 0, J = 0$.

246.2 2. Recurrence, interval propagation, and finite matrix signs

The retained entire-function identity $F(z^2) = g(1/2 + iz)$, together with the even moments, gives $a_j = (-1)^j M_{2j}/(2j)!$. Extracting the coefficient of w^n from

$$F(w) \sum_{n \geq 0} b_n w^n = -F'(w)$$

gives

$$a_0 b_n + \sum_{j=1}^n a_j b_{n-j} = -(n+1)a_{n+1}, \quad (\text{TR9})$$

which is TC11. The original $a_0 = M_0 > 0$ is retained in this recurrence.

Each interval operation encloses its exact operation. Induction on n , with a positive lower bound for a_0 , therefore encloses every exact b_n . Repeated use of an interval variable can increase enclosure width, but cannot invalidate containment.

For the exact symmetric matrix, the recursion TC13 successively constructs a unit lower triangular L and diagonal D . At stage j , the formula for L_{ij} gives

$$A_{ij} = \sum_{k < j} L_{ik} D_k L_{jk} + L_{ij} D_j \quad (i > j),$$

and the pivot formula gives the same equality for $i = j$, since $L_{jj} = 1$. Symmetry supplies the other half. Thus $A = LDL^\top$. If all interval pivots have positive lower endpoints, induction proves that the exact pivots are positive and every division is defined. Hence for $v \neq 0$

$$v^* A v = \sum_j D_j |(L^\top v)_j|^2 > 0. \quad (\text{TR10})$$

The vector is nonzero after the invertible triangular map. This proves the stated positive-definiteness implication.

For a real rational vector c , let $\gamma_n = \sum_{r+s=n} c_r c_s$. Expanding the complete quadratic form gives

$$c^\top H_d c = \sum_{n=0}^{2d} \gamma_n b_n. \quad (\text{TR11})$$

This includes off-diagonal cross terms twice in the coefficient convolution when their indices differ. If $b_n \in [\ell_n, u_n]$, choosing u_n when $\gamma_n \geq 0$ and ℓ_n when $\gamma_n < 0$ maximizes that term. Therefore TC15 is a valid rational upper bound. No square-modulus form is substituted for this real bilinear form.

246.3 3. Reciprocal-zero map and the full polynomial form

The source imports the workbench's absolutely summable reciprocal-zero expansion TC16. All deductions below retain those exact multiplicities and its full infinite tail. The coordinate map in TC17 is verified directly:

$$-\frac{1}{(\delta + i\gamma)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \quad (\text{TR12})$$

Replacing ρ by $1 - \rho$ changes $\rho - 1/2$ to its negative and leaves the square unchanged. At a zero, $\rho - 1/2 \neq 0$, because the source value $g(1/2) = M_0$ is positive. The map $w = -(\rho - 1/2)^2$ has nonzero derivative there. It preserves the local zero order, so the two reflected source zeros correspond to one zero of F of that same order. This is the exact multiplicity map used in TC16–TC17.

The finiteness of $S_* = \sum_j m_j |\beta_j|$ implies $\sup_j |\beta_j| < \infty$, since $m_j \geq 1$. Put $R_* = \max\{1, \sup_j |\beta_j|\}$. For $P(X) = \sum p_r X^r$,

$$|P(\beta_j)| \leq \sum_r |p_r| R_*^r = \|P\|_{R_*}. \quad (\text{TR13})$$

Thus $\mathcal{Q}(P) = \sum_j m_j \beta_j P(\beta_j)^2$ converges absolutely. It is real for real P because the reciprocal set and multiplicities are conjugation stable. Finite expansion of P^2 gives TC19 exactly.

For real polynomials P, E , the identity $(P + E)^2 - P^2 = 2PE + E^2$, applied termwise in that absolutely convergent series, gives

$$|\mathcal{Q}(P + E) - \mathcal{Q}(P)| \leq S_* \|E\|_{R_*} (2\|P\|_{R_*} + \|E\|_{R_*}). \quad (\text{TR14})$$

This verifies TC20 without dropping any reciprocal zero or multiplicity.

246.4 4. Separator, explicit cutoff, and all edge cases

Fix a nonreal reciprocal zero β , its conjugate, and their common multiplicity m . Let $r = |\beta|/2 > 0$. In TC21, “all other” excludes both selected points. Every element of $\mathcal{A}$ contributes at least r to S_* , so $\#\mathcal{A} \leq S_*/r$ and the set is finite. It is conjugation stable. Therefore B has real coefficients and $B(\beta) \neq 0$. The empty product is one if $\mathcal{A}$ is empty.

Either choice of c with $\beta c^2 = -1$ is nonzero. Set w_N, u_N, v_N as in TC24. They satisfy

$$u_N + v_N \beta = \Re w_N - (\Re \beta) v_N + v_N (\Re \beta + i \Im \beta) = w_N. \quad (\text{TR15})$$

Consequently $P_N(\beta) = c$, and real coefficients give $P_N(\bar{\beta}) = \bar{c}$. The two selected summands of $\mathcal{Q}(P_N)$ are each exactly $-m$. Every other reciprocal zero of modulus at least r is a root of B , so its entire weighted summand is zero, including its multiplicity.

For $|X| \leq r$, the literal product gives $|B(X)| \leq B_0$. Also

$$\begin{aligned} |u_N + v_N X| &\leq |\Re w_N| + (|\Re \beta| + r)|v_N| \\ &\leq |w_N| \left(1 + \frac{|\Re \beta| + r}{|\Im \beta|} \right). \end{aligned} \quad (\text{TR16})$$

Multiplication by $|X|^N |B(X)|$, and using $|w_N| = |c|/(|\beta|^N |B(\beta)|)$, gives

$$|P_N(X)| \leq C_* 2^{-N}. \quad (\text{TR17})$$

This includes $N = 0$, with X^0 the constant polynomial one. The full remaining reciprocal tail therefore has absolute contribution at most $C_*^2 T_* 4^{-N}$, yielding

$$\mathcal{Q}(P_N) \leq -2m + C_*^2 T_* 4^{-N}. \quad (\text{TR18})$$

If $T_* = 0$, the last term is zero for every $N \geq 0$. If $T_* > 0$, set $A = C_*^2 T_*/m > 0$ and $x = \log A / \log 4$. When $x \geq 0$, $1 + \lfloor x \rfloor > x$, so every $N \geq N_*$ satisfies $N > x$. When $x < 0$, $N \geq 0 > x$, regardless of whether the inner integer in TC23 is zero or negative. Hence in both cases

$$A 4^{-N} < 1, \quad C_*^2 T_* 4^{-N} < m. \quad (\text{TR19})$$

In particular TC23 gives the claimed weak bound $\mathcal{Q}(P_N) \leq -m$. At $A = 4^s$ with integer $s \geq 0$, the chosen cutoff is $s + 1$, not s ; the boundary case is therefore safe. The formula is conservative by one degree at that boundary, and is mathematically valid as printed.

246.5 5. Rational coefficient accuracy and interval detection

The polynomial P_N has real coefficients and degree at most $D = N + \#\mathcal{A} + 1$. Retain all $D + 1$ coefficient slots, inserting exact zeros above its actual degree if necessary. For a positive integer H , round each coefficient to a nearest element of $H^{-1}\mathbb{Z}$. Either nearest element is allowed at a tie. The error polynomial $E = \tilde{P}_N - P_N$ then has

$$|e_r| \leq \frac{1}{2H}, \quad \eta := \|E\|_{R_*} \leq \frac{\sum_{r=0}^D R_*^r}{2H}. \quad (\text{TR20})$$

Every denominator in TC25 is positive: $m > 0$, $S_* > 0$ because it includes the selected pair, and $2\|P_N\|_{R_*} + 1 > 0$. The finite geometric sum is positive. Thus a sufficiently large positive integer H always satisfies TC25.

The two bounds in TC25 give $\eta \leq 1$ and $\eta \leq m/[2S_*(2\|P_N\|_{R_*} + 1)]$. By (TR14),

$$\begin{aligned} |\mathcal{Q}(\tilde{P}_N) - \mathcal{Q}(P_N)| &\leq S_* \eta (2\|P_N\|_{R_*} + \eta) \\ &\leq S_* \eta (2\|P_N\|_{R_*} + 1) \leq m/2. \end{aligned} \quad (\text{TR21})$$

Together with (TR18)–(TR19), this proves $\mathcal{Q}(\tilde{P}_N) \leq -m/2 < 0$. The error estimate includes a coefficient that rounds from a nonzero value to zero; such rounding is permitted and does not invalidate the proof.

For clarity, the eventual finite-interval claim has a direct quantitative margin. Write c for the rational coefficient vector of $\tilde{P}_N$, define γ_n as in (TR11), and set $A_c = \sum_n |\gamma_n|$. The polynomial is nonzero because its form is negative, so its square is nonzero and $A_c > 0$. If rational enclosing intervals for every required b_n have width at most $m/(4A_c)$, then their signed upper bound satisfies

$$U(c) - \mathcal{Q}(\tilde{P}_N) \leq \sum_n |\gamma_n| (u_n - \ell_n) \leq m/4, \quad U(c) \leq -m/4 < 0. \quad (\text{TR22})$$

This proves an explicit sufficient finite enclosure width. There is no need to assume cancellation or independent errors.

For finitely many required coefficients, shrinking the moment intervals shrinks the resulting b_n intervals: TC11 uses finitely many additions, products, and reciprocals, and one can first restrict the a_0 interval to have lower endpoint at least $M_0/2 > 0$. Every such interval operation is continuous at those exact inputs; induction in n proves convergence of enclosure widths to zero. The positive tail bounds can be made arbitrarily small by the sequential choice of L, N established above.

The finite integral itself admits arbitrarily small validated errors. One direct existence proof uses the entire finite integrand f on $[0, L]$. On all unit complex discs about that interval, a finite bound for $|f|$ is

$$K = (L + 2)^J e^{(L+1)/2} \sum_{n=1}^{N-1} \left(16\pi^2 n^4 e^{4(L+1)} + 24\pi n^2 e^{2(L+1)} \right) e^{\pi n^2 e^{2(L+1)}}. \quad (\text{TR23})$$

Indeed $\Re y \leq L + 1$, $|y| \leq L + 2$, and $|e^{-\pi n^2 e^{2y}}| \leq e^{\pi n^2 e^{2\Re y}}$. Cauchy's derivative estimate on each unit circle gives $|f'(x)| \leq K$ for $x \in [0, L]$. Composite midpoint quadrature on a uniform mesh of width $h = L/s$ has error at most $KLh/4$, by integrating $K|x - x_{\text{mid}}|$ on each interval. Finite evaluations of the explicit exponential expression can be enclosed to arbitrary accuracy; outward rational bounds for K suffice. If $N = 1$, the finite integrand is zero and this step is immediate. This proves the finite integration approximation needed for (TR22), independently of any fixed evaluation limit of the particular integration run.

246.6 6. Relation to the recorded finite certificate

For dimension 16, $d = 15$. The unshifted matrix uses $b_0, \dots, b_{30}$, and the shifted one uses $b_1, \dots, b_{31}$. TC11 computes these from $a_0, \dots, a_{32}$, hence exactly the stated 33 even moments $M_0, M_2, \dots, M_{64}$. The index ranges are consistent. Positivity of both sets of sixteen exact pivots proves TC27 by (TR10); every smaller leading matrix is positive definite by restriction of the same positive quadratic form.

The parent performs the complete rational endpoint and replay audit. This subreview supplies the independent mathematical derivation and does not turn a finite-dimensional sign result into a claim at larger dimensions. Likewise the separator theorem constructs and bounds the witness associated with each specified nonreal reciprocal zero; it asserts no universal cutoff and supplies no nonreal zero of the actual zeta function.

247 Root acceptance of the original theta certificate and complete proof closure

13 September 2026. This is the final local mathematical and execution review for the current finite certificate. It does not close the wider Split-Zero/RH research goal.

247.1 Exact accepted calculation

Both original real symmetric matrices

$$H_{15} = (b_{i+j})_{0 \leq i,j \leq 15}, \quad H_{15}^{(1)} = (b_{i+j+1})_{0 \leq i,j \leq 15}$$

are positive definite. The coefficients are those of the actual $-F'/F$, with $F(z^2) = 2\xi(1/2 + iz)$. All 33 moments $M_0, M_2, \dots, M_{64}$ are calculated from the original theta integral. All 32 pivots and all 32 leading determinants have positive rigorous lower endpoints. The original mass and all monomial coordinates remain unchanged.

The complete integrator, CH.1–19 method proof, TR1–33 analytic/code review, and TC1–28 root proof have been read. The two omitted regions are disjoint, their integrands are nonnegative, and their upper bounds retain the factor $16\pi^2$, the original exponential factors and both exact denominators. The complex integration callback is entire. Its real ball contains the real integral; the positive tail interval is added with outward rounding. The dyadic exporter uses exact outward lower/upper endpoints and their integer mantissa/exponent, with no decimal conversion in the certificate.

247.2 Independent integer arithmetic and actual replay

The saved independent script `theta_certified_hankel_exact_rational_review_20260913.py` was read completely. It imports no flint or floating-point arithmetic. It uses a fixed grid $Q = 2^{4096}$: an interval is the exact pair $[\ell/Q, u/Q]$ of integers. Addition and negation are exact; multiplication takes the minimum and maximum of all four integer endpoint products and rounds outward after division by Q . Squaring uses zero as its lower endpoint when the interval crosses zero. For a positive interval its reciprocal is enclosed by

$$\left[\frac{\lfloor Q^2/u \rfloor}{Q}, \frac{\lceil Q^2/\ell \rceil}{Q} \right].$$

Negative reciprocals are obtained by the exact negation map. These formulas prove containment of every operation, including all rounded operations.

The script reconstructs each moment from the eight exported finite-piece intervals and the sum of the two exported tail upper bounds. It does not use the saved moment, coefficient or pivot values to calculate its result. It then propagates the original factorial/sign formula and logarithmic recurrence and independently performs both unscaled LDL recursions. Every denominator is checked positive before division. The complete matrix identity and positive-definiteness proof are TC13 and CH.15–17.

Root ran a fresh validated integration after the independent review worker hit its usage limit. The fresh certificate is `reviewer_replay_dimension16_1024.json`, SHA-256 `0fcac21e20680a6d6a3fab1671b0782b56de005ce749ee33373b1193f20a0af7`. The complete numerical payload agrees exactly with the original certificate; elapsed time and platform fields are excluded from that comparison. Root then executed the independent integer script successfully in ordinary Python. It validated 1,400 exact intervals, reconstructed both matrix certificates, and proved all 32 positive pivots and determinants. It also proved by exact rational comparison that the readable TC28 interval contains the full exported determinant interval.

The original certificate SHA-256 is `869ebacd3f82e46c3fe5fac1f07bcbaae7f4c333e1a5ba2e0ffc b1f91884a0fc`. The reviewed integrator SHA-256 is `a97486edc2fc1f05f155a9539ad280433be39a`

8f6924efbc8c1c1c4d71033950. The independent integer script SHA-256 is a181ef220578eaf232b3e01398d61b1a6f36625f056c5e6bd365453b9ebfa885. Its successful complete rational output is supplied alongside this proof. The script uses Python assertions for validation and must be run without the optimization switch. No optimized-mode verifier claim is made.

247.3 Written-proof corrections and closure

The complete TC mathematical subreview accepted every displayed bound and reported two prose clarifications: coefficient rounding must mean a nearest multiple, with either tie choice allowed; every coefficient slot enters the error bound even when a nonzero coefficient rounds to zero. Both are now explicit in the accepted source. Its current SHA-256 is 8dab2193a7ef3287f58ff5038fc34988883d8899dc4a2a9868098856b48772a5. The earlier subreview remains a record of the reviewed predecessor and its resolved findings. The bounds and their indices were unchanged.

The complete workbench NOTE.tex and WBR1–41 independent proof were read. The archived separator defines u_N and then writes ν_N in its boxed formula. The exact correction is the identity $\nu_N := u_N$; the archive is preserved and the consistent full formula appears in WBR16 and TC24. The rational functional calculus has the exact full local inverse WBR29, and its two reflected modules remain separate through WBR33. The half-trace, nilradical and critical-observation defect are completely proved in WBR34–40.

The workbench originally invokes the classical genus-zero factorization without including its proof. The additional complete source tau_genus_zero_product_proof_20260913.tex, GF1–13, now proves the precise case used. Root wrote and read back that whole proof. It derives the circle-mean zero formula, the $O(r^\alpha)$ zero count, absolute reciprocal summability, uniform product convergence, a lower bound for the product on explicitly selected zero-avoiding circles, and constancy of the entire zero-free factor from its Fourier coefficients. It then derives the full logarithmic-derivative expansion. Taking $\alpha = 3/4$ applies it to WBR9's explicit theta growth bound. Its SHA-256 is 9bade2c098f1b03ba4ec4e526f4dffe4ced36f70bd7a533931a16335b162c394. This is an included proof of a classical theorem, not a novelty claim. It is root-reviewed; no separate independent acceptance of GF is claimed.

The finite dimension-16 certificate does not require factorization or any assumed zero location. Factorization belongs to the separate proof that positivity of the original unshifted matrices in every dimension is equivalent to RH. The complete finite tails and rational rounding formula give rigorous finite witnesses without choosing a replacement metric. The actual computation produced positive finite matrices and no negative witness. It proves no sign statement at larger dimensions.

248 Independent final review: the full genus-zero proof and the positive theta tails

Date: 2026-09-13.

The complete files `tau_genus_zero_product_proof_20260913.tex` (GF1–GF13) and `theta_certified_tail_subreview_20260913.md` (TT1–TT12) were read and checked. The original workbench `NOTE.tex` was also read in full to check the exact function, coefficient convention, and invocation of the product theorem. A separate reviewer read the entire retained `certify_theta_hankel.py` and matched its callback and tail expressions to TT1–TT12. No source file was modified and no integration, Lean run, or remote publication was performed in this review.

Acceptance: GF1–GF13 and TT1–TT12 are mathematically valid with their original constants, signs, multiplicities, and domains. No correction is required. The explicit derivation below supplies the original-theta growth estimate invoked in the first paragraph of GF, and the remainder checks every step of the product argument and of the two tail estimates. The finite computation’s exact arithmetic and numerical integration certification are outside this review; accepting the tail formulas does not by itself certify those separate operations.

248.1 1. Exact sources and the original analytic input

The reviewed bytes are:

Source	Bytes	SHA-256
<code>tau_genus_zero_product_proof_20260913.tex</code>	7008	9bade2c098f1b03ba4ec4e526f4dffe4ced36f70bd7a533931a16335b162c394
<code>theta_certified_tail_subreview_20260913.md</code>	6761	4e5c6ad421b6d893e551c1746e05a08be3bf3b0753fae69c16feb5b7025dedf
<code>certify_theta_hankel.py</code>	7472	a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950
Original workbench <code>NOTE.tex</code>	20408	df7854f73ecb396c21165f8c032722a851a2a4c893f2553c6da96f88014b62f6

The original source in the workbench is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \psi(y) = e^{y/2} f_0(e^y). \quad (\text{GFR1})$$

The supplied Mellin identity and reflection give

$$g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy, \quad \psi(-y) = \psi(y), \quad g = 2\xi. \quad (\text{GFR2})$$

The workbench’s coefficients specify the entire function by

$$F(w) = \sum_{j \geq 0} \frac{(-1)^j M_{2j}}{(2j)!} w^j, \quad M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy, \quad F(z^2) = g(1/2 + iz). \quad (\text{GFR3})$$

All factors of two in (GFR1)–(GFR3) are retained. In particular, the original value is $F(0) = g(1/2) = \int_{\mathbb{R}} \psi(y) dy > 0$.

Here is a completely explicit route from the source to GF1. Put

$$\rho = \exp(4 - 2\pi), \quad C_\theta = \frac{8\pi^2}{1 - \rho}. \quad (\text{GFR4})$$

Because $\pi > 3, 0 < \rho < 1$. For $n \geq 1$,

$$\frac{(n+1)^4 e^{-\pi((n+1)^2-1)}}{n^4 e^{-\pi(n^2-1)}} = (1+1/n)^4 e^{-\pi(2n+1)} \leq e^{4/n-2\pi n} \leq \rho.$$

The first term of the sequence is one, so its sum is at most $(1-\rho)^{-1}$. For $y \geq 0$, the literal summand of ψ is positive and bounded above by its positive fourth-power term. Consequently

$$\begin{aligned} 0 < \psi(y) &\leq 8\pi^2 e^{9y/2-\pi e^{2y}} \sum_{n \geq 1} n^4 e^{-\pi(n^2-1)e^{2y}} \\ &\leq C_\theta e^{9y/2-\pi e^{2y}}. \end{aligned} \tag{GFR5}$$

Reflection gives the same estimate with $|y|$. For every fixed $s \geq 0$, this bound times $e^{s|y|}$ is integrable. It justifies the power-series expansion and differentiations in (GFR2) by domination on every compact set of z . The evenness of ψ removes exactly the odd coefficients. It also proves convergence of (GFR3) on every compact set of w , since one can compare the absolute series with the even exponential series at $|z| = \sqrt{|w|}$.

For $R \geq 1$ and every $|w| \leq R$, choose one number z with $z^2 = w$. This pointwise choice uses no holomorphic square-root branch; it only gives $|z| \leq \sqrt{R}$. Let $a = 9/4 + \sqrt{R}/2$. Equation (GFR5) gives

$$\begin{aligned} |F(w)| &\leq 2C_\theta \int_0^\infty e^{(9/2+\sqrt{R})y-\pi e^{2y}} dy \\ &= C_\theta \int_1^\infty t^{a-1} e^{-\pi t} dt \leq C_\theta \pi^{-a} \Gamma(a). \end{aligned} \tag{GFR6}$$

The substitution is exactly $t = e^{2y}$, so $dy = dt/(2t)$. The factor two from the two half-lines cancels this denominator two. For $a > 1$, the positive function $t^{a-1} e^{-t/2}$ has its maximum at $t = 2(a-1)$; this follows from its logarithmic derivative $(a-1)/t - 1/2$. Thus

$$\Gamma(a) = \int_0^\infty (t^{a-1} e^{-t/2}) e^{-t/2} dt \leq 2 \left(\frac{2(a-1)}{e} \right)^{a-1}. \tag{GFR7}$$

Write $K_0 = \max\{0, \log(2C_\theta)\}$. Since $a-1 = 5/4 + \sqrt{R}/2 \leq 7\sqrt{R}/4$, equations (GFR6)–(GFR7) imply

$$\log M_F(R) \leq K_0 + \frac{7}{4}\sqrt{R} \log \frac{7}{2} + \frac{7}{8}\sqrt{R} \log R. \tag{GFR8}$$

In deriving this upper bound, the terms $-(a-1) - a \log \pi$ were discarded only because they are negative; the original function and its mass remain unchanged. For $R \geq 1$, $\log R/R^{1/4} \leq 4/e$: putting $u = \log R \geq 0$, the derivative of $ue^{-u/4}$ is $e^{-u/4}(1-u/4)$, whose maximum value is $4/e$. Hence a concrete choice in GF1 is

$$\alpha = \frac{3}{4}, \quad R_0 = 1, \quad C = K_0 + \frac{7}{4} \log \frac{7}{2} + \frac{7}{2e} > 0. \tag{GFR9}$$

This derivation proves the growth hypothesis for precisely the original F . It does not use zero ordinates, a product formula, or an assumed location of zeros.

248.2 2. Counting all zeros, with their multiplicities

For the general entire function in GF1, $F(0) \neq 0$ ensures that the function is not identically zero and that all zeros have positive modulus. Zeros in any compact disk are finite, counting multiplicities. Choose a radius R whose circle has no zero and divide F by the factors $z - a_j$ for all its zeros in that disk. The quotient extends holomorphically through those zeros and is nonvanishing on a slightly larger disk: the boundary circle is compact and nonvanishing, so an annular neighborhood of it contains no zero. Its logarithmic derivative has a primitive on this disk, obtained from its power series. If L' is that logarithmic derivative, differentiating the quotient times e^{-L} gives zero, so this quotient equals a nonzero constant times e^L . A choice of a complex logarithm of the constant supplies its holomorphic logarithm.

The mean of the real part of a holomorphic logarithm equals its central value by its uniformly convergent Taylor series on the circle. For a removed zero $|a_j| < R$, the uniformly convergent expansion

$$\log(1 - a_j e^{-i\theta}/R) = - \sum_{k \geq 1} \frac{(a_j/R)^k e^{-ik\theta}}{k}$$

has real part of mean zero. Thus the mean of $\log |R e^{i\theta} - a_j|$ is exactly $\log R$. Subtracting the central value of the removed linear factors gives GF3, including the term $\log |F(0)|$ and each repeated zero.

For sufficiently large r , the set of excluded radii in $(2r, 3r)$ is finite. Choose any other R there. Every zero with $|a_j| \leq r$ contributes at least $\log 2$, whereas every other term in GF3 is nonnegative. Therefore

$$n(r) \log 2 \leq C(3r)^\alpha - \log |F(0)|. \quad (\text{GFR10})$$

This proves $n(r) \leq C_1 r^\alpha$ for all sufficiently large r . Choose $r_0 > 0$ strictly below the modulus of every zero. Such a radius exists by nonvanishing at zero; if the zero set is empty any positive radius works. On the remaining bounded interval, $n(r)/r^\alpha$ is bounded above by a fixed finite count divided by r_0^α . Enlarging C_1 therefore proves the estimate for every $r \geq r_0$.

Partition the zeros into the half-open annuli $2^k r_0 < |a_j| \leq 2^{k+1} r_0$, $k \geq 0$. There are no zeros omitted below or at r_0 . Each term in the k -th annulus is at most $(2^k r_0)^{-1}$, and that annulus contains at most $n(2^{k+1} r_0)$ zeros. Consequently

$$\sum_j |a_j|^{-1} \leq C_1 2^\alpha r_0^{\alpha-1} \sum_{k \geq 0} 2^{k(\alpha-1)} < \infty. \quad (\text{GFR11})$$

This verifies GF4 without a density assumption and with every multiplicity included.

248.3 3. Construction of the literal product and the selected-circle bound

Let $P(z) = \prod_j (1 - z/a_j)$. For a fixed disk $|z| \leq A$, only finitely many zeros have modulus at most $2A$. For every remaining factor, the power-series logarithm obeys

$$|\log(1 - z/a_j)| \leq \sum_{k \geq 1} |z/a_j|^k / k \leq \frac{|z/a_j|}{1 - |z/a_j|} \leq 2A/|a_j|. \quad (\text{GFR12})$$

By (GFR11) these logarithms have an absolutely and uniformly convergent sum. Its exponential has no zero on the disk. Multiplication by the preceding finite factors therefore gives a holomorphic function with exactly the prescribed zeros and orders there. The constructions agree on intersections because they use the same absolutely convergent tail and finite factors. They define the entire P , independent of the enumeration of zeros, with $P(0) = 1$.

At each zero, F and P have the same order and a nonzero leading coefficient, so $H = F/P$ extends as a nonvanishing holomorphic function. Thus H is nonvanishing entire. Integrating the entire power series of H'/H and choosing the constant at zero gives an entire h satisfying $H = e^h$. No bound on $1/P$ has been used in this construction.

For sufficiently large t , set $N_t = n(4t)$ and $\delta_t = t/[4(N_t + 1)]$. The union of the open intervals of radius δ_t about the at most N_t zero moduli at most $4t$ has total length at most $2N_t\delta_t = tN_t/[2(N_t + 1)] < t/2$. The interval $[t, 2t]$ has length t , so a radius R_t outside the union exists. For every $|z| = R_t$ and every zero $|a_j| \leq 4t$, the reverse triangle inequality yields

$$|1 - z/a_j| = \frac{|a_j - z|}{|a_j|} \geq \frac{|R_t - |a_j||}{4t} \geq \frac{1}{16(N_t + 1)}. \quad (\text{GFR13})$$

Repeated zeros merely repeat factors; counting their excluded intervals more than once only weakens the estimate on the union length. For every remaining zero, $|a_j| > 4t$ and $|z| \leq 2t$, so $|z/a_j| < 1/2$. Equation (GFR12) implies $\log |1 - z/a_j| \geq -2|z|/|a_j|$. Using annuli $2^k 4t < |a_j| \leq 2^{k+1} 4t$, exactly the counting argument above gives

$$\sum_{|a_j| > 4t} |a_j|^{-1} \leq \frac{C_1 2^\alpha 4^{\alpha-1}}{1 - 2^{\alpha-1}} t^{\alpha-1} = C_2 t^{\alpha-1}. \quad (\text{GFR14})$$

Both the finite log sum and its uniformly convergent infinite tail can now be evaluated on the full circle, giving

$$\log |P(z)| \geq -N_t \log(16(N_t + 1)) - 2R_t C_2 t^{\alpha-1}. \quad (\text{GFR15})$$

For completeness, put $A = C_1 4^\alpha$. For $t \geq e$, $N_t \leq At^\alpha$, $N_t + 1 \leq (A + 1)t^\alpha$, and $R_t \leq 2t$. Thus

$$\begin{aligned} N_t \log(16(N_t + 1)) &\leq At^\alpha [\log(16(A + 1)) + \alpha \log t], \\ 2R_t C_2 t^{\alpha-1} &\leq 4C_2 t^\alpha. \end{aligned}$$

Taking $C_3 = A[\log(16(A + 1)) + \alpha] + 4C_2$ proves GF9 for all these sufficiently large t . The proof covers infinitely many, finitely many, or no zeros; an empty finite log sum contributes zero.

248.4 4. The exact argument forcing the zero-free quotient to be constant

On the selected circle, F and P are nonzero. Therefore $\Re h = \log |H| = \log |F| - \log |P|$. For large t , GF1 applies at R_t and yields

$$\Re h(z) \leq C 2^\alpha t^\alpha + C_3 t^\alpha \log t \leq B_t := (C 2^\alpha + C_3) t^\alpha \log t \quad (|z| = R_t), \quad (\text{GFR16})$$

where $t \geq e$. The product $C 2^\alpha$ retains the original growth constant C ; the tail constant C_2 has already entered through C_3 .

Write $h(z) = \sum_{k \geq 0} d_k z^k$. On any fixed circle this series and its conjugate converge uniformly. For every integer $\ell \geq 1$, multiplying

$$\Re h(R_t e^{i\theta}) = \frac{1}{2} \sum_{k \geq 0} d_k R_t^k e^{ik\theta} + \frac{1}{2} \sum_{k \geq 0} \overline{d_k} R_t^k e^{-ik\theta}$$

by $e^{-i\ell\theta}$ and integrating gives exactly

$$\frac{1}{\pi} \int_0^{2\pi} \Re h(R_t e^{i\theta}) e^{-i\ell\theta} d\theta = d_\ell R_t^\ell. \quad (\text{GFR17})$$

Only the first sum's term $k = \ell$ survives. In particular, the coefficient $1/\pi$ in GF10 is correct. The integral of the constant B_t times $e^{-i\ell\theta}$ is zero. Since $B_t - \Re h \geq 0$, the triangle inequality and the mean-value identity give

$$|d_\ell| R_t^\ell \leq \frac{1}{\pi} \int_0^{2\pi} (B_t - \Re h(R_t e^{i\theta})) d\theta = 2(B_t - \Re h(0)). \quad (\text{GFR18})$$

For fixed $\ell \geq 1$, divide by $R_t^\ell \geq t^\ell$. The upper bound tends to zero because $t^{\alpha-\ell} \log t \rightarrow 0$ and $\alpha < 1 \leq \ell$. Hence $d_\ell = 0$ for every positive ℓ . This proves that h , and therefore H , is constant. Its value is determined without modifying the original source: $H(0) = F(0)/P(0) = F(0)$. The result is the literal factorization

$$F(z) = F(0) \prod_j (1 - z/a_j). \quad (\text{GFR19})$$

This proof does not invoke the conclusion of a factorization theorem as an input. The counting, convergence, selected-circle lower bound, and Fourier-coefficient bound together prove the entire special case used by the workbench.

248.5 5. The logarithmic coefficient map retains every multiplicity

Group repeated zeros and put $\beta_j = 1/a_j$, now indexing distinct zeros with multiplicities m_j . Equation (GFR11) gives $S = \sum_j m_j |\beta_j| < \infty$, and the β_j are bounded because there is a zero-free neighborhood of zero. Choose $r > 0$ with $r|\beta_j| \leq 1/2$ for all j ; if there are no zeros any positive r works. For $|z| \leq r$,

$$\sum_j \left| \frac{m_j \beta_j}{1 - \beta_j z} \right| \leq 2S. \quad (\text{GFR20})$$

The uniform logarithm convergence used to construct P , together with this uniform derivative bound, permits termwise differentiation there. Equivalently, integrate this uniformly convergent derivative series from zero and compare the analytic logarithms, whose values at zero are fixed. In either construction,

$$-\frac{F'(z)}{F(z)} = \sum_j \frac{m_j \beta_j}{1 - \beta_j z}.$$

The double geometric series is absolutely summable because

$$\sum_j \sum_{n \geq 0} m_j |\beta_j|^{n+1} r^n = \sum_j \frac{m_j |\beta_j|}{1 - r|\beta_j|} \leq 2S. \quad (\text{GFR21})$$

Thus its sums can be interchanged, and the coefficient of z^n is exactly $\sum_j m_j \beta_j^{n+1}$. This proves GF13 and the workbench's logarithmic coefficients in their original scalar coordinate. It leaves the reflected packet modules, their nilpotents, and their independent operator constructions unchanged upstream of that scalar map.

248.6 6. Verification of the positive tails and the exact omitted regions

For the half-line moment integrand define, exactly as in TT1,

$$x_n(y) = \pi n^2 e^{2y}, \quad f_{J,n}(y) = y^J e^{y/2} (16x_n(y)^2 - 24x_n(y)) e^{-x_n(y)}. \quad (\text{GFR22})$$

Here $y \geq 0$, $n \geq 1$, and J is a nonnegative integer. At $J = 0$, y^J is the constant polynomial one. For even J , summing and integrating (GFR22) is precisely M_J in (GFR3): the source factor two and the reflection factor two are both already present in the coefficients 16 and -24 .

Because $x_n(y) \geq \pi > 3/2$, $16x_n^2 - 24x_n = 8x_n(2x_n - 3) > 0$. Therefore

$$0 \leq f_{J,n}(y) \leq 16\pi^2 n^4 y^J e^{9y/2 - \pi n^2 e^{2y}}. \quad (\text{GFR23})$$

The finite callback keeps $1 \leq n < N$ on $[0, L]$. Up to their measure-zero common boundary, the omitted regions are exactly $n \geq N$, $0 \leq y \leq L$ and $n \geq 1$, $y \geq L$. They are disjoint and their union is the full complement of the retained region. Positivity permits the sum and integral interchanges by Tonelli's theorem; the bounds below prove their total finiteness.

On $[0, L]$, the majorant in (GFR23) is at most $16\pi^2 n^4 L^J e^{9L/2 - \pi n^2}$. Integrating the interval of length L and setting $a_n = n^4 e^{-\pi n^2}$ gives the upper bound $16\pi^2 L^{J+1} e^{9L/2} \sum_{n \geq N} a_n$. For every $n \geq N$,

$$\frac{a_{n+1}}{a_n} = (1 + 1/n)^4 e^{-\pi(2n+1)} \leq e^{4/n - \pi(2n+1)} \leq e^{4/N - 2\pi N} =: \rho_n. \quad (\text{GFR24})$$

The final symbol is constant for the selected first omitted index N ; its subscript does not introduce a summation over ratios. For $\rho_n < 1$, induction and the geometric sum give the full upper bound

$$T_n = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}}. \quad (\text{GFR25})$$

For the omitted y -interval, apply the same exact ratio at fixed $y \geq 0$:

$$\frac{(n+1)^4 e^{-\pi(n+1)^2 e^{2y}}}{n^4 e^{-\pi n^2 e^{2y}}} \leq e^{4/n - 2\pi n e^{2y}} \leq e^{4 - 2\pi e^{2y}} \leq \rho_y := e^{4 - 2\pi} < 1. \quad (\text{GFR26})$$

The first term is $e^{-\pi e^{2y}}$; hence the full n -sum in (GFR23) is bounded by it divided by $1 - \rho_y$. Put $V_J(y) = y^J e^{9y/2 - \pi e^{2y}}$, which is positive on $[L, \infty)$ when $L > 0$. Its logarithmic derivative is

$$\frac{V'_J(y)}{V_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq \frac{J}{L} + \frac{9}{2} - 2\pi e^{2L} = -\delta, \quad \delta = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{GFR27})$$

When $\delta > 0$, integration from L to y gives $V_J(y) \leq V_J(L) e^{-\delta(y-L)}$. Its integral is at most $V_J(L)/\delta$. Thus the second omitted region is bounded above by

$$T_y = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4 - 2\pi})(2\pi e^{2L} - 9/2 - J/L)}. \quad (\text{GFR28})$$

These are exactly TT6 and TT10. At the actual parameters $N = 20$, $L = 4$, $0 \leq J \leq 64$, the elementary bounds $\pi > 3$ and $e^8 > 1 + 8 = 9$ give

$$4/N - 2\pi N < -599/5, \quad 4 - 2\pi < -2, \quad \delta > 54 - 9/2 - 16 = 67/2 > 0. \quad (\text{GFR29})$$

All denominators are strictly positive. Consequently the literal omitted integral belongs to $[0, T_n + T_y]$. The negative summand $-24x_n$ remains part of the exact integral and finite callback; its removal in (GFR23) is solely the proved inequality bounding the omitted tails.

248.7 7. Analytic callback and source-to-proof comparison

The retained callback extends to every complex y as

$$y^J e^{y/2} \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{4y} - 24\pi n^2 e^{2y}) e^{-\pi n^2 e^{2y}}. \quad (\text{GFR30})$$

It is a finite sum of products and compositions of entire functions, and y^J is a polynomial. Thus it is entire, including at zero for every allowed J . It has no branch choice or hidden singularity. Treating π as its exact real constant, or any fixed constant value in an enclosing ball, does not alter this analyticity statement. The callback's complex ball value must enclose the exact expression; a possibly wide or nonfinite enclosure is a numerical issue, not an analytic singularity. The retained caller separately rejects a nonfinite integration result.

The separate full code comparison verified the following literal correspondence: the callback loops through $n = 1, \dots, N - 1$; tn is (GFR25); ty and decay are (GFR28) and (GFR27); the default values are $N = 20, L = 4$; dimension 16 requests exactly $J = 0, 2, \dots, 64$; the retained integral is partitioned into the eight half-unit intervals with endpoints $j/2, (j+1)/2, 0 \leq j \leq 7$. The denominator comparisons must succeed before the code proceeds. The code forms `arb(0).union((tn+ty).upper())`, which encloses the proved interval $[0, T_n + T_y]$ whenever the computed balls enclose the stated expressions. Ignoring the callback's analytic flag is consistent with the entire formula (GFR30). This establishes the exact code-to-tail-formula match without asserting an integration replay.

248.8 8. Acceptance and scope

Every numbered mathematical step GF1–GF13 and TT1–TT12 is accepted. The original-theta specialization of GF1 has the explicit constants in (GFR4) and (GFR9), and its product and logarithmic coefficients follow from the complete proof above. The original half-line tail formulas have the exact constants and denominator signs in (GFR25)–(GFR29). No alteration of the original input, source mass, coefficient coordinate, multiplicity, finite integration interval, or first omitted index is needed.

This review establishes these analytic statements and their exact callback correspondence. It makes no claim that a finite collection of positive Hankel matrices proves RH or that the broader absolute-base geometric program fails. Those conclusions do not follow from the statements proved here.

249 SplitZero: maps, support fibres, ideals, and exact corrections

12 September 2026. The construction and existing manuscript results are attributed to the owner's research programme. This note gives proofs for the bounded structural package; it does not assert global priority or a completed analytic programme. Validation is tied to exact source hashes in `VALIDATION.json`.

249.1 Conventions

For a commutative ring R , write $G(R)$ for a supported copy of R together with one extra point τ . Supported elements add and multiply as in R . The point τ is the additive identity and multiplicative absorber. Write e for supported 0_R . Thus e is not τ , $e+e=e$, $e*r=e$ for every supported r , and $1+e=1$. None of the following statements requires R to be nontrivial.

249.2 1. Exact universal morphisms

Let $p:G(R) \rightarrow R$ send τ to 0 and supported r to r . Composition with p gives a bijection

$$\text{Hom_ring}(R,A) \cong \text{Hom_semiring}(G(R),A)$$

for every commutative ring A . Indeed, for a semiring homomorphism h , the equation $e+e=e$ implies $h(e)+h(e)=h(e)$, hence $h(e)=0$ by cancellation. Therefore $r \mapsto h(r)$ is a unital ring map and factors h through p . Uniqueness follows from surjectivity of p . In particular, no semiring map $G(R) \rightarrow A$ to a ring is injective: it identifies e and τ .

Let B be the Boolean semiring. The support character $\text{chi}:G(R) \rightarrow B$ sends τ to 0 and every supported r to 1 . It is unique: $e=1+(-1)$ gives $h(e)=1$ for every unital semiring map h to B ; then $re=e$ gives $h(r)=1$. The map $(p,\text{chi}):G(R) \rightarrow R \times B$ is injective with image exactly $\{(0,0)\} \cup (R \times \{1\})$. The second coordinate distinguishes absence from a supported zero; the first distinguishes supported amplitudes.

Every ring map $f:R \rightarrow S$ lifts to $G(f)$, fixing τ and acting by f on supported elements. Conversely, if $H:G(R) \rightarrow G(S)$ is a semiring map, $p_S \circ H$ factors through p_R as a unique ring map f . The support coordinate of H equals chi_R by uniqueness of the Boolean character. Thus H and $G(f)$ have equal amplitude and support and are equal by joint injectivity. Consequently

$$\text{Hom_semiring}(G(R),G(S)) \cong \text{Hom_ring}(R,S).$$

This identifies the morphisms without asserting $G(R) \cong R$. In Lean the exact statements are `ringHomEquiv`, `coordinates_injective`, `coordinates_image`, and `splitHomEquiv`.

249.3 2. Intrinsic support and fibre modules

For any $G(R)$ -semimodule M , put $s(m)=e*m$. The scalar identities imply

$$s(m+n)=s(m)+s(n), s(s(m))=s(m), m+s(m)=m, s(m)+s(m)=s(m).$$

Conversely suppose $y+y=y$. Put $v=(-1_R)*y$. Since $e=1_R+(-1_R)$, $s(y)=y+v$. Therefore $y+s(y)=(y+y)+v=y+v=s(y)$, while absorption gives $y+s(y)=y$. Hence $s(y)=y$. Thus the support set $L=\{l:s(l)=l\}$ is exactly the additive-idempotent locus. Addition gives its join, $l \leq k$ iff $l+k=k$, and its least element is 0_M .

For $l \in L$, define $M_l=\{m:s(m)=l\}$. This is an actual R -module: addition is inherited, its zero is l , inverse is $m \mapsto (-1_R)*m$, and r acts as supported r . The essential identities are

$$s(m+n)=l+l=l, m+l=m, m+(-1_R)m=l, er=e.$$

For $l \leq k$, $T_{lk}(m)=m+k$ lies in M_k . It is additive because $k+k=k$. Every supported scalar fixes k , since $rk=r(ek)=(re)k=ek=k$; hence T_{lk} is R -linear. Also $T_{ll}=\text{id}$ and $T_{kn}T_{lk}=T_{ln}$ by absorption and $k+n=n$.

A $G(R)$ -linear map F commutes with s . Its support restriction preserves zero and joins by $F(0)=0$ and $F(l+k)=F(l)+F(k)$; its fibre restriction is R -linear. Naturality is the literal equation $F(m+k)=F(m)+F(k)$. Lean implements the fibre restrictions as `LinearMaps` and checks transport identity, composition and this naturality square.

Finally $m \mapsto (s(m), m)$ is a bijection with the disjoint union of the intrinsic fibres. If $u=s(m)+s(n)=s(m+n)$, the reconstructed sum is

$$(m+u)+(n+u)=m+n+u=m+n.$$

The Lean package supplies the forward fibre construction and reconstruction for an existing semimodule. It does not yet construct the semimodule associated with every arbitrary abstract diagram or the complete categorical equivalence with natural isomorphisms. These are separate remaining implementation steps; the manuscript already proves the equivalence on paper.

249.4 3. Complete ideal classification

For $I \triangleleft R$ let $I^G = p^{(-1)}(I) = I \cup \{\tau\}$. This contains e , so it is not the zero semiring ideal $\{\tau\}$. Conversely, if $J \triangleleft G(R)$ differs from $\{\tau\}$, choose a supported $r \in J$. Multiplication by e gives $e \in J$. Therefore $J \cap R$ contains 0_R and is closed under addition and multiplication by arbitrary R -elements, hence is a ring ideal. The two constructions are mutually inverse and preserve inclusion. Thus

$$\text{Ideal}(R) \cong \{J \in \text{Ideal}(G(R)) : J \neq \{\tau\}\}$$

as ordered sets. Equivalently every J is either $\{\tau\}$ or uniquely I^G . The whole ideal poset has one additional bottom element below $(0_R)^G = \{\tau, e\}$. Lean constructs `nonzeroIdealOrderIso`, not merely a conditional theorem assuming the classification.

249.4.1 Empty joins

The isomorphism preserves joins computed in the non-bottom ideal poset. In the ambient $\text{Ideal}(G(R))$, however, the empty join is $\{\tau\}$, whereas the lift of the empty join from $\text{Ideal}(R)$ is $\{\tau, e\}$. Hence ambient empty joins are not preserved. Nonempty ambient joins are preserved because their supremum contains a lifted ideal and therefore e ; it is still in the non-bottom subposet.

249.5 4. Correcting the monoid-semiring presentation

The printed presentation in chapter 14 takes the ordinary free monoid semiring $N[M_R]$, where M_R is the multiplicative monoid of $G(R)$, modulo

$$[a]+[b]=[a+b] \text{ for supported } a, b, [x]+[\tau]=[x] \text{ for multiplicative generators } x.$$

These relations do not imply $[\tau]=0$. Evaluate every multiplicative generator at 1 in the Boolean semiring. This is unital and multiplicative; every displayed relation becomes $1+1=1$, but $[\tau]$ maps to $1 \neq 0$. The free monoid-semiring universal property therefore gives a model of the relations in which the missing equation fails.

The repair is to add $[\tau]=0$, or explicitly use a free construction already identifying the absorber with semiring zero. With the extra relation, the proposed inverse $G(R) \rightarrow Q$ preserves zero: τ maps to 0 and supported r to $[r]$. It preserves addition by the supported relation and the zero law, and multiplication and one by the monoid presentation. This inverse and evaluation $Q \rightarrow G(R)$ are identities on generators, hence mutual inverses.

The Lean file checks the multiplicative-generator countermodel and the corrected semiring-lift universal property. It does not construct the entire free monoid-semiring congruence quotient. No existing frozen reader text is silently rewritten.

249.6 5. Scope within the Zeta programme

Chapter 14 already gives the semimodule/linear-join-diagram result and ideal classification. PR #3's Rees continuation explicitly begins above that common support. The present work supplies reusable algebraic structure for that continuation; it does not certify its analytic, arithmetic-trace, purity or positivity claims. No RH statement is asserted.

The source acknowledges classical semiring/semimodule terminology and strong semilattices of groups. No exhaustive priority comparison has been completed. Attribution of this construction to the owner's programme, a new Lean implementation, and a claim of new mathematics are distinct assertions.

249.7 Source records

KokunoYumeto. (2026a). *Split-zero support and shifted Dedekind sheets* [Research manuscript, chapter 14]. In *Riemann Zeta Function: Research Reader*. Commit 42d00e359b16d52ca71568ce5e3db5341949d929; blob 49318a578b96462384000afa16794cfd5fe492c1. https://github.com/KokunoYumeto/zeta-function-research-reader/blob/42d00e359b16d52ca71568ce5e3db5341949d929/satellites/14_globalization_cue_split_zero_and_shifted_sheets.tex

KokunoYumeto. (2026b). *Split support, Rees defects, and the completed arithmetic trace* [Draft research note, PR #3]. Commit 37b2cc9b9baee3ffe25a7649314adb025302509f. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/3>

KokunoYumeto. (2026c). *SplitZero.lean* [Lean source]. Blob ff991f7383922e71cdf0e4a3bc85e89e18f808ef. https://github.com/KokunoYumeto/modern-latex-manuscripts/blob/f7ff59b176c7dc3941babd4cb9272dffc653070d/formalization/lean/classical_candidates_20260626/split_support_sidecar/SplitZero.lean

Part V

Exact original-source metric and recursive propagation proofs

250 Source enclosures and convergent signed projector control

This chapter incorporates the complete written derivations in the user-supplied continuation following merged PR29. The canonical secant and quotient-variation identities already occur in that PR. The four-projector pairing is the same pairing as AW and SP, with the exact coordinate correspondence proved below. The source sandwich, restriction-loss expression for its Hilbert–Schmidt size, and centered signed enclosure are retained in their full forms. We additionally construct the approximating operator in that enclosure and prove an explicit convergent residual bound. The supplied Lean draft is preserved separately as uncompiled; the mathematical scope here is the written proof.

250.1 The unchanged polynomial and arithmetic source

Fix the actual full packet h , tensor order $k \geq 1$, and its monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Put $E = \mathbb{C}[S]/(\chi)$, $H = \mathcal{P}_{2q}$, and $n = 2q + 1$. All coordinates are the original ascending monomial coordinates. The original analytic source and observation are

$$\begin{aligned} g &= 2\xi, & \mathcal{M}F_h &= g/h, & \mathcal{V}P &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, \\ J^{(k)}\mathcal{V}P &= \eta_{h,k}\pi_\chi P, & \eta_{h,k}[P] &= v_h^{\otimes k}P(A_h^{(k)})1, & v_h &= j_h(g/h), & A_h^{(k)} &= \sum_{i=1}^k A_{h,i}. \end{aligned} \quad (\text{ISM1})$$

Here A_h is multiplication by the original coordinate on $E_h = \mathbb{C}[s]/(h)$. Thus the domain of $\eta_{h,k}$ is E , its codomain is $E_h^{\otimes k}$, and its full unit is retained. The minimal polynomial definition of χ on the cyclic vector 1 proves that $[P] \mapsto P(A_h^{(k)})1$ is injective; multiplication by the invertible $v_h^{\otimes k}$ preserves injectivity.

For $N \in \{q-1, q, 2q-1, 2q\}$, let $I_N : \mathcal{P}_N \hookrightarrow H$ be coefficient inclusion, $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ followed by inclusion, and $C : E \rightarrow H$ the degree-less-than- q remainder lift in the retained quotient basis. The relation space is zero when $N = q-1$. Polynomial division by the monic χ proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{ISM2})$$

Indeed division writes every polynomial uniquely as $\chi Q + r$, $\deg r < q$; if its class is zero, $r = 0$. This also proves injectivity of the relation map and every dimension used below.

Let M_0, M_1 be the actual positive definite Hermitian forms on this same H . In the arithmetic comparison these are the moments of $r_{1/4}^{*k}(y) dy$ and $w_h^{*k}(y) dy$ in $S = k/2 + iy$, with $w_h(y) = |(g/h)(1/2 + iy)|^2/(2\pi)$. The finite identities below also hold for any two positive forms on this unchanged presentation. The canonical section, its relation coefficient, and its positive quotient Gram are

$$\begin{aligned} Z_N(M) &= (B_N^* M B_N)^{-1} B_N^* M C, \\ \rho_N(M) &= C - B_N Z_N(M) : E \longrightarrow H, \\ G_N(M) &= \rho_N(M)^* M \rho_N(M), & V_N(M) &= \det G_N(M) > 0. \end{aligned} \quad (\text{ISM3})$$

At $N = q-1$ the first map is the unique map to the zero space and $B_N Z_N = 0$; no preceding inverse is used. For every other degree, injectivity of B_N and positivity of M make its relation Gram positive definite. Direct substitution gives $B_N^* M \rho_N = 0$ and $\pi_\chi \rho_N = 1_E$. Hence ρ_N is injective, which proves positivity of G_N .

250.2 Both scaled secants, with the original boundary terms

Fix one admitted N , write $\rho_a = \rho_N(M_a)$, and put

$$\Delta = \rho_1 - \rho_0 = B_N(Z_N(M_0) - Z_N(M_1)). \quad (\text{ISM4})$$

For every coefficient map $Z : E \rightarrow \mathcal{P}_{N-q}$, the two cross terms in the Gram expansion vanish by $B_N^* M_a \rho_a = 0$, giving

$$(\rho_a + B_N Z)^* M_a (\rho_a + B_N Z) = G_N(M_a) + (B_N Z)^* M_a (B_N Z). \quad (\text{ISM5})$$

Taking $a = 0$, $B_N Z = \Delta$, and then $a = 1$, $B_N Z = -\Delta$, proves

$$\rho_1^* M_0 \rho_1 = G_0 + \Delta^* M_0 \Delta, \quad \rho_0^* M_1 \rho_0 = G_1 + \Delta^* M_1 \Delta. \quad (\text{ISM6})$$

Here $G_a = G_N(M_a)$. Subtracting the displayed matrices without deleting either boundary term yields, for real a, b ,

$$\begin{aligned} G_1 - aG_0 &= \rho_1^* (M_1 - aM_0) \rho_1 + a\Delta^* M_0 \Delta, \\ bG_0 - G_1 &= \rho_0^* (bM_0 - M_1) \rho_0 + \Delta^* M_1 \Delta. \end{aligned} \quad (\text{ISM7})$$

In particular both boundary terms have the positive signs shown. For $0 < a \leq b$ and $aM_0 \preceq M_1 \preceq bM_0$, congruence of positive matrices and (ISM7) give

$$aG_N(M_0) \preceq G_N(M_1) \preceq bG_N(M_0) \quad \text{at every one of the four degrees.} \quad (\text{ISM8})$$

Such finite endpoints always exist: they may be taken to be the smallest and largest eigenvalues of $M_0^{-1} M_1$, which is positive self-adjoint in the original M_0 metric.

Conjugating (ISM8) by $G_N(M_0)^{-1/2}$ places every eigenvalue of the conjugated positive matrix in $[a, b]$. The determinant is its product and logarithm is the sum of the real logarithms, so

$$q \log a \leq \log \frac{V_N(M_1)}{V_N(M_0)} \leq q \log b. \quad (\text{ISM9})$$

Set

$$\mathcal{B}(M) = \log \frac{V_{q-1}(M) V_q(M)}{V_{2q-1}(M) V_{2q}(M)}, \quad \mathcal{C} = \mathcal{B}(M_1) - \mathcal{B}(M_0). \quad (\text{ISM10})$$

There are exactly two positive and two negative terms in $\mathcal{C}$. Applying the upper endpoint in (ISM9) to the positive terms and the lower endpoint to the negative ones, and then reversing these choices, proves

$$|\mathcal{C}| \leq 2q \log(b/a). \quad (\text{ISM11})$$

Consequently a certified common source enclosure $(1 - \varepsilon)\widehat{M} \preceq M \preceq (1 + \varepsilon)\widehat{M}$, $0 \leq \varepsilon < 1$, gives

$$|\mathcal{B}(M) - \mathcal{B}(\widehat{M})| \leq 2q \log \frac{1 + \varepsilon}{1 - \varepsilon}. \quad (\text{ISM12})$$

These are the supplied source-error estimates in their original forms. At these same use sites the stronger previously proved AW/SP estimate must also be retained. For all ordered eigenvalues $\beta_1 \leq \dots \leq \beta_{2q+1}$ of $D = M_0^{-1} M_1$, define its full existing expression

$$\mathcal{L}_q(D) = \log \frac{\beta_{q+2}}{\beta_q} + 2 \sum_{j=1}^{q-1} \log \frac{\beta_{2q+2-j}}{\beta_j}.$$

The sum is empty at $q = 1$. AW's two nested-flag trace estimates, with the exact operator identification (ISM16)–(ISM19), give

$$|\mathcal{C}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log(b/a).$$

For the last inequality every displayed eigenvalue ratio is at most b/a , and its total coefficient is $1 + 2(q - 1) = 2q - 1$. Thus the same source enclosure in (ISM12) also gives the stronger $(2q - 1) \log((1 + \varepsilon)/(1 - \varepsilon))$ bound.

For the actual moment paths of RMT, let J_q be its integral of the pointwise minimum of the full-spectrum, exact trace-norm, overlap, centered Hilbert–Schmidt, and angle bounds. Each bounds the identical derivative (ISM19); the Hilbert–Schmidt member is (ISM26). Taking their pointwise minimum before integrating gives the recursively refined source-error statement

$$|\mathcal{C}| \leq \min\{J_q, \mathcal{L}_q(D), (2q - 1) \log(b/a)\},$$

$$\mathcal{C} \in [J_L - \epsilon_L, J_L + \epsilon_L] \cap [-J_q, J_q] \quad (L \geq 0).$$

Here J_L, ϵ_L are the fully constructed signed center and residual in (ISM31)–(ISM37). Their interval therefore retains its original orientation in this earlier source-error calculation. For generic positive coefficient forms, the full-spectrum, trace-norm, overlap, Hilbert–Schmidt and signed bounds still apply, but the moment-specific angle improvement and its strict moment-positivity proof are used only on the original moment paths. The exact primitive and coefficient support are established in (ISM38)–(ISM43). Scaling a source by $c > 0$ leaves Z_N, ρ_N unchanged and multiplies G_N, V_N by c, c^q , respectively, directly by (ISM3). Thus its four $q \log c$ terms cancel in (ISM10); this proves the scalar invariance without altering any source form.

250.3 The canonical projector is exactly the original difference

Use $x \in [0, 1]$ for the comparison path, retaining the frequency y and all other deformation parameters. Write

$$M_x = (1 - x)M_0 + xM_1, \quad \dot{M} = M_1 - M_0, \quad X_x = M_x^{-1} \dot{M}. \quad (\text{ISM13})$$

Since $M_x X_x = \dot{M} = \dot{M}^*$, this is self-adjoint in the original M_x inner product. Define

$$\begin{aligned} P_N &= I_N (I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= B_N (B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0, \\ \Pi_N &= \rho_N G_N^{-1} \rho_N^* M_x. \end{aligned} \quad (\text{ISM14})$$

Each formula squares to itself and has M_x -self-adjoint adjoint law. The ranges of the first two are $\mathcal{P}_N$ and $B_N \mathcal{P}_{N-q}$; the range of the third is $\rho_N E$. By (ISM2)–(ISM3),

$$\mathcal{P}_N = \rho_N E \perp_{M_x} B_N \mathcal{P}_{N-q}. \quad (\text{ISM15})$$

To prove the equality, subtract $\rho_N \pi_\chi p$ from any $p \in \mathcal{P}_N$; (ISM2) puts the result in the relation range, and (ISM3) proves orthogonality. Therefore the original P_N, Q_N commute with product Q_N , and their orthogonal range decomposition proves the exact operator identity

$$\Pi_N = P_N - Q_N, \quad \text{Tr } \Pi_N = q. \quad (\text{ISM16})$$

The inverse of the range parametrization $\rho_N : E \rightarrow \text{ran } \Pi_N$ is the literal π_χ restricted to that range; both composites are identities by (ISM3). Thus (ISM16) identifies actual maps and their domains, not only their traces.

Define

$$\begin{aligned} \mathcal{A}_x &= \Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{ISM17})$$

Substitution of (ISM16), with $Q_{q-1} = 0$, gives $\mathcal{A}_x = U_x - W_x$ and $\text{Tr } \mathcal{A}_x = 0$. These U_x, W_x are AW's U, W and SP's $\mathcal{R}, \mathcal{P}$, respectively. The source relation matrix is related to AW's unpadded matrix by $B_N = I_N B_N^{\text{AW}}$, so (ISM14) reproduces its actual projections with no sign or metric change.

Differentiating $\pi_X \rho_N = 1$ shows that ρ'_N has image in the original relation space. More explicitly, differentiation of $B_N^* M_x \rho_N = 0$ gives

$$\rho'_N = -B_N (B_N^* M_x B_N)^{-1} B_N^* \dot{M} \rho_N, \quad G'_N = \rho_N^* \dot{M} \rho_N. \quad (\text{ISM18})$$

The second identity follows because the two cross terms containing ρ'_N vanish by the same orthogonality. All these matrices are smooth on the closed interval, since their positive Gram inverses exist there. Jacobi differentiation and cyclicity of finite trace prove

$$\frac{d}{dx} \log V_N = \text{Tr}(X_x \Pi_N), \quad \mathcal{B}'(x) = \text{Tr}(X_x \mathcal{A}_x), \quad \mathcal{C} = \int_0^1 \text{Tr}(X_x \mathcal{A}_x) dx. \quad (\text{ISM19})$$

For the original row $v(y) = (1, S, \dots, S^{2q})$, $\Pi_N M_x^{-1} = \rho_N G_N^{-1} \rho_N^*$. Thus the signed density in PR29 is exactly

$$\kappa_x(y) = v(y) \mathcal{A}_x M_x^{-1} v(y)^*, \quad \int \kappa_x d\mu_x = \text{Tr} \mathcal{A}_x = 0. \quad (\text{ISM20})$$

The first equality keeps the inverse metric that is required by the original coordinate kernel. Integrating against the original signed measure $\mu_1 - \mu_0$ gives precisely (ISM19).

250.4 Restriction losses and the full cross trace

Fix $i \leq j$ among admitted degrees. The map $\rho_i - \rho_j : E \rightarrow \mathcal{P}_j$ has zero remainder, so (ISM2) makes it an original degree- j relation. Orthogonality of ρ_j therefore gives the cross-Grams

$$\rho_j^* M_x \rho_i = G_j, \quad \rho_i^* M_x \rho_j = G_j, \quad G_i - G_j = (\rho_i - \rho_j)^* M_x (\rho_i - \rho_j). \quad (\text{ISM21})$$

The exact reverse restriction on E is $T_{i,j} = G_i^{-1} G_j$. It is positive self-adjoint in G_i , with spectrum in $(0, 1]$, by the last identity. Cyclic trace and the first two identities give

$$\begin{aligned} \text{Tr}(\Pi_i \Pi_j) &= \text{Tr}(G_i^{-1} (\rho_i^* M_x \rho_j) G_j^{-1} (\rho_j^* M_x \rho_i)) \\ &= \text{Tr}(G_i^{-1} G_j), \\ \text{Tr}((\Pi_i - \Pi_j)^2) &= 2 \text{Tr}(1 - T_{i,j}). \end{aligned} \quad (\text{ISM22})$$

All traces are finite traces of the original coordinate operators. In particular their spectra are not declared to be arithmetic invariant summands.

Set

$$\begin{aligned} A_0 &= \Pi_{q-1} - \Pi_{2q-1}, \quad A_1 = \Pi_q - \Pi_{2q}, \\ \ell_0 &= \text{Tr}(1 - T_{q-1,2q-1}), \quad \ell_1 = \text{Tr}(1 - T_{q,2q}), \\ Z_x &= \text{Tr}(\mathcal{A}_x^2) = 2\ell_0 + 2\ell_1 + 2 \text{Tr}(A_0 A_1). \end{aligned} \quad (\text{ISM23})$$

This retains the cross term with its actual sign. On the full algebra $\text{End}(H)$, the map $L \mapsto M_x^{1/2} L M_x^{-1/2}$ is an algebra isomorphism with inverse $K \mapsto M_x^{-1/2} K M_x^{1/2}$. It sends the M_x -self-adjoint operators to Hermitian operators, and cyclicity preserves traces of every product. This proves $Z_x \geq 0$ and, from $\|A_0 + A_1\|_{\text{HS}}^2 \leq 2\|A_0\|_{\text{HS}}^2 + 2\|A_1\|_{\text{HS}}^2$, proves

$$0 \leq Z_x \leq 4(\ell_0 + \ell_1). \quad (\text{ISM24})$$

Here and below the Hilbert–Schmidt norm is computed after this exact metric isomorphism; no new source metric is used.

Put

$$c_x = \frac{\text{Tr} X_x}{n}, \quad \Sigma_x = \text{Tr}(X_x^2) - \frac{(\text{Tr} X_x)^2}{n}. \quad (\text{ISM25})$$

The trace is real by self-adjointness. Expanding $\text{Tr}((X_x - c_x I)^2)$ proves its equality with Σ_x , hence its nonnegativity. Trace zero of $\mathcal{A}_x$, and Cauchy–Schwarz for the two Hermitian images in the preceding isomorphism, give

$$|\mathcal{B}'(x)| \leq S_q(x) := \sqrt{\Sigma_x Z_x} \leq 2\sqrt{\Sigma_x(\ell_0 + \ell_1)}. \quad (\text{ISM26})$$

This bound can be combined pointwise with the existing full-spectrum, exact trace-norm, overlap, and angle bounds on the same $\mathcal{B}'$. It does not replace any smaller available bound.

250.5 A specified signed approximation and its exact residual

For an M_x -self-adjoint, continuous operator Y_x define $E_x = X_x - c_x I - Y_x$. It is self-adjoint too. Expanding the trace and applying the same Cauchy–Schwarz inequality proves the two signed inequalities

$$\text{Tr}(Y_x \mathcal{A}_x) - \sqrt{\text{Tr}(E_x^2) Z_x} \leq \mathcal{B}'(x) \leq \text{Tr}(Y_x \mathcal{A}_x) + \sqrt{\text{Tr}(E_x^2) Z_x}. \quad (\text{ISM27})$$

Every term is real. Integration is legitimate by continuity on the compact interval. We now construct such a Y_x , with its residual controlled from the actual two endpoint forms.

Let $D = M_0^{-1} M_1$, with ordered positive eigenvalues $b_1 \leq \dots \leq b_n$, counted with multiplicity. For this subsection write $B_x^\circ = (1-x)I + xD$, to distinguish it from every original relation matrix. Since $M_x = M_0 B_x^\circ$, direct inversion gives

$$X_x = (B_x^\circ)^{-1}(D - I), \quad \lambda_j(x) = 1 - x + x b_j, \quad \gamma_j(x) = \frac{b_j - 1}{1 - x + x b_j}. \quad (\text{ISM28})$$

Take an M_0 -orthogonal eigenbasis of D . It is also M_x -orthogonal, since on its j -th line M_x multiplies M_0 by $\lambda_j(x) > 0$. Therefore (ISM28) gives the full spectrum of X_x in the original varying metric.

Define the positive scalar and commuting operator

$$a_x = \frac{\lambda_1(x) + \lambda_n(x)}{2}, \quad H_x = I - B_x^\circ / a_x. \quad (\text{ISM29})$$

Their use does not rescale M_x , its moments, or its quotient Grams. The eigenvalues of H_x are $\zeta_j(x) = 1 - \lambda_j(x)/a_x$, and

$$\|H_x\|_{M_x} = \frac{x(b_n - b_1)}{2(1-x) + x(b_1 + b_n)} \leq \vartheta := \frac{b_n - b_1}{b_n + b_1} < 1. \quad (\text{ISM30})$$

Indeed the extrema of $1 - \lambda/a_x$ are equal and opposite at the two endpoints. The displayed scalar function of x has derivative $2(b_n - b_1)/(2(1-x) + x(b_1 + b_n))^2 \geq 0$. This proves the uniform bound, including $x = 0$.

For every integer $L \geq 0$, form the finite matrix polynomial and its centered version

$$\begin{aligned} X_x^{[L]} &= a_x^{-1} \sum_{r=0}^L H_x^r (D - I), \\ Y_x^{[L]} &= X_x^{[L]} - \frac{\text{Tr } X_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM31})$$

All factors commute with D and are real functions of its self-adjoint operator. They are consequently self-adjoint in both M_0 and M_x . The exact finite geometric-series identity $(I - H_x) \sum_{r=0}^L H_x^r = I - H_x^{L+1}$, together with $B_x^\circ = a_x(I - H_x)$, proves

$$\begin{aligned} R_x^{[L]} &:= X_x - X_x^{[L]} = H_x^{L+1} X_x, \\ E_x^{[L]} &:= X_x - c_x I - Y_x^{[L]} = R_x^{[L]} - \frac{\text{Tr } R_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM32})$$

No infinite-series interchange is used in this identity. In the original eigenbasis the residual eigenvalues are $\delta_j(x) = \zeta_j(x)^{L+1} \gamma_j(x)$. Thus its exact centered squared size is

$$\mathrm{Tr}((E_x^{[L]})^2) = \sum_{j=1}^n \delta_j(x)^2 - \frac{(\sum_{j=1}^n \delta_j(x))^2}{n}. \quad (\text{ISM33})$$

Both sides are nonnegative, and all multiplicities and signs of the residual eigenvalues are retained. The signed center is $\mathrm{Tr}(Y_x^{[L]} \mathcal{A}_x) = \mathrm{Tr}(X_x^{[L]} \mathcal{A}_x)$ by $\mathrm{Tr} \mathcal{A}_x = 0$. Equations (ISM27), (ISM31)–(ISM33) therefore provide a finite constructed signed center and its exact error.

There is also a uniform elementary error bound for this fixed original pair of forms. Define

$$K_D = \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \quad (\text{ISM34})$$

The denominator of $\gamma_j(x)$ is at least $\min\{1, b_j\}$, so (ISM30)–(ISM33) imply $\mathrm{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$. The nested relation and source flags in (ISM17) give both U_x, W_x eigenvalues $0^{[q]}, 1^{[2]}, 2^{[q-1]}$: the first relation line contributes one, its next $q - 1$ increments contribute two, and the final increment contributes one; the source increments give the same list. In particular both have rank $q + 1$, and each dominates its range projection. Their ranges in dimension $2q + 1$ intersect in dimension at least one. The trace product of the two range projections is at least that intersection dimension, by summing squared projected lengths in an orthonormal basis. For positive operators, increasing a factor cannot decrease the product trace, since $\mathrm{Tr}(T^{1/2}(S' - S)T^{1/2}) \geq 0$. It follows that $\mathrm{Tr}(U_x W_x) \geq 1$. Using their complete spectra, we obtain

$$Z_x = 8q - 4 - 2 \mathrm{Tr}(U_x W_x) \leq 8q - 6. \quad (\text{ISM35})$$

This proof includes $q = 1$, where the multiplicity $q - 1$ is zero.

Set

$$\begin{aligned} J_L &= \int_0^1 \mathrm{Tr}(X_x^{[L]} \mathcal{A}_x) dx, \\ \epsilon_L &= \int_0^1 \sqrt{\mathrm{Tr}((E_x^{[L]})^2) Z_x} dx. \end{aligned} \quad (\text{ISM36})$$

Every integrand is a specified continuous finite matrix expression on the original source path. Integrating (ISM27) with the constructed (ISM31) and the established uniform bound gives

$$J_L - \epsilon_L \leq \mathcal{C} \leq J_L + \epsilon_L, \quad 0 \leq \epsilon_L \leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \quad (\text{ISM37})$$

Thus the supplied signed enclosure now has an explicit convergent residual on every fixed original finite source. It retains the arithmetic correlation in its center. The evaluation of the center integral still uses the specified arithmetic entries and a controlled integration method; this calculation supplies no uniform estimate in k . If $b_n = b_1$, then $D = b_1 I$, $H_x = 0$, the residual and $\mathcal{C}$ vanish exactly, and every formula above remains valid.

250.6 The original primitive and the residual at a proper support

We finally prove the source comparison underlying all these metric changes. In the original one-factor complex $V \xrightarrow{\Theta} \mathcal{B}$, retain $h(D)F_h = \Theta \phi_*$, with the original D action commuting with Θ . For $v \in E$, the polynomial $(\rho_1 - \rho_0)v = \chi Q_v$ has the coefficient $Q_v = (Z_N(M_0) - Z_N(M_1))v$ by (ISM4). In the multivariate polynomial ring put

$$f_v(s_1, \dots, s_k) = \chi(s_1 + \dots + s_k) Q_v(s_1 + \dots + s_k). \quad (\text{ISM38})$$

Its image in $E_h^{\otimes k}$ is zero, because $\chi(A_h^{(k)})1 = 0$. Successive monic division in $s_1, \dots, s_k$ constructs specific polynomials $Q_{v,i}$ as follows. Let $f_{v,0} = f_v$ and write $f_{v,i-1} = h(s_i)Q_{v,i} + f_{v,i}$, where $\deg_{s_i} f_{v,i} < \deg h$. Earlier remainder degree bounds are preserved because $h(s_i)$ uses only s_i . The final remainder $f_{v,k}$ has degree below $\deg h$ in every variable and zero class in the quotient, so it is zero by the tensor monomial basis. Telescoping proves

$$f_v = \sum_{i=1}^k h(s_i)Q_{v,i}. \quad (\text{ISM39})$$

This constructs a primitive using precisely the original relations, without replacing them by a larger observation kernel.

In the tensor complex $\mathbf{C}^\bullet = [V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ define

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}) \in \mathbf{C}^{k-1}. \quad (\text{ISM40})$$

The displayed source tensor has exactly $i-1$ preceding factors of degree one. Its differential therefore has sign $(-1)^{i-1}$; multiplying by the sign in (ISM40) gives $+1$. The source operators commute with the differential. Applying $h(D_i)F_h = \Theta\phi_*$ to (ISM39) now proves

$$dK_v = \mathcal{V}((\rho_1 - \rho_0)v). \quad (\text{ISM41})$$

Consequently the two canonical sections have the same class in the original full-source quotient; their common arithmetic observation is exactly $\eta_{h,k}v$ by (ISM1). Composing with the original full-source quotient $q^{(k)}$ gives the already constructed map $\sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}$. Here no map was made natural merely by changing its metric; the actual primitive (ISM40) is the comparison.

To express the full two-leg source without deleting its diagonal, retain the labelled Fourier copy $\widehat{V}$, with the original isomorphism $\mathcal{F} : V \rightarrow \widehat{V}$, and the complex $\mathbf{T}^\bullet = [V \oplus \widehat{V} \rightarrow \mathcal{B}]$, whose differential is $(\phi, \widehat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\widehat{\psi})$. There are exact cochain maps from the one-leg complex to this complex with degree-one map identity and degree-zero maps

$$j_+(v) = (v, 0), \quad j_-(v) = (0, -\mathcal{F}v).$$

Indeed either differential equals Θv . The reverse cochain map has degree-one map identity and degree-zero map $p(\phi, \widehat{\psi}) = \phi - \mathcal{F}^{-1}\widehat{\psi}$. Thus $pj_+ = pj_- = 1$, whereas $(j_+ - j_-)(v) = (v, \mathcal{F}v)$ is the full degree-zero diagonal kernel. These formulas prove both composites and retain that kernel explicitly. Tensoring specified choices of these degree-zero maps with their degree-one identities gives cochain maps, since the displayed differential identities preserve each original tensor sign. Apply one such map to (ISM40) to obtain $\widetilde{K}_v \in (\mathbf{T}^{\otimes k})^{k-1}$. Its top differential is still (ISM41).

For precision at a proper declared support $\lambda = (W, S; A)$, let $\mathcal{J}$ be the specified finite carrier of coefficient positions with $A \subseteq \mathcal{J}$. Put $\mathbf{C}_{\text{full}}^\bullet = (\mathbf{T}^{\otimes k})^{\oplus \mathcal{J}}$, with the original differential on each coordinate. Insert the active A -coordinates by literal zero insertion into this fixed full carrier. Let $\mathbf{C}_\lambda^\bullet \subseteq \mathbf{C}_{\text{full}}^\bullet$ be its declared subcomplex, retaining its independent outer label and coefficient mask. The same source primitive is taken in the indicated coefficient copy. When the compared top source representatives lie in $\mathbf{C}_\lambda^k$, their difference has the exact residual

$$[d\widetilde{K}_v]_\lambda \in \frac{d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_\lambda^k}{d\mathbf{C}_\lambda^{k-1}}. \quad (\text{ISM42})$$

This quotient is the kernel of the natural map $\mathbf{C}_\lambda^k / d\mathbf{C}_\lambda^{k-1} \rightarrow \mathbf{C}_{\text{full}}^k / d\mathbf{C}_{\text{full}}^{k-1}$: a class maps to zero exactly when its representative is a full-source boundary. The inclusion of the displayed numerator therefore induces the kernel isomorphism, with inverse obtained by using the same representative. These two maps are well-defined since differences in $d\mathbf{C}_\lambda^{k-1}$ are precisely the equivalence relation on both sides.

Thus (ISM41) kills the class at this support exactly when $d\tilde{K}_v \in d\mathcal{C}_\lambda^{k-1}$. Membership $\tilde{K}_v \in \mathcal{C}_\lambda^{k-1}$ is one explicit sufficient witness, not an assumption made for arbitrary W .

First specialize to $k = 1$ and the single coefficient carrier $\mathcal{I} = \{i_0\}$, with active face $A = \{i_0\}$. The coordinate projection onto i_0 and its inverse coordinate insertion identify this one-coordinate carrier with the original one-factor complex. In these specified coordinates put $\mathcal{B}_\lambda = \mathcal{C}_\lambda^1 \subseteq \mathcal{B}$. Suppose the admitted source image under p is $W \subseteq V$, so its boundary range is exactly ΘW , and retain $\Theta V \subseteq \mathcal{B}_\lambda$. The precise kernel map in this single coordinate is

$$V/W \longrightarrow \mathcal{B}_\lambda/\Theta W, \quad [v] \longmapsto [\Theta v]_W. \quad (\text{ISM43})$$

It is injective because Θ is injective: $\Theta v = \Theta w$, $w \in W$, forces $v = w$. Its range is the kernel of the map to the full quotient $\mathcal{B}/\Theta V$, since being in that kernel means being represented by Θv . In this case (ISM42) is exactly the class of the original one-factor primitive in V/W .

For an arbitrary admitted face $A \subseteq \mathcal{I}$, restore the original coordinate injections $\iota_a : \mathcal{B} \rightarrow \mathcal{B}^{\oplus \mathcal{I}}$ and projections $\text{pr}_a : \mathcal{B}^{\oplus \mathcal{I}} \rightarrow \mathcal{B}$. They satisfy $\text{pr}_b \iota_a = \delta_{ab} 1$ and $\sum_{a \in A} \iota_a \text{pr}_a = 1$ on the A -supported subspace. Let the degree-one source at this face be $\mathcal{B}_\lambda^{\oplus A}$, inserted in the fixed carrier by these ι_a , and retain the degree-zero source image $W^{\oplus A}$ and boundary $(\Theta W)^{\oplus A}$. Its kernel map is

$$(V/W)^{\oplus A} \longrightarrow \mathcal{B}_\lambda^{\oplus A}/(\Theta W)^{\oplus A}, \quad ([v_a])_{a \in A} \longmapsto \left[\sum_{a \in A} \iota_a \Theta v_a \right].$$

Replacing v_a by $v_a + w_a$, $w_a \in W$, changes the displayed sum by the specified boundary, so the map is well-defined. If its image is zero, projection by each pr_a gives $\Theta v_a \in \Theta W$, whence $v_a \in W$ by injectivity of Θ . Thus the map is injective. A representative in its codomain maps to zero in $(\mathcal{B}/\Theta V)^{\oplus \mathcal{I}}$ exactly when each active coordinate is Θv_a for some $v_a \in V$. Its inverse on that kernel takes these unique preimages modulo W . Changing a representative by an element of $(\Theta W)^{\oplus A}$ changes each preimage by an element of W . The coordinate identities above then prove both inverse composites. This is the exact $(V/W)^{\oplus A}$ residual, with its full ambient carrier specified.

For $A \subseteq A'$, zero insertion sends the tuple $([v_a])_{a \in A}$ to the same tuple extended by zero; both orders of this insertion and the kernel map give $\sum_{a \in A} \iota_a \Theta v_a$. A signed coefficient permutation sends that sum to $\sum_{a \in A} \epsilon_a \iota_{\sigma(a)} \Theta v_a = \sum_{a \in A} \iota_{\sigma(a)} \Theta(\epsilon_a v_a)$, which proves the same commuting identity with every sign retained. When $A = \emptyset$, the two displayed direct sums and their unique map are zero vector spaces and a zero map at the retained outer label $(W, S; \emptyset)$; the coordinate calculation does not identify that label with external absence. This proves the typed relation to the existing mixed-support quotients while retaining every proper-source residual.

All source identities, projector equalities, scaled comparison signs, and signed estimates above use the same original finite polynomial presentation. Their full primitive and proper support kernel have now been exhibited, so backward and forward propagation can replace weaker metric bounds at their earlier use sites without erasing the older sources or claiming that the arbitrary proper-support residual vanishes.

251 Recursive propagation of the original signed metric bounds

This proof combines the earlier full-relative-spectrum estimate AW16, the exact trace norm and actual-overlap estimates of SP, the supplied centered Hilbert–Schmidt estimate, and the all-degree angle estimate EP17–EP25 on their common original source. Each of those complete proofs is retained in this edition. The construction below carries their pointwise minimum through both the original volume and the exact nonlinear endpoint coordinate, without changing either endpoint form.

251.1 The original source and all four quotient volumes

Fix the original packet h , tensor order $k \geq 1$, monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, and

$$H = \mathcal{P}_{2q}, \quad S = k/2 + iy, \quad v(y) = (1, S, \dots, S^{2q}). \quad (\text{RMT1})$$

The coordinates are the original ascending monomial coordinates. The original reference and arithmetic measures are

$$d\mu_0 = r_{1/4}^{*k}(y) dy, \quad d\mu_1 = w_h^{*k}(y) dy, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}. \quad (\text{RMT2})$$

All their original mass and Taylor-unit factors remain. In particular the exact reference convolution identity is

$$r_{1/4}^{*k} = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{RMT3})$$

Thus these are the same measures in AW and SP. Every argument below also applies to the actual finite-circle comparison when one of the measures is its specified positive atomic measure on this same line and its retained degree- $2q$ Gram is positive definite. That positivity is already proved by the original sampled-source bounds. This statement does not assign moment structure to an arbitrary positive coefficient matrix.

For $0 \leq x \leq 1$ let

$$M_x = \int v(y)^* v(y) d((1-x)\mu_0 + x\mu_1)(y), \quad \dot{M} = M_1 - M_0, \quad C_x = M_x^{-1} \dot{M}. \quad (\text{RMT4})$$

The two positive endpoint forms make M_x positive definite throughout the closed interval. For every original cutoff N , let $I_N : \mathcal{P}_N \rightarrow H$ be coefficient inclusion and $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ , followed by inclusion. Negative relation degree means the zero space. Retain the orthogonal projections

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{RMT5})$$

AW's relation matrix has target $\mathcal{P}_N$, so its exact relation to this one is $B_N = I_N B_N^{\text{AW}}$. Equation (RMT5), after that substitution, is precisely AW4 and SP2. The quotient map remains $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[S]/(\chi)$, and its canonical positive quotient Gram is $G_N(x)$. Put $V_N(x) = \det G_N(x)$ and retain the four signs:

$$\mathcal{B}(x) = \log \frac{V_{q-1}(x)V_q(x)}{V_{2q-1}(x)V_{2q}(x)}, \quad U_x = Q_{2q-1} + Q_{2q} - Q_q, \quad W_x = P_{2q-1} - P_{q-1} + P_{2q} - P_q. \quad (\text{RMT6})$$

The relation-first determinant-line formula and differentiation of each of these four determinants give, without commuting their operators,

$$\mathcal{B}'(x) = \text{Tr}((U_x - W_x)C_x). \quad (\text{RMT7})$$

This is the same actual integrand as AW6 and SP7.

251.2 The simultaneous bounds on that integrand

Let $b_1 \leq \dots \leq b_{2q+1}$ be every eigenvalue, with multiplicity, of $D = M_0^{-1} M_1$ in the original M_0 metric. They are positive. Retain

$$c_j(x) = \frac{b_j - 1}{1 - x + x b_j}, \quad d_x = c_{2q+1}(x) - c_1(x), \quad \kappa = \frac{b_{2q+1}}{b_1}. \quad (\text{RMT8})$$

These are the ordered eigenvalues of C_x . Indeed $C_x = ((1-x)I + xD)^{-1}(D-I)$, and differentiation with respect to b_j gives the positive quantity $(1-x+xb_j)^{-2}$. In particular $\int_0^1 d_x dx = \log \kappa$.

Set $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two ranks are $q+1$ in dimension $2q+1$, so $s_x \geq 1$. All traces and trace norms below are in the same original M_x metric. Define the two calculated functions

$$\begin{aligned} E_q(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ O_q(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}. \end{aligned} \quad (\text{RMT9})$$

The sum is empty when $q=1$. The complete nested-flag calculation gives both U_x, W_x the spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. Decomposing each as the sum of its spectral projections of ranks $q+1$ and $q-1$, and applying the ordered projection-trace inequalities, gives $|\mathcal{B}'| \leq E_q$, exactly as in AW15–AW16. For clarity the subtraction of the two bounds cancels the common c_{q+1} only after both sums have been written; it leaves every term displayed in E_q .

Both positive operators dominate the original orthogonal projection onto their common range. Subtracting that same projection from both leaves positive operators with traces $2q - s_x$, whence $\|U_x - W_x\|_1 \leq 4q - 2s_x$. Also their computed spectra give

$$\text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{RMT10})$$

Cauchy–Schwarz on all $2q+1$ eigenvalues of $U_x - W_x$ gives the second trace-norm bound in (RMT9). Since $\text{Tr}(U_x - W_x) = 0$, subtract $(c_1 + c_{2q+1})I/2$ from C_x in (RMT7). In an M_x -orthonormal eigenbasis of $U_x - W_x$, the absolute value of each diagonal entry of this centered C_x is at most $d_x/2$. Multiplying by the corresponding absolute eigenvalues and summing gives the retained exact trace-norm bound

$$|\mathcal{B}'| \leq T_q(x) := \frac{d_x}{2} \|U_x - W_x\|_1 \leq O_q(x). \quad (\text{RMT10a})$$

The second inequality follows from the two trace-norm estimates just proved. Thus neither of those majorants replaces the original trace norm.

The user's additional supplied projector calculation gives another bound on this same integrand. Define

$$\Sigma_x = \text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{2q+1}, \quad S_q(x) = \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}. \quad (\text{RMT10b})$$

The nonnegative number Σ_x is the squared Hilbert–Schmidt norm of $C_x - \text{Tr}(C_x)I/(2q+1)$ in the original M_x Hilbert space. The trace of $U_x - W_x$ is zero, so Hilbert–Schmidt Cauchy–Schwarz gives $|\mathcal{B}'| \leq S_q$. Equivalently one may apply the isometry $M_x^{1/2} : (H, M_x) \rightarrow (H, I)$ simultaneously to both operators; conjugation preserves their traces and products and proves the same inequality. This is a calculation with the original metric, not a change of either endpoint source form.

The original crossed quotient pairs are $(q-1, 2q)$ and $(q, 2q-1)$. Retain their full canonical minimum-section angle lists $(\theta_1, \dots, \theta_q)$ and $(\varphi_1, \dots, \varphi_q)$. In the second list, ordered with a smallest angle first, the original ambient dimension forces $\varphi_1 = 0$. The count is proved in EP17–EP25. For the following scalar sums put $a = 2q-1$ and define the explicitly indexed list

$$\lambda_i = -\log \cos^2 \theta_i \quad (1 \leq i \leq q), \quad \lambda_{q+j-1} = -\log \cos^2 \varphi_j \quad (2 \leq j \leq q).$$

Then $\mathcal{B} = -\log \cos^2 \varphi_1 + \sum_{\nu=1}^a \lambda_\nu = 0 + \sum_{\nu=1}^a \lambda_\nu$. Every additional zero remains in the λ -list, while the designated geometric entry φ_1 is still displayed in the full second list. The exact projection-difference spectra give

$$|\mathcal{B}'(x)| \leq d_x \left(\sin \varphi_1 + \sum_{\nu=1}^a \sqrt{1 - e^{-\lambda_\nu(x)}} \right) \leq a \sqrt{1 - e^{-\mathcal{B}(x)/a}} d_x. \quad (\text{RMT11})$$

The second inequality is concavity of $z \mapsto \sqrt{1 - e^{-z}}$ on $[0, \infty)$, applied to the full a -term list, including its zeros, together with the displayed $\sin \varphi_1 = 0$. Both q -entry lists remain the geometric data; the known zero contribution is kept explicitly in these scalar identities.

The needed strict positivity of $\mathcal{B}$ follows on the actual measures rather than being an additional assumption. If $\mathcal{B} = 0$, both nested quotient pairs have equal minimum spaces. To see every implication, their canonical minimum sections satisfy $G_i - G_j = (R_i - R_j)^* M_x (R_i - R_j) \succeq 0$. Thus each determinant ratio in the crossed pairs is at least one. A zero sum of their nonnegative logarithms makes both ratios one. All eigenvalues of $G_j^{-1/2} G_i G_j^{-1/2}$ are at least one, so determinant one makes every eigenvalue one and $G_i = G_j$. The displayed squared norm then gives $R_i = R_j$ as maps to the same original source H . In the first pair R_{q-1} is the literal degree-less-than- q remainder section. Its equality with R_{2q} puts the constant polynomial 1 in the latter minimum space, which is orthogonal to $\ker J_{2q} = \chi \mathcal{P}_q$. For $\chi(S) = \sum a_j S^j$, put $\chi^{\#k}(S) = \sum \bar{a}_j (k - S)^j$. This belongs to $\mathcal{P}_q$ and equals $\overline{\chi(S)}$ on $S = k/2 + iy$. The asserted orthogonality would therefore give

$$0 = \int \chi(S) \chi^{\#k}(S) d\mu_x = \int |\chi(S)|^2 d\mu_x, \quad (\text{RMT12})$$

contradicting the positive Gram of the nonzero polynomial χ . This proof also covers the specified positive atomic comparison. Thus $\mathcal{B}(x) > 0$ on the entire closed interval.

251.3 Carrying the refined bound through both endpoint coordinates

Define the positive function

$$z_x = a \sqrt{1 - e^{-\mathcal{B}(x)/a}} > 0.$$

With every preceding control retained, set

$$\begin{aligned} A_q(x) &= z_x d_x, \\ J_q &= \int_0^1 \min\{E_q(x), T_q(x), O_q(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\left\{d_x, \frac{E_q(x)}{z_x}, \frac{T_q(x)}{z_x}, \frac{O_q(x)}{z_x}, \frac{S_q(x)}{z_x}\right\} dx. \end{aligned} \quad (\text{RMT13})$$

These are actual quantities of the original finite source path. The projections are continuous matrix functions of x . Explicitly $s_x = 2(q+1) - \text{rank}[U_x W_x]$. A rank superlevel set is a finite union of sets where a specified minor is nonzero, hence is Borel. This proves measurability of s_x and of every displayed integrand. Equation (RMT12) and compactness give a positive lower bound for $\mathcal{B}$ on this fixed path. Hence all denominators in I_q are nonzero, and the integrands are finite and bounded. This is a fixed-source statement and asserts no bound uniform in the tensor order.

All five preceding estimates bound the same $|\mathcal{B}'|$. Pointwise comparison followed by the triangle inequality for the integral therefore proves

$$\begin{aligned} |\mathcal{B}(1) - \mathcal{B}(0)| &\leq J_q, \\ J_q &\leq \min\left\{\mathcal{L}_q(D), \int_0^1 T_q(x) dx, \int_0^1 O_q(x) dx, \int_0^1 S_q(x) dx, \int_0^1 A_q(x) dx\right\}. \end{aligned} \quad (\text{RMT14})$$

where the earlier AW13 quantity is retained in full:

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \kappa. \quad (\text{RMT15})$$

Indeed integration of each c_j gives $\log b_j$, which proves the first entry of the minimum in (RMT14).

For $B > 0$ let

$$\begin{aligned}\mathcal{H}_a(B) &= 2 \operatorname{arcosh}(e^{B/(2a)}), & \mathcal{H}'_a(B) &= \frac{1}{a\sqrt{1 - e^{-B/a}}}, \\ \mathcal{H}_a^{-1}(z) &= 2a \log \cosh(z/2) \quad (z > 0).\end{aligned}\tag{RMT16}$$

The derivative follows by differentiating the positive real inverse branch of $\cosh$; substitution proves both inverse composites. Applying this exact derivative to each of the five bounds on $\mathcal{B}'$ gives

$$|\mathcal{H}_a(\mathcal{B}(1)) - \mathcal{H}_a(\mathcal{B}(0))| \leq I_q \leq \int_0^1 d_x dx = \log \kappa.\tag{RMT17}$$

Thus the exact trace norm, supplied variance estimate, overlap and full-relative-spectrum information propagate into the nonlinear bound, not only into its linear consequence.

Writing $H_0 = \mathcal{H}_a(\mathcal{B}(0))$, the inverse in (RMT16), extended continuously by value zero at zero, and the additive estimate in (RMT14) give the simultaneous interval

$$\begin{aligned}\mathcal{B}(1) &\geq \max \left\{ 0, \mathcal{B}(0) - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right) \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ \mathcal{B}(0) + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right) \right\}.\end{aligned}\tag{RMT18}$$

The actual $\mathcal{B}(1)$ is strictly positive although the lower bound can equal zero. If $M_1 = cM_0$ with $c > 0$, all the c_j coincide, so $d_x, E_q, T_q, O_q, S_q, A_q, I_q, J_q$ vanish. Each original quotient determinant is multiplied by the same power q of $1 - x + xc$; the four signs then give $\mathcal{B}(1) = \mathcal{B}(0)$, as (RMT18) also states. This includes $q = 1$, for which the ordered sums above are empty and $a = 1$.

Every projection and metric in this argument uses the original J_N, B_N , coefficient basis and source measure. The arithmetic class map $\sigma_h^{\otimes k} \eta_{h,k} J_N$, its complete Taylor unit, and the original theta primitives of differences of minimum sections are unchanged. The proof refines numerical bounds on those exact maps; it introduces no identification between their source kernels or between distinct support labels. The earlier complete AW, SP and EP proofs, together with the explicit correspondences (RMT3)–(RMT7), provide the full input proofs used in this propagation.

251.4 Carrying the constructed signed residual into the endpoint

The complete supplied projector correspondence and its explicit finite approximation are proved in Section 250, (ISM14)–(ISM37). Their notation has the exact identities

$$X_x = C_x, \quad \mathcal{A}_x = U_x - W_x, \quad \Pi_N = P_N - Q_N.$$

For completeness retain the actual finite construction here. For each integer $L \geq 0$, put

$$\begin{aligned}B_x^\circ &= (1 - x)I + xD, & \alpha_x &= \frac{2(1 - x) + x(b_1 + b_{2q+1})}{2}, \\ H_x^\circ &= I - B_x^\circ/\alpha_x, & C_x^{[L]} &= \alpha_x^{-1} \sum_{r=0}^L (H_x^\circ)^r (D - I), \\ R_x^{[L]} &= (H_x^\circ)^{L+1} C_x, & E_x^{[L]} &= R_x^{[L]} - \frac{\operatorname{Tr} R_x^{[L]}}{2q+1} I.\end{aligned}\tag{RMT19}$$

The symbols $B_x^\circ, H_x^\circ, \alpha_x$ label operators and the scalar in this calculation; they leave every original relation matrix B_N , source metric M_x , and coordinate in (RMT1) in place. Multiplication of the finite geometric sum by $I - H_x^\circ$ gives $C_x - C_x^{[L]} = R_x^{[L]}$ exactly. Every factor is a real function of the same

M_0 -self-adjoint D , hence is self-adjoint in $M_x = M_0 B_x^\circ$ as well. Since the trace of $U_x - W_x$ vanishes, (RMT7) becomes

$$\mathcal{B}'(x) = \text{Tr}(C_x^{[L]}(U_x - W_x)) + \text{Tr}(E_x^{[L]}(U_x - W_x)).$$

Apply Hilbert–Schmidt Cauchy–Schwarz in the original M_x metric to the last term and integrate. Define the specified real numbers

$$\begin{aligned} j_L &= \int_0^1 \text{Tr}(C_x^{[L]}(U_x - W_x)) dx, \\ e_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) \text{Tr}((U_x - W_x)^2)} dx, \\ \vartheta &= \frac{b_{2q+1} - b_1}{b_{2q+1} + b_1}, \quad K_D = \sum_{j=1}^{2q+1} \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \end{aligned} \tag{RMT20}$$

These are $J_L, \epsilon_L, \vartheta, K_D$ of (ISM34)–(ISM37) with j_L, e_L used to avoid collision with J_q in (RMT13). The eigenvalues of H_x° have absolute value at most $\vartheta < 1$; those of C_x are the full $c_j(x)$. Thus $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$, because subtracting the trace mean subtracts a nonnegative square from the full sum of squared residual eigenvalues. The two positive operators U_x, W_x dominate their range projections. Those ranges intersect in dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$, by positivity of the trace product and the squared-length formula for the product of the range projections. Equation (RMT10) gives $\text{Tr}((U_x - W_x)^2) \leq 8q - 6$. The signed estimate and its computed residual are therefore

$$\begin{aligned} j_L - e_L &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_L + e_L, \\ 0 \leq e_L &\leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \end{aligned} \tag{RMT21}$$

Every integral is a continuous finite matrix expression. This proves convergence for the fixed actual two source forms; no uniformity in k is assumed or inferred.

Let $B_0 = \mathcal{B}(0)$ and retain H_0 from (RMT18). Intersecting the proved intervals, for every $L \geq 0$, gives the propagated endpoint bounds

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, B_0 - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right), B_0 + j_L - e_L \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ B_0 + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right), B_0 + j_L + e_L \right\}. \end{aligned} \tag{RMT22}$$

Indeed each lower entry has separately been proved no larger than the actual endpoint, and each upper entry no smaller. Taking their maximum and minimum therefore preserves both inequalities and yields a nonempty interval containing that endpoint. A numerically integrated center must retain its own integration error before use; (RMT20)–(RMT22) state the exact integrals. The scalar source case has $H_x^\circ = E_x^{[L]} = 0$ and $j_L = e_L = 0$, so it also satisfies (RMT22) exactly.

Part VI**The original phase graph, mixed residual,
relation rows and signed arithmetic return**

252 Original full-quartet critical values and every ordered tensor sum

Fix the original offcritical point and the original equal full order

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 0, \quad m \in \mathbb{Z}_{\geq 1}. \quad (\text{QCV.1})$$

The order is the full multiplicity in the arithmetic divisor; no simple-root replacement is made. Keep the ordered roots

$$(r_1, r_2, r_3, r_4) = (\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}), \quad h(s) = \prod_{j=1}^4 (s - r_j)^m, \quad \Phi_h(s) = \int_0^s h(t) dt. \quad (\text{QCV.2})$$

The last integral means the unique polynomial antiderivative with $\Phi_h(0) = 0$. All paths give the same polynomial. In particular, the constant in this phase is fixed, and is not removed below.

252.1 The exact polynomial, primitive and local leading coefficients

The translation $z = s - \frac{1}{2}$ gives the explicit identity

$$\begin{aligned} f(z) &= h(z + \tfrac{1}{2}) = [((z - \delta)^2 + \gamma^2)((z + \delta)^2 + \gamma^2)]^m \\ &= [z^4 + 2(\gamma^2 - \delta^2)z^2 + (\delta^2 + \gamma^2)^2]^m, \quad F(z) = \int_0^z f(x) dx, \quad A = F(\tfrac{1}{2}). \end{aligned} \quad (\text{QCV.3})$$

These equations preserve the original s -coordinate through the inverse map $s = z + \frac{1}{2}$. The polynomial f is even with real coefficients. It is strictly positive on the real axis: each quadratic factor in its first expression is positive because $\gamma > 0$. Thus F is odd with real coefficients, and $A > 0$. Direct differentiation and evaluation at zero prove

$$\Phi_h(s) = F(s - \tfrac{1}{2}) + A, \quad \Phi_h(1) = 2A, \quad \Phi_h(1 - s) = 2A - \Phi_h(s), \quad \Phi_h(\bar{s}) = \overline{\Phi_h(s)}. \quad (\text{QCV.4})$$

In particular the positive constant A remains in every critical value and becomes kA in an ordered k -fold tensor phase.

For a completely finite expression put

$$a_j = \sum_{\substack{r, t \geq 0 \\ r+t \leq m \\ r+2t=j}} \frac{m!}{r!t!(m-r-t)!} [2(\gamma^2 - \delta^2)]^r (\delta^2 + \gamma^2)^{2(m-r-t)} \quad (0 \leq j \leq 2m). \quad (\text{QCV.5})$$

The multinomial theorem, followed by termwise polynomial integration, gives

$$f(z) = \sum_{j=0}^{2m} a_j z^{2j}, \quad F(z) = \sum_{j=0}^{2m} \frac{a_j z^{2j+1}}{2j+1}, \quad A = \sum_{j=0}^{2m} \frac{a_j}{(2j+1)2^{2j+1}}. \quad (\text{QCV.6})$$

There is no convergence issue in these finite sums, including when $\gamma < \delta$ and some a_j have negative signs.

Define the real numbers U, V by the exact evaluation

$$U + iV = F(\delta + i\gamma) = \sum_{j=0}^{2m} \frac{a_j (\delta + i\gamma)^{2j+1}}{2j+1}. \quad (\text{QCV.7})$$

Oddness and conjugation now give, in the unchanged order (QCV.2),

$$(\Phi_h(r_1), \Phi_h(r_2), \Phi_h(r_3), \Phi_h(r_4)) = (A + U + iV, A + U - iV, A - U - iV, A - U + iV). \quad (\text{QCV.8})$$

The local coefficient at the first root, and all its partners, are

$$b_1 = \frac{h(s)}{(s - r_1)^m} \Big|_{s=r_1} = [8i\delta\gamma(\delta + i\gamma)]^m \neq 0, \quad b_2 = \overline{b_1}, \quad b_3 = (-1)^m b_1, \quad b_4 = (-1)^m \overline{b_1},$$

$$\Phi_h(r_j + y) = \Phi_h(r_j) + \frac{b_j}{m+1} y^{m+1} + y^{m+2} R_j(y), \quad R_j \in \mathbb{C}[y]. \quad (\text{QCV.9})$$

Indeed the three differences at r_1 are $2i\gamma, 2\delta + 2i\gamma, 2\delta$, whose product gives b_1 . Conjugation gives b_2 ; the substitution $s \mapsto 1 - s$ gives the sign $(-1)^m$ in b_3, b_4 . Expanding the polynomial h and integrating proves the last line. This is the local coefficient of the full polynomial (QCV.2), and not of the smaller polynomial $(s - r_j)^m$.

252.2 A positive sum of squares proves that the rectangle does not collapse

For $d > 0$ put $P_d(x) = (d^2 - x^2)^m$. The exact identities are

$$\boxed{\begin{aligned} 2U &= \sum_{j=0}^m \frac{(2\gamma)^{2j}}{(2j)!} \int_{-\delta}^{\delta} (P_{\delta}^{(j)}(x))^2 dx, \\ 2V &= \sum_{j=0}^m \frac{(2\delta)^{2j}}{(2j)!} \int_{-\gamma}^{\gamma} (P_{\gamma}^{(j)}(x))^2 dx. \end{aligned}} \quad U > 0, \quad V > 0. \quad (\text{QCV.10})$$

Here every derivative is the ordinary derivative of the displayed real polynomial. These are finite identities, with all factorials and interval lengths retained.

To prove the first identity, integrate f along the segment from $-\delta + i\gamma$ to $\delta + i\gamma$. Its primitive difference is $F(\delta + i\gamma) - F(-\delta + i\gamma) = 2U$. On this segment

$$\begin{aligned} f(x + i\gamma) &= [(x^2 - \delta^2)((x + 2i\gamma)^2 - \delta^2)]^m = P_{\delta}(x)P_{\delta}(x + 2i\gamma), \\ P_{\delta}(x + 2i\gamma) &= \sum_{r=0}^{2m} \frac{(2i\gamma)^r}{r!} P_{\delta}^{(r)}(x). \end{aligned} \quad (\text{QCV.11})$$

An odd derivative of the even polynomial P_{δ} is odd, so its product with P_{δ} has zero integral on the symmetric interval. For $0 \leq j \leq m$, integrate by parts j times:

$$\int_{-\delta}^{\delta} P_{\delta} P_{\delta}^{(2j)} dx = (-1)^j \int_{-\delta}^{\delta} (P_{\delta}^{(j)})^2 dx. \quad (\text{QCV.12})$$

At the r -th integration, every boundary term contains a derivative of P_{δ} of order at most $j - 1 < m$. It vanishes at both endpoints because P_{δ} has a zero of order m there. The factor $(2i\gamma)^{2j} = (-1)^j (2\gamma)^{2j}$ cancels the sign in (QCV.12), proving the first identity in (QCV.10). Its $j = 0$ integral is strictly positive because $P_{\delta}(0) > 0$.

Write $F_{d,e}$ for the polynomial in (QCV.3) with parameters (d, e) and the same m . Its defining polynomials satisfy

$$f_{\delta,\gamma}(iw) = f_{\gamma,\delta}(w), \quad F_{\delta,\gamma}(iw) = iF_{\gamma,\delta}(w). \quad (\text{QCV.13})$$

The first identity follows by substituting iw in (QCV.3); the second follows by differentiating and comparing the zero constant terms. At $w = \gamma - i\delta$, it gives $U_{\delta,\gamma} + iV_{\delta,\gamma} = iU_{\gamma,\delta} + iV_{\gamma,\delta}$. Thus $V_{\delta,\gamma} = U_{\gamma,\delta} > 0$, and the first identity with exchanged parameters proves the second one in (QCV.10). No restriction on γ/δ was used.

If desired each integral in (QCV.10) has the fully rational finite evaluation

$$\int_{-d}^d (P_d^{(j)}(x))^2 dx = 2d^{4m-2j+1} \sum_{r,t=\lceil j/2 \rceil}^m \frac{(-1)^{r+t} \binom{m}{r} \binom{m}{t} (2r)!(2t)!}{(2r-j)!(2t-j)![2(r+t-j)+1]}. \quad (\text{QCV.14})$$

Expand $P_d(x) = \sum_{r=0}^m (-1)^r \binom{m}{r} d^{2(m-r)} x^{2r}$, differentiate j times, multiply, and integrate the even monomials. The exponent of d is $4m - 2j + 1$ in every term. Positivity is proved by the integral in (QCV.10); no sign assertion about the individual summands of (QCV.14) is needed.

252.3 An exact real-part bound, retaining the original constant

The following bound on the same critical values is elementary:

$$0 < U < \delta [\delta^2(\delta^2 + 4\gamma^2)]^m, \quad \gamma \geq \delta \implies A > \frac{1}{2}(\delta^2 + \gamma^2)^{2m}, \quad \frac{U}{A} < 2\delta \left(\frac{\delta^2(\delta^2 + 4\gamma^2)}{(\delta^2 + \gamma^2)^2} \right)^m. \quad (\text{QCV.15})$$

For its proof $F(i\gamma)$ is purely imaginary, since F is odd with real coefficients. Therefore

$$U = \operatorname{Re} \int_0^\delta f(x + i\gamma) dx. \quad (\text{QCV.16})$$

Put $t = \delta^2 - x^2$, so $0 \leq t \leq \delta^2$. Direct multiplication gives the exact squared modulus of the expression inside the m -th power:

$$\begin{aligned} B(x) &= (x^2 - \delta^2)[x^2 - \delta^2 - 4\gamma^2 + 4i\gamma x], \\ |B(x)|^2 &= t^2[t^2 - 8\gamma^2 t + 16\gamma^4 + 16\gamma^2 \delta^2], \\ \frac{d}{dt}|B|^2 &= 4t[t^2 - 6\gamma^2 t + 8\gamma^4 + 8\gamma^2 \delta^2] > 0 \quad (0 < t \leq \delta^2). \end{aligned} \quad (\text{QCV.17})$$

The strict sign follows from $8\gamma^2 \delta^2 - 6\gamma^2 t \geq 2\gamma^2 \delta^2 > 0$. Thus $|B(x)| \leq \delta^2(\delta^2 + 4\gamma^2)$, with equality only at $x = 0$. Integrating its m -th power on a positive-length interval proves the strict upper bound for U in (QCV.15), using the already proved $U > 0$. When $\gamma \geq \delta$, the second expression for $f(x)$ in (QCV.3) is at least $(\delta^2 + \gamma^2)^{2m}$, strictly larger for $x > 0$. Integrating from 0 to $1/2$ proves the bound for A , and division proves the ratio bound.

For the original range $0 < \delta < 1/2$ and $\gamma \geq 2$, put $r = \delta^2/\gamma^2 < 1/16$. Since

$$q(r) = \frac{r(r+4)}{(1+r)^2}, \quad q'(r) = \frac{4-2r}{(1+r)^3} > 0 \quad (0 \leq r \leq 1/16), \quad q(1/16) = \frac{65}{289}, \quad (\text{QCV.18})$$

equation (QCV.15), including its 2δ factor, implies

$$0 < U < A \left(\frac{65}{289} \right)^m < A, \quad A \left[1 - \left(\frac{65}{289} \right)^m \right] < \operatorname{Re} \Phi_h(r_j) < A \left[1 + \left(\frac{65}{289} \right)^m \right] \quad (1 \leq j \leq 4). \quad (\text{QCV.19})$$

This estimate has not assumed that an arithmetic zero has height at least two. That fact is proved from the original theta kernel in (LHT.1)–(LHT.18) below. Those equations give $\operatorname{Re} g(s) > 97/2052$ for $0 \leq \operatorname{Re} s \leq 1, |\operatorname{Im} s| \leq 2$. Consequently every actual zero (QCV.1) of the original $g = 2\xi$ has $\gamma > 2$, and (QCV.19) applies to every actual offcritical quartet with its full order. This use of the low-height calculation does not assert that offcritical quartets exist.

252.4 Every ordered tensor critical point, collision and full local rank

For $k \geq 1$, keep the ordered tensor phase on the original coordinates

$$\Phi_{h,k}(s_1, \dots, s_k) = \sum_{i=1}^k \Phi_h(s_i), \quad E_{h,k} = \mathbb{C}[s_1, \dots, s_k]/(h(s_1), \dots, h(s_k)). \quad (\text{QCV.20})$$

Its critical points are precisely the ordered tuples $\mathbf{r} = (r_{j_1}, \dots, r_{j_k})$, $j_i \in \{1, 2, 3, 4\}$, since its i -th derivative is $h(s_i)$. Write n_j for the number of occurrences of r_j ; thus $n_j \geq 0$ and $\sum_j n_j = k$. Evaluation gives exactly

$$\Phi_{h,k}(\mathbf{r}) = kA + (n_1 + n_2 - n_3 - n_4)U + i(n_1 - n_2 - n_3 + n_4)V. \quad (\text{QCV.21})$$

With $a = n_1 + n_2$ and $b = n_1 + n_4$, these values are

$$c_{k,a,b} = kA + (2a - k)U + i(2b - k)V, \quad 0 \leq a, b \leq k. \quad (\text{QCV.22})$$

Every pair (a, b) occurs: choose a set of a positions for a positive U -sign and, independently, a set of b positions for a positive V -sign. At each position the pair of signs picks exactly one of r_1, r_2, r_3, r_4 in (QCV.8). This also proves that the number of ordered tuples giving $c_{k,a,b}$ is exactly

$$N_{k,a,b} = \binom{k}{a} \binom{k}{b} = \sum_{x=\max(0, a+b-k)}^{\min(a,b)} \frac{k!}{x!(a-x)!(k-a-b+x)!(b-x)!}. \quad (\text{QCV.23})$$

The sum follows by setting $(n_1, n_2, n_3, n_4) = (x, a-x, k-a-b+x, b-x)$ and counting permutations. Since $U, V > 0$, two values in (QCV.22) are equal if and only if both their indices a, b agree. Thus there are exactly $(k+1)^2$ distinct critical values. All collisions among different ordered tuples are the ones counted in (QCV.23).

At an individual tuple set $y_i = s_i - r_{j_i}$. Polynomial Chinese remaindering gives the exact decomposition

$$E_{h,k} \simeq \bigoplus_{(j_1, \dots, j_k) \in \{1, 2, 3, 4\}^k} \mathbb{C}[y_1, \dots, y_k] / (y_1^m, \dots, y_k^m). \quad (\text{QCV.24})$$

For completeness, the ideals $((s - r_j)^m)$ are pairwise comaximal: distinct r_j give coprime polynomials, and the Euclidean algorithm supplies polynomials expressing 1 as a sum of multiples of each pair. The Chinese-remainder map is evaluation of every full Taylor jet $s = r_j + y$ modulo y^m ; its kernel is the product ideal (h) , and the same Bezout identities construct the inverse idempotents. Tensoring these finite-dimensional isomorphisms in the unchanged order gives (QCV.24). The monomials $\prod_i y_i^{e_i}$, $0 \leq e_i < m$, form a basis of each summand, so its dimension is m^k .

The summand belonging to a coincident critical value $c_{k,a,b}$ therefore has dimension

$$d_{k,a,b} = m^k \binom{k}{a} \binom{k}{b}, \quad \sum_{a,b=0}^k d_{k,a,b} = (4m)^k. \quad (\text{QCV.25})$$

Grouping by equal value is a direct sum of the original full local algebras, and does not identify or quotient their jets. In each individual summand the original arithmetic tensor sum is still

$$A_{h,k} = \left(\sum_{i=1}^k r_{j_i} \right) I + \sum_{i=1}^k M_{y_i}, \quad \exp(tA_{h,k}) = e^{t \sum_i r_{j_i}} \prod_{i=1}^k \sum_{e=0}^{m-1} \frac{t^e M_{y_i}^e}{e!}. \quad (\text{QCV.26})$$

All nilpotent powers remain, and the scalar $\sum_i r_{j_i}$ in this arithmetic action is distinct data from the phase value $\sum_i \Phi_h(r_{j_i})$. Both are calculated on the same unchanged tuple; no equality between them is asserted.

For an actual zero of g , (QCV.19) and (QCV.22) finally give

$$\boxed{\begin{aligned} \operatorname{Re} c_{k,a,b} &\geq k(A - U) > kA \left[1 - \left(\frac{65}{289} \right)^m \right] > 0, \\ |c_{k,a,b}| &\geq \operatorname{Re} c_{k,a,b} > \frac{k}{2} (\delta^2 + \gamma^2)^{2m} \left[1 - \left(\frac{65}{289} \right)^m \right]. \end{aligned}} \quad (\text{QCV.27})$$

Every inequality uses the original constant A , with the explicit lower bound for A inserted only in the second line. In particular no original ordered tensor critical value is zero. This proves the non-vanishing of the phase constants required at those precise vertices of the programme; it does not identify the arithmetic tensor action with a Frobenius action, and it proves neither the existence nor the absence of an offcritical zero by itself.

253 The calculated phase operator on the original full jets and theta source

Use precisely $h, \Phi_h, r_j, m, A, U, V$ from (QCV.1)–(QCV.27). Here h is an actual full arithmetic quartet divisor of $g = 2\xi$. Put $E_h = \mathbb{C}[s]/(h)$, $A_h = M_s$, and

$$E_{h,k} = E_h^{\otimes k}, \quad C_h = M_{[\Phi_h(s)]}, \quad C_{h,k} = \sum_{i=1}^k I^{\otimes(i-1)} \otimes C_h \otimes I^{\otimes(k-i)}. \quad (\text{QPS.1})$$

Thus $C_h = \Phi_h(A_h)$ is an exact polynomial of the original arithmetic operator. No new comparison is postulated.

253.1 The complete jet map and its inverse

The original monomial basis of E_h is $1, s, \dots, s^{4m-1}$. The full jet map has entries

$$\mathcal{J}_h : E_h \longrightarrow \bigoplus_{j=1}^4 \mathbb{C}[y]/(y^m), \quad (\mathcal{J}_h p)_j = p(r_j + y) \bmod y^m, \quad (\mathcal{J}_h)_{(j,a),b} = \binom{b}{a} r_j^{b-a}, \quad (\text{QPS.2})$$

where the entry is zero if $a > b$, and $0 \leq a < m, 0 \leq b < 4m$. This includes every Taylor coefficient with its binomial factor.

Here is an explicit inverse, including all full-order idempotents. Write

$$H_j(s) = h(s)/(s - r_j)^m, \quad v_{j,\ell} = H_j^{(\ell)}(r_j)/\ell! \quad (0 \leq \ell < m),$$

$$w_{j,0} = v_{j,0}^{-1}, \quad w_{j,\ell} = -v_{j,0}^{-1} \sum_{a=1}^{\ell} v_{j,a} w_{j,\ell-a}. \quad (\text{QPS.3})$$

The nonzero value $v_{j,0} = b_j$ is calculated in (QCV.9). Define $W_j(y) = \sum_{\ell=0}^{m-1} w_{j,\ell} y^\ell$, and let r_h denote the unique remainder of degree less than $4m$ upon division by the original monic h . Then

$$e_j = r_h(H_j(s)W_j(s - r_j)), \quad \mathcal{J}_h^{-1}(p_1, \dots, p_4) = r_h \left(\sum_{j=1}^4 e_j(s) p_j(s - r_j) \right). \quad (\text{QPS.4})$$

Each p_j in this formula is its unique representative of degree less than m . Using any other lift gives the same class: the difference is a multiple of y^m , and multiplication by e_j has zero full jets both at r_j and at all other roots. The recurrence (QPS.3) says $H_j(r_j + y)W_j(y) \equiv 1 \bmod y^m$. At another root r_ℓ the polynomial H_j is divisible by $(s - r_\ell)^m$. Consequently $\mathcal{J}_h(e_j)$ has value 1 at the j -th full block and 0 at the others. This proves that (QPS.4) is a right inverse. If all full jets of a polynomial p vanish, each $(s - r_j)^m$ divides p ; distinct factors are coprime, so their product h divides p . This proves injectivity and hence the inverse assertion. In particular

$$e_j e_\ell = \mathbf{1}_{j=\ell} e_j, \quad \sum_{j=1}^4 e_j = 1 \quad \text{in } E_h. \quad (\text{QPS.5})$$

253.2 The phase operator is invertible on every original tensor block

Equation (QCV.9), reduced modulo y^m , gives the exact operator identities

$$\mathcal{J}_h C_h \mathcal{J}_h^{-1} = \bigoplus_{j=1}^4 \Phi_h(r_j) I_m, \quad C_h = \sum_{j=1}^4 \Phi_h(r_j) M_{e_j},$$

$$\mathcal{J}_h A_h \mathcal{J}_h^{-1} = \bigoplus_{j=1}^4 (r_j I_m + M_y). \quad (\text{QPS.6})$$

The disappearance of higher powers in $\Phi_h(r_j + y) \bmod y^m$ changes neither the local vector space nor its vectors: the phase acts as its nonzero scalar on all m jets. In particular the full nilpotent action M_y in the second line remains in the same object.

Put $\mathcal{J}_{h,k} = \mathcal{J}_h^{\otimes k}$ in the unchanged tensor order, and $e_j = \bigotimes_i e_{j_i}$. For $\mathbf{j} = (j_1, \dots, j_k)$, set $c_j = \sum_i \Phi_h(r_{j_i})$, calculated in (QCV.21)–(QCV.23). By (QCV.27) every $c_j \neq 0$. It follows, with an explicit inverse, that

$$C_{h,k} = \sum_{\mathbf{j}} c_j M_{e_j}, \quad C_{h,k}^{-1} = M_{p_{h,k}}, \quad p_{h,k} = \sum_{\mathbf{j}} c_j^{-1} e_j, \quad (\text{QPS.7})$$

$$C_{h,k} C_{h,k}^{-1} = C_{h,k}^{-1} C_{h,k} = I_{E_{h,k}}.$$

All four-to-the- k full idempotents are retained, including those sharing a critical value. The last equalities follow by multiplying the orthogonal idempotents in (QPS.5). Multiplication by a unit is a module automorphism here; it is not being called a unital ring automorphism.

For clarity the arithmetic scalar and the phase scalar on the same tuple are related by a completely specified map. At the grid point (a, b) of (QCV.22), the arithmetic tensor centre and its phase centre are respectively

$$\lambda_{k,a,b} = \frac{k}{2} + (2a - k)\delta + i(2b - k)\gamma, \quad c_{k,a,b} = kA + (2a - k)U + i(2b - k)V. \quad (\text{QPS.8})$$

The real affine map

$$x + iy \mapsto kA + \frac{U}{\delta}(x - k/2) + i\frac{V}{\gamma}y \quad (\text{QPS.9})$$

maps the first finite grid bijectively to the second. Its inverse is $X + iY \mapsto k/2 + (\delta/U)(X - kA) + i(\gamma/V)Y$. Both inverses exist by (QCV.10). Equation (QPS.9) is a map of the scalar grids. The full operator map retaining nilpotents is the polynomial construction (QPS.1) and its exact full-jet action (QPS.6), not an extension of (QPS.9) asserted without proof.

253.3 The original full Taylor unit and the literal theta boundary

Retain the original source notation and all its scalars:

$$\begin{aligned} \phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad D = -x\partial_x, \quad \Theta\phi = 2 \sum_{n \geq 1} \phi(nx), \\ v_h &= g/h, \quad v_h = j_h(v_h) \in E_h^\times, \quad \varepsilon_h = v_h^{-1}, \quad R_h(w) = r_h(\varepsilon_h w), \\ s_h(w) &= S_h\Theta(R_h(w)(D)\phi_*), \quad \mathcal{M}s_h(w) = v_h R_h(w), \quad j_h \mathcal{M}s_h(w) = w. \end{aligned} \quad (\text{QPS.10})$$

These are precisely the original maps of (HCA.5)–(HCA.6); S_h is the programme's Euler inverse on the zero-full-jet range. At r_j every Taylor coefficient

$$(v_h)_{j,\ell} = \frac{1}{\ell!} v_h^{(\ell)}(r_j) \quad (0 \leq \ell < m) \quad (\text{QPS.11})$$

remains in the multiplication operator v_h . Its constant coefficient is nonzero because h uses the full order of g . The reciprocal coefficients are given by the same exact finite convolution recurrence as (QPS.3), with $v_{j,\ell}$ replaced by $(v_h)_{j,\ell}$. Thus no scalar-only replacement of this unit has entered (QPS.10).

The numerator in the following expression has zero class in E_h , so the displayed quotient is an actual polynomial:

$$K_{\Phi,h}(w; s) = \frac{\Phi_h(s)R_h(w; s) - R_h(C_h w; s)}{h(s)} \in \mathbb{C}[s]. \quad (\text{QPS.12})$$

Indeed modulo h its numerator is $[\Phi_h]\varepsilon_h w - \varepsilon_h[\Phi_h]w = 0$. Mellin transformation of the original theta source gives the exact function identity

$$\Phi_h(D)s_h(w) - s_h(C_h w) = \Theta(K_{\Phi,h}(w; D)\phi_*). \quad (\text{QPS.13})$$

Both sides have Mellin transform $g(s)K_{\Phi,h}(w; s)$: use (QPS.10) on the left and $\mathcal{M}\Theta(P(D)\phi_*) = gP$ on the right. The original Mellin injectivity on the stated source spaces proves the equality. All polynomials in D preserve those spaces.

For a pure tensor $w = w_1 \otimes \cdots \otimes w_k$, let $\mathcal{S}_k = s_h^{\otimes k}$, and form the degree- $(k-1)$ cochain

$$\mathcal{H}_k(w) = \sum_{i=1}^k (-1)^{i-1} s_h(w_1) \otimes \cdots \otimes K_{\Phi,h}(w_i; D)\phi_* \otimes \cdots \otimes s_h(w_k). \quad (\text{QPS.14})$$

Each $s_h(w_j)$ lies in degree one of the original complex $[V \xrightarrow{\Theta} \mathcal{B}]$; the displayed middle factor lies in V , in degree zero. Consequently the tensor differential on its i -th factor has sign $(-1)^{i-1}$, and all other differentials in that summand vanish. Applying (QPS.13) gives

$$\left(\sum_{i=1}^k \Phi_h(D_i) \right) \mathcal{S}_k(w) - \mathcal{S}_k(C_{h,k}w) = d\mathcal{H}_k(w). \quad (\text{QPS.15})$$

Linearity proves this equality for every tensor, not only pure ones. This supplies the literal homotopy connecting the calculated phase operator to the original theta source, with its full Taylor unit and the tensor signs. Proper-source kernels and independently supported coefficients are not quotiented out in this calculation: the identity takes place in the original full complex.

253.4 The original norm retains its complete comparison factor

No orthogonality of the full jet idempotents is assumed. The original marked theta Gram matrix in the monomial basis is exactly

$$(G_h)_{a,b} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{v_h(\frac{1}{2} + it)R_h(s^a; \frac{1}{2} + it)} v_h(\frac{1}{2} + it)R_h(s^b; \frac{1}{2} + it) dt, \quad (\text{QPS.16})$$

$$0 \leq a, b < 4m, \quad G_{h,k} = G_h^{\otimes k}.$$

This is the original Mellin-line norm and its ordered Hilbert tensor norm. The integral converges by the original source estimates. It is positive definite: a vector of zero norm has the entire Mellin representative $v_h R_h(w)$ zero on the line, hence identically zero, hence $w = j_h(v_h R_h(w)) = 0$ by (QPS.10). The full factors v_h and R_h remain in every entry. The norm of the inverse in this original metric is exactly

$$\|C_{h,k}^{-1}\|_{G_{h,k}} = \left\| G_{h,k}^{1/2} \mathcal{J}_{h,k}^{-1} \left(\bigoplus_j c_j^{-1} I_{m^k} \right) \mathcal{J}_{h,k} G_{h,k}^{-1/2} \right\|_2. \quad (\text{QPS.17})$$

To verify it, the map from a vector v with norm $(v^* G_{h,k} v)^{1/2}$ to $G_{h,k}^{1/2} v$ preserves that norm exactly. Conjugating the operator by this map computes its norm in Euclidean coordinates. Substitution of (QPS.7) and (QPS.2) then gives the displayed equality. These are explicit coordinate maps for the original norm, not a change to its definition.

Submultiplicativity and (QCV.27) give the useful bound with its entire metric factor:

$$\kappa_{h,k} = \|G_{h,k}^{1/2} \mathcal{J}_{h,k}^{-1}\|_2 \|\mathcal{J}_{h,k} G_{h,k}^{-1/2}\|_2, \quad (\text{QPS.18})$$

$$\|C_{h,k}^{-1}\|_{G_{h,k}} < \frac{\kappa_{h,k}}{kA[1 - (65/289)^m]}.$$

The factor itself satisfies the exact tensor formula

$$\kappa_{h,k} = \kappa_{h,1}^k. \quad (\text{QPS.19})$$

Indeed $G_{h,k}^{1/2} = (G_h^{1/2})^{\otimes k}$ and likewise for the inverse square root: the displayed positive definite tensor matrix squares to $G_h^{\otimes k}$, so uniqueness of the positive square root proves the claim. Also $\mathcal{J}_{h,k} =$

$\mathcal{J}_h^{\otimes k}$. Singular-value decompositions show $\|X \otimes Y\|_2 = \|X\|_2 \|Y\|_2$: tensor the two unitary changes of basis, after which the singular values are the pairwise products of the original ones. Applying this fact to the two factors in (QPS.18) proves (QPS.19). This proves the precise metric consequence of the critical-value bound. It does not remove $\kappa_{h,k}$, assert that it is uniform in k , or turn the phase bound into an arithmetic purity bound. The inverse, the full jet comparison and the original theta boundary connecting them have all been constructed above.

253.5 Exact phase cross-pairing on that same original source

Since F is odd with real coefficients, $F(it)$ is purely imaginary for real t . The original constant in (QCV.4) therefore gives the exact line identity

$$\operatorname{Re} \Phi_h(\tfrac{1}{2} + it) = A \quad (t \in \mathbb{R}). \quad (\text{QPS.20})$$

Let $\mathcal{T}_k = \sum_i \Phi_h(D_i)$, with its domain on the original k -variable Mellin source. The representatives $\mathcal{S}_k w$ belong to this domain: their Mellin transforms are the finite linear combinations of products $v_h(s_i) R_h(w_i; s_i)$ from (QPS.10), and the original decay estimates remain integrable after multiplication by any fixed polynomial $\Phi_h(s_i)$. Mellin transformation of each D_i is multiplication by s_i . Using the original product measure $dt_1 \cdots dt_k / (2\pi)^k$ and (QPS.20) thus proves

$$\operatorname{Re} \langle \mathcal{S}_k w, \mathcal{T}_k \mathcal{S}_k w \rangle = kA \|\mathcal{S}_k w\|^2. \quad (\text{QPS.21})$$

The pairing is conjugate linear in its first argument. On a full phase block with $C_{h,k} w = c_{k,a,b} w$, subtract the scalar term in (QPS.15) to obtain

$$\begin{aligned} \operatorname{Re} \langle \mathcal{S}_k w, d\mathcal{H}_k(w) \rangle &= -(2a - k)U \|\mathcal{S}_k w\|^2, \\ \|d\mathcal{H}_k(w)\| &\geq |2a - k|U \|\mathcal{S}_k w\|. \end{aligned} \quad (\text{QPS.22})$$

The first equality uses $\operatorname{Re} c_{k,a,b} = kA + (2a - k)U$. For nonzero w , the second follows by bounding the absolute value of the real pairing by $\|\mathcal{S}_k w\| \|d\mathcal{H}_k(w)\|$ and dividing by the nonzero source norm; for $w = 0$ it holds directly. In particular, on the original all- ρ block $a = b = k$, the lower bound is $kU \|\mathcal{S}_k w\|$, with the exact strictly positive U in (QCV.10). This is the same full-unit theta boundary as (QPS.15). It is not a scalar replacement for the source norm or its kernel.

There is also an exact norm equality with every remaining term. Define the real odd polynomial $q_\Phi(t) = \operatorname{Im} F(it)$; it has degree $4m + 1$ and leading coefficient $1/(4m + 1)$. Let $\Psi_w(t_1, \dots, t_k) = \mathcal{M}^{\otimes k}(\mathcal{S}_k w)(\tfrac{1}{2} + it_1, \dots, \tfrac{1}{2} + it_k)$, including all factors v_h and R_h in (QPS.10). On the same phase block, Mellin transformation of (QPS.15) gives

$$\begin{aligned} \|d\mathcal{H}_k(w)\|^2 &= (2a - k)^2 U^2 \|\mathcal{S}_k w\|^2 \\ &+ \frac{1}{(2\pi)^k} \int_{\mathbb{R}^k} \left[\sum_{i=1}^k q_\Phi(t_i) - (2b - k)V \right]^2 |\Psi_w(t_1, \dots, t_k)|^2 dt_1 \cdots dt_k. \end{aligned} \quad (\text{QPS.23})$$

Indeed its multiplier is exactly $-(2a - k)U + i[\sum_i q_\Phi(t_i) - (2b - k)V]$, whose squared modulus is the sum displayed. For nonzero w , the integral is strictly positive. Section injectivity gives nonzero norm and hence a point where the continuous Ψ_w is nonzero. It stays nonzero on an open neighborhood. The nonconstant polynomial $\sum_i q_\Phi(t_i) - (2b - k)V$ cannot vanish throughout that neighborhood: fix all but one coordinate in an open box and use the nonzero leading coefficient in the remaining coordinate. It is therefore nonzero on a smaller open set there, where the nonnegative integrand is positive. This proves strict positivity of its integral, including when $a = k/2$.

For the exact original terminal class at r_j , put

$$\begin{aligned} z_j &= e_j(s - r_j)^{m-1} \in E_h, \quad f_j = s_h(z_j) \neq 0, \quad c_j = \Phi_h(r_j), \\ g_j &= (\Phi_h(D) - c_j)f_j, \quad \mu_j = \frac{\langle f_j, g_j \rangle}{\|f_j\|^2}, \quad \nu_j = \frac{\|g_j\|^2}{\|f_j\|^2}. \end{aligned} \quad (\text{QPS.24})$$

The jet of z_j is y^{m-1} at its original full block and zero elsewhere, so nonvanishing and section injectivity prove $f_j \neq 0$. Write the signs in (QCV.8) as $c_j = A + \epsilon_j U + i\eta_j V$, where $(\epsilon_j, \eta_j) = (+, +), (+, -), (-, -), (-, +)$ in the original root order. Equation (QPS.21) for $k = 1$ gives

$$\operatorname{Re} \mu_j = -\epsilon_j U, \quad \nu_j \geq |\mu_j|^2 \quad (\text{QPS.25})$$

by the same inner-product convention and Cauchy–Schwarz. On the original tensor representative $f_j^{\otimes k}$ the literal defect is

$$\mathcal{B}_{j,k} = \sum_{i=1}^k f_j^{\otimes(i-1)} \otimes g_j \otimes f_j^{\otimes(k-i)}. \quad (\text{QPS.26})$$

Its k diagonal pairings are $\|g_j\|^2 \|f_j\|^{2k-2}$. Each of the $k(k-1)$ ordered off-diagonal pairings is $|\langle f_j, g_j \rangle|^2 \|f_j\|^{2k-4}$, since the two distinguished factors contribute conjugate inner products. Consequently

$$\frac{\|\mathcal{B}_{j,k}\|^2}{\|f_j^{\otimes k}\|^2} = k\nu_j + k(k-1)|\mu_j|^2 = k^2|\mu_j|^2 + k(\nu_j - |\mu_j|^2) \geq k^2 U^2. \quad (\text{QPS.27})$$

Any original nonzero scalar coefficient multiplying the class $z_j^{\otimes k}$ multiplies both squared norms by its squared absolute value; equations (QPS.26)–(QPS.27) remain exact with that coefficient retained on both sides. No coefficient has been set to one by altering the original source.

Finally the full phase-boundary map $w \mapsto d\mathcal{H}_k(w)$ is injective even on mixtures of different phase blocks. If it were zero for a nonzero w , put $P = R_h^{\otimes k}(w) \neq 0$, a polynomial of degree less than $d = 4m$ in each variable. Dividing its Mellin equation by the nonzero entire product $\prod_i v_h(s_i)$ gives the polynomial identity

$$\left(\sum_{i=1}^k \Phi_h(s_i) \right) P(s_1, \dots, s_k) = R_h^{\otimes k}(C_{h,k}w). \quad (\text{QPS.28})$$

If b is the degree of P in s_1 , its left side has degree $b + d + 1$ in s_1 : the nonzero leading coefficient comes from $\Phi_h(s_1)P$, and the other summands have degree at most b . The right side has degree less than d in s_1 , a contradiction. This proves the injectivity using the same full representatives and their actual theta boundary.

253.6 The full quartet connection and its exact lattice obstruction

Put $d = 4m$, $\mathcal{A} = \mathbb{C}[u]$, and retain the original polynomial differential $\mathcal{L}_u = u\partial_s + h(s)$. Its cokernel over $\mathcal{A}$ is free with ordered basis $e = (1, s, \dots, s^{d-1})$. To see this directly, applying $\mathcal{L}_u$ to s^n gives s^{n+d} plus terms of strictly lower s -degree. Successive reduction therefore produces a representative of degree less than d . If a nonzero $q(s) \in \mathcal{A}[s]$ has degree n , then $\mathcal{L}_u q$ has degree $n + d$, with the same leading coefficient as q ; hence it cannot be a nonzero representative of degree less than d . This proves existence and uniqueness of the reduction. The ordered k -variable reductions commute, giving the free original lattice

$$L_{h,k} = \mathcal{A} \otimes_{\mathbb{C}} E_{h,k}, \quad M_{h,k} = L_{h,k}[u^{-1}]. \quad (\text{QPS.29})$$

This is the same marked lattice and its localization used in the programme’s localization–Tor calculation.

For each original basis column s^b , perform the exact monic polynomial division

$$\Phi_h(s)s^b = h(s)q_b(s) + r_b(s), \quad \deg r_b < d, \quad C_{a,b} = [s^a]r_b, \quad B_{a,b} = [s^a]q'_b, \quad 0 \leq a, b < d. \quad (\text{QPS.30})$$

Here $\deg q_b \leq b + 1 \leq d$, so $\deg q'_b < d$. The matrix C is exactly C_h from (QPS.1), with the original $\Phi_h(0) = 0$, and B is a specified constant matrix. On the localized cokernel the original connection $\partial_u - \Phi_h(s)/u^2$ acts in this row basis as

$$\nabla e = e \left(-\frac{C}{u^2} + \frac{B}{u} \right). \quad (\text{QPS.31})$$

Indeed $[\mathcal{L}_u q_b] = 0$ gives $[h q_b] = -u[q'_b]$. Substituting (QPS.30) in $[-\Phi_h s^b/u^2]$ proves (QPS.31) with its displayed signs. The connection descends after localization because

$$[\partial_u - \Phi_h/u^2, \mathcal{L}_u] = u^{-1} \mathcal{L}_u. \quad (\text{QPS.32})$$

This follows by applying both sides to an arbitrary polynomial: $[\partial_u, u\partial_s] = \partial_s$ and $[-\Phi_h/u^2, u\partial_s] = h/u$. For the ordered tensor lattice and row basis e_k , the same calculation adds the k factor connections:

$$\nabla e_k = e_k \left(-\frac{C_{h,k}}{u^2} + \frac{B_{h,k}}{u} \right), \quad B_{h,k} = \sum_{i=1}^k I^{\otimes(i-1)} \otimes B \otimes I^{\otimes(k-i)}. \quad (\text{QPS.33})$$

No commutation of $C_{h,k}$ with $B_{h,k}$ is used.

Let $\mathcal{D}_A$ be the Weyl algebra generated by u, ∂ with $\partial u - u\partial = 1$, acting on $M_{h,k}$ through the displayed connection. For every integer $r \geq 0$, write $f_r = u^{-r} e_k$, a row of all original basis vectors in that layer. The exact recurrence is

$$\begin{aligned} u\nabla f_r &= -f_{r+1} C_{h,k} + f_r B_{h,k} - r f_r, \\ f_{r+1} &= -(u\nabla f_r - f_r B_{h,k} + r f_r) C_{h,k}^{-1}. \end{aligned} \quad (\text{QPS.34})$$

The inverse exists by (QPS.7), proved from the actual theta exclusion and the original critical-value bound. If every entry of f_r belongs to $\mathcal{D}_A L_{h,k}$, the same holds for its derivative, its product by u , and every constant linear combination of those entries. Thus (QPS.34) puts every entry of f_{r+1} in that module. The initial row $f_0 = e_k$ lies in $L_{h,k}$; induction proves

$$\mathcal{D}_A L_{h,k} = M_{h,k}. \quad (\text{QPS.35})$$

This is an induction on entire ordered layers. It makes no assumption that an arbitrary intermediate submodule is stable under multiplication by $B_{h,k}$ or $C_{h,k}$.

Finally set $Q_{h,k} = M_{h,k}/L_{h,k}$, as an $\mathcal{A}$ -module, and define $\Omega : L_{h,k} \rightarrow Q_{h,k}$ by $\Omega(v) = [\nabla v]$. It is $\mathcal{A}$ -linear: for $a \in \mathcal{A}$, the difference $\nabla(av) - a\nabla v = a'v$ is in $L_{h,k}$. In column coefficients a vector $v = v_0 + uv_1 + u^2 v_2(u)$ has

$$\Omega(v) = \left[-\frac{C_{h,k} v_0}{u^2} + \frac{B_{h,k} v_0 - C_{h,k} v_1}{u} \right]. \quad (\text{QPS.36})$$

Terms from the coefficient derivative and from $u^2 v_2(u)$ are in $L_{h,k}$. Every element of $Q_{h,k}$ annihilated by u^2 has a unique representative $x/u^2 + y/u$, with $x, y \in E_{h,k}$. Solving the two equations in (QPS.36) in the unchanged, possibly noncommuting matrix order gives

$$\begin{aligned} v_0 &= -C_{h,k}^{-1} x, & v_1 &= -C_{h,k}^{-1} B_{h,k} C_{h,k}^{-1} x - C_{h,k}^{-1} y, \\ \ker \Omega &= u^2 L_{h,k}, & \bar{\Omega} : L_{h,k}/u^2 L_{h,k} &\xrightarrow{\sim} Q_{h,k}[u^2]. \end{aligned} \quad (\text{QPS.37})$$

The displayed v_0, v_1 provide the inverse, and substitution proves it. If $\Omega(v) = 0$, first $C_{h,k} v_0 = 0$ and then $C_{h,k} v_1 = 0$; invertibility proves the stated kernel. The two modules are therefore related by this exact isomorphism. The quotient $Q_{h,k}$ is not being declared a connection module: the original lattice is not preserved by ∇ . Equations (QPS.34)–(QPS.37) extend the scalar-primary localization calculation to the full original quartet and every ordered tensor power through the inverse already constructed on the original full jets.

254 The full mixed phase boundary on the original arithmetic source

Retain the actual quartet packet, its full order and its original primitive from QCV and QPS: Throughout, $m, k \in \mathbb{Z}_{>0}$.

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ h(s) = \prod_{r \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-r)^m, \quad d = 4m, \quad \Phi' = h, \quad \Phi(0) = 0. \quad (\text{MBG.1})$$

The condition that this is an actual full divisor of $g = 2\xi$ is the counterfactual packet under investigation. Every map below is calculated on that packet; none of its zeros is asserted to have been found. Write $\Phi(s) = A + F(s - \frac{1}{2})$ as in QCV, with the retained real constant $A > 0$, and let

$$q_\Phi(t) = \text{Im } F(it), \quad \Phi(\frac{1}{2} + it) = A + iq_\Phi(t), \quad q'_\Phi(t) = [((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2)]^m > 0. \quad (\text{MBG.2})$$

Indeed $dF(it)/dt = ih(\frac{1}{2} + it)$; substituting it into the quartet polynomial gives the displayed real expression. Since F is odd and real, q_Φ is real and odd, of degree $d + 1$ with leading coefficient $1/(d + 1)$. This proves all claims in (MBG.2) without a change of the primitive constant.

254.1 One finite matrix for the whole boundary, including mixed pairings

Use $E_h = \mathbb{C}[s]/(h)$ in its original ordered monomial basis, $C = \Phi(A_h)$, $v_h = g/h$, and the entire original Taylor unit $v_h = j_h(v_h)$. For a coefficient vector w , set

$$R_h(w) = \text{rem}_h(v_h^{-1}w), \quad (Sw)(t) = v_h(\frac{1}{2} + it)R_h(w; \frac{1}{2} + it). \quad (\text{MBG.3})$$

Thus S is the Mellin transform of the original theta section s_h , with no altered coefficient, unit, or measure. Its Hilbert target is $\mathcal{H} = L^2(\mathbb{R}, dt/(2\pi))$. The tensor target is $\mathcal{H}^{\otimes k} = L^2(\mathbb{R}^k, \prod_i dt_i/(2\pi))$. All adjoints below use these measures and the displayed coefficient frames; inner products are conjugate linear in their first entry. Let Q be multiplication by $q_\Phi(t)$, with domain $\{f : q_\Phi f \in \mathcal{H}\}$, and let $T = AI + iQ$. All columns of S are in the domains of every power used here, by the original polynomial Mellin decay estimates. Define the complete matrices

$$G = S^*S, \quad J = S^*QS, \quad Z = S^*Q^2S, \quad H = S^*TS = AG + iJ. \quad (\text{MBG.4})$$

For example, putting $r_a(s) = R_h(s^a; s)$, their entries are

$$G_{ab} = \frac{1}{2\pi} \int |v_h(\frac{1}{2} + it)|^2 \overline{r_a(\frac{1}{2} + it)} r_b(\frac{1}{2} + it) dt, \\ J_{ab} = \frac{1}{2\pi} \int q_\Phi(t) |v_h(\frac{1}{2} + it)|^2 \overline{r_a(\frac{1}{2} + it)} r_b(\frac{1}{2} + it) dt, \quad (\text{MBG.5}) \\ Z_{ab} = \frac{1}{2\pi} \int q_\Phi(t)^2 |v_h(\frac{1}{2} + it)|^2 \overline{r_a(\frac{1}{2} + it)} r_b(\frac{1}{2} + it) dt, \quad 0 \leq a, b < d.$$

These matrices retain pairings between different full root blocks. In particular no CRT idempotents have been declared orthogonal. The injectivity of S , equivalently $j_h(v_h R_h(w)) = w$, proves $G > 0$.

Here are explicit coordinate maps for that original positive norm:

$$F = SG^{-1/2}, \quad F^*F = I_d, \\ J_0 = G^{-1/2}JG^{-1/2}, \quad X = AI_d + iJ_0, \quad (\text{MBG.6}) \\ C_0 = G^{1/2}CG^{-1/2}, \quad L = X - C_0.$$

The map $w \mapsto G^{1/2}w$ has exactly the original norm $(w^*Gw)^{1/2}$; these coordinates do not redefine that norm. The orthogonal projection onto SE_h is $P = FF^*$. Define

$$R = (I - P)TF, \quad K = R^*R = G^{-1/2}(Z - JG^{-1}J)G^{-1/2}. \quad (\text{MBG.7})$$

For the last equality, $\mathbf{TF} = \mathbf{F}X + \mathbf{R}$ and $\mathbf{F}^*\mathbf{R} = 0$. The A -part lies entirely in $\mathcal{P}\mathcal{H}$, while the orthogonal part is $i(I - \mathbf{P})\mathbf{QF}$. Expanding its squared norm gives exactly (MBG.7).

In fact $K > 0$, not merely $K \geq 0$. If $\mathbf{R}x = 0$, then $\mathbf{QF}x \in SE_h$. For $x \neq 0$, the polynomial $p = R_h(G^{-1/2}x)$ is nonzero and has degree less than d . Cancelling v_h on an open interval where it is nonzero would say that $q_\Phi(t)p(\frac{1}{2} + it)$ equals a polynomial in $\frac{1}{2} + it$ of degree less than d . Replacing $q_\Phi(t)$ by $(\Phi(\frac{1}{2} + it) - A)/i$ makes this an identity of polynomials in s , by infinitely many points in that interval. Its left degree is $d + 1 + \deg p$, a contradiction. Hence $\mathbf{R}$ is injective and $K > 0$. Set

$$\kappa_- = \lambda_{\min}(K) > 0, \quad \kappa_+ = \lambda_{\max}(K), \quad \ell = \|\mathbf{L}\|_2. \quad (\text{MBG.8})$$

They are the eigenvalues and norm of the explicit finite matrices (MBG.4)–(MBG.7), not constants assigned by a purity hypothesis.

For any d -dimensional endomorphism Y , write $Y^{[k]} = \sum_{i=1}^k I^{\otimes(i-1)} \otimes Y \otimes I^{\otimes(k-i)}$. On the original full tensor space let $\mathbf{S}_k = \mathbf{S}^{\otimes k}$, $\mathbf{G}_k = \mathbf{G}^{\otimes k}$, $\mathbf{T}_k = \sum_i \mathbf{T}_i$, $\mathbf{C}_k = \mathbf{C}^{[k]}$, and

$$\mathbf{B}_k = \mathbf{T}_k \mathbf{S}_k - \mathbf{S}_k \mathbf{C}_k. \quad (\text{MBG.9})$$

This is exactly the Mellin image of QPS.15, including its original theta primitive and tensor differential signs. The entire boundary Gram, for arbitrary mixtures of all 4^k full blocks, is

$$\boxed{G_k^{-1/2} \mathbf{B}_k^* \mathbf{B}_k G_k^{-1/2} = (L^{[k]})^* L^{[k]} + K^{[k]}.} \quad (\text{MBG.10})$$

To prove it, put $\mathbf{F}_k = \mathbf{F}^{\otimes k}$ and

$$\mathbf{R}_{i,k} = \mathbf{F}^{\otimes(i-1)} \otimes \mathbf{R} \otimes \mathbf{F}^{\otimes(k-i)}.$$

The tensor product rule and (MBG.6)–(MBG.7) give $\mathbf{B}_k G_k^{-1/2} = \mathbf{F}_k L^{[k]} + \sum_i \mathbf{R}_{i,k}$. Here $\mathbf{F}_k^* \mathbf{R}_{i,k} = 0$. For $i \neq j$, $\mathbf{R}_{i,k}^* \mathbf{R}_{j,k} = 0$, because at position i one tensor factor is $\mathbf{R}^* \mathbf{F} = 0$. For $i = j$ that product is the i -th copy of K . Expanding the full squared operator norm proves (MBG.10), with no omitted cross term: all remaining cross terms are in $(L^{[k]})^* L^{[k]}$. Consequently

$$k\kappa_- G_k \preceq \mathbf{B}_k^* \mathbf{B}_k \preceq (k^2 \ell^2 + k\kappa_+) G_k. \quad (\text{MBG.11})$$

The upper bound uses $\|L^{[k]}\| \leq k\ell$; the lower bound uses the positive summands $K_i \geq \kappa_- I$. This proves full mixed-boundary injectivity quantitatively.

The actual graph map and its induced norm are

$$\text{Gr}_k(w) = (\mathbf{S}_k w, \mathbf{B}_k w) \in \mathcal{H}^{\otimes k} \oplus \mathcal{H}^{\otimes k}, \quad G_k^{\text{gr}} = G_k + \mathbf{B}_k^* \mathbf{B}_k. \quad (\text{MBG.12})$$

Projection onto the first summand followed by the inverse of $\mathbf{S}_k : E_h^{\otimes k} \rightarrow \text{im } \mathbf{S}_k$ is a specified inverse on this graph. Both maps are bounded on their finite dimensional domains. Its relative condition number satisfies

$$\text{cond}\left(G_k^{-1/2} G_k^{\text{gr}} G_k^{-1/2}\right) \leq \frac{1 + k^2 \ell^2 + k\kappa_+}{1 + k\kappa_-}. \quad (\text{MBG.13})$$

Thus this exact graph/source comparison has at most linear condition growth in k , with the displayed original finite constants retained. For an actual injective cyclic inclusion $\eta : E_{\text{cyc}} \rightarrow E_h^{\otimes k}$, congruence by η in (MBG.11) proves the same two-sided constants for $\eta^* G_k^{\text{gr}} \eta$ relative to $\eta^* G_k \eta$. This is a proof about this graph and that actual inclusion. The original period metric and the four polynomial source windows are calculated separately below; they are not identified with this graph metric.

254.2 The arithmetic and phase boundary on the identical terminal class

For each original root r put

$$z_r = e_r(s - r)^{m-1}, \quad a_r = \frac{m!}{g^{(m)}(r)}, \quad f_r = \mathcal{M}s_h(z_r) = a_r \frac{g(s)}{s - r}. \quad (\text{MBG.14})$$

The last equality follows in the full original remainder frame. Writing $H_r = h/(s - r)^m$, the degree- $(d - 1)$ representative of z_r is $H_r(s)(s - r)^{m-1}/H_r(r)$: its full jet at r is y^{m-1} , and all other full jets vanish. Multiplication by v_h^{-1} multiplies this terminal jet by $v_h(r)^{-1}$. Since $v_h(r)H_r(r) = g^{(m)}(r)/m!$, multiplying its representative by $v_h(s)$ proves (MBG.14). No Taylor coefficient in the surrounding module was discarded in obtaining this terminal identity.

For $r = \rho$ retain $c = \Phi(\rho) = A + U + iV$, with $U, V > 0$. On the original source define the two literal defects

$$b_{\text{ar}} = (D - \rho)s_h(z_\rho), \quad b_{\text{ph}} = (\Phi(D) - c)s_h(z_\rho). \quad (\text{MBG.15})$$

Their Mellin transforms are respectively $a_\rho g(s)$ and $a_\rho Q_\rho(s)g(s)$, where the complete polynomial is

$$Q_\rho(s) = \frac{\Phi(s) - \Phi(\rho)}{s - \rho}, \quad b_{\text{ph}} = Q_\rho(D)b_{\text{ar}}. \quad (\text{MBG.16})$$

Polynomial division proves that quotient is a polynomial of degree d . The equality follows by factoring $\Phi(D) - \Phi(\rho)$ on the stated common polynomial domain, and preserves the original scalar a_ρ .

The exact scalar cross-pairings can be evaluated as integrals with the unchanged source mass. Put $N_\rho = \|s_h(z_\rho)\|^2 > 0$ and use the probability measure only as notation for these ratios:

$$\begin{aligned} d\nu_\rho(t) &= \frac{|a_\rho|^2 |g(\frac{1}{2} + it)|^2}{2\pi N_\rho [\delta^2 + (t - \gamma)^2]} dt, & m_t &= \int t d\nu_\rho, & m_q &= \int q_\Phi(t) d\nu_\rho, \\ v_t &= \int (t - m_t)^2 d\nu_\rho, & v_q &= \int (q_\Phi - m_q)^2 d\nu_\rho, \\ c_{tq} &= \int (t - m_t)(q_\Phi - m_q) d\nu_\rho. \end{aligned} \quad (\text{MBG.17})$$

Every norm formula below retains N_ρ ; introducing ν_ρ does not rescale a source vector. All moments exist by the same decay. The density is positive on an open interval and almost everywhere, since nonzero entire g has isolated zeros. In particular

$$v_t > 0, \quad v_q > 0, \quad c_{tq} > 0, \quad v_t v_q - c_{tq}^2 > 0. \quad (\text{MBG.18})$$

For c_{tq} , integrate $(t - u)(q_\Phi(t) - q_\Phi(u))/2$ against $d\nu_\rho(t)d\nu_\rho(u)$. Expansion gives c_{tq} ; strict monotonicity in (MBG.2) makes the integrand positive off the diagonal, a set of positive measure. Equality in Cauchy-Schwarz for the last inequality would make $q_\Phi(t) - m_q$ a scalar multiple of $t - m_t$ almost everywhere. Continuity on an interval and polynomial identity would then make q_Φ affine, contrary to degree $d + 1$. This proves all the strict signs. The exact two means are

$$\mu_{\text{ar}} = -\delta + i(m_t - \gamma), \quad \mu_{\text{ph}} = -U + i(m_q - V). \quad (\text{MBG.19})$$

These are $\langle f, b \rangle / N_\rho$ in the original inner-product convention. In particular $\mu_{\text{ph}} \neq 0$.

For $f = s_h(z_\rho)$, define $\mathcal{B}_{\text{ar},k}$ and $\mathcal{B}_{\text{ph},k}$ by summing over the k ordered tensor positions with respectively b_{ar} and b_{ph} in that position and f in every other position. The entire joint Gram is

$$\frac{1}{N_\rho^k} \begin{pmatrix} \langle \mathcal{B}_{\text{ar},k}, \mathcal{B}_{\text{ar},k} \rangle & \langle \mathcal{B}_{\text{ar},k}, \mathcal{B}_{\text{ph},k} \rangle \\ \langle \mathcal{B}_{\text{ph},k}, \mathcal{B}_{\text{ar},k} \rangle & \langle \mathcal{B}_{\text{ph},k}, \mathcal{B}_{\text{ph},k} \rangle \end{pmatrix} = k^2 \begin{pmatrix} |\mu_{\text{ar}}|^2 & \bar{\mu}_{\text{ar}}\mu_{\text{ph}} \\ \bar{\mu}_{\text{ph}}\mu_{\text{ar}} & |\mu_{\text{ph}}|^2 \end{pmatrix} + k \begin{pmatrix} v_t & c_{tq} \\ c_{tq} & v_q \end{pmatrix}. \quad (\text{MBG.20})$$

In a diagonal tensor pairing the centered multipliers are $i(t - m_t)$ and $i(q_\Phi - m_q)$, giving the displayed real covariance. In two different positions the measure factors, giving the product of means. There

are k diagonal and $k(k-1)$ ordered off-diagonal pairs. This proves (MBG.20) with every term. For the same nonzero class there is consequently the exact comparison

$$\begin{aligned} \beta &= \mu_{\text{ar}}/\mu_{\text{ph}}, & \mathcal{E}_k &= \mathcal{B}_{\text{ar},k} - \beta \mathcal{B}_{\text{ph},k}, \\ \langle f^{\otimes k}, \mathcal{E}_k \rangle &= 0, & \frac{\|\mathcal{E}_k\|^2}{N_\rho^k} &= k(v_t + |\beta|^2 v_q - 2c_{tq} \operatorname{Re} \beta). \end{aligned} \quad (\text{MBG.21})$$

The coefficient in parentheses is strictly positive by (MBG.18). The centered means cancel by the stated value of β , and expansion of (MBG.20) proves the norm equality. This is a calculated square-root error between the two actual tensor boundaries; their nonzero mean terms are still of order k . In particular it does not discard the original discrepancy:

$$\operatorname{Re} \frac{\langle f^{\otimes k}, \mathcal{B}_{\text{ar},k} \rangle}{N_\rho^k} = -k\delta, \quad \operatorname{Re} \frac{\langle f^{\otimes k}, \mathcal{B}_{\text{ph},k} \rangle}{N_\rho^k} = -kU. \quad (\text{MBG.22})$$

Multiplication by the original nonzero coefficient $c_{h,k} = [k(m-1)]!(m-1)!^{-k}((g/h)(\rho))^k$ multiplies each vector by that coefficient and every pairing and squared norm by $|c_{h,k}|^2$. Thus (MBG.20)–(MBG.22) are exact on the specified amplified arithmetic class of MAI, with its coefficient retained on both sides.

254.3 The exact attachment to the four original polynomial windows

The preceding finite jet graph has a concrete map to the original polynomial source, including its relations. Retain

$$\begin{aligned} \chi &= \chi_{h,k}, & q &= \deg \chi = [1 + k(m-1)](k+1)^2, & \mathcal{P}_N &= \mathbb{C}[S]_{\leq N}, \\ \Sigma &= s_1 + \cdots + s_k, & \mathfrak{h} &= (h(s_1), \dots, h(s_k)). \end{aligned} \quad (\text{MBG.23})$$

All windows below have $N \geq q-1$, with $\mathcal{P}_{-1} = 0$. Their maps have the precise domains

$$B_N : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad Q \longmapsto \chi Q, \quad J_N : \mathcal{P}_N \longrightarrow E_{\text{cyc}} = \mathbb{C}[S]/(\chi). \quad (\text{MBG.23a})$$

The map $s_N : E_{\text{cyc}} \rightarrow \mathcal{P}_N$ selects the unique representative of degree less than q . Monic division proves $\ker J_N = \operatorname{im} B_N$, injectivity of B_N , and $J_N s_N = I$, including the stated empty relation space. The degree formula applies to the full quartet and its original cyclic vector; all $(k+1)^2$ distinct arithmetic centres have nilpotence length $1 + k(m-1)$. Distinctness follows from $\delta, \gamma > 0$; at every tuple the highest nilpotent coefficient $[k(m-1)]!/((m-1)!)^k$ is nonzero. Thus no factor in this annihilator has been silently removed. Let $\mathfrak{r}$ denote successive monic remainders in each variable modulo $h(s_i)$; the remainder operations commute. For $P \in \mathcal{P}_N$ put

$$\begin{aligned} \mathcal{V}_N P &= \left(\prod_i v_h(s_i) \right) P(\Sigma), \\ \eta[P] &= v_h^{\otimes k} P(A_h^{[k]}) 1, & J_N P &= [P] \pmod{\chi}, \\ \mathcal{D}_N P &= \left(\prod_i v_h(s_i) \right) \left[\left(\sum_i \Phi(s_i) \right) P(\Sigma) - \mathfrak{r} \left(\left(\sum_i \Phi(s_i) \right) P(\Sigma) \right) \right]. \end{aligned} \quad (\text{MBG.24})$$

These are Mellin representatives on the original product line, with the same product measure as before. In particular $\|\mathcal{V}_N P\|^2$ is exactly the arithmetic convolution norm with density w_h^{*k} used in AT. The full unit cancels only in the proved composite of maps

$$\begin{aligned} S_k \eta J_N P &= \left(\prod_i v_h(s_i) \right) \mathfrak{r}(P(\Sigma)), \\ S_k C_k \eta J_N P &= \left(\prod_i v_h(s_i) \right) \mathfrak{r} \left(\left(\sum_i \Phi(s_i) \right) P(\Sigma) \right), \\ \mathcal{D}_N &= \mathfrak{T}_k \mathcal{V}_N - S_k C_k \eta J_N. \end{aligned} \quad (\text{MBG.25})$$

For example $R_h^{\otimes k} v_h^{\otimes k}$ is the identity on full tensor classes, whose unique polynomial representative is $\mathfrak{r}$; this proves the first line. Multiplication by the original phase gives the second. These equalities prove all types and the last identity.

The literal difference $\mathcal{V}_N - S_k \eta J_N$ is a theta relation with an explicit polynomial primitive. Its contribution to the full graph map and all cross-pairings is also exact. Put

$$\begin{aligned} \mathfrak{J}_N &= \eta J_N, & \Delta_N &= \mathcal{V}_N - S_k \mathfrak{J}_N, & \mathbf{E}_N &= (\Delta_N, \mathbf{T}_k \Delta_N), \\ & & (\mathcal{V}_N, \mathbf{D}_N) &= \mathbf{Gr}_k \mathfrak{J}_N + \mathbf{E}_N. \end{aligned} \quad (\text{MBG.25a})$$

The second component follows by subtracting (MBG.9) composed with $\mathfrak{J}_N$ from (MBG.25); the first is the definition of Δ_N . Therefore, in the unchanged coefficient frames,

$$\begin{aligned} Y_N &= S_k^* \Delta_N + \mathbf{B}_k^* \mathbf{T}_k \Delta_N, & Z_N^{\text{rel}} &= \Delta_N^* \Delta_N + (\mathbf{T}_k \Delta_N)^* \mathbf{T}_k \Delta_N, \\ \widehat{M}_N &= \mathfrak{J}_N^* G_k^{\text{gr}} \mathfrak{J}_N + \mathfrak{J}_N^* Y_N + Y_N^* \mathfrak{J}_N + Z_N^{\text{rel}}. \end{aligned} \quad (\text{MBG.25b})$$

This is the expansion of the squared norm of the preceding exact sum of maps. In particular neither cross term is zero by a declared splitting. Each is an actual finite polynomial integral in the same original product measure; (MBG.28)–(MBG.29) below give the complete entry formula for their sum. On a relation χQ the first summand of (MBG.25a) vanishes, while its second is retained.

To construct the promised primitive on any polynomial F , set $F_0 = F$ and divide successively

$$F_{i-1} = h(s_i) Q_i + F_i, \quad \deg_{s_i} F_i < d \quad (1 \leq i \leq k). \quad (\text{MBG.26})$$

Earlier reduced variables remain reduced by this division, so $F_k = \mathfrak{r}F$ and $F - \mathfrak{r}F = \sum_i h(s_i) Q_i$. After multiplication by $\prod_j v_h(s_j)$, the i -th term is $g(s_i) \prod_{j \neq i} v_h(s_j) Q_i$. Each monomial of Q_i is the Mellin transform of a tensor with D_i -polynomial applied to the original seed ϕ_* in degree zero at position i and the original Euler-inverse theta sources in degree one elsewhere. Give that cochain sign $(-1)^{i-1}$. Its tensor differential contributes the second sign $(-1)^{i-1}$, and hence exactly the displayed term. Summing proves the primitive for $F = P(\Sigma)$ and for the phase polynomial in (MBG.24). This retains the actual relation source before passing to cohomology. In particular on the full original relation space one has

$$\mathbf{D}_N(\chi Q) = \mathbf{T}_k \mathcal{V}_N(\chi Q), \quad \|\mathbf{D}_N(\chi Q)\|^2 \geq k^2 A^2 \|\mathcal{V}_N(\chi Q)\|^2. \quad (\text{MBG.27})$$

Here $\mathfrak{r}(\chi(\Sigma) Q(\Sigma)) = 0$, since $\chi(A_h^{[k]})1 = 0$, and multiplication by the phase preserves that zero class. On the product line the multiplier of $\mathbf{T}_k$ has real part kA , proving the inequality. Thus the kernel directions in the original four windows retain their source and their controlled nonzero phase energy.

All entries of the resulting graph Gram are explicit finite sums of original one-variable moments. For $0 \leq j \leq N$, expand

$$\begin{aligned} P_j(s_1, \dots, s_k) &= \left(\sum_i \Phi(s_i) \right) \Sigma^j - \mathfrak{r} \left(\left(\sum_i \Phi(s_i) \right) \Sigma^j \right) \\ &= \sum_{\alpha} d_{\alpha j} s_1^{\alpha_1} \cdots s_k^{\alpha_k}, \\ M_{ab}^{(h)} &= \frac{1}{2\pi} \int \bar{s}^a s^b |v_h(s)|^2 dt, \quad s = \frac{1}{2} + it. \end{aligned} \quad (\text{MBG.28})$$

Every coefficient $d_{\alpha j}$ is specified by the finite monic division (MBG.26), and each exponent satisfies $\alpha_i \leq N + d + 1$. Absolute integrability and the finite sum give the complete formula

$$(D_N)_{ij} := \langle \mathbf{D}_N S^i, \mathbf{D}_N S^j \rangle = \sum_{\alpha, \beta} \bar{d}_{\alpha i} d_{\beta j} \prod_{\ell=1}^k M_{\alpha_{\ell} \beta_{\ell}}^{(h)}, \quad \widehat{M}_N = M_N + D_N. \quad (\text{MBG.29})$$

Here M_N is precisely the arithmetic source Gram of AT, and $\widehat{M}_N$ is the norm of $P \mapsto (\mathcal{V}_N P, D_N P)$. The source $\mathcal{V}_N$ is injective: a nonzero polynomial in Σ cannot vanish on every product-line point. Thus both Grams are positive definite, including the relation-source restrictions. The boundary Gram D_N is itself positive definite. For a nonzero P of degree n , the polynomial $(\sum_i \Phi(s_i))P(\Sigma)$ has s_1 -degree $n + d + 1$ with nonzero leading coefficient. Its complete remainder has s_1 -degree less than d . Their difference is therefore a nonzero polynomial. After multiplication by the nonzero entire product $\prod_i v_h(s_i)$, it cannot vanish on a product of open intervals of the Mellin line: fixing all but one variable successively would force every polynomial coefficient to vanish. Its squared integral is consequently strictly positive.

Retain multiplication $B_N : Q \mapsto \chi Q$, the literal remainder J_N , and its original degree-less-than- q section s_N . Write $H_N = B_N^* M_N B_N$ for the original arithmetic relation Gram before adding the phase graph. The graph norm induces the exact canonical section and quotient Gram

$$\begin{aligned}\widehat{H}_N &= B_N^* \widehat{M}_N B_N, \\ \widehat{R}_N &= s_N - B_N \widehat{H}_N^{-1} B_N^* \widehat{M}_N s_N, \\ \widehat{G}_N &= (J_N \widehat{M}_N^{-1} J_N^*)^{-1}, \quad \det \widehat{G}_N = \det \widehat{M}_N / \det \widehat{H}_N.\end{aligned}\tag{MBG.30}$$

Indeed any lift of x is $s_N x + B_N y$; completing its positive quadratic form gives the displayed $\widehat{R}_N$ and $B_N^* \widehat{M}_N \widehat{R}_N = 0$. Define its induced quotient form first as $\widetilde{G}_N = \widehat{R}_N^* \widehat{M}_N \widehat{R}_N$. It follows that $\widehat{M}_N \widehat{R}_N = J_N^* \widetilde{G}_N$: decompose every coefficient vector into its relation component and its $\widehat{R}_N$ component, and pair with a column of $\widehat{R}_N$. Multiply by $\widehat{M}_N^{-1}$ and apply J_N to obtain $I = (J_N \widehat{M}_N^{-1} J_N^*) \widetilde{G}_N$. Thus $\widetilde{G}_N = \widehat{G}_N$, proving the displayed inverse formula without circularly defining its induced form. In the ordered monomial frame $[s_N \ B_N]$ has determinant one because χ is monic; the Schur determinant formula gives the last identity. When $N = q - 1$, the relation domain is zero, its determinant is one and all maps through it are zero. These facts prove (MBG.30) at all four endpoints without an added hypothesis.

For the actual signed four-endpoint expression retain the signs

$$\mathcal{B}(M) = \log V_{q-1} + \log V_q - \log V_{2q-1} - \log V_{2q}, \quad V_N = \det G_N.\tag{MBG.31}$$

Here $\mathcal{B}$ is a functional of the compatible family of source Grams and their same remainder quotient: M denotes $(M_N)_{N \geq q-1}$, and similarly for $\widehat{M}$. The original reference family M^Γ below is exactly AT.4–AT.5 at its reference endpoint $t = 0$, with density

$$r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad S = k/2 + iu.\tag{MBG.31a}$$

Its original mass is $(2\pi)^{k/2}$, and its source Gram entries are the integrals of $\bar{S}^i S^j$ against that density. Its exact additional phase-graph term, on these same arithmetic windows, is the following finite expression:

$$\begin{aligned}\mathfrak{a}_N &= \log \det(I + M_N^{-1/2} D_N M_N^{-1/2}), \\ \mathfrak{b}_N &= \log \det(I + H_N^{-1/2} B_N^* D_N B_N H_N^{-1/2}), \\ \mathcal{B}(\widehat{M}) - \mathcal{B}(M) &= (\mathfrak{a}_{q-1} - \mathfrak{b}_{q-1}) + (\mathfrak{a}_q - \mathfrak{b}_q) \\ &\quad - (\mathfrak{a}_{2q-1} - \mathfrak{b}_{2q-1}) - (\mathfrak{a}_{2q} - \mathfrak{b}_{2q}).\end{aligned}\tag{MBG.32}$$

The expression retains the ordered list of four terms, so for $q = 1$ the two occurrences of endpoint 1 have opposite signs and cancel. The zero-dimensional determinant convention is as in (MBG.30). Factor $M_N + D_N$ and $H_N + B_N^* D_N B_N$ by their original positive square roots in (MBG.30), and subtract the four logarithms; this proves (MBG.32). It connects the computed mixed theta boundary, including its relation directions, to the original signed control. It changes neither the arithmetic-Gamma term already proved in AT nor its source mass: its exact composition is

$$\mathcal{B}(\widehat{M}) - \mathcal{B}(M^\Gamma) = [\mathcal{B}(M) - \mathcal{B}(M^\Gamma)] + [\mathcal{B}(\widehat{M}) - \mathcal{B}(M)].\tag{MBG.33}$$

All entries of the second bracket are computed in (MBG.28)–(MBG.32); no sign is assigned to that four-term difference by positivity of the individual graph Grams. Formula (MBG.13) concerns the canonical full-jet graph and its restrictions. Formula (MBG.29), including $\chi \mathcal{P}_{N-q}$, is the actual polynomial-window graph. Their precise source discrepancy and primitive are (MBG.25)–(MBG.27); they have not been equated by dropping those relations.

254.4 Original mixed support and the existing analytic period comparison

Every displayed linear map lifts at an original outer label (W, S) and each independent active coefficient mask B by applying it separately to each coordinate:

$$(W, S; B, (x_i)_{i \in B}) \mapsto (W, S; B, (f(x_i))_{i \in B}), \quad \tau_{\text{out}} \mapsto \tau_{\text{out}}. \quad (\text{MBG.34})$$

At a present label its supported-zero fibre is $(\ker f)^B$; this follows coordinate by coordinate from $f(x_i) = 0$. If $B \subseteq B'$, insertion of supported zeros commutes with f , because $f(0) = 0$. At the empty mask the unique zero tuple remains a present labelled vertex. Thus none of these maps identifies it with τ_{out} , nor removes proper-source kernels. These are the existing MCF/CFA module maps, not claims that Gram matrices or linear sections are ring maps. Here is the precise coefficient domain of this analytic attachment. At an original arithmetic realization $\alpha : R \rightarrow \mathbb{C}$, regard its complex source as an R -module through α . The displayed maps are R -linear because $f(\alpha(a)x) = \alpha(a)f(x)$. For an original coefficient arrow $\lambda : R \rightarrow R'$ with $\alpha'\lambda = \alpha$, scalar extension commutes with these maps: on each pure tensor both composites send $b \otimes x$ to $b \otimes f(x)$, and the displayed scalar identity proves this is balanced. Composing with the actual product, quadratic, negacyclic and ideal-chain source arrows of MFC thus uses their specified domain and arithmetic realization, retaining any coefficient kernel as its original fibre. This does not manufacture a ring map from a positive-characteristic residue field into $\mathbb{C}$. Those residue vertices and their original coefficient maps remain in the total diagram at their established TPF/TCL types. The present norm formulas evaluate the complex arithmetic leg of that diagram. Explicitly the realization of the extended source is

$$\mu_X : R' \otimes_R X \longrightarrow X, \quad b \otimes x \mapsto \alpha'(b)x, \quad \mu_Y(1 \otimes f) = f\mu_X. \quad (\text{MBG.34a})$$

Balancing follows from $\alpha'\lambda = \alpha$; evaluating a pure tensor proves the commuting square. The kernel is exactly the set of finite sums

$$\ker \mu_X = \left\{ \sum_i (b_i \otimes x_i - 1 \otimes \alpha'(b_i)x_i) \right\}. \quad (\text{MBG.34b})$$

Every displayed difference maps to zero. Conversely, subtract these differences from any element $\sum_i b_i \otimes x_i$ in the kernel; the remaining vector is $1 \otimes \sum_i \alpha'(b_i)x_i = 0$. Thus the fibre over x is precisely $1 \otimes x + \ker \mu_X$. The map $x \mapsto 1 \otimes x$ is a section over R , not generally over R' . This explicit fibre is retained at every active coefficient mask in (MBG.34).

There is also an exact coordinate transport for the already constructed period comparison; it leaves the above original source and defect forms measurable in every admitted sector. At an admitted u , let $\mathcal{P}_\theta(u)$ be the full period map on the original theta jets, with all primitive, gamma, contour, Jacobian and unit factors as explicitly constructed in FPB and the analytic sectorial source. For two sectors its proved transition is $\mathcal{P}_+ = \mathfrak{S}\mathcal{P}_-$. The source and boundary forms in their period coordinates are

$$G_\pm^{\text{per}} = \mathcal{P}_\pm^* G \mathcal{P}_\pm^{-1}, \quad D_\pm^{\text{per}} = \mathcal{P}_\pm^* \mathbf{B}_1^* \mathbf{B}_1 \mathcal{P}_\pm^{-1}. \quad (\text{MBG.35})$$

Direct substitution, with the inverse order preserved, proves

$$\mathfrak{S}^* G_+^{\text{per}} \mathfrak{S} = G_-^{\text{per}}, \quad \mathfrak{S}^* D_+^{\text{per}} \mathfrak{S} = D_-^{\text{per}}. \quad (\text{MBG.36})$$

The same formula holds for the entire tensor boundary by replacing $\mathcal{P}_\pm$ with $\mathcal{P}_\pm^{\otimes k}$, G with G_k , and $\mathbf{B}_1^* \mathbf{B}_1$ with the original coefficient matrix

$$\mathbf{B}_k^* \mathbf{B}_k = G_k^{1/2} ((L^{[k]})^* L^{[k]} + K^{[k]}) G_k^{1/2}. \quad (\text{MBG.37})$$

Both original Gram factors are retained when passing from (MBG.10) to these period coordinates. Thus changes of sector preserve the exact original source norm and its boundary energy in these transported forms. Using Euclidean norms in every sector instead would introduce the actual factors $\mathfrak{S}^* \mathfrak{S}$; those are not set to one by (MBG.36). This proves the exact map from analytic gluing to the original boundary form. It supplies no unproved replacement of arithmetic dilation by geometric Frobenius.

This calculation retains the arithmetic coefficient complex over the absolute pointed base $\mathbf{F}_{1,\tau} = \{\tau, 1\}$. It computes the signed arithmetic/reference determinant term on the original finite theta source. The new formulas are a complete first and second variation, an exact four-window projection trace, and a quantitative bound in terms of the actual common source Gram. No arithmetic source multiplier is assumed bounded above or bounded away from zero.

255 Original objects and the fixed comparison

The retained tau-base construction has the four-point chart $+, - < \eta < \sigma$ and the coefficient sheaf $\mathcal{T}$ with $(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0)$. Its two nonzero restrictions are Θ and $J\Theta = \Theta\mathcal{F}$. The actual projective resolution gives

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi-\psi)} \mathcal{B}], \quad H^1\mathbf{A}_\tau(\mathcal{T}) = Q = \mathcal{B}/\Theta V. \quad (\text{AT1})$$

These are the fixed source objects of the original Tau Base note, equations (18)–(25), supplied in the source closure. The tensor resolution in its equations (46)–(48) retains the original differential signs and identifies its top cohomology with $Q^{\otimes k}$. The operation here uses that pushforward, with its arithmetic classes. Its separately computed restriction to σ remains an available comparison; it is not substituted for (AT1).

Fix an original actual packet h of full zero orders of $g = 2\xi$, and a positive integer k . Retain the full tensor-sum operator, its original cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $E = \mathbb{C}[S]/(\chi)$ in the increasing-power remainder frame. The original construction supplies

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}P = \eta_{h,k}J_N P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad (\text{AT2})$$

where $v_h = j_h(g/h)$ is the full invertible Taylor unit, J_N is literal remainder modulo χ , and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the original injective full-jet observation, $E_h = \mathbb{C}[s]/(h)$. The original section $\sigma_h : E_h \hookrightarrow Q$ gives the separate arithmetic cohomology injection and equality

$$q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}J_N P, \quad \sigma_h^{\otimes k}\eta_{h,k} : E \hookrightarrow Q^{\otimes k}. \quad (\text{AT2a})$$

The supplied arithmetic-input and cyclic-sum proofs construct these maps; all operations below keep them fixed. In particular the sum coordinate is $S = k/2 + iu$, with no division by k .

For $N \geq q - 1$, set $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and let $B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ be multiplication by the literal monic χ . Write $s_N : E \rightarrow \mathcal{P}_N$ for the degree-less-than- q remainder section. Polynomial division proves

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad J_N s_N = I_E. \quad (\text{AT3})$$

The arithmetic norm has density $m_1(u) = m_{h,k}(u) = w_h^{*k}(u)$, $w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi)$. The retained Gamma reference has density

$$m_0(u) = r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a). \quad (\text{AT4})$$

Its mass is $(2\pi)^{k/2}$; the arithmetic mass is the original μ_h^k . The source proofs included with this calculation establish finite moments of every degree and positivity almost everywhere for both densities. The calculation below only integrates polynomials against these two actual measures and their sum. It needs no estimate on the pointwise ratio m_1/m_0 .

For $0 \leq t \leq 1$ define, in the unchanged coefficient frame,

$$\begin{aligned} m_t &= (1-t)m_0 + tm_1, \quad \Delta m = m_1 - m_0, \quad v_N(u) = (1, S(u), \dots, S(u)^N), \\ M_N(t) &= \int v_N(u)^* v_N(u) m_t(u) du, \quad \Delta M_N = M_N(1) - M_N(0), \\ H_N(t) &= B_N^* M_N(t) B_N. \end{aligned} \quad (\text{AT5})$$

Every $M_N(t)$ is positive definite: a nonzero polynomial has finitely many real zeros, and m_t is positive almost everywhere. B_N is injective, so $H_N(t)$ is positive definite on its actual relation domain. When $N = q - 1$, that domain is zero; its determinant is one, its unique endomorphism is its inverse, and all compositions through that domain are zero. The real parameter t interpolates norms on (AT3); it changes neither the absolute base nor a cochain map.

256 Canonical sections, determinant lines and full variations

Theorem 256.1. *On the fixed exact sequence (AT3), its least-norm section and quotient Gram are*

$$\begin{aligned} R_N(t) &= s_N - B_N H_N(t)^{-1} B_N^* M_N(t) s_N, \\ J_N R_N(t) &= I_E, \quad B_N^* M_N(t) R_N(t) = 0, \\ G_N(t) &= R_N(t)^* M_N(t) R_N(t) = (J_N M_N(t)^{-1} J_N^*)^{-1}, \\ V_N(t) &:= \det G_N(t) = \frac{\det M_N(t)}{\det H_N(t)}. \end{aligned} \tag{AT6}$$

In the original polynomial frame the exact first and second variations are

$$\begin{aligned} \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \quad \dot{G}_N = R_N^* \Delta M_N R_N, \\ \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N \preceq 0. \end{aligned} \tag{AT7}$$

The exact logarithmic variation, with both trace terms retained, is

$$\begin{aligned} \frac{d}{dt} \log V_N &= \text{Tr}(M_N^{-1} \Delta M_N) - \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N) \\ &= \text{Tr}(G_N^{-1} R_N^* \Delta M_N R_N). \end{aligned} \tag{AT8}$$

Its second derivative is

$$\begin{aligned} \frac{d^2}{dt^2} \log V_N &= -2 \text{Tr}(G_N^{-1} R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N) \\ &\quad - \text{Tr}(G_N^{-1} \dot{G}_N G_N^{-1} \dot{G}_N) \leq 0. \end{aligned} \tag{AT9}$$

Proof. Every lift of $x \in E$ is $s_N x + B_N y$. Expanding its squared norm and completing its finite quadratic expression gives the unique minimizer $y = -H_N^{-1} B_N^* M_N s_N$. This proves the first two lines of (AT6). For the inverse expression, every polynomial vector p decomposes as $B_N y + R_N J_N p$. Orthogonality therefore gives $R_N^* M_N p = G_N J_N p$ for every p . Taking adjoints yields $M_N R_N = J_N^* G_N$. Substituting this into $J_N R_N = I_E$ gives $(J_N M_N^{-1} J_N^*) G_N = I_E$, which proves the inverse expression.

The determinant calculation preserves orientation. Put $r = N - q + 1$ and $L_N = [B_N \ s_N]$. In the ordered monomial bases, $[s_N \ B_N]$ has determinant one, since χ is monic and the lower diagonal block is triangular with diagonal one. Thus $\det L_N = (-1)^{qr}$. The upper-left block of $L_N^* M_N L_N$ is H_N , and its Schur complement is G_N . Consequently $\det H_N \det G_N = |(-1)^{qr}|^2 \det M_N = \det M_N$. Equivalently the map

$$\begin{aligned} \det \mathcal{P}_{N-q} \otimes \det E &\longrightarrow \det \mathcal{P}_N, \\ (y_1 \wedge \cdots \wedge y_r) \otimes (x_1 \wedge \cdots \wedge x_q) &\longmapsto B_N y_1 \wedge \cdots \wedge B_N y_r \wedge s_N x_1 \wedge \cdots \wedge s_N x_q \end{aligned} \tag{AT10}$$

is the fixed determinant-line isomorphism. Replacing any lift by a relation does not change its wedge. Orthogonal lifts give the product metric just computed. This proves the determinant identity including the empty relation space, without changing any scalar mass.

Differentiate $J_N R_N = I$; then $\dot{R}_N$ lies in $\text{im } B_N$. Differentiating $B_N^* M_N R_N = 0$ determines its unique coefficient and proves the first identity of (AT7). On differentiating $R_N^* M_N R_N$, the two

terms involving $\dot{R}_N$ vanish by that orthogonality. A second differentiation gives the two equal negative terms in (AT7), since ΔM_N is Hermitian. This proves the claimed second derivative and its sign. Applying the finite determinant derivative to (AT6) gives the first equality of (AT8); applying it to G_N gives the second. One more derivative gives (AT9). The first trace there is the squared Hilbert–Schmidt norm of $H_N^{-1/2} B_N^* \Delta M_N R_N G_N^{-1/2}$; the second is the squared Hilbert–Schmidt norm of $G_N^{-1/2} \dot{G}_N G_N^{-1/2}$. These identities establish their signs and retain the original forms. All inverse matrices are smooth on a neighborhood of the compact interval $[0, 1]$, by positivity and continuity. No infinite-dimensional inverse has been used. $\square$

The derivative $\dot{R}_N$ in (AT7) is a literal polynomial relation. Applying the original $\mathcal{V}$ in (AT2) therefore retains its original theta primitive and sends its arithmetic class to zero; $J^{(k)} \mathcal{V} R_N(t) = \eta_{h,k}$ for every t . At each nonempty original support label A , the cochain identities lift by $(A, P) \mapsto (A, fP)$, while $\tau \mapsto \tau$. Thus a relation-valued derivative has a supported zero class at label A . Its source vector and its full nonnegative energy in (AT7)–(AT9) remain present. These formulas compute their contribution rather than deleting it after taking cohomology.

For every independent coefficient mask $B_c \subseteq I_n$, retain the separate outer source and leg label (W, S) and apply each of these linear maps in every active coordinate. The supported-zero fibre of $f[B_c]$ is precisely $(\ker f)^{B_c}$ at that same label. For $B_c \subseteq C_c$, zero insertion commutes with every such map: both composites use $f(x_i)$ in an old coordinate and $f(0) = 0$ in a new coordinate. It is a labelled face arrow $e_{B_c} \mapsto e_{C_c}$, not a collapse of the two faces. At empty coefficient support the zero amplitude space retains $(W, S; \emptyset, 0)$, distinct from global external τ_{out} . In each active coordinate the relation-valued derivative is the original $\dot{R}_N$ and has the same theta primitive. The equation $(J^{(k)} \mathcal{V} R_N)[B_c] = \eta_{h,k}[B_c]$ holds because it is the already proved equality in every coordinate. The source derivative and the forms (AT7)–(AT9) remain the original ones on those vectors.

257 The signed four-endpoint term on one retained source

Let $N_* = 2q$. Every space in (AT3) for the four endpoints sits in $\mathcal{P}_{N_*}$ by its actual coefficient inclusion. Give this common space the form $M_*(t) = M_{N_*}(t)$. Write $P_N(t)$ for its orthogonal projection onto $\mathcal{P}_N$, and $Q_N(t)$ for its orthogonal projection onto the relation subspace $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$. The latter is zero at $N = q - 1$. For the coefficient inclusion $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$, the formulas, with their domains retained, are

$$P_N = I_N M_N^{-1} I_N^* M_*, \quad Q_N = I_N B_N H_N^{-1} B_N^* I_N^* M_*, \quad C(t) = M_*(t)^{-1} \Delta M_*. \quad (\text{AT11})$$

Both P_N and Q_N are $M_*(t)$ -self-adjoint idempotents with their displayed images, verified by substitution. Since $\mathcal{R}_N \subset \mathcal{P}_N$, $P_N Q_N = Q_N P_N = Q_N$. Their difference is the orthogonal projection onto the finite canonical quotient section and has trace q . Cyclicity of finite matrix trace in (AT11) proves the exact common-source version of (AT8):

$$\frac{d}{dt} \log V_N(t) = \text{Tr}((P_N(t) - Q_N(t))C(t)). \quad (\text{AT12})$$

Put $i_0 = q - 1$, $i_1 = q$, $j_0 = 2q - 1$, and $j_1 = 2q$. Define the two positive operators

$$\begin{aligned} U(t) &= (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \\ W(t) &= (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \end{aligned} \quad (\text{AT13})$$

Each difference in this display is an orthogonal projection, because the two spaces in its pair are nested in the same Hilbert space. Each has rank q . The sums need not be projections and are not asserted to be orthogonal sums; they are positive operators with $\text{Tr } U = \text{Tr } W = 2q$.

Write

$$\begin{aligned}\mathcal{B}(t) &= \log \frac{V_{i_0}(t)V_{i_1}(t)}{V_{j_0}(t)V_{j_1}(t)}, \quad T_N = V_N(1)/V_N(0), \\ \Delta\mathcal{B} &= \mathcal{B}(1) - \mathcal{B}(0) = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}}.\end{aligned}\tag{AT14}$$

Subtracting the four exact formulas (AT12), with their original signs, and integrating gives

$$\Delta\mathcal{B} = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt.\tag{AT15}$$

This is the actual arithmetic correction in the audit's formula (54). It compares the two added relation windows with the two added polynomial windows. Concavity of each individual $\log V_N(t)$ in (AT9) does not assign a sign to their signed combination in (AT15); that combination has now been computed.

There is also an exact pointwise integral, using no division by the arithmetic density. Define

$$\begin{aligned}k_N(t, u) &= v_N(u)M_N(t)^{-1}v_N(u)^*, \\ r_N(t, u) &= |\chi(S(u))|^2 v_{N-q}(u)H_N(t)^{-1}v_{N-q}(u)^*, \\ d_N(t, u) &= k_N(t, u) - r_N(t, u), \quad r_{q-1}(t, u) = 0, \\ d_N(t, u) &= v_N(u)R_N(t)G_N(t)^{-1}R_N(t)^*v_N(u)^* \geq 0.\end{aligned}\tag{AT16}$$

The last equality follows by writing the orthogonal splitting of M_N^{-1} as $B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*$; multiplying by M_N and evaluating on $\text{im } B_N \oplus \text{im } R_N$ proves that inverse identity. In particular $\int d_N(t, u)m_t(u) du = q$. Consequently (AT8) also reads

$$\frac{d}{dt} \log V_N(t) = \int d_N(t, u) \Delta m(u) du, \quad \Delta\mathcal{B} = \int_0^1 \int (d_{i_0} + d_{i_1} - d_{j_0} - d_{j_1})(t, u) \Delta m(u) du dt.\tag{AT17}$$

All coefficients of these fixed-degree polynomials in u and $\bar{S}(u)$ are bounded for $t \in [0, 1]$. Since $|\Delta m| \leq m_0 + m_1$ and these measures have the required finite moments, absolute integrability proves the exchange of integrals. This remains valid at every zero of m_1 .

258 An exact finite-source bound for the correction

Let $b_1 \leq \dots \leq b_{2q+1}$ be every positive eigenvalue of $D_* = M_*(0)^{-1}M_*(1)$ in the original $M_*(0)$ metric, and put $\kappa_* = b_{2q+1}/b_1$. These data are on the unchanged common source $H = \mathcal{P}_{2q}$; neither packet nor tensor order is suppressed. For the exact U, W, C in (AT11)–(AT15), let $c_j(t)$ be the ordered eigenvalues of $C(t)$, $d(t) = c_{2q+1}(t) - c_1(t)$, and $s(t) = \dim(\text{ran } U \cap \text{ran } W)$. All norms and projections in the following formulas use $M_*(t)$:

$$\begin{aligned}S_{\text{spec}}(t) &= c_{q+2}(t) - c_q(t) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(t) - c_j(t)), \\ S_{\text{tr}}(t) &= \frac{1}{2} d(t) \|U(t) - W(t)\|_{1,t}, \\ S_{\text{ov}}(t) &= \frac{1}{2} d(t) \min\{4q - 2s(t), \sqrt{(2q+1)(8q-4-2\text{Tr}(U(t)W(t)))}\}, \\ \mathfrak{C}_* &= \int_0^1 \min\{S_{\text{spec}}(t), S_{\text{tr}}(t)\} dt, \\ \mathfrak{C}_*^{\text{ov}} &= \int_0^1 \min\{S_{\text{spec}}(t), S_{\text{ov}}(t)\} dt, \\ L_q(D_*) &= \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}.\end{aligned}\tag{AT18a}$$

The trace norm is the sum of absolute eigenvalues of the displayed self-adjoint endomorphism. For $q = 1$ the sums are empty. Under $x = t$, these are literally the ACM/SP operators: their relation matrix is $I_N B_N$, hence their projector is exactly (AT11); their density $r_{1/4}^{*k}$ equals (AT4), with $c_{1/4} = \sqrt{2\pi}$ and its full mass. Thus $\mathfrak{C}_*^{\text{ov}}$ is the original ACM pointwise minimum. The sharper $\mathfrak{C}_*$ also retains SP's already proved exact trace norm; no new source form is chosen.

Theorem 258.1. *The actual signed correction satisfies*

$$|\Delta \mathcal{B}| \leq J_* \leq \mathfrak{C}_* \leq \mathfrak{C}_*^{\text{ov}} \leq L_q(D_*) \leq (2q - 1) \log \kappa_*. \quad (\text{AT18})$$

Here the full actual-moment proof of the additional J_* entry is given in (AT21a)–(AT21e). Consequently its original endpoint quantities obey

$$\mathcal{B}_{h,k}^\Gamma - J_* \leq \mathcal{B}_{h,k}^{\text{ar}} \leq \mathcal{B}_{h,k}^\Gamma + J_*. \quad (\text{AT19})$$

Proof. Multiplication by the same monic χ gives the nested relation $\text{flag } \chi \mathcal{P}_0 \subset \chi \mathcal{P}_{q-1} \subset \chi \mathcal{P}_q$ of dimensions $1, q, q + 1$. On its orthogonal increments U acts by $1, 2, 1$, and acts by zero on the q -dimensional complement. The polynomial flag $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q}$ gives the same full spectrum for W : $0^q, 1^2, 2^{q-1}$. Repeated spaces when $q = 1$ give zero-dimensional increments, not omitted endpoint maps. In particular

$$\text{Tr } U = \text{Tr } W = 2q, \quad \text{Tr } U^2 = \text{Tr } W^2 = 4q - 2, \quad \text{Tr}((U - W)^2) = 8q - 4 - 2 \text{Tr}(UW).$$

For a rank- r orthogonal projection P , its diagonal entries in an orthonormal eigenbasis of C lie in $[0, 1]$ and sum to r . Moving this fixed nonnegative mass to the r smallest or largest eigenvalues proves $\sum_{j=1}^r c_j \leq \text{Tr}(PC) \leq \sum_{j=2q+2-r}^{2q+1} c_j$. Each of U, W is its range projection of rank $q + 1$ plus its eigenvalue-two projection of rank $q - 1$. Applying these two bounds and subtracting the complete sums gives $|\text{Tr}((U - W)C)| \leq S_{\text{spec}}$.

Since $\text{Tr}(U - W) = 0$, subtracting the midpoint of the extremes of C leaves the trace pairing unchanged. In an orthonormal eigenbasis of $U - W$ every resulting diagonal entry of C has absolute value at most $d/2$. Therefore $|\text{Tr}((U - W)C)| \leq S_{\text{tr}}$. The two ranges have dimension $q + 1$ in dimension $2q + 1$, so $s \geq 1$. Let E be their intersection projection. The inequalities $U \succeq E, W \succeq E$ follow from their nonzero eigenvalues being at least one. The two positive differences have trace $2q - s$; their difference is $U - W$. Thus $\|U - W\|_1 \leq 4q - 2s$. Cauchy-Schwarz on its $2q + 1$ real eigenvalues proves the second trace-norm upper bound in S_{ov} . We have proved pointwise

$$|\text{Tr}((U - W)C)| \leq \min\{S_{\text{spec}}, S_{\text{tr}}\} \leq \min\{S_{\text{spec}}, S_{\text{ov}}\}. \quad (\text{AT20})$$

All the matrix entries are smooth on the compact interval. Eigenvalues and trace norms are continuous. The intersection dimension is Borel, since rank strata are specified by vanishing and nonvanishing minors. The bounded nonnegative functions above are consequently integrable.

In the fixed coefficient space, $C(t) = ((1 - t)I + tD_*)^{-1}(D_* - I)$, so

$$c_j(t) = \frac{b_j - 1}{1 - t + tb_j}, \quad \int_0^1 c_j(t) dt = \log b_j, \quad \int_0^1 d(t) dt = \log \kappa_*. \quad (\text{AT21})$$

The derivative of the fraction with respect to b_j is $(1 - t + tb_j)^{-2} > 0$, proving its order through both endpoints. Integrating (AT20) in (AT15) proves the first inequalities of (AT18). Integrating S_{spec} gives $L_q(D_*)$ exactly. Its logarithmic ratios are each between zero and $\log \kappa_*$; their total multiplicity is $1 + 2(q - 1) = 2q - 1$. This proves the last inequality. Substitution of (AT14) gives the additive endpoint consequence; the stronger J_* version of (AT19) is proved in (AT21a)–(AT21e). The preceding coarser $2q \log \kappa_*$ estimate is a corollary because $\log \kappa_* \geq 0$; its weaker coefficient is not used in the current endpoint consequences. $\square$

258.1 The same control in the original nonlinear endpoint coordinate

The additional moment structure is retained in this step. The two endpoint forms are the original positive measures on $S = k/2 + iy$; the proof also applies to a specified positive atomic measure on that line when its degree- $2q$ moment Gram is positive definite. No arbitrary coefficient form is assigned this moment structure. Write $a = 2q - 1$, $B(t) = \mathcal{B}(t)$, and $f_a(B) = a\sqrt{1 - e^{-B/a}}$ for $B > 0$. The exact additional functions are

$$\begin{aligned}
 S_{\sin}(t) &= d(t) \sum_{\ell=1}^a \sqrt{1 - e^{-\lambda_{\ell}(t)}}, & S_{\text{ang}}(t) &= f_a(B(t))d(t), \\
 \Sigma(t) &= \text{Tr}(C(t)^2) - \frac{(\text{Tr } C(t))^2}{2q+1}, \\
 S_{\text{HS}}(t) &= \sqrt{\Sigma(t) \text{Tr}((U(t) - W(t))^2)}, \\
 J_* &= \int_0^1 \min\{S_{\text{spec}}, S_{\text{tr}}, S_{\text{ov}}, S_{\text{HS}}, S_{\sin}, S_{\text{ang}}\} dt, \\
 I_* &= \int_0^1 \min\{d(t), S_{\text{spec}}/f_a(B(t)), S_{\text{tr}}/f_a(B(t)), \\
 &\quad S_{\text{ov}}/f_a(B(t)), S_{\text{HS}}/f_a(B(t)), S_{\sin}/f_a(B(t))\} dt, \\
 H_a(B) &= 2 \text{arcosh}(e^{B/(2a)}), & \Psi_a(z) &= 2a \log \cosh(z/2) \quad (z \geq 0).
 \end{aligned} \tag{AT21a}$$

The λ_{ℓ} are the complete crossed-pair losses specified below, with the forced zero contribution separately retained. The additional centered Hilbert–Schmidt estimate is proved on the same original $M_*(t)$ metric as follows. Since $\text{Tr}(U - W) = 0$, subtract $\text{Tr}(C)I/(2q+1)$ from C inside the trace pairing. Its squared Hilbert–Schmidt norm is exactly $\Sigma(t) \geq 0$. For Hermitian matrices in an $M_*(t)$ -orthonormal basis, $\text{Tr}(AB) = \sum_{i,j} A_{ij} \overline{B_{ij}}$; Cauchy–Schwarz on those finitely many entries proves $|\text{Tr}(AB)| \leq \sqrt{\text{Tr}(A^2) \text{Tr}(B^2)}$. Applying this to the centered C and $U - W$ proves $|B'(t)| \leq S_{\text{HS}}(t)$. The metric is retained on both operators, so this proof changes neither the source nor its arithmetic observation. We prove the complete moment and projection facts used here. For cutoffs $i < j$, regard the original minimum sections R_i, R_j as maps to H . Their difference lies in $\ker J_j$, and R_j is orthogonal to that kernel, so

$$R_i^* M R_j = G_j, \quad G_i - G_j = (R_i - R_j)^* M (R_i - R_j) \succeq 0.$$

The orthonormal frames $X_i = R_i G_i^{-1/2}$, $X_j = R_j G_j^{-1/2}$ have overlap $X_i^* M X_j = G_i^{-1/2} G_j^{1/2}$. Its singular values σ_{ℓ} are positive and at most one, and $\prod_{\ell} \sigma_{\ell}^2 = \det G_j / \det G_i$. To compute the projection difference exactly, choose paired unit vectors with overlap σ_{ℓ} . When $\sigma_{\ell} < 1$, the unit vector $(y_{\ell} - \sigma_{\ell} x_{\ell}) / \sqrt{1 - \sigma_{\ell}^2}$ is orthogonal to the first space. In this two-dimensional plane the two projection matrices differ by a trace-zero matrix of determinant $-(1 - \sigma_{\ell}^2)$, with eigenvalues $\pm \sqrt{1 - \sigma_{\ell}^2}$. An overlap one gives the identical direction and zero difference. Orthogonal complements give further zeros. Thus its trace norm is exactly $2 \sum_{\ell} \sqrt{1 - \sigma_{\ell}^2}$, without changing the source norm.

Apply this to the crossed pairs $(q-1, 2q)$ and $(q, 2q-1)$. Their differences sum to $U - W$ by (AT11)–(AT13). The second pair's two q -dimensional section spaces lie in $\mathcal{P}_{2q-1} \cap (\chi \mathcal{P}_0)^{\perp}$, of dimension $2q-1$; therefore they share a direction. Retain its overlap one in the full second list. The remaining $a = 2q-1$ logarithmic losses $\lambda_{\ell} = -\log \sigma_{\ell}^2$ include every further zero, and their sum is $B(t)$. The trace-norm triangle inequality gives

$$S_{\text{tr}}(t) \leq d(t) \sum_{\ell=1}^a \sqrt{1 - e^{-\lambda_{\ell}}} \leq f_a(B(t))d(t) = S_{\text{ang}}(t). \tag{AT21b}$$

The last inequality follows from concavity of $z \mapsto \sqrt{1 - e^{-z}}$: its second derivative for $z > 0$ is $-e^{-z}/(2\sqrt{1 - e^{-z}}) - e^{-2z}/(4(1 - e^{-z})^{3/2}) < 0$, and continuity includes zero. Jensen's inequality on the full a -entry list proves the displayed bound.

The number $B(t)$ is strictly positive. If it were zero, both nonnegative logarithmic determinant ratios would vanish. All relative Gram eigenvalues are at least one, so each ratio one forces $G_i = G_j$, hence $R_i = R_j$ by the squared norm above. In particular $1 \in \text{im } R_{q-1} = \text{im } R_{2q}$ would be orthogonal to $\chi\mathcal{P}_q$. For $\chi(S) = \sum a_j S^j$ let $\chi^{\#_k}(S) = \sum \bar{a}_j(k - S)^j$. On the actual line this is $\overline{\chi(S)}$. The asserted orthogonality would give $0 = \int |\chi(S)|^2 d\mu_t$, contradicting the positive moment Gram of the nonzero polynomial χ . This proof also covers the specified positive atomic source. Smoothness and compactness now give a positive minimum for $B(t)$, so all denominators in (AT21a) are bounded away from zero on this fixed path.

Since $S_{\text{tr}} \leq S_{\text{ov}}$ and (AT21b) holds, pointwise comparison proves

$$J_* = \int_0^1 \min\{S_{\text{spec}}, S_{\text{tr}}, S_{\text{HS}}\} dt \leq \mathfrak{C}_*, \quad I_* = \int_0^1 \frac{\min\{S_{\text{spec}}, S_{\text{tr}}, S_{\text{HS}}\}}{f_a(B(t))} dt \leq \int_0^1 d(t) dt = \log \kappa_*. \quad (\text{AT21c})$$

Differentiation gives $H'_a(B) = 1/f_a(B) > 0$; substitution gives $\Psi_a(H_a(B)) = B$ and $H_a(\Psi_a(z)) = z$ for $z > 0$. Intersecting the original (AT20) estimate with the proved S_{HS} bound gives $|B'| \leq \min(S_{\text{spec}}, S_{\text{tr}}, S_{\text{HS}})$. It proves $|B(1) - B(0)| \leq J_*$. Applying the exact derivative to that same pointwise bound and integrating proves

$$|H_a(B(1)) - H_a(B(0))| \leq I_*. \quad (\text{AT21d})$$

This proves the recursive nonlinear refinement from the original source maps rather than assuming a stronger endpoint bound. Put $H_0 = H_a(B(0))$ and define the actual current endpoints

$$\begin{aligned} \mathfrak{b}_*^- &= \max\{0, B(0) - J_*, \Psi_a(\max\{0, H_0 - I_*\})\}, \\ \mathfrak{b}_*^+ &= \min\{B(0) + J_*, \Psi_a(H_0 + I_*)\}, \\ \mathfrak{d}_*^- &= \mathfrak{b}_*^- - B(0), \quad \mathfrak{d}_*^+ = \mathfrak{b}_*^+ - B(0). \end{aligned} \quad (\text{AT21e})$$

Monotonicity of Ψ_a on $[0, \infty)$, the proved additive J_* bound, and (AT21d) prove $\mathfrak{b}_*^- \leq B(1) \leq \mathfrak{b}_*^+$ and $\mathfrak{d}_*^- \leq \Delta\mathcal{B} \leq \mathfrak{d}_*^+$. These endpoints refine the earlier additive interval while retaining every mass, angle and spectrum. The case $q = 1$ has $a = 1$, one angle, and the displayed zero second-pair angle. No estimate uniform in k follows from these fixed-path integrals.

When $M_*(1) = cM_*(0)$ with its literal $c > 0$, the exact source form at time t is $(1 - t + tc)M_*(0)$. Each quotient determinant is $(1 - t + tc)^q V_N(0)$, so all four factors in (AT14) cancel and $\Delta\mathcal{B} = 0$. This is also forced by $\kappa_* = 1$ in (AT18). The individual masses and volume factors remain as written; cancellation occurs only in this specified ratio. For an arbitrary fixed source-coordinate change, write old common-source coordinates as T times new coordinates, and include each original section in that source as $\mathbf{R}_N = I_N R_N$. With the quotient frame fixed, the exact transported maps and forms are

$$\begin{aligned} M'_* &= T^* M_* T, & J'_N &= J_N T, \\ \mathbf{R}'_N &= T^{-1} \mathbf{R}_N, & G'_N &= G_N, & V'_N &= V_N. \end{aligned}$$

Here each endpoint source is its transported subspace of the common source, and $J_N T$ denotes the restriction to that subspace. The transported section remains the minimum section because the change is an isometry between the two displayed forms and preserves each affine remainder fibre. Direct substitution then gives $(\mathbf{R}'_N)^* M'_* \mathbf{R}'_N = \mathbf{R}_N^* M_* \mathbf{R}_N = G_N$. If, in addition, old quotient coordinates equal A times new quotient coordinates, then $J'_N = A^{-1} J_N T$, $\mathbf{R}'_N = T^{-1} \mathbf{R}_N A$, and $G'_N = A^* G_N A$, $V'_N = |\det A|^2 V_N$. The same quotient-frame multiplier at all four endpoints cancels in (AT14). The source change conjugates D_* and C , and sends each projection to $T^{-1} P T$; its trace and spectrum therefore remain unchanged. These exact maps preserve (AT15)–(AT21).

For the reflection-stable quartet calculation of the supplied Gamma audit, let its actual displacement be $\delta > 0$ and retain $q = [1 + k(m - 1)](k + 1)^2$. Its fully written Gamma theorem gives $\mathcal{B}_{h,k}^\Gamma \geq 4q \log(\delta k / (2\sqrt{5}))$. Substitution in (AT19) gives the new quantitative comparison on the same source:

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{ar}} &\geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - J_*, \\ \mathfrak{C}_* &\geq J_* \geq \max\left\{0, 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \mathcal{B}_{h,k}^{\text{ar}}\right\}, \\ \log \kappa_* &\geq \frac{1}{2q-1} \max\left\{0, 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \mathcal{B}_{h,k}^{\text{ar}}\right\}. \end{aligned} \tag{AT22}$$

The same original Gamma lower bound also propagates through the nonlinear endpoint coordinate. Set $L_\Gamma = \max\{0, 4q \log(\delta k / (2\sqrt{5}))\}$. The full original Gamma moment argument gives $B(0) \geq L_\Gamma$, and (AT21e) then implies

$$\mathcal{B}_{h,k}^{\text{ar}} \geq \max\{0, L_\Gamma - J_*, \Psi_a(\max\{0, H_a(L_\Gamma) - I_*\})\}, \quad a = 2q - 1. \tag{AT22a}$$

Here $H_a(0) = 0$ is its continuous extension. Each function of the initial value in (AT21e) is increasing, so substituting the proved lower bound proves every displayed entry without replacing the actual path integral I_* . The simpler first-line estimate (AT22) is retained as its weaker corollary.

The quartet and actual packet restriction are retained in this application; a reference test polynomial has not been declared to divide 2ξ . Equations (AT15) and (AT17) give the full signed quantity behind this bound. In particular they provide the exact integrand that an arithmetic cancellation calculation must evaluate, with both relation windows and both polynomial windows present. No subthreshold arithmetic bound or weight purity is asserted by this source comparison alone.

259 Source dependencies and precise status

The complete retained Tau Base note supplies (AT1), its tensor pushforward, right adjoint and supported maps. The original arithmetic-input and cyclic-sum proofs supply (AT2), full packet units and the theta primitives. The Gamma/source-density notes supply the actual measures and finite moments in (AT4)–(AT5). The new Deligne-bound note supplies the full quartet Gamma lower bound used only in (AT22). Each is preserved in the accompanying source closure, with its actual edition and hash.

The new proved calculations here are (AT6)–(AT21) on those unchanged input objects, and their substitution (AT22). They give an analytic calculation on the canonical finite complex mapping to $A_{\tau,k}(\mathcal{T}^{\boxtimes k})$. They neither replace that pushforward by a generic stalk nor deduce a verdict about the entire tau-based program from an auxiliary model.

260 The exact gamma reference and the arithmetic Gram determinants

The norm sequence in (KL.24) is attached to the actual measure

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi. \quad (\text{TG.1})$$

We construct an explicit gamma reference for this measure, prove its polynomial norms, and retain the full arithmetic multiplier in finite Gram determinants. This provides another exact formula for the original boundary allowance; no asymptotic estimate for an arithmetic determinant is assumed.

260.1 The original measure, phase, and packet cancellation

The defining identity for the completed function gives, on the original line $s = 1/2 + it$,

$$\begin{aligned} g(\tfrac{1}{2} + it) &= \Gamma(\tfrac{1}{4} + it/2) a(t), \\ a(t) &= -\pi^{-1/4} (t^2 + \tfrac{1}{4}) e^{-it \log \pi/2} \zeta(\tfrac{1}{2} + it). \end{aligned} \quad (\text{TG.2})$$

Indeed $s(s-1) = -(t^2 + 1/4)$ and $\pi^{-s/2} = \pi^{-1/4} e^{-it \log \pi/2}$. Both the minus sign and this phase remain in the multiplier.

For $\lambda > 0$, define

$$d\sigma_\lambda(t) = |\Gamma(\lambda + it/2)|^2 \frac{dt}{2\pi}. \quad (\text{TG.3})$$

In particular

$$d\mu = |a|^2 d\sigma_{1/4}, \quad d\nu_h = |a_h|^2 d\sigma_{1/4}, \quad a_h(t) = \frac{g(1/2 + it)/h(1/2 + it)}{\Gamma(1/4 + it/2)}. \quad (\text{TG.4})$$

The last expression defines the values at every zero of the original full packet h ; away from those points it equals $a(t)/h(1/2 + it)$. The function g/h is entire because the complete order of each selected zero is removed. Gamma has no zero and has no pole on the real t axis. Thus (TG.4) retains the actual cancellation at each critical packet centre, without introducing a pole into the measure.

Multiplication gives surjective Hilbert isometries

$$\begin{aligned} U_a : L^2(\mu) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto aF, \\ U_{a_h} : L^2(\nu_h) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto a_h F. \end{aligned} \quad (\text{TG.5})$$

The integral identity proves the isometry. The zero set of each multiplier on the real axis is discrete, since it comes from a nonzero analytic function. Given $H \in L^2(\sigma_{1/4})$, define $F = H/a$, respectively H/a_h , off this null set and give F arbitrary finite values on it. Then $\int |F|^2 d\mu = \int |H|^2 d\sigma_{1/4}$, respectively the packet identity, proving surjectivity and the stated inverse. These are maps between the indicated Hilbert spaces. Their images of polynomial subspaces are exactly $a\mathbb{C}[t]$ and $a_h\mathbb{C}[t]$, with the displayed multipliers retained. Moreover the packet isometry from (KL.23) satisfies the exact diagram

$$U_{a_h}(h(1/2 + it)P(t)) = U_a P(t). \quad (\text{TG.6})$$

260.2 Gamma norms from a generating integral

We give the integral derivation to fix all constants independently of different conventions for Meixner–Pollaczek polynomials. The beta integral, with $r = e^x$, gives

$$\int_{\mathbb{R}} \frac{e^{\lambda x}}{(1 + e^x)^{2\lambda}} e^{iux} dx = \frac{\Gamma(\lambda + iu)\Gamma(\lambda - iu)}{\Gamma(2\lambda)}. \quad (\text{TG.7})$$

For completeness the beta integral follows by applying the change of variables $v = r_1 + r_2$, $y = r_1/(r_1 + r_2)$ to the product of the two absolutely convergent gamma integrals; the Jacobian is v . The conditions $\operatorname{Re}(\lambda \pm iu) > 0$ hold for real u . The function on the left before Fourier transformation and its second derivative are integrable: at each end they are bounded by a constant times $e^{-\lambda|x|}$. Two integrations by parts therefore make its Fourier transform integrable. Fourier inversion applies. One direct justification is to multiply the transform by $e^{-\varepsilon u^2}$, apply Fubini, and use the Gaussian approximate identity; dominated convergence removes that factor because the transform is integrable. Using its evenness and then $t = 2u$, we obtain

$$\int_{\mathbb{R}} e^{iyt} d\sigma_{\lambda}(t) = \frac{2\Gamma(2\lambda)}{(2 \cosh y)^{2\lambda}} = 2^{1-2\lambda} \Gamma(2\lambda) (\cosh y)^{-2\lambda}. \quad (\text{TG.8})$$

The factor 2 in the numerator is the Jacobian of the original change $t = 2u$.

The following integral proves the required exponential integrability.

The function $(2 \cosh(x/2))^{-2\lambda}$ extends holomorphically to $|\operatorname{Im} x| < \pi$, using the branch positive on the real axis. On each closed narrower strip it is bounded by $Ce^{-\lambda|\operatorname{Re} x|}$. In (TG.7), for $u > 0$, shift the contour to $\mathbb{R} + ic$, where $0 < c < \pi$. The vertical sides of the truncated rectangle tend to zero by that bound. For $u < 0$, shift down instead. It follows that

$$|\Gamma(\lambda + iu)|^2 \leq C_{\lambda,c} e^{-c|u|}.$$

Consequently $\int e^{v|t|} d\sigma_{\lambda}(t) < \infty$ for every $0 < v < \pi/2$, by choosing $2v < c < \pi$. This proves holomorphic continuation of both sides of (TG.8) to $|\operatorname{Im} y| < \pi/2$, where they agree by the identity theorem and use the branch inherited from the real axis.

For z near zero use the logarithms that vanish at $z = 0$ and set

$$F_{\lambda}(z, t) = (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} = \sum_{n \geq 0} P_n^{(\lambda)}(t/2; \pi/2) z^n. \quad (\text{TG.9})$$

For real z near zero this is $(1 + z^2)^{-\lambda} \exp(t \arctan z)$. Apply (TG.8) at $y = -i(\arctan z + \arctan w)$, where z, w are real and sufficiently small. The identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

then proves

$$\int_{\mathbb{R}} F_{\lambda}(z, t) F_{\lambda}(w, t) d\sigma_{\lambda}(t) = 2^{1-2\lambda} \Gamma(2\lambda) (1 - zw)^{-2\lambda}. \quad (\text{TG.10})$$

All derivatives in z, w at zero can be passed through the integral: on a smaller closed bidisc their integrands are bounded by a polynomial in $|t|$ times $e^{c|t|}$, with $c < \pi/2$. Thus comparison of the Taylor coefficients in (TG.10) is justified, not formal.

Define the monic polynomial and its unchanged squared norm by

$$b_n^{(\lambda)}(t) = n! P_n^{(\lambda)}(t/2; \pi/2), \quad \gamma_n^{(\lambda)} = 2^{1-2\lambda} n! \Gamma(n + 2\lambda). \quad (\text{TG.11})$$

The coefficient of t^n in (TG.9) is $1/n!$, since $\arctan z = z + O(z^3)$. Consequently $b_n^{(\lambda)}$ is monic. Equation (TG.10) gives the full orthogonality identity

$$\int_{\mathbb{R}} b_n^{(\lambda)}(t) b_j^{(\lambda)}(t) d\sigma_{\lambda}(t) = \gamma_n^{(\lambda)} \delta_{nj}. \quad (\text{TG.12})$$

All these polynomials have real coefficients, as is also clear from the real form of the generating function. Differentiating (TG.9) gives $(1 + z^2) \partial_z F_{\lambda} = (t - 2\lambda z) F_{\lambda}$. Its coefficient of z^n proves

$$\begin{aligned} b_0^{(\lambda)} &= 1, & b_1^{(\lambda)} &= t, \\ b_{n+1}^{(\lambda)} &= t b_n^{(\lambda)} - n(n + 2\lambda - 1) b_{n-1}^{(\lambda)}, & n &\geq 1, \\ \gamma_n^{(\lambda)} / \gamma_{n-1}^{(\lambda)} &= n(n + 2\lambda - 1), & n &\geq 1. \end{aligned} \quad (\text{TG.13})$$

For the actual reference $\lambda = 1/4$, abbreviate $b_n = b_n^{(1/4)}$ and $\gamma_n = \gamma_n^{(1/4)}$. Thus

$$\boxed{\gamma_n = \sqrt{2} n! \Gamma(n + \tfrac{1}{2}), \quad \gamma_{n+1}/\gamma_n = (n+1)(n + \tfrac{1}{2}).} \quad (\text{TG.14})$$

These formulas agree, with the displayed Jacobian and monic factors, with NIST DLMF §18.19, equations (18.19.8)–(18.19.9), and §18.22, equation (18.22.8).

260.3 Finite arithmetic determinants in the retained metrics

For $N \geq 0$ let $N = \text{diag}(\gamma_0, \dots, \gamma_N)$, and keep both matrices

$$\begin{aligned} \mathbf{B}_N &= \left(\int_{\mathbb{R}} b_i(t) b_j(t) |a(t)|^2 d\sigma_{1/4}(t) \right)_{0 \leq i, j \leq N}, \\ \mathbf{J}_N &= {}_N^{-1/2} \mathbf{B}_N {}_N^{-1/2}, \quad D_N = \det \mathbf{J}_N, \quad D_{-1} = 1. \end{aligned} \quad (\text{TG.15})$$

The diagonal factors are explicit invertible coordinate maps; $\mathbf{B}_N = {}_N^{1/2} \mathbf{J}_N {}_N^{1/2}$ recovers every original inner product. The integrals converge with no unproved growth estimate. Indeed, integration of the step function $\lfloor x \rfloor$ gives, initially for $\text{Re } s > 1$, and then by analytic continuation for $\text{Re } s > 0$, $s \neq 1$,

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx.$$

The initial equality follows by integrating each constant step on $[n, n+1)$, or by summation by parts of the absolutely convergent Dirichlet series. The last integral converges absolutely and locally uniformly in the asserted half-plane since $0 \leq \{x\} < 1$. On $s = 1/2 + it$, it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4}$, because $|s/(s-1)| = 1$. Equation (TG.2) therefore bounds $|a(t)|$ by a polynomial of degree three. The exponential integrability just proved gives all its polynomial moments. The packet multiplier has the same property on the tail because h is monic, and is bounded on each compact interval by its cancelled analytic expression in (TG.4). Each matrix is positive definite: the integral of the squared modulus of a nonzero polynomial is positive, because a vanishes only at isolated real points.

Theorem 260.1 (The actual theta norms as arithmetic determinants). *For all $n \geq 0$, with the initial value included,*

$$\boxed{\mathfrak{h}_n = \gamma_n \frac{D_n}{D_{n-1}}, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} = (n+1)(n + \tfrac{1}{2}) \frac{D_{n+1} D_{n-1}}{D_n^2}.} \quad (\text{TG.16})$$

Proof. Let $Q_n(t)$ be the monic polynomial orthogonal to degrees below n in $L^2(\mu)$. Its existence and uniqueness follow from the positive definite Gram matrix. Expressing $Q_0, \dots, Q_N$ in $b_0, \dots, b_N$ gives an upper triangular matrix with diagonal entries one. The determinant of the Gram matrix is therefore unchanged, and orthogonality gives $\det \mathbf{B}_N = \prod_{j=0}^N \|Q_j\|_\mu^2$.

The algebra isomorphism $\Psi : \mathbb{C}[s] \rightarrow \mathbb{C}[t]$, $(\Psi p)(t) = p(1/2 + it)$, preserves the original integral inner product. On the affine space of degree- n monic polynomials the map $p \mapsto i^{-n} \Psi p$ is its monic version and preserves the norm because $|i^{-n}| = 1$. It takes all lower-degree polynomial subspaces onto themselves. Thus the original monic theta residual has squared norm $\mathfrak{h}_n = \|Q_n\|_\mu^2$. The degree-dependent phase is used on this affine space; it is not asserted to define a linear map on the entire polynomial algebra.

Finally $D_N = \det \mathbf{B}_N / \prod_{j=0}^N \gamma_j$. Divide this equality at $N = n$ by the equality at $N = n-1$, taking an empty product to be one. This proves the first formula. Dividing consecutive formulas and using (TG.14) proves the second. $\square$

Define $\mathbf{B}_N^{(h)}, \mathbf{J}_N^{(h)}, D_N^{(h)}$ by (TG.15) with the entire-cancelled multiplier a_h of (TG.4). The same proof, with no restriction on whether the packet centres are critical, gives

$$\kappa_n^{(h)} = \gamma_n \frac{D_n^{(h)}}{D_{n-1}^{(h)}}. \quad (\text{TG.17})$$

Combining (TG.17) and (TG.16) with (KL.24) yields the exact packet-to-reference determinant map, for every $m \geq 0$,

$$r_m^{(h)} = \frac{(d+m)! \Gamma(d+m+1/2)}{m! \Gamma(m+1/2)} \frac{D_{d+m}^{(h)}}{D_{d+m-1}^{(h)}} \frac{D_{m-1}}{D_m}. \quad (\text{TG.18})$$

Every determinant is strictly positive and the full degree $d = \sum_{\rho} m_{\rho}$ is retained.

For a packet stable under both ordinary conjugation and $\rho \mapsto 1 - \bar{\rho}$, and for every $m \geq 0$, (KL.32) becomes

$$\epsilon_m^2 = (m+1)(m+\frac{1}{2}) \frac{D_{m+1} D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{TG.19})$$

This is the exact same finite-stage allowance. Its factor $(m+1)(m+1/2)$ is explicit; the arithmetic determinant quotient remains in the formula.

260.4 Full confluent kernel and the original coordinates

Set $\hat{b}_j(s) = i^j b_j((s-1/2)/i)$. These are monic polynomials in the original variable s . With $\Delta_N = \text{diag}(1, i, \dots, i^N)$, their actual packet Gram matrix is

$$\mathbf{B}_{N,s}^{(h)} = \Delta_N^* \mathbf{B}_N^{(h)} \Delta_N. \quad (\text{TG.20})$$

Indeed $\hat{b}_j(1/2 + it) = i^j b_j(t)$; taking the conjugate in the first argument gives exactly the displayed congruence, including every phase.

Let $\mathbf{C}_N$ have columns $[\hat{b}_j]_h \in E_h = \mathbb{C}[s]/(h)$, in the original monomial remainder coordinates. For $N \geq d-1$,

$$K_N = \mathbf{C}_N (\mathbf{B}_{N,s}^{(h)})^{-1} \mathbf{C}_N^*, \quad \mathbf{G}_{N-d} = U_{\varepsilon_h}^* K_N^{-1} U_{\varepsilon_h}. \quad (\text{TG.21})$$

Here $\mathbf{G}_{-1}$ is the fixed initial-section metric, and $\varepsilon_h = (j_h(g/h))^{-1}$ is the same full local unit as before. To prove the first identity, choose the actual monic orthogonal polynomial basis through degree N . If T is its invertible coefficient matrix in the $\hat{b}_j$ basis, then

$$T^* \mathbf{B}_{N,s}^{(h)} T = \text{diag}(\kappa_0, \dots, \kappa_N).$$

Inverting gives $(\mathbf{B}_{N,s}^{(h)})^{-1} = T \text{diag}(\kappa_j^{-1}) T^*$. The columns of $\mathbf{C}_N T$ are the full remainders $[p_j]_h$. Substitution proves exactly $K_N = \sum_{j=0}^N [p_j]_h [p_j]_h^* / \kappa_j$, the original interpolation kernel. The first d monic polynomials span all remainders, so $\mathbf{C}_N$ is onto and K_N is positive definite. The second identity is the fixed-unit minimum-lift formula already proved in (KL.27), with the exact index $N = d+m$. No single remainder column is required to be nonzero or invertible.

For a centre ρ , put $t_{\rho} = (\rho - 1/2)/i$ and write $f^{[\ell]} = f^{(\ell)}/\ell!$. The full Taylor coordinates of these columns are explicitly

$$\hat{b}_j^{[\ell]}(\rho) = i^{j-\ell} b_j^{[\ell]}(t_{\rho}), \quad 0 \leq \ell < m_{\rho}. \quad (\text{TG.22})$$

This follows by differentiating the affine argument ℓ times. CRT converts these full Taylor coordinates to the monomial remainder coordinates, and multiplication by U_{v_h} retains all derivatives of g/h . Thus (TG.21) computes the entire confluent kernel, including nilpotent orders and critical centres.

260.5 The finite map to Romik's gamma parameter

Romik's expansion uses $P_n^{(3/4)}(t/2; \pi/2)$, with the unchanged variable $t/2$; see his §3 and Appendix A.2. The exact finite change to the parameter in (TG.3) is

$$\begin{aligned} b_n^{(1/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{1/2}{k} b_{n-2k}^{(3/4)}(t), \\ b_n^{(3/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{-1/2}{k} b_{n-2k}^{(1/4)}(t). \end{aligned} \quad (\text{TG.23})$$

To prove these formulas, divide the two generating functions: $F_{1/4}(z, t) = (1 + z^2)^{1/2} F_{3/4}(z, t)$. The binomial series at zero and coefficient comparison prove the first formula. Multiplication by $(1 + z^2)^{-1/2}$ proves the inverse. Each finite coefficient matrix is triangular with diagonal one, preserves parity, and is invertible. Applied to the same measure, it transports Gram matrices by its full congruence and transports every Taylor column by the same coefficient matrix. Consequently the kernel expression (TG.21) is invariant under this change of polynomial basis.

The two reference measures themselves have the exact Hilbert-space map

$$L^2(\sigma_{1/4}) \longrightarrow L^2(\sigma_{3/4}), \quad F(t) \longmapsto \frac{\Gamma(1/4 + it/2)}{\Gamma(3/4 + it/2)} F(t). \quad (\text{TG.24})$$

The gamma quotient has neither zeros nor poles for real t . Cancellation of its squared modulus proves the isometry and its inverse; this isometry has a multiplier, whereas (TG.23) is the finite polynomial coordinate map. Their respective domains and metrics have both been given explicitly.

Romik's coefficient asymptotics concern the expansion of Ξ in a fixed gamma-orthogonal basis. Inoue's §2 similarly treats expansion coefficients and convergence. In the present construction the arithmetic norms are the determinants (TG.16). Equations (TG.20)–(TG.24) are the exact maps connecting those polynomial presentations to these determinants; no assertion that their coefficient asymptotics already estimate D_n is used.

260.6 Exact scope of the exponential-weight theorem

The original theorem of Lubinsky–Mhaskar–Saff does include tail order one. In their 1988 paper, Definition 2.1 requires $W = e^{-Q}$, with Q even and continuous, Q' existing on $(0, \infty)$, $xQ'(x)$ bounded at zero, and, eventually,

$$Q' > 0, \quad x^2|Q'''|/Q' \leq C, \quad 1 + xQ''/Q' \longrightarrow \alpha > 0. \quad (\text{TG.25})$$

Theorem 2.3 admits a finite generalized Jacobi factor, a lower-degree polynomial exponential, and a bounded nonnegative factor $\Psi(t) \rightarrow 1$. Its Theorem 2.4 replaces the polynomial exponential by V , but requires $\log V(t)/Q(t) \rightarrow 0$, as well as bounded reciprocal approximation on the expanding interval by polynomials of degree $o(n)$. The precise statements are on printed pages 67–70; their proof is on pages 80–81. These hypotheses concern the full weight, not its gamma factor alone.

We prove a concrete obstruction to inserting the actual arithmetic multiplier into those displayed hypotheses. Hardy's theorem supplies real numbers $T_j \rightarrow \infty$ with $g(1/2 + iT_j) = 0$. Removing a fixed full polynomial packet leaves an infinite subsequence of these zeros. At each remaining T_j , the multiplier a_h of (TG.4) has the local form

$$a_h(t) = (t - T_j)^{\ell_j} v_j(t), \quad \ell_j \geq 1, \quad v_j(T_j) \neq 0, \quad (\text{TG.26})$$

where v_j is analytic in a neighbourhood of T_j . This follows from the Taylor series at an isolated zero, with its entire order ℓ_j retained.

Let $B(t) > 0$ be any continuous comparison amplitude near all sufficiently large T_j , and suppose it has no zeros there. For every j and every $\eta > 0$, continuity and (TG.26) give a nonempty interval $I_{j,\eta}$ about T_j on which

$$|a_h(t)|/B(t) < \eta. \quad (\text{TG.27})$$

Thus a decomposition with this ratio as the bounded multiplier Ψ cannot satisfy $\Psi \rightarrow 1$. The obstruction holds on intervals of positive measure, so changing the value only at each zero does not repair it. A finite generalized Jacobi factor has only finitely many real zeros or singularities; choosing the subsequence beyond them leaves (TG.27) unchanged.

For the logarithmic hypothesis choose any eventually positive continuous Q . At a remaining T_j , shrink the neighbourhood so that $Q(t) \leq Q(T_j) + 1$, and choose $\eta = \exp(-j(Q(T_j) + 1))$ in (TG.27). On the resulting positive-measure interval,

$$\frac{\log(|a_h(t)|/B(t))}{Q(t)} < -j \quad (t \neq T_j). \quad (\text{TG.28})$$

Hence this ratio cannot be V with $\log V/Q \rightarrow 0$. An additional factor $\Psi \rightarrow 1$ is bounded away from zero and infinity on the tail and does not remove the contradiction. This disproves the indicated application of Theorems 2.3–2.4, including almost-everywhere modifications of the weight. It makes no claim that other asymptotic methods are unavailable. The exact positive Gram matrices (TG.15)–(TG.21) retain the arithmetic information at all those zeros and are available for such further analysis.

260.7 Primary source locators

The content-reading route and source hashes are recorded in `work/theta_norm_literature_route_20260912.json` and `work/theta_norm_literature_sources/download_receipt.json`. The principal public locators are:

- D. S. Lubinsky, H. N. Mhaskar and E. B. Saff, *A proof of Freud's conjecture for exponential weights*, Constructive Approximation 4 (1988), 65–83, Definition 2.1, Theorems 2.3–2.5 and §4: <https://math.vanderbilt.edu/saffeb/texts/81.pdf>.
- Dan Romik, *Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases*, §3.1–3.3 and Appendix A.2: <https://www.math.ucdavis.edu/~romik/data/uploads/papers/riemannxi-final.pdf>.
- Hiroto Inoue, *Expansion of Dirichlet L-function on the critical line in Meixner–Pollaczek polynomials*, arXiv:1412.1220, §2.1–2.2: <https://arxiv.org/abs/1412.1220>.
- NIST DLMF, equations (18.19.8)–(18.19.9), (18.22.8): <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22>.
- G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, Comptes rendus 158 (1914), 1012–1014, §1–3. The exact page images and transcriptions are linked in the machine-readable source route.

261 1. Result and exact source boundary

The formalization session's endpoint criterion selects an actual member of the canonical arithmetic family. This continuation proves one of its previously unproved analytic inputs. For the **unchanged arithmetic measure**, a fixed finite packet h , and integers $n \geq k \geq 3$, there is a finite constant C_h , independent of n and k , such that

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n. \quad (1)$$

The norms are the original monic polynomial norms for $m_{h,k} = w_h^{*k}$; the total mass remains μ_h^k . No root-location hypothesis is used in this bound. The proof retains a compact-source mass, a lower convolution constant, and both exponential moments. These constants are defined explicitly below; they have not been numerically enclosed.

For the packet consisting exactly of a hypothetical nonreal off-line quartet, let

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (2)$$

The existing exterior theorem gives $L_{h,k} \leq \epsilon_{h,k,N}$ at every admitted degree. Combining (1) with the other session's endpoint theorem gives

$$\min_{q_k \leq N < 2q_k} \epsilon_{h,k,N} \leq C_h q_k \sinh\left(\frac{\mathcal{B}_{h,k}}{2q_k}\right), \quad \mathcal{B}_{h,k} = \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}}. \quad (3)$$

Consequently such a quartet forces

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh}\left(\frac{\delta k}{2C_h}\right), \quad \liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 2. \quad (4)$$

The upper bound for this four-volume quantity is not proved here. In particular, (1) does not establish RH. It completes the polynomial-norm input in (3) on a specific admissible window and leaves the source/relation-volume comparison with its exact maps. Section 9 expresses that remaining quantity as two positive block log-determinants of actual new relation layers.

The finite quotient, canonical source, residue action and rank-two control are inputs from the preceding analytic notes. The new proofs in Sections 3–7 use only their explicitly defined arithmetic measure. The endpoint theorem is the formalizer's written result, not attributed to this continuation. Its finite telescopes are reported kernel-checked; the analytic estimates below have written proofs, not new Lean certificates.

262 2. The base, source, and both kinds of quotient

Retain the original marked coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (5)$$

Here $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$, $e_R = 0_R^\bullet$, $p_R(\tau) = 0_R$, and $p_R(r^\bullet) = r$. The target of $p_{\mathbb{Z}}$ is infinite. The construction remains over the original pointed base $\mathfrak{b}_\tau$; no new scalar zero is adjoined at a tensor degree or at a polynomial relation layer.

Keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $Q = \mathcal{B}/\Theta V$, and $D = -x\partial_x$. The source spaces and the Fourier transform remain those of the earlier theta construction. In particular,

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}\Theta\phi_* = g. \quad (6)$$

Let h be a fixed finite product of zeros of g , with the full order of each selected zero retained, and write $d = \deg h$. Define

$$v_h = g/h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (7)$$

The quotient v_h is entire, including at the selected centres. It is not the meromorphic expression evaluated before removing its actual removable factors. For $h = 1$, the same analytic definitions are meaningful, but the finite arithmetic packet is zero. Only the nonempty case is used for the finite Gram inverses below.

The source map is

$$\mathcal{V}_{h,k} : \mathbb{C}[S] \longrightarrow \mathcal{B}^{\widehat{\otimes} k}, \quad P \longmapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h. \quad (8)$$

The original cyclic module is $E_{h,k} = \mathbb{C}[S]/(\chi_{h,k})$. Its arithmetic inclusion keeps the Taylor unit:

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h), \quad A_k = \sum_i M_{s_i}. \quad (9)$$

The supplied source identities are

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_{\chi}, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_{\chi}. \quad (10)$$

The actual norm of (8) is

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du. \quad (11)$$

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$, the monic norm is the quotient norm of the leading-coefficient map

$$0 \longrightarrow \mathcal{P}_{n-1} \longrightarrow \mathcal{P}_n \xrightarrow{\text{lc}_n} \mathbb{C} \longrightarrow 0, \quad \omega_{h,k,n} = \inf_{\text{lc}_n P=1} \|\mathcal{V}_{h,k}P\|^2. \quad (12)$$

The arithmetic quotient is the other specified map from the same source:

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_{\chi}} E_{h,k} \longrightarrow 0, \quad q = \deg \chi, \quad (13)$$

for $n \geq q - 1$, with the preceding relation space zero at $n = q - 1$. Multiplication by the monic χ carries $\mathcal{P}_{n-q}/\mathcal{P}_{n-q-1}$ isomorphically to $\mathcal{P}_n/\mathcal{P}_{n-1}$, with leading coefficient unchanged. The map π_{χ} kills that very relation. Thus (12), (13), and multiplication by χ specify the relation between the degree-line norm and the arithmetic quotient metric. One norm is not substituted for the other.

At each admitted original support label λ , the lifted quotient is $(\lambda, P) \mapsto (\lambda, [P]_{\chi})$. An original relation has image $(\lambda, 0)$; external absence has image τ . Its norm before this projection remains available through (11).

263 3. A quantitative local arithmetic mass lemma

263.1 3.1 Uniform polynomial growth on a fixed strip

There is a constant $C_0 \geq 1$ such that, for $|T| \geq 10$, the function

$$f_T(z) = \zeta(1/2 + iT + iz) \quad (14)$$

is holomorphic on and inside the ellipse

$$z = \frac{1}{2}(w + w^{-1}), \quad |w| = 8, \quad (15)$$

and has modulus at most $M_T = C_0(1 + |T|)^6$ there.

The ellipse has real and imaginary semiaxes $65/16$ and $63/16$. Hence its image in the s -plane lies in $-55/16 \leq \Re s \leq 71/16$ and $|\Im s - T| \leq 65/16$. The pole of ζ is outside this set when $|T| \geq 10$.

For $\Re s \geq 1/4$, use the identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (16)$$

It follows by summing the steps of $\lfloor x \rfloor$ for $\Re s > 1$ and then by analytic continuation for $\Re s > 0$, $s \neq 1$. Its integral is bounded by $1/\Re s$, giving $O(1 + |T|)$ on the indicated part of the ellipse. For $\Re s \leq 1/4$, the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s) \quad (17)$$

and the vertical-strip gamma estimate give $O((1 + |T|)^{3/2 - \Re s})$. On the stated strip this exponent is below 5. The common power 6 in (14) is therefore a valid upper bound with one fixed constant. The gamma estimate and functional equation are the classical inputs recorded in DLMF §§5.11 and 25.4; none assumes RH.

Every disc of radius one about a point of $[-1, 1]$ lies inside the ellipse. The maximum principle and Cauchy's derivative formula therefore also give

$$\sup_{x \in [-1, 1]} |f'_T(x)| \leq M_T. \quad (18)$$

263.2 3.2 Propagate a nonzero value to the critical-line interval

Put

$$r_* = (3 + \sqrt{13})/2 < 4, \quad z_* = -3i/2 = \frac{1}{2}(-ir_* + (-ir_*)^{-1}), \quad c_* = \zeta(2)^{-1}. \quad (19)$$

The absolutely convergent reciprocal Dirichlet series gives $|1/\zeta(2 + iT)| \leq \zeta(2)$, so $|f_T(z_*)| \geq c_* > 0$. Let $m_T = \max_{[-1, 1]} |f_T|$.

Apply the maximum principle on $1 \leq |w| \leq 8$ to $f_T((w + w^{-1})/2)$. Equivalently subtract from its logarithmic modulus the harmonic function linear in $\log |w|$ with boundary values $\log m_T$ and $\log M_T$. Zeros are handled by the usual subharmonic maximum principle. At $w = -ir_*$ this gives

$$c_* \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}. \quad (20)$$

Here $0 < c_* < 1$ and $M_T \geq 1$. Consequently

$$m_T \geq c_*^3 M_T^{-2}. \quad (21)$$

This is the three-circles argument with every coordinate map and radius fixed. No polynomial approximation to ζ or assumption about its zeros is used.

Choose a point attaining m_T . By (18), on a one-sided subinterval of length $m_T/(2M_T) \leq 1/2$ inside $[-1, 1]$, the modulus is at least $m_T/2$. Thus

$$\int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq \frac{m_T^3}{8M_T} \geq \frac{c_*^9}{8C_0^7} (1 + |T|)^{-42}. \quad (22)$$

For $T \in [-10, 10]$, the integral in (22) is continuous and strictly positive: otherwise ζ would vanish on an interval and hence identically on its holomorphic domain. Its minimum on that compact interval is positive. Combining it with (22) proves the all-real statement

$$\boxed{\exists c_\zeta > 0 \quad \int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq c_\zeta (1 + |T|)^{-42} \quad (T \in \mathbb{R}).} \quad (23)$$

The exponent 42 is deliberately an explicit sufficient exponent, not an optimal local mean-square estimate. The proof is a classical analytic propagation argument specialized here to the needed arithmetic interval.

264 4. A lower envelope for the actual convolution, after three factors

264.1 4.1 Local mass of the unchanged g/h

The gamma estimate, with a positive lower constant valid also on compact sets, implies

$$|\Gamma(1/4 + it/2)|^2 \geq c_\Gamma (1 + |t|)^{-1/2} e^{-\pi|t|/2}. \quad (24)$$

Let $H_h = \prod_{\rho \in Z} (1 + |\rho - 1/2|)^{m_\rho}$, so $|h(1/2 + it)| \leq H_h (1 + |t|)^d$. The entire identity $g = hv_h$ makes this bound valid at selected centres as well: $|g| \leq H_h (1 + |t|)^d |v_h|$. The factors in (6), including $|s(s-1)|^2 = (t^2 + 1/4)^2$, then give a positive constant $c_h^{(0)}$ with

$$w_h(t) \geq c_h^{(0)} e^{-\alpha_0|t|} (1 + |t|)^{-2d} |\zeta(1/2 + it)|^2, \quad \alpha_0 = \pi/2. \quad (25)$$

Equation (25) is a weaker bound than the available factor $(1 + |t|)^{7/2-2d}$; it follows by an inequality, not by changing the density. On $[T-1, T+1]$, its exponential and polynomial weights are bounded below by $e^{-\alpha_0} e^{-\alpha_0|T|}$ and $2^{-2d} (1 + |T|)^{-2d}$, respectively. Hence (23) proves

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\alpha_0|T|} (1 + |T|)^{-B_h}, \quad B_h = 42 + 2d. \quad (26)$$

264.2 4.2 The positive compact convolution factor

The function w_h is continuous, bounded, integrable, and positive outside a discrete set. Thus $m_{h,2} = w_h * w_h$ is continuous and strictly positive at every real point. Indeed its integrand is positive almost everywhere; continuity follows from translation continuity in L^1 and boundedness of the other factor. Define the **actual** compact quantities

$$b_h = \min_{|v| \leq 1} m_{h,2}(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0. \quad (27)$$

Both are determined by the original density; neither is assigned the value one. Restricting a non-negative convolution integral gives

$$m_{h,3}(u) \geq b_h \int_{u-1}^{u+1} w_h(t) dt \geq c_h e^{-\alpha_0|u|} (1 + |u|)^{-B_h}, \quad c_h = b_h c_h^{(1)} > 0. \quad (28)$$

The omitted part is the nonnegative integral over $|v| > 1$ and remains part of $m_{h,3}$. We use a lower bound, not a new compactly supported density.

For $k \geq 3$, restrict the remaining $k - 3$ variables to $[-1, 1]$ and use (28). Their sum has absolute value at most $k - 3$. This gives the central analytic estimate

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(|u|+k-3)} (1 + |u| + k - 3)^{-B_h} \quad (k \geq 3, u \in \mathbb{R}). \quad (29)$$

All mass and tensor factors are explicit. For $k = 3$ the factor ϑ_h^0 is the empty product. No statement for $k = 1, 2$ is inferred by using a negative exponent. The amplification below only needs $k \geq 3$.

265 5. Upper moments and exact lower monic norms

Fix $0 < b < \pi/2$ and retain both one-factor Laplace values

$$M_h(\pm b) = \int e^{\pm bt} w_h(t) dt > 0. \quad (30)$$

They are finite by the original gamma decay. Fubini and positivity give

$$\int e^{b|u|} m_{h,k}(u) du \leq M_h(b)^k + M_h(-b)^k. \quad (31)$$

The elementary exponential-series inequality $x^j \leq j! b^{-j} e^{bx}$ for $x \geq 0$ yields

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k). \quad (32)$$

The trial polynomial is $(S - k/2)^j$, whose value on $S = k/2 + iu$ is $i^j u^j$. Its phase has modulus one and remains in the source map; the generator $A = M_S$ has not been shifted to define a different spectral problem.

For every leading coefficient of modulus one, the exact least squared norm of a degree- j polynomial on $[-L, L]$ with Lebesgue measure is

$$\mathfrak{l}_j(L) = \frac{2L^{2j+1}}{2j+1} \left(\frac{2^j (j!)^2}{(2j)!} \right)^2. \quad (33)$$

To check all constants, Rodrigues' formula gives the Legendre polynomial $P_j(x)$ with leading coefficient $(2j)!/(2^j (j!)^2)$ and squared norm $2/(2j+1)$. Its scaled monic version on $[-L, L]$ is the unique minimizer: subtract it from any competing monic polynomial, and the difference has lower degree and is orthogonal to it. For the source coordinate $S = k/2 + iu$, multiply this minimizer by the retained leading phase i^j . This proves (33) at the original type. See also DLMF §§18.3 and 18.5.

Restricting (11) to $[-L, L]$ and using (29) proves

$$\omega_{h,k,j} \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(L+k-3)} (1 + L + k - 3)^{-B_h} \mathfrak{l}_j(L) \quad (k \geq 3, L > 0). \quad (34)$$

The observation has the explicit isometric splitting into the inside and outside interval components. Addition recovers the full source function, and their squared norms add. In the split lift each

zero component is the supported zero at its label. The outside component has not been declared absent.

266 6. The norm-window estimate is proved

Set $j = n$ and $L = n$ in (34), and use $j = 2n$ in (32). The exact finite bound is

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq \left[\frac{(4n)! b^{-4n} (M_h(b)^k + M_h(-b)^k) e^{\alpha_0(n+k-3)} (1+n+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2n)}. \quad (35)$$

For $n \geq k \geq 3$, this is at most $C_h n$. Here is one explicit sufficient constant, with $M_* = \max\{1, M_h(b), M_h(-b)\}$:

$$C_h = 64b^{-2} \exp(\alpha_0) 2^{B_h/2} \exp\left(\frac{B_h}{2 \exp(1)}\right) \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}. \quad (36)$$

Proof of the uniform estimate: $\binom{2n}{n} \leq 4^n$ in (33) gives $\mathfrak{l}_n(n) \geq 2n^{2n+1}/((2n+1)4^n)$. Also $(4n)! \leq (4n)^{4n}$, $M_h(b)^k + M_h(-b)^k \leq 2M_*^k$, $n+k-3 \leq 2n$, and $1+n+k-3 \leq 2n$. After taking the $2n$ -th root the factorial and monic factors contribute at most $64n$. The mass factors contribute at most the square root in (36); the exponential contributes at most $\exp(\alpha_0)$. Finally $n^{1/n} \leq \exp(1/\exp(1))$ gives $(2n)^{B_h/(2n)} \leq 2^{B_h/2} \exp(B_h/(2 \exp(1)))$. This proves (1).

The constant is intentionally not optimized. Its dependence on the fixed packet is allowed; its independence of the polynomial and tensor degrees is what is needed. Defining it via positive compact minima does not claim to have provided a numerical interval enclosure for those minima.

For a general source window the same two inequalities give, without asymptotic notation,

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b)^k + M_h(-b)^k) e^{\alpha_0(L+k-3)} (1+L+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(L)} \right]^{1/(2r)}. \quad (37)$$

267 7. Compose the proved bound with the formalizer's endpoint theorem

For a nonempty cyclic packet $E_{h,k}$, let G_N be its original canonical quotient metric, $V_N = \det G_N > 0$, and $\epsilon_{h,k,N}$ the original rank-two control allowance. The new endpoint theorem supplied by PR #23 is

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \sinh\left(\frac{1}{2r} \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}\right), \quad n \geq q. \quad (38)$$

Its proof multiplies the local Toda bounds, applies concavity of $\log \sinh x$, and uses the two exact telescopes. The zero-contraction case is handled before taking logarithms. This result is credited to the parallel formalization/research session. The proof and the distinction between checked finite components and written Jensen assembly are retained in its source.

Choose $n \geq \max(q, k)$ and $r = n$. Equations (1) and (38) yield the actual canonical selection estimate

$$\min_{n \leq N < 2n} \epsilon_{h,k,N} \leq C_h n \sinh \left(\frac{1}{2n} \log \frac{V_{n-1} V_n}{V_{2n-1} V_{2n}} \right). \quad (39)$$

The chosen index is a member of the existing canonical family. The source representative, Taylor unit, kernel, action and trace are not chosen again after selecting this index.

For the packet consisting exactly of the quartet in (2), $q_k \geq k$ and $L_{h,k}/q_k \geq \delta k/2$. The earlier exterior bound holds at every admitted index, including the one selected by (39). Since $\sinh$ is increasing on $[0, \infty)$, (4) follows. Moreover $\operatorname{arsinh}(ak) = \log k + O_a(1)$ for every fixed $a > 0$, which proves the stated threshold 2.

A proved upper bound with $\limsup \mathcal{B}_{h,k}/(q_k \log k) < 2$ would contradict (4). Such an upper bound is **not** an output of this note. The previous sufficient target $\mathcal{B}_{h,k} = o(q_k \log k)$ remains sufficient, but (4) specifies the larger quantitative threshold against which a future upper estimate can be compared. No repeated cost-free exterior operation is used: the multiplicity correction in the new formalization is retained.

268 8. The remaining four volumes retain their finite jet maps

The previous source/relation determinant identity remains

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (40)$$

In the parallel full-jet transfer coordinates (PR #24), let $a = N - q + 1$, let F_a be its $2q$ by $2q$ raw confluent-jet matrix, and let V be its raw-derivative Vandermonde matrix, with every factorial retained. Its established identity is

$$V_N = \left(\prod_{j=a}^{a+q-1} \omega_{h,k,j} \right) \frac{\det V}{\det F_a}. \quad (41)$$

For the window $n = r = q$, the four transfer indices are exactly $0, 1, q, q + 1$. Put $\Omega_j = \prod_{i=j}^{j+q-1} \omega_{h,k,i}$. Then

$$\exp \mathcal{B}_{h,k} = \frac{\Omega_0 \Omega_1}{\Omega_q \Omega_{q+1}} \frac{\det F_q \det F_{q+1}}{\det F_0 \det F_1}. \quad (42)$$

The common Vandermonde cancels in this explicit ratio, not inside any underlying map. The transfer matrices and their inverses still need their actual arithmetic control; the fixed width alone gives no such bound.

The empty packet has $q = 0$, $V_N = 1$ as an empty determinant, and no positive-rank inverse or quartet estimate. The analytic inequalities (23)–(37) still apply to its theta seed, but are not statements about a nonexistent finite packet.

269 9. The volume window is also two explicit retained-boundary costs

There is a second useful representation of the same remaining quantity, with no reference-density change. Fix $i < j$, both admissible. In the original monic polynomial basis put

$$F_{i,j} = [b_{i+1}, \dots, b_j], \quad \Omega_{i,j} = \operatorname{diag}(\omega_{i+1}, \dots, \omega_j), \quad b_a = [p_a]_\chi. \quad (43)$$

The kernel update is the actual sum $K_j = K_i + F_{i,j}\Omega_{i,j}^{-1}F_{i,j}^*$, where $K_i = G_i^{-1}$. The matrix determinant lemma gives

$$\boxed{\frac{V_i}{V_j} = \det\left(I_{j-i} + \Omega_{i,j}^{-1}F_{i,j}^*G_iF_{i,j}\right)}. \quad (44)$$

This determinant has positive real value. Its factors are $1 + \lambda_a$, where $\lambda_a \geq 0$ are the generalized eigenvalues of the Hermitian pair $(F_{i,j}^*G_iF_{i,j}, \Omega_{i,j})$. No non-Hermitian matrix is asserted positive in an unspecified Euclidean metric.

Let $T_{i,j}$ have columns $\mathcal{V}_{h,k}p_{i+1}, \dots, \mathcal{V}_{h,k}p_j$. The original boundary-layer isomorphism is

$$\mathfrak{b}_{i,j} = T_{i,j} - R_iF_{i,j} : \mathbb{C}^{j-i} \xrightarrow{\sim} \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathfrak{b}_{i,j}^*\mathfrak{b}_{i,j} = \Omega_{i,j} + F_{i,j}^*G_iF_{i,j}. \quad (45)$$

Its jets vanish; its highest polynomial coefficients give injectivity. Any new boundary can be reduced by these highest coefficients to an old boundary, and orthogonality then gives surjectivity. These statements prove the displayed map with the same original relation spaces, not only a dimension count. The representative update is

$$R_j = R_i + \mathfrak{b}_{i,j}(\Omega_{i,j} + F_{i,j}^*G_iF_{i,j})^{-1}F_{i,j}^*G_i. \quad (46)$$

Every correction in (46) is an original theta boundary. Its split quotient image is e at the receiving fibre, and the input relation remains available in (45). Its cochain primitives are the same fixed-order divisions and tensor signs as in the previous cyclic source construction.

Apply (44) to $(i, j) = (q-1, 2q-1)$ and $(q, 2q)$. This proves

$$\boxed{\mathcal{B}_{h,k} = \sum_a \log(1 + \lambda_a^{(0)}) + \sum_a \log(1 + \lambda_a^{(1)})}. \quad (47)$$

In particular

$$\mathcal{B}_{h,k} \leq \text{Tr}(\Omega_{q-1,2q-1}^{-1}F^*G_{q-1}F) + \text{Tr}(\Omega_{q,2q}^{-1}F^*G_qF), \quad (48)$$

where each F has its own indicated source and target from (43). When this trace bound is too large, (47) retains the full logarithmic cost rather than replacing it by its linear upper bound. The analytic norm estimate in (1) does not bound (47); it supplies the other factor in (39).

270 10. Deligne reading and what has actually been transferred

The six supplied S20 parts were reassembled in this turn. Their individual chunk hashes matched, and the assembled archive has 216,580,466 bytes and SHA-256 e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8. This verifies the supplied bytes, not every transcription claim in them.

I read the TeX for Definition 1.3.5, Lemmas 3.2.10–3.2.12, the tensor argument 3.2.13, and Corollary 3.3.6 in the nested source files. I compared those exact passages with the supplied top-level French/English frozen page records at printed pages 158, 203, and 206. No PDF text extraction or OCR was used. The source review records a genuine inherited transcription fault: old `fr_seg2.tex` writes a weight bound ≤ 2 at 3.2.10, whereas the controlling page record says < 2 ; only the latter, with integrality, implies ≤ 1 . The old file also corrupts the explicit squared-eigenvalue step in 3.2.13. Those old files remain unmodified; they are not used as authority for these comparisons. No claim of a full independent audit of Weil II is made.

Deligne’s determinant/rank operation and the upper/dual-lower architecture remain attached to the programme’s specified exterior and residue maps. Here the additional estimate is a characteristic-zero analytic estimate for the actual source, (1), followed by the endpoint selection (39). This is not a statement that the new base satisfies Deligne’s finite-field hypotheses. The remaining volume estimate has not been inferred from purity, positivity of a Gram, or the mere existence of the support.

The new result is therefore a completed analytic component of the current route: its two endpoint monic norms have the required $O_h(q_k)$ scale on the actual quartet window. The other session’s four-volume theorem now leaves the explicitly specified relation-volume cost as the remaining quantitative input on that window.

271 11. Proof and computation status

Sections 3–7 contain the new written analytic argument. Sections 8–9 contain explicit finite comparisons and a handoff for checking them using the existing quotient and matrix library. The standard complex-analysis, gamma and Legendre facts used in the argument are proved to the extent needed or referenced below. No general novelty claim is made for these classical ingredients.

The accompanying checker tests finite exact identities: coordinate factors, Legendre extremal norms, the local algebra in the interval argument, convolution moments on declared Laplace models, source quotients, block updates, raw confluent-jet endpoint factors, the volume telescope, and retained support cases. These tests do not certify the analytic inequalities by sampling them. There is no new Lean certificate and no interval-certified value of C_h .

This continuation makes no remote changes. The integration files are add-only, and the parallel formalization branches and the reader’s existing sources are left untouched.

272 References and pinned inputs

- Deligne, P. *La conjecture de Weil. II*. Publications Mathématiques de l’IHÉS **52** (1980), 137–252. Supplied S20 source and current page-local records; the selected reading and discrepancies are itemized in SOURCE_REVIEW.md.
- NIST Digital Library of Mathematical Functions: §§25.4 (zeta functional equation), 5.11 (gamma estimates), 18.3 and 18.5 (Legendre polynomials). Public source pages consulted: <https://dlmf.nist.gov/25.4>, <https://dlmf.nist.gov/5.11>, <https://dlmf.nist.gov/18.3>, <https://dlmf.nist.gov/18.5>.
- Original programme: the delivered *Tau Gamma Convolution Descent*, *Tau Toda Volume Control*, *Tau Cyclic Sum Control*, and *Tau Exterior Trace Amplification* notes. Their definitions of the source, action, unit and canonical quotient are retained; this is not a complete replay of all their upstream analytic claims.
- Parallel endpoint criterion: PR #23, c720f40530eed2f5969dbabe94dfd3fddc0f507f, workbenches/tau-toda-volume-formal/ENDPOINT_CRITERION.md.
- Parallel confluent transfer: PR #24, dfcbba5cbf7fec8c9301fe242c13e741d762013e, workbenches/tau-confluent-transfer/RESEARCH_NOTE.md.
- Inspected main: b32ca2128e0deb0eec6eafb860776a5d6a28dcbb, after its merge of the original-Gram exterior/trace contribution.

273 Actual boundary tilts and uniform convolution tails

This calculation retains the original arithmetic density and the complete hypothetical actual quartet. It does not replace that density with a reference measure. The estimates do not assume RH or a pointwise lower bound for zeta.

273.1 1. Objects and the exact retained expression

Fix $0 < \delta < 1/2$, $\gamma > 0$, and an integer $m \geq 1$, and suppose the selected full-order quartet is a quartet of zeros of the original $g = 2\zeta$. Its monic polynomial is

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - \tfrac{1}{2} - \epsilon\delta - i\eta\gamma)^m, \quad \deg h = 4m. \quad (\text{BT1})$$

The entire function g/h , its original Taylor unit, source maps, and measure remain unchanged:

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad \alpha = \frac{\pi}{2}, \quad R_0 = \sqrt{\delta^2 + \gamma^2}. \quad (\text{BT2})$$

The original completion formula gives, exactly,

$$\begin{aligned} w_h(t) &= \frac{(t^2 + 1/4)^2}{\sqrt{\pi}} \frac{|\zeta(1/2 + it)|^2}{|h(1/2 + it)|^2} \sigma(t), \\ \sigma(t) &= \frac{|\Gamma(1/4 + it/2)|^2}{2\pi}, \\ |h(1/2 + it)|^2 &= [((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2)]^{2m}. \end{aligned} \quad (\text{BT3})$$

The two factors with each fixed imaginary sign contribute a minus sign before multiplication, so their product is the positive expression in the last line. Every denominator in (BT3) is positive. Conjugation of the original g and the real coefficients of h give $w_h(-t) = w_h(t)$.

273.2 2. A centered partial-sum bound from the original floor formula

For every integer $N \geq 1$, initially for $\Re s > 1$,

$$\zeta(s) = \sum_{n=1}^N n^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty \{x\} x^{-s-1} dx. \quad (\text{BT4})$$

To verify (BT4), integrate the constant value $\{x\}$ on each interval $[n, n+1)$, or telescope integration by parts of the absolutely convergent tail. The last integral is absolutely and locally uniformly convergent for $\Re s > 0$, so the identity continues to that half-plane with $s \neq 1$.

Let $B_1(x) = \{x\} - 1/2$, so $|B_1(x)| \leq 1/2$. The identity $\int_N^\infty x^{-s-1} dx = N^{-s}/s$ gives exactly

$$\zeta(s) = \sum_{n=1}^{N-1} n^{-s} + \frac{1}{2} N^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty B_1(x) x^{-s-1} dx. \quad (\text{BT5})$$

No asymptotic remainder is substituted here.

On $s = 1/2 + it$, put $\varrho = \sqrt{t^2 + 1/4}$, which is both $|s|$ and $|s-1|$. For $N \geq 2$,

$$\sum_{n=1}^{N-1} n^{-1/2} + \frac{1}{2} N^{-1/2} \leq 2\sqrt{N-1} - 1 + \frac{1}{2\sqrt{N}} \leq 2\sqrt{N} - 1.$$

The first inequality follows from the integral bound for the decreasing function $x^{-1/2}$. For the second, $2(\sqrt{N} - \sqrt{N-1}) = 2/(\sqrt{N} + \sqrt{N-1}) \geq 1/\sqrt{N}$. At $N = 1$, the same asserted inequality holds directly, with left side $1/2$. The absolute integral in (BT5), before its multiplier s , is at most $\frac{1}{2} \int_N^\infty x^{-3/2} dx = N^{-1/2}$. Therefore

$$|\zeta(1/2 + it)| \leq 2\sqrt{N} - 1 + \frac{\sqrt{N}}{\varrho} + \frac{\varrho}{\sqrt{N}} \quad (t \in \mathbb{R}, N \geq 1). \quad (\text{BT6})$$

The infimum over these explicitly specified integer choices is also a bound.

For $x = |t| \geq 1$, choose $N = \lfloor x \rfloor$. Then

$$x/2 \leq N \leq x, \quad \varrho \leq \frac{\sqrt{5}}{2}x, \quad \frac{\sqrt{N}}{\varrho} \leq x^{-1/2}.$$

Since $-1 + x^{-1/2} \leq 0$, (BT6) yields

$$|\zeta(1/2 + it)| \leq Z_* \sqrt{|t|} \quad (|t| \geq 1), \quad Z_* = 2 + \sqrt{5/2}. \quad (\text{BT7})$$

This improves the polynomial exponent in the uncentered estimate used in TG.15. It is not a claim that (BT7) is the strongest known zeta bound. On the whole real line the original $N = 1$ uncentered formula gives independently

$$|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4}. \quad (\text{BT8})$$

273.3 3. Explicit two-sided Gamma bounds

The complete Binet calculation included below proves

$$c_\Gamma e^{-\alpha|t|} (1 + |t|)^{-1/2} \leq \sigma(t) \leq C_\Gamma e^{-\alpha|t|} (1 + |t|)^{-1/2} \quad (t \in \mathbb{R}), \quad (\text{BT9})$$

where

$$c_\Gamma = \sqrt{2} e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}} e^{2/3}. \quad (\text{BT10})$$

For completeness, its exact starting identity for $\Re z > 0$ is

$$\log \Gamma z = (z - \tfrac{1}{2}) \log z - z + \tfrac{1}{2} \log(2\pi) + R(z), \quad R(z) = \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}.$$

The logarithmic branch is the one real on the positive real axis. The kernel is $(\coth x - 1/x)/(4x)$, with $x = u/2 > 0$. Define

$$F(x) = x \cosh x - \sinh x, \quad G(x) = (x^2 + 3) \sinh x - 3x \cosh x.$$

Both vanish at zero, and $F'(x) = x \sinh x \geq 0$, $G'(x) = xF(x) \geq 0$. Dividing their nonnegative values by the appropriate positive factors gives $0 \leq \coth x - 1/x \leq x/3$. Thus the kernel lies between zero and $1/12$. At $z = 1/4 + it/2$, integration of $e^{-u/4}/12$ proves $|R(z)| \leq 1/3$.

Twice the real part of Binet's identity, with $x = |t|$, gives exactly

$$\sigma(t) = |z|^{-1/2} \exp[-x \arctan(2x) - 1/2 + 2\Re R(z)].$$

For $x > 0$,

$$-x \arctan(2x) - \tfrac{1}{2} = -\alpha x + x \arctan(1/(2x)) - \tfrac{1}{2} \leq -\alpha x,$$

because $\arctan v \leq v$ for $v \geq 0$. At $x = 0$, the same inequality is $-1/2 \leq 0$. The lower bound uses $\arctan(2x) \leq \pi/2$. Hence the full exponent is between $-\alpha x - 7/6$ and $-\alpha x + 2/3$, including zero.

Finally $|z|^2 = 1/16 + x^2/4$, and the exact squared differences

$$\frac{(1+x)^2}{4} - |z|^2 = \frac{3+8x}{16}, \quad |z|^2 - \frac{(1+x)^2}{20} = \frac{(4x-1)^2}{80}$$

give

$$\sqrt{2}(1+x)^{-1/2} \leq |z|^{-1/2} \leq \sqrt{2\sqrt{5}}(1+x)^{-1/2}.$$

These prove (BT9)–(BT10) with no unproved Stirling remainder and no unquantified compact Gamma minimum.

273.4 4. A global explicit actual boundary-tilt bound

Set

$$p = 8m - \frac{9}{2}, \quad \beta = p - \frac{1}{2} = 8m - 5, \quad T_h = \max(1, 2R_0). \quad (\text{BT11})$$

Then $p \geq 7/2 > 1$ and $\beta > 0$. Define

$$\begin{aligned} C_{\text{comp}} &= \frac{C_\Gamma}{\sqrt{\pi}} (T_h^2 + 1/4)^2 [1 + 2\sqrt{T_h^2 + 1/4}]^2 \delta^{-8m} (1 + T_h)^\beta, \\ C_{\text{tail}} &= \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} \left(\frac{4}{3}\right)^{4m} \left(1 + \frac{1}{T_h}\right)^\beta, \\ C_h^{\text{tilt}} &= \max(C_{\text{comp}}, C_{\text{tail}}). \end{aligned} \quad (\text{BT12})$$

These constants are explicit in the fixed quartet and elementary constants. None is an unevaluated zeta supremum, and none changes the source mass.

For $|t| \leq T_h$, (BT3) has $|h(1/2 + it)|^2 \geq \delta^{8m}$. Apply (BT8) and the upper inequality of (BT9). Multiplying by $e^{\alpha|t|}(1+|t|)^p$ leaves the factor $(1+|t|)^\beta$; bounding $|t|$ by T_h proves the constant C_{comp} .

For $x = |t| \geq T_h$, retain the exact identity

$$((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2) = (t^2 - R_0^2)^2 + 4\delta^2 t^2 \geq \frac{9}{16} t^4. \quad (\text{BT13})$$

The last inequality uses $t^2 \geq 4R_0^2$. Thus

$$|h(1/2 + it)|^2 \geq (3/4)^{4m} x^{8m}, \quad (t^2 + 1/4)^2 \leq (25/16)x^4.$$

Applying (BT7) and (BT9) to (BT3) gives

$$e^{\alpha x} w_h(t) \leq \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} \left(\frac{4}{3}\right)^{4m} x^{5-8m} (1+x)^{-1/2}.$$

Multiplication by $(1+x)^p$ leaves $((1+x)/x)^\beta \leq (1+1/T_h)^\beta$. This proves the global bound

$$\boxed{0 \leq e^{\alpha|t|} w_h(t) \leq C_h^{\text{tilt}} (1+|t|)^{-p} \quad (t \in \mathbb{R}).} \quad (\text{BT14})$$

The exact expression (BT3), including its actual zeta numerator and complete quartet denominator, is the object bounded throughout.

273.5 5. Boundary moments and finite derivative orders

Define the original two boundary tilts and their actual masses:

$$f_\pm(t) = e^{\pm\alpha t} w_h(t), \quad M_\pm = \int_{\mathbb{R}} f_\pm(t) dt. \quad (\text{BT15})$$

The nonzero entire function g/h has only isolated zeros on the integration line, so both masses are positive. Evenness gives $M_+ = M_- =: M_\alpha$. Integration of (BT14) proves

$$0 < M_\alpha = M_h(\alpha) = M_h(-\alpha) \leq \frac{2C_h^{\text{tilt}}}{p-1} < \infty. \quad (\text{BT16})$$

For every real $j \geq 0$ with $j < p-1$, the inequality $|t|^j \leq (1+|t|)^j$ gives

$$\int_{\mathbb{R}} |t|^j e^{\pm \alpha t} w_h(t) dt \leq \frac{2C_h^{\text{tilt}}}{p-j-1}. \quad (\text{BT17})$$

More precisely, for $R \geq 0$, the part $|t| \geq R$ is at most

$$\frac{2C_h^{\text{tilt}}}{p-j-1} (1+R)^{j-p+1}. \quad (\text{BT18})$$

Thus every integer order $0 \leq j \leq 8m-6$ is absolutely integrable at both boundary tilts. For a simple quartet, this includes orders zero, one and two.

For the actual Laplace function $M_h(z) = \int e^{zt} w_h(t) dt$, differentiation in $|\Re z| < \alpha$ gives $\int t^j e^{zt} w_h(t) dt$. For each stated order, (BT14) dominates the absolute integrand uniformly on $|\Re z| \leq \alpha$ by the integrable function $C_h^{\text{tilt}} |t|^j (1+|t|)^{-p}$. Dominated convergence therefore extends those derivatives continuously to the closed strip with exactly the boundary moments above. This does not assert holomorphic continuation through the boundary or finiteness of every endpoint derivative.

273.6 6. A uniform convolution envelope with the literal tilted mass

If a nonnegative integrable function f has actual mass $M > 0$ and $f(x) \leq C(1+|x|)^{-p}$, then for every integer $k \geq 1$,

$$f^{*k}(u) \leq kCM^{k-1}(1+|u|/k)^{-p}. \quad (\text{BT19})$$

For $k=1$ this is the hypothesis. In the convolution integral with $x_1 + \cdots + x_k = u$, at least one index has $|x_i| \geq |u|/k$. Bound its integrand by the sum of the k restrictions to these events. On the event for index i , the factor $f(x_i)$ is at most $C(1+|u|/k)^{-p}$. Parametrize the convolution hyperplane by the other $k-1$ coordinates; the change from any such convolution parametrization to another has absolute determinant one. Dropping the event now leaves the full product integral M^{k-1} . Summing proves (BT19), without a probability normalization.

Multiplication inside the original convolution gives exactly

$$f_{\pm}^{*k}(u) = e^{\pm \alpha u} m_{h,k}(u), \quad m_{h,k} = w_h^{*k}. \quad (\text{BT20})$$

Use the plus tilt for $u \geq 0$, the minus tilt for $u < 0$, and their same actual mass M_α . The result is

$$m_{h,k}(u) \leq kC_h^{\text{tilt}} M_\alpha^{k-1} e^{-\alpha|u|} (1+|u|/k)^{-p} \quad (k \geq 1, u \in \mathbb{R}). \quad (\text{BT21})$$

All dependence on k is displayed. The constants $C_h^{\text{tilt}}, p, M_\alpha$ belong to fixed h .

For every integer $n \geq 0$ with $2n > p-1$, use $(1+u/k)^{-p} \leq k^p u^{-p}$ for $u > 0$. The Gamma integral gives

$$\int_{\mathbb{R}} u^{2n} m_{h,k}(u) du \leq 2C_h^{\text{tilt}} k^{p+1} M_\alpha^{k-1} \frac{\Gamma(2n-p+1)}{\alpha^{2n-p+1}}. \quad (\text{BT22})$$

The condition $2n > p-1$ is precisely what makes this comparison integrable at zero. The actual raw moments exist also when the condition fails, but (BT22) is not used there.

For the retained source comparison, one direct corollary is

$$m_{h,k}(u) \leq b_k \sigma(u), \quad b_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1} > 0. \quad (\text{BT23})$$

Indeed $1 + |u|/k \geq (1 + |u|)/k$; (BT21) and the lower Gamma bound leave the additional factor $(1 + |u|)^{-p+1/2} \leq 1$. This is a proved identity-map norm comparison; it does not assert equality of the measures.

274 Internal derivation of the exact Binet identity

This note proves the exact identity used in the actual boundary-tilt calculation from the defining Gamma integral, elementary integration, locally uniform products, and the Beta substitution. No Stirling formula, Gamma asymptotic, or cited Binet identity is assumed.

All complex powers of positive real integration variables use their real logarithms. The notation $\text{Log } z$ on $\Re z > 0$ denotes the principal holomorphic logarithm. The holomorphic logarithm of Gamma will be constructed below and will be real on the positive real axis.

274.1 1. Defining integrals and the finite Euler formula

For $\Re z > 0$, define

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx. \quad (\text{BI1})$$

This is absolutely convergent. On a compact set with $0 < a \leq \Re z \leq b$, its integrand and each of its first finitely many derivatives in z are bounded in absolute value by

$$(x^{a-1} + x^{b-1})(1 + |\log x|^j)e^{-x},$$

with the appropriate fixed integer j . This function is integrable at zero and infinity. Differentiation under the integral is therefore valid, and Γ is holomorphic on the right half-plane. Integration by parts gives

$$\Gamma(z+1) = z\Gamma(z), \quad \Gamma(1) = 1, \quad (\text{BI2})$$

because $x^z e^{-x}$ tends to zero at both endpoints for $\Re z > 0$.

For $\Re z, \Re w > 0$, let

$$B(z, w) = \int_0^1 t^{z-1} (1-t)^{w-1} dt. \quad (\text{BI3})$$

For an integer $N \geq 1$, integration by parts gives

$$B(z, N+1) = \frac{N}{z} B(z+1, N).$$

Iterating, and using $B(z+N, 1) = 1/(z+N)$, proves the exact finite formula

$$B(z, N+1) = \frac{N!}{z(z+1) \cdots (z+N)}. \quad (\text{BI4})$$

In particular none of these denominators vanishes in the right half-plane.

Set

$$E_N(z) = N^z B(z, N+1) = \frac{N^z N!}{z(z+1) \cdots (z+N)}, \quad N \geq 1. \quad (\text{BI5})$$

The substitution $t = x/N$ gives

$$E_N(z) = \int_0^N x^{z-1} (1 - x/N)^N dx. \quad (\text{BI6})$$

For $0 \leq x < N$, $\log(1 - x/N) \leq -x/N$, so

$$0 \leq (1 - x/N)^N \leq e^{-x}.$$

For each fixed $x > 0$, the expression on the left tends to e^{-x} , with the indicator of $x < N$ understood.

On a compact set as above, therefore

$$\begin{aligned} \sup_z |E_N(z) - \Gamma(z)| &\leq \int_0^\infty (x^{a-1} + x^{b-1}) \\ &\quad \cdot |(1 - x/N)^N \mathbf{1}_{\{x < N\}} - e^{-x}| dx \longrightarrow 0 \end{aligned}$$

by dominated convergence. The same argument with an additional factor $|\log x|^j$, for $j = 1, 2$, proves locally uniform convergence of those two derivatives as well. Thus the Euler limit and the derivative limits are established directly from their integrals.

274.2 2. Nonvanishing and the holomorphic Gamma logarithm

Write $H_N = \sum_{j=1}^N 1/j$. The real numbers

$$\gamma_N = H_N - \log N$$

decrease and are bounded below by zero. Indeed

$$\gamma_{N+1} - \gamma_N = \frac{1}{N+1} - \log(1 + 1/N) < 0$$

by integrating $1/x > 1/(N+1)$ on $[N, N+1)$, while $H_N \geq \int_1^{N+1} dx/x = \log(N+1)$. Let $\gamma_E = \lim_N \gamma_N$.

For $\Re z > 0$, every $1 + z/j$ has positive real part. Its principal logarithm is holomorphic. Define

$$Q_N(z) = \text{Log } z + z(H_N - \log N) + \sum_{j=1}^N (\text{Log}(1 + z/j) - z/j). \quad (\text{BI7})$$

Exponentiating this literal sum and using (BI5) gives exactly

$$e^{Q_N(z)} = zN^{-z} \prod_{j=1}^N (1 + z/j) = \frac{1}{E_N(z)}. \quad (\text{BI8})$$

The logarithmic series in (BI7) converges locally uniformly. To prove this explicitly, for $|w| \leq 1/2$,

$$\text{Log}(1 + w) - w = - \int_0^1 \frac{tw^2}{1 + tw} dt,$$

whence $|\text{Log}(1 + w) - w| \leq |w|^2$. On a compact set with $|z| \leq R$, this bounds the j -th summand by R^2/j^2 whenever $j \geq 2R$. The finite initial part is holomorphic and the summable bound proves uniform convergence of the tail. Hence

$$Q(z) = \text{Log } z + \gamma_E z + \sum_{j=1}^{\infty} (\text{Log}(1 + z/j) - z/j) \quad (\text{BI9})$$

is holomorphic and $Q_N \rightarrow Q$ locally uniformly.

Taking limits in $E_N e^{Q_N} = 1$, using Section 1, proves

$$\Gamma(z) e^{Q(z)} = 1. \quad (\text{BI10})$$

This proves that Gamma is nowhere zero on the entire right half-plane. It also constructs the required holomorphic logarithm:

$$\mathcal{G}(z) := -Q(z), \quad e^{\mathcal{G}(z)} = \Gamma(z). \quad (\text{BI11})$$

For $x > 0$, every term in $Q(x)$ is real and the defining Gamma integral is positive. Thus $\mathcal{G}(x)$ is precisely the real logarithm of $\Gamma(x)$; there is no unspecified additive $2\pi i$ constant.

The second derivatives of the series in (BI9) converge locally uniformly, since $\sum_j |z + j|^{-2}$ does. Termwise differentiation gives

$$\mathcal{G}''(z) = \sum_{j=0}^{\infty} \frac{1}{(z + j)^2} \quad (\Re z > 0). \quad (\text{BI12})$$

For justification without assuming a derivative-series theorem, on any slightly larger compact subset of the same half-plane the series converges uniformly; Cauchy's integral formula on small circles then gives uniform convergence of its derivatives. Direct estimates of the derivatives of its summands give the same conclusion.

274.3 3. Beta product, duplication, and the factor $\sqrt{\pi}$

The absolutely convergent product of the two defining Gamma integrals admits the change of variables

$$x = rt, \quad y = r(1 - t), \quad r > 0, \quad 0 < t < 1.$$

Its Jacobian matrix has determinant $-r$, so the positive integration Jacobian is r . Consequently

$$\Gamma(z)\Gamma(w) = \Gamma(z + w)B(z, w) \quad (\Re z, \Re w > 0). \quad (\text{BI13})$$

All powers here are powers of positive real bases. Fubini and the change of variables are justified by the same integrals with real parts of the exponents.

In $B(z, z)$, first substitute $t = (1 + u)/2$, and then use symmetry and $v = u^2$. Every factor is retained:

$$\begin{aligned} B(z, z) &= 2^{1-2z} \int_{-1}^1 (1 - u^2)^{z-1} du \\ &= 2^{2-2z} \int_0^1 (1 - u^2)^{z-1} du \\ &= 2^{1-2z} B(1/2, z). \end{aligned} \quad (\text{BI14})$$

Finally $x = t^2$ in (BI1) gives $\Gamma(1/2) = 2 \int_0^\infty e^{-t^2} dt$. The latter is the positive integral over the whole real line. Its square is $\int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$. Polar coordinates, whose positive Jacobian is r , evaluate this square as

$$2\pi \int_0^\infty r e^{-r^2} dr = \pi.$$

Thus $\Gamma(1/2) = \sqrt{\pi}$. Inserting (BI13) into (BI14), and using the nonvanishing established in (BI10), proves

$$\Gamma(z)\Gamma(z + 1/2) = 2^{1-2z} \sqrt{\pi} \Gamma(2z) \quad (\Re z > 0). \quad (\text{BI15})$$

On the positive real axis all these factors are positive, so their real logarithms obey the corresponding additive identity without any branch ambiguity.

274.4 4. The bounded integral remainder

For $u > 0$, define

$$K(u) = \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{1}{u}. \quad (\text{BI16})$$

We prove its exact bounds rather than assuming a Bernoulli expansion. With $x = u/2 > 0$,

$$K(u) = \frac{\coth x - 1/x}{4x}.$$

The two functions

$$F_0(x) = x \cosh x - \sinh x, \quad F_1(x) = (x^2 + 3) \sinh x - 3x \cosh x$$

vanish at zero and satisfy $F_0'(x) = x \sinh x \geq 0$ and $F_1'(x) = xF_0(x) \geq 0$. Dividing their nonnegative values by positive factors gives

$$0 \leq \coth x - \frac{1}{x} \leq \frac{x}{3}.$$

Therefore

$$\boxed{0 \leq K(u) \leq \frac{1}{12} \quad (u > 0).} \quad (\text{BI17})$$

Define the actual remainder candidate

$$R(z) = \int_0^\infty e^{-zu} K(u) du, \quad \Re z > 0. \quad (\text{BI18})$$

It is absolutely convergent and holomorphic. Differentiating j times is justified on every compact set with $\Re z \geq a > 0$ by $u^j e^{-au}/12$. In particular

$$|R(z)| \leq \frac{1}{12\Re z}, \quad 0 \leq R(x) \leq \frac{1}{12x} \quad (x > 0). \quad (\text{BI19})$$

274.5 5. Equality of second derivatives

Put $c = \frac{1}{2} \log(2\pi)$, and define

$$\mathcal{F}(z) = (z - \tfrac{1}{2}) \text{Log } z - z + c + R(z). \quad (\text{BI20})$$

Differentiation and the exact algebraic identity

$$u^2 K(u) = \frac{u}{e^u - 1} - 1 + \frac{u}{2}$$

give

$$\begin{aligned} \mathcal{F}''(z) &= \frac{1}{z} + \frac{1}{2z^2} + \int_0^\infty e^{-zu} \left(\frac{u}{e^u - 1} - 1 + \frac{u}{2} \right) du \\ &= \frac{1}{z^2} + \int_0^\infty \frac{ue^{-zu}}{e^u - 1} du. \end{aligned} \quad (\text{BI21})$$

Here $\int_0^\infty e^{-zu} du = 1/z$ and $\int_0^\infty ue^{-zu} du = 1/z^2$ follow by elementary integration on $\Re z > 0$; all cancellation factors are displayed.

The geometric series $1/(e^u - 1) = \sum_{j=1}^\infty e^{-ju}$ is valid for $u > 0$. Its absolute integral after multiplication by ue^{-zu} is bounded by the same expression with $\Re z$, which is integrable at zero and infinity. Hence termwise integration is valid, and

$$\mathcal{F}''(z) = \frac{1}{z^2} + \sum_{j=1}^\infty \int_0^\infty ue^{-(z+j)u} du = \sum_{j=0}^\infty \frac{1}{(z+j)^2} = \mathcal{G}''(z). \quad (\text{BI22})$$

The right half-plane is connected, so the holomorphic function $\mathcal{F} - \mathcal{G}$ has zero second derivative and consequently equals $az + b$ there, for constants $a, b \in \mathbb{C}$.

274.6 6. The recurrence fixes the linear constant

For positive real x , (BI2) and the real-axis branch in (BI11) give

$$\mathcal{G}(x+1) - \mathcal{G}(x) = \log x.$$

The corresponding exact difference of the explicit function (BI20) is

$$\mathcal{F}(x+1) - \mathcal{F}(x) - \log x = (x + \tfrac{1}{2}) \log(1 + 1/x) - 1 + R(x+1) - R(x). \quad (\text{BI23})$$

The left side equals a . The elementary integral bounds

$$\frac{y}{1+y} \leq \log(1+y) \leq y \quad (y > 0)$$

show that $x \log(1 + 1/x) \rightarrow 1$ and $\frac{1}{2} \log(1 + 1/x) \rightarrow 0$. The remainder bound (BI19) makes the remaining difference tend to zero. Thus $a = 0$, and $\mathcal{F} - \mathcal{G} = b$ throughout the right half-plane.

274.7 7. Duplication fixes the additive constant

For $x > 0$, the positive-real logarithm of (BI15) is

$$\mathcal{G}(x) + \mathcal{G}(x+1/2) - \mathcal{G}(2x) = (1-2x) \log 2 + \tfrac{1}{2} \log \pi. \quad (\text{BI24})$$

Since $\mathcal{F} = \mathcal{G} + b$, the difference between the same expression using $\mathcal{F}$ and the right side of (BI24) equals exactly $b + b - b = b$.

Insert (BI20) without dropping any constant. The terms containing $\log x$ cancel, and the resulting difference is

$$\begin{aligned} b &= x \log(1 + 1/(2x)) - \tfrac{1}{2} \\ &\quad + (c - \tfrac{1}{2} \log(2\pi)) + R(x) + R(x+1/2) - R(2x). \end{aligned} \quad (\text{BI25})$$

Our displayed $c = \frac{1}{2} \log(2\pi)$ cancels exactly. The same elementary logarithm bounds show $x \log(1 + 1/(2x)) \rightarrow 1/2$, and each remainder tends to zero by (BI19). Thus $b = 0$.

Combining Sections 5–7 proves the complete exact identity

$$\boxed{\log \Gamma(z) = (z - \tfrac{1}{2}) \text{Log } z - z + \tfrac{1}{2} \log(2\pi) + \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}, \quad \Re z > 0.} \quad (\text{BI26})$$

The logarithm on the left is the uniquely specified holomorphic branch from (BI11). The integral is absolutely convergent, and its bound and all differentiations have been justified. This is the identity used by the actual boundary-tilt proof.

275 Uniform original source comparison and the two growing norm windows

Retain an actual complete off-line quartet $1/2 \pm \delta \pm i\gamma$ of common order m , with $0 < \delta < 1/2$ and $\gamma > 0$. The polynomial h , the entire quotient $v_h = (2\xi)/h$, its full Taylor unit, and the original source are unchanged. Set

$$w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \mu_h = \int_{\mathbb{R}} w_h(u) du, \quad \alpha = \pi/2. \quad (\text{TW1})$$

The proved actual boundary-tilt calculation supplies

$$p = 8m - \frac{9}{2} > 1, \quad e^{\alpha|u|} w_h(u) \leq C_h(1 + |u|)^{-p}, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha u} w_h(u) du \in (0, \infty). \quad (\text{TW2})$$

Here C_h denotes exactly the explicit constant C_h^{tilt} of (BT12), with its original compact and tail expressions. Its proof retains the exact factor $-\pi^{-1/4}(u^2 + 1/4)e^{-iu \log \pi/2} \zeta(1/2 + iu)$ and the full quartet denominator. In particular (TW2) does not use a positive lower bound for ζ or an assumption about its zero locations.

The comparison measure used below is the fixed original one-factor Gamma measure

$$d\sigma(u) = \frac{|\Gamma(1/4 + iu/2)|^2}{2\pi} du, \quad \int d\sigma = \sqrt{2\pi}. \quad (\text{TW3})$$

Its proved two-sided boundary bounds are

$$c_\Gamma e^{-\alpha|u|}(1 + |u|)^{-1/2} \leq \frac{d\sigma}{du} \leq C_\Gamma e^{-\alpha|u|}(1 + |u|)^{-1/2}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{TW4})$$

This fixed measure has mass $\sqrt{2\pi}$ for every k . It is distinct from the k -fold Gamma convolution of mass $(2\pi)^{k/2}$ in (AT4). Both comparisons, with their exact connecting constants, will be retained.

275.1 A uniform convolution bound at the actual exponential boundary

For every integer $k \geq 1$ and every $u \in \mathbb{R}$,

$$\boxed{m_{h,k}(u) \leq kC_h M_\alpha^{k-1} (1 + |u|/k)^{-p} e^{-\alpha|u|}}. \quad (\text{TW5})$$

To prove this, put $f_+(x) = e^{\alpha x} w_h(x)$. Its mass is exactly M_α , and $f_+(x) \leq C_h(1 + |x|)^{-p}$. The original product integral gives $f_+^{*k}(u) = e^{\alpha u} m_{h,k}(u)$. On the hyperplane $x_1 + \dots + x_k = u$, at least one coordinate satisfies $|x_i| \geq |u|/k$. Split this integral into the union of those k sets. On its i th set the i th factor is at most $C_h(1 + |u|/k)^{-p}$. Eliminate $x_i = u - \sum_{j \neq i} x_j$; the change from the original convolution coordinates has absolute Jacobian one. Integrate the other $k - 1$ nonnegative factors over all their original domains; the resulting bound is $C_h(1 + |u|/k)^{-p} M_\alpha^{k-1}$. Summing proves the bound with $e^{-\alpha u}$. Conjugation of the original real-entire 2ξ and of the complete quartet polynomial proves $w_h(-x) = w_h(x)$. Thus $f_-(x) = e^{-\alpha x} w_h(x)$ has the same mass, and the same argument with f_- proves the bound with $e^{\alpha u}$. Taking the smaller of these two proved expressions gives exactly (TW5). This argument also covers $k = 1$ directly. Tonelli's theorem applies because all integrands are nonnegative and the displayed masses are finite.

For integers n with $2n > p - 1$, this yields the literal raw moment bound

$$\int_{\mathbb{R}} u^{2n} m_{h,k}(u) du \leq 2C_h k^{p+1} M_\alpha^{k-1} \frac{\Gamma(2n - p + 1)}{\alpha^{2n - p + 1}}. \quad (\text{TW6})$$

Indeed $(1 + u/k)^{-p} \leq k^p u^{-p}$ for $u > 0$, and substitution $y = \alpha u$ evaluates the remaining positive half-line integral. No arithmetic density has been assigned the value of a Gamma density.

275.2 Pointwise comparison of the original convolution

For $k \geq 3$, retain the proved original convolution lower envelope

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (k-2+|u|)^{-B}, \quad B = 42 + 8m, \quad \vartheta_h = \int_{-1}^1 w_h(u) du > 0. \quad (\text{TW7})$$

Its complete proof is the original arithmetic envelope (AU.10), also retained as (CJ.15); it concerns this same g/h convolution. Define the following explicit constants, without altering either measure:

$$M = B/2 = 21 + 4m, \\ a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_\Gamma 2^{B/2}}, \quad A_k = \frac{C_h}{c_\Gamma} k^{p+1} M_\alpha^{k-1}. \quad (\text{TW8})$$

Then

$$\boxed{a_k (1+u^2)^{-M} \frac{d\sigma}{du} \leq m_{h,k}(u) \leq A_k \frac{d\sigma}{du}}. \quad (\text{TW9})$$

For the upper bound, $1 + |u|/k \geq (1 + |u|)/k$ in (TW5), and the lower inequality of (TW4) leaves the factor $(1 + |u|)^{-(p-1/2)} \leq 1$. This gives precisely A_k . For the lower bound, $k-2 \geq 1$ gives $k-2+|u| \leq (k-2)(1+|u|)$. The upper inequality of (TW4) implies $e^{-\alpha|u|} \geq C_\Gamma^{-1} d\sigma/du$. Finally $(1+|u|)^B \leq 2^{B/2} (1+u^2)^{B/2}$. Substitution of these three inequalities in (TW7) proves its lower counterpart in (TW9), with its displayed constant and direction.

275.3 Full finite polynomial forms, with the original coordinate maps

Let b_j be the fixed Gamma monic polynomials in u . The original generating-integral calculation (TG.7)–(TG.14) proves

$$\gamma_j = \|b_j\|_\sigma^2 = \sqrt{2} j! \Gamma(j+1/2) = \sqrt{2\pi} \frac{(2j)!}{4^j}, \quad ub_j = b_{j+1} + j(j-1/2)b_{j-1}. \quad (\text{TW10})$$

The last term is absent at $j = 0$. The second equality follows by induction from $\Gamma(1/2) = \sqrt{\pi}$ and the Gamma recurrence; no asymptotic formula is used there.

For $N \geq 0$ put

$$D_N = [1 + 4(N+M)^2]^M.$$

Every polynomial P of degree at most N satisfies

$$\int |P(u)|^2 (1+u^2)^M d\sigma(u) \leq D_N \int |P(u)|^2 d\sigma(u). \quad (\text{TW11})$$

Here is the full multiplication estimate. The orthonormal vectors $e_j = b_j/\sqrt{\gamma_j}$ obey

$$ue_j = \sqrt{(j+1)(j+1/2)} e_{j+1} + \sqrt{j(j-1/2)} e_{j-1}.$$

On degrees at most N , the first weighted shift has norm at most $N+1$ and the second has norm at most N ; this follows by taking squared norms of their orthogonal images term by term. The triangle inequality therefore gives $\|uP\|_\sigma \leq 2(N+1)\|P\|_\sigma$. After each multiplication the degree increases by one. Iteration gives $\|u^j P\|_\sigma \leq [2(N+M)]^j \|P\|_\sigma$ for $0 \leq j \leq M$. Expand $(1+u^2)^M$, integrate its nonnegative summands, and use the binomial theorem; the result is exactly (TW11).

Apply Cauchy–Schwarz with the two functions $|P|(1+u^2)^{-M/2}$ and $|P|(1+u^2)^{M/2}$ in $L^2(\sigma)$:

$$\left(\int |P|^2 d\sigma \right)^2 \leq \left(\int |P|^2 (1+u^2)^{-M} d\sigma \right) \left(\int |P|^2 (1+u^2)^M d\sigma \right).$$

For $P \neq 0$, divide by its positive σ norm and apply (TW11). For $P = 0$ the resulting inequality is immediate. Combining with (TW9) proves the full finite-form comparison

$$\boxed{\frac{a_k}{D_N} \int |P|^2 d\sigma \leq \int |P|^2 m_{h,k}(u) du \leq A_k \int |P|^2 d\sigma \quad (\deg P \leq N).} \quad (\text{TW12})$$

It holds for every complex coefficient vector, not just monic polynomials. In particular $A_k D_N / a_k \geq 1$ by applying (TW12) to any nonzero P .

The original sum coordinate is $S = c + iu$, $c = k/2$. Its exact algebra map is $\Psi_k : \mathbb{C}[S] \rightarrow \mathbb{C}[u]$, $\Psi_k P(u) = P(c + iu)$, with inverse $f \mapsto f((S - c)/i)$. In degree at most N its coefficient matrix has diagonal $1, i, \dots, i^N$. Thus (TW12) is a congruence statement in the original S coordinates as well. On the monic degree- n affine space, $P \mapsto i^{-n} \Psi_k P$ is a bijection to the monic u polynomials and preserves the norm because $|i^{-n}| = 1$. No degree-dependent phase is asserted to be a single algebra map. Consequently, writing $H_{k,N}$ for the actual original polynomial Gram and $H_N^{\sigma,k}$ for the fixed σ Gram evaluated at $S = c + iu$, the fully typed matrix form is

$$\boxed{(a_k/D_N) H_N^{\sigma,k} \preceq H_{k,N} \preceq A_k H_N^{\sigma,k}.} \quad (\text{TW13})$$

The identity on the original polynomial source commutes with remainder modulo the same $\chi_{h,k}$ and hence with its full unitful jet observation. Neither this comparison nor the coordinate congruence changes $\chi_{h,k}$, S , its full jets, or the original source norm.

275.4 Uniform arithmetic monic norms

Let $\omega_{h,k,n}$ be the minimum original squared norm of a monic degree- n polynomial. Minimizing (TW12) on that same affine space, using (TW10) for its fixed comparison minimum, gives

$$\boxed{\frac{a_k \gamma_n}{D_n} \leq \omega_{h,k,n} \leq A_k \gamma_n, \quad k \geq 3, n \geq 0.} \quad (\text{TW14})$$

The upper bound may equivalently be verified on the explicit trial polynomial b_n ; the lower bound applies to every monic polynomial. Thus no minimizer was transferred between different metrics without the stated inequality.

Here and below constants in O_h may depend on the fixed complete quartet and the fixed proven envelope constants, but not on k, n . Taking logarithms of the exact constants in (TW8) gives

$$\log a_k = (k-3)(\log \vartheta_h - \alpha) - B \log(k-2) + \log c_h - \log C_\Gamma - (B/2) \log 2,$$

$$\log A_k = (k-1) \log M_\alpha + (p+1) \log k + \log(C_h/c_\Gamma).$$

Also $\log D_n = M \log(1 + 4(n+M)^2) = O_h(1 + \log(n+1))$. These displayed equalities prove, uniformly for $k \geq 3, n \geq 0$,

$$\boxed{\log \omega_{h,k,n} = \log \left(\sqrt{2\pi} \frac{(2n)!}{4^n} \right) + O_h(k + \log(n+1)).} \quad (\text{TW15})$$

This is a high-degree norm estimate for the actual arithmetic convolution. It is not an assertion that consecutive norm ratios have an asymptotic; the error in (TW15) does not justify that differentiation in n .

The same all-polynomial comparison also controls the original raw even moments $I_{k,2n} = \int_{\mathbb{R}} u^{2n} m_{h,k}(u) du$. For $k \geq 3$ and $n \geq 1$ it gives the explicit two-sided bounds

$$\boxed{\frac{a_k \sqrt{2} c_\Gamma e^{-\alpha} (2n-1)!}{D_n \alpha^{2n}} \leq I_{k,2n} \leq 2 A_k C_\Gamma \frac{(2n)!}{\alpha^{2n+1}}.} \quad (\text{TW15a})$$

For the lower comparison moment, restrict (TW4) to $u \geq 1$ and use $(1+u)^{-1/2} \geq 1/(\sqrt{2}u)$. Both half-lines give $\int u^{2n} d\sigma \geq \sqrt{2}c_\Gamma \alpha^{-2n} \Gamma(2n, \alpha)$. Repeated integration by parts gives $\Gamma(2n, \alpha) = (2n-1)!e^{-\alpha} \sum_{j=0}^{2n-1} \alpha^j/j! \geq (2n-1)!e^{-\alpha}$. For the upper comparison moment, drop the factor $(1+|u|)^{-1/2} \leq 1$ in (TW4) and evaluate the Gamma integral. Applying (TW12) to the original degree- n polynomial u^n proves (TW15a). Its logarithms give, uniformly on these stated degrees,

$$\log I_{k,2n} = \log((2n)!\alpha^{-2n}) + O_h(k + \log(n+1)). \quad (\text{TW15b})$$

The full one-factor zeta numerator and actual tilted mass in (TW2) remain in a_k, A_k ; (TW15a) is a bound on the original moments.

275.5 The two original windows with full finite errors

Put $q = [1 + k(m-1)](k+1)^2$. For $n = q-1$ or $n = q$, division of (TW14) at the two exact degrees gives

$$\log \frac{\gamma_{n+q}}{\gamma_n} - \log \frac{A_k D_{n+q}}{a_k} \leq \log \frac{\omega_{h,k,n+q}}{\omega_{h,k,n}} \leq \log \frac{\gamma_{n+q}}{\gamma_n} + \log \frac{A_k D_n}{a_k}. \quad (\text{TW16})$$

All logarithmic error bounds here are nonnegative by (TW12). The exact factorial formula (TW10) and monotonicity of $\log x$ give

$$\log \frac{\gamma_{2q}}{\gamma_q} = 2q \log(4q/e) + \eta_q, \quad 0 \leq \eta_q \leq \log 2. \quad (\text{TW17})$$

Indeed the logarithm of $(4q)!/(2q)!$ is $\sum_{j=2q+1}^{4q} \log j$. Its excess over $\int_{2q}^{4q} \log x dx$ is nonnegative and at most $\sum_{j=2q+1}^{4q} (\log j - \log(j-1)) = \log 2$. Subtracting $q \log 4$ gives precisely (TW17). The other original window satisfies the exact identity

$$\frac{\gamma_{2q-1}}{\gamma_{q-1}} = \frac{\gamma_{2q}}{\gamma_q} \frac{q-1/2}{4q-1}, \quad \frac{1}{6} \leq \frac{q-1/2}{4q-1} < \frac{1}{4}.$$

It follows in particular that

$$4q \log(4q/e) - \log 6 \leq \log \frac{\gamma_{2q-1} \gamma_{2q}}{\gamma_{q-1} \gamma_q} \leq 4q \log(4q/e). \quad (\text{TW18})$$

For fixed h , one has $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Thus each finite error in (TW16), divided by $2q$, tends to zero. Equations (TW16)–(TW18) prove both limits on their stated windows:

$$\lim_{k \rightarrow \infty} \frac{1}{q} \left(\frac{\omega_{h,k,2q}}{\omega_{h,k,q}} \right)^{1/(2q)} = \lim_{k \rightarrow \infty} \frac{1}{q} \left(\frac{\omega_{h,k,2q-1}}{\omega_{h,k,q-1}} \right)^{1/(2q)} = \frac{4}{e}. \quad (\text{TW19})$$

No limit with an unspecified ratio n/q is asserted.

275.6 The resulting actual four-window lower estimate

Retain the original relative controls ε_N , phases ϕ_N , and contraction penalties P_- , P_+ of (HC1)–(HC2). Their exact first boundary and regular products, inherited from (CJ.9)–(CJ.13), give

$$\mathcal{B}_k = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{h,k,2q-1} \omega_{h,k,2q}}{\omega_{h,k,q-1} \omega_{h,k,q}} + P_- + P_+,$$

where the endpoint weights are $1, 2, \dots, 2, 1$. Set

$$E_k = 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q \geq 0, \quad E_k = O_h(k + \log q).$$

Adding the upper sides of (TW16) and applying (TW18) gives the actual sum-norm upper bound $4q \log(4q/e) + E_k$. Consequently, with the full original trace $\mathcal{L} = 2\delta\ell_k(k+1)\lfloor(k+1)^2/4\rfloor$,

$$\begin{aligned} \mathcal{B}_k &\geq 4q \log(\delta ek/8) - E_k + P_- + P_+ \\ &\quad + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} + 4q \log \frac{2\mathcal{L}}{\delta kq}. \end{aligned} \quad (\text{TW20})$$

All terms after $-E_k$ are the displayed original nonnegative costs. The sums retain both endpoint degrees and all their multiplicities. To verify the constant, $4q \log \mathcal{L} - 4q \log(4q/e)$ equals $4q \log(\delta ek/8) + 4q \log(2\mathcal{L}/(\delta kq))$. This proves the identity used in the bound with no rescaling of S . Since $E_k/q \rightarrow 0$, the quantified refinement includes

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_k - 4q \log k}{q} \geq 4 \log(\delta e/8). \quad (\text{TW21})$$

The leading threshold four is inherited. The new content is the actual boundary-tilt comparison, its uniform finite-degree errors, and the sharper literal window constant. These lower estimates alone do not assign a sign to the signed arithmetic correction in (AT14)–(AT17).

276 An arithmetic lower-envelope comparison for central quotient volumes: full jets, finite determinant bounds, and an exact quadratic benchmark

This calculation constructs upper bounds for the original central volume $\log(V_q/V_{2q-1})$. Its input is the proved lower envelope and Laplace estimate of the complete *Arithmetic Endpoint Bounds* source, equations (29)–(32). The comparison acts on the original polynomial and quotient maps, including all multiplicities and phases. The strongest displayed bound retains an explicit positive matrix: its arithmetic moments stop at degree $2q$, while its comparison moments continue to $4q - 2$. A second bound uses only the supplied envelope constants and two Laplace values. The explicit scalar estimate has quadratic, rather than logarithmic, growth per quotient direction. An exact auxiliary calculation determines that scale and supplies a monic family relating its full jets to the original quartet jets. No bound contradicting the quartet lower estimate is asserted.

276.1 Original source and every jet coordinate

Fix a complete quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$, and common complete zero order $m \geq 1$ of $g = 2\xi$. Keep its monic polynomial h , its entire quotient $v_h = g/h$, and

$$w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad k \geq 3. \quad (\text{AU.1})$$

Write $c = k/2$ and keep the original observation $S = c + iu$. The source form is $\int \overline{P(c + iu)} Q(c + iu) m_k(u) du$, conjugate-linear in its first argument. Its constant norm is μ_h^k . The coefficient map and its inverse are

$$\Psi : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad P \longmapsto P(c + iu), \quad f \longmapsto f((S - c)/i). \quad (\text{AU.2})$$

These are inverse ring homomorphisms by substitution. Multiplication by S is carried to $c + iM_u$, and $\partial_u^d \Psi(P) = i^d \Psi(P^{(d)})$. On the ordered degree- N coefficient spaces the matrix is triangular with diagonal $1, i, \dots, i^N$, so its determinant has modulus one. The original quotient volumes therefore have exactly the same values in the transported remainder basis; no mass or norm is divided out.

Let $\ell = 1 + k(m - 1)$ and use the ordered distinct grid

$$\lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \zeta_{ab} = (\lambda_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{AU.3})$$

The original cyclic relation and its transported monic polynomial are

$$q = \ell(k + 1)^2, \quad \chi(S) = \prod_{a,b} (S - \lambda_{ab})^\ell, \quad \psi(u) = i^{-q} \chi(c + iu) = \prod_{a,b} (u - \zeta_{ab})^\ell. \quad (\text{AU.4})$$

The grid is invariant under complex conjugation in the u chart, so ψ is real monic. Equivalently the original coefficient identity is $\overline{\chi}(k - S) = (-1)^q \chi(S)$. In particular the original norm multiplier is $\overline{\chi}(c - iu) \chi(c + iu) = \psi(u)^2$, with the source multiplication still carrying the unit i^q .

For clarity the preceding cyclic degree also follows from the original tensor algebra. In a tensor of k length- m local factors, the sum of the k nilpotent generators has power $k(m - 1)$ nonzero: its coefficient on the product of their top powers is $(k(m - 1))!/((m - 1)!)^k$, nonzero in $\mathbb{C}$. Its next power is zero by total degree. Every grid sum in (AU.3) occurs, since the two signs in each quartet factor can be chosen independently. Direct products of local factors give the same maximal length ℓ at every grid sum. Distinct pairs have distinct real and imaginary parts. The minimal polynomial of the original sum is therefore exactly (AU.4), with no lost multiplicity.

Put $\mathcal{E} = \mathbb{C}[u]/(\psi)$, with ordered remainder coefficients $1, u, \dots, u^{q-1}$. For $N \geq q - 1$ let J_N be remainder and B_N multiplication by ψ in the ordered monomials. Monic division gives the exact sequence

$$0 \longrightarrow \mathbb{C}[u]_{\leq N-q} \xrightarrow{B_N} \mathbb{C}[u]_{\leq N} \xrightarrow{J_N} \mathcal{E} \longrightarrow 0. \quad (\text{AU.5})$$

The source on the left is zero if $N = q - 1$. Precisely, the coefficient isomorphisms are $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}[u]_{\leq N}$, $a \mapsto \sum_{j=0}^N a_j u^j$, and $\kappa : \mathbb{C}^q \rightarrow \mathcal{E}$, $x \mapsto [\sum_{j < q} x_j u^j]$. The polynomial quotient map is $\kappa J_N \iota_N^{-1}$ and the polynomial multiplication map is $\iota_N B_N \iota_N^{-1}$. When a matrix formula is evaluated on a polynomial below, these displayed coefficient isomorphisms are the specified identifications. The original S chart uses the same coefficient isomorphisms with u^j replaced by S^j . The complete raw derivative map is

$$\mathbf{E}_N f = (f^{(d)}(\zeta_{ab}))_{a,b, 0 \leq d < \ell}, \quad (\mathbf{E}_N)_{(a,b,d),j} = \begin{cases} j! \zeta_{ab}^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d. \end{cases} \quad (\text{AU.6})$$

Taylor expansion at each distinct centre shows that its kernel is $(\psi) \cap \mathbb{C}[u]_{\leq N}$. Let $\mathbf{Z} = \mathbf{E}_{q-1}$. Then $\mathbf{E}_N = \mathbf{Z} J_N$ and $\mathbf{Z}$ is invertible. Indeed a degree below q polynomial with all those zero raw jets is divisible by degree- q ψ and is zero. For the fixed row order,

$$\det \mathbf{Z} = \left(\prod_{a,b} \prod_{d=0}^{\ell-1} d! \right) \prod_{(a,b) < (a',b')} (\zeta_{a'b'} - \zeta_{ab})^{\ell^2}. \quad (\text{AU.7})$$

One proof first regards all roots as distinct variables in the ordinary Vandermonde determinant, groups each cluster, and replaces its rows by successive divided differences. Each row operation divides by exactly the corresponding within-cluster difference. Sending the nodes in a cluster to its centre replaces its divided-difference row of order d by $f^{(d)}/d!$, by the finite Taylor expansion of each monomial. Multiplying those rows by $d!$ proves (AU.7). This is an identity of polynomials after removal of the indicated factors; no source norm or zero multiplicity is approximated. In the original S coordinates the same raw jet of order d has its literal factor i^d from (AU.2).

For any positive source matrix H_N , the least quotient form is

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \quad R_N = H_N^{-1} J_N^* G_N. \quad (\text{AU.8})$$

Surjectivity proves $K_N > 0$. Multiplication proves $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Thus every lift $R_N x + B_N y$ has norm $x^* G_N x + y^* B_N^* H_N B_N y$, proving the minimum property. In raw jets the kernel is $\mathbf{K}_N = \mathbf{E}_N H_N^{-1} \mathbf{E}_N^*$, and $\det \mathbf{K}_N = |\det \mathbf{Z}|^2 / V_N$. Hence

$$C_k := \log(V_q / V_{2q-1}) = \log \det \mathbf{K}_{2q-1} - \log \det \mathbf{K}_q. \quad (\text{AU.9})$$

All factorials in (AU.7) cancel only in this specified ratio.

276.2 The lower envelope as an explicit positive comparison form

The supplied arithmetic proof establishes $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, $B = 42 + 2 \deg h = 42 + 8m$, and

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad \alpha = \pi/2. \quad (\text{AU.10})$$

Set $a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}$ and $d_k = k - 2 > 0$. Define the comparison measure, retaining its literal mass, by

$$d\nu_k(u) = a_k e^{-\alpha|u|} (d_k + |u|)^{-B} du. \quad (\text{AU.11})$$

The identity map $L^2(m_k du) \rightarrow L^2(\nu_k)$ is well defined and a contraction, since (AU.10) gives the squared-norm inequality for every measurable function. It restricts to the identity on each polynomial space and commutes with multiplication by $c + iu$, remainder (AU.5), and every raw jet (AU.6).

The comparison is therefore a map on the same quotient, not a declaration that the measures coincide.

Every moment of ν_k is finite, and the following integral evaluates it with positive integrands:

$$\nu_{k,j} := \int_{\mathbb{R}} u^j d\nu_k(u) = \begin{cases} \frac{2a_k j!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-d_k t}}{(\alpha+t)^{j+1}} dt, & j \text{ even}, \\ 0, & j \text{ odd}. \end{cases} \quad (\text{AU.12})$$

To prove it, repeated integration by parts gives $\int_0^\infty t^{B-1} e^{-yt} dt = (B-1)! y^{-B}$ for $y > 0$. Insert $y = d_k + u$ on the positive half-line. Tonelli applies because each integrand is nonnegative there. Integrating $u^j e^{-(\alpha+t)u}$ gives $j!(\alpha+t)^{-j-1}$ by the same integration by parts. Evenness supplies the factor two or the vanishing odd moment. The bound $\nu_{k,j} \leq 2a_k d_k^{-B} j! \alpha^{-j-1}$ for even j also proves finiteness directly. In particular $\nu_{k,0} > 0$ is retained, and generally differs from μ_h^k .

Let $H_N^\nu = (\nu_{k,i+j})_{0 \leq i,j \leq N}$. Its quadratic form is the ν_k integral of the polynomial's squared modulus. The weight is strictly positive everywhere, so a nonzero polynomial has positive norm. Thus $0 < H_N^\nu \preceq H_N$. Minimizing on the identical affine fibres in (AU.5) proves

$$0 < G_N^\nu \preceq G_N, \quad K_N \preceq K_N^\nu, \quad K_N^\nu = E_N (H_N^\nu)^{-1} E_N^*. \quad (\text{AU.13})$$

The inverse inequality follows also by conjugating H_N by $(H_N^\nu)^{-1/2}$ and inverting its positive eigenvalues.

Theorem 276.1 (Actual low-degree data control the higher quotient volume). *For the original quartet source and $k \geq 3$, the finite upper bound is*

$$0 \leq C_k \leq \log \det K_{2q-1}^\nu - \log \det K_q. \quad (\text{AU.14})$$

Its actual arithmetic inputs are the moments through degree $2q$ and the stated envelope constants. The comparator moments through $4q-2$ are the explicit positive integrals (AU.12).

Proof. The first inequality follows by minimizing the same quotient norm in successively larger polynomial spaces. For the second use (AU.13) at $N = 2q-1$ and (AU.9). Determinant is increasing on positive Hermitian forms: conjugate by the square root of the smaller one and multiply eigenvalues at least one. H_q uses precisely the original moments $0, \dots, 2q$, and all the matrices in the second term of (AU.14) have already specified finite algebraic maps from them. No higher actual moment enters. Equivalently, if $\mathfrak{D}_j^\nu = \det H_{j-1}^\nu$ and $\mathfrak{B}_j^\nu$ is the Gram determinant of the original ψ times $1, u, \dots, u^{j-1}$, the quotient-first coefficient matrix $[1, u, \dots, u^{q-1}, \psi, u\psi, \dots]$ is triangular with diagonal one. Orthogonal elimination of its relation block gives

$$V_{2q-1}^\nu = \mathfrak{D}_{2q}^\nu / \mathfrak{B}_q^\nu, \quad C_k \leq \log V_q + \log \mathfrak{B}_q^\nu - \log \mathfrak{D}_{2q}^\nu. \quad (\text{AU.15})$$

This proves the same upper bound in the exact source/relation determinants. Both the source mass and relation multiplication factors remain in their respective determinants. $\square$

The existing Gamma coefficient map also makes the cutoff in (AU.14) explicit. With $\lambda > 0$, $\alpha_\lambda = 2\lambda$, $c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda)$ and $C_\lambda = c_\lambda^k$, define the original monic Gamma polynomials by

$$\sum_{j \geq 0} b_j^{(\lambda)}(u) z^j / j! = (1+z^2)^{-\lambda} e^{u \arctan z}, \quad c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha_\lambda)_j}.$$

Multiplying the k generating functions and taking finite derivatives at zero gives

$$d_l = \sum_{j_1 + \dots + j_k = l} \prod_a (\alpha_\lambda)_{j_a} c_{h,j_a}, \quad \int b_l^{(k\lambda)} m_k = C_\lambda l! d_l. \quad (\text{AU.16})$$

All finite terms are integrable by the supplied exponential moments. Expand $u^{i+j} = \sum_{l \leq i+j} L_{ij,l} b_l^{(k\lambda)}(u)$ by monic subtraction. Then $(H_q)_{ij} = C_\lambda \sum_l l! L_{ij,l} d_l$. Thus the actual coefficient list $c_{h,0}, \dots, c_{h,2q}$ suffices for (AU.14). At zero degree it gives $C_\lambda c_{h,0}^k = \mu_h^k$, retaining the source mass.

276.3 An upper determinant depending only on the two Laplace values

Fix $0 < b < \pi/2$ and set $M_k = M_h(b)^k + M_h(-b)^k$, with the original $M_h(\pm b) = \int e^{\pm bu} w_h(u) du$. Positivity and convolution give $\int e^{b|u|} m_k(u) du \leq M_k$. The exponential series therefore gives $\int |u|^{2j} m_k(u) du \leq M_k (2j)! b^{-2j}$. For arbitrary $\tau_0, \dots, \tau_q > 0$ define

$$D_q(\tau) = M_k \left(\sum_{j=0}^q \tau_j \right) \text{diag} \left(\frac{(2j)!}{b^{2j} \tau_j} \right)_{j=0}^q. \quad (\text{AU.17})$$

For $f = \sum a_j u^j$, Cauchy–Schwarz applied to $\sqrt{\tau_j}$ and $a_j u^j / \sqrt{\tau_j}$ proves

$$|f(u)|^2 \leq \left(\sum \tau_j \right) \sum |a_j|^2 |u|^{2j} / \tau_j.$$

Integration proves $H_q \preceq D_q(\tau)$. Inverting and applying the same full jet map proves

$$\mathbf{K}_q \succeq \frac{1}{M_k \sum \tau_j} \mathbf{A}_q(\tau), \quad \mathbf{A}_q(\tau) = \mathbf{E}_q \text{diag} \left(\frac{\tau_j b^{2j}}{(2j)!} \right) \mathbf{E}_q^* > 0. \quad (\text{AU.18})$$

Consequently an upper bound without any unevaluated actual polynomial moment is

$$C_k \leq q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det \mathbf{K}_{2q-1}^\nu - \log \det \mathbf{A}_q(\tau). \quad (\text{AU.19})$$

Every positive choice of τ gives a proved bound, and the infimum over such choices remains a bound. Multiplying all τ_j by the same positive number cancels between the two written determinant factors; it changes neither measure. No optimizer or numerical enclosure is assumed in (AU.19).

The lower determinant has the exact positive expansion

$$\det \mathbf{A}_q(\tau) = \sum_{\substack{I \subset \{0, \dots, q\} \\ |I|=q}} |\det(\mathbf{E}_q)_{[:,I]}|^2 \prod_{j \in I} \frac{\tau_j b^{2j}}{(2j)!}. \quad (\text{AU.20})$$

Expand each determinant multilinearly in columns in the product $\mathbf{E}_q D \mathbf{E}_q^*$: terms with repeated column indices vanish by alternation, and grouping permutations of the same subset gives (AU.20). Its subset $I = \{0, \dots, q-1\}$ is strictly positive by (AU.7). This expression retains full raw-jet minors and factorials without estimating an inverse by its individual entries.

276.4 The interval kernel and the actual comparison map

For $T > 0$, set

$$a_k(T) = a_k e^{-\alpha T} (d_k + T)^{-B}. \quad (\text{AU.21})$$

Restriction to $[-T, T]$ followed by the identity on polynomials maps the actual source to $L^2([-T, T], a_k(T) du)$ contractively. The same is true from $L^2(\nu_k)$, by the pointwise lower bound on that interval. Let $H_N^I(T)$ be the Lebesgue interval Gram. Then $a_k(T) H_N^I(T) \preceq H_N^\nu \preceq H_N$. Let P_j denote the Legendre polynomial specified by Rodrigues' formula, $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$.

Integration by parts j times proves its orthogonality to lower degrees; direct evaluation of the resulting beta integral gives its squared norm $2/(2j+1)$. Its highest coefficient is $(2j)!/(2^j(j!)^2)$. The endpoint boundary terms vanish because $(x^2-1)^j$ has zeros of order j at both endpoints. Applying these constants to $P_j(u/T)$ gives the full interval jet kernel

$$(\mathbf{K}_N^I(T))_{(a,b,d),(a',b',e)} = \sum_{j=0}^N \frac{2j+1}{2T^{1+d+e}} P_j^{(d)}(\zeta_{ab}/T) \overline{P_j^{(e)}(\zeta_{a'b'}/T)}. \quad (\text{AU.22})$$

This follows by writing the inverse Gram in the orthogonal Legendre basis and applying (AU.6); each derivative contributes its stated T^{-d} . In particular

$$\mathbf{K}_{2q-1} \preceq \mathbf{K}_{2q-1}^\vee \preceq a_k(T)^{-1} \mathbf{K}_{2q-1}^I(T). \quad (\text{AU.23})$$

Equations (AU.14) and (AU.19) remain valid if their $\log \det \mathbf{K}_{2q-1}^\vee$ is replaced by $\log \det \mathbf{K}_{2q-1}^I(T) - q \log a_k(T)$. The global envelope version is at least as strong as each such interval version by the proved positive-matrix inequality (AU.23).

The restriction map has the exact orthogonal decomposition $f \mapsto (f1_{[-T,T]}, f1_{\mathbb{R} \setminus [-T,T]})$ in the original weighted L^2 source; its two squared norms add to the original norm. The interval bound drops a nonnegative term in that equality. It does not assign the outside component to external absence.

276.5 The explicit scalar bound and its finite gap

The following explicit scalar estimate is the independent full calculation VC.1–43, composed here with the preceding original quotient maps. Put $R_0 = \sqrt{\delta^2 + \gamma^2}$ and define

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = kR_0/T \leq 1/2, \quad N = 2q-1, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \quad A_N = \sqrt{\frac{2N+1}{2}} \frac{4^{N+1}-1}{3}, \\ B_q(r) &= \left(\frac{1+r}{1-r}\right)^q, \quad C_{k,q} = \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{AU.24})$$

All values are positive. In particular $q \geq k$ and $D \geq 2R_0$ give the stated bound on r . These choices bound constants; they do not change the original density.

Theorem 276.2 (A full finite Hermitian upper bound). *In the transported ordered quotient coordinates, and by the explicit congruence below also in the original S coordinates,*

$$\boxed{K_q \preceq K_{2q-1} \preceq C_{k,q} K_q, \quad 0 \leq C_k \leq q \log C_{k,q}.} \quad (\text{AU.25})$$

Proof. Let $\mathbf{T}_j$ be the coefficient matrix of the original map $\Psi(P) = P(c+iu)$ in degree j . Write $H_j^S, J_j^S, K_j^S, G_j^S$ for matrices in the original S monomials. Then

$$H_j^S = \mathbf{T}_j^* H_j \mathbf{T}_j, \quad J_j \mathbf{T}_j = \mathbf{T}_{q-1} J_j^S, \quad K_j = \mathbf{T}_{q-1} K_j^S \mathbf{T}_{q-1}^*.$$

The first identity is the source substitution, the second is monic division, and substitution in (AU.8) gives the third. Each $\det \mathbf{T}_j$ has modulus one, as proved in (AU.2). All source norms on S polynomials in this proof use H_j^S . The exact auxiliary coordinate map used in this proof is $\Psi_T P(x) = P(c+iTx)$, with inverse $f((S-c)/(iT))$. Its target measure is $T m_k(Tx) dx$ and retains mass μ_h^k by substitution. The target monic relation is $\chi_T(x) = (iT)^{-q} \chi(c+iTx)$, so $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$. Monic division therefore proves $\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_\chi P$. At every original centre λ , the full order- d raw jet is $(\Psi_T P)^{(d)}((\lambda-c)/(iT)) = (iT)^d P^{(d)}(\lambda)$. The coefficient matrix has diagonal $(iT)^j$ at degree j ; its quotient determinant is $(iT)^{q(q-1)/2}$. Consequently the target quotient form is the original form

transported by inverse congruence, and its determinant is $T^{-q(q-1)}V_j$ at every source degree j . This factor cancels in the central ratio, with all phases and masses still present in the map.

For a polynomial $f(x) = \sum f_j x^j$ put $\|f\|_{\text{coeff},1} = \sum |f_j|$. The Rodrigues formula above, expanded by the product rule, gives

$$P_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a.$$

The coefficient norm of a product is at most the product of the two coefficient norms by convolution and the triangle inequality. Since $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$, this proves $\|P_j\|_{\text{coeff},1} \leq \sum_a \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j$. The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$. In the orthogonal expansion $f = \sum_{j=0}^N f_j^L P_j$, Cauchy-Schwarz and the squared norm $2/(2j+1)$ give $|f_j^L| \leq \sqrt{(2j+1)/2} \|f\|_{L^2([-1,1])}$. Summing 4^j therefore proves

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}, \quad \deg f \leq N. \quad (\text{AU.26})$$

List all roots $z_1, \dots, z_q$ of χ_T , with repetitions; each has modulus at most r . Define e_a, h_j by

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j.$$

The second identity is formal multiplication of geometric series. It gives $|e_a| \leq \binom{q}{a} r^a$ and $|h_j| \leq \binom{q+j-1}{j} r^j$: the latter counts exponent lists of total degree j by their $q-1$ separating bars. For $0 \leq s \leq q-1$, multiply those inverse series through order s . It proves that the quotient of x^{q+s} by χ_T is exactly $Q_s = \sum_{j=0}^s h_j x^{s-j}$ and its remainder is $x^{q+s} - \chi_T Q_s$, of degree below q . Removing the leading term of $\chi_T Q_s$ removes precisely one from its coefficient norm. All its remaining coefficients are the negative remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \\ &\leq ((1+r)/(1-r))^q = B_q(r). \end{aligned}$$

At degrees below q the remainder is the original monomial and has coefficient norm one, also bounded by $B_q(r)$. Linearity proves

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{AU.27})$$

Every root was retained in this proof. Its smaller literal finite constant $\max\{1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1\}$ can also replace $B_q(r)$.

For v of degree at most $q-1$, each monomial is bounded by $\max\{1, |x|^{q-1}\}$. Using $\max(1, y) \leq 1+y$ for $y \geq 0$ and the proved Laplace moment bound gives

$$\int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2. \quad (\text{AU.28})$$

On the other hand, restriction to $[-T, T]$ and (AU.21) prove $\|P\|_{H_N^S}^2 \geq T a_k(T) \|\Psi_T P\|_{L^2([-1,1])}^2$. Take $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (AU.26)–(AU.28) and the exact division map now give

$$\|\text{rem}_{\chi} P\|_{H_{q-1}^S}^2 \leq C_{k,q} \|P\|_{H_N^S}^2. \quad (\text{AU.29})$$

The map estimated here is $s\pi_{\chi,N} : \mathcal{P}_N^S \rightarrow \mathcal{P}_{q-1}^S$, where $\pi_{\chi,N}$ is the polynomial quotient map and $s : \mathbb{C}[S]/(\chi) \rightarrow \mathcal{P}_{q-1}^S$ selects the unique remainder representative. Their exact identities are $\pi_{\chi,q-1} s =$

I and $\pi_{\chi, q-1} s \pi_{\chi, N} = \pi_{\chi, N}$. The composite $s \pi_{\chi, N}$ equals the identity on polynomials of degree below q .

Apply (AU.29) to the least lift $R_N^S x$ from the original matrices. Its remainder lifts the same x already at degree $q-1$, hence $x^* G_q^S x \leq x^* G_{q-1}^S x \leq C_{k,q} x^* G_N^S x$. Conversely degree inclusion gives $G_N^S \preceq G_q^S$. Conjugating by $(G_q^S)^{-1/2}$, bounding its positive eigenvalues and then inverting proves (AU.25), including the determinant exponent q , first for K_j^S and then by the displayed congruence for K_j . It also proves $C_{k,q} \geq 1$. $\square$

Retain the fixed constants

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_{\text{env}} = \max\{1, 2/(3c_h D)\}.$$

Because $bT \geq 2q$, the factorial ratio in (AU.24) is at most one. Also $M_k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that $U_k(T) \leq 3X^k$. The displayed A_N satisfies $A_{2q-1}^2 \leq (2q/9)256^q$. On $[0, 1/2]$ the derivative of $\log((1+r)/(1-r))$ is $2/(1-r^2) \leq 8/3$, so

$$\log B_q(r)^2 \leq \frac{16}{3}qr = \frac{16R_0}{3D}k. \quad (\text{AU.30})$$

Use also $\vartheta_h^{-(k-3)} \leq K^k$, $1+Dq+k-3 \leq 1+(D+1)q$, and $e^{-3\alpha} \leq 1$. Substitution in (AU.24)–(AU.25) proves the finite scalar bound

$$\boxed{0 \leq C_k \leq \mathcal{U}_k := (\alpha D + \log 256)q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}.} \quad (\text{AU.31})$$

All of its constants are independent of k and q for fixed h, b . The exact finite bounds (AU.14), (AU.19), and $q \log C_{k,q}$ can be retained together by taking their minimum; each independently bounds the same C_k with the proved maps.

The complete central lower estimate in the balanced-window/factor-four source is $C_k \geq \mathcal{L}_k = 2(q-1)(\log k + \log(\delta/(2C_h^{\text{bal}})))$. The current explicitly computed gap is the literal function

$$\begin{aligned} \mathcal{U}_k - \mathcal{L}_k &= (\alpha D + \log 256)q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &\quad + Bq \log(1 + (D+1)q) + q \log C_{\text{env}} \\ &\quad - 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \end{aligned} \quad (\text{AU.32})$$

This quantity concerns the two displayed upper and lower estimates. It is not an asserted exact value of the unknown C_k . Since $q = \ell(k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Dividing every term of (AU.31)–(AU.32) by q^2 proves

$$\lim_{k \rightarrow \infty} \frac{\mathcal{U}_k - \mathcal{L}_k}{q_k^2} = \alpha D + \log 256 > 0, \quad \limsup_{k \rightarrow \infty} C_k/q_k^2 \leq \alpha D + \log 256. \quad (\text{AU.33})$$

Dividing the upper estimate by $q \log k$ gives the exact leading factor $(\alpha D + \log 256)q/\log k$. It grows as $(\alpha D + \log 256)k^2/\log k$ for $m=1$ and as $(\alpha D + \log 256)(m-1)k^3/\log k$ for $m>1$. The division step itself has only the linear-in- k logarithmic cost (AU.30). The two quadratic contributions in this scalar proof come from the full coefficient estimate (AU.26) and the least density over the interval in (AU.21). The positive matrices in (AU.14), (AU.19), and (AU.22) retain the directional information discarded in those scalar estimates, with their exact source and quotient maps already proved. No upper asymptotic for those sharper determinants is inserted.

276.6 A full-jet quadratic benchmark and its exact relation family

Here the auxiliary source is explicitly Lebesgue measure on $[-1, 1]$ and the auxiliary relation is u^q . Its mass is two. These data are inserted into the same sequence (AU.5), kernel (AU.8), and central ratio (AU.9). No assertion that they are an arithmetic density is made. Write D_d for its d by d monomial determinant, $D_0 = 1$, and $B_j(q)$ for the j by j determinant of the relation measure $u^{2q}du$ on that interval. The quotient-first triangular determinant calculation proving (AU.15) gives exactly

$$\frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} = \frac{D_{q+1}B_q(q)}{B_1(q)D_{2q}}, \quad B_1(q) = \frac{2}{2q+1}. \quad (\text{AU.40})$$

The orthogonal Legendre calculation gives

$$D_d = \prod_{j=0}^{d-1} \frac{2}{2j+1} \left(\frac{2^j}{\binom{2j}{j}} \right)^2. \quad (\text{AU.41})$$

For $q = 2s$, parity splits the relation Gram into two s by s blocks, and

$$B_{2s}(2s) = \mathcal{C}_s(2s+1/2)\mathcal{C}_s(2s+3/2), \quad \mathcal{C}_s(x) = \frac{\prod_{j=0}^{s-1} (j!)^2}{\prod_{a,b=0}^{s-1} (x+a+b)}. \quad (\text{AU.42})$$

Indeed the even block has entries $1/(2s+a+b+1/2)$ and the odd block entries $1/(2s+a+b+3/2)$. The Cauchy determinant formula used here has numerator $\prod_{a<b} (b-a)^2 = \prod_j (j!)^2$. It follows by multiplying the determinant by all its denominators: the numerator is alternating separately in its two lists of variables, so contains their two Vandermonde factors. Its degree is their total degree, and comparing one residue inductively gives constant one. Substitution gives (AU.42) with no absolute-value or sign ambiguity.

Theorem 276.3 (An explicit quadratic scale in the full-origin jet fibre). *For every even integer $q \geq 2$,*

$$\left| \log \frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} - \kappa_* q^2 \right| \leq 12q \log(8q), \quad \kappa_* := \frac{9}{2} \log 3 - 6 \log 2 = \frac{3}{2} \log(27/16) > 0. \quad (\text{AU.43})$$

Proof. The elementary binomial bounds $4^j/(2j+1) \leq \binom{2j}{j} \leq 4^j$ hold because the central coefficient is the largest of the $2j+1$ coefficients summing to 4^j . Therefore (AU.41) gives

$$\log D_d = -d(d-1) \log 2 + E_d, \quad |E_d| \leq d \log(4d) \quad (d \geq 1). \quad (\text{AU.44})$$

At $s \geq 1$, expand the numerator of $\mathcal{C}_s(2s)$ as $2 \sum_{i=1}^{s-1} (s-i) \log i$. The function $f(x) = (1-x) \log x$ is increasing on $(0, 1]$, has integral $-3/4$, and is nonpositive. Its right Riemann sum differs from its integral by a number between zero and $(\log s + 1)/s$: the difference is bounded above by $-\int_0^{1/s} f(x) dx \leq (\log s + 1)/s$, using $-f(x) \leq -\log x$. It follows that this numerator is $s^2 \log s - 3s^2/2$ plus an error of absolute value at most $s \log s + 2s$.

The denominator's logarithm is $s^2 \log s + \sum_{a,b=0}^{s-1} \log(2+a/s+b/s)$. Both partial derivatives of the latter function lie in $[0, 1/2]$. Its left-square Riemann sum differs from its integral by at most $1/s$, hence the sum by at most s . Two integrations give

$$\int_0^1 \int_0^1 \log(2+x+y) dx dy = 18 \log 2 - 9 \log 3 - 3/2.$$

For example use the second antiderivative $F(t) = t^2 \log t/2 - 3t^2/4$ and calculate $F(4) - 2F(3) + F(2)$. Thus

$$|\log \mathcal{C}_s(2s) - 9s^2 \log(3/4)| \leq s \log s + 3s.$$

For $c > 0$, the change to $\mathcal{C}_s(2s + c)$ has logarithm $-\sum_{a,b} \log(1 + c/(2s + a + b))$, between $-cs/2$ and zero. Using $c = 1/2, 3/2$ in (AU.42) proves

$$\left| \log B_q(q) - \frac{9}{2}q^2 \log(3/4) \right| \leq q \log(q/2) + \frac{7}{2}q. \quad (\text{AU.45})$$

Insert (AU.44)–(AU.45) into (AU.40). The quadratic coefficient is $3 \log 2 + (9/2) \log(3/4) = \kappa_*$. The remaining absolute error is at most

$$3q \log 2 + (q + 1) \log(4(q + 1)) + 2q \log(8q) \\ + q \log(q/2) + \frac{7}{2}q + \log((2q + 1)/2).$$

For $q \geq 2$, use $q + 1 \leq 3q/2$, $4(q + 1) \leq 8q$, $\log(q/2) \leq \log(8q)$ and $\log((2q + 1)/2) \leq \log(8q)$. The two constant multiples of q are bounded together by $3q \log(8q)$, since $3 \log(16) > 3 \log 2 + 7/2$. Their sum is then at most $8q \log(8q)$, hence certainly the displayed $12q \log(8q)$. This proves (AU.43) for every indicated finite q , not only along a limit. $\square$

There is an exact algebraic relation of this benchmark's quotient to every original grid quotient. Define

$$\psi_t(u) = \prod_{a,b} (u - t\zeta_{ab})^\ell, \quad \mathcal{A} = \mathbb{C}[t, u]/(\psi_t(u)). \quad (\text{AU.46})$$

Monic division over $\mathbb{C}[t]$ proves that $1, u, \dots, u^{q-1}$ form a free $\mathbb{C}[t]$ basis: division gives spanning and a nonzero multiple of a monic degree- q polynomial cannot have smaller degree. Every fibre has dimension q . The fibre at $t = 1$ is the original transported quotient; at $t = 0$ it is $\mathbb{C}[u]/u^q$. For $t \neq 0$,

$$\mathcal{A}_t \xrightarrow{\sim} \mathcal{A}_1, \quad [f(u)] \mapsto [f(tu)], \quad [g(u)] \mapsto [g(u/t)] \quad (\text{AU.47})$$

are inverse maps, since $\psi_t(tu) = t^q \psi_1(u)$. Their full raw jet equation is $(d/du)^d f(tu)|_{u=\zeta_{ab}} = t^d f^{(d)}(t\zeta_{ab})$, so the coefficient d in multiplication's raw-jet Jordan blocks and each order ℓ remain for $t \neq 0$. At $t = 0$ monic division gives the complete merged order q , rather than an assertion that separate evaluation rows stay invertible at coincident points.

For any fixed positive polynomial source H_N , remainder matrices $J_N(t)$ are polynomial in t by the monic subtraction algorithm. They contain I_q in their first q columns. Thus $K_N(t) = J_N(t)H_N^{-1}J_N(t)^*$ is positive for every real t , including zero, and the exact central function obeys

$$C(t) = \log \det K_{2q-1}(t) - \log \det K_q(t), \quad C'(t) = \text{Tr}(K_{2q-1}^{-1}K'_{2q-1} - K_q^{-1}K'_q), \quad (\text{AU.48})$$

where $K'_N = J'_N H_N^{-1} J_N^* + J_N H_N^{-1} (J'_N)^*$. The determinant derivative follows by its multilinear expansion. This gives both the free family and the exact finite source-dependent map along it. The estimate (AU.43) belongs to its stated Lebesgue special fibre. Equations (AU.46)–(AU.48) do not transfer that estimate to $t = 1$ without the displayed change in the full kernel.

276.7 Source representatives and support

The source polynomial maps throughout are the original $\mathcal{V}_{h,k}P = P(D_1 + \dots + D_k)F_h^{\otimes k}$, $D = -x\partial_x$, $\mathcal{M}F_h = g/h$. The arithmetic inclusion is the original $\eta[P] = [g/h]_h^{\otimes k} P(\sum s_i)$, after the explicit chart (AU.2). Each local Taylor constant of g/h at a selected complete zero is nonzero. Recursively solving multiplication of truncated Taylor series therefore proves this factor is a unit. It does not alter the original cyclic kernel (AU.4). The comparison to ν_k changes the norm used in an estimate, while its identity polynomial map leaves η , every raw jet, and the original relations unchanged.

For two polynomial representatives with the same quotient, monic division gives their difference χQ . In the tensor variables the original annihilator identity supplies $\chi(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$, with Q_i obtained by successive fixed-order monic divisions. The final remainder is zero because

the monomial tensor basis is independent and the left side annihilates the sum. If $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$ and $h(D)F_h = \Theta\phi_0$, the actual primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) Q(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{AU.49})$$

The tensor differential at position i contributes the second $(-1)^{i-1}$, cancelling the displayed one and giving exactly $\mathcal{V}_{h,k}(\chi Q)$. Thus every quotient comparison retains the same cohomology class with an explicit original primitive. On the supported carrier a fibre map sends (λ, v) to (λ, Lv) and sends external τ to τ . The killed relation has image $(\lambda, 0) = e_\lambda$. No comparison identifies that supported zero with external absence.

276.8 Verification and provenance

The complete arithmetic source (including the proof of its envelope), the exact endpoint-window source, the full PR24 note and scope correction/review, and the owner's complete balanced-window/factor-four proof were read for this calculation. Its new work is (AU.11)–(AU.23), the scalar estimate (AU.24)–(AU.33), and the full calculation (AU.40)–(AU.48). The constants $c_h, \vartheta_h, M_h(\pm b)$ retain their supplied definitions; no numerical enclosure of actual quartet data is claimed. The accompanying finite rational checker verifies the displayed maps and determinant formulas on declared auxiliary positive sources. It does not certify the analytic source by sampled values and makes no Lean claim.

277 The first canonical window, its full phase product, and both arithmetic block budgets

This chapter joins the original first-degree source calculation (CV.11)–(CV.12) and (TVB.37)–(TVB.38) to the consecutive-window proof of PR 25 at the exact head 66fac5e7885a40cd4873e74b754907320f05b067. The complete original and proposed corrected sources accompany this chapter. Their correction retains all zero generalized eigenvalues. The source radius, original norm envelope and finite propagation are proved below with their original domains. The full first-window phase product has a single terminal contraction and two copies of each interior contraction; its first step contains no preceding quotient inverse.

277.1 Original source and the actual first-degree step

Retain $g = 2\xi$, the complete fixed packet h , integer tensor degree $k \geq 1$, and the original density

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad S = c + iu, \quad c = k/2. \quad (\text{CJ.1})$$

Its source inner product is the integral against $m_{h,k}(u)du$; the original monic orthogonal polynomials p_n have squared norms $\omega_n > 0$, including $\omega_0 = \mu_h^k$. Use the original dagger-stable monic cyclic polynomial χ of degree $q \geq 1$, the full quotient $E = \mathbb{C}[S]/(\chi)$ and the original remainder basis. Its columns, kernels and metrics are

$$\begin{aligned} b_n &= [p_n]_\chi, \quad B_N = [b_0, \dots, b_N], \quad O_N = \text{diag}(\omega_0, \dots, \omega_N), \\ K_N &= B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N \quad (N \geq q-1). \end{aligned} \quad (\text{CJ.2})$$

Surjectivity of B_N and positivity of O_N prove $K_N > 0$. The section $R_N = O_N^{-1} B_N^* G_N$ satisfies $B_N R_N = I$ and $R_N^* O_N R_N = G_N$; its orthogonality to $\ker B_N$ proves that it is the unique minimum lift. Source inclusion proves $G_{N+1} \preceq G_N$ and $0 < V_{N+1} \leq V_N$.

Write the original expansion, retaining every coefficient,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle}{\omega_j}, \quad \nu_0 = \|\chi\|^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j. \quad (\text{CJ.3})$$

The first q remainder columns form an invertible triangular matrix, so $b_a^* G_{q-1} b_b = \delta_{ab} \omega_b$ for $a, b < q$. Reduction of (CJ.3) gives $b_q = -\sum_{j < q} \gamma_j b_j$ and hence $b_q^* G_{q-1} b_q = \nu_0 - \omega_q$. The exact rank-one determinant formula therefore gives

$$\delta_q := \frac{V_q}{V_{q-1}} = \left(1 + \frac{b_q^* G_{q-1} b_q}{\omega_q} \right)^{-1} = \frac{\omega_q}{\nu_0}. \quad (\text{CJ.4})$$

The rank-one determinant identity follows by taking a basis whose first vector is the update vector u when it is nonzero. In that basis uv^* has entries only in its first row; $I + uv^*$ is therefore upper triangular with diagonal entries $1 + v^*u, 1, \dots, 1$. Its determinant is $1 + v^*u$. The zero update gives the same identity directly.

Let ϕ_N denote the retained real phase of the original theta deformation, with the sign convention of (TVB.26): $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. Its reality follows in the literal chart $S = c + iu$ as follows. Write $Q_n(u) = i^{-n} p_n(c + iu)$ and $\hat{\chi}(u) = i^{-q} \chi(c + iu)$. Dagger stability gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$, so $\hat{\chi}$ is real monic. The source measure is real and positive; the monic orthogonality equations therefore give real Q_n . If E is the exact coefficient map $P(S) \mapsto P(c + iu)$, then $E b_n = i^n [Q_n]_{\hat{\chi}}$, while the transported kernel and metric are $E K_N E^*$ and $E^{-*} G_N E^{-1}$. The latter has real entries because the phases in each kernel summand occur together with their conjugates. Thus the original cross pairing is i times a real number, with its original factor ω_N retained. This proves the stated reality and

specifies the exact chart map rather than changing the metric. At the first degree that identity reads $\gamma_{q-1} = -i\phi_{q-1}$. The original two-column Hermitian control formula (CV.4), before specialization, is

$$\epsilon_N^2 = \frac{a_N d_N - |z_N|^2}{\omega_N^2}, \quad a_N = b_N^* G_N b_N, \quad z_N = b_N^* G_N b_{N+1}, \quad d_N = b_{N+1}^* G_N b_{N+1}. \quad (\text{CJ.5})$$

Here z_N is purely imaginary in the stated dagger-stable source. The original boundary-control identity on (E, G_N) is

$$H_N = G_N^{-1}(A^* G_N + G_N A - k G_N) = \frac{b_{N+1} b_N^* G_N + b_N b_{N+1}^* G_N}{\omega_N}, \quad A = M_S.$$

For completeness this identity follows directly from the original monic recurrence. Self-adjointness of multiplication by u in the integral pairing and orthogonality give $uQ_n = Q_{n+1} + \beta_n Q_n + \alpha_n Q_{n-1}$, where $\beta_n \in \mathbb{R}$ and $\alpha_n = \omega_n/\omega_{n-1}$ for $n \geq 1$; the lower term is zero at $n = 0$. Indeed all coefficients below degree $n - 1$ vanish by pairing with uQ_j of degree at most $n - 1$, and pairing with Q_{n-1} gives the displayed norm ratio. In the original S coordinates this is $Sp_n = p_{n+1} + (c + i\beta_n)p_n - \alpha_n p_{n-1}$. Reduce it modulo χ and insert it into $AK_N + K_N A^* - kK_N$. The diagonal coefficients are $c + i\beta_n + c - i\beta_n - k = 0$. For each internal adjacent pair the coefficient is $\omega_n^{-1} - \alpha_{n+1}\omega_{n+1}^{-1} = 0$. The remaining two terms are $(b_{N+1}b_N^* + b_N b_{N+1}^*)/\omega_N$. Multiplication on the right by the original G_N proves the written boundary-control formula with its signs. It has rank at most two, is self-adjoint in G_N , and has trace $(z_N + \bar{z}_N)/\omega_N = 0$. Squaring its two summands and taking the trace gives $[z_N^2 + \bar{z}_N^2 + 2a_N d_N]/\omega_N^2 = 2(a_N d_N - |z_N|^2)/\omega_N^2$. For $q \geq 2$ its full spectrum is $\epsilon_N, -\epsilon_N$ and $q - 2$ zeros, with $\epsilon_N \geq 0$; these first two entries may themselves be zero. For $q = 1$ trace zero makes the entire control zero. This proves (CJ.5), including the cross-pairing. At $N = q - 1$ the three entries are respectively $\omega_{q-1}, -\gamma_{q-1}\omega_{q-1}, \nu_0 - \omega_q$. Substitution proves

$$\epsilon_{q-1}^2 + \phi_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}. \quad (\text{CJ.6})$$

For $q = 1$, the lower projection in (CV.12) is the zero source and $\epsilon_{q-1} = 0$; the retained phase identity and (CJ.6) still hold. No G_{q-2} occurs in this calculation.

Define $\Lambda_N = \omega_N/V_N$. Combining (CJ.4)–(CJ.6) gives the exact first step

$$(\epsilon_{q-1}^2 + \phi_{q-1}^2)\Lambda_{q-1} = \Lambda_q(1 - \delta_q). \quad (\text{CJ.7})$$

Indeed its left side is $(\nu_0 - \omega_q)/V_{q-1}$, and $\nu_0/V_{q-1} = \omega_q/V_q$ by (CJ.4). This proves the identity with its single contraction and full original phase.

277.2 The complete first-window product and both finite budgets

At each regular degree $N \geq q$, the proved original radius identity (EP.11), equivalently (TVB.27), is

$$(\epsilon_N^2 + \phi_N^2)\Lambda_N = \Lambda_{N+1}(1 - \delta_N)(1 - \delta_{N+1}), \quad \delta_N = V_N/V_{N-1}. \quad (\text{CJ.8})$$

Every δ_N lies in $(0, 1]$. Multiply (CJ.7) by the consecutive identities (CJ.8) at $N = q, \dots, j - 1$. All Λ_N are positive, so canceling their identical intermediate factors is valid. It yields, for every $j \geq q$,

$$\prod_{N=q-1}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_{q-1}}{\omega_{q-1} V_j} (1 - \delta_j) \prod_{N=q}^{j-1} (1 - \delta_N)^2. \quad (\text{CJ.9})$$

For $j = q$ the interior product is empty and equal to one, so this is precisely the first step. For $j > q$, each $\delta_N, q \leq N < j$, occurs once from the step ending at N and once from the step beginning there.

The terminal δ_j occurs once. This proves every exponent in (CJ.9). For a regular initial degree $i \geq q$, the identical multiplication instead gives

$$\prod_{N=i}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_i}{\omega_i V_j} (1 - \delta_i)(1 - \delta_j) \prod_{N=i+1}^{j-1} (1 - \delta_N)^2, \quad j > i. \quad (\text{CJ.10})$$

These two expressions explicitly relate the first and regular windows to the same source, phase and quotient volumes.

For the original off-line quartet let its common multiplicity be $m \geq 1$, its real displacement be $\delta > 0$ and its positive imaginary coordinate be $\gamma > 0$. The complete tensor calculation gives, with no omitted repeated root,

$$q = [1 + k(m-1)](k+1)^2, \quad L = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2. \quad (\text{CJ.11})$$

The original Hermitian compression proof gives $0 < L \leq \epsilon_N$ at every admitted degree: on the full sum of generalized eigenspaces with $\Re \lambda > k/2$, its trace is L . An orthogonal projection onto that subspace in the original metric has trace pairing at most ϵ_N with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N$ and zeros. This is the same source metric as (CJ.2). Thus each contraction occurring in (CJ.9)–(CJ.10) is strictly between zero and one. A value one for any δ_N would make a positive radius product zero.

For either original length- q block $(i, j) = (q-1, 2q-1)$ or $(q, 2q)$, taking logarithms of its exact product now gives

$$\log(V_i/V_j) = \sum_{N=i}^{j-1} \log(\epsilon_N^2 + \phi_N^2) - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j}, \quad (\text{CJ.12})$$

where the entire contraction penalty is

$$\begin{aligned} \mathcal{P}_{q-1, 2q-1} &= -\log(1 - \delta_{2q-1}) - 2 \sum_{N=q}^{2q-2} \log(1 - \delta_N), \\ \mathcal{P}_{q, 2q} &= -\log(1 - \delta_q) - \log(1 - \delta_{2q}) - 2 \sum_{N=q+1}^{2q-1} \log(1 - \delta_N). \end{aligned} \quad (\text{CJ.13})$$

Both sums retain their literal first and last degrees and are nonnegative. Since $\epsilon_N \geq L$, the exact finite bound is

$$\log(V_i/V_j) \geq 2q \log L - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (\text{CJ.14})$$

Every phase term remains; none is assigned the value zero.

277.3 Actual comparable norm windows with complete constants

The original arithmetic lower-envelope proof gives, for $k \geq 3$,

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h. \quad (\text{CJ.15})$$

Its complete analytic proof is retained in the arithmetic source appendix and AU.1–AU.12. Fix $0 < b < \pi/2$, and keep $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$. Set the explicit positive constants

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ M &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{M}/b. \end{aligned} \quad (\text{CJ.16})$$

For every integer $n \geq k \geq 3$ the literal monic norms obey

$$(\ell_h n)^{2n} \leq \omega_n \leq (U_h n)^{2n}. \quad (\text{CJ.17})$$

Here is the full envelope calculation. The exact minimum squared norm of a monic polynomial on $[-n, n]$ is

$$\frac{2n^{2n+1}}{2n+1} \left[\frac{2^n (n!)^2}{(2n)!} \right]^2 \geq \frac{2}{3} n^{2n} 4^{-n}.$$

The last inequality uses $(2n)!/(n!)^2 \leq 4^n$ and $2n/(2n+1) \geq 2/3$. The original chart's factor i^n has modulus one; its monic polynomial space is exactly this extremal space through the inverse substitution. On $[-n, n]$, (CJ.15) is at least $c_h \vartheta_h^{k-3} e^{-\alpha(k-3+n)} (k-2+n)^{-B}$. Use, in their stated directions,

$$\vartheta_h^{k-3} \geq t_h^n, \quad e^{-\alpha(k-3+n)} \geq e^{-2\alpha n}, \quad (k-2+n)^{-B} \geq (2n)^{-B} \geq 2^{-nB}, \quad 2c_h/3 \geq c_0^n.$$

The elementary $2n \leq 2^n$ holds at $n = 1$ and follows by induction from $2(n+1) \leq 4n$ for $n \geq 1$. Multiplication proves the lower bound (CJ.17). For the upper bound use the monic trial polynomial $(S - c)^n$, the nonnegative exponential series, and convolution to obtain $\omega_n \leq (2n)! b^{-2n} (M_h(b)^k + M_h(-b)^k)$. This is at most $(2n)^{2n} b^{-2n} 2M^n$, which is at most $(4\sqrt{M}n/b)^{2n}$ since $2 \leq 4^n$. This proves both inequalities with the original mass retained.

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq 2n$, divide the two instances of (CJ.17) and take the positive $2r$ -th root:

$$\left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)} \leq n U_h^{1+n/r} \ell_h^{-n/r} (1 + r/n)^{1+n/r} \leq C_h^{\text{win}} n, \quad C_h^{\text{win}} = 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2. \quad (\text{CJ.18})$$

Indeed $1 + n/r \leq 3$, $n/r \leq 2$, and $1 + r/n \leq 3$. For each block above, $r = q$ and $n = q$ or $q - 1$. Equation (CJ.11) gives $q - 1 \geq k$ for $k \geq 3$, and $q/2 \leq q \leq 2(q - 1)$ for these q . Both norm factors are therefore at most $C_h^{\text{win}} q$. Combining (CJ.11), (CJ.14) and (CJ.18), with $D_h = \delta/(2C_h^{\text{win}}) > 0$, proves for each block

$$\log(V_i/V_j) \geq 2q \log(D_h k) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (\text{CJ.19})$$

Adding the two bounds and dividing by $q \log k$ proves the four-volume lower limit at least four. This includes the complete finite penalties before taking that consequence.

277.4 The full relation cost and its retained zero directions

For any admitted $i < j$, set $F = [b_{i+1}, \dots, b_j]$ and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. The exact update is $K_j - K_i = F\Omega^{-1}F^*$, so

$$V_i/V_j = \det(I + F\Omega^{-1}F^*G_i) = \det(I + \Omega^{-1}F^*G_iF). \quad (\text{CJ.20})$$

The rectangular determinant equality follows by eliminating either off-diagonal block of the original block matrix

$$\begin{pmatrix} I & F \\ -\Omega^{-1}F^*G_i & I \end{pmatrix}.$$

On the new-column source $\mathbb{C}^{j-i}$ with metric Ω , the second cost is positive self-adjoint. Its eigenvalues are nonnegative, including their zero multiplicities. Write $L_{i,j} : \mathbb{C}^{i+1} \rightarrow \mathbb{C}^{j+1}$ for padding in the same monic orthogonal basis, and $T_{\text{new}} : \mathbb{C}^{j-i} \rightarrow \mathbb{C}^{j+1}$ for the coordinate inclusion into positions $i+1, \dots, j$. The original relation map from $(\mathbb{C}^{j-i}, \Omega)$ to $(\mathcal{P}_j, O_j)$ is $\mathfrak{b} = T_{\text{new}} - L_{i,j}R_iF$. The new columns and the lower source are orthogonal, and $R_i^*O_iR_i = G_i$, so

$$B_j\mathfrak{b} = 0, \quad \mathfrak{b}^*O_j\mathfrak{b} = \Omega + F^*G_iF. \quad (\text{CJ.21})$$

Set $\mathcal{D}_N = \ker B_N = \chi \mathcal{P}_{N-q}$, with the zero space when $N = q - 1$. The new-coordinate component of $\mathfrak{b}$ is the identity; it is therefore injective. Its image lies in $\mathcal{D}_j \cap (L\mathcal{D}_i)^\perp$, whose dimension is $j - i$, by the dimensions of the two nested relation spaces. Thus it is an isomorphism onto that entire relation layer. Reduction sends its labelled vectors to their receiving supported zero; the lifted reduction fixes external absence τ .

At zero tilt the full quartet gives an even original density. Its u -polynomial $\psi = i^{-q} \chi(c + iu)$ satisfies $\psi(-u) = (-1)^q \psi(u)$ because its complete root multiset is stable under negation. Uniqueness of the monic orthogonal polynomial gives $Q_n(-u) = (-1)^n Q_n(u)$: substituting $-u$ preserves all original integral pairings and the adjusted leading coefficient. Monic division modulo ψ preserves parity. When k is even, q in (CJ.11) is odd. In the first block the q new polynomials $Q_q, \dots, Q_{2q-1}$ contain $(q+1)/2$ odd polynomials, while the odd quotient subspace has dimension $(q-1)/2$. Their reduction columns are dependent. The original $S = c + iu$ basis map and the nonzero column phases i^n preserve rank. Thus at least one of the q cost eigenvalues in that block is zero. It remains in the full determinant as a factor one; (CJ.19)–(CJ.21) remain valid with every original dimension and nilpotent direction retained.

For either length- q block let $\lambda_a \geq 0$ be all its cost eigenvalues. Equations (CJ.19)–(CJ.20) imply

$$\prod_{a=1}^q (1 + \lambda_a) \geq (D_h k)^{2q}, \quad \sum_{a=1}^q (1 + \lambda_a)^s \geq q(D_h k)^{2s} \quad (s > 0). \quad (\text{CJ.22})$$

The second assertion is the arithmetic-geometric mean inequality for the q positive numbers $(1 + \lambda_a)^s$; it also obeys the trivial lower bound q . Here is its exact restriction map. Give the same coefficient space E its two original metrics and let $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$ be the identity coefficient map. Its Hilbert adjoint is $T_{i,j} : (E, G_j) \rightarrow (E, G_i)$ with matrix $G_i^{-1} G_j$, because $x^* G_j y = x^* G_i (G_i^{-1} G_j) y$ for every x, y . Thus $Q = T_{i,j} I_{i,j}$ acts on (E, G_i) and satisfies

$$Q^{-1} = G_j^{-1} G_i = K_j G_i = I + F \Omega^{-1} F^* G_i.$$

It is positive self-adjoint in that original metric; its quadratic form is $x^* G_i K_j G_i x > 0$ for $x \neq 0$. Consequently its eigenvalues are exactly the $1 + \lambda_a$ in (CJ.22), with all zero cost directions giving eigenvalue one. Hence (CJ.22) is also an inverse-restriction trace bound in its specified original metric. No individual positive cost lower bound or strict-positive rank claim is inferred.

277.5 Provenance and verification scope

The complete cumulative sources CV and TVB prove the radius and phase map used here, and the complete arithmetic source and AU prove the actual density envelope (CJ.15). PR 25 supplies the comparable-window constants and first-degree propagation. The full first-window phase product (CJ.9), its finite penalties (CJ.13)–(CJ.19), and their connection to the retained relation costs are written out here. The owner review at the cited head requires a nonnegative-eigenvalue correction and workflow-trigger corrections; this chapter preserves that status and the complete proposed parity proof. It performs no local Lean execution and makes no new formal-verification claim. Its finite algebra and analytic envelope calculations belong to this written proof.

278 Result and scope

The original arithmetic source is recovered exactly by the complete family of twisted circle periodizations. The new calculation retains what happens when minimization over the original theta relations is performed in that family. There is a positive, explicitly calculated difference between the original quotient metric and the average of the fibre quotient metrics. It is the norm of a family of original relations, not an unspecified error.

If every phase-source Gram lies between $(1 - \eta)M$ and $(1 + \eta)M$, where M is their exact mean and $0 \leq \eta < 1$, the calculated comparison is

$$(1 - \eta^2)G \preceq \overline{G} \preceq G, \quad G - \overline{G} = \int (C - C_\theta)^* M_\theta (C - C_\theta) \frac{d\theta}{2\pi}. \quad (\text{H1})$$

Here C and C_θ are minimum-norm sections of the **same** original polynomial quotient. Every difference $C - C_\theta$ has its actual theta primitive. The factor η^2 is sharp for the stated matrix hypotheses. This is an application of positive-matrix variational methods, not a priority claim for operator concavity.

The family is joined to finite cyclic covers, the new residue-constituent coefficient, and the original endpoint-volume criterion. All these constructions remain over the marked split base. No finite-field weight theorem is applied to a Hilbert circle quotient. No uniform arithmetic upper bound contradicting the inherited quartet threshold is proved in this note.

279 1. Fixed inputs, maps, and the marked split base

The owner's scalar object, its support observation, and its arithmetic observation are retained:

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r, \quad b_R(\tau) = 0_{\mathbb{B}}, \quad b_R(r^\bullet) = 1_{\mathbb{B}}. \quad (\text{H2})$$

The map (p_R, b_R) is injective. The arithmetic coefficient square is

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (\text{H3})$$

The target of $p_{\mathbb{Z}}$ is infinite. The pointed-monoid base has structure map $\{\tau_0, 1\} \rightarrow U_{\times} G(R)$, taking τ_0 to τ and 1 to $1^\bullet$. Direct image to its point retains the coefficient object; it is not replaced by its Boolean localization.

On a fixed original support label ℓ , a coefficient-linear map f lifts by

$$(\ell, v) \mapsto (\ell, fv), \quad \tau \mapsto \tau. \quad (\text{H4})$$

For a relation submodule B_ℓ , the quotient is $(\ell, v) \mapsto (\ell, [v])$. Its equality criterion is $v - v' \in B_\ell$. Thus $f(v) = 0$ gives the receiving fibre zero, not external absence. Every map constructed below has this lift. Products of phase fibres use the original product support diagram; on a common label the scalar Fourier matrices give linear equivalences. For unequal labels, sums first use the stated joins and transports. No injectivity of arbitrary support transports is assumed.

Retain the original theta data

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int \phi = 0\}, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \\ Q &= \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi. \end{aligned} \quad (\text{H5})$$

The prior full-jet analytic construction supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$, for a selected finite packet h with every selected order complete. The present finite arithmetic module is

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k} > 0, \quad A = M_S, \quad S = s_1 + \cdots + s_k. \quad (\text{H6})$$

Use its same source map and full inclusion:

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi, \quad (\text{H7})$$

$$\eta[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h). \quad (\text{H8})$$

The unit in (H8), every nilpotent jet, and $g = 2\xi$ stay in these equations. For $N \geq q - 1$, let $P_N = \mathbb{C}[S]_{\leq N}$, $J_N = \pi_\chi|_{P_N}$, and $B_N = \ker J_N = \chi P_{N-q}$, where $P_{-1} = 0$. Fix their coefficient bases. A matrix denoted B below is the injective coefficient matrix for multiplication by this literal χ ; it is not a replacement of a higher contracted relation ideal by a power of χ .

The new source corrections are incorporated as follows. The supplied primary-source correction resolves the reference at Weil II 3.7.4 as Roman I.8.11 in Weil I. The strict bound in 3.2.10, the square in 3.2.13, and the lisse/mixed hypotheses in 1.8.12 remain. The supplied translation note keeps its scalar $\exp((-\Phi(-a) - ta)/u)$; the marked arithmetic fibre remains $a = 0$. The supplied residue note's exact rank-one and curvature identities are inherited inputs in Section 8, not discoveries of this continuation.

280 2. All circle holonomies recover the full source

Retain the predecessor's exact logarithmic coordinates

$$r = \log x_k, \quad z_i = \log x_i - \log x_k \ (i < k), \quad (\mathcal{U}_k F)(r, z) = e^{(kr + \sum z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r). \quad (\text{H9})$$

Their logarithmic Jacobian is one. With $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$, this is an isometry into $L^2(\mathbb{R}, dr; \mathcal{K})$, and

$$\mathcal{U}_k D^{(k)} = (-\partial_r + k/2)\mathcal{U}_k. \quad (\text{H10})$$

No relative variable is integrated out of the vector-valued source. Integrals in this note are defined inside each specified coefficient fibre, then lifted by (H4); no unconstructed integration operation on arbitrary mixed-support carriers is assumed. For $L > 0$ and $0 \leq \theta < 2\pi$, define

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad 0 \leq r < L. \quad (\text{H11})$$

The original finite sources have sufficient weighted Schwartz bounds to differentiate and sum this expression on compact sets. It extends to the Hilbert-space transform by density. Its boundary and Fourier equations are

$$\mathcal{Z}_{L,\theta}\psi(r + L) = e^{-i\theta} \mathcal{Z}_{L,\theta}\psi(r), \quad \widehat{\mathcal{Z}_{L,\theta}\psi}(n) = L^{-1/2} \widehat{\psi}((2\pi n + \theta)/L), \quad (\text{H12})$$

where $\widehat{\psi}(u) = \int \psi(r) e^{iur} dr$, and the target basis is $L^{-1/2} e^{-i(2\pi n + \theta)r/L}$. To check the factor, substitute $y = r + aL$ in the coefficient integral. The two phase factors give $e^{ia\theta} e^{-ia(2\pi n + \theta)} = 1$.

Exact isometry and inverse. The full transform is

$$\mathcal{Z}_L : L^2(\mathbb{R}; \mathcal{K}) \xrightarrow{\sim} \int_{[0, 2\pi]}^{\oplus} L^2([0, L]; \mathcal{K}) \frac{d\theta}{2\pi}. \quad (\text{H13})$$

For $0 \leq r < L$ and $a \in \mathbb{Z}$, its inverse is

$$\psi(r + aL) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ia\theta} (\mathcal{Z}_L \psi)(r, \theta) d\theta. \quad (\text{H14})$$

For each r , these are precisely the Fourier-series map and inverse on the sequence $(\psi(r + aL))_a$. Parseval followed by integration in r proves (H13). The measure $d\theta/(2\pi)$ is the actual Fourier inversion measure; no arithmetic mass has been assigned the value one.

For the fixed source-column map $\Psi_N = \mathcal{U}_k \mathcal{V}|_{P_N}$ put

$$M_N(\theta) = (\mathcal{Z}_{L,\theta} \Psi_N)^* (\mathcal{Z}_{L,\theta} \Psi_N), \quad M_N = \Psi_N^* \Psi_N. \quad (\text{H15})$$

Then, exactly,

$$\int_0^{2\pi} M_N(\theta) \frac{d\theta}{2\pi} = M_N. \quad (\text{H16})$$

The original norm is $\int |P(k/2 + iu)|^2 m_{h,k}(u) du$, where $m_{h,k} = w_h^{*k}$ and $w_h = |(g/h)(1/2 + iu)|^2/(2\pi)$. Partial Plancherel therefore gives

$$M_N(\theta)_{ab} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}((2\pi n + \theta)/L) \overline{P_a(k/2 + i(2\pi n + \theta)/L)} P_b(k/2 + i(2\pi n + \theta)/L). \quad (\text{H17})$$

For $k \geq 2$ all these scalar weights are positive. Infinitely many distinct sample points then imply positive definiteness on every finite polynomial source. The uniform quantitative statements below instead use the displayed relative source bound, which by itself implies positive definiteness, including other admitted sources.

At $\theta = 0, n = 0$, retain the term $L^{-1} \widehat{\Psi}_N(0)^* \widehat{\Psi}_N(0)$. It is not discarded in (H15)–(H17). The phase variable is a character of the translation lattice. It is not the coefficient translation a of the exponential note: χ, A, η , and g are fixed throughout (H11)–(H17).

281 3. Every finite cyclic cover is an exact character decomposition

Let $m \geq 1$ and $\theta_j = 2\pi j/m$. For a function f on the circle of length mL , define

$$Q_j f(r) = \frac{1}{m} \sum_{a=0}^{m-1} e^{ia\theta_j} f(r + aL). \quad (\text{H18})$$

Finite character sums prove $Q_j Q_l = \delta_{jl} Q_j$, $\sum_j Q_j = I$, and $Q_j f(r + L) = e^{-i\theta_j} Q_j f(r)$. The actual unitary map and inverse are

$$f \mapsto (\sqrt{m} Q_j f|_{[0,L]})_{j=0}^{m-1}, \quad f(r + aL) = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} e^{-ia\theta_j} g_j(r). \quad (\text{H19})$$

The norm follows either by the finite Fourier identity or orthogonality of the Q_j on the long circle. Applied to the same original source,

$$Q_j \mathcal{Z}_{mL,0} \psi = \frac{1}{m} \mathcal{Z}_{L,\theta_j} \psi, \quad M_N(mL, 0) = \frac{1}{m} \sum_{j=0}^{m-1} M_N(L, \theta_j). \quad (\text{H20})$$

The second equation is exact at every finite degree; the individual masses are not reset. For a general phase θ , the compatible refinement is

$$M_N(L, \theta) = \frac{1}{n} \sum_{a=0}^{n-1} M_N(L/n, (\theta + 2\pi a)/n). \quad (\text{H21})$$

It follows equally by partitioning the sampled frequency lattice into congruence classes. Repeated refinements compose because these are literal partitions of the same lattice.

281.1 The finite locally free coefficient cover

Here is the algebraic cover that gives the finite characters, rather than an identification with an arithmetic Frobenius. Let

$$A_{\text{cov}} = \mathbb{C}[z, z^{-1}], \quad B_{\text{cov}} = \mathbb{C}[w, w^{-1}], \quad z \mapsto w^m. \quad (\text{H22})$$

The pushforward coefficient module is free with its full basis $1, w, \dots, w^{m-1}$. Division of an integer exponent by m proves this also for negative powers. For $\zeta = e^{2\pi i/m}$, the deck action is $\gamma_a(w) = \zeta^{-a}w$. The projector $m^{-1} \sum_a \zeta^{ja} \gamma_a$ has image $A_{\text{cov}} w^j$; multiplication of those summands retains the extra factor z when indices cross m .

The derivative mw^{m-1} is invertible. This proves this particular cover is finite étale, hence finite locally free. EGA I, 1.6.3 and 9.2, and EGA II, 6.1 describe the affine direct-image and finite-morphism setting; here the module and projectors are written explicitly. The base-change map is

$$B_{\text{cov}} \otimes_{A_{\text{cov}}} A' \xrightarrow{\sim} A'[w, w^{-1}]/(w^m - z'), \quad b \otimes a' \mapsto a'b(w). \quad (\text{H23})$$

The circle restriction is $w = e^{-2\pi i r/(mL)}$, $z = e^{-2\pi i r/L}$ and yields (H18). Applying G to the ring map in (H22) commutes with (H3), and fixes τ and e . Character projectors act linearly on the coefficient fibres. A coefficient projected to zero is a represented fibre zero. These idempotent endomorphisms are not relabelled as the supported-zero scalar e .

This is a use of the finite-cover and same-image architecture, not a proof of arithmetic weight bounds from finite étaleness. No six-functor theorem for the split semiring category is asserted.

282 4. Common arithmetic representatives and the exact descent cost

Suppress N for this section. Let $J : P_N \rightarrow E$ be the original surjection, let $B : \mathbb{C}^r \hookrightarrow P_N$ be the coefficient matrix of $\chi_{P_{N-q}}$, and retain $M = \int M_\theta$. For each phase and the common source, define

$$G = (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \quad G_\theta = (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta. \quad (\text{H24})$$

These formulas hold at $N = q - 1$ too: the relation column space then has dimension zero and the corresponding correction is zero. The empty-package coefficient module has empty Gram determinant one and is handled separately; it does not erase the nonzero analytic seed.

The defining equations are

$$JC = JC_\theta = I_E, \quad B^*MC = 0, \quad B^*M_\theta C_\theta = 0. \quad (\text{H25})$$

Consequently there is a unique coefficient map $X_\theta : E \rightarrow \mathbb{C}^r$ with

$$C - C_\theta = BX_\theta, \quad X_\theta = (B^*M_\theta B)^{-1}B^*M_\theta C. \quad (\text{H26})$$

The last formula follows by applying B^*M_θ to the first. Thus every discrepancy is in the original relation submodule, and monic division followed by the existing tensor theta primitive realizes it in the original cochain complex.

Set

$$\overline{G} = \int G_\theta \frac{d\theta}{2\pi}, \quad \mathcal{D} = \int X_\theta^*(B^*M_\theta B)X_\theta \frac{d\theta}{2\pi}. \quad (\text{H27})$$

The fundamental identity is

$$\boxed{G = \overline{G} + \mathcal{D}.} \quad (\text{H28})$$

Proof. Write $C = C_\theta + BX_\theta$ and expand $C^*M_\theta C$. Both cross terms vanish by (H25), leaving $G_\theta + X_\theta^*B^*M_\theta BX_\theta$. Integrating gives $C^*MC = G$. Every integrand is positive semidefinite. Uniform source bounds justify the finite-dimensional integrals and their inverses; for continuous positive definite matrices on the compact phase circle these bounds also exist without a specified value.

Equation (H28) is the exact difference between first taking the common-source quotient and first minimizing separately in each phase. It replaces a proposed interchange of those operations with a calculated relation norm.

282.1 The cochain comparison and its full contractible complement

For m phases put

$$\mathcal{P}_{N,m}^{\text{eq}} = \{(P_j) \in P_N^m : JP_j \text{ is independent of } j\}. \quad (\text{H29})$$

There are actual complexes in degrees 0, 1,

$$[B_N \hookrightarrow P_N] \xrightarrow{\Delta_m} [B_N^m \hookrightarrow \mathcal{P}_{N,m}^{\text{eq}}], \quad (\text{H30})$$

where Δ_m repeats every coefficient. The reverse cochain map is $\text{av}_m(P_j) = m^{-1} \sum_j P_j$ in each degree. It satisfies $\text{av}_m \Delta_m = I$. The complement is

$$\mathcal{B}_{N,m}^0 = \{(b_j) \in B_N^m : \sum_j b_j = 0\}, \quad [\mathcal{B}_{N,m}^0 \xrightarrow{I} \mathcal{B}_{N,m}^0]. \quad (\text{H31})$$

The inverse coefficient coordinates are $P_j \mapsto (\overline{P}, (P_j - \overline{P})_j)$ and $(P, (b_j)) \mapsto (P + b_j)_j$. Since all jets agree, each difference lies in B_N . They prove the asserted direct-sum decomposition of complexes, and the identity is the explicit contraction on the complement. Both complexes have the same degree-one E , with original arithmetic inclusion η . In original arithmetic degree k the complement has degrees $k-1, k$, so an endomorphism preserving this decomposition has its two complement traces with opposite cohomological signs.

The maps are strict for the stated polynomial filtration: repeating a polynomial does not lower its degree, and averaging a degree- N tuple remains degree at most N . More explicitly, $\Delta_m(P_N) = \text{im } \Delta_m \cap P_N^m$, and averaging the equal-jet tuples at degree N is onto P_N . The multiplication map

$M_S : P_N \rightarrow P_{N+1}$ commutes with both operations; its quotient is the original A . At a fixed cutoff it is a degree-raising map, not a presumed endomorphism of P_N .

The complement's cohomology is zero, but its source norm is present in (H27)–(H28). Its contraction is not asserted isometric. Split reconstruction retains its represented vectors, the corresponding supported-zero maps, and external absence. Thus the exactness on the common image does not supply a missing norm estimate by itself.

283 5. Sharp quadratic control of the descent cost

Theorem. Let $M_\theta \succ 0$ be the specified source matrices with mean M , and suppose

$$(1 - \eta)M \preceq M_\theta \preceq (1 + \eta)M, \quad 0 \leq \eta < 1. \quad (\text{H32})$$

Then for the original J and (H24),

$$\boxed{(1 - \eta^2)G \preceq \overline{G} \preceq G, \quad 0 \leq \mathcal{D} \preceq \eta^2 G.} \quad (\text{H33})$$

Proof of the upper inequality. It is (H28).

Proof of the lower inequality. For $x \in [1 - \eta, 1 + \eta]$, the complete scalar identity is

$$\frac{2 - x}{1 - \eta^2} - \frac{1}{x} = \frac{\eta^2 - (x - 1)^2}{x(1 - \eta^2)} \geq 0. \quad (\text{H34})$$

The endomorphism $X_\theta = M^{-1}M_\theta$ is positive self-adjoint in the retained M -metric and has that spectral interval. Applying (H34) to this finite operator and composing with the displayed metric-dual map M^{-1} proves

$$M_\theta^{-1} \preceq \frac{2M^{-1} - M^{-1}M_\theta M^{-1}}{1 - \eta^2}. \quad (\text{H35})$$

Averaging retains $\int M_\theta = M$ and gives

$$\overline{K} := \int JM_\theta^{-1}J^* \frac{d\theta}{2\pi} \preceq (1 - \eta^2)^{-1}K, \quad K = JM^{-1}J^* = G^{-1}. \quad (\text{H36})$$

For each phase the block matrix

$$\begin{pmatrix} K_\theta & I_E \\ I_E & G_\theta \end{pmatrix} \succeq 0 \quad (\text{H37})$$

is positive: its quadratic form is $(x + G_\theta y)^* K_\theta (x + G_\theta y)$. Averaging and taking the Schur complement gives $\overline{G} \succeq \overline{K}^{-1}$. Inverting (H36) proves the lower inequality of (H33). This proof does not change the original metric, discard the mean condition, or choose a new section to fit a desired form.

More generally, $aM \preceq M_\theta \preceq bM$, $0 < a \leq 1 \leq b$, gives

$$\overline{G} \succeq \frac{ab}{a + b - 1}G, \quad G - \overline{G} \preceq \frac{(1 - a)(b - 1)}{a + b - 1}G. \quad (\text{H38})$$

The inverse chord is $x^{-1} \leq (a + b - x)/(ab)$ on $[a, b]$; the same proof applies.

These are variational positive-operator calculations related to the classical shorted-operator construction. The contribution here is the exact supported source, common jet map, family of theta primitives, and phase recovery to which they are applied.

283.1 The arithmetic hypothesis is supplied by the existing correlation bound

For $a > 0$, retain the two actual weighted Grams

$$M_{a,\pm} = (e^{\pm ar} \Psi_N)^* (e^{\pm ar} \Psi_N), \quad \kappa_{a,N} = \sup_{v \neq 0} \frac{v^*(M_{a,+} + M_{a,-})v}{v^* M v}. \quad (\text{H39})$$

Their original coordinate factors are $x_k^{\pm 2a}$. With $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r+s) dr$, unfolding gives

$$M_N(\theta) = \sum_{b \in \mathbb{Z}} e^{ib\theta} \mathcal{C}(bL). \quad (\text{H40})$$

Weighted Cauchy–Schwarz bounds each quadratic correlation by $e^{-a|s|} \sqrt{(v^* M_{a,+} v)(v^* M_{a,-} v)}$. Summing both signs proves, uniformly in phase,

$$-\delta_L M \preceq M_N(\theta) - M \preceq \delta_L M, \quad \delta_L = \frac{\kappa_{a,N}}{e^{aL} - 1}. \quad (\text{H41})$$

Thus any specified L with $\delta_L < 1$ supplies (H32). The passage from source error to the averaged quotient error is now quadratic: $\mathcal{D} \preceq \delta_L^2 G$. Growth of $\kappa_{a,N}$ with the arithmetic packet is not bounded by (H41).

284 6. Endpoints and a genuinely finite phase construction

Apply the uniform phase bound to one finite source containing all four endpoint spaces, for example the original degree $2q+2$ source. Restriction gives the same η on every smaller source. Write $V_N = \det G_N$ and $\bar{V}_N = \det \bar{G}_N$.

Equation (H33) proves

$$q \log(1 - \eta^2) \leq \log \bar{V}_N - \log V_N \leq 0. \quad (\text{H42})$$

For $\mathcal{B} = \log(V_{q-1} V_q / (V_{2q-1} V_{2q}))$ and its averaged-metric analogue $\bar{\mathcal{B}}$,

$$|\bar{\mathcal{B}} - \mathcal{B}| \leq 2q \log \frac{1}{1 - \eta^2}. \quad (\text{H43})$$

The complete equality remains available:

$$V_N = \bar{V}_N \det(I + \bar{G}_N^{-1} \mathcal{D}_N). \quad (\text{H44})$$

The endomorphism $\bar{G}_N^{-1} \mathcal{D}_N$ is positive self-adjoint for $\bar{G}_N$. Both its source map and its determinant contribution are specified. In particular, averaging the quotient metrics without the correction in (H44) is not an asserted exact reconstruction.

At m equally spaced phases define

$$M^{(m)} = \frac{1}{m} \sum_j M(L, \theta_j) = M(mL, 0), \quad G^{(m)} = (J(M^{(m)})^{-1} J^*)^{-1}, \quad \bar{G}^{(m)} = \frac{1}{m} \sum_j G(L, \theta_j). \quad (\text{H45})$$

Every formula (H24)–(H28) holds exactly with this finite mean and its actual canonical section. To compare this finite construction with the original M , put

$$\delta_m = \frac{\kappa_{a,N}}{e^{amL} - 1}, \quad \nu_m = \frac{\delta_L + \delta_m}{1 - \delta_m}. \quad (\text{H46})$$

For $\delta_m < 1$ and $\nu_m < 1$, triangle inequalities relative to M and $M^{(m)}$ give

$$(1 - \nu_m)M^{(m)} \preceq M(L, \theta_j) \preceq (1 + \nu_m)M^{(m)}. \quad (\text{H47})$$

The theorem and the original quotient transport then prove

$$(1 - \nu_m^2)(1 - \delta_m)G \preceq \overline{G}^{(m)} \preceq (1 + \delta_m)G. \quad (\text{H48})$$

Consequently the finite-phase four-volume error is at most

$$2q \log \frac{1 + \delta_m}{(1 - \delta_m)(1 - \nu_m^2)}. \quad (\text{H49})$$

For fixed sufficiently large L , δ_m tends to zero exponentially with the number of phases. The remaining comparison cost tends to the proved quadratic bound, rather than to an unproved zero.

284.1 Exponentially convergent quadrature for the quotient itself

The exact finite-cover identity controls the source mean. There is also a direct, quantitative convergence estimate for the mean of the quotient metrics, with the original matrix inversions included.

Let $\kappa_{\pm}$ be the individual largest generalized eigenvalues of $(M_{a,\pm}, M)$. Weighted Cauchy-Schwarz on two distinct vectors proves

$$\|M^{-1}\mathcal{C}(nL)\|_M \leq \sqrt{\kappa_+\kappa_-} e^{-a|n|L}. \quad (\text{H49a})$$

Here $\|\cdot\|_M$ is the operator norm in the original source metric. Choose $s > 0$ with $s < aL$ and retain

$$d_s = \frac{2\sqrt{\kappa_+\kappa_-}}{e^{aL-s} - 1} < 1. \quad (\text{H49b})$$

The correlation series defines a holomorphic matrix $\widetilde{M}(z)$ on $|\Im z| < aL$, agreeing with M_θ on the real axis, and $\|M^{-1}(\widetilde{M}(z) - M)\|_M \leq d_s$ on $|\Im z| \leq s$. This is a holomorphic extension of the real-axis Gram coefficients. It is not defined by conjugating the complex parameter in a new Gram formula.

Retain the common section C and original relation columns B from (H24), and define its holomorphic Schur complement

$$\widetilde{G}(z) = C^* \widetilde{M}(z) C - C^* \widetilde{M}(z) B (B^* \widetilde{M}(z) B)^{-1} B^* \widetilde{M}(z) C. \quad (\text{H49c})$$

The inverse exists throughout that strip: its difference from $B^* M B$ has relative operator norm at most $d_s < 1$, so the Neumann series converges. On the real axis (H49c) is exactly G_θ . Orthogonality $B^* M C = 0$ gives diagonal bound $1 + d_s$, cross bounds d_s , and inverse bound $(1 - d_s)^{-1}$ in the stated source/quotient metrics. Consequently

$$\|G^{-1} \widetilde{G}(z)\|_G \leq 1 + d_s + \frac{d_s^2}{1 - d_s} = \frac{1}{1 - d_s}. \quad (\text{H49d})$$

Shift the contour in its periodic Fourier coefficient to $\Im z = -s$ for positive frequency and $+s$ for negative frequency. The two vertical sides cancel by periodicity. This proves the coefficient bound

$(1 - d_s)^{-1}e^{-s|n|}$. Root-of-unity averaging keeps only multiples of m , so summing those remaining coefficients gives

$$\left\| G^{-1}(\overline{G}^{(m)} - \overline{G}) \right\|_G \leq \varepsilon_{m,s} := \frac{2}{(1 - d_s)(e^{sm} - 1)}. \quad (\text{H49e})$$

Both real matrices are Hermitian. Combining this norm inequality with the proved quadratic comparison gives the improved finite statement

$$(1 - \eta^2 - \varepsilon_{m,s})G \preceq \overline{G}^{(m)} \preceq (1 + \varepsilon_{m,s})G. \quad (\text{H49f})$$

It is useful whenever its displayed lower factor is positive. Unlike a bare claim of convergence of phase sampling, (H49b)–(H49f) specify the source-weight cost, the analytic strip, the matrix inverse domain and the complete geometric-series tail. The finite four-volume error is then at most

$$2q \log \frac{1 + \varepsilon_{m,s}}{1 - \eta^2 - \varepsilon_{m,s}}. \quad (\text{H49g})$$

The same maximal finite source supplies constants for all four endpoint restrictions. No off-axis value of ξ has been replaced by a phase character; this is a quadrature estimate for the same canonical quotient construction.

284.2 Finite frequency truncation with the phase retained

Write $0 \leq \theta < 2\pi$, $p \geq 1$, and

$$H_{p,N} = \int (1 + r^2)(\partial_r^p \Psi_N)^* \partial_r^p \Psi_N dr. \quad (\text{H50})$$

Integration by parts and Cauchy–Schwarz prove $\|\widehat{\Psi_N v}(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For an integer $J \geq 2$, retain the frequencies $|n| \leq J$. Since $|n + \theta/(2\pi)| \geq |n| - 1$ on the omitted indices,

$$0 \preceq M(L, \theta) - M(L, \theta; J) \preceq \frac{1}{2p-1} \left(\frac{L}{2\pi(J-1)} \right)^{2p-1} H_{p,N}. \quad (\text{H51})$$

The two tails and the factor π/L combine to give exactly this coefficient. A source-relative upper bound $H_{p,N} \preceq h_{p,N} M$ therefore supplies a uniform omitted error ϵ_J . If $\epsilon_J < 1 - \delta_L$, put $\xi_J = \epsilon_J/(1 - \delta_L)$. Then

$$(1 - \xi_J)G(L, \theta) \preceq G(L, \theta; J) \preceq G(L, \theta). \quad (\text{H52})$$

Combining either (H48) or (H49f) with (H52) multiplies that lower factor by $1 - \xi_J$ and leaves its upper factor unchanged. The four-volume error is respectively (H49) or (H49g) with the extra term $2q \log(1/(1 - \xi_J))$. Rounded arithmetic samples require an additional supplied enclosure. No unevaluated sample or high-degree matrix has been certified by writing these bounds.

285 7. The original action and relation primitive descend together

Multiplication by S maps P_N to P_{N+1} . If $C_{N,\theta}$ is the actual phase section, monic division defines the unique $Q_{N,\theta}$ by

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}. \quad (\text{H53})$$

Applying J_{N+1} proves divisibility; this is a finite polynomial identity. For the actual theta map the right-hand side has the inherited original tensor primitive. Denote its image by $\mathcal{B}_{N,\theta}$. The half-density and twisted periodization commute with $D^{(k)}$ as in (H10), so

$$D_{L,\theta}R_{N,\theta} - R_{N,\theta}A = \mathcal{B}_{N,\theta}, \quad D_{L,\theta} = -\partial_r + k/2. \quad (\text{H54})$$

On the quasi-periodic domain both arguments have boundary multiplier $e^{-i\theta}$, whose modulus is one. Integration by parts therefore has no residual endpoint term. It gives

$$A^*G_{N,\theta} + G_{N,\theta}A - kG_{N,\theta} = -(R_{N,\theta}^*\mathcal{B}_{N,\theta} + \mathcal{B}_{N,\theta}^*R_{N,\theta}). \quad (\text{H55})$$

The direct integral and finite character maps carry these identities with all phases and source primitives present. They do not make the theta boundary orthogonal to the arithmetic representative. The circle spectrum alone does not settle (H55).

The period differential from the uploaded residue note is compatible with this same A : $\Pi' = -\Pi(A + t\mathcal{R})/u$, $\mathcal{R} = 1_E \otimes \ell$, and

$$u^2\Pi'' + ku\Pi' + u\Pi\mathcal{R} = \Pi((A + t\mathcal{R})^2 - k(A + t\mathcal{R})). \quad (\text{H56})$$

The extra $u\Pi\mathcal{R}$ is retained. The new phase parameter does not replace t or introduce a translated cyclic polynomial.

286 8. Quantitative transport of the new residue-constituent coefficient

The uploaded note proves the following on the marked cyclic fibre. Put

$$\ell([P]) = [S^{q-1}] \text{rem}_\chi P, \quad \beta(x, y) = \ell(xy), \quad \mathcal{R} = 1_E \otimes \ell. \quad (\text{H57})$$

The residue form β is perfect, with determinant $(-1)^{q(q-1)/2}$. The maps $A^i\mathcal{R}A^j = (A^i1) \otimes (\ell A^j)$ span all endomorphisms. Thus for a fixed nonzero proper A -invariant inclusion $\iota : F \hookrightarrow E$, both $\pi 1_E$ and $\ell_F = \ell\iota$ are nonzero, and

$$\pi\mathcal{R}\iota = (\pi 1_E) \otimes \ell_F : F \rightarrow E/F \quad (\text{H58})$$

has rank exactly one. These are inherited results, including repeated-root primary algebras.

For a positive form M on this same E , retain the coefficient

$$\mathfrak{c}_F(M) = \left(\min_{f \in F} \|1_E - \iota f\|_M^2 \right) \ell_F(\iota^* M \iota)^{-1} \ell_F^*. \quad (\text{H59})$$

Apply (H33), without averaging this nonlinear coefficient. Minimizing the first factor and inverting the restriction in the second factor prove

$$\boxed{(1 - \eta^2)\mathfrak{c}_F(G_N) \leq \mathfrak{c}_F(\overline{G}_N) \leq (1 - \eta^2)^{-1}\mathfrak{c}_F(G_N).} \quad (\text{H60})$$

In logarithms the absolute error is at most $-\log(1 - \eta^2)$, with no factor q . For the finite construction, if $aG \preceq M_0 \preceq bG$ are the explicit bounds from (H48), (H52), the factors are a/b and b/a .

For $H(t) = \Pi(t)^*\Pi(t)$ the inherited period curvature is

$$\partial_t \partial_{\bar{t}} \log \det(\iota^* H(t) \iota) = \frac{|t|^2}{|u|^2} \mathfrak{c}_F(H(t)). \quad (\text{H61})$$

Its original first real derivative, for real $u > 0$, is

$$-u \frac{d}{dt} \log \frac{\det(\iota^* H(t) \iota)}{\det(\iota^* G_N \iota)} \Big|_{t=0} - k \dim F = 2\Re \operatorname{Tr} A_F - k \dim F. \quad (\text{H62})$$

Equations (H60)–(H62) connect the new computable source average with the inherited curvature coefficient and the same marked trace observable. They do not convert control of a single coefficient into an upper bound for every endpoint volume. The original period/source condition number is still needed to compare $H(t)$ and G_N .

The translation note's exact map is retained:

$$\Pi_a = f_a \Pi C_a, \quad f_a = \exp((- \Phi(-a) - ta)/u), \quad C_a A_a C_a^{-1} = A + aI. \quad (\text{H63})$$

Its nonzero holomorphic factor leaves the curvature in (H61) unchanged but changes the additive trace in (H62) by $2 \dim F \Re a$. No such translation is performed here; the marked value is $a = 0$. On a finite-field specialization the separate constant character line is the one in the supplied note, not an identification with the additive shift.

287 9. What premature completion would forget

For a fixed phase, the predecessor's discrete-completion argument gives the map

$$E \longrightarrow \ell^2(\nu_{L,\theta}) / \overline{\chi \mathbb{C}[S]} \simeq \bigoplus_{\chi(k/2+i(2\pi n+\theta)/L)=0} \mathbb{C}, \quad (\text{H64})$$

with kernel $(p_{L,\theta})/(\chi)$, where $p_{L,\theta}$ is the squarefree product over the sampled roots. This follows from polynomial density for the exponentially weighted discrete measure and its $|\chi|^2$ multiple; it is a further Hilbert quotient, not (H24).

Only finitely many phases modulo 2π can sample a root of the fixed polynomial. Thus the direct integral of the **completed quotient targets** in (H64) is zero almost everywhere, even when the original E is nonzero. Its map from E has kernel all of E . In split coordinates that map sends represented inputs to the zero of their receiving support; it still fixes external absence. At every finite degree in (H24), $J C_\theta = I_E$ and the averaged Gram is positive definite.

The typed order of operations is therefore part of the result: the exact source isometry (H13), the original finite quotient (H24), and the later completion (H64) are accompanied by their actual maps and kernels. None is silently exchanged with another. The contractible complement in (H31) and the positive norm (H28) likewise coexist.

288 10. Exact calibrations and the remaining arithmetic estimate

The constant in (H33) is sharp. Take a coefficient source with the literal mass 7,

$$M_\pm = 7 \begin{pmatrix} 1 & \pm\eta \\ \pm\eta & 1 \end{pmatrix}, \quad M = \frac{1}{2}(M_+ + M_-) = 7I_2, \quad J = (1 \ 0), \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (\text{H65})$$

These are also moment matrices of the two measures on the points $-1, 1$ with masses $7(1 \mp \eta)/2$ and $7(1 \pm \eta)/2$. Direct minimization gives

$$C = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C_+ = \begin{pmatrix} 1 \\ -\eta \end{pmatrix}, \quad C_- = \begin{pmatrix} 1 \\ \eta \end{pmatrix}, \quad G = 7, \quad G_\pm = 7(1 - \eta^2), \quad \mathcal{D} = 7\eta^2. \quad (\text{H66})$$

Every difference is in the original kernel of J . This is a finite positive-source calibration, not a zeta packet. It proves that the quadratic loss cannot be reduced under only (H32) and the mean condition.

The new exact checker includes repeated-root polynomial quotient fixtures, complex Gram cross terms, finite character projectors, the cochain complement and contraction, source-mean equality, and the precise residue-coefficient comparison. It uses exact rational or Gaussian-rational arithmetic. Its finite checks do not certify the infinite theta integrals, the period determinant argument, or a general Lean theorem.

The arithmetic endpoint issue remains the inherited threshold-four problem for the fixed hypothetical quartet: the original source would require $\mathcal{B}_{h,k} \geq 4q_k \log(D_h k)$. Equation (H43) and its finite version transport that threshold with an explicit $O(q_k)$ allowance when the supplied error factors are fixed. They do not prove a contradicting upper bound for $\bar{\mathcal{B}}_{h,k}$.

The progress here is a stronger, source-preserving descent: all holonomies recover the source exactly; finite covers assemble those holonomies coherently; the failure of independently minimized local representatives to glue is an explicit theta-relation norm; its effect is quadratically bounded; and the same bound reaches the new residue-curvature coefficient. The original complex, infinite arithmetic quotient, support labels, arithmetic action, and full jets remain throughout.

289 Source basis and attribution

The owner supplies the SplitZero, theta, canonical full-jet, periodization, and residue-constituent programme. This note supplies the displayed all-holonomy comparison, exact common-versus-local minimum correction, its sharp quadratic enclosure, and the explicit joins to that programme. No global novelty claim is made for Fourier-series decomposition, finite-cover character projectors, positive-matrix concavity, or Schur complements.

The new uploaded primary-source correction is accepted at its stated review scope. This session read its TeX, not the underlying scan anew. The selected uploaded EGA readings were I, 1.6.3 and 9.2.1–9.2.2, and II, 6.1.1–6.1.5; neither the entire EGA corpus nor the complete Deligne corpus has been audited. The explanatory Deligne sidebar’s comparison, cone, dual, and split-lift sections were read. The standalone `Deligne_Split_Cohomology(2).tex` is a wrapper; its sibling sidebar was read as the intended substantive input without silently editing its historical include path.

Primary external attribution:

- Deligne, P. (1980). *La conjecture de Weil II*. Publications mathématiques de l’IHÉS 52, 137–252. The supplied correction retains the precise roles of 1.8.12, 3.2.10, 3.2.13, 3.3.4–3.3.6 and 3.7.2–3.7.4.
- Grothendieck, A., and Dieudonné, J. *Éléments de géométrie algébrique*, the selected uploaded editable sections identified above. For the finite locally free definition, see also Stacks Project, Tag 02K9; for affine morphisms and their coefficient algebra, Tag 01S5.
- Zak, J. (1967). Finite translations in solid-state physics. *Physical Review Letters* 19, 1385–1387. DOI 10.1103/PhysRevLett.19.1385. The publisher abstract was inspected for historical attribution; (H11)–(H21) have proofs in this note rather than a claim to have audited the full paper.
- Anderson, W. N., Jr., and Trapp, G. E. (1975). Shorted operators II. *SIAM Journal on Applied Mathematics* 28, 60–71. DOI 10.1137/0128007. The publisher abstract and bibliographic page were inspected; the exact finite inequality used here is proved in (H34)–(H38).

Source files, byte hashes, exact selected ranges, execution receipts, and current-repository inspection are recorded separately. Literature source files and the owner’s unrelated corpus are not republished in this delivery.

290 Density and exact completion of the original periodized arithmetic source

290.1 1. The unchanged theta amplitude and its canceled zeros

Use the original Mellin transform

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx, \quad D = -x\partial_x, \quad g(s) = 2\zeta(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{PSD1})$$

The source is the supplied strong-Schwartz function satisfying

$$v_h(s) := \mathcal{M}F_h(s) = \frac{g(s)}{h(s)}, \quad h(D)F_h = \Theta\phi_*, \quad \phi_*(x) = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}. \quad (\text{PSD2})$$

Here the quotient is its entire extension, retaining the given polynomial h . Every chosen root occurs in h with its full multiplicity in g ; no location assumption is imposed on those roots. Writing a possible leading coefficient explicitly, let

$$h(s) = a_h \prod_{\rho \in Z_h} (s - \rho)^{m_\rho}, \quad a_h \neq 0, \quad d = \sum_{\rho} m_\rho. \quad (\text{PSD3})$$

For the original monic packet, $a_h = 1$; retaining it in these formulas shows where that convention enters.

The theta factor is exact. For $\text{Re}(s) > 1$, absolute convergence gives

$$\begin{aligned} \mathcal{M}\phi_*(s) &= \pi^{-s/2}(2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) \\ &= \frac{s(s-1)}{2}\pi^{-s/2}\Gamma(s/2), \\ \mathcal{M}(\Theta\phi_*)(s) &= 2\zeta(s)\mathcal{M}\phi_*(s) = g(s). \end{aligned} \quad (\text{PSD4})$$

The factor two in Θ and the factor one-half in the Mellin transform of ϕ_* remain visible. Logarithmic integration by parts gives $\mathcal{M}(DF) = s\mathcal{M}F$ for the given strong-Schwartz source; the boundary terms vanish at both ends. Thus (PSD2) has the same Mellin transform as its original theta right-hand side. The conditions $\phi_*(0) = 0$ and $\int_{\mathbb{R}} \phi_* = 2\mathcal{M}\phi_*(1) = 0$ also retain the two source moments.

At a selected zero ρ of order m , put

$$g(s) = (s - \rho)^m a(s), \quad h(s) = (s - \rho)^m b(s), \quad a(\rho)b(\rho) \neq 0.$$

The canceled value is therefore

$$v_h(\rho) = \frac{a(\rho)}{b(\rho)} = \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)} \neq 0. \quad (\text{PSD5})$$

This value applies when a selected critical zero lies exactly on a sampled Fourier frequency. It must not be evaluated as an uncanceled $0/0$ or assigned zero. The entire jet is fixed as well: with

$$a_r = \frac{g^{(m+r)}(\rho)}{(m+r)!}, \quad b_r = \frac{h^{(m+r)}(\rho)}{(m+r)!}, \quad v_h(\rho + z) = \sum_{j \geq 0} c_j z^j,$$

one has

$$c_0 = a_0/b_0, \quad c_j = \frac{a_j - \sum_{r=1}^j b_r c_{j-r}}{b_0} \quad (j \geq 1). \quad (\text{PSD6})$$

The arithmetic density used throughout the periodized note is exactly

$$w_h(t) = \frac{1}{2\pi} |v_h(1/2 + it)|^2, \quad m_{h,k} = \underbrace{w_h * \cdots * w_h}_{k \text{ factors}}. \quad (\text{PSD7})$$

Because v_h is entire and not identically zero, its zeros on the vertical line form a discrete subset of the real parameter t . Consequently w_h is continuous, nonnegative, and strictly positive outside that discrete subset. Formula (PSD5) determines the values at every canceled critical root. No zero-location theorem is used.

290.2 2. A proved exponential tail, with the original constants

An elementary contour estimate supplies the needed gamma decay without substituting a numerical approximation. If $a > 0$, y is real, and $0 < \theta < \pi/2$, rotating the gamma-integral ray by $\text{sign}(y)\theta$ gives

$$|\Gamma(a + iy)| \leq \Gamma(a)(\cos \theta)^{-a} e^{-\theta|y|}. \quad (\text{PSD8})$$

For completeness, take the branch of z^{a+iy-1} in the sector between the positive ray and that rotated ray. Integrate $z^{a+iy-1} e^{-z}$ around its boundary between radii ϵ and R . For each fixed y , the small arc is bounded by a constant times ϵ^a and tends to zero. On the large arc $\text{Re}(z) \geq R \cos(\theta) > 0$, so exponential decay dominates the power of R . The two ray integrals agree in the limit. Parametrizing the rotated ray by $z = e^{i \text{sign}(y)\theta} r$ contributes the exact factor $e^{i \text{sign}(y)\theta(a+iy)}$, whose modulus is $e^{-\theta|y|}$. Taking the modulus inside the remaining integral gives $\int_0^\infty r^{a-1} e^{-r \cos \theta} dr = \Gamma(a)(\cos \theta)^{-a}$. At $y = 0$ the same bound follows from either ray.

On the original critical line, an elementary bound for zeta follows from the alternating Dirichlet series. For $\text{Re}(s) > 0$, its partial summation formula is

$$\eta(s) = s \int_1^\infty A(x) x^{-s-1} dx, \quad A(x) = \sum_{1 \leq n \leq x} (-1)^{n-1} \in \{0, 1\}.$$

The integral converges locally uniformly there and agrees with the alternating series; the finite partial-summation boundary has modulus at most $N^{-\text{Re}(s)}$ and tends to zero. For $s = 1/2 + it$,

$$|\eta(s)| \leq 2|s|, \quad |1 - 2^{1-s}| \geq \sqrt{2} - 1, \quad |\zeta(s)| \leq \frac{2|s|}{\sqrt{2} - 1}. \quad (\text{PSD9})$$

The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ first follows by absolutely convergent rearrangement for $\text{Re}(s) > 1$, then by analytic continuation. Its denominator on the line has the displayed strictly positive lower bound.

Apply (PSD8) with $a = 1/4$, $y = t/2$, and retain $|\pi^{-s/2}|^2 = \pi^{-1/2}$. For every real t with $h(1/2 + it) \neq 0$, (PSD1), (PSD7), and (PSD9) give

$$w_h(t) \leq \frac{2\Gamma(1/4)^2}{\pi^{3/2}(\sqrt{2} - 1)^2 \sqrt{\cos \theta}} \frac{(t^2 + 1/4)^3}{|h(1/2 + it)|^2} e^{-\theta|t|}. \quad (\text{PSD10})$$

The power $(t^2 + 1/4)^3$ comes from $|s(s-1)|^2|s|^2$; no asymptotic equality is asserted. Let

$$R = \max(\{1\} \cup \{2|\rho - 1/2| : \rho \in Z_h\}).$$

This definition also gives $R = 1$ for the retained analytic seed $h = 1$, whose selected-root set is empty; all product and tail formulas then use $d = 0$ and $a_h = 1$.

For $|t| \geq R$, each factor satisfies

$$|1/2 + it - \rho| \geq |t| - |\rho - 1/2| \geq |t|/2, \quad |h(1/2 + it)| \geq |a_h|(|t|/2)^d.$$

Thus, with the exact constant in front of the fraction in (PSD10) denoted by C_θ ,

$$w_h(t) \leq C_\theta \frac{4^d}{|a_h|^2} \frac{(t^2 + 1/4)^3}{|t|^{2d}} e^{-\theta|t|} \quad (|t| \geq R). \quad (\text{PSD11})$$

On the compact interval $[-R, R]$, the entire canceled quotient in (PSD2) is continuous, so $B_R = \max_{|t| \leq R} w_h(t) < \infty$. For any $0 < b < \pi/2$, choose $b < \theta < \pi/2$. The compact bound together with (PSD11) proves

$$A_b := \|e^{b|\cdot|} w_h\|_\infty < \infty, \quad B_b := \|e^{b|\cdot|} w_h\|_1 < \infty. \quad (\text{PSD12})$$

For example, source-dependent upper bounds are the maximum of $e^{bR} B_R$ and the supremum of the right side of (PSD11) multiplied by $e^{b|t|}$ on the two tails, and

$$B_b \leq 2R e^{bR} B_R + 2C_\theta \frac{4^d}{|a_h|^2} \int_R^\infty \frac{(T^2 + 1/4)^3}{T^{2d}} e^{-(\theta-b)T} dT. \quad (\text{PSD13})$$

The remaining integral is finite because $\theta - b > 0$. These are analytical constants of the original source; this audit has not computed their values or interval enclosures.

290.3 3. Pointwise convolution, strict positivity, and the full scalar-circle map

Convolution is taken with ordinary Lebesgue measure, with the factors $1/(2\pi)$ already contained in every w_h . The triangle inequality for $u = t_1 + \dots + t_k$ gives pointwise

$$e^{b|u|} m_{h,k}(u) \leq (e^{b|\cdot|} w_h)^{*k}(u) \leq A_b B_b^{k-1}. \quad (\text{PSD14})$$

For two factors this follows directly by bounding one factor by A_b and integrating the other. Induction gives the displayed expression for every k . The integrals are absolutely convergent for every u . The same bounds and continuity under translation give a continuous representative of the convolution. For instance, w_h is bounded, continuous, and tends to zero at infinity by (PSD11), hence uniformly continuous; dominated convergence against the integrable other factor proves continuity at every u .

For $k = 2$ and fixed u , the sets where $w_h(t) = 0$ or $w_h(u - t) = 0$ are discrete. There is a t_0 outside their union. The continuous nonnegative integrand is strictly positive on a neighborhood of t_0 , and therefore

$$(w_h * w_h)(u) > 0 \quad \text{for every real } u. \quad (\text{PSD15})$$

For the induction step, a continuous everywhere-positive convolution is integrated against w_h , which is positive on some open interval. The resulting convolution is strictly positive at every u . Consequently all (P53) weights are strictly positive for $k \geq 2$, including $n = 0$.

Retain the original diagonal coordinate isometry $\mathcal{U}_k$, the space $\mathcal{K}$, and the plus-sign Fourier transform from (P11)–(P14). Put

$$\psi_0 = \mathcal{U}_k F_h^{\otimes k}, \quad u_n = 2\pi n/L, \quad S_n = k/2 + iu_n, \quad b_n = \widehat{\psi}_0(u_n) \in \mathcal{K}.$$

In the simultaneous Fourier variables for $(r, z_1, \dots, z_{k-1})$, the amplitude is exactly

$$\prod_{j < k} v_h(1/2 + it_j) v_h \left(1/2 + i \left(u - \sum_{j < k} t_j \right) \right).$$

For each fixed u , partial Plancherel in the retained z variables therefore gives

$$\|\widehat{\psi}_0(u)\|_{\mathcal{K}}^2 = (2\pi)^{-(k-1)} \int_{\mathbb{R}^{k-1}} \prod_{j < k} |v_h(1/2 + it_j)|^2 \left| v_h \left(1/2 + i \left(u - \sum_{j < k} t_j \right) \right) \right|^2 d\mathbf{t} = 2\pi m_{h,k}(u). \quad (\text{PSD16})$$

The equality holds pointwise, not just almost everywhere: the original transformed source is Schwartz in the Hilbert-valued variable, and both sides are continuous; alternatively the preceding absolutely convergent integral computes the partial transform at each u . Thus no choice of an almost-everywhere convolution version can alter a lattice sample.

Let $H_{\text{orig}, \text{cyc}, L}$ denote the closure in the original $L^2(\mathbb{R}/L\mathbb{Z}; \mathcal{K})$ of the periodized polynomial source. Define the weighted scalar space

$$H_\nu = \ell^2(\mathbb{Z}, \nu), \quad \nu_n = \frac{2\pi}{L} m_{h,k}(u_n) = \frac{\|b_n\|_{\mathcal{K}}^2}{L}. \quad (\text{PSD17})$$

There is an exact isometry into the original Hilbert-valued circle space:

$$I_L : H_\nu \longrightarrow L^2(\mathbb{R}/L\mathbb{Z}; \mathcal{K}), \quad \widehat{I_L a}(n) = L^{-1/2} a_n b_n. \quad (\text{PSD18})$$

The hats on the right are coefficients in the unchanged orthonormal basis $L^{-1/2} \exp(-2\pi i n r / L)$. Parseval gives $\|I_L a\|^2 = \sum_n \nu_n |a_n|^2$. For a polynomial P , (P13) and (P16) give

$$I_L((P(S_n))_n) = \mathcal{P}_L \mathcal{U}_k \mathcal{V}_{h,k} P. \quad (\text{PSD19})$$

Polynomial density, proved next, identifies the range of (PSD18) with $H_{\text{orig}, \text{cyc}, L}$. Its inverse on that range is

$$a_n = \sqrt{L} \frac{\langle b_n, \widehat{f}(n) \rangle_{\mathcal{K}}}{\|b_n\|_{\mathcal{K}}^2}, \quad (\text{PSD20})$$

with the inner product conjugate-linear in its first variable. Every denominator is positive for $k \geq 2$. These maps retain the amplitude b_n , the actual source measure, and every Fourier constant.

290.4 4. Polynomial density with an explicit analytic strip and uniqueness

Fix $k \geq 2$ and $L > 0$. By (PSD14), for each $0 < b < \pi/2$,

$$\nu_n \leq \frac{2\pi}{L} A_b B_b^{k-1} e^{-b|u_n|}. \quad (\text{PSD21})$$

For every fixed polynomial Q and $0 < c < b$,

$$\sum_{n \in \mathbb{Z}} e^{c|u_n|} |Q(S_n)|^2 \nu_n < \infty. \quad (\text{PSD22})$$

Indeed $|Q(k/2 + iu)|^2$ is bounded by a finite polynomial in $|u|$, and the remaining factor is $\exp(-(b-c)|u_n|)$ on a lattice with spacing $2\pi/L$. All polynomial moments exist.

Here is the density proof for any positive submeasure μ_n on a subset I of this lattice satisfying a finite exponential moment $\sum_{n \in I} e^{c|u_n|} \mu_n < \infty$ for some $c > 0$. Suppose $f \in \ell^2(I, \mu)$ is orthogonal to all polynomial sequences $P(S_n)$. For $0 < a < c/2$, Cauchy–Schwarz gives

$$\sum_{n \in I} |f_n| \mu_n e^{a|u_n|} \leq \left(\sum_{n \in I} |f_n|^2 \mu_n \right)^{1/2} \left(\sum_{n \in I} e^{2a|u_n|} \mu_n \right)^{1/2} < \infty. \quad (\text{PSD23})$$

Consequently

$$F(z) = \sum_{n \in I} \overline{f_n} \mu_n e^{iz u_n} \quad (\text{PSD24})$$

is holomorphic on $|\text{Im}(z)| < c/2$. On every smaller closed strip, the series and all its derivative series converge normally: choose a further positive margin inside $c/2$, and absorb each fixed power of $|u_n|$ into the exponential margin in (PSD23). The half-width is justified by the square in Cauchy–Schwarz; no full-width extension is needed.

For every $j \geq 0$, the exact polynomial $P_j(S) = ((S - k/2)/i)^j$ evaluates to u_n^j , so orthogonality gives

$$F^{(j)}(0) = i^j \sum_{n \in I} \overline{f_n} \mu_n u_n^j = 0. \quad (\text{PSD25})$$

The Taylor expansion and the identity theorem on the connected strip imply $F = 0$ there. Uniqueness can be proved directly with the actual lattice, rather than invoked abstractly: extend the coefficients by zero outside I . Since (PSD23) gives absolute summability, termwise integration is valid and

$$\frac{1}{L} \int_0^L F(x) e^{-iu_m x} dx = \sum_{n \in I} \overline{f_n} \mu_n \frac{1}{L} \int_0^L e^{2\pi i(n-m)x/L} dx = \begin{cases} \overline{f_m} \mu_m, & m \in I, \\ 0, & m \notin I. \end{cases} \quad (\text{PSD26})$$

The left side is zero. Positivity of each μ_m on I implies $f_m = 0$. The orthogonal complement of the polynomials is zero, proving their density.

Apply this first to $\mu = \nu$ and $I = \mathbb{Z}$. It proves the assertion used in (PSD19). Apply it again to

$$I = \{n : \chi(S_n) \neq 0\}, \quad \mu_n = |\chi(S_n)|^2 \nu_n. \quad (\text{PSD27})$$

The exponential moment follows from (PSD22) with $Q = \chi$. Zero atoms are removed rather than divided by. This establishes exactly the weighted density needed for (P56).

290.5 5. The exact closed relation space, quotient metric, and kernel

Let χ be the original nonconstant packet-sum polynomial of degree q , and set

$$\mathcal{Z}_L = \{n \in \mathbb{Z} : \chi(S_n) = 0\}, \quad p_L(S) = \prod_{n \in \mathcal{Z}_L} (S - S_n). \quad (\text{PSD28})$$

The set is finite because χ is a nonzero polynomial and the S_n are distinct. The empty product is one. These are exact sampled roots; no tolerance rule is introduced.

Every χP vanishes on $\mathcal{Z}_L$. Coordinate evaluation is continuous on H_ν , since $|f_n| \leq \nu_n^{-1/2} \|f\|_\nu$. Therefore the closure of $\chi \mathbb{C}[S]$ is contained in the closed subspace

$$V_L = \{f \in H_\nu : f_n = 0 \ (n \in \mathcal{Z}_L)\}.$$

Conversely take $f \in V_L$ and define $g_n = f_n/\chi(S_n)$ on its complement I . Then

$$\sum_{n \in I} |g_n|^2 |\chi(S_n)|^2 \nu_n = \sum_{n \in I} |f_n|^2 \nu_n < \infty.$$

By (PSD27), polynomials P_j approximate g in that weighted space. Multiplication by χ gives

$$\|\chi P_j - f\|_\nu^2 = \sum_{n \in I} |P_j(S_n) - g_n|^2 |\chi(S_n)|^2 \nu_n \longrightarrow 0. \quad (\text{PSD29})$$

Thus $\overline{\chi \mathbb{C}[S]} = V_L$. The orthogonal complement is the sequence space supported on $\mathcal{Z}_L$, and the Hilbert quotient is isometric to that subspace by restriction. Its metric is exactly $\sum_{n \in \mathcal{Z}_L} \nu_n |v_n|^2$.

The resulting algebraic comparison is the surjection

$$\theta_L : \mathbb{C}[S]/(\chi) \longrightarrow H_\nu/\overline{\chi \mathbb{C}[S]} \simeq \bigoplus_{n \in \mathcal{Z}_L} \mathbb{C}, \quad [P] \longmapsto (P(S_n))_{n \in \mathcal{Z}_L}. \quad (\text{PSD30})$$

It is well-defined because $\chi(S_n) = 0$. Surjectivity follows from the explicit Lagrange polynomials

$$\ell_n(S) = \prod_{m \in \mathcal{Z}_L, m \neq n} \frac{S - S_m}{S_n - S_m}, \quad \ell_n(S_m) = \delta_{nm}. \quad (\text{PSD31})$$

For an empty set the target is the zero vector space, and the map is the unique surjection onto it. Since the sampled roots are distinct, a polynomial vanishes on them if and only if it is divisible by p_L . Moreover p_L divides χ . Hence

$$\ker \theta_L = (p_L)/(\chi), \quad \text{rank } \theta_L = |\mathcal{Z}_L|, \quad \dim \ker \theta_L = q - |\mathcal{Z}_L|. \quad (\text{PSD32})$$

These formulas include the empty set, when $(p_L)/(\chi) = \mathbb{C}[S]/(\chi)$.

The generator intertwines exactly. With $A[P] = [SP]$ on the original algebra and $A_L(v_n) = (S_n v_n)$ on the finite target,

$$\theta_L A = A_L \theta_L. \quad (\text{PSD33})$$

The kernel is therefore an invariant submodule; it has not been removed before forming the comparison.

There is a precise cyclic presentation of this invariant kernel. Put $t_L = \chi/p_L$ and define

$$\mu_{p_L} : \mathbb{C}[S]/(t_L) \longrightarrow E, \quad [Q]_{(t_L)} \longmapsto [p_L Q]_{(\chi)}. \quad (\text{PSD33a})$$

This is well-defined because $p_L t_L = \chi$. Its image is $(p_L)/(\chi)$. If $p_L Q$ is divisible by $\chi = p_L t_L$, polynomial cancellation gives divisibility of Q by t_L , so the map is injective. No coprimality between p_L and t_L is used. Consequently the entire comparison is the exact sequence

$$0 \longrightarrow \mathbb{C}[S]/(\chi/p_L) \xrightarrow{\mu_{p_L}} \mathbb{C}[S]/(\chi) \xrightarrow{\theta_L} \mathbb{C}^{\mathcal{Z}_L} \longrightarrow 0. \quad (\text{PSD33b})$$

Multiplication commutes in the polynomial ring, hence $A\mu_{p_L} = \mu_{p_L} A_{t_L}$, with A_{t_L} the original multiplication by S on the source algebra. Together with (PSD33), this proves equivariance of both arrows.

The Fourier-side generator has the same typed action. Define

$$T_L a = (S_n a_n)_n, \quad \mathcal{D}(T_L) = \left\{ a \in H_\nu : \sum_n \nu_n |S_n a_n|^2 < \infty \right\}. \quad (\text{PSD33c})$$

It is closed: convergence of a and $T_L a$ in H_ν implies convergence of each coordinate, giving the displayed pointwise relation in the limit and its square summability. The coordinate projection Q_{Z_L} commutes with T_L on this domain and has range in it, because its support is finite. On polynomials, $T_L(P(S_n)) = ((SP)(S_n))$; under (PSD18) it is exactly the circle operator $-\partial_r + k/2$. It induces A_L on the quotient. Therefore the quadratic generator also intertwines exactly:

$$\theta_L(A^2 - kA) = (A_L^2 - kA_L)\theta_L, \quad S_n(S_n - k) = -k^2/4 - (2\pi n/L)^2. \quad (\text{PSD33d})$$

290.6 6. Full repeated-root blocks and the canonical metric limit

Factor the actual polynomial as $\chi(S) = a_\chi \prod_\lambda (S - \lambda)^{m_\lambda}$. The exact Chinese-remainder map is

$$\mathbb{C}[S]/(\chi) \xrightarrow{\sim} \bigoplus_\lambda \mathbb{C}[\epsilon_\lambda]/(\epsilon_\lambda^{m_\lambda}), \quad [P] \mapsto \left(\sum_{j=0}^{m_\lambda-1} \frac{P^{(j)}(\lambda)}{j!} \epsilon_\lambda^j \right)_\lambda. \quad (\text{PSD34})$$

Under this map, A acts on each summand by $\lambda + \epsilon_\lambda$. At an unsampled root the whole summand lies in $\ker(\theta_L)$. At a sampled root $\lambda = S_n$, the comparison takes the constant coefficient of this full Taylor class, and its kernel is exactly

$$\epsilon_\lambda \mathbb{C}[\epsilon_\lambda]/(\epsilon_\lambda^{m_\lambda}). \quad (\text{PSD35})$$

In particular every nonconstant generalized-jet direction is retained in the explicit kernel. The induced action on the remaining constant coefficient is multiplication by λ ; (PSD33) follows in these coordinates as well. This proves the full multiplicity statement rather than inferring it from a list of sampled eigenvalues.

The inverse in (PSD34) is explicit. Set $H_\lambda(S) = \chi(S)/(S - \lambda)^{m_\lambda}$. Its value at λ is nonzero. Let $V_\lambda(\epsilon)$ be the Taylor polynomial of $1/H_\lambda(\lambda + \epsilon)$ through degree $m_\lambda - 1$, and put $e_\lambda(S) = H_\lambda(S)V_\lambda(S - \lambda)$. Modulo the λ primary factor this polynomial equals one; modulo every other primary factor it equals zero. The inverse sends a tuple $F_\lambda(\epsilon)$ to $[\sum_\lambda e_\lambda(S)F_\lambda(S - \lambda)]_{(\chi)}$. Both composites are the identity: their differences are divisible by every pairwise coprime primary factor, hence by their product, which differs from χ only by its original nonzero leading coefficient.

At a sampled root, write $p_L(\lambda + \epsilon) = \epsilon u_\lambda(\epsilon)$, where $u_\lambda(0) = \prod_{\substack{\mu \text{ sampled} \\ \mu \neq \lambda}} (\lambda - \mu) \neq 0$. Multiplication by p_L identifies the t_L block $\mathbb{C}[\epsilon]/(\epsilon^{m_\lambda-1})$ with (PSD35) by multiplication by ϵu_λ . At an unsampled root, $p_L(\lambda + \epsilon)$ is a unit and identifies the entire t_L block with the original full block. Thus the kernel generator has block length $m_\lambda - 1$ at each sampled root and length m_λ at each unsampled root, with the original eigenvalues retained.

Now retain the note's fixed remainder-coordinate basis on $E = \mathbb{C}[S]/(\chi)$, the degree- N polynomial source $\mathcal{P}_N$, and the canonical constrained minimum $G_N(L)$. For every $N \geq q - 1$, its source Gram is positive definite: the summands have positive weights and a nonzero polynomial has only finitely many zeros, whereas this lattice is infinite. The finite-dimensional minimum therefore exists uniquely and is precisely

$$e^* G_N(L) e = \min \{ \|P\|_\nu^2 : P \in \mathcal{P}_N, [P] = e \}. \quad (\text{PSD36})$$

Choose the unique degree-less-than- q remainder R_e of e . The permitted representatives are $R_e + \chi \mathcal{P}_{N-q}$, with the relation space understood to be zero when $N = q - 1$. These relation spaces

increase with N , so the forms $G_N(L)$ decrease in Loewner order. Their union is $\chi\mathbb{C}[S]$; their closure is the space computed in (PSD29). For a closed-subspace distance, approximation in that closure proves

$$\begin{aligned} \lim_{N \rightarrow \infty} e^* G_N(L) e &= \inf_{Q \in \mathbb{C}[S]} \|R_e + \chi Q\|_\nu^2 \\ &= \text{dist}(R_e, V_L)^2 = \sum_{n \in \mathcal{Z}_L} \nu_n |R_e(S_n)|^2. \end{aligned} \quad (\text{PSD37})$$

The equality with the infimum is exact: every polynomial Q appears at a finite degree, so the decreasing finite-degree infima converge to the infimum over the union; the norm is continuous, so replacing the union by its closure does not change that infimum.

Let $J_{\mathcal{Z}_L}$ be evaluation at the sampled roots in the same remainder basis. Equation (PSD37) gives the matrix limit

$$G_N(L) \downarrow G_\infty(L) := J_{\mathcal{Z}_L}^* \text{diag}(\nu_n)_{n \in \mathcal{Z}_L} J_{\mathcal{Z}_L}. \quad (\text{PSD38})$$

Polarization of (PSD37) gives every matrix entry; finite dimension then gives convergence in matrix norm. Since all sampled weights are positive,

$$\text{rad } G_\infty(L) = \ker J_{\mathcal{Z}_L} = (p_L)/(\chi). \quad (\text{PSD39})$$

Thus a nonzero repeated-jet class in (PSD35) has positive norm for every finite N , while its minimum norm tends to zero as N tends to infinity at this fixed L . More explicitly, choose such a class e and its minimizing representatives $P_{N,e}$. They retain the complete prescribed remainder, hence every original Taylor coefficient, and (PSD37) proves $\|P_{N,e}\|_\nu \rightarrow 0$. Any Taylor functional that is nonzero on e is consequently unbounded for this completed polynomial norm. This supplies the actual relation between retained finite-degree generalized jets and their fixed-circle completion.

There is a strict finite-degree strengthening. For $e \neq 0$, let $P_{N,e}$ be its nonzero minimizing polynomial in (PSD36). It has the prescribed values $R_e(S_n)$ on $\mathcal{Z}_L$. Hence

$$e^*(G_N(L) - G_\infty(L))e = \sum_{n \notin \mathcal{Z}_L} \nu_n |P_{N,e}(S_n)|^2 > 0. \quad (\text{PSD40})$$

The complement of the finite set $\mathcal{Z}_L$ is infinite, all its weights are positive, and a nonzero polynomial cannot vanish at every node there. Thus $G_N(L) - G_\infty(L)$ is positive definite for every finite $N \geq q - 1$, although it converges to zero. The proof gives no positive lower bound uniform in N . The limit has rank $|\mathcal{Z}_L|$; in particular it is positive definite exactly when all roots of χ are simple and sampled. When $\mathcal{Z}_L$ is empty, the exact sequence has kernel equal to all of E and $G_N(L)$ tends to zero.

For $k = 1$, the density and closed-relation quotient proofs apply to the positive support $\mathcal{I}_{h,L} = \{n : w_h(2\pi n/L) > 0\}$. The Hilbert space, all evaluations, and the definition of $\mathcal{Z}_L$ must be restricted to that support. On this support, the formula (PSD20) also gives the inverse of the source-to-sequence isometry. The all-integer positivity, the positive-definite matrix assertions for every finite degree at arbitrary fixed L , and the strict difference in (PSD40) have been proved here for $k \geq 2$. For $k = 1$, the source's separately proved large-period comparison (P34)–(P37) supplies the finite-degree injectivity used there; no assertion that every fixed circle has all integer atoms with positive weight is inserted into that result.

For the actual one-fold packet, $\chi_{h,1} = h$. If $h(S_n) = 0$ with $S_n = 1/2 + 2\pi in/L$, the full-order cancellation formula (PSD5) gives

$$w_h(2\pi n/L) = \frac{1}{2\pi} \left| \frac{g^{(m)}(S_n)}{h^{(m)}(S_n)} \right|^2 > 0, \quad m = \text{ord}_{S_n} h = \text{ord}_{S_n} g. \quad (\text{PSD41})$$

Consequently the sampled-root set $\mathcal{Z}_L = \{n : h(1/2 + 2\pi in/L) = 0\}$ is already contained in $\mathcal{I}_{h,L}$. Restricting the Hilbert space to its positive support therefore retains every sampled root of this original one-fold packet. Its reduced evaluation quotient is $\mathbb{C}^{\mathcal{Z}_L}$, with the original weights, and its algebraic kernel is $(p_L)/(h)$ by the same proof as (PSD28)–(PSD32). Each factor of p_L is a critical-line factor of h , occurring once, so p_L divides the full-multiplicity critical factor $\chi_{c,1}$. This is exactly the cancellation fact used by the accompanying critical-jet comparison (DC1)–(DC30). It asserts positivity at the selected sampled roots; it does not assert positivity at every other integer atom or invertibility of every high-degree Gram at arbitrary fixed L .

291 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R , while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m , including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

291.1 Source spaces and fixed arithmetic factors

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x.$$

Use the original Fourier, theta and Mellin transforms

$$\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{A1})$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 291.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \rightarrow \mathcal{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\hat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \text{Re } s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{A2})$$

For the exact original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \quad H_{P(D)\phi_*} = P, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP. \quad (\text{A3})$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\hat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) \phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k)|\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\operatorname{Re} s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2}\{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. $\square$

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Let $Q = \mathcal{B}/\Theta V$ and $q : \mathcal{B} \rightarrow Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \quad (\text{A4})$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q . This follows from injectivity of theta and $\operatorname{im} d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-} , with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

291.2 The exact operator range, including its inverse

Proposition 291.2 (Source and target Euler inverses). *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Then

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)S_\rho = 1, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{A5})$$

The operator $D - \rho$ is injective. For $0 < \operatorname{Re} \rho < 1$, the same formula, extended evenly, gives

$$(D - \rho)V = \ker(H(\cdot)(\rho)). \quad (\text{A6})$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \operatorname{Re} \rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N - \sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D - \rho)S_\rho \phi = \phi$ on $\mathbb{R}$ gives $(1 - \rho) \int S_\rho \phi = \int \phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V . $\square$

For any polynomial $h(t) = \prod_i (t - \rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathcal{B} = \ker(j_h\mathcal{M}), \quad h(D)V = \ker(j_hH), \quad E_h = \mathbb{C}[t]/(h). \quad (\text{A7})$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho_i y}/k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_hH \downarrow \wr & & \wr \downarrow j_h\mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \quad (\text{A8})$$

Theorem 291.3 (Finite arithmetic kernel, quotient and packet section). *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{A9})$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu , and set

$$R_{\text{ref}}u = S_h^{\mathcal{B}}\Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M}F, \quad \sigma = qR_{\text{ref}}.$$

Then

$$\mathcal{M}R_{\text{ref}}u = (g/h)R_Z(u), \quad J_Z R_{\text{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \quad (\text{A10})$$

The projector is $\sigma \bar{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h\Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\text{ref}} = j_h v \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ . Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. $\square$

291.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E , define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$DR_{\text{ref}} - R_{\text{ref}}A = f_0\ell, \quad f_0 = \Theta\phi_*. \quad (\text{A11})$$

Indeed $tR_Z(u) - R_Z(tu)$ is divisible by the monic h , and its degree is at most d ; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathcal{B}_Z = \Theta V + R_{\text{ref}}E$ fits into the actual extension

$$0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J_Z} E \rightarrow 0. \quad (\text{A12})$$

It splits complex-linearly through R_{ref} ; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_* \ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathcal{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\simeq} [V \rightarrow \mathcal{B}_Z] \longrightarrow [V \rightarrow \mathcal{B}]. \quad (\text{A13})$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

291.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below uses these specified metrics. Put $\phi_j = D^j \phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j \phi_j$. For $F_m = \Theta \Phi_m : \mathbb{C}^{m+1} \rightarrow \mathcal{H}$, define

$$\begin{aligned} H_m &= F_m^* F_m, & C_m &= F_m^* R_{\text{ref}}, & B_m &= -H_m^{-1} C_m, \\ \Pi_m &= F_m H_m^{-1} F_m^*, & R_m &= R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{aligned} \quad (\text{A14})$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \rightarrow \mathbb{C}^{m+1}$,

$$\begin{aligned} (R_{\text{ref}} + F_m B)^* (R_{\text{ref}} + F_m B) &= G_m + (B - B_m)^* H_m (B - B_m), \\ G_m &= R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{aligned} \quad (\text{A15})$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0 \ell + S_m B_m - I_m B_m A, \quad D R_m - R_m A = F_{m+1} K_m. \quad (\text{A16})$$

Integration by parts in dx , with the source boundary term $[x \bar{F} H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \quad (\text{A17})$$

In particular $\text{rank } W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \bar{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy–Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \quad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \quad (\text{A18})$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{A19})$$

The next sections compute the actual arithmetic moment problem behind G_m , sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

292 An exact low-height exclusion from the original theta boundary

This calculation keeps the original boundary $f_0 = \Theta\phi_*$, its Mellin transform $g = 2\xi$, and its unscaled positive kernel ψ . In particular, the constant $g(1) = 1$ below is a value calculated from that boundary, not a normalization imposed on g . Every estimate is an inequality of actual integrals. The numerical lower bound at the end is a rational number proved positive by exact arithmetic.

292.1 The original source and its exact endpoint mass

Use the functions and coordinates in (HCA.1)–(HCA.2):

$$\begin{aligned} \vartheta(x) &= \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x^2} = 1 + 2 \sum_{n \geq 1} e^{-\pi n^2 x^2}, \\ \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \\ f_0(x) &= \Theta\phi_*(x) = 2 \sum_{n \geq 1} \phi_*(nx), \\ \psi(y) &= e^{y/2} f_0(e^y), \\ g(s) &= \mathcal{M}f_0(s) = \int_0^\infty f_0(x) x^s \frac{dx}{x} = \int_{\mathbb{R}} e^{(s-1/2)y} \psi(y) dy. \end{aligned} \tag{LHT.1}$$

Here $x > 0$ and $y = \log x \in \mathbb{R}$. The original theta identity is $\vartheta(x) = x^{-1}\vartheta(x^{-1})$. This is the same Gaussian theta identity used in the source proof of $\psi(-y) = \psi(y)$ in (HCA.1); no different theta function is introduced here. Its series and every differentiated series converge uniformly on compact subsets of $x > 0$. Differentiating them gives

$$\begin{aligned} \vartheta'(x) &= -4\pi x \sum_{n \geq 1} n^2 e^{-\pi n^2 x^2}, \\ (x^2 \vartheta'(x))' &= 8\pi^2 x^4 \sum_{n \geq 1} n^4 e^{-\pi n^2 x^2} - 12\pi x^2 \sum_{n \geq 1} n^2 e^{-\pi n^2 x^2} = f_0(x). \end{aligned} \tag{LHT.2}$$

For $x \geq 1$, every term of $f_0(x)$ is positive, since $4\pi^2 n^4 x^4 - 6\pi n^2 x^2 = 2\pi n^2 x^2(2\pi n^2 x^2 - 3) > 0$. The evenness of ψ therefore gives $\psi(y) > 0$ for every real y . On $y \geq 0$ we retain the termwise upper bound

$$0 < \psi(y) \leq 8\pi^2 \sum_{n \geq 1} n^4 \exp\left(\frac{9}{2}y - \pi n^2 e^{2y}\right). \tag{LHT.3}$$

Together with evenness, this is an integrable majorant after multiplication by any fixed polynomial in y and any fixed $e^{b|y|}$. For example, for $y \geq 0$ the sum is bounded by $(\sum_{n \geq 1} n^4 e^{-\pi(n^2-1)})e^{9y/2-\pi e^{2y}}$ times $8\pi^2$. All Mellin integrals and differentiations below consequently converge absolutely; termwise upper bounds on their nonnegative integrands also follow directly from monotone convergence.

The limits in (LHT.2) can be evaluated before integrating. At $x \rightarrow \infty$, the differentiated theta series gives $x^2 \vartheta'(x) \rightarrow 0$. Differentiating the theta identity itself gives the exact equation

$$x^2 \vartheta'(x) = -\vartheta(x^{-1}) - x^{-1} \vartheta'(x^{-1}). \tag{LHT.4}$$

As $x \rightarrow 0^+$ the two terms on the right tend to -1 and 0 , respectively. Integrating (LHT.2) first on a compact interval and then taking its endpoints to 0 and ∞ is justified by the absolute convergence just proved. It follows that

$$g(1) = \int_0^\infty f_0(x) dx = [x^2 \vartheta'(x)]_{0^+}^\infty = 1. \tag{LHT.5}$$

This calculation does not interchange the integrals of the signed individual theta terms on the whole positive axis.

Define two actual, unscaled source moments

$$\begin{aligned} M_0 &= \int_{\mathbb{R}} \psi(y) dy = g(1/2), \\ G_2 &= \int_{\mathbb{R}} y^2 e^{y/2} \psi(y) dy = 2 \int_0^\infty y^2 \cosh(y/2) \psi(y) dy = g''(1). \end{aligned} \quad (\text{LHT.6})$$

Both numbers are strictly positive. For real z , $\cosh z - 1 \leq \frac{1}{2} z^2 \cosh z$: for $z \geq 0$ this follows from $\cosh z - 1 = \int_0^z (z-v) \cosh v dv \leq \cosh z \int_0^z (z-v) dv$, and both sides are even. Equations (LHT.5)–(LHT.6) therefore imply

$$1 - M_0 = 2 \int_0^\infty (\cosh(y/2) - 1) \psi(y) dy \leq \frac{1}{8} G_2, \quad M_0 \geq 1 - \frac{1}{8} G_2. \quad (\text{LHT.7})$$

292.2 A rational upper bound on the unscaled second moment

We now bound G_2 without a floating-point computation or a zero-location theorem. For $y \geq 0$ use $\cosh(y/2) \leq e^{y/2}$ in (LHT.3), and then put $x = e^y$. The resulting nonnegative sum and integrals give

$$\begin{aligned} G_2 &\leq 16\pi^2 \sum_{n \geq 1} n^4 \int_0^\infty y^2 e^{5y - \pi n^2 e^{2y}} dy \\ &= 16\pi^2 \sum_{n \geq 1} n^4 \int_1^\infty (\log x)^2 x^4 e^{-\pi n^2 x^2} dx. \end{aligned} \quad (\text{LHT.8})$$

For $x \geq 1$, $\log x \leq x - 1$. Set $v = x - 1 \geq 0$ and use $(1+v)^2 \geq 1 + 2v$. Expanding the remaining polynomial, with no change to the coordinate or the original moment, yields

$$\begin{aligned} G_2 &\leq 16\pi^2 \sum_{n \geq 1} n^4 e^{-\pi n^2} \int_0^\infty v^2 (1+v)^4 e^{-2\pi n^2 v} dv \\ &= \sum_{n \geq 1} e^{-\pi n^2} \sum_{r=0}^4 \frac{16 \binom{4}{r} (r+2)!}{2^{r+3} \pi^{r+1} n^{2r+2}}. \end{aligned} \quad (\text{LHT.9})$$

The integral identity used here is $\int_0^\infty v^j e^{-av} dv = j! a^{-j-1}$ for $a > 0$ and nonnegative integer j ; it follows by j integrations by parts, with base integral $1/a$ and zero endpoint terms.

The elementary inequality $\pi > 3$ and $n \geq 1$ bound the five terms of the inner sum, respectively, by

$$\frac{4}{3}, \quad \frac{8}{3}, \quad \frac{8}{3}, \quad \frac{40}{27}, \quad \frac{10}{27}; \quad \frac{4}{3} + \frac{8}{3} + \frac{8}{3} + \frac{40}{27} + \frac{10}{27} = \frac{230}{27}. \quad (\text{LHT.10})$$

For the exponential bound, the exact finite Taylor sum gives

$$e^3 > \sum_{j=0}^8 \frac{3^j}{j!} = \frac{89641}{4480} > \frac{89600}{4480} = 20. \quad (\text{LHT.11})$$

Thus $e^{-\pi n^2} < 20^{-n^2} \leq 20^{-n}$ for $n \geq 1$, and

$$\sum_{n \geq 1} e^{-\pi n^2} < \sum_{n \geq 1} 20^{-n} = \frac{1}{19}. \quad (\text{LHT.12})$$

Combining (LHT.9)–(LHT.12) proves the entirely rational bound

$$0 < G_2 = g''(1) < \frac{230}{27 \cdot 19} = \frac{230}{513}. \quad (\text{LHT.13})$$

In particular this proof includes every $n \geq 1$ and the entire positive integration axis. There is no unestimated theta tail.

292.3 Uniform exclusion in the closed low-height rectangle

Write $s = \frac{1}{2} + a + it$, where $a, t \in \mathbb{R}$. Evenness and reality of ψ in (LHT.1) give

$$\operatorname{Re} g(\tfrac{1}{2} + a + it) = 2 \int_0^\infty \psi(y) \cosh(ay) \cos(ty) dy. \quad (\text{LHT.14})$$

The real inequality $\cos z \geq 1 - z^2/2$ follows from $1 - \cos z = 2 \sin^2(z/2) \leq z^2/2$. If $|a| \leq 1/2$, then $1 \leq \cosh(ay) \leq \cosh(y/2)$ for $y \geq 0$. Since every factor $\psi(y) \cosh(ay)$ is nonnegative, substitution of this cosine inequality in (LHT.14) gives

$$\begin{aligned} \operatorname{Re} g(\tfrac{1}{2} + a + it) &\geq 2 \int_0^\infty \psi(y) \cosh(ay) dy - t^2 \int_0^\infty y^2 \psi(y) \cosh(ay) dy \\ &\geq M_0 - \frac{t^2}{2} G_2. \end{aligned} \quad (\text{LHT.15})$$

For $|t| \leq 2$, equations (LHT.7) and (LHT.13) now imply

$$\boxed{\operatorname{Re} g(s) \geq 1 - \frac{17}{8} G_2 > 1 - \frac{17 \cdot 230}{8 \cdot 513} = \frac{97}{2052} > 0, \quad 0 \leq \operatorname{Re} s \leq 1, \quad |\operatorname{Im} s| \leq 2.} \quad (\text{LHT.16})$$

Consequently $g(s) = 2\xi(s)$ has no zero in that closed rectangle. This is proved directly from the actual theta source, without assuming the Riemann hypothesis or using numerical verification of its low zeros.

The resulting map on the programme's original local packet objects is also explicit. At every point s_0 of this rectangle, $g(s_0) \neq 0$, so multiplication by g on the local holomorphic ring $\mathcal{O}_{\mathbb{C}, s_0}$ is an isomorphism with inverse multiplication by the holomorphic germ $1/g$. On the rectangle itself the preceding calculation proves

$$|1/g(s)| \leq \frac{1}{\operatorname{Re} g(s)} < \frac{2052}{97}. \quad (\text{LHT.17})$$

It follows that $\mathcal{O}_{\mathbb{C}, s_0}/(g) = 0$ at every such s_0 . Under the programme's already constructed Mellin-jet isomorphism to the actual packet summands, each of these local summands therefore vanishes. No proper-source kernel is quotiented out by this argument. In particular an actual upper-half-plane offcritical centre obeys

$$g(\rho) = 0, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 0 \implies \gamma > 2. \quad (\text{LHT.18})$$

Its reflected and conjugate centres retain their original orders, jets, and Taylor units.

293 The certified theta moments as observations of the total original object

The two certified matrices attach to the same original theta seed and the same finite arithmetic packets in $\mathbf{C}_U$. The global moment sum includes critical and offcritical packets in the ambient $Q = \mathcal{B}/\Theta V$ of (T0.5). Its offcritical part is exactly the already constructed $T_U Q$ of (T0.7), not a replacement for that carrier. We construct the attachment and calculate what the finite certificate excludes.

293.1 From the original theta boundary to the finite moment matrices

Retain the original operators and seed, with every scalar unchanged:

$$\begin{aligned} D &= -x\partial_x, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V, \quad f_0 = \Theta\phi_* = 2 \sum_{n \geq 1} \phi_*(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x)x^s \frac{dx}{x}, \quad \mathcal{M}f_0 = g = 2\xi, \quad \psi(y) = e^{y/2} f_0(e^y). \end{aligned} \quad (\text{HCA.1})$$

The complete theta calculation (WBR1)–(WBR8) gives $\psi(-y) = \psi(y) > 0$ and, for $y \geq 0$, $\psi(y) \leq C_\theta e^{9y/2 - \pi e^{2y}}$ with $C_\theta = 8\pi^2 \sum_{n \geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. It follows by differentiation under this integrable majorant that

$$M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy = g^{(2j)}(1/2), \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad F(w) = \sum_{j \geq 0} a_j w^j, \quad F(z^2) = g(1/2 + iz). \quad (\text{HCA.2})$$

The map $z \mapsto z^2$ and the even entire Taylor series define F without choosing a square-root branch. Its mass is exactly $a_0 = M_0 = g(1/2) > 0$.

The original moment observation on a general $F_1 \in \mathcal{B}$ is the linear map

$$F_1 \mapsto \left(\int_{\mathbb{R}} y^{2j} e^{y/2} F_1(e^y) dy \right)_{j=0}^{32}. \quad (\text{HCA.3})$$

The two-ended defining bounds of $\mathcal{B}$ dominate each integrand by an integrable exponential times a polynomial, so every component is defined. At the actual boundary $f_0 = \Theta\phi_*$, this map has first component $M_0 > 0$. Thus it is an observation of the actual degree-one source functions, and does not factor through their theta quotient: factoring would send the class of f_0 to zero. This explicit map is permitted by the marked source vertices of (T0.1)–(T0.4), which retain ϕ_* , f_0 and their differential.

On the open set of coefficient vectors with $a_0 \neq 0$, define the finite rational recurrence

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0} \quad (0 \leq n \leq 31), \quad H_{15}^{(\nu)} = (b_{r+s+\nu})_{0 \leq r, s \leq 15}, \quad \nu = 0, 1. \quad (\text{HCA.4})$$

Multiplication of $-F'/F = \sum_{n \geq 0} b_n w^n$ by F proves each coefficient formula. Thus the source observation (HCA.3), the factorial/sign map (HCA.2), the recurrence and the two matrix assemblies form a single explicit composite. The recurrence is a rational observation of its coefficients; it is not asserted to be a linear or cochain map. This is exactly the typed convention for numerical observations in (T0.4)–(T0.6).

293.2 The reciprocal action on every full original packet

Let Z be any finite actual zero packet in the open strip, stable under reflection and conjugation, with all full orders m_ρ . Critical points are allowed in this ambient packet. Put

$$\begin{aligned} h(s) &= \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A_h = M_s, \quad v_h = g/h, \\ v_h &= j_h(v_h), \quad \varepsilon_h = v_h^{-1}, \quad R_h(u) = r_h(\varepsilon_h u), \\ s_h u &= S_h \Theta(R_h(u)(D)\phi_*), \quad \sigma_h u = [s_h u] \in Q, \quad J_h = j_h \mathcal{M}. \end{aligned} \quad (\text{HCA.5})$$

Here r_h is the unique degree-below-deg h remainder and S_h is the original iterated Euler inverse on zero full Mellin jets. These are precisely (AG6)–(AG8) and (WBR22)–(WBR25): the workbench's η_h is the same element as ε_h here. In particular

$$\mathcal{M} s_h u = v_h R_h(u), \quad J_h s_h = I, \quad J_h \Theta = 0, \quad D_Q \sigma_h = \sigma_h A_h. \quad (\text{HCA.6})$$

The first equality follows by dividing $gR_h(u)$ by h ; the second is $j_h(v_h)\varepsilon_h = 1$ with the entire Taylor unit retained. Thus σ_h is injective. Polynomial division gives $Ds_h - s_h A_h = \Theta\phi_*[s^{\deg h-1}]R_h$, proving the last equality before passing to the quotient, including its boundary. The range and inverse proofs used here apply to every root in the open strip and therefore include the ambient critical packets.

Since $g(1/2) > 0$, the polynomial $s - 1/2$ is a unit on every E_h . Define the actual finite-packet operator

$$T_h = -(A_h - \tfrac{1}{2}I)^{-2}, \quad \beta_\rho = -\frac{1}{(\rho - 1/2)^2}. \quad (\text{HCA.7})$$

No inverse on the whole quotient Q is required. On the full ρ -block, write $A_h = \rho I + N_\rho$ and $\lambda_\rho = \rho - 1/2$. The finite coordinate maps are

$$\begin{aligned} T_h &= \sum_{j=0}^{m_\rho-1} (-1)^{j+1} (j+1) \lambda_\rho^{-j-2} N_\rho^j, \quad Y_\rho = T_h - \beta_\rho I, \\ N_\rho &= \lambda_\rho \sum_{j=1}^{m_\rho-1} \binom{-1/2}{j} (Y_\rho/\beta_\rho)^j. \end{aligned} \quad (\text{HCA.8})$$

Multiplication of the first polynomial by $(\lambda_\rho + N_\rho)^2$ gives -1 modulo $N_\rho^{m_\rho}$. For the inverse, the element with constant term one solving $U^2(1 + Y_\rho/\beta_\rho) = 1$ is unique by successive coefficient comparison, whose leading scalar is two. The binomial polynomial and $1 + N_\rho/\lambda_\rho$ both solve it. This proves (HCA.8) with every nilpotent coefficient and with no analytic branch choice.

For $h \mid H$ with full orders, let ι_{hH} insert zero jets at added centres, let π_{Hh} restrict to old jets, and let $e_{hH} = \iota_{hH}(1)$ be the complete old-packet idempotent. The original arrows (AG10)–(AG12) give

$$\begin{aligned} \pi \iota &= I, \quad \iota \pi = M_{e_{hH}}, \quad s_H \iota = s_h, \quad \sigma_H \iota = \sigma_h, \\ \pi T_H &= T_h \pi, \quad T_H \iota = \iota T_h. \end{aligned} \quad (\text{HCA.9})$$

To prove the last two identities, the maps commute with $A - 1/2$; its invertibility on both domains makes them commute with its inverse, and then with its inverse square. Thus the reciprocal action is natural for the actual divisor arrows, including their full units and literal representatives. Through σ_h it acts on the actual finite image in Q .

The representative of each polynomial of this action is also fixed. Let $t_h(s) = r_h(-(s - 1/2)^{-2})$, let $p_h(s) = r_h(P(t_h(s)))$, and put

$$K_{P,h}(u; s) = \frac{p_h(s)R_h(u)(s) - R_h(P(T_h)u)(s)}{h(s)}. \quad (\text{HCA.10})$$

The numerator is zero modulo h because multiplication by the unit ε_h commutes with $P(T_h)$; thus the quotient is a literal polynomial. Mellin transformation gives

$$p_h(D)s_h u - s_h P(T_h)u = \Theta(K_{P,h}(u; D)\phi_*). \quad (\text{HCA.11})$$

Both sides have transform $g(s)K_{P,h}(u; s)$, so Mellin injectivity proves this equality. This is the full original theta boundary, not only equality of its classes. Tensoring (HCA.11) with the other source factors gives its boundaries in (TO.3) with the original tensor differential and its Koszul signs.

The Mellin isomorphism of (TO.7), (CAU.9) sends $\sigma_h u$ on its offcritical part to precisely its prescribed local jets u_ρ . Equations (HCA.7)–(HCA.8) therefore give the following endomorphism of the already constructed offcritical direct sum:

$$\bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_{\mathbb{C},\rho}/(g) \longrightarrow \bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_{\mathbb{C},\rho}/(g), \quad (u_\rho)_\rho \longmapsto (-(s-1/2)^{-2}u_\rho)_\rho. \quad (\text{HCA.12})$$

Each denominator is a local analytic unit. Finite support is unchanged. Thus this is exactly the same reciprocal action as the one defined in the ambient finite images in Q .

293.3 The full half-trace and the exact global moment limit

For the packet Z , let $\mathcal{B}_Z$ be its distinct reciprocal nodes. Each fibre of $\rho \mapsto \beta_\rho$ is the pair $\rho, 1-\rho$, since equality of nonzero reciprocal squares is equality up to sign. Reflection gives the same full order m_β at both points. Define

$$f_Z(X) = \prod_{\beta \in \mathcal{B}_Z} (X - \beta)^{m_\beta}, \quad \mathcal{A}_Z = \mathbb{C}[X]/(f_Z). \quad (\text{HCA.13})$$

The inverse local map (HCA.8) proves that the kernel of $P \mapsto P(T_h)$ is exactly (f_Z) . On choosing one point in each reflected pair it also gives the full module isomorphism $E_h \simeq \mathcal{A}_Z \oplus \mathcal{A}_Z$. A different choice interchanges the corresponding two factors; both remain present. Triangular multiplication in each local basis gives, for $\nu = 0, 1$,

$$\mathcal{L}_{Z,\nu}(PQ) := \frac{1}{2} \text{Tr}_{E_h}(T_h^{\nu+1} P(T_h) Q(T_h)) = \sum_{\beta \in \mathcal{B}_Z} m_\beta \beta^{\nu+1} P(\beta) Q(\beta). \quad (\text{HCA.14})$$

Indeed every diagonal entry on a length- m_β block equals $\beta^{\nu+1} P(\beta) Q(\beta)$, and there are two reflected blocks. This proves the factor $1/2$ and the retained multiplicity. The radical of this bilinear form on $\mathcal{A}_Z$ is exactly its nilradical $(\text{rad } f_Z)/(f_Z)$: vanishing at every node is sufficient by the sum, and Lagrange interpolation at a node where a value is nonzero proves necessity, since $m_\beta \beta^{\nu+1} \neq 0$. Thus (HCA.8) retains all local jets before the trace and (HCA.14) specifies its exact scalar kernel.

Packet growth has a specific trace law. From (HCA.9) and the idempotent splitting of the larger packet,

$$\begin{aligned} \frac{1}{2} \text{Tr}_{E_H}(M_{e_{hH}} T_H^{\nu+1} P(T_H) Q(T_H)) &= \mathcal{L}_{Z,\nu}(PQ), \\ \mathcal{L}_{Z',\nu}(PQ) &= \mathcal{L}_{Z,\nu}(PQ) + \mathcal{L}_{Z' \setminus Z,\nu}(PQ) \quad (Z \subseteq Z'). \end{aligned} \quad (\text{HCA.15})$$

The first is a trace on the old direct summand, conjugate by ι and π to the original endomorphism. The second is additivity of trace on the two complementary full summands. Thus uncompressed traces on growing packets are not asserted constant. Their precise change is the added original packet.

The growth calculation (WBR9) and the complete product proof (GF1)–(GF13) for this same F give distinct nonzero reciprocal nodes β_j , their full orders m_j , and

$$\sum_j m_j |\beta_j| < \infty, \quad F(w) = a_0 \prod_j (1 - \beta_j w)^{m_j}, \quad b_n = \sum_j m_j \beta_j^{n+1}. \quad (\text{HCA.16})$$

That proof applies the explicit bound $\log M_F(R) = O(\sqrt{R} \log R)$ with exponent $3/4 < 1$. Its circle-mean calculation bounds the zero count by $CR^{3/4}$; dyadic annuli then prove reciprocal summability. The uniformly convergent product has exactly the original zeros. Dividing F by it gives a nonvanishing entire function e^H . The selected circles in (GF7)–(GF9) bound $\Re H$ by $CR^{3/4} \log R$; the circle Fourier formula (GF10)–(GF11) forces every positive Taylor coefficient of H to vanish. Evaluation at zero fixes the remaining constant to a_0 . Thus no unproved factorization input, exponential factor, or change of mass enters (HCA.16). The complete product proof is included with the attachment. The nodes are bounded, since absolute summability excludes infinitely many nodes outside any disc about zero. Hence, for any fixed P, Q , the sum in (HCA.14) over all nodes is absolutely convergent, also for $\nu = 1$. Exhausting by finite full reflection- and conjugation-stable packets proves the exact limit

$$\begin{aligned} \mathcal{L}_\nu(PQ) &= \lim_Z \mathcal{L}_{Z,\nu}(PQ) = \sum_j m_j \beta_j^{\nu+1} P(\beta_j) Q(\beta_j), \\ \mathcal{L}_\nu(X^r X^s) &= b_{r+s+\nu}, \quad \mathcal{Q}_\nu(P) := \mathcal{L}_\nu(P^2) = c^\top H_{15}^{(\nu)} c \quad \left(P(X) = \sum_{r=0}^{15} c_r X^r \right). \end{aligned} \quad (\text{HCA.17})$$

Absolute convergence proves independence of the exhaustion and justifies the finite polynomial expansion. For real coefficients the value is real by conjugation of the full node set. The square in this identity is the algebraic square $P(\beta)^2$.

Equations (HCA.3)–(HCA.4) and (HCA.5)–(HCA.17) are therefore two equal evaluations of the same original data: the first through the theta integral and coefficient recurrence, the second through the full packet action and absolutely convergent half-trace. Their equality is proved by (HCA.16), not imposed as a new relation between independently adjustable matrices. Adjoining these specified observation vertices and the displayed maps to the path representation $\mathbf{C}_U$ of (T0.4) gives one represented extension: every path is evaluated by composition of the actual maps, and (HCA.9), (HCA.11), (HCA.15), (HCA.17) prove the new path relations. The old represented diagram is retained through its identity inclusion. No quotient identifies its offcritical component with a scalar or with a zero target.

In particular the global trace splits into the absolutely convergent critical and offcritical sums. The critical-only projection of (WBR37)–(WBR40) kills every full offcritical packet but need not kill its trace: its reciprocal primary idempotent e_β has $\text{Tr}_{\mathcal{A}_Z} M_{X^{\nu+1}e_\beta} = m_\beta \beta^{\nu+1} \neq 0$. Thus the full functional has this explicit omitted summand under critical projection. At the split support level all linear maps above send $(\mathfrak{a}, u)$ to $(\mathfrak{a}, Lu)$ and τ to τ ; a killed class lands at $(\mathfrak{a}, 0)$. The moment and quadratic evaluations remain the marked observations in their actual linear or polynomial types.

293.4 The certified finite restriction on this attached observation

The certified theta calculation (TC1)–(TC28) encloses exactly the moments (HCA.2). It retains theta terms $1 \leq n < 20$ on $0 \leq y \leq 4$, and, for $J = 0, 2, \dots, 64$, adds the proved nonnegative tails

$$\begin{aligned} & \frac{16\pi^2 4^{J+1} e^{18} 20^4 e^{-400\pi}}{1 - e^{4/20-40\pi}}, \\ & \frac{16\pi^2 4^J e^{18-\pi e^8}}{(1 - e^{4-2\pi})(2\pi e^8 - 9/2 - J/4)}. \end{aligned} \quad (\text{HCA.18})$$

These are exactly (TC5) at $N = 20, L = 4$, with N the first omitted integer. All denominators are strictly positive as proved there. The finite integrands, quadrature subdivisions and dyadic interval endpoints are preserved in the certificate. The interval recurrence (TC11)–(TC13) encloses the exact coefficients in (HCA.4) and their $LDL^\top$ decompositions in the unchanged coordinate order. Every one of its thirty-two pivot intervals has a positive lower endpoint, proving

$$H_{15}^{(0)} > 0, \quad H_{15}^{(1)} > 0. \quad (\text{HCA.19})$$

The certificate is the actual file `certificate_dimension16_1024.json`, with SHA-256 `869ebacd3f82e46c3fe5fac1f07bcbaae7f4c333e1a5ba2e0ffcb1f91884a0fc`. Its exact endpoint checks and the full analytic tail and validated quadrature proof accompany this statement.

The exact rational endpoint check also gives a uniform quantitative margin in the original monomial coefficient frame. Write the exact decompositions as $H_{15}^{(\nu)} = L_\nu D_\nu L_\nu^\top$. The sum of the absolute upper endpoint bounds in every off-diagonal row and column of $L_\nu - I$ is strictly less than $1/64$. Every diagonal pivot of D_0 is greater than $6 \cdot 10^{-107}$, and every pivot of D_1 is greater than $7 \cdot 10^{-111}$. These comparisons use exact dyadic fractions in the certificate. For a matrix E , Cauchy-Schwarz with row weights proves $\|Ex\|_2^2 \leq \|E\|_\infty \|E\|_1 \|x\|_2^2$: bound the square of row i by $(\sum_j |E_{ij}|)(\sum_j |E_{ij}| |x_j|^2)$ and sum in i . Thus $\|L_\nu - I\|_2 < 1/64$ and $\|L_\nu^\top x\|_2 > (63/64)\|x\|_2$ for $x \neq 0$. Substitution in the exact decomposition proves

$$H_{15}^{(\nu)} \succ \gamma_\nu I_{16}, \quad \gamma_0 = (63/64)^2 6 \cdot 10^{-107}, \quad \gamma_1 = (63/64)^2 7 \cdot 10^{-111}. \quad (\text{HCA.19a})$$

No polynomial basis or mass was changed in obtaining this bound.

Consequently, every nonzero real polynomial of degree at most fifteen satisfies $Q_\nu(P) > 0$ for both $\nu = 0, 1$. Padding its coefficient vector with zeros identifies its form with the corresponding leading principal submatrix; positivity of (HCA.19) proves the assertion. More generally, on the explicitly specified finite cone

$$\mathcal{C}_{15} = \left\{ \sum_j A_j(X)^2 + X \sum_j B_j(X)^2 : A_j, B_j \in \mathbb{R}[X]_{\leq 15}, \text{ finite sums} \right\}, \quad (\text{HCA.20})$$

one has $\mathcal{L}_0(C) > 0$ for every nonzero C in that cone. Indeed it is the sum of the positive forms in (HCA.19), and a nonzero polynomial represented in the cone has at least one nonzero summand. This uses the shifted matrix through $\mathcal{L}_0(XB^2) = \mathcal{Q}_1(B)$; no characterization of larger polynomial cones or positivity at higher degrees is presumed.

293.5 The precise exclusion for an offcritical separator

Take any actual offcritical pair $\beta, \bar{\beta}$ of multiplicity m in (HCA.16). In original zero coordinates it is

$$\beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}, \quad \rho = 1/2 + \delta + i\gamma. \quad (\text{HCA.21})$$

It is nonreal because $\delta\gamma \neq 0$. Put $r = |\beta|/2$, let $\mathcal{A}$ be the set of all other distinct nodes of modulus at least r , and define

$$B_\beta(X) = \prod_{\alpha \in \mathcal{A}} (X - \alpha), \quad B_0 = \prod_{\alpha \in \mathcal{A}} (r + |\alpha|), \quad a = \#\mathcal{A}. \quad (\text{HCA.22})$$

The set is finite by (HCA.16), conjugation stable, and excludes $\beta, \bar{\beta}$, so B_β is real and $B_\beta(\beta) \neq 0$. For $\nu = 0, 1$ choose either c_ν with $\beta^{\nu+1} c_\nu^2 = -1$, and retain the actual constants

$$C_\nu = \frac{|c_\nu| B_0}{|B_\beta(\beta)|} \left(1 + \frac{|\Re\beta| + r}{|\Im\beta|} \right), \quad T_\nu = \sum_{|\beta_j| < r} m_j |\beta_j|^{\nu+1}. \quad (\text{HCA.23})$$

Both tails converge and $T_1 \leq rT_0$. Define for every integer $N \geq 0$

$$\begin{aligned} w_{\nu,N} &= \frac{c_\nu}{\beta^N B_\beta(\beta)}, \quad v_{\nu,N} = \frac{\Im w_{\nu,N}}{\Im \beta}, \quad u_{\nu,N} = \Re w_{\nu,N} - (\Re \beta) v_{\nu,N}, \\ P_{\nu,N}(X) &= X^N B_\beta(X) (u_{\nu,N} + v_{\nu,N} X), \quad D_{\nu,N} = \deg P_{\nu,N} \leq N + a + 1. \end{aligned} \quad (\text{HCA.24})$$

This is precisely (TC24) for $\nu = 0$. The shifted version uses the original shifted form and its exponent $\nu + 1 = 2$. The polynomial is real and nonzero, because its value at β is $c_\nu \neq 0$. Its actual degree is $N + a + 1$ when $v_{\nu,N} \neq 0$, and $N + a$ otherwise. Evaluation proves $P_{\nu,N}(\beta) = c_\nu$ and conjugate equality at $\bar{\beta}$. Hence its selected two contributions to the actual scalar form are exactly $-2m$.

Every other node of modulus at least r is a zero of the polynomial. For $|X| \leq r$, the elementary estimates $|B_\beta(X)| \leq B_0$ and $|u_{\nu,N} + v_{\nu,N}X| \leq |w_{\nu,N}| + (|\Re\beta| + r)|w_{\nu,N}|/|\Im\beta|$ give $|P_{\nu,N}(X)| \leq C_\nu 2^{-N}$. Absolute summation proves the exact identity and tail bound

$$\begin{aligned} \mathcal{Q}_\nu(P_{\nu,N}) &= -2m + \mathcal{T}_{\nu,N}, & \mathcal{T}_{\nu,N} &= \sum_{|\beta_j| < r} m_j \beta_j^{\nu+1} P_{\nu,N}(\beta_j)^2 \in \mathbb{R}, \\ |\mathcal{T}_{\nu,N}| &\leq C_\nu^2 T_\nu 4^{-N}. \end{aligned} \quad (\text{HCA.25})$$

No tail is assigned a sign before applying the certificate. If $D_{\nu,N} \leq 15$, the nonzero polynomial in (HCA.24) is within (HCA.19), and therefore

$$\boxed{\mathcal{T}_{\nu,N} = 2m + \mathcal{Q}_\nu(P_{\nu,N}) > 2m, \quad C_\nu^2 T_\nu 4^{-N} > 2m.} \quad (\text{HCA.26})$$

This is the finite exclusion on the actual packet together with its entire reciprocal tail. In particular, when $a \leq 14$, choosing $N = 14 - a$ makes its degree at most fifteen and gives

$$C_\nu^2 T_\nu > 2m 4^{14-a} \quad (\nu = 0, 1). \quad (\text{HCA.27})$$

If the affine factor has zero leading coefficient, the exact degree condition in (HCA.26) additionally retains that case. There is also an explicit strictly positive margin. For a polynomial $P(X) = \sum_{j=0}^D p_j X^j$ with $D \leq 15$, Cauchy-Schwarz gives $|P(\beta)|^2 \leq (\sum_j |p_j|^2)(\sum_{j=0}^D |\beta|^{2j})$. For $P = P_{\nu,N}$ its value is c_ν and $|c_\nu|^2 = |\beta|^{-\nu-1}$. Applying (HCA.19a) to its coefficient vector padded with zeros strengthens (HCA.26) to

$$\boxed{C_\nu^2 T_\nu 4^{-N} \geq \mathcal{T}_{\nu,N} > 2m + \frac{\gamma_\nu |\beta|^{-\nu-1}}{\sum_{j=0}^{D_{\nu,N}} |\beta|^{2j}} \quad (D_{\nu,N} \leq 15).} \quad (\text{HCA.27a})$$

The inequality on the left uses the actual tail's strict positivity already proved in (HCA.26), followed by its absolute upper bound.

The shifted certificate also tests the identical original polynomial $P_{0,N}$, without changing c_0 or any coefficient. Its two selected terms in $\mathcal{Q}_1$ are $m\beta^2 c_0^2 + m\bar{\beta}^2 \bar{c}_0^2 = -2m\Re\beta$. The absolute value of its remaining sum is at most $C_0^2 T_1 4^{-N} \leq r C_0^2 T_0 4^{-N}$. If $D_{0,N} \leq 15$ and p is its original coefficient vector padded with zeros through degree fifteen, (HCA.19a) consequently proves the simultaneous exact bounds

$$\begin{aligned} C_0^2 T_0 4^{-N} &> \max \left\{ 2m + \gamma_0 \|p\|_2^2, \frac{2m\Re\beta + \gamma_1 \|p\|_2^2}{r} \right\} \\ &> \max \left\{ 2m, \frac{4m\Re\beta}{|\beta|} \right\}. \end{aligned} \quad (\text{HCA.27b})$$

The second entry remains valid even when its numerator is negative: the absolute upper tail bound is still an upper bound for the signed tail. Both positive coefficient margins are strict. Substitution $\|p\|_2^2 \geq |\beta|^{-1} / \sum_{j=0}^{D_{0,N}} |\beta|^{2j}$ from the preceding evaluation argument retains an entirely explicit margin in both entries of the first maximum. The elementary shifted entry improves $2m$ exactly when $\Re\beta/|\beta| > 1/2$. In the original coordinates (HCA.21) that inequality is $\gamma^2 > 3\delta^2$, since $\Re\beta/|\beta| = (\gamma^2 - \delta^2)/(\gamma^2 + \delta^2)$. This additional comparison uses the same original unshifted separator, as well as the separately constructed shifted family.

There is a further comparison between the two separately phased families. Set $U_{\nu,N} = C_\nu^2 T_\nu 4^{-N}$. The exact modulus $|c_\nu|^2 = |\beta|^{-\nu-1}$ in (HCA.23) gives $C_1^2 = C_0^2/|\beta|$. Since $T_1 \leq rT_0$ and $r = |\beta|/2$,

one obtains $U_{1,N} \leq U_{0,N}/2$. For $D_{1,N} \leq 15$, apply (HCA.19a) to the original coefficient vector p_1 of $P_{1,N}$, padded through degree fifteen, and use (HCA.25). The resulting exact inequalities are

$$U_{0,N} \geq 2U_{1,N} > 4m + 2\gamma_1 \|p_1\|_2^2 \geq 4m + \frac{2\gamma_1 |\beta|^{-2}}{\sum_{j=0}^{D_{1,N}} |\beta|^{2j}}, \quad (\text{HCA.27c})$$

$$a \leq 14 \implies C_0^2 T_0 > 4m 4^{14-a} = m 4^{15-a}.$$

Here the last line chooses $N = 14 - a$, so that $D_{1,N} \leq 15$; the coefficient estimate is the same evaluation argument as in (HCA.27a). This estimate uses the shifted polynomial $P_{1,N}$ to bound the original unshifted tail constant. It is distinct from the same-polynomial comparison (HCA.27b), which retains $P_{0,N}$.

For the original unshifted quantitative separator of (TC23), put

$$N_* = \begin{cases} 0, & T_0 = 0, \\ \max \left\{ 0, 1 + \left\lfloor \frac{\log(C_0^2 T_0 / m)}{\log 4} \right\rfloor \right\}, & T_0 > 0. \end{cases} \quad (\text{HCA.28})$$

Its definition gives $C_0^2 T_0 4^{-N_*} \leq m$ and thus $\mathcal{Q}_0(P_{0,N_*}) \leq -m < 0$ by (HCA.25). Comparison with (HCA.19) proves the concrete restriction

$$\boxed{\deg P_{0,N_*} \geq 16, \quad N_* + a + 1 \geq 16.} \quad (\text{HCA.29})$$

The shifted comparison strengthens the nominal bound for the same original threshold. If $a \leq 14$, (HCA.27c) implies $T_0 > 0$ and $\log(C_0^2 T_0 / m) / \log 4 > 15 - a$. Because $15 - a$ is an integer, its floor is at least $15 - a$, and (HCA.28) gives

$$\boxed{a \leq 14 \implies N_* \geq 16 - a, \quad N_* + a + 1 \geq 17.} \quad (\text{HCA.29a})$$

This is a bound on the nominal degree. Its actual degree remains $N_* + a + 1$ if $v_{0,N_*} \neq 0$ and $N_* + a$ otherwise; the latter case can still have degree sixteen. There is a further exact boundary case: if $a \leq 15$ and $v_{1,15-a} = 0$, the shifted polynomial with $N = 15 - a$ has degree exactly fifteen. The first line of (HCA.27c) then gives $C_0^2 T_0 > m 4^{16-a}$, and the identical integer-floor argument yields $N_* \geq 17 - a$. No assertion on a matrix of degree sixteen is needed for either conclusion.

The rational rounding of (TC25) keeps the actual quadratic form strictly negative and has degree at most $N_* + a + 1$. Its actual degree is therefore at least sixteen as well. Coefficient rounding cannot produce a smaller-degree negative witness against the certified matrices. The same conclusion holds for any rational negative polynomial in either shifted family, by (HCA.19).

The full finite packet alone may already contribute $-2m$ under the exact half-trace (HCA.14); its remaining original arithmetic nodes still enter (HCA.17). The certificate therefore excludes the low-degree global separators specified above, rather than deleting that offcritical packet from $T_U Q$. It imposes no positivity assertion on H_{16} or any larger matrix. Every source block, its full reciprocal coordinate, its scalar radical, its receiving supported zero and its exact ambient trace remain the explicitly constructed parts of this attachment.

294 The original full packet at its formal boundary

This calculation concerns the original polynomial at its original marked parameter $t = 0$. It does not replace that polynomial by a single-primary polynomial. All its primary pieces occur simultaneously in the target of a proved map. The completion used below is in the deformation variable u ; its coefficient, differential, and marked fibre maps are specified explicitly.

294.1 The fixed polynomial and the two complete complexes

Fix the ordered nonempty finite set $Z = (\rho_1, \dots, \rho_r)$ of selected actual zeros of $g = 2\xi$, with their complete orders $m_\rho \geq 1$. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho} = s^d + \sum_{b=0}^{d-1} h_b s^b, \quad d = \sum_{\rho} m_\rho, \quad \Phi(s) = \frac{s^{d+1}}{d+1} + \sum_{b=0}^{d-1} \frac{h_b s^{b+1}}{b+1}. \quad (\text{FPB.1})$$

Thus $\Phi' = h$ and $\Phi(0) = 0$. In particular the constants $c_\rho := \Phi(\rho)$ are fixed by the original polynomial; they are not the constants of separately reassigned primitives.

Let $A = \mathbb{C}[[u]]$, $K = \mathbb{C}((u))$, and define

$$R = \mathbb{C}[s][[u]] = \varprojlim_n \mathbb{C}[s, u]/(u^n), \quad \widehat{\mathcal{C}} = [R \xrightarrow{L} R ds], \quad Lf = (u\partial_s f + hf) ds. \quad (\text{FPB.2})$$

Degrees are zero and one, respectively. A coefficient of each power of u is a polynomial in s , with no common degree bound required. Thus R is not being identified with $A[s]$. It is complete and separated for the u -adic topology. Its differential is continuous for that topology.

For each ρ , write $y = s - \rho$, $m = m_\rho$, and $N = m + 1$. Define the entire polynomial unit germ

$$a_\rho(y) = \frac{N(\Phi(\rho + y) - \Phi(\rho))}{y^N}, \quad a_\rho(0) = \prod_{\sigma \neq \rho} (\rho - \sigma)^{m_\sigma} \neq 0. \quad (\text{FPB.3})$$

The first equality is polynomial because $\Phi(\rho + y) - \Phi(\rho)$ vanishes to order N . For the second, integrate $h(\rho + y) = y^m \prod_{\sigma \neq \rho} (\rho - \sigma + y)^{m_\sigma}$ coefficient by coefficient. Choose and retain a specified number b_ρ with $b_\rho^N = a_\rho(0)$. The branch

$$A_\rho(y) = b_\rho \sum_{j \geq 0} \binom{1/N}{j} \left(\frac{a_\rho(y)}{a_\rho(0)} - 1 \right)^j, \quad w = y A_\rho(y), \quad y = \psi_\rho(w) \quad (\text{FPB.4})$$

is a formal coordinate with a uniquely specified formal inverse. Indeed its linear coefficient is $b_\rho \neq 0$; equating the coefficient of w^n in $\psi A_\rho(\psi) = w$ determines the coefficient of w^n in ψ uniquely by division by b_ρ , after the preceding coefficients have been determined. The binomial identity gives $A_\rho^N = a_\rho$. Consequently

$$\Phi(\rho + \psi_\rho(w)) = c_\rho + \frac{w^N}{N}, \quad h(\rho + \psi_\rho(w)) \psi'_\rho(w) = w^m. \quad (\text{FPB.5})$$

These are equalities of formal series retaining the original constant.

Put $R_\rho = \mathbb{C}[[w]][[u]]$, equipped with its u -adic topology and the coefficientwise formal derivative in w . Although its elements can also be written as double formal series, only the stated topology is used in the completion arguments. Set

$$\mathcal{C}_\rho = [R_\rho \xrightarrow{L_\rho} R_\rho dw], \quad L_\rho f = (u\partial_w f + w^m f) dw. \quad (\text{FPB.6})$$

The actual map of complexes is

$$F_\rho^0 f = f(\rho + \psi_\rho(w)), \quad F_\rho^1(f ds) = f(\rho + \psi_\rho(w))\psi'_\rho(w) dw, \quad F = \bigoplus_{\rho \in Z} F_\rho. \quad (\text{FPB.7})$$

Substitution is performed on each polynomial coefficient of u . The chain rule and (FPB.5) prove $F_\rho^1 L = L_\rho F_\rho^0$. The Jacobian in degree one is part of this identity. The map F is not asserted to be surjective on cochains.

294.2 Complete remainder, inverse, and chain homotopies

For $f = \sum_{n \geq 0} u^n f_n(s) \in R$, recursively divide

$$f_n - \partial_s q_{n-1} = h q_n + r_n, \quad q_{-1} = 0, \quad \deg r_n < d. \quad (\text{FPB.8})$$

Each division is the original monic polynomial division. Then $f = Lq + r$, where $q = \sum u^n q_n$, $r = \sum u^n r_n$, and the one-form symbol is understood. Conversely this identity determines every coefficient by (FPB.8). In particular L is injective, and

$$R = L(R) \oplus \mathbb{C}[s]_{<d}[[u]], \quad H^0(\widehat{\mathcal{C}}) = 0, \quad \mathcal{H} := H^1(\widehat{\mathcal{C}}) = \bigoplus_{b=0}^{d-1} A[s^b ds]. \quad (\text{FPB.9})$$

For injectivity one can also take the first nonzero coefficient of u in $Lq = 0$, obtaining $h q_n = 0$. Since $\mathbb{C}[s]$ is a domain, this is impossible.

In the local ring, divide every coefficient of $f_n - \partial_w q_{n-1} \in \mathbb{C}[[w]]$ uniquely by w^m :

$$f_n - \partial_w q_{n-1} = w^m q_n + r_n, \quad \deg r_n < m. \quad (\text{FPB.10})$$

The same coefficient recursion proves

$$H^0(\mathcal{C}_\rho) = 0, \quad \mathcal{H}_\rho := H^1(\mathcal{C}_\rho) = \bigoplus_{a=0}^{m_\rho-1} A[w^a dw]. \quad (\text{FPB.11})$$

No convergence of an infinite sum of fixed u -order is needed: each recurrence is a single formal division in w . The resulting quotient and remainder maps, denoted Q, R_h globally and $Q_\rho, R_\rho^{\text{rem}}$ locally, are continuous and A -linear. Their constructions agree with the polynomial-exponential remainder when the input is a polynomial, because that remainder has the same unique decomposition. Thus (FPB.9) is also the completion of the original global lattice, rather than a differently presented quotient.

In local coordinates the remainder has an explicit coefficient formula. Define

$$p_{a,q} = \prod_{\ell=0}^{q-1} (a + 1 + \ell N), \quad p_{a,0} = 1.$$

For $0 \leq a < m$ and $q \geq 0$,

$$R_\rho^{\text{rem}}(w^{a+qN}) = (-u)^q p_{a,q} w^a, \quad R_\rho^{\text{rem}}(w^{m+qN}) = 0. \quad (\text{FPB.12})$$

To prove this, use $L_\rho(w^{j-m}) = w^j + u(j-m)w^{j-N}$ for $j \geq m$. The case $j = m$ gives zero remainder. Repeatedly reducing a monomial in the other congruence classes gives precisely the displayed product. For an infinite series in w , its coefficient of $u^q w^a$ uses just the coefficient of w^{a+qN} . For a series already containing u , the coefficient of any specified power of u uses a finite sum. This proves (FPB.12) at the stated complete types.

Write the matrix induced by F in the original ascending monomial columns and the ordered local rows (ρ, a) as $T(u)$. Then

$$T_{(\rho,a),b}(u) = \sum_{q \geq 0} (-u)^q p_{a,q} [y^{a+qN}] (\rho + y)^b a_\rho(y)^{-(a+qN+1)/N}, \quad (\text{FPB.13})$$

$$0 \leq b < d, \quad 0 \leq a < m_\rho.$$

Every fractional power uses the branch in (FPB.4), so its constant is $b_\rho^{-(a+qN+1)}$. For proof of this formula, formal change of variable in a residue gives, for any formal germ $f(\rho + y) \in \mathbb{C}[[y]]$,

$$[w^j] f(\rho + \psi(w)) \psi'(w) = [y^j] f(\rho + y) a_\rho(y)^{-(j+1)/N}. \quad (\text{FPB.14})$$

Indeed the left side is the coefficient of $w^{-1} dw$ in $f(\rho + \psi) \psi' w^{-j-1} dw$; substituting $w = y A_\rho(y)$ gives $f(\rho + y) y^{-j-1} A_\rho(y)^{-j-1} dy$. Formal residues are unchanged by this substitution: for monomials other than exponent -1 , this follows by differentiating a formal Laurent primitive; for exponent -1 , $w'/w = 1/y + A'_\rho/A_\rho$ has residue one. Combining (FPB.14) and (FPB.12) proves (FPB.13).

At $u = 0$, let $E_h = \mathbb{C}[s]/h$ and $E_\rho^w = \mathbb{C}[w]/w^{m_\rho}$. Then

$$T_0 : E_h ds \longrightarrow \bigoplus_{\rho} E_\rho^w dw, \quad [f] \longmapsto ([f(\rho + \psi_\rho) \psi'_\rho] dw)_\rho \quad (\text{FPB.15})$$

is invertible. Here is its explicit inverse. Given local remainders $q_\rho(w)$, set

$$f_\rho(y) = q_\rho(y A_\rho(y)) \frac{d}{dy} (y A_\rho(y)) \pmod{y^{m_\rho}}. \quad (\text{FPB.16})$$

Set $H_\rho = h/(s - \rho)^{m_\rho}$ and let $B_\rho(y)$ be the truncated inverse of $H_\rho(\rho + y)$ modulo y^{m_ρ} , calculated by the recursion for the inverse of a series with nonzero constant. The original Hermite idempotent is the degree-below- d remainder $e_\rho = \text{rem}_h(H_\rho B_\rho(s - \rho))$. The inverse of (FPB.15) is

$$(q_\rho)_\rho \longmapsto \text{rem}_h \sum_{\rho} e_\rho(s) f_\rho(s - \rho). \quad (\text{FPB.17})$$

Each e_ρ is one modulo its own primary factor and zero modulo every other one. The product of the distinct primary factors divides a polynomial divisible by each: repeatedly apply the Bezout identity to pairs of coprime factors. Thus their simultaneous jets determine a unique remainder modulo h . Finally $\psi'(w) w'(\psi(w)) = 1$ verifies (FPB.16) in both directions. This proves (FPB.17), without assuming the desired comparison.

With $T(u) = \sum_{n \geq 0} u^n T_n$, define

$$V_0 = T_0^{-1}, \quad V_n = -T_0^{-1} \sum_{i=1}^n T_i V_{n-i} \quad (n \geq 1). \quad (\text{FPB.18})$$

For $V(u) = \sum_{n \geq 0} u^n V_n$, it follows coefficient by coefficient that $TV = 1$. The same recursion or the adjugate formula over A , since $\det T_0 \neq 0$, gives $VT = 1$. Therefore

$$F_* : \mathcal{H} \xrightarrow[T(u)]{\sim} \bigoplus_{\rho} \mathcal{H}_\rho \quad (\text{FPB.19})$$

is the required isomorphism of complete original lattices.

There are explicit chain homotopies as well. Let i_h and i_{loc} include remainder representatives into degree one, and let p_h, p_{loc} denote their remainder projections, with zero maps in degree zero. The quotient maps Q, Q_{loc} are degree-minus-one maps and satisfy

$$1 - i_h p_h = dQ + Qd, \quad 1 - i_{\text{loc}} p_{\text{loc}} = dQ_{\text{loc}} + Q_{\text{loc}}d. \quad (\text{FPB.20})$$

Define $J = i_h T^{-1} p_{\text{loc}}$ as a map from the direct sum of local complexes to the global complex. Since remainder vanishes on every differential, this is a chain map. The identity $p_{\text{loc}} F = T p_h$ gives $JF = i_h p_h$. Moreover $p_{\text{loc}}(1 - FJ) = 0$, and hence

$$1 - JF = dQ + Qd, \quad 1 - FJ = d(Q_{\text{loc}}(1 - FJ)) + (Q_{\text{loc}}(1 - FJ))d. \quad (\text{FPB.21})$$

For the second identity, in degree one use zero remainder to write $(1 - FJ)f = L_{\text{loc}} Q_{\text{loc}}(1 - FJ)f$; in degree zero, $FJ = 0$ and $Q_{\text{loc}} L_{\text{loc}} = 1$. This also proves that these formulas retain the cochain information through a specific homotopy equivalence. The homotopies are A -linear; no assertion that the chosen remainders are horizontal is needed.

294.3 The exact connection and the constant determinant

Localize the displayed complete complexes at u . On the global degrees zero and one respectively the original connection is

$$\nabla^0 = \partial_u - \frac{\Phi(s)}{u^2} + \frac{1}{u}, \quad \nabla^1 = \partial_u - \frac{\Phi(s)}{u^2}. \quad (\text{FPB.22})$$

All operators are defined on $R[u^{-1}]$: each element has a finite lower bound on its powers of u , and differentiation preserves that property. Direct calculation gives $[u\partial_s + h, \partial_u] = -\partial_s$ and $[u\partial_s + h, -\Phi/u^2] = -h/u$, proving $\nabla^1 L = L\nabla^0$. Under (FPB.7) the phase becomes exactly $c_\rho + w^N/N$. The change of variables and its Jacobian are independent of u , so F intertwines both degree connections on these localized complexes.

In the unchanged global monomial basis, divide

$$\Phi(s)s^b = h(s)q_b(s) + r_b(s), \quad \deg r_b < d, \quad 0 \leq b < d.$$

Let C have columns r_b and B have columns q'_b . The derivative already has degree below d . Subtracting Lq_b gives the original connection matrix

$$G(u) = -C/u^2 + B/u. \quad (\text{FPB.23})$$

For the local column w^a , $(c_\rho + w^N/N)w^a = w^m w^{a+1}/N + c_\rho w^a$. The same reduction gives

$$D(u) = \bigoplus_{\rho \in Z} \text{diag}_{0 \leq a < m_\rho} \left(-\frac{c_\rho}{u^2} + \frac{a+1}{N_\rho u} \right), \quad T' + DT = TG. \quad (\text{FPB.24})$$

The second identity is equality of actual matrices over K , obtained by applying the proved chain intertwining to each original basis column. It is also an all-order identity satisfied by the coefficient formula (FPB.13).

The determinant is explicit and independent of u :

$$\det T(u) = \det T_0 = \left(\prod_{\rho \in Z} b_\rho^{-m_\rho(m_\rho+1)/2} \right) \prod_{i < j} (\rho_j - \rho_i)^{m_{\rho_i} m_{\rho_j}}. \quad (\text{FPB.25})$$

First consider T_0 . The ordinary jet matrix has entries $\binom{b}{a} \rho^{b-a}$. Its determinant is the confluent Vandermonde product in (FPB.25), with coefficient one and the displayed order. One elementary verification starts with the Vandermonde determinant at distinct variables $x_{i,a}$, whose determinant is $\prod_{(i,a) < (j,b)} (x_{j,b} - x_{i,a})$: it is alternating, divisible by every displayed difference, and has that leading coefficient by the identity permutation in the determinant. Apply successive divided differences within each ordered cluster and then set every variable in cluster i to ρ_i . The divided-difference row of order a becomes the Taylor-coefficient row $f^{(a)}(\rho_i)/a!$. The denominators cancel precisely the within-cluster Vandermonde factors; each cross-cluster difference becomes $\rho_j - \rho_i$. This proves

the asserted coefficient and sign. The map of a local jet $y^a dy$ to $\psi(w)^a \psi'(w) dw$ is triangular with diagonal $b_\rho^{-(a+1)}$, proving the formula for $\det T_0$.

For independence of u , C is multiplication by Φ modulo h . On the ρ -primary quotient, $\Phi(\rho + y) = c_\rho + O(y^{m_\rho+1})$, so it is exactly scalar c_ρ , with multiplicity m_ρ . Hence $\text{Tr } C = \sum_\rho m_\rho c_\rho$. The quotient q_b has degree $b+1$ and leading coefficient $1/(d+1)$, so B is upper triangular with diagonal $(b+1)/(d+1)$ and trace $d/2$. Also $\sum_{a=0}^{m-1} (a+1)/(m+1) = m/2$. Thus $\text{Tr } G = \text{Tr } D$. Taking the trace of (FPB.24) after multiplication by T^{-1} gives $(\det T)' = 0$. In characteristic zero a formal series with zero derivative is constant, proving (FPB.25).

Replacing the chosen branch by $b_\rho \epsilon$, $\epsilon^{N_\rho} = 1$, replaces w by ϵw . Since $\epsilon^{a+qN+1} = \epsilon^{a+1}$, it replaces the local rows of T by exactly $\text{diag}(\epsilon^{-(a+1)})T_\rho$. This is the complete branch-change map; the branch is not suppressed.

294.4 The original arithmetic source and its Jacobian

Keep the original $\mathcal{B}$, V , $D = -x\partial_x$, Θ , seed ϕ_* , and Euler inverse F_h , with $\mathcal{M}\Theta\phi_* = g$ and $\mathcal{M}F_h = g/h$. Explicitly, V consists of the even Schwartz functions on $\mathbb{R}$ satisfying $\phi(0) = \int_{\mathbb{R}} \phi = 0$, and $\mathcal{B}$ consists of smooth functions on $\mathbb{R}_{>0}$ with $\sup_{x>0} x^b |D^n F(x)| < \infty$ for every integer b and every $n \geq 0$. The maps are $\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx)$ and $\mathcal{M}F(s) = \int_0^\infty F(x) x^s dx/x$; $\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. The original source maps are

$$\begin{aligned} \alpha_h(P) &= P(D)\phi_*, & \mathcal{T}_h(P) &= P(D)F_h, \\ \Theta\alpha_h(P) &= \mathcal{T}_h(hP), & J_h\mathcal{T}_h(P) &= v_h[P], & v_h &= [g/h] \in E_h^\times. \end{aligned} \quad (\text{FPB.26})$$

These are the inherited maps, with their original spaces and norms; the coordinate calculation does not replace those spaces. In each local primary factor retain every coefficient

$$v_\rho(y) = \sum_{j=0}^{m_\rho-1} \frac{1}{j!} \left(\frac{d^j}{ds^j} \frac{g(s)}{h(s)} \right)_{s=\rho} y^j \pmod{y^{m_\rho}}. \quad (\text{FPB.27})$$

Its retained constant is exactly

$$v_\rho(0) = \frac{g^{(m_\rho)}(\rho)}{m_\rho! \prod_{\sigma \neq \rho} (\rho - \sigma)^{m_\sigma}} \neq 0,$$

because the selected order is complete. This equality follows by factoring the leading Taylor term of g and the original factor $(s - \rho)^{m_\rho}$ of h . The ordinary marked arithmetic operator is $A_h = M_s$, and the ordinary full-unit operator is $U_h = M_{v_h}$. Under (FPB.15), their exact marked images are

$$T_0 A_h T_0^{-1} = \bigoplus_\rho M_{\rho+\psi_\rho(w)}, \quad T_0 U_h T_0^{-1} = \bigoplus_\rho M_{v_\rho(\psi_\rho(w))}. \quad (\text{FPB.28})$$

These multiplications are on $E_\rho^w dw$, hence truncated at the complete original order. To check them, compose the formulas for T_0 and its inverse; the two Jacobians cancel in a conjugation, while the full Taylor series remains.

Let $z_\rho = e_\rho(s)(s - \rho)^{m_\rho-1} \in E_h$. Then

$$T_0 z_\rho = b_\rho^{-m_\rho} w^{m_\rho-1} dw \quad \text{in the } \rho\text{-factor, zero elsewhere,} \quad T_0 U_h z_\rho = v_\rho(0) b_\rho^{-m_\rho} w^{m_\rho-1} dw. \quad (\text{FPB.29})$$

Indeed $\psi(w) = b_\rho^{-1} w + O(w^2)$ and $\psi'(w) = b_\rho^{-1} + O(w)$; all higher terms vanish at degree m_ρ . Both displayed terminal coefficients are nonzero.

Away from the marked fibre, extend the original matrices A_h, U_h over A in the unchanged global monomial basis. Their exact local operators are the explicitly computable matrices

$$\mathcal{A}(u) = T(u) A_h T(u)^{-1}, \quad \mathcal{U}(u) = T(u) U_h T(u)^{-1}. \quad (\text{FPB.30})$$

Equations (FPB.13) and (FPB.18) determine every coefficient of these operators. They are not identified with naive multiplication on a twisted de Rham quotient. The exact obstruction to that identification is visible already on representatives: for a polynomial v ,

$$vLq = L(vq) - uv'q, \quad R_h(vLq) = -uR_h(v'q). \quad (\text{FPB.31})$$

At $u = 0$ this defect vanishes. For an original representative p of degree below d , write $vp = hq + r$, $\deg r < d$. Then the full deformed remainder is

$$R_h(vp) = r - uR_h(q'), \quad TR_h(vp) = Tr - uTR_h(q'). \quad (\text{FPB.32})$$

This proves the precise comparison with coefficientwise multiplication by $[v]$ in E_h , including its derivative term. The same identity in the local complex has v' replaced by the derivative of the actual pulled-back coefficient $v(\rho + \psi(w))$. For $v = s$ and $\deg p < d$, the quotient q is constant, so the defect in (FPB.32) is zero on that section; (FPB.31) still proves why this is not a multiplication map on arbitrary cochains.

294.5 The precise attachment to the mixed support carriers

The marked comparison contains two different, explicitly related maps. The algebra map is

$$S_0 : E_h \xrightarrow{\sim} \prod_{\rho} E_{\rho}^w, \quad [f] \mapsto ([f(\rho + \psi_{\rho})])_{\rho}. \quad (\text{FPB.33})$$

It is a unital algebra isomorphism: the same CRT proof applies, with the Jacobian omitted in both directions. Put $j = (\psi'_{\rho}(w))_{\rho}$, an invertible element of its target. Then the actual one-form map is $T_0 = M_j S_0$, and it satisfies

$$T_0(ax) = S_0(a)T_0(x) \quad (a, x \in E_h). \quad (\text{FPB.34})$$

Thus (S_0, T_0) is an exact isomorphism of a coefficient algebra and its rank-one module of marked one-forms. It is this pair that attaches to the mixed carriers. In general T_0 itself is not a unital algebra map: it sends $1 \, ds$ to $j \, dw$. If multiplication by an invertible j were multiplicative, its value on $1 \cdot 1$ would force $j = j^2$, hence $j = 1$.

For completeness, the split extension of a unital ring map $f : R \rightarrow R'$ is $G(f)(\tau) = \tau$ and $G(f)(r^{\bullet}) = f(r)^{\bullet}$. The operations are the original ones: τ is additive zero and multiplicatively absorbing; supported addition and multiplication are those of the ring. These formulas prove directly that $G(f)$ is a unital semiring map and that the supported zero $e = 0^{\bullet}$ stays distinct from τ . In particular $G(S_0)$ retains the original absolute-base point and the supported zero. The synchronized map

$$G\left(\prod_{\rho} E_{\rho}^w\right) \longrightarrow \prod_{\rho} G(E_{\rho}^w)$$

sends τ to the all- τ tuple and a supported tuple to the tuple with every coordinate supported. It does not introduce or delete the other independent support masks in the target.

More generally every original independent coefficient-slot mask and every original source-leg mask is retained as an external label. On the module attached to a fixed coefficient face, apply F , its remainder maps, and its homotopies coefficientwise; on the label use the identity. Inclusions, projections, and zero insertions of labelled slots commute with this operation by linearity. A newly inserted supported-zero slot remains supported-zero because its label is unchanged, while a missing slot remains missing. The same statement holds for a proper-source kernel carried as an additional labelled summand: the map acts on its coefficient module and never quotients out the summand. For multiplicative carrier arrows it is S_0 , not the Jacobian module map, that supplies the coefficient-ring arrow. On the deformed lattices the exact coefficient action is its original action transported by $T(u)$, as in (FPB.30); no unwarranted multiplicativity of the remainder projection is used.

294.6 All ordered tensor factors and formal character blocks

For $k \geq 1$, use the original ordered complex on $\mathbb{C}[s_1, \dots, s_k][[u]]$ with differential

$$\mathcal{D} = \sum_{i=1}^k (u\partial_{s_i} + h(s_i)) ds_i \wedge (-). \quad (\text{FPB.35})$$

This is the completed tensor product over A of the one-factor complexes: modulo u^n , the ordinary tensor product has polynomial coefficients in all the s_i ; taking the inverse limit gives precisely the displayed ring. The one-factor split homotopies extend to these completed tensor products. On a tensor summand with a contractible factor, apply its degree-minus-one homotopy with the sign from all preceding degrees; the two cross terms cancel in the tensor differential. Iterating leaves only the ordered top remainder summand. Thus its cohomology is concentrated in degree k , with basis $\prod_i s_i^{b_i} ds_1 \wedge \dots \wedge ds_k$, $0 \leq b_i < d$, and rank d^k .

Apply (FPB.7) independently in every ordered factor. For clarity, the local complex at a tuple $\rho \in Z^k$ is taken on $\mathbb{C}[[w_1, \dots, w_k]][[u]]$, with its u -adic topology and the differential $\sum_i (u\partial_{w_i} + w_i^{m_{\rho_i}}) dw_i \wedge (-)$. This ring is not identified with the mere u -adic completion of an algebraic tensor product of one-variable formal-series rings. Its cohomology is computed by successive one-variable versions of (FPB.10), with coefficient ring the formal series in the remaining variables. Each division and derivative in one variable commutes with those in another. The same split tensor homotopies therefore leave the top remainder spanned over A by $\prod_i w_i^{a_i}$, $0 \leq a_i < m_{\rho_i}$, and no other cohomology. On an original monomial the separate reductions are the product of the separate one-variable reductions. The target is the direct sum over every tuple, and its top lattice map and phase are exactly

$$\begin{aligned} T_k(u) &= T(u)^{\otimes k}, \\ ds_1 \wedge \dots \wedge ds_k &\mapsto \prod_i \psi'_{\rho_i}(w_i) dw_1 \wedge \dots \wedge dw_k, \\ \sum_i \Phi(s_i) &= C_\rho + \sum_i \frac{w_i^{N_{\rho_i}}}{N_{\rho_i}}, \quad C_\rho = \sum_i c_{\rho_i}. \end{aligned} \quad (\text{FPB.36})$$

No permutation of the ordered one-forms was made, so no additional sign occurs. The target connection on the column indexed by $(\rho, \mathbf{a})$ is

$$\partial_u - \frac{C_\rho}{u^2} + \frac{\beta_{\rho, \mathbf{a}}}{u}, \quad \beta_{\rho, \mathbf{a}} = \sum_i \frac{a_i + 1}{N_{\rho_i}}, \quad 0 \leq a_i < m_{\rho_i}. \quad (\text{FPB.37})$$

Tensoring (FPB.24), or differentiating $T^{\otimes k}$ in each factor, proves the exact intertwining with the original tensor-sum connection. In degree j , the original chain connection retains the correction $(k-j)/u$. All tuples with the same value of C_ρ assert as distinct summands. Their direct sum is the full block with that exponential constant; no assertion that these sums are distinct enters the proof.

The original arithmetic tensor action is $A^{(k)} = \sum_i A_{h,i}$, with ordered full unit $U_h^{\otimes k}$. Under the complete map these become $T_k A^{(k)} T_k^{-1}$ and $T_k U_h^{\otimes k} T_k^{-1}$. At $u = 0$, the images are the actual multiplications by $\sum_i (\rho_i + \psi_{\rho_i}(w_i))$ and $\prod_i v_{\rho_i}(\psi_{\rho_i}(w_i))$, respectively. In particular

$$T_{k,0} U_h^{\otimes k} \bigotimes_i z_{\rho_i} = \left(\prod_i v_{\rho_i}(0) b_{\rho_i}^{-m_{\rho_i}} \right) \prod_i w_i^{m_{\rho_i}-1} dw_1 \wedge \dots \wedge dw_k \neq 0. \quad (\text{FPB.38})$$

For the original cyclic terminal in a tuple, put $K_\rho = \sum_i (m_{\rho_i} - 1)$. The multinomial formula gives, in that actual primary corner,

$$\left(\sum_i (s_i - \rho_i) \right)^{K_\rho} = \frac{K_\rho!}{\prod_i (m_{\rho_i} - 1)!} \prod_i (s_i - \rho_i)^{m_{\rho_i}-1}. \quad (\text{FPB.39})$$

Every other monomial has an exponent exceeding one of the retained orders and vanishes. Its theta-weighted image under $T_{k,0}$ therefore has the coefficient in (FPB.38) multiplied by exactly the factorial ratio in (FPB.39). This coefficient is nonzero over the original complex numbers. No factorial, Taylor unit, or Jacobian has been absorbed into a new choice of terminal vector.

Here are the precise formal monodromy conventions. For one local factor put $\zeta_\rho = e^{2\pi i/N_\rho}$, $\beta_{\rho,a} = (a+1)/N_\rho$, and

$$W_{\rho,ja} = \zeta_\rho^{j(a+1)} - 1 \quad (1 \leq j \leq m_\rho), \quad d_{\rho,a} = N_\rho^{\beta_{\rho,a}-1} e^{\pi i \beta_{\rho,a}} \Gamma(\beta_{\rho,a}). \quad (\text{FPB.40})$$

These are the original homogeneous-model ordered contour constants: the ray of argument $(\arg u + \pi)/N_\rho + 2\pi j/N_\rho$ gives $d_{\rho,a} u^{\beta_{\rho,a}} \zeta_\rho^{j(a+1)}$ after the positive radial substitution $r^{N_\rho}/(N_\rho|u|)$, and subtraction of the zeroth ray gives W_ρ . The columns are independent because a polynomial $\sum_{a=0}^{m-1} x_a(z^{a+1} - 1)$ vanishing at all nontrivial N -th roots also vanishes at 1 and has degree below N , so is zero. The gamma factors are nonzero since $0 < \beta < 1$.

Adjoin formal symbols $e^{c_\rho/u}$ and the fractional powers $u^{\beta_{\rho,a}}$, with the derivative specified by their displayed exponents. A full formal period matrix for the original global connection is

$$\Pi^{\text{form}}(u) = \left(\bigoplus_\rho e^{c_\rho/u} W_\rho \text{diag}_a(d_{\rho,a} u^{\beta_{\rho,a}}) \right) T(u). \quad (\text{FPB.41})$$

Equations (FPB.24) and (FPB.40) verify $(\Pi^{\text{form}})' = \Pi^{\text{form}} G$ directly. Its exact determinant, in the specified block and monomial orders, is

$$\det \Pi^{\text{form}} = \det T_0 \left(\prod_\rho \det W_\rho \prod_{a=0}^{m_\rho-1} d_{\rho,a} \right) u^{d/2} \exp \left(\frac{\sum_\rho m_\rho c_\rho}{u} \right). \quad (\text{FPB.42})$$

Formal continuation $\log u \mapsto \log u + 2\pi i$ fixes the integer-power matrix $T(u)$ and the exponential symbols. It therefore has period monodromy

$$M^{\text{form}} = \bigoplus_\rho W_\rho \text{diag}_a(\zeta_\rho^{a+1}) W_\rho^{-1}. \quad (\text{FPB.43})$$

For the differential module's horizontal coefficient columns the solutions of $(\partial_u + D)v = 0$ instead have factors $e^{-c_\rho/u} u^{-\beta_{\rho,a}}$. Thus their monodromy characters are the inverses of the period characters in (FPB.43). This records the dual convention explicitly.

The ordered tensor period matrix is $(\Pi^{\text{form}})^{\otimes k}$, and its period character on (ρ, a) is $\exp(2\pi i \beta_{\rho,a})$. These matrices are diagonalizable. On the formal regular factors the unipotent logarithm is zero, and the dimension of the character-one space inside the block with constant C is exactly

$$\# \left\{ (\rho, a) : \sum_i c_{\rho_i} = C, \quad \sum_i \frac{a_i + 1}{N_{\rho_i}} \in \mathbb{Z}, \quad 0 \leq a_i < m_{\rho_i} \right\}. \quad (\text{FPB.44})$$

Repeated equal characters contribute their full multiplicity because all their ordered summands were retained. Formula (FPB.44) is about the formal regular character factors of this proved decomposition; it does not identify actual analytic inertia invariants or finite-field boundary stalks.

294.7 Horizontal formal projectors and the unchanged source form

Let P_ρ be the constant block projector on the ρ -summand of the local direct sum. The actual formal projector in the original global lattice is

$$\mathcal{E}_\rho(u) = T(u)^{-1} P_\rho T(u), \quad \mathcal{E}_\rho(0) = M_{e_\rho}, \quad \mathcal{E}'_\rho + [G, \mathcal{E}_\rho] = 0. \quad (\text{FPB.45})$$

The first identity defines all coefficients by (FPB.13) and (FPB.18). The marked identity follows from the weighted CRT map, whose Jacobian preserves each primary summand. Orthogonality, idempotence, and sum one follow by conjugating the corresponding identities for P_ρ . Here algebraic orthogonality means $\mathcal{E}_\rho \mathcal{E}_\sigma = 0$ when $\rho \neq \sigma$; it makes no assertion of orthogonality in the original source metric. Differentiating the definition and using $T' = TG - DT$, with $[D, P_\rho] = 0$, gives the displayed horizontal identity. Thus the retained primary labels have explicitly horizontal projectors over the original complete lattice, including when critical values coincide; they have not merely been recovered from a dimension count.

There is also a marked constraint on equal-critical-value off-diagonal blocks. If $\rho \neq \sigma$ and $c_\rho = c_\sigma = c$, then

$$M_{e_\sigma} B M_{e_\rho} = 0 \quad \text{on } E_h. \quad (\text{FPB.46})$$

Take the degree-below- d representative f of an element supported on the ρ -factor. Multiplication by Φ modulo h is cf , so the original quotient defining Bf is exactly $q = (\Phi - c)f/h$, and $Bf = q'$. At σ , f vanishes to order at least m_σ , while $\Phi - c$ vanishes to order at least $m_\sigma + 1$. Therefore q vanishes to order at least $m_\sigma + 1$, and q' vanishes to order at least m_σ , proving (FPB.46). This proves the exact compressed first-order relation without assuming that all original critical values are different.

The original source form on the polynomial coefficient section is

$$(G_\theta)_{ab} = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{(1/2 + it)^a} (1/2 + it)^b \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 dt, \quad 0 \leq a, b < d. \quad (\text{FPB.47})$$

At a removable zero this integrand uses the entire quotient g/h . To verify its source type, put $x = e^z$. The original $L^2(dx)$ norm of $\mathcal{T}_h(P)$ becomes the $L^2(dz)$ norm of $e^{z/2} \mathcal{T}_h(P)(e^z)$. Its Fourier transform with the positive exponential convention is $P(1/2 + it)(g/h)(1/2 + it)$. Fourier Plancherel, with inverse factor $1/(2\pi)$, gives (FPB.47). The original $\mathcal{B}$ bounds imply square integrability and decay of all needed derivatives, so the displayed integrals are finite. If the norm is zero, the continuous original source function is zero; its Mellin transform $P(g/h)$ is zero. Since g/h is a nonzero entire function, a polynomial P with that property is zero. Hence G_θ is positive definite.

On a phase-coordinate vector representing the polynomial section $q = T_0 p$, the very same form is

$$G_{\text{phase}} = T_0^{-*} G_\theta T_0^{-1}. \quad (\text{FPB.48})$$

On the actual theta jet $q_\theta = T_0 U_h p$, the same source form instead has matrix

$$G_{\text{theta jet}} = (T_0 U_h)^{-*} G_\theta (T_0 U_h)^{-1}. \quad (\text{FPB.49})$$

Here $X^{-*} = (X^{-1})^*$. Substitution in the two quadratic forms proves both identities, including the full unit and every Jacobian factor. Their ordered tensor versions use the k -fold Kronecker products of the displayed maps and G_θ . There is no assertion that a possibly divergent formal $T(u)$ has a positive analytic Gram matrix at a nonzero numerical u .

The distinction is exact at the map level. Equations (FPB.7), (FPB.19), and (FPB.24) give a chain homotopy equivalence of the specified u -adic complexes and an isomorphism of their localized formal differential modules. No convergence of $T(u)$, no map from its formal local model rays to the original global rapid-decay cycles, and no Stokes transition matrices has been proved here. Consequently (FPB.41) is the specified formal period solution, whose relation to the original global connection is the proved identity $(\Pi^{\text{form}})' = \Pi^{\text{form}} G$; it is not declared to be an evaluated global analytic contour integral. The full original theta operator is related to it by (FPB.26)–(FPB.32) and their ordered tensors, rather than by an assumed Frobenius or weight intertwiner. In particular no arithmetic purity or Riemann-hypothesis conclusion is inserted into the boundary comparison.

295 The actual sectorial comparison of the original full packet

Retain the original nonempty packet $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, its degree d , the primitive $\Phi' = h$, $\Phi(0) = 0$, and the ordered roots and monomial columns of FPB. Put $D = d + 1$. For each root retain $N_\rho = m_\rho + 1$, $c_\rho = \Phi(\rho)$, the specified phase-root branch b_ρ , the coordinate $w = yA_\rho(y)$, and its inverse $y = \psi_\rho(w)$. Thus

$$\Phi(\rho + \psi_\rho(w)) = c_\rho + w^{N_\rho}/N_\rho, \quad h(\rho + \psi_\rho(w))\psi'_\rho(w) = w^{m_\rho}. \quad (\text{SPG.1})$$

The matrices C, B are the original polynomial-division matrices, so $G = -C/u^2 + B/u$. The formal comparison of FPB is $T \in \text{GL}_d(\mathbb{C}[[u]])$, with $T' + D_{\text{loc}}T = TG$ and its explicitly retained constant determinant $\det T = \det T_0 \neq 0$. Here D_{loc} is the diagonal matrix with entry $-c_\rho/u^2 + \beta_{\rho,a}/u$, where $\beta_{\rho,a} = (a + 1)/N_\rho$, $0 \leq a < m_\rho$.

295.1 A specified algebraic Borel germ for every original column

Fix a row (ρ, a) , abbreviate $N = N_\rho$, $m = m_\rho$, $\beta = (a + 1)/N$, and keep the original column index $0 \leq b < d$. Define the holomorphic germ

$$H_b(w) = (\rho + \psi_\rho(w))^b \psi'_\rho(w) = \sum_{n \geq 0} H_{b,n} w^n. \quad (\text{SPG.2})$$

The phase coordinate is holomorphic on a neighbourhood of zero: the chosen binomial root of the nonzero polynomial germ is holomorphic on a sufficiently small disc, and the holomorphic inverse theorem applies because its derivative at zero is $b_\rho \neq 0$. FPB's exact remainder gives, with $(\beta)_q = \prod_{j=0}^{q-1} (\beta + j)$,

$$(T_q)_{(\rho,a),b} = (-N)^q (\beta)_q H_{b,a+qN}. \quad (\text{SPG.3})$$

Consequently the following transform is an actual convergent algebraic germ, whose coefficients have all been specified:

$$\mathcal{B}_{\rho,a;b}(\xi) := \sum_{q \geq 0} \frac{(T_q)_{(\rho,a),b}}{(\beta)_q} \xi^q = \sum_{q \geq 0} H_{b,a+qN} (-N\xi)^q. \quad (\text{SPG.4})$$

Indeed, for $\zeta = e^{2\pi i/N}$, $z^N = -N\xi$, it is the exact root-of-unity filter

$$\mathcal{B}_{\rho,a;b}(\xi) = \frac{z^{-a}}{N} \sum_{j=0}^{N-1} \zeta^{-aj} H_b(\zeta^j z). \quad (\text{SPG.5})$$

Expand the right side in powers of z ; the sum is one on exponents congruent to a modulo N , and zero on the other exponents. It therefore is single valued and holomorphic at $\xi = 0$, regardless of the temporary choice of the root z . All its expressions are algebraic because $\rho + \psi_\rho(w)$ satisfies the original polynomial equation in (SPG.1), and $\psi'_\rho = w^m/h(\rho + \psi_\rho)$.

For a disc on which $|H_b(w)| \leq M$ on $|w| = r$, the Cauchy coefficient estimate and $0 < \beta < 1$ give

$$|(T_q)_{(\rho,a),b}| \leq Mr^{-a} (N/r^N)^q (\beta)_q \leq Mr^{-a} (N/r^N)^q q!. \quad (\text{SPG.6})$$

This is a bound for the original formal coefficients, with their factor $(-N)^q (\beta)_q$ retained in the defining formula.

Write the row of germs $\mathcal{B}_{\rho,a;b}$ as $\mathcal{B}_{\rho,a}$. Set $A_\rho^c = C - c_\rho I_d$, an abbreviation for this specified matrix, not a new arithmetic operator. Comparing coefficients of the original equation $T' + D_{\text{loc}}T = TG$ gives

$$T_{0,\rho a} A_\rho^c = 0, \quad T_{q,\rho a} A_\rho^c + T_{q-1,\rho a} ((q-1+\beta)I_d - B) = 0 \quad (q \geq 1). \quad (\text{SPG.7})$$

Here each $T_{q,\rho a}$ is a row with all original columns. Divide by $(\beta)_{q-1}$, sum the convergent germs, and differentiate term by term. The result is the exact identity

$$\frac{d}{d\xi} \left\{ \xi^{\beta-1} \mathcal{B}_{\rho,a}(\xi) (A_\rho^c + \xi I_d) \right\} = \xi^{\beta-1} \mathcal{B}_{\rho,a}(\xi) B. \quad (\text{SPG.8})$$

For a direct coefficient check, the left coefficient of $\xi^{q+\beta-2}$ for $q \geq 1$ is $(q + \beta - 1) \{ \mathcal{B}_q A_\rho^c + \mathcal{B}_{q-1} \}$, and the corresponding right coefficient is $\mathcal{B}_{q-1} B$. Equation (SPG.7) proves equality. The lowest coefficient vanishes because $\mathcal{B}_0 A_\rho^c = 0$. A branch of $\xi^{\beta-1}$ is used consistently on both sides.

295.2 Continuation along the original critical-value rays

Retain the finite set of actual differences

$$\mathcal{A} = \{c_\rho - c_\sigma : \rho, \sigma \in Z\}. \quad (\text{SPG.9})$$

An admissible direction $\theta \in \mathbb{R}$ means that no nonzero member of $\mathcal{A}$ lies on $L_\theta = \{te^{i\theta} : t > 0\}$. Zero differences remain in the data and do not have an argument. No claim that distinct roots have distinct critical values is required here.

For $\xi = te^{i\theta}$, use the retained lift of the argument and set

$$z(\xi) = \exp \left(\frac{\log N + \log \xi + i\pi}{N} \right), \quad s_{\rho,j}(\xi) = \rho + \psi_\rho(\zeta^j z(\xi)), \quad 0 \leq j < N, \quad (\text{SPG.10})$$

initially for small positive t . Continue these branches by the same original polynomial equation

$$\Phi(s_{\rho,j}(\xi)) = c_\rho - \xi. \quad (\text{SPG.11})$$

They continue uniquely along the entire open ray. To prove this, a multiple root of (SPG.11) would satisfy $h(s) = 0$, hence $s = \sigma \in Z$ and $\xi = c_\rho - c_\sigma$. Such a point does not occur on an admissible open ray. The implicit function theorem therefore continues each simple root locally. No root can escape to infinity above a compact part of the ray, because the usual elementary polynomial root bound follows from the triangle inequality for its fixed leading coefficient and bounded remaining coefficients. Cover a compact segment by finitely many such local neighbourhoods. This proves the stated continuation. It also proves continuation on a sufficiently thin open wedge about the ray, using the original germ near zero. Equation (SPG.5) then continues $\mathcal{B}$ on that same wedge, and (SPG.8) remains true by analytic continuation.

There is an explicit bound at infinity. All roots of (SPG.11) have $|s|$ bounded above and below by positive multiples of $|\xi|^{1/D}$ for sufficiently large $|\xi|$. One way to see both inequalities is to substitute $s = \xi^{1/D} v$: the equation tends to $v^D/D = -1$, whose D roots are distinct and nonzero. The implicit function theorem at each of those roots, in the parameter $\xi^{-1/D}$, supplies all D branches for large $|\xi|$. Since $h(s) = s^d + O(s^{d-1})$, $|h(s)|$ is at least a positive multiple of $|\xi|^{d/D}$. Using (SPG.1) in (SPG.5) proves, for each original column,

$$|\mathcal{B}_{\rho,a;b}(\xi)| \leq C_{\rho,a,b,\theta} |\xi|^{(b+1)/D-\beta} \quad (|\xi| \text{ sufficiently large on the wedge}). \quad (\text{SPG.12})$$

In detail the factor $z^{-a} \psi'(\zeta^j z)$ has order $|\xi|^{(m-a)/N-d/D}$, and s^b contributes $|\xi|^{b/D}$. Since $(m-a)/N = 1 - \beta$, their sum is the exponent in (SPG.12). The bound is uniform on smaller closed wedges. This retains the derivative factor responsible for the decay exponent; it is not a bound for an unweighted inverse root.

295.3 The exact sectorial gauge, not only its asymptotic series

On the logarithmic sector $\mathfrak{S}_\theta = \{u \neq 0 : |\arg u - \theta| < \pi/2\}$, with these arguments as real lifts, define the complete row

$$T_{\theta, \rho a}(u) = \frac{u^{-\beta}}{\Gamma(\beta)} \int_{L_\theta} e^{-\xi/u} \xi^{\beta-1} \mathcal{B}_{\rho, a}(\xi) d\xi. \quad (\text{SPG.13})$$

The branch $\log \xi = \log t + i\theta$ and the branch of $u^{-\beta}$ from $\mathfrak{S}_\theta$ are part of the definition. Near zero, the integrand is bounded by a multiple of $|\xi|^{\beta-1}$, which is integrable because $\beta > 0$. At infinity, (SPG.12) bounds the nonexponential part by a power of $|\xi|$, while $\text{Re}(e^{i\theta}/u) > 0$. The integral and its u -derivatives converge uniformly on compact subsets of the sector. Thus (SPG.13) is holomorphic there.

It has precisely the original asymptotic expansion

$$T_\theta(u) \sim \sum_{q \geq 0} u^q T_q = T(u) \quad \text{on every smaller closed subsector of } \mathfrak{S}_\theta. \quad (\text{SPG.14})$$

Here is a direct proof of the required all-orders statement. Subtract the Taylor polynomial of $\mathcal{B}$ through order $q - 1$ on a disc of radius R inside its holomorphic germ. Its remainder is bounded by $C_q |\xi|^q$ on a smaller disc. On a closed subsector the exponential satisfies $|e^{-\xi/u}| \leq e^{-c|\xi|/|u|}$ for a fixed $c > 0$. The integral of the small-disc remainder, multiplied by $|u|^{-\beta}$, is therefore $O(|u|^q)$. On the ray outside that disc the algebraic bound, and the finitely many polynomial terms subtracted, make the remaining integral $O(|u|^{-M_q} e^{-cR/(2|u|)})$, which is $O(|u|^q)$ as well. Integrating a Taylor monomial gives exactly

$$\frac{u^{-\beta}}{\Gamma(\beta)} \int_{L_\theta} e^{-\xi/u} \xi^{q+\beta-1} d\xi = (\beta)_q u^q.$$

For u on the central ray this is the positive real gamma integral after $\xi = ut$; elsewhere it follows by holomorphy on the same connected sector. This proves every coefficient and the estimate (SPG.14).

The integral satisfies the exact original gauge equation

$$T'_\theta + D_{\text{loc}} T_\theta = T_\theta G. \quad (\text{SPG.15})$$

To prove this rather than deduce it from an asymptotic expansion, put $F(\xi) = \xi^{\beta-1} \mathcal{B}_{\rho, a}(\xi) (A_\rho^c + \xi I_d)$. At zero, (SPG.7) gives $F(\xi) = O(|\xi|^\beta)$, so its endpoint is zero. At infinity it has algebraic growth by (SPG.12), so $e^{-\xi/u} F(\xi)$ tends to zero. Integrate (SPG.8) by parts on a truncated ray and take its two endpoints to their limits. The result is

$$u \mathcal{L}_\beta(\mathcal{B}_{\rho, a}) B = \mathcal{L}_\beta(\mathcal{B}_{\rho, a} (A_\rho^c + \xi I_d)), \quad (\text{SPG.16})$$

where $\mathcal{L}_\beta$ denotes the integral in (SPG.13). Differentiating that integral directly also gives $u^2 T'_{\theta, \rho a} + \beta u T_{\theta, \rho a} = \mathcal{L}_\beta(\xi \mathcal{B}_{\rho, a})$. Substitution in (SPG.16) is exactly the row of (SPG.15).

The determinant is the original constant:

$$\det T_\theta(u) = \det T_0 \neq 0 \quad (u \in \mathfrak{S}_\theta). \quad (\text{SPG.17})$$

The multilinear derivative of a determinant and (SPG.15) give $(\det T_\theta)' = (\text{Tr } G - \text{Tr } D_{\text{loc}}) \det T_\theta = 0$, without assuming invertibility in deriving this equality. Along the central ray, (SPG.14) gives its limiting value $\det T_0$. Since the sector is connected, the determinant has that value throughout. In particular the constructed analytic gauge is invertible everywhere in its stated sector.

295.4 Evaluation as actual lifted contours, with all gamma factors

Retain exactly the homogeneous constants of FPB:

$$d_{\rho,a} = N_{\rho}^{\beta_{\rho,a}-1} e^{\pi i \beta_{\rho,a}} \Gamma(\beta_{\rho,a}), \quad (W_{\rho})_{la} = \zeta_{\rho}^{l(a+1)} - 1. \quad (\text{SPG.18})$$

Set $W_{\text{loc}} = \bigoplus_{\rho} W_{\rho}$ and define

$$P_{\theta}(u) = W_{\text{loc}} \text{diag}_{\rho,a} (e^{c_{\rho}/u} d_{\rho,a} u^{\beta_{\rho,a}}) T_{\theta}(u). \quad (\text{SPG.19})$$

It is an actual invertible analytic matrix, and (SPG.15) proves $P'_{\theta} = P_{\theta} G$.

Let $\gamma_{\rho,j,\theta}$ be the oriented path $s_{\rho,j}(te^{i\theta})$, $0 < t < \infty$, with its initial endpoint ρ . All its periods converge. Near the endpoint, $s - \rho = O(t^{1/N})$ and $ds/dt = O(t^{1/N-1})$, which is integrable. At infinity, $\Phi(s) = c_{\rho} - te^{i\theta}$ gives exponential decay for $u \in \mathfrak{S}_{\theta}$, and (SPG.11) bounds all remaining polynomial and derivative factors.

Before multiplication by W_{loc} , row (ρ, a) of (SPG.19) is exactly

$$\frac{1}{N} \sum_{j=0}^{N-1} \zeta^{-j(a+1)} \int_{\gamma_{\rho,j,\theta}} e^{\Phi(s)/u} s^b ds \quad \text{in column } b. \quad (\text{SPG.20})$$

The factors in this identity can be checked without an unspecified phase convention. Insert (SPG.5) into (SPG.13), multiply by $e^{c_{\rho}/u} d_{\rho,a} u^{\beta}$, and substitute $\xi = -z^N/N$ on its specified branch. Then

$$N^{\beta-1} e^{\pi i \beta} \xi^{\beta-1} \frac{z^{-a}}{N} d\xi = \frac{1}{N} dz.$$

Indeed $\log \xi = N \log z - \log N - i\pi$ and $d\xi = -z^{N-1} dz$; the two minus signs cancel and the power of z is zero because $N\beta = a + 1$. Finally $ds = \zeta^j \psi'_{\rho}(\zeta^j z) dz$, giving the extra ζ^{-j} and hence the coefficient in (SPG.20). The exponential becomes exactly $e^{\Phi(s)/u}$. This proves the identity with the original Jacobian, gamma constants, primitive constant, and branch orientation.

Multiplying the Fourier rows by the original W_{ρ} gives the especially concrete exact formula

$$(P_{\theta})_{(\rho,l),b} = \int_{\gamma_{\rho,l,\theta} - \gamma_{\rho,0,\theta}} e^{\Phi(s)/u} s^b ds, \quad 1 \leq l \leq m_{\rho}. \quad (\text{SPG.21})$$

For verification, the coefficient of $\gamma_{\rho,j,\theta}$ after this multiplication is $N^{-1} \sum_{a=0}^{m-1} (\zeta^{l(a+1)} - 1) \zeta^{-j(a+1)} = \delta_{jl} - \delta_{j0}$. Thus the formal local ray labels have become actual global rapid-decay paths through a proved analytic integral, rather than through an assumed convergence of $T(u)$.

295.5 The exact integer comparison to the original global rays

Let Π_h be the original ordered global ray matrix of GRM, whose rows are $\Gamma_n = -\ell_0 + \ell_n$, $1 \leq n \leq d$, and whose ray arguments are $(\arg u + \pi + 2\pi n)/D$. Each lifted path has a unique label $n_{\rho,j}(\theta) \in \{0, \dots, d\}$ determined by

$$\frac{s_{\rho,j}(te^{i\theta})}{(Dt)^{1/D}} \longrightarrow \exp\left(\frac{i(\theta + \pi + 2\pi n_{\rho,j}(\theta))}{D}\right) \quad (t \longrightarrow \infty). \quad (\text{SPG.22})$$

The large-parameter implicit-function argument used for (SPG.12) proves existence and uniqueness of this limit, with precisely these labels on the retained lift of θ .

Define the actual integer matrix, with the original row orders,

$$(E_{\theta})_{(\rho,l),n} = \delta_{n_{\rho,l}(\theta),n} - \delta_{n_{\rho,0}(\theta),n}, \quad 1 \leq n \leq d. \quad (\text{SPG.23})$$

Its entries are determined by continuation of the original polynomial inverse along the explicitly given rays; there is no free choice of an incidence pattern. The exact contour comparison is

$$P_\theta = E_\theta \Pi_h. \quad (\text{SPG.24})$$

To prove it, attach the same finite path from 0 to ρ to both paths in (SPG.21); its integral cancels. A path from 0 to infinity whose endpoint is in the labelled decaying sector has the same integral as the corresponding original ray. For a complete justification, truncate the paths at large radius and join their endpoints inside that decaying sector. Cauchy's theorem applies because the integrand is entire in s . Along the joining arc the leading term of the real phase is at most a negative constant times the radius to the power D , while the lower terms are $O(R^{D-1})$, uniformly on any closed subsector in the stated u -domain. Its polynomial factor and the arc length cannot overcome this decay. The joining integral tends to zero. For the initial comparison one may use the ray at argument $(\theta + \pi + 2\pi n)/D$; it and the original ray defined by $\arg u$ lie in the same strictly decaying sector because $|\arg u - \theta| < \pi/2$. The same arc argument relates them. Taking the difference of the two endpoint labels gives precisely (SPG.23)–(SPG.24).

This matrix is unimodular:

$$E_\theta \in \text{GL}_d(\mathbb{Z}). \quad (\text{SPG.25})$$

Indeed the nonzero determinants of P_θ and Π_h prove that E_θ has full rank. Regard each row as an oriented edge joining its two endpoint labels among $d + 1$ vertices, with column zero deleted. A loop or a disconnected component would give a zero row or a nonzero vector in the kernel: in the latter case take the indicator of a component not containing vertex zero. Thus the graph is connected. It has exactly d edges on $d + 1$ vertices, so it is a tree. One proof is to grow a connected set from a vertex, needing an edge for each added vertex; a cycle would allow removal of an edge while preserving connectedness, contradicting that minimum count. A leaf other than vertex zero has exactly one nonzero entry, equal to 1 or -1 , in its column. Expanding the determinant in that column and removing the leaf proves $\det E_\theta = \pm 1$ by induction. This also proves that all the actual lifted cycles in (SPG.21) form a full integral basis in the original global ray lattice.

All constants in this comparison can be checked through determinants:

$$\det E_\theta = \frac{\det T_0 \prod_\rho (\det W_\rho \prod_{a=0}^{m_\rho-1} d_{\rho,a})}{K_d}, \quad (\text{SPG.26})$$

where K_d is the unchanged global gamma and contour constant in GRM.5. The factors $u^{d/2}$ and $\exp(\sum_\rho m_\rho c_\rho / u)$ cancel between the two actual determinants, because both primitives and both bases have already been fixed. Formula (SPG.25) proves that the displayed retained expression is the actual orientation sign; it is not a normalization imposed on the period matrices.

295.6 Actual transitions and their exact critical-value support

Within an interval of admissible directions containing no argument of a nonzero action, the algebraic branches in (SPG.10) continue continuously with the direction. Their distinct endpoint labels cannot change there. Hence E_θ is constant in that interval. Equivalently one can rotate the Laplace contour inside the action-free wedge; its large arc vanishes by the same exponential estimate and its small arc vanishes by $\beta > 0$. The two analytic gauges then agree on their common domain.

For two sufficiently close directions on opposite sides of an action direction θ_0 , let the superscripts $+$ and $-$ denote increasing and decreasing arguments on the same logarithmic cover. Their actual transition matrix is

$$\mathfrak{S}_{\theta_0} = E_{\theta_0+} E_{\theta_0-}^{-1}, \quad P_{\theta_0+} = \mathfrak{S}_{\theta_0} P_{\theta_0-}, \quad \mathfrak{S}_{\theta_0} \in \text{GL}_d(\mathbb{Z}). \quad (\text{SPG.27})$$

Here the matrices are the actual endpoint-incidence matrices, and (SPG.24) proves the equality. No matrix is inferred merely from a characteristic polynomial. The full data needed to evaluate every entry are the original inverse-polynomial paths and their endpoint labels in (SPG.10)–(SPG.23).

There is exact control on which original critical blocks can mix:

$$\begin{cases} (\mathfrak{S}_{\theta_0})_{\rho\sigma} = 0, & \rho \neq \sigma, \quad c_\rho - c_\sigma \notin e^{i\theta_0}\mathbb{R}_{>0}, \\ (\mathfrak{S}_{\theta_0})_{\rho\sigma} = \delta_{\rho\sigma}I, & c_\rho = c_\sigma. \end{cases} \quad (\text{SPG.28})$$

In the second line the rectangular off-diagonal block is zero and the square diagonal block is the identity of its original size. To prove the assertion without assuming a Stokes theorem, conjugate by the fixed block matrix W_{loc} , and write $\mathfrak{S}^\chi = W_{\text{loc}}^{-1}\mathfrak{S}W_{\text{loc}}$. If $\mathcal{D}_0(u) = \text{diag}_{\rho,a}(e^{c_\rho/u}d_{\rho,a}u^{\beta_{\rho,a}})$, then

$$\mathfrak{S}^\chi - I = \mathcal{D}_0(u)(T_{\theta_0+}T_{\theta_0-}^{-1} - I)\mathcal{D}_0(u)^{-1}. \quad (\text{SPG.29})$$

The middle matrix is smaller than every fixed power of $|u|$ on every smaller closed subsector of the overlap, because both gauges have the same full asymptotic expansion and their inverses tend to T_0^{-1} . If its corresponding constant transition entry were nonzero, (SPG.29) would therefore require $u^{\beta_\sigma - \beta_\rho}e^{(c_\sigma - c_\rho)/u}$ to be smaller than every power on the whole overlap. Equal constants make that impossible. For unequal constants it requires $\text{Re}((c_\sigma - c_\rho)e^{-i\varphi}) < 0$ throughout the open interval of overlap arguments φ . The matrices $E_{\theta_0\pm}$ do not depend on how close the two directions are taken within their adjacent intervals. Letting them approach θ_0 , that interval exhausts $(\theta_0 - \pi/2, \theta_0 + \pi/2)$. The only nonzero complex number with the required strict negative real part on this whole interval is a negative real multiple of $e^{i\theta_0}$. Thus $c_\rho - c_\sigma \in e^{i\theta_0}\mathbb{R}_{>0}$. This proves the entrywise assertion in the character basis and hence the stated block assertion, since W_{loc} acts within each original root block. It follows in particular that the transition is unipotent after ordering the critical-value blocks along the specified action direction. Equal-value blocks remain intact.

295.7 The exact original source and ordered tensors

Keep the complete original unit matrix $U_h = M_{[g/h]}$ in the original monomial basis. On the actual theta-jet coordinate $q = U_h p$, the analytic comparison is

$$P_\theta^\theta = P_\theta U_h^{-1} = E_\theta \Pi_h U_h^{-1}, \quad P_{\theta_0+}^\theta = \mathfrak{S}_{\theta_0} P_{\theta_0-}^\theta. \quad (\text{SPG.30})$$

The two uses of the letter θ in P_θ^θ mean respectively the ray direction (subscript) and the inherited theta source (superscript). The matrices U_h and U_h^{-1} retain all Taylor coefficients and their original nonzero constants. Every equality follows by multiplying the proved analytic equality on the right by this same original inverse unit. The arithmetic operator on the sectorial period coordinates is exactly $P_\theta A_h P_\theta^{-1}$, or $P_\theta^\theta A_h (P_\theta^\theta)^{-1}$ on theta jets; the two are equal because A_h commutes with its full unit. No intertwining with Frobenius is inserted here.

For the original source Gram G_θ of FPB.47, the exact sectorial period form is

$$P_\theta^* P_\theta = \Pi_h^* E_\theta^* E_\theta \Pi_h, \quad (G_\theta)^{-1} P_\theta^* P_\theta = (G_\theta)^{-1} \Pi_h^* E_\theta^* E_\theta \Pi_h. \quad (\text{SPG.31})$$

This is equality in the unchanged original coefficient coordinates. On theta-jet coordinates the source form is $U_h^{-*} G_\theta U_h^{-1}$, and the period form is $U_h^{-*} \Pi_h^* E_\theta^* E_\theta \Pi_h U_h^{-1}$. These follow by substituting $p = U_h^{-1} q$ in the quadratic forms. In particular the generally nonidentity integer Gram $E_\theta^* E_\theta$ has not been omitted from the comparison.

All the constructions tensor in the original factor order and common parameter u :

$$T_{\theta,k} = T_\theta^{\otimes k}, \quad P_{\theta,k} = P_\theta^{\otimes k}, \quad E_{\theta,k} = E_\theta^{\otimes k}, \quad P_{\theta,k} = E_{\theta,k} \Pi_h^{\otimes k}, \quad \mathfrak{S}_{\theta_0,k} = \mathfrak{S}_{\theta_0}^{\otimes k}. \quad (\text{SPG.32})$$

The connection is the original tensor sum because differentiating the Kronecker product differentiates one retained factor at a time. The form is $ds_1 \wedge \cdots \wedge ds_k$, with no permutation and hence no extra sign. In the local block every ordered tuple has exponential constant $\sum_i c_{\rho_i}$ and fractional exponent $\sum_i \beta_{\rho_i, a_i}$. The complete theta factor is $U_h^{\otimes k}$, and the original source Gram is $G_\theta^{\otimes k}$. Every tuple remains, including all tuples sharing a critical sum. Formula (SPG.32) is an exact amplified analytic comparison, not a replacement by the single-primary polynomial or by formal character counts.

295.8 The signed path of every actual transition entry

For the two actual adjacent trees of (SPG.23), keep the vertex labels $0, \dots, d$, the original row orders, and all edge orientations. The row indexed by a new edge f of E_+ is the difference of the vertex at its original head $h_+(f)$ and its original tail $t_+(f)$. The subscript $+$ here denotes that tree; it does not change the given orientation. Root the old tree E_- at vertex zero. For an old edge e , let $p_-(e), c_-(e)$ be the parent and child vertices, and let ϵ_e equal 1 if its original orientation points from parent to child and -1 otherwise. Let $\mathcal{P}_-(v)$ be the unique old-tree path from zero to v , with $\mathcal{P}_-(0) = \emptyset$. Then every entry of the original transition in (SPG.27) is exactly

$$(\mathfrak{S}_{\theta_0})_{f,e} = \epsilon_e (\mathbf{1}_{e \in \mathcal{P}_-(h_+(f))} - \mathbf{1}_{e \in \mathcal{P}_-(t_+(f))}). \quad (\text{SPG.33})$$

Indeed the old inverse incidence matrix is $(E_-^{-1})_{v,e} = \epsilon_e \mathbf{1}_{e \in \mathcal{P}_-(v)}$: the sum of original edge differences along a root path telescopes to its endpoint value, with value zero fixed at the root. Subtract the two endpoint rows of that matrix to obtain (SPG.33). The intersection of the two root paths is their common initial part; those terms cancel exactly as integers. Their symmetric difference is the unique path from $t_+(f)$ to $h_+(f)$. Its coefficient is 1 when traversing an old edge in its original orientation and -1 when traversing it against that orientation. Thus

$$(\mathfrak{S}_{\theta_0})_{f,e} \in \{-1, 0, 1\}, \quad \sum_e |(\mathfrak{S}_{\theta_0})_{f,e}|^2 = \text{dist}_{\mathcal{T}_-}(t_+(f), h_+(f)). \quad (\text{SPG.34})$$

All distances and signs concern the actual endpoint trees of the original polynomial. Combining (SPG.33) with (SPG.28) proves that an old edge can occur on this path only in its permitted critical-value block, except for the identity entry. In particular no unspecified integer coefficient remains in the transition formula.

Let $L_{+-} = \sum_f \text{dist}_{\mathcal{T}_-}(t_+(f), h_+(f))$, and define L_{-+} by exchanging the two trees. The exact Frobenius norm, the operator bounds, and the two directions of the original period comparison are

$$\begin{aligned} \|\mathfrak{S}_{\theta_0}\|_{\mathbb{F}}^2 &= L_{+-}, & \|\mathfrak{S}_{\theta_0}^{-1}\|_{\mathbb{F}}^2 &= L_{-+}, & \|\mathfrak{S}_{\theta_0}\|_2 &\leq \sqrt{L_{+-}} \leq d, & \|\mathfrak{S}_{\theta_0}^{-1}\|_2 &\leq \sqrt{L_{-+}} \leq d, \\ L_{-+}^{-1} \|P_{-p}\|_2^2 &\leq \|P_{+p}\|_2^2 \leq L_{+-} \|P_{-p}\|_2^2. \end{aligned} \quad (\text{SPG.35})$$

The operator norm of a matrix is bounded by its Frobenius norm, as follows by applying Cauchy-Schwarz to each row and summing. Every path has at most d edges and there are d rows, giving the displayed upper bounds. Apply the same estimate to the inverse and use $P_+ = \mathfrak{S}P_-$ to prove both period inequalities. These bounds retain the actual Euclidean period norms.

295.9 The mixed split-support comparison and its complete fibres

Apply the oriented-tree calculation TSI.1–12 to this actual E_{θ} and its original root zero. For precision, its coefficient semiring is the original $\mathbb{G} = G(\mathbb{Z}) = \{\tau\} \sqcup \mathbb{Z}^{\bullet}$. Here τ is the absent-support zero, $0^{\bullet}$ is the supported zero, and supported addition and multiplication retain the integer operations. The amplitude together with the support is injective. No finite residue field replaces this integer target. Lift each nonzero entry of $E = E_{\theta}$ and $F = E^{-1}$ to its supported integer, and lift a zero entry to τ , obtaining the exact semiring matrices $\widehat{E}, \widehat{F}$. Their products are

$$P = \widehat{E}\widehat{F}, \quad Q = \widehat{F}\widehat{E}, \quad (\text{SPG.36})$$

with $1^{\bullet}$ on the diagonal, $0^{\bullet}$ at strict ancestor positions, and τ elsewhere. On the edge module P uses edge ancestry; on the nonroot vertex module Q uses vertex ancestry. The proof is exact cancellation of two supported opposite integers at an ancestor, one supported unit on the diagonal, and no contribution off those positions. The absence of a contribution and the cancellation of contributions are different elements of $\mathbb{G}$.

Both products preserve every amplitude and close a support mask under descendants. Write this action generically as H , on its stated edge or nonroot vertex module, and write $\text{Desc}(A)$ for the descendants of a mask A , including its own elements. Then

$$\text{amp}(Hx) = \text{amp}(x), \quad \text{supp}(Hx) = \text{Desc}(\text{supp } x), \quad H^2 = H. \quad (\text{SPG.37})$$

The entire fibre of this map, rather than only its image, can be constructed. Let z have descendant-closed mask B , and let $a_i = \text{amp}(z_i)$, using $a_i = 0$ outside B . Denote by $\text{Min}(B)$ the vertices or edges in B with no strict ancestor in B , and put $\text{NZ}(a) = \{i : a_i \neq 0\}$. The exact parametrization is

$$H^{-1}(z) = \{x_A : \text{Min}(B) \cup \text{NZ}(a) \subseteq A \subseteq B\},$$

$$(x_A)_i = \begin{cases} a_i^\bullet, & i \in A, \\ \tau, & i \notin A, \end{cases} \quad |H^{-1}(z)| = 2^{|B \setminus (\text{Min}(B) \cup \text{NZ}(a))|}. \quad (\text{SPG.38})$$

For necessity, (SPG.37) requires $A \subseteq B$, and every minimal element of B must itself be in A : no other element of B can generate it by descendant closure. Every nonzero amplitude must also have been supported at the input. For sufficiency, a finite descendant-closed set is precisely the descendant closure of its minimal elements. Thus the displayed inclusions give $\text{Desc}(A) = B$, and the definition of x_A gives all the required amplitudes. The injective amplitude/support observation proves $Hx_A = z$ and proves that different masks give different fibre elements. This proves the parametrization and the count. It includes $B = \emptyset$, whose fibre is the single all- τ vector. For a target whose mask is not descendant closed, (SPG.37) proves that its fibre is empty. Thus both image and nonimage targets have their complete fibres specified.

There are exact mutually inverse semimodule maps

$$\widehat{E} : \text{im } Q \xrightarrow{\sim} \text{im } P, \quad \widehat{F} : \text{im } P \xrightarrow{\sim} \text{im } Q. \quad (\text{SPG.39})$$

For example, $x = Qx$ implies $P\widehat{E}x = \widehat{E}Qx = \widehat{E}x$ and $\widehat{F}\widehat{E}x = Qx = x$; the other direction is the same associativity identity. The image masks are exactly the descendant-closed masks. Formula (SPG.38) records all the fibres of the maps into those images, so no equality on the images discards the distinct missing-support inputs that preceded closure.

There is a second support comparison for the actual Stokes matrix. Retain $\widehat{E}_+ \widehat{F}_-$ as its specified two-stage split composite, and distinguish it from the entrywise minimal lift $\widehat{\mathfrak{S}}$. For a new edge f and old edge e , the complete value of that composite is

$$(\widehat{E}_+ \widehat{F}_-)_{f,e} = \begin{cases} (\mathfrak{S}_{f,e})^\bullet, & e \in \mathcal{P}_-(h_+(f)) \triangle \mathcal{P}_-(t_+(f)), \\ 0^\bullet, & e \in \mathcal{P}_-(h_+(f)) \cap \mathcal{P}_-(t_+(f)), \\ \tau, & \text{otherwise.} \end{cases} \quad (\text{SPG.40})$$

The first case has exactly one nonzero product. The second has two opposite supported products. The last has no product. This proves each entry, retaining the common initial path which cancelled in the integer formula (SPG.33). In particular the root-path stages and their support masks remain explicit even where the final integer amplitude is zero. Neither this comparison nor (SPG.39) asserts a ring homomorphism: these are semimodule maps with the specified split coefficients, and ordinary linear maps under the amplitude observation.

On an original independent coefficient face, apply these maps to its coefficient module while retaining that face's original label. On a separate original source-leg mask, retain the identity on the label. The scalar map $\mathbb{Z} \rightarrow R$ to any original coefficient ring gives $G(\mathbb{Z}) \rightarrow G(R)$, so the same displayed matrix actions are defined on each such coefficient module. This map sends supported zero to supported zero and τ to τ ; it commutes with all the displayed sums and products. Any proper-source kernel carried as a separate labelled summand is retained under this coefficient action. No quotient of the original theta complex or removal of an original source leg occurs in the support comparison.

295.10 Every original source metric factor under the transition

In the actual rooted tree define $\ell_v = |\mathcal{P}(v)|$, $s_\theta = \sum_{v=1}^d \ell_v$, and $t_\theta = 2d - \deg(0)$. The exact matrix $L_\theta = E_\theta^* E_\theta$ is the original Dirichlet tree Laplacian, including root-edge contributions in its diagonal. Its inverse has entries

$$(L_\theta^{-1})_{vw} = |\mathcal{P}(v) \cap \mathcal{P}(w)|, \quad \frac{1}{s_\theta} I \leq L_\theta \leq t_\theta I, \quad s_\theta \leq \frac{d(d+1)}{2}. \quad (\text{SPG.41})$$

The inverse formula follows from $F = E^{-1}$ and its signed path entries: each edge in the intersection contributes its squared orientation sign, hence one. The trace of this inverse is s_θ ; the trace of L_θ is t_θ . Both are positive definite, so their largest eigenvalues are at most their traces. The bound on s_θ counts equal ancestor pairs and at most one ordered ancestor pair for each unordered pair of distinct vertices. These arguments prove all the inequalities with the actual tree fixed.

For every original coefficient column p , with all its jets, the source and both period norms therefore obey exactly

$$\begin{aligned} \|\mathcal{T}_h(p)\|_{L^2(dx)}^2 &= p^* G_\theta p, & \|P_\theta p\|_2^2 &= p^* \Pi_h^* L_\theta \Pi_h p, \\ \frac{1}{s_\theta} \|\Pi_h p\|_2^2 &\leq \|P_\theta p\|_2^2 \leq t_\theta \|\Pi_h p\|_2^2. \end{aligned} \quad (\text{SPG.42})$$

Here p denotes its unchanged degree-below- d polynomial representative in $\mathcal{T}_h(p)$; thus the first form is precisely the original section norm, not a new quotient norm. The middle identity is the actual period equality, and the two bounds are substitutions into (SPG.41).

There is also an exact metric on the sectorial period coordinates which carries the same original source norm. Put

$$H_\theta^{\text{src}}(u) = E_\theta^{-*} \Pi_h(u)^{-*} G_\theta \Pi_h(u)^{-1} E_\theta^{-1}. \quad (\text{SPG.43})$$

It is positive definite since it is the pullback of the positive original G_θ by an invertible map. On theta-jet coordinates $q = U_h p$, both the source form and the period matrix contain the same retained U_h^{-1} . Explicitly,

$$\begin{aligned} H_\theta^{\text{src}} &= (P_\theta U_h^{-1})^{-*} (U_h^{-*} G_\theta U_h^{-1}) (P_\theta U_h^{-1})^{-1}, \\ (P_\theta p)^* H_\theta^{\text{src}} (P_\theta p) &= (P_\theta U_h^{-1} q)^* H_\theta^{\text{src}} (P_\theta U_h^{-1} q) = p^* G_\theta p = q^* U_h^{-*} G_\theta U_h^{-1} q. \end{aligned} \quad (\text{SPG.44})$$

These identities result by multiplying the displayed matrices; the inverse-unit factors cancel only through this exact coordinate square, after being retained on both sides. Under the actual transition,

$$\mathfrak{S}_{\theta_0}^* H_+^{\text{src}} \mathfrak{S}_{\theta_0} = H_-^{\text{src}}. \quad (\text{SPG.45})$$

Indeed $P_+ = \mathfrak{S} P_-$, and substitution into $H_\pm^{\text{src}} = P_\pm^{-*} G_\theta P_\pm^{-1}$ proves the identity. This is an isometry between the two explicitly transported source forms. The Euclidean norms in (SPG.35) and (SPG.42) remain their original Euclidean norms and are not identified with these source forms.

For the ordered k -fold product all these form identities are Kronecker products of the displayed matrices. In particular

$$\begin{aligned} s_\theta^{-k} \|\Pi_h^{\otimes k} p\|_2^2 &\leq \|P_\theta^{\otimes k} p\|_2^2 \leq t_\theta^k \|\Pi_h^{\otimes k} p\|_2^2, \\ (\mathfrak{S}^{\otimes k})^* (H_+^{\text{src}})^{\otimes k} \mathfrak{S}^{\otimes k} &= (H_-^{\text{src}})^{\otimes k}, \quad \|\mathfrak{S}^{\otimes k}\|_2 = \|\mathfrak{S}\|_2^k. \end{aligned} \quad (\text{SPG.46})$$

For the inequalities, diagonalize the positive matrices in (SPG.41); each eigenvalue of a Kronecker power is a product of the original eigenvalues, proving exactly the stated bounds. The norm equality follows by diagonalizing $\mathfrak{S}^* \mathfrak{S}$ in the same way. The constants therefore have their proved multiplicative growth under this particular tensor operation. No uniform-in- k error bound or arithmetic purity is substituted for that calculation.

296 Recursive propagation of the original signed metric bounds

This proof combines the earlier full-relative-spectrum estimate AW16, the exact trace norm and actual-overlap estimates of SP, the supplied centered Hilbert–Schmidt estimate, and the all-degree angle estimate EP17–EP25 on their common original source. Each of those complete proofs is retained in this edition. The construction below carries their pointwise minimum through both the original volume and the exact nonlinear endpoint coordinate, without changing either endpoint form.

296.1 The original source and all four quotient volumes

Fix the original packet h , tensor order $k \geq 1$, monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, and

$$H = \mathcal{P}_{2q}, \quad S = k/2 + iy, \quad v(y) = (1, S, \dots, S^{2q}). \quad (\text{RMT1})$$

The coordinates are the original ascending monomial coordinates. The original reference and arithmetic measures are

$$d\mu_0 = r_{1/4}^{*k}(y) dy, \quad d\mu_1 = w_h^{*k}(y) dy, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}. \quad (\text{RMT2})$$

All their original mass and Taylor-unit factors remain. In particular the exact reference convolution identity is

$$r_{1/4}^{*k} = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{RMT3})$$

Thus these are the same measures in AW and SP. Every argument below also applies to the actual finite-circle comparison when one of the measures is its specified positive atomic measure on this same line and its retained degree- $2q$ Gram is positive definite. That positivity is already proved by the original sampled-source bounds. This statement does not assign moment structure to an arbitrary positive coefficient matrix.

For $0 \leq x \leq 1$ let

$$M_x = \int v(y)^* v(y) d((1-x)\mu_0 + x\mu_1)(y), \quad \dot{M} = M_1 - M_0, \quad C_x = M_x^{-1} \dot{M}. \quad (\text{RMT4})$$

The two positive endpoint forms make M_x positive definite throughout the closed interval. For every original cutoff N , let $I_N : \mathcal{P}_N \rightarrow H$ be coefficient inclusion and $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ , followed by inclusion. Negative relation degree means the zero space. Retain the orthogonal projections

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{RMT5})$$

AW's relation matrix has target $\mathcal{P}_N$, so its exact relation to this one is $B_N = I_N B_N^{\text{AW}}$. Equation (RMT5), after that substitution, is precisely AW4 and SP2. The quotient map remains $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[S]/(\chi)$, and its canonical positive quotient Gram is $G_N(x)$. Put $V_N(x) = \det G_N(x)$ and retain the four signs:

$$\mathcal{B}(x) = \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \quad U_x = Q_{2q-1} + Q_{2q} - Q_q, \quad W_x = P_{2q-1} - P_{q-1} + P_{2q} - P_q. \quad (\text{RMT6})$$

The relation-first determinant-line formula and differentiation of each of these four determinants give, without commuting their operators,

$$\mathcal{B}'(x) = \text{Tr}((U_x - W_x)C_x). \quad (\text{RMT7})$$

This is the same actual integrand as AW6 and SP7.

296.2 The simultaneous bounds on that integrand

Let $b_1 \leq \dots \leq b_{2q+1}$ be every eigenvalue, with multiplicity, of $D = M_0^{-1}M_1$ in the original M_0 metric. They are positive. Retain

$$c_j(x) = \frac{b_j - 1}{1 - x + xb_j}, \quad d_x = c_{2q+1}(x) - c_1(x), \quad \kappa = \frac{b_{2q+1}}{b_1}. \quad (\text{RMT8})$$

These are the ordered eigenvalues of C_x . Indeed $C_x = ((1-x)I + xD)^{-1}(D - I)$, and differentiation with respect to b_j gives the positive quantity $(1 - x + xb_j)^{-2}$. In particular $\int_0^1 d_x dx = \log \kappa$.

Set $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two ranks are $q + 1$ in dimension $2q + 1$, so $s_x \geq 1$. All traces and trace norms below are in the same original M_x metric. Define the two calculated functions

$$\begin{aligned} E_q(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ O_q(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}. \end{aligned} \quad (\text{RMT9})$$

The sum is empty when $q = 1$. The complete nested-flag calculation gives both U_x, W_x the spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. Decomposing each as the sum of its spectral projections of ranks $q + 1$ and $q - 1$, and applying the ordered projection-trace inequalities, gives $|\mathcal{B}'| \leq E_q$, exactly as in AW15–AW16. For clarity the subtraction of the two bounds cancels the common c_{q+1} only after both sums have been written; it leaves every term displayed in E_q .

Both positive operators dominate the original orthogonal projection onto their common range. Subtracting that same projection from both leaves positive operators with traces $2q - s_x$, whence $\|U_x - W_x\|_1 \leq 4q - 2s_x$. Also their computed spectra give

$$\text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{RMT10})$$

Cauchy–Schwarz on all $2q + 1$ eigenvalues of $U_x - W_x$ gives the second trace-norm bound in (RMT9). Since $\text{Tr}(U_x - W_x) = 0$, subtract $(c_1 + c_{2q+1})I/2$ from C_x in (RMT7). In an M_x -orthonormal eigenbasis of $U_x - W_x$, the absolute value of each diagonal entry of this centered C_x is at most $d_x/2$. Multiplying by the corresponding absolute eigenvalues and summing gives the retained exact trace-norm bound

$$|\mathcal{B}'| \leq T_q(x) := \frac{d_x}{2} \|U_x - W_x\|_1 \leq O_q(x). \quad (\text{RMT10a})$$

The second inequality follows from the two trace-norm estimates just proved. Thus neither of those majorants replaces the original trace norm.

The user's additional supplied projector calculation gives another bound on this same integrand. Define

$$\Sigma_x = \text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{2q+1}, \quad S_q(x) = \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}. \quad (\text{RMT10b})$$

The nonnegative number Σ_x is the squared Hilbert–Schmidt norm of $C_x - \text{Tr}(C_x)I/(2q+1)$ in the original M_x Hilbert space. The trace of $U_x - W_x$ is zero, so Hilbert–Schmidt Cauchy–Schwarz gives $|\mathcal{B}'| \leq S_q$. Equivalently one may apply the isometry $M_x^{1/2} : (H, M_x) \rightarrow (H, I)$ simultaneously to both operators; conjugation preserves their traces and products and proves the same inequality. This is a calculation with the original metric, not a change of either endpoint source form.

The original crossed quotient pairs are $(q-1, 2q)$ and $(q, 2q-1)$. Retain their full canonical minimum-section angle lists $(\theta_1, \dots, \theta_q)$ and $(\varphi_1, \dots, \varphi_q)$. In the second list, ordered with a smallest angle first, the original ambient dimension forces $\varphi_1 = 0$. The count is proved in EP17–EP25. For the following scalar sums put $a = 2q - 1$ and define the explicitly indexed list

$$\lambda_i = -\log \cos^2 \theta_i \quad (1 \leq i \leq q), \quad \lambda_{q+j-1} = -\log \cos^2 \varphi_j \quad (2 \leq j \leq q).$$

Then $\mathcal{B} = -\log \cos^2 \varphi_1 + \sum_{\nu=1}^a \lambda_\nu = 0 + \sum_{\nu=1}^a \lambda_\nu$. Every additional zero remains in the λ -list, while the designated geometric entry φ_1 is still displayed in the full second list. The exact projection-difference spectra give

$$|\mathcal{B}'(x)| \leq d_x \left(\sin \varphi_1 + \sum_{\nu=1}^a \sqrt{1 - e^{-\lambda_\nu(x)}} \right) \leq a \sqrt{1 - e^{-\mathcal{B}(x)/a}} d_x. \quad (\text{RMT11})$$

The second inequality is concavity of $z \mapsto \sqrt{1 - e^{-z}}$ on $[0, \infty)$, applied to the full a -term list, including its zeros, together with the displayed $\sin \varphi_1 = 0$. Both q -entry lists remain the geometric data; the known zero contribution is kept explicitly in these scalar identities.

The needed strict positivity of $\mathcal{B}$ follows on the actual measures rather than being an additional assumption. If $\mathcal{B} = 0$, both nested quotient pairs have equal minimum spaces. To see every implication, their canonical minimum sections satisfy $G_i - G_j = (R_i - R_j)^* M_x (R_i - R_j) \succeq 0$. Thus each determinant ratio in the crossed pairs is at least one. A zero sum of their nonnegative logarithms makes both ratios one. All eigenvalues of $G_j^{-1/2} G_i G_j^{-1/2}$ are at least one, so determinant one makes every eigenvalue one and $G_i = G_j$. The displayed squared norm then gives $R_i = R_j$ as maps to the same original source H . In the first pair R_{q-1} is the literal degree-less-than- q remainder section. Its equality with R_{2q} puts the constant polynomial 1 in the latter minimum space, which is orthogonal to $\ker J_{2q} = \chi \mathcal{P}_q$. For $\chi(S) = \sum a_j S^j$, put $\chi^{\#k}(S) = \sum \bar{a}_j (k - S)^j$. This belongs to $\mathcal{P}_q$ and equals $\overline{\chi(S)}$ on $S = k/2 + iy$. The asserted orthogonality would therefore give

$$0 = \int \chi(S) \chi^{\#k}(S) d\mu_x = \int |\chi(S)|^2 d\mu_x, \quad (\text{RMT12})$$

contradicting the positive Gram of the nonzero polynomial χ . This proof also covers the specified positive atomic comparison. Thus $\mathcal{B}(x) > 0$ on the entire closed interval.

296.3 Carrying the refined bound through both endpoint coordinates

Define the positive function

$$z_x = a \sqrt{1 - e^{-\mathcal{B}(x)/a}} > 0.$$

With every preceding control retained, set

$$\begin{aligned} A_q(x) &= z_x d_x, \\ J_q &= \int_0^1 \min\{E_q(x), T_q(x), O_q(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\left\{d_x, \frac{E_q(x)}{z_x}, \frac{T_q(x)}{z_x}, \frac{O_q(x)}{z_x}, \frac{S_q(x)}{z_x}\right\} dx. \end{aligned} \quad (\text{RMT13})$$

These are actual quantities of the original finite source path. The projections are continuous matrix functions of x . Explicitly $s_x = 2(q+1) - \text{rank}[U_x W_x]$. A rank superlevel set is a finite union of sets where a specified minor is nonzero, hence is Borel. This proves measurability of s_x and of every displayed integrand. Equation (RMT12) and compactness give a positive lower bound for $\mathcal{B}$ on this fixed path. Hence all denominators in I_q are nonzero, and the integrands are finite and bounded. This is a fixed-source statement and asserts no bound uniform in the tensor order.

All five preceding estimates bound the same $|\mathcal{B}'|$. Pointwise comparison followed by the triangle inequality for the integral therefore proves

$$\begin{aligned} |\mathcal{B}(1) - \mathcal{B}(0)| &\leq J_q, \\ J_q &\leq \min\left\{\mathcal{L}_q(D), \int_0^1 T_q(x) dx, \int_0^1 O_q(x) dx, \int_0^1 S_q(x) dx, \int_0^1 A_q(x) dx\right\}. \end{aligned} \quad (\text{RMT14})$$

where the earlier AW13 quantity is retained in full:

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \kappa. \quad (\text{RMT15})$$

Indeed integration of each c_j gives $\log b_j$, which proves the first entry of the minimum in (RMT14).

For $B > 0$ let

$$\begin{aligned} \mathcal{H}_a(B) &= 2 \operatorname{arcosh}(e^{B/(2a)}), & \mathcal{H}'_a(B) &= \frac{1}{a\sqrt{1 - e^{-B/a}}}, \\ \mathcal{H}_a^{-1}(z) &= 2a \log \cosh(z/2) \quad (z > 0). \end{aligned} \quad (\text{RMT16})$$

The derivative follows by differentiating the positive real inverse branch of cosh; substitution proves both inverse composites. Applying this exact derivative to each of the five bounds on $\mathcal{B}'$ gives

$$|\mathcal{H}_a(\mathcal{B}(1)) - \mathcal{H}_a(\mathcal{B}(0))| \leq I_q \leq \int_0^1 d_x dx = \log \kappa. \quad (\text{RMT17})$$

Thus the exact trace norm, supplied variance estimate, overlap and full-relative-spectrum information propagate into the nonlinear bound, not only into its linear consequence.

Writing $H_0 = \mathcal{H}_a(\mathcal{B}(0))$, the inverse in (RMT16), extended continuously by value zero at zero, and the additive estimate in (RMT14) give the simultaneous interval

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, \mathcal{B}(0) - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right) \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ \mathcal{B}(0) + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right) \right\}. \end{aligned} \quad (\text{RMT18})$$

The actual $\mathcal{B}(1)$ is strictly positive although the lower bound can equal zero. If $M_1 = cM_0$ with $c > 0$, all the c_j coincide, so $d_x, E_q, T_q, O_q, S_q, A_q, I_q, J_q$ vanish. Each original quotient determinant is multiplied by the same power q of $1 - x + xc$; the four signs then give $\mathcal{B}(1) = \mathcal{B}(0)$, as (RMT18) also states. This includes $q = 1$, for which the ordered sums above are empty and $a = 1$.

Every projection and metric in this argument uses the original J_N , B_N , coefficient basis and source measure. The arithmetic class map $\sigma_h^{\otimes k} \eta_{h,k} J_N$, its complete Taylor unit, and the original theta primitives of differences of minimum sections are unchanged. The proof refines numerical bounds on those exact maps; it introduces no identification between their source kernels or between distinct support labels. The earlier complete AW, SP and EP proofs, together with the explicit correspondences (RMT3)–(RMT7), provide the full input proofs used in this propagation.

296.4 Carrying the constructed signed residual into the endpoint

The complete supplied projector correspondence and its explicit finite approximation are proved in Section 297, (ISM14)–(ISM37). Their notation has the exact identities

$$X_x = C_x, \quad \mathcal{A}_x = U_x - W_x, \quad \Pi_N = P_N - Q_N.$$

For completeness retain the actual finite construction here. For each integer $L \geq 0$, put

$$\begin{aligned} B_x^\circ &= (1-x)I + xD, & \alpha_x &= \frac{2(1-x) + x(b_1 + b_{2q+1})}{2}, \\ H_x^\circ &= I - B_x^\circ/\alpha_x, & C_x^{[L]} &= \alpha_x^{-1} \sum_{r=0}^L (H_x^\circ)^r (D - I), \\ R_x^{[L]} &= (H_x^\circ)^{L+1} C_x, & E_x^{[L]} &= R_x^{[L]} - \frac{\operatorname{Tr} R_x^{[L]}}{2q+1} I. \end{aligned} \quad (\text{RMT19})$$

The symbols $B_x^\circ, H_x^\circ, \alpha_x$ label operators and the scalar in this calculation; they leave every original relation matrix B_N , source metric M_x , and coordinate in (RMT1) in place. Multiplication of the finite geometric sum by $I - H_x^\circ$ gives $C_x - C_x^{[L]} = R_x^{[L]}$ exactly. Every factor is a real function of the same M_0 -self-adjoint D , hence is self-adjoint in $M_x = M_0 B_x^\circ$ as well. Since the trace of $U_x - W_x$ vanishes, (RMT7) becomes

$$\mathcal{B}'(x) = \text{Tr}(C_x^{[L]}(U_x - W_x)) + \text{Tr}(E_x^{[L]}(U_x - W_x)).$$

Apply Hilbert–Schmidt Cauchy–Schwarz in the original M_x metric to the last term and integrate. Define the specified real numbers

$$\begin{aligned} j_L &= \int_0^1 \text{Tr}(C_x^{[L]}(U_x - W_x)) dx, \\ e_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) \text{Tr}((U_x - W_x)^2)} dx, \\ \vartheta &= \frac{b_{2q+1} - b_1}{b_{2q+1} + b_1}, \quad K_D = \sum_{j=1}^{2q+1} \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \end{aligned} \tag{RMT20}$$

These are J_L, e_L, ϑ, K_D of (ISM34)–(ISM37) with j_L, e_L used to avoid collision with J_q in (RMT13). The eigenvalues of H_x° have absolute value at most $\vartheta < 1$; those of C_x are the full $c_j(x)$. Thus $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$, because subtracting the trace mean subtracts a nonnegative square from the full sum of squared residual eigenvalues. The two positive operators U_x, W_x dominate their range projections. Those ranges intersect in dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$, by positivity of the trace product and the squared-length formula for the product of the range projections. Equation (RMT10) gives $\text{Tr}((U_x - W_x)^2) \leq 8q - 6$. The signed estimate and its computed residual are therefore

$$\begin{aligned} j_L - e_L &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_L + e_L, \\ 0 \leq e_L &\leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \end{aligned} \tag{RMT21}$$

Every integral is a continuous finite matrix expression. This proves convergence for the fixed actual two source forms; no uniformity in k is assumed or inferred.

Let $B_0 = \mathcal{B}(0)$ and retain H_0 from (RMT18). Intersecting the proved intervals, for every $L \geq 0$, gives the propagated endpoint bounds

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, B_0 - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right), B_0 + j_L - e_L \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ B_0 + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right), B_0 + j_L + e_L \right\}. \end{aligned} \tag{RMT22}$$

Indeed each lower entry has separately been proved no larger than the actual endpoint, and each upper entry no smaller. Taking their maximum and minimum therefore preserves both inequalities and yields a nonempty interval containing that endpoint. A numerically integrated center must retain its own integration error before use; (RMT20)–(RMT22) state the exact integrals. The scalar source case has $H_x^\circ = E_x^{[L]} = 0$ and $j_L = e_L = 0$, so it also satisfies (RMT22) exactly.

297 Source enclosures and convergent signed projector control

This chapter incorporates the complete written derivations in the user-supplied continuation following merged PR29. The canonical secant and quotient-variation identities already occur in that PR. The four-projector pairing is the same pairing as AW and SP, with the exact coordinate correspondence proved below. The source sandwich, restriction-loss expression for its Hilbert–Schmidt size, and centered signed enclosure are retained in their full forms. We additionally construct the approximating operator in that enclosure and prove an explicit convergent residual bound. The supplied Lean draft is preserved separately as uncompiled; the mathematical scope here is the written proof.

297.1 The unchanged polynomial and arithmetic source

Fix the actual full packet h , tensor order $k \geq 1$, and its monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Put $E = \mathbb{C}[S]/(\chi)$, $H = \mathcal{P}_{2q}$, and $n = 2q + 1$. All coordinates are the original ascending monomial coordinates. The original analytic source and observation are

$$\begin{aligned} g &= 2\xi, & \mathcal{M}F_h &= g/h, & \mathcal{V}P &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, \\ J^{(k)}\mathcal{V}P &= \eta_{h,k}\pi_\chi P, & \eta_{h,k}[P] &= v_h^{\otimes k}P(A_h^{(k)})1, & v_h &= j_h(g/h), & A_h^{(k)} &= \sum_{i=1}^k A_{h,i}. \end{aligned} \quad (\text{ISM1})$$

Here A_h is multiplication by the original coordinate on $E_h = \mathbb{C}[s]/(h)$. Thus the domain of $\eta_{h,k}$ is E , its codomain is $E_h^{\otimes k}$, and its full unit is retained. The minimal polynomial definition of χ on the cyclic vector 1 proves that $[P] \mapsto P(A_h^{(k)})1$ is injective; multiplication by the invertible $v_h^{\otimes k}$ preserves injectivity.

For $N \in \{q-1, q, 2q-1, 2q\}$, let $I_N : \mathcal{P}_N \hookrightarrow H$ be coefficient inclusion, $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ followed by inclusion, and $C : E \rightarrow H$ the degree-less-than- q remainder lift in the retained quotient basis. The relation space is zero when $N = q-1$. Polynomial division by the monic χ proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{ISM2})$$

Indeed division writes every polynomial uniquely as $\chi Q + r$, $\deg r < q$; if its class is zero, $r = 0$. This also proves injectivity of the relation map and every dimension used below.

Let M_0, M_1 be the actual positive definite Hermitian forms on this same H . In the arithmetic comparison these are the moments of $r_{1/4}^{*k}(y) dy$ and $w_h^{*k}(y) dy$ in $S = k/2 + iy$, with $w_h(y) = |(g/h)(1/2 + iy)|^2/(2\pi)$. The finite identities below also hold for any two positive forms on this unchanged presentation. The canonical section, its relation coefficient, and its positive quotient Gram are

$$\begin{aligned} Z_N(M) &= (B_N^* M B_N)^{-1} B_N^* M C, \\ \rho_N(M) &= C - B_N Z_N(M) : E \longrightarrow H, \\ G_N(M) &= \rho_N(M)^* M \rho_N(M), & V_N(M) &= \det G_N(M) > 0. \end{aligned} \quad (\text{ISM3})$$

At $N = q-1$ the first map is the unique map to the zero space and $B_N Z_N = 0$; no preceding inverse is used. For every other degree, injectivity of B_N and positivity of M make its relation Gram positive definite. Direct substitution gives $B_N^* M \rho_N = 0$ and $\pi_\chi \rho_N = 1_E$. Hence ρ_N is injective, which proves positivity of G_N .

297.2 Both scaled secants, with the original boundary terms

Fix one admitted N , write $\rho_a = \rho_N(M_a)$, and put

$$\Delta = \rho_1 - \rho_0 = B_N(Z_N(M_0) - Z_N(M_1)). \quad (\text{ISM4})$$

For every coefficient map $Z : E \rightarrow \mathcal{P}_{N-q}$, the two cross terms in the Gram expansion vanish by $B_N^* M_a \rho_a = 0$, giving

$$(\rho_a + B_N Z)^* M_a (\rho_a + B_N Z) = G_N(M_a) + (B_N Z)^* M_a (B_N Z). \quad (\text{ISM5})$$

Taking $a = 0$, $B_N Z = \Delta$, and then $a = 1$, $B_N Z = -\Delta$, proves

$$\rho_1^* M_0 \rho_1 = G_0 + \Delta^* M_0 \Delta, \quad \rho_0^* M_1 \rho_0 = G_1 + \Delta^* M_1 \Delta. \quad (\text{ISM6})$$

Here $G_a = G_N(M_a)$. Subtracting the displayed matrices without deleting either boundary term yields, for real a, b ,

$$\begin{aligned} G_1 - aG_0 &= \rho_1^* (M_1 - aM_0) \rho_1 + a\Delta^* M_0 \Delta, \\ bG_0 - G_1 &= \rho_0^* (bM_0 - M_1) \rho_0 + \Delta^* M_1 \Delta. \end{aligned} \quad (\text{ISM7})$$

In particular both boundary terms have the positive signs shown. For $0 < a \leq b$ and $aM_0 \preceq M_1 \preceq bM_0$, congruence of positive matrices and (ISM7) give

$$aG_N(M_0) \preceq G_N(M_1) \preceq bG_N(M_0) \quad \text{at every one of the four degrees.} \quad (\text{ISM8})$$

Such finite endpoints always exist: they may be taken to be the smallest and largest eigenvalues of $M_0^{-1} M_1$, which is positive self-adjoint in the original M_0 metric.

Conjugating (ISM8) by $G_N(M_0)^{-1/2}$ places every eigenvalue of the conjugated positive matrix in $[a, b]$. The determinant is its product and logarithm is the sum of the real logarithms, so

$$q \log a \leq \log \frac{V_N(M_1)}{V_N(M_0)} \leq q \log b. \quad (\text{ISM9})$$

Set

$$\mathcal{B}(M) = \log \frac{V_{q-1}(M) V_q(M)}{V_{2q-1}(M) V_{2q}(M)}, \quad \mathcal{C} = \mathcal{B}(M_1) - \mathcal{B}(M_0). \quad (\text{ISM10})$$

There are exactly two positive and two negative terms in $\mathcal{C}$. Applying the upper endpoint in (ISM9) to the positive terms and the lower endpoint to the negative ones, and then reversing these choices, proves

$$|\mathcal{C}| \leq 2q \log(b/a). \quad (\text{ISM11})$$

Consequently a certified common source enclosure $(1 - \varepsilon)\widehat{M} \preceq M \preceq (1 + \varepsilon)\widehat{M}$, $0 \leq \varepsilon < 1$, gives

$$|\mathcal{B}(M) - \mathcal{B}(\widehat{M})| \leq 2q \log \frac{1 + \varepsilon}{1 - \varepsilon}. \quad (\text{ISM12})$$

These are the supplied source-error estimates in their original forms. At these same use sites the stronger previously proved AW/SP estimate must also be retained. For all ordered eigenvalues $\beta_1 \leq \dots \leq \beta_{2q+1}$ of $D = M_0^{-1} M_1$, define its full existing expression

$$\mathcal{L}_q(D) = \log \frac{\beta_{q+2}}{\beta_q} + 2 \sum_{j=1}^{q-1} \log \frac{\beta_{2q+2-j}}{\beta_j}.$$

The sum is empty at $q = 1$. AW's two nested-flag trace estimates, with the exact operator identification (ISM16)–(ISM19), give

$$|\mathcal{C}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log(b/a).$$

For the last inequality every displayed eigenvalue ratio is at most b/a , and its total coefficient is $1 + 2(q - 1) = 2q - 1$. Thus the same source enclosure in (ISM12) also gives the stronger $(2q - 1) \log((1 + \varepsilon)/(1 - \varepsilon))$ bound.

For the actual moment paths of RMT, let J_q be its integral of the pointwise minimum of the full-spectrum, exact trace-norm, overlap, centered Hilbert–Schmidt, and angle bounds. Each bounds the identical derivative (ISM19); the Hilbert–Schmidt member is (ISM26). Taking their pointwise minimum before integrating gives the recursively refined source-error statement

$$|\mathcal{C}| \leq \min\{J_q, \mathcal{L}_q(D), (2q-1)\log(b/a)\},$$

$$\mathcal{C} \in [J_L - \epsilon_L, J_L + \epsilon_L] \cap [-J_q, J_q] \quad (L \geq 0).$$

Here J_L, ϵ_L are the fully constructed signed center and residual in (ISM31)–(ISM37). Their interval therefore retains its original orientation in this earlier source-error calculation. For generic positive coefficient forms, the full-spectrum, trace-norm, overlap, Hilbert–Schmidt and signed bounds still apply, but the moment-specific angle improvement and its strict moment-positivity proof are used only on the original moment paths. The exact primitive and coefficient support are established in (ISM38)–(ISM43). Scaling a source by $c > 0$ leaves Z_N, ρ_N unchanged and multiplies G_N, V_N by c, c^q , respectively, directly by (ISM3). Thus its four $q \log c$ terms cancel in (ISM10); this proves the scalar invariance without altering any source form.

297.3 The canonical projector is exactly the original difference

Use $x \in [0, 1]$ for the comparison path, retaining the frequency y and all other deformation parameters. Write

$$M_x = (1-x)M_0 + xM_1, \quad \dot{M} = M_1 - M_0, \quad X_x = M_x^{-1}\dot{M}. \quad (\text{ISM13})$$

Since $M_x X_x = \dot{M} = \dot{M}^*$, this is self-adjoint in the original M_x inner product. Define

$$\begin{aligned} P_N &= I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0, \\ \Pi_N &= \rho_N G_N^{-1} \rho_N^* M_x. \end{aligned} \quad (\text{ISM14})$$

Each formula squares to itself and has M_x -self-adjoint adjoint law. The ranges of the first two are $\mathcal{P}_N$ and $B_N \mathcal{P}_{N-q}$; the range of the third is $\rho_N E$. By (ISM2)–(ISM3),

$$\mathcal{P}_N = \rho_N E \perp_{M_x} B_N \mathcal{P}_{N-q}. \quad (\text{ISM15})$$

To prove the equality, subtract $\rho_N \pi_\chi p$ from any $p \in \mathcal{P}_N$; (ISM2) puts the result in the relation range, and (ISM3) proves orthogonality. Therefore the original P_N, Q_N commute with product Q_N , and their orthogonal range decomposition proves the exact operator identity

$$\Pi_N = P_N - Q_N, \quad \text{Tr } \Pi_N = q. \quad (\text{ISM16})$$

The inverse of the range parametrization $\rho_N : E \rightarrow \text{ran } \Pi_N$ is the literal π_χ restricted to that range; both composites are identities by (ISM3). Thus (ISM16) identifies actual maps and their domains, not only their traces.

Define

$$\begin{aligned} \mathcal{A}_x &= \Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{ISM17})$$

Substitution of (ISM16), with $Q_{q-1} = 0$, gives $\mathcal{A}_x = U_x - W_x$ and $\text{Tr } \mathcal{A}_x = 0$. These U_x, W_x are AW's U, W and SP's $\mathcal{R}, \mathcal{P}$, respectively. The source relation matrix is related to AW's unpadded matrix by $B_N = I_N B_N^{\text{AW}}$, so (ISM14) reproduces its actual projections with no sign or metric change.

Differentiating $\pi_\chi \rho_N = 1$ shows that ρ'_N has image in the original relation space. More explicitly, differentiation of $B_N^* M_x \rho_N = 0$ gives

$$\rho'_N = -B_N(B_N^* M_x B_N)^{-1} B_N^* \dot{M} \rho_N, \quad G'_N = \rho_N^* \dot{M} \rho_N. \quad (\text{ISM18})$$

The second identity follows because the two cross terms containing ρ'_N vanish by the same orthogonality. All these matrices are smooth on the closed interval, since their positive Gram inverses exist there. Jacobi differentiation and cyclicity of finite trace prove

$$\frac{d}{dx} \log V_N = \text{Tr}(X_x \Pi_N), \quad \mathcal{B}'(x) = \text{Tr}(X_x \mathcal{A}_x), \quad \mathcal{C} = \int_0^1 \text{Tr}(X_x \mathcal{A}_x) dx. \quad (\text{ISM19})$$

For the original row $v(y) = (1, S, \dots, S^{2q})$, $\Pi_N M_x^{-1} = \rho_N G_N^{-1} \rho_N^*$. Thus the signed density in PR29 is exactly

$$\kappa_x(y) = v(y) \mathcal{A}_x M_x^{-1} v(y)^*, \quad \int \kappa_x d\mu_x = \text{Tr} \mathcal{A}_x = 0. \quad (\text{ISM20})$$

The first equality keeps the inverse metric that is required by the original coordinate kernel. Integrating against the original signed measure $\mu_1 - \mu_0$ gives precisely (ISM19).

297.4 Restriction losses and the full cross trace

Fix $i \leq j$ among admitted degrees. The map $\rho_i - \rho_j : E \rightarrow \mathcal{P}_j$ has zero remainder, so (ISM2) makes it an original degree- j relation. Orthogonality of ρ_j therefore gives the cross-Grams

$$\rho_j^* M_x \rho_i = G_j, \quad \rho_i^* M_x \rho_j = G_j, \quad G_i - G_j = (\rho_i - \rho_j)^* M_x (\rho_i - \rho_j). \quad (\text{ISM21})$$

The exact reverse restriction on E is $T_{i,j} = G_i^{-1} G_j$. It is positive self-adjoint in G_i , with spectrum in $(0, 1]$, by the last identity. Cyclic trace and the first two identities give

$$\begin{aligned} \text{Tr}(\Pi_i \Pi_j) &= \text{Tr}(G_i^{-1} (\rho_i^* M_x \rho_j) G_j^{-1} (\rho_j^* M_x \rho_i)) \\ &= \text{Tr}(G_i^{-1} G_j), \\ \text{Tr}((\Pi_i - \Pi_j)^2) &= 2 \text{Tr}(1 - T_{i,j}). \end{aligned} \quad (\text{ISM22})$$

All traces are finite traces of the original coordinate operators. In particular their spectra are not declared to be arithmetic invariant summands.

Set

$$\begin{aligned} A_0 &= \Pi_{q-1} - \Pi_{2q-1}, \quad A_1 = \Pi_q - \Pi_{2q}, \\ \ell_0 &= \text{Tr}(1 - T_{q-1,2q-1}), \quad \ell_1 = \text{Tr}(1 - T_{q,2q}), \\ Z_x &= \text{Tr}(\mathcal{A}_x^2) = 2\ell_0 + 2\ell_1 + 2 \text{Tr}(A_0 A_1). \end{aligned} \quad (\text{ISM23})$$

This retains the cross term with its actual sign. On the full algebra $\text{End}(H)$, the map $L \mapsto M_x^{1/2} L M_x^{-1/2}$ is an algebra isomorphism with inverse $K \mapsto M_x^{-1/2} K M_x^{1/2}$. It sends the M_x -self-adjoint operators to Hermitian operators, and cyclicity preserves traces of every product. This proves $Z_x \geq 0$ and, from $\|A_0 + A_1\|_{\text{HS}}^2 \leq 2\|A_0\|_{\text{HS}}^2 + 2\|A_1\|_{\text{HS}}^2$, proves

$$0 \leq Z_x \leq 4(\ell_0 + \ell_1). \quad (\text{ISM24})$$

Here and below the Hilbert-Schmidt norm is computed after this exact metric isomorphism; no new source metric is used.

Put

$$c_x = \frac{\text{Tr} X_x}{n}, \quad \Sigma_x = \text{Tr}(X_x^2) - \frac{(\text{Tr} X_x)^2}{n}. \quad (\text{ISM25})$$

The trace is real by self-adjointness. Expanding $\text{Tr}((X_x - c_x I)^2)$ proves its equality with Σ_x , hence its nonnegativity. Trace zero of $\mathcal{A}_x$, and Cauchy-Schwarz for the two Hermitian images in the preceding isomorphism, give

$$|\mathcal{B}'(x)| \leq S_q(x) := \sqrt{\Sigma_x Z_x} \leq 2\sqrt{\Sigma_x(\ell_0 + \ell_1)}. \quad (\text{ISM26})$$

This bound can be combined pointwise with the existing full-spectrum, exact trace-norm, overlap, and angle bounds on the same $\mathcal{B}'$. It does not replace any smaller available bound.

297.5 A specified signed approximation and its exact residual

For an M_x -self-adjoint, continuous operator Y_x define $E_x = X_x - c_x I - Y_x$. It is self-adjoint too. Expanding the trace and applying the same Cauchy–Schwarz inequality proves the two signed inequalities

$$\mathrm{Tr}(Y_x \mathcal{A}_x) - \sqrt{\mathrm{Tr}(E_x^2) Z_x} \leq \mathcal{B}'(x) \leq \mathrm{Tr}(Y_x \mathcal{A}_x) + \sqrt{\mathrm{Tr}(E_x^2) Z_x}. \quad (\text{ISM27})$$

Every term is real. Integration is legitimate by continuity on the compact interval. We now construct such a Y_x , with its residual controlled from the actual two endpoint forms.

Let $D = M_0^{-1} M_1$, with ordered positive eigenvalues $b_1 \leq \dots \leq b_n$, counted with multiplicity. For this subsection write $B_x^\circ = (1-x)I + xD$, to distinguish it from every original relation matrix. Since $M_x = M_0 B_x^\circ$, direct inversion gives

$$X_x = (B_x^\circ)^{-1}(D - I), \quad \lambda_j(x) = 1 - x + x b_j, \quad \gamma_j(x) = \frac{b_j - 1}{1 - x + x b_j}. \quad (\text{ISM28})$$

Take an M_0 -orthogonal eigenbasis of D . It is also M_x -orthogonal, since on its j -th line M_x multiplies M_0 by $\lambda_j(x) > 0$. Therefore (ISM28) gives the full spectrum of X_x in the original varying metric.

Define the positive scalar and commuting operator

$$a_x = \frac{\lambda_1(x) + \lambda_n(x)}{2}, \quad H_x = I - B_x^\circ / a_x. \quad (\text{ISM29})$$

Their use does not rescale M_x , its moments, or its quotient Grams. The eigenvalues of H_x are $\zeta_j(x) = 1 - \lambda_j(x)/a_x$, and

$$\|H_x\|_{M_x} = \frac{x(b_n - b_1)}{2(1-x) + x(b_1 + b_n)} \leq \vartheta := \frac{b_n - b_1}{b_n + b_1} < 1. \quad (\text{ISM30})$$

Indeed the extrema of $1 - \lambda/a_x$ are equal and opposite at the two endpoints. The displayed scalar function of x has derivative $2(b_n - b_1)/(2(1-x) + x(b_1 + b_n))^2 \geq 0$. This proves the uniform bound, including $x = 0$.

For every integer $L \geq 0$, form the finite matrix polynomial and its centered version

$$\begin{aligned} X_x^{[L]} &= a_x^{-1} \sum_{r=0}^L H_x^r (D - I), \\ Y_x^{[L]} &= X_x^{[L]} - \frac{\mathrm{Tr} X_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM31})$$

All factors commute with D and are real functions of its self-adjoint operator. They are consequently self-adjoint in both M_0 and M_x . The exact finite geometric-series identity $(I - H_x) \sum_{r=0}^L H_x^r = I - H_x^{L+1}$, together with $B_x^\circ = a_x(I - H_x)$, proves

$$\begin{aligned} R_x^{[L]} &:= X_x - X_x^{[L]} = H_x^{L+1} X_x, \\ E_x^{[L]} &:= X_x - c_x I - Y_x^{[L]} = R_x^{[L]} - \frac{\mathrm{Tr} R_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM32})$$

No infinite-series interchange is used in this identity. In the original eigenbasis the residual eigenvalues are $\delta_j(x) = \zeta_j(x)^{L+1} \gamma_j(x)$. Thus its exact centered squared size is

$$\mathrm{Tr}((E_x^{[L]})^2) = \sum_{j=1}^n \delta_j(x)^2 - \frac{(\sum_{j=1}^n \delta_j(x))^2}{n}. \quad (\text{ISM33})$$

Both sides are nonnegative, and all multiplicities and signs of the residual eigenvalues are retained. The signed center is $\mathrm{Tr}(Y_x^{[L]} \mathcal{A}_x) = \mathrm{Tr}(X_x^{[L]} \mathcal{A}_x)$ by $\mathrm{Tr} \mathcal{A}_x = 0$. Equations (ISM27), (ISM31)–(ISM33) therefore provide a finite constructed signed center and its exact error.

There is also a uniform elementary error bound for this fixed original pair of forms. Define

$$K_D = \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \quad (\text{ISM34})$$

The denominator of $\gamma_j(x)$ is at least $\min\{1, b_j\}$, so (ISM30)–(ISM33) imply $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$. The nested relation and source flags in (ISM17) give both U_x, W_x eigenvalues $0^{[q]}, 1^{[2]}, 2^{[q-1]}$: the first relation line contributes one, its next $q - 1$ increments contribute two, and the final increment contributes one; the source increments give the same list. In particular both have rank $q + 1$, and each dominates its range projection. Their ranges in dimension $2q + 1$ intersect in dimension at least one. The trace product of the two range projections is at least that intersection dimension, by summing squared projected lengths in an orthonormal basis. For positive operators, increasing a factor cannot decrease the product trace, since $\text{Tr}(T^{1/2}(S' - S)T^{1/2}) \geq 0$. It follows that $\text{Tr}(U_x W_x) \geq 1$. Using their complete spectra, we obtain

$$Z_x = 8q - 4 - 2 \text{Tr}(U_x W_x) \leq 8q - 6. \quad (\text{ISM35})$$

This proof includes $q = 1$, where the multiplicity $q - 1$ is zero.

Set

$$\begin{aligned} J_L &= \int_0^1 \text{Tr}(X_x^{[L]} \mathcal{A}_x) dx, \\ \epsilon_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) Z_x} dx. \end{aligned} \quad (\text{ISM36})$$

Every integrand is a specified continuous finite matrix expression on the original source path. Integrating (ISM27) with the constructed (ISM31) and the established uniform bound gives

$$J_L - \epsilon_L \leq \mathcal{C} \leq J_L + \epsilon_L, \quad 0 \leq \epsilon_L \leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \quad (\text{ISM37})$$

Thus the supplied signed enclosure now has an explicit convergent residual on every fixed original finite source. It retains the arithmetic correlation in its center. The evaluation of the center integral still uses the specified arithmetic entries and a controlled integration method; this calculation supplies no uniform estimate in k . If $b_n = b_1$, then $D = b_1 I$, $H_x = 0$, the residual and $\mathcal{C}$ vanish exactly, and every formula above remains valid.

297.6 The original primitive and the residual at a proper support

We finally prove the source comparison underlying all these metric changes. In the original one-factor complex $V \xrightarrow{\Theta} \mathcal{B}$, retain $h(D)F_h = \Theta\phi_*$, with the original D action commuting with Θ . For $v \in E$, the polynomial $(\rho_1 - \rho_0)v = \chi Q_v$ has the coefficient $Q_v = (Z_N(M_0) - Z_N(M_1))v$ by (ISM4). In the multivariate polynomial ring put

$$f_v(s_1, \dots, s_k) = \chi(s_1 + \dots + s_k) Q_v(s_1 + \dots + s_k). \quad (\text{ISM38})$$

Its image in $E_h^{\otimes k}$ is zero, because $\chi(A_h^{(k)})1 = 0$. Successive monic division in $s_1, \dots, s_k$ constructs specific polynomials $Q_{v,i}$ as follows. Let $f_{v,0} = f_v$ and write $f_{v,i-1} = h(s_i)Q_{v,i} + f_{v,i}$, where $\deg_{s_i} f_{v,i} < \deg h$. Earlier remainder degree bounds are preserved because $h(s_i)$ uses only s_i . The final remainder $f_{v,k}$ has degree below $\deg h$ in every variable and zero class in the quotient, so it is zero by the tensor monomial basis. Telescoping proves

$$f_v = \sum_{i=1}^k h(s_i) Q_{v,i}. \quad (\text{ISM39})$$

This constructs a primitive using precisely the original relations, without replacing them by a larger observation kernel.

In the tensor complex $\mathbf{C}^\bullet = [V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ define

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}) \in \mathbf{C}^{k-1}. \quad (\text{ISM40})$$

The displayed source tensor has exactly $i-1$ preceding factors of degree one. Its differential therefore has sign $(-1)^{i-1}$; multiplying by the sign in (ISM40) gives $+1$. The source operators commute with the differential. Applying $h(D_i)F_h = \Theta\phi_*$ to (ISM39) now proves

$$dK_v = \mathcal{V}((\rho_1 - \rho_0)v). \quad (\text{ISM41})$$

Consequently the two canonical sections have the same class in the original full-source quotient; their common arithmetic observation is exactly $\eta_{h,k}v$ by (ISM1). Composing with the original full-source quotient $q^{(k)}$ gives the already constructed map $\sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}$. Here no map was made natural merely by changing its metric; the actual primitive (ISM40) is the comparison.

To express the full two-leg source without deleting its diagonal, retain the labelled Fourier copy $\widehat{V}$, with the original isomorphism $\mathcal{F} : V \rightarrow \widehat{V}$, and the complex $\mathbf{T}^\bullet = [V \oplus \widehat{V} \rightarrow \mathcal{B}]$, whose differential is $(\phi, \widehat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\widehat{\psi})$. There are exact cochain maps from the one-leg complex to this complex with degree-one map identity and degree-zero maps

$$j_+(v) = (v, 0), \quad j_-(v) = (0, -\mathcal{F}v).$$

Indeed either differential equals Θv . The reverse cochain map has degree-one map identity and degree-zero map $p(\phi, \widehat{\psi}) = \phi - \mathcal{F}^{-1}\widehat{\psi}$. Thus $pj_+ = pj_- = 1$, whereas $(j_+ - j_-)(v) = (v, \mathcal{F}v)$ is the full degree-zero diagonal kernel. These formulas prove both composites and retain that kernel explicitly. Tensoring specified choices of these degree-zero maps with their degree-one identities gives cochain maps, since the displayed differential identities preserve each original tensor sign. Apply one such map to (ISM40) to obtain $\tilde{K}_v \in (\mathbf{T}^{\otimes k})^{k-1}$. Its top differential is still (ISM41).

For precision at a proper declared support $\lambda = (W, S; A)$, let $\mathcal{J}$ be the specified finite carrier of coefficient positions with $A \subseteq \mathcal{J}$. Put $\mathbf{C}_{\text{full}}^\bullet = (\mathbf{T}^{\otimes k})^{\oplus \mathcal{J}}$, with the original differential on each coordinate. Insert the active A -coordinates by literal zero insertion into this fixed full carrier. Let $\mathbf{C}_\lambda^\bullet \subseteq \mathbf{C}_{\text{full}}^\bullet$ be its declared subcomplex, retaining its independent outer label and coefficient mask. The same source primitive is taken in the indicated coefficient copy. When the compared top source representatives lie in $\mathbf{C}_\lambda^k$, their difference has the exact residual

$$[d\tilde{K}_v]_\lambda \in \frac{d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_\lambda^k}{d\mathbf{C}_\lambda^{k-1}}. \quad (\text{ISM42})$$

This quotient is the kernel of the natural map $\mathbf{C}_\lambda^k / d\mathbf{C}_\lambda^{k-1} \rightarrow \mathbf{C}_{\text{full}}^k / d\mathbf{C}_{\text{full}}^{k-1}$: a class maps to zero exactly when its representative is a full-source boundary. The inclusion of the displayed numerator therefore induces the kernel isomorphism, with inverse obtained by using the same representative. These two maps are well-defined since differences in $d\mathbf{C}_\lambda^{k-1}$ are precisely the equivalence relation on both sides. Thus (ISM41) kills the class at this support exactly when $d\tilde{K}_v \in d\mathbf{C}_\lambda^{k-1}$. Membership $\tilde{K}_v \in \mathbf{C}_\lambda^{k-1}$ is one explicit sufficient witness, not an assumption made for arbitrary W .

First specialize to $k = 1$ and the single coefficient carrier $\mathcal{J} = \{i_0\}$, with active face $A = \{i_0\}$. The coordinate projection onto i_0 and its inverse coordinate insertion identify this one-coordinate carrier with the original one-factor complex. In these specified coordinates put $\mathcal{B}_\lambda = \mathbf{C}_\lambda^1 \subseteq \mathcal{B}$. Suppose the admitted source image under p is $W \subseteq V$, so its boundary range is exactly ΘW , and retain $\Theta V \subseteq \mathcal{B}_\lambda$. The precise kernel map in this single coordinate is

$$V/W \longrightarrow \mathcal{B}_\lambda / \Theta W, \quad [v] \longmapsto [\Theta v]_W. \quad (\text{ISM43})$$

It is injective because Θ is injective: $\Theta v = \Theta w, w \in W$, forces $v = w$. Its range is the kernel of the map to the full quotient $\mathcal{B}/\Theta V$, since being in that kernel means being represented by Θv . In this case (ISM42) is exactly the class of the original one-factor primitive in V/W .

For an arbitrary admitted face $A \subseteq \mathcal{I}$, restore the original coordinate injections $\iota_a : \mathcal{B} \rightarrow \mathcal{B}^{\oplus \mathcal{I}}$ and projections $\text{pr}_a : \mathcal{B}^{\oplus \mathcal{I}} \rightarrow \mathcal{B}$. They satisfy $\text{pr}_b \iota_a = \delta_{ab} 1$ and $\sum_{a \in A} \iota_a \text{pr}_a = 1$ on the A -supported subspace. Let the degree-one source at this face be $\mathcal{B}_\lambda^{\oplus A}$, inserted in the fixed carrier by these ι_a , and retain the degree-zero source image $W^{\oplus A}$ and boundary $(\Theta W)^{\oplus A}$. Its kernel map is

$$(V/W)^{\oplus A} \longrightarrow \mathcal{B}_\lambda^{\oplus A}/(\Theta W)^{\oplus A}, \quad ([v_a])_{a \in A} \longmapsto \left[\sum_{a \in A} \iota_a \Theta v_a \right].$$

Replacing v_a by $v_a + w_a, w_a \in W$, changes the displayed sum by the specified boundary, so the map is well-defined. If its image is zero, projection by each pr_a gives $\Theta v_a \in \Theta W$, whence $v_a \in W$ by injectivity of Θ . Thus the map is injective. A representative in its codomain maps to zero in $(\mathcal{B}/\Theta V)^{\oplus \mathcal{I}}$ exactly when each active coordinate is Θv_a for some $v_a \in V$. Its inverse on that kernel takes these unique preimages modulo W . Changing a representative by an element of $(\Theta W)^{\oplus A}$ changes each preimage by an element of W . The coordinate identities above then prove both inverse composites. This is the exact $(V/W)^{\oplus A}$ residual, with its full ambient carrier specified.

For $A \subseteq A'$, zero insertion sends the tuple $([v_a])_{a \in A}$ to the same tuple extended by zero; both orders of this insertion and the kernel map give $\sum_{a \in A} \iota_a \Theta v_a$. A signed coefficient permutation sends that sum to $\sum_{a \in A} \epsilon_a \iota_{\sigma(a)} \Theta v_a = \sum_{a \in A} \iota_{\sigma(a)} \Theta(\epsilon_a v_a)$, which proves the same commuting identity with every sign retained. When $A = \emptyset$, the two displayed direct sums and their unique map are zero vector spaces and a zero map at the retained outer label $(W, S; \emptyset)$; the coordinate calculation does not identify that label with external absence. This proves the typed relation to the existing mixed-support quotients while retaining every proper-source residual.

All source identities, projector equalities, scaled comparison signs, and signed estimates above use the same original finite polynomial presentation. Their full primitive and proper support kernel have now been exhibited, so backward and forward propagation can replace weaker metric bounds at their earlier use sites without erasing the older sources or claiming that the arbitrary proper-support residual vanishes.

298 Exact relation-window spectral transport on the original tau complex

This calculation continues arithmetic determinant transport on its original finite source over the absolute tau base. It computes the full spectra of the two retained window operators, constructs an isometry intertwining them in the actual source metric, and strengthens the signed determinant bound. The source measures, their masses, the cyclic polynomial, the Taylor unit, and the cohomology observation remain fixed.

298.1 The original source and its four windows

Use a full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and the original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Let $E = \mathbb{C}[S]/(\chi)$, $E_h = \mathbb{C}[s]/(h)$, and $v_h = j_h(g/h) \in E_h^\times$. The original tensor-sum observation is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1. \quad (\text{AW1})$$

The original section $\sigma_h : E_h \hookrightarrow Q = \mathcal{B}/\Theta V$ gives the separate injection $\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}$. Here $Q^{\otimes k}$ is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau} = \{\tau, 1\}$; the one-factor complex is $[V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}]$. The complete Tau Base and cyclic source proofs accompany the preceding arithmetic determinant transport calculation (AT).

For $N \geq q - 1$, retain the increasing-power spaces and maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad \mathcal{P}_{-1} = 0. \quad (\text{AW2})$$

The common source is $H = \mathcal{P}_{2q}$, of dimension $n = 2q + 1$. Its original coordinate on the integration line is $S = k/2 + iu$. Set $v_N(u) = (1, S(u), \dots, S(u)^N)$ and retain the two densities

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, & c_a &= 2^{1-2a}\Gamma(2a). \end{aligned} \quad (\text{AW3})$$

Their actual masses are μ_h^k and $(2\pi)^{k/2}$. For the real parameter $0 \leq t \leq 1$, put $m_t = (1-t)m_0 + tm_1$, $M_N(t) = \int v_N^* v_N m_t$, and $M(t) = M_{2q}(t)$. The source density proofs give all finite moments and positivity almost everywhere, so all these finite Grams are positive definite. We use the original form $\langle x, y \rangle_t = x^* M(t) y$ on H .

Let $P_N(t)$ be the orthogonal projection onto $\mathcal{P}_N \subset H$, and $Q_N(t)$ the projection onto $\chi \mathcal{P}_{N-q} \subset H$. Set $Q_{q-1} = 0$. For the coefficient inclusion $I_N : \mathcal{P}_N \rightarrow H$, their fully typed coefficient formulas are

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M, \\ Q_N &= I_N B_N (B_N^* M_N B_N)^{-1} B_N^* I_N^* M. \end{aligned} \quad (\text{AW4})$$

The zero relation domain at $N = q - 1$ contributes the zero operator. Let $G_N(t)$ be the minimum-lift Gram on E from (AW2) and $V_N(t) = \det G_N(t)$. Retain

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}, \\ C(t) &= M(t)^{-1} (M(1) - M(0)), \\ U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{AW5})$$

The AT proof, included in full, establishes $V_N = \det M_N / \det(B_N^* M_N B_N)$ with the original orientation $(-1)^{q(N-q+1)}$, and

$$\Delta B = B(1) - B(0) = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \quad (\text{AW6})$$

Thus the following computation evaluates the operators in the already proved signed arithmetic correction.

298.2 Full spectra and an explicit source isometry

Theorem 298.1. *In the original $M(t)$ metric, both $U(t)$ and $W(t)$ have spectrum*

$$0 \text{ with multiplicity } q, \quad 1 \text{ with multiplicity } 2, \quad 2 \text{ with multiplicity } q - 1. \quad (\text{AW7})$$

The last multiplicity is zero when $q = 1$. There is an explicitly constructed smooth $M(t)$ -unitary $T(t) : H \rightarrow H$ such that $U(t) = T(t)W(t)T(t)^{-1}$.

Proof. Every subspace in either of the following flags is the literal displayed polynomial subspace, with its original coefficient inclusion. For the relation flag use

$$F_0 = \chi \mathcal{P}_0 \subset F_1 = \chi \mathcal{P}_{q-1} \subset F_2 = \chi \mathcal{P}_q \subset H. \quad (\text{AW8})$$

Their dimensions are 1, q , $q+1$, since multiplication by χ is injective. When $q = 1$, $F_0 = F_1$ is retained as that equality. Projections of nested subspaces commute and their products equal the projection of the smaller subspace: each larger space is the orthogonal direct sum of the smaller space and its orthogonal complement within the larger space. On the original orthogonal decomposition $F_0 \oplus (F_1 \cap F_0^\perp) \oplus (F_2 \cap F_1^\perp) \oplus F_2^\perp$, the operator $U = Q_{F_1} + Q_{F_2} - Q_{F_0}$ acts respectively by 1, 2, 1, 0. The dimensions are respectively 1, $q - 1$, 1, q .

For W use $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q} = H$. On the four corresponding orthogonal pieces, W acts by 0, 1, 2, 1, with dimensions q , 1, $q - 1$, 1. This proves (AW7), including the repeated middle subspaces at $q = 1$.

For an explicit intertwiner retain every Gram factor as follows. For any specified ordered independent list $z_0, \dots, z_d$, apply the recursion

$$y_j = z_j - \sum_{i < j} e_i \langle e_i, z_j \rangle_t, \quad a_j = (y_j^* M(t) y_j)^{1/2} > 0, \quad e_j = y_j / a_j. \quad (\text{AW9})$$

The positive real square root fixes its phase. No source density or mass is changed; each a_j is the norm in the original form. Independence proves $y_j \neq 0$ at every step, so this recursion is defined and smooth on the entire parameter interval.

First use $z_j = S^j$, $0 \leq j \leq 2q$, obtaining $p_0, \dots, p_{2q}$. Next use $z_j = \chi S^j$, $0 \leq j \leq q$, obtaining $r_0, \dots, r_q$. These lie in the successive relation spaces in (AW8). Finally let $R_{2q}(t) : E \rightarrow H$ be the original minimum section from AT6 and apply (AW9) to $R_{2q}(t)[1], \dots, R_{2q}(t)[S^{q-1}]$, obtaining $c_0, \dots, c_{q-1}$. The relation orthogonality in AT6 puts this entire list in $F_2^\perp$. Since $J_{2q} R_{2q} = I_E$, the list is independent and spans that q -dimensional complement.

Define T on the first orthonormal basis by

$$\begin{aligned} Tp_j &= c_j & (0 \leq j \leq q-1), \\ Tp_q &= r_0, & Tp_{2q} = r_q, \\ Tp_{q+j} &= r_j & (1 \leq j \leq q-1). \end{aligned} \quad (\text{AW10})$$

The middle range is empty for $q = 1$, with p_q and p_{2q} still distinct. The target list is an orthonormal basis. Thus this linear map is invertible and $T^*MT = M$. The spectral actions already calculated agree on each indicated basis vector, proving $UT = TW$. The recursion, the smooth minimum section and the positive norms prove the asserted smoothness. This completes the construction. $\square$

The exact isometry gives a commutator form of the signed integrand:

$$\mathrm{Tr}((U - W)C) = \mathrm{Tr}(WT^{-1}[C, T]), \quad [C, T] = CT - TC. \quad (\text{AW11})$$

Indeed cyclicity moves T in $\mathrm{Tr}(TWT^{-1}C)$ to the right, and subtraction then gives the displayed product. This T is an isometry of the original source; its cohomology observation is computed, rather than silently assigned an intertwining property. The unchanged arithmetic observation $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ satisfies

$$\mathcal{O}T = \sigma_h^{\otimes k} \eta J_{2q} T, \quad \mathcal{O}T - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} (T - I), \quad \ker(\mathcal{O}T) = T^{-1}F_2. \quad (\text{AW12})$$

The last identity follows from both injections in (AW1) and $\ker J_{2q} = F_2$. By (AW10), $T^{-1}F_2$ is exactly $\mathrm{span}\{p_q, \dots, p_{2q}\} = \mathcal{P}_{q-1}^\perp$ in H . At any fixed nonempty support label A , lift every displayed linear map by $(A, x) \mapsto (A, fx)$ and send τ to τ . Consequently the supported-zero inverse image of the lifted $\mathcal{O}T$ at A is the full set $\{A\} \times \mathcal{P}_{q-1}^\perp$. The Taylor unit in (AW1) and the defect in (AW12) remain present.

298.3 The spectral bound with every eigenvalue retained

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues, counted with multiplicity, of $D = M(0)^{-1}M(1)$ on its original $M(0)$ Hilbert space. All are positive. Define

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \kappa(D) = b_n/b_1. \quad (\text{AW13})$$

For $q = 1$ the sum is empty and $\mathcal{L}_1(D) = \log(b_3/b_1)$.

Retain also the actual polynomial-space overlap. At each t , let $s(t) = \dim(\mathrm{ran} U \cap \mathrm{ran} W)$ and $d(t) = c_n(t) - c_1(t)$. With the actual $c_j(t)$ of (AW17), set

$$\begin{aligned} S_{\mathrm{AW}}(t) &= c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j), \\ S_{\mathrm{SP}}(t) &= \frac{d(t)}{2} \min \left\{ 4q - 2s(t), \sqrt{(2q+1)(8q-4-2\mathrm{Tr}(UW))} \right\}, \\ \mathcal{J}_{h,k} &= \int_0^1 \min\{S_{\mathrm{AW}}(t), S_{\mathrm{SP}}(t)\} dt. \end{aligned} \quad (\text{AW13a})$$

The overlap refinement from SP8–SP10 is transported to this earlier calculation by the exact arrows $x = t, y = u, B_N^{\mathrm{SP}} = I_N B_N^{\mathrm{AW}}$. Indeed $M_N = I_N^* M I_N$ gives $(B_N^{\mathrm{SP}})^* M B_N^{\mathrm{SP}} = B_N^* M_N B_N$, so its projectors are precisely (AW4), and $(\mathcal{R}^{\mathrm{SP}}, \mathcal{P}^{\mathrm{SP}}, T^{\mathrm{SP}}) = (U, W, C)$. The original Gamma densities agree because $r_{1/4}^{*k} = c_{1/4}^k r_{k/4}/c_{k/4}$ with $c_{1/4} = \sqrt{2\pi}$, exactly (AW3). The arithmetic densities already coincide as w_h^{*k} . Thus this comparison preserves the moments, masses, polynomial coefficients and all four endpoint signs.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\mathrm{tr}}(t) &= \frac{d(t)}{2} \|U(t) - W(t)\|_1, \\ \Sigma(t) &= \mathrm{Tr}(C(t)^2) - \frac{(\mathrm{Tr} C(t))^2}{2q+1}, \\ S_{\mathrm{HS}}(t) &= \sqrt{\Sigma(t) \mathrm{Tr}((U(t) - W(t))^2)}. \end{aligned} \quad (\text{AW13c})$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral

diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(C(t))I/(2q+1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(t)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(t)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

The original-moment angle calculation supplies a further simultaneous bound at this same claim site. Put $a_q = 2q - 1$ and

$$\begin{aligned} A_{h,k}(t) &= a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}} d(t), \\ \mathcal{J}_{h,k}^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(t), S_{\text{tr}}(t), S_{\text{SP}}(t), S_{\text{HS}}(t), A_{h,k}(t)\} dt, \\ \mathcal{I}_{h,k} &= \int_0^1 \min\left\{d(t), \frac{S_{\text{AW}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{SP}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}}, \frac{S_{\text{HS}}(t)}{a_q \sqrt{1 - e^{-\mathcal{B}(t)/a_q}}} \right\} dt. \end{aligned} \quad (\text{AW13b})$$

These specialize RMT1–RMT18 through $a_{\text{RMT}} = a_q$, $x = t$, $C_x = C(t)$ and $(U_x, W_x) = (U(t), W(t))$. The density and relation arrows were proved in (AW3)–(AW4) and (AW13a), so every source moment and original cohomology map agrees. This application uses the actual moment measures (AW3). It does not impose moment structure on an arbitrary positive coefficient Gram.

Theorem 298.2. *On the unchanged arithmetic source,*

$$|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)} \leq \mathcal{J}_{h,k} \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa(D). \quad (\text{AW14})$$

The previously proved coefficient $2q - 1$ and full relative-spectrum bound are retained. The pointwise minimum additionally preserves the actual polynomial-space overlap, and is at most $\int_0^1 S_{\text{SP}}(t) dt$.

Proof. We first prove the needed finite spectral trace inequality. Let C be self-adjoint in a finite Hilbert space with eigenvalues $c_1 \leq \dots \leq c_n$. If P is an orthogonal projection of rank r , let e_j be an orthonormal eigenbasis of C and $d_j = \langle e_j, P e_j \rangle$. Then $0 \leq d_j \leq 1$ and $\sum_j d_j = r$, since $d_j = \|P e_j\|^2$ and $\text{Tr } P = r$. Thus $\text{Tr}(PC) = \sum_j c_j d_j$. Its upper bound by $\sum_{j=n-r+1}^n c_j$ follows by subtracting the two expressions: the mass $\sum_{j \leq n-r} d_j$ equals $\sum_{j > n-r} (1 - d_j)$, and every coefficient in the latter group is at least every coefficient in the former. Pairing the same nonnegative mass, or bounding both groups by c_{n-r+1} , proves the inequality. Applying this argument to $-C$ proves the lower bound $\sum_{j=1}^r c_j$. This proof includes repeated eigenvalues and the ranks zero and n .

For an operator Z with spectrum (AW7), its spectral projection A_Z onto eigenvalues 1, 2 has rank $q + 1$, and its spectral projection B_Z onto eigenvalue 2 has rank $q - 1$. The exact decomposition is $Z = A_Z + B_Z$. Applying the proved projection bounds gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AW15})$$

Each bound is attained for an abstract operator of this spectrum by assigning its eigenspaces to the appropriate eigenvectors of C . This attainment statement is only about the displayed spectral constraint; it assigns no freely chosen orientation to the actual polynomial relation spaces. Both U and W satisfy (AW15). Subtract their bounds and cancel the common c_{q+1} term only after writing both complete sums:

$$|\text{Tr}((U - W)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{AW16})$$

No mutual commutation of U, W, C is used.

The range dimensions in (AW7) give $s(t) \geq 1$. Let E_L be the orthogonal projector onto their intersection. Each $Z = U, W$ is at least its range projection, since its nonzero eigenvalues are 1, 2, and therefore $Z - E_L$ is positive of trace $2q - s(t)$. Writing $A = U - W = (U - E_L) - (W - E_L)$ proves

$$\|A\|_1 \leq 4q - 2s(t), \quad \text{Tr } A = 0, \quad \text{Tr } A^2 = 8q - 4 - 2 \text{Tr}(UW), \quad \|A\|_1 \leq \sqrt{(2q+1) \text{Tr } A^2}. \quad (\text{AW16a})$$

The square identity uses $\text{Tr } U^2 = \text{Tr } W^2 = 4q - 2$ and cyclicity. The last inequality is Cauchy-Schwarz on every real eigenvalue of the self-adjoint A . Subtracting $(c_1 + c_n)I/2$ from C does not change its trace pairing with A . In an orthonormal eigenbasis of A , that pairing is bounded in absolute value by $d(t)\|A\|_1/2$. This proves $|\text{Tr}((U - W)C)| \leq S_{\text{SP}}(t)$ in addition to (AW16), and hence bounds it by their pointwise minimum. Smooth positive Grams give bounded continuous eigenvalues and overlaps; $s(t)$ is Borel because its rank strata are specified by matrix minors. Therefore the minimum is integrable. The signed identity (AW6) now proves $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}$ and $\mathcal{J}_{h,k} \leq \int S_{\text{SP}}$. For the original moment path, the exact original-source identification above also applies RMT11–RMT12: its full crossed angle lists give $|\mathcal{B}'| \leq A_{h,k}$, and $\mathcal{B} > 0$ follows because otherwise $1 \perp \chi \mathcal{P}_q$ would force $\int |\chi|^2 m_t = 0$ by the reflected polynomial $\chi^{\#k}(S) = \sum_j \bar{\chi}_j(k-S)^j$. Taking the pointwise minimum of this additional proved inequality with (AW16), (AW16a), and both complete bounds in (AW13c) gives $|\Delta\mathcal{B}| \leq \mathcal{J}_{h,k}^{(5)}$. The complete proof of the original angle estimate and its exact chain-rule continuation is retained in EP and RMT; it uses AW's earlier projection identities, not the final claim (AW14).

For the actual interpolated metric, $C(t) = ((1-t)I + tD)^{-1}(D - I)$, so its eigenvalues in increasing order are

$$c_j(t) = \frac{b_j - 1}{1 - t + tb_j}. \quad (\text{AW17})$$

The denominator is positive and the derivative with respect to b_j is $(1 - t + tb_j)^{-2} > 0$. Thus the ordering is valid for the entire closed interval. Each $c_j(t)$ is the derivative of $\log(1 - t + tb_j)$ and has integral $\log b_j$. Integrating (AW16) proves $\mathcal{J}_{h,k} \leq \mathcal{L}_q(D)$; the preceding pointwise-minimum argument proves the first inequality of (AW14). Every ratio in (AW13) lies between 1 and b_n/b_1 , so retaining its coefficient gives $1 + 2(q-1) = 2q-1$ times $\log \kappa(D)$ for the second bound. This proves the theorem for all $q \geq 1$. $\square$

All bounds retain the literal source masses. If $M(1) = cM(0)$, every $b_j = c$, so (AW13) is zero. Directly each original quotient determinant is multiplied at time t by $(1 - t + tc)^q$, and the four displayed factors in (AW5) cancel. For a fixed invertible source-coordinate arrow F , the transported forms are $M' = F^*MF$, all subspaces have their F^{-1} images, and $D' = F^{-1}DF$. Hence the eigenvalues, (AW6) and (AW13) are unchanged under this explicit map.

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Here $\mathbf{M}_z = M(t)|_{t=z}$, $\mathbf{C}_z = C(t)|_{t=z}$ and $\mathbf{A}_z = (U(t) - W(t))|_{t=z}$, with $\mathbf{D} = D$. The exact RMT/ISM source arrow is the identity, and $\mathcal{B}$ is the function (AW5). Write $\mathbf{D} = (\mathbf{M}_0)^{-1}\mathbf{M}_1$, with its complete ordered positive eigenvalues $b_1, \dots, b_{2q+1}$, and let $\mathbf{A}_z$ be this same signed projection difference. Define

$$\begin{aligned} \alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, & H_z^\circ &= I - \frac{(1-z)I + z\mathbf{D}}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (\mathbf{D} - I), & R_z^{[\ell_N]} &= (H_z^\circ)^{\ell_N+1} \mathbf{C}_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I. \end{aligned} \quad (\text{AW17aa})$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1 - z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned} j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}. \end{aligned} \tag{AW17ab}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q - 6)} \longrightarrow 0. \tag{AW17ac}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

The original quartet Gamma theorem, when applied to the actual full quartet packet with displacement $\delta > 0$ and common order m , retains $q = [1 + k(m - 1)](k + 1)^2$ and gives

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) + \max\{-\mathcal{J}_{h,k}^{(5)}, j_{\ell_N} - e_{\ell_N}\}. \tag{AW18}$$

This substitutes the complete supplied Gamma theorem into the lower branches of (AW14) and the constructed signed inequality (AW17ab), then takes their maximum. It retains the stronger actual overlap bound, and also implies the earlier lower bounds with $-\mathcal{L}_q(D)$ and $-(2q - 1) \log \kappa(D)$ respectively. It proves no uniform estimate for the eigenvalues b_j , and asserts no existence of an off-line packet. The new computed objects are the exact window spectra, their source isometry, its arithmetic observation and commutator, and the strengthened finite bound.

299 The one-dimensional quotient: exact angle and nonlinear transport

When $q = 1$, the same four endpoints are 0, 1, 1, 2. Their middle factors cancel in the specified ratio, leaving $\mathcal{B}(t) = \log(V_0(t)/V_2(t))$ with both original masses still present. Write $\chi = S - a$. The actual operators now are $U = Q_2$ and $W = I - P_0$. If $\mathcal{N} = (\chi \mathcal{P}_1)^\perp$ and $P_{\mathcal{N}} = I - Q_2$, then $U - W = P_0 - P_{\mathcal{N}}$. Both projections have rank one.

Let $e = 1 \in H$, $z = R_2(t)[1]$, and retain their actual norms $\langle e, e \rangle_t = V_0(t)$ and $\langle z, z \rangle_t = V_2(t)$. The source splitting gives $z = P_{\mathcal{N}} e$, since $J_2 e = 1$. Thus $\langle e, z \rangle_t = V_2(t)$, and the angle between the two lines has the exact value

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)}. \tag{AW19}$$

Both Grams are positive, so the cosine is positive. It is strictly less than one for the actual measure (AW3). Otherwise $e \in \mathcal{N}$, implying $\int \chi(S)m_t = 0$ and $\int S\chi(S)m_t = 0$. On the unchanged line $\bar{S} = k - S$, and consequently $|\chi(S)|^2 = ((k - \bar{a}) - S)\chi(S)$. The two equalities would force $\int |\chi(S)|^2 m_t = 0$, contrary to the positive Gram of this nonzero polynomial. Hence $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the whole closed interval.

For completeness, take the unit vector $p = e/\sqrt{V_0}$ and write $z/\sqrt{V_2} = \cos \theta p + \sin \theta r$, where r is the resulting unit vector orthogonal to p . The phase is already fixed since $\langle e, z \rangle_t = V_2 > 0$. In this orthonormal pair the exact difference of projections is

$$(P_0 - P_{\mathcal{N}})|_{\text{span}(p,r)} = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & -\sin^2 \theta \end{pmatrix}. \quad (\text{AW20})$$

Its trace is zero and its determinant is $-\sin^2 \theta$. It vanishes on the remaining one-dimensional orthogonal complement. Thus its full eigenvalues are $\sin \theta, 0, -\sin \theta$. Applying the proved rank-one spectral trace bounds to its positive and negative eigenprojections yields

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{AW21})$$

This retains the actual alignment of the two lines.

The function $\Phi(B) = 2 \operatorname{arcosh}(e^{B/2})$ for $B > 0$ has derivative $\Phi'(B) = 1/\sqrt{1 - e^{-B}}$, by differentiating the inverse of the hyperbolic cosine on its positive branch. The preceding strict positivity on the compact interval justifies its smooth composition with $\mathcal{B}(t)$. Divide (AW21) by its positive square-root factor. The four independent generic inequalities (AW16), (AW16a), and the trace-norm and Hilbert–Schmidt inequalities (AW13c) can all be divided by that same factor because $q = 1$ gives $a_q = 1$ and $\mathcal{B} > 0$. Their pointwise minimum integrates to $\mathcal{I}_{h,k}$ in (AW13b); integration of (AW17) bounds it by $\log(b_3/b_1)$. The exact nonlinear endpoint bound is

$$\left| 2 \operatorname{arcosh}(e^{\mathcal{B}(1)/2}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/2}) \right| \leq \mathcal{I}_{h,k} \leq \log(b_3/b_1). \quad (\text{AW22})$$

Combining (AW22) with the signed interval (AW17ab), with the original values $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/2})$ and $L = \mathcal{I}_{h,k}/2 \leq \frac{1}{2} \log(b_3/b_1)$, monotonicity of $2 \log \cosh A$ for $A \geq 0$ gives

$$\begin{aligned} \mathcal{B}(1) &\geq \max\{2 \log \cosh(\max\{0, A_0 - L\}), \mathcal{B}(0) + j_{\ell_N} - e_{\ell_N}\}, \\ \mathcal{B}(1) &\leq \min\{2 \log \cosh(A_0 + L), \mathcal{B}(0) + j_{\ell_N} + e_{\ell_N}\}. \end{aligned} \quad (\text{AW23})$$

This last lower bound can be zero although the actual $\mathcal{B}(1)$ is strictly positive, as proved above. Every step concerns the original $q = 1$ quotient and its degree-two source. No extension of that dimension restriction to a quartet quotient is asserted.

299.1 Complete proof dependencies

The complete AT calculation accompanies this note, together with its 29 complete source dependencies and independent proof review. Those sources prove the original pushforward, tensor signs, full Taylor-unit injections, theta primitive, positive arithmetic and Gamma moments, and the quartet application. Every new assertion (AW7)–(AW23) is proved above on those unchanged finite objects. A separate independent review supplies the spectral calculation and its edge-case checks.

300 Joint trace control on the original source and relation spaces

Retain $h, k, \chi = \chi_{h,k}$ from the marked-product note. Here $q = \deg \chi \geq 1$ denotes the polynomial degree. Use the original coordinate $S = k/2 + iy$, the densities $m = w_h^{*k}$ and $r = r_{1/4}^{*k}$ with their literal masses, and $\mu_x = ((1-x)r + xm) dy$ for $0 \leq x \leq 1$. The common source is $H = \mathcal{P}_{2q}$ in the retained monomial basis. Its positive definite Gram and its original metric derivative are

$$M_x = M_0 + x\dot{M}, \quad M_0 = M_{2q}^\Gamma, \quad \dot{M} = M_{2q}^{\text{ar}} - M_{2q}^\Gamma, \quad T_x = M_x^{-1}\dot{M}. \quad (\text{SP1})$$

The equality $M_x T_x = \dot{M} = \dot{M}^*$ proves that T_x is self-adjoint in the M_x inner product. Its exact relation to the positive M_0 -self-adjoint operator $R = M_0^{-1}M_1$ is $T_x = (I + x(R - I))^{-1}(R - I)$, obtained by factoring $M_x = M_0(I + x(R - I))$. Thus its eigenvalues are $(r - 1)/(1 + x(r - 1))$ for the positive eigenvalues r of R , with their actual signs retained.

The precise arrow to the earlier AW source keeps H and every coefficient fixed, sends t_{AW} to x and u_{AW} to y , and sends the relation map by $B_N^{\text{SP}} = I_N B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow H$. The Gamma convolution constant is $c_a = 2^{1-2a}\Gamma(2a)$, so

$$r_{1/4}^{*k}(y) = \frac{c_{1/4}^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}, \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{SP1a})$$

This is exactly AW3, including its mass $(2\pi)^{k/2}$; both arithmetic densities are w_h^{*k} with mass μ_h^k . Thus $M_x = M_{\text{AW}}(x)$, and $(B_N^{\text{SP}})^* M_x B_N^{\text{SP}} = (B_N^{\text{AW}})^* M_N B_N^{\text{AW}}$ with $M_N = I_N^* M_x I_N$.

300.1 All four endpoints as operators on one source

Let $I_N : \mathcal{P}_N \rightarrow H$ be literal coefficient inclusion, and let $B_N : \mathcal{P}_{N-q} \rightarrow H$ be multiplication by the original monic χ , followed by coefficient inclusion. Define

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{SP2})$$

The second formula is used for $N \geq q$; its Gram is invertible because multiplication by χ is injective on polynomials. Direct multiplication proves that these are the M_x -orthogonal projections onto the original source and relation subspaces. Within either family the subspaces are nested, so their orthogonal projections commute and their product is the smaller projection. Indeed the smaller subspace, its orthogonal complement inside the larger one, and the orthogonal complement of the larger one give three invariant summands on which this statement is immediate.

Let $V_N(x)$ be the original canonical quotient volume. In the fixed quotient-lift and monic-relation coordinates, block determinant gives

$$\log V_N(x) = \log \det(I_N^* M_x I_N) - \log \det(B_N^* M_x B_N). \quad (\text{SP3})$$

The relation determinant is one when its dimension is zero. The fixed monic change-of-basis determinant has modulus one, so SP3 retains every volume factor. Differentiation and cyclicity of finite matrix trace give

$$(\log V_N)' = \text{Tr}(T_x(P_N - Q_N)). \quad (\text{SP4})$$

For example $\text{Tr}(T_x P_N) = \text{Tr}((I_N^* M_x I_N)^{-1} I_N^* \dot{M} I_N)$; the relation formula uses precisely the same calculation. Set

$$F(x) = \log \frac{V_{q-1}(x)V_q(x)}{V_{2q-1}(x)V_{2q}(x)}, \quad \Delta_{h,k} = F(1) - F(0), \quad (\text{SP5})$$

and define two positive operators on this same source:

$$\mathcal{R}_x = Q_{2q-1} + Q_{2q} - Q_q, \quad \mathcal{P}_x = (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q), \quad \mathcal{A}_x = \mathcal{R}_x - \mathcal{P}_x. \quad (\text{SP6})$$

The four original signs in SP4 give

$$F'(x) = \text{Tr}(T_x \mathcal{A}_x). \quad (\text{SP7})$$

For the original row $v = v_{2q}(y)$, the density identity has the exact metric factor

$$\mathbf{R}_x(y) - \mathbf{P}_x(y) = v \mathcal{A}_x M_x^{-1} v^*.$$

Indeed $P_N M_x^{-1} = I_N (I_N^* M_x I_N)^{-1} I_N^*$, and the same identity holds with B_N, Q_N . This recovers the marked-product kernels and proves the common-source matrix comparison.

300.2 Ranks, multiplicities and the retained overlap

The following complete spectra and the final condition-number coefficient were already established in AW7 and AW14. They are retained here with the exact source correspondence above; the actual cross-trace and intersection terms supply the added refinement.

The nested relation spaces in SP6 have dimensions $1, q, q+1$. On the first space $\mathcal{R}_x$ has eigenvalue one; on the next orthogonal increment it has eigenvalue two with multiplicity $q-1$; on the last increment it has eigenvalue one; elsewhere it is zero. The nested source increments in $\mathcal{P}_x$ give the same list. When $q=1$, the multiplicity $q-1$ is zero. Consequently

$$\begin{aligned} \text{rank } \mathcal{R}_x &= \text{rank } \mathcal{P}_x = q+1, & \text{Tr } \mathcal{R}_x &= \text{Tr } \mathcal{P}_x = 2q, \\ \text{Tr } \mathcal{R}_x^2 &= \text{Tr } \mathcal{P}_x^2 = 4q-2, & \text{Tr } \mathcal{A}_x &= 0, \\ \text{Tr } \mathcal{A}_x^2 &= 8q-4-2\text{Tr}(\mathcal{R}_x \mathcal{P}_x). \end{aligned} \quad (\text{SP8})$$

The equal kernel masses $2q$ are traces. The two positive operators have rank $q+1$, with their multiplicity-two directions retained.

The overlap in SP8 is at least one. Each positive operator is at least the orthogonal projection onto its range, since its nonzero eigenvalues are one and two. The two ranges each have dimension $q+1$ in dimension $2q+1$, so their intersection has dimension at least one. Choose a unit vector in this intersection and complete it to an orthonormal basis of the first range. The trace of the product of the two range projections is the sum of squared norms of the second projection on this basis, hence at least one. For positive operators, replacing one factor by a larger positive factor cannot decrease the trace of the product: the difference is $\text{Tr}(B^{1/2}(A' - A)B^{1/2}) \geq 0$. Applying this twice proves the stated bound for the actual operators.

300.3 A bound that uses cancellation before estimating

Write $d_x = \lambda_{\max}(T_x) - \lambda_{\min}(T_x)$. Since $\text{Tr } \mathcal{A}_x = 0$, for every real c

$$F'(x) = \text{Tr}((T_x - cI)\mathcal{A}_x). \quad (\text{SP9})$$

Choose c to be the midpoint of the two extreme eigenvalues. In an M_x -orthonormal eigenbasis e_j of $\mathcal{A}_x$, with real eigenvalues α_j , one obtains

$$|F'(x)| \leq \sum_j |\alpha_j| |\langle e_j, (T_x - cI)e_j \rangle_{M_x}| \leq \frac{1}{2}d_x \sum_j |\alpha_j|.$$

This proves the trace-norm bound directly. Let L_x be the intersection of the two ranges in SP6, with $s_x = \dim L_x \geq 1$, and let E_{L_x} be its orthogonal projection in M_x . Each of $\mathcal{R}_x, \mathcal{P}_x$ is at least E_{L_x} . Thus $\mathcal{A}_x = (\mathcal{R}_x - E_{L_x}) - (\mathcal{P}_x - E_{L_x})$ is a difference of two positive operators of trace $2q - s_x$. The triangle inequality for trace norm gives $\|\mathcal{A}_x\|_1 \leq 4q - 2s_x \leq 4q - 2$. Cauchy-Schwarz on its at most $2q+1$ eigenvalues gives the refinement

$$|F'(x)| \leq \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(\mathcal{R}_x \mathcal{P}_x))} \right\}. \quad (\text{SP10})$$

The overlap is calculated from the unchanged source and relation Grams in SP2. Every cross-pairing remains in that expression.

The complete earlier AW13–AW16 bound applies to this same source through $(U, W, C) = (\mathcal{R}_x, \mathcal{P}_x, T_x)$. Write $c_1(x) \leq \cdots \leq c_{2q+1}(x)$ for all eigenvalues of T_x in M_x and define

$$S_{AW}(x) = c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \quad S_{SP}(x) = \text{the right-hand side of (SP10)}. \quad (\text{SP10a})$$

Here is the complete retained spectral proof. For an orthogonal projection P of rank r , its diagonal values a_j in an orthonormal eigenbasis of T_x satisfy $0 \leq a_j \leq 1$ and $\sum_j a_j = r$. Thus $\text{Tr}(PT_x) = \sum_j c_j a_j$ is at least the sum of the smallest r and at most the sum of the largest r eigenvalues: moving a fixed diagonal mass from a smaller coefficient to a larger one increases this sum. Either $Z = \mathcal{R}_x$ or $Z = \mathcal{P}_x$ equals its range projection, of rank $q+1$, plus its eigenvalue-two projection, of rank $q-1$. Therefore

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZT_x) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{SP10b})$$

Subtracting the two extrema with their exact multiplicities gives $|F'(x)| \leq S_{AW}(x)$. Combining this bound with SP10 before integration proves

$$|F'(x)| \leq \min\{S_{AW}(x), S_{SP}(x)\}. \quad (\text{SP10c})$$

Positive smooth Gram matrices make the eigenvalues and overlaps bounded on $[0, 1]$; s_x is Borel since its rank strata are given by matrix minors. Hence the right side is integrable. The original source isometry of AW10 also yields the exact commutator identity $F' = \text{Tr}(W(T^{\text{iso}})^{-1}[C, T^{\text{iso}}])$. It follows from $U = T^{\text{iso}}W(T^{\text{iso}})^{-1}$ and cyclicity, with T^{iso} the explicit isometry and $C = T_x$ the metric derivative. Its full original observation remains $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}$ and has exact defect $\mathcal{O}T^{\text{iso}} - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}(T^{\text{iso}} - I)$ and kernel $\ker(\mathcal{O}T^{\text{iso}}) = (T^{\text{iso}})^{-1}(\chi \mathcal{P}_q)$. The defect follows by expansion and the kernel follows from injectivity of $\sigma_h^{\otimes k} \eta$ and $\ker J_{2q} = \chi \mathcal{P}_q$. Thus no additional cohomology-intertwining property of the metric isometry is used.

There is a closed integrated bound. Set $R = M_0^{-1}M_{2q}^{\text{ar}}$. It is positive and self-adjoint in the M_0 inner product, since $\langle v, Rv \rangle_{M_0} = \langle v, v \rangle_{M_{2q}^{\text{ar}}} > 0$. Let its smallest and largest eigenvalues be $r_{\min}, r_{\max} > 0$. The literal factorization of M_x gives

$$T_x = (I + x(R - I))^{-1}(R - I), \quad d_x = \frac{r_{\max} - 1}{1 + x(r_{\max} - 1)} - \frac{r_{\min} - 1}{1 + x(r_{\min} - 1)}. \quad (\text{SP11})$$

For fixed x , the function $(r - 1)/(1 + x(r - 1))$ is increasing on $r > 0$, its derivative being $(1 + x(r - 1))^{-2} > 0$. This proves the extreme-eigenvalue formula even though the inner product varies. Integration, including the identically zero integrand when an eigenvalue equals one, yields

$$\int_0^1 d_x dx = \log(r_{\max}/r_{\min}). \quad (\text{SP12})$$

Write $r_1 \leq \cdots \leq r_{2q+1}$ for all positive eigenvalues of R , and retain

$$\mathcal{L}_q(R) = \log \frac{r_{q+2}}{r_q} + 2 \sum_{j=1}^{q-1} \log \frac{r_{2q+2-j}}{r_j}.$$

The ordered form of SP11 gives $c_j(x) = (r_j - 1)/(1 + x(r_j - 1))$ and $\int_0^1 c_j(x) dx = \log r_j$. Therefore SP7–SP12 and the full spectral proof above give both original-source controls

$$\boxed{|\Delta_{h,k}| \leq \int_0^1 \min\{S_{AW}(x), S_{SP}(x)\} dx \leq \mathcal{L}_q(R) \leq (2q - 1) \log \frac{r_{\max}}{r_{\min}}}, \quad (\text{SP13})$$

$$|\Delta_{h,k}| \leq \frac{1}{2} \int_0^1 d_x \|\mathcal{R}_x - \mathcal{P}_x\|_1 dx \leq (2q - 1) \log \frac{r_{\max}}{r_{\min}}.$$

The first integral is also at most $\int_0^1 S_{\text{SP}}(x) dx$. Every logarithmic ratio in $\mathcal{L}_q$ is bounded by $\log(r_{\max}/r_{\min})$, with coefficient total $1 + 2(q - 1) = 2q - 1$; this proves the last inequality explicitly. The two displayed integral bounds each remain available: no ordering between the exact trace norm and the spectral refinement is assumed. When the Grams are proportional, every c_j agrees, both bounds are zero, and the correction is exactly zero without rescaling their original masses.

The exact trace norm and the centered Hilbert–Schmidt estimate are also retained before taking a minimum. On this same source define

$$\begin{aligned} S_{\text{tr}}(x) &= \frac{d_x}{2} \|\mathcal{R}_x - \mathcal{P}_x\|_1, \\ \Sigma(x) &= \text{Tr}(T_x^2) - \frac{(\text{Tr } T_x)^2}{2q + 1}, \\ S_{\text{HS}}(x) &= \sqrt{\Sigma(x) \text{Tr}((\mathcal{R}_x - \mathcal{P}_x)^2)}. \end{aligned} \tag{SP13d}$$

The trace-norm inequality follows by subtracting the midpoint of the extreme eigenvalues from the metric derivative, since the trace of the signed projection difference is zero. In an orthonormal eigenbasis of that difference, every diagonal derivative entry is bounded in modulus by half the spectral diameter; summing with the absolute eigenvalues proves the assertion. For the other bound subtract the trace mean $\text{Tr}(T_x)I/(2q + 1)$ instead. The square of its Hilbert–Schmidt norm in the unchanged source metric is exactly $\Sigma(x)$. Hilbert–Schmidt Cauchy–Schwarz with the signed difference proves that its trace pairing is bounded by $S_{\text{HS}}(x)$. Positivity of both factors under the square root follows from their self-adjointness. Thus both functions bound the same absolute signed derivative on general positive source forms; the angle term below still uses the specified moment measures. These are RMT10a–10b and ISM25–26 through the already proved original coefficient and projection maps.

On the original moment source (SP1a), the later angle calculation propagates further at this endpoint. With $a_q = 2q - 1$ define

$$\begin{aligned} A(x) &= a_q \sqrt{1 - e^{-F(x)/a_q}} d_x, \\ J^{(5)} &= \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{tr}}(x), S_{\text{SP}}(x), S_{\text{HS}}(x), A(x)\} dx, \\ I &= \int_0^1 \min \left\{ d_x, \frac{S_{\text{AW}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \frac{S_{\text{SP}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \right. \\ &\quad \left. \frac{S_{\text{tr}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}}, \frac{S_{\text{HS}}(x)}{a_q \sqrt{1 - e^{-F(x)/a_q}}} \right\} dx. \end{aligned} \tag{SP13a}$$

The exact map to RMT is $a_{\text{RMT}} = a_q$, $F = \mathcal{B}$, $T_x = C_x$ and $(\mathcal{R}_x, \mathcal{P}_x) = (U_x, W_x)$, with the same original moment measures by SP1a. RMT11–RMT12 proves $F > 0$ and $|F'| \leq A$: its full two angle lists keep the designated zero angle, and vanishing F would imply $1 \perp \chi \mathcal{P}_q$, contradicting $\int |\chi|^2 d\mu_x > 0$ through $\chi^{\#k}(S) = \sum_j \overline{\chi_j}(k - S)^j$. Thus the two-bound proof above remains valid for general positive forms, while on these actual moment forms its displayed first bound in SP13 strengthens to

$$|\Delta_{h,k}| \leq J^{(5)} \leq \int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx. \tag{SP13b}$$

For every integer $\ell_N \geq 0$ the constructed signed approximation is applied on these exact endpoint forms as well. Use $M_z = M_x|_{x=z}$, $C_z = T_x|_{x=z}$, $A_z = (\mathcal{R}_x - \mathcal{P}_x)|_{x=z}$, and $D = R$. The same-source RMT/ISM function $\mathcal{B}$ is exactly F in SP5. Write $D = (M_0)^{-1}M_1$, with its complete ordered positive

eigenvalues $b_1, \dots, b_{2q+1}$, and let A_z be this same signed projection difference. Define

$$\begin{aligned}\alpha_z^\circ &= \frac{2(1-z) + z(b_1 + b_{2q+1})}{2}, & H_z^\circ &= I - \frac{(1-z)I + zD}{\alpha_z^\circ}, \\ C_z^{[\ell_N]} &= (\alpha_z^\circ)^{-1} \sum_{r=0}^{\ell_N} (H_z^\circ)^r (D - I), & R_z^{[\ell_N]} &= (H_z^\circ)^{\ell_N+1} C_z, \\ E_z^{[\ell_N]} &= R_z^{[\ell_N]} - \frac{\text{Tr } R_z^{[\ell_N]}}{2q+1} I.\end{aligned}\tag{SP13ea}$$

The symbols $\alpha_z^\circ, H_z^\circ$ preserve every original source parameter and relation map; ℓ_N is only the finite series truncation. Multiplication of the finite geometric sum gives $C_z - C_z^{[\ell_N]} = R_z^{[\ell_N]}$. All factors are real functions of the positive relative operator D , hence self-adjoint in the original $M_z = M_0((1-z)I + zD)$ metric. Since $\text{Tr } A_z = 0$, the derivative of the same four-volume function is its signed center $\text{Tr}(C_z^{[\ell_N]} A_z)$ plus $\text{Tr}(E_z^{[\ell_N]} A_z)$. Consequently

$$\begin{aligned}j_{\ell_N} &= \int_0^1 \text{Tr}(C_z^{[\ell_N]} A_z) dz, \\ e_{\ell_N} &= \int_0^1 \sqrt{\text{Tr}((E_z^{[\ell_N]})^2) \text{Tr}(A_z^2)} dz, \\ j_{\ell_N} - e_{\ell_N} &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_{\ell_N} + e_{\ell_N}.\end{aligned}\tag{SP13eb}$$

For this block $\mathcal{B}$ denotes precisely the four-volume function identified above. The signed inequalities follow by Hilbert–Schmidt Cauchy–Schwarz in that unchanged metric and integration of the exact derivative. The full residual calculation ISM29–37, equivalently RMT19–21, uses $\vartheta = (b_{2q+1} - b_1)/(b_{2q+1} + b_1) < 1$ and $K_D = \sum_j (|b_j - 1|/\min\{1, b_j\})^2$ to prove

$$0 \leq e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_D(8q-6)} \longrightarrow 0.\tag{SP13ec}$$

Indeed every eigenvalue of H_z° has modulus at most ϑ , centering the residual subtracts a nonnegative square from its squared Hilbert–Schmidt norm, and $\text{Tr}(A_z^2) \leq 8q - 6$ by the full flag spectra and the common range. This proves the bound with all eigenvalue multiplicities retained. The displayed integrals are exact; a numerical evaluation must retain its own integration error. Intersecting this proved signed interval with the already proved absolute and nonlinear intervals supplies the additional endpoint entries below, for every specified ℓ_N .

For $H_0 = 2 \text{arcosh}(e^{F(0)/(2a_q)})$, the derivative $[2 \text{arcosh}(e^{B/(2a_q)})]' = 1/(a_q \sqrt{1 - e^{-B/a_q}})$ and its inverse $2a_q \log \cosh(z/2)$ give the actual endpoint interval

$$\begin{aligned}F(1) &\geq \max \left\{ 0, F(0) - J^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - I\}}{2}, F(0) + j_{\ell_N} - e_{\ell_N} \right\}, \\ F(1) &\leq \min \left\{ F(0) + J^{(5)}, 2a_q \log \cosh \frac{H_0 + I}{2}, F(0) + j_{\ell_N} + e_{\ell_N} \right\}, \quad I \leq \log(r_{\max}/r_{\min}).\end{aligned}\tag{SP13c}$$

The nonlinear branches follow by dividing all five derivative bounds by the same positive factor, integrating, and applying the increasing inverse. Intersecting with the constructed signed interval (SP13eb) gives the additional signed branches; compactness and $F > 0$ justify every denominator. It is precisely RMT13–RMT22 on the stated unchanged original source, including the full local signed construction (SP13ea)–(SP13ec).

SP1–SP13 give finite two-sided control on every original packet. They do not bound the displayed condition number or projection overlap along the growing arithmetic family. The next calculation can use these exact quantities, their original moment formulae, and the newly retained singular-boundary geometry; none has been replaced by an assumed asymptotic estimate or an RH verdict.

301 Exact comparison of inherited EW transport with AGT and AA

13 September 2026. This add-only proof and attribution note compares the complete inherited `gamma_endpoint_window_bridge.tex` (EW.1–EW.59, including EW.35a–c and EW.46a) with the complete sealed AGT1–AGT19 and AA1–AA44 sources. All three bodies were read. No sealed source or package is changed by this note.

The inherited EW source already proves the general canonical projection and principal-angle calculation, nonlinear arcosh transport, strict positivity for the original positive polynomial moments, and the explicit signed tensor theta primitive. The later AGT calculation explicitly sharpens the chosen window by changing the pairing of the same four projections: its guaranteed scalar angle count is $2q - 1$, whereas EW.35a–c use $2q$ at that window. The proof below establishes precisely this refinement and its attribution. Strict positivity and existence of the theta primitive are not additional results of AGT or AA relative to EW.

301.1 1. Exact source pins and reading scope

The base directory in this table is `source-workspace`.

Complete source records

Source

EW

Relative path

`output/tau_split_zero_counterfactual_reconstruction_20260913/original_programme/tex/gamma_endpoint_window_bridge.tex`

SHA-256

`400541cdcb35d44c0f7e68f5db1107df4fcf707d0be83b5a8a970d8ec9c73160`

Reading scope

Entire source, EW.1–59 and all prose, including appended-letter equations

Source

Identical inherited reader copy

Relative path

`output/tau_split_zero_counterfactual_reconstruction_20260913/original_programme/sources/reader_preparation/gamma_endpoint_window_bridge.tex`

SHA-256

`400541cdcb35d44c0f7e68f5db1107df4fcf707d0be83b5a8a970d8ec9c73160`

Reading scope

Byte equality to the fully read EW source

Source

AGT

Relative path

`work/tau_all_degree_angle_transport_20260913.tex`

SHA-256

cc0c0c008b0532c5f120d1cea76b94e2d08153c87a694ce18775a5189c72d5f8

Reading scope

Entire corrected AGT1–19 source

Source

AA

Relative path

work/tau_all_degree_angle_independent_review_20260913.md

SHA-256

b0634428d69f80dedd4f68766ef9e6468c390eaac86dfe58b08c21c8fb896e03

Reading scope

Entire AA1–44 proof and acceptance scope

The EW introduction identifies its input criterion as PR 23 source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f. That is an internal provenance statement in the read source; this comparison does not assign an independently verified publication date or external mathematical priority to it. In particular, “additional” below means additional explicit content relative to these pinned bodies. The inherited source also attributes EW.35a–c to its independent projection review. This attribution is retained.

The rational calculations EW.49–59 and their written derivations were read. Their reported checker executions were not rerun in this comparison, and their counts are not transferred to AGT or AA. The other archived EW copies and checker programs are not silently substituted for the pinned source above.

301.2 2. Original source, parameters, metrics and maps

Use EW’s original full packet $h, g = 2\xi$, integer $k \geq 1$, original line $S = k/2 + iu$, and actual cyclic relation χ of degree $q \geq 1$. Its quotient $\mathcal{C} = \mathbb{C}[S]/(\chi)$ and AGT/AA’s E have precisely the same ordered remainder coordinates. The coefficient map

$$I_E : \mathcal{C} \longrightarrow E, \quad \sum_{j=0}^{q-1} x_j[S^j] \longmapsto \sum_{j=0}^{q-1} x_j[S^j] \quad (\text{EP1})$$

is the identity algebra isomorphism, with inverse the same coefficient map. It retains all multiplicities of χ . For the ambient source specialize EW’s parameters by

$$n = q, \quad r = q, \quad m_{\text{EW}} = n + r = 2q, \quad s_{\text{EW}} = \min(q, r) = q, \quad \lambda = \frac{1}{4}. \quad (\text{EP2})$$

The symbol m_{EW} is an ambient polynomial cutoff; the symbol m in AGT and AA is instead the count $2q - 1$ in a later scalar inequality. They do not denote the same integer. The exact relation between them at EP2 is $m_{\text{AGT}} = m_{\text{EW}} - 1$, while the ambient vector-space dimension is $m_{\text{EW}} + 1 = 2q + 1$.

The gamma densities coincide literally under EP2. Indeed EW.5 gives

$$c_{1/4} = 2^{1-2/4} \Gamma(2/4) = \sqrt{2\pi}, \quad r_{1/4}^{*k}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi} = m_0(u), \quad (\text{EP3})$$

using EW.7. The gamma mass is $(2\pi)^{k/2}$, and EW.1's arithmetic density is AGT2/AA2's $m_1 = w_h^{*k}$, with mass μ_h^k . Therefore the common identity coefficient map on $\mathcal{P}_{2q}$ identifies the two measures at both endpoints without a mass rescaling. Along the segment $m_t = (1-t)m_0 + tm_1$, define $M(t)$ as its Gram in the fixed monomial frame. EW's general positive-family theorem applies with

$$H_{m_{EW}}(t) = M(t), \quad H_N(t) = \iota_N^* M(t) \iota_N, \quad \dot{H}_{m_{EW}} = \dot{M} = M(1) - M(0), \quad (\text{EP4})$$

where $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}^{2q+1}$ is the original coefficient inclusion. The scalar relation χ , the monic remainder J_N , and multiplication $B_N = \times \chi$ are fixed along this same path.

EW.11 supplies

$$K_N^{EW} = J_N H_N^{-1} J_N^*, \quad G_N = (K_N^{EW})^{-1}, \quad R_N^{EW} = H_N^{-1} J_N^* G_N, \quad R_N^{AGT} = R_N^{AA} = \iota_N R_N^{EW}. \quad (\text{EP5})$$

The last equality follows both by the displayed matrix formula and by uniqueness of the minimum section: the included EW vector has remainder x , is orthogonal to $B_N \mathcal{P}_{N-q}$, and any other lift differs by a vector in that space. The squared norm of the latter is added orthogonally, so it cannot be another minimum unless the difference is zero. Hence every G_N and $V_N = \det G_N$ is the same original quantity. In particular, EW's K_N^{EW} is the inverse quotient Gram; AGT's $K_{ij} = G_i^{-1/2} G_j G_i^{-1/2}$ is a two-cutoff relative Gram. Their exact relation is the displayed substitution of $G_N = (K_N^{EW})^{-1}$, with this factor order preserved.

There is a further operator-type distinction requiring an explicit map. EW.30 defines a contravariant matrix

$$Q_N^{EW} = R_N G_N^{-1} R_N^*, \quad \Pi_N = Q_N^{EW} M = Z_N^{AGT} = N_N^{AA}. \quad (\text{EP6})$$

Here and below R_N on the common ambient space includes ι_N . To prove EP6, compute $\Pi_N^2 = R_N G_N^{-1} (R_N^* M R_N) G_N^{-1} R_N^* M = \Pi_N$. Its adjoint for M is itself, and its range is $R_N E$. Thus it is exactly the orthogonal projection that AGT denotes by $Z_N = P_N - Q_N$ and AA by N_N . EW's Q_N^{EW} is converted to that endomorphism by right multiplication by the original M ; it is not identified with AGT's relation projection Q_N^{AGT} .

For completeness EW's P_j^{EW} is the contravariant matrix of the single monic orthogonal line. Those lines span the cutoff source, so the remaining exact conversions are

$$P_N^{AGT} = \sum_{j=0}^N P_j^{EW} M, \quad Q_N^{AGT} = P_N^{AGT} - Q_N^{EW} M. \quad (\text{EP7})$$

They follow by decomposing a polynomial in the complete monic orthogonal family and then subtracting its minimum-section component. Every factor M remains in the written map.

Set $C = M^{-1} \dot{M}$, the AGT4/AA21 operator. The map $v \mapsto M^{1/2} v$ is an isometry from $(\mathcal{P}_{2q}, M)$ to its displayed Euclidean coefficient space, and

$$M^{1/2} C M^{-1/2} = M^{-1/2} \dot{M} M^{-1/2}, \quad \mathfrak{o}_H(\dot{H}) = c_{\max}(C) - c_{\min}(C). \quad (\text{EP8})$$

The trace also has the exact original-coordinate conversion

$$\text{Tr}(Q_N^{EW} \dot{M}) = \text{Tr}(\Pi_N C). \quad (\text{EP9})$$

Indeed $\Pi_N C = Q_N^{EW} M M^{-1} \dot{M}$. Thus EW.31–32 and AGT4/AA23 have identical signed determinant derivatives. The positive relative operators $M(0)^{-1} M(1)$ and $M(0)^{-1/2} M(1) M(0)^{-1/2}$ are conjugate by $M(0)^{1/2}$, so EW.34 and AGT/AA use exactly the same positive relative eigenvalues and κ .

301.3 3. Existing angle proof and the exact changed pairing

The equation-level crosswalk is as follows.

Inherited EW statement	AGT statement	AA statement	Mathematical scope
EW.1–7 and EW.3–4	AGT1–3	AA1–5	Same source at EP2–4, full packet, unit, exact quotient and masses
EW.25–27 and proof of EW.28	AGT5–8	AA6, AA8–16	Same canonical projection, Gram cosines, all common directions, and signed 2×2 projection blocks
EW.31–32	AGT4	AA21–24	Same four signed derivatives after EP6–9
EW.28 rank bound and EW.35a concavity	AGT9–11, AGT15–16	AA17–26	Later crossed pairing gives $2q - 1$ instead of EW.35a's $2q$ at $n = r = q$
EW.35b–c	AGT14, AGT17–19	AA32–37	Same integration and inverse-cosh construction, with that changed count
EW.46a and its proof	AGT12–13	AA27–31	Strict positivity already proved; later direct involution proof supplies another calculation at the selected cutoffs
EW.46–48, with EW.3	AGT paragraph after AGT8	AA38–44	Same exact theta primitive; AA adds expanded division details and the explicit energy matrix coordinate

To establish the shared angle calculation without presuming a change of presentation is an identity, put $L_i = R_i E$. EW.26 proves $R_j = \Pi_j R_i$ for $i \leq j$, since both vectors have remainder x and their difference is in the relation space orthogonal to L_j . Thus

$$R_i^* M R_j = G_j, \quad G_i - G_j = (R_i - R_j)^* M (R_i - R_j). \quad (\text{EP10})$$

For isometries $F_i = R_i G_i^{-1/2}$, the two cross matrices in AGT7 and AA9 are actual adjoints:

$$F_j^* M F_i = G_j^{1/2} G_i^{-1/2}, \quad F_i^* M F_j = G_i^{-1/2} G_j^{1/2}, \quad K_{ij} = G_i^{-1/2} G_j G_i^{-1/2}. \quad (\text{EP11})$$

Their squared singular values are the eigenvalues $c_a^2 \in (0, 1]$ of K_{ij} . If $x_a \in L_i$ are orthonormal eigenvectors for the squared projection restriction and $y_a = \Pi_j x_a / c_a$, then y_a form an orthonormal basis of L_j , with $\langle x_a, y_b \rangle = c_b \delta_{ab}$. For $c_a = 1$, equality in orthogonal Pythagoras gives $x_a = y_a \in L_i \cap L_j$, and conversely every common vector contributes eigenvalue one. For $s_a = \sqrt{1 - c_a^2} > 0$, put $z_a = (y_a - c_a x_a) / s_a$. Computing these inner products gives orthonormal, mutually orthogonal two-dimensional planes, on which

$$(\Pi_i - \Pi_j)|_{\text{span}(x_a, z_a)} = \begin{pmatrix} s_a^2 & -s_a c_a \\ -s_a c_a & -s_a^2 \end{pmatrix}. \quad (\text{EP12})$$

This has eigenvalues $s_a, -s_a$; every remaining direction has eigenvalue zero. Taking determinants in EP11 yields $\log(V_i/V_j) = \sum_a -\log c_a^2$. These are precisely the calculations already written in the complete EW.26–28 proof.

EW also proves that at most $\min(q, j - i)$ of the angles move. Indeed L_i is already orthogonal to the relation columns up through degree i . The extra $j - i$ independent monic boundary columns impose at most $j - i$ linear equations on L_i , and their joint kernel is exactly $L_i \cap L_j$. Hence the common-subspace dimension is at least $q - (j - i)$ if positive. This exact rank bound includes all common vectors, and for $i = j$ all angles are zero.

At EP2, the original loss and derivative are

$$B = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}} = \mathcal{L}_{n=q, r=q}, \quad B' = \text{Tr}((\Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q})C). \quad (\text{EP13})$$

EW.33–35a use the pairs $(q-1, 2q-1)$ and $(q, 2q)$. Each has gap q , hence its displayed estimate uses up to q moving angles and a list of length $2q$. AGT instead uses

$$(q-1, 2q), \quad (q, 2q-1), \quad (\text{EP14})$$

whose gaps are $q+1$ and $q-1$. EW's already proved rank bound gives q and $q-1$ moving slots respectively. In AGT9 the latter is also proved directly by placing both q -dimensional spaces in the common $2q-1$ -dimensional hyperplane. It is the same incidence bound. The reordered sum in EP13 is exactly unchanged. One zero angle from the full second list can therefore be omitted from the scalar sum, retaining every original vector space and every other zero contribution, for a count $2q-1$.

For a self-adjoint C with width $w = c_{\max} - c_{\min}$, each positive/negative spectral pair in EP12 contributes to the trace by $s_a(\langle v_a, Cv_a \rangle - \langle w_a, Cw_a \rangle)$, in absolute value at most $s_a w$, since both Rayleigh quotients are in the same interval. Summing gives the shared angle estimate. For $f(x) = \sqrt{1 - e^{-x}}$, its full second derivative is

$$f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} = -\frac{e^{-x}(2 - e^{-x})}{4(1 - e^{-x})^{3/2}} < 0. \quad (\text{EP15})$$

The two displayed formulas in EW/AGT and AA are therefore identical. Concavity extends to zero by continuity, and finite Jensen on any retained list of a slots with total cost B gives

$$|B'| \leq a\sqrt{1 - e^{-B/a}} w. \quad (\text{EP16})$$

EW.35a gives this with $a = 2q$ at EP2, while AGT16/AA26 give it with $a = 2q-1$. The general theorem and concavity method predate the latter specialization in the pinned source graph.

301.4 4. Full general-window consequence of the crossed pairing

The same refinement can be expressed directly in EW's general domain $n \geq q \geq 1, r \geq 1, m_{\text{EW}} = n + r$. Retain its exact $\mathcal{L} = \log(V_{n-1}V_n/(V_{n+r-1}V_{n+r}))$. Use the two nested pairs

$$(n-1, n+r), \quad (n, n+r-1), \quad a(q, r) = \min(q, r+1) + \min(q, r-1). \quad (\text{EP17})$$

For $r = 1$ the second pair is identical, contributes its full list of unit cosines, and contributes zero moving slots. In every case $a(q, r) \geq 1$. The two gap rank bounds just proved show that their nonzero angle costs fit in $a(q, r)$ slots. Append zero scalar terms if there are fewer. Their sum is the same original $\mathcal{L}$, because the two log determinant ratios telescope to the four original endpoints. The trace signs are also the same. The preceding spectral-pair estimate and EP15 therefore prove

$$|\mathcal{L}'| \leq a(q, r)\sqrt{1 - e^{-\mathcal{L}/a(q, r)}} \mathfrak{o}_H(\dot{H}). \quad (\text{EP18})$$

The count has the exact elementary form

$$a(q, r) = \begin{cases} 2\min(q, r) - 1 = 2q - 1, & r = q, \\ 2\min(q, r), & r \neq q. \end{cases} \quad (\text{EP19})$$

For $r < q$, the two terms are $r + 1, r - 1$; for $r = q$, they are $q, q - 1$; and for $r > q$, both are q . This proves all cases, including $q = r = 1$. In particular the explicit improvement is available for every $n \geq q$ when $r = q$, while AGT states it at $n = q$. This paragraph is an add-only consequence derived here from EW.28 and its transport proof, and is not attributed as an extra assertion already present in the AGT body.

For any positive integer a , define the exact increasing map and its inverse by

$$\Psi_a(B) = 2 \operatorname{arcosh}(e^{B/(2a)}), \quad \Psi_a^{-1}(v) = 2a \log \cosh(v/2), \quad B, v \geq 0. \quad (\text{EP20})$$

The real branch is $\operatorname{arcosh}(z) = \log(z + \sqrt{z^2 - 1})$ for $z \geq 1$. Differentiating $\cosh(\Psi_a(B)/2) = e^{B/(2a)}$ gives $\Psi'_a(B) = [a\sqrt{1 - e^{-B/a}}]^{-1}$ for $B > 0$. For possible zero losses along a general positive C^1 family, apply the derivative to $\Psi_a(\mathcal{L} + \varepsilon)$. EP18 bounds the absolute derivative by

$$\mathfrak{o}_H(\dot{H}) \sqrt{\frac{1 - e^{-\mathcal{L}/a}}{1 - e^{-(\mathcal{L} + \varepsilon)/a}}} \leq \mathfrak{o}_H(\dot{H}). \quad (\text{EP21})$$

Integrate and let $\varepsilon \downarrow 0$ by continuity. For the original straight positive segment, put $D = (H^{(0)})^{-1}H^{(1)}$. It is self-adjoint positive for $H^{(0)}$, and every positive relative eigenvalue b gives exactly $c_b(t) = (b - 1)/(1 - t + tb)$. Since its derivative with respect to b is $(1 - t + tb)^{-2} > 0$, the same extreme b 's determine the width throughout the path. Its integral is

$$\int_0^1 \mathfrak{o}_{H(t)}(\dot{H}) dt = \int_0^1 \frac{d}{dt} \log \frac{1 - t + tb_{\max}}{1 - t + tb_{\min}} dt = \log \kappa. \quad (\text{EP22})$$

Thus the crossed-pair general-window consequence is

$$|\Psi_{a(q,r)}(\mathcal{L}^{(1)}) - \Psi_{a(q,r)}(\mathcal{L}^{(0)})| \leq \log \kappa. \quad (\text{EP23})$$

It holds including zero loss for every such positive matrix pair. At $n = r = q$, it is AGT14/AA35; with $a = 2s_{\text{EW}}$, EP20 is exactly EW's Φ_s , and the proof is EW.35b's existing integration argument. This identifies both the inherited map and its changed parameter rather than giving a new name to the inherited result.

The endpoint interval is equally the same exact inverse construction. Put $A_0 = \operatorname{arcosh}(e^{\mathcal{L}^{(0)}/(2a)})$ and $d = \frac{1}{2} \log \kappa$. Equation EP23 is equivalent to

$$2a \log \cosh(\max\{0, A_0 - d\}) \leq \mathcal{L}^{(1)} \leq 2a \log \cosh(A_0 + d). \quad (\text{EP24})$$

Exponentiating and writing $z_0 = e^{A_0}$, $z_- = \max\{1, z_0/\sqrt{\kappa}\}$, $z_+ = \sqrt{\kappa} z_0$ gives

$$\left(\frac{z_- + z_-^{-1}}{2}\right)^{2a} \leq \mathcal{R}^{(1)} \leq \left(\frac{z_+ + z_+^{-1}}{2}\right)^{2a}. \quad (\text{EP25})$$

For $a = 2s$, the exponent $2a = 4s$ is exactly EW.35c. The replacement $a = 2q - 1$ is therefore a coefficient refinement of that inherited endpoint certificate. The original source-norm factor $U = \omega_{n+r}/\omega_n$ remains present in EW.20's actual endpoint $U^{1/(2r)} \sinh(\mathcal{L}/(2r))$; AGT's estimate concerns its four-volume argument and does not replace the complete EW endpoint construction.

The improvement is strict at the level of these scalar constraints for distinct positive losses. To prove it, for fixed $B > 0$ let $F_a(B) = a\sqrt{1 - e^{-B/a}}$ and set $x = B/a > 0$. Direct differentiation gives

$$\frac{\partial F_a(B)}{\partial a} = \frac{2 - (2 + x)e^{-x}}{2\sqrt{1 - e^{-x}}} > 0. \quad (\text{EP26})$$

The numerator is positive because $e^x > 1 + x/2$ for $x > 0$: the difference has value zero at zero and derivative $e^x - 1/2 > 0$. Therefore $F_{2q-1}(B) < F_{2q}(B)$. Since $\Psi'_a(B) = 1/F_a(B)$, for $0 < B_0 < B_1$

$$\Psi_{2q-1}(B_1) - \Psi_{2q-1}(B_0) > \Psi_{2q}(B_1) - \Psi_{2q}(B_0). \quad (\text{EP27})$$

Integration proves this inequality; continuity gives the non-strict comparison when endpoints coincide or a zero endpoint is admitted. Thus the $2q-1$ constraint implies the inherited $2q$ constraint and is tighter for a nonzero change of positive losses. No claim of sharpness for actual arithmetic measures or any bound uniform in growing k, q, n is inferred from this comparison.

301.5 5. Strict positivity: prior theorem and alternate later proof

EW.46a already proves $\mathcal{L} > 0$ for every admitted nonempty actual positive polynomial window. Its proof uses the actual real- u moment form. Here is its full link to the later one.

Let π_j be its real monic orthogonal polynomials, with positive squared norms ω_j . The matrix of multiplication by real u is self-adjoint. Degree and orthogonality imply

$$u\pi_j = \pi_{j+1} + b_j\pi_j + a_j\pi_{j-1}, \quad b_j \in \mathbb{R}, \quad a_j = \omega_j/\omega_{j-1} > 0 \quad (j \geq 1). \quad (\text{EP28})$$

Indeed the coefficient against π_l , $l \leq j-2$, vanishes by moving u to π_l , whose degree is at most $j-1$. The coefficient of π_{j-1} is $\langle \pi_{j-1}, u\pi_j \rangle / \omega_{j-1} = \omega_j/\omega_{j-1}$, since the leading coefficient of $u\pi_{j-1}$ against π_j is one. The diagonal coefficient is real by self-adjointness. The $j=0$ equation lacks its last term.

On the original $S = c + iu$ line, the exact monic phase map $p_j(S) = i^j \pi_j((S - c)/i)$ has modulus-one factor and retains ω_j . Multiplying EP28 by its full factor gives

$$Sp_j = p_{j+1} + (c + ib_j)p_j - a_j p_{j-1}. \quad (\text{EP29})$$

If nonconstant χ divided consecutive p_j, p_{j+1} , EP29 and $a_j \neq 0$ force divisibility of p_{j-1} , and descending induction forces divisibility of $p_0 = 1$, impossible. The two residual monomial columns at degrees $N, N+1$, after projecting away $\mathcal{P}_{N-1}$, are p_N and $p_{N+1} + zp_N$, with $z = \langle p_N, S^{N+1} \rangle / \omega_N$. Their quotient residual map F_2 cannot vanish, or both consecutive p 's would have zero remainder and be divisible by χ . Thus $F_2^* G_{N-1} F_2$ is nonzero positive semidefinite. EW.17 then gives $V_{N-1}/V_{N+1} > 1$, because the determinant of identity plus a nonzero positive semidefinite matrix is strictly greater than one. Multiplying the exact overlapping factors gives

$$\frac{V_{n-1}V_n}{V_{n+r-1}V_{n+r}} = \prod_{j=0}^{r-1} \frac{V_{n+j-1}}{V_{n+j+1}} > 1. \quad (\text{EP30})$$

This is precisely EW.46a. It applies to every positive measure along the AGT convex moment path as well, since that measure remains positive on nonzero polynomials and has every needed moment. Consequently AGT/AA's strict positivity statement already follows from the inherited proof.

AGT12-13 and AA28-30 give a different direct proof for the specific first pair in EP14. If $B = 0$, all its angles vanish, and $L_{q-1} = L_{2q}$. The first space is $\mathcal{P}_{q-1}$; the second is orthogonal to $\chi\mathcal{P}_q$. Thus $1 \perp \chi\mathcal{P}_q$. For $\chi(S) = \sum_{j=0}^q a_j S^j$, define the full antilinear involution

$$\chi^{\#_k}(S) = \sum_{j=0}^q \overline{a_j}(k - S)^j. \quad (\text{EP31})$$

It has degree q , leading coefficient $(-1)^q \overline{a_q}$, and equals $\overline{\chi(S)}$ on $S = k/2 + iu$, because $\overline{S} = k - S$. Hence $1 \perp \chi\mathcal{P}_q$ would imply $0 = \langle 1, \chi\chi^{\#_k} \rangle = \int |\chi(k/2 + iu)|^2 m_t(u) du > 0$. This contradiction is a

shorter alternate proof at the chosen cutoffs. Its exact involution calculation is additional proof detail relative to EW's recurrence proof; the strict positivity conclusion is already inherited, in a broader window domain.

301.6 6. The theta primitive is the same original cochain

EW.46's coefficient κ_x and AA7's $X_{ij}x$ are equal under EP5. To prove this with its full coefficient metric, put $H_j = B_j^* M B_j$, using the original inclusion of B_j in the common source. Then

$$X_{ij} = H_j^{-1} B_j^* M R_i : E \rightarrow \mathcal{P}_{j-q}, \quad R_i - R_j = B_j X_{ij}, \quad G_i - G_j = X_{ij}^* H_j X_{ij}. \quad (\text{EP32})$$

The orthogonal projection onto the relation space is $B_j H_j^{-1} B_j^* M$. Applying it to R_i gives $R_i - R_j$, by EP10. This proves the second equality and then its Gram energy. For the empty relation domain the maps are its unique zero maps and no nonzero inverse is needed. Monic division makes multiplication by χ injective, so both the EW relation $R_i x - R_j x = \chi \kappa_x$ and EP32 force $\kappa_x = X_{ij} x$. In AA the name H_j denotes H_j here; EW's name H_j denotes the full source Gram from EP4. EP32 specifies the exact relationship and avoids identifying those two forms.

Retain the original $D = -x\partial_x$, $\mathcal{M}F_h = g/h$, and seed

$$\phi_0 = \phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad h(D)F_h = \Theta\phi_0. \quad (\text{EP33})$$

EW explicitly supplies the same successive divisions as AA40. Start with $F_0 = \chi(s_1 + \cdots + s_k)$, and divide successively in the fixed order $s_1, \dots, s_k$: $F_{j-1} = h(s_j)L_j + F_j$, $\deg_{s_j} F_j < \deg h$. The polynomial $h(s_j)$ is monic with complex coefficients, so each division is unique over the ring in the other variables and does not increase an earlier restricted degree. The final remainder is in the tensor monomial basis of $E_h^{\otimes k}$. It has zero quotient image because χ annihilates the original sum operator on the unit; thus that remainder is zero. Therefore

$$\chi(\sum s_j) = \sum_{j=1}^k h(s_j)L_j(\mathbf{s}). \quad (\text{EP34})$$

EW's Q_j and AA's L_j are the same coefficient polynomials if the same stated successive-division order is used. This follows inductively from the uniqueness of each monic division; every F_{j-1} is then the same polynomial in the two constructions. In particular no simple-root replacement is made and every full-order nilpotent relation remains in EP34.

For the one-leg complex $[V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, both sources consequently write the identical degree- $k-1$ cochain

$$\mathcal{P}_\chi(P) = \mathcal{K}_\chi(P) = \sum_{j=1}^k (-1)^{j-1} L_j(\mathbf{D}) P(\sum D_i) (F_h^{\otimes(j-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-j)}). \quad (\text{EP35})$$

The only factor on which the tensor differential acts in the j -th summand is ϕ_0 in degree zero. There are exactly $j-1$ preceding degree-one factors, so the differential contributes $(-1)^{j-1}$, cancelling the written coefficient. Polynomial D_i 's have degree zero and commute with the differential through $D\Theta = \Theta D$. Equations EP33–34 then give, term by term,

$$d\mathcal{P}_\chi(P) = P(\sum D_i) \sum_j h(D_j)L_j(\mathbf{D}) F_h^{\otimes k} = \mathcal{V}_{h,k}(\chi P). \quad (\text{EP36})$$

This is EW.47–48 and AA42–43, with the same seed, signs, coefficients, and factor $g = 2\xi$. Substituting $P = \kappa_x = X_{ij}x$ gives precisely the same boundary of the change of canonical representative. For $k = 1$, the cyclic annihilator is the original monic h , the division gives $L_1 = 1$, and the identity reduces to EP33 with the polynomial factor $P(D)$.

EW.3 and AGT1/AA44 also preserve the exact same full-unit observation $\eta[P] = [g/h]_h^{\otimes k} P(s_1 + \dots + s_k)$. The further supported cohomology observation is the previously supplied $\sigma_h^{\otimes k} \eta$. The one-leg inclusion used in AA maps $\phi \mapsto (\phi, 0)$ into the original two-leg complex and is the identity in degree one, so the original difference differential still sends this pair to $\Theta\phi$. It therefore carries EP35–36 to the original tau-base complex with no sign change. At every retained support label A , the map $(A, v) \mapsto (A, fv), \tau \mapsto \tau$, sends a zero coefficient to the represented zero at A and external absence to external absence. This exact support rule is already stated after EW.48.

The explicitly displayed additional coordinate in AA7/AA44 is the energy $X_{ij}^* H_j X_{ij}$ in EP32 and its simultaneous presentation with the primitive. That energy is already an immediate orthogonal Pythagoras consequence of EW.11 and EW.26, as proved in EP10 and EP32. It must not be described as a new existence theorem for a theta primitive or a new connection of the angles to cohomology. Its metric is $M(t)$ along the interpolation, and the ordinary original arithmetic norm at $t = 1$; the two are related by the specified segment, not silently identified at other parameters.

301.7 7. Attribution text and cumulative integration scope

The following paragraph is suitable for R74 and the cumulative text:

The inherited endpoint-window calculation already proves the canonical principal-angle comparison, nonlinear arcosh transport, strict positivity for the actual polynomial moments, and the signed tensor theta primitive (EW.25–28, EW.35a–c, EW.46a–48). The later AGT/AA calculation independently rederives these structures and sharpens the selected window $n = r = q$: crossing the same four projection terms changes the guaranteed moving-angle count from $2q$ to $2q - 1$, yielding the corresponding stronger arcosh bound. Its direct reflected-polynomial positivity proof and explicit relation-energy coordinates provide additional proof detail. The original masses, full jets, theta primitive, and tau support remain those of the inherited construction.

If the add-only general-window consequence EP17–25 is included, append:

Applying the same crossed pairing to the inherited arbitrary-window theorem gives the count $\min(q, r + 1) + \min(q, r - 1)$. It improves the earlier count $2 \min(q, r)$ by one exactly when $r = q$, for every $n \geq q$. This is an explicit refinement derived from EW's rank bound and transport argument.

These attribution statements replace any implication that AGT or AA first supplied an all-degree principal-angle method, first obtained nonlinear arcosh transport, first proved strict moment positivity, or first constructed the tensor theta primitive. The original source has broader retained content: arbitrary windows, the norm factor U , coefficient-to-Gram maps, rational enclosures, and zero-loss positive-matrix families. That content remains part of the cumulative program; the selected-window refinement does not supersede it. The preserved global analytic limitation is the absence here of a uniform growing-packet bound on the original relative Gram or its κ . Neither source claims such a bound from the finite transport formula alone.

The companion receipt pins the compared bodies and this add-only note. The mathematical derivations above are written proofs. No numerical, Lean, remote-publication, or sealed-package mutation is claimed by this comparison.

302 The off-line counterfactual inside the original holonomy construction

This calculation uses the original theta source in the final analytic continuation of the programme. In particular, the scalar measure below is computed from g/h with $g = 2\xi$; it is not a Gamma reference measure. The source equations being continued are (H5)–(H17), (H24)–(H28), and (H53)–(H55) of *All-holonomy descent of the original theta quotient*. The complete source record and locators are recorded in the accompanying audit. The conclusions concern the stated counterfactual and do not assert the existence of an off-line zero.

302.1 The retained arithmetic objects and every phase

Suppose $1/2 + \delta + i\gamma$ is a zero of g of its full order m , where $0 < \delta < 1/2$ and $\gamma > 0$. Reflection and conjugation give the complete quartet, with the same order. Retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - 1/2 - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}. \quad (\text{HT1})$$

The division is entire and its full jet $v_h = j_h(v_h)$ is a unit. For an integer $k \geq 2$, put

$$c = k/2, \quad \ell_k = 1 + k(m-1), \quad q = \ell_k(k+1)^2, \quad m_{h,k} = w_h^{*k},$$

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^{\ell_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \quad (\text{HT2})$$

These are the original cyclic sum quotient, its full multiplicities, and its multiplication operator. For completeness, in each tensor local factor the nilpotent sum is $n_1 + \cdots + n_k$ with $n_i^m = 0$. Its power $k(m-1)$ has nonzero coefficient $(k(m-1))! / ((m-1)!)^k$ on $n_1^{m-1} \cdots n_k^{m-1}$, whereas the next power vanishes. Thus its nilpotency index is exactly ℓ_k . Every pair (a, b) is realized by choosing independently the real and imaginary signs in the k tensor factors. Distinct pairs give distinct complex sums, so the annihilator in the sum variable is exactly χ .

The original analytic construction provides F_h with Mellin transform v_h , and the original source and jet maps are

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{HT3})$$

Here η is an injective complex linear module map. Its unit factor is retained. The norm of this source is exactly $\int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du$. The half-density map is

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad r = \log x_k, \quad z_i = \log x_i - \log x_k. \quad (\text{HT4})$$

It maps into $L^2(\mathbb{R}, dr; \mathcal{K})$ with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$ and intertwines $D_1 + \cdots + D_k$ with $-\partial_r + c$. No relative variable is removed.

Fix any $L > 0$ and $\theta \in [0, 2\pi)$. The source of phase θ is

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{j \in \mathbb{Z}} e^{ij\theta} \psi(r + jL), \quad \psi(r + L) = e^{-i\theta} \psi(r) \quad \text{in the target domain.} \quad (\text{HT5})$$

On the original finite source, its Gram form is exactly the polynomial norm for the measure

$$\nu_{L,\theta} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}, \quad u_n = \frac{2\pi n + \theta}{L}, \quad S = c + iu. \quad (\text{HT6})$$

The factor $2\pi/L$ is unchanged. The measure has positive mass at every u_n . Indeed the nonzero entire function v_h has only isolated zeros on the line; hence $w_h > 0$ almost everywhere. For every u , the product $w_h(v)w_h(u-v)$ is positive almost everywhere in v , so its integral is positive. Iterated convolution preserves strict positivity. The original weighted Schwartz estimates give convergence and all moments in (HT6). A nonzero polynomial cannot vanish at every lattice point; consequently every finite polynomial Gram matrix is positive definite.

For $N \geq q-1$, let $P_N = \mathbb{C}[S]_{\leq N}$ and $J_N = \pi_\chi|_{P_N}$. With the fixed coefficient bases define

$$M_{N,\theta} = (\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N})^*(\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N}), \quad K = J_N M_{N,\theta}^{-1} J_N^*, \quad G = K^{-1},$$

$$C_{N,\theta} = M_{N,\theta}^{-1} J_N^* G, \quad R = \mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V} C_{N,\theta}. \quad (\text{HT7})$$

Thus $J_N C_{N,\theta} = I_E$ and $R^* R = G$. All inverses in (HT7) exist: $M_{N,\theta}$ is positive definite and J_N is surjective. No inverse at degree $q-2$ is used.

302.2 A uniform phasewise lower bound for the original control

Define

$$H = G^{-1/2}(A^*G + GA - kG)G^{-1/2}, \quad \mathcal{L}_{h,k} = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{HT8})$$

For every $k \geq 2$, $L > 0$, $\theta \in [0, 2\pi)$, and $N \geq q-1$, the nonzero eigenvalues of H are ε , $-\varepsilon$, and

$$\boxed{\varepsilon \geq \mathcal{L}_{h,k} > 0, \quad \frac{\mathcal{L}_{h,k}}{q} \geq \frac{\delta k}{2}.} \quad (\text{HT9})$$

Proof. Let $p_j(S)$ be the monic orthogonal polynomials for (HT6), with squared norms $\omega_j > 0$, and let $b_j = [p_j] \in E$. In the real coordinate u , write $p_j(c+iu) = i^j r_j(u)$, where r_j is real monic. Orthogonality and the symmetry of multiplication by u show

$$ur_j = r_{j+1} + a_j r_j + \frac{\omega_j}{\omega_{j-1}} r_{j-1}, \quad a_j \in \mathbb{R},$$

with the last term omitted at $j=0$. To verify that no earlier term occurs, pair ur_j with r_i , $i \leq j-2$, and move u to r_i ; its degree is at most $j-1$. Pairing with r_{j-1} gives precisely ω_j/ω_{j-1} . In the S coordinate this is

$$Sp_j = p_{j+1} + (c + ia_j)p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}. \quad (\text{HT10})$$

Since the source Gram is diagonal in this basis, $K = \sum_{j=0}^N b_j b_j^* / \omega_j$. Apply A on the left and A^* on the right and use (HT10). The imaginary diagonal terms cancel, and each adjacent term except the last cancels its predecessor. This gives, including $N = q-1$,

$$AK + KA^* - kK = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{HT11})$$

Congruence by $G^{1/2}$ identifies the left side with H . Thus its rank is at most two. Also $\text{Tr } H = 2 \text{Re } \text{Tr } A - kq = 0$ by the complete grid (HT2).

Let E_+ be the direct sum of the complete primary spaces with $a > k/2$, and let P be the orthogonal projection onto $G^{1/2}E_+$ in the standard Hermitian space. The latter subspace is invariant under $G^{1/2}AG^{-1/2}$. Taking traces in an orthonormal basis of this subspace gives

$$\text{Tr}(PH) = 2 \sum_{a > k/2} \sum_{b=0}^k \ell_k(2a-k)\delta = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = \mathcal{L}_{h,k}. \quad (\text{HT12})$$

The finite sum is r^2 if $k = 2r-1$, and $r(r+1)$ if $k = 2r$. A Hermitian rank-at-most-two matrix of trace zero has eigenvalues ε , $-\varepsilon$ and zeroes. For every orthogonal projection, $\text{Tr}(PH) \leq \varepsilon$: in an eigenbasis the positive contribution is at most ε and the negative one is nonpositive. Equation (HT12) proves the first inequality of (HT9), and $4\lfloor (k+1)^2/4 \rfloor \geq k(k+1)$ proves the second. $\square$

302.3 The exact energy forced into the retained theta boundary

In $\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K})$ put $D_{L,\theta} = -\partial_r + c$, with domain $\{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}$. The domain of its square is $\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), f'(L) = e^{-i\theta} f'(0)\}$. The original smooth periodized source columns belong to these domains. Monic division in the original coefficient source gives the unique polynomial map $Q_{N,\theta}$ with

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}.$$

Indeed applying J_{N+1} gives zero. Define the actual boundary

$$\mathcal{B} = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}(\chi Q_{N,\theta}) = D_{L,\theta}R - RA. \quad (\text{HT13})$$

This retains its theta primitive. Explicitly, tensor monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}),$$

because its image in the full tensor jet algebra vanishes. In the original theta complex, ϕ_h has degree zero, F_h has degree one, and $h(D)F_h = \Theta\phi_h$. The degree- $(k-1)$ tensor primitive is exactly

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_h \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes a further sign $(-1)^{i-1}$ when it acts on ϕ_h , because the preceding $i-1$ factors have degree one. The two signs cancel, so its differential is precisely $\mathcal{V}(\chi Q)$. Applying the same half-density and phase maps gives precisely (HT13).

Set

$$U = RG^{-1/2}, \quad T = \mathcal{B}G^{-1/2}, \quad Z = U^*T, \quad T_\perp = (I - UU^*)T.$$

Then the following exact energy identity, with no omitted source term, holds:

$$\boxed{\text{Tr}(G^{-1}\mathcal{B}^*\mathcal{B}) = \frac{\varepsilon^2}{2} + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2 \geq \frac{\mathcal{L}_{h,k}^2}{2} \geq \frac{\delta^2 k^2 q^2}{8}.} \quad (\text{HT14})$$

All quantities on the left are calculated from g/h and its original theta representatives. In particular the same estimate holds after integration with the original Fourier inversion measure:

$$\int_0^{2\pi} \text{Tr}(G_{N,\theta}^{-1}\mathcal{B}_{N,\theta}^*\mathcal{B}_{N,\theta}) \frac{d\theta}{2\pi} \geq \frac{\mathcal{L}_{h,k}^2}{2}. \quad (\text{HT15})$$

Proof. The two arguments in integration by parts have the same boundary multiplier $e^{-i\theta}$, so the endpoint products cancel. Hence $D_{L,\theta}^* + D_{L,\theta} = 2cI$ on the stated domain. Inserting $D_{L,\theta}R = RA + \mathcal{B}$ and $R^*R = G$ gives

$$A^*G + GA - kG = -(R^*\mathcal{B} + \mathcal{B}^*R), \quad H = -Z - Z^*.$$

Since $U^*U = I$, the decomposition $T = UZ + T_\perp$ is orthogonal in Hilbert-Schmidt norm. The Hermitian and skew-Hermitian parts of Z are orthogonal for the real Hilbert-Schmidt scalar product. Therefore

$$\|T\|_{\text{HS}}^2 = \frac{1}{4}\|H\|_{\text{HS}}^2 + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2.$$

By (HT9), $\|H\|_{\text{HS}}^2 = 2\varepsilon^2$, which proves (HT14). For fixed k, L, N the weighted source sums and their derivatives converge uniformly in θ . Thus $M_{N,\theta}$ is continuous. Its smallest eigenvalue is positive on the compact phase circle, so its inverse, $G_{N,\theta}$, and the boundary columns are continuous there. Integration of (HT14) proves (HT15). $\square$

302.4 The full lattice test for the original Laplacian boundary

Put $\mathcal{L}_{L,\theta} = D_{L,\theta}^2 - kD_{L,\theta} = \partial_r^2 - c^2$. Direct substitution of (HT13) gives the exact map

$$\mathcal{L}_{L,\theta}R - R(A^2 - kA) = (D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A. \quad (\text{HT16})$$

If $Av = \lambda v$ with $v \neq 0$, write $\lambda = c + x + iy$. Then

$$\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq \min_{n \in \mathbb{Z}} \sqrt{(u_n^2 + x^2 - y^2)^2 + 4x^2y^2}. \quad (\text{HT17})$$

In particular every phase and every admitted degree satisfy, on an eigenvector for $\lambda = k(1/2 + \delta + i\gamma)$,

$$\boxed{\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq 2k^2\delta\gamma.} \quad (\text{HT18})$$

For an eigenvalue with $x \neq 0$ and $y = 0$, (HT17) also gives the positive lower bound $x^2 + \min_n u_n^2$, even though its imaginary part is zero.

Proof. The Fourier basis $L^{-1/2}e^{-iu_nr}$ diagonalizes the quasi-periodic Laplacian with eigenvalues $-u_n^2 - c^2$, each acting identically on $\mathcal{K}$. The source vectors lie in its domain by the original weighted differentiability estimates. Writing Rv in that basis and using Parseval gives

$$\|(\mathcal{L}_{L,\theta} - (\lambda^2 - k\lambda))Rv\|^2 = \sum_n ((u_n^2 + x^2 - y^2)^2 + 4x^2y^2) \|\widehat{Rv}(n)\|_{\mathcal{K}}^2.$$

The lattice is discrete and the expression tends to infinity as $|n| \rightarrow \infty$, so its infimum is a minimum. Also R is injective because $R^*R = G \succ 0$. Divide the displayed identity by $\|Rv\|^2$; (HT16) proves (HT17). The extreme primary factor in (HT2) is nonzero and has an eigenvector, so substituting $x = k\delta, y = k\gamma$ proves (HT18). If $y = 0$, the square root in (HT17) is $u_n^2 + x^2$. $\square$

302.5 What these computations decide

The original off-line counterfactual survives passage to every finite holonomy quotient only with the explicit boundary energy (HT14) and Laplacian residual (HT17). These quantities belong to the original arithmetic source. The statement is compatible with the exact all-phase identity $\int M_{N,\theta} d\theta / (2\pi) = M_N$ and with the original small gluing-norm estimate. That estimate concerns $C_N - C_{N,\theta}$; (HT13) applies S and subtracts right multiplication by A , preserving a different, explicitly displayed source quantity. There is no claim here that averaging the $G_{N,\theta}$ preserves the rank-two identity (HT11): the lower bound (HT15) integrates the proved phasewise energies themselves.

The transcript's threshold-four volume lower bound was already present before its final Deligne comparison. Its last counterexamples retain many generic identities but do not satisfy the actual divisor condition $h \mid 2\xi$. Consequently they disprove the reduced generic inference stated there; they do not disprove the entire arithmetic programme or provide an RH counterexample. Equations (HT9), (HT14), and (HT17) give further quantified consequences within the original programme without replacing its theta metric or its represented relations.

303 Exact endpoint windows: Gamma coefficients, paired Schur volumes, and intrinsic sensitivity

This continuation evaluates the original endpoint quantity of the complete written criterion at source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f of PR 23. The source criterion and the supplied *Endpoint degree selection in the actual gamma-corrected coordinates* were read completely, together with GC.1–59 and GMT.1–42, including GMT.29a–b. The present work proves the arbitrary-window Schur map, its exact determinant cancellations, a finite rational interval certificate, and a new sensitivity estimate through the actual angles between canonical least-lift spaces. The finite input list grows through degree $2(n + r)$. Every statement below is finite. No estimate for that growing list or its arithmetic conditioning is presumed.

303.1 The original source and the complete finite input map

Let $h(s)$ be the given monic packet of actual zeros, with their complete orders, of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s).$$

Retain its entire quotient $v_h = g/h$ and original positive density

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (\text{EW.1})$$

The complete analytic construction in GC.2–9 proves that w_h has every polynomial moment and exponential moments of every order less than $\pi/2$. Its mechanism is the theta seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, the equality $\mathcal{M}\Theta\phi_0 = g$, contour rotation through every angle $|\beta| < \pi/4$, and exact local division by h . In particular $|v_h(\sigma + it)|$ is bounded, uniformly for σ in each fixed compact real interval, by $C_{j,\beta,h}(1 + |t|)^{-j}e^{-\beta|t|}$ for every $j \geq 0$ and $0 < \beta < \pi/4$. These proofs and their original constants are retained in the accompanying complete GC source. The zero set of $v_h(1/2 + it)$ is discrete, so $w_h > 0$ almost everywhere. Its convolution has positive mass μ_h^k .

Throughout this paper

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k, \quad (\text{EW.2})$$

$$\langle P, Q \rangle_H = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du.$$

The pairing is conjugate-linear in its first argument. The sum chart $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Fubini therefore proves both the convolution in (EW.1) and the full mass μ_h^k , with no average coordinate or mass division. Every nonzero polynomial has finitely many real-line zeros, so its squared norm in (EW.2) is strictly positive.

Fix the original monic relation $\chi \in \mathbb{C}[S]$, of degree $q \geq 1$. For the packet source it generates the exact intersection $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ under $S = \sum s_i$. Define $\mathcal{C} = \mathbb{C}[S]/(\chi)$ in the ordered coordinates $1, S, \dots, S^{q-1}$. The quotient map is the actual remainder map. Its injection into the tensor arithmetic jets is

$$\eta : \mathcal{C} \longrightarrow (\mathbb{C}[s]/(h))^{\otimes k}, \quad [P] \longmapsto [v_h]_h^{\otimes k} P(s_1 + \cdots + s_k). \quad (\text{EW.3})$$

Complete local cancellation makes the constant term of v_h nonzero at every selected centre. Each local truncated power series is invertible by recursively solving its product with an inverse to equal one. The Chinese remainder isomorphism then proves that $[v_h]_h$ is a unit. The unfactored polynomial map $\mathbb{C}[S] \rightarrow (\mathbb{C}[s]/(h))^{\otimes k}$ in (EW.3) consequently has kernel equal to the stated intersection ideal (χ) . Factoring by that ideal gives the displayed η with kernel zero, proving injectivity without discarding a jet.

For $N \geq q - 1$, put $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{P}_{-1} = 0$. Let H_N be its source matrix in the original monomials. Let $J_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}^q$ be monic remainder and $B_N : \mathbb{C}^{N-q+1} \rightarrow \mathbb{C}^{N+1}$ be multiplication by χ . Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0. \quad (\text{EW.4})$$

In particular B_N is injective, J_N is surjective, and $\ker J_N = \text{im } B_N$.

For later coefficient evaluation keep all Gamma constants:

$$\lambda > 0, \quad \alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{EW.5})$$

Let the real monic polynomials $b_j^{(\lambda)}$ be specified by

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 + z^2)^{-\lambda} e^{t \arctan z}, \quad |z| < 1, \quad b_{j+1} = t b_j - j(j + \alpha - 1) b_{j-1}, \quad (\text{EW.6})$$

where the logarithms defining the generating function are the branches from zero, $b_0 = 1$, and $b_1 = t$. Differentiation of the displayed generating function proves the recurrence and monicity. The beta integral and Fourier inversion give the literal convolution mass and Laplace transform

$$r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \int e^{\theta u} r_{\lambda,k}(u) du = C_k (\cos \theta)^{-\beta}, \quad |\Re \theta| < \pi/2. \quad (\text{EW.7})$$

Their complete beta-integral proof with the changes $t = 2v$ and $u = \sum t_i$ is GC.10–15. In particular the reference monic norms at $\theta = 0$ are $C_k j! (\beta)_j$. They also follow directly by integrating the product of two functions (EW.6), which gives $C_k (1 - zw)^{-\beta}$, and comparing its finite derivatives.

Define the given one-factor coefficients and their finite powers by

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_\lambda j! (\alpha)_j}, \quad d_l = \sum_{j_1 + \dots + j_k = l} \prod_{a=1}^k (\alpha)_{j_a} c_{h,j_a}. \quad (\text{EW.8})$$

Multiplying the k generating functions (EW.6) and comparing the coefficient of z^l gives the finite polynomial identity

$$b_l^{(k\lambda)}(\sum t_i) = \sum_{j_1 + \dots + j_k = l} \frac{l!}{\prod j_i!} \prod_i b_{j_i}^{(\lambda)}(t_i).$$

Integrating every term against the full product measure proves

$$\int b_l^{(k\lambda)}(u) m_{h,k}(u) du = C_k l! d_l. \quad (\text{EW.9})$$

All sums here are finite and all their terms are integrable. For any specified polynomials $p_a \in \mathcal{P}_M$, perform monic polynomial subtraction in the actual u variable to find the unique coefficients $L_{ab,l}$ in

$$\overline{p_a(c + iu)} p_b(c + iu) = \sum_{l=0}^{2M} L_{ab,l} b_l^{(k\lambda)}(u).$$

The complete coefficient-to-Gram map is therefore

$$H(p)_{ab} = C_k \sum_{l=0}^{2M} l! L_{ab,l} d_l. \quad (\text{EW.10})$$

It is a degree- k real polynomial map from the actual coefficient vector to Hermitian forms. The precise tests contain $c = k/2$, the two phases $\pm i$, and, when a relation column is used, both $\chi(c + iu)$ and $\overline{\chi}(c - iu)$. In particular (EW.10) constructs H_M from $c_{h,0}, \dots, c_{h,2M}$ and the stated k, λ . At $l = 0$, $c_{h,0} = \mu_h / c_\lambda$, so the same map gives $H_{00} = \mu_h^k$.

303.2 Canonical lifts and one arbitrary-window block

For each admitted degree define the exact matrices and maps

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : \mathbb{C}^q \longrightarrow \mathbb{C}^{N+1}, \quad V_N = \det G_N. \quad (\text{EW.11})$$

Positivity of H_N and surjectivity of J_N prove $K_N > 0$. Multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. For every $x \in \mathbb{C}^q$, any lift is $R_N x + B_N y$, and its norm is exactly $x^* G_N x + y^* B_N^* H_N B_N y$. This proves the unique minimum property and $R_N^* H_N R_N = G_N$.

Let E_N include the first q monomials. The square coefficient matrix $[E_N, B_N]$ has determinant one: its columns are ordered by successive degree and each leading coefficient is one. Subtracting boundary columns to replace E_N by R_N retains determinant one. Its Gram is block diagonal, so

$$V_N = \frac{\det H_N}{\det(B_N^* H_N B_N)}. \quad (\text{EW.12})$$

Empty determinants are one. This fixes the relation determinant and its index $N - q + 1$ in the original quotient coordinates.

Take integers

$$n \geq q, \quad r \geq 1, \quad m = n + r, \quad d = r + 1. \quad (\text{EW.13})$$

In $\mathcal{P}_m$ split the ordered monomials into degrees $0, \dots, n-1$ and the d appended columns $S^n, \dots, S^m$. Write

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad J_m = [J_{n-1}, J_D], \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EW.14})$$

The map

$$\Psi : \mathbb{C}^n \oplus \mathbb{C}^d \longrightarrow \mathbb{C}^{m+1}, \quad (x, y) \longmapsto (x - H_{n-1}^{-1} C_D y, y) \quad (\text{EW.15})$$

is invertible with determinant one. Direct multiplication gives $\Psi^* H_m \Psi = \text{diag}(H_{n-1}, W)$ and $J_m \Psi = [J_{n-1}, F]$. Thus $W > 0$; F maps the appended orthogonal residual polynomials to their original quotient classes. No quotient is independently chosen.

Let W_j be the first j rows and columns of W , F_j its first j quotient columns, and $Z_j = F_j^* G_{n-1} F_j$ for $1 \leq j \leq d$. The same map (EW.15), restricted to these columns, gives the exact positive-kernel update

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*. \quad (\text{EW.16})$$

Eliminating the two diagonal blocks of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in the two orders proves $\det(I + AB) = \det(I + BA)$. Use this identity in (EW.16) to obtain

$$\mathcal{R}_j := \frac{V_{n-1}}{V_{n+j-1}} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad \mathcal{R}_0 = 1. \quad (\text{EW.17})$$

All the determinants on the right are of size j , whereas its equivalent kernel calculation has size q . The matrices Z_j are positive semidefinite. Diagonalizing $W_j^{-1/2} Z_j W_j^{-1/2}$ proves $\mathcal{R}_j \geq 1$. Nested minimization also proves $G_{N+1} \preceq G_N$, so $\mathcal{R}_j$ is nondecreasing in j .

Let $p_N(S)$ be the unique monic polynomial of degree N orthogonal to $\mathcal{P}_{N-1}$, and let ω_N be its original squared norm. Existence and uniqueness follow by solving the positive lower-degree projection problem; its leading coefficient remains one. The first diagonal entry $w = W_1$ is ω_n . Eliminating the first r residual columns from the final one gives $\omega_m = \det W_d / \det W_r$. Consequently

$$U := \frac{\omega_m}{\omega_n} = \frac{\det W_d}{\det W_r w}, \quad \mathcal{R} := \frac{V_{n-1} V_n}{V_{m-1} V_m} = \frac{\mathcal{R}_r \mathcal{R}_d}{\mathcal{R}_1} \geq 1. \quad (\text{EW.18})$$

The last inequality also follows by multiplying $V_{n-1}/V_{m-1} \geq 1$ and $V_n/V_m \geq 1$. Both numerator and denominator norms carry their literal masses.

For the Gamma source (EW.7), the same J_m and χ give matrices $W^\Gamma, F^\Gamma, Z_j^\Gamma$ by the identical maps. For their original source determinants put $\mathfrak{D}_j = \det H_{j-1}$ for $j \geq 1$ and $\mathfrak{D}_0 = 1$. For the relation determinant on exactly j columns put $\mathfrak{B}_j = \det(B_{q+j-1}^* H_{q+j-1} B_{q+j-1})$ for $j \geq 1$ and $\mathfrak{B}_0 = 1$. The reference determinants use the same columns and carry the superscript Γ . Writing the supplied corrections as $X_j = \mathfrak{D}_j/\mathfrak{D}_j^\Gamma, Y_j = \mathfrak{B}_j/\mathfrak{B}_j^\Gamma$ and $T_N = X_{N+1}/Y_{N-q+1}$, the relation (EW.12) proves

$$\begin{aligned} U &= (n+1)_r (n+\beta)_r \frac{X_{m+1} X_n}{X_m X_{n+1}}, \\ \mathcal{R} &= \frac{V_{n-1}^\Gamma V_n^\Gamma}{V_{m-1}^\Gamma V_m^\Gamma} \frac{T_{n-1} T_n}{T_{m-1} T_m}. \end{aligned} \quad (\text{EW.19})$$

Indeed $\omega_j = C_{kj}!(\beta)_j X_{j+1}/X_j$ follows by taking successive source determinants in the original monic basis. Taking the ratio gives the first line with its exact rising factorials. The new symbol $\mathcal{R}_j$ in (EW.17) has no effect on the original positive arithmetic correction T_N .

303.3 Endpoint roots and exact paired-volume cancellations

Put $a = 1/(2r)$. The original endpoint expression is

$$\mathcal{E}_{n,r} = U^a \sinh(a \log \mathcal{R}) = \frac{(U\mathcal{R})^a - (U/\mathcal{R})^a}{2}. \quad (\text{EW.20})$$

All roots are the unique positive real roots. The equality follows from the definition of the real hyperbolic sine and positivity. It evaluates the existing quantity, preserving its two original norm endpoints and all four original volume endpoints.

There is a further exact cancellation inside its two root arguments. Set

$$p = \det W_r, \quad s = \det W_d, \quad e = w + Z_1, \quad D = \det(W_r + Z_r) \det(W_d + Z_d).$$

The scalar Z_1 is real nonnegative, and all four quantities p, s, e, D are strictly positive. Substituting (EW.17)–(EW.18) without changing a coordinate gives

$$U = \frac{s}{pw}, \quad \mathcal{R} = \frac{Dw}{pse}, \quad \boxed{U\mathcal{R} = \frac{D}{p^2e}, \quad \frac{U}{\mathcal{R}} = \frac{s^2e}{w^2D}}. \quad (\text{EW.21})$$

Thus the common highest source determinant cancels from the first argument; the common prefix source determinant cancels from the second. The first and last original monic norms have not been set equal or removed from the underlying source. Formula (EW.21) is the exact rational map explaining these cancellations.

It is sometimes useful to retain the small difference in (EW.20) as one positive numerator. With $P = (U\mathcal{R})^a$ and $Q = (U/\mathcal{R})^a$, the finite factorization $P^{2r} - Q^{2r} = (P - Q) \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j$ proves

$$\boxed{\mathcal{E}_{n,r} = \frac{U(\mathcal{R}^2 - 1)}{2\mathcal{R} \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j}}. \quad (\text{EW.22})$$

The denominator remains positive at $\mathcal{R} = 1$, and the numerator is then exactly zero. This representation preserves the sign of the endpoint quantity and avoids subtracting two enclosures before their common positive factor has been identified.

Every entry in (EW.14) is determined by (EW.10) through degree $2m = 2(n + r)$. Each J_N and B_N is computed by monic division or multiplication by the same χ . Equations (EW.11)–(EW.22) therefore prove the complete finite map

$$(c_{h,0}, \dots, c_{h,2(n+r)}; k, \lambda, \chi) \longmapsto (H_m, J_m, W, F, G_{n-1}) \longmapsto \mathcal{E}_{n,r}. \quad (\text{EW.23})$$

It consists of stated polynomial operations, inverses of matrices whose positivity has been proved, determinants, and positive roots.

For reference the endpoint selection follows from the original degree-wise estimate by a finite argument. Write $x_j = \frac{1}{2} \log(V_{n+j-1}/V_{n+j+1}) \geq 0$ and $A_j = \sqrt{\omega_{n+j+1}/\omega_{n+j}}$. The local estimate is $0 \leq \epsilon_{n+j} \leq A_j \sinh x_j$. If an x_j is zero, the minimum allowance is zero. Otherwise $(\log \sinh x)'' = -1/\sinh^2 x < 0$ proves finite Jensen, hence $\prod_j \sinh x_j \leq \sinh(\sum_j x_j/r)^r$. The actual products give $\prod_j A_j = \sqrt{U}$ and $2 \sum_j x_j = \log \mathcal{R}$. Multiplying the local estimates and taking the positive r th root therefore proves

$$\min_{0 \leq j < r} \epsilon_{n+j} \leq \mathcal{E}_{n,r}. \quad (\text{EW.24})$$

The local arithmetic trace inequality and its retained loss terms are proved in the complete source TVB.32–34a; (EW.24) is their exact finite selection, not an added local estimate or an asymptotic claim.

303.4 Exact projection between the canonical representative spaces

The following result supplies more information than the separate determinant values. Use the common ambient space $(\mathcal{P}_m, \langle \cdot, \cdot \rangle_H)$, with its original source inner product. Let $\iota_j : \mathbb{C}^{j+1} \rightarrow \mathbb{C}^{m+1}$ be monomial inclusion. The least-lift space and its exact isometry are

$$L_j = \iota_j R_j \mathbb{C}^q \subset \mathcal{P}_m, \quad \mathcal{J}_j : (\mathbb{C}^q, G_j) \longrightarrow L_j, \quad x \longmapsto \iota_j R_j x. \quad (\text{EW.25})$$

Here $\mathcal{J}_j^* H_m \mathcal{J}_j = G_j$ proves the word isometry; its inverse on L_j is the original remainder map. The original arithmetic correction T_N retains its meaning in (EW.19).

Theorem 303.1 (Canonical projection and all its moving directions). *For $q - 1 \leq a < b \leq m$, let $\Pi_b : \mathcal{P}_m \rightarrow \mathcal{P}_m$ be the ambient orthogonal projection with range L_b , including that range by its fixed polynomial inclusion. It obeys*

$$\Pi_b \iota_a R_a x = \iota_b R_b x \quad (x \in \mathbb{C}^q). \quad (\text{EW.26})$$

Its restriction $L_a \rightarrow L_b$ is invertible. Its squared singular values, with precisely the source inner products in (EW.25), are the eigenvalues σ_i^2 of

$$G_a^{-1/2} G_b G_a^{-1/2}, \quad 0 < \sigma_i \leq 1. \quad (\text{EW.27})$$

At most $s_{a,b} = \min(q, b - a)$ of these values are less than one, and

$$\begin{aligned} \prod_i \sigma_i^2 &= V_b/V_a, \\ \|\Pi_a - \Pi_b\|_1 &= 2 \sum_i \sqrt{1 - \sigma_i^2}, \\ \|\Pi_a - \Pi_b\|_1 &\leq \min\{2\sqrt{s_{a,b} \log(V_a/V_b)}, 2s_{a,b}\}. \end{aligned} \quad (\text{EW.28})$$

The trace norm is for operators in the stated ambient inner product.

Proof. Both representatives in (EW.26) have remainder x . Their difference belongs to $\chi \mathcal{P}_{b-q}$ and is therefore orthogonal to L_b by (EW.11). The second representative belongs to L_b ; this proves the orthogonal projection identity. The same identity and injectivity of each lift show that the restriction is

invertible. In the quotient coordinates it is the identity map from metric G_a to metric G_b . The explicit maps from Euclidean coordinates are $u \mapsto \iota_a R_a G_a^{-1/2} u$ and $v \mapsto \iota_b R_b G_b^{-1/2} v$. Their matrices give the squared singular values (EW.27), and no original form or mass has been changed. A projection is a contraction, so these values lie in $(0, 1]$. Taking its determinant proves the first line of (EW.28).

The boundary space at degree b extends the one at degree a by exactly $b - a$ independent monic columns. Vectors in L_a are already orthogonal to the older boundary. Imposing orthogonality to these extra $b - a$ columns has rank at most $b - a$. Its kernel is $L_a \cap L_b$: any vector in that kernel is in $\mathcal{P}_b$ and orthogonal to its full boundary, hence is a minimum representative. The intersection therefore has dimension at least $q - (b - a)$ when this number is positive. A unit vector in L_a is unchanged in norm by Π_b precisely when it belongs to L_b , by orthogonal Pythagoras. Thus at most $s_{a,b}$ singular values are less than one.

For a complete trace-norm calculation choose orthonormal eigenvectors $x_i \in L_a$ for the squared restriction and put $y_i = \Pi_b x_i / \sigma_i \in L_b$. Orthogonality gives $\langle x_i, y_j \rangle = \sigma_j \delta_{ij}$ and the y_i form an orthonormal basis of L_b . If $\sigma_i = 1$, then $y_i = x_i$. If $\sigma_i < 1$, set $z_i = (y_i - \sigma_i x_i) / \sqrt{1 - \sigma_i^2}$. Directly taking all these inner products proves that the planes $\text{span}(x_i, z_i)$ are mutually orthogonal and orthogonal to the common directions. In such a plane the matrix of $\Pi_a - \Pi_b$ is exactly

$$\begin{pmatrix} 1 - \sigma_i^2 & -\sigma_i \sqrt{1 - \sigma_i^2} \\ -\sigma_i \sqrt{1 - \sigma_i^2} & -(1 - \sigma_i^2) \end{pmatrix}.$$

Its eigenvalues are $\pm \sqrt{1 - \sigma_i^2}$. On the common directions and the orthogonal complement of $L_a + L_b$ the difference is zero. This proves the exact trace-norm sum. Finally $1 - x \leq -\log x$ for $0 < x \leq 1$, because the derivative of $x - 1 - \log x$ is $1 - 1/x$ and its value at one is zero. Cauchy-Schwarz on the at most $s_{a,b}$ nonzero terms gives the first bound in (EW.28); bounding each square root by one gives the second bound independently. $\square$

303.5 Window sensitivity and exact cancellation of mass changes

The next calculation holds for every continuously differentiable family of positive source matrices $H_m(t)$, using all their principal submatrices and the fixed χ, J_j . Thus it applies both to the actual coefficient forms and to a segment between their certified finite enclosures. Write $E = \dot{H}_m$ and define the real spectral spread

$$\sigma_H(E) = \lambda_{\max}(H_m^{-1/2} E H_m^{-1/2}) - \lambda_{\min}(H_m^{-1/2} E H_m^{-1/2}) \geq 0. \quad (\text{EW.29})$$

The two congruences in this expression specify its exact coordinate map; the source H_m remains in every evaluation.

Embed the coefficient vector of p_j in $\mathcal{P}_m$ and set

$$P_j = \frac{\iota_j p_j p_j^* \iota_j^*}{\omega_j}, \quad Q_j = \iota_j R_j G_j^{-1} R_j^* \iota_j^*. \quad (\text{EW.30})$$

These are contravariant matrices; their products $P_j H_m$ and $Q_j H_m$ are respectively the ambient orthogonal projections onto the monic orthogonal line and L_j . To verify this, multiply each by itself and use $p_j^* H_j p_j = \omega_j$ or $R_j^* H_j R_j = G_j$. The resulting operators are idempotent and self-adjoint in H_m and have the stated ranges. The matrices $H_m^{1/2} P_j H_m^{1/2}$ and $H_m^{1/2} Q_j H_m^{1/2}$ are their exact Euclidean isometric conjugates.

Differentiating the norm of the monic polynomial gives $\dot{\omega}_j = p_j^* \dot{H}_j p_j$: its derivative has degree less than j , and both cross terms vanish by its original orthogonality. Differentiating $J_j R_j = I$ makes $\dot{R}_j$ a boundary, so the same expansion gives $\dot{G}_j = R_j^* \dot{H}_j R_j$. The determinant derivative follows

by expanding the determinant multilinearly in its columns, or by differentiating $\det(G_j + t\dot{G}_j) = \det G_j \det(I + tG_j^{-1}\dot{G}_j)$. Thus

$$\frac{d}{dt} \log \omega_j = \text{Tr}(P_j E), \quad \frac{d}{dt} \log V_j = \text{Tr}(Q_j E). \quad (\text{EW.31})$$

All these formulas use the original coordinates before the final cyclic trace.

Put $\mathcal{L} = \log \mathcal{R} \geq 0$ and $s = \min(q, r)$. The endpoint derivatives are exactly

$$\begin{aligned} (\log U)' &= \text{Tr}((P_m - P_n)E), \\ \mathcal{L}' &= \text{Tr}((Q_{n-1} + Q_n - Q_{m-1} - Q_m)E), \\ \mathcal{E}'_{n,r} &= \frac{1}{2r} \left\{ \mathcal{E}_{n,r} (\log U)' + U^{1/(2r)} \cosh(\mathcal{L}/(2r)) \mathcal{L}' \right\}. \end{aligned} \quad (\text{EW.32})$$

These equalities also hold at $\mathcal{R} = 1$. They identify the exact paired sensitivity rather than separate errors of large source and relation determinants.

Theorem 303.2 (Intrinsic finite sensitivity). *Throughout every such positive continuously differentiable family,*

$$|(\log U)'| \leq \mathfrak{o}_H(E), \quad |\mathcal{L}'| \leq \mathfrak{o}_H(E) \min\{2s, \sqrt{2s\mathcal{L}}\}. \quad (\text{EW.33})$$

For any two positive source matrices $H^{(0)}, H^{(1)}$ in the same original coordinates, let

$$\kappa = \frac{\lambda_{\max}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})}{\lambda_{\min}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})} \geq 1. \quad (\text{EW.34})$$

Their original window values satisfy

$$|\log(U^{(1)}/U^{(0)})| \leq \log \kappa, \quad |\sqrt{\mathcal{L}^{(1)}} - \sqrt{\mathcal{L}^{(0)}}| \leq \sqrt{s/2} \log \kappa. \quad (\text{EW.35})$$

In particular these sensitivity bounds contain no independent source-size determinant exponent. They vanish for any scalar change $H^{(1)} = tH^{(0)}$, $t > 0$, exactly as the original endpoint does.

Proof. Let $\Lambda = H_m^{-1/2} E H_m^{-1/2}$. Subtracting $\frac{1}{2}(\lambda_{\max}\Lambda + \lambda_{\min}\Lambda)I$ leaves operator norm $\mathfrak{o}_H(E)/2$. This scalar subtraction does not change either trace in (EW.32): the two monic-line projections each have trace one and the four least-lift projections each have trace q , with the displayed signs summing to zero.

The two monic orthogonal lines at degrees $n < m$ are orthogonal in the common ambient source. Their projection difference has eigenvalues 1, -1 and zeros, hence trace norm two. For any self-adjoint A and B , diagonalizing A gives $|\text{Tr}(AB)| \leq \sum_i |\lambda_i(A)| \|B\|$. This proves the first bound in (EW.33).

For the second trace split its four projections as $(\Pi_{n-1} - \Pi_{m-1}) + (\Pi_n - \Pi_m)$. Both index gaps are exactly r . Put $\ell_1 = \log(V_{n-1}/V_{m-1})$ and $\ell_2 = \log(V_n/V_m)$, so $\ell_1 + \ell_2 = \mathcal{L}$. Equation (EW.28), the triangle inequality for the trace norm, and $\sqrt{\ell_1} + \sqrt{\ell_2} \leq \sqrt{2\mathcal{L}}$ give

$$\|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 2\sqrt{2s\mathcal{L}}, \quad \|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 4s.$$

Multiplying either bound by $\mathfrak{o}_H(E)/2$ proves the second bound of (EW.33).

For a continuously differentiable path and $\varepsilon > 0$, the exact bound $|\mathcal{L}'| \leq \mathfrak{o}_H(E)\sqrt{2s\mathcal{L}}$ implies

$$\left| \frac{d}{dt} \sqrt{\mathcal{L} + \varepsilon} \right| \leq \sqrt{s/2} \mathfrak{o}_H(E).$$

Integrate on the compact path and let ε decrease to zero. This proves the square-root Lipschitz assertion along a path, including points at which $\mathcal{L} = 0$; no division by zero is used.

Use the straight positive segment $H(t) = (1 - t)H^{(0)} + tH^{(1)}$, $0 \leq t \leq 1$. Let $\ell_i > 0$ be the eigenvalues of $(H^{(0)})^{-1/2}H^{(1)}(H^{(0)})^{-1/2}$. Congruence by $(H^{(0)})^{1/2}$ and then its displayed orthonormal eigenbasis gives the generalized eigenvalues of $(E, H(t))$ exactly as

$$\frac{\ell_i - 1}{1 + t(\ell_i - 1)}.$$

These are the eigenvalues of $H(t)^{-1/2}EH(t)^{-1/2}$: both follow from the same equation $Ex = \lambda H(t)x$. The displayed function is increasing in ℓ_i , with derivative $[1 + t(\ell_i - 1)]^{-2}$. Hence

$$\int_0^1 \mathfrak{o}_{H(t)}(E) dt = \log \ell_{\max} - \log \ell_{\min} = \log \kappa.$$

Integrating the already proved differential inequalities proves (EW.35). Finally direct substitution in (EW.11) gives $R_N(tH) = R_N(H)$, $G_N(tH) = tG_N(H)$, and $\omega_j(tH) = t\omega_j(H)$. Thus U and $\mathcal{R}$ are exactly unchanged by that explicitly stated common scaling. The source mass remains $t\mu_h^k$ in the scaled comparison; it was never reassigned the value one. $\square$

The independent projection review supplied a stronger exact version from the same angles. For $s \geq 1$, it is

$$|\mathcal{L}'| \leq 2s \mathfrak{o}_H(E) \sqrt{1 - e^{-\mathcal{L}/(2s)}}. \quad (\text{EW.35a})$$

To prove it, list $x_i = -\log \sigma_i^2$ for each of the two projection pairs used above, omitting their zero terms and then appending zeros to make exactly $2s$ entries. Their sum is $\mathcal{L}$. The exact trace-norm formula and the scalar cancellation prove $|\mathcal{L}'| \leq \mathfrak{o}_H(E) \sum_i f(x_i)$, where $f(x) = \sqrt{1 - e^{-x}}$. For $x > 0$,

$$f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} < 0.$$

Continuity at zero extends concavity there: apply the concavity inequality to positive arguments increased by ε and then let $\varepsilon \downarrow 0$. Finite Jensen on the full $2s$ -term list proves (EW.35a). Since $1 - e^{-x} \leq x$ and $1 - e^{-x} \leq 1$, it implies both bounds on $|\mathcal{L}'|$ in (EW.33).

Define the explicit increasing map on the actual loss variable by

$$\Phi_s : [0, \infty) \longrightarrow [0, \infty), \quad \Phi_s(L) = 2 \operatorname{arcosh}(e^{L/(4s)}), \quad \Phi_s^{-1}(v) = 4s \log \cosh(v/2).$$

Here $\operatorname{arcosh} x = \log(x + \sqrt{x^2 - 1})$ for $x \geq 1$, so both maps and their inverses have specified real branches. For $L > 0$, $\Phi'_s(L) = [2s\sqrt{1 - e^{-L/(2s)}}]^{-1}$. Apply (EW.35a) to $\Phi_s(\mathcal{L} + \varepsilon)$. Its derivative in absolute value is at most $\mathfrak{o}_H(E) \sqrt{(1 - e^{-\mathcal{L}/(2s)})/(1 - e^{-(\mathcal{L}+\varepsilon)/(2s)})} \leq \mathfrak{o}_H(E)$. Integrating and letting $\varepsilon \downarrow 0$, then using the exact straight-segment length already proved, gives

$$|\Phi_s(\mathcal{L}^{(1)}) - \Phi_s(\mathcal{L}^{(0)})| \leq \log \kappa. \quad (\text{EW.35b})$$

This conclusion includes zero losses by continuity. For $s = 0$, the only loss is zero and define the one-point map $\Phi_0(0) = 0$; the projection and endpoint comparisons are their zero identities.

For $s \geq 1$, the stronger enclosure itself has an exact algebraic form. Given any $\rho \geq \kappa$ and the original $\mathcal{R}^{(0)} \geq 1$, put

$$z_0 = (\mathcal{R}^{(0)})^{1/(4s)} + \sqrt{(\mathcal{R}^{(0)})^{1/(2s)} - 1}, \quad z_- = \max\{1, z_0/\sqrt{\rho}\}, \quad z_+ = \sqrt{\rho} z_0, \quad (\text{EW.35c})$$

$$\left(\frac{z_- + z_-^{-1}}{2} \right)^{4s} \leq \mathcal{R}^{(1)} \leq \left(\frac{z_+ + z_+^{-1}}{2} \right)^{4s}.$$

Indeed $z_0 = \exp(\Phi_s(\mathcal{L}^{(0)})/2)$, (EW.35b) encloses its counterpart between z_- and z_+ , and $z \mapsto (z + z^{-1})/2$ is increasing for $z \geq 1$. The identity $\mathcal{R} = \cosh(\Phi_s(\log \mathcal{R})/2)^{4s}$ proves (EW.35c). At the

same time $U^{(0)}/\rho \leq U^{(1)} \leq \rho U^{(0)}$ by (EW.35). Thus monotonicity in (EW.20) supplies an endpoint enclosure using only nonnegative real roots, rational operations and integer powers when the central U , $\mathcal{R}$ and a bound ρ are rational. For example any proved rational Loewner comparison $aH^{(0)} \preceq H^{(1)} \preceq bH^{(0)}$, $0 < a \leq b$, gives $\rho = b/a$: the extreme Rayleigh quotients lie between a and b . This is the exact map from a positive-matrix certificate to the sharper window certificate, with the original mass and relation unchanged.

The square-root assertion is a Lipschitz assertion, and does not assert ordinary differentiability at every zero of $\mathcal{L}$. For an exact auxiliary illustration take $q = 1$, $\chi = S$, $n = 1$, $r = 1$, and the positive original-coordinate matrix

$$H_2(t) = \begin{pmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ t & 0 & 1 \end{pmatrix}, \quad |t| < 1.$$

Its leading pivots are $1, 1, 1 - t^2$, its remainder is evaluation at $S = 0$, and (EW.11) gives $V_0 = V_1 = 1$, $V_2 = 1 - t^2$. Thus $\mathcal{L} = -\log(1 - t^2)$ and $\sqrt{\mathcal{L}}/|t| \rightarrow 1$ as $t \rightarrow 0$. Its square root has a corner. This positive-form example belongs exactly to the finite family domain of (EW.33)–(EW.35); it is not claimed to be an arithmetic moment source. The regularized proof above includes it.

For completeness, (EW.35) has a directly usable endpoint enclosure. Writing $d_H = \log \kappa$ and $b = \sqrt{s/2} d_H$, it proves

$$\begin{aligned} e^{-d_H} U^{(0)} &\leq U^{(1)} \leq e^{d_H} U^{(0)}, \\ (\max\{0, \sqrt{\mathcal{L}^{(0)}} - b\})^2 &\leq \mathcal{L}^{(1)} \leq (\sqrt{\mathcal{L}^{(0)}} + b)^2. \end{aligned} \tag{EW.36}$$

The function $U^{1/(2r)} \sinh(\mathcal{L}/(2r))$ is increasing in each positive U and nonnegative $\mathcal{L}$, as direct differentiation shows. Substitution of the two endpoints in (EW.36) therefore bounds the actual endpoint expression. This is a finite stability theorem, including its exact $s = \min(q, r)$ dependence.

303.6 Rational interval certificates for the original input

Here is an explicit finite algorithm which uses enclosures of the original coefficients; it does not replace them by an arbitrary moment matrix. For clarity first suppose the constants in (EW.5) and the polynomial tests in (EW.10) are exact. Given real intervals $|c_{h,j} - \hat{c}_j| \leq e_j$ for $0 \leq j \leq 2m$, set

$$\begin{aligned} \hat{A}(z) &= \sum_{j=0}^{2m} (\alpha)_j \hat{c}_j z^j, \\ A_+(z) &= \sum_{j=0}^{2m} (\alpha)_j |\hat{c}_j| z^j, \quad E_+(z) = \sum_{j=0}^{2m} (\alpha)_j e_j z^j, \\ \hat{d}_l &= [z^l] \hat{A}(z)^k, \quad e_l^{(k)} = [z^l] \{(A_+(z) + E_+(z))^k - A_+(z)^k\}. \end{aligned} \tag{EW.37}$$

The multinomial expansion groups each difference of products by the nonempty set of factors carrying an error. Taking absolute values term by term proves $|d_l - \hat{d}_l| \leq e_l^{(k)}$. Thus the Hermitian central matrix and entry radii

$$\hat{H}_{ab} = C_k \sum_l l! L_{ab,l} \hat{d}_l, \quad \varepsilon_{ab} = C_k \sum_l l! |L_{ab,l}| e_l^{(k)} \tag{EW.38}$$

enclose the actual source. Rational upper bounds for the moduli give rational radii. An uncertain real constant, including C_k , is propagated by interval multiplication before the final matrix is reported. For $|x - \hat{x}| \leq e_x$ and $|y - \hat{y}| \leq e_y$, the exact elementary rule is $|xy - \hat{x}\hat{y}| \leq |\hat{x}|e_y + |\hat{y}|e_x + e_x e_y$. Repeated use handles every finite product in (EW.8)–(EW.10).

The original relation need not have rational coefficients. To retain it exactly while producing rational certificates, use its quotient-first coordinate map at each endpoint degree N :

$$A_N = [E_N, B_N] : \mathbb{C}^q \oplus \mathbb{C}^{N-q+1} \longrightarrow \mathbb{C}^{N+1}, \quad \det A_N = 1, \quad \tilde{H}_N = A_N^* H_N A_N. \quad (\text{EW.39})$$

The first component of $A_N^{-1}P$ is exactly $J_N P$ by monic division. Supply rational enclosures of the real and imaginary parts of the actual χ coefficients when they are not exact rational numbers. Ordinary complex interval addition, multiplication and conjugation in (EW.39) give rational entry enclosures of the actual $\tilde{H}_N$. For a complex interval a modulus radius is bounded rationally, for example by the sum of its two coordinate radii. Conjugation changes the sign of the imaginary coordinate; it is applied to the actual first factor. This procedure never changes χ , its degree, its monicity or the original quotient coordinates. Intervals for λ and its positive constants, if required, are propagated in the same finite operations and in the specified recurrence (EW.6).

Here is a rational matrix certificate valid for either H_m or any of these $\tilde{H}_N$. Choose a Hermitian rational centre $\widehat{M}$ and symmetric rational radii ε_{ab} enclosing its actual error. Choose any positive rational v_a and form

$$\Delta = \text{diag} \left(\sum_b \varepsilon_{ab} \frac{v_b}{v_a} \right), \quad M^- = \widehat{M} - \Delta, \quad M^+ = \widehat{M} + \Delta. \quad (\text{EW.40})$$

The pairwise inequality $2|x_a||x_b| \leq (v_b/v_a)|x_a|^2 + (v_a/v_b)|x_b|^2$ proves $M^- \preceq M \preceq M^+$. Rational complex arithmetic now eliminates each leading pivot by the congruence

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \mapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \mapsto (x - a^{-1}b^*y, y). \quad (\text{EW.41})$$

Recording strictly positive rational pivots proves $M^- > 0$ by induction and hence also $M^+ > 0$. A nonpositive pivot rejects that particular proposed certificate. Every actual matrix is already positive by its source integral and, for (EW.39), its invertible congruence. At a fixed finite degree, convergent valid entry intervals eventually give positive lower matrices: the least unit-sphere value of the actual quadratic form is positive by compactness, and the finite sum of entry errors tends to zero. This proves the finite certificate's convergence without assuming or claiming that a growing-degree arithmetic input has already been enclosed.

For the two norm endpoints take the corresponding principal submatrices of the certified H_m^-, H_m^+ . Define ω_j^-, ω_j^+ by their exact monic Schur complements. The squared norm of any fixed monic polynomial is between its two enclosing forms. Minimizing over the same affine monic set proves

$$0 < \omega_j^- \leq \omega_j \leq \omega_j^+, \quad j = n, m. \quad (\text{EW.42})$$

For each of $N = n-1, n, m-1, m$, use (EW.39)–(EW.41) and split its first q and last $N-q+1$ coordinates. The Schur complements of the latter block in $\tilde{H}_N^-, \tilde{H}_N^+$ give rational positive forms G_N^-, G_N^+ . Their affine fibre has fixed first coordinate x , by the exact map (EW.39). Minimization therefore proves

$$G_N^- \preceq G_N \preceq G_N^+, \quad 0 < V_N^- := \det G_N^- \leq V_N \leq \det G_N^+ =: V_N^+. \quad (\text{EW.43})$$

For the determinant inequality, conjugate by the square root of the smaller positive form and multiply the resulting eigenvalues, each at least one. At $N = q-1$ the boundary block is empty and the Schur complement is the whole q -dimensional form.

All numbers in the following enclosure are rational and positive:

$$\begin{aligned} U_- &= \omega_m^-/\omega_n^+, & U_+ &= \omega_m^+/\omega_n^-, \\ \mathcal{R}_- &= \max \left\{ 1, \frac{V_{n-1}^- V_n^-}{V_{m-1}^+ V_m^+} \right\}, & \mathcal{R}_+ &= \frac{V_{n-1}^+ V_n^+}{V_{m-1}^- V_m^-}. \end{aligned} \quad (\text{EW.44})$$

They satisfy $U_- \leq U \leq U_+$ and $1 \leq \mathcal{R}_- \leq \mathcal{R} \leq \mathcal{R}_+$. The use of the lower bound one follows from the proved nested minimization, rather than an interval rounding choice.

Positive roots require no floating logarithm certificate. For any positive rational x and integer $2r$, find rational numbers $l, u > 0$ such that $l^{2r} \leq x \leq u^{2r}$. Doubling upper or lower powers of two supplies an initial interval, and ordinary rational bisection makes it arbitrarily narrow. Each decision is a positive-denominator integer comparison. Strict monotonicity of $t \mapsto t^{2r}$ for $t > 0$ proves the enclosure and convergence. Apply this algorithm to

$$U_- \mathcal{R}_-, \quad U_- / \mathcal{R}_-, \quad U_+ \mathcal{R}_+, \quad U_+ / \mathcal{R}_+.$$

Write their certified root intervals as $[l_1, u_1], \dots, [l_4, u_4]$. The function in (EW.20) is increasing in $U > 0$ and $\mathcal{R} \geq 1$: its U derivative is its nonnegative value times $1/(2rU)$, and its $\mathcal{R}$ derivative is $U^{1/(2r)} \cosh(\log \mathcal{R}/(2r)) / (2r\mathcal{R}) > 0$. Therefore the complete rational endpoint certificate is

$$\boxed{\max\{0, (l_1 - u_2)/2\} \leq \mathcal{E}_{n,r} \leq (u_3 - l_4)/2.} \quad (\text{EW.45})$$

Alternatively (EW.22) gives a positive numerator and denominator certificate from the same root brackets. Every matrix pivot, integer power comparison, relation coefficient and source mass used in this algorithm is inspectable. It makes no claim for coefficient intervals which have not been provided.

303.7 Boundary cases and the retained cochain map

At $r = 1$, the block has two columns, $W_r = w$, and $\mathcal{R} = \mathcal{R}_2$ because the two copies of $\mathcal{R}_1$ cancel. Equations (EW.18)–(EW.22) reduce exactly to the two-degree formula GMT.24–27. At $n = q$, the base source degree is $q - 1$ and its boundary dimension is zero; its quotient metric is simply the source Gram. All formulas above still use positive matrices of their specified sizes. No degree $q - 2$ full-quotient form is inserted.

The actual positive polynomial source has an additional strictness property for $q \geq 1$: every two-degree quotient-volume ratio is strictly greater than one. Here is the complete algebraic reason. Let $\pi_j(u)$ be the real monic orthogonal family of the original positive real- u measure. Multiplication by u is self-adjoint. Pairing $u\pi_j$ with lower orthogonal polynomials and using their degrees gives

$$u\pi_j = \pi_{j+1} + b_j\pi_j + a_j\pi_{j-1}, \quad b_j \in \mathbb{R}, \quad a_j = \omega_j/\omega_{j-1} > 0 \quad (j \geq 1).$$

For $j = 0$ the last term is absent. The exact monic phase map $p_j(S) = i^j \pi_j((S - c)/i)$ has the same squared norm, because $|i^j| = 1$, and gives

$$Sp_j = p_{j+1} + (c + ib_j)p_j - a_j p_{j-1}.$$

If a nonconstant χ divided p_j and p_{j+1} , this recurrence and $a_j \neq 0$ would make it divide p_{j-1} . Induction would make it divide $p_0 = 1$, a contradiction. For the two appended columns at degrees $N, N + 1$ over $\mathcal{P}_{N-1}$, the residuals are p_N and $p_{N+1} + zp_N$, where the original coefficient is exactly $z = \langle p_N, S^{N+1} \rangle_H / \omega_N$. Indeed their difference from those polynomials has smaller degree, and both residuals are orthogonal to $\mathcal{P}_{N-1}$. If their full quotient residual map F_2 were zero, both p_N and p_{N+1} would have zero remainder, contradicting the divisibility argument. Thus $Z_2 = F_2^* G_{N-1} F_2$ is a nonzero positive semidefinite matrix, and (EW.17) gives $V_{N-1}/V_{N+1} > 1$ for every $N \geq q$. The finite product

$$\mathcal{R} = \prod_{j=0}^{r-1} \frac{V_{n+j-1}}{V_{n+j+1}} > 1 \quad (q \geq 1, n \geq q, r \geq 1) \quad (\text{EW.46a})$$

is its exact overlapping telescope. This proves strict positivity of the actual $\mathcal{E}_{n,r}$ and of $\mathcal{L}$ at each admitted nonempty arithmetic window; it supplies no uniform lower bound as degrees or packets

grow. The more general positive-form sensitivity theorem includes the zero-loss boundary separately.

The empty packet is a separate declared case: $h = 1$, $\chi = 1$, $q = 0$, and $\mathcal{C} = 0$. The analytic density and mass remain those of (EW.1). Put $V_N = 1$ even at the empty degree $N = -1$; all quotient matrices, projections and F_j have their empty dimensions. For $n \geq 0$, $r \geq 1$ the source norms exist, $\mathcal{R} = 1$, and $\mathcal{E}_{n,r} = 0$. If $n = 0$, use the empty base source H_{-1} , with determinant one and no inverse operation; the block $W = H_r$ then has its original monomial coordinates. Its first entry is the full mass. The projection and volume sensitivity statements become their zero-map identities, while the source-norm sensitivity remains the proved rank-two calculation.

Finally the projection comparison (EW.26) has an explicit original theta representative. For $a < b$ and $x \in \mathcal{C}$, monic division gives a unique polynomial $\kappa_x \in \mathcal{P}_{b-q}$ with

$$\iota_a R_a x - \iota_b R_b x = \chi \kappa_x. \quad (\text{EW.46})$$

Its coefficients are obtained by the same exact division, so the map $x \mapsto \kappa_x$ is linear. Let $\mathcal{V}_{h,k} P = P(D_1 + \cdots + D_k) F_h^{\otimes k}$, where $D = -x \partial_x$, $\mathcal{M} F_h = v_h$, and $h(D) F_h = \Theta \phi_0$. Successive division in $s_1, \dots, s_k$ supplies specific Q_i with $\chi(\sum s_i) = \sum_i h(s_i) Q_i(\mathbf{s})$. The final tensor remainder vanishes because its monomials form a basis of $(\mathbb{C}[s]/(h))^{\otimes k}$ and χ annihilates the original sum. Thus the exact cochain of degree $k - 1$

$$\mathcal{P}_\chi(\kappa_x) = \sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) \kappa_x(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}) \quad (\text{EW.47})$$

satisfies

$$d_{\text{tot}} \mathcal{P}_\chi(\kappa_x) = \mathcal{V}_{h,k} (\iota_a R_a x - \iota_b R_b x). \quad (\text{EW.48})$$

Indeed the tensor differential at position i contributes $(-1)^{i-1}$ because its preceding factors have cochain degree one; this cancels the written sign, and $h(D_i) F_h = \Theta \phi_0$ then proves the equality term by term. The arithmetic jet of both representatives is the same ηx of (EW.3). In the original support carrier $\{\tau\} \sqcup \bigsqcup_\ell (\{\ell\} \times E_\ell)$, a fibrewise linear map sends (ℓ, v) to $(\ell, L_\ell v)$ and τ to τ . Thus the displayed boundary maps to $e_\ell = (\ell, 0)$ while external absence remains τ . The endpoint's representative changes, projection angles, and volume changes have therefore all been related to the original typed maps and their full theta primitives.

303.8 Verification scope and exact auxiliary calibration

The companion checker uses the explicitly auxiliary weight

$$\begin{aligned} \lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = 1/2, \quad C_2 = 1/4, \\ B(t) = 1 + t^2/4, \quad w = B r_1, \quad \chi(S) = S^2 - 2S - 1. \end{aligned} \quad (\text{EW.49})$$

This is a specified positive $L^2(r_1)$ multiplier and the same finite coefficient-to-Gram domain as (EW.8)–(EW.10). It is not asserted to be an actual zeta packet. Its exact connection to the arithmetic problem is the complete sequence of maps (EW.10), (EW.11), (EW.14)–(EW.23), and (EW.39)–(EW.45), with its auxiliary inputs written explicitly.

For this fixture $b_2^{(1)} = t^2 - 2$, so $c_{B,0} = 3/2$, $c_{B,2} = 1/4$, and every other coefficient is zero. Consequently $\sum_j (2)_j c_{B,j} z^j = 3/2 + 3z^2/2$ and its square has coefficients $d_0 = 9/4$, $d_2 = 9/2$, $d_4 = 9/4$. Its one-factor mass is $3/4$, the literal sum mass is $9/16$, and the reference sum mass is $1/4$. The checker independently constructs Gamma moments from the monic recurrence, inserts $1 + t^2/4$ in each factor, and expands every power of $t_1 + t_2$ with its full binomial coefficient. It also constructs the coefficient-side moments from (EW.9); agreement compares two exact finite calculations, including the tensor cross terms.

In this auxiliary example gamma recurrence gives a further exact all-degree check. Since $|\Gamma(2 + it/2)|^2 = (1 + t^2/4)|\Gamma(1 + it/2)|^2$, one has $Br_1 = r_2$. The displayed masses give $c_2 = 3/4$, $c_4 = 315/8$, and hence

$$m = r_2 * r_2 = r_4/70, \quad r_1 * r_1 = r_2/3, \quad \omega_j = \frac{9}{16}j!(8)_j, \quad U = (n+1)_r(n+8)_r. \quad (\text{EW.50})$$

The first two equalities follow from the literal convolution law (EW.7), with coefficients $(3/4)^2/(315/8) = 1/70$ and $(1/2)^2/(3/4) = 1/3$. The monic original-coordinate map is $p_j(S) = i^j b_j^{(4)}((S-1)/i)$, whose squared norm is exactly the third quantity by (EW.7). Taking the stated endpoint ratio proves the final formula.

The exact selected windows computed in the accompanying rational receipt are

(n, r)	U	$\mathcal{R}$
(2, 1)	30	20191/4860
(2, 2)	1320	3073295611/216513000
(3, 2)	2640	295913127219469/36104825790000

(EW.51)

Here is a full independent finite derivation of the entries in that table, using the original quotient kernel rather than a Schur block. Let $x = S - 1$ solely in the stated quotient-coordinate map

$$A : \mathbb{C}_{(1,x)}^2 \longrightarrow \mathbb{C}_{(1,S)}^2, \quad (a, b) \longmapsto (a - b, b), \quad A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \det A = 1.$$

It takes the identical relation to $x^2 - 2$, so its residue equation is exactly $x^2 = 2$. It preserves the determinant of the quotient form by the congruence $G_N^x = A^* G_N A$. The monic source family in (EW.50) obeys $p_{j+1} = x p_j + j(j+7)p_{j-1}$, by applying the exact phase map to (EW.6). Orthogonal expansion of the original source inverse in (EW.11) proves the finite quotient kernel formula

$$(G_N^x)^{-1} = \sum_{j=0}^N \frac{[p_j]_{(1,x)} [p_j]_{(1,x)}^*}{\omega_j}. \quad (\text{EW.52})$$

Indeed the matrix of the monic orthogonal columns in the original source is invertible triangular with diagonal one, takes H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, and sends the reduction columns to their written remainders. Substituting that inverse congruence into $J_N H_N^{-1} J_N^*$ gives (EW.52) term by term. The recurrence and $x^2 = 2$ give the complete required list

j	$[p_j]_{(1,x)}$	ω_j
0	(1, 0)	9/16
1	(0, 1)	9/2
2	(10, 0)	81
3	(0, 28)	2430
4	(356, 0)	106920
5	(0, 1588)	6415200

(EW.53)

For example the successive remainders are obtained from $x^2 + 8 = 10$, $10x + 18x = 28x$, $28x^2 + 300 = 356$, and $356x + 44 \cdot 28x = 1588x$. The positive kernels from (EW.52) are diagonal in this explicitly transported quotient basis, with

N	$((G_N^x)^{-1})_{00}$	$((G_N^x)^{-1})_{11}$	V_N
1	16/9	2/9	81/32
2	244/81	2/9	729/488
3	244/81	662/1215	98415/161528
4	56102/13365	662/1215	16238475/37139524
5	56102/13365	376069/400950	2679348375/10549111519

(EW.54)

Each diagonal entry is the indicated finite sum of rational terms in (EW.52), and V_N is the reciprocal of their product because $\det A = 1$. Substituting (EW.53)–(EW.54) in (EW.18) gives all three rows of (EW.51) by positive-denominator multiplication.

There is also an executed coefficient-box certificate, before any matrix error is chosen. Put $\varepsilon = 10^{-12}$ and retain $n = 2, r = 1, m = 3$, the same reference mass and the same χ . Allow the complete specified one-factor input box

$$|c_{B,0} - 3/2| \leq \varepsilon, \quad |c_{B,2} - 1/4| \leq \varepsilon, \quad c_{B,j} = 0 \quad (j = 1, 3, 4, 5, 6). \quad (\text{EW.55})$$

Every point of this box is realized by the specified positive multiplier $B_{a,b}(t) = a + bt^2$, where $a = c_{B,0} - 2c_{B,2}$ and $b = c_{B,2}$. They obey $a \geq 1 - 3\varepsilon > 0$ and $b \geq 1/4 - \varepsilon > 0$. Its coefficient formula follows by $a + bt^2 = (a + 2b)b_0^{(1)} + bb_2^{(1)}$. The interior input $a = 1 + \varepsilon, b = 1/4 - \varepsilon/4$ has one-factor mass $3/4 + \varepsilon/4$ and sum mass $(3/4 + \varepsilon/4)^2$. Thus neither the box nor its test input replaces the zeroth coefficient by a unit mass.

For (EW.55), the complete coefficient-power error formula (EW.37) is exactly

$$\begin{aligned} e_0^{(2)} &= 3\varepsilon + \varepsilon^2, & e_2^{(2)} &= 21\varepsilon + 12\varepsilon^2, & e_4^{(2)} &= 18\varepsilon + 36\varepsilon^2, \\ e_1^{(2)} &= e_3^{(2)} = e_5^{(2)} = e_6^{(2)} &= 0. \end{aligned} \quad (\text{EW.56})$$

Indeed the central polynomial is $3/2 + (3/2)z^2$ and its error polynomial is $\varepsilon + 6\varepsilon z^2$; expanding the full square difference gives the three displayed coefficients. The upper corner of the coefficient box attains all three positive differences, so its quadratic error terms cannot be omitted.

Expanding all sixteen original tests $(1 - iu)^a(1 + iu)^b, 0 \leq a, b \leq 3$, in the monic Gamma basis of (EW.6), then using (EW.38)–(EW.40) with four weights equal to one, gives

$$\begin{aligned} \hat{H}_3 &= \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix}, \\ \Delta_{00} &= 54\varepsilon + 28\varepsilon^2, & \Delta_{11} &= (1029\varepsilon + 873\varepsilon^2)/2, \\ \Delta_{22} &= (2109\varepsilon + 1773\varepsilon^2)/2, & \Delta_{33} &= 29754\varepsilon + 32878\varepsilon^2. \end{aligned} \quad (\text{EW.57})$$

These diagonal sums retain the absolute value of every test coefficient. The real coefficients occur at even degrees and the imaginary coefficients at odd degrees; the latter pair exactly to zero against every member of the specified even coefficient box. The complete complex expansions, including those zero pairings, are recorded in the certificate.

Here is an elementary exact positive-definiteness certificate for the whole box. Rational multiplication verifies

$$\hat{H}_3^{-1} = \frac{1}{2430} \begin{pmatrix} 8019 & -1863 & 351 & -27 \\ -1863 & 1501 & -147 & 29 \\ 351 & -147 & 39 & -3 \\ -27 & 29 & -3 & 1 \end{pmatrix}, \quad \text{Tr } \hat{H}_3^{-1} = 956/243 < 4.$$

The central form is positive by its actual measure, or by the four original monic norms in (EW.53). Its inverse is positive, so each of its eigenvalues is at most its trace. Consequently $\hat{H}_3 \succ I_4/4$. Substitution of $\varepsilon = 10^{-12}$ into the four full polynomials (EW.57) gives $0 < \Delta_{jj} < 10^{-6}$ by integer comparison. Hence

$$\hat{H}_3 - \Delta \succ (1/4 - 10^{-6})I_4 \succ 0, \quad \hat{H}_3 - \Delta \preceq H_3 \preceq \hat{H}_3 + \Delta \quad (\text{EW.58})$$

for every actual multiplier in (EW.55). This proves the positive matrix enclosure before its inverses are evaluated. The accompanying receipt also records each exact rational elimination pivot separately.

Use the original J_N and χ at $N = 1, 2, 3$ in (EW.42)–(EW.45). The resulting rational interval for every endpoint value in this whole coefficient box is

$$\boxed{\frac{10247872526420120792621067}{2^{81}} \leq \mathcal{E}_{2,1} \leq \frac{10247872551311149445602506}{2^{81}}}. \quad (\text{EW.59})$$

The four endpoint arguments in (EW.45) are computed from the written rational matrices $\hat{H}_3 \pm \Delta$ by their monic and quotient Schur formulas; the receipt gives every one of these exact rational values and its two squared-root comparisons. Thus (EW.59) is certified by integer powers and the minimum principle, with no rounding assertion about a transcendental logarithm. The interior positive multiplier and all four box corners are independently reconstructed by the literal one-factor moment recurrence and tensor binomial expansion, and pass the same original-coordinate entry bounds.

All original- S Gram entries are computed from $(1 - iu)^a(1 + iu)^b$. The checker's window certificates include the exact reduction matrix, source norms, quotient metrics, the $r + 1$ Schur block and all its prefixes, both cancellations in (EW.21), positive rational matrix envelopes, and the integer power comparisons for (EW.45). The written proof establishes the all-degree statements; each executed finite check is reported separately in the accompanying receipt. The base fixture executes 416 distinct mathematical checks across fifteen windows; its normal and optimized runs pass and six changed-formula runs in each mode are rejected after executing the complete suite. The separate supplement executes 329 mathematical checks and two dependency-hash checks; its four changed-formula runs in each mode are also rejected after the complete suite, including the omitted quadratic coefficient error at an attained box corner. The independent remainder-kernel calibration executes the six scalar comparisons in (EW.51). These counts describe their separate scopes; repeated modes and changed-formula runs do not add new mathematical checks. Numerical examples do not stand in for the arithmetic input or a growing-window bound.

304 Complete Gamma-reference lower bound and arithmetic determinant comparison

Source role. Complete written proof, with original source bytes and reversible notation/publication mapping retained.

Exact revision. 830fbfe19d688b1e138b41ea4a784c662629dfec71812627b66927f91ce468f

Complete source. `sources/next_edition/actual_tau_conversions_v1/sources/gamma_review/deligne_bound_gamma_independent_review_20260913.md`

Source SHA-256. 830fbfe19d688b1e138b41ea4a784c662629dfec71812627b66927f91ce468f

304.0.1 Independent proof audit of the Gamma-reference lower bound

Date: 13 September 2026. Status: the Gamma theorem and the signed arithmetic comparison are accepted with the domains proved below. This review gives the complete calculation for delivered equations (28)–(43) and (51)–(56). Its appended Gamma derivation is part of this proof body.

304.0.2 Sources and reading scope

Paths below are relative to the mathematical workspace. The delivered files have not been changed.

Source	Bytes	SHA-256	Reading scope
output/split_zero_rh_t andem_2026-09-12/ sources/web_deligne_ bound_audit_delivery /Tau_Deligne_Bound _Audit/NOTE.tex The same delivery's verify.py	35701 11159	bfe63094ec360b6059 f2936e5e456c6c0fd0 8db144bbabc29621a5 27b89c13c6 fb39ffeeec2de6002c d477e92ab2db1e52ed dfdd945f8596f15702 63cf9c2cb0	Complete note Complete code; no execution of this old suite is claimed here
The same delivery's SOURCES.json	3582	6edb542a72730aa6ba 05db9f37700b65a50e 345e96fdc5f48bf623 e9e47b2286	Complete source record
output/split_zero_rh_t andem_2026-09-12/te x/arithmetic_input.tex	14644	a50b0fb587b644c4ea 94dba45b2b423d038f 2bfbe7fef188543939 ba850e58d2	Complete A1–A19 and proofs
output/split_zero_rh_t andem_2026-09-12/te x/toda_cv_exact_brid ge.tex	45603	c9629b11d4b4a173dd 52bdf75fafb437883a 85aba05e598c637f39 0386d45512	Lines 682–1098: separate first degree, phase term, spectral projection, consecutive estimates, and scope
output/split_zero_rh_t andem_2026-09-12/ sources/web_gamma_ convolution_delivery /Tau_Gamma_Convolu tion_Descent/NOTE.te x	34859	7cf984d87de405b815 250b44454c4ee27c8a b46574da245b377ebc 6e49327fa2	Opening and Sections 2–6.1: literal measure, sum coordinate, amplitude, conditional multiplier and bounds
work/deligne_bound _gamma_generating_ subreview_20260913 .md	11155	d4dfa59e923e7c7223 b31396e84dc17b595f 2fdd5eeddbadb63f3d 11aec971cb	Complete independent Gamma derivation; appended verbatim below

This is an all-degree written proof. No fresh Lean execution, complete Deligne scan audit, interval certification of arithmetic moments, or remote publication is claimed by this review. The purity/operator branch is being audited separately.

304.0.3 1. Literal norms and their coordinate map

The complete Beta/Fourier calculation in the appended proof derives, for $a > 0$,

$$r_a(x) = \frac{|\Gamma(a + ix/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \quad \int e^{iyx} r_a(x) dx = c_a (\cosh y)^{-2a}. \quad (\text{GR1})$$

In particular c_a is the original mass. The literal substitution $t = e^{2y}$ in the Beta integral gives the factor 2^{1-2a} . Exponential moments exist for $|\theta| < \pi/2$, and the same calculation gives $c_a (\cos \theta)^{-2a}$. Fourier uniqueness for finite measures then gives

$$r_a^{*k}(x) = \frac{c_a^k}{c_{ka}} r_{ka}(x). \quad (\text{GR2})$$

The coordinate is the original sum $x = x_1 + \cdots + x_k$. At $a = 1/4$, $c_{1/4} = \sqrt{2\pi}$, so the mass is $(2\pi)^{k/2}$.

The printed generating function, with branches equal to one at zero, is

$$F_a(z, x) = (1 - iz)^{-a+ix/2} (1 + iz)^{-a-ix/2} = \sum_{n \geq 0} b_n^{(a)}(x) \frac{z^n}{n!}. \quad (\text{GR3})$$

For real small z , it is $(1 + z^2)^{-a} \exp(x \arctan z)$. Thus $b_1 = x$, with the positive sign. The appended proof shows

$$\int F_a(z, x) F_a(w, x) r_a(x) dx = c_a (1 - zw)^{-2a}. \quad (\text{GR4})$$

All coefficient extractions are justified by exponential moments on a larger neighborhood. Thus the real, monic polynomials have orthogonal norms $c_a n! (2a)_n$. Their recurrence follows from $(1 + z^2) \partial_z F_a = (x - 2az) F_a$:

$$b_{n+1}^{(a)} = x b_n^{(a)} - n(n + 2a - 1) b_{n-1}^{(a)}, \quad b_0 = 1, \quad b_1 = x. \quad (\text{GR5})$$

Keep $S = k/2 + ix$ and define

$$p_n(S) = i^n b_n^{(k/4)}((S - k/2)/i). \quad (\text{GR6})$$

The leading coefficient is $i^n i^{-n} = 1$. On the integration line the factor i^n has squared modulus one. Substituting in GR5 and using $i^2 = -1$ gives, with the actual original Gamma sum measure,

$$\begin{aligned} \omega_{k,n}^\Gamma &= \int |p_n(k/2 + ix)|^2 r_{1/4}^{*k}(x) dx = (2\pi)^{k/2} n! (k/2)_n, \\ S p_n &= p_{n+1} + \frac{k}{2} p_n - n(n + k/2 - 1) p_{n-1}. \end{aligned} \quad (\text{GR7})$$

At $n = 0$ the last term is absent. Thus delivered (28), (29), and (35) have the correct masses, monic convention, and negative recurrence sign.

304.0.4 2. The complete cyclic polynomial for every multiplicity

Write

$$h_*(S) = \prod_{\epsilon, \eta \in \{\pm 1\}} (S - 1/2 - \epsilon/4 - i\eta). \quad (\text{GR8})$$

For $Z = S - 1/2$, multiplying conjugate factors gives $h_* = Z^4 + (15/8)Z^2 + 289/256$. It is monic, real, and fixed by $S \mapsto 1 - S$.

Fix integers $m \geq 1, k \geq 1$. Consider the full tensor algebra of k copies of $\mathbb{C}[S]/(h_*^m)$, the sum multiplication operator, and cyclic vector $1^{\otimes k}$. Chinese remainders decompose each factor into four local algebras $\mathbb{C}[n]/(n^m)$. Every tensor occupation therefore has local algebra

$$\mathbb{C}[n_1, \dots, n_k]/(n_1^m, \dots, n_k^m)$$

and sum operator equal to its center plus $n_1 + \dots + n_k$. Its power of degree $k(m-1) + 1$ vanishes because every monomial then has some exponent at least m . Its power of degree $k(m-1)$ is exactly

$$\frac{(k(m-1))!}{((m-1)!)^k} \prod_{j=1}^k n_j^{m-1} \neq 0. \quad (\text{GR9})$$

For $m = 1$ this is the zeroth power 1, giving the same length-one result. The coefficient is nonzero in characteristic zero. The cyclic vector projects to the unit in every local algebra, so this full nilpotency length also occurs in its cyclic minimal polynomial.

For margins $0 \leq a, b \leq k$, the center is

$$\lambda_{a,b} = k/2 + (2a - k)/4 + i(2b - k). \quad (\text{GR10})$$

All margin pairs occur. Indeed, $t = \max(0, a + b - k)$ gives nonnegative occupations $t, a - t, b - t, k - a - b + t$, with total k and the required two margins. Distinct margins have distinct centers because both coordinate coefficients are nonzero. Repeated occurrences of the same center take the maximum of their identical nilpotency lengths in the minimal polynomial, not their sum. Therefore exactly

$$\ell_k = 1 + k(m-1), \quad \chi_k(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^{\ell_k}, \quad q_k = \ell_k(k+1)^2 \quad (\text{GR11})$$

is the cyclic minimal polynomial. Each primary summand in $\mathbb{C}[S]/(\chi_k)$ has dimension and block length ℓ_k . This proves delivered (30), including all $m > 1$.

304.0.5 3. Canonical quotient metric and the complete boundary identity

Fix k, m and abbreviate $q = q_k, \chi = \chi_k, c = k/2$. Use the ordered remainder basis $1, S, \dots, S^{q-1}$ of $E = \mathbb{C}[S]/(\chi)$, and let $A : E \rightarrow E$ be multiplication by S . Write $b_n = [p_n]_\chi$.

For $N \geq q - 1$, the first q monic remainder columns are a basis. In the source basis $p_0, \dots, p_N$, the source Gram is $O_N = \text{diag}(\omega_0, \dots, \omega_N)$, and the remainder matrix is $J_N = [b_0 \cdots b_N]$. Hence

$$K_N = J_N O_N^{-1} J_N^* = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n} > 0. \quad (\text{GR12})$$

The coefficient matrix of the canonical minimum section is

$$C_N = O_N^{-1} J_N^* K_N^{-1}. \quad (\text{GR13})$$

One has $J_N C_N = I_E$. If $z \in \ker J_N$, then $z^* O_N C_N v = z^* J_N^* K_N^{-1} v = 0$. Every other representative is $C_N v + z$, whose squared norm is $\|C_N v\|^2 + \|z\|^2$. Thus the actual minimum quotient Gram and volume are

$$G_N = K_N^{-1}, \quad V_N = \det G_N > 0. \quad (\text{GR14})$$

Put $\alpha_n = n(n + k/2 - 1)$. GR7 gives $\omega_n/\omega_{n-1} = \alpha_n$, and reduction gives $Ab_n = b_{n+1} + cb_n - \alpha_n b_{n-1}$. Substitution in $AK_N + K_N A^* - kK_N$ cancels each interior pair: the term $(b_{n+1}b_n^* + b_n b_{n+1}^*)/\omega_n$ is cancelled by the next negative term because $\alpha_{n+1}/\omega_{n+1} = 1/\omega_n$. No lower term occurs at zero. Therefore

$$AK_N + K_N A^* - kK_N = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{GR15})$$

Multiplication by G_N on both sides gives

$$W_N := A^* G_N + G_N A - k G_N = G_N \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N} G_N. \quad (\text{GR16})$$

Thus $H_N = G_N^{-1} W_N$ is self-adjoint in the original G_N inner product and has rank at most two. Its trace is $2\Re \text{Tr } A - kq = 0$: reflection $(a, b) \leftrightarrow (k - a, b)$ pairs real parts with sum k , retaining equal full multiplicities. Its possible nonzero eigenvalues are $+\epsilon_N, -\epsilon_N$, with $\epsilon_N \geq 0$.

To preserve the whole complex cross term, apply the exact isometry $G_N^{1/2} : (E, G_N) \rightarrow (\mathbb{C}^q, I)$. Put $u = G_N^{1/2} b_N, v = G_N^{1/2} b_{N+1}$, and $z_N = u^* v = b_N^* G_N b_{N+1}$. The conjugated matrix is $(vu^* + uv^*)/\omega_N$. Its zero trace says $\Re z_N = 0$. Direct multiplication gives trace of its square equal to

$$\frac{2\|u\|^2\|v\|^2 + 2\Re(z_N^2)}{\omega_N^2} = \frac{2(\|u\|^2\|v\|^2 - |z_N|^2)}{\omega_N^2}.$$

This trace is also $2\epsilon_N^2$, including the zero-rank case. Consequently

$$\epsilon_N^2 = \frac{(b_N^* G_N b_N)(b_{N+1}^* G_N b_{N+1}) - |z_N|^2}{\omega_N^2}. \quad (\text{GR17})$$

The isometry does not assign a different metric to either source. The phase term remains present before every ensuing inequality.

304.0.6 4. The reflected generalized trace and its exact count

Let $E_{>} \subset E$ be the sum of all primary spaces with $\Re \lambda > k/2$, retaining every power in GR11. It is A -invariant. Let $P_{>}$ be its G_N -orthogonal projection. In $E_{>} \oplus E_{>}^{\perp G_N}$, the lower-left block of A vanishes. Taking the first diagonal block of $A^{\dagger G_N} + A - kI$ proves

$$\text{Tr}(P_{>} H_N) = 2\Re \text{Tr}(A|_{E_{>}}) - k \dim E_{>}. \quad (\text{GR18})$$

If $\epsilon_N > 0$, choose unit eigenvectors u_+, u_- at its two nonzero eigenvalues. Its spectral expansion gives $\text{Tr}(P_{>} H_N) = \epsilon_N \|P_{>} u_+\|^2 - \epsilon_N \|P_{>} u_-\|^2 \leq \epsilon_N$. If $\epsilon_N = 0$, the same trace is zero. This uses the actual orthogonal projection and does not silently identify it with a generally nonorthogonal algebraic spectral projector.

The exact count is

$$\sum_{a > k/2} (2a - k) = \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{GR19})$$

For $k = 2r$ it is $2 + 4 + \dots + 2r = r(r+1)$; for $k = 2r+1$ it is $1 + 3 + \dots + (2r+1) = (r+1)^2$. There are $k+1$ choices of imaginary margin and multiplicity ℓ_k at each center. Thus

$$L_k := \text{Tr}(P_{>} H_N) = \frac{1}{2} \ell_k (k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_N. \quad (\text{GR20})$$

The two parity formulas also prove $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$, so

$$L_k \geq kq_k/8 > 0. \quad (\text{GR21})$$

The trace retains the full generalized dimensions for every m .

304.0.7 5. Consecutive degrees and the separate first degree

For $N \geq q$, the update is $K_N = K_{N-1} + b_N b_N^*/\omega_N$. If $t = b_N^* G_{N-1} b_N/\omega_N \geq 0$, the determinant lemma gives $\delta_N := V_N/V_{N-1} = (1+t)^{-1} \in (0, 1]$. Multiplying the rank-one inverse formula verifies $b_N^* G_N b_N = \omega_N t/(1+t)$. The next determinant update uses this same G_N . Hence

$$\begin{aligned} b_N^* G_N b_N &= \omega_N (1 - \delta_N), \\ b_{N+1}^* G_N b_{N+1} &= \omega_{N+1} (\delta_{N+1}^{-1} - 1). \end{aligned} \quad (\text{GR22})$$

Substitution in GR17 yields

$$\epsilon_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1) - \frac{|z_N|^2}{\omega_N^2}. \quad (\text{GR23})$$

Because $0 \leq 1 - \delta_N \leq 1$ and $0 \leq \delta_{N+1}^{-1} - 1 \leq \delta_{N+1}^{-1}$, this gives

$$\epsilon_N^2 \leq \frac{\omega_{N+1}}{\omega_N} \frac{V_N}{V_{N+1}} = \frac{\Lambda_{N+1}}{\Lambda_N}, \quad \Lambda_N = \omega_N / V_N. \quad (\text{GR24})$$

At $N = q - 1$, do not invert a preceding rank-deficient Gram. Expand the original monic relation

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \nu_0 = \|\chi\|_\Gamma^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j. \quad (\text{GR25})$$

The degree- $(q - 1)$ remainder map is an isomorphism, so its canonical metric satisfies $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Reducing GR25 gives $b_q = -\sum_{j < q} \gamma_j b_j$. Therefore

$$\begin{aligned} b_{q-1}^* G_{q-1} b_{q-1} &= \omega_{q-1}, \\ b_q^* G_{q-1} b_q &= \nu_0 - \omega_q, \\ z_{q-1} &= -\gamma_{q-1} \omega_{q-1}, \\ V_q / V_{q-1} &= \omega_q / \nu_0. \end{aligned} \quad (\text{GR26})$$

The final equality is the forward determinant lemma at K_q . Substitution in GR17 proves the stronger exact endpoint identity

$$\begin{aligned} \epsilon_{q-1}^2 &= \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} \\ &= \frac{\nu_0 - \omega_q}{\omega_{q-1}} - |\gamma_{q-1}|^2 \\ &\leq \frac{\nu_0}{\omega_{q-1}} = \frac{\Lambda_q}{\Lambda_{q-1}}. \end{aligned} \quad (\text{GR27})$$

Thus delivered (41) is valid. No G_{q-2} , K_{q-2}^{-1} , or V_{q-2} is used. The endpoint phase term is present before the inequality.

304.0.8 6. Both windows and the literal constant

For either $i = q - 1$ or $i = q$, let $j = i + q$. For each $n = i, \dots, j - 1$, GR20 gives $L_k^2 \leq \epsilon_n^2$, and GR24 or its one initial instance GR27 gives $\epsilon_n^2 \leq \Lambda_{n+1} / \Lambda_n$. Multiplication of these q inequalities telescopes:

$$L_k^{2q} \leq \frac{\Lambda_j}{\Lambda_i} = \frac{\omega_j}{\omega_i} \frac{V_i}{V_j}. \quad (\text{GR28})$$

The indices in ω_j / ω_i are $n = i + 1, \dots, j$, all at most $2q$. Also $k \leq q$ by GR11. Every positive recurrence factor therefore satisfies

$$n(n + k/2 - 1) \leq 2q(2q + q/2 - 1) \leq 5q^2. \quad (\text{GR29})$$

Thus, cancelling the common mass only in the displayed ratio,

$$\frac{\omega_j}{\omega_i} = \prod_{n=i+1}^j n(n + k/2 - 1) \leq (5q^2)^q. \quad (\text{GR30})$$

Combining GR21, GR28, and GR30 yields, at each of the two windows,

$$\frac{V_i}{V_j} \geq \left(\frac{k}{8\sqrt{5}} \right)^{2q}, \quad \log(V_i / V_j) \geq 2q \log \left(\frac{k}{8\sqrt{5}} \right). \quad (\text{GR31})$$

Their final degrees are respectively $2q - 1$ and $2q$. Addition proves

$$\mathcal{B}_k^\Gamma = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}} \geq 4q_k \log \left(\frac{k}{8\sqrt{5}} \right). \quad (\text{GR32})$$

When the right side is negative the inequality remains valid; the separate monotonicity $V_i/V_j \geq 1$ does not conflict with it. For $k > 1$, division by $q_k \log k$ gives $4 - 4 \log(8\sqrt{5})/\log k$, tending to four. This proves the claimed liminf for every fixed $m \geq 1$.

For centers $1/2 \pm \delta \pm i\gamma$ with fixed $\delta > 0, \gamma \neq 0$, the same occupation calculation is valid because both margins remain distinct. Its trace is $L = 2\delta\ell_k(k+1)\lfloor(k+1)^2/4\rfloor \geq \delta kq/2$. The identical two windows therefore give

$$\mathcal{B}_{h,k}^\Gamma \geq 4q_k \log \left(\frac{\delta k}{2\sqrt{5}} \right). \quad (\text{GR33})$$

This proves delivered (55) for its stipulated quartet geometry. It does not assert that an off-line zeta quartet exists.

The optional Gaussian comparison also checks exactly. Its mass and monic norms are 7^k and $7^k k^n n!$, so the ratio at index $n \leq 2q$ is $kn \leq 2kq$. The same $L \geq kq/8$ gives each window at least $q \log(kq/128)$. At $m = 1$, $q = (k+1)^2$, making this $3q \log k + q(2 \log(1 + 1/k) - \log 128)$. The combined liminf is at least six. The mass remains in each norm and cancels only from the ratio. This calibration does not change either the Gamma source or the original zeta source.

304.0.9 7. The actual arithmetic source and the exact determinant correction

Now fix an actual finite packet h of zeros of $g = 2\xi$ with its complete selected orders. Retain its actual cyclic polynomial $\chi = \chi_{h,k}$, quotient E , and full observation

$$\begin{aligned} \mathcal{V}_{h,k}P &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, & \mathcal{M}F_h &= g/h, \\ J^{(k)}\mathcal{V}_{h,k} &= \eta_{h,k}\pi_\chi, & \eta_{h,k}[P] &= v_h^{\otimes k}P(A_h^{(k)})1, & v_h &= j_h(g/h). \end{aligned} \quad (\text{GR34})$$

The actual amplitude and scalar norm observation are

$$\begin{aligned} A_h(t) &= \frac{(g/h)(1/2 + it)}{\Gamma(1/4 + it/2)}, & B_h(t) &= |A_h(t)|^2, \\ w_h(t) &= B_h(t)r_{1/4}(t), & B_{h;k}(u) &= C_\Sigma(B_h^{\otimes k})(u), \\ m_{h,k}(u) &= r_{1/4,k}(u)B_{h;k}(u). \end{aligned} \quad (\text{GR35})$$

Here C_Σ is the original conditional integral on $u = t_1 + \cdots + t_k$, with denominator the actual fibre mass $r_{1/4,k}(u)$. Fubini proves the last equality by testing against an arbitrary nonnegative measurable function of that sum. The full complex amplitude multiplies the Gamma tensor amplitude before taking its squared norm; GR35 therefore does not replace that source vector or its phase by a real scalar.

At every degree $N \geq q - 1$, use the same monomial coefficient space $\mathcal{P}_N$, remainder map J_N , and relation injection

$$B_N : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad Q \longmapsto \chi Q. \quad (\text{GR36})$$

At $N = q - 1$, the domain is zero. Let $M_N^\Gamma, M_N^{\text{ar}}$ be the original two source Grams. Each is positive: Gamma has positive density; the nonzero entire function g/h vanishes at a discrete subset of the line, so no nonzero polynomial has zero arithmetic norm. The convolution retains positivity. Every moment is finite in the original source; for a quartet the proved $B_{h;k} \leq C_h^k$ and Gamma moments also prove finiteness.

The coefficient identity $I_N : P \mapsto P$, from $(\mathcal{P}_N, M_N^\Gamma)$ to $(\mathcal{P}_N, M_N^{\text{ar}})$, intertwines both J_N and B_N . Its induced quotient map is the identity on E with the two stated metrics.

Let R_N^Γ be the Gamma minimum section. Every arithmetic representative of a class v is uniquely $R_N^\Gamma v + B_N z$. Orthogonality to the arithmetic relation subspace gives $(B_N^* M_N^{\text{ar}} B_N)z = -B_N^* M_N^{\text{ar}} R_N^\Gamma v$. The relation Gram is positive on its actual domain. Thus

$$R_N^{\text{ar}} = R_N^\Gamma - B_N(B_N^* M_N^{\text{ar}} B_N)^{-1} B_N^* M_N^{\text{ar}} R_N^\Gamma. \quad (\text{GR37})$$

At zero relation dimension, the inverse is the unique automorphism of the zero space; the composite is the zero map $E \rightarrow \mathcal{P}_N$. Applying the actual source map $\mathcal{V}_{h,k}$ to the difference gives an original relation and its theta primitive, while GR34 keeps the full arithmetic jet unchanged. This proves delivered (53) with its exact source, codomain, and original unit.

There is a full determinant formula. Define

$$L_N : E \oplus \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad ([P], Q) \longmapsto \text{rem}_\chi(P) + \chi Q. \quad (\text{GR38})$$

In the ordered monomial bases, the first columns are $1, S, \dots, S^{q-1}$; each later χS^j has coefficient one at degree $q+j$ and zero above. The matrix is triangular with determinant one. Its inverse is monic polynomial division into the remainder and quotient. For either source Gram M , $L_N^* M L_N$ has lower-right block $H = B_N^* M B_N$. The triangular change of coefficients determined by GR37 completes the square, with determinant one, and its other diagonal block is the minimum quotient Gram G . Hence

$$\det M = (\det G)(\det H), \quad V_N = \frac{\det M_N}{\det(B_N^* M_N B_N)}. \quad (\text{GR39})$$

At $N = q - 1$, the relation determinant is the empty determinant one. Therefore the positive quotient-volume correction $T_{k,N} = V_{k,N}^{\text{ar}} / V_{k,N}^\Gamma$ is exactly

$$T_{k,N} = \frac{\det M_N^{\text{ar}} \det(B_N^* M_N^\Gamma B_N)}{\det M_N^\Gamma \det(B_N^* M_N^{\text{ar}} B_N)}. \quad (\text{GR40})$$

Taking the four literal endpoint ratios gives

$$\mathcal{B}_{h,k}^{\text{ar}} - \mathcal{B}_{h,k}^\Gamma = \log T_{k,q-1} + \log T_{k,q} - \log T_{k,2q-1} - \log T_{k,2q}. \quad (\text{GR41})$$

This proves delivered (54). The final two minus signs are essential. An individual upper bound on each positive $T_{k,N}$ provides no lower bound for those denominator factors. On the reference itself, $B_{h,k} = 1$ is realized, all $T_{k,N} = 1$, and the correction is zero while GR32 holds. Positivity and the assertion that the multiplier is bounded, without other restrictions, include this reference calibration.

For the stipulated actual off-line quartet displacement $\delta > 0$, an arithmetic upper estimate $(4 - \varepsilon)q \log k$, together with the proved GR33, would force exactly

$$\log \frac{T_{k,q-1} T_{k,q}}{T_{k,2q-1} T_{k,2q}} \leq -\varepsilon q \log k - 4q \log \left(\frac{\delta}{2\sqrt{5}} \right). \quad (\text{GR42})$$

The last term supplies the complete constant represented by the delivered $O_h(q)$. This is the necessary consequence of the displayed estimates, not an assertion that the arithmetic upper estimate has been proved.

304.0.10 8. What this establishes about the broader program

For any monic test polynomial h_0 with centers in the open critical strip, the original theta source proves the exact quotient square

$$\begin{array}{ccc} V/h_0(D)V & \xrightarrow{\Theta} & \mathcal{B}/h_0(D)\mathcal{B} \\ j_{h_0} H \downarrow \wr & & \wr \downarrow j_{h_0} \mathcal{M} \\ E_{h_0} & \xrightarrow{\times j_{h_0} g} & E_{h_0}. \end{array} \quad (\text{GR43})$$

These are the full Taylor-jet maps. The original Euler inverses prove their kernels to be $h_0(D)V$ and $h_0(D)\mathcal{B}$; the seed $H_{P(D)\phi_*} = P$ gives source surjectivity, and the inverse Gram of the independent functions $y^j e^{py}/j!$ against an interval bump gives target surjectivity. The literal identity $\mathcal{M}\Theta\phi = gH_\phi$ proves commutativity.

The associated exact sequence is

$$0 \longrightarrow Q[h_0(D)] \longrightarrow E_{h_0} \xrightarrow{\times j_{h_0} g} E_{h_0} \longrightarrow Q/h_0(D)Q \longrightarrow 0. \quad (\text{GR44})$$

Its first arrow has an explicit inverse on the kernel. If $h_0(D)[F] = 0$, write $h_0(D)F = \Theta\phi$, uniquely by theta injectivity, and send the class to $j_{h_0}H_\phi$. A change by a theta relation changes ϕ by $h_0(D)V$, proving well-definedness. Zero image permits $\phi = h_0(D)\psi$; injectivity of $h_0(D)$ on $\mathcal{B}$ gives $F = \Theta\psi$. Conversely, a polynomial representative P killed by $j_{h_0}g$ gives the class of $S_{h_0}^{\mathcal{B}}\Theta(P(D)\phi_*)$, defined by the original Euler inverse and the range identity. Changing P by h_0T changes this by $\Theta(T(D)\phi_*)$, so its class is independent and maps back to P . Quotienting the square proves the final cokernel statement.

For an actual full packet h , $j_h g = 0$, g/h is entire, and $j_h(g/h)$ is a unit because the exact zero orders were used. These are additional properties of the original source. They have not been established for h_* , and none is assumed in GR8–GR32. Choosing an invertible observation on a calibration model preserves its feasible coefficient sets; it does not identify that observation with the actual Taylor unit.

On each fixed support fibre, all the linear arrows above lift by $(\ell, v) \mapsto (\ell, fv)$, with $\tau \mapsto \tau$. The preimage of the supported zero is exactly $\{(\ell, v) : v \in \ker f\}$. In particular the support lift records the actual kernel of multiplication by $j_{h_0}g$; it does not replace that map by zero.

The Gamma obstruction invalidates the universal sub-threshold estimate using only the generic Gamma source and displayed quotient identities. When the same auxiliary purity is available for these calibration polynomials, adjoining that fact cannot restore this universal implication. The original arithmetic restrictions, full theta primitive, and signed correction GR41 remain present through the proved maps GR34–GR44. No result here states that all F1 geometry or split-base constructions fail, that the desired estimate for the actual arithmetic source is false, or that an off-critical zero of 2ξ exists.

304.0.11 9. Audit findings and verification scope

No correction is required to the Gamma constants, monic recurrence, full multiplicity q_k , reflected rank-two trace, first admitted degree, two consecutive windows, the $8\sqrt{5}$ constant, or the signs of the arithmetic correction. GR27 and GR40 expand portions of the supplied argument into explicit identities.

The delivered verifier's complete canonical-volume calculations are at $m = 1, k = 1$. Its separate repeated-primary check verifies a nonzero nilpotent with square zero. The all- m , all- k statement rests on the written proof GR9–GR33. Its negative-control flag rejects the scalar equality $3/4 = 1/2$; that flag alone does not replay or certify the full auxiliary-purity argument. These scope limits do not affect the proof above.

304.0.12 Appendix: Complete independent Gamma derivation

The following complete proof is appended from the pinned generating-function subreview. It preserves its equation labels G1–G28 and original measures.

304.0.13 Independent Gamma generating-function calculation

Date: 2026-09-13. Scope: the Gamma source, its exact mass and convolution, and the polynomial generating function and recurrence in equations (28), (29), (31), and (35) of the delivered `Tau_Deligne_Bound_Audit/NO TE.tex`. The source SHA-256 is `bfe63094ec360b6059f2936e5e456c6c0fd08db144bbabc29621a527b89c13c6`. The source remains unchanged. The quotient, trace, and determinant estimates are reviewed separately.

Finding. The printed generating function has linear coefficient $+x$, and its exponential generating coefficients are literally monic polynomials in x . The printed substitution produces literally monic polynomials in S , and produces the negative last term of recurrence (35). The constants and convolution mass in (28)–(29) are also correct. No sign or mass correction is required for these formulas.

The proof below retains the original density, coordinate $S = k/2 + ix$, positive integer k , and all factors of 2, i , and π . Euler's Beta integral is recorded in [DLMF 5.12.1](#) and [DLMF 5.12.3](#). The Fourier, convolution, and polynomial calculations here are derived explicitly from that integral.

304.0.14 1. Fourier transform of the unchanged Gamma density

Let $a > 0$, and define

$$r_a(x) = \frac{|\Gamma(a + ix/2)|^2}{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \quad x \in \mathbb{R}. \quad (\text{G1})$$

We use the convention

$$\mathcal{F}_+ f(t) = \int_{\mathbb{R}} f(x) e^{itx} dx. \quad (\text{G2})$$

The Beta identity with arguments $a + iu, a - iu$, followed by $v = e^y$, gives

$$\begin{aligned} \frac{\Gamma(a + iu)\Gamma(a - iu)}{\Gamma(2a)} &= \int_0^\infty \frac{v^{a+iu-1}}{(1+v)^{2a}} dv \\ &= \int_{\mathbb{R}} \frac{e^{ay}}{(1+e^y)^{2a}} e^{iuy} dy \\ &= \int_{\mathbb{R}} f_a(y) e^{iuy} dy, \quad f_a(y) = (2 \cosh(y/2))^{-2a}. \end{aligned} \quad (\text{G3})$$

The function f_a and all its derivatives decay exponentially as $y \rightarrow \pm\infty$. In particular, $f_a, f_a'' \in L^1(\mathbb{R})$; integration by parts twice shows that its Fourier transform is $O((1+|u|)^{-2})$. Fourier inversion therefore applies as an absolutely convergent integral. Write the left side of (G3) as $F_a(u)$. Then

$$f_a(y) = \frac{1}{2\pi} \int_{\mathbb{R}} F_a(u) e^{-iuy} du. \quad (\text{G4})$$

Since $r_a(x) = \Gamma(2a)F_a(x/2)/(2\pi)$, the change of variables $x = 2u$ in the original density gives

$$\begin{aligned} \mathcal{F}_+ r_a(t) &= \frac{\Gamma(2a)}{\pi} \int_{\mathbb{R}} F_a(u) e^{2itu} du \\ &= 2\Gamma(2a)f_a(-2t) \\ &= 2\Gamma(2a)(2 \cosh t)^{-2a} = c_a(\cosh t)^{-2a}. \end{aligned} \quad (\text{G5})$$

Every integral used in (G5) is absolutely convergent. In particular, its value at $t = 0$ proves the original mass

$$\int_{\mathbb{R}} r_a(x) dx = c_a. \quad (\text{G6})$$

For later coefficient extraction we also prove the exponential moments, rather than assuming them. The expression $f_a(z) = (2 \cosh(z/2))^{-2a}$ has the analytic branch positive on the real axis throughout $|\operatorname{Im} z| < \pi$. For $|\sigma| < \pi$,

$$|\cosh((y + i\sigma)/2)|^2 = \sinh^2(y/2) + \cos^2(\sigma/2) \geq \cos^2(\sigma/2) \cosh^2(y/2). \quad (\text{G7})$$

Consequently $f_a(y + i\sigma)$ is integrable in y , uniformly when σ ranges in a closed smaller strip. The vertical sides of the contour rectangle in (G3) tend to zero because f_a decays exponentially in the real direction. Shifting the contour upward for $u \geq 0$ and downward for $u < 0$ therefore proves, for each $0 < \sigma < \pi$,

$$|F_a(u)| \leq e^{-\sigma|u|} \int_{\mathbb{R}} |f_a(y + i\sigma)| dy. \quad (\text{G8})$$

For $u < 0$ the integral along $-i\sigma$ has the same bound by conjugation. Inserting $u = x/2$ proves that $r_a(x)$ is bounded by a constant times $e^{-\sigma|x|/2}$. Thus every exponential moment of order strictly less than $\pi/2$ is finite. Both sides of (G5) extend holomorphically to $|\operatorname{Im} t| < \pi/2$; the right side uses the branch equal to the positive real value on the real axis. Equality on the real axis and the identity theorem imply

$$\int_{\mathbb{R}} e^{vx} r_a(x) dx = c_a(\cos v)^{-2a}, \quad v \in \mathbb{R}, \quad |v| < \pi/2. \quad (\text{G9})$$

This is the exact moment generating function of the density with mass c_a ; no mass has been divided out.

304.0.15 2. The k -fold convolution and its mass

Fix the original value $a = 1/4$. Since $\Gamma(1/2) = \sqrt{\pi}$, (G1) gives

$$c_{1/4} = 2^{1/2} \Gamma(1/2) = \sqrt{2\pi}. \quad (\text{G10})$$

For every positive integer k , the convolution theorem for the integrable functions in (G1) and formula (G5) give

$$\begin{aligned} \mathcal{F}_+(r_{1/4}^{*k})(t) &= c_{1/4}^k (\cosh t)^{-k/2} \\ &= \mathcal{F}_+\left(\frac{c_{1/4}^k}{c_{k/4}} r_{k/4}\right)(t). \end{aligned} \quad (\text{G11})$$

Fourier uniqueness proves equality almost everywhere. The densities here are continuous: each r_a is continuous, bounded, and integrable, and convolution of an integrable function with a bounded uniformly continuous function is continuous. The required uniform continuity of r_a follows either from (G4) and the exponential estimate (G8), or directly by dominated differentiation there. Hence the equality holds at every real x :

$$r_{1/4,k}(x) = r_{1/4}^{*k}(x) = \frac{c_{1/4}^k}{c_{k/4}} \frac{|\Gamma(k/4 + ix/2)|^2}{2\pi}. \quad (\text{G12})$$

At $t = 0$, or by Tonelli's theorem in the original k variables,

$$\int_{\mathbb{R}} r_{1/4,k}(x) dx = c_{1/4}^k = (2\pi)^{k/2}. \quad (\text{G13})$$

These are exactly equations (28)–(29) of the delivered note, including the coefficient $c_{1/4}^k/c_{k/4}$.

304.0.16 3. The printed generating function has $+x$, not $-x$

The printed function is, for z near zero,

$$F_a(z, x) = (1 - iz)^{-a+ix/2} (1 + iz)^{-a-ix/2} = \sum_{n=0}^{\infty} b_n^{(a)}(x) \frac{z^n}{n!}. \quad (\text{G14})$$

Use the logarithms vanishing at $z = 0$. Their exact derivatives give

$$\begin{aligned} \partial_z \log F_a(z, x) &= \frac{(-a + ix/2)(-i)}{1 - iz} + \frac{(-a - ix/2)i}{1 + iz} \\ &= \frac{x - 2az}{1 + z^2}. \end{aligned} \quad (\text{G15})$$

In particular $F_a(0, x) = 1$ and $\partial_z F_a(0, x) = x$. For real z near zero the same branches give the exact expression

$$F_a(z, x) = (1 + z^2)^{-a} \exp(x \arctan z). \quad (\text{G16})$$

The sign in the exponential is positive because

$$\log(1 - iz) - \log(1 + iz) = -2i \arctan z, \quad \frac{ix}{2} (-2i \arctan z) = x \arctan z. \quad (\text{G17})$$

Multiply the differential equation (G15) by F_a and compare the coefficient of $z^n/n!$. For $n \geq 1$, the coefficient contributed by $z^2 \partial_z F_a$ is $n(n-1)b_{n-1}^{(a)}$, and that contributed by $2azF_a$ is $2anb_{n-1}^{(a)}$. Therefore

$$b_0^{(a)} = 1, \quad b_1^{(a)} = x, \quad b_{n+1}^{(a)}(x) = xb_n^{(a)}(x) - n(n+2a-1)b_{n-1}^{(a)}(x). \quad (\text{G18})$$

For $n = 0$, the same equation reads $b_1^{(a)} = xb_0^{(a)}$, with the last term omitted. Induction in (G18) proves that $b_n^{(a)} \in \mathbb{R}[x]$ has degree n and leading coefficient exactly 1. For example, the first values are

$$\begin{aligned} b_0^{(a)} &= 1, & b_1^{(a)} &= x, \\ b_2^{(a)} &= x^2 - 2a, & b_3^{(a)} &= x^3 - (6a + 2)x. \end{aligned} \quad (\text{G19})$$

Thus no replacement of z , x , or either exponent is needed to obtain monicity.

304.0.17 4. Exact orthogonality and norms

Take real z, w in a neighborhood of zero sufficiently small that $|\arctan z + \arctan w| < \pi/2$. Equations (G9) and (G16) give

$$\begin{aligned} \int_{\mathbb{R}} F_a(z, x) F_a(w, x) r_a(x) dx &= c_a (1 + z^2)^{-a} (1 + w^2)^{-a} (\cos(\arctan z + \arctan w))^{-2a} \\ &= c_a (1 - zw)^{-2a}. \end{aligned} \quad (\text{G20})$$

The last equality uses the exact identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}. \quad (\text{G21})$$

On a sufficiently small closed complex bidisc around $z = w = 0$, the absolute value of the integrand in (G20) is bounded by a constant times $r_a(x)e^{v_0|x|}$ for some $v_0 < \pi/2$. This follows directly from the analytic logarithms in (G14). That dominating function is integrable by (G8). Consequently the integral is holomorphic on the open bidisc. The real identity (G20) extends there by applying the one-variable identity theorem first in z , then in w . Differentiation under the integral at zero is justified by Cauchy's formula and the same domination.

The coefficient of $z^n w^m$ on the left side of (G20) is

$$\frac{1}{n!m!} \int_{\mathbb{R}} b_n^{(a)}(x) b_m^{(a)}(x) r_a(x) dx.$$

The binomial series on the right side is $c_a \sum_{j \geq 0} (2a)_j (zw)^j / j!$. Comparing coefficients yields

$$\int_{\mathbb{R}} b_n^{(a)}(x) b_m^{(a)}(x) r_a(x) dx = \mathbf{1}_{n=m} c_a n! (2a)_n. \quad (\text{G22})$$

The polynomials have real coefficients, so (G22) is also their Hermitian orthogonality relation. In particular it is the norm of the original monic polynomial, with mass c_a retained.

304.0.18 5. Exact transport to $S = k/2 + ix$

Set $a = k/4$ and $c = k/2$. Define the polynomial in the original complex variable S by

$$p_n(S) = i^n b_n^{(k/4)}((S - c)/i). \quad (\text{G23})$$

The coefficient of S^n is $i^n i^{-n} = 1$; hence this is literally monic in S . The lower coefficients are obtained by the displayed affine substitution, with no change to the source norm or its mass. On the source line, $S = c + ix$, equation (G23) gives $p_n(c + ix) = i^n b_n^{(k/4)}(x)$. Therefore (G12) and (G22) imply

$$\begin{aligned} \int_{\mathbb{R}} \overline{p_n(c + ix)} p_m(c + ix) r_{1/4, k}(x) dx &= \frac{c_{1/4}^k}{c_{k/4}} (-i)^n i^m \int_{\mathbb{R}} b_n^{(k/4)}(x) b_m^{(k/4)}(x) r_{k/4}(x) dx \\ &= \mathbf{1}_{n=m} (2\pi)^{k/2} n! (k/2)_n. \end{aligned} \quad (\text{G24})$$

In the surviving term $m = n$, the phase is $(-i)^n i^n = 1$. This proves the exact norm $\omega_{k, n}^\Gamma$ printed in (35).

To transport the recurrence, multiply (G18), in its form $xb_n = b_{n+1} + n(n + 2a - 1)b_{n-1}$, by i^{n+1} . Since $S - c = ix$, the result is

$$\begin{aligned}(S - c)p_n(S) &= p_{n+1}(S) + n(n + k/2 - 1)i^{n+1}b_{n-1}^{(k/4)}((S - c)/i) \\ &= p_{n+1}(S) - n(n + k/2 - 1)p_{n-1}(S).\end{aligned}\tag{G25}$$

The minus sign is exactly $i^{n+1}/i^{n-1} = i^2 = -1$. Restoring $c = k/2$ proves

$$Sp_n = p_{n+1} + \frac{k}{2}p_n - n(n + k/2 - 1)p_{n-1}, \quad \omega_{k,n}^\Gamma = (2\pi)^{k/2}n!(k/2)_n.\tag{G26}$$

The last term is omitted at $n = 0$. The first polynomials in the original coordinate are

$$p_0 = 1, \quad p_1 = S - k/2, \quad p_2 = (S - k/2)^2 + k/2, \quad p_3 = (S - k/2)^3 + (3k/2 + 2)(S - k/2).\tag{G27}$$

Finally, consecutive norm ratios retain the same constants before their displayed cancellation:

$$\frac{\omega_{k,n}^\Gamma}{\omega_{k,n-1}^\Gamma} = \frac{(2\pi)^{k/2}n!(k/2)_n}{(2\pi)^{k/2}(n-1)!(k/2)_{n-1}} = n(n + k/2 - 1), \quad n \geq 1.\tag{G28}$$

This supplies precisely the coefficient needed for the interior cancellation following equation (36). The derivation establishes the Gamma formulas and their original polynomial coordinates; it does not by itself assert the later determinant bound or a comparison with the arithmetic theta source.

305 The original source and relation Toda flows with their exact control maps

This is a standalone continuation of the identities CV.1–CV.12. It retains the original polynomial quotient, arithmetic measure, monic coordinates, arithmetic unit, source relations and complex phase. The delivered `Tau_Toda_Volume_Control/NOTE.tex` was read in full from its original archive; its SHA256 is

027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375.

The calculations below prove the exact maps underlying the determinant quotient, its first and second tilt derivatives, the omitted-direction determinant, the positive minor sum and the finite upper bound. No asymptotic arithmetic estimate is asserted.

305.1 Original arithmetic objects and the strip of real tilts

The original spaces and transforms are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx. \end{aligned}$$

The series and its Euler derivatives converge uniformly at large x with every power of decay. For small x , Poisson summation with Fourier kernel $e^{-2\pi ixy}$ gives $\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x)$; both zero terms vanish because $\phi(0) = \hat{\phi}(0) = 0$. The Fourier transform preserves the stated two zero constraints and evenness. This formula proves every small- x seminorm after Euler differentiation, hence $\Theta : V \rightarrow \mathcal{B}$. It also commutes with D on these spaces by termwise differentiation. Mellin transforms of elements of $\mathcal{B}$ are entire, and integration by parts gives $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$; both endpoint terms vanish with their original coefficients.

Let h be a monic, nonempty packet of actual nontrivial zeros of

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with their complete orders. Assume the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Set $d = \deg h$ and retain

$$v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = [v_h]_h \in E_h^\times, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}. \quad (\text{TVB.1})$$

Here the jet of an entire function f means its unique polynomial Hermite remainder of degree less than d : in the local factor $\mathbb{C}[t]/(t^{m_\rho})$, it is $\sum_{j < m_\rho} f^{(j)}(\rho)t^j/j!$, with $t = s - \rho$. The coprime factors of h give the isomorphism to their product by monic division and Bezout's identity, which supplies the unique global remainder. The constant $v_h(\rho)$ is nonzero at each selected zero because its full order was divided out. If the local jet is $\sum a_j t^j$, its inverse coefficients are $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$ for $n < m_\rho$. Multiplication verifies this inverse in every local factor, and the product isomorphism gives the inverse of v_h .

The original F_h has a construction within $\mathcal{B}$. For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy.$$

The equality is the vanishing Mellin value. For any desired power bound at infinity choose $A > \Re \rho$ with $|F(y)| \leq C_A y^{-A}$ for $y \geq 1$; the first integral is bounded by $C_A x^{-A}/(A - \Re \rho)$ there. At zero choose B with $B + \Re \rho > 0$ and $|F(y)| \leq C_B y^B$ for $0 < y \leq 1$; the second integral is bounded by $C_B x^B/(B + \Re \rho)$. The exact identity $(D - \rho)S_\rho F = F$ now gives all Euler derivative bounds inductively, so $S_\rho F \in \mathcal{B}$. Mellin integration by parts proves $\mathcal{M}(S_\rho F) = \mathcal{M}F/(s - \rho)$, with its removable value at $s = \rho$ supplied by the derivative of the numerator. For the original seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V$, direct substitution $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \Re s > -2.$$

For $\Re s > 1$, absolute convergence permits the two-sided theta sum to be integrated termwise, yielding $\mathcal{M}(\Theta\phi_0) = 2\zeta(s)\mathcal{M}\phi_0(s) = g(s)$; both sides continue to the same entire function. Apply S_ρ successively to $\Theta\phi_0$ over the full list of roots of h , including each repeated occurrence. At every stage the next vanishing Mellin value holds by its remaining order. This constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$ and $h(D)F_h = \Theta\phi_0$; the latter also follows by Mellin injectivity, which is Fourier injectivity after $x = e^y$. Fix $k \geq 1$, set $c = k/2$, $S = s_1 + \cdots + s_k$, and put $I = (h(s_1), \dots, h(s_k)) \subset \mathbb{C}[s_1, \dots, s_k]$. Let χ be the original monic generator of $\{P \in \mathbb{C}[S] : P(S) \in I\}$. Its degree is $q \geq 1$ and

$$C = \mathbb{C}[S]/(\chi), \quad A = M_S : C \longrightarrow C. \quad (\text{TVB.2})$$

The coordinate basis of C is always $1, S, \dots, S^{q-1}$. The polynomial exists because multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$ has an annihilating polynomial. Division by its least-degree monic annihilator proves that its entire annihilating ideal is (χ) . In particular the injection

$$\eta : C \longrightarrow (E_h^{\otimes k})^{\mathfrak{S}_k}, \quad [P] \longmapsto v_h^{\otimes k}[P(S)]_I \quad (\text{TVB.3})$$

has exactly zero kernel: multiplication by the displayed unit is invertible and the preceding annihilator ideal describes the other factor's kernel. The original tensor source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad D = -x\partial_x, \quad \mathcal{M}F_h = v_h.$$

On the original Mellin line its scalar observation has squared norm

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad \int m_k = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{TVB.4})$$

Indeed the tensor Mellin–Plancherel density is $\prod_i w_h(t_i) d^k t$. In coordinates $u = \sum_i t_i$, $y = (t_1, \dots, t_{k-1})$, the inverse is $(y, u) \mapsto (y_1, \dots, y_{k-1}, u - \sum y_i)$. With (y, u) in this order its determinant is 1; with (u, y) its determinant is $(-1)^{k-1}$. Integrating the original density over y gives (TVB.4), with no scalar divisor.

For every compact substrip $|\Re \theta| \leq r < \pi/2$, all integrals

$$M_h(\theta) = \int e^{\theta t} w_h(t) dt, \quad Z(\theta) = M_h(\theta)^k = \int e^{\theta u} m_k(u) du \quad (\text{TVB.5})$$

and all their polynomially weighted derivatives converge locally uniformly. Here is an elementary sufficient bound. Summation by parts for the alternating Dirichlet series, whose partial sums have modulus at most one, gives $|\eta(s)| \leq |s|/\Re s$ when $\Re s > 0$. On $\Re s = 1/2$, the denominator in $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ has modulus at least $\sqrt{2} - 1$. For $0 < \varphi < \pi/2$, rotate the defining gamma integral to the ray with argument $\text{sgn}(t)\varphi$. The large arc vanishes by $e^{-R \cos \varphi}$ and the small arc by the positive real part $1/4$ of its exponent. Its absolute value is bounded by

$$|\Gamma(1/4 + it/2)| \leq \Gamma(1/4)(\cos \varphi)^{-1/4} e^{-\varphi|t|/2}.$$

For sufficiently large $|t|$, the monic polynomial h satisfies $|h(1/2 + it)| \geq C_h(1 + |t|)^d$ with $C_h > 0$. The two factors $s, s - 1$, the preceding zeta bound and the gamma bound consequently give

$$w_h(t) \leq C_{h,\varphi}(1 + |t|)^{6-2d} e^{-\varphi|t|}$$

outside a compact interval. On that interval v_h is entire, since the selected orders are complete. Choose $r < \varphi < \pi/2$ and absorb every fixed polynomial in the remaining exponential decay. This proves the asserted domination and differentiation under the integral. Fubini gives the exact product M_h^k in (TVB.5). The nonzero entire function v_h has a discrete real-line zero set; therefore $w_h > 0$ almost everywhere. Its convolutions are positive almost everywhere (in fact at every u for $k \geq 2$), since their fibre integrands are positive off a finite union of measure-zero inverse images of this discrete set. Thus the norm of every nonzero polynomial is positive.

In what follows θ is real, $|\theta| < \pi/2$, and the source pairing, conjugate-linear in its first argument, is exactly

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (\text{TVB.6})$$

The map $f(u) \mapsto e^{\theta u/2} f(u)$ is an onto isometry from $L^2(e^{\theta u} m_k du)$ to $L^2(m_k du)$, with inverse $f \mapsto e^{-\theta u/2} f$. It commutes with multiplication by $c + iu$. This specifies the exact Hilbert comparison for the tilt. The polynomial algebra, its reduction and η are unchanged. When the target of $\mathcal{V}$ is observed at this tilt, its declared tensor norm is

$$\|F\|_{k,\theta}^2 = \int_{\mathbb{R}^k} e^{\theta \sum_i t_i} |\mathcal{M}_{\text{unit}} F(t_1, \dots, t_k)|^2 d^k t,$$

where $\mathcal{M}_{\text{unit}}$ is the original Mellin–Plancherel map with a factor $(2\pi)^{-1/2}$ in each coordinate. The map $f(t) \mapsto e^{\theta \sum_i t_i/2} f(t)$ is an onto isometry from this weighted Mellin Hilbert space to the original one, with the opposite exponential as inverse. The restriction to $\mathcal{V}P$ is exactly (TVB.6), by the same sum-fibre integral as (TVB.4). Thus every tilted exterior norm below has a specified target norm; at $\theta = 0$ it is the original tensor-source norm.

305.2 The source relation and its Laplace observation

Write $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$. For $N \geq q - 1$, put $n = N + 1$, $m = N - q + 1$. Literal monic division gives the exact sequence

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{T_N} C \longrightarrow 0, \quad M_\chi Q = \chi Q, \quad T_N P = [P]_\chi. \quad (\text{TVB.7})$$

Its kernel is precisely the indicated image, and its first q monomial columns prove surjectivity. Denote that kernel by $\mathcal{D}_N = \chi \mathcal{P}_{m-1}$.

This sequence has an explicit map to the original theta boundaries. To see every sign, choose the quotients Q_i from successive monic division, in the order $s_1, \dots, s_k$, so that

$$\chi(S) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{TVB.8})$$

The remainder is zero by the definition of χ and uniqueness of the tensor remainder basis. Put $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. It is even, vanishes at zero and has integral zero. Direct gamma integration of its two terms gives $\mathcal{M}(\Theta \phi_0) = g$, hence $h(D)F_h = \Theta \phi_0$ by Mellin injectivity on the original rapidly decreasing source. For every polynomial P , an actual primitive of $\mathcal{V}(\chi P)$ in the tensor complex $(V \xrightarrow{\Theta} \mathcal{B})^{\otimes k}$ is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{TVB.9})$$

The preceding $i - 1$ factors have cochain degree one. The tensor differential therefore contributes $(-1)^{i-1}$ at the i th factor, cancelling the written sign and yielding (TVB.8). Euler differentiation preserves the source conditions and commutes with Θ , so all terms have the indicated domains. Conversely the full jet of $\mathcal{V}P$ is $\eta T_N P$; injectivity in (TVB.3) shows that the admitted polynomial source

has no further zero-jet elements. In particular every boundary vector in (TVB.7) retains its positive source norm before the quotient sends it to represented zero.

For any polynomial P , define its Laplace observation by

$$\mathcal{L}(P)(\theta) = \int P(c + iu)e^{\theta u} m_k(u) du.$$

Differentiation under (TVB.5) proves the typed intertwining $\mathcal{L}(SP) = (c + i\partial_\theta)\mathcal{L}(P)$. Coefficient-wise conjugation gives the exact Gram observation

$$Z_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta)Z = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0. \quad (\text{TVB.10})$$

Thus the two signs and both polynomial factors arise from $\langle M_\chi 1, M_\chi 1 \rangle_\theta$. This is a map on source functions, with no replacement of the higher contracted ideal by a power of χ .

305.3 Both monic determinants and their exact determinant-line map

Set

$$\mathfrak{D}_j = \det[Z^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{B}_j = \det[Z_\chi^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{D}_0 = \mathfrak{B}_0 = 1. \quad (\text{TVB.11})$$

All nonempty determinants are positive by polynomial positivity. The coefficient map from $1, S, \dots, S^{j-1}$ to $1, u, \dots, u^{j-1}$ is explicitly

$$S^a = (c + iu)^a = \sum_{b=0}^a \binom{a}{b} c^{a-b} i^b u^b. \quad (\text{TVB.12})$$

Its determinant is $i^{j(j-1)/2}$ and its Gram determinant multiplier is exactly one. Therefore $\mathfrak{D}_j$ is also the original S -monomial Gram determinant. The same calculation applies to the relation-weight pairing $\langle P, Q \rangle_{\chi, \theta} = \langle \chi P, \chi Q \rangle_\theta$ and $\mathfrak{B}_j$. This proves the coordinate correspondence without changing the original monic polynomial.

Let $p_j(S; \theta)$ be the monic source orthogonal polynomials, and $r_j(S; \theta)$ the monic polynomials for this relation-weight pairing. Retain their original squared norms

$$\omega_j = \|p_j\|_\theta^2 = \mathfrak{D}_{j+1}/\mathfrak{D}_j, \quad \nu_j = \|\chi r_j\|_\theta^2 = \mathfrak{B}_{j+1}/\mathfrak{B}_j. \quad (\text{TVB.13})$$

The equations follow by triangular monic Gram elimination, whose diagonal change is one. In particular $\omega_0 = M_h(\theta)^k$. The full mass has not been divided out.

Give C its written coordinate Euclidean pairing only when specifying adjoints. Let M_N be the original monomial source Gram, and J_N its reduction matrix. The minimum lift is

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = M_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N. \quad (\text{TVB.14})$$

Here the last expression specifies polynomial coefficients. It satisfies $T_N R_N = I$, $\text{im } R_N \perp \mathcal{D}_N$ and $R_N^* R_N = G_N$ in the original source pairing. Indeed the first equality is matrix multiplication, and pairing with any vector in $\ker J_N$ gives zero. Every other lift is $R_N x + B$ for $B \in \mathcal{D}_N$, so its squared norm is $x^* G_N x + \|B\|^2$. This proves the minimum assertion and uniqueness.

Use the source basis $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$. Its change from the n source monomials is triangular with diagonal one. Replacing its first q columns by $R_N[1], \dots, R_N[S^{q-1}]$ adds boundary columns and again has determinant one. The resulting Gram is the direct sum of G_N and the relation monomial Gram. Consequently

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_m}, \quad D_N^{\text{CV}} := \det K_N = V_N^{-1}. \quad (\text{TVB.15})$$

More precisely the map $\det C \otimes \det \mathcal{P}_{m-1} \rightarrow \det \mathcal{P}_N$ sends the written coordinate wedges to the wedge of these quotient lifts followed by M_χ of the relation monomials; its coordinate determinant is 1. Putting the kernel wedge first instead contributes the explicit sign $(-1)^{qm}$. Its norm identity is $\det G_N \mathfrak{B}_m = \mathfrak{D}_n$. Composing the wedge with $\mathcal{V}^{\otimes n} \text{Alt}_n$ multiplies its squared norm by exactly $n!$: expanding the unscaled alternating tensor sum gives $n!$ copies of the Gram determinant. The original tensor target is $\mathcal{B}^{\otimes kn}$ in cochain degree kn .

305.4 Toda derivatives and the full curvature operators

For either positive real u -weight in (TVB.11), write $Q_j(u)$ for its real monic orthogonal polynomials and h_j for their norms. Orthogonality and leading coefficients prove

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1} \ (j \geq 1), \quad a_0 = 0.$$

For $i \leq j-2$, differentiating $\langle Q_i, Q_j \rangle = 0$ shows that $\partial_\theta Q_j$ is orthogonal to Q_i , because uQ_i and $\partial_\theta Q_i$ have degree at most $j-1$. For $i = j-1$ the same identity gives its coefficient $-a_j$. Thus $\partial_\theta Q_j = -a_j Q_{j-1}$ and $h'_j = b_j h_j$. To compute b'_j , differentiate $h_j b_j = \langle Q_j, uQ_j \rangle$. The two polynomial derivative terms contribute $-2a_j h_j$. The derivative of the measure contributes $\|uQ_j\|^2 = h_j(a_{j+1} + b_j^2 + a_j)$. After subtracting $h'_j b_j$ this gives $b'_j = a_{j+1} - a_j$. The monic Gram determinant is $\prod_{j < n} h_j$, hence

$$(\log \mathfrak{D}_n)'' = \frac{\omega_n}{\omega_{n-1}} \quad (n \geq 1), \quad (\log \mathfrak{B}_m)'' = \frac{\nu_m}{\nu_{m-1}} \quad (m \geq 1). \quad (\text{TVB.16})$$

Both 0th logarithms have second derivative zero. Multiplication by the square of the determinant proves each Hankel-Toda equation with coefficient one. Set

$$\alpha_{N+1} = \omega_{N+1}/\omega_N, \quad \beta_m = \nu_m/\nu_{m-1} \ (m \geq 1), \quad \beta_0 = 0, \quad \ell_N = \log V_N.$$

Then $\ell''_N = \alpha_{N+1} - \beta_m$.

There is a finer exact source-map identity. Let $X = (S - c)/i$, which is multiplication by the real variable u in (TVB.6). Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ to $\mathcal{D}_N$ in that pairing. Differentiating $T_N R_N = I$ gives $R'_N x \in \mathcal{D}_N$. Differentiating orthogonality against each fixed boundary polynomial gives

$$R'_N = -P_{\mathcal{D}_N} X R_N =: -I_N, \quad O_N := (1 - P_N) X R_N. \quad (\text{TVB.17})$$

Every derivative is an actual boundary with the primitive (TVB.9). Differentiating its norm, including the derivative of the measure, gives

$$G'_N = R_N^* X R_N, \quad G''_N = R_N^* X^2 R_N - 2R_N^* X P_{\mathcal{D}_N} X R_N.$$

The projection to $\text{im } R_N$ is $R_N G_N^{-1} R_N^*$ and, together with $P_{\mathcal{D}_N}$, sums to P_N . Subtracting the square of the first derivative therefore proves

$$G''_N - G'_N G_N^{-1} G'_N = O_N^* O_N - I_N^* I_N. \quad (\text{TVB.18})$$

This is an equality of maps $C \rightarrow C^{\text{anti}*}$, where $C^{\text{anti}*}$ consists of conjugate-linear functionals on C . For example $O_N^* O_N$ sends x to the functional $y \mapsto \langle O_N y, O_N x \rangle_\theta$; this assignment is complex-linear in x . The fixed coordinate Euclidean pairing identifies that antidual with the written matrix coordinates.

Write $b_j = [p_j]_\chi$, and $a = b_N^* G_N b_N$, $d_+ = b_{N+1}^* G_N b_{N+1}$. The canonical next boundary has the particularly exact factorization

$$b_N := p_{N+1} - R_N b_{N+1} = \chi r_m, \quad \nu_m = \omega_{N+1} + d_+. \quad (\text{TVB.19})$$

To prove it, χr_m is monic of degree $N+1$ and orthogonal to $\mathcal{D}_N$. Subtract p_{N+1} ; the difference lies in $\mathcal{P}_N \cap \mathcal{D}_N^\perp$ and has reduction $-b_{N+1}$. Uniqueness of R_N proves the first equality; the two terms

in its definition are orthogonal, giving the second. The coefficient of p_N in $R_N x$ is $b_N^* G_N x / \omega_N$. The leading coefficient of X as an S -polynomial is $1/i = -i$. Consequently the full maps are

$$O_N x = -i p_{N+1} \frac{b_N^* G_N x}{\omega_N}, \quad I_N x = -i \chi r_{m-1} \frac{b_{N+1}^* G_N x}{\nu_{m-1}} \quad (m \geq 1), \quad I_N = 0 \quad (m = 0). \quad (\text{TVB.20})$$

For the second equation, $X \chi r_j$ is an old boundary if $j \leq m - 2$. For $j = m - 1$ its coefficient of χr_m is $-i$. Moving X across the pairing conjugates this coefficient to i , and $\langle \chi r_m, R_N x \rangle = -b_{N+1}^* G_N x$ by (TVB.19). Division by the original norm ν_{m-1} gives exactly the second sign in (TVB.20).

Both maps have rank at most one. For $N \geq q$, put $\delta_N = V_N / V_{N-1}$. The rank-one calculation in the next section gives $a / \omega_N = 1 - \delta_N$ and $\delta_N = \omega_N / \nu_{m-1}$. Taking traces in (TVB.20) gives the separate, positive source costs

$$\begin{aligned} \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1} (1 - \delta_N), \\ \text{Tr}(G_N^{-1} I_N^* I_N) &= \beta_m - \alpha_{N+1} \delta_N = \frac{\nu_m - \omega_{N+1}}{\nu_{m-1}}. \end{aligned} \quad (\text{TVB.21})$$

Their difference is $\alpha_{N+1} - \beta_m$ as required. Thus the individual positive map costs each retain the common quantity $\alpha_{N+1} \delta_N$; assigning them separately the two Toda coefficients would lose this quantity. At $N = q - 1$, the outer cost is α_q and the inner cost is zero. If a single notation for the cost identities is wanted at this endpoint, define only the scalar convention $\delta_{q-1}^{\text{end}} := \det K_{q-2} / \det K_{q-1} = 0$; the included column has leverage $a / \omega_{q-1} = 1$. Together with $\beta_0 = 0$ the cost identities are $\alpha_q (1 - \delta_{q-1}^{\text{end}}) = \alpha_q$ and $\beta_0 - \alpha_q \delta_{q-1}^{\text{end}} = 0$. This convention is not a quotient involving V_{q-2} and does not introduce an inverse for K_{q-2} or a norm ν_{-1} . All statements concern the original polynomial source and its original boundary subspace.

305.5 The rank-one update, trace phase and original rank-two radius

The source-orthogonal basis gives

$$K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad K_{N+1} = K_N + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}. \quad (\text{TVB.22})$$

The first q monic remainder columns are a triangular basis; K_N is positive definite precisely in the admitted range. Block elimination of $\begin{pmatrix} K & U \\ -V^* & I \end{pmatrix}$ in either order proves $\det(K + UV^*) = \det K \det(I + V^* K^{-1} U)$. Applying it to the positive update and the subtraction of b_N from K_N yields, for $N \geq q$,

$$\frac{a}{\omega_N} = 1 - \delta_N, \quad \frac{d_+}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1, \quad \delta_N = \frac{\omega_N}{\nu_{m-1}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_m}. \quad (\text{TVB.23})$$

The last two identities also follow directly from (TVB.15). Each δ is positive and at most one, since increasing the source space weakly decreases every minimum squared norm. The exact inverse update is

$$G_{N+1} = G_N - \frac{G_N b_{N+1} b_{N+1}^* G_N}{\omega_{N+1} + d_+}. \quad (\text{TVB.24})$$

Multiplying this expression by K_{N+1} proves it directly; the denominator is $\nu_m > 0$ by (TVB.19).

Reflection of the tensor algebra sends S to $k - \bar{S}$ and preserves its annihilator ideal. Monicity gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic. The exact algebra isomorphism

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P(S)] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q(u)] = [Q((S - c)/i)] \quad (\text{TVB.25})$$

transports the original ideal by its leading unit i^q . Its coefficient matrix E is (TVB.12) in dimension q . Thus A transports to $cI + iT$, where $T = M_u$ in the real quotient. Its Gram transports to $E^{-*} G_N E^{-1}$, its kernel to $E K_N E^*$, and $\det G_N$ is unchanged because $|\det E| = 1$. Write

$$\sigma = \text{Tr } T = \frac{\text{Tr } A - cq}{i} \in \mathbb{R}, \quad \psi_N = \sigma - \ell'_N.$$

Reality of $\widehat{\chi}$, the weight and its monic orthogonal polynomials makes all the u -coordinate source and quotient Grams real. Let $Q_j(u) = i^{-j}p_j(c + iu)$ and $d_j = [Q_j]$. Then $Eb_j = i^j d_j$ and both source norms are ω_j .

The leading coefficient of the u -polynomial minimum lift of x is $d_N^* G_N x / \omega_N$, where in this calculation G_N denotes the transported real-coordinate Gram. The difference $X R_N - R_N T$ has zero reduction, and subtracting $(Q_{N+1} - R_N d_{N+1}) d_N^* G_N / \omega_N$ cancels its highest coefficient. Its remainder is an old boundary. Pair with R_N ; old boundaries vanish and $R_N^* (Q_{N+1} - R_N d_{N+1}) = -G_N d_{N+1}$. Taking the trace after multiplication by G_N^{-1} proves

$$\ell'_N = \sigma - \frac{d_N^* G_N d_{N+1}}{\omega_N}, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i \psi_N. \quad (\text{TVB.26})$$

The second equation is back in the fixed original S coordinates: the phase is $\overline{i^N} i^{N+1} = i$. Both signs are retained.

For completeness, the u recurrence proved above transports to $S p_j = p_{j+1} + (c + i b_j^{\text{rec}}) p_j - \alpha_j p_{j-1}$ in the original source coordinates. Inserting this in (TVB.22) telescopes all interior terms and gives

$$A K_N + K_N A^* - k K_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}.$$

For $W_N = A^* G_N + G_N A - k G_N$, multiplication on both sides by G_N gives the corresponding identity for W_N . Its Hermitian relative operator is $H_N = G_N^{-1/2} W_N G_N^{-1/2}$. It has rank at most two and trace $2\Re \text{Tr } A - kq = 0$. The sum of the squares of its eigenvalues is $2(ad_+ - |z_N|^2)/\omega_N^2$, where $z_N = b_N^* G_N b_{N+1} = i \omega_N \psi_N$. This last assertion follows by expanding the square of $(xy^* + yx^*)/\omega_N$ for $x = G_N^{1/2} b_N$, $y = G_N^{1/2} b_{N+1}$; its trace is $[2ad_+ + 2\Re(z_N^2)]/\omega_N^2$. Its nonzero eigenvalues are therefore $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2. \quad (\text{TVB.27})$$

The first expression is valid already at $N = q - 1$.

305.6 Exact maps to the CV omitted determinant and positive minors

Retain $D_j^{\text{CV}} = \det K_j$, put $K_{-1} = 0$, and form

$$\tilde{K}_N = K_{N-1} + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad \Delta_N = D_{N+1}^{\text{CV}} + D_{N-1}^{\text{CV}} - D_N^{\text{CV}} - \tilde{D}_N.$$

This is the actual quotient kernel of the source with p_N omitted and p_{N+1} included. No inverse of it is used. The two-column determinant formula preceding (TVB.23), with one negative and one positive update, proves

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} = \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d_+}{\omega_{N+1}}\right) + \frac{|z_N|^2}{\omega_N \omega_{N+1}}, \quad \epsilon_N^2 = \alpha_{N+1} \frac{\Delta_N}{D_N^{\text{CV}}}. \quad (\text{TVB.28})$$

In particular, for $N \geq q$, the complete phase map is the exact nonnegative difference

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} - \frac{\delta_N}{\delta_{N+1}} = \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.29})$$

The sign of ψ_N is retained by the first derivative in (TVB.26); the omitted determinant observes its square.

For any finite column set L , determinant multilinearity gives

$$\det \left(\sum_{j \in L} \frac{b_j b_j^*}{\omega_j} \right) = \sum_{\substack{J \subseteq L \\ |J|=q}} \frac{|\det B_J|^2}{\prod_{j \in J} \omega_j}, \quad B_J = [b_j]_{j \in J \text{ increasing}}.$$

Terms with repeated indices vanish. The $q!$ remaining permutations of a fixed increasing index set yield its determinant times its conjugate, which proves the displayed identity including singular kernels. Apply it to the four column sets defining Δ_N . A set containing neither of the two last indices has coefficient $1 + 1 - 1 - 1 = 0$; one containing exactly one has coefficient $1 - 1 = 0$; one containing both has coefficient one. Hence

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \quad (\text{TVB.30})$$

For $q = 1$ it is zero. This proof applies at $N = q - 1$. For $q \geq 2$ define the actual source subspace $\mathcal{E}_N = (\bigwedge^{q-2} \mathcal{P}_{N-1}) \wedge p_N \wedge p_{N+1}$ and $\Lambda_N = (\bigwedge^q T_{N+1})|_{\mathcal{E}_N}$. For the fixed coordinate wedge $e_C = 1 \wedge S \wedge \dots \wedge S^{q-1}$, whose coordinate Euclidean squared norm is one, the adjoint is

$$\zeta_N := \Lambda_N^* e_C = \sum_{|I|=q-2} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \left(\bigwedge_{i \in I} p_i \right) \wedge p_N \wedge p_{N+1}. \quad (\text{TVB.31})$$

The displayed source wedges are orthogonal with the written full product norms. Therefore $\|\zeta_N\|_\theta^2 = \Delta_N$ and $\Lambda_N \zeta_N = \Delta_N e_C$. At $\theta = 0$ its literal original tensor-source image has squared norm $q! \Delta_N$ under $\mathcal{V}^{\otimes q} \text{Alt}_q$. At other real tilts use the onto Hilbert comparison after (TVB.6) on each scalar source factor; the same determinant norm and factorial hold in that specified tilted observation. For $q = 1$ set $\mathcal{E}_N = 0$, $\Lambda_N = 0$, $\zeta_N = 0$.

305.7 Both exact losses, their equality cases and the telescope

For $N \geq q$ define

$$v_N = -\log \delta_N, \quad s_N = (v_N + v_{N+1})/2, \quad t_N = (v_{N+1} - v_N)/2, \quad \mathcal{R}_N = e^{2s_N} = \frac{V_{N-1}}{V_{N+1}} \geq 1.$$

Expanding $(1 - e^{-v_N})(e^{v_{N+1}} - 1)$ gives $2e^{t_N} \cosh s_N - e^{2t_N} - 1$. Completing this scalar square, with both terms retained, yields

$$\boxed{\epsilon_N^2 = \alpha_{N+1} \sinh^2 s_N - \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 - \psi_N^2.} \quad (\text{TVB.32})$$

Thus the exact upper bound and its equality conditions are

$$\epsilon_N \leq \sqrt{\alpha_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}, \quad \text{equality} \iff \psi_N = 0 \text{ and } \delta_N(2 - \delta_{N+1}) = 1. \quad (\text{TVB.33})$$

Indeed $\alpha_{N+1} > 0$ and both lost squares in (TVB.32) are nonnegative. Multiplication of $e^{t_N} = \cosh s_N$ by $2\sqrt{\delta_N \delta_{N+1}}$ gives precisely the displayed equation. This includes the case $\mathcal{R}_N = 1$, when both contractions equal one and $\epsilon_N = \psi_N = 0$ by (TVB.27). The corresponding equality with the full positive minor source is

$$\frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N} - \frac{\|\zeta_N\|_\theta^2}{D_N^{\text{CV}}} = (e^{t_N} - \cosh s_N)^2 + \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.34})$$

It follows from (TVB.28), (TVB.31) and (TVB.32), so its minor equality has the same two necessary and sufficient conditions. In particular equality is a fully specified source-volume equation, not a consequence of positivity alone.

There is a direct formulation in the actual volumes, without logarithmic variables. Put $U_N = \sqrt{\alpha_{N+1}}(\mathcal{R}_N - 1)/(2\sqrt{\mathcal{R}_N})$. Since

$$e^{t_N} = \frac{V_N}{\sqrt{V_{N-1} V_{N+1}}}, \quad \cosh s_N = \frac{V_{N-1} + V_{N+1}}{2\sqrt{V_{N-1} V_{N+1}}},$$

equation (TVB.32) gives the exact two losses

$$U_N^2 - \epsilon_N^2 = \alpha_{N+1} \frac{(2V_N - V_{N-1} - V_{N+1})^2}{4V_{N-1}V_{N+1}} + \frac{|z_N|^2}{\omega_N^2}. \quad (\text{TVB.34a})$$

Thus the volume equality is the arithmetic midpoint $2V_N = V_{N-1} + V_{N+1}$ together with $z_N = 0$. Equal consecutive logarithmic contractions alone do not give this midpoint: their extra equation $V_N^2 = V_{N-1}V_{N+1}$ implies it only if all three positive volumes are equal, by the square $(\sqrt{V_{N-1}} - \sqrt{V_{N+1}})^2$. In the original inverse-volume coordinates $D_- = D_{N-1}^{\text{cv}}$, $D_0 = D_N^{\text{cv}}$, $D_+ = D_{N+1}^{\text{cv}}$ and $E_N = \tilde{D}_N$, the same identities become

$$\begin{aligned} E_N D_0 - D_- D_+ &= \frac{D_0^2 |z_N|^2}{\omega_N \omega_{N+1}}, \\ \frac{D_0(D_+ - D_-)^2}{4D_- D_+} - \Delta_N &= \frac{(2D_- D_+ - D_0(D_- + D_+))^2}{4D_0 D_- D_+} + E_N - \frac{D_- D_+}{D_0}. \end{aligned} \quad (\text{TVB.34b})$$

Substitute $D_j = V_j^{-1}$ in (TVB.34a), or multiply out its two squares, to verify every denominator. Both terms on the right are nonnegative by the first equation. Equality is equivalently $D_0 = 2D_- D_+ / (D_- + D_+)$ and $E_N = (D_- + D_+)/2$; it then gives $\Delta_N = E_N - D_0$. In particular phase zero at these degrees records a positive omitted-direction determinant $E_N = D_- D_+ / D_0$, with its exact map specified in (TVB.28), rather than a zero determinant.

The algebraic exterior lower bound applies at each real tilt as well as at zero. Here is a direct proof. In the Hermitian matrix H_N above let P be the orthogonal projection to the image, under $G_N^{1/2}$, of the sum of all full generalized eigenspaces of A with $\Re \lambda > c$. Invariance of that subspace makes the transformed A upper triangular in its orthogonal direct sum with the complement, without asserting that the spectral projection itself is orthogonal. Taking the trace of its first diagonal block gives

$$\text{Tr}(PH_N) = L := \sum_{\Re \lambda > c} \ell_\lambda (2\Re \lambda - k).$$

Since H_N has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros, $\text{Tr}(PH_N) \leq \epsilon_N$: its positive eigenline contributes at most ϵ_N and its negative eigenline contributes a nonpositive number. All algebraic multiplicities ℓ_λ are retained. Thus $L \leq \epsilon_N$ for the same tilted minimum metric, without choosing another quotient metric.

Combining this with (TVB.33), using monotonicity of $\sinh$ on $[0, \infty)$ and its inverse, proves

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.35})$$

For $N = N_0 + 2j$, $0 \leq j < r$, with $N_0 \geq q$, every denominator in one logarithm is the numerator in the next. Summing the exact logs and then their lower bounds proves

$$\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N_0+2j+1}}} \right), \quad r \geq 1. \quad (\text{TVB.36})$$

Every term is at the same fixed real tilt, packet and tensor degree. No relation between different k or a limiting value of either determinant is used. Under $\mathcal{V}$ the minimum representatives at all degrees retain jets η and their original quotient classes; each difference of two such lifts is the explicit boundary in (TVB.7)–(TVB.9).

305.8 The first admitted degree and the empty quotient

At $N = q - 1$, the first q monic remainder columns are a basis, so $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Expand the original polynomial, with its coefficients retained,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle_\theta}{\omega_j}. \quad (\text{TVB.37})$$

Reduction gives $b_q = -\sum_{j<q} \gamma_j b_j$. Hence

$$a = \omega_{q-1}, \quad z_{q-1} = -\gamma_{q-1} \omega_{q-1}, \quad d_+ = \sum_{j<q} |\gamma_j|^2 \omega_j = \nu_0 - \omega_q, \quad \gamma_{q-1} = -i\psi_{q-1}.$$

The last equality follows from (TVB.26), which does not require K_{q-2}^{-1} . Substitution gives the exact CV.12-Toda endpoint

$$\epsilon_{q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - \psi_{q-1}^2. \quad (\text{TVB.38})$$

There is no V_{q-2} here. The omitted determinant instead records $\tilde{D}_{q-1}/D_{q-1}^{\text{CV}} = \psi_{q-1}^2/\alpha_q$. The polynomial projection $\mathbf{P}_{q-2}\chi = \sum_{j<q-1} \gamma_j p_j$ has the actual jet $\eta \sum_{j<q-1} \gamma_j b_j$, whereas χ itself has zero jet and primitive (TVB.9). Its squared source norm in (TVB.6) is the numerator in (TVB.38). This is the exact map from the whole relation to its lower orthogonal projection, including the possibly nonzero resulting observation. For $q = 1$, the sum is empty, the lower source is $\mathcal{P}_{-1} = 0$, and $\epsilon_N = 0$ at every admitted degree because its one-dimensional relative control is trace zero.

The endpoint projection and the minor vector have a stronger exact typed correspondence. Assume $q \geq 2$, let $D = D_{q-1}^{\text{CV}} = \prod_{j<q} \omega_j^{-1}$, and set $I_j = \{0, \dots, q-2\} \setminus \{j\}$ for $0 \leq j \leq q-2$. The determinant of the first q monic remainder columns is 1. Substituting $b_q = -\sum_{l<q} \gamma_l b_l$ in the minor with columns $I_j \cup \{q-1, q\}$ leaves only $l = j$. Moving b_j from its last position to position j takes $q-1-j$ transpositions; with the preceding minus sign this proves

$$\det B_{I_j \cup \{q-1, q\}} = (-1)^{q-j} \gamma_j. \quad (\text{TVB.38a})$$

Write $e_j = (\bigwedge_{i \in I_j} p_i) \wedge p_{q-1} \wedge p_q$. The explicit conjugate-linear isomorphism

$$\mathcal{H} : \mathcal{P}_{q-2} \longrightarrow \mathcal{E}_{q-1}, \quad \mathcal{H} \left(\sum_{j<q-1} a_j p_j \right) = \sum_{j<q-1} (-1)^{q-j} \frac{\omega_j D}{\omega_q} e_j \quad (\text{TVB.38b})$$

has inverse $\sum t_j e_j \mapsto \sum (-1)^{q-j} \omega_q \overline{t_j} p_j / (\omega_j D)$. The source and target basis vectors are orthogonal and $\|e_j\|_\theta^2 = \omega_q / (D \omega_j)$. Consequently

$$\|\mathcal{H}P\|_\theta^2 = \frac{D}{\omega_q} \|P\|_\theta^2, \quad \zeta_{q-1} = \mathcal{H}(\mathbf{P}_{q-2}\chi), \quad \Delta_{q-1} = \frac{D}{\omega_q} \|\mathbf{P}_{q-2}\chi\|_\theta^2. \quad (\text{TVB.38c})$$

This follows term by term from (TVB.31) and (TVB.38a). The actual alternating tensor image has squared norm $q! D \|\mathbf{P}_{q-2}\chi\|_\theta^2 / \omega_q$ in the specified tilted tensor observation, and in the original source norm at $\theta = 0$. The top coefficient is retained separately: $\tilde{D}_{q-1} = D |\gamma_{q-1}|^2 \omega_{q-1} / \omega_q$. For $q = 1$ the map $\mathcal{H}$ is the unique map between zero spaces, with the same zero identities; the top-coefficient formula still records the entire omitted determinant.

For completeness the finite upper bound is strict when $q = 1$ and $N \geq 1$. Consecutive source polynomials have no common root: their three-term recurrence with its nonzero coefficient $\alpha_j > 0$ would propagate such a root backwards to $p_0 = 1$. For any nonzero quotient $q \geq 1$, both b_N, b_{N+1} therefore cannot vanish, since a root of χ would then be common to both polynomials. When $N \geq q$, K_{N-1} is positive definite and $K_{N+1} - K_{N-1}$ is a nonzero positive semidefinite matrix. Conjugating by $K_{N-1}^{-1/2}$ and multiplying its eigenvalues proves $D_{N+1}^{\text{CV}} > D_{N-1}^{\text{CV}}$, hence $\mathcal{R}_N > 1$ and $U_N > 0$. At $q = 1$ this is strictly larger than the already proved $\epsilon_N = 0$.

Finally let $h = 1$, so $\chi = 1$, $q = 0$, and $C = 0$. The analytic function g , w_1 and all source norms still exist. For every $N \geq 0$ the source sequence is multiplication by one from $\mathcal{P}_N$ onto itself followed by the zero quotient. Then $m = N + 1$, $Z_\chi = Z$, $\mathfrak{B}_m = \mathfrak{D}_m$, and the determinant of the zero-dimensional Gram is the empty determinant $V_N = 1$. Its logarithm and derivatives are zero. The minimum lift, its curvature maps, and the relative control are the unique zero maps, with $\epsilon_N = L = 0$. The two Toda coefficients in their curvature difference are equal, so that difference is zero. No positive-dimensional inverse is introduced. The support lift retains $G(0) = \{\tau, e\}$: every represented polynomial, including each nonzero source relation, maps to e , while τ maps to τ .

305.9 The complete join with the original spectral coupling

This section is the exact join supplied in PARENT_EXACT_JOIN.md, SHA256

71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370.

Its two inputs are the delivered Toda equation (28), proved here as (TVB.32), and the complete original-metric exterior identity E8. The latter source is retained in the cyclic companion at commit 161089942cce70a09fbe441d08ba4d36c1fd810c. We reproduce the finite identity and its typed coupling map here to make the join complete; no novelty or limiting estimate is claimed.

All quantities in this section are at the original observation $\theta = 0$, with $N \geq q$. Write $A^\sharp = G_N^{-1} A^* G_N$ and $\mathcal{H}_N = A^\sharp + A - kI$, acting on the original (C, G_N) . Let C_+ be the full generalized spectral subspace with $\Re \lambda > c$, let $C_- = C_+^{\perp G_N}$, and let $P : C \rightarrow C$ be their orthogonal projection onto C_+ . Suppose first $C_+ \neq 0$. Its trace L from (TVB.35) is positive, so $\epsilon_N > 0$. Choose G_N -unit eigenvectors u_+, u_- of $\mathcal{H}_N$ at $\epsilon_N, -\epsilon_N$; selfadjointness makes them orthogonal. They specify only the coefficient injection

$$E : \mathbb{C}^2 \longrightarrow C, \quad E(z_+, z_-) = z_+ u_+ + z_- u_-, \quad E^\sharp E = I_2, \quad M = E^\sharp P E. \quad (\text{TVB.39})$$

Both M and $I_2 - M$ are positive semidefinite, since their quadratic forms are respectively the squared original norms of PEz and $(1 - P)Ez$. In the original orthogonal decomposition define

$$B = A|_{C_+}, \quad D^{\text{op}} = (1 - P)A|_{C_-}, \quad C = PA|_{C_-} : C_- \longrightarrow C_+.$$

Invariance of C_+ gives $A = \begin{pmatrix} B & C \\ 0 & D^{\text{op}} \end{pmatrix}$. Its adjoint has the conjugate-transpose blocks in the original restricted metrics, so $P\mathcal{H}_N|_{C_-} = C$ exactly.

Put $a_+ = \|Pu_+\|^2$, $a_- = \|Pu_-\|^2$, and $z = \langle Pu_+, Pu_- \rangle_{G_N}$. The compression M has these two diagonal entries and off-diagonal entries $z, \bar{z}$. The original rank-two spectral expansion gives $L = \epsilon_N(a_+ - a_-)$ and

$$\begin{aligned} \|C\|_{\text{HS}, G_N}^2 &= \text{Tr}(P\mathcal{H}_N^2 P) - \text{Tr}((P\mathcal{H}_N P)^2) \\ &= \epsilon_N^2(a_+ + a_- - a_+^2 - a_-^2 + 2|z|^2). \end{aligned}$$

The cross trace in the second equation is $|\langle v, w \rangle|^2$ for the two rank-one maps $v\langle v, \cdot \rangle$ and $w\langle w, \cdot \rangle$, as follows by composing them. Subtract the displayed norm and $\epsilon_N^2(a_+ - a_-)^2$ from ϵ_N^2 ; direct expansion proves

$$\epsilon_N^2 - L^2 = \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}. \quad (\text{TVB.40})$$

Both determinants are nonnegative by the quadratic forms after (TVB.39). Adding (TVB.40) to (TVB.32) cancels precisely the intermediate ϵ_N^2 and gives all four nonnegative losses

$$\boxed{\alpha_{N+1} \sinh^2 s_N - L^2 = \alpha_{N+1}(e^{t_N} - \cosh s_N)^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}.} \quad (\text{TVB.41})$$

Every adjoint, projection, norm and coefficient is for the same original minimum metric as the Toda volumes.

The spectral projector has an exact map to the coupling. Factor the minimal polynomial of A as $m_+ m_-$ over the selected and remaining full generalized blocks, with their original exponents. Bezout polynomials a, b with $am_+ + bm_- = 1$ give $Q = b(A)m_-(A)$: it is identity on each selected block and zero on each other block. Thus $Q^2 = Q$, $AQ = QA$, and in the preceding orthogonal decomposition $Q = \begin{pmatrix} I & Y \\ 0 & 0 \end{pmatrix}$ for a unique map $Y : C_- \rightarrow C_+$. Block multiplication proves

$$\mathcal{S} : \text{Hom}(C_-, C_+) \longrightarrow \text{Hom}(C_-, C_+), \quad \mathcal{S}(Y) = BY - YD^{\text{op}} = C, \quad Q - P = \begin{pmatrix} 0 & Y \\ 0 & 0 \end{pmatrix}. \quad (\text{TVB.42})$$

This morphism is invertible with its full original nilpotent parts:

$$\mathcal{S}^{-1}(Z) = \int_0^\infty e^{-tB} Z e^{tD^{\text{op}}} dt. \quad (\text{TVB.43})$$

Indeed every selected eigenvalue has real part greater than c and every remaining one has real part at most c . On a full block at λ the first exponential is exactly $e^{-t\lambda} \sum_{j < \ell_\lambda} (-t)^j (B - \lambda I)^j / j!$, with its corresponding projector, and the second has the same formula with positive t on its remaining blocks. Summing these finite expressions gives a polynomial times a strictly decaying exponential for their product in (TVB.43). This proves convergence in the original operator norm. Differentiating the integrand gives $-B e^{-tB} Z e^{tD^{\text{op}}} + e^{-tB} Z e^{tD^{\text{op}}} D^{\text{op}}$. Its integral has lower endpoint Z and zero upper endpoint, proving the inverse identity. If $BY - YD^{\text{op}} = 0$, the same product with $Z = Y$ is constant and decays to zero, hence $Y = 0$. This proves uniqueness as well. The space C_- is nonzero: otherwise $C_+ = C$ and $L = \text{Tr } \mathcal{H}_N = 0$, contradicting $L > 0$. Together with $C_+ \neq 0$ this makes the unit sphere in $\text{Hom}(C_-, C_+)$ nonempty. The actual smallest singular value $s_{B,D} > 0$ of $\mathcal{S}$ in the original Hilbert–Schmidt pairing exists by compactness of its unit sphere and injectivity. Consequently $\|C\|_{\text{HS}, G_N} \geq s_{B,D} \|Q - P\|_{\text{HS}, G_N}$. No bound on $s_{B,D}$ independent of the arithmetic inputs is used.

Retaining the phase and coupling in (TVB.41) gives the finite strengthening

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{\sqrt{L^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2}}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.44})$$

It follows by subtracting only the nonnegative imbalance and compression-determinant terms and applying the same inverse sinh as in (TVB.35). Summation over the disjoint indices $N = N_0 + 2j$ has the same exact endpoints as (TVB.36), with this degree-dependent square root in each summand. If $C_+ = 0$, use (TVB.32) directly; no positive-eigenline compression is asserted when $\epsilon_N = 0$. At $N = q - 1$, the scalar term instead uses the separate endpoint expression (TVB.38). These identities supply no asymptotic estimate and change no arithmetic class.

305.10 Crosswalk and exact continuing arithmetic quantity

CV.1–CV.3 are (TVB.22)–(TVB.24), with the additional proved identification $D_N^{\text{CV}} = \mathfrak{B}_m / \mathfrak{D}_{N+1}$. CV.4–CV.5 become (TVB.27)–(TVB.29), where the whole complex cross term is $i\omega_N(\sigma - \ell'_N)$. CV.6 is the minimum lift (TVB.14) and its determinant-line map after (TVB.15). CV.10–CV.10a are (TVB.30)–(TVB.31), now with the exact two-loss identity (TVB.34). CV.11–CV.12 are precisely (TVB.37)–(TVB.38). The new operator refinement is (TVB.20)–(TVB.21): it computes both individual rank-one curvature costs, including their shared retained quantity $\alpha_{N+1} \delta_N$.

For an actual quartet polynomial the exterior multiplicities and their lower trace L remain those of that exact polynomial; for a larger packet they remain a lower contribution to its actual full trace. All admission thresholds here use its actual $q = \deg \chi$. The finite formulas prove an upper estimate by the exact source and relation determinants and prove the telescope (TVB.36). They do not estimate the growth of α_{N+1} or the joint determinant ratio as k and N grow. That continuing arithmetic question is the original expression (TVB.27), equivalently the positive-minor source norm in (TVB.28)–(TVB.31), with both losses (TVB.32) available without changing any of its inputs.

306 Combining the existing spectral and overlap controls on the original source

This attachment records the exact comparison already established in the coordinating source intake. The complete earlier AW proof is retained in this reader; SP reproduces its spectra and final condition-number bound and adds the stated overlap refinement. Neither bound is discarded.

Fix the actual full packet h , tensor order $k \geq 1$, and its original cyclic polynomial χ of degree $q \geq 1$. The common coefficient space is $H = \mathcal{P}_{2q}$. Its integration coordinate is $S = k/2 + iy$. Write $x \in [0, 1]$ for metric interpolation and reserve u for the geometric deformation parameter of BC. The exact AW/SP coordinate map is

$$y = u_{AW}, \quad x = t_{AW}, \quad H_{AW} \xrightarrow{\text{id}} H_{SP}. \quad (\text{ACM1})$$

This identity leaves every polynomial coefficient unchanged. The arithmetic density is w_h^{*k} in both calculations. For the reference density, the original constant $c_a = 2^{1-2a}\Gamma(2a)$ gives $c_{1/4} = \sqrt{2\pi}$, so the exact convolution formula is

$$r_{1/4}^{*k}(y) = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}(y) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}. \quad (\text{ACM2})$$

Consequently the densities, their masses and all monomial integrals coincide. In particular $M_{AW}(x) = M_{SP}(x) = M_x$ on H , and $M_{N,AW} = I_N^* M_x I_N$ on the original subspace $I_N : \mathcal{P}_N \hookrightarrow H$.

AW's relation map has codomain $\mathcal{P}_N$, whereas SP incorporates its inclusion into H . Their literal equality is

$$B_N^{SP} = I_N B_N^{AW}. \quad (\text{ACM3})$$

Substitution into the orthogonal projector formula gives

$$\begin{aligned} P_N &= I_N (I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= I_N B_N^{AW} ((B_N^{AW})^* I_N^* M_x I_N B_N^{AW})^{-1} (B_N^{AW})^* I_N^* M_x. \end{aligned} \quad (\text{ACM4})$$

The two definitions of each projector therefore agree, including $Q_{q-1} = 0$ for the zero relation domain. Put

$$\begin{aligned} U_x &= Q_{2q-1} + Q_{2q} - Q_q = \mathcal{R}_x^{SP}, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q) = \mathcal{P}_x^{SP}, \\ C_x &= M_x^{-1} \dot{M}_x = T_x^{SP}. \end{aligned} \quad (\text{ACM5})$$

These are AW's original U, W, C . Its independently constructed source isometry T_x^{iso} satisfies

$$(T_x^{\text{iso}})^* M_x T_x^{\text{iso}} = M_x, \quad U_x = T_x^{\text{iso}} W_x (T_x^{\text{iso}})^{-1}. \quad (\text{ACM6})$$

Thus cyclicity of the same finite trace proves

$$\text{Tr}((U_x - W_x)C_x) = \text{Tr}(W_x (T_x^{\text{iso}})^{-1} [C_x, T_x^{\text{iso}}]). \quad (\text{ACM7})$$

The original observation is still $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}$. Its exact difference and kernel after the isometry remain

$$\begin{aligned} \mathcal{O} T_x^{\text{iso}} - \mathcal{O} &= \sigma_h^{\otimes k} \eta J_{2q} (T_x^{\text{iso}} - I), \\ \ker(\mathcal{O} T_x^{\text{iso}}) &= (T_x^{\text{iso}})^{-1} (\chi \mathcal{P}_q). \end{aligned} \quad (\text{ACM8})$$

The first equality is expansion. The second follows because $\sigma_h^{\otimes k}$ and η are the original injections and $\ker J_{2q} = \chi \mathcal{P}_q$. This retains the actual cohomology defect of the isometry rather than assigning it an intertwining property.

Let $F(x)$ be the original logarithmic four-endpoint ratio and $\Delta_{h,k} = F(1) - F(0)$. Its signed identity is $F'(x) = \text{Tr}((U_x - W_x)C_x)$ by (AW6), or equivalently (SP4)–(SP5) and (ACM5). Both U_x, W_x have the complete spectrum 0 with multiplicity q , 1 with multiplicity 2, and 2 with multiplicity $q - 1$. All statements below also cover $q = 1$.

Write $c_1(x) \leq \dots \leq c_{2q+1}(x)$ for the eigenvalues of C_x in the original M_x metric and $d_x = c_{2q+1}(x) - c_1(x)$. Let $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two already proved pointwise bounds, on this identical source, are

$$\begin{aligned} S_{\text{AW}}(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ S_{\text{SP}}(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}. \end{aligned} \quad (\text{ACM9})$$

For completeness, their simultaneous application follows directly as follows. An orthogonal projection P of rank r satisfies $\sum_{j=1}^r c_j \leq \text{Tr}(PC_x) \leq \sum_{j=2q+2-r}^{2q+1} c_j$: in an orthonormal eigenbasis of C_x , its diagonal entries lie in $[0, 1]$ and have sum r , so moving their mass to the lowest or highest eigenvalues proves each inequality. Each of U_x, W_x is the sum of its range projection of rank $q + 1$ and its eigenvalue-two projection of rank $q - 1$. Apply these bounds to both sums and subtract their extrema to obtain $|F'(x)| \leq S_{\text{AW}}(x)$, with every multiplicity retained.

The range dimensions give $s_x \geq 1$. If E_x projects onto the intersection, then $U_x - E_x$ and $W_x - E_x$ are positive, each of trace $2q - s_x$. Therefore $\|U_x - W_x\|_1 \leq 4q - 2s_x$. The same complete spectra give

$$\text{Tr}(U_x - W_x) = 0, \quad \text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{ACM10})$$

Cauchy–Schwarz on all $2q + 1$ real eigenvalues gives the other trace-norm bound in (ACM9). Subtract the midpoint $(c_1 + c_{2q+1})I/2$ from C_x in the trace pairing; the first identity in (ACM10) leaves F' unchanged. Diagonalizing $U_x - W_x$ then bounds the pairing by $d_x \|U_x - W_x\|_1/2$. This proves $|F'(x)| \leq S_{\text{SP}}(x)$ in the same original metric.

Every subsequent bound is now evaluated on this identical path, including the centered Hilbert–Schmidt, angle and signed finite-residual controls. Retain the original monomial row $v(y) = (1, S, \dots, S^{2q})$, $S = k/2 + iy$, in the same H , and the original interpolation measure $d\mu_x(y) = ((1-x)r_{1/4}^{*k}(y) + xw_h^{*k}(y)) dy$. For compact notation put $n = 2q + 1$, $a_q = 2q - 1$, and define

$$\begin{aligned} T_q(x) &= \frac{1}{2} d_x \|U_x - W_x\|_{1, M_x}, \\ S_q(x) &= \sqrt{\left(\text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{n} \right) \text{Tr}((U_x - W_x)^2)}, \\ z_x &= a_q \sqrt{1 - e^{-F(x)/a_q}}, \quad A_q(x) = z_x d_x, \\ J_q &= \int_0^1 \min\{S_{\text{AW}}(x), T_q(x), S_{\text{SP}}(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\{d_x, S_{\text{AW}}(x)/z_x, T_q(x)/z_x, S_{\text{SP}}(x)/z_x, S_q(x)/z_x\} dx. \end{aligned} \quad (\text{ACM11a})$$

These symbols are exactly $E_q, T_q, O_q, S_q, A_q, J_q, I_q$ of (RMT9)–(RMT13), with $E_q = S_{\text{AW}}$ and $O_q = S_{\text{SP}}$. Equations (ACM1)–(ACM5) prove the operator and measure identities needed for that substitution. In particular they retain the actual inverse metric in the signed density $v(U_x - W_x)M_x^{-1}v^*$.

Here is the proof of the additional terms at this earlier use site. Trace zero in (ACM10) allows subtraction of $\text{Tr } C_x/n$ from C_x ; Hilbert–Schmidt Cauchy–Schwarz gives $|F'| \leq S_q$. The midpoint argument already proved gives $|F'| \leq T_q \leq S_{\text{SP}}$. Retain both complete q -entry angle lists of the minimum-section pairs $(q - 1, 2q)$ and $(q, 2q - 1)$. The second pair has a designated zero angle by the original ambient dimension. Each additional zero remains in its list. The exact projection-difference spectra give $|F'| \leq d_x \sum \sin \theta$ over the two lists. Their cosine-square determinants give

$F = \sum -\log \cos^2 \theta$; applying concavity of $t \mapsto \sqrt{1 - e^{-t}}$ to the remaining $2q - 1$ entries gives $|F'| \leq A_q$, exactly as proved with every multiplicity in (RMT11). The designated zero still contributes its displayed zero to both scalar sums.

The denominator in (ACM11a) is strictly positive on this actual path. Indeed equality $F = 0$ makes both crossed minimum sections equal: their Gram difference is the squared norm of their difference, and every relative eigenvalue is at least one. Equality of their determinants makes every such eigenvalue one. In particular the constant remainder 1 would be orthogonal to $\chi \mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \overline{\chi(k - \overline{S})}$ has degree q and equals $\overline{\chi(\overline{S})}$ on $S = k/2 + iy$. That orthogonality would imply $\int |\chi|^2 d\mu_x = 0$, contradicting the positive Gram of the nonzero polynomial χ . This proves $F(x) > 0$ without inserting a new hypothesis. Compactness gives a positive minimum. All remaining functions are continuous except the intersection dimension, whose rank strata are Borel sets defined by matrix minors. Thus every displayed integral is finite.

The connection to the retained (J_*, I_*) is an exact identity, not an additional choice of estimate. Put $\Pi_N = P_N - Q_N$, the orthogonal projection onto the original minimum-section space at cutoff N . Substitution in (ACM5), including $Q_{q-1} = 0$, gives

$$U_x - W_x = (\Pi_{q-1} - \Pi_{2q}) + (\Pi_q - \Pi_{2q-1}).$$

For each pair of rank- q projections in H of dimension $2q + 1$, the complete difference spectrum consists of the positive and negative sine of each of its q principal angles and one additional ambient zero. All zero angles still contribute both of their zero entries. The triangle inequality for the trace norm therefore gives

$$\frac{1}{2} \|U_x - W_x\|_{1, M_x} \leq \sum_{j=1}^q \sin \theta_j + \sum_{j=1}^q \sin \varphi_j \leq a_q \sqrt{1 - e^{-F(x)/a_q}} = z_x. \quad (\text{ACM11b})$$

Here (θ_j) and (φ_j) are respectively the two complete angle lists just used; $\varphi_1 = 0$ is the designated forced zero. The last inequality is the same concavity calculation, and no other zero entry is removed. Consequently $T_q \leq A_q$ and the already proved $T_q \leq S_{\text{SP}}$ gives, pointwise and then by integration,

$$\begin{aligned} J_q &= \int_0^1 \min\{S_{\text{AW}}, T_q, S_q\} dx = J_*, \\ I_q &= \int_0^1 \frac{\min\{S_{\text{AW}}, T_q, S_q\}}{z_x} dx = I_*. \end{aligned} \quad (\text{ACM11c})$$

These are precisely the three terms in (AT21a)–(AT21c), through the already proved identical coefficients, measures and operators. The exact path dictionary is $B(x) = F(x)$, $a = a_q$ and $f_a(B(x)) = z_x$; it preserves the original initial value and every measure factor. For I_q , the omitted entry d_x is at least T_q/z_x by (ACM11b); the omitted overlap term is also at least T_q/z_x . Thus both equalities retain the same original nonlinear denominator.

Both bounds are finite and measurable on $[0, 1]$. Indeed the positive Grams and their inverses vary smoothly, the eigenvalues vary continuously, and the dimension s_x is a Borel function: in any fixed coefficient frame it is determined by the ranks of matrices of continuous functions, whose rank strata are given by vanishing and nonvanishing minors. Thus the triangle inequality for integration gives the combined, original-source control

$$\boxed{|\Delta_{h,k}| \leq J_* \leq \underbrace{\int_0^1 \min\{S_{\text{AW}}(x), \frac{1}{2} d_x \|U_x - W_x\|_{1, M_x}\} dx}_{\mathfrak{C}_*} \leq \underbrace{\int_0^1 \min\{S_{\text{AW}}(x), S_{\text{SP}}(x)\} dx}_{\mathfrak{C}_*^{\text{ov}}}} \quad (\text{ACM11})$$

The first bound preserves the exact trace norm already present in (SP13); its proof above gives $d_x \|U_x - W_x\|_{1/2} \leq S_{\text{SP}}(x)$. The identity maps (ACM1)–(ACM5) identify these quantities with

(AT18a), including the source density and its mass. If $b_1 \leq \dots \leq b_{2q+1}$ are the positive eigenvalues of $D = M_0^{-1}M_1$ in M_0 , the original interpolation gives

$$c_j(x) = \frac{b_j - 1}{1 - x + xb_j}, \quad \int_0^1 c_j(x) dx = \log b_j. \quad (\text{ACM12})$$

The order is preserved since differentiation of the displayed fraction with respect to b_j gives $(1 - x + xb_j)^{-2} > 0$. Integration of (ACM9) consequently yields

$$|\Delta_{h,k}| \leq J_* \leq \mathfrak{C}_* \leq \mathfrak{C}_*^{\text{ov}} \leq \min \left\{ \mathcal{L}_q(D), \int_0^1 S_{\text{SP}}(x) dx \right\}, \quad (\text{ACM13})$$

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}.$$

The same pointwise minimum also retains the individually integrated trace, centered variance and angle bounds. Termwise integration in (ACM12), with the original order and all multiplicities, proves

$$J_q \leq \min \left\{ \mathcal{L}_q(D), \int_0^1 T_q(x) dx, \int_0^1 S_{\text{SP}}(x) dx, \int_0^1 S_q(x) dx, \int_0^1 A_q(x) dx \right\} \quad (\text{ACM13a})$$

$$\mathcal{L}_q(D) \leq (2q - 1) \log(b_{2q+1}/b_1).$$

The last inequality bounds each of the $1 + 2(q - 1)$ logarithmic factors in the unchanged (ACM13) by $\log(b_{2q+1}/b_1)$. It is therefore valid also for $q = 1$, where the original sum is empty.

The empty sum at $q = 1$ is zero. No universal ordering of the two refinements is used: each bounds the same signed integrand, hence their pointwise minimum does. Their full original arithmetic moments, relative eigenvalues and overlap remain the actual terms to be evaluated along the growing packet family.

The actual moment interpolation above also satisfies the complete proof (AT21a)–(AT21e). With $a = 2q - 1$ its recursive additive minimum also retains the independently proved centered HS term S_{HS} in (AT21a), so $J_* \leq \mathfrak{C}_*$. The exact trace norm is at most both the overlap and full angle bounds. The nonlinear radius $I_* = \int_0^1 \min(S_{\text{spec}}, S_{\text{tr}}, S_{\text{HS}}) / [a \sqrt{1 - e^{-B(x)/a}}] dx$ is defined on the same strictly positive four-volume path, with $I_* \leq \log \kappa$. Thus the current endpoint consequence here is the simultaneous interval $\mathfrak{b}_*^- \leq B(1) \leq \mathfrak{b}_*^+$ of (AT21e), not only its additive corollary. The identity of the coefficient maps and original measures proved above verifies its hypotheses; no purity or tensor-uniform bound is supplied by this finite transport.

The following signed construction further intersects this same retained additive/nonlinear interval. By (ACM11c) its (J_q, I_q) are literally (J_*, I_*) ; hence no earlier endpoint or overlap bound is discarded.

The nonlinear coordinate also receives every one of these controls. For $B > 0$ put

$$\mathcal{H}_{a_q}(B) = 2 \operatorname{arcosh}(e^{B/(2a_q)}), \quad \Psi_{a_q}(t) = 2a_q \log \cosh(t/2) \quad (t \geq 0). \quad (\text{ACM14})$$

Their two inverse composites on the positive half-lines are identities. Direct differentiation gives

$$\mathcal{H}_{a_q}'(B) = \frac{1}{a_q \sqrt{1 - e^{-B/a_q}}}.$$

Dividing the five proved derivative bounds by z_x and integrating therefore gives $|\mathcal{H}_{a_q}(F(1)) - \mathcal{H}_{a_q}(F(0))| \leq I_q$. Moreover $I_q \leq \int dx = \log(b_n/b_1)$.

For every integer $p \geq 0$ retain the following specified finite signed construction, on the same source operators:

$$\begin{aligned}
B_x^\circ &= (1-x)I + xD, & \alpha_x &= \frac{1}{2}(2(1-x) + x(b_1 + b_n)), & H_x^\circ &= I - B_x^\circ/\alpha_x, \\
C_x^{[p]} &= \alpha_x^{-1} \sum_{r=0}^p (H_x^\circ)^r (D - I), \\
R_x^{[p]} &= (H_x^\circ)^{p+1} C_x, & E_x^{[p]} &= R_x^{[p]} - \text{Tr}(R_x^{[p]})I/n, \\
j_p &= \int_0^1 \text{Tr}(C_x^{[p]}(U_x - W_x)) dx, \\
e_p &= \int_0^1 \sqrt{\text{Tr}((E_x^{[p]})^2) \text{Tr}((U_x - W_x)^2)} dx.
\end{aligned} \tag{ACM15}$$

The finite geometric identity proves $C_x - C_x^{[p]} = R_x^{[p]}$. Every factor is a real function of D , hence self-adjoint in M_x . Trace zero removes the centered scalar, so Hilbert–Schmidt Cauchy–Schwarz proves $j_p - e_p \leq \Delta_{h,k} \leq j_p + e_p$. Writing $\vartheta = (b_n - b_1)/(b_n + b_1) < 1$ and $K_D = \sum_{j=1}^n (|b_j - 1|/\min\{1, b_j\})^2$, the full residual eigenvalues give $\text{Tr}((E_x^{[p]})^2) \leq \vartheta^{2p+2} K_D$. Both U_x, W_x dominate their range projections, whose intersection has dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$. Equation (ACM10) then gives $e_p \leq \vartheta^{p+1} \sqrt{K_D(8q-6)} \rightarrow 0$. This retains the exact signed center and its controlled residual, with no assumption of uniformity in the tensor order.

Put $F_0 = F(0)$ and $H_0 = \mathcal{H}_{a_q}(F_0)$. For every $p \geq 0$ define the actual simultaneous endpoint functions

$$\begin{aligned}
\beta_p^- &= \max\{0, F_0 - J_q, \Psi_{a_q}(\max\{0, H_0 - I_q\}), F_0 + j_p - e_p\}, \\
\beta_p^+ &= \min\{F_0 + J_q, \Psi_{a_q}(H_0 + I_q), F_0 + j_p + e_p\}.
\end{aligned} \tag{ACM16}$$

The preceding additive, nonlinear and signed inequalities prove $\beta_p^- \leq F(1) \leq \beta_p^+$ by taking their intersection. Both original angle lists, source masses, relation maps and arithmetic Taylor unit remain fixed. At $D = cI$ all diameter and residual terms vanish: each quotient volume is multiplied by the same scalar power q , which cancels with the four signs. The formulas then give $F(1) = F_0$ exactly. Numerical integration of a signed center must retain its own integration error; (ACM15) specifies exact integrals.

306.1 The literal graph augmentation on this finite coefficient source

The following additional calculation uses the full quartet domain of PGB.1: $\gamma > 2, 0 < \delta < 1/2, m, k \geq 1$. It leaves the preceding Gamma-to-arithmetic formulas on their entire stated domain. Here χ is the literal monic polynomial

$$\chi(S) = \prod_{a,b=0}^k (S - k/2 - (2a - k)\delta - i(2b - k)\gamma)^{1+k(m-1)}. \tag{ACM17}$$

Thus (HT2) and PGB.1 have identical roots, multiplicities, leading coefficient, increasing remainder frame and remainder map; their coefficient identification is the identity, including every nilpotent primary coordinate. Retain $g = 2\xi, v_h = g/h, \Phi' = h, \Phi(0) = 0, s_i = 1/2 + it_i, \Sigma = \sum_i s_i$ and $\tau_k = \sum_i \Phi(s_i)$. The integration measure in the full source is $\prod_i dt_i/(2\pi)$, and

$$\begin{aligned}
(\mathcal{V}P)(\mathbf{t}) &= \left(\prod_i v_h(s_i) \right) P(\Sigma), \\
L_{\text{obs}}x &= \left(\prod_i v_h(s_i) \right) \sum_{b=0}^{q-1} x_b \text{rem}_{h(s_1), \dots, h(s_k)}(\tau_k \Sigma^b), \\
D_{\text{def}}P &= \tau_k \mathcal{V}P - L_{\text{obs}}J_{2q}P, & H &= \mathcal{P}_{2q}, & M^{\text{ar}} &= \mathcal{V}^* \mathcal{V}.
\end{aligned} \tag{ACM18}$$

The symbols L_{obs} and D_{def} denote exactly L and D_{2q} of PGB.2; neither is the circle circumference or the positive relative matrix below. PGB.2 and its leading-coefficient argument prove that these columns lie in the stated Hilbert source and that D_{def} is injective. In particular every following finite Gram exists and is positive. On the unchanged coefficient space define the actual affine path

$$M_x^{\text{gr}} = M^{\text{ar}} + x D_{\text{def}}^* D_{\text{def}}, \quad 0 \leq x \leq 1, \quad \mathcal{V}_x^{\text{gr}} P = (\mathcal{V}P, \sqrt{x} D_{\text{def}} P). \quad (\text{ACM19})$$

Its Gram is exactly M_x^{gr} , by the orthogonal direct-sum norm; no mixed term was omitted. At $x = 1$ this is the unchanged graph of PGB.2. The restriction to $\mathcal{P}_N$ uses the literal I_N and has defect $D_{\text{def}} I_N = D_N$: monic remainders satisfy $J_{2q} I_N = J_N$, proving this equality directly in (ACM18). Its relation columns are $I_N B_N^{\text{AW}}$, exactly as in (ACM3).

Use (ACM4)–(ACM5) with this new, explicitly specified form and put $U_x^{\text{gr}} = Q_{2q-1} + Q_{2q} - Q_q$ and $W_x^{\text{gr}} = P_{2q-1} - P_{q-1} + P_{2q} - P_q$. Let $G_N(x)$ denote the quotient Gram of its restriction and set

$$\begin{aligned} F_{\text{gr}}(x) &= \log \det G_{q-1}(x) + \log \det G_q(x) - \log \det G_{2q-1}(x) - \log \det G_{2q}(x), \\ C_x^{\text{gr}} &= (M_x^{\text{gr}})^{-1} D_{\text{def}}^* D_{\text{def}}, \quad A_x^{\text{gr}} = U_x^{\text{gr}} - W_x^{\text{gr}}, \\ \Gamma_k^{\text{gr}} &= F_{\text{gr}}(1) - F_{\text{gr}}(0) = \widehat{\mathcal{B}}_k - \mathcal{B}_k. \end{aligned} \quad (\text{ACM20})$$

Here $\widehat{\mathcal{B}}_k$ is the four-endpoint logarithm of $\widehat{G}_N$ in PGB.26, and $\mathcal{B}_k$ is that of the original arithmetic source. To prove the derivative at this use site, the minimum section $R_N(x)$ satisfies $J_N R_N = I$ and $B_N^* M_N(x) R_N = 0$. Differentiating the first equality puts $\dot{R}_N$ in the relation space. The two cross terms in $d(R_N^* M_N R_N)/dx$ therefore vanish by the second equality, and $\dot{G}_N = R_N^* \dot{M}_N R_N$. Jacobi's determinant formula and finite trace cyclicity give $d \log \det G_N/dx = \text{Tr}((P_N - Q_N) C_x^{\text{gr}})$. Taking the four signs proves

$$F'_{\text{gr}}(x) = \text{Tr}(A_x^{\text{gr}} C_x^{\text{gr}}). \quad (\text{ACM21})$$

The flag inclusions imply that each difference of nested orthogonal projections is an orthogonal projection. In particular Q_q has rank one and is contained in both Q_{2q-1} and Q_{2q} . The original dimensions give the complete spectra $0^{[q]}, 1^{[2]}, 2^{[q-1]}$ for $U_x^{\text{gr}}, W_x^{\text{gr}}$. Their traces agree, so $\text{Tr } A_x^{\text{gr}} = 0$ and

$$\text{Tr}((A_x^{\text{gr}})^2) = 8q - 4 - 2 \text{Tr}(U_x^{\text{gr}} W_x^{\text{gr}}). \quad (\text{ACM22})$$

These arguments use positive finite forms and the same flags; they do not use a scalar moment representation of the graph norm. Hence the proofs at (ACM9)–(ACM11a) apply to their spectral, trace-norm, overlap and centered Hilbert–Schmidt terms. Explicitly let $c_1^{\text{gr}} \leq \dots \leq c_n^{\text{gr}}$ be the full eigenvalue list of C_x^{gr} , $n = 2q + 1$, and let $d_x^{\text{gr}} = c_n^{\text{gr}} - c_1^{\text{gr}}$ and $s_x^{\text{gr}} = \dim(\text{ran } U_x^{\text{gr}} \cap \text{ran } W_x^{\text{gr}})$. Set

$$\begin{aligned} E_{\text{gr}}(x) &= c_{q+2}^{\text{gr}} - c_q^{\text{gr}} + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}^{\text{gr}} - c_j^{\text{gr}}), \\ T_{\text{gr}}(x) &= \frac{1}{2} d_x^{\text{gr}} \|A_x^{\text{gr}}\|_{1, M_x^{\text{gr}}}, \\ O_{\text{gr}}(x) &= \frac{1}{2} d_x^{\text{gr}} \min \left\{ 4q - 2s_x^{\text{gr}}, \sqrt{n \text{Tr}((A_x^{\text{gr}})^2)} \right\}, \\ S_{\text{gr}}(x) &= \sqrt{\left(\text{Tr}((C_x^{\text{gr}})^2) - \frac{(\text{Tr } C_x^{\text{gr}})^2}{n} \right) \text{Tr}((A_x^{\text{gr}})^2)}, \\ J_{\text{gr}} &= \int_0^1 \min\{E_{\text{gr}}, T_{\text{gr}}, O_{\text{gr}}, S_{\text{gr}}\} dx. \end{aligned} \quad (\text{ACM23})$$

Rank-stratum measurability and continuity of the other entries prove integrability exactly as before. The four pointwise trace proofs give $|\Gamma_k^{\text{gr}}| \leq J_{\text{gr}}$.

For a specified integer $p \geq 0$, this same path also has an actual signed finite approximation. Put

$$\begin{aligned}
 D^{\text{gr}} &= (M^{\text{ar}})^{-1}(M^{\text{ar}} + D_{\text{def}}^* D_{\text{def}}), & 1 < b_1^{\text{gr}} \leq \dots \leq b_n^{\text{gr}}, \\
 B_x^{\text{gr}} &= (1-x)I + xD_x^{\text{gr}}, & \alpha_x^{\text{gr}} &= 1-x + \frac{x}{2}(b_1^{\text{gr}} + b_n^{\text{gr}}), \\
 H_x^{\text{gr}} &= I - B_x^{\text{gr}}/\alpha_x^{\text{gr}}, & C_x^{[p],\text{gr}} &= (\alpha_x^{\text{gr}})^{-1} \sum_{r=0}^p (H_x^{\text{gr}})^r (D^{\text{gr}} - I), \\
 R_x^{[p],\text{gr}} &= (H_x^{\text{gr}})^{p+1} C_x^{\text{gr}}, & E_x^{[p],\text{gr}} &= R_x^{[p],\text{gr}} - \text{Tr}(R_x^{[p],\text{gr}})I/n, \\
 j_{p,\text{gr}} &= \int_0^1 \text{Tr}(C_x^{[p],\text{gr}} A_x^{\text{gr}}) dx, \\
 e_{p,\text{gr}} &= \int_0^1 \sqrt{\text{Tr}((E_x^{[p],\text{gr}})^2) \text{Tr}((A_x^{\text{gr}})^2)} dx.
 \end{aligned} \tag{ACM24}$$

The strict lower bound for b_1^{gr} follows from injectivity of D_{def} in this finite dimension. The finite geometric identity gives $C_x^{\text{gr}} - C_x^{[p],\text{gr}} = R_x^{[p],\text{gr}}$. All these factors are real functions of the same D^{gr} , so they are self-adjoint for M_x^{gr} . Pairing the last identity with the trace-zero A_x^{gr} and applying Hilbert-Schmidt Cauchy-Schwarz proves

$$j_{p,\text{gr}} - e_{p,\text{gr}} \leq \Gamma_k^{\text{gr}} \leq j_{p,\text{gr}} + e_{p,\text{gr}}. \tag{ACM25}$$

For completeness set

$$\vartheta_{\text{gr}} = \frac{b_n^{\text{gr}} - b_1^{\text{gr}}}{b_n^{\text{gr}} + b_1^{\text{gr}}}, \quad K_{\text{gr}} = \sum_{j=1}^n \left(\frac{|b_j^{\text{gr}} - 1|}{\min\{1, b_j^{\text{gr}}\}} \right)^2.$$

The eigenvalue formula $c_j^{\text{gr}}(x) = (b_j^{\text{gr}} - 1)/(1 - x + x b_j^{\text{gr}})$ shows $\|H_x^{\text{gr}}\| \leq \vartheta_{\text{gr}} < 1$ and $\text{Tr}((E_x^{[p],\text{gr}})^2) \leq \vartheta_{\text{gr}}^{2p+2} K_{\text{gr}}$. The range intersection has dimension at least one. Since each of $U_x^{\text{gr}}, W_x^{\text{gr}}$ dominates its range projection, the positive-product trace inequality gives $\text{Tr}(U_x^{\text{gr}} W_x^{\text{gr}}) \geq 1$. Together with (ACM22), this proves

$$e_{p,\text{gr}} \leq \vartheta_{\text{gr}}^{p+1} \sqrt{K_{\text{gr}}(8q-6)} \rightarrow 0. \tag{ACM26}$$

Thus the two actual signed endpoints

$$g_{p,\text{gr}}^- := \max\{-J_{\text{gr}}, j_{p,\text{gr}} - e_{p,\text{gr}}\}, \quad g_{p,\text{gr}}^+ := \min\{J_{\text{gr}}, j_{p,\text{gr}} + e_{p,\text{gr}}\} \tag{ACM27}$$

satisfy $g_{p,\text{gr}}^- \leq \Gamma_k^{\text{gr}} \leq g_{p,\text{gr}}^+$. This calculation uses no division by the angle coordinate of the new graph form. The moment-path nonlinear formulas (ACM14)–(ACM16) retain their original scope. All integrals in (ACM23)–(ACM27) are exact integrals of the displayed finite matrices; numerical evaluation must add its own certified integration error.

306.2 The complete original relation rows in the graph endpoint control

Keep exactly the domain and coefficient identity of (ACM17)–(ACM20). The symbols in this paragraph include a superscript rel so that the monic relation polynomial is never confused with a consecutive quotient-volume ratio. They are the literal objects of (PGRT.2)–(PGRT.8):

$$\begin{aligned}
 r_k(u) &= \int_{\sum t_i=u} (1 + |\tau_k|^2) \prod_i w_h(t_i), & F_b &= \text{rem}_{h(s_1), \dots, h(s_k)}(\tau_k \Sigma^b), \\
 z_b(u) &= \int_{\sum t_i=u} \bar{\tau}_k F_b \prod_i w_h(t_i), & g_x &= z(u)x/r_k(u), & F^0 x &= (sx)(k/2 + iu) - g_x(u), \\
 d_j^{\text{rel}} &= \chi p_{j+1-q}, & a_j^{\text{rel}} &= \int |d_j^{\text{rel}}(k/2 + iu)|^2 r_k(u) du > 0, \\
 (\alpha_j^{\text{rel}})_b &= \int \overline{d_j^{\text{rel}}(k/2 + iu)} ((k/2 + iu)^b r_k(u) - z_b(u)) du, & j &\geq q-1, \quad 0 \leq b < q.
 \end{aligned} \tag{ACM28}$$

Here $s : E \rightarrow \mathcal{P}_{q-1}$ is the original monic remainder section, p_n is the unique monic polynomial of degree n orthogonal to lower polynomials for $|\chi|^2 r_k du$, and the fibre integral eliminates t_k with its original Lebesgue measure. For $k = 1$ it is evaluation. The constant factors are $w_h(t) = |v_h(1/2 + it)|^2 / (2\pi)$ throughout. Each coefficient is absolutely integrable by the proved polynomial exponential moments and Cauchy-Schwarz applied to F^0 . Expanding the full polynomials in these integrals gives the finite products of the original one-factor moments in (PGS.35) and (PGS.38), without dropping their conjugate or cross terms. The coefficients of p_n are obtained from its positive finite lower-polynomial Gram system. This is the exact finite arithmetic evaluation of each row.

The complete weighted density proof (PGRT.3)–(PGRT.6) gives

$$\widehat{G}_N - \Omega_k = \sum_{j=N}^{\infty} (a_j^{\text{rel}})^{-1} (\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}, \quad \Omega_k = K_{\text{low}} + D_{\text{mix}}^* D_{\text{mix}} > 0. \quad (\text{ACM29})$$

The series converges in the original finite matrix operator norm. Specifically, the relation columns are a complete orthogonal family in $L^2(r_k du)$; the coefficient of $F^0 x$ on its j th column is $(a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} x$. Removing the first $N - q + 1$ columns is precisely minimization over the original relations $\chi \mathcal{P}_{N-q}$. Parseval gives the displayed quadratic forms, and polarization gives their matrices. The trace of the remaining positive tail is the sum of the squared tails of the q fixed frame vectors, hence tends to zero and bounds its operator norm. The positive completed term and its mixed realization are the same as (HC22), with full proof in PGMB.

For $q - 1 \leq j \leq 2q - 1$ put

$$\begin{aligned} l_j^{\text{rel}} &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \widehat{G}_{q-1}^{-1} (\alpha_j^{\text{rel}})^*, \\ u_j^{\text{rel}} &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \widehat{G}_{2q}^{-1} (\alpha_j^{\text{rel}})^*, \\ Z_{2q}^{\text{mix}} x &= (D_{\text{mix}} x, f_{2q} x), \\ u_j^{\text{rel}} &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \\ &\quad - (a_j^{\text{rel}})^{-1} \left\| (I + Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (Z_{2q}^{\text{mix}})^*)^{-1/2} Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \right\|^2. \end{aligned} \quad (\text{ACM30})$$

The domain of Z_{2q}^{mix} is the original E , and its ordered target is the mixed weighted Hilbert space of (HC22) followed by $L^2(r_k du)$. In particular it retains the finite tail as well as the mixed component. To prove the subtraction, write $K = K_{\text{low}}$ and $Z = Z_{2q}^{\text{mix}}$. The identity $\widehat{G}_{2q} = K + Z^* Z$ gives

$$(K + Z^* Z)^{-1} = K^{-1} - K^{-1} Z^* (I + Z K^{-1} Z^*)^{-1} Z K^{-1}.$$

Multiplication by $K + Z^* Z$ cancels the leftover factor $Z^* Z K^{-1} - Z^* (I + Z K^{-1} Z^*) (I + Z K^{-1} Z^*)^{-1} Z K^{-1}$. The target inverse exists because it is the identity off the finite-dimensional range of Z and is positive with eigenvalues at least one on that range. Pairing with the specified row proves (ACM30). If $k \geq 2$ and $\alpha_j^{\text{rel}} \neq 0$, the subtracted squared norm is strictly positive: D_{mix} is injective by the full fixed-sum calculation PGM.6–9, and the displayed inverse square root has zero kernel. For $k = 1$ its mixed component is zero but its finite tail is retained. A zero row gives zero in both terms. These statements retain the original strictness domain and the whole negative term.

Use the endpoint weights $\omega_{q-1} = \omega_{2q-1} = 1$ and $\omega_j = 2$ for $q \leq j \leq 2q - 2$, and define

$$\begin{aligned} L_k^{\text{rel}} &= \sum_{j=q-1}^{2q-1} \omega_j \log(1 + l_j^{\text{rel}}), \quad U_{k,2q}^{\text{rel}} = \sum_{j=q-1}^{2q-1} \omega_j \log(1 + u_j^{\text{rel}}), \\ L_k^{\text{rel}} &\leq \widehat{\mathcal{B}}_k \leq U_{k,2q}^{\text{rel}}, \\ L_k^{\text{rel}} - g_{s,\text{gr}}^+ &\leq \mathcal{B}_k \leq U_{k,2q}^{\text{rel}} - g_{s,\text{gr}}^- \quad (s \geq 0). \end{aligned} \quad (\text{ACM31})$$

To verify every index, (ACM29) gives $\widehat{G}_j = \widehat{G}_{j+1} + (a_j^{\text{rel}})^{-1} (\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}$. The rank-one determinant identity therefore gives the j th decrement $\log(1 + (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \widehat{G}_{j+1}^{-1} (\alpha_j^{\text{rel}})^*)$. Since $\widehat{G}_{2q} \preceq \widehat{G}_{j+1} \preceq$

$\widehat{G}_{q-1}$, inverse order bounds its argument between $1 + l_j^{\text{rel}}$ and $1 + u_j^{\text{rel}}$. Inverse order follows by congruence with the smaller positive form, inversion of its positive eigenvalues, and reversal of that congruence. Summing the decrements from $q - 1$ to $2q - 2$ and from q to $2q - 1$ gives precisely the displayed weights. The full quartet in (ACM17) has $q \geq 4$, so the listed endpoints and interior have these literal ranges. Finally (ACM20) says $\mathcal{B}_k = \widehat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$. Subtracting (ACM27) yields the last line, with its ordered negative $g_{s,\text{gr}}^-$ term intact. This is a finite relation-row bound on the same arithmetic scalar; it does not replace the exact value of $\widehat{\mathcal{B}}_k$ when that value is retained.

All retained relation correlations in the same inverse. The complete calculation PGRC.1–18 now sharpens the row enclosure (ACM31) by evaluating its full cross terms. Keep $H = \widehat{G}_{2q}$, $I = \{q - 1, \dots, 2q - 1\}$ and put $C_{ij} = \alpha_i^{\text{rel}} H^{-1} (\alpha_j^{\text{rel}})^*$. For an ordered subset $J \subset I$ let A_J have the corresponding original rows, let $D_J = \text{diag}(a_i^{\text{rel}} : i \in J)$, and define

$$\begin{aligned} H_J &= H + A_J^* D_J^{-1} A_J, & B_J &= D_J + C_{JJ}, \\ V_J : (E, H_J) &\longrightarrow (E, H) \oplus (\mathbb{C}^J, D_J^{-1}), & V_J x &= (x, A_J x), \\ V_J^\dagger(y, z) &= H_J^{-1} (Hy + A_J^* D_J^{-1} z), \\ V_J^\dagger V_J &= I_E, & \ker V_J &= 0, & \ker V_J^\dagger &= \{(y, z) : Hy + A_J^* D_J^{-1} z = 0\}. \end{aligned} \tag{ACM32}$$

The norm expansion proves the Gram H_J and the adjoint; multiplication proves the inverse on the image and both kernel descriptions. Empty subsets mean the zero-dimensional coordinate space and its unique empty inverses. For every subset, direct multiplication also proves $H_J^{-1} = H^{-1} - H^{-1} A_J^* B_J^{-1} A_J H^{-1}$: the extra middle factor is $D_J^{-1} - B_J^{-1} - D_J^{-1} C_{JJ} B_J^{-1} = 0$ because $I + D_J^{-1} C_{JJ} = D_J^{-1} B_J$.

Let $J_j = \{j + 1, \dots, 2q - 1\}$. For $J \subset J_j$ put $r_j(J) = C_{jJ} B_{J_j}^{-1} C_{Jj}$ and $v_j(J) = (C_{jj} - r_j(J)) / a_j^{\text{rel}}$. The inverse identity proves $v_j(J) = \alpha_j^{\text{rel}} H_J^{-1} (\alpha_j^{\text{rel}})^* / a_j^{\text{rel}} \geq 0$. Telescoping (ACM29) identifies $H_{J_j} = \widehat{G}_{j+1}$. Thus the exact decrement parameter is

$$\eta_j = \frac{C_{jj} - C_{jJ_j} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j j}}{a_j^{\text{rel}}}. \tag{ACM33}$$

For disjoint $J, K \subset J_j$, keep the remaining block and row

$$\begin{aligned} S_{K|J} &= D_K + C_{KK} - C_{KJ} B_J^{-1} C_{JK} = D_K + A_K H_J^{-1} A_K^* \succeq D_K > 0, \\ e_{j,K|J} &= C_{jK} - C_{jJ} B_J^{-1} C_{JK} = \alpha_j^{\text{rel}} H_J^{-1} A_K^*, \\ r_j(J \cup K) - r_j(J) &= e_{j,K|J} S_{K|J}^{-1} e_{j,K|J}^*, \\ 0 \leq r_j(J \cup K) - r_j(J) &\leq \sum_{i \in K} |(e_{j,K|J})_i|^2 / a_i^{\text{rel}}. \end{aligned} \tag{ACM34}$$

Substitution of the proved inverse gives the first two lines. For interleaved disjoint subsets, first use the exact coordinate map $P_{J,K} : \mathbb{C}^{J \cup K} \rightarrow \mathbb{C}^J \oplus \mathbb{C}^K$, $z \mapsto (z|_J, z|_K)$, with inverse reassembling the coordinates in their inherited order. This is an isometry for the retained diagonal lengths: its two squared norms sum to the original sum over $J \cup K$. On coefficient matrices it is a permutation, so $P_{J,K} (D_{J \cup K} + C_{J \cup K, J \cup K}) P_{J,K}^{-1}$ is exactly the displayed two-block matrix. Conjugating the inverse and applying the same restriction to the row leaves the scalar $r_j(J \cup K)$ unchanged. To prove the third line, solve the resulting two block equations $B_J x + C_{JK} y = C_{Jj}$ and $C_{KJ} x + (D_K + C_{KK}) y = C_{Kj}$. The first gives $x = B_J^{-1} (C_{Jj} - C_{JK} y)$ and the second then gives $y = S_{K|J}^{-1} e_{j,K|J}^*$. Pairing with (C_{jJ}, C_{jK}) proves exactly the displayed quadratic correction. Inverse order from $S_{K|J} \succeq D_K$ gives the final inequality. In particular all cross subtractions are retained even when they vanish.

Choose actual finite subsets $J(j) \subset J_j$, let $K(j) = J_j \setminus J(j)$, and set

$$\begin{aligned} \bar{\eta}_j &= v_j(J(j)), \quad \underline{\eta}_j = \max \left\{ l_j^{\text{rel}}, v_j(J(j)) - \sum_{i \in K(j)} \frac{|(e_{j,K(j)|J(j)})_i|^2}{a_j^{\text{rel}} a_i^{\text{rel}}} \right\}, \\ L_k^{\text{corr}}(J) &= \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \underline{\eta}_j), \quad U_k^{\text{corr}}(J) = \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \bar{\eta}_j), \\ L_k^{\text{rel}} &\leq L_k^{\text{corr}}(J) \leq \hat{\mathcal{B}}_k \leq U_k^{\text{corr}}(J) \leq U_{k,2q}^{\text{rel}}. \end{aligned} \tag{ACM35}$$

The last line of (ACM34) with $J \cup K = J_j$ proves $0 \leq \underline{\eta}_j \leq \eta_j \leq \bar{\eta}_j$. Moreover $\underline{\eta}_j \geq l_j^{\text{rel}}$ and $\bar{\eta}_j \leq C_{jj}/a_j^{\text{rel}} = u_j^{\text{rel}}$. Increasing logarithms and the unchanged endpoint weights prove all the inequalities in (ACM35). If every $J(j) = J_j$, the remaining block is empty and both new bounds equal the exact graph volume. Insertion of one further index b subtracts exactly

$$\frac{|C_{jb} - C_{jJ} B_J^{-1} C_{Jb}|^2}{a_j^{\text{rel}} (a_b^{\text{rel}} + C_{bb} - C_{bJ} B_J^{-1} C_{Jb})}$$

from $v_j(J)$, by (ACM34). Its denominator is positive. Thus successive upper bounds decrease and become exact after at most q later-row insertions for each row; retaining the maximum of all previous valid lower bounds makes that retained sequence increasing as well.

In particular the exact $a_j^{\text{rel}} \eta_j$ is the nonnegative $\alpha_j^{\text{rel}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^*$ minus the full squared norm in (ACM30) minus $C_{jJ} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j j}$. The latter vanishes exactly when $C_{J_j j} = 0$, because its inverse matrix is positive. This proves that both negative corrections occur in the same original inverse quantity. Their signs and equality cases are retained. The matrix C in the correlation subtraction is computed using $\hat{G}_{2q}^{-1}$, with its full mixed term; it is never replaced there by K_{low}^{-1} .

At every independent coefficient face $(W, S; A)$ the coefficient inclusions and zero entries act componentwise on these same linear source maps. Retain the original source F_h with $\mathcal{M}F_h = g/h$, $D_i = -x_i \partial_{x_i}$, $h(D)F_h = \Theta \phi_*$, and $\mathcal{V}P = P(D_1 + \cdots + D_k) F_h^{\otimes k}$ from (ISM1). The full Taylor unit in that observation is unchanged. The original minimum-section difference is a relation χQ ; successive monic division constructs $\chi(\sum s_i) Q(\sum s_i) = \sum_i h(s_i) Q_i(\mathbf{s})$. Its original primitive is $\sum_i (-1)^{i-1} Q_i(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)})$. The tensor differential contributes the same sign and therefore recovers the relation. The differential operators commute with the original tensor differential, and $h(D_i)F_h = \Theta \phi_*$, proving that its image is exactly $\mathcal{V}(\chi Q)$. For the original two-leg complex, use the degree-zero map $j_+(v) = (v, 0)$ and degree-one identity into $[V \oplus \hat{V} \rightarrow \mathcal{B}]$, whose differential is $\Theta(\phi - \mathcal{F}^{-1} \hat{\psi})$. The tensor cochain map preserves every sign and gives the same top boundary. The one-factor two-leg differential has the entire diagonal kernel $(v, \mathcal{F}v)$: its zero equation is equivalent to $\phi = \mathcal{F}^{-1} \hat{\psi}$ by injectivity of Θ , and every such diagonal pair has zero differential. When the two compared top representatives lie in an admitted proper subcomplex $\mathbf{C}_{W,S;A}^\bullet$ of the fixed full coefficient carrier, their residual is the class in $(d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_{W,S;A}^k) / d\mathbf{C}_{W,S;A}^{k-1}$, with its complete two-leg primitive retained. The inclusion-induced map sends a numerator representative to its class in $\mathbf{C}_{W,S;A}^k / d\mathbf{C}_{W,S;A}^{k-1}$. Its image is exactly the kernel of the map to the full top quotient: mapping to zero there means being a full boundary. Its inverse uses that same representative, and both are well-defined because their denominator is the identical $d\mathbf{C}_{W,S;A}^{k-1}$. This proves the stated kernel isomorphism, as in the complete coefficient and source maps (ISM38)–(ISM43). Consequently none of the metric estimates deletes this residual, the full diagonal degree-zero source, or an empty mask's outer label.

307 Actual bounds for the retained four-window costs

Continue with (HT1)–(HT18), now with $k \geq 3$. All unadorned source norms and quotient volumes in this section are those of the original $m_{h,k}(u) du$, with $g = 2\xi$ and the complete hypothetical quartet. Write $\mathcal{L} = \mathcal{L}_{h,k}$ and retain $q = \ell_k(k+1)^2$. The threshold-four lower bound is inherited from *Consecutive identities and the first window*, (CJ.9)–(CJ.19); it is already explicitly present in the recovered transcript at lines 4202–4228. It is not a new claim of the present continuation.

307.1 Every term of the inherited lower bound

Put $d_j = V_j/V_{j-1}$ for $j \geq q$. For the two original windows $I_- = (q-1, 2q-1)$ and $I_+ = (q, 2q)$, retain the contraction penalties

$$\begin{aligned} P_- &= -\log(1 - d_{2q-1}) - 2 \sum_{j=q}^{2q-2} \log(1 - d_j), \\ P_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - d_j). \end{aligned} \tag{HC1}$$

The source identities prove $0 < d_j < 1$ in these ranges. The real phase ϕ_N is defined by the retained source pairing $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. The first window uses its original boundary identity; no inverse at degree $q-2$ occurs. Define weights

$$w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q-2), \quad \sum_{N=q-1}^{2q-1} w_N = 2q.$$

Adding the two exact source products gives

$$\mathcal{B}_k := \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + P_- + P_+. \tag{HC2}$$

Here every $\varepsilon_N \geq \mathcal{L}$, including the first degree.

For clarity, retain all constants in the inherited comparable-window bound. With $\alpha = \pi/2$, $B = 42 + 8m$, and $\vartheta_h = \int_{-1}^1 w_h$, put

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ X &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{X}/b, \\ C_h^{\text{win}} &= 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2, \quad D_h = \frac{\delta}{2C_h^{\text{win}}}. \end{aligned} \tag{HC3}$$

Here $0 < b < \pi/2$, and $c_h > 0$ is the original proved convolution constant. The source proves $(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{win}} n$ for $n \geq k \geq 3$, $n/2 \leq r \leq 2n$. Thus it applies to both literal windows, and $0 < D_h < 1/108$.

The exact residual in (HC2) has the following five nonnegative pieces:

$$\begin{aligned} \mathcal{R}_k &= P_- + P_+ + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) \\ &\quad + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} \\ &\quad + 4q \log(C_h^{\text{win}} q) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + 4q \log \frac{2\mathcal{L}}{\delta k q}. \end{aligned} \tag{HC4}$$

Both sums use exactly $q - 1 \leq N \leq 2q - 1$. Nonnegativity follows, respectively, from $0 < d_j < 1$, the squared phases, (HT9), the two comparable-window estimates, and $\mathcal{L}/q \geq \delta k/2$. Expanding each logarithm in (HC4), with $D_h = \delta/(2C_h^{\text{win}})$, proves the exact identity

$$\boxed{\mathcal{B}_k = 4q \log(D_h k) + \mathcal{R}_k, \quad \mathcal{R}_k \geq 0.} \quad (\text{HC5})$$

This retains, rather than discards, all the contraction, phase, norm, and trace savings in the original lower bound.

307.2 A proved upper bound for these same four original volumes

The original convolution estimate and Laplace moments are

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bu} w_h(u) du.$$

Write $\mu_h = \int_{\mathbb{R}} w_h(u) du$ for the original source mass. Put

$$\begin{aligned} R_0 &= \sqrt{\delta^2 + \gamma^2}, \quad D = \max(1, 2/b, 2R_0), \quad T = Dq, \quad r = kR_0/T \leq 1/2, \\ a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B}, \\ M_k &= M_h(b)^k + M_h(-b)^k, \quad U_k(T) = \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{HC6})$$

The capital D in the interval scale is distinct from the inherited lower constant D_h and from the source differential. All masses, coordinates, and root multiplicities remain as in (HT1)–(HT7). One has the actual finite Hermitian comparisons

$$G_{q-1} \preceq C_{2q-1} G_{2q-1}, \quad G_q \preceq G_{q-1} \preceq C_{2q} G_{2q}.$$

Consequently, with the exact ratio

$$\rho_q = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1} - 1}{4^{2q} - 1} \right)^2,$$

$$\boxed{0 \leq \mathcal{B}_k \leq q \log C_{2q-1} + q \log C_{2q} = 2q \log C_{2q-1} + q \log \rho_q.} \quad (\text{HC7})$$

Proof of the degree extension and its maps. Use the exact substitution $f(x) = P(c + iTx)$ with inverse $P(S) = f((S-c)/(iT))$. Its source measure is $Tm_{h,k}(Tx) dx$, with mass μ_h^k , and its monic relation is $\chi_T(x) = (iT)^{-q} \chi(c + iTx)$. The roots $z_1, \dots, z_q$, listed with their repetitions, have modulus at most r . The raw jet of order d acquires exactly $(iT)^d$; the quotient determinant acquires $T^{-q(q-1)}$. This common factor cancels in each displayed quotient-volume ratio.

The Legendre expansion on $[-1, 1]$ gives $\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}$ for $\deg f \leq N$. Indeed Rodrigues' polynomial has coefficient norm at most $\binom{2j}{j} \leq 4^j$ and squared integral $2/(2j+1)$. Pairing the expansion with this polynomial bounds its coefficient by $\sqrt{(2j+1)/2} \|f\|_2$; summing the geometric series gives exactly A_N in (HC6).

Write $\prod_a (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j$. Geometric-series multiplication gives $|h_j| \leq \binom{q+j-1}{j} r^j$ and $\|\chi_T\|_{\text{coeff},1} \leq (1+r)^q$. For every integer $s \geq 0$, monic division of x^{q+s} has quotient $\sum_{j=0}^s h_j x^{s-j}$. The inverse-series identity cancels all other terms of degree at least q . Removing the leading coefficient one therefore gives

$$\|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} \leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq B_q(r).$$

The bound is valid also for $s = q$, the extra degree required here. At degrees below q the remainder is the original monomial, whose coefficient norm is one. Linearity gives the same remainder-map bound for every degree at most $2q$.

For a remainder v of degree below q ,

$$\int |v(u/T)|^2 m_{h,k}(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2.$$

This follows by bounding each monomial by $\max(1, |u/T|^{q-1})$ and using $\int u^{2q-2} m_{h,k} \leq (2q-2)! b^{-(2q-2)} M_k$. Restriction of the original norm to $[-T, T]$ gives the lower bound $T a_k(T) \|f\|_{L^2([-1,1])}^2$. Combining these three estimates proves

$$\|\text{rem}_\chi P\|_{H_{q-1}}^2 \leq C_N \|P\|_{H_N}^2 \quad (N = 2q - 1, 2q).$$

The map is the literal remainder representative of the original jet. Apply it to the canonical least lift of $x \in E$ at degree N . Its norm at degree $q-1$ is at least $x^* G_{q-1} x$; this proves the Hermitian comparisons preceding (HC7). Inclusion of polynomial degrees gives $G_q \preceq G_{q-1}$ and monotonicity of all volumes. Taking determinants proves (HC7). Applying the same remainder inequality to any nonzero polynomial below degree q proves $C_N \geq 1$ without an additional assumption. $\square$

For an entirely scalar upper bound retain the original constants

$$K = \max(1, \vartheta_h^{-1}), \quad C_{\text{env}} = \max(1, 2/(3c_h D)), \quad a = \alpha D + \log 256,$$

and the original expression (AU.31)

$$\mathcal{U}_k = a q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}.$$

Then the proved four-window upper bound is

$$\boxed{\mathcal{B}_k \leq \widehat{\mathcal{U}}_k := 2\mathcal{U}_k + q \log 20.} \quad (\text{HC8})$$

To verify the constants, $bT \geq 2q$ implies the factorial ratio in (HC6) is at most one; also $\mu_h \leq X$ and $M_k \leq 2X^k$, so $U_k(T) \leq 3X^k$. Directly from (HC6),

$$A_{2q-1}^2 \leq (2q/9) 256^q, \quad A_{2q}^2 \leq (40q/9) 256^q.$$

The logarithmic derivative $2/(1-r^2) \leq 8/3$ on $[0, 1/2]$ gives $\log B_q(r)^2 \leq 16kR_0/(3D)$. Substitution into C_N yields $q \log C_{2q-1} \leq \mathcal{U}_k$ and $q \log C_{2q} \leq \mathcal{U}_k + q \log 20$. Adding proves (HC8). This argument does not use the false inequality $\rho_2 < 20$; in the actual quartet domain $q \geq 16$ anyway.

307.3 The exact compatible interval for the signed arithmetic correction

Let V_N^Γ denote the inherited Gamma-reference quotient volume for the *same* original polynomial χ , and define

$$T_N = V_N/V_N^\Gamma, \quad \mathcal{B}_k^\Gamma = \log \frac{V_{q-1}^\Gamma V_q^\Gamma}{V_{2q-1}^\Gamma V_{2q}^\Gamma}, \quad \Delta_k^\Gamma = \log \frac{T_{q-1} T_q}{T_{2q-1} T_{2q}}.$$

Cancellation of the four literal volume factors gives $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$. No auxiliary polynomial is introduced and no sign is imposed on Δ_k^Γ . From (HC5) and (HC8) the actual retained residuals must belong to the following explicit intervals. Let $\mathfrak{C}_*$ be (AT18a) on the same arithmetic and tensor Gamma moment matrices. Its proof uses the original full packet and applies to exactly these four determinants, so $|\Delta_k^\Gamma| \leq J_* \leq \mathfrak{C}_*$. Write $T_k = 4q \log(D_h k)$ for this displayed scalar only:

$$\boxed{\max\{0, -T_k, \mathfrak{b}_*^- - T_k\} \leq \mathcal{R}_k \leq \min\{\widehat{\mathcal{U}}_k, \mathfrak{b}_*^+\} - T_k,} \quad (\text{HC9})$$

$$\boxed{\max\{\mathfrak{d}_*^-, \max\{0, 4q \log(D_h k)\} - \mathcal{B}_k^\Gamma\} \leq \Delta_k^\Gamma \leq \min\{\mathfrak{d}_*^+, \widehat{\mathcal{U}}_k - \mathcal{B}_k^\Gamma\}}. \quad (\text{HC10})$$

The new entries follow by substituting $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$ and $\mathcal{R}_k = \mathcal{B}_k - T_k$ into (AT18)–(AT19). Here $\mathfrak{b}_*^\pm, \mathfrak{d}_*^\pm$ are the simultaneous original Gamma endpoints (AT21e); its complete actual moment proof supplies both the additive and nonlinear coordinate. Their intersection width has not been evaluated for an actual offcritical packet. Removing the simultaneous (AT21e) endpoint entries $\mathfrak{b}_*^\pm$ and $\mathfrak{d}_*^\pm$ gives the preceding coarse intervals, whose width is strictly positive for every $k \geq 3$. Indeed all nonquadratic terms of $\mathcal{U}_k$ are nonnegative, $a > 1$, $q \geq (k+1)^2 > 2k$, and $D_h < 1$. Thus

$$\widehat{\mathcal{U}}_k \geq 2aq^2 > 4qk \geq \max\{0, 4q \log(D_h k)\}.$$

Moreover, keeping the fixed packet constants, each lower-order term divided by q^2 tends to zero, so

$$\lim_{k \rightarrow \infty} \frac{\widehat{\mathcal{U}}_k - \max\{0, 4q \log(D_h k)\}}{q^2} = 2(\alpha D + \log 256) > 0. \quad (\text{HC11})$$

This proves numerical compatibility of those coarse bound functions. It does not prove compatibility of the refined (HC9)–(HC10), which retain the source-dependent (AT21e) endpoints computed from J_* and I_* . It does not construct a packet realizing an arbitrarily chosen point of either interval, and it does not assert the existence of the hypothetical zero. In particular the interval may lie entirely at negative values of Δ_k^Γ ; determinant-ratio positivity does not imply positivity of its signed four-window logarithm.

The constructed signed refinement on this identical source. The exact coordinate map to (ACM1)–(ACM5) is $y = u$, $x = t$, with $S = k/2 + iu$ and every ascending coefficient unchanged. The reference density (ACM2) is exactly the tensor Gamma density of (AT4), including its mass $(\sqrt{2\pi})^k$; the arithmetic density is the original $m_{h,k} = w_h^{*k}$. The relation map in the common source is $I_N B_N^{\text{AW}}$ in both presentations. Consequently the source Grams, their quotient minima, and all four signed determinants agree. In particular $F(0) = \mathcal{B}_k^\Gamma$ and $F(1) = \mathcal{B}_k$ for the F of (ACM14)–(ACM16).

Here is the exact dictionary for all pointwise controls:

$$(S_{\text{AW}}, T_q, S_{\text{SP}}, S_q, A_q) = (S_{\text{spec}}, S_{\text{tr}}, S_{\text{ov}}, S_{\text{HS}}, S_{\text{ang}}), \quad \mathcal{H}_{2q-1} = H_{2q-1}.$$

The original crossed-pair proof (AT21b) gives $S_{\text{tr}} \leq S_{\text{sin}} \leq S_{\text{ang}}$, and the original overlap proof gives $S_{\text{tr}} \leq S_{\text{ov}}$. Therefore the full pointwise minima in (ACM11a) and (AT21a)–(AT21c) are identical, as are those minima divided by $(2q-1)\sqrt{1 - e^{-F(x)/(2q-1)}}$. Thus $J_q = J_*$ and $I_q = I_*$ on this actual path. These equalities retain both complete angle lists, including their designated and additional zero entries. They compare the calculated expressions without removing any source vector or changing either endpoint form.

For every finite-series degree $p \geq 0$, let $j_{p,\Gamma}, e_{p,\Gamma}$ be the exact signed center and residual constructed in (ACM15) on this Gamma-to-arithmetic path. Their actual finite geometric identity and centered Hilbert–Schmidt proof give $j_{p,\Gamma} - e_{p,\Gamma} \leq \Delta_k^\Gamma \leq j_{p,\Gamma} + e_{p,\Gamma}$. The explicit residual in (ACM15) tends to zero for this fixed source pair, with every eigenvalue and the factor $8q-6$ retained. Substituting the preceding dictionary into (ACM16) gives exactly

$$\begin{aligned} \beta_{p,\Gamma}^- &= \max\{\mathfrak{b}_*^-, \mathcal{B}_k^\Gamma + j_{p,\Gamma} - e_{p,\Gamma}\}, \\ \beta_{p,\Gamma}^+ &= \min\{\mathfrak{b}_*^+, \mathcal{B}_k^\Gamma + j_{p,\Gamma} + e_{p,\Gamma}\}, \\ \mathfrak{b}_*^- &\leq \beta_{p,\Gamma}^- \leq \mathcal{B}_k \leq \beta_{p,\Gamma}^+ \leq \mathfrak{b}_*^+. \end{aligned} \quad (\text{HC9a})$$

Every lower entry is at most the same actual $\mathcal{B}_k$ and every upper entry is at least it, proving the displayed intersection and its order. Their signed entries also prove $0 \leq \beta_{p,\Gamma}^+ - \beta_{p,\Gamma}^- \leq 2e_{p,\Gamma}$. Indeed the upper endpoint is at most $\mathcal{B}_k^\Gamma + j_{p,\Gamma} + e_{p,\Gamma}$ and the lower endpoint is at least $\mathcal{B}_k^\Gamma + j_{p,\Gamma} - e_{p,\Gamma}$. Thus

both endpoints converge to the actual $\mathcal{B}_k$ on this fixed source as $p \rightarrow \infty$; no nesting in p is required. Inserting (HC5) and subtracting the same original Gamma value yields the stronger intervals

$$\begin{aligned} \max\{0, -T_k, \beta_{p,\Gamma}^- - T_k\} &\leq \mathcal{R}_k \leq \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\} - T_k, \\ \max\{0, T_k, \beta_{p,\Gamma}^- - \mathcal{B}_k^\Gamma\} &\leq \Delta_k^\Gamma \\ &\leq \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\} - \mathcal{B}_k^\Gamma. \end{aligned} \quad (\text{HC10a})$$

The first line uses $\mathcal{R}_k = \mathcal{B}_k - T_k \geq 0$, and the last two use $\Delta_k^\Gamma = \mathcal{B}_k - \mathcal{B}_k^\Gamma$. The nesting in (HC9a), together with $\mathfrak{d}_*^\pm = \mathfrak{b}_*^\pm - \mathcal{B}_k^\Gamma$, proves that these intervals lie inside (HC9)–(HC10). Those original bounds and their full proofs are consequently retained corollaries. Removing all source-dependent endpoint entries gives the original coarse intervals in (HC11); its positive width is not an evaluation of the refined width. All signed centers here are the specified exact integrals. Any numerical evaluation must retain its integration error; fixed-source convergence in p supplies no uniform estimate in k .

The graph calculation carried back into these arithmetic intervals. On the additional full-quartet domain $\gamma > 2$ of (ACM17)–(ACM27), retain $k \geq 3$ as in this section. The coefficients, polynomial χ , original $v_h = 2\xi/h$ and quotient maps are identified there literally with (HT1)–(HT3). The graph computation therefore controls these same arithmetic quantities by $\mathcal{B}_k = \widehat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$. For any specified integers $p, s \geq 0$, define

$$\begin{aligned} b_{p,s}^{\text{gr},-} &= \max\{\beta_{p,\Gamma}^-, \widehat{\mathcal{B}}_k - g_{s,\text{gr}}^+\}, \\ b_{p,s}^{\text{gr},+} &= \min\{\beta_{p,\Gamma}^+, \widehat{\mathcal{B}}_k - g_{s,\text{gr}}^-\}. \end{aligned} \quad (\text{HC9g})$$

The $\beta_{p,\Gamma}^\pm$ retain the original Gamma-to-arithmetic path; $g_{s,\text{gr}}^\pm$ are the separately computed arithmetic-to-graph endpoints (ACM27). Subtracting the two bounds on Γ_k^{gr} from this same $\widehat{\mathcal{B}}_k$ gives

$$\beta_{p,\Gamma}^- \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq \beta_{p,\Gamma}^+.$$

Inserting this exact intersection at the use sites (HC9)–(HC10) and keeping their original $T_k = 4q \log(D_h k)$ proves

$$\begin{aligned} \max\{0, -T_k, b_{p,s}^{\text{gr},-} - T_k\} &\leq \mathcal{R}_k \leq \min\{\widehat{\mathcal{U}}_k, b_{p,s}^{\text{gr},+}\} - T_k, \\ \max\{0, T_k, b_{p,s}^{\text{gr},-} - \mathcal{B}_k^\Gamma\} &\leq \Delta_k^\Gamma \\ &\leq \min\{\widehat{\mathcal{U}}_k, b_{p,s}^{\text{gr},+}\} - \mathcal{B}_k^\Gamma. \end{aligned} \quad (\text{HC10g})$$

Indeed $\mathcal{R}_k = \mathcal{B}_k - T_k \geq 0$ by (HC5), $\mathcal{B}_k \geq 0$ by (HC7), and $\Delta_k^\Gamma = \mathcal{B}_k - \mathcal{B}_k^\Gamma$ by its definition. Subtracting these unchanged constants gives every entry and proves nesting into the preceding intervals. The additional interval width obeys

$$0 \leq b_{p,s}^{\text{gr},+} - b_{p,s}^{\text{gr},-} \leq \min\{2e_{p,\Gamma}, 2e_{s,\text{gr}}\}. \quad (\text{HC10h})$$

For the first bound, (ACM16) places the two Gamma endpoints inside the interval centered at $\mathcal{B}_k^\Gamma + j_{p,\Gamma}$ with radius $e_{p,\Gamma}$. For the second, (ACM27) places the graph-corrected endpoints inside the interval centered at $\widehat{\mathcal{B}}_k - j_{s,\text{gr}}$ with radius $e_{s,\text{gr}}$. Taking an intersection cannot increase either width. These are actual finite-source enclosures. Their fixed-source residual limits do not estimate the separate limit in the growing tensor order. All bounds preceding this paragraph retain their original domain $\gamma > 0$; (HC9g)–(HC10h) assert the additional calculated refinement on the explicitly stated graph-source domain.

The original relation-row enclosure at these same use sites. Retain every equation (HC9g)–(HC10h), and keep its domain $k \geq 3$, $\gamma > 2$, $0 < \delta < 1/2$, $m \geq 1$. Equations (ACM28)–(ACM31) compute the additional lower and upper values $L_k^{\text{rel}}, U_{k,2q}^{\text{rel}}$ from the full monic relation rows and the original finite arithmetic graph Grams. Here d_j^{rel} is a relation polynomial as defined there; the original $d_j = V_j/V_{j-1}$ of (HC1) remains its stated scalar. Likewise the form $T_N^{\text{rel}} := f_N^* f_N$ below retains its superscript and does not rename either $T_N = V_N/V_N^\Gamma$ or the scalar $T_k = 4q \log(D_h k)$. With $g_s^\pm = g_{s,\text{gr}}^\pm$ exactly as in (ACM27), put

$$\begin{aligned} b_{p,s}^{\text{rel},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{rel}} - g_{s,\text{gr}}^+\}, \\ b_{p,s}^{\text{rel},+} &= \min\{\beta_{p,\Gamma}^+, U_{k,2q}^{\text{rel}} - g_{s,\text{gr}}^-\}, \\ \beta_{p,\Gamma}^- &\leq b_{p,s}^{\text{rel},-} \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq b_{p,s}^{\text{rel},+} \leq \beta_{p,\Gamma}^+. \end{aligned} \quad (\text{HC9r})$$

The coefficient identity (ACM17)–(ACM20) identifies the graph and arithmetic remainder frames by the literal identity, including the full primary nilpotents. Consequently (ACM31) bounds the same $\mathcal{B}_k$ as (HC9a). Taking their interval intersection proves the displayed relation enclosure. For its comparison with (HC9g), $L_k^{\text{rel}} \leq \hat{\mathcal{B}}_k$ gives $\max\{\beta^-, L_k^{\text{rel}} - g^+\} \leq \max\{\beta^-, \hat{\mathcal{B}}_k - g^+\}$, and $\hat{\mathcal{B}}_k \leq U_{k,2q}^{\text{rel}}$ gives the corresponding order for the two upper minima. These two elementary monotonicities prove every remaining order in (HC9r). The exact graph-volume interval is contained in the relation-row interval; its intersection with the latter is itself. Thus (HC9g) retains its full exact input, while (HC9r) gives the separately calculated finite row enclosure for that input.

At the two receiving residuals define

$$\begin{aligned} L_{k;p,s}^{\text{ar,rel}} &= \max\{0, T_k, b_{p,s}^{\text{rel},-}\}, \\ U_{k;p,s}^{\text{ar,rel}} &= \min\{\hat{U}_k, b_{p,s}^{\text{rel},+}\}, \\ L_{k;p,s}^{\text{ar,rel}} - T_k &\leq \mathcal{R}_k \leq U_{k;p,s}^{\text{ar,rel}} - T_k, \\ L_{k;p,s}^{\text{ar,rel}} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \leq U_{k;p,s}^{\text{ar,rel}} - \mathcal{B}_k^\Gamma. \end{aligned} \quad (\text{HC10r})$$

To prove this complete substitution, (HC5) gives $\mathcal{B}_k \geq T_k$ and $\mathcal{R}_k = \mathcal{B}_k - T_k \geq 0$, (HC7)–(HC8) give $0 \leq \mathcal{B}_k \leq \hat{U}_k$, and (HC9r) gives its third lower and second upper entries. Their intersection contains this same scalar. The two maps are the translations $x \mapsto x - T_k$ and $x \mapsto x - \mathcal{B}_k^\Gamma$, with inverse maps adding the displayed unchanged constant. They are order-preserving bijections of the real line, so they preserve interval widths and give the two complete receiving intervals. The first lower endpoint is exactly $\max\{0, -T_k, b_{p,s}^{\text{rel},-} - T_k\}$. No sign is assigned to $g_{s,\text{gr}}^-$, to T_k , or to Δ_k^Γ before these explicit operations. Monotonicity of the maximum and minimum also proves that (HC10g) is contained in (HC10r), which is contained in (HC10a).

The relation-row width has the precise bound

$$\begin{aligned} 0 \leq b_{p,s}^{\text{rel},+} - b_{p,s}^{\text{rel},-} &\leq \min\{2e_{p,\Gamma}, U_{k,2q}^{\text{rel}} - L_k^{\text{rel}} + g_{s,\text{gr}}^+ - g_{s,\text{gr}}^-\} \\ &\leq \min\{2e_{p,\Gamma}, U_{k,2q}^{\text{rel}} - L_k^{\text{rel}} + 2e_{s,\text{gr}}\}. \end{aligned} \quad (\text{HC10s})$$

The first lower sign follows because the interval contains $\mathcal{B}_k$. Its Gamma width is at most $2e_{p,\Gamma}$ by (HC9a). Its other interval has lower endpoint $L_k^{\text{rel}} - g^+$ and upper endpoint $U_{k,2q}^{\text{rel}} - g^-$, so direct subtraction gives the second entry. Equation (ACM27) bounds $g^+ - g^-$ by $2e_{s,\text{gr}}$. Taking an intersection cannot increase either of these two widths, which proves the stated order. The complete negative mixed squared norm in (ACM30) reduces the row upper comparison relative to its K_{low}^{-1} term. It does not alter the exact-graph nesting just proved. All fixed-source limits in p and s retain their stated fixed h, k domain.

The full correlation calculation in the retained signed intervals. Keep the exact row subsets $J(j)$ of (ACM35), on the same receiving domain as (HC9r). Their calculated bounds now give

$$\begin{aligned} b_{p,s}^{\text{corr},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{corr}}(J) - g_{s,\text{gr}}^+\}, \\ b_{p,s}^{\text{corr},+} &= \min\{\beta_{p,\Gamma}^+, U_k^{\text{corr}}(J) - g_{s,\text{gr}}^-\}, \\ b_{p,s}^{\text{rel},-} &\leq b_{p,s}^{\text{corr},-} \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq b_{p,s}^{\text{corr},+} \leq b_{p,s}^{\text{rel},+}. \end{aligned} \quad (\text{HC9rc})$$

The exact identity $\mathcal{B}_k = \widehat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$ and the proved bounds (ACM27), (ACM35) show that the interval contains $\mathcal{B}_k$. Comparing the arguments of the lower maximum and upper minimum, using each inequality in (ACM35), proves the complete order in the last line. All later rows included gives $L_k^{\text{corr}} = U_k^{\text{corr}} = \widehat{\mathcal{B}}_k$, so both endpoints become exactly (HC9g). For any specified subsets their complete width obeys

$$\begin{aligned} 0 \leq b_{p,s}^{\text{corr},+} - b_{p,s}^{\text{corr},-} &\leq \min\{2e_{p,\Gamma}, U_k^{\text{corr}}(J) - L_k^{\text{corr}}(J) + g_{s,\text{gr}}^+ - g_{s,\text{gr}}^-\} \\ &\leq \min\{2e_{p,\Gamma}, U_k^{\text{corr}}(J) - L_k^{\text{corr}}(J) + 2e_{s,\text{gr}}\}. \end{aligned} \quad (\text{HC10rcw})$$

The Gamma interval lies in its radius- $e_{p,\Gamma}$ signed interval, and the graph correction lies in its radius- $e_{s,\text{gr}}$ signed interval by their actual finite remainder proofs. Taking an intersection cannot increase either width. Subtracting the displayed graph lower endpoint from its upper endpoint gives the middle expression, proving every term of this bound.

The complete residual substitution at (HC9)–(HC10) is

$$\begin{aligned} L_{k;p,s}^{\text{ar,corr}} &= \max\{0, T_k, b_{p,s}^{\text{corr},-}\}, \\ U_{k;p,s}^{\text{ar,corr}} &= \min\{\widehat{\mathcal{U}}_k, b_{p,s}^{\text{corr},+}\}, \\ L_{k;p,s}^{\text{ar,corr}} - T_k &\leq \mathcal{R}_k \leq U_{k;p,s}^{\text{ar,corr}} - T_k, \\ L_{k;p,s}^{\text{ar,corr}} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \leq U_{k;p,s}^{\text{ar,corr}} - \mathcal{B}_k^\Gamma. \end{aligned} \quad (\text{HC10rc})$$

Indeed intersect (HC9rc) with the original inequalities $\mathcal{B}_k \geq 0$, $\mathcal{B}_k \geq T_k$ and $\mathcal{B}_k \leq \widehat{\mathcal{U}}_k$ proved in (HC5), (HC7), (HC8). The order-preserving translations already proved in (HC10r) give the two displayed receiving intervals with their exact inverse additions. Their widths do not exceed (HC10rcw). The inclusion of intervals is now exact-graph inside correlation inside relation-row inside Gamma; monotonicity of the maximum, minimum and the two translations proves the same inclusion at both residuals. All original lower-bound pieces and all negative signed corrections remain present.

307.4 The holonomy energy is tied to the same endpoint contractions

Fix an actual holonomy θ and period L . In this paragraph all metrics, lifts and volumes carry that same phase. Put

$$\begin{aligned} e_N &= \log(V_N/V_{N+1}), \quad \beta_N = \lambda_{\max}(G_{N+1}^{-1/2} G_N G_{N+1}^{-1/2}), \\ J_N^D &= \|D_{L,\theta} R_N G_N^{-1/2}\|_{\text{HS}}^2, \quad \mathcal{E}_N = \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}}^2. \end{aligned}$$

These use original source moments through degree $2N + 2$. They obey

$$\boxed{\frac{\mathcal{L}^2}{2} \leq \mathcal{E}_N \leq (1 + \sqrt{\beta_N})^2 J_N^D, \quad e_N \geq 2 \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}.} \quad (\text{HC12})$$

The denominator is positive: integration by parts on the stated twisted domain gives $\|D_{L,\theta} f\|^2 = c^2 \|f\|^2 + \|\partial_r f\|^2$, whence $J_N^D \geq c^2 q > 0$.

To prove the upper inequality, $D_{L,\theta}R_N v$ is an original degree- $(N+1)$ representative of Av . The same quotient minimum therefore gives

$$A^*G_{N+1}A \preceq R_N^*D_{L,\theta}^*D_{L,\theta}R_N.$$

Use $G_N \preceq \beta_N G_{N+1}$, sandwich by $G_N^{-1/2}$, and take traces. This gives $\|R_N A G_N^{-1/2}\|_{\text{HS}}^2 \leq \beta_N J_N^D$. The triangle inequality in $\mathcal{B}_N = D_{L,\theta}R_N - R_N A$ proves the upper inequality in (HC12); (HT14) proves the lower. All eigenvalues in the definition of β_N are at least one by degree monotonicity. In fact the original rank-one kernel update proves the exact identity

$$e_N = \log \beta_N, \quad \beta_N = 1 + \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}}. \quad (\text{HC12a})$$

Here b_{N+1} and ω_{N+1} are the remainder column and monic squared norm for this same original phase measure. Indeed $K_{N+1} = K_N + b_{N+1}b_{N+1}^*/\omega_{N+1}$, so

$$K_N^{-1/2}K_{N+1}K_N^{-1/2} = I + vv^*, \quad v = K_N^{-1/2}b_{N+1}/\sqrt{\omega_{N+1}}.$$

The relative matrix defining β_N is $K_{N+1}^{1/2}K_N^{-1}K_{N+1}^{1/2} = X^*X$ for the invertible $X = K_N^{-1/2}K_{N+1}^{1/2}$. The displayed matrix $I + vv^*$ is XX^* , and $X(X^*X)X^{-1} = XX^*$ proves they have identical eigenvalues. These are 1 with multiplicity at least $q-1$ and $1 + \|v\|^2$. Taking the determinant gives $V_N/V_{N+1} = \det K_{N+1}/\det K_N = 1 + \|v\|^2 = \beta_N$, which proves (HC12a), including a zero update if one occurs. Rearranging the two energy bounds now proves the second inequality of (HC12) with this exact endpoint identity.

The same weights as in (HC2) reconstruct the actual phase volume:

$$\mathcal{B}_{k,\theta} = \sum_{N=q-1}^{2q-1} w_N e_N \geq 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}. \quad (\text{HC13})$$

Thus a small original derivative-source energy at a degree has the explicit endpoint contraction cost (HC12); that cost enters the same four-window budget, with its actual multiplicity.

Finally the supplied holonomy construction allows any fixed $0 < \eta < 1$ by choosing the original period from its computed weighted source comparison at degree $2q+2$. It proves simultaneously $(1-\eta)G_N \preceq G_{N,\theta} \preceq (1+\eta)G_N$ at all these degrees. For completeness, apply the comparison before quotient minimization on the common coefficient space $H = \mathcal{P}_{2q}$. The supplied weighted source estimate, written explicitly in (PSC.33), is $(1-\eta)M_{2q} \preceq M_{2q,\theta} \preceq (1+\eta)M_{2q}$. Set $H_\theta(t) = (1-t)M_{2q} + tM_{2q,\theta}$, $C^\theta = H_\theta(t)^{-1}(M_{2q,\theta} - M_{2q})$, and define P_N^θ, Q_N^θ by (AT11) with $M_* = H_\theta(t)$ and the unchanged inclusions I_N, B_N . Then $U^\theta = Q_{2q-1}^\theta + Q_{2q}^\theta - Q_q^\theta$ and $W^\theta = P_{2q-1}^\theta - P_{q-1}^\theta + P_{2q}^\theta - P_q^\theta$. Substitution proves all projector identities and the exact signed trace formula (AT15) on this distinct interpolation. Both positive forms have the same original remainder fibres. Thus (AT18a)–(AT21) defines its own control $\mathfrak{C}_\theta$ and proves

$$\begin{aligned} |\mathcal{B}_{k,\theta} - \mathcal{B}_k| &\leq J_\theta \leq \mathfrak{C}_\theta := \int_0^1 \min\{S_{\text{spec}}^\theta(t), \tfrac{1}{2}d_\theta(t)\|U^\theta - W^\theta\|_{1,H_\theta(t)}\} dt \\ &\leq (2q-1) \log \frac{1+\eta}{1-\eta}. \end{aligned} \quad (\text{HC14})$$

Indeed every eigenvalue of $M_{2q}^{-1}M_{2q,\theta}$ lies in $[1-\eta, 1+\eta]$ by the source sandwich, and the full spectra and trace norms in the finite AT proof are on the displayed polynomial flags. This is an explicit source comparison; taking determinants only after quotient minimization gives the preceding $2q$ allowance as a coarser consequence. The interpolated forms are not identified with either Gamma reference. On these same phase-source operators define S_{HS}^θ by the exact centered trace formula (AT21a), and

$J_\theta = \int_0^1 \min\{S_{\text{spec}}^\theta, S_{\text{tr}}^\theta, S_{\text{HS}}^\theta\} dt$. The same finite Hermitian Cauchy–Schwarz proof gives the first entry in (HC14); hence $J_\theta \leq \mathfrak{C}_\theta$. The positive atomic phase measure also meets the moment hypothesis proved above. Define I_θ by (AT21a) on the exact path $H_\theta(t)$; its original $B_\theta(t)$ is strictly positive by the same nonzero moment $\int |\chi|^2 d\mu_t$. Put

$$\begin{aligned} \mathfrak{U}_{\text{ar}} &= \min\{\widehat{\mathcal{U}}_k, \mathfrak{b}_*^+\}, \\ \mathfrak{U}_\theta &= \min\{\mathfrak{U}_{\text{ar}} + J_\theta, \Psi_a(H_a(\mathfrak{U}_{\text{ar}}) + I_\theta)\}, \quad a = 2q - 1. \end{aligned} \quad (\text{HC14a})$$

Then (HC13)’s right side is at most $\mathfrak{U}_\theta$. Indeed $0 < \mathcal{B}_k \leq \mathfrak{U}_{\text{ar}}$ by the original upper estimates. Apply (AT21e) on $H_\theta(t)$, then use monotonicity of $\Psi_a \circ (H_a + I_\theta)$ in its positive initial value. This proves both upper entries and their minimum. This substitution uses (AT19) on the original arithmetic form before the explicit source-phase trace calculation; it does not identify the sampled phase metric with the original one.

The original source-phase map and its signed refinement. We now refine the same bound (HC14a), retaining its original $J_\theta, I_\theta, \mathfrak{C}_\theta$ and both upper entries. Let $I_{2q}^{2q+2} : \mathcal{P}_{2q} \hookrightarrow \mathcal{P}_{2q+2}$ be literal coefficient inclusion. The two original forms on $H = \mathcal{P}_{2q}$ are

$$\begin{aligned} M_0^\theta &= (I_{2q}^{2q+2})^* M_{2q+2} I_{2q}^{2q+2} = \int_{\mathbb{R}} v(u)^* v(u) m_{h,k}(u) du, \\ M_1^\theta &= (I_{2q}^{2q+2})^* M_{2q+2,\theta} I_{2q}^{2q+2} = \frac{2\pi}{L} \sum_{r \in \mathbb{Z}} m_{h,k}(u_r) v(u_r)^* v(u_r), \\ v(u) &= (1, c + iu, \dots, (c + iu)^{2q}), \quad c = k/2, \quad u_r = (2\pi r + \theta)/L. \end{aligned} \quad (\text{HC14b})$$

By (HT3)–(HT7), these are exactly the squared norms of $\mathcal{U}_k \mathcal{V}|_H$ and $\mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}|_H$. Here $\mathcal{U}_k$ is the half-density map (HT4), whose source and target include every relative variable; the earlier scalar upper bound with this notation is still the expression in (HC8). The original relative variables, full Taylor unit, coordinate, phase, and factor $2\pi/L$ are present in these two source maps. To specify the period cost without colliding with $a = 2q - 1$, write $a_{\text{wt}} > 0$ for the weighted-source exponent of (PSC.32). Its actual degree- $2q + 2$ cost is

$$\kappa_{a_{\text{wt}}, 2q+2} = \sup_{v \neq 0} \frac{v^*(M_{a_{\text{wt}}, +, 2q+2} + M_{a_{\text{wt}}, -, 2q+2})v}{v^* M_{2q+2} v}.$$

The weighted Grams retain the squared weights $x_k^{\pm 2a_{\text{wt}}}$. Retain the originally fixed period L whenever $\kappa_{a_{\text{wt}}, 2q+2}/(e^{a_{\text{wt}}L} - 1) \leq \eta$. One exact admissible choice is $L = a_{\text{wt}}^{-1} \log(1 + \kappa_{a_{\text{wt}}, 2q+2}/\eta)$; the comparison below holds for every retained period satisfying the displayed inequality. Equation (PSC.33), restricted by I_{2q}^{2q+2} , proves $(1 - \eta)M_0^\theta \preceq M_1^\theta \preceq (1 + \eta)M_0^\theta$. For $q - 1 \leq N \leq 2q$ the remainder J_N is surjective; taking the two infima over its identical nonempty fibre $J_N P = v$ also proves the quotient sandwiches used above.

Put $M_x^\theta = (1 - x)M_0^\theta + xM_1^\theta$ for $0 \leq x \leq 1$. At $x = t$ this is precisely $H_\theta(t)$ in (HC14). Use $I_N : \mathcal{P}_N \rightarrow H$ and the unpadded original relation $B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N, P \mapsto \chi P$. Its included map is $B_N^\theta = I_N B_N^{\text{AW}}$, and the exact projections and operators are

$$\begin{aligned} P_N^\theta &= I_N (I_N^* M_x^\theta I_N)^{-1} I_N^* M_x^\theta, \\ Q_N^\theta &= B_N^\theta ((B_N^\theta)^* M_x^\theta B_N^\theta)^{-1} (B_N^\theta)^* M_x^\theta, \\ U_x^\theta &= Q_{2q-1}^\theta + Q_{2q}^\theta - Q_q^\theta, \\ W_x^\theta &= P_{2q-1}^\theta - P_{q-1}^\theta + P_{2q}^\theta - P_q^\theta, \\ C_x^\theta &= (M_x^\theta)^{-1} (M_1^\theta - M_0^\theta), \quad D_\theta = (M_0^\theta)^{-1} M_1^\theta. \end{aligned} \quad (\text{HC14c})$$

At $N = q - 1$ the relation domain is zero and the second formula is the unique zero map $Q_{q-1}^\theta = 0$. These are exactly (ACM4)–(ACM5) after the stated substitution, and exactly the projections already

used in (HC14). The source and relation ranges are the same nested polynomial flags, so both U_x^θ, W_x^θ have full spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. Differentiating the actual canonical section as in (ISM18) and taking the four determinant signs gives

$$F'_\theta(x) = \text{Tr}(C_x^\theta(U_x^\theta - W_x^\theta)), \quad F_\theta(0) = \mathcal{B}_k, \quad F_\theta(1) = \mathcal{B}_{k,\theta}.$$

The polynomial χ has positive norm for the continuous measure and for the infinite positive lattice measure (HT6). Their convex mixture has the same property. The original $\chi^{\#_k}$ proof (AT21a)–(AT21c) therefore gives $F_\theta(x) > 0$ on this closed path. In particular every nonlinear denominator is bounded away from zero for this fixed source comparison.

Apply the pointwise dictionary preceding (HC9a) on this path. The additional trace inequalities $S_{\text{tr}}^\theta \leq S_{\text{ov}}^\theta$ and $S_{\text{tr}}^\theta \leq S_{\text{sin}}^\theta \leq S_{\text{ang}}^\theta$ prove that the complete (ACM11a) minima are precisely the J_θ, I_θ already defined above. In particular both complete angle lists, the overlap majorant and the centered Hilbert–Schmidt expression are retained in these identities. Let $j_{p,\theta}, e_{p,\theta}$ be the (ACM15) construction with

$$(M_x, C_x, U_x, W_x, D, F(x)) \mapsto (M_x^\theta, C_x^\theta, U_x^\theta, W_x^\theta, D_\theta, F_\theta(x)).$$

The integer $p \geq 0$ is its finite-series degree, while the original period is still L . If $b_1^\theta \leq \dots \leq b_n^\theta$ are every eigenvalue of D_θ in M_0^θ , $n = 2q + 1$, write

$$\vartheta_\theta = \frac{b_n^\theta - b_1^\theta}{b_n^\theta + b_1^\theta}, \quad K_{D_\theta} = \sum_{j=1}^n \left(\frac{|b_j^\theta - 1|}{\min\{1, b_j^\theta\}} \right)^2.$$

The finite identity $C_x^\theta - (C_x^\theta)^{[p]} = (H_x^{\circ,\theta})^{p+1} C_x^\theta$ is the identical (ACM15) geometric sum on these specified operators. Subtracting the trace mean of this residual leaves its pairing with $U_x^\theta - W_x^\theta$ unchanged. All its factors are real functions of D_θ , hence self-adjoint in M_x^θ . The Hilbert–Schmidt inequality therefore gives

$$\begin{aligned} j_{p,\theta} - e_{p,\theta} &\leq \mathcal{B}_{k,\theta} - \mathcal{B}_k \leq j_{p,\theta} + e_{p,\theta}, \\ 0 \leq e_{p,\theta} &\leq \vartheta_\theta^{p+1} \sqrt{K_{D_\theta}(8q-6)} \xrightarrow{p \rightarrow \infty} 0. \end{aligned} \tag{HC14d}$$

For completeness the first factor bounds every residual eigenvalue because $\|H_x^{\circ,\theta}\|_{M_x^\theta} \leq \vartheta_\theta < 1$ and $|c_j^\theta(x)| \leq |b_j^\theta - 1| / \min\{1, b_j^\theta\}$. Centering subtracts the nonnegative square of the trace mean. Both positive operators dominate their range projections, whose ranges intersect in dimension at least one. Thus $\text{Tr}(U_x^\theta W_x^\theta) \geq 1$, and the full spectra give $\text{Tr}((U_x^\theta - W_x^\theta)^2) = 8q - 4 - 2 \text{Tr}(U_x^\theta W_x^\theta) \leq 8q - 6$. This proves the displayed constant on the original metric.

The source sandwich places every b_j^θ in $[1 - \eta, 1 + \eta]$. It consequently gives $\vartheta_\theta \leq \eta$ and $K_{D_\theta} \leq (2q+1)\eta^2/(1-\eta)^2$: the first follows by increasing the larger endpoint and decreasing the smaller one in their positive ratio, and each summand of the second has numerator at most η^2 and denominator at least $(1 - \eta)^2$. Hence the explicit further bound is

$$e_{p,\theta} \leq \frac{\eta^{p+2}}{1 - \eta} \sqrt{(2q+1)(8q-6)}.$$

This holds for every phase at each retained period satisfying the displayed original degree- $2q + 2$ source-cost inequality, retaining the complete $q = [1 + k(m-1)](k+1)^2$ dependence. Retaining their entire ordered list, the same signed derivative and the exact derivative of H_a give

$$\begin{aligned} |\mathcal{B}_{k,\theta} - \mathcal{B}_k| &\leq J_\theta \leq \min\{\mathfrak{C}_\theta, \mathcal{L}_q(D_\theta)\} \leq (2q-1) \log \frac{1+\eta}{1-\eta}, \\ |H_{2q-1}(\mathcal{B}_{k,\theta}) - H_{2q-1}(\mathcal{B}_k)| &\leq I_\theta \leq \log \frac{1+\eta}{1-\eta}, \\ \mathcal{L}_q(D_\theta) &= \log \frac{b_{q+2}^\theta}{b_q^\theta} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}^\theta}{b_j^\theta}. \end{aligned} \tag{HC14e}$$

Indeed integration of each ordered $c_j^\theta(x) = (b_j^\theta - 1)/(1 - x + xb_j^\theta)$ gives $\log b_j^\theta$. The total coefficient of the resulting logarithmic ratios is $1 + 2(q-1) = 2q-1$. Dividing the original pointwise minimum by $(2q-1)\sqrt{1 - e^{-F_\theta(x)/(2q-1)}}$ and integrating proves the second line. These are the same additive and nonlinear bounds as (HC14)–(HC14a), now accompanied by the constructed signed center (HC14d).

For any integers $p, r \geq 0$, define

$$\begin{aligned} U_{k,p}^{\text{ar}} &= \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\}, \\ U_{k,\theta;p,r} &= \min\{U_{k,p}^{\text{ar}} + J_\theta, \Psi_{2q-1}(H_{2q-1}(U_{k,p}^{\text{ar}}) + I_\theta), \\ &\quad U_{k,p}^{\text{ar}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \quad (\text{HC14f})$$

The actual $\mathcal{B}_k$ is positive and at most $U_{k,p}^{\text{ar}}$ by (HC9a)–(HC10a). The additive and signed expressions increase with this initial scalar. For the fixed computed I_θ , both H_{2q-1} and Ψ_{2q-1} are increasing on the displayed domains. Applying the three proved phase upper bounds and substituting this initial upper value proves

$$2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} \leq \mathcal{B}_{k,\theta} \leq U_{k,\theta;p,r} \leq \mathfrak{U}_\theta. \quad (\text{HC13a})$$

For the final inequality (HC9a) gives $U_{k,p}^{\text{ar}} \leq \mathfrak{U}_{\text{ar}}$, and monotonicity of the first two upper expressions bounds their minimum by the original (HC14a). The third entry can only lower it. This proof holds the actual phase-path integrals fixed during the scalar comparison; it does not replace their source path. The weights w_N , derivative-source energies J_N^D , and every boundary term in (HC12)–(HC13) remain in their original phase metric. No fixed-source residual limit is promoted to a uniform estimate in the growing tensor order.

The source comparison also retains its exact cochain map. For two minimum sections ρ_0, ρ_1 on this common source, $(\rho_1 - \rho_0)v = \chi Q_v$ by monic division. Successive division in $s_1, \dots, s_k$ gives the original polynomials $Q_{v,i}$ with $\chi(\sum s_i)Q_v(\sum s_i) = \sum_i h(s_i)Q_{v,i}(\mathbf{s})$. With the original $h(D)F_h = \Theta\phi_*$, the actual primitive is

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes the same $(-1)^{i-1}$, so $dK_v = \mathcal{V}((\rho_1 - \rho_0)v)$, as in (ISM38)–(ISM41). For the two-leg complex with differential $(\phi, \widehat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\widehat{\psi})$, the degree-zero map $j_+(\phi) = (\phi, 0)$ and degree-one identity form a cochain map because their differential is exactly $\Theta\phi$. Tensor these specified maps and use the original coefficient insertions into the fixed full carrier of (ISM42). This defines $\widetilde{K}_v$ and preserves its displayed top differential. At a declared support $\lambda = (W, S; A)$, apply these maps coordinatewise. With the original coefficient injections ι_i , the exact top maps and their common inverse-image domain are

$$\begin{aligned} F_{a,A} : E^{\oplus A} &\longrightarrow \mathcal{C}_{\text{full}}^k, & F_{a,A}((v_i)_{i \in A}) &= \sum_{i \in A} \iota_i \mathcal{V}_{\rho_a} v_i \quad (a = 0, 1), \\ E_\lambda &= F_{0,A}^{-1}(\mathcal{C}_\lambda^k) \cap F_{1,A}^{-1}(\mathcal{C}_\lambda^k). \end{aligned}$$

This is a linear subspace. For $\mathbf{v} = (v_i)_{i \in A} \in E_\lambda$, put $\widetilde{K}_\mathbf{v} = \sum_{i \in A} \iota_i j_+^{\otimes k} K_{v_i}$, where the tensor cochain map has the degree-one identities already specified. Its top differential is $F_{1,A}(\mathbf{v}) - F_{0,A}(\mathbf{v})$, which lies in $\mathcal{C}_\lambda^k$ and is a boundary in the full carrier. The exact residual map on this specified domain is

$$E_\lambda \longrightarrow \frac{d\mathcal{C}_{\text{full}}^{k-1} \cap \mathcal{C}_\lambda^k}{d\mathcal{C}_\lambda^{k-1}}, \quad \mathbf{v} \longmapsto [d\widetilde{K}_\mathbf{v}]_\lambda.$$

Its inclusion-induced map to the kernel of $\mathbb{C}_\lambda^k/d\mathbb{C}_\lambda^{k-1} \rightarrow \mathbb{C}_{\text{full}}^k/d\mathbb{C}_{\text{full}}^{k-1}$ and its inverse both use the same representative, proving the exact isomorphism (ISM42). The full two-leg degree-zero diagonal and the source images are those of (ISM41)–(ISM43). For independent coefficient masks the linear maps act in each active coordinate; zero insertion commutes because every new coordinate maps 0 to 0 with its original outer label. This calculation retains the proper-source residual and the empty mask's outer label throughout.

The graph refinement in this original phase-energy budget. Retain the domain $\gamma > 2$ of (HC9g), the actual original period and phase above, and every previously calculated $J_\theta, I_\theta, j_{r,\theta}, e_{r,\theta}$. For $p, s, r \geq 0$ put

$$\begin{aligned} U_{k;p,s}^{\text{ar,gr}} &= \min\{\widehat{U}_k, b_{p,s}^{\text{gr},+}\}, \\ U_{k,\theta;p,s,r}^{\text{gr}} &= \min\{U_{k;p,s}^{\text{ar,gr}} + J_\theta, \\ &\quad \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k;p,s}^{\text{ar,gr}}) + I_\theta), \\ &\quad U_{k;p,s}^{\text{ar,gr}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \quad (\text{HC14g})$$

The original arithmetic moment proof at (ACM11a) gives strict positivity of $\mathcal{B}_k$; (HC10g) and (HC9g) then give $0 < \mathcal{B}_k \leq U_{k;p,s}^{\text{ar,gr}} \leq U_{k,p}^{\text{ar}}$. For each of the three fixed phase controls the upper expression is increasing in its initial scalar: the first and third are affine of slope one, and the second is the composite of the two increasing functions already differentiated in (ACM14). Applying their actual phase inequalities and this scalar comparison proves the propagated bound at (HC13a):

$$\begin{aligned} 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} &\leq \mathcal{B}_{k,\theta} \\ &\leq U_{k,\theta;p,s,r}^{\text{gr}} \leq U_{k,\theta;p,r}. \end{aligned} \quad (\text{HC13g})$$

The original w_N, J_N^D , phase source, twisted derivative domain and boundary residual in (HC12)–(HC13) are unchanged. The graph data enter through the proved signed bound for the initial arithmetic volume. The inequality does not identify the two-component graph source with the phase source.

The relation-row interval propagated into the phase budget. Keep the original phase, circumference, source-path matrices and the exact integrals $J_\theta, I_\theta, j_{r,\theta}, e_{r,\theta}$ from (HC14b)–(HC14g). Use $U_{k;p,s}^{\text{ar,rel}}$ defined in (HC10r), and define the complete three-entry upper value

$$\begin{aligned} U_{k,\theta;p,s,r}^{\text{rel}} &= \min\{U_{k;p,s}^{\text{ar,rel}} + J_\theta, \\ &\quad \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k;p,s}^{\text{ar,rel}}) + I_\theta), \\ &\quad U_{k;p,s}^{\text{ar,rel}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \quad (\text{HC14r})$$

By (HC9r)–(HC10r) and the strict original-moment positivity,

$$0 < \mathcal{B}_k \leq U_{k;p,s}^{\text{ar,gr}} \leq U_{k;p,s}^{\text{ar,rel}} \leq U_{k,p}^{\text{ar}}.$$

For this fixed phase path the first and third functions of an initial scalar x in (HC14r) are affine of slope one. The second is the composite of increasing $\mathcal{H}_{2q-1}$ on $x > 0$ and increasing Ψ_{2q-1} on its nonnegative argument; their derivatives and inverse domains were computed in (ACM14). The actual phase inequalities (HC14d)–(HC14e) use $x = \mathcal{B}_k$. Substituting the larger scalar upper endpoints and applying each monotonicity proves

$$\begin{aligned} 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} &\leq \mathcal{B}_{k,\theta} \leq U_{k,\theta;p,s,r}^{\text{gr}} \\ &\leq U_{k,\theta;p,s,r}^{\text{rel}} \leq U_{k,\theta;p,r}. \end{aligned} \quad (\text{HC13r})$$

The left inequality is the unchanged original phase-energy identity (HC13), with the complete multiplicities, twisted derivative domain and boundary terms. The comparison holds the actual path controls fixed; the scalar substitution does not change their integration path or measure. It also retains the entire cochain map and proper-source residual following (HC13a): the only new maps here are the two real translations in (HC10r) and these three functions of their proved arithmetic upper endpoint. The exact-graph phase bound (HC13g) remains the smaller one at the same p, s, r , as the displayed inequalities prove.

The correlation interval in the three original phase functions. With the same fixed original phase controls as (HC14r), set

$$\begin{aligned} U_{k;\theta;p,s,r}^{\text{corr}} = \min\{ & U_{k;p,s}^{\text{ar,corr}} + J_\theta, \\ & \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k;p,s}^{\text{ar,corr}}) + I_\theta), \\ & U_{k;p,s}^{\text{ar,corr}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \quad (\text{HC14rc})$$

Equation (HC9rc) and the arithmetic minimum in (HC10rc) give $0 < \mathcal{B}_k \leq U_{k;p,s}^{\text{ar,gr}} \leq U_{k;p,s}^{\text{ar,corr}} \leq U_{k;p,s}^{\text{ar,rel}}$. The three increasing functions proved in (HC14r) therefore give

$$\begin{aligned} 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} & \leq \mathcal{B}_{k,\theta} \leq U_{k;\theta;p,s,r}^{\text{gr}} \\ & \leq U_{k;\theta;p,s,r}^{\text{corr}} \leq U_{k;\theta;p,s,r}^{\text{rel}} \leq U_{k;\theta;p,r}. \end{aligned} \quad (\text{HC13rc})$$

This proves the full propagation into the original phase-energy budget by applying each actual phase inequality before taking their minimum. The source-path integrals, circumference condition, twisted derivative domain, complete weights, boundary terms and proper-source cochain residual are those already proved above. Including all later rows makes the correlation upper bound equal to the exact-graph bound (HC14g) at the same p, s, r . No tensor uniform rate is inferred from this finite exactness statement.

For the averaged original phase metrics the already proved stronger comparison $(1 - \eta^2)G_N \preceq \overline{G}_N \preceq G_N$ gives instead $|\overline{\mathcal{B}}_k - \mathcal{B}_k| \leq 2q \log(1/(1 - \eta^2))$. These changes are of order q for a fixed η , whereas the compatible width (HC11) has positive quadratic order. The exact Laplacian residual (HT17) remains present at every phase. No displayed upper estimate for its norm, or for the original derivative moments in (HC12), contradicts these lower bounds.

The graph-to-average cost and its complete mixed remainder. On the same graph-source domain $\gamma > 2$, fix one original circumference for all four endpoints. The inverse-form statements below use the already proved degree- $2q + 2$ original source comparison above. For the graph map L_{obs} of (ACM18), write L_{circ} for this circumference to avoid a collision. At each $N \in (q - 1, q, 2q - 1, 2q)$ use the original arithmetic minimum section C_N and the relation map B_N of PGB.3. Put $H_N = B_N^* M_N B_N$, $T_N^{\text{def}} = D_N B_N$, $H_{+,N} = H_N + (T_N^{\text{def}})^* T_N^{\text{def}}$, and retain

$$Q_{D,N} = I - T_N^{\text{def}} H_{+,N}^{-1} (T_N^{\text{def}})^*, \quad \mathcal{E}_N^{\text{aug}} = C_N^* D_N^* Q_{D,N} D_N C_N, \quad \mathcal{D}_N^{\text{hol}} = G_N - \overline{G}_N. \quad (\text{HC16})$$

For zero relation dimension the maps through that domain are zero and $Q_{D,N} = I$. PGB.3–PGB.10 proves, by the displayed completed squares and the original phase section difference,

$$\begin{aligned} \widehat{G}_N &= G_N + \mathcal{E}_N^{\text{aug}}, \quad \mathcal{E}_N^{\text{aug}} > 0, \\ \mathcal{D}_N^{\text{hol}} &= \int (C_N - C_{N,\theta})^* M_{N,\theta} (C_N - C_{N,\theta}) \frac{d\theta}{2\pi} \succeq 0, \\ \widehat{G}_N - \overline{G}_N &= \mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}}. \end{aligned} \quad (\text{HC17})$$

The exact map whose Gram is the last difference is

$$x \mapsto \left(Q_{D,N}^{1/2} D_N C_N x, [\theta \mapsto (C_N - C_{N,\theta})x] \right) \in \mathcal{H}_k \oplus L^2(\Theta; \text{im } B_N, M_{N,\theta} d\theta / (2\pi)). \quad (\text{HC18})$$

Its squared norm is (HC17), and its kernel is zero by the injectivity of $D_N C_N$ proved from the full leading coefficient in PGB.6. The cochain map inducing the identity contraction $(E, \hat{G}_N) \rightarrow (E, \overline{G}_N)$ sends each polynomial and each relation to its constant phase field. It commutes with the inclusion differential and preserves J_N ; its norm identity is $\int p^* M_{N,\theta} p d\theta / (2\pi) = p^* M_N p \leq p^* \hat{M}_N p$. Thus the finite comparison is accompanied by its original source map and its complete nonzero metric loss.

To carry this calculation into the four-volume use site define the exact individual determinant costs

$$\begin{aligned} a_N^{\text{aug}} &= \log \frac{\det \hat{G}_N}{\det G_N} = \log \det(I + G_N^{-1/2} \mathcal{E}_N^{\text{aug}} G_N^{-1/2}) > 0, \\ h_N^{\text{hol}} &= \log \frac{\det \overline{G}_N}{\det G_N} = \log \det(I + \overline{G}_N^{-1/2} \mathcal{D}_N^{\text{hol}} \overline{G}_N^{-1/2}) \geq 0, \\ a_N^{\text{av}} &= \log \frac{\det \hat{G}_N}{\det \overline{G}_N} = a_N^{\text{aug}} + h_N^{\text{hol}}. \end{aligned} \quad (\text{HC19})$$

These equalities follow by determinant multiplication; no commutation of their positive matrices is required. List the four endpoints as $(N_1, N_2, N_3, N_4) = (q-1, q, 2q-1, 2q)$ with $\epsilon = (1, 1, -1, -1)$, retaining repeated entries if they occur. The exact signed identities are

$$\begin{aligned} \Gamma_k^{\text{gr}} &= \sum_{j=1}^4 \epsilon_j a_{N_j}^{\text{aug}}, \\ H_k^{\text{hol}} &:= \sum_{j=1}^4 \epsilon_j h_{N_j}^{\text{hol}} = \mathcal{B}_k - \overline{\mathcal{B}}_k, \\ \hat{\mathcal{B}}_k - \overline{\mathcal{B}}_k &= \Gamma_k^{\text{gr}} + H_k^{\text{hol}} = \sum_{j=1}^4 \epsilon_j a_{N_j}^{\text{av}}. \end{aligned} \quad (\text{HC20})$$

For the original source sandwich chosen above, $(1 - \eta^2)G_N \preceq \overline{G}_N \preceq G_N$ implies $0 \leq h_N^{\text{hol}} \leq q \log(1/(1 - \eta^2))$. There are two positive and two negative signs, so

$$|H_k^{\text{hol}}| \leq 2q \log \frac{1}{1 - \eta^2}, \quad \left| \hat{\mathcal{B}}_k - \overline{\mathcal{B}}_k - \Gamma_k^{\text{gr}} \right| \leq 2q \log \frac{1}{1 - \eta^2}. \quad (\text{HC21})$$

This proves exactly where the earlier order- q allowance enters. The augmentation's signed term is retained through (ACM23)–(ACM27), rather than assigned that allowance. The separate positivity of the four costs in (HC19) does not give a sign to their complete signed sum.

The graph volume appearing in (HC9g) also has its full mixed-source expression. Retain $\Phi(1/2 + it) = A + iq\Phi(t)$, the density r_k , function $g_x = z(u)x/r_k(u)$, and maps $Y_\perp, \mathcal{J}_k, \hat{R}_N$ exactly as in PGB.14–PGB.22. In particular $\Omega_k = Y_\perp^* Y_\perp > 0$ and $f_N x = \iota \hat{R}_N x - g_x$, with $\iota P(u) = P(k/2 + iu)$. The complete squared-norm identity PGMB.5 is

$$\begin{aligned} \hat{G}_N &= K_{\text{low}} + (\mathcal{D}_k^{\text{mix}})^* \mathcal{D}_k^{\text{mix}} + f_N^* f_N = \Omega_k + f_N^* f_N, \\ (\mathcal{D}_k^{\text{mix}} x)(\mathbf{t}) &= \frac{\overline{\tau}_k F_x(\mathbf{s})}{1 + |\tau_k|^2} - g_x(u). \end{aligned} \quad (\text{HC22})$$

The target norm of the second line is the full weighted space $L^2((1 + |\tau_k|^2) \prod_i w_h(t_i) dt_i)$. Thus it retains $1 + k^2 A^2 + (\sum_i q_\Phi(t_i))^2$, the conjugate phase and every tensor variable. PGMB.2 evaluates its kernel: it is zero for $k \geq 2$, while the map vanishes for $k = 1$. Its first term has the original

realization $L_{\text{low}}x = (\prod_i v_h(s_i))F_x/\sqrt{1+|\tau_k|^2}$, with Gram K_{low} . The map $x \mapsto (L_{\text{low}}x, \mathcal{D}_k^{\text{mix}}x, f_Nx)$ has precisely the Gram $\widehat{G}_N$ in (HC22), proving the source realization of every term.

Define its finite positive tail costs by

$$t_N = \log \det(I + \Omega_k^{-1/2} f_N^* f_N \Omega_k^{-1/2}). \quad (\text{HC23})$$

Equation (HC22) gives $\log \det \widehat{G}_N = \log \det \Omega_k + t_N$. The minimum over the nested original relation spaces proves $\widehat{G}_{N+1} \preceq \widehat{G}_N$, hence $t_{N+1} \leq t_N$; determinant monotonicity follows by congruence with the smaller positive form and its eigenvalues. PGB.22's affine approximation proves $f_N^* f_N \rightarrow 0$ at fixed h, k , so $t_N \downarrow 0$ in that limit. The actual four endpoints therefore give

$$\begin{aligned} \widehat{\mathcal{B}}_k &= (t_{q-1} - t_{2q-1}) + (t_q - t_{2q}), \\ 0 &\leq \widehat{\mathcal{B}}_k \leq t_{q-1} + t_q, \\ \mathcal{B}_k &= t_{q-1} + t_q - t_{2q-1} - t_{2q} - \Gamma_k^{\text{gr}}. \end{aligned} \quad (\text{HC24})$$

Substitution into (HC9g), (HC10g) and then (HC14g) is the explicit original-source tail calculation for their graph entry. In particular it retains all four tails and the signed augmentation correction. Although the common $\log \det \Omega_k$ cancels with $1 + 1 - 1 - 1 = 0$, that full matrix remains in all four congruences (HC23). Its strictly positive mixed contribution at every $k \geq 2$ has not been removed. The fixed- k degree limit of t_N does not supply a rate for these four growing endpoints as k varies.

Finally the completed receiving maps remain explicit. PGB.20–PGB.21 gives $\Xi(f, x) = \mathcal{J}_k f - Y_{\perp}x$ and $j_{\infty}\Xi(f, x) = x$, where $j_{\infty} = -\Omega_k^{-1}Y_{\perp}^*$ and $\ker j_{\infty} = \mathcal{J}_k L^2(r_k du)$. The first-coordinate projection to the original unweighted completed source is $\pi_1\Xi(f, x) = \mathcal{J}_k^0 I_{r,m}(f + g_x)$. It maps the closed graph relation into the original closed unweighted relation, inducing $(E, \Omega_k) \rightarrow 0$ with full kernel E , exactly as proved in PGB.21b. On the graph quotient, PGMB.7–PGMB.8 instead gives $\mathfrak{C}_k\Xi(f, x) = (L_{\text{low}}x, \mathcal{D}_k^{\text{mix}}x)$ with kernel $\mathcal{J}_k L^2(r_k du)$, and, for $k \geq 2$,

$$(\Omega_k - K_{\text{low}})^{-1}(\mathcal{D}_k^{\text{mix}})^* \text{pr}_2 \mathfrak{C}_k = j_{\infty}. \quad (\text{HC25})$$

The equality follows by the exact Gram in (HC22). These are the maps from the same completed source to its unweighted and mixed targets, with their complete kernels. On each fixed coefficient mask they act coordinatewise; represented zero and an empty present tuple keep their original label, and external absence keeps its singleton inverse image. The finite graph-to-average contraction (HC18) and these completed maps use their stated different domains. Every proper-source residual retained above continues to use its original inverse-image domain and boundary quotient; none of these Hilbert quotient calculations declares that residual zero.

307.5 The sharper finite comparison retains its exact error

The original lower-envelope measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du.$$

Let V_N^{ν} be its quotient volume for the identical full remainder map. The actual finite upper expression for the four-window volume is

$$\mathcal{U}_{\nu,k} = \log(V_{q-1}V_q) - \log(V_{2q-1}^{\nu}V_{2q}^{\nu}),$$

with the exact nonnegative error

$$\boxed{\mathcal{U}_{\nu,k} - \mathcal{B}_k = \log \frac{V_{2q-1}}{V_{2q-1}^{\nu}} + \log \frac{V_{2q}}{V_{2q}^{\nu}} \geq 0.} \quad (\text{HC15})$$

Indeed $m_{h,k}(u) du \geq d\nu_k$ bounds every source polynomial quadratic form. Minimizing on each identical affine remainder fibre gives $G_N \succeq G_N^\nu \succ 0$. Conjugation by $(G_N^\nu)^{-1/2}$ shows that each volume ratio in (HC15) is at least one. Cancellation proves the equality. The low-degree factors require only original moments through $2q$, and the comparison moments in the last factors go through $4q$. These are well-defined finite expressions, with the actual full jet map and source mass. The supplied proof records do not evaluate their errors at the hypothetical arithmetic quartet or bound them at a rate opposing (HC5). Retaining this sharper finite expression therefore supplies no extra contradictory inequality.

308 Continuous holonomy cochains and the original relation metric

This continuation proves the continuous-phase version of H29–H31 and the exact maps to the completed targets in H64 of the delivered *All-holonomy descent of the original theta quotient*. The delivered NOTE.tex was read completely, all 1152 lines; its SHA-256 is

444f82a8fa420e6f471ef448442085f3a93acb137069e1505d155faab051b881.

The archive SHA-256 is

f0506a27cd4c9012f43fa70c3d1f8f996bbb6db01b0eea2944b34dcf290ca8f0.

All source coordinates, masses, phases, relation columns, and full primary multiplicities below are retained. No finite test or Lean execution is used as a proof. The sampled-density argument uses the full analytic proof *Periodized source density review*, PSD8–PSD14, whose TeX SHA-256 is

51102189bcc7a45610464143a5c0bee21c08059c317024e6e856bc7de993fe74.

The completed-fibre calculation develops H64 and is also proved in the companion completed-quotient and holonomy-intake derivations. Its repetition here supplies the maps needed for this continuous common-image complex, including the mean-zero contraction and the precise weighted relation gap.

308.1 Original coefficient spaces and actual phase norms

Fix the original monic polynomial χ , its full degree $q \geq 1$, and an integer $N \geq q - 1$. Put

$$\begin{aligned} E &= \mathbb{C}[S]/(\chi), & P_N &= \mathbb{C}[S]_{\leq N}, & J &= J_N : P_N \longrightarrow E, \\ B_N &= \ker J = \chi P_{N-q}, & A &= M_S : E \longrightarrow E, \end{aligned} \quad (\text{HCD1})$$

where $P_{-1} = 0$. Use the original increasing-power coefficient bases. Let $B : \mathbb{C}^r \rightarrow P_N$, $r = N - q + 1$, be multiplication by the literal χ , so that $\text{im } B = B_N$. At $r = 0$ it is the unique map from the zero coefficient space.

Let $\Theta = [0, 2\pi)$ carry its actual Fourier phase measure $d\mu(\theta) = d\theta/(2\pi)$. Retain the measurable Hermitian phase source matrices M_θ and their mean

$$M = \int_{\Theta} M_\theta d\mu, \quad 0 < aI_{P_N} \preceq M_\theta \preceq bI_{P_N} < \infty \quad \text{almost everywhere,} \quad a, b > 0. \quad (\text{HCD2})$$

The original source isometry H13–H17 identifies M with the original source Gram, including its full constant mass. For the actual source, the bounds in (HCD2) are obtained from H39–H41, also proved in HG3, as follows. Let $\Psi_N : P_N \rightarrow L^2(\mathbb{R}; \mathcal{K})$ be the original finite source columns after the half-density map, with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$. Fix $\beta > 0$ for the retained weighted source Grams, whose finite entries are supplied by the original source regularity:

$$M_{\beta, \pm} = (e^{\pm\beta r} \Psi_N)^* (e^{\pm\beta r} \Psi_N), \quad \kappa_{\beta, N} = \max_{v \neq 0} \frac{v^* (M_{\beta, +} + M_{\beta, -}) v}{v^* M v}. \quad (\text{HCD2a})$$

These are the original weights $x_k^{\pm 2\beta}$; all original source measures remain present. The maximum is finite and positive because $M > 0$ and the two weighted Grams are finite positive forms. For $\mathcal{C}(s) = \int \Psi_N(r)^* \Psi_N(r+s) dr$ and $s \geq 0$, write the integrand at v as the inner product of $e^{-\beta r} \Psi_N(r)v$ and $e^{\beta(r+s)} \Psi_N(r+s)v$, multiplied by $e^{-\beta s}$. Cauchy–Schwarz, followed by translation in the second integral, gives

$$|v^* \mathcal{C}(s) v| \leq e^{-\beta s} \sqrt{(v^* M_{\beta, -} v)(v^* M_{\beta, +} v)}.$$

For $-s$ the same bound follows from $\mathcal{C}(-s) = \mathcal{C}(s)^*$. Unfolding the literal Zak sum gives $M_\theta = \sum_{j \in \mathbb{Z}} e^{ij\theta} \mathcal{C}(jL)$. The correlation bound and polarization give absolute convergence of each matrix entry, uniformly in phase. Pairing j and $-j$, using $2\sqrt{uv} \leq u + v$, and summing the geometric series prove

$$-\delta_L M \preceq M_\theta - M \preceq \delta_L M, \quad \delta_L = \frac{\kappa_{\beta,N}}{e^{\beta L} - 1}. \quad (\text{HCD2b})$$

For any chosen $0 < \eta < 1$, the explicit finite circumference

$$L \geq \frac{1}{\beta} \log(1 + \kappa_{\beta,N}/\eta) \implies \delta_L \leq \eta < 1 \quad (\text{HCD2c})$$

therefore supplies (HCD2), with $a = (1 - \delta_L)\lambda_{\min}(M)$ and $b = (1 + \delta_L)\lambda_{\max}(M)$ in the original coefficient frame. The phase mean is still exactly M : every nonzero Fourier mode integrates to zero against $d\theta/(2\pi)$. The concrete choice $\eta = 1/2$ gives $L = \beta^{-1} \log(1 + 2\kappa_{\beta,N})$ and $\delta_L = 1/2$. These are actual finite-source bounds at the stated cutoff; no bound uniform in N or the arithmetic packet is asserted. The argument below also applies to any fixed L for which (HCD2) is already established.

Define the Hilbert space

$$\mathcal{P} = L^2(\Theta; P_N, M_\theta d\mu), \quad \|P\|_{\mathcal{P}}^2 = \int_{\Theta} P(\theta)^* M_\theta P(\theta) d\mu.$$

By (HCD2), its elements are exactly coefficientwise L^2 functions, with equivalent norms. Equality means equality almost everywhere. The coefficientwise Bochner integral $\text{av } P = \int P d\mu$ exists, since coefficientwise L^2 implies L^1 on this finite phase measure. Every fixed coefficient map commutes with this integral, entry by entry.

The closed relation space and common-image space are

$$\mathcal{B} = L^2(\Theta; B_N), \quad \mathcal{P}^{\text{eq}} = \{P \in \mathcal{P} : JP(\theta) = x \text{ a.e. for a fixed } x \in E\}. \quad (\text{HCD3})$$

They carry the restricted source norms. They are closed: J is continuous on coefficientwise L^2 , $B_N = \ker J$ is closed, and constant E -valued functions form a closed subspace. For the last assertion, averaging followed by repetition is a bounded idempotent with precisely that image.

308.2 The continuous cochain splitting and its full contraction

Use complexes in degrees 0, 1 with actual inclusion differentials:

$$\mathcal{C}_N = [B_N \hookrightarrow P_N], \quad \mathcal{C}_N = [\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}]. \quad (\text{HCD4})$$

Repetition $\Delta v(\theta) = v$ and averaging av are cochain maps. In degree one, $J \text{av } P = \int JP d\mu = x$; in degree zero the mean belongs to B_N . The equality $\text{av } \Delta = I$ uses the literal phase measure. Repetition is an isometry:

$$\|\Delta v\|_{\mathcal{P}}^2 = v^* \left(\int M_\theta d\mu \right) v = v^* M v.$$

Averaging is bounded with the explicit estimate

$$\|\text{av } P\|_M^2 \leq b \|\text{av } P\|_{\text{coeff}}^2 \leq b \int \|P(\theta)\|_{\text{coeff}}^2 d\mu \leq (b/a) \|P\|_{\mathcal{P}}^2. \quad (\text{HCD5})$$

This uses scalar Cauchy-Schwarz for the coefficient vector and does not replace the arithmetic mass by one.

Put $\mathcal{B}^0 = \{b \in \mathcal{B} : \text{av } b = 0\}$. The inverse continuous coefficient maps of complexes are

$$\begin{aligned} \mathcal{C}_N &\simeq \mathcal{C}_N \oplus [\mathcal{B}^0 \xrightarrow{I} \mathcal{B}^0], \\ P &\longmapsto (\text{av } P, P - \Delta \text{av } P), \quad (v, b^0) \longmapsto \Delta v + b^0. \end{aligned} \quad (\text{HCD6})$$

In degree one, $J(P - \Delta \text{av } P) = x - x = 0$ and its mean is zero. In degree zero every value already belongs to B_N . Both composites restore the original vector. Inclusion carries these components to inclusion and identity, respectively, proving the isomorphism of complexes.

The degree-minus-one map

$$h : \mathcal{P}^{\text{eq}} \longrightarrow \mathcal{B}, \quad hP = P - \Delta \text{av } P \quad (\text{HCD7})$$

satisfies $dh + hd = I - \Delta \text{av}$. In degree one this is $dhP = P - \Delta \text{av } P$; in degree zero it is $hdb = b - \Delta \text{av } b$. On the complementary complex it is the identity from degree one to degree zero. Boundedness follows from (HCD5) and the isometry of repetition. It is not asserted orthogonal or isometric.

The common quotient map is

$$j : \mathcal{P}^{\text{eq}} \longrightarrow E, \quad P \longmapsto JP(\theta) \text{ a.e.} \quad (\text{HCD8})$$

It is onto by repeating a right inverse of J , and its kernel is exactly $\mathcal{B}$. Inclusion is injective. Therefore $H^0(\mathcal{C}_N) = 0$ and $H^1(\mathcal{C}_N) = E$, with the isomorphism induced by the original full J . Repetition and averaging induce the identity on this E .

Multiplication by S maps P_N to P_{N+1} and B_N to B_{N+1} . Applied pointwise, it sends common image x to common image Ax and commutes with repetition, mean and h . These are maps between the stated cutoffs. On degree-one cohomology the action is the full companion A , including its nilpotent primary parts. No trace of the infinite-dimensional contractible complement is assumed.

308.3 The minimum norm of the common image

Retain the original and phasewise minimum sections

$$\begin{aligned} G &= (JM^{-1}J^*)^{-1}, \quad C = M^{-1}J^*G, \\ G_\theta &= (JM_\theta^{-1}J^*)^{-1}, \quad C_\theta = M_\theta^{-1}J^*G_\theta, \\ \overline{G} &= \int G_\theta d\mu. \end{aligned} \quad (\text{HCD9})$$

Surjectivity and positivity make every inverse defined. Continuity of matrix inversion proves measurability. For any fixed coefficient right inverse C_{ref} of J ,

$$\frac{a}{\|J\|_{\text{coeff}}^2} I_E \preceq G_\theta \preceq b \|C_{\text{ref}}\|_{\text{coeff}}^2 I_E. \quad (\text{HCD10})$$

Indeed any lift p of x has $\|x\| \leq \|J\| \|p\|$, proving the lower bound, and the particular lift $C_{\text{ref}}x$ proves the upper bound. The formula for C_θ then also gives boundedness. All following integrals and sections are consequently defined.

Direct multiplication gives $JC_\theta = I$, and for $b_0 \in B_N$,

$$b_0^* M_\theta C_\theta = b_0^* J^* G_\theta = 0. \quad (\text{HCD11})$$

The section $\mathcal{S}x(\theta) = C_\theta x$ has common image x . Any other P with that image has $P - \mathcal{S}x \in \mathcal{B}$. Phasewise orthogonality gives

$$\|P\|_{\mathcal{P}}^2 = x^* \overline{G} x + \|P - \mathcal{S}x\|_{\mathcal{P}}^2. \quad (\text{HCD12})$$

Thus the quotient metric of the continuous complex is exactly $\overline{G} > 0$ on its full E . For the constant-source complex the identical argument gives G . Repetition is isometric on sources, and its induced identity $(E, G) \rightarrow (E, \overline{G})$ has the additional continuous relation directions available in the target.

308.4 The positive relation gap and its mean-zero decomposition

For $r > 0$, put $X_\theta = (B^*M_\theta B)^{-1}B^*M_\theta C$. Both sections have right-inverse image I_E , so their difference lies in B_N and equals BX_θ ; multiplication by B^*M_θ proves the formula and uniqueness. At $r = 0$, both sections equal the unique inverse of J and X_θ is the zero map into the zero space. Expanding $C = C_\theta + BX_\theta$ and using (HCD11) proves

$$\boxed{G - \overline{G} = \mathcal{D} = \int (C - C_\theta)^* M_\theta (C - C_\theta) d\mu = \int X_\theta^* (B^* M_\theta B) X_\theta d\mu \succeq 0.} \quad (\text{HCD13})$$

The integral of $C^*M_\theta C$ is $C^*MC = G$. Every term is the norm of a field in the original relation image. Multiplication by the literal χ , followed by the inherited finite tensor theta primitive, realizes that field in the original cochain complex.

The relation to the particular mean-zero contraction is also exact. Define

$$\overline{C} = \int C_\theta d\mu, \quad a_C = C - \overline{C} : E \rightarrow B_N, \quad h_\theta = C_\theta - \overline{C} : E \rightarrow B_N. \quad (\text{HCD14})$$

Then $\int h_\theta d\mu = 0$, and h_θ is precisely the field h_θ . The original discrepancy is $C - C_\theta = a_C - h_\theta$ and need not have mean zero. Since a_C has relation-valued columns, phase and common-source orthogonality give

$$\begin{aligned} \int a_C^* M_\theta h_\theta d\mu &= -a_C^* M \overline{C} \\ &= -a_C^* M (C - a_C) = a_C^* M a_C. \end{aligned} \quad (\text{HCD15})$$

The adjoint cross term is the same Hermitian matrix. Expanding all terms of (HCD13) therefore proves

$$\boxed{\mathcal{D} = \int h_\theta^* M_\theta h_\theta d\mu - a_C^* M a_C.} \quad (\text{HCD16})$$

This retains the positive relation norm and the complete weighted cross terms. A mean-zero coefficient field is not presumed orthogonal to constant relations for the varying metric. At $N = q - 1$, $B_N = 0$, $C_\theta = C = J^{-1}$, $a_C = h_\theta = 0$ and $\overline{G} = G$.

308.5 Actual sampled Hilbert spaces, including zero weights

Fix the original circumference $L > 0$, tensor degree $k \geq 1$, and $c = k/2$. Retain

$$\begin{aligned} u_n(\theta) &= (2\pi n + \theta)/L, & s_n(\theta) &= c + iu_n(\theta), \\ w_n(\theta) &= \frac{2\pi}{L} m_{h,k}(u_n(\theta)) \geq 0, & n &\in \mathbb{Z}. \end{aligned} \quad (\text{HCD17})$$

The Fourier coefficient is literally $2\pi/L$, with $m_{h,k} = w_h^{*k}$, $w_h = |(g/h)(1/2 + iu)|^2/(2\pi)$ and $g = 2\xi$. Sampling gives

$$\tau_\theta P = (P(s_n(\theta)))_n, \quad \|\tau_\theta P\|_\theta^2 = \sum_n w_n(\theta) |P(s_n(\theta))|^2 = P^* M_\theta P. \quad (\text{HCD18})$$

This is the actual phase source Gram H17.

Define $\mathcal{H}_\theta$ to be weighted square-summable sequences modulo sequences supported where $w_n(\theta) = 0$. Thus zero weights, possible when $k = 1$, have an explicit null subspace. For $k \geq 2$,

positivity of the original convolution makes every sample weight positive. At $k = 1$, retain the exact ideal equality $(\chi_{h,1}) = (h)$. If $a_h \neq 0$ is the original leading coefficient of h , then $\chi_{h,1} = h/a_h$ is its monic generator, and $\chi_{h,1} = h$ in the original monic packet convention. Indeed a polynomial annihilates multiplication by S on $\mathbb{C}[S]/(h)$ if and only if its action on $[1]$ is zero, which is equivalent to belonging to (h) . The analytic source remains the original g/h , with the factor a_h retained in that quotient and its norm. Now take a complete selected root $\rho = 1/2 + iu_n(\theta)$ whose common order in g and h is m . The two Taylor expansions have leading coefficients $g^{(m)}(\rho)/m!$ and $h^{(m)}(\rho)/m!$, so the canceled entire quotient has the exact nonzero value

$$(g/h)(\rho) = \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)}, \quad w_n(\theta) = \frac{1}{L} \left| \frac{g^{(m)}(\rho)}{h^{(m)}(\rho)} \right|^2 > 0. \quad (\text{HCD18a})$$

This retains both cancellation factorials and the full Fourier mass $(2\pi/L)(1/(2\pi))$. It is the selected-root calculation PSD5 and PSD41. Other $k = 1$ coordinates where g/h vanishes still give the explicit null subspace above. For a concrete measurable realization use the isometry $v \mapsto (\sqrt{w_n(\theta)}v_n)_n$ into ordinary $\ell^2(\mathbb{Z})$, with zero coordinates at zero weights. Its inverse on its range is $v_n = y_n/\sqrt{w_n(\theta)}$ at positive weights; other coordinates give the null class. This specifies the norm map without changing a mass. The range projections are measurable diagonal matrices; their images of the countable standard basis give a measurable field.

The original gamma-contour and alternating-zeta calculation PSD8–PSD13 proves, for every $0 < \gamma < \pi/2$, the finite constants

$$A_\gamma = \|e^{\gamma|\cdot|}w_h\|_\infty, \quad B_\gamma = \|e^{\gamma|\cdot|}w_h\|_1.$$

That calculation retains the leading coefficient of h , its full degree, and the continuous canceled values on the compact interval containing its selected roots. With $f_\gamma(v) = e^{\gamma|v|}w_h(v)$, the ordinary triangle inequality on the convolution variables and integration of the remaining factors give, for every real u ,

$$e^{\gamma|u|}m_{h,k}(u) \leq f_\gamma^{*k}(u) \leq A_\gamma B_\gamma^{k-1}. \quad (\text{HCD19a})$$

For the second inequality bound one factor by A_γ and integrate each of the other $k - 1$ factors with its literal Lebesgue measure. For $k = 1$ the expression is A_γ , with $B_\gamma^0 = 1$. For any $0 < 2\epsilon < \gamma$,

$$\sup_{0 \leq \theta < 2\pi} \sum_n w_n(\theta) e^{2\epsilon|u_n(\theta)|} < \infty. \quad (\text{HCD19})$$

Indeed $|u_n(\theta)| \geq (2\pi/L)(|n| - 1)$, so with $d = \gamma - 2\epsilon > 0$ the sum is at most

$$\frac{2\pi}{L} A_\gamma B_\gamma^{k-1} e^{2\pi d/L} \left(1 + \frac{2e^{-2\pi d/L}}{1 - e^{-2\pi d/L}} \right).$$

For every fixed polynomial Q , the constant $\sup_{u \in \mathbb{R}} |Q(c + iu)|^2 e^{-d|u|/2}$ is finite. Inserting it into the same geometric estimate, with $d/2$ in place of d , proves uniform exponential summability of $w_n(\theta)|Q(s_n(\theta))|^2$. This includes $Q = \chi$ and every fixed polynomial relation used below. All original constants in the weights are unchanged. The full source proof of PSD8–PSD14 is a dependency of this continuation; the remaining discrete density argument is given explicitly here with its shifted Fourier phase.

308.6 Density and the complete fibrewise Hilbert quotient

Here is a direct density proof with the zero weights retained. If $f \in \mathcal{H}_\theta$ is orthogonal to all polynomials, put $a_n = w_n(\theta)\overline{f_n}$. Cauchy–Schwarz and (HCD19) give $\sum |a_n|e^{\epsilon|u_n|} < \infty$. Hence $F(z) = \sum_n a_n e^{zu_n}$ is holomorphic on $|\Re z| < \epsilon$, with termwise derivatives on smaller closed strips.

Every derivative at zero is zero, since polynomials in S and in $(S - c)/i$ span the same space. The Taylor series and the identity theorem give $F = 0$ on the connected strip. On the imaginary axis,

$$F(it) = e^{it\theta/L} \sum_n a_n e^{2\pi i n t/L}.$$

The series is absolutely uniformly convergent. Multiplication by $e^{-it\theta/L} e^{-2\pi i n t/L}$ and integration over $0 \leq t \leq L$ give $a_m = 0$. At positive weights this implies $f_m = 0$; at zero weights the class is already null. Thus polynomials are dense. The identical argument proves polynomial density for the weights $w_n |\chi(s_n)|^2$.

Multiplication by the original polynomial is the isometry

$$U_\chi : \ell^2(w_n |\chi(s_n)|^2) \longrightarrow \mathcal{H}_\theta, \quad f \longmapsto (\chi(s_n) f_n)_n. \quad (\text{HCD20})$$

Its image is precisely the sequences vanishing at the positive-weight sampled roots of χ . To prove the converse inclusion, take such a sequence v and set $f_n = v_n / \chi(s_n)$ when $\chi(s_n) \neq 0$, and zero elsewhere. Its squared domain norm equals $\|v\|_\theta^2$. Thus the image is closed. Polynomial density in the domain now proves

$$\mathcal{R}_\theta := \overline{\chi \mathbb{C}[S]}^{\mathcal{H}_\theta} = \{v : v_n = 0 \text{ if } w_n(\theta) > 0, \chi(s_n(\theta)) = 0\}. \quad (\text{HCD21})$$

The receiving Hilbert quotient is exactly

$$\begin{aligned} \mathcal{Q}_\theta &= \mathcal{H}_\theta / \mathcal{R}_\theta \simeq \bigoplus_{\substack{n: \chi(s_n(\theta))=0 \\ w_n(\theta) > 0}} \mathbb{C}, \\ \|v\|_{\mathcal{Q}_\theta}^2 &= \sum_{\text{displayed } n} w_n(\theta) |v_n|^2. \end{aligned} \quad (\text{HCD22})$$

The quotient map is orthogonal coordinate restriction. No weight equal to zero is inverted.

308.7 The complete primary kernel before phase integration

Let Λ_θ^+ be the distinct sampled roots of χ with positive weight, and set $p_\theta(S) = \prod_{\lambda \in \Lambda_\theta^+} (S - \lambda)$, using $p_\theta = 1$ for an empty product. The finite map is

$$\rho_\theta : E \longrightarrow \mathcal{Q}_\theta, \quad [P]_\chi \longmapsto (P(\lambda))_{\lambda \in \Lambda_\theta^+}. \quad (\text{HCD23})$$

It is well defined, and is onto by Lagrange interpolation at these distinct points, using degree less than $|\Lambda_\theta^+| \leq q$. Vanishing at all those points is equivalent to divisibility by p_θ . Since $p_\theta \mid \chi$,

$$\ker \rho_\theta = (p_\theta) / (\chi) \subset E. \quad (\text{HCD24})$$

This specifies the zero-weight-corrected kernel in H64: a root at a zero-mass coordinate contributes no receiving Hilbert line.

Write the full original factorization as $\chi = \prod_{\lambda \in \Lambda} (S - \lambda)^{m_\lambda}$. The complete Taylor map is

$$\begin{aligned} E &\longrightarrow \prod_{\lambda \in \Lambda} \mathbb{C}[\varepsilon_\lambda] / (\varepsilon_\lambda^{m_\lambda}), \\ [P] &\longmapsto \left(\sum_{j=0}^{m_\lambda-1} \frac{P^{(j)}(\lambda)}{j!} \varepsilon_\lambda^j \right)_\lambda. \end{aligned} \quad (\text{HCD25})$$

For an explicit inverse, let $q_\lambda = \chi / (S - \lambda)^{m_\lambda}$ and let u_λ be the Taylor polynomial of $1/q_\lambda$ at λ through degree $m_\lambda - 1$. Then $e_\lambda = [u_\lambda q_\lambda]_\chi$ is one in the λ factor and zero in every other factor, by divisibility

by each original prime power. It follows that $\sum e_\lambda = 1$ and the factors are orthogonal idempotents. The inverse is $\sum_\lambda e_\lambda \sum_j a_{\lambda,j} (S - \lambda)^j$.

In these full primary coordinates, ρ_θ takes the constant coefficient of each active sampled factor. Its kernel in that factor is (ε_λ) , retaining every order $1, \dots, m_\lambda - 1$. For an unsampled factor or a zero-weight sampled factor, the kernel is the entire primary algebra, including its constant term. The original action is multiplication by $\lambda + \varepsilon_\lambda$; the receiving action is the simple scalar λ . Evaluation intertwines these actions with the displayed nilpotent kernel.

308.8 The measurable completed field and its vanishing map

Every sampled root has real part c . For a fixed root λ with $\Re \lambda = c$, its only possible phase is

$$\theta_\lambda = L\Im \lambda \pmod{2\pi}, \quad 0 \leq \theta_\lambda < 2\pi, \quad n_\lambda = (L\Im \lambda - \theta_\lambda)/(2\pi) \in \mathbb{Z}.$$

Thus the set Θ_χ of possible phases is finite. For $\theta \notin \Theta_\chi$, $\mathcal{Q}_\theta = 0$. At an exceptional phase the positive-weight condition in (HCD22) is still retained.

In the measurable realization after (HCD18), the projection onto $\mathcal{Q}_\theta$ has diagonal entries

$$\mathbf{1}_{\{w_n(\theta) > 0\}} \mathbf{1}_{\{\chi(s_n(\theta)) = 0\}}, \quad n \in \mathbb{Z}.$$

These are measurable and give an explicit field of projections. The direct integral satisfies

$$\mathcal{Q} = \int_{\Theta}^{\oplus} \mathcal{Q}_\theta d\mu = 0, \quad (\text{HCD26})$$

because every measurable section has squared norm supported on the finite measure-zero set Θ_χ . This is an almost-everywhere assertion; each exceptional nonzero receiving line is still exactly the weighted line in (HCD22).

For $x \in E$, the section $\theta \mapsto \rho_\theta x$ is measurable: it is given by polynomial evaluations and the displayed coordinate masks. It defines

$$\rho : E \longrightarrow \mathcal{Q}, \quad x \longmapsto [\theta \mapsto \rho_\theta x]_{\text{a.e.}}, \quad \ker \rho = E. \quad (\text{HCD27})$$

Every pointwise exceptional value has zero direct-integral class. This does not assert that those pointwise maps vanish, or that the original nonreduced coefficient algebra was zero.

308.9 The source-to-completion cochain map

The direct-integral source is the weighted space with squared norm $\int_{\Theta} \sum_n w_n(\theta) |v_n(\theta)|^2 d\mu$. Let $\mathcal{H}$ denote this space, and let $\mathcal{R}$ be its closed subspace of sections belonging to $\mathcal{R}_\theta$ almost everywhere. The coordinate projection above has kernel $\mathcal{R}$.

The measurable evaluation map is

$$\mathbb{T} : \mathcal{P} \longrightarrow \mathcal{H}, \quad P \longmapsto (P(\theta, s_n(\theta)))_{\theta, n}.$$

The integrated identity (HCD18) proves it is an isometry. Evaluation of a relation-valued section is in $\mathcal{R}$, because the original section is a multiple of the literal χ . Hence there is a cochain map

$$[\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}] \longrightarrow [\mathcal{R} \hookrightarrow \mathcal{H}], \quad (\text{HCD28})$$

using evaluation in both degrees. On a common image x , the receiving quotient value is $\rho_\theta x$, since representatives differ by an original relation. The induced degree-one map is exactly (HCD27).

The global relation subspace is also the closure of measurable finite polynomial relation sections. To prove this without a measurable choice of approximating polynomials, suppose $v \in \mathcal{H}$ is orthogonal to every section $\mathbf{1}_U(\theta) \chi(S) S^j$, for measurable U and $j \geq 0$. Its fibre inner product with

χS^j is integrable by Cauchy–Schwarz, and its integral over every U is zero, so that inner product is zero almost everywhere. The original exponential sample bounds make every fixed polynomial section square-integrable. Intersecting the countably many resulting full-measure sets, (HCD21) gives $v(\theta) \in \mathcal{Q}_\theta$ there. Conversely every such section is orthogonal to the relations. The orthogonal-complement characterization proves that their closed span is exactly $\mathcal{R}$.

Since $\mathcal{Q} = 0$, $\mathcal{R} = \mathcal{H}$ as Hilbert spaces, and the completed complex is the identity complex. Its degree-one cohomology is zero. The finite continuous common-image complex has positive Gram $\overline{G}$ on its full E by (HCD8)–(HCD12). The intervening map and its whole kernel are precisely (HCD23)–(HCD28).

308.10 The receiving supported zero and all marked fibres

Integrate in the stated coefficient fibre, then apply the original marked support construction. At a fixed original label ℓ , a linear map f has the lift

$$(\ell, x) \mapsto (\ell, f(x)), \quad \tau \mapsto \tau. \quad (\text{HCD29})$$

For (HCD27), every represented input maps to the receiving supported zero $(\ell, 0)$, and external absence maps to external absence τ . Their inverse images are respectively the full represented fibre and the singleton external absence. At a fixed phase, the represented kernel is the entire ideal (HCD24), with its full primary decomposition (HCD25). No integral on arbitrary mixed-support totals is presumed.

If the empty-packet case $q = 0$ is separately admitted, $E = 0$, $B_N = P_N$, and the finite and continuous coefficient complexes are identity complexes. The quotient Gram determinant is the empty determinant one. The analytic source and its literal norm remain present before the quotient. Its marked receiving object still has τ and the represented zero, with the same maps (HCD29).

308.11 Exact outcome and scope

The continuous common-image complex has the explicit bounded cochain contraction (HCD7) on its complement and retains the whole nonreduced quotient in degree one. Its minimum metric is $\overline{G}$, and its difference from the constant-source metric is the positive original relation norm (HCD13). The mean-zero decomposition retains the weighted cross term (HCD15), giving (HCD16).

The subsequent completion has receiving weighted value spaces (HCD22), primary kernels (HCD24)–(HCD25), measurable field (HCD26), and actual cochain map (HCD28). Its receiving coefficient direct integral is zero because only finitely many phases can sample a fixed polynomial root. Every exceptional phase, nilpotent component and zero-weight coordinate remains in the stated maps and kernels. No arithmetic volume asymptotic or RH conclusion is inferred.

309 The literal phase graph as a weighted source and its exact Schur correction

Keep the actual counterfactual full quartet, its multiplicities and its original primitive from QCV, QPS and MBG:

$$\begin{aligned} m, k \in \mathbb{Z}_{>0}, \quad d = 4m, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ h(s) = \prod_{r \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-r)^m, \quad \Phi' = h, \quad \Phi(0) = 0, \quad v_h = g/h, \quad g = 2\xi, \\ v_h = j_h(v_h), \quad E_h = \mathbb{C}[s]/(h), \quad C = \Phi(A_h), \quad C_k = C^{[k]}, \quad \Phi(\frac{1}{2} + it) = A + iq_\Phi(t). \end{aligned} \quad (\text{PGS.1})$$

Here $A > 0$ is the retained primitive constant, and q_Φ is the original real odd polynomial of degree $d+1$, with leading coefficient $1/(d+1)$. Its derivative is exactly $[((t-\gamma)^2 + \delta^2)((t+\gamma)^2 + \delta^2)]^m > 0$. No replacement of h , its unit or its primitive is made. An actual off-line zero is not asserted to have been located. The present calculation evaluates the original source attached to this actual counterfactual packet, using the full-divisor hypothesis defining that packet.

The Hilbert space is $\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i dt_i/(2\pi))$, and inner products are conjugate linear in the first variable. Put

$$\begin{aligned} s_i = \frac{1}{2} + it_i, \quad \Sigma = \sum_i s_i, \quad \tau_k = \sum_i \Phi(s_i) = kA + i \sum_i q_\Phi(t_i), \\ \mathsf{T}_k f = \tau_k f, \quad \text{dom } \mathsf{T}_k = \{f : \tau_k f \in \mathcal{H}_k\}, \\ Sw = v_h(s) \text{rem}_h(v_h^{-1}w; s), \quad S_k = S^{\otimes k}, \quad G_k = S_k^* S_k. \end{aligned} \quad (\text{PGS.2})$$

All polynomial columns used below are in the indicated multiplication domains by the original Mellin decay estimates. In particular every adjoint below is the adjoint of a map with finite dimensional source; an expression such as $V^* \mathsf{T}_k^* L$ denotes its literal integral pairing, whose columns are in the required domains.

Retain the original full cyclic annihilator and observation:

$$\begin{aligned} \chi = \chi_{h,k}, \quad q = \deg \chi = [1 + k(m-1)](k+1)^2, \quad E = \mathbb{C}[S]/(\chi), \\ \eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{[k]})1, \\ \mathcal{P}_N = \mathbb{C}[S]_{\leq N}, \quad N \geq q-1, \quad n = N+1, \quad r = N-q+1, \quad \mathcal{P}_{-1} = 0, \\ B_N : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad Q \longmapsto \chi Q, \quad J_N : \mathcal{P}_N \longrightarrow E, \quad P \longmapsto [P], \\ s_N : E \longrightarrow \mathcal{P}_N, \quad [P] \longmapsto \text{rem}_\chi P. \end{aligned} \quad (\text{PGS.3})$$

All finite matrices use increasing monomial frames, including the original increasing remainder frame of E . Monic division proves $\ker J_N = \text{im } B_N$, B_N injective and $J_N s_N = I_q$. The full length and injectivity of η are those of MBG.23: each ordered tuple has nilpotence length $1 + k(m-1)$, its terminal coefficient is $[k(m-1)!]/((m-1)!)^k \neq 0$, and the $(k+1)^2$ arithmetic centres are distinct. This retains all ordered tuple contributions before passing to the stated cyclic subspace.

Write r for the successive remainders modulo the original $h(s_i)$. Define the following maps with their full units:

$$\begin{aligned} \mathcal{V}_N : \mathcal{P}_N \longrightarrow \mathcal{H}_k, \quad \mathcal{V}_N P = \left(\prod_i v_h(s_i) \right) P(\Sigma), \\ L : E \longrightarrow \mathcal{H}_k, \quad L = S_k C_k \eta, \quad L[S^b] = \left(\prod_i v_h(s_i) \right) F_b(s_1, \dots, s_k), \\ F_b = \mathsf{r}(\tau_k \Sigma^b), \quad 0 \leq b < q, \quad \mathsf{D}_N = \mathsf{T}_k \mathcal{V}_N - L J_N. \end{aligned} \quad (\text{PGS.4})$$

The formula for F_b is the exact composite $\text{rem}_h v_h^{-\otimes k} C_k v_h^{\otimes k}$, proved in MBG.25; every factor $v_h(s_i)$ remains in the source. The map $\mathcal{V}_N$ is injective, since a nonzero polynomial in Σ cannot vanish on

a product of open intervals where all $v_h(s_i) \neq 0$. The map L is injective as well. Indeed η and S_k are injective, and the already proved original critical-value calculation gives $\operatorname{Re} c_{k,a,b} \geq kA[1 - (65/289)^m] > 0$. The actual inverse is explicit on the full ordered tensor blocks. Let e_r be the original CRT idempotent of r in E_h , and for every ordered tuple $\mathbf{r} = (r_1, \dots, r_k)$ put $e_{\mathbf{r}} = e_{r_1} \otimes \dots \otimes e_{r_k}$ and $c_{\mathbf{r}} = \sum_i \Phi(r_i)$. Since $\Phi' = h$, at a root of order m one has $\Phi(s) - \Phi(r) \in (s - r)^{m+1}$, which vanishes in its full local quotient modulo $(s - r)^m$. Therefore multiplication by the phase is exactly scalar on every entire primary block, and

$$C_k = \sum_{\mathbf{r}} c_{\mathbf{r}} M_{e_{\mathbf{r}}}, \quad C_k^{-1} = \sum_{\mathbf{r}} c_{\mathbf{r}}^{-1} M_{e_{\mathbf{r}}}.$$

Here M_e is multiplication by the stated original idempotent. The idempotents are pairwise orthogonal and sum to one, so composing the two displayed matrices proves their inverse identity directly. Every ordered tuple is retained, including distinct tuples with the same critical sum. The arithmetic nilpotents of $A_h^{[k]}$ remain those retained in (PGS.3). This proves the injectivity used here on the full original tensor object.

309.1 The graph factorization and its finite-rank correction

Define maps into the same ordered two-component Hilbert space:

$$\begin{aligned} U_N &= (\mathcal{V}_N, \mathbf{T}_k \mathcal{V}_N) : \mathcal{P}_N \longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k, & Y &= (0, L) : E \longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k, \\ \mathcal{A}_N &= (\mathcal{V}_N, \mathbf{D}_N) = U_N - Y J_N, \\ W_N &= U_N^* U_N = \mathcal{V}_N^* (I + \mathbf{T}_k^* \mathbf{T}_k) \mathcal{V}_N \in \operatorname{Mat}_{n \times n}(\mathbb{C}), \\ Z_N &= U_N^* Y = \mathcal{V}_N^* \mathbf{T}_k^* L \in \operatorname{Mat}_{n \times q}(\mathbb{C}), & K &= L^* L = \eta^* C_k^* G_k C_k \eta \in \operatorname{Mat}_{q \times q}(\mathbb{C}). \end{aligned} \tag{PGS.5}$$

This K denotes the displayed cyclic phase-source Gram; it is not the one-factor covariance matrix called K in MBG.7. Both W_N and K are positive definite by the injectivities just proved. Taking the adjoint of the displayed literal graph map gives, with every cross term and its sign retained,

$$\widehat{M}_N = \mathcal{A}_N^* \mathcal{A}_N = W_N - Z_N J_N - J_N^* Z_N^* + J_N^* K J_N. \tag{PGS.6}$$

This is precisely $M_N + \mathbf{D}_N^* \mathbf{D}_N$, where $M_N = \mathcal{V}_N^* \mathcal{V}_N$ is the original AT source Gram. The correction to W_N factors as

$$\widehat{M}_N - W_N = [Z_N \ J_N^*] \begin{pmatrix} 0 & -I_q \\ -I_q & K \end{pmatrix} \begin{bmatrix} Z_N^* \\ J_N \end{bmatrix}. \tag{PGS.7}$$

Its rank is at most $2q$; no assertion that this difference has a sign is made. The exact relation square is

$$\mathcal{A}_N B_N = U_N B_N, \quad J_N B_N = 0, \quad \widehat{H}_N := B_N^* \widehat{M}_N B_N = B_N^* W_N B_N =: H_N^W. \tag{PGS.8}$$

It follows directly by applying the maps to each Q , rather than discarding the relations. The first components are the same nonzero original source $\mathcal{V}_N(\chi Q)$; the second components equal $\mathbf{T}_k \mathcal{V}_N(\chi Q)$. Their original theta primitives and the two cancelling tensor signs $(-1)^{i-1}(-1)^{i-1} = +1$ are the exact successive divisions of MBG.26. Thus (PGS.8) is a statement about the original relation arrows of AT.3 and MBG, including their source norms.

Here every matrix entry is a finite expression in the original moments. Put $M_{ab}^{(h)} =$

$\int \bar{s}^a s^b |v_h(s)|^2 dt / (2\pi)$, $P_{\alpha j} = [s^\alpha] \Sigma^j$, $T_{\alpha j} = [s^\alpha] \tau_k \Sigma^j$, and $f_{\alpha b} = [s^\alpha] F_b$. Then

$$\begin{aligned} (W_N)_{ij} &= \sum_{\alpha, \beta} (\bar{P}_{\alpha i} P_{\beta j} + \bar{T}_{\alpha i} T_{\beta j}) \prod_{a=1}^k M_{\alpha_a \beta_a}^{(h)}, \\ (Z_N)_{ib} &= \sum_{\alpha, \beta} \bar{T}_{\alpha i} f_{\beta b} \prod_{a=1}^k M_{\alpha_a \beta_a}^{(h)}, \\ K_{bc} &= \sum_{\alpha, \beta} \bar{f}_{\alpha b} f_{\beta c} \prod_{a=1}^k M_{\alpha_a \beta_a}^{(h)}. \end{aligned} \quad (\text{PGS.9})$$

All sums are finite polynomial expansions. The coefficients $f_{\alpha b}$ are obtained by monic division in the original variables and satisfy $\alpha_a < d$. Fubini applies because each integrand is a polynomial times the original product density with all moments finite. This proves the entry formulas including the cross source matrix Z_N , rather than replacing it by a norm.

309.2 The two exact least-norm sections

For conciseness suppress N within the next identities. Define

$$R_W = s - B(H^W)^{-1} B^* W s, \quad G_W = R_W^* W R_W = (JW^{-1} J^*)^{-1}. \quad (\text{PGS.10})$$

Every lift of $x \in E$ is $sx + By$. Completing its positive W -quadratic form gives the displayed minimizer, so $JR_W = I_q$, $B^* W R_W = 0$. These equalities imply $W R_W = J^* G_W$: decompose each vector as a relation plus $R_W Jp$ and pair against $R_W x$. Multiplying by W^{-1} and then J proves the inverse expression in (PGS.10). The inverse also has the exact decomposition

$$W^{-1} = B(H^W)^{-1} B^* + R_W G_W^{-1} R_W^*. \quad (\text{PGS.11})$$

To prove it, use the invertible ordered frame $[R_W \ B]$. Its W -Gram is $\text{diag}(G_W, H^W)$. Inverting that congruence yields (PGS.11).

For the literal phase graph write a lift as $p = R_W x + By$. Using (PGS.6), (PGS.8) and $B^* W R_W = 0$, its squared norm is

$$\begin{aligned} \|\mathcal{A}p\|^2 &= x^* (G_W - R_W^* Z - Z^* R_W + K) x + y^* H^W y - y^* B^* Z x - x^* Z^* B y \\ &= (y - (H^W)^{-1} B^* Z x)^* H^W (y - (H^W)^{-1} B^* Z x) + x^* \hat{G} x, \end{aligned} \quad (\text{PGS.12})$$

where consequently the unique graph section and its quotient Gram are

$$\begin{aligned} \hat{R} &= R_W + B(H^W)^{-1} B^* Z, \\ \hat{G} &= G_W - R_W^* Z - Z^* R_W + K - Z^* B(H^W)^{-1} B^* Z. \end{aligned} \quad (\text{PGS.13})$$

The signs come from the literal subtraction $U - YJ$. In particular $J\hat{R} = I_q$, $B^* \hat{M} \hat{R} = 0$ and $\hat{G} = \hat{R}^* \hat{M} \hat{R} = (J\hat{M}^{-1} J^*)^{-1}$, by the same argument as for (PGS.10). Thus these are exactly the MBG.30 section and quotient, not a new quotient chosen for this calculation.

The determinant orientation also stays fixed. In monomial frames $\det[s \ B] = 1$; replacing s by either section adds only relation columns and preserves that determinant. The order $[B \ s]$ has determinant $(-1)^{qr}$, as in AT.10. Hence

$$\det G_W = \frac{\det W}{\det H^W}, \quad \det \hat{G} = \frac{\det \hat{M}}{\det H^W}, \quad \frac{\det \hat{G}}{\det G_W} = \frac{\det \hat{M}}{\det W}. \quad (\text{PGS.14})$$

For $N = q - 1$, the relation domain has dimension zero, its determinant is one, and every composition through it is zero. All identities (PGS.8)–(PGS.14) therefore hold there literally.

309.3 The positive residual and its exact source meaning

Put

$$F_N^{\text{obs}} = J_N W_N^{-1} Z_N, \quad \Delta_N = K - Z_N^* W_N^{-1} Z_N, \quad \Pi_N = U_N W_N^{-1} U_N^*. \quad (\text{PGS.15})$$

The operator Π_N is the orthogonal projection onto $\text{im } U_N$: its adjoint equals itself, its square equals itself because $U_N^* U_N = W_N$, and it acts as the identity on $U_N p$. It follows that

$$\Delta_N = Y^*(I - \Pi_N)Y, \quad x^* \Delta_N x = \min_{p \in \mathcal{P}_N} (\|\mathcal{V}_N p\|^2 + \|\mathbf{T}_k \mathcal{V}_N p - Lx\|^2) > 0 \quad (x \neq 0). \quad (\text{PGS.16})$$

The unique unrestricted minimizing coefficient is $W_N^{-1} Z_N x$, obtained by completing this quadratic form. For strict positivity, equality would put $Yx = U_N p$; its first component gives $\mathcal{V}_N p = 0$, hence $p = 0$, and its second then gives $Lx = 0$, hence $x = 0$. This proves the strict sign for every actual finite window.

There is also an explicit positive lower bound independent of N . Define the full-source form

$$K_{\text{low}} = L^*(I + \mathbf{T}_k \mathbf{T}_k^*)^{-1} L, \quad (K_{\text{low}})_{bc} = \int_{\mathbb{R}^k} \frac{\overline{F_b(s)} F_c(s) \prod_i |v_h(s_i)|^2}{1 + k^2 A^2 + (\sum_i q_\Phi(t_i))^2} \prod_i \frac{dt_i}{2\pi}. \quad (\text{PGS.17})$$

For arbitrary $f \in \text{dom } \mathbf{T}_k$ and $y = Lx$, the pointwise identity is

$$|f|^2 + |\tau_k f - y|^2 = (1 + |\tau_k|^2) \left| f - \frac{\overline{\tau_k} y}{1 + |\tau_k|^2} \right|^2 + \frac{|y|^2}{1 + |\tau_k|^2}. \quad (\text{PGS.18})$$

Its minimizing function lies in the multiplication domain because $|\tau_k|/(1 + |\tau_k|^2) \leq 1/2$ and $|\tau_k|^2/(1 + |\tau_k|^2) \leq 1$. Integrate and restrict to $f = \mathcal{V}_N p$ to obtain

$$0 < K_{\text{low}} \preceq \Delta_N \preceq K. \quad (\text{PGS.19})$$

The lower matrix is strictly positive because L is injective and its scalar denominator is finite and strictly positive almost everywhere. The upper inequality follows from $\Pi_N \geq 0$. Thus the residual has a fully specified original source form; it has not been assigned a bound by arithmetic purity.

Equations (PGS.11) and (PGS.15) turn the quotient into the second exact completed square

$$\begin{aligned} \widehat{R}_N &= W_N^{-1} Z_N + R_{W,N}(I_q - F_N^{\text{obs}}), \\ \widehat{G}_N &= (I_q - F_N^{\text{obs}})^* G_{W,N}(I_q - F_N^{\text{obs}}) + \Delta_N. \end{aligned} \quad (\text{PGS.20})$$

Indeed $R_{W,N}^* Z_N = G_{W,N} F_N^{\text{obs}}$ and $Z_N^* W_N^{-1} Z_N = Z_N^* B_N (H_N^W)^{-1} B_N^* Z_N + (F_N^{\text{obs}})^* G_{W,N} F_N^{\text{obs}}$. Substitution into (PGS.13) proves both lines with their matrix order.

For the actual commuting source diagram, define $\mathcal{E}_N = (I - \Pi_N)Y$. Then

$$\mathcal{A}_N \widehat{R}_N = U_N R_{W,N}(I_q - F_N^{\text{obs}}) - \mathcal{E}_N, \quad (U_N R_{W,N})^* \mathcal{E}_N = 0, \quad \mathcal{E}_N^* \mathcal{E}_N = \Delta_N. \quad (\text{PGS.21})$$

Use $U_N W_N^{-1} Z_N = \Pi_N Y$ and $J_N \widehat{R}_N = I$ to prove its first identity; the remaining two are projection identities. This is an exact linear map of the cyclic quotient into two specified orthogonal parts of the original graph Hilbert target. It proves the morphism underlying (PGS.20) even when $I_q - F_N^{\text{obs}}$ is singular. No invertibility of that matrix has been presumed or needed.

All coefficient maps just displayed extend to each original active mask by applying the same linear map coordinate by coordinate. For example $\ker J_N = (\chi \mathcal{P}_{N-q})$, $\ker B_N = \ker \widehat{R}_N = 0$, and $\ker \mathcal{E}_N = 0$ by $\Delta_N > 0$. For F_N^{obs} the supported-zero fibre is its literal kernel; it is not asserted injective. Every present empty-mask tuple stays present and maps to the empty tuple; τ_{out} maps to τ_{out} . At an original compatible realization $R \rightarrow \mathbb{C}$, these are R -linear maps and commute with scalar extension on pure tensors. The realization and its full kernel are exactly MBG.34a–34b. Thus none of the new norm or section maps identifies a supported zero with absence or adds a residue-field map into $\mathbb{C}$.

309.4 Literal transitions between polynomial windows

For $M \geq N$, let $e_{N,M} : \mathcal{P}_N \hookrightarrow \mathcal{P}_M$ be the original coefficient inclusion, and let $e_{N-q,M-q}$ be the inclusion of relation coefficient spaces. Polynomial evaluation proves all four commuting identities

$$\begin{aligned} U_M e_{N,M} &= U_N, & J_M e_{N,M} &= J_N, & B_M e_{N-q,M-q} &= e_{N,M} B_N, & \mathcal{A}_M e_{N,M} &= \mathcal{A}_N, \\ W_N &= e_{N,M}^* W_M e_{N,M}, & Z_N &= e_{N,M}^* Z_M, & \widehat{M}_N &= e_{N,M}^* \widehat{M}_M e_{N,M}. \end{aligned} \quad (\text{PGS.22})$$

In particular $\Pi_N \preceq \Pi_M$, since their ranges are nested. Orthogonal projection onto the range difference gives the exact residual drop

$$\Delta_N - \Delta_M = Y^*(\Pi_M - \Pi_N)Y \succeq 0. \quad (\text{PGS.23})$$

For the quotient sections the full discrepancy, including its relation source, is

$$\begin{aligned} T_{N,M} &= (H_M^W)^{-1} B_M^* \widehat{M}_M e_{N,M} \widehat{R}_N, \\ e_{N,M} \widehat{R}_N - \widehat{R}_M &= B_M T_{N,M}, & \widehat{G}_N - \widehat{G}_M &= T_{N,M}^* H_M^W T_{N,M} \succeq 0. \end{aligned} \quad (\text{PGS.24})$$

The first right side is the relation-orthogonal projection of the given lift onto $\text{im } B_M$; subtracting it yields the unique least-norm lift in the larger window. Its graph is orthogonal to the relation graph, so Pythagoras proves the last identity. The same identities with $\widehat{M}, \widehat{R}, \widehat{G}$ replaced by W, R_W, G_W follow from the same projection, with no changes to B_M, J_M , or their labels.

Here is an explicit rank-one version of every consecutive transition. In $\mathcal{P}_{N+1}$ set

$$\begin{aligned} b_N &= \chi S^{N+1-q}, & d_N &= b_N - e_{N,N+1} B_N (H_N^W)^{-1} B_N^* e_{N,N+1}^* W_{N+1} b_N, \\ a_N &= d_N^* W_{N+1} d_N > 0, & \beta_N &= d_N^* W_{N+1} e_{N,N+1} R_{W,N}, \\ \alpha_N &= \beta_N - d_N^* Z_{N+1}. \end{aligned} \quad (\text{PGS.25})$$

Here d_N is a column of length $N+2$, and α_N, β_N are rows of length q . The polynomial d_N has leading coefficient one in degree $N+1$, so it is nonzero, and it is W_{N+1} -orthogonal to all old relation columns. New and old relation graphs are consequently orthogonal by (PGS.8). Pairing $U_{N+1} d_N$ against $\mathcal{A}_N \widehat{R}_N$ gives exactly α_N : the term $B_N (H_N^W)^{-1} B_N^* Z_N$ in (PGS.13) pairs to zero, leaving $\beta_N - d_N^* Z_{N+1}$. Subtracting this single projection proves

$$\begin{aligned} \widehat{R}_{N+1} &= e_{N,N+1} \widehat{R}_N - d_N a_N^{-1} \alpha_N, & \widehat{G}_{N+1} &= \widehat{G}_N - a_N^{-1} \alpha_N^* \alpha_N, \\ R_{W,N+1} &= e_{N,N+1} R_{W,N} - d_N a_N^{-1} \beta_N, & G_{W,N+1} &= G_{W,N} - a_N^{-1} \beta_N^* \beta_N. \end{aligned} \quad (\text{PGS.26})$$

Both proposed section corrections are relations since $J_{N+1} d_N = 0$. They kill exactly the remaining new relation pairing, and all old pairings already vanish. This proves the section and Gram formulas without an assumed canonical transport.

309.5 All four signed endpoints, with the finite correction evaluated

Set

$$\widehat{\sigma}_N = a_N^{-1} \alpha_N \widehat{G}_N^{-1} \alpha_N^*, \quad \sigma_N^W = a_N^{-1} \beta_N G_{W,N}^{-1} \beta_N^*. \quad (\text{PGS.27})$$

These are real numbers in $[0, 1)$. Nonnegativity is the positive inverse Gram. The actual congruent matrix $\widehat{G}_N^{-1/2} \widehat{G}_{N+1} \widehat{G}_N^{-1/2}$ equals $I_q - a_N^{-1} (\widehat{G}_N^{-1/2} \alpha_N^*) (\alpha_N \widehat{G}_N^{-1/2})$, and the weighted matrix has the same expression with $\alpha_N, \widehat{G}_N$ replaced by $\beta_N, G_{W,N}$. These congruent matrices have eigenvalues one except possibly $1 - \widehat{\sigma}_N$, or $1 - \sigma_N^W$; their strict positivity proves the strict upper bound. The rank-one determinant identity, proved by extending its nonzero update vector to a unitary basis, gives

$$\frac{\det \widehat{G}_{N+1}}{\det \widehat{G}_N} = 1 - \widehat{\sigma}_N, \quad \frac{\det G_{W,N+1}}{\det G_{W,N}} = 1 - \sigma_N^W. \quad (\text{PGS.28})$$

If the update vector is zero both equalities read one, so the proof includes that case.

For any of the compatible source families H , retain exactly the original AT.14 convention

$$\mathcal{B}(H) = \log \det G_{q-1}^H + \log \det G_q^H - \log \det G_{2q-1}^H - \log \det G_{2q}^H. \quad (\text{PGS.29})$$

Define weights $\omega_{q-1} = \omega_{2q-1} = 1$, and $\omega_j = 2$ for $q \leq j \leq 2q - 2$; for $q = 1$ this interior is empty. Telescoping (PGS.28) along the two intervals $[q - 1, 2q - 1]$ and $[q, 2q]$ proves the literal finite evaluation

$$\begin{aligned} \mathcal{B}(\widehat{M}) &= - \sum_{j=q-1}^{2q-1} \omega_j \log(1 - \widehat{\sigma}_j) \geq 0, \\ \mathcal{B}(W) &= - \sum_{j=q-1}^{2q-1} \omega_j \log(1 - \sigma_j^W) \geq 0, \\ \mathcal{B}(\widehat{M}) - \mathcal{B}(W) &= \sum_{j=q-1}^{2q-1} \omega_j \log \frac{1 - \sigma_j^W}{1 - \widehat{\sigma}_j}. \end{aligned} \quad (\text{PGS.30})$$

Thus its inputs are the explicit moment matrices (PGS.9), the weighted monic relation columns (PGS.25), and the computed cross rows $\alpha_j = \beta_j - d_j^* Z_{j+1}$. No sign of their difference is inferred from the separate nonnegative expressions. For $q = 1$, (PGS.29) cancels its two occurrences of endpoint one, and (PGS.30) is exactly the sum of the steps zero-to-one and one-to-two. The same formulas handle the original quartet, for which q takes its full value in (PGS.3).

Equivalently, the quotient-sized matrices

$$X_N = G_{W,N}^{-1/2} ((I - F_N^{\text{obs}})^* G_{W,N} (I - F_N^{\text{obs}}) + \Delta_N) G_{W,N}^{-1/2} > 0 \quad (\text{PGS.31})$$

give the same answer as

$$\begin{aligned} \mathcal{B}(\widehat{M}) - \mathcal{B}(W) &= \log \det X_{q-1} + \log \det X_q - \log \det X_{2q-1} - \log \det X_{2q}, \\ \det X_N &= \det \widehat{M}_N / \det W_N. \end{aligned} \quad (\text{PGS.32})$$

These follow from (PGS.14) and (PGS.20); the second is also the exact finite-rank determinant correction in (PGS.7).

The original Gamma source is still

$$m_k^\Gamma(t) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + it/2)|^2}{2\pi}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad S = k/2 + it. \quad (\text{PGS.33})$$

Its mass is $(2\pi)^{k/2}$, and the arithmetic mass remains μ_h^k , $\mu_h = \int |v_h(1/2 + it)|^2 dt / (2\pi)$. With exactly those families, the full three-part arithmetic identity is

$$\begin{aligned} \mathcal{B}(\widehat{M}) - \mathcal{B}(M^\Gamma) &= \mathcal{B}(M) - \mathcal{B}(M^\Gamma) \\ &\quad + \mathcal{B}(W) - \mathcal{B}(M) \\ &\quad + \sum_{j=q-1}^{2q-1} \omega_j \log \frac{1 - \sigma_j^W}{1 - \widehat{\sigma}_j}. \end{aligned} \quad (\text{PGS.34})$$

The first part is the original AT arithmetic/reference transport. The second is evaluated from the original weighted scalar density in the accompanying phase-density calculation. The final part is the literal phase observation correction calculated here. This equality retains all relation determinants, full units, reference masses, ordered endpoints and coefficient maps.

309.6 The exact scalar source, its fibre residual and all cross convolutions

The scalar reduction has an explicit isometry on the full Hilbert source, including the phase observation. Absorb only the original measure factor into the notation $w(t) = |v_h(1/2 + it)|^2/(2\pi)$; this changes no source vector. For an integrable function on the product line write its fibre integral over $t_1 + \dots + t_k = t$ by substituting $t_k = t - \sum_{i < k} t_i$ and integrating in $dt_1 \dots dt_{k-1}$. At $k = 1$ this means evaluation at t . Put $m_k = w^{*k}$. Expanding the actual scalar multiplier gives

$$\begin{aligned} r_k(t) &= \int_{\sum t_i=t} (1 + k^2 A^2 + (\sum_i q_\Phi(t_i))^2) \prod_i w(t_i), \\ r_1(t) &= (1 + A^2 + q_\Phi(t)^2) w(t), \\ r_k &= (1 + k^2 A^2) w^{*k} + k(q_\Phi^2 w) * w^{*(k-1)} \\ &\quad + k(k-1)(q_\Phi w) * (q_\Phi w) * w^{*(k-2)} \quad (k \geq 2). \end{aligned} \tag{PGS.35}$$

Here $w^{*0} = \delta_0$ is the convolution identity measure, of mass one; thus the displayed last term at $k = 2$ is exactly $2(q_\Phi w) * (q_\Phi w)$. The k diagonal terms and $k(k-1)$ ordered distinct pairs in the square give exactly these coefficients by Fubini and permutation of the original tensor variables. The last convolution is a signed function, whereas the fibre formula proves $r_k \geq (1 + k^2 A^2) m_k > 0$ almost everywhere. Every term is absolutely integrable, including every fixed polynomial moment, by the original one-variable moments and the triangle inequality. Both densities are finite almost everywhere, which suffices for the following specified Hilbert spaces and maps.

The map

$$\begin{aligned} \mathcal{J}_k : L^2(\mathbb{R}, r_k(t) dt) &\longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k, \\ \mathcal{J}_k f &= \left(\left(\prod_i v_h(s_i) \right) f\left(\sum_i t_i\right), \tau_k \left(\prod_i v_h(s_i) \right) f\left(\sum_i t_i\right) \right), \\ \mathcal{J}_k^* \mathcal{J}_k &= I, \quad U_N = \mathcal{J}_k \iota_N, \quad (\iota_N P)(t) = P(k/2 + it). \end{aligned} \tag{PGS.36}$$

is well-defined on equivalence classes: a null set for $r_k dt$ has zero product integral of both displayed squared components, by (PGS.35). The same formula gives equality of squared norms; polarization gives equality of pairings. Its range is therefore closed and its stated adjoint identity holds. In particular $W_N = \iota_N^* \iota_N$ in this exact weighted scalar source.

The full cyclic observation has explicit fibre data too. Define the row $z(t)$ and the positive semidefinite matrix $\Lambda(t)$ by

$$z_b(t) = \int_{\sum t_i=t} \overline{\tau_k F_b(s)} \prod_i w(t_i), \quad \Lambda_{bc}(t) = \int_{\sum t_i=t} \overline{F_b(s)} F_c(s) \prod_i w(t_i). \tag{PGS.37}$$

These are concrete finite sums of one-variable convolutions. For example, with the same f_{ab} as in (PGS.9), set $u_a(t) = s(t)^a w(t)$, $b_a(t) = (A - iq_\Phi(t)) s(t)^a w(t)$, and $u_{a,c}(t) = \bar{s}(t)^a s(t)^c w(t)$. Then

$$\begin{aligned} z_b &= \sum_\alpha f_{ab} \sum_{i=1}^k u_{\alpha_1} * \dots * u_{\alpha_{i-1}} * b_{\alpha_i} * u_{\alpha_{i+1}} * \dots * u_{\alpha_k}, \\ \Lambda_{bc} &= \sum_{\alpha, \beta} \bar{f}_{ab} f_{\beta c} u_{\alpha_1, \beta_1} * \dots * u_{\alpha_k, \beta_k}, \\ (Z_N)_{ab} &= \int \overline{(k/2 + it)^a} z_b(t) dt, \quad K = \int \Lambda(t) dt. \end{aligned} \tag{PGS.38}$$

At $k = 1$ each convolution list is its one factor. Expansion of the original polynomials, followed by the defining fibre integral, proves every identity. No phase source coordinate is replaced by an independently chosen scalar function.

For $x \in E$, fibre Cauchy–Schwarz applied to τ_k and $\sum_b F_b x_b$ gives

$$|z(t)x|^2 \leq (r_k(t) - m_k(t))x^* \Lambda(t)x. \quad (\text{PGS.39})$$

Indeed the first squared factor integrates to $\int_{\sum t_i=t} |\tau_k|^2 \prod_i w = r_k - m_k$, and the second to $x^* \Lambda x$. Thus $g_x(t) = z(t)x/r_k(t)$ is a specified element of $L^2(r_k dt)$, setting it to zero on the negligible exceptional set where division is not defined. Pair against any f in this scalar Hilbert space, first on bounded truncations and then by Cauchy–Schwarz: $\langle \mathcal{J}_k f, Yx \rangle = \int \bar{f} z x = \langle f, g_x \rangle_{r_k}$. It follows that $g = \mathcal{J}_k^* Y$, as a linear map from E .

Let $\mathbf{P}_N = \iota_N W_N^{-1} \iota_N^*$ be the orthogonal projection onto the actual scalar polynomial space and define $Y_\perp = (I - \mathcal{J}_k \mathcal{J}_k^*)Y$. The complete original observation and the finite-window residual now decompose as

$$\begin{aligned} Y &= \mathcal{J}_k g + Y_\perp, & \mathcal{J}_k^* Y_\perp &= 0, & \Omega_k &= Y_\perp^* Y_\perp = K - \int \frac{z(t)^* z(t)}{r_k(t)} dt, \\ \Delta_N &= \Omega_k + ((I - \mathbf{P}_N)g)^* ((I - \mathbf{P}_N)g), \\ 0 &< \int \frac{m_k(t)}{r_k(t)} \Lambda(t) dt \preceq \Omega_k \preceq \Delta_N. \end{aligned} \quad (\text{PGS.40})$$

All adjoints in the second line use the original scalar density $r_k dt$. The first line is an orthogonal projection identity. Since $U_N = \mathcal{J}_k \iota_N$, its projection is $\Pi_N = \mathcal{J}_k \mathbf{P}_N \mathcal{J}_k^*$; substituting the first decomposition into (PGS.16) proves the second line. Inequality (PGS.39), divided by r_k and integrated, proves the last lower bound. It is strict because $m_k/r_k > 0$ almost everywhere and $\int x^* \Lambda x = x^* K x > 0$ for every nonzero x . This gives the full map and positive fibre contribution before any limit of polynomial windows is considered. It does not assume that the finite observation F_N^{obs} is an identity.

In particular the exact finite minimizer in (PGS.16) is the scalar projection $\mathbf{P}_N g_x$, represented in the original coefficient frame by $W_N^{-1} Z_N x$. Combining it with the specified jet correction $R_{W,N}(I - F_N^{\text{obs}})x$ gives $\hat{R}_N x$ exactly as in (PGS.20). Equations (PGS.35)–(PGS.40) therefore construct the scalar weighted source, the full mixed observation, their projection, every fibre residual and their return to the original quotient in a single commuting diagram of explicitly given maps.

The two proved positive source bounds are themselves related by an exact morphism and norm equality. For $x \in E$ write $F_x = \sum_b F_b x_b$, $u = \sum_i t_i$, and $g_x(u) = z(u)x/r_k(u)$. The scalar-source minimizing vector is $\prod_i v_h(s_i) g_x(u)$, whereas the whole multiplication-graph minimizer from (PGS.18) is $\prod_i v_h(s_i) \overline{\tau_k} F_x(s) / (1 + |\tau_k|^2)$. Both belong to the same original multiplication domain. Substituting the first into (PGS.18) and using (PGS.40) gives

$$\begin{aligned} x^* (\Omega_k - K_{\text{low}}) x &= \int_{\mathbb{R}^k} (1 + |\tau_k|^2) \left| g_x\left(\sum_i t_i\right) - \frac{\overline{\tau_k} F_x(s)}{1 + |\tau_k|^2} \right|^2 \prod_i w(t_i) dt_i \geq 0, \\ 0 &< K_{\text{low}} \preceq \Omega_k \preceq \Delta_N, & \Omega_1 &= K_{\text{low}} \quad (k=1). \end{aligned} \quad (\text{PGS.41})$$

Indeed the squared distance from Yx to the closed scalar graph is $x^* \Omega_k x$; the minimum over the entire multiplication graph is $x^* K_{\text{low}} x$. The pointwise completed square proves their displayed difference, including the unchanged product measure. At $k=1$, (PGS.37) and (PGS.35) give $g_x(t) = \overline{\Phi(1/2 + it)} F_x(1/2 + it) / (1 + |\Phi(1/2 + it)|^2)$ almost everywhere, so the integrand vanishes. This proves the last equality without identifying the two minimizers for higher tensor powers.

310 The scalar density of the original phase graph source

Retain the actual quartet, its full order, its full arithmetic unit and its original primitive from MBG.1–MBG.3. In particular

$$\begin{aligned} h(s) &= \prod_{r \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-r)^m, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m \in \mathbb{Z}_{>0}, \\ g &= 2\xi, \quad v_h = g/h, \quad \Phi' = h, \quad \Phi(0) = 0, \quad \Phi(\tfrac{1}{2} + it) = A + iq_\Phi(t), \quad A > 0, \\ q'_\Phi(t) &= [((t-\gamma)^2 + \delta^2)((t+\gamma)^2 + \delta^2)]^m, \quad q_\Phi(0) = 0. \end{aligned} \quad (\text{PGD.1})$$

Thus the actual polynomial q_Φ is real and odd of degree $p_\Phi = 4m+1$, with leading coefficient $1/(4m+1)$. These facts follow by differentiating the retained primitive on its original line, not by choosing a different phase. Put

$$\begin{aligned} w(t) &= \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad m_k = w^{*k}, \quad \mu_h = \int_{\mathbb{R}} w(t) dt > 0, \\ s_j &= \frac{1}{2} + it_j, \quad S = \frac{k}{2} + iu, \quad u = \sum_j t_j, \\ (\mathcal{V}_N P)(t_1, \dots, t_k) &= \prod_j v_h(s_j) P\left(\sum_j s_j\right), \quad \mathsf{T}_k = \sum_j \Phi(s_j) = kA + i \sum_j q_\Phi(t_j), \\ M_N &= \mathcal{V}_N^* \mathcal{V}_N, \quad W_N = \mathcal{V}_N^* (I + \mathsf{T}_k^* \mathsf{T}_k) \mathcal{V}_N, \quad \mathcal{P}_N = \mathbb{C}[S]_{\leq N}. \end{aligned} \quad (\text{PGD.2})$$

The Hilbert measure before multiplication by the displayed full units is $\prod_j dt_j/(2\pi)$. Consequently all following scalar convolutions use the already displayed density $w(t)dt$; they introduce no further factor of 2π . The adjoints use the original increasing-power coefficient frame. These are the full phase graph forms; the subtracted finite-jet boundary in MBG.24 is a different, explicitly related map.

310.1 Exact fibre and three-convolution formulas

For $k \geq 2$, eliminate the last variable by $t_k = u - \sum_{j < k} t_j$, and define

$$r_k(u) = \int_{\mathbb{R}^{k-1}} \left(1 + k^2 A^2 + \left[\sum_{j=1}^k q_\Phi(t_j) \right]^2 \right) \prod_{j=1}^k w(t_j) dt_1 \cdots dt_{k-1}. \quad (\text{PGD.3})$$

The linear substitution from $(t_1, \dots, t_k)$ to $(t_1, \dots, t_{k-1}, u)$ has determinant one. Tonelli's theorem on the nonnegative diagonal integrand, followed by polarization for cross-pairings, therefore proves

$$\langle P, W_N Q \rangle = \int_{\mathbb{R}} \overline{P(k/2 + iu)} Q(k/2 + iu) r_k(u) du. \quad (\text{PGD.4})$$

Here $\langle P, W_N Q \rangle$ denotes the original coefficient pairing. All polynomial moments converge absolutely by the retained exponential decay of w , proved quantitatively below. This also justifies each finite expansion used next. In the original density notation the exact formula is

$$r_k = (1 + k^2 A^2) w^{*k} + k(q_\Phi^2 w) * w^{*(k-1)} + k(k-1)(q_\Phi w) * (q_\Phi w) * w^{*(k-2)}. \quad (\text{PGD.5})$$

For this displayed formula $k \geq 2$, and w^{*0} is the unit Dirac measure δ_0 for convolution, with mass one. For $k = 1$ there is no third term and the complete formula is explicitly

$$r_1(u) = [1 + A^2 + q_\Phi(u)^2] w(u). \quad (\text{PGD.6})$$

Indeed the square in (PGD.3) has exactly k diagonal terms and $k(k-1)$ ordered terms with unequal indices. Fubini and the same determinant-one substitution give their three convolutions. The third convolution in (PGD.5) is a signed function: $q_\Phi w$ is odd, and it must not be replaced by its absolute value in that equality.

Conjugation and the full quartet symmetry give $w(-t) = w(t)$, whereas q_Φ is odd. Hence $\int q_\Phi w = 0$. Integrating all three terms retains the exact source mass:

$$\int r_k(u) du = (1 + k^2 A^2) \mu_h^k + k \mu_h^{k-1} \int q_\Phi(t)^2 w(t) dt. \quad (\text{PGD.7})$$

All functions r_k are even. The fibre representation proves

$$(1 + k^2 A^2) m_k(u) \leq r_k(u) \leq (1 + k^2 A^2) m_k(u) + k^2 z_k(u), \quad z_k = (q_\Phi^2 w) * w^{*(k-1)}. \quad (\text{PGD.8})$$

The upper inequality follows from $(\sum_j x_j)^2 \leq k \sum_j x_j^2$ for real x_j , followed by permutation of the k original integration variables.

The equality cases at fixed ordinate can also be calculated. For two factors and $u = 0$, oddness gives $q_\Phi(t) + q_\Phi(-t) = 0$ at every point of the fibre, and hence

$$r_2(0) = (1 + 4A^2) m_2(0), \quad [(q_\Phi w) * (q_\Phi w)](0) = -[(q_\Phi^2 w) * w](0). \quad (\text{PGD.9})$$

For $k = 2$ and $u \neq 0$, the polynomial $q_\Phi(x) + q_\Phi(u-x)$ has leading term $ux^{p_\Phi-1}$, so is nonconstant. For $k \geq 3$ and every u , the fibre polynomial is nonconstant: set $t_1 = x, t_2 = y, t_3 = u - x - y, t_j = 0$ for $j > 3$. In its homogeneous part of degree p_Φ , the coefficient of $x^{p_\Phi-1}y$ is -1 . No lower phase coefficient can cancel it. The product density on the fibre is positive almost everywhere, since each factor is the squared modulus of a nonzero entire function evaluated on a real affine coordinate. A nonzero polynomial cannot vanish on an open set; moreover its zero set has Lebesgue measure zero, as follows inductively by viewing it as a one-variable polynomial with polynomial coefficients and applying Fubini. Its squared integral against this density is therefore strictly positive. This proves

$$r_k(u) > (1 + k^2 A^2) m_k(u) \quad \begin{cases} k = 2, u \neq 0, \\ k \geq 3, u \in \mathbb{R}. \end{cases} \quad (\text{PGD.10})$$

For one factor, (PGD.6) states the exact equality cases, including zeros of the actual w . The preceding statements are about the whole phase-square fibre, not a sign assertion for its individual mixed convolution in (PGD.5).

310.2 Uniform actual tail constants and the phase insertion

The original BT12–BT16 proof supplies the following constants and bound. They are retained explicitly to specify the estimate being used:

$$\begin{aligned} \alpha &= \pi/2, \quad p_0 = 8m - 9/2 > 1, \quad R_0 = (\delta^2 + \gamma^2)^{1/2}, \quad T_h = \max(1, 2R_0), \\ Z_* &= 2 + \sqrt{5/2}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \quad \beta_0 = 8m - 5, \\ C_{\text{comp}} &= \frac{C_\Gamma}{\sqrt{\pi}} (T_h^2 + 1/4)^2 [1 + 2\sqrt{T_h^2 + 1/4}]^2 \delta^{-8m} (1 + T_h)^{\beta_0}, \\ C_{\text{tail}} &= \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} (4/3)^{4m} (1 + 1/T_h)^{\beta_0}, \quad C_h = \max(C_{\text{comp}}, C_{\text{tail}}), \\ w(t) &\leq C_h e^{-\alpha|t|} (1 + |t|)^{-p_0}, \quad M_\alpha = \int e^{\alpha t} w(t) dt = \int e^{-\alpha t} w(t) dt \in (0, \infty). \end{aligned} \quad (\text{PGD.11})$$

These are the original arithmetic bounds proved from the floor formula for zeta and the complete Binet identity in BT. In particular their use does not assume a positive lower bound at individual zeta

zeros. They imply $M_\alpha \leq 2C_h/(p_0 - 1)$. Write the complete phase as $q_\Phi(t) = \sum_{j=0}^{p_\Phi} q_j t^j$, and set

$$Q_h = \sum_{j=0}^{p_\Phi} |q_j| > 0, \quad a = 13/2, \quad C_\Phi = C_h Q_h^2 H_a, \quad (\text{PGD.12})$$

$$H_a = \sup_{x \geq 0} (1+x)^a e^{-2\alpha x} = \begin{cases} 1, & a \leq 2\alpha, \\ e^{2\alpha-a}(a/(2\alpha))^a, & a > 2\alpha. \end{cases}$$

Every q_j here is the coefficient of the original polynomial in (PGD.1). The phase has not been replaced by the polynomial with absolute coefficients; that polynomial is used only for the bound $|q_\Phi(t)| \leq Q_h(1+|t|)^{p_\Phi}$. Because $2p_\Phi - p_0 = 13/2$, (PGD.11) gives

$$q_\Phi(t)^2 w(t) \leq C_h Q_h^2 e^{-\alpha|t|} (1+|t|)^a. \quad (\text{PGD.13})$$

The formula for H_a follows by differentiating $a \log(1+x) - 2\alpha x$, whose derivative is strictly decreasing. Let $t_- := \max(-t, 0)$, and retain the actual finite one-sided moment

$$I_a^- := \int (1+t_-)^a e^{\alpha t} w(t) dt > 0. \quad (\text{PGD.14})$$

It is finite with the explicit bound

$$I_a^- \leq C_h \left[\frac{1}{p_0 - 1} + \sum_{j=0}^7 \binom{7}{j} \frac{j!}{(2\alpha)^{j+1}} \right]. \quad (\text{PGD.15})$$

Indeed on the positive half-line the integrand is at most $C_h(1+t)^{-p_0}$. On the negative half-line put $x = -t$; its bound is $C_h e^{-2\alpha x} (1+x)^{a-p_0}$, which is at most $C_h e^{-2\alpha x} (1+x)^7$. Expanding this last polynomial and integrating each monomial gives (PGD.15). No positive-part moment of order a at the exponential boundary is required.

For every $k \geq 2$, the actual inserted convolution obeys

$$\boxed{z_k(u) \leq C_\Phi (k-1)^a M_\alpha^{k-2} I_a^- e^{-\alpha|u|} (1+|u|)^a.} \quad (\text{PGD.16})$$

Here is its proof with the one-sided tilt preserved. Put $b(t) = e^{\alpha t} w(t)$, of mass M_α , and $f(t) = e^{\alpha t} q_\Phi(t)^2 w(t)$. For $t \geq 0$, (PGD.13) gives $f(t) \leq C_\Phi (1+t)^a$. For $t < 0$, it gives $f(t) \leq C_h Q_h^2 H_a$. Thus on both half-lines $f(t) \leq C_\Phi (1+t_+)^a$. For $u \geq 0$ and every v , $1 + (u-v)_+ \leq (1+u)(1+v_-)$. Consequently, with $n = k-1$,

$$\begin{aligned} e^{\alpha u} z_k(u) &= (f * b^{*n})(u) \\ &\leq C_\Phi (1+u)^a \int (1+v_-)^a b^{*n}(v) dv \\ &\leq C_\Phi (1+u)^a n^a M_\alpha^{n-1} I_a^-. \end{aligned} \quad (\text{PGD.17})$$

For the last inequality, in the product integral use $(\sum_j t_j)_- \leq \sum_j (t_j)_-$ and

$$(1 + \sum_{j=1}^n (t_j)_-)^a \leq \left(\sum_{j=1}^n [1 + (t_j)_-] \right)^a \leq n^{a-1} \sum_{j=1}^n [1 + (t_j)_-]^a.$$

The last inequality is convexity of x^a , with $a > 1$, applied to the arithmetic mean of these n non-negative numbers. Integrating each of the n summands gives exactly $n^a M_\alpha^{n-1} I_a^-$, including $n = 1$. Evenness of $q_\Phi^2 w$ and of w proves $z_k(-u) = z_k(u)$, so the positive half-line inequality proves (PGD.16) on the whole line.

The original Gamma density $\sigma(u) = |\Gamma(1/4 + iu/2)|^2 / (2\pi)$ has mass $\sqrt{2\pi}$ and satisfies the retained pointwise lower bound $\sigma(u) \geq c_\Gamma e^{-\alpha|u|} (1 + |u|)^{-1/2}$. It follows from (PGD.16) that

$$z_k(u) \leq F_k(1 + |u|)^7 \sigma(u), \quad F_k = \frac{C_\Phi I_a^-}{c_\Gamma} (k-1)^a M_\alpha^{k-2}, \quad k \geq 2. \quad (\text{PGD.18})$$

The measure $\sigma(u)du$ is used through this stated inequality; it is not the k -fold Gamma reference of AT.4, and its mass has not been raised to a different tensor power.

310.3 All-vector bounds on the original growing windows

Let σ retain that same density. Its complete monic orthogonality calculation TG.7–TG.14 gives

$$\gamma_j = \sqrt{2\pi}(2j)!/4^j, \quad ue_j = \sqrt{(j+1)(j+1/2)}e_{j+1} + \sqrt{j(j-1/2)}e_{j-1}, \quad (\text{PGD.19})$$

where e_j are orthonormal, and the second term is absent at $j = 0$. On degree at most n , the two weighted shifts have norms at most $n+1$ and n , as is seen by summing squared coefficients in their orthogonal target frames. Hence $\|uf\|_\sigma \leq 2(n+1)\|f\|_\sigma$. For every integer $r \geq 0$, iterating through the degree increases and expanding the nonnegative polynomial $(1 + u^2)^r$ proves

$$\int (1 + u^2)^r |f(u)|^2 \sigma(u) du \leq [1 + 4(N+r)^2]^r \int |f(u)|^2 \sigma(u) du, \quad \deg f \leq N. \quad (\text{PGD.20})$$

For $r = 0$ this is equality. For $r \geq 1$, the estimate $\|u^j f\|_\sigma \leq [2(N+r)]^j \|f\|_\sigma$, $0 \leq j \leq r$, followed by the binomial theorem proves the displayed constant. In particular, Holder's inequality with exponents $8/7$ and 8 , followed by the case $r = 4$, gives

$$L_N = 2^{7/2} [1 + 4(N+4)^2]^{7/2}, \quad \int (1 + |u|)^7 |f(u)|^2 \sigma(u) du \leq L_N \int |f(u)|^2 \sigma(u) du. \quad (\text{PGD.21})$$

Explicitly $(1 + |u|)^7 \leq 2^{7/2} (1 + u^2)^{7/2}$, and

$$\int (1 + u^2)^{7/2} |f|^2 \sigma \leq \left(\int (1 + u^2)^4 |f|^2 \sigma \right)^{7/8} \left(\int |f|^2 \sigma \right)^{1/8}.$$

Composition $P \mapsto f(u) = P(k/2 + iu)$ is the exact invertible degree-preserving algebra map, with inverse $f \mapsto f((S - k/2)/i)$. Therefore these estimates apply to every original polynomial vector, in the original sum coordinate.

Retain the original positive convolution lower envelope TW7:

$$\begin{aligned} B = 42 + 8m, \quad M = B/2 = 21 + 4m, \quad \vartheta_h = \int_{-1}^1 w(t) dt > 0, \\ m_k(u) &\geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (k-2 + |u|)^{-B}, \quad k \geq 3, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_\Gamma 2^M}, \quad D_N = [1 + 4(N+M)^2]^M. \end{aligned} \quad (\text{PGD.22})$$

Here $c_h > 0$ is exactly the source constant of the proved arithmetic envelope AU.10, retained in CJ.15 and TW7. It depends on the fixed full arithmetic packet and not on k, u, N . The original proof of that envelope, rather than a new lower bound on zeta, is the input in (PGD.22). Its all-vector implication can be proved directly here. The upper Gamma bound and $(k-2 + |u|)^B \leq (k-2)^B 2^M (1 + u^2)^M$ imply $m_k \geq a_k (1 + u^2)^{-M} \sigma$. Cauchy–Schwarz, followed by (PGD.20) at the integer $r = M$, gives

$$\begin{aligned} \left(\int |f|^2 \sigma \right)^2 &\leq \left(\int (1 + u^2)^{-M} |f|^2 \sigma \right) \left(\int (1 + u^2)^M |f|^2 \sigma \right) \\ &\leq D_N \left(\int (1 + u^2)^{-M} |f|^2 \sigma \right) \left(\int |f|^2 \sigma \right). \end{aligned}$$

Division for $f \neq 0$, with the zero vector handled separately, proves

$$\int |P(k/2 + iu)|^2 \sigma(u) du \leq \frac{D_N}{a_k} \int |P(k/2 + iu)|^2 m_k(u) du, \quad \deg P \leq N. \quad (\text{PGD.23})$$

Combining (PGD.8), (PGD.18), (PGD.21), and (PGD.23) now gives the full finite-source comparison

$$\boxed{\begin{aligned} c_k M_N &\preceq W_N \preceq (c_k + E_{k,N}) M_N, \\ c_k &= 1 + k^2 A^2, \quad E_{k,N} = \frac{k^2 F_k L_N D_N}{a_k}, \\ k &\geq 3, N \geq 0. \end{aligned}} \quad (\text{PGD.24})$$

It applies to arbitrary mixtures of all polynomial coefficients. Restriction by multiplication by the literal χ gives the same constants on its full relation space. For $N \geq q - 1$, minimization over each unchanged affine remainder fibre gives the same constants for the two induced full q -dimensional quotient Grams. For smaller N , the same assertion concerns the image of the restricted remainder map, with its actual dimension. None of these steps drops a relation source or its phase energy. The original monomial matrices themselves are retained; c_k is a proved lower comparison scalar, not a rescaling of them.

The logarithmic comparison constant and its full dependence are

$$\begin{aligned} \Lambda_{k,N}^{\text{ph}} &= \log \left(1 + \frac{E_{k,N}}{c_k} \right), \quad d_h = \alpha + \log(M_\alpha / \vartheta_h) > \alpha, \\ \log \frac{E_{k,N}}{c_k} &= 2 \log k - \log(1 + k^2 A^2) + a \log(k - 1) + (k - 2) \log M_\alpha \\ &\quad - (k - 3) \log \vartheta_h + \alpha(k - 3) + B \log(k - 2) \\ &\quad + \log \left(\frac{C_\Phi I_a^- C_\Gamma 2^M}{c_\Gamma c_h} \right) + \log L_N + \log D_N. \end{aligned} \quad (\text{PGD.25})$$

The strict inequality in d_h follows from $M_\alpha = \int \cosh(\alpha t) w(t) dt > \int_{-1}^1 w(t) dt$: the complement has positive measure with positive density. All terms in the logarithm have been retained. For fixed h , $\log L_N + \log D_N = O_h(1 + \log(N + 1))$, with polynomial degree $B + 7 = 49 + 8m$ in N . The other displayed terms prove, uniformly for $k \geq 3, N \geq 0$,

$$\Lambda_{k,N}^{\text{ph}} = k d_h + O_h(\log k + \log(N + 1)). \quad (\text{PGD.26})$$

To justify using $\log(1 + x)$ here, the displayed exact expression for $\log(E_{k,N}/c_k)$, together with $A > 0$ and the positive lower bounds of L_N, D_N , tends uniformly to $+\infty$ as $k \rightarrow \infty$. Thus $\log(1 + x) - \log x = \log(1 + x^{-1})$ is uniformly bounded in the asserted range; finitely many smaller k are absorbed by the same fixed constant. This is a rate for the proved upper envelope, not an assertion that the actual condition number has that growth.

310.4 The exact signed four-endpoint consequence

Retain $\chi = \chi_{h,k}$, $q = [1 + k(m - 1)](k + 1)^2$, and the original four endpoints $q - 1, q, 2q - 1, 2q$, with signs $+, +, -, -$. For any compatible positive source family X_N , let

$$\begin{aligned} G_N(X) &= (J_N X_N^{-1} J_N^*)^{-1}, \\ \mathcal{B}(X) &= \log \det G_{q-1}(X) + \log \det G_q(X) - \log \det G_{2q-1}(X) - \log \det G_{2q}(X). \end{aligned} \quad (\text{PGD.27})$$

The quotient is always by the same literal monic χ . Its exact determinant formula is $\det G_N(X) = \det X_N / \det(B_N^* X_N B_N)$, including determinant one for the zero-dimensional relation space. It follows by completing the positive quadratic form on $s_N x + B_N y$; the frame $[s_N B_N]$ has determinant one by monic division.

For completeness the sharp dimension coefficient in the comparison can be recovered on this very source. Interpolate the compatible families by $X_N(t) = (1 - t)M_N + tW_N$. On the common space $\mathcal{P}_{2q}$, let $P_N(t)$ be orthogonal projection onto $\mathcal{P}_N$, $Q_N(t)$ projection onto $\chi\mathcal{P}_{N-q}$, and $C(t) = X_{2q}(t)^{-1}(W_{2q} - M_{2q})$. Finite matrix differentiation of the preceding determinant identity gives the exact, signed formula

$$\mathcal{B}(W) - \mathcal{B}(M) = \int_0^1 \text{Tr}((U(t) - V(t))C(t)) dt, \quad (\text{PGD.28})$$

$$U = (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \quad V = (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q).$$

All projections and adjoints here use the same actual $X_{2q}(t)$ metric. Each difference comes from nested subspaces. Reading their orthogonal increments shows that both U and V have the complete spectrum $0^q, 1^2, 2^{q-1}$. If $c_1(t) \leq \dots \leq c_{2q+1}(t)$ are the eigenvalues of $C(t)$, the trace against either positive operator with this spectrum is between the sums obtained by pairing its increasing eigenvalues with the decreasing or increasing lists c_j . One direct proof writes the operator as its range projection of rank $q+1$ plus its eigenvalue-two projection of rank $q-1$. In an eigenbasis of C , a rank- r projection has diagonal entries in $[0, 1]$ summing to r . Moving this fixed mass to the r smallest or largest c_j proves the requisite two trace bounds. Subtracting the complete sums therefore gives

$$|\text{Tr}((U - V)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{PGD.29})$$

Let $b_1 \leq \dots \leq b_{2q+1}$ be the positive generalized eigenvalues of W_{2q} relative to M_{2q} . Then $c_j(t) = (b_j - 1)/(1 - t + tb_j)$. This follows by diagonalizing $M_{2q}^{-1}W_{2q}$ in its original M_{2q} metric. The fraction is increasing in b_j and integrates to $\log b_j$. Consequently (PGD.28)–(PGD.29) prove the exact finite-spectrum and uniform bounds

$$|\mathcal{B}(W) - \mathcal{B}(M)| \leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q - 1)\Lambda_{k,2q}^{\text{ph}}. \quad (\text{PGD.30})$$

The last inequality uses (PGD.24), so $c_k \leq b_j \leq c_k + E_{k,2q}$. Its total multiplicity is $1 + 2(q - 1) = 2q - 1$. In particular the common scalar c_k has cancelled in these ratios; none of the four original determinant factors was changed before taking the ratio.

This source is itself a positive moment source. In fact $\mathcal{B}(W) > 0$. To prove this directly, minimum sections at two nested cutoffs satisfy $G_i(W) - G_j(W) = (R_i - R_j)^*W(R_i - R_j) \succeq 0$, because $R_i - R_j$ is a relation and R_j is orthogonal to the larger relation space. Thus each of the two logarithmic determinant ratios in $\mathcal{B}(W)$ is nonnegative. If their sum vanished, both Gram differences, and hence both section differences, would vanish. The ordered chain $G_{q-1} \succeq G_q \succeq G_{2q-1} \succeq G_{2q}$ then forces all four Grams to agree, and the squared section-difference identity forces all four sections to agree in the common source. In particular the constant polynomial, which belongs to $\text{im } R_{q-1} = \mathcal{P}_{q-1}$, would be orthogonal to $\chi\mathcal{P}_q$ at cutoff $2q$. For $\chi(S) = \sum a_j S^j$, the polynomial $\chi^{\#k}(S) = \sum \bar{a}_j (k - S)^j$ belongs to $\mathcal{P}_q$ and equals $\overline{\chi(S)}$ on the original line $S = k/2 + iu$. The asserted orthogonality would give $\int |\chi(k/2 + iu)|^2 r_k(u) du = 0$, contrary to positivity of the moment norm of this nonzero polynomial. This proof assigns no sign to the difference $\mathcal{B}(W) - \mathcal{B}(M)$.

Finally $q = [1 + k(m - 1)](k + 1)^2$ is the original unaltered degree, so $\log q = O_h(\log k)$. Equations (PGD.25)–(PGD.26) give

$$\frac{\Lambda_{k,2q}^{\text{ph}}}{k} \longrightarrow d_h > \alpha, \quad \frac{(2q - 1)\Lambda_{k,2q}^{\text{ph}}}{kq} \longrightarrow 2d_h. \quad (\text{PGD.31})$$

These are proved rates of this explicit estimate. They do not assert that the actual signed correction is nonzero or attains the bound. For the retained quartet spectral quantity $\mathcal{L}_k = 2\delta[1 + k(m -$

$1)](k+1)\lfloor(k+1)^2/4\rfloor$, one has $\mathcal{L}_k/(kq) \rightarrow \delta/2$. Thus the upper envelope in (PGD.30), measured on that actual scale, tends to $4d_h/\delta$. Measured against $4q \log k$, the same upper envelope diverges as $d_h k/(2 \log k)$. This calculation preserves the exact phase graph and all mixed terms while identifying the actual size of the control furnished by its present decay estimate; it supplies neither a shrinking exponent error nor a contradiction to the counterfactual packet.

311 The strict mixed-fibre component of the original phase observation

Retain every object and coordinate of PGS.1–PGS.41. In particular $d = 4m, m \geq 1, k \geq 1, a = kA > 0$,

$$s_i = \frac{1}{2} + it_i, \quad u = \sum_{i=1}^k t_i, \quad Q(t) = \sum_{i=1}^k q_\Phi(t_i), \quad \tau_k = a + iQ(t), \quad c = 1 + a^2. \quad (\text{PGM.1})$$

The real odd polynomial q_Φ has degree $d + 1$ and leading coefficient $1/(d + 1)$. The full original cyclic source is $E = \mathbb{C}[S]/(\chi_{h,k})$ of dimension $q = [1 + k(m - 1)](k + 1)^2$. Its actual observation is

$$Lx = \left(\prod_i v_h(s_i) \right) F_x(s_1, \dots, s_k),$$

$$F_x = \sum_{b=0}^{q-1} x_b \text{rem}_{h(s_1), \dots, h(s_k)} \left(\left[\sum_i \Phi(s_i) \right] \left[\sum_i s_i \right]^b \right). \quad (\text{PGM.2})$$

Every variable degree of F_x is at most $d - 1$. The full unit $v_h = 2\xi/h$ is retained in L , which is injective by PGS.4 and its proof: the original cyclic map and section are injective and the actual ordered phase sums are nonzero. Thus $F_x = 0$ implies $x = 0$. No tensor tuple, repeated critical value or arithmetic nilpotent has been removed.

Write $w(t) = |v_h(1/2 + it)|^2/(2\pi)$. Define the two original weighted Hilbert spaces and the sum-coordinate inclusion by

$$\mathcal{H}_k = L^2 \left(\mathbb{R}^k, (c + Q(t)^2) \prod_i w(t_i) dt_i \right), \quad \mathcal{L}_k = L^2(\mathbb{R}, r_k(u) du),$$

$$j_k : \mathcal{L}_k \longrightarrow \mathcal{H}_k, \quad (j_k f)(t) = f \left(\sum_i t_i \right). \quad (\text{PGM.3})$$

The fibre density r_k is exactly PGS.35, including all three convolutions and the original measure factors. Its fibre integral proves $\|j_k f\|_{\mathcal{H}_k} = \|f\|_{\mathcal{L}_k}$, so j_k has closed range and $\mathbf{P}_k = j_k j_k^*$ is its orthogonal projection. Define the full multiplication-graph minimizer and its mixed part:

$$B_k : E \longrightarrow \mathcal{H}_k, \quad B_k x = \frac{(a - iQ)F_x(s)}{c + Q^2}, \quad g_x(u) = \frac{z(u)x}{r_k(u)},$$

$$j_k^* B_k x = g_x, \quad \mathcal{D}_k = (I - \mathbf{P}_k)B_k, \quad \mathcal{D}_k x = B_k x - j_k g_x. \quad (\text{PGM.4})$$

Here B_k is a new map with the explicitly displayed domain and target, not the relation-column map B_N of PGS.3. It is well-defined, since

$$\|B_k x\|_{\mathcal{H}_k}^2 = \int \frac{a^2 + Q^2}{c + Q^2} |F_x(s)|^2 \prod_i w(t_i) dt_i \leq \|Lx\|_{\mathcal{H}_k}^2.$$

Pairing $B_k x$ against $j_k f$ cancels exactly the positive factor $c + Q^2$; the resulting fibre integral is $\int \overline{f(u)} z(u) x du$. This proves the adjoint formula in PGM.4, with its conjugate phase and its original masses. The cancellation does not change either source vector.

The two residuals of PGS.17 and PGS.40 satisfy the exact matrix identity

$$\boxed{\mathcal{D}_k^* \mathcal{D}_k = \Omega_k - K_{\text{low}}.} \quad (\text{PGM.5})$$

Indeed $B_k^* B_k = K - K_{\text{low}}$ by the preceding integral, where $K = L^* L$ is the actual cyclic observation Gram. The adjoint formula gives $(j_k^* B_k)^* (j_k^* B_k) = \int z(u)^* z(u)/r_k(u) du = K - \Omega_k$. Subtracting these equalities proves PGM.5, with all cross entries retained. It also proves the norm identity in PGS.41 directly.

311.1 The exact kernel for every mixed tensor

For $k \geq 2$ the map $\mathcal{D}_k$ is injective. We prove this on its actual measurable source, not on an auxiliary polynomial model. Suppose $\mathcal{D}_k x = 0$. The nonzero entire function v_h has only isolated zeros on the original line. Hence $\prod_i w(t_i) > 0$ for Lebesgue-almost every $t \in \mathbb{R}^k$, and $c + Q^2$ is everywhere strictly positive. The asserted Hilbert-space equality therefore gives

$$(a - iQ(t))F_x(s) = g_x(u)(c + Q(t)^2) \quad \text{for almost every } t \in \mathbb{R}^k. \quad (\text{PGM.6})$$

The determinant-one change of variables $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ preserves null sets. By Fubini, for almost every u at which the finite representative $g_x(u)$ is defined, PGM.6 holds for almost every $(t_1, \dots, t_{k-1})$.

Fix such a u and, when $k > 2$, fix almost every $(t_2, \dots, t_{k-1})$ to which Fubini applies. Put $b = u - \sum_{i=2}^{k-1} t_i$, where the empty sum at $k = 2$ is zero, and set $t_k = b - t_1$. Exclude $b = 0$; this removes only a null set, since it is $u = 0$ for $k = 2$, and a proper affine hyperplane in the remaining variables for $k > 2$. The polynomial

$$Q_b(X) = q_\Phi(X) + q_\Phi(b - X) + \sum_{i=2}^{k-1} q_\Phi(t_i) \quad (\text{PGM.7})$$

has degree exactly d and leading coefficient $b \neq 0$. To verify this, d is even and the leading phase terms give $[X^{d+1} + (b - X)^{d+1}]/(d+1) = bX^d + \text{terms of lower degree}$. Every lower term of the odd phase has degree at most $d-1$; its substitution cannot supply an X^d coefficient. Meanwhile the substituted polynomial $F_b(X)$, obtained from the very same F_x in PGM.2, has degree at most $2d-2$: only its first and last variables depend on X , each with degree at most $d-1$.

The almost-everywhere equality in X is an identity of complex polynomials, because a nonzero one-variable polynomial has only finitely many real roots. Set $g = g_x(u) \in \mathbb{C}$. Its exact polynomial consequence is

$$\begin{aligned} (a - iQ_b)F_b &= g(c + Q_b^2), \\ (c + Q_b^2) - (a + iQ_b)(a - iQ_b) &= 1, \\ F_b &= (c + Q_b^2)[F_b - g(a + iQ_b)]. \end{aligned} \quad (\text{PGM.8})$$

The second identity uses the original constant $c = 1 + a^2$; the 1 is essential. The third follows by multiplying the second by F_b and using the first. The polynomial $c + Q_b^2$ has degree $2d$ and nonzero leading coefficient b^2 . If the bracket were nonzero, its product would have degree at least $2d$, exceeding $2d-2$. Therefore the bracket and F_b are both zero.

It follows for almost every fixed fibre and remaining coordinate choice that $F_x(s) = 0$ for every t_1 . Fubini and the same invertible linear change of variables give $F_x(s) = 0$ for almost every $t \in \mathbb{R}^k$. A polynomial vanishing almost everywhere on $\mathbb{R}^k$ is identically zero: apply the one-variable fact iteratively to its coefficient polynomials, using Fubini. The invertible affine substitutions $s_i = 1/2 + it_i$ then prove $F_x = 0$ in its original variables. Injectivity of L proves $x = 0$, completing the kernel proof.

For $k = 1$, the sum-coordinate inclusion in PGM.3 is the identity between the identical weighted spaces $r_1(t)dt = (1 + A^2 + q_\Phi(t)^2)w(t)dt$. Consequently $\mathcal{D}_1 = 0$. Combining this with PGM.5 and the preceding injectivity gives the complete assertion

$$\begin{aligned} \ker \mathcal{D}_k &= \begin{cases} E, & k = 1, \\ \{0\}, & k \geq 2, \end{cases} \quad \text{rank } \mathcal{D}_k = q \quad (k \geq 2), \\ \Omega_1 &= K_{\text{low}}, \quad \Omega_k - K_{\text{low}} > 0 \quad (k \geq 2), \quad 0 < K_{\text{low}} < \Omega_k \preceq \Delta_N \quad (k \geq 2, N \geq q-1). \end{aligned} \quad (\text{PGM.9})$$

Strict matrix inequalities use the original coefficient frame of E ; for every nonzero x , their value is the strictly positive squared norm of the actual $\mathcal{D}_k x$.

On each retained coefficient mask A_{mask} , the same map acts coordinatewise from $E^{A_{\text{mask}}}$ into $\mathcal{K}_k^{A_{\text{mask}}}$. Its kernel is zero for $k \geq 2$. It commutes with zero insertion between masks because it is linear. The present empty tuple retains its outer support label; absence τ_{out} remains a separate point and maps to absence. For the already specified compatible coefficient maps $\alpha : R \rightarrow \mathbb{C}$, $\lambda : R \rightarrow R'$, and $\alpha' : R' \rightarrow \mathbb{C}$ with $\alpha'\lambda = \alpha$, set

$$\begin{aligned} \mu_E : R' \otimes_R E &\longrightarrow E, & b \otimes x &\longmapsto \alpha'(b)x, \\ \mathcal{D}_k^{\text{real}} &= \mathcal{D}_k \mu_E, & \mathcal{D}_k^{\text{real}}(b \otimes x) &= \alpha'(b) \mathcal{D}_k x, \\ \ker \mathcal{D}_k^{\text{real}} &= \ker \mu_E = \left\{ \sum_j (b_j \otimes x_j - 1 \otimes \alpha'(b_j)x_j) \right\} & (k \geq 2). \end{aligned} \quad (\text{PGM.10})$$

Balancedness follows from the specified equality of coefficient actions. Every generator in the last display has zero image under μ_E . Conversely, for a tensor $z = \sum_j b_j \otimes x_j$, subtracting $1 \otimes \mu_E(z)$ gives exactly the displayed sum. The map $x \mapsto 1 \otimes x$ is an R -linear section, so this also proves surjectivity and identifies every fibre as $1 \otimes x + \ker \mu_E$. Injectivity of $\mathcal{D}_k$ proves the equality of kernels. This constructs the mixed component on the existing carrier, with the full supported-zero realization fibre retained. It does not assert an unprovided residue-field map to $\mathbb{C}$.

The original polynomial and theta relation arrows have an equally explicit comparison. For $k \geq 2$ and $N \geq q - 1$, put $\mathcal{D}_{k,N} = \mathcal{D}_k J_N$. Then

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{\mathcal{D}_{k,N}} \mathcal{D}_k(E) \longrightarrow 0, \quad \mathcal{D}_{k,N} s_N = \mathcal{D}_k \quad (\text{PGM.11})$$

is exact. Indeed $B_N Q = \chi Q$ is injective by monic division, J_N is onto with kernel $\chi \mathcal{P}_{N-q}$, and $\mathcal{D}_k$ is injective. These statements prove each exactness assertion and the section equality, including $\mathcal{P}_{-1} = 0$ at $N = q - 1$. On the literal Hilbert source $\mathcal{V}_N(\mathcal{P}_N)$, the same arrow is $\mathcal{D}_{k,N} \mathcal{V}_N^{-1}$, where the inverse is defined on that actual image by the proved injectivity of $\mathcal{V}_N$. Its kernel is precisely $\mathcal{V}_N(\chi \mathcal{P}_{N-q})$. Thus all nonzero original relation vectors, their theta primitives and their original norms remain present. The arrow factors through the stated polynomial cyclic quotient, not through an unidentified proper-source quotient.

Finally, for $k \geq 2$, its inverse on the mixed image is explicitly

$$\begin{aligned} \mathcal{R}_k^{\text{mix}} : \mathcal{K}_k &\longrightarrow E, & \mathcal{R}_k^{\text{mix}} &= (\Omega_k - K_{\text{low}})^{-1} \mathcal{D}_k^*, \\ \mathcal{R}_k^{\text{mix}} \mathcal{D}_k &= I_E, & \mathcal{D}_k \mathcal{R}_k^{\text{mix}} &= \text{orthogonal projection onto } \mathcal{D}_k(E). \end{aligned} \quad (\text{PGM.12})$$

The first composite follows from PGM.5 and PGM.9. The second operator is self-adjoint, its square equals itself, and it acts identically on $\mathcal{D}_k(E)$; its range is contained in that image. These facts prove the projection assertion with the unchanged weighted Hilbert metric. All adjoints and matrix inverses refer to the original cyclic coefficient frame specified above.

The result is a strict, explicitly calculated mixed-source residual. Its matrix entries are the integrals in PGM.4–PGM.5 with their unchanged units and domains. It does not give a uniform estimate for its eigenvalues as k grows. In particular it does not replace the signed four-window identity PGS.34 by an asserted sign or by a shrinking exponent error.

312 The exact phase-graph map to the original averaged quotient and its completion

Fix the original full quartet and all the data of PGS.1–PGS.41. Thus $m, k \geq 1, d = 4m, \rho = 1/2 + \delta + i\gamma, 0 < \delta < 1/2, \gamma > 2$,

$$h(s) = \prod_{r \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-r)^m, \quad v_h(s) = \frac{2\xi(s)}{h(s)}, \quad \Phi' = h, \quad \Phi(0) = 0.$$

Retain $s_i = 1/2 + it_i, \Sigma = \sum_i s_i = k/2 + iu, u = \sum_i t_i, \Phi(1/2 + it) = A + iq_\Phi(t), A > 0$, and $\tau_k = \sum_i \Phi(s_i) = kA + i \sum_i q_\Phi(t_i)$. The polynomial q_Φ has degree $4m+1$ and leading coefficient $1/(4m+1)$. These are the original objects, and the full-divisor hypothesis of the original counterfactual packet is unchanged. Let

$$\begin{aligned} \mathcal{H}_k &= L^2 \left(\mathbb{R}^k, \prod_i \frac{dt_i}{2\pi} \right), \quad \mathsf{T}_k f = \tau_k f, \quad \text{dom } \mathsf{T}_k = \{f \in \mathcal{H}_k : \tau_k f \in \mathcal{H}_k\}, \\ \chi &= \chi_{h,k}, \quad q = [1 + k(m-1)](k+1)^2, \quad E = \mathbb{C}[S]/(\chi), \\ \mathcal{P}_N &= \mathbb{C}[S]_{\leq N}, \quad N \geq q-1, \quad B_N : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad Q \longmapsto \chi Q, \\ J_N : \mathcal{P}_N &\longrightarrow E, \quad P \longmapsto [P], \quad s : E \longrightarrow \mathcal{P}_{q-1}, \quad [P] \longmapsto \text{rem}_\chi P. \end{aligned} \tag{PGB.1}$$

Here $\mathcal{P}_{-1} = 0$. Every finite matrix below uses the original increasing monomial frame, including the increasing remainder frame of E . All Hilbert inner products are conjugate linear in their first argument. Monic division proves $\ker J_N = \text{im } B_N$, injectivity of B_N , and $J_N s = I_E$.

Write the actual polynomial observation and defect as

$$\begin{aligned} (\mathcal{V}_N P)(t) &= \left(\prod_i v_h(s_i) \right) P(\Sigma), \quad Lx = \left(\prod_i v_h(s_i) \right) F_x(s_1, \dots, s_k), \\ F_x &= \sum_{b=0}^{q-1} x_b \text{rem}_{h(s_1), \dots, h(s_k)}(\tau_k \Sigma^b), \quad D_N = \mathsf{T}_k \mathcal{V}_N - L J_N, \\ \mathcal{A}_N &= (\mathcal{V}_N, D_N) : \mathcal{P}_N \longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k, \quad M_N = \mathcal{V}_N^* \mathcal{V}_N, \quad \widehat{M}_N = \mathcal{A}_N^* \mathcal{A}_N = M_N + D_N^* D_N. \end{aligned} \tag{PGB.2}$$

The definition of F_x retains every original ordered tensor remainder and every factor v_h . Its equality with $\mathsf{S}_k C_k \eta x$ and the injectivity of L are the fully proved PGS.4 construction. In particular $F_x = 0$ implies $x = 0$. The source decay of PGS and PGD places all the displayed columns in the multiplication domain. The injectivity of $\mathcal{V}_N$ makes both finite Grams positive definite.

312.1 The positive cost of adding the literal defect

Suppress N only inside the next calculation. Define

$$\begin{aligned} H &= B^* M B, \quad C = s - B H^{-1} B^* M s, \quad G = C^* M C = (J M^{-1} J^*)^{-1}, \\ T &= D B : \mathcal{P}_{N-q} \longrightarrow \mathcal{H}_k, \quad H_+ = H + T^* T = B^* \widehat{M} B, \\ Q_D &= I_{\mathcal{H}_k} - T H_+^{-1} T^*. \end{aligned} \tag{PGB.3}$$

If $N = q - 1$, every map through the zero relation space is zero, both relation determinants are one, and $Q_D = I$. For positive relation dimension, H, H_+ are positive definite. The formula for C is proved by writing a lift as $sx + By$ and completing its M square. It gives $JC = I$ and $B^* M C = 0$. To verify the inverse formula for G , decompose every polynomial as $CJp + By$; pairing against Cx proves $MC = J^* G$. Multiplication first by M^{-1} and then by J gives the formula.

The bounded operator Q_D is positive and has the exact inverse

$$Q_D = (I_{\mathcal{H}_k} + TH^{-1}T^*)^{-1}, \quad 0 < Q_D \preceq I_{\mathcal{H}_k}. \quad (\text{PGB.4})$$

Indeed $I + TH^{-1}T^*$ is self-adjoint and at least I . The product of this operator with $I - TH_+^{-1}T^*$ is I : the coefficient between T and T^* in the remaining term is

$$H^{-1} - H_+^{-1} - H^{-1}T^*TH_+^{-1} = H^{-1} - (I + H^{-1}T^*T)H_+^{-1} = 0.$$

Taking adjoints proves the product in the opposite order. On the finite-dimensional range of T positivity follows by diagonalizing the positive operator $TH^{-1}T^*$; on its orthogonal complement the inverse is I . This also proves $Q_D \succeq (1 + \|TH^{-1}T^*\|)^{-1}I$ and its bounded positive square root without requiring an infinite-dimensional spectral assertion.

Every lift is $Cx + By$. Its unchanged graph norm is

$$\begin{aligned} \|\mathcal{A}(Cx + By)\|^2 &= x^*Gx + y^*Hy + \|DCx + Ty\|^2 \\ &= x^*(G + C^*D^*Q_DDC)x + (y + H_+^{-1}T^*DCx)^*H_+(y + H_+^{-1}T^*DCx). \end{aligned} \quad (\text{PGB.5})$$

Expansion proves this identity with its two positive cross terms in the first square and the negative minimized contribution $-C^*D^*TH_+^{-1}T^*DC$ retained. Therefore

$$\begin{aligned} \widehat{R}_N &= C_N - B_N(H_{+,N})^{-1}T_N^*D_NC_N, \\ \widehat{G}_N &= G_N + \mathcal{E}_N^{\text{aug}}, \quad \mathcal{E}_N^{\text{aug}} = C_N^*D_N^*Q_{D,N}D_NC_N > 0, \\ \ker \mathcal{E}_N^{\text{aug}} &= \ker(D_NC_N) = \{0\}. \end{aligned} \quad (\text{PGB.6})$$

The section has $J_N\widehat{R}_N = I$ and is orthogonal to the relation columns in $\widehat{M}_N$. Thus uniqueness of a minimum identifies it with the original section in PGS.13 and PGS.20; the displayed $\widehat{G}_N$ is that same graph quotient. The kernel assertion follows from the strict bounded positivity of $Q_{D,N}$ proved above. The actual map whose Gram is the new positive term is $x \mapsto Q_{D,N}^{1/2}D_NC_Nx$ into $\mathcal{H}_k$.

Here the kernel can be completely evaluated on the actual source. If $P \neq 0$ has degree n and leading coefficient p_n , the polynomial $\tau_k P(\Sigma)$, as a polynomial in s_1 , has degree $n + d + 1$ and leading coefficient $p_n/(d + 1)$. Indeed the original monic h gives Φ leading coefficient $1/(d + 1)$, while the terms $\Phi(s_j)P(\Sigma)$ with $j > 1$ have s_1 degree only n . On the other hand every variable degree of F_{JP} is at most $d - 1$ by its original successive remainders. Consequently $\tau_k P(\Sigma) - F_{JP}$ is a nonzero polynomial. Its restriction under $s_i = 1/2 + it_i$ is nonzero almost everywhere outside its polynomial zero set; the product $\prod_i v_h(s_i)$ is nonzero almost everywhere because its one-variable factors are nonzero entire functions. Thus $D_N P \neq 0$ in $\mathcal{H}_k$. This proves injectivity of D_N . Since $J_NC_N = I$, it also proves injectivity of D_NC_N , all asserted zero kernels, and the strict sign in PGB.6. In particular $G_N < \widehat{G}_N$ even at the empty relation window.

There is a second realization retaining both original graph components. Put $\mathbf{a}_N = \widehat{R}_N - C_N : E \rightarrow \text{im } B_N$. Orthogonality of $\mathcal{V}_N C_N$ and $\mathcal{V}_N \mathbf{a}_N$ gives

$$\begin{aligned} \mathcal{A}_N \widehat{R}_N &= (\mathcal{V}_N C_N, 0) + (\mathcal{V}_N \mathbf{a}_N, D_N \widehat{R}_N), \\ \mathcal{E}_N^{\text{aug}} &= \mathbf{a}_N^* M_N \mathbf{a}_N + (D_N \widehat{R}_N)^*(D_N \widehat{R}_N). \end{aligned} \quad (\text{PGB.7})$$

The two displayed summands in the first line are orthogonal. Taking their Grams proves the second line and also proves its equality with PGB.6, rather than merely comparing two presentations.

312.2 Composition with the accepted all-holonomy quotient

At each fixed N , retain the original HCD circumference, denoted here by L_{circ} to distinguish that scalar from the observation map $L : E \rightarrow \mathcal{H}_k$. Thus L_{circ} is exactly the quantity called L in HCD,

with the same value. Retain the original holonomy matrices of HCD.2–HCD.2c. Concretely choose the already computed original $\beta > 0$, weighted source constant $\kappa_{\beta,N}$ and any $0 < \eta < 1$, and take $L_{\text{circ}} \geq \beta^{-1} \log(1 + \kappa_{\beta,N}/\eta)$. With $d\mu = d\theta/(2\pi)$ the HCD proof then gives

$$\begin{aligned} \int_0^{2\pi} M_{N,\theta} d\mu &= M_N, & (1-\eta)M_N &\preceq M_{N,\theta} \preceq (1+\eta)M_N, \\ C_{N,\theta} &= M_{N,\theta}^{-1} J_N^* G_{N,\theta}, & G_{N,\theta} &= (J_N M_{N,\theta}^{-1} J_N^*)^{-1}, & \overline{G}_N &= \int_0^{2\pi} G_{N,\theta} d\mu. \end{aligned} \quad (\text{PGB.8})$$

No phase graph matrix is substituted into this accepted definition. The full relative Hilbert fibre and the original half-density source used to construct $M_{N,\theta}$ remain those of HCD. Uniform finite-dimensional equivalence makes every following coefficient field measurable and square integrable.

For each phase set

$$X_{N,\theta} = (B_N^* M_{N,\theta} B_N)^{-1} B_N^* M_{N,\theta} C_N, \quad C_N - C_{N,\theta} = B_N X_{N,\theta}. \quad (\text{PGB.9})$$

At zero relation dimension $X_{N,\theta}$ is the zero map. Both sections lift the same vector, so their difference lies in $\text{im } B_N$. Pairing the difference with B_N in the phase metric, and using $B_N^* M_{N,\theta} C_{N,\theta} = 0$, proves exactly the displayed formula. Squaring this orthogonal decomposition of C_N and integrating proves

$$\mathcal{D}_N^{\text{hol}} := G_N - \overline{G}_N = \int_0^{2\pi} (C_N - C_{N,\theta})^* M_{N,\theta} (C_N - C_{N,\theta}) d\mu \succeq 0. \quad (\text{PGB.10})$$

This is the exact accepted HCD.13 relation cost. In particular the graph augmentation does not remove it.

Define

$$\mathcal{B}_N = L^2(\Theta; \text{im } B_N, M_{N,\theta} d\mu), \quad \mathfrak{F}_N x = \left(Q_{D,N}^{1/2} D_N C_N x, [\theta \mapsto (C_N - C_{N,\theta})x] \right) \in \mathcal{H}_k \oplus \mathcal{B}_N. \quad (\text{PGB.11})$$

Its complete matrix identity and kernel are

$$\begin{aligned} \widehat{G}_N - \overline{G}_N &= \mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}} = \mathfrak{F}_N^* \mathfrak{F}_N > 0, \\ \ker \mathfrak{F}_N &= \{x : D_N C_N x = 0, C_N x = C_{N,\theta} x \text{ a.e.}\} = \{0\}. \end{aligned} \quad (\text{PGB.12})$$

Indeed the two orthogonal direct-sum target norms are precisely PGB.6 and PGB.10. A sum of two nonnegative real norms vanishes exactly when both vanish; the positivity in PGB.4 and that of the phase source metric give the stated full kernel. The actual injectivity proved after PGB.6 evaluates that kernel as zero and makes the displayed matrix difference strictly positive.

The identity on coefficients consequently defines the actual contraction $(E, \widehat{G}_N) \rightarrow (E, \overline{G}_N)$. It is induced by an explicit map of the original two-term complexes: send p in $(\mathcal{P}_N, \widehat{M}_N)$ to the constant field $\theta \mapsto p$ in the HCD common-image space, and send a relation $B_N y$ to that same constant relation field. The inclusion differentials commute, $J_N p$ is unchanged, and

$$\int p^* M_{N,\theta} p d\mu = p^* M_N p \leq p^* \widehat{M}_N p.$$

Thus the source map is a contraction and its induced quotient map is the displayed identity. More precisely, the complete isometric dilation of this quotient contraction is

$$x \mapsto (x, Q_{D,N}^{1/2} D_N C_N x, [\theta \mapsto (C_N - C_{N,\theta})x]) \quad \text{from } (E, \widehat{G}_N) \text{ to } (E, \overline{G}_N) \oplus \mathcal{H}_k \oplus \mathcal{B}_N. \quad (\text{PGB.13})$$

Its squared norm identity is PGB.12; polarization proves every cross pairing. This gives the precise relation between the graph quotient, the accepted average, and the common finite-source relation obstruction on the unchanged coefficient object E .

312.3 Exponential moments and the complete graph quotient

For clarity the density argument needed for the degree limit is proved here. If a finite positive Borel measure ν on $\mathbb{R}$ has $\int e^{\epsilon|u|} d\nu < \infty$ for some $\epsilon > 0$, then polynomials are dense in $L^2(\nu)$. To prove this, let f be orthogonal to them and put $d\lambda = \bar{f} d\nu$. Cauchy-Schwarz gives

$$\int e^{\epsilon|u|/2} |d\lambda| \leq \|f\|_{L^2(\nu)} \left(\int e^{\epsilon|u|} d\nu \right)^{1/2} < \infty.$$

The Fourier integral $F(z) = \int e^{izu} d\lambda(u)$ is holomorphic on $|\operatorname{Im} z| < \epsilon/2$: on a smaller closed strip every differentiated integrand is dominated by a constant times $e^{\epsilon|u|/2} |d\lambda|$. Every derivative at zero vanishes by polynomial orthogonality. The Taylor expansion and then overlapping analytic discs give $F = 0$ throughout this connected strip, in particular on its real axis.

Here is the needed uniqueness on that axis. Let $g_t(x) = (4\pi t)^{-1/2} \exp(-x^2/(4t))$, $t > 0$. The elementary Gaussian integral gives

$$g_t(v) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-t\zeta^2} e^{i\zeta v} d\zeta.$$

For completeness, differentiating the latter integral in v and integrating by parts in ζ gives the differential equation $I'(v) = -vI(v)/(2t)$; its value at zero is $I(0) = \sqrt{\pi/t}$, obtained by squaring the Gaussian integral and using polar coordinates. These statements prove the formula with its constants. Absolute integrability and Fubini now give $\lambda * g_t = 0$ because $F(-\zeta) = 0$. For $\varphi \in C_c(\mathbb{R})$ it follows that $\int (\varphi * g_t) d\lambda = 0$. The functions $\varphi * g_t$ converge uniformly to φ : split the unit-mass Gaussian integral into $|y| < a$, where uniform continuity controls the difference, and $|y| \geq a$, where its mass tends to zero. Thus $\int \varphi d\lambda = 0$ for every such φ . Regularity of a finite Borel measure on the real line, followed by continuous approximations to indicators of compact sets and open sets, gives $\lambda = 0$. It follows that $f = 0$ ν -almost everywhere. The orthogonal complement criterion for closed subspaces proves the density claim.

Apply this to the original two densities

$$\begin{aligned} w(t) &= |v_h(1/2 + it)|^2 / (2\pi), \quad m_k = w^{*k}, \\ r_k(u) &= \int_{\sum t_i = u} (1 + |\tau_k|^2) \prod_i w(t_i), \quad \mathcal{L}_k = L^2(\mathbb{R}, r_k(u) du). \end{aligned} \tag{PGB.14}$$

All fibre integrals use elimination of t_k and measure $dt_1 \cdots dt_{k-1}$, with determinant-one change of variables. For $k = 1$ the fibre integral is evaluation. The proved PGD.11 bound is $w(t) \leq C_h e^{-\alpha|t|} (1 + |t|)^{-p_0}$, $\alpha = \pi/2$, $p_0 = 8m - 9/2 > 1$. For fixed k, h and $0 < \epsilon < \alpha$, both

$$\int e^{\epsilon|u|} r_k(u) du < \infty, \quad \int e^{\epsilon|u|} |\chi(k/2 + iu)|^2 r_k(u) du < \infty \tag{PGB.15}$$

follow directly on the product source. Indeed $|u| \leq \sum_i |t_i|$, while $1 + |\tau_k|^2$ and $|\chi(k/2 + iu)|^2$ are bounded by fixed products of powers of $(1 + |t_i|)$. Their product with $\prod_i e^{-(\alpha-\epsilon)|t_i|}$ is integrable. The same proof holds with r_k replaced by m_k . Every constant here depends on the fixed original packet and k ; no uniform statement in k has been inserted.

The density is positive almost everywhere by PGS.35, and the polynomial $\chi(k/2 + iu)$ has only finitely many real zeros. Consequently multiplication by that unchanged polynomial is the onto isometry

$$L^2(\mathbb{R}, |\chi(k/2 + iu)|^2 r_k(u) du) \longrightarrow \mathcal{L}_k, \quad f(u) \longmapsto \chi(k/2 + iu) f(u). \tag{PGB.16}$$

Its inverse is division by $\chi(k/2 + iu)$ away from its finite zero set, with arbitrary values on that null set. Equality of the two squared norms proves both the isometry and the inverse domain assertion. PGB.15 and the density proof therefore imply

$$\overline{\{\chi(k/2 + iu) Q(k/2 + iu) : Q \in \mathbb{C}[S]\}}^{\mathcal{L}_k} = \mathcal{L}_k. \tag{PGB.17}$$

Polynomials in u and in $S = k/2 + iu$ have exactly the same span: the forward substitution has inverse $u = (S - k/2)/i$. This proves the density assertion on the original coordinate, not on an independently substituted source.

Recall the actual isometry and observation from PGS.36–PGS.40:

$$\begin{aligned} \mathcal{J}_k : \mathcal{L}_k &\longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k, & \mathcal{J}_k f &= \left(\prod_i v_h(s_i) f(u), \tau_k \prod_i v_h(s_i) f(u) \right), \\ Y &= (0, L), & g &= \mathcal{J}_k^* Y, & g_x(u) &= z(u)x/r_k(u), & z_b(u) &= \int_{\sum t_i=u} \overline{\tau_k} F_b(s) \prod_i w(t_i), \\ Y_\perp &= (I - \mathcal{J}_k \mathcal{J}_k^*) Y, & \Omega_k &= Y_\perp^* Y_\perp > 0. \end{aligned} \quad (\text{PGB.18})$$

These maps are the original isometry and mixed observation; the adjoint is exactly z/r_k with its conjugate phase. To recall why the strict sign holds, fibre Cauchy–Schwarz gives $|z(u)x|^2 \leq (r_k(u) - m_k(u))x^* \Lambda(u)x$, where $\Lambda_{bc}(u) = \int_{\sum t_i=u} \overline{F_b} F_c \prod_i w$. It follows that $\Omega_k \succeq \int (m_k/r_k) \Lambda du > 0$; for nonzero x , the integral of $x^* \Lambda x$ is $\|Lx\|^2 > 0$, and $m_k/r_k > 0$ almost everywhere. This argument also proves $g_x \in \mathcal{L}_k$ and the stated adjoint by direct pairing. Thus the preceding uses of these objects involve actual bounded maps on the specified domains.

Put $\iota P(u) = P(k/2 + iu)$. For every original polynomial P write $\mathcal{A}P$ and JP for the compatible union of the finite-window maps, whose definitions agree on every inclusion. Then the exact source identity is

$$\mathcal{A}P = \mathcal{J}_k(\iota P - gJP) - Y_\perp JP, \quad \|\mathcal{A}P\|^2 = \|\iota P - gJP\|_{\mathcal{L}_k}^2 + (JP)^* \Omega_k (JP). \quad (\text{PGB.19})$$

It follows by substituting $Y = \mathcal{J}_k g + Y_\perp$ into $\mathcal{A} = U - YJ$. The two summands lie in orthogonal closed subspaces. Define

$$\begin{aligned} \mathcal{H}_\infty &= \mathcal{J}_k \mathcal{L}_k \oplus^\perp Y_\perp E, & \Xi : \mathcal{L}_k \oplus (E, \Omega_k) &\longrightarrow \mathcal{H}_\infty, & (f, x) &\longmapsto \mathcal{J}_k f - Y_\perp x, \\ \Xi^{-1}(v) &= (\mathcal{J}_k^* v, -\Omega_k^{-1} Y_\perp^* v). \end{aligned} \quad (\text{PGB.20})$$

The space is closed because the first summand is the range of an isometry and the second is finite dimensional and orthogonal. The formulas $\mathcal{J}_k^* Y_\perp = 0$ and $Y_\perp^* Y_\perp = \Omega_k$ prove both inverse identities, and prove the exact squared norm $\|\Xi(f, x)\|^2 = \|f\|^2 + x^* \Omega_k x$. This is a complete isometric parametrization with its minus sign determined by the original subtraction in D .

In fact $\mathcal{H}_\infty$ is exactly the closure of the original polynomial graph image. Inclusion follows from PGB.19. Conversely take (f, x) in the domain of Ξ . The affine set

$$\{\iota(sx + \chi Q) - g_x : Q \in \mathbb{C}[S]\}$$

is dense in $\mathcal{L}_k$ by PGB.17. Choose Q_j for which these functions converge to f and put $P_j = sx + \chi Q_j$. Then $JP_j = x$ for every j , and PGB.19 proves $\mathcal{A}P_j \rightarrow \Xi(f, x)$. This proves surjectivity onto the completed graph from limits of original polynomials, retaining the prescribed quotient coordinate throughout. In the same manner

$$\overline{\mathcal{A}(\chi \mathbb{C}[S])} = \mathcal{J}_k \mathcal{L}_k, \quad \mathcal{H}_\infty / \overline{\mathcal{A}(\chi \mathbb{C}[S])} \simeq (E, \Omega_k). \quad (\text{PGB.21})$$

The quotient map on the whole completed source is the bounded map $v \mapsto -\Omega_k^{-1} Y_\perp^* v$; it sends $\mathcal{A}P$ to JP and has precisely the displayed closed relation space as kernel. Its section is $x \mapsto -Y_\perp x$. This explicitly computes the cochain completion and every map.

This completed source is a closed linear relation in $\mathcal{H}_k \oplus \mathcal{H}_k$, with its vertical part fully determined. Indeed $Y = \mathcal{J}_k g + Y_\perp$ gives

$$\Xi(f, x) = \mathcal{J}_k(f + g_x) - Yx, \quad \mathcal{H}_\infty \cap (\{0\} \oplus \mathcal{H}_k) = \{(0, -Lx) : x \in E\}. \quad (\text{PGB.21a})$$

If the first component vanishes, its squared norm is $\int |f + g_x|^2 m_k du = 0$. Since $m_k > 0$ almost everywhere, $f = -g_x$ almost everywhere, also as elements of $\mathcal{L}_k$; the source vector is therefore $-Yx$. Conversely $\Xi(-g_x, x) = -Yx$ proves the reverse inclusion. In particular the original densely defined defect operator $\mathcal{V}P \mapsto DP$ on the closure of the polynomial observation source is not closable. For any fixed $x \neq 0$, PGB.17 supplies $P_j = sx + \chi Q_j$ with $\iota P_j \rightarrow 0$ in $\mathcal{L}_k$. The original first components tend to zero, while $DP_j = \tau_k \mathcal{V}P_j - Lx \rightarrow -Lx \neq 0$. The two convergence assertions follow from the two component norms of $\mathcal{J}_k \iota P_j$. This witnesses the entire vertical subspace without asserting that the completed relation is the graph of a single-valued operator.

There is also an explicit map to the accepted unweighted completion. Put $\mathcal{L}_k^0 = L^2(m_k du)$ and

$$\mathcal{J}_k^0 : \mathcal{L}_k^0 \rightarrow \mathcal{H}_k, \quad f \mapsto \prod_i v_h(s_i) f(u), \quad I_{r,m} : \mathcal{L}_k \rightarrow \mathcal{L}_k^0, \quad f \mapsto f.$$

The first is an isometry by the original convolution integral; the second is a contraction because $r_k \geq m_k$. The first-coordinate projection π_1 obeys $\pi_1 \Xi(f, x) = \mathcal{J}_k^0 I_{r,m}(f + g_x)$ and $\pi_1 \mathcal{J}_k f = \mathcal{J}_k^0 I_{r,m} f$. Since the same exponential-moment argument proves that the unweighted relation closure is all of $\mathcal{J}_k^0 \mathcal{L}_k^0$, these formulas define a bounded cochain map

$$[\mathcal{J}_k \mathcal{L}_k \hookrightarrow \mathcal{H}_\infty] \xrightarrow{\pi_1} [\mathcal{J}_k^0 \mathcal{L}_k^0 \xrightarrow{I} \mathcal{J}_k^0 \mathcal{L}_k^0]. \quad (\text{PGB.21b})$$

It agrees on the original polynomial complex with $\mathcal{A}P \mapsto \mathcal{V}P$ and $\mathcal{A}(\chi Q) \mapsto \mathcal{V}(\chi Q)$. Its degree-one map is $(E, \Omega_k) \rightarrow 0$ with kernel all of E . Thus the positive completed graph quotient maps exactly to the zero unweighted completion in the accepted R66 source; its nonzero metric and the receiving zero object are connected by this specific cochain map and its full kernel.

312.4 The degree limit and the exact obstruction to a uniform comparison

For any $x \in E$, minimizing PGB.19 over $P \in \mathcal{P}_N$ with $JP = x$ gives

$$x^* \hat{G}_N x = x^* \Omega_k x + \inf_{Q \in \mathcal{P}_{N-q}} \|\iota(sx + \chi Q) - g_x\|_{\mathcal{L}_k}^2. \quad (\text{PGB.22})$$

At $N = q - 1$ the infimum contains exactly $Q = 0$. The nested relation spaces and their proved density show that the nonnegative second term decreases to zero for every x . Using polarization on the fixed finite remainder frame gives entrywise convergence, hence operator-norm convergence of the finite matrices. Thus, at fixed h, k ,

$$\hat{G}_N \downarrow \Omega_k > 0, \quad G_{W,N} \downarrow 0, \quad \Delta_N \downarrow \Omega_k, \quad G_N \downarrow 0. \quad (\text{PGB.23})$$

For $G_{W,N}$ repeat the distance calculation with $g = 0$ and without the Ω_k term, using the same r_k and PGB.17. For Δ_N , its PGS.40 expression is $\Omega_k + ((I - P_N)g)^*((I - P_N)g)$; polynomial density in $\mathcal{L}_k$ makes the latter term decrease to zero. For G_N repeat the affine approximation argument in $L^2(m_k du)$, whose exponential moments and dense relation image were proved by the same PGB.15–PGB.17 argument. These are matrix limits in the unchanged frame, not estimates in the separate tensor variable k .

The original averaged minimum form is defined at every $L_{\text{circ}} > 0$, including a phase with a singular finite matrix, by its actual source map T_θ into the sampled Hilbert space of HCD.18. Let P_θ^{rel} be orthogonal projection onto the finite-dimensional closed subspace $T_\theta(\text{im } B_N)$. Then

$$x^* G_{N,\theta} x = \|(I - P_\theta^{\text{rel}})T_\theta C_N x\|^2, \\ x^*(G_N - \overline{G}_N)x = \int \|P_\theta^{\text{rel}} T_\theta C_N x\|^2 d\mu.$$

The first equality follows by minimizing over the actual affine source image; the second follows from its orthogonal decomposition and $\int \|T_\theta C_N x\|^2 d\mu = x^* G_N x$. The projections are measurable: for

the map $T_\theta B_N$, their finite-rank formula is the pointwise limit as $t \downarrow 0$ of $(T_\theta B_N)(B_N^* M_{N,\theta} B_N + tI)^{-1}(T_\theta B_N)^*$. Diagonalization of the finite Gram proves this limit, including its zero eigenvalues. The displayed integrands are bounded above by the integrable full source norm, so both minimum forms and integrals are defined. Consequently $0 \preceq \overline{G}_N \preceq G_N$ at every original circumference, with no inverse imposed on a singular phase. Positivity of the finite $\overline{G}_N$ follows directly: for $k \geq 2$ every original sampled convolution weight is positive; for $k = 1$ all weights are positive outside a countable set of phases determined by the isolated real zeros of v_h . At such a phase a nonzero polynomial cannot vanish on infinitely many distinct sample coordinates, so its finite source Gram and its quotient Gram are positive definite. Integrating any nonzero quotient vector gives a strictly positive norm.

The identity $G_N \rightarrow 0$ is therefore precisely consistent with the accepted R66/CR conclusion. For these actual averaged minimum forms the augmentation identity gives

$$\mathcal{E}_N^{\text{aug}} = \widehat{G}_N - G_N \longrightarrow \Omega_k, \quad \widehat{G}_N - \overline{G}_N \longrightarrow \Omega_k \quad (\text{PGB.24})$$

at every fixed $L_{\text{circ}} > 0$. This conclusion also holds for any choices $L_{\text{circ}} = L_N$ satisfying PGB.8 at each finite cutoff; no interchange of the limit with a changing phase source was used, since the order bound alone gives $\|\overline{G}_N\| \leq \|G_N\| \rightarrow 0$.

Where $\overline{G}_N > 0$, which the accepted source proves at every finite N , the comparison has the explicit divergence

$$\inf_{x \neq 0} \frac{x^* \widehat{G}_N x}{x^* \overline{G}_N x} \geq \frac{\lambda_{\min}(\Omega_k)}{\|G_N\|} \longrightarrow +\infty. \quad (\text{PGB.25})$$

Indeed the numerator is at least $\lambda_{\min}(\Omega_k)\|x\|^2$, while the denominator is at most $\|G_N\|\|x\|^2$. Both constants refer to the same original remainder frame. Hence no finite constant independent of N can bound $\widehat{G}_N$ above by that constant times the accepted $\overline{G}_N$. The direction $\overline{G}_N \preceq \widehat{G}_N$ remains the exact cochain-induced contraction PGB.12–PGB.13. This proves an actual relation and its precise obstruction, rather than inferring lack of relation from the obstruction.

Finally fix the same original circumference at all four endpoints. For the inverse-form realization PGB.8, choose it using the original common maximal source $N_* = 2q + 2$ as in R63 and restrict its bound to every endpoint source. Alternatively use the proved minimum-form realization at any fixed circumference. The literal endpoint convention remains $(q - 1, q, 2q - 1, 2q)$ with signs $(+, +, -, -)$. Set $K_N^{\text{av}} = \overline{G}_N^{-1/2} \widehat{G}_N \overline{G}_N^{-1/2} > I$ at these finite endpoints. Direct determinant multiplication gives the exact signed bridge

$$\begin{aligned} & [\log \det \widehat{G}_{q-1} + \log \det \widehat{G}_q - \log \det \widehat{G}_{2q-1} - \log \det \widehat{G}_{2q}] \\ & - [\log \det \overline{G}_{q-1} + \log \det \overline{G}_q - \log \det \overline{G}_{2q-1} - \log \det \overline{G}_{2q}] \\ & = \log \det K_{q-1}^{\text{av}} + \log \det K_q^{\text{av}} - \log \det K_{2q-1}^{\text{av}} - \log \det K_{2q}^{\text{av}}. \end{aligned} \quad (\text{PGB.26})$$

Each separate logarithm on the right is nonnegative; their signed sum has no sign consequence from that fact alone. The identity retains all four original endpoints, including their possible repeated appearance when $q = 1$.

All the maps on E and polynomial coefficient spaces above extend to a fixed original active mask by applying the same linear map in each coordinate. For PGB.12 the represented-zero kernel is exactly the stated kernel in each coordinate; for the completed quotient PGB.21 it is the stated closed relation space. The empty present tuple stays present. External absence maps to external absence. At any of the already specified compatible realizations $R \rightarrow \mathbb{C}$, these complex-linear maps are R -linear with the given action, and their scalar extension sends $b \otimes x$ to $b \otimes f(x)$. No new residue-field realization, elimination of arithmetic nilpotents, or identification of supported zero with absence is made.

313 The sealed strict mixed residual inside the completed graph quotient

Retain the entire original packet, measures, coordinates and maps of PGS.1–41, PGM.1–12 and PGB.1–26, including PGB.21a–21b. The PGM source used here has SHA-256 105a4a7fc7a4f1853aea5884f58fc45a6dae74aab52a86784401e3662f310b40. Write $\mathcal{D}_k^{\text{mix}}$ for exactly the map called $\mathcal{D}_k$ in that source. Its original domain is $E = \mathbb{C}[S]/(\chi_{h,k})$, and its original target is

$$\mathcal{K}_k^{\text{mix}} = L^2 \left(\mathbb{R}^k, \left[1 + k^2 A^2 + \left(\sum_i q_\Phi(t_i) \right)^2 \right] \prod_i w(t_i) dt_i \right), \quad w(t) = |v_h(1/2 + it)|^2 / (2\pi). \quad (\text{PGMB.1})$$

This target is the PGM weighted scalar space, distinct from the closed two-component relation $\mathcal{K}_\infty$ of PGB.20. With $u = \sum_i t_i$, $\tau_k = kA + i \sum_i q_\Phi(t_i)$, F_x the full original successive remainder observation, and $g_x = z(u)x/r_k(u)$ the actual PGS observation adjoint, the map is

$$(\mathcal{D}_k^{\text{mix}}x)(t) = \frac{\overline{\tau_k} F_x(s)}{1 + |\tau_k|^2} - g_x(u), \quad (\mathcal{D}_k^{\text{mix}})^* \mathcal{D}_k^{\text{mix}} = \Omega_k - K_{\text{low}}. \quad (\text{PGMB.2})$$

All adjoints use the displayed original weighted target and the original increasing remainder frame of E . The second equality is the sealed PGM.5 exact Gram, with its complete proof from the orthogonal sum-coordinate projection. PGM.6–9 proves

$$\ker \mathcal{D}_k^{\text{mix}} = \{0\}, \quad \Omega_k - K_{\text{low}} > 0 \quad (k \geq 2), \quad \mathcal{D}_1^{\text{mix}} = 0, \quad \Omega_1 = K_{\text{low}}.$$

In particular this is not the finite polynomial defect $D_N = \tau_k \mathcal{V}_N - L J_N$ of PGB.2: its domain and target differ, and the latter is injective even when $k = 1$ by the original leading-coefficient proof after PGB.6.

The completed quotient of PGB.21 has the exact isometric map

$$\mathcal{I}_k : (E, \Omega_k) \longrightarrow (E, K_{\text{low}}) \oplus \mathcal{K}_k^{\text{mix}}, \quad x \longmapsto (x, \mathcal{D}_k^{\text{mix}}x). \quad (\text{PGMB.3})$$

Indeed the squared target norm is $x^* K_{\text{low}} x + \|\mathcal{D}_k^{\text{mix}}x\|^2 = x^* \Omega_k x$ by PGMB.2. Polarization proves equality of all inner products; the first coordinate proves injectivity. Projection to that coordinate is the actual identity contraction $(E, \Omega_k) \rightarrow (E, K_{\text{low}})$. Its loss is exactly the mixed map Gram, with zero loss kernel for $k \geq 2$ and full loss kernel E for $k = 1$. The quotient map itself has kernel zero in both cases. Its composition with the completed source quotient is explicitly $v \mapsto -\Omega_k^{-1} Y_\perp^* v$, now regarded as a vector of (E, K_{low}) ; its source kernel remains precisely $\mathcal{J}_k L^2(r_k du)$. Thus the strict PGM contribution is embedded in the existing completed morphism without altering its sign, quotient coordinate or relation kernel.

There is a compatible finite-window identity. Let $\hat{R}_N : E \rightarrow \mathcal{P}_N$ be the same actual minimum section of PGS.13 and PGB.6, and define the bounded map

$$f_N : E \longrightarrow L^2(r_k du), \quad f_N x = \iota \hat{R}_N x - g_x, \quad (\iota P)(u) = P(k/2 + iu). \quad (\text{PGMB.4})$$

Since $J_N \hat{R}_N = I_E$, PGB.19 evaluated on that section gives $\hat{G}_N = \Omega_k + f_N^* f_N$. Inserting PGMB.2 gives the full original Gram and its isometric realization

$$\begin{aligned} \hat{G}_N &= K_{\text{low}} + (\mathcal{D}_k^{\text{mix}})^* \mathcal{D}_k^{\text{mix}} + f_N^* f_N, \\ (E, \hat{G}_N) &\longrightarrow (E, K_{\text{low}}) \oplus \mathcal{K}_k^{\text{mix}} \oplus L^2(r_k du), \quad x \longmapsto (x, \mathcal{D}_k^{\text{mix}}x, f_N x). \end{aligned} \quad (\text{PGMB.5})$$

Each term is the norm of its displayed original map, so expanding the direct-sum norm proves this identity, and polarization proves every cross pairing. PGB.22–23 proves $f_N^* f_N = \hat{G}_N - \Omega_k \rightarrow 0$ at fixed original h, k . It follows that $\|f_N\| \rightarrow 0$, since its squared operator norm in the fixed coefficient

frame equals $\|f_N^* f_N\|$. The limiting isometry is exactly PGMB.3 with zero third component. Therefore the strict mixed contribution survives the polynomial degree limit for every $k \geq 2$, whereas the finite affine-approximation contribution tends to zero. The original relation $0 < K_{\text{low}} < \Omega_k \leq \hat{G}_N$ is now strict in its first comparison for every $k \geq 2$. No uniform eigenvalue lower bound in the distinct parameter k is inferred from this fixed-packet argument.

The first summand also has its exact original Hilbert-source realization. Define

$$\begin{aligned} L_{\text{low}} : E &\longrightarrow \mathcal{H}_k, & L_{\text{low}} x &= \frac{(\prod_i v_h(s_i)) F_x(s)}{\sqrt{1 + |\tau_k|^2}}, & L_{\text{low}}^* L_{\text{low}} &= K_{\text{low}}, \\ \mathfrak{U}_k : E &\longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k^{\text{mix}}, & \mathfrak{U}_k x &= (L_{\text{low}} x, \mathcal{D}_k^{\text{mix}} x), & \mathfrak{U}_k^* \mathfrak{U}_k &= \Omega_k. \end{aligned} \quad (\text{PGMB.6})$$

The square root is the positive real square root of the unchanged scalar $1 + k^2 A^2 + (\sum_i q_\Phi(t_i))^2$. The map lies in $\mathcal{H}_k$ since this reciprocal multiplier has absolute value at most one and $Lx \in \mathcal{H}_k$. Taking its original product integral gives exactly the entries of PGS.17, proving the first Gram identity. PGMB.2 proves the second. Thus $\mathfrak{U}_k$ is an isometry from (E, Ω_k) onto its actual closed finite-dimensional image. Its bounded left inverse on the whole displayed target is

$$(a, b) \longmapsto \Omega_k^{-1} (L_{\text{low}}^* a + (\mathcal{D}_k^{\text{mix}})^* b).$$

Composition with $\mathfrak{U}_k$ gives I_E by the computed Gram. The reverse composite is self-adjoint, idempotent, and has exactly $\mathfrak{U}_k(E)$ as its range, so it is the orthogonal projection in the original direct-sum Hilbert metric.

Let $j_\infty : \mathcal{H}_\infty \rightarrow E$ be the exact completed quotient map $j_\infty v = -\Omega_k^{-1} Y_\perp^* v$ of PGB.21. The complete original-source realization is

$$\begin{aligned} \mathfrak{C}_k &= \mathfrak{U}_k j_\infty : \mathcal{H}_\infty \longrightarrow \mathcal{H}_k \oplus \mathcal{H}_k^{\text{mix}}, \\ \mathfrak{C}_k \Xi(f, x) &= (L_{\text{low}} x, \mathcal{D}_k^{\text{mix}} x), \quad \ker \mathfrak{C}_k = \mathcal{J}_k L^2(r_k du), \quad \|\mathfrak{C}_k \Xi(f, x)\|^2 = x^* \Omega_k x. \end{aligned} \quad (\text{PGMB.7})$$

The identities follow from $j_\infty \Xi(f, x) = x$ and PGMB.6. Injectivity of $\mathfrak{U}_k$ and the computed kernel of j_∞ prove the complete kernel assertion. This is a map on the actual completed polynomial graph source; it factors through its specifically proved cyclic quotient. It asserts no factorization through an unidentified larger or proper-source quotient. Its norm equals the distance to the completed relation subspace by the orthogonal coordinates in PGB.20. In particular it is a contraction on that whole completed source and an isometry on its quotient.

For $k \geq 2$ the second component alone detects the entire quotient. Its inverse on the mixed image is exactly the already computed PGM.12 map

$$\mathcal{R}_k^{\text{mix}} = (\Omega_k - K_{\text{low}})^{-1} (\mathcal{D}_k^{\text{mix}})^*, \quad \mathcal{R}_k^{\text{mix}} \mathcal{D}_k^{\text{mix}} = I_E, \quad \mathcal{R}_k^{\text{mix}} \text{pr}_2 \mathfrak{C}_k = j_\infty. \quad (\text{PGMB.8})$$

The first composition follows by substituting PGMB.2; the second follows by composing that identity with j_∞ . For $k = 1$ the second component is identically zero by PGM.9; the first component still has positive Gram K_{low} and therefore retains the complete quotient.

The exact relative Gram and determinant cost can be written without changing the original coefficient frame. Set

$$\begin{aligned} H_k^{\text{mix}} &= \Omega_k - K_{\text{low}}, & S_k^{\text{mix}} &= K_{\text{low}}^{-1/2} H_k^{\text{mix}} K_{\text{low}}^{-1/2}, & q &= q_k = [1 + k(m-1)](k+1)^2, \\ 0 &\leq \lambda_{k,1} \leq \dots \leq \lambda_{k,q_k} & \text{the eigenvalues of } S_k^{\text{mix}}. \end{aligned}$$

These are generalized eigenvalues of the actual two original forms $H_k^{\text{mix}}, K_{\text{low}}$; the positive congruence specifies their computation and does not replace either form. Every $\lambda_{k,j} > 0$ for $k \geq 2$ and every $\lambda_{1,j} = 0$. Finite-dimensional Hermitian diagonalization and determinant multiplication give

$$\begin{aligned} K_{\text{low}}^{-1/2} \Omega_k K_{\text{low}}^{-1/2} &= I + S_k^{\text{mix}}, \\ (1 + \lambda_{k,1}) K_{\text{low}} &\preceq \Omega_k \preceq (1 + \lambda_{k,q_k}) K_{\text{low}}, \\ c_k^{\text{mix}} &:= \log \frac{\det \Omega_k}{\det K_{\text{low}}} = \sum_{j=1}^{q_k} \log(1 + \lambda_{k,j}), \quad \begin{cases} c_1^{\text{mix}} = 0, \\ c_k^{\text{mix}} > 0 \end{cases} \quad (k \geq 2). \end{aligned} \quad (\text{PGMB.9})$$

The extremal constants are exact: test the original vector $x = K_{\text{low}}^{-1/2}v$ for a corresponding eigenvector v . All k dependence is displayed through the unchanged degree q_k , the full matrices (PGS.17), (PGS.40), and (PGMB.2), and their stated eigenvalues. No bound for these eigenvalues uniform in k follows merely from their strict positivity.

At finite N put $S_{k,N}^{\text{tail}} = \Omega_k^{-1/2} f_N^* f_N \Omega_k^{-1/2}$ and let its nonnegative eigenvalues be $\nu_{k,N,1}, \dots, \nu_{k,N,q_k}$. Then

$$\log \frac{\det \widehat{G}_N}{\det K_{\text{low}}} = \sum_{j=1}^{q_k} \log(1 + \lambda_{k,j}) + \sum_{j=1}^{q_k} \log(1 + \nu_{k,N,j}), \quad \nu_{k,N,j} \rightarrow 0 \quad (N \rightarrow \infty; h, k \text{ fixed}).$$

(PGMB.10)

Indeed PGMB.5 gives $\widehat{G}_N = \Omega_k + f_N^* f_N$, so multiplying determinants after the positive congruence yields $\det \widehat{G}_N / \det \Omega_k = \det(I + S_{k,N}^{\text{tail}})$. The stated limit follows from the proved norm convergence of f_N and the fixed positive matrix Ω_k . Taking the four original signed endpoints cancels the same $\log \det \Omega_k$ or c_k^{mix} with coefficients $1+1-1-1=0$; their residual expression is exactly the signed sum of these four tail determinant costs. This cancellation does not remove Ω_k from the tail congruences and gives no estimate at the distinct growing endpoints $N \asymp q_k$ as k varies.

On each retained coordinate mask all these maps act coordinatewise. The represented-zero kernel of $\mathfrak{C}_k$ is the product of its displayed completed relation kernels, and the represented kernel of its second component is that same product for $k \geq 2$ and the whole completed fibre for $k = 1$. The present empty tuple remains present, and external absence maps to external absence with its singleton inverse image. Compatible coefficient realizations are exactly those specified in PGM.10, with the full kernel of their realized tensor map; composition with the injective quotient realizations above does not remove that earlier supported-zero fibre.

The finite graph-to-average morphism PGB.12–13 and the completed first-coordinate cochain map PGB.21b retain their original domains and kernels. In particular the former still has its strictly positive two-term loss relative to $\overline{G}_N$, and the latter still induces $(E, \Omega_k) \rightarrow 0$ with full coefficient kernel. The added mixed isometry computes the interior structure of that same Ω_k and supplies no identification of either target with the other. The signed four endpoints in PGB.26 and PGS.34 retain their complete differences; none receives a new sign from PGMB.2 or PGMB.5.

314 Exact generator control on the finite graph and its strict mixed quotient

Keep every source, coordinate and full-unit factor of PGS.1–41, PGB.1–26 and PGMB.1–10. In the original increasing remainder frame of $E = \mathbb{C}[S]/(\chi_{h,k})$, retain

$$A = M_S, \quad e_0 = [1], \quad \ell[P] = [S^{q-1}] \text{rem}_\chi P, \quad R = e_0 \ell, \quad T_t = A + tR - kI/2, \quad t \in \mathbb{C}. \quad (\text{PGGT.1})$$

The multiplication A retains every original primary nilpotent. The parameter t is the original period parameter; it is not the real metric interpolation parameter, the holonomy coordinate, or one of the original t_i variables.

There are three actual positive-form comparisons on this same coefficient space. The first two fix the original packet and finite cutoff $N \geq q - 1$; the averaged comparison also retains the common original circumference domain of PGB.8. The third uses the calculated completed quotient, with its original two weighted observation spaces from PGMB.1 and PGMB.6:

G	H	$B = G^{-1}(H - G)$	(PGGT.2)
G_N	$\widehat{G}_N$	$G_N^{-1} \mathcal{E}_N^{\text{aug}}$	
$\overline{G}_N$	$\widehat{G}_N$	$\overline{G}_N^{-1} (\mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}})$	
K_{low}	Ω_k	$K_{\text{low}}^{-1} (\Omega_k - K_{\text{low}})$	

All three rows contain the forms already calculated on their original sources. The complete PGB.6 and PGB.12 injectivity proofs make the first two B strictly positive in their displayed G metrics. The PGM.5–9 proof makes the third strictly positive for $k \geq 2$ and exactly zero for $k = 1$. Directly $GB = H - G$ is Hermitian, so $B^{\dagger G} = B$; and $x^*GBx = x^*(H - G)x$ proves each asserted sign in that same original metric. None of the three rows identifies a different source or completion with its neighbour.

For each row write the entire G -orthogonal spectral decomposition as $B = \sum_{i=1}^r b_i P_i$, where $0 \leq b_1 < \dots < b_r$, each P_i retains its full multiplicity, $\sum_i P_i = I$, and $P_i P_j = \delta_{ij} P_i$. These statements follow by diagonalizing the Hermitian matrix $G^{1/2} B G^{-1/2}$ and transporting its complete eigenspaces by $G^{-1/2}$. The transport changes no form: the projections satisfy $P_i^* G = G P_i$. Define

$$s_i = \sqrt{1 + b_i}, \quad S = \sum_i s_i P_i, \quad S^{-1} = \sum_i s_i^{-1} P_i, \quad \alpha = 1 + b_1 > 0. \quad (\text{PGGT.3})$$

Then $S^*GS = G(I + B) = H$, by multiplying the displayed sums. Thus this specific S is an isometry $(E, H) \rightarrow (E, G)$ with the displayed inverse. It need not commute with the original generator. That failure is computed next rather than omitted.

Put $X_H = T_t^{\dagger H} + T_t$, $X_G = T_t^{\dagger G} + T_t$ and $\epsilon_H(t) = \|X_H\|_H$, $\epsilon_G(t) = \|X_G\|_G$. The full transported error is

$$\begin{aligned} Z &= [S, T_t] S^{-1} = \sum_{i,j} \frac{s_i - s_j}{s_j} P_i T_t P_j, \\ SX_H S^{-1} - X_G &= Z + Z^{\dagger G}, \\ [B, T_t] &= [B, A] + t[B, R] = \sum_{i,j} (b_i - b_j) P_i T_t P_j. \end{aligned} \quad (\text{PGGT.4})$$

For the first line multiply the complete sums in PGGT.3 on both sides of T_t . For the second, $H = GS^2$ gives $T_t^{\dagger H} = S^{-2} T_t^{\dagger G} S^2$. Therefore the left side is $S^{-1} T_t^{\dagger G} S + S T_t S^{-1} - T_t^{\dagger G} - T_t$, whose two ordered differences are respectively $Z^{\dagger G}$ and Z . The last line follows from $[B, -kI/2] = 0$ and the complete spectral multiplication. This retains the original complex t ; taking the adjoint in the second line consequently retains $\bar{t}$ as well.

The exact G -operator norm of the Hermitian matrix in PGGT.4 is one control with all of these cancellations. A second directly evaluated control is

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \frac{\|[B, A] + t[B, R]\|_G}{1 + b_1} \leq \frac{b_r - b_1}{1 + b_1} \|T_t\|_G. \quad (\text{PGGT.5})$$

Here every denominator is the actual spectral minimum of the corresponding row of PGGT.2, not an additional hypothesis. To prove the middle inequality, the absolutely convergent finite-dimensional integral

$$Y = \int_0^\infty e^{-uS} [B, T_t] e^{-uS} du$$

has (i, j) block $(b_i - b_j)/(s_i + s_j) P_i T_t P_j$ and hence equals $[S, T_t]$, since $b_i - b_j = s_i^2 - s_j^2$. The full repeated eigenspaces remain present; their equal-eigenvalue blocks are zero. Because $s_i \geq \sqrt{\alpha}$, the integral norm is at most $\|[B, T_t]\|_G/(2\sqrt{\alpha})$. Multiplication by S^{-1} and adding its adjoint give the middle inequality, including its factor one. The first inequality is the reverse triangle inequality after the exact isometry PGGT.3. For the last, put $Q_j = \sum_{i>j} P_i$. Then

$$B = b_1 I + \sum_{j=1}^{r-1} (b_{j+1} - b_j) Q_j.$$

In the original G -orthogonal image/kernel decomposition of Q_j , the commutator $[Q_j, T_t]$ has off-diagonal blocks T_{12} and $-T_{21}$ and zero diagonal blocks. Its norm is $\max(\|T_{12}\|, \|T_{21}\|) \leq \|T_t\|_G$. The triangle inequality and the telescoping sum of the positive gaps prove the last bound. When $r = 1$, the sum is empty and every commutator and error above is zero.

The whole rank-one residue term remains explicit even before taking its norm. Write $x = e_0$, $y = G^{-1}\ell^*$; then $\ell(v) = \langle y, v \rangle_G$. Self-adjointness of B gives

$$[B, R]v = (Bx)\langle y, v \rangle_G - x\langle By, v \rangle_G, \quad [B, R] = [Bx, x][y, -By]^{\dagger G}. \quad (\text{PGGT.6})$$

The displayed expression proves the factorization on each original vector. Thus PGGT.4–5 contains its negative term and its full interference with $[B, A]$ and the original complex period parameter; no separate positive bound replaces that actual sum.

For the completed row the b_i in PGGT.3 are exactly the distinct values among the full $\lambda_{k,j}$ in PGMB.9. Thus its concrete last constant is $(\lambda_{k,q_k} - \lambda_{k,1})/(1 + \lambda_{k,1})$, and its earlier whole-error and commutator controls retain every eigenspace. At $k = 1$, $B = 0$, $S = I$ and the two Hermitian generators agree exactly. At $k \geq 2$ strict positivity supplies the displayed positive denominator; it supplies no unstated smallness of the spectral diameter as k varies.

Finally the completed generator has an exact realization on the original Hilbert targets, rather than only on a coefficient form. Retain $\mathfrak{U}_k = (L_{\text{low}}, \mathscr{D}_k^{\text{mix}})$ and $\mathfrak{U}_k^* \mathfrak{U}_k = \Omega_k$ from PGMB.6. On its original direct-sum target define

$$\tilde{T}_t = \mathfrak{U}_k T_t \Omega_k^{-1} \mathfrak{U}_k^*, \quad \tilde{T}_t \mathfrak{U}_k = \mathfrak{U}_k T_t, \quad \ker \tilde{T}_t = \mathfrak{U}_k(\ker T_t) \oplus^\perp \mathfrak{U}_k(E)^\perp. \quad (\text{PGGT.7})$$

Every map is bounded because its intermediate coefficient space is finite dimensional. The Gram identity proves the intertwining. Every target vector decomposes uniquely as $\mathfrak{U}_k x + v_\perp$, with $x = \Omega_k^{-1} \mathfrak{U}_k^* v$; the formula then sends it to $\mathfrak{U}_k T_t x$, proving the complete kernel. Taking adjoints in the original direct sum gives

$$\tilde{T}_t + \tilde{T}_t^* = \mathfrak{U}_k (T_t + T_t^{\dagger \Omega_k}) \Omega_k^{-1} \mathfrak{U}_k^*, \quad \|\tilde{T}_t + \tilde{T}_t^*\| = \epsilon_{\Omega_k}(t). \quad (\text{PGGT.8})$$

Indeed $T_t^{\dagger \Omega_k} = \Omega_k^{-1} T_t^* \Omega_k$, so both sides expand to the same two ordered terms. The isometry from (E, Ω_k) onto $\mathfrak{U}_k(E)$ identifies their restrictions, and both vanish on its orthogonal complement. This proves equality of the operator norms. PGGT.5 therefore controls this specified realized Hermitian generator through the original K_{low} generator with its exact mixed discrepancy. The calculation uses the actually computed finite-dimensional image. It asserts no identification with an unbounded ambient multiplication operator outside that image.

315 The original relation coefficients, their complete tail, and mixed source subtraction

Retain the full quartet and every original map of PGS.1–41, PGB.14–23 and PGMB.1–10. In particular

$$\begin{aligned}
 h(s) &= \prod_{r \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-r)^m, \quad \rho = \frac{1}{2} + \delta + i\gamma, \\
 0 &< \delta < \frac{1}{2}, \quad \gamma > 2, \quad m, k \geq 1, \\
 v_h &= 2\xi/h, \quad \Phi' = h, \quad \Phi(0) = 0, \quad \Phi(\frac{1}{2} + it) = A + iq\Phi(t), \\
 s_i &= \frac{1}{2} + it_i, \quad u = \sum_i t_i, \quad S(u) = k/2 + iu, \\
 \tau_k &= \sum_i \Phi(s_i), \quad w(t) = |v_h(\frac{1}{2} + it)|^2/(2\pi), \\
 \chi(S) &= \prod_{a,b=0}^k (S - k/2 - (2a-k)\delta - i(2b-k)\gamma)^{1+k(m-1)}, \\
 q &= [1 + k(m-1)](k+1)^2, \quad E = \mathbb{C}[S]/(\chi).
 \end{aligned} \tag{PGRT.1}$$

The frame of E is $1, S, \dots, S^{q-1}$. The map $s : E \rightarrow \mathbb{C}[S]$ is the unique remainder of degree less than q and $\iota P(u) = P(S(u))$. The measure and full observation are

$$\begin{aligned}
 r_k(u) &= \int_{\sum t_i=u} (1 + |\tau_k|^2) \prod_i w(t_i), \quad \mathcal{L}_k = L^2(r_k(u) du), \\
 F_b &= \text{rem}_{h(s_1), \dots, h(s_k)}(\tau_k \Sigma^b), \quad \Sigma = \sum_i s_i, \quad z_b(u) = \int_{\sum t_i=u} \bar{\tau}_k F_b \prod_i w(t_i), \\
 g_x(u) &= z(u)x/r_k(u), \quad F^0 x = \iota s x - g_x.
 \end{aligned} \tag{PGRT.2}$$

The fibre integral eliminates t_k and uses $dt_1 \cdots dt_{k-1}$; at $k=1$ it is evaluation. Null exceptional sets in z/r_k have arbitrary values. PGS.39 proves $g : E \rightarrow \mathcal{L}_k$ is bounded. All polynomial columns belong to $\mathcal{L}_k$ by the proved exponential moments PGB.15. Inner products are conjugate linear in their first variable. The original masses and full units remain in every integral in this section.

315.1 An exact analysis map for the unchanged relation source

For each integer $j \geq q-1$ put $n = j+1-q$. There is a unique monic polynomial p_n of degree n orthogonal to all lower polynomials for the measure $|\chi(S(u))|^2 r_k(u) du$. Indeed the Gram of $1, S, \dots, S^{n-1}$ is positive definite: a nonzero polynomial cannot vanish almost everywhere for a density positive almost everywhere. Solving that Gram system for the lower coefficients of S^n gives existence and uniqueness, including $p_0 = 1$. Define

$$d_j = \chi p_{j+1-q}, \quad a_j = \int |d_j(S(u))|^2 r_k(u) du > 0, \quad \alpha_j x = \langle \iota d_j, F^0 x \rangle_{\mathcal{L}_k}. \tag{PGRT.3}$$

These d_j are exactly PGS.25: subtracting the old relation projection from χS^{j+1-q} leaves a monic relation of the same degree, orthogonal to $\chi \mathcal{P}_{j-q}$; uniqueness just proved identifies the two constructions. In particular there is no change of the monic relation columns or their lengths.

The complete orthogonal family ιd_j spans a dense subspace of $\mathcal{L}_k$. For specificity, PGB.15 proves an exponential moment for $|\chi|^2 r_k du$. Orthogonality to all its polynomials forces the Fourier transform of the resulting finite complex measure to have all derivatives zero on a nonempty analytic strip; Fourier uniqueness, proved there by convolution with the exact unit-mass Gaussian, forces that measure to vanish. Thus polynomials are dense for this measure. Multiplication by $\chi(S(u))$ is the onto

isometry to $\mathcal{L}_k$, with inverse division off its finite zero set, by the direct norm identity PGB.16. These facts prove the asserted completeness of precisely these d_j .

Keep their norms in the sequence space

$$\ell^2(a^{-1}) = \{(c_j)_{j \geq q-1} : \sum_{j \geq q-1} |c_j|^2 / a_j < \infty\}.$$

The exact analysis and synthesis maps are

$$\mathcal{F} : \mathcal{L}_k \longrightarrow \ell^2(a^{-1}), \quad f \longmapsto (\langle \iota d_j, f \rangle)_j, \quad \mathcal{F}^{-1}c = \sum_{j=q-1}^{\infty} a_j^{-1} c_j \iota d_j. \quad (\text{PGRT.4})$$

Orthogonality proves that the squared norm of each finite sum is $\sum |c_j|^2 / a_j$, hence proves convergence for every displayed sequence. Its coefficients are c_j , first for finite sequences and then by continuity of each inner product. Conversely subtract the synthesis of the coefficients of f ; finite orthogonal projection gives the Bessel bound, and the difference has zero inner product with the complete family, hence is zero. This proves both inverse identities, norm equality and zero kernels. No orthogonal column has been rescaled in this construction.

Let P_N^{rel} denote orthogonal projection onto $\iota(\chi \mathcal{P}_{N-q})$, with zero image for $N = q - 1$. The projection has the exact expression

$$P_N^{\text{rel}} f = \sum_{j=q-1}^{N-1} a_j^{-1} \langle \iota d_j, f \rangle \iota d_j, \quad f_N = (I - P_N^{\text{rel}}) F^0. \quad (\text{PGRT.5})$$

To identify f_N with PGMB.4, minimize the PGB.19 norm over $P = sx + \chi Q$, $\deg P \leq N$. Its scalar component is $F^0 x + \iota \chi Q$, so its unique minimum is obtained by subtracting $P_N^{\text{rel}} F^0 x$. The component has exactly the form $\iota \hat{R}_N x - g_x$. This proves the displayed identity for the original section rather than selecting another lift.

The complete residual form is therefore

$$T_N := \hat{G}_N - \Omega_k = f_N^* f_N = \sum_{j=N}^{\infty} Q_j, \quad Q_j = a_j^{-1} \alpha_j^* \alpha_j, \quad \Omega_k = Y_{\perp}^* Y_{\perp} > 0. \quad (\text{PGRT.6})$$

For each x , PGRT.4–5 prove equality of the associated squared norms. Polarization gives every matrix entry in the stated frame. The series converges in operator norm: its positive tail has trace equal to the sum of the squared tail norms of the q fixed frame vectors, which tends to zero by PGRT.4; its operator norm is at most that trace. The exact kernel, without imposing additional analytic injectivity, is

$$\ker f_N = \bigcap_{j \geq N} \ker \alpha_j = \{x \in E : F^0 x \in \iota(\chi \mathcal{P}_{N-q})\} = \{x : g_x = \iota P \text{ for some } \deg P \leq N, [P] = x\}. \quad (\text{PGRT.7})$$

The first equality is the positive norm sum. The second is the projection kernel, and the third follows by subtracting the stated relation from sx . This is the full supported-zero fibre of the tail map, while the analysis isometry itself has zero kernel.

Every row in this expansion has its original arithmetic integral:

$$(\alpha_j)_b = \int_{\mathbb{R}} \overline{d_j(S(u))} (S(u)^b r_k(u) - z_b(u)) du. \quad (\text{PGRT.8})$$

It is absolutely convergent by Cauchy–Schwarz. Expanding the polynomials and inserting the PGS.35 and PGS.38 convolution formulas expresses every term as a finite sum of products of $M_{ab}^{(h)} = \int \bar{s}^a s^b |v_h(s)|^2 dt / (2\pi)$. The coefficients of d_j are obtained by the positive Gram system in PGRT.3, whose entries are those same finite moments. Since $\hat{R}_j - s$ is an old relation, its scalar component pairs to zero with d_j . Thus this row is exactly $\beta_j - d_j^* Z_{j+1}$ in PGS.25, including its negative cross term. This proves an explicit finite arithmetic route to each coefficient, even though the complete tail is infinite.

315.2 A strict mixed subtraction in the endpoint upper bound

Retain the original Hilbert maps $D_{\text{mix}} = \mathcal{D}_k^{\text{mix}} : E \rightarrow \mathcal{H}_k^{\text{mix}}$ and $L_{\text{low}} : E \rightarrow \mathcal{H}_k$ of PGMB.2 and PGMB.6. For a fixed integer $M \geq 2q$ define

$$H_M = \widehat{G}_M = K_{\text{low}} + Z_M^{\text{mix}*} Z_M^{\text{mix}}, \quad Z_M^{\text{mix}} x = (D_{\text{mix}} x, f_M x). \quad (\text{PGRT.9})$$

Here Z_M^{mix} maps into the ordered Hilbert direct sum $\mathcal{H}_k^{\text{mix}} \oplus \mathcal{L}_k$; it is not the finite cross-moment matrix Z_M of PGS.5. The identity is exactly PGMB.5. The operator $I + Z_M^{\text{mix}} K_{\text{low}}^{-1} Z_M^{\text{mix}*}$ is invertible: on the orthogonal complement of the finite-dimensional range of Z_M^{mix} it is the identity, and on that range it is positive with all eigenvalues at least one. Direct multiplication of both sides by H_M proves

$$H_M^{-1} = K_{\text{low}}^{-1} - K_{\text{low}}^{-1} Z_M^{\text{mix}*} (I + Z_M^{\text{mix}} K_{\text{low}}^{-1} Z_M^{\text{mix}*})^{-1} Z_M^{\text{mix}} K_{\text{low}}^{-1}. \quad (\text{PGRT.10})$$

For the multiplication, write $K = K_{\text{low}}$ and $Z = Z_M^{\text{mix}}$; the leftover terms factor as $Z^* Z K^{-1} - Z^* (I + Z K^{-1} Z^*) (I + Z K^{-1} Z^*)^{-1} Z K^{-1} = 0$. Thus all ordered products in the inverse formula are retained.

For $q - 1 \leq j \leq 2q - 1$ its scalar consequence is the exact nonnegative value

$$\begin{aligned} u_{j,M} &= a_j^{-1} \alpha_j H_M^{-1} \alpha_j^* \\ &= a_j^{-1} \alpha_j K_{\text{low}}^{-1} \alpha_j^* - a_j^{-1} \left\| (I + Z_M^{\text{mix}} K_{\text{low}}^{-1} Z_M^{\text{mix}*})^{-1/2} Z_M^{\text{mix}} K_{\text{low}}^{-1} \alpha_j^* \right\|^2. \end{aligned} \quad (\text{PGRT.11})$$

If $k \geq 2$ and $\alpha_j \neq 0$, the subtracted term is strictly positive. Indeed PGM.6–9 proves D_{mix} injective, so $Z_M^{\text{mix}} K_{\text{low}}^{-1} \alpha_j^* \neq 0$, and the displayed positive inverse square root has zero kernel. When $k = 1$ the mixed component is zero, but its finite tail component is still present; no strictness is asserted from the vanishing component. If $\alpha_j = 0$, both terms are exactly zero. Consequently the strict mixed contribution gives a calculated subtraction at each nonzero relation row, not merely a positive addition to an unrelated norm.

Let $\widehat{B}_k$ denote exactly the four signs of PGS.29, and retain $\omega_{q-1} = \omega_{2q-1} = 1$ and $\omega_j = 2$ for $q \leq j \leq 2q - 2$. Set

$$l_j = a_j^{-1} \alpha_j \widehat{G}_{q-1}^{-1} \alpha_j^*, \quad \eta_j = a_j^{-1} \alpha_j \widehat{G}_{j+1}^{-1} \alpha_j^*.$$

PGRT.6 gives $\widehat{G}_j = \widehat{G}_{j+1} + Q_j$. Congruence with $\widehat{G}_{j+1}^{-1/2}$ gives a rank-one addition with eigenvalues $1 + \eta_j, 1, \dots, 1$; extending the nonzero update vector to an orthonormal basis proves this statement, and the zero vector gives the identity matrix. Therefore

$$\begin{aligned} \log \frac{\det \widehat{G}_j}{\det \widehat{G}_{j+1}} &= \log(1 + \eta_j), \quad \widehat{B}_k = \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \eta_j), \\ L_k^{\text{rel}} &:= \sum_j \omega_j \log(1 + l_j) \leq \widehat{B}_k \leq U_{k,M}^{\text{rel}} := \sum_j \omega_j \log(1 + u_{j,M}). \end{aligned} \quad (\text{PGRT.12})$$

For the inequalities use $H_M \preceq \widehat{G}_{j+1} \preceq \widehat{G}_{q-1}$. For positive forms inverse order reverses: conjugate by the square root of the smaller form, diagonalize the resulting matrix with eigenvalues at least one, invert them, and undo the congruence. Thus $l_j \leq \eta_j \leq u_{j,M}$, and increasing logarithms prove PGRT.12. Telescoping along the two exact endpoint intervals proves the weights. For $q = 1$ the interior is empty and the two steps remain distinct, so no repeated endpoint is dropped before the cancellation. The actual quartet has its full q in PGRT.1.

The choice $M = 2q$ gives the strongest upper bound among these common later-window comparisons, because H_M decreases as M increases. All H_M in PGRT.11–12 are finite arithmetic Gram matrices computed by PGS.9–13. At $M \rightarrow \infty$ they tend to Ω_k , and $f_M \rightarrow 0$, so PGRT.10–11 also prove the exact limiting subtraction with $Z_{\infty}^{\text{mix}} = D_{\text{mix}}$ and $H_{\infty} = \Omega_k$. No rate in the separate variable k has been presumed. The strongest available local upper bound can retain the minimum of PGRT.12 and the proved PGMB.10 value $t_{q-1} + t_q$, where $t_N = \log \det(I + \Omega_k^{-1/2} T_N \Omega_k^{-1/2})$.

315.3 Return to the original arithmetic endpoint

Write $\Gamma_k^{\text{gr}} = \widehat{\mathcal{B}}_k - \mathcal{B}_k$ for the literal signed augmentation. Choose integers $p, s \geq 0$ and set $g_s^\pm := g_{s,\text{gr}}^\pm$ exactly as defined in ACM27. Keep its previously proved finite original-source enclosure $g_s^- \leq \Gamma_k^{\text{gr}} \leq g_s^+$. This enclosure is the output of its full signed projector calculation and its proved finite Neumann remainder, not an assumed estimate. Subtracting those two inequalities from PGRT.12 yields

$$L_k^{\text{rel}} - g_s^+ \leq \mathcal{B}_k \leq U_{k,2q}^{\text{rel}} - g_s^-. \quad (\text{PGRT.13})$$

Intersecting this interval with the already proved Gamma-to-arithmetic interval $[\beta_{p,\Gamma}^-, \beta_{p,\Gamma}^+]$ gives

$$\begin{aligned} b_{p,s}^{\text{rel},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{rel}} - g_s^+\}, \\ b_{p,s}^{\text{rel},+} &= \min\{\beta_{p,\Gamma}^+, U_{k,2q}^{\text{rel}} - g_s^-\}, \quad b_{p,s}^{\text{rel},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{rel},+}. \end{aligned} \quad (\text{PGRT.14})$$

Each input is a defined original arithmetic matrix or a bound already proved for it. The interval is nonempty because it contains the same $\mathcal{B}_k$. At HC9–10 use the unchanged $T_k = 4q \log(D_h k)$: subtract T_k from both endpoints for $\mathcal{R}_k$, and subtract $\mathcal{B}_k^\Gamma$ for Δ_k^Γ . Intersect with the earlier lower and upper inequalities before these subtractions. Their validity follows directly from equality of the retained scalar and does not discard the negative g_s^- term. The phase propagation HC14g then uses the resulting arithmetic upper endpoint in each of its three increasing scalar functions. This is the exact return map of the new relation calculation into the existing endpoint program. For these HC substitutions retain its receiving domain $k \geq 3$, $\gamma > 2$, and $0 < b < \pi/2$ in its original constants. In HC14g also retain its original phase θ and circumference $L > 0$, with the proved source comparison $\kappa_{a_{\text{wt}},2q+2}/(e^{a_{\text{wt}}L} - 1) \leq \eta$, $a_{\text{wt}} > 0$ and $0 < \eta < 1$, and its nonnegative integer phase series degree. The tail calculation PGRT.1–12 itself continues to hold on its stated domain $k \geq 1$.

The exact graph-volume input of HC9g is retained. The relation interval in PGRT.14 lies inside the Gamma-only interval, and contains the HC9g interval, because $L_k^{\text{rel}} \leq \widehat{\mathcal{B}}_k \leq U_{k,2q}^{\text{rel}}$. This follows by taking the maximum of the respective lower endpoints and the minimum of their upper endpoints. It specifies the exact order of the two evaluation routes; using a bound for an already known exact graph volume does not improve that exact input. The mixed subtraction in PGRT.11 improves the same relation-row upper comparison made with K_{low}^{-1} .

All analysis, projection, synthesis and quotient maps in this section act coordinatewise on every fixed original support mask. Their represented-zero fibres are products of the displayed linear kernels. A present empty tuple maps to a present empty tuple, and external absence maps to external absence with singleton inverse image. None of the Hilbert completions identifies those two fibres. The unresolved growing- k analytic calculation is now the actual size of the rows PGRT.8 in the full inverses PGRT.11–12 together with the signed arithmetic return PGRT.13; the complete identities here neither remove that calculation nor substitute an assumed bound.

316 The full correlations of the original relation rows

Retain the complete quartet, full Taylor unit, original theta measure, ordered tensor coordinates and monic relation columns of PGRT.1–8. In particular $k, m \geq 1$, $0 < \delta < 1/2$, $\gamma > 2$, $q = [1 + k(m - 1)](k + 1)^2$, and $E = \mathbb{C}[S]/(\chi)$ has its original remainder frame $1, S, \dots, S^{q-1}$. Every $a_j > 0$ is the original squared length of $d_j = \chi p_{j+1-q}$, and $\alpha_j : E \rightarrow \mathbb{C}$ is the exact integral PGRT.8. No relation column or measure is rescaled below. For finite coefficient matrices, stars denote conjugate transposes in the indicated frames. For maps into a stated Hilbert target, including Z below, stars denote adjoints for that target and the stated coefficient metric. All Hilbert inner products are conjugate linear in the first variable.

316.1 A finite source map for all the retained correlations

Fix the original last endpoint and the complete row index set

$$H = \widehat{G}_{2q}, \quad I = \{q - 1, q, \dots, 2q - 1\}, \quad C_{ij} = \alpha_i H^{-1} \alpha_j^* \quad (i, j \in I). \quad (\text{PGRC.1})$$

Here $H > 0$ is the actual graph quotient form computed by PGS.9–13. Thus each entry of C is determined by the original finite theta moments, the full tensor remainder, and the inverse of that form. For $J \subset I$, in the inherited order, let A_J have rows α_j , let $D_J = \text{diag}(a_j : j \in J)$ and put

$$H_J = H + A_J^* D_J^{-1} A_J, \quad B_J = D_J + C_{JJ}. \quad (\text{PGRC.2})$$

The empty set means the zero-dimensional coordinate space, zero row map, and $H_\emptyset = H$. The determinants of $B_\emptyset$ and $D_\emptyset$ are one, while $\det H_\emptyset = \det H$. Both H_J and B_J are positive definite on their respective spaces: their quadratic forms are the sum of a positive original form and a nonnegative squared norm.

The precise graph realization, including its inverse on the image, is

$$\begin{aligned} V_J : (E, H_J) &\longrightarrow (E, H) \oplus (\mathbb{C}^J, D_J^{-1}), & V_J x &= (x, A_J x), \\ V_J^\dagger(y, z) &= H_J^{-1}(Hy + A_J^* D_J^{-1} z), & V_J^\dagger V_J &= I_E, \\ \ker V_J &= \{0\}, & \ker V_J^\dagger &= \{(y, z) : Hy + A_J^* D_J^{-1} z = 0\}. \end{aligned} \quad (\text{PGRC.3})$$

Indeed expanding the norm of $(x, A_J x)$ gives $x^* H_J x$, and pairing this vector with (y, z) proves the stated adjoint. Multiplication proves the left inverse and the kernel formula. Consequently $V_J V_J^\dagger$ is the orthogonal projection onto this image; it is self-adjoint and its square equals itself. This construction retains each original relation length in the coordinate-space metric D_J^{-1} .

Direct multiplication, with every factor in its stated order, gives

$$H_J^{-1} = H^{-1} - H^{-1} A_J^* B_J^{-1} A_J H^{-1}. \quad (\text{PGRC.4})$$

To verify it, multiply the right side on the left by H_J . The extra terms, after factoring A_J^* on the left and $A_J H^{-1}$ on the right, have middle factor $D_J^{-1} - B_J^{-1} - D_J^{-1} C_{JJ} B_J^{-1} = 0$, because $I + D_J^{-1} C_{JJ} = D_J^{-1} B_J$. The positive forms are invertible, so this is the claimed inverse.

For $j \in I$ and a subset of its strictly later rows $J \subset J_j := \{j + 1, \dots, 2q - 1\}$ define

$$r_j(J) = C_{jJ} B_J^{-1} C_{Jj}, \quad v_j(J) = \frac{C_{jj} - r_j(J)}{a_j} = \frac{\alpha_j H_J^{-1} \alpha_j^*}{a_j} \geq 0. \quad (\text{PGRC.5})$$

The empty-set correction is zero. The second expression follows by PGRC.4 and proves the sign without discarding a cross term. The correction $r_j(J)$ is zero exactly when $C_{Jj} = 0$, since $B_J^{-1} > 0$. In particular its strictness is determined by the actual original-metric correlations of the rows, not by their mere presence as formal coordinates.

PGRT.6 gives the precise identification for all later rows:

$$H_{J_j} = \widehat{G}_{j+1}, \quad \eta_j = v_j(J_j) = \frac{C_{jj} - C_{jJ_j}(D_{J_j} + C_{J_jJ_j})^{-1}C_{J_jj}}{a_j}. \quad (\text{PGRC.6})$$

The first identity telescopes the finite sum $\widehat{G}_{j+1} - \widehat{G}_{2q} = \sum_{i=j+1}^{2q-1} \alpha_i^* \alpha_i / a_i$. Inserting it into the proved inverse calculation gives the second. This evaluates exactly the same coefficient η_j as PGRT.12.

316.2 Every omitted correlation and an explicit finite remainder

For disjoint $J, K \subset J_j$, define the full remaining block and row, retaining their subtraction:

$$\begin{aligned} S_{K|J} &= D_K + C_{KK} - C_{KJ}B_J^{-1}C_{JK}, \\ e_{j,K|J} &= C_{jK} - C_{jJ}B_J^{-1}C_{JK}. \end{aligned} \quad (\text{PGRC.7})$$

They have exact source expressions

$$S_{K|J} = D_K + A_K H_J^{-1} A_K^* \succeq D_K > 0, \quad e_{j,K|J} = \alpha_j H_J^{-1} A_K^*. \quad (\text{PGRC.8})$$

These follow by substituting PGRC.4. The block calculation uses the exact coordinate restriction permutation $P_{J,K} : \mathbb{C}^{J \cup K} \rightarrow \mathbb{C}^J \oplus \mathbb{C}^K, z \mapsto (z|_J, z|_K)$, whose inverse reassembles the coordinates in the original inherited order. Thus $P_{J,K} B_{J \cup K} P_{J,K}^*$ has diagonal blocks $B_J, D_K + C_{KK}$ and off-diagonal blocks C_{JK}, C_{KJ} . The row $C_{j,J \cup K} P_{J,K}^*$ is (C_{jJ}, C_{jK}) . Since $P_{J,K}^* P_{J,K} = I$, their inverse quadratic pairing $r_j(J \cup K)$ is unchanged. This explicitly covers interleaved indices. Block elimination in these specified coordinates gives

$$r_j(J \cup K) - r_j(J) = e_{j,K|J} S_{K|J}^{-1} e_{j,K|J}^*, \quad (\text{PGRC.9})$$

which also holds if either block is empty. Here is the elimination calculation. The equations $B_J x + C_{JK} y = C_{jJ}$ and $C_{KJ} x + (D_K + C_{KK}) y = C_{jK}$ first give $x = B_J^{-1}(C_{jJ} - C_{JK} y)$ and then $S_{K|J} y = e_{j,K|J}^*$. Pair the first solution and this second solution with (C_{jJ}, C_{jK}) ; the resulting scalar is $r_j(J) + e_{j,K|J} S_{K|J}^{-1} e_{j,K|J}^*$, which is the left side's full block inverse expression.

Since $S_{K|J} \succeq D_K$, inverse order yields the fully explicit estimate

$$0 \leq r_j(J \cup K) - r_j(J) \leq \sum_{i \in K} \frac{|(e_{j,K|J})_i|^2}{a_i}. \quad (\text{PGRC.10})$$

For completeness, conjugating the positive inequality by $D_K^{-1/2}$ produces a positive matrix with eigenvalues at least one. Its inverse has eigenvalues at most one; undoing the congruence proves $S_{K|J}^{-1} \preceq D_K^{-1}$. Pairing with the original row proves PGRC.10. No row's length has changed during this argument.

For a single additional index b , PGRC.9 is the exact update

$$r_j(J \cup \{b\}) = r_j(J) + \frac{|C_{jb} - C_{jJ} B_J^{-1} C_{Jb}|^2}{a_b + C_{bb} - C_{bJ} B_J^{-1} C_{Jb}}. \quad (\text{PGRC.11})$$

Its denominator is at least $a_b > 0$ by PGRC.8. The update is strictly positive exactly when its displayed numerator is nonzero. Thus ordered row insertion monotonically decreases $v_j(J)$; every equality case is given by the stated cross row.

Choose any subsets $J(j) \subset J_j$, and let $K(j) = J_j \setminus J(j)$. Set

$$\begin{aligned} \bar{\eta}_j &= v_j(J(j)), \\ \underline{\eta}_j &= \max \left\{ l_j, v_j(J(j)) - \sum_{i \in K(j)} \frac{|(e_{j,K(j)|J(j)})_i|^2}{a_j a_i} \right\}, \quad l_j = \frac{\alpha_j \widehat{G}_{q-1}^{-1} \alpha_j^*}{a_j}. \end{aligned} \quad (\text{PGRC.12})$$

PGRT.12 gives $0 \leq l_j \leq \eta_j$. Equations PGRC.9–10 with $J \cup K = J_j$ prove $0 \leq \underline{\eta}_j \leq \eta_j \leq \bar{\eta}_j$. Using the original weights $\omega_{q-1} = \omega_{2q-1} = 1$ and $\omega_j = 2$ in between, one obtains the finite original-source bounds

$$\begin{aligned} L_k^{\text{corr}}(J) &= \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \underline{\eta}_j), \\ U_k^{\text{corr}}(J) &= \sum_{j=q-1}^{2q-1} \omega_j \log(1 + \bar{\eta}_j), \\ L_k^{\text{rel}} &\leq L_k^{\text{corr}}(J) \leq \hat{B}_k \leq U_k^{\text{corr}}(J) \leq U_{k,2q}^{\text{rel}}. \end{aligned} \quad (\text{PGRC.13})$$

The first inequality uses $\underline{\eta}_j \geq l_j$; the last uses $r_j(J) \geq 0$ and $C_{jj}/a_j = u_{j,2q}$ in PGRT.11. The middle inequalities follow from the increasing logarithm and the exact weighted sum PGRT.12. Taking all $J(j) = J_j$ makes both bounds equal to $\hat{B}_k$, since then every remainder is zero. Taking the first $\min(n, |J_j|)$ later rows gives decreasing upper bounds, exact after at most q insertions per row. Each step has the original finite dimensions and can be evaluated using PGRC.11. Lower bounds may be retained by taking the maximum over all previous insertions, which preserves their validity and makes that retained sequence increasing.

The excess of the partial upper bound is itself evaluated by

$$U_k^{\text{corr}}(J) - \hat{B}_k = \sum_j \omega_j \log \left(1 + \frac{e_{j,K(j)|J(j)} S_{K(j)|J(j)}^{-1} e_{j,K(j)|J(j)}^*}{a_j(1 + \eta_j)} \right). \quad (\text{PGRC.14})$$

Indeed PGRC.9 gives $\bar{\eta}_j - \eta_j = e S^{-1} e^* / a_j$; divide $(1 + \bar{\eta}_j)/(1 + \eta_j)$ and take logarithms. This also proves a finite computable bound by replacing the numerator with PGRC.10 and η_j with $\underline{\eta}_j$, retaining each sign and each row length.

316.3 Both negative corrections in the same original inverse

Keep $Z = Z_{2q}^{\text{mix}} = (D_{\text{mix}}, f_{2q})$ from PGRT.9, mapping into its original ordered Hilbert sum. PGRT.10–11 and PGRC.6 together give the exact double subtraction

$$\begin{aligned} a_j \eta_j &= \alpha_j K_{\text{low}}^{-1} \alpha_j^* \\ &- \left\| (I + Z K_{\text{low}}^{-1} Z^*)^{-1/2} Z K_{\text{low}}^{-1} \alpha_j^* \right\|^2 \\ &- C_{jJ_j} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j j}. \end{aligned} \quad (\text{PGRC.15})$$

Every C on the last line is computed using the same full $H = \hat{G}_{2q}$; it is not replaced by a Gram in K_{low} . The first subtracted squared norm is positive for $k \geq 2$ and $\alpha_j \neq 0$, by the proved injectivity of D_{mix} . The second is positive exactly when $C_{J_j j} \neq 0$, by PGRC.5. Both can vanish in their explicitly stated cases. Their sum remains at most the first nonnegative term, because the left side is the nonnegative inverse quadratic value of PGRC.6. Thus the mixed residual and the relation correlations have been carried into one original quantity with their full negative signs.

The arithmetic receiving calculation retains the exact graph route HC9g. For integers $p, s \geq 0$ use the proved ACM27 $g_s^\pm = g_{s,\text{gr}}^\pm$ and define

$$\begin{aligned} b_{p,s}^{\text{corr},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{corr}}(J) - g_s^+\}, \\ b_{p,s}^{\text{corr},+} &= \min\{\beta_{p,\Gamma}^+, U_k^{\text{corr}}(J) - g_s^-\}. \end{aligned} \quad (\text{PGRC.16})$$

Subtraction of the two proved bounds on the literal signed augmentation gives an interval containing $\mathcal{B}_k$. The endpoints obey the exact inclusion order

$$[b_{p,s}^{\text{gr},-}, b_{p,s}^{\text{gr},+}] \subset [b_{p,s}^{\text{corr},-}, b_{p,s}^{\text{corr},+}] \subset [b_{p,s}^{\text{rel},-}, b_{p,s}^{\text{rel},+}] \subset [\beta_{p,\Gamma}^-, \beta_{p,\Gamma}^+]. \quad (\text{PGRC.17})$$

For the first inclusion use $L_k^{\text{corr}} \leq \widehat{\mathcal{B}}_k \leq U_k^{\text{corr}}$; for the second use PGRC.13; and for the third use the maximum and minimum in the definitions. With all later rows included, the first two endpoints equal the exact graph endpoints HC9g. The computed interval width is at most

$$\min\{2e_{p,\Gamma}, U_k^{\text{corr}}(J) - L_k^{\text{corr}}(J) + 2e_{s,\text{gr}}\}, \quad (\text{PGRC.18})$$

because ACM16 and ACM27 place their endpoints inside the stated centered intervals. Intersection cannot increase either width.

For HC9–10 and HC14g preserve their domain $k \geq 3$, the original $0 < b < \pi/2$, $\gamma > 2$, and the phase and circumference comparison specified in PGRT.14's receiving paragraph. Subtract exactly $T_k = 4q \log(D_h k)$ and $\mathcal{B}_k^\Gamma$ for the respective residuals, intersect with their retained earlier endpoints, and insert the arithmetic upper value into the three increasing phase functions. This proves the same exact receiving maps as PGRT.13–14 with the more precise finite row enclosure PGRC.16. It leaves the full exact-graph path available, rather than replacing it by a weaker evaluation.

All maps in PGRC.3 and all inverse-image kernel descriptions act coordinatewise on a fixed original coefficient mask. The empty present mask retains its outer label, external absence retains its distinct label and singleton inverse image, and every represented-zero fibre retains the displayed linear kernel in each active coordinate. The full finite calculation has at most $q + 1$ relation rows with $q = [1 + k(m - 1)](k + 1)^2$; these dimensions remain in the source integrals and inverse blocks. No fixed-packet row convergence has been used to assert an unevaluated uniform bound in the separate variable k .

317 The finite generator converges to the realized completed generator

Retain the original increasing remainder frame of $E = \mathbb{C}[S]/(\chi_{h,k})$, every primary nilpotent, and $T_t = A + tR - kI/2$ from PGGT.1, with $t \in \mathbb{C}$. PGMB.4–5 and PGRT.6 prove

$$\hat{G}_N = \Omega_k + T_N, \quad T_N = f_N^* f_N = \sum_{j=N}^{\infty} a_j^{-1} \alpha_j^* \alpha_j, \quad T_N \longrightarrow 0 \quad (N \rightarrow \infty; h, k \text{ fixed}). \quad (\text{PGGL.1})$$

The symbol T_N denotes this positive form; the original period generator is always written T_t . Their indices and domains fix which object is meant. Define the Ω_k -self-adjoint operator $B_N = \Omega_k^{-1} T_N \succeq_{\Omega_k} 0$. Its full spectral decomposition is $\sum_i b_{N,i} P_{N,i}$ with $P_{N,i}^* \Omega_k = \Omega_k P_{N,i}$. Transporting the complete eigenspaces of $\Omega_k^{-1/2} T_N \Omega_k^{-1/2}$ proves this decomposition in the original frame. Put

$$S_N = \sum_i \sqrt{1 + b_{N,i}} P_{N,i}, \quad S_N^{-1} = \sum_i (1 + b_{N,i})^{-1/2} P_{N,i}, \quad S_N^* \Omega_k S_N = \hat{G}_N. \quad (\text{PGGL.2})$$

Multiplication of the orthogonal projections proves the inverse and Gram identities, including all repeated eigenspaces. Thus S_N is the specified isometry from $(E, \hat{G}_N)$ to (E, Ω_k) ; the original form is not replaced by an identity.

For $G > 0$ let $X_G(t) = T_t^{\dagger G} + T_t$ and $\epsilon_G(t) = \|X_G(t)\|_G$. Applying the actual isometry gives the complete error

$$S_N X_{\hat{G}_N}(t) S_N^{-1} - X_{\Omega_k}(t) = Z_N(t) + Z_N(t)^{\dagger \Omega_k}, \quad Z_N(t) = [S_N, T_t] S_N^{-1},$$

$$\|Z_N(t) + Z_N(t)^{\dagger \Omega_k}\|_{\Omega_k} \leq \frac{\|[B_N, A] + t[B_N, R]\|_{\Omega_k}}{1 + \min_i b_{N,i}}. \quad (\text{PGGL.3})$$

For the identity, use $T_t^{\dagger \hat{G}_N} = S_N^{-2} T_t^{\dagger \Omega_k} S_N^2$ and expand its two ordered terms. For the bound, the convergent integral $\int_0^\infty e^{-uS_N} [B_N, T_t] e^{-uS_N} du$ has (i, j) block $(b_{N,i} - b_{N,j})/(\sqrt{1 + b_{N,i}} + \sqrt{1 + b_{N,j}}) P_{N,i} T_t P_{N,j}$, hence equals $[S_N, T_t]$. Its norm is at most the commutator norm divided by $2\sqrt{1 + \min b_{N,i}}$. Multiplication by S_N^{-1} and addition of the adjoint prove the stated factor one. Both t and $\bar{t}$ remain in the error through its adjoint.

For every specified finite radius $r \geq 0$, the elementary operator-norm bounds on the two commutators now give

$$\sup_{|t| \leq r} |\epsilon_{\hat{G}_N}(t) - \epsilon_{\Omega_k}(t)| \leq 2\|B_N\|_{\Omega_k} (\|A\|_{\Omega_k} + r\|R\|_{\Omega_k})$$

$$\leq 2(\|A\|_{\Omega_k} + r\|R\|_{\Omega_k}) \sum_{j=N}^{\infty} \frac{\alpha_j \Omega_k^{-1} \alpha_j^*}{a_j} \longrightarrow 0. \quad (\text{PGGL.4})$$

The first inequality uses the reverse triangle inequality after PGGL.2, and $1 + \min b_{N,i} \geq 1$. For the second, a positive matrix has largest eigenvalue at most its trace. In the original coordinates this trace is $\text{tr}(\Omega_k^{-1} T_N)$, whose complete convergent tail is PGRT.6. The fixed positive matrix Ω_k and the fixed operators A, R make the last limit zero. This proves local uniformity in the original complex period parameter; it asserts no uniformity in the separate tensor degree k .

The receiving operator is the already constructed actual Hilbert realization, not an unspecified limiting action. Namely let $\mathfrak{U}_k = (L_{\text{low}}, D_{\text{mix}})$ with $\mathfrak{U}_k^* \mathfrak{U}_k = \Omega_k$. Then $\tilde{T}_t = \mathfrak{U}_k T_t \Omega_k^{-1} \mathfrak{U}_k^*$ intertwines T_t on $\mathfrak{U}_k(E)$, vanishes on that image's orthogonal complement, and has kernel $\mathfrak{U}_k(\ker T_t) \oplus^\perp \mathfrak{U}_k(E)^\perp$. The Gram identity proves these statements by decomposing every target vector as $\mathfrak{U}_k x + v_\perp$ with $x = \Omega_k^{-1} \mathfrak{U}_k^* v$. Taking adjoints shows $\|\tilde{T}_t + \tilde{T}_t^*\| = \epsilon_{\Omega_k}(t)$. Consequently PGGL.4 is a quantitative comparison from the finite original graph generators to this specified mixed-source generator, with its full kernel, period parameter, and relation-tail error. Every coefficient map retains its coordinate-wise lift to the original fixed support masks and keeps external absence separate from each present zero vector.

318 The current arithmetic route and its next calculation

The source chain now supplies an actual $\mathbb{C}[t]$ -module cohomology Q , a finite invariant packet inside that cohomology, a specified nonzero extension class, an original theta boundary for its representative defect, and an exact arithmetic formula for the relative weight form. The local continuation makes that last formula an explicit confluent interpolation problem. These constructions are logically prior to any purity conclusion. The subsequent continuation now supplies certified entries of the original arithmetic moment matrix, the exact finite-compression and infinite-tail resolvent comparison, and a convergence rate for the initial resolvent including its complete off-line jets.

318.1 A single chain of exact comparison maps

For each actual finite packet, retaining every full order, the maps proved in this reader are

$$E = \mathbb{C}[t]/h \xrightarrow{R_{\text{ref}}} \mathcal{B} \xrightarrow{q} Q, \quad J_Z R_{\text{ref}} = 1, \quad D R_{\text{ref}} - R_{\text{ref}} A = \Theta \phi_* \ell.$$

Every admitted polynomial correction keeps these same quotient and jet maps. At its distinguished orthogonal level let $B_{N,u}(t) = (p_{N,u}(t) - R_Z(u)(t))/h(t)$, a polynomial of degree at most m . The complete calculation is

$$\begin{aligned} d\nu_Z(t) &= \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad N = d + m, \quad C_N P = j_h P, \quad K_N = C_N (H_N^\nu)^{-1} C_N^*, \\ p_{N,u} &= (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad R_m u = R_{\text{ref}} u + \Theta(B_{N,u}(D) \phi_*), \\ G_m &= U_\varepsilon^* K_N^{-1} U_\varepsilon, \quad W_m = U_\varepsilon^* K_N^{-1} (A K_N + K_N A^* - K_N) K_N^{-1} U_\varepsilon. \end{aligned} \quad (\text{R1})$$

The middle line evaluates the specified polynomial at the original D ; every source vector lies in V .

The quotient in this statement is the full-source quotient q . At a proper original source $W \subseteq V$, the same correction $R = s_h + \Theta T$ has the exact class

$$[Ru]_W - [s_h u]_W = \eta_W([Tu]_W), \quad \eta_W : V/W \longrightarrow \mathcal{B}_\Omega / \Theta W, \quad [v] \longmapsto [\Theta v]_W. \quad (\text{R1a})$$

Injectivity of Θ proves injectivity of η_W , and subtracting the original source functions proves the equality. Thus every independent coefficient face retains the kernel $(V/W)^a$ before passing to q , including its specified outer leg label and its empty-coefficient endpoint. The complete quotient and support maps are (D9)–(D13).

For the finite sources $W_m^{\text{src}} = \text{span}\{\phi_*, D\phi_*, \dots, D^m \phi_*\}$, which are the spaces denoted W_m in (TB.37)–(TB.39), the generator has type

$$\overline{D} : \mathcal{B}_\Omega / \Theta W_m^{\text{src}} \longrightarrow \mathcal{B}_\Omega / \Theta W_{m+1}^{\text{src}}. \quad (\text{R1b})$$

It is well-defined because $DW_m^{\text{src}} \subseteq W_{m+1}^{\text{src}}$ and $D\Theta = \Theta D$. The full primitive (TB.35) belongs to W_{m+1}^{src} , and $R_{m+1} - R_m \in \Theta W_{m+1}^{\text{src}}$. Taking their original quotient classes therefore gives

$$\overline{D}[R_m u]_{W_m^{\text{src}}} = [R_{m+1} A u]_{W_{m+1}^{\text{src}}}, \quad [K_{R_m} u]_{W_m^{\text{src}}} = -[D^{m+1} \phi_*]_{W_m^{\text{src}}} d_m u. \quad (\text{R1c})$$

The second equality follows from the monic Q_{m+1} and degree- m Q_m in the full source formula (TB.35). These exact source-transition identities accompany the unchanged Grams in (R1); the actual lower source term and full two-row observation remain in (TB.35)–(TB.36).

Write $v_n = j_h q_n$ and $\kappa_n = \|q_n\|_{\nu_Z}^2$. The original arithmetic measure now determines the exact quantity

$$\epsilon_m = \frac{|\text{Re}(v_N^* K_N^{-1} v_{N+1})| + \sqrt{(v_{N+1}^* K_N^{-1} v_{N+1})(v_N^* K_N^{-1} v_N) - (\text{Im}(v_N^* K_N^{-1} v_{N+1}))^2}}{\kappa_N}. \quad (\text{R2})$$

The proofs in Section 9 establish (R1)–(R2) before passing to cohomology. For a reflection-stable packet the first absolute-value term is zero by the exact trace identity. The same sequence obeys

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}|. \quad (\text{R3})$$

Thus the rank-two boundary detects the combined displacement of the entire packet, not only its largest individual displacement. The estimate concerns the distinguished R_m sequence. The larger family of all polynomial source corrections has the separately proved metric image and sharp infimum in Section 8; those results do not identify its arbitrary member with R_m .

The full two-row source observation in Section 9 is injective after the original A -iterates are retained. It therefore offers a finite arithmetic calculation preserving every nilpotent jet: compute the source moments, the adjacent monic polynomials and their complete residue vectors, rather than discard a component whose scalar trace vanishes.

318.2 Topology, duality and the two sides of a weight estimate

The actual source moments now have the absolutely convergent incomplete-gamma expression and explicit double-tail bound of Section 7. Its certified six-by-six Gram matrix concerns the original $|2\xi|^2 dt/(2\pi)$ measure. The exact identity (AM.19) supplies the original monic recurrence norms through degree five, with interval enclosures.

For the $|g/h|^2$ kernel, Section 10 proves a universal polynomial quadrature formula, including centres that coincide with a quadrature node. On the precise nonintersection domain its alternate expression is

$$K_{n-1} = \kappa_{n-1}^{-1} q_n(A) S_n q_n(A)^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*, \quad r_n = F_n(A)c.$$

The exact Schur comparison (KR.15)–(KR.20) then joins the initial and terminal observations to the actual infinite-dimensional tail. Section 11 proves the explicit initial-resolvent rate (GC.8) and its complete finite-algebra bound (GC.23). The projection to that off-line domain retains the critical summand as its stated kernel. All factors in the Schur identity remain in converting the initial estimate to the terminal form that determines ϵ_m .

In particular, Section 13 identifies exactly the two contributions in a reflection-stable packet:

$$\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).$$

Its full-jet inequalities also quantify the required metric ratios along each retained nilpotent chain. These identities do not infer a relative estimate from absolute matrix shrinkage.

The theta-polynomial span is dense in every finite differential graph norm. The completed jet graph, however, is $\mathcal{H}^{(r)} \oplus E$, with quotient E , not a quotient of $\mathcal{H}^{(r)}$ in which the arithmetic component has vanished. This supplies the exact comparison between the shrinking Hilbert representative and its unchanged cohomological class.

The exponential observation norm supplies a second exact comparison. A Mellin jet at ρ is bounded precisely in the strip

$$|\operatorname{Re} \rho - \tfrac{1}{2}| < a,$$

with the factorial norm in Theorem 104.3. The original differential then has the energy term $2a \operatorname{sgn}(\log x)$ in Theorem 104.5. Consequently stronger topology and weight control are related by that explicit term; choosing a positive observation norm does not remove it.

For any packet Z , reflection gives the conjugate-linear ring isomorphism

$$\dagger_Z : E_Z \longrightarrow E_{Z^\dagger}, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad Z^\dagger = \{1 - \bar{\rho} : \rho \in Z\},$$

with all multiplicities retained. In fact $h_Z^\dagger = (-1)^d h_{Z^\dagger}$; this proves well-definedness, and applying the same operation twice is the identity. Restrict the following internal residue pairing to reflection-stable Z , so that this map is an involution on E_Z . The arithmetic residue form is the source form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger u}{g} ds.$$

Its Jacobian contraction is

$$\mathcal{R}_Z(f, g'u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho). \quad (\text{R4})$$

Indeed locally $g = z^{m_\rho} w(z)$ gives $g'/g = m_\rho/z + w'/w$, proving (R4) with multiplicities. The trace form pairs reflected centres and can retain nontrivial null directions from nilpotents. In the retained basis $1, t, \dots, t^{d-1}$, define

$$S_{ij} = \sum_{\rho \in Z} \text{Res}_\rho \frac{(s^i)^\dagger s^j}{g(s)} ds, \quad 0 \leq i, j < d, \quad J_g = M_{j_h g'}.$$

Changing representatives by h makes the local quotient by g holomorphic, proving that this matrix represents the intrinsic form. By (R4) the trace pairing has matrix SJ_g . Its exact comparison to the positive source metric is therefore $G_m^{-1} SJ_g$, since $f^* G_m (G_m^{-1} SJ_g) u = f^* SJ_g u$. Its positivity is an arithmetic question about this operator, not a consequence of the positive Gram construction.

The finite-field two-sided weight theorem remains a source for methods, with its compact-support comparison, duality and tensor hypotheses. This reader has not constructed a realization meeting those geometric hypotheses for the entire arithmetic quotient. It has instead calculated the current characteristic-zero objects and their original maps, including the local-cohomology spectral extraction and meromorphic complement documented in the complete finite-comparison appendix.

318.3 Provenance and mathematical status

The coherent interpolation now has the full coefficient identity $J_{h,N} = U_{v_h} C_N$, with $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$. Equations (CT.3)–(CT.7) prove equality with (R1), including the original representative, and compatibility with the complete CRT injections between finite packets. The actual reflection preserves the unique minimum and proves the same-metric signed pairing of (CT.10) and (CT.16a).

On the original tensor complex, the complete calculation is

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)), \\ R_M &= (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}, \\ W_M &= -(Y_M^* C_M + C_M^* Y_M), \quad G_M - G_{M+1} = Y_M^* Y_M, \quad Y_M, C_M : E^{(k)} \longrightarrow \mathcal{E}_M, \\ \dim \mathcal{E}_M &= \binom{M+k}{k-1}. \end{aligned} \quad (\text{R5})$$

Section 14 proves the original ordered division, both cochain signs, every layer map, and the extra orthogonal projection required when tensor packets are enlarged. Its formula (CT.18) calculates the resulting change in both G and W .

For a reflection-stable packet, let p_M be either signed inertia of W_M , and let $\mathfrak{d}_M^{(k)}$ be its complete tensor departure from (CT.20). The same source layers give

$$\begin{aligned} \mathcal{D}_k &\leq p_M \epsilon_M, \quad p_M \epsilon_M^2 \geq 2kd^{k-1} \sum_{\rho \in Z} m_\rho (\text{Re } \rho - 1/2)^2 + \mathfrak{d}_M^{(k)}, \\ p_M &\leq \min\{\text{rank } Y_M, \text{rank } C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k^2 \leq \text{Tr}(G_M^{-1}(G_M - G_{M+1})) \text{Tr}(G_M^{-1} C_M^* C_M). \end{aligned} \quad (\text{R6})$$

The zero-inertia case is proved directly in (CT.25). The full ordered sum $\mathcal{D}_k$ and all multiplicities are retained in (CT.20). The joint density and exact graph closure in Section 15 show $G_M \downarrow 0$ for this actual total-degree sequence and identify every retained jet in the vertical fibre of (JD.11). The same section proves the coercive inverse-interpolation limit with its arithmetic unit, the supported-zero limit morphism, and the accumulated determinant cost (JD.14)–(JD.15). These formulas connect the relative layer energy to the exact rate of determinant loss through the same original matrices.

The September 11 support, Rees, comparison and audit packages are retained in the companion source directory. The current analytic baseline is the September 12 sequence: finite operator comparison (PR 8), chain descent (PR 9), global retraction (PR 10), homotopy control (PR 11), formal tau recovery (PR 12) orthogonal boundary control (PR 13), and coherent interpolation with joint tensor boundaries (PR 14). Part II includes the complete selected source proof notes with their exact revisions. Historical statements that a map remains unconstructed belong to their stated source dates; a later explicit construction in this reader or the source chain supersedes that particular statement.

The new local proofs are the full support-index equivalence, the corrected carrier and sector comparisons, the arithmetic confluent-kernel minimum and complete two-row source observation, the classification of source-metric images and aggregate control, and the graph/weighted-jet comparison. The next tranche adds the certified original theta-moment series and its full truncation bound, the confluent quadrature and finite/terminal/tail resolvent identities, the explicit initial convergence rate, and the metric-departure and nilpotent-chain identities. The recovered June 24 historical packet calculation is reproduced with a full-jet comparison in Section 2. Its exact A -invariant saturation is every off-axis local factor; the quotient, involutions, source-metric transport and split-zero endomorphism descent are proved there. The original transcript excerpt and its response locators are retained in the source package. This is an explicitly recorded historical recovery, with its new arithmetic saturation proof distinguished from the original permutation calculation. The subsequent historical reconstruction in Section 3 proves the inverse map from a complete local Hurwitz zero-cluster germ to every moment of its original compact probability measure, followed by the explicit Bernstein recovery of that measure. The monic cluster polynomial retains arbitrary zero multiplicity. The same section proves the complete Laurent period maps and their collision kernels. For two distinct primes, the joint Laurent map is surjective onto the full finite spectral algebra; Theorem 3.1 gives every primary kernel and the inverse original spectral coordinate, including its truncated logarithm on each nilpotent factor. The factorwise Split-Zero lifts and their exact comparison with the synchronized global lift are proved in (S4)–(S5). Their elementary algebraic and analytic steps are written here. No claim of global mathematical priority follows from finding or proving an identity in this research programme.

The finite validation records distinguish exact symbolic identities, rational calibration models, numerical quadratures and intentionally rejected negative controls. They do not count test methods as theorems or imply analytic certification from an L^2 numerical experiment. The supplied formalization records are also kept separate: successful exact-head CI metadata does not turn the unformalized analytic theta arguments into Lean proofs. No Lean run was performed for this continuation.

The current research calculation includes (R2) on the actual $|g/h|^2$ -moments, with its determinant update and two-row recurrence, together with the topology constants of Theorem 104.3. No uniform arithmetic estimate forcing (R2) to vanish has been established in this reader. RH, GRH and a global purity theorem are not conclusions of it. The exact maps and source constraints above specify the live route on which tandem work can now continue. The next estimate must propagate (GC.8) through (KR.16), retaining q_n^2 , κ_n , a_n , the terminal ratio F_n and the actual tail denominator. The arithmetic moment formulas give an executable source for those finite data. The exact departure identity and the full-jet metric bounds specify what a resulting relative estimate would control in the original packet; no component is removed by changing observation.

The next step of that propagation is now calculated in Section 12. For $u_n = q_n(A)c$, $w_n = q_{n-1}(A)c$, and $M_n = \kappa_{n-1}^{-1}K_{n-1}^{-1}$, let $\beta_n = u_n^* M_n u_n$, and let P_n be the exact M_n -orthogonal projection

with kernel $\mathbb{C}u_n$ when $\beta_n > 0$. Put $p_n = P_n w_n$, $z_n = P_n A u_n$. The universal step is

$$\begin{aligned} M_{n+1} &= a_n^{-1} \left(M_n - \frac{M_n u_n u_n^* M_n}{a_n + \beta_n} \right), & \frac{\det G_{m+1}}{\det G_m} &= \frac{a_n}{a_n + \beta_n}, \\ \Delta_{n+1} &= \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}, & \epsilon^{(n)} &= |g_Z| + \sqrt{g_Z^2 + \Delta_n}, \quad g_Z = \operatorname{Re} \operatorname{tr} A - d/2. \end{aligned} \quad (\text{R7})$$

Here $m = n - d - 1$. The zero-vector exceptional step is given completely in (KT.14), so critical centres and every polynomial coincidence remain in (R7)'s source calculation. The alternate terminal Gram follows by the exact isometry (KT.15) and recurrence (KT.16). The full scalar-to-matrix extraction uses the doubled off-line algebra and the explicit sequence (KT.23); at a reflected pair its matrix coefficient is exactly $(-1)^l F_n^{[k+l+1]}(\rho)$. Finally (KT.24)–(KT.25) recover these coefficients from the original initial error and tail, with both reciprocal series and all polynomial/norm factors displayed. The remaining energy estimate is therefore attached to the three explicit terms of (KT.11), including their actual cross pairing, on the same arithmetic moment sequence.

For a packet closed under both conjugation and reflection, the original real-coefficient calculation in (KT.28) closes this step in terms of consecutive determinant ratios. With the fixed section metric G_{-1} retained, put $r_m = \det G_m / \det G_{m-1}$ for $m \geq 0$. Then the actual arithmetic allowance obeys exactly

$$\epsilon_m^2 = a_{d+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{R8})$$

The coefficient reality and full reflection trace prove $\gamma_n = 0$ in the original coordinates; (KT.7)–(KT.8) then prove (R8), including zero-step cases. This places the relative quartet control directly on the same monic norm ratios and consecutive Gram determinants as the certified moment problem.

This also gives a direct arithmetic bound for the original tesserine violation channel. Retain the original quartet element

$$Q(\rho) = \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa, \quad \rho = \beta + i\gamma,$$

and its character $c_{-+}(\rho) = 2(2\beta - 1)$, as calculated in the carrier section. If $\gamma \neq 0$, $\beta \neq 1/2$, and Z is this entire four-point zero orbit with common full order μ , then $d = 4\mu$ and for every $m \geq 0$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = a_{4\mu+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{R9})$$

To prove every coefficient, conjugation and the functional equation preserve the full order, and the four real displacements are $\beta - 1/2, \beta - 1/2, -(\beta - 1/2), -(\beta - 1/2)$. The proved aggregate bound (R3) is therefore $\epsilon_m \geq 4\mu|\beta - 1/2| = \mu|2(2\beta - 1)|$. Squaring and using (R8) gives (R9). The sector morphism on the original supported quartet is precisely

$$G(\chi_{-+}) : G(\mathbb{C}[K_4]) \longrightarrow G(\mathbb{C}), \quad Q(\rho)^\bullet \longmapsto c_{-+}(\rho)^\bullet, \quad \tau \longmapsto \tau.$$

Here $\chi_{-+} = \operatorname{ev}_{-1,+1}$ is the original group character extended linearly; it preserves multiplication because the character does, and hence its displayed G -lift is a semiring morphism. At $\beta = 1/2$ this morphism takes the supported value e ; the same original quartet has phase channel $c_{--} = 4i\gamma$. The orbit then has two distinct points when $\gamma \neq 0$, so its degree is 2μ and (R8) applies with that exact degree; the left-hand side of the sector bound is zero. Thus the arithmetic volume comparison and the supported critical value use the same typed character map, with the phase coordinate retained.

318.4 The original theta norm and the positive arithmetic frontier

The two newly delivered source notes, reproduced completely in the appendices, admit the following exact common formula with (R8). Keep $u_j = (I - P_{L_{j-1}})D^j f_0$, $\mathfrak{h}_j = \|u_j\|^2$, and the original measure

$|g|^2 dt/(2\pi)$. The polynomial division and original minimum in (KL.23)–(KL.27) prove, at every full packet,

$$\kappa_{d+j} = \mathfrak{h}_j r_j \quad (j \geq 0), \quad \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2 \quad (\text{R10})$$

for reflection-stable packets and $m \geq 0$, where $c_m = d_m G_m^{-1} d_{m+1}^*$ uses the original theta projection rows. The first identity includes $j = 0$ and the fixed section G_{-1} : the actual new-layer vector at that endpoint is exactly f_0 . The row dictionary (KL.30) retains its separate domain $m \geq 0$; its cross term is $\mathfrak{h}_{m+1} c_m = \overline{\gamma_n}$, with $n = d + m + 1$. Both signs and the conjugation are obtained by multiplying those rows against the original inverse metric, including both local-unit matrices. Equation (KL.25) substitutes the first identity in (R10) into the complete terminal energy, proving its second identity. Under conjugation closure the original coefficients are real, so γ_n is real; the reflection trace makes its real part zero. Thus $c_m = 0$ in this case, by precisely the same calculation, with no discarded cross term.

For the full quartet in (R9), this gives the proved chain

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{R11})$$

Indeed the order-preserving symmetries establish the hypotheses of (R10) and its vanishing cross term; the lower bound was proved with every multiplicity in (R9). The norms $\mathfrak{h}_j$ are independent of the packet. For $h \mid H$, the original polynomial map $P \mapsto (H/h)P$ is an isometry because $(g/H)(H/h)P = (g/h)P$. Its jets are the old complete jets and zero jets at each added centre. Its fixed representative and all its corrected representatives agree under this insertion, as proved in (KL.28). Thus the common theta sequence in (R11) connects the actual nested-packet maps; the packet dependence remains in the two displayed determinant ratios.

For the tensor frontier, retain the original full column maps $F, E_+ : \mathbb{C}^r \rightarrow E_h^{\otimes k}$ and its original norm diagonal Ω in (AP.4). With G_M the actual total-degree minimum, define

$$\begin{aligned} \mathbf{A} &= \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, & \mathbf{B} &= \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}, \\ \Gamma_{k,M} &= \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), & \pi_M &= \frac{\det G_{M+1}}{\det G_M}. \end{aligned} \quad (\text{R12})$$

All complete conjugation-and-reflection packets satisfy

$$\epsilon_{k,M}^2 = \lambda_{\max}(\mathbf{A}^{1/2} \mathbf{B} \mathbf{A}^{1/2}) \leq \Gamma_{k,M} \lambda_{\max}(\mathbf{A}) \leq \Gamma_{k,M} (\pi_M^{-1} - 1). \quad (\text{R13})$$

Here the square roots are the explicit metric isometries, with their inverses in (AP.12); the original matrices in (R12) remain part of the formula. The full proof in (AP.7)–(AP.21) starts from the actual linear reflection of the original Mellin source. It places the two frontiers in opposite orthogonal eigenspaces and gives $U^*V = 0$. The relative weight is consequently the two off-diagonal blocks UV^* and its adjoint. Its complete singular-vector decomposition identifies its norm squared with the positive operator in (R13), including all zero vectors. The explicit incidence row sum gives $VV^* \preceq \Gamma I$; the inverse-metric update gives $\pi_M^{-1} = \det(I + UU^*)$. Multiplication of its nonnegative eigenvalue factors proves the two inequalities. For $k = 1, M = d + m$, (R13), (R11), and (AP.23) calculate the same original tesserine bound through its norm, volume, and positive-operator presentations.

318.5 The symmetric summand, the actual relation layer, and support

Every factorial in the symmetric comparison is retained. The literal orbit-sum injection I , its diagonal $D_o = I^*I$, and $L = D_o^{-1}I^*$ in (AP.24) give $G_M^s = I^*G_M I$ and the exact identity $(G_M^s)^{-1} = L G_M^{-1} L^*$. The proof uses permutation invariance of the actual minimum, so that its metric commutes with the

projection IL . Hence the original redundant frontier has symmetric maps $F_s = LF$, $E_s = LE_+$ with the unchanged Ω , and

$$\begin{aligned} A_s &= \Omega^{-1/2} F_s^* G_M^s F_s \Omega^{-1/2}, & B_s &= \Omega^{-1/2} E_s^* G_M^s E_s \Omega^{-1/2}, \\ (\epsilon_M^s)^2 &= \lambda_{\max}(A_s^{1/2} B_s A_s^{1/2}) \leq \Gamma_{k,M}((\pi_M^s)^{-1} - 1), & \pi_M^s &= \frac{\det G_{M+1}^s}{\det G_M^s}. \end{aligned} \quad (R14)$$

Equations (AP.25)–(AP.36) prove all these maps, the full positive and zero spectra, the opposite reflection parities, and the orbit-factorial grouping. In particular the isometry $Q_M = G_M^{1/2} I (G_M^s)^{-1/2}$ satisfies $Q_M^* Q_M = I$ and $Q_M Q_M^* = IL$; it carries the symmetric relative operator to the invariant restriction of the original one, retaining the complementary operator on $\ker L$. The continuous cochain projector in (SS.1)–(SS.9) has exactly this cohomology image. Its finite packet version uses the original section and full jet, and all Koszul signs in that construction remain in its primitive.

The actual next relation layer underlying these matrices is proved in (KL.7)–(KL.20). Its complete graph frame, Gram, fixed-order division primitive, derivative image, and derivative cokernel are given there. The comparison with a homogeneous shell is the exact polynomial identity (KPC.7), with every lower coefficient in (KPC.6). Its full graph is the fibre product (KPC.12), and its two projection kernels are specified in (KPC.11). For two tensor coordinates the original ratio algebra embeds by (KPC.16), with exponent $\max_{\rho/\sigma=r}(m_\rho + m_\sigma - 1)$ at each coincident ratio. Its adjacent-shell generator action is (KPC.22a); the missing second-coordinate multiplier remains in that formula. The finite example (KPC.23)–(KPC.28a) calculates both original shell ranks, all nine node values, and the actual minimum, establishing the exact scope of that comparison.

For every linear map in these graph, kernel, quotient and cochain comparisons, the fixed-support lift at the end of the projective-comparison section maps (λ, v) to (λ, Tv) and maps external absence to absence. Its scalar and additive identities are proved there, including supported zero. For measured support, the historical branch chapter proves the separate full connection from ordinary branches to frame points through $\text{Idl}(L) \rightarrow L$, with its exact kernel congruence and descent criterion. Its L^∞ correspondence identifies every Boolean branch with a multiplicative state and its normal branches with actual measure atoms. The factor $\mu(a)^{-1} \int_a f d\mu$ is retained. These explicit maps specify the support observations available to the analytic construction and preserve the nonatomic support component alongside its atomic quotient.

318.6 The original derivative cost and both positive-half-line kernels

The same original derivative layer has the stronger arithmetic identity proved in (QD.6)–(QD.15). In its exact metric coordinates, $-\hat{Y}^* \hat{C} = UV^*$ occupies one off-diagonal reflection block. Thus its norm is the full relative allowance, and the original trace cost χ_M , operator cost ζ_M and largest update eigenvalue λ_M satisfy

$$\epsilon_{k,M}^2 \leq \frac{\lambda_M}{1 + \lambda_M} \zeta_M \leq \frac{\lambda_M}{1 + \lambda_M} \chi_M, \quad \epsilon_{1,M}^2 = (1 - \pi_M) \chi_M. \quad (R15)$$

The first two inequalities follow from the norm of the product of the two actual layer maps and the eigenvalues of $UU^*(I + UU^*)^{-1}$. For one factor, the layer is one-dimensional, so that product is the outer product of its two original rows and its norm is exactly their norm product. Its rank-one determinant update is $\pi_M = (1 + \lambda_M)^{-1}$, proving the last equality. This argument includes either row zero. The full lower bounds $4k^2 \delta_*^2$ and all finite accumulated volume constraints follow in (QD.10)–(QD.13), with their positive denominators established before division.

For one factor, put $z = s - 1/2$, $x_h = -z^2$, and retain the monic polynomial H defined by $h(s) = (-1)^D H(x_h)$, $d = 2D$. The exact algebra is $A_H \oplus zA_H = \mathbb{C}[x_h, z]/(H, z^2 + x_h)$, with its two local length- m_ρ branches explicitly inverted in (SP.8)–(SP.9). The original source is the Hilbert direct sum for $d\lambda_h = |V(x_h)|^2 dx_h / (2\pi \sqrt{x_h})$ and $x_h d\lambda_h$. Here $v_h(s) = V(x_h)$ is the entire original quotient and

its full unit modulo H enters both kernels. With those kernels and their inverse metrics, (SP.23)–(SP.35) give

$$\begin{aligned} \epsilon_{N-d} &= \left\| G_e^{-1/2} (G_o - G_e X) G_o^{-1/2} \right\|, \\ K_e - X K_o &= \begin{cases} f_n e_n^* / p_n, & N = 2n, \\ -f_{n+1} e_n^* / u_n, & N = 2n + 1. \end{cases} \end{aligned} \quad (\text{R16})$$

The first line follows by substituting the exact multiplication matrix $\frac{1}{2}I + \begin{pmatrix} 0 & -X \\ I & 0 \end{pmatrix}$ in the original weight form. The second is its full inverse-metric recurrence: all interior adjacent terms cancel with their original norm ratios, and the last pair has the two displayed parity signs. The finite kernels have indices $(n, n-1)$ and (n, n) , respectively. Hence the norm is given by the full quadratic contractions in (SP.33), with at most one nonzero singular value. Every nilpotent jet remains inside f_n, e_n, X and both metrics. The exact derivative cost (SP.34) identifies (R16) with (R15), including the two zero-frontier cases.

318.7 The gamma determinant and the retained spectral-sum fibre

The original completed function has the exact factorization $g(1/2 + it) = \Gamma(1/4 + it/2)a(t)$, with $a(t) = -\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)$. The beta integral and generating-function calculation (TG.7)–(TG.14) prove the reference monic norms $\gamma_n = \sqrt{2} n! \Gamma(n + 1/2)$. Keeping the actual multiplier in the full positive arithmetic Gram matrix gives its determinant D_n , with $D_{-1} = 1$. Monic triangular Gram congruence proves $\mathfrak{h}_n = \gamma_n D_n / D_{n-1}$, so the actual quartet formula is exactly

$$\epsilon_m^2 = (m+1)(m+\frac{1}{2}) \frac{D_{m+1} D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}), \quad m \geq 0. \quad (\text{R17})$$

This follows by substituting the consecutive norm ratio into (R10), whose cross term vanishes by the proved quartet symmetry. The full packet determinants are independently expressed in (TG.17)–(TG.18), and the original confluent kernel is the complete Gram congruence (TG.20)–(TG.22). The explicit triangular maps to the parameter $3/4$ family and their inverses retain every phase and degree. The theorem-hypothesis analysis (TG.25)–(TG.28) proves the obstruction from the infinitely many retained critical zeros on positive-measure intervals; it also supplies these exact Gram maps for continuing beyond that obstruction.

The web spectral-sum construction retains an additional relative coordinate $\Delta = (s_1 - s_2)^2$. Write $S = s_1 + s_2$, $Z = S - 1$ and $x_\Sigma = -Z^2$. Its complete reflection components, proved in Section 24, are

$$B_e = \frac{\mathbb{C}[x_\Sigma, \Delta]}{(A_0, x_\Sigma \Delta A_1)}, \quad B_o = \frac{\mathbb{C}[x_\Sigma, \Delta]}{(A_0, \Delta A_1)}. \quad (\text{R18})$$

Here the original polynomial identities are $H_0(S, \Delta) = A_0(x_\Sigma, \Delta)$ and $H_1(S, \Delta) = Z A_1(x_\Sigma, \Delta)$; all coefficients come from $h((S+r)/2) = H_0 + r H_1$, $r^2 = \Delta$. The map $\alpha : B_o \rightarrow B_e$ multiplies by x_Σ , and $\beta : B_e \rightarrow B_o$ is the quotient. They satisfy $\alpha\beta = x_\Sigma$ and $\beta\alpha = x_\Sigma$. The original generator is $S = I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}$. Their dimensions are $D(D+1)$ and D^2 ; both values follow by taking the symmetric square of the original two D -dimensional reflection eigenspaces. The exact maps, rather than an identification of these unequal spaces, enter the weight.

The full kernel and cokernel of $S - 1$ both recover A_H . Indeed $A_0(0, \Delta) = (-1)^D H(-\Delta/4) \neq 0$ makes α injective, while β is onto. Its cokernel map evaluates $x_\Sigma = 0$ and sends Δ to $-4x_h$. The kernel injection sends $q(x_h)$ to $\Delta A_1 q(-\Delta/4)$ in B_e , with the full polynomial ΔA_1 retained before taking the quotient. The ideal calculation and the dimension count prove both exact sequences in the cited section, including all orders. On both defect modules the actual tensor unit acts as $V(x_h)^2$, because at $S = 1$ the two original arguments are s and $1 - s$. Thus the complete paired packet, its nilpotents and its arithmetic unit remain in the relative fibre.

The exact matrix-weight pushforward retains all powers of Δ , the Jacobian $dx_\Sigma/\sqrt{x_\Sigma}$, and both vector Hilbert measures. Its actual least-norm quotient gives the two metrics in (R18); the upper block of the original weight is $\beta^*G_o - G_e\alpha$, with its full rectangular singular spectrum. All maps preserve external absence and carry a vanishing amplitude to its supported zero through the displayed fixed-support lifts. These finite identities and complete source maps are established without an asymptotic estimate for (R17).

318.8 The sum derivative, its strict normal energy and the full conormal algebra

For the actual sum amplitude retain $a_h^{\text{SC}}(t) = v_h(1/2 + it)/\sqrt{2\pi}$, $w = |a_h^{\text{SC}}|^2$, $\mu_h = \int w$, and $I_h = 4 \int |(a_h^{\text{SC}})'|^2$. In the original coordinates $t_i = u/k + y_i$, $y_k = -\sum_{i < k} y_i$, the full product amplitude is $\Psi = \prod_i a_h^{\text{SC}}(t_i)$ and its squared fibre norm is $m = w^{*k}$. The exact Jacobian has absolute value one. Differentiating before integration gives $\partial_u \Psi = k^{-1} \sum_i a'_i \prod_{j \neq i} a_j$. The mixed integrated terms vanish because $\int \bar{a} a' = 0$. Orthogonal projection onto the actual product-amplitude line gives

$$\int_{\mathbb{R}} \frac{|m'|^2}{m} du + 4 \int_{\mathbb{R}} \|n(u)\|^2 du = \frac{\mu_h^{k-1}}{k} I_h, \quad n = \partial_u \Psi - \frac{m'}{2m} \Psi. \quad (\text{R19})$$

The proof in (FC.1)–(FC.4) retains the original mass and phase, and establishes every required integrability statement. For every finite independent relative polynomial frame, $j_u c = \Psi \sum_\alpha \theta_\alpha(i\mathbf{y}) c_\alpha$, the corresponding exact derivative is

$$\partial_u(j_u c) = j_u(c' + \Gamma_u c) + N_u c, \quad \Gamma = W^{-1} j^* j', \quad N = (I - j W^{-1} j^*) j', \quad j^* N = 0. \quad (\text{R20})$$

This identity follows by inserting the orthogonal projection onto $\text{im } j_u$, with $W = j^* j > 0$. Its full normal Gram is $\mathcal{N} = N^* N$.

Every finite packet leaves infinitely many critical-line zeros of the original amplitude. At such a zero T of complete order ℓ_T , the transverse expansion in (FC.25) has coefficients $c_T A_b \delta^{\ell_T}$ for Ψ and $(\ell_T/k) c_T A_b \delta^{\ell_T-1}$ for $\partial_u \Psi$, with $c_T A_b \neq 0$. Thus an equality $N_u c = 0$ forces the relative polynomial to vanish on the entire hyperplane $y_1 = T - u/k$ whenever the other factors can be chosen nonzero. For $k \geq 3$ this holds at every u ; for $k = 2$ it holds outside the countable sumset $\mathcal{Z}_h + \mathcal{Z}_h$. Infinitely many such hyperplanes force the finite polynomial to vanish. Consequently (FC.21)–(FC.27) prove

$$\begin{aligned} \mathcal{N}(u) &> 0 \text{ everywhere for } k \geq 3, \\ \mathcal{N}(u) &> 0 \text{ almost everywhere for } k = 2, \quad \int \frac{|m'|^2}{m} < \frac{\mu_h^{k-1}}{k} I_h. \end{aligned} \quad (\text{R21})$$

The full proof retains the orders, coefficient domains and the quartet two-factor zero fibre $N_0 = 0$.

The two-branch transformation is equally exact for a matrix weight which is not even. At $x > 0$, put $v = \sqrt{x}$ and

$$M = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}, \quad H = \frac{M^* \text{diag}(W(v), W(-v)) M}{2v}.$$

The original coefficient is $c(\pm v) = a(x) \pm ivb(x)$. Equations (FC.10)–(FC.18) prove the full unitary correspondence, the covariant derivative $\nabla_x = \partial_x + \mathcal{A}$, and its energy

$$\begin{aligned} \mathcal{A} &= M^{-1} \text{diag}\left(\frac{\Gamma(v)}{2v}, -\frac{\Gamma(-v)}{2v}\right) M + \text{diag}(0, I/(2x)), \\ \int \|\partial_u(jc)\|^2 du &= 4 \int_0^\infty x(\nabla_x q)^* H \nabla_x q dx + \int_0^\infty q^* Q q dx, \quad q = (a, b). \end{aligned} \quad (\text{R22})$$

Here $Q = M^* \text{diag}(\mathcal{N}(v), \mathcal{N}(-v)) M / (2v)$. The retained Jacobian drift is $\mathcal{A}^* H + H \mathcal{A} = H' + H/(2x)$. The two endpoint traces $a + i\sqrt{x}b$ and $a - i\sqrt{x}b$ must agree at $x = 0$; this is the proved joining condition for the original derivative domain.

The arithmetic derivative has the unital algebra bridge

$$\Phi : P/I^2 \longrightarrow (P/I)[\epsilon]/(\epsilon^2), \quad [p] \longmapsto [p] + \epsilon[\partial_\Sigma p], \quad I = (h(s_1), \dots, h(s_k)), \quad \partial_\Sigma = k^{-1} \sum_i \partial_{s_i}. \quad (\text{R23})$$

The product rule modulo I proves multiplicativity. Its conormal restriction is the row $(a_i) \mapsto k^{-1} \sum_i h'_i a_i$ on $I/I^2 \cong (P/I)^k$. On an ordered local block of lengths m_i , multiplication by h'_i sends the input constant in variable i to $m_i c_i(0) z_i^{m_i-1}$ times that constant. Counting the monomials having at least one top exponent proves its full rank $d^k - (d-r_0)^k$, r_0 the number of distinct packet roots. The kernel and image of the algebra map, including its linear cokernel, are proved in (CW.6)–(CW.11).

For the original monic remainder section $j : P/I \rightarrow P/I^2$, $Z = S - k/2$ and $D_0 = \delta j$, the exact section defect and correction are

$$[Z, D_0] = -I + J, \quad J = \frac{1}{k} \sum_i M_{h'_i c_i} \ell_i = \frac{d}{k} \sum_i P_i, \quad P_i = \frac{1}{d} M_{h'_i c_i} \ell_i. \quad (\text{R24})$$

Here ℓ_i extracts the original s_i^{d-1} coefficient and c_i inserts the constant polynomial in that variable. The identity $\ell_i M_{h'_i c_i} = dI$ proves that the P_i are commuting idempotents. Their complete tensor decomposition gives eigenvalues dj/k with multiplicities $\binom{k}{j}(d-1)^{k-j}$ for $d > 1$, and $J = I$ for $d = 1$. Its trace is d^k and its rank $d^k - (d-1)^k$. The explicit factorization (CW.22) through the full conormal row proves the relationship between these two rank formulas.

The complete arithmetic unit $\widehat{U} = [\prod_i v_h(s_i)]_{I^2}$ gives $j_U = \widehat{U} j U^{-1}$, $D_U = M_{U^{-1} \delta \widehat{U}} + M_U D_0 M_U^{-1}$, and $[Z, D_U] = -I + U J U^{-1}$. For the actual minimum section j_M , the difference $a_M = j_M - j_U$ is its admitted original relation in I/I^2 ; the correction is exactly $U J U^{-1} + [Z, \delta a_M]$, with trace d^k . These are the full source maps proved in (CW.23)–(CW.30). In one variable they also give

$$j_U u = (j_h g') \mathcal{R}_Z(1, u), \quad \mathcal{R}_Z(1, u) = \ell(U^{-1} u), \quad \mathcal{R}_Z(1, j_h g') = \sum_\rho m_\rho = d. \quad (\text{R25})$$

The partial-fraction proof in (CW.32)–(CW.36) retains the full poles before taking residues. Finally $G(\Phi)$ is the proved semiring lift; the derivative is its infinitesimal coefficient map. Thus every relation killed by the quotient retains its supported zero and its explicit derivative observation.

318.9 The full relative limit and the original arithmetic observation

Retain $k \geq 2$ and the nonempty original packet $d = \deg h \geq 1$ throughout this subsection. For two nested relative frames, keep the literal coefficient inclusion E_{LM} and the original matrices of (NT.1)–(NT.37). Their exact connection difference $K_{LM} = \Gamma_L E_{LM} - E_{LM} \Gamma_M$ obeys

$$N_M = j_L K_{LM} + N_L E_{LM}, \quad N_M^* N_M = K_{LM}^* W_L K_{LM} + E_{LM}^* N_L^* N_L E_{LM}. \quad (\text{R26})$$

The cross terms vanish because $j_L^* N_L = 0$. Thus enlarging a frame transfers the first displayed component into its tangent part, with its complete positive Gram retained. The arithmetic exponential bounds and weighted polynomial density proved in (ND.1)–(ND.34) give $\Pi_M \rightarrow I$ strongly on every original relative fibre. For each fixed original source polynomial P , (NN.1)–(NN.22) prove $r_M(P) = (I - \Pi_M) \partial_u(\Psi P) \rightarrow 0$ in the original global L^2 norm. Their normal-image transitions are the explicit contractions $r_M(P) \mapsto (I - \Pi_L) r_M(P) = r_L(P)$; their full derivative-graph transitions keep $\partial_u(\Psi P)$ unchanged and are isometric.

The same proofs retain the actual logarithmic arithmetic observation. The invariant relation $P_{\text{rel}} = \sum_i h(s_i)$ has zero ordinary jet but derivative jet $(U_k/k) \sum_i [h'(s_i)] \neq 0$. For $d = 1$ the sum is k ; for $d > 1$ its distinct highest standard monomials have coefficients d , so it cannot vanish. Its normal seminorm tends to zero, while its algebraic observation remains this specified nonzero vector. The exact split quotient of that zero limiting seminorm sends every supported source to e and external

absence to τ ; the nonzero observation therefore cannot factor through that quotient. Its original finite maps and full derivative graph remain. For the changing canonical target metrics the complete comparison is

$$G_M \preceq G_*, \quad B_M = G_*^{-1/2} G_M G_*^{-1/2}, \quad x^* G_M x = (G_*^{1/2} x)^* B_M (G_*^{1/2} x). \quad (\text{R27})$$

The fixed-source limit has not estimated this changing B_M . For $h = 1$ the finite coefficient quotient is zero, so the arithmetic observation above is its zero map; the original analytic source and the fixed-source density and derivative-graph maps remain as in (NN.22). The cyclic/full-frame derivative split and canonical-section difference are proved with all four metric terms in (CM.1)–(CM.19).

318.10 Every relation depth and its full cyclic complement

For the same monic h and $I = (h(s_1), \dots, h(s_k))$, the complete depth calculation (CC.1)–(CC.52) proves

$$\begin{aligned} I^r \cap \mathbb{C}[S] &= (\chi_r), \quad \chi_r(T) = \prod_{\lambda} (T - \lambda)^{\ell_{\lambda,r}}, \\ \ell_{\lambda,r} &= \max_{\sum_i \rho_i = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) + (r - 1) \max_i m_{\rho_i} \right), \\ C_r &= \mathbb{C}[T] / \chi_r \xrightarrow{\eta_r = U_r \alpha_r} A_r = (P/I^r)^{\mathfrak{S}_k}, \quad \Pi_r \eta_r = I, \quad A_r \cong C_r \oplus \ker \Pi_r. \end{aligned} \quad (\text{R28})$$

The coefficient-extraction formula (CC.13) constructs this retraction, and (CC.10) counts every invariant orbit before taking its trace. The quotient derivative $\delta_r : P/I^{r+1} \rightarrow P/I^r$ and its scalar counterpart satisfy the commuting squares (CC.17). Their full unit term has the proved decomposition $\delta_r \eta_{r+1} = \eta_r (\partial_r + M_{b_r} \rho_r) + \mathcal{N}_r \rho_r$ of (CC.23); its complementary quotient map vanishes exactly when $U_r^{-1} \delta_r U_{r+1} \in \alpha_r C_r$. The first-jet maps, their full kernels, the product-Jacobian factorization, and the exact two-factor parity ideals at every depth are proved in (CC.29)–(CC.51b). An arbitrary weighted multiplier has all four blocks (CC.45); its lower diagonal block is a genuine quotient operator precisely when the lower left block is zero. Thus the earlier weighted-trace language has the exact operator interpretation (CC.45)–(CC.49), with no complementary direction lost.

318.11 Exterior amplification and the arithmetic volume to estimate

For the actual cyclic quotient, keep $A = M_S$, its canonical G_N and $H = G_N^{-1}(A^* G_N + G_N A - k G_N)$. The complete exterior source note proves that H has spectrum $\epsilon, -\epsilon, 0, \dots, 0$. The additive exterior action $A^{[p]}$ and the original unscaled representative $R_N^{\otimes p} \text{Alt}_p$ have source Gram $p!(\det(G_N)_{I,J})_{I,J}$; their control has the same literal factorial and allowance ϵ . On the determinant of the full positive generalized spectral subspace this gives

$$L_{h,k} = \sum_{\Re \lambda > k/2} \ell_{\lambda}(2\Re \lambda - k) \leq \epsilon_{h,k,N}^{\text{cyc}}. \quad (\text{R29})$$

The full cochain primitive carries the sign $(-1)^{k(j-1)}$ in block j , which the tensor differential repeats; their product is one. Hence this exterior comparison uses the original theta boundary. The cochain idempotent splits full cohomology, while the ordinary trace in (R29) is on the specified finite arithmetic packet.

For the exact complete nonreal quartet polynomial of (E33), with common order m , let $\ell_k = 1 + k(m-1)$. The counted sum grid yields

$$q = \ell_k(k+1)^2, \quad L_{h,k} = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad N \geq q-1. \quad (\text{R30})$$

The explicit four occupation counts in (E33)–(E34) prove this formula without a noncollision assumption on ordered tuples. In a larger packet it remains a lower bound and the threshold is its actual $\deg \chi_{h,k} - 1$. The equality analysis (E1)–(E36) further proves, in the original metric, with positive spectral projector Q , orthogonal range projection P , and $T = PA(1 - P)$,

$$\begin{aligned} \epsilon^2 - L^2 &= \|T\|_{\text{HS}, G_N}^2 + \epsilon^2 \{\det M + \det(I_2 - M)\}, \\ BX - XD &= T, \quad X = (Q - P)|_{\ker P}, \quad X = \int_0^\infty e^{-tB} T e^{tD} dt. \end{aligned} \quad (\text{R31})$$

Here M is the explicit positive compression in (E5)–(E7), and B, D are the original diagonal blocks of A in the orthogonal sum. The finite nilpotent exponential formulas (E15) prove convergence of the integral and that this Sylvester map is invertible. For $L > 0$, equality $L = \epsilon$ is equivalent to $P = Q$, or to $T = 0$. A retained critical Jordan block forces strict inequality. The canonical Gaussian equality calibration (E35)–(E36) proves why positivity of a moment measure alone supplies no further strictness.

Finally the arithmetic matrices themselves give the exact calculation (CV.1)–(CV.12). Set $D_N = \det K_N$ and $\tilde{K}_N = K_{N-1} + b_{N+1}b_{N+1}^*/\omega_{N+1}$. Then

$$\begin{aligned} \epsilon^2 &= \frac{\omega_{N+1}}{\omega_N D_N} \Delta_N, \quad \Delta_N = D_{N+1} + D_{N-1} - D_N - \det \tilde{K}_N, \\ \Delta_N &= \sum_{\substack{I \subset \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det[b_I, b_N, b_{N+1}]|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \end{aligned} \quad (\text{R32})$$

Determinant multilinearity proves the positive sum, including the singular first-degree boundary, and (CV.10a) constructs its original source vector with norm squared Δ_N and tensor norm squared $q! \Delta_N$. At $N = q - 1$, (CV.11)–(CV.12) give the same control as $\sum_{j < q-1} |\gamma_j|^2 \omega_j / \omega_{q-1}$ in the unchanged orthogonal expansion of χ . Thus the current quantitative expression is $\omega_{N+1} \Delta_N / (k^6 \omega_N D_N)$ along admitted growing degrees. Its limit has not been estimated. The complete lower bound (R30), the positive exterior data (R32), and the changing metric (R27) are retained together for that calculation.

318.12 The original Toda volumes and the complete four-term loss

The Toda continuation now calculates this same quantity from the original source and relation volumes. In the notation proved in (TVB.1)–(TVB.38c), use $k \geq 1$, a nonempty dagger-stable packet with its full zero orders, $q \geq 1$, real $|\theta| < \pi/2$, and $N \geq q$, except at the separately stated $N = q - 1$ endpoint, throughout (R33)–(R34). We retain $c = k/2$, $Z(\theta) = M_h(\theta)^k$ and $Z_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta)Z$. The Hankel determinants $\mathfrak{D}_n = \det[Z^{(a+b)}]_{a,b < n}$ and $\mathfrak{B}_m = \det[Z_\chi^{(a+b)}]_{a,b < m}$ are the actual source and relation Gram determinants. With $m = N - q + 1$ their exact determinant-line map and the rank-one update give

$$\begin{aligned} V_N &= \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}, \quad \delta_N = \frac{V_N}{V_{N-1}}, \quad \alpha_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \\ \psi_N &= \frac{\text{Tr } A - cq}{i} - \partial_\theta \log V_N, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i\psi_N, \\ \epsilon_N^2 &= \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2 = \alpha_{N+1} \frac{\Delta_N}{\det K_N}, \quad N \geq q. \end{aligned} \quad (\text{R33})$$

Every equality has its complete source, phase and determinant proof in the cited bridge. At $N = q - 1$, the exact replacement remains $\epsilon_{q-1}^2 = (\nu_0 - \omega_q)/\omega_{q-1} - \psi_{q-1}^2$, with $\nu_0 = \|\chi\|^2$; no preceding inverse occurs. The explicit conjugate-linear map (TVB.38b) sends the lower orthogonal projection of χ to the actual positive-minor exterior source vector, with the full squared-norm factor $\det K_{q-1}/\omega_q$.

The two scalar Toda curvatures have full source maps. For $X = (S - c)/i$, the actual least-norm lift satisfies $R'_N = -I_N$, where $I_N = P_{D_N} X R_N$ and $O_N = (1 - P_N) X R_N$ in the current tilted pairing, where P_N is the tilted source projection onto $\mathcal{P}_N$. With $\beta_m = \nu_m/\nu_{m-1}$ for $m \geq 1$, the proofs of (TVB.17)–(TVB.21) give

$$\begin{aligned} G''_N - G'_N G_N^{-1} G'_N &= O_N^* O_N - I_N^* I_N, \\ \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1}(1 - \delta_N), \quad \text{Tr}(G_N^{-1} I_N^* I_N) = \beta_m - \alpha_{N+1} \delta_N. \end{aligned} \quad (\text{R34})$$

The common subtracted term is present in both positive costs. At the first admitted degree the two costs are α_q and zero, with no negative-index relation norm. Every derivative boundary has the literal cochain primitive (TVB.9), including its tensor signs.

The coordinating Zeta contribution joins the two volume losses with the original spectral coupling. At $\theta = 0$, $N \geq q$ and $L > 0$, let $\mathbf{P}$ be the G_N -orthogonal projection onto the full positive spectral subspace, $\mathbf{C} = \mathbf{P}A(1 - \mathbf{P})$ its restricted coupling, and $M = E^\# \mathbf{P}E$, where E embeds the two G_N -unit eigenvectors of $A^\# + A - kI$ at $\pm \epsilon_N$. Set $v_N = -\log \delta_N$, $s_N = (v_N + v_{N+1})/2$ and $t_N = (v_{N+1} - v_N)/2$. The complete derivation in (TVB.39)–(TVB.43), using the same metric on both sides, proves

$$\begin{aligned} \alpha_{N+1} \sinh^2 s_N - L^2 &= \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 + \psi_N^2 \\ &\quad + \|\mathbf{C}\|_{\text{HS}, G_N}^2 + \epsilon_N^2 \{\det M + \det(I_2 - M)\}. \end{aligned} \quad (\text{R35})$$

All four terms are nonnegative. In particular the two-step volume lower bound retains $L^2 + \psi_N^2 + \|\mathbf{C}\|_{\text{HS}, G_N}^2$ inside its inverse-hyperbolic-sine expression. The exact Sylvester inverse maps $\mathbf{C}$ to the difference between spectral and orthogonal projectors; its full original nilpotent factors and singular value remain specified. The complete supplied join is also reproduced as a source appendix.

318.13 Certified theta input and a calculated comparison of the two gaps

The theta continuation proves the sector construction and the exact maps

$$h(D)F_h = f_0, \quad w_1(t) = |h(1/2 + it)|^2 w_h(t), \quad M_h(\theta) = 2 \int_1^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (\text{R36})$$

The absolute-square integral is for real $|\theta| < \pi/2$; its holomorphic strip continuation is proved in the theta input chapter. The dagger sign, complex Fourier phase, endpoint Jacobian and source factor $g = 2\xi$ are all retained in (TI.1)–(TI.16). The exact incomplete-gamma double series and its explicit infinite-square-tail bounds (TI.19)–(TI.30) supply rigorous interval enclosures for $M_1(0)$, $M_1(1/10)$, $M_1''(0)$ and $\alpha_1^{(k)}/k = M_1''(0)/M_1(0)$. Each displayed rational interval in that proof has width $3 \cdot 10^{-40}$. The certificate adds the proved infinite tail to outward-rounded ball arithmetic; it does not infer an enclosure from precision agreement. The source norm is positive for $h = 1$ while its arithmetic cyclic quotient is zero. The density identity in (R36), and the globally constructed Mellin division, specify the exact passage to packet input without division of a measured zero value by zero. No packet determinant estimate follows from these four seed intervals alone. For the empty packet $h = 1$, the arithmetic quotient is zero and $V_N = 1$ by the empty-determinant convention. For $q = 1$, $\epsilon_N = 0$ at every admitted degree, whereas the upper Toda expression is strictly positive for $N \geq 1$, as proved in the bridge.

Finally (TC1)–(TC13) calculate both gaps in the original Gaussian calibration with mass μ , variance σ^2 , $c = 1/2$ and $\chi = (S - c)^2 - \sigma^2$. The source and quotient coordinate maps are fully displayed there. They give

$$\begin{aligned} (V_1, V_2, V_3) &= \mu^2 \sigma^2 (1, 1/3, 1/11), \quad \epsilon_1^2 = L_{\text{spec}}^2 = 4\sigma^2, \quad \epsilon_2^2 = 16\sigma^2/3, \\ \alpha_3 \frac{(\mathcal{R}_2 - 1)^2}{4\mathcal{R}_2} - \epsilon_2^2 &= 49\sigma^2/33, \quad \epsilon_2^2 - L_{\text{spec}}^2 = L_{\text{spec}}^2 \|Q - P\|_{\text{HS}, G_2}^2 = 4\sigma^2/3. \end{aligned} \quad (\text{R37})$$

The first gap is precisely the Toda imbalance square; the second has the explicitly calculated spectral/orthogonal projector map. This calibration uses declared polynomial coefficients. The actual growing packet calculation continues with (R33)–(R36), retaining its arithmetic measure and all four terms of (R35).

318.14 Gamma convolution, certified higher input, and the complete quotient comparison

Result R38: an exact analytic and finite coefficient interface. Keep the original $w_h(t) = |g(1/2 + it)/h(1/2 + it)|^2/(2\pi)$ and $r_\lambda(t) = |\Gamma(\lambda + it/2)|^2/(2\pi)$, with $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda)$. The full Gamma source appendix and the coefficient-join proof construct

$$r_{\lambda,k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad m_{h,k} = w_h^{*k} = r_{\lambda,k} C_\Sigma(B_{h,\lambda}^{\otimes k}), \quad B_{h,\lambda} = |v_h/\Gamma(\lambda + it/2)|^2.$$

The Hilbert maps $U_\Sigma f = f(\sum t_i)$ and its conditional adjoint C_Σ have the inverse paired decomposition $F \leftrightarrow (C_\Sigma F, (I - U_\Sigma C_\Sigma)F)$. All masses and the relative vector remain. With $\alpha = 2\lambda$,

$$c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha)_j}, \quad \mathcal{A}_h(z) = \sum_{j \geq 0} (\alpha)_j c_{h,j} z^j, \quad d_{h,k,n} = [z^n] \mathcal{A}_h(z)^k,$$

the exact finite moment is $\int b_n^{(k\lambda)} m_{h,k} = c_\lambda^k n! d_{h,k,n}$. The original degree- N source and relation Grams at theta zero use $c_{h,0}, \dots, c_{h,2N}$; their derivative of order r uses the list through $2N+r$, through the fully written original- S polynomial test. The analytic join supplied by the Zeta task and proved in the coefficient chapter is

$$\mathcal{A}_h(z) = c_\lambda^{-1} (1 + z^2)^{-\lambda} M_h(\arctan z), \quad |z| < 1, \quad M_h(\theta) = \int e^{\theta t} w_h(t) dt.$$

The specified branches at zero and exponential domination justify all coefficient derivatives. Its inverse uses $M_h(\theta) = c_\lambda \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta)$ for $|\operatorname{Re} \theta| < \pi/4$, with the analytic branches continued from zero. The complete source appendix is preserved alongside the explicit cochain correction; applying $\mathcal{V}$ to a section difference gives the boundary, and (GP.26)–(GP.27) construct its primitive.

Result R39: certified original moments and signed coefficients. Equations (GS.1)–(GS.40) prove the original theta derivative recurrence

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y),$$

$$\mu_{2r} = 2 \int_1^\infty \left| \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2} \right|^2 dx.$$

Their fully specified positive square tail (GS.18) bounds the omitted infinite integer sum at every fixed order. At order two and three, the rigorous rational enclosures are

$$\begin{aligned} \mu_4 &\in [383.2743417173642425544072374791308324300288, \\ &\quad 383.2743417173642425544072374791308324300291], \\ \mu_6 &\in [17946.5273016052076235667646919834204772848931, \\ &\quad 17946.5273016052076235667646919834204772848934]. \end{aligned}$$

The original reference $c_{13/4} = 10395\sqrt{\pi}/(2048\sqrt{2})$ is enclosed separately from the actual mass μ_0 . The complete invertible moment/coefficient map (GS.34)–(GS.35) certifies c_0, c_2, c_4, c_6 and proves $c_4 < 0, c_6 < 0$. Its tensor coefficient of degree six retains all three terms

$$d_{k,6} = k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3, \quad a_j = (13/2)_j c_j.$$

The accompanying source-cost continuation (GS.41)–(GS.50) proves the first four monic norms h_0, h_1, h_2, h_3 and all three positive ratios $a_j^{(k)} = h_j/h_{j-1}$ from these same moments. In particular, with $\nu = \mu_2/\mu_0$ and the original sum moments $M_j(k)$,

$$a_1^{(k)} = k\nu, \quad a_2^{(k)} = \frac{\mu_4}{\mu_2} + (2k-3)\nu, \quad a_3^{(k)} = \frac{M_6 - M_4^2/M_2}{M_4 - M_2^2/M_0}.$$

The exact polynomial isometry $P(S) \mapsto P(k/2 + iu)$ and its monic phase i^j preserve these norms and prove the original spectral recurrence $SP_j = P_{j+1} + (k/2)P_j - a_j^{(k)}P_{j-1}$. For the source determinant corrections, put

$$Q_N = \frac{X_{N+2}X_N}{X_{N+1}^2}, \quad Q_0 = \frac{2\nu}{13}, \quad Q_1(k) = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \frac{\mu_4}{\mu_2} - \frac{43\nu}{13}.$$

The certified interval for ρ is strictly negative; hence $0 < Q_1(1) \leq Q_1(k) < Q_0$ and $Q_1(k+1) - Q_1(k) = -13\rho/((13k+2)(13k+15)) > 0$. These are exact statements at every tensor degree for the specified source polynomial degrees. The analytic seed has the original zero arithmetic quotient $\mathbb{C}[S]/(1)$ throughout.

Result R40: finite error propagation to the actual consecutive ratio. The exact polynomial coefficient power and test expansion in (GMT.12)–(GMT.17) propagate finite coefficient enclosures into Hermitian source envelopes. The full boundary-resolvent formula (GMT.19)–(GMT.20) carries those actual forms to their least lifts, including the signed quadratic loss and its original theta primitive. Equations (GMT.29a)–(GMT.29b) also certify every coefficient of the least lift and the full representative loss $\widehat{R}^*H\widehat{R} - G(H)$ by an explicit positive q -dimensional rational upper form. The positive finite envelopes induce quotient-dimensional determinant enclosures by minimizing on the same affine space of representatives. For $N \geq q$, retain the exact two-degree source block

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^*H_-^{-1}C_D, \quad F_D = J_D - J_-H_-^{-1}C_D, \quad Z_D = F_D^*G_-F_D.$$

Equations (GMT.22)–(GMT.27) prove, in the original coordinates,

$$\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W}, \quad a_{N+1} = \frac{\det W}{W_{11}^2}.$$

Applying the same maps to the Gamma reference gives the exact relative correction T_{N-1}/T_{N+1} as the quotient of these two positive ratios. The fully proved auxiliary bounded multiplier $\min(1 + t^2/4, 10001)$ with $\lambda = 1$, $k = 2$, and the declared $\chi(S) = (S-1)^2 - 2$ has $T_1/T_3 < 1$. Its exact finite tail, rational envelopes, relation matrices, and source/quotient maps are (GMT.32)–(GMT.42). Thus positivity and boundedness alone do not imply relative correction monotonicity. This explicit counterexample is connected to the arithmetic problem by those same coefficient, least-lift, and determinant maps; its auxiliary weight is not assigned the value of an actual zeta packet weight.

Result R41: the complete phase and a strict relative contribution. For the actual complex product $a = \prod_i A_{h,\lambda}(t_i)$, put $a_k = C_{\Sigma}a$, $b_k = C_{\Sigma}|a|^2$, and $d_k = b_k - |a_k|^2$. Equations (GP.6)–(GP.11) prove the exact unitary transport of the arithmetic sum projection and its fibre defect. Equations (GP.12)–(GP.13) use the original quartet tail, with its full constant $25 \cdot 2^{2d}/\sqrt{\pi}$, to prove $d_k(u) > 0$ at every real u for $\lambda = 1/4$, $d \geq 4$, and $k \geq 2$. The full complex coefficient transform and its Parseval identity preserve the factors c_{λ}^k and $n!$. The source matrices and the original quotient satisfy

$$\begin{aligned} M_h &= M_{\text{coh}} + M_{\text{rel}}, & R_h &= R_{\text{coh}} - B_{\chi}K, \\ K &= (B_{\chi}^*M_hB_{\chi})^{-1}B_{\chi}^*M_{\text{rel}}R_{\text{coh}}, \\ G_h &= G_{\text{coh}} + K^*B_{\chi}^*M_{\text{coh}}B_{\chi}K + R_h^*M_{\text{rel}}R_h. \end{aligned}$$

For every admitted nonzero quartet quotient the last difference is positive definite. The empty boundary at $N = q - 1$ and zero quotient at $q = 0$ retain their explicit separate values. The graph-to-source addition, gamma multiplication, inverse Mellin map, and full jet unit in (GP.24)–(GP.27) reconstruct the original arithmetic observation and its exact supported zero.

The new finite coefficients, strict fibre contribution, and interval certificates therefore enter the same original Toda and exterior quantities in R33–R37. In particular the lower spectral cost and the upper expression are compared in the same G_N , with its volume-imbalance square, phase square, spectral coupling, and projection terms retained. The following arithmetic endpoint continuation proves a norm bound for the original measure and a finite upper comparison at the full amplified degree. Results R42–R46 below give their complete maps and the explicitly calculated gap between the resulting upper and lower volume estimates.

318.15 Arithmetic endpoint estimates and the full reflected relation map

Result R42: the actual monic norm and its exact leading line. The complete arithmetic endpoint source and its analytic audit prove, for the original density and every integer $n \geq k \geq 3$,

$$(\omega_{h,k,2n}/\omega_{h,k,n})^{1/(2n)} \leq C_h n.$$

Here the literal constant C_h is (EP.40), and the original lower convolution envelope is (AU.10), with $B = 42 + 2 \deg h$ and the compact mass v_h^{k-3} unchanged. The complete balanced-window proof strengthens the admitted norm comparison to

$$(\omega_{h,k,n+r}/\omega_{h,k,n})^{1/(2r)} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n,$$

with every factor of C_h^{bal} stated in (EP.50b). The analytic proof derives the local zeta mass from its exact ellipse, the entire identity $g = hv_h$ at the selected centres, the three-factor positive convolution, and the monic Legendre norm. It includes the two Laplace values and their factor two before the final root.

Equations (AL.3)–(AL.10) prove the endpoint and metric of the underlying degree map. Multiplication by the original monic χ gives $\mathcal{L}_{n-q} \simeq \mathcal{L}_n$ for $n \geq q$, with squared operator norm ω_n/ω_{n-q} and adjoint coefficient equal to that same positive ratio. At $n = q - 1$ its source is zero and its target is the original one-dimensional leading line. For $n \geq q$, its representative has the complete Pythagorean decomposition

$$\|\mathcal{V}_{h,k}(\chi p_{n-q})\|^2 = \omega_n + \|\mathcal{V}_{h,k}(\chi p_{n-q} - p_n)\|^2.$$

The correction reduces to $-[p_n]_\chi$ and the relation reduces to zero. Thus the degree-line norm, the original source cost and the supported arithmetic zero are connected by the explicit commutative diagrams of the leading-line chapter.

Result R43: exact products, their optimizer, and full window transport. Keep $n \geq q$, $r \geq 1$, $m = n + r$ and $d_j = V_{n+j}/V_{n+j-1}$. The full radius calculation in (EP.1)–(EP.11), including the original complex phase $b_N^* G_N b_{N+1} = i\omega_N \phi_N$, gives

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_m}{\omega_n} \frac{V_n}{V_m} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2.$$

For the combined budget $\mathcal{B} = -\log(d_0 d_r \prod_{j=1}^{r-1} d_j^2) > 0$, the unique maximizing contraction sequence for this scalar product is

$$d_0 = (1 + 2x)^{-1}, \quad d_r = 1 - 2x, \quad d_j = (1 - x)/(1 + x) \quad (1 \leq j < r),$$

where $0 < x < 1/2$ solves $\mathcal{B} = 2 \operatorname{artanh}(2x) + 4(r-1) \operatorname{artanh} x$. The proof in (EP.22)–(EP.31) establishes existence, uniqueness, the exact maximum, its strict improvement over the earlier hyperbolic-sine bound when $r > 1$, and rational root certificates. The fixed-four-endpoint optimum, the phase polynomial and the quantitative concavity loss are separately proved in (EP.15)–(EP.21). No assertion that the scalar optimizer is realized by an arithmetic source is needed for the proved bound on the original evaluation map.

The full Hermitian window chapter constructs its Schur matrices W_j, F_j, Z_j and their determinant ratios from the same source coefficients. Its finite metric transport retains the eigenvalue spread and the exact hyperbolic coordinate for the four-volume logarithm. Equations (EP.32)–(EP.38) give the literal evaluation map from those matrices to every d_j , the budget and the product. With H_+, p, e defined in (EP.34), the complete cancellation is

$$\frac{\omega_m}{\omega_n} \frac{V_n}{V_m} = \frac{H_+}{pe} = \frac{\mathfrak{B}_{m-q+1} \mathfrak{D}_n}{\mathfrak{D}_m \mathfrak{B}_{n-q+1}}.$$

The confluent transfer proof (EP.38a)–(EP.38k) constructs the raw-derivative matrices for every node and order, proves their invertibility, and proves $\det F_a / \det V = \mathfrak{B}_a / \mathfrak{D}_a$. The sign of the column shift, all derivative factorials and every power of i remain in that construction. It also proves the theta derivative giving the phase and the exact Gamma-reference ratio $\det F_a / \det F_a^\Gamma = Y_a / X_a$.

Result R44: the necessary central and four-volume costs. For an actual off-line quartet with full common multiplicity m_0 , the retained degree and allowance are

$$q_k = [1 + k(m_0 - 1)](k + 1)^2, \\ L_{h,k} = 2\delta[1 + k(m_0 - 1)](k + 1) \lfloor (k + 1)^2 / 4 \rfloor.$$

The complete local tensor calculation and Hermitian compression proof in (EP.41) give $\epsilon_{h,k,N} \geq L_{h,k}$ and $L_{h,k}/q_k \geq \delta k/2$. At $n = q_k, r = q_k - 1$, write $C_k = \log(V_{q_k}/V_{2q_k-1})$. Combining the just-proved balanced norm estimate with the exact product yields

$$C_k \geq 2(q_k - 1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q_k}$$

and the stronger full phase and contraction penalties of (EP.50c). The exact four-volume identity is

$$\log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}} = 2C_k + \log \frac{V_{q_k-1}}{V_{q_k}} + \log \frac{V_{2q_k-1}}{V_{2q_k}}.$$

Both additional logarithms are nonnegative by minimization over nested lift spaces. Substitution of $L_{h,k}/q_k \geq \delta k/2$, division by $q_k \log k$, and $(q_k - 1)/q_k \rightarrow 1$ prove the necessary four-volume lower limit 4 in (EP.50e). The new signed control is now substituted into this same identity. Write $q = q_k$ and retain the arithmetic endpoint losses $E_0 = \log(V_{q-1}/V_q)$, $E_1 = \log(V_{2q-1}/V_{2q})$, and put $\mathcal{B}_{h,k} = 2C_k + E_0 + E_1$. On the original Gamma-to-arithmetic moment path, (FV.M1)–(FV.M18) proves the exact identity $\mathcal{B}_{h,k} = \mathcal{B}_{h,k}^\Gamma + \Delta_{h,k}$ and $|\Delta_{h,k}| \leq \mathfrak{J}_{h,k}$, where

$$\mathfrak{J}_{h,k} = \int_0^1 \min\{S_{\text{AW}}(x), T_q(x), S_{\text{SP}}(x), S_{\text{HS}}(x), (2q-1)\sqrt{1 - e^{-F(x)/(2q-1)}}\} dx.$$

Here $F(x)$ is the unchanged four-volume function and d_x the full original relative spectral diameter. The full formulas for $S_{\text{AW}}, S_{\text{SP}}$, their exact source correspondence and the nonlinear coordinate are proved at that original endpoint site. Equation (FV.M15a) retains $T_q = d_x \|U_x - W_x\|_1/2$ and $S_{\text{HS}} = \sqrt{\Sigma_x \operatorname{Tr}((U_x - W_x)^2)}$, where $\Sigma_x = \operatorname{Tr}(C_x^2) - (\operatorname{Tr} C_x)^2/(2q+1)$, and proves both bounds on the original metric by midpoint centering and centered Hilbert–Schmidt Cauchy–Schwarz. The last minimum entry is the actual-moment specialization of RMT11. Every entry is retained, including

the exact T_q and its majorant S_{Sp} . Let $a = 2q - 1$, $H_0 = 2 \operatorname{arcosh}(e^{\mathcal{B}_{h,k}^\Gamma/(2a)})$, and retain the complete $\mathfrak{J}_{h,k}^{\text{nl}}$ of (FV.M15). For every finite truncation $p_N \geq 0$, the exact source dictionary and finite construction in (FV.M16a)–(FV.M16c) give the signed center and residual j_{p_N}, e_{p_N} : these are exactly J_L, ϵ_L of ISM36 with $L = p_N$, and $X_x^{\text{ISM}} = C_x, \mathcal{A}_x^{\text{ISM}} = U_x - W_x$. They obey $j_{p_N} - e_{p_N} \leq \Delta_{h,k} \leq j_{p_N} + e_{p_N}$ and $e_{p_N} \leq \vartheta^{p_N+1} \sqrt{K_D(8q-6)} \rightarrow 0$ for the fixed original pair of forms, with the unchanged ϑ, K_D in (FV.M16b). Their integrals and the source's own integration errors remain explicit. The simultaneous endpoints are

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{lo}} &= \max \left\{ 0, \mathcal{B}_{h,k}^\Gamma - \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{\max\{0, H_0 - \mathfrak{J}_{h,k}^{\text{nl}}\}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} - e_{p_N} \right\}, \\ \mathcal{B}_{h,k}^{\text{hi}} &= \min \left\{ \mathcal{B}_{h,k}^\Gamma + \mathfrak{J}_{h,k}, 2a \log \cosh \left(\frac{H_0 + \mathfrak{J}_{h,k}^{\text{nl}}}{2} \right), \right. \\ &\quad \left. \mathcal{B}_{h,k}^\Gamma + j_{p_N} + e_{p_N} \right\}. \end{aligned}$$

Monotonicity of the exact inverse coordinate in (FV.M16)–(FV.M18) and the signed bound give

$$\begin{aligned} \mathcal{B}_{h,k}^{\text{lo}} &\leq \mathcal{B}_{h,k} \leq \mathcal{B}_{h,k}^{\text{hi}}, \\ 4(q-1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q} + E_0 + E_1 &\leq \mathcal{B}_{h,k}^{\text{hi}}, \quad C_k \leq \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}. \end{aligned}$$

The last two inequalities substitute the retained central lower bound and the exact four-volume identity; neither arithmetic endpoint loss is omitted. The proper-source primitive residual is the exact tensor intersection quotient (FV.M13), with the one-leg V/W map proved in (FV.M14), so this metric comparison does not erase any proper-source class.

Equations (EP.44)–(EP.48) construct the original relation layer and the exact change of least lift. Its central Gram has largest relative eigenvalue at least $\delta^2 k^2 / (4(C_h^{\text{bal}})^2) - 1$ by (EP.50f). The cochain primitive and supported quotient image are written there for the same representative.

Result R45: an actual arithmetic upper comparison and its calculated gap. The new arithmetic upper-volume chapter constructs the positive measure

$$d\nu_k(u) = a_k e^{-(\pi/2)|u|} (k-2+|u|)^{-B} du, \quad a_k = c_h \vartheta_h^{k-3} e^{-(\pi/2)(k-3)},$$

which is dominated by the original $m_{h,k}(u) du$ for $k \geq 3$. The identity on polynomials is contractive and commutes with every original raw jet, remainder and spectral multiplication map. Its moments are the complete positive Gamma-mixture integrals (AU.12). Writing K_N for the original raw-jet kernel, minimization and positive inversion prove

$$0 \leq C_k \leq \log \det \mathsf{K}_{2q_k-1}^\nu - \log \det \mathsf{K}_{q_k}.$$

The actual arithmetic moments needed by this bound stop at degree $2q_k$; the comparator moments through $4q_k - 2$ have the explicit positive integral in (AU.12). The exact finite Gamma coefficient map (AU.16) gives the same cutoff on $c_{h,j}$. The two-Laplace-value majorant, positive Cauchy–Binet minor sum, and interval Legendre jet kernel in (AU.17)–(AU.23) supply further proved upper expressions for this same C_k .

The full scalar proof (AU.24)–(AU.33) gives

$$\begin{aligned} C_k &\leq \mathcal{U}_k = (\alpha D + \log 256) q_k^2 \\ &\quad + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k q_k \\ &\quad + B q_k \log(1 + (D+1)q_k) + q_k \log C_{\text{env}}, \end{aligned}$$

where $\alpha = \pi/2$, $R_0 = \sqrt{\delta^2 + \gamma^2}$, $D = \max(1, 2/b, 2R_0)$, $0 < b < \pi/2$, $X = \max(1, M_h(b), M_h(-b))$, $K = \max(1, \vartheta_h^{-1})$, and $C_{\text{env}} = \max(1, 2/(3c_h D))$. All constants and the literal source mass are retained. For $\mathcal{L}_k = 2(q_k - 1) \log(\delta k / (2C_h^{\text{bal}}))$, (AU.32) computes every term of $\mathcal{U}_k - \mathcal{L}_k$, and (AU.33) proves that its quotient by q_k^2 tends to $\alpha D + \log 256 > 0$. This upper estimate leaves the stated gap to the quartet lower estimate. The propagated original-source estimate is the additional bound

$$C_k \leq \mathcal{U}_k^{\text{signed}} := \frac{\mathcal{B}_{h,k}^{\text{hi}} - E_0 - E_1}{2}.$$

Its complete proof (AU.31a)–(AU.31b) transports the unchanged monomial source and original relation by $P(S) \mapsto P(k/2 + iu)$, retaining the factor i^q and every raw derivative factor i^d . This signed entry uses actual moments through $4q$; the original first matrix bound continues to require actual moments only through $2q$. Let $\hat{\mathcal{U}}_k$ be the minimum of this signed entry and all the full bounds in (AU.31b), including the original $\mathcal{U}_k$ and the actual low-degree matrix entry. They bound the same C_k , so

$$\mathcal{L}_k \leq C_k \leq \hat{\mathcal{U}}_k \leq \mathcal{U}_k, \quad 0 \leq \hat{\mathcal{U}}_k - \mathcal{L}_k \leq \mathcal{U}_k - \mathcal{L}_k.$$

The positive limit in (AU.33) belongs to the explicitly displayed old scalar majorant; it is not assigned to this refined minimum. The complete angle, spectral, and overlap integrands remain the actual quantities controlling the newly narrowed finite interval.

The complete auxiliary Lebesgue calculation proves

$$|\log(V_q^{\text{aux}}/V_{2q-1}^{\text{aux}}) - (\frac{9}{2} \log 3 - 6 \log 2) q^2| \leq 12q \log(8q)$$

for every even $q \geq 2$. Its full-origin-jet quotient is connected to the original grid by the free monic family $\mathbb{C}[t, u] / \prod_{a,b} (u - t\zeta_{ab})^\ell$ in (AU.46)–(AU.48). The original full jets at $t \neq 0$, the merged jet at zero and the source-dependent kernel derivative are all explicit. The displayed auxiliary estimate belongs to its Lebesgue source; the exact derivative gives the change along the family, without assigning the same asymptotic to the arithmetic source.

Result R46: reflected norm, conormal algebra and their complete kernels. For any original monic χ of degree $q \geq 1$, set $\chi^\dagger = (-1)^q \bar{\chi}(k - S)$, $N_\chi = \chi\chi^\dagger$ and $g = \gcd(\chi, \chi^\dagger)$. The reflected relation chapter proves the algebra isomorphism

$$\mathbb{C}[S]/(\chi^2\chi^\dagger/g) \simeq \mathbb{C}[S]/(\chi^2) \times_{\mathbb{C}[S]/(\chi g)} \mathbb{C}[S]/(\chi\chi^\dagger),$$

with its exact sequence, inverse residue construction and original M_S action. The norm map retains $\chi(k/2 + iu) = i^q \psi(u)$ and $N_\chi(k/2 + iu) = i^{2q} \psi(u) \bar{\psi}(u)$. Thus the full union of reflected and original roots, with added orders at overlaps, is the precise general norm-transfer domain.

The kernel coefficient modules have annihilators (χ) and $(\chi^\dagger)$. Equations (RF.19)–(RF.21) prove their conjugate-linear bijection $F([\chi f]_{\chi^2}) = [\chi f^*]_{N_\chi}$ with inverse, $FM_S = (kI - M_S)F$, and the exact transported norm-kernel product. If a_λ, b_λ are the respective root orders,

$$\dim I_N^t = \sum_{\lambda} \max(b_\lambda - (t-1)a_\lambda, 0), \quad t \geq 1.$$

Equations (RF.22)–(RF.26) prove the exact nilpotence index, the persistent reflected-only CRT algebra, and the fully worked mixed example of kernel dimensions 3, 1, 0. Projection sends each specified kernel vector, including the persistent sector's idempotent when present, to supported zero in its original fibre. The same lifted map fixes the external point τ .

Result R47: restriction spectra, actual arithmetic gaps and finite certificates. For every $q - 1 \leq i \leq j$, retain the original quotient kernels K_N , metrics $G_N = K_N^{-1}$ and minimum source lifts R_N . In the actual monic orthogonal coordinates, let $L_{i,j}$ pad source coefficients and let $P_{i,j}$ be the

orthogonal endomorphism projection onto its range in $\mathcal{P}_j$. The complete restriction source and the arithmetic join prove

$$T_{i,j} = K_i G_j : (E, G_j) \longrightarrow (E, G_i), \quad L_{i,j} R_i T_{i,j} = P_{i,j} R_j.$$

Let $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$ denote the original identity transport, and let A_i, A_j denote the same original action A on these two specified metric spaces. The return is its metric adjoint. The actual endomorphism $U_{i,j} = T_{i,j} I_{i,j}$ of (E, G_i) has coordinate matrix $K_i G_j$. Its eigenvalues $0 < g_a \leq 1$ are the squared singular values of the source restriction between the two canonical minimum-lift ranges. For the orthogonal projections Π_i, Π_j onto those ranges in the same original source, the join constructs both restricted-product similarities and proves

$$\mathrm{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2m} = 2 \mathrm{Tr}_E ((I - U_{i,j})^m), \quad m \geq 1.$$

Its explicit two-dimensional principal-angle blocks retain all common-range and ambient-zero components.

The original arithmetic action A has the exact reverse-map defect

$$A_i T_{i,j} - T_{i,j} A_j = H_i T_{i,j} - T_{i,j} H_j, \quad H_N = K_N (A^* G_N + G_N A - k G_N).$$

The change of representatives is the complete relation

$$\mathfrak{d}_{i,j} = L_{i,j} R_i - R_j I_{i,j} = L_{i,j} R_i (I - U_{i,j}) - (I - P_{i,j}) R_j I_{i,j}, \quad \mathfrak{d}_{i,j}^* O_j \mathfrak{d}_{i,j} = G_i - G_j.$$

Here $\mathfrak{d}_{i,j} : (E, G_i) \rightarrow \mathcal{P}_j$ has the original theta image. Both terms on the right have the same remainder $I_{i,j}(I - U_{i,j}) : (E, G_i) \rightarrow (E, G_j)$: indeed $B_j L_{i,j} R_i = I_{i,j}$ and $B_j P_{i,j} R_j I_{i,j} = I_{i,j} U_{i,j}$. The full division primitive and theta image of their difference are supplied in the join. Reduction sends its labelled relation vector to the supported zero at that receiving label; the lifted map sends external τ to τ .

The arithmetic remainder estimate is extended to every admitted degree with its entire monic quotient formula. In the original notation of the upper-comparison chapter, it gives

$$G_{q-1} \preceq C_N G_N, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}, \quad N \geq q - 1,$$

with $T = Dq, r = kR_0/T < 1$, the unchanged original density mass, and all constants defined in the complete proof. Thus the actual endpoint gaps C_{2q-1}^{-1} and C_{2q}^{-1} are proved arithmetic inequalities. The exact ratio satisfies $C_{2q}/C_{2q-1} \leq 147/5 < 32$, so the same construction gives $\mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32$ with $\mathcal{U}_k$ from R45. The same-source metric refinement now gives the stronger literal joint budget

$$\mathcal{B}_{h,k}^{\mathrm{lo}} \leq \mathcal{B}_{h,k} \leq U_{h,k}^{\mathrm{joint}} := \min\{q \log C_{2q-1} + q \log C_{2q}, \mathcal{B}_{h,k}^{\mathrm{hi}}\}.$$

Equation (ERJ.32a) proves the required equality of the monic-source and monomial-source quotient kernels before taking this minimum. The complete five-estimate pointwise minimum and its nonlinear transport therefore apply to these exact original restriction returns. The two finite trace-certificate upper endpoints below may also be summed and included in the same minimum; their original entry errors and series tails remain attached to their actual matrices.

The finite matrix construction improves these directional bounds. For rational Hermitian enclosures of the actual kernels,

$$0 < L_i \preceq K_i \preceq U_i, \quad 0 < L_j \preceq K_j \preceq U_j,$$

define

$$g = (\mathrm{Tr}(L_i^{-1} U_j) - q + 1)^{-1}, \quad r_* = 1 - g, \quad H_* = I - L_i U_j^{-1}, \quad t_m^* = \mathrm{Tr}(H_*^m),$$

$$d = \text{Tr}(U_i L_j^{-1}) - \text{Tr}(L_i U_j^{-1}).$$

The complete ordered generalized-eigenvalue proof gives $0 < g \leq 1$, $d \geq 0$ and the rational intervals

$$\max\{0, t_m^* - m r_*^{m-1} d\} \leq \text{Tr}((I - U_{i,j})^m) \leq t_m^*.$$

The exact quotient-first triangular basis constructs these brackets from certified source and relation-coefficient intervals, retaining the actual polynomial rather than replacing it by an interval centre. For $r_* > 0$, summing the moment intervals and bounding the positive logarithmic tail gives a certificate for $\log(V_i/V_j)$ with width at most

$$\frac{d(1 - r_*^p)}{g} + \frac{q r_*^{p+1}}{(p+1)g}.$$

The source's sharper coefficient and a rational logarithm enclosure give the further proved refinement. At $g = 1$ the loss is exactly zero. Entry refinement followed by increasing p proves convergence at each fixed pair with every error term displayed.

The balanced central product of R44 now has a consequence for each of the two original endpoint pairs. Put $X = L_{h,k}/(C_h^{\text{bal}} q)$ and retain the full nonnegative phase/contraction penalty P_{mid} from the central product. Since $\text{rank}(K_{2q-1} - K_q) \leq q - 1$, at least one central return eigenvalue is exactly one. The remaining $q - 1$ eigenvalues have product at most $X^{-2(q-1)} e^{-P_{\text{mid}}}$. The ordered Rayleigh-quotient comparison with each endpoint therefore proves

$$\min \text{Spec } U_{q-1, 2q-1}, \quad \min \text{Spec } U_{q, 2q} \leq \min \left\{ 1, X^{-2} e^{-P_{\text{mid}}/(q-1)} \right\} \leq \min \left\{ 1, \frac{4(C_h^{\text{bal}})^2}{\delta^2 k^2} \right\}.$$

Every such eigenvalue is the squared norm fraction of the actual higher-degree minimum representative retained by the lower-degree source projection. The inverse return recovers its original coordinate, with precisely the relation correction proved above.

The whole inverse spectrum has the corresponding finite lower bound. To retain endomorphism types, write $U_{i,j} = T_{i,j} I_{i,j}$ on (E, G_i) , where $I_{i,j}$ is the original identity transport. Its coordinate matrix is $K_i G_j$. For every real $s > 0$, each endpoint pair separately obeys

$$\text{Tr } U_{i,j}^{-s} \geq \max \left\{ q, 1 + (q-1) X^{2s} e^{s P_{\text{mid}}/(q-1)} \right\}.$$

Indeed the central trace is one plus the inverse-power sum of its remaining $q - 1$ eigenvalues. The tangent inequality $e^y \geq 1 + y$ at their logarithmic mean bounds this sum from below by their geometric mean, and the ordered comparison transfers every eigenvalue. These same inverse observations also acquire the finite upper control proved in (ERJ.32b):

$$\text{Tr } U_{q-1, 2q-1}^{-s} + \text{Tr } U_{q, 2q}^{-s} \leq 2q - 1 + e^{s \mathcal{B}_{h,k}} \leq 2q - 1 + e^{s U_{h,k}^{\text{joint}}}.$$

Indeed all $2q$ return eigenvalues $g_{a,j}$ lie in $(0, 1]$ and their negative logarithms sum exactly to $\mathcal{B}_{h,k}$. Writing $z_{a,j} = g_{a,j}^{-s} - 1 \geq 0$, expansion gives $\prod (1 + z_{a,j}) \geq 1 + \sum z_{a,j}$, which is precisely the displayed bound. Each individual eigenvalue also satisfies $g_{a,j} \geq e^{-\mathcal{B}_{h,k}} \geq e^{-U_{h,k}^{\text{joint}}}$. The original lower bounds and central unit eigenvalues above are retained simultaneously, with their full phase/contraction penalty.

If the exact central update rank is $\rho = \text{rank}(K_{2q-1} - K_q) > 0$, put $A_+ = \max\{0, 2(q-1) \log X + P_{\text{mid}}\}$. Retaining all $q - \rho$ unit returns gives the stronger bound

$$\text{Tr } U_{i,j}^{-s} \geq q - \rho + \rho e^{s A_+/\rho}.$$

At $\rho = 0$ the central trace is exactly q and both endpoint traces are at least q . The complete proof and rank comparison are (ERJ.42)–(ERJ.42a). Under the canonical isometries the source restriction is the bijection $\mathcal{R} = \Pi_i|_{\ell_j E}$ between the two canonical images. Its inverse singular-value-power sum is exactly $\sum_a \sigma_a(\mathcal{R}^{-1})^{2s} = \text{Tr } U_{i,j}^{-s}$ by (ERJ.11a). The ambient zero eigenspaces of the full projection products remain outside this restricted inverse and are retained by (ERJ.9).

Result R48: the complete arithmetic weight optimizer and a stronger moment family. For the original relation χ , coordinate $S = k/2 + iu$ and full raw-jet matrix E_q , put $\psi(u) = i^{-q}\chi(k/2 + iu)$, $Z = E_{q-1}$ and $\beta_j = b^{2j}/(2j)!$ for $0 < b < \pi/2$. The weight-optimization chapter proves the signed cofactor map

$$\det E_q[\cdot, \{0, \dots, q\} \setminus \{j\}] = (-1)^{q-j} \psi_j \det Z, \quad \psi_j = \frac{i^{j-q}}{j!} \chi^{(j)}(k/2).$$

In particular every vanishing coefficient has its corresponding zero minor, with the full jet space retained. For auxiliary weights $\tau_j \geq 0$ of sum one, set

$$\eta_j = \frac{(2j)!}{(j!)^2 b^{2j}} |\chi^{(j)}(k/2)|^2, \quad D_\eta(\tau) = \sum_{j=0}^q \eta_j \prod_{l \neq j} \tau_l, \quad d(\eta) = \max_{\tau \geq 0, \sum \tau_j = 1} D_\eta(\tau).$$

Equations (OW.4)–(OW.13) prove the precise determinant identity

$$\det(E_q \operatorname{diag}(\tau_j \beta_j) E_q^*) = |\det Z|^2 \left(\prod_{j=0}^q \beta_j \right) D_\eta(\tau)$$

and its original phase-preserving congruence. The chapter proves every maximizer: a largest coefficient at least the sum of the others gives the corresponding zero weight and the remaining weights $1/q$; exactly two equal positive coefficients give the explicit segment of maximizers. Otherwise the maximizer is unique and interior. Its zero-coefficient weights are $1/q$, and its positive-coefficient weights are $(1 - y_j)/q$, where

$$\sum_{\eta_j > 0} y_j = 1, \quad \eta_j = c_* y_j (1 - y_j), \quad 0 < y_j < 1.$$

The two monotone scalar constructions (OW.16)–(OW.19) determine the unique solution, including the branch with one large root. Equations (OW.21)–(OW.22a) prove the rational three-coefficient formula, coefficient-error bounds and terminating rational value certificates. The linear case $q = 1$ and all boundary cases are included in the same complete proof.

With the literal original $M_k = M_h(b)^k + M_h(-b)^k$ and comparator volume V_{2q-1}^ν from R45, the resulting arithmetic bound is

$$C_k \leq \mathcal{U}_{\text{opt}} = q \log M_k - \log \left(\prod_{j=0}^q \beta_j \right) - \log d(\eta) - \log V_{2q-1}^\nu.$$

The limit of the original positive raw-kernel inequality proves this bound at boundary optima; no reciprocal zero weight is used. For $\mathcal{U}_{\text{eq}}$ obtained from the same inequality with equal weights, (OW.23)–(OW.29) prove

$$0 \leq \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} = \log \frac{d(\eta)}{\sum_j \eta_j / (q+1)^q} \leq q \log(1 + 1/q) < 1.$$

Both non-strict inequalities are strict for the admitted complete quartet: its parity gives a zero coefficient, while its nonzero roots and monicity give at least two nonzero coefficients. Thus the finite optimization within this specified family is solved, and its improvement divided by $q_k \log k$ or q_k^2 tends to zero.

The continuation (OW.30)–(OW.40) proves a stronger source inequality. Write $\mu_{2j} = \int u^{2j} m_{h,k}(u) du$, retaining $\mu_0 = \mu_h^k$, and set $S_q(b) = \sum_{j=0}^q \beta_j \mu_{2j}$. Pointwise Cauchy–Schwarz and the full positive even exponential series, integrated against the original source, give

$$H_q \preceq S_q(b) \operatorname{diag}(\beta_j^{-1}) \prec \frac{M_k}{2} \operatorname{diag}(\beta_j^{-1}), \quad S_q(b) < M_k/2.$$

The corresponding upper expression $\mathcal{U}_{\cosh}$ improves the earlier $\mathcal{U}_{\text{eq}}$ by exactly $q \log(2(q+1))$. Using $S_q(b)$ gives the further strict improvement $q \log(M_k/(2S_q(b)))$. Every constant follows from the integrated finite sum and both original Laplace values.

For every positive auxiliary list r_j , the same calculation proves

$$H_q \preceq \left(\sum_{j=0}^q r_j \mu_{2j} \right) \text{diag}(r_j^{-1}).$$

The exact positive-cone map $t_j = r_j \mu_{2j} / \sum_l r_l \mu_{2l}$, with inverse $r_j = st_j / \mu_{2j}$, transforms its determinant optimization to the already solved polynomial D_κ , where $\kappa_j = |\psi_j|^2 \mu_{2j}$. Thus the entire stronger diagonal family has the proved optimum

$$C_k \leq \mathcal{U}_{\text{mom}} = \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log V_{2q-1}^\nu.$$

All $q+1$ moment factors, every zero coefficient and the boundary raw-kernel limit are retained. The ray $r_j = \beta_j$ maps to $t_j^\beta = \beta_j \mu_{2j} / S_q(b)$, and the exact total gain is

$$\mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{mom}} = q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} + \log \frac{d(\kappa)}{D_\kappa(t^\beta)}.$$

The last term is nonnegative and is evaluated by the same complete optimizer. The removed term $q_k \log(2(q_k+1))$, divided by $q_k \log k$, tends to 2 for a simple quartet and to 3 for fixed common multiplicity greater than one.

Finally the original quotient-first source Gram proves $V_q = \det H_q / (\psi^* H_q \psi)$. Equation (OW.40) therefore calculates the exact remaining comparison loss against the full low-degree kernel:

$$\mathcal{U}_{\text{mom}} - (\log V_q - \log V_{2q-1}^\nu) = \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log \det H_q + \log(\psi^* H_q \psi) \geq 0.$$

These identities retain all actual moments through degree $2q$. They determine the finite improvement and its precise comparison map without assigning an unproved growth rate to the last loss or the comparator determinant. The original source, support label and quotient action remain connected through the same cofactor and kernel maps.

Result R49: the first-degree phase product and each full block budget. The complete first-window chapter joins the original first-degree radius to the comparable-window theorem of PR 25, retaining the nonnegative generalized-eigenvalue correction. At $N = q - 1$, expand $\chi = p_q + \sum_{j < q} \gamma_j p_j$ and keep $\nu_0 = \|\chi\|^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j$. The rank-one update and the original phase identity give

$$\frac{V_q}{V_{q-1}} = \frac{\omega_q}{\nu_0}, \quad \gamma_{q-1} = -i\phi_{q-1}, \quad \epsilon_{q-1}^2 + \phi_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}.$$

Equations (CJ.3)–(CJ.7) prove each equality in the actual source. For $\Lambda_N = \omega_N / V_N$, the first step consequently has exactly one contraction:

$$(\epsilon_{q-1}^2 + \phi_{q-1}^2) \Lambda_{q-1} = \Lambda_q (1 - \delta_q).$$

Multiplication by the regular consecutive identities gives the full first-window formula (CJ.9), for every $j \geq q$:

$$\prod_{N=q-1}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_{q-1}}{\omega_{q-1} V_j} (1 - \delta_j) \prod_{N=q}^{j-1} (1 - \delta_N)^2.$$

The original source contains no quotient inverse below degree $q - 1$. The complete two-sided monic norm envelope (CJ.15)–(CJ.18) supplies the explicit C_h^{win} for $n/2 \leq r \leq 2n$. For $D_h = \delta/(2C_h^{\text{win}})$ and each original block $(i, j) = (q - 1, 2q - 1)$ or $(q, 2q)$, it proves the finite bound

$$\log(V_i/V_j) \geq 2q \log(D_h k) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L_{h,k}^2).$$

The penalties $\mathcal{P}_{i,j}$ are written completely in (CJ.13), including the single terminal factor for the first block and both boundary factors for the regular block. All terms added to the per-block $2q$ budget are nonnegative.

The original relation map in (CJ.20)–(CJ.21) has Gram $\Omega + F^*G_iF$ and is an isomorphism onto the whole newly admitted relation layer. At zero tilt and even tensor degree, the exact quartet's first increment matrix has a kernel by the full parity calculation. Every zero generalized eigenvalue stays in the q -dimensional cost spectrum, contributing the factor one to the determinant. For all $s > 0$, (CJ.22) also proves

$$\text{Tr } Q_{i,j}^{-s} \geq \max\{q, q(D_h k)^{2s}\}, \quad Q_{i,j} = T_{i,j}I_{i,j} \text{ on } (E, G_i).$$

These are traces of the same original metric endomorphism. The full cost, source relation, and its supported-zero quotient image remain connected by the maps in that chapter.

Result R50: the two relation traces and their actual comparison source. The complete relation-moment chapter proves the sharp finite determinant majorant from the two traces of the original positive cost operator $Z = (K_j - K_i)G_i$, keeping its full dimension and every zero eigenvalue. Writing $t_1 = \text{Tr } Z$, $t_2 = \text{Tr } Z^2$, for $q \geq 2$ define

$$d = \sqrt{\frac{qt_2 - t_1^2}{q - 1}}, \quad a = \frac{t_1 + (q - 1)d}{q}, \quad b = \frac{t_1 - d}{q}.$$

The complete rational Hermite remainder in that chapter gives

$$\log(V_i/V_j) = \log \det(I + Z) \leq \log(1 + a) + (q - 1) \log(1 + b).$$

For dimension one the exact expression is $\log(1 + t_1)$, and for dimension zero it is zero. The proof identifies both traces with the original relation map and its exterior square, including every cross pairing. For the arithmetic upper kernel U_j , the positive space $\mathcal{C}_+ = (E^\vee / \ker(U_j - K_i), U_j - K_i)$ and its map $J_+[\alpha] = (U_j - K_i)\alpha$ realize the comparison cost $Z_+ = (U_j - K_i)G_i$. The constructed contraction from the original relation source has the complete positive defect

$$J_+(I - CC^\dagger)J_+^\dagger = (U_j - K_j)G_i.$$

This retains the precise map and the excess in the bound, together with the original arithmetic action defect and all unit return factors. The relation-moment chapter contains the full definitions and proofs.

Result R51: the constructed Deligne comparison maps and constituent coupling. The detailed Deligne sidebar proves the image, kernel and cone maps for the same compact-to-ordinary comparison, and the complementary degree duality with its actual boundary line. For every specified invariant $F \subseteq E$ and divisor depth $m \geq 1$, its original curve has the logarithmic complex

$$\mathcal{H}_{F,m} = [F\mathcal{O} + E\mathcal{O}(-m\infty) \xrightarrow{d} (F\mathcal{O} + E\mathcal{O}(-m\infty)) \otimes \Omega^1(\log \infty)].$$

Its complete cohomology is F in degree zero, $(E/F) \otimes \mathbb{T}$ in degree two and zero in degree one, with $\mathbb{T} = H^1(\Omega^1)$ and the retained trace $\text{tr}[dz/z] = 1$. The higher-depth quotient contracts by division

by the displayed nonzero integer coefficients; its dual retains all meromorphic powers and the twist by $\mathcal{O}(-\infty)$. Each split lift distinguishes the represented kernel from external absence through their disjoint inclusions into the full kernel carrier.

The same original metric, $G(t) = G_j + t(G_i - G_j)$, gives the invariant quotient metric and coupling map proved in (DS53)–(DS65). With those specified maps, its endpoint identity is

$$\log(V_i/V_j) = \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}.$$

The series has nonnegative terms before subtraction, and the sidebar proves its finite remainder. It also proves the exact original theta source defect $dK(z)H_t + r(z)Ia_t$, so the theta boundary and the retained constituent pass through their specified successive quotients. The two positive determinants can use R50 on their respective compressed matrices; the coupling subtraction remains in the same upper estimate.

Result R52: a degree-one exponential family with exact periods. For the original cyclic polynomial χ and its specified primitive Φ , the delivered exponential note proves

$$\mathcal{H} = \text{coker}(u\partial_S + \chi(S) - t), \quad \mathcal{H} \cong \mathbb{C}[u, t]^q, \quad \mathcal{H}/(u, t)\mathcal{H} \cong E, \quad (-u\nabla_t) \bmod (u, t) = A.$$

Its coefficient-module division, original theta observation, finite-rank source deformation, infinity chart and dual residue pairing are all retained with complete proofs in the exponential note and sidebar. The new determinant proof (XD1)–(XD26) computes the entire period determinant for the stated oriented rays:

$$\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp\left(\frac{\text{Tr} \Phi_t(A(t))}{u}\right).$$

All Gamma factors and monomial phases are given in (XD16)–(XD18). The proof uses the actual twisted-reduction matrices, keeps their off-diagonal corrections, and proves that their trace corrections vanish. It establishes nonvanishing across every repeated-root parameter. On the original canonical metric it gives

$$\log \det B_N^{\text{per}} = \log C_q(u) + 2 \text{Re} \frac{\text{Tr} \Phi_t(A(t))}{u} - \log V_N.$$

Subtracting at the same parameters retains the exact original endpoint cost. The translation companion (DT1)–(DT13) further proves the complete maps under $\chi_a(S) = \chi(S - a)$: the coefficient action shifts by aI , the period gains the scalar $\exp[(-\Phi(-a) - ta)/u]$ and the full Pascal matrix, and the finite-field realization retains the character sheaf $\mathcal{L}_\psi((-\Phi(-a) - ta)/u)$ on the original (u, t) -base; its point restriction is the stated constant character line. The complete derived direct-image map and its projections are proved in (DT.14)–(DT.16). The original arithmetic fibre remains $a = 0$. These explicit maps preserve the canonical metric comparison and the full inherited endpoint inequalities.

The full connection in the same cyclic coefficient frame is now (XD.u1)–(XD.u7). Define its columns by the literal monic division

$$\Phi_t(S)S^b = (\chi(S) - t)Q_b^\Phi + R_b^\Phi, \quad (C_\chi(t))_b = R_b^\Phi, \quad (B_\chi)_b = (Q_b^\Phi)'.$$

The differential remainder is $R_b^\Phi - u(Q_b^\Phi)'$. With the original branch and all oriented contours retained,

$$\begin{aligned} \Omega_u &= -C_\chi(t)/u^2 + B_\chi/u, & \Pi_u &= \Pi\Omega_u, \\ \Omega_t &= -A(t)/u, & \Pi_t &= \Pi\Omega_t, \\ \partial_u \Omega_t - \partial_t \Omega_u + [\Omega_u, \Omega_t] &= 0. \end{aligned}$$

The degree-zero connection is $\partial_u - \Phi_t/u^2 + 1/u$; the degree-one connection is $\partial_u - \Phi_t/u^2$. Their cochain identity, complete division, and locally uniform contour differentiation are proved in (XD.u1)–(XD.u6). Here

$$C_\chi(t) = \Phi_t(A(t)), \quad (B_\chi)_{bb} = (b+1)/(q+1), \quad \partial_t B_\chi = 0, \quad \text{Tr } B_\chi = q/2.$$

Thus the preceding determinant retains its full matrix connection, with the same original constant

$$K_q = (q+1)^{-q/2} e^{\pi i q/2} \det W \prod_{b=0}^{q-1} \Gamma((b+1)/(q+1)), \quad W_{jb} = e^{2\pi i j(b+1)/(q+1)} - 1.$$

On the exact primary slice $\chi = h = (s - \rho)^m$, $t = 0$, set $n = m + 1$, $c = -(-\rho)^n/n$, $\beta_b = (b+1)/n$ and $d_b = n^{\beta_b-1} e^{\pi i \beta_b} \Gamma(\beta_b)$. The original Pascal matrix has $P_{ab} = \binom{b}{a} \rho^{b-a}$ for $a \leq b$, and the ordered period matrix is exactly $\Pi_s = e^{c/u} W D(u) P$, where $D(u) = \text{diag}(d_b u^{\beta_b})$. For an original p -dimensional ideal inclusion $I = J_{m-p} C_F$, whose J_{m-p} columns are $(s - \rho)^{m-p}, \dots, (s - \rho)^{m-1}$ in the unchanged s -basis, (SC.u5)–(SC.u7) compute

$$\det(I^* \Pi_s^* \Pi_s I) = |\det C_F|^2 n^p (p+1) e^{2p \text{Re}(c/u)} \prod_{b=m-p}^{m-1} |d_b|^2 |u|^{2\beta_b}.$$

The full Gram is $e^{2 \text{Re}(c/u)} C_F^* D_{m-p}^* (W_{m-p}^* W_{m-p}) D_{m-p} C_F$, with $W_{m-p}^* W_{m-p} = n(I_p + \mathbf{1}\mathbf{1}^*)$. Its comparison with the original theta Gram divides the determinant by $\det(I^* G_N I)$, and uses $(I^* G_N I)^{-1}$ on that complete matrix. The proof retains the original source unit and every frame determinant. Its image plane is constant in u , so its normal u -map and u -curvature vanish on precisely this primary $t = 0$ slice. The general two-parameter normal map remains $N \Pi \Omega_a I$ for $a = u, t$, as in (SC.u1)–(SC.u4).

For AP's original packet polynomial h , full fixed Taylor representative ν , and fixed poles $p \in P_\nu$, the localized complex now has the entire connection (APU1)–(APU23). Put $f = \Phi_h - ts$ and

$$Q_p = \frac{f(s, t) - f(p, t)}{s - p} - \frac{h(s) - t}{\deg h + 1}.$$

Its ordinary quotient is the constant $1/(\deg h + 1)$, so its derivative correction is exactly zero. In AP6's original polynomial and simple-pole columns,

$$\Omega_u^{\text{loc}} = \begin{bmatrix} \Omega_u^{\text{pol}} & -Q/u^2 \\ 0 & -\text{diag}(f(p, t))/u^2 \end{bmatrix}.$$

The weighted-residue matrix is $\mathcal{R} = [0, W_{\text{res}}]$, with $W_{\text{res}} = \text{diag}(e^{f(p, t)/u})$; the positively oriented loop map is exactly $(2\pi i)\mathcal{R}$. Both maps are horizontal in u, t . The same complete ν intertwines these connections on the transported gauge frame, with target weight w/ν . The original endpoint-source map is $\sigma_h[\nu] J_0$, with every inverse Taylor coefficient retained, as proved in (APU18)–(APU22). Finally the original entire transport $C_a = \Pi_{\text{loc}}(u, t)^{-1} \Pi_{\text{loc}}(u, t + ua)$ has the exact derivative

$$(C_a)_u = -\Omega_u^{\text{loc}}(u, t) C_a + C_a (\Omega_u^{\text{loc}}(u, t + ua) + a \Omega_t^{\text{loc}}(u, t + ua)).$$

Its t -derivative and the differentiated weighted-residue and loop diagrams are proved in (APU23). The extra $a \Omega_t^{\text{loc}}$ is the derivative of the original parameter shift. The identification $h = \chi_{h,1}$ is the literal variable isomorphism (PC4); at higher tensor degree the cyclic and product families retain their specified η_k and source maps rather than being assigned a common coefficient rank.

Result R53: the residue trace and the complete collision map. For the unchanged monic polynomial $H(S) = \chi(S) - t$, the complete SGA trace chapter constructs the perfect residue functional $\lambda_t[P] = [S^{q-1}] \text{rem}_H P$ and the actual diagonal coevaluation

$$\mathcal{C}_H(X, Y) = \frac{H(X) - H(Y)}{X - Y}, \quad \text{Tr}_{E_t/R[t]} M_P = \lambda_t(\chi' P).$$

The fraction is a polynomial, its multiplication image is χ' , and the chapter proves the inverse pairing identities without dividing by a discriminant. The original logarithmic source relation maps to $[\chi' P]_\chi$ before this trace is taken. Its complete periodic diagonal resolution retains both odd and positive even degrees.

The new comparison (SC26)–(SC30) proves the precise specialization of those degrees. Over $T = \mathbb{C}[t]$, put $\mathcal{E} = T[S]/(\chi - t)$ and $M = \mathcal{E}/\chi'\mathcal{E}$. The isomorphism $\mathcal{E} \simeq \mathbb{C}[S]$, with $t \mapsto \chi(S)$, makes $0 \rightarrow \mathcal{E} \xrightarrow{\chi'} \mathcal{E} \rightarrow M \rightarrow 0$ a free T -resolution. For $\kappa = \mathbb{C}[t]/(t - t_0)$,

$$\text{Tor}_1^T(M, \kappa) \xrightarrow{\sim} \text{ann}_{E_{t_0}}(\chi'), \quad [v]_{\chi'} \mapsto [w]_{\chi - t_0}, \quad (\chi - t_0)v = \chi'w.$$

The proof gives both inverse directions and the exact S -action. Thus every positive even group of the family is zero, while the positive even groups at a collision are these explicit Tor modules; the odd groups are $M \otimes_T \kappa$. Each length- m local factor retains both $m - 1$ -dimensional modules and their nilpotent actions. The Gamma calculation also evaluates the entire period Gram determinant as

$$\det(\Pi^* \Pi) = (2\pi|u|)^q \exp \left(2 \text{Re} \frac{\text{Tr} \Phi_t(A + t\mathcal{R})}{u} \right),$$

with the complete phase-containing determinant still given by R52.

Result R54: exact constituent curvature and neighboring volumes. For an actual invariant injection $I : F \hookrightarrow E$, $0 < p = \dim F < q$, put $\ell[P] = [S^{q-1}] \text{rem}_\chi P$, $\mathcal{R} = e_0 \ell$, $L = \ell I$, $Y = \Pi I$, $H = Y^* Y$, $N = 1 - YH^{-1}Y^*$, and $z = N\Pi e_0$. The complete proof (SC9)–(SC25) identifies the invariant image with the unchanged polynomial ideal $(g)/(\chi)$, proves $L \neq 0$ and $z(t) \neq 0$ for every parameter, and computes

$$NY' = -\frac{t}{u}zL, \quad \partial_{\bar{t}}\partial_t \log \det H = \frac{|t|^2}{|u|^2} \|z\|^2 L H^{-1} L^*.$$

The normal derivative has rank one at every nonzero parameter; the normal acceleration has rank one at zero. The curvature vanishes precisely quadratically at zero and is positive elsewhere. Its factorization through the original section $r_N e_0 L$, quotient j_E , and period observation $N\Pi$ retains the source kernel and its supported zero.

For the literal polynomial frames $J_g = [g, Sg, \dots, S^{p-1}g]$, its prefix J_- , and $J_+ = [J_g, e_0]$, put $h_J = \det(J^* \Pi^* \Pi J)$. The full frame map, Schur-complement and cofactor proof in (SC31)–(SC37) gives

$$\partial_{\bar{t}}\partial_t \log h_g = \frac{|t|^2 h_+ h_-}{|u|^2 h_g^2}.$$

Each volume factors exactly as $h_J = v_{N,J} b_{N,J}$, where $v_{N,J} = \det(J^* G_N J)$ retains the original theta metric and $b_{N,J}$ is its specified period-comparison determinant. The empty prefix has determinant one. In codimension one, $\det J_+ = (-1)^{q-1}$, so h_+ is the full explicitly evaluated period determinant above. These are the exact maps and weights joining this curvature to the neighboring-volume calculation.

Result R55: the retained Hochschild and critical-line comparisons. The complete Connes–Consani comparison proves the actual restricted factorization

$$V \xrightarrow{j} j(V) \xrightarrow{\text{Tr}} \mathcal{B}, \quad \text{Tr } j = \Theta, \quad \mathcal{E} = \frac{1}{2}\mathcal{U}\Theta, \quad (D^2 - D)e^{-\pi u^2} = \phi_*.$$

The half-density map $\mathcal{U}F = u^{1/2}F$ is the specified half-line isometry. The cochain isomorphism uses the complete two-moment source and the Laplacian’s continuous inverse. The critical-line map $\Gamma : Q \rightarrow Q_{\text{crit}}$ has, on each original finite packet, the exact kernel $\ker(\Gamma\sigma_Z) = E_{Z,\text{off}}$, including every generalized block. The critical recovery is the explicit jet

$$\Xi_\rho[f] = \sum_{j=0}^{m-1} \frac{i^{-j}}{j!} f^{(j)}(\gamma) \epsilon^j, \quad \rho = \frac{1}{2} + i\gamma.$$

The full note proves the reciprocal multiplier argument, the recovery composite and the supported image of every killed vector.

The raw circle zero mode is retained as a separate smooth coordinate; the specified Gaussian average of the original theta seed proves that this line is in the closed source-generated module. Its exact local flat-seminorm estimate and the printed large-circle limit correction remain with that proof. Finally the original-metric Laplacian identities

$$\mathbb{L}_k^* G - G \mathbb{L}_k = A^* W_k - W_k A, \quad G \mathbb{L}_k = W_k A - A^* G A$$

and their complete numerical-range argument are retained with the actual metric control parameter. Neither the trace construction, the finite-kernel calculation, nor the period determinant supplies an unproved uniform bound for that parameter.

Result R56: the actual critical-comparison kernel through periods and tensors. The full proof (PC1)–(PC41) identifies the single-packet polynomial $\chi_{h,1} = h$ by its explicit variable isomorphism $\iota_1 : [P(S)] \mapsto [P(s)]$. The original arithmetic inclusion is $\eta_h = M_{v_h} \iota_1$, with $v_h = j_h(2\xi/h)$ and every inverse jet coefficient retained. It proves

$$J_h r_N = \eta_h, \quad j_E = \eta_h^{-1} J_h, \quad j_E r_N = I, \quad q r_N = \sigma_h \eta_h.$$

The critical comparison is consequently the actual weighted map $\mathcal{C}_h = \Gamma \sigma_h \eta_h$; its kernel is the full off-critical primary summand. Each surviving critical jet is recovered with the original $i^{-j}/j!$ factor and the inverse of the displayed local arithmetic unit.

For every original integer tensor degree $k \geq 1$, retain $E_k = \mathbb{C}[S]/(\chi_{h,k})$, its actual weighted inclusion η_k , and the factorwise completed-projective comparison

$$\mathcal{C}_k = \Gamma^{\widehat{\otimes} k} \sigma_h^{\widehat{\otimes} k} \eta_k.$$

Let $\chi_{c,k}$ be the monic sum polynomial obtained from the all-critical tuples, with the complete maximum nilpotent length at each literally equal sum; for no critical factors set it to 1. The full proof constructs the variable isomorphism on the tensor carriers, preserves the original unit, and gives a continuous jet retraction proving that the finite image survives completion. It yields

$$\boxed{\ker \mathcal{C}_k = (\chi_{c,k})/(\chi_{h,k}).}$$

At a sum λ with full multiplicity m_λ and critical-tuple multiplicity r_λ , its exact local kernel is the image of $[b] \mapsto [x^{r_\lambda} a_\lambda(x)b]$ from $\mathbb{C}[x]/(x^{m_\lambda - r_\lambda})$, where $x = S - \lambda$ and the local unit in $\chi_{c,k}$ is retained. This includes the additional jets when different tuples share a sum.

For the period matrix of this same original polynomial, $\mathcal{C}_{k,t} = \mathcal{C}_k \Pi_k^{-1}$ has kernel $\Pi_k \ker \mathcal{C}_k$. Its exact sum-action defect is

$$\mathcal{C}_{k,t} A_{k,t}^{\text{per}} - M^{[k]} \mathcal{C}_{k,t} = t \mathcal{C}_k(e_{0,k}) \ell_k \Pi_k^{-1}.$$

For a proper nonzero kernel it has the proved rank-one restriction, with kernel $\Pi_k(\ker \mathcal{C}_k \cap \ker \ell_k)$. The curvature and neighboring-minor identities in R54 therefore measure this actual critical-comparison kernel. The original theta metric is connected by the complete congruence and derivative cancellation. All-critical and no-critical packets retain their different empty and full kernels, even though both endpoint curvatures vanish.

The same weighted observer now has its full singular-parameter equation (PC.u1)–(PC.u3):

$$\mathcal{C}_{k,u,t} = \mathcal{C}_k \Pi_\chi(u, t)^{-1}, \quad \partial_u \mathcal{C}_{k,u,t} = \mathcal{C}_k (C_\chi(t)/u^2 - B_\chi/u) \Pi_\chi^{-1}, \quad \ker \mathcal{C}_{k,u,t} = \Pi_\chi \ker \mathcal{C}_k.$$

Here the polynomial is literally $\chi = \chi_{h,k}$; the product-period restriction on $E_h^{\otimes k}$ is instead the actual rectangular form $\eta_k^* \Pi_{\text{prod}}^* \Pi_{\text{prod}} \eta_k$. The source identity and parameter cyclic relation following (XD.u7) are the exact maps between those two domains. For the original kernel inclusion I_k , its complete normal maps are $Z_a = N \Pi_\chi \Omega_a I_k$, $a = u, t$, and all four curvature coefficients are

$$\partial_a \partial_b \log \det H = \text{Tr}(H^{-1} Z_a^* Z_b).$$

At $t = 0$, the order-two coefficient preserves this ideal, so $Z_u = u^{-1} N \Pi_\chi B_\chi I_k$, $Z_t = 0$. These are calculated matrix entries on the same kernel; neither the full ideal nor its nilpotent jets have been suppressed.

The actual tensor coefficient map is also now retained in (PC30a)–(PC30g). On an ordered local tuple let $x_i = s_i - \rho_i$, $\lambda = \sum_i \rho_i$, $D = \sum_i (m_i - 1)$, and $U_\rho = \prod_i j_{\rho_i}(v_h)$. Over the specified coefficient localization $\mathcal{R} = A_{\ker \alpha}$, for an actual map to characteristic $p > \max_i m_i$, the exact map and its saturation basis are

$$V_j = \sum_{\substack{0 \leq b_i < m_i \\ |\mathbf{b}|=j}} \frac{x^{\mathbf{b}}}{\prod_i b_i!}, \quad W_j = U_\rho V_j, \quad \eta_\rho((S - \lambda)^j) = j! W_j, \quad \Lambda_U = \bigoplus_{j=0}^D \mathcal{R} W_j.$$

Its full containing tensor module retains the explicitly split free complement, and

$$\Lambda_U / \eta_\rho \mathcal{C}_{\mathcal{R}} \simeq \bigoplus_{j=0}^D \mathcal{R} / (p^{v_p(j!)}), \quad \ker \bar{\eta}_\rho = ((S - \bar{\lambda})^p), \quad \text{im } \bar{\eta}_\rho = \bigoplus_{j=0}^{\min(D, p-1)} B \bar{W}_j.$$

Here B is the specified target k_0 , or the explicit quotient $\mathcal{R}/p\mathcal{R}$. The kernel is zero when $D < p$. The original source index is $D + 1$, while the index on the actual specialized image is $\min(p, D + 1)$; the full free resolution and Tor connecting map in (PC30e) prove their exact relation. When the specified map includes all one-factor CRT data and $p > m_\rho$ for every retained primary order, (PC30f)–(PC30g) also retain every new residual collision:

$$\psi_{h,k}(X) = \prod_{\mu} (X - \mu)^{\max_{\alpha}(\sum_i \rho_i) = \mu \min(p, 1 + \sum_i (m_i - 1))},$$

$$\ker \bar{\eta}_k = (\psi_{h,k})/(\bar{\chi}_{h,k}), \quad \text{im } \bar{\eta}_k \simeq k_0[X]/(\psi_{h,k}).$$

The product is over attained residual sums; if the carrier is empty, its product is 1 and both modules are zero. The last arrow is the proved module map $[P] \mapsto \bar{U}_k P(\bar{S})$; its unweighted counterpart is the corresponding algebra map.

For the complete primary packet at $t_i = 0$, the ordered product boundary is explicitly calculated in (SPBT.1)–(SPBT.3), (BPC.1)–(BPC.14), (BTP.1)–(BTP.11), and (BTF.1)–(BTF.11). Writing $n = m + 1$, its ordinary complex invariant dimension is

$$d_{0,k} = \frac{m^k + m(-1)^k}{m + 1}.$$

With E_0 selecting $\sum_i (b_i + 1) \equiv 0 \pmod{n}$, the exact coefficient, source and period projectors satisfy

$$Q_0 = P_k^{-1} E_0 P_k, \quad \widehat{Q}_0 = U_k Q_0 U_k^{-1}, \quad \mathcal{P}_0 \Pi_{\text{prod}} U_k^{-1} = \Pi_{\text{prod}} U_k^{-1} \widehat{Q}_0.$$

The terminal original monomial survives precisely when $n \mid k$, and retains its arithmetic eigenvalue $a^{k\rho}$. The exact compression and its outgoing nilpotent map are (BTP.7)–(BTP.8). On the actual phase specialization with $p > m + 1$, the common additive factor remains $\mathcal{L}_\psi(kc/u)$: the invariant dimension is $d_{0,k}$ when $kc = 0$ and zero otherwise, and the Swan conductor is respectively zero or m^k . Every surviving boundary orbit retains its full product of negative Gauss constants and weight k . The common cyclotomic projector maps in (BTF.6) give their exact two base changes; they do not assign a characteristic- p coefficient scalar to a characteristic-zero Gauss line. The amplified coefficient still contains its original $K!$, $K = k(m - 1)$, with image zero exactly when $K \geq p$ on a coefficient stage retaining the inverse unit. The case $m = 1$ has $K = 0$ and no factorial loss. Moreover, in the retained off-critical quartet CRT corner for $k \geq p$, (BTF.9) proves

$$1 = p(\rho + \bar{\rho} - 1)\mathcal{D}, \quad p^{-1} = (\rho + \bar{\rho} - 1)\mathcal{D}$$

from the actual separating idempotent's divided difference. That larger stage has no unital characteristic- p map; the original local coefficient map, its precise saturation, kernel, Tor class and character remain as displayed above.

Result R57: computed Laplacian control and its curvature comparison. The full original-metric calculation (LC1)–(LC16) uses $A_t = A + tR$, $R = e_0\ell$, and $W_t = A_t^* G_N + G_N A_t - kG_N$. For $q > 1$, $\ell e_0 = 0$, so $R^2 = 0$. The two actual G_N -orthogonal columns $e_0, G_N^{-1}\ell^*$ give the exact additional control spectrum

$$\text{spec}_{\neq 0}(G_N^{-1}(W_t - W_0)) = \{|t|\sigma_N, -|t|\sigma_N\}, \quad \sigma_N^2 = (e_0^* G_N e_0)(\ell G_N^{-1} \ell^*),$$

when $t \neq 0$. For $q = 1$ the same difference is exactly $2 \operatorname{Re} t$ times identity. For the retained reflection-stable packet, the complete earlier source-minor calculation evaluates $\epsilon_N^2 = \alpha_{N+1} \Delta_N / \det K_N$, so the actual quantity

$$E_N(t) = \epsilon_N + \begin{cases} |t|\sigma_N, & q > 1, \\ 2|\operatorname{Re} t|, & q = 1 \end{cases}$$

bounds the control on the same metric. With this computed value, every point z of the original-metric numerical range of $A_t^2 - kA_t$ satisfies

$$\operatorname{Re} z \leq \frac{E_N(t)^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq E_N(t)^2 \left(\frac{E_N(t)^2 - k^2}{4} - \operatorname{Re} z \right).$$

The proof retains every original input vector in homogeneous coordinates and transports the complete inequality through Π with metric $\Pi^{-*} G_N \Pi^{-1}$.

The same source factors also control the calculated constituent curvature. For its actual G_N -projection P_F , set $a_\perp = \|(I - P_F)e_0\|_{G_N}^2$ and $b_F = \|P_F G_N^{-1} \ell^*\|_{G_N}^2$. Let $m(t), M(t)$ be the actual least and greatest eigenvalues of $G_N^{-1} \Pi^* \Pi$. Its positive curvature coefficient obeys

$$\frac{m(t)}{M(t)} \frac{a_\perp b_F}{|u|^2} \leq c_F(t) \leq \frac{M(t)}{m(t)} \frac{a_\perp b_F}{|u|^2} \leq \frac{M(t)}{m(t)} \frac{\sigma_N^2}{|u|^2}.$$

The complete proof supplies the exact nonnegative projection losses between $a_\perp b_F$ and σ_N^2 . These statements retain the actual period-comparison eigenvalues and original theta masses; no uniform packet-size bound is inserted.

The same numerical-range calculation is now transported in both original parameters by (LC.u1), with the unchanged $E_N(t)$ and G_N :

$$\begin{aligned}\partial_u \tilde{G}_N &= -\Pi^{-*} G_N \Omega_u \Pi^{-1}, & \partial_{\tilde{u}} \tilde{G}_N &= -\Pi^{-*} \Omega_u^* G_N \Pi^{-1}, \\ \partial_u \tilde{L}_t &= \Pi[\Omega_u, L_t] \Pi^{-1}, & \partial_u \tilde{A}_t &= \Pi[\Omega_u, A_t] \Pi^{-1}.\end{aligned}$$

Here $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$ is the full original cyclic connection. The exact original source commutator retains $r_N[A, \Omega_u]j_E + dK_N^{\text{prim}}\Omega_u j_E$, as proved in (DS74) and carried through (LC.u1). Thus the preceding original-metric parabola holds for every retained $u \neq 0$, with its same control value and the displayed moving metric, rather than an uncomputed target norm. The comparison numbers in (LC14)–(LC16) are now the actual $m(u, t)$, $M(u, t)$ of $G_N^{-1}\Pi(u, t)^*\Pi(u, t)$. For each constituent and each neighboring inclusion, (SC.u1)–(SC.u4) give the complete Hermitian Hessian $\text{Tr}(H^{-1}Z_a^*Z_b)$; at nonzero t , both terms of Ω_u and their cross pairing remain. On the precise single-primary ideal slice the full matrix and ideal determinant are evaluated by (SC.u5)–(SC.u7), including the original comparison Gram.

Result R58: the original arithmetic quotient from a finite circle. The complete periodized-source proof (P1)–(P64), together with the full source calculations (PSA1)–(PSA29) and (PSD1)–(PSD41), retains $v_h = 2\xi/h$, $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$ and $m_{h,k} = w_h^{*k}$. Its logarithmic coordinate map keeps the density $e^{kr + \sum z_i}$ and the ordered-coordinate orientation $(-1)^{k-1}$. The scalar circle sequence is related to the actual relative Hilbert fibre by Fourier coefficients $L^{-1/2}a_n \widehat{\mathcal{U}_k F_h^{\otimes k}}(2\pi n/L)$, whose squared norm is $\sum_n \nu_n |a_n|^2$ with $\nu_n = (2\pi/L)m_{h,k}(2\pi n/L)$. The zero mode remains in this sum.

For the original degree- N polynomial Gram M_N and unchanged remainder map J_N , define the actual finite sampled Gram $M_N(L, J)$ by retaining $|n| \leq J$. The explicit original-source quantities in (FC3)–(FC3a) produce $\alpha = 1 - \eta_P - \eta_T > 0$ and $\beta = 1 + \eta_P$, and prove

$$\alpha M_N \preceq M_N(L, J) \preceq \beta M_N, \quad \alpha G_N \preceq G_N(L, J) \preceq \beta G_N, \quad G_N = (J_N M_N^{-1} J_N^*)^{-1}.$$

The second assertion follows by taking minima over the same original remainder fibres. The two minimizing representatives retain their explicit difference in $\chi \mathcal{P}_{N-q}$ and its complete theta primitive. On the common degree $D = 2q + 2$, all four endpoint degrees and both generator raises are covered. On the literal common source $\mathcal{H} = \mathcal{P}_{2q}$, restrict those degree- D forms to $M^{(0)} = M_{2q}$ and $M^{(1)} = M_{2q}(L, J)$. The full signed derivation (PSA24a)–(PSA25) now gives the propagated error at the original claim site:

$$\begin{aligned}|\mathcal{B}_{h,k}(L, J) - \mathcal{B}_{h,k}| &\leq \mathcal{J}_{L,J}^{(5)} \leq \mathcal{J}_{L,J} \\ &\leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log(\beta/\alpha).\end{aligned}$$

Here $b_1 \leq \dots \leq b_{2q+1}$ are the eigenvalues of $D_{L,J} = (M^{(0)})^{-1}M^{(1)}$, all in $[\alpha, \beta]$. For $M(x) = (1-x)M^{(0)} + xM^{(1)}$, let P_N, Q_N be its orthogonal projections onto $\mathcal{P}_N$ and $\chi \mathcal{P}_{N-q}$ inside $\mathcal{H}$, and put $U = Q_{2q-1} + Q_{2q} - Q_q$, $W = P_{2q-1} - P_{q-1} + P_{2q} - P_q$. Then the actual signed derivative is $\text{Tr}((U-W)M(x)^{-1}(M^{(1)} - M^{(0)}))$. Writing $c_j(x) = (b_j - 1)/(1 - x + xb_j)$, $d(x) = c_{2q+1}(x) - c_1(x)$ and $s(x) = \dim(\text{ran } U \cap \text{ran } W)$, the retained full control is

$$\begin{aligned}\mathcal{J}_{L,J} &= \int_0^1 \min \left\{ c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j), \right. \\ &\quad \left. \frac{d(x)}{2} \min \{ 4q - 2s(x), \sqrt{(2q+1)(8q-4-2\text{Tr}(UW))} \} \right\} dx.\end{aligned}$$

For the actual positive atomic moment measure in this construction, put $C(x) = M(x)^{-1}(M^{(1)} - M^{(0)})$, the exact metric derivative already displayed. The complete five-entry bound in (PSA24f)–(PSA24g) additionally retains the exact trace term $S_{\text{tr}}(x) = d(x)\|U - W\|_1/2$, the centered Hilbert-Schmidt term $S_{\text{HS}}(x) = \sqrt{(\text{Tr } C^2 - (\text{Tr } C)^2/(2q+1)) \text{Tr}((U - W)^2)}$, and the original-angle term $(2q-1)\sqrt{1 - e^{-\mathcal{B}(x)/(2q-1)}}d(x)$. Thus $\mathcal{J}_{L,J}^{(5)}$ is the integral of the minimum of these three functions and the two original functions in $\mathcal{J}_{L,J}$. All five bound the same signed derivative. Strict $\mathcal{B}(x) > 0$ is proved on this actual moment measure by the reflected polynomial $\chi^{\#k}$ and positivity of its Gram. The constructed finite Neumann control in (PSA25ba)–(PSA25bc) also gives $j_{\ell_N} - e_{\ell_N} \leq \mathcal{B}(L, J) - \mathcal{B} \leq j_{\ell_N} + e_{\ell_N}$, with the complete finite matrix center and $e_{\ell_N} \leq \vartheta^{\ell_N+1} \sqrt{K_{D_{L,J}}(8q-6)} \rightarrow 0$ on this fixed original pair of forms. Let $a_q = 2q - 1$ and $H_0 = 2 \text{arcosh}(e^{\mathcal{B}/(2a_q)})$. The five-entry nonlinear radius $\mathcal{I}_{L,J}$ is proved in (PSA24f) and includes all four linear-control numerators divided by $a_q \sqrt{1 - e^{-\mathcal{B}(x)/a_q}}$. Consequently the earlier finite-circle endpoint now has the full simultaneous enclosure

$$\begin{aligned} \mathcal{B}(L, J) &\geq \max \left\{ 0, \mathcal{B} - \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{\max\{0, H_0 - \mathcal{I}_{L,J}\}}{2}, \mathcal{B} + j_{\ell_N} - e_{\ell_N} \right\}, \\ \mathcal{B}(L, J) &\leq \min \left\{ \mathcal{B} + \mathcal{J}_{L,J}^{(5)}, 2a_q \log \cosh \frac{H_0 + \mathcal{I}_{L,J}}{2}, \mathcal{B} + j_{\ell_N} + e_{\ell_N} \right\}. \end{aligned}$$

This is precisely (PSA25a) and (FC13b), with the unchanged continuous and positive atomic measures explicitly identified. The exact signed center retains its integration error in any numerical evaluation. The complete proof at (PSA24a)–(PSA25a) establishes these source maps, ranks $q+1$, traces $2q$, and the exact metric comparison, including the common-range and cross-trace terms. Its endpoint substitution is also made in (FC13a). The two generator raises still use the original degree $D = 2q + 2$. At fixed positive error budgets this is $O(q_k)$; its ratio to $q_k \log k$ tends to zero. The required L, J are explicit functions of the original weighted and derivative Grams. Their growth and the arithmetic sample enclosures are retained as actual quantities, rather than replaced by unproved uniform estimates.

Result R59: the joint critical-jet and circle observation. The full density proof obtains the original exponential tails from the gamma ray integral and the alternating Dirichlet series, including every canceled value $v_h(\rho) = g^{(m)}(\rho)/h^{(m)}(\rho) \neq 0$. It proves density on the required analytic strip, with its Cauchy-Schwarz half-width retained, and computes the completed circle quotient exactly. If $b = p_L$ is the squarefree product of the sampled roots of $\chi = \chi_{h,k}$, then

$$\ker \theta_L = (b)/(\chi), \quad G_N(L) \downarrow G_\infty(L), \quad \text{rad } G_\infty(L) = (b)/(\chi).$$

For $k \geq 2$ the complete difference $G_N(L) - G_\infty(L)$ is positive definite at every finite admitted degree: a nonzero minimizing polynomial cannot vanish on the infinite complement of the selected sample atoms. The limiting kernel retains the full unsampled primary blocks and all nonconstant jets at sampled roots.

The proof (DC1)–(DC30) compares this with the actual weighted tensor-critical map C_k , whose kernel is $(a)/(\chi)$ for $a = \chi_{c,k}$. Put $d = \gcd(a, b)$ and $l = \text{lcm}(a, b)$. Using the complete critical inverse jets and the original weighted circle coordinates gives

$$\text{im}(C_k, \theta_L) \simeq \text{im } C_k \times_{\mathbb{C}[S]/(d)} \text{im } \theta_L \simeq \mathbb{C}[S]/(l).$$

The joint kernel is $(l)/(\chi)$ and the common quotient is $\mathbb{C}[S]/(d)$. At each original sum root, their quotient lengths are respectively the maximum and minimum of the critical cyclic length and the sampled length in $\{0, 1\}$; all local unit cofactors are retained in the kernel inclusions. A factorization $\theta_L = TC_k$ exists exactly when $b \mid a$, with its explicit inverse-unit and reduction formula. The reverse factorization exists exactly when $a \mid b$. At $k = 1$, full-order cancellation proves that every sampled

selected root has positive weight and $b \mid a$. For general tensor degree, the extra circle summand has defining polynomial b/d and the explicit original polynomial source lift in (DC29). Its existence criterion asserts no existence of an off-critical zero.

All four observations transport through the same original period matrix. For the role $s \in \{a, b, d, l\}$, retain its actual map T_s and multiplier M_s . Then

$$T_{s,t} = T_s \Pi^{-1}, \quad u \partial_t T_{s,t} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}.$$

Their kernels, intersection, sum, crossed observations and complete rank-one action defects are calculated in (DC23)–(DC30). Their Split-Zero lifts retain each support and its receiving supported zero.

The entire same diamond now has its exact u -equation (DC31), for all four formal roles $s \in \{a, b, d, l\}$:

$$T_{s,u,t} = T_s \Pi(u, t)^{-1}, \quad \partial_u T_{s,u,t} = T_s (C_\chi(t)/u^2 - B_\chi/u) \Pi(u, t)^{-1}.$$

Its kernels are ΠK_s , and its specified right inverses, intersection and sum diagrams are the same ones in (DC23)–(DC30). Multiplication by the original common quotient maps proves their differential compatibility, retaining all four roles and every inverse-unit factor. For the actual circle weights put $D_\nu = \text{diag}(\nu_n)_{n \in \mathcal{Z}_L}$. The exact observed form and its derivatives are (DC32):

$$\begin{aligned} H_b^{\text{obs}} &= \Pi^{-*} T_b^* D_\nu T_b \Pi^{-1}, & \ker H_b^{\text{obs}} &= \Pi K_b, \\ \partial_u H_b^{\text{obs}} &= -\Pi^{-*} T_b^* D_\nu T_b \Omega_u \Pi^{-1}, & \partial_{\bar{u}} H_b^{\text{obs}} &= -\Pi^{-*} \Omega_u^* T_b^* D_\nu T_b \Pi^{-1}. \end{aligned}$$

Every retained atom has its original positive weight; an empty sample set gives the empty observation and its full kernel. The shared and additional circle sums, polynomial source lift and receiving support labels therefore retain their original maps while the parameter varies.

Result R60: universal residue endomorphisms and metric variation. The full calculation (RCX1)–(RCX40) extends the residue insertion $R = e_0 \ell$ over every commutative coefficient ring B with monic χ . With the original residue matrix $\mathcal{B}_{ij} = \ell(S^{i+j})$, its determinant remains $(-1)^{q(q-1)/2}$. Every coefficient endomorphism has the exact inverse expansion

$$T = \sum_{i,j} (T \mathcal{B}^{-1})_{ij} A^i R A^j.$$

The full proof gives matrix units, their multiplication and base change. Their original source lifts $r_N T j_E$ form an injective representation in the corner with identity $r_N j_E$ and kill the same original theta boundaries.

For a proper nonzero invariant constituent $I : F \hookrightarrow E$, retain the residual $r_M = (1 - P_M) e_0$ and the residue-dual vector $w_M = I(I^* M I)^{-1} (\ell I)^*$. They are M -orthogonal, with squared norms $q_M, d_M > 0$, and $\mathfrak{c}_F(M) = q_M d_M$. Along any actual differentiable positive-form path,

$$\frac{d}{ds} \log \mathfrak{c}_F(M) = \text{Tr} \left[\left(\frac{r_M r_M^* M}{q_M} - \frac{w_M w_M^* M}{d_M} \right) M^{-1} \dot{M} \right].$$

The parenthesized operator has eigenvalues $1, -1, 0$ with the stated multiplicities, yielding the exact spectral-spread bound. For the original endpoint forms $G_i \succeq G_j$, with extreme relative eigenvalues $0 < \theta \leq \eta \leq 1$, this proves

$$\left| \log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} \right| \leq \log(\eta/\theta) \leq -\log \theta \leq \log \frac{\det G_i}{\det G_j}.$$

The complete neighboring-minor formula $\mathfrak{c}_F(M) = v_{J_+}(M) v_{J_-}(M) / v_{J_g}(M)^2$ retains every original frame determinant and both constituent dimensions. The finite sampling comparison from R58 gives the dimension-free coefficient interval $\alpha/\beta \leq \mathfrak{c}_F(G_N(L, J)) / \mathfrak{c}_F(G_N) \leq \beta/\alpha$.

Result R61: sampled control with its exact commutator error. The full proof (FC1)–(FC29), including (FC3a), keeps the two actual forms $G = G_N$, $H = G_N(L, J)$, the original $T_t = A + tR - k/2$, and $B = G^{-1}(H - G)$. It first proves

$$(I + B)X_H = X_G + T_t^{\dagger G} B + BT_t, \quad X_G = T_t^{\dagger G} + T_t, \quad X_H = H^{-1}(T_t^* H + HT_t).$$

The coordinating Zeta calculation sharpens the transfer through the explicit isometry $\mathcal{S} = (I + B)^{1/2} : (E, H) \rightarrow (E, G)$. Set $Z = [\mathcal{S}, T_t]\mathcal{S}^{-1}$. Its complete Hermitian error and the convergent Sylvester integral give

$$\|X_H\|_H - \|X_G\|_G \leq \|Z + Z^{\dagger G}\|_G \leq \alpha^{-1} \| [B, A] + t[B, R] \|_G, \quad [B, R]v = (Be_0)\ell(v) - e_0\ell(Bv).$$

Both original metrics, the original unit and the ordered residue terms remain. Commuting discrepancies give exactly equal control radii. For the actual Fourier-projected representative $\mathcal{R}$ and its theta boundary $\mathcal{B}$, the full identities are

$$W_H(0) = -(\mathcal{R}^* \mathcal{B} + \mathcal{B}^* \mathcal{R}), \\ (D_L^2 - kD_L)\mathcal{R} - \mathcal{R}(A^2 - kA) = (D_L - k)\mathcal{B} + \mathcal{B}A.$$

The resulting original-metric control radius gives both inequalities of the full Laplacian numerical-range parabola (FC18), including the zero-radius case. These statements preserve the actual arithmetic commutator and boundary that a numerical endpoint estimate must evaluate.

Result R62: quotient curvature and the complete fourth derivative. Retain the exact original coefficient sequence $0 \rightarrow F \xrightarrow{I} E \rightarrow E/F \rightarrow 0$ and a constant quotient lift J_F . Its least-period-norm lift and quotient Gram $K(t)$ are the full Schur formulas (FC21). With $C = [I, J_F]$ they satisfy

$$\det(I^* \Pi^* \Pi I) \det K = |\det C|^2 |\det \Pi|^2, \quad \partial_t \partial_t \log \det K = -\frac{|t|^2}{|u|^2} \mathfrak{c}_F(\Pi^* \Pi).$$

The positive constituent curvature is therefore compensated in this same coefficient quotient, with both curvatures vanishing at zero and with opposite nonzero leading coefficients. The full proof includes quotient-lift independence, basis covariance and the literal comparison with the original theta quotient metric.

The residue continuation evaluates the pure jets as well. If $\psi = \log \det(I^* \Pi(t)^* \Pi(t) I)$ and $\phi(t) = \log \det(I^* \Pi(0)^* \Pi(t) I)$ near zero, then

$$\left. \frac{d^4}{dx^4} \psi(x) \right|_0 = 2 \operatorname{Re} \phi^{(4)}(0) + 6 \mathfrak{c}_F(\Pi(0)^* \Pi(0)) / |u|^2.$$

The full ordered recursion for $\Pi^{(n)} = \Pi Q_n$, its explicit Q_4 including $3R^2/u^2$, and the complete trace-log formula in (RCX30)–(RCX32) evaluate the remaining pure term. The forced period equation retains $u\Pi R$, and its original source primitive retains tKR . The exact Pascal translation preserves the t -derivatives of order at least two at fixed u , while its scalar factor retains the first-derivative trace shift $2p \operatorname{Re} a$. Its pure u -derivatives are the explicitly retained terms in (RCX.u5)–(RCX.u6) below.

The same constant quotient lift and the complete two-parameter period matrix give the exact compensating Hessian

$$\partial_a \partial_b \log \det K = -\operatorname{Tr}(H^{-1} Z_a^* Z_b), \quad a, b \in \{u, t\}, \quad H = I^* \Pi^* \Pi I, \quad Z_a = N \Pi \Omega_a I.$$

Indeed the displayed Schur determinant identity holds at each (u, t) , its constant frame factor has zero derivative, and $\log |\det \Pi|^2$ is pluriharmonic where Π is invertible. Applying (SC.u2) proves the equality for both diagonal and both mixed entries, with the original quotient metric and frame

preserved. The earlier quadratic vanishing statement is its t entry at $t = 0$; the u entry retains the B_χ/u normal term of (SC.u3).

The complete ordered derivative recursion now is (RCX.u1)–(RCX.u2):

$$Q_{0,0} = I, \quad Q_{r+1,s} = \partial_u Q_{r,s} + \Omega_u Q_{r,s}, \quad Q_{r,s+1} = \partial_t Q_{r,s} + \Omega_t Q_{r,s}, \quad \partial_u^r \partial_t^s \Pi = \Pi Q_{r,s}.$$

Flatness proves compatibility of all derivative orders. In particular,

$$\begin{aligned} Q_{2,0} &= C_\chi^2/u^4 + (2C_\chi - C_\chi B_\chi - B_\chi C_\chi)/u^3 + (B_\chi^2 - B_\chi)/u^2, \\ Q_{1,1} &= C_\chi A_t/u^3 + (A_t - B_\chi A_t)/u^2. \end{aligned}$$

All earlier pure t jets, including $3R^2/u^2$ in the fourth derivative, are the same $r = 0$ slice. Along a specified curve $(u(s), t(s))$, the exact metric variation inserted into the original residue Rayleigh-quotient formula is

$$\dot{H}_{\text{full}} = \Omega_{\text{path}}^* H_{\text{full}} + H_{\text{full}} \Omega_{\text{path}}, \quad \Omega_{\text{path}} = \dot{u} \Omega_u + \dot{t} \Omega_t, \quad H_{\text{full}} = \Pi^* \Pi.$$

The full forced-equation descendant also remains, by (RCX.u4):

$$2u\Pi_{tt} + u^2\Pi_{utt} + k\Pi_t + ku\Pi_{ut} + \Pi R + u\Pi_u R = \Pi\Omega_u L_t.$$

It composes with the unchanged theta-source map j_E and retains its original boundary primitive.

Finally, with $p = \dim F$ and $h_a(t) = -\Phi(-a) - ta$, the original translated phase gives

$$\begin{aligned} \psi_a - \psi &= 2p \operatorname{Re}(h_a(t)/u), \\ \partial_u(\psi_a - \psi) &= -ph_a(t)/u^2, \\ \partial_u^2(\psi_a - \psi) &= 2ph_a(t)/u^3, \\ \partial_u \partial_t(\psi_a - \psi) &= pa/u^2. \end{aligned}$$

These are the exact derivatives proved in (RCX.u5)–(RCX.u6), through the full translated connection (DT.14)–(DT.16). Their Hermitian mixed derivatives vanish because the retained scalar difference is pluriharmonic, while these pure derivatives remain in the ordered recursion.

Result R63: all holonomies reconstruct the original source. The complete proof (H1)–(H66), including (H49a)–(H49g), and the typed continuation (HDI1)–(HDI24) retain the original half-density source $\Psi_N = \mathcal{U}_k \mathcal{V}|_{\mathcal{P}_N}$, the full relative Hilbert fibre, and $g = 2\xi$. For $L > 0$ the exact map and inverse are

$$\begin{aligned} \mathcal{Z}_{L,\theta} \psi(r) &= \sum_{a \in \mathbb{Z}} e^{ia\theta} \psi(r + aL), \quad \psi(r + aL) = \int_0^{2\pi} e^{-ia\theta} \mathcal{Z}_{L,\theta} \psi(r) \frac{d\theta}{2\pi}, \\ \int_0^{2\pi} M_{N,\theta} \frac{d\theta}{2\pi} &= M_N, \quad M_{N,\theta} = (\mathcal{Z}_{L,\theta} \Psi_N)^* (\mathcal{Z}_{L,\theta} \Psi_N). \end{aligned}$$

Fourier series in the original translation index prove the isometry. The quasi-periodic basis has coefficients $L^{-1/2} \widehat{\psi}((2\pi n + \theta)/L)$; the full zero mode remains. The length- mL source has the exact finite character decomposition $M_N(mL, 0) = m^{-1} \sum_j M_N(L, 2\pi j/m)$, with the isometry factors $\sqrt{m}$ and $1/\sqrt{m}$ retained. Its coefficient cover is $\mathbb{C}[z, z^{-1}] \rightarrow \mathbb{C}[w, w^{-1}]$, $z \mapsto w^m$, with the full basis $1, w, \dots, w^{m-1}$ and the multiplication carry z .

For the same original remainder map J_N , let $C_N, C_{N,\theta}$ be the actual least-norm sections, and retain the literal relation columns B_{rel} . The complete relation identity is

$$\begin{aligned} C_N - C_{N,\theta} &= B_{\text{rel}} X_{N,\theta}, \quad X_{N,\theta} = (B_{\text{rel}}^* M_{N,\theta} B_{\text{rel}})^{-1} B_{\text{rel}}^* M_{N,\theta} C_N, \\ G_N &= \overline{G}_N + \mathcal{D}_N, \quad \mathcal{D}_N = \int (C_N - C_{N,\theta})^* M_{N,\theta} (C_N - C_{N,\theta}) \frac{d\theta}{2\pi}. \end{aligned}$$

Every relation has its original theta primitive. Empty relation columns give $\mathcal{D}_N = 0$. With the actual weighted source constant $\kappa_{a,N}$ and $\delta_L = \kappa_{a,N}/(e^{aL} - 1) < 1$, the full inverse-chord and block-matrix proof gives

$$(1 - \delta_L^2)G_N \preceq \overline{G}_N \preceq G_N.$$

For the four-volume comparison, compute the common bound $\delta_L = \kappa_{a,N_*}/(e^{aL} - 1) < 1$ on the retained maximal source $N_* = 2q + 2$ and restrict it to each of the four endpoint spaces, as in (H42)–(H43). Its error is then at most $2q \log(1/(1 - \delta_L^2))$. This bound is attached to the arithmetic average of the canonical quotient forms. The exact comparison is $G_N = \overline{G}_N + \mathcal{D}_N$, with the displayed positive integral of original relation corrections; the continuous equal-image complex and its cochain contraction realizing $\overline{G}_N$ are given in R64 and the full HCD proof. The individual phase-source comparisons have their stronger current common-polynomial-source controls in (HC13)–(HC14). Their averaging retains this separate positive correction and all of its original entries. The supplied two-point sharpness is strengthened by the explicit compactly supported full-circle source in (HDI22)–(HDI24), with original calibration mass seven and phase factors $e^{\pm i\theta}$.

Exact graph-to-average continuation of R63. Retain the same $\mathcal{P}_N, E, J_N, B_N, M_N, M_{N,\theta}$ and the original graph $\mathcal{A}_N = (\mathcal{V}_N, D_N)$, where $D_N = \tau_k \mathcal{V}_N - LJ_N$. The complete proof (PGB.1)–(PGB.13), with the original all-vector source constants and the circumference domain (PGB.8), gives the actual additional positive form

$$\begin{aligned} H_N &= B_N^* M_N B_N, \quad T_N = D_N B_N, \quad Q_{D,N} = (I + T_N H_N^{-1} T_N^*)^{-1}, \\ \mathcal{E}_N^{\text{aug}} &= C_N^* D_N^* Q_{D,N} D_N C_N > 0, \quad \widehat{G}_N - \overline{G}_N = \mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}}. \end{aligned}$$

The relation-space inverse is the unique empty inverse when $N = q - 1$, so the composite through that space is zero. Here $\mathcal{D}_N^{\text{hol}}$ is exactly the already displayed original relation gap $G_N - \overline{G}_N$. The proof computes the actual defect map

$$x \mapsto \left(Q_{D,N}^{1/2} D_N C_N x, [\theta \mapsto (C_N - C_{N,\theta})x] \right)$$

into the direct sum of the original defect Hilbert target and the original relation-field space. Its Gram is the displayed difference and its kernel is zero. The zero kernel is proved by the actual leading coefficient: for P of degree n with leading coefficient $p_n \neq 0$, $\tau_k P(\Sigma)$ has s_1 degree $n + 4m + 1$ with leading coefficient $p_n/(4m + 1)$, while $F_{J_N P}$ has s_1 degree at most $4m - 1$. The full-unit source consequently makes D_N injective. Thus $\overline{G}_N \preceq G_N < \widehat{G}_N$ with a strict finite graph cost. Repetition of the same polynomial coefficients is the contracting cochain map inducing the identity $(E, \widehat{G}_N) \rightarrow (E, \overline{G}_N)$; PGB.13 supplies its complete isometric dilation.

For the four endpoints use one original circumference, chosen from the common maximal source $N_* = 2q + 2$ whenever the inverse formulas are used. PGB.26 gives the exact signed difference as the four logarithmic determinants of $\overline{G}_N^{-1/2} \widehat{G}_N \overline{G}_N^{-1/2}$, with signs $(+, +, -, -)$ at $(q - 1, q, 2q - 1, 2q)$. The original averaged-quotient bound above stays attached to $\overline{G}_N$ and its original relation gap. The new signed graph-to-average expression retains every extra term; positivity of each individual logarithm does not give a sign for that four-term expression.

Result R64: continuous cochain descent and the exact completion map. The full proof (HCD1)–(HCD29), including the actual source bounds (HCD2a)–(HCD2c), considers relation fields $\mathcal{B}$ and the common-image fields $\mathcal{P}^{\text{eq}}$ over $d\theta/(2\pi)$. At the fixed cutoff N , choose $0 < \eta < 1$ and retain the actual weighted source constant $\kappa_{\beta,N}$. The explicit circumference $L \geq \beta^{-1} \log(1 + \kappa_{\beta,N}/\eta)$ gives the uniform equivalence in (HCD2a)–(HCD2c) used in the following bounded maps. Repetition and Bochner mean give the bounded coefficient isomorphism

$$[\mathcal{B} \hookrightarrow \mathcal{P}^{\text{eq}}] \simeq [\chi \mathcal{P}_{N-q} \hookrightarrow \mathcal{P}_N] \oplus [\mathcal{B}^0 \xrightarrow{I} \mathcal{B}^0], \quad hP = P - \Delta \text{av } P.$$

The full calculation proves $dh + hd = I - \Delta \text{av}$, and the quotient is the original $E = \mathbb{C}[S]/(\chi)$ with metric $\overline{G}_N$. Its relation gap retains the weighted cross term. If $\overline{C}_N = \int C_{N,\theta} d\theta/(2\pi)$, $a_C = C_N - \overline{C}_N$ and $h_\theta = C_{N,\theta} - \overline{C}_N$, then

$$\int a_C^* M_{N,\theta} h_\theta \frac{d\theta}{2\pi} = a_C^* M_N a_C, \quad \mathcal{D}_N = \int h_\theta^* M_{N,\theta} h_\theta \frac{d\theta}{2\pi} - a_C^* M_N a_C.$$

For every $L > 0$, the full shifted-density proofs (CQ1)–(CQ29) and (HDI11)–(HDI21) identify the subsequent completed phase quotient with the weighted values at positive-mass sampled roots. Its exact map from E has kernel $(p_\theta)/(\chi)$, where p_θ is the squarefree sampled product and χ retains every primary multiplicity. The original one-fold packet satisfies $(\chi_{h,1}) = (h)$; its selected-root mass is $|g^{(r)}(\rho)/h^{(r)}(\rho)|^2/L > 0$, with the original leading coefficient retained. The complete primary-coordinate inverse gives the nilpotent kernel at every sampled root and the full block at every unsampled root.

The diagonal root projections give an explicit measurable field. Only finitely many phases can sample a fixed root of χ , so its completed direct integral is zero. The evaluation maps form the actual cochain map from the continuous common-image complex to this completed complex; its map on degree-one cohomology has kernel all of E . On an original split label λ , its inverse image of the receiving represented zero is the full represented coefficient kernel, while the inverse image of τ is the singleton $\{\tau\}$. This specifies the operation and its kernel with the original support and all finite classes present.

The corrected-graph completion and its exact map in R64. The full proof (PGB.14)–(PGB.21b) retains the original scalar graph density r_k , full-unit isometry $\mathcal{J}_k$, mixed observation $Y = (0, L)$, its actual adjoint $g = \mathcal{J}_k^* Y$, and $Y_\perp = (I - \mathcal{J}_k \mathcal{J}_k^*) Y$. The density proof uses the original exponential moments and proves the onto multiplication isometry for the literal χ ; it does not presume polynomial density. The completed source, its inverse coordinate map, and its completed relation space are

$$\begin{aligned} \mathcal{H}_\infty &= \mathcal{J}_k L^2(r_k du) \oplus^\perp Y_\perp E, \quad \Omega_k = Y_\perp^* Y_\perp > 0, \\ \Xi(f, x) &= \mathcal{J}_k f - Y_\perp x, \quad \Xi^{-1}(v) = (\mathcal{J}_k^* v, -\Omega_k^{-1} Y_\perp^* v), \\ \overline{\mathcal{A}(\chi \mathbb{C}[S])} &= \mathcal{J}_k L^2(r_k du), \quad \mathcal{H}_\infty / \overline{\mathcal{A}(\chi \mathbb{C}[S])} \simeq (E, \Omega_k). \end{aligned}$$

The quotient map is the displayed second coordinate, so its kernel is precisely that closed relation subspace. Its vertical part in the original ordered two-component target is exactly $\{(0, -Lx) : x \in E\}$. This is a closed linear relation; the full proof constructs original polynomials with fixed nonzero remainder, observation tending to zero and defect tending to $-Lx$, proving that the original defect operator on the polynomial observation source is not closable.

The first-coordinate projection has the exact formula $\pi_1 \Xi(f, x) = \mathcal{J}_k^0 I_{r,m}(f + g_x)$, where $\mathcal{J}_k^0 f = \prod_i v_h(s_i) f(u)$ and $I_{r,m} : L^2(r_k du) \rightarrow L^2(m_k du)$ is the unchanged-function contraction. It defines the bounded cochain map

$$[\mathcal{J}_k L^2(r_k du) \hookrightarrow \mathcal{H}_\infty] \longrightarrow [\mathcal{J}_k^0 L^2(m_k du) \xrightarrow{I} \mathcal{J}_k^0 L^2(m_k du)].$$

On original polynomials it is literally $\mathcal{A}P \mapsto \mathcal{V}P$. Its degree-one map is $(E, \Omega_k) \rightarrow 0$, with kernel all of E , matching the original unweighted completion. The finite contraction of R63 and this completed map have their distinct, fully specified domains and kernels. Applying a fixed retained support label preserves the represented kernel; external absence retains its singleton preimage.

The strict mixed source inside the same completed quotient. The complete original-source calculation (PGMB.1)–(PGMB.8) now computes the interior of Ω_k . With the unchanged positive scalar

$1 + |\tau_k|^2$ define

$$L_{\text{low}}x = \frac{(\prod_i v_h(s_i))F_x(s)}{\sqrt{1 + |\tau_k|^2}}, \quad L_{\text{low}} : E \longrightarrow \mathcal{H}_k,$$

$$\mathcal{D}_k^{\text{mix}}x = \frac{\overline{\tau_k}F_x(s)}{1 + |\tau_k|^2} - g_x(u), \quad \mathcal{D}_k^{\text{mix}} : E \longrightarrow L^2(\mathbb{R}^k, (1 + |\tau_k|^2) \prod_i w(t_i) dt_i).$$

The first original product integral has Gram K_{low} ; the orthogonal sum-coordinate projection in PGM.3–5 gives $(\mathcal{D}_k^{\text{mix}})^* \mathcal{D}_k^{\text{mix}} = \Omega_k - K_{\text{low}}$. Thus the specified map $\mathfrak{U}_kx = (L_{\text{low}}x, \mathcal{D}_k^{\text{mix}}x)$ is an isometry from (E, Ω_k) into that original direct sum. Its complete left inverse is $\Omega_k^{-1}(L_{\text{low}}^*a + (\mathcal{D}_k^{\text{mix}})^*b)$; the reverse composite is the orthogonal projection onto $\mathfrak{U}_k(E)$. Writing the already computed quotient as $j_\infty v = -\Omega_k^{-1}Y_\perp^*v$, its composition is

$$\mathfrak{C}_k = \mathfrak{U}_k j_\infty, \quad \mathfrak{C}_k \Xi(f, x) = (L_{\text{low}}x, \mathcal{D}_k^{\text{mix}}x),$$

$$\ker \mathfrak{C}_k = \mathcal{J}_k L^2(r_k du), \quad \|\mathfrak{C}_k \Xi(f, x)\|^2 = x^* \Omega_k x.$$

This is the actual quotient contraction and its whole relation kernel, since $\mathfrak{U}_k$ is injective and j_∞ has the displayed kernel. The complete fixed-sum polynomial argument PGM.6–9 proves $\ker \mathcal{D}_k^{\text{mix}} = 0$ for every $k \geq 2$ and $\mathcal{D}_1^{\text{mix}} = 0$. Consequently for $k \geq 2$

$$(\Omega_k - K_{\text{low}})^{-1}(\mathcal{D}_k^{\text{mix}})^* \text{pr}_2 \mathfrak{C}_k = j_\infty,$$

by the computed mixed Gram. For $k = 1$, the second component vanishes and the positive first Gram retains the full quotient. All maps act coordinatewise on each retained split label; their represented-zero kernels are the products of the exact displayed coefficient kernels. The empty present tuple and the singleton inverse image of external τ are unchanged. Composition from a larger or proper-source realization retains the whole earlier realization kernel specified in PGM.10; this calculation asserts no descent through another quotient.

Result R65: quadratic generator transfer with an exact residue norm. The complete proof (HG1)–(HG25), including (HG11a), uses the actual relation-cost endomorphism $D = G_N^{-1} \mathcal{D}_N$, the original $T_t = A + tR - kI/2$, and $R = e_0 \ell$. It proves

$$|\epsilon_{G_N}(t) - \epsilon_{G_N}(t)| \leq \frac{\|[D, A] + t[D, R]\|_{G_N}}{1 - \delta_L^2} \leq \frac{\delta_L^2}{1 - \delta_L^2} \|T_t\|_{G_N}.$$

The exact square-root isometry and Sylvester integral preserve the full Hermitian error. The sharper scalar constant follows from the complete spectral-projection decomposition of the original self-adjoint discrepancy: its commutator norm is at most its spectral diameter times the original generator norm.

For any actual self-adjoint discrepancy $B = G^{-1}(H - G)$, let $x = e_0$, $y = G^{-1}\ell^*$ and retain

$$a_0 = \|Bx\|_G^2, \quad c_0 = \|x\|_G^2, \quad d_0 = \|y\|_G^2, \quad f_0 = \|By\|_G^2, \quad p_0 = \langle x, Bx \rangle_G, \quad r_0 = \langle y, By \rangle_G,$$

$$T_{\text{res}} = a_0 d_0 + c_0 f_0 - 2p_0 r_0, \quad \Delta_{\text{res}} = (a_0 c_0 - p_0^2)(d_0 f_0 - r_0^2),$$

$$\|[B, R]\|_G^2 = \frac{T_{\text{res}} + \sqrt{T_{\text{res}}^2 - 4\Delta_{\text{res}}}}{2}.$$

The full coordinating residue proof is reproduced in (HG12)–(HG15), including both negative cross terms, singular Gram matrices, rank zero, rank one, repeated singular values and $q = 1$. The primary quantity remains $\|[B, A] + t[B, R]\|_G$ with its complete cancellation.

Equations (HG16)–(HG20) give explicit L, s, m, J from the actual weighted and derivative source Grams. The holomorphic Schur inverse is certified on the specified smaller strip, with its neighborhood, and the finite mean obeys

$$\alpha G_N \preceq H_{m,J} \preceq \beta G_N, \quad \alpha = (1 - \xi_J)(1 - \delta_L^2 - \varepsilon) > 0, \quad \beta = 1 + \varepsilon.$$

The exact discrepancy is $B_{m,J} = -D + E_{m,J}$, where the finite-tail spectrum lies in $[-\rho, \varepsilon]$, $\rho = \xi_J + (1 - \xi_J)\varepsilon$. The proof retains its entire commutator and gives the corresponding spectral-diameter bounds. The finite representative retains both ordered theta boundary terms in (HG23). The resulting original-metric radius enters the complete Laplacian numerical-range parabola (HG25); the same proved metric comparison gives the residue-coefficient bounds (HG24).

Exact discrepancy inputs supplied to R65 by the graph continuation. The original estimates above use $D = G_N^{-1} \mathcal{D}_N^{\text{hol}}$ and retain their specified source and circumference constants. The exact graph continuation computes, rather than omits, the further form $\mathcal{E}_N^{\text{aug}}$ in PGB.6. Relative to the original metric, the graph discrepancy is the actual self-adjoint endomorphism

$$B_N^{\text{aug}} = G_N^{-1}(\hat{G}_N - G_N) = G_N^{-1} \mathcal{E}_N^{\text{aug}}.$$

It is positive because $\langle x, B_N^{\text{aug}} x \rangle_{G_N} = x^* \mathcal{E}_N^{\text{aug}} x > 0$ for $x \neq 0$; $G_N B_N^{\text{aug}} = \mathcal{E}_N^{\text{aug}}$ is Hermitian, which proves self-adjointness in precisely that original metric. For the comparison based at the averaged metric the endomorphism is instead the explicitly typed

$$B_N^{\text{graph,av}} = \bar{G}_N^{-1}(\hat{G}_N - \bar{G}_N) = \bar{G}_N^{-1}(\mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}}).$$

Multiplication by $\bar{G}_N$ proves its self-adjointness and positivity in $\bar{G}_N$. Thus the generator terms for these comparisons are respectively $[B_N^{\text{aug}}, A] + t[B_N^{\text{aug}}, R]$ and $[B_N^{\text{graph,av}}, A] + t[B_N^{\text{graph,av}}, R]$, with the unchanged A, R, T_t . This identifies all entries needed by the existing transfer calculation. The present continuation supplies no new small bound for these additional commutators; the existing bound on D remains a bound for its stated original term.

The calculated finite and completed generator controls in R65. The complete proof PGGT.1–8 now evaluates the transfer for three actual rows: $(G, H) = (G_N, \hat{G}_N)$, $(\bar{G}_N, \hat{G}_N)$ and $(K_{\text{low}}, \Omega_k)$. Their respective discrepancies are exactly

$$B = G_N^{-1} \mathcal{E}_N^{\text{aug}}, \quad B = \bar{G}_N^{-1}(\mathcal{E}_N^{\text{aug}} + \mathcal{D}_N^{\text{hol}}), \quad B = K_{\text{low}}^{-1}(\Omega_k - K_{\text{low}}).$$

Each is self-adjoint and nonnegative in its displayed original G metric, by $GB = H - G$ and the preceding positive Gram proofs. Keep its complete original spectral decomposition $B = \sum_i b_i P_i$ with all repeated eigenspaces, and set $s_i = \sqrt{1 + b_i}$ and $S = \sum_i s_i P_i$. Multiplying proves $S^*GS = H$ and $S^{-1} = \sum_i s_i^{-1} P_i$. Thus with the original $T_t = A + tR - kI/2$ the full error is

$$Z = \sum_{i,j} \frac{s_i - s_j}{s_j} P_i T_t P_j, \quad SX_H S^{-1} - X_G = Z + Z^{\dagger G},$$

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \frac{\|[B, A] + t[B, R]\|_G}{1 + b_1} \leq \frac{b_r - b_1}{1 + b_1} \|T_t\|_G.$$

The displayed denominator is the actual spectral minimum. PGGT.4–5 proves the entire identity by ordered multiplication, then proves the middle estimate by integrating $e^{-uS}[B, T_t]e^{-uS}$ with its exact $1/(2\sqrt{1 + b_1})$ factor. The full spectral projection decomposition supplies the last bound; the residue term remains $[B, R]v = (Be_0)\langle G^{-1}\ell^*, v \rangle_G - e_0\langle BG^{-1}\ell^*, v \rangle_G$, including the negative term and its full cancellation with $[B, A]$.

For the third row the spectral endpoints are precisely $b_1 = \lambda_{k,1}$ and $b_r = \lambda_{k,q_k}$ of PGMB.9. At $k = 1$ their common zero gives exact equality of the two Hermitian generators. The original Hilbert-source map $\mathfrak{U}_k = (L_{\text{low}}, \mathcal{D}_k^{\text{mix}})$ realizes the completed generator by

$$\begin{aligned} \tilde{T}_t &= \mathfrak{U}_k T_t \Omega_k^{-1} \mathfrak{U}_k^*, & \tilde{T}_t \mathfrak{U}_k &= \mathfrak{U}_k T_t, \\ \ker \tilde{T}_t &= \mathfrak{U}_k(\ker T_t) \oplus^\perp \mathfrak{U}_k(E)^\perp, & \|\tilde{T}_t + \tilde{T}_t^*\| &= \epsilon_{\Omega_k}(t). \end{aligned}$$

PGGT.7–8 proves these on the original target by decomposing each vector into $\mathfrak{U}_k x + v_\perp$ and using $\mathfrak{U}_k^* \mathfrak{U}_k = \Omega_k$. Thus the new estimate controls that actual realized Hermitian generator; its constants retain the entire mixed spectrum and make no unproved claim of smallness as k grows.

The complete relation tail now supplies a fourth generator row in R65. Keep the original $T_t = A + tR - kI/2$, $t \in \mathbb{C}$, and all primary coordinates. At fixed full h, k , (PGRT.6) gives the actual positive form and its complete arithmetic coefficients

$$\begin{aligned} T_N^{\text{rel}} &= \widehat{G}_N - \Omega_k = f_N^* f_N = \sum_{j=N}^{\infty} (a_j^{\text{rel}})^{-1} (\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}, \\ B_N^{\text{rel}} &= \Omega_k^{-1} T_N^{\text{rel}} = \sum_i b_{N,i} P_{N,i}, \quad b_{N,i} \geq 0, \\ S_N &= \sum_i \sqrt{1 + b_{N,i}} P_{N,i}, \quad S_N^{-1} = \sum_i (1 + b_{N,i})^{-1/2} P_{N,i}, \\ S_N^* \Omega_k S_N &= \widehat{G}_N. \end{aligned} \tag{R65r1}$$

The names $a_j^{\text{rel}}, d_j^{\text{rel}}, \alpha_j^{\text{rel}}$ are exactly (ACM28), which supplies their full original moment integrals and positive Gram construction. The form T_N^{rel} retains this superscript and does not rename the period generator T_t . The matrix $\Omega_k B_N^{\text{rel}}$ is the positive Hermitian form T_N^{rel} , proving self-adjointness in the original Ω_k metric. Transport of the complete eigenspaces of $\Omega_k^{-1/2} T_N^{\text{rel}} \Omega_k^{-1/2}$ proves the displayed spectral decomposition, including repeated eigenspaces. Multiplication of its projections proves the inverse and Gram identities. Thus $S_N : (E, \widehat{G}_N) \rightarrow (E, \Omega_k)$ is the specified isometry, with the displayed inverse and zero kernel.

Put $X_G(t) = T_t + T_t^{\dagger G}$ and $\epsilon_G(t) = \|X_G(t)\|_G$, retaining both t and $\bar{t}$. The complete error on this fourth row is

$$\begin{aligned} Z_N(t) &= [S_N, T_t] S_N^{-1}, \\ S_N X_{\widehat{G}_N}(t) S_N^{-1} - X_{\Omega_k}(t) &= Z_N(t) + Z_N(t)^{\dagger \Omega_k}, \\ |\epsilon_{\widehat{G}_N}(t) - \epsilon_{\Omega_k}(t)| &\leq \|Z_N(t) + Z_N(t)^{\dagger \Omega_k}\|_{\Omega_k} \leq \frac{\|[B_N^{\text{rel}}, A] + t[B_N^{\text{rel}}, R]\|_{\Omega_k}}{1 + \min_i b_{N,i}}. \end{aligned} \tag{R65r2}$$

For the identity use $T_t^{\dagger \widehat{G}_N} = S_N^{-2} T_t^{\dagger \Omega_k} S_N^2$ and expand the two ordered terms. For the inequality, the convergent integral

$$\int_0^{\infty} e^{-u S_N} [B_N^{\text{rel}}, T_t] e^{-u S_N} du = [S_N, T_t]$$

has (i, j) block $\frac{b_{N,i} - b_{N,j}}{\sqrt{1 + b_{N,i}} + \sqrt{1 + b_{N,j}}} P_{N,i} T_t P_{N,j}$, equal to the block on the right by factoring the difference of squares. Its norm is bounded by the commutator norm times $\int_0^{\infty} e^{-2u \sqrt{1 + \min b_{N,i}}} du = 1/(2\sqrt{1 + \min b_{N,i}})$. The inverse has norm $1/\sqrt{1 + \min b_{N,i}}$. Multiplication and addition of the adjoint give exactly the factor in (R65r2). The first inequality is the reverse triangle inequality under the specified isometry. This retains the cancellation between the two original commutators.

For every fixed $r \geq 0$ the finite-frame tail yields the quantitative local uniform conclusion

$$\begin{aligned} \sup_{|t| \leq r} |\epsilon_{\widehat{G}_N}(t) - \epsilon_{\Omega_k}(t)| &\leq 2 \|B_N^{\text{rel}}\|_{\Omega_k} (\|A\|_{\Omega_k} + r \|R\|_{\Omega_k}) \\ &\leq 2 (\|A\|_{\Omega_k} + r \|R\|_{\Omega_k}) \sum_{j=N}^{\infty} \frac{\alpha_j^{\text{rel}} \Omega_k^{-1} (\alpha_j^{\text{rel}})^*}{a_j^{\text{rel}}} \\ &\longrightarrow 0 \quad (N \rightarrow \infty; h, k, r \text{ fixed}). \end{aligned} \tag{R65r3}$$

Indeed $\|[B, A]\| \leq 2\|B\|\|A\|$ and the analogous inequality for R prove the first line from (R65r2). The largest eigenvalue of a positive matrix is at most its trace. Here that trace is $\text{tr}(\Omega_k^{-1} T_N^{\text{rel}})$, equal to the displayed convergent series by (R65r1). It tends to zero by the complete relation expansion, which proves the limit. No statement of uniformity in k is used.

This limit has the actual target already constructed in R65. Writing $\mathfrak{U}_k = (L_{\text{low}}, D_{\text{mix}})$, its Gram is Ω_k and the target operator is $\tilde{T}_t = \mathfrak{U}_k T_t \Omega_k^{-1} \mathfrak{U}_k^*$. Every target vector decomposes uniquely as $\mathfrak{U}_k x + v_\perp$ with $x = \Omega_k^{-1} \mathfrak{U}_k^* v$ and $v_\perp \in \mathfrak{U}_k(E)^\perp$. Substitution proves $\tilde{T}_t \mathfrak{U}_k = \mathfrak{U}_k T_t$ and its complete kernel $\mathfrak{U}_k(\ker T_t) \oplus^\perp \mathfrak{U}_k(E)^\perp$. Taking adjoints proves $\|\tilde{T}_t + \tilde{T}_t^*\| = \epsilon_{\Omega_k}(t)$. Consequently (R65r3) compares the finite graph generator to this fully realized mixed-source generator with its entire relation-tail error. Every stated coefficient map retains its coordinatewise action on fixed masks and the singleton inverse image of external absence.

The same finite row window retained in the fourth R65 discrepancy. For $q - 1 \leq N \leq 2q$, the complete relation series telescopes to the exact ordered operator identity

$$\begin{aligned} B_N^{\text{rel}} &= B_{2q}^{\text{rel}} + \sum_{j=N}^{2q-1} \frac{\Omega_k^{-1}(\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}}{a_j^{\text{rel}}}, \\ [B_N^{\text{rel}}, A] + t[B_N^{\text{rel}}, R] &= [B_{2q}^{\text{rel}}, A] + t[B_{2q}^{\text{rel}}, R] \\ &\quad + \sum_{j=N}^{2q-1} \frac{1}{a_j^{\text{rel}}} \left([\Omega_k^{-1}(\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}, A] + t[\Omega_k^{-1}(\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}, R] \right). \end{aligned} \tag{R65rc}$$

For $N = 2q$ the sum is empty. Subtracting the two full series in (R65r1) proves the first line; linearity of the commutator proves the remaining ordered terms, retaining every negative product inside each bracket. These are the same actual rows whose original $\widehat{G}_{2q}^{-1}$ correlations are evaluated in (ACM32)–(ACM35). The generator discrepancy remains in its specified Ω_k metric, so its entries here use Ω_k^{-1} in their stated positions. This gives their exact return into (R65r2), without replacing that entire commutator by a scalar endpoint determinant or deleting the remaining B_{2q}^{rel} term.

Result R66: the original degree cost and full continuous closure. The complete proof (HD1)–(HD10) fixes h, k, L and a real sample u_0 , with $S_0 = k/2 + iu_0$ and $a_N = (1, S_0, \dots, S_0^N)$. For the unchanged original polynomial Gram it constructs the polynomial P_N with its displayed coefficient vector c_N :

$$\Lambda_N = a_N M_N^{-1} a_N^* \uparrow \infty, \quad c_N = M_N^{-1} a_N^* / \Lambda_N, \quad P_N(S_0) = 1, \quad \|\mathcal{V} P_N\|^2 = 1 / \Lambda_N.$$

The divergence is proved by the original exponential moments, Hilbert representation, holomorphic Fourier uniqueness and the Gaussian measure argument. At an actually positive sampled mass $\omega_0 = (2\pi/L)m_{h,k}(u_0)$ this gives the exact degree cost

$$\kappa_{a,N} \geq (e^{aL} - 1)(\omega_0 \Lambda_N - 1) \longrightarrow \infty.$$

The same columns have full holonomy mean squared norm tending to zero, while their selected phase squared norm remains at least ω_0 . Evaluation factors through the original E exactly when $\chi(S_0) = 0$, with the fully calculated primary kernel.

The coordinating complete proof (CR1)–(CR14), including (CR12a)–(CR12c), extends this to the entire original source. Multiplication by the literal χ is the onto isometry

$$L^2(|\chi|^2 m \, du) \xrightarrow{\times \chi} L^2(m \, du), \quad f \longmapsto \chi f,$$

with its division inverse defined almost everywhere. Polynomial density therefore proves that the continuous relation closure is the whole Hilbert source. If Π_N is its actual finite relation projection and $s : E \rightarrow \mathcal{P}_{q-1}$ its fixed remainder section, then

$$R_N = (I - \Pi_N)s, \quad J_N R_N = I_E, \quad G_N = R_N^* R_N \downarrow 0, \quad \overline{\text{graph } J} = L^2(m \, du) \times E.$$

All limits here fix the original h, k and retain the complete finite coefficient frame. At singular phase Grams the exact receiving space is $T_\theta(\mathcal{P}_N)/T_\theta(\mathcal{B}_N)$, with kernel $J_N(\ker T_\theta)$ on E . Its measurable minimum form gives $0 \preceq \overline{G}_N \preceq G_N$ by testing the original R_N . The independent full completion proof further gives the relation gap as the integral of the squared projection of $T_\theta R_N$ onto $T_\theta \mathcal{B}_N$. Every finite $\overline{G}_N$ is positive for each original $k \geq 1$: outside a countable set of one-fold phases all sampled weights are positive. At an actual positive sampled root, its fixed-phase quotient/source ratio diverges by (CR14). These exact maps and limits impose no unstated rate in the distinct tensor parameter k and do not identify the relative gap with its absolute limit.

Exact simultaneous degree limits and morphisms extending R66. The complete affine-distance formula in PGB.22 is

$$x^* \widehat{G}_N x = x^* \Omega_k x + \inf_{Q \in \mathcal{P}_{N-q}} \|\iota(sx + \chi Q) - g_x\|_{L^2(r_k du)}^2, \quad (\iota P)(u) = P(k/2 + iu).$$

The original weighted relation density, proved in PGB.15–17, therefore gives at fixed original h, k

$$\widehat{G}_N \downarrow \Omega_k > 0, \quad G_{W,N} \downarrow 0, \quad \Delta_N \downarrow \Omega_k, \quad G_N \downarrow 0, \quad \overline{G}_N \rightarrow 0.$$

The last two limits are the original unweighted limits already stated above. For every circumference the phase quotient is defined by minimum source images, including singular finite Grams, and its projected relation gap proves $0 \preceq \overline{G}_N \preceq G_N$ without inverting a null coordinate. The full proof also establishes strict positivity of every finite $\overline{G}_N$ from the original sampled weights. Consequently the graph augmentation tends to Ω_k and the exact comparison obstruction is

$$\widehat{G}_N - \overline{G}_N \longrightarrow \Omega_k, \quad \inf_{x \neq 0} \frac{x^* \widehat{G}_N x}{x^* \overline{G}_N x} \geq \frac{\lambda_{\min}(\Omega_k)}{\|G_N\|} \longrightarrow \infty.$$

The inequality follows by bounding numerator and denominator in the same original remainder frame. It excludes a constant upper comparison independent of the polynomial cutoff; the exact contracting identity in the other direction remains PGB.12–13. At completion, the bounded first-coordinate cochain map in PGB.21b sends (E, Ω_k) to the original zero unweighted quotient with full coefficient kernel. None of these fixed- h, k degree limits asserts a rate in the separate tensor parameter or deletes any retained primary coordinate before its specified quotient map.

Finite tail costs and strict mixed persistence in R66. For the same actual graph minimum $\widehat{R}_N$ and the same adjoint g_x , set $f_N x = \iota \widehat{R}_N x - g_x \in L^2(r_k du)$. Substitution in PGB.19 and orthogonality prove the full identity

$$\widehat{G}_N = K_{\text{low}} + (\mathcal{D}_k^{\text{mix}})^* \mathcal{D}_k^{\text{mix}} + f_N^* f_N = \Omega_k + f_N^* f_N.$$

Its actual isometry is $x \mapsto (L_{\text{low}} x, \mathcal{D}_k^{\text{mix}} x, f_N x)$, with all three original Hilbert targets as in PGMB.5–6. The preceding degree limit gives $\|f_N\| \rightarrow 0$ at fixed h, k . For every $k \geq 2$, PGM proves the strictly positive mixed Gram $\Omega_k - K_{\text{low}}$, so this entire mixed part survives in the quotient while the third component tends to zero. At $k = 1$ its vanishing is the exact PGM identity.

Let $\lambda_{k,j}$ be the full generalized eigenvalues of $(\Omega_k - K_{\text{low}}, K_{\text{low}})$ in the retained coefficient frame, and let $\nu_{k,N,j}$ be the eigenvalues of $\Omega_k^{-1/2} f_N^* f_N \Omega_k^{-1/2}$. The complete determinant calculation PGMB.9–10 yields

$$\log \frac{\det \widehat{G}_N}{\det K_{\text{low}}} = \sum_{j=1}^{q_k} \log(1 + \lambda_{k,j}) + \sum_{j=1}^{q_k} \log(1 + \nu_{k,N,j}), \quad \nu_{k,N,j} \longrightarrow 0.$$

Every $\lambda_{k,j}$ is positive for $k \geq 2$ and zero for $k = 1$. The exact extremal comparison constants are $1 + \lambda_{k,1}$ and $1 + \lambda_{k,q_k}$, as proved by testing the corresponding original-frame generalized eigenvectors.

The four unchanged endpoints cancel the entire first sum with its literal coefficients $1 + 1 - 1 - 1 = 0$. Their graph-volume expression is therefore the full signed sum of the four tail costs, with Ω_k still present in every tail congruence. These fixed-packet limits do not assign any rate when k and the endpoints grow together.

The complete relation coefficients and signed endpoint return in R66. Retain $\mathcal{L}_k = L^2(r_k du)$ with its original density, $F^0 x = \iota s x - g_x$ and the monic columns, lengths and rows $d_j^{\text{rel}}, a_j^{\text{rel}}, \alpha_j^{\text{rel}}$ of (ACM28). The relation-column index is $j \geq q - 1$ and its degree is $j + 1$; in particular $d_{q-1}^{\text{rel}} = \chi$. The full exponential-moment and Fourier-uniqueness proof of (PGRT.3)–(PGRT.4) makes these columns complete in this precise weighted space. Define the retained-length sequence space and the actual coefficient morphisms by

$$\begin{aligned} \ell^2((a^{\text{rel}})^{-1}) &= \left\{ c : \sum_{j=q-1}^{\infty} |c_j|^2 / a_j^{\text{rel}} < \infty \right\}, \\ \mathcal{F} : \mathcal{L}_k &\longrightarrow \ell^2((a^{\text{rel}})^{-1}), \quad f \longmapsto (\langle \iota d_j^{\text{rel}}, f \rangle)_j, \\ \mathcal{F}^{-1} c &= \sum_{j=q-1}^{\infty} (a_j^{\text{rel}})^{-1} c_j \iota d_j^{\text{rel}}. \end{aligned} \tag{R66r1}$$

The inner product is conjugate linear in its first entry. Orthogonality shows that every finite synthesis has squared norm $\sum |c_j|^2 / a_j^{\text{rel}}$, which proves norm convergence for every displayed sequence. Each coefficient functional is continuous, so the limit has coefficients c_j . Conversely, finite orthogonal projection bounds the squared coefficients of any f by $\|f\|^2$; after synthesizing them the difference has all coefficients zero, and completeness makes it zero. These are both inverse identities and the exact norm equality. Both linear kernels are zero, and no monic relation column is rescaled.

Let $\mathbf{E}_{\geq N}$ be the coordinate projection retaining exactly $j \geq N$ in this weighted sequence space. Then

$$\begin{aligned} f_N &= \mathcal{F}^{-1} \mathbf{E}_{\geq N} \mathcal{F} F^0, \quad T_N^{\text{rel}} = f_N^* f_N = \sum_{j=N}^{\infty} (a_j^{\text{rel}})^{-1} (\alpha_j^{\text{rel}})^* \alpha_j^{\text{rel}}, \\ \ker f_N &= \bigcap_{j \geq N} \ker \alpha_j^{\text{rel}} = \{x : F^0 x \in \iota(\chi \mathcal{P}_{N-q})\} \\ &= \{x : g_x = \iota P, \deg P \leq N, [P] = x\}, \quad N \geq q - 1. \end{aligned} \tag{R66r2}$$

For $N = q - 1$ the relation space in this formula is zero. At arbitrary N , the complementary finite coordinate projection is orthogonal projection onto precisely $\iota(\chi \mathcal{P}_{N-q})$. Minimizing $\|F^0 x + \iota \chi Q\|^2$ over that space therefore gives the original $f_N x = \iota \hat{R}_N x - g_x$, proving the first map identity. Norm equality in (R66r1) gives the positive series, and polarization gives all its entries in the retained frame. The kernel of the coordinate projection gives the first two kernel identities. For the last, write $F^0 x = \iota \chi Q$ and take $P = s x - \chi Q$; the reverse direction follows by monic division of $s x - P$, which has zero remainder and degree at most N . These statements specify the full coefficient fibres without assuming injectivity of a finite tail. The tail operator tends to zero because its trace is the sum of the squared tails of the q fixed frame vectors.

Every coefficient used here is the full integral (ACM28), including the negative z_b term. The positive graph source retains the exact mixed inverse calculation (ACM30) at $M = 2q$:

$$\begin{aligned} (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} \hat{G}_{2q}^{-1} (\alpha_j^{\text{rel}})^* &= (a_j^{\text{rel}})^{-1} \alpha_j^{\text{rel}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \\ &\quad - (a_j^{\text{rel}})^{-1} \left\| (I + Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (Z_{2q}^{\text{mix}})^*)^{-1/2} Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \right\|^2, \\ Z_{2q}^{\text{mix}} &= (D_{\text{mix}}, f_{2q}). \end{aligned} \tag{R66r3}$$

Multiplication of the exact inverse identity in (ACM30) proves this expression with its ordered factors. At every $k \geq 2$ the mixed map is injective, so its subtracted term is strictly positive for each

nonzero row. Zero rows retain their zero value; at $k = 1$ the finite tail remains even though the mixed component vanishes. Thus this calculation carries the positive mixed source through the inverse that controls each relation decrement, with its complete negative correction.

The arithmetic endpoint return is on the receiving HC domain $k \geq 3$ and $\gamma > 2$. By the complete decrement calculation (ACM31), the row sums $L_k^{\text{rel}}, U_{k,2q}^{\text{rel}}$ bound $\mathcal{B}_k$. Here the decrements sum over the unchanged two intervals $[q-1, 2q-2]$ and $[q, 2q-1]$, giving weight one at $q-1, 2q-1$ and weight two at every interior index. Retain the separately computed signed $g_{s,\text{gr}}^\pm$ of (ACM27) and the Gamma endpoints $\beta_{p,\Gamma}^\pm$ of (HC9a). The current use-site calculation is

$$\begin{aligned} b_{p,s}^{\text{rel},-} &= \max\{\beta_{p,\Gamma}^-, L_k^{\text{rel}} - g_{s,\text{gr}}^+\}, & b_{p,s}^{\text{rel},+} &= \min\{\beta_{p,\Gamma}^+, U_{k,2q}^{\text{rel}} - g_{s,\text{gr}}^-\}, \\ \beta_{p,\Gamma}^- &\leq b_{p,s}^{\text{rel},-} \leq b_{p,s}^{\text{gr},-} \leq \mathcal{B}_k \leq b_{p,s}^{\text{gr},+} \leq b_{p,s}^{\text{rel},+} \leq \beta_{p,\Gamma}^+, \\ \mathcal{R}_k &\in [L_{k;p,s}^{\text{ar,rel}} - T_k, U_{k;p,s}^{\text{ar,rel}} - T_k], & \Delta_k^\Gamma &\in [L_{k;p,s}^{\text{ar,rel}} - \mathcal{B}_k^\Gamma, U_{k;p,s}^{\text{ar,rel}} - \mathcal{B}_k^\Gamma], \end{aligned} \quad (\text{R66r4})$$

where the full endpoints are $L_{k;p,s}^{\text{ar,rel}} = \max\{0, T_k, b_{p,s}^{\text{rel},-}\}$ and $U_{k;p,s}^{\text{ar,rel}} = \min\{\widehat{U}_k, b_{p,s}^{\text{rel},+}\}$, with $T_k = 4q \log(D_h k)$ unchanged. To prove the first interval subtract the actual inequalities for Γ_k^{gr} from $L_k^{\text{rel}} \leq \widehat{\mathcal{B}}_k \leq U_{k,2q}^{\text{rel}}$, using the exact identity $\mathcal{B}_k = \widehat{\mathcal{B}}_k - \Gamma_k^{\text{gr}}$. Then intersect with the original Gamma interval. Comparing the lower maximum and upper minimum with their exact-graph versions proves the displayed nesting. Intersecting also with (HC5), (HC7) and (HC8) gives the two arithmetic endpoints just stated. The two translations in (HC10r), with their inverse additions, prove the two receiving residual intervals and preserve their widths. In particular their width is at most $\min\{2e_{p,\Gamma}, U_{k,2q}^{\text{rel}} - L_k^{\text{rel}} + 2e_{s,\text{gr}}\}$ by the exact width proof (HC10s).

Finally (HC14r) inserts this same positive arithmetic upper endpoint into each of the three fixed original phase functions. Their proved monotonicities give $\mathcal{B}_{k,\theta} \leq U_{k,\theta;p,s,r}^{\text{gr}} \leq U_{k,\theta;p,s,r}^{\text{rel}} \leq U_{k,\theta;p,r}$, while the entire phase-energy lower sum (HC13) remains unchanged. This carries the new coefficients backward to both original signed residuals and forward to the phase-energy use site. Its fixed-packet degree limit feeds the actual fourth generator row (R65r1)–(R65r3); no rate in the distinct growing k variable is supplied by that limit. Coordinatewise support maps preserve the full linear kernels in (R66r2), every represented-zero label, each present empty tuple and the singleton fibre of external absence. Proper-source residuals retain the inverse-image domain and boundary quotient already proved in the source comparison.

The complete relation cross terms refine the R66 endpoint enclosure. For $H = \widehat{G}_{2q}$ put $C_{ij} = \alpha_i^{\text{rel}} H^{-1} (\alpha_j^{\text{rel}})^*$, and retain the later index set $J_j = \{j+1, \dots, 2q-1\}$. The full calculation (ACM32)–(ACM34), proved independently in PGRC.1–11, keeps all entries of C , including their complex conjugates. In particular

$$\begin{aligned} a_j^{\text{rel}} \eta_j &= \alpha_j^{\text{rel}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \\ &\quad - \left\| (I + Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (Z_{2q}^{\text{mix}})^*)^{-1/2} Z_{2q}^{\text{mix}} K_{\text{low}}^{-1} (\alpha_j^{\text{rel}})^* \right\|^2 \\ &\quad - C_{jJ_j} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j j}, \quad D_{J_j} = \text{diag}(a_i^{\text{rel}} : i \in J_j). \end{aligned} \quad (\text{R66rc1})$$

Indeed (R66r3) gives the first two terms as C_{jj} , and telescoping gives $\widehat{G}_{j+1} = H + A_{jJ_j}^* D_{J_j}^{-1} A_{J_j}$. Multiplication of the inverse identity in (ACM32) and pairing with α_j^{rel} gives the third term and its exact left side. The first subtraction is strictly positive for $k \geq 2$ and a nonzero row. The second is strictly positive exactly when $C_{J_j j} \neq 0$, since the retained block inverse is positive. Their sum is at most the first term because $\eta_j = \alpha_j^{\text{rel}} \widehat{G}_{j+1}^{-1} (\alpha_j^{\text{rel}})^* / a_j^{\text{rel}} \geq 0$. The last row has an empty later block and zero correlation subtraction, with its mixed subtraction unchanged.

For actual retained subsets $J(j) \subset J_j$, the full remaining row is $e = C_{jK} - C_{jJ_j} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j K}$, $K = J_j \setminus J$, and the full remaining positive block is $S = D_K + C_{KK} - C_{KJ_j} (D_{J_j} + C_{J_j J_j})^{-1} C_{J_j K}$. Block elimination in (ACM34) proves that the omitted correction is exactly $eS^{-1}e^*$ and is bounded above

by $\sum_{i \in K} |e_i|^2 / a_i^{\text{rel}}$. Thus the explicit lower and upper decrement arguments in (ACM35) give the finite bounds $L_k^{\text{corr}}(J), U_k^{\text{corr}}(J)$ with

$$L_k^{\text{rel}} \leq L_k^{\text{corr}}(J) \leq \widehat{\mathcal{B}}_k \leq U_k^{\text{corr}}(J) \leq U_{k,2q}^{\text{rel}}. \quad (\text{R66rc2})$$

These inequalities follow by inserting the exact nonnegative omitted correction in each η_j , applying its explicit upper estimate for the lower argument, and then applying $\log(1+x)$ with the original endpoint weights. All later rows included makes both bounds equal the exact graph volume. Each one-row insertion subtracts the precise nonnegative fraction in (ACM35); its numerator retains the full cross term and its denominator is at least the original positive row length. Thus there are at most q insertions per row before this finite exactness, with every original dimension and norm unchanged.

The complete receiving formulas are now (HC9rc), (HC10rc) and (HC14rc). In detail, taking $b^{\text{corr},-} = \max\{\beta_{p,\Gamma}^-, L_k^{\text{corr}} - g_{s,\text{gr}}^+\}$ and $b^{\text{corr},+} = \min\{\beta_{p,\Gamma}^+, U_k^{\text{corr}} - g_{s,\text{gr}}^-\}$ gives the exact chain

$$[b_{p,s}^{\text{gr},-}, b_{p,s}^{\text{gr},+}] \subseteq [b_{p,s}^{\text{corr},-}, b_{p,s}^{\text{corr},+}] \subseteq [b_{p,s}^{\text{rel},-}, b_{p,s}^{\text{rel},+}] \subseteq [\beta_{p,\Gamma}^-, \beta_{p,\Gamma}^+]. \quad (\text{R66rc3})$$

Subtracting the same signed augmentation from (R66rc2) and then taking the common maximum and minimum proves this chain. Intersect next with $\mathcal{B}_k \geq \max\{0, T_k\}$ and $\mathcal{B}_k \leq \widehat{\mathcal{U}}_k$ to obtain precisely $L_{k;p,s}^{\text{ar,corr}}, U_{k;p,s}^{\text{ar,corr}}$ in (HC10rc). The translations $x \mapsto x - T_k$ and $x \mapsto x - \mathcal{B}_k^\Gamma$, with the stated inverse additions, carry these intervals to both original residuals. Their width is at most $\min\{2e_{p,\Gamma}, U_k^{\text{corr}} - L_k^{\text{corr}} + 2e_{s,\text{gr}}\}$ by the complete signed-width calculation (HC10rcw). Finally each of the three original phase functions is increasing on the positive arithmetic domain, so their unchanged actual integrals give

$$\mathcal{B}_{k,\theta} \leq U_{k,\theta;p,s,r}^{\text{gr}} \leq U_{k,\theta;p,s,r}^{\text{corr}} \leq U_{k,\theta;p,s,r}^{\text{rel}} \leq U_{k,\theta;p,r}. \quad (\text{R66rc4})$$

The entire lower energy sum of (HC13) is still at the left of this chain, as proved in (HC13rc). This is the backward and forward transport of both exact negative corrections to the original arithmetic and phase quantities. The coefficient maps and kernels of (ACM32), (R66r1)–(R66r2) retain their original maskwise lifts and proper-source residuals; no support class is removed in taking these finite row subsets.

Result R67: the original pushforward over the absolute tau base. The complete original Tau Base construction works over the pointed blueprint $\mathbf{F}_{1,\tau} = \{\tau, 1\}$ and its four-point chart $+, - < \eta < \sigma$. In its declared coefficient-diagram category the projective resolution of the constant object is

$$0 \longrightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u,v) \mapsto u+v} \mathbb{C} \longrightarrow 0.$$

At the two lower points the last map is the identity; at the two upper points its kernel is the displayed anti-diagonal. This proves exactness pointwise. Applying Hom gives the actual derived pushforward

$$\mathbf{A}_\tau(F) = [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta].$$

For the retained theta diagram $\mathcal{T} = (V, V, \mathcal{B}, 0)$ the restrictions are Θ and $\Theta\mathcal{F}$, so its first cohomology is $Q = \mathcal{B}/\Theta V$. Tensoring the pointwise resolution, with the original signs $(-1)^{\sum_{j < i} |x_j|}$, gives top cohomology $Q^{\otimes k}$. Its numerator is $\mathcal{B}^{\otimes k}$ and its boundary subspace is the sum of tensors having a factor ΘV ; tensoring the quotient maps proves that identification. The right adjoint is the full injective complex $K_\tau(W) = [I_\eta W \rightarrow I_+ W \oplus I_- W]$, in degrees $-1, 0$, with the two signed restrictions. The complete Tau Base proof and (DB1)–(DB13) give the adjunction, tensor signs and residue-dual arrows, including the original full Taylor units.

For a full actual packet h of $g = 2\xi$, keep $E_h = \mathbb{C}[s]/(h)$, $v_h = j_h(g/h) \in E_h^\times$, and its original section $\sigma_h : E_h \hookrightarrow Q$. For the cyclic tensor-sum quotient $E = \mathbb{C}[S]/(\chi_{h,k})$, the two separately typed injections are

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_h^{(k)})1 \quad \text{in } E_h^{\otimes k}, \quad \sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}.$$

Their injectivity follows from the defining cyclic annihilator, the invertible full tensor unit and the tensor of the original left inverse of σ_h . These are the original arithmetic classes in the pushforward over τ . Restriction to σ is a separate cochain map from that pushforward, defined by its original restrictions; its homotopy and fibre are retained in the full original and transcript-continuation proofs.

The full-source pushforward also has the proved maps from every declared mixed support $\lambda = (W, S; \mathfrak{a})$. Its coefficient copies retain the independent outer label and the coefficient mask, including an empty mask at a present label. For one original leg the exact map is

$$\eta_W : V/W \longrightarrow \mathcal{B}_\Omega/\Theta W, \quad [v] \longmapsto [\Theta v]_W.$$

Injectivity follows from injectivity of Θ . Its image is the kernel of the map to $\mathcal{B}_\Omega/\Theta V$: a representative maps to zero precisely when it is Θv . These maps applied to each admitted coefficient copy give the full kernel $(V/W)^{\mathfrak{a}}$, without altering the outer label. The full coefficient masks, action intertwiners and zero-coordinate insertion maps are proved in (MCF42)–(MCF51) and (D9)–(D14).

For tensor sources retain the declared subcomplex $\mathcal{C}_\lambda^\bullet \subseteq \mathcal{C}_{\text{full}}^\bullet$. The exact comparison on top quotient classes has kernel

$$\frac{d\mathcal{C}_{\text{full}}^{k-1} \cap \mathcal{C}_\lambda^k}{d\mathcal{C}_\lambda^{k-1}}.$$

The isomorphism sends a represented boundary to its original class; its inverse chooses the same representative. Both are well-defined because their equivalence relation is the displayed denominator. The ordered polynomial divisions and signed original tensor primitive in (ISM38)–(ISM43) compute the actual member $[d\tilde{K}_v]_\lambda$ of this kernel for each canonical metric secant whose compared top representatives lie in $\mathcal{C}_\lambda^k$. It vanishes at this label exactly when $d\tilde{K}_v \in d\mathcal{C}_\lambda^{k-1}$.

The original finite source tower also retains its actual type: D sends W_m^{src} to W_{m+1}^{src} , giving the quotient arrow and residual of (R1b)–(R1c). The source-level preimages of the arithmetic length-two quotient in (MCF47)–(MCF51), together with the shifted filtrations proved in MW and MRE, carry those adjacent-level arrows. Their low-degree one-leg tail is acyclic at the full source $W = V$; the proper-source tail retains $H^1 = V/W$, and the two-leg complex retains its degree-zero diagonal kernel. Thus the original τ pushforward and its arithmetic quotient are connected through the actual source kernel and filtered preimages throughout this edition.

Result R68: the arithmetic cyclic image inside the full period family. For $d = \deg h$, the original one-factor exponential family over $\mathbb{C}[u, t]$ is $[\mathbb{C}[u, t][s] \xrightarrow{u\partial_s + h(s)-t} \mathbb{C}[u, t][s] ds]$. Use the original primitive $\Phi'_h = h$, its fixed additive constant, and the oriented rapid-decay contours in the admitted sector of $u \neq 0$. Their full matrix $\Pi_h(u, t)$ has columns $\int_{\Gamma_j} s^a e^{(\Phi_h(s)-ts)/u} ds$, $0 \leq a < d$. The complete period proof retains its determinant, ray phases, contour orientations and all coefficients. With independent parameters $t_1, \dots, t_k$, the unchanged rectangular matrix

$$\mathcal{P}(u, \mathbf{t}) = \left(\bigotimes_{i=1}^k \Pi_h(u, t_i) \right) \eta_{h,k}$$

is injective, because each period factor is invertible and $\eta_{h,k}$ is injective. The exact derivative at the undeformed parameter is

$$-u \sum_{i=1}^k \partial_{t_i} \mathcal{P}(u, \mathbf{0}) = \mathcal{P}(u, \mathbf{0}) A_{h,k}.$$

Differentiating each exponential inserts s_i ; the cyclic intertwining and commutation of the full tensor unit with $\sum A_i$ prove the formula. It uses the original sum coordinate, with no division by k . The full (DP23)–(DP33) proof supplies the coefficient-ring reductions, exact left inverse on this image,

original metric pullback and separately defined finite-field external product. All nonsemisimple jets and root collisions have their stated domains there.

The complete singular-parameter connection acts on this same rectangular image. Write $f_i(s_i) = \Phi_h(s_i) - t_i s_i$ and retain the literal monic divisions

$$f_i(s_i)s_i^b = (h(s_i) - t_i)q_{i,b} + r_{i,b}, \quad (C_h(t_i))_b = r_{i,b}, \quad (B_h)_b = q'_{i,b}.$$

The differential remainder is $r_{i,b} - uq'_{i,b}$, with $C_h(t_i) = f_i(A_h(t_i))$. The original one-factor matrix is $\Omega_{u,h}(u, t_i) = -C_h(t_i)/u^2 + B_h/u$, and $\partial_u \Pi_h = \Pi_h \Omega_{u,h}$. The complete division and contour proof is in the boundary connection chapter and (XD.u1)–(XD.u6), applied to the displayed original h -variable. On the ordered product complex the degree- j cochain operator is exactly

$$\nabla_u^{(j)} = \partial_u - \frac{\sum_i f_i}{u^2} + \frac{k-j}{u}.$$

Its cochain identity follows one slot at a time from the degree-zero/degree-one identity: increasing exterior degree removes exactly one $1/u$ term. Every exterior sign and the original ordered form are retained.

Set $\Pi_{\text{prod}} = \bigotimes_i \Pi_h(u, t_i)$; let $\Omega_{u,\text{prod}}$ be the sum of the k matrices $\Omega_{u,h}(u, t_i)$ acting in their respective tensor slots. The product rule gives the exact original map

$$\partial_u \mathcal{P} = \Pi_{\text{prod}} \Omega_{u,\text{prod}} \eta_{h,k}.$$

For the full original positive tensor coefficient Gram G at the specified finite source cutoff, retain the explicit left inverse

$$\ell_\eta = (\eta_{h,k}^* G \eta_{h,k})^{-1} \eta_{h,k}^* G, \quad \ell_\eta \eta_{h,k} = I.$$

The inverse exists because $\eta_{h,k}$ is injective and G is positive on the original full tensor source. The complete parameter defect is therefore

$$\partial_u \mathcal{P} - \mathcal{P}(\ell_\eta \Omega_{u,\text{prod}} \eta_{h,k}) = \Pi_{\text{prod}}(I - \eta_{h,k} \ell_\eta) \Omega_{u,\text{prod}} \eta_{h,k}.$$

Substitution and factoring prove this equality, retaining the actual transverse component and full unit in $\eta_{h,k}$. The observed form remains $\eta_{h,k}^* \Pi_{\text{prod}}^* \Pi_{\text{prod}} \eta_{h,k}$, with its original source comparison Gram. The cyclic $\chi_{h,k}$ -potential has its own connection (XD.u1)–(XD.u7); this injection, left inverse and transverse map give the precise comparison with the product potential.

On the complete primary slice $h = (s - \rho)^m$, $t_i = 0$, the exact product period matrix, source-unit character projectors, terminal arithmetic eigenclass and finite-field boundary are proved in (BTP.1)–(BTP.11) and (BTF.1)–(BTF.11). The complex invariant dimension is $(m^k + m(-1)^k)/(m+1)$; the actual finite-field inertia dimension is that number when $kc = 0$ and zero otherwise, with $c = -(-\rho)^{m+1}/(m+1)$. The original unit-weighted coefficient map and all its factorial losses are (PC30a)–(PC30g) and (BPC.1)–(BPC.14), on precisely their stated coefficient-map domains. These full proofs retain the one-factor vanishing as its $k = 1$ case and carry each surviving tensor character to its actual boundary stalk.

Result R69: actual theta dilation and the moving-parameter flow. On the original theta source use $U_a F(x) = F(x/a)$ for positive real a ; its Mellin action is a^s . The two-leg action is $(\phi, \psi) \mapsto (U_a \phi, a U_{1/a} \psi)$. The complete scaling proof computes the original source boundary of its packet section and the finite cocycle, so the quotient section intertwines this action with the full exponential of the original jet operator, including all nilpotents. For the h exponential family the regular connection operator is

$$L = u \partial_t - s, \quad [u \partial_s + h - t, L] = 0, \quad \frac{dC_b}{db} = -C_b A_h(t + ub), \quad C_0 = I.$$

The two nonzero commutators are $-u$ and $+u$. The finite matrix ODE defines the entire transport; on the admitted period domain it is exactly $\Pi_h(u, t)^{-1} \Pi_h(u, t + ub)$. Its special-fibre matrix is $C_b(0, t) = e^{-bA_h(t)}$. At $b = \log p$ this gives the original p^{-A_h} , corresponding to $U_{1/p}$, for every positive real p . The separate global algebraic curve power map $z \mapsto z^p$ has the additional domain $p \in \mathbb{Z}_{>0}$; there its pullback is $dz/z \mapsto p dz/z$ and the divisor depth is multiplied by that integer. The corrected full source states these distinct parameter domains explicitly.

For the retained reflection-stable cyclic polynomial, put $\varepsilon = (-1)^q$, $q = \deg \chi$, and $RP(S) = P(k - S)$. The full reflection proof gives

$$RA_\chi(t)R = kI - A_\chi(\varepsilon t), \quad RC_b^\chi(u, t)R = e^{-kb}C_{-b}^\chi(-\varepsilon u, \varepsilon t).$$

Here the cyclic potential is $\Phi'_\chi = \chi$, with the original endpoint $\Phi_\chi(0) = 0$. Its reflection gauge is $\exp((\Phi_\chi(k) - kt)/u)R$ on $u \neq 0$; the unsquared contour sign remains. The matrix identity extends through $u = 0$ by the entire ODE, rather than by evaluating that singular gauge. The same proof retains the inverse-transpose dual and the original moment-line factor e^{-kb} .

The same finite transport has its complete original u -derivative. Keep $\Omega_{u,h} = -C_h(t)/u^2 + B_h/u$, $\Omega_{t,h} = -A_h(t)/u$, and $t_b = t + ub$. Differentiating the exact matrix formula at fixed b gives

$$\begin{aligned} \partial_u C_b &= -\Omega_{u,h}(u, t)C_b + C_b(\Omega_{u,h}(u, t_b) + b\Omega_{t,h}(u, t_b)), \\ \partial_t C_b &= -\Omega_{t,h}(u, t)C_b + C_b\Omega_{t,h}(u, t_b). \end{aligned}$$

Indeed $\partial_a \Pi^{-1} = -\Omega_{a,h} \Pi^{-1}$; the target derivative is $\Pi(u, t_b)(\Omega_{u,h} + b\Omega_{t,h})(u, t_b)$. This proves the pull-back connection under the actual base map $(u, t) \mapsto (u, t + ub)$, including $b\partial_t$. It is the polynomial restriction of the fully proved (APU23) transport. These equations hold on the original nonzero- u contour domain; the same entire finite-matrix ODE supplies the regular transport through $u = 0$. The reflection and inverse-transpose dual retain the stated parameter pullbacks and original moment-line factor; their derivatives follow by these same inverse and chain rules on the unchanged maps.

Result R70: the original Taylor unit is an exact formal gauge. Let ν be the degree-less-than- d polynomial representing the full v_h , so $\gcd(h, \nu) = 1$. The exact rings are $\widehat{B} = \mathbb{C}[s][[u, t]]$ and $\widehat{B}_\nu = \mathbb{C}[s, \nu^{-1}][[u, t]]$. The coefficientwise pole quotient is $M[[u, t]]$, $M = \mathbb{C}[s, \nu^{-1}]/\mathbb{C}[s]$. Multiplication by h on M has an exact inverse H : Bezout coefficients for h, ν^n prove surjectivity on each principal part and coprimality proves injectivity. With $E_0 = u\partial_s - t$ the inverse on the pole complex is

$$J = \sum_{n \geq 0} (-HE_0)^n H.$$

The displayed order is essential. Each factor HE_0 raises parameter order, making every coefficient a finite sum. The proper-rational section of the quotient gives the explicit contraction and retraction in (UG8)–(UG11). Multiplication by the full ν then gives

$$\nu(u\partial_s + h - t)\nu^{-1} = u\partial_s + h - t - u\nu'/\nu.$$

It commutes with the same $L = u\partial_t - s$. Its endpoint on E_h is exactly multiplication by v_h ; the signed completed external product recovers exactly $\eta_{h,k}$ after the original cyclic inclusion. The complete (UG1)–(UG19) proof includes every tensor homotopy and the inverse-unit residue dual on its original reflection-stable domain.

The formal gauge carries the complete regular form of the singular-parameter operator. Put $f = \Phi_h - ts$. On the original two formal complex terms define

$$H_j = u^2 \partial_u - f + (1 - j)u, \quad j = 0, 1, \quad H_1 D = D H_0, \quad D = u\partial_s + h - t.$$

The commutator of $u^2\partial_u$ with D is $u^2\partial_s$, and that of $-f$ with D is $u(h-t)$. Their sum is uD , canceled by the extra u on the degree-zero term. This proves the cochain identity. These operators act coefficientwise on the stated completed rings. After the specified localization in u , they are $u^2\nabla_u^{(j)}$, with the same degree terms as (APU1)–(APU4). Since ν is independent of u, t ,

$$\nu H_j \nu^{-1} = H_j, \quad H_1(\nu D \nu^{-1}) = (\nu D \nu^{-1}) H_0.$$

These are operators on the transported gauge terms; multiplication by f commutes with ν . The polynomial-to-localized inclusion commutes with H_j . Its cohomology isomorphism, proved by the full contraction J , consequently has inverse commuting with the induced H : multiply the intertwining identity by that inverse on both sides. This proves compatibility without assigning strict horizontality to an uncomputed chain retraction. The exact homotopy and full gauge endpoint remain (UG8)–(UG19).

The existing $L = u\partial_t - s$ satisfies $[H_j, L] = uL$, because its two nonzero terms are $u^2\partial_t$ and $-us$. Including the derivative of $1/u$ therefore gives $[H_j/u^2, L/u] = 0$ on the localized complex. At the original formal fibre H_1 is multiplication by $-\Phi_h$, commuting with the entire $[\nu] = v_h$. On the ordered k -fold complex the same calculation gives $u^2\partial_u - \sum_i(\Phi_h - t_i s_i) + (k-j)u$; the original signed tensor gauge and its source endpoint retain their full degree terms and Taylor units.

Result R71: analytic pole residues, their connection and their endpoint. Return to the original h -potential $\Phi'_h = h$ with its specified additive constant. For fixed $u \neq 0$, write $w = e^{(\Phi_h(s)-ts)/u}$ and let $\mathcal{P}_\nu$ be the finite set of distinct zeros of ν , of size r . The exact pole sequence now is

$$0 \longrightarrow M \xrightarrow{u\partial_s + h - t} M \xrightarrow{F \mapsto (\text{Res}_p(wF ds))_p} \mathbb{C}^{\mathcal{P}_\nu} \longrightarrow 0.$$

The highest pole term $c(s-p)^{-n}$ maps to $-unc(s-p)^{-n-1}$, proving injectivity. A zero-residue Laurent series wF has a meromorphic primitive without a logarithm; taking the principal part of that primitive divided by uw proves exactness in the middle. The lift $1/(w(p)(s-p))$ proves surjectivity. The original polynomial cohomology therefore injects into the analytic localization, with precisely this residue cokernel. A full basis consists of the original d polynomial classes and the simple-pole classes $1/(s-p)$.

Let $P = \text{diag}(p)$ and let B_0 have each column the original constant-polynomial vector e_0 . Their exact connection and transport are

$$L = u\partial_t - \begin{pmatrix} A_h(t) & B_0 \\ 0 & P \end{pmatrix}, \quad C_{\text{ext},b} = \begin{pmatrix} C_b & K_b \\ 0 & e^{-bP} \end{pmatrix}, \quad K_b = - \int_0^b C_c B_0 e^{-(b-c)P} dc.$$

The off-diagonal term follows from $L(1/(s-p)) = -p/(s-p) - 1$ and the displayed ODE. The full routed period determinant is $\det \Pi_{\text{ext}} = \det \Pi_h (2\pi i)^r \prod_p w(p)$, with positive small-loop orientations. Under the actual unit gauge the weight is w/ν ; all periods agree because $(w/\nu)(\nu F) = wF$.

At the original formal endpoint let Q_0 have columns $-[(h(s) - h(p))/(s-p)h(p)]$ in E_h . Then

$$A_h Q_0 = B_0 + Q_0 P, \quad J_0 = [I \ Q_0], \quad \ker J_0 = \{(-Q_0 c, c) : c \in \mathbb{C}^r\}, \quad J_0 C_{\text{ext},b}(0,0) = e^{-bA_h} J_0.$$

Multiplication by the full v_h after J_0 is the gauge endpoint. These are explicit comparison maps between the analytic extension and the original formal fibre. The complete (AP1)–(AP26) proof retains all higher poles, the ordinary dual, the zero-pole case and the exceptional $p = 0$ phase; it proves no evaluation of the formal inverse at a nonzero complex u .

The full connection of this analytic extension now includes its u direction, as proved in (APU1)–(APU23). Write $f(s, t) = \Phi_h(s) - ts$, and let Q have columns

$$Q_p(s, t) = \frac{f(s, t) - f(p, t)}{s - p} - \frac{h(s) - t}{d + 1}, \quad p \in \mathcal{P}_\nu.$$

The first polynomial has degree d and leading coefficient $1/(d+1)$. Its division by $h-t$ has that constant quotient; the derivative correction is zero and the displayed remainder has degree less than d . In the original polynomial and simple-pole columns the exact matrix is

$$\Omega_{u,\text{ext}} = \begin{pmatrix} -C_h(t)/u^2 + B_h/u & -Q/u^2 \\ 0 & -\text{diag}(f(p,t))/u^2 \end{pmatrix}, \quad \partial_u \Pi_{\text{ext}} = \Pi_{\text{ext}} \Omega_{u,\text{ext}}.$$

Apply $\nabla_u = \partial_u - f/u^2$ to $1/(s-p)$ and separate $f(p)/(s-p)$ from its complete polynomial divided difference to prove this formula. The original t -matrix remains $-u^{-1} \begin{pmatrix} A_h(t) & B_0 \\ 0 & P \end{pmatrix}$. The full block flatness, including its off-diagonal block, is proved in (APU8)–(APU12) using these literal remainders.

Put $W_{\text{res}} = \text{diag}(e^{f(p,t)/u})$. The weighted-residue row matrix is $\mathcal{R} = [0, W_{\text{res}}]$; the positively oriented loop row matrix is $(2\pi i)\mathcal{R}$. Differentiation gives

$$\partial_u W_{\text{res}} = -\text{diag}(f(p,t))W_{\text{res}}/u^2, \quad \partial_t W_{\text{res}} = -PW_{\text{res}}/u.$$

Thus $\partial_a \mathcal{R} = \mathcal{R} \Omega_{a,\text{ext}}$ for $a = u, t$; both residue and loop maps are horizontal, with their original $2\pi i$ factors retained. The entire transport has the two derivative identities in R69 with the full extension matrices. Equation (APU23) proves the resulting shifted endpoint and residue diagrams, including $b\Omega_{t,\text{ext}}(u, t+ub)$.

Multiplication by the complete ν commutes with the parameter operators and transports the entire frame; its weight is still w/ν . Its inverse-transpose dual and unchanged endpoint J_0 are proved in (APU18)–(APU22). The original theta-source observation is exactly $\sigma_h[\nu]J_0$, retaining the entire unit. When ν is constant, $r = 0$ and every pole block, row and additional loop is empty; the same equations reduce to the full original polynomial connection and its unit gauge. These maps retain the analytic family, the specified formal endpoint and the formal coefficient operations of R70.

Result R72: the signed arithmetic determinant variation. On the unchanged common source $\mathcal{P}_{2q}$ interpolate the original Gamma and arithmetic densities as $m_t = (1-t)m_\Gamma + tm_{\text{ar}}$, and write M_N, H_N, R_N, G_N for the original source Gram, relation Gram, minimum section and quotient Gram. The complete (AT1)–(AT22) proof preserves the determinant-line sign and gives

$$\begin{aligned} V_N &= \det G_N = \det M_N / \det H_N, & \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \\ \dot{G}_N &= R_N^* \Delta M_N R_N, & \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N. \end{aligned}$$

Both negative squared norms in the logarithmic second derivative remain. The full (AV1)–(AV32) calculation proves that equality holds exactly when $\Delta M_N R_N = 0$. With P_N, Q_N the original same-metric polynomial and relation projections, retain

$$\begin{aligned} U &= Q_{2q-1} + Q_{2q} - Q_q, & W &= P_{2q-1} - P_{q-1} + P_{2q} - P_q, \\ C(t) &= M_{2q}(t)^{-1} \Delta M_{2q}, & \mathcal{B}(t) &= \log \frac{V_{q-1}(t)V_q(t)}{V_{2q-1}(t)V_{2q}(t)}. \end{aligned}$$

Subtracting the four exact determinant derivatives proves

$$\mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U - W)C) dt.$$

The full proof also gives its absolutely convergent pointwise kernel integral against $m_{\text{ar}} - m_\Gamma$, including zeros of the arithmetic density. It assumes no bounded pointwise ratio of the two densities.

The newly supplied canonical-projector formula has the exact operator correspondence

$$\Pi_N = R_N G_N^{-1} R_N^* M_{2q} = P_N - Q_N, \quad \mathcal{A} = \Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q} = U - W.$$

Here every R_N is padded by the original coefficient inclusion into the same $\mathcal{P}_{2q}$. The decomposition $\mathcal{P}_N = R_N E \perp_M \chi \mathcal{P}_{N-q}$ proves the first equality: subtracting $R_N \pi_\chi p$ from p gives an original relation, and canonical orthogonality annihilates its cross Gram. The restricted remainder map is the inverse of $R_N : E \rightarrow \text{ran } \Pi_N$. Thus the signed trace in this result is precisely the supplied trace, with $\text{Tr } \mathcal{A} = 0$.

For two source forms M_0, M_1 , let $\Delta = R_1 - R_0 = B_N(Z_0 - Z_1)$, where $Z_i = (B_N^* M_i B_N)^{-1} B_N^* M_i C_{\text{rem}}$. The complete original remainder lift is C_{rem} . Orthogonality expands both secants with their actual signs:

$$\begin{aligned} G_1 - aG_0 &= R_1^*(M_1 - aM_0)R_1 + a\Delta^* M_0 \Delta, \\ bG_0 - G_1 &= R_0^*(bM_0 - M_1)R_0 + \Delta^* M_1 \Delta. \end{aligned}$$

At the first endpoint the relation space is zero and $\Delta = 0$. Equations (ISM4)–(ISM12) prove these identities and their full source enclosure; the revised coefficient $2q - 1$ and all smaller common-source controls are retained at that same proof site.

For admitted degrees $i \leq j$, the canonical reverse restriction $T_{ij} = G_i^{-1} G_j$ has the exact cross trace $\text{Tr}(\Pi_i \Pi_j) = \text{Tr } T_{ij}$. Indeed $R_j^* M R_i = G_j$, by the original relation-valued section difference, and substitution in the trace proves the identity. The full proof (ISM21)–(ISM26) retains the cross term between the two projector differences in $\text{Tr}(\mathcal{A}^2)$. Centering C by its actual trace mean gives

$$|\mathcal{B}'| \leq \sqrt{\left(\text{Tr}(C^2) - \frac{(\text{Tr } C)^2}{2q+1} \right) \text{Tr}(\mathcal{A}^2)}.$$

Weighted self-adjointness and Hilbert–Schmidt Cauchy–Schwarz prove this on the original metric. Furthermore (ISM28)–(ISM37) constructs a finite matrix polynomial $C^{[L]}$, with exact residual, whose signed integral j_L encloses the correction:

$$j_L - e_L \leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_L + e_L, \quad e_L \leq \vartheta^{L+1} \sqrt{K_D(8q-6)} \rightarrow 0.$$

All of $C^{[L]}$, j_L , e_L , ϑ , K_D are explicitly defined and proved in (RMT19)–(RMT21), with their identity to the ISM notation. This supplies a constructed residual on the actual finite source, preserving the signed center and all arithmetic entries.

Result R73: full spectral data and the exact source isometry. The nested original flags give both U and W the full spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. The explicit original-metric isometry T in (AW9)–(AW10) intertwines them and gives

$$U = TWT^{-1}, \quad \text{Tr}((U - W)C) = \text{Tr}(WT^{-1}[C, T]).$$

Its original arithmetic observation is explicitly $\sigma_h^{\otimes k} \eta J_{2q} T$, with defect $\sigma_h^{\otimes k} \eta J_{2q} (T - I)$ and kernel $\mathcal{P}_{q-1}^\perp$. Thus its action on original classes is retained rather than presumed to be identity. For all eigenvalues $b_1 \leq \dots \leq b_{2q+1}$ of the actual relative source Gram, counted with multiplicity, the complete spectral proof gives

$$|\mathcal{B}(1) - \mathcal{B}(0)| \leq \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \frac{b_{2q+1}}{b_1}.$$

The full (FS1)–(FS35) proof gives strict pointwise inequality for nonscalar C on the literal polynomial flags, and an exact limiting positive-coefficient-metric family. Its moment obstruction is explicitly nonzero; the constructed limiting family is not assigned an arithmetic density. The complete (AW1)–(AW23), (WS1)–(WS39) and independent spectral proofs retain the precise equality scope and all relative eigenvalues.

The complete full-spectrum, trace-norm, overlap, centered Hilbert–Schmidt and angle controls now apply jointly to this same integrand. With their full definitions (RMT8)–(RMT13), the current stronger bound at this site is

$$|\mathcal{B}(1) - \mathcal{B}(0)| \leq J_q = \int_0^1 \min\{E_q, T_q, O_q, S_q, A_q\} dx \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa.$$

Here $T_q = d_x \|U_x - W_x\|_1 / 2$ retains the exact trace norm, O_q its two proved overlap majorants, and S_q the supplied centered variance bound. Their complete proofs and the exact source correspondence are in (RMT3)–(RMT15). Every entry bounds the identical signed trace, so taking its pointwise minimum and integrating proves the displayed inequality. The angle entry A_q uses the actual common moment path and its proved positivity in R74; for a general positive coefficient form the other four entries and the finite signed certificate retain their proved scope.

Result R74: all-degree nonlinear control from the original moments. The inherited endpoint-window proof already gives the canonical principal-angle comparison, nonlinear arcosh transport, strict moment positivity and signed tensor theta primitive (EW.25)–(EW.28), (EW.35a)–(EW.35c), (EW.46a)–(EW.48). The complete (AGT1)–(AGT19) and (AA1)–(AA44) proofs independently re-derive these structures and sharpen the selected window $n = r = q$: crossing its same four projection terms changes the guaranteed moving-angle count from $2q$ to $2q - 1$. The full comparison (EP1)–(EP36) proves every coefficient, metric and operator identification and retains the inherited proof’s broader window domain. For two nested cutoffs the actual relation-valued difference of minimum sections satisfies

$$D_{ij} = R_i - R_j, \quad G_i - G_j = D_{ij}^* M D_{ij}, \quad K_{ij} = G_i^{-1/2} G_j G_i^{-1/2}.$$

The eigenvalues of K_{ij} are the squared principal cosines of the original minimum-section subspaces, with their full multiplicities. The pairs $(q - 1, 2q)$ and $(q, 2q - 1)$ give the original $\mathcal{B}$ as their combined negative log determinant. The second pair has at least one exact zero angle in an ambient space of dimension $2q - 1$. Retaining the full angle lists thus leaves $m = 2q - 1$ possibly nonzero contributions.

For $\chi(S) = \sum_{j=0}^q a_j S^j$ define $\chi^{\#k}(S) = \sum_{j=0}^q \bar{a}_j (k - S)^j$. The actual line $\bar{S} = k - S$ gives $\chi^{\#k}(S) = \overline{\chi(\bar{S})}$ there, with $\chi^{\#k} \in \mathcal{P}_q$. If $\mathcal{B}$ vanished, its first pair would coincide, forcing $1 \perp \chi \mathcal{P}_q$ and hence $0 = \int |\chi(k/2 + iu)|^2 m_t(u) du > 0$. Thus $\mathcal{B}(t) > 0$ throughout the actual interpolation. The exact projection-difference spectra and concavity of $\sqrt{1 - e^{-x}}$ now yield

$$|\dot{\mathcal{B}}| \leq m \sqrt{1 - e^{-\mathcal{B}/m}} (c_{\max} - c_{\min}), \quad \left| 2 \operatorname{arcosh}(e^{\mathcal{B}(1)/(2m)}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/(2m)}) \right| \leq I_q \leq \log \kappa,$$

where $\kappa = b_{2q+1}/b_1$ is the original finite relative Gram ratio. The proof includes the explicit hyperbolic-cosine endpoint interval, the $q = 1$ case, every zero angle, the original theta primitive of D_{ij} and its full source energy. It applies in particular to the original full quartet value $q = [1 + k(m_h - 1)](k + 1)^2$, where m_h denotes the quartet’s common zero order; this last m_h is distinct from the angle count $2q - 1$.

In the preceding formula the refined radius is the actual I_q of (RMT13), with $a = m = 2q - 1$; it divides each of the five same-source pointwise controls by the exact derivative denominator of the nonlinear coordinate. Differentiating $\mathcal{H}_m(B) = 2 \operatorname{arcosh}(e^{B/(2m)})$ gives $1/(m \sqrt{1 - e^{-B/m}})$, so that integration proves this replacement of the former radius $\log \kappa$. Its inverse is $2m \log \cosh(z/2)$, with continuous value zero at zero. Write $B_0 = \mathcal{B}(0)$, $H_0 = \mathcal{H}_m(B_0)$. For every constructed finite order L , the resulting current interval is

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, B_0 - J_q, 2m \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right), B_0 + j_L - e_L \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ B_0 + J_q, 2m \log \cosh \left(\frac{H_0 + I_q}{2} \right), B_0 + j_L + e_L \right\}. \end{aligned}$$

Each entry is a bound proved in R72–R73 or obtained by applying the increasing inverse just calculated; their intersection therefore contains the same actual endpoint. The complete proof (RMT19)–(RMT22) retains the computed geometric residual and original theta comparison; it also proves the scalar and $q = 1$ cases without dropping any coefficient or source mass. The same new upper endpoint is substituted into the central arithmetic and inverse-return estimates at R44–R45 and R47 and their complete earlier proofs.

For the original arbitrary window $n \geq q, r \geq 1$, the same crossed pairing $(n - 1, n + r)$ and $(n, n + r - 1)$ gives the proved count $a(q, r) = \min(q, r + 1) + \min(q, r - 1)$ in (EP17)–(EP25). It improves the inherited count $2 \min(q, r)$ by one exactly when $r = q$. Its entire loss, norm factor, full Taylor units and supported theta primitive retain the original definitions.

The complete proofs following these results retain the original arithmetic Grams as actual data. Their uniform control in tensor degree is the continuing analytic calculation; no missing estimate has been introduced as an assumption in the bounds just proved.

The complete preceding proofs now carry the corrected source kernels, full two-parameter connections, actual tensor specializations and current signed metric controls through their earlier dependent calculations. The original arithmetic norm and endpoint lower products remain in the same estimates. The finite signed residual is explicitly constructed for each fixed source; its constants remain the original arithmetic spectral data as the tensor order varies. The complete source provenance and earlier editions accompany this revised reader.

Part VII**Complete source proof notes and chronology**

S1 PR 8: Finite operator comparison

Source attribution. Web contribution, pull request #8, [Zeta research repository](#).

Exact revision. 25689c5376166df37801229bc275ff5a896007d9

Source file. workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 57502c7fe6a7e1de239c94c3479abbb6a70b4718b551ad1bb1e26e2d3e4f4c22

S1.1 Exact finite operator quotients of the tau-base theta cohomology

Owner-directed continuation, 12 September 2026. Mathematical derivation and research draft; no new analytic theorem below is claimed to have been checked by Lean.

S1.1.1 0. Inputs, provenance, and the new result

This note continues TAU_BASE_MODEL.md, rather than moving the programme back to a finite-field variety. Its arithmetic coefficient object is the original split semiring

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

The geometric base remains the one-point absolute pointed base described there. The quotient

$$p_{\mathbb{Z}} : G(\mathbb{Z}) \longrightarrow \mathbb{Z}$$

and its square with $G(\mathbb{C}) \rightarrow \mathbb{C}$ remain part of the marked object. No ideal norm, Euler factor, scaling parameter, Gaussian, or completion factor is changed here.

The supplied derived formalization (PR #6, 5f6ee6575550d8a20c6fc3990359cb4f67855f80) provides the actual reconstruction, internal quotient, homology-kernel equivalence, and compatible kernel actions. Its finite-jet module supplies the balanced tensor-quotient comparison with the split-reflection square. Those results are reused; this note does not duplicate their Lean implementation.

The new analytic step computes the previously only surjective finite Mellin-jet observation as an exact operator quotient. For every polynomial h whose roots lie in the open critical strip, it constructs

$$Q/h(D)Q \xrightarrow{\sim} E_h/g_h E_h, \quad Q[h(D)] \xrightarrow{\sim} \ker(g_h : E_h \rightarrow E_h),$$

with $E_h = \mathbb{C}[t]/(h)$ and $g = 2\xi$. It then constructs finite spectral projectors on the original Q , not only quotients of Q , and identifies the torsion subsheaf of the balanced analytic realization with $\mathcal{O}/(2\xi)$. Their kernels, original supported zeros, nilpotent actions, and trace-producing residue pairing are all retained.

The general existence of spectral realizations of zeta zeros is prior literature (not a novelty claim here): see R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. The contribution of this note is the explicit comparison in the specified workbench, together with its original SplitZero quotients and its complete finite kernel calculation.

S1.1.2 1. The exact arithmetic complex and its operator

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}.$$

The source maps, with their original factors, are

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The Fourier transform is $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx$. Poisson and the two moment conditions put ΘV in $\mathcal{B}$. Also $D\Theta = \Theta D$, and D preserves V .

Set

$$A = \mathbb{C}[t], \quad t \cdot F = DF, \quad t \cdot \phi = D\phi,$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1, \quad Q = \mathcal{B}/\Theta V.$$

The existing injectivity proof of Θ gives $H^0(C_+) = 0$ and $H^1(C_+) = Q$.

This is the one-leg subcomplex of the actual tau-base chart complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta\phi - \Theta\widehat{\psi}} \mathcal{B}].$$

Its inclusion is $\phi \mapsto (\phi, 0)$ in degree zero and the identity in degree one. The quotient complex is the second copy of V in degree zero. The explicit section is $\psi \mapsto (\psi, \psi)$. The operator on that second copy is $1 - D$, because $D\widehat{\psi} = (1 - D)\psi$. Thus the additional joint-support degree-zero cohomology is retained in this split exact sequence; it is not included in Q and then silently discarded.

The actual source multiplier is

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$H_\phi(s) = c(s)M_+\phi(s), \quad c(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

On the open strip $\mathcal{S} = \{0 < \Re s < 1\}$, $c(s)$ is holomorphic and nowhere zero. The source identity is

$$\mathcal{M}\Theta\phi = gH_\phi.$$

The specific function

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$$

has $H_{\phi_*} = 1$. Consequently

$$H_{P(D)\phi_*}(s) = P(s), \quad \mathcal{M}\Theta(P(D)\phi_*) = g(s)P(s).$$

These are identities with the stated Gaussian and $g = 2\xi$; no rescaling is performed.

S1.1.3 2. A continuous inverse to the actual Euler differential

For any $\rho \in \mathbb{C}$, define on the subspace $\{H \in \mathcal{B} : \mathcal{M}H(\rho) = 0\}$

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1} dt.$$

All powers use the real logarithm on $x, t > 0$. Direct differentiation gives

$$(D - \rho)S_\rho H = H.$$

Endpoint control. Write $x = e^y$, $h(y) = H(e^y)$, and $\sigma = \Re \rho$. At $y \geq 0$, the displayed tail and a bound $|h(t)| \leq C_N e^{-Nt}$ give

$$|S_\rho H(e^y)| \leq \frac{C_N}{N - \sigma} e^{-Ny} \quad (N > \sigma).$$

At $y \leq 0$, the zero moment changes the same expression into the negative lower tail. A bound $|h(t)| \leq C'_N e^{Nt}$ gives

$$|S_\rho H(e^y)| \leq \frac{C'_N}{N + \sigma} e^{Ny} \quad (N + \sigma > 0).$$

Differentiating uses $(S_\rho H)'_y = -\rho S_\rho H - H(e^y)$, so every derivative has the same family of estimates. This proves membership in $\mathcal{B}$ and continuity in its stated seminorms. The homogeneous solution $x^{-\rho}$ fails the required rapid decrease unless its coefficient is zero. Thus $D - \rho$ is injective, and

$$(D - \rho)\mathcal{B} = \ker(F \mapsto \mathcal{M}F(\rho)).$$

Mellin intertwining gives

$$\mathcal{M}(S_\rho H)(s) = \frac{\mathcal{M}H(s)}{s - \rho},$$

with the removable value filled at $s = \rho$.

There is also a source inverse. For $\rho \in \mathcal{S}$ and $\phi \in V$ with $H_\phi(\rho) = 0$, use the same tail for $x > 0$ and extend evenly. The nonvanishing of $c(\rho)$ gives $M_+\phi(\rho) = 0$. Near zero the formula is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Differentiation under this integral proves smoothness at zero in every order. The function is even and its value at zero is zero. The upper-tail formula proves Schwartz decay at infinity. Finally

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi(x) dx = \int_{\mathbb{R}} \phi(x) dx = 0,$$

so the second moment condition is preserved. This proves the source range identity

$$(D - \rho)V = \ker(\phi \mapsto H_\phi(\rho)) \quad (\rho \in \mathcal{S}).$$

At the level of Taylor coefficients $\phi(x) = \sum_{j \geq 1} a_{2j} x^{2j}$, the local expression is $-\sum_{j \geq 1} a_{2j} x^{2j} / (\rho + 2j)$, interpreted to each finite Taylor order with its smooth remainder. All denominators and the minus sign are retained. Endpoints 0, 1 are not included in this source range theorem.

S1.1.4 3. Every finite operator test, without choosing zeros first

Choose any finite set of distinct centres $\rho_1, \dots, \rho_r \in \mathcal{S}$ and positive integers m_i . Define the particular polynomial

$$h(t) = \prod_{i=1}^r (t - \rho_i)^{m_i}, \quad d = \deg h, \quad E_h = A/(h).$$

This defines the test; it does not presume that any centre is a zeta zero. The Chinese-remainder map is

$$\theta_h : E_h \xrightarrow{\sim} \prod_i \mathbb{C}[z_i]/(z_i^{m_i}), \quad [P(t)] \mapsto [P(\rho_i + z_i)].$$

For a holomorphic germ family f at the centres, $j_h f$ denotes its Taylor jets transported back by θ_h^{-1} . In particular $g_h = j_h g$ retains the original local units of g .

Iterate Section 2 through the factors of h . Uniqueness makes the inverse independent of their chosen order. This gives the continuous inverse

$$S_h : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B}, \quad h(D)S_h = \text{id}.$$

The exact range identities are

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H).$$

Surjectivity on the source follows from $H_{P(D)\phi_*} = P$. Surjectivity of the target jet map follows directly from finite Mellin interpolation: on an interval in $y = \log x$, the functions $e^{\rho_i y} y^j / j!$ are linearly independent. For a nonnegative smooth bump η positive on a subinterval, the Gram matrix

$$G_{(i,j),(k,l)} = \int \eta(y) e^{(\rho_i + \bar{\rho}_k)y} \frac{y^{j+l}}{j!l!} dy$$

is positive definite. Multiplying its inverse into the prescribed jet vector produces a compactly supported smooth preimage. This is only an interpolation device; it does not define or alter the arithmetic Weil pairing.

Consequently the vertical arrows in

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\bar{\theta}} & \mathcal{B}/h(D)\mathcal{B} \\ \alpha_h \downarrow \wr & & \wr \downarrow \beta_h \\ E_h & \xrightarrow{\times g_h} & E_h \end{array}$$

are actual A -module isomorphisms:

$$\alpha_h[\phi] = j_h H_\phi, \quad \beta_h[F] = j_h \mathcal{M}F.$$

An inverse to α_h sends the unique polynomial representative P of degree less than d to $[P(D)\phi_*]$. An inverse to β_h is any of the above explicit interpolation lifts, taken modulo $h(D)\mathcal{B}$; the class is independent of that choice by the range theorem.

S1.1.5 4. The kernel and quotient of the global arithmetic operator

Both V and $\mathcal{B}$ are torsion-free A -modules: every factor $D - \rho$ is injective, since its homogeneous solution cannot be Schwartz at infinity. Over the PID A , they are flat. The bounded complex C_+ is therefore a valid flat model for derived tensoring. The displayed isomorphism computes

$$C_+ \otimes_A^{\mathbf{L}} E_h \simeq [E_h \xrightarrow{\times g_h} E_h]$$

in degrees zero and one. It is the base-change of the actual theta complex, not a newly declared spectral complex with an assigned differential.

The original exact sequence $0 \rightarrow V \xrightarrow{\Theta} \mathcal{B} \rightarrow Q \rightarrow 0$ also gives

$$0 \rightarrow Q[h(D)] \xrightarrow{\kappa_h} E_h \xrightarrow{\times g_h} E_h \rightarrow Q/h(D)Q \rightarrow 0.$$

Here $Q[h(D)] = \ker(h(D) : Q \rightarrow Q)$. The maps are explicit:

- Given $[F] \in Q[h(D)]$, uniquely write $h(D)F = \Theta\phi$. Then $\kappa_h[F] = j_h H_\phi$.
- The quotient observation sends $[F]$ to $j_h \mathcal{M}F$ modulo $g_h E_h$.
- For $P \in \ker(g_h : E_h \rightarrow E_h)$, choose its polynomial representative of degree less than d and set

$$\partial_h(P) = [S_h \Theta(P(D)\phi_*)] \in Q[h(D)].$$

The last expression is defined because its theta input has zero jets modulo h . Replacing P by $P + hR$ changes the result by $\Theta(R(D)\phi_*)$. Thus it is independent of representatives and is the inverse of κ_h onto its kernel. The two injectivity and surjectivity assertions also follow directly by lifting representatives in the preceding exact sequence; no dimension-only argument is needed.

All maps commute with D , with $U_a F(x) = F(x/a)$, and with compactly supported multiplicative convolution. On E_h , the scaling action is multiplication by $j_h(a^s)$, including its entire finite nilpotent Taylor polynomial.

At one centre, retain $g(\rho + z) = u(z)z^r$, where $u(0) \neq 0$ (and $r = 0$ is allowed). For test order m , the two cohomology modules have dimension $\min(m, r)$. When $r = 0$, multiplication by g_h is invertible. The derived amplitude complex is then acyclic, while its reconstructed supported homology still has each labelled internal zero.

The degree-zero term is $Q[h(D)]$; the degree-one term is $Q/h(D)Q$. Their finite alternating traces cancel. The arithmetic packet trace below is the degree-one action, or equivalently its explicitly constructed direct summand in the original Q . The canonical truncation triangle retains the degree-zero term rather than relabelling it as zero.

S1.1.6 5. Exact finite spectral projectors on the original cohomology

Now let Z be any finite set of actual nontrivial zeros and retain their full multiplicities r_ρ . Put

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{r_\rho}, \quad E_Z = A/(h_Z), \quad v_Z(s) = g(s)/h_Z(s).$$

The quotient v_Z is entire and its germs are nonzero at every point of Z . Hence $v_{Z,h} = j_{h_Z} v_Z$ is an invertible element of E_Z . This inverse retains the actual Taylor coefficients of g/h_Z ; it is not set to one.

Here $g_{h_Z} = 0$, and the exact sequence becomes

$$Q[h_Z(D)] \xrightarrow[\kappa_Z]{\sim} E_Z, \quad Q/h_Z(D)Q \xrightarrow[\beta_Z]{\sim} E_Z.$$

There is a further map that must be calculated, not assumed: the inclusion of the kernel into Q , followed by the quotient. For $P \in E_Z$,

$$\mathcal{M}(S_{h_Z}\Theta(P(D)\phi_*)) = \frac{g^P}{h_Z} = v_Z P.$$

Therefore this kernel-to-quotient comparison is precisely

$$\boxed{\beta_Z q_Z \partial_Z = M_{v_{Z,h}} : E_Z \xrightarrow{\sim} E_Z.}$$

Define

$$\sigma_Z = \partial_Z \circ M_{v_{Z,h}}^{-1} : E_Z \longrightarrow Q,$$

$$\boxed{\Pi_Z = \sigma_Z J_Z : Q \longrightarrow Q, \quad J_Z[F] = j_{h_Z} \mathcal{M}F.}$$

Then

$$\boxed{J_Z \sigma_Z = \text{id}, \quad \Pi_Z^2 = \Pi_Z, \quad \text{im } \Pi_Z = Q[h_Z(D)], \quad \ker \Pi_Z = h_Z(D)Q.}$$

This gives the actual equivariant direct sum

$$\boxed{Q = Q[h_Z(D)] \oplus h_Z(D)Q.}$$

A representative formula is completely determined. For $F \in \mathcal{B}$, let $P_{Z,F}$ be the unique polynomial of degree less than $\deg h_Z$ representing $v_{Z,h}^{-1} j_{h_Z} \mathcal{M}F$. Then

$$\boxed{\Pi_Z[F] = [S_{h_Z}\Theta(P_{Z,F}(D)\phi_*)].}$$

For a simple actual zero ρ , this specializes to

$$\Pi_{\{\rho\}}[F] = \frac{\mathcal{M}F(\rho)}{g'(\rho)} \left[x^{-\rho} \int_x^\infty \Theta\phi_*(t) t^{\rho-1} dt \right].$$

The displayed denominator is the actual $g'(\rho) = 2\xi'(\rho)$; it is not omitted. The bracketed class is an actual nonzero eigenclass in Q , proved by the exact sequence, even though its representative is only an eigenvector modulo the theta relation.

For finite packets the projectors satisfy

$$\Pi_Z \Pi_W = \Pi_{Z \cap W}, \quad \Pi_{Z \cup W} = \Pi_Z + \Pi_W \quad (Z \cap W = \emptyset).$$

These follow from the direct sums, coprimality of disjoint packet polynomials, and commutation with D . Inclusions of finite packets therefore give compatible invariant subspaces and quotient observations. No convergence of an infinite sum of projectors is asserted.

S1.1.7 6. Carry every projector through the original split quotient

For any vector space M the single-active-fibre lift is $M^\tau = \{\tau\} \sqcup \{m^\bullet : m \in M\}$. For a submodule N use exactly

$$q_N^{\text{sp}} : M^\tau \rightarrow (M/N)^\tau, \quad m^\bullet \mapsto [m]^\bullet, \quad \tau \mapsto \tau.$$

This is the original coequalizer of the relation inclusion and its internal-zero companion. Its universal property is the one implemented in PR #6, including arbitrary noncancellative target semi-modules.

Consequently the finite observation has the split-linear equivalence

$$(Q/h(D)Q)^\tau \xrightarrow{\sim} G(E_h/g_h E_h), \quad [F]^\bullet \mapsto (j_h \mathcal{M} F \bmod g_h E_h)^\bullet.$$

The domain is being used as a split semimodule, not as an unconstructed ring. A class in $h(D)Q$ maps to the supported zero, and external absence alone maps to external absence.

Lift the projector as

$$\tilde{\Pi}_Z(\lambda, [F]) = (\lambda, \Pi_Z[F]).$$

This formula applies to the actual nonempty support faces of the tau-base theta model; their degree-one transports are the stated identities of Q . At bottom it is the original zero map of the bottom module. The full reconstruction interface permits a nonzero bottom fibre when a subsequent diagram requires one; it is not silently specialized away on an opposite-index dual diagram.

For disjoint packets,

$$\tilde{\Pi}_Z \tilde{\Pi}_W = \varepsilon, \quad \varepsilon(\lambda, x) = (\lambda, 0).$$

This is the supported lift of the zero linear operator, not the constant external-zero map. More uniformly, take $\Pi_\emptyset = 0_Q$ and lift it to ε . The scalar representation

$$G(\text{End}_{\mathbb{C}} Q) \longrightarrow \text{End}_{G(\mathbb{C})}(Q^\tau)$$

sends $f^\bullet$ to its split lift and sends τ to the constant- τ map. It is a unital semiring morphism and sends the supported scalar zero to ε .

Retain smaller admitted source labels For a subspace $W \subset V$, first use its specified enlargement $W_D = \mathbb{C}[D]W$ and keep the map

$$\mathcal{B}/\Theta W \longrightarrow Q_{W_D} := \mathcal{B}/\Theta W_D, \quad \ker = \Theta W_D/\Theta W.$$

For a D -stable W , define the ideal

$$J_{W,h} = \{j_h H_\phi : \phi \in W\} \subseteq E_h.$$

It is an ideal because $H_{P(D)\phi} = PH_\phi$ and every element of E_h has a polynomial representative. The same target range calculation gives

$$Q_W/h(D)Q_W \cong E_h/(g_h J_{W,h}).$$

Enlarging to $W + \mathbb{C}[D]\phi_*$ makes $J_{W,h} = E_h$. The exact finite comparison kernel is

$$\boxed{g_h E_h / (g_h J_{W,h}).}$$

Thus the finite equality for the full arithmetic relation space has a precise smaller-label comparison. It is not asserted for every original W without this kernel.

S1.1.8 7. The Hausdorff observation and the remaining global kernel

The projectors just constructed have continuous finite-rank representatives on $\mathcal{B}$: J_Z is continuous, E_Z is finite dimensional, and S_{h_Z} is continuous on its zero-jet domain. They annihilate ΘV , hence also its closure. They therefore descend to

$$Q^H = \mathcal{B} / \overline{\Theta V}.$$

The finite observation still has kernel exactly $h_Z(D)Q^H$. Indeed

$$\ker J_Z = h_Z(D)\mathcal{B} + \Theta V,$$

and continuity gives $\overline{\Theta V} \subset \ker J_Z$. Hence adding the closure to the right side leaves it unchanged. The same finite section yields a continuous direct sum

$$Q^H = \sigma_Z(E_Z) \oplus h_Z(D)Q^H.$$

The algebraic packet $Q[h_Z(D)]$ injects into this Hausdorff quotient: its composite with J_Z is already an isomorphism onto E_Z . No conclusion about vanishing of the entire algebraic-to-Hausdorff kernel follows from this finite statement.

Let

$$\mathcal{R}_\infty = \bigcap_{Z \text{ finite}} \ker \Pi_Z = \bigcap_{Z \text{ finite}} h_Z(D)Q.$$

The map $Q \rightarrow \prod_\rho \mathcal{O}_\rho / (g)$ has exactly this kernel. Every nonzero polynomial whose roots are in $\mathcal{S}$ acts bijectively on $\mathcal{R}_\infty$. To verify this, its kernel on Q lies in the finite packet at its zero centres; its cokernel is supported there as calculated in Section 4. On the complementary summand it is bijective, and its inverse preserves all other commuting finite projectors. This proves injectivity and surjectivity on the intersection.

The continuation retains $\mathcal{R}_\infty$ and its scaling action. Finite packets do not prove it zero. A topological spectral-synthesis theorem or a global cohomological trace argument must address that specific invariant module rather than infer its disappearance from finite-jet isomorphisms.

7.1 The global balanced kernel has a computed local-cohomology complement Work now on the analytic strip $\mathcal{S}$, with its sheaf $\mathcal{O}$ and sheaf $\mathcal{K}$ of meromorphic functions. Form the sheafified balanced realization

$$\mathcal{M}_Q = \mathcal{O} \otimes_A Q, \quad \beta : \mathcal{M}_Q \longrightarrow \mathcal{O}/(g), \quad b \otimes [F] \longmapsto [bMF].$$

Let $\text{tors}_{\mathcal{O}}(\mathcal{M}_Q)$ denote its torsion subsheaf: a germ is in it when some nonzero holomorphic germ annihilates it. This is torsion over the specified analytic coefficient ring; its connection to the original split support is the reconstruction of the inclusion and quotient maps below.

Theorem. The actual Mellin map restricts to an isomorphism

$$\boxed{\beta|_{\text{tors}} : \text{tors}_{\mathcal{O}}(\mathcal{M}_Q) \xrightarrow{\sim} \mathcal{O}/(g).}$$

Moreover, $\mathcal{N} = \ker \beta$ has a canonical $\mathcal{K}$ -module structure, and the displayed map canonically splits the sequence:

$$\boxed{\mathcal{M}_Q \cong \mathcal{O}/(g) \oplus \mathcal{N}, \quad \mathcal{K} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{N}.}$$

Proof at every stalk. Fix $\rho \in \mathcal{S}$ and set $x = s - \rho$. If $g(\rho) \neq 0$, Section 4 with $h = t - \rho$ proves that $D - \rho$ is bijective on Q . Consequently x is invertible on $\mathcal{M}_{Q,\rho}$, and this module is a module over the meromorphic field $\mathcal{O}_\rho[1/x]$. Its torsion is zero, as is $\mathcal{O}_\rho/(g)$.

If $r = \text{ord}_\rho g > 0$, the finite projector for this one full packet gives

$$Q = Q[(D - \rho)^r] \oplus (D - \rho)^r Q.$$

On the second summand, $(D - \rho)^r$ is bijective: it is injective by the direct-sum intersection, and

$$(D - \rho)^r Q = (D - \rho)^{2r} Q$$

follows by applying $(D - \rho)^r$ to the direct sum. Hence $D - \rho$ itself is bijective there. Tensoring the direct sum with $\mathcal{O}_\rho$ gives a first summand $\mathcal{O}_\rho/(x^r)$, on which β is the isomorphism calculated in Section 5, and a second summand on which x is invertible. The map β kills the second summand because it is in $x^r \mathcal{M}_{Q,\rho}$, while x^r annihilates its target. Every nonzero germ of $\mathcal{O}_\rho$ is a unit times a power of x , so that second summand is a module over $\mathcal{K}_\rho$ and is torsion-free.

This proves the stated torsion isomorphism at every stalk. It glues without choosing neighbourhood projectors: both the torsion subsheaf and the restricted map β are intrinsic. The inverse of that isomorphism, followed by the torsion inclusion, is the global section of β . Local inverses to nonzero holomorphic scalars on $\mathcal{N}$ are unique, so they glue to its $\mathcal{K}$ -module structure. This proves the theorem.

7.2 The extraction is an actual local-cohomology map At the same stalk the principal-ideal local-cohomology complex is the explicit localization complex

$$R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) = [\mathcal{M}_{Q,\rho} \longrightarrow \mathcal{M}_{Q,\rho}[1/x]], \quad \text{degrees } 0, 1.$$

The decomposition just proved computes it:

$$\boxed{R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) \xrightarrow{\sim} (\mathcal{O}_\rho/(g))[0].}$$

The finite torsion summand maps to zero upon localization; the meromorphic summand maps isomorphically to itself. The map to the displayed target is β in degree zero and zero in degree one. Thus the supported arithmetic divisor is the actual cohomological image of the localization kernel, not a prescribed deletion of a list of spectral points.

Before analytic extension there is the corresponding operator-localization map

$$Q \longrightarrow A[h_Z^{-1}] \otimes_A Q,$$

with kernel $Q[h_Z(D)]$ and cokernel zero, because $h_Z(D)$ is invertible on its complementary summand. Its supported lift sends that kernel to the internal supported zero, not to external absence.

At every original label λ , reconstruct the localization complex using

$$d(\lambda, m) = (\lambda, m/1), \quad \varepsilon(\lambda, m) = (\lambda, 0).$$

Its internal degree-zero cycles are exactly the reconstructed torsion module. Its internal first cohomology is the support skeleton, since the ordinary first cohomology vanishes. The map $p_{\mathbb{Z}}$ remains

in the structural coefficient square throughout; x is the analytic image of the operator $t - \rho$, not a replacement for the scalar e or τ .

The theorem does not assert $\mathcal{N} = 0$. It computes what local cohomology does to it: localization is an isomorphism on that module, so it contributes no cohomology to the stated local-support complex. Its possible contribution to a different global trace functor has not been declared absent.

S1.1.9 8. The residue duality is now inside an actual derived finite comparison

Assume the centres and their orders in h are stable under $\iota(s) = 1 - \bar{s}$. For germs define

$$f^\dagger(s) = \overline{f(1 - \bar{s})}.$$

The chosen monic polynomial satisfies $h^\dagger = (-1)^d h$. Retain that factor. Define the perfect finite algebra pairing

$$R_h(f, u) = \sum_i \operatorname{Res}_{s=\rho_i} \frac{f^\dagger(s)u(s)}{h(s)} ds.$$

It satisfies $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$. No phase is inserted to change this sign. Since $g^\dagger = g$, multiplication by g_h is self-adjoint for this pairing. Therefore it induces a perfect pairing

$$\boxed{\overline{\operatorname{coker} g_h} \otimes \ker g_h \longrightarrow \mathbb{C}.$$

Well-definedness is the identity $R_h(ga, u) = R_h(a, gu)$; nondegeneracy follows by taking the annihilator of the image in the finite perfect algebra pairing. Through Section 4 this pairs the two actual cohomology groups of $C_+ \otimes_A^{\mathbf{L}} E_h$.

For an exact packet h_Z , set $v_{Z,h} = j_h(g/h_Z)$ as above and

$$j_Z : A_Z \rightarrow \ker g_h = E_Z, \quad [f] \mapsto [j_h(h_Z/g)f].$$

Here $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g) \cong E_Z$ by the specified local quotient maps; h_Z/g is holomorphic at every centre. The local units are not replaced by one. The exact contraction is

$$\boxed{R_h(f, j_Z k) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger k}{g} ds =: R_Z(f, k).$$

Thus the previously constructed arithmetic residue form is the duality pairing of the actual finite operator comparison after the displayed unit morphism.

The Jacobian map is still $J_g[k] = [g'k]$. Its contraction is

$$\boxed{R_h(f, j_Z J_g k) = R_Z(f, J_g k) = \sum_{\rho \in Z} r_\rho \overline{f(1 - \bar{\rho})} k(\rho) = \operatorname{Tr}(M_{f^\dagger k} | A_Z).$$

In particular the factor 2 in $g = 2\xi$ is retained in both the denominator and the Jacobian. The trace contraction is invariant under their combined explicitly specified transformation, not by deleting that factor in only one place.

Every map lifts to the original support fibres. Dual support arrows are precomposition maps on the opposite index category. A vanishing pairing value is the supported scalar $e_{\mathbb{C}}$; an absent argument gives τ .

S1.1.10 9. Cohomological placement and arithmetic traces

On the tau-base chart use the one-leg sheaf $\mathcal{T}_+$ and its A -action. The bounded projective resolution from TAU_BASE_MODEL.md gives the canonical map

$$\mathbf{A}_\tau(\mathcal{T}_+ \otimes_A^{\mathbf{L}} E_h) \xrightarrow{\sim} \mathbf{A}_\tau(\mathcal{T}_+) \otimes_A^{\mathbf{L}} E_h \xrightarrow{\sim} [E_h \xrightarrow{g_h} E_h].$$

These isomorphisms use finite evaluation sums and the displayed free/flat resolutions. They are not compact-support base-change assertions for a different arithmetic scheme.

The packet projector is now an actual finite-rank endomorphism of the original $H_\tau^1 = Q$ (and of each of its retained support fibres). For an admissible smooth compactly supported multiplicative test k ,

$$H_k F(x) = \int_0^\infty k(a) F(x/a) \frac{da}{a}$$

commutes with the projector. Its ordinary finite-rank trace is

$$\boxed{\mathrm{Tr}(H_k \Pi_Z | Q) = \sum_{\rho \in Z} r_\rho \mathcal{M}k(\rho).}$$

This is a finite-rank trace on the original cohomology, using its explicit finite invariant image. It is stronger than declaring that a surjective finite observation has a trace. Infinite trace summation still requires the specified topology and admissible test class.

On r -fold products over the tau-point, the already calculated external tensor map gives $H^r = Q^{\otimes r}$. The operator $\Pi_Z^{\otimes r}$ is a finite projector onto $Q[h_Z]^{\otimes r}$. The full nilpotent scaling action there is

$$a^{\rho_1 + \cdots + \rho_r} \exp \left((\log a) \sum_{j=1}^r N_{\rho_j, j} \right).$$

For a repeated exponent the exact modulus comparison remains

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re\rho - 1) \log a.$$

This supplies actual finite invariant objects on which a Deligne-style tensor estimate would have to act. It does not prove that estimate. The source's opposite weight bounds, its global dualizing theory, and the admissibility of its geometric hypotheses have not been inferred from these finite isomorphisms.

S1.1.11 10. Status and publication boundary

Established in this note, with written proofs: the differential range identities on the stated spaces; the exact finite operator quotient and torsion kernel; the derived finite comparison; the analytic torsion-subsheaf and local-cohomology identification with $\mathcal{O}/(2\xi)$; finite spectral direct summands with explicit projectors; the original split lifts and smaller-label kernels; the residue/Jacobian contraction on that comparison; and compatible finite-rank arithmetic traces and tensor actions.

Not claimed: a global injectivity theorem for the full balanced analytic realization; vanishing of $\mathcal{R}_\infty$; identification of the entire four-point chart category with an arithmetic curve; a full compact-support/ordinary-cohomology realization satisfying the hypotheses of Weil II; positivity of the unshifted Weil form; RH or GRH. The finite tests are not a Lean or analytic proof certificate.

The prior base note and this continuation are published as additive research material beside PR #6. The original scalar core, formal modules, frozen reader and all existing branches remain unchanged. New derivations are AI-assisted work under the owner's programme; the structural source and its Lean implementation retain their own attribution. No copyrighted source corpus or private conversation is included.

References and source interfaces

- Owner's SplitZero programme: the original scalar, support fibre and quotient constructions; PR #6 at 5f6ee6575550d8a20c6fc3990359cb4f67855f80, especially SplitZeroHomology.lean, SplitZeroComplex.lean, SplitZeroInternalQuotient.lean, and SplitZeroFiniteJets.lean.
- TAU_BASE_MODEL.md: the actual characteristic-zero chart, cohomology, right-adjoint and tensor constructions used here. It remains an authored research model, not an accepted geometric RH realization.
- Deligne, P., *La conjecture de Weil. II* (1980), sections 3.3.4-3.3.6 and 6.2.1-6.2.7. Selected supplied TeX passages were reread for the intended comparison/duality architecture; no source proof is republished or claimed fully audited.
- Meyer, R., *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. Prior spectral-realization context; no theorem from it is silently used to identify the test spaces here.
- The Stacks Project, Derived tensor product, Tag 06XY. The general bounded-flat-complex construction used in Section 4.

S2 PR 9: Chain descent

Source attribution. Web contribution, pull request #9, [Zeta research repository](#).

Exact revision. 8f99b94d306c3b1aaa817d52c57fa6fd298d511c

Source file. workbenches/tau-chain-descent/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 1ab370cf2db9fce70020bebbab76d3dbd3bd37d39e28c83114eeb0eccfa2f0b3

S2.1 Chain-level arithmetic descent over the tau base

Owner-directed research continuation, 12 September 2026. Draft proofs for review; no new Lean certification or RH claim.

S2.1.1 0. Fixed inputs and publication scope

This is an additive continuation of Zeta PR #8 at 25689c5376166df37801229bc275ff5a896007d9, particularly workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md (Git blob 68f59d098d60708f8e899bb960abbd1135ebb3b4). Its sections 1–5 supply the stated analytic range theorem and the finite packet maps. The new result here is their chain-level descent, its actual extension class, and the connection of that class to residue duality and the arithmetic trace.

The source programme retains

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The absolute pointed base and the coefficient square remain

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the first vertical map remains the infinite ring of integers. Every new arrow below is a coefficient or cochain construction over this marked base. No base field, ideal norm, Euler factor, or completion constant is changed.

The supplied derived note and PR #6 provide `Recovery.recoverEquiv`, `Relations.coequalizer_universal`, the internal-homology coequalizer, `ChainMap.homologyKernelEquiv`, and `homology_int_ertwines`. Their algebraic conclusions are used at their stated types. No existing formal module is modified. Analytic inputs from PR #8 remain written research arguments; they are not promoted to Lean theorems.

S2.1.2 1. The actual arithmetic source and packet

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Retain

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad A = \mathbb{C}[t], \quad t \cdot F = DF.$$

The one-leg tau-chart complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad |V| = 0, \quad |\mathcal{B}| = 1, \quad Q = H^1(C_+).$$

The injectivity of theta gives $H^0(C_+) = 0$. The two-leg complex is

$$C_{+-} = [V \oplus V \longrightarrow \mathcal{B}], \quad (\phi, \psi) \longmapsto \Theta\phi - \Theta\widehat{\psi}.$$

Its source operator is $(D, 1 - D)$; the inclusion $C_+ \rightarrow C_{+-}$ is $\phi \mapsto (\phi, 0)$ and the identity in degree one. It leaves the extra degree-zero copy of V in C_{+-} present.

Use the unchanged arithmetic multiplier

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The input function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

and the established identities are

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s).$$

Fix a nonempty finite packet Z of actual nontrivial zeros, with full orders m_ρ . Set

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = A/(h), \quad v(s) = g(s)/h(s).$$

The specific monic polynomial defines this finite test; it does not rescale g . Write j_h for the complete Taylor-jet/Chinese-remainder map at these centres, and retain the invertible element

$$v_h = j_h v \in E^\times, \quad \varepsilon_Z = v_h^{-1}.$$

For $u \in E$, let $R_Z(u) \in \mathbb{C}[t]_{<d}$ be the unique polynomial representing $\varepsilon_Z u$.

The input range theorem supplies the inverse $S_h^{\mathcal{B}}$ of $h(D)$ on the zero-jet range, and the analogous source inverse S_h^V . For one factor,

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

The stated source inverse uses $0 < \Re \rho < 1$ and the zero moment. Iteration supplies S_h . Define the actual maps

$$J : \mathcal{B} \rightarrow E, \quad JF = j_h \mathcal{M}F,$$

$$s : E \rightarrow \mathcal{B}, \quad s(u) = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*).$$

The inverse is defined because $gR_Z(u)$ has all required zero jets. Directly,

$$\mathcal{M}s(u) = vR_Z(u), \quad Js = 1_E, \quad J\Theta = 0, \quad h(D)s(u) = \Theta(R_Z(u)(D)\phi_*). \quad (1)$$

These equations fix the representative map, including all derivatives of g/h . On Q , the existing projector is $\Pi_Z = \bar{s}\bar{J}$, with image $Q[h(D)]$ and kernel $h(D)Q$.

S2.1.3 2. The exact equivariant extension and its class

Define an actual subspace of the original test-function space:

$$\mathcal{B}_Z = \{F \in \mathcal{B} : [F] \in Q[h(D)]\}.$$

Then

$$\mathcal{B}_Z = \Theta V \oplus s(E) \quad \text{as complex vector spaces,}$$

with maps

$$(\phi, u) \mapsto \Theta\phi + s(u), \quad F \mapsto (\Theta^{-1}(F - sJF), JF). \quad (2)$$

For the inverse, $F - sJF$ has zero class in Q by the packet projector theorem; injectivity of theta makes its source unique. The intersection is zero by $Js = 1$ and $J\Theta = 0$.

The original operator D preserves $\mathcal{B}_Z$, and there is a short exact sequence of A -modules

$$0 \longrightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J} E \longrightarrow 0. \quad (3)$$

The arrows in (3), not just the three modules, specify its extension class. Compute it using the resolution

$$0 \longrightarrow A \xrightarrow{\times h} A \longrightarrow E \longrightarrow 0.$$

The resulting isomorphism $\text{Ext}_A^1(E, V) \cong V/h(D)V$ sends an extension to $h(D)b$ in the source, where b is any lift of $1 \in E$. Fixing this explicit rule fixes its sign. Under the existing map $[\phi] \mapsto j_h H_\phi$, the class of (3) is

$$[\mathcal{B}_Z] = \varepsilon_Z = j_h(h/g) \in E^\times. \quad (4)$$

Proof: use the lift $s(1)$ and (1). Its source boundary is $R_Z(1)(D)\phi_*$, whose H -jet is ε_Z .

In particular, (3) has no A -linear section. This also follows immediately from the injectivity of $h(D)$ on $\mathcal{B}$: an A -linear map $E \rightarrow \mathcal{B}$ has image killed by $h(D)$ and is therefore zero. The map s in (2) is the explicit complex-linear section, and the next section computes exactly how it fails to intertwine the arithmetic action. This non-splitting occurs for every nonempty packet, including critical-line packets; it is not an RH-failure criterion.

The general use of extension classes is standard homological algebra (Stacks, Tags 010I and 06XP). Formula (4) calculates the specific extension arising from this theta complex.

S2.1.4 3. The scaling defect is an original theta boundary

For $P \in A$ and $u \in E$, define

$$q_P(u) = \frac{P(t)R_Z(u) - R_Z(Pu)}{h(t)} \in A, \quad k_P(u) = q_P(u)(D)\phi_* \in V. \quad (5)$$

The numerator is divisible by h , since both of its residues are $P\varepsilon_Z u$. Using (1) and the injectivity of $h(D)$ on $\mathcal{B}$ gives

$$\boxed{P(D)s - sM_P = \Theta k_P.} \quad (6)$$

No quotient has yet been applied in (6). The right side is an actual theta function. The composition law is

$$\boxed{k_{PQ} = P(D)k_Q + k_P M_Q.} \quad (7)$$

It follows either by expanding the numerator of (5), or by expanding the commutator in (6) and using injectivity of theta.

For $P = t$, the numerator of (5) has degree at most d . Hence

$$\ell_Z(u) = [t^{d-1}]R_Z(u), \quad q_t(u) = \ell_Z(u), \quad \boxed{Ds(u) - s(tu) = \ell_Z(u)\Theta\phi_*} \quad (8)$$

Thus, in the exact coordinates (2), the original operator is

$$\boxed{D_{\mathcal{B}_Z} = \begin{pmatrix} D_V & \phi_* \ell_Z \\ 0 & M_t \end{pmatrix}.} \quad (9)$$

Its off-diagonal map is rank one and its coefficient is determined by the retained local arithmetic unit. It is not chosen to improve a spectral estimate.

For one simple zero,

$$h = t - \rho, \quad \varepsilon_Z = 1/g'(\rho),$$

$$s(1) = \frac{1}{g'(\rho)} x^{-\rho} \int_x^\infty \Theta\phi_*(y) y^{\rho-1} dy,$$

$$\boxed{(D - \rho)s(1) = \frac{1}{g'(\rho)} \Theta\phi_*} \quad (10)$$

The coefficient is $1/(2\xi'(\rho))$, with the original factor two retained.

The full dilation group For $a > 0$, retain $U_a F(x) = F(x/a)$ and $A_a = M_{j_h(a^s)}$ on E . Define

$$k_a(u) = S_h^V (U_a(R_Z(u)(D)\phi_*) - R_Z(A_a u)(D)\phi_*). \quad (11)$$

The source of S_h^V has zero H -jets because $a^s \varepsilon_Z u$ and $\varepsilon_Z A_a u$ have equal jets. All powers use the real logarithm of a . The exact identities are

$$U_a s - s A_a = \Theta k_a, \quad k_{ab} = U_a k_b + k_a A_b, \quad k_1 = 0. \quad (12)$$

The same construction applies to a smooth compactly supported multiplicative test f : replace U_a by $H_f = \int f(a) U_a da/a$ and A_a by $M_{j_h \mathcal{M} f}$. The source integral belongs to V ; its H -jets give the same cancellation. These are actual linear maps, not a choice of a formal matrix on E alone.

S2.1.5 4. A derived equivariant inclusion with its full homotopy

Put $P_Z = sJ : \mathcal{B} \rightarrow \mathcal{B}$. It is a finite-rank idempotent, and $P_Z\Theta = 0$. Consequently

$$\mathfrak{p}_Z = (0, P_Z) : C_+ \rightarrow C_+ \quad (13)$$

is a cochain idempotent. Its image is the two-term complex $[0 \rightarrow s(E)]$, complex-linearly isomorphic to $E[-1]$.

For the operator $P(D)$ on C_+ , define a degree-minus-one map with sole component

$$\mathbf{H}_P^1 = k_P J : \mathcal{B} \rightarrow V.$$

Equation (6) gives

$$\boxed{P(D)\mathfrak{p}_Z - \mathfrak{p}_Z P(D) = d\mathbf{H}_P + \mathbf{H}_P d.} \quad (14)$$

In degree one this is $\Theta k_P J$; in degree zero it is zero because $J\Theta = 0$. For dilation use $\mathbf{H}_a^1 = k_a J$. Equation (12) is the coherence law of those homotopies.

There is also a strictly A -linear derived presentation. Set

$$C_Z = [V \xrightarrow{\Theta} \mathcal{B}_Z].$$

In the coordinates (2) its differential is $\phi \mapsto (\phi, 0)$ and its action is (9). The maps

$$\iota : C_Z \hookrightarrow C_+, \quad \pi : C_Z \rightarrow E[-1], \quad \pi^0 = 0, \pi^1 = J \quad (15)$$

are strictly A -linear cochain maps, and π is a quasi-isomorphism. Thus the actual equivariant inclusion is the roof

$$\boxed{E[-1] \xleftarrow[\simeq]{\pi} C_Z \xrightarrow{\iota} C_+.} \quad (16)$$

The complex-linear deformation retraction also has an exact contracting homotopy: on $\mathcal{B}_Z$, send $\Theta\phi + s(u)$ to $\phi \in V$. Its equation is $1 - s\pi = dH + Hd$ (with s interpreted as the degree-one cochain section). This homotopy is not claimed A -linear; the A -linear replacement is the roof (16).

The strictly A -linear map $j : C_+ \rightarrow E[-1]$ is J in degree one and zero in degree zero. It satisfies $j\iota = \pi$. This gives a split retract in $D(A)$ without falsely asserting an A -linear section into $\mathcal{B}$.

On the two-leg complex use $\mathfrak{p}_Z^0 = 0$ on $V \oplus V$, $\mathfrak{p}_Z^1 = P_Z$, and $\mathbf{H}_a^1(F) = (k_a JF, 0)$. The source dilation is $(U_a, aU_{1/a})$, so (14) holds with its original two-leg differential. The inclusion $C_+ \rightarrow C_{+-}$ commutes with these cochain projectors. The extra degree-zero cohomology of C_{+-} is killed by this projector; it remains in the complementary complex.

S2.1.6 5. The mixed-support homotopies require the actual synchronization map

Let $L = \mathcal{P}(\{+, -\})$ be the original chart-support lattice. Its three active complexes are

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad C_- = [V \xrightarrow{-\Theta\mathcal{F}} \mathcal{B}], \quad C_{+-} = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}].$$

Here $\mathcal{F}$ is the specified Fourier transform, with $\mathcal{F}^2 = 1$ on even functions. The source inclusions into the joint fibre are $v \mapsto (v, 0)$ and $v \mapsto (0, v)$; degree-one transports are identities. The bottom complex is zero in this particular model. The new formalization allows a nonzero bottom fibre in the general reconstruction; no assertion about all diagrams is obtained by specializing this model.

The reconstructed operations are

$$(\lambda, x) + (\mu, y) = (\lambda \vee \mu, \rho_{\lambda, \lambda \vee \mu} x + \rho_{\mu, \lambda \vee \mu} y),$$

$$e(\lambda, x) = (\lambda, 0), \quad \tau(\lambda, x) = (\perp, 0). \quad (17)$$

The cochain projector $\tilde{p}_Z$ is zero-in-its-fibre in degree zero and P_Z in degree one. It commutes with the support transports. So does the actual action: U_a on the plus source, $aU_{1/a}$ on the minus source, their direct sum on the joint source, and U_a at every active overlap.

The objectwise homotopies are, however, different maps. With $b = k_a JF$, they are

$$H_{a,+}^1(F) = b, \quad H_{a,-}^1(F) = -\hat{b}.$$

Their images in the joint source are $(b, 0)$ and $(0, -\hat{b})$. The difference is the explicit cycle

$$(b, \hat{b}) \in \ker d_{+-} = H^0(C_{+-}). \quad (18)$$

For nonzero k_a , there is no label-preserving family of homotopies compatible with both source inclusions: the joint component would have to equal both $(b, 0)$ and $(0, -\hat{b})$. This says exactly which naturality square fails. It does not discard the cohomology or the arithmetic action.

The replacement is the programme's actual synchronization. Define the support map

$$r_L(\perp) = \perp, \quad r_L(\lambda) = \{+, -\} \quad (\lambda \neq \perp).$$

In degree zero use literal multiplication by $E = (1, e)$ on the original two-coordinate support module:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea).$$

In degree one define $\mathfrak{r}^1(\lambda, F) = (r_L \lambda, F)$ for active labels and preserve global zero. An absent source coordinate becomes its supported vector zero, so the source differential has unchanged amplitude. Hence

$$d\mathfrak{r}^0 = \mathfrak{r}^1 d, \quad \mathfrak{r}^2 = \mathfrak{r}. \quad (19)$$

This uses the operation verified in the synchronization attachment, now on the actual complex. The fixed active fibre is exactly C_{+-} . On degree-one homology, the map is

$$(\lambda, [F]) \mapsto (\{+, -\}, [F]).$$

It has no nonzero amplitude kernel, because all degree-one transports are the identity of Q . The original mask is retained through

$$x \mapsto (\text{supp}(x), \mathfrak{r}x).$$

This map is injective in each degree of this chart complex: in degree zero the coordinate inclusions are injective, and in degree one the transport is the identity. This injectivity is a proved property of these maps, not an assumption for general diagrams with noninjective transports.

There are now two actual split-linear homotopies with the same support map r_L . Define

$$\tilde{H}_a^{\text{left},1}(\lambda, F) = (\{+, -\}, (k_a JF, 0)),$$

$$\tilde{H}_a^{\text{right},1}(\lambda, F) = (\{+, -\}, (0, -\mathcal{F}k_a JF)),$$

and send global zero to global zero. Both satisfy

$$\tau(\tilde{U}_a \tilde{p}_Z + (-1)^{\bullet} \tilde{p}_Z \tilde{U}_a) = \tilde{d} \tilde{H}_a + \tilde{H}_a \tilde{d}. \quad (20)$$

In degree one this is (6); in degree zero it follows from $Jd = 0$. The two choices differ by the retained cycle (18). No averaging or removal of that cycle is performed. This is the precise role of synchronization here: it permits a common supported comparison fibre for the homotopy, while the original mask and the difference of homotopies remain recorded.

In the simple-zero case (10), the defect in any source support is the nonzero chain amplitude $\Theta(\phi_*/g'(\rho))$. Its internal homology class is the zero of that support, not external absence.

For every chain comparison $f : C \rightarrow C'$, the supplied formalized kernel equivalence remains

$$\{z \in Z(C) : f(z) \in B(C')\} / B(C) \xrightarrow{\sim} \ker H(f).$$

Applied to π in (15), it computes the zero kernel in degree one from its actual exact sequence. Applied to support synchronization, it yields the degree-one assertion just proved and the separately retained degree-zero data. It is not applied to a different global comparison without carrying that comparison's maps.

For any smaller D -stable admitted source $W \subset V$, the map $Q_W \rightarrow Q$ has kernel $\Theta V / \Theta W$. A homotopy value may not already lie in W . The explicit replacement is enlargement by the actual source values, with this quotient map and its kernel retained. The present chart uses full V , where every displayed homotopy is defined.

S2.1.7 6. The same extension class produces the arithmetic duality factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$. Then

$$f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad h^\dagger = (-1)^d h.$$

The perfect finite pairing is

$$R_h(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) u(s)}{h(s)} ds,$$

with exact symmetry $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$.

Identify E with the actual divisor algebra $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g)$ by polynomial evaluation and the complete Chinese-remainder jets. The nonzero local units in g/h make this an isomorphism. The operator represented by the extension class is

$$M_{\varepsilon_Z} : E \rightarrow E.$$

Substituting (4) into the residue formula gives

$$R_h(f, M_{\varepsilon_Z} u) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger u}{g} ds = R_Z(f, u). \quad (21)$$

This is independent of representatives: changing ε_Z by h times a germ adds a holomorphic differential. Thus the actual extension class is precisely the unit morphism converting the polynomial-resolution residue form into the arithmetic residue form.

Let $J_g = M_{j_h g'}$. Then

$$R_h(f, M_{\varepsilon_Z} J_g u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho) = \text{Tr}(M_{f^\dagger u} \mid E). \quad (22)$$

Locally $g = z^m w(z)$ gives $g'/g = m/z + w'/w$. This proves the formula and retains every multiplicity. The product $\varepsilon_Z J_g$ carries both factors associated to $g = 2\xi$.

The pairing-valued support map sends a zero value to $e_{\mathbb{C}}$ and an absent argument to τ . Its original input support and relation fibre remain available upstream. Formula (22) is a calculated trace, not a positivity assertion.

Reflection is strict for this specified section Define conjugate-linear involutions

$$j_B F(x) = x^{-1} \overline{F(1/x)}, \quad j_V \phi = \widehat{\phi}, \quad j_E u = u^\dagger.$$

They satisfy $\Theta j_V = j_B \Theta$ and $J j_B = j_E J$. The actual source function is Fourier-fixed: writing $\phi_* = D(D-1)e^{-\pi x^2}$, Fourier sends the two factors to $(1-D)(-D) = D(D-1)$ and fixes the Gaussian. It is real, so $j_V \phi_* = \phi_*$.

Since $\varepsilon_Z^\dagger = (-1)^d \varepsilon_Z$ and reflection preserves polynomial degree,

$$R_Z(j_E u) = (-1)^d (R_Z(u))^\dagger.$$

Uniqueness of the inverse of $h(D)$ on its range gives

$$j_B S_h^{\mathcal{B}} = (-1)^d S_h^{\mathcal{B}} j_B.$$

Both factors $(-1)^d$ are retained in the substitution, and their product is one. Therefore

$$j_B s = s j_E, \quad j_B P_Z = P_Z j_B. \quad (23)$$

This is an actual equality for this section; it would not follow for an arbitrary choice of representatives. Its compatibility with the nonzero dilation homotopy is

$$j_V k_a = a k_{1/a} j_E. \quad (24)$$

Indeed, apply j_B to (12), use $j_B U_a = a U_{1/a} j_B$ and (23), then use injectivity of theta. The split lift is semilinear through $r^\bullet \mapsto \bar{r}^\bullet$, which fixes both e and τ .

S2.1.8 7. A finite-rank Lefschetz trace on the actual chain complex

For $f \in C_c^\infty(\mathbb{R}_{>0})$, let U_f be its convolution action on C_+ . The cochain endomorphism

$$p_Z U_f = (0, s J H_f)$$

has finite-rank components. Its cochain supertrace is therefore defined without an infinite-dimensional regularization:

$$\text{Str}(p_Z U_f \mid C_+) = -\text{Tr}(s J H_f \mid \mathcal{B}) = -\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (25)$$

The finite-rank identity $\text{Tr}(AB) = \text{Tr}(BA)$ reduces the middle expression to $\text{Tr}(J H_f s \mid E)$; this is multiplication by $j_h \mathcal{M}f$. Equation (25) equals the supertrace of the induced operator on homology.

On C_{+-} its degree-zero contribution is still zero, since the same cochain projector has zero degree-zero component.

The boundary representatives in (6), (14), and (20) are retained even though their contribution to homology is zero. The finite-rank trace does not delete their complex or assign its zero to external absence.

For the n -fold tensor complex over the tau-base product, use the original coefficient tensor over $\mathbb{C}$ and its differential

$$d(x_1 \otimes \cdots \otimes x_n) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_n.$$

The projector $\mathfrak{p}_Z^{\otimes n}$ has image $E^{\otimes n}[-n]$. For independently specified tests $f_1, \dots, f_n$,

$$\text{Str} \left(\bigotimes_j (\mathfrak{p}_Z \cup_{f_j}) \right) = (-1)^n \prod_j \sum_{\rho \in Z} m_\rho \mathcal{M} f_j(\rho). \quad (26)$$

Let $A = U_a \mathfrak{p}_Z$ and $B = \mathfrak{p}_Z U_a$ as cochain maps. A homotopy for $A^{\otimes n} - B^{\otimes n}$ is

$$\sum_{j=1}^n B^{\otimes(j-1)} \otimes H_a \otimes A^{\otimes(n-j)},$$

with the displayed Koszul sign on homogeneous inputs when the degree-minus-one map crosses the preceding factors. The telescoping identity and (14) prove it. The full support mask maps to the product of its input masks; vanishing products of represented amplitudes stay supported.

S2.1.9 8. What now enters the Weil II programme

The structural base, the infinite quotient, and the actual theta complex are unchanged. The new objects are the extension (3), its unit class (4), its synchronized homotopy (20), its strict equivariant derived roof (16), and the chain trace (25). The arithmetic residue unit in (21) is now supplied by that very extension rather than appended without its source.

Deligne's inspected comparison-image argument (Weil II 3.3.4–3.3.6) requires two weight bounds on the same cohomological image. The inspected derived formulation (6.2.1–6.2.7) retains the dualizing operation, its degree and Tate shifts, and its exchange of direct-image functors. This continuation supplies coherent arithmetic maps and finite trace operations for the tau model; it does not transfer the finite-field hypotheses or assert those two bounds.

The retained error is explicit: it is the original boundary Θ_{k_a} on representatives, with the nonzero Ext class (4). It is removed at homology by the original supported quotient, not by making its input absent. A geometric comparison can now be tested for compatibility with this class and the residue map (21). A complex-linear section that ignores (6) fails that test.

No infinite sum of packet projectors is asserted to converge. The existing meromorphic/kernel summand from PR #8 remains in its own exact sequence, and a different global trace functor must retain or analyze its contribution. The source's finite-jet isomorphisms and the new finite-rank chain trace do not by themselves set that summand to zero.

S2.1.10 9. Verification and references

The new finite checker tests polynomial remainder identities, the extension class, rank-one scaling defects, cochain homotopies, dilation cocycles, residue contraction, tensor signs, trace identities, and

original support operations. These finite algebraic models are not numerical tests of zeta zeros and not certificates for the analytic range theorem. No Lean execution is claimed for this continuation.

Read source interfaces:

- PR #8 at the pinned revision above, `FINITE_OPERATOR_COMPARISON.md`, sections 1–10; `TAU_BASE_MODEL.md` is its already-published model.
- Owner's `SplitZero_Derived_Mathematics_2026-09-12(1).md`, sections 1–7. In particular its allowance of a nonzero bottom fibre and the exact homology-kernel/action theorems are retained.
- `formal/splitzero/SplitZeroHomology.lean`, lines 175–end, read at the same revision; blob `7b9e5c49a494dd0c751133e0234b3e01048f7b67`.
- Supplied Deligne TeX excerpts: French segment 2, original lines 882–950; segment 3, original lines 675–756. The used equations are the image bounds and duality operations, not every assertion in the historical transcription. No source corpus is republished with this contribution.
- The Stacks Project, Tags 010I and 06XP, for extension classes and their projective-resolution interpretation.
- R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, `arXiv:math/0412277`, for prior spectral-realization context. No global priority claim is made here.

New calculations are AI-assisted work under the owner's programme. Existing mathematical files, formalizations, licensing, and source attribution remain untouched.

S3 PR 10: Global retraction

Source attribution. Web contribution, pull request #10, [Zeta research repository](#).

Exact revision. d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7

Source file. workbenches/tau-global-retraction/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859

S3.1 A global retraction of the original theta complex over the tau base

Owner-directed continuation of Zeta PR #9, 12 September 2026. New derivations are submitted as research proofs for review, not as Lean-certified analytic theorems.

S3.1.1 0. Fixed source, result, and provenance

The base is PR #9 at 8f99b94d306c3b1aaa817d52c57fa6fd298d511c; its chain-descent note has Git blob 86db838e79d10a8fe7d4376303706c4ea64570a1. Its source and the owner's derived SplitZero attachment remain the inputs. This continuation constructs an inverse on the entire original theta image, not only on a chosen finite spectral packet.

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The first vertical target is still infinite. The geometric base, the original additive laws, and the arithmetic quotient are unchanged. The auxiliary cutoffs below are parameters of an explicit continuous representative map; they do not change an Euler factor, the theta relation space, or the completed zeta function.

Result. For every pair of the cutoffs specified in section 3 there is an explicitly constructed continuous complex-linear map

$$\Lambda_{\alpha,\beta} : \mathcal{B} \longrightarrow V, \quad \boxed{\Lambda_{\alpha,\beta}\Theta = \operatorname{id}_V}. \quad (2)$$

Consequently the actual image ΘV is closed and complemented in the stated strong-Schwartz topology. The original quotient $Q = \mathcal{B}/\Theta V$ is Hausdorff. The note constructs its global section, the full supported cochain homotopy, the exact change of the finite sections in #9, its continuous dual sequence, and the completed tensor powers. The proof does not assume RH, simplicity of any zero, or vanishing of the separate balanced-Mellin kernel.

Spectral realizations on nuclear Frechet spaces are prior work, notably R. Meyer, arXiv:math/0412277. That paper's abstract was consulted for scope, but no theorem from an unread body of that paper is imported. The proof below is given for the workbench's precise spaces. No global novelty claim is made.

S3.1.2 1. Spaces and all transforms

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) := \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \quad (3)$$

These seminorms give $\mathcal{B}$ its Frechet topology; under $x = e^y$ they are the weighted uniform derivative seminorms on smooth functions of y . Completeness follows by uniform convergence on compact sets of every derivative, followed by the weighted bounds. V has its closed-subspace Schwartz topology.

The original maps are

$$\begin{aligned} \Theta\phi(x) &= \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{F}\phi(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \mathcal{F}^{-1}u(x) = \int_{\mathbb{R}} u(\xi) e^{2\pi i x \xi} d\xi, \\ JF(x) &= x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (4)$$

Poisson summation, with both source moments zero, gives

$$\Theta\mathcal{F} = J\Theta, \quad J^2 = 1, \quad DJ = J(1 - D). \quad (5)$$

The theta sum and each derivative converge absolutely on closed subsets of $x > 0$. At infinity, Schwartz estimates give all $p_{b,k}$ bounds. At zero use (5) and the same estimates on $\hat{\phi}$. This proves continuity $\Theta : V \rightarrow \mathcal{B}$. Also J is continuous on $\mathcal{B}$: $p_{b,k}(JF)$ is bounded by the binomial sum of $p_{1-b,j}(F)$, $0 \leq j \leq k$.

The original complex is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, over $A = \mathbb{C}[t]$ with t acting as D . The two-leg tau chart retains $C_{+-} = [V^2 \rightarrow \mathcal{B}]$ with differential $\Theta(\phi - \hat{\psi})$ and source action $(D, 1 - D)$.

S3.1.3 2. Arithmetic inversion on the two exterior regions

Let μ be the ordinary Mobius function, characterized by

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For $F \in \mathcal{B}$ define, for $x \neq 0$,

$$\boxed{\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|)}. \quad (6)$$

The factor $1/2$ is the inverse of the full nonzero-integer theta sum in (4). The map is complex-linear into smooth even functions on $\mathbb{R} \setminus \{0\}$. It is not asserted to be smooth at zero for arbitrary F .

Here are the required estimates. For $j \geq 0$, let $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, with $P_0 = 1$. Then $x^j \partial_x^j = P_j(D)$. If $C_{N,j}(F)$ is the sum of the $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (7)$$

Indeed each summand is bounded by $C_{N,j}(F) n^{-N} x^{-N-j}/2$. This proves convergence of every derivative uniformly for $|x| \geq \delta > 0$, and gives rapid decay there. It also proves continuity of the cut-off exterior maps used below.

For every $\phi \in V$,

$$\boxed{\mathcal{U}\Theta\phi(x) = \phi(x), \quad \mathcal{U}J\Theta\phi(x) = \widehat{\phi}(x) \quad (x \neq 0).} \quad (8)$$

For the first equality the double series is absolutely convergent for each $x > 0$: its absolute value is bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which is finite by Schwartz decay. Rearrange by $k = mn$ and use the displayed divisor identity. Evenness handles negative x . The second equality follows from (5) and the first equality applied to $\widehat{\phi} \in V$.

In the split lift, every $\mu(n)$ is the supported integer $\mu(n)^\bullet$, including $\mu(n) = 0$. A convergent sum within a specified active fibre begins at that fibre's zero and is taken in that fibre's vector-space topology. Formula (8) thus lifts to a support-preserving equality. A computed cancellation is not external absence.

S3.1.4 3. The reconstruction operator, with the strict contraction proved

Choose and retain real, even $\alpha, \beta \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq \alpha, \beta \leq 1$ and equal to 1 on neighborhoods of zero. Their supports and all values are part of the auxiliary data. Work on $\mathcal{H} = L^2(\mathbb{R}, dx)$ with the Fourier convention in (4).

Define bounded operators

$$A_\alpha = M_\alpha, \quad B_\beta = \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad T_{\alpha,\beta} = A_\alpha B_\beta. \quad (9)$$

Lemma. $T_{\alpha,\beta}$ is compact and $c_{\alpha,\beta} := \|T_{\alpha,\beta}\| < 1$.

Proof. Its kernel is $\alpha(x)\check{\beta}(x-y)$, whose squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$. Hence it is Hilbert-Schmidt. Both factors are contractions. If the norm were 1, compactness would supply a unit vector f attaining it. Equality in

$$\|A_\alpha B_\beta f\| \leq \|B_\beta f\| \leq \|f\|$$

forces $\widehat{f}$ to be supported where $\beta = 1$; there $B_\beta f = f$. The first equality forces f to be supported where $\alpha = 1$. Both supports are compact. The inverse Fourier transform of compactly supported $\widehat{f} \in L^2$ is entire, by differentiation under its finite-interval integral. Its continuous restriction agrees with the compactly supported f , so it vanishes on an open real interval and hence identically. This contradicts $\|f\| = 1$. Thus $c_{\alpha,\beta} < 1$.

Consequently the exact Neumann inverse exists:

$$\boxed{(1 - T_{\alpha,\beta})^{-1} = \sum_{k \geq 0} T_{\alpha,\beta}^k, \quad \|(1 - T_{\alpha,\beta})^{-1}\| \leq \frac{1}{1 - c_{\alpha,\beta}}.} \quad (10)$$

No uniform bound as the cutoffs vary is claimed. In particular the gap $1 - c_{\alpha,\beta}$ is retained, not replaced by a positive universal constant.

The operator $T_{\alpha,\beta}$ maps L^2 continuously to the Schwartz space. For each derivative, compact Fourier support and Cauchy-Schwarz bound $\partial^j B_\beta f$ uniformly by a constant times $\|f\|_2$; multiplication by the smooth compactly supported α gives every Schwartz seminorm.

Define the two exterior data and their combination:

$$a_F = (1 - \alpha)\mathcal{U}F, \quad b_F = (1 - \beta)\mathcal{U}JF,$$

$$Y_{\alpha,\beta}F = a_F + \alpha\mathcal{F}^{-1}b_F. \quad (11)$$

The exterior terms are extended by zero near the origin, where their cutoff is identically zero. Estimates (7) prove that both are even Schwartz functions and depend continuously on F . Therefore $Y : \mathcal{B} \rightarrow \mathcal{S}(\mathbb{R})^{\text{even}}$ is continuous.

Put

$$\Phi_{\alpha,\beta}F = (1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta}F. \quad (12)$$

This initially defines an L^2 vector, but the identity $\Phi F = YF + T\Phi F$ proves it is Schwartz. If q is any Schwartz seminorm and $q(Tu) \leq C_q\|u\|_2$, then

$$q(\Phi F) \leq q(YF) + \frac{C_q}{1 - c_{\alpha,\beta}}\|YF\|_2. \quad (13)$$

This is an explicit continuity estimate. Both sides are even because the cutoff operators preserve parity.

For $F = \Theta\phi$, equations (8) give

$$Y\Theta\phi = (1 - A_\alpha)\phi + A_\alpha(1 - B_\beta)\phi = (1 - T_{\alpha,\beta})\phi,$$

hence

$$\boxed{\Phi_{\alpha,\beta}\Theta\phi = \phi.} \quad (14)$$

S3.1.5 4. Keep both original moment conditions

For arbitrary F , (12) need not satisfy the two source moments. The correction is the following specified projection, not an alteration of V .

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int \phi)$ are $(1, 1)$ and $(0, 1)$. Define

$$\mathbf{m}(\phi) = (\phi(0), \int_{\mathbb{R}} \phi), \quad j_0(a, b) = a\gamma_0 + (b - a)\gamma_2,$$

$$P_V\phi = \phi - j_0\mathbf{m}(\phi) = \phi - \phi(0)\gamma_0 - \left(\int \phi - \phi(0)\right)\gamma_2. \quad (15)$$

Then $\mathbf{m}j_0 = 1$, $P_V^2 = P_V$, $\text{im } P_V = V$, and $P_V|_V = 1$. All maps are continuous.

The promised map is

$$\boxed{\Lambda_{\alpha,\beta} = P_V(1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta} : \mathcal{B} \rightarrow V.} \quad (16)$$

Equation (14) and $P_V|_V = 1$ prove (2). The discarded moment vector has the recorded map $m\Phi : \mathcal{B} \rightarrow \mathbb{C}^2$ and the explicit correction $j_0 m\Phi$. It vanishes on the actual theta image. No endpoint constant was absorbed into the zeta function.

S3.1.6 5. The full quotient is closed and has a continuous section

Write $\Lambda = \Lambda_{\alpha,\beta}$, and define

$$P_\Theta = \Theta\Lambda, \quad K = 1_{\mathcal{B}} - \Theta\Lambda. \quad (17)$$

Direct substitution gives

$$P_\Theta^2 = P_\Theta, \quad K^2 = K, \quad \ker K = \Theta V, \quad \text{im } K = \ker \Lambda. \quad (18)$$

Since K is continuous, its kernel is closed. Thus $Q = \mathcal{B}/\Theta V$ with its actual quotient topology is Frechet and Hausdorff. If $q : \mathcal{B} \rightarrow Q$ is the original quotient map, the continuous section is

$$\boxed{s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0.} \quad (19)$$

The explicit topological direct-sum isomorphism is

$$\boxed{V \oplus Q \xrightarrow{\sim} \mathcal{B}, \quad (\phi, u) \mapsto \Theta\phi + su, \quad F \mapsto (\Lambda F, qF).} \quad (20)$$

For a convergent sequence $\Theta\phi_n \rightarrow F$ in $\mathcal{B}$, one can check the closedness without any abstract theorem: $\phi_n = \Lambda\Theta\phi_n \rightarrow \Lambda F$ in V , so $F = \Theta\Lambda F$ by continuity of theta.

This computes the earlier Hausdorff arrow:

$$Q^{\text{alg}} = \mathcal{B}/\Theta V \longrightarrow Q^{\text{H}} = \mathcal{B}/\overline{\Theta V}, \quad [F] \mapsto [F], \quad (21)$$

as an isomorphism with zero amplitude kernel. It does not say the map from Q into its product of spectral jets is injective. The latter map and the balanced analytic map still have their own named kernels:

$$Q \rightarrow \prod_{\rho} \mathcal{O}_{\rho}/(g), \quad \mathcal{O} \otimes_{\mathbb{C}[t]} Q \xrightarrow{\beta} \mathcal{O}/(g).$$

No claim that $\mathcal{R}_{\infty}$ or $\ker \beta$ vanishes is inferred from (21).

Smaller admitted relations For any specified subspace $W \subset V$, retain

$$Q_W = \mathcal{B}/\Theta W \longrightarrow Q, \quad \ker = \Theta V/\Theta W \cong V/W. \quad (22)$$

The isomorphism sends $[\phi] \mapsto [\Theta\phi]$. It is algebraic for arbitrary W ; when W is closed, (20) makes it a topological isomorphism onto the kernel. The pullback of the global section has the representative relation kernel in (22); the full-source result is not silently asserted with W in place of V .

S3.1.7 6. Perform the original internal quotient and synchronization

On the single-leg map, or a fixed-index diagram of that same map, the split lifts are

$$\tilde{\Theta}(\lambda, \phi) = (\lambda, \Theta\phi), \quad \tilde{\Lambda}(\lambda, F) = (\lambda, \Lambda F), \quad \tilde{K}(\lambda, F) = (\lambda, KF). \quad (23)$$

They preserve external bottom and commute with scalar reflection. In particular

$$\tilde{K}\tilde{\Theta}(\lambda, \phi) = (\lambda, 0_{\mathcal{B}}) = z_{V, \mathcal{B}}(\lambda, \phi).$$

Here $z_{V, \mathcal{B}}$ is the label-preserving zero map from the source fibre to the target fibre. The mixed chart, whose minus restriction is $-\Theta\mathcal{F}$, is treated separately by the explicit maps (25)-(28), without changing that restriction. The actual chart has zero bottom fibre; a general nonzero bottom fibre is acted on by its stated linear map, rather than identified with the single global zero.

The supported image projection is idempotent, and the quotient map is exactly the coequalizer of the theta relation inclusion and its supported-zero companion. The implementation's universal property applies against all split-linear targets. A theta boundary is sent to $e_\lambda = (\lambda, 0)$; it is not sent to external absence.

For the single-leg complex there is an actual continuous contraction

$$C_+ \xrightleftharpoons[s]{q} Q[-1], \quad H^1 = \Lambda,$$

$$\boxed{qs = 1, \quad 1_{C_+} - sq = dH + Hd.} \quad (24)$$

In degree zero this is $\Lambda\Theta = 1_V$; in degree one it is $\Theta\Lambda = 1 - K$. In split notation the subtraction means addition of the supported scalar $(-1)^\bullet$ times the second map. It yields the correct labelled zero wherever the amplitude vanishes.

For the two-leg chart set $H_0 = V[0] \oplus Q[-1]$ with zero differential, and define

$$p_L^0(\phi, \psi) = \psi, \quad p_L^1 = q, \quad i^0(\psi) = (\hat{\psi}, \psi), \quad i^1 = s,$$

$$H_L^1(F) = (\Lambda F, 0). \quad (25)$$

Then $p_L i = 1$, and $1 - ip_L = dH_L + H_L d$. The degree-zero identity is $(\phi, \psi) - (\hat{\psi}, \psi) = (\phi - \hat{\psi}, 0)$. The other leg gives

$$p_R^0(\phi, \psi) = \hat{\phi}, \quad p_R^1 = q, \quad H_R^1(F) = (0, -\widehat{\Lambda F}). \quad (26)$$

It obeys $1 - ip_R = dH_R + H_R d$. Their homotopies differ by

$$\boxed{H_L^1 F - H_R^1 F = (\Lambda F, \widehat{\Lambda F}),} \quad (27)$$

the actual joint degree-zero cycle, which is retained.

On the complete original support diagram, use literal multiplication by $E_{\text{sync}} = (1, e)$ in degree zero and the source's active-label-to-joint map in degree one:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea), \quad \mathfrak{r}^1(\lambda, F) = (\{+, -\}, F) \quad (\lambda \neq \perp).$$

The absent input remains absent. The original differential obeys $d\mathfrak{r} = \mathfrak{r}d$ and $\mathfrak{r}^2 = \mathfrak{r}$. Consequently (25) lifts to the fully typed equation

$$\mathfrak{r} - \tilde{i}\tilde{p}_L\mathfrak{r} = \tilde{d}\tilde{H}_L + \tilde{H}_L\tilde{d}, \quad (28)$$

where $\tilde{H}_L(\lambda, F) = (\{+, -\}, (\Lambda F, 0))$ for each active label. The right-leg equation is analogous with (26). The original mask is recorded with the injection $x \mapsto (\text{supp } x, \mathfrak{r}x)$ already proved for this chart. Neither synchronization nor the contraction identifies the three supported origins with τ .

S3.1.8 7. Global scaling defects, and their agreement with the finite extensions

Let $U_a F(x) = F(x/a)$, $a > 0$, and denote its induced continuous action on Q by $\overline{U}_a$. The section (19) gives a global continuous boundary map

$$k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\overline{U}_a = \Theta k_a. \quad (29)$$

The proof applies $K + \Theta\Lambda = 1$ to $U_a s$, then uses $qU_a s = \overline{U}_a$. Therefore

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (30)$$

The same calculation works for D and compactly supported multiplicative convolution, with $k_D = \Lambda D s$. In the explicit coordinates (20),

$$U_a|_{\mathcal{B}} = \begin{pmatrix} U_a|_V & k_a \\ 0 & \overline{U}_a \end{pmatrix}, \quad D|_{\mathcal{B}} = \begin{pmatrix} D|_V & k_D \\ 0 & D_Q \end{pmatrix}. \quad (31)$$

Thus the topological splitting is not substituted for an equivariant splitting. Equation (29) is the precise comparison between them.

For two cutoff choices, let s_0, s_1 be their sections and set $b_{10} = \Lambda_0 s_1 : Q \rightarrow V$. Then

$$s_1 - s_0 = \Theta b_{10}, \quad k_a^{(1)} = k_a^{(0)} + U_a b_{10} - b_{10} \overline{U}_a. \quad (32)$$

All dependence on the chosen representative map is carried by this displayed coboundary. No canonical cutoff or average over choices is declared.

Restrict back to PR #9 without changing its extension class Use its existing maps $s_Z : E_Z \rightarrow \mathcal{B}$, $\sigma_Z = q s_Z : E_Z \rightarrow Q$, and $J_Z : \mathcal{B} \rightarrow E_Z$. Put $b_Z = \Lambda s_Z$. The new continuous representative of the same finite arithmetic class is

$$s_Z^{\text{new}} = s \sigma_Z = K s_Z = s_Z - \Theta b_Z. \quad (33)$$

If $P(D)s_Z - s_Z M_P = \Theta k_{P,Z}$, direct substitution gives

$$k_{P,Z}^{\text{new}} = k_{P,Z} - P(D)b_Z + b_Z M_P. \quad (34)$$

For $P = h_Z$, this changes the source boundary by $-h_Z(D)b_Z$. It is zero in $V/h_Z(D)V$, so the extension class remains exactly

$$\varepsilon_Z = j_{h_Z}(h_Z/g), \quad g = 2\xi. \quad (35)$$

The polynomial residue form still becomes the arithmetic one by M_{ε_Z} ; its Jacobian contraction remains $M_{\varepsilon_Z} M_{j_h g'}$. No unit, derivative, sign, or factor of 2 is changed.

The two finite cochain projectors differ by the explicit homotopy $-b_Z J_Z$ in degree minus one. Their finite-rank trace difference is zero because $J_Z \Theta = 0$. Hence their cochain Lefschetz trace is the unchanged

$$-\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (36)$$

S3.1.9 8. Continuous duality and completed tensor powers of the full object

Use continuous duals, denoted E'_b with the strong topology. Transpose the actual topological splitting (20). It gives

$$\boxed{0 \rightarrow Q'_b \xrightarrow{q'} \mathcal{B}'_b \xrightarrow{\Theta'} V'_b \rightarrow 0, \quad \Theta' \Lambda' = 1.} \quad (37)$$

The continuous cochain-dual complex is $[\mathcal{B}'_b \xrightarrow{\Theta'} V'_b]$ in degrees $-1, 0$. Its cohomology is Q'_b in degree -1 and zero in degree 0 . The transpose contraction proves this directly; no general exactness theorem for arbitrary locally convex quotients is assumed.

The actual finite residue map into that continuous dual is

$$\overline{E_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto (u \mapsto R_Z(f, j_Z u)), \quad (38)$$

where $j_Z q = J_Z$. Continuity follows from the finite Mellin jets and finite dimensionality of E_Z . It is injective by their surjectivity and perfect residue duality. Multiplication by g' then gives the arithmetic finite trace pairing, with its previously calculated radical retained.

Let $\widehat{\otimes}_\pi$ denote completed projective tensor product. The comparison from the previous algebraic tensor is its canonical completion map. All chain operators in (24) are continuous. Thus their finite tensor products and homotopies extend to the completed products. With $P = sq$ on C_+ , the homotopy

$$H_n = \sum_{j=1}^n P^{\widehat{\otimes}(j-1)} \widehat{\otimes} H \widehat{\otimes} 1^{\widehat{\otimes}(n-j)} \quad (39)$$

includes $(-1)^{\sum_{i < j} |x_i|}$ on homogeneous inputs. It satisfies

$$dH_n + H_n d = 1 - P^{\widehat{\otimes} n}.$$

Consequently, for the completed external coefficient complex on the same tau-base chart product,

$$\boxed{H^k(C_+^{\widehat{\otimes} n}) = \begin{cases} Q^{\widehat{\otimes} n}, & k = n, \\ 0, & k \neq n. \end{cases}} \quad (40)$$

This follows from a split continuous contraction, not from a blanket claim that completed tensor product preserves every exact sequence.

For the joint chart use (25). Its full cohomology in degree k is the direct sum, indexed by subsets $J \subseteq \{1, \dots, n\}$ with $|J| = k$, of the ordered completed tensors having Q in the slots of J and V in the other slots. In those V slots the scaling action is $aU_{1/a}$, exactly as in the original minus-coordinate representative. The degree-zero contribution is not discarded. Split reconstruction retains each product mask and each internal zero.

S3.1.10 9. The pairing calculation is now a specified global boundary problem

The section $s : Q \rightarrow \mathcal{B} \subset L^2(dx)$ supplies a positive definite auxiliary form

$$\mathcal{H}_s(u, v) = \langle su, sv \rangle_{L^2(dx)}.$$

This is not assigned to the arithmetic Weil form. The explicit weight-covariance defect is calculated using (29) and $\langle U_a f, U_a g \rangle = a \langle f, g \rangle$:

$$\boxed{\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a \mathcal{H}_s(u, v) = -\langle U_a s u, \Theta k_a v \rangle - \langle \Theta k_a u, U_a s v \rangle + \langle \Theta k_a u, \Theta k_a v \rangle.} \quad (41)$$

The original source boundary and both of its cross-pairings remain. Being a boundary in Q does not assert these pairings are zero in the representative Hilbert space. In a split lift each computed zero is its supported value e .

For a reduced eigenline of the actual dilation representation with exponent ρ , the left side at $u = v$ is $(a^{2\rho} - a) \mathcal{H}_s(u, u)$. Formula (41) locates that defect in the full global boundary cocycle, with the original operator retained.

A concrete comparison explaining which topology must be kept In fact ΘV is dense in $L^2(\mathbb{R}_{>0}, dx)$, although it is closed in $\mathcal{B}$. Here is a proof retaining the map between them. Set $b_* = \Theta \phi_*$ and

$$W : L^2(dx) \xrightarrow{\sim} L^2(dy), \quad W F(y) = e^{y/2} F(e^y).$$

The Fourier transform in y , with kernel $e^{-2\pi i \nu y}$, gives

$$\mathcal{F}_y W b_*(\nu) = g(1/2 - 2\pi i \nu).$$

This is nonzero almost everywhere, because g is a nonzero entire function. The span of all translates of $W b_*$ is dense in $L^2(dy)$: a vector orthogonal to all translates has Fourier transform whose product with $\overline{\mathcal{F}_y W b_*}$ is an L^1 function with zero Fourier transform, hence vanishes almost everywhere. The nonvanishing just stated then makes that vector zero. Since $W U_a b_* = a^{1/2} (W b_*)(y - \log a)$ and $U_a b_* \in \Theta V$, density follows.

The exact quotient comparison is therefore

$$\boxed{Q = \mathcal{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2} = 0.} \quad (42)$$

Its kernel is all of Q . Its split lift sends every supported input to the supported zero in $G(0) = \mathbb{B}$, while keeping external absence separate. The constructive replacement is the Frechet quotient and the explicit global section (19), not a claim that the ordinary Hilbert quotient secretly retains those classes.

S3.1.11 10. Scope of the progress toward the Deligne comparison

The selected supplied Weil II passages 3.3.4-3.3.6 put two bounds on the same compact-to-ordinary image; 6.2 retains the dualizing operation and its shifts. This continuation does not apply those bounds over a new base without their geometric proof. It makes the existing characteristic-zero arithmetic complex globally split in its precise topological category, gives its continuous dual and completed products, and computes the operator defects on those same objects.

The formerly unresolved algebraic-to-Hausdorff kernel is now zero by (16)-(21). The finite extension class and its residue/Jacobian trace are carried by (33)-(36). The full scaling boundary is (29), its dependence on the auxiliary cutoffs is (32), and its metric defect is (41). These are concrete maps available for a global comparison.

Still retained: the common kernel of all finite spectral observations, the meromorphic summand of the balanced analytic realization, and the requirement to identify the correct arithmetic polarization/weight control on the global geometric image. The density map (42) explains why discarding the stronger topology would remove the entire arithmetic cohomology, not why the supported programme should be abandoned.

The reconstruction estimate contains $1 - c_{\alpha,\beta}$; no asymptotic uniform estimate for it is proved. No global positivity, RH, GRH, or infinite trace-class theorem is claimed. No theorem from the earlier finite-range work is upgraded to a Lean certificate. The new global retraction proof itself uses only the original theta/Poisson identities, absolute Mobius inversion, the displayed Fourier compactness argument, and the two retained moment maps.

S3.1.12 11. Verification and references

The written analytic proofs are the evidence for the new analytic statements. The accompanying unittest script tests exact finite identities: divisor cancellation with supported zeros, cutoff-resolvent identities in a finite unitary model, moment corrections, contractions, synchronization masks, operator cocycles, changes of sections, preservation of extension classes, finite trace changes, and completed-tensor algebra before completion. It does not numerically certify a Fourier operator norm, an analytic convergence statement, or a Riemann zero.

Source interfaces read:

- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1-7; the source permits nonzero bottom fibres in its general interface.
- PR #9, workbenches/tau-chain-descent/RESEARCH_NOTE.md, at the commit above; local bytes match its Git blob.
- PR #8's finite comparison as retained and invoked in #9; the sections using s_Z and J_Z are explicitly dependencies of (33)-(38), not inputs to the proof of (2).
- Supplied Weil II TeX: the actual 3.3.1-3.3.6 excerpt, including its finite-type base and duality assumptions. It was read as TeX, not obtained by OCR. No full source-proof audit is claimed.
- Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277: prior-context abstract only in this continuation. A source-archive retrieval failed; no unread detailed theorem is cited as a premise.

The contribution is AI-assisted under the owner's mathematical direction. Original mathematical files, formal sources, attribution and licenses are unchanged. Only authored exposition, test code and validation records are proposed for public addition; no private transcript or source-literature corpus is published.

S4 PR 11: Homotopy control

Source attribution. Web contribution, pull request #11, [Zeta research repository](#).

Exact revision. d37eade4f9e6a82c1a88aeff710a3b851a34eb49

Source file. workbenches/tau-homotopy-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9924b358c0365a8cbdcddb3a7496f610c65d64617cd1860679b42e371d2fe88d

S4.1 Homotopy choices and source-preserving arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #10, d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7. This integration note records the new derivations; the conversation delivery also contains a longer equation-by-equation exposition. No original scalar definitions or formal modules are changed.

S4.1.1 1. Reviewed theorem and analytic specialization

The supplied other-session proof classifies all homotopies for

$$C = [V \oplus V \xrightarrow{d} B], \quad d(v, w) = \Theta(v - \mathcal{F}w), \quad Q = B/\Theta V,$$

where Θ is injective and $\mathcal{F}$ is an R -linear automorphism. For a fixed $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, every solution of BOTH equations

$$dH = \Theta \kappa, \quad Hd = 0$$

has the unique form

$$H_\alpha(b) = (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b), \quad \alpha : Q \rightarrow_R V.$$

Proof: writing $H=(u,v)$, injectivity gives $u-\mathcal{F}v=\kappa b$. Since $d(x,0)=\Theta x$, $Hd=0$ gives $H \Theta = 0$. Hence $\mathcal{F}v$ factors uniquely through q , and supplies α . Conversely substitution proves both equations. The reference solution is $H_0=(\kappa, 0)$; the family is an affine space under $\text{Hom}_R(Q, V)$, relative to that reference.

Let $\gamma(v)=(v, \mathcal{F}^{-1} v)$. Then

$$H_\alpha - H_\beta = \gamma(\alpha - \beta)q.$$

The difference is a map $H_1(C) \rightarrow H_0(C)$. There are no degree-minus-two endomorphisms of this two-term coefficient complex. Thus it is a genuine degree-minus-one Hom-complex class, not an erased difference.

The supplied theorem is mathematically correct. Its Lean draft explicitly remains uncompiled, and its earlier failed chart build is not relabelled successful. No Lean or Lake executable is installed in the review environment.

For the actual analytic spaces instantiate $R=C$. With the quotient topology on Q , the continuous version follows: $\alpha q = \mathcal{F} \text{pr}_2 H$ is continuous, so the quotient property makes α continuous. Conversely the displayed formula sends continuous α to continuous H .

For the polynomial action $A_0 = \mathbb{C}[t]$, Fourier has the exact type

$$\mathcal{F} : V^j \xrightarrow{\sim} V, \quad t \cdot_{V^j} v = (1 - D)v, \quad j(t) = 1 - t,$$

because $F D = (1 - D)F$. Thus d is A_0 -linear from $V + V^\wedge j$ to B . A complex-linear homotopy is not automatically A_0 -linear. The reflected module above is the actual comparison, not a change to the Fourier definition.

S4.1.2 2. Original split support on every homotopy

Retain the base b_τ and the square $G(Z) \rightarrow G(C)$ over $Z > C$. The quotient $G(Z) \rightarrow Z$ remains infinite. Both lifted coefficient maps preserve τ and the supported scalar e separately.

Use the original support lattice $L = \{\text{bottom}, +, -, J\}$, $J = \{+, -\}$. Degree-zero fibres are $0, V, V, V + V$ with the coordinate inclusions. Degree-one active fibres are all B , with identity transports. Their differentials are Θ , $-\Theta$, F and d . Retain zero modules in the other degrees; their reconstructed zero map is $z(\lambda, x) = (\lambda, 0)$.

Literal multiplication by $E_{\text{sync}} = (1, e)$ induces

$$r^0(a, b) = (a + eb, b + ea), \quad r_L(\perp) = \perp, \quad r_L(\lambda) = J(\lambda \neq \perp).$$

Every homotopy lifts by

$$\tilde{H}_\alpha(\lambda, b) = (J, (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b)), \quad \lambda \neq \perp; \quad \tilde{H}_\alpha(\tau) = \tau.$$

This is split-linear: two active inputs add at their joined label, both outputs add at J , and their amplitudes obey the two displayed linear formulas. A supported scalar zero gives $(J, (0, 0))$; external τ gives the global zero. Therefore

$$\tilde{d} \tilde{H}_\alpha + \tilde{H}_\alpha \tilde{d} = r \tilde{\Delta}.$$

The difference is $(J, \gamma(\alpha - \beta)qb)$, retaining its degree-zero class. For this chart, the pair $(\text{support}(x), r(x))$ is injective. No such injectivity is assumed for arbitrary diagrams with noninjective transports. For a smaller relation space W , keep $Q_W \rightarrow Q$ with kernel $\Theta V / \Theta W$; q in the formulas means the displayed quotient by the full admitted V .

S4.1.3 3. Coherence for all dilations

Let $S_a = U_a$ on V , $T_a = U_a$ on B , $U_a F(x) = F(x/a)$, and T_{bar}_a the quotient action. The actual cochain action is

$$\mathsf{T}_a^0 = (S_a, \mathcal{F}^{-1} S_a \mathcal{F}) = (U_a, a U_{1/a}), \quad \mathsf{T}_a^1 = T_a.$$

It satisfies $T_a^0 \gamma = \gamma S_a$ and $q T_a = T_{\text{bar}}_a q$. With the PR #10 section s , take $P = (0, sq)$, and write

$$[\mathsf{T}_a, P]^1 = \Theta \kappa_a, \quad \kappa_a = k_a q, \quad T_a s - s \overline{T}_a = \Theta k_a.$$

The commutator product identity gives $\kappa_{ab} = S_a \kappa_b + \kappa_a T_b$. The family of homotopies is composition-coherent,

$$H_{ab} = \mathsf{T}_a H_b + H_a \mathsf{T}_b,$$

exactly when

$$\alpha_{ab} = S_a \alpha_b + \alpha_a \bar{T}_b.$$

Proof: the reference family $(\kappa_a, 0)$ is coherent. Subtract it and use injectivity of γ and surjectivity of q . No action or boundary is omitted.

For a smoothly parameterized continuous-linear family with $L = (d/dt)|_0 \alpha(e^t)$, the exact formula is

$$\alpha_{e^t} = \int_0^t S_{e^{t-r}} L \bar{T}_{e^r} dr.$$

This is a vectorwise integral in the complete Frechet source, with the original oriented integral for negative t . Differentiating the composition equation gives the formula; splitting the integral proves its converse. In particular $\alpha_a = S_a b - b \bar{T}_a$ has infinitesimal parameter $D_V b - b D_Q$.

S4.1.4 4. A homotopy parameter leaves the control form fixed

Fix an actual finite arithmetic packet with full multiplicities,

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

Use the existing $R: E_Z \rightarrow B$ and $J_Z: B \rightarrow E_Z$ with

$$qR = \sigma_Z, \quad J_Z R = 1, \quad J_Z \Theta = 0, \quad J_Z D = A J_Z, \quad DR - RA = \Theta k.$$

The cochain projector $P_R = (0, R|_Z)$ has commutator $(0, \Theta k|_Z)$, so the reviewed theorem applies with $\kappa = k|_Z$. The quotient $\text{jet } J_{\text{bar}}_Z$ satisfies $J_{\text{bar}}_Z \sigma_Z = 1$, hence any $\alpha_E: E_Z \rightarrow V$ extends to $\alpha_E J_{\text{bar}}_Z$ on Q . Restriction to R gives

$$H_{\alpha_E} R = (k + \alpha_E, \mathcal{F}^{-1} \alpha_E), \quad dH_{\alpha_E} R = \Theta k.$$

Take $H = L^2(R_+, dx)$ with the original inner product, conjugate-linear in the first variable. Define

$$G_R = R^* R, \quad B_R = \Theta k, \quad W_R = -(R^* B_R + B_R^* R).$$

The composite of actual maps

$$\text{Hom}_{\mathbb{C}}(E_Z, V) \longrightarrow \text{Hom}(E_Z, V^2) \xrightarrow{d \circ (-)} \text{Hom}(E_Z, H) \xrightarrow{B \mapsto -(R^* B + B^* R)} \text{Herm}(E_Z)$$

is constant at W_R . Its last map is real-linear. The retained H^0 ambiguity does not change this boundary contraction.

Integration by parts on the actual source functions gives

$$A^* G_R + G_R A - G_R = W_R.$$

The boundary term $[x \text{ conjugate}(F)H]_0^\infty$ vanishes by the stated decay. No claim that a θ boundary is orthogonal to a representative is used.

S4.1.5 5. The actual representative-change family

Every representative of the same inclusion σ_Z is uniquely

$$R_b = R + \Theta b, \quad b : E_Z \rightarrow_{\mathbb{C}} V.$$

Indeed $q(R'-R)=0$ gives values in ΘV , and the PR #10 retraction gives the unique $b=\Lambda(R'-R)$. Finite source dimension gives continuity. Global continuous sections have the identical formula with $b:Q \rightarrow V$ continuous.

The complete changes are

$$k_b = k + D_V b - bA,$$

$$\delta G_b = R^* \Theta b + (\Theta b)^* R + (\Theta b)^* \Theta b,$$

$$\boxed{W_{R_b} - W_R = A^* \delta G_b + \delta G_b A - \delta G_b.}$$

Their extension-coordinate map is $U_b(v,u)=(v+bu,u)$, and

$$U_b^{-1} \begin{pmatrix} D_V & k \\ 0 & A \end{pmatrix} U_b = \begin{pmatrix} D_V & k + D_V b - bA \\ 0 & A \end{pmatrix}.$$

Lift that automorphism on the common active fibre $(V+E_Z)^{\tau}$. For the original masks, first keep the pair $(\text{support}(x), r(x))$; apply the shear on the joint fibre, and recover the original amplitudes by U_b^{-1} and the original mask by its recorded component. No inverse to the unrecorded support collapse is asserted.

The projectors obey $P_{(R_b)} - P_R = \text{partial } L_b$ with $L_b^{-1}(F)=(b|_Z F, 0)$. Their finite cochain traces agree because

$$\text{Tr}(\Theta b J_Z H_f) = \text{Tr}(J_Z H_f \Theta b) = 0.$$

The cochain supertrace sign remains -1 in degree one. The arithmetic extension class changes by $h_Z(D)b(1)$, so its class in $V/h_Z(D)V$ remains $\epsilon_Z = j_h(h_Z/(2\pi))$. The residue/Jacobian trace form is consequently unchanged.

S4.1.6 6. Finite actual source ansatz and one master Gram matrix

Retain $\phi = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$, $\phi_j = D^j \phi$, and

$$\Phi_m : \mathbb{C}^{m+1} \rightarrow V, \quad (c_j) \mapsto \sum_{j=0}^m c_j \phi_j.$$

Their actual Mellin identities are $M \Theta \phi_j(s) = 2\pi i(s)^j$. They remain in the original two-moment source and are linearly independent. The source section $R_0 = s_Z$ has the exact rank-one defect $DR_0 - R_0 A = \Theta \phi^* \ell_Z$, with the unchanged arithmetic remainder functional ℓ_Z .

Define maps, including the target enlargement,

$$I_m e_j = e_j, \quad S_m e_j = e_{j+1}, \quad I_m, S_m : \mathbb{C}^{m+1} \rightarrow \mathbb{C}^{m+2}, \quad D\Phi_m = \Phi_{m+1} S_m.$$

For B in $\text{Mat}_{((m+1) \times d)}(\mathbb{C})$, $d = \dim E_Z$, put

$$R_B = R_0 + \Theta \Phi_m B, \quad K_B = e_0 \ell_Z + S_m B - I_m B A.$$

Then

$$DR_B - R_B A = \Theta \Phi_{m+1} K_B, \quad q R_B = \sigma_Z, \quad J_Z R_B = 1.$$

Take the actual column map and its Gram matrix

$$\mathcal{T}_m : E_Z \oplus \mathbb{C}^{m+2} \rightarrow L^2(\mathbb{R}_+, dx), \quad (u, c) \mapsto R_0 u + \Theta \Phi_{m+1} c, \quad M_m = \mathcal{T}_m^* \mathcal{T}_m.$$

M_m is positive definite. Applying J_Z to a vanishing combination gives $u=0$; Mellin then gives $c=0$. Its entries are the convergent integrals of the displayed arithmetic functions, with the original measure.

Define

$$X_B = \begin{pmatrix} I_d \\ I_m B \end{pmatrix}, \quad Y_B = \begin{pmatrix} 0 \\ K_B \end{pmatrix}.$$

The complete control calculation is

$$\boxed{G_B = X_B^* M_m X_B, \quad W_B = -(X_B^* M_m Y_B + Y_B^* M_m X_B) = A^* G_B + G_B A - G_B.}$$

This restricts representative choice to an actual finite family of theta boundaries. It is not an arbitrary positive matrix ansatz, and it is not asserted to exhaust all continuous source corrections. All packet coordinates, multiplicities, local units and finite source-degree changes are retained.

S4.1.7 7. Error transport, duality, and the remaining estimate

With actual enclosures $||M\text{-}\hat{M}|| \leq \eta_M$, $||X\text{-}\hat{X}|| \leq \eta_X$, $||Y\text{-}\hat{Y}|| \leq \eta_Y$, put $x = ||\hat{X}||$, $y = ||\hat{Y}||$, $m_0 = ||\hat{M}||$. Product estimates give

$$\eta_G = \eta_M (x + \eta_X)^2 + m_0 (2x\eta_X + \eta_X^2),$$

$$\eta_W = 2[\eta_M (x + \eta_X)(y + \eta_Y) + m_0 (x\eta_Y + y\eta_X + \eta_X \eta_Y)].$$

These include coefficient errors, not just quadrature errors. Exact certificates for $G\text{-}\hat{\eta}_G \geq 0$ and

$$\epsilon \hat{G} \pm \hat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0$$

imply the actual two-sided control. This is an error-transfer theorem; arithmetic enclosures have not been computed here.

The unchanged residue matrix S satisfies $S^* = -S$ and $A^* S + S A = S$. Its exact dual transport is

$$G_B^D = S^* G_B^{-1} S, \quad W_B^D = -S^* G_B^{-1} W_B G_B^{-1} S.$$

Thus an upper certificate gives a lower certificate on the same arithmetic packet. The Weil form stays $S \text{ } J\text{-}g$, with metric comparison $G_B^{-1} S \text{ } J\text{-}g$; no positivity is assigned by choosing G_B .

For tensor degree n , retain $A_n = \sum_j 1 \text{ tensor } A \text{ tensor } 1$ and $G_{(B,n)} = G_B^{-1} (\text{tensor } n)$. The product control is $\sum_j G_B \text{ tensor } W_B \text{ tensor } G_B$. Changing a homotopy by $\gamma \alpha \text{ } q$ changes

the tensor homotopy by its Koszul-signed insertions; its differential is zero, hence it does not change that control form.

The existing orbit construction on this actual tensor image gives

$$G_{B,n,T} = \int_0^T e^{-nt} e^{tA_n^*} G_{B,n} e^{tA_n} dt,$$

$$A_n^* G_{B,n,T} + G_{B,n,T} A_n - n G_{B,n,T} = e^{-nT} e^{TA_n^*} G_{B,n} e^{TA_n} - G_{B,n}.$$

Every occurrence of $G_{B,n}$ now has the explicitly constrained source in section 6.

A nonzero boundary need not have nonzero control. The exact calibration $D_{\text{amb}} = \text{diag}(1/2+i, 1/2+2i)$, $\Theta(v) = (v, 0)$, $A = 1/2+2i$, $R(u) = (u, u)$ has boundary $\Theta(-i)$, Gram 2, and control zero. It is an algebraic calibration, not a zeta packet. The real-linear map $B \mapsto (R^*B + B^*R)$ explains which nonzero boundaries can have zero control.

Conversely every eigenvector $Av = \rho v$ obeys

$$\frac{v^* W_B v}{v^* G_B v} = 2\Re \rho - 1.$$

This ensures that changing representatives has not changed a spectral coordinate. The analytic target is now to bound the actual matrices from section 6, or their tensor-orbit endpoints, with all packet and tensor dependence retained. No uniform purity estimate, RH, GRH, or global spectral-kernel vanishing is proved here.

S4.1.8 8. Validation and attribution

The attachment supplies the homotopy-classification theorem and the uncompiled draft. This note verifies its proof and adds the continuous/twisted interface, supported reconstruction, coherent-family condition, and the connection to the actual control matrices. The original owner-directed SplitZero construction remains the source.

The independent checker has sixteen exact unittest methods, including finite parameter families. They passed normally and under python -O with identical successful JSON records. Intentional negative controls failed in both modes. The previous control checker also passed its sixteen methods. These finite checks cover algebraic models and polynomial-times-exponential integral calibrations, not evaluated zeta packets or analytic/Lean certification.

General references: The Stacks Project, Tags 0A8H and 0115 for the Hom-complex differential and homotopy, and 010I and 06XP for extension classes. The formulas above are proved on the specified programme objects; those references are not imported as a number-field weight theorem. Only authored notes, test code and validation records are published; no private transcript, source-literature corpus, or font files are included.

S5 PR 13: Orthogonal boundary control

Source attribution. Web contribution, pull request #13, [Zeta research repository](#).

Exact revision. e7ba4c29d4503bf20d06cb70c93c3347f199f1b4

Source file. workbenches/tau-orthogonal-boundary-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9d364d9cc2f6840af8eb39555e4b89622acd7714fbcab7ca135e8ec0ce25ab0b

S5.1 Orthogonal theta boundaries and rank-two arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #11, d37eade4f9e6a82c1a88aeff710a3b851a34eb49. This is an additive research contribution with written proofs for review, not a new Lean certificate for the analytic results. The conversation delivery contains a longer numbered LaTeX/HTML version, the full reading record, and exploratory arithmetic moments.

S5.1.1 1. Reviewed input and exact types

The supplied other-session theorem is correct: for injective $\Theta: V \rightarrow B$, an R -linear automorphism F , $Q = B/\Theta V$, and $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, all homotopies satisfying BOTH $dH = \Theta \kappa$ and $Hd = 0$, for $d(v, w) = \Theta(v - Fw)$, are

$$H_\alpha(b) = (\kappa b + \alpha \kappa b, F^{-1} \alpha \kappa b), \quad \alpha: Q \rightarrow R V.$$

Writing $H = (u, v)$ gives $u - Fv = \kappa$. Since $d(x, 0) = \Theta x$, $Hd = 0$ gives $H \Theta = 0$; hence Fv descends uniquely to α . Substitution proves the converse. The difference is $\gamma(\alpha - \beta)q$ with $\gamma(v) = (v, F^{-1}v)$. The solution set is affine relative to H_0 , not generally a vector space with the zero homotopy as origin.

The attachment's uncompiled-draft status has now been superseded by the other session's successful PR #12 at 5d2772ea0a16d177ac3e01ff70394b90ba95a232. The actual SplitZeroTauHomotopy.lean, completed job 103535438942, and its full log were read. Both cochain equations and the inverse equivalence passed the specified kernel checks. Retraction and quotient-topology interfaces also passed with their explicit hypotheses. This is the other session's formalization; neither the actual analytic theta construction nor the new Gram/control results below are certified by it. No local Lean/Lake execution is claimed. See COORDINATION_UPDATE.md for exact scope. PR #11 already supplies the continuous specialization, the Fourier twist over $C[t]$, support-changing lifts, coherent dilations and representative-change formulas. Those are retained inputs, not new results here.

Keep the original tau base and the square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$, with $p: G(Z) \rightarrow Z$ still infinite, $e = \text{ofR}(0)$, and tau the external scalar zero. No Boolean localization replaces the base quotient. All quotient relations below use the original internal coequalizer and preserve the support label.

Retain V, B and the actual operators of PRs #8-#11:

```
V={even Schwartz phi on R: phi(0)=0 and integral(phi)=0};
B={smooth F on R+: sup x^b |D^k F(x)| < infinity for all integer b and k>=0};
D=-x d/dx; Theta phi(x)=2 sum_(n>=1) phi(nx);
M F(s)=integral_0^infinity F(x)x^s dx/x;
phi_*= (4 pi^2 x^4 - 6 pi x^2)exp(-pi x^2);
phi_j=D^j phi_*; f_j=Theta phi_j=D^j f_0;
M f_j(s)=s^j g(s), g=2 xi.
```

The Hilbert observation is $H=L^2(\mathbb{R}_+, dx)$, conjugate-linear in the first argument. Polynomial packet coordinates are retained, not changed to orthonormal coordinates.

For an actual finite packet with all multiplicities, let $E=C[t]/h_Z$, $A=M_t$. The previous analytic constructions supply

$$R_\theta: E \rightarrow B, \quad J_Z: B \rightarrow E, \quad qR_\theta = \sigma_Z, \quad J_{ZR_\theta} = 1, \quad J_Z \theta = 0, \\ DR_\theta - R_\theta A = f_\theta \text{ ell}_Z.$$

The exact ell_Z comes from $\epsilonpsilon_Z = j_h(h/(2\xi))$; at a simple zero its value on 1 is $1/(2\xi'(\rho))$. These analytic constructions remain the cited research-proof dependencies, not theorems certified by the finite tests here.

S5.1.2 2. An actual minimum over the admitted theta relations

For $m \geq 0$ define

$$W_m = \text{span}(\phi_0, \dots, \phi_m); \\ \phi_m: C^{(m+1)} \rightarrow V, \quad c \mapsto \sum c_j \phi_j; \\ F_m = \theta \phi_m: C^{(m+1)} \rightarrow H; \\ H_m = F_m^* F_m; \quad C_m = F_m^* R_\theta; \quad G_\theta = R_\theta^* R_\theta.$$

The ϕ_j are independent: Mellin sends a relation to $g(s)P(s)=0$. Thus H_m is positive definite. Here C_m is a cross-Gram matrix $E \rightarrow C^{(m+1)}$, not the complex of that name in the endpoint workbench.

Set

$$B_m = -H_m^{-1} C_m; \\ \pi_m = F_m H_m^{-1} F_m^*; \\ R_m = R_\theta + \theta \phi_m B_m = (1 - \pi_m) R_\theta.$$

The correction is a finite original theta boundary, so R_m lands in B , $qR_m = \sigma_Z$, and $J_{ZR_m} = 1$ at every m . Direct multiplication proves

$$F_m^* R_m = 0; \\ G_m = R_m^* R_m = G_\theta - C_m^* H_m^{-1} C_m > 0.$$

Positive definiteness follows from $J_{ZR_m} = 1$. For any $B: E \rightarrow C^{(m+1)}$,

$$(R_\theta + F_m B)^* (R_\theta + F_m B) \\ = G_\theta + (B - B_m)^* H_m (B - B_m).$$

This is the minimum of the Hilbert Gram within the stated source family, not yet a minimum of its relative weight error. Also $(1 - \pi_m)(R_\theta + F_m B') = R_m$: changing the initial representative by a relation already in W_m leaves the result unchanged.

S5.1.3 3. The complete control form has rank at most two

Let $I_m, S_m: C^{(m+1)} \rightarrow C^{(m+2)}$ be $I_m e_j = e_j$ and $S_m e_j = e_{(j+1)}$. Define

$$K_m = e_\theta \text{ ell}_Z + S_m B_m - I_m B_m A.$$

Then

$$DR_m - R_m A = F_{(m+1)} K_m = \theta (\phi_{(m+1)} K_m).$$

No boundary component is discarded in this identity. Put $b_m = e_m^* B_m: E \rightarrow C$, and $z_m = R_m^* f_{(m+1)}: C \rightarrow E$. Integration by parts on the stated source functions gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

with vanishing boundary term $[x \text{ conjugate}(F)H]_0^\infty$. Hence

$$\begin{aligned} W_m &= A^* G_m + G_m A - G_m \\ &= -(R_m^* F_{(m+1)} K_m + K_m^* F_{(m+1)}^* R_m) \\ &= -z_m b_m - b_m^* z_m^*. \end{aligned}$$

The last equality uses $F_m^* R_m = 0$ and the exact last row of K_m , which is b_m . Thus $\text{rank}(W_m) \leq 2$ for every packet dimension and every finite m . The pairings with the first $m+1$ theta directions vanish; those supported vectors themselves are retained.

Define the actual scalar quantities

$$\begin{aligned} u_m &= z_m^* G_m^{-1} z_m \geq 0; \\ v_m &= b_m^* G_m^{-1} b_m \geq 0; \\ c_m &= b_m^* G_m^{-1} z_m. \end{aligned}$$

The nonzero spectrum of $G_m^{-1} W_m$ is the nonzero spectrum of

$$-\begin{bmatrix} c_m & v_m \\ u_m & \text{conjugate}(c_m) \end{bmatrix}.$$

This follows by factoring through the two-column map $[z_m, b_m^*]$ and using the equality of nonzero spectra of the two compositions. Its trace is $-2 \text{Re}(c_m)$, determinant $|c_m|^2 - u_m v_m$. Therefore the least symmetric control allowance is

$$\begin{aligned} \epsilonpsilon_m &= |\text{Re}(c_m)| + \sqrt{u_m v_m - (\text{Im}(c_m))^2}, \\ -\epsilonpsilon_m G_m &\leq W_m \leq \epsilonpsilon_m G_m. \end{aligned}$$

Cauchy-Schwarz makes the radicand nonnegative. The formula also covers rank one. A reflection-stable full packet has

$$\text{trace}(G_m^{-1} W_m) = 2 \text{Re}(\text{trace } A) - \dim E = 0,$$

so $\text{Re}(c_m) = 0$. This is an exact trace identity, not a choice of coordinates. For every eigenvector $Av = \rho v$,

$$(v^* W_m v) / (v^* G_m v) = 2 \text{Re}(\rho) - 1.$$

Thus the relative control still detects precisely the original spectral weight.

S5.1.4 4. The next relation is a rank-one update with its own retained coordinate

Set

$$\begin{aligned} a_{(m+1)} &= (1 - \Pi_m) f_{(m+1)} = \Theta \psi_{(m+1)}, \\ \psi_{(m+1)} &= \phi_{(m+1)} - \Pi_m H_m^{-1} F_m^* f_{(m+1)}, \\ \eta_{(m+1)} &= \|a_{(m+1)}\|^2 > 0. \end{aligned}$$

Then $R_m^* a_{(m+1)} = z_m$, and orthogonal decomposition gives

$$\begin{aligned} \Pi_{(m+1)} &= \Pi_m + a_{(m+1)} \eta_{(m+1)}^{-1} a_{(m+1)}^*; \\ R_{(m+1)} &= R_m - a_{(m+1)} \eta_{(m+1)}^{-1} z_m^*; \\ G_{(m+1)} &= G_m - \eta_{(m+1)}^{-1} z_m z_m^*. \end{aligned}$$

The control update is obtained by applying the same real-linear operator $X \mapsto A^*X + XA - X$ to the last identity.

The original quotient transport is

$$Q_W = B/\Theta W_m \rightarrow Q_{W(m+1)} = B/\Theta W_{(m+1)}, \\ \text{kernel} = \Theta W_{(m+1)} / \Theta W_m \simeq W_{(m+1)} / W_m.$$

That kernel is one-dimensional. The square $E \rightarrow Q_W$ by R_m and $E \rightarrow Q_{W(m+1)}$ by $R_{(m+1)}$ commutes. The new relation becomes the supported zero only through the actual next-level quotient; it is not another scalar adjunction at the bottom.

There is a form on the entire quotient:

$$H_{\text{cal}_m}([F], [G]) = \langle (1 - \Pi_m)F, (1 - \Pi_m)G \rangle.$$

It is positive definite since the kernel of $1 - \Pi_m$ in B is exactly ΘW_m . Let $\lambda_{(m+1)}[F] = \langle a_{(m+1)}, F \rangle$. This is well defined on Q_W and

$$H_{\text{cal}_m}(x, y) = H_{\text{cal}_{(m+1)}}(qx, qy) \\ + \eta_{(m+1)}^{-1} \text{conjugate}(\lambda_{(m+1)}(x)) \lambda_{(m+1)}(y).$$

The map $(q, \lambda_{(m+1)}): Q_W \rightarrow Q_{W(m+1)}$ plus C is a linear isomorphism. Its inverse is

$$([F], c) \mapsto [F - a \eta_{(m+1)}^{-1} \langle a, F \rangle + a \eta_{(m+1)}^{-1} c]_W.$$

Changing F by a relation at level $m+1$ changes this expression only by a relation at level m . This supplies the exact metric comparison between levels instead of treating the quotient transports as isometries.

Every displayed linear map has its original fixed-support lift. On the full source chain, reconstructed addition uses the join $\max(m, n)$ and the above transports. The scalar e sends $(m, [F])$ to $(m, [0])$; τ sends it to external absence. The two-leg primitives are $(k_m u, 0)$ and $(0, \text{Fourier}^{-1} k_m u)$, where $k_m = \Phi_{(m+1)} K_m$. Synchronization by the original $E_{\text{sync}} = (1, e)$ supplies their common joint support. Their difference remains the H_0 cycle $(k_m u, \text{Fourier}^{-1} k_m u)$. The other session's $\text{Hom}(Q, H_0)$ freedom leaves their differential and hence W_m unchanged.

S5.1.5 5. Actual arithmetic moments and their truncation allowance

The matrices H_m come from the original arithmetic functions. Since $JF(x) = x^{-1} F(1/x)$ is unitary on $L_2(dx)$, $Jf_0 = f_0$ and $JD = (1 - D)J$,

$$H_{ij} = \int_0^1 \text{conjugate}(f_i) f_j \\ + \text{conjugate}(f_i^{\text{ref}}) f_j^{\text{ref}} dx, \\ f_j^{\text{ref}} = (1 - D)^j f_0.$$

This retains both multiplicative ends and permits absolutely convergent Gaussian sums on $x \geq 1$. Write

$$\phi_j(x) = P_j(x) \exp(-\pi x^2), \\ P_0 = 4 \pi^2 x^4 - 6 \pi x^2, \\ P_{(j+1)} = 2 \pi x^2 P_j - x P_j', \\ P_j = \sum_r c_{(j, r)} x^{(2r)}, \\ c_{(j+1, r)} = 2 \pi c_{(j, r-1)} - 2r c_{(j, r)}.$$

For $f_j^{[N]}(x) = 2 \sum_{(n=1)}^N P_j(nx) \exp(-\pi n^2 x^2)$, put

$$C_j = \sum_r |c(j, r)| \sqrt{\Gamma(2r+1/2) / (2 \pi^{(2r+1/2)})}.$$

Then

$$\begin{aligned} & \|f_j - f_j^N\|_{L^2([1, \infty))} \\ & \leq 2 C_j (N+1)^{-1/2} \exp(-\pi(N+1)^2/2) \\ & \quad / [1 - \exp(-\pi(2N+3)/2)]. \end{aligned}$$

Proof: on $x \geq 1$, $2 \pi n^2 x^2 \geq \pi n^2 + \pi n^2 x^2$. Integrating each monomial's squared bound gives $n^{-1/2} \exp(-\pi n^2/2)$ times the displayed gamma constant. Minkowski and a geometric bound on consecutive n terms give the formula. The reflected derivative retains the absolute binomial sum of these allowances.

The finite integral entries use exactly

$$\begin{aligned} & \int_1^\infty x^{2k} \exp(-c x^2) dx \\ & = \Gamma(k+1/2, c) / (2 c^{(k+1/2)}), \quad c = \pi(n^2 + n'^2). \end{aligned}$$

For truncation tail norms δ_i, δ_j , each inner-product error is bounded by $\delta_i \|f_j^N\| + \|f_i^N\| \delta_j + \delta_i \delta_j$, plus its reflected contribution. Numerical evaluation errors for the finite expression still require their own enclosure. The optional conversation computation is explicitly exploratory, not interval-certified.

The exact consistency check is $H_{(i+1, j)} + H_{(i, j+1)} = H_{ij}$. Cross-Gram columns $F_m \cdot R_0$ remain their actual source integrals and are not replaced by arbitrary matrices.

S5.1.6 6. The same source proves a strong topology comparison

For this actual f_0 ,

$$\text{closure span}\{D^j f_0 : j \geq 0\} \text{ in } L^2(\mathbb{R}_+, dx) = H.$$

Proof. The explicit unitary Mellin-line map sends F to $M F(1/2+it)$ in $L^2(\mathbb{R}, dt/(2\pi))$. It sends D to $1/2+it$ and f_j to $(1/2+it)^j g(1/2+it)$. The log-space function $h(z) = \exp(z/2) f_0(\exp z)$ is holomorphic in $|\text{Im } z| < \pi/4$ by its Gaussian series. For $|\text{Im } z| \leq a < \pi/4$ it decays superexponentially as $\text{Re } z$ tends to positive infinity. Poisson gives $h(z) = h(-z)$, so the same holds at the other end. Fourier contour displacement therefore gives $|g(1/2+it)| \leq C_a \exp(-a|t|)$.

The measure $d\mu = |g(1/2+it)|^2 dt/(2\pi)$ has a positive exponential moment. If an $L^2(\mu)$ function is orthogonal to all polynomials, multiplying μ by its conjugate gives a finite complex measure with a smaller positive exponential moment. Its Fourier transform is analytic in a strip and all derivatives at zero vanish. Analyticity and uniqueness of the Fourier transform imply that measure is zero. Thus polynomials are dense in $L^2(\mu)$. Multiplication by g is unitary from this weighted space onto $L^2(dt/(2\pi))$, because the nonzero analytic function g on the line is nonzero almost everywhere. This proves the assertion without RH.

As a result, for a fixed finite packet,

$$\begin{aligned} & R_m \rightarrow 0 \text{ in } \text{Hom}(E, H), \quad G_m \rightarrow 0 \text{ and } W_m \rightarrow 0, \\ & \text{while } q_{R_m} = \sigma_Z \text{ and } J_{ZR_m} = 1 \text{ for every } m. \end{aligned}$$

This specifies why unscaled smallness of W_m cannot be used alone as weight control: its denominator G_m is also shrinking. The exact relative formula in section 3 must be retained.

The corrective map is the Hilbert/jet graph:

$$\begin{aligned} & \text{closure}\{(F, J_{ZF}) : F \in B\} = H \text{ plus } E; \\ & \text{closure}\{(\Theta \phi, 0) : \phi \in V\} = H \text{ plus } 0. \end{aligned}$$

Indeed finite theta polynomials give every $(h,0)$ in the closure, and the sequence $(R_m u, u)$ gives $(0, u)$. For a specified positive metric on E , completion in $||F||_2^2 + ||J_Z F||_E^2$ is this graph closure, and the quotient by the boundary closure is E by $[(h, u)] \rightarrow u$. The added metric is retained supplied data, not an implied positivity theorem for the Weil pairing.

Its split observation is

$$((R_m u)^{\bullet}, u^{\bullet}) \rightarrow (e_H, u^{\bullet})$$

in the product of the stated support-fibre topologies. The arithmetic component survives above a supported Hilbert zero. External absence is (τ, τ) .

S5.1.7 7. Duality, trace and scope

The original residue map has matrix S with $S^* = -S$ and $A^*S + SA = S$. Its pullback of the dual metric is

$$\begin{aligned} G_m^* D &= S^* G_m^{-1} S, \\ W_m^* D &= -S^* G_m^{-1} W_m G_m^{-1} S. \end{aligned}$$

Thus the upper-to-lower comparison is retained on the same arithmetic packet. The Weil pairing stays $S J_g$, with explicit metric comparison $G_m^{-1} S J_g$. Its positivity is not imposed by choosing a positive Gram matrix.

All representative changes above are original theta boundaries. Their extension class $j_h(h/(2\xi))$ is unchanged; the induced finite cochain projectors differ by the homotopy $(b)_Z, 0$. The trace difference is zero because $J_Z \Theta = 0$. Degree-one and tensor Koszul signs remain those of the source complex. On a raw tensor metric, the inserted W_m terms give the copied allowance $n \epsilon_m$; tensoring alone does not improve it.

The new results are the actual orthogonal source choice, rank-two boundary formula, exact relative certificate, rank-one source-kernel recurrence, arithmetic moment tail bound, and the full Hilbert/jet topology comparison. No uniform arithmetic purity estimate, global spectral-kernel vanishing, RH or GRH is asserted.

The thirteen finite exact tests check the encoded algebra and declared calibration models, not infinite analytic arguments or actual zeta zeros. The complete cochain-homotopy theorem remains the other session's result, now supported by the independently inspected successful PR #12 run. Its certificate does not extend to the new analytic control results. Existing Lean modules and earlier validation receipts are untouched.

References: live PR #11 at the pinned head; successful PR #12 and source/job references in COORDINATION_UPDATE.md; the supplied `Tau_Deligne_Control` and `Tau_Global_Comparison` source notes; the supplied other-session homotopy proof; the previously retained Weil II 3.3.4-3.3.6 and 6.2 duality excerpts. Standard orthogonal-polynomial context is NIST DLMF 18.2; the density argument needed here is proved above for the actual arithmetic measure.

S6 PR 17: Spectral-sum descent and the complete relative arithmetic fibre

Source attribution. Web contribution, pull request #17, [Zeta research repository](#).

Exact revision. 33b29f706008124886614ba4bd55bffc489df9e2

Source file. workbenches/tau-spectral-sum/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 551e3fe89b4409d5ed1eeacfaac4cb7d9c1291978127de1b4f989b272bd24a47

S6.1 Spectral-sum descent with the relative arithmetic fibre retained

Owner-directed continuation, 12 September 2026. Draft for review; no purity or RH theorem is claimed. Base: PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc.

S6.1.1 1. Original source and scope

Keep $G(R) = \{\tau\} \text{ disjoint-union } \{r^\bullet\}$, $e_R = 0_R^\bullet$, and the original square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$. The first arithmetic quotient remains infinite. Fixed-label linear maps lift by $(\lambda, v) \rightarrow (\lambda, Tv)$ and fix external absence; a vanishing value remains the original labelled supported zero.

Use the existing $C_+ = [V \rightarrow \Theta_B]$, $Q = B/\Theta \backslash V$, $D = -x \backslash d/dx$, $g = 2 \backslash \xi$, and the actual source function $\phi_* = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$. For a finite packet h of actual zeros, including full orders, the prior analytic construction supplies F_h with Mellin transform g/h . Retain

$$T_h(P) = P(D)F_h, \quad T_h(hP) = \Theta(P(D)\phi_*), \quad J_h T_h(P) = v_h[P]_h,$$

where $\text{upsilon}_h = j_h(g/h)$ is the full invertible jet, not the scalar e . These analytic range statements are inputs from the previous work, not newly Lean-certified here.

For k factors, the actual numerator is $P_M = C[s_1, \dots, s_k]_{(\text{degree} \leq M)}$, the relation space is its intersection with $(h(s_1), \dots, h(s_k))$, and its theta realization is $B_-(k, M)$. The canonical representative $R_-(k, M)$ is the unique least-norm lift with exactly these relations allowed. Its full jets, arithmetic class, Gram G and control W remain those of #14.

The supplied Kernel Layer Integration supplement agrees with the Symmetric Frontier note under $K = C_{\text{inverse}}$, $U = F$, $D = \Omega$, $E_{\text{layer}} = b_M$, and $F_{\text{derivative}} = \Omega^{(-1)} E_{\text{plus}}$. Its sharper rank and original-layer identities are retained as source results, not counted again as new work.

S6.1.2 2. An actual one-parameter pushforward and its resolution

Define

$$\iota_k : \mathbb{C}[S] \rightarrow E_h^{\otimes k}, \quad S \mapsto [s_1 + \dots + s_k].$$

Restriction of modules through this map is the exact quasi-coherent pushforward along the finite scheme morphism. Its adjunction maps are $f \rightarrow (1 \rightarrow f(1 \otimes 1))$ and $a \otimes 1 \rightarrow a f(1)$. Applying G gives $p \backslash G(\iota) = \iota \backslash p$, with e sent to e and τ to τ ; the morphism is over the same absolute pointed base. No finite field is substituted.

For the actual finite image E , with sum operator A , there is the free resolution

$$0 \rightarrow \mathbb{C}[S] \otimes E \xrightarrow{SI-A} \mathbb{C}[S] \otimes E \xrightarrow{\text{ev}_A} E_A \rightarrow 0.$$

Its augmentation sends $P \backslash \text{tensor} \backslash v$ to $P(A)v$. Injectivity follows from the highest polynomial coefficient. Exactness follows from

$$S^j v - A^j v = (SI - A) \sum_{i=0}^{j-1} S^{j-1-i} A^i v.$$

Dualizing with $\mathbb{C}[S] dS$ gives $SI - A^{\vee}$ in degrees 0,1, where $A^{\vee}(\lambda) = \lambda$ compose A is the algebraic transpose, not a Hermitian adjoint. Thus

$$\text{Ext}_{\mathbb{C}[S]}^1(E_A, \mathbb{C}[S]dS) \cong E^{\vee},$$

with $[\sum S^j \lambda_j dS] \rightarrow \sum (A^{\vee})^j \lambda_j$ and inverse the constant class. With $\mathbb{C}[b]^i = \mathbb{C}^{i+b}$, the exact shifted statement is

$$R\text{Hom}(E_A[-k], \mathbb{C}[S]dS[1]) \simeq E^{\vee}[k].$$

The right adjoint is the specified module functor $R\text{Hom}_{\mathbb{C}[S]}(E_h^{\text{tensor } k}, -)$. This is not an assertion about pushforward on all sheaves of modules. See Stacks Tags 0AVV and 0AWZ.

The same duality has the concrete pairing

$$\langle v, [P(S)dS] \rangle = \sum_{\lambda \in \text{Spec } A} \text{Res}_{S=\lambda} P(S)((SI - A)^{-1}v)dS.$$

Expansion at infinity gives $\sum \lambda_j (A^j v)$, proving the formula and perfectness. On a length- l block,

$$(SI - A)^{-1} = \sum_{j=0}^{l-1} \frac{N^j}{(S - \lambda)^{j+1}}.$$

All poles are retained before taking the trace. The further observation is

$$\text{Tr}(\psi(A)) = \sum_{\lambda} \text{Res}_{S=\lambda} \psi(S) \frac{\chi'_A(S)}{\chi_A(S)} dS.$$

Here $\chi_A = \det(SI - A)$ is the finite characteristic polynomial, not a replacement for $g=2x$. With $\psi(S) = a^S$ this is the finite trace of the same diagonal scaling correspondence; its cochain sign is $(-1)^k$.

S6.1.3 3. Exact symmetric-square relation algebra

Retain the coordinate morphisms

$$S = s_1 + s_2, \quad r = s_1 - s_2, \quad \Delta = r^2, \quad s_1 = (S + r)/2, \quad s_2 = (S - r)/2.$$

Swap fixes S and negates r . Even/odd polynomial expansion gives the filtered algebra isomorphism $\Phi: \mathbb{C}[S, \Delta] \rightarrow \mathbb{C}[s_1, s_2]^{\text{swap}}$, with $\deg(S)=1$, $\deg(\Delta)=2$.

Define H_0, H_1 without localizing away from $r=0$:

$$h((S+r)/2) = H_0(S, r^2) + rH_1(S, r^2), \quad h((S-r)/2) = H_0(S, r^2) - rH_1(S, r^2).$$

Then

$$(E_h \otimes E_h)^{S_2} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1).$$

Proof: the original ideal in $\mathbb{C}[S, r]$ is (H_0, rH_1) . Averaging an expression $H_0 \setminus a + r \setminus H_1 \setminus b$ under $r \rightarrow -r$ gives exactly $H_0 \setminus a_{\text{even}} + \Delta \setminus H_1 \setminus b_{\text{odd}}$. Conversely both generators lie in the original ideal. Reynolds averaging is exact over $\mathbb{C}$, so the invariant quotient has precisely this ideal, including all nilpotents.

The unscaled orbit sums of $s_1^{i_1} \dots s_d^{i_d}$, $0 \leq i_1 \leq \dots \leq i_d \leq d$, give dimension $d(d+1)/2$. The associated graded Hilbert series for the original degree filtration is

$$\sum_{0 \leq i_1 \leq \dots \leq i_d \leq d} z^{i_1 + \dots + i_d} = \frac{(1-z^d)(1-z^{d+1})}{(1-z)(1-z^2)}.$$

No grading of a nonhomogeneous h is asserted.

The original primitive map is explicit:

$$H_0 A_0 + \Delta H_1 A_1 = h(s_1) \frac{A_0 + r A_1}{2} + h(s_2) \frac{A_0 - r A_1}{2}.$$

Its two cochain primitives use $\phi_* \setminus \text{tensor} \setminus F_h$ and $F_h \setminus \text{tensor} \setminus \phi_*$, respectively. The second primitive has a minus sign, matched by the second tensor-differential sign. For a relation of degree M , ordered division followed by swap averaging gives $\deg A_0 \leq M-d$, $\deg A_1 \leq M-d-1$.

The unit $\epsilon_h \setminus \text{tensor} \setminus \epsilon_h$ descends to the invariant algebra and remains in its full jet map. All finite relation quotients and the transitions $M \rightarrow M+1$ commute with Φ . A class killed by the next relation maps to the receiving e -zero, never to external absence.

S6.1.4 4. Push the actual norm to a matrix weight, not a scalar observation

Set

$$w_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{1}{2\pi}.$$

The exact real coordinate map is

$$u = t_1 + t_2, \quad v = t_1 - t_2, \quad dt_1 dt_2 = \frac{1}{2} du dv.$$

It gives $S=1+iu$, $\Delta=-v^2$. For

$$P(S, \Delta) = \sum_{a=0}^{\lfloor M/2 \rfloor} \Delta^a f_a(S), \quad \deg f_a \leq M - 2a,$$

define

$$W_{ac}(u) = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv.$$

Then the original arithmetic norm is exactly

$$\|T_h^{(2)} \Phi P\|^2 = \int_{\mathbb{R}} f(1+iu)^* \mathbf{W}(u) f(1+iu) du.$$

Proof: substitute the coordinate map and expand the finite polynomial. Exponential moments justify Fubini. Each finite matrix $W(u)$ is strictly positive: the integrand is the squared modulus of a nonzero polynomial in $-v^2$ against a density positive almost everywhere. The entire nonzero g/h has only discrete zeros on these real lines. The source exponential bound also gives convergence for each fixed u .

Its scalar entry is $(w_h^* w_h)(u)$. Its other entries retain the relative-coordinate moments; they are not replaced by that scalar entry.

For general k retain $S = \sum s_i$, $x_i = s_i$ for $i < k$ and $s_k = S - \sum_{i < k} s_i$. This is a polynomial isomorphism and the real Jacobian is one. Writing $P = \sum x^\alpha f_\alpha(S)$ gives

$$W_{\alpha\beta}(u) = \int_{\mathbb{R}^{k-1}} \prod_{i < k} (1/2 + it_i)^{\alpha_i} \prod_{i < k} (1/2 + it_i)^{\beta_i} \prod_{i < k} w_h(t_i) w_h(u - \sum_{i < k} t_i) d^{k-1}t.$$

The outer coordinate remains $S = k/2 + iu$, and $\deg f_\alpha \leq M - |\alpha|$. This supplies an all- k isometry retaining the entire filtered source. Constants in exponential-moment bounds depend on h, k and the degrees; no uniform tensor bound is inferred.

S6.1.5 5. Canonical control commutes with the descent

For the explicit source-coordinate matrix C and quotient-coordinate matrix U , keep

$$M' = C^* M C, \quad J' = U^{-1} J C, \quad K' = U^{-1} K (U^{-1})^*, \quad G' = U^* G U,$$

$$A' = U^{-1} A U, \quad W' = U^* W U, \quad R' = R U.$$

These follow by substituting into $K = J M^* (-1) J^*$ and the exact constrained-minimum formula. Thus no positive metric has been freely selected. The same isometry carries the actual next relation layer and both boundary maps Y, C_{rel} ; their cross-pairing is unchanged.

There is a further exact base/relative map. Write $w(u) = [m_0, b], [b^*, C_{\text{rel}}]$, with $m_0 > 0$. Then

$$f \mapsto (f_0 + m_0^{-1} b f_{\text{rel}}, f_{\text{rel}})$$

is an isometry into the direct sum weighted by m_0 and $C_{\text{rel}} - b^* m_0^{-1} b$. Its inverse is $(a, z) \mapsto (a - m_0^{-1} b^* z, z)$. The precise image of the polynomial source is retained; the ratio need not be polynomial. Multiplication by S commutes with this map. Neither the scalar mass nor positivity of this representative norm is substituted for the arithmetic Weil pairing.

S6.1.6 6. Full nilpotent fibres for every tensor degree

For an ordered tuple of roots with orders m_i , the local algebra is

$$L = \mathbb{C}[z_1, \dots, z_k] / (z_i^{m_i}), \quad A = \rho_\Sigma I + N, \quad N = M_{\sum z_i}, \quad D_0 = \sum_i (m_i - 1).$$

Let $c_r = [t^r] \prod (1+t+\dots+t^{(m_i-1)})$, $c_{-1} = 0$. Then the complete module pushforward is

$$\pi_* L \cong \bigoplus_{r=0}^{\lfloor D_0/2 \rfloor} (\mathbb{C}[S]/(S - \rho_\Sigma)^{D_0-2r+1})^{c_r - c_{r-1}}.$$

Proof and maps: on $z_i \wedge j$ define $F_i = j(m_i - j)z_i \wedge (j-1)$ and $H_i = (2j - (m_i - 1))z_i \wedge j$. Direct calculation gives $[N, F] = H$, $[H, N] = 2N$, $[H, F] = -2F$. The explicit auxiliary diagonal Gram

$$\|z^\alpha\|_0^2 = \prod_i \frac{\alpha_i! (m_i - 1)!}{(m_i - 1 - \alpha_i)!}$$

makes $F = N^*$. A degree- r primitive p with $Fp = 0$ has ladder length $w+1$, $w = D_0 - 2r$, since

$$FN^j p = j(w - j + 1)N^{j-1}p, \quad \|N^j p\|_0^2 = \frac{j!w!}{(w - j)!} \|p\|_0^2.$$

Orthogonal invariant complements construct all ladders; degree counts give $c_r - c_{(r-1)}$. The actual module map is $[P(S)] \rightarrow P(\rho_\Sigma + N)p$ on each chosen primitive. This auxiliary Gram is related to the arithmetic one by the explicit operator $G_{\text{arithmetic}}^{(-1)} G_0$; it is used for the algebraic decomposition only. All cross-ladder entries $U^* G_{\text{arithmetic}} U$ remain in control estimates.

For an occupation list n_ρ with sum k , the symmetric component has graded dimensions given by

$$\prod_{\rho} \prod_{j=1}^{n_{\rho}} \frac{1 - t^{m_{\rho} + j - 1}}{1 - t^j}.$$

These are polynomial orbit-count identities, not evaluations of a rational function at roots of unity. Apply the same primitive construction to invariants, with $D_0 = \sum n_\rho (m_\rho - 1)$ and $\rho_\Sigma a = \sum n_\rho \rho$. Central occupation idempotents are retained even when different occupations have the same spectral sum.

For an order-three local input taken twice, the symmetric fibre has ladders

$$1, X, X^2, X^3, X^4, \quad p = z_1^2 - z_1 z_2 + z_2^2, \quad X = z_1 + z_2, \quad Xp = 0.$$

It is $\mathbb{C}[S]/(S - 2\rho)^5$ direct-sum $\mathbb{C}[S]/(S - 2\rho)$, dimension six. This is a local algebra calculation, not an assertion that ζ has a triple zero. The full arithmetic unit commutes with A and carries the ladder coordinates to the original jets.

Each occupation trace is its full dimension times $a^{(\sum n_\rho \rho)}$, with dimension product $\text{binomial}(n_\rho + m_\rho - 1, n_\rho)$. The cochain trace retains $(-1)^k$. Quotienting a top ladder vector, when explicitly requested as a module quotient, makes it the supported zero; the full pushforward itself keeps it.

S6.1.7 7. Exact calibration and scope

For the declared measure $\exp(-t^2/2)dt$, not the arithmetic measure, the three relative directions $1, \Delta, \Delta^2$ give

$$W(u) = \sqrt{\pi} e^{-u^2/4} \begin{pmatrix} 1 & -2 & 12 \\ -2 & 12 & -120 \\ 12 & -120 & 1680 \end{pmatrix}.$$

The unscaled coordinate columns $1, \Delta + 2, \Delta^2 + 12\Delta + 12$ turn this into $\sqrt{\pi} \exp(-u^2/4) \text{diag}(1, 8, 384)$. The full two-variable mass is still 2π . The published core

checker verifies these signs, constants, algebraic primitives, sum resolution and nilpotent blocks. The longer conversation package has a separate 24-method suite, plus independent reruns of the source 18- and 22-method suites; those are not all claimed to be tests in this smaller repository checker.

The new reduction isolates the convolution mass, the relative covariance, and full-jet interpolation on one explicit filtered image. It does not prove their required uniform sublinear tensor estimate. The finite pushforward and matrix-weight model are not asserted to be lisse sheaves on Deligne's Lefschetz pencil. They supply a precisely defined product-to-one-parameter operation to study next, with its adjoint, cohomology, arithmetic trace and original supported relations retained.

References: Stacks Tags 0AVV and 0AWZ; NIST DLMF 18.2; Deligne, La conjecture de Weil II, IHES 52 (1980), 137-252. A bounded supplied French TeX window around 3.2.11–3.2.15 was reread. Its historical line 872 contains an inconsistent numerical implication and is not used as a theorem. No complete new audit of Weil II, analytic integral certificate, Lean execution, or global novelty claim is made.

S7 Derived Split-Zero mathematics

Source attribution. Web contribution retained on the pre-integration main revision.

Exact revision. 14c69d604044b848d64318751a7bee19edfd9aba

Source file. formal/splitzero/DERIVED_MATHEMATICS.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 6a290e193161eb09f49b6faba578193fdaf28b5ae3a97340d1c41a40aa99e695

S7.1 SplitZero: reconstruction, internal homology, and balanced finite jets

Owner-directed formalization continuation, 12 September 2026.

S7.1.1 Scope and provenance

This contribution extends formal/splitzero on the owner's Zeta workbench. Its base is the synchronization PR #5, commit 236f0a8370477991e2d0cf334ad1ebfe4f3f8b17, itself based on the structural PR #4. The original scalar core and all five preceding extension sources remain unchanged. There is no Sneed-side submission or integration in this continuation.

The mathematical source is the owner's SplitZero programme, including the supplied continuation's internal homology and balanced-Mellin sections. The implementation below distinguishes source-derived structure, ordinary reusable algebra, and the new explicit comparison example. It makes no global priority claim and does not treat the number of Lean declarations as a number of mathematical discoveries. No private conversation or source-literature corpus is republished.

The exact build certificate for a revision is the completed SplitZero\derived\constructions workflow on that commit, including direct source checks and its generated AuditDerived.lean. The PR records the accepted execution revision and run. A failed earlier development run is not a certificate for later source.

S7.1.2 1. Reverse reconstruction, with actual scalar action

Let R be a commutative ring, L a join-semilattice with bottom, and let D consist of R -modules V_i with linear transition maps

$$\begin{aligned} \text{rho_ij} &: V_i \rightarrow V_j \quad (i \leq j), \\ \text{rho_ii} &= \text{id}, \quad \text{rho_jk} \text{ rho_ij} = \text{rho_ik}. \end{aligned}$$

No finiteness, field, injectivity, or nontriviality assumption is imposed. Construct

$$\text{Total}(D) = \text{disjoint union of } V_i \text{ over } i \text{ in } L.$$

The global zero is $(\text{bottom}, 0)$, and

$$\begin{aligned} (i, x) + (j, y) &= (i \text{ join } j, \text{rho_i, i-join-j}(x) + \text{rho_j, i-join-j}(y)). \\ \text{tau} \cdot (i, x) &= (\text{bottom}, 0). \\ \text{ofR}(r) \cdot (i, x) &= (i, r \cdot x). \end{aligned}$$

`LinearDiagram.totalAddCommMonoid` and `LinearDiagram.totalModule` construct the actual instances and prove every law. For associativity, transport both sides to $i \text{ join } j \text{ join } k$. Linearity and transition composition reduce both to the same three-term sum. The scalar laws follow in the supported case from the R -module laws and in the absent cases from the chosen global zero. The supported-zero scalar sends (i,x) to $(i,0)$.

The bottom fibre is allowed to be nonzero. Specializing it to the zero module recovers the source's carrier consisting of one absent point and the non-bottom fibres. This is an explicit generalization, not an assumption that the bottom fibre in every later dual diagram vanishes. Opposite-index duality itself is not constructed here.

Fixed-index natural families of linear maps produce actual $G(R)$ -linear maps between reconstructed totals. Identity and composition are proved. More substantially, `Recovery.recoverEquiv` reconstructs the intrinsic support fibres of **any** existing $G(R)$ -semimodule M and gives a full $G(R)$ -linear equivalence back to M :

$$(1,x) \mapsto x, \quad m \mapsto (e\ m, m).$$

Its addition proof uses the original absorption identity $m + e\ m = m$. It is not merely a bijection of carriers. A bundled equivalence of the two full categories, including arbitrary changes of support index and both natural inverse comparisons, is a separate library interface not claimed by this implementation.

S7.1.3 2. The internal quotient is characterized by a universal property

Let B_i be transition-stable submodules of V_i . The quotient diagram has fibres V_i/B_i and the genuinely induced quotient transition maps. The projection

$$q : \text{Total}(D) \rightarrow \text{Total}(D/B), \quad q(i,x) = (i,[x])$$

is a surjective $G(R)$ -linear map. The code proves the exact equality criterion

$$q(i,x)=q(i,y) \text{ iff } x-y \text{ is in } B_i.$$

Different labels cannot become equal under q . In particular a relation maps to its own fibre zero.

The main point is stronger than that observation. Let N be **any** $G(R)$ -semimodule, not assumed additively cancellative, and let $f:\text{Total}(D)\rightarrow N$ be split-linear. If

$$f(i,b)=f(i,0) \text{ for every } b \text{ in } B_i,$$

then there exists a unique split-linear $g:\text{Total}(D/B)\rightarrow N$ with $g\ q=f$. The proof does not subtract elements in N . If $x-y$ is in B_i , use the identity $y+(x-y)=x$ inside V_i and then apply additivity of f . This proves constancy on q -fibres. Surjectivity constructs the descended function; linearity and uniqueness follow by lifting representatives.

`Relations.coequalizer_universal` states this against the actual inclusion of the relation diagram and its supported-zero companion. Thus the universal property is proved against arbitrary semimodule targets, not only against a selected class of fixed-support maps. The implementation gives the complete unbundled universal property; it does not merely define the desired quotient and label it a coequalizer.

S7.1.4 3. Actual homology and the exact transported-class kernel

At a degree i of a cochain complex, use the window

$C^{(i-1)} \xrightarrow{d_{\text{Prev}}} C^i \xrightarrow{d_{\text{Next}}} C^{(i+1)}, \quad d_{\text{Next}} d_{\text{Prev}} = 0.$

Window.boundaryToCycles constructs the incoming map into $\ker(d_{\text{Next}})$. Boundaries are its actual linear range, and H is the module quotient of cycles by that range. Window.ofCochain imports these data at every integer degree of a Mathlib CochainComplex; ChainMap.ofCochain imports an actual cochain map without replacing its differentials.

For a chain map $f:C \rightarrow D$, the code constructs

$f_Z : Z(C) \rightarrow Z(D),$
 $H(f) : Z(C)/B(C) \rightarrow Z(D)/B(D).$

It proves identity, composition, and

$H(f)([z]) = 0 \iff \text{there exists } y \text{ with } d_{\text{Prev}_D}(y) = f_i(z).$

Define the module of killed representatives and its original relation submodule:

$\text{Krep} = \{z \text{ in } Z(C) : f_Z(z) \text{ is in } B(D)\},$
 $\text{Bold} = B(C), \text{ viewed as a submodule of Krep.}$

There is a constructed linear equivalence

$\text{Krep}/\text{Bold} \simeq \ker H(f).$

Forward map: the class of z goes to the kernel element $[z]$. It is well-defined because Bold consists exactly of original source boundaries. It is injective because equality of source classes is exactly a difference in Bold. It is surjective because every kernel class has a cycle representative whose image is a boundary. This is ChainMap.homologyKernelEquiv, obtained from a general quotient-kernel equivalence proved in the same file.

The calculation retains actions as well. Given chain endomorphisms a of C and b of D satisfying $f \circ a = b \circ f$, homology_intertwines proves $H(f) \circ H(a) = H(b) \circ H(f)$. kernelAction constructs the restriction of $H(a)$ to $\ker H(f)$. Thus passing to the comparison kernel does not discard a compatible operator or require asserting that the kernel is zero.

S7.1.5 4. Internal homology on the complete support diagram

For coherent support-indexed chain windows, the code constructs the diagrams of preceding terms, middle terms, following terms, cycles, boundaries, and homology. Let R_L denote the reverse reconstruction.

The differential identity is typed as

$R_L(d_{\text{Next}}) \circ R_L(d_{\text{Prev}}) = z_L : R_L(C^{(i-1)}) \rightarrow R_L(C^{(i+1)}),$
 $z_L(1, x) = (1, 0).$

The source and target degrees are explicit. A theorem shows that z_L differs from the constant global-zero map whenever L has a non-bottom element. Cycles satisfy the actual equalizer equation $d(x) = e.d(x)$.

The internal homology projection

$q : R_L(Z(C)) \rightarrow R_L(H(C))$

is the coequalizer of the **original incoming differential into cycles** and its supported-zero companion. `homology_coequalizes` and `homology_coequalizer` establish both parts of that universal property against every split-linear target. This is not an assumed homology compatibility of reconstruction.

For every transition $i \leq j$,

$\text{rho}^{\wedge} H_{ij}([z]) = 0$ iff $\text{rho}_{ij}(z)$ is a boundary at j .

If z is not a boundary at i , its retained class $(i, [z])$ is not $(i, 0)$, regardless of whether that transition kills it. The quotient-kernel equivalence of section 3 computes all such classes as a module, not just a yes/no predicate.

S7.1.6 5. A concrete comparison that detects more than the support labels

Fix any nontrivial commutative ring k . In the displayed degree let

$C: k \xrightarrow{-\theta} k \xrightarrow{-\theta} k,$
 $D: k \xrightarrow{-a} (a, \theta) \xrightarrow{-} k^2 \xrightarrow{-\theta} k.$

Use zero maps in the outside degrees. The two middle chain maps are

$f(a) = (a, \theta),$ $g(a) = (\theta, a).$

The code proves that the source class $[1]$ is nonzero. Its f -image is the **nonzero vector** $(1, 0)$, but that vector is a boundary in D , so $H(f)([1]) = 0$. In fact $H(f) = 0$ on every source class. The g -image $(0, 1)$ cannot be a boundary, because every boundary has second coordinate zero. Hence $H(g)([1])$ is nonzero and $H(f)$ is not $H(g)$.

The ordinary calculation gives $H(C) = k$ and $H(D) = k$ (the latter through second-coordinate projection). These identifications explain the example; the Lean acceptance tests directly prove the nonzero source class, the nonzero chain image, the zero induced map, the surviving class, and the distinction of induced maps rather than relying on a dimension computation.

Consequently the source and target complexes, individual fibre homology, and terminal homology can be identical while the transition kernel differs. This example is parametric over all nontrivial commutative rings, not a finite numerical experiment. Together with the general assembly theorem it supplies an explicit test of the information retained by support-indexed homology. It is an algebraic test case, not a claimed model of the global theta comparison.

S7.1.7 6. Balanced finite jets, including the original split quotient square

Let k and B be commutative rings and fix a $k[t]$ -algebra structure on B . Let $u: k[t] \rightarrow B$ denote that map and

$I_{-}(\text{rho}, m) = ((t - \text{rho})^{\wedge} m), \quad m \geq 0.$

There is an explicit B -algebra equivalence

$B \otimes_{k[t]} (k[t] / I_{-}(\text{rho}, m)) \simeq B / ((u(t) - u(\text{rho}))^{\wedge} m).$

It maps $b \otimes [p]$ to $[u(p)b]$, with inverse $[b]$ to $b \otimes 1$. No flatness assumption is used for this quotient base-change theorem. This is a specialization of the existing Mathlib theorem `Algebra.tensorProduct.quotIdealMapEquivTensorQuot`, not a claim to have independently developed general tensor-product quotient theory.

splitBaseChange lifts this equivalence to the original G-construction on both sides and proves the commutative square with reflect. It preserves τ and e separately and is a genuine semiring equivalence, not just an amplitude comparison. Thus every element of the finite-jet quotient has an inverse image before forgetting split support.

For any $k[t]$ -module V and v in V , the exact balancing calculation is

$$(1-u(t)) \text{ tensor } (t.v) - u(t) \text{ tensor } (v-t.v) = 0.$$

Expanding gives $1 \text{ tensor } (t.v) - u(t) \text{ tensor } v$, which vanishes by the tensor balancing relation. This is the algebraic equation of the source's Mellin repair. The choice $u(t)=s$ and $t.v=Dv$ is an application once those analytic module structures and intertwining maps have been constructed. It does not assert that the entire global balanced kernel vanishes, or that analytic scalar extension is flat without proof.

S7.1.8 7. Where Deligne enters the intended RH route

None of sections 1-6 imports a Deligne purity theorem. They provide the algebraic maps with which a proposed realization can now be specified and its kernel retained.

For this programme, the subsequent dependency diagram should distinguish the following mathematical obligations:

1. Construct the actual operator-balanced theta complex at the relevant support labels, and its action-compatible analytic comparison β to the spectral sheaf $\mathcal{O}/(2\xi)$. Identify its kernel or cone using the same maps; a finite-jet isomorphism is not a global injectivity argument.
2. Construct the geometric compact-support and ordinary complexes, their actual comparison c , their correspondence action, and a compatible realization identifying the required spectral sheaf with the image of $H^1(c)$. Any additional kernel/cone contributions to the trace must be kept or proved to vanish.
3. Establish two-sided weight control on that **same image**. In Deligne's geometric setting the compact-support upper estimate and the duality-derived ordinary-cohomology lower estimate meet on the image. The dualizing operation, degree and Tate twists, tensor-power maps, and hypotheses of the estimates are part of this step, not consequences stipulated in the definition of support.
4. Transport the spectral normalization without changing the arithmetic function. For a dilation parameter $a>1$, an eigenvalue a^ρ has modulus $a^{(\text{Re } \rho)}$. A proved weight-one modulus $a^{(1/2)}$, for the representation identified with every relevant spectral zero, then forces $\text{Re } \rho=1/2$. This last calculation does not supply the missing weight estimate; it states the precise final implication of this route.

These are dependencies of the specified route, not a claim that every possible proof of RH must follow it. The newly implemented kernel action is useful in steps 1-2 precisely because it keeps the part killed by a comparison available to the operator and trace calculation.

The supplied Connes-Consani article has an actual quotient-sheaf realization with a specified closure and scaling action. Its Theorem 1.2 realizes **critical zeros**. That theorem alone cannot be substituted for an identification representing all nontrivial zeros in the desired purity argument. This is a concrete source/target distinction for a future comparison map, not a dismissal of the construction.

S7.1.9 8. Reading record and references

The six supplied S20 chunks were checked against their individual declared hashes and concatenated in order. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256

e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The current top-level readers/english_standalone.html and readers/source_language.html were used for the selected readings, not the nested zero-accepted salvage corpus. The inspected windows include the tensor-square improvement in 3.2.13, the upper-bound statement in 3.3.1, and the duality/image argument in 3.3.4-3.3.6. This is not an independent audit of every page or of the entire historical proof. The Connes-Consani introduction and Theorem 1.2 were read in the supplied volume, printed pages 83-85 (physical PDF pages 94-96). No copyrighted source corpus is included in this contribution.

Deligne, P. (1980). La conjecture de Weil. II. *Publications Mathematiques de l'IHES*, 52, 137-252. <https://doi.org/10.1007/BF02684780>

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and zeta-cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83-102). American Mathematical Society.

Mathlib contributors. (2026). *Mathlib4*, revision fabf563a7c95a166b8d7b6efca11c8b4dc9d911f. In particular, LinearAlgebra/Quotient/{Defs,Basic}.lean, RingTheory/TensorProduct/Quotient.lean, Algebra/Homology/HomologicalComplex.lean, and Algebra/Category/ModuleCat/Basic.lean.

S7.1.10 9. Reproduction

From formal/splitzero on this branch, with elan and Python installed:

```
python3 prepare.py
python3 check_derived.py --prepare
python3 -m unittest -v test_derived_checker.py
lake update
# Verify the resolved Mathlib revision before fetching its cache.
test "$(git -C .lake/packages/mathlib rev-parse HEAD)" = fabf563a7c95a166b8d7b6efca11c8b4dc9d911f
lake exe cache get
lake -KmaxJobs=1 build
for name in SplitZeroReconstruction SplitZeroRecovery SplitZeroHomology SplitZeroInternalQuotient
↪ SplitZeroComplex SplitZeroHomologyExample SplitZeroFiniteJets; do
  lake env lean --trust=0 -DwarningAsError=true "$name.lean"
done
lake env lean --trust=0 -DwarningAsError=true AuditDerived.lean > AuditDerived.log
python3 check_derived.py AuditDerived.log
```

The manifest enumerates the selected declarations, including the principal structures, universal properties, equivalences, and example proofs. Missing, duplicate, unexpected, or nonstandard axiom reports fail. Source checks reject unfinished proofs and native-reduction escapes. The Python tests validate this checking machinery; the mathematical certificate is the successful Lean execution. Build-job totals include dependencies and are not theorem counts.

S8 PR 14: Coherent arithmetic interpolation and joint tensor boundary control

Source attribution. Complete TeX witness delivered with [web pull request #14](#); the associated Git revision is recorded in its delivery receipt.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Coherent_Interpolation_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 01a4fa76072ca4f3d8266c9fdbf61c816e53d09c673f8b8776b981b736c09e26

Original archive. Tau_Coherent_Interpolation_Control_2026-09-12.zip

Archive SHA-256. a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26

S8.1 Coherent arithmetic interpolation and joint tensor boundary control

Owner-directed continuation, 12 September 2026. Written mathematical arguments for review; the new analytic and matrix results in this note have not been Lean-checked.

S8.1.1 0. Accepted inputs and the change in the continuation

The new formalization receipt reports—and the repository metadata and workflow step summaries confirm—a successful check at 5d2772ea0a16d177ac3e01ff70394b90ba95a232 in PR #12. That certificate covers the specified retraction’s algebraic/topological consequences, the complete coefficient homotopy family, and the stated finite-chart dual maps. It does not certify the analytic construction of the theta retraction, support-changing synchronization homotopies, or the matrix estimates developed here. Its branch is left untouched.

The second attachment proves, for an unrestricted finite correction matrix B , the exact formula

$$G_B = G_{\perp,m} + (B + H_m^{-1}C_m)^* H_m (B + H_m^{-1}C_m). \quad (0.1)$$

Once $m+1 \geq \dim E_Z$, every positive metric ray is realized by an actual theta-boundary correction, with the arithmetic jets unchanged. Its further conclusion

$$\inf_B \epsilon_A(G_B) = \max_{\rho \in \mathbb{Z}} |2\Re \rho - 1| \quad (0.2)$$

therefore has to be retained. This note does not count unrestricted metric optimization as a new arithmetic estimate. It instead uses the unique least-norm representative for a fixed, increasing space of the original arithmetic boundaries. It proves compatibility of these representatives for all finite packets, an exact full-jet moment-matrix formula, and a joint total-degree tensor construction.

All conclusions are derived on the existing characteristic-zero theta complex over the marked tau-base. None is asserted to be an application of finite-field purity to a space merely renamed tau.

S8.1.2 1. Fixed scalar, cochain, and arithmetic maps

Retain the original scalar construction

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the entire square

$$\boxed{\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}} \quad (1.1)$$

The target of $p_{\mathbb{Z}}$ is infinite. This quotient remains part of the marked geometry over $\mathfrak{b}_\tau$. Coefficient maps below fix external absence and carry a represented scalar zero to the represented scalar zero of the target.

Use the unchanged spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \forall b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{m \neq 0} \phi(mx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Write $q : \mathcal{B} \rightarrow Q$. Its action is by the original D , not a shifted operator. Let

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}f_0 = g. \quad (1.2)$$

The established range identities are used with these exact constants. They give $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ entire. In particular every finite jet map used below annihilates ΘV .

For $n \geq 0$, retain

$$V_n = \text{span}\{\phi_*, D\phi_*, \dots, D^n\phi_*\}, \quad L_n = \Theta V_n,$$

$$Q_n = \mathcal{B}/L_n \longrightarrow Q_{n+1}, \quad \ker = L_{n+1}/L_n. \quad (1.3)$$

Their split reconstructions have labels V_n . The transport of a class in L_{n+1}/L_n is the supported zero at V_{n+1} ; it is not external absence. The derivative has type $Q_n \rightarrow Q_{n+1}$ because $DL_n \subseteq L_{n+1}$.

S8.1.3 2. A polynomial presentation of the actual finite representatives

Take a nonempty finite set Z of actual zeros of g , with their complete orders m_ρ , and put

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[t]/(h), \quad v_h(s) = g(s)/h(s). \quad (2.1)$$

The polynomial variable t acts on the source by D and is evaluated at s by Mellin transformation. Full orders make v_h nonzero at all selected centres. Let j_h be the complete Hermite-jet/Chinese-remainder map into E_h and retain

$$v_h = j_h v_h \in E_h^\times, \quad \varepsilon_h = v_h^{-1}. \quad (2.2)$$

The unit ε_h is not the scalar e . Its split scalar is $\varepsilon_h^\bullet$.

There is an actual $F_h \in \mathcal{B}$ with $h(D)F_h = f_0$ and $\mathcal{M}F_h = v_h$. It can be obtained successively from

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad (2.3)$$

whenever the required Mellin value vanishes. At zero, replace the integral by its negative from 0 to x , using that vanishing; at infinity use the displayed integral. The defining estimates of $\mathcal{B}$ prove rapid decay of every scaling derivative at both ends. Repeated zero jets permit iteration. Uniqueness follows from injectivity of Mellin and $h\mathcal{M}F = 0 \Rightarrow F = 0$.

Define the linear map

$$\boxed{\mathcal{T}_h : \mathbb{C}[t] \longrightarrow \mathcal{B}, \quad P \longmapsto P(D)F_h.} \quad (2.4)$$

It is injective and satisfies

$$\mathcal{M}\mathcal{T}_h(P) = \frac{g}{h}P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (2.5)$$

Here $J_h F = j_h \mathcal{M}F$. The first equation retains the whole completed function, not its values only on the critical line.

For $N = n + d$, let $\mathcal{P}_N = \mathbb{C}[t]_{\leq N}$, and define

$$\mathcal{J}_{h,N} : \mathcal{P}_N \twoheadrightarrow E_h, \quad P \longmapsto v_h[P]_h.$$

Its exact sequence is

$$\boxed{0 \longrightarrow \mathcal{P}_n \xrightarrow{\times h} \mathcal{P}_{n+d} \xrightarrow{\mathcal{J}_{h,n+d}} E_h \longrightarrow 0.} \quad (2.6)$$

Applying $\mathcal{T}_h$ sends the left map to the original theta relations L_n . Thus its split internal quotient has exactly the relation $P - P' \in h\mathcal{P}_n$, with the support label retained.

The original source section is

$$s_h(u) = \mathcal{T}_h(\text{rem}_h(\varepsilon_h u)), \quad J_h s_h = 1, \quad q s_h = \sigma_h. \quad (2.7)$$

No arbitrary metric or section has entered (2.6).

S8.1.4 3. Exact arithmetic moment matrices and the full-jet interpolation formula

The Hilbert observation is $\mathcal{H} = L^2((0, \infty), dx)$. The precise transform map is

$$\mathcal{H} \xrightarrow{\mathcal{L}} L^2(\mathbb{R}, dy) \xrightarrow{\mathcal{F}_+} L^2(\mathbb{R}, dt/(2\pi)),$$

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{F}_+\psi(t) = \int_{\mathbb{R}} \psi(y)e^{ity}dy, \quad (\mathcal{F}_+\mathcal{L}F)(t) = \mathcal{M}F(\tfrac{1}{2} + it). \quad (3.1)$$

Both arrows are unitary with these measures. The inverse Fourier factor is $1/(2\pi)$; no mass of a measure below is set equal to one. The same map carries D to multiplication by $s_t = 1/2 + it$.

Define the finite positive measure

$$d\nu_h(t) = \left| \frac{g(\frac{1}{2} + it)}{h(\frac{1}{2} + it)} \right|^2 \frac{dt}{2\pi}. \quad (3.2)$$

Any apparent singularity at a selected critical-line zero is removed by the entire function v_h . All moments are finite. In the fixed basis $1, t, \dots, t^N$ of $\mathcal{P}_N$, put

$$(M_{h,N})_{ij} = \int_{\mathbb{R}} \overline{s_t^i} s_t^j d\nu_h(t), \quad 0 \leq i, j \leq N. \quad (3.3)$$

In the integrand t is the real integration variable; the polynomial basis is evaluated at s_t . This prevents the polynomial variable from being confused with the spectral-height coordinate.

The matrix $M_{h,N}$ is positive definite. A polynomial with zero norm vanishes almost everywhere on a line on which v_h is nonzero almost everywhere, hence is zero. Write $J_{h,N}$ for the matrix of $\mathcal{J}_{h,N}$ in the fixed quotient basis $1, t, \dots, t^{d-1}$. It includes multiplication by the actual unit v_h .

Set

$$\mathcal{C}_{h,N} = J_{h,N} M_{h,N}^{-1} J_{h,N}^*, \quad C_{h,N} = M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1}. \quad (3.4)$$

These are finite matrices on declared coordinate spaces. $J_{h,N} C_{h,N} = 1$. The actual test-function map is

$$R_{h,n} = \mathcal{T}_h C_{h,n+d} : E_h \longrightarrow \mathcal{B}. \quad (3.5)$$

Theorem. This is precisely the preceding orthogonal representative $(1 - P_n)s_h$, where P_n is the $L^2(dx)$ orthogonal projection onto L_n . In particular,

$$qR_{h,n} = \sigma_h, \quad J_h R_{h,n} = 1, \quad G_{h,n} := R_{h,n}^* R_{h,n} = \mathcal{C}_{h,n+d}^{-1}. \quad (3.6)$$

Proof. Every polynomial with jet u is $C_{h,N}u + z$ with $J_{h,N}z = 0$. The cross term vanishes because

$$z^* M_{h,N} C_{h,N} u = z^* J_{h,N}^* \mathcal{C}_{h,N}^{-1} u = 0.$$

Its norm is therefore $u^* \mathcal{C}_{h,N}^{-1} u + z^* M_{h,N} z$. This proves the unique minimizer. By (2.6), its difference from the degree-below- d lift (2.7) is exactly a member of L_n . It must be $(1 - P_n)s_h$. This also proves (3.6).

Thus the subtraction-of-nearby-Grams formula of the previous note has the exact alternative (3.6): invert the full-jet interpolation matrix. This is a matrix-valued Christoffel construction with the arithmetic local unit retained. The finite least-norm identity itself is standard linear algebra; its application here identifies the actual original representative, source relation, and completed multiplier.

For one simple zero ρ , the same formula is

$$G_{\rho,n} = \frac{1}{|g'(\rho)|^2 \mathcal{K}_{\nu_{\rho,n+1}}(\rho, \rho)}, \quad \mathcal{K}_{\nu,N}(z, w) = (1, z, \dots, z^N) M_N^{-1} (1, w, \dots, w^N)^*. \quad (3.7)$$

The complex conjugate in the second evaluation is part of the displayed kernel. Higher orders use (3.4), not just point evaluations. The factor $g' = 2\xi'$ is retained.

S8.1.5 4. All finite packets fit the same source filtration

Let $h \mid H$ be polynomials for nested packets with full orders and set $m = H/h$. Since the newly added centres are disjoint, h and m are coprime. There is a $\mathbb{C}[t]$ -linear CRT injection

$$\iota_{hH} : E_h \hookrightarrow E_H,$$

which inserts zero jets at the new centres. It is multiplication by the appropriate CRT idempotent after lifting a residue; it sends the unit of E_h to that idempotent, so it is not being asserted to be a unital ring map.

The polynomial map $P \mapsto mP$ has the exact properties

$$\mathcal{T}_H(mP) = \mathcal{T}_h(P), \quad J_H \mathcal{T}_h(P) = \iota_{hH} \mathcal{J}_h(P), \quad (4.1)$$

and is isometric from ν_h to ν_H because $|v_H mP| = |v_h P|$. The CRT equations and degree bounds prove

$$s_H \iota_{hH} = s_h.$$

The original L_n is independent of the packet, so

$$\boxed{R_{H,n} \iota_{hH} = R_{h,n}, \quad \iota_{hH}^* G_{H,n} \iota_{hH} = G_{h,n}, \quad \iota_{hH}^* W_{H,n} \iota_{hH} = W_{h,n}.} \quad (4.2)$$

The action also intertwines: $A_H \iota_{hH} = \iota_{hH} A_h$, where $A_h = M_t$. Thus a certificate on the larger packet restricts by this actual congruence to the smaller packet. The inverse interpolation matrices satisfy the corresponding inverse-compression formula; inverse and compression are not commuted.

The finite sections consequently glue on the algebraic direct limit of the actual packet injections. This statement does not assert convergence of an infinite sum of packet projectors or identify the whole Q with that direct limit.

At every support label, lift these maps as $(\lambda, u) \mapsto (\lambda, \iota_{hH} u)$ and $(\lambda, u) \mapsto (\lambda, R_{h,n} u)$. Added zero jets remain represented zeros. The label map in (1.3) is still the map that admits one more source relation.

S8.1.6 5. A global density result, with its precise cohomological consequence

The single derivative family $\{D^n f_0 : n \geq 0\}$ is already dense in $\mathcal{H}$.

Proof. Put $d\nu(t) = |g(1/2 + it)|^2 dt / (2\pi)$. It has an exponential moment of every order $c < \pi/2$. To verify this, the alternating-series formula and partial summation give

$$|\zeta(\tfrac{1}{2} + it)| \leq \frac{2|\tfrac{1}{2} + it|}{\sqrt{2} - 1}.$$

The gamma vertical-line asymptotic gives, for sufficiently large $|t|$,

$$|g(\tfrac{1}{2} + it)|^2 \leq C(1 + |t|)^{11/2} e^{-\pi|t|/2}.$$

The constant C is a fixed bound, not a numerical enclosure claimed here. The gamma fact used is NIST DLMF 5.11.9. The same exponential-moment property holds for ν_h , since v_h is entire at the finitely many possible denominator zeros and has polynomially modified tails.

If $f \in L^2(\nu)$ is orthogonal to every polynomial, the finite measure $\bar{f} d\nu$ has a holomorphic Fourier transform in a nonempty horizontal strip, by Cauchy–Schwarz and the exponential moment. All

derivatives at the origin vanish. Analytic uniqueness makes its transform zero; Fourier uniqueness for finite measures makes the measure zero. Thus polynomials are dense in $L^2(\nu)$.

Multiplication by the actual function $g(1/2 + it)$ is a unitary map $L^2(\nu) \rightarrow L^2(dt/(2\pi))$: its inverse divides by g away from its measure-zero set of real-line zeros. The inverse is a map of weighted L^2 spaces, not holomorphic division through an arithmetic zero. Combining it with (3.1) sends polynomials to $P(D)f_0$. This proves the assertion.

It follows for each fixed packet that

$$\boxed{G_{h,n} \longrightarrow 0 \quad \text{in every matrix entry and in operator norm.}} \quad (5.1)$$

Indeed $P_n \rightarrow 1$ strongly on $\mathcal{H}$, and the finite number of columns of s_h is fixed. Since each $G_{h,n}$ is positive definite, the least eigenvalue of its inverse tends to infinity. Every nonzero arithmetic row functional $\ell : E_h \rightarrow \mathbb{C}$ therefore has

$$\boxed{\|\ell\|_{G_{h,n}^\vee}^2 = \ell \mathcal{C}_{h,n+d} \ell^* \longrightarrow \infty.} \quad (5.2)$$

These facts do not kill a cohomology class. The actual maps are

$$\mathcal{B}/L_n \longrightarrow \mathcal{H}/L_n, \quad Q = \mathcal{B}/\Theta V \longrightarrow \mathcal{H}/\overline{\Theta V}^{\mathcal{H}} = 0. \quad (5.3)$$

The first map is injective because L_n is finite-dimensional and consists of the same smooth functions in both spaces. The second map forgets the stronger topology and has the whole Q as kernel. Its split lift lands at the represented zero in $G(0)$, not at τ . $G(0)$ retains both elements. An active class may therefore tend to zero under the L^2 observation while remaining fixed by $qR_{h,n} = \sigma_h$ and $J_h R_{h,n} = 1$.

The constructive consequence is to estimate relative boundary flux against the complete interpolation cost (3.4), not to use the absolute smallness of $G_{h,n}$ as a weight bound.

S8.1.7 6. Tail representation and same-metric reflection control

Retain the unscaled orthogonal vectors from the previous calculation:

$$u_0 = f_0, \quad u_n = (1 - P_{n-1})D^n f_0, \quad \mathfrak{h}_n = \|u_n\|^2, \\ Du_n = u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1). \quad (6.1)$$

The diagonal $1/2$ and the negative last coefficient follow from $D^* = 1 - D$ on these functions. Neither the u_n nor their norms are rescaled. Define $r_n = \mathfrak{h}_n^{-1}u_n^* s_h$. The new density result makes the entire tail formula exact:

$$\boxed{G_{h,n} = \sum_{j>n} \mathfrak{h}_j r_j^* r_j.} \quad (6.2)$$

This series converges in finite matrix norm. Its source map is $u \mapsto (r_j u)_{j>n}$ into the weighted sequence space with norm $\sum_{j>n} \mathfrak{h}_j |c_j|^2$. The map retains every coefficient; it is an isometry for $G_{h,n}$.

The previous boundary calculation, or telescoping (6.1), gives

$$\boxed{W_{h,n} = A_h^* G_{h,n} + G_{h,n} A_h - G_{h,n} = \mathfrak{h}_{n+1}(r_{n+1}^* r_n + r_n^* r_{n+1}).} \quad (6.3)$$

It pairs two explicit maps into L_{n+1}/L_n . At the original support V_n the derivative defect is $-[u_{n+1}]r_n$; at V_{n+1} it becomes the supported zero. This is also the input to the now-checked coefficient homotopy family, with $\kappa_n = K_n J_h$ and $\kappa_n \Theta = 0$.

For a reflection-stable packet define the actual antilinear involution

$$j_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}.$$

On functions use $j_B F(x) = x^{-1} \overline{F(1/x)}$. It is antiunitary on $\mathcal{H}$ and satisfies $Dj_B = j_B(1 - D)$. The specific section (2.7) satisfies $j_B s_h = s_h j_h$: since $h^\dagger = (-1)^d h$, both the unit and the numerator have the corresponding $(-1)^d$ factor. Also $j_B L_n = L_n$, so uniqueness of orthogonal projection gives $j_B R_{h,n} = R_{h,n} j_h$.

Write $j_h u = C_h \bar{u}$ in the retained coordinates. Then

$$\boxed{C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}.} \quad (6.4)$$

Consequently the generalized spectrum of $(W_{h,n}, G_{h,n})$ is symmetric under $\lambda \mapsto -\lambda$ **for this same canonical metric**, with full multiplicities retained. An upper inequality $W_{h,n} \preceq \epsilon G_{h,n}$ therefore implies the lower inequality $W_{h,n} \succeq -\epsilon G_{h,n}$ by applying j_h to the input vector.

The residue duality is still the separate explicit morphism with matrix S_h , and its pulled-back metric is $S_h^* G_{h,n}^{-1} S_h$. The relation to the current metric is the linear map $G_{h,n}^{-1} S_h$ into its antilinear dual coordinates. The arithmetic Weil form remains $S_h J_g$; no positivity is supplied merely by (6.4).

For completeness, retain the full rank-two formula

$$a_n = r_n G_{h,n}^{-1} r_n^*, \quad b_n = r_{n+1} G_{h,n}^{-1} r_{n+1}^*, \quad c_n = r_n G_{h,n}^{-1} r_{n+1}^*.$$

The least control excess is

$$\epsilon_{h,n} = \mathfrak{h}_{n+1} \left(|\Re c_n| + \sqrt{a_n b_n - (\Im c_n)^2} \right). \quad (6.5)$$

For reflection-stable Z , the additional proved identity is $\Re c_n = 0$: take the trace of $G_{h,n}^{-1} W_{h,n}$, which is $2\Re \operatorname{Tr} A_h - d = 0$. Thus (6.5) also equals $\mathfrak{h}_{n+1} \sqrt{a_n b_n - (\Im c_n)^2}$. Both equations are retained. Equation (3.6) supplies every occurrence of $G_{h,n}^{-1}$ without a difference of nearly equal Gram matrices.

S8.1.8 7. Joint total-degree control on the actual product complex

The tensor degree will be denoted k , independently of the relation degree n and the packet multiplicities. Put

$$E_h^{(k)} = E_h^{\otimes k}, \quad A_h^{(k)} = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A_h \otimes I^{\otimes(k-j)}.$$

The existing product complex over $\mathfrak{b}_\tau$ has differential

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k.$$

Define $\mathcal{P}_M^{(k)} = \mathbb{C}[t_1, \dots, t_k]_{\text{total degree} \leq M}$ for $M \geq k(d-1)$ and

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) (F_h^{\otimes k}). \quad (7.1)$$

Its Mellin transform is $P(s_1, \dots, s_k) \prod_j g(s_j)/h(s_j)$. Its full jet map $J_{k,M}$ is reduction modulo each $h(t_j)$ followed by multiplication by $v_h^{\otimes k}$. The target is $E_h^{(k)}$ and this map is surjective. The tensor remainder basis has total degree at most $k(d-1)$.

The exact relation space is

$$\mathcal{I}_{k,M} = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)). \quad (7.2)$$

Polynomial division in the fixed order $t_1, \dots, t_k$ represents each member as $\sum_j h(t_j)P_j$, with $\deg P_j \leq M-d$. It terminates because every h is monic and replacing a leading power lowers total degree. Under (7.1), each term is an actual top cochain boundary: in the j -th source factor use ϕ_* in place of F_h and give its primitive the factor $(-1)^{j-1}$. The differential contributes the second $(-1)^{j-1}$. The resulting boundary is $+h(t_j)P_j$ with no removed sign.

Thus these joint corrections are maps in the original product complex, not freely chosen positive tensor metrics. Set $\mathcal{B}_{k,M} = \mathcal{T}_h^{(k)} \mathcal{I}_{k,M}$.

The moment matrix on the monomials of total degree at most M is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s_t^{\alpha_j}} s_t^{\beta_j} d\nu_h(t). \quad (7.3)$$

Its positive definiteness follows because a polynomial cannot vanish almost everywhere on the full real product unless it is zero. It is a principal submatrix of the stated product moment matrix, not a declaration that the total-degree space is a tensor product of smaller univariate spaces.

The actual joint least-norm representative and its metric are

$$R_{k,M} = \mathcal{T}_h^{(k)} M_{k,M}^{-1} J_{k,M}^* (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1},$$

$$G_{k,M} = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (7.4)$$

The proof is the same explicit constrained-minimum calculation as (3.6), now with the relation ideal (7.2). In particular its arithmetic class is exactly $\sigma_h^{\otimes k}$ and every jet is fixed.

For a rectangular degree budget N one has the actual inclusion

$$\mathcal{P}_N^{\otimes k} \hookrightarrow \mathcal{P}_{kN}^{(k)}.$$

It carries the old product representative into the feasible set of the new problem. Therefore

$$G_{k,kN} \preceq G_{h,N-d}^{\otimes k} \quad (N \geq d). \quad (7.5)$$

This is an inequality from an inclusion of admissible original boundaries. It is not asserted to imply the same ordering of control forms, whose dependence on A is retained separately.

S8.1.9 8. The product error is one new total-degree layer

Let $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, and let $P_{\mathcal{B}_{k,M}}$ be its orthogonal projection onto the finite-dimensional actual boundary space. With the fixed low-degree section $s_h^{\otimes k}$,

$$R_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) s_h^{\otimes k}.$$

Put

$$\mathcal{E}_{k,M} = (1 - P_{\mathcal{B}_{k,M}})\mathcal{B}_{k,M+1}.$$

Because the jet map is already onto at both degrees,

$$\dim \mathcal{E}_{k,M} = \binom{M+k}{k-1}. \quad (8.1)$$

This is $\dim \mathcal{P}_{M+1}^{(k)} - \dim \mathcal{P}_M^{(k)}$; the d^k jet constraints cancel from the dimension difference.

Let $D^{(k)} = \sum_j D_j$. Its induced action on jets is $A_h^{(k)}$. The actual derivative defect lies in $\mathcal{B}_{k,M+1}$:

$$B_{k,M} = D^{(k)} R_{k,M} - R_{k,M} A_h^{(k)}.$$

Define the two explicitly typed maps into the same new relation layer

$$Y_{k,M} = P_{\mathcal{E}_{k,M}} R_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M},$$

$$C_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) B_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M}.$$

Integration by parts gives $(D^{(k)})^* + D^{(k)} = kI$ on these functions. Orthogonality then proves

$$W_{k,M} := (A_h^{(k)})^* G_{k,M} + G_{k,M} A_h^{(k)} - k G_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad (8.2)$$

$$G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}, \quad \text{rank } W_{k,M} \leq 2 \binom{M+k}{k-1}. \quad (8.3)$$

For (8.3), use the orthogonal decomposition $\mathcal{B}_{k,M+1} = \mathcal{B}_{k,M} \oplus \mathcal{E}_{k,M}$. Its projection sends $R_{k,M}$ to $Y_{k,M}$ and the new representative is $R_{k,M} - Y_{k,M}$.

The source quotient map is the same degree transport as before:

$$\mathcal{B}_{k,M+1}/\mathcal{B}_{k,M} \xrightarrow{\sim} \mathcal{E}_{k,M}, \quad [z] \mapsto (1 - P_{\mathcal{B}_{k,M}})z.$$

A derivative class in this kernel becomes the supported zero only at the next total-degree label. The primitive retains which tensor factor supplied the theta boundary and its cochain sign. Synchronization by $(1, e)$ in each required two-leg factor supplies the common joint support, while the original mask remains recorded.

Thus the product estimate need not simply copy k identical one-factor errors: (8.2) gives a new jointly optimized boundary-layer expression with completely specified moment data and maps. Its rank bound alone does not bound its generalized eigenvalues.

S8.1.10 9. Same-metric reflection and residue duality on the product

For a reflection-stable packet, $j_h^{\otimes k}$ is the antilinear involution on $E_h^{(k)}$, defined on pure tensors by applying j_h to every factor. The corresponding function reflection is $F(x_1, \dots, x_k) \mapsto \prod_j x_j^{-1} \overline{F(x_1^{-1}, \dots, x_k^{-1})}$.

It preserves both the total-degree numerator space and its relation ideal. Since $(g/h)^\dagger = (-1)^d (g/h)$, its action on a numerator is $(-1)^{kd} P^\dagger$; this factor is retained. Uniqueness of the minimum in (7.4) proves reflection compatibility of $R_{k,M}$. Therefore the analogue of (6.4) gives symmetry of the generalized spectrum of $(W_{k,M}, G_{k,M})$ on the same metric.

Residue duality remains the explicitly specified map with matrix $S_h^{\otimes k}$ and target line $\mathcal{L}_k$, on which dilation acts by a^k . It satisfies

$$(A_h^{(k)})^* S_h^{\otimes k} + S_h^{\otimes k} A_h^{(k)} = k S_h^{\otimes k}.$$

Any fixed degree-orientation scalar in the cochain evaluation is retained separately; the ordered product of the scalar residue pairings here has matrix exactly $S_h^{\otimes k}$.

Its dual metric and control are

$$G_{k,M}^D = (S_h^{\otimes k})^* G_{k,M}^{-1} S_h^{\otimes k},$$

$$W_{k,M}^D = -(S_h^{\otimes k})^* G_{k,M}^{-1} W_{k,M} G_{k,M}^{-1} S_h^{\otimes k}. \quad (9.1)$$

The Jacobian contraction is still the original multiplication trace, with every multiplicity, and is unchanged by the source boundaries used to construct (7.4). Neither (9.1) nor reflection alone establishes an arithmetic upper bound.

S8.1.11 10. Quantitative certificate and the Deligne comparison

For one or several variables write M for the positive moment matrix, J for its fixed full-jet map, and $\mathcal{C} = JM^{-1}J^*$. These actual matrices now replace an unrestricted metric search.

A useful deterministic enclosure is as follows. Suppose a Hermitian approximation $\widehat{M}$ has a verified bound $\|M - \widehat{M}\| \leq \eta$ and $\widehat{M} \succeq \mu I$ for a retained $\mu > \eta$. Then

$$\|M^{-1} - \widehat{M}^{-1}\| \leq \frac{\eta}{\mu(\mu - \eta)}, \quad \|\mathcal{C} - J\widehat{M}^{-1}J^*\| \leq \|J\|^2 \frac{\eta}{\mu(\mu - \eta)}. \quad (10.1)$$

This is the resolvent identity for inverses. The positive lower bound on $\mathcal{C}$ must also be retained before a second inversion. With $\widehat{G}$ and $\widehat{W}$ and verified errors η_G, η_W , the sufficient matrix certificate is

$$\widehat{G} - \eta_G I \succ 0, \quad \epsilon \widehat{G} \pm \widehat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0. \quad (10.2)$$

No numerical interval enclosure for the actual infinite arithmetic moments is claimed merely from a floating-point calculation. The previous exterior-theta sums give another exact representation of their one-variable inputs; all tails and roundoff must be supplied to (10.1).

Deligne's retained Corollaries 3.3.4–3.3.6 place upper and dual lower bounds on the same compact-to-ordinary image; the tensor step in 3.2.13 improves an estimate only after additional geometric control. This note supplies a canonical coherent family, its exact source relation ideal on products, and the same-metric reflection/dual operations with which such an upper estimate can now be tested. It does not import finite-field hypotheses into the tau chart.

For an actual eigenvector $A_h v = \rho v$, every constructed tensor metric still has the exact diagnostic

$$\frac{(v^{\otimes k})^* W_{k,M} v^{\otimes k}}{(v^{\otimes k})^* G_{k,M} v^{\otimes k}} = k(2\Re \rho - 1). \quad (10.3)$$

The quantity to bound is therefore the generalized boundary form (8.2) relative to (7.4). A dimension/rank estimate, a decrease of a Gram matrix, or unrestricted metric realizability is not substituted for that relative estimate. The new constructive route is the exact finite full-jet moment problem together with its next-layer flux, not another homotopy choice.

S8.1.12 11. Reading, provenance, and validation

Newly read: both supplied formalization/control notes in full; the preceding local `Tau_Homotopy_Boundary_Control/NOTE.tex`, especially sections 4–10; the retained Deligne English TeX excerpt 3.3.4–3.3.6 and French 6.2.4–6.2.7. This is not a new audit of every page of Weil II. No OCR or PDF extraction was used. Source hashes are in the package reading record.

External context checked: NIST DLMF 5.11.9 for the gamma bound; Connes–Consani–Moscovici, arXiv:2403.01247, abstract, for prior moment/Jacobi constructions from squared absolute local factors; Romik, arXiv:1902.06330, abstract, for prior orthogonal-polynomial expansions of Ξ ; Lasserre–Pauwels, arXiv:1606.03858, abstract, for inverse-moment/Christoffel context. Only the gamma asymptotic is used as an external estimate; the other papers are priority/context references, not sources of an unread theorem. The arguments above prove the specific maps used here.

The accompanying exact checker uses declared finite discrete calibration measures and rational complex polynomial data. It checks the general finite matrix identities, joint relation ranks, signs, full repeated-root reductions, reflection, naturality, and comparison metrics. Its model polynomial is never labelled an arithmetic zeta counterexample. Analytic density, asymptotic gamma bounds, and the original theta range identities are written arguments, not certified by those finite tests. No new Lean execution is claimed. No RH or GRH conclusion is claimed.

S9 Full-jet interpolation kernels and the actual next relation layer

Source attribution. Complete TeX witness from the dated kernel-layer supplement, delivered 12 September 2026. Its handoff identifies PR 14 as the arithmetic source and the existing PR 15 revision as its formal baseline. The supplement’s add-only patch was verified locally; its receipt records no remote branch modification. These NOTE bytes are identified by the source and archive hashes below; neither baseline revision is represented as their Git head.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Kernel_Layer_Integration/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Arithmetic source baseline PR 14

Cited existing PR 15 formal baseline. 21970bbf4760d0bbca512a4a7996a2e38f968947

Source SHA-256. be8485471fc7dac41d97e4a9fd52413da14e83fbc42c6c55c01a5f0908caa06c

Original archive. Tau_Kernel_Layer_Integration_2026-09-12(1).zip

Archive SHA-256. ebd927c8a9afa90600f5320bd4189ecb48085889835205057b1bfafdb68d23d2

S9.1 Full-jet interpolation kernels and the actual next relation layer

S9.1.1 Integration supplement — 12 September 2026

This supplement reviews the supplied **Coherent arithmetic interpolation and joint tensor boundary control** package and derives an additional finite-stage bridge. The source’s canonical representative, original theta boundaries, full multiplicities, and support transport are retained. The new formulas below have written proofs and exact finite regression tests; they are **not new Lean certificates**.

The supplied ZIP has SHA-256 a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26. All 29 entries of its manifest verified. Its note, handoff, and checker match the published Git blobs at PR #14, commit 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc. The source checker was independently rerun: nineteen tests passed normally and under optimized Python, with identical result records; the intentionally false twentieth test failed in both modes. See SOURCE_REVIEW_RECEIPT.json.

The comparison with PR #15 is limited precisely: its existing five-module receipt certifies its stated support-changing, relation-layer, finite-Krylov and rank-two interfaces. This supplement does not extend that receipt to the source’s theta analytics, all-degree orthogonal-polynomial construction, or the new tensor-kernel calculations.

S9.1.2 1. Source results retained, rather than reconstructed differently

Write the source polynomial degree as N , to distinguish it from a moment matrix. Fix tensor degree $k \geq 1$, a nonempty packet polynomial h of degree d , and $N \geq k(d - 1)$. The packet includes the complete orders of the selected actual zeros. Retain

$$g = 2\xi, \quad E = E_h^{\otimes k}, \quad A = A_h^{(k)}, \quad \mathcal{J}_N : \mathcal{P}_N^{(k)} \twoheadrightarrow E. \quad (1)$$

The last map is reduction modulo the original ideal $(h(t_1), \dots, h(t_k))$, followed by multiplication by the original local unit $v_h^{\otimes k}$. Its kernel is $\mathcal{I}_{k,N}$. The source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{MT}(P) = P(s_1, \dots, s_k) \prod_i \frac{g(s_i)}{h(s_i)}. \quad (2)$$

Let $\mathcal{L}_N = \mathcal{TI}_{k,N}$, $P_N = P_{\mathcal{L}_N}$, and let $\sigma : E \hookrightarrow Q^{\widehat{\otimes} k}$ be the source arithmetic inclusion. Its canonical representative and forms are

$$R_N = (I - P_N)s_h^{\otimes k}, \quad G_N = R_N^* R_N \succ 0, \quad q^{(k)} R_N = \sigma, \quad J^{(k)} R_N = I, \quad (3)$$

$$B_N = D^{(k)} R_N - R_N A, \quad W_N = A^* G_N + G_N A - k G_N. \quad (4)$$

The supplied note proves $B_N(E) \subseteq \mathcal{L}_{N+1}$ and, with $\mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp$,

$$Y_N = P_{\mathcal{E}_N} R_N, \quad C_N = (I - P_N) B_N, \quad W_N = -(Y_N^* C_N + C_N^* Y_N). \quad (5)$$

These are source results (source note, §§2–4 and 7–9), not claimed as discoveries of this supplement. Their finite-dimensional derivations check: the constrained minimum is over precisely the original ideal, the derivative increases total degree by one, and orthogonal projection removes only the admitted relation space. The source's exact sequence and packet naturality require full orders. Its total-degree statements require the threshold in (1). Its analytic function-space identities remain the stated analytic inputs, not consequences of a finite fixture.

The original coefficient square $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ over $\mathbb{Z} \rightarrow \mathbb{C}$, with projections $p_{\mathbb{Z}}, p_{\mathbb{C}}$, is unchanged. A vector in an admitted relation maps to the receiving fibre's supported zero, never to external absence τ .

S9.1.3 2. Calculate the inverse interpolation metric by full-jet columns

Use exactly the source measure

$$d\nu_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}. \quad (6)$$

Let $p_j^{(h)}(s)$ be its monic orthogonal polynomials, with the polynomial variable evaluated at $s = 1/2 + it$, and retain their positive squared norms

$$\kappa_j^{(h)} = \int |p_j^{(h)}(1/2 + it)|^2 d\nu_h(t) > 0. \quad (7)$$

These are **not** the earlier theta-Krylov vectors u_j , nor their norms $\mathfrak{h}_j$. In particular they depend on h ; they are not assigned norm one. Existence through any finite degree follows from the source measure's positive-definite moment matrices. No limiting assertion is needed here.

Suppress h in notation. For a multi-index α , define

$$p_\alpha = \prod_i p_{\alpha_i}, \quad \kappa_\alpha = \prod_i \kappa_{\alpha_i}, \quad a_j = v_h[p_j]_h \in E_h, \quad a_\alpha = \bigotimes_i a_{\alpha_i} \in E. \quad (8)$$

Every a_j is the full jet, including the local unit. Product orthogonality gives $\langle \mathcal{T}p_\alpha, \mathcal{T}p_\gamma \rangle = \kappa_\alpha \delta_{\alpha\gamma}$. These polynomials form a basis of $\mathcal{P}_N^{(k)}$, because the monic triangular basis change preserves total degree.

Set

$$K_N := G_N^{-1} = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha}. \quad (9)$$

Proof. In this basis the source moment matrix is $\text{diag}(\kappa_\alpha)$ and the full-jet matrix has columns a_α . Substitute into the source identity $G_N^{-1} = J_N M_N^{-1} J_N^*$. A change of polynomial basis does not change that product. This proves (9) for the same canonical metric, not a newly selected metric.

Its representative has the explicit coefficient formula

$$R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (10)$$

In particular $R_N K_N$ is an actual map from finite antidual coordinates into the original test-function space. It satisfies $J^{(k)} R_N K_N = K_N$ and $q^{(k)} R_N K_N = \sigma K_N$. The invertible coordinate change K_N is retained, not identified with the identity.

S9.1.4 3. An explicit basis of the original next relation layer

Let $\Sigma_{N+1} = \{\beta \in \mathbb{N}^k : |\beta| = N + 1\}$. Define actual test functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \beta \in \Sigma_{N+1}. \quad (11)$$

They are in $\mathcal{E}_N$. First, their full jets vanish by (3), so their numerator polynomials lie in the original ideal at degree at most $N + 1$; thus $e_\beta \in \mathcal{L}_{N+1}$. Second, $\mathcal{T}p_\beta$ is orthogonal to all degree-at-most- N polynomials and $R_N a_\beta$ is orthogonal to $\mathcal{L}_N$. Hence $e_\beta \perp \mathcal{L}_N$.

Define

$$\mathcal{E}_N : \mathbb{C}^{\Sigma_{N+1}} \longrightarrow \mathcal{E}_N, \quad c \longmapsto \sum_{\beta} c_\beta e_\beta, \quad (12)$$

$$U_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad D_N = \text{diag}(\kappa_\beta)_{\beta \in \Sigma_{N+1}}. \quad (13)$$

The map $\mathcal{E}_N$ is an isomorphism. Independence follows by taking the degree- $N + 1$ polynomial part of (11). For spanning, subtract from a relation its degree- $N + 1$ combination of the e_β . The remainder is a relation of degree at most N , hence belongs to $\mathcal{L}_N$. An element of $\mathcal{E}_N$ has no such remaining component.

Its Gram matrix and pairing with the representative are exactly

$$H_N := \mathcal{E}_N^* \mathcal{E}_N = D_N + U_N^* G_N U_N, \quad \mathcal{E}_N^* R_N = -U_N^* G_N. \quad (14)$$

The cross terms vanish because the new-degree polynomials are orthogonal to the old polynomial space. In particular $H_N \succ 0$, and no original norm has been suppressed.

This identifies the existing quotient equivalence in coordinates:

$$\mathbb{C}^{\Sigma_{N+1}} \xrightarrow{\mathcal{E}_N} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1} / \mathcal{L}_N. \quad (15)$$

The inverse quotient-to-layer map is the source's $[z] \mapsto (I - P_N)z$. For each nonzero coordinate vector, its class in this relative quotient is nonzero, and its image in the next admitted quotient is zero. At the split level this remains the next supported zero.

For an explicit theta primitive, form $p_\beta - \sum_{|\alpha| \leq N} p_\alpha a_\alpha^* G_N a_\beta / \kappa_\alpha$. Its full jet is zero. Fixed-order monic division expresses it as $\sum_i h(t_i) Q_{i,\beta}$, with $\deg Q_{i,\beta} \leq N + 1 - d$. In the i -th cochain factor use the original ϕ_* in place of F_h , and multiply its primitive by $(-1)^{i-1}$. The differential supplies the same sign, producing the displayed relation with positive sign. This supplies the actual primitive, not just a dimension count.

S9.1.5 4. Both boundary maps now have explicit coordinates

Define the matrix $F_N : E \rightarrow \mathbb{C}^{\Sigma_{N+1}}$ by its rows

$$(F_N)_\beta = \sum_{i: \beta_i > 0} \frac{a_{\beta - \mathbf{e}_i}^*}{\kappa_{\beta - \mathbf{e}_i}}. \quad (16)$$

Then

$$\boxed{Y_N = -\mathcal{E}_N H_N^{-1} U_N^* G_N, \quad C_N = \mathcal{E}_N F_N G_N.} \quad (17)$$

The first equation is the actual orthogonal projection formula using (14). For the second, multiply the finite expansion (10) by $s_1 + \cdots + s_k$. Monicity implies that its degree- $N + 1$ coefficient at p_β is exactly $(F_N)_\beta G_N u$. The term $R_N A u$ has degree at most N . Subtract $\mathcal{E}_N F_N G_N u$ from $B_N u$: the result has degree at most N and zero full jets, hence is in $\mathcal{L}_N$. Projecting by $I - P_N$ proves the equation.

Consequently the old quotient records

$$\boxed{[B_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N} = [\mathcal{E}_N F_N G_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N}.} \quad (18)$$

It becomes the supported zero under the next relation transition. The symbol $\mathcal{B}^{(k)}$ denotes the original product test-function space used by the source; no derivative on all of its L^2 completion is defined here.

S9.1.6 5. The control form is a top-degree full-jet endpoint

Define

$$\boxed{V_N := A K_N + K_N A^* - k K_N.} \quad (19)$$

Substituting (17) into the source boundary identity (5), and using $\mathcal{E}_N^* \mathcal{E}_N = H_N$, gives

$$\boxed{W_N = G_N (U_N F_N + F_N^* U_N^*) G_N.} \quad (20)$$

Meanwhile direct multiplication of (4) by K_N on both sides gives $K_N W_N K_N = V_N$. Thus

$$\boxed{V_N = U_N F_N + F_N^* U_N^* = \sum_{|\alpha|=N} \sum_{i=1}^k \frac{a_{\alpha + \mathbf{e}_i} a_\alpha^* + a_\alpha a_{\alpha + \mathbf{e}_i}^*}{\kappa_\alpha}.} \quad (21)$$

This is a finite endpoint identity derived through the actual theta relation layer. It does not assume a small boundary, discard mixed terms, or replace the packet operator. It can also be obtained by finite Christoffel–Darboux telescoping. The standard scalar orthogonal-polynomial kernel and telescoping identities are recorded by NIST (n.d., §18.2(v)); the full-jet, original-layer identification (11)–(21) is proved here rather than imported from that scalar formula.

For this alternative telescoping proof, multiplication satisfies $s p_j = p_{j+1} + \beta_j p_j - (\kappa_j / \kappa_{j-1}) p_{j-1}$, with $\operatorname{Re} \beta_j = 1/2$. The imaginary part need not vanish for the packet-dependent measure. Only $\beta_j + \overline{\beta_j} = 1$ is needed. Interior adjacent terms cancel; (21) is the uncanceled total-degree boundary. The new tests include an asymmetric measure with a nonzero imaginary diagonal coefficient.

Regroup by α and put $b_\alpha = \sum_i a_{\alpha + \mathbf{e}_i}$. Equation (21) factors through the two sets of columns $(a_\alpha)_{|\alpha|=N}$ and $(b_\alpha)_{|\alpha|=N}$. Therefore

$$\boxed{\text{rank } W_N = \text{rank } V_N \leq \min \left\{ d^k, 2 \binom{N+k-1}{k-1} \right\}.} \quad (22)$$

This sharpens the source's general next-layer bound $2 \binom{N+k}{k-1}$ for this product-polynomial construction. It does **not** change the actual dimension of $\mathcal{E}_N$, which remains $\binom{N+k}{k-1}$. The sharpening comes from the scalar summed derivative and regrouping its top-degree columns. For $k = 2, d = 4, N = 6$, the two structural bounds are 14 and 16 on a 16-dimensional packet; the exact regression tests cover this nonvacuous case. Neither rank bound is a small-eigenvalue estimate.

S9.1.7 6. Estimate the same form without a second inversion

Since K_N is Hermitian positive definite,

$$\boxed{-\epsilon G_N \preceq W_N \preceq \epsilon G_N \iff -\epsilon K_N \preceq V_N \preceq \epsilon K_N.} \quad (23)$$

This is congruence by the specified invertible map K_N . More explicitly, for $u = K_N \lambda$, $u^* G_N u = \lambda^* K_N \lambda$ and $u^* W_N u = \lambda^* V_N \lambda$. This identifies the generalized Rayleigh quotients, including multiplicities of generalized eigenvalues. The original generator A is not replaced.

The source's same-metric reflection therefore still transfers an upper estimate to a lower estimate for precisely the original G_N . Under the natural weight- k antidual action $A^D = kI - A^*$, the defect of the antidual metric K_N is $-V_N$, not V_N . The dual sign and weight are both retained.

The reflection map itself can be retained explicitly. If the source involution is $j(u) = C\bar{u}$, its antidual involution is $j^\#(\lambda) = C^T \bar{\lambda}$. From $C\bar{C} = I$ and $C^* G_N C = \overline{G_N}$, inversion gives $(C^T)^* K_N C^T = \overline{K_N}$. The control identity transports to $(C^T)^* V_N C^T = -\overline{V_N}$. Thus the two-sided equivalence retains the actual involution, not only spectral symmetry.

Equation (23) avoids the *second* inversion $G_N = K_N^{-1}$ in the final inequality test. It does not eliminate the need to enclose moments, full-jet coefficients, or the conditioning of the first construction of K_N .

A deterministic finite certificate can therefore work directly with $\hat{K}$, $\hat{A}$ and error bounds $\|K_N - \hat{K}\| \leq \eta_K$, $\|A - \hat{A}\| \leq \eta_A$, where $\hat{K}$ is Hermitian. Put

$$\hat{V} = \hat{A}\hat{K} + \hat{K}\hat{A}^* - k\hat{K}, \quad \eta_V = (2\|\hat{A}\| + k)\eta_K + 2\eta_A(\|\hat{K}\| + \eta_K). \quad (24)$$

Expanding both perturbations proves $\|V_N - \hat{V}\| \leq \eta_V$. Sufficient conditions, for $\epsilon \geq 0$, are

$$\hat{K} - \eta_K I \succ 0, \quad \epsilon \hat{K} \pm \hat{V} - (\epsilon \eta_K + \eta_V) I \succeq 0. \quad (25)$$

No enclosure of the actual infinite arithmetic moments is asserted in this supplement.

S9.1.8 7. The metric update and the univariate connection

The positive update from (9) and its inverse are

$$\boxed{K_{N+1} = K_N + U_N D_N^{-1} U_N^*,} \quad (26)$$

$$\boxed{G_{N+1} = G_N - G_N U_N (D_N + U_N^* G_N U_N)^{-1} U_N^* G_N.} \quad (27)$$

Equation (27) follows either by direct multiplication with (26), or from $R_{N+1} = R_N - Y_N$ and the orthogonal decomposition. The middle matrix is exactly the actual layer Gram matrix H_N in (14), not an unrelated auxiliary matrix. The lost Gram form is $Y_N^* Y_N$.

For $k = 1$ and $N = n + d$, (11) recovers the earlier monic theta vector:

$$\boxed{e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1}.} \quad (28)$$

Indeed the numerator of (11) is monic of degree $n + d + 1$, divisible by monic h , and its quotient has degree $n + 1$ and leading coefficient one. Applying $\mathcal{T}_h$ gives f_{n+1} plus an element of L_n , orthogonal to L_n . This uniquely identifies u_{n+1} . Thus the packet-dependent polynomial construction is connected to the packet-independent original theta norm by an exact identity; those two norm families were not conflated.

S9.1.9 8. Degree recursion and the remaining estimate

For $A_j = a_j a_j^* / \kappa_j$, define finite matrix sums

$$Q_{k,m} = \sum_{|\alpha|=m} \bigotimes_i A_{\alpha_i}, \quad Q_{k,m} = \sum_{j=0}^m A_j \otimes Q_{k-1,m-j}, \quad K_{k,N} = \sum_{m=0}^N Q_{k,m}. \quad (29)$$

These are finite degree convolutions; they do not assert convergence of an infinite generating series. They compute the product full-jet interpolation matrix using one-variable polynomial norms and jets, without inverting a total-degree moment matrix of dimension $\binom{N+k}{k}$. The target dimension remains d^k ; no polynomial-time complexity or unconditional numerical stability is claimed.

In univariate packet inclusions, the source's exact equality $R_{H,n} \iota_{hH} = R_{h,n}$ remains in force. The packet-dependent polynomial bases used here need not themselves commute under inclusion. No extension of this univariate equality to different joint total-degree budgets is presumed.

Finally, on $Av = k\rho v$, with $v = v_0^{\otimes k}$, and $\lambda = G_N v$,

$$\frac{\lambda^* V_N \lambda}{\lambda^* K_N \lambda} = k(2 \operatorname{Re} \rho - 1). \quad (30)$$

The additional work is therefore the explicit layer basis, derivative-coordinate map, endpoint identity, and direct relative certificate for the same source-controlled object. It is not a proof of a uniform sublinear excess. No arithmetic block or nilpotent jet has been removed.

S9.1.10 Verification and literature scope

The independent new checker contains eighteen exact test methods. They cover the kernel basis change, repeated jets, two-variable layers, actual derivative-class coordinates, Gram updates, the original univariate theta vector, endpoint factorization, a nonvacuous improved rank bound, antidual signs, generalized characteristic polynomials, nonzero off-line *polynomial model* diagnostics, nonreal recurrence diagonals, fixed-order theta primitives, and deterministic matrix perturbation certificates. Normal and optimized result records agree. Both intentional-failure modes return failure.

These are finite regression tests, not proofs of the general analytic inputs and not Lean execution. The general finite identities have the written proofs above. The source's nineteen tests and this supplement's eighteen tests are separate suites, not counts of novel theorems.

The matrix-kernel and Christoffel–Darboux methods have established antecedents. NIST's general orthogonal-polynomial formulas supply the standard scalar comparison. Lasserre and Pauwels (2016) connect inverse moment matrices with orthogonal polynomials and the Christoffel function; only their abstract was examined here, so no theorem from that paper is used as a premise. No global priority claim is made for standard kernel or inverse-update identities.

S9.1.11 References

Coherent arithmetic interpolation and joint tensor boundary control. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #14, commit 3bff1b2bd87dfa2592f65c5cc231f6ee9541dfc. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/14>

Formalize support-changing relation layers and the Krylov boundary-control calculation. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #15, verified source commit 21970bbf4760d0bbca512a4a7996a2e38f968947. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/15>

Lasserre, J.-B., & Pauwels, E. (2016). *Sorting out typicality with the inverse moment matrix SOS polynomial* [Preprint]. arXiv. <https://arxiv.org/abs/1606.03858>

National Institute of Standards and Technology. (n.d.). *Digital Library of Mathematical Functions*, §18.2(v), Christoffel–Darboux formula. <https://dlmf.nist.gov/18.2#v>

S10 Arithmetic frontier estimates and symmetric tensor control

Source attribution. Complete TeX witness from the dated symmetric-frontier continuation, delivered 12 September 2026. Its delivery receipt records an add-only local patch based on PR 14, with no remote write. The exact revision below is that patch base; it is not a new remote head for this NOTE. The source and archive hashes identify the complete delivered witness.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Symmetric_Frontier_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Recorded add-only patch base, PR 14

Source SHA-256. 734b6c935892ebd9d18b506f6b592f1ca3489afcb95d5c669a489f4ca04656cf

Original archive. Tau_Symmetric_Frontier_Control_2026-09-12(1).zip

Archive SHA-256. a024ee4bbb4df9555bc6f94c2d74a2e90c077533ad2ce312863342b758539796

S10.1 Arithmetic frontier estimates and symmetric tensor control

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. New written deductions from the precise arithmetic source of PR #14, submitted for mathematical review. The supplied Lean receipt certifies its stated coefficient, quotient, homotopy, and topology interfaces; it does not certify the analytic inputs or the results in this note. No purity estimate is assumed. No claim of global novelty is made for the underlying orthogonal-polynomial, finite-group, or matrix identities.

S10.1.1 0. Source and purpose

The source is PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, research-note Git blob 4ec40c15397128552bc276ec903fe59c89657d4e. Its actual total-degree representative, full-jet map, and theta relation ideal are used without changing them.

The source obtained two maps into the next relation layer, with

$$W_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}. \quad (0.1)$$

This note constructs coordinates on that *actual relation layer*, computes the relative control in them, proves a finite arithmetic upper bound, and constructs a smaller symmetric tensor cohomology summand that still contains every repeated eigenline used in amplification. Its exact complement and its trace are retained.

The decisive new coordinate map is

$$b_M(q) = \mathcal{T}_h^{(k)} q - R_{k,M} \mathcal{J} q. \quad (0.2)$$

It converts the highest orthogonal polynomial degree into an original theta boundary and has a calculated inverse and Gram matrix. A highest-degree polynomial alone need not be a boundary; its full-jet value is subtracted by the specific source representative in (0.2).

S10.1.2 1. Unchanged split base, arithmetic source, and quotients

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.1)$$

The base is the marked absolute point $\mathfrak{b}_\tau$. The first arithmetic target in (1.1) is still infinite. Every fixed-label linear map below is reconstructed by $(\lambda, x) \mapsto (\lambda, Tx)$; its value on external absence is external absence. A vanishing amplitude at an active label remains that label's supported zero.

Use the original $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, where

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= \sum_{n \neq 0} \phi(nx), \quad q : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V. \end{aligned}$$

The input is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_*(s) = g(s) = 2\xi(s).$$

Fix a nonempty finite packet of actual nontrivial zeros of g , with their full orders, and define

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[t]/(h), \quad v_h = g/h, \quad v = j_h v_h \in E^\times. \quad (1.2)$$

No squarefree assumption is made. The source constructs $F_h \in \mathcal{B}$ with $h(D)F_h = \Theta\phi_*$ and $\mathcal{M}F_h = v_h$. It supplies

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{M}\mathcal{T}_h(P) = v_h P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v[P]_h. \quad (1.3)$$

For tensor degree k , put $E_k = E^{\otimes k}$ and $A_k = \sum_{j=1}^k A_j$, where $A = M_t$. Let

$$\begin{aligned} \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, \quad M \geq k(d-1), \\ \mathcal{T}^{(k)}P &= P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{J}P = v^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \quad (1.4)$$

The relation space and its realization are

$$\mathcal{I}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)}\mathcal{I}_M. \quad (1.5)$$

The source's least-norm representative $R_M : E_k \rightarrow \mathcal{B}^{\otimes k} \subset \mathcal{H}_k$, with $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, satisfies

$$J^{(k)}R_M = 1, \quad q^{\otimes k}R_M = \sigma_h^{\otimes k}, \quad R_M(E_k) \perp \mathcal{B}_M. \quad (1.6)$$

Its finite Gram is $G_M = R_M^* R_M \succ 0$. Every adjoint here is of a finite-domain map into the declared Hilbert observation; no bounded Hilbert-space extension of the full theta map is assumed.

S10.1.3 2. Unscaled polynomial coordinates with every moment retained

The arithmetic measure is exactly

$$d\nu_h(y) = |v_h(\tfrac{1}{2} + iy)|^2 \frac{dy}{2\pi}. \quad (2.1)$$

The source's log-coordinate and Fourier maps give $\|\mathcal{T}_h P\|_{L^2(dx)}^2 = \int |P(\tfrac{1}{2} + iy)|^2 d\nu_h(y)$. The mass of this measure is not set to one.

Starting from $1, t, t^2, \dots$, subtract its orthogonal projections onto the preceding span:

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathcal{P}_{n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \quad (2.2)$$

No polynomial is divided by its norm. Let T_N be the triangular matrix with columns $p_0, \dots, p_N$ in the original power basis. The exact coordinate relation is

$$T_N^* M_{h,N} T_N = \text{diag}(\omega_0, \dots, \omega_N). \quad (2.3)$$

The earlier seed-boundary vectors $u_n = P_n(D)\Theta\phi_*$ are related to these coordinates by $\mathcal{T}_h(hP_n) = u_n$. Their measure is $|g|^2 dy/(2\pi)$; equation (1.3) is the map to the present $|g/h|^2$ measure. The two orthogonalizations have not been interchanged.

Recurrence. Define $a_0 = 0$ and $a_n = \omega_n/\omega_{n-1}$ for $n \geq 1$. Then

$$tp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad b_n = \frac{\int (\tfrac{1}{2} + iy) |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, \quad \Re b_n = \tfrac{1}{2}. \quad (2.4)$$

Here $p_{-1} = 0$. The possibly nonzero $\Im b_n$ remains. The packet measure need not be even in the height coordinate y .

Proof. Multiplication by t has adjoint multiplication by $1 - t$ for this polynomial inner product. Pairing tp_n with $p_j, j < n-1$, gives zero. Its coefficient of p_{n+1} is one by its leading term; pairing with p_{n-1} gives $-\omega_n/\omega_{n-1}$. The diagonal is the displayed integral. This proves every coefficient. These are the usual orthogonal-polynomial operations, applied through the specified coordinate map rather than changing the arithmetic generator.

For $\alpha \in \mathbb{N}^k$, define

$$p_\alpha = \prod_j p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}p_\alpha \in E_k. \quad (2.5)$$

The columns z_α contain the complete unit $v^{\otimes k}$ and all nilpotent jets.

Inverse-metric formula. In the fixed original basis of E_k ,

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}. \quad (2.6)$$

Indeed (2.3) applied to the tensor polynomial basis transforms $JM^{-1}J^*$ exactly into this sum. The representative coefficient on p_α is

$$c_\alpha(u) = \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (2.7)$$

These formulas avoid inversion of a large raw monomial moment matrix, without imposing a new metric.

S10.1.4 3. The highest polynomial degree is isomorphic to the actual new relation layer

Let $\mathcal{U}_M = \text{span}\{p_\beta : |\beta| = M + 1\}$ and retain its ordered basis. Put

$$\Omega = \text{diag}(\omega_\beta)_{|\beta|=M+1}, \quad F = [z_\beta]_{|\beta|=M+1} : \mathbb{C}^r \rightarrow E_k, \quad r = \binom{M+k}{k-1}. \quad (3.1)$$

Write $T_+ : \mathbb{C}^r \rightarrow \mathcal{H}_k$ for the map with columns $\mathcal{T}^{(k)} p_\beta$. Its image is orthogonal to the entire degree- M polynomial image, including $R_M(E_k)$.

The prior actual new relation layer is

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M})\mathcal{B}_{M+1}.$$

Theorem: relation-graph isomorphism. The map

$$b_M = T_+ - R_M F : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M \quad (3.2)$$

is an isomorphism and has

$$H_M = b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (3.3)$$

Proof. Its full jet is $F - F = 0$, so (1.5) makes it an actual boundary of degree at most $M + 1$. Both terms are orthogonal to $\mathcal{B}_M$. If $b_M u = 0$, its component in the highest orthogonal polynomial degree is $T_+ u = 0$, so $u = 0$. The source calculation gives $\dim \mathcal{E}_M = r$, proving surjectivity. Alternatively every degree- $M + 1$ boundary has a unique highest orthogonal component; subtract its b_M image and the result lies in $\mathcal{B}_M$. This also constructs the inverse. Orthogonality $T_+^* R_M = 0$ proves (3.3).

A primitive is fully specified: form the numerator $p_\beta - c_M(z_\beta)$, where c_M is (2.7), divide it in the fixed variable order by $h(t_1), \dots, h(t_k)$, and use the original theta primitive in the factor where $h(t_j)$ appears. Give that primitive coefficient $(-1)^{j-1}$; the tensor differential supplies the same sign. No relation is inferred solely from a dimension count.

The comparison with the original internal quotient is

$$\mathbb{C}^r \xrightarrow{b_M} \mathcal{E}_M \xrightarrow{v \mapsto [v]} \mathcal{B}_{M+1}/\mathcal{B}_M \hookrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_M \longrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_{M+1}. \quad (3.4)$$

The last composite is zero. Its split lift sends (λ_M, u) to $(\lambda_{M+1}, 0)$, not to τ . At λ_M , $u \neq 0$ represents a nonzero kernel class. The map b_M describes the entire class that is about to become a supported zero.

Representative update. Orthogonal projection onto $b_M(\mathbb{C}^r)$ gives

$$Y_M = -b_M H_M^{-1} F^* G_M, \quad R_{M+1} = R_M + b_M H_M^{-1} F^* G_M. \quad (3.5)$$

Consequently

$$G_{M+1} = G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \quad (3.6)$$

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (3.7)$$

The complete volume identity is

$$\frac{\det \mathcal{C}_{M+1}}{\det \mathcal{C}_M} = \frac{\det(\Omega + F^* G_M F)}{\det \Omega}. \quad (3.8)$$

The full matrices in (3.6)–(3.7) remain available; their determinant in (3.8) is an additional observation, not their replacement.

S10.1.5 4. Relative control is a highest-degree Christoffel–Darboux identity

For each $|\beta| = M + 1$, put

$$w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (4.1)$$

The name E_+ denotes this finite matrix, not the scalar e or the arithmetic quotient E_h ; its type is $\mathbb{C}^r \rightarrow E_k$.

Theorem: exact displacement.

$$\boxed{Z_M := A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*}. \quad (4.2)$$

Proof. Apply (2.4) in each factor of (2.6). A pair of adjacent indices inside $|\alpha| \leq M$ cancels because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, precisely the term subtracted in (4.2). The remaining ordered pairs have $|\beta| = M + 1$ and predecessor $\beta - e_j$; collecting them gives (4.1)–(4.2). The imaginary diagonal coefficients cancel with their conjugates here; they were retained in the recurrence and are not assigned zero.

This identity is the tensor total-degree version of the Christoffel–Darboux cancellation. The standard one-variable formula is background; the maps to the full arithmetic quotient and its source boundaries are specified above.

The derivative defect also has an exact expression in the same relation-graph coordinates:

$$B_M = D^{(k)} R_M - R_M A_k, \quad \boxed{C_M^{\text{rel}} := (1 - P_{\mathcal{B}_M}) B_M = b_M \Omega^{-1} E_+^* G_M}. \quad (4.3)$$

Proof. The coefficient of p_β in $D^{(k)} R_M u$ is $\sum_j \omega_{\beta-e_j}^{-1} z_{\beta-e_j}^* G_M u = \omega_\beta^{-1} w_\beta^* G_M u$. The defect is an original boundary. The inverse of (3.2), followed by projection away from $\mathcal{B}_M$, therefore gives (4.3).

Substituting (3.5) and (4.3) into the previous cross-pairing formula proves

$$\boxed{W_M = A_k^* G_M + G_M A_k - k G_M = G_M Z_M G_M = -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M)}. \quad (4.4)$$

Thus the new highest-degree computation is connected to both maps in the preceding note, including their minus sign and their support-valued boundary primitive.

S10.1.6 4.1 Exact finite certificate and a proved upper estimate

For every real ϵ , congruence by the actual invertible G_M gives

$$\boxed{-\epsilon G_M \preceq W_M \preceq \epsilon G_M \iff -\epsilon C_M \preceq Z_M \preceq \epsilon C_M}. \quad (4.5)$$

This transports the inequality; no eigenvalue or generator has been rescaled.

Put $\mathcal{B} = [F, E_+]$ and

$$J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

The nonzero generalized eigenvalues of (Z_M, C_M) are the nonzero eigenvalues of

$$\boxed{J \mathcal{B}^* G_M \mathcal{B}}. \quad (4.6)$$

The proof is the pair of maps $\mathcal{B} : \mathbb{C}^{2r} \rightarrow E_k$ and $J\mathcal{B}^* G_M : E_k \rightarrow \mathbb{C}^{2r}$ and the equality of nonzero spectra of their two composites. These eigenvalues are real because the first composite is the operator of the Hermitian pencil in (4.5). The $2r$ -matrix itself need not be Hermitian in its displayed coordinates.

There is also a direct upper bound. Let

$$\lambda_{k,M} = \max_{v \neq 0} \frac{v^* F \Omega^{-1} F^* v}{v^* \mathcal{C}_M v},$$

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (4.7)$$

Then the actual arithmetic control excess satisfies

$$\boxed{\epsilon_{k,M} \leq 2\sqrt{\Gamma_{k,M} \lambda_{k,M}}}. \quad (4.8)$$

Proof. Let Z_M be the columns z_α , $|\alpha| = M$, and $\Omega_M = \text{diag}(\omega_\alpha)$. The lowering-incidence matrix L has $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, summed over matching j . Then $E_+ = Z_M L$. The matrix $\Omega_M L \Omega^{-1} L^*$ is similar to a positive semidefinite matrix and has nonnegative entries. Its row sum at α is exactly the sum in (4.7), so its spectral radius is at most $\Gamma_{k,M}$. This proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z_M \Omega_M^{-1} Z_M^* \preceq \Gamma_{k,M} \mathcal{C}_M.$$

Cauchy–Schwarz in the same Ω^{-1} pairing now gives

$$|v^* \mathcal{Z}_M v| \leq 2\sqrt{v^* F \Omega^{-1} F^* v} \sqrt{v^* E_+ \Omega^{-1} E_+^* v} \leq 2\sqrt{\lambda_{k,M} \Gamma_{k,M}} v^* \mathcal{C}_M v.$$

This proves (4.8). The phase-sensitive formula (4.6) is retained for a sharper estimate. Equation (4.8) is not claimed sublinear in k ; its two arithmetic factors are now fully specified.

S10.1.7 5. A symmetric cohomology summand that retains the amplification witnesses

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$ with its original tensor differential. For $\pi \in S_k$, the chain permutation is

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i| |x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (5.1)$$

The exponent uses the original inputs crossed by π . Adjacent transpositions verify the cochain equation and the permutation relations. In top degree every input has degree one, so $T_\pi = \text{sgn}(\pi) P_\pi$, where P_π is ordinary factor permutation.

Define the actual idempotent

$$\boxed{S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi}. \quad (5.2)$$

The unscaled numerator squares to $k!$ times itself. The displayed $1/k!$ is necessary to make the specified morphism idempotent and is not absorbed into a pairing or an arithmetic function. In degree k the two signs multiply to $+1$, hence

$$\boxed{H^k(S_k \mathcal{C}_k) \cong (Q^{\widehat{\otimes} k})^{S_k}}. \quad (5.3)$$

On a finite packet this contains $(E^{\otimes k})^{S_k}$ with its entire nilpotent action. If $Av = \rho v$, then $v^{\otimes k}$ is nonzero, lies in this summand, and has generator eigenvalue $k\rho$. Using the unsigned chain average instead would give the alternating top-degree summand and would kill a repeated vector for $k > 1$; equation (5.2) is the precise replacement.

The complement is retained through the actual cochain decomposition

$$\mathcal{C}_k \xrightarrow{x \mapsto (S_k x, (1-S_k)x)} \text{im } S_k \oplus \text{im}(1 - S_k), \quad (u, v) \mapsto u + v. \quad (5.4)$$

At an invariant joint support this is reconstructed fibrewise. On all support masks the lift of (5.2) has the specified support map $\lambda \mapsto \bigvee_{\pi} \pi \lambda$. Its formula is the sum of the actual supported scalars $(\text{sgn } \pi / k!) \bullet T_{\pi}$. A vanishing projected amplitude lands at the orbit-joined supported zero, not at τ . Original transports into that joint fibre remain recorded. No injectivity of arbitrary support transports is assumed.

S10.1.8 5.1 The canonical representatives commute with this operation

Permutation of variables preserves the total-degree numerator, its relation ideal, the product arithmetic measure, and the full unit $v^{\otimes k}$. Uniqueness of the constrained minimum gives

$$R_M P_{\pi} = P_{\pi} R_M. \quad (5.5)$$

Consequently R_M restricts to the invariant arithmetic module

$$E_k^{\text{sym}} = (E^{\otimes k})^{S_k}, \quad \boxed{\dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}}. \quad (5.6)$$

This is not a change of the spectral packet. The inclusion $I_k : E_k^{\text{sym}} \hookrightarrow E_k$ gives

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad W_M^{\text{sym}} = I_k^* W_M I_k, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (5.7)$$

All repeated eigenlines remain, so this smaller family still contains every witness needed by the tensor-amplification upper estimate. For example, $d = 3, k = 8$ gives dimension 45 rather than $3^8 = 6561$. This is a dimension reduction, not a bound on W_M^{sym} .

S10.1.9 6. Orbit sums give the symmetric matrices with all factorials retained

For a nondecreasing tuple $\lambda \in \mathbb{N}^k$, let $\text{Orb } \lambda$ denote its distinct permutations and

$$n_{\lambda} = \frac{k!}{\prod_{j \geq 0} m_j(\lambda)!}, \quad p_{[\lambda]} = \sum_{\alpha \in \text{Orb } \lambda} p_{\alpha}, \quad \omega_{[\lambda]} = n_{\lambda} \prod_j \omega_{\lambda_j}. \quad (6.1)$$

Distinct product basis vectors are orthogonal, proving the last formula directly. The orbit sum is not divided by its norm or its orbit size.

The target basis is likewise the unscaled orbit sums of $e_{a_1} \otimes \cdots \otimes e_{a_k}$, $0 \leq a_j < d$, where $e_i = [t^i]$. Let I_k be their column inclusion and

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_i n_i!} \right), \quad L_k = D_k^{-1} I_k^*. \quad (6.2)$$

The explicit L_k extracts the common coefficient on an orbit. For each orbit polynomial, define

$$z_{[\lambda]} = L_k \mathcal{J} p_{[\lambda]}.$$

Then

$$\mathcal{C}_M^{\text{sym}} = \sum_{|\lambda| \leq M} \frac{z^{[\lambda]} z_{[\lambda]}^*}{\omega_{[\lambda]}}, \quad G_M^{\text{sym}} = (\mathcal{C}_M^{\text{sym}})^{-1}. \quad (6.3)$$

The comparison with full coordinates is

$$\mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad G_M^{\text{sym}} = I_k^* G_M I_k. \quad (6.4)$$

Inversion and compression have not been commuted without the displayed D_k factors. Equation (6.4) follows because the full form commutes with the orthogonal permutation projection and by substituting $L_k = D_k^{-1} I_k^*$.

The symmetric polynomial numerator has one basis vector per partition of its degree into at most k parts. For $M \geq k(d-1)$ the jet map is onto, because a tensor remainder has total degree at most $k(d-1)$ and averaging makes a symmetric lift of every invariant jet. Thus

$$\dim(\mathcal{E}_M^{S_k}) = p_{\leq k}(M+1), \quad (6.5)$$

where $p_{\leq k}(n)$ is the number of partitions of n into at most k parts. The maps $b_M, Y_M, C_M^{\text{rel}}$ are permutation-equivariant and therefore restrict to that very subspace. Fixed-order polynomial division yields a primitive; applying S_k to it gives a symmetric cochain primitive because $dS_k = S_k d$.

The highest-layer formulas in sections 3–4 hold on the invariant coordinates by grouping F, E_+ into orbit sums, applying L_k , and using the diagonal entries $\omega_{[\lambda]}$. This computes the symmetric bound directly, instead of building a d^k -dimensional matrix first. The upper-bound row-sum estimate can still use the fully retained $\Gamma_{k,M}$; the exact grouped incidence matrix can give a smaller constant.

S10.1.10 6.1 Reflection and the original trace

For a reflection-stable packet the source involution j_h induces $j_h^{\otimes k}$ on both summands in (5.4), commutes with permutations, and preserves the canonical representatives. Hence on the symmetric metric an upper control implies the opposite lower control by the same involution. The residue pairing and its Jacobian map are still transported through their specified inclusions; no auxiliary positive Gram is declared to be the Weil form.

For a finite packet action $U(a) = \exp((\log a)A)$, the exact trace on the invariant summand is

$$\text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|} | E). \quad (6.6)$$

The proof expands the permutation operator in the tensor basis; each cycle forces the corresponding contracted indices and gives the indicated trace. It holds for nonsemisimple U . The cohomological supertrace is $(-1)^k$ times (6.6). The complement trace remains

$$(-1)^k \left(\text{Tr}(U(a) | E)^k - \text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) \right). \quad (6.7)$$

These are traces of finite-rank cochain projectors on the original complex. An infinite packet sum and its trace topology are not supplied by these formulas.

S10.1.11 7. An exact joint-control calculation with the nilpotent direction retained

This section is a declared algebraic calibration, not a packet of zeta zeros. Take the polynomial quotient

$$E = \mathbb{C}[s]/((s - \frac{1}{2})^2), \quad d\nu(y) = e^{-y^2/2} dy, \quad c = \sqrt{2\pi}.$$

The measure has its literal mass c , and the unit in the calibration jet map is 1. The polynomial basis comparison $x = s - 1/2$ is given by

$$B : \text{span}(1, x) \xrightarrow{\sim} \text{span}(1, s), \quad B = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}. \quad (7.1)$$

The operator remains multiplication by $s = 1/2 + x$. In the symmetric basis

$$1 \otimes 1, \quad 1 \otimes x + x \otimes 1, \quad x \otimes x,$$

its two-fold generator is

$$A_2^{\text{sym}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}. \quad (7.2)$$

The two-fold total-degree-4 calculation gives

$$\boxed{G_{2,4}^{\text{sym}} = c^2 \text{diag}(1/3, 2/3, 1/4),} \quad (7.3)$$

$$\boxed{W_{2,4}^{\text{sym}} = c^2 \begin{pmatrix} 0 & 2/3 & 0 \\ 2/3 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{pmatrix}.} \quad (7.4)$$

Its generalized characteristic polynomial is

$$\frac{\det(zG_{2,4}^{\text{sym}} - W_{2,4}^{\text{sym}})}{\det G_{2,4}^{\text{sym}}} = z(z^2 - 7/2),$$

so the exact two-sided excess is $\sqrt{7/2}$. The independent rectangular degree-2-per-factor construction gives $\sqrt{6}$ at the same containing total degree. This proves an actual improvement for the specified joint construction in this calibration; the nilpotent operator in (7.2) is unchanged.

For comparison, the smaller joint total degree $M = 2$ has excess $\sqrt{6}$, whereas the rectangular degree-1 construction contained in it has excess 2. Consequently a larger correction space alone does not monotonically improve the control constant. The typed comparison is inclusion of the two polynomial numerators and the resulting Gram compression; the direct formula (4.2) supplies the actual control value. Both calibrations are retained so that the first improvement is not promoted into a false monotonicity theorem.

S10.1.12 8. Exact target, counterfactuals, and handoff

The next estimate is now an arithmetic highest-layer estimate, either by the phase-sensitive pencil (4.5)–(4.6), or by bounding the two factors in (4.8). On the symmetric cohomology family it concerns the same G_M^{sym} and W_M^{sym} , with all relation primitives retained. The source of every matrix entry is the actual measure $|2\xi/h|^2 dy/(2\pi)$, not a fitted metric.

A sublinear excess in tensor degree has not been proved. For every actual eigenvector $Av = \rho v$, the retained witness in the symmetric summand satisfies

$$\frac{(v^{\otimes k})^* W_M(v^{\otimes k})}{(v^{\otimes k})^* G_M(v^{\otimes k})} = k(2\Re\rho - 1). \quad (8.1)$$

The symmetric reduction therefore retains the witness needed by Deligne-style amplification; it does not assume its disappearance. The input Deligne passages use tensoring *and* a separate geometric estimate. Our new work supplies a computable smaller arithmetic target and a proved finite bound, not that geometric estimate by renaming a category.

Constructive continuations are recorded with their exact maps:

- Raw moment inversion: use (2.3)–(2.7), retaining every ω_j and the full jet unit.
- Highest polynomials that are not relations: use $b_M = T_+ - R_M F$, not an assertion that high degree is zero.
- Absolute Gram decay: use the congruence (4.5), retaining $\mathcal{C}_M = G_M^{-1}$ and the whole highest-layer update.
- Excessive tensor dimension: use the chain projector (5.2) and the symmetric numerator/quotient (6.1)–(6.5), while retaining the complementary cohomology and trace.
- A Cauchy–Schwarz bound too large: use the exact signed matrix (4.6); its phases and cross terms have not been replaced by their absolute values in the stored data.
- Ordinary zero observations: reconstruct the maps and kernels in (3.4). A new relation is *e*-zero at the next label; the original class and the full kernel remain upstream.

The supplied quotient/homotopy Lean results can formalize the finite algebra of (3.2)–(4.6) and the chain-permutation calculation. The original analytic range theorem, arithmetic moment bounds, and any uniform tensor estimate remain separate written-analysis inputs.

S10.1.13 References and evidence scope

The current input was read in the mounted NOTE.tex and checked against the pinned public mark-down (especially sections 7–9). The prior formalization receipt was read at its recorded scope: its section-change and internal-quotient statements are used, without treating its analytic assumptions as proved Lean theorems.

The classical orthogonal-polynomial recurrence and Christoffel–Darboux formula are documented in NIST DLMF sections 18.2(iv)–(v), <https://dlmf.nist.gov/18.2>. The tensor complex and its signed commutativity are documented in the Stacks Project, Tag 0GWN, <https://stacks.math.columbia.edu/tag/0GWN>. These are background mechanisms; sections 3–6 above prove their specific maps on this arithmetic source.

The retained Deligne reading for the intended amplification is the supplied account of Weil II 3.2.13 and 3.3.4–3.3.6. No new full rereading of Weil II is claimed in this turn. No external source corpus is redistributed in this package.

The checker uses exact finite Gaussian moment fixtures, including repeated-root quotients, non-constant units, a nonzero imaginary recurrence diagonal, and both signed chain degrees. They are not arithmetic measurements. All derived infinite-dimensional statements rest on the written arguments and the explicitly named prior analytic inputs, not on the finite tests.

S11 Complete spectral-sum descent: relative arithmetic fibres, dual resolution and all nilpotents

Source attribution. Complete spectral-sum TeX witness retained inside the sum-connection delivery archive. The earlier public PR 17 Markdown note is separately preserved in this reader. This longer witness includes the original sum-action resolution, linear dual and antidual comparison, Ext and resolvent calculations, full matrix weight and constrained minimum. The associated public revision below is not a Git identity assigned to these longer TeX bytes; the source and archive hashes identify them.

Exact revision. 33b29f706008124886614ba4bd55bffc489df9e2

Source file. Tau_Sum_Connection_Control/sources/Tau_Spectral_Sum_Descent/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Associated public PR 17 note revision; full delivered TeX identified by its separate hash

Source SHA-256. a3becf8455ddf96e8e382762acbc0e9d0548ea7342c01d70aecdad744ae79552

Original archive. Tau_Sum_Connection_Control_2026-09-12.zip

Archive SHA-256. f2de513041b7336b495a338e90325e84c5ce53cfa98799e1b89c53af308356de

S11.1 Spectral-sum descent with the relative arithmetic fibre retained

Owner-directed continuation of the split-zero programme. 12 September 2026.

Status. Written deductions for review. These results use the actual theta range and representative constructions stated in the preceding packages. Their analytic inputs are not newly Lean-certified. The finite polynomial, quotient, and matrix statements below have independent proofs and executable finite tests. No RH, GRH, or sublinear arithmetic weight bound is asserted.

S11.1.1 1. Integration and purpose

The new Kernel-Layer Integration supplement and the preceding Symmetric Frontier note overlap in their explicit relation basis and inverse-metric control. We retain the dictionary

$$K_N = \mathcal{C}_M, \quad U_N = F, \quad D_N = \Omega, \quad \mathcal{E}_N = b_M, \quad F_N = \Omega^{-1}E_+^*, \quad V_N = \mathcal{Z}_M, \quad N = M. \quad (1.1)$$

The supplement's sharper rank bound follows by grouping the summed derivative at degree M rather than counting all degree- $M+1$ coordinates. Its identities are not counted again as new work. In particular its one-variable map

$$e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1} \quad (1.2)$$

is retained: it identifies the packet-dependent interpolation layer with the original packet-independent theta relation by an actual polynomial-divisibility map.

The new step is to push the arithmetic tensor system along its **sum coordinate**, retaining the relative coordinates in a matrix-valued arithmetic measure. We calculate the entire symmetric-square relation algebra, its original theta primitives, the exact norm under the pushforward, and the full

nilpotent fibres of the sum action. This is a product-to-one-parameter construction on the existing characteristic-zero object over the tau base. It is not a use of finite-field purity with relabelled coefficients.

S11.1.2 2. Unchanged source, split quotient, and tensor cohomology

Keep

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (2.1)$$

The first quotient remains infinite. All spaces and morphisms are marked over the existing absolute pointed base $\mathfrak{b}_\tau$. At an active coefficient label λ , the lift of a linear map T is $(\lambda, x) \mapsto (\lambda, Tx)$; it fixes external absence. Its amplitude-zero value is $(\lambda, 0)$, the image of the original supported-zero action, not τ .

Use precisely

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad \mathcal{B} = \{F : \sup_{x>0} x^b |D^j F(x)| < \infty (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \\ \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g, \quad g(s) = 2\xi(s). \quad (2.2)$$

For a finite nonempty packet of actual zeros with their full orders, put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times.$$

The prior analytic construction supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. Retain its maps

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h. \quad (2.3)$$

For tensor degree k , use $\mathcal{P}_M = \mathbb{C}[s_1, \dots, s_k]_{\deg \leq M}$ and

$$E_k = E_h^{\otimes k}, \quad A_k = M_{s_1 + \dots + s_k}, \quad \mathcal{T}_k(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \\ I_{k,M} = \mathcal{P}_M \cap (h(s_1), \dots, h(s_k)), \quad B_{k,M} = \mathcal{T}_k I_{k,M}. \quad (2.4)$$

The jet map $\mathcal{J}_k P = v_h^{\otimes k}[P]$ has this exact kernel and is onto when $M \geq k(d-1)$. This map is linear, with an invertible unit multiplication after the algebra quotient; it is not called a unital algebra homomorphism. The source representative $R_{k,M}$ is its unique least-norm lift with the specified theta relations allowed. Thus

$$J^{(k)} R_{k,M} = 1, \quad q^{(k)} R_{k,M} = \sigma_h^{\otimes k}, \quad G_{k,M} = R_{k,M}^* R_{k,M}, \quad W_{k,M} = A_k^* G_{k,M} + G_{k,M} A_k - k G_{k,M}. \quad (2.5)$$

The tensor differential retains $(-1)^{j-1}$ when it applies the differential in the j th factor to a degree- $(k-1)$ primitive with all other factors of degree one. A primitive for the positive relation $h(s_j)P_j$ has the matching sign $(-1)^{j-1}$.

In top degree the signed chain projector

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi \quad (2.6)$$

selects ordinary symmetric tensors: the Koszul sign in T_π supplies the second permutation sign. The factor $1/k!$ is retained. We work on this image while retaining the complementary chain projector $1 - S_k$ and its trace. A source mask is transported to the union of its permutation orbit before averaging; its cancelled coordinates remain supported at that joined label.

S11.1.3 3. The spectral-sum direct image is an exact functor with its action fixed

Define the actual ring map

$$\iota_k : \mathbb{C}[S] \longrightarrow E_k, \quad S \longmapsto [s_1 + \cdots + s_k]. \quad (3.1)$$

For an E_k -module M , let $\pi_{k*}M$ be that same module with the $\mathbb{C}[S]$ action restricted through ι_k . The functor is exact because its maps, kernels, and underlying abelian groups are unchanged. It has the explicit adjunction

$$\text{Hom}_{E_k}(E_k \otimes_{\mathbb{C}[S]} L, M) \cong \text{Hom}_{\mathbb{C}[S]}(L, \pi_{k*}M), \quad (3.2)$$

with forward map $f \mapsto (l \mapsto f(1 \otimes l))$ and inverse $a \otimes l \mapsto af(l)$. On quasi-coherent sheaves this is pushforward along the finite morphism $\text{Spec } E_k \rightarrow \text{Spec } \mathbb{C}[S]$. The affine pushforward statement concerns quasi-coherent complexes, not all sheaves of modules (Stacks, Tag 0AVV).

The split map is $G(\iota_k)$, with

$$p_{E_k} G(\iota_k) = \iota_k p_{\mathbb{C}[S]}, \quad G(\iota_k)(e_{\mathbb{C}[S]}) = e_{E_k}, \quad G(\iota_k)(\tau) = \tau. \quad (3.3)$$

Consequently the geometric arrow is over the same $\mathfrak{b}_\tau$. It does not replace that base by a finite field. The additional line $\text{Spec } \mathbb{C}[S]$ is the recipient of a specified arithmetic correspondence.

At the original cochain level, S acts by $D_1 + \cdots + D_k$. Its finite representative equation remains

$$D^{(k)} R_{k,M} - R_{k,M} A_k = dK_{k,M} \in B_{k,M+1}. \quad (3.4)$$

The jet and cohomology maps intertwine that action exactly. Restriction of scalars preserves their kernels and this boundary equation. It does not declare the chosen representative $R_{k,M}$ equivariant.

On the symmetric summand the ring is $B_k = (E_k)^{S_k}$, its inclusion is the original invariant inclusion, and S acts by the same sum. Exactness of taking invariants here follows from the specified Reynolds projector $k!^{-1} \sum P_\pi$ over $\mathbb{C}$.

S11.1.4 3.1 The pushforward has a computed free resolution and dualizing comparison

Let E denote the actual finite ordered or symmetric image, with its exact sum operator A . Put $R = \mathbb{C}[S]$ and $n = \dim_{\mathbb{C}} E$. There is the actual free resolution

$$0 \longrightarrow R \otimes_{\mathbb{C}} E \xrightarrow{SI-A} R \otimes_{\mathbb{C}} E \xrightarrow{\text{ev}_A} \pi_* E \longrightarrow 0, \quad \text{ev}_A(P \otimes v) = P(A)v. \quad (3.5)$$

Injectivity of $SI - A$ follows either from its monic nonzero determinant or by inspecting the highest coefficient of a polynomial vector. To identify the kernel of ev_A , use

$$S^j v - A^j v = (SI - A) \sum_{i=0}^{j-1} S^{j-1-i} A^i v.$$

Subtracting these displayed relations leaves precisely the constant vector $\text{ev}_A(f)$. This proves exactness with both original arrows retained.

Let $E^\vee = \text{Hom}_{\mathbb{C}}(E, \mathbb{C})$ and $A^\vee \lambda = \lambda \circ A$. This is the algebraic dual; its matrix is a transpose. The coefficient conjugation $\overline{E^\vee} \rightarrow E^\#$, $\bar{\lambda} \mapsto (v \mapsto \overline{\lambda(v)})$, connects it explicitly to the antidual used in the control matrices.

Applying $\text{Hom}_R(-, R dS)$ to (3.5) gives

$$R \text{Hom}_R(\pi_* E, R dS) \simeq [R \otimes E^\vee dS \xrightarrow{SI - A^\vee} R \otimes E^\vee dS], \quad (3.6)$$

in degrees 0, 1. Its degree-zero group is zero, and

$$\boxed{\text{Ext}_R^1(\pi_* E, R dS) \cong E^\vee, \quad [P(S) dS] \mapsto P(A^\vee).} \quad (3.7)$$

Here $P(A^\vee)$ means $\sum_j (A^\vee)^j \lambda_j$ for a polynomial dual vector $P(S) = \sum_j S^j \lambda_j$. The inverse is the constant class $\lambda \mapsto [\lambda dS]$.

For the explicit cohomological shift, $C[b]^i = C^{i+b}$. Thus

$$R \text{Hom}_R(\pi_* E[-k], R dS[1]) \simeq E^\vee[k]. \quad (3.8)$$

The degree- k arithmetic image becomes degree $-k$ under the stated dualizing object of the sum line. These shifts are obtained from (3.6), not inferred by calling the finite packet a curve.

The adjoint behind this calculation is the actual module-category map

$$\text{Hom}_R(\pi_* M, L) \cong \text{Hom}_{B_k}(M, \text{Hom}_R(B_k, L)), \quad f \mapsto (m \mapsto (b \mapsto f(bm))). \quad (3.9)$$

The inverse evaluates the inner map at 1. Passing to derived Hom gives the corresponding derived right adjoint. This concerns the exact restriction functor on module categories; no exactness of pushforward on all sheaves is asserted (Stacks, Tag 0AWZ).

S11.1.5 3.2 Resolve the finite arithmetic trace through that same duality

The pairing in (3.7) is equivalently the concrete resolvent-residue pairing

$$\boxed{\langle v, [P(S) dS] \rangle = \sum_{\lambda \in \text{Spec } A} \text{Res}_{S=\lambda} P(S) ((SI - A)^{-1} v) dS.} \quad (3.10)$$

Relations from $SI - A^\vee$ produce polynomial expressions with no finite residue. Expanding the resolvent at infinity proves that the sum is $P(A^\vee)(v)$: the coefficient of S^{-1} is $\sum_j \lambda_j (A^j v)$. This proves perfectness without deleting nilpotents.

On a block $A = \lambda I + N$ the retained resolvent is

$$(SI - A)^{-1} = \sum_{j=0}^{\ell-1} \frac{N^j}{(S - \lambda)^{j+1}}, \quad N^\ell = 0. \quad (3.11)$$

The trace observation of that resolvent is a further specified map. For a function ψ holomorphic at the finite spectrum,

$$\boxed{\mathrm{Tr}(\psi(A)) = \sum_{\lambda} \mathrm{Res}_{S=\lambda} \psi(S) \mathrm{Tr}((SI - A)^{-1}) dS = \sum_{\lambda} \mathrm{Res}_{S=\lambda} \psi(S) \frac{\chi'_A(S)}{\chi_A(S)} dS,} \quad (3.12)$$

where $\chi_A(S) = \det(SI - A)$ is the exact finite characteristic polynomial. It is not substituted for the original $g = 2\xi$. With $\psi(S) = a^S$, (3.12) is the finite Lefschetz contribution of the same diagonal scaling correspondence. The cochain image is in degree k , so its supertrace includes $(-1)^k$.

The constant-class maps in (3.5)–(3.7), and the linear functional in (3.10), lift fibrewise to the original split objects. Every zero functional or zero residue remains supported. At fixed label the equations forming the resolution have composite the supported zero map. The residue pairing is an ordinary bilinear pairing in that coefficient fibre, with its split lift separately additive, not a linear map on its Cartesian product.

S11.1.6 4. The symmetric tensor-square has an explicit two-generator relation algebra

Use three distinguished polynomials

$$S = s_1 + s_2, \quad r = s_1 - s_2, \quad \Delta = r^2. \quad (4.1)$$

The inverse map before taking invariants is

$$s_1 = (S + r)/2, \quad s_2 = (S - r)/2.$$

Swap fixes S and sends r to $-r$. Therefore the algebra map

$$\Phi : \mathbb{C}[S, \Delta] \xrightarrow{\sim} \mathbb{C}[s_1, s_2]^{S_2}, \quad S \mapsto s_1 + s_2, \quad \Delta \mapsto (s_1 - s_2)^2 \quad (4.2)$$

is an isomorphism. Its proof is even/odd expansion in r , using the displayed inverse with both factors $1/2$. Give S degree one and Δ degree two; then Φ preserves the entire original total-degree filtration.

Define actual polynomials H_0, H_1 by

$$\begin{aligned} h((S + r)/2) &= H_0(S, r^2) + rH_1(S, r^2), \\ h((S - r)/2) &= H_0(S, r^2) - rH_1(S, r^2). \end{aligned} \quad (4.3)$$

In particular H_0 is the half-sum and H_1 is the difference divided by $2r$. That difference is an odd polynomial, so this is polynomial division, not localization away from $r = 0$.

Theorem (complete symmetric relation ideal).

$$\boxed{B_2 = (E_h \otimes E_h)^{S_2} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1).} \quad (4.4)$$

Proof. In $\mathbb{C}[S, r]$, the original ideal $(h(s_1), h(s_2))$ is (H_0, rH_1) . Its even part is exactly (H_0, r^2H_1) : average an expression $H_0a + rH_1b$ under $r \mapsto -r$. Writing $a = a_0 + ra_1$, $b = b_0 + rb_1$ gives the even expression $H_0a_0 + \Delta H_1b_1$. Conversely both generators are in the original ideal. Invariants of the quotient equal the quotient of invariants because Reynolds averaging is an exact projector. This proves (4.4), including coincident roots and all nilpotents.

The quotient has dimension $d(d+1)/2$. Indeed the unscaled orbit sums of the basis $s_1^i s_2^j$, $0 \leq i, j < d$, give that many independent invariant vectors. The associated graded Hilbert series is

$$\sum_{0 \leq i \leq j < d} z^{i+j} = \frac{(1-z^d)(1-z^{d+1})}{(1-z)(1-z^2)}. \quad (4.5)$$

Here associated graded refers to the original degree filtration; (4.5) is not a claim that a nonhomogeneous h defines a graded algebra. Polynomial remainder lowers degree, and Reynolds averaging respects the filtration, proving the formula for every monic h of degree d .

S11.1.7 4.1 All original primitives and their degrees survive

Let a symmetric relation of degree at most M have, after original ordered division, the expression $h(s_1)Q_1 + h(s_2)Q_2$ with $\deg Q_i \leq M-d$. Averaging yields paired coefficients

$$C_1 = (Q_1 + Q_2 \circ \text{swap})/2, \quad C_2 = C_1 \circ \text{swap}.$$

Put $A_0 = C_1 + C_2$ and $A_1 = (C_1 - C_2)/r$. Then A_0, A_1 are polynomials in S, Δ and

$$H_0 A_0 + \Delta H_1 A_1 = h(s_1) \frac{A_0 + r A_1}{2} + h(s_2) \frac{A_0 - r A_1}{2}. \quad (4.6)$$

Their weighted degree bounds are $\deg A_0 \leq M-d$ and $\deg A_1 \leq M-d-1$. The two source primitives are respectively

$$\frac{A_0 + r A_1}{2} (D_1, D_2) (\phi_* \otimes F_h), \quad -\frac{A_0 - r A_1}{2} (D_1, D_2) (F_h \otimes \phi_*). \quad (4.7)$$

The negative sign in the second primitive meets the negative cochain sign of the second differential. Applying d gives exactly (4.6) through $\mathcal{T}_2$.

The unit $v_h \otimes v_h$ is symmetric. Under (4.4) denote its full class by $v_{2,h}$. The jet map in these coordinates is exactly

$$\mathcal{J}'_{2,M} : \mathbb{C}[S, \Delta]_{\deg S + 2 \deg \Delta \leq M} \longrightarrow B_2, \quad P \longmapsto v_{2,h}[P]. \quad (4.8)$$

Its kernel is the full weighted-degree part of $(H_0, \Delta H_1)$. It is onto for $M \geq 2(d-1)$. This is the original full-jet map conjugated by Φ , not a scalar evaluation at the sum coordinate.

On the split totals, $\tilde{\Phi}(\lambda_M, P) = (\lambda_M, \Phi P)$ and $\tilde{\Phi}(\tau) = \tau$. The induced isomorphism of relation quotients commutes with every $M \rightarrow M+1$. A nonzero new relation therefore becomes $(\lambda_{M+1}, 0)$ under exactly the same degree transition as before.

S11.1.8 5. The arithmetic norm becomes a one-variable matrix weight

Let

$$w_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{1}{2\pi}, \quad d\nu_h(t) = w_h(t) dt. \quad (5.1)$$

No mass is assigned the value one. The source norm of a two-variable polynomial is

$$\|\mathcal{T}_2 P\|^2 = \int_{\mathbb{R}^2} |P(1/2 + it_1, 1/2 + it_2)|^2 w_h(t_1) w_h(t_2) dt_1 dt_2.$$

Make the exact real coordinate change

$$u = t_1 + t_2, \quad v = t_1 - t_2, \quad (t_1, t_2) = ((u+v)/2, (u-v)/2), \quad dt_1 dt_2 = \frac{1}{2} du dv. \quad (5.2)$$

Then $S = 1 + iu$, $r = iv$, and $\Delta = -v^2$. If

$$P(S, \Delta) = \sum_{a=0}^b \Delta^a f_a(S), \quad b = \lfloor M/2 \rfloor, \quad \deg f_a \leq M - 2a,$$

define the full matrix

$$W_{h,b}(u)_{ac} = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv. \quad (5.3)$$

Theorem (isometric spectral-sum realization).

$$\|\mathcal{T}_2 \Phi P\|^2 = \int_{\mathbb{R}} f(1 + iu)^* W_{h,b}(u) f(1 + iu) du. \quad (5.4)$$

This follows by (5.2), finite polynomial expansion, and Fubini. All integrals converge by the source's exponential moments. For every real u , the matrix $W_{h,b}(u)$ is positive definite: its quadratic form is the integral of the squared modulus of a nonzero polynomial in $-v^2$ against a density positive almost everywhere. Zeros of the nonzero entire g/h are discrete and cannot remove a positive-measure set of the v -line. The exponential bounds prove convergence also for each fixed u .

For $b = 0$ the scalar coefficient is the convolution

$$W_{h,0}(u) = (w_h * w_h)(u). \quad (5.5)$$

For $b > 0$, its additional rows are the actual relative-coordinate moments. They have not been replaced by (5.5).

The map in (5.4) sends a polynomial to its vector of coefficient functions; its inverse is $f \mapsto \sum \Delta^a f_a(S)$ on the specified filtered image. Multiplication by $s_1 + s_2$ is exactly multiplication by S on every component. Hence its generator is still A_2 , its degree increase is still one, and its boundary is still (3.4).

S11.1.9 5.1 General tensor degree

For any k , set $S = \sum s_i$, retain $x_i = s_i$ for $i < k$, and put $s_k = S - \sum_{i < k} x_i$. This is a polynomial algebra isomorphism with determinant one on the real variables $(t_1, \dots, t_k) \leftrightarrow (t_1, \dots, t_{k-1}, u = \sum t_i)$.

Write $P = \sum_{\alpha} x^{\alpha} f_{\alpha}(S)$, with $\deg f_{\alpha} \leq M - |\alpha|$. Its matrix weight is

$$W_{k,M}(u)_{\alpha\beta} = \int_{\mathbb{R}^{k-1}} \prod_{i < k} \overline{(1/2 + it_i)^{\alpha_i}} \prod_{i < k} (1/2 + it_i)^{\beta_i} \prod_{i < k} w_h(t_i) w_h(u - \sum_{i < k} t_i) d^{k-1}t. \quad (5.6)$$

The outer coordinate is $S = k/2 + iu$. The same proof gives an isometry on the actual degree- M source, with every original relation substituted through this algebra isomorphism. Its entries have exponential moments in u for every exponent below the source threshold $\pi/2$: use $|\sum t_i| \leq \sum |t_i|$, the polynomial factors, and the one-variable exponential moment. Constants depend on the retained packet, k , and degrees; no uniform-in- k estimate is inferred.

S11.1.10 6. The canonical representative, kernel control and relation layers commute with the descent

Let C_M be the explicit coefficient matrix of Φ from $S^a \Delta^b$ to the original symmetric polynomial basis. It is invertible between those two filtered spaces. Let U be the exact coordinate map of (4.4) to the original unscaled invariant quotient basis. Retain its orbit coefficients.

For the source moment and full-jet matrices,

$$M'_M = C_M^* M_M C_M, \quad J'_M = U^{-1} J_M C_M. \quad (6.1)$$

The matrix in (5.3) gives exactly M'_M by one-dimensional integration of its entries against the retained powers of S .

The inverse interpolation kernel and its metric satisfy

$$K'_M = J'_M (M'_M)^{-1} (J'_M)^* = U^{-1} K_M (U^{-1})^*, \quad G'_M = U^* G_M U. \quad (6.2)$$

These identities follow by multiplying the stated inverses, not by selecting a new metric. The least-norm representatives obey $R'_M = R_M U$ after their polynomial-source identification. Their action and control are

$$A'_2 = U^{-1} A_2 U, \quad W'_M = U^* W_M U, \quad \mathcal{Z}'_M = U^{-1} \mathcal{Z}_M (U^{-1})^*. \quad (6.3)$$

Thus every control inequality is exactly transported. The same isometry sends the actual next boundary layer, its Gram matrix, and its two maps Y_M, C_M^{rel} to their images in (5.4). It preserves the finer rank bound supplied by the new attachment.

The arithmetic representative defect remains a relation with an explicit primitive (4.7). Its cross-pairing is evaluated by (5.3); it is not erased when the relation becomes a supported zero in the next internal quotient.

S11.1.11 6.1 Retain the base and relative parts of the norm separately through a triangular map

Write the finite matrix weight as

$$W(u) = \begin{pmatrix} m_0(u) & b(u) \\ b(u)^* & C(u) \end{pmatrix}, \quad m_0(u) > 0.$$

There is the exact isometric map into the stated direct-sum weighted space

$$\mathcal{A} : f \mapsto (f_0 + m_0^{-1} b f_{\text{rel}}, f_{\text{rel}}), \quad (6.4)$$

with inverse $(a, z) \mapsto (a - m_0^{-1} b z, z)$ and weights

$$m_0(u), \quad C(u) - b(u)^* m_0(u)^{-1} b(u). \quad (6.5)$$

Completing the square proves the isometry and positivity of the second weight. No measure is renormalized: m_0 and its reciprocal are both in the formulas. The image $\mathcal{A}(\mathcal{P}_M^{\text{sym}})$ is retained, not enlarged to arbitrary functions in the target. The ratio $m_0^{-1} b$ need not be polynomial. Multiplication by $S = 1 + iu$ commutes with this pointwise map, so the derivative action is still specified without a new connection term.

This supplies two concrete analytic quantities for the next estimates: the convolution mass and the full relative covariance matrix. Neither positivity of this representative norm nor the scalar mass alone is being substituted for the arithmetic Weil pairing.

S11.1.12 7. Calculate every nilpotent fibre of the sum map

The pushforward must not replace a multiple jet by its residue. We now give its complete Jordan lengths.

For an ordered root tuple $(\rho_1, \dots, \rho_k)$ with retained orders m_i , its local algebra is

$$L_\rho = \mathbb{C}[z_1, \dots, z_k]/(z_1^{m_1}, \dots, z_k^{m_k}), \quad A_k = \rho_\Sigma I + N, \quad \rho_\Sigma = \sum \rho_i, \quad N = M_{\sum z_i}. \quad (7.1)$$

Let $D_0 = \sum (m_i - 1)$ and $c_r = [t^r] \prod_i (1 + t + \dots + t^{m_i-1})$, with $c_{-1} = 0$.

Theorem (full ordered sum-fibre decomposition). As a $\mathbb{C}[S]$ -module,

$$\pi_{k*} L_\rho \cong \bigoplus_{r=0}^{\lfloor D_0/2 \rfloor} (\mathbb{C}[S]/(S - \rho_\Sigma)^{D_0-2r+1})^{\oplus (c_r - c_{r-1})}. \quad (7.2)$$

Here the isomorphism is constructed by primitive vectors, as follows. On z_i^j set

$$F_i z_i^j = j(m_i - j) z_i^{j-1}, \quad H_i z_i^j = (2j - (m_i - 1)) z_i^j.$$

For $F = \sum F_i$ and $H = \sum H_i$, direct substitution gives

$$[N, F] = H, \quad [H, N] = 2N, \quad [H, F] = -2F. \quad (7.3)$$

An auxiliary positive inner product making $F = N^*$ is explicitly diagonal:

$$\langle z^\alpha, z^\beta \rangle_0 = \delta_{\alpha\beta} \prod_i \frac{\alpha_i! (m_i - 1)!}{(m_i - 1 - \alpha_i)!}. \quad (7.4)$$

These are specified positive numbers. For the actual arithmetic Gram $G_{k,M}$ and the displayed diagonal Gram G_0 , their exact comparison is the invertible operator $C_0 = G_{k,M}^{-1} G_0$, with inverse $G_0^{-1} G_{k,M}$:

$$\langle v, w \rangle_0 = \langle v, C_0 w \rangle_{G_{k,M}}. \quad (7.4a)$$

Equation (7.4) is used only to prove the algebraic decomposition. All arithmetic control remains with $G_{k,M}$. In the primitive-ladder basis its entries are $(N^i p)^* G_{k,M} (N^j q)$, including all cross-ladder terms. They are not discarded because the auxiliary Gram admits orthogonal ladders.

A primitive vector p has degree r and satisfies $Fp = 0$. Put $w = D_0 - 2r$. The commutators prove

$$F N^j p = j(w - j + 1) N^{j-1} p, \quad \|N^j p\|_0^2 = \frac{j! w!}{(w - j)!} \|p\|_0^2 \quad (0 \leq j \leq w). \quad (7.5)$$

Thus its ladder has length $w + 1$ and $N^{w+1} p = 0$. Conversely a nonzero lowest vector has $r \leq D_0/2$, since $\|Np\|_0^2 = (D_0 - 2r) \|p\|_0^2$. Orthogonal complements of the constructed ladders are invariant under N, F, H , so induction decomposes the full finite space. Counting degree- r vectors before the middle gives $c_r - c_{r-1}$ new primitive ladders. The isomorphism in (7.2) maps $[q(S)]$ in a chosen primitive copy to $q(\rho_\Sigma + N)p$.

S11.1.13 7.1 All symmetric packets, including collisions of sums

For an occupation list $\mathbf{n} = (n_\rho)_\rho$, $\sum n_\rho = k$, the corresponding invariant local component is

$$B_{\mathbf{n}} = \bigotimes_{\rho} \left(\mathbb{C}[z_1, \dots, z_{n_\rho}] / (z_j^{m_\rho}) \right)^{S_{n_\rho}}.$$

It retains its own canonical central idempotent even when two different occupation lists have the same sum $\sum n_\rho \rho$. Its graded dimensions c_r are counted by unscaled orbit monomials; equivalently their generating polynomial is

$$\prod_{\rho} \prod_{j=1}^{n_\rho} \frac{1 - t^{m_\rho+j-1}}{1 - t^j}. \quad (7.6)$$

For $n_\rho = 0$ the factor is one. This is a polynomial identity for the orbit counts; no division at a root of unity is needed. The same operators preserve invariants and the same primitive proof gives (7.2) with

$$D_0 = \sum n_\rho (m_\rho - 1), \quad \rho_\Sigma = \sum n_\rho \rho$$

and these invariant c_r . It includes all k and all finite multiplicities.

For one order-three zero taken twice, write $z_i = s_i - \rho$, $X = z_1 + z_2$. The symmetric component has the exact ladders

$$1, X, X^2, X^3, X^4, \quad p = z_1^2 - z_1 z_2 + z_2^2, \quad Xp = 0, \quad (7.7)$$

with $X^3 = 3(z_1^2 z_2 + z_1 z_2^2)$ and $X^4 = 6z_1^2 z_2^2$. Its pushforward is

$$\mathbb{C}[S]/(S - 2\rho)^5 \oplus \mathbb{C}[S]/(S - 2\rho). \quad (7.8)$$

The actual total dimension is six. Merely retaining one evaluation at 2ρ would lose five dimensions; the displayed injective primitive maps and direct sum retain all six.

The original $v_h^{\otimes k}$ acts by an invertible multiplication map on every component and commutes with A_k . It carries these coordinates to the original full-jet coordinates. Its coefficients are not assigned one.

S11.1.14 8. Pairings, traces, and the support-layer meaning of the ladders

The algebraic direct image in (3.1) keeps the full underlying vector space. Therefore its trace of a^{A_k} is unchanged. On the occupation component,

$$\mathrm{Tr}(a^{A_k} \mid B_{\mathbf{n}}) = \left[\prod_{\rho} \binom{n_\rho + m_\rho - 1}{n_\rho} \right] a^{\sum n_\rho \rho}. \quad (8.1)$$

This is valid with all nonzero nilpotents, since their exponential is triangular with diagonal one. The cochain summand has degree k , so its supertrace is $(-1)^k$ times (8.1). The complementary symmetry summands retain their previously calculated traces. No packet is deleted by pushing to the sum line.

Each ladder map in (7.2) has a split lift fixing τ . In the module $\mathbb{C}[S]/(S - \rho_\Sigma)^\ell$, the transition to the quotient of length $\ell - 1$ takes the nonzero top jet to the supported zero. On its original side the quotient map is the corresponding quotient by $\mathbb{C}N^{\ell-1}p$; it is a $\mathbb{C}[S]$ -module quotient, not a declaration

that this line is an algebra ideal. The full direct-image theorem retains the line before that optional observation.

The residue pairing and arithmetic Jacobian trace form are transported by the same full coordinate matrices. For an invertible coordinate map U , retain

$$S' = U^* S U, \quad J'_g = U^{-1} J_g U, \quad G' = U^* G U, \quad S' J'_g = U^* S J_g U. \quad (8.2)$$

On product cohomology one additionally retains the original tensor-duality signs. No auxiliary inner product from (7.4) is substituted in (8.2). A supported zero of the trace and a full vanishing jet are connected only through the stated Jacobian/trace map.

S11.1.15 9. A complete unscaled arithmetic-moment calibration

For a declared Gaussian calibration, not an asserted zeta packet, take the one-variable measure $e^{-t^2/2} dt$. The change (5.2) gives

$$\mathbf{W}_2(u) = \sqrt{\pi} e^{-u^2/4} \begin{pmatrix} 1 & -2 & 12 \\ -2 & 12 & -120 \\ 12 & -120 & 1680 \end{pmatrix} \quad (9.1)$$

on coefficient functions of $1, \Delta, \Delta^2$. This uses

$$\frac{1}{2} \int v^{2j} e^{-v^2/4} dv = 2^j (2j-1)!! \sqrt{\pi},$$

with $(-1)!! = 1$. The actual triangular coordinate matrix with columns $1, \Delta + 2, \Delta^2 + 12\Delta + 12$ is

$$T = \begin{pmatrix} 1 & 2 & 12 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{pmatrix}, \quad T^* \mathbf{W}_2(u) T = \sqrt{\pi} e^{-u^2/4} \text{diag}(1, 8, 384). \quad (9.2)$$

All three fibre directions and their norms survive. For the first two directions, the exact base/relative decomposition is

$$\|f_0 + \Delta f_1\|^2 = \sqrt{\pi} \int e^{-u^2/4} (|f_0 - 2f_1|^2 + 8|f_1|^2) du. \quad (9.3)$$

The source metric has total two-variable mass 2π , retained by these formulas. Exact symbolic tests compare the moments of $S^a \Delta^b$ computed directly in (t_1, t_2) and through (9.1), and compare the resulting constrained representatives, jet maps, Grams and controls.

For the actual arithmetic measure, (5.3) supplies the corresponding nonconstant matrix. Its relative covariance need not have the constant Gaussian form. Neither the calibration's triangular coefficients nor its control are imported into the arithmetic case.

S11.1.16 10. Position in the Deligne-style programme

The source's tensor-square step is followed by geometric pushforward and control on that pushforward, not by tensoring alone. The construction here now supplies a **specified sum-coordinate pushforward** of the actual finite arithmetic image and its original filtered representatives. Its metric realization is the exact one-dimensional matrix weight (5.3) or (5.6), with the relative directions retained. The symmetric relation presentation (4.4) and original primitives (4.7) say exactly which zero-amplitude relations are present in that fibre.

The result does not identify this finite spectral scheme or matrix-weight model with a lisse sheaf on a Lefschetz pencil. The map actually constructed is the quasi-coherent finite pushforward (3.1), connected to the original analytic representatives by (2.3), (4.8), and (5.4). Deligne’s geometric hypotheses and estimates have not thereby been imported.

The immediate control task now has an explicit base/relative split: estimate the actual convolution mass, the full relative covariance, and the full-jet interpolation on the **image** of the filtered polynomial family in (6.4). These determine the same $K_M, \mathcal{Z}_M$ as the preceding work. The sum map retains A_k and every repeated eigenline, so a bound sublinear in k would still have the intended force. No such bound is proved in this note.

The concrete advances are the exact filtered symmetric relation algebra, an all- k spectral-sum norm realization, and a full nilpotent direct-image decomposition that retains multiplicities and colliding occupations. They are new applications to this specified programme; the reusable algebra and integration tools are standard and are not asserted to be globally novel.

S11.1.17 References and reading scope

1. Supplied **Tau Kernel Layer Integration**, RESEARCH_NOTE.md and NOTE.tex; read in full in this continuation. Its finite-layer identifications and sharper rank are retained as source results.
2. Supplied **Tau Symmetric Frontier Control**, RESEARCH_NOTE.md; read in full. Its signed projector, orbit factors, next-layer map, and finite bound are retained.
3. Supplied **Tau Coherent Interpolation Control**, especially polynomial/jet definitions and joint total-degree sections; used as stated analytic input.
4. The Stacks Project, Tag **0AVV**, affine pushforward of quasi-coherent complexes; Tag **0AWZ**, the precise categories for the finite-pushforward adjoint. <https://stacks.math.columbia.edu/tag/0AVV> and <https://stacks.math.columbia.edu/tag/0AWZ>.
5. NIST DLMF, §18.2, polynomial moment and kernel background. <https://dlmf.nist.gov/18.2>.
6. Deligne, *La conjecture de Weil II*, IHES 52 (1980), 137–252. The retained French TeX window around §§3.2.11–3.2.15 was reread. Its line 872 contains an inconsistent numerical implication in the historical transcription; that literal line is not used as a proof step here. The source-based tensor/pushforward architecture is used, not a certification of the entire 116-page proof. Original bibliographic record: <https://archive.numdam.org/articles/10.1007/BF02684780/>.

S12 PR 20: Arithmetic sum density, the full relative connection and conormal derivatives

Source attribution. Complete TeX proof witness delivered in the sum-connection archive on 12 September 2026. The verified PR 20 head contains its shorter public note and checker. This full NOTE is identified by its own source and archive hashes, and is not assigned the public note's Git blob. The local conormal continuation proves the input-constant to socle map with its full multiplicities. The local connection continuation supplies the exact projected derivative pair for the graph map. The reproduced source witness is preserved with its original statements.

Exact revision. 2144c358c2b41aa235b15cf0fa472289aec8673f

Source file. Tau_Sum_Connection_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Verified public PR 20 head; the complete delivered witness has the separate hash below

Source SHA-256. 11870208e3a08df186e42629ae1a6c956faaf28f79dfce0dc2f2d44dc64b4687

Original archive. Tau_Sum_Connection_Control_2026-09-12.zip

Archive SHA-256. f2de513041b7336b495a338e90325e84c5ce53cfa98799e1b89c53af308356de

S12.1 Quantitative sum-fibre connection and the conormal arithmetic trace

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. Written analytic and algebraic deductions for review. The original theta source, its asserted range theorem, and the canonical finite representatives are inputs from the preceding work. No new Lean execution or arithmetic purity theorem is claimed. The new analytic estimate is a mass-retaining contraction for the actual spectral-sum density. The full matrix connection, its normal component, and the conormal derivative of the original theta relations are constructed as well; the contraction is not assigned to the full Weil control form without a comparison.

S12.1.1 1. Source intake and the unchanged marked object

The newly uploaded Kernel Layer Integration archive is byte-identical to its earlier uploaded copy. Its thirty-one manifest entries and the fifty-three entries in the Spectral Sum Descent archive were independently verified. The kernel-layer note's identities are used as source results, not counted as new deductions:

$$K_M = G_M^{-1}, \quad V_M = A_k K_M + K_M A_k^* - k K_M, \quad W_M = G_M V_M G_M. \quad (1.1)$$

In particular the same control problem is transported by the actual map $v = K_M \lambda$. The source's next theta-relation layer, its full-jet columns, and its map into B_{M+1}/B_M remain attached. Its formalization interface is the existing internal quotient, not a replacement quotient.

Retain the marked absolute base $\mathfrak{b}_\tau$, the scalar semiring, and the original infinite quotient:

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.2)$$

The target of $p_{\mathbb{Z}}$ is infinite. Every ring map below is lifted through G and this square. A complex-linear map $T : V \rightarrow W$ is lifted instead by the explicitly defined split-linear map $T^\tau(v^\bullet) = (Tv)^\bullet$, $T^\tau(\tau) = \tau$. These are different types of construction, connected by their respective arithmetic projection squares. In the full support diagram the same formula is $(\lambda, v) \mapsto (\varphi(\lambda), T_\lambda v)$, with the displayed support transport φ . In particular amplitude zero is the receiving fibre's zero, not external absence.

The original test spaces and operators are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ C_+ &= [V \xrightarrow{\Theta} \mathcal{B}], \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \\ \phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad g(s) = 2\xi(s), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1. \end{aligned} \quad (1.3)$$

Fix a finite packet of actual zeros with full multiplicities, stable under $\rho \mapsto 1 - \bar{\rho}$. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h v_h \in E_h^\times. \quad (1.4)$$

The source supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. It also supplies

$$\mathcal{T}_h P = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (1.5)$$

The statements below also apply to $h = 1$ as the actual theta-seed calculation, but that case has zero-dimensional arithmetic quotient and is not substituted for a nonempty packet.

S12.1.2 2. The sum coordinate and its conjugate logarithmic operator

Let $k \geq 2$, $S = \sum_{i=1}^k s_i$, and use the explicit centered relative coordinates

$$z_i = s_i - S/k \quad (i < k), \quad z_k = -\sum_{i < k} z_i, \quad s_i = S/k + z_i. \quad (2.1)$$

On the original integration lines set

$$s_i = \frac{1}{2} + it_i, \quad S = \frac{k}{2} + iu, \quad u = \sum_i t_i, \quad y_i = t_i - u/k \ (i < k), \quad y_k = -\sum_{i < k} y_i. \quad (2.2)$$

The inverse is $t_i = u/k + y_i$. The real Jacobian has absolute value one. For $k = 2$, the earlier relative coordinate was $v = t_1 - t_2 = 2y_1$, so $dt_1 dt_2 = du dy_1 = \frac{1}{2} du dv$. Thus the earlier factor $1/2$ remains exactly in that coordinate comparison. Its Δ is $4z_1^2 = -4y_1^2$.

Write

$$a_h(t) = \frac{v_h(\frac{1}{2} + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt = \|F_h\|_{L^2(dx)}^2. \quad (2.3)$$

The factor $\sqrt{2\pi}$ is the stated Mellin–Fourier Plancherel factor. No mass has been set to one. Define the actual fibre amplitude

$$\Psi_k(u, \mathbf{y}) = \prod_{i=1}^k a_h(u/k + y_i). \quad (2.4)$$

Mellin–Fourier followed by (2.2) is the unitary map

$$\mathcal{U}_k : L^2(\mathbb{R}_{>0}^k, d^k x) \longrightarrow L^2(\mathbb{R}_u \times \mathbb{R}_{\mathbf{y}}^{k-1}, du d\mathbf{y}).$$

On the given test functions it satisfies

$$\begin{aligned} \mathcal{U}_k(D^{(k)}F) &= (\tfrac{k}{2} + iu)\mathcal{U}_kF, \quad D^{(k)} = \sum_i D_i, \\ \partial_u \mathcal{U}_kF &= i\mathcal{U}_k(\mathcal{L}_kF), \quad \mathcal{L}_k = \frac{1}{k} \sum_{i=1}^k \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -1. \end{aligned} \quad (2.5)$$

Here $\mathcal{L}_k$ is multiplication by the displayed function; $1/k$ comes from the inverse coordinate map keeping the relative z_i fixed. It does not change the scaling generator $D^{(k)}$. The identities follow by differentiating the Mellin integral. Logarithmic multiplication preserves $\mathcal{B}$ and the corresponding tensor test space.

For the polynomial source,

$$\mathcal{U}_k \mathcal{T}_h^{(k)} P = \Psi_k(u, \mathbf{y}) P(S/k + iy_1, \dots, S/k + iy_k). \quad (2.6)$$

Thus the operator whose new estimate is computed is related to the original scaling operator by the precise commutator (2.5), not equated with the generator or its weight defect.

S12.1.3 3. A quantitative contraction on the actual arithmetic density

Define the convolution mass and its logarithmic-derivative energy by

$$\begin{aligned} m_k(u) &= \int_{\mathbb{R}^{k-1}} |\Psi_k(u, \mathbf{y})|^2 d\mathbf{y} = w_h^{*k}(u), \quad \int m_k = \mu_h^k, \\ \mathcal{I}_h &= \int_{\mathbb{R}} \frac{|w'_h(t)|^2}{w_h(t)} dt, \quad \mathcal{I}_{h,k} = \int_{\mathbb{R}} \frac{|m'_k(u)|^2}{m_k(u)} du. \end{aligned} \quad (3.1)$$

At isolated zeros of w_h , the integrand is assigned its limiting value where applicable, or any value on that null set. The construction and proofs use the smooth amplitude a_h , so no division at a zero is required. For $k \geq 2$, $m_k(u) > 0$ for every real u : each factor w_h is positive almost everywhere, and its convolution integral has a positive integrand almost everywhere on its fibre.

The original functional equation gives

$$g^\dagger = g, \quad h^\dagger = (-1)^d h, \quad v_h^\dagger = (-1)^d v_h, \quad \overline{a_h(t)} = (-1)^d a_h(t). \quad (3.2)$$

Accordingly a_h and a'_h have the same fixed real/imaginary phase and

$$\boxed{\mathcal{I}_h = 4 \int_{\mathbb{R}} |a'_h(t)|^2 dt = 4 \int_0^\infty (\log x)^2 |F_h(x)|^2 dx < \infty, \quad \int \overline{a_h} a'_h = 0.} \quad (3.3)$$

Proof: outside the discrete zero set, $w'_h = 2\overline{a_h}a'_h$ is real, giving $w_h'^2/w_h = 4|a'_h|^2$. Mellin Plancherel and $\partial_t v_h(\frac{1}{2} + it) = i v'_h(\frac{1}{2} + it)$ give the second equality. The zero integral is the integral of $w'_h/2$. The decay needed at both endpoints follows from $F_h \in \mathcal{B}$: $e^{y/2} F_h(e^y)$ is Schwartz, as are its polynomially weighted versions. Reflection (3.2) is essential for the equality rather than just an upper bound.

Let $\mathcal{H}_u = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$. The line inclusion and projection are

$$j_{0,u} : \mathbb{C} \rightarrow \mathcal{H}_u, \quad c \mapsto c\Psi_k(u, \cdot), \quad \Pi_{0,u}Z = \Psi_k \frac{\langle \Psi_k, Z \rangle}{m_k(u)}.$$

Put

$$n_k(u) = (1 - \Pi_{0,u})\partial_u \Psi_k = \partial_u \Psi_k - \frac{m'_k}{2m_k} \Psi_k. \quad (3.4)$$

The inner product is antilinear in its first argument. Equation (3.2) makes $\langle \Psi_k, \partial_u \Psi_k \rangle = m'_k/2$ real. Therefore (3.4) is precisely an orthogonal decomposition, including at zeros of individual amplitude factors.

Theorem 3.1 (mass-retaining sum contraction and exact relative term).

$$\mathcal{I}_{h,k} + 4 \int_{\mathbb{R}} \|n_k(u)\|_{\mathcal{H}_u}^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h. \quad (3.5)$$

In particular

$$\mathcal{I}_{h,k} \leq \frac{\mu_h^{k-1}}{k} \mathcal{I}_h. \quad (3.6)$$

Proof. Differentiate the actual product amplitude in the centered coordinates:

$$\partial_u \Psi_k = \frac{1}{k} \sum_{i=1}^k a'_h(t_i) \prod_{j \neq i} a_h(t_j).$$

Integrating its squared modulus over all original t_i , the k diagonal terms each give $\mu_h^{k-1} \|a'_h\|^2$. Every distinct-index cross term is $\mu_h^{k-2} |\int \overline{a_h} a'_h|^2 = 0$. Thus

$$\int \|\partial_u \Psi_k\|^2 du = \frac{\mu_h^{k-1}}{4k} \mathcal{I}_h.$$

The pointwise orthogonal decomposition (3.4) gives

$$\|\partial_u \Psi_k\|^2 = \frac{m_k'^2}{4m_k} + \|n_k\|^2.$$

Integrate and multiply by four. This proves both statements and the finiteness of $\mathcal{I}_{h,k}$. All masses and the factor four remain.

This is an application of the projection mechanism underlying Fisher-information convolution inequalities (Johnson–Barron), with the actual arithmetic amplitude, its phase, and its residual relative vector kept explicitly. It does not rely on log-concavity, a Gaussian approximation, or locations of the zeros beyond the stipulated reflection-stable packet and the source construction of F_h .

The pair $(\Pi_{0,u}\partial_u \Psi_k, n_k)$ retains the full derivative by addition. Its split lift can have an active zero in either component without converting the other component or its source to τ . Dropping n_k would replace equality (3.5) by an inequality and lose the quantified relative contribution; the actual projection and its complementary map remain available.

S12.1.4 4. The full relative matrix has a computed connection and normal variation

Take any finite linearly independent family of relative polynomials $\theta_\alpha(\mathbf{z})$, independent of S . For the original degree- M source use the actual coefficient functions

$$P(\mathbf{s}) = \sum_{\alpha} \theta_{\alpha}(\mathbf{z}) f_{\alpha}(S), \quad \deg f_{\alpha} \leq M - \deg \theta_{\alpha}. \quad (4.1)$$

Choose a homogeneous relative monomial basis, or the stated unscaled invariant basis on a symmetric summand, so that these degree conditions specify the exact original filtered image. This is the centered-coordinate version of the preceding matrix weight; substitution $s_i = S/k + z_i$ is triangular by total degree and invertible.

Define the actual fibre map, its derivative, and its three Gram matrices:

$$\begin{aligned} j_u : \mathbb{C}^r &\rightarrow \mathcal{H}_u, & j_u c &= \Psi_k(u, \mathbf{y}) \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}, \\ W(u) &= j_u^* j_u, & B(u) &= j_u^* j'_u, & T(u) &= (j'_u)^* j'_u. \end{aligned} \quad (4.2)$$

These are integrals in the same arithmetic measure. $W(u)$ is strictly positive because the relative polynomials are independent and Ψ_k is nonzero almost everywhere. Every entry is smooth by differentiated Schwartz estimates on compact u intervals. Its domain has its stated finite size r ; no assertion is made that r remains fixed as the source degree grows.

In the fixed-phase case (3.2),

$$B(u) = \frac{1}{2} W'(u). \quad (4.3)$$

Indeed $\overline{\Psi_k} \partial_u \Psi_k$ is real almost everywhere, hence B is Hermitian; differentiate W to get $W' = B + B^*$. The integrals remain valid at the zero set because the smooth product amplitude, not its logarithm, is used.

Let

$$\Pi_u = j_u W^{-1} j_u^*, \quad \Gamma_u = W^{-1} B = \frac{1}{2} W^{-1} W', \quad N_u = (1 - \Pi_u) j'_u. \quad (4.4)$$

The exact differential map is

$$\boxed{\partial_u(j_u c(u)) = j_u(c'(u) + \Gamma_u c(u)) + N_u c(u), \quad j_u^* N_u = 0.} \quad (4.5)$$

Thus $\nabla_u c = c' + \Gamma_u c$ is a metric-compatible connection on this finite coefficient bundle:

$$\Gamma_u^* W + W \Gamma_u = W'. \quad (4.6)$$

The complete normal form is

$$\boxed{\mathcal{N}(u) := N_u^* N_u = T - B^* W^{-1} B = T - \frac{1}{4} W' W^{-1} W' \succeq 0.} \quad (4.7)$$

In particular no commutation of W and W' is assumed. Multiplication proves the identity and positivity follows from $1 - \Pi_u$ being an orthogonal projection. The norm decomposition is

$$\boxed{\|\partial_u(j_u c)\|^2 = (\nabla_u c)^* W (\nabla_u c) + c^* \mathcal{N} c.} \quad (4.8)$$

For the one-column constant family, (4.8) is exactly (3.5) before integration. For the full relative family it retains every matrix entry in both terms. Scalar inequality (3.6) is not inserted as a bound

on this larger matrix: its precise connection is the inclusion of the constant column and the pair of projections $\Pi_{0,u}, \Pi_u$ with $\text{im } \Pi_{0,u} \subset \text{im } \Pi_u$.

S12.1.5 Change of relative frame

For an explicitly given invertible smooth matrix $C(u)$, replace the fibre map by $j_u^C = j_u C(u)$. The exact transformed connection and normal map are

$$W^C = C^* W C, \quad \Gamma^C = C^{-1} \Gamma C + C^{-1} C', \quad N^C = N C. \quad (4.12)$$

These follow by differentiating j^C and applying the same orthogonal projection. This is the additional derivative term needed when using the earlier pointwise triangular square-completion map, whose coefficients can depend on u . Multiplication by S still intertwines without such a term; the derivative's relationship to it is the explicit commutator (2.5). The exact filtered image of the original coefficient functions is retained after this change of frame.

S12.1.6 4.1 The differentiated filtered image is retained

For the actual source coefficient function $c_P(u) = (f_\alpha(k/2 + iu))_\alpha$, define the linear graph map

$$\mathcal{D}_M : P \mapsto (j_u \nabla_u c_P, N_u c_P). \quad (4.9)$$

Its values lie in the specified graph image $\mathcal{G}_M$, not in two freely chosen function spaces. Addition of the two components recovers

$$\text{Add } \mathcal{D}_M(P) = \partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P = i \mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} P. \quad (4.10)$$

The two summands are pointwise orthogonal, so this addition is an isometric injection on the graph into the derivative's actual image. This supplies a precise way to recover its arithmetic jet:

$$\mathcal{G}_M \xrightarrow{\text{Add}} i \mathcal{U}_k (\mathcal{B}^{\widehat{\otimes} k}) \xrightarrow{-i \mathcal{U}_k^{-1}} \mathcal{B}^{\widehat{\otimes} k} \xrightarrow{J^{(k)}} E_h^{\otimes k}. \quad (4.11)$$

No separate arithmetic-jet map on an arbitrary measurable normal section is asserted. The combined graph is exactly the domain on which the formula is defined. The original support label is retained in both components and in (4.11). The cohomological control still uses A_k, R_M, G_M, W_M and (1.1); (2.5), (4.10), and (4.11) are the actual comparison to this new analytic derivative.

S12.1.7 5. Differentiating a supported relation requires its first infinitesimal layer

The logarithmic observable in (2.5) has a calculated effect on the original theta relation, even in one variable:

$$\boxed{J_h(\log x \Theta \phi) = j_h(g') j_h(H_\phi)}. \quad (5.1)$$

Proof: logarithmic multiplication differentiates the actual Mellin transform, so

$$\mathcal{M}(\log x \Theta \phi) = g' H_\phi + g H'_\phi.$$

Taking full jets modulo h removes the second term, but not the first. For $\phi = P(D)\phi_*$, $H_\phi = P$; hence all E_h values of $j_h H_\phi$ are supplied by actual original source functions.

This is the map connecting the original supported boundary to the Jacobian trace contraction. In the original quotient $q\Theta = 0$, while $J_h(\log x \Theta)$ has the explicitly displayed image. The typed resolution of that effect is a first thickening of the *same* relation ideal.

Put

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)), \quad E = \mathcal{P}/I, \quad E^{[2]} = \mathcal{P}/I^2.$$

There is the actual exact sequence

$$0 \longrightarrow I/I^2 \longrightarrow E^{[2]} \xrightarrow{\pi} E \longrightarrow 0. \quad (5.2)$$

The coefficient conormal map is the isomorphism

$$\iota_N : E^k \xrightarrow{\sim} I/I^2, \quad (a_1, \dots, a_k) \mapsto \left[\sum_i h(s_i) \tilde{a}_i \right]. \quad (5.3)$$

A direct proof uses monic division in each variable. The polynomial ring is free over $\mathbb{C}[h(s_1), \dots, h(s_k)]$ with basis $\prod s_i^{a_i}$, $0 \leq a_i < d$. Modulo the square of the ideal of these latter variables, only coefficient degrees zero and one survive. This proves uniqueness and surjectivity in (5.3), retaining all nilpotents in E .

Define the sum derivative at fixed relative coordinates

$$\partial_S^{\text{rel}} = \frac{1}{k} \sum_i \partial_{s_i}. \quad (5.4)$$

It maps I^2 into I , so it induces a *complex-linear* map

$$\delta_S : E^{[2]} \longrightarrow E, \quad [P]_{I^2} \mapsto [\partial_S^{\text{rel}} P]_I. \quad (5.5)$$

It is a derivation along π :

$$\delta_S(ab) = \pi(a)\delta_S(b) + \pi(b)\delta_S(a), \quad \delta_S(S) = 1.$$

Its restriction to the retained conormal fibre is the E -linear row

$$\delta_S \iota_N(a_1, \dots, a_k) = \frac{1}{k} \sum_i h'(s_i) a_i. \quad (5.6)$$

The full conormal differential, before contraction by ∂_S^{rel} , is

$$I/I^2 \xrightarrow{d_I} \bigoplus_i E \, ds_i, \quad (a_i)_i \mapsto \sum_i h'(s_i) a_i \, ds_i. \quad (5.7)$$

This is the two-term cotangent presentation for the complete intersection, in cohomological degrees $-1, 0$. Its universal differential and (5.5) are related by contraction $\sum b_i ds_i \mapsto k^{-1} \sum b_i$. This is the established conormal/cotangent construction applied to this exact packet ideal (Stacks, Tags 00S0 and 08SH).

The original split diagram is

$$G(E^{[2]}) \xrightarrow{G(\pi)} G(E), \quad \delta_S^\tau : G(E^{[2]}) \longrightarrow G(E) \quad \text{as } G(\mathbb{C})\text{-semimodules.} \quad (5.8)$$

The first is a semiring morphism; the second is a split-linear map of coefficient modules, not a semiring morphism. Both send external absence to external absence and supported zero to supported zero. A supported conormal class has image e_E under the first map, while its image under the second is the supported scalar specified in (5.6). The larger domain retains the datum needed to differentiate that relation.

For one local block $h = (s - \rho)^m$ times a retained unit, (5.6) detects the z^{m-1} coefficient with its factor m . In an ordered product of local block lengths m_i , the image of (5.6) has dimension

$$\prod_i m_i - \prod_i (m_i - 1), \quad (5.9)$$

because it is the ideal generated by the unit multiples of $z_i^{m_i-1}$. Each unit is invertible in its actual local algebra. Monomials missing all top exponents form the complementary vector count; they are not discarded from E .

S12.1.8 5.1 Carry the arithmetic jet unit through the derivative

Let $U(\mathbf{s}) = \prod_i v_h(s_i)$, and retain its full Taylor class $\hat{U} \in E^{[2]}$. This class is invertible. Define

$$\hat{\mathcal{J}}P = \hat{U}[P]_{I^2}, \quad \mathcal{J}P = U[P]_I, \quad \beta_h = U^{-1}\delta_S\hat{U} \in E. \quad (5.10)$$

The finite Taylor realization here uses the entire functions on neighbourhoods of the finite zero set; polynomial Hermite interpolation supplies their classes. $E^{[2]}$ is *not* being called a new zeta-zero packet: it also retains first-order source relations not annihilated by g .

The exact commuting equation is

$$\boxed{J^{(k)}\mathcal{L}_k\mathcal{T}_h^{(k)}P = \delta_S\hat{\mathcal{J}}P = \mathcal{J}(\partial_S^{\text{rel}}P) + \beta_h\mathcal{J}P.} \quad (5.11)$$

Proof: its Mellin transform is $\partial_S^{\text{rel}}(UP)$. Apply the product rule before taking jets. Thus the coefficient-connection term is $\beta_h = k^{-1}\sum_i j(v'_h/v_h)(s_i)$, with no discarded local unit.

For an original relation $P = \sum_i h(s_i)P_i$, the last term vanishes and

$$\boxed{J^{(k)}\mathcal{L}_k\mathcal{T}_h^{(k)}P = \frac{1}{k}U\sum_i h'(s_i)[P_i]_I.} \quad (5.12)$$

Modulo I , $v_h(s_i)h'(s_i) = g'(s_i)$, so the same equation can be written with $g'(s_i)$ and the other retained v_h factors. This is the precise relationship between the analytic sum derivative in (4.10) and the conormal row (5.6).

In one variable the source's residue pairing therefore obeys

$$\boxed{\mathcal{R}_Z(f, J_h(\log x \Theta(P(D)\phi_*))) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho).} \quad (5.13)$$

Indeed (5.1) makes the second argument $g'P$, and g'/g has residue m_ρ . The residue pairing, its Jacobian contraction, and the logarithmic derivative of an actual original theta relation have now been connected by a single formula. No new positivity assertion is inferred from that equality.

S12.1.9 6. All higher derivative layers and the old cochain signs remain available

For $r \geq 1$, the derivation gives actual maps

$$\partial_S^{\text{rel}} : \mathcal{P}/I^{r+1} \longrightarrow \mathcal{P}/I^r. \quad (6.1)$$

They commute with the quotient maps wherever their sources and targets match. The associated-graded algebra is explicitly

$$\text{gr}_I \mathcal{P} \cong E[\eta_1, \dots, \eta_k], \quad \eta_i \longleftrightarrow [h(s_i)]. \quad (6.2)$$

On it, the degree-minus-one derivative is

$$\boxed{\text{gr}(\partial_S^{\text{rel}}) = \frac{1}{k} \sum_i h'(s_i) \partial_{\eta_i}.} \quad (6.3)$$

The proof expands $\partial(h^\alpha P)$. The term differentiating P stays in $I^{|\alpha|}$ and disappears only in the explicitly specified quotient $I^{|\alpha|-1}/I^{|\alpha|}$. The other terms give (6.3) with every factor α_i retained. Consequently this layer has rank $\binom{r+k-1}{k-1}$ over E , not one new scalar support.

There is also a full mixed derivative formula retaining all coordinate directions:

$$\boxed{\left[\partial_{s_1} \cdots \partial_{s_k} \left(UP \prod_i h(s_i) \right) \right]_I = \left[UP \prod_i h'(s_i) \right]_I = \left[P \prod_i g'(s_i) \right]_I.} \quad (6.4)$$

Every other product-rule term contains an undifferentiated $h(s_i)$ and maps to the supported zero in E . At the test-function level these derivatives are multiplication by $\prod_i \log x_i$. At the residue level (6.4) is the product Jacobian in the existing tensor arithmetic trace. The ordered volume form $ds_1 \wedge \cdots \wedge ds_k$ and the previously specified tensor-duality sign are retained; no new identification of the iterated residue with an unsigned cochain supertrace is made.

The original primitive of $h(s_i)P_i$ still has the sign $(-1)^{i-1}$ from the prior product complex. Its cochain differential supplies the second $(-1)^{i-1}$. Conormal reduction records the resulting coefficient $[P_i]_I$. It does not replace that source primitive or the higher syzygies of its Koszul complex.

For a finite total-degree relation space, (6.1) is applied to the image of that exact space in $\mathcal{P}/I^{r+1}$. It does not enlarge the set of admissible representatives. The degree- M next-relation map from the attachment feeds (5.12) by its actual polynomial numerator. This supplies a differentiated observation of those very same e -relations.

S12.1.10 7. Exact calibrations and a bounded arithmetic seed evaluation

S12.1.11 7.1 Gaussian fibre: the relative connection can be flat without deleting it

Take the declared calibration $a(t) = e^{-t^2/4}$, $w(t) = e^{-t^2/2}$, $\mu = \sqrt{2\pi}$. This is a test of the formula, not the arithmetic input. For two factors,

$$\Psi_2(u, y) = e^{-u^2/8} e^{-y^2/2}, \quad m_2(u) = \sqrt{\pi} e^{-u^2/4}.$$

For every fixed relative polynomial frame, $j'_u = -(u/4)j_u$, so

$$\Gamma_u = -\frac{u}{4}I, \quad \mathcal{N}(u) = 0, \quad \mathcal{I}_2 = \frac{\mu \mathcal{I}_1}{2} = \pi. \quad (7.1)$$

All relative directions and their Gram entries remain; zero normal variation is a proved consequence of this particular product, not an instruction to erase the relative fibre in the arithmetic case.

S12.1.12 7.2 A zero-bearing calibration has nonzero retained normal variation

Take instead $a(t) = te^{-t^2/4}$ and $w(t) = t^2e^{-t^2/2}$. Then

$$\begin{aligned}\mu &= \sqrt{2\pi}, & \mathcal{I}_1 &= 3\sqrt{2\pi}, \\ \Psi_2(u, y) &= (u^2/4 - y^2)e^{-u^2/8 - y^2/2}, \\ m_2(u) &= \frac{\sqrt{\pi}}{16}e^{-u^2/4}(u^4 - 4u^2 + 12).\end{aligned}\tag{7.2}$$

These identities follow from the complete Gaussian even moments; the polynomial has positive minimum 8, so m_2 has no zero. For $u \neq 0$, $\partial_u \Psi_2$ is not a multiple of Ψ_2 as a polynomial in y , so $n_2(u) \neq 0$. The exact identity is

$$\mathcal{I}_2 + 4 \int \|n_2\|^2 = 3\pi, \quad \mathcal{I}_2 < 3\pi.\tag{7.3}$$

The checker calculates the two-column matrix connection and nonzero positive normal Gram at $u = 1$, retaining the relative basis and its mass factors. This also tests the handling of zeros without assuming logarithmic scores are bounded.

S12.1.13 7.3 The actual arithmetic seed, with no selected zero coordinates

For $h = 1$, $F_h = f_0 = \Theta\phi_*$. Its exact inversion is $f_0(x) = x^{-1}f_0(1/x)$. Hence

$$\mu_1 = 2 \int_1^\infty f_0(x)^2 dx, \quad \mathcal{I}_1 = 8 \int_1^\infty (\log x)^2 f_0(x)^2 dx.\tag{7.4}$$

On that half-line use the actual absolutely convergent sum

$$f_0(x) = 2 \sum_{n \geq 1} (4\pi^2 n^4 x^4 - 6\pi n^2 x^2) e^{-\pi n^2 x^2}.$$

The numerical evaluations at 45 digits with six integer terms and at 65 digits with eight integer terms agree on the displayed 35-digit values:

$$\mu_1 \approx 1.2790072478464851404795335922671933, \quad \mathcal{I}_1 \approx 0.12575246821348846545509759756793817.$$

The optional computation evaluates (7.4) at two stated decimal precisions and two integer cutoffs. Its output is labeled numerical, not an interval certificate. The theorem uses the exact integrals; no numerical value or cutoff is imported into a purity assertion.

S12.1.14 8. What the new estimate controls in the programme

The earlier arithmetic defect remains

$$W_M = A_k^* G_M + G_M A_k - k G_M, \quad V_M = A_k K_M + K_M A_k^* - k K_M, \quad W_M = G_M V_M G_M. \tag{8.1}$$

The new operations retain the path

$$\mathcal{T}_k(P) \xrightarrow{\mathcal{U}_k} \Psi_{kCP} \xrightarrow{\partial_u} \mathcal{G}_M \xrightarrow{\text{Add}, -i\mathcal{U}_k^{-1}} \mathcal{L}_k \mathcal{T}_k(P) \xrightarrow{J^{(k)}} \delta_S \hat{\mathcal{J}} P. \tag{8.2}$$

This is the explicit connection between the positive differential estimate, the matrix-valued relative fibre, and the first conormal layer of the original arithmetic quotient. It does not replace the scaling generator A_k with differentiation. Their relation is the commutator (2.5), and the source's original representative relation $D^{(k)}R_M - R_MA_k \in B_{M+1}$ continues to determine (8.1).

The result of this continuation is a genuine $1/k$ factor in (3.5), with the literal mass μ_h^{k-1} , and an exact decomposition of the full matrix derivative. Uniform relative interpolation control is still needed to transfer that integrated smoothing to the extremal finite-packet estimate (8.1). The required comparison cannot be inferred from the scalar density alone; (4.9)–(4.11) and (5.11) provide the actual objects on which to work. All normal terms and the conormal Jacobian remain in that comparison.

Deligne's source architecture requires a further estimate after the product-to-one-parameter operation, on the same arithmetic image and with its duality. This note constructs and estimates a differential of that new-base pushforward, and identifies its exact interaction with the trace-producing Jacobian. It does not assert that a Fisher inequality is already Deligne purity, or that the analytic matrix bundle is a finite-field lisse sheaf. The finite-point cohomology, spectral-sum restriction functor, kernel-layer calculation, and first infinitesimal quotient remain connected by the typed arrows above.

S12.1.15 References and reading record

1. Supplied *Tau Kernel Layer Integration*, NOTE.tex and RESEARCH_NOTE.md, read for the original relation-layer and antidual comparisons; the archive matched the earlier copy exactly and all manifest entries verified. The source's eighteen-method exact suite was rerun normally and with optimized Python. This is not a new formal certificate.
2. Supplied *Tau Spectral Sum Descent*, NOTE.tex, read in full, especially §§2, 5, 6, 8 and 10. The same sum coordinate, matrix density, relative polynomial image, full jets, and residue comparison are used.
3. Supplied *SplitZero: reconstruction, internal homology, and balanced finite jets*, §§1–4, for the existing original internal quotient and supported cohomology interfaces. No new scalar implementation is introduced.
4. Oliver Johnson and Andrew Barron, *Fisher Information inequalities and the Central Limit Theorem*, arXiv:math/0111020, published in Probability Theory and Related Fields 129 (2004), 391–409. Their projection method is established background; Theorem 3.1 is proved here directly with the unaltered arithmetic mass and phase. <https://arxiv.org/abs/math/0111020>.
5. The Stacks Project, Tags 00S0 and 08SH, the conormal presentation and cotangent complex of a local complete intersection. These are reusable algebraic constructions; the explicit arithmetic conormal row and the trace connection are computed in §§5–6. <https://stacks.math.columbia.edu/tag/00S0> and <https://stacks.math.columbia.edu/tag/08SH>.
6. Deligne, *La conjecture de Weil II*, IHES 52 (1980), 137–252. This continuation relies on the previously identified tensor/pushforward/dual-control architecture, not a fresh reading or independent audit of all pages. No finite-field theorem is asserted on the new base merely from terminology.

S13 Cyclic arithmetic sum control and the complete contracted relation tower

Source attribution. Complete TeX witness supplied in the Cyclic Sum Control archive. Its exact source and archive hashes identify the delivered text. The seven retained prior source files are identified separately; the earlier 48-entry validation claim concerns the separate Sum Connection archive. The new 24-method finite checker has a separate independent replay record. No Lean execution, remote CI or uniform arithmetic estimate is inferred from this source witness.

Exact revision. 2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c

Source file. Tau_Cyclic_Sum_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Source witness SHA-256; no remote Git revision is assigned to this delivery

Source SHA-256. 2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c

Original archive. Tau_Cyclic_Sum_Control_2026-09-12.zip

Archive SHA-256. e0e4d0ecda69e4738dcb4cdd3b750e623d863eac8b44f353737393e720a840a7

S13.1 Sum-generated arithmetic cohomology, scalar control, and exact conormal depth

Owner-directed continuation of the split-zero programme — 12 September 2026.

Status. Written deductions for mathematical review. This note uses the original theta source and its stated finite-packet realization from the preceding work. It does not claim a new Lean certificate, a global comparison-kernel theorem, or a uniform arithmetic purity estimate. The new constructions are (i) an explicitly injected sum-generated arithmetic submodule containing every amplified spectral value, (ii) its scalar convolution norm and rank-two control formula, (iii) raw-moment formulas for the complete differentiated polynomial image, and (iv) the exact contraction of every original conormal thickening to that sum coordinate. The full relative quotient, its action, and its trace are retained.

S13.1.1 1. Source intake and fixed objects

The two new transcript files were searched for changes to the arithmetic/control and formalization interfaces. They preserve the earlier full-jet kernel/layer calculations and the formalized internal quotient; they do not supply a later analytic proof of the remaining estimate. The previous Tau_Sum_Connection_Control archive was opened directly, its LaTeX and mathematical note were read, and all 48 entries of its supplied manifest verified. Its scalar contraction and matrix connection are inputs [S1], not attributed to the present continuation. The previous transcript's checked algebraic interfaces remain within their reported scope [S2–S4].

The original base is still b_τ . Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\begin{array}{ccc}
G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\
p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\
\mathbb{Z} & \xrightarrow{j} & \mathbb{C}.
\end{array} \tag{1.1}$$

The target of $p\mathbb{Z}$ is infinite. A ring homomorphism is lifted through G . A complex-linear map $T : V \rightarrow W$ has instead the explicitly defined $G(\mathbb{C})$ -linear lift $T^\tau(v^\bullet) = (Tv)^\bullet$, $T^\tau(\tau) = \tau$. In a support diagram the formula is $(\lambda, v) \mapsto (\varphi(\lambda), T_\lambda v)$, with its stated support map φ . These formulas commute with the respective amplitude observations.

Use exactly

$$\begin{aligned}
V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\
\mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\
C_+ &= [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \\
\phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad g(s) = 2\xi(s), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1.
\end{aligned} \tag{1.2}$$

Let $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$ be a **nonempty** finite packet of actual zeros of g , with their full orders. Assume it is stable under $\rho \mapsto 1 - \bar{\rho}$. Put

$$d = \deg h, \quad v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(v_h) \in E_h^\times. \tag{1.3}$$

The preceding source provides the actual test function F_h and map

$$\begin{aligned}
\mathcal{M}F_h &= v_h, \quad h(D)F_h = \Theta\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \\
\mathcal{T}_h(hP) &= \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h, \quad q\mathcal{T}_h(P) = \sigma_h(v_h[P]_h).
\end{aligned} \tag{1.4}$$

For $k \geq 2$, define

$$\begin{aligned}
\mathcal{P}_k &= \mathbb{C}[s_1, \dots, s_k], \quad I_k = (h(s_1), \dots, h(s_k)), \quad B_k = \mathcal{P}_k/I_k = E_h^{\otimes k}, \\
S &= s_1 + \dots + s_k, \quad A_k = M_S, \quad U_k = v_h^{\otimes k} \in B_k^\times.
\end{aligned} \tag{1.5}$$

The ordinary invariant submodule is $B_k^{\text{sym}} = B_k^{S_k}$. Its cochain realization is the existing signed chain projector $\frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi$: in top degree the Koszul sign in T_π cancels the displayed sign. The complementary cochain summand and both traces remain in the programme. No change to that projector is made here.

S13.1.2 2. The sum-generated submodule and its exact spectrum

Index the distinct roots by $\rho_1, \dots, \rho_a$, with orders $m_1, \dots, m_a$. For an occupation vector $\mathbf{n} = (n_1, \dots, n_a)$ with $\sum n_i = k$, set

$$\lambda_{\mathbf{n}} = \sum_i n_i \rho_i, \quad L_{\mathbf{n}} = 1 + \sum_i n_i (m_i - 1). \tag{2.1}$$

Let Λ_k be the set of distinct values of $\lambda_{\mathbf{n}}$, and put

$$\ell_\lambda = \max_{\mathbf{n}: \lambda_{\mathbf{n}} = \lambda} L_{\mathbf{n}}, \quad \chi_{h,k}(S) = \prod_{\lambda \in \Lambda_k} (S - \lambda)^{\ell_\lambda}. \tag{2.2}$$

Equal sums are grouped by their exact equality; no assumption of linear independence or of absent collisions is used. The distinct occupation idempotents stay in B_k^{sym} even when their sums agree.

Theorem 2.1. The polynomial (2.2) is the minimal polynomial of A_k both on B_k and on B_k^{sym} . There are an injective unital algebra map and a separate injective module map

$$\begin{aligned} \alpha_{h,k} : C_{h,k} := \mathbb{C}[S]/(\chi_{h,k}) &\hookrightarrow B_k^{\text{sym}}, \quad [P] \mapsto P(A_k)1, \\ \eta_{h,k} = M_{U_k} \alpha_{h,k} : C_{h,k} &\hookrightarrow B_k^{\text{sym}}, \quad [P] \mapsto U_k P(A_k)1. \end{aligned} \quad (2.3)$$

The map M_{U_k} is an invertible B_k^{sym} -module map on its carrier. It is not asserted to be a unital algebra homomorphism; its inverse is $M_{U_k^{-1}}$. The full jet unit is retained by this pair of maps rather than assigned the value one.

Proof. At an ordered root tuple the local factor is

$$L_\rho = \mathbb{C}[z_1, \dots, z_k]/(z_1^{m_{i_1}}, \dots, z_k^{m_{i_k}}), \quad A_k = \lambda I + N, \quad N = M_{z_1 + \dots + z_k}.$$

Every monomial of degree greater than $\sum_j (m_{i_j} - 1)$ vanishes. In precisely that degree, $N^D 1$ contains the nonzero monomial $\prod z_j^{m_{i_j} - 1}$ with coefficient $D! / \prod (m_{i_j} - 1)!$. Thus its nilpotence index is $D + 1$. The minimal polynomial on the product takes the maximum index over factors at the same eigenvalue, which is (2.2). For the invariant algebra, the vector 1 is invariant and its annihilator under $\mathbb{C}[S]$ is already the same polynomial; indeed $P(A_k)1 = 0$ is equivalent to the algebra element $P(S)$ being zero. Thus restriction to invariants does not reduce this minimal polynomial. Multiplication by the actual unit U_k has zero kernel and commutes with A_k . This proves all assertions.

In particular, for every $\rho \in Z$, the value $k\rho$ occurs in Λ_k . The cyclic module has a nonzero eigenvector at every $\lambda \in \Lambda_k$: take the local class $(S - \lambda)^{\ell_\lambda - 1}$ in its Chinese-remainder factor, and zero in its other factors. Its image under $\eta_{h,k}$ is still nonzero and has eigenvalue λ .

This retains every exponent required for tensor amplification. It does not claim that a chosen repeated pure tensor is literally the same vector as this cyclic eigenvector when different occupations have the same sum. The original repeated-local-factor projection is the explicit map

$$C_{h,k} \xrightarrow{\eta_{h,k}} B_k^{\text{sym}} \longrightarrow L_{(\rho, \dots, \rho)}; \quad (2.4)$$

its complete nilpotent image is generated by the local unit times $1, N, \dots, N^{k(m_\rho - 1)}$. A cyclic eigenvector supported at a collided sum can have zero projection to a shorter local block. The spectral value is nevertheless present by the proved injection (2.3).

S13.1.3 The remainder of the arithmetic module remains as a quotient

Let

$$T_{h,k} = B_k^{\text{sym}} / \eta_{h,k} C_{h,k}.$$

Then

$$0 \rightarrow C_{h,k} \xrightarrow{\eta_{h,k}} B_k^{\text{sym}} \rightarrow T_{h,k} \rightarrow 0 \quad (2.5)$$

is an exact sequence of $\mathbb{C}[S]$ -modules. This is a module quotient, not a new algebra quotient of B_k . Its split lift is the original $G(\mathbb{C})$ -linear internal quotient. An element of the retained submodule maps to the supported zero; τ remains τ .

The sequence also has a constructed $\mathbb{C}[S]$ -linear retraction. On the generalized λ -part let $N_\lambda = A_k - \lambda I$, $w_\lambda = e_\lambda(A_k)U_k 1$, and $\ell = \ell_\lambda$. Choose the fixed-order root tuple attaining ℓ and let ϑ_λ extract

one of the top monomial coefficients of $N_\lambda^{\ell-1}w_\lambda$ there. That coefficient is its nonzero local arithmetic unit times the corresponding multinomial coefficient. In particular

$$\vartheta_\lambda(N_\lambda^{\ell-1}w_\lambda) \neq 0.$$

Define

$$\pi_\lambda^0(v) = \sum_{j=0}^{\ell-1} \vartheta_\lambda(N_\lambda^{\ell-1-j}v)X^j \pmod{X^\ell}, \quad p_\lambda(X) = \pi_\lambda^0(w_\lambda).$$

Its constant term is nonzero, so p_λ has a unique multiplicative inverse modulo X^ℓ . The actual retraction is

$$\boxed{\pi_\lambda(v) = p_\lambda(X)^{-1}\pi_\lambda^0(v).} \quad (2.5a)$$

Indeed $\pi_\lambda^0(N_\lambda v) = X\pi_\lambda^0(v)$, and $\pi_\lambda(w_\lambda) = 1$. Combining these maps with the CRT idempotents gives $\pi : B_k^{\text{sym}} \rightarrow C_{h,k}$ with $\pi\eta_{h,k} = 1$. Thus the inverse pair is

$$\begin{aligned} C_{h,k} \oplus \ker \pi &\longrightarrow B_k^{\text{sym}}, & (c, t) &\mapsto \eta_{h,k}c + t, \\ v &\longmapsto (\pi v, v - \eta_{h,k}\pi v). \end{aligned} \quad (2.5b)$$

The quotient $T_{h,k}$ is linked to $\ker \pi$ by $[v] \mapsto v - \eta\pi v$, with inverse given by inclusion followed by the quotient. This retains every remaining nilpotent block; at each sum precisely one maximal-length block is assigned to the cyclic summand. The choice of top coefficient is recorded. This splitting is not asserted orthogonal: an arithmetic Gram H pulls back to the full block matrix containing $\eta^*H\eta$, $\eta^*H|_{\ker \pi}$, and its adjoint cross term, not just their diagonal blocks.

On a fixed sum λ , the dimension of the symmetric generalized eigenspace is

$$a_\lambda = \sum_{\mathbf{n}:\lambda_{\mathbf{n}}=\lambda} \prod_i \binom{n_i + m_i - 1}{m_i - 1}. \quad (2.6)$$

Thus $T_{h,k}$ has dimension $a_\lambda - \ell_\lambda$ there. For every entire f , trace additivity in (2.5) proves

$$\text{Tr}(f(A_k)|B_k^{\text{sym}}) = \sum_\lambda \ell_\lambda f(\lambda) + \sum_\lambda (a_\lambda - \ell_\lambda) f(\lambda). \quad (2.7)$$

The first term is the cyclic trace; the second is the retained quotient trace. All terms get the same cohomological factor $(-1)^k$ in top-degree supertrace. This calculation is valid with nilpotents; it does not replace their modules by their residue values.

For example, the algebraic fixture with simple roots $1/4, 1/2, 3/4$ and $k = 2$ has a six-dimensional symmetric module but a five-dimensional cyclic module. The two occupations with sum 1 give a one-dimensional quotient at that sum. The fixture with one root of order three and $k = 2$ has symmetric dimension six, cyclic length five, and quotient length one at 2ρ . These are fixtures of the formulas, not assertions of zeta zeros at those locations or multiplicities.

S13.1.4 3. An original source realization whose norm is the scalar sum density

Define the linear map

$$\boxed{\mathcal{V}_{h,k} : \mathbb{C}[S] \rightarrow \mathcal{B}^{\otimes k}, \quad P \mapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k}).} \quad (3.1)$$

Its Mellin transform is

$$\mathcal{M}^{(k)}\mathcal{V}_{h,k}(P) = \left(\prod_i v_h(s_i) \right) P(s_1 + \cdots + s_k).$$

It is injective: if this analytic function vanishes, it vanishes on an open set where all v_h are nonzero, forcing the polynomial P to vanish identically. Its actual arithmetic map is

$$\boxed{J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi,} \quad (3.2)$$

where $\pi_\chi(P) = [P]_\chi$. Thus its arithmetic kernel is exactly (χ) and not a selected subset of the original relations.

Let $q_k = \deg \chi_{h,k}$ and $N \geq q_k - 1$. At finite source degree,

$$\boxed{0 \rightarrow \mathbb{C}[S]_{\leq N-q_k} \xrightarrow{\times \chi} \mathbb{C}[S]_{\leq N} \xrightarrow{\pi_\chi} C_{h,k} \rightarrow 0.} \quad (3.3)$$

A negative bound means the zero space. Define the actual boundary space

$$\mathcal{D}_{h,k,N} = \mathcal{V}_{h,k}(\chi \mathbb{C}[S]_{\leq N-q_k}).$$

Because $\chi(s_1 + \cdots + s_k) \in I_k$, fixed-order monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}). \quad (3.4)$$

Each term is an original top cochain boundary: replace F_h in factor i by ϕ_* , apply the polynomial $Q_i(D_1, \dots, D_k)$, and give the primitive $(-1)^{i-1}$. The tensor differential gives the second $(-1)^{i-1}$. Their product is $+1$. This supplies the original primitive rather than defining a new vanishing relation by fiat.

The split quotient of (3.3) has equality criterion $P \sim P'$ precisely when $P - P' \in \chi \mathbb{C}[S]_{\leq N-q_k}$. A relation maps to the supported zero at label λ_N . Degree transport retains the preceding label and its killed-representative kernel.

S13.1.5 The exact norm

Retain from [S1]

$$a_h(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h(t)|^2, \quad \mu_h = \int w_h, \quad m_{h,k} = w_h^{*k}.$$

In centered sum coordinates $u = \sum t_i$, $y_i = t_i - u/k$, the amplitude is $\Psi_{h,k}(u, \mathbf{y}) = \prod_i a_h(u/k + y_i)$. Its relative integral is $m_{h,k}(u)$. Therefore

$$\boxed{\|\mathcal{V}_{h,k}P\|_{L^2(d^k x)}^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du.} \quad (3.5)$$

This is the restriction of the preceding full matrix-weight realization to the **specific source family** $P(S)$, through (3.1) and (3.2). It does not identify every relative coefficient with the constant relative coefficient.

The full relative module remains in (2.5). For comparison with the previous least-norm representative, at the same ambient degree N the maps obey

$$\mathcal{V}_{h,k}(\mathbb{C}[S]_{\leq N}) \hookrightarrow \mathcal{T}_h^{(k)}(\mathcal{P}_{k,N}), \quad \mathcal{D}_{h,k,N} \hookrightarrow \mathcal{B}_{k,N}.$$

Let $R_{k,N}^{\text{full}}$ be the old joint minimum, and $\mathcal{R}_N$ the cyclic minimum constructed next. Their exact relationship is

$$\Delta_N = \mathcal{R}_N - R_{k,N}^{\text{full}} \eta_{h,k} \in \text{Hom}(C_{h,k}, \mathcal{B}_{k,N}),$$

$$\boxed{\mathcal{G}_N = \eta_{h,k}^* G_{k,N}^{\text{full}} \eta_{h,k} + \Delta_N^* \Delta_N.} \quad (3.6)$$

Both maps represent the same full jets; their difference is in the original relation ideal, and the full minimum is orthogonal to it. This proves (3.6). The corresponding control difference is

$$\mathcal{W}_N - \eta_{h,k}^* W_{k,N}^{\text{full}} \eta_{h,k} = A_\chi^* \Delta_N^* \Delta_N + \Delta_N^* \Delta_N A_\chi - k \Delta_N^* \Delta_N. \quad (3.7)$$

This explicitly retains why the Gram inequality does not itself order the control forms.

The scalar source must reach degree $q_k - 1$ to realize all cyclic jets. This can exceed $k(d - 1)$. The reduction in relative rank has a degree cost which remains in every estimate.

S13.1.6 4. Canonical scalar representatives and rank-two arithmetic control

Apply Gram-Schmidt by subtracting lower-degree projections from $1, S, S^2, \dots$ for the exact measure in (3.5). Let $p_j^{[h,k]}$ be the resulting polynomial with leading coefficient one, and retain

$$\omega_j^{[h,k]} = \int |p_j^{[h,k]}(k/2 + iu)|^2 m_{h,k}(u) du > 0. \quad (4.1)$$

No polynomial is divided by its norm. In particular $\omega_0^{[h,k]} = \mu_h^k$. For this section suppress the indices h, k and write

$$b_j = [p_j]_\chi \in C_{h,k}, \quad \mathcal{K}_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad \mathcal{G}_N = \mathcal{K}_N^{-1}. \quad (4.2)$$

The coordinates on $C_{h,k}$ are the fixed remainder basis $1, S, \dots, S^{q_k-1}$. Its inclusion into the original full jets is $\eta_{h,k}$, carrying U_k ; b_j is not an assertion that $U_k = 1$.

The unique least-norm representative modulo the specified $\mathcal{D}_{h,k,N}$ is

$$\boxed{\mathcal{R}_N u = \mathcal{V}_{h,k} \left(\sum_{j=0}^N p_j \frac{b_j^* \mathcal{G}_N u}{\omega_j} \right).} \quad (4.3)$$

Its identities are

$$J^{(k)} \mathcal{R}_N = \eta_{h,k}, \quad q^{(k)} \mathcal{R}_N = \sigma_h^{\otimes k} \eta_{h,k}, \quad \mathcal{R}_N^* \mathcal{R}_N = \mathcal{G}_N. \quad (4.4)$$

Proof. The coefficient map in (4.3) is the weighted adjoint of the full remainder map followed by $(JJ^*)^{-1}$ in the displayed orthogonal coordinates. Multiplication verifies its right-inverse identity. If Z is another polynomial with the same remainder, their difference lies in the exact kernel (3.3); the cross term with (4.3) is zero by the adjoint formula. The norm is therefore the displayed minimum plus the squared norm of that difference. This proves uniqueness and all equations, before any completion of the arithmetic quotient.

Let $A_\chi = M_S$ on $C_{h,k}$. Multiplication by S has adjoint multiplication by $k - S$ on the integration line. Orthogonality and leading coefficients give

$$Sp_j = p_{j+1} + \beta_j p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}, \quad \Re \beta_j = k/2, \quad (4.5)$$

with the last term absent at $j = 0$. This is the ordinary orthogonal-polynomial recurrence in the retained complex coordinate; its general framework is recorded in DLMF §18.2 [E1].

Theorem 4.1 (two-row control on a spectrum-complete submodule).

$$\mathcal{Z}_N := A_\chi \mathcal{K}_N + \mathcal{K}_N A_\chi^* - k \mathcal{K}_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (4.6)$$

Consequently

$$\mathcal{W}_N := A_\chi^* \mathcal{G}_N + \mathcal{G}_N A_\chi - k \mathcal{G}_N = \mathcal{G}_N \mathcal{Z}_N \mathcal{G}_N, \quad \text{rank } \mathcal{W}_N \leq 2. \quad (4.7)$$

Proof. Substitute (4.5) into the finite sum (4.2). Each adjacent pair internal to $0, \dots, N$ cancels because the negative lower coefficient is exactly $-\omega_j/\omega_{j-1}$. Each diagonal pair is $k b_j b_j^*/\omega_j$. The only remaining adjacent pair is the one in (4.6). Congruence gives (4.7). This is a Christoffel–Darboux cancellation on the actual cyclic source, not a statement that the full relative module’s control has rank two.

S13.1.7 The original next relation is retained

Define

$$\mathfrak{b}_N = \mathcal{V}_{h,k}(p_{N+1}) - \mathcal{R}_N b_{N+1}.$$

Its full jets vanish and its polynomial degree is $N + 1$. It gives the explicit isomorphism

$$\mathbb{C} \xrightarrow{c \mapsto c \mathfrak{b}_N} \mathcal{D}_{h,k,N+1} \cap \mathcal{D}_{h,k,N}^\perp \xrightarrow{z \mapsto [z]} \mathcal{D}_{h,k,N+1} / \mathcal{D}_{h,k,N}. \quad (4.8)$$

The leading term proves injectivity; the one-dimensional degree increment proves surjectivity. Its norm and pairing are

$$\|\mathfrak{b}_N\|^2 = \omega_{N+1} + b_{N+1}^* \mathcal{G}_N b_{N+1}, \quad \mathfrak{b}_N^* \mathcal{R}_N = -b_{N+1}^* \mathcal{G}_N. \quad (4.9)$$

The class is nonzero at degree N and becomes the receiving supported zero at degree $N + 1$. Its theta primitive is obtained from the original division (3.4). The scalar support itself is not adjoined again.

The exact update is

$$\begin{aligned} \mathcal{K}_{N+1} &= \mathcal{K}_N + b_{N+1} b_{N+1}^* / \omega_{N+1}, \\ \mathcal{G}_{N+1} &= \mathcal{G}_N - \frac{\mathcal{G}_N b_{N+1} b_{N+1}^* \mathcal{G}_N}{\omega_{N+1} + b_{N+1}^* \mathcal{G}_N b_{N+1}}. \end{aligned} \quad (4.10)$$

S13.1.8 Same-metric reflection and exact excess

Let $P^{\dagger k}(S) = \overline{P(k - \bar{S})}$. The polynomial χ obeys $\chi^{\dagger k} = (-1)^{q_k} \chi$. The induced antilinear involution on the **weighted** cyclic module is

$$j_C[P] = (-1)^{kd} [P^{\dagger k}]. \quad (4.11)$$

This sign is forced by $U_k^\dagger = (-1)^{kd}U_k$ in the original embedding. It intertwines with the actual function reflection and preserves the source degree and norm. Hence the unique minimum obeys reflection on the same $\mathcal{G}_N$.

Uniqueness of the monic orthogonal polynomial gives $p_j^{\dagger k} = (-1)^j p_j$. In particular

$$c_N = b_N^* \mathcal{G}_N b_{N+1} \quad \text{satisfies} \quad \bar{c}_N = -c_N.$$

Write $a_N = b_N^* \mathcal{G}_N b_N$ and $d_N = b_{N+1}^* \mathcal{G}_N b_{N+1}$. The nonzero generalized control eigenvalues are the eigenvalues of

$$\frac{1}{\omega_N} \begin{pmatrix} \bar{c}_N & d_N \\ a_N & c_N \end{pmatrix}.$$

Therefore the exact least two-sided excess is

$$\epsilon_{h,k,N}^{\text{cyc}} = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}. \quad (4.12)$$

The determinant is nonnegative by the Gram inequality. Zero and rank-one degeneracies are covered by the same formula; in the reflection-stable case a nonzero rank-one signed defect cannot occur. No phase is dropped.

For the actual eigenvector at $k\rho$ constructed in Section 2,

$$k|2\Re\rho - 1| \leq \epsilon_{h,k,N}^{\text{cyc}}. \quad (4.13)$$

This is a proved diagnostic of this explicit family. A sublinear bound on the calculated right side as k grows is still to be established from the arithmetic input. The rank-two identity is not such a bound by itself.

S13.1.9 5. Keep packet changes through their actual source seeds

Let $h \mid H$ include complete multiplicities, and let $m = H/h$. Then

$$F_h = m(D)F_H, \quad \mathcal{V}_{h,k}P = \mathcal{T}_H^{(k)} \left(\prod_i m(s_i) P(S) \right). \quad (5.1)$$

The full CRT inclusion on jets is the $\mathbb{C}[S]$ -linear map which inserts zero jets at the added centres. Its old unit maps to the associated idempotent, so it is not described as a unital algebra inclusion into the entire larger packet.

The right side of (5.1) has the fixed seed $a_{hH}(\mathbf{s}) = \prod_i m(s_i)$. Its source norm is exactly the old norm, since $(g/H)m = g/h$ in every factor. It therefore defines an isometric map of the marked polynomial sources and their internal quotients, with the old cyclic annihilator $\chi_{h,k}$.

This is the correct packet-coherence comparison. The new seed is generally not a polynomial in S alone. For example, if $m(s) = s - 1/2$ and $k = 2$, then

$$m(s_1)m(s_2) = \frac{(S-1)^2 - (s_1 - s_2)^2}{4}. \quad (5.2)$$

The relative square and its coefficient are retained. The explicit source-seed map (5.1), followed by the larger packet's full relative source inclusion, replaces any unjustified identification with that larger packet's unmarked constant-seed cyclic submodule. No original packet, supported relation, or canonical scalar norm is changed by this marked comparison.

S13.1.10 6. The scalar Fisher estimate now acts on this source, and all polynomial derivative moments are calculable

From [S1],

$$\mathcal{I}_{h,k} + 4 \int \|n_{h,k}(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h, \quad \mathcal{I}_h = 4 \int |a'_h(t)|^2 dt. \quad (6.1)$$

The cyclic source (3.5) uses **that very density** $m_{h,k}$, not another scalar replacement for the full matrix norm. The full derivative still has the specified two components:

$$\partial_u(\Psi_{h,k}P(k/2 + iu)) = \Psi_{h,k} \left(iP'(k/2 + iu) + \frac{m'_{h,k}}{2m_{h,k}} P(k/2 + iu) \right) + n_{h,k}P(k/2 + iu). \quad (6.2)$$

They are pointwise orthogonal. The second component is retained even though the original coefficient polynomial depends only on the sum. Its arithmetic interpretation is the graph map and logarithmic observable from [S1]. The generator remains A_k ; the derivative corresponds to $i\mathcal{L}_k$, with $[D^{(k)}, \mathcal{L}_k] = -1$.

S13.1.11 Raw moments give the complete differentiated finite source without multidimensional quadrature

Define the finite moments

$$\mu_j = \int t^j |a_h(t)|^2 dt, \quad \nu_j = \int t^j |a'_h(t)|^2 dt, \quad \theta_j = \int t^j \overline{a_h(t)} a'_h(t) dt.$$

Fixed arithmetic phase and integration by parts give

$$\theta_0 = 0, \quad \theta_j = -\frac{j}{2} \mu_{j-1} \quad (j \geq 1). \quad (6.3)$$

Use the **formal** exponential generating series

$$\mathcal{M}(z) = \sum_{j \geq 0} \mu_j z^j / j!, \quad \mathcal{N}(z) = \sum_{j \geq 0} \nu_j z^j / j!.$$

No convergence of these series is needed for the finite coefficient identities. Their constant terms remain the actual mass and derivative energy.

Let

$$M_{j,k} = \int u^j m_{h,k}(u) du, \quad B_{j,k} = \int u^j \|\partial_u \Psi_{h,k}(u, \cdot)\|^2 du.$$

Theorem 6.1 (all polynomial derivative moments).

$$M_{j,k} = j! [z^j] \mathcal{M}(z)^k, \quad (6.4)$$

$$B_{j,k} = j! [z^j] \left[\frac{1}{k} \mathcal{N}(z) \mathcal{M}(z)^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}(z)^k \right]. \quad (6.5)$$

Proof. Expand $u^j = (t_1 + \cdots + t_k)^j$ and use Fubini; all integrals are absolutely finite. For (6.5), the diagonal terms of $|k^{-1} \sum_i a'_i \prod_{r \neq i} a_r|^2$ give $k^{-1} \mathcal{N} \mathcal{M}^{k-1}$. The distinct-index terms give

$(k-1)\Theta(z)^2\mathcal{M}^{k-2}/k$, where $\Theta(z) = \sum \theta_j z^j/j! = -z\mathcal{M}(z)/2$ by (6.3). Substitution gives the displayed expression. The second term is not discarded: the cross terms vanish only for the unweighted zeroth moment, not for arbitrary polynomial moments.

In particular,

$$\begin{aligned} M_{0,k} &= \mu_0^k, & M_{1,k} &= k\mu_1\mu_0^{k-1}, \\ M_{2,k} &= k\mu_2\mu_0^{k-1} + k(k-1)\mu_1^2\mu_0^{k-2}, \\ B_{0,k} &= \frac{1}{k}\nu_0\mu_0^{k-1} = \frac{\mu_h^{k-1}}{4k}\mathcal{I}_h. \end{aligned} \quad (6.6)$$

The first orthogonal norm and diagonal coefficient on the sum line are accordingly

$$\omega_0 = \mu_0^k, \quad \beta_0 = \frac{k}{2} + i\frac{k\mu_1}{\mu_0}, \quad \omega_1 = k\mu_0^{k-2}(\mu_0\mu_2 - \mu_1^2). \quad (6.7)$$

No symmetry assumption setting $\mu_1 = 0$ is made.

For any two polynomials P, Q , let $c_P(u) = P(k/2 + iu)$ and $c_Q(u) = Q(k/2 + iu)$. Their complete graph-energy form is

$$\begin{aligned} \mathfrak{D}_k(P, Q) &= \int \overline{c_P}c_Q \|\Psi'_k\|^2 du + \int \overline{c'_P}c'_Q m_k du \\ &\quad + \frac{1}{2} \int \left(\overline{c_P}c'_Q + \overline{c'_P}c_Q \right) m'_k du. \end{aligned} \quad (6.8)$$

All four terms remain. Equations (6.4)–(6.5) and $\frac{1}{2} \int u^j m'_k = -jM_{j-1,k}/2$ evaluate every matrix entry on a fixed polynomial degree using only μ_j, ν_j . This is the complete derivative graph norm from the existing sum-fibre construction, not merely the scalar score norm. It is positive definite on every nonzero polynomial source: a zero derivative would make $\Psi_k P$ constant in u for almost every relative coordinate, while its decay forces that constant and then P to be zero.

For a finite canonical representative, its graph-energy matrix is obtained by applying its actual coefficient matrix from (4.3) on both sides of (6.8). Thus there are now explicit moment matrices for both the original metric and the differentiated representative. No continuous jet functional on arbitrary L^2 classes is assumed.

S13.1.12 7. Exact sum-coordinate conormal depth at every original thickening

The derivative graph still needs the class **before** an original relation is sent to e . The required domains can now be computed exactly on the sum line.

For $r \geq 1$, retain the original thickened ring

$$B_{k,r} = \mathcal{P}_k/I_k^r.$$

Define, for an ordered root tuple $\rho = (\rho_{i_1}, \dots, \rho_{i_k})$,

$$\begin{aligned} L_{\rho,r} &= 1 + \sum_{j=1}^k (m_{i_j} - 1) + (r-1) \max_j m_{i_j}, \\ \ell_{\lambda,r} &= \max_{\rho: \sum \rho_{i_j} = \lambda} L_{\rho,r}, \end{aligned} \quad \boxed{\chi_{h,k,r}(S) = \prod_{\lambda \in \Lambda_k} (S - \lambda)^{\ell_{\lambda,r}}.} \quad (7.1)$$

Theorem 7.1. The exact contracted ideal is

$$\boxed{I_k^r \cap \mathbb{C}[S] = (\chi_{h,k,r}).} \quad (7.2)$$

Consequently there is an injective unital algebra map

$$\boxed{\alpha_r : C_{h,k,r} := \mathbb{C}[S]/(\chi_{h,k,r}) \hookrightarrow B_{k,r}^{S_k}, \quad [P] \mapsto [P(S)].} \quad (7.3)$$

Proof. At a root tuple, the units in $h(\rho_i + z_i) = u_i(z_i)z_i^{m_i}$ make the original local ideal equal to $(z_1^{m_1}, \dots, z_k^{m_k})$, with its r th power retained. A monomial z^α survives precisely when

$$\sum_i \left\lfloor \frac{\alpha_i}{m_i} \right\rfloor \leq r - 1. \quad (7.4)$$

The largest surviving total degree is $\sum_i (m_i - 1) + (r - 1) \max_i m_i$. It is attained by assigning all $r - 1$ full blocks to an index of maximal m_i , and taking remainders $m_i - 1$ in every index. The corresponding term of $(\sum z_i)^D$ has coefficient $D! / \prod \alpha_i! \neq 0$. All terms of degree $D + 1$ vanish. Thus the exact nilpotence index is (7.1). Take the maximum over collided sums and then the product over different sums. This gives precisely the annihilator of the algebra element S , proving (7.2). The same cyclic image is in the invariants because S and 1 are invariant. No local unit has been discarded: it enters the specified invertible change of generators of the ideal, not an alteration of the physical coordinate S .

S13.1.13 Quotient and derivative maps on the same tower

There are commuting quotient squares

$$\begin{array}{ccc} C_{h,k,r+1} & \xrightarrow{\alpha_{r+1}} & B_{k,r+1} \\ \downarrow & & \downarrow \\ C_{h,k,r} & \xrightarrow{\alpha_r} & B_{k,r}. \end{array} \quad (7.5)$$

Each quotient is a ring map, hence has the original split lift. Its kernel is retained. On the sum side,

$$0 \rightarrow \mathbb{C}[S]/(\chi_{r+1}/\chi_r) \xrightarrow{\times \chi_r} C_{r+1} \rightarrow C_r \rightarrow 0. \quad (7.6)$$

The first arrow is explicitly a module map. Under α_{r+1} its image lies in I_k^r/I_k^{r+1} ; this is the actual relation fibre sent to the supported zero by the right vertical quotient.

The exact exponent satisfies $\ell_{\lambda,r+1} \geq \ell_{\lambda,r} + 1$. Hence χ_r divides χ'_{r+1} , and the linear derivative map is well-defined:

$$\boxed{\partial_r : C_{r+1} \rightarrow C_r, \quad [P] \mapsto [P']}. \quad (7.7)$$

It commutes with the original relative-sum derivative

$$\delta_r : B_{k,r+1} \rightarrow B_{k,r}, \quad [P] \mapsto \left[\frac{1}{k} \sum_i \partial_{s_i} P \right],$$

$$\boxed{\delta_r \alpha_{r+1} = \alpha_r \partial_r.} \quad (7.8)$$

The derivatives are coefficient-linear maps and are lifted as such, not called ring homomorphisms. For example,

$$h(s) = (s - \rho)^2, \quad k = 2, \quad X = S - 2\rho \implies \chi_r = X^{2r+1}. \quad (7.9)$$

The class X^3 is nonzero modulo X^5 , maps to the supported zero modulo X^3 , and its derivative maps to $3X^2$, which is nonzero there.

This also calculates the relationship to a naive scalar power thickening:

$$\mathbb{C}[S]/(\chi_1^r) \twoheadrightarrow C_r \hookrightarrow B_{k,r}. \quad (7.10)$$

For the fixture in (7.9) at $r = 2$, the first map is $\mathbb{C}[X]/(X^6) \twoheadrightarrow \mathbb{C}[X]/(X^5)$, with kernel the retained class of X^5 . The original joint thickening therefore carries a different sum-coordinate depth, and the precise quotient above gives their relationship. This is a calculation about powers of the original relation ideal; it is not iteration of the scalar zero adjunction.

S13.1.14 The arithmetic Taylor unit remains through differentiation

Let $U_{k,r}$ be the full Taylor class of $\prod v_h(s_i)$ in $B_{k,r}$. It is invertible and compatible under the quotient tower. Define the coefficient-linear inclusion

$$\eta_r = M_{U_{k,r}} \alpha_r.$$

The full equation is

$$\boxed{\delta_r \eta_{r+1}(P) = U_{k,r} \alpha_r(P') + (\delta_r U_{k,r+1}) \alpha_r(P).} \quad (7.11)$$

The second term remains in the relative ring $B_{k,r}$. It is not presumed to be in the cyclic image. Its exact map is $M_{\delta_r U_{k,r+1}} \alpha_r$, and the complementary quotient $B_{k,r}/\eta_r C_r$ records its noncyclic part. This is the algebraic counterpart of the retained normal derivative in (6.2).

At $r = 1$, the quotient and derivative thus give two actual maps out of the same lifted thickening. A conormal relation can map to e under the quotient and to a nonzero supported value under the derivative. Differentiating the original theta relation, not the scalar e itself, is what produces that value.

S13.1.15 8. The conormal derivative reaches both cyclic and full arithmetic traces, with their difference retained

Choose an original division

$$\chi_1(S) = \sum_i h(s_i) B_i(\mathbf{s}).$$

Applying the actual derivative and then the original quotient yields

$$\boxed{\alpha_1(\chi_1') = \frac{1}{k} \sum_i h'(s_i) [B_i]_{I_k}.} \quad (8.1)$$

The right side is the source's conormal row. For the weighted relation $U \chi_1 P$, the term containing δU has an undifferentiated χ_1 and vanishes in B_k . Therefore

$$\boxed{\delta_1[U \chi_1 P]_{I_k^2} = U_k \alpha_1(\chi_1' P).} \quad (8.2)$$

It has the actual analytic representative

$$J^{(k)} \mathcal{L}_k \mathcal{V}_{h,k}(\chi_1 P) = U_k \alpha_1(\chi'_1 P), \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i. \quad (8.3)$$

Here the left source is an original theta boundary, supplied by (3.4). Its first differentiated arithmetic value is read before it is collapsed to its supported zero.

For $C_1 = \mathbb{C}[S]/(\chi_1)$, the explicitly chosen finite residue pairing is

$$R_\chi(f, P) = \sum_\lambda \text{Res}_{S=\lambda} \frac{f^{\dagger k}(S) P(S)}{\chi_1(S)} dS.$$

Its Jacobian contraction is

$$R_\chi(f, \chi'_1 P) = \sum_\lambda \ell_\lambda \overline{f(k - \bar{\lambda})} P(\lambda) = \text{Tr}(M_{f^{\dagger k} P} | C_1). \quad (8.4)$$

This is the cyclic trace. The original full symmetric trace has a_λ instead of ℓ_λ and is their sum with the quotient trace (2.7). Equivalently, let $\chi_{\text{full}}(S) = \prod_\lambda (S - \lambda)^{a_\lambda}$. Its difference is the explicit residue functional

$$\text{Tr}(f(A_k) | T_{h,k}) = \sum_\lambda \text{Res}_{S=\lambda} f(S) \left(\frac{\chi'_{\text{full}}}{\chi_{\text{full}}} - \frac{\chi'_1}{\chi_1} \right) dS. \quad (8.5)$$

There is a further explicit comparison for the original arithmetic vector pairing. Its pullback through the **weighted** inclusion is

$$\text{Tr}_{B_k^{\text{sym}}} M_{(\eta f)^{\dagger} \eta P} = \text{Tr}_{B_k^{\text{sym}}} M_{U_k^{\dagger} U_k \alpha_1(f^{\dagger k} P)}. \quad (8.6)$$

The factor $U_k^{\dagger} U_k$ remains. It need not lie in the sum-generated subalgebra. The trace decomposition (2.7) applies to the invariant operator $f(A_k)$; it is not an assertion that this weighted vector pairing is automatically the cyclic residue form (8.4).

Neither polynomial is substituted for the original $g = 2\xi$. Their finite trace comparison is (8.5), while $U_{k,r}$ and the map (8.3) retain their arithmetic source. The original product residue/Jacobian trace can still be evaluated on the full module and its retained quotient. No trace contribution is declared absent.

S13.1.16 9. Checks, counterfactual tests, and the next estimate

The exact checker implements finite polynomial, matrix, and Gaussian-amplitude fixtures. Its tests cover exact thickening dimensions and nilpotence; collision grouping; cyclic injections and trace complements; the non-naive thickening map; the derivative square; supported-zero versus absence; the original conormal row; full unit differentiation; scalar minimum and updates; rank-two control; reflection with a nonzero imaginary cross term; amplified off-line test exponents; raw convolution and complete polynomial derivative moments; and the marked source-seed transition. These are regression checks, not tests asserting that a fixture's roots are zeta zeros.

In particular an off-line reflected polynomial fixture remains off-line under the new construction: its exact eigenline identity is still $2\Re\lambda - k$. The scalar Fisher contraction does not annul this value. This guards against an inference from convolution smoothing alone to purity. The constructive object supplied instead is (4.12): two explicitly computed full-jet rows relative to the exact cyclic interpolation metric, with every scalar moment obtainable from the original one-factor moments by (6.4).

The degree cost $N \geq \deg \chi_{h,k} - 1$ and the reciprocal metric in (4.12) are retained as k grows. The derivative matrix (6.8) likewise keeps the polynomially weighted normal contribution. Controlling those quantities on the actual arithmetic inputs, rather than just the constant polynomial, is the remaining quantitative step of this branch.

The connection to the Deligne amplification programme is now a smaller, exactly included cohomological image with every amplified exponent represented; its source norm is the scalar sum-density norm for which (6.1) was proved. The original multi-direction module, its complementary trace, and all powers of its conormal ideal remain as explicit surrounding objects. This is not a transfer of finite-field purity by notation.

No remote repository is modified in this delivery. The user has assigned GitHub integration and Lean work to the parallel session. The authored note, checker, and a bounded handoff are ready for that session without changes to its existing files.

S13.1.17 Source record and references

- **[S1]** Owner programme, `Tau_Sum_Connection_Control/NOTE.tex` and `RESEARCH_NOTE.md`, archive `Tau_Sum_Connection_Control_2026-09-12.zip`. Selected analytic source, scalar information identity, full matrix connection, original first and higher conormal derivative. Exact hashes and source-manifest verification are in `checks/source_manifest.json` and `SOURCE_READING.json`.
- **[S2]** User upload `Untitled\1765(1).md`: retained progression from globalization to coherent interpolation, highest relation layer and spectral-sum descent. Read as a research transcript, not as an independent certificate for every embedded claim.
- **[S3]** User upload `Untitled\1766(1).md`: formalization progression, retained bounded scopes, and the kernel-layer integration. No new Lean run is claimed in this note.
- **[S4]** `SplitZero_Derived_Mathematics_2026-09-12(1).md` and retained source-derived interfaces: internal quotient, transported-class kernel and tensor base-change. Used within their stated hypotheses.
- **[E1]** NIST, *Digital Library of Mathematical Functions*, §18.2, particularly recurrence, Christoffel–Darboux, and moments. <https://dlmf.nist.gov/18.2>. General framework only; the arithmetic source maps and cyclic calculation are proved here.
- **[E2]** The Stacks Project, Tag 00S0, *The naive cotangent complex*. <https://stacks.math.columbia.edu/tag/00S0>. Background for the conormal map; (7.1)–(7.11) compute the exact present ideals and coordinate derivative.
- **[E3]** O. Johnson and A. Barron, *Fisher Information inequalities and the Central Limit Theorem*, arXiv:math/0111020. <https://arxiv.org/abs/math/0111020>. General convolution-projection mechanism; the mass-retaining arithmetic identity is the earlier source [S1].
- **[D]** P. Deligne, *La conjecture de Weil. II*, supplied TeX/source excerpts in the programme, especially §3.2.13 and §3.3.4–3.3.6. This turn uses the previously read amplification target; it does not claim to have reread or independently verified the whole paper.

S14 Exterior trace amplification with the original arithmetic Gram and full quartet multiplicities

Source attribution. Complete TeX witness supplied in the Exterior Trace Amplification archive. The original Gram, additive exterior action, factorial tensor inclusion, full generalized eigenspaces and quartet multiplicities are retained. The source supplies an amplification lower bound and explicitly leaves the actual subcubic arithmetic upper estimate unproved. Its finite fixtures and source-reported verification have their recorded scopes; the separate PR 21 formalization is identified independently.

Exact revision. c7050e1c2f76b31b72af1d237efd29f656fd34a92492e5a11f76d8bc55b2acd9

Source file. Tau_Exterior_Trace_Amplification/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Complete source witness SHA-256; cited repository main is a source baseline

Source SHA-256. c7050e1c2f76b31b72af1d237efd29f656fd34a92492e5a11f76d8bc55b2acd9

Original archive. Tau_Exterior_Trace_Amplification_2026-09-12.zip

Archive SHA-256. 168f977a9b54cf461f564e099fe55dc7b8feffbc09c563f541e0be2bb5746a94

S14.1 Result and scope

The previous cyclic-sum construction has a Hermitian control form of rank at most two. The continuation here uses that specific property, rather than changing its metric. Passing to an exterior power of the same arithmetic module does **not** multiply the control allowance by the exterior degree. It makes it possible to control a determinant line containing the sum of the positive real-part defects.

For the actual cyclic packet and its specified least-norm representative, the new inequality is

$$\sum_{\Re \lambda > k/2} \ell_{\lambda}(2\Re \lambda - k) \leq \epsilon_{h,k,N}^{\text{cyc}}.$$

For a complete nonreal off-line quartet with real displacement $\delta > 0$ and common multiplicity m , it gives the exact lower bound

$$\epsilon_{h,k,N}^{\text{cyc}} \geq 2\delta [1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad N \geq [1 + k(m-1)](k+1)^2 - 1.$$

For simple roots this grows as $\delta k^3/2$. The corresponding sufficient analytic target is therefore a **subcubic** upper bound on this same explicitly defined allowance, rather than the earlier sublinear target obtained from one repeated eigenvector. This note proves the lower bound and constructs the exterior/cohomological maps. It does not prove that analytic upper bound, and it does not claim RH.

The finite-dimensional extremal-trace ingredient is classical: it is a rank-two case of the Ky Fan principle. The contribution here is its application through the existing arithmetic representatives, their original theta boundaries, their split-support lifts, and the exact sum-set and multiplicity calculation. General finite-dimensional linear algebra is not presented as a new classical theorem.

S14.2 1. GitHub corrections integrated at a pinned revision

The repository read for this continuation is KokunoYumeto/zeta-function-research-reader, with pinned main revision

a4494fba4968db837a8af6fcb952c8688cd5d1da.

The inspection was bounded: the latest PR metadata, the main revision, the current sum-connection note, selected parts of the new Stieltjes/conormal continuation, and the formal frontier note. It was not an independent audit of the entire 292-page edition.

S14.2.1 1.1 The empty packet

The current PR #20 source explicitly corrects the scope of its monic-freeness argument. For a nonempty packet $d = \deg h \geq 1$, the original conormal calculation applies. For $h = 1$ and $k \geq 2$,

$$I = (h(s_1), \dots, h(s_k)) = \mathbb{C}[s_1, \dots, s_k], \quad \mathbb{C}[\mathbf{s}]/I^r = 0 \quad (r \geq 1).$$

The algebraic spectral module is then zero. Its split lift $G(0)$ still contains the external absence τ and the supported zero e . The analytic amplitude g and its mass do not vanish. No Gram inverse on a nonzero spectral space is inferred from that analytic amplitude.

Every inverse matrix below is on a **nonempty** actual spectral packet. The empty case has the unique maps between zero coefficient modules and the corresponding original split lifts.

The reviewed public note blob is 797d8bc298aec06b112b53a03654a9664323e748. Its historical validation file records an earlier note blob; that historical receipt is not relabelled as a new proof check.

S14.2.2 1.2 Metric restriction retains its hypotheses

The current formal frontier note requires, for rectangular inclusion and extraction matrices,

$$LI = 1, \quad P = IL = P^*, \quad PK = KP, \quad GK = 1$$

before concluding

$$(I^*GI)(LKL^*) = 1.$$

The spectral projection used later need not be orthogonal for the arithmetic metric. Accordingly this note uses the actual compressed Gram $C_+^*GC_+$ and its inverse. It does not replace that inverse by $C_+^*G^{-1}C_+$.

S14.2.3 1.3 Formalization and source boundaries

The source library already constructs the original internal quotient, cohomology maps, comparison kernels, support-changing morphisms, and homology of a cochain-idempotent image. These are inputs. This continuation neither reruns Lean nor extends the scope of their published certificates to the analytic theta construction.

The new tandem note also constructs the conormal algebra map into dual numbers and keeps the variable-frame connection term. Those results remain compatible inputs; they are not recounted as new results here.

S14.3 2. The original scalar and arithmetic source

Retain the original split scalar

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet.$$

Its original quotient and coefficient maps remain

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

Here j is the specified unital embedding. The target of $p_{\mathbb{Z}}$ is infinite. Every lifted coefficient map sends e to e and τ to τ before the arithmetic observation.

Keep the source spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0 \right\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = H^1(C_+) = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Retain

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

These definitions and the continuous/finite source comparisons are inherited from the supplied cyclic-sum note. Their analytic proofs are not newly re-established in this note.

For a nonempty finite reflection-stable actual zero packet with complete orders, set

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h, \quad v_h = j_h(v_h) \in E_h^\times.$$

For $k \geq 1$, the full tensor arithmetic algebra, generator and unit are

$$B_k = E_h^{\otimes k}, \quad S = s_1 + \cdots + s_k, \quad A_k = M_S, \quad U_k = v_h^{\otimes k}.$$

At a sum λ , its maximal local nilpotent length is

$$\ell_\lambda = \max_{\rho_1 + \cdots + \rho_k = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) \right).$$

Use the original cyclic algebra and its arithmetic inclusion

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{\ell_\lambda}, \quad C = \mathbb{C}[S]/(\chi), \quad q_k = \deg \chi,$$

$$\alpha : C \hookrightarrow B_k^{S_k}, \quad [P] \mapsto P(A_k)1, \quad \eta = M_{U_k} \alpha : C \hookrightarrow B_k^{S_k}.$$

The first arrow is a unital algebra map. The map M_{U_k} and its inverse $M_{U_k}^{-1}$ are explicit module automorphisms. Thus η retains the arithmetic unit without declaring it to be a unital algebra inclusion.

The source already provides an equivariant retraction $\pi : B_k^{S_k} \rightarrow C$, $\pi\eta = 1$, and retains $\ker \pi$. Together with the full finite jet map and the original signed symmetrization this gives a left inverse to the cyclic inclusion in the top cohomology of $C_+^{\widehat{\otimes} k}$. No cyclic complement is discarded in this continuation.

S14.4 3. The existing canonical metric and its rank-two operator

The actual source map is

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = g/h.$$

It obeys

$$J^{(k)}\mathcal{V}_{h,k} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta\pi_\chi.$$

The norm remains

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du,$$

where

$$a_h(t) = \frac{(g/h)(1/2 + it)}{\sqrt{2\pi}}, \quad w_h = |a_h|^2, \quad \mu_h = \int_{\mathbb{R}} w_h, \quad m_{h,k} = w_h^{*k}.$$

The total mass is μ_h^k . Construct the monic orthogonal polynomials $p_j(S)$ by subtracting lower-degree projections from S^j ; retain their squared norms ω_j , including $\omega_0 = \mu_h^k$.

For $N \geq q_k - 1$, in the fixed remainder coordinates on C , put

$$b_j = [p_j]_\chi, \quad K = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad G = K^{-1},$$

$$Ru = \mathcal{V}_{h,k} \left(\sum_{j=0}^N p_j \frac{b_j^* G u}{\omega_j} \right).$$

These are the actual canonical representative and metric:

$$R^*R = G, \quad J^{(k)}R = \eta, \quad q^{(k)}R = \sigma_h^{\otimes k}\eta.$$

For $A = M_S$ on C , the source's exact boundary control is

$$W = A^*G + GA - kG = G \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N} G.$$

Introduce the metric raising map $G : C \rightarrow C^\#$, $v \mapsto (w \mapsto w^*Gv)$, where $C^\#$ is the space of conjugate-linear functionals. Its inverse is K . Define the corresponding Gram adjoint and control endomorphism by the typed composites

$$A^{\sharp G} = G^{-1} A^* G : C \rightarrow C, \quad H = G^{-1} W = A^{\sharp G} + A - kI_C.$$

Then $H^{\sharp G} = H$ and $\text{rank } H \leq 2$. Reflection pairs λ with $k - \bar{\lambda}$, with equal full length. Hence

$$\text{Tr } H = 2\Re \text{Tr } A - kq_k = 0.$$

Thus the nonzero spectrum of H is $\{+\epsilon, -\epsilon\}$ for the actual nonnegative number

$$\epsilon = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}, \quad a_N = b_N^* G b_N, \quad d_N = b_{N+1}^* G b_{N+1}, \quad c_N = b_N^* G b_{N+1}.$$

The source reflection gives $\bar{c}_N = -c_N$. Equivalently $\epsilon^2 = \text{Tr}(H^2)/2$. The case $\epsilon = 0$ means $H = 0$ by its G -self-adjointness. No orthonormal replacement of the retained polynomial coordinates is made.

S14.5 4. Exterior control: the allowance does not multiply by exterior degree

Let $1 \leq p \leq q_k$. On $\bigwedge^p C$, define the **additive exterior generator**

$$A^{[p]}(v_1 \wedge \cdots \wedge v_p) = \sum_{j=1}^p v_1 \wedge \cdots \wedge A v_j \wedge \cdots \wedge v_p.$$

Its relation to the multiplicative exterior functor is

$$A^{[p]} = \left. \frac{d}{dt} \right|_{t=0} \bigwedge^p (e^{tA}), \quad \bigwedge^p (e^{tA}) = e^{tA^{[p]}}.$$

In the unscaled increasing-index wedge basis, the Gram matrix is

$$G_{I,J}^{[p]} = \det G_{I,J}.$$

It is positive definite and is the determinant inner product induced from the original G .

Theorem 4.1. The exact exterior control identity is

$$(A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} = G^{[p]} H^{[p]}.$$

For $1 \leq p \leq q_k - 1$,

$$-\epsilon G^{[p]} \preceq (A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} \preceq \epsilon G^{[p]}.$$

If $\epsilon > 0$, this is the exact optimal allowance. For $p = q_k$ the control form is zero.

Proof. The induced Gram adjoint satisfies

$$(A^{[p]})^{\sharp G^{[p]}} = (A^{\sharp G})^{[p]}.$$

This follows by applying the product rule to the determinant pairing of decomposable vectors. Since $(kI_C)^{[p]} = kpI$, add the two terms to obtain the identity.

For the order bound, keep the G -orthogonal eigenspace decomposition

$$C = L_+ \oplus Z_0 \oplus L_-, \quad H|_{L_+} = \epsilon, \quad H|_{Z_0} = 0, \quad H|_{L_-} = -\epsilon.$$

Both L_+ and L_- have dimension one, and $\dim Z_0 = q_k - 2$. The exterior product of each direct-summand inclusion gives the complete decomposition

$$\begin{aligned} \bigwedge^p C = \bigwedge^p Z_0 \oplus (L_+ \wedge \bigwedge^{p-1} Z_0) \\ \oplus (L_- \wedge \bigwedge^{p-1} Z_0) \oplus (L_+ \wedge L_- \wedge \bigwedge^{p-2} Z_0). \end{aligned}$$

The eigenvalues of $H^{[p]}$ on these four summands are respectively $0, \epsilon, -\epsilon, 0$. The multiplicity of each nonzero eigenvalue is $\binom{q_k-2}{p-1}$; summands with impossible degree are zero modules. This proves every assertion, including the top-degree case. When $\epsilon = 0$, the identity gives the conclusion directly. $\square$

This is the mechanism that makes a determinant-line estimate stronger than testing one eigenvector: a wedge cannot contain the same one-dimensional positive control direction twice.

S14.6 5. The exterior maps live on the original cochain source

Define the unscaled alternating map

$$\text{Alt}_p : \bigwedge^p C \hookrightarrow C^{\otimes p}, \quad v_1 \wedge \cdots \wedge v_p \mapsto \sum_{\pi \in S_p} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(p)}.$$

If $\text{wed} : C^{\otimes p} \rightarrow \bigwedge^p C$ is the exterior quotient, then

$$\text{wed} \text{Alt}_p = p! I.$$

The actual source representative is

$$R_p = R^{\otimes p} \text{Alt}_p : \bigwedge^p C \longrightarrow \mathcal{B}^{\widehat{\otimes} kp}.$$

It retains

$$R_p^* R_p = p! G^{[p]}, \quad J^{(kp)} R_p = \eta^{\otimes p} \text{Alt}_p,$$

$$q^{(kp)} R_p = (\sigma_h^{\otimes k} \eta)^{\otimes p} \text{Alt}_p.$$

The last map is injective: compose with the existing finite-jet/retraction map in each block, then with $\text{wed} / p!$. The factorial is part of this inverse composite; the representative itself is not divided by its norm.

The actual source control is therefore

$$W_p^{\text{source}} = p! \left((A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} \right).$$

Both its Gram and its control retain the same literal $p!$. Its relative allowance is the original ϵ .

S14.6.1 5.1 The cochain idempotent and both permutation signs

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$. Let T_π be the actual Koszul permutation on $\mathcal{C}_k^{\widehat{\otimes} p}$. Its top-degree action is $(\text{sgn } \pi)^k P_\pi$, with P_π ordinary block permutation. The chain idempotent for the exterior top-degree component is

$$A_{p,k} = \frac{1}{p!} \sum_{\pi \in S_p} (\text{sgn } \pi)^{k+1} T_\pi.$$

The two top-degree signs multiply to $\text{sgn } \pi$. Its square equals itself by the group-algebra calculation, with the factor $1/p!$ retained. The complementary idempotent $1 - A_{p,k}$ remains, with its own cohomology and trace. The existing projected-homology theorem supplies inclusion and retraction on the original cycle/boundary quotient.

These sign rules are the standard symmetric-monoidal structure on complexes, instantiated here on the retained arithmetic degree k ; they are not an assumption that degree-one vectors commute without sign.

S14.6.2 5.2 Propagate the actual theta primitive

The previous source construction supplies a map

$$K_N^{\text{prim}} : C \rightarrow \mathcal{C}_k^{k-1}, \quad D^{(k)} R - RA = dK_N^{\text{prim}}.$$

Define

$$K_{N,p}^{\text{prim}} = \sum_{j=1}^p (-1)^{k(j-1)} R^{\otimes(j-1)} \otimes K_N^{\text{prim}} \otimes R^{\otimes(p-j)} \text{Alt}_p.$$

The source and target are $\bigwedge^p C$ and $\mathcal{C}_k^{\widehat{\otimes} p, kp-1}$. Applying the tensor differential contributes the same $(-1)^{k(j-1)}$ on the displayed summand. Differentials on the top-degree R factors vanish. Consequently

$$D^{(kp)} R_p - R_p A^{[p]} = dK_{N,p}^{\text{prim}}.$$

This is the exact boundary producing the exterior control form on the original source.

S14.6.3 5.3 Original support and the alternative full-product minimum

Every coefficient map above has its original $G(\mathbb{C})$ -linear reconstructed lift. At a chosen active support label, a term with zero amplitude remains at that label's zero. The sums of differently located primitives use the joins of their actual tensor support masks; the character average uses the corresponding orbit join. External scalar τ maps every input to external absence, while e maps it to its fibre zero. In particular the boundary in the preceding equation is sent by the next internal quotient to the receiving supported zero, not to τ .

There is also an exact comparison with the earlier full-product minimum. Set

$$I_{k,p} = \eta^{\otimes p} \text{Alt}_p, \quad M \geq \max\{pN, kp(\deg h - 1)\}.$$

Let $R_{kp,M}^{\text{full}}$ be the already-constructed canonical full-jet representative. Both it and R_p lie in the stated degree- M source after applying $I_{k,p}$. Hence

$$\Delta = R_p - R_{kp,M}^{\text{full}} I_{k,p}$$

has zero full jets, and its numerator belongs to the original ideal $(h(s_1), \dots, h(s_{kp}))$. Thus it is an original theta boundary with the source's polynomial-division primitive. Orthogonality of the full minimum gives

$$p!G^{[p]} = I_{k,p}^* G_{kp,M}^{\text{full}} I_{k,p} + \Delta^* \Delta.$$

The complete control comparison is

$$W_p^{\text{source}} = I_{k,p}^* W_{kp,M}^{\text{full}} I_{k,p} + (A^{[p]})^* \Delta^* \Delta + \Delta^* \Delta A^{[p]} - kp \Delta^* \Delta.$$

This connects the two representative choices without identifying their metrics or discarding their boundary difference.

S14.7 6. A determinant line measures the sum of the positive defects

Let

$$C_{>} = \bigoplus_{\Re \lambda > k/2} C_{\lambda} \subseteq C$$

be the sum of the **full generalized eigenspaces**, of total dimension p . Here $C_{\lambda} \cong \mathbb{C}[S]/(S - \lambda)^{\ell_{\lambda}}$. Write its inclusion as $I_{>} : C_{>} \hookrightarrow C$. Its determinant line has the actual injection

$$\det(C_{>}) = \bigwedge^p C_{>} \xrightarrow{\bigwedge^p I_{>}} \bigwedge^p C \xrightarrow{R_p} \mathcal{B}^{\widehat{\otimes} kp}.$$

The additive exterior generator acts on this line by

$$\zeta_{>} = \text{Tr}(A|_{C_{>}}) = \sum_{\Re \lambda > k/2} \ell_{\lambda} \lambda.$$

Indeed expansion of A on a wedge of a basis keeps only its diagonal coefficients: every off-diagonal replacement repeats one basis vector and has zero wedge. The full nilpotent modules are retained before this determinant map; their dimensions appear in the trace. No simplicity assertion follows: a critical-line Jordan block can have nonzero numerical control while contributing zero real-part defect.

Define

$$L_{h,k} := 2\Re \zeta_{>} - kp = \sum_{\Re \lambda > k/2} \ell_{\lambda} (2\Re \lambda - k).$$

Theorem 6.1. For every admitted N ,

$$0 \leq L_{h,k} \leq \epsilon_{h,k,N}^{\text{cyc}}.$$

Proof. If $p = 0$, the left side is zero. Otherwise reflection implies $p < q_k$. Let v be the nonzero determinant vector of an unscaled basis of $C_{>}$. On this actual eigenline,

$$\frac{v^* W_p^{\text{source}} v}{v^* (p!G^{[p]}) v} = 2\Re \zeta_{>} - kp = L_{h,k}.$$

Theorem 4.1 bounds the same quotient by ϵ . $\square$

The old eigenline argument gave the maximum individual defect. This theorem controls their entire positive sum, with their full lengths. Reflection also gives

$$L_{h,k} = \frac{1}{2} \sum_{\lambda} \ell_{\lambda} |2\Re \lambda - k| = \sum_{\lambda} \ell_{\lambda} |\Re \lambda - k/2|.$$

S14.7.1 6.2 The exact arithmetic volume equation

Let $B_{>} : \mathbb{C}^p \rightarrow C$ be the column map of a fixed unscaled basis of $C_{>}$, and set $G_{>} = B_{>}^* G B_{>}$. Its determinant vector $v_{>}$ has squared source norm

$$\|R_p v_{>}\|^2 = p! \det G_{>}.$$

For the original parameter $a > 0$, let $U_{>}(a) = a^{A|C_{>}}$. Then

$$\det(U_{>}(a)^* G_{>} U_{>}(a)) = a^{kp+L_{h,k}} \det G_{>}.$$

The entire exterior control integrates for $a \geq 1$ to

$$a^{kp-\epsilon} p! \det G_{>} \leq \|R_p \bigwedge^p U_{>}(a) v_{>}\|^2 \leq a^{kp+\epsilon} p! \det G_{>}.$$

Differentiating the norm along $a = e^t$ and applying the exact quadratic inequality proves the bound. Comparing with the exact determinant equality again gives $L_{h,k} \leq \epsilon$. The determinant, its norm, and its weight are all produced by maps from the original arithmetic module.

S14.8 7. Spectral and metric projections are linked explicitly

The determinant argument has an equivalent trace calculation useful for computation. Let $Q_{>} = e_{>}(A)$ be the actual CRT spectral idempotent, where $e_{>}$ is one modulo each positive-defect factor and zero modulo every other factor. It commutes with A .

Choose the original CRT column map $B_{>} : \mathbb{C}^p \rightarrow C$, including every jet. Its G -orthogonal projection is

$$P_{>} = B_{>} (B_{>}^* G B_{>})^{-1} B_{>}^* G.$$

Their exact relationship is

$$Q_{>} P_{>} = P_{>}, \quad P_{>} Q_{>} = Q_{>}, \quad P_{>}^{\sharp G} = P_{>}, \quad (P_{>} - Q_{>})^2 = 0.$$

The first two identities follow from their common range and identity restrictions there. Expanding the square gives the last identity. Invariance of that range gives $P_{>} A P_{>} = A P_{>}$ and

$$\mathrm{Tr}(P_{>} A) = \mathrm{Tr}(Q_{>} A) = \zeta_{>}.$$

Since $\mathrm{Tr}(P_{>} A^{\sharp G})$ is the conjugate of $\mathrm{Tr}(P_{>} A)$,

$$\mathrm{Tr}(P_{>} H) = L_{h,k}.$$

For $\epsilon > 0$, the two G -orthogonal rank-one control projections are the exact polynomials

$$F_+ = \frac{H^2 + \epsilon H}{2\epsilon^2}, \quad F_- = \frac{H^2 - \epsilon H}{2\epsilon^2}.$$

They give the full slack formula

$$L_{h,k} = \epsilon(1 - \eta_{h,k,N}), \quad \eta_{h,k,N} = 1 - \text{Tr}(P_{>}F_+) + \text{Tr}(P_{>}F_-) \geq 0.$$

With $\|T\|_{\text{HS},G}^2 = \text{Tr}(T^\sharp G T)$, this is

$$\eta_{h,k,N} = \|(1 - P_{>})F_+\|_{\text{HS},G}^2 + \|P_{>}F_-\|_{\text{HS},G}^2.$$

Thus the gap has a specified projection meaning rather than an omitted term. This is the elementary rank-two instance of the classical extremal-trace principle.

The retained finite trace has its resolvent description as well:

$$\zeta_{>} = \frac{1}{2\pi i} \int_{\Gamma_{>}} S \text{Tr}((SI - A)^{-1}) dS,$$

where the positively oriented contour encloses exactly the indicated finite blocks. The complementary trace remains

$$\text{Tr } f(A) = \text{Tr}(Q_{>}f(A)) + \text{Tr}((1 - Q_{>})f(A)).$$

Multiplication by $(-1)^k$ gives the inherited cohomological sign. The determinant line occurs in degree kp , retaining $(-1)^{kp}$ for its trace. Split lifts of either projector send an existing class with zero projected amplitude to a supported zero, while the paired projector retains the complementary component.

S14.9 8. Exact sum counting for a nonreal quartet

Take a complete actual nontrivial quartet

$$\rho_{\varepsilon,\eta} = \frac{1}{2} + \varepsilon\delta + i\eta\gamma, \quad \varepsilon, \eta \in \{+1, -1\}, \quad \delta > 0, \quad \gamma > 0,$$

with common full multiplicity m . This is a specification for the following theorem, not an assertion that an off-line zero has been found. The two standard symmetries of $g = 2\xi$ give the full quartet and equal orders from any such actual zero.

The distinct k -fold sums are precisely

$$\lambda_{a,b} = \frac{k}{2} + \delta(2a - k) + i\gamma(2b - k), \quad 0 \leq a, b \leq k.$$

Proof of existence. Let a count positive real signs and b positive imaginary signs. Choose an integer t with

$$\max(0, a + b - k) \leq t \leq \min(a, b).$$

The four occupation counts

$$(t, a - t, b - t, k - a - b + t)$$

are nonnegative and sum to k , with the required two margins. Conversely every tuple has such margins. Since δ and γ are nonzero real numbers, equality of two sums forces equality of both margins. No hypothesis of algebraic independence of zero ordinates is used.

All ordered tuple blocks have the same maximal length

$$\ell_k = 1 + k(m-1).$$

The nilpotency proof is the original one: the top monomial in $(z_1 + \cdots + z_k)^{k(m-1)}$ has coefficient $(k(m-1))!/((m-1)!)^k \neq 0$. It survives before the next power vanishes. Hence

$$\chi_{h,k}(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^{\ell_k}, \quad q_k = \ell_k(k+1)^2.$$

For the positive-defect subspace,

$$p = \ell_k(k+1) \left\lceil \frac{k}{2} \right\rceil.$$

Summing its real defects gives

$$\begin{aligned} L_{h,k} &= 2\delta\ell_k(k+1) \sum_{a=\lfloor k/2 \rfloor+1}^k (2a-k) \\ &= 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \end{aligned}$$

For $k = 2n$, the last sum is $n(n+1)$; for $k = 2n+1$, it is $(n+1)^2$. The imaginary parts cancel in the determinant trace because $\sum_{b=0}^k (2b-k) = 0$.

Combining this calculation with Theorem 6.1 proves

$$2\delta[1+k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}^{\text{cyc}}, \quad N \geq [1+k(m-1)](k+1)^2 - 1.$$

Every multiplicity is retained. For simple roots the leading term is $\delta k^3/2$; for $m > 1$ it is $\delta(m-1)k^4/2$.

For the two-point reflected pair alone, the analogous exact lower bound is

$$\epsilon_{h,k,N}^{\text{cyc}} \geq 2\delta[1+k(m-1)] \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

The quartet supplies the additional, rigorously counted factor $k+1$ through its distinct imaginary sums.

S14.10 9. The quantitative target after this construction

The current source formula is still

$$\epsilon_{h,k,N}^{\text{cyc}} = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}.$$

A contradiction to any actual off-line quartet now follows from a bound

$$\frac{\sqrt{a_N d_N - |c_N|^2}}{k^3 \omega_N} \rightarrow 0$$

along a permitted choice $N = N(k) \geq [1+k(m-1)](k+1)^2 - 1$, proved for its actual arithmetic measure and complete jet map. For this contradiction the packet h is fixed: constants may depend

on h , and uniformity over all zero locations is not required. The expression itself has not been shown to tend to zero here. The result is a stronger amplification theorem and a weaker sufficient growth target for the next analytic estimate.

For a simple quartet, the source degree is at least quadratic in k . This cost is retained; a fixed-degree central-limit estimate cannot simply be inserted at that growing degree. The previously proved scalar Fisher contraction, with its factor μ_h^{k-1}/k , is not assigned to every polynomial moment.

The existing polynomial derivative formulas supply those actual moments:

$$\begin{aligned}\mathcal{M}_h(z) &= \sum_{j \geq 0} \mu_j \frac{z^j}{j!}, & \mathcal{N}_h(z) &= \sum_{j \geq 0} \nu_j \frac{z^j}{j!}, \\ \int u^j m_{h,k}(u) du &= j! [z^j] \mathcal{M}_h(z)^k, \\ \int u^j \|\partial_u \Psi_{h,k}\|^2 du &= j! [z^j] \left[\frac{1}{k} \mathcal{N}_h \mathcal{M}_h^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}_h^k \right].\end{aligned}$$

All masses, polynomial degrees and cross terms remain. The new determinant bound uses their already-defined Gram and control matrices rather than changing the source in order to obtain a favourable value.

The real interval $0 < s < 1$ causes no missing quartet case: the alternating zeta series there is positive when grouped into consecutive pairs, while $1 - 2^{1-s} < 0$, so $\zeta(s) < 0$. An actual non-trivial off-line zero is therefore nonreal and is covered by its complete quartet. This argument only identifies the case split; it supplies no upper estimate.

S14.11 10. Deligne architecture and the boundary retained in this step

The inherited Weil-II reading identifies an amplification pattern: carry the same arithmetic spectral information into a cohomological tensor image, obtain a further estimate there, and transport dual control without changing the representation. The finite-field geometric theorem is not being applied on a renamed base.

Here the new characteristic-zero operation is

$$\begin{aligned}& \text{original theta complex over } \mathfrak{b}_\tau \\ & \downarrow \text{existing } k\text{-fold cyclic sum image} \\ & (C, A, R_N, G_N, W_N) \\ & \downarrow \text{signed exterior cochain map} \\ & \left(\bigwedge^p C, A^{[p]}, R_p, p!G^{[p]} \right) \\ & \uparrow \det(C_{>}) \\ & \text{exact sum of positive arithmetic defects.}\end{aligned}$$

The exterior operation controls that determinant line with the **same** allowance as the cyclic image. The original e -relation primitives are carried by $K_{N,p}^{\text{prim}}$; the scalar τ remains external absence. The spectral and metric projectors, their difference, the cyclic complement, the exterior complement and their finite traces are all explicit.

No direct implication from the cardinality of the infinite quotient to positivity is used. Its role is to remain the structural arithmetic quotient throughout the construction, not to be replaced by a finite-field scalar or by the Boolean support observation.

S14.12 11. Verification and reproducibility

The supplied cyclic-sum archive was safely extracted and all 42 entries of its SHA-256 manifest verified. Its 24-method checker was rerun normally and under Python optimization, with matching success records. The new checker contains 22 exact finite test methods; its normal, optimized and deliberately failing results are recorded in checks/.

The Gaussian calculations in the checker are declared algebraic calibrations. Their literal measure is

$$\frac{7^k}{\sqrt{2\pi k}} e^{-u^2/(2k)} du.$$

They test companion matrices, full repeated-root modules, the canonical inverse-moment Gram, exterior factorials, the additive generator, the trace bound and the quartet counting. The mass is 7^k , not one. These examples do not claim that the calibration roots are zeta zeros.

The written arguments are general. Finite tests are regression evidence, not a Lean certificate, interval arithmetic for the actual spectral integrals, or independent certification of the whole analytic source corpus. No repository or other-session branch is changed by this continuation.

S14.13 References and provenance

1. Owner programme, **Tau Cyclic Sum Control**, delivered NOTE.tex, 12 September 2026. The original packet, norm, canonical section, rank-two identity, reflection and thickened relation maps are source inputs. Archive and member hashes are in checks/source_manifest.json.
2. Owner repository, pinned main a4494fba4968db837a8af6fcb952c8688cd5d1da. Reviewed workbenches/tau-sum-connection/RESEARCH_NOTE.md, current blob 797d8bc298aec06b112b53a03654a9664323e748; workbenches/tau-frontier-formal/RESEARCH_NOTE.md; selected sum_connection_stieltjes.tex and conormal_finite_weyl.tex. Exact reading scope is in SOURCE_REVIEW.md.
3. Original SplitZero derived/reconstruction note and the actual universal property of its internal quotient; retained in the owner's supplied formalization material.
4. Overton, M. L., and Womersley, R. S. (1992), **On the Sum of the Largest Eigenvalues of a Symmetric Matrix**, SIAM Journal on Matrix Analysis and Applications 13(1), 41–45, doi:10.1137/0613006. Used only as attribution for the classical extremal-trace ingredient. The rank-two proof used here is fully written above.
5. The Stacks Project, Tag **OGWN**, tensor products of complexes. Used for the standard tensor differential and Koszul permutation types; the specific arithmetic signs and primitive are calculated above.
6. NIST DLMF §25.4, equations 25.4.3–25.4.4, the functional equation and the literal definition of ξ . The original arithmetic function in this program remains 2ξ .
7. Deligne, P. (1980), **La conjecture de Weil II**, Publications Mathématiques de l'IHÉS 52, 137–252. The supplied TeX/reading record at §3.2.13 motivates the amplification architecture. This note does not claim a new full reading or a finite-field purity theorem on the τ -base.

S15 Source and relation volume dynamics for the original arithmetic control

Source attribution. Complete source note from the original Toda Volume Control archive. The source and relation Hankel determinants, tilt derivatives, imaginary cross term, first-degree endpoint, both loss squares and original theta input are retained. Its endpoint proof uses a vector symbol for a scalar norm; the explicit typed correction is proved in the integrated Toda bridge, while these original source bytes remain unchanged. The delivered floating evaluations are historical computations without interval or tail certification. The new local theta-tail proof and its separately scoped interval certificate supply that additional evidence for the analytic seed. Neither source asserts the growing-degree arithmetic estimate, and no local Lean execution is attributed to this edition.

Exact revision. 027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375

Source file. Tau_Toda_Volume_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Complete delivered source witness SHA-256; no new remote revision is assigned

Source SHA-256. 027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375

Original archive. Tau_Toda_Volume_Control_2026-09-12.zip

Archive SHA-256. 80b365233ba5ebaf1b52aeea369a7771d74bf774dbb4416f10a56503321b8f41

S15.1 Results and scope

The preceding exterior construction proves an aggregate lower bound on the canonical arithmetic control. This continuation constructs a scalar route to its upper estimate without selecting another metric on the arithmetic quotient.

The actual canonical quotient determinant is

$$V_N(\theta) = \det G_N(\theta) = \frac{\mathfrak{D}_{N+1}(\theta)}{\mathfrak{B}_{N-q+1}(\theta)}. \quad (1)$$

The numerator is the Gram determinant of the original polynomial source. The denominator is the Gram determinant of the original theta relations in that source. Both determinants come from one retained arithmetic Laplace transform and an explicitly given differential filter. Both obey the classical Hankel-Toda identity. No new general integrable-system theorem is claimed.

For $N \geq q$, put

$$v_N = \log \frac{V_{N-1}}{V_N}, \quad a_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \quad \ell_N = \log V_N, \quad \sigma = \frac{\text{Tr } A - (k/2)q}{i} \in \mathbb{R}.$$

The following is an exact formula, at every real tilt in the stated domain:

$$\boxed{\epsilon_N^2 = a_{N+1}(1 - e^{-v_N})(e^{v_{N+1}} - 1) - (\sigma - \partial_\theta \ell_N)^2.} \quad (2)$$

At $\theta = 0$, ϵ_N is the original allowance in the exterior-amplification note. The new formula retains the cross-term phase through a derivative of a scalar determinant. It is not an estimate that forgets that term.

In particular, with the actual two-step volume ratio

$$\mathcal{R}_N = \frac{V_{N-1}}{V_{N+1}},$$

one obtains the proved finite upper bound

$$\epsilon_N \leq \sqrt{a_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}. \quad (3)$$

Both nonnegative losses in this upper estimate are computed below. The previous aggregate spectral lower bound therefore forces an explicit minimum two-step contraction of the source/boundary volume ratio. This turns the remaining asymptotic question into a comparison between two specified scalar determinant flows. It does not prove that this comparison has the required subcubic bound.

S15.2 1. Integration of the new formalization report

The supplied report identifies verified source commit 2abc351424ba87aeda948a5cfb846e15ed9373d1 in draft PR #21. The current PR metadata and its complete mathematical note were read. The report's successful execution is used with its stated scope; this session did not run Lean or replay the workflow log.

The checked coefficient interfaces are the original quotient derivative

$$\delta_r : A_0/I^{r+1} \longrightarrow A_0/I^r,$$

its square with quotient transitions, the cyclic injection obtained from the **literal** pullback

$$\mathcal{K}_r = \{P : P(S) \in I^r\},$$

and the full multiplier identity

$$\delta_r([uP(S)]) = [uP'(S)] + [(\partial u)P(S)].$$

The report separately identifies the local quotient-basis and CRT interpretation as written proofs. That separation is maintained. Its relation-depth maximum includes attainment and permits the tuple maximizing a collided sum to change with depth.

Here the scalar relation determinant uses the first cyclic relation $\chi = \chi_{h,k,1}$. The weight $|\chi|^2$ is the **squared Hilbert norm of the multiplication map** M_χ ; it is not a declaration that the next contracted ideal is (χ^2) . The map used is displayed in (11). At higher original depth the relevant polynomial remains $\chi_{h,k,r}$, through the checked pullback $\mathcal{K}_r$, not an assumed power of χ .

The empty-packet correction is also kept. For $h = 1$, I is the whole coefficient ring and every positive-depth quotient is zero. The analytic seed g and its mass remain. The split lift $G(0)$ retains two elements. In that case the determinant quotient below is the determinant of the zero-dimensional module, namely 1, and no positive-dimensional inverse or spectral estimate is inferred.

S15.3 2. Original base, coefficient square, and arithmetic source

Keep

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the original square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (4)$$

The target of $p\mathbb{Z}$ remains infinite. The base remains $\mathfrak{b}_\tau$. Every lifted coefficient map fixes external absence and carries represented zero to represented zero.

The analytic source remains

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = \int \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \quad \forall b \in \mathbb{Z}, j \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Let h be a nonempty reflection-stable packet of actual zeros with complete orders. Set

$$E_h = \mathbb{C}[s]/h, \quad v_h = g/h, \quad v_h = j_h(v_h) \in E_h^\times, \quad \mathcal{M}F_h = v_h.$$

Fix tensor degree $k \geq 1$. Use the cyclic algebra already constructed:

$$C = \mathbb{C}[S]/\chi, \quad q = \deg \chi \geq 1, \quad c = k/2, \quad A = M_S,$$

$$\eta : C \hookrightarrow (E_h^{\otimes k})^{S_k}, \quad \eta[P] = v_h^{\otimes k} P \left(\sum_i M_{s_i} \right) 1.$$

The source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}).$$

It retains

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi. \quad (5)$$

Here $q^{(k)}$ denotes the original top cohomology quotient; $q = \deg \chi$ denotes a dimension. The full arithmetic unit is still in η .

The scalar norm is

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c+iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad w_h(t) = \left| \frac{v_h(1/2+it)}{\sqrt{2\pi}} \right|^2. \quad (6)$$

Its mass is μ_h^k , with $\mu_h = \int w_h$. No measure is divided by that mass.

S15.4 3. One arithmetic function generates both determinant flows

S15.4.1 3.1 Analytic domain

Define

$$M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad Z_{h,k}(\theta) = M_h(\theta)^k. \quad (7)$$

These are analytic on $|\Re \theta| < \pi/2$. Here and below positivity statements concern real θ in that interval.

A direct bound suffices. At $\Re s = 1/2$, bounded partial sums of the alternating series and summation by parts give $|\eta(s)| \leq C(1 + |s|)$. The denominator $|1 - 2^{1-s}|$ is bounded below by $\sqrt{2} - 1$. Stirling's vertical-strip estimate then gives, beyond a fixed compact set,

$$w_h(t) \leq C_h(1 + |t|)^{11/2 - 2 \deg h} e^{-\pi|t|/2}.$$

On the compact set, g/h is entire because the selected zero orders are complete. Thus every differentiated integral is locally uniformly dominated on the displayed strip. This is a sufficient domain, not a claim that it is the maximal analytic continuation of M_h .

Fubini and the exact sum coordinate give

$$Z_{h,k}(\theta) = \int e^{\theta u} m_k(u) du, \quad Z_{h,k}^{(j)}(\theta) = \int u^j e^{\theta u} m_k(u) du. \quad (8)$$

In particular $Z_{h,k}(0) = \mu_h^k$.

S15.4.2 3.2 The source and the relation filter

Let $\bar{\chi}$ mean coefficientwise conjugation, and define the differential operator and its output

$$\mathcal{F}_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta), \quad Z_\chi(\theta) = \mathcal{F}_\chi Z_{h,k}(\theta). \quad (9)$$

Differentiation under the integral proves

$$\boxed{Z_\chi(\theta) = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0.} \quad (10)$$

Both factors, their signs, and the coefficient $c = k/2$ remain. This filter is multiplication by a source polynomial under the Laplace transform. It is not the quotient-level derivation δ_r .

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$ and $N \geq q - 1$, the actual source sequence is

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad (11)$$

with $\mathcal{P}_{-1} = 0$. Applying $\mathcal{V}$ identifies its first image with the original admitted theta boundaries.

The Laplace observation gives the typed identity

$$\mathcal{L}_\theta(SP) = (c + i\partial_\theta)\mathcal{L}_\theta(P), \quad \mathcal{L}_\theta(P) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Equation (10) is the Gram observation of M_χ . A source relation has nonzero positive norm before the quotient sends it to a fibre zero.

S15.4.3 3.3 The two determinants

Put

$$\mathfrak{D}_n(\theta) = \det[Z_{h,k}^{(i+j)}(\theta)]_{i,j=0}^{n-1}, \quad \mathfrak{B}_n(\theta) = \det[Z_\chi^{(i+j)}(\theta)]_{i,j=0}^{n-1}, \quad (12)$$

with the actual empty determinants $\mathfrak{D}_0 = \mathfrak{B}_0 = 1$. Every positive-order determinant is positive. A zero norm polynomial would vanish on a full-measure subset of the real line.

The use of u -moments is linked to the original S -basis by

$$S^j = (c + iu)^j = \sum_{r=0}^j \binom{j}{r} c^{j-r} i^r u^r.$$

This triangular coordinate map has determinant $i^{n(n-1)/2}$ on dimension n . Its Gram determinant multiplier is exactly 1, because that determinant has modulus one. No polynomial coefficient or factorial has been dropped.

The determinant has the exact integral representation

$$\mathfrak{D}_n(\theta) = \frac{1}{n!} \int_{\mathbb{R}^n} \prod_{i < j} (u_i - u_j)^2 \prod_{j=1}^n e^{\theta u_j} m_k(u_j) d^n u. \quad (13)$$

For $\mathfrak{B}_n$, insert $\prod_j |\chi(c + iu_j)|^2$. Expanding both Vandermonde determinants and integrating proves the formula, including $1/n!$. The total mass in the first expression scales as $(\mu_h^k)^n$, not as one.

S15.5 4. The quotient determinant is the source/boundary volume ratio

At real θ equip the same polynomial source with

$$\langle P, Q \rangle_\theta = \int \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (14)$$

Let $H_\theta = L^2(\mathbb{R}, e^{\theta u} m_k(u) du)$. The map $f \mapsto e^{\theta u/2} f$ is an isometry $H_\theta \rightarrow H_0$ with inverse $f \mapsto e^{-\theta u/2} f$. The source in each space is exactly the image of $\mathcal{P}_N$ under evaluation. This is an observation of the original source, not an enlargement of its admitted polynomials.

Let J_N be its quotient matrix and $M_N(\theta)$ its source Gram. Define

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad \hat{R}_N = M_N^{-1} J_N^* G_N, \quad R_N = \mathcal{V} \hat{R}_N. \quad (15)$$

The adjoints in this calculation use (14). At $\theta = 0$ these are the preceding canonical arithmetic representatives and their original norm. At every θ ,

$$J^{(k)} R_N = \eta, \quad q^{(k)} R_N = \sigma_h^{\otimes k} \eta.$$

Use the source basis

$$1, S, \dots, S^{q-1}, \quad \chi, \chi S, \dots, \chi S^{N-q}.$$

Its change from $1, S, \dots, S^N$ is triangular with diagonal one because the original χ is monic. The Schur complement of its boundary Gram is G_N . This proves

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (16)$$

Equivalently, the determinant line of (11) is mapped by the wedge of the **specified** kernel columns and quotient lifts. Orthogonal quotient representatives supply the Schur complement. This is the exact determinant-line map behind the ratio.

The original support lift is

$$(\lambda_N, P) \mapsto (\lambda_N, [P]_\chi), \quad (\lambda_N, \chi Q) \mapsto (\lambda_N, 0), \quad \tau \mapsto \tau. \quad (17)$$

The determinant $\mathfrak{B}_{N-q+1}$ retains the norm-volume of those nonzero source relations. Replacing that source by its quotient value would incorrectly replace the positive denominator by information-free zeros. The construction evaluates its Gram before applying (17).

For $h = 1$, $\chi = 1$ and $C = 0$. Then $Z_\chi = Z_{h,k}$ and (16) is 1 identically. This is the empty determinant, not an inverse on a nonzero spectral module.

S15.6 5. The two classical Toda identities have their actual arithmetic input

The source determinants satisfy

$$\mathfrak{D}_n \mathfrak{D}_n'' - (\mathfrak{D}_n')^2 = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1}, \quad n \geq 1. \quad (18)$$

The relation determinants satisfy the same equation with $\mathfrak{D}$ replaced by $\mathfrak{B}$.

Here is a proof with the constants retained. For a real positive weight $e^{\theta u} d\mu(u)$, let Q_j be the monic orthogonal polynomial obtained by subtracting lower-degree projections, with norm h_j . Multiplication has the recurrence

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1}, \quad a_0 = 0,$$

The variable remains u . Differentiating orthogonality gives

$$\partial_\theta Q_j = -a_j Q_{j-1}, \quad \partial_\theta \log h_j = b_j, \quad b_j' = a_{j+1} - a_j.$$

For the last identity, differentiate $h_j b_j = \int u Q_j^2 e^{\theta u} d\mu$, substitute the derivative of Q_j , and use the recurrence. Since $\mathfrak{D}_n = \prod_{j < n} h_j$, the sum telescopes:

$$\partial_\theta^2 \log \mathfrak{D}_n = a_n = \frac{\mathfrak{D}_{n+1} \mathfrak{D}_{n-1}}{\mathfrak{D}_n^2}.$$

This proves (18). The relation weight is the same positive weight multiplied by $|\chi(c+iu)|^2$, which is independent of θ , so the proof applies without alteration to its determinants.

The original S -monic polynomial norms are exactly

$$\omega_j = \frac{\mathfrak{D}_{j+1}}{\mathfrak{D}_j}, \quad \nu_j = \frac{\mathfrak{B}_{j+1}}{\mathfrak{B}_j}. \quad (19)$$

The phases linking monic polynomials in S and u are i^j and i^{-j} ; their norms agree because the phase has modulus one.

For $m = N - q + 1 \geq 0$ this gives the exact quotient-volume curvature

$$\partial_\theta^2 \log V_N = \frac{\mathfrak{D}_{N+2} \mathfrak{D}_N}{\mathfrak{D}_{N+1}^2} - \frac{\mathfrak{B}_{m+1} \mathfrak{B}_{m-1}}{\mathfrak{B}_m^2}. \quad (20)$$

The second term is defined to be zero for $m = 0$, because $\log \mathfrak{B}_0 = 0$. It is not a quotient involving $\mathfrak{B}_{-1}$.

S15.6.1 5.1 The curvature comes from two actual source maps

Let X be multiplication by u on the polynomial realization in H_θ . Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ project to the original boundary subspace $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$. Write R for its least-norm polynomial lift.

Differentiating the quotient condition and its orthogonality proves

$$\partial_\theta R = -P_{\mathcal{D}_N} X R. \quad (21)$$

Thus its derivative is an original boundary, with a primitive obtained by literal monic division by χ and the existing theta division maps. Its quotient class and arithmetic jets stay fixed.

Define

$$O_N = (1 - P_N) X R : C \rightarrow \mathcal{P}_{N+1} \cap \mathcal{P}_N^\perp, \quad I_N = P_{\mathcal{D}_N} X R : C \rightarrow \mathcal{D}_N.$$

Direct differentiation, with the derivative of the measure included, yields

$$\boxed{G_N'' - G_N' G_N^{-1} G_N' = O_N^* O_N - I_N^* I_N.} \quad (22)$$

Indeed $G_N' = R^* X R$, $G_N'' = R^* X^2 R - 2R^* X P_{\mathcal{D}_N} X R$, and $R G_N^{-1} R^*$ is the projection to the orthogonal quotient representatives. Their orthogonal direct sum with $\mathcal{D}_N$ is $\mathcal{P}_N$, which proves (22).

Both maps have rank at most one. For O_N this follows from the highest polynomial degree. For I_N^* , multiplication sends every boundary of degree below its highest degree back into $\mathcal{D}_N$, orthogonal to R ; only the highest boundary direction can contribute.

Taking $\text{Tr}(G_N^{-1} \cdot)$ gives (20). The positive source contribution and the negative boundary contribution are retained. Positivity of each Gram separately is not assigned to their difference.

S15.7 6. The phase in the rank-two control is a scalar volume derivative

Put $T = (A - cI)/i$ on C , with $c = k/2$. This is the coefficient operator corresponding to multiplication by u under the explicit ring map $S \mapsto c + iu$. The actual scaling generator is still $A = cI + iT$.

Reflection makes the polynomial $i^{-q} \chi(c + iu)$ real. This assertion follows by substituting $S = c + iu$ into $\chi^\dagger = (-1)^q \chi$. The retained leading phase of $\chi(c + iu)$ is i^q . Its remainder ideal and coefficient extraction are transported through that scalar unit, rather than altering χ in the arithmetic source.

In those real coefficient coordinates let $Q_j(u)$ be the monic source polynomials and $d_j = [Q_j]$. Their norm is ω_j . The metric and representatives have the same quotient definition. Define

$$\ell_N = \log V_N, \quad \sigma = \text{Tr } T \in \mathbb{R}.$$

The exact identity is

$$\boxed{\frac{d_N^* G_N d_{N+1}}{\omega_N} = \sigma - \ell_N'.} \quad (23)$$

To prove it, let $\mathfrak{b}_N = Q_{N+1} - R_N d_{N+1}$ in the polynomial source. It is the actual next boundary and

$$R_N^* \mathfrak{b}_N = -G_N d_{N+1}.$$

Multiplying the least-norm polynomial representative by u , its highest coefficient is $d_N^* G_N / \omega_N$. Therefore, modulo old boundaries,

$$X R_N - R_N T = \mathfrak{b}_N \frac{d_N^* G_N}{\omega_N}.$$

Multiplication by R_N^* eliminates precisely those old boundary terms. Consequently

$$\mathrm{Tr}(G_N^{-1} R_N^* X R_N) = \mathrm{Tr} T - \frac{d_N^* G_N d_{N+1}}{\omega_N}.$$

The left side is ℓ'_N by (21) and differentiation of the norm. This proves (23).

In the original S -polynomial coordinates, $p_j(S)$ maps to $i^j Q_j(u)$. Hence the preceding source cross term satisfies

$$\boxed{\frac{b_N^* G_N b_{N+1}}{\omega_N} = i(\sigma - \ell'_N).} \quad (24)$$

Both the sign and the factor i are retained. In particular an auxiliary real tilt generally creates a nonzero imaginary cross term even from an initially even measure.

S15.8 7. The exact scalar formula and a two-step upper bound

For $N \geq q$, define

$$\delta_N = \frac{V_N}{V_{N-1}}, \quad v_N = -\log \delta_N \geq 0, \quad a_{N+1} = \frac{\omega_{N+1}}{\omega_N}.$$

The nonnegativity follows from the same minimum-norm comparison at nested source degrees. The rank-one kernel update and the matrix determinant lemma give

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - \delta_N, \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1. \quad (25)$$

Substitute (24)–(25) into the inherited exact rank-two radius. This proves

$$\boxed{\epsilon_N^2 = a_{N+1}(1 - e^{-v_N})(e^{v_{N+1}} - 1) - (\sigma - \ell'_N)^2.} \quad (26)$$

All scalar inputs come from (7), (9), and (12). In particular

$$\delta_N = \frac{\omega_N}{\nu_{N-q}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_{N-q+1}}.$$

This does not assert that Hankel determinant evaluation is well-conditioned; the exact formula still requires error enclosures for numerical certification.

S15.8.1 7.1 The first admitted degree

At $N = q - 1$, K_{N-1} is not invertible. The displayed formula is replaced by its direct endpoint calculation:

$$\boxed{\epsilon_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - (\sigma - \ell'_{q-1})^2.} \quad (27)$$

Here $\nu_0 = Z_\chi$ is the actual norm of the first source relation. To prove (27), the first q remainder vectors are a basis, so the final included column has leverage one; the next determinant update gives $d_N = \nu_0 - \omega_q$. Formula (24) supplies the phase. No V_{q-2}^{-1} is introduced.

S15.8.2 7.2 Both losses in the finite upper bound

For $N \geq q$, put

$$s_N = \frac{v_N + v_{N+1}}{2}, \quad t_N = \frac{v_{N+1} - v_N}{2}.$$

Direct scalar multiplication gives

$$(1 - e^{-v_N})(e^{v_{N+1}} - 1) = \sinh^2 s_N - (e^{t_N} - \cosh s_N)^2.$$

Thus

$$\boxed{\epsilon_N^2 = a_{N+1} \sinh^2 s_N - a_{N+1} (e^{t_N} - \cosh s_N)^2 - (\sigma - \ell'_N)^2.} \quad (28)$$

In particular, with $\mathcal{R}_N = e^{2s_N} = V_{N-1}/V_{N+1} \geq 1$,

$$\boxed{\epsilon_N^2 \leq a_{N+1} \frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N}.} \quad (29)$$

This upper estimate retains two computable losses: imbalance between consecutive volume contractions and the original cross-term phase. When the bound is too coarse, (26) or (28) is the exact replacement within the same construction.

S15.9 8. What the exterior lower bound forces on these arithmetic volumes

The preceding exterior theorem gives

$$L_{h,k} := \sum_{\Re \lambda > k/2} \ell_\lambda (2\Re \lambda - k) \leq \epsilon_{h,k,N}.$$

It uses the original representative on its determinant line, not a newly chosen scalar metric. Combining it with (29), at $\theta = 0$, proves

$$\boxed{\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L_{h,k}}{\sqrt{a_{N+1}}} \right), \quad N \geq q.} \quad (30)$$

The equivalent inequality without logarithms is

$$4\mathcal{R}_N L_{h,k}^2 \leq a_{N+1} (\mathcal{R}_N - 1)^2.$$

For disjoint successive two-degree steps the logarithms telescope. For every $N_0 \geq q$ and integer $r \geq 1$,

$$\boxed{\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L_{h,k}}{\sqrt{a_{N_0+2j+1}}} \right).} \quad (31)$$

This is a source-volume consequence for the actual nested relation quotients, not a claim that arithmetic classes disappear when their representative norms decrease. At every degree their jets and original classes remain fixed by (5) and (15).

For a full off-line quartet with displacement $\delta > 0$, height $\gamma > 0$, and common multiplicity m , the previous count is

$$q = [1 + k(m - 1)](k + 1)^2,$$

$$L_{h,k} = 2\delta[1 + k(m - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor. \quad (32)$$

The quotient (16) and its derivative therefore give an entirely scalar test of the exterior obstruction. For this full quartet m_k and $|\chi(c + iu)|^2$ are even, so $\sigma = 0$ and $\ell'_N(0) = 0$. The phase term vanishes at zero tilt by this **additional** symmetry; it has not been removed from the general formula.

The sufficient subcubic target can now be tested from the quantities

$$\frac{\sqrt{a_{N(k)+1}}}{k^3} \frac{\mathcal{R}_{N(k)} - 1}{2\sqrt{\mathcal{R}_{N(k)}}}, \quad N(k) \geq q, \quad (33)$$

or, more sharply, from the full expression (26). An upper limit of zero for (33) would contradict (32) for that fixed quartet. This continuation proves the formula, the finite upper inequality, and the forced volume lower inequalities. It does not prove that limit.

S15.10 9. The Laplace input can be evaluated from the original theta source

The inverse Mellin representation makes F_h holomorphic in $|\arg z| < \pi/4$. Uniform vertical-strip bounds for v_h permit differentiation and shifts on closed subsectors, and shifting the Mellin contour gives the required rapid endpoint bounds there.

The exact analytic-continuation map is

$$E_\theta F_h(y) = e^{(y+i\theta/2)/2} F_h(e^{y+i\theta/2}).$$

For $|\theta| < \pi/2$, its Fourier transform, with kernel e^{ity} , is

$$\mathcal{F}_+ E_\theta F_h(t) = e^{\theta t/2} v_h(1/2 + it).$$

Contour shift proves the equality; Plancherel yields

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (34)$$

The phase $e^{i\theta/4}$ in E_θ has modulus one and is retained before taking the norm.

Dagger stability gives the analytic identity

$$F_h(1/\bar{z}) = (-1)^{\deg h} \bar{z} \overline{F_h(z)}.$$

Substitute $z = e^{i\theta/2}x$ and change variables on $(0, 1)$. This proves

$$M_h(\theta) = 2 \int_1^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (35)$$

Evenness in θ requires conjugation stability as well; it is not inferred just from this endpoint identity.

For the **analytic seed** $h = 1$, keep

$$f_0(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}.$$

Let $c_1 = -6, c_2 = 4$ and

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}).$$

Substitution into (35), where $\Re C_{mn} > 0$, gives

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(r+s+1/2, C_{mn}(\theta))}{C_{mn}(\theta)^{r+s+1/2}}. \quad (36)$$

The complex power is the analytic branch on the open right half-plane agreeing with the positive real power on its positive axis. The sum converges locally uniformly for $|\theta| < \pi/2$, by the exterior Gaussian estimates. Thus all derivatives used by (12) can be taken there. The factor 4 includes both the original two-sided theta sum and the change from the integral to the incomplete gamma function.

The accompanying numerical script evaluated (36) at 50 and 70 digits, with integer cutoffs 6 and 8. Their printed values agree:

$$M_1(0) \approx 1.279007247846485140479533592267193274492,$$

$$M_1(0.1) \approx 1.345907178458249796968220592357629805161.$$

An independent source integral gives

$$M_1''(0) = \|(D - 1/2)f_0\|^2 \approx 13.05554930257055843539268465812319368243.$$

Here $M_1'(0) = 0$. Consequently the first source Toda coefficient for tensor degree k is exactly

$$a_1^{(k)} = k \frac{M_1''(0)}{M_1(0)} \approx 10.207564753485721688865761252237555175 k. \quad (37)$$

This evaluates initial analytic data of the determinant flow. It is not a finite-packet control estimate: for $h = 1$ the arithmetic cyclic quotient is zero. The numerical results have neither directed-rounding enclosures nor a numerical omitted-tail certificate.

S15.11 10. Exact finite checks, an essential phase example, and current endpoint

The checker independently constructs source Gram matrices, quotient maps, representatives, action matrices, and scalar determinants in rational Gaussian fixtures. It compares the scalar result against $\text{Tr}((G^{-1}W)^2)/2$.

For the declared coefficient fixture with weight $7e^{-u^2/2} du / \sqrt{2\pi}$, relation $u^2 + u + 1$, and degree $N = 3$, it obtains

$$\ell'_N(0) = -52/111, \quad \sigma = -1, \quad \epsilon_N^2 = 1156/333.$$

The phase subtraction is $(59/111)^2$, which is nonzero. Omitting it changes the answer. A tilted even-relation fixture gives the same warning: parity at zero tilt is not imposed at another tilt.

Twenty-four exact test methods passed in normal and optimized Python, with identical successful records. Both deliberate-failure controls failed. Repeated roots, source-coordinate phases, mass factors, the minimal-degree endpoint, both Toda equations, the curvature difference, the complete scalar radius, and the supported-zero relation are included. These are finite regression tests, not Lean certificates or verification of infinite arithmetic integrals.

All 34 members named in the preceding exterior package's manifest verified. Its 22-method checker passed a fresh normal run. A combined command attempting a subsequent optimized rerun timed out; no completed optimized rerun is claimed for that inherited suite. The new suite passed in both modes.

The present advance is an upper-estimate mechanism on the exact original family: its input is the one-factor arithmetic Laplace transform; its relation filter is the original cyclic annihilator; its output is the canonical quotient volume and complete control radius. The source-boundary curvature and the positive volume retained before an e -quotient are explicit maps. No uniform subcubic upper estimate, RH/GRH proof, additional Lean execution, or remote repository modification is claimed.

S15.12 References and source boundary

The original source, support, and cohomological conventions are those in the supplied `Tau_Exterior_Trace_Amplification` package and its predecessor `Tau_Cyclic_Sum_Control`. The new formalization report and pinned PR #21 note supply the checked coefficient interfaces, not the analytic estimates in this note.

The general determinant, norm, and Christoffel–Darboux identities are classical. NIST DLMF §18.2(iv), (v), and (ix) records them, including the caution that raw Hankel determinant formulas are often poorly conditioned numerically: <https://dlmf.nist.gov/18.2>. The proof in §5 derives the needed Toda identity directly for the actual exponential deformation, without an isomonodromic or rational-logarithmic-derivative hypothesis. Gamma vertical-strip estimates used in §3 are recorded in DLMF §5.11: <https://dlmf.nist.gov/5.11>.

The finite exterior trace bound remains the theorem proved in the supplied preceding note; this continuation changes its analytic upper-estimate realization, not its algebraic proof or the scope of the parallel formalization.

S16 Supplied conormal derivative, symmetric restriction and uncompiled tower draft

Source attribution. This appendix reproduces the complete supplied mathematical continuation. Its first-person reports about PR 16, PR 18, combined CI, Mathlib inspection and 161 Wolfram checks are historical source statements, not additional verification performed for this edition. Both Lean blocks are retained; the proposed continuation is explicitly uncompiled. The monic-division coordinates and displayed dimension formulas have positive packet degree and tensor degree at least one; the empty packet is handled separately in the integrated proof. Full arithmetic unit and reflection formulas retain the cited source packet hypotheses.

Exact revision. 18f86ad411d0dfafc003aabddf25bccae2d38dcc4006c4d27e5be8aece9347b8

Source file. sources/web_conormal_tower_paste/CONORMAL_TOWER_SOURCE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 18f86ad411d0dfafc003aabddf25bccae2d38dcc4006c4d27e5be8aece9347b8

Formal status. Both Lean blocks are source text; the draft has not been compiled for this edition.

Codex's integration has advanced: PR #16 is merged, and the combined frontier/Codex run has now passed. I checked the current repository state and the completed job: it rebuilt the inherited sources, checked the frontier files, imported both developments into one Lean environment, and completed their selected axiom audits. The “combined run pending” sentence in PR #18's description is therefore out of date.

I continued with the new sum-connection note rather than duplicating that work. **The additional result is a complete calculation of what its conormal derivative detects, what it misses, and how that calculation restricts to the symmetric tensor summand.** It also packages the quotient and derivative together as a genuine algebra homomorphism.

The new results below have complete mathematical proofs and exact finite algebra checks. The Lean continuation at the end is an **uncompiled draft**: the local execution service is timing out, and the currently exposed GitHub actions provide no write operation. No remote files were changed in this turn.

S16.0.1 1. The derivative now has a complete quotient construction

Retain the source's notation, with tensor degree $k \geq 1$:

$$P = \mathbb{C}[s_1, \dots, s_k], \quad h_i = h(s_i), \quad I = (h_1, \dots, h_k),$$

$$E = P/I, \quad E^{[2]} = P/I^2, \quad \partial_S = \frac{1}{k} \sum_{i=1}^k \partial_{s_i}.$$

Here h is the original monic packet polynomial, including all multiplicities. These are precisely the relation ideal and derivative in the new attachment.

For any derivation D of a commutative algebra and any ideal I ,

$$\boxed{D(I^{r+1}) \subseteq I^r \quad (r \geq 0).}$$

Proof. An element of I^{r+1} is a sum of multiples of products of $r+1$ elements of I . The product rule differentiates one factor at a time, leaving at least r factors in I . Differentiating an exterior coefficient leaves all $r+1$ factors. Additivity completes the proof.

Consequently the source's maps exist at every order:

$$\delta_r : P/I^{r+1} \longrightarrow P/I^r, \quad [F] \longmapsto [\partial_S F].$$

They commute with the quotient transitions. In particular,

$$\delta_1 : E^{[2]} \longrightarrow E$$

is a derivation **along** the quotient $\pi : E^{[2]} \rightarrow E$:

$$\boxed{\delta_1(xy) = \pi(x)\delta_1(y) + \pi(y)\delta_1(x).}$$

This is the appropriate product rule; δ_1 is not an endomorphism of the arithmetic algebra E .

Monic division constructs all the retained layers There is a unique finite expansion

$$F = \sum_{\alpha \in \mathbb{N}^k} h_1^{\alpha_1} \cdots h_k^{\alpha_k} r_\alpha, \quad \deg_{s_i} r_\alpha < d := \deg h.$$

Existence follows by repeated monic division in the separate variables. For uniqueness, the leading monomials of

$$h^\alpha s^\beta, \quad 0 \leq \beta_i < d,$$

have exponent vectors $d\alpha + \beta$, which are distinct. A finite dependence therefore has a unique largest leading monomial and cannot cancel.

This gives explicit coordinates—not just an existence statement—for

$$I^r/I^{r+1} \cong E[\eta_1, \dots, \eta_k]_r, \quad \eta_i \longleftrightarrow [h_i].$$

In particular,

$$I/I^2 \cong E^k, \quad (a_i) \longmapsto \left[\sum_i h_i \tilde{a}_i \right],$$

and

$$\dim_{\mathbb{C}} P/I^{r+1} = d^k \binom{k+r}{k}.$$

On the associated graded, the derivative is exactly

$$\boxed{\text{gr}(\partial_S) = \frac{1}{k} \sum_i [h'_i] \partial_{\eta_i} .}$$

Indeed, differentiating a coefficient of $h^\alpha r_\alpha$ stays in $I^{|\alpha|}$, while differentiating one h_i -factor lowers the ideal order and contributes the literal exponent α_i . This proves the higher-layer formula in the attachment without deleting those exponent factors.

The standard conormal construction supplies the map

$$I/I^2 \longrightarrow \Omega_{P/\mathbb{C}} \otimes_P E.$$

The calculation here identifies its coordinates for this particular ideal and then contracts it by the specified sum direction; those are distinct operations (The Stacks Project Authors, n.d., Tag 00S0). (The Stacks Project)

S16.0.2 2. The exact information detected by the sum derivative

Let

$$A_h = \mathbb{C}[s]/(h), \quad E = A_h^{\otimes k},$$

and write

$$c_i = [h'(s_i)] \in E.$$

Under the preceding conormal coordinates, the restriction of δ_1 is the E -linear map

$$\partial_N : E^k \longrightarrow E, \quad (a_1, \dots, a_k) \longmapsto \frac{1}{k} \sum_i c_i a_i.$$

The attachment gives this row. The following kernel, cokernel, and rank calculations are additional deductions.

Let

$$b = \deg \gcd(h, h').$$

Over $\mathbb{C}$, if h has q distinct roots, then

$$b = d - q = \sum_{\rho} (m_{\rho} - 1).$$

The image and cokernel are

$$\operatorname{im} \partial_N = (c_1, \dots, c_k) \subseteq E,$$

$$\operatorname{coker} \partial_N \cong (\mathbb{C}[s]/(h, h'))^{\otimes k}.$$

Proof. The image is the ideal generated by the row entries because $1/k$ is invertible. Quotienting E by those entries imposes $h(s_i) = h'(s_i) = 0$ in each separate variable, giving the displayed tensor product.

Therefore

$$\operatorname{rank}_{\mathbb{C}} \partial_N = d^k - b^k, \quad \dim_{\mathbb{C}} \ker \partial_N = (k-1)d^k + b^k.$$

These formulas separate two sources of information lost by this particular observation.

When h is squarefree, $b = 0$, so the contracted derivative is onto. Nevertheless, for $k > 1$ its kernel still has dimension $(k-1)d^k$: **contracting to the sum direction does not observe every relative conormal direction, even without repeated roots.**

When roots are repeated, the additional term b^k records the failure of the Jacobian entries to generate the unit ideal.

For example, take the declared algebraic calibration

$$h(s) = s^2, \quad k = 2, \quad E = \mathbb{C}[x, y]/(x^2, y^2).$$

Then

$$\partial_N(a, b) = xa + yb.$$

Its image is

$$\text{span}\{x, y, xy\},$$

so its rank is three and its kernel has dimension five. The conormal vector $(x, 0)$ is nonzero but lies in that kernel. Its representative is x^3 modulo

$$I^2 = (x^4, x^2y^2, y^4),$$

so the retained relation is genuinely nonzero before observation.

These are ranks of the **conormal restriction**, not of the whole map $\delta_1 : E^{[2]} \rightarrow E$. The latter is onto: in coordinates

$$S = \sum_i s_i, \quad z_i = s_i - S/k,$$

one has $\partial_S = \partial/\partial S$, and a polynomial primitive is obtained by integrating its powers of S . The same argument makes every δ_r onto.

Package the quotient and derivative as one algebra map Introduce an ordinary formal square-zero parameter ϵ , distinct from the supported scalar e . Define

$$\boxed{\mathcal{J}_S^1 : E^{[2]} \longrightarrow E[\epsilon]/(\epsilon^2), \quad x \longmapsto \pi(x) + \epsilon \delta_1(x).}$$

This is a unital $\mathbb{C}$ -algebra homomorphism. Multiplication follows from the relative product rule:

$$(\pi x + \epsilon \delta x)(\pi y + \epsilon \delta y) = \pi(xy) + \epsilon \delta(xy).$$

Thus the source's two differently typed maps can be retained together in a single multiplicative observation. Its kernel is exactly

$$\ker \mathcal{J}_S^1 = \{n \in I/I^2 : \partial_N n = 0\},$$

and consequently

$$\boxed{\dim \ker \mathcal{J}_S^1 = (k-1)d^k + b^k, \quad \dim \text{im } \mathcal{J}_S^1 = 2d^k - b^k.}$$

The proof of the image dimension is the exact sequence

$$0 \longrightarrow \epsilon \text{ im } \partial_N \longrightarrow \text{im } \mathcal{J}_S^1 \longrightarrow E \longrightarrow 0.$$

This also describes the precise quotient of the first thickening seen by the sum derivative. It does not silently identify that quotient with the entire first thickening.

Retaining every first derivative is a different observation If all coordinate derivatives are retained, the map is

$$\mathcal{J}_{\text{all}}^1 : E^{[2]} \longrightarrow E \oplus E^k, \quad [F] \longmapsto ([F], ([\partial_i F])_i),$$

where E^k is a square-zero ideal in the target.

On $I/I^2 \cong E^k$, its derivative component is

$$(a_i) \longmapsto (c_i a_i)_i.$$

Each multiplication map M_{c_i} has kernel dimension $b d^{k-1}$. Hence

$$\dim \ker \mathcal{J}_{\text{all}}^1 = k b d^{k-1}.$$

In the squarefree case this full first-jet map is an algebra isomorphism onto the square-zero extension $E \oplus E^k$. With repeated roots it still loses the explicitly computed Jacobian-annihilator directions, which remain in the higher source layers.

The split lifts retain their original meaning throughout:

$$G(\pi)(n^\bullet) = e_E, \quad G(\mathcal{J}_S^1)(n^\bullet) = (\epsilon \partial_N n)^\bullet.$$

External absence maps to external absence. This is a calculation of a retained relation before quotienting, not a rule that differentiates e into a nonzero scalar. The attachment makes that distinction explicitly.

S16.0.3 3. This calculation restricts to the symmetric tensor construction

The preceding maps commute with simultaneous permutation of variables and conormal indices. Let

$$N = I/I^2 \cong E^k, \quad T = \mathbb{C}[s]/(h, h').$$

The cokernel sequence is equivariant:

$$N \xrightarrow{\partial_N} E \longrightarrow T^{\otimes k} \longrightarrow 0.$$

Taking S_k -invariants remains exact: given an invariant element in an image, average any preimage with the literal factor $1/k!$. Thus

$$\text{coker}(\partial_N : N^{S_k} \rightarrow E^{S_k}) \cong (T^{\otimes k})^{S_k}.$$

The invariant conormal domain has the explicit description

$$N^{S_k} \cong A_h \otimes (A_h^{\otimes(k-1)})^{S_{k-1}}.$$

An invariant tuple is determined by its first component, which is invariant under permutations fixing the first index; the other components are its prescribed translates. This constructs both directions of the identification.

Consequently,

$$\dim N^{S_k} = d \binom{d+k-2}{k-1},$$

$$\text{rank } \partial_N^{S_k} = \binom{d+k-1}{k} - \binom{b+k-1}{k},$$

and

$$\dim \ker \partial_N^{S_k} = d \binom{d+k-2}{k-1} - \binom{d+k-1}{k} + \binom{b+k-1}{k}.$$

For $b = 0$, the final binomial is zero.

For $h = s^2$ and $k = 2$, the symmetric conormal domain has dimension four, the symmetric arithmetic target has dimension three, and the restricted derivative has rank two. Its kernel still has dimension two.

These are ordinary variable-permutation invariants of the relation algebra. To place them in the original cochain construction, use the source's **signed chain idempotent**

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi.$$

Its two top-degree signs produce precisely the ordinary symmetric tensors used here. Replacing it by the unsigned chain average would change the selected summand.

This gives the smaller cohomological calculation its own exact derivative kernel; selecting the symmetric summand does not amount to presuming that the remaining relative information vanishes.

S16.0.4 4. The arithmetic unit and trace remain attached

The new note retains

$$U = \prod_i (g/h)(s_i)$$

and its Taylor class $\widehat{U} \in E^{[2]}$. The relative Leibniz rule proves

$$\delta_1(\widehat{U}x) = U \delta_1(x) + \delta_1(\widehat{U})\pi(x).$$

On a conormal class n , the second term vanishes:

$$\delta_1(\widehat{U}n) = U \partial_N n.$$

Because U is a unit, this arithmetic multiplication changes neither the kernel nor the ranks calculated above. It is nevertheless retained in the actual map, rather than assigned the value one. This is the source's specified unit correction.

There is also an important distinction between the **sum-Jacobian response** and the **product Jacobian used by the full product trace**.

Multiplication by

$$\prod_i h'(s_i)$$

on $E = A_h^{\otimes k}$ has rank

$$\boxed{q^k},$$

where q is the number of distinct roots of h . Its kernel has dimension

$$d^k - q^k$$

and is the nilradical of E .

Proof. At a root ρ of multiplicity m , write

$$h(\rho + z) = z^m u_\rho(z), \quad u_\rho(0) \neq 0.$$

Modulo z^m ,

$$h'(\rho + z) = m u_\rho(0) z^{m-1}.$$

Multiplication by this element has one-dimensional image and kernel (z) . At a k -tuple of roots, the product Jacobian similarly sends the constant coefficient to a nonzero multiple of

$$z_1^{m_1-1} \cdots z_k^{m_k-1}$$

and annihilates the maximal ideal. There is one image dimension per root tuple, giving q^k .

This is not the rank $d^k - b^k$ of the contracted conormal row. The two operators use the same derivative factors but perform different observations.

The one-variable trace identity remains exact. Let

$$\lambda_h([P]) = \sum_{\rho} \operatorname{Res}_{s=\rho} \frac{P(s)}{h(s)} ds.$$

Then

$$\boxed{\lambda_h(h'P) = \operatorname{Tr}(M_P \mid A_h).}$$

Locally, $h'/h = m_\rho/(s - \rho) + u'_\rho/u_\rho$, so its residue against P is $m_\rho P(\rho)$. Multiplication by P on the local algebra has that same trace because its nilpotent terms have zero diagonal.

Writing $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$, the actual arithmetic pairing gives

$$\begin{aligned} \mathcal{R}_Z(f, j_h(g')P) &= \lambda_h(\varepsilon_h f^\dagger v_h h'P) \\ &= \lambda_h(f^\dagger h'P) \\ &= \operatorname{Tr}(M_{f^\dagger P} \mid A_h). \end{aligned}$$

Both unit factors are present before cancellation. This is the algebraic form of the attachment's observation of

$$(\log x)\Theta(P(D)\phi_*),$$

with the original $g = 2\xi$, reflection, and multiplicities retained.

S16.0.5 5. Lean continuation against the integrated library

The merged Codex source already supplies

`SplitZero.RelationLayer.derivative`

as the quotient map induced by a linear map that carries the source relations into the target relations. I read its implementation and reused that interface below rather than introducing another quotient.

The following proposed `SplitZeroConormalTower.lean` proves the ideal-power containment from the derivation law, uses it to construct every descended derivative, proves compatibility between levels, and computes the derivative of an original relation. Its proof scripts are complete, but **compilation has not been checked**.

```
import SplitZeroRelationLayerIntegration
import Mathlib.RingTheory.Derivation.Basic
import Mathlib.RingTheory.Ideal.Operations

/-!
Derivative maps between the original ideal-power quotients.

This draft reuses RelationLayer.derivative. It does not assume that
the ideal is derivative-stable or that the packet is squarefree.
-/

noncomputable section

namespace SplitZero.ConormalTower

variable {R A : Type*}
  [CommRing R] [CommRing A] [Algebra R A]

/-- The original ideal-power relation, viewed over the coefficient ring. -/
def level (I : Ideal A) (r : ℕ) : Submodule R A :=
  (I ^ r).restrictScalars R

/-- A derivation lowers ideal order by at most one. -/
theorem deriv_mem_pow
  (D : Derivation R A A) (I : Ideal A) (r : ℕ)
  {x : A} (hx : x ∈ I ^ (r + 1)) :
  D x ∈ I ^ r := by
  induction r generalizing x with
  | zero =>
    rw [Submodule.pow_zero, Ideal.one_eq_top]
    exact Submodule.mem_top
  | succ r ih =>
    rw [Submodule.pow_succ] at hx
    refine Submodule.mul_induction_on hx ?_ ?_
    · intro a ha b hb
      rw [D.leibniz]
      change a * D b + b * D a ∈ I ^ (r + 1)
      have hd : D a * b ∈ I ^ (r + 1) := by
        rw [Submodule.pow_succ]
        exact Ideal.mul_mem_mul (ih ha) hb
      exact (I ^ (r + 1)).add_mem
        ((I ^ (r + 1)).mul_mem_right (D b) ha)
        (by simpa only [mul_comm] using hd)
    · intro a b ha hb
      rw [map_add]
      exact (I ^ (r + 1)).add_mem ha hb

/-- The derivative on A/I^(r+1), with values in A/I^r. -/
```

```

def descended
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) :
  (A / level (R := R) I (r + 1)) →1[R]
  (A / level (R := R) I r) :=
  SplitZero.RelationLayer.derivative
  (level I (r + 1)) (level I r) D.toLinearMap
  (fun _ hx => deriv_mem_pow D I r hx)

@[simp]
theorem descended_mk
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) (x : A) :
  descended D I r ((level I (r + 1)).mkQ x) =
  (level I r).mkQ (D x) :=
  rfl

/-- Forget one ideal-adic layer without differentiating. -/
def projection (I : Ideal A) (r : ℕ) :
  (A / level (R := R) I (r + 1)) →1[R]
  (A / level (R := R) I r) :=
  SplitZero.RelationLayer.transport
  (level I (r + 1)) (level I r)
  (fun _ hx => Ideal.pow_le_pow_right (Nat.le_succ r) hx)

@[simp]
theorem projection_mk (I : Ideal A) (r : ℕ) (x : A) :
  projection (R := R) I r ((level I (r + 1)).mkQ x) =
  (level I r).mkQ x :=
  rfl

/-- Derivative and quotient transition form the actual commuting square. -/
theorem tower_square
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) :
  (projection I r).comp (descended D I (r + 1)) =
  (descended D I r).comp (projection I (r + 1)) := by
  apply LinearMap.ext
  intro z
  obtain ⟨x, rfl⟩ :=
    Submodule.Quotient.mk_surjective (level I (r + 2)) z
  rfl

/-- On a retained relation z*a, only a*D(z) survives modulo I. -/
theorem conormal_product
  (D : Derivation R A A) (I : Ideal A)
  (z a : A) (hz : z ∈ I) :
  descended D I 1 ((level I 2).mkQ (z * a)) =
  (level I 1).mkQ (a * D z) := by
  rw [descended_mk, D.leibniz]
  change
    (level I 1).mkQ (z * D a + a * D z) =
    (level I 1).mkQ (a * D z)
  rw [map_add]
  have hzero : (level (R := R) I 1).mkQ (z * D a) = 0 := by
    apply (Submodule.Quotient.mk_eq_zero _).mpr
    change z * D a ∈ I ^ 1
    rw [Submodule.pow_one]
    exact I.mul_mem_right (D a) hz
  rw [hzero, zero_add]

end SplitZero.ConormalTower

```

The derivation and ideal-product APIs used here were checked against the repository's existing Mathlib revision, rather than assumed from a different release. That source inspection does not substitute for running Lean.

S16.0.6 Verification and the remaining quantitative connection

The fresh Wolfram calculations passed **161 exact checks**: conormal ranks and invariant restrictions, full first-jet ranks, ideal-square annihilation, residue traces, product-Jacobian ranks, higher derivative identities, and retained-unit trace identities. The fixtures include squarefree polynomials, repeated roots, mixed multiplicities, and tensor degrees through three. An initial variable-binding error in my checker was corrected before the successful runs. These are finite algebra checks, not a new Lean certificate or evaluations of arithmetic zero locations.

The new result does not identify the Hilbert-space normal map N_u with the algebraic conormal module I/I^2 . The attachment's filtered graph supplies the stated route from the full derivative to arithmetic jets. Its exact scalar identity

$$\mathcal{I}_{h,k} + 4 \int \|n_k(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h$$

retains the Hilbert normal contribution; the conormal calculations above now identify the algebraic kernel encountered by the subsequent sum-direction observation. Those two pieces must remain separately typed when estimating the original weight-control form.

The continuation now provides an exact first-jet algebra map, its full retained kernel, the corresponding symmetric restriction, and all-order quotient derivatives. Codex can build those on the merged quotient library without recreating the scalar, homology, or boundary-control implementations.

Reference The Stacks Project Authors. (n.d.). *The naive cotangent complex* (Tag 00S0). *The Stacks Project*. ([The Stacks Project](#))

S17 PR 21: conormal tower and cyclic depth formalization with its complete mathematical note

Source attribution. Complete mathematical note at PR 21 implementation commit 2abc351424ba87aeda948a5cfb846e15ed9373d1. The staged source provenance identifies all eight added Git blobs. The historical pasted draft, these repository modules and the observed remote CI evidence retain their distinct revisions and verification scopes.

Exact revision. 2abc351424ba87aeda948a5cfb846e15ed9373d1

Source file. `sources/web_conormal_cyclic_formal_delivery/repository/workbenches/tau-conormal-cyclic-formal/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 202d7e9111c0a88fdd30c15554f3dd1e0907b2fc4fb2e11da129c97b10a053fd

Formal status. The associated two Lean modules and observed remote CI evidence are archived separately. No local Lean execution is attributed to this edition.

S17.1 Original conormal and cyclic relation towers

S17.1.1 Scope and source reconciliation

This add-only continuation starts from main at a4494fba4968db837a8af6fcb952c8688cd5d1da. It retains the merged Codex quotient/homotopy/boundary implementation, the merged finite-frontier implementation, and the reviewed sum-connection note. It uses the owner's attached **Handoff: sum-generated arithmetic control**, especially targets 1, 2, 4 and 8. The current upload is an analytic/algebraic continuation with a separately reported 24-method finite checker; it is not a Lean certificate. The local execution service failed before its ZIP could be opened, so neither the ZIP manifest nor that checker has been independently rerun here.

Two actual GitHub corrections are incorporated, not merely acknowledged:

- Commit 600ab7a527f7a626c162bf0199d093d454b86c14 makes the nonempty hypothesis explicit in the monic-freeness argument and calculates $h=1$ separately. Every positive-depth relation ideal for $h=1$ is the whole polynomial ring. The corresponding underlying quotients are zero, while the split lift still retains tau and supported zero. The analytic theta seed and its mass do not disappear. `ConormalTower.top_level_zero` checks the full-ideal quotient at every level.
- Commit 7aec03a93b66d3d97fbf6f7042f58bb2c1019543 corrects the completed integration audit to 68 selected declarations and enables the existing frontier workflow on main and relevant pull requests. This branch inherits those files unchanged. The current 68-target audit, not the historical 66-target description, is rerun by the added workflow.

The original scalar G , e , tau, arithmetic projection to the infinite ring, reconstructed carriers, quotient transports and theta normalization $g=2$ xi are unchanged. No claim about a global arithmetic small-control estimate follows from the present algebraic results.

S17.1.2 1. An actual derivative between consecutive original quotients

Let R and A be commutative rings, A an R -algebra, I an ideal of A , and D an R -derivation of A . Then

$D(I^{(r+1)})$ is contained in I^r , $r \geq 0$.

Proof: use induction on r and the additive generation of an ideal product by products. For a in $I^{(r+1)}$, b in I , the Leibniz terms $a D(b)$ and $b D(a)$ both lie in $I^{(r+1)}$: the first by the ideal property, the second by the induction hypothesis. At $r=0$ the target ideal is the whole ring.

SplitZeroConormalTower.lean implements that proof as `deriv_mem_pow`. It then uses the merged `SplitZero.RelationLayer.derivative` to construct

$\text{delta}_r : A/I^{(r+1)} \rightarrow_R A/I^r, [x] \mapsto [D x]$.

It separately constructs the quotient projection p_r between the same types and proves

$p_r \text{delta}_{(r+1)} = \text{delta}_r p_{(r+1)}$.

The derivative is not an endomorphism of A/I . These are coefficient-linear maps on the actual original quotient types, not new quotient carriers. At a retained relation z in I ,

$\text{delta}_1([z a]) = [a D(z)]$.

For the full arithmetic multiplier u , the implemented formula is

$\text{delta}_r([u x]) = [u D(x)] + [x D(u)]$.

It applies in particular to the retained Taylor unit, without assigning the unit the value one or deleting its derivative. The definitions work for arbitrary commutative rings and do not assume characteristic zero or squarefreeness.

S17.1.3 2. The exact cyclic pullback, before a polynomial is used to describe it

Fix S in A . At each depth define the actual coefficient submodule

$K_r = \{P \text{ in } R[X] : P(S) \text{ belongs to } I^r\}$.

The implementation calls this `cyclicLevel`; it is literally the comap of the original relation submodule along `Polynomial.aeval\S`. Its constructed quotient map is

$\text{alpha}_r : R[X]/K_r \rightarrow_R A/I^r, [P] \mapsto [P(S)]$.

`evaluation_injective` proves injectivity by the two original quotient equality criteria. No assumed injectivity of polynomial evaluation is required. This implementation is a linear map; its usual algebra-homomorphism packaging is not claimed as a new bundled declaration.

If $D(S)=1$, Mathlib's formal polynomial chain rule gives

$D(P(S)) = P'(S)$.

Thus P in $K_{(r+1)}$ implies P' in K_r . The code constructs the cyclic derivative and proves the full square

$$\begin{array}{ccc} R[X]/K_{(r+1)} & \xrightarrow{-d/dX-} & R[X]/K_r \\ \downarrow \text{alpha}_{(r+1)} & & \downarrow \text{alpha}_r \\ A/I^{(r+1)} & \xrightarrow{\quad D \quad} & A/I^r. \end{array}$$

The actual unit comparison is also proved:

$\text{delta}_r([u P(S)]) = [u P'(S)] + [D(u) P(S)]$.

The final term need not be in the cyclic image. The equation retains its larger target. Most importantly, K_r is not replaced by K_1^r . The explicit annihilator calculation below explains the difference.

S17.1.4 3. Exact local maximum, including an attaining monomial

Here the base field is $\mathbb{C}$. Let $k \geq 1$, $d \geq 1$, $r \geq 1$ and write the complete packet polynomial as

$$h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}.$$

For a root tuple $\mathbf{\rho} = (\rho_1, \dots, \rho_k)$, put $m_i = m_{(\rho_i)}$ and $y_i = s_i - \rho_i$. Locally the ideal $(h(s_i))$ is $(y_i^{m_i})$, because each remaining factor is a unit. In the quotient by its r -th power the monomial basis is precisely

$$y^{\mathbf{a}} \text{ such that } \sum_i \lfloor a_i / m_i \rfloor < r.$$

Write $a_i = m_i q_i + t_i$ with $0 \leq t_i < m_i$. Its total degree satisfies

$$\sum_i a_i \leq \sum_i (m_i - 1) + (r - 1) \max_i m_i.$$

The proof uses $\sum_i q_i \leq r - 1$ and bounds each m_i by its maximum. This bound is attained: choose j with $m_j = \max_i m_i$, let every $t_i = m_i - 1$, let $q_j = r - 1$, and let every other $q_i = 0$.

SplitZeroCyclicDepth.lean formalizes the quotient/remainder predicate, the degree bound, that explicit exponent vector, and attainment. The floor-sum formulation is linked through Euclidean division. This is an all-index, all-depth result, not a finite calibration. The additional algebraic identification of these monomials with the polynomial quotient and the characteristic-zero multinomial step below have written proofs, not an additional Lean certificate in these two modules.

Let $N = y_1 + \dots + y_k$. Its exact nilpotency index is

$$L_r(\mathbf{\rho}) = 1 + \sum_i (m_i - 1) + (r - 1) \max_i m_i.$$

Indeed every monomial of total degree L_r is killed. The attaining monomial at degree $L_r - 1$ has coefficient

$$(L_r - 1)! / \prod_i a_i!$$

in $\mathbb{N}^{L_r - 1}$, a nonzero positive integer in characteristic zero. Distinct surviving monomials cannot cancel. This proves both vanishing and minimality.

The proof also counts the local dimension as

$$(\prod_i m_i) * \text{binomial}(k + r - 1, k).$$

The q_i with sum less than r give the binomial factor, and their remainders give the product factor.

S17.1.5 4. Collisions require a maximum at every depth

Set $S = \sum_i s_i$. At a collided sum λ , take the maximum over all root tuples with that sum:

$$\begin{aligned} \text{ell}(\lambda, r) &= \max_{\sum \rho_i = \lambda} L_r(\mathbf{\rho}). \\ \chi_r(X) &= \prod_{\lambda} (X - \lambda)^{(\text{ell}(\lambda, r))}. \end{aligned}$$

The Chinese-remainder decomposition and the exact local nilpotency prove

$$I^{\wedge r} \text{ intersect } \mathbb{C}[S] = (\chi_r(S)).$$

The same annihilator is obtained in the variable-permutation invariant subalgebra because every polynomial in S is already invariant. Thus α_r identifies the cyclic algebra $C[X]/(\chi_r)$ with its actual image, with no squarefree simplification.

The maximizing tuple can change with depth. In the declared calibration packet with roots 0,1,2 of multiplicities 5,4,1, respectively, the sum $\lambda=2$ in tensor degree two has

$$\text{ell}_r(2, r) = \max(5r, 4r+3).$$

At depths 1,...,5 the orders are 7,11,15,20,25. A tuple chosen only at depth one is insufficient for the whole tower. This is an algebraic test packet, not a statement about actual zeta zeros.

For every λ ,

$$\begin{aligned} \text{ell}_r(\lambda, r+1) &\geq \text{ell}_r(\lambda, r)+1, \\ \text{ell}_r(\lambda, r) &\leq r \text{ell}_r(\lambda, 1). \end{aligned}$$

The first follows by retaining a maximizing tuple from depth r ; its next slope is a positive multiplicity. For the second, note $\max_i m_i \leq 1 + \sum_i (m_i - 1)$ for every tuple. Hence

$$\begin{aligned} \chi_r &\text{ divides } \chi_{r+1}, \\ \chi_r &\text{ divides } \chi_{r+1}', \\ \chi_r &\text{ divides } \chi_1^r. \end{aligned}$$

The last divisibility supplies the stated surjection $C[X]/(\chi_1^r) \rightarrow C[X]/(\chi_r)$, not an equality. For $h(s)=(s-\rho)^2$ and $k=2$,

$$\begin{aligned} \chi_r(X) &= (X-2\rho)^{(2r+1)}, \\ \chi_1(X)^r &= (X-2\rho)^{(3r)}. \end{aligned}$$

They differ as soon as $r>1$. The derivative maps $C[X]/\chi_{r+1} \rightarrow C[X]/\chi_r$ are onto over C : integrate any polynomial representative, then take its source class. Their kernel dimensions are $\deg(\chi_{r+1}) - \deg(\chi_r)$.

For the empty packet $h=1$, handled separately as required by the GitHub correction, $I=P$, $K_r=C[X]$, and $\chi_r=1$ for all positive depths. The cyclic arithmetic quotient is zero. No maximum over an empty tuple set or inverse of a zero-dimensional Gram matrix is used to define it. The original nonzero analytic seed remains available as an analytic object.

S17.1.6 5. Explicit equivariant retraction of the cyclic summand (written proof)

This checks the handoff's retraction formula, including its inverse factor. On a λ -primary part M of the symmetric finite algebra, let $N=S-\lambda$, $N^{\text{ell}}=0$, and let w be the λ -component of the retained arithmetic unit U . Exact maximality gives $N^{(\text{ell}-1)}w \neq 0$. Choose the stated top-monomial coefficient functional θ with $\theta(N^{(\text{ell}-1)}w) \neq 0$. If it is realized first on an ordered root component, it can be restricted to the invariant subspace; the component of the symmetric vector is still detected.

Define

$$\pi_0(v) = \sum_{j=0}^{(\text{ell}-1)} \theta(N^{(\text{ell}-1-j)}v) X^j \bmod X^{\text{ell}}.$$

Coefficient comparison gives $\pi_0(Nv)=X \pi_0(v)$: the constant term on the left is $\theta(N^{\text{ell}}v)=0$ and the other coefficients shift exactly. Put $c=\pi_0(w)$. Its constant coefficient is nonzero, so it is invertible modulo X^{ell} . Explicitly, if $c=c_0(1+n)$, the inverse is $c_0^{-1} \sum_{j=0}^{(\text{ell}-1)} (-n)^j$.

Then $\pi=c^{-1}\pi_0$ satisfies

$$\pi(P(N)w) = P(X) \bmod X^{e11}.$$

Therefore the unit-weighted cyclic injection η has an S -equivariant left inverse. Summing over λ constructs the full retraction, and $\ker(\pi)$ is an invariant complement. This is a module splitting, not an assertion that π is an algebra map or that the summands are orthogonal for the canonical arithmetic Gram. All cross terms under that splitting remain.

This cyclic summand retains each amplified value $k \rho$; collided repeated pure tensors need not equal the particular cyclic eigenvector. The cyclic residue trace and the full symmetric trace have their distinct multiplicities. No part of the present calculation identifies those traces or drops the U -dagger U factor in the ambient arithmetic pairing.

S17.1.7 6. Formal versus analytic scope

The new formal outputs are the ideal-power derivative, actual quotient transitions, exact cyclic pull-back/injection, derivative square, unit rule, full-ideal case, and the all-degree attained monomial ceiling. Existing SplitZero support maps lift their coefficient maps at the original labels. No duplicate scalar construction or alternative homology core is introduced.

The exact χ_r annihilator, local basis/multinomial identification and the retraction in Sections 3-5 are supplied here as complete written arguments. Their finite calibrations are separate Python tests. The new upload's analytic cyclic convolution measure, canonical Gram, differentiated moments, rank-two control and full trace comparison remain its separately stated inputs and targets. This work does not independently certify that archive, an analytic moment enclosure, or a tensor-uniform small-control bound.

S17.1.8 Reproduction

Run the added SplitZero\conormal\cyclic\tower workflow on this branch. It installs unchanged Lean 4.31.0 and Mathlib fabf563a7c95a166b8d7b6efca11c8b4dc9d911f, rebuilds all inherited modules, checks every new source with `- \/-trust=0 \-DwarningAsError=true`, imports the combined sources, and rejects any selected transitive axiom outside `propept`, `Classical.choice` and `Quot.sound`. It reuses the previously tested axiom parser. The original core's byte pin is checked by the inherited prepare script.

The new finite checker directly multiplies the nilpotent sum in explicit local monomial quotients. It tests attainment and its literal multinomial coefficient, collided maxima, derivative divisibility, local dimensions and the $h=1$ branch, in normal and optimized Python with deliberate-failure controls. These finite tests do not replace the general Lean proofs or the written polynomial arguments. The final PR record identifies the exact completed run; no pending run is a certificate.

S18 The coordinating Zeta contribution: determinant losses and spectral coupling

Source attribution. Complete mathematical note supplied by the coordinating Zeta task on 13 September 2026. The direct sum of the two exact identities retains the original quotient metric, phase, spectral coupling and compression remainder. Its full formulas, domains, endpoint scope and Sylvester comparison are reproduced as supplied. The integrated bridge gives the same join with its complete derivation. No asymptotic estimate or novelty claim is inferred.

Exact revision. 71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370

Source file. sources/web_toda_parent_join/PARENT_EXACT_JOIN.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370

Formal status. Complete written join of Toda (28) and frozen exterior identity E8; no new Lean execution.

S18.1 Exact join: determinant losses and spectral coupling

This short continuation joins two already written identities on the same finite arithmetic packet and its original least-norm metric. It does not replace that metric or claim an asymptotic estimate. It is a direct algebraic consequence of the supplied Toda note (28) and the frozen companion reader's exterior equality calculation (E8); no novelty claim or Lean execution.

S18.1.1 Original objects and notation

Fix the existing nonempty packet, its tensor degree k , cyclic algebra $C = C[S]/\chi$, $q = \deg \chi$, $A = M_S$, and an admitted degree $N \geq q$. All quantities below are at the original observation $\theta = 0$. Retain G_N , the least-norm quotient Gram, and the adjoint $A^\sharp = G_N^{-1} A^* G_N$. Write $H_N = A^\sharp + A - kI$. Its eigenvalues are $+\epsilon_N$, $-\epsilon_N$, and zero, with the multiplicities from the rank-two calculation.

Let V_+ be the full generalized right-half-plane spectral subspace of A , and P_N its G_N -orthogonal projector. Its positive excess is

$$L = \sum_{\text{Re } \lambda > k/2} \ell_\lambda (2 \text{Re } \lambda - k).$$

Consider the nonempty V_+ case, so $L > 0$ and $\epsilon_N > 0$ by the proved exterior trace bound. Let u_+ , u_- be G_N -unit eigenvectors of H_N at $+\epsilon_N$ and $-\epsilon_N$. The coordinate injection $E_N : C^2 \rightarrow C$ sends (z_+, z_-) to $z_+ u_+ + z_- u_-$. Set $M_N = E_N^\sharp P_N E_N$. Thus $0 \leq M_N \leq I_2$. These eigen-coordinates describe a compression only; they do not replace the arithmetic metric.

Use $C_N = P_N A (1 - P_N) : V_+^\perp \rightarrow V_+$ for the actual off-diagonal coupling. The symbol C_N is used here to avoid confusing this coupling with the Toda note's operator $T = (A - kI/2)/i$. Every Hilbert-Schmidt norm below uses the original restricted G_N metrics.

For the scalar quantities retain

$$V_N = \det G_N = D_{(N+1)}/B_{(N-q+1)}, \quad v_N = \log(V_{(N-1)}/V_N), \quad s_N = (v_N + v_{(N+1)})/2, \quad t_N = (v_{(N+1)} - v_N)/2, \quad a_{(N+1)} = \omega_{(N+1)}/\omega_N, \quad \phi_N = \sigma - \partial_\theta \log V_N|_{\theta=0}, \quad \sigma = \text{Tr}((A - kI/2)/i).$$

Here D and B are exactly the source and relation Hankel determinant sequences in the supplied note, not alternative free sequences.

S18.1.2 Complete loss identity

The Toda identity gives

$$a_{(N+1)} \sinh(s_N)^2 - \epsilon_N^2 = a_{(N+1)} (\exp(t_N) - \cosh(s_N))^2 + \phi_N^2.$$

The companion's E8 gives, on the same original metric,

$$\epsilon_N^2 - L^2 = \|C_N\|_-(HS, G_N)^2 + \epsilon_N^2 (\det M_N + \det(I_2 - M_N)).$$

Adding these equalities cancels precisely ϵ_N^2 and proves

$$a_{(N+1)} \sinh(s_N)^2 - L^2 = a_{(N+1)} (\exp(t_N) - \cosh(s_N))^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2 + \epsilon_N^2 (\det M_N + \det(I_2 - M_N)).$$

All four terms on the right are nonnegative. The first two are the successive-volume imbalance and the retained phase. The last two are the original spectral-subspace coupling and the compression remainder. There is no cancellation between them.

The exact Sylvester morphism is also retained. In the G_N -orthogonal decomposition $C = V_+ \text{ plus } V_+^{\perp}$, write $A = [[B_N, C_N], [0, D_N^{\text{op}}]]$, and the CRT spectral idempotent as $Q = [[I, X_N], [0, 0]]$. The identity $AQ = QA$ is exactly

$$B_N X_N - X_N D_N^{\text{op}} = C_N.$$

Its inverse is $\int_0^{\infty} \exp(-t B_N) Y \exp(t D_N^{\text{op}}) dt$; the strict real spectral separation and the complete finite nilpotent polynomial factors give convergence. Consequently the coupling term controls $\|Q - P_N\|$ by the actual smallest singular value of this Sylvester map, as in E14-E18. No uniform bound for that singular value is inserted here.

S18.1.3 What this adds to the two-step inequality

Let $R_N = V_{(N-1)}/V_{(N+1)} = \exp(2s_N)$. Dropping only the imbalance square and the nonnegative compression remainder gives the stronger finite inequality

$$a_{(N+1)} (R_N - 1)^2 / (4 R_N) \geq L^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2.$$

Since $s_N \geq 0$, this is equivalently

$$\log R_N \geq 2 \operatorname{arsinh}(\sqrt{L^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2} / \sqrt{a_{(N+1)}}).$$

For a full quartet at zero tilt, the supplied symmetry proves $\phi_N = 0$; the coupling term is still present. If that coupling is nonzero, the original lower bound with L alone is strictly sharpened at that N . Nothing here supplies uniform strictness or large- k decay. The retained endpoint $N = q-1$ uses the source note's separate endpoint formula rather than these preceding-volume ratios.

The empty selected subspace has $L = 0$ and should use the Toda identity directly; no positive-eigenline construction is asserted when $\epsilon_N = 0$.

S18.1.4 Evidence and reading scope

The parent read the complete supplied NOTE.tex, including its sections 1-10, references and explicit first-degree calculation, directly from `Tau_Toda_Volume_Control_2026-09-12.zip` (SHA-256 80b365233ba5ebaf1b52aeea369a7771d74bf774dbb4416f10a56503321b8f41). The companion E8/E14-E18 source is pinned by GitHub commit 161089942cce70a09fbe441d08ba4d36c1fd810c, under `workbenches/splitzero-tandem/continuations/20260912-cyclic/tex/` `exterior_trace_equality_continuation.tex`. The displayed join was checked by literal addition of the two equalities. The independent delivery/checker review is separate in `toda_review/`.

Primary-source context checked in this turn: DLMF 5.11.9 for gamma vertical-strip decay, and DLMF 18.2(ix), especially 18.2.30 and the following numerical-conditioning warning. The supplied 1997 Toda paper was verified at publisher abstract/metadata scope; its full proof was not fetched. These references identify the classical ingredients, not a prior source for this arithmetic specialization.

S19 Gamma convolution descent and the original arithmetic coefficient correction

Source attribution. Complete original Gamma Convolution Descent Markdown note delivered on 13 September 2026. Its archive and alternate complete TeX note are retained byte-for-byte. The full local coefficient chapter proves the exact coordinate maps, finite derivative indices, cochain primitive and transform domains alongside this unmodified source witness. Source reports about remote revisions and computations retain the verification scope of their original receipts.

Exact revision. 886c82e2562d9b4846adb44640d28a34552e38eb2518a3cd47f90dbbedd95e5b

Source file. `sources/web_gamma_convolution_delivery/Tau_Gamma_Convolution_Descent/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 886c82e2562d9b4846adb44640d28a34552e38eb2518a3cd47f90dbbedd95e5b

Formal status. Complete source witness. Finite checker records are separately retained. No new local Lean execution is claimed.

S19.1 1. Scope and the inspected collaboration state

This is a mathematical continuation of **Tau Toda Volume Control** and **Tau Exterior Trace Amplification**. It retains their arithmetic source, quotient, scaling action, full packet multiplicities, and canonical minimum-norm metric. The new calculation links those objects to the gamma reference that is already in the repository. The gamma orthogonal polynomials and their general convolution identities are classical. The contribution here is the explicit arithmetic coefficient transform, its source and quotient maps, its quantitative bounds, and the resulting exact substitution into the existing Toda/exterior estimate.

The GitHub read was pinned to main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f. The separate formalization was read at PR #22 head 811210d24b80813a08972ca23f919db015383137. At that inspected revision the PR explicitly said strict verification was still in progress. No successful certificate for that revision is inferred here. Its invariant-filtration trace budget is retained as the other session's result, not presented as a new deduction of this note.

The main-branch gamma source read was `workbenches/splitzero-tandem/continuations/20260912-stieltjes/tex/theta_gamma_reference.tex`, equations TG.2–TG.14. It already constructs the exact one-factor gamma measure, its constants, its monic polynomials and its arithmetic multiplier. We extend that specified reference through the sum pushforward at every tensor degree.

There is no new RH proof or arithmetic asymptotic assertion here. Theorems below have written proofs. The checker supplies finite exact regression evidence, not analytic interval certification and not Lean execution. No inherited file or other-session branch is modified.

S19.2 2. The original arithmetic source and its split maps

Retain

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

with the original maps

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The target of the first quotient is infinite. Its structural base is the same absolute pointed base $\mathfrak{b}_\tau$. The supported scalar and the external absorber have not changed.

The source complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x,$$

where V consists of the original even Schwartz functions with both original moments zero, and $\mathcal{B}$ has the original rapid decay at both multiplicative ends with all D -derivatives. Keep

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Fix a nonempty finite packet $h(s) = \prod_\rho (s - \rho)^{m_\rho}$ of actual zeros, with every selected order complete. For reflection statements require the same dagger stability as the source. Write

$$v_h = g/h, \quad \mathcal{M}F_h = v_h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int w_h(t) dt. \quad (2)$$

The cyclic tensor source and its maps remain

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad C_{h,k} = \mathbb{C}[S]/\chi_{h,k}, \quad A = M_S, \quad (3)$$

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi,$$

where $S = s_1 + \cdots + s_k$ and $\eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1$, with $v_h = j_h(g/h)$. All local units remain in this map.

At polynomial degree N , the original relation map is

$$M_\chi : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad P \mapsto \chi P, \quad q = \deg \chi, \quad (4)$$

and its image under $\mathcal{V}_{h,k}$ consists of the already constructed theta boundaries. Every norm construction below concerns this same map and its quotient. Its squared-norm weight $|\chi|^2$ does not mean that the next conormal ideal is (χ^2) : that ideal remains the exact pullback of the original I^r .

For $h = 1$, the analytic source (2) still exists, but the finite arithmetic quotient is the zero module. Its split lift has e and τ . No nonzero packet or Gram inverse is inferred from that analytic branch.

S19.3 3. Carry the repository's gamma reference through every tensor power

For a fixed real $\lambda > 0$ define exactly

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad d\sigma_\lambda(t) = r_\lambda(t)dt, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda). \quad (5)$$

The repository proves, by the beta integral and Fourier inversion,

$$\int e^{iyt} d\sigma_\lambda(t) = c_\lambda (\cosh y)^{-2\lambda}. \quad (6)$$

Its factor 2 comes from $t = 2u$ in the beta integral. Thus c_λ , rather than 1, is the mass. Fourier uniqueness for finite measures gives the all-tensor identity

$$r_{\lambda,k} := r_\lambda^{*k} = \kappa_{\lambda,k} r_{k\lambda}, \quad \kappa_{\lambda,k} = \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (7)$$

In particular the mass of the sum measure is c_λ^k . For $\lambda = 1/4$ this reads

$$r_{1/4,k}(u) = \frac{2^{k-1} \pi^{k/2}}{\Gamma(k/2)} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad \int r_{1/4,k} = (2\pi)^{k/2}. \quad (8)$$

The literal coordinate map is $(t_1, \dots, t_k) \mapsto u = \sum_i t_i$; u is not replaced by an average or a rescaled variable. The original arithmetic scaling coordinate is still $S = k/2 + iu$.

For real $|\theta| < \pi/2$, analytic continuation of (6) gives

$$Z_{\lambda,k}^\Gamma(\theta) := \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}. \quad (9)$$

The exponential integrability follows either from the original beta-contour estimate or the gamma vertical-strip bound. This justifies all finite differentiated moment integrals used below.

S19.4 4. The sum projection and its complete polynomial complement

Define actual Hilbert spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du).$$

Their isometric inclusion is

$$U_\Sigma : \mathcal{H}_\Sigma \rightarrow \mathcal{H}_k, \quad (U_\Sigma f)(\mathbf{t}) = f(t_1 + \dots + t_k). \quad (10)$$

For $k \geq 2$, its adjoint has the explicit disintegration formula

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (11)$$

Its denominator is the actual fibre mass. Fubini proves $C_\Sigma U_\Sigma = 1$ and $C_\Sigma = U_\Sigma^*$. For $k = 1$ both maps are the identity. Put $P_\Sigma = U_\Sigma C_\Sigma$ and retain

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n. \quad (12)$$

These are inverse Hilbert isometries onto $\mathcal{H}_\Sigma \oplus \ker C_\Sigma$. The second component is not discarded. At each original support label ℓ , their reconstructed maps are $(\ell, F) \mapsto (\ell, C_\Sigma F)$ and $(\ell, F) \mapsto (\ell, (1 - P_\Sigma)F)$. A nonzero normal vector has first image $(\ell, 0)$, while its second image retains that vector. Both maps send external absence to external absence; their paired inverse is still (12). Norm and Gram observations have their actual quadratic types, not the type of a linear quotient: $\text{Gram} : \text{Hom}_\mathbb{C}(E, \mathcal{H}) \rightarrow \text{Herm}(E), R \mapsto R^* R$.

S19.4.1 4.1 Exact action on every product-polynomial degree

Let $b_n^{(\lambda)}(t)$ be the repository's monic gamma polynomial, with

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}.$$

Set $\alpha = 2\lambda$. Its complete norm and recurrence are

$$\|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \quad (13)$$

The polynomials are not divided by their norms. Multiplying the k generating functions proves the literal addition formula

$$b_n^{(k\lambda)}\left(\sum_i t_i\right) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \quad (14)$$

For a multi-index $\mathbf{n}$, put $n = \sum_i n_i$. The exact sum projection is

$$C_\Sigma \left(\prod_i b_{n_i}^{(\lambda)}(t_i) \right) = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (15)$$

Proof. Pair the product against (14), in the actual product measure. Orthogonality makes the answer zero unless the sum degree is n , and at degree n the answer is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$. The squared norm of the sum polynomial is $c_\lambda^k n! (k\alpha)_n$. Their quotient gives (15).

These tests identify the whole conditional projection, not merely its finite polynomial moments: the gamma sum measure has exponential moments, so its polynomials are dense in L^2 . To verify that last statement, an L^2 vector orthogonal to all polynomials defines a finite measure whose Fourier transform is analytic on a strip by Cauchy-Schwarz and exponential integrability. All derivatives at zero vanish; analytic uniqueness and uniqueness of Fourier transforms give the zero vector.

The retained normal polynomial is

$$n_{\mathbf{n}} = \prod_i b_{n_i}^{(\lambda)}(t_i) - \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}\left(\sum_i t_i\right).$$

Its exact squared norm is

$$\|n_{\mathbf{n}}\|^2 = c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right]. \quad (16)$$

Thus every degree, every factorial, and the relative component have an explicit map and norm. This is the classical Meixner-Pollaczek addition mechanism, applied with the source's exact scale and masses; it is not a claim to have discovered a new general family of orthogonal polynomials.

S19.5 5. Insert the complete arithmetic multiplier, not a substituted gamma measure

Define the complex amplitude and its squared modulus by

$$A_{h,\lambda}(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda}(t) = |A_{h,\lambda}(t)|^2, \quad w_h = B_{h,\lambda} r_\lambda. \quad (17)$$

The entire quotient g/h supplies values at the selected zeros. No pole is inserted into the multiplier at those locations. The full complex amplitude, not only its modulus, remains in the source comparison

$$P \mapsto \prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} A_{h,\lambda}(t_i) P(k/2 + i \sum_i t_i). \quad (18)$$

This is the original Mellin image of $\mathcal{V}_{h,k}P$.

The scalar multiplier on the sum line is

$$B_{h,\lambda;k} = C_\Sigma(B_{h,\lambda}^{\otimes k}).$$

The exact pushforward identity is

$$\boxed{m_{h,k}(u) = w_h^{*k}(u) = r_{\lambda,k}(u) B_{h,\lambda;k}(u)}. \quad (19)$$

This is a norm observation of the original source; it does not project the original finite arithmetic module or change its trace. The maps in (3), including the full unit in η , remain attached to every polynomial. The full field $B_{h,\lambda}^{\otimes k}$ is recoverable from (12), so its relative complement is retained too.

S19.5.1 5.1 A bounded multiplier for every full quartet

For $\lambda = 1/4$, the original formula is

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (20)$$

The minus sign and phase are still in (18). The elementary continuation formula

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx$$

gives, on this line,

$$|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|).$$

Let $d = \deg h \geq 3$ and take $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. For $|t| \geq T_h$, $|h(1/2 + it)| \geq (|t|/2)^d$, and direct substitution gives

$$B_{h,1/4}(t) \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d}.$$

Consequently the finite, specified constant

$$C_h = \max \left\{ \sup_{|t| \leq T_h} |A_{h,1/4}(t)|^2, \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} T_h^{6-2d} \right\}$$

satisfies $0 \leq B_{h,1/4} \leq C_h$. The compact supremum uses the analytic quotient, not the expression with a removable denominator left untreated. A full nonreal quartet has $d = 4m \geq 4$, so this applies to the entire packet used in the exterior amplification, with constants allowed to depend on that fixed packet.

The positive integral in (11) now proves

$$0 < B_{h,1/4;k}(u) \leq C_h^k \quad (k \geq 2), \quad 0 \leq m_{h,k}(u) \leq C_h^k r_{1/4,k}(u). \quad (21)$$

Strict positivity for $k \geq 2$ follows because the one-factor multiplier only vanishes at a discrete set; its excluded hyperplanes have zero fibre measure. For $k = 1$ retain its actual isolated zeros.

S19.5.2 5.2 The other packet sizes and the empty analytic source

For a chosen reference below, C_h always denotes an explicitly specified upper bound for that reference's $B_{h,\lambda}$. An explicit alternate reference is available without changing the arithmetic source. For integer $L \geq 0$,

$$A_{h,1/4+L}(t) = \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}. \quad (22)$$

Its product is a known nonvanishing gamma-shift factor on the integration line. Taking $L \geq \max(0, 3 - d)$ makes its squared modulus bounded by the same compact/tail argument. Equations (17)–(19) connect both references to the identical original w_h ; no coordinate or original mass is reassigned.

For the actual seed $h = 1$, choosing $L = 3$ gives the explicit all-real estimate

$$B_{1,13/4}(t) \leq 1024/\sqrt{\pi}. \quad (23)$$

Indeed the gamma-shift denominator is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, and $|\zeta(1/2 + it)| \leq 4\sqrt{t^2 + 1/4}$. The remaining fraction is bounded by one since $(t^2 + 1/4)^2 \leq (t^2 + 25/4)(t^2 + 81/4)$.

This is an analytic estimate for a nonzero seed. Its finite spectral quotient is still zero, as required by the merged correction.

S19.6 6. Exact arithmetic coefficients at every tensor degree

Use a reference for which $B = B_{h,\lambda}$ is bounded, as constructed above. Define its actual coefficients

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_\lambda j! (\alpha)_j}, \quad \mathcal{A}_h(z) = \sum_{j \geq 0} (\alpha)_j c_{h,j} z^j. \quad (24)$$

The series is used coefficientwise; no convergence radius for this formal series is stipulated. Each coefficient is an actual one-factor arithmetic integral. Define

$$d_{h,k,n} = [z^n] \mathcal{A}_h(z)^k.$$

Then the complete $L^2(r_{\lambda,k} du)$ expansion is

$$B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (25)$$

Proof. Pair $B^{\otimes k}$ with the finite addition formula (14). Each one-factor pairing is $c_\lambda j! (\alpha)_j c_{h,j}$. The multinomial coefficient cancels the displayed factorials and gives

$$\int B_{h,\lambda;k} b_n^{(k\lambda)} r_{\lambda,k} = c_\lambda^k n! [z^n] \mathcal{A}_h(z)^k.$$

Divide by the exact norm $c_\lambda^k n! (k\alpha)_n$. Completeness gives (25). No exchange of an infinite product of formal series with an integral was needed: every coefficient used a finite polynomial identity.

At degree zero,

$$c_{h,0} = \mu_h / c_\lambda, \quad d_{h,k,0} = (\mu_h / c_\lambda)^k,$$

so (25) recovers exactly $\int m_{h,k} = \mu_h^k$.

The retained relative residual is

$$\mathcal{N}_{h,k} = B^{\otimes k} - U_\Sigma B_{h,\lambda;k}.$$

Its whole norm is calculated by Pythagoras:

$$\|\mathcal{N}_{h,k}\|^2 = \left(\int B(t)^2 d\sigma_\lambda(t) \right)^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (26)$$

The difference is nonnegative because it is the norm of the explicitly retained vector. It is not set to zero by calling the sum map a pushforward.

S19.6.1 6.1 Finite exactness and approximation errors

Let $B_L = \sum_{j=0}^L c_{h,j} b_j^{(\lambda)}$ and $B_{k,L} = C_\Sigma(B_L^{\otimes k})$. Projection contraction and the tensor telescoping identity give

$$\|B_{h,\lambda;k} - B_{k,L}\| \leq k \|B\|^{k-1} \|B - B_L\|. \quad (27)$$

These norms have the respective original reference measures. The proof uses $\|B_L\| \leq \|B\|$ and expands the tensor difference into k terms. Both sides retain the full mass factors.

More strongly, the moments of $B_{k,L} r_{\lambda,k}$ against every polynomial of degree at most L are **exactly** those of $m_{h,k}$, since coefficient $n \leq L$ in (25) uses only $c_{h,j}$ with $j \leq n$.

It follows that the entire degree- N source Gram, and its degree- N relation Gram, are determined exactly by $c_{h,0}, \dots, c_{h,2N}$. The relation integrands are also degree at most $2N$. Their tensor dependence is the explicit finite coefficient power in (24), not a fresh k -dimensional integral.

A truncated $B_{k,L}$ need not be nonnegative pointwise. Its relationship to the true positive metric is the exact finite-moment equality just proved when $L \geq 2N$, or the error estimate (27) and Cauchy-Schwarz for the specified polynomial tests when $L < 2N$. Positivity is not inferred from a plotted truncated density.

S19.6.2 6.2 Actual seed coefficients, and their tensor cross term

The supplied numerical driver evaluates the original theta seed on $x \geq 1$, using its inversion symmetry. With $L = D - 1/2$, the even spectral moments are $\mu_{2r} = 2 \int_1^\infty |L^r f_0(x)|^2 dx$. It constructs their source polynomials by $q_0(z) = 4z^2 - 6z$ and $q_{r+1} = 2z(q_r - q'_r) - q_r/2$, and integrates the integer double sums using upper incomplete gamma integrals. This is a numerical check, not a validated interval computation.

For $h = 1$ and the bounded-multiplier reference $\lambda = 13/4$, the 50-digit, cutoff-6 and 70-digit, cutoff-8 calculations agree in the displayed values:

Quantity	Numerical value
μ_0	1.279007247846485140479533592267
μ_2	13.055549302570558435392684658123
μ_4	383.274341717364242554407237479131
μ_6	17946.5273016052076235667646919834
$c_{h,0}$	0.201056688692785865809916163220
$c_{h,2}$	0.007645442999486401188133851481
$c_{h,4}$	-0.000030516920803762043746724840

The reference mass is $c_{13/4} = 6.361426004587219292629853548733\dots$, so it has not been assigned the actual mass μ_0 or the value one. The negative fourth expansion coefficient is retained; a positive density need not have positive coefficients in this orthogonal basis.

For every even arithmetic multiplier, the first tensor terms from (25) are exactly

$$d_{h,k,2} = k c_{h,0}^{k-1}(\alpha)_2 c_{h,2},$$

$$d_{h,k,4} = k c_{h,0}^{k-1}(\alpha)_4 c_{h,4} + \binom{k}{2} c_{h,0}^{k-2}((\alpha)_2 c_{h,2})^2.$$

Both terms in the fourth coefficient remain. These are coefficients of the exact arithmetic source norm; the empty packet still supplies no nonzero finite spectral quotient. Agreement of two numerical truncations is not a bound on their shared truncation or rounding error.

S19.7 7. Feed the result into the two original Toda determinants

Put $\alpha_k = 2k\lambda$. The full gamma source determinants are explicit:

$$\mathfrak{D}_n^\Gamma(\theta) = c_\lambda^{kn} \prod_{j=0}^{n-1} j! (\alpha_k)_j (\cos \theta)^{-n(\alpha_k+n-1)}. \quad (28)$$

At $\theta = 0$ this is the product of the monic norms from (13) and (7). One direct proof for all real θ starts with $\mathfrak{D}_0^\Gamma = 1$ and $\mathfrak{D}_1^\Gamma = c_\lambda^k \cos^{-\alpha_k} \theta$. Apply the already-proved Hankel-Toda identity recursively; the second derivative of the logarithm of the right side is $n(\alpha_k + n - 1) \sec^2 \theta$. Substitution proves the next determinant with all constants. Strict positivity makes the recurrence division valid.

In particular

$$\omega_n^\Gamma(\theta) = c_\lambda^k n! (\alpha_k)_n (\cos \theta)^{-(\alpha_k+2n)}, \quad a_n^\Gamma(\theta) = n(n + \alpha_k - 1) \sec^2 \theta. \quad (29)$$

The arithmetic source determinant is the previous $\mathfrak{D}_n$. The reference relation determinant is

$$\mathfrak{B}_n^\Gamma(\theta) = \det \left[\partial_\theta^{i+j} \left\{ \bar{\chi}(k/2 - i\partial_\theta) \chi(k/2 + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \right\} \right]_{i,j < n}. \quad (30)$$

It is the Gram of the **same** multiplication map M_χ . This is an explicit finite expression in gamma constants, $\cos \theta$, $\tan \theta$, and the original coefficients of χ .

Define positive scalar corrections, including the empty-index value one,

$$X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_n = \mathfrak{B}_n / \mathfrak{B}_n^\Gamma.$$

The source-boundary determinant theorem now factors as

$$\boxed{V_N = V_N^\Gamma T_N, \quad V_N^\Gamma = \frac{\mathfrak{D}_{N+1}^\Gamma}{\mathfrak{B}_{N-q+1}^\Gamma}, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}.} \quad (31)$$

This retains the entire arithmetic correction in both numerator and denominator. It is not an assertion that the arithmetic quotient metric equals the reference metric.

S19.7.1 7.1 The representative map between the two metrics

Use the same finite polynomial source. Let B_χ be the injective matrix of its original relation columns, and let R_N^Γ be the least-norm lift for the gamma observation. Let M_h be the actual arithmetic source Gram. Then the actual lift is

$$\boxed{R_N = R_N^\Gamma - B_\chi (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_N^\Gamma.} \quad (32)$$

For an empty boundary block the correction is the zero map. The proof is direct: the correction is an old relation, so the quotient remains the identity; multiplication by $B_\chi^* M_h$ gives zero. These two properties characterize the arithmetic least-norm lift.

Applying $\mathcal{V}_{h,k}$ to (32) supplies the original theta primitive of the difference. In the reconstructed support carrier, the difference maps to the supported zero in its fibre. It does not disappear from the source metric or from its boundary primitive. The comparison keeps the complete unit in (3).

S19.7.2 7.2 A genuine source-side upper bound

From (21), for every admitted polynomial,

$$\|P\|_h^2 \leq C_h^k \|P\|_\Gamma^2.$$

Evaluate this inequality on $R_N^\Gamma u$ and then minimize on the arithmetic side. This proves

$$\boxed{G_N \preceq C_h^k G_N^\Gamma, \quad 0 < T_N \leq C_h^{kq}.} \quad (33)$$

The second assertion is the determinant inequality in dimension q . It has the quotient dimension, rather than the full polynomial-source dimension, in its exponent. The corresponding source estimate is $0 < X_n \leq C_h^{kn}$, and similarly $0 < Y_n \leq C_h^{kn}$.

These are one-sided estimates. They do not bound the consecutive ratio T_{N-1}/T_{N+1} , which is the quantity required by the Toda upper bound. Its exact map to that estimate is the following.

S19.8 8. The current upper-control expression, with all reference costs exposed

Define

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

At the original observation $\theta = 0$, (29)–(31) give

$$\boxed{a_{N+1} = (N+1)(N+\alpha_k)Q_N, \quad \mathcal{R}_N = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}.} \quad (34)$$

The preceding exact phase identity is also retained through

$$\ell'_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N.$$

Thus its finite upper bound becomes exactly

$$\epsilon_{h,k,N} \leq \sqrt{(N+1)(N+\alpha_k)Q_N} \frac{\mathcal{R}_N^\Gamma T_{N-1}/T_{N+1} - 1}{2\sqrt{\mathcal{R}_N^\Gamma T_{N-1}/T_{N+1}}}. \quad (35)$$

The nonnegative numerator uses the original monotonicity of the actual quotient minimum, not a presumed monotonicity of T_N . When this inequality is coarse, the inherited equality with the volume-imbalance square and the phase square remains the sharper calculation.

For the full quartet one can keep $\lambda = 1/4$, so $\alpha_k = k/2$. The gamma source recurrence is now explicit at every degree; the remaining arithmetic dependence is in Q_N and the relative source-boundary correction T_{N-1}/T_{N+1} . The reference relation term still contains the full spectral annihilator χ . It is not bounded by pretending its roots are on the integration line.

The exterior theorem remains

$$L_{h,k} \leq \epsilon_{h,k,N}, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

Neither (33) nor (35) proves a subcubic bound. Equation (35) identifies exactly which degree-to-degree arithmetic corrections must be controlled next. All entries of those finite determinants can now be generated by the all- k coefficient transform (25), with the source-boundary comparison map (32) retained.

S19.9 9. A proved all-degree tail estimate for the actual arithmetic moments

The multiplier bound gives more than a change of coordinates. For $a, b > 0$ with $a + b < \pi/2$, every integer $r \geq 0$ and $T \geq 0$,

$$\int_{|u|>T} |u|^r m_{h,k}(u) du \leq 2C_h^k c_\lambda^k r! a^{-r} e^{-bT} (\cos(a+b))^{-2k\lambda}. \quad (36)$$

Proof. For $x \geq 0$, $x^r \leq r! a^{-r} e^{ax}$. On the indicated tail, $1 \leq e^{b(x-T)}$. Apply (21) and integrate. Since the reference density is even,

$$\int e^{(a+b)|u|} r_{\lambda,k}(u) du \leq Z_{\lambda,k}^\Gamma(a+b) + Z_{\lambda,k}^\Gamma(-a-b).$$

Equation (9) gives (36). This estimate retains the original mass and the complete tensor cost. It is a theorem with specified constants, not a claim that a floating-point quadrature has validated them numerically.

For a source or relation moment whose integrand is a polynomial $P(u) = \sum_j p_j u^j$, sum $|p_j|$ times (36). In particular this controls every entry of a degree- N source and relation Gram, including the full $|\chi|^2$ factor. It is then combined with an independently certified compact-interval quadrature and the existing matrix inverse/error inequalities; it does not certify that quadrature by itself.

This is an explicit tail interface for the actual arithmetic input at all degrees and tensor powers. For the empty analytic seed, (23) gives a completely stated bound without locating any zeta zero. For a nonempty packet the compact constant in C_h must be enclosed using the actual analytic quotient.

S19.10 10. What has and has not been concluded

The constructed chain is

- original split-supported polynomial source
- full gamma product fibre with arithmetic amplitude
- sum density and retained relative component
- exact all-tensor arithmetic coefficients
- the same source–boundary Toda determinant ratio.

The finite coefficient projection, polynomial degree, all masses, all gamma factors, the arithmetic unit, and the original theta primitives are explicit. The same finite quotient is used throughout; an existing relation becomes e only under its original quotient map, while the source norm and all derivative data remain available before that map. No new scalar-zero adjunction is performed at a tensor or relation level.

There is now a proved bound for the full arithmetic sum density, explicit closed-form reference determinants at every tensor degree, an exact finite coefficient algorithm, and a rigorous moment-tail bound. There is not yet a uniform small bound for the consecutive arithmetic determinant correction. The upper-estimate problem is sharpened to that specific comparison rather than being claimed solved by gamma convolution or by positivity of individual source Grams.

S19.11 References and exact source roles

- Repository main at `fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f`, `tex/theta_gamma_reference.tex` in the 20260912-stieltjes continuation, TG.2–TG.14: original one-factor gamma reference, arithmetic amplitude, constants and polynomial norms. Only the stated TeX window was read in this review; no complete reader audit is claimed.
- PR #22 head `811210d24b80813a08972ca23f919db015383137`, `workbenches/tau-exterior-trace-formal/RESEARCH_NOTE.md`: current exterior and filtration formalization work, with its separate verification scope. Its reported in-progress status is not promoted to a completed certificate.
- Delivered `Tau_Toda_Volume_Control/NOTE.tex`: original source and relation determinants, quotient metric, phase derivative and finite upper inequality. Its supplied manifest was independently checked here.
- NIST Digital Library of Mathematical Functions, §§18.22 and 18.23: classical Meixner–Pollaczek recurrence and generating function. The formulas here retain the repository’s original $t/2$ scale and all factorials.
- Koelink, H. T., and Van der Jeugt, J. (1998). Convolutions for orthogonal polynomials from Lie and quantum algebra representations. *SIAM Journal on Mathematical Analysis*, 29(3), 794–822. arXiv:q-alg/9607010. Background for the classical addition/convolution mechanism, not a claim that this article treats the present arithmetic quotient.

The proofs in this note establish the stated specialized maps directly. The regression checker tests finite exact polynomial and matrix consequences; it does not replace the analytic proofs or establish the outstanding uniform estimate.

S20 Arithmetic endpoint bounds: complete original norm proof

Source role. Complete original delivered witness; historical statements retain their source revision.

Exact revision. 8fa97624a55c2f1baad54e465e4637c584329f24f6c198bb5d168a443811a280

Complete source. sources/web_arithmetic_endpoint_delivery/Tau_Arithmetic_Endpoint_Bounds/RESEARCH_NOTE.md

Source SHA-256. 8fa97624a55c2f1baad54e465e4637c584329f24f6c198bb5d168a443811a280

Source metadata: author (MetaInlines)

Owner-directed SplitZero analytic continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

A uniform polynomial-norm estimate; integration with the four-volume criterion

Source metadata: title (MetaInlines)

Arithmetic endpoint bounds on the original τ -base source

S20.1 1. Result and exact source boundary

The formalization session's endpoint criterion selects an actual member of the canonical arithmetic family. This continuation proves one of its previously unproved analytic inputs. For the **unchanged arithmetic measure**, a fixed finite packet h , and integers $n \geq k \geq 3$, there is a finite constant C_h , independent of n and k , such that

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n. \quad (1)$$

The norms are the original monic polynomial norms for $m_{h,k} = w_h^{*k}$; the total mass remains μ_h^k . No root-location hypothesis is used in this bound. The proof retains a compact-source mass, a lower convolution constant, and both exponential moments. These constants are defined explicitly below; they have not been numerically enclosed.

For the packet consisting exactly of a hypothetical nonreal off-line quartet, let

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (2)$$

The existing exterior theorem gives $L_{h,k} \leq \epsilon_{h,k,N}$ at every admitted degree. Combining (1) with the other session's endpoint theorem gives

$$\min_{q_k \leq N < 2q_k} \epsilon_{h,k,N} \leq C_h q_k \sinh\left(\frac{\mathcal{B}_{h,k}}{2q_k}\right), \quad \mathcal{B}_{h,k} = \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}}. \quad (3)$$

Consequently such a quartet forces

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh}\left(\frac{\delta k}{2C_h}\right), \quad \liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 2. \quad (4)$$

The upper bound for this four-volume quantity is not proved here. In particular, (1) does not establish RH. It completes the polynomial-norm input in (3) on a specific admissible window and leaves the source/relation-volume comparison with its exact maps. Section 9 expresses that remaining quantity as two positive block log-determinants of actual new relation layers.

The finite quotient, canonical source, residue action and rank-two control are inputs from the preceding analytic notes. The new proofs in Sections 3–7 use only their explicitly defined arithmetic measure. The endpoint theorem is the formalizer's written result, not attributed to this continuation. Its finite telescopes are reported kernel-checked; the analytic estimates below have written proofs, not new Lean certificates.

S20.2 2. The base, source, and both kinds of quotient

Retain the original marked coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (5)$$

Here $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$, $e_R = 0_R^\bullet$, $p_R(\tau) = 0_R$, and $p_R(r^\bullet) = r$. The target of $p_{\mathbb{Z}}$ is infinite. The construction remains over the original pointed base $\mathfrak{b}_\tau$; no new scalar zero is adjoined at a tensor degree or at a polynomial relation layer.

Keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $Q = \mathcal{B}/\Theta V$, and $D = -x\partial_x$. The source spaces and the Fourier transform remain those of the earlier theta construction. In particular,

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}\Theta\phi_* = g. \quad (6)$$

Let h be a fixed finite product of zeros of g , with the full order of each selected zero retained, and write $d = \deg h$. Define

$$v_h = g/h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (7)$$

The quotient v_h is entire, including at the selected centres. It is not the meromorphic expression evaluated before removing its actual removable factors. For $h = 1$, the same analytic definitions are meaningful, but the finite arithmetic packet is zero. Only the nonempty case is used for the finite Gram inverses below.

The source map is

$$\mathcal{V}_{h,k} : \mathbb{C}[S] \longrightarrow \mathcal{B}^{\widehat{\otimes} k}, \quad P \longmapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h. \quad (8)$$

The original cyclic module is $E_{h,k} = \mathbb{C}[S]/(\chi_{h,k})$. Its arithmetic inclusion keeps the Taylor unit:

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h), \quad A_k = \sum_i M_{s_i}. \quad (9)$$

The supplied source identities are

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi. \quad (10)$$

The actual norm of (8) is

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du. \quad (11)$$

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$, the monic norm is the quotient norm of the leading-coefficient map

$$0 \longrightarrow \mathcal{P}_{n-1} \longrightarrow \mathcal{P}_n \xrightarrow{\text{lc}_n} \mathbb{C} \longrightarrow 0, \quad \omega_{h,k,n} = \inf_{\text{lc}_n} \|\mathcal{V}_{h,k}P\|^2. \quad (12)$$

The arithmetic quotient is the other specified map from the same source:

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_\chi} E_{h,k} \longrightarrow 0, \quad q = \deg \chi, \quad (13)$$

for $n \geq q - 1$, with the preceding relation space zero at $n = q - 1$. Multiplication by the monic χ carries $\mathcal{P}_{n-q}/\mathcal{P}_{n-q-1}$ isomorphically to $\mathcal{P}_n/\mathcal{P}_{n-1}$, with leading coefficient unchanged. The map π_χ kills that very relation. Thus (12), (13), and multiplication by χ specify the relation between the degree-line norm and the arithmetic quotient metric. One norm is not substituted for the other.

At each admitted original support label λ , the lifted quotient is $(\lambda, P) \mapsto (\lambda, [P]_\chi)$. An original relation has image $(\lambda, 0)$; external absence has image τ . Its norm before this projection remains available through (11).

S20.3 3. A quantitative local arithmetic mass lemma

S20.3.1 3.1 Uniform polynomial growth on a fixed strip

There is a constant $C_0 \geq 1$ such that, for $|T| \geq 10$, the function

$$f_T(z) = \zeta(1/2 + iT + iz) \quad (14)$$

is holomorphic on and inside the ellipse

$$z = \frac{1}{2}(w + w^{-1}), \quad |w| = 8, \quad (15)$$

and has modulus at most $M_T = C_0(1 + |T|)^6$ there.

The ellipse has real and imaginary semiaxes $65/16$ and $63/16$. Hence its image in the s -plane lies in $-55/16 \leq \Re s \leq 71/16$ and $|\Im s - T| \leq 65/16$. The pole of ζ is outside this set when $|T| \geq 10$.

For $\Re s \geq 1/4$, use the identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (16)$$

It follows by summing the steps of $\lfloor x \rfloor$ for $\Re s > 1$ and then by analytic continuation for $\Re s > 0$, $s \neq 1$. Its integral is bounded by $1/\Re s$, giving $O(1 + |T|)$ on the indicated part of the ellipse. For $\Re s \leq 1/4$, the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s) \quad (17)$$

and the vertical-strip gamma estimate give $O((1 + |T|)^{3/2 - \Re s})$. On the stated strip this exponent is below 5. The common power 6 in (14) is therefore a valid upper bound with one fixed constant. The gamma estimate and functional equation are the classical inputs recorded in DLMF §§5.11 and 25.4; none assumes RH.

Every disc of radius one about a point of $[-1, 1]$ lies inside the ellipse. The maximum principle and Cauchy's derivative formula therefore also give

$$\sup_{x \in [-1, 1]} |f'_T(x)| \leq M_T. \quad (18)$$

S20.3.2 3.2 Propagate a nonzero value to the critical-line interval

Put

$$r_* = (3 + \sqrt{13})/2 < 4, \quad z_* = -3i/2 = \frac{1}{2}(-ir_* + (-ir_*)^{-1}), \quad c_* = \zeta(2)^{-1}. \quad (19)$$

The absolutely convergent reciprocal Dirichlet series gives $|1/\zeta(2 + iT)| \leq \zeta(2)$, so $|f_T(z_*)| \geq c_* > 0$. Let $m_T = \max_{[-1, 1]} |f_T|$.

Apply the maximum principle on $1 \leq |w| \leq 8$ to $f_T((w + w^{-1})/2)$. Equivalently subtract from its logarithmic modulus the harmonic function linear in $\log |w|$ with boundary values $\log m_T$ and $\log M_T$. Zeros are handled by the usual subharmonic maximum principle. At $w = -ir_*$ this gives

$$c_* \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}. \quad (20)$$

Here $0 < c_* < 1$ and $M_T \geq 1$. Consequently

$$m_T \geq c_*^3 M_T^{-2}. \quad (21)$$

This is the three-circles argument with every coordinate map and radius fixed. No polynomial approximation to ζ or assumption about its zeros is used.

Choose a point attaining m_T . By (18), on a one-sided subinterval of length $m_T/(2M_T) \leq 1/2$ inside $[-1, 1]$, the modulus is at least $m_T/2$. Thus

$$\int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq \frac{m_T^3}{8M_T} \geq \frac{c_*^9}{8C_0^7} (1 + |T|)^{-42}. \quad (22)$$

For $T \in [-10, 10]$, the integral in (22) is continuous and strictly positive: otherwise ζ would vanish on an interval and hence identically on its holomorphic domain. Its minimum on that compact interval is positive. Combining it with (22) proves the all-real statement

$$\boxed{\exists c_\zeta > 0 \quad \int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq c_\zeta (1 + |T|)^{-42} \quad (T \in \mathbb{R}).} \quad (23)$$

The exponent 42 is deliberately an explicit sufficient exponent, not an optimal local mean-square estimate. The proof is a classical analytic propagation argument specialized here to the needed arithmetic interval.

S20.4 4. A lower envelope for the actual convolution, after three factors

S20.4.1 4.1 Local mass of the unchanged g/h

The gamma estimate, with a positive lower constant valid also on compact sets, implies

$$|\Gamma(1/4 + it/2)|^2 \geq c_\Gamma (1 + |t|)^{-1/2} e^{-\pi|t|/2}. \quad (24)$$

Let $H_h = \prod_{\rho \in Z} (1 + |\rho - 1/2|)^{m_\rho}$, so $|h(1/2 + it)| \leq H_h (1 + |t|)^d$. The entire identity $g = hv_h$ makes this bound valid at selected centres as well: $|g| \leq H_h (1 + |t|)^d |v_h|$. The factors in (6), including $|s(s-1)|^2 = (t^2 + 1/4)^2$, then give a positive constant $c_h^{(0)}$ with

$$w_h(t) \geq c_h^{(0)} e^{-\alpha_0|t|} (1 + |t|)^{-2d} |\zeta(1/2 + it)|^2, \quad \alpha_0 = \pi/2. \quad (25)$$

Equation (25) is a weaker bound than the available factor $(1 + |t|)^{7/2-2d}$; it follows by an inequality, not by changing the density. On $[T-1, T+1]$, its exponential and polynomial weights are bounded below by $e^{-\alpha_0} e^{-\alpha_0|T|}$ and $2^{-2d} (1 + |T|)^{-2d}$, respectively. Hence (23) proves

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\alpha_0|T|} (1 + |T|)^{-B_h}, \quad B_h = 42 + 2d. \quad (26)$$

S20.4.2 4.2 The positive compact convolution factor

The function w_h is continuous, bounded, integrable, and positive outside a discrete set. Thus $m_{h,2} = w_h * w_h$ is continuous and strictly positive at every real point. Indeed its integrand is positive almost everywhere; continuity follows from translation continuity in L^1 and boundedness of the other factor. Define the **actual** compact quantities

$$b_h = \min_{|v| \leq 1} m_{h,2}(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0. \quad (27)$$

Both are determined by the original density; neither is assigned the value one. Restricting a nonnegative convolution integral gives

$$m_{h,3}(u) \geq b_h \int_{u-1}^{u+1} w_h(t) dt \geq c_h e^{-\alpha_0|u|} (1 + |u|)^{-B_h}, \quad c_h = b_h c_h^{(1)} > 0. \quad (28)$$

The omitted part is the nonnegative integral over $|v| > 1$ and remains part of $m_{h,3}$. We use a lower bound, not a new compactly supported density.

For $k \geq 3$, restrict the remaining $k-3$ variables to $[-1, 1]$ and use (28). Their sum has absolute value at most $k-3$. This gives the central analytic estimate

$$\boxed{m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(|u|+k-3)} (1 + |u| + k - 3)^{-B_h} \quad (k \geq 3, u \in \mathbb{R}).} \quad (29)$$

All mass and tensor factors are explicit. For $k = 3$ the factor ϑ_h^0 is the empty product. No statement for $k = 1, 2$ is inferred by using a negative exponent. The amplification below only needs $k \geq 3$.

S20.5 5. Upper moments and exact lower monic norms

Fix $0 < b < \pi/2$ and retain both one-factor Laplace values

$$M_h(\pm b) = \int e^{\pm bt} w_h(t) dt > 0. \quad (30)$$

They are finite by the original gamma decay. Fubini and positivity give

$$\int e^{b|u|} m_{h,k}(u) du \leq M_h(b)^k + M_h(-b)^k. \quad (31)$$

The elementary exponential-series inequality $x^j \leq j! b^{-j} e^{bx}$ for $x \geq 0$ yields

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k). \quad (32)$$

The trial polynomial is $(S - k/2)^j$, whose value on $S = k/2 + iu$ is $i^j u^j$. Its phase has modulus one and remains in the source map; the generator $A = M_S$ has not been shifted to define a different spectral problem.

For every leading coefficient of modulus one, the exact least squared norm of a degree- j polynomial on $[-L, L]$ with Lebesgue measure is

$$\mathfrak{l}_j(L) = \frac{2L^{2j+1}}{2j+1} \left(\frac{2^j (j!)^2}{(2j)!} \right)^2. \quad (33)$$

To check all constants, Rodrigues' formula gives the Legendre polynomial $P_j(x)$ with leading coefficient $(2j)!/(2^j (j!)^2)$ and squared norm $2/(2j+1)$. Its scaled monic version on $[-L, L]$ is the unique minimizer: subtract it from any competing monic polynomial, and the difference has lower degree and is orthogonal to it. For the source coordinate $S = k/2 + iu$, multiply this minimizer by the retained leading phase i^j . This proves (33) at the original type. See also DLMF §§18.3 and 18.5.

Restricting (11) to $[-L, L]$ and using (29) proves

$$\omega_{h,k,j} \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(L+k-3)} (1+L+k-3)^{-B_h} \mathfrak{l}_j(L) \quad (k \geq 3, L > 0). \quad (34)$$

The observation has the explicit isometric splitting into the inside and outside interval components. Addition recovers the full source function, and their squared norms add. In the split lift each zero component is the supported zero at its label. The outside component has not been declared absent.

S20.6 6. The norm-window estimate is proved

Set $j = n$ and $L = n$ in (34), and use $j = 2n$ in (32). The exact finite bound is

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq \left[\frac{(4n)! b^{-4n} (M_h(b)^k + M_h(-b)^k) e^{\alpha_0(n+k-3)} (1+n+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2n)}. \quad (35)$$

For $n \geq k \geq 3$, this is at most $C_h n$. Here is one explicit sufficient constant, with $M_* = \max\{1, M_h(b), M_h(-b)\}$:

$$C_h = 64b^{-2} \exp(\alpha_0) 2^{B_h/2} \exp\left(\frac{B_h}{2 \exp(1)}\right) \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}. \quad (36)$$

Proof of the uniform estimate: $\binom{2n}{n} \leq 4^n$ in (33) gives $\mathfrak{l}_n(n) \geq 2n^{2n+1}/((2n+1)4^n)$. Also $(4n)! \leq (4n)^{4n}$, $M_h(b)^k + M_h(-b)^k \leq 2M_*^k$, $n+k-3 \leq 2n$, and $1+n+k-3 \leq 2n$. After taking the $2n$ -th root the factorial and monic factors contribute at most $64n$. The mass factors contribute at most the square root

in (36); the exponential contributes at most $\exp(\alpha_0)$. Finally $n^{1/n} \leq \exp(1/\exp(1))$ gives $(2n)^{B_h/(2n)} \leq 2^{B_h/2} \exp(B_h/(2\exp(1)))$. This proves (1).

The constant is intentionally not optimized. Its dependence on the fixed packet is allowed; its independence of the polynomial and tensor degrees is what is needed. Defining it via positive compact minima does not claim to have provided a numerical interval enclosure for those minima.

For a general source window the same two inequalities give, without asymptotic notation,

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b))^k + M_h(-b)^k e^{\alpha_0(L+k-3)} (1+L+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(L)} \right]^{1/(2r)}. \quad (37)$$

S20.7 7. Compose the proved bound with the formalizer's endpoint theorem

For a nonempty cyclic packet $E_{h,k}$, let G_N be its original canonical quotient metric, $V_N = \det G_N > 0$, and $\epsilon_{h,k,N}$ the original rank-two control allowance. The new endpoint theorem supplied by PR #23 is

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \sinh \left(\frac{1}{2r} \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}} \right), \quad n \geq q. \quad (38)$$

Its proof multiplies the local Toda bounds, applies concavity of $\log \sinh x$, and uses the two exact telescopes. The zero-contraction case is handled before taking logarithms. This result is credited to the parallel formalization/research session. The proof and the distinction between checked finite components and written Jensen assembly are retained in its source.

Choose $n \geq \max(q, k)$ and $r = n$. Equations (1) and (38) yield the actual canonical selection estimate

$$\boxed{\min_{n \leq N < 2n} \epsilon_{h,k,N} \leq C_h n \sinh \left(\frac{1}{2n} \log \frac{V_{n-1} V_n}{V_{2n-1} V_{2n}} \right)}. \quad (39)$$

The chosen index is a member of the existing canonical family. The source representative, Taylor unit, kernel, action and trace are not chosen again after selecting this index.

For the packet consisting exactly of the quartet in (2), $q_k \geq k$ and $L_{h,k}/q_k \geq \delta k/2$. The earlier exterior bound holds at every admitted index, including the one selected by (39). Since $\sinh$ is increasing on $[0, \infty)$, (4) follows. Moreover $\operatorname{arsinh}(ak) = \log k + O_a(1)$ for every fixed $a > 0$, which proves the stated threshold 2.

A proved upper bound with $\limsup \mathcal{B}_{h,k}/(q_k \log k) < 2$ would contradict (4). Such an upper bound is **not** an output of this note. The previous sufficient target $\mathcal{B}_{h,k} = o(q_k \log k)$ remains sufficient, but (4) specifies the larger quantitative threshold against which a future upper estimate can be compared. No repeated cost-free exterior operation is used: the multiplicity correction in the new formalization is retained.

S20.8 8. The remaining four volumes retain their finite jet maps

The previous source/relation determinant identity remains

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (40)$$

In the parallel full-jet transfer coordinates (PR #24), let $a = N - q + 1$, let F_a be its $2q$ by $2q$ raw confluent-jet matrix, and let V be its raw-derivative Vandermonde matrix, with every factorial retained. Its established identity is

$$V_N = \left(\prod_{j=a}^{a+q-1} \omega_{h,k,j} \right) \frac{\det V}{\det F_a}. \quad (41)$$

For the window $n = r = q$, the four transfer indices are exactly $0, 1, q, q+1$. Put $\Omega_j = \prod_{i=j}^{j+q-1} \omega_{h,k,i}$. Then

$$\boxed{\exp \mathcal{B}_{h,k} = \frac{\Omega_0 \Omega_1}{\Omega_q \Omega_{q+1}} \frac{\det F_q \det F_{q+1}}{\det F_0 \det F_1}}. \quad (42)$$

The common Vandermonde cancels in this explicit ratio, not inside any underlying map. The transfer matrices and their inverses still need their actual arithmetic control; the fixed width alone gives no such bound.

The empty packet has $q = 0$, $V_N = 1$ as an empty determinant, and no positive-rank inverse or quartet estimate. The analytic inequalities (23)–(37) still apply to its theta seed, but are not statements about a nonexistent finite packet.

S20.9 9. The volume window is also two explicit retained-boundary costs

There is a second useful representation of the same remaining quantity, with no reference-density change. Fix $i < j$, both admissible. In the original monic polynomial basis put

$$F_{i,j} = [b_{i+1}, \dots, b_j], \quad \Omega_{i,j} = \text{diag}(\omega_{i+1}, \dots, \omega_j), \quad b_a = [p_a]_\chi. \quad (43)$$

The kernel update is the actual sum $K_j = K_i + F_{i,j}\Omega_{i,j}^{-1}F_{i,j}^*$, where $K_i = G_i^{-1}$. The matrix determinant lemma gives

$$\boxed{\frac{V_i}{V_j} = \det(I_{j-i} + \Omega_{i,j}^{-1}F_{i,j}^*G_iF_{i,j})}. \quad (44)$$

This determinant has positive real value. Its factors are $1 + \lambda_a$, where $\lambda_a \geq 0$ are the generalized eigenvalues of the Hermitian pair $(F_{i,j}^*G_iF_{i,j}, \Omega_{i,j})$. No non-Hermitian matrix is asserted positive in an unspecified Euclidean metric.

Let $T_{i,j}$ have columns $\mathcal{V}_{h,kp_{i+1}}, \dots, \mathcal{V}_{h,kp_j}$. The original boundary-layer isomorphism is

$$\mathbf{b}_{i,j} = T_{i,j} - R_iF_{i,j} : \mathbb{C}^{j-i} \xrightarrow{\sim} \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathbf{b}_{i,j}^*\mathbf{b}_{i,j} = \Omega_{i,j} + F_{i,j}^*G_iF_{i,j}. \quad (45)$$

Its jets vanish; its highest polynomial coefficients give injectivity. Any new boundary can be reduced by these highest coefficients to an old boundary, and orthogonality then gives surjectivity. These statements prove the displayed map with the same original relation spaces, not only a dimension count. The representative update is

$$R_j = R_i + \mathbf{b}_{i,j}(\Omega_{i,j} + F_{i,j}^*G_iF_{i,j})^{-1}F_{i,j}^*G_i. \quad (46)$$

Every correction in (46) is an original theta boundary. Its split quotient image is e at the receiving fibre, and the input relation remains available in (45). Its cochain primitives are the same fixed-order divisions and tensor signs as in the previous cyclic source construction.

Apply (44) to $(i, j) = (q-1, 2q-1)$ and $(q, 2q)$. This proves

$$\boxed{\mathcal{B}_{h,k} = \sum_a \log(1 + \lambda_a^{(0)}) + \sum_a \log(1 + \lambda_a^{(1)})}. \quad (47)$$

In particular

$$\mathcal{B}_{h,k} \leq \text{Tr}(\Omega_{q-1,2q-1}^{-1}F_{q-1}^*G_{q-1}F_{q-1}) + \text{Tr}(\Omega_{q,2q}^{-1}F_q^*G_qF_q), \quad (48)$$

where each F has its own indicated source and target from (43). When this trace bound is too large, (47) retains the full logarithmic cost rather than replacing it by its linear upper bound. The analytic norm estimate in (1) does not bound (47); it supplies the other factor in (39).

S20.10 10. Deligne reading and what has actually been transferred

The six supplied S20 parts were reassembled in this turn. Their individual chunk hashes matched, and the assembled archive has 216,580,466 bytes and SHA-256 e2005bf31e1fc362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8. This verifies the supplied bytes, not every transcription claim in them.

I read the TeX for Definition 1.3.5, Lemmas 3.2.10–3.2.12, the tensor argument 3.2.13, and Corollary 3.3.6 in the nested source files. I compared those exact passages with the supplied top-level French/English frozen page records at printed pages 158, 203, and 206. No PDF text extraction or OCR was used. The source review

records a genuine inherited transcription fault: old `fr_seg2.tex` writes a weight bound ≤ 2 at 3.2.10, whereas the controlling page record says < 2 ; only the latter, with integrality, implies ≤ 1 . The old file also corrupts the explicit squared-eigenvalue step in 3.2.13. Those old files remain unmodified; they are not used as authority for these comparisons. No claim of a full independent audit of Weil II is made.

Deligne’s determinant/rank operation and the upper/dual-lower architecture remain attached to the programme’s specified exterior and residue maps. Here the additional estimate is a characteristic-zero analytic estimate for the actual source, (1), followed by the endpoint selection (39). This is not a statement that the new base satisfies Deligne’s finite-field hypotheses. The remaining volume estimate has not been inferred from purity, positivity of a Gram, or the mere existence of the support.

The new result is therefore a completed analytic component of the current route: its two endpoint monic norms have the required $O_h(q_k)$ scale on the actual quartet window. The other session’s four-volume theorem now leaves the explicitly specified relation-volume cost as the remaining quantitative input on that window.

S20.11 11. Proof and computation status

Sections 3–7 contain the new written analytic argument. Sections 8–9 contain explicit finite comparisons and a handoff for checking them using the existing quotient and matrix library. The standard complex-analysis, gamma and Legendre facts used in the argument are proved to the extent needed or referenced below. No general novelty claim is made for these classical ingredients.

The accompanying checker tests finite exact identities: coordinate factors, Legendre extremal norms, the local algebra in the interval argument, convolution moments on declared Laplace models, source quotients, block updates, raw confluent-jet endpoint factors, the volume telescope, and retained support cases. These tests do not certify the analytic inequalities by sampling them. There is no new Lean certificate and no interval-certified value of C_h .

This continuation makes no remote changes. The integration files are add-only, and the parallel formalization branches and the reader’s existing sources are left untouched.

S20.12 References and pinned inputs

- Deligne, P. *La conjecture de Weil. II*. Publications Mathématiques de l’IHÉS **52** (1980), 137–252. Supplied S20 source and current page-local records; the selected reading and discrepancies are itemized in SOURCE_REVIEW.md.
- NIST Digital Library of Mathematical Functions: §§25.4 (zeta functional equation), 5.11 (gamma estimates), 18.3 and 18.5 (Legendre polynomials). Public source pages consulted: <https://dlmf.nist.gov/25.4>, <https://dlmf.nist.gov/5.11>, <https://dlmf.nist.gov/18.3>, <https://dlmf.nist.gov/18.5>.
- Original programme: the delivered *Tau Gamma Convolution Descent*, *Tau Toda Volume Control*, *Tau Cyclic Sum Control*, and *Tau Exterior Trace Amplification* notes. Their definitions of the source, action, unit and canonical quotient are retained; this is not a complete replay of all their upstream analytic claims.
- Parallel endpoint criterion: PR #23, `c720f40530eed2f5969dbabe94dfd3fddc0f507f`, `workbenches/ta u-toda-volume-formal/ENDPOINT_CRITERION.md`.
- Parallel confluent transfer: PR #24, `dfcbb5cbf7fec8c9301fe242c13e741d762013e`, `workbenches/ta u-confluent-transfer/RESEARCH_NOTE.md`.
- Inspected main: `b32ca2128e0deb0eec6eafb860776a5d6a28dcbb`, after its merge of the original-Gram exterior/trace contribution.

S21 Complete supplied endpoint window product continuation

Source role. Complete original mathematical paste, with every formula and calculation.

Exact revision. f34647902fd4f6a913b5adae421c3df20bb1773902a797ec934750b02ed90ddb

Complete source. sources/endpoint_user_sources/ARITHMETIC_ENDPOINT_USER_SOURCE_1_20260913.txt

Source SHA-256. f34647902fd4f6a913b5adae421c3df20bb1773902a797ec934750b02ed90ddb

The attached four-endpoint criterion can be sharpened using its own exact radius identity. The improvement comes from multiplying the consecutive-step identities *before* discarding their imbalance terms. This preserves the fact that each interior volume contraction occurs in two adjacent control estimates.

The resulting four-endpoint bound is never weaker than the attachment's bound. It also has a simpler corollary:

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} \right)^{1/(2r)}. \quad (1)$$

This corollary uses only two quotient volumes and two monic norms. It is useful for controlling large contractions, although it need not outperform the original bound when the contractions are small.

These are deductions below from the supplied exact-radius, positivity, and monotonicity hypotheses—not an assertion that the remaining arithmetic endpoint estimates have been established. The attachment itself distinguishes that analytic obligation from the existing finite formalizations.

S21.0.1 1. An exact product identity for a whole degree window

Fix the original packet h , tensor degree k , and cyclic dimension $q \geq 1$. Work at the original observation $\theta = 0$, or at a fixed admissible tilt for which the supplied identities hold. Suppress h, k, θ from the notation.

Retain

$$V_N = \det G_N > 0, \quad \omega_N > 0, \quad \delta_N = \frac{V_N}{V_{N-1}} \in (0, 1],$$

and write

$$\phi_N = \sigma - \partial_\theta \log V_N$$

for the original phase. The source identity is

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1), \quad N \geq q. \quad (2)$$

The phase is therefore retained on the left, not set to zero.

Take a consecutive window

$$N = n, \dots, n + r - 1, \quad n \geq q, \quad r \geq 1,$$

and define

$$d_j = \delta_{n+j} \quad (0 \leq j \leq r), \quad \Omega_{n,r} = \frac{\omega_{n+r}}{\omega_n}.$$

Multiplication of (2) gives

$$\begin{aligned} \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) &= \Omega_{n,r} \prod_{j=0}^{r-1} \frac{(1 - d_j)(1 - d_{j+1})}{d_{j+1}} \\ &= \Omega_{n,r} \frac{(1 - d_0)(1 - d_r)}{\prod_{j=1}^r d_j} \prod_{j=1}^{r-1} (1 - d_j)^2. \end{aligned}$$

Since

$$\prod_{j=1}^r d_j = \frac{V_{n+r}}{V_n},$$

the exact window formula is

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (3)$$

The squared interior factors are essential: each interior contraction belongs to two consecutive radius identities. The first and last contractions occur only once.

Put

$$E_{n,r} = \min_{0 \leq j < r} \epsilon_{n+j}.$$

Then

$$E_{n,r}^{2r} \leq \prod_{j=0}^{r-1} \epsilon_{n+j}^2 \leq \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2).$$

All factors $1 - d_j$ lie in $[0, 1)$. Dropping them from the right of (3) proves (1).

This proof does not require a supremum over recurrence coefficients, an estimate for every intermediate inverse matrix, or a choice of a different Gram.

S21.0.2 2. The sharper bound from the same four endpoint volumes

For $r \geq 2$, define the three endpoint ratios

$$\alpha = \frac{V_n}{V_{n-1}}, \quad \beta = \frac{V_{n+r}}{V_{n+r-1}}, \quad t = \frac{V_{n+r-1}}{V_n}.$$

Thus

$$d_0 = \alpha, \quad d_r = \beta, \quad \prod_{j=1}^{r-1} d_j = t,$$

with $0 < \alpha, \beta, t \leq 1$. Also retain the attachment's norm factor

$$\mathcal{A}_{n,r} = \left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)}.$$

The sharpened theorem is

$$E_{n,r} \leq \mathcal{U}_{n,r}, \quad (4)$$

where

$$\mathcal{U}_{n,r} = \mathcal{A}_{n,r} \left(\frac{(1-\alpha)(1-\beta)}{\beta t} \right)^{1/(2r)} \left(1 - t^{1/(r-1)} \right)^{(r-1)/r}. \quad (5)$$

Only

$$V_{n-1}, \quad V_n, \quad V_{n+r-1}, \quad V_{n+r}, \quad \omega_n, \quad \omega_{n+r}$$

are needed—the same endpoint data used in the attachment. Its existing criterion combines the four volumes into one logarithmic budget; (5) retains their separate contributions.

Proof Let $s = r - 1$. For positive contractions $d_1, \dots, d_s \leq 1$ with product t ,

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (6)$$

If a $d_j = 1$, the left side is zero, so the inequality is immediate. Otherwise set $v_j = -\log d_j > 0$. The function

$$f(v) = \log(1 - e^{-v})$$

has

$$f''(v) = -\frac{e^{-v}}{(1 - e^{-v})^2} < 0.$$

Jensen's inequality gives

$$\sum_{j=1}^s \log(1 - e^{-v_j}) \leq s \log \left(1 - e^{-(v_1 + \dots + v_s)/s} \right).$$

Exponentiating and using $e^{-\sum v_j} = t$ proves (6). This is an application of the ordinary concavity form of Jensen's inequality, not a new general inequality (National Institute of Standards and Technology [NIST], n.d.-b, §1.7(iv)). (DLMF)

Insert (6) into (3):

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) \leq \Omega_{n,r} \frac{(1 - \alpha)(1 - \beta)}{\beta t} (1 - t^{1/s})^{2s}.$$

Taking the nonnegative $2r$ -th root proves (4)–(5). $\square$

For $r = 1$, there is no interior product and no fractional exponent involving $r - 1$. The separate formula is simply

$$\epsilon_n \leq \sqrt{\frac{\omega_{n+1}}{\omega_n}} \sqrt{(1 - \alpha)(\beta^{-1} - 1)}. \quad (7)$$

It retains the exact contraction contribution at that single step.

Comparison with the attached criterion The attachment proves

$$E_{n,r} \leq \mathcal{A}_{n,r} \sinh \left(\frac{\mathcal{B}_{n,r}}{2r} \right), \quad \mathcal{B}_{n,r} = \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}.$$

Its proof first replaces each exact contraction product by a hyperbolic-sine square and then applies Jensen. The new endpoint expression satisfies

$$\mathcal{U}_{n,r} \leq \mathcal{A}_{n,r} \sinh \left(\frac{\mathcal{B}_{n,r}}{2r} \right). \quad (8)$$

To see this, fix the four endpoints. Equation (6) shows that the product of the exact local contraction terms is maximized when all interior ratios equal $t^{1/(r-1)}$. Apply the attachment's local upper inequality and its Jensen inequality to that maximizing equal sequence. Their endpoint budget is still exactly $\mathcal{B}_{n,r}$, so their upper bound dominates the maximum used in (5). Degenerate cases follow directly or by continuity.

A useful explicit comparison occurs for geometric volume contraction:

$$V_{n+j} = V_{n-1} d^{j+1}, \quad d = e^{-v} \in (0, 1).$$

Then

$$\alpha = \beta = d, \quad t = d^{r-1},$$

and (5) becomes

$$\mathcal{U}_{n,r} = \mathcal{A}_{n,r} \frac{1-d}{\sqrt{d}} = 2\mathcal{A}_{n,r} \sinh(v/2). \quad (9)$$

The previous bound is $\mathcal{A}_{n,r} \sinh v$. Their ratio is

$$\frac{\text{previous bound}}{\mathcal{U}_{n,r}} = \cosh(v/2).$$

The improvement can therefore be substantial, rather than merely a rearrangement of the same estimate.

Exact scope of sharpness Equation (5) is sharp for the **relaxed scalar problem** consisting of positive norm sequences, nonincreasing positive volume sequences, and identity (2).

For strictly contracting endpoints, choose all interior $d_j = t^{1/(r-1)}$. Write

$$f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0.$$

Set

$$\frac{\omega_{n+j+1}}{\omega_{n+j}} = \frac{\mathcal{U}_{n,r}^2}{f_j}, \quad \phi_{n+j} = 0, \quad \epsilon_{n+j} = \mathcal{U}_{n,r}.$$

The product of the specified norm ratios is exactly $\Omega_{n,r}$, so the prescribed norm endpoints are respected and equality holds.

This is **not** an assertion that every such extremizing scalar sequence arises from the arithmetic convolution measure. The additional moment, recurrence, and interpolation constraints of that measure remain relevant.

Neither the phase nor the interior imbalance needs to be forgotten With all interior contractions strictly below one, define the nonnegative concavity loss

$$\mathcal{J}_{n,r} = (r-1) \log(1 - t^{1/(r-1)}) - \sum_{j=1}^{r-1} \log(1 - d_j).$$

Equation (3) becomes the exact identity

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \mathcal{U}_{n,r}^{2r} e^{-2\mathcal{J}_{n,r}}. \quad (10)$$

Thus the additional loss in passing to the endpoint bound is explicitly recorded.

For example, when the phases have been calculated, the sharper conclusion is

$$\prod_{j=0}^{r-1} (E_{n,r}^2 + \phi_{n+j}^2) \leq \mathcal{U}_{n,r}^{2r} e^{-2\mathcal{J}_{n,r}}. \quad (11)$$

The left side is an increasing polynomial in $E_{n,r}^2$. Its unique nonnegative threshold gives a phase-sensitive bound. No phase is presumed to vanish at a general auxiliary tilt.

S21.0.3 3. The simpler corollary reduces to a ratio of two determinant sequences

The quantity in (1) admits a cancellation that reduces the number of endpoint objects further. Define

$$\Lambda_N = \frac{\omega_N}{V_N}.$$

The supplied source formulas are

$$\omega_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{D}_N}, \quad V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}.$$

They use the original source and relation Gram determinants. The ordinary monic-norm and moment-determinant framework is also recorded in DLMF §18.2(iv), (ix); the arithmetic identification of the relation determinant is the supplied construction. (DLMF)

Consequently,

$$\Lambda_N = \frac{\mathfrak{B}_{N-q+1}}{\mathfrak{D}_N}. \quad (12)$$

The source determinant $\mathfrak{D}_{N+1}$ cancels literally. Equation (1) is therefore

$$E_{n,r} \leq \left(\frac{\Lambda_{n+r}}{\Lambda_n} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{n+r-q+1} \mathfrak{D}_n}{\mathfrak{D}_{n+r} \mathfrak{B}_{n-q+1}} \right)^{1/(2r)}. \quad (13)$$

This uses two source determinants and two relation determinants. It does not identify the relation space with its quotient value.

The mass bookkeeping remains exact. Rescaling the fixed source measure by $c > 0$ would give

$$\omega_N \mapsto c\omega_N, \quad V_N \mapsto c^q V_N, \quad \Lambda_N \mapsto c^{1-q} \Lambda_N.$$

Hence Λ_{n+r}/Λ_n is unchanged. No rescaling is performed; this calculation explains the cancellation while retaining the original mass.

Connection to the confluent-transfer interface The attachment supplies

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det V}{\det F_a}, \quad a = N - q + 1,$$

with the full confluent-jet conventions retained.

It follows that

$$\Lambda_N = \frac{\det F_a}{\det V \prod_{j=a}^{N-1} \omega_j}.$$

For $a = n - q + 1$, equation (13) becomes

$$E_{n,r} \leq \left[\frac{\det F_{a+r}}{\det F_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}. \quad (14)$$

At $q = 1$, the product is empty and equals one.

Thus the simpler criterion requires **two transfer determinants**, rather than four, together with the displayed norm bands. The common confluent Vandermonde factor cancels through an explicit ratio. No modulus is inserted to conceal a phase: positivity of the total expression follows from its equality with $\Lambda_{n+r}/\Lambda_n > 0$.

A quantitative target with an explicit threshold The supplied exterior result gives

$$L_{h,k} \leq \epsilon_{h,k,N}$$

at every admitted degree. For the packet consisting exactly of the hypothetical quartet,

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} \geq \frac{\delta}{2} k q_k.$$

The exact-quartet qualification is necessary.

Therefore

$$\boxed{L_{h,k}^{2r} \leq \frac{\Lambda_{n+r}}{\Lambda_n}.} \quad (15)$$

A sufficient asymptotic estimate is now

$$\boxed{\frac{1}{2r(k)} \log \frac{\Lambda_{n(k)+r(k)}}{\Lambda_{n(k)}} - \log(k q_k) \rightarrow -\infty.} \quad (16)$$

For a more concrete sufficient condition, suppose

$$\mathcal{A}_{n,r} \leq C_h q_k$$

and, for some fixed $0 < \eta < 2$,

$$\frac{1}{r} \log \frac{V_n}{V_{n+r}} \leq (2 - \eta) \log k.$$

Then (1) gives

$$E_{n,r} \leq C_h q_k k^{1-\eta/2} = o(k q_k). \quad (17)$$

These are still estimates to prove for the actual arithmetic family. The deduction specifies their required scale; it does not derive them from determinant positivity.

S21.0.4 4. Independent calculation on a canonical polynomial fixture

I checked the product identities and the numerical improvement using a declared Gaussian fixture—not an actual zeta-zero packet.

Take the measure

$$d\mu(u) = \frac{7}{\sqrt{2\pi}} e^{-u^2/2} du,$$

of literal mass seven, and the quotient

$$C = \mathbb{C}[u]/(u^2 + 1).$$

Its monic orthogonal polynomials satisfy

$$Q_0 = 1, \quad Q_1 = u, \quad Q_{j+1} = uQ_j - jQ_{j-1},$$

with

$$\omega_j = 7j!.$$

In the quotient basis $(1, u)$, let $b_j = [Q_j]$, and construct

$$K_N = \sum_{j=0}^N \frac{b_j b_j^T}{\omega_j}, \quad G_N = K_N^{-1}.$$

These are constrained-minimum Grams calculated from the source, not assigned matrices. The resulting values include

$$G_2 = \text{diag} \left(\frac{7}{3}, 7 \right),$$

$$G_3 = \text{diag} \left(\frac{7}{3}, \frac{21}{11} \right), \quad G_4 = \text{diag} \left(\frac{42}{43}, \frac{21}{11} \right).$$

Retain the coordinate generator

$$A = \frac{1}{2}I + iT, \quad T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

For $G = \text{diag}(g_0, g_1)$, direct calculation of

$$A^*G + GA - G = i(GT - T^T G)$$

gives

$$\epsilon^2 = \frac{(g_0 + g_1)^2}{g_0 g_1}.$$

Hence

$$(\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = \left(\frac{16}{3}, \frac{400}{99}, \frac{4225}{946} \right).$$

The five required volumes are

$$(V_1, V_2, V_3, V_4, V_5) = \left(49, \frac{49}{3}, \frac{49}{11}, \frac{882}{473}, \frac{980}{1333} \right).$$

For the window $n = 2, r = 3$,

$$\Omega_{2,3} = 60, \quad \alpha = \frac{1}{3}, \quad \beta = \frac{110}{279}, \quad t = \frac{54}{473}.$$

The comparison is

$$\begin{aligned} E_{2,3} &= 2.0100756305\dots, \\ (\epsilon_2 \epsilon_3 \epsilon_4)^{1/3} &= 2.1407199197\dots, \\ \mathcal{U}_{2,3} &= 2.1666731523\dots, \\ \text{attached endpoint bound} &= 2.5178436564\dots \end{aligned}$$

Thus

$$\boxed{2.0100756305 \leq 2.1407199197 \leq 2.1666731523 < 2.5178436564.} \quad (18)$$

A Wolfram kernel independently verified the rational Grams, all three exact radius identities, and symbolic versions of the window factorization for $r = 1, \dots, 8$. The general proof is the algebra above; these finite calculations are checks, not arithmetic interval certificates or additional Lean certificates.

S21.0.5 5. What remains attached to the selected degree

For the fixed arithmetic packet and window, let N_* be the first index attaining the finite minimum. Its representative remains the existing

$$R_{N_*} : C_{h,k} \longrightarrow \mathcal{B}^{\widehat{\otimes} k},$$

with

$$J^{(k)} R_{N_*} = \eta_{h,k}, \quad q^{(k)} R_{N_*} = \sigma_h^{\otimes k} \eta_{h,k}.$$

These are the supplied source and quotient maps. Selecting a degree does not alter their arithmetic unit, nilpotent jets, or cohomology classes.

At the corresponding support label, the lifted map is still

$$(\lambda_{N_*}, c) \longmapsto (\lambda_{N_*}, R_{N_*} c), \quad \tau \longmapsto \tau.$$

An original relation is sent by the quotient to the supported zero at that label. Its pre-quotient norm remains recorded by the relation determinant in (12)–(14).

There is also a computational qualification: **endpoint data certify the existence of a suitable degree; they do not generally identify which degree minimizes the allowance.** Locating N_* still requires evaluating candidate allowances or obtaining additional information about their distribution.

The completed continuation is the exact window product (3), its sharper four-endpoint consequence (5), and the reduced determinant/transfer criteria (13)–(14). No GitHub changes or new Lean verification were performed for these deductions. The remaining arithmetic task is now expressible as the specific growth estimate (16), with the full phase and interior-contraction loss available whenever the coarse endpoint observation is insufficient.

References

National Institute of Standards and Technology. (n.d.-a). General orthogonal polynomials (§18.2). In *NIST Digital Library of Mathematical Functions*. (DLMF)

National Institute of Standards and Technology. (n.d.-b). Inequalities (§1.7). In *NIST Digital Library of Mathematical Functions*. (DLMF)

S22 Complete supplied arithmetic norm continuation

Source role. Complete original mathematical paste, with every formula and calculation.

Exact revision. abdacca551cc7178170128c522beef8ac07d3948bf9ceea792177d0e0e8a21a8

Complete source. sources/endpoint_user_sources/ARITHMETIC_ENDPOINT_USER_SOURCE_2_20260913.txt

Source SHA-256. abdacca551cc7178170128c522beef8ac07d3948bf9ceea792177d0e0e8a21a8

One of the analytic estimates in the new endpoint criterion is now proved for the actual arithmetic measure. For a fixed packet h , retaining all its selected zero orders, I obtain

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n, \quad n \geq k \geq 3,$$

where $C_h < \infty$ depends on h , but **not** on the tensor degree k or polynomial degree n .

This establishes the $O_h(q_k)$ polynomial-norm input proposed in the formalization session's handoff, on the explicit window $q_k \leq N < 2q_k$. That handoff had correctly left this arithmetic estimate unproved.

Combining the bound with its endpoint-selection theorem gives a sharper statement about the remaining source-relation volumes: **an off-line quartet would force**

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 2.$$

The upper bound excluding that volume growth is not proved here. The norm factor in this particular criterion, however, is no longer an assumed estimate.

Full derivation in the HTML reader · LaTeX · Formalization handoff · Complete package

S22.0.1 1. The estimate concerns the original arithmetic source

I read the current main revision, which now includes the exterior/trace formalization, and the endpoint and confluent-transfer developments in PRs #23 and #24. Their source maps remain the inputs here; I have not replaced them with another positive metric.

The base remains $\mathfrak{b}_\tau$, carrying

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of $p\mathbb{Z}$ remains infinite. The maps preserve $e = 0^\bullet$ and external absence τ separately before the arithmetic observation.

Retain

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi.$$

For the fixed packet,

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad v_h = \frac{g}{h},$$

define the same density and convolution as before:

$$w_h(t) = \frac{|v_h(\frac{1}{2} + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \int_{\mathbb{R}} m_{h,k} = \mu_h^k.$$

The source map is

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h.$$

Its norm is exactly

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du.$$

The arithmetic observation still satisfies

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_{\chi},$$

where

$$\eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h).$$

Thus the full arithmetic Taylor unit stays in the map.

The norm $\omega_{h,k,n}$ is the infimum over **monic degree- n polynomials in this source**. It belongs to the leading-coefficient quotient

$$\mathcal{P}_n/\mathcal{P}_{n-1}.$$

The arithmetic quotient is connected through the other exact sequence

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_{\chi}} \mathbb{C}[S]/(\chi) \longrightarrow 0.$$

Multiplication by monic χ carries the leading-coefficient line at degree $n - q$ to the one at degree n . The arithmetic quotient sends that same relation to its supported zero. These maps connect the two norm calculations without identifying them. The original internal quotient retains labels and identifies precisely differences lying in the specified relation submodule.

S22.0.2 2. The new analytic input is a lower bound after three actual convolutions

The earlier gamma comparison supplied an upper envelope. For the endpoint norm ratio, I constructed a lower envelope for the **unchanged arithmetic density**.

The starting lemma is

$$\int_{T-1}^{T+1} |\zeta(\tfrac{1}{2} + it)|^2 dt \geq c_{\zeta}(1 + |T|)^{-42}, \quad T \in \mathbb{R},$$

for a fixed positive constant c_{ζ} . The exponent 42 is a sufficient explicit exponent; no optimal local mean-square estimate is claimed.

The local-mass proof For $|T| \geq 10$, put

$$f_T(z) = \zeta(\tfrac{1}{2} + iT + iz).$$

On the ellipse

$$z = \frac{1}{2}(w + w^{-1}), \quad |w| = 8,$$

the functional equation and the vertical-strip gamma estimate give

$$|f_T(z)| \leq M_T = C_0(1 + |T|)^6.$$

The relevant strip is fixed, and the zeta pole lies outside the ellipse. For the part with positive real component, the direct identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx$$

gives the needed polynomial growth; reflection handles the remaining part. These use the classical zeta functional equation and gamma estimates, without RH. (DLMF)

The absolutely convergent reciprocal Dirichlet series gives

$$|\zeta(2 + iT)| \geq \zeta(2)^{-1}.$$

Let

$$r_* = \frac{3 + \sqrt{13}}{2} < 4.$$

Then

$$\frac{1}{2} \left(-ir_* + \frac{1}{-ir_*} \right) = -\frac{3i}{2},$$

so f_T at this point is $\zeta(2 + iT)$.

Write

$$m_T = \max_{x \in [-1, 1]} |f_T(x)|.$$

The maximum principle on the annulus $1 \leq |w| \leq 8$, applied through the displayed coordinate map, gives

$$\zeta(2)^{-1} \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}.$$

Hence

$$m_T \geq \zeta(2)^{-3} M_T^{-2}.$$

Cauchy's estimate also gives

$$\sup_{[-1, 1]} |f'_T| \leq M_T.$$

Around a point attaining m_T , there is a one-sided interval of length $m_T/(2M_T)$ on which $|f_T| \geq m_T/2$. Therefore

$$\int_{T-1}^{T+1} |\zeta(\tfrac{1}{2} + it)|^2 dt \geq \frac{m_T^3}{8M_T} \geq \frac{\zeta(2)^{-9}}{8C_0^7} (1 + |T|)^{-42}.$$

On the remaining compact range of T , the integral is continuous and strictly positive. Its minimum supplies the remaining positive constant.

Carry the bound through g/h The original gamma factor and the elementary polynomial bound for h imply

$$w_h(t) \geq c_h^{(0)} e^{-\pi|t|/2} (1 + |t|)^{-2 \deg h} |\zeta(\tfrac{1}{2} + it)|^2.$$

This inequality is valid at selected centres through the entire identity $g = hv_h$; it does not divide by an unevaluated zero.

Consequently,

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\pi|T|/2} (1 + |T|)^{-B_h}, \quad B_h = 42 + 2 \deg h.$$

Now w_h is continuous, integrable, and positive outside a discrete set. Thus $w_h * w_h$ is continuous and strictly positive everywhere. Define the actual compact quantities

$$b_h = \min_{|v| \leq 1} (w_h * w_h)(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0.$$

Restricting a nonnegative convolution integral gives

$$m_{h,3}(u) \geq b_h \int_{u-1}^{u+1} w_h(t) dt.$$

For the remaining $k - 3$ factors, restrict their integration variables to $[-1, 1]$. Their sum has absolute value at most $k - 3$. This proves

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\frac{\pi}{2}(|u|+k-3)} (1 + |u| + k - 3)^{-B_h}, \quad k \geq 3.$$

The original mass has not been changed. The complementary integration regions remain part of $m_{h,k}$; this is a lower bound obtained from one positive contribution to that same density.

S22.0.3 3. This proves the required endpoint norm bound

Fix $0 < b < \pi/2$, and retain the two original Laplace values

$$M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bt} w_h(t) dt.$$

The exponential-series inequality and convolution identity give

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k).$$

The trial polynomial is $(S - k/2)^j$, whose value on the original integration line is $i^j u^j$. The operator remains multiplication by S .

For the lower estimate, the exact least squared norm of a monic polynomial of degree n on $[-L, L]$ is

$$\mathfrak{l}_n(L) = \frac{2L^{2n+1}}{2n+1} \left(\frac{2^n (n!)^2}{(2n)!} \right)^2.$$

This follows from the leading coefficient and orthogonality of the Legendre polynomial. Under $S = k/2 + iu$, the retained leading phase is i^n , whose squared modulus is one. (DLMF)

Applying the lower convolution envelope on $[-L, L]$ yields

$$\omega_{h,k,n} \geq c_h \vartheta_h^{k-3} e^{-\frac{\pi}{2}(L+k-3)} (1 + L + k - 3)^{-B_h} \mathfrak{l}_n(L).$$

Set $L = n$, and apply the upper estimate at degree $2n$. The exact finite bound is

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq \left[\frac{(4n)! b^{-4n} (M_h(b)^k + M_h(-b)^k) e^{\frac{\pi}{2}(n+k-3)} (1 + n + k - 3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2n)}.$$

For $n \geq k \geq 3$, this is at most $C_h n$.

The degree dependence follows directly from

$$\binom{2n}{n} \leq 4^n, \quad (4n)! \leq (4n)^{4n}, \quad n + k - 3 \leq 2n.$$

The factorial and monic-norm factors contribute $O(n)$ after taking the $2n$ -th root. The retained mass factors contribute bounded factors of the form $\exp(O_h(k/n))$. The polynomial lower-envelope factor contributes

$$(2n)^{B_h/(2n)},$$

which is uniformly bounded.

Equation (36) of the note gives a literal sufficient C_h , retaining all these constants. It is deliberately not optimized or presented as a numerically certified constant.

This establishes a bound on the actual arithmetic norm sequence, uniformly in the required tensor and polynomial degrees.

S22.0.4 4. The endpoint criterion now leaves a precise four-volume threshold

The formalization session's theorem is

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \sinh \left(\frac{1}{2r} \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}} \right).$$

It selects a member of the existing canonical family. Its proof retains both telescopes and the zero-contraction case; the report distinguishes its checked finite components from the written Jensen assembly.

Taking $r = n \geq \max(q, k)$ now gives

$$\min_{n \leq N < 2n} \epsilon_{h,k,N} \leq C_h n \sinh \left(\frac{1}{2n} \log \frac{V_{n-1} V_n}{V_{2n-1} V_{2n}} \right).$$

For the packet **consisting exactly of** an off-line quartet

$$\frac{1}{2} \pm \delta \pm i\gamma, \quad \delta, \gamma > 0,$$

with common multiplicity m , retain

$$q_k = [1 + k(m-1)](k+1)^2$$

and the existing aggregate defect

$$L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

Here $q_k \geq k$, and

$$\frac{L_{h,k}}{q_k} \geq \frac{\delta k}{2}.$$

The exterior theorem bounds $L_{h,k}$ by every admitted allowance, including the degree selected above.

Define

$$\mathcal{B}_{h,k} = \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}}.$$

Then

$$\frac{\delta k}{2} \leq C_h \sinh \left(\frac{\mathcal{B}_{h,k}}{2q_k} \right),$$

and therefore

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right).$$

Since h is fixed,

$$\operatorname{arsinh}\left(\frac{\delta k}{2C_h}\right) = \log k + O_h(1).$$

This proves the threshold

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 2.$$

The remaining upper estimate in this endpoint criterion is therefore sharply specified. The previously proposed

$$\mathcal{B}_{h,k} = o_h(q_k \log k)$$

would suffice; an upper limit strictly below 2 would already contradict the displayed lower threshold. **Neither upper estimate is being claimed here.**

No second cost-free exterior iteration is used. The formalization session's correction to that iteration's multiplicity cost remains intact.

S22.0.5 5. Those four volumes are two explicit costs of retained e -relations

I also expressed the remaining volume quantity directly through the original relation layers.

For admitted degrees $i < j$, put

$$F_{i,j} = [b_{i+1}, \dots, b_j], \quad \Omega_{i,j} = \operatorname{diag}(\omega_{i+1}, \dots, \omega_j),$$

where $b_a = [p_a]_\chi$ retains all quotient coordinates.

The existing full-jet kernel has the exact update

$$K_j = K_i + F_{i,j} \Omega_{i,j}^{-1} F_{i,j}^*, \quad K_i = G_i^{-1}.$$

Thus

$$\frac{V_i}{V_j} = \det(I + \Omega_{i,j}^{-1} F_{i,j}^* G_i F_{i,j}).$$

The source map behind this determinant is

$$\mathfrak{b}_{i,j} = T_{i,j} - R_i F_{i,j} : \mathbb{C}^{j-i} \xrightarrow{\sim} \mathcal{D}_j \cap \mathcal{D}_i^\perp,$$

where $T_{i,j}$ has the actual source columns $\mathcal{V}_{h,kp_{i+1}}, \dots, \mathcal{V}_{h,kp_j}$.

Its jets vanish, its highest polynomial coefficients give the inverse onto the new relation layer, and its Gram is

$$\mathfrak{b}_{i,j}^* \mathfrak{b}_{i,j} = \Omega_{i,j} + F_{i,j}^* G_i F_{i,j}.$$

The representative update remains

$$R_j = R_i + \mathfrak{b}_{i,j} (\Omega_{i,j} + F_{i,j}^* G_i F_{i,j})^{-1} F_{i,j}^* G_i.$$

Every correction is an original theta boundary. The complete transport is

$$\mathbb{C}^{j-i} \xrightarrow{\mathfrak{b}_{i,j}} \mathcal{D}_j / \mathcal{D}_i \hookrightarrow \mathcal{B}^{\widehat{\otimes} k} / \mathcal{D}_i \longrightarrow \mathcal{B}^{\widehat{\otimes} k} / \mathcal{D}_j.$$

Its final composite is zero. In the original split lift, it is the receiving fibre's **supported zero**, not τ .

Applying this to

$$(i, j) = (q_k - 1, 2q_k - 1), \quad (i, j) = (q_k, 2q_k)$$

gives

$$\mathcal{B}_{h,k} = \sum_a \log(1 + \lambda_a^{(0)}) + \sum_a \log(1 + \lambda_a^{(1)}),$$

where each $\lambda_a^{(\ell)} \geq 0$ is a generalized eigenvalue of the corresponding explicit pair

$$(F_{i,j}^* G_i F_{i,j}, \Omega_{i,j}).$$

This is the remaining volume cost in terms of the relations that become e -zero. Their source norms and cross-pairings are still present before that quotient.

The PR #24 transfer coordinates provide another exact calculation of the same four endpoints. Only its matrices at indices

$$0, \quad 1, \quad q_k, \quad q_k + 1$$

are needed, together with the specified products of ω_j . The common confluent Vandermonde cancels only in the displayed endpoint ratio; its raw-derivative factorials remain in the underlying maps. The note spells out that formula.

S22.0.6 6. The Deligne reading is now tied to this estimate

I reassembled the six supplied S20 parts and verified their chunk hashes and the full archive hash. I then read the TeX passages for Definition 1.3.5, Lemmas 3.2.10–3.2.12, the tensor step 3.2.13, and Corollary 3.3.6, comparing them with the archive’s current page-local French and English records.

There is a genuine inherited transcription trap: the older French TeX gives ≤ 2 in Lemma 3.2.10, while the current page record gives < 2 . The subsequent use of integral weights requires the strict inequality to conclude ≤ 1 . The older TeX also corrupts the explicit squared-eigenvalue step in 3.2.13. Both discrepancies are recorded; the old files were not overwritten.

The relevant operation remains the one identified in the other session’s reading: **an additional estimate before tensor amplification improves the weight bound**. The determinant/exterior map itself does not supply that estimate.

This continuation supplies a characteristic-zero estimate for the actual source—the norm-window bound above—and composes it with the endpoint criterion. It does not import finite-field purity hypotheses onto $\mathfrak{b}_\tau$.

Source reading and discrepancy record

Verification and handoff The new checker passed **18 exact finite test methods** normally and under python -0, with identical successful records; deliberately failing controls failed in both modes. All **37 entries** of the preceding gamma package’s manifest verified, and its **17-method suite** passed fresh normal and optimized runs.

The analytic theorem has a written proof. These finite checks are not a Lean certificate, a numerical enclosure of C_h , or a proof of the remaining volume estimate.

The [add-only integration patch](#) adds fifteen files and passed local application and byte-for-byte checks. No remote branch was changed.

The completed advance is the required $O_h(q_k)$ norm estimate on the actual quartet window. The remaining endpoint-volume problem is now a concrete two-block cost of the original supported relations, with a proved off-line threshold of $2q_k \log k$.

S23 PR 23: complete original endpoint and multiplicity criterion

Source role. Original pinned PR23 source; later results have their own complete proofs.

Exact revision. 61dd03044ad56b2baf3671fa217497b0f4749c1259f841cdc8491846abbd2e46

Complete source. sources/local_endpoint_continuations/endpoint_criterion_pr23_c720f405_20260913.md

Source SHA-256. 61dd03044ad56b2baf3671fa217497b0f4749c1259f841cdc8491846abbd2e46

S23.1 Sharper endpoint-only criterion and the scope of exterior iteration

This is a mathematical continuation of the supplied Toda and exterior notes. The general Jensen argument and exterior-multiplicity count below have written proofs; they are not additional Lean declarations. The norm-ratio product and overlapping-volume telescope are included in the finite formalization target list. No arithmetic asymptotic bound is claimed.

S23.1.1 1. Eliminate the supremum over recurrence coefficients

Retain a nonempty fixed packet and tensor degree k , cyclic dimension q , the actual positive monic norm sequence ω_N , the actual quotient volume $V_N = \det G_N$, and the source's exact control allowance ϵ_N . Choose consecutive admitted degrees $N = n, \dots, n+r-1$ with $n \geq q$ and integer $r \geq 1$. Put

$$B_{(n,r)} = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})),$$

$$A_{(n,r)} = (\omega_{(n+r)} / \omega_n)^{1/(2r)}.$$

These are four endpoint quotient volumes and two endpoint polynomial norms of the existing family. Since V is positive and nonincreasing, $B_{(n,r)} \geq 0$. The source's finite upper bound implies

$$\min_{(0 \leq j < r)} \epsilon_{(n+j)} \leq A_{(n,r)} \sinh(B_{(n,r)}/(2r)). \quad (E1)$$

Unlike the preceding maximum-recurrence criterion, E1 needs no pointwise upper bound for every recurrence coefficient in the window.

Complete proof For $N = n+j$ set

$$x_j = (1/2) \log(V_{(N-1)} / V_{(N+1)}) \geq 0,$$

$$A_j = \sqrt{\omega_{(N+1)} / \omega_N} > 0.$$

The existing local estimate is $\epsilon_{(n+j)} \leq A_j \sinh(x_j)$. If any $x_j = 0$, the nonnegative allowance at that step is zero, and E1 follows. Otherwise all $x_j > 0$. The function $f(x) = \log(\sinh x)$ has

$$f''(x) = -1/\sinh(x)^2 < 0.$$

Finite Jensen therefore gives

$$\text{product}_j \sinh(x_j) \leq \sinh((\sum_j x_j)/r)^r.$$

Let E be the minimum of the nonnegative allowances. Multiplication of the local bounds gives

$$E^r \leq \text{product}_j \epsilon_{(n+j)} \leq (\text{product}_j A_j) \text{product}_j \sinh(x_j).$$

The actual products telescope:

$$\text{product}_j A_j = (\omega_{(n+r)} / \omega_n)^{1/2},$$

$$2 \sum_j x_j = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})).$$

Taking the nonnegative r -th root proves E1. A finite minimum is attained, so E1 selects an actual original degree N inside the specified window. No favorable metric is selected independently, and its original theta representative, quotient and full jets remain attached.

The recurrence product is `VolumeWindow.norm_ratio_product`; the second telescope is `VolumeWindow.overlapping_log_telescope`, applied to $V(j)=V_{(n-1+j)}$. The Jensen and nonnegative-root assembly is this written proof rather than a claim about an unimplemented exterior/analytic declaration.

S23.1.2 2. Every input is an endpoint of the two original determinant sequences

The source's $\omega_j = D_{(j+1)}/D_j$ gives

$$A_{(n,r)} = [D_{(n+r+1)} D_{(n)} / (D_{(n+r)} D_{(n+1)})]^{1/(2r)}.$$

The four V -values in B are evaluated by $V_N = D_{(N+1)}/B_{(N-q+1)}$, with the source's literal relation determinant B . All indices are nonnegative because $n \geq q$. No negative-index determinant or inverse of a rank-deficient earlier kernel is used. The common mass factors cancel only in these displayed ratios; the source $\text{norm } \omega_0 = \mu_h^k$ has not been reset.

For the original quartet,

$$q_k = [1 + k(m-1)](k+1)^2, \\ L_{(h,k)} \geq \delta k * q_k / 2.$$

The exterior theorem gives $L_{(h,k)} \leq \epsilon_N$ at every admitted N . Thus E1 yields

$$L_{(h,k)} \leq A_{(n,r)} \sinh(B_{(n,r)}) / (2r). \quad (E2)$$

A proof that the right side is $o(k * q_k)$ along admitted $n=n(k)$, $r=r(k)$ would exclude that fixed off-line quartet. One sufficient set of estimates is

$$r(k) \geq c_h q_k, \quad A_{(n,r)} = O_h(q_k), \\ B_{(n,r)} = O_h(q_k \log k),$$

for $c_h > 0$. Indeed $B/(2r) = o(\log k)$, so $\sinh(B/(2r)) \leq \exp(o(\log k)) = k^{o(1)}$, and the ratio to $k * q_k$ tends to zero. These are candidate arithmetic estimates, not consequences of the formal telescope, positivity, the Toda equation, or Deligne's finite-field hypotheses.

The weighted endpoint criterion is sharp as an abstract sequence inequality: equality can occur when all x_j are equal and every $A_j \sinh(x_j)$ equals the same allowance. For arbitrary recurrence sequences it is strictly preferable to replacing $A_{(n,r)}$ by the largest A_j .

S23.1.3 3. A useful boundary on further exterior amplification

The supplied single exterior theorem is correct. It must not be reused with a rank-two hypothesis that no longer holds after exterior powering.

Suppose the original self-adjoint control H has spectrum

$$(+\epsilon, 0 \text{ repeated } q-2 \text{ times}, -\epsilon), \quad \epsilon > 0.$$

On the actual p -th exterior power, $1 \leq p \leq q$, the additive control has $+\epsilon$ and $-\epsilon$ each with multiplicity

$$r_p = \text{binomial}(q-2, p-1),$$

and zero with multiplicity $\text{binomial}(q-2, p) + \text{binomial}(q-2, p-2)$. This follows by the unscaled wedge basis: select the positive direction but not the negative, the negative but not the positive, or both/neither. In particular its operator norm is still ϵ , but its rank is $2r_p$, not generally two.

Apply an s -th exterior power to that whole space, whose dimension is $D_p = \text{binomial}(q, p)$. Its largest absolute additive-control eigenvalue is exactly

$$\epsilon * \min(s, r_p, D_p - s), \quad 0 \leq s \leq D_p.$$

Proof: select positive eigenvectors first, zero eigenvectors next, and negative eigenvectors only when the degree forces them. There are r_p available positive directions and r_p negative ones. This attains the stated piecewise maximum; reversing the choices attains its negative. At $s=D_p$ the sum is zero. For example $q=4, p=2$ gives spectrum $(+\epsilon, +\epsilon, 0, 0, -\epsilon, -\epsilon)$, and the next exterior square has allowance 2ϵ .

This does not weaken the original determinant-line amplification or remove its nilpotent arithmetic action. It identifies the precise cost of iterating the operation on the entire exterior space, and prevents a spurious second cost-free amplification. The source's chosen determinant line has its own one-dimensional action and must remain distinguished from the complete first exterior power.

S23.1.4 4. Collaboration boundary

The formal work carries the exact source/quotient determinant and the two loss terms into the original-Gram trace certificate. The analytic collaborator can now estimate the two endpoint quantities in E1, or retain the exact phase and imbalance when the coarse upper bound is too large. The six-part Deligne archive and uploaded ZIP test suites were not opened in this execution environment; the source statements used here came from the complete pasted notes. The current GitHub correction and original source maps are unchanged.

S24 The complete original endpoint and Gamma join

Source role. Complete original EG.1-5 mathematical witness.

Exact revision. ce4af7908adc2f8003b6fe7e7f62068a1ee125c45412048159fde67f2cfb1b50

Complete source. sources/local_endpoint_continuations/owner_endpoint_gamma_join/ENDPOINT_GAMMA_JOIN.md

Source SHA-256. ce4af7908adc2f8003b6fe7e7f62068a1ee125c45412048159fde67f2cfb1b50

S24.1 Endpoint degree selection in the actual gamma-corrected coordinates

This short calculation joins the complete written endpoint criterion in PR #23 (reported source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f) to the gamma convolution descent supplied on 13 September 2026. It uses the original arithmetic source, the original polynomial relation, and the same canonical quotient metric. It is a written substitution and finite-dependence proof, not an additional Lean theorem or a uniform arithmetic estimate.

S24.1.1 Objects and the estimate being evaluated

Fix the original packet h and tensor degree k . Put $\alpha_k = 2k$ and keep the source and reference Hankel determinants D_j and D_j^Γ of GD.28. Their positive quotient is $X_j = D_j / D_j^\Gamma$. For the original relation χ , of degree $q \geq 1$, let $Y_j = B_j / B_j^\Gamma$ and

$$T_N = \frac{X_{N+1}}{Y_{N-q+1}}, \quad V_N = \det G_N = V_N^\Gamma T_N, \quad N \geq q - 1.$$

Here T_N is a positive scalar correction ratio, not the coordinate operator also denoted T in the Toda calculation. Let $n \geq q$ and $r \geq 1$ be integers. The endpoint criterion uses the actual quantities

$$\mathcal{A}_{n,r} = \left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)}, \quad \mathcal{B}_{n,r} = \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}.$$

Its complete Jensen and telescoping proof gives

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \mathcal{A}_{n,r} \sinh \left(\frac{\mathcal{B}_{n,r}}{2r} \right). \quad (\text{EG.1})$$

No metric is selected here: the finite minimum is over the pre-existing canonical representatives at degrees n through $n+r-1$.

S24.1.2 Exact reference and arithmetic factors at the endpoints

At the original observation $\theta = 0$, GD.28–29 gives

$$\omega_j = c_\lambda^k j! (\alpha_k)_j \frac{X_{j+1}}{X_j}. \quad (\text{EG.2})$$

Consequently,

$$\mathcal{A}_{n,r} = \left((n+1)_r (n+\alpha_k)_r \right)^{1/(2r)} \left(\frac{X_{n+r+1} X_n}{X_{n+r} X_{n+1}} \right)^{1/(2r)}. \quad (\text{EG.3})$$

The mass c_λ^k is retained in EG.2 and cancels between the two specified norms. The arithmetic mass μ_h^k was never assigned the value one. Each rising factorial in EG.3 is the exact r -factor product, not a large-degree approximation.

Substitution of $V_N = V_N^\Gamma T_N$ gives a second exact identity:

$$\mathcal{B}_{n,r} = \log \frac{V_{n-1}^\Gamma V_n^\Gamma}{V_{n+r-1}^\Gamma V_{n+r}^\Gamma} + \log \frac{T_{n-1} T_n}{T_{n+r-1} T_{n+r}}. \quad (\text{EG.4})$$

The reference quotient volumes in this expression are formed with the same χ , using its two conjugate differential factors in GD.30. The roots, their orders and the centre $k/2$ have not been changed. All ratios are positive, so the real logarithms and the indicated positive roots exist.

Equations EG.3–4, inserted in EG.1, express the endpoint selection estimate through an explicit reference factor, four source corrections X , and four source-to-relation corrections T . A pointwise upper bound $T_N \leq C_h(kq)$ does not control the denominator in EG.4. Neither of these exact identities supplies that missing quantitative estimate.

S24.1.3 Exact finite input count for the whole endpoint expression

The maximum source degree occurring in EG.1–4 is $n+r$: ω_{n+r} is its highest monic polynomial norm, and V_{n+r} is its highest quotient metric. Each defining source or relation Gram entry is the integral of a polynomial of degree at most $2(n+r)$ against the original arithmetic sum measure. The gamma coefficient theorem therefore proves that the complete endpoint expression is determined by

$$c_{h,0}, c_{h,1}, \dots, c_{h,2(n+r)}, \quad k, \lambda, \chi. \quad (\text{EG.5})$$

The zeroth coefficient carries the original total mass. The higher coefficients may have either sign. The finite coefficient-power formula and the Schur complement give the actual endpoint values; an arbitrary positive moment matrix cannot be inserted in their place. EG.5 gives a finite route for each chosen window. It does not bound the conditioning of the matrices or permit truncating the growing input count uniformly.

S24.1.4 Source roles

- [PR #23 endpoint proof at its supplied revision](#): finite Jensen selection, four-volume telescope, two-norm telescope. The user supplied its complete proof, which was read and the telescopes checked. Current CI/source review is recorded separately.
- Tau Gamma Convolution Descent, GD.25 and GD.28–35: exact coefficient transform and source/reference determinant ratios. The full original note and independent finite/analytic review are retained in this edition.
- GJ.1–8 in the gamma chapter: the explicit analytic map from the same one-factor moment function to the coefficient series, and finite source/quotient derivative jets.

The new content here is their literal endpoint-coordinate substitution and the maximum-degree input count. No additional theorem about zeta-zero locations or a completed growing-window bound is claimed.

S25 PR 24: complete original confluent transfer proof

Source role. Original historical PR24 proof, retained separately from the correction.

Exact revision. d1046b75762e4248ded8f0b901c1ac12dc01065e4e684ca2d166b6a535425763

Complete source. sources/owner_endpoint_review/review_pr24/files/workbenches/tau-confluent-transfer/RESEARCH_NOTE.md

Source SHA-256. d1046b75762e4248ded8f0b901c1ac12dc01065e4e684ca2d166b6a535425763

S25.1 Confluent transfer for the original source-relation volume

S25.1.1 Scope and current-main integration

This add-only analytic continuation starts from main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f, after the merge of PR #21 and its workflow-trigger correction. The PR #22 mathematical note was read at 811210d24b80813a08972ca23f919db015383137; its strict verification was in progress at that reading. This contribution neither modifies that branch nor treats pending checks as a certificate.

The source is the delivered Toda-volume continuation, following the cyclic and exterior-trace notes. Its full LaTeX was read directly. The new result is a fixed-width, full-jet transfer for the SAME canonical source/relation quotient metric. It applies the classical Christoffel transformation; it is not a claim to a new general orthogonal-polynomial theorem or to RH.

Keep $G(\mathbb{Z}) \rightarrow \mathbb{Z}$, with its infinite target, and its coefficient square to $G(\mathbb{C}) \rightarrow \mathbb{C}$. The structural base remains b_τ . Every supported coefficient map below retains $e=0^\bullet$ and external τ separately. The original theta complex is $C_+ = [V \rightarrow \Theta \rightarrow B]$, with $D = -x \, d/dx$, $Q = B/\Theta \, V$ and $g=2 \, \xi$. No scalar, source function or measure is assigned a new normalization.

For a nonempty finite packet h of actual zeros with full orders, retain F_h with Mellin transform g/h . At tensor degree k , keep the actual cyclic sum module

$$\begin{aligned} C &= C[S]/(\chi), \quad q = \deg \chi \geq 1, \quad c = k/2, \quad A = M_S, \\ \eta[P] &= \epsilon_h^{(\text{tensor } k)} P(\sum_i M(s_i)), \\ \epsilon_h &= j_h(g/h), \\ \text{Vcal } P &= P(D_1 + \dots + D_k)(F_h^{(\text{tensor } k)}). \end{aligned}$$

The original equations are

$$\begin{aligned} J^{(k)} \text{Vcal} &= \eta \, \pi_\chi, \\ q^{(k)} \text{Vcal} &= \sigma_h^{(\text{tensor } k)} \eta \, \pi_\chi. \end{aligned}$$

Its norm is the actual integral against $m_k = w_h^{\{k\}}$, where

$$w_h(t) = |(g/h)(1/2 + i \, t)|^{2/(2 \, \pi)}, \quad \text{integral } m_k = \mu_h^k.$$

The auxiliary measure is $d\mu_\theta(u) = \exp(\theta u) m_k(u) du$, $|\theta| < \pi/2$. Its mass is $M_h(\theta)^k$. θ in this expression is the real tilt parameter, not the theta-summation map. The exponential-moment bound from the source makes all finite derivatives below valid. At tilt zero the metric is the original arithmetic metric.

For $h=1$ the finite arithmetic quotient is zero. Its determinant is the empty determinant 1; the analytic seed and its mass remain, and $G(0)$ still has e and τ . The positive-dimensional inverses below require $q \geq 1$.

S25.1.2 1. Exact coordinate and raw-jet maps

Use the ring isomorphism

$$\text{Psi}: C[S] \rightarrow C[u], \quad P \mapsto P(c + i \, u), \quad u \mapsto (S - c)/i.$$

Thus $d_u \text{Psi}(P) = i \, \text{Psi}(P')$, and multiplication by S remains $c + i \, M_u$. Define the original norm polynomial

$$\Pi(u) = \text{conj}(\chi)(c - i \, u) \, \chi(c + i \, u).$$

Its leading coefficient is literally $(-i)^q i^q = 1$. It is real and nonnegative on $\mathbb{R}$. Reflection gives $\chi^\dagger = (-1)^q \chi$, hence

$\psi(u) = i^{(-q)} \chi(c + i u)$ is real and monic,
 $\Pi = \psi^2$, $\Psi(\chi) = i^q \psi$.

The factor i^q remains in the source multiplication map. For distinct roots λ_a of χ , of complete lengths ℓ_a , put

$\zeta_a = (\lambda_a - c)/i$, $r_a = 2 \ell_a$, $r = \sum r_a = 2q$.

Define raw, NOT factorial-divided, derivative coordinates

$J_2 f = (f^{(d)}(\zeta_a))_{(a, 0 \leq d < r_a)}$.

Their kernel is (Π) . Their product uses the full binomial rule. Multiplication by u has matrix

$(X_{\Pi} v)_{(a,d)} = \zeta_a v_{(a,d)} + d v_{(a,d-1)}$, $v_{(a,-1)} = 0$.

All nilpotent orders and the coefficient d are retained.

Let Q_j be the real monic orthogonal source polynomials for $d\mu_\theta$, obtained by subtracting lower-degree projections, and write

$\omega_j = \int Q_j^2 d\mu_\theta > 0$,
 $a_j = \omega_j / \omega_{(j-1)}$, $a_0 = 0$,
 $b_j = \int u Q_j^2 d\mu_\theta / \omega_j$.

The source recurrence is

$u Q_j = Q_{(j+1)} + b_j Q_j + a_j Q_{(j-1)}$, $Q_{(-1)} = 0$.

No Q_j is divided by its norm. Set

$v_j = J_2 Q_j$,
 $F_n = [v_n, \dots, v_{(n+r-1)}]$,
 $V = [J_2 1, \dots, J_2 u^{(r-1)}]$.

Both are r by r . The raw confluent determinant is

$\det V = (\prod_{d=0}^{r_a-1} d!) \prod_{a < b} (\zeta_b - \zeta_a)^{(r_a - r_b)}$.

The fixed row order is part of this formula. $F_0 = V$ times the actual unit-triangular polynomial coefficient matrix; only their determinants are equal.

S25.1.3 2. Invertibility and the original relation polynomial

Invertibility proof. Suppose F lies in $\text{span}(Q_n, \dots, Q_{(n+r-1)})$ and $J_2 F = 0$. Then $F = \Pi H$ with $\deg H < n$. Orthogonality gives

$0 = \langle H, F \rangle = \int \Pi |H|^2 d\mu_\theta$.

The integrand is nonnegative and its weight is positive almost everywhere except finitely many points. Thus $H=0$, then $F=0$. For $n=0$ the degree argument suffices. Hence every F_n is invertible. This does NOT estimate its inverse uniformly.

Solve

$d_n = F_n^{(-1)} v_{(n+r)}$, $\vartheta_n = -(d_n)_0$.

The polynomial

$$Q_{\hat{n}} = (Q_{n+r} - \sum_{j=0}^{r-1} (d_n)_j Q_{n+j}) / \Pi$$

exists by exact raw-jet divisibility and is monic of degree n . Pairing its numerator with every polynomial of degree below n proves that $Q_{\hat{n}}$ is precisely the monic orthogonal polynomial for the ORIGINAL relation weight $\Pi \, d\mu_{\theta}$. Its norm is

$$\begin{aligned} \nu_n &= \int Q_{\hat{n}}^2 \Pi \, d\mu_{\theta} \\ &= -(d_n)_0 \, \omega_n = \omega_n \, \vartheta_n > 0. \end{aligned} \quad (1)$$

Indeed only the Q_n column pairs with the degree- n quotient; its coefficient is $-(d_n)_0$. This proves $\vartheta_n > 0$ with arbitrary nonreal and repeated nodes.

Let $K_n = [e_1, \dots, e_{r-1}, d_n]$. Then

$$\begin{aligned} F_{n+1} &= F_n K_n, \\ \det K_n &= (-1)^{r-1} (d_n)_0 = \vartheta_n. \end{aligned} \quad (2)$$

The sign uses the displayed even $r=2q$. On the same jet algebra,

$$\begin{aligned} T_n &= F_n^{(-1)} X_{\Pi} F_n, \\ T_{n+1} &= K_n^{(-1)} T_n K_n, \\ d_n &= (T_n - b_{n+r-1} I) e_{r-1} - a_{n+r-1} e_{r-2}. \end{aligned} \quad (3)$$

Thus the full operator and its jets have an explicit discrete transport, rather than only a determinant observation.

S25.1.4 3. Fixed-width determinant and quotient-volume formulas

Let D_n be the n by n source Hankel determinant, and B_n the corresponding determinant for $\Pi \, d\mu_{\theta}$. $D_0 = B_0 = 1$. Orthogonality gives

$$D_n = \prod_{j < n} \omega_j, \quad B_n = \prod_{j < n} \nu_j.$$

Multiplying (1) and (2) proves the complete confluent Christoffel identity

$$B_n / D_n = \prod_{j < n} \vartheta_j = \det F_n / \det V. \quad (4)$$

This is a proof from the source division and norm, not a distinct-root approximation. Raw derivative factors remain in $\det V$.

For $N \geq q-1$ set $n = N - q + 1$. The source sequence is still

$$0 \rightarrow P_{N-q} \xrightarrow{\chi} P_N \rightarrow C[S] / \chi \rightarrow 0.$$

Its canonical least-norm quotient Gram G_N satisfies the existing Schur identity $\det G_N = D_{N+1} / B_n$. Consequently

$$V_N = \det G_N = \left(\prod_{j=n}^{n+q-1} \omega_j \right) \det V / \det F_n. \quad (5)$$

The constant coefficient map $S = c + i \, u$ has determinant of modulus one on each remainder basis, so these are the original quotient determinants, with that coordinate map retained.

For $n \geq 1$,

$$\begin{aligned} \delta_N &= V_N / V_{N-1} = \omega_N / \nu_{N-1} \\ &= \omega_N / (\omega_{N-1} \, \vartheta_{N-1}), \end{aligned} \quad (6)$$

$$\begin{aligned} R_N &= V_{N-1} / V_{N+1} \\ &= \omega_{N-1} \omega_n \, \vartheta_{N-1} \vartheta_n \\ &\quad / (\omega_N \, \omega_{N+1}). \end{aligned} \quad (7)$$

The transfer width $2q$ does not grow with N . The source recurrence and high-degree polynomial evaluation are still required; no computational-complexity or conditioning claim is inferred from the width alone.

S25.1.5 4. One additional solve retains the full phase

Differentiating actual source orthogonality under $\exp(\theta u)$ gives

$$d_{\theta} Q_j = -a_j Q_{(j-1)}, \quad d_{\theta} \log \omega_j = b_j.$$

For the first identity, every coefficient below $j-1$ vanishes by differentiated orthogonality; pairing with $Q_{(j-1)}$ gives $-\omega_j$. Let L_n have entries $(L_n)(j-1,j) = -a(n+j)$ for $1 \leq j \leq r$, all other entries zero. Then

$$F_n' = F_n L_n - a_n v_{(n-1)} e_{\theta}^T. \quad (8)$$

At $n=0$ the last term is zero. Trace of L_n is zero. Hence

$$\begin{aligned} (\log \det F_n)' &= -a_n e_{\theta}^T F_n^{-1} v_{(n-1)}, \\ (\log V_N)' &= \sum_{(j=n)}^{(n+q-1)} b_j \\ &\quad + a_n e_{\theta}^T F_n^{-1} v_{(n-1)}. \end{aligned} \quad (9)$$

Retain $T = (A - cI)/i$ and $\sigma = \text{Tr } T$, which is real. The previous source cross term is exactly

$$(b_N * G_N b_{(N+1)}) / \omega_N = i(\sigma - (\log V_N)').$$

Here b_N in that formula denotes the remainder vector of the S -monic source polynomial, not the scalar recurrence b_j . The factor i and the distinction of types remain. The phase is $\phi_N = \sigma - (\log V_N)'$. At zero tilt for a full quartet, evenness gives $\phi_N = 0$; it is not set to zero at a general tilt.

S25.1.6 5. The positive gaps are norms of actual source vectors

Put

$$Z_j = \psi \hat{Q}_j - Q_{(j+q)} \text{ in } P_{(j+q-1)}.$$

The leading terms cancel. Source orthogonality therefore proves

$$\Delta_j := ||Z_j||^2 = \nu_j - \omega_{(j+q)} \geq 0. \quad (10)$$

The first term $\psi \hat{Q}_j$ is, through the explicit unit $i^{(-q)}$, an original χ relation. Its quotient is its supported zero. The difference Z_j instead has class $-[Q_{(j+q)}]$ in the original quotient. Its norm is not assigned to an absent element.

The SAME canonical rank-two control is $W_N = A * G_N + G_N A - k G_N$. Combining its previously proved radius identity with (6) gives, for $n \geq 1$,

$$\epsilon_N^2 = \Delta_{(n-1)} \Delta_n / (\nu_{(n-1)} \omega_N) - \phi_N^2. \quad (11)$$

Thus every factor in the upper bound

$$\epsilon_N \leq \sqrt{\Delta_{(n-1)} \Delta_n / (\nu_{(n-1)} \omega_N)}$$

comes from a specified positive source norm. Both gaps are computed by the transfer:

$$\begin{aligned} \Delta_{(n-1)} &= \omega_{(n-1)} \vartheta_{(n-1)} - \omega_N, \\ \Delta_n &= \omega_n \vartheta_n - \omega_{(N+1)}. \end{aligned}$$

The previous two-step bound is now

$$\epsilon_N \leq \sqrt{\omega_{(N+1)} / \omega_N} \cdot \frac{(R_N - 1)}{(2 \sqrt{R_N})}, \quad (12)$$

with R_N from (7). At the first degree $N = q-1$, use the separate expression

$$\epsilon_{(q-1)}^2 = (\nu_0 - \omega_q) / \omega_{(q-1)} - \phi_{(q-1)}^2.$$

No inverse at the previous rank-deficient degree is introduced.

S25.1.7 6. Keep the actual conormal quotient, not a repeated first ideal

The helper algebra for the norm computation is $C[u]/\Pi$, isomorphic through $S=c+iu$ to $C[S]/\chi_1^2$. The actual depth-two object is $C[S]/\chi_2$, defined using the literal pullback of the original I^2 . The merged conormal work gives $\chi_2 \mid \chi_1^2$. The actual maps are

$$C[S]/\chi_1^2 \twoheadrightarrow C[S]/\chi_2 \twoheadrightarrow C[S]/\chi_1. \quad (13)$$

Their kernels are respectively $(\chi_2)/(\chi_1^2)$ and $(\chi_1)/(\chi_2)$. In raw jets they truncate at the indicated full orders and intertwine multiplication by S . Differentiation into depth one also commutes, since $\chi_1 \mid \chi_2$. The coordinate relation $d_u \Psi = i \Psi d_S$ is retained.

For $h=(s-\rho)^2$ and tensor degree two, $X=S-2\rho$ gives

$$C[X]/X^6 \twoheadrightarrow C[X]/X^5 \twoheadrightarrow C[X]/X^3.$$

The first kernel is the class of X^5 . The retained depth-two class X^3 has derivative $3X^2$ at depth one. These maps use one original scalar support. Their split lifts send quotient-killed classes to e at the receiving label and fix external τ .

The full multiplier $U=\text{product}_i(g/h)(s_i)$ retains the checked equation $\delta(U\Psi)=U \delta(\Psi)+(\delta U)\Psi$. The second term is not discarded or presumed cyclic. The 2q-jet norm calculation is not counted as construction of new actual spectral multiplicities.

S25.1.8 7. Residual-controlled numerical interface

For an actual F and a specified candidate inverse Y , suppose a justified bound gives $\|I-YF\|_\infty \leq \eta < 1$. For candidate $\hat{d}$ and residual $r=v-F \hat{d}$, the exact Neumann argument proves

$$\|\hat{d}-d\|_\infty \leq \|Y\|_\infty \|r\|_\infty / (1-\eta). \quad (14)$$

This bounds the first coordinates used in (1) and (9). Arithmetic application requires certified input enclosures for the moments, source polynomials and zeros. Floating-point output alone is not a certificate.

Positive enclosures of ω and ϑ bound (7). Since $(R-1)^2/(4R)$ is increasing for $R \geq 1$, (12) yields a valid upper enclosure of ϵ^2 . The full phase can improve it through (11). Finite positivity/invertibility is not substituted for a uniform estimate of these residuals or inverse norms.

S25.1.9 8. Exact calibration and analytic target

For a declared Gaussian measure of literal mass seven and variance one, $\psi=u^2+1$, $q=2, n=2, N=3$, the calculation gives

```
(omega_0,...,omega_4)=(7,7,14,42,168),
vartheta_1=22, vartheta_2=86/3,
Qhat_1=u, Qhat_2=u^2-11/3,
Delta_1=112, Delta_2=700/3,
G_3=diag(7/3,21/11), epsilon_3^2=400/99.
```

An independent quotient-Gram calculation with $A=(1/2)I+i M_u$ gives the same result. The nonreal u -roots are algebraic fixtures, not zeros of ξ ; the nonzero allowance explicitly tests that the method does not manufacture purity. Repeated nodes, complex quartets, asymmetric atomic tilt derivatives and mass changes are covered by the expanded checks.

For the actual quartet retain

$$q_k=[1+k(m-1)](k+1)^2, \\ L_{-}(h,k)=2 \delta [1+k(m-1)](k+1) \lfloor (k+1)^2/4 \rfloor.$$

The source exterior/filtration result gives $L_{-}(h,k) \leq \epsilon_{-}(h,k,N)$, and $L_{-}(h,k)/(k q_k) \geq \delta/2$. Thus $\epsilon_{-} = o(k q_k)$, with fixed h and admitted $N \geq q_k$, is sufficient for the intended contradiction. The new representation attaches that exact target to TWO consecutive source-to-relation transfer coefficients and ONE phase solve. It has NOT proved that asymptotic bound.

No new Lean execution or arithmetic interval certificate is claimed. All new mathematical claims have the written proofs above at the retained source inputs. The original arithmetic function, unit, action, theta relations, residue pairing and trace remain unchanged.

S25.1.10 References and checks

Classical background: C. Krattenthaler (2021), ‘A determinant identity for moments of orthogonal polynomials that implies Uvarov’s formula for the orthogonal polynomials of rationally related densities’, arXiv:2103.03969; NIST DLMF Section 18.2. The proofs above supply the particular confluent and arithmetic interfaces rather than attributing them wholesale to those references.

The expanded conversation delivery includes the complete LaTeX/HTML reader and twenty exact finite regression methods; both normal and optimized runs passed with matching records, and both deliberate-failure controls failed. The preceding Toda archive’s 35 manifest entries verified and its 24-method checker passed fresh normal and optimized runs. These are stated verification scopes, not new Lean certificates. The public core checker in this directory independently tests the principal raw-jet, transfer, norm, determinant, volume and phase interfaces.

S26 Complete scalar volume comparison and arithmetic maps

Source role. Full VC.1-43 and VC.15a-b proof invoked by the AU continuation.

Exact revision. 47ac2b43904ec43f3ce19d7c74ba760a29ce4d7a3e411132fe81ec0b27af6104

Complete source. sources/local_endpoint_continuations/volume_comparator_review_20260913.md

Source SHA-256. 47ac2b43904ec43f3ce19d7c74ba760a29ce4d7a3e411132fe81ec0b27af6104

S26.1 An explicit scalar comparison for the original arithmetic quotient volumes

Date: 13 September 2026. This is the independent scalar comparison requested by the arithmetic volume upper-estimate lane. All inequalities below concern the original source and its original monic quotient. The displayed coordinate maps are invertible maps used in the proof; their Jacobians, phases, and the quotient units are retained.

S26.1.1 1. Original data and the finite answer

Let the fixed packet be a complete off-line quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$ and common order $m \geq 1$. Put

$$c = k/2, \quad \ell = 1 + k(m-1), \quad q = \ell(k+1)^2, \quad R_0 = \sqrt{\delta^2 + \gamma^2}, \quad k \geq 3. \quad (\text{VC.1})$$

Keep the given monic annihilator $\chi \in \mathbb{C}[S]$ of degree q , with all its original local lengths. Its roots in the chart $S = c + iu$ have absolute value at most $\rho = kR_0$. The source is

$$w_h(t) = \frac{|(2\xi/h)(1/2 + it)|^2}{2\pi}, \quad m_k = w_h^*, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad M_h(a) = \int_{\mathbb{R}} e^{at} w_h(t) dt. \quad (\text{VC.2})$$

The proved arithmetic density estimate supplies its actual constants $c_h > 0$, $\vartheta_h > 0$, $B_h = 42 + 2 \deg h$, and $\alpha = \pi/2$:

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B_h}. \quad (\text{VC.3})$$

Choose any $0 < b < \alpha$. These are the established analytic inputs of the supplied Arithmetic Endpoint Bounds note, rather than added estimates on its unknown coefficients. Define the following literal finite numbers:

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = \rho/T, \quad N = 2q - 1, \\ \lambda_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B_h}, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} (M_h(b)^k + M_h(-b)^k), \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r} \right)^q, \\ C_{k,q} &= \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{T \lambda_k(T)}. \end{aligned} \quad (\text{VC.4})$$

Here $q \geq k$, so $0 \leq r \leq 1/2$, and all denominators are strictly positive. The formula for U_k uses the usual $0! = 1$ convention, although the quartet has $q \geq 16$.

For every source degree $j \geq q-1$, let H_j be the positive definite Gram matrix in the original ordered monomials $1, S, \dots, S^j$, let J_j be the coefficient matrix of remainder modulo the original χ , and set

$$K_j = J_j H_j^{-1} J_j^*, \quad G_j = K_j^{-1}, \quad V_j = \det G_j. \quad (\text{VC.5})$$

The finite Hermitian comparison is

$$K_q \preceq K_{2q-1} \preceq C_{k,q} K_q, \quad 0 \leq \log \frac{V_q}{V_{2q-1}} \leq q \log C_{k,q}. \quad (\text{VC.6})$$

The proof below establishes every algebraic and comparison step in (VC.6). In particular it proves $C_{k,q} \geq 1$, since the matrices in (VC.6) are positive definite.

S26.1.2 2. Exact source, quotient, and raw-jet maps

For each finite degree define the complex-linear isomorphism

$$\Psi_T : \mathbb{C}[S]_{\leq j} \longrightarrow \mathbb{C}[x]_{\leq j}, \quad (\Psi_T P)(x) = P(c + iTx). \quad (\text{VC.7})$$

Its inverse is $f \mapsto f((S - c)/(iT))$. A change of real variable gives the equality of the original norms

$$\int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du = \int_{\mathbb{R}} |(\Psi_T P)(x)|^2 T m_k(Tx) dx. \quad (\text{VC.8})$$

Thus the target measure in this equality has mass exactly μ_h^k . There is no mass divisor in the map. Define

$$\chi_T(x) = (iT)^{-q} \chi(c + iTx). \quad (\text{VC.9})$$

Its leading coefficient is one, its roots are the original u -chart roots divided by T , with the same multiplicities, and their absolute values are at most r . For every polynomial Q ,

$$\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q. \quad (\text{VC.10})$$

The nonzero scalar $(iT)^q$ is the exact unit connecting the two displayed generators. Consequently (VC.7) induces a quotient isomorphism

$$\bar{\Psi}_T : \mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[x]/(\chi_T), \quad [P] \longmapsto [P(c + iTx)]. \quad (\text{VC.11})$$

Division is compatible with this map. Indeed, if $P = \chi Q + R$ with $\deg R < q$, then (VC.10) gives $\Psi_T P = (iT)^q \chi_T \Psi_T Q + \Psi_T R$, and $\deg \Psi_T R < q$; uniqueness of division proves

$$\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_{\chi} P. \quad (\text{VC.12})$$

If λ is a root in the original S chart and $z = (\lambda - c)/(iT)$, the raw derivative of order d obeys

$$(\Psi_T P)^{(d)}(z) = (iT)^d P^{(d)}(\lambda), \quad 0 \leq d < \text{ord}_{\lambda} \chi. \quad (\text{VC.13})$$

This follows by differentiating the affine composition d times. In local Taylor coordinates $P^{(d)}(\lambda)/d!$, the multiplier is still $(iT)^d$, while the factorial stays $d!$. The derivative phase has therefore been specified for every retained local length.

For precision at the matrix level, let $\mathbf{A}_j$ be the coefficient matrix of Ψ_T in increasing monomial degrees. Its columns are the coefficient vectors of $(c + iTx)^a$, $0 \leq a \leq j$; its diagonal is $(iT)^a$. Thus it is invertible and

$$\det \mathbf{A}_j = (iT)^{j(j+1)/2}. \quad (\text{VC.14})$$

Let tildes denote the target-coordinate Gram and remainder matrices. Equations (VC.8) and (VC.12) give

$$H_j = \mathbf{A}_j^* \tilde{H}_j \mathbf{A}_j, \quad \tilde{J}_j \mathbf{A}_j = \mathbf{A}_{q-1} J_j, \quad \tilde{K}_j = \mathbf{A}_{q-1} K_j \mathbf{A}_{q-1}^*. \quad (\text{VC.15})$$

The third equality follows by inserting the first two into $\tilde{J}_j \tilde{H}_j^{-1} \tilde{J}_j^*$. Hence $\tilde{G}_j = \mathbf{A}_{q-1}^{-*} G_j \mathbf{A}_{q-1}^{-1}$ and $\det \tilde{G}_j = T^{-q(q-1)} V_j$. The exact determinant factor is independent of j and cancels in the ratio in (VC.6). All subsequent norm estimates can therefore be pulled back to the original matrices by this proved congruence.

To state the same comparison in the original raw-jet coordinates, list the distinct roots of χ as λ_{ν} with full lengths d_{ν} , $\sum_{\nu} d_{\nu} = q$. In the fixed ordering of pairs (ν, a) , $0 \leq a < d_{\nu}$, define the square matrix

$$E_{(\nu,a),j} = \begin{cases} \frac{j!}{(j-a)!} \lambda_{\nu}^{j-a}, & j \geq a, \\ 0, & j < a, \end{cases} \quad 0 \leq j < q. \quad (\text{VC.15a})$$

Its action is the raw-derivative map on polynomials of degree below q . The exact Taylor formula shows that a polynomial has all these derivatives zero if and only if it is divisible by each $(S - \lambda_\nu)^{d_\nu}$, hence by their product χ . For degree below q , such a polynomial is zero. Thus E is injective, and equality of its domain and codomain dimensions proves that it is invertible. Every polynomial and its remainder differ by a multiple of χ , so their listed raw derivatives agree. Therefore the degree- j raw-jet matrix is exactly $F_j = EJ_j$, and

$$K_j^{\text{raw}} = F_j H_j^{-1} F_j^* = EK_j E^*, \quad G_j^{\text{raw}} = E^{-*} G_j E^{-1}, \quad \det G_j^{\text{raw}} = |\det E|^{-2} V_j. \quad (\text{VC.15b})$$

The multiplier $|\det E|^{-2}$ is independent of source degree. Consequently (VC.6) holds with the same constant in raw-jet coordinates and has the same determinant ratio. Equation (VC.13) gives the exact phase for each component when those raw jets are expressed in the affine chart. This is the direct typed relation between the remainder-kernel estimate and the raw-jet determinant calculation.

S26.1.3 3. The Legendre coefficient bound, proved in full

For a polynomial $f(x) = \sum a_j x^j$, write $\|f\|_{\text{coeff},1} = \sum |a_j|$. We prove

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1],dx)}, \quad \deg f \leq N. \quad (\text{VC.16})$$

Define

$$L_j(x) = \frac{1}{2^j j!} \frac{d^j}{dx^j} (x^2 - 1)^j. \quad (\text{VC.17})$$

The leading coefficient is $\kappa_j = (2j)!/(2^j(j!)^2)$. Repeated integration by parts, with every boundary term zero because $(x^2 - 1)^j$ has zeros of order j at both endpoints, proves that L_j is orthogonal to every polynomial of degree below j . The same integration by parts gives

$$\int_{-1}^1 L_j(x)^2 dx = \frac{\kappa_j}{2^j} \int_{-1}^1 (1 - x^2)^j dx = \frac{2}{2j+1}. \quad (\text{VC.18})$$

Here the intermediate integral is $2^{2j+1}(j!)^2/(2j+1)!$. To verify that expression without any special-function identity, substitute $x = 2t - 1$, obtaining $2 \cdot 4^j \int_0^1 t^j (1-t)^j dt$. Repeated integration by parts proves $\int_0^1 t^a (1-t)^b dt = a!b!/(a+b+1)!$ for nonnegative integers a, b : integration by parts with antiderivative $t^{a+1}/(a+1)$ reduces b by one and increases a by one, and the case $b = 0$ is $1/(a+1)$. This proves (VC.18).

Leibniz's rule applied to $(x-1)^j(x+1)^j$ proves the exact finite formula

$$L_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a. \quad (\text{VC.19})$$

The coefficient norm of a product is at most the product of its coefficient norms, by the triangle inequality for each convolution coefficient. The binomial theorem gives $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$. Therefore

$$\|L_j\|_{\text{coeff},1} \leq \sum_{a=0}^j \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j. \quad (\text{VC.20})$$

The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$; the last inequality bounds one coefficient by the sum of all positive coefficients. The degrees and nonzero leading coefficients show that $L_0, \dots, L_N$ are a basis. Write $f = \sum_{j=0}^N a_j L_j$. By orthogonality and Cauchy-Schwarz,

$$|a_j| = \frac{2j+1}{2} \left| \int_{-1}^1 L_j(x) f(x) dx \right| \leq \sqrt{\frac{2j+1}{2}} \|f\|_{L^2([-1,1])}. \quad (\text{VC.21})$$

This formula is valid for complex coefficients because L_j is real. Combining (VC.20)–(VC.21), bounding $2j+1$ by $2N+1$, and summing the geometric series $\sum_{j=0}^N 4^j = (4^{N+1} - 1)/3$ proves (VC.16).

S26.1.4 4. The complete monic-division bound

Write the roots of χ_T , with repetitions, as $z_1, \dots, z_q$, where $|z_a| \leq r < 1$. Define the elementary symmetric polynomials e_a and complete homogeneous polynomials h_j by the finite coefficient identities

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j. \quad (\text{VC.22})$$

The second identity is a formal power-series identity: h_j is the sum of all monomials $z_1^{a_1} \cdots z_q^{a_q}$ with non-negative a_i and $\sum a_i = j$. There are $\binom{q+j-1}{j}$ such monomials: placing $q-1$ separators among $j+q-1$ positions gives a bijection with the exponent lists. Consequently

$$|e_a| \leq \binom{q}{a} r^a, \quad |h_j| \leq \binom{q+j-1}{j} r^j. \quad (\text{VC.23})$$

For $0 \leq s \leq q-1$, set $Q_s(x) = \sum_{j=0}^s h_j x^{s-j}$. Multiplying the two series in (VC.22) shows that every coefficient of degrees $q+s-1, q+s-2, \dots, q$ in $\chi_T Q_s$ is zero, while its coefficient of degree $q+s$ is one. It follows that

$$\text{rem}_{\chi_T} x^{q+s} = x^{q+s} - \chi_T Q_s, \quad \deg(x^{q+s} - \chi_T Q_s) < q. \quad (\text{VC.24})$$

The leading coefficient of $\chi_T Q_s$ is exactly one, so removing its leading term reduces its coefficient norm by exactly one. All remaining terms are the negative of the remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &= \|\chi_T Q_s\|_{\text{coeff},1} - 1 \\ &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq \left(\frac{1+r}{1-r}\right)^q. \end{aligned} \quad (\text{VC.25})$$

For the last inequality, the convergent geometric-series product at real $r < 1$ gives $\sum_{j \geq 0} \binom{q+j-1}{j} r^j = (1-r)^{-q}$. For $0 \leq j < q$, the remainder of x^j is x^j and its coefficient norm is one, also at most $B_q(r)$. Linearity and the triangle inequality now give the exact finite operator estimate

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{VC.26})$$

Multiplicity was retained throughout the root list; no distinct-root interpolation or inverse Vandermonde estimate occurs here. If desired, the smaller literal finite constant

$$\max \left\{ 1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1 \right\} \quad (\text{VC.27})$$

can replace $B_q(r)$ in (VC.4), by the preceding proof.

S26.1.5 5. Original moment control and the remainder's source norm

For every nonnegative integer a and real u , the nonnegative Taylor terms of $e^{b|u|}$ imply

$$|u|^{2a} \leq (2a)! b^{-2a} e^{b|u|} \leq (2a)! b^{-2a} (e^{bu} + e^{-bu}). \quad (\text{VC.28})$$

Tonelli's theorem applied to the nonnegative product density and the identity $e^{b(t_1 + \cdots + t_k)} = \prod e^{bt_i}$ gives

$$\int e^{\pm bu} m_k(u) du = M_h(\pm b)^k, \quad \int m_k(u) du = \mu_h^k. \quad (\text{VC.29})$$

Consequently the actual $2a$ -th moment is at most the right side of (VC.28) integrated using (VC.29). If v has degree at most $q-1$, each monomial has absolute value at most $\max\{1, |x|^{q-1}\}$. This gives

$$\begin{aligned} \int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du &\leq \|v\|_{\text{coeff},1}^2 \int \max\{1, |u/T|^{2q-2}\} m_k(u) du \\ &\leq U_k(T) \|v\|_{\text{coeff},1}^2. \end{aligned} \quad (\text{VC.30})$$

The second inequality uses $\max\{1, t\} \leq 1 + t$ for $t \geq 0$, and the proved moment bound with $a = q - 1$. On the other hand, (VC.3) on $[-T, T]$ gives

$$\|P\|_{H_N}^2 \geq T \lambda_k(T) \|\Psi_T P\|_{L^2([-1,1])}^2. \quad (\text{VC.31})$$

Set $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (VC.8), (VC.12), (VC.16), (VC.26), (VC.30), and (VC.31), in that order, prove

$$\|\text{rem}_{\chi} P\|_{H_{q-1}}^2 \leq U_k(T) B_q(r)^2 A_N^2 \|\Psi_T P\|_{L^2([-1,1])}^2 \leq C_{k,q} \|P\|_{H_N}^2. \quad (\text{VC.32})$$

To specify the map's type, let $\iota_j : \mathbb{C}^{j+1} \rightarrow \mathcal{P}_j$ be $\iota_j(a) = \sum_{d=0}^j a_d S^d$, and let $\kappa : \mathbb{C}[S]/(\chi) \rightarrow \mathbb{C}^q$ send a class to the coefficient vector of its unique degree-below- q representative. Write $\mathcal{J}_j = J_j \iota_j^{-1} : \mathcal{P}_j \rightarrow \mathbb{C}^q$, and $s : \mathbb{C}^q \rightarrow \mathcal{P}_{q-1}$, $s(y) = \sum_{d=0}^{q-1} y_d S^d$. Then $\mathcal{J}_{q-1} s = I_{\mathbb{C}^q}$, and the original abstract quotient map π satisfies $\pi s = \kappa^{-1}$. The operator estimated in (VC.32) is exactly $s \mathcal{J}_N : \mathcal{P}_N \rightarrow \mathcal{P}_{q-1}$. Its restriction to $\mathcal{P}_{q-1}$ is the identity. In the formulas that follow, coefficient vectors and polynomials are identified only by the displayed ι_j .

S26.1.6 6. From the bounded remainder to the kernel inequality

The positive density in (VC.3) makes every polynomial Gram matrix positive definite. The remainder map J_j is surjective because its restriction to degrees below q is the identity. Thus K_j is positive definite. For completeness, define

$$R_j = H_j^{-1} J_j^* K_j^{-1}. \quad (\text{VC.33})$$

Then $J_j R_j = I$ by the definition of K_j . For $J_j z = 0$, $z^* H_j R_j = z^* J_j^* K_j^{-1} = 0$. Every lift of a quotient vector y is uniquely $R_j y + z$, $J_j z = 0$, so

$$\|R_j y + z\|_{H_j}^2 = y^* K_j^{-1} y + z^* H_j z. \quad (\text{VC.34})$$

The minimum lift norm is therefore exactly $y^* G_j y$. Degree inclusion enlarges the set of lifts and gives $G_{j+1} \preceq G_j$. Taking $P = R_N y$ in (VC.32), the degree- $(q-1)$ remainder is a lift of the same y , so

$$y^* G_q y \leq y^* G_{q-1} y \leq \|\text{rem}_{\chi} R_N y\|_{H_{q-1}}^2 \leq C_{k,q} y^* G_N y. \quad (\text{VC.35})$$

Together with $G_N \preceq G_q$, this gives $C_{k,q}^{-1} G_q \preceq G_N \preceq G_q$. Congruence by $G_q^{-1/2}$ bounds every eigenvalue of the middle matrix between $C_{k,q}^{-1}$ and one. Inversion reverses these positive eigenvalue inequalities; conjugating back proves the first inequality chain in (VC.6). Taking products of the same q eigenvalues proves its determinant statement. This proves the finite answer.

S26.1.7 7. The complete growth calculation and its exact gap

Let

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_0 = \max\{1, 2/(3c_h D)\}. \quad (\text{VC.36})$$

These are bounding constants for the unchanged source. Since $bT \geq 2q$, $(2q-2)!/(bT)^{2q-2} \leq 1$. Also $M_h(b)^k + M_h(-b)^k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that

$$U_k(T) \leq \frac{3}{2} (M_h(b)^k + M_h(-b)^k) \leq 3X^k. \quad (\text{VC.37})$$

The constants in (VC.4) further satisfy

$$A_{2q-1}^2 \leq \frac{2q}{9} 256^q, \quad \vartheta_h^{-(k-3)} \leq K^k. \quad (\text{VC.38})$$

On $[0, 1/2]$, the derivative of $\log((1+t)/(1-t))$ is $2/(1-t^2) \leq 8/3$. Integration from zero to r gives

$$\log B_q(r)^2 \leq \frac{16}{3} qr = \frac{16R_0}{3D} k. \quad (\text{VC.39})$$

Finally $1 + Dq + k - 3 \leq 1 + (D+1)q$, and $e^{-3\alpha} \leq 1$. Substituting (VC.37)–(VC.39) into the literal constant in (VC.4), and using $T = Dq$, yields

$$\begin{aligned} \log C_{k,q} &\leq (\alpha D + \log 256)q + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k \\ &\quad + B_h \log(1 + (D+1)q) + \log C_0. \end{aligned} \quad (\text{VC.40})$$

Thus an explicit upper bound on the requested original central volume is

$$\begin{aligned} 0 \leq \log \frac{V_q}{V_{2q-1}} &\leq (\alpha D + \log 256)q^2 \\ &\quad + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &\quad + B_h q \log(1 + (D+1)q) + q \log C_0. \end{aligned} \quad (\text{VC.41})$$

All constants in (VC.41) depend only on the fixed packet and the chosen b ; no constant depends on k or q . Since $q = [1 + k(m-1)](k+1)^2$, one has $k/q \rightarrow 0$ and $(\log q)/q \rightarrow 0$. In particular

$$\limsup_{k \rightarrow \infty} \frac{1}{q_k^2} \log \frac{V_{q_k}}{V_{2q_k-1}} \leq \alpha D + \log 256. \quad (\text{VC.42})$$

The proved central-window lower estimate in the companion factor-four note is

$$\log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (\text{VC.43})$$

The precise comparison between (VC.41) and (VC.43) is therefore an interval of admissible growth, not an RH contradiction. After division by $q \log k$, the leading term of the upper bound is $(\alpha D + \log 256)q / \log k$. This diverges like $(\alpha D + \log 256)k^2 / \log k$ when $m = 1$, and like $(\alpha D + \log 256)(m-1)k^3 / \log k$ when $m > 1$. The upper bound is larger than the lower scale by those explicit factors. This calculation locates exactly the loss in this scalar route: (VC.16) controls every coefficient of a degree- $(2q-1)$ polynomial, and (VC.31) replaces the full source by its least value on a length- $2Dq$ interval. The remainder step itself has only $O(k)$ logarithmic cost by (VC.39).

The sharper raw-jet matrix determinant comparison in the parent lane retains the directional dependence discarded by the scalar estimates (VC.16) and (VC.31). Equations (VC.7)–(VC.15) and (VC.33)–(VC.35) give the exact maps through which the present scalar estimate bounds that same original quotient. No claim about the sharper determinant's asymptotic size is made here.

S26.1.8 8. Provenance and verification boundary

The inherited analytic input is (VC.3) and the finiteness of $M_h(\pm b)$, proved in the complete supplied Tau_Arithmetic_Endpoint_Bounds/NOTE.tex. The central lower bound (VC.43) is proved in the owner's complete FOUR_VOLUME_THRESHOLD.md continuation. This note proves (VC.4)–(VC.42) directly from those established input theorems, with full finite calculations. It does not replace the arithmetic weight, discard the mass, erase a raw-jet phase, shorten a local length, postulate the desired upper estimate, or assert RH. Any finite symbolic test accompanying this note tests the algebraic maps and constants; the written argument proves the estimates for all the indicated degrees.

S27 Complete all-degree division and affine raw-jet estimates

Source role. Full RB.1-37 proof with all original derivative factors.

Exact revision. 56bee5eba5f951570428939169f69020a5c2b71c2ffc5ceaa18d26a2fe43049d

Complete source. sources/local_endpoint_continuations/volume_remainder_bound_review_20260913.md

Source SHA-256. 56bee5eba5f951570428939169f69020a5c2b71c2ffc5ceaa18d26a2fe43049d

S27.1 Exact finite remainder bound and affine raw-jet transport

This note proves the finite coefficient estimate requested for the volume-comparator calculation. All roots are counted with their original multiplicities. The degree, affine unit, complex phase, domains, and codomains remain explicit throughout.

S27.1.1 1. Polynomial spaces and the remainder map

Let $q \geq 1$ be an integer and write

$$\chi_x(x) = \prod_{a=1}^q (x - \lambda_a), \quad |\lambda_a| \leq R, \quad R \geq 0. \quad (\text{RB.1})$$

Repeated entries of $(\lambda_1, \dots, \lambda_q)$ record root multiplicity. Set

$$\mathcal{P}_d(x) = \left\{ \sum_{j=0}^d u_j x^j : u_j \in \mathbb{C} \right\}, \quad \|P\|_{1,x} = \sum_{j=0}^d |u_j|. \quad (\text{RB.2})$$

The exact linear map in this note is

$$\mathcal{R}_{\chi_x} : \mathcal{P}_{2q-1}(x) \longrightarrow \mathcal{P}_{q-1}(x), \quad P = \chi_x Q + \mathcal{R}_{\chi_x} P, \quad Q \in \mathcal{P}_{q-1}(x). \quad (\text{RB.3})$$

Existence follows by subtracting the leading coefficient times the corresponding monomial multiple of the monic polynomial χ_x , decreasing the degree at each step until it is below q . For uniqueness, the difference of two decompositions satisfies $\chi_x(Q_1 - Q_2) = r_2 - r_1$, where the right side has degree below q . A nonzero left side has degree at least q , so both sides vanish. Applying this uniqueness to linear combinations proves linearity. In particular $\mathcal{R}_{\chi_x} x^j = x^j$ for $0 \leq j < q$.

S27.1.2 2. The complete homogeneous division formula

Let e_a be the elementary symmetric polynomial of degree a in the listed roots, with $e_0 = 1$, and define the complete homogeneous polynomials by

$$h_\ell(\lambda_1, \dots, \lambda_q) = \sum_{\substack{\alpha_1, \dots, \alpha_q \geq 0 \\ \alpha_1 + \dots + \alpha_q = \ell}} \lambda_1^{\alpha_1} \cdots \lambda_q^{\alpha_q}, \quad h_0 = 1. \quad (\text{RB.4})$$

As an identity of formal power series,

$$\left(\sum_{a=0}^q (-1)^a e_a t^a \right) \left(\sum_{\ell=0}^{\infty} h_\ell t^\ell \right) = \prod_{a=1}^q (1 - \lambda_a t) \prod_{a=1}^q \left(\sum_{b=0}^{\infty} (\lambda_a t)^b \right) = 1. \quad (\text{RB.5})$$

There is no convergence requirement: every coefficient uses only finitely many factors and terms. Thus, for every integer $s \geq 1$,

$$\sum_{a=0}^{\min(q,s)} (-1)^a e_a h_{s-a} = 0. \quad (\text{RB.6})$$

For $0 \leq m \leq q-1$, put

$$Q_m(x) = \sum_{\ell=0}^m h_\ell x^{m-\ell}. \quad (\text{RB.7})$$

The coefficient of x^{q+m} in $\chi_x Q_m$ is one. For $1 \leq s \leq m$, the coefficient of x^{q+m-s} is the left side of (RB.6), hence zero. Therefore

$$x^{q+m} - \chi_x(x) Q_m(x) \in \mathcal{P}_{q-1}(x).$$

The uniqueness in (RB.3) now gives the exact division identity

$$\boxed{\mathcal{R}_{\chi_x} x^{q+m} = x^{q+m} - \chi_x Q_m.} \quad (\text{RB.8})$$

For completeness its individual coefficients are

$$[x^r] \mathcal{R}_{\chi_x} x^{q+m} = - \sum_{\substack{0 \leq a \leq q, 0 \leq \ell \leq m \\ a+\ell=q+m-r}} (-1)^a e_a h_\ell, \quad 0 \leq r < q. \quad (\text{RB.9})$$

This expression remains valid for repeated and zero roots without taking limits.

S27.1.3 3. Explicit coefficient-norm bound

The coefficient norm is submultiplicative. Indeed, if $U = \sum u_j x^j$ and $V = \sum v_k x^k$, then

$$\|UV\|_{1,x} = \sum_r \left| \sum_{j+k=r} u_j v_k \right| \leq \sum_{j,k} |u_j| |v_k| = \|U\|_{1,x} \|V\|_{1,x}. \quad (\text{RB.10})$$

Consequently

$$\|\chi_x\|_{1,x} \leq \prod_{a=1}^q (1 + |\lambda_a|) \leq (1 + R)^q. \quad (\text{RB.11})$$

The number of weak compositions of ℓ into q nonnegative parts is $\binom{q+\ell-1}{\ell}$: place $q-1$ separators among a row of ℓ markers and those separators. Thus (RB.4) yields

$$|h_\ell| \leq \binom{q+\ell-1}{\ell} R^\ell, \quad \|Q_m\|_{1,x} \leq \sum_{\ell=0}^m \binom{q+\ell-1}{\ell} R^\ell. \quad (\text{RB.12})$$

In (RB.8), the leading monomial x^{q+m} and the remainder have disjoint degree supports. In particular

$$\|\chi_x Q_m\|_{1,x} = 1 + \|\mathcal{R}_{\chi_x} x^{q+m}\|_{1,x}. \quad (\text{RB.13})$$

Combining (RB.10)–(RB.13) proves, with the original radius retained,

$$\boxed{\|\mathcal{R}_{\chi_x} x^{q+m}\|_{1,x} \leq (1+R)^q \sum_{\ell=0}^m \binom{q+\ell-1}{\ell} R^\ell - 1, \quad 0 \leq m \leq q-1.} \quad (\text{RB.14})$$

When $R \leq 1$, the finite sum of binomial coefficients is

$$\sum_{\ell=0}^m \binom{q+\ell-1}{\ell} = \binom{q+m}{m}. \quad (\text{RB.15})$$

To verify this identity, its $m = 0$ case is $1 = 1$, and its induction step is Pascal's identity $\binom{q+m-1}{m-1} + \binom{q+m-1}{m} = \binom{q+m}{m}$. The right side of (RB.15) increases with m , as is also immediate from its sum expression. Define the exact integer

$$C_q = 2^q \binom{2q-1}{q-1} - 1. \quad (\text{RB.16})$$

Then $C_q \geq 1$, with $C_1 = 1$, and every column of the remainder map satisfies

$$\max_{0 \leq j \leq 2q-1} \|\mathcal{R}_{\chi_x} x^j\|_{1,x} \leq C_q. \quad (\text{RB.17})$$

The two central coefficients $\binom{2q-1}{q-1}$ and $\binom{2q-1}{q}$ are equal. Their sum is at most the sum of all coefficients of $(1+t)^{2q-1}$ at $t = 1$, which is 2^{2q-1} . Therefore

$$C_q \leq 2^{3q-2} - 1 < 8^q. \quad (\text{RB.18})$$

For an arbitrary $P = \sum_{j=0}^{2q-1} u_j x^j$, linearity and the triangle inequality give

$$\|\mathcal{R}_{\chi_x} P\|_{1,x} \leq C_q \|P\|_{1,x} < 8^q \|P\|_{1,x} \quad (P \neq 0). \quad (\text{RB.19})$$

The non-strict version with 8^q holds also for $P = 0$. In fact the induced operator norm between these coefficient ℓ^1 spaces equals the maximum column norm: the displayed triangle inequality proves the upper bound, and applying the operator to a monomial attaining the finite maximum proves the reverse inequality.

S27.1.4 4. Raw jets and the exact quotient section

Write the distinct roots as $\lambda_1^*, \dots, \lambda_s^*$, with multiplicities $m_1, \dots, m_s$, so $\sum_{\nu} m_{\nu} = q$. Define the raw-jet space and map by

$$\mathcal{J}_x = \bigoplus_{\nu=1}^s \mathbb{C}^{m_{\nu}}, \quad J_{\chi_x, d} : \mathcal{P}_d(x) \longrightarrow \mathcal{J}_x, \quad P \longmapsto \left(P^{(r)}(\lambda_{\nu}^*) \right)_{1 \leq \nu \leq s, 0 \leq r < m_{\nu}}. \quad (\text{RB.20})$$

The exact Taylor identity at a root is

$$P(x) = \sum_{r=0}^d \frac{P^{(r)}(\lambda_{\nu}^*)}{r!} (x - \lambda_{\nu}^*)^r.$$

It shows that the first m_{ν} raw derivatives vanish precisely when $(x - \lambda_{\nu}^*)^{m_{\nu}}$ divides P . Simultaneous divisibility by these distinct-root factors is equivalent to divisibility by their product χ_x : in the factorization of a nonzero polynomial over $\mathbb{C}$, the exponent at each distinct root must be at least the listed multiplicity. The zero polynomial is divisible by every factor. Hence, for $d \geq q$,

$$\ker J_{\chi_x, d} = \chi_x \mathcal{P}_{d-q}(x). \quad (\text{RB.21})$$

The restriction $J_{\chi_x, q-1}$ has zero kernel by degree, and both its domain and codomain have dimension q . It is therefore an isomorphism. Equation (RB.3) gives

$$J_{\chi_x, q-1} \mathcal{R}_{\chi_x} = J_{\chi_x, 2q-1}, \quad \mathcal{R}_{\chi_x} = J_{\chi_x, q-1}^{-1} J_{\chi_x, 2q-1}. \quad (\text{RB.22})$$

Thus this remainder map preserves every specified raw jet, including every derivative order required by multiplicity. Equivalently, the inclusion of $\mathcal{P}_{q-1}(x)$ followed by the quotient map is an isomorphism onto $\mathbb{C}[x]/(\chi_x)$, and (RB.3) is the unique representative of degree below q in that quotient class.

S27.1.5 5. Original S variable, affine unit, and phase

Let the original monic annihilator be

$$\chi(S) = \prod_{\nu=1}^s (S - \zeta_{\nu})^{m_{\nu}}, \quad c \in \mathbb{C}, \quad T > 0. \quad (\text{RB.23})$$

Use the requested transformed-polynomial notation with its full definition:

$$\bar{\chi}(x) = (iT)^{-q} \chi(c + iTx) = \prod_{\nu=1}^s \left(x - \frac{\zeta_{\nu} - c}{iT} \right)^{m_{\nu}}. \quad (\text{RB.24})$$

Here the bar symbol is defined by (RB.24); complex conjugation of a polynomial, if used elsewhere, has its separate coefficientwise definition. The displayed unit $(iT)^{-q}$ proves monicity and retains the full complex phase. The transformed root radius is exactly bounded by

$$\left| \frac{\zeta_{\nu} - c}{iT} \right| = \frac{|\zeta_{\nu} - c|}{T}; \quad \text{in particular } \max_{\nu} |\zeta_{\nu} - c| \leq T \implies R \leq 1. \quad (\text{RB.25})$$

For every $d \geq 0$, define the invertible linear map

$$\begin{aligned} A_{c,T,d} : \mathcal{P}_d(S) &\longrightarrow \mathcal{P}_d(x), & (A_{c,T,d}P)(x) &= P(c + iTx), \\ A_{c,T,d}^{-1}F(S) &= F\left(\frac{S - c}{iT}\right). \end{aligned} \quad (\text{RB.26})$$

Substitution also defines an algebra isomorphism on the whole polynomial rings. In particular it sends the principal ideal (χ) onto $(\bar{\chi})$, because $A_{\chi} = (iT)^q \bar{\chi}$ and the factor $(iT)^q$ is a nonzero scalar unit. Thus the induced quotient map is exactly

$$\hat{A}_{c,T} : \mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[x]/(\bar{\chi}), \quad [P] \longmapsto [P(c + iTx)], \quad (\text{RB.27})$$

with inverse induced by the second formula in (RB.26).

Apply (RB.26) to the original division $P = \chi Q + \mathcal{R}_{\chi}P$. One obtains

$$A_{c,T,2q-1}P = \bar{\chi}((iT)^q A_{c,T,q-1}Q) + A_{c,T,q-1}\mathcal{R}_{\chi}P. \quad (\text{RB.28})$$

The last term has degree below q . Uniqueness of division gives the exact commuting identity

$$\boxed{\mathcal{R}_{\bar{\chi}} A_{c,T,2q-1} = A_{c,T,q-1} \mathcal{R}_{\chi}}. \quad (\text{RB.29})$$

For raw jets, set $\lambda_{\nu}^* = (\zeta_{\nu} - c)/(iT)$. Let $J_{\chi,d}$ be defined using derivatives in S at ζ_{ν} , and let $J_{\bar{\chi},d}$ use derivatives in x at λ_{ν}^* , with the same ordering and multiplicities. Repeated application of the ordinary chain rule gives, for every integer $r \geq 0$,

$$\frac{d^r}{dx^r} P(c + iTx) = (iT)^r P^{(r)}(c + iTx).$$

Define the invertible diagonal map

$$D_T : \bigoplus_{\nu} \mathbb{C}^{m_{\nu}} \longrightarrow \bigoplus_{\nu} \mathbb{C}^{m_{\nu}}, \quad (v_{\nu,r}) \longmapsto ((iT)^r v_{\nu,r}). \quad (\text{RB.30})$$

The complete raw-jet transport identity is therefore

$$\boxed{J_{\bar{\chi},d} A_{c,T,d} = D_T J_{\chi,d}}. \quad (\text{RB.31})$$

Every scalar $(iT)^r$, including its phase, appears in (RB.30)–(RB.31); the jet coordinates throughout are raw derivatives. Combining (RB.22), (RB.29), and (RB.31) proves preservation of the original raw jets as well as the transformed raw jets.

S27.1.6 6. Exact norm transfer back to the original coordinates

Define a norm on the original polynomial space by the displayed affine transport,

$$\|P\|_{c,T,1} := \|A_{c,T,d}P\|_{1,x}. \quad (\text{RB.32})$$

This definition is independent of the choice of $d \geq \deg P$, because adding zero coefficients does not change the sum. Invertibility of A proves that it is a norm. Equations (RB.19) and (RB.29), under the root-radius hypothesis in (RB.25), yield

$$\|\mathcal{R}_\chi P\|_{c,T,1} \leq C_q \|P\|_{c,T,1}, \quad P \in \mathcal{P}_{2q-1}(S). \quad (\text{RB.33})$$

The same estimate expressed using the untransformed coefficient norm also has fully explicit constants. For $0 \leq j \leq d$, the binomial theorem gives

$$\|A_{c,T,d}S^j\|_{1,x} = (|c| + T)^j, \quad \|A_{c,T,d}^{-1}x^j\|_{1,S} = \left(\frac{1 + |c|}{T}\right)^j. \quad (\text{RB.34})$$

The maximum-column characterization of the coefficient ℓ^1 operator norm, proved after (RB.19), consequently gives the exact operator norms

$$\|A_{c,T,d}\|_{1 \rightarrow 1} = \max\{1, (|c| + T)^d\}, \quad \|A_{c,T,d}^{-1}\|_{1 \rightarrow 1} = \max\left\{1, \left(\frac{1 + |c|}{T}\right)^d\right\}. \quad (\text{RB.35})$$

Thus the original-coordinate estimate is

$$\|\mathcal{R}_\chi P\|_{1,S} \leq C_q \max\left\{1, \left(\frac{1 + |c|}{T}\right)^{q-1}\right\} \max\{1, (|c| + T)^{2q-1}\} \|P\|_{1,S}. \quad (\text{RB.36})$$

Finally, for any $x \in \mathbb{C}$ and any polynomial r of degree below q , the coefficient triangle inequality gives $|r(x)| \leq \|r\|_{1,x} \max\{1, |x|^{q-1}\}$. Applying this to the exact representative in (RB.29) yields

$$|(\mathcal{R}_\chi P)(c + iTx)| \leq C_q \|P\|_{c,T,1} \max\{1, |x|^{q-1}\}. \quad (\text{RB.37})$$

In particular the maximum on the segment $S = c + iTx$, $-1 \leq x \leq 1$, is at most $C_q \|P\|_{c,T,1}$. Equations (RB.24), (RB.27), (RB.29), and (RB.31) specify the exact relation to the original annihilator, quotient, representative, and raw-jet coordinates; no measure or quotient metric is replaced in this coefficient calculation.

S28 The complete corrected endpoint and multiplicity criterion

Source role. Complete corrected PR23 witness with the full multiplicity proof.

Exact revision. a9a8b080b3779f82388a8bd4338ada2564ceb2410a0734bd58700f65c7789c62

Complete source. sources/owner_endpoint_review/review_pr24/join_inputs/PR23_ENDPOINT_CORRECTED.md

Source SHA-256. a9a8b080b3779f82388a8bd4338ada2564ceb2410a0734bd58700f65c7789c62

S28.1 Sharper endpoint-only criterion and the scope of exterior iteration

This is a mathematical continuation of the supplied Toda and exterior notes. The general Jensen argument and exterior-multiplicity count below have written proofs; they are not additional Lean declarations. The norm-ratio product and overlapping-volume telescope are included in the finite formalization target list. No arithmetic asymptotic bound is claimed.

S28.1.1 1. Eliminate the supremum over recurrence coefficients

Retain a nonempty fixed packet and tensor degree k , cyclic dimension q , the actual positive monic norm sequence ω_N , the actual quotient volume $V_N = \det G_N$, and the source's exact control allowance ϵ_N . Choose consecutive admitted degrees $N=n, \dots, n+r-1$ with $n \geq q$ and integer $r \geq 1$. Put

$$B_{(n,r)} = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})),$$

$$A_{(n,r)} = (\omega_{(n+r)} / \omega_n)^{1/(2r)}.$$

These are four endpoint quotient volumes and two endpoint polynomial norms of the existing family. Since V is positive and nonincreasing, $B_{(n,r)} \geq 0$. The source's finite upper bound implies

$$\min_{(0 \leq j < r)} \epsilon_{(n+j)} \leq A_{(n,r)} \sinh(B_{(n,r)}/(2r)). \quad (E1)$$

Unlike the preceding maximum-recurrence criterion, E1 needs no pointwise upper bound for every recurrence coefficient in the window.

Complete proof For $N=n+j$ set

$$x_j = (1/2) \log(V_{(N-1)}/V_{(N+1)}) \geq 0,$$

$$A_j = \sqrt{\omega_{(N+1)}/\omega_N} > 0.$$

The existing local estimate is $\epsilon_{(n+j)} \leq A_j \sinh(x_j)$. If any $x_j = 0$, the nonnegative allowance at that step is zero, and E1 follows. Otherwise all $x_j > 0$. The function $f(x) = \log(\sinh x)$ has

$$f''(x) = -1/\sinh(x)^2 < 0.$$

Finite Jensen therefore gives

$$\text{product}_j \sinh(x_j) \leq \sinh((\sum_j x_j)/r)^r.$$

Let E be the minimum of the nonnegative allowances. Multiplication of the local bounds gives

$$E^r \leq \text{product}_j \epsilon_{(n+j)} \leq (\text{product}_j A_j) \text{product}_j \sinh(x_j).$$

The actual products telescope:

$$\text{product}_j A_j = (\omega_{(n+r)}/\omega_n)^{1/2},$$

$$2 \sum_j x_j = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})).$$

Taking the nonnegative r -th root proves E1. A finite minimum is attained, so E1 selects an actual original degree N inside the specified window. No favorable metric is selected independently, and its original theta representative, quotient and full jets remain attached.

The recurrence product is $\text{VolumeWindow.norm_ratio_product}$; the second telescope is $\text{VolumeWindow.overlapping_log_telescope}$, applied to $V(j)=V_{(n-1+j)}$. The Jensen and nonnegative-root assembly is this written proof rather than a claim about an unimplemented exterior/analytic declaration.

S28.1.2 2. Every input is an endpoint of the two original determinant sequences

The source's $\omega_j=D_{(j+1)}/D_j$ gives

$$A_{(n,r)}=[D_{(n+r+1)} D_{(n)} / (D_{(n+r)} D_{(n+1)})]^{(1/(2r))}.$$

The four V -values in B are evaluated by $V_N=D_{(N+1)}/B_{(N-q+1)}$, with the source's literal relation determinant B . All indices are nonnegative because $n \geq q$. No negative-index determinant or inverse of a rank-deficient earlier kernel is used. The common mass factors cancel only in these displayed ratios; the source $\text{norm } \omega_0 = \mu_h^k$ has not been reset.

For the original quartet,

$$q_k = [1+k(m-1)](k+1)^2, \\ L_{(h,k)} \geq \delta k^2 q_k / 2.$$

The exterior theorem gives $L_{(h,k)} \leq \epsilon_N$ at every admitted N . Thus E1 yields

$$L_{(h,k)} \leq A_{(n,r)} \sinh(B_{(n,r)}) / (2r). \quad (E2)$$

A proof that the right side is $o(k^2 q_k)$ along admitted $n=n(k)$, $r=r(k)$ would exclude that fixed off-line quartet. One sufficient set of estimates is

$$r(k) \geq c_h q_k, \quad A_{(n,r)} = O_h(q_k), \\ B_{(n,r)} = O_h(q_k \log k),$$

for $c_h > 0$. Indeed $B/(2r) = o(\log k)$, so $\sinh(B/(2r)) \leq \exp(o(\log k)) = k^{o(1)}$, and the ratio to $k^2 q_k$ tends to zero. These are candidate arithmetic estimates, not consequences of the formal telescope, positivity, the Toda equation, or Deligne's finite-field hypotheses.

The weighted endpoint criterion is sharp as an abstract sequence inequality: equality can occur when all x_j are equal and every $A_j \sinh(x_j)$ equals the same allowance. For arbitrary recurrence sequences the bound is never larger, and potentially strictly smaller, than the bound obtained by replacing $A_{(n,r)}$ by the largest A_j .

S28.1.3 3. A useful boundary on further exterior amplification

The supplied single exterior theorem is correct. It must not be reused with a rank-two hypothesis that no longer holds after exterior powering.

Suppose the original self-adjoint control H has spectrum

$$(+\epsilon, 0 \text{ repeated } q-2 \text{ times}, -\epsilon), \quad \epsilon > 0.$$

On the actual p -th exterior power, $1 \leq p \leq q$, the additive control has $+\epsilon$ and $-\epsilon$ each with multiplicity

$$r_p = \text{binomial}(q-2, p-1),$$

and zero with multiplicity $\text{binomial}(q-2, p) + \text{binomial}(q-2, p-2)$. This follows by the unscaled wedge basis: select the positive direction but not the negative, the negative but not the positive, or both/neither. In particular its operator norm is at most ϵ : it equals ϵ for $1 \leq p \leq q-1$ and is zero for $p=q$. Its rank is $2r_p$, not generally two.

Apply an s -th exterior power to that whole space, whose dimension is $D_p = \text{binomial}(q, p)$. Its largest absolute additive-control eigenvalue is exactly

$\epsilon \cdot \min(s, r_p, D_p - s), \quad 0 \leq s \leq D_p.$

Proof: select positive eigenvectors first, zero eigenvectors next, and negative eigenvectors only when the degree forces them. There are r_p available positive directions and r_p negative ones. This attains the stated piecewise maximum; reversing the choices attains its negative. At $s=D_p$ the sum is zero. For example $q=4, p=2$ gives spectrum $(+\epsilon, +\epsilon, 0, 0, -\epsilon, -\epsilon)$, and the next exterior square has allowance 2ϵ .

This does not weaken the original determinant-line amplification or remove its nilpotent arithmetic action. It identifies the precise cost of iterating the operation on the entire exterior space, and prevents a spurious second cost-free amplification. The source's chosen determinant line has its own one-dimensional action and must remain distinguished from the complete first exterior power.

S28.1.4 4. Collaboration boundary

The formal work carries the exact source/quotient determinant and the two loss terms into the original-Gram trace certificate. The analytic collaborator can now estimate the two endpoint quantities in E1, or retain the exact phase and imbalance when the coarse upper bound is too large. The six-part Deligne archive and uploaded ZIP test suites were not opened in this execution environment; the source statements used here came from the complete pasted notes. The current GitHub correction and original source maps are unchanged.

S29 Complete original endpoint window product proof

Source role. Complete terminal owner WP.1-13 witness, all six sections and every mathematical source map.

Exact revision. c97c831a5e1c2ef3685910c648bce0f9fbc54230bc8a1bec7e521bd808a1209

Complete source. sources/owner_endpoint_terminal/published/workbenches/tau-arithmetic-endpoint-bounds/window/WINDOW_PRODUCT_PROOF.md

Source SHA-256. c97c831a5e1c2ef3685910c648bce0f9fbc54230bc8a1bec7e521bd808a1209

S29.1 Exact consecutive-window control and endpoint determinants

This note proves the consecutive product, its sharper four-endpoint consequence, and its two-transfer reduction for the original canonical quotient metrics. Composed with the independently reviewed arithmetic norm bound on a balanced window, it gives a necessary four-volume growth threshold of **4**, improving the older threshold of 2. It proves no opposing arithmetic upper estimate and no Riemann-hypothesis conclusion. It belongs to the next integration cut, not the frozen Toda-Gamma edition.

S29.1.1 1. Original identity and exact window product

Fix a nonempty reflection-stable packet, tensor degree, cyclic dimension $q \geq 1$, and one admissible observation. Write $V_N = \det G_N > 0$, retain the original monic squared norms $\omega_N > 0$, and set $\delta_N = V_N/V_{N-1} \in (0, 1]$. The original Toda source, TVB.13, TVB.15 and TVB.22–27, supplies

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N)(\delta_{N+1}^{-1} - 1), \quad \phi_N = \sigma - \partial_\theta \log V_N, \quad N \geq q. \quad (\text{WP1})$$

The real phase is not set to zero. At a general tilt all norms, volumes and derivatives in WP1 are evaluated at that same tilt. The arithmetic norm and asymptotic composition below use the original observation $\theta = 0$.

For integers $n \geq q, r \geq 1$, put $d_j = \delta_{n+j}$ for $0 \leq j \leq r$, $\Omega = \omega_{n+r}/\omega_n$, and $E = \min_{0 \leq j < r} \epsilon_{n+j}$. Multiplying WP1 first, the norm ratios telescope, every interior $1 - d_j$ occurs twice, and $\prod_{j=1}^r d_j = V_{n+r}/V_n$. Therefore

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (\text{WP2})$$

Since $E^{2r} \leq \prod_j \epsilon_{n+j}^2 \leq \prod_j (\epsilon_{n+j}^2 + \phi_{n+j}^2)$, dropping only the displayed nonnegative factors, each at most one, gives

$$E \leq \left(\frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} \right)^{1/(2r)}. \quad (\text{WP3})$$

No inverse at degree $q - 2$, reassigned Gram, or recurrence supremum occurs. The empty quotient $q = 0$ is outside this positive-dimensional theorem; its determinant and control are separately the empty determinant and zero.

S29.1.2 2. Sharper four-endpoint bound and comparison

For $r \geq 2$, define

$$\alpha = V_n/V_{n-1}, \quad \beta = V_{n+r}/V_{n+r-1}, \quad t = V_{n+r-1}/V_n, \quad A = \Omega^{1/(2r)}.$$

These are positive and at most one, and the $r - 1$ interior contractions have product t . For $s = r - 1$,

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (\text{WP4})$$

If any interior $d_j = 1$, the left side is zero. Otherwise set $v_j = -\log d_j > 0$. The second derivative of $f(v) = \log(1 - e^{-v})$ is $-e^{-v}/(1 - e^{-v})^2 < 0$, so finite concavity proves WP4. This is ordinary Jensen concavity, obtainable from the integral form by a step function; see [DLMF §1.7\(iv\)](#). Inserting WP4 into WP2 gives

$$E \leq U := A \left(\frac{(1 - \alpha)(1 - \beta)}{\beta t} \right)^{1/(2r)} \left(1 - t^{1/(r-1)} \right)^{(r-1)/r}. \quad (\text{WP5})$$

For $r = 1$ use the separate expression

$$E = \epsilon_n \leq \sqrt{\frac{\omega_{n+1}}{\omega_n} (1 - \alpha)(\beta^{-1} - 1)}. \quad (\text{WP6})$$

No $1/(r - 1)$ is evaluated in WP6. For any r , a unit contraction $d_j = 1$ makes an adjacent local radius sum zero; hence the corresponding allowance and phase both vanish and $E = 0$. At an interior unit contraction both neighboring local sums vanish. If $t = 1$, all interior contractions are one and WP5 is zero.

The corrected PR23 endpoint note proves the older bound

$$E \leq A \sinh\left(\frac{\mathcal{B}}{2r}\right), \quad \mathcal{B} = \log \frac{V_{n-1}V_n}{V_{n+r-1}V_{n+r}} = -\log \alpha - \log \beta - 2 \log t.$$

In fact $U \leq A \sinh(\mathcal{B}/(2r))$. To prove it without assuming an arithmetic extremizer, replace just the interior scalar contractions by their common geometric mean. This maximizes the exact product by WP4 while leaving the four endpoints fixed. For that scalar sequence put $v_j = -\log d_j$. The original local completion of the square is

$$(1 - d_j)(d_{j+1}^{-1} - 1) = \sinh^2 \frac{v_j + v_{j+1}}{2} - \left(e^{(v_{j+1} - v_j)/2} - \cosh \frac{v_j + v_{j+1}}{2} \right)^2.$$

Drop this square and apply concavity of $\log \sinh x$, whose second derivative is $-1/\sinh^2 x$, to the r nonnegative half-sums. Their sum is $\mathcal{B}/2$, proving the comparison. A zero half-sum is handled before logarithms. WP5 also dominates the simpler WP3 because all factors it retains are at most one; WP3 alone need not dominate the older bound for small contractions.

For constant $d = e^{-v} < 1$, WP5 becomes $2A \sinh(v/2)$, whereas the older expression is $A \sinh v$; their ratio is $\cosh(v/2)$.

The sharpness claim is only for the relaxed scalar data, not the arithmetic moment family. In its strictly positive case require **all three** $\alpha < 1, \beta < 1, t < 1$. Set the interior ratios equal, put $f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0$, choose successive norm ratios U^2/f_j , phases zero and all allowances U . The norm-ratio product is exactly Ω , so the prescribed endpoints and equality hold. Zero-product cases give a zero bound directly. Strict contraction of only the two outside ratios would not justify claiming $f_j > 0$ when $t = 1$.

S29.1.3 3. Retained phase and interior loss

When every interior contraction is strictly below one, define

$$J = (r - 1) \log(1 - t^{1/(r-1)}) - \sum_{j=1}^{r-1} \log(1 - d_j) \geq 0.$$

WP2 is precisely

$$\prod_j (\epsilon_{n+j}^2 + \phi_{n+j}^2) = U^{2r} e^{-2J}, \quad \prod_j (E^2 + \phi_{n+j}^2) \leq U^{2r} e^{-2J}. \quad (\text{WP7})$$

The polynomial $P(x) = \prod_j (x + \phi_{n+j}^2)$ is strictly increasing for $x \geq 0$. For consistent input, the right side R is at least $P(0)$, so $P(x) = R$ has a unique nonnegative solution x_* and $E \leq \sqrt{x_*}$. If proposed data instead have $R < P(0)$, they are inconsistent with the original radius identities; a nonnegative threshold cannot then be asserted. Unit-contraction cases are handled directly, without using an infinite or undefined J .

S29.1.4 4. Literal determinant and full-jet transfer cancellation

To distinguish this scalar from the existing exterior map named Λ_N , write $\Lambda_N^{\text{sc}} = \omega_N / V_N$. With the unchanged source and relation determinants, TVB.13 and TVB.15 give

$$\omega_N = \mathfrak{D}_{N+1} / \mathfrak{D}_N, \quad V_N = \mathfrak{D}_{N+1} / \mathfrak{B}_{N-q+1}, \quad \boxed{\Lambda_N^{\text{sc}} = \mathfrak{B}_{N-q+1} / \mathfrak{D}_N}. \quad (\text{WP8})$$

The ordinary monic recurrence and moment-determinant background is [DLMF §18.2\(iv\)](#) and [§18.2\(ix\)](#); the specific arithmetic relation determinant is the supplied construction, not a conclusion attributed to DLMF. Thus WP3 equals

$$E \leq \left(\frac{\Lambda_{n+r}^{\text{sc}}}{\Lambda_n^{\text{sc}}} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{n+r-q+1} \mathfrak{D}_n}{\mathfrak{D}_{n+r} \mathfrak{B}_{n-q+1}} \right)^{1/(2r)}. \quad (\text{WP9})$$

PR24's complete transfer note, equations (4)–(5), defines F_a using raw derivatives of the real monic source polynomials at every doubled root multiplicity, with fixed row order. It proves

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det V}{\det F_a}, \quad a = N - q + 1.$$

Consequently, setting now $a = n - q + 1$,

$$E \leq \left[\frac{\det F_{a+r}}{\det F_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}. \quad (\text{WP10})$$

This uses two transfer determinants; at $q = 1$ the norm product is empty. The common raw confluent Vandermonde, including every derivative factorial and its coordinate phase, cancels through the displayed ratio. No absolute value is inserted: positivity follows from equality with the positive scalar ratio in WP9. The relation map still contains the unit i^q , and F_0 need not equal V , although their determinants agree.

Literal mass bookkeeping confirms the cancellation: rescaling the fixed measure by $c > 0$ would give $\omega_N \mapsto c\omega_N$, $V_N \mapsto c^q V_N$, $\Lambda_N^{\text{sc}} \mapsto c^{1-q} \Lambda_N^{\text{sc}}$. Only the endpoint ratio is invariant. No such rescaling is performed here.

S29.1.5 5. Immediate arithmetic consequence: threshold four

The complete Arithmetic Endpoint Bounds note proves its original lower convolution envelope and exact general norm-window bound, equations (29), (33), (34), (37). Its independently checked balanced extension yields

$$A_{h,k,n,r} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n,$$

where, with $b \in (0, \pi/2)$, $B_h = 42 + 2 \deg h$, $X_h = \max(1, M_h(b), M_h(-b))$, $K_h = \max(1, \vartheta_h^{-1})$,

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) X_h K_h \max(1, b^{-3}) 6^{B_h/3}. \quad (\text{WP11})$$

The constant is finite and depends only on the fixed packet and chosen b ; it is not a numerically enclosed constant. A direct check of the extension uses $\mathfrak{l}_n(n) \geq 2n^{2n+1}/((2n+1)4^n)$, $(2(n+r))! \leq (4n)^{2(n+r)}$, and the two moment masses before taking roots. The factorial/Legendre contribution becomes $4n 8^{n/r} ((2n+1)/(c_h n))^{1/(2r)} \leq 256n \max(1, 3/c_h)$. The remaining factors are bounded by $X_h K_h \max(1, b^{-3}) e^\pi 6^{B_h/3}$, using $2r \geq n$ and decreasing $\log(2n)/n$ for $n \geq 3$.

For a packet consisting exactly of a hypothetical full off-line quartet of common multiplicity m , the original exterior lower allowance satisfies

$$q = q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} \geq \delta k q / 2 > 0, \quad L_{h,k} \leq \epsilon_N.$$

Choose $n = q$, $r = q - 1$. Here $q \geq k \geq 3$, the allowance indices are precisely $q, \dots, 2q - 2$, and the last norm/volume endpoint is $2q - 1$. WP3 and WP11 imply

$$C_k := \log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \log \frac{\delta k}{2C_h^{\text{bal}}}. \quad (\text{WP12})$$

The old four-volume budget obeys the exact decomposition

$$\mathcal{B}_{h,k} = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}} = 2C_k + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}} \geq 2C_k.$$

Therefore

$$\boxed{\mathcal{B}_{h,k} \geq 4(q_k - 1) \log \frac{\delta k}{2C_h^{\text{bal}}}, \quad \liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4.} \quad (\text{WP13})$$

The diagonal window also gives $\liminf \log(V_q/V_{2q})/(q_k \log k) \geq 2$. These lower bounds follow from the actual arithmetic norm estimate and original radius identities, not from scalar-fixture sampling. The old threshold 2 remains a valid weaker result. A separately proved $\limsup \mathcal{B}_{h,k}/(q_k \log k) < 4$ would now contradict the hypothetical quartet; **no such upper bound is established**. Equivalently, WP9 identifies the sufficient target $(2r)^{-1} \log(\Lambda_{n+r}^{\text{sc}}/\Lambda_n^{\text{sc}}) - \log(kq_k) \rightarrow -\infty$, but does not prove that target for the arithmetic family.

S29.1.6 6. Exact fixture and verification scope

The independently recomputed fixture is $d\mu = 7e^{-u^2/2} du/\sqrt{2\pi}$, $\chi = u^2 + 1$, $Q_{j+1} = uQ_j - jQ_{j-1}$, $\omega_j = 7j!$, and $G_N = (\sum_{j \leq N} [Q_j][Q_j]^T/\omega_j)^{-1}$. It gives exactly

$$G_2 = \text{diag}(7/3, 7), \quad G_3 = \text{diag}(7/3, 21/11), \quad G_4 = \text{diag}(42/43, 21/11), \\ (\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = (16/3, 400/99, 4225/946), \quad (V_1, \dots, V_5) = (49, 49/3, 49/11, 882/473, 980/1333).$$

For $n = 2, r = 3, \Omega = 60, \alpha = 1/3, \beta = 110/279, t = 54/473$. The minimum, geometric mean, sharper bound and older bound are respectively approximately 2.0100756305184, 2.1407199197265, 2.1666731523012 and 2.5178436563897. Their exact underlying quantities, not these decimal strings, enter the identity checks.

A separate original exponential tilt $\theta = 1/2$ retains mass $7e^{1/8}$. At degree 2 it gives $\phi_2 = 7/129 \neq 0$ and $\epsilon_2^2 = 18769/4128$; further nonzero phases, the independent derivative $\sigma - \partial_\theta \log V_N$, and the whole window product are checked. The polynomial $u^2 - 3$ supplies an actual interior unit contraction. A one-dimensional quotient tests zero control with potentially nonzero phase and the empty norm band in WP10. Raw derivatives at doubled nonreal nodes independently verify WP8–WP10, while the literal moment matrices verify the source/relation determinants.

The receipt separates exact symbolic/rational/data checks from four approximate decimal-display comparisons. Symbolic product factorizations were checked for $r = 1, \dots, 8$; rational phases and degeneracies, exact powered endpoint comparisons, mass scaling, full raw-jet transfers and balanced-window index/algebra checks are retained. These finite checks do not replace the written general proofs, are not arithmetic zero packets, and are not interval or Lean certificates. No Wolfram session was run or its reported execution adopted as independent evidence.

The complete submitted calculation, original TVB source sections used above, corrected PR23 endpoint note, complete PR24 transfer note, complete Arithmetic Endpoint Bounds note, balanced-window proof, and parent four-volume synthesis were read. Their immutable hashes are in EXACT_VERIFICATION_RECEIPT.json; no private attachment path occurs in the public derivative or receipt. Source attachments remain unchanged. The parent's FOUR_VOLUME_THRESHOLD.md was independently approved at SHA-256 9ff55c84d74db85306e7a68c70fe7b08d8e642e8ac1a3e6137649f68e102fc3a.

Selecting the finite minimizing degree keeps the existing representative $R_N, J^{(k)}R_N = \eta_{h,k}$, and $q^{(k)}R_N = \sigma_h^{\otimes k} \eta_{h,k}$, with the same arithmetic unit, jets, cohomology classes and supported-zero/absence distinction. Endpoint data certify existence, not identification of the minimizing degree. No owner scalar-budget optimization, canonical source, frozen edition, or remote publication was changed by this review.

S30 Complete original endpoint restriction control

Source role. Complete delivered proof witness. Its historical claims retain their source revision; corrections and full current maps are proved in the corresponding integrated chapter.

Exact revision. 905ad0dcd20a2630eb6338488206a387265223590a527f867c1266b85bb27491

Complete source. sources/web_endpoint_restriction_delivery/Tau_Endpoint_Restriction_Control/RESEARCH_NOTE.md

Source SHA-256. 905ad0dcd20a2630eb6338488206a387265223590a527f867c1266b85bb27491

Source metadata: author (MetaInlines)

Owner-directed SplitZero analytic continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

The adjoint of the original support transition, exterior minor weights, and a logarithmic control certificate

Source metadata: title (MetaInlines)

Canonical restriction and the arithmetic endpoint volume

S30.1 1. Result, inherited input, and scope

The remaining quantity in the current endpoint criterion is a ratio of four canonical arithmetic quotient volumes. This note constructs its underlying restriction operator, proves composition and action identities for that operator, and gives two further ways to calculate and bound the same volume. No new arithmetic metric is selected.

The principal map is

$$T_{i,j} : (E, G_j) \longrightarrow (E, G_i), \quad T_{i,j} = K_i G_j, \quad K_n = G_n^{-1}, \quad i \leq j. \quad (1)$$

It is the **metric adjoint of the original arithmetic identity transport** from degree i to degree j . It is also the map read by restricting the actual degree- j canonical polynomial representative to degrees at most i :

$$R_i T_{i,j} = \text{pr}_{\leq i} R_j. \quad (2)$$

The source coordinate inclusions in (2) are made explicit below. The return maps compose in the reverse direction,

$$T_{i,j} T_{j,l} = T_{i,l}, \quad \det T_{i,j} = V_j / V_i, \quad V_n = \det G_n. \quad (3)$$

Their eigenvalues $g_{i,j,a}$ are in $(0, 1]$ and have an exact source meaning: they are the squared singular values of the original identity transport equipped with its two actual quotient metrics, and also of the polynomial restriction on the canonical image. Hence

$$\log(V_i / V_j) = - \sum_{a=1}^{\dim E} \log g_{i,j,a}. \quad (4)$$

No arithmetic vector is killed by $T_{i,j}$ at a finite admitted degree. Its small eigenvalues quantify the cost of recovering the original arithmetic coordinates from a lower-degree portion of their representatives.

The new finite upper bounds express (4) through positive traces of powers of $I - T_{i,j}$, with an explicitly certified lower bound on $g_{i,j,a}$. A Cauchy–Binet expansion gives another exact representation, by nonnegative full-jet minor weights. The expansion retains the literal mass of the arithmetic source and does not assign its total weight the value one.

The preceding analytic continuation proved, for a fixed full packet h and $n \geq k \geq 3$,

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n. \quad (5)$$

We use this as an inherited written analytic theorem, not as a newly kernel-checked result. Composing the new upper certificates with that theorem and the formalizer's endpoint criterion gives an explicit control certificate for the original family.

For a packet consisting exactly of an off-line quartet, the already-established endpoint lower bound additionally implies the following new representative-level consequence: among the two indicated degree restrictions, at least one squared singular value is at most

$$\exp \left[-\operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right) \right]. \quad (6)$$

This is a necessary consequence of an off-line quartet, not an assertion that small singular values imply one. The calibration below retains a nilpotent on-line block with nontrivial restriction loss.

The general matrix and exterior mechanisms are classical. The contribution is their explicit propagation through this programme's actual source, full-jet quotient, support transitions, and arithmetic action. No upper estimate uniform in k for the logarithmic loss is proved in this note.

S30.2 2. Preserve the original base and source

Retain the marked coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (7)$$

Here $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$, $e_R = 0_R^\bullet$, and $p_R(\tau) = 0_R$, $p_R(r^\bullet) = r$. The target of $p_{\mathbb{Z}}$ is infinite. The structural base remains $\mathfrak{b}_\tau$. Polynomial degree and relation depth do not adjoin another scalar zero.

Keep

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi. \quad (8)$$

For a fixed finite packet, retain every selected zero with its complete order:

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad v_h = g/h, \quad \mathcal{M}F_h = v_h. \quad (9)$$

The arithmetic integration density at tensor degree k is

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \mu_h = \int w_h, \quad \int m_{h,k} = \mu_h^k. \quad (10)$$

On the existing cyclic sum module use

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k}, \quad A = M_S, \quad S = s_1 + \cdots + s_k, \quad c = k/2. \quad (11)$$

The canonical arithmetic source is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad (12)$$

and its norm is

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du. \quad (13)$$

The full arithmetic coefficient maps remain

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h). \quad (14)$$

All derivatives and arithmetic units in these jets remain. The quotient coordinate $[P]_\chi$ and its arithmetic image $\eta[P]_\chi$ are connected by this specified injective module map.

Let $p_0, p_1, \dots$ be the original monic S -polynomials orthogonal for (13), with squared norms $\omega_n > 0$. In particular $\omega_0 = \mu_h^k$. No polynomial is divided by its norm.

For $N \geq q - 1$, let

$$\mathcal{P}_N = \mathbb{C}[S]_{\leq N}, \quad \mathcal{D}_N = \chi \mathcal{P}_{N-q}, \quad \mathcal{P}_{-1} = 0. \quad (15)$$

The exact source sequence is

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (16)$$

Monicity proves surjectivity and the displayed kernel, including at $N = q - 1$. Under $\mathcal{V}$, the relation χP is the same original theta boundary whose primitive was constructed by fixed-order division into the tensor factors. Its support label and both occurrences of every Koszul sign remain in that primitive.

All finite Gram inverses in the note require $q \geq 1$. For $h = 1$ the finite arithmetic packet is zero, all displayed determinant-line observations have empty determinant one, and no positive-rank inverse is introduced. Its analytic source and mass remain as in (10).

S30.3 3. Construct the canonical degree transport and its adjoint

Use the ordered, unscaled basis $(p_0, \dots, p_N)$ of $\mathcal{P}_N$. Define

$$O_N = \text{diag}(\omega_0, \dots, \omega_N), \quad B_N = [b_0, \dots, b_N], \quad b_n = [p_n]_\chi. \quad (17)$$

Here $B_N : \mathbb{C}^{N+1} \rightarrow E$ is the remainder map in the fixed quotient basis $1, S, \dots, S^{q-1}$. Put

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N : E \rightarrow \mathcal{P}_N. \quad (18)$$

The formula gives

$$B_N R_N = I_E, \quad R_N^* O_N R_N = G_N, \quad \text{im } R_N = \mathcal{D}_N^\perp. \quad (19)$$

Indeed B_N is onto, so K_N is positive definite. For $z \in \ker B_N$, $z^* O_N R_N = z^* B_N^* G_N = 0$. Every representative is $R_N u + z$; the two summands are orthogonal. Thus (18) is the unique minimum-norm representative for the original relation space, not an independently selected form.

The actual test-function representative is $\mathcal{V} R_N$. Every identity involving R_N below passes to that source by (12)–(14).

For $i \leq j$, let $L_{i,j} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$ be coefficient padding and let $P_{i,j} : \mathcal{P}_j \rightarrow \mathcal{P}_i$ keep exactly the first $i + 1$ monic coordinates. Since the full monic family is orthogonal, this is the orthogonal projection onto the original lower-degree source. One has

$$B_j L_{i,j} = B_i, \quad L_{i,j}^* O_j L_{i,j} = O_i. \quad (20)$$

Let $\mathcal{E}_N$ denote E with its actual quotient metric G_N . The original quotient transport is

$$I_{i,j} : \mathcal{E}_i \rightarrow \mathcal{E}_j, \quad u \mapsto u, \quad B_j L_{i,j} = I_{i,j} B_i. \quad (21)$$

It intertwines A exactly. Its Gram decreases, because the larger source admits the old representative and additional original relations:

$$G_j \preceq G_i. \quad (22)$$

Its metric adjoint is the map in (1). To prove it, compute

$$\langle I_{i,j} u, v \rangle_{G_j} = u^* G_j v = u^* G_i (K_i G_j v) = \langle u, T_{i,j} v \rangle_{G_i}. \quad (23)$$

This is an adjoint of the **existing** transport. It is not asserted to be a unital algebra homomorphism.

Direct substitution into (18) proves the representative restriction identity

$$\boxed{L_{i,j}R_iT_{i,j} = P_{i,j}R_j.} \quad (24)$$

For example, the first $i + 1$ coordinates of R_j are $O_i^{-1}B_i^*G_j$; the right coordinates of $R_iT_{i,j}$ are $O_i^{-1}B_i^*G_iK_iG_j$, which are identical.

The adjoints compose exactly:

$$T_{i,j}T_{j,l} = K_iG_jK_jG_l = K_iG_l = T_{i,l}. \quad (25)$$

This agrees with contravariance of metric adjoints applied to the original identity transports. There is no intervening choice of a metric.

S30.4 4. Record the arithmetic action and the actual boundary correction

Set

$$H_N = G_N^{-1}(A^*G_N + G_NA - kG_N), \quad W_N = A^*G_N + G_NA - kG_N, \quad V_N = AK_N + K_NA^* - kK_N. \quad (26)$$

The adjoint-action identity gives

$$\boxed{AT_{i,j} - T_{i,j}A = H_iT_{i,j} - T_{i,j}H_j = V_iG_j - K_iW_j.} \quad (27)$$

For a direct proof, expand the rightmost expression and cancel the two occurrences of $K_iA^*G_j$ and of kK_iG_j . This records the full equivariance defect of the return map. The forward transport (21) is equivariant. The return need not be equivariant for the arithmetic action.

If ϵ_i, ϵ_j are the actual operator norms of H_i, H_j in their respective metrics, (22)–(23) imply the finite bound

$$\|AT_{i,j} - T_{i,j}A\|_{\mathcal{E}_j \rightarrow \mathcal{E}_i} \leq \epsilon_i + \epsilon_j. \quad (28)$$

No spectral subspace of the return map is therefore declared invariant under A merely because it is a useful metric diagnostic. Equation (27) is the explicit comparison.

Define the degree-restriction loss endomorphism

$$H_{i,j} = I_E - T_{i,j} = (K_j - K_i)G_j. \quad (29)$$

The original relation correction is

$$\boxed{\mathfrak{d}_{i,j} = L_{i,j}R_i - R_j = L_{i,j}R_iH_{i,j} - (I - P_{i,j})R_j.} \quad (30)$$

Both terms on the right have arithmetic remainder $H_{i,j}$. Their difference has zero remainder. Consequently

$$B_j\mathfrak{d}_{i,j} = 0, \quad \text{im } \mathfrak{d}_{i,j} \subseteq \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathfrak{d}_{i,j}^*O_j\mathfrak{d}_{i,j} = G_i - G_j. \quad (31)$$

The Gram identity follows because R_j is orthogonal to every relation in $\mathcal{D}_j$. Thus the lower-degree canonical representative is the higher-degree representative plus the explicit original relation (30).

The split map of a single active fibre is

$$\begin{aligned} I_{i,j}^\tau : \mathcal{E}_i^\tau &\rightarrow \mathcal{E}_j^\tau, & u^\bullet &\mapsto u^\bullet, & \tau &\mapsto \tau, \\ T_{i,j}^\tau : \mathcal{E}_j^\tau &\rightarrow \mathcal{E}_i^\tau, & u^\bullet &\mapsto (T_{i,j}u)^\bullet, & \tau &\mapsto \tau. \end{aligned} \quad (32)$$

These are $G(\mathbb{C})$ -linear single-fibre maps. The forward maps embed into the original directed support diagram. The reverse maps are the metric-adjoint diagram; an additional naturality across the original forward transports is not presumed. Their actual reverse-composition identity is (25). Inclusion of a single-fibre lift into the reconstructed carrier sends $u^\bullet$ to (ℓ_N, u) and its external τ to the carrier's external zero.

Under the original degree- j internal quotient, a supported value of $\mathfrak{d}_{i,j}$ becomes $(\ell_j, 0)$; external absence remains external absence. The relation before that map remains available in (30)–(31), including its original theta primitive.

S30.5 5. Exact volume as a restriction spectrum

The endomorphism $T_{i,j}$ is positive self-adjoint in both G_i and G_j :

$$G_i T_{i,j} = G_j, \quad G_j T_{i,j} = G_j K_i G_j. \quad (33)$$

Also $0 \prec T_{i,j} \preceq I$ in either metric. For G_i , this is (22); for G_j , use $K_i \preceq K_j$. Let its q eigenvalues, with multiplicities, be $g_a \in (0, 1]$.

The singular values of $I_{i,j} : \mathcal{E}_i \rightarrow \mathcal{E}_j$ are exactly $\sqrt{g_a}$ because $I_{i,j}^\dagger I_{i,j} = T_{i,j}$. The same nonzero singular values belong to $T_{i,j}$ in the reverse Hilbert types and to the source restriction (24). This is the concrete principal-angle construction for the canonical image and the lower-degree source; it uses their actual inner products.

Taking determinants gives

$$\boxed{\det T_{i,j} = \frac{\det K_i}{\det K_j} = \frac{V_j}{V_i}, \quad \mathcal{L}_{i,j} := \log \frac{V_i}{V_j} = - \sum_a \log g_a.} \quad (34)$$

The top exterior map $\bigwedge^q T_{i,j}$ acts by this literal determinant in the fixed determinant coordinates. With the source's unscaled alternating tensor representative, the determinant vector has squared norm $q! V_N$ at degree N . Both factorials remain in the ratio. This is determinant-line accounting through the same source-relation sequence as in the previous Deligne-oriented construction, not an imported purity conclusion.

An equivalent exact count is

$$\boxed{\mathcal{L}_{i,j} = \int_0^1 \frac{\#\{a : g_a < t\}}{t} dt = \int_0^\infty \#\{a : g_a < e^{-s}\} ds.} \quad (35)$$

For one $g \in (0, 1]$, the first integral is $\int_g^1 dt/t$; summing proves the formula. The second uses $t = e^{-s}$. Values at equality do not affect the integrals.

For the quartet endpoint window, (34) is applied twice, to

$$(i_0, j_0) = (q-1, 2q-1), \quad (i_1, j_1) = (q, 2q). \quad (36)$$

Their losses add to the original four-volume quantity $\mathcal{B}_{h,k}$.

S30.6 6. An exterior expansion retains every full-jet basis weight

For a q -element subset $D = \{d_1 < \dots < d_q\} \subseteq \{0, \dots, j\}$ define

$$a_D = \frac{|\det[b_{d_1}, \dots, b_{d_q}]|^2}{\prod_{d \in D} \omega_d}. \quad (37)$$

Cauchy–Binet applied to (18) gives

$$\boxed{\det K_j = \sum_{|D|=q} a_D.} \quad (38)$$

This is an identity of positive finite scalar observations, with no assignment of the sum to one. The actual exterior map giving the numerator is

$$\bigwedge^q B_j : \bigwedge^q \mathcal{P}_j \rightarrow \det E, \quad p_{d_1} \wedge \dots \wedge p_{d_q} \mapsto \det[b_{d_1}, \dots, b_{d_q}] e_0 \wedge \dots \wedge e_{q-1}. \quad (39)$$

The squared norm of the source wedge is $\prod_{d \in D} \omega_d$. The determinant-coordinate line used to read its coefficient is specified; it is not the canonical arithmetic Gram reset to the identity.

Introduce the finite scalar polynomial

$$Z_{i,j}(z) = \det(K_i + z(K_j - K_i)). \quad (40)$$

Multilinearity and Cauchy–Binet prove the full coefficient identity

$$Z_{i,j}(z) = \sum_{|D|=q} a_D z^{\#(D \cap \{i+1, \dots, j\})}. \quad (41)$$

In particular $Z(0) = \det K_i > 0$, $Z(1) = \det K_j$, and

$$\frac{Z_{i,j}(z)}{\det K_j} = \det(T_{i,j} + zH_{i,j}) = \prod_a (g_a + z(1 - g_a)). \quad (42)$$

The scalar division is the displayed observation map from the retained pair $(Z, \det K_j)$; the original total and mass remain in (38)–(41). This determinant generating mechanism is classical in projection determinantal measures. Its present interpretation concerns the original degree-admission and full-jet maps, not a randomly selected alternative arithmetic source.

The first two occupation ratios are

$$\frac{Z'(1)}{Z(1)} = \text{Tr } H_{i,j}, \quad \frac{Z''(1) + Z'(1)}{Z(1)} - \left(\frac{Z'(1)}{Z(1)} \right)^2 = \text{Tr}(H_{i,j} - H_{i,j}^2). \quad (43)$$

These identities follow by differentiating (42); every denominator is the original positive scalar total. The weighted counts in (41) remain available before taking those ratios.

There are two explicit ways a term can be missing from a scalar total. An admitted subset with dependent full-jet columns has $a_D = 0$, a supported zero. A subset containing an index above the admitted degree is outside that finite indexing object, and its extension to a larger indexing object is assigned external absence. For example, in the calibration $\chi = (S - 1)^2$, b_0 and b_2 are dependent: the subset $\{0, 2\}$ is admitted and has supported zero weight. The subset $\{0, 4\}$ is absent from degree three. The algebraic specialization $z = 0$ in (41) sends each positive-degree monomial to a supported scalar zero; its support record is not retroactively deleted.

Full derivatives are retained under a change to raw confluent jet coordinates. For roots λ_a of orders m_a , let

$$C_{\chi(a,r),b} = \partial_S^r S^b|_{S=\lambda_a} = \begin{cases} \frac{b!}{(b-r)!} \lambda_a^{b-r}, & b \geq r, \\ 0, & b < r. \end{cases} \quad (44)$$

Its determinant, in the stated block order, is

$$\det C_\chi = \left(\prod_a \prod_{r=0}^{m_a-1} r! \right) \prod_{a < b} (\lambda_b - \lambda_a)^{m_a m_b}. \quad (45)$$

For any explicitly retained invertible arithmetic unit matrix M_u , the quotient-coordinate change is $C_\chi M_u$. Every weight (37) acquires the same factor $|\det(C_\chi M_u)|^2$, both totals acquire it, and (34), (42) are unchanged through that specified ratio. The action and return are conjugated by this matrix, and the canonical representative is precomposed with its inverse. No nilpotent derivative order is removed.

For the actual cyclic inclusion $\eta : E \hookrightarrow B_k^{S_k}$, the ambient matrix can be rectangular. Its correct exterior comparison is

$$\bigwedge^q (\eta B_D) = (\det B_D) \left(\bigwedge^q \eta \right) (e_0 \wedge \cdots \wedge e_{q-1}).$$

The fixed nonzero image vector is retained. There is no determinant of a rectangular matrix being inverted. A square coordinate change $C_\chi M_u$ applies only on a specified coordinate model of the finite image, or for an actual unit endomorphism of E . In the ambient symmetric module the original map remains η , including the complete factor $v_h^{\otimes k}$.

S30.7 7. A certified finite logarithmic upper bound

Let $H = H_{i,j}$ and $s_m = \text{Tr}(H^m)$. Positivity in the original metric implies $s_m \geq 0$, and the eigenvalues lie in $[0, 1)$. Therefore

$$\mathcal{L}_{i,j} = \sum_{m=1}^{\infty} \frac{s_m}{m}. \quad (46)$$

A valid rational or real lower gap $g_0 > 0$ is supplied by the **actual matrix inequality**

$$K_i \succeq g_0 K_j. \quad (47)$$

It gives $H \preceq (1 - g_0)I$ in G_j . Let $r = 1 - g_0$. For integer $p \geq 1$ and $0 < r < 1$, set

$$c_p(r) = \frac{-\log(1-r) - \sum_{m=1}^p r^m/m}{r^{p+1}}. \quad (48)$$

The finite enclosure is

$$\sum_{m=1}^p \frac{s_m}{m} \leq \mathcal{L}_{i,j} \leq U_p(H, g_0) := \sum_{m=1}^p \frac{s_m}{m} + c_p(r)s_{p+1}. \quad (49)$$

For $r = 0$, (47) and $K_i \preceq K_j$ give $H = 0$ and both bounds are zero.

Proof. For $0 \leq x \leq r$, the tail of $-\log(1-x)$ is

$$x^{p+1} \sum_{m=p+1}^{\infty} \frac{x^{m-p-1}}{m} \leq x^{p+1} \sum_{m=p+1}^{\infty} \frac{r^{m-p-1}}{m} = c_p(r)x^{p+1}.$$

Apply this scalar inequality to the positive self-adjoint H and take its trace. For every fixed admitted pair and fixed certified g_0 , the lower and upper bounds converge to the same loss as $p \rightarrow \infty$. This is fixed-pair convergence, not a uniform assertion in tensor degree.

Already $p = 1$ gives

$$\mathcal{L}_{i,j} \leq s_1 + \frac{-\log g_0 - (1 - g_0)}{(1 - g_0)^2} s_2. \quad (50)$$

It retains a second moment rather than replacing every loss by the maximum one. Higher p retain the successive moments of the same operator. No eigenvalue of A is inferred from an eigenvalue of H ; their actual commutator is (27).

For the two windows in (36), verified gaps $g_0^{(0)}, g_0^{(1)}$ give

$$\mathcal{B}_{h,k} \leq U_p(H^{(0)}, g_0^{(0)}) + U_p(H^{(1)}, g_0^{(1)}). \quad (51)$$

Combining (5), the formalizer's endpoint theorem, and monotonicity of $\sinh$ gives the finite canonical-family estimate

$$\min_{q \leq N < 2q} \epsilon_{h,k,N} \leq C_h q \sinh \left(\frac{U_p(H^{(0)}, g_0^{(0)}) + U_p(H^{(1)}, g_0^{(1)})}{2q} \right), \quad q \geq k \geq 3. \quad (52)$$

This theorem uses the explicit matrix certificate (47). Existence of some positive gap for each finite pair does not imply the uniform gap or moment control that would finish the arithmetic argument. Formula (52) states exactly the quantity a certified finite or uniform analytic estimate must bound.

S30.7.1 Error-aware certificate types

The checker supplies exact rational lower and upper bounds for $\log x$ on positive rational inputs. It uses $x = 2^e y$, $1 \leq y < 2$, and

$$\log y = 2 \sum_{a=0}^{m-1} \frac{t^{2a+1}}{2a+1} + \mathcal{E}_m, \quad t = \frac{y-1}{y+1}, \quad 0 \leq \mathcal{E}_m \leq \frac{2t^{2m+1}}{(2m+1)(1-t^2)}. \quad (53)$$

The same formula encloses $\log 2$; reciprocation handles $x < 1$. Finite rational LDL or principal-minor checks certify (47). No floating-point eigenvalue is accepted as proof of that inequality.

For actual analytic matrix enclosures, a sufficient input is $K_i^- \preceq K_i$ and $K_j \preceq K_j^+$ together with $K_i^- \succeq g_0 K_j^+$. Then (47) follows by transitivity. The trace moments in (49) also need certified input enclosures, or exact arithmetic; they are not automatically exact because a gap was enclosed. Independently,

$$\mathcal{L}_{i,j} \leq \log \det K_j^+ - \log \det K_i^- \quad (54)$$

when both bounding matrices are positive definite. The implemented test fixtures have exact rational entries. This note supplies no interval certificate for the actual arithmetic integrals or root coordinates.

S30.8 8. The off-line alternative forces a precise restriction phenomenon

For the packet consisting exactly of a quartet with displacement $\delta > 0$, height $\gamma > 0$, and common multiplicity m , retain

$$q_k = [1 + k(m-1)](k+1)^2. \quad (55)$$

The inherited endpoint theorem and its proved norm input give

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right). \quad (56)$$

There are $2q_k$ positive eigenvalues g_a across the two return maps in (36). Equations (34) and (56) imply that at least one satisfies (6): otherwise every $-\log g_a$ would be smaller than the common lower average in (56), contradicting their sum.

An eigenvector v for that g_a is nonzero in the original arithmetic module. Equation (24) and the actual Gram calculation give

$$\boxed{\frac{\|P_{i,j} R_j v\|_{\mathcal{O}_j}^2}{\|R_j v\|_{\mathcal{O}_j}^2} = g_a, \quad \frac{\|(I - P_{i,j}) R_j v\|_{\mathcal{O}_j}^2}{\|R_j v\|_{\mathcal{O}_j}^2} = 1 - g_a.} \quad (57)$$

The old portion has arithmetic coordinate $g_a v$, not v . The exact original representative of v at degree i is recovered by applying $T_{i,j}^{-1}$ in (24). The difference from the full degree- j representative is the actual relation (30). Thus no arithmetic class is called absent, and no numerical near-zero is set equal to e .

The exact distributional statement is stronger than one selected direction:

$$\sum_{a=0}^1 \int_0^\infty \#\{b : g_b^{(a)} < e^{-s}\} ds = \mathcal{B}_{h,k}. \quad (58)$$

The growing-degree task can now be stated as control of this cumulative small-restriction spectrum, or as the positive trace certificates (51). A proved upper limit below 2 for $\mathcal{B}_{h,k}/(q_k \log k)$ would contradict (56). Such a uniform upper bound is not an output of the present finite identities.

S30.9 9. Exact calibration retaining nilpotence and mass

Use the explicitly declared algebraic fixture

$$E = \mathbb{C}[S]/((S-1)^2), \quad S = 1 + ix, \quad (59)$$

with the Gaussian source $7e^{-x^2/2}dx/\sqrt{2\pi}$, of literal mass seven and variance one. This fixture is not asserted to be a zeta packet. Its source monic polynomials and norms are

$$p_0 = 1, \quad p_1 = S - 1, \quad p_{n+1} = (S-1)p_n + np_{n-1}, \quad \omega_n = 7n!. \quad (60)$$

The coordinate basis $(1, S-1)$ is connected to $(1, S)$ by

$$C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad (1, S)C = (1, S - 1). \quad (61)$$

In the $(1, S - 1)$ coordinates, the kernel matrices are

$$\begin{aligned} K_1 &= \text{diag}(1/7, 1/7), & K_2 &= \text{diag}(3/14, 1/7), \\ K_3 &= \text{diag}(3/14, 5/14), & K_4 &= \text{diag}(15/56, 5/14). \end{aligned} \quad (62)$$

The action remains $A = I + N$ with $N \neq 0$ and $N^2 = 0$. It is not replaced by its reduced one-dimensional quotient.

The two return spectra are exactly

$$\text{Spec } T_{1,3} = \{2/3, 2/5\}, \quad \text{Spec } T_{2,4} = \{4/5, 2/5\}. \quad (63)$$

Consequently

$$\mathcal{B} = \log(15/4) + \log(25/8) = \log(375/32) \approx 2.46119012317068427. \quad (64)$$

For $H_{1,3}$ the first two trace moments are $14/15$ and $106/225$; for $H_{2,4}$ they are $4/5$ and $2/5$. Using $g_0 = 2/5$ and $p = 2$ in (49) gives the two rigorously enclosed upper bounds whose displayed decimals are

$$1.32854908781965490, \quad 1.14133853675838303. \quad (65)$$

Their sum is below 2.469887624578038 . The earlier direct bound by $\text{Tr}(K_i^{-1}(K_j - K_i))$ gives $2 + 7/4 = 3.75$ on these two windows. Thus the new finite certificate makes a strict, verified improvement on the same source and same metrics. The exact logarithmic enclosures, rather than the printed rounded values, certify the comparison.

The dependent subset $\{0, 2\}$ in (37) has a supported zero minor because $[p_2] = [p_0]$ in this quotient. Nevertheless the first derivative coordinate at $S = 1$ is retained and makes the quotient dimension two. A distinct-root replacement would not give the same matrices.

The test suite also contains off-line algebraic fixtures, nonconstant invertible jet units, and repeated confluent derivatives. It does not infer a zeta-zero statement from these fixtures.

S30.10 10. Integration with the Deligne-oriented programme

The same cohomological transition now carries an explicit positive adjoint and a determinant-line norm. Its action covariance is (27), its theta-boundary correction is (30), and its original support quotient is retained in (32). This is a metric calculation on the existing characteristic-zero realization over b_r .

Deligne's use of determinant weights and the compact-to-ordinary image requires an actual estimate on the same image. This note supplies additional calculation and finite upper certificates for that estimate; it does not invoke his finite-field purity hypotheses for the theta object. The inherited source review's strict inequality in Lemma 3.2.10 and its squared-eigenvalue step in 3.2.13 are retained. This turn did not newly reassemble the Deligne archive or claim a complete rereading.

Current repository input was checked at main 7ea0a49945390eae14d3160a5730858899768b5f. The current endpoint criterion retains the corrected top-exterior case and the actual second-exterior multiplicity cost. No second cost-free exterior amplification is used here. The proof status of the original norm bound (5) remains the inherited written analytic status; the present finite tests do not promote it to a Lean certificate.

The principal next quantity is no longer only an unnamed ratio of large determinants. It is the explicit log-loss of the original identity transports, with the canonical representative restriction, all original relations, full jets, arithmetic action, and complementary polynomial degrees attached. The uniform bound still needed is precisely the one on that constructed family.

S30.11 References and provenance

1. Owner-directed *Arithmetic endpoint bounds on the original tau-base source*, supplied TeX and archive, 13 September 2026. Equations (1), (35)–(39), and (43)–(48) are the inherited analytic input, endpoint composition, and original relation-layer formulas.

2. Parallel formalization session, `workbenches/tau-toda-volume-formal/ENDPOINT_CRITERION.md`, read at main `7ea0a49945390eae14d3160a5730858899768b5f`; fetched Git blob `09d658b1c50245b9e340722817bc9487f1ce409a`. Endpoint selection is its written Jensen theorem; the finite telescope components have their separately reported formal scope.
3. Original SplitZero reconstruction/internal quotient and supplied full-jet kernel-layer note. The present construction reuses their quotient and support maps.
4. Lyons, R. (2003). *Determinantal probability measures*. Publications mathématiques de l’IHES 98, 167–212; arXiv:math/0204325. Background for the exterior/Cauchy–Binet generating mechanism, not a source for the arithmetic objects constructed here.
5. Drmac, Z. (2000). *On principal angles between subspaces of Euclidean space*. SIAM Journal on Matrix Analysis and Applications 22, 173–194, doi:10.1137/S0895479897320824. Background for principal angles and singular values. All weighted source maps needed here are proved explicitly.
6. Deligne, P. (1980). *La conjecture de Weil. II*. Publications mathématiques de l’IHES 52, 137–252. Used through the inherited precise reading record; no new finite-field purity application is claimed.

S31 Complete original two-trace relation control

Source role. Complete delivered proof witness. Its historical claims retain their source revision; corrections and full current maps are proved in the corresponding integrated chapter.

Exact revision. 17e14c11ca60906604f66ddf9e6a758412a16143d5742736f3a5708658270795

Complete source. sources/web_relation_moment_delivery/Tau_Relation_Moment_Control/RESEARCH_NOTE.md

Source SHA-256. 17e14c11ca60906604f66ddf9e6a758412a16143d5742736f3a5708658270795

Source metadata: author (MetaInlines)

Owner-directed SplitZero analytic continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

An upper determinant certificate without a separately supplied restriction gap

Source metadata: title (MetaInlines)

Two-moment control of the original arithmetic relation blocks

S31.0.1 1. Result and the current inherited theorem

This note continues the canonical endpoint-restriction construction. It does not choose a different positive metric or replace the arithmetic source by a comparison measure. Its new application is an upper bound for each original relation-block volume from two source-derived traces, together with a complete positive remainder and a source interpretation of the second trace as exterior area.

The general sharp determinant optimization with prescribed dimension, trace, and squared trace is classical; Grone, Johnson, Marques de Sá, and Wolkowicz (1984) study precisely that optimization. The proof below is included in full, using a scalar Hermite interpolation identity. No priority claim is made for the general inequality. The work here is its composition with the actual arithmetic representative, internal quotient, inverse restriction, exterior source maps, and consecutive-window theorem.

For one original block, the output is

$$\boxed{\frac{V_i}{V_j} \leq (1 + a_{i,j})(1 + b_{i,j})^{q-1}.} \quad (1)$$

Both $a_{i,j}$ and $b_{i,j}$ are calculated from the first two traces of a specified endomorphism. There is no assumption that the arithmetic action is normal, no assumption that a restriction eigenspace is invariant under it, and no independently supplied lower gap for the restriction.

Integration of the two new reports The attachment reporting PR #25 supplies the phase-retaining consecutive product, the separate first-admissible-degree step, a two-sided written arithmetic norm envelope, and its comparable-window consequence. Its resulting off-line-quartet lower budgets are

$$\log(V_q/V_{2q}) \geq 2q \log(D_h k), \quad \log(V_{q-1}/V_{2q-1}) \geq 2q \log(D_h k), \quad (2)$$

for a fixed positive D_h , hence the four-volume lower threshold is **four**, not merely the earlier sufficient lower threshold two. This is inherited from the parallel contributions, not claimed as a discovery of this note. The source explicitly distinguishes its checked finite implications from the written analytic estimates and the asymptotic conclusion.

Its regularity clarification is retained: the actual arithmetic density is in $L^1 \cap L^\infty$, and

$$\|(\tau_a w_h - w_h) * w_h\|_\infty \leq \|\tau_a w_h - w_h\|_1 \|w_h\|_\infty \longrightarrow 0.$$

Here τ_a denotes translation of a function, not the split scalar τ ; its definition is $(\tau_a f)(t) = f(t - a)$. This supplies continuity of the convolution used in the source's lower-envelope argument.

The second attachment is a working transcript reporting further Gamma, coefficient-error, endpoint, and reflected-algebra work. Its reported moment certificates and unpublished upper-bound details are not independently reproduced here or silently made premises. In particular, this note does not duplicate its claimed fixed-budget extremizer or claim to have executed its local Windows commands.

GitHub main was read at 952ef9fee1e1419b6858d920354de8fa99430b7d. PR #25 was read at its reported verified head 66fac5e7885a40cd4873e74b754907320f05b067, including its full RESEARCH_NOT E.md. Its CI status is a source report; this session did not execute Lean.

S31.0.2 2. Original arithmetic source and support

Retain the original scalar object and amplitude map

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The original coefficient square remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (3)$$

Its arithmetic target $\mathbb{Z}$ is infinite. The geometric base remains $\mathfrak{b}_\tau$. The square does not replace the internal support fibres by the ordinary amplitude observation.

Keep the original theta complex and generator

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi.$$

For a nonempty full packet h of actual zeros and tensor degree k , retain the cyclic module and its action

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k}, \quad A = M_S, \quad S = s_1 + \cdots + s_k.$$

The source and arithmetic inclusion are the constructed maps

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = g/h,$$

$$\eta[P] = v_h^{\otimes k} P(A_k) 1, \quad v_h = j_h(g/h), \quad (4)$$

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k} \eta\pi_\chi.$$

The original norm is the integral against $m_{h,k} = w_h^{*k}$ on $S = k/2 + iu$, where

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad \int m_{h,k} = \mu_h^k.$$

No mass is assigned a new value. All nilpotent jet orders and the unit in (4) remain in the arithmetic maps.

For $h = 1$ the finite packet is the zero coefficient module. Its determinant is the empty determinant one, its log-volume is zero, and $G(0)$ still contains the two split elements. The analytic theta seed remains nonzero. Positive-dimensional formulas below use $q \geq 1$; the divisions by $q - 1$ use $q \geq 2$.

Fixed polynomial coordinates Let $p_0, p_1, \dots$ be the original monic orthogonal polynomials and $\omega_n > 0$ their actual squared norms. Keep $\omega_0 = \mu_h^k$. In fixed remainder coordinates on E , set

$$b_n = [p_n]_\chi, \quad B_N = [b_0, \dots, b_N], \quad O_N = \text{diag}(\omega_0, \dots, \omega_N).$$

For every admitted $N \geq q - 1$,

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad V_N = \det G_N. \quad (5)$$

Thus $B_N R_N = I$ and $R_N^* O_N R_N = G_N$. The actual test-function representative is obtained by the specified polynomial-column realization $\mathcal{V}$, not by treating a coefficient vector as a function without that map.

S31.0.3 3. The positive cost operator comes from the original restriction

Fix $q - 1 \leq i \leq j$. Set $r = j - i$ and retain the new columns and their norms

$$F = F_{i,j} = [b_{i+1}, \dots, b_j] : \mathbb{C}^r \rightarrow E, \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j),$$

$$D_{i,j} = K_j - K_i = F \Omega^{-1} F^*. \quad (6)$$

The Hilbert spaces $E_i = (E, G_i)$ and $E_j = (E, G_j)$ are connected by the actual forward transport $I_{i,j} : E_i \rightarrow E_j$, identity on the fixed coefficient space. Its adjoint is

$$T_{i,j} = I_{i,j}^\dagger = K_i G_j : E_j \rightarrow E_i.$$

To form an endomorphism of one specified Hilbert object, compose first:

$$Q_{i,j} = T_{i,j} I_{i,j} : E_i \rightarrow E_i.$$

The coefficient matrix of $Q_{i,j}$ is $K_i G_j$. Define

$$Z_{i,j} = Q_{i,j}^{-1} - I_E = (K_j - K_i) G_i = D_{i,j} G_i. \quad (7)$$

It is positive self-adjoint on E_i , because

$$G_i Z_{i,j} = G_i D_{i,j} G_i \succeq 0.$$

Its eigenvalues λ_a are nonnegative. The restriction eigenvalues in the preceding note are precisely

$$g_a = \frac{1}{1 + \lambda_a}, \quad \frac{V_i}{V_j} = \det(I_E + Z_{i,j}), \quad \log(V_i/V_j) = \sum_{a=1}^q \log(1 + \lambda_a). \quad (8)$$

No inverse-square-root transformation changes the canonical arithmetic metric. Equation (7) is a typed inverse of the already-constructed restriction endomorphism.

Source and action maps remain attached The source map $F : (\mathbb{C}^r, \Omega) \rightarrow (E, G_i)$ has adjoint

$$F^\dagger = \Omega^{-1} F^* G_i, \quad Z_{i,j} = F F^\dagger. \quad (9)$$

Let $H_N = G_N^{-1}(A^* G_N + G_N A - k G_N)$ be the inherited control endomorphism. Put $\hat{H}_j = I_{i,j}^{-1} H_j I_{i,j}$ on E_i . The previous action defect and inversion give

$$[A, Z_{i,j}] = \hat{H}_j (I + Z_{i,j}) - (I + Z_{i,j}) H_i. \quad (10)$$

Indeed $[A, Q] = H_i Q - Q \hat{H}_j$ and $[A, Q^{-1}] = -Q^{-1} [A, Q] Q^{-1}$. This is the actual relation between the cost operator and the arithmetic generator. A cost eigenspace is not presumed invariant for A .

S31.0.4 4. The two input traces are norms of actual source maps

Write

$$t_1 = \text{Tr } Z_{i,j}, \quad t_2 = \text{Tr}(Z_{i,j}^2), \quad \mathcal{A}_2 = \frac{t_1^2 - t_2}{2}. \quad (11)$$

The scalar $\mathcal{A}_2$ is the second exterior coefficient of this cost operator. Its split observation is $\mathcal{A}_2^\bullet \in G(\mathbb{R})$; a vanishing value maps to the represented scalar e .

From (6) and cyclicity of the finite trace,

$$t_1 = \sum_{n=i+1}^j \frac{b_n^* G_i b_n}{\omega_n},$$

$$t_2 = \sum_{n,m=i+1}^j \frac{|b_n^* G_i b_m|^2}{\omega_n \omega_m}, \quad (12)$$

$$\mathcal{A}_2 = \sum_{i < n < m \leq j} \frac{(b_n^* G_i b_n)(b_m^* G_i b_m) - |b_n^* G_i b_m|^2}{\omega_n \omega_m}. \quad (13)$$

The off-diagonal pairings in (12)–(13) are essential.

Let $\Phi = R_i F$ with its actual degree- i source norm. Then $t_1 = \|\Phi\|_{\text{HS}}^2$. Equip unscaled exterior bases with determinant Grams. The precise second identity is

$$\mathcal{A}_2 = \left\| \bigwedge^2 \Phi \right\|_{\text{HS}}^2. \quad (14)$$

Proof: in the coordinate wedge indexed by $n < m$, the squared source norm is $\omega_n \omega_m$. Its image squared norm is the two-by-two Gram determinant in (13). Summing proves (14). An alternating-tensor realization instead has its literal factor $2!$ in each wedge Gram; this statement uses the specified exterior quotient, so no factorial is dropped.

Let T_{new} denote the actual newly admitted orthogonal polynomial columns in the degree- j source. The source relation is

$$\mathbf{b}_{i,j} = T_{\text{new}} - L_{i,j} R_i F, \quad B_j \mathbf{b}_{i,j} = 0,$$

$$\mathbf{b}_{i,j}^* O_j \mathbf{b}_{i,j} = \Omega + F^* G_i F. \quad (15)$$

Thus (12)–(14) measure the old representative components needed to cancel the new columns' jets. The relation itself maps to the receiving supported zero through the original coequalizer. Neither component is declared absent.

An admitted pair with dependent images has zero exterior area in (13), a represented zero. The pair and its original support label remain available. A pair outside the degree-indexing object is absent. No statement that these two cases have the same upstream datum is used.

S31.0.5 5. A sharp two-trace upper bound, with a complete proof

Suppose $q \geq 2$. Define

$$d = \sqrt{\frac{qt_2 - t_1^2}{q-1}}, \quad a = \frac{t_1 + (q-1)d}{q}, \quad b = \frac{t_1 - d}{q}. \quad (16)$$

Both discriminants and signs come from the actual positive spectrum:

$$\begin{aligned} \frac{t_1^2}{q} \leq t_2 \leq t_1^2, \quad 0 \leq b \leq a, \\ a + (q-1)b = t_1, \quad a^2 + (q-1)b^2 = t_2. \end{aligned} \quad (17)$$

The spectral cap is proved, not supplied For any eigenvalue x of Z , Cauchy–Schwarz on the other $q-1$ eigenvalues gives

$$t_2 \geq x^2 + \frac{(t_1 - x)^2}{q-1}.$$

Solving this quadratic yields $x \leq a$. Hence the same two traces also produce the actual restriction-gap bound

$$Q_{i,j} \succeq \frac{1}{1+a} I_E \quad \text{in the original metric } G_i. \quad (18)$$

This is a finite derived bound, not a uniform estimate in k .

The optimal determinant bound from these two traces The result is

$$\log \det(I + Z) \leq U_2(Z) := \log(1+a) + (q-1) \log(1+b). \quad (19)$$

For $a > b$, define the quadratic

$$P_{a,b}(x) = \log(1+b) + \frac{x-b}{1+b} + c_{a,b}(x-b)^2,$$

$$c_{a,b} = \frac{\log(1+a) - \log(1+b) - (a-b)/(1+b)}{(a-b)^2} < 0. \quad (20)$$

It matches $\log(1+x)$ and its derivative at b , and the value at a . For $0 \leq x \leq a$ the exact remainder is

$$P_{a,b}(x) - \log(1+x) = (a-x)(x-b)^2 \int_1^\infty \frac{ds}{(s+x)(s+b)^2(s+a)} \geq 0. \quad (21)$$

Proof of (21): use $\log(1+x) = \int_1^\infty (1/s - 1/(s+x)) ds$ and the exact third divided difference of $1/(s+x)$. Interpolation at the repeated point b and at a gives the displayed rational denominator. The integral converges as s^{-4} ; all factors have their shown positive signs. This also directly proves the Hermite remainder without an unspecified intermediate point.

Sum (21) over the actual eigenvalues. Since a quadratic trace depends only on q, t_1, t_2 , equations (17) give

$$\text{Tr } P_{a,b}(Z) = P_{a,b}(a) + (q-1)P_{a,b}(b) = U_2(Z).$$

This proves (19) and (1).

If $a = b$, equality in $qt_2 \geq t_1^2$ forces every eigenvalue to equal b , so (19) is equality without the quotient in (20). If $q = 1$, the exact answer is $\log(1+t_1)$. If $t_1 = 0$, positivity gives $Z = 0$ and the volume is zero. If $q = 0$, use the empty determinant, without forming (16).

For $a > b$, the integrand in (21) is strictly positive except at $x = a, b$. Equality in (19) therefore holds exactly when the spectrum consists of one a and $q-1$ copies of b . These values have the original two traces by (17), so the bound is best possible given only those traces and the dimension. This does not assert that every such extremizer arises from an arithmetic packet.

The general optimization is classical. The source application uses precisely $Z = (K_j - K_i)G_i$, not an independently chosen Hermitian matrix.

S31.0.6 6. Retain the complete remainder; a third trace improves the bound

Let

$$C_3 = \text{Tr}((aI - Z)(Z - bI)^2) = -t_3 + (a+2b)t_2 - (2ab + b^2)t_1 + qab^2,$$

where $t_3 = \text{Tr } Z^3$. Positivity on the spectrum proves $C_3 \geq 0$.

The full loss in the upper bound is

$$U_2(Z) - \log \det(I + Z) = \int_1^\infty \frac{\text{Tr}((aI - Z)(Z - bI)^2(sI + Z)^{-1})}{(s + b)^2(s + a)} ds. \quad (22)$$

All four operator factors commute because they are functions of this same G_i -self-adjoint Z . The expression is positive in that same metric. No claim about commutation with A replaces (10).

Since $0 \leq x, b \leq a$, the integrand denominator in (21) is at most $(s + a)^4$. Therefore

$$\log \det(I + Z) \leq U_2(Z) - \frac{C_3}{3(1 + a)^3}. \quad (23)$$

This retains a concrete part of the higher-spectrum correction. It requires one further trace of the same source cost, not a new spectral gap or arithmetic realization.

Equation (19) is always at least as sharp as the trace-only bound

$$U_2(Z) \leq q \log(1 + t_1/q), \quad (24)$$

by scalar concavity and (17). Equality in (24), apart from a zero-dimensional case, requires $a = b$. It is not asserted to dominate every higher-moment certificate from the previous note; those can be retained and their certified upper bounds compared.

S31.0.7 7. The simplest bound exposes independent pair growth

For $q \geq 2$ and $t_1 > 0$, (17) implies

$$\mathcal{A}_2 = (q - 1)b \left(t_1 - \frac{q}{2}b \right).$$

Because $0 \leq b \leq t_1/q$,

$$\frac{\mathcal{A}_2}{(q - 1)t_1} \leq b \leq \frac{2\mathcal{A}_2}{(q - 1)t_1}, \quad a \leq t_1.$$

Consequently there is a radical-free finite upper bound

$$\frac{V_i}{V_j} \leq (1 + t_1) \left(1 + \frac{2\mathcal{A}_2}{(q - 1)t_1} \right)^{q-1}. \quad (25)$$

The quantity $\mathcal{A}_2$ is exactly the sum of the original two-column exterior areas (13), including every cross-pairing. The first trace alone cannot distinguish nearly parallel old representative corrections from many independent ones; (13) does make that distinction quantitatively.

If $\mathcal{A}_2 = 0$ and $t_1 > 0$, the positive operator has rank one, and the sharper formula (19) is the exact value $\log(1 + t_1)$. This is a statement about the rank of this specified comparison map, not a claim about uniqueness of the original scalar support or about the number of higher support fibres.

S31.0.8 8. Compose with the new consecutive-window theorem

The inherited phase-retaining product implies for every original admitted window $i, \dots, j - 1$, $r = j - i$, including $i = q - 1$ through its separately proved first-degree step,

$$\min_{i \leq N < j} \epsilon_N \leq \left(\frac{\omega_j}{\omega_i} \right)^{1/(2r)} \left(\frac{V_i}{V_j} \right)^{1/(2r)}. \quad (26)$$

This is credited to PR #25 and its source continuation. It uses the original norm and volume sequence. It does not require constructing an inverse before the first admissible degree.

Combining (1) and (26) proves the finite upper certificate

$$\min_{i \leq N < j} \epsilon_N \leq \left(\frac{\omega_j}{\omega_i} \right)^{1/(2r)} ((1+a)(1+b)^{q-1})^{1/(2r)}. \quad (27)$$

For either quartet block $(i, j) = (q, 2q)$ or $(q-1, 2q-1)$, the reviewed comparable-window norm bound supplies an inherited constant C_h^{win} independent of k . Enlarge it once to account for $i = q-1$. Then

$$\min_{i \leq N < j} \epsilon_N \leq C_h^{\text{win}} q(1+a)^{1/(2q)} (1+b)^{(q-1)/(2q)}. \quad (28)$$

The selected member is still a canonical degree in the original window. Its source relation, arithmetic unit, and action are not chosen again after selecting the degree.

Quantitative consequence of the already established off-line lower bound For the packet consisting exactly of an off-line quartet with full common multiplicity, retain

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} \geq \delta k q_k / 2.$$

Let $D_h > 0$ be the fixed constant in the inherited per-block lower inequality (2). The new upper bound implies, on each original block,

$$(D_h k)^{2q} \leq (1+a)(1+b)^{q-1}. \quad (29)$$

In the source area quantities it implies

$$\frac{2\mathcal{A}_2}{(q-1)t_1} \geq \left(\frac{(D_h k)^{2q}}{1+t_1} \right)^{1/(q-1)} - 1, \quad (30)$$

whenever that hypothetical lower budget is positive, so $t_1 > 0$. When the right side is negative the inequality remains valid but has no positive force.

This strengthens the earlier statement that the mean relation cost must grow: unless the total cost has an extremely large individual scale, there must also be substantial *two-direction exterior area* in the original correction map. It is not a lower bound on every correction eigenvalue.

For example, along such a family, a proved estimate $\log(1+t_1) = o(q \log k)$ would turn (30) into the necessary lower exponent

$$\liminf \frac{\log(1 + 2\mathcal{A}_2/((q-1)t_1))}{\log k} \geq 2.$$

This last statement is a consequence with that displayed premise, not a claimed arithmetic estimate. The actual unresolved calculation is to bound the explicit sums (12)–(13) along the required growing-degree quartet family. Their uniform size has not been established in this note.

For the two-block total, (19) yields

$$\mathcal{B}_{h,k} \leq U_2(Z^{(0)}) + U_2(Z^{(1)}).$$

The inherited lower threshold remains four. No new purity conclusion is inferred from finite positivity or the optimality of a generic determinant inequality.

S31.0.9 9. A finite certificate robust to moment enclosures

The sharp formula (16) uses exact traces. One must not insert uncertified decimal traces into it and call the result an upper certificate.

There is a direct error-aware alternative. Suppose proved enclosures give $t_1 \in [t_1^-, t_1^+]$ and $t_2 \in [t_2^-, t_2^+]$. A rational cap $A_0 \geq t_1^+$ bounds every nonnegative eigenvalue. Choose a rational $0 \leq B_0 < A_0$ and use the explicit polynomial P_{A_0, B_0} in (20).

The same positive remainder proves

$$\log \det(I + Z) \leq q \log(1 + B_0) + \frac{t_1 - qB_0}{1 + B_0} + c_{A_0, B_0}(t_2 - 2B_0 t_1 + qB_0^2). \quad (31)$$

Every coefficient in (31) is a rational combination of two logarithms of positive rationals. Rational interval arithmetic, including all products, gives a valid upper enclosure from the two trace intervals. It needs no separately certified bound $K_i \geq g_0 K_j$.

The checker encloses $\log x$ by writing $x = 2^e y$, $1 \leq y < 2$, and retaining both factors in

$$\log y = 2 \sum_{n=0}^{M-1} \frac{z^{2n+1}}{2n+1} + R_M, \quad z = \frac{y-1}{y+1}, \quad 0 \leq R_M \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

The same formula encloses $\log 2$; reciprocation handles $x < 1$. These finite interval checks certify their supplied rational data only. An arithmetic application must supply certified moment/jet enclosures and hence valid trace intervals.

Arithmetic input size remains explicit For a direct construction through degree j , the source monic coefficients and norms are determined by the original convolution moments through degree $2j$, and the quotient columns retain the full coefficients of χ and the specified arithmetic inclusion. The sums (12)–(13) use G_i ; they do not require a second inverse G_j or an independent spectral-gap proof. This is not a claim that the required source moments are low-order or uniformly bounded. In the two length- q endpoint blocks, this direct route can require moments through $4q$. The parallel session's separate lower-input upper-volume comparison is not being replaced or credited as proved here.

Coordinate and mass propagation Under a fixed invertible quotient-coordinate map C , the actual data become

$$F \mapsto CF, \quad G_i \mapsto C^{-*} G_i C^{-1}, \quad Z \mapsto CZC^{-1}.$$

Thus both traces, $\mathcal{A}_2$, and the bound remain unchanged through that similarity. Raw confluent derivative coordinates keep their factorials in C ; the map is not silently made unitary.

A common source mass multiplier $c > 0$ gives

$$\Omega \mapsto c\Omega, \quad K_N \mapsto c^{-1} K_N, \quad G_N \mapsto cG_N,$$

so $Z, t_1, t_2, \mathcal{A}_2$ are unchanged by these displayed composites. The original mass has not been reset. An ambient rectangular arithmetic inclusion η is retained as an injection, not used as an invertible square coordinate map.

S31.0.10 10. Exact source calibrations

These are declared Gaussian polynomial fixtures, not asserted zeta packets. Use original coordinate $S = 1 + ix$, literal Gaussian mass seven, variance one, and quotient $\chi = (S - 1)^q$. The monic source recurrence and norms are

$$p_0 = 1, \quad p_1 = S - 1, \quad p_{n+1} = (S - 1)p_n + np_{n-1}, \quad \omega_n = 7n!.$$

All quotient matrices below are computed in the original remainder basis $1, S, \dots, S^{q-1}$.

For $q = 2$, the blocks (1, 3) and (2, 4) give

$$(t_1, t_2) = \left(2, \frac{5}{2}\right), \quad (t_1, t_2) = \left(\frac{7}{4}, \frac{37}{16}\right).$$

Their exact two-trace factors are $(a, b) = (3/2, 1/2)$ and $(3/2, 1/4)$, so

$$\frac{V_1}{V_3} = \frac{15}{4}, \quad \frac{V_2}{V_4} = \frac{25}{8}, \quad B = \log(375/32).$$

The two-moment upper bound is exact in dimension two. This matches the preceding restriction example without supplying a spectral gap.

For $q = 3$, the original block $(3, 6)$ has

$$t_1 = 11, \quad t_2 = \frac{3365}{32}, \quad \mathcal{A}_2 = \frac{507}{64}, \quad \frac{V_3}{V_6} = \frac{5145}{256},$$

$$a = \frac{11}{3} + \frac{7\sqrt{127}}{12}, \quad b = \frac{11}{3} - \frac{7\sqrt{127}}{24}.$$

Numerical evaluations of these exact expressions are

$$\log(V_3/V_6) \approx 3.000603203789, \quad U_2 \approx 3.063328308618, \quad 3 \log(1 + t_1/3) \approx 4.621335122841.$$

The exact checks use rational enclosures and polynomial identities, not these decimals as proof. The original operator has its full nonzero nilpotent jets. No inference from a restriction eigenvalue to an off-line arithmetic zero is made.

S31.0.11 11. What is established and what is handed onward

Established here are the source cost map (7)–(10), the first and second exterior source norms (12)–(15), the sharp two-trace upper bound with a proved cap (16)–(21), its positive remainder and third-trace refinement, the area-only upper certificate, and their composition with the current first-degree-correct consecutive theorem. These are written finite theorems; exact algebraic regression checks support their implementation. They are not additional Lean certificates.

The same canonical arithmetic volume is now bounded without a separate restriction-gap premise. Estimating the growing arithmetic values of (12)–(13) remains a real analytic task. This note does not assert a uniform bound completing RH, does not erase a non-radical packet, and does not turn an e -valued quotient observation into absence.

No remote branch, publication cut, or formalization source was changed. The handoff lists bounded reusable targets for the parallel session and leaves its own interval-moment and fixed-budget work untouched.

S31.0.12 References and source locators

- Owner-supplied *Tau Endpoint Restriction Control*, NOTE.tex and original checker in the 13 September 2026 ZIP. Read as source; its finite map definitions are inputs here.
- Owner-supplied Pasted markdown(20260913-004825).md: PR #25 report, norm-review clarification, exact product, first-degree step, and threshold-four result. Source claims are distinguished from this session's executions.
- Owner-supplied Pasted markdown (2)(5).md: ongoing collaboration transcript; partial reported results are not substituted for their missing full proofs.
- GitHub KokunoYumeto/zeta-function-research-reader, main 952ef9fee1e1419b6858d920354de8fa99430b7d; PR #25 66fac5e7885a40cd4873e74b754907320f05b067, workbenches/tau-consecutive-window-formal/RESEARCH_NOTE.md.
- Grone, B., Johnson, C. R., Marques de Sá, E., and Wolkowicz, H. (1984). *Improving Hadamard's inequality*. Linear and Multilinear Algebra 16(1–4), 305–322. DOI: 10.1080/03081088408817634. The publisher abstract and author-hosted bibliography were inspected; the present proof is given explicitly rather than attributed to an unread theorem number.
- NIST Digital Library of Mathematical Functions, §3.3, especially the divided-difference and confluent interpolation formulas. <https://dlmf.nist.gov/3.3>

S32 Complete original polynomial-exponential cohomology comparison

Source role. Complete delivered proof witness. Its historical claims retain their source revision; corrections and full current maps are proved in the corresponding integrated chapter.

Exact revision. c2246c9f6046347bbd246890db88b1676b1a53487d66583a13b5883528bc0402

Complete source. sources/web_deligne_exponential_delivery/Tau_Deligne_Exponential_Comparison/RESEARCH_NOTE.md

Source SHA-256. c2246c9f6046347bbd246890db88b1676b1a53487d66583a13b5883528bc0402

Source metadata: author (MetaInlines)

Owner-directed SplitZero research continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

A flat rank- q deformation, its exact operator connection, and the logarithmic image of the preceding comparison curve

Source metadata: title (MetaInlines)

Deligne's exponential-family step and the original arithmetic packet

S32.0.1 1. What is new, and what the source actually supplies

This continuation reads further into Deligne's *La conjecture de Weil II*, especially §§3.7.2–3.7.4, while completing the cohomological calculation of the preceding representative curve. There are two new applications, with different roles.

First, the logarithmic de Rham complex of that curve is calculated, rather than inferred from its underlying coherent bundle. Its arithmetic coefficients occur in degrees zero and two; its degree-one middle image is zero. This identifies it as a constant-coefficient comparison curve, not as a nonconstant constituent in Deligne's pencil argument. The full coefficients and the source-induced metric are retained by explicit maps.

Second, an actual polynomial-exponential family is built from the existing cyclic relation polynomial. Its characteristic-zero cohomology is a free module of exactly the original packet rank, including at repeated roots. Its special fibre is the original cyclic quotient, and the original sum operator is the exact polar coefficient of its parameter connection. Its finite-field realizations have precisely the form to which Deligne applies his exponential-sum theorem. The two realizations are exhibited as base changes of the same polynomial data, not declared isomorphic and not assigned the same operator.

The new algebraic statements below have proofs. The finite-field purity statement is an application of Deligne's theorem under the hypotheses verified in Section 9. No arithmetic norm comparison between these exponential periods and the original theta metric is proved here. In particular, the finite-field Frobenius has not been identified with the original sum operator.

S32.0.2 2. Reading Deligne, rather than using the word purity

The supplied S20 archive was reassembled in this turn. All six binary parts were checked against their declared lengths and SHA-256 hashes. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256 e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The retained French and English LaTeX were read at the selected passages recorded in SOURCE_REVIEW.md, alongside the current page-local French records. No PDF extraction or OCR was used. The historical TeX is retained as a witness, not automatically treated as an error-free edition.

On printed pages 202–203, Deligne's argument has three separate inputs: the exceptional-fibre exact sequence, the integer weights of nonconstant constituents, and the independent bound strictly below two. Constant constituents have zero first cohomology on the projective line. The tensor-square step then improves the bound on the same Frobenius eigenvalue. On page 206, compact-support and ordinary-cohomology bounds meet on the image of their actual comparison.

The additional reading is printed pages 215–216. For a degree- d polynomial with d prime to the

residue characteristic and nonsingular leading projective hypersurface, Deligne proves concentration of the exponential-sheaf cohomology in degree n , dimension $(d-1)^n$, and weight n . His next step is to put the polynomials into a parameter space, prove lissity by local acyclicity including at infinity, and use connectedness and the weight results to reduce to a particular polynomial. The purity propagation used in that argument is explicitly stated on page 178, Corollary 1.8.12: on a connected base, a lisse mixed sheaf pure at one point is pointwise pure. Its proof passes through the filtration into pure constituents, rather than inferring purity merely from constant rank. This supplies an actual geometric family to build from our relation polynomial. The supplied reference for the final monomial case has an internal numbering mismatch, recorded in SOURCE_REVIEW.md; the displayed exponential theorem is used with its stated scope, not represented as a newly verified entire historical proof.

The source does not say that the critical points of a polynomial have real part one half. The operator singled out by its purity theorem is finite-field Frobenius. The connection coefficient calculated below is a different output of the same potential data, linked through the displayed realization diagram rather than an assumed equality.

S32.0.3 3. Original data and their supported realization

Retain the scalar construction and its specified maps:

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r. \quad (1)$$

The original coefficient square remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (2)$$

The first target is infinite. Every lifted scalar specialization below fixes external absence and sends a represented scalar zero to the represented zero of the receiving ring.

Keep the original theta complex, source function, full packet, and sum coordinate:

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi, \quad h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}. \quad (3)$$

At tensor degree k , write

$$E = \mathbb{C}[S]/(\chi), \quad \chi = \chi_{h,k}, \quad q = \deg \chi > 0, \quad A = M_S, \quad (4)$$

where $S = s_1 + \cdots + s_k$, not its average. The existing cyclic inclusion is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_k) 1, \quad v_h = j_h(g/h). \quad (5)$$

Its characteristic polynomial as a cyclic sum module need not be the characteristic polynomial of the entire tensor module. The actual injection (5), the previously constructed equivariant retraction, and its complementary module retain that distinction.

For an admitted degree N , retain the original canonical representative R_N and the source map $\mathcal{V}$:

$$J^{(k)} \mathcal{V} R_N = \eta, \quad q^{(k)} \mathcal{V} R_N = \sigma_h^{\otimes k} \eta, \quad R_N^* O_N R_N = G_N. \quad (6)$$

The monic norms in O_N have their original mass $\omega_0 = \mu_h^k$. No metric is chosen in this note to make an operator unitary. The empty-packet case has $E = 0$ and is handled outside all formulas requiring q positive.

S32.0.4 4. First compute the logarithmic cohomology of the preceding curve

For an original degree pair $i \leq j$, retain

$$\Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j, \quad R(z) = R_j + z\Delta, \quad G(z) = G_j + |z|^2 C. \quad (7)$$

The predecessor constructs the image lattice on $X = \mathbb{P}_{\mathbb{C}}^1$, with boundary $D_{\infty} = \{\infty\}$ and $w = 1/z$:

$$\mathcal{M}_{E_0} = E_0\mathcal{O} + wE\mathcal{O}, \quad E_0 = \ker \Delta. \quad (8)$$

Its smallest A-stable enlargement is $\mathcal{M}_F = F\mathcal{O} + wE\mathcal{O}$, where

$$F = E_0 + AE_0 + \cdots + A^{q-1}E_0. \quad (9)$$

All statements below concern this original target E. Let $Q_F = E/F$ denote this finite quotient, not the theta quotient Q.

4.1 The actual differential

For any positive integer m, define the lattice

$$\mathcal{M}_{F,m} = F\mathcal{O} + \mathcal{O}(-mD_{\infty})E \subset E\mathcal{O}, \quad (10)$$

and its two-term logarithmic complex, in degrees zero and one,

$$\mathcal{K}_{F,m} = [\mathcal{M}_{F,m} \xrightarrow{d} \mathcal{M}_{F,m} \otimes \Omega_X^1(\log D_{\infty})]. \quad (11)$$

The differential is the restriction of the constant coefficient connection: in the local frame w^me its coefficient is $w\partial_w + m$. Consequently it preserves (10) after tensoring with logarithmic forms. Because F is A-stable, A acts on the entire complex. A choice of complement $E = F \oplus J$ gives the coefficient description $\mathcal{M}_{F,m} = F\mathcal{O} \oplus J\mathcal{O}(-m)$; the action can have an off-diagonal block and is not asserted to split with that choice.

Computed cohomology. The canonical identifications as A-modules are

$$\boxed{\mathbb{H}^0(X, \mathcal{K}_{F,m}) = F, \quad \mathbb{H}^1(X, \mathcal{K}_{F,m}) = 0, \quad \mathbb{H}^2(X, \mathcal{K}_{F,m}) = Q_F.} \quad (12)$$

Here is a direct calculation retaining the coefficients. For the compactly supported component, use the two affine charts and the finite frame at z. The Cech groups are

$$H^1(\mathcal{O}(-mD_{\infty})) = \text{span}\{z^{-1}, \dots, z^{-(m-1)}\},$$

$$H^1(\mathcal{O}(-mD_{\infty}) \otimes \Omega^1(\log D_{\infty})) = \text{span}\{z^{-1}dz, \dots, z^{-m}dz\}.$$

The induced differential is exactly

$$z^{-a} \mapsto -a z^{-(a+1)}dz, \quad 1 \leq a < m. \quad (13)$$

It is injective, and its cokernel is the line generated by $z^{-1}dz$. For the ordinary logarithmic component the complex is $[\mathcal{O} \rightarrow \mathcal{O}(-1)]$, with only the constant H^0 . The exact sequence of complexes with subobject F and quotient Q_F then gives (12) canonically, without an A-invariant complement. In particular the degree-two identification is residue in the displayed coordinate.

If the original arithmetic cochain degree k is restored by the shift $[-k]$, the coefficient groups are in degrees k and k+2. For every polynomial f,

$$\boxed{\text{Str}(f(A) \mid R\Gamma\mathcal{K}_{F,m}[-k]) = (-1)^k (\text{Tr } f(A|F) + \text{Tr } f(\bar{A}|Q_F)) = (-1)^k \text{Tr } f(A|E).} \quad (14)$$

The quotient retains all induced Jordan data. This is trace additivity on the original exact sequence, not deletion of its second term.

4.2 Compact, middle, and ordinary comparisons

Put

$$\mathcal{K}_! = [E\mathcal{O}(-D_{\infty}) \rightarrow E\Omega_X^1], \quad \mathcal{K}_{\text{mid}} = [E\mathcal{O} \rightarrow E\Omega_X^1], \quad \mathcal{K}_* = [E\mathcal{O} \rightarrow E\Omega_X^1(\log D_{\infty})]. \quad (15)$$

Their natural inclusions compute the usual compact-support, smooth middle, and ordinary comparisons for the constant coefficient system on $\mathbb{A}^1$. At m=1 the mixed complex fits $\mathcal{K}_! \rightarrow \mathcal{K}_{F,1} \rightarrow \mathcal{K}_*$. The cohomology groups are

$$H^a(\mathcal{K}_!) = \begin{cases} E & a = 2, \\ 0 & a \neq 2, \end{cases} \quad H^a(\mathcal{K}_*) = \begin{cases} E & a = 0, \\ 0 & a \neq 0, \end{cases} \quad (16)$$

and $\mathcal{K}_{\text{mid}}$ has E in degrees zero and two. In particular,

$$\boxed{\text{im}(H_c^1(\mathbb{A}^1, E) \rightarrow H^1(\mathbb{A}^1, E)) = 0.} \quad (17)$$

This is an application of Deligne's constant-constituent test to the actual constructed curve. It locates its role: it is a metric comparison with retained boundary data, not yet the nonconstant cohomological image carrying E in degree one. The constructive continuation to such a cohomological family begins in Section 5.

The two boundary comparisons remain explicit. For $m=1$, the termwise quotient $\mathcal{K}_*/\mathcal{K}_{F,1}$ is the punctual complex $[Q_F \xrightarrow{0} Q_F]$ in degrees zero and one, while $\mathcal{K}_{F,1}/\mathcal{K}_! = [F \xrightarrow{0} F]$. For general m the first quotient has local coefficient differential

$$w^a \mapsto aw^a dw/w, \quad 0 \leq a < m. \quad (18)$$

Every positive a is paired with its exact inverse coefficient $1/a$; the constant class survives in both degrees. Thus the extra coherent length is kept before its logarithmic differential is taken. In the split construction those cycles, boundaries, and their quotients use the original fibrewise operations. External absence is never used as the image of a represented boundary.

4.3 Ramification and saturation retain the metric cost The ramified parameter map $f_m : \mathbb{P}^1 \rightarrow \mathbb{P}^1$, $z \mapsto z^m$, gives the actual representative $R_m(z) = R_j + z^m \Delta$. Its pulled-back determinant curvature has radial density

$$\kappa_m(t) = m^2 t^{m-1} \text{Tr}(Z(I + t^m Z)^{-2}), \quad Z = G_j^{-1} C. \quad (19)$$

Substitution $v = t^m$ proves

$$\boxed{\int_0^\infty \kappa_m(t) dt = m \text{rank } C, \quad \int_0^1 \log(1/t) \kappa_m(t) dt = \log(V_i/V_j).} \quad (20)$$

The pullback on the surviving degree-two de Rham class is $dz/z \mapsto m dz/z$. It is not an identity disguised by dividing the trace by m .

In the A -stable enlargement, let $s_A = \dim F - \dim E_0$. Its determinant metric in the infinity chart is the old one multiplied by $|w|^{-2ms_A}$. With the retained curvature definition $(i/2\pi)\partial\bar{\partial} \log \det G$, the current is

$$\boxed{c_1((\det \mathcal{M}_{F,m})^\vee, h^\vee) = \mu_m - ms_A \delta_\infty.} \quad (21)$$

Indeed $(i/2\pi)\partial\bar{\partial} \log |w|^2 = \delta_\infty$. Its total mass is $m(q - \dim F)$, the degree of the dual determinant. Full saturation subtracts $m \text{rank } C$ at infinity and has total degree zero, but leaves the endpoint logarithmic integral (20) unchanged. The negative atom is a geometric current; it is not a negative support label.

S32.0.5 5. An actual rank- q cohomological deformation of the cyclic packet

Write the original monic polynomial in its original coordinate:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a. \quad (22)$$

Introduce two new parameters u and t . They are independent of the scalar τ , the arithmetic coordinate S , the logarithmic comparison parameter z , and the tensor degree k . Put $R = \mathbb{C}[u, t]$ and define

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1}, \quad \Phi_t(S) = \Phi(S) - tS. \quad (23)$$

This is the specified primitive with $\Phi(0) = 0$, with every coefficient $1/(a+1)$ written. It changes no source coefficient in $g = 2\xi$. Define the two-term complex

$$\mathcal{C}_{u,t} = [R[S] \xrightarrow{\mathcal{D}_{u,t}} R[S]dS], \quad \mathcal{D}_{u,t}(P) = (uP'(S) + (\chi(S) - t)P(S))dS, \quad (24)$$

in degrees zero and one.

Flatness and exact rank. There is a direct sum of R-modules

$$R[S] = \mathcal{D}_{u,t}(R[S])/dS \oplus R[S]_{<q}. \quad (25)$$

In particular,

$$H^0(\mathcal{C}_{u,t}) = 0, \quad \mathcal{H} := H^1(\mathcal{C}_{u,t}) \cong R^q, \quad \mathcal{H}/(u,t)\mathcal{H} \cong E. \quad (26)$$

Proof. If P has degree a and leading coefficient b, then $uP' + (\chi - t)P$ has degree $a+q$ and leading coefficient b. It cannot be zero. For any polynomial F of degree at least q, subtract $\mathcal{D}_{u,t}(bS^{\deg F - q})/dS$ to remove exactly its highest term. Repeating terminates with a remainder of degree below q. Uniqueness follows from the leading-degree statement. The coefficients stay in R throughout. The same proof survives arbitrary base change of R, so (25) is a split exact presentation, not only a dimension count over its fraction field. Specializing $u=t=0$ turns (24) into multiplication by the literal χ , proving the final isomorphism.

Let $\text{red}_{u,t}$ denote the resulting coefficient-linear remainder map and $s_{<q}$ its specified polynomial section. The fibre map in (26) is $[P]_{\mathcal{D}} \mapsto [P|_{u=t=0}]_{\chi}$, with inverse induced by the chosen remainder basis.

There is no ring quotient asserted for u nonzero. Its exact relationship to multiplication is the commutator

$$[\mathcal{D}_{u,t}/dS, M_S] = uI. \quad (27)$$

For example, $\mathcal{D}(1)/dS = \chi - t$, while $\mathcal{D}(S)/dS = u + S(\chi - t)$; hence $[S(\chi - t)] = -u[1]$ in $\mathcal{H}$. The image of the differential is generally not an ideal. Formula (25) supplies the coefficient module and its section; the next paragraph supplies the operator that really descends.

S32.0.6 6. The parameter connection contains the original sum operator

On $\mathcal{H}[u^{-1}]$, define

$$\nabla_t[P] = [\partial_t P - SP/u]. \quad (28)$$

The exact relation

$$[\mathcal{D}_{u,t}/dS, \partial_t] = I, \quad [\mathcal{D}_{u,t}/dS, -M_S/u] = -I \quad (29)$$

proves that their sum commutes with the differential. Thus (28) descends without asserting that multiplication by S separately descends.

In the specified remainder basis $1, S, \dots, S^{q-1}$, the connection is

$$\nabla_t = \partial_t - \frac{A(t)}{u}, \quad A(t) = A + t\mathcal{R}, \quad \mathcal{R} = e_0 e_{q-1}^T. \quad (30)$$

Here $Ae_a = e_{a+1}$ for $a < q-1$ and

$$A(t)e_{q-1} = (-c_0 + t)e_0 - c_1 e_1 - \dots - c_{q-1} e_{q-1}. \quad (31)$$

Only a degree-q polynomial has to be reduced in deriving (30); $\mathcal{D}(1) = \chi - t$ gives exactly (31), including the sign of t.

The operator $-u\nabla_t = -u\partial_t + A(t)$ is regular in u and obeys the exact coefficient rule $(-u\nabla_t)(fv) = f(-u\nabla_t)v - u(\partial_t f)v$. It is a parameter-scaled connection in the t direction, not a derivative in u. Its reduction at $u=0$ is the honest endomorphism $A(t)$, and

$$\boxed{(-u\nabla_t) \bmod (u, t) = A = M_S \text{ on } E.} \quad (32)$$

This is the precise operator bridge. It contains a minus sign fixed by (24) and (28). The original arithmetic scaling is not renamed Frobenius.

Carry the parameter through the original theta representative Let $r_N = \mathcal{V}R_N : E \rightarrow \mathcal{B}^{\widehat{\otimes} k}$. On its specified test-function domain, use the earlier full jet map followed by the fixed equivariant cyclic retraction to obtain j_E with

$$j_E r_N = I_E, \quad j_E D^{(k)} = A j_E, \quad j_E \text{ kills the original theta boundaries.} \quad (33)$$

The construction of j_E uses the retained equivariant retraction, not an arbitrary quotient of the full tensor packet. Define the concrete finite-rank perturbation

$$\boxed{D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E.} \quad (34)$$

Then

$$j_E D_t^{(k)} = A(t) j_E, \quad \boxed{D_t^{(k)} r_N - r_N A(t) = D^{(k)} r_N - r_N A = dK_N.} \quad (35)$$

Both identities follow by substitution. The added term vanishes on every original boundary because j_E does. Thus the deformation has an explicit source realization preserving the original relation primitive. At $t=0$ it is exactly the original generator. At t nonzero it is the displayed finite-rank deformation, not a new assertion about the zero set of g .

The canonical arithmetic metric remains G_N . Its parameter control is explicitly

$$\boxed{W_N(t) = W_N(0) + t G_N \mathcal{R} + \bar{t} \mathcal{R}^* G_N.} \quad (36)$$

No positive metric has been selected from the exponential family to replace G_N .

6.1 The exponential periods are constructed and proved to span Fix a nonzero complex u , write $d=q+1$, and fix an argument of u . Set

$$\theta_0 = \frac{\arg u + \pi}{d}, \quad \theta_j = \theta_0 + \frac{2\pi j}{d} \quad (0 \leq j < d). \quad (36a)$$

Let ℓ_j be the oriented ray from zero to infinity with angle θ_j , and set $\Gamma_j = -\ell_0 + \ell_j$ for $1 \leq j \leq q$. The minus sign makes the initial endpoint contributions cancel. For a polynomial P define

$$\mathcal{P}_j(P) = \int_{\Gamma_j} e^{\Phi_t(S)/u} P(S) dS, \quad \Pi_{jb}(u, t) = \mathcal{P}_j(S^b), \quad 0 \leq b < q. \quad (36b)$$

These are absolutely convergent contour integrals. Along each ray the leading exponent is $-r^d/(d|u|)$. Lower-degree terms are dominated by that negative term outside a sufficiently large compact interval, uniformly when the lower coefficients and t range over a compact set. The same estimate after multiplying by any polynomial proves holomorphic dependence on those parameters and justifies their differentiation under the integral.

Integration of the actual total derivative gives

$$\mathcal{P}_j(uP' + (\chi - t)P) = u \int_{\Gamma_j} d(e^{\Phi_t/u} P) = 0. \quad (36c)$$

Both ends at infinity have zero boundary value, and the two values at zero cancel. Consequently the periods descend through the very differential (24), not through a quotient chosen afterwards.

The period matrix is invertible for every choice of the lower coefficients and every t , at this fixed nonzero u . Here is a direct proof, including the singular critical-point cases.

Temporarily regard the coefficients $c_0, \dots, c_{q-1}$ as independent parameters. The coefficient-direction connections are

$$\nabla_{c_a} = \partial_{c_a} + \frac{S^{a+1}}{(a+1)u}, \quad [\mathcal{D}/dS, \nabla_{c_a}] = 0. \quad (36d)$$

The two commutator terms are $-S^a$ and $+S^a$. Reducing $S^{a+b+1}/((a+1)u)$ by (25) gives the b-th column of a q-by-q matrix B_a . Every entry is a polynomial in the lower coefficients and t, with powers of the fixed u^{-1} . Thus

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u. \quad (36e)$$

At the monomial potential $\Phi = S^d/d$, $t=0$, direct substitution in the ray integral gives

$$\Pi_{jb}^{\text{mon}} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} \left(e^{2\pi i j(b+1)/d} - 1\right). \quad (36f)$$

All Gamma factors, all powers, and all phases remain. The final q-by-q matrix is nonsingular: a dependence among its columns would give a polynomial $\sum_{l=1}^q a_l(z^l - 1)$ of degree at most q vanishing at every d-th root of unity, including 1. It has $d=q+1$ distinct zeros, so it is the zero polynomial and every a_l vanishes.

For the phase matrix $V_{jl} = e^{2\pi i j l/d} - 1$, $1 \leq j, l < d$, finite root-of-unity summation also gives $V^*V = d(I_q + \mathbf{1}\mathbf{1}^T)$ and $\det(V^*V) = d^d$. This evaluates its nonzero size without absorbing any of the Gamma factors in (36f).

Join this monomial parameter to the actual parameter by a straight line in the coefficient space, keeping the rays fixed. Equations (36e) give $\Pi' = \Pi B$ with an everywhere continuous, explicitly reduced matrix B. Differentiation of the determinant, valid through the adjugate formula even at a putative singular matrix, yields

$$(\det \Pi)' = (\text{Tr } B) \det \Pi. \quad (36g)$$

Its initial value is nonzero by (36f). The scalar equation has solution given by that value times the exponential of the integral of $\text{Tr } B$, and never vanishes. This proves the stated invertibility without assuming simple critical points or a basis of unproved cycles.

It follows that the actual period map is the coefficient-linear isomorphism

$$\boxed{\mathcal{H}_{u,t} \xrightarrow{(\mathcal{P}_1, \dots, \mathcal{P}_q)} \mathbb{C}^q, \quad [P] \mapsto \Pi(u, t) \text{red}_{u,t}(P),} \quad (36h)$$

with inverse given by $\Pi(u, t)^{-1}$ followed by the specified degree-below-q section. Polynomial-exponential de Rham cohomology and rapid-decay period pairings are established machinery; (36a)–(36h) give an elementary proof of the particular one-variable comparison used here, with its contours and factors fixed.

6.2 Transport the period norm through the original source map Use the explicit coefficient identification

$$j_{u,t} : E \xrightarrow{\sim} \mathcal{H}_{u,t}, \quad [P]_{\chi} \mapsto [\text{rem}_{\chi} P]_{\mathcal{D}_{u,t}}. \quad (36i)$$

This is a complex-linear basis comparison. It is not a ring morphism when the target differential image is not an ideal. Equations (30)–(35) are the operator comparisons attached to it.

The original representative image now has an actual period isomorphism and inverse:

$$\boxed{r_N(E) \xrightarrow{\Pi(u,t)j_E} \mathbb{C}^q, \quad y \mapsto r_N \Pi(u, t)^{-1} y.} \quad (36j)$$

Their two composites are identities on the specified domains, by $j_E r_N = 1$. The same map applied to a retained theta boundary gives zero because j_E kills that boundary; its split lift gives the receiving supported zero, not external absence.

The operator comparison also holds at the level of actual period functions. For a t-independent F in the admitted theta test-function domain, (33), (35), and (36e) give

$$\boxed{-u \partial_t (\Pi(u, t) j_E F) = \Pi(u, t) j_E D_t^{(k)} F.} \quad (36j.1)$$

At $t=0$ the right side uses the original $D^{(k)}$. On the full test-function domain the kernel of this period observation is exactly $\ker j_E$, because Π is invertible. It remains an explicit retained kernel; the isomorphism (36j) concerns only the stated representative image $r_N(E)$.

Give the period-coordinate target its explicitly chosen ordinary coordinate Gram I_q and write

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (36k)$$

Then B_N^{per} is positive self-adjoint in the original G_N , and

$$\|\Pi v\|^2 = v^* G_N B_N^{\text{per}} v, \quad \boxed{\log \frac{V_i}{V_j} = \log \frac{\det B_j^{\text{per}}}{\det B_i^{\text{per}}}.} \quad (36l)$$

The latter follows from $\det B_N^{\text{per}} = \det H_\Pi / \det G_N$ at the same (u, t) . Thus the endpoint cost has been carried into two explicit period-to-source norm comparisons; it has not been set to zero by choosing period coordinates. The degree of the cyclic packet and the fixed contour-coordinate metric remain in the formula. Equation (36l) is a comparison identity, not a uniform bound on those determinants.

For $q=1$, $\chi(S) = S - \lambda$, real $u > 0$, and the contour $i\mathbb{R}$ oriented from $-i\infty$ to $+i\infty$, the integral is exactly

$$\boxed{\Pi(u, t) = i\sqrt{2\pi u} \exp\left(-\frac{(\lambda+t)^2}{2u}\right), \quad -u \partial_t \log \Pi = \lambda + t.} \quad (36m)$$

This uses the displayed orientation; it is the negative of the $q=1$ ray-difference cycle in (36a). The positive metric value for real λ, t is $2\pi u \exp(-(\lambda+t)^2/u)$. Its exponentially singular u -behaviour is retained. Thus even in rank one, passage through $u=0$ is a substantive norm-comparison question, not a consequence of finite-field purity of a different operator.

S32.0.7 7. A constant residue pairing gives the exact dual connection

On the remainder basis define the complex-bilinear form

$$\mathcal{S}_t(P, Q) = \sum_{\chi(\lambda)=t} \text{Res}_{S=\lambda} \frac{P(S)Q(S)}{\chi(S)-t} dS. \quad (37)$$

Multiplicity is handled by the residue, not by an assumption that the roots are simple. The form can equivalently be evaluated as the coefficient of S^{-1} in the Laurent expansion at infinity of the displayed rational function. This is minus the residue at infinity, so the sign agrees with the sum of the finite residues.

For degree-below- q P, Q ,

$$\boxed{\mathcal{S}_t(P, Q) = \mathcal{S}_0(P, Q).} \quad (38)$$

Indeed,

$$\frac{1}{\chi-t} - \frac{1}{\chi} = \frac{t}{\chi(\chi-t)},$$

and multiplication by PQ , whose degree is at most $2q-2$, gives a Laurent expansion beginning at S^{-2} . Its S^{-1} coefficient is zero. Let S be the constant matrix. Its entries below the first nonzero antidiagonal vanish and that antidiagonal has entries one, so

$$\det S = (-1)^{q(q-1)/2}, \quad \boxed{A(t)^T S = S A(t).} \quad (39)$$

The latter equation follows because multiplication by S is self-adjoint for the residue pairing on the algebra $\mathbb{C}[S]/(\chi-t)$; (30) is precisely its multiplication matrix in this basis.

Let ι_u be the base involution $(u, t) \mapsto (-u, t)$. The actual dual map is

$$\boxed{\mathcal{H}[u^{-1}] \longrightarrow (\iota_u^* \mathcal{H}[u^{-1}])^\vee, \quad v \longmapsto (w \mapsto v^T S w).} \quad (40)$$

It is horizontal:

$$\partial_t(v^T S w) = (\nabla_t^u v)^T S w + v^T S (\nabla_t^{-u} w). \quad (41)$$

This is a bilinear duality; the original Hermitian arithmetic pairing is connected on the special fibre by its retained antilinear reflection $f \mapsto f^{\dagger k}$. In particular the Jacobian contraction is still

$$\boxed{\mathcal{S}_0(f^{\dagger k}, \chi' v) = \text{Tr}_E M_{f^{\dagger k} v}.} \quad (42)$$

The proof is the local equality $\chi'/\chi = \ell_\lambda/(S-\lambda) + \text{a holomorphic term}$. The trace on the full original tensor or symmetric packet is still obtained through η and its retained complement; (42) is not silently substituted for that larger trace. Likewise the original factor $v_h^{\otimes k}$ stays in (5) and (6).

S32.0.8 8. The split coefficient maps and the original derivative tower

The family is over $R = \mathbb{C}[u, t]$, with scalar map $G(\mathbb{C}) \rightarrow G(R)$. Its reconstruction at any original support label ℓ has classes (ℓ, v) , and

$$e_R(\ell, v) = (\ell, 0), \quad \tau(\ell, v) = \tau. \quad (43)$$

The specialization $R \rightarrow \mathbb{C}, u, t \mapsto 0$, lifts to a semiring map $G(R) \rightarrow G(\mathbb{C})$, fixing τ and carrying both $u^\bullet, t^\bullet$ to the receiving e . The induced coefficient map is

$$(\ell, [P]_{\mathcal{D}}) \longmapsto (\ell, [P(0, 0, S)]_{\chi}). \quad (44)$$

For every P , the exact family relation is

$$[\chi P] = t[P] - u[P'] \quad \text{in } \mathcal{H}. \quad (45)$$

At $u=t=0$, this becomes the original supported zero. This family identity must not be substituted for the already formalized derivative between powers of the original relation ideal. Their precise polynomial relationship is

$$\partial_S(\chi P) = \chi' P + \chi P', \quad [\partial_S(\chi P)]_{\chi} = [\chi' P]_{\chi}, \quad (46)$$

whereas the u -term of (45) contains $-P'$. Both are obtained from the same polynomial and the displayed maps. The conormal response $\chi' P$ remains exactly the Jacobian in (42). None of the new parameters is another scalar zero-adjunction, and no original ideal-power level is replaced by a power of a contracted first ideal.

S32.0.9 9. This potential has actual Deligne exponential-sheaf fibres

Let R_0 be the finitely generated subring of $\mathbb{C}$ generated by the coefficients c_a and $1/(q+1)!$. Keep its specified embedding into $\mathbb{C}$. Equations (23) and (24) are defined over $R_0[u, t]$.

For every specified ring map $R_0 \rightarrow \mathbb{F}_Q$ with residue characteristic $p > q+1$, and every $u_0 \in \mathbb{F}_Q^*$ and $t_0 \in \mathbb{F}_Q$, the same potential gives the explicit Artin-Schreier cover

$$\boxed{Y^p - Y = \frac{\Phi(S) - t_0 S}{u_0}.} \quad (47)$$

The infinity calculation in Deligne's lissity proof can be made explicitly on this family. Put $d = q+1$ and $w = 1/S$. Then

$$\frac{\Phi(S) - tS}{u} = w^{-d}b(w, u, t), \quad b(w, u, t) = \frac{1}{u} \left[\frac{1}{d} + \sum_{a=0}^{q-1} \frac{c_a}{a+1} w^{q-a} - tw^q \right]. \quad (47a)$$

At $w=0$ its value is the unit $1/(du)$. On the neighborhood where b is invertible, adjoin the unit v by $bv^d = 1$ and define $x_\infty = wv$. This is an explicit finite étale cover: the derivative in v is dbv^{d-1} , whose inverse on the cover is v/d . The new relative coordinate has invertible derivative at $w=0$. Direct substitution gives

$$\boxed{\frac{\Phi(S) - tS}{u} = x_\infty^{-d}, \quad x_\infty = wv, \quad bv^d = 1.} \quad (47b)$$

Every coefficient in b is retained, as is the cover's root-of-unity action $v \mapsto \zeta v, \zeta^d = 1$. This is the local, étale source map used to see a constant exponential model at infinity. Together with the locally constant sheaf on the smooth affine part, it is exactly the local-acyclicity mechanism used in Deligne's §3.7.3. The open u nonzero remains necessary; this map makes no claim at $u=0$.

Choose a nontrivial additive character $\psi : \mathbb{F}_p \rightarrow \overline{\mathbb{Q}}_\ell^*$, with $\ell \neq p$, and let $\mathcal{L}_\psi((\Phi - t_0S)/u_0)$ be the corresponding eigensheaf. The polynomial on the right has degree $q+1$, prime to p , and leading coefficient $1/((q+1)u_0)$. In one variable its leading projective zero locus in $\mathbb{P}^0$ is empty, and is smooth. Thus the precise hypotheses of Deligne §3.7.2 hold.

His theorem supplies

$$\boxed{H_c^a(\mathbb{A}_{\mathbb{F}_Q}^1, \mathcal{L}_\psi((\Phi - t_0S)/u_0)) = 0 \ (a \neq 1), \quad \dim H_c^1 = q, \quad H_c^1 \text{ is pure of weight } 1.} \quad (48)$$

The universal-polynomial argument in §§3.7.3–3.7.4 supplies a lisse family on the parameter open $u \neq 0$, after that finite-field specialization. This is a direct application of Deligne's result, not a new proof of it.

The exact realization diagram is a span of specified base changes:

$$(R_0, \Phi, u, t) \longrightarrow \left\{ \begin{array}{l} (\mathbb{C}, \mathcal{C}_{u,t}, \nabla_t), \\ (\mathbb{F}_Q, \mathcal{L}_\psi((\Phi - tS)/u), R^1p_!). \end{array} \right. \quad (49)$$

Its characteristic-zero branch has the proved specialization (26) and operator formula (32). Its finite-field branch has Frobenius and (48). No field map $\mathbb{C} \rightarrow \mathbb{F}_Q$ is asserted; both arrows start at the actual ring R_0 . Nor is a comparison equating the two operators supplied. The arithmetic theta metric G_N is not part of Deligne's finite-field conclusion until an additional comparison theorem carries it there.

This places the cyclic relation in a genuine family of the kind Deligne studies, instead of attributing his nonconstant-constituent argument to the constant representative curve. It leaves an explicit operator/norm comparison to investigate.

A calibration of the operator distinction Take the declared algebraic polynomial $\chi(S) = S - 3/4$; this is not an asserted zeta packet. Then

$$\Phi(S) = S^2/2 - 3S/4, \quad A(t) = 3/4 + t, \quad \nabla_t = \partial_t - (3/4 + t)/u. \quad (50)$$

All the rank, specialization, and duality results above hold. At every finite-field realization of characteristic greater than two where the coefficient is defined, (48) still gives weight-one Frobenius. The characteristic-zero special coefficient remains $3/4$. This exact example shows why a purity theorem for the displayed Artin-Schreier family does not, by itself, locate the spectrum of (32). The additional comparison must use the specific original arithmetic source, not only the fact that a polynomial-exponential realization exists.

S32.0.10 10. What this changes in the programme

The predecessor's nonconcentration problem remains

$$\mathcal{B}_{h,k} = \sum_{\ell=0}^1 \int_{|z|<1} \log(1/|z|^2) d\mu_\ell. \quad (51)$$

The reviewed source-norm theorem and consecutive-window result still force the inherited off-line threshold of four. Neither log extension, finite ramification, nor saturation changes its original endpoint metric cost; (20) and (21) state precisely how those operations act.

What is added is a cohomological family with the exact original cyclic special fibre, an explicit connection containing A , an explicit dual connection, and finite-field exponential fibres covered by a theorem actually read in Deligne. The source-side realization (34)–(36) specifies the finite-rank deformation and keeps the theta boundary. The contour periods now give actual isomorphisms at u nonzero and the matrices $B_N^{\text{per}} = G_N^{-1} \Pi^* \Pi$ give their exact comparison to the fixed arithmetic metric. Equation (36l) carries the endpoint volume into their determinant ratio. The remaining estimate must control those specified matrices through singular $u=0$ and along the original tensor/degree windows. The existence of the period isomorphism or of scalar support is not itself that uniform estimate.

This continuation proves no new upper bound contradicting the endpoint threshold. It does provide the actual family and the exact maps on which a Deligne-style comparison can be attempted, after calculating and retaining the limitations of the earlier constant curve.

S32.0.11 References and verification scope

Deligne, P. (1980), *La conjecture de Weil II*, Publications Mathématiques de l’IHÉS 52, 137–252. Selected current source records: printed pages 201–203, 206, 215–216, 248. Retained French and English TeX windows are listed in SOURCE_REVIEW.md.

The Stacks Project, Section 50.15, Tag 0FMU, *Log poles along a divisor*. Its residue exact sequence fixes the complexes in (15); the actual calculations in (12), (13), and (18) are given here. The comparison is made in complexes of complex vector-space sheaves, since their connections are coefficient-linear, not $\mathcal{O}_X$ -linear.

The new exact tests are finite regression checks of the displayed polynomial, matrix, quotient, connection, residue, and Čech formulas. They do not certify infinite arithmetic integrals, compile new Lean modules, or prove a realization comparison absent from this note. The frozen predecessor is preserved. No remote branch is modified by this delivery.

Marco Hien (2009), *Periods for flat algebraic connections*, Inventiones Mathematicae 178, 1–22, DOI 10.1007/s00222-009-0185-7, is primary background for de Rham/rapid-decay duality. Only its bibliographic record and abstract were consulted here; no general theorem from it is needed in the direct contour proof (36a)–(36h).

S33 The original complete comparable-window norm proof

Source role. Complete original pinned PR25 source. Proposal and adopted status are recorded separately.

Exact revision. 10b89fa667c5fe88ada6631b67af81754773aeeb6efe377c3f3d6b26d4bc86c3

Complete source. sources/owner_pr25/original_66/workbenches/tau-consecutive-window-formal/NORM_REV IEW.md

Source SHA-256. 10b89fa667c5fe88ada6631b67af81754773aeeb6efe377c3f3d6b26d4bc86c3

S33.1 Arithmetic norm review and comparable-degree windows

The owner supplied the 2026-09-13 arithmetic endpoint note. Its principal input is the actual density $w_h = |2\chi/h|^2/(2\pi)$, its k -fold convolution, and the unscaled monic squared norms $\omega_{h,k,j}$. This is a review and extension of that written proof, not a Lean certificate for zeta estimates or arithmetic integrals. The locally linked delivery archive was not accessible in this runtime; its reported test and manifest results are not counted as fresh executions.

S33.1.1 1. Audit of the analytic steps

The Joukowski ellipse with $|w|=8$ has semiaxes $65/16$ and $63/16$. For $f_T(z) = \zeta(1/2 + iT + iz)$, $|T| \geq 10$ keeps the pole outside its closure. Its image lies in the fixed real strip $[-55/16, 71/16]$, with imaginary magnitude between $|T| - 65/16$ and $|T| + 65/16$. Split that strip at $\text{Re}(s) = 1/4$. Above this value, the fractional-part integral gives polynomial growth $O((1+|T|))$; below it the functional equation and vertical-strip gamma estimate give $O((1+|T|)^{(1/2 - \text{Re}(s) + 1)})$. The largest exponent is $79/16 < 6$. Thus the claimed envelope $M_T = C_0(1+|T|)^6$ is sufficient, uniformly on the ellipse. Increase C_0 so $M_T \geq 1$.

The point $z = -3i/2$ corresponds to $|w| = (3 + \sqrt{13})/2 < 4$ and hence to $\zeta(2 + iT)$. The reciprocal Dirichlet series gives modulus at least $\zeta(2)^{-1}$. The annular maximum principle gives $m_T \geq \zeta(2)^{-3} M_T^{-2}$. Every closed unit disk centred on $[-1, 1]$ lies strictly within the ellipse: its surrounding rectangle $[-2, 2] \times [-1, 1]$ already satisfies the ellipse inequality. The maximum principle and Cauchy estimate therefore give $\sup |f_T| \leq M_T$. At a maximizer use the longer one-sided interval in $[-1, 1]$, of length at least one; $m_T/(2M_T) \leq 1/2$. Integration gives $m_T^3/(8M_T)$ and the stated exponent 42. The compact remaining T -range is handled by continuity and analytic nonvanishing on an interval. No nonvanishing on the critical line is assumed.

For the passage to $|g/h|$, use $g = h(g/h)$ before taking absolute values. The polynomial upper bound on $|h|$ avoids division at a selected zero. Gamma asymptotics yield the claimed interval-mass bound. One regularity detail should be explicit: w_h is bounded as well as integrable, by its continuous entire quotient on compact sets and exponential tail. Thus $w_h * w_h$ is continuous, using L^1 translation continuity against the bounded factor. Its integrand is positive almost everywhere, so the convolution is strictly positive everywhere. This justifies its positive minimum on $[-1, 1]$. Restricting the remaining factors to that interval proves the note's lower envelope with the original factor $\vartheta_h(k-3)$.

The monic Legendre norm on $[-L, L]$ is exactly

$$1_n(L) = 2^{-L} \frac{L^{2n+1}}{(2n+1)!} [2^n n! / (2n)!]^2.$$

The trial upper bound at degree j is $(2j)! b^{(-2j)} (M_h(b)^k + M_h(-b)^k)$, for fixed $0 < b < \pi/2$. Both are used on the original shifted line $S = k/2 + iu$; multiplication by i^j has modulus one and does not change the leading coefficient norm. These arguments validate the supplied $O_h(n)$ doubling-window result as a written theorem, not as a floating-point or Lean assertion.

S33.1.2 2. Stronger, two-sided envelope

Write $a = \pi/2$ and retain the lower-envelope constants $c > 0$, $\vartheta > 0$, $B = 42 + 2 \deg(h)$. Put

$$\begin{aligned} t &= \min(1, \vartheta), \quad c_0 = \min(1, 2c/3), \\ \text{ell} &= (1/2) \exp(-a) \sqrt{c_0 t} 2^{(-B/2)}, \\ M &= \max(1, M_h(b), M_h(-b)), \quad U = 4 \sqrt{M}/b. \end{aligned}$$

Then, for every $n \geq k \geq 3$,

$$(e11\ n)^{(2n)} \leq \omega_{(h,k,n)} \leq (U\ n)^{(2n)}. \quad (A)$$

Proof of the lower bound: $l_n(n) \geq (2/3)n^{(2n)4^{(-n)}}$; $\vartheta^{(k-3)} \geq t^n$; $\exp(-a(n+k-3)) \geq \exp(-2an)$; $(1+n+k-3)^{(-B)} \geq (2n)^{(-B)} \geq 2^{(-nB)}$, since $2n \leq 2^n$ for integer $n \geq 1$; finally $2c/3 \geq c0^n$. Multiply. For the upper bound use $(2n)! \leq (2n)^{(2n)}$, $k \leq n$ and $M_h(b)^k + M_h(-b)^k \leq 2M^n$, then $2 \leq 4^n$. This proves (A), retaining all masses in constants rather than resetting them.

For $n \geq k \geq 3$ and $n/2 \leq r \leq 2n$, let $m = n + r$. From (A),

$$\begin{aligned} & (\omega_m/\omega_n)^{(1/(2r))} \\ & \leq n\ U^{(1+n/r)}\ e11^{(-n/r)}(1+r/n)^{(1+n/r)} \\ & \leq C_h\ n, \\ C_h = & 27\ \max(1,U)^3\ \max(1,e11^{(-1)})^2. \end{aligned} \quad (B)$$

This extends the doubling-window estimate to the overlapping windows needed by the volume theorem. Constants depend only on the fixed h and chosen b , not k, n, r . No monotonicity of the monic norms is assumed. The inequality is valid at $r = n$ and at $r = n + 1$ when $n \geq 1$.

References: NIST DLMF 25.4 (functional equation), 5.11.9 (uniform bounded-real-part gamma asymptotic), 18.3 (Legendre normalization). The general arguments above are written proofs; the present Lean contribution formalizes their finite norm-to-volume consequences, not their analytic hypotheses.

S34 The original complete consecutive-window proof

Source role. Complete original pinned PR25 source. Proposal and adopted status are recorded separately.

Exact revision. cc61ab74ee33eeb43f0bd968a9825be35b3eb3475fa2ff0678f833ba22daa332

Complete source. sources/owner_pr25/original_66/workbenches/tau-consecutive-window-formal/RESEARCH_NOTE.md

Source SHA-256. cc61ab74ee33eeb43f0bd968a9825be35b3eb3475fa2ff0678f833ba22daa332

S34.1 Consecutive losses and the arithmetic endpoint threshold

S34.1.1 Source, scope, and notation

This add-only continuation starts from main 7ea0a49945390eae14d3160a5730858899768b5f. The source is the owner's arithmetic endpoint note dated 2026-09-13 and the previously supplied exact Toda and exterior formulas. NORM_REVIEW.md audits its analytic proof and establishes the comparable-window extension used below. Those analytic results have written proofs, not a new Lean certificate. The Lean files prove the finite propagation independently of assuming RH, normality of the arithmetic action, or an estimate for the quotient volume.

Keep the original packet h , $g=2x_i$, theta source, full Taylor unit, canonical representatives R_N , and arithmetic cyclic module $C[S]/\chi(h,k)$. Write $q=\dim C$. The norms ω_N are the unscaled monic source norms; $V_N=\det G_N$ are its canonical quotient volumes. The phase is $\phi_N=\sigma\text{-partial_theta} \log V_N$. Λ_N below is ω_N/V_N , NOT the global theta retraction denoted Λ in earlier notes.

S34.1.2 1. Exact consecutive product

For $N \geq q$ the existing identity is

$$\begin{aligned} \epsilon_N^{2+\phi_N^2} &= (\omega_{N+1}/\omega_N)(1-\delta_N)(\delta_{N+1})^{(-1)-1}, \\ \delta_N &= V_N/V_{N-1} \text{ in } (0,1]. \end{aligned}$$

Rearrange it as

$$\begin{aligned} (\epsilon_N^{2+\phi_N^2}) \Lambda_N &= \Lambda_{N+1}(1-\delta_N)(1-\delta_{N+1}). \end{aligned} \quad (1)$$

For $n \geq q$ and $r \geq 1$, multiplication and cancellation of positive Λ -values gives

$$\begin{aligned} \text{product}_{(j < r)} (\epsilon_{n+j}^{2+\phi_{n+j}^2}) &= (\omega_{n+r} V_n) / (\omega_n V_{n+r}) \\ &\quad (1-\delta_n)(1-\delta_{n+r}) \\ &\quad \text{product}_{(0 < j < r)} (1-\delta_{n+j})^2. \end{aligned} \quad (2)$$

Every interior contraction occurs twice. This is an exact phase-retaining identity; the Lean theorem `exact_product` proves its general algebraic form. In particular,

$$\begin{aligned} \min_{(j < r)} \epsilon_{n+j} &\leq [(\omega_{n+r} V_n) / (\omega_n V_{n+r})]^{1/(2r)}. \end{aligned} \quad (3)$$

The finite minimum exists. Equation (3) follows by bounding each factor $1-\delta$ by one, comparing the product with the minimum's $2r$ -th power, and taking a nonnegative root. This proof does not require Jensen or the earlier four-endpoint hyperbolic-sine bound. It need not be sharper when contractions are small.

The local coarse bound $\epsilon_N^2 \Lambda_N \leq \Lambda_{N+1}$ also holds at $N=q-1$. Use the separate first-degree identity

$$\begin{aligned} \epsilon_{q-1}^{2+\phi_{q-1}^2} &= (\nu_0 - \omega_q) / \omega_{q-1}, \\ V_q/V_{q-1} &= \omega_q/\nu_0. \end{aligned}$$

It follows that $\epsilon_{(q-1)}^2 \Lambda_{(q-1)} \leq \Lambda_q$. No inverse at degree $q-2$ is introduced. Thus (3) and the lower-allowance consequence below apply to windows starting at $q-1$, although the exact two-contraction identity (2) is used only from q onward.

S34.1.3 2. The actual spectral lower allowance forces each single block

The existing original-Gram exterior/trace theorem gives $L_{(h,k)} \leq \epsilon_N$ for every admitted N . Iterating the coarse local inequality proves

$$L_{(h,k)}^{(2r)} \Lambda_n \leq \Lambda_{(n+r)}. \quad (4)$$

If the actual norm bound gives $\omega_{(n+r)}/\omega_n \leq C^{(2r)}$, then

$$(L_{(h,k)}/C)^{(2r)} V_{(n+r)} \leq V_n. \quad (5)$$

For $L > 0, C > 0$ this is

$$\log(V_n/V_{(n+r)}) \geq 2r \log(L_{(h,k)}/C). \quad (6)$$

The formal propagation, norm comparison, and log conversion have separate declarations, so the analytic premise is explicit.

For the exact hypothetical quartet, retain

$$\begin{aligned} q_k &= [1+k(m-1)](k+1)^2, \\ L_{(h,k)} &\geq \delta k^2 q_k/2, \quad \delta > 0. \end{aligned}$$

By NORM_REVIEW.md, both windows $(n,r)=(q_k,q_k)$ and (q_{k-1},q_k) have norm factors at most $C_h q_k$. Indeed $q_{k-1} \geq k$ for $k \geq 3$, and r is between $n/2$ and $2n$. The first window uses only regular local degrees; the second begins at the separately checked first degree. Let $D_h = \delta/(2C_h) > 0$. Then

$$\begin{aligned} \log(V_{(q_k)}/V_{(2q_k)}) &\geq 2q_k \log(D_h k), \\ \log(V_{(q_{k-1})}/V_{(2q_{k-1})}) &\geq 2q_k \log(D_h k). \end{aligned} \quad (7)$$

Therefore the source's SAME four-volume budget satisfies the stronger implication

$$\begin{aligned} B_{(h,k)} &= \log[V_{(q_{k-1})}V_{(q_k)}/(V_{(2q_{k-1})}V_{(2q_k)})] \\ &\geq 4q_k \log(D_h k), \\ \liminf B_{(h,k)}/(q_k \log k) &\geq 4. \end{aligned} \quad (8)$$

The source's threshold 2 remains correct but is not optimal after retaining the adjacency of the exact radius equations. This is a new deduction, not a correction falsely attributed to its original proof. No upper estimate below 4 is claimed. Either one-block upper limsup below 2 would already suffice for contradiction, while a total upper limsup below 4 would also suffice. Multiplicities remain in q_k .

S34.1.4 3. Relation-layer meaning

For the original block (i,j) , put $F=[b_{(i+1)}, \dots, b_j]$, $\Omega = \text{diag}(\omega_{(i+1)}, \dots, \omega_j)$. The supplied map $b = T \cdot R_{iF}$ is an isomorphism onto $D_j \cap D_i^\perp$, with Gram $\Omega + F^*G_iF$, and original quotient value zero. Hence

$$V_i/V_j = \det(I + \Omega^{-1}F^*G_iF).$$

Take the positive generalized eigenvalues λ_a of (F^*G_iF, Ω) . The two blocks $(q-1, 2q-1)$ and $(q, 2q)$ each have q source coordinates. Equations (7) imply, for each block separately,

$$\begin{aligned} \text{product}_a (a=1..q) (1+\lambda_a) &\geq (D_h k)^{(2q)}, \\ \text{geometric_mean}(1+\lambda_a) &\geq (D_h k)^2, \\ (1/q) \sum_a \lambda_a &\geq (D_h k)^2 - 1. \end{aligned} \quad (9)$$

The arithmetic-geometric mean inequality proves the last line. This does NOT force every λ_a to grow. Their multiplicity and rank profile remain. On the split lift the actual final quotient sends each retained relation to the receiving label's supported zero, not to external τ . Its source Gram is measured before that map.

S34.1.5 4. Constant and source accounting

No measure is normalized. The original mass $\omega_0 = \mu_h^k$ remains, and its effect is in the explicit constants of NORM_REVIEW.md. The Taylor multiplier and its derivative are not omitted. The norm leading-coefficient quotient is not identified with the arithmetic quotient. No complementary arithmetic block, nilpotent direction, or cochain sign is removed.

The analytic proof and the asymptotic consequence (8) are written mathematics. The new Lean run, when successful, certifies the finite local/product/forcing interfaces named in AuditConsecutiveWindow.lean; it does not certify the complex-analytic zeta argument, the arithmetic integrals, or a new weight theorem. Consult the PR for the exact completed run rather than treating this source note as an execution receipt.

S35 Complete supplied specialization and curvature continuation

Source role. Complete supplied mathematical text, retained with its original formulas and source chronology. The present integrated chapters state and prove the subsequent corrections and exact maps.

Exact revision. ed20dd4c189eb741d666edbaa964a03f616279a90e88b171bb9f31bba8652106

Complete source. sources/endpoint_user_sources/USER_SPECIALIZATION_CURVATURE_PROOF_20260913.txt

Source SHA-256. ed20dd4c189eb741d666edbaa964a03f616279a90e88b171bb9f31bba8652106

I added an actual compactification, specialization calculation, and dual boundary to the existing arithmetic representatives. The main result is that the endpoint volume we have been trying to control is exactly a logarithmic curvature integral on that compactified curve:

$$\log \frac{V_i}{V_j} = \int_{|z|<1} \log \frac{1}{|z|^2} c_1((\det \mathcal{E}_{ij})^\vee).$$

The curve is constructed from the **original theta relation**

$$\Delta_{ij} = L_{ij}R_i - R_j.$$

Its exceptional fibre, the directions killed by specialization, and their duals are calculated. The previous two trace inputs become the first two curvature coefficients of this same object.

There is also an additional repair: when the original arithmetic action develops a pole at the exceptional fibre, I construct its **smallest action-stable enlargement** and calculate the metric contribution that enlargement retains.

[Full derivation in the HTML reader](#) · [LaTeX](#) · [Formalization handoff](#) · [Complete package](#)

S35.0.1 1. What the Deligne rereading adds

I reassembled and hash-checked the six supplied S20 parts, then read the relevant retained French and English LaTeX against the current page-local French records. The [reading record](#) identifies the exact files, ranges, and transcription discrepancies. This is a selected reading of *La conjecture de Weil II*, not a new audit of its entire proof. ([Numdam](#))

The decisive addition is **not just another tensor operation**.

In §§3.2.8–3.2.13, Deligne first controls what occurs at the exceptional fibres. The resulting calculation gives integral weights for the relevant nonconstant constituents. The independent bound is **strictly below two**, so those integral weights are at most one. Constant constituents contribute zero to the specified first cohomology of the projective line. Only after those steps do the Leray and tensor-square maps improve the bound on the same eigenvalue.

The older French transcription loses two essential details: it writes a non-strict inequality in 3.2.10 and loses the square on the tensor eigenvalue in 3.2.13. The current page record retains both.

The later purity statement also concerns a particular comparison:

$$H_c^i(X, \mathcal{F}) \twoheadrightarrow \text{im } c_i \hookrightarrow H^i(X, \mathcal{F}).$$

The two already-proved bounds meet on that image; the mixed source and target are not removed. The degree and dualizing twist are part of the argument.

What I added here is the corresponding concrete work on our side: construct the curve, calculate the exceptional comparison and its complete fibre, retain the action, and calculate the determinant metric on that object.

S35.0.2 2. Build the curve from the existing representatives

Keep the original coefficient square over the marked τ -base:

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The first arithmetic target remains infinite. The split coefficients retain $e = 0^\bullet$ and external absence τ before applying p .

For the current finite cyclic object,

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad A = M_S,$$

retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N.$$

Here B_N is the original full remainder matrix, O_N contains the actual monic source norms, and the arithmetic inclusion still contains $v_h^{\otimes k}$. The canonical metric is determined by that source and its admitted relations.

For $i \leq j$, pad the earlier source through

$$L_{ij} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$$

and define

$$\Delta = L_{ij} R_i - R_j : E \longrightarrow \mathcal{P}_j.$$

Both representatives have remainder identity, and the later representative is orthogonal to its relation space. Therefore

$$B_j \Delta = 0, \quad R_j^* O_j \Delta = 0,$$

and

$$C := \Delta^* O_j \Delta = G_i - G_j \succeq 0.$$

Now form the actual family

$$R(z) = R_j + z\Delta : E \longrightarrow \mathcal{P}_j.$$

Direct expansion gives

$$B_j R(z) = I_E, \quad G(z) := R(z)^* O_j R(z) = G_j + |z|^2 C.$$

Every finite z therefore retains the same arithmetic jets and cohomology class:

$$J^{(k)} \mathcal{V} R(z) = \eta, \quad q^{(k)} \mathcal{V} R(z) = \sigma_h^{\otimes k} \eta.$$

The marked endpoints are

$$R(0) = R_j, \quad R(1) = L_{ij} R_i.$$

The new coordinate z parametrizes this comparison. The original spectral coordinate remains S , with its original action A .

S35.0.3 3. Compactify it and compute the exceptional fibre

Set

$$E_0 = \ker \Delta, \quad E_1 = E_0^{\perp G_j}, \quad r = \dim E_1.$$

Let P_0, P_1 be the corresponding specified projections.

On $\mathbb{P}_{\mathbb{C}}^1$, with homogeneous coordinates $[u : v]$, define

$$\mathcal{E}_{ij} = (E_0 \otimes \mathcal{O}) \oplus (E_1 \otimes \mathcal{O}(-1)).$$

The map into the original source is

$$\begin{aligned} \Psi : \mathcal{E}_{ij} &\longrightarrow \mathcal{P}_j \otimes \mathcal{O}, \\ x_0 + x_1 \otimes (u, v) &\longmapsto R_j x_0 + u R_j x_1 + v \Delta x_1. \end{aligned}$$

This is injective on every fibre. For $u \neq 0$, applying B_j forces $x_0 = x_1 = 0$. At $u = 0$, it first forces $x_0 = 0$, and injectivity of $\Delta|_{E_1}$ forces $x_1 = 0$.

On the finite chart $z = v/u$, this is exactly $R(z)$. On the infinity chart $w = u/v$, it is

$$\Psi_{\infty}(w) = R_j P_0 + w R_j P_1 + \Delta P_1.$$

The original arithmetic observation becomes

$$\alpha = B_j \Psi : \mathcal{E}_{ij} \longrightarrow E \otimes \mathcal{O},$$

with local expression

$$\alpha_{\infty} = P_0 + w P_1.$$

Consequently,

$$0 \longrightarrow \mathcal{E}_{ij} \xrightarrow{\alpha} E \otimes \mathcal{O} \longrightarrow i_{\infty*} E_1 \longrightarrow 0.$$

This is the effective-divisor exact sequence on the modified columns, with its actual source map supplied above. ([The Stacks Project](#))

The lost fibre is retained in the derived comparison Specializing $w = 0$ gives P_0 . The derived fibre of the displayed cokernel is the actual complex

$$[E \xrightarrow{P_0} E] \quad \text{in degrees } -1, 0.$$

Its cohomology is

$$H^{-1} = E_1, \quad H^0 = E_1.$$

Intrinsically, the kernel carries

$$E_1 \otimes \mathcal{O}(-1)|_{\infty};$$

the chosen infinity frame gives the displayed coefficient identification.

In the split lift,

$$w^{\bullet} \longmapsto e$$

under specialization. A represented vector killed by P_0 maps to the receiving supported zero, **and remains present in the derived kernel**. External absence maps to τ . This is precisely the original internal-homology treatment of a class whose image becomes a boundary.

The global comparison is also calculated:

$$\boxed{H^0(\mathbb{P}^1, \mathcal{E}_{ij}) = E_0, \quad H^1(\mathbb{P}^1, \mathcal{E}_{ij}) = 0.}$$

Its source image is $E_0 \subseteq E$. The full E remains the target and the saturation of the image lattice; it is not silently assigned to the smaller global image.

S35.0.4 4. Duality and the arithmetic action now have explicit boundary maps

Dualizing the comparison gives

$$\boxed{0 \rightarrow E^* \otimes \mathcal{O} \rightarrow \mathcal{E}_{ij}^\vee \rightarrow i_{\infty*}(E_1^* \otimes \mathcal{O}(1)|_\infty) \rightarrow 0.}$$

The two boundary factors pair through

$$\boxed{(E_1 \otimes \mathcal{O}(-1)|_\infty) \otimes (E_1^* \otimes \mathcal{O}(1)|_\infty) \longrightarrow \mathbb{C}_\infty.}$$

Both line factors remain.

The connection transported through α is

$$\nabla_\infty = d + P_1 \frac{dw}{w}.$$

Its residue is P_1 , and its horizontal monodromy is

$$\exp(-2\pi i P_1) = I.$$

The original arithmetic operator transports through the same map:

$$\boxed{\hat{A}_\infty = (P_0 + w^{-1}P_1)A(P_0 + wP_1) = \begin{pmatrix} A_{00} & wA_{01} \\ w^{-1}A_{10} & A_{11} \end{pmatrix}.$$

Thus the precise possible pole is

$$w^{-1}P_1AP_0.$$

The kernel of a metric comparison has not been presumed A -invariant.

Construct the smallest enlargement that repairs that pole Take the actual invariant hull

$$\boxed{F_A = E_0 + AE_0 + \cdots + A^{q-1}E_0, \quad q = \dim E.}$$

Cayley–Hamilton proves that this is the smallest A -invariant subspace containing E_0 .

Let $\mathcal{M} = \alpha(\mathcal{E}_{ij})$. At infinity,

$$\mathcal{M}_\infty = E_0\mathcal{O} + wE\mathcal{O}.$$

Its smallest coherent A -stable enlargement inside the original target is

$$\boxed{\mathcal{M}_A = \sum_{a=0}^{q-1} A^a \mathcal{M}, \quad (\mathcal{M}_A)_\infty = F_A\mathcal{O} + wE\mathcal{O}.$$

The exact sequences are

$$0 \rightarrow \mathcal{M} \rightarrow \mathcal{M}_A \rightarrow i_{\infty*}(F_A/E_0) \rightarrow 0,$$

and

$$0 \rightarrow \mathcal{M}_A \rightarrow E \otimes \mathcal{O} \rightarrow i_{\infty*}(E/F_A) \rightarrow 0.$$

The second sequence now carries the regular original action throughout. It gives the equivariant global sequence

$$0 \rightarrow F_A \rightarrow E \rightarrow E/F_A \rightarrow 0$$

and retains the trace

$$\mathrm{Tr} f(A) = \mathrm{Tr}(f(A)|F_A) + \mathrm{Tr}(f(\bar{A})|E/F_A).$$

The enlargement has a calculated source cost. Put

$$s_A = \dim F_A - \dim E_0.$$

The inclusion of the old lattice into the enlarged one has determinant w^{s_A} . Pulling through the **original** source metric gives

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w).$$

Regularizing the arithmetic action transfers the retained boundary contribution into a specified metric pole. It does not make that contribution disappear.

S35.0.5 5. The endpoint cost is the determinant curvature of this curve

Let

$$\mathcal{L}_{ij} = (\det \mathcal{E}_{ij})^\vee.$$

Its metric in the finite chart is

$$(\det G(z))^{-1}.$$

Define the curvature with its complete coefficient:

$$c_1(\mathcal{L}_{ij}) = \frac{i}{2\pi} \partial \bar{\partial} \log \det G(z).$$

The operator in this formula is exactly the preceding relation-cost operator:

$$Z = G_j^{-1}(G_i - G_j) = (K_j - K_i)G_i.$$

It is positive self-adjoint for G_j .

Differentiation gives

$$c_1(\mathcal{L}_{ij}) = \frac{i}{2\pi} \mathrm{Tr} [Z(I + |z|^2 Z)^{-2}] dz \wedge d\bar{z}.$$

Write

$$\kappa_{ij}(t) = \mathrm{Tr} [Z(I + tZ)^{-2}].$$

The coordinate identity

$$i dz \wedge d\bar{z} = 2 dx \wedge dy$$

and polar integration give

$$\int_{\mathbb{P}^1(\mathbb{C})} c_1(\mathcal{L}_{ij}) = \int_0^\infty \kappa_{ij}(t) dt = r.$$

That agrees with the independently computed divisor:

$$\det \alpha = u^r, \quad \deg \mathcal{L}_{ij} = r.$$

Recover the exact endpoint volume Put

$$f(t) = \log \det(I + tZ), \quad M(t) = t f'(t).$$

Then

$$M'(t) = \kappa_{ij}(t), \quad M(0) = 0.$$

Fubini gives

$$\begin{aligned} \int_0^1 \log(1/t) \kappa_{ij}(t) dt &= \int_0^1 \frac{1}{s} \int_0^s \kappa_{ij}(t) dt ds \\ &= \int_0^1 f'(s) ds \\ &= \log \det(I + Z). \end{aligned}$$

Since $G(1) = G_i$ and $G(0) = G_j$,

$$\log \frac{V_i}{V_j} = \int_0^1 \log(1/t) \kappa_{ij}(t) dt.$$

This is the same endpoint ratio as before, now computed on a compactified, marked curve with its boundary sheaf and metric.

Its second fundamental map retains the arithmetic cancellation The actual source projection is

$$P_z = R(z)G(z)^{-1}R(z)^*O_j.$$

Define

$$\mathcal{N}_z = (I - P_z)\Delta.$$

Then

$$\mathcal{N}_z^* O_j \mathcal{N}_z = C - |z|^2 C G(z)^{-1} C,$$

and

$$\mathrm{Tr}[G(z)^{-1} \mathcal{N}_z^* O_j \mathcal{N}_z] = \kappa_{ij}(|z|^2).$$

The individual arithmetic observations are

$$B_j P_z \Delta = \bar{z} G(z)^{-1} C,$$

$$B_j \mathcal{N}_z = -\bar{z} G(z)^{-1} C.$$

Their sum is the original relation. **Neither projected part is declared a boundary separately.** Their cancellation is retained before its quotient value becomes e .

S35.0.6 6. The previous trace certificate is now a curvature-jet estimate

Differentiating the curvature density at the marked point gives

$$\kappa_{ij}^{(n)}(0) = (-1)^n (n+1)! \operatorname{Tr}(Z^{n+1}).$$

Therefore the last note's quantities satisfy

$$t_1 = \kappa_{ij}(0), \quad t_2 = -\frac{1}{2} \kappa'_{ij}(0).$$

Its independent-pair quantity becomes

$$\mathcal{A}_2 = \frac{1}{2} \left(\kappa_{ij}(0)^2 + \frac{1}{2} \kappa'_{ij}(0) \right).$$

Thus the already-proved two-trace bound is now a bound on the logarithmic curvature moment of this particular determinant line:

$$\int_{|z|<1} \log(1/|z|^2) c_1(\mathcal{L}_{ij}) \leq \log(1+a) + (q-1) \log(1+b).$$

The a, b on the right are precisely the earlier functions of q, t_1, t_2 . No new estimate is being attributed to Deligne here; the geometric object underlying our existing estimate has been constructed.

S35.0.7 7. The construction also follows the actual arithmetic constituents

Deligne's determinant-weight bookkeeping concerns constituents, so I carried this construction through an actual invariant inclusion

$$I_F : F \hookrightarrow E, \quad A I_F = I_F A_F.$$

Set

$$R_F(z) = R(z) I_F, \quad G_F(z) = I_F^* G(z) I_F, \quad \Delta_F = \Delta I_F.$$

The resulting constituent bundle maps into $\mathcal{E}_{ij}$. On its modified part, the map is explicitly

$$x \otimes (u, v) \longmapsto u P_0 I_F x + P_1 I_F x \otimes (u, v).$$

Its composite with Ψ is exactly the constituent source map, including at infinity.

The original control compresses through the same inclusion:

$$W_F(z) = I_F^* W(z) I_F = A_F^* G_F(z) + G_F(z) A_F - k G_F(z).$$

Its determinant action satisfies

$$2\Re \operatorname{Tr} A_F - k \dim F = \operatorname{Tr} [G_F(z)^{-1} W_F(z)].$$

The original theta primitive also varies explicitly:

$$K(z) = (1 - z)K_j + zK_i,$$

$$D^{(k)}\mathcal{V}R(z) - \mathcal{V}R(z)A = dK(z).$$

Consequently,

$$W(z) = W_j + |z|^2(W_i - W_j) = -[(\mathcal{V}R(z))^* dK(z) + (dK(z))^* \mathcal{V}R(z)].$$

This connects the constituent curve, its determinant metric, and its **actual spectral action**. The integer curve degree is not substituted for the trace on the left.

S35.0.8 8. What is now available for the remaining estimate

For the two original endpoint blocks, let their determinant-curvature measures be μ_0, μ_1 . Then

$$\mathcal{B}_{h,k} = \sum_{\ell=0}^1 \int_{|z|<1} \log(1/|z|^2) d\mu_\ell, \quad \int d\mu_\ell = r_\ell \leq q_k.$$

The inherited off-line-quartet implication remains

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4.$$

I checked the pinned GitHub proof and retained its stated scope.

So the remaining upper estimate is now a **nonconcentration estimate for two specified source-induced curvature measures**, with their exceptional fibres and arithmetic action still available.

There is an exact calibration showing why the total degree alone cannot supply it. The family

$$G_j = I_r, \quad C = LI_r, \quad L > 0$$

has total curvature r for every L , but its logarithmic moment is

$$r \log(1 + L).$$

Each component has density

$$\frac{L}{(1 + Lt)^2},$$

and half its mass lies before $t = 1/L$. This is a declared algebraic family, not an asserted arithmetic packet.

The arithmetic work is therefore to control that concentration **using the original theta source and its relation maps**. Its curvature jets, second fundamental map, action-stable enlargement, and metric-pole contribution are now explicit quantities for doing so.

Deligne's finite-field argument supplies a model for the architecture—exceptional-fibre calculation, constituent control, then amplification—but its arithmetic integrality and strict weight estimate have not been proved for this new coherent-sheaf realization. Likewise, the calculated global image E_0 or F_A is retained rather than being relabelled as the entire arithmetic E .

S35.0.9 Verification and handoff

The new checker passed **23 exact finite test methods** normally and under `python -0`, with identical successful records. Both deliberately false controls were rejected in both modes. The tests include the derived special-fibre kernel, dual line factors, curvature sign, constituent maps, the minimal action-stable enlargement, and its determinant-metric pole.

All **36 manifest entries** of the preceding relation-moment archive verified, and its unchanged **22-method suite** passed fresh normal and optimized runs. The HTML reader was rendered and inspected at desktop and mobile widths; its displayed equation labels and local equation scrolling are retained.

The package includes a [tested eighteen-file add-only integration patch](#). No remote branch was changed, no new Lean execution is claimed, and no uniform arithmetic nonconcentration estimate is claimed.

The added machinery is now geometric in a concrete sense: a curve of the original representatives, its complete specialization and dual boundary, the minimal repair of its arithmetic action, and a determinant curvature whose logarithmic moment is exactly the endpoint cost already under study.

S36 Complete supplied logarithmic constituent and coupling calculation

Source role. Complete supplied mathematical text, retained with its original formulas and source chronology. The present integrated chapters state and prove the subsequent corrections and exact maps.

Exact revision. 501decfd268d2e9a382196f61169e9cd3fb27484b5ff03119f4855955328c864

Complete source. sources/endpoint_user_sources/LOGARITHMIC_CONSTITUENT_USER_SOURCE.txt

Source SHA-256. 501decfd268d2e9a382196f61169e9cd3fb27484b5ff03119f4855955328c864

I have now carried the previous curve through its logarithmic de Rham complex and used the resulting subobject-quotient comparison to improve the upper-volume calculation. The new quantitative term is a **subtraction computed from the coupling between an invariant arithmetic subspace and its quotient**:

$$\log \frac{V_i}{V_j} \leq \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det(Q_0 + C_Q)}{\det Q_0} - \sum_{r=1}^p \frac{\text{Tr}(\Xi_F^r)}{r}.$$

Every matrix here comes from the **same original canonical representatives**. The final term is nonnegative before subtraction, and its full infinite sum recovers the exact volume. It is not a freely chosen correction or a consequence stipulated under the name “purity.”

The cohomological calculation also identifies exactly which degrees carry the arithmetic information on the auxiliary curve. That matters for using Deligne’s comparison correctly.

[Full derivation in the HTML reader](#) · [LaTeX](#) · [Formalization handoff](#) · [Complete package](#)

S36.0.1 1. The Deligne operation being applied

The operative passage remains the actual comparison

$$H_c^a(X, \mathcal{F}) \twoheadrightarrow \text{im } c_a \hookrightarrow H^a(X, \mathcal{F}).$$

Deligne’s upper and dual lower estimates meet on that **same image**, with the cohomological degree and dualizing twist retained. Neither the source nor its mixed boundary contribution is erased.

I used that structure in two concrete ways here: first, by constructing the compact-support and ordinary complexes of the connection already present in our curve; second, by carrying the original metric and arithmetic action through an invariant inclusion and its quotient.

The new argument starts with the existing coefficient square:

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of $p_{\mathbb{Z}}$ remains infinite. The supported scalar $e = 0^\bullet$ and external absence τ remain separately represented before applying p .

For the current cyclic arithmetic module,

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad A = M_S, \quad q = \dim E,$$

retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N.$$

For an admitted degree pair $i \leq j$, put

$$G_0 = G_j, \quad \Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j.$$

The original relation and orthogonality equations are

$$B_j \Delta = 0, \quad R_j^* O_j \Delta = 0.$$

They give precisely the previous family

$$R(z) = R_j + z\Delta, \quad G(z) = G_0 + |z|^2 C.$$

Every finite z retains the same full arithmetic jets and cohomology class. These are inherited source identities, not a new choice of metric.

S36.0.2 2. The compact-support and ordinary complexes are now calculated

Let

$$X = \mathbb{P}_{\mathbb{C}}^1, \quad w = z^{-1},$$

and take a specified A -invariant subspace $F \subseteq E$. In particular, the previous action-stable repair supplies

$$F_A = \sum_{a=0}^{q-1} A^a \ker \Delta.$$

Define the actual lattice

$$\mathcal{M}_F = F\mathcal{O}_X + \mathcal{O}_X(-\infty)E \subseteq E\mathcal{O}_X$$

and its logarithmic complex

$$\mathcal{K}_F = \left[\mathcal{M}_F \xrightarrow{\nabla} \mathcal{M}_F \otimes \Omega_X^1(\log \infty) \right],$$

in degrees 0, 1.

The differential is the restriction of the trivial coefficient connection. It is a map of sheaves of $\mathbb{C}$ -vector spaces satisfying the Leibniz rule. In a fixed local complement to F , its two pieces are

$$d, \quad d + \frac{dw}{w}.$$

The logarithmic construction and its residue map are standard; here the lattice, differential, and comparison maps are specified explicitly. ([The Stacks Project](#))

There are actual inclusions of complexes

$$\mathcal{K}_0 \longrightarrow \mathcal{K}_F \longrightarrow \mathcal{K}_E,$$

where

$$\mathcal{K}_0 = [E\mathcal{O}(-1) \longrightarrow E\Omega_X^1]$$

is the compact-support model, and

$$\mathcal{K}_E = [E\mathcal{O} \longrightarrow E\Omega_X^1(\log \infty)]$$

is the ordinary model.

The global result Let

$$\mathbb{T} = H^1(X, \Omega_X^1)$$

with its specified trace

$$\mathrm{tr}[dz/z] = 1, \quad dz/z = -dw/w.$$

The full calculation gives

$$\mathbb{H}^a(X, \mathcal{K}_F) = \begin{cases} F, & a = 0, \\ 0, & a = 1, \\ (E/F) \otimes \mathbb{T}, & a = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. A chosen coefficient complement writes the underlying complex as

$$F[\mathcal{O} \rightarrow \mathcal{O}(-1)] \oplus (E/F)[\mathcal{O}(-1) \rightarrow \mathcal{O}(-2)].$$

The two-chart Laurent calculation gives

$$H^0(\mathcal{O}) = \mathbb{C}, \quad H^a(\mathcal{O}(-1)) = 0, \quad H^1(\mathcal{O}(-2)) = \mathbb{T}.$$

These yield the displayed groups. The identifications are compatible with the original action even when the chosen complement is not invariant.

The action is therefore

$$A_F \quad \text{on degree zero}, \quad A_{E/F} \otimes I_{\mathbb{T}} \quad \text{on degree two},$$

and

$$\mathrm{Str}(f(A) \mid R\Gamma(X, \mathcal{K}_F)) = \mathrm{Tr} f(A_F) + \mathrm{Tr} f(A_{E/F}) = \mathrm{Tr} f(A).$$

Placing the coefficient object in the existing arithmetic degree k puts these contributions in degrees k and $k + 2$, both with sign $(-1)^k$.

This adds a term the degree-zero bundle calculation alone did not contain: **the quotient E/F survives in degree two.**

The compact-to-ordinary maps are now explicitly

$$0 \longrightarrow F \longrightarrow E$$

in degree zero, and

$$E \otimes \mathbb{T} \longrightarrow (E/F) \otimes \mathbb{T} \longrightarrow 0$$

in degree two.

For this constant-coefficient parameter curve, the degree-one comparison image is zero. That is a statement about this particular auxiliary connection, not about the arithmetic theta quotient Q . The full arithmetic E remains in the outer comparison complexes and their original action.

The distinction also preserves extensions. For example,

$$A = \begin{pmatrix} \rho & 1 \\ 0 & \rho \end{pmatrix}, \quad F = \mathbb{C}e_1,$$

induces multiplication by ρ on both one-dimensional constituents. Their two induced actions alone would not reconstruct the nonzero map $e_2 \mapsto e_1$. That map remains in the original coefficient sequence and in the action comparison calculated below.

The dual boundary is retained For

$$F^{\text{ann}} = \{\lambda \in E^\vee : \lambda(F) = 0\},$$

evaluation gives the actual map

$$\mathcal{M}_F \otimes \mathcal{M}_{F^{\text{ann}}} \longrightarrow \mathcal{O}_X(-\infty).$$

Its constant term vanishes at infinity, so it induces

$$\boxed{\mathcal{K}_F(E) \otimes \mathcal{K}_{F^{\text{ann}}}(E^\vee) \longrightarrow \mathcal{K}_0(\mathbb{C}).}$$

The resulting degree- a , degree- $(2-a)$ pairings are the perfect pairings

$$F \otimes (E^\vee / F^{\text{ann}}) \rightarrow \mathbb{C}, \quad (E/F) \otimes F^{\text{ann}} \rightarrow \mathbb{C},$$

with $\mathbb{T}$ and its residue orientation retained.

The note also gives the explicit conjugation map from the linear dual to the metric antidual. Thus the original $kI - A^*$ antidual action is not substituted for a linear transpose without its coefficient map.

S36.0.3 3. Higher layers remain above the same split support

For every positive integer r , the same construction gives

$$\mathcal{M}_{F,r} = F\mathcal{O}_X + \mathcal{O}_X(-r\infty)E.$$

Its modified connection has residue r :

$$d + r \frac{dw}{w}.$$

The quotient comparing level r with level one has the actual coefficient basis

$$w, w^2, \dots, w^{r-1}$$

in both degrees, with differential

$$\boxed{\text{diag}(1, 2, \dots, r-1).}$$

Its inverse is the explicit contracting homotopy. The higher coefficient classes are present; their vanishing in cohomology follows from this differential.

This divisor-depth calculation is connected to the original conormal-depth calculation by

$$\boxed{\frac{\mathbb{C}[w] \otimes \mathcal{P}}{(w^r) + I^a(\mathbb{C}[w] \otimes \mathcal{P})} \xrightarrow{\sim} (\mathbb{C}[w]/w^r) \otimes (\mathcal{P}/I^a),}$$

where

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)).$$

The map sends $[f(w)P]$ to $[f] \otimes [P]$; its inverse multiplies the representatives.

Restriction and differentiation in the two directions therefore have their actual commuting maps. The two depth indices are retained independently, over the same scalar G -construction.

When a relation vanishes, its split lift lands at its receiving fibre's zero. This is the original internal-homology operation, which keeps both the source class and the comparison kernel available.

S36.0.4 4. Carry the original metric through an invariant arithmetic quotient

This is the new quantitative part.

Keep the actual invariant maps

$$I : F \hookrightarrow E, \quad AI = IA_F,$$

$$\pi : E \twoheadrightarrow E/F, \quad \pi A = A_Q \pi.$$

Write $t = |z|^2$, so

$$G(t) = G_0 + tC.$$

Choose any fixed coefficient lift $\overline{H} : E/F \rightarrow E$. Its **specified G_0 -orthogonal lift** is

$$H_0 = \overline{H} - I(I^*G_0I)^{-1}I^*G_0\overline{H}.$$

It satisfies

$$\pi H_0 = I, \quad I^*G_0H_0 = 0.$$

Define

$$G_{F,0} = I^*G_0I, \quad C_F = I^*CI,$$

$$D_F = I^*CH_0, \quad Q_0 = H_0^*G_0H_0, \quad C_Q = H_0^*CH_0.$$

In the actual coefficient coordinates $[I, H_0]$, the original metric is

$$\begin{pmatrix} G_{F,0} + tC_F & tD_F \\ tD_F^* & Q_0 + tC_Q \end{pmatrix}.$$

Now set

$$G_F(t) = G_{F,0} + tC_F,$$

$$H_t = H_0 - tIG_F(t)^{-1}D_F.$$

Then

$$\pi H_t = I, \quad I^*G(t)H_t = 0,$$

and the actual quotient metric is

$$Q(t) = H_t^*G(t)H_t = Q_0 + tC_Q - t^2D_F^*G_F(t)^{-1}D_F.$$

The correction $H_t - H_0$ lies in F . The full map $[I, H_t]$ remains invertible and retains that subspace. It is not a declaration that F is a theta boundary.

The coupling has an exact sign and size Define

$$\beta_t = G_F(t)^{-1} I^* C H_t : E/F \rightarrow F.$$

Differentiating the displayed maps gives

$$H'_t = -I\beta_t, \quad Q'(t) = H_t^* C H_t, \quad Q''(t) = -2\beta_t^* G_F(t) \beta_t.$$

For the first metric derivative, the terms containing H'_t vanish by orthogonality. Differentiating once more gives the last equation, with its minus sign and factor two.

Integrating with the original endpoints retained proves

$$D_F^* G_F(1)^{-1} D_F = 2 \int_0^1 (1-t) \beta_t^* G_F(t) \beta_t dt.$$

The subtraction in the quotient metric is therefore exactly an integrated squared coupling map. Its finite matrix formula defines it without requiring a new numerical quadrature.

S36.0.5 5. The same arithmetic action and theta relation reach this quotient

The actual lift defect is the map $a_t : E/F \rightarrow F$ defined by

$$A H_t - H_t A_Q = I a_t.$$

Differentiation gives

$$a'_t = \beta_t A_Q - A_F \beta_t.$$

Thus the metric coupling is related to the original arithmetic action through a concrete Sylvester map. On distinct primary blocks, the note constructs its finite inverse, retaining every power of the nilpotent parts. At coincident primary values, it retains the map with its kernel and image instead of using a nonexistent inverse.

For the original control form

$$W(t) = A^* G(t) + G(t) A - k G(t),$$

both comparisons are exact:

$$I^* W(t) I = A_F^* G_F(t) + G_F(t) A_F - k G_F(t),$$

$$H_t^* W(t) H_t = A_Q^* Q(t) + Q(t) A_Q - k Q(t).$$

The second equation uses $H_t^* G(t) I = 0$, not invariance of the orthogonal complement.

At the original test-function level,

$$D^{(k)} \mathcal{V} R(z) H_t - \mathcal{V} R(z) H_t A_Q = dK(z) H_t + \mathcal{V} R(z) I a_t.$$

The first term is the original theta boundary. The second belongs to the retained arithmetic subspace F , and becomes zero only under the further quotient by that included subspace. Both terms, and their different quotient maps, remain in the calculation.

This is the source-level connection between the constituent calculation and the original e -valued relation.

S36.0.6 6. The resulting upper bound keeps the extension contribution

At the endpoint, put

$$Q_+ = Q_0 + C_Q, \quad \Pi_F = D_F^* G_F(1)^{-1} D_F, \quad \boxed{\Xi_F = Q_+^{-1} \Pi_F.}$$

The source proves

$$Q_+ \succ 0, \quad Q(1) = Q_+ - \Pi_F \succ 0.$$

Hence Ξ_F is positive self-adjoint in Q_+ , with eigenvalues in $[0, 1)$.

The determinant of the coefficient transition $[I, H_0] \rightarrow [I, H_t]$ is independent of t . Taking determinants of the original metric therefore gives

$$\boxed{\log \frac{V_i}{V_j} = \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} + \log \det(I - \Xi_F).}$$

Now

$$-\log \det(I - \Xi_F) = \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r},$$

with nonnegative terms. Retaining the first p proves

$$\boxed{\log \frac{V_i}{V_j} \leq \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^p \frac{\text{Tr}(\Xi_F^r)}{r}.}$$

The upper bounds decrease to the exact volume. No separately assumed uniform restriction gap is needed to justify a finite subtraction.

The two positive terms can themselves be bounded by the previous two-trace certificate, applied to the actual compressed operators

$$Z_F = G_{F,0}^{-1} C_F, \quad Z_Q = Q_0^{-1} C_Q.$$

Writing that established majorant as U , the combined result is

$$\boxed{\log \frac{V_i}{V_j} \leq U(Z_F) + U(Z_Q) - \sum_{r=1}^p \frac{\text{Tr}(\Xi_F^r)}{r}.}$$

This uses the invariant subobject and quotient without estimating them as independent, uncoupled objects.

An exact improvement on the same source fixture For the preceding mass-seven Gaussian calibration,

$$\chi = (S-1)^3, \quad \omega_n = 7n!, \quad (i, j) = (3, 6),$$

take the actual invariant line generated by $(S-1)^2$.

The calculated matrices give

$$\frac{\det G_F(1)}{\det G_F(0)} = \frac{63}{8}, \quad \frac{\det Q_+}{\det Q_0} = \frac{245}{32},$$

and

$$\Xi_F = \begin{pmatrix} 2/3 & 2/3 \\ 0 & 0 \end{pmatrix}.$$

This matrix is self-adjoint in Q_+ , with eigenvalues $2/3, 0$.

The exact original loss is

$$\log(V_3/V_6) = \log(5145/256) = 3.000603203788587506 \dots$$

Four terms of the new subtraction give

$$\log(V_3/V_6) \leq \log(15435/256) - \frac{28}{27} = 3.062178455419660160 \dots$$

The preceding full two-trace bound was approximately 3.063328308618.

This is a strict improvement on that **same canonical source and metric**. It is a declared finite calibration, not an asserted zeta packet or an asymptotic estimate.

S36.0.7 7. The bound is composed with the existing endpoint theorem

Let $\mathcal{U}_{ij}$ be either new upper expression. The inherited consecutive-window theorem gives

$$\min_{i \leq N < j} \epsilon_N \leq \left(\frac{\omega_j}{\omega_i} \right)^{1/(2(j-i))} \exp \left(\frac{\mathcal{U}_{ij}}{2(j-i)} \right).$$

The selected degree belongs to the existing canonical family. No metric is selected again after estimating it.

For either original quartet block, the supplied arithmetic norm estimate bounds the first factor by $C_h^{\text{win}} q_k$. The other session's finite propagation and written analytic composition retain the lower cost

$$2q_k \log(D_h k)$$

for each block, and the combined limiting threshold four.

The present calculation therefore gives a constituent-level upper side of those **same inequalities**, including a negative coupling contribution that was previously unavailable in that form.

The regular extension at infinity is accounted for as well. Passing to the residue-zero extension gives, with the original metric retained,

$$c_1((\det E\mathcal{O}_X)^\vee) = \mu_{ij} - \text{rank}(C) \delta_\infty.$$

The positive curvature measure and its logarithmic moment remain. The boundary atom makes the total degree zero; it does not make that logarithmic moment small.

That is why this continuation uses the actual subquotient estimate rather than inferring arithmetic purity from the compactification's integer degree.

S36.0.8 Verification and current result

I checked GitHub main at 1f7e7c02343884a17df9873566e81a9083c70951; its current four-volume source retains the same threshold and proof scope.

The new checker passed **16 exact finite test methods**, normally and under python -0, with identical successful records. Both deliberately false controls failed in both modes. The parent specialization-curvature archive's **41 manifest entries** verified, and its unchanged **23-method suite** passed fresh normal and optimized runs.

The supplied Deligne archive was reassembled and hash-checked; the relevant LaTeX and page-local text passages were read. No PDF extraction or OCR was used. The HTML reader was rendered and inspected at desktop and mobile widths.

The package includes a **tested 23-file add-only integration patch**. No remote branch was changed, and no new Lean certificate is claimed.

The completed advance is the actual logarithmic comparison, its retained degree-two and dual boundary information, and an upper-volume estimate that uses the calculated coupling between arithmetic constituents. A uniform bound below the inherited quartet threshold is still not established. The quantitative work is now focused on the original compressed source matrices and that explicit subtraction—not on assigning purity to a curve merely because it has been compactified.

S37 Complete exact restriction fixture and its source maps

Source role. Full finite-calibration derivation with original masses, every map and the exact scope of its executed checks.

Exact revision. f5ef0746d22ba641aa2fd0d50c0f19954fe9c367d8cacd0206aa1e28f6702b29

Complete source. sources/local_endpoint_continuations/endpoint_restriction_join_fixture_20260913.md

Source SHA-256. f5ef0746d22ba641aa2fd0d50c0f19954fe9c367d8cacd0206aa1e28f6702b29

S37.1 Exact lower-source gap and two-pair restriction certificates

This is a complete finite calculation for the explicitly declared auxiliary source below. It joins the lower-source comparison in AU.11–29 to the restriction and trace construction in equations (24), (34), and (47)–(51) of the supplied restriction note. The original coordinate is $S=1+iu$, the relation is $\chi(S)=S^2-2S-1$, the quotient has dimension two, and the source mass is exactly $9/16$.

The result is an explicit lower gap $108/763$ obtained from a positive source comparison and an exact remainder operator norm. The actual sharp gap is $135/331$. Both apply to the two pairs (1,3) and (2,4). Their trace certificates are computed with exact rational logarithm enclosures. These are finite conclusions for this specified source; no estimate for a growing arithmetic quartet family is inferred.

S37.1.1 1. Source, lower source, and literal masses

For positive integer λ define

$$r_\lambda(u) = \frac{|\Gamma(\lambda + iu/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda).$$

The measures used here are

$$w(t) = \left(1 + \frac{t^2}{4}\right) r_1(t) = r_2(t), \quad m = w * w = \frac{1}{70} r_4, \quad v = r_1 * r_1 = \frac{1}{3} r_2, \quad d\nu = \frac{54}{35} v(u) du. \quad (\text{F.1})$$

Here is the measure identity with its constants. Euler's gamma integral gives $\Gamma(z+1)=z \Gamma(z)$ by integration by parts when $\Re z > 0$. Multiplying this equality by its conjugate proves the first equality in (F.1). The change from two positive gamma-integral variables to their sum and ratio, with Jacobian equal to the sum, gives

$$\Gamma(a)\Gamma(b) = \Gamma(a+b) \int_0^1 x^{a-1}(1-x)^{b-1} dx, \quad \Re a, \Re b > 0.$$

Set $a=\lambda+iu/2$, $b=\lambda-iu/2$, and then $x=e^t/(1+e^t)$. This gives

$$|\Gamma(\lambda + iu/2)|^2 = \Gamma(2\lambda) \int_{\mathbb{R}} (2 \cosh(t/2))^{-2\lambda} e^{iut/2} dt.$$

After $t=2y$, Fourier inversion, with convention $\hat{f}(t)=\int_{\mathbb{R}} f(u)e^{-itu}du$, proves

$$\hat{r}_\lambda(t) = 2\Gamma(2\lambda)(2 \cosh t)^{-2\lambda} = c_\lambda(\cosh t)^{-2\lambda}. \quad (\text{F.2})$$

All inversions and moment differentiations used here are legitimate. The integrand $(2 \cosh y)^{-2\lambda}$ and every derivative decay exponentially on the real axis. For each $0 < a < \pi/2$, it is holomorphic on the closed strip $|\Im y| < a$; shifting the Fourier integral up or down inside that strip bounds $r_\lambda(u)$ by a constant times $\exp(-a|u|)$. The vertical sides of a rectangular contour tend to zero because the integrand decays exponentially in $|\Re y|$, uniformly in that strip. Thus every polynomial moment exists, the Fourier transform extends to imaginary arguments near zero, and its moment-generating function is

$$\int e^{tu} r_\lambda(u) du = c_\lambda(\cos t)^{-2\lambda}, \quad |t| < \pi/2.$$

Multiplying (F.2) proves

$$r_a * r_b = \frac{c_a c_b}{c_{a+b}} r_{a+b}.$$

The constants are $c_1=1/2$, $c_2=3/4$, and $c_4=315/8$. Consequently the masses of w, m, v, ν are respectively

$$\left[\frac{3}{4}, \frac{9}{16}, \frac{1}{4}, \frac{27}{70} \right] \quad (\text{F.3})$$

Two applications of the gamma recurrence give the pointwise identity

$$m(u) = \left(\frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120} \right) v(u). \quad (\text{F.4})$$

Indeed $r_4/r_2=(4+u^2/4)(9+u^2/4)$ and $(r_4/70)/(r_2/3)$ multiplies that product by $3/70$. Every coefficient in (F.4) is retained. Since $v(u) > 0$, this proves $d\nu \leq m(u)du$ and hence the contraction

$$\mathbb{C}[S]_{\leq j} \longrightarrow \mathbb{C}[S]_{\leq j}, \quad P \longmapsto P, \quad \|P\|_\nu^2 \leq \|P\|_m^2. \quad (\text{F.5})$$

The forms in (F.5) integrate $|p(1+iu)|^2$. The polynomial map, quotient map, and original coordinate are unchanged.

The even moments of m through degree eight are

$$(9/16, 9/2, 117, 5472, 385272),$$

and those of v are

$$(1/4, 1, 14, 376, 16064). \quad (\text{F.6})$$

The ν moments are the latter multiplied by $54/35$; all odd moments are zero. These values follow by differentiating the functions in (F.2). The executable constructs them by the equivalent finite multiplication recurrence

$$ub_j = b_{j+1} + j(j + \alpha - 1)b_{j-1}, \quad \alpha = 2\lambda,$$

starting from $b_0=1$, retaining the coefficient of b_0 and multiplying by the literal mass. On the finite degrees needed here it also verifies them by two independent constructions: binomial convolution of $(1+t^2/4)r_1$ moments, and the full three-term density (F.4). The monic-polynomial orthogonality needed below is verified directly against every Gram entry, so no unchecked infinite recurrence is used as a finite matrix certificate.

S37.1.2 2. Complete original-coordinate matrices

The polynomial space has ordered basis $1, S, \dots, S^N$. Its conjugate-linear-first Gram is

$$(H_N)_{ab} = \sum_{x=0}^a \sum_{y=0}^b \binom{a}{x} \binom{b}{y} (-1)^x i^{x+y} m_{x+y}. \quad (\text{F.7})$$

The imaginary terms vanish for the stated even source and are separately accumulated and checked in the executable. The full largest matrices are

$$H_4 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 & 1449 \\ 9 & 81 & 81 & -1863 & -5751 \\ -63 & 81 & 2025 & 2025 & -89343 \\ -207 & -1863 & 2025 & 93393 & 93393 \\ 1449 & -5751 & -89343 & 93393 & 6526089 \end{pmatrix}, \quad (\text{F.8})$$

$$H_4^\vee = \frac{1}{70} \begin{pmatrix} 27 & 27 & -81 & -297 & 891 \\ 27 & 135 & 135 & -1485 & -4725 \\ -81 & 135 & 1755 & 1755 & -41985 \\ -297 & -1485 & 1755 & 45495 & 45495 \\ 891 & -4725 & -41985 & 45495 & 1906875 \end{pmatrix}. \quad (\text{F.9})$$

Every smaller source form is the corresponding leading principal submatrix. The polynomial relation yields

$$J_1 = I_2, \quad J_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{pmatrix}, \quad J_4 = \begin{pmatrix} 1 & 0 & 1 & 2 & 5 \\ 0 & 1 & 2 & 5 & 12 \end{pmatrix}. \quad (\text{F.10})$$

These are the maps to coefficients in $E = \mathbb{C}[S]/(\chi)$ with basis $(1, S)$. The recurrence for their columns is $J_{a+2} = 2J_{a+1} + J_a$, exactly the relation $S^2 = 2S + 1$.

For each degree define

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N, \quad V_N = \det G_N. \quad (\text{F.11})$$

Positivity of the density proves $H_N > 0$. Surjectivity of J_N proves $K_N > 0$. Direct multiplication gives $J_N R_N = I$ and $R_N^* H_N R_N = G_N$. For any z with $J_N z = 0$, $z^* H_N R_N = z^* J_N^* G_N = 0$. Therefore every representative $R_N x + z$ has squared norm $x^* G_N x + z^* H_N z$, proving its minimum property on the original quotient fibre.

For transparent scalar calculations introduce the explicitly typed coordinate isomorphism

$$\mathbb{C}[x]/(x^2 - 2) \xrightarrow{\sim} E, \quad [x] \mapsto [S - 1], \quad C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad (1, S)C = (1, S - 1). \quad (\text{F.12})$$

Its inverse substitutes $S = x + 1$. Its determinant is one. The source observation of x is still iu , with its sign and phase retained. A quotient form transforms by $G_x = C^* G_S C$; its inverse transforms by $K_x = C^{-1} K_S C^{-*}$. No measure is changed by these identities.

The following finite polynomials and norms follow by substitution into (F.6)–(F.9). They are monic in S ; writing them in $x = S - 1$ is only (F.12).

Degree	Original source polynomial	Squared norm	Remainder in $(1, x)$
0	1	9/16	1
1	x	9/2	x
2	$x^2 + 8$	81	10
3	$x^3 + 26x$	2430	28x
4	$x^4 + 56x^2 + 240$	106920	356

Degree	Lower-source polynomial	Squared norm	Remainder in $(1, x)$
0	1	27/70	1
1	x	54/35	x
2	$x^2 + 4$	108/7	6
3	$x^3 + 14x$	1944/7	16x
4	$x^4 + 32x^2 + 72$	7776	140

Let P_N have these original S coefficient columns and let O_N be the diagonal of their squared norms. The checked identity is the full matrix equation $P_N^* H_N P_N = O_N$. Its inverse and (F.11) give

$$K_N = (J_N P_N) O_N^{-1} (J_N P_N)^*. \quad (\text{F.13})$$

Thus the inverse quotient form is the sum of each listed remainder outer product divided by its original norm. Explicitly,

$$\begin{aligned}
K_{1,x} &= \text{diag}(16/9, 2/9), & K_{2,x} &= \text{diag}(244/81, 2/9), \\
K_{3,x} &= \text{diag}(244/81, 662/1215), & K_{4,x} &= \text{diag}(56102/13365, 662/1215), \\
K_{3,x}^\nu &= \text{diag}(133/27, 763/486), & K_{4,x}^\nu &= \text{diag}(3619/486, 763/486).
\end{aligned} \tag{F.14}$$

The full original-basis matrices and least representatives at all four degrees are included as rational entries in the JSON. Equations (F.8)–(F.11) are their complete finite construction. Their quotient volumes are

$$V_1 = 81/32, \quad V_2 = 729/488, \quad V_3 = 98415/161528, \quad V_4 = 16238475/37139524. \tag{F.15}$$

S37.1.3 3. A proved gap from the exact remainder operator

Let $B=H_1$, the norm on the unique representative of degree below two. In (F.12),

$$B_x = C^*BC = \text{diag}(9/16, 9/2).$$

Consider the actual remainder operator

$$\mathcal{R}_j : (\mathbb{C}[S]_{\leq j}, H_j^\nu) \longrightarrow (\mathbb{C}[S]_{\leq 1}, B), \quad P \longmapsto \text{rem}_\chi P, \quad j = 3, 4. \tag{F.16}$$

Its coefficient matrix is exactly J_j . Its squared norm is the greatest eigenvalue of $B^{\wedge\{1/2\}} J_j (H_j^\nu)^{\wedge\{-1\}} J_j^* B^{\wedge\{1/2\}}$. To see this, the substitutions $a = (H_j^\nu)^{\wedge\{1/2\}} v$ and $b = B^{\wedge\{1/2\}} J_j v$ transport its norm ratio to the Euclidean operator $B^{\wedge\{1/2\}} J_j (H_j^\nu)^{\wedge\{-1\}}$. For a matrix L , its greatest squared singular value is the greatest eigenvalue of LL^* : the nonzero spectra of L^*L and LL^* coincide by the maps L and L^* , and maximization of the positive Hermitian quadratic form is diagonalization by an orthonormal eigenbasis.

By (F.14), these positive eigenvalues are exactly

j	first	second
3	133/48	763/108
4	3619/864	763/108.

Both first entries are smaller than the second: the differences are 1855/432 and 2485/864. Consequently

$$\boxed{\|\mathcal{R}_3\|^2 = \|\mathcal{R}_4\|^2 = \frac{763}{108}.} \tag{F.17}$$

This norm is attained. In the original S coefficient basis use

$$f_3(S) = -\frac{245}{36} + \frac{791}{108}S - \frac{7}{9}S^2 + \frac{7}{27}S^3, \quad f_4 = f_3 \in \mathbb{C}[S]_{\leq 4}. \tag{F.18}$$

This is $f_j = (H_j^\nu)^{\wedge\{-1\}} J_j^* B(-1, 1)^{\wedge\tau}$; the two expressions agree by direct multiplication of the matrices (F.8)–(F.10). The norm and remainder norm are

$$\|f_j\|_\nu^2 = \frac{763}{24}, \quad \|\text{rem}_\chi f_j\|_B^2 = \frac{582169}{2592},$$

whose ratio is 763/108. This supplies a concrete extremizing polynomial, in addition to the complete matrix certificate

$$\frac{763}{108} H_j^\nu - J_j^* B J_j \succeq 0. \tag{F.19}$$

The JSON records every principal minor of (F.19), including its zero minors. For a Hermitian matrix, non-negativity of all principal minors implies positive semidefiniteness: the coefficients of $\det(tI+A)$ are the sums of its principal minors of each size, so this polynomial is strictly positive for $t > 0$; a negative eigenvalue of A would give a positive root. This proves the exact PSD validation criterion used here without a floating-point eigenvalue.

The source contraction (F.5) now gives

$$\|\text{rem}_x P\|_B^2 \leq \frac{763}{108} \|P\|_\nu^2 \leq \frac{763}{108} \|P\|_m^2. \quad (\text{F.20})$$

Apply this to the actual least representative R_{jx} . Its remainder is x , so

$$G_j \succeq \frac{108}{763} B.$$

The degree-one remainder is an admitted representative for every $i \geq 1$, hence $G_i \preceq B$ by the minimum property. Therefore for $(i,j)=(1,3),(2,4)$,

$$\boxed{G_j \succeq \frac{108}{763} G_i, \quad K_i \succeq \frac{108}{763} K_j.} \quad (\text{F.21})$$

The inverse inequality follows by conjugating the first by $G_i^{-1/2}$, inverting its positive eigenvalues, and transporting back. The gap $108/763$ is thus obtained from the explicitly proved source comparison and exact remainder norm. Its construction uses the low-degree actual form B and the declared comparison moments through degree eight.

S37.1.4 4. Restriction in the original polynomial coordinates

Define the typed restriction on the same quotient by

$$T_{i,j} : (E, G_j) \longrightarrow (E, G_i), \quad T_{i,j} = K_i G_j. \quad (\text{F.22})$$

For x, y in E , $x^* G_j y = x^* G_i T_{i,j} y$; this proves it is the adjoint of identity transport $(E, G_i) \rightarrow (E, G_j)$. In the original $(1, S)$ coordinates the two return maps are

$$T_{1,3} = \begin{pmatrix} 36/61 & 3681/20191 \\ 0 & 135/331 \end{pmatrix}, \quad T_{2,4} = \begin{pmatrix} 20130/28051 & 2876145/9284881 \\ 0 & 135/331 \end{pmatrix}. \quad (\text{F.23})$$

The coordinate matrices in (F.12) diagonalize these endomorphisms by $T_x = C^{-1} T_{SC}$, giving spectra

$$\{36/61, 135/331\}, \quad \{20130/28051, 135/331\}. \quad (\text{F.24})$$

Both eigenvalues in each pair lie in $(0,1)$. The smallest is $135/331$ for both pairs, and hence this is the exact best scalar in $\kappa_i \geq g_k_j$ for these actual source forms. It is stronger than (F.21), through the strictly positive difference $135/331 - 108/763$.

Let $L_{\{i,j\}}$ pad original polynomial coefficients with zeros. Since $P_j^* H_j P_j = 0_j$, the orthogonal projection from degree j to degrees at most i is the endomorphism

$$Q_{i,j} = P_j \text{diag}(\underbrace{1, \dots, 1}_{i+1}, 0, \dots, 0) P_j^{-1}. \quad (\text{F.25})$$

Its range is $L_{\{i,j\}} P_i$; direct multiplication proves $Q^2 = Q$ and $Q^* H_j = H_j Q$. In the two original S coordinate spaces these are

$$Q_{1,3} = \begin{pmatrix} 1 & 0 & -9 & 0 \\ 0 & 1 & 2 & -23 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$Q_{2,4} = \begin{pmatrix} 1 & 0 & 0 & 27 & -189 \\ 0 & 1 & 0 & -29 & 0 \\ 0 & 0 & 1 & 3 & -50 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{F.26})$$

In monic polynomial coordinates the first $i+1$ rows of the degree- j least lift are $0_i^{-1} (J_i P_i)^* G_j$; multiplication of the degree- i lift by $T_{\{i,j\}}$ gives precisely that expression because $G_i K_i = I$. Transporting by the original coefficient matrices therefore proves

$$\boxed{L_{i,j}R_iT_{i,j} = Q_{i,j}R_j.} \quad (\text{F.27})$$

Its Gram is $G_jK_iG_j$. For an eigenvector v of T with eigenvalue g , this becomes

$$\frac{\|Q_{i,j}R_jv\|_{H_j}^2}{\|R_jv\|_{H_j}^2} = g, \quad \frac{\|(I - Q_{i,j})R_jv\|_{H_j}^2}{\|R_jv\|_{H_j}^2} = 1 - g. \quad (\text{F.28})$$

The second equation follows from orthogonal decomposition by (F.25). The same original representative determines both quantities.

Writing $D = L_{\{i,j\}}R_i - R_j$ and $H = I - T_{\{i,j\}}$, equation (F.27) gives the exact correction

$$D = L_{i,j}R_iH - (I - Q_{i,j})R_j, \quad J_jD = 0, \quad D^*H_jD = G_i - G_j. \quad (\text{F.29})$$

The remainder equation follows from $J_jL_{\{i,j\}}R_i = J_jR_j = I$. For the Gram identity, R_j is orthogonal to every relation in $\ker J_j$ by (F.11), so $L_{\{i,j\}}R_i = R_j + D$ is an orthogonal sum. The return maps compose by $K_iG_jK_jG_i = K_iG_i$.

The quotient multiplication action is retained as

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

For $W_N = A^*G_N + G_N - 2G_N$ and $H_N = K_NW_N$, expansion gives

$$AT - TA = H_iT - TH_j. \quad (\text{F.30})$$

Indeed the two terms $K_iA^*G_j$ and the two terms $2K_iG_j$ cancel. This records the exact action defect of the return map. Each finite matrix equation (F.27), (F.29), and (F.30) is checked in the literal original coordinates.

S37.1.5 5. Full finite logarithmic certificates

The determinant identity in (F.22) is

$$\det T_{i,j} = V_j/V_i.$$

Since the two positive eigenvalues are (F.24), put $s_a = \text{Tr}(I - T)^a$. Expansion of $-\log g = \sum_{a=1}^{\infty} (1-g)^a/a$ proves

$$\mathcal{L}_{i,j} = \log(V_i/V_j) = \sum_{a \geq 1} \frac{s_a}{a}. \quad (\text{F.31})$$

For any of the proved lower gaps g_0 , let $r = 1 - g_0$. Each loss eigenvalue $x = 1 - g$ satisfies $0 < x < r < 1$. The exact tail comparison is

$$\sum_{a=p+1}^{\infty} \frac{x^a}{a} = x^{p+1} \sum_{a=p+1}^{\infty} \frac{x^{a-p-1}}{a} \leq x^{p+1} \frac{-\log g_0 - \sum_{a=1}^p r^a/a}{r^{p+1}}.$$

Thus

$$\sum_{a=1}^p \frac{s_a}{a} \leq \mathcal{L}_{i,j} \leq U_p(H, g_0) := \sum_{a=1}^p \frac{s_a}{a} + \frac{-\log g_0 - \sum_{a=1}^p r^a/a}{r^{p+1}} s_{p+1}. \quad (\text{F.32})$$

For these finite pairs, the executable computes every trace through power six directly by matrix multiplication and independently from the rational spectrum. Its exact first three values are:

Pair	s_1	s_2	s_3
(1,3)	20231/20191	211421561/407676481	2275697407691/8231395827871
(2,4)	8119847/9284881	37102062682217/86209015184161	184216076205697661987/800440447112127969841

The two determinant ratios and their product are

$$\frac{V_1}{V_3} = \frac{20191}{4860}, \quad \frac{V_2}{V_4} = \frac{9284881}{2717550}, \quad \boxed{\frac{V_1 V_2}{V_3 V_4} = \frac{3073295611}{216513000}}. \quad (\text{F.33})$$

All logarithms in (F.31)–(F.32) have rational arguments. The executable uses exact rational enclosures. For $1 < y < 2$, put $t = (y-1)/(y+1)$. Integrating the geometric series $2/(1-t^2) = 2 \sum_{a \geq 0} t^{2a}$ from zero gives

$$2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} \leq \log y \leq 2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} + \frac{2t^{2M+1}}{(2M+1)(1-t^2)}. \quad (\text{F.34})$$

The upper tail follows by replacing all positive denominators by the first denominator and summing the geometric series. For $z \geq 1$, finite multiplication writes $z = 2^e y$ with $1 < y < 2$, and $\log z = e \log 2 + \log y$ transports the two intervals. For $z < 1$, $\log z = -\log(1/z)$ reverses interval endpoints. These are exact scalar logarithm identities; the source measures remain (F.1). The script uses $M=90$, checks loss widths below 10^{-70} , and checks certificate widths below 10^{-65} . Every enclosure endpoint and exact tail coefficient is retained in the JSON.

For a compact rational display, all following intervals have denominator 10^{15} . Their endpoints were obtained by outward integer rounding of the saved exact fractions, retaining the correct inequalities:

Quantity	Lower numerator	Upper numerator
$\log(V_1/V_3 - V_2/V_4)$	2652855161899890	2652855161899891
Sum of p=2 upper certificates using 108/763	2934134222167925	2934134222167926
Sum of p=2 upper certificates using 135/331	2666680039877747	2666680039877748

The exact old direct trace bound is

$$\text{Tr}(G_1(K_3 - K_1)) + \text{Tr}(G_2(K_4 - K_2)) = \frac{1159}{540} + \frac{334321}{181170} = \frac{1446331}{362340}. \quad (\text{F.35})$$

To prove it bounds the loss, diagonalize $G_i^{-1/2} K_j G_i^{-1/2}$, whose eigenvalues are $1/g$; the scalar inequality $\log x \leq x-1$ for $x \geq 1$ follows by integrating $1/t \leq 1$ from one. Summing gives (F.35). Exact rational comparison verifies that each new p=2 total upper bound is strictly below (F.35), and that the sharp-gap certificate is strictly below the lower-source-gap certificate.

The cost of using the explicit lower-source gap can also be inspected pair by pair. The intervals below again have denominator 10^{15} and enclose the excess $U_{2-\text{mathcal{L}}}$:

Pair and gap	Lower numerator	Upper numerator
(1,3), lower-source 108/763	155314375058502	155314375058503
(2,4), lower-source 108/763	125964685209532	125964685209533
(1,3), actual 135/331	9359847652264	9359847652265
(2,4), actual 135/331	4465030325591	4465030325592

The script additionally computes the exact (F.32) enclosures for $p=1,3,5$. Each is checked against the independently enclosed determinant logarithm. These finitely many calculations make no claim about a uniform gap in a tensor-degree limit.

S37.1.6 6. Complete executable replay and provenance

Only these three files were authored for this lane:

- `endpoint_restriction_join_fixture_20260913.py`: complete standard-library rational calculation and subprocess driver.
- `endpoint_restriction_join_fixture_20260913.json`: all matrices, principal minors, witnesses, trace coefficients, logarithm enclosures, check labels, full stdout/stderr, actual commands, timings, return codes, and hashes for every subprocess.
- This Markdown proof and local continuity record.

The complete positive run contains **303 distinct mathematical checks**. The driver ran the positive calculation and seven changed calculations in both ordinary Python and python -O: **16 actual complete subprocess jobs**. Every job reaches all 303 checks. Assertions are not used as tests, and no changed computation is rejected by a startup guard or deliberately raised exception. The changed computations fail at the following counts, identically in both modes:

Changed calculation	Failed checks
Set the actual source mass to one	20
Set the lower-source mass to one	30
Replace K_{iG_j} by the reverse K_{jG_i}	36
Truncate original monomial coefficients in place of the orthogonal projection	8
Use the smaller comparison eigenvalue as the remainder norm cap	24
Replace trace of a power by power of the trace	23
Use denominator r^p in place of $r^{(p+1)}$ in the logarithmic tail coefficient	24

Each unchanged run has zero failures and exit code zero; each changed run has the listed positive failure count and exit code one. The driver separately checks full counts, unique labels, outcomes, return codes, empty stderr, and equality of complete result objects between ordinary and optimized modes. These execution/provenance checks and repeated runs do not increase the count of distinct mathematical checks.

The mathematical inputs were read completely before construction:

- `arithmetic_volume_upper_route_20260913.tex`, SHA256 21aaedc3585887fa7d2cb3e37e68d14415829a4b213566f7a08f4619835ef38b.
- `Supplied Tau_Endpoint_Restriction_Control/NOTE.tex`, SHA256 ff4cc53cb4d3b10a9547a8a2365db7fd5d874bf1ecf7dd8542b8d1f05f433ef6.
- Earlier auxiliary-source proof `gamma_endpoint_window_bridge_fixture_20260913.md`, SHA256 05108b93cfd1e15e782f171f37569e3d77c8c4775b1bf8349f4021a0eb4315ed.

No existing executable was imported into this fixture. An independent mathematical reviewer derived the same source moments, remainder kernels, comparison eigenvalues, both original-coordinate return matrices, and exact trace values without importing this script or writing files.

The completed independent read-only audit found no defect. It read all 415 script lines and inspected the normal result and all 16 replay outcomes. Its separate calculation passed **170 audit predicates** without importing or running this fixture script. It reconstructed the original S Grams, all four kernels and least lifts, and the projections directly from $L_{H_i}^{-1} L^{*} H_j$, independently of the monic-basis implementation. It verified the extremizer (F.18), all six trace powers, and the four $p=2$ certificates. A separate 1,024-term rational logarithm series without range reduction produced intervals contained in every audited saved logarithm/certificate interval. Every stdout/stderr hash, Windows CRLF reconstruction, unique check label, complete job count, return code, and ordinary/optimized equality matched; all 88 driver provenance checks passed. These 170 audit predicates remain separate from the fixture's 303 mathematical checks. The reviewer wrote no files.

Final executable SHA256: 88bcee664a3812dd914e5d164de37d48f4388307e2359062ace360e988d9df14.

Final replay JSON SHA256: a8205b83417feb94baf903f53f9bdd0d2e38cf325fc28ceaac6a098530b4075d.

Assigned work, verbatim

New bounded independent task while I derive restriction-AU join: read full work/arithmetic_volume_upper_route_20260913.tex and new restriction full NOTE once intake path available (ask root/arithmetic_endpoint_intake). Construct meaningful exact rational fixtures validating two-pair restriction $T=KiGj$, projection, spectral trace finite upper certificate, explicit remainder-comparator g_0 for $(q-1, 2q-1), (q, 2q)$, retaining literal source mass. Prefer existing EW auxiliary source $S=1+iu$ $\chi=S^2-2S-1$, mass9/16, or actual new Gaussian nilpotent mass7. Do full derivation of finite g_0 from exact polynomial remainder norm and source lower form, and compare bound's gap and trace certificate. Send formulas early; only new work/endpoint_restriction_join_fixture_20260913.{py,md,json} files. Full exact output, actual complete mutations not assertions, independent ordinary+optimized if apt. No edits to frozen files, no Lean or publication. Do not assume a uniform gap.

The selected source is the first explicit option. The parent received early exact comparator and spectral formulas, followed by the replay counts and upper-bound comparison. Existing frozen files, inherited intake checks, Lean calculations, and publication are outside this independent lane and were not changed.

S38 SGA trace, the retained diagonal, and the arithmetic period determinant

Source role. Complete delivered proof note, equations 1-62 including 21a, 28a and 54a-54c; apply the separately recorded visible coefficient-ring and primary-scope corrections. Original display tags retain their complete literal source tokens; the local typesetting namespace SGA identifies this source. The supplied inline metric paragraph after equation50 has a recorded Markdown/TeX embedding defect; SGAC7 gives its complete correctly typed sentence and maps. Its single bare tilde command and exact following G_N argument are rendered inside math delimiters with the original raw payload unchanged and the complete original syntax tree recoverable.

Exact revision. 96061b31f6f99ce3d268bc4411b680bfe01592fdbfe0e879de9316ea81d0142b

Complete source. sources/web_sga_trace_period_delivery/Tau_SGA_Trace_Period_Control/RESEARCH_NOTE.md

Source SHA-256. 96061b31f6f99ce3d268bc4411b680bfe01592fdbfe0e879de9316ea81d0142b

Source metadata: author (MetaInlines)

Owner-directed SplitZero research continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: lang (MetaInlines)

en

Source metadata: title (MetaInlines)

SGA trace, the retained diagonal, and the arithmetic period determinant

S38.1 1. The completed calculation and its role

This continuation uses the **actual inline text in the supplied SGA 5 master**, especially Exposé III, §6.8, together with the preceding polynomial-exponential construction. It calculates, rather than postulates, the relative duality and diagonal class of the current cyclic arithmetic module. That class then evaluates the determinant of the entire exponential period matrix.

The principal formulas are

$$\mathrm{Tr}_{E_t/R}(M_P) = \mathrm{res}_{\chi-t}(\chi'P), \quad \det \Pi(\mathbf{c}, u, t) = \det \Pi_{\mathrm{mon}}(u) \exp\left(\frac{\mathcal{F}(\mathbf{c}, t)}{u}\right), \quad (\text{SGA } 1)$$

where every symbol and map is specified below. In particular,

$$\mathcal{F}(\mathbf{c}, t) = \mathrm{Tr}_{E_t/\mathbb{C}}(M_{\Phi_t}), \quad \det(\Pi^*\Pi) = (2\pi|u|)^q \exp\left(2\Re\frac{\mathcal{F}(\mathbf{c}, t)}{u}\right). \quad (\text{SGA } 2)$$

This evaluates a previously unspecified analytic determinant. Its coefficient derivatives recover **all the first q power traces of the original companion operator**, and hence its whole characteristic polynomial, including multiplicities. The original theta metric is not changed. Its comparison with this calculated determinant is carried through in §§10–12.

These are explicit applications and extensions of classical residue/trace and irregular-period methods; no global priority claim is made. SGA 5 supplies the residue/trace operation, not a Hermitian positivity theorem for the theta quotient. Bloch–Esnault study determinants and periods for irregular connections, including the $\exp(f)$ case; their abstract was inspected, but their full article was not successfully retrieved in this execution. The proof here is supplied in full at the types used here.

S38.2 2. Source intake and mathematical categories

The nine uploaded files are LaTeX masters. SGA 5 contains 15,818 lines of substantive inline material. The other masters have differing amounts of inline material and refer to further component files; for example the SGA 4½ master consists of its preamble and eight component inputs. The absent component contents are not counted as read. SOURCE_REVIEW.md and checks/source-inventory.json give the exact reading and dependency inventory.

The key supplied statement is SGA 5, Exposé III, (6.8.5): the residue of a function multiplied by the regular-immersion class equals the trace of its multiplication endomorphism on a finite flat algebra. The preceding paragraph constructs the class from the conormal differential. The source explicitly records a qualification about the proof/reference of this compatibility. We retain that qualification and prove its one-variable instance directly below.

The appendix works with coherent complexes over a noetherian scheme, with the indicated finite Tor-dimension, properness, and Tor-independence hypotheses. Here the finite map is finite free, so properness and flatness are actual properties of the displayed algebra. Its dual is computed directly as a finite module dual. No general six-operation formalism on semiring schemes is assumed.

Keep the programme's one scalar construction

$$G(R) = \{\tau\} \sqcup \{a^\bullet : a \in R\}, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0_R, \quad p_R(a^\bullet) = a. \quad (\text{SGA } 3)$$

Its support observation is $b_R(\tau) = 0$ and $b_R(a^\bullet) = 1$. The pair (p_R, b_R) retains the whole scalar. For every ring map f , its split lift fixes τ and sends $a^\bullet$ to $f(a)^\bullet$. In particular

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C} \end{array} \quad (\text{SGA } 4)$$

remains the original arithmetic square with an infinite left target. All calculations below first specify coefficient maps and then lift them fibrewise through the original support reconstruction. The ordinary coherent realization is connected to that richer object by p , not substituted for the complete split object.

At a label ℓ , an R -linear map f lifts as $(\ell, x) \mapsto (\ell, f(x))$; a coefficient zero remains $(\ell, 0)$. External absence maps to absence. Kernels and quotients below are the original coefficient kernels and quotients before that reconstruction. No new scalar zero is adjoined for a diagonal homology degree or a deformation parameter.

S38.3 3. The arithmetic object that is retained

Fix the original finite packet h , with complete orders, and tensor degree k . Retain

$$g = 2\xi, \quad v_h = g/h, \quad \mathcal{M}F_h = v_h, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S, \quad \chi = \chi_{h,k}, \quad q = \deg \chi > 0. \quad (\text{SGA } 5)$$

The source and actual weighted inclusion are

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h). \quad (\text{SGA } 6)$$

The retained source identities are

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k} \eta\pi_\chi. \quad (\text{SGA } 7)$$

The original scalar measure is

$$w_h(y) = \frac{|v_h(1/2 + iy)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k, \quad (\text{SGA } 8)$$

and the norm is

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iy)|^2 m_{h,k}(y) dy. \quad (\text{SGA } 9)$$

For $N \geq q - 1$, the canonical matrix and source section remain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad V_N = \det G_N. \quad (\text{SGA } 10)$$

Write $r_N = \mathcal{V}R_N$. The established cyclic retraction composed with the full jet map gives j_E on the stated test-function domain, with $j_E r_N = 1$ and $j_E D^{(k)} = A j_E$. The source relation is

$$D^{(k)}r_N - r_N A = dK_N^{\text{prim}}. \quad (\text{SGA } 11)$$

These are inherited interfaces, not analytic theorems re-proved by the finite tests of this note. The empty packet $E=0$ has the empty determinant one; all q -positive constructions below concern $q>0$.

S38.4 4. SGA 5's relative duality, calculated on this polynomial algebra

Let R be a commutative ring, let χ be monic of degree q , and put

$$H(S) = \chi(S) - t, \quad E_t = R[t, S]/(H), \quad f : \text{Spec } E_t \longrightarrow \text{Spec } R[t]. \quad (\text{SGA } 12)$$

The basis $1, S, \dots, S^{q-1}$ proves f finite free of degree q . Set

$$\lambda_t([P]) = [S^{q-1}] \text{rem}_H P. \quad (\text{SGA } 13)$$

Here $[S^{q-1}]$ means coefficient extraction in the actual monic remainder. Over $\mathbb{C}$ it equals the sum of finite residues of $P(S)dS/H(S)$, with the positive finite-residue orientation. Equivalently it is the coefficient of S^{q-1} in P/H , which is the negative of the residue at infinity.

The bilinear form

$$\beta_t(x, y) = \lambda_t(xy), \quad \beta_t^b : E_t \longrightarrow \text{Hom}_{R[t]}(E_t, R[t]) \quad (\text{SGA } 14)$$

is perfect. Its Gram in the unchanged power basis has zeros below total exponent $q-1$ and ones on that antidiagonal. Thus

$$\det \beta_t = (-1)^{q(q-1)/2}. \quad (\text{SGA } 15)$$

This proves perfection over the original coefficient ring, without dividing by a discriminant. The finite duality functor is

$$f^! R[t] = \text{Hom}_{R[t]}(E_t, R[t]), \quad (\text{SGA } 16)$$

in degree zero. Equation (14) is its explicit trivialization; evaluation at 1 is its counit. The factor dS/H and the conormal generator $[H]$ account for the smooth dimension-one and regular codimension-one shifts, which cancel in this finite relative dimension-zero calculation.

S38.4.1 4.1 The actual diagonal class

In $B_t = E_t \otimes_{R[t]} E_t$, with coordinates X, Y , define

$$\mathcal{C}_H(X, Y) = \frac{H(X) - H(Y)}{X - Y} = \sum_{n=1}^q c_n \sum_{a=0}^{n-1} X^{n-1-a} Y^a, \quad c_q = 1. \quad (\text{SGA } 17)$$

The fraction denotes polynomial division with zero remainder; there is no localization at $X-Y$. Its defining identities are

$$(X - Y)\mathcal{C}_H = 0 \text{ in } B_t, \quad \mu(\mathcal{C}_H) = \chi'(S), \quad (\text{SGA } 18)$$

where μ is multiplication $E_t \otimes E_t \rightarrow E_t$.

The first identity shows the class is supported on the diagonal. Moreover

$$(\lambda_t \otimes 1)((P \otimes 1)\mathcal{C}_H) = P \text{ in } E_t. \quad (\text{SGA } 19)$$

To prove it, first take $P=1$: the coefficient of X^{q-1} in (17) is 1. The identity $(X-Y)\mathcal{C}_H=0$ then gives $(P(X)-P(Y))\mathcal{C}_H=0$ for every polynomial P , proving (19). This also proves that $\mathcal{C}_H$ is the coevaluation tensor inverse to (14).

Taking the categorical trace of multiplication gives the complete composite

$$R[t] \xrightarrow{1 \mapsto \mathcal{C}_H} E_t \otimes E_t \xrightarrow{M_P \otimes 1} E_t \otimes E_t \xrightarrow{\mu} E_t \xrightarrow{\lambda_t} R[t]. \quad (\text{SGA } 20)$$

Consequently

$$\boxed{\text{Tr}_{E_t/R[t]} M_P = \lambda_t(\chi' P)}. \quad (\text{SGA } 21)$$

This proves the one-variable instance of the supplied SGA 5 III (6.8.5). In that statement the regular-immersion class is the conormal differential $[H] \mapsto \chi' dS$. Its trace map is exactly (20), not an additional sign-free identification.

The bilinear trace form is connected to the perfect residue form by the exact map

$$\tau_t^b = \beta_t^b \circ M_{\chi'} : E_t \longrightarrow E_t^\vee, \quad \tau_t(x, y) = \text{Tr } M_{xy}, \quad \ker \tau_t^b = \text{ann}_{E_t}(\chi'). \quad (\text{SGA } 21a)$$

This proves the complete radical statement for this finite trace form. At a complex root of multiplicity m its radical has dimension $m-1$; the perfect residue form itself remains nondegenerate. It is not the assertion of positivity for the original sesquilinear Weil form. Their connection is the retained reflection and Jacobian composite (24).

All maps (12)–(21) commute with coefficient base change: their entries are polynomial expressions in the coefficients and monic remainders. For tensor products, the coevaluation is the external product of the displayed tensors; the diagonal Jacobian is the product of the χ_i 's. If the coefficient object is placed in degree k , its supertrace is $(-1)^k$ times (21). No geometric weight is inferred from that cohomological sign.

S38.4.2 4.2 The source's supported relation maps to this same class

At $t=0$, the original sum-direction logarithm operator is

$$\mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad \partial_S^{\text{rel}} = \frac{1}{k} \sum_i \partial_{s_i}, \quad \partial_S^{\text{rel}} S = 1. \quad (\text{SGA } 22)$$

Differentiating the full Mellin amplitude $\mathcal{U}(s)P(S)$, where $\mathcal{U} = \prod [v_h(s_i)]$, gives both $\mathcal{U}'$ and $(\partial \mathcal{U})P$. Applied to an original relation χP , the latter contains χ and is killed by the original arithmetic quotient. Therefore

$$\boxed{j_E \mathcal{L}_k \mathcal{V}(\chi P) = [\chi' P]_\chi}. \quad (\text{SGA } 23)$$

The arithmetic unit is retained in the map before j_E ; it is removed only by the specified inverse/retraction in j_E . Formula (23) does not differentiate an already-formed scalar e . It differentiates an actual source relation before that relation is sent to its supported quotient zero.

The composition of (23) with λ_0 is the trace (21). More generally, with the original reflection $f^\dagger \text{dagger}_k(S) = \overline{f(k - \overline{\lambda} S)}$,

$$\lambda_0(\chi' f^{\dagger k} P) = \text{Tr}_E M_{f^{\dagger k} P} = \sum_\lambda \ell_\lambda \overline{f(k - \overline{\lambda})} P(\lambda). \quad (\text{SGA } 24)$$

The induced trace in the full arithmetic module still uses η and its retained complement. It is not equated with the cyclic trace without that inclusion and complement.

The conormal lift $C[S]/(\chi^2)$ used to compute (23) maps surjectively to the actual contracted level $C[S]/(\chi_2)$, because $(\chi^2) \subseteq I^2 \cap C[S] = (\chi_2)$. No equality of these ideals is asserted. This keeps the formalization session's correction to higher relation depth.

S38.5 5. The whole derived diagonal is retained

There is a complete explicit resolution behind (18). Put $d=X-Y$ and $b=C_H$. Over B_t it is

$$\cdots \xrightarrow{d} B_t \xrightarrow{b} B_t \xrightarrow{d} B_t \xrightarrow{\mu} E_t \longrightarrow 0. \quad (\text{SGA } 25)$$

Exactness. Work first in $T=(R[t,Y]/(H(Y)))[X]$. Then $B_t=T/(db)$. Both d and b are monic polynomials in X , hence non-zero-divisors in T even when the coefficient algebra has nilpotents. If $da \in (db)$, cancellation of

d in T gives $a \in (b)$. If $ba \in (db)$, cancellation of b gives $a \in (d)$. These are precisely the alternating kernel–image identities in (25); $B_t/(d) = E_t$ supplies the augmentation.

Pullback to the diagonal is tensoring (25) by μ . The two differentials become 0 and multiplication by χ' . It follows that

$$HH_0(E_t/R[t]) = E_t, \quad HH_{2a+1}(E_t/R[t]) = E_t/(\chi'), \quad HH_{2a+2}(E_t/R[t]) = \text{ann}_{E_t}(\chi'), \quad a \geq 0. \quad (\text{SGA } 26)$$

These are module identifications; no multiplication or shuffle structure is silently included. Multiplication by X commutes with every map of (25) and specializes to the original S -action on (26).

At a complex root of multiplicity m , χ' is the product of $m(S-\lambda)^{(m-1)}$ with the actual nonzero local unit. Both positive-degree modules have dimension $m-1$. For $m=1$ they vanish. For $m>1$ the higher classes are retained as distinct cohomological degrees over the same split scalar support.

This is a classical hypersurface calculation performed here on the exact polynomial. It computes the derived diagonal; it does not assert that these HH degrees equal the separately indexed original nullity ladder. The map linking this calculation to that ladder is the shared conormal differential (23), together with the displayed quotient from the original contracted relation levels.

S38.6 6. The preceding exponential family and its exact coefficient connections

Now work over $\mathbb{C}$ and write

$$\chi_{\mathbf{c}}(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad \Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS. \quad (\text{SGA } 27)$$

The original polynomial is the marked coefficient value $\mathbf{c} = \mathbf{c}_{h,k}$. None of its coefficients is shifted or rescaled.

The previous two-term coefficient complex is

$$\mathcal{C}_{u,t} = [\mathbb{C}[u, t, S] \xrightarrow{D} \mathbb{C}[u, t, S] dS], \quad D(P) = (uP' + (\chi_{\mathbf{c}} - t)P) dS. \quad (\text{SGA } 28)$$

Its unique degree- $<q$ remainder gives a free degree-one module. At $u=t=0$ it is E . The classical finite algebra and this differential quotient have the explicit coefficient-linear inverse maps

$$\rho_{u,t} : E_t \longrightarrow \mathcal{H}_{u,t}, \quad [P]_H \longmapsto [\text{rem}_H P]_D, \quad \rho_{u,t}^{-1}[Q]_D = [\text{rem}_D Q]_H. \quad (\text{SGA } 28a)$$

Both normal-form operations fix every degree- $<q$ polynomial, proving the inverse laws. The map is a coefficient-module isomorphism, not a declaration that multiplication descends through D . For $u \neq 0$ its connections are

$$\nabla_t = \partial_t - S/u, \quad \nabla_{c_a} = \partial_{c_a} + S^{a+1}/((a+1)u). \quad (\text{SGA } 29)$$

The commutators with D vanish term by term. The quotient is by the linear range of D , not an algebra ideal: $[S(\chi-t)] = -u[1]$. The previous note's source deformation

$$D_t^{(k)} = D^{(k)} + tr_N \mathcal{R} j_E, \quad \mathcal{R}[P] = [S^{q-1}] \text{rem}_{\chi}(P) [1] \quad (\text{SGA } 30)$$

satisfies $j_{ED,t}^{(k)}(k) = A(t)j_E$ and $D_t^{(k)}(k)r_N - r_{NA}(t) = dK_N \wedge^{\text{prim}}$, with $A(t) = A + t\mathcal{R}$. Its original source relation is unchanged.

Let C_a be the degree- $<q$ matrix for reduction of $S^{(a+1)}$ times a representative by D/dS , and let $A(t)$ be the usual companion matrix for $\chi-t$. The period matrix obeys

$$\partial_{c_a} \Pi = \Pi C_a / ((a+1)u), \quad \partial_t \Pi = -\Pi A(t) / u. \quad (\text{SGA } 31)$$

S38.6.1 6.1 The trace of the reduced multiplication is exactly classical

Although C_a need not equal $A(t)^{(a+1)}$,

$$\boxed{\operatorname{Tr} C_a = \operatorname{Tr} A(t)^{a+1}.} \quad (\text{SGA } 32)$$

Here is the explicit comparison. Reducing $S^{(b+a+1)}$, with $0 \leq b \leq q-1$ and $0 \leq a \leq q-1$, starts with degree at most $2q-1$. Every term introduced by the uP' part has degree at most $b+a-q \leq b-1$. Later reductions only lower degree. Thus $C_a - A(t)^{(a+1)}$ is strictly upper triangular in the fixed increasing-power frame: its row indices are strictly below its column index. Its diagonal entries vanish, proving (32). No u -dependent coefficient is discarded from the full matrix.

S38.7 7. Complete evaluation of the determinant of periods

Set $d=q+1$. Fix $u \neq 0$ and a choice of $\arg u$ on a sector. Retain the rays

$$\theta_j = (\arg u + \pi)/d + 2\pi j/d, \quad \ell_j(r) = r e^{i\theta_j}, \quad r \geq 0, \quad \Gamma_j = -\ell_0 + \ell_j, \quad 1 \leq j \leq q. \quad (\text{SGA } 33)$$

The periods are

$$\Pi_{jb}(\mathbf{c}, u, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 0 \leq b < q. \quad (\text{SGA } 34)$$

The leading exponent on each ray is $-r^d/(d|u|)$. Lower-degree terms are uniformly dominated on compact coefficient sets. This proves convergence and all coefficient differentiations in (31). Integrating $D(P)e^{(\Phi_t/u)}$ gives a total derivative whose endpoints at infinity vanish and whose two origin values cancel with the stated orientations.

Define the polynomial

$$\boxed{\mathcal{F}(\mathbf{c}, t) = \operatorname{Tr}_{E_t/\mathbb{C}} M_{\Phi_t} = \lambda_t(\chi' \Phi_t).} \quad (\text{SGA } 35)$$

Both descriptions are finite coefficient calculations. At a complex fibre it is the sum of all critical values $\Phi_t(\lambda)$, counted with the multiplicity of each root of $\chi - t$.

Its gradients are

$$\boxed{\partial_{c_a} \mathcal{F} = \frac{1}{a+1} \operatorname{Tr} A(t)^{a+1}, \quad \partial_t \mathcal{F} = -\operatorname{Tr} A(t).} \quad (\text{SGA } 36)$$

Proof. On the open set of simple roots α_j , differentiate $\sum \Phi_t(\alpha_j)$. The α_j derivatives are multiplied by $\Phi_t'(\alpha_j) = \chi'(\alpha_j) - t = 0$. The remaining terms are exactly (36). Both sides of (36) are polynomials in the coefficients, obtained by companion traces. Their equality on the nonempty simple-root open set proves the polynomial identity everywhere, including all collisions. This extension does not replace the fibres by their reductions: their ranks and traces retain multiplicities.

Equations (31), (32), and Jacobi's determinant derivative give

$$\partial_{c_a}(\det \Pi) = \frac{\partial_{c_a} \mathcal{F}}{u} \det \Pi, \quad \partial_t(\det \Pi) = \frac{\partial_t \mathcal{F}}{u} \det \Pi. \quad (\text{SGA } 37)$$

These determinant equations hold even before invertibility is known, by the adjugate identity. At the monomial member $\mathbf{c}=0, t=0, \mathbb{F}=0$ and direct integration gives

$$(\Pi_{\text{mon}})_{jb} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/d} - 1). \quad (\text{SGA } 38)$$

The last matrix F_d has Gram $F_d^* F_d = d(I_q + J_q)$, where J_q is the all-ones matrix. Indeed expansion of the finite geometric sums gives diagonal $2d$ and off-diagonal d . Hence $|\det F_d|^2 = d^{q+1}$, in particular it is nonzero.

Along the straight coefficient path from the monomial member to any chosen $(\mathbf{c}, t)$, (37) is a scalar differential equation. Integrating it with its actual initial value proves

$$\det \Pi(\mathbf{c}, u, t) = \det \Pi_{\text{mon}}(u) \exp(\mathcal{F}(\mathbf{c}, t)/u). \quad (\text{SGA } 39)$$

There is no path-dependent logarithm in this equation. In particular the period matrix is invertible for every $u \neq 0$, at repeated as well as simple critical points.

The complete unsquared constant is retained by (38), its determinant, the ray ordering, and the choice of $\arg u$. Taking squared moduli and using the gamma multiplication identity

$$\prod_{l=1}^{d-1} \Gamma(l/d) = (2\pi)^{(d-1)/2} d^{-1/2} \quad (\text{SGA } 40)$$

gives

$$\det(\Pi^* \Pi) = (2\pi|u|)^q \exp(2\Re(\mathcal{F}/u)). \quad (\text{SGA } 41)$$

The factor $(2\pi|u|)^q$ is calculated from (38)–(40); it is not assigned a value by choosing a new basis. Formula (38) remains the authoritative phase-containing version.

This explicitly finishes the determinant calculation for the previously constructed exponential comparison. It is consistent with the classical $\exp(f)$ determinant theory; it is not a claim that the full period matrix, or its arithmetic condition number, is determined by its determinant.

S38.8 8. Recover the complete cyclic spectrum from that determinant family

For $1 \leq r \leq q$, equation (37) gives the logarithmic derivative without choosing a logarithm branch:

$$p_r := ru \frac{\partial_{c_{r-1}} \det \Pi}{\det \Pi} = \text{Tr } A(t)^r. \quad (\text{SGA } 42)$$

Put $d_0=1$ and recursively define

$$d_n = -\frac{1}{n} \sum_{r=1}^n d_{n-r} p_r, \quad 1 \leq n \leq q. \quad (\text{SGA } 43)$$

Then

$$\det(XI - A(t)) = X^q + d_1 X^{q-1} + \cdots + d_q = \chi_{\mathbf{c}}(X) - t. \quad (\text{SGA } 44)$$

Proof. In the formal variable z ,

$$\frac{d}{dz} \log \det(I - zA(t)) = -\text{Tr}(A(t)(I - zA(t))^{-1}) = -\sum_{r \geq 1} p_r z^{r-1}. \quad (\text{SGA } 45)$$

Coefficient extraction gives (43), and the determinant is a degree- q polynomial. Because $A(t)$ is the specified cyclic companion action, its characteristic polynomial is also its minimal polynomial. Thus this recovery retains one complete length- m block at every root of multiplicity m . For a general noncyclic larger arithmetic module, traces do not classify all Jordan blocks; its inclusion η and complementary module stay in the source programme.

The precise map chain is

$$(\mathcal{H}, \nabla_{c_a}) \longrightarrow (\det \mathcal{H}, \det \nabla_{c_a}) \xrightarrow{(42)} (p_1, \dots, p_q) \xrightarrow{(43)} \chi - t. \quad (\text{SGA } 46)$$

Only the full coefficient-family derivatives in (42), not a single determinant value, give the recovery.

S38.9 9. Two explicit calibrations

For $\chi(S)=S-\lambda$,

$$\Phi_t(S) = S^2/2 - (\lambda + t)S, \quad A(t) = \lambda + t, \quad \mathcal{F} = -(\lambda + t)^2/2. \quad (\text{SGA } 47)$$

For $u>0$, with the original contour oriented upwards on $i\mathbb{R}$,

$$\Pi(u, t) = i\sqrt{2\pi u} \exp(-(\lambda + t)^2/(2u)). \quad (\text{SGA } 48)$$

In the ordering (33), $q=1$ gives Γ_1 oriented downwards, so $\Gamma_{\text{up}} = -\Gamma_1$ and the period in (48) is the negative of the sole entry of (34). This is the exact orientation map; the squared determinant is unchanged. The full nonzero λ remains; a cohomological dimension or pure finite-field weight does not set it to a critical-line value.

For $\chi(S)=S^2+c_1S+c_0$,

$$\mathcal{F} = c_1^3/6 - c_1(c_0 - t), \quad \det(\Pi^*\Pi) = (2\pi|u|)^2 \exp\left(2\Re \frac{c_1^3/6 - c_1(c_0 - t)}{u}\right). \quad (\text{SGA } 49)$$

At the discriminant $c_1^2-4(c_0-t)=0$ the formula stays unchanged and the fibre retains its double-root algebra. This is tested symbolically. Independent ray quadrature for small declared complex coefficient fixtures is reported separately, without an interval-certification claim.

S38.10 10. Transport the original metric and control through the same periods

At the actual marked coefficients, use the original arithmetic G_N of (10). On the period-coordinate target define the **transported original metric**

$$\tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}, \quad \Pi^{-*} := (\Pi^{-1})^*. \quad (\text{SGA } 50)$$

The map $\Pi:(E, G_N) \rightarrow (\mathbb{C}^q, \tilde{G}_N)$ is an isometry by this explicit formula, not by an assignment of the original metric to L_q . Conversely $\Pi^* \Pi$ is the Gram for the separately specified coordinate metric L_q , and its comparison operator is $G_N \wedge (-1) \Pi^* \Pi$.

The exact determinant formula is

$$\det \tilde{G}_N = \frac{V_N}{(2\pi|u|)^q} \exp\left(-2\Re \frac{\mathcal{F}}{u}\right). \quad (\text{SGA } 51)$$

At a common (c, u, t) , the complete marked endpoint ratio satisfies

$$\frac{\det \tilde{G}_i}{\det \tilde{G}_j} = \frac{V_i}{V_j}. \quad (\text{SGA } 52)$$

Thus every common period determinant factor in that ratio is now explicitly known and cancels by (51), with no remaining unknown proportionality constant. This identifies exactly which information the period determinant does and does not add to the endpoint estimate.

For fixed $u>0$ and real t , differentiate (50), retaining the original G_N fixed and using $\Pi' = -\Pi A(t)/u$. One obtains

$$u \frac{d}{dt} \tilde{G}_N = \Pi^{-*} (A(t)^* G_N + G_N A(t)) \Pi^{-1}, \quad (\text{SGA } 53)$$

and therefore

$$u \frac{d}{dt} \tilde{G}_N - k \tilde{G}_N = \Pi^{-*} W_N(t) \Pi^{-1}, \quad W_N(t) = A(t)^* G_N + G_N A(t) - k G_N. \quad (\text{SGA } 54)$$

At $t = 0$, $W_N(t)$ is the original arithmetic control. For any vector x , the coordinate change $x = \Pi v$ carries both numerator and denominator of its generalized Rayleigh quotient exactly. Equations (30) and (11) show that the source theta primitive remains the same along this specific t -deformation.

This is a differential-metric reformulation of the same problem. The finite-field pure cohomology of the auxiliary family controls its Frobenius, not automatically the right side of (54). The required relation between those operators has not been supplied by taking a determinant.

S38.10.1 10.1 The constituent period minor detects the arithmetic defect

The full determinant is sensitive to all coefficient derivatives by (42), but along the original t-direction it sees only $\text{Tr } A$. In a reflection-stable arithmetic packet of dimension q , $2\Re \text{Tr } A = kq$. For the full conjugation-stable quartet packet, $\text{Tr } A = kq/2$ exactly. This is already the total weight balance; an off-line pair can retain that same real trace.

The correct retained subobject is an actual A -invariant inclusion $I_F : F \hookrightarrow E$, $AI_F = I_F A_F$. Let $p = \dim F$ and define the **rectangular** period map and its two Grams

$$P_F(t) = \Pi(u, t)I_F : F \longrightarrow \mathbb{C}^q, \quad H_F(t) = P_F(t)^* P_F(t), \quad G_{N,F} = I_F^* G_N I_F. \quad (\text{SGA } 54a)$$

The determinant here is the norm of the actual exterior map $\bigwedge^p P_F$ on $\det F$; no determinant of a rectangular matrix is taken. The domain Gram retains the theta-source inclusion $r_N I_F$ and all its jets. For real t and $u > 0$,

$$P'_F(0) = -P_F(0)A_F/u, \quad \boxed{-u \frac{d}{dt} \log \frac{\det H_F(t)}{\det G_{N,F}} \Big|_{t=0} - kp = 2\Re \text{Tr } A_F - kp.} \quad (\text{SGA } 54b)$$

This follows by differentiating the displayed Gram and taking its trace. It is not a purity assumption on F . For the full positive-real-defect generalized eigenspace $F = E_{>}$, the right side is the exact aggregate $L_{h,k}$ already used by the exterior theorem. Thus the period family has an explicit observable for the original off-line defect, not just for its already-balanced total trace.

The original extension to the other constituents is present as well. Let P be the orthogonal projection in the specified period-coordinate metric onto $\text{im } P_F(0)$. Since $A(t) = A + t\mathcal{R}$,

$$\boxed{(1 - P)P'_F(0) = 0, \quad (1 - P)P''_F(0) = -\frac{1}{u}(1 - P)\Pi(u, 0)\mathcal{R}I_F.} \quad (\text{SGA } 54c)$$

Both identities are direct differentiations. The first derivative is internal to F ; the first possible normal acceleration is the exact rank-at-most-one map on the right. Its kernel, image, and norm remain available. A repeated-root primary subspace is allowed; no eigenspace basis or semisimple replacement is used.

Equations (54a)–(54c) specify the constituent period object that can be estimated next. Formula (41) finishes the full determinant, but does not by itself bound these minor norms or their comparison with $G_{N,F}$. The preceding logarithmic-constituent extension and its coupling subtraction remain the source-specific metric input.

S38.11 11. The source-specific remaining term stays explicit

For the current programme, the original endpoint cost is

$$\mathcal{B}_{h,k} = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}}. \quad (\text{SGA } 55)$$

The preceding logarithmic-constituent calculation already supplies a genuine source upper certificate. For an A -invariant inclusion $I:F \rightarrow E$ and its G_j -orthogonal coefficient lift H_0 of E/F , let $C = G_i - G_j$ and

$$G_{F,0} = I^* G_j I, \quad C_F = I^* C I, \quad D_F = I^* C H_0, \quad Q_0 = H_0^* G_j H_0, \quad C_Q = H_0^* C H_0. \quad (\text{SGA } 56)$$

Put $Q_+ = Q_0 + C_Q$ and

$$\Xi_F = Q_+^{-1} D_F^* (G_{F,0} + C_F)^{-1} D_F. \quad (\text{SGA } 57)$$

That construction proves the exact decomposition

$$\log(V_i/V_j) = \log \frac{\det(G_{F,0} + C_F)}{\det G_{F,0}} + \log \frac{\det Q_+}{\det Q_0} - \sum_{a \geq 1} \frac{\text{Tr}(\Xi_F^a)}{a}. \quad (\text{SGA } 58)$$

These are the source matrices that still require uniform control. Under the coordinate change Π , all inclusions, quotient lifts, metrics and D_F are transported together; (58) retains the same values. A period basis does not eliminate the coupling or the theta mass.

The current GitHub statement retains the already derived off-line-quartet consequence

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4. \quad (\text{SGA } 59)$$

This note does not establish a contradictory upper estimate. It finishes the diagonal trace and the determinant-period calculation, recovers all cyclic power traces from that determinant family, and gives the exact unchanged theta-metric comparison (51)–(54). The quantitative focus remains a bound on (58), or equivalently on the same original generalized control in (54), rather than another uncomputed determinant or an assumption that local support entails metric purity.

S38.12 12. Relation to Deligne’s finite-field family

The previous construction takes the finitely generated coefficient ring R_0 containing the actual c_a and $1/(q+1)!$, and for a specified finite-field specialization of characteristic $p > q+1$ obtains

$$Y^p - Y = \frac{\Phi_t(S)}{u}, \quad u \neq 0. \quad (\text{SGA } 60)$$

Its infinity chart is the original one: with $w=1/S$,

$$\Phi_t/u = w^{-d}b(w), \quad b(0) = 1/(du), \quad bw^d = 1, \quad x_\infty = wv, \quad \Phi_t/u = x_\infty^{-d}. \quad (\text{SGA } 61)$$

The derivative in v has the specified inverse v/d . This is the retained local model for applying Deligne’s exponential-sum/lissity theorem over $u \neq 0$. The generic polynomial has rank- q degree-one compact cohomology and weight-one Frobenius under those hypotheses. Those are the inherited cited Deligne results, not fresh analytic conclusions of this note.

The maps are the span

$$(R_0, \Phi, u, t) \longrightarrow \begin{cases} (\mathbb{C}, \mathcal{C}_{u,t}, \nabla, \Pi), \\ (\mathbb{F}_Q, \mathcal{L}_\psi(\Phi_t/u), R^1 f_!). \end{cases} \quad (\text{SGA } 62)$$

The SGA trace formalism applies in each specified coefficient realization with its own endomorphism. It does not supply a homomorphism $\mathbb{C} \rightarrow \mathbb{F}_Q$ or an intertwiner from the auxiliary Frobenius to the original A . The actual complex source/operator comparison is (30), (31), and (54). This keeps the two realizations connected at their common coefficient model without filling the missing operator comparison with an unsupported equality.

S38.13 13. Verification and reproducibility

check_trace_period.py contains eighteen finite exact regression methods: residue duality, the diagonal tensor, the trace/Jacobian composite, repeated-root fibres, periodic diagonal resolution, coefficient-base-change identities, quantum-reduction traces, critical-value gradients, Newton recovery, source conormal response, full metric congruences, and split quotient labels. Separate negative controls alter the Jacobian, discard the quantum derivative, or erase multiplicity; each must fail in both normal and optimized Python.

The period convergence, determinant formula for arbitrary q , and all-general resolution exactness have written proofs above. Finite symbolic or numerical tests are not replacements for those proofs. period_numerics.py performs additional explicitly non-certified contour comparisons on declared small fixtures. No new Lean execution, arithmetic zero location, interval certificate, or uniform purity theorem is claimed.

The package does not redistribute the uploaded SGA corpus. The source inventory retains hashes, original paths and read ranges. The GitHub read was pinned to c05c709d707bacda335deff70fbb5b841a751d1f. No remote file is changed by building this package.

S38.14 References and source status

1. A. Grothendieck, written up by L. Illusie, *SGA 5, Exposé III: Lefschetz Formula*, §§4.1, 4.7–4.8, and coherent appendix §§6.1–6.8, especially (6.8.5). Basis: the user’s supplied English master, read directly as LaTeX. Source wording and qualifications are recorded separately.
2. P. Deligne, *La conjecture de Weil II*, Publications Mathématiques de l’IHÉS 52 (1980), 137–252. The previous source-reviewed construction uses §§3.7.2–3.7.4. This execution reads that predecessor’s TeX and preserves its stated scope, rather than claiming a new complete scan verification.
3. S. Bloch and H. Esnault, *Gauß–Manin determinant connections and periods for irregular connections*, arXiv:math/9912095 (1999). The abstract was inspected for attribution; its TeX download failed, so no detailed theorem of that article is asserted as independently read here.
4. NIST Digital Library of Mathematical Functions, §5.5(iii), gamma multiplication formula. It is used only to evaluate the explicit product in (38); the full product and phases remain available.
5. Parent programme sources: *Tau_Deligne_Exponential_Comparison* and *Tau_Deligne_Logarithmic_Constituent_Control*, 13 September 2026. Their original sections, arithmetic source maps, metric definitions, and verification scopes are retained. General source and analytic constructions are inherited where expressly marked above.

S39 The original theta complex inside the Hochschild trace realization

Source role. Complete public PR27 proof note H1-H42, including all source qualifications, critical-kernel maps and the zero-line heat-family proof. Original display tags retain their complete literal source tokens; the local typesetting namespace CC-H identifies this source.

Exact revision. 0f5707281ee46ec411b1af5d25361cf23bb88c3689a0f3cd75f7fac5b951be13

Complete source. `sources/sga_connes_complete_closure/final_proof_acceptance/PC/dependencies/HOCHSCHILD_COMPARISON.md`

Source SHA-256. 0f5707281ee46ec411b1af5d25361cf23bb88c3689a0f3cd75f7fac5b951be13

S39.1 The original theta complex inside the Hochschild trace realization

Research continuation, 13 September 2026. Source attribution: the split-support arithmetic programme and its representative maps are the owner's construction. The literature comparison uses Connes and Consani (2023), printed pp. 83–101. This note gives written proofs of the comparison and its boundary qualification. It does not claim a formalization of the adèle crossed product, Schwartz analysis, sheaf cohomology, or a uniform arithmetic weight estimate.

S39.1.1 1. Fix the objects before comparing them

Use the original spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad \mathcal{B} = \{F : \sup_{x>0} x^b |D^m F(x)| < \infty \forall b \in \mathbb{Z}, m \geq 0\}, \quad D = -x\partial_x. \quad (\text{CC-H H1})$$

The arithmetic map and its original quotient are

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V. \quad (\text{CC-H H2})$$

The factor two, the quotient topology, and the original supported coefficient square remain

$$\begin{array}{ccc} G(\mathbb{Z}) & \longrightarrow & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \longrightarrow & \mathbb{C}. \end{array} \quad (\text{CC-H H3})$$

Here the supported scalar zero is $e = 0^\bullet$, not the absent scalar τ . Every linear comparison below is a comparison of coefficient fibres; its split lift sends (λ, v) to (λ, fv) , and sends external absence to external absence. In particular an amplitude killed by a comparison remains $(\lambda, 0)$. We do not apply ordinary Hochschild homology directly to the split semiring and assert that additive completion preserves its support masks.

S39.1.2 2. The theta map is the actual Hochschild trace

Let

$$\mathcal{A} = \mathcal{S}(\mathbb{A}_{\mathbb{Q}}) \rtimes \mathbb{Q}^\times, \quad j(f) = [(1_{\mathbb{Z}} \otimes f)U(1)] \in HH_0(\mathcal{A}). \quad (\text{CC-H H4})$$

Connes and Consani, Proposition 3.1, identify the kernel of the inclusion-induced map $\pi_!$ with the span of the differences $f - f^q$. Proposition 3.3 identifies even Schwartz functions with the specified finite-unit-invariant **image of** $\pi_!$. This is not an identification with every sector of $HH_0(\mathcal{A})$.

Their orbit-trace calculation, Lemma 3.2(ii), gives

$$\text{Tr } j(f)(u) = 2 \sum_{n \geq 1} f(nu). \quad (\text{CC-H H5})$$

Consequently the following is a literal factorization of the original arithmetic map:

$$V \xrightarrow{j} HH_0(\mathcal{A}) \xrightarrow{\text{Tr}} \mathcal{B}, \quad \text{Tr} \circ j|_V = \Theta. \quad (\text{CC-H H6})$$

The last arrow in (H6) is understood on the displayed subspace, where its trace has the stated strong-Schwartz range; an unrestricted trace on all crossed-product classes is not asserted.

The book's normalized map is

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu) = \frac{1}{2} \mathcal{U} \Theta f(u), \quad \mathcal{U}F(u) = u^{1/2} F(u). \quad (\text{CC-H H7})$$

The map $\mathcal{U} : L^2(\mathbb{R}_{>0}, du) \rightarrow L^2(\mathbb{R}_{>0}, du/u)$ is an isometry, since its squared norm is exactly $\int |F|^2 du$. This assertion uses the positive half-line. An even full-line realization has its separately retained factor two.

S39.1.3 3. The source Gaussian is exactly a Laplacian preimage

Write the book's generator as $H = x\partial_x = -D$, and distinguish its Laplacian from a representative increment:

$$\mathcal{L} = H(1 + H) = D^2 - D. \quad (\text{CC-H H8})$$

For the unchanged Gaussian $g_0(x) = e^{-\pi x^2}$, direct differentiation gives

$$\boxed{\mathcal{L}g_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} = \phi_*}. \quad (\text{CC-H H9})$$

Thus the original source function has the exact homological expression

$$f_0 = \Theta\phi_* = \text{Tr } j(\mathcal{L}g_0), \quad \mathcal{M}f_0(s) = 2\xi(s). \quad (\text{CC-H H10})$$

For example, Mellin integration by parts gives

$$\mathcal{M}\phi_*(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad (\text{CC-H H11})$$

initially in a common convergence half-plane and then by continuation. Multiplication by $2\zeta(s)$ gives $2\xi(s)$, not $\xi(s)$. No change of seed or leading coefficient is involved.

Lemma 4.1(iii) of the chapter gives a bijection $\mathcal{L} : \mathcal{S}(\mathbb{R})_{\text{ev}} \rightarrow V$. The differential operator is continuous. The two moment constraints define a closed Frechet subspace, so the open mapping theorem also gives a continuous inverse. We obtain a topological cochain isomorphism

$$[\mathcal{S}(\mathbb{R})_{\text{ev}} \xrightarrow{\Theta\mathcal{L}} \mathcal{B}] \xrightarrow{(\mathcal{L}, 1)} [V \xrightarrow{\Theta} \mathcal{B}]. \quad (\text{CC-H H12})$$

It commutes with the polynomial action of D . Moreover $\mathcal{F}D = (1 - D)\mathcal{F}$, so $\mathcal{F}\mathcal{L} = \mathcal{L}\mathcal{F}$. Hence the same comparison on both ordered legs gives

$$[\mathcal{S}_{\text{ev}}^2 \xrightarrow{\Theta\mathcal{L}(1, -\mathcal{F})} \mathcal{B}] \xrightarrow{(\mathcal{L} \oplus \mathcal{L}, 1)} [V^2 \xrightarrow{\Theta(1, -\mathcal{F})} \mathcal{B}]. \quad (\text{CC-H H13})$$

The degree-zero cycle $(v, \mathcal{F}^{-1}v)$ is transported, not removed. The fibrewise lifts of (H12)–(H13) preserve the original labels and commute with synchronization into the joint label.

S39.1.4 4. Keep the closure comparison as an arrow

The quotient in Proposition 4.2 is, on the specified trace image,

$$Q_{\text{CC}} = \mathcal{B} / \overline{\text{Tr } j(\mathcal{L}\mathcal{S}_{\text{ev}})} = \mathcal{B} / \overline{\Theta V}. \quad (\text{CC-H H14})$$

There is always a canonical continuous map

$$Q \longrightarrow Q_{\text{CC}}, \quad \ker = \overline{\Theta V} / \Theta V. \quad (\text{CC-H H15})$$

Under the prior global-retraction theorem $\Lambda\Theta = 1_V$, the image ΘV is closed: it is the range of the continuous idempotent $\Theta\Lambda$, equivalently the kernel of $1 - \Theta\Lambda$. Then (H15) is a topological isomorphism. This use of the prior analytic theorem is explicit; a finite matrix computation does not prove it.

The chapter's full spectral statement concerns Q_{cc} : the Laplacian eigenvalues are $\rho(\rho - 1)$ for all nontrivial zeros, with their jets. Its critical-only construction later in the chapter is a different quotient. The following calculation supplies a map between the two kinds of coefficient object without presuming injectivity.

S39.1.5 5. A specified critical-line comparison and its exact finite kernel

Choose the plus-sign Mellin restriction

$$J_{\text{crit}} : \mathcal{B} \longrightarrow \mathcal{S}(\mathbb{R}), \quad J_{\text{crit}} F(t) = \mathcal{M}F(\tfrac{1}{2} + it). \quad (\text{CC-H H16})$$

This is continuous: in logarithmic coordinates, $e^{y/2}F(e^y)$ and all derivatives have faster-than-polynomial decay, with bounds controlled by finitely many defining seminorms of $\mathcal{B}$. Fourier transformation therefore lands continuously in Schwartz space. The plus sign is deliberate; the chapter's displayed Fourier transform uses the opposite sign, related by $t \mapsto -t$.

Put

$$z_c(t) = \zeta(\tfrac{1}{2} + it), \quad Q_{\text{crit}} = \mathcal{S}(\mathbb{R}) / \overline{z_c \mathcal{S}(\mathbb{R})}. \quad (\text{CC-H H17})$$

For $\phi \in V$, Mellin factorization gives

$$J_{\text{crit}}(\Theta\phi)(t) = 2z_c(t)\mathcal{M}\phi(\tfrac{1}{2} + it). \quad (\text{CC-H H18})$$

The remaining Mellin factor is Schwartz: at the origin $\phi(u) = O(u^2)$, and at infinity it is Schwartz; repeated logarithmic differentiation and integration by parts control every Fourier seminorm. The zeta factor is a Schwartz multiplier, as used in the chapter's Section 5. Thus (H18) induces an actual continuous linear map

$$\boxed{\Gamma : Q \longrightarrow Q_{\text{crit}}, \quad [F] \longmapsto [J_{\text{crit}}F].} \quad (\text{CC-H H19})$$

It also factors through (H15), by continuity and the closed target relation space. Its operator equations are

$$\Gamma D_Q = M_{1/2+it}\Gamma, \quad \Gamma(a^{-1/2}\overline{U}_a) = M_{a^{it}}\Gamma. \quad (\text{CC-H H20})$$

Theorem 5.4 identifies the target coefficient quotient with global sections of the Scaling Site quotient sheaf. Section 7 below records a zero-mode qualification needed before using the printed raw-section formula at the boundary $L = 0$.

Now let Z be a finite packet of actual zeros, with the retained multiplicities, and use the previously constructed inclusion

$$\sigma_Z : E_Z = \mathbb{C}[s]/(h_Z) \hookrightarrow Q, \quad D_Q\sigma_Z = \sigma_Z A_Z, \quad A_Z = M_s. \quad (\text{CC-H H21})$$

Chinese remainders give the specified decomposition $E_Z = E_{Z,\text{crit}} \oplus E_{Z,\text{off}}$, according to whether $\Re\rho = 1/2$. Then

$$\boxed{\ker(\Gamma\sigma_Z) = E_{Z,\text{off}}.} \quad (\text{CC-H H22})$$

Proof on the off-critical blocks. For $\Re\rho \neq 1/2$, the function $(1/2 + it - \rho)^{-1}$ is smooth on the real line and all its derivatives are polynomially bounded. It is a continuous Schwartz multiplier, commutes with z_c , and preserves both the ideal and its closure. Therefore $M_{1/2+it} - \rho$ is invertible on Q_{crit} . If $(A_Z - \rho)^m v = 0$, (H20) gives

$$(M_{1/2+it} - \rho)^m \Gamma\sigma_Z v = 0,$$

and hence $\Gamma\sigma_Z v = 0$. This kills the full generalized block under this particular map, not just its reduced eigenline.

Proof on a critical block. Let $\rho = 1/2 + i\gamma$ have order m . The continuous derivative evaluations descend to

$$\Xi_\rho : Q_{\text{crit}} \longrightarrow \mathbb{C}[\epsilon]/(\epsilon^m), \quad \Xi_\rho[f] = \sum_{j=0}^{m-1} \frac{i-j}{j!} f^{(j)}(\gamma) \epsilon^j. \quad (\text{CC-H H23})$$

Indeed these functionals vanish on $z_c \mathcal{S}$, and hence on its closure. The chain rule gives

$$\Xi_\rho \Gamma[F] = \sum_{j < m} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} \epsilon^j. \quad (\text{CC-H H24})$$

The right side is exactly the original Mellin-jet observation. Its composite with σ_Z is the Chinese-remainder projection to the ρ -block, by $J_Z \sigma_Z = 1$. Taking all critical blocks provides a left inverse on $E_{Z,\text{crit}}$, proving (H22).

In split coordinates, (H22) says

$$\tilde{\Gamma}(\lambda, \sigma_Z v) = (\lambda, 0) \quad (v \in E_{Z,\text{off}}), \quad (\text{CC-H H25})$$

not external absence. The critical quotient's spectral location does not establish injectivity of (H19). Formula (H22) calculates the exact finite failure of injectivity that a comparison argument would have to address.

S39.1.6 6. The original metric supplies the Laplacian defect, not an assumed Hodge star

Fix an original representative $R : E_Z \rightarrow \mathcal{B}$ with

$$DR - RA = \Theta K, \quad G = R^* R, \quad W = A^* G + GA - G. \quad (\text{CC-H H26})$$

The normalized representative $\hat{R} = \mathcal{U}R$ has the same Gram matrix. With $D_\times = -u\partial_u$ acting in multiplicative coordinates,

$$D_\times \hat{R} - \hat{R}(A - \tfrac{1}{2}I) = \mathcal{U}\Theta K = 2\mathcal{E}K, \quad W = (A - \tfrac{1}{2}I)^* G + G(A - \tfrac{1}{2}I). \quad (\text{CC-H H27})$$

Thus the shift and the source boundary are both accounted for. In particular applying $\mathcal{U}$ to the entire relation curve $R_j + z\delta R$ leaves its Gram matrix, restriction return, determinant ratio, and curvature cost unchanged.

For the tensor-weight algebraic formula, let $k \in \mathbb{R}$, keep the original A, G , and put

$$L_k = A^2 - kA, \quad W_k = A^* G + GA - kG. \quad (\text{CC-H H28})$$

Expansion proves

$$\boxed{L_k^* G - GL_k = A^* W_k - W_k A}, \quad \boxed{GL_k = W_k A - A^* G A}. \quad (\text{CC-H H29})$$

Writing $B = A - kI/2$ gives the equivalent centered energy identity

$$\boxed{GL_k = W_k B - B^* G B - \frac{k^2}{4} G}. \quad (\text{CC-H H30})$$

For (H29), expand $A^*(A^*G + GA - kG) - (A^*G + GA - kG)A$; the middle A^*GA terms cancel and the remaining expression is $L_k^* G - GL_k$. For (H30), substitute $GA = W_k - A^*G + kG$, or expand with $A = B + kI/2$. These are exact identities before any estimate. The real part of the first term on the right of (H30) is the retained error in the expected negative energy inequality.

The existing rank-two source formula for W_n consequently gives a rank-at-most-four formula for $L^*G - GL$. This rank fact alone bounds neither its relative norm nor its effect on the growing arithmetic packet.

An actual constituent inclusion J with $AJ = JA_F$ satisfies $L_k J = J(A_F^2 - kA_F)$. No commutation of A with a metric restriction-loss operator is required.

The full jet formula is

$$L_k|_{E_\rho} = \rho(\rho - k)I + (2\rho - k)N_\rho + N_\rho^2. \quad (\text{CC-H H31})$$

When $2\rho - k \neq 0$, the nilpotent factor $(2\rho - k)I + N_\rho$ is invertible and commutes with N_ρ . Therefore the powers of the nilpotent part have exactly the same kernels as N_ρ^j : Jordan lengths remain unchanged. In particular negative real diagonal spectrum is not the same as self-adjointness on a positive Hilbert metric. Requiring exact self-adjointness on a multiple-zero packet would also force semisimplicity. The chapter itself retains nontrivial Jordan actions in Remark 5.5.

S39.1.7 7. The boundary zero mode needs to be retained separately

There is a concrete qualification in the printed Section 5 construction. Proposition 5.2(iii), p. 94, places the raw family $\Sigma_{eL}\mathcal{E}\phi$ in the smooth-section sheaf vanishing at $L = 0$. But the same proof, p. 95, explicitly gives its zeroth normalized Fourier coefficient as

$$\widehat{\Sigma_{eL}\mathcal{E}\phi}(0) = L^{-1/2}a_\phi, \quad a_\phi = \int_0^\infty \mathcal{E}\phi(u) \frac{du}{u} = \zeta(\tfrac{1}{2})\mathcal{M}\phi(\tfrac{1}{2}). \quad (\text{CC-H H32})$$

For the original seed (H9) this is

$$a_{\phi_*} = \xi(\tfrac{1}{2}) \neq 0. \quad (\text{CC-H H33})$$

The nonvanishing is exact, not inferred from numerics. In (H11) the gamma factor is nonzero at $1/2$; and $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$ for real $0 < s < 1$, since the paired alternating series for $\eta(s)$ is strictly positive. Thus the raw coefficient $\xi(1/2)L^{-1/2}$ does not extend continuously to zero, let alone vanish to all orders. This calculation concerns the literal raw-section assertion, not the validity of the preceding Hochschild trace and all-zero quotient statements.

A support-retaining repair is to use the augmented Fourier-coordinate object

$$\mathcal{H}^\#(U) = C_{\text{flat}}^\infty(U, \ell^2(\mathbb{Z} \setminus \{0\})) \oplus C^\infty(U), \quad (\text{CC-H H34})$$

where flatness is imposed at zero when it belongs to U . Map a source function to

$$\Sigma^\# \mathcal{E}\phi = (\Pi_{\neq 0} \widehat{\Sigma \mathcal{E}\phi}, a_\phi). \quad (\text{CC-H H35})$$

For $L > 0$, reconstruct the full raw Fourier vector by

$$(\eta(L), a(L)) \mapsto \eta(L) + L^{-1/2}a(L)e_0. \quad (\text{CC-H H36})$$

In the circle's function coordinates, the added constant function has value $a(L)/L$. The different powers in these two coordinate systems are forced by the unitary normalization of the Fourier basis.

The covering transfer satisfies

$$\widehat{\sigma_n \xi}(L, j) = \sqrt{n} \widehat{\xi}(nL, nj). \quad (\text{CC-H H37})$$

Accordingly the renormalized zero-line coordinate transforms by $a(L) \mapsto a(nL)$. The two components in (H34) therefore have genuine compatible $\mathbb{N}^\times$ -actions. Normalized scaling is trivial on the zero line. Nothing has been silently discarded.

The extra zero line is an actual closed boundary Here is an explicit source proof of its fate in the quotient. For $T > 0$, form in the original Frechet space

$$\phi_T(x) = \frac{1}{\sqrt{4\pi T}} \int_{\mathbb{R}} e^{-r^2/(4T)} e^{-r/2} \phi_*(e^{-r}x) dr. \quad (\text{CC-H H38})$$

For every fixed Schwartz seminorm the seminorm of the integrand's dilated factor is bounded by a constant times $e^{C|r|}$. Hence the Gaussian integral converges in Schwartz space. Parity and both original moment conditions persist, so $\phi_T \in V$. This does not assert uniform source seminorm bounds as $T \rightarrow \infty$.

The normalized dilation identity

$$\mathcal{E}(e^{-r/2}\phi_*(e^{-r}\cdot))(u) = \mathcal{E}\phi_*(e^{-r}u)$$

and Fourier transformation give

$$\widehat{\mathcal{E}\phi_T}(s) = e^{-Ts^2}\widehat{\mathcal{E}\phi_*}(s), \quad a_{\phi_T} = a_{\phi_*}. \quad (\text{CC-H H39})$$

On every compact interval $0 \leq L \leq M$, all nonzero frequencies have magnitude at least $2\pi/M$. Differentiating their Fourier coefficients produces only finite sums of powers of T , frequency, and L^{-1} , times the Gaussian and derivatives of a Schwartz function. Splitting the Gaussian into two equal factors yields, for each flat-section seminorm of derivative order m , a bound

$$p_{N,m,[0,M]}(\Pi_{\neq 0}\widehat{\Sigma\mathcal{E}\phi_T}) \leq C_{N,m,M}(1+T)^m e^{-2\pi^2 T/M^2} \longrightarrow 0 \quad (T \geq 1). \quad (\text{CC-H H40})$$

To see the uniformity at $L = 0$, replace every negative power of L by a corresponding power of $|s|/(2\pi|n|)$, use the second Gaussian to absorb frequency powers, and sum an arbitrarily high negative power of $|n|$. The required constants depend on finitely many Schwartz seminorms of $\widehat{\mathcal{E}\phi_*}$.

Thus

$$\Sigma^\# \mathcal{E}\phi_T \longrightarrow (0, \xi(1/2)). \quad (\text{CC-H H41})$$

Since the last scalar is nonzero, the closed source-generated module in (H34) contains the entire renormalized zero line. Multiplication by local smooth functions gives the corresponding sheaf statement. Once this line is retained, its removal from the quotient is justified by a displayed family of original source relations. Algebraically, if $N \subset M \oplus \mathcal{O}$ is the closed source module and $0 \oplus \mathcal{O} \subset N$, then $N = \text{pr}_M \bar{N} \oplus \mathcal{O}$, and the quotient is the nonzero-mode quotient. The split lift makes the zero line a supported boundary, not absence.

This is a written repair and comparison, not a completed Lean proof of the heat-limit or sheaf assertion. It supplies the missing zero-mode convention without adding an artificial third vanishing condition to the original V .

A separate local typo on p. 100 is also relevant when reading the quotient argument: $\zeta(1/2 \mp 2\pi i/L)$ tends to $\zeta(1/2) \neq 0$ as $L \rightarrow \infty$, not as $L \rightarrow 0$. We use the former limit. The published page and the supplied scan were inspected; this is not an OCR substitution.

S39.1.8 8. What has and has not been identified

The established chain is

$$\text{original theta source} \xleftarrow{\mathcal{L}} \text{specified Hochschild trace image} \longrightarrow Q_{\text{CC}} \xrightarrow{\Gamma} Q_{\text{crit}}, \quad (\text{CC-H H42})$$

where the first comparison is the actual two-term-complex isomorphism (H12), not a degree-forgetting isomorphism of every displayed space. On finite arithmetic packets the last arrow has the calculated kernel (H22), and every surviving critical jet has the explicit recovery (H23).

The polynomial Laplacian uses the actual arithmetic action A . The curvature restriction-loss operator uses the actual source metrics. They are not identified with one another, and their eigenspaces are not presumed to agree. Equations (H27)–(H31) give the actual connection through the original boundary form W .

The chapter's Hodge dictionary is explicitly heuristic on p. 91. Its Corollary 4.3 makes a negative-spectrum statement equivalent to RH; it does not prove that statement by declaring the displayed star to be the adjoint for the original Gram metric. Here (H29) computes that adjoint defect instead.

S39.1.9 9. Formal and computational scope

SplitZeroLaplacianControl.lean formalizes the matrix identities (H29), constituent intertwining, the full quadratic jet (H31), and the general lemma that an intertwiner sends a nilpotent block to zero when the corresponding shifted target operator is injective. The actual target multiplier's invertibility and the critical-jet recovery are the analytic written proof in Section 5, not assumed custom Lean axioms.

The four preceding new modules formalize the relation Gram curve, polynomial local specialization sequence, least invariant hull, and curvature/restriction scalar identities. Their scope and the cyclic gcd/metric-pole consequences are recorded in RESEARCH_NOTE.md. The strict workflow checks individual sources at `--trust=0`, treats warnings as errors, and audits the selected 29+9 new declarations against `propext`, `Classical.choice`, and `Quot.sound`. Actual run status is recorded separately after execution; source text alone is not a successful certificate.

The exact Python checks include the original Gaussian differentiation, all factors in (H11), the non-Euclidean polynomial-source Gram metric, generalized jets, the derivative phase $i^{-j}/j!$, and the renormalized covering coefficient. The finite critical/off-critical polynomial calibration is expressly not a set of asserted zeta zeros. Neither those checks nor the Lean algebra prove a uniform arithmetic nonconcentration bound.

S39.1.10 References and reading scope

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and ζ -cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83–101). American Mathematical Society. <https://doi.org/10.1090/pspum/105/01896>

The complete nineteen-page chapter was read in the uploaded volume. The exact trace factor, Laplacian lemma, Hodge dictionary, and zero-mode passage were also inspected as rendered pages. The p. 100 limit and Jordan remark were checked in the author's published offprint. This does not claim reading the full proceedings volume or formal verification of the chapter.

Prior programme inputs: the owner's `Tau_Chain_Descent`, `Tau_Global_Comparison`, `Tau_Homotopy_Boundary_Control`, and the 13 September 2026 specialization/curvature note; the existing reader's restriction formalization at PR26 commit `e382e4f61aa4a61b621a8f064dfefc1ab80640d8`. The present branch also retains main commit `1f7e7c02343884a17df9873566e81a9083c70951` without changing the frozen publication sources.

S40 A quantitative Laplacian bound from the original control form

Source role. Complete public PR27 numerical-range proof P1-P8, with the actual original metric and the explicit unproved tower-growth statement retained. Original display tags retain their complete literal source tokens; the local typesetting namespace CC-P identifies this source.

Exact revision. f127b4ed560fbe5215d24a9337a70be4ae316f518ac0f2d05f2919f4ae491fdb

Complete source. sources/sga_connes_complete_closure/work/connes_pr27_intake_20260913/source/workbenches/tau-specialization-curvature-formal/LAPLACIAN_PARABOLA.md

Source SHA-256. f127b4ed560fbe5215d24a9337a70be4ae316f518ac0f2d05f2919f4ae491fdb

S40.1 A quantitative Laplacian bound from the original control form

13 September 2026. This is a written consequence of the source-preserving identities in HOCHSCHILD_COMPARISON.md, equations (H29)–(H30). It concerns the original Gram metric and does not require diagonalizability. The scalar enclosure below is not included in the selected Lean audit of this package.

Let $G \succ 0$, $k \in \mathbb{R}$, and let the actual arithmetic action be A . Set

$$B = A - \frac{k}{2}I, \quad W = A^*G + GA - kG, \quad L = A^2 - kA. \quad (\text{CC-P P1})$$

Suppose a verified two-sided control is available on that same metric:

$$-\varepsilon G \preceq W \preceq \varepsilon G, \quad \varepsilon \geq 0. \quad (\text{CC-P P2})$$

The source's rank-two formula supplies an explicit way to evaluate the least such ε ; (P2) is not asserted to be uniformly small on growing arithmetic packets.

S40.1.1 Numerical-range enclosure

For every $v \neq 0$, put $z = \langle v, Lv \rangle_G / \|v\|_G^2$. Then

$$\Re z \leq \frac{\varepsilon^2 - k^2}{4}, \quad (\Im z)^2 \leq \varepsilon^2 \left(\frac{\varepsilon^2 - k^2}{4} - \Re z \right). \quad (\text{CC-P P3})$$

In particular every eigenvalue of L lies in the displayed closed left-facing parabola. This is stronger than a statement only about the eigenvalues: it controls the entire numerical range in the original G -inner product.

Proof. Normalize $\|v\|_G = 1$, put $r = \|Bv\|_G$, and define

$$a + ib = \langle v, G^{-1}WBv \rangle_G. \quad (\text{CC-P P4})$$

The operator $G^{-1}W$ is self-adjoint in the G -metric, and (P2) bounds its operator norm by ε . Cauchy-Schwarz gives

$$a^2 + b^2 \leq \varepsilon^2 r^2. \quad (\text{CC-P P5})$$

The exact energy identity $GL = WB - B^*GB - k^2G/4$ gives

$$x := \Re z = a - r^2 - k^2/4, \quad \Im z = b. \quad (\text{CC-P P6})$$

First $a \leq \varepsilon r$, and completing the square proves

$$x \leq \varepsilon r - r^2 - k^2/4 = \frac{\varepsilon^2 - k^2}{4} - (r - \varepsilon/2)^2. \quad (\text{CC-P P7})$$

For the second assertion, compute without discarding the cross term:

$$\varepsilon^2 \left(\frac{\varepsilon^2 - k^2}{4} - x \right) - b^2 = (a - \varepsilon^2/2)^2 + \varepsilon^2 r^2 - a^2 - b^2 \geq 0. \quad (\text{CC-P P8})$$

This proves both assertions, including $\varepsilon = 0$. No spectral simplicity, Euclidean normality, or commutation with the restriction-loss matrix is used.

S40.1.2 What it says about the Connes–Consani comparison

At weight one, the chapter’s arithmetic Laplacian acts as $A(A - I)$. Formula (P3) therefore turns a source-level rank-two control certificate into a quantitative spectral enclosure for that Laplacian. The positivity and adjoint belong to the original $G = R^*R$, whose failure to be invariant was calculated rather than assumed away.

If $0 < \varepsilon < |k|$, the numerical range has strictly negative **real part**. This does not make its spectrum real; it is not the negative-real-spectrum conclusion in Connes and Consani’s Corollary 4.3. At a finite positive error, the parabola retains non-real possibilities.

For a repeated block $A = \rho I + N$, the action remains $L = \rho(\rho - k)I + (2\rho - k)N + N^2$. Nothing in the proof deletes the nilpotent term. An exact positive self-adjoint realization on a full multiple-zero block would impose extra semisimplicity; (P3) avoids making that assumption.

For an actual tensor tower with action containing the character $k\rho$, division by k^2 gives the same enclosure for the fixed scalar $\rho(\rho - 1)$, with error ratio ε_k/k . A separately proved limit $\varepsilon_k/k \rightarrow 0$ would collapse these enclosures to the negative real half-line. This records a precise quantitative contract for the original tower; the limit is not proved by the enclosure itself.

Reference: Connes, A., & Consani, C. (2023). Hochschild homology, trace map and ζ -cycles. In *Cyclic cohomology at 40: Achievements and future prospects* (Vol. 105, pp. 83–101). American Mathematical Society. <https://doi.org/10.1090/pspum/105/01896>

S41 Specialization, invariant hulls, and the original curvature cost

Source role. Complete public PR27 proof note S1-S28, including all local chart, derived-fibre, gcd-hull, signed-current and curvature-certificate proofs. Original display tags retain their complete literal source tokens; the local typesetting namespace CC-S identifies this source.

Exact revision. 1afdd51aff38ffb086f11569a295a914fe3eae7c20c2dca042a3c04017b78239

Complete source. `sources/sga_connes_complete_closure/work/connes_pr27_intake_20260913/source/workbenches/tau-specialization-curvature-formal/RESEARCH_NOTE.md`

Source SHA-256. 1afdd51aff38ffb086f11569a295a914fe3eae7c20c2dca042a3c04017b78239

S41.1 Specialization, invariant hulls, and the original curvature cost

13 September 2026. This note continues the owner's supplied specialization/curvature construction and the verified restriction work of PR26. It records the written geometric consequences and the precise scope of the four new coefficient-level Lean modules. The companion HOCHSCHILD_COMPARISON.md uses the newly supplied Connes–Consani chapter without replacing these source metrics.

S41.1.1 1. Source and finite observation

Fix one of the original blocks $i < j$, its full quotient $E = \mathbb{C}[S]/(\chi)$, and its original action $A = M_S$. Work in the specified polynomial-source coordinates, with positive Gram matrix O_j . Pad the earlier representative into the later source and call it X . Keep

$$B_j X = B_j R_j = 1_E, \quad G_i = X^* O_j X, \quad R_j^* O_j = G_j B_j. \quad (\text{CC-S S1})$$

Put $\delta R = X - R_j$. The original observation gives $B_j \delta R = 0$, while the dual equation gives

$$R_j^* O_j \delta R = 0, \quad C := \delta R^* O_j \delta R = G_i - G_j \succeq 0. \quad (\text{CC-S S2})$$

For $R(z) = R_j + z \delta R$, expansion therefore yields

$$B_j R(z) = 1_E, \quad R(z)^* O_j R(z) = G_j + |z|^2 C. \quad (\text{CC-S S3})$$

The original action defect obeys

$$W(z) = A^* G(z) + G(z) A - k G(z) = W_j + |z|^2 (W_i - W_j). \quad (\text{CC-S S4})$$

All of (S2)–(S4) are composed with the previously verified relation-Gram theorem in `SplitZeroRelationCurve.lean`. A specified constituent $I_F : F \rightarrow E$ is carried by $R(z)I_F$, not by an unrelated spectral model.

S41.1.2 2. The actual local equation and the derived fibre

Let $E_0 = \ker \delta R$ and let E_1 be its G_j -orthogonal complement. Write P_0, P_1 for these projections and $r = \dim E_1 = \text{rank } C$. The source bundle on $\mathbb{P}^1$ is

$$\mathcal{M} = (E_0 \otimes \mathcal{O}) \oplus (E_1 \otimes \mathcal{O}(-1)). \quad (\text{CC-S S5})$$

On the infinity chart $w = 1/z$, its inclusion into the trivial quotient bundle is

$$\alpha(w) = P_0 + w P_1. \quad (\text{CC-S S6})$$

The polynomial local presentation is not merely a pointwise matrix: on free coefficient modules over $K[w]$,

$$(a(w), b(w)) \longmapsto (a(w), w b(w)), \quad (a(w), b(w)) \longmapsto b(0) \quad (\text{CC-S S7})$$

is an injective map followed by a surjective map, with the image of the first equal to the kernel of the second. The proof divides every zero-constant-coefficient polynomial by w . `SplitZeroSpecializationChart.lean` constructs these linear maps and proves all three assertions.

After specialization at $w = 0$, the two-term map is $\text{diag}(1, 0)$, not an injective map. Its kernel is the full modified summand E_1 , and second projection identifies its cokernel with E_1 . Globally, the degree-minus-one fibre is $E_1 \otimes \mathcal{O}(-1)|_\infty$; choosing the local generator in (S7) trivializes that line, but does not erase its transition factor. The local Lean statements prove the kernel/image calculations; they do not construct the derived category of coherent sheaves or an Ext duality functor.

The actual source map on this chart is

$$\Psi_\infty(w) = R_j P_0 + w R_j P_1 + \delta R P_1, \quad B_j \Psi_\infty(w) = P_0 + w P_1. \quad (\text{CC-S S8})$$

It is injective. For $w \neq 0$, apply the observation and the invertibility of (S6). At $w = 0$, a relation $R_j x_0 + \delta R x_1 = 0$ first gives $x_0 = 0$ under B_j , then $x_1 = 0$ because $\delta R|_{E_1}$ is injective. The induced metric on (S5) is thus smooth and positive at infinity even though the quotient comparison has the exceptional fibre above.

S41.1.3 3. The action pole and the least invariant repair

On the overlap, every action block is retained:

$$\alpha(w)^{-1} A \alpha(w) = P_0 A P_0 + w P_0 A P_1 + w^{-1} P_1 A P_0 + P_1 A P_1. \quad (\text{CC-S S9})$$

The polar coefficient vanishes exactly when $A E_0 \subseteq E_0$. This is not assumed. The formal chart theorem proves (S9) and its invariance criterion.

The least action-invariant subspace containing E_0 is

$$F_A = \text{span}\{A^n v : n \geq 0, v \in E_0\}. \quad (\text{CC-S S10})$$

`SplitZeroActionHull.lean` constructs (S10) as a submodule and proves containment, invariance, its universal minimality, monotonicity, idempotence, and the criterion $F_A = E_0$ iff E_0 is invariant. These assertions do not require finite dimension.

When $\dim E = q$, Cayley–Hamilton expresses A^q as a linear combination of the preceding powers. Multiplying this relation by further powers and inducting gives

$$F_A = E_0 + A E_0 + \cdots + A^{q-1} E_0. \quad (\text{CC-S S11})$$

This finite truncation is a written consequence, not an additional claimed Lean target in this package.

Compute the repair directly in the original cyclic quotient Let $p_1, \dots, p_m$ be the polynomial representatives of any basis of E_0 , retaining all multiplicities in χ . Define the monic polynomial

$$d = \gcd(\chi, p_1, \dots, p_m). \quad (\text{CC-S S12})$$

The full inverse image of F_A under $\mathbb{C}[S] \rightarrow E$ is the ideal generated by $\chi, p_1, \dots, p_m$. Indeed invariance under $A = M_S$ is precisely closure under polynomial multiplication. The principal ideal theorem in $\mathbb{C}[S]$ identifies this ideal with (d) . Therefore

$$\boxed{F_A = (d)/(\chi), \quad E/F_A \cong \mathbb{C}[S]/(d).} \quad (\text{CC-S S13})$$

Multiplication by d gives the additional module isomorphism

$$\mathbb{C}[S]/(\chi/d) \xrightarrow{\sim} F_A. \quad (\text{CC-S S14})$$

Surjectivity is immediate from the ideal description. If $df \in (\chi) = (d(\chi/d))$, cancellation in the polynomial domain gives $f \in (\chi/d)$, proving injectivity. The formula also handles $E_0 = 0$, where $d = \chi$, and $d = 1$, where the repair is all of E .

In particular, with $s_A = \dim F_A - \dim E_0$,

$$\boxed{s_A = r - \deg d, \quad \dim(E/F_A) = \deg d.} \quad (\text{CC-S S15})$$

This computes which arithmetic constituents remain at the boundary of the action-stable repair. It does not identify that boundary with the full original quotient.

S41.1.4 4. The repair's metric pole has a signed curvature contribution

The repaired lattice is the preimage of F_A under specialization of $E \otimes \mathcal{O}$ at infinity. It contains $\mathcal{M}$, with torsion quotient of length s_A . In compatible local frames the inclusion determinant vanishes to exactly order s_A . Hence the metric transported from the same source satisfies

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w). \quad (\text{CC-S S16})$$

The determinant metric was not replaced by a newly selected smooth metric on the larger lattice. With $dd^c = (i/2\pi)\partial\bar{\partial}$, the dual determinant's curvature current is

$$\boxed{\mu_A = \mu_{\mathcal{M}} - s_A \delta_{\infty}.} \quad (\text{CC-S S17})$$

To verify the sign, take logarithms in (S16) and use $dd^c \log |w|^2 = \delta_0$. The latter normalization follows from Green's identity on a punctured disk: its shrinking boundary integral is the test function's value at the origin. The smooth part is unchanged off infinity, and the polar logarithm contributes a negative atom to the dual determinant curvature.

The total degree is correspondingly

$$\deg \det(\mathcal{M}_A)^\vee = r - s_A = \deg d. \quad (\text{CC-S S18})$$

Thus an action-stable repair can have degree zero while retaining a nonzero positive curvature density on the affine chart, compensated by a negative atom at infinity. Degree zero does not force the original curvature cost to vanish. This is an exact account of the metric pole, not a counterexample to an unproved arithmetic estimate.

For the declared Gaussian polynomial fixture $\psi(u) = u^3 + u$, $(i, j) = (2, 5)$, the retained zero mode generates the whole cyclic module: its polynomial is coprime to ψ . Then $r = s_A = 2$ and $d = 1$. The initial curvature has degree two, while the repaired current has a negative atom of mass two and total degree zero. This fixture uses the monic Hermite norms with mass seven and the literal phase i^n ; it is not asserted to be an arithmetic zero packet.

S41.1.5 5. Curvature and the existing restriction certificate are the same cost

Let $K_i = G_i^{-1}$, $K_j = G_j^{-1}$, and retain the two distinct operators

$$Z = K_j(G_i - G_j) = (K_j - K_i)G_i, \quad T = K_i G_j, \quad H = 1 - T. \quad (\text{CC-S S19})$$

The original inverse equations give

$$(1 + Z)T = T(1 + Z) = 1, \quad H = Z(1 + Z)^{-1}. \quad (\text{CC-S S20})$$

These identities are formalized directly, without imposing Euclidean Hermitian symmetry on either operator. The cost spectrum of Z is nonnegative because Z is similar, via the original positive Gram matrix, to $G_j^{-1/2} C G_j^{-1/2}$. If $\lambda \geq 0$ is a cost eigenvalue, its return-loss coordinate is

$$x = \lambda/(1 + \lambda) \in [0, 1). \quad (\text{CC-S S21})$$

Every zero mode remains in this finite spectrum. For $t = |z|^2$, radial determinant curvature and cumulative mass are

$$\kappa(t) = \sum_a \frac{\lambda_a}{(1 + t\lambda_a)^2}, \quad M(t) = \sum_a \frac{t\lambda_a}{1 + t\lambda_a}, \quad M'(t) = \kappa(t). \quad (\text{CC-S S22})$$

The scalar derivative and its first derivative are checked in Lean. Changing to the bounded spectrum gives

$$\kappa(t) = \sum_a \frac{x_a(1 - x_a)}{[1 - (1 - t)x_a]^2}. \quad (\text{CC-S S23})$$

The original logarithmic moment is

$$\mathcal{L} = \int_0^1 \log(1/t) \kappa(t) dt = \sum_a \log(1 + \lambda_a) = \sum_a -\log(1 - x_a). \quad (\text{CC-S S24})$$

For completeness, integrate by parts using (S22). At zero the boundary term $M(t) \log(1/t)$ vanishes since $M(t) = O(t)$ for each fixed finite block; it also vanishes at one. The remaining integral is $\sum_a \int_0^1 \lambda_a / (1 + t\lambda_a) dt$, giving (S24). This integral argument is written analysis; the equality between the two finite logarithmic sums is formalized.

Write $s_m = \text{Tr}(H^m) = \sum_a x_a^m$. At $t = 1$,

$$M(1) = s_1, \quad \kappa(1) = s_1 - s_2. \quad (\text{CC-S S25})$$

More generally the rational normalized coefficients are

$$b_n = \sum_a \frac{\lambda_a^{n+1}}{(1 + \lambda_a)^{n+2}} = s_{n+1} - s_{n+2}. \quad (\text{CC-S S26})$$

Direct repeated differentiation gives $b_n = (-1)^n \kappa^{(n)}(1)/(n+1)!$. The all-order differentiation interpretation is written, whereas (S26) itself and its telescoping consequence are formalized:

$$\boxed{s_{p+1} = M(1) - \sum_{n=0}^{p-1} b_n.} \quad (\text{CC-S S27})$$

Thus the curvature data feed the already verified trace-only certificate. Put $P_p = \sum_{m=1}^p s_m/m$. Once $s_p < 1$,

$$\boxed{P_p \leq \mathcal{L} \leq P_p + \frac{P_{2p} - P_p}{1 - s_p}.} \quad (\text{CC-S S28})$$

The upper formula is evaluated on actual complex matrix traces via the previously formalized spectral witness, not by pretending that the original H is Euclidean Hermitian. Every fixed finite block has some successful stopping degree, since $s_p \rightarrow 0$. This is not a uniform stopping-degree or arithmetic upper-growth theorem as the packet grows.

S41.1.6 6. Integration with the new source

The normalized Hochschild trace comparison $\mathcal{U}$ in HOCHSCHILD_COMPARISON.md is isometric on the original half-line source. Applying it to (S3) therefore leaves (S19)–(S28) unchanged. Its arithmetic Laplacian is $A^2 - kA$, not the metric loss H . The exact connection is through $W = A^*G + GA - kG$, whose Laplacian adjoint defect is $A^*W - WA$. The two spectra have different meanings and have not been identified.

The original supported lifts of all coefficient maps are retained. The ordinary local chart $w = 0$, an amplitude killed by critical-line observation, and the structural absence scalar τ are three different data. None is substituted for another.

S41.1.7 Verification boundary

The workflow compiles five new Lean modules individually at `--trust=0` with warnings as errors. Its selected new transitive audits have 29 specialization/curvature declarations and nine Laplacian/intertwiner declarations. It also replays the inherited restriction declarations and regressions. Actual status is recorded in STATUS.md after observing the run; this note is not itself a compiler certificate.

The general source construction, coherent-sheaf globalizations, metric current (S17), cyclic gcd description (S13), all-order derivative formula, and integral (S24) have written proofs above. They are not silently included in the Lean scope. The original source note's 23-method archive was not re-executed in the unavailable local runtime; the new repository's 17-method exact checker is a separate suite, and the companion Laplacian checker has ten methods.

The publication sources in main 1f7e7c02343884a17df9873566e81a9083c70951 and every PR26 addition at e382e4f61aa4a61b621a8f064dfefc1ab80640d8 remain intact on the integration branch. No frozen reader, dependency pin, inherited mathematical source, or main branch is changed by this work.